

7 **Draft Standard for**
8 **Local and Metropolitan Area Networks—**
9 **Bridges and Bridged Networks**

10 Prepared by the
11 **IEEE 802.1 Working Group**

12 Sponsor
13 **LAN/MAN Standards Committee**
14 **of the**
15 **IEEE Computer Society**

16 This and the following cover pages are not part of the draft. They provide revision and other information for
17 IEEE 802.1 Working Group members and will be updated as convenient. New participants: Please read these
18 cover pages, they contain information that should help you contribute effectively to this standards
19 development project. The [Introduction to the current draft](#) should be useful to all readers.

20 The text proper of this draft begins with the [Title page](#).

Important Notice

This document is an unapproved draft of a proposed IEEE Standard. IEEE hereby grants the named IEEE SA Working Group or Standards Committee Chair permission to distribute this document to participants in the receiving IEEE SA Working Group or Standards Committee, for purposes of review for IEEE standardization activities. No further use, reproduction, or distribution of this document is permitted without the express written permission of IEEE Standards Association (IEEE SA). Prior to any review or use of this draft standard, in part or in whole, by another standards development organization, permission must first be obtained from IEEE SA (stds-copyright@ieee.org). This page is included as the cover of this draft, and shall not be modified or deleted.

IEEE Standards Association
445 Hoes Lane
Piscataway, NJ 08854, USA

1 Introduction to the current draft¹

2 This document is a revision in progress of IEEE Std 802.1Q-2022, incorporating published amendments, draft
3 amendments (if, and as, noted on the Title page), the agreed or proposed resolution of [Maintenance items](#)
4 (see below), and additional [Technical corrections](#) and [Editorial corrections](#) to the description of existing
5 functionality.

6 These cover pages also include an introduction to [Participation in 802.1 standards development](#), a summary
7 of the [PAR \(Project Authorization Request\)](#), and a general discussion of [Draft development](#).

8 Maintenance items

9 This draft includes the following proposed or agreed resolutions of maintenance items that were not
10 addressed by rolled-up amendments.

- 11 — Maintenance item [0312](#). See below under [YANG modules](#).
- 12 NOTE—The ieee802-types.yang module also contains a typedef for MAC addresses (typedef mac-address) that
seems to have the same issue. Needs to be discussed in YANGsters if that also needs to be updated.
- 14 — Maintenance item [0366](#). See also
15 [.../docs2024/yangsters-kauer-maintenance-request-0366-0624-v01.pdf](#) and below under [YANG](#)
16 [modules](#).

17 Technical corrections

18 This draft does not include any technical corrections beyond those addressed by the rolled-up amendments.

19 Editorial corrections

20 Pre-publication staff editing of amendments addressed a number of inconsistencies in terminology,
21 capitalisation, etc. That editing was necessarily restricted to the text of each published amendment, though
22 the choice between alternatives was largely determined by predominant use and general rules in the
23 standard as a whole. This draft carries that resolution into the remaining text of the clauses touched by each
24 amendment or the rest of the standard. In some cases the cost benefit of comprehensive change favours
25 devoting time to more significant issues:

26 This draft also includes the following corrections that do not appear in the amendments:

- 27 — Tables in Clause 12 after Table 12-38 were mis-numbered.
- 28 — Clause 50 included “This is a” or “This is a variable” as a repetitive introduction to items in list of
29 variables. Where this can be trivially accomplished, the unnecessary text has been
30 removed/replaced.
- 31 — Figure 25-4 included some unwanted parentheses, and has been updated to remove these.

¹The whole or parts of the introduction, possibly updated, to past drafts may be retained at the Editor’s discretion, with the most recent introduction first. The introduction to each draft may solicit input on specific subjects.

1 Keywords

2 The following Keywords have been added or removed to the Keywords in the frontmatter of the standard
3 (following the Title Page proper which lists the base standard for this revision plus incorporated amendments),
4 following the abstract. The starting point for the Keyword list was IEEE Std 802.1Q-2022, and the list omits
5 some keywords found in amendments prior to that revision and additional Keywords to be found in the
6 subsequent amendments incorporated in this revision. A complete list of past and present keywords would be
7 extensive, and thus possibly undesirable.

8 — D1.6 added “MIB” and “YANG”.

9 MIB modules

10 The MIB modules specified by this standard are attached to the draft pdf as plain text (UTF-8) .mib files.

11 YANG modules

12 The MIB modules specified by this standard are attached to the draft pdf as plain text (UTF-8) .mib files.

13 As noted above, the following YANG modules in this draft have been updated as a result of maintenance
14 activity not addressed by the included amendments:

15 — ieee802-dot1q-tsn-types@2024-03-21 ([48.6.3](#))

16 — ieee802-dot1q-sched@2024-10-22 ([48.6.17](#))

17 — ieee802-dot1q-psfp@2024-10-22 ([48.6.21](#))

18 A Revision statement has been added to each of these modules indicating that they are “TO BE Published ..”
19 as part of the Revision, together with an indication of the change. The “TO BE ” is to be removed and the
20 revision date updated at pre-publication time [the revision statements in published standard reflects only the
21 historical derivation of each published (i.e. standardized) module version from its published predecessors.
22 The description of the change from the prior published version should be retained. Note that these updated
23 modules are not ‘the standard’ at this time.

24 Sources

25 This draft has been prepared from a set of Framemaker files with conditional text that supports the production
26 of an amendment draft and a preliminary rollup of that amendment draft into the text of the base standard,
27 IEEE Std 802.1Q-2022 as amended by prior amendments as of the close of their successful SA ballots.

28 For a description of the use of conditional text and other FrameMaker and IEEE Std 802.1Q Style
29 considerations applicable to this draft see the EDITOR-PLEASE-READ-ME file in the FrameMaker books
30 used to generate this draft.

¹ **This page intentionally left blank**

¹ **This page intentionally left blank**

¹ **This page intentionally left blank**

1 Participation in 802.1 standards development

2 All participants in IEEE 802.1 activities should be aware of the Working Group Policies and Procedures, and
3 their obligations under the IEEE Patent Policy, the IEEE Standards Association (SA) Copyright Policy, and the
4 IEEE SA Participation Policy. For information on these policies see 1.ieee802.org/rules/ and the slides
5 presented at the beginning of each of our Working Group and Task Group meeting.

6 The IEEE SA [PAR \(Project Authorization Request\)](#) is summarized in these cover pages and a link provided to
7 the full text. As part of the IEEE 802® process, the PAR of each project is reviewed regularly to ensure its
8 continued validity. A vote of “Approve” on this draft is also an affirmation that the PAR and CSD is still valid.

9 Comments on this draft are encouraged. NOTE: All issues related to IEEE standards presentation style,
10 formatting, spelling, etc. are routinely handled between the 802.1 Editor and the IEEE Staff Editors prior to
11 publication, after balloting and the process of achieving agreement on the technical content of the standard is
12 complete. Readers are urged to devote their valuable time and energy only to comments that materially affect
13 either the technical content of the document or the clarity of that technical content. Comments should not
14 simply state what is wrong, but also what might be done to fix the problem.

15 Full participation in the work of IEEE 802.1 requires attendance at IEEE 802 meetings. Information on 802.1
16 activities, working papers, and email distribution lists etc. can be found on the 802.1 Website:

17 <http://ieee802.org/1/>

18 Use of the email distribution list is not presently restricted to 802.1 members, and the working group has a
19 policy of considering comments from all who are interested and willing to contribute to the development of the
20 draft. Individuals not attending meetings have helped to identify sources of misunderstanding and ambiguity
21 in past projects. The email lists exist primarily to allow the members of the working group to develop
22 standards, and are not a general forum. All contributors to the work of 802.1 should familiarize themselves
23 with the IEEE patent policy and anyone using the email distribution list will be assumed to have done so.
24 Information can be found at <http://standards.ieee.org/db/patents/>

25 Comments on this draft may be sent to the 802.1 email exploder, to the Editors, or to the Chairs of the 802.1
26 Working Group and Time-Sensitive Networking (TSN) Task Group.

27 Mick Seaman
28 Editor, IEEE Std 802.1Q
29 Email: mickseaman@gmail.com

30 Janos Farkas
31 Chair, 802.1 TSN Task Group
32
33 Email: Janos.Farkas@ericsson.com

Glenn Parsons
Chair, 802.1 Working Group
+1 514-379-9037
Email: glenn.parsons@ericsson.com

34 NOTE: Comments whose distribution is restricted in any way cannot be considered, and may not be
35 acknowledged.

36 **All participants in IEEE standards development have responsibilities under the IEEE patent policy and**
37 **should familiarize themselves with that policy, see**
38 <http://standards.ieee.org/about/sasb/patcom/materials.html>

39 As part of our IEEE 802 process, the text of the PAR and CSD (Criteria for Standards Development, formerly
40 referred to as the 5 Criteria or 5C's) is reviewed on a regular basis in order to ensure their continued validity.
41 A vote of “Approve” on this draft is also an affirmation by the balloter that the PAR is still valid.

1 PAR (Project Authorization Request)

2 The PAR for the Revision of IEEE Std 802.1Q-2022 was approved by IEEE NesCom, June 5th, 2023:

3 <https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/10606>

4 This is a maintenance PAR so did not require an IEEE 802 CSD (Criteria for Standards Development).

5 The Scope and Purpose remains unchanged from that for IEEE Std 802.1Q-2022.

6 The Need for the Project is as follows:

7 “This revision project is needed in order to incorporate approved amendments and corrigenda, to incorporate
8 technical and editorial corrections to existing functionality, and to ensure that consistency is maintained in the
9 consolidated text.”

10 Draft development

11 During the early stages of draft development, 802.1 editors have a responsibility to attempt to craft technically
12 coherent drafts from the resolutions of ballot comments and from the other discussions that take place in the
13 working group meetings. Preparation of drafts often exposes inconsistencies in editor's instructions or
14 exposes the need to make choices between approaches that were not fully apparent in the meeting. Choices
15 and requests by the editors' for contributions on specific issues will be found in the editors' [Introduction to the](#)
16 [current draft](#) and at appropriate points in the draft.

17 Any text with a Cyan background (as in this sentence) is temporary, with conditional tag 'Editor comment',
18 inserted by the Editors to solicit comment, suggest a future change, or act simply as an aide memoire. Text
19 can also be highlighted to draw it to the readers' attention, using conditional tag 'Editor highlight'. In both
20 these cases conditional tagging helps location, and eventual removal, of text or highlighting and can control
21 whether or not it is displayed.

22 The ballot comments received on each draft, and the editors' proposed and final disposition of comments on
23 working group drafts, are part of the audit trail of the development of the standard and are available, along
24 with all the revisions of the draft on the 802.1 website (for address see above).

25 During the early stages of draft development the proposed text can be moved around a great deal, and even
26 minor rearrangement can lead to a lot of 'change', not all of which is noteworthy from the point of the reviewer,
27 so the use of automatic change bars is not very effective. In early drafts change bars may be omitted or
28 applied manually, with a view to drawing the readers attention to the most significant areas of change.
29 Readers interested in viewing every change are encouraged to use Adobe Acrobat to compare the document
30 with their selected prior draft. Note that the FrameMaker change bar feature is useless when it comes to
31 indicating changes to Figures.

32 This draft has been prepared from a set of Framemaker files with conditional text that supports the production
33 of an amendment draft and a preliminary roll up of that amendment draft into the text of the base standard, i.e.
34 IEEE Std 802.1Q as of the last Revision as amended by prior amendments (usually as of the close of their
35 successful SA ballots) as noted on the Title Page and the first Cover Page. The editor may make preliminary
36 roll ups available to check consistency with the base standard and cross-references to text that does not
37 appear in this amendment. Roll ups may also be recorded as part of the approved P802.1Q Revision project.

38 For a description of the use of conditional text and other FrameMaker and IEEE Std 802.1Q Style
39 considerations applicable to this draft see the EDITOR-PLEASE-READ-ME file in the FrameMaker books
40 used to generate these drafts.

41 There are generally multiple amendments under development at any time, and while they will add or amend
42 different clauses in the base standard, there are some clauses (notably Clauses 12, 48, and the PICS
43 Annexes that all are likely to change). They will need to be fully integrated before or during SA Ballot, and
44 complete that ballot in serial order to avoid future problems.

45 Records of participants in the development of the standard are added after SA Ballot, as part of
46 pre-publication editing by IEEE Staff.

(IEEE Std 802.1Q™-2022 as amended by IEEE Std 802.1Qcz™-2023, IEEE Std 802.1Qcw™-2023,
IEEE Std 802.1Qcj™-2023, IEEE Std 802.1Qdj™-2024, IEEE Std 802.1Qdx™-2024, and
IEEE Std 802.1Qdy™-2025)

Draft Standard for Local and Metropolitan Area Networks— Bridges and Bridged Networks

Prepared by the

IEEE 802.1 Working Group

Sponsor

LAN/MAN Standards Committee

of the

IEEE Computer Society

Copyright © 2025 by the IEEE.

Three Park Avenue

New York, New York 10016-5997, USA

All rights reserved.

This document is an unapproved draft of a proposed IEEE Standard. As such, this document is subject to change. USE AT YOUR OWN RISK! IEEE copyright statements SHALL NOT BE REMOVED from draft or approved IEEE standards, or modified in any way. Because this is an unapproved draft, this document must not be utilized for any conformance/compliance purposes. Permission is hereby granted for officers from each IEEE Standards Working Group or Committee to reproduce the draft document developed by that Working Group for purposes of international standardization consideration. IEEE Standards Department must be informed of the submission for consideration prior to any reproduction for international standardization consideration (stds.ipr@ieee.org). Prior to adoption of this document, in whole or in part, by another standards development organization, permission must first be obtained from the IEEE Standards Department (stds.ipr@ieee.org). When requesting permission, IEEE Standards Department will require a copy of the standard development organization's document highlighting the use of IEEE content. Other entities seeking permission to reproduce this document, in whole or in part, must also obtain permission from the IEEE Standards Department.

IEEE Standards Department

445 Hoes Lane

Piscataway, NJ 08854, USA

1

2 **Abstract:** This standard specifies how the Media Access Control (MAC) Service is supported by
3 Bridged Networks, the principles of operation of those networks, and the operation of MAC Bridges
4 and VLAN Bridges, including management, protocols, and algorithms.

5 **Keywords:** Bridged Network, IEEE 802.1Q™, LAN, local area network, MAC Bridge, metropolitan
6 area network, MIB, MSTP, Multiple Spanning Tree Protocol, PBN, Provider Bridged Network, Rapid
7 Spanning Tree Protocol, RSTP, Shortest Path Bridging Protocol, SPB Protocol, Time-Sensitive
8 Networking, TSN, Virtual Bridged Network, virtual LAN, VLAN Bridge, YANG.

9

The Institute of Electrical and Electronics Engineers, Inc.
3 Park Avenue, New York, NY 10016-5997, USA

Copyright © 2025 by the Institute of Electrical and Electronics Engineers, Inc.
All rights reserved. Unapproved draft.

IEEE and 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by the Institute of Electrical and Electronics Engineers, Incorporated.

PDF: ISBN 978-X-XXX-XXX-X STDXXXXX
Print: ISBN 978-X-XXX-XXX-X STDPDXXXXX

IEEE prohibits discrimination, harassment, and bullying.

For more information, visit <http://www.ieee.org/web/aboutus/whatis/policies/p9-26.html>.

No part of this publication may be reproduced in any form, in an electronic retrieval system or otherwise, without the prior written permission of the publisher.

1 Important Notices and Disclaimers Concerning IEEE Standards 2 Documents

3 IEEE Standards documents are made available for use subject to important notices and legal disclaimers.
4 These notices and disclaimers, or a reference to this page (<https://standards.ieee.org/ipr/disclaimers.html>),
5 appear in all IEEE standards and may be found under the heading “Important Notices and Disclaimers
6 Concerning IEEE Standards Documents.”

7 Notice and Disclaimer of Liability Concerning the Use of IEEE Standards 8 Documents

9 IEEE Standards documents are developed within IEEE Societies and subcommittees of IEEE Standards
10 Association (IEEE SA) Board of Governors. IEEE develops its standards through an accredited consensus
11 development process, which brings together volunteers representing varied viewpoints and interests to
12 achieve the final product. IEEE Standards are documents developed by volunteers with scientific, academic,
13 and industry-based expertise in technical working groups. Volunteers involved in technical working groups
14 are not necessarily members of IEEE or IEEE SA and participate without compensation from IEEE. While
15 IEEE administers the process and establishes rules to promote fairness in the consensus development
16 process, IEEE does not independently evaluate, test, or verify the accuracy of any of the information or the
17 soundness of any judgments contained in its standards.

18 IEEE makes no warranties or representations concerning its standards, and expressly disclaims all
19 warranties, express or implied, concerning all standards, including but not limited to the warranties of
20 merchantability, fitness for a particular purpose and non-infringement. IEEE Standards documents do not
21 guarantee safety, security, health, or environmental protection, or compliance with law, or guarantee against
22 interference with or from other devices or networks. In addition, IEEE does not warrant or represent that the
23 use of the material contained in its standards is free from patent infringement. IEEE standards documents are
24 supplied “AS IS” and “WITH ALL FAULTS.”

25 Use of an IEEE standard is wholly voluntary. The existence of an IEEE Standard does not imply that there
26 are no other ways to produce, test, measure, purchase, market, or provide other goods and services related to
27 the scope of the IEEE standard. Furthermore, the viewpoint expressed at the time a standard is approved and
28 issued is subject to change brought about through developments in the state of the art and comments
29 received from users of the standard.

30 In publishing and making its standards available, IEEE is not suggesting or rendering professional or other
31 services for, or on behalf of, any person or entity, nor is IEEE undertaking to perform any duty owed by any
32 other person or entity to another. Any person utilizing any IEEE Standards document should rely upon their
33 own independent judgment in the exercise of reasonable care in any given circumstances or, as appropriate,
34 seek the advice of a competent professional in determining the appropriateness of a given IEEE standard.

35 IN NO EVENT SHALL IEEE BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL,
36 EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO: THE
37 NEED TO PROCURE SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR
38 BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY,
39 WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR
40 OTHERWISE) ARISING IN ANY WAY OUT OF THE PUBLICATION, USE OF, OR RELIANCE UPON
41 ANY STANDARD, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE AND
42 REGARDLESS OF WHETHER SUCH DAMAGE WAS FORESEEABLE.

43 Translations

44 The IEEE consensus balloting process involves the review of documents in English only. In the event that an
45 IEEE standard is translated, only the English language version published by IEEE is the approved IEEE
46 standard.

1 Use by artificial intelligence systems

2 In no event shall material in any IEEE Standards documents be used for the purpose of creating, training,
3 enhancing, developing, maintaining, or contributing to any artificial intelligence systems without the
4 express, written consent of the IEEE SA in advance. “Artificial intelligence” refers to any software,
5 application, or other system that uses artificial intelligence, machine learning, or similar technologies, to
6 analyze, train, process, or generate content. Requests for consent can be submitted using the [Contact Us](#)
7 form.¹

8 Official statements

9 A statement, written or oral, that is not processed in accordance with the IEEE SA Standards Board
10 Operations Manual is not, and shall not be considered or inferred to be, the official position of IEEE or any
11 of its committees and shall not be considered to be or be relied upon as, a formal position of IEEE or IEEE
12 SA. At lectures, symposia, seminars, or educational courses, an individual presenting information on IEEE
13 standards shall make it clear that the presenter’s views should be considered the personal views of that
14 individual rather than the formal position of IEEE, IEEE SA, the Standards Committee, or the Working
15 Group. Statements made by volunteers may not represent the formal position of their employer(s) or
16 affiliation(s). News releases about IEEE standards issued by entities other than IEEE SA should be
17 considered the view of the entity issuing the release rather than the formal position of IEEE or IEEE SA.

18 Comments on standards

19 Comments for revision of IEEE Standards documents are welcome from any interested party, regardless of
20 membership affiliation with IEEE or IEEE SA. However, **IEEE does not provide interpretations,**
21 **consulting information, or advice pertaining to IEEE Standards documents.**

22 Suggestions for changes in documents should be in the form of a proposed change of text, together with
23 appropriate supporting comments. Since IEEE standards represent a consensus of concerned interests, it is
24 important that any responses to comments and questions also receive the concurrence of a balance of interests.
25 For this reason, IEEE and the members of its Societies and subcommittees of the IEEE SA Board of
26 Governors are not able to provide an instant response to comments or questions except in those cases where
27 the matter has previously been addressed. For the same reason, IEEE does not respond to interpretation
28 requests. Any person who would like to participate in evaluating comments or revisions to an IEEE standard is
29 welcome to join the relevant IEEE SA working group. You can indicate interest in a working group using the
30 Interests tab in the Manage Profile & Interests area of the [IEEE SA myProject system](#).² An IEEE Account is
31 needed to access the application.

32 Comments on standards should be submitted using the [Contact Us](#) form.¹

33 Laws and regulations

34 Users of IEEE Standards documents should consult all applicable laws and regulations. Compliance with the
35 provisions of any IEEE Standards document does not constitute compliance to any applicable regulatory
36 requirements. Implementers of the standard are responsible for observing or referring to the applicable
37 regulatory requirements. IEEE does not, by the publication of its standards, intend to urge action that is not
38 in compliance with applicable laws, and these documents may not be construed as doing so.

39 Data privacy

40 Users of IEEE Standards documents should evaluate the standards for considerations of data privacy and
41 data ownership in the context of assessing and using the standards in compliance with applicable laws and
42 regulations.

¹ Available at: <https://standards.ieee.org/content/ieee-standards/en/about/contact/>.

² Available at: <https://development.standards.ieee.org/myproject-web/public/view.html#landing>.

1 Copyrights

2 IEEE draft and approved standards are copyrighted by IEEE under U.S. and international copyright laws.
3 They are made available by IEEE and are adopted for a wide variety of both public and private uses. These
4 include both use, by reference, in laws and regulations, and use in private self-regulation, standardization,
5 and the promotion of engineering practices and methods. By making these documents available for use and
6 adoption by public authorities and private users, neither IEEE nor its licensors waive any rights in copyright
7 to the documents.

8 Photocopies

9 Subject to payment of the appropriate licensing fees, IEEE will grant users a limited, non-exclusive license
10 to photocopy portions of any individual standard for company or organizational internal use or individual,
11 non-commercial use only. To arrange for payment of licensing fees, please contact Copyright Clearance
12 Center, Customer Service, 222 Rosewood Drive, Danvers, MA 01923 USA; +1 978 750 8400;
13 <https://www.copyright.com/>. Permission to photocopy portions of any individual standard for educational
14 classroom use can also be obtained through the Copyright Clearance Center.

15 Updating of IEEE Standards documents

16 Users of IEEE Standards documents should be aware that these documents may be superseded at any time
17 by the issuance of new editions or may be amended from time to time through the issuance of amendments,
18 corrigenda, or errata. An official IEEE document at any point in time consists of the current edition of the
19 document together with any amendments, corrigenda, or errata then in effect.

20 Every IEEE standard is subjected to review at least every 10 years. When a document is more than 10 years
21 old and has not undergone a revision process, it is reasonable to conclude that its contents, although still of
22 some value, do not wholly reflect the present state of the art. Users are cautioned to check to determine that
23 they have the latest edition of any IEEE standard.

24 In order to determine whether a given document is the current edition and whether it has been amended
25 through the issuance of amendments, corrigenda, or errata, visit [IEEE Xplore](#) or [contact IEEE](#).³ For more
26 information about the IEEE SA or IEEE's standards development process, visit the IEEE SA Website.

27 Errata

28 Errata, if any, for all IEEE standards can be accessed on the [IEEE SA Website](#).⁴ Search for standard number
29 and year of approval to access the web page of the published standard. Errata links are located under the
30 Additional Resources Details section. Errata are also available in [IEEE Xplore](#). Users are encouraged to
31 periodically check for errata.

32 Patents

33 IEEE Standards are developed in compliance with the [IEEE SA Patent Policy](#).⁵

34 Attention is called to the possibility that implementation of this standard may require use of subject matter
35 covered by patent rights. By publication of this standard, no position is taken by the IEEE with respect to the
36 existence or validity of any patent rights in connection therewith. If a patent holder or patent applicant has
37 filed a statement of assurance via an Accepted Letter of Assurance, then the statement is listed on the
38 IEEE SA Website at <https://standards.ieee.org/about/sasb/patcom/patents.html>. Letters of Assurance may
39 indicate whether the Submitter is willing or unwilling to grant licenses under patent rights without
40 compensation or under reasonable rates, with reasonable terms and conditions that are demonstrably free of
41 any unfair discrimination to applicants desiring to obtain such licenses.

³ Available at: <https://ieeexplore.ieee.org/browse/standards/collection/ieee>.

⁴ Available at: <https://standards.ieee.org/standard/index.html>.

⁵ Available at: <https://standards.ieee.org/about/sasb/patcom/materials.html>.

1 Essential Patent Claims may exist for which a Letter of Assurance has not been received. The IEEE is not
2 responsible for identifying Essential Patent Claims for which a license may be required, for conducting
3 inquiries into the legal validity or scope of Patents Claims, or determining whether any licensing terms or
4 conditions provided in connection with submission of a Letter of Assurance, if any, or in any licensing
5 agreements are reasonable or non-discriminatory. Users of this standard are expressly advised that
6 determination of the validity of any patent rights, and the risk of infringement of such rights, is entirely their
7 own responsibility. Further information may be obtained from the IEEE Standards Association.

8 **IMPORTANT NOTICE**

9 IEEE Standards do not guarantee or ensure safety, security, health, or environmental protection, or ensure
10 against interference with or from other devices or networks. Technologies, application of technologies, and
11 recommended procedures in various industries evolve over time. The IEEE standards development process
12 allows participants to review developments in industries, technologies, and practices, and to determine what,
13 if any, updates should be made to the IEEE standard. During this evolution, the technologies and
14 recommendations in IEEE standards may be implemented in ways not foreseen during the standards
15 development. IEEE Standards development activities consider research and information presented to the
16 standards development group in developing any safety recommendations. Other information about safety
17 practices, changes in technology or technology implementation, or impact by peripheral systems also may
18 be pertinent to safety considerations during implementation of the standard. Implementers and users of IEEE
19 Standards documents are responsible for determining and complying with all appropriate safety, security,
20 environmental, health, data privacy, and interference protection practices and all applicable laws and
21 regulations.

1 Participants

2 <<The following lists will be updated in the usual way prior to publication>>

3 At the time this standard was submitted to the IEEE-SA Standards Board for approval, the IEEE 802.1
4 Working Group had the following membership:

5 **Glenn Parsons**, *Chair*
6 **Jessy V. Rouyer**, *Vice Chair*
7 **Mick Seaman**, *Editor; Chair, Security Task Group*
8 **János Farkas**, *Chair, Time-Sensitive Networking Task Group*
9 **David McCall**, *Vice Chair, Time-Sensitive Networking Task Group*
10 **Mark Hantel**, *Chair, Maintenance Task Group*
11

<<TBA>>

¹ The following members of the individual balloting committee voted on this standard. Balloters may have
² voted for approval, disapproval, or abstention.

<<TBA>>

³ When the IEEE-SA Standards Board approved this standard on XX Month 20xx, it had the following
⁴ membership:

⁵ <<TBA>>

<<TBA>>

⁶
⁷ *Member Emeritus

1 Historical participants

2 Since the initial publication, many IEEE standards have added functionality or provided updates to material
3 included in this standard. The following is a historical list of participants who have dedicated their valuable
4 time, energy, and knowledge to the creation of this material:

IEEE 802.1Q Standard	Date approved by IEEE	Officers at the time of Working Group Ballot
IEEE Std 802.1Q-1998	8 December 1998	William P. Lidinsky , <i>Chair</i> Mick Seaman , <i>Chair, Interworking Task Group</i> Tony Jeffree , <i>Coordinating Editor</i> Anil Rijasinghani, Richard Hausmann, Michele Wright, Paul Langille, P. J. Singh , <i>Editorial Team</i>
IEEE Std 802.1u-2001	17 March 2001	Tony Jeffree , <i>Chair</i> Neil Jarvis , <i>Vice Chair</i> Mick Seaman , <i>Chair, Interworking Task Group</i>
IEEE Std 802.1v-2001	17 March 2001	Tony Jeffree , <i>Chair</i> Neil Jarvis , <i>Vice Chair</i> Mick Seaman , <i>Chair, Interworking Task Group</i> David Delany , <i>Editor</i> Andrew Smith , <i>Editor</i>
IEEE Std 802.1s-2002	11 December 2002	Tony Jeffree , <i>Chair</i> Neil Jarvis , <i>Vice Chair</i> Mick Seaman , <i>Chair, Interworking Task Group</i> Norman W. Finn , <i>Editor</i>
IEEE Std 802.1ad-2005	28 March 2005	Tony Jeffree , <i>Chair</i> Paul Congdon , <i>Vice Chair</i> Mick Seaman , <i>Chair, Interworking Task Group</i> Stephen R. Haddock , <i>Editor</i>
IEEE Std 802.1Q-2005	7 December 2005	Tony Jeffree , <i>Chair and Editor</i> Paul Congdon , <i>Vice Chair</i> Mick Seaman , <i>Chair, Interworking Task Group</i>
IEEE Std 802.1ak-2007	22 March 2007	Tony Jeffree , <i>Chair and Editor</i> Paul Congdon , <i>Vice Chair</i> Mick Seaman , <i>Chair, Interworking Task Group</i>
IEEE Std 802.1ag-2007	27 September 2007	Tony Jeffree , <i>Chair</i> Paul Congdon , <i>Vice Chair</i> Stephen R. Haddock , <i>Chair, Interworking Task Group</i> Norman W. Finn , <i>Editor-in-Chief</i> David V. Elie-Dit-Cosaque, Dinesh Mohan, Oscar Rodriguez, Ali Sajassi , <i>Assistant Editors</i>

1

IEEE 802.1Q Standard	Date approved by IEEE	Officers at the time of Working Group Ballot
IEEE Std 802.1ah-2008	12 June 2008	Tony Jeffree, Chair Paul Congdon, Vice Chair Stephen R. Haddock, Chair, Interworking Task Group Paul Bottorff, Stephen Haddock, and Muneyoshi Suzuki, Editors
IEEE Std 802.1Q-2005/Cor-1-2008	26 September 2008	Tony Jeffree, Chair and Editor Paul Congdon, Vice Chair Stephen R. Haddock, Chair, Interworking Task Group
IEEE Std 802.1ap-2008	10 December 2008	Tony Jeffree, Chair Paul Congdon, Vice Chair Stephen R. Haddock, Chair, Interworking Task Group Glenn Parsons, Editor David Levi, Assistant Editor
IEEE Std 802.1Qaw-2009	17 June 2009	Tony Jeffree, Chair Paul Congdon, Vice Chair Stephen R. Haddock, Chair, Interworking Task Group Linda Dunbar, Editor
IEEE Std 802.1Qay-2009	17 June 2009	Tony Jeffree, Chair Paul Congdon, Vice Chair Stephen R. Haddock, Chair, Interworking Task Group Panagiotis Saltsidis, Editor
IEEE Std 802.1aj-2009	9 December 2009	Tony Jeffree, Chair Paul Congdon, Vice Chair Stephen R. Haddock, Chair, Interworking Task Group John Messenger, Editor Brian Hassink, MIB Editor
IEEE Std 802.1Qav-2009	9 November 2009	Tony Jeffree, Chair and Editor Paul Congdon, Vice Chair Michael Johas Teener, Chair, Audio Video Bridging Task Group
IEEE Std 802.1Qau-2010	25 March 2010	Tony Jeffree, Chair Paul Congdon, Vice Chair Patricia Thaler, Chair, Data Center Bridging Task Group Norman W. Finn, Editor
IEEE Std 802.1Qat-2010	30 September 2010	Tony Jeffree, Chair Paul Congdon, Vice Chair Michael Johas Teener, Chair, Audio Video Bridging Task Group Craig Gunther, Editor

1

IEEE 802.1Q Standard	Date approved by IEEE	Officers at the time of Working Group Ballot
IEEE Std 802.1Q-2011	16 May 2011	Tony Jeffree , <i>Chair and Editor</i> Paul Congdon , <i>Vice Chair</i> Stephen Haddock , <i>Chair, Interworking Task Group</i>
IEEE Std 802.1Qbc-2011	16 June 2011	Tony Jeffree , <i>Chair</i> Paul Congdon , <i>Vice Chair</i> Stephen Haddock , <i>Chair, Interworking Task Group</i> Norman Finn , <i>Editor</i>
IEEE Std 802.1Qbc-2011	16 June 2011	Tony Jeffree , <i>Chair</i> Paul Congdon , <i>Vice Chair</i> Stephen Haddock , <i>Chair, Interworking Task Group</i> Thomas Mack-Crane , <i>Editor</i>
IEEE Std 802.1Qbb-2011	16 June 2011	Tony Jeffree , <i>Chair</i> Paul Congdon , <i>Vice Chair</i> Patricia Thaler , <i>Chair, Data Center Bridging Task Group</i> Claudio DeSanti , <i>Editor</i>
IEEE Std 802.1Qaz-2011	16 June 2011	Tony Jeffree , <i>Chair</i> Paul Congdon , <i>Vice Chair</i> Patricia Thaler , <i>Chair, Data Center Bridging Task Group</i> Craig W. Carlson , <i>Editor</i>
IEEE Std 802.1Qbf-2011	7 December 2011	Tony Jeffree , <i>Chair</i> Paul Congdon , <i>Vice Chair</i> Stephen Haddock , <i>Chair, Interworking Task Group</i> Robert Sultan , <i>Editor</i>
IEEE Std 802.1aq-2012	29 March 2012	Tony Jeffree , <i>Chair</i> Glenn Parsons , <i>Vice Chair</i> Stephen Haddock , <i>Chair, Interworking Task Group</i> Donald Fedyk , Mick Seaman , <i>Editors</i>
IEEE Std 802.1Qbg-2012	14 May 2012	Tony Jeffree , <i>Chair and Editor</i> Paul Congdon , <i>Vice Chair</i> Patricia Thaler , <i>Chair, Data Center Bridging Task Group</i> Paul Bottorff , <i>Editor, Clauses 12 and 17</i>
IEEE Std 802.1Q-2011/Cor-2-2012	19 October 2012	Tony Jeffree , <i>Chair and Editor</i> Glenn Parsons , <i>Vice Chair and Chair, Maintenance Task Group</i>
IEEE Std 802.1Qbp-2014	27 March 2014	Tony Jeffree , <i>Chair</i> Glenn Parsons , <i>Vice Chair</i> Stephen Haddock , <i>Chair, Interworking Task Group</i> Ben Mack-Crane , <i>Editor</i>

1

IEEE 802.1Q Standard	Date approved by IEEE	Officers at the time of Working Group Ballot
IEEE Std 802.1D-2004	9 February 2004	Tony Jeffree , <i>Chair and Editor</i> Paul Congdon , <i>Vice Chair</i> Mick Seaman , <i>Chair, Interworking Task Group and Editor</i>
IEEE Std 802.1Q-2014	3 November 2014	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair</i> Tony Jeffree , <i>Editor</i> Stephen Haddock , <i>Chair, Interworking Task Group</i>
IEEE Std 802.1Qcd-2015	16 February 2015	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice Chair</i> Patricia Thaler , <i>Chair, Data Center Bridging Task Group</i> Eric Multanen , <i>Editor</i>
IEEE Std 802.1Qca-2015	3 September 2015	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair</i> János Farkas , <i>Editor</i> Stephen Haddock , <i>Chair, Interworking Task Group</i>
IEEE Std 802.1Q-2014/Cor 1-2015	5 December 2015	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair and Chair, Maintenance Task Group</i> Tony Jeffree , <i>Editor</i>
IEEE Std 802.1Qbv-2015	5 December 2015	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair</i> Michael Johas Teener , <i>Chair, Time-Sensitive Networking Task Group</i> Tony Jeffree , <i>Editor</i>
IEEE Std 802.1Qbu-2016	30 June 2016	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair</i> Michael Johas Teener , <i>Chair, Time-Sensitive Networking Task Group</i> Tony Jeffree , <i>Editor</i>
IEEE Std 802.1Qbz-2016	30 June 2016	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair</i> Michael Johas Teener , <i>Chair, Time-Sensitive Networking Task Group</i> Norm Finn , <i>Editor</i>
IEEE Std 802.1Qci-2017	14 February 2017	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Tony Jeffree , <i>Editor</i>
IEEE Std 802.1Qch-2017	15 May 2017	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice-Chair</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Tony Jeffree , <i>Editor</i>

1

IEEE 802.1Q Standard	Date approved by IEEE	Officers at the time of Working Group Ballot
IEEE Std 802.1Q-2018	7 May 2018	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice Chair, Chair of Maintenance Task Group, Editor</i>
IEEE Std 802.1Qcc-2018	14 June 2018	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice Chair and Acting Chair</i> Jessy V. Rouyer , <i>Acting Vice Chair</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Rodney Cummings , <i>Editor</i>
IEEE Std 802.1Qcp-2018	14 June 2018	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice Chair and Acting Chair</i> Jessy V. Rouyer , <i>Acting Vice Chair</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Marc Holness , <i>Editor</i>
IEEE Std 802.1Qcy-2019	21 March 2019	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice Chair and Acting Chair</i> Jessy V. Rouyer , <i>Acting Vice Chair</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Yizhou Li , <i>Editor</i> Paul Bottorff , <i>Editor</i>
IEEE Std 802.1Qcx-2020	4 June 2020	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice Chair</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Marc Holness , <i>Editor</i>
IEEE Std 802.1Qcr-2020	24 September 2020	Glenn Parsons , <i>Chair</i> John Messenger , <i>Vice Chair</i> Jessy V. Rouyer , <i>Secretary</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Craig Gunther , <i>Vice Chair, Time-Sensitive Networking Task Group</i> Johannes Specht , <i>Editor</i>
IEEE Std 802.1Q-2022	21 September 2022	Glenn Parsons , <i>Chair</i> Jessy V. Rouyer , <i>Vice Chair</i> John Messenger , <i>Editor</i> Mick Seaman , <i>Editor</i>
IEEE Std 802.1Qcz-2023	5 June 2023	Glenn Parsons , <i>Chair</i> Jessy V. Rouyer , <i>Vice Chair</i> János Farkas , <i>Chair, Time-Sensitive Networking Task Group</i> Craig Gunther , <i>Vice Chair, Time-Sensitive Networking Task Group</i> Paul Congdon , <i>Editor</i>

1

IEEE 802.1Q Standard	Date approved by IEEE	Officers at the time of Working Group Ballot
IEEE Std 802.1Qcw-2023	17 November 2023	Glenn Parsons, <i>Chair</i> Jessy V. Rouyer, <i>Vice Chair</i> János Farkas, <i>Chair, Time-Sensitive Networking Task Group</i> Craig Gunther, <i>Vice Chair, Time-Sensitive Networking Task Group</i> Marina Gutiérrez, <i>Editor</i>
IEEE Std 802.1Qcj-2023	17 November 2023	Glenn Parsons, <i>Chair</i> Jessy V. Rouyer, <i>Vice Chair</i> János Farkas, <i>Chair, Time-Sensitive Networking Task Group</i> Craig Gunther, <i>Vice Chair, Time-Sensitive Networking Task Group</i> Paul Borttorff, <i>Editor</i>
IEEE Std 802.1Qdj-2024	20 May 2024	Glenn Parsons, <i>Chair</i> Jessy V. Rouyer, <i>Vice Chair</i> János Farkas, <i>Chair, Time-Sensitive Networking Task Group</i> Craig Gunther, <i>Vice Chair, Time-Sensitive Networking Task Group</i> Stephan Kehrer, <i>Editor</i>
IEEE Std 802.1Qdx-2024	6 June 2024	Glenn Parsons, <i>Chair</i> Jessy V. Rouyer, <i>Vice Chair</i> János Farkas, <i>Chair, Time-Sensitive Networking Task Group</i> Craig Gunther, <i>Vice Chair, Time-Sensitive Networking Task Group</i> Abdul Jabbar, <i>Editor</i>
IEEE Std 802.1Qdy-2025	12 February 2025	Glenn Parsons, <i>Chair</i> Jessy V. Rouyer, <i>Vice Chair</i> János Farkas, <i>Chair, Time-Sensitive Networking Task Group</i> David McCall, <i>Vice Chair, Time-Sensitive Networking Task Group</i> Martin Mittelberger, <i>Editor</i>

1

Osama Aboul-Magd
Steve Adams
Stephen Ades
Fumio Akashi
Zehavit Alon
Ken Alonge
Ann Ambler
Paul D. Amer
Yafan An
Ting Ao
Charles Arnold
Peter Ashwood-Smith
Siamack Ayandeh
Floyd Backes
Ann Ballard
Richard Bantel
John Bartlett
Sy Bederman
Alexei Beliaev
Les Bell
Amatzia Ben-Artzi
Avner Ben-Dor
Michael Berger
Caitlin Bestler
Jan Bialkowski
James S. Binder
Robert Bledsoe
Rob Boatright
Kwami Boakaye
Christian Boiger
Brad Booth
Jean-Michel Bonnamy
Mike Borza
Paul Bottorff
David Brady
Rudolf Brandner
Martin Brewer
Frank Bruns
Juan Bulnes
Bill Bunch
Jim Burns
Peter Carbone
Bob Cardinal
Craig W. Carlson
Paul Carroll
Marco Carugi
Jeffrey Catlin
Dennis Cave
Dirceu Cavendish
Alan Chambers
Steve Chan
David W. Chang
Xin Chang
Frank Chao
Ken Chapman
Alice Chen
David Chen
Feng Chen
Weiyang Cheng
Rao Cherukuri
Taesik Cheung
Jade Chien
Hon Wah Chin

Chi Chong
Jin-Seek Choi
Chris Christ
Marc Cochran
Paul Congdon
Glenn Connery
Alex Conta
Jim Corrigan
Diego Crupnicoff
David Cullerot
Rodney Cummings
Uri Cummings
Ted Davies
Andy Davis
Peter Dawe
Stan Degen
Arjan de Heer
Frank Deignan
David Delaney
Prakash Desai
Claudio DeSanti
Ron Dhondy
Patrick Diamond
Aboubacar Kader Diarra
Jeffrey Dietz
Russell Dietz
Zheming Ding
Kurt Dobbins
Eiji Doi
Barbara J. Don Carlos
Linda Dunbar
Craig Easley
Donald Eastlake, III
Peter Ecclesine
J. J. Ekstrom
Anush Elangovan
Hesham M. Elbakoury
Walter Eldon
David Elie-Dit-Cosaque
János Farkas
Donald W. Fedyk
Eldon D. Feist
Felix Feifei Feng
Norman Finn
Len Fishler
Kevin Flanagan
Yishai Fraenkel
Paul Frantz
David Frattura
Robert Frazier
Lars Henrik Frederiksen
Andre Fredette
John Fuller
Ilango Ganga
Geoffrey Garner
Anoop Ghanwani
Franz Goetz
Pat Gonia
Gerrard Goubert
Richard Graham
Mark Gravel
Michael A. Gravel
Eric W. Gray

Karanvir Grewal
John Grinham
Yingjie Gu
Craig Gunther
Mitch Gusat
Stephen Haddock
Sharam Hakimi
Mogens Hansen
Mark Hantel
Harold Harrington
John Hart
Scott Harvell
Mike Harvey
Takashi Hasegawa
Brian Hassink
Wayne Hathaway
Brian Hausauer
Richard Hausman
Hitoshi Hayakawa
Vic Hayes
Asif Hazarika
David Head
Gaby Hecht
Deepak Hegde
Ariel Hendel
Bob Herbst
John Hickey
Jeremy Hitt
David Hollender
Marc Holness
Steve Horowitz
Bob Hott
Michelle Hsiung
Charles Hudson
Jack R. Hung
Rita Hunt
David Husak
Altaf Hussain
Rahil Hussain
Thomas Hytry
Romain Insler
Ran Ish-Shalom
Jay Israel
Atsushi Iwata
Vipin K. Jain
Neil Jarvis
Tony Jeffree
Paul Hongkyu Jeong
Pankaj Jha
Markus Jochim
Michael Johas Teener
Girault Jones
Peter Jones
Shyam Kaluve
Daya Kamath
Abhay Karandikar
Allen Kasey
Prakash Kashyap
Toyayuki Kato
Manu Kaycee

1

Hal Keen	Bruce McClure	Karen Randall
Srikanth Keesara	Tom McGowan	Shlomo Reches
Stephan Kehrer	Alan McGuire	Frank Reichstein
Daniel Kelley	James McIntosh	Dick Reohr
Kevin Ketchum	Martin McNealis	Trudy Reusser
Keti Kilcrease	Menucher Menuchery	James Richmond
Doyeon Kim	Milan Merhar	Maximilian Riegel
Yongbum Kim	Margaret A. Merrick	Anil Rijasinghani
Alan Kirby	John Messenger	Robert Roden
Kimberly Kirkpatrick	Colin Mick	Edouard Rocher
Keith Klamm	Amol Mitra	Gunter Roeck
Steve Kleiman	Dinesh Mohan	John J. Roese
Philippe Klein	Gabriel Montenegro	Josef Roese
Oliver Kleinberg	Jim Montrose	Derek J. Rohde
Marcel Kiessling	Matthew Mora	Allyn Romanow
Yongbum Kim	John Morris	Dan Romascanu
Philippe Klein	Bob Moskowitz	Paul Rosenblum
Bruce Kling	Eric Multanen	Moran Roth
Walter Knitl	Yaron Nachman	Jessy V. Rouyer
Mike Ko	Yukihiro Nakagawa	Doug Ruby
Raghu Kondapalli	Hiroki Nakano	Eric Ryu
Jouni Korhonen	Krishna Narayanaswamy	Jonathan Sadler
Michael Krause	Lawrence Ng	Ali Sajassi
Dan Krent	Henry Ngai	Dolors Sala
James Kristof	Paul Nikolich	Joseph Salowey
Vinod Kumar	Kevin Nolith	John Salter
Paul Kummer	Bob Noseworthy	Panagiotis Saltsidis
Bruce Kwan	Don O'Connor	Sam Sambasivan
Paul Lachapelle	Karen O'Donoghue	Ray Samora
Kari Laihonon	Jerry O'Keefe	Behcet Sarikaya
Ashvin Lakshmikantha	Eugene O'Neil	Alan Sarsby
Bill Lane	Satoshi Obara	Satish Sathe
Paul Langille	Hiroshi Ohta	John M. Sauer
Roger Lapuh	David Olsen	Ayman Sayed
H. Eugene Latham	Toshio Ooka	Susan Schanning
Loren Larsen	Jörg Ottensmeyer	Ted Schroeder
Yannick Le Goff	Shlomo Ovardia	Benjamin Schultz
Marcus Leech	Vijoy Pandey	Mick Seaman
John Lemon	Don Pannell	Gerry Segal
Michael Lerer	Luc Pariseau	Rich Seifert
Lin Li	Glenn W. Parsons	Lee Sendelbach
Bing Liao	Richard Patti	Koichiro Seto
George Lin	Ken Patton	Daniel Sexton
William P. Lidinsky	Mark Pearson	Himanshu Shah
Johann Lindmeyr	Joseph Pelissier	Rakesh Sharma
Marina Lipshteyn	Yonadav Perry	Ravi Shenoy
Gary Littleton	David Peterson	Howard Sherry
Robert D. Love	Roger Pfister	K. Karl Shimada
Yuanqui Luo	Thomas L. Phinney	Fred Shu
Andy Luque	John Pickens	Wu-Shi Shung
Jeff Lynch	Walter Pienciak	Taeshi Shimizu
Gael Mace	Daniel Pitt	Phil Simmons
Thomas Mack-Crane	Hayim Porat	Curtis Simonson
Phillip Magnuson	Gideon Prat	Paramjeet Singh
Christophe Mangin	Kirk Preiss	Rosemary V. Slager
Mahalingam Mani	Ron L. G. Prince	Alexander Smith
David Martin	Max Pritikin	Andrew Smith
Peter Martini	Ray Qiu	
Riccardo Martinotti	Rene Raeber	
Marco Mascitto	Ananda Rajagopal	
Tom McBeath	Steve Ramberg	
Keith McCloghrie	Nigel Ramsden	

1

Michel Soerensen
M. Soha
Stuart Soloway
Johannes Specht
Nurit Sprecher
Kevin B. Stanton
Larry Stefani
Wilfried Steiner
Dan Stokesberry
Sundar Subramaniam
Robert Sultan
Muneyoshi Suzuki
Yoshihiro Suzuki
George Swallow
Lennart Swartz
Richard Sweatt
Vahid Tabatabaee
Attila Takacs
Kenta Takumi
Francois Tallet
Robin Tasker
Angus Telfer
John Terry
Patricia A. Thaler
Jonathan Thatcher
Dave Thompson
Geoff Thompson
David Thornburg
Oliver Thorp
Michel Thorsen

Fouad Tobagi
Nathan Tobol
Jeremy Touve
Naoki Tsukutari
Fred Tuck
Chait Tumuluri
Wendell Turner
Paul Unbehagen
Dhadesugoor Vaman
Steve Van Seters
Dono van-Mierop
Peter Videcranz
John Viega
Maarten Vissers
Dennis Volpano
Manoj Wadekar
Paul Wainwright
Scott Wasson
Daniel Watts
Yuehua Wei
John Wakerly
Peter Wang
Philip Wang
Y. C. Wang
Yan Wang
Trevor Warwick
Bob Watson
Karl Weber
Yuehua Wei
Brian Weis

Alan Weissberger
Glenn Wenig
Martin White
Bert Wijnen
Deborah Wilbert
Keith Willette
Robert Williams
Val Wilson
Ludwig Winkel
Robert Winter
Michael Witkowski
Edward Wong
Jordon Woods
Michael D. Wright
Michele Wright
Chien-Hsien Wu
Min Xiao
Ken Young
Allen Yu
Wayne Zakowski
Igor Zhovnirovsky
Carolyn Zimmer
Helge Zinner
Glen Zorn
Nick Zuccherro
Juan-Carlos Zuniga

1 Introduction

This introduction is not part of IEEE Std 802.1Q™-20XX, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks.

2 IEEE Std 802.1Q™-20XX incorporated the text of the following amendments into IEEE Std 802.1Q™.

3 IEEE Std 802.1Qdy™-2025: YANG for Multiple Spanning Trees addresses requirements arising from
4 industrial automation networks, specifying YANG for bridge and bridge component MSTP configuration
5 and status reporting.

6 IEEE Std 802.1Qdx™-2024: YANG Data Models for the Credit-Based Shaper addressed the need to
7 manage the credit-based shaper algorithm via a YANG model compatible with modern management
8 systems.

9 IEEE Std 802.1Qdj™-2024: Configuration Enhancements for Time-Sensitive Networks described the
10 relationships and division of responsibilities between Centralized User Configuration (CUC) components,
11 that can be used to configure end stations' use of Time-Sensitive Networking (TSN) capabilities, and a
12 Centralized Network Configuration (CNC) component that can be used to configure network resources
13 within an administrative Configuration Domain. This standard also included a YANG model and modules
14 that can be used by a network configuration protocol, such as NETCONF, to provide communication
15 between the CNC and a CUC.

16 IEEE Std 802.1Qcj™-2023: Automatic Attachment to Provider Backbone Bridging (PBB) Services
17 specified protocols, procedures and management objects for auto attachment of network devices to Provider
18 Backbone service instances using Type, Length, Value (TLVs) within the Link Layer Discovery Protocol
19 (LLDP).

20 IEEE Std 802.1Qcw™-2023: YANG Data Models for Scheduled Traffic, Frame Preemption, and Per-Stream
21 Filtering and Policing support configuration and status reporting for bridges and bridge components with the
22 capabilities of scheduled traffic, frame preemption, and per-stream filtering and policing.

23 IEEE Std 802.1Qcz™-2023: Congestion Isolation specified protocols, procedures and managed objects that
24 support the isolation of congesting data flows within data center environments. Congestion isolation capable
25 systems identify flows creating congestion, adjust transmission selection for packets of those flows, and
26 signal to neighbors. This mechanism reduces head-of-line blocking for non-congesting flows sharing a
27 traffic class in lossless networks. Congestion Isolation is intended to be used with higher layer protocols that
28 utilize end-to-end congestion control in order to reduce packet loss and latency.

29 IEEE Std 802.1Q™-2022 incorporated the text of the following amendments into IEEE Std 802.1Q™-2018.

30 IEEE Std 802.1Qcc™-2018	Stream Reservation Protocol (SRP) Enhancements and
31	Performance Improvements
32 IEEE Std 802.1Qcp™-2018	YANG Data Model
33 IEEE Std 802.1Qcy™-2019	Virtual Station Interface (VSI) Discovery and Configuration
34	Protocol (VDP) Extension to Support Network Virtualization
35	Overlays Over Layer 3 (NVO3)
36 IEEE Std 802.1Qcx™-2020	YANG Data Model for Connectivity Fault Management
37 IEEE Std 802.1Qcr™-2020	Asynchronous Traffic Shaping

1 The 2018 revision of this standard incorporated the text of the following amendments into IEEE Std
2 802.1Q-2014.

3 IEEE Std 802.1Qcd TM -2015	Application Virtual Local Area Network (VLAN) Type,
4	Length, Value (TLV)
5 IEEE Std 802.1Qca TM -2015	Path Control and Reservation
6 IEEE Std 802.1Q-2014 Cor 1-2015	Technical and editorial corrections
7 IEEE Std 802.1Qbv TM -2015	Enhancements for scheduled traffic
8 IEEE Std 802.1Qbu TM -2016	Frame preemption
9 IEEE Std 802.1Qbz TM -2016	Enhancements to Bridging of IEEE 802.11 Media
10 IEEE Std 802.1Qci TM -2017	Per-Stream Filtering and Policing
11 IEEE Std 802.1Qch TM -2017	Cyclic Queuing and Forwarding

12 The 2014 revision of this standard incorporated the text of the following amendments into IEEE Std
13 802.1Q-2011.

14 IEEE Std 802.1Qbe TM -2011	Multiple I-SID Registration Protocol
15 IEEE Std 802.1Qbc TM -2011	Provider Bridging—Remote Customer Service Interfaces
16 IEEE Std 802.1Qbb TM -2011	Priority-based Flow Control
17 IEEE Std 802.1Qaz TM -2011	Enhanced Transmission Selection for Bandwidth Sharing
18	Between Traffic Classes
19 IEEE Std 802.1Qbf TM -2011	PBB-TE Infrastructure Segment Protection
20 IEEE Std 802.1Qbg TM -2012	Edge Virtual Bridging
21 IEEE Std 802.1aq TM -2012	Shortest Path Bridging
22 IEEE Std 802.1Q-2011/Cor 2-2012	Technical and editorial corrections
23 IEEE Std 802.1Qbp TM -2014	Equal Cost Multiple Paths (ECMP)

24 The 2011 revision of this standard incorporated the text of the following amendments into
25 IEEE Std 802.1Q-2005.

26 IEEE Std 802.1ad TM -2005	Provider Bridges
27 IEEE Std 802.1ak TM -2007	Multiple Registration Protocol
28 IEEE Std 802.1ag TM -2007	Connectivity Fault Management
29 IEEE Std 802.1ah TM -2008	Provider Backbone Bridges
30 IEEE Std 802-1Q-2005/Cor-1-2008	Corrections to the Multiple Registration Protocol
31 IEEE Std 802.1ap TM -2008	Management Information Base (MIB) Definitions for
32	VLAN Bridges
33 IEEE Std 802.1Qaw TM -2009	Management of Data Driven and Data Dependent
34	Connectivity Faults
35 IEEE Std 802.1Qay TM -2009	Provider Backbone Bridge Traffic Engineering
36 IEEE Std 802.1aj TM -2009	Two-Port Media Access Control (MAC) Relay
37 IEEE Std 802.1Qav TM -2009	Forwarding and Queuing Enhancements for
38	Time-Sensitive Streams
39 IEEE Std 802.1Qau TM -2010	Congestion Notification
40 IEEE Std 802.1Qat TM -2010	Stream Reservation Protocol

41 Clause 13 of IEEE Std 802.1Q-2011 was also revised to include an updated specification of the Rapid
42 Spanning Tree Algorithm and Protocol (RSTP), superseding references to IEEE Std 802.1DTM-2004 [B14].⁶

⁶The numbers in brackets correspond to those of the bibliography in Annex X.

1 The 2005 revision of this standard incorporated the text of the following amendments into
2 IEEE Std 802.1Q-1998.

3 IEEE Std 802.1u TM -2001	Technical and Editorial Corrections
4 IEEE Std 802.1v TM -2001	VLAN Classification by Protocol and Port
5 IEEE Std 802.1s TM -2002	Multiple Spanning Trees

6 This standard was first published as IEEE Std 802.1Q-1998, making use of the concepts and mechanisms of
7 LAN Bridging that were introduced by IEEE Std 802.1D and defining additional mechanisms to allow the
8 implementation of Virtual Bridged Networks.

9 For an introduction to this standard that details each of the provisions introduced by amendments and
10 revisions throughout its development, refer to 1.3.

11 This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution.
12 Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and
13 to incorporate new related material. Information on the current revision state of this and other IEEE 802
14 standards may be obtained from

15 Secretary, IEEE-SA Standards Board
16 445 Hoes Lane
17 Piscataway, NJ 08854-4141
18 USA

1 Contents

2	1.	Overview.....	88
3	1.1	Scope.....	88
4	1.2	Purpose.....	88
5	1.3	Introduction.....	89
6	2.	Normative references.....	98
7	3.	Definitions	102
8	4.	Abbreviations.....	125
9	5.	Conformance.....	132
10	5.1	Requirements terminology.....	132
11	5.2	Conformant components and equipment	132
12	5.3	Protocol Implementation Conformance Statement (PICS).....	133
13	5.4	VLAN Bridge component requirements.....	133
14	5.4.1	VLAN Bridge component options	134
15	5.4.2	Multiple VLAN Registration Protocol (MVRP) requirements	140
16	5.4.3	VLAN Bridge requirements for congestion notification	140
17	5.4.4	Multiple Stream Registration Protocol (MSRP) requirements	141
18	5.4.5	Shortest Path Bridging (SPB) operation (optional)	141
19	5.4.6	Path Control and Reservation (PCR) (optional)	142
20	5.4.7	VLAN Bridge requirements for congestion isolation (optional)	143
21	5.5	C-VLAN component conformance.....	143
22	5.5.1	C-VLAN component options	144
23	5.5.2	TE-MSTID (optional)	144
24	5.6	S-VLAN component conformance	144
25	5.6.1	S-VLAN component options	145
26	5.6.2	S-VLAN component requirements for Provider Backbone Bridge Traffic Engineering (PBB-TE)	145
27	5.6.3	S-VLAN component requirements for PBB-TE IPS	145
28	5.6.4	S-VLAN component requirements for ECMP with flow filtering	145
29	5.7	I-component conformance	145
30	5.7.1	I-component options	146
31	5.8	B-component conformance.....	146
32	5.8.1	B-component options	146
33	5.8.2	B-component requirements for PBB-TE	147
34	5.8.3	B-component requirements for PBB-TE IPS	147
35	5.8.4	B-component requirements for ECMP with flow filtering	147
36	5.9	C-VLAN Bridge conformance.....	147
37	5.9.1	C-VLAN Bridge options	147
38	5.9.2	C-VLAN Bridge requirements for PBBN Auto Attach (optional)	148
39	5.10	Provider Bridge conformance.....	148
40	5.10.1	S-VLAN Bridge conformance	148
41	5.10.2	Provider Edge Bridge conformance	148
42	5.10.3	S-VLAN Bridge requirements for PBBN Auto Attach (optional)	149
43	5.11	System requirements for Priority-based Flow Control (PFC)	149
44	5.12	Backbone Edge Bridge (BEB) conformance.....	149
45	5.12.1	BEB requirements for PBB-TE	150
46	5.12.2	Backbone Edge Bridge requirements for PBBN Auto Attach (optional)	150
47			

1	5.13	MAC Bridge component requirements	150
2	5.13.1	MAC Bridge component options	151
3	5.14	MAC Bridge conformance.....	152
4	5.14.1	MAC Bridge options	152
5	5.14.2	MAC Bridge requirements for PBBN Auto Attach (optional)	152
6	5.15	TPMR component conformance.....	152
7	5.15.1	TPMR component options	153
8	5.16	TPMR conformance.....	153
9	5.16.1	TPMR options	153
10	5.17	T-component conformance	154
11	5.17.1	T-component options	154
12	5.18	End station requirements for MMRP, MVRP, and MSRP	154
13	5.18.1	MMRP requirements and options	154
14	5.18.2	MVRP requirements and options	155
15	5.18.3	MSRP requirements and options	155
16	5.19	VLAN-aware end station requirements for CFM.....	156
17	5.20	End station requirements—FQTSS	156
18	5.21	End station requirements for congestion notification	157
19	5.22	MAC-specific bridging methods	157
20	5.23	EVB Bridge requirements.....	157
21	5.24	EVB station requirements.....	158
22	5.24.1	Edge relay (ER) requirements	159
23	5.25	End station requirements—enhancements for scheduled traffic	160
24	5.26	End station requirements—enhancements for frame preemption.....	161
25	5.27	End station requirements—PSFP	161
26	5.28	End station requirements—Cyclic queuing and forwarding.....	161
27	5.29	TSN CNC station requirements	161
28	5.30	VDP-NVO3 requirements.....	162
29	5.30.1	VDP-NVO3 nNVE requirements	162
30	5.30.2	VDP-NVO3 tNVE requirements	162
31	5.31	End station requirements—ATS.....	162
32	5.32	End station requirements for congestion isolation.....	163
33	5.33	End station requirements for PBBN Auto Attach (optional).....	163
34	6.	Support of the MAC Service	164
35	6.1	Basic architectural concepts and terms	165
36	6.2	Provision of the MAC Service.....	165
37	6.2.1	Point-to-point, multipoint-to-multipoint, and rooted-multipoint connectivity ...	166
38	6.3	Support of the MAC Service	166
39	6.4	Preservation of the MAC Service	167
40	6.5	Quality of service (QoS) maintenance.....	167
41	6.5.1	Service availability	167
42	6.5.2	Frame loss	168
43	6.5.3	Frame misordering	168
44	6.5.4	Frame duplication	169
45	6.5.5	Transit delay	170
46	6.5.6	Frame lifetime	170
47	6.5.7	Undetected frame error rate	171
48	6.5.8	Maximum Service Data Unit Size	171
49	6.5.9	Priority	171
50	6.5.10	Throughput	172
51	6.6	Internal Sublayer Service (ISS)	173

1	6.7	Support of the ISS by specific MAC procedures.....	173
2	6.7.1	Support of the ISS by IEEE Std 802.3 (Ethernet)	173
3	6.7.2	Frame preemption	173
4	6.8	Enhanced Internal Sublayer Service (EISS)	174
5	6.8.1	Service primitives	174
6	6.8.2	Status parameters	175
7	6.8.3	Point-to-point parameters	175
8	6.8.4	Control primitives and parameters	175
9	6.9	Support of the EISS	176
10	6.9.1	Data indications	177
11	6.9.2	Data requests	178
12	6.9.3	Priority Code Point encoding	178
13	6.9.4	Regenerating priority	180
14	6.10	Support of the ISS/EISS by PIPs	181
15	6.10.1	Data indications	183
16	6.10.2	Data requests	184
17	6.10.3	Priority Code Point encoding	184
18	6.11	Support of the EISS by CBPs	185
19	6.11.1	Data indications	186
20	6.11.2	Data requests	187
21	6.11.3	Priority Code Point decoding	188
22	6.11.4	Regenerating priority	188
23	6.12	Protocol VLAN classification.....	188
24	6.12.1	Protocol Templates	190
25	6.12.2	Protocol Group Identifiers	190
26	6.12.3	Protocol Group Database	190
27	6.13	Support of the ISS for attachment to a PBN	191
28	6.13.1	Data requests	192
29	6.13.2	Data indications	192
30	6.14	Support of the ISS within a system.....	193
31	6.15	Support of the ISS by additional technologies.....	193
32	6.16	Filtering services in Bridged Networks	193
33	6.16.1	Purpose(s) of filtering service provision	194
34	6.16.2	Goals of filtering service provision	194
35	6.16.3	Users of filtering services	194
36	6.16.4	Basis of service	194
37	6.16.5	Categories of service	195
38	6.16.6	Service configuration	195
39	6.16.7	Service definition for Extended Filtering Services	195
40	6.17	EISS Multiplex Entity.....	197
41	6.18	Backbone Service Instance Multiplex Entity.....	198
42	6.18.1	Demultiplexing direction	199
43	6.18.2	Multiplexing direction	200
44	6.18.3	Priority Code Point encoding	200
45	6.18.4	Status parameters	200
46	6.19	TESI Multiplex Entity	201
47	6.20	Support of the ISS with signaled priority	202
48	6.20.1	Data indications	202
49	6.20.2	Data requests	203
50	6.21	Infrastructure Segment Multiplex Entity	203
51	6.22	PDU and protocol discrimination and media.....	204

1	7.	Principles of Virtual Bridged Network operation.....	205
2	7.1	Network overview.....	205
3	7.2	Use of VLANs	206
4	7.3	Active topology.....	206
5	7.4	VLAN topology	207
6	7.5	Locating end stations	208
7	7.6	Ingress, forwarding, and egress rules.....	208
8	8.	Principles of Bridge operation	210
9	8.1	Bridge operation	210
10	8.1.1	Relay	210
11	8.1.2	Filtering and relaying information	211
12	8.1.3	Duplicate frame prevention	211
13	8.1.4	Traffic segregation	211
14	8.1.5	Traffic reduction	212
15	8.1.6	Traffic expediting	212
16	8.1.7	Conversion of frame formats	212
17	8.2	Bridge architecture.....	213
18	8.3	Model of operation.....	215
19	8.4	Active topologies, learning, and forwarding	218
20	8.5	Bridge Port Transmit and Receive.....	220
21	8.5.1	Bridge Port connectivity	220
22	8.5.2	TPMR Port connectivity	221
23	8.5.3	Support of Higher Layer Entities	222
24	8.6	The Forwarding Process	222
25	8.6.1	Active topology enforcement	223
26	8.6.2	Ingress filtering	225
27	8.6.3	Frame filtering	225
28	8.6.4	Egress filtering	229
29	8.6.5	Flow classification and metering	229
30	8.6.6	Queuing frames	240
31	8.6.7	Queue management	241
32	8.6.8	Transmission selection	242
33	8.6.9	Scheduled traffic state machines	248
34	8.6.10	Stream gate control state machines	255
35	8.6.11	ATS Scheduler state machines	257
36	8.7	The Learning Process.....	262
37	8.7.1	Default filtering utility criteria	262
38	8.7.2	Enhanced filtering utility criteria	262
39	8.7.3	Ageing of Dynamic Filtering Entries	263
40	8.8	The Filtering Database (FDB)	263
41	8.8.1	Static Filtering Entries	266
42	8.8.2	Static VLAN Registration Entries	267
43	8.8.3	Dynamic Filtering Entries	268
44	8.8.4	MAC Address Registration Entries	269
45	8.8.5	Dynamic VLAN Registration Entries	269
46	8.8.6	Default Group filtering behavior	270
47	8.8.7	Dynamic Reservation Entries	271
48	8.8.8	Allocation of VIDs to FIDs	271
49	8.8.9	Querying the FDB	272
50	8.8.10	Determination of the member set for a VID	276
51	8.8.11	Permanent Database	277
52	8.8.12	Connection_Identifier	277

1	8.9	MST, SPB, and ESP configuration information	277
2	8.9.1	MST Configuration Table	278
3	8.9.2	MST configuration identification	278
4	8.9.3	FID to MSTI Allocation Table	279
5	8.9.4	SPT Configuration Identification	279
6	8.10	Spanning Tree Protocol Entity.....	279
7	8.11	MRP entities	280
8	8.12	Bridge Management Entity	280
9	8.13	Addressing	280
10	8.13.1	End stations	280
11	8.13.2	Bridge Ports	281
12	8.13.3	Use of LLC by Spanning Tree Protocol Entities	281
13	8.13.4	Reserved MAC addresses	281
14	8.13.5	Group MAC addresses for spanning tree entity	282
15	8.13.6	Group MAC addresses for MRP Applications	283
16	8.13.7	Bridge Management Entities	284
17	8.13.8	Unique identification of a Bridge	284
18	8.13.9	Points of attachment and connectivity for Higher Layer Entities	284
19	8.13.10	VLAN attachment and connectivity for Higher Layer Entities	288
20	8.13.11	CFM entities	289
21	9.	Tagged frame format	291
22	9.1	Purpose of tagging	291
23	9.2	Representation and encoding of tag fields	291
24	9.3	Tag format.....	292
25	9.4	TPID formats	292
26	9.5	Tag Protocol identification	292
27	9.6	VLAN Tag Control Information (TCI).....	293
28	9.7	Backbone Service Instance Tag Control Information (I-TAG TCI).....	294
29	10.	Multiple Registration Protocol (MRP) and Multiple MAC Registration Protocol (MMRP)	296
30	10.1	MRP overview	296
31	10.2	MRP architecture	299
32	10.3	MRP Attribute Propagation (MAP).....	300
33	10.3.1	MAP Context	301
34	10.4	Requirements to be met by MRP	302
35	10.5	Requirements for interoperability between MRP Participants	303
36	10.6	Protocol operation.....	304
37	10.7	Protocol specification	308
38	10.7.1	Notational conventions and abbreviations	309
39	10.7.2	Registrar Administrative Controls	311
40	10.7.3	Applicant Administrative Controls	311
41	10.7.4	Protocol timers	311
42	10.7.5	Protocol event definitions	312
43	10.7.6	Protocol Action definitions	315
44	10.7.7	Applicant state machine	316
45	10.7.8	Registrar state machine	316
46	10.7.9	LeaveAll state machine	318
47	10.7.10	PeriodicTransmission state machine	318
48	10.7.11	Timer values	319
49	10.7.12	Operational reporting and statistics	320
50	10.7.13	Interoperability considerations	320
51	10.7.14	External control	321

1	10.8	Structure and encoding of Multiple Registration Protocol Data Units (MRPDUs)	321
2	10.8.1	Structure	321
3	10.8.2	Encoding of MRPDU parameters	323
4	10.8.3	Packing and parsing MRPDUs	326
5	10.9	Multiple MAC Registration Protocol (MMRP)—Purpose	328
6	10.10	MMRP Model of operation.....	329
7	10.10.1	Propagation of Group Membership information	330
8	10.10.2	Propagation of Group service requirement information	331
9	10.10.3	Source pruning	331
10	10.10.4	Use of Group service requirement registration by end stations	331
11	10.11	Default Group filtering behavior and MMRP propagation	331
12	10.12	Definition of the MMRP application	333
13	10.12.1	Definition of MRP elements	333
14	10.12.2	Provision and support of Extended Filtering Services	335
15	10.12.3	Use of “new” declaration capability	337
16	10.12.4	Attribute value support requirements	337
17	10.12.5	Registrar Administrative Controls	337
18	11.	VLAN topology management.....	338
19	11.1	Static and dynamic VLAN configuration	338
20	11.2	Multiple VLAN Registration Protocol (MVRP)	339
21	11.2.1	MVRP overview	339
22	11.2.2	VLAN registration service definition	341
23	11.2.3	Definition of the MVRP application	342
24	11.2.4	VID translation table	345
25	11.2.5	Use of “new” declaration capability	345
26	11.2.6	New-Only Participant and Registrar Administrative Controls	345
27	11.2.7	Attribute value support requirements	345
28	12.	Bridge management	346
29	12.1	Management functions.....	346
30	12.1.1	Configuration Management	346
31	12.1.2	Fault Management	347
32	12.1.3	Performance Management	347
33	12.1.4	Security Management	347
34	12.1.5	Accounting Management	347
35	12.2	VLAN Bridge objects	347
36	12.3	Data types	348
37	12.4	Bridge Management Entity	349
38	12.4.1	Bridge Configuration	349
39	12.4.2	Port configuration	352
40	12.5	MAC entities	354
41	12.5.1	ISS Port Number table managed object (optional)	354
42	12.6	Forwarding process	355
43	12.6.1	The Port Counters	355
44	12.6.2	Priority handling	356
45	12.6.3	Traffic Class Table	363

1	12.7	Filtering Database (FDB).....	364
2	12.7.1	The Filtering Database object	364
3	12.7.2	A Static Filtering Entry object	365
4	12.7.3	A Dynamic Filtering Entry object	365
5	12.7.4	A MAC Address Registration Entry object	366
6	12.7.5	A VLAN Registration Entry object	366
7	12.7.6	Permanent Database object	366
8	12.7.7	General FDB operations	367
9	12.8	Bridge Protocol Entity	369
10	12.8.1	The Protocol Entity	369
11	12.8.2	Bridge Port	372
12	12.9	MRP Entities.....	376
13	12.9.1	The MRP Timer object	376
14	12.9.2	The MRP Attribute Type object	377
15	12.9.3	Periodic state machine objects	378
16	12.10	Bridge VLAN managed objects.....	378
17	12.10.1	Bridge VLAN Configuration managed object	379
18	12.10.2	VLAN Configuration managed object	384
19	12.10.3	The VID to FID allocation managed object	385
20	12.11	MMRP entities.....	387
21	12.11.1	MMRP Configuration managed object	387
22	12.12	MST configuration entities	388
23	12.12.1	The MSTI List	389
24	12.12.2	The FID to MSTID Allocation Table	390
25	12.12.3	The MST Configuration Table	391
26	12.13	Provider Bridge management	393
27	12.13.1	Provider Bridge Port Type managed object	394
28	12.13.2	Customer Edge Port Configuration managed object	395
29	12.13.3	Remote Customer Access managed object	398
30	12.14	CFM entities	400
31	12.14.1	Maintenance Domain list managed object	400
32	12.14.2	CFM Stack managed object	402
33	12.14.3	Default MD Level managed object	403
34	12.14.4	Configuration Error List managed object	404
35	12.14.5	Maintenance Domain managed object	404
36	12.14.6	Maintenance Association managed object	407
37	12.14.7	Maintenance association Endpoint managed object	409
38	12.15	Backbone Core Bridge (BCB) management.....	417
39	12.16	Backbone Edge Bridge (BEB) management	417
40	12.16.1	BEB configuration managed object	419
41	12.16.2	BEB/PB/VLAN Bridge Port configuration managed object	423
42	12.16.3	VIP configuration managed object	423
43	12.16.4	PIP configuration managed object	424
44	12.16.5	CBP Configuration managed object	431
45	12.17	DDCFM entities.....	434
46	12.17.1	DDCFM Stack managed object	434
47	12.17.2	Reflection Responder managed object	434
48	12.17.3	RFM Receiver managed object	438
49	12.17.4	Decapsulator Responder managed object	439
50	12.17.5	SFM Originator managed object	441
51	12.18	PBB-TE Protection Switching managed objects	444
52	12.18.1	TE protection group list managed object	444
53	12.18.2	TE protection group managed object	445

1	12.19	TPMR managed objects	447
2	12.19.1	TPMR management entity	448
3	12.19.2	MAC and PHY entities	449
4	12.19.3	Forwarding Process	449
5	12.19.4	MAC Status Propagation Entity (MSPE)	454
6	12.20	Management entities for FQTSS	456
7	12.20.1	The Bandwidth Availability Parameter Table	456
8	12.20.2	The Transmission Selection Algorithm Table	457
9	12.20.3	The Priority Regeneration Override Table	457
10	12.20.4	SR Class to Priority Mapping Table	458
11	12.21	Congestion Notification managed objects	458
12	12.21.1	CN component managed object	459
13	12.21.2	CN component priority managed object	459
14	12.21.3	CN Port priority managed object	460
15	12.21.4	Congestion Point managed object	461
16	12.21.5	Reaction Point port priority managed object	462
17	12.21.6	Reaction Point group managed object	462
18	12.22	Stream Reservation Protocol (SRP) entities	463
19	12.22.1	SRP Bridge Base Table	463
20	12.22.2	SRP Bridge Port Table	464
21	12.22.3	SRP Latency Parameter Table	464
22	12.22.4	SRP Stream Table	464
23	12.22.5	SRP Reservations Table	465
24	12.22.6	SRP Stream Preload Table	466
25	12.22.7	SRP Reservations Preload Table	466
26	12.23	Priority-based Flow Control objects	467
27	12.24	1:1 PBB-TE IPS managed objects	468
28	12.24.1	IPG list managed object	468
29	12.24.2	IPG managed object	470
30	12.25	Shortest Path Bridging managed objects	472
31	12.25.1	The SPB System managed object	473
32	12.25.2	The SPB MTID Static managed object	475
33	12.25.3	The SPB Topology Instance Dynamic managed object	477
34	12.25.4	The SPB ECT Static Entry managed object	477
35	12.25.5	The SPB ECT Dynamic Entry managed object	479
36	12.25.6	The SPB Adjacency Static Entry managed object	480
37	12.25.7	The SPB Adjacency Dynamic Entry managed object	481
38	12.25.8	The SPBM BSI Static Entry managed object	481
39	12.25.9	The SPB Topology Node Table managed object	483
40	12.25.10	The SPB Topology ECT Table managed object	484
41	12.25.11	The SPB Topology Edge Table managed object	484
42	12.25.12	The SPBM Topology Service Table managed object	485
43	12.25.13	The SPBV Topology Service Table managed object	486
44	12.25.14	The ECMP ECT Static Entry managed object	487
45	12.26	Edge Virtual Bridging (EVB) management.....	488
46	12.26.1	EVB system base table	491
47	12.26.2	SBP table entry	493
48	12.26.3	VSI table entry	494
49	12.26.4	S-channel configuration and management	495
50	12.26.5	ER management	497
51	12.27	Edge Control Protocol (ECP) management.....	498
52	12.27.1	ECP table entry	498

1	12.28	Path Control and Reservation (PCR) management.....	499
2	12.28.1	The PCR ECT Static Entry managed object	500
3	12.28.2	The PCR Topology ECT Table managed object	502
4	12.29	Managed objects for scheduled traffic	503
5	12.29.1	The Gate Parameter Table	503
6	12.29.2	Timing points for scheduled traffic	505
7	12.30	Managed objects for frame preemption	506
8	12.30.1	Frame Preemption Parameter Table	506
9	12.31	Managed objects for per-stream classification and metering	509
10	12.31.1	The Stream Parameter Table	509
11	12.31.2	The Stream Filter Instance Table	510
12	12.31.3	The Stream Gate Instance Table	512
13	12.31.4	The Flow Meter Instance Table	515
14	12.31.5	The Scheduler Instance Table	515
15	12.31.6	The Scheduler Group Instance Table	517
16	12.31.7	The Scheduler Port Parameter Table	517
17	12.31.8	The Scheduler Timing Characteristics Table	518
18	12.32	Stream reservation remote management.....	520
19	12.32.1	Bridge Delay	520
20	12.32.2	Propagation Delay	522
21	12.32.3	Static Trees	522
22	12.32.4	MRP External Control	523
23	12.33	Congestion Isolation managed objects	526
24	12.33.1	CI entity managed object	526
25	12.33.2	CI Peer Table	527
26	12.33.3	CI Stream Table	527
27	12.33.4	CIP entity managed object	527
28	12.34	Managed objects for PBBN Auto Attach	529
29	12.34.1	PBBN Auto Attach System objects	529
30	12.34.2	PBBN Auto Attach port table	530
31	12.34.3	PBBN Discovered Auto Attach Systems table	531
32	12.34.4	PBBN Auto Attach assignment table	532
33	12.34.5	PBBN Auto Attach statistics table	533
34	13.	Spanning tree protocols	534
35	13.1	Protocol design requirements.....	535
36	13.2	Protocol support requirements	536
37	13.2.1	MSTP support requirements	536
38	13.2.2	SPB support requirements	536
39	13.3	Protocol design goals	537
40	13.4	RSTP overview	537
41	13.4.1	Computation of the active topology	538
42	13.4.2	Example topologies	539
43	13.5	MSTP overview	542
44	13.5.1	Example topologies	543
45	13.5.2	Relationship of MSTP to RSTP	546
46	13.5.3	Modeling an MST or SPT Region as a single Bridge	546
47	13.6	SPB overview	547
48	13.7	Compatibility and interoperability	548
49	13.7.1	Designated Port selection	548
50	13.7.2	Force Protocol Version	548
51	13.8	MST Configuration Identifier (MCID).....	549
52	13.9	Spanning tree priority vectors.....	550

1	13.10	CIST Priority Vector calculations.....	552
2	13.11	MST Priority Vector calculations	554
3	13.12	Port Role assignments.....	556
4	13.13	Stable connectivity	557
5	13.14	Communicating spanning tree information	557
6	13.15	Changing spanning tree information.....	559
7	13.16	Changing Port States with RSTP or MSTP	560
8	13.16.1	Subtree connectivity and priority vectors	560
9	13.16.2	Root Port transition to Forwarding	561
10	13.16.3	Designated Port transition to Forwarding	561
11	13.16.4	Master Port transition to Forwarding	563
12	13.17	Changing Port States with SPB	565
13	13.17.1	Agreement Digest	567
14	13.18	Managing spanning tree topologies	568
15	13.19	Updating learned station location information	569
16	13.20	Managing reconfiguration.....	571
17	13.21	Partial and disputed connectivity	572
18	13.22	In-service upgrades	572
19	13.23	Fragile Bridges.....	573
20	13.24	Spanning tree protocol state machines.....	574
21	13.25	State machine timers.....	574
22	13.25.1	edgeDelayWhile	577
23	13.25.2	fdWhile	577
24	13.25.3	helloWhen	577
25	13.25.4	mdelayWhile	577
26	13.25.5	rbWhile	577
27	13.25.6	rcvdInfoWhile	577
28	13.25.7	rrWhile	577
29	13.25.8	tcDetected	577
30	13.25.9	tcWhile	577
31	13.25.10	pseudoInfoHelloWhen	577
32	13.26	Per Bridge variables.....	578
33	13.26.1	agreementDigest	578
34	13.26.2	BridgeIdentifier	578
35	13.26.3	BridgePriority	579
36	13.26.4	BridgeTimes	579
37	13.26.5	ForceProtocolVersion	579
38	13.26.6	MigrateTime	579
39	13.26.7	MstConfigId	579
40	13.26.8	AuxMstConfigId	579
41	13.26.9	rootPortId	579
42	13.26.10	rootPriority	580
43	13.26.11	rootTimes	580
44	13.26.12	TxHoldCount	580

1	13.27	Per port variables	580
2	13.27.1	AdminEdge	582
3	13.27.2	ageingTime	582
4	13.27.3	agree	582
5	13.27.4	agreed	582
6	13.27.5	agreedAbove	582
7	13.27.6	agreedDigest	583
8	13.27.7	agreedDigestValid	583
9	13.27.8	agreeDigest	583
10	13.27.9	agreeDigestValid	583
11	13.27.10	agreedMisorder	583
12	13.27.11	agreedN	583
13	13.27.12	agreedND	583
14	13.27.13	agreedPriority	583
15	13.27.14	agreedTopology	583
16	13.27.15	agreementOutstanding	583
17	13.27.16	agreeN	583
18	13.27.17	agreeND	584
19	13.27.18	AutoEdge	584
20	13.27.19	AutoIsolate	584
21	13.27.20	designatedPriority	584
22	13.27.21	designatedTimes	584
23	13.27.22	disputed	584
24	13.27.23	enableBPDURx	584
25	13.27.24	enableBPDUTx	584
26	13.27.25	ExternalPortPathCost	585
27	13.27.26	isL2gp	585
28	13.27.27	isolate	585
29	13.27.28	fdbFlush	585
30	13.27.29	forward	585
31	13.27.30	forwarding	585
32	13.27.31	infoInternal	585
33	13.27.32	infoIs	585
34	13.27.33	InternalPortPathCost	586
35	13.27.34	learn	586
36	13.27.35	learning	586
37	13.27.36	master	586
38	13.27.37	mastered	586
39	13.27.38	mcheck	586
40	13.27.39	msgPriority	586
41	13.27.40	msgTimes	586
42	13.27.41	neighbourPriority	587
43	13.27.42	newInfo	587
44	13.27.43	newInfoMsti	587
45	13.27.44	operEdge	587
46	13.27.45	portEnabled	587
47	13.27.46	portId	587
48	13.27.47	portPriority	587
49	13.27.48	portTimes	587
50	13.27.49	proposed	588
51	13.27.50	proposing	588
52	13.27.51	pseudoRootId	588
53	13.27.52	rcvdBPDU	588
54	13.27.53	rcvdInfo	588

1	13.27.54	rcvdInternal	588
2	13.27.55	rcvdMsg	588
3	13.27.56	rcvdRSTP	588
4	13.27.57	rcvdSTP	588
5	13.27.58	rcvdTc	588
6	13.27.59	rcvdTcAck	588
7	13.27.60	rcvdTen	588
8	13.27.61	reRoot	588
9	13.27.62	reselect	588
10	13.27.63	restrictedDomainRole	589
11	13.27.64	restrictedRole	589
12	13.27.65	restrictedTen	589
13	13.27.66	role	589
14	13.27.67	selected	589
15	13.27.68	selectedRole	589
16	13.27.69	sendRSTP	589
17	13.27.70	sync	589
18	13.27.71	synced	589
19	13.27.72	tcAck	589
20	13.27.73	tcProp	590
21	13.27.74	tick	590
22	13.27.75	txCount	590
23	13.27.76	updtInfo	590
24	13.28	State machine conditions and parameters	590
25	13.28.1	allSptAgree	590
26	13.28.2	allSynced	591
27	13.28.3	allTransmitReady	591
28	13.28.4	BestAgreementPriority	591
29	13.28.5	cist	591
30	13.28.6	cistRootPort	591
31	13.28.7	cistDesignatedPort	591
32	13.28.8	EdgeDelay	591
33	13.28.9	forwardDelay	591
34	13.28.10	FwdDelay	591
35	13.28.11	HelloTime	591
36	13.28.12	MaxAge	591
37	13.28.13	msti	592
38	13.28.14	mstiDesignatedOrTCpropagatingRootPort	592
39	13.28.15	mstiMasterPort	592
40	13.28.16	operPointToPoint	592
41	13.28.17	rcvdAnyMsg	592
42	13.28.18	rcvdCistMsg	592
43	13.28.19	rcvdMstiMsg	592
44	13.28.20	reRooted	592
45	13.28.21	rstpVersion	592
46	13.28.22	spt	592
47	13.28.23	stpVersion	592
48	13.28.24	updtCistInfo	592
49	13.28.25	updtMstiInfo	592

1	13.29	State machine procedures	593
2	13.29.1	betterorsameInfo(newInfoIs)	594
3	13.29.2	clearAllRcvdMsgs()	594
4	13.29.3	clearReselectTree()	594
5	13.29.4	disableForwarding()	594
6	13.29.5	disableLearning()	594
7	13.29.6	enableForwarding()	594
8	13.29.7	enableLearning()	594
9	13.29.8	fromSameRegion()	594
10	13.29.9	newTcDetected()	594
11	13.29.10	newTcWhile()	595
12	13.29.11	pseudoRcvMsgs()	595
13	13.29.12	rcvInfo()	595
14	13.29.13	rcvMsgs()	596
15	13.29.14	rcvAgreements()	596
16	13.29.15	recordAgreement()	596
17	13.29.16	recordDispute()	597
18	13.29.17	recordMastered()	597
19	13.29.18	recordPriority()	597
20	13.29.19	recordProposal()	597
21	13.29.20	recordTimes()	597
22	13.29.21	setReRootTree()	597
23	13.29.22	setSelectedTree()	597
24	13.29.23	setSyncTree()	597
25	13.29.24	setTcFlags()	598
26	13.29.25	setTcPropTree()	598
27	13.29.26	syncMaster()	598
28	13.29.27	txConfig()	598
29	13.29.28	txRstp()	598
30	13.29.29	txTcn()	599
31	13.29.30	updtAgreement()	599
32	13.29.31	updtBPDUVersion()	600
33	13.29.32	updtDigest()	600
34	13.29.33	updtRcvdInfoWhile()	601
35	13.29.34	updtRolesTree()	601
36	13.29.35	uptRolesDisabledTree()	603
37	13.30	The Port Timers state machine	603
38	13.31	Port Receive state machine	604
39	13.32	Port Protocol Migration state machine	605
40	13.33	Bridge Detection state machine	605
41	13.34	Port Transmit state machine	606
42	13.35	Port Information state machine.....	607
43	13.36	Port Role Selection state machine	608
44	13.37	Port Role Transitions state machine	608
45	13.38	Port State Transition state machine	613
46	13.38.1	Port State transitions for the CIST and MSTIs	614
47	13.38.2	Port State transitions for SPTs	614
48	13.39	Topology Change state machine.....	615
49	13.40	Layer 2 Gateway Port Receive state machine	616

1	13.41	CEP spanning tree operation.....	616
2	13.41.1	PEP operPointToPointMAC and operEdge	616
3	13.41.2	updtRolesTree()	617
4	13.41.3	setReRootTree(), setSyncTree(), setTcPropTree()	617
5	13.41.4	allSynced, reRooted	617
6	13.41.5	Configuration parameters	617
7	13.42	Virtual Instance Port (VIP) spanning tree operation	618
8	14.	Encoding of Bridge Protocol Data Units (BPDUs)	619
9	14.1	BPDUs Structure	619
10	14.1.1	Transmission and representation of octets	619
11	14.1.2	Common BPDU fields	619
12	14.2	Encoding of parameter types	621
13	14.2.1	Encoding of Protocol Identifiers	621
14	14.2.2	Encoding of Protocol Version Identifiers	621
15	14.2.3	Encoding of BPDU types	621
16	14.2.4	Encoding of flags	621
17	14.2.5	Encoding of Bridge Identifiers	621
18	14.2.6	Encoding of External Root Path Cost and Internal Root Path Cost	622
19	14.2.7	Encoding of Port Identifiers	622
20	14.2.8	Encoding of Timer Values	622
21	14.2.9	Encoding of Port Role values	622
22	14.2.10	Encoding of Length Values	622
23	14.2.11	Encoding of Hop Counts	622
24	14.3	Transmission of BPDUs	623
25	14.4	Encoding and decoding of STP Configuration, RST, MST, and SPT BPDUs.....	623
26	14.4.1	MSTI Configuration Messages	625
27	14.5	Validation of received BPDUs	626
28	14.6	Validation and interoperability	627
29	15.	Support of the MAC Service by PBNs	628
30	15.1	Service transparency	628
31	15.2	Customer service interfaces	629
32	15.3	Port-based service interface	629
33	15.4	C-tagged service interface	630
34	15.5	S-tagged service interface	631
35	15.6	Remote customer service interfaces (RCSIs)	631
36	15.7	Service instance segregation	635
37	15.8	Service instance selection and identification	635
38	15.9	Service priority selection	636
39	15.10	Service access protection	636
40	16.	Principles of Provider Bridged Network (PBN) operation	637
41	16.1	PBN overview	637
42	16.2	Provider Bridged Network (PBN)	637
43	16.3	Service instance connectivity.....	641
44	16.4	Service provider learning of customer end station addresses	642
45	16.5	Detection of connectivity loops through attached networks.....	642
46	16.6	Network management	643
47	17.	Management Information Base (MIB)	644
48	17.1	Internet Standard Management Framework	644

1	17.2	Structure of the MIB	644
2	17.2.1	Structure of the IEEE8021-TC-MIB	646
3	17.2.2	Structure of the IEEE8021-BRIDGE-MIB	647
4	17.2.3	Structure of the IEEE8021-SPANNING-TREE MIB	651
5	17.2.4	Structure of the IEEE8021-Q-BRIDGE-MIB	654
6	17.2.5	Structure of the IEEE8021-PB-MIB	659
7	17.2.6	Structure of the IEEE8021-MSTP-MIB	661
8	17.2.7	Structure of the IEEE8021-CFM-MIB	664
9	17.2.8	Structure of the IEEE8021-PBB-MIB	671
10	17.2.9	Structure of the IEEE8021-DDCFM-MIBs	673
11	17.2.10	Structure of the IEEE8021-PBBTE-MIB	676
12	17.2.11	Structure of the TPMR MIB	679
13	17.2.12	Structure of the IEEE8021-FQTSS-MIB	681
14	17.2.13	Structure of the IEEE8021-CN-MIB	682
15	17.2.14	Structure of the IEEE8021-SRP-MIB	684
16	17.2.15	Structure of the IEEE8021-MVRPX-MIB	686
17	17.2.16	Structure of the IEEE8021-MIRP-MIB	686
18	17.2.17	Structure of the IEEE8021-PFC-MIB	687
19	17.2.18	Structure of the IEEE8021-TEIPS-MIB	687
20	17.2.19	Structure of the IEEE8021-SPB-MIB	689
21	17.2.20	Structure of the IEEE8021-EVB-MIB	693
22	17.2.21	Structure of the IEEE8021-ECMP-MIB	697
23	17.2.22	Structure of the IEEE8021-ST-MIB	698
24	17.2.23	Structure of the IEEE8021-Preemption-MIB	699
25	17.2.24	Structure of the IEEE8021-PSFP-MIB	700
26	17.2.25	Structure of the IEEE8021-TSN-REMOTE-MANAGEMENT-MIB	703
27	17.2.26	Structure of the IEEE8021-PBBN-AA-MIB	704
28	17.3	MIB module relationships	706
29	17.3.1	Relationship of the IEEE8021-TC-MIB to other MIB modules	706
30	17.3.2	Relationship of the IEEE8021-BRIDGE-MIB to other MIB modules	706
31	17.3.3	Relationship of the IEEE8021-RSTP MIB to other MIB modules	709
32	17.3.4	Relationship of the IEEE8021-Q-BRIDGE-MIB to other MIB modules	709
33	17.3.5	Relationship of the IEEE8021-PB-BRIDGE MIB to other MIB modules	710
34	17.3.6	Relationship of the IEEE8021-MSTP-MIB to other MIB modules	711
35	17.3.7	Relationship of the IEEE8021-CFM-MIB to other MIB modules	711
36	17.3.8	Relationship of the IEEE8021-PBB-MIB to other MIB modules	712
37	17.3.9	Relationship of the IEEE8021-DDCFM to other MIB modules	713
38	17.3.10	Relationship of the IEEE8021-PBBTE-MIB to other MIB modules	713
39	17.3.11	Relationship of the IEEE8021-TPMR MIB to other MIB modules	714
40	17.3.12	Relationship of the IEEE8021-FQTSS-MIB to other MIB modules	714
41	17.3.13	Relationship of the IEEE802-CN-MIB to other MIB modules	714
42	17.3.14	Relationship of the IEEE8021-SRP-MIB to other MIB modules	715
43	17.3.15	Relationship of the IEEE8021-MVRPX-MIB to other MIB modules	715
44	17.3.16	Relationship of the IEEE8021-MIRP-MIB to other MIB modules	715
45	17.3.17	Relationship of the IEEE8021-PFC-MIB to other MIB modules	715
46	17.3.18	Relationship of the IEEE8021-TEIPS-MIB to other MIB modules	715
47	17.3.19	Relationship of the IEEE8021-SPB-MIB to other MIB modules	716
48	17.3.20	Relationship of the IEEE8021-EVB-MIB to other MIB modules	716
49	17.3.21	Relationship of the IEEE8021-ECMP-MIB to other MIB modules	716
50	17.3.22	Relationship of the IEEE8021-ST-MIB to other MIB modules	716
51	17.3.23	Relationship of the IEEE8021-Preemption-MIB to other MIB modules	716
52	17.3.24	Relationship of the IEEE8021-PSFP-MIB to other MIB modules	716
53	17.3.25	Relationship of the IEEE8021-TSN-REMOTE-MANAGEMENT-MIB to other	

1		MIB modules	717
2	17.3.26	Relationship of the IEEE8021-PBBN-AA-MIB to other MIB modules	717
3	17.4	Security considerations	717
4	17.4.1	Security considerations of the IEEE8021-TC-MIB	717
5	17.4.2	Security considerations of the IEEE8021-BRIDGE-MIB	717
6	17.4.3	Security considerations of the IEEE8021-SPANNING-TREE MIB	718
7	17.4.4	Security considerations of the IEEE8021-Q-BRIDGE-MIB	719
8	17.4.5	Security considerations of the IEEE8021-PB-MIB	720
9	17.4.6	Security considerations of the IEEE8021-MSTP-MIB	720
10	17.4.7	Security considerations of the IEEE8021-CFM-MIB	721
11	17.4.8	Security considerations of the IEEE8021-PBB-MIB	722
12	17.4.9	Security considerations of the IEEE8021-DDCFM-MIB	723
13	17.4.10	Security considerations of the IEEE8021-PBBTE-MIB	724
14	17.4.11	Security considerations of the IEEE8021-TPMR-MIB	725
15	17.4.12	Security considerations of the IEEE8021-FQTSS-MIB	725
16	17.4.13	Security considerations of the IEEE8021-CN-MIB	726
17	17.4.14	Security considerations of the IEEE8021-SRP-MIB	727
18	17.4.15	Security considerations of the IEEE8021-MVRPX-MIB	728
19	17.4.16	Security considerations of the IEEE8021-MIRP-MIB	728
20	17.4.17	Security considerations of the IEEE8021-PFC-MIB	729
21	17.4.18	Security considerations of the IEEE8021-TEIPS-MIB	729
22	17.4.19	Security considerations of the IEEE8021-SPB-MIB	729
23	17.4.20	Security considerations of the IEEE8021-EVB-MIB	730
24	17.4.21	Security considerations of the IEEE8021-ECMP-MIB	732
25	17.4.22	Security considerations of the IEEE8021-ST-MIB	732
26	17.4.23	Security considerations of the IEEE8021-Preemption-MIB	733
27	17.4.24	Security considerations of the IEEE8021-PSFP-MIB	733
28	17.4.25	Security considerations of the	
29		IEEE8021-TSN-REMOTE-MANAGEMENT-MIB	734
30	17.4.26	Security considerations of the IEEE8021-PBBN-AA-MIB	735
31	17.5	Dynamic component and Port creation.....	736
32	17.5.1	Overview of the dynamically created Bridge entities	736
33	17.5.2	Component creation	737
34	17.5.3	Port creation	738
35	17.6	MIB operations for service interface configuration.....	749
36	17.6.1	Provisioning PBN service interfaces	749
37	17.6.2	Provisioning Backbone Bridged Network service interfaces	752

1	17.7	MIB modules	758
2	17.7.1	Definitions for the IEEE8021-TC-MIB module	758
3	17.7.2	Definitions for the IEEE8021-BRIDGE-MIB module	767
4	17.7.3	Definitions for the IEEE8021-SPANNING-TREE-MIB module	800
5	17.7.4	Definitions for the IEEE8021-Q-BRIDGE-MIB module	815
6	17.7.5	Definitions for the IEEE8021-PB-MIB module	853
7	17.7.6	Definitions for the IEEE8021-MSTP-MIB module	868
8	17.7.7	Definitions for the CFM MIB modules	892
9	17.7.8	Definitions for the IEEE8021-PBB-MIB module	960
10	17.7.9	Definitions for the IEEE8021-DDCFM-MIB module	979
11	17.7.10	Definitions for the IEEE8021-PBBTE-MIB module	994
12	17.7.11	Definitions for the IEEE8021-TPMR-MIB module	1008
13	17.7.12	Definitions for the IEEE8021-FQTSS-MIB module	1020
14	17.7.13	Definitions for the IEEE8021-CN-MIB module	1032
15	17.7.14	Definitions for the IEEE8021-SRP-MIB module	1062
16	17.7.15	Definitions for the IEEE8021-MVRPX-MIB module	1080
17	17.7.16	Definitions for the IEEE8021-MIRP-MIB module	1084
18	17.7.17	Definitions for the IEEE8021-PFC-MIB module	1089
19	17.7.18	Definitions for the IEEE8021-TEIPS-V2-MIB module	1092
20	17.7.19	Definitions for the IEEE8021-SPB-MIB module	1104
21	17.7.20	Definitions for the IEEE8021-EVB-MIB module	1140
22	17.7.21	Definitions for the IEEE8021-ECMP-MIB module	1164
23	17.7.22	Definitions for the IEEE8021-ST-MIB module	1171
24	17.7.23	Definitions for the IEEE8021-Preemption-MIB module	1182
25	17.7.24	Definitions for the IEEE8021-PSFP-MIB module	1187
26	17.7.25	Definitions for the IEEE8021-TSN-REMOTE-MANAGEMENT-MIB module	1207
27			
28	17.7.26	Definitions for the IEEE8021-PBBN-AA-MIB module	1216
29	18.	Principles of Connectivity Fault Management operation	1225
30	18.1	Maintenance Domains and DoSAPs.....	1227
31	18.2	Service instances and MAs	1228
32	18.3	Maintenance Domain Levels	1229
33	19.	CFM entity operation.....	1233
34	19.1	Maintenance Points (MPs).....	1233

1	19.2	MA Endpoints (MEPs)	1233
2	19.2.1	MEP identification	1234
3	19.2.2	MEP functions	1235
4	19.2.3	MEP architecture	1235
5	19.2.4	MP Type Demultiplexer	1237
6	19.2.5	MP Multiplexer	1237
7	19.2.6	MP Level Demultiplexer	1237
8	19.2.7	MP OpCode Demultiplexer	1237
9	19.2.8	MEP Continuity Check Receiver	1237
10	19.2.9	MEP Continuity Check Initiator	1238
11	19.2.10	MP Loopback Responder	1238
12	19.2.11	MEP Loopback Initiator	1239
13	19.2.12	MEP Linktrace Initiator	1239
14	19.2.13	MEP LTI SAP	1239
15	19.2.14	MEP Linktrace SAP	1239
16	19.2.15	MEP CCM Database	1239
17	19.2.16	MEP Fault Notification Generator	1239
18	19.2.17	MEP Decapsulator Responder (DR)	1239
19	19.2.18	MEP RFM Receiver	1240
20	19.3	MIP Half Function	1240
21	19.3.1	MHF identification	1240
22	19.3.2	MHF functions	1240
23	19.3.3	MHF architecture	1241
24	19.3.4	MHF Level Demultiplexer	1241
25	19.3.5	MHF Type Demultiplexer	1242
26	19.3.6	MHF OpCode Demultiplexer	1242
27	19.3.7	MHF Multiplexer	1242
28	19.3.8	MHF Loopback Responder	1242
29	19.3.9	MHF Continuity Check Receiver	1242
30	19.3.10	MIP CCM Database	1242
31	19.3.11	MHF Linktrace SAP	1242
32	19.3.12	MHF DR	1242
33	19.3.13	MHF RFM Receiver	1242
34	19.4	MP addressing.....	1243
35	19.5	Linktrace Output Multiplexer (LOM).....	1243
36	19.6	Linktrace Responder	1244
37	20.	CFM protocols	1246
38	20.1	Continuity Check protocol.....	1247
39	20.1.1	MAC status reporting in the CCM	1249
40	20.1.2	Defects and Fault Alarms	1249
41	20.1.3	CCM reception	1250
42	20.2	Loopback protocol	1250
43	20.2.1	LBM transmission	1251
44	20.2.2	LBM reception and LBR transmission	1251
45	20.2.3	LBR reception	1252
46	20.3	Linktrace protocol.....	1252
47	20.3.1	LTM origination	1253
48	20.3.2	LTM reception, forwarding, and replying	1254
49	20.3.3	LTR reception	1255
50	20.4	CFM state machines.....	1256

1	20.5	CFM state machine timers	1256
2	20.5.1	LTFwhile	1256
3	20.5.2	CCIwhile	1258
4	20.5.3	errorCCMwhile	1258
5	20.5.4	xconCCMwhile	1258
6	20.5.5	LBIwhile	1258
7	20.5.6	FNGwhile	1258
8	20.5.7	mmCCMwhile	1258
9	20.5.8	mmLocwhile	1258
10	20.5.9	mmFNGwhile	1258
11	20.5.10	rMEPwhile	1258
12	20.6	CFM procedures	1259
13	20.6.1	CCMtime()	1259
14	20.7	Maintenance Domain variable	1259
15	20.7.1	mdLevel	1259
16	20.8	MA variables.....	1259
17	20.8.1	CCMinterval	1259
18	20.9	MEP variables.....	1259
19	20.9.1	MEPactive	1260
20	20.9.2	enableRmepDefect	1260
21	20.9.3	MAdefectIndication	1261
22	20.9.4	allRMEPsDead	1261
23	20.9.5	lowestAlarmPri	1261
24	20.9.6	presentRDI	1261
25	20.9.7	MEPprimaryVID	1261
26	20.9.8	presentTraffic	1261
27	20.9.9	presentmmLoc	1261
28	20.9.10	ISpresentTraffic	1262
29	20.9.11	ISpresentmmLoc	1262
30	20.9.12	EpMEP	1262
31	20.10	MEP Continuity Check Initiator variables.....	1262
32	20.10.1	CCIenabled	1262
33	20.10.2	CCIsentCCMs	1262
34	20.10.3	MACstatusChanged	1262
35	20.10.4	Npaths	1262
36	20.10.5	flowHash[]	1263
37	20.10.6	pathN	1263
38	20.10.7	CCMcnt	1263
39	20.11	MEP Continuity Check Initiator procedures	1263
40	20.11.1	xmitCCM()	1263
41	20.12	MEP Continuity Check Initiator state machine	1264
42	20.13	MHF Continuity Check Receiver variables.....	1264
43	20.13.1	MHFrecvDCCM	1265
44	20.13.2	MHFCCMPDU	1265
45	20.14	MHF Continuity Check Receiver procedures.....	1265
46	20.14.1	MHFprocessCCM()	1265
47	20.15	MHF Continuity Check Receiver state machine	1265

1	20.16	MEP Continuity Check Receiver variables	1265
2	20.16.1	CCMreceivedEqual	1266
3	20.16.2	CCMequalPDU	1266
4	20.16.3	CCMreceivedLow	1266
5	20.16.4	CCMlowPDU	1266
6	20.16.5	recvdMacAddress	1266
7	20.16.6	recvdRDI	1266
8	20.16.7	recvdInterval	1266
9	20.16.8	recvdPortState	1267
10	20.16.9	recvdInterfaceStatus	1267
11	20.16.10	recvdSenderId	1267
12	20.16.11	recvdFrame	1267
13	20.16.12	CCMsequenceErrors	1267
14	20.16.13	rcvdTrafficBit	1267
15	20.17	MEP Continuity Check Receiver procedures	1267
16	20.17.1	MEPprocessEqualCCM()	1267
17	20.17.2	MEPprocessLowCCM()	1268
18	20.18	MEP Continuity Check Receiver state machine.....	1268
19	20.19	Remote MEP variables	1269
20	20.19.1	rMEPCCMdefect	1269
21	20.19.2	rMEPlastRDI and rMEPlastRDI[i]	1269
22	20.19.3	rMEPlastPortState	1270
23	20.19.4	rMEPlastInterfaceStatus	1270
24	20.19.5	rMEPlastSenderId	1270
25	20.19.6	rCCMreceived	1270
26	20.19.7	rMEPmacAddress	1270
27	20.19.8	rMEPportStatusDefect	1270
28	20.19.9	rMEPinterfaceStatusDefect	1270
29	20.19.10	lastPathN	1270
30	20.20	Remote MEP state machine.....	1271
31	20.21	Remote MEP Error variables.....	1271
32	20.21.1	errorCCMreceived	1272
33	20.21.2	errorCCMlastFailure	1272
34	20.21.3	errorCCMdefect	1272
35	20.22	Remote MEP Error state machine	1272
36	20.23	MEP Cross Connect variables	1272
37	20.23.1	xconCCMreceived	1273
38	20.23.2	xconCCMlastFailure	1273
39	20.23.3	xconCCMdefect	1273
40	20.24	MEP Cross Connect state machine.....	1273
41	20.25	MEP Mismatch variables.....	1273
42	20.25.1	mmCCMreceived	1274
43	20.25.2	mmCCMdefect	1274
44	20.25.3	mmCCMTime	1274
45	20.25.4	disableLocdefect	1274
46	20.25.5	mmLocdefect	1274
47	20.26	MEP Mismatch state machines.....	1274
48	20.27	MP Loopback Responder variables.....	1276
49	20.27.1	LBMreceived	1276
50	20.27.2	LBMPDU	1276
51	20.28	MP Loopback Responder procedures.....	1276
52	20.28.1	ProcessLBM()	1276
53	20.28.2	xmitLBR()	1277
54	20.29	MP Loopback Responder state machine.....	1277

1	20.30	MEP Loopback Initiator variables	1277
2	20.30.1	LBMstosend	1278
3	20.30.2	nextLBMtransID	1278
4	20.30.3	expectedLBRtransID	1278
5	20.30.4	LBIactive	1278
6	20.30.5	xmitReady	1278
7	20.30.6	LBRreceived	1278
8	20.30.7	LBRPDU	1279
9	20.31	MEP Loopback Initiator transmit procedures	1279
10	20.31.1	xmitLBM()	1279
11	20.32	MEP Loopback Initiator transmit state machine	1279
12	20.33	MEP Loopback Initiator receive procedures	1280
13	20.33.1	ProcessLBR()	1280
14	20.34	MEP Loopback Initiator receive state machine	1281
15	20.35	MEP Fault Notification Generator variables	1281
16	20.35.1	fngPriority	1281
17	20.35.2	fngDefect	1281
18	20.35.3	fngAlarmTime	1282
19	20.35.4	fngResetTime	1282
20	20.35.5	someRMEPCCMdefect	1282
21	20.35.6	someMACstatusDefect	1282
22	20.35.7	someRDId defect	1282
23	20.35.8	highestDefectPri	1282
24	20.35.9	highestDefect	1282
25	20.36	MEP Fault Notification Generator procedures	1283
26	20.36.1	xmitFaultAlarm()	1283
27	20.37	MEP Fault Notification Generator state machine	1283
28	20.38	MEP Mismatch Fault Notification Generator variables	1284
29	20.38.1	mfngAllowed	1284
30	20.38.2	mmdefectIndication	1284
31	20.38.3	mfngAlarmTime	1284
32	20.38.4	mfngResetTime	1284
33	20.39	MEP Mismatch Fault Notification Generator procedures	1284
34	20.39.1	xmitFaultAlarm()	1284
35	20.40	MEP Mismatch Fault Notification Generator state machine	1284
36	20.41	MEP Linktrace Initiator variables	1285
37	20.41.1	nextLTMtransID	1285
38	20.41.2	ltmReplyList	1286
39	20.42	MEP Linktrace Initiator procedures	1287
40	20.42.1	xmitLTM()	1287
41	20.43	MEP Linktrace Initiator receive variables	1288
42	20.43.1	LTRreceived	1288
43	20.43.2	LTRPDU	1288
44	20.44	MEP Linktrace Initiator receive procedures	1289
45	20.44.1	ProcessLTR()	1289
46	20.45	MEP Linktrace Initiator receive state machine	1289
47	20.46	Linktrace Responder variables	1289
48	20.46.1	nPendingLTRs	1289
49	20.46.2	LTMreceived	1290
50	20.46.3	LTMPDU	1290

1	20.47	LTM Receiver procedures	1290
2	20.47.1	ProcessLTM()	1290
3	20.47.2	clearPendingLTRs()	1294
4	20.47.3	ForwardLTM()	1294
5	20.47.4	enqueueLTR()	1295
6	20.48	LTM Receiver state machine	1297
7	20.49	LTR Transmitter procedure	1297
8	20.49.1	xmitOldestLTR()	1297
9	20.50	LTR Transmitter state machine	1297
10	20.51	CFM PDU validation and versioning	1297
11	20.51.1	Goals of CFM PDU versioning	1298
12	20.51.2	PDU transmission	1298
13	20.51.3	PDU validation	1299
14	20.51.4	Validation pass	1299
15	20.51.5	Execution pass	1300
16	20.51.6	Future extensions	1301
17	20.52	PDU identification	1301
18	20.53	Use of transaction IDs and sequence numbers	1301
19	21.	Encoding of CFM PDUs.....	1303
20	21.1	Structure, representation, and encoding.....	1303
21	21.2	CFM encapsulation	1303
22	21.3	CFM request and indication parameters	1304
23	21.3.1	destination_address parameter	1304
24	21.3.2	source_address parameter	1304
25	21.4	Common CFM Header.....	1304
26	21.4.1	MD Level	1304
27	21.4.2	Version	1304
28	21.4.3	OpCode	1305
29	21.4.4	Flags	1305
30	21.4.5	First TLV Offset	1306
31	21.5	TLV format.....	1306
32	21.5.1	General format for CFM TLVs	1306
33	21.5.2	Organization-Specific TLV	1306
34	21.5.3	Sender ID TLV	1308
35	21.5.4	Port Status TLV	1309
36	21.5.5	Interface Status TLV	1310
37	21.5.6	Data TLV	1311
38	21.5.7	End TLV	1311
39	21.6	CCM format.....	1311
40	21.6.1	Flags	1312
41	21.6.2	First TLV Offset	1313
42	21.6.3	Sequence Number	1313
43	21.6.4	Maintenance association Endpoint Identifier	1313
44	21.6.5	Maintenance Association Identifier	1313
45	21.6.6	Defined by ITU-T G.8013/Y.1731	1316
46	21.6.7	Optional CCM TLVs	1316
47	21.7	LBM and LBR formats.....	1316
48	21.7.1	Flags	1317
49	21.7.2	First TLV Offset	1317
50	21.7.3	Loopback Transaction Identifier	1317
51	21.7.4	Additional LBM/LBR TLVs	1317
52	21.7.5	PBB-TE MIP TLV	1317

1	21.8	LTM format	1318
2	21.8.1	Flags	1319
3	21.8.2	First TLV Offset	1319
4	21.8.3	LTM Transaction Identifier	1319
5	21.8.4	LTM TTL	1319
6	21.8.5	Original MAC Address	1319
7	21.8.6	Target MAC Address	1319
8	21.8.7	Additional LTM TLVs	1320
9	21.8.8	LTM Egress Identifier TLV	1320
10	21.9	LTR format	1320
11	21.9.1	Flags	1320
12	21.9.2	First TLV Offset	1321
13	21.9.3	LTR Transaction Identifier	1321
14	21.9.4	Reply TTL	1321
15	21.9.5	Relay Action	1322
16	21.9.6	Additional LTR TLVs	1322
17	21.9.7	LTR Egress Identifier TLV	1322
18	21.9.8	Reply Ingress TLV	1323
19	21.9.9	Reply Egress TLV	1324
20	22.	CFM in systems	1326
21	22.1	CFM shims in Bridges	1326
22	22.1.1	Preliminary positioning of MPs	1326
23	22.1.2	CFM and the Forwarding Process	1327
24	22.1.3	Up/Down separation of MPs	1329
25	22.1.4	Service instances over multiple Bridges	1331
26	22.1.5	Multiple VID service instances	1333
27	22.1.6	Untagged CFM PDUs	1333
28	22.1.7	MPs and non-VLAN-aware Bridges	1333
29	22.1.8	MPs and other standards	1334
30	22.1.9	CFM and IEEE 802.3 OAM	1336
31	22.2	Maintenance Entity creation	1336
32	22.2.1	Creating Maintenance Domains and MAs	1337
33	22.2.2	Creating MEPs	1337
34	22.2.3	Creating MIPs	1339
35	22.2.4	CFM configuration errors	1340
36	22.3	MPs, Ports, and MD Level assignment.....	1341
37	22.4	Stations and CFM	1342
38	22.5	Scalability of CFM.....	1342
39	22.6	CFM in Provider Bridges.....	1343
40	22.6.1	MPs and C-VLAN components	1343
41	22.6.2	Maintenance C-VLAN on a Port-based service interface	1343
42	22.6.3	Maintenance C-VLAN on a C-tagged service interface	1345
43	22.6.4	MPs and Port-mapping S-VLAN components	1345
44	22.7	Management Port MEPs and CFM in the enterprise environment.....	1347
45	22.8	Implementing CFM on Bridges that implement earlier revisions of IEEE Std 802.1Q	1349
46	23.	MAC status propagation	1350
47	23.1	Model of operation.....	1352
48	23.1.1	MAC Status Shim (MSS)	1353
49	23.1.2	Relationship of CFM to the MSS	1353
50	23.2	MAC Status Protocol (MSP) overview	1353
51	23.3	MSP state machines.....	1358

1	23.4	State machine timers	1359
2	23.4.1	linkNotifyWhen	1359
3	23.4.2	linkNotifyWhile	1359
4	23.4.3	macNotifyWhile	1359
5	23.4.4	macRecoverWhile	1359
6	23.5	MSP performance parameters	1359
7	23.5.1	LinkNotify	1360
8	23.5.2	LinkNotifyWait	1360
9	23.5.3	LinkNotifyRetry	1360
10	23.5.4	MACNotify	1360
11	23.5.5	MACNotifyTime	1360
12	23.5.6	MACRecoverTime	1360
13	23.6	State machine variables	1360
14	23.6.1	BEGIN	1360
15	23.6.2	addConfirmed	1360
16	23.6.3	disableMAC	1360
17	23.6.4	disabledMAC	1360
18	23.6.5	disableMSS	1360
19	23.6.6	lossConfirmed	1360
20	23.6.7	macOperational	1361
21	23.6.8	mssOperational	1361
22	23.6.9	prop	1361
23	23.6.10	rxAck	1361
24	23.6.11	rxAdd	1361
25	23.6.12	rxAddConfirm	1361
26	23.6.13	rxLoss	1361
27	23.6.14	rxLossConfirm	1361
28	23.6.15	txAck	1361
29	23.6.16	txAdd	1361
30	23.6.17	txAddConfirm	1361
31	23.6.18	txLoss	1361
32	23.6.19	txLossConfirm	1361
33	23.7	State machine procedures	1362
34	23.8	Status Transition state machine (STM)	1362
35	23.9	Status Notification state machine (SNM)	1362
36	23.10	Receive Process	1362
37	23.11	Transmit Process	1362
38	23.12	Management of MSP	1363
39	23.13	MSPDU transmission, addressing, and protocol identification	1364
40	23.13.1	Destination MAC Address	1364
41	23.13.2	Source MAC Address	1364
42	23.13.3	Priority	1364
43	23.13.4	EtherType use and encoding	1364
44	23.14	Representation and encoding of octets	1365
45	23.15	MSPDU structure	1365
46	23.15.1	Protocol Version	1365
47	23.15.2	Packet Type	1365
48	23.16	Validation of received MSPDUs	1366
49	23.17	Other MSP participants	1366
50	24.	Bridge performance	1367
51	24.1	Guaranteed Port Filtering Rate	1367
52	24.2	Guaranteed Bridge Relaying Rate	1367

1	24.3	RSTP performance requirements	1368
2	25.	Support of the MAC Service by PBBNs	1369
3	25.1	Service transparency	1371
4	25.2	Customer service interface.....	1371
5	25.3	Port-based service interface.....	1372
6	25.4	S-tagged service interface.....	1373
7	25.5	I-tagged service interface.....	1375
8	25.6	Service instance segregation.....	1377
9	25.7	Service instance selection and identification.....	1377
10	25.8	Service priority and drop eligibility selection.....	1377
11	25.9	Service access protection	1378
12	25.9.1	Class II redundant LANs access protection	1380
13	25.9.2	Class III simple redundant LANs and nodes access protection	1380
14	25.10	Support of the MAC Service by a PBB-TE Region	1381
15	25.10.1	Provisioning TESI.....	1382
16	25.10.2	ESP forwarding behavior	1384
17	25.11	Transparent service interface.....	1385
18	26.	Principles of Provider Backbone Bridged Network (PBBN) operation	1387
19	26.1	PBBN overview	1387
20	26.2	PBBN example	1388
21	26.3	B-VLAN connectivity.....	1390
22	26.4	Backbone addressing	1390
23	26.4.1	Learning individual backbone addresses at a PIP	1391
24	26.4.2	Translating backbone destination addresses at a CBP	1392
25	26.4.3	Backbone addressing considerations for CFM MPs	1392
26	26.5	Detection of connectivity loops through attached networks.....	1393
27	26.6	Scaling of PBBs	1393
28	26.6.1	Hierarchical PBBNs	1393
29	26.6.2	Peer PBBNs	1394
30	26.7	Network management.....	1394
31	26.8	CFM in PBBs.....	1394
32	26.8.1	CFM over Port-based and S-tagged service interfaces	1399
33	26.8.2	CFM over I-tagged Service Interfaces	1400
34	26.8.3	CFM over hierarchical E-NNI	1400
35	26.8.4	CFM over peer E-NNI	1401
36	26.9	CFM in a PBB-TE Region.....	1401
37	26.9.1	Addressing PBB-TE MEPs	1402
38	26.9.2	TESI identification	1402
39	26.9.3	PBB-TE MEP placement in a Bridge Port	1402
40	26.9.4	PBB-TE MIP placement in a Bridge Port	1403
41	26.9.5	TESI Maintenance Domains	1403
42	26.9.6	PBB-TE enhancements of the CFM protocols	1404
43	26.9.7	Addressing Infrastructure Segment MEPs	1406
44	26.9.8	Infrastructure Segment identification	1406
45	26.9.9	Infrastructure Segment MEP placement in a Bridge Port	1406
46	26.9.10	Infrastructure Segment Maintenance Domains	1408
47	26.9.11	IPS extensions to Continuity Check operation	1408
48	26.10	Protection switching for point-to-point TESI.....	1408
49	26.10.1	Introduction	1408
50	26.10.2	1:1 point-to-point TESI protection switching	1409
51	26.10.3	Protection Switching state machines	1412

1	26.11	IPS in PBB-TE Region	1418
2	26.11.1	Infrastructure Segment monitoring	1419
3	26.11.2	1:1 IPS	1420
4	26.11.3	IPS Control entity	1423
5	26.11.4	1:1 IPS state machines	1424
6	26.11.5	M:1 IPS	1425
7	26.12	Mismatch defect.....	1430
8	26.13	Signaling VLAN registrations among I-components	1431
9	27.	Shortest Path Bridging (SPB)	1432
10	27.1	Protocol design requirements.....	1434
11	27.2	Protocol support.....	1435
12	27.3	Protocol design goals	1436
13	27.4	ISIS-SPB VLAN configuration	1436
14	27.4.1	SPT Region and ISIS-SPB adjacency determination	1437
15	27.5	ISIS-SPB information	1439
16	27.6	Calculating CIST connectivity.....	1440
17	27.7	Connectivity between regions in the same domain	1441
18	27.8	Calculating SPT connectivity	1441
19	27.8.1	ISIS-SPB overload	1442
20	27.9	Loop prevention	1442
21	27.10	SPVID and SPSourceID allocation.....	1442
22	27.11	Allocation of VIDs to FIDs	1444
23	27.12	SPBV SPVID translation	1445
24	27.13	VLAN topology management.....	1445
25	27.14	Individual addresses and SPBM	1446
26	27.14.1	Loop mitigation	1447
27	27.14.2	Loop prevention	1447
28	27.15	SPBM group addressing	1447
29	27.16	Backbone service instance topology management	1449
30	27.17	Equal cost shortest paths, ECTs, and load spreading	1450
31	27.18	Connectivity Fault Management for SPBM	1450
32	27.18.1	SPBM MA types	1450
33	27.18.2	SPBM MEP placement in a Bridge Port	1451
34	27.18.3	SPBM MIP placement in a Bridge Port	1452
35	27.18.4	SPBM modifications of the CFM protocols	1452
36	27.19	Using SPBV and SPBM modes	1453
37	27.19.1	Shortest Path Bridging—VID	1453
38	27.19.2	Shortest Path Bridging—MAC	1454
39	27.20	Security considerations	1456
40	28.	ISIS-SPB Link State Protocol.....	1457
41	28.1	ISIS-SPB control plane MAC.....	1457
42	28.2	Formation and maintenance of ISIS-SPB adjacencies	1458
43	28.3	Loop prevention	1459
44	28.4	The Agreement Digest	1459
45	28.4.1	Agreement Digest Format Identifier	1459
46	28.4.2	Agreement Digest Format Capabilities	1460
47	28.4.3	Agreement Digest Convention Identifier	1460
48	28.4.4	Agreement Digest Convention Capabilities	1460
49	28.4.5	Agreement Digest Edge Count	1461
50	28.4.6	The Computed Topology Digest	1461
51	28.5	Symmetric shortest path tie breaking.....	1462

1	28.6	Symmetric ECT framework.....	1463
2	28.7	Symmetric ECT	1464
3	28.8	Symmetric ECT Algorithm details	1465
4	28.9	ECT Migration.....	1466
5	28.9.1	Use of a new ECT Algorithm in SPBV	1466
6	28.9.2	Use of a new ECT Algorithm in SPBM	1467
7	28.10	MAC address registration	1468
8	28.11	Circuit IDs and Port Identifiers.....	1468
9	28.12	ISIS-SPB TLVs.....	1468
10	28.12.1	MT-Capability TLV	1469
11	28.12.2	SPB MCID sub-TLV	1469
12	28.12.3	SPB Digest sub-TLV	1470
13	28.12.4	SPB Base VLAN-Identifiers sub-TLV	1471
14	28.12.5	SPB Instance sub-TLV	1472
15	28.12.6	SPB Instance Opaque ECT Algorithm sub-TLV	1473
16	28.12.7	SPB Link Metric sub-TLV	1475
17	28.12.8	SPB Adjacency Opaque ECT Algorithm sub-TLV	1475
18	28.12.9	SPBV MAC address sub-TLV	1476
19	28.12.10	SPBM Service Identifier and Unicast Address (ISID-ADDR) sub-TLV	1477
20	29.	DDCFM operations and protocols.....	1480
21	29.1	Principles of DDCFM operation.....	1480
22	29.1.1	Data-driven and data-dependent faults (DDFs)	1480
23	29.1.2	Basic principle to diagnose and isolate DDFs	1480
24	29.2	DDCFM Entity operation	1483
25	29.2.1	DDCFM implementation	1483
26	29.2.2	FPT RR	1484
27	29.2.3	RR-related parameters	1485
28	29.2.4	Reflection Target and RFM Receiver	1486
29	29.2.5	RPT-related parameters	1486
30	29.2.6	Decapsulator Responder (DR)	1487
31	29.2.7	SFM Originator	1488
32	29.3	DDCFM protocols	1488
33	29.3.1	RR variables	1488
34	29.3.2	RR Filter procedures	1490
35	29.3.3	RR Encapsulation procedures	1491
36	29.3.4	RR Transmit procedure	1492
37	29.3.5	RR-related state machines	1493
38	29.3.6	RFM Receiver variables	1494
39	29.3.7	RFM Receiver procedure	1495
40	29.3.8	DR variables	1496
41	29.3.9	DR procedures	1496
42	29.3.10	Decapsulator Responder state machine	1498
43	29.4	Encoding of DDCFM PDUs.....	1498
44	29.4.1	RFM and SFM Header	1498
45	29.4.2	RFM format	1498
46	29.4.3	SFM format	1499
47	30.	Principles of congestion notification	1501
48	30.1	Congestion notification design requirements	1501

1	30.2	Quantized Congestion Notification protocol (QCN)	1503
2	30.2.1	The CP algorithm	1504
3	30.2.2	Basic RP algorithm	1505
4	30.2.3	RP algorithm with timer	1506
5	30.3	Congestion Controlled Flow (CCF).....	1507
6	30.4	Congestion Notification Priority Value (CNPV).....	1508
7	30.5	Congestion Notification tag (CN-TAG)	1508
8	30.6	Congestion Notification Domain (CND).....	1508
9	30.7	Multicast data.....	1509
10	30.8	Congestion notification and additional tags.....	1509
11	31.	Congestion notification entity operation.....	1511
12	31.1	Congestion-aware Bridge Forwarding Process.....	1511
13	31.1.1	Congestion Point (CP)	1512
14	31.1.2	CP ingress multiplexer	1512
15	31.2	Congestion-aware end station functions.....	1512
16	31.2.1	Output flow segregation	1513
17	31.2.2	Per-CNPV station function	1514
18	31.2.3	Flow Select Database	1516
19	31.2.4	Flow multiplexer	1516
20	31.2.5	CNM demultiplexer	1516
21	31.2.6	Input flow segregation	1516
22	31.2.7	End station input queue	1517
23	31.2.8	Reception selection	1517
24	32.	Congestion notification protocol	1518
25	32.1	CND operations	1518
26	32.1.1	CND defense	1518
27	32.1.2	Automatic CND recognition	1520
28	32.1.3	Variables controlling CND defense	1520
29	32.2	CN component variables.....	1521
30	32.2.1	cngMasterEnable	1522
31	32.2.2	cngCnmTransmitPriority	1522
32	32.2.3	cngDiscardedFrames	1522
33	32.2.4	cngErroredPortList	1522
34	32.3	Congestion notification per-CNPV variables	1522
35	32.3.1	cncpDefModeChoice	1522
36	32.3.2	cncpAlternatePriority	1523
37	32.3.3	cncpAutoAltPri	1523
38	32.3.4	cncpAdminDefenseMode	1523
39	32.3.5	cncpCreation	1523
40	32.3.6	cncpLdpInstanceChoice	1523
41	32.3.7	cncpLdpInstanceSelector	1523

1	32.4	CND defense per-Port per-CNPV variables	1524
2	32.4.1	cnpdDefModeChoice	1524
3	32.4.2	cnpdAdminDefenseMode	1524
4	32.4.3	cnpdAutoDefenseMode	1525
5	32.4.4	cnpdLldpInstanceChoice	1525
6	32.4.5	cnpdLldpInstanceSelector	1525
7	32.4.6	cnpdAlternatePriority	1525
8	32.4.7	cnpdXmitCnpvCapable	1525
9	32.4.8	cnpdXmitReady	1525
10	32.4.9	cncpDoesEdge	1526
11	32.4.10	cnpdAcceptsCnTag	1526
12	32.4.11	cnpdRcvdCnpv	1526
13	32.4.12	cnpdRcvdReady	1526
14	32.4.13	cnpdIsAdminDefMode	1526
15	32.4.14	cnpdDefenseMode	1526
16	32.5	CND defense procedures	1527
17	32.5.1	DisableCnpvRemapping()	1527
18	32.5.2	TurnOnCnDefenses()	1527
19	32.5.3	TurnOffCnDefenses()	1527
20	32.6	CND defense state machine.....	1527
21	32.7	Congestion notification protocol	1528
22	32.8	CP variables	1529
23	32.8.1	cpMacAddress	1530
24	32.8.2	cpId	1530
25	32.8.3	cpQSp	1530
26	32.8.4	cpQLen	1530
27	32.8.5	cpQLenOld	1530
28	32.8.6	cpW	1530
29	32.8.7	cpQOffset	1530
30	32.8.8	cpQDelta	1530
31	32.8.9	cpFb	1530
32	32.8.10	cpEnqueued	1531
33	32.8.11	cpSampleBase	1531
34	32.8.12	cpDiscardedFrames	1531
35	32.8.13	cpTransmittedFrames	1531
36	32.8.14	cpTransmittedCnms	1531
37	32.8.15	cpMinHeaderOctets	1531
38	32.9	CP procedures	1531
39	32.9.1	Random	1531
40	32.9.2	NewCpSampleBase()	1531
41	32.9.3	EM_UNITDATA.request (parameters)	1532
42	32.9.4	GenerateCnmPdu()	1532
43	32.10	RP per-Port per-CNPV variables	1533
44	32.10.1	rpppMaxRps	1533
45	32.10.2	rpppCreatedRps	1533
46	32.10.3	rpppRpCentiseconds	1534

1	32.11	RP group variables.....	1534
2	32.11.1	rpgEnable	1534
3	32.11.2	rpgTimeReset	1534
4	32.11.3	rpgByteReset	1534
5	32.11.4	rpgThreshold	1535
6	32.11.5	rpgMaxRate	1535
7	32.11.6	rpgAiRate	1535
8	32.11.7	rpgHaiRate	1535
9	32.11.8	rpgGd	1535
10	32.11.9	rpgMinDecFac	1535
11	32.11.10	rpgMinRate	1535
12	32.12	RP timer.....	1535
13	32.12.1	RpWhile	1535
14	32.13	RP variables.....	1536
15	32.13.1	rpEnabled	1536
16	32.13.2	rpByteCount	1536
17	32.13.3	rpByteStage	1536
18	32.13.4	rpTimeStage	1536
19	32.13.5	rpTargetRate	1536
20	32.13.6	rpCurrentRate	1536
21	32.13.7	rpFreeze	1536
22	32.13.8	rpLimiterRate	1537
23	32.13.9	rpFb	1537
24	32.14	RP procedures.....	1537
25	32.14.1	ResetCnm	1537
26	32.14.2	TestRpTerminate	1537
27	32.14.3	TransmitDataFrame	1537
28	32.14.4	ReceiveCnm	1538
29	32.14.5	ProcessCnm	1538
30	32.14.6	AdjustRates	1538
31	32.15	RP rate control state machine	1539
32	32.16	Congestion notification and encapsulation interworking function.....	1541
33	33.	Encoding of congestion notification PDUs	1543
34	33.1	Structure, representation, and encoding.....	1543
35	33.2	CN-TAG format.....	1543
36	33.2.1	Flow Identifier	1544
37	33.3	Congestion Notification Message (CNM).....	1544
38	33.4	Congestion Notification Message PDU format	1544
39	33.4.1	Version	1544
40	33.4.2	ReservedV	1544
41	33.4.3	Quantized Feedback	1545
42	33.4.4	Congestion Point Identifier	1545
43	33.4.5	cnmQOffset	1545
44	33.4.6	cnmQDelta	1545
45	33.4.7	Encapsulated priority	1546
46	33.4.8	Encapsulated destination MAC address	1546
47	33.4.9	Encapsulated MSDU length	1546
48	33.4.10	Encapsulated MSDU	1546
49	33.4.11	CNM Validation	1546
50	34.	Forwarding and Queuing Enhancements for time-sensitive streams (FQTSS).....	1547
51	34.1	Overview.....	1547

1	34.2	Detection of SRP domains	1547
2	34.3	The bandwidth availability parameters	1548
3	34.3.1	deltaBandwidth when lockClassBandwidth is false	1548
4	34.3.2	deltaBandwidth when lockClassBandwidth is true	1549
5	34.3.3	Bandwidth availability parameter management	1549
6	34.4	Deriving actual bandwidth requirements from the size of the MSDU	1550
7	34.5	Default SR class configuration	1551
8	34.6	Transmission selection	1552
9	34.6.1	Credit-based shaper	1553
10	34.6.2	Strict priority	1554
11	34.6.3	Scheduled traffic	1555
12	35.	Stream Reservation Protocol (SRP)	1556
13	35.1	Multiple Stream Registration Protocol (MSRP)	1557
14	35.1.1	MSRP and Shared Media	1558
15	35.1.2	Behavior of end stations	1559
16	35.1.3	Behavior of Bridges	1561
17	35.1.4	SRP domains and status parameters	1561
18	35.2	Definition of the MSRP application	1562
19	35.2.1	Definition of internal state variables	1562
20	35.2.2	Definition of MRP elements	1564
21	35.2.3	Provision and support of Stream registration service	1588
22	35.2.4	MSRP Attribute Propagation	1593
23	35.2.5	Operational reporting and statistics	1602
24	35.2.6	Encoding	1602
25	35.2.7	Attribute value support requirements	1603
26	36.	Priority-based Flow Control (PFC)	1604
27	36.1	PFC operation	1604
28	36.1.1	Overview	1604
29	36.1.2	PFC primitives	1605
30	36.1.3	Detailed specification of PFC operation	1606
31	36.2	PFC-aware system queue functions	1607
32	36.2.1	PFC Initiator	1607
33	36.2.2	PFC Receiver	1607
34	37.	Enhanced Transmission Selection (ETS)	1610
35	37.1	Overview	1610
36	37.1.1	Relationship to other transmission selection algorithms	1610
37	37.2	ETS configuration parameters	1610
38	37.3	ETS algorithm	1610
39	37.4	Legacy configuration	1611
40	38.	Data Center Bridging eXchange protocol (DCBX)	1612
41	38.1	Overview	1612
42	38.2	Goals	1612
43	38.3	Types of DCBX attributes	1612
44	38.3.1	Informational attributes	1612
45	38.4	DCBX and LLDP	1612
46	38.4.1	Asymmetric attribute passing	1613
47	38.4.2	Symmetric attribute passing	1614

1	39.	Multiple I-SID Registration Protocol (MIRP)	1616
2	39.1	MIRP overview	1616
3	39.1.1	Behavior of I-components	1618
4	39.1.2	Behavior of B-components	1618
5	39.2	Definition of the MIRP application	1618
6	39.2.1	Definition of MRP elements	1618
7	39.2.2	Alternate MIRP model for B-components	1621
8	39.2.3	Use of “new” declaration capability	1623
9	39.2.4	Attribute value support requirements	1623
10	39.2.5	MRP Message filtering	1623
11	40.	Edge Virtual Bridging (EVB)	1624
12	40.1	EVB architecture without S-channels	1625
13	40.2	EVB architecture with S-channels	1626
14	40.3	Asymmetric EVB architecture without S-channels	1628
15	40.4	EVB status parameters	1630
16	40.4.1	EVBMode = Not supported	1630
17	40.4.2	EVBMode = EVB Bridge	1630
18	40.4.3	EVBMode = EVB station	1630
19	40.4.4	EVBMode = NVO3 Mode	1631
20	40.5	EVB Status Parameter for NVO3 Mode Support	1631
21	40.5.1	NVERole = nNVE	1631
22	40.5.2	NVERole = tNVE	1631
23	41.	VSI Discovery and Configuration Protocol (VDP)	1632
24	41.1	VSI manager ID TLV definition	1632
25	41.1.1	TLV type	1633
26	41.1.2	TLV information string length	1633
27	41.1.3	VSI Manager ID	1633
28	41.2	VDP association TLV definitions	1633
29	41.2.1	TLV type	1634
30	41.2.2	TLV information string length	1634
31	41.2.3	Status	1634
32	41.2.4	VSI Type ID (VTID)	1635
33	41.2.5	VSI Type Version	1635
34	41.2.6	VSIID format	1635
35	41.2.7	VSIID	1636
36	41.2.8	Filter Info format	1636
37	41.2.9	Filter Info field	1637
38	41.2.10	VDP TLV type and status semantics	1641
39	41.3	Organizationally defined TLV definitions	1642
40	41.3.1	TLV type	1642
41	41.3.2	TLV information string length	1642
42	41.3.3	Organizationally unique identifier (OUI) or Company ID (CID)	1642
43	41.3.4	Organizationally defined information	1643
44	41.4	Validation rules for VDP TLVs	1643

1	41.5	VDP state machines	1643
2	41.5.1	State machine conventions	1643
3	41.5.2	Bridge VDP state machine	1644
4	41.5.3	Station VDP state machine	1645
5	41.5.4	VDP state machine timers	1646
6	41.5.5	VDP state machine variables and parameters	1646
7	41.5.6	Command-Response TLV field references in state machines	1648
8	41.5.7	VDP state machine procedures	1649
9	42.	S-Channel Discovery and Configuration Protocol (CDCP)	1651
10	42.1	CDCP discovery and configuration	1651
11	42.2	CDCP state machine overview	1651
12	42.3	CDCP configuration state machine.....	1652
13	42.4	CDCP configuration variables	1653
14	42.4.1	AdminChnCap	1653
15	42.4.2	AdminRole	1654
16	42.4.3	AdminSVIDWants	1654
17	42.4.4	LastLocalSVIDPool	1654
18	42.4.5	LastRemoteSVIDList	1654
19	42.4.6	LastSVIDWants	1654
20	42.4.7	LocalSVIDPool	1654
21	42.4.8	OperChnCap	1654
22	42.4.9	OperRole	1654
23	42.4.10	OperSVIDList	1655
24	42.4.11	RemoteChnCap	1655
25	42.4.12	RemoteRole	1655
26	42.4.13	RemoteSVIDList	1655
27	42.4.14	schState	1655
28	42.5	CDCP configuration procedures.....	1655
29	42.5.1	SetSVIDRequest (OperRole, AdminSVIDWants, OperSVIDList)	1655
30	42.5.2	RxSVIDConfig (OperSVIDList, LastRemoteSVIDList)	1656
31	42.5.3	TxSVIDConfig (OperChnCap, RemoteChnCap, LastLocalSVIDPool, RemoteSVIDList, OperSVIDList)	1656
32			
33	43.	Edge Control Protocol (ECP)	1657
34	43.1	ECP operation.....	1657
35	43.2	Edge Control Sublayer Service (ECSS).....	1658
36	43.3	ECP state machines.....	1658
37	43.3.1	State machine conventions	1658
38	43.3.2	Overview	1658
39	43.3.3	Edge Control Protocol Data Unit (ECPDU)	1659
40	43.3.4	ECP transmit state machine	1660
41	43.3.5	ECP receive state machine	1661
42	43.3.6	ECP state machine timers	1661
43	43.3.7	ECP state machine variables and parameters	1662
44	43.3.8	ECP state machine procedures	1663
45	44.	Equal Cost Multiple Paths (ECMP).....	1664
46	44.1	SPBM ECMP.....	1664
47	44.1.1	ECMP Operation	1664
48	44.1.2	ECMP ECT Algorithm	1665
49	44.1.3	Loop prevention for ECMP	1667

1	44.2	Support for Flow Filtering	1667
2	44.2.1	Flow filtering tag (F-TAG)	1668
3	44.2.2	F-TAG processing	1669
4	44.2.3	Forwarding process extension for flow filtering	1670
5	44.2.4	TTL Loop mitigation	1671
6	44.2.5	CFM for ECMP with flow filtering	1671
7	44.2.6	Operation with selective support for flow filtering	1673
8	45.	Path Control and Reservation (PCR)	1674
9	45.1	Explicit trees	1674
10	45.1.1	Tree structures	1678
11	45.1.2	Explicit ECT Algorithms	1679
12	45.1.3	ISIS-PCR VLAN configuration	1681
13	45.1.4	Use of VIDs for strict explicit trees	1685
14	45.1.5	MAC addresses and ISIS-PCR	1686
15	45.1.6	Filtering Database entries for explicit trees	1686
16	45.1.7	ISIS-PCR support	1687
17	45.1.8	Attributes for path computation	1687
18	45.1.9	Topology sub-TLV	1689
19	45.1.10	Hop sub-TLV	1692
20	45.1.11	Administrative Group sub-TLV	1696
21	45.1.12	Bandwidth Constraint sub-TLV	1696
22	45.2	Reservation	1697
23	45.2.1	Bandwidth Assignment sub-TLV	1697
24	45.2.2	Timestamp sub-TLV	1698
25	45.2.3	Precedence ordering	1699
26	45.3	Redundancy	1699
27	45.3.1	Loop-free alternates for unicast data flows	1699
28	45.3.2	Static redundant trees	1700
29	45.3.3	Maximally Redundant Trees (MRTs)	1701
30	45.3.4	MRTs with centralized GADAG computation	1703
31	46.	Time-Sensitive Networking (TSN) configuration	1708
32	46.1	Overview of TSN configuration	1708
33	46.1.1	Streams, Talkers, and Listeners	1708
34	46.1.2	TSN configuration protocols	1708
35	46.1.3	TSN configuration models	1708
36	46.1.4	Stream identification and transformation	1712
37	46.1.5	Centralized User Configuration	1714
38	46.1.6	Centralized Network Configuration	1715
39	46.1.7	Configuration Domain	1716
40	46.2	User/network configuration information	1716
41	46.2.1	Data types	1716
42	46.2.2	Protocol integration	1717
43	46.2.3	Talker	1718
44	46.2.4	Listener	1731
45	46.2.5	Status	1731
46	46.2.6	Protocol operations	1738
47	46.2.7	Remote Procedure Calls	1738
48	46.2.8	Actions	1740
49	46.2.9	Notifications	1740
50	46.3	YANG for TSN user/network configuration	1741

1	47.	Asynchronous Traffic Shaping (ATS) in end stations	1742
2	47.1	Talker transmission behavior	1742
3	47.1.1	ATS traffic class model in Talkers	1742
4	47.1.2	Simplified ProcessFrame(frame) procedure	1742
5	47.1.3	System clock functions and processing delays	1742
6	47.2	Scheduler parameter consistency	1743
7	48.	YANG Data Models	1744
8	48.1	YANG Framework	1744
9	48.1.1	Interface Management (IETF RFC 8343) Model	1745
10	48.2	IEEE 802.1Q YANG models	1747
11	48.2.1	VLAN Bridge components model	1747
12	48.2.2	Two-Port MAC Relay (TPMR) model	1750
13	48.2.3	Customer VLAN Bridge model	1752
14	48.2.4	Provider Bridge model	1753
15	48.2.5	CFM Model	1756
16	48.2.6	Stream filters and stream gates model	1760
17	48.2.7	Asynchronous Traffic Shaping (ATS) model	1761
18	48.2.8	Congestion Isolation (CI) model	1762
19	48.2.9	Scheduled Traffic model	1763
20	48.2.10	Frame Preemption model	1764
21	48.2.11	Per-Stream Filtering and Policing model	1764
22	48.2.12	CNC-configuration model	1766
23	48.2.13	Credit-based shaper algorithm model	1768
24	48.2.14	Rapid Spanning Tree Protocol (RSTP) model	1769
25	48.2.15	Multiple Spanning Tree Protocol (MSTP) model	1770
26	48.3	Structure of the YANG models	1771
27	48.3.1	VLAN Bridge components model	1773
28	48.3.2	Two-Port MAC Relay model	1773
29	48.3.3	Customer VLAN Bridge model	1773
30	48.3.4	Provider Bridge model	1773
31	48.3.5	CFM model	1774
32	48.3.6	Stream filters and stream gates model	1774
33	48.3.7	Asynchronous Traffic Shaping (ATS) model	1775
34	48.3.8	Congestion Isolation (CI) model	1775
35	48.3.9	Scheduled Traffic model	1775
36	48.3.10	Frame Preemption model	1776
37	48.3.11	Per-Stream Filtering and Policing model	1776
38	48.3.12	User/Network Interface model	1777
39	48.3.13	Credit-based shaper algorithm model	1777
40	48.3.14	RSTP model	1777
41	48.3.15	MSTP model	1778

1	48.4	Security considerations	1779
2	48.4.1	Security considerations of the VLAN Bridge components model	1779
3	48.4.2	Security considerations of the Two-Port MAC Relay model	1780
4	48.4.3	Security considerations of the Customer VLAN Bridge model	1780
5	48.4.4	Security considerations of the Provider Bridge model	1780
6	48.4.5	Security considerations of the CFM model	1780
7	48.4.6	Security considerations of the Stream filters and stream gates model	1780
8	48.4.7	Security considerations of the Asynchronous Traffic Shaping model	1781
9	48.4.8	Security considerations of the Congestion Isolation model	1781
10	48.4.9	Security considerations of the Scheduled Traffic model	1781
11	48.4.10	Security considerations of the Frame Preemption model	1781
12	48.4.11	Security considerations of the Per-Stream Filtering and Policing model	1782
13	48.4.12	Security considerations of the CNC-configuration model	1782
14	48.4.13	Security considerations of the credit-based shaper algorithm model	1782
15	48.4.14	Security considerations of the RSTP model	1782
16	48.4.15	Security considerations of the Multiple Spanning Trees model	1782
17	48.5	YANG schema tree definitions	1783
18	48.5.1	Schema for the ieee802-types YANG module	1783
19	48.5.2	Schema for the ieee802-dot1q-types YANG module	1783
20	48.5.3	Schema for the ieee802-dot1q-tsn-types YANG module	1783
21	48.5.4	Schema for the ieee802-dot1q-bridge YANG module	1783
22	48.5.5	Schema for the ieee802-dot1q-tpmr YANG module	1787
23	48.5.6	Schema for the ieee802-dot1q-pb YANG module	1787
24	48.5.7	Schema for the ieee802-dot1q-cfm-types YANG module	1788
25	48.5.8	Schema for the ieee802-dot1q-cfm YANG module	1788
26	48.5.9	Schema for the ieee802-dot1q-cfm-bridge YANG module	1790
27	48.5.10	Schema for the ieee802-dot1q-cfm-alarm YANG module	1791
28	48.5.11	Schema for the ieee802-dot1q-stream-filters-gates YANG module	1791
29	48.5.12	Schema for the ieee802-dot1q-stream-filters-gates-bridge YANG module	1791
30	48.5.13	Schema for the ieee802-dot1q-ats YANG module	1792
31	48.5.14	Schema for the ieee802-dot1q-ats-bridge YANG module	1794
32	48.5.15	Schema for the ieee802-dot1q-congestion-isolation YANG module	1795
33	48.5.16	Schema for the ieee802-dot1q-congestion-isolation-bridge YANG module	1796
34	48.5.17	Schema for the ieee802-dot1q-sched YANG module	1797
35	48.5.18	Schema for the ieee802-dot1q-sched-bridge YANG module	1797
36	48.5.19	Schema for the ieee802-dot1q-preemption YANG module	1798
37	48.5.20	Schema for the ieee802-dot1q-preemption-bridge YANG module	1798
38	48.5.21	Schema for the ieee802-dot1q-psfp YANG module	1798
39	48.5.22	Schema for the ieee802-dot1q-psfp-bridge YANG module	1798
40	48.5.23	Schema for the ieee802-dot1q-cnc-config YANG module	1799
41	48.5.24	Schema for the ieee802-dot1q-cbsa YANG module	1803
42	48.5.25	Schema for the ieee802-dot1q-cbsa-bridge YANG module	1803
43	48.5.26	Schema for the ieee802-dot1q-rstp YANG module	1803
44	48.5.27	Schema for the ieee802-dot1q-rstp-bridge YANG module	1803
45	48.5.28	Schema for the ieee802-dot1q-mstp YANG module	1804
46	48.5.29	Schema for the ieee802-dot1q-mstp-bridge YANG module	1804

1	48.6	YANG modules	1806
2	48.6.1	The ieee802-types YANG module	1806
3	48.6.2	The ieee802-dot1q-types YANG module	1811
4	48.6.3	The ieee802-dot1q-tsn-types YANG module	1827
5	48.6.4	The ieee802-dot1q-bridge YANG module	1847
6	48.6.5	The ieee802-dot1q-tpmr YANG module	1873
7	48.6.6	The ieee802-dot1q-pb YANG module	1878
8	48.6.7	The ieee802-dot1q-cfm-types YANG module	1882
9	48.6.8	The ieee802-dot1q-cfm YANG module	1893
10	48.6.9	The ieee802-dot1q-cfm-bridge YANG module	1912
11	48.6.10	The ieee802-dot1q-cfm-alarm YANG module	1920
12	48.6.11	The ieee802-dot1q-stream-filters-gates YANG module	1922
13	48.6.12	The ieee802-dot1q-stream-filters-gates-bridge YANG module	1928
14	48.6.13	The ieee802-dot1q-ats YANG module	1929
15	48.6.14	The ieee802-dot1q-ats-bridge YANG module	1934
16	48.6.15	The ieee802-dot1q-congestion-isolation YANG module	1936
17	48.6.16	The ieee802-dot1q-congestion-isolation-bridge YANG module	1944
18	48.6.17	The ieee802-dot1q-sched YANG module	1946
19	48.6.18	The ieee802-dot1q-sched-bridge YANG module	1952
20	48.6.19	The ieee802-dot1q-preemption YANG module	1953
21	48.6.20	The ieee802-dot1q-preemption-bridge YANG module	1957
22	48.6.21	The ieee802-dot1q-psfp YANG module	1958
23	48.6.22	The ieee802-dot1q-psfp-bridge YANG module	1968
24	48.6.23	The ieee802-dot1q-cnc-config YANG module	1969
25	48.6.24	The ieee802-dot1q-cbsa YANG module	1977
26	48.6.25	The ieee802-dot1q-cbsa-bridge YANG module	1979
27	48.6.26	The ieee802-dot1q-rstp YANG module	1980
28	48.6.27	The ieee802-dot1q-rstp-bridge YANG module	1989
29	48.6.28	The ieee802-dot1q-mstp YANG module	1990
30	48.6.29	The ieee802-dot1q-mstp-bridge YANG module	1997
31	49.	Congestion Isolation	1999
32	49.1	Congestion isolation objectives	2000
33	49.2	Principles of congestion isolation	2001
34	49.2.1	Congesting flow identification	2002
35	49.2.2	IEEE Std 802.1CB stream identification	2002
36	49.2.3	Flow priority modification	2002
37	49.2.4	Priority-based Flow Control interaction	2002
38	49.2.5	Congestion isolation signaling	2003
39	49.2.6	Congesting to non-congesting status change	2003
40	49.2.7	System topology and port orientation	2004
41	49.2.8	Comparison to Congestion Notification	2004
42	49.3	The Congestion Isolation Aware Forwarding Process	2005
43	49.3.1	CIP Congestion Detection	2006
44	49.3.2	CIP transmission gates	2006
45	49.3.3	CIM Demultiplexer	2006
46	49.3.4	Congesting flow identification	2006
47	49.3.5	CIM Multiplexer	2006
48	49.3.6	CI Peer Table	2007
49	49.3.7	CI Stream Table	2007

1	49.4	Congestion Isolation Protocol.....	2007
2	49.4.1	Variables controlling operation	2007
3	49.4.2	CIP procedures	2011
4	49.4.3	Encoding of the CIM PDU	2018
5	49.4.4	LLDP Congestion Isolation TLV	2021
6	49.5	Topology Recognition	2022
7	49.5.1	TR theory of operation	2022
8	49.5.2	TR variables controlling operation	2023
9	49.5.3	TR procedures	2023
10	50.	PBBN Auto Attach	2025
11	50.1	Overview.....	2025
12	50.2	Service interfaces.....	2026
13	50.2.1	S-tagged service interface	2026
14	50.2.2	C-tagged service interface	2027
15	50.3	State machine overview	2028
16	50.3.1	System inactivity	2029
17	50.3.2	Multiple AAD systems	2029
18	50.3.3	AAD assignment requests	2029
19	50.3.4	AAB assignment request processing	2030
20	50.3.5	Assignment updates	2030
21	50.4	State machine variables	2031
22	50.4.1	Structure type definitions	2031
23	50.4.2	Per AAS variables	2032
24	50.4.3	Per port variables	2033
25	50.5	State machine functions	2036
26	50.5.1	aaBuildSysTlv(SYSstring tlv)	2036
27	50.5.2	aaBuildAsgnsTlv(ASGNSstring tlv, VIDISIDpair vidIsidArray[], ASGNSstring operTlv)	2036
28	50.5.3	aaConfigAabAsgn(unsigned vid, unsigned isid)	2037
29	50.5.4	aaDisableAabVidIsidAsgns(ASGNSstring tlvNew, ASGNSstring tlvOld) ...	2037
30	50.5.5	aaDisableLldpAaTlvTx()	2038
31	50.5.6	aaDisableVidIsidAsgns()	2038
32	50.5.7	aaDiscAssocAabTblCheck(unsigned port, SYSstring tlv)	2038
33	50.5.8	aaDiscAssocAadTblCheck(unsigned port, SYSstring tlv)	2039
34	50.5.9	aaDiscAssocTblDelete(unsigned port)	2039
35	50.5.10	aaDiscAssocTblInsert(unsigned port, SYSstring tlv)	2039
36	50.5.11	aaEnableAabVidIsidAsgns(ASGNSstring tlv)	2039
37	50.5.12	aaEnableAadVidIsidAsgns(ASGNSstring tlv)	2040
38	50.5.13	aaPortClearStats()	2040
39	50.5.14	aaSetAsgnsTlv(ASGNSstring tlv)	2040
40	50.5.15	aaSetSysTlv(SYSstring tlv)	2040
41	50.5.16	aaSomethingChangedRemote()	2041
42	50.5.17	aaSomethingChangeRemoteInit()	2041
43	50.5.18	aaUpdateReqDelVidIsidStats(ASGNSstring tlvNew, ASGNSstring tlvOld)	2042
44	50.5.19	aaUpdateAcptRejVidIsidStats(ASGNSstring tlvNew, ASGNSstring tlvOld)	2043
45	50.5.20	SomethingChangedLocal()	2043
46	50.6	AAD state machine	2044
47	50.7	AAB state machine	2045
48			
49	Annex A	(normative) PICS proforma—Bridge implementations	2046
50	A.1	Introduction.....	2046

1	A.2	Abbreviations and special symbols.....	2046
2	A.2.1	Status symbols	2046
3	A.2.2	General abbreviations	2046
4	A.3	Instructions for completing the PICS proforma.....	2047
5	A.3.1	General structure of the PICS proforma	2047
6	A.3.2	Additional information	2047
7	A.3.3	Exception information	2047
8	A.3.4	Conditional status	2048
9	A.4	PICS proforma for IEEE Std 802.1Q—Bridge implementations	2049
10	A.4.1	Implementation identification	2049
11	A.4.2	Protocol summary, IEEE Std 802.1Q	2049
12	A.5	Major capabilities	2050
13	A.6	Media access control methods	2056
14	A.7	Relay and filtering of frames	2057
15	A.8	Basic Filtering Services	2058
16	A.9	Addressing	2060
17	A.10	Rapid Spanning Tree Protocol (RSTP).....	2062
18	A.12	Implementation parameters.....	2064
19	A.11	BPDU encoding	2064
20	A.13	Performance.....	2065
21	A.14	Bridge management.....	2066
22	A.15	Remote management.....	2077
23	A.16	Expedited traffic classes	2078
24	A.17	Extended Filtering Services	2078
25	A.18	Multiple Spanning Tree Protocol (MSTP).....	2079
26	A.19	VLAN support	2081
27	A.20	Multiple MAC Registration Protocol (MMRP).....	2084
28	A.21	Multiple VLAN Registration Protocol (MVRP)	2085
29	A.22	Multiple Registration Protocol (MRP)	2086
30	A.23	Connectivity Fault Management (CFM).....	2087
31	A.24	Management Information Base (MIB)	2093
32	A.25	Protection Switching (PS).....	2095
33	A.26	Data-driven and data-dependent connectivity fault management (DDCFM).....	2096
34	A.27	Two-Port MAC Relay (TPMR)	2096
35	A.28	MAC Status Protocol (MSP)	2097
36	A.29	Forwarding and Queuing Enhancements for time-sensitive streams (FQTSS).....	2098
37	A.30	Congestion notification.....	2098
38	A.31	Stream Reservation Protocol (SRP).....	2099
39	A.32	Multiple I-SID Registration Protocol (MIRP)	2103
40	A.34	Enhanced Transmission Selection (ETS)	2104
41	A.33	Priority-based Flow Control (PFC).....	2104
42	A.35	Data Center Bridging eXchange protocol (DCBX).....	2105
43	A.36	Infrastructure Protection Switching (IPS).....	2105
44	A.38	EVB Bridge.....	2106
45	A.37	Shortest Path Bridging (SPB)	2106
46	A.39	EVB station.....	2107
47	A.40	Edge relay (ER)	2108
48	A.42	VDP, CDCP, and ECP	2110
49	A.41	VEB and VEPA ER components	2110
50	A.43	Path Control and Reservation	2111
51	A.44	Scheduled traffic	2112
52	A.45	Frame preemption	2112
53	A.46	Per-Stream Filtering and Policing.....	2113
54	A.47	YANG	2114

1	A.48	Stream reservation remote management (SRRM)	2116
2	A.49	TSN Centralized Network Configuration (CNC) station	2117
3	A.50	VDP for NVO3 nNVE Devices	2118
4	A.51	VDP for NVO3 tNVE Devices	2119
5	A.52	Asynchronous Traffic Shaping	2120
6	A.53	Congestion Isolation	2121
7	A.54	PBBN auto attach.....	2122
8	A.55	Credit-based shaper algorithm	2122
9	Annex B	(normative) PICS proforma—End station implementations	2123
10	B.1	Introduction.....	2123
11	B.2	Abbreviations and special symbols.....	2123
12	B.2.1	Status symbols	2123
13	B.2.2	General abbreviations	2123
14	B.3	Instructions for completing the PICS proforma.....	2124
15	B.3.1	General structure of the PICS proforma	2124
16	B.3.2	Additional information	2124
17	B.3.3	Exception information	2124
18	B.3.4	Conditional status	2125
19	B.4	PICS proforma for IEEE Std 802.1Q—End station implementations	2126
20	B.4.1	Implementation identification	2126
21	B.4.2	Protocol summary, IEEE Std 802.1Q	2126
22	B.5	Major capabilities	2127
23	B.6	Multiple MAC Registration Protocol (MMRP).....	2128
24	B.7	Multiple VLAN Registration Protocol (MVRP)	2129
25	B.8	Multiple Registration Protocol (MRP)	2130
26	B.9	Forwarding and Queuing Enhancements for time-sensitive streams (FQTSS).....	2130
27	B.10	Stream Reservation Protocol (SRP).....	2131
28	B.11	Congestion notification.....	2134
29	B.13	Enhanced Transmission Selection (ETS)	2136
30	B.14	Data Center Bridging eXchange protocol (DCBX).....	2136
31	B.12	Priority-based Flow Control (PFC).....	2136
32	B.15	Scheduled traffic.....	2137
33	B.16	Frame Preemption.....	2137
34	B.17	Per-Stream Filtering and Policing.....	2138
35	B.19	Congestion Isolation	2139
36	B.18	Asynchronous Traffic Shaping	2139
37	B.20	PBBN Auto Attach	2140
38	B.21	Credit-based shaper algorithm	2140
39	Annex C	(normative) Designated MSRP Node (DMN) Implementations	2141
40	C.1	DMNs on CSNs	2141
41	C.1.1	CSN characteristics	2141
42	C.1.2	DMN handling on CSN	2142
43	C.1.3	MSRPDU handling on a CSN	2143
44	C.1.4	CSN bandwidth fluctuations	2144
45	C.2	DMN on MoCA	2144
46	C.2.1	DMN Selection on MoCA Network	2144
47	C.2.2	MoCA network bandwidth management	2148
48	C.3	DMNs on IEEE 802.11 media	2149
49	C.3.1	MSRP handling	2150
50	C.3.2	BSS DMN selection	2153
51	C.3.3	BSS network bandwidth management	2154

1	Annex D	(normative) IEEE 802.1 Organizationally Specific TLVs	2157
2	D.1	Requirements of the IEEE 802.1 Organizationally Specific TLV sets.....	2157
3	D.2	Organizationally Specific TLV definitions.....	2158
4	D.2.1	Port VLAN ID TLV	2158
5	D.2.2	Port And Protocol VLAN ID TLV	2159
6	D.2.3	VLAN Name TLV	2160
7	D.2.4	Protocol Identity TLV	2160
8	D.2.5	VID Usage Digest TLV	2161
9	D.2.6	Management VID TLV	2161
10	D.2.7	Congestion Notification TLV	2162
11	D.2.8	ETS Configuration TLV	2163
12	D.2.9	ETS Recommendation TLV	2165
13	D.2.10	Priority-based Flow Control Configuration TLV	2166
14	D.2.11	Application Priority TLV	2167
15	D.2.12	EVB TLV	2168
16	D.2.13	CDCP TLV	2173
17	D.2.14	Application VLAN TLV	2174
18	D.2.15	Congestion Isolation TLV	2176
19	D.2.16	Topology Recognition TLV	2177
20	D.2.17	PBBN Auto Attach System TLV	2179
21	D.2.18	PBBN Auto Attach Assignment TLV	2181
22	D.3	IEEE 802.1 Organizationally Specific TLV management.....	2183
23	D.3.1	IEEE 802.1 Organizationally Specific TLV selection management	2183
24	D.3.2	IEEE 802.1 managed objects—TLV variables	2183
25	D.4	PICS proforma for IEEE 802.1 Organizationally Specific TLV extensions	2185
26	D.4.1	Implementation identification	2185
27	D.4.2	Protocol summary, IEEE Std 802.1Q	2185
28	D.4.3	Major capabilities and options	2186
29	D.5	IEEE 802.1/LLDP extension MIB.....	2188
30	D.5.1	Internet Standard Management Framework	2188
31	D.5.2	Structure of the IEEE 802.1/LLDP extension MIB	2188
32	D.5.3	Relationship to other MIBs	2197
33	D.5.4	Security considerations for IEEE 802.1 LLDP extension MIB module	2197
34	D.5.5	IEEE 802.1 LLDP extension MIB module—version 2	2200
35	D.5.6	EVB extensions to the IEEE 802.1 LLDP extension MIB module	2272
36	D.6	IEEE 802.1/LLDP extension YANG.....	2283
37	D.6.1	YANG framework	2283
38	D.6.2	IEEE 802.1 Organizationally Specific TLV YANG data models	2283
39	D.6.3	Structure of the IEEE 802.1/LLDP extension YANG models	2289
40	D.6.4	Security considerations	2289
41	D.6.5	Definition of the IEEE 802.1/LLDP extension YANG modules	2292
42	D.6.6	IEEE 802.1/LLDP extension YANG modules	2296
43	Annex E	(normative) Notational conventions used in state diagrams.....	2325
44	Annex F	(informative) Shared and Independent VLAN Learning (SVL and IVL)	2327
45	F.1	Requirements for Shared and Independent Learning	2327
46	F.1.1	Connecting independent VLANs	2328
47	F.1.2	Duplicate MAC addresses	2329
48	F.1.3	Asymmetric VLANs and Rooted-Multipoint connectivity	2330
49	F.1.4	Shared learning and Shortest Path Bridging VID (SPBV) mode	2333
50	F.1.5	Generic constraints on SVL and IVL use	2335

1	Annex G	(informative) MAC method-dependent aspects of VLAN support.....	2336
2	G.1	Example tagged IEEE 802.3 EtherType-encoded frame format	2336
3	G.2	Padding and frame size considerations	2336
4	G.2.1	Treatment of PAD fields in IEEE 802.3 frames	2336
5	G.2.2	Maximum PDU size	2337
6	G.2.3	Minimum PDU size	2337
7	G.3	Tag insertion and removal for LLC media	2338
8	G.4	IEEE 802.11 and PMPN media	2338
9	G.4.1	IEEE 802.11 Portal convergence	2338
10	G.4.2	Point-to-Multipoint Network convergence: multiple connections	2338
11	G.4.3	Point-to-Multipoint Network convergence: single connection	2339
12	Annex H	(informative) Interoperability considerations.....	2340
13	H.1	Requirements for interoperability	2340
14	H.1.1	Static filtering requirements	2340
15	H.1.2	Configuration requirements for VLAN-tagging	2340
16	H.2	Homogeneous VLAN-aware networks.....	2341
17	H.2.1	Consistency of static VLAN filtering	2341
18	H.2.2	Consistent view of the “untagged VLAN(s)” on a given LAN	2342
19	H.3	Heterogeneous networks: Intermixing MAC Bridges (M) and VLAN Bridges (V)	2343
20	H.3.1	Example: Adding a VLAN Bridge to provide filtering to a MAC Bridged Network	2343
21			
22	H.3.2	Example: Adding a MAC Bridge to a (previously) Homogeneous VLAN Bridged	2344
23		Network	
24	H.4	Intermixing Port-based classification and Port-and-Protocol-based classification or future	2344
25		enhancements in VLAN Bridges	
26	H.4.1	Example: Intermixing Protocol-based ingress rules	2345
27	H.4.2	Differing views of untagged traffic on a given LAN	2345
28	Annex I	(informative) Priority and drop precedence.....	2346
29	I.1	Traffic types.....	2346
30	I.2	Managing latency and throughput	2347
31	I.3	Traffic type to traffic class mapping.....	2347
32	I.4	Traffic types and priority values.....	2349
33	I.5	Supporting the credit-based shaper algorithm	2350
34	I.6	Supporting drop precedence	2351
35	I.7	Priority Code Point allocation.....	2351
36	I.8	Interoperability.....	2352
37	Annex J	(informative) CFM protocol design and use.....	2354
38	J.1	Origin of CFM	2354
39	J.2	Deployment of CFM	2354
40	J.3	MD Level allocation alternative	2355
41	J.4	Relationship of IEEE Std 802.1Q CFM to other standards	2355
42	J.5	Interpreting Linktrace results.....	2356
43	J.6	MP addressing: Individual and Shared MP addresses	2357
44	J.6.1	Individual MP address model	2358
45	J.6.2	Shared MP address model and the CFM Port	2358
46	Annex K	(informative) TPMR use cases.....	2361
47	K.1	Use case 1—TPMR as User to Network Interface (UNI) demarcation device	2361

1	K.2	Use case 2—TPMRs with aggregated links	2362
2	K.3	Use case 3—Multiple TPMRs	2362
3	K.4	Special cases	2363
4	Annex L	(informative) Operation of the credit-based shaper algorithm	2366
5	L.1	Overview of credit-based shaper operation	2366
6	L.2	“Class measurement intervals” in Bridges.....	2371
7	L.3	Determining worst-case latency contribution and buffering requirements	2372
8	L.3.1	Interference delay	2373
9	L.3.2	Maximum interference delay and maximum buffer requirement	2381
10	L.4	Operation of credit-based shaper in Coordinated Shared Network (CSN).....	2382
11	Annex M	(normative) Support for PFC in link layers without MAC Control	2383
12	M.1	Overview.....	2383
13	M.2	PFC PDU format.....	2383
14	Annex N	(informative) Buffer requirements for PFC	2384
15	N.1	Overview.....	2384
16	N.2	Delay model.....	2384
17	N.3	Interface Delay.....	2387
18	N.4	Cable Delay.....	2387
19	N.5	Higher Layer Delay	2387
20	N.6	Computation example	2388
21	Annex O	(informative) Preserving the integrity of FCS fields in MAC Bridges	2389
22	O.1	Background.....	2389
23	O.2	Basic mathematical ideas behind CRC and FCS	2390
24	O.3	Detection Lossless Circuit approach.....	2391
25	O.4	Algorithmic modification of an FCS	2392
26	O.4.1	Data changed, length unchanged	2392
27	O.4.2	Length changed, original data unchanged	2393
28	O.4.3	Preservation of detectability	2394
29	O.5	Conclusions.....	2395
30	Annex P	(informative) Frame duplication and misordering	2396
31	P.1	Background.....	2396
32	P.2	Frame duplication	2396
33	P.3	Frame misordering	2397
34	P.4	Other considerations	2398
35	Annex Q	(informative) Traffic scheduling	2399
36	Q.1	Motivation.....	2399
37	Q.2	Using gate operations to create protected windows.....	2400
38	Q.3	Availability of PTP	2401
39	Q.4	Scheduled traffic and end stations	2401
40	Q.5	CycleTimeExtension variables	2401

1	Annex R	(informative) Preemption and IEEE 802.1AE MAC Security	2402
2	Annex S	(informative) Preemption and scheduled traffic	2404
3	S.1	Scheduling used in isolation	2404
4	S.2	Preemption used in isolation.....	2404
5	S.3	Scheduling and preemption used in combination, no HOLD/RELEASE	2405
6	S.4	Scheduling and preemption used in combination with HOLD/RELEASE	2405
7	S.5	Bandwidth allocation and express traffic.....	2405
8	Annex T	(informative) Cyclic queuing and forwarding	2407
9	T.1	Overview of CQF.....	2407
10	T.2	An approach to CQF implementation	2408
11	T.3	Use of Per-Stream Filtering and Policing for CQF.....	2409
12	T.3.1	Stream filter configuration	2409
13	T.3.2	Stream gate configuration	2409
14	T.4	Use of traffic scheduling for CQF	2410
15	T.5	Timing considerations.....	2411
16	T.5.1	Choice of T	2411
17	T.5.2	Cycle interleaving	2412
18	T.5.3	Cycle alignment between adjacent Ports	2414
19	Annex U	(informative) TSN configuration examples	2415
20	U.1	Examples for time-aware talker.....	2415
21	U.1.1	Using enhancements for scheduled traffic	2416
22	U.1.2	Using strict priority	2417
23	U.1.3	Using per-stream scheduling	2418
24	U.2	Example of workflow for fully centralized models.....	2419
25	Annex V	(informative) Asynchronous Traffic Shaping delay analysis framework	2423
26	V.1	General assumptions	2423
27	V.2	End-to-end delay modeling approach	2423
28	V.3	Buffering delays.....	2424
29	V.4	Media-dependent delays	2426
30	V.5	Bridge—Internal arrival time recognition delays	2426
31	V.6	Bridge—Internal processing delays.....	2426
32	V.7	Bridge—Internal clock offset variations.....	2427
33	V.8	Inter-device clock rate deviations	2427
34	V.9	Combined delay bounds.....	2428
35	Annex W	(informative) Maintaining frame order with Congestion Isolation	2429
36	W.1	Queue markers for order preservation	2431
37	W.2	Congestion Isolation queuing and Priority-based Flow Control	2432
38	Annex X	(informative) Bibliography	2436

1 Figures

2	Figure 6-1	Internal organization of the MAC sublayer	164
3	Figure 6-2	Provider Instance Ports (PIPs)	182
4	Figure 6-3	B-Component CBP	185
5	Figure 6-4	Example of operation of Port-and-Protocol-based classification	188
6	Figure 6-5	Service access priority selection	191
7	Figure 6-6	Two back-to-back EISS Multiplex Entities	197
8	Figure 6-7	Two back-to-back Backbone Service Instance Multiplex Entities	198
9	Figure 6-8	Backbone Service Instance Multiplex Entities with example CFM shims	198
10	Figure 6-9	Two back-to-back Up and Down TESI Multiplex Entities	201
11	Figure 6-10	Supporting the ISS with signaled priority	202
12	Figure 6-11	Two back-to-back Up and Down Infrastructure Segment Multiplex Entities	203
13	Figure 7-1	VLAN Bridging overview	205
14	Figure 8-1	A Bridged Network	211
15	Figure 8-2	VLAN Bridge architecture	213
16	Figure 8-3	MAC Bridge architecture	214
17	Figure 8-4	Relaying MAC frames	216
18	Figure 8-5	Observation of network traffic	216
19	Figure 8-6	Operation of Spanning Tree Protocol Entity	217
20	Figure 8-7	Operation of MRP	217
21	Figure 8-8	Management Port transmission and reception	218
22	Figure 8-9	Infrastructure Segment MEP placement in a PNP	218
23	Figure 8-10	Bridge Port Transmit and Receive	221
24	Figure 8-11	TPMR Port Transmit and Receive	221
25	Figure 8-12	Forwarding process functions	223
26	Figure 8-13	Flow classification and metering	229
27	Figure 8-14	Per-stream classification for PSFP	231
28	Figure 8-15	Per-stream classification and metering for ATS	233
29	Figure 8-16	Per-stream classification and assignment for CI	234
30	Figure 8-17	Transmission selection with gates	245
31	Figure 8-18	Frame timing at gate-close events	247
32	Figure 8-19	Scheduled traffic state machines—overview and relationships	249
33	Figure 8-20	Cycle Timer state machine	250
34	Figure 8-21	List Execute state machine	251
35	Figure 8-22	List Config state machine	252
36	Figure 8-23	Logical points of attachment of the Higher Layer and Relay Entities	285
37	Figure 8-24	Effect of control information on the forwarding path	285
38	Figure 8-25	Per-Port points of attachment	286
39	Figure 8-26	Single point of attachment—relay permitted	286
40	Figure 8-27	Single point of attachment—relay not permitted	287
41	Figure 8-28	Effect of Port State	287
42	Figure 8-29	Controlled and Uncontrolled Port connectivity	288
43	Figure 8-30	Ingress/egress control information in the forwarding path	288
44	Figure 9-1	VLAN TCI format	293
45	Figure 9-2	I-TAG TCI format	294
46	Figure 10-1	Example—Attribute value propagation from one station	297
47	Figure 10-2	Example—Attribute value propagation from two stations	298
48	Figure 10-3	Example—Registrations as pointers to the sources of declarations	298
49	Figure 10-4	MRP architecture	300
50	Figure 10-5	Format of the major components of an MRPDU	323
51	Figure 10-6	Operation of MMRP for a single VLAN Context	329
52	Figure 10-7	Example Directed Graph	330
53	Figure 10-8	Example of MMRP propagation in a VLAN Context	332

1	Figure 11-1	Operation of MVRP.....	340
2	Figure 12-1	Relationships among CFM managed objects.....	401
3	Figure 12-2	Relationship among BEB managed objects.....	418
4	Figure 12-3	SPB managed objects (MOs).....	473
5	Figure 12-4	Relationships among EVB Bridge managed objects	489
6	Figure 12-5	Relationship among EVB station managed objects.....	489
7	Figure 12-6	Timing points for scheduled traffic	506
8	Figure 12-7	Timing points for PSFP	515
9	Figure 13-1	Diagrammatic conventions for spanning tree topologies	539
10	Figure 13-2	Physical topology and active topology	540
11	Figure 13-3	Port Roles and Port States.....	540
12	Figure 13-4	A Backup Port.....	541
13	Figure 13-5	“Ring Backbone” example.....	541
14	Figure 13-6	An MST Bridge network	543
15	Figure 13-8	MSTI Active Topology in Region 2	544
16	Figure 13-7	CIST Priority Vectors, Port Roles, and MST Regions	545
17	Figure 13-9	CIST and MSTI active topologies in Region 1 of the example network.....	558
18	Figure 13-10	Agreements and Proposals.....	562
19	Figure 13-11	CIST and MSTI Active Topologies in Region 2 of Figure 13-6	563
20	Figure 13-12	Enhanced Agreements	564
21	Figure 13-13	Spanning tree protocol state machines—overview and relationships	575
22	Figure 13-14	MSTP overview notation	576
23	Figure 13-15	Port Timers state machine.....	603
24	Figure 13-16	Port Receive state machine	604
25	Figure 13-17	Port Protocol Migration state machine	605
26	Figure 13-18	Bridge Detection state machine	605
27	Figure 13-19	Port Transmit state machine	606
28	Figure 13-20	Port Information state machine.....	607
29	Figure 13-21	Port Role Selection state machine	608
30	Figure 13-22	Disabled Port role transitions.....	609
31	Figure 13-23	Port Role Transitions state machine—MasterPort.....	610
32	Figure 13-24	Port Role Transitions state machine—RootPort.....	611
33	Figure 13-25	Port Role Transitions state machine—DesignatedPort.....	612
34	Figure 13-26	Port Role Transitions state machine—AlternatePort and BackupPort	613
35	Figure 13-27	Port State Transition state machine	613
36	Figure 13-28	Topology Change state machine.....	615
37	Figure 13-29	L2 Gateway Port Receive state machine	616
38	Figure 14-1	RST, MST, SPT, and STP Configuration BPDU format.....	620
39	Figure 14-2	STP TCN BPDU format	620
40	Figure 14-3	MSTI Configuration Message parameters and format	625
41	Figure 15-1	Internal organization of the MAC sublayer in a PBN	628
42	Figure 15-2	Port-based service interface to a PBN	629
43	Figure 15-3	Port-based service interface to a PBN	630
44	Figure 15-4	C-tagged service interface to a PBN.....	630
45	Figure 15-5	C-tagged service interface to a PBN.....	631
46	Figure 15-6	Customer Edge Ports (CEPs).....	631
47	Figure 15-7	S-tagged service interface to a PBN	632
48	Figure 15-8	S-tagged interface to a PBN.....	632
49	Figure 15-9	RCSIs to a PBN	632
50	Figure 15-10	Remote Customer Access Ports (RCAPs)	633
51	Figure 15-11	C-tagged RCSI to a PBN	634
52	Figure 15-12	Port-based RCSI to a PBN.....	634
53	Figure 15-13	Provider Network Port (PNP) interface	635
54	Figure 16-1	PBN with interface examples	638

1	Figure 16-2	Examples of remote customer service access via a second PBN	640
2	Figure 16-3	Access service separation and “Hairpin Switching”	641
3	Figure 17-1	C-VLAN component internal LAN managed system	708
4	Figure 17-2	I/B-component internal LAN managed system	713
5	Figure 18-1	One Maintenance Domain: operator’s view	1227
6	Figure 18-2	One service instance: operator’s view	1228
7	Figure 18-3	One service instance: customer’s view	1229
8	Figure 18-4	MEP and MIP Symbols	1229
9	Figure 18-5	MAs: one service instance in a provider network	1231
10	Figure 18-6	MAs: Expansion of Figure 18-5	1231
11	Figure 18-7	MEPs, MIPs, and MD Levels	1232
12	Figure 19-1	CFM Protocol shims	1233
13	Figure 19-2	MA Endpoint (MEP)	1236
14	Figure 19-3	MIP Half Function (MHF)	1241
15	Figure 19-4	LOM shim	1243
16	Figure 19-5	LOM architecture	1244
17	Figure 20-1	MEP state machines—overview and relationships	1257
18	Figure 20-2	MEP Continuity Check Initiator state machine	1264
19	Figure 20-3	MHF Continuity Check Receiver state machine	1265
20	Figure 20-4	MEP Continuity Check Receiver state machine	1269
21	Figure 20-5	Remote MEP state machine	1271
22	Figure 20-6	Remote MEP Error state machine	1272
23	Figure 20-7	MEP Cross Connect state machine	1273
24	Figure 20-8	MEP Traffic Field Mismatch state machine	1275
25	Figure 20-9	MEP Local Mismatch state machine	1275
26	Figure 20-10	MP Loopback Responder state machine	1278
27	Figure 20-11	MEP Loopback Initiator transmit state machine	1280
28	Figure 20-12	MEP Loopback Initiator receive state machine	1281
29	Figure 20-13	MEP Fault Notification Generator state machine	1283
30	Figure 20-14	MEP Mismatch Fault Notification Generator state machine	1285
31	Figure 20-15	MEP Linktrace Initiator receive state machine	1289
32	Figure 20-16	Linktrace Responder, MEPs, MHFs, and LOMs	1291
33	Figure 20-17	LTM Receiver state machine	1297
34	Figure 20-18	LTR Transmitter state machine	1298
35	Figure 22-1	MEPs and MIPs distinguished by VID (incomplete picture)	1327
36	Figure 22-2	Alternate view of Forwarding process	1328
37	Figure 22-3	Combining per-VLAN MPs into two shims	1329
38	Figure 22-4	More complete picture of MP placement in a Bridge Port	1330
39	Figure 22-5	Service instance spanning two Bridges protected by Up MPs	1332
40	Figure 22-6	Service instance spanning two Bridges protected by Down MPs	1332
41	Figure 22-7	MP placement in a non-VLAN-aware Bridge Port	1334
42	Figure 22-8	MP placement relative to other standards	1335
43	Figure 22-9	Creating MEPs and MIPs	1338
44	Figure 22-10	CFM in a Provider Edge Bridge C-tagged service interface	1344
45	Figure 22-11	CFM in a Provider Edge Bridge C-tagged RCSI	1346
46	Figure 22-12	Up MEPs in a Management Port	1347
47	Figure 22-13	CFM in the enterprise environment	1348
48	Figure 22-14	CFM on a Bridge that implements IEEE Std 802.1Q-2005	1349
49	Figure 23-1	TPMR connecting two Bridge Ports	1350
50	Figure 23-2	TPMR chain connecting Bridge Ports	1350
51	Figure 23-3	MSSs and the MSPE	1352
52	Figure 23-4	Adding connectivity	1354
53	Figure 23-5	Losing connectivity	1355
54	Figure 23-6	TPMR recovery	1356

1	Figure 23-7	Notification from one end of the link to the other	1357
2	Figure 23-8	Immediate MAC status notification at the end of a link.....	1357
3	Figure 23-9	MSPE state machine overview	1358
4	Figure 23-10	Status Transition state machine (STM)	1362
5	Figure 23-11	Status Notification state machine (SNM)	1363
6	Figure 23-12	MSPDU structure.....	1365
7	Figure 25-1	Internal organization of the MAC sublayer in a PBBN.....	1369
8	Figure 25-2	PBB terminology	1370
9	Figure 25-3	Customer service interface types	1371
10	Figure 25-4	Port-based service interface	1372
11	Figure 25-5	Port-based interface equipment	1373
12	Figure 25-6	Encapsulated service frames at ISS	1374
13	Figure 25-7	S-tagged service interface.....	1374
14	Figure 25-8	S-tagged service interface equipment	1375
15	Figure 25-9	I-tagged service interface	1376
16	Figure 25-10	I-tagged service interface equipment.....	1376
17	Figure 25-11	S-tagged and Port-based service interface access classifications	1378
18	Figure 25-12	I-tagged service interface access protection classifications.....	1379
19	Figure 25-13	Internal organization of the MAC sublayer in a PBB-TE Region.....	1382
20	Figure 25-14	PBB-TE Region	1384
21	Figure 25-15	Transparent service interface	1385
22	Figure 25-16	Transparent service interface equipment	1386
23	Figure 26-1	PBBN example	1388
24	Figure 26-2	CFM shim model	1395
25	Figure 26-3	CFM example applied to a Port-based and S-tagged service interface	1396
26	Figure 26-4	CFM example applied to an I-tagged Service Interface	1397
27	Figure 26-5	CFM example applied to a hierarchal E-NNI, CBP-PIP Demarc.....	1398
28	Figure 26-6	CFM example applied to a peer E-NNI, CBP-PIP	1399
29	Figure 26-7	Independent ESPs using the same ESP-DAs and ESP-VIDs	1402
30	Figure 26-8	PBB-TE MEP placement in a CBP.....	1403
31	Figure 26-9	Independent Infrastructure Segments distinguished by SMP-SA.....	1406
32	Figure 26-10	Infrastructure Segment MEP placement in a PNP	1407
33	Figure 26-11	Protection switching architecture.....	1408
34	Figure 26-12	PBB-TE point-to-point protection switching.....	1410
35	Figure 26-13	Mapping data traffic to the protection entity	1412
36	Figure 26-14	Relationships of the Protection switching state machines—overview	1412
37	Figure 26-15	Hold-off state machine.....	1416
38	Figure 26-16	Clear Manual Switch state machine.....	1416
39	Figure 26-17	Service Mapping state machine	1417
40	Figure 26-18	Segment terminology and properties	1418
41	Figure 26-19	Infrastructure Segment monitoring	1419
42	Figure 26-20	Working Segment and Protection Segment.....	1420
43	Figure 26-21	Nested IPGs	1421
44	Figure 26-22	IPS Control entity	1423
45	Figure 26-23	M:1 IPS	1425
46	Figure 26-24	M:1 IPS state machines.....	1426
47	Figure 26-25	M:1 Hold-off state machine.....	1429
48	Figure 26-26	Protection Segment Selection state machine	1430
49	Figure 27-1	Configuring VLAN support in an SPT Region (example)	1437
50	Figure 27-2	SPBM group MAC address—general format.....	1448
51	Figure 27-3	SPBM group MAC addresses—source rooted SPT	1448
52	Figure 27-4	SPBM group MAC addresses—shared tree.....	1449
53	Figure 27-5	SPBM MEP placement in a CBP.....	1451
54	Figure 27-6	SPBV campus network example.....	1454

1	Figure 27-7	SPT Bridge Network using SPBM example	1455
2	Figure 28-1	Agreement Digest field format	1460
3	Figure 28-2	MT-Capability TLV	1469
4	Figure 28-3	SPB MCID sub-TLV	1469
5	Figure 28-4	SPB Digest sub-TLV	1470
6	Figure 28-5	SPB Base VLAN-Identifiers sub-TLV	1471
7	Figure 28-6	SPB Instance sub-TLV	1472
8	Figure 28-7	SPB Instance Opaque ECT-ALGORITHM sub-TLV	1474
9	Figure 28-8	ECMP ECT-ALGORITHM sub-TLV	1474
10	Figure 28-9	SPB Link Metric sub-TLV	1475
11	Figure 28-10	SPB Adjacency Opaque ECT-ALGORITHM sub-TLV	1476
12	Figure 28-11	SPBV MAC Address sub-TLV	1476
13	Figure 28-12	SPBM Service Identifier and Unicast Address sub-TLV	1478
14	Figure 29-1	Forward path test (FPT)	1481
15	Figure 29-2	Return path test (RPT)	1482
16	Figure 29-3	Combination of FPT and RPT	1483
17	Figure 29-4	Detailed functions of RR	1484
18	Figure 29-5	RFM Receiver on an non-MP	1487
19	Figure 29-6	Return Path DR	1488
20	Figure 29-7	RR Filter state machine.....	1493
21	Figure 29-8	RR Encapsulation state machine.....	1494
22	Figure 29-9	RR Transmit state machine.....	1494
23	Figure 29-10	RFM Receiver state machine.....	1495
24	Figure 29-11	Decapsulator Responder state machine	1498
25	Figure 30-1	Congestion detection in QCN CP	1504
26	Figure 30-2	Sampling (reflection) probability in QCN CP as a function of $ Fb $	1504
27	Figure 30-3	QCN RP operation	1505
28	Figure 30-4	Byte Counter and Timer interaction with Rate Limiter.....	1507
29	Figure 30-5	CP–RP peering in VLAN Bridged Network.....	1509
30	Figure 30-6	CP–RP peering in PBBN	1510
31	Figure 31-1	CPs and congestion-aware queues in a Bridge	1511
32	Figure 31-2	Congestion-aware queue functions in an end station.....	1513
33	Figure 31-3	Per-CNPV station function	1515
34	Figure 32-1	CND defense state machine.....	1528
35	Figure 32-2	RP rate control state machine	1540
36	Figure 32-3	CP–RP peering in any hierarchical Bridged Network	1541
37	Figure 34-1	Queuing model for a Talker station	1553
38	Figure 35-1	Operation of MSRP	1558
39	Figure 35-2	Format of the components of the reservation FirstValue fields.....	1569
40	Figure 35-3	Format of the components of the Domain FirstValue	1574
41	Figure 35-4	Value of StreamID TLV	1579
42	Figure 35-5	Value of StreamRank TLV	1579
43	Figure 35-6	Value of InterfaceID TLV	1579
44	Figure 35-7	Value of IEEE802-MacAddresses TLV	1580
45	Figure 35-8	Value of IEEE802-VlanTag TLV	1580
46	Figure 35-9	Value of IPv4-tuple TLV	1581
47	Figure 35-10	Value of IPv6-tuple TLV	1581
48	Figure 35-11	Value of TrafficSpecification TLV.....	1584
49	Figure 35-12	Value of TSpecTimeAware TLV	1584
50	Figure 35-13	Value of UserToNetworkRequirements TLV.....	1585
51	Figure 35-14	Value of InterfaceCapabilities TLV	1586
52	Figure 35-15	Value of StatusInfo TLV	1586
53	Figure 35-16	Value of AccumulatedLatency TLV.....	1587
54	Figure 35-17	Value of TimeAwareOffset TLV	1588

1	Figure 36-1	PFC peering	1604
2	Figure 36-2	PFC Receiver state diagram for priority n	1606
3	Figure 36-3	PFC-aware system queue functions.....	1608
4	Figure 36-4	PFC-aware system queue functions with Link Aggregation	1609
5	Figure 38-1	DCBX Asymmetric state machine.....	1614
6	Figure 38-2	Symmetric state machine	1615
7	Figure 39-1	Operation of MIRP in an I-component.....	1617
8	Figure 39-2	Operation of MIRP in a B-component.....	1617
9	Figure 39-3	Alternate model for MIRP in a B-component	1622
10	Figure 40-1	EVB architecture overview.....	1624
11	Figure 40-2	EVB architecture without S-channels.....	1626
12	Figure 40-3	EVB architecture with S-channel	1626
13	Figure 40-4	EVB components and internal LANs with S-channels.....	1627
14	Figure 40-5	EVB architecture without S-channels, with EVB Bridge S-VLAN component	1629
15	Figure 40-6	EVB architecture without S-channels, with EVB station S-VLAN component.....	1629
16	Figure 41-1	VSI manager ID TLV	1632
17	Figure 41-2	VDP association TLV	1633
18	Figure 41-3	VID Filter Info format	1638
19	Figure 41-4	MAC/VID filter format.....	1638
20	Figure 41-5	GroupID/VID filter format	1639
21	Figure 41-6	GroupID/MAC/VID filter format	1639
22	Figure 41-7	GroupID/VID/IPv4 filter format.....	1639
23	Figure 41-8	GroupID/MAC/VID/IPv4 filter format	1640
24	Figure 41-9	GroupID/VID/IPv6 filter format.....	1640
25	Figure 41-10	GroupID/MAC/VID/IPv6 filter format	1641
26	Figure 41-11	Organizationally defined TLV.....	1642
27	Figure 41-12	Bridge VDP state machine.....	1644
28	Figure 41-13	Station VDP state machine	1645
29	Figure 42-1	CDCP state machine—Station role.....	1652
30	Figure 42-2	CDCP state machine—Bridge role	1653
31	Figure 43-1	Example ECP exchange.....	1657
32	Figure 43-2	ECPDU structure	1659
33	Figure 43-3	ECP transmit state machine.....	1660
34	Figure 43-4	ECP receive state machine.....	1661
35	Figure 44-1	Flow Filtering TCI format	1668
36	Figure 44-2	SPBM VID MEP and ECMP path MEP placement in a CBP.....	1672
37	Figure 45-1	An SPT Region controlled by a single PCE	1675
38	Figure 45-2	An SPT Region controlled by multiple PCEs.....	1676
39	Figure 45-3	The use of the SPB Instance sub-TLV for MRT	1684
40	Figure 45-4	Shared Risk Link Group (SRLG) TLV.....	1688
41	Figure 45-5	Topology sub-TLV	1689
42	Figure 45-6	A strict tree and its descriptor Topology sub-TLV.....	1690
43	Figure 45-7	Topology sub-TLV of a loose tree.....	1691
44	Figure 45-8	Hop sub-TLV.....	1693
45	Figure 45-9	Administrative Group sub-TLV.....	1696
46	Figure 45-10	Bandwidth Constraint sub-TLV.....	1696
47	Figure 45-11	Bandwidth Assignment sub-TLV	1697
48	Figure 45-12	Timestamp sub-TLV	1698
49	Figure 45-13	A GADAG and its descriptor Topology sub-TLV	1704
50	Figure 45-14	MRT-Blue and MRT-Red for MRT Root 55.....	1705
51	Figure 45-15	A GADAG for a topology with multiple blocks.....	1706
52	Figure 46-1	Fully distributed model.....	1709
53	Figure 46-2	Centralized network/distributed user model	1710
54	Figure 46-3	Fully centralized model	1711

1	Figure 46-4	Example of Stream transformation in Talker end station	1713
2	Figure 46-5	Example of IEEE 802.1CB functions in Talker end station	1714
3	Figure 46-6	Example of IEEE 802.1CB functions in Listener end station	1714
4	Figure 48-1	General YANG hierarchy	1745
5	Figure 48-2	YANG root hierarchy with IEEE 802.1Q YANG modules.....	1745
6	Figure 48-3	Interface YANG model.....	1746
7	Figure 48-4	VLAN Bridge components model (MAC Relay Entities).....	1748
8	Figure 48-5	Bridge Port model.....	1749
9	Figure 48-6	TPMR model (MAC Relay Entity).....	1750
10	Figure 48-7	TPMR port model.....	1751
11	Figure 48-8	Provider Bridge model.....	1753
12	Figure 48-9	Provider Edge Bridge C-VLAN Interface model	1754
13	Figure 48-10	Provider Edge Bridge S-VLAN interface model	1755
14	Figure 48-11	Bridge to CFM YANG model	1757
15	Figure 48-12	CFM MEP model relationships	1758
16	Figure 48-13	CFM MEP model.....	1759
17	Figure 48-14	CFM operations structure	1759
18	Figure 48-15	Stream filters and stream gates model MEP model.....	1760
19	Figure 48-16	Asynchronous Traffic Shaping model	1761
20	Figure 48-17	Congestion Isolation model	1762
21	Figure 48-18	Scheduled Traffic model.....	1763
22	Figure 48-19	Frame Preemption model.....	1764
23	Figure 48-20	PSFP model.....	1765
24	Figure 48-21	CNC-configuration model A	1766
25	Figure 48-22	CNC-configuration model B.....	1767
26	Figure 48-23	Credit-based shaper algorithm model	1768
27	Figure 48-24	RSTP model.....	1769
28	Figure 48-25	MSTP model.....	1770
29	Figure 49-1	Congestion Isolation example operation.....	1999
30	Figure 49-2	Congestion Isolation reference diagram	2005
31	Figure 49-3	Layer-2 CIM Encapsulation.....	2019
32	Figure 49-4	IPv4 layer-3 CIM Encapsulation	2019
33	Figure 49-5	IPv6 layer-3 CIM Encapsulation	2020
34	Figure 49-6	CIM PDU	2020
35	Figure 50-1	PBBN Auto Attach model	2025
36	Figure 50-2	PBBN Auto Attach functions diagram	2026
37	Figure 50-3	PBBN Auto Attach for customer networks	2027
38	Figure 50-4	PBBN Auto Attach Device (AAD) state machine.....	2044
39	Figure 50-5	PBBN Auto Attach BEB (AAB) state machine.....	2045
40	Figure C-1	CSN backbone	2141
41	Figure C-2	Bridge's CSN model for bandwidth reservation.....	2142
42	Figure C-3	Talker MSRPDU flow	2143
43	Figure C-4	Listener MSRPDU flow.....	2143
44	Figure C-5	IEEE DMN Device Attribute IE.....	2145
45	Figure C-6	DMN Confirmation Transaction.....	2147
46	Figure C-7	Bandwidth reservation—bridge model for IEEE 802.11 BSS (STA downstream Port)	2149
47			
48	Figure C-8	Bandwidth reservation—bridge model for IEEE 802.11 BSS (STA upstream Port)	2150
49			
50	Figure C-9	Bandwidth reservation—bridge model for IEEE 802.11 BSS (direct link setup)	2150
51			
52	Figure C-10	MSRP/IEEE 802.11 query flows	2151
53	Figure C-11	MSRP/802.11 Talker STA to Listener STA reservation flows	2151
54	Figure C-12	MSRP/802.11 “Bridged” Listener to Talker STA reservation flows	2152

1	Figure C-13	MSRP/802.11 Listener STA to “Bridged” Talker reservation flows	2152
2	Figure D-1	Port VLAN ID TLV format.....	2158
3	Figure D-2	Port And Protocol VLAN ID TLV format.....	2159
4	Figure D-3	VLAN Name TLV format	2160
5	Figure D-4	Protocol Identity TLV format	2160
6	Figure D-5	VID Usage Digest TLV format	2161
7	Figure D-6	Management VID TLV format	2162
8	Figure D-7	Congestion Notification TLV format	2162
9	Figure D-8	ETS Configuration TLV format	2163
10	Figure D-9	ETS Recommendation TLV format.....	2165
11	Figure D-10	Priority-based Flow Control Configuration TLV format	2166
12	Figure D-11	Application Priority TLV format.....	2167
13	Figure D-12	EVb TLV format	2169
14	Figure D-13	CDCP TLV structure	2173
15	Figure D-14	Application VLAN TLV format.....	2174
16	Figure D-15	Congestion Isolation TLV format.....	2176
17	Figure D-16	Topology Recognition TLV Format	2177
18	Figure D-17	PBBN Auto Attach System TLV Format	2179
19	Figure D-18	PBBN Auto Attach Assignment TLV format.....	2181
20	Figure D-19	YANG hierarchy with IEEE 802.1Q Extension TLV YANG.....	2283
21	Figure D-20	basicSet TLV model	2284
22	Figure D-21	cnSet TLV model.....	2285
23	Figure D-22	dcbxSet TLV model.....	2286
24	Figure D-23	evbSet TLV model.....	2287
25	Figure D-24	ciSet TLV model.....	2287
26	Figure D-25	trSet TLV model	2288
27	Figure D-26	PBBN Auto Attach aaSet TLV model.....	2289
28	Figure F-1	Connecting independent VLANs—1	2328
29	Figure F-2	Connecting independent VLANs—2	2329
30	Figure F-3	Duplicate MAC addresses	2329
31	Figure F-4	Asymmetric VID use: “multi-netted server”	2330
32	Figure F-5	Asymmetric VLAN use: “Rooted-Multipoint”.....	2332
33	Figure F-6	Rooted-Multipoint with tagged interfaces	2333
34	Figure F-7	SPBV VLAN Shared Learning and VID Translation.....	2334
35	Figure G-1	Example of IEEE 802.3 MAC frame format	2336
36	Figure G-2	Methods for Bridge access to IEEE 802.11 and PMPN media: example.....	2339
37	Figure H-1	Static filtering inconsistency.....	2342
38	Figure H-2	Interoperability with MAC Bridges: example 1	2343
39	Figure H-3	Interoperability with MAC Bridges: example 2	2344
40	Figure H-4	Interoperability between Port-based and	
41		Port-and-Protocol-based classification	2345
42	Figure J-1	Up MPs in a CFM Port	2359
43	Figure K-1	TPMR as UNI demarcation device	2361
44	Figure K-2	TPMRs with aggregated links.....	2362
45	Figure K-3	Multiple TPMRs	2362
46	Figure K-4	Recovery at the end of a chain.....	2363
47	Figure K-5	Near simultaneous recoveries	2364
48	Figure K-6	Near simultaneous failure and recovery	2364
49	Figure K-7	Loss with quick recovery	2365
50	Figure L-1	Credit-based shaper operation—no conflicting traffic	2368
51	Figure L-2	Credit-based shaper operation—conflicting traffic	2369
52	Figure L-3	Credit-based shaper operation—burst traffic.....	2370
53	Figure L-4	Interference and latency	2374
54	Figure L-5	Burst behavior and credit	2374

1	Figure L-6	Fan-in scenario.....	2378
2	Figure L-7	Permanent delay scenario	2379
3	Figure L-8	Building up buffer occupancy—1.....	2380
4	Figure L-9	Building up buffer occupancy—2.....	2380
5	Figure L-10	Building up buffer occupancy—3.....	2380
6	Figure L-11	Building up buffer occupancy—4.....	2381
7	Figure M-1	PFC PDU format.....	2383
8	Figure N-1	PFC delays	2384
9	Figure N-2	Delay model.....	2385
10	Figure N-3	Worst-case delay.....	2386
11	Figure O-1	Converting a CRC to an FCS.....	2391
12	Figure O-2	Detection Lossless Circuit	2391
13	Figure O-3	Field change adjustment	2393
14	Figure O-4	Field insertion adjustment.....	2394
15	Figure P-1	Frame duplication scenario	2397
16	Figure P-2	Frame misordering scenario.....	2398
17	Figure Q-1	Establishing a guard band	2400
18	Figure Q-2	Using gate operations.....	2401
19	Figure T-1	Example Stream Filter and Stream Gate configuration for CQF.....	2410
20	Figure T-2	Traffic scheduling example for CQF	2411
21	Figure T-3	Example Stream Filter and Stream Gate configuration with two values of T	2412
22	Figure T-4	Traffic scheduling example with two values of T	2412
23	Figure T-5	Interleaving example—factor of 2	2413
24	Figure U-1	Example of enhancements for scheduled traffic.....	2417
25	Figure V-1	Path of frames along a single hop with index k with two Bridges	2424
26	Figure W-1	Isolation out-of-order frame example	2429
27	Figure W-2	De-isolation out-of-order frame example	2430
28	Figure W-3	Using queue markers and counters to preserve order when isolating.....	2431
29	Figure W-4	Using queue markers and counters to preserve order when de-isolating	2432
30	Figure W-5	Example Bridge buffering supporting PFC and CI	2433
31	Figure W-6	Example CI initiation by downstream peer	2433
32	Figure W-7	Example CI in process	2434
33	Figure W-8	Example PFC request for congesting queue	2434
34	Figure W-9	Example PFC request to avoid packet loss with CI enabled	2435

1 Tables

2	Table 6-1	Bridge transit delay	171
3	Table 6-2	Priority Code Point encoding.....	179
4	Table 6-3	Priority Code Point decoding.....	179
5	Table 6-4	Priority regeneration	180
6	Table 6-5	Default SRP domain boundary port priority regeneration override values	181
7	Table 6-6	Service Access Priority	192
8	Table 6-7	Encapsulated Addresses EtherType.....	199
9	Table 8-1	C-VLAN and MAC Bridge component Reserved addresses.....	227
10	Table 8-2	S-VLAN component Reserved addresses.....	228
11	Table 8-3	TPMR component Reserved addresses.....	228
12	Table 8-4	Stream gate control operations	237
13	Table 8-5	Recommended priority to traffic class mappings	241
14	Table 8-6	Transmission selection algorithm identifiers.....	243
15	Table 8-7	Gate operations	246
16	Table 8-8	Scheduled Traffic and Stream Gate procedures/variables	256
17	Table 8-9	Ageing time parameter value.....	263
18	Table 8-10	Combining Static and Dynamic Filtering Entries for an individual MAC address.....	273
19	Table 8-11	Combining Static Filtering Entry and MAC Address Registration Entry for	
20		“All Group Addresses” and “All Unregistered Group Addresses”	274
21	Table 8-12	Forwarding or Filtering for specific group MAC addresses.....	275
22	Table 8-13	Forwarding or Filtering with Dynamic Reservation Entries	276
23	Table 8-14	Determination of whether a Port is in a VID’s member set.....	276
24	Table 8-15	Standard LLC address assignment.....	281
25	Table 8-17	ISIS-SPB Recommended Address Usage.....	283
26	Table 8-16	ISIS-SPB reserved addresses.....	283
27	Table 8-18	CCM group destination MAC addresses	290
28	Table 8-19	LTM group destination MAC addresses.....	290
29	Table 9-2	Reserved VID values	293
30	Table 9-1	IEEE 802.1Q™ EtherType allocations.....	293
31	Table 9-3	Reserved I-SID values	295
32	Table 10-1	MRP application addresses	303
33	Table 10-2	MRP EtherType values	304
34	Table 10-3	Applicant state table.....	317
35	Table 10-4	Registrar state table.....	318
36	Table 10-5	LeaveAll state table	319
37	Table 10-6	PeriodicTransmission state table	319
38	Table 10-7	MRP timer parameter default values	320
39	Table 12-1	Component table entry managed object.....	351
40	Table 12-2	Port table entry	353
41	Table 12-3	ISS Port Number table entry	354
42	Table 12-5	Transmission Selection Algorithm Table row elements.....	457
43	Table 12-4	Bandwidth Availability Parameter Table row elements.....	457
44	Table 12-7	SR Class to Priority Mapping Table row elements.....	458
45	Table 12-6	Priority Regeneration Override Table row elements	458
46	Table 12-8	CN component managed object row elements	459
47	Table 12-9	CN component priority managed object row elements	460
48	Table 12-11	Congestion Point managed object row elements	461
49	Table 12-10	CN Port priority managed object row elements.....	461
50	Table 12-12	Reaction Point port priority managed object row elements.....	462
51	Table 12-13	Reaction Point group managed object row elements.....	462
52	Table 12-14	SRP Bridge Base Table row elements	463
53	Table 12-15	SRP Bridge Port Table row elements	464

1	Table 12-16	SRP Latency Parameter Table row elements.....	464
2	Table 12-18	SRP Reservations Table row elements	465
3	Table 12-17	SRP Stream Table row elements	465
4	Table 12-19	SRP Stream Preload Table row elements	466
5	Table 12-20	SRP Reservations Preload Table row elements	466
6	Table 12-21	Priority-based Flow Control objects	468
7	Table 12-22	EVB system base table	492
8	Table 12-24	SBP table entry	493
9	Table 12-23	EVB system parameter defaults.....	493
10	Table 12-25	VSI table entry	494
11	Table 12-27	UAP table entry parameters.....	495
12	Table 12-26	VSI MAC/VLAN table entry.....	495
13	Table 12-28	UAP table entry	496
14	Table 12-29	S-channel interface table entry	497
15	Table 12-31	ECP table entry	498
16	Table 12-30	URP table entry.....	498
17	Table 12-32	The Gate Parameter Table	503
18	Table 12-33	Frame Preemption Parameter table.....	507
19	Table 12-34	The Stream Parameter Table.....	509
20	Table 12-35	Stream Filter Instance Table	511
21	Table 12-36	The Stream Gate Instance Table.....	512
22	Table 12-37	The Flow Meter Instance Table	516
23	Table 12-38	The Scheduler Instance Table.....	516
24	Table 12-39	The Scheduler Group Instance Table.....	517
25	Table 12-40	The Scheduler Port Parameter Table	517
26	Table 12-41	The Timing Characteristics Table.....	518
27	Table 12-42	Bridge Delay attributes	520
28	Table 12-43	Propagation Delay attributes.....	522
29	Table 12-44	Static Trees attributes.....	522
30	Table 12-45	MRP External Control attributes	524
31	Table 12-43	CI Peer Table entry	527
32	Table 12-42	CI entity managed object	527
33	Table 12-44	CI Stream Table entry	528
34	Table 12-45	CIP entity managed object.....	528
35	Table 12-46	PBBN Auto Attach System objects	529
36	Table 12-47	PBBN Auto Attach port table	530
37	Table 12-48	PBBN Discovered Auto Attach Systems table.....	531
38	Table 12-49	PBBN Auto Attach assignment table.....	532
39	Table 12-50	PBBN Auto Attach statistics table.....	533
40	Table 13-2	Sample Configuration Digest Signature Keys	550
41	Table 13-1	Configuration Digest Signature Key.....	550
42	Table 13-3	Bridge and Port Priority values.....	568
43	Table 13-4	Port Path Cost values	569
44	Table 13-5	Timer and related parameter values.....	576
45	Table 17-1	IEEE 802.1Q MIB modules.....	644
46	Table 17-2	IEEE8021-TC-MIB structure	646
47	Table 17-3	IEEE8021-BRIDGE-MIB structure.....	647
48	Table 17-4	IEEE 802.1D objects not in the IEEE8021-BRIDGE-MIB.....	651
49	Table 17-5	IEEE8021-SPANNING-TREE MIB structure	652
50	Table 17-6	Clause 12 objects not in the IEEE8021-SPANNING-TREE MIB	654
51	Table 17-7	IEEE8021-Q-BRIDGE MIB structure	654
52	Table 17-8	Clause 12 management not in IEEE8021-Q-BRIDGE-MIB	659
53	Table 17-9	IEEE8021-PB-MIB structure.....	660
54	Table 17-10	IEEE8021-MSTP-MIB structure	662

1	Table 17-11	IEEE8021-CFM-MIB structure	665
2	Table 17-12	IEEE8021-CFM-V2-MIB structure	669
3	Table 17-13	IEEE8021-PBB-MIB structure	671
4	Table 17-14	IEEE8021-DDCFM-MIB structure	674
5	Table 17-15	IEEE8021-PBBTE-MIB structure	676
6	Table 17-16	Example of ieee8021PbbTeTeSiEspTable	678
7	Table 17-17	IEEE8021-TPMR-MIB structure.....	679
8	Table 17-18	IEEE8021-FQTSS-MIB structure.....	681
9	Table 17-19	IEEE8021-CN-MIB structure	682
10	Table 17-20	IEEE8021-SRP-MIB structure	684
11	Table 17-21	IEEE8021-MVRPX-MIB structure	686
12	Table 17-22	IEEE8021-MIRP-MIB structure.....	686
13	Table 17-23	PFC-MIB structure	687
14	Table 17-24	IEEE8021-TEIPS MIB structure	687
15	Table 17-25	IEEE8021-SPB-MIB structure	689
16	Table 17-26	IEEE8021-EVB-MIB structure.....	693
17	Table 17-27	IEEE8021-ECMP-MIB structure.....	697
18	Table 17-28	IEEE8021-ST-MIB structure.....	698
19	Table 17-29	IEEE8021-Preemption-MIB structure	699
20	Table 17-30	IEEE8021-PSFP-MIB structure.....	700
21	Table 17-31	IEEE8021-TSN-REMOTE-MANAGEMENT-MIB structure	703
22	Table 17-32	IEEE8021-PBBN-AA-MIB Structure	704
23	Table 17-1	PBB-TE required MIB compliances.....	714
24	Table 17-2	Sensitive managed objects: tables and notifications.....	722
25	Table 17-3	Sensitive managed objects: variables in dot1agCfmMdTable.....	723
26	Table 17-5	Sensitive managed objects (of DDCFM) for read	724
27	Table 17-4	Sensitive managed objects (of DDCFM): tables and notifications.....	724
28	Table 17-6	Sensitive managed objects (of EVB): tables and notifications.....	731
29	Table 17-7	Sensitive managed objects (of EVB) for read	731
30	Table 17-1	Provider Bridge service interface parameters	749
31	Table 17-2	PBB service interface parameters	753
32	Table 19-1	Actions taken by MP OpCode Demultiplexers.....	1238
33	Table 19-2	SAP use for LTMs and LTRs	1245
34	Table 20-1	Fault Alarm defects and priorities	1250
35	Table 20-2	Deriving enableRmepDefect and Port Status TLV in a Bridge.....	1260
36	Table 21-1	CFM PDU Encapsulation EtherType	1303
37	Table 21-3	OpCode Field range assignments	1305
38	Table 21-2	Common CFM Header format	1305
39	Table 21-4	TLV format	1306
40	Table 21-5	Type Field values.....	1307
41	Table 21-6	Organization-Specific TLV format.....	1307
42	Table 21-7	Sender ID TLV format.....	1308
43	Table 21-8	Port Status TLV format.....	1309
44	Table 21-10	Interface Status TLV format	1310
45	Table 21-11	Interface Status TLV values	1310
46	Table 21-9	Port Status TLV values	1310
47	Table 21-12	Data TLV format	1311
48	Table 21-13	End TLV format.....	1311
49	Table 21-14	CCM format.....	1312
50	Table 21-15	CCM Interval field encoding.....	1313
51	Table 21-16	CCM Maintenance Association Identifier field format:	
52		Maintenance Domain present	1314
53	Table 21-17	CCM Maintenance Association Identifier field format:	
54		Maintenance Domain not present	1314

1	Table 21-19	Short MA Name Format	1315
2	Table 21-18	Maintenance Domain Name Format	1315
3	Table 21-20	LBM and LBR formats	1316
4	Table 21-21	PBB-TE MIP TLV format	1317
5	Table 21-22	LTM format	1318
6	Table 21-23	LTM Flags field	1319
7	Table 21-24	LTM Egress Identifier TLV format	1320
8	Table 21-25	LTR format	1321
9	Table 21-26	LTR Flags field	1321
10	Table 21-27	Relay Action field values	1322
11	Table 21-28	LTR Egress Identifier TLV format	1322
12	Table 21-29	Reply Ingress TLV format	1323
13	Table 21-30	Ingress Action field values	1324
14	Table 21-31	Reply Egress TLV format	1325
15	Table 21-32	Egress Action field values	1325
16	Table 22-1	MEP creation	1338
17	Table 22-2	MIP creation	1339
18	Table 22-3	Bandwidth required for CCMs for 1 MA	1342
19	Table 22-4	Bandwidth required for CCMs for 1000 MAs	1343
20	Table 23-1	Time sequence diagram symbols	1354
21	Table 23-2	MSP performance parameters	1359
22	Table 23-3	MSP EtherType assignment	1364
23	Table 23-4	MSP Packet Types	1365
24	Table 24-1	Transmission and reception delays	1368
25	Table 26-1	Backbone Service Instance Group address OUI	1391
26	Table 26-2	Protection Requests Hierarchy	1414
27	Table 27-1	Allocation of VIDs to FIDs and FIDs to MSTIDs in an SPT Region (example)	1437
28	Table 28-1	Bridge Priority Masking	1465
29	Table 29-1	RFM format	1499
30	Table 29-2	SFM format	1500
31	Table 32-1	LLDP instance selection managed object overrides	1521
32	Table 32-2	CND defense mode selection managed object overrides	1521
33	Table 32-3	Determining cnpdIsAdminDefMode and cnpdDefenseMode	1527
34	Table 32-4	Correspondence of QCN and CCF message fields	1529
35	Table 32-5	NewCpSampleBase() return value as a function of cpFb	1532
36	Table 33-2	CNM Encapsulation	1544
37	Table 33-1	CN-TAG Encapsulation	1544
38	Table 33-3	Congestion Notification Message PDU	1545
39	Table 34-1	Default priority to traffic class mappings for SR classes A and B	1551
40	Table 34-2	Default priority to traffic class mappings for SR class B only	1552
41	Table 35-1	AttributeType Values	1566
42	Table 35-2	AttributeLength Values	1566
43	Table 35-3	FourPackedEvent Values	1567
44	Table 35-4	MSRP FirstValue NumberOfValues example	1568
45	Table 35-5	TSpec components examples	1571
46	Table 35-6	SR class ID	1574
47	Table 35-7	TLV types	1577
48	Table 35-8	Summary of Talker primitives	1590
49	Table 35-9	Summary of Listener primitives	1590
50	Table 35-11	Translation of Talker attributes	1595
51	Table 35-10	Talker attribute propagation per port	1595
52	Table 35-12	Incoming Listener attribute propagation per port	1598
53	Table 35-13	Updating Dynamic Reservation Entries	1599
54	Table 35-14	Updating operIdleSlope(N)	1599

1	Table 35-15	Listener Declaration Type Summation	1600
2	Table 35-16	Translation of Listener attributes	1601
3	Table 41-1	VDP TLV types	1633
4	Table 41-2	Flag values in VDP requests	1634
5	Table 41-3	Error types in VDP responses	1635
6	Table 41-4	Flag values in VDP responses	1635
7	Table 41-6	Filter Info format values	1636
8	Table 41-5	VSID format values	1636
9	Table 43-1	ECP subtypes	1659
10	Table 44-1	ECMP ECT-ALGORITHM values	1667
11	Table 44-2	F-TAG EtherType.....	1668
12	Table 45-1	ECT-ALGORITHM values for explicit trees	1679
13	Table 45-2	Bridge Priority Masking for the LT and LTS ECT Algorithms	1680
14	Table 45-3	Hop sub-TLV flags	1694
15	Table 46-1	StreamID elements.....	1719
16	Table 46-2	StreamRank elements	1720
17	Table 46-3	InterfaceID elements.....	1721
18	Table 46-4	IEEE802-MacAddresses elements.....	1722
19	Table 46-5	IEEE802-VlanTag elements	1723
20	Table 46-6	IPv4-tuple elements	1723
21	Table 46-7	IPv6-tuple elements	1724
22	Table 46-8	TrafficSpecification elements	1725
23	Table 46-9	TSpecTimeAware elements.....	1725
24	Table 46-10	UserToNetworkRequirements elements.....	1727
25	Table 46-11	InterfaceCapabilities elements.....	1729
26	Table 46-12	StreamStatus enumeration	1731
27	Table 46-13	StatusInfo elements.....	1733
28	Table 46-14	TalkerStatus enumeration	1733
29	Table 46-15	ListenerStatus enumeration.....	1733
30	Table 46-16	TSN Failure Codes.....	1734
31	Table 46-17	AccumulatedLatency elements	1736
32	Table 48-1	Summary of the YANG modules.....	1771
33	Table 48-2	VLAN Bridge component model YANG modules.....	1773
34	Table 48-3	Two-Port MAC Relay (TPMR) model YANG modules	1773
35	Table 48-4	Customer VLAN Bridge model YANG modules.....	1773
36	Table 48-6	CFM model YANG modules.....	1774
37	Table 48-7	Stream filters and stream gates model YANG modules	1774
38	Table 48-5	Provider Bridge model YANG modules.....	1774
39	Table 48-8	ATS model YANG modules.....	1775
40	Table 48-9	YANG module dependencies for the CI model.....	1775
41	Table 48-11	Frame Preemption model YANG modules.....	1776
42	Table 48-12	Per-Stream Filtering and Policing model YANG modules	1776
43	Table 48-10	Scheduled Traffic model YANG modules.....	1776
44	Table 48-13	CNC-configuration model YANG modules	1777
45	Table 48-14	Credit-based shaper algorithm model YANG modules	1777
46	Table 48-15	RSTP model YANG modules.....	1777
47	Table 48-16	MSTP model YANG modules	1778
48	Table 49-1	Congestion Isolation Message EtherType	2019
49	Table C-1	SRP to MoCA PQoS Transaction mapping.....	2148
50	Table C-2	SRP TSpec to MoCA TSPEC mapping.....	2148
51	Table C-3	SRP StreamID to MoCA PQoS Flow transaction mapping	2149
52	Table C-4	SRP to MLME QoS Services mapping.....	2154
53	Table C-5	EDCA-AC for AV Streams	2155
54	Table C-6	HCCA for AV Streams	2156

1 Table D-1	IEEE 802.1 Organizationally Specific TLVs	2157
2 Table D-2	Port and protocol capability/status.....	2159
3 Table D-3	Priority assignment table	2164
4 Table D-4	Traffic class bandwidth assignment table.....	2164
5 Table D-5	TSA Assignment Table.....	2164
6 Table D-6	PFC Enable bit vector	2167
7 Table D-7	Application Priority Table Entry format.....	2168
8 Table D-8	Sel field values.....	2168
9 Table D-9	RRSAT flag values and meanings	2171
10 Table D-10	EVB Mode values.....	2172
11 Table D-11	NVE Role values	2173
12 Table D-12	Application VLAN Table Entry format.....	2175
13 Table D-13	Sel field values.....	2175
14 Table D-14	Device Type field values	2178
15 Table D-15	Port Orientation field values	2178
16 Table D-18	PBBN Auto Attach Tagging field values	2180
17 Table D-16	PBBN Auto Attach association states.....	2180
18 Table D-17	PBBN Auto Attach System Type values	2180
19 Table D-19	PBBN Auto Attach PortNetId values	2181
20 Table D-20	PBBN Auto Attach Assignment Status values	2182
21 Table D-21	IEEE 802.1 extension MIB object group conformance requirements	2188
22 Table D-22	IEEE 802.1/LLDP extension MIB object cross reference.....	2189
23 Table D-23	Summary of the YANG modules.....	2290
24 Table E-1	State machine symbols.....	2326
25 Table I-1	Traffic type to traffic class mapping	2348
26 Table I-2	Traffic type acronyms	2349
27 Table I-3	Defining traffic types	2349
28 Table I-4	Defining traffic types—Credit-based shaper support of SR class B only	2350
29 Table I-5	Defining traffic types—Credit-based shaper support of SR classes A and B	2351
30 Table I-6	Priority Code Point encoding.....	2353
31 Table I-7	Priority Code Point decoding.....	2353
32 Table J-1	Provider MD Level allocation	2355
33 Table J-2	IEEE / ITU-T terminology differences	2355
34 Table N-1	IEEE 802.3 Interface Delays	2387

1

2 **IEEE Standard for**
3 **Local and metropolitan area networks—**

4 **Bridges and Bridged Networks**

5 **1. Overview**

6 IEEE 802® Local Area Networks (LANs, 3.119)⁶ of all types can be connected together with Media Access
7 Control (MAC) Bridges (3.159) or Virtual Local Area Network (VLAN) Bridges (3.306), collectively
8 known as Bridges (3.28). This standard defines the operation of Bridges and Bridged Networks. VLANs
9 facilitate the administration of logical groups of stations. Stations in the same VLAN communicate as if they
10 were on the same LAN, while traffic between VLANs is restricted. Management of VLAN Bridges and
11 stations allows stations to be added to, removed from, or moved between VLANs.

12 This standard further extends the specification of VLAN Bridges to enable a service provider organization to
13 use a common infrastructure of Bridges and LANs to offer the equivalent of separate LANs, Bridged, or
14 Virtual Bridged Networks to independent customer organizations.

15 This standard specifies protocols and protocol entities within the architecture of Bridges that provide
16 capabilities for detecting, verifying, and isolating connectivity failures in Bridged Networks. These
17 capabilities can be used in networks operated by multiple independent organizations, each with restricted
18 management access to each other's equipment.

19 **1.1 Scope**

20 This standard specifies Bridges that interconnect individual LANs, each supporting the IEEE 802 MAC
21 Service using a different or identical media access control method, to provide Bridged Networks and
22 VLANs.

23 **1.2 Purpose**

24 Bridges, as specified by this standard, allow the compatible interconnection of information technology
25 equipment attached to separate individual LANs.

⁶ IEEE and IEEE 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by The Institute of Electrical and Electronics Engineers, Incorporated.

1.3 Introduction

For the purpose of compatible interconnection of information technology equipment using the IEEE 802 MAC Service supported by interconnected IEEE 802 standard LANs using different or identical media access control methods, this standard specifies the operation of MAC Bridges and VLAN Bridges. To this end, it:

- a) Positions the support of VLANs within an architectural description of the MAC Sublayer.
- b) Defines the principles of operation of the MAC Bridge and VLAN Bridge in terms of the support and preservation of the MAC Service, and the maintenance of quality of service (QoS).
- c) Specifies an Enhanced Internal Sublayer Service (EISS) provided to the Media Access-Independent functions that provide frame relay in a VLAN Bridge.
- d) Establishes the principles and a model of Virtual Bridged Network operation.
- e) Identifies the functions to be performed by Bridges, and provides an architectural model of the operation of a Bridge in terms of processes and entities that provide those functions.
- f) Specifies a frame format that allows a VLAN Identifier (VID) and priority information to be carried by VLAN-tagged user data frames.
- g) Specifies the rules that govern the addition or removal of VLAN tags to and from user data frames.
- h) Establishes the requirements for automatic configuration of VLAN topology.
- i) Establishes the requirements for VLAN Bridge Management in a Virtual Bridged Network, identifying managed objects and defining management operations.
- j) Defines SMIPv2 (IETF STD 58)⁷ Management Information Based (MIB) modules for the management of VLAN Bridge capabilities including spanning tree protocols and Provider Bridges.
- k) Define YANG configuration and operational state models (Clause 48) in support of Two-Port MAC Relays, Customer VLAN Bridges, and Provider Bridges, including Connectivity Fault Management (CFM) for those Bridges.
- l) Defines the operation of the Multiple Spanning Tree Algorithm and Protocol (MSTP).
- m) Describes the protocols and procedures necessary to support interoperation between Multiple Spanning Tree (MST) and Single Spanning Tree (SST) Bridges in the same Virtual Bridged Networks.
- n) Specifies the requirements to be satisfied by equipment claiming conformance to this standard.

To enable a service provider to use a Virtual Bridged Network to provide separate instances of the IEEE 802 MAC Service, MAC Internal Sublayer Service (ISS), and EISS to multiple independent customers, in a manner that does not require cooperation among the customers and that requires a minimum of cooperation between the customers and the provider of the MAC Service, this standard further specifies the operation of Provider Bridges. To this end, it:

- o) Differentiates Customer VLANs (C-VLANs) that are under the administrative control of a single customer of a service provider, from the Service VLANs (S-VLANs) that are used by a service provider to support different customers.
- p) Specifies VLAN tag formats for both C-VLANs and S-VLANs, allowing each to be distinguished and separately applied and administered by customers and by a service provider.
- q) Specifies the functionality of a generic VLAN Bridge component within a system and the specific requirements of derived C-VLAN and S-VLAN components.
- r) Specifies a C-VLAN Bridge as comprising a single C-VLAN component, and a Provider Bridge as encompassing Bridges that comprise a single S-VLAN component and no C-VLAN components (S-VLAN Bridge) or a single S-VLAN component and one or more C-VLAN components (Provider Edge Bridge).

⁷ Information on references can be found in Clause 2.

- 1 s) Specifies parameters and mappings that allow the EISS to support traffic classes that comprise
2 distinct aggregate flows supporting different QoS characteristics and provide independent
3 guarantees to different customers, through support of priority and drop precedence marking.
- 4 t) Specifies the incorporation of flow metering, transmission queue management, and transmission
5 selection algorithms within the forwarding process of a Bridge.
- 6 u) Positions the support of S-VLANs within the architectural description of the MAC Sublayer and
7 specifies their relationship to media access method-dependent functions and to the
8 media-independent functions used by customers to administer their networks, including the support
9 of C-VLANs.
- 10 v) Allocates the reserved multicast addresses to media access method-dependent, provider network,
11 and customer network functions, specifying the filtering to be applied in each type of VLAN Bridge
12 component.
- 13 w) Defines the principles of network operation in terms of the support and preservation of the MAC
14 Service, and the maintenance of QoS for each service instance, including the segregation of data
15 belonging to different organizations.
- 16 x) Specifies customer interfaces to a Provider Bridged Network (PBN) in terms of the operation and
17 configuration of the VLAN Bridge components of Provider Bridges, including interfaces that
 - 18 1) Provide access to a single service instance through a Bridge Port.
 - 19 2) Allow a customer to select among and identify service instances by Customer VLAN Identifier
20 (C-VID).
 - 21 3) Allow a customer to select among and identify service instances by Service VLAN Identifier
22 (S-VID).
 - 23 4) Support customer signaling of priority information on a frame by frame basis.
 - 24 5) Multiplex service instances over LANs that provide access to a provider network.
 - 25 6) Support fault tolerance through redundant provision of access LANs and equipment.
- 26 y) Describes the functions to be performed within the PBN in order to support and maintain the
27 connectivity provided to customer service instances.
- 28 z) Establishes the requirements for Bridge Management in the PBN, identifying the managed objects
29 and defining the management operations.
- 30 aa) Specifies performance requirements, and recommends default values and applicable ranges for the
31 operational parameters of a Provider Bridge.

32 This standard specifies protocols, procedures, and managed objects to support Connectivity Fault
33 Management (CFM). These allow discovery and verification of the path, through Bridges and LANs, taken
34 for frames addressed to and from specified network users, and support detection and isolation of a
35 connectivity fault to a specific Bridge or LAN. To this end, it:

- 36 ab) Defines Maintenance Domains, Maintenance Associations (MAs), their constituent Maintenance
37 Points (MPs), and the managed objects required to create and administer them.
- 38 ac) Describes the protocols and procedures used by MPs to detect and diagnose connectivity faults
39 within a Maintenance Domain.

1 This standard specifies protocols, procedures, and managed objects to allow support of provisioning systems
2 that explicitly select traffic engineered paths within Provider Backbone Bridged Networks (PBBNs) by
3 allowing a network operator to disable unknown destination address forwarding, source address learning and
4 spanning tree protocols for administratively selected VIDs, while allowing other network control protocols
5 to dynamically determine active topologies for other services. These interoperable capabilities are supported
6 by management of individual Bridges by Simple Network Management Protocol (SNMP) using an SMIPv2
7 MIB, by extensions to the other control protocols specified in this standard, by the use of CFM with the
8 addresses and VIDs that specify traffic engineered connections, and by 1:1 path protection switching
9 capable of load sharing. To this end, it:

- 10 ad) Enables construction of active topologies by an external agent that is responsible for setting up
11 Ethernet Switched Paths (ESPs) by splitting the B-VID space between distributed spanning tree
12 protocols and provisioned control.
- 13 ae) Supports discard of frames with unknown destination addresses for B-VIDs under provisioned
14 control.
- 15 af) Supports the operation of Continuity Check, Loopback, and Linktrace protocols on provisioned
16 traffic engineered paths.
- 17 ag) Supports 1:1 protection switching capable of load sharing for Traffic Engineering service instances
18 (TESIs).
- 19 ah) Supports protection of a group of TESIS that traverses a sequence of LANs and intervening Bridges
20 using a method that does not require the modification of data or control frames.
- 21 ai) Provides required extension to SNMP management by SMIPv2 MIB modules.

22 This standard does not specify operation of ESPs through multiple Provider Backbone Bridge Traffic
23 Engineering (PBB-TE) Regions. All the Backbone Edge Bridges (BEBs) specified for use in a PBB-TE
24 Region are combined I type and B type Backbone Edge Bridges (IB-BEBs).

25 This standard specifies protocols, procedures, and managed objects to support the Multiple Registration
26 Protocol (MRP). MRP allows participants in an MRP Application to register attributes with other
27 participants in a Bridged Network. Four applications are defined—one to register VIDs [Multiple VLAN
28 Registration Protocol (MVRP)], one to register MAC addresses [Multiple MAC Registration Protocol
29 (MMRP)], one to register Streams and configure associated network resources [Multiple Stream
30 Registration Protocol (MSRP)], and one that provides the ability to flush learned MAC Address Entries held
31 in the Filtering Database (FDB) of an I-component on a per-I-SID basis [Multiple I-SID Registration
32 Protocol (MIRP)]. MVRP will furthermore provide for the rapid healing of network failures without
33 interrupting services to unaffected VLANs. To this end, it specifies the following:

- 34 aj) MRP and the operation of MRP entities.⁸
- 35 ak) The generic frame formats used in MRP exchanges.
- 36 al) The MMRP application of MRP, and the frame formats that it uses.
- 37 am) The MVRP application of MRP, and the frame formats that it uses.

38 To allow scaling of Provider Networks to at least 2^{24} S-VLANs, this standard further specifies the operation
39 of Provider Backbone Bridges (PBBs) by means of an architecture and Bridge protocols compatible and
40 interoperable with PBN protocols and equipment, allowing interconnection of multiple PBNs. To this end,
41 it:

- 42 an) Introduces BEBs that, by exchanging backbone frames that encapsulate the addresses, VLAN tags,
43 and data of customer frames, support the virtual, media-independent equivalent of a number of
44 independent instances of the service provided by media-dependent frame transmission procedures.

⁸ MRP replaces the Generic Attribute Registration Protocol (GARP), defined in IEEE Std 802.1D™-2004 [B14], that was used to support GVRP and GMRP in earlier revisions of IEEE Std 802.1Q. Similarly, GVRP and GMRP are replaced by MVRP and MMRP, respectively.

- 1 ao) Extends the parameters of the ISS and EISS to include a connection identifier, capable of
2 referencing the backbone addresses and other parameters, used to convey customer frames from one
3 BEB to all, or one of, the other BEBs supporting a particular backbone service instance.
- 4 ap) Specifies the format of the Backbone Service Instance tag (I-TAG) that encapsulates the customer
5 addresses, and introduces a Backbone Service Instance Identifier (I-SID) that allows each BEB to
6 support a number of backbone service instances and permits the unambiguous identification of up to
7 2^{24} backbone service instances within a single PBBN.
- 8 aq) Provides a model of BEB operation in terms of VLAN Bridge components that allows the use of
9 Provider Bridges as Backbone Core Bridges (BCBs), with PBBN traffic carried as frames
10 containing I-TAGs on particular Backbone VLANs (B-VLANs) potentially coexisting with PBN
11 traffic carried as frames without I-TAGs on other B-VLANs.
- 12 ar) Specifies the interfaces that a PBBN can provide to transport service frames. These comprise a
13 Port-based service interface that assigns all received untagged and priority-tagged frames to a single
14 S-VLAN transported over a single backbone service instance, an S-tagged service interface capable
15 of mapping individual S-VLANs to different backbone service instances, and an I-tagged service
16 interface capable of mapping frames from one set of backbone service instances to another.
- 17 as) Describes the use of redundant Bridges and access LANs to protect backbone service access against
18 failure of any of those systems or components.
- 19 at) Specifies the management of BEBs in terms of the model of operation [item ap) above], making use
20 of defined management objects for the individual VLAN Bridge components, and adding managed
21 objects to facilitate service creation.
- 22 au) Describes the use of CFM to detect and isolate faults in the connectivity provided to individual
23 S-VLANs across the PBBN, in the connectivity provided to the group of S-VLANs supported by a
24 single backbone service instance (identified by an I-SID), and in the connectivity provided to
25 individual B-VLANs within the backbone itself.
- 26 av) Specifies extensions to MSTP to allow network administrators to protect against loops through
27 peered PBBNs without requiring coupling of spanning trees that operate independently for each
28 PBBN.

29 This standard specifies CFM protocols, procedures, and managed objects that provide confirmation of
30 successful transmission of frames conveying specified data. This capability supports diagnosis of faults
31 sensitive to, or caused by, particular data patterns, and their isolation to part of the transmission path.
32 Connectivity verification can be carried out from any single point with bridged connectivity to MPs on the
33 path, can isolate failures to communicate in a specific direction, and can be carried out while service is being
34 provided to other users of the data path. To this end, it:

- 35 aw) Defines the extensions to CFM capabilities defined by Clause 18 through Clause 22 to facilitate
36 diagnosis and isolation of faults sensitive to, or caused by, particular data patterns in frames
37 transmitted by a service user.
- 38 ax) Describes the protocols and procedures for data-driven and data-dependent connectivity fault
39 management (DDCFM).

40 This standard specifies the function of a Two-Port MAC Relay (TPMR), along with protocols and
41 procedures that support its operation. A TPMR is a type of Bridge that has only two externally accessible
42 Bridge Ports, and supports a subset of the functionality of a MAC Bridge. A TPMR is transparent to all
43 frame-based media-independent protocols, except those explicitly addressed to it and those that are destined
44 for reserved MAC addresses that the relay function of the TPMR is defined not to forward. It is remotely
45 manageable through at least one of its external MACs, and signals a failure of either MAC's LAN through
46 the other MAC. A TPMR should only be attached to point-to-point LANs. The conformance requirements
47 for a TPMR are stated in 5.13 and 5.15.

1 This standard allows Bridges to provide performance guarantees for time-sensitive (i.e., bounded latency
2 and latency variation) loss-sensitive real-time audio/video (AV) data stream transmission (AV traffic). It
3 specifies priority regeneration and controlled bandwidth queue draining algorithms. VLAN tag encoded
4 priority values are allocated, in aggregate, to segregate frames among queues that support AV traffic and
5 queues that support non-AV traffic, allowing simultaneous support of both AV traffic and other bridged
6 traffic over and between wired and wireless Local Area Networks (LANs). To this end, it:

7 ay) Defines status parameters that allow the boundaries of a Stream Reservation Protocol (SRP—see
8 Clause 35) domain (35.1.4) to be identified and maintained.

9 az) Specifies how the priority information in frames received at SRP domain boundary ports is
10 regenerated.

11 NOTE 1—The priorities in frames transmitted from outside an SRP domain to a Bridge inside an SRP domain
12 are remapped in order to ensure that traffic that is not associated with a reservation does not disrupt traffic that
13 is associated with a reservation. Hence, traffic entering an SRP domain that uses Priority Code Point values
14 associated with reserved traffic classes will be remapped to Priority Code Point values that are not associated
15 with reserved traffic classes.⁹

16 ba) Specifies how priority information is used to determine the traffic classes to be used for
17 time-sensitive streams.

18 bb) Defines a credit-based shaper algorithm to shape traffic in accordance with stream reservations.

19 NOTE 2—The credit-based shaper algorithm operates on the outbound queues; the mechanisms specified for
20 the support of time-sensitive AV traffic do not involve any form of ingress metering or policing.

21 This standard specifies protocols, procedures, and managed objects to support congestion notification. These
22 allow a Virtual Bridged Network or a portion thereof, with a limited bandwidth-delay product, to transfer
23 long-lived data flows with a significantly reduced chance of frame loss compared to a network without
24 congestion notification. To this end, it:

25 bc) Defines a means for VLAN Bridges that support congestion notification to form Congestion
26 Managed Domains within a Virtual Bridged Network.

27 bd) Defines a means for detecting congested queues in end stations and VLAN Bridges, for signaling
28 such congestion to the end stations sourcing the frames causing the congestion, and for those end
29 stations to control the rate of transmission of those frames.

30 To enable the end-to-end management of resource reservation for QoS guaranteed streams, this standard
31 further specifies protocols, procedures, and managed objects, usable by existing higher layer mechanisms,
32 that allow network resources to be reserved for specific traffic streams traversing a Bridged Network. To this
33 end, it:

34 be) Specifies the use of Dynamic Reservation Entries (8.8.7) in the FDB to control the forwarding of
35 frames associated with a particular Stream.

36 bf) Specifies a Stream Reservation Protocol (SRP). SRP facilitates the registration, deregistration, and
37 maintenance of stream reservation information in relevant Bridges to establish end-to-end stream
38 paths.

39 This standard specifies protocols, procedures, and managed objects to support topology change signaling to
40 alter the binding (held in an I-Component) of Customer addresses to backbone addresses on a per-I-SID
41 basis. This is accomplished by extending the use of MRP. To this end, it specifies the MIRP application of
42 MRP and the frame formats that it uses.

43 NOTE 3—MIRP can only trigger the flushing of learned MAC address information; it does not propagate the
44 registration of I-SIDs. The name Multiple I-SID Registration Protocol is chosen because MIRP is a Multiple
45 Registration Protocol (MRP) application and can be extended to perform I-SID registrations.

⁹ Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

1 This standard allows an S-tagged service interface connecting two independently administered PBNs to be
2 used to handle traffic (identified by a single S-VID) for a given customer attached to one PBN as if the
3 customer were directly attached to the other PBN using a Port-based or C-tagged service interface. To this
4 end, it:

- 5 bg) Specifies the use of a Port-mapping S-VLAN component to associate selected S-VIDs registered on
6 an external port with distinct internal ports, each of which supports a separate service interface.

7 This standard specifies protocols, procedures, and managed objects to support Priority-based Flow Control
8 (PFC). These allow a Virtual Bridged Network, or a portion thereof, to enable flow control per traffic class
9 on IEEE 802 point-to-point full-duplex links. To this end, it:

- 10 bh) Defines a means for a system to inhibit transmission of data frames on certain priorities from the
11 remote system on the link.

12 This standard specifies protocols, procedures, and managed objects for enhancement of transmission
13 selection to support allocation of bandwidth among traffic classes. When the offered load in a traffic class
14 does not use its allocated bandwidth, Enhanced Transmission Selection (ETS) will allow other traffic classes
15 to use the available bandwidth. Bandwidth is used by traffic classes subject to ETS when there are no frames
16 to be transmitted for traffic classes subject to strict priority or credit-based shaper algorithms. It defines the
17 Data Center Bridging eXchange protocol (DCBX), which controls the application of ETS and PFC.

18 This standard specifies Shortest Path Bridging (SPB) of unicast and multicast frames, specifying protocols
19 to calculate multiple active topologies that can share learned station information, and support of a VLAN by
20 multiple, per-topology, Shortest Path VLAN Identifiers (SPVIDs). To this end, it:

- 21 bi) Describes the use of shortest paths to increase throughput and minimize transit delay, while
22 introducing a negligible rate of frame misordering.

- 23 bj) Requires that active topologies calculated by spanning tree protocols and Shortest Path Tree (SPT)
24 protocols be stable, predictable, and reproducible to maintain the characteristics of the MAC Service
25 provided.

- 26 bk) Requires, except in the case of SPB using Equal Cost Multiple Paths (ECMP), active topologies that
27 are reverse path congruent and unicast-multicast congruent to permit learning of station location
28 from the source addresses of all frames and simplify the detection and management of faults.

29 NOTE 4—ECMP operation does not provide (nor does this standard attempt to define for ECMP VLANs)
30 reverse path congruence and unicast-multicast congruence as these concepts cease to have utility in an ECMP
31 context.

- 32 bl) Specifies the calculation of symmetric sets of SPTs, each rooted at a Bridge within an SPT Region
33 comprising Bridges operating compatible protocols and configurations.

- 34 bm) Specifies the use of Bridge Protocol Data Units (BPDUs) to identify and bound SPT Regions and to
35 ensure loop-free interoperability with regions using the Rapid Spanning Tree Algorithm and
36 Protocol (RSTP) and MSTP.

- 37 bn) Specifies both Shortest Path Bridging VID (SPBV) and Shortest Path Bridging MAC (SPBM)
38 modes:

- 39 1) for SPBV, identifying each SPT by SPVID and locating end stations by source MAC address
40 learning.

- 41 2) for SPBM, identifying each SPT by VID and source MAC address and distributing end station
42 location information explicitly.

- 43 bo) Supports management selection of the Common Spanning Tree (CST), a Multiple Spanning Tree
44 Instance (MSTI), or SPB for support of any given VLAN within an SPT Region.

- 45 bp) Specifies a protocol that automatically assigns SPVIDs for each VLAN supported by SPBV.

- 46 bq) Supports load sharing by Equal Cost Trees (ECTs) through the calculation of multiple SPT Sets,
47 with each shortest path VLAN being assigned to one SPT Set.

- 1 br) Specifies Intermediate System to Intermediate System Protocol for Shortest Path Bridging
2 (ISIS-SPB): the use of and extensions to the Intermediate System to Intermediate System (IS-IS)
3 Protocol to calculate SPTs for both SPBV and SPBM.
- 4 bs) Describes the addressing of ISIS-SPB entities and specifies the group MAC addresses they use.
- 5 bt) Specifies the use of loop prevention (for SPBV and for multicast frames for SPBM) and loop
6 mitigation (for unicast frames for SPBM).
- 7 bu) Specifies an Agreement Protocol that prevents loops, specifying the necessary state information and
8 computation as part of ISIS-SPB and communicating agreement information for the CIST and (as a
9 compact Digest) for SPTs in each BPDUs.

10 This standard specifies protocols, procedures, and managed objects that:

- 11 bv) Provide for the discovery, configuration, and control of a pair of direct-attached Port-mapping
12 S-VLAN components to extend the operation of a Customer Bridge to remote ports and enable
13 coexistence of multiple services on station-resident ports (e.g., embedded bridging).
- 14 bw) Provide for discovery, configuration, and operation of reflective relay (8.6.1) for a Bridge Port.
- 15 bx) Provide for discovery of, and coordinated configuration of, edge relays (ERs) and other devices that
16 utilize the reflective relay service.
- 17 by) Provide for dynamic profile-driven port configuration.
- 18 bz) Specifies load spreading by distributing unicast traffic over the set of available equal cost paths and
19 assigning multicast traffic flows to a variety of trees.
- 20 ca) Specifies a flow filtering tag (F-TAG) containing a flow hash used in unicast ECMP traffic
21 distribution and a TTL (time-to-live) field used to mitigate the effects of traffic loops resulting from
22 transient conditions or control software errors or faults.

23 This standard also specifies further protocol extensions, procedures, and managed objects to IS-IS for
24 providing capabilities beyond Shortest Path Bridging (SPB) for Bridged Networks. These extensions
25 involve explicit path control, bandwidth reservation, and redundancy (protection, restoration) for data flows.
26 Thus, this standard specifies bridging on explicit paths for unicast and multicast frames, specifying protocols
27 to determine multiple active topologies. To this end, it:

- 28 cb) Describes the use of explicit trees, e.g., to improve resiliency and decrease the probability of
29 congestion.
- 30 cc) Requires that active topologies calculated by one or multiple entities external to the routing protocol
31 are such that the characteristics of the MAC Service are provided.
- 32 cd) Supports management selection of explicit trees for support of any given VLAN within an SPT
33 Region.
- 34 ce) Specifies Intermediate System to Intermediate System Path Control and Reservation (ISIS-PCR):
35 the use of and extensions to the Intermediate System to Intermediate System (IS-IS) protocol to
36 establish explicit trees.
- 37 cf) Specifies the use of ISIS-PCR for recording bandwidth assignments.
- 38 cg) Specifies redundancy for ISIS-SPB and ISIS-PCR.

39 This standard also:

- 40 ch) Provides for the use of IEEE 802.11™ media as links internal to, as well as links providing access
41 to, a Bridged Network or Virtual Bridged Network.
- 42 ci) Defines enhancements for scheduled traffic to allow transmissions scheduled relative to a known
43 timescale.
- 44 cj) Defines frame preemption to interrupt transmission of preemptable frames by express frames.

1 This standard specifies protocols, procedures, and managed objects that:

2 ck) Allow for the filtering and policing of individual traffic streams.

3 cl) Allow for Asynchronous Traffic Shaping (ATS) over full-duplex links with constant bit data rates.

4 This standard specifies enhancements to protocols, procedures, and managed objects for the configuration of
5 network resources for time-sensitive applications that require timely, high probability, delivery of frames
6 without end station retransmission. To this end, it:

7 cm) Describes three approaches to network configuration: fully distributed, centralized
8 network/distributed user, and fully centralized.

9 cn) Specifies managed objects for forwarding and queuing enhancements for time-sensitive streams
10 (FQTSS).

11 co) Describes Centralized User Configuration (CUC) and Centralized Network Configuration (CNC)
12 entities.

13 cp) Specifies managed objects for configuration of Bridges by a CNC.

14 cq) Specifies YANG configuration and operational state models (Clause 48) in support of Scheduled
15 Traffic, Frame Preemption, Per-Stream Filtering and Policing, and CUC configuration.

16 This standard specifies protocols, procedures, and managed objects that support the isolation of congesting
17 data flows within data center environments. This is achieved by enabling systems to individually identify
18 flows creating congestion, isolate those flows to the congesting queue, and signal to neighbors. This
19 mechanism reduces head-of-line blocking for non-congesting flows sharing the same traffic class.
20 Congestion Isolation is used with higher layer protocols that utilize end-to-end congestion control in order to
21 reduce packet loss and latency. To this end, it:

22 cr) Defines a means for VLAN-aware Bridges that support congestion isolation to identify flows that
23 are creating congestion.

24 cs) Defines a means for adjusting transmission selection for frames of congesting flows.

25 ct) Provides a means for discovering peer VLAN-aware Bridges and stations that support congestion
26 isolation.

27 cu) Defines a means for signaling congestion isolation to supporting peer Bridges and stations.

28 cv) Defines a means for recognizing a system's level and port orientation within the topology relative to
29 the edge.

30 This standard specifies the protocols, procedures, and management objects for auto attachment of network
31 devices to backbone service instances (BSIs) by adding an Auto Attach Protocol (AAP) over the Link Layer
32 Discovery Protocol (LLDP). The AAP simplifies the deployment and administration of PBB networks by
33 automatically coupling end stations or Bridges to BSIs.

34 To this end, it:

35 cz) Introduces the concepts of Auto Attach Protocol (AAP), Auto Attach System (AAS) on devices
36 supporting Auto Attach¹⁰; non-PBB-based Auto Attach Devices (AADs) and PBB-based Auto
37 Attach BEBs (AABs).

38 da) Describes the process of discovery and advertising of capabilities for AADs and AABs supporting
39 the AAP acting as an AAS.

40 db) Describes the bindings between Backbone Service Instance Identifiers (I-SIDs) and VLAN
41 Identifiers.

42 dc) Specifies IEEE 802.1 Organizationally Specific LLDP TLVs for discovery and capabilities
43 advertising, and attachment of C-VLANs to Backbone Service Instances.

44 dd) Specifies SNMP MIB support for PBBN Auto Attach.

¹⁰ The term "Auto Attach" is used interchangeably with "PBBN Auto Attach" where it is clear from the context that "PBBN Auto Attach" is discussed.

- 1 de) Specifies YANG extensions to the LLDP YANG modules for the PBBN Auto Attach LLDP TLVs.
- 2 df) Specifies SNMP MIB extensions to the LLDP EVB extension MIB module for the PBBN Auto
- 3 Attach LLDP TLVs.

4

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

ANSI X3.159, American National Standards for Information Systems—Programming Language—C.¹¹

IEEE Std 802[®], IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.^{12, 13}

IEEE Std 802dTM-2017, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture—Amendment 1: Allocation of Uniform Resource Name (URN) Values in IEEE 802[®] Standards.

IEEE Std 802.1ABTM, IEEE Standard for Local and metropolitan area networks—Station and Media Access Control Connectivity Discovery.

IEEE Std 802.1ACTM, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.1AETM, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Security.

IEEE Std 802.1ASTM, IEEE Standard for Local and metropolitan area networks—Timing and Synchronization for Time-Sensitive Applications in Bridged Local Area Networks.

IEEE Std 802.1AXTM, IEEE Standard for Local and metropolitan area networks—Link Aggregation.

IEEE Std 802.1BRTM, IEEE Standard for Local and metropolitan area networks—Virtual Bridged Local Area Networks—Bridge Port Extension.

IEEE Std 802.1CBTM, IEEE Standard for Local and metropolitan area networks—Frame Replication and Elimination for Reliability.

IEEE Std 802.1CSTM, IEEE Standard for Local and Metropolitan Area Networks—Link-local Registration Protocol.

IEEE Std 802.1XTM, IEEE Standard for Local and Metropolitan Area Networks—Port-Based Network Access Control.

IEEE Std 802.3TM, IEEE Standard for Ethernet.

IEEE Std 802.11TM, Standard for Information Technology—Telecommunications and Information Exchange between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications.

IEEE Std 802.20TM, IEEE Standard for Local and metropolitan area networks—Part 20: Air Interface for Mobile Broadband Wireless Access Systems Supporting Vehicular Mobility—Physical and Media Access Control Layer Specification.

IETF RFC 768 (STD0006), User Datagram Protocol, August 1980.

IETF RFC 791 (STD0005), Internet Protocol—DARPA Internet Program Protocol Specification, September 1981.¹⁴

¹¹ ANSI publications are available from the IHS Standards Store (<https://global.ihs.com/>).

¹² The IEEE standards or products referred to in Clause 2 are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

¹³ IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>).

¹⁴ IETF RFCs are available from the Internet Engineering Task Force (<https://www.ietf.org/>).

- 1 IETF RFC 1035 (STD 13), Domain Names—Implementation and Specification, November 1987.
- 2 IETF RFC 1042, A Standard for the Transmission of IP Datagrams over IEEE 802 Networks, February
3 1988.
- 4 IETF RFC 1390 (STD 36), Transmission of IP and ARP over FDDI Networks, January 1993.
- 5 IETF RFC 2104, HMAC: Keyed-Hashing for Message Authentication, February 1997.
- 6 IETF RFC 2119 (BCP 14), Key words for use in RFCs to Indicate Requirement Levels, March 1997.
- 7 IETF RFC 2205, Resource ReSerVation Protocol (RSVP)—Version 1 Functional Specification, September
8 1997.
- 9 IETF RFC 2271, An Architecture for Describing SNMP Management Frameworks, January 1998.
- 10 IETF RFC 2474, Definition of the Differentiated Services Field (DS Field) in the IPv4 and IPv6 Headers,
11 December 1998.
- 12 IETF RFC 2578 (STD 58), Structure of Management Information Version 2 (SMIv2), April 1999.
- 13 IETF RFC 2579 (STD 58), Textual Conventions for SMIv2, April 1999.
- 14 IETF RFC 2580 (STD 58), Conformance Statements for SMIv2, April 1999.
- 15 IETF RFC 2685, Virtual Private Networks Identifier, September 1999.
- 16 IETF RFC 2737, Entity MIB (Version 2), December 1999.
- 17 IETF RFC 2750, RSVP Extensions for Policy Control, January 2001.
- 18 IETF RFC 2863, The Interfaces Group MIB, June 2000.
- 19 IETF RFC 3046, DHCP Relay Agent Information Option, January 2000.
- 20 IETF RFC 3168, The Addition of Explicit Congestion Notification (ECN) to IP, September 2001.
- 21 IETF RFC 3232, Assigned Numbers: RFC 1700 is Replaced by an On-line Database, January 2002.
- 22 IETF RFC 3410, Introduction and Applicability Statements for Internet Standard Management Framework,
23 December 2002.
- 24 IETF RFC 3411, An Architecture for Describing Simple Network Management Protocol (SNMP)
25 Management Frameworks, December 2002.
- 26 IETF RFC 3413 (STD 62), Simple Network Management Protocol (SNMP) Applications, December 2002.
- 27 IETF RFC 3414 (STD 62), User-based Security Model (USM) for version 3 of the Simple Network
28 Management Protocol (SNMPv3), December 2002.
- 29 IETF RFC 3415 (STD 62), View-based Access Control Model (VACM) for the Simple Network
30 Management Protocol (SNMP), December 2002.
- 31 IETF RFC 3417 (STD 62), Transport Mappings for the Simple Network Management Protocol (SNMP),
32 December 2002.
- 33 IETF RFC 3418 (STD 62), Management Information Base (MIB) for the Simple Network Management
34 Protocol (SNMP), December 2002.
- 35 IETF RFC 3419, Textual Conventions for Transport Addresses, December 2002.
- 36 IETF RFC 4122, A Universally Unique Identifier (UUID) URN Namespace, July 2005.
- 37 IETF RFC 4188, Definitions of Managed Objects for Bridges, September 2005.
- 38 IETF RFC 4291, IP Version 6 Addressing Architecture, February 2006.

- 1 IETF RFC 4318, Definitions of Managed Objects for Bridges with Rapid Spanning Tree Protocol,
2 December 2005.
- 3 IETF RFC 4363, Definitions of Managed Objects for Bridges with Traffic Classes, Multicast Filtering, and
4 Virtual LAN Extensions, January 2006.
- 5 IETF RFC 4789, Simple Network Management Protocol (SNMP) over IEEE 802 Networks, November
6 2006.
- 7 IETF RFC 5120, M-ISIS: Multi Topology (MT) Routing in Intermediate System to Intermediate Systems
8 (IS-ISs), February 2008.
- 9 IETF RFC 5303, Three-Way Handshake for IS-IS Point-to-Point Adjacencies, October 2008.
- 10 IETF RFC 5305, IS-IS Extensions for Traffic Engineering, October 2008.
- 11 IETF RFC 5307, IS-IS Extensions in Support of Generalized Multi-Protocol Label Switching (GMPLS),
12 October 2008.
- 13 IETF RFC 6165, Extensions to IS-IS for Layer-2 Systems, April 2011.
- 14 IETF RFC 6335, Internet Assigned Numbers Authority (IANA) Procedures for the Management of the
15 Service Name and Transport Protocol Port Number Registry, August 2011.
- 16 IETF RFC 7365, Framework for Data Center (DC) Network Virtualization, October 2014.
- 17 IETF RFC 7810, IS-IS Traffic Engineering (TE) Metric Extensions, May 2016.
- 18 IETF RFC 7811, An Algorithm for Computing IP/LDP Fast Reroute Using Maximally Redundant Trees
19 (MRT-FRR) , June 2016.
- 20 IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.
- 21 IETF RFC 8200 (STD0086), Internet Protocol, Version 6 (IPv6) Specification, July 2017.
- 22 IETF RFC 8343, A YANG Data Model for Interface Management, March 2018.
- 23 IETF RFC 8394, Split Network Virtualization Edge (Split-NVE) Control-Plane Requirements, May 2018.
- 24 ISO/IEC 7498-1, Information technology—Open Systems Interconnection—Basic Reference Model: The
25 Basic Model.¹⁵
- 26 ISO/IEC 8802-2, Information technology—Telecommunications and information exchange between
27 systems—Local and metropolitan area networks—Specific requirements—Part 2: Logical link control.
- 28 ISO/IEC 8802-11, Telecommunications and information exchange between systems—Specific requirements
29 for local and metropolitan area networks—Part 11: Wireless LAN medium access control (MAC) and
30 physical layer (PHY) specifications.
- 31 ISO/IEC TR 9577:1999, Information technology—Protocol identification in the network layer.
- 32 ISO/IEC 10589:2002, Information technology—Telecommunications and information exchange between
33 systems—Intermediate System to Intermediate System intra-domain routing information exchange
34 protocol for use in conjunction with the protocol for providing the connectionless-mode network service
35 (ISO 8473).
- 36 ISO/IEC TR 11802-5:1997, Information technology—Telecommunications and information exchange
37 between systems—Local and metropolitan area networks—Technical reports and guidelines—Part 5: Media
38 Access Control (MAC) Bridging of Ethernet V2.0 in Local Area Networks.

¹⁵ ISO/IEC publications are available from the International Organization for Standardization (<https://www.iso.org/>) and the International Electrotechnical Commission (<https://www.iec.ch/>). ISO/IEC publications are also available from the American National Standards Institute (<https://www.ansi.org/>).

- ¹ ITU-T Recommendation X.690 (2002), Information technology—ASN.1 encoding rules: Specification of
² Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules
³ (DER).¹⁶
- ⁴ ITU-T Recommendation G.8013/Y.1731, Operation, administration and maintenance (OAM) functions and
⁵ mechanisms for Ethernet-based networks.
- ⁶ MEF Technical Specification 10.3 (MEF 10.3), Ethernet Services Attributes Phase 3, October 2013.¹⁷

¹⁶ ITU-T publications are available from the International Telecommunications Union (<https://www.itu.int/>).

¹⁷ MEF publications are available from the MEF Forum (<https://www.mef.net/>).

3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.¹⁸

This standard makes use of the following terms defined in IEEE Std 802:

- end station
- station

The following terms are specific to this standard:

3.1 Active Segment: The Infrastructure Segment, either the Working Segment or the Protection Segment, that is currently carrying traffic of Traffic Engineering service instances (TESIs) associated with the Infrastructure Protection Group (IPG).

3.2 Active Service Access Point [SAP] (Active SAP): The SAP, bounding a Maintenance Point (MP), that is on the side of the MP toward the monitored Maintenance Association (MA).

3.3 Adjacency: A neighbor in Intermediate System to Intermediate System (IS-IS) Protocol. From section 3.6 of ISO/IEC 10589:2002.

3.4 Agreement Protocol: A protocol that uses control plane synchronization between adjacent Bridges to guarantee loop prevention, so enabling rapid transition to forwarding after network topology changes.

NOTE—See 13.16 and 13.17.

3.5 Almost Directed Acyclic Graph (ADAG): A directed graph that can be transformed into a Directed Acyclic Graph (DAG) by removing all arcs incoming to the ADAG Root.

3.6 audio/video (AV) traffic: Traffic associated with audio and/or video applications that is sensitive to transmission latency and latency variation, and, for some applications, packet loss.

3.7 Auto Attach BEB (AAB) Bridge component: A Backbone Edge Bridge having at least one I-component that uses the AAP to respond to requests from AADs to couple LANs or VLANs to backbone service instances within a PBBN.

3.8 Auto Attach Device (AAD): An end station, MAC Bridge, C-VLAN Bridge, or S-VLAN Bridge that uses the AAP to request attachment to a backbone service instance within a PBBN.

3.9 Auto Attach Protocol (AAP): A protocol operating over LLDP used by an Auto Attach System (AAS) which provides auto attachment to a PBBN.

3.10 Auto Attach System (AAS): An Auto Attach Device (AAD) or Auto Attach BEB (AAB).

3.11 B-component (B-Comp): A Service Virtual Local Area Network (S-VLAN) component with one or more Customer Backbone Ports (CBPs).

3.12 B-DEI field: A field of a Backbone Virtual Local Area Network (B-VLAN) tag (B-TAG) that identifies the drop eligibility of the frame.

3.13 B-PCP field: A field of a Backbone Virtual Local Area Network (B-VLAN) tag (B-TAG) that indicates the B-VLAN priority of the frame.

¹⁸ *IEEE Standards Dictionary Online* is available at <https://dictionary.ieee.org>.

1 **3.14 Backbone Core Bridge (BCB):** A Service Virtual Local Area Network (S-VLAN) Bridge used within
2 the core of a Provider Backbone Bridged Network (PBBN).

3 **3.15 Backbone Edge Bridge (BEB):** A system that encapsulates customer frames for transmission across a
4 Provider Backbone Bridged Network (PBBN).

5 **3.16 Backbone Media Access Control (B-MAC) address:** An individual Media Access Control (MAC)
6 address associated with a Provider Instance Port (PIP) and used in creating the MAC header of I-tagged
7 frames transmitted across a Provider Backbone Bridged Network (PBBN).

8 **3.17 Backbone Media Access Control (B-MAC) Frame:** A local area network (LAN) frame addressed
9 with B-MAC addresses.

10 **3.18 backbone service instance:** An instance of the Media Access Control (MAC) Service in a Provider
11 Backbone Bridged Network (PBBN), provided between two or more Virtual Instance Ports (VIPs) in
12 Backbone Edge Bridges (BEBs).

13 **3.19 Backbone Service Instance tag (I-TAG):** A tag with an EtherType value allocated for “IEEE 802.1Q
14 Backbone Service Instance Tag EtherType.”

15 **3.20 Backbone Virtual Local Area Network (B-VLAN):** A Virtual Local Area Network (VLAN)
16 identified by a B-VLAN Identifier (B-VID).

17 **3.21 Backbone Virtual Local Area Network [B -VLAN] Identifier (B-VID):** A Virtual Local Area
18 Network (VLAN) Identifier (VID) conveyed in a B-VLAN tag (B-TAG).

19 **3.22 Backbone Virtual Local Area Network [B -VLAN] tag (B-TAG):** An Service Virtual Local Area
20 Network (S-VLAN) tag (S-TAG) used in conjunction with Backbone Media Access Control (B-MAC)
21 addresses.

22 **3.23 Backbone Virtual Local Area Network (B-VLAN) tagged frames:** Frames that contain a B-VLAN
23 tag (B-TAG) immediately following the Source MAC Address field.

24 **3.24 base time:** A time from which future cycles of gate close and open events are calculated.

25 **3.25 Base Virtual Local Area Network [VLAN] Identifier [VID] (Base VID):** The identifier of a VLAN
26 in management operations in both Shortest Path Bridging VID (SPBV) mode and Shortest Path Bridging
27 Media Access Control (MAC) (SPBM) mode.

28 **3.26 bit time:** The duration of one bit as transferred to and from the Media Access Control (MAC).

29 **3.27 Boundary Port:** A Bridge Port attaching a Multiple Spanning Tree (MST) Bridge to a Local Area
30 Network (LAN) that is not in the same region.

31 **3.28 Bridge:** A system that includes Media Access Control (MAC) Bridge or Virtual Local Area Network
32 (VLAN) Bridge component functionality.

33 **3.29 Bridge Local Computation Engine (BLCE):** A computation engine in a Bridge that performs path
34 and routing computations. The BLCE implements, e.g., Shortest Path First (SPF), Constrained Shortest Path
35 First (CSPF), or the Maximally Redundant Trees Algorithm (IETF RFC 7811).

1 **3.30 Bridge Port:** A Service Access Point (SAP) that provides the Enhanced Internal Sublayer Service
2 (EISS) to the Media Access Control (MAC) Relay Entity of a Virtual Local Area Network (VLAN) Bridge
3 component, or that provides the Internal Sublayer Service (ISS) to the MAC Relay Entity of a MAC Bridge,
4 and the interface stack that supports that SAP.

5 NOTE—See 6.1, 8.5, and IEEE Std 802.1AC.

6 **3.31 Bridged Network:** A concatenation of individual IEEE 802 Local Area Networks (LANs)
7 interconnected by Bridges.

8 **3.32 Centralized Network Configuration (CNC):** A centralized component that configures network
9 resources on behalf of TSN applications (users).

10 **3.33 Centralized User Configuration (CUC):** A centralized entity that discovers end stations, retrieves end
11 station capabilities and user requirements, and configures TSN features in end stations.

12 **3.34 C-tagged service interface:** The interface provided by a Provider Edge Bridge (PEB) to allow the
13 attached customer to select between and identify service instances using the C-VID associated with
14 transmitted and received frames.

15 **3.35 candidate Protection Segment:** In the case of M:1 Infrastructure Protection Switching (IPS), an
16 Infrastructure Segment that can assume the role of Protection Segment.

17 **3.36 Common and Internal Spanning Tree (CIST):** The single spanning tree calculated by the Rapid
18 Spanning Tree Algorithm and Protocol (RSTP) and the logical continuation of that connectivity through
19 Multiple Spanning Tree (MST) Bridges and Regions, calculated by the Multiple Spanning Tree Algorithm
20 and Protocol (MSTP) to ensure that all Local Area Networks (LANs) in the Bridged Network are simply and
21 fully connected.

22 **3.37 Common Spanning Tree (CST):** The single spanning tree calculated by the Rapid Spanning Tree
23 Algorithm and Protocol (RSTP) and Multiple Spanning Tree Algorithm and Protocol (MSTP) to connect
24 Multiple Spanning Tree (MST) Regions.

25 **3.38 Configuration Domain:** A set of stations that are under a common configuration and management
26 scheme, and a single administration.

27 **3.39 congestion-aware system:** A Bridge component or end station that is capable of detecting congestion
28 and sending congestion notifications.

29 **3.40 Congestion Controlled Flow (CCF):** A sequence of frames with the same priority, that the
30 transmitting end station treats as belonging to a single flow, using a single Reaction Point (RP) to control the
31 transmission rate of all those frames.

32 **3.41 congesting flow:** A sequence of frames the end-to-end congestion controlled higher layer protocol
33 treats as belonging to a single flow that is experiencing congestion within a congestion isolation aware
34 system.

35 **3.42 congestion isolation aware system:** A Bridge component conforming to the congestion isolation
36 provisions of this standard.

37 **3.43 Congestion Isolation Message (CIM):** A message transmitted by a congestion isolation aware system,
38 conveying congesting flow information used by the upstream peer congestion isolation aware system.

1 **3.44 Congestion Isolation Point (CIP):** A Virtual Local Area Network (VLAN) Bridge or end station Port
2 function that monitors a set of queues for congesting flows, isolates congesting flows to a congesting queue,
3 and can generate Congestion Isolation Messages.

4 **3.45 Congestion Notification Domain (CND):** A connected subset of the end stations and Virtual Local
5 Area Network (VLAN) Bridges in a Virtual Bridged Network that are compatibly configured to support a
6 Congestion Notification Priority Value (CNPV).

7 **3.46 Congestion Notification Message (CNM):** A message transmitted by a Congestion Point (CP) to a
8 Reaction Point (RP), in response to a frame received from that RP, conveying congestion information used
9 by the RP to reduce its transmission rate.

10 **3.47 Congestion Notification Priority Value (CNPV):** A value of the priority parameter that a
11 congestion-aware system uses to support congestion notification.

12 **3.48 Congestion Notification Tag (CN-TAG):** A tag that conveys a Flow Identifier (ID), that a Reaction
13 Point (RP) can add to transmitted Congestion Controlled Flow (CCF) frames, and that a Congestion Point
14 (CP) includes in a Congestion Notification Message (CNM).

15 **3.49 Congestion Point (CP):** A Virtual Local Area Network (VLAN) Bridge or end station Port function
16 that monitors a single queue serving one or more Congestion Notification Priority Values (CNPVs),
17 can generate Congestion Notification Messages (CNMs), and can remove Congestion Notification Tags
18 (CN-TAGs).

19 NOTE—See 31.1.1.

20 **3.50 congruent:** (A) An adjective to describe two or more paths that coincide at all points when
21 superimposed. (B) An adjective to describe a set of trees specifying paths that coincide at all points when
22 superimposed, for all Virtual Local Area Networks (VLANs). *See also:* **reverse path congruent, unicast**
23 **multicast congruent, downstream congruent, and symmetric.**

24 **3.51 Connectivity Fault Management (CFM):** Management capabilities for detecting, verifying, and
25 isolating connectivity failures in Virtual Bridged Networks.

26 **3.52 constrained routing (CR):** Shortest path routing on a topology pruned to only contain the links
27 meeting a given constraint.

28 **3.53 Constrained Shortest Path First (CSPF):** The phrase for the Dijkstra Shortest Path First Algorithm
29 when it is run after pruning the links not meeting a given constraint.

30 **3.54 constrained tree:** A tree meeting a certain constraint, e.g., providing a minimal available bandwidth.

31 **3.55 Continuity Check Message (CCM):** A multicast Connectivity Fault Management (CFM) protocol
32 data unit (PDU) transmitted periodically by a Maintenance Association (MA) Endpoint (MEP) in order to
33 ensure continuity over the MA to which the transmitting MEP belongs.

34 **3.56 current Protection Segment:** In the case of M:1 Infrastructure Protection Switching (IPS), the
35 candidate Protection Segment that is currently assuming the role of Protection Segment.

36 **3.57 customer:** The purchaser of a service instance from a service provider.

37 NOTE—Certain terms in this standard, including customer and service provider, reflect common business relationships
38 that exist among the users of equipment implemented in conformance with this standard. The use of these terms does not
39 imply or impose any requirement on such business relationships. A customer is often a business.

1 **3.58 Customer Backbone Port (CBP):** A Backbone Edge Bridge (BEB) Port that can receive and transmit
2 I-tagged frames for multiple customers and can assign Backbone Virtual Local Area Network (B-VLAN)
3 Identifiers (B-VIDs) and translate Backbone Service Instance Identifier (I-SID) on the basis of the received
4 I-SID.

5 **3.59 Customer Bridge:** A Media Access Control (MAC) Bridge or a Customer Virtual Local Area Network
6 (C-VLAN) Bridge.

7 **3.60 Customer Bridged Network (CBN):** A network of Customer Bridges.

8 **3.61 Customer Edge Port (CEP):** A Customer Virtual Local Area Network (C-VLAN) component Port on
9 a Provider Edge Bridge (PEB) that is connected to customer-owned equipment and receives and transmits
10 frames for a single customer.

11 **3.62 customer equipment:** The physical embodiment of one or more customer systems.

12 **3.63 Customer Media Access Control (C-MAC) address:** A Media Access Control (MAC) address within
13 a Customer Bridged Network (CBN) or Provider Bridged Network (PBN). C-MAC addresses are carried
14 within an Backbone Service Instance tag (I-TAG) in a Provider Backbone Bridged Network (PBBN).

15 **3.64 Customer Network Port (CNP):** A Service Virtual Local Area Network (S-VLAN) component Port
16 on a Provider Bridge or within a Provider Edge Bridge (PEB) that receives and transmits frames for a single
17 customer.

18 **3.65 customer system:** A system attached to a Provider Bridged Network (PBN) but not intended by the
19 service provider to be under the control of the service provider.

20 **3.66 Customer Virtual Local Area Network (C-VLAN):** A Virtual Local Area Network (VLAN)
21 identified by a C-VLAN Identifier (C-VID).

22 **3.67 Customer Virtual Local Area Network (C-VLAN) Bridge:** A system comprising a single C-VLAN
23 component.

24 **3.68 Customer Virtual Local Area Network (C-VLAN) component:** A Virtual Local Area Network
25 (VLAN) Bridge component with each Port supported by an instance of the Enhanced Internal Sublayer
26 Service (EISS) that can recognize, insert, and remove C-VLAN tags (C-TAGs).

27 **3.69 Customer Virtual Local Area Network [C-VLAN] Identifier (C-VID):** A Virtual Local Area
28 Network (VLAN) Identifier (VID) conveyed in a C-VLAN tag (C-TAG).

29 **3.70 Customer Virtual Local Area Network [C-VLAN] tag (C-TAG):** A Virtual Local Area Network
30 (VLAN) tag with a Tag Protocol Identification value allocated for “802.1Q Tag Protocol EtherType.”

31 **3.71 cut-bridge:** A Bridge whose removal partitions the network.

32 **3.72 cut-link:** A link whose removal partitions the network.

33 **3.73 Data Center Bridging (DCB):** A set of protocols and capabilities for use in a data center environment.

34 **3.74 data-driven and data-dependent connectivity fault management (DDCFM):** Connectivity Fault
35 Management (CFM) capabilities that facilitate detecting and isolating data-driven and data-dependent faults
36 (DDFs) in Virtual Bridged Networks.

1 **3.75 data-driven and data-dependent fault (DDF):** A fault that is sensitive to, or caused by, particular data
2 patterns in frames transmitted by a service user.

3 **3.76 Decapsulator Responder (DR):** An entity for decapsulating Send Frame Messages (SFM) and
4 sending the decapsulated frames toward destinations specified by their own destination_address field.

5 **3.77 Designated Multiple Stream Registration Protocol [MSRP] Node (DMN):** A single station on a
6 shared medium (e.g., IEEE 802.11 or a Coordinated Shared Network (CSN)) that controls MSRP access for
7 all other stations on that shared medium.

8 **3.78 Directed Acyclic Graph (DAG):** A directed graph containing no directed cycle.

9 **3.79 Domain Service Access Point (DoSAP):** A member of a set of Service Access Points (SAPs) at which
10 a Maintenance Domain is capable of offering connectivity to systems outside the Maintenance Domain.
11 Each DoSAP provides access to an instance either of the Enhanced Internal Sublayer Service (EISS) or of
12 the Internal Sublayer Service (ISS).

13 **3.80 Down Maintenance Association [MA] Endpoint [MEP] (Down MEP):** A MEP residing in a Bridge
14 that receives Connectivity Fault Management (CFM) protocol data units (PDUs) from, and transmits them
15 toward, the direction of the Local Area Network (LAN).

16 **3.81 Down Maintenance Point [MP] (Down MP):** A Maintenance Association (MA) Endpoint (MEP) or a
17 Maintenance Domain Intermediate Point (MIP) Half Function (MHF) residing in a Bridge that receives
18 Connectivity Fault Management (CFM) protocol data units (PDUs) from, and transmits them toward, the
19 direction of the Local Area Network (LAN).

20 **3.82 downlink relay port (DRP):** A port of an edge relay (ER) that is capable of supporting at least one
21 Virtual Station Interface (VSI). For NVO3 a port of a tNVE that is capable of supporting at least one NVO3
22 Tenant Station Interface (TSI) (see IETF RFC 8394).

23 **3.83 downstream congruent:** A phrase to describe the path for a given frame relayed by a first Bridge when
24 that path passes through a second Bridge and is thereafter the same as the path for any other frame (assigned
25 to the same Virtual Local Area Network (VLAN) and destined to the same station) relayed by the second
26 Bridge.

27 **3.84 Edge Control Protocol (ECP):** A protocol that provides reliable delivery of control service data units
28 (SDUs).

29 **3.85 edge relay (ER):** A Bridge supporting the transfer of frames between one or more downlink relay ports
30 (DRPs) and one uplink relay port (URP).

31 **3.86 Edge Virtual Bridging (EVB):** The set of functions supporting Virtual Station Interfaces (VSIs) in
32 Bridges and attached end stations.

33 **3.87 Edge Virtual Bridging (EVB) Bridge:** A Customer Virtual Local Area Network (C-VLAN) Bridge
34 that supports the Virtual Station Interface (VSI) Discovery and Configuration Protocol (VDP).

35 **3.88 Edge Virtual Bridging (EVB) station:** An end station containing one or more edge relays (ERs).

36 **3.89 Enhanced Internal Sublayer Service [EISS] Service Access Point [SAP] (EISS-SAP):** An instance
37 of the EISS.

38 NOTE—See 6.8.

1 **3.90 Equal Cost Multiple Paths (ECMP):** The use of multiple possible next hop choices for frames within
2 a single service in Shortest Path Bridging Media Access Control (MAC) (SPBM) mode networks.

3 **3.91 Equal Cost Multiple Paths [ECMP] path Maintenance Association [MA] (ECMP path MA):** An
4 MA that monitors a set, which can be complete, of equal connectivity paths between two specific endpoints
5 as constructed by ECMP.

6 **3.92 Equal Cost Tree (ECT):** One of the shortest path trees from the root Bridge selected from alternatives
7 with the same path costs.

8 **3.93 Equal Cost Tree (ECT) Algorithm:** A Shortest Path Tree (SPT) algorithm yielding a consistent result
9 for a given root Bridge when run by all Bridges within an SPT Region.

10 **3.94 Ethernet:** A term that is used to refer either to the IEEE 802.3 media access method or to the
11 IEEE 802.3 frame format.

12 NOTE—In many places, “Ethernet” is also used as part of a reserved term with its own specific meaning.

13 **3.95 Ethernet Switched Path (ESP):** A provisioned traffic engineered unidirectional connectivity path
14 among two or more Customer Backbone Ports (CBPs) that extends over a Provider Backbone Bridged
15 Network (PBBN). The path is identified by a 3-tuple <ESP-DA, ESP-SA, ESP-VID>, where ESP-DA and
16 ESP-SA are Media Access Control (MAC) addresses and ESP-VID is a Virtual Local Area Network
17 (VLAN) Identifier (VID) allocated to Traffic Engineering Multiple Spanning Tree Instance Identifier
18 (TE-MSTID).

19 NOTE—The use of “Ethernet” in this definition does not imply that an ESP is limited to use in conjunction with
20 IEEE Std 802.3™.

21 **3.96 Ethernet Switched Path [ESP] Virtual Local Area Network [VLAN] Identifier [VID] (ESP-VID):**
22 A VLAN Identifier (VID) associated with a special (Traffic Engineering) value of the Multiple Spanning
23 Tree Instance Identifier (MSTID) in the Multiple Spanning Tree (MST) Configuration Table, the
24 TE-MSTID, indicating that the VID is under the control of an external agent responsible for setting up ESPs
25 and that learning is disabled and forwarding is enabled.

26 **3.97 Expedited traffic:** Traffic that requires preferential treatment as a consequence of jitter, latency, or
27 throughput constraints or as a consequence of management policy.

28 **3.98 Explicit Tree (ET):** An explicitly defined tree, which is specified by its Edge Bridges and the paths
29 among the Edge Bridges. If only the Edge Bridges are specified but the paths are not, then it is a loose
30 explicit tree. If the paths are also specified, then it is a strict explicit tree.

31 **3.99 Explicit Tree Database (ETDB):** A database storing explicit trees.

32 **3.100 Fault Alarm:** An out-of-band signal, which can be a Simple Network Management Protocol (SNMP)
33 Notification, that notifies a system administrator of a connectivity failure.

34 **3.101 flow filtering tag (F-TAG):** A tag with a Tag Protocol Identification value allocated for “IEEE
35 802.1Q Flow Filtering Tag EtherType.”

36 **3.102 flow hash:** An arbitrary value that serves to distinguish flows that may be subject to different
37 treatment in a bridged network, for example, flows subject to independent treatment by Equal Cost Multiple
38 Paths (ECMP).

39 **3.103 Flow Identifier [ID] (Flow ID):** An ID assigned by a congestion-aware end station, unique within the
40 scope of one or more of the source Media Access Control (MAC) addresses used by that system to transmit
41 Congestion Controlled Flow (CCF) frames.

1 **3.104 forced switch (FS):** An administrative command to force the backbone service instances assigned to a
2 Traffic Engineering (TE) protection group to be carried by this group's protection entity.

3 **3.105 forward path test (FPT):** A test to determine whether a data frame that matches the specified filter
4 condition can reach a specific location within a network.

5 **3.106 Frame:** A unit of data transmission on an IEEE 802 Local Area Network (LAN) that conveys a Media
6 Access Control (MAC) Protocol Data Unit (MPDU).

7 **3.107 Frame relay:** Forwarding of frames between the Ports of a Bridge.

8 **3.108 gate-close event:** An event that occurs when the transmission gate associated with a queue transitions
9 from the Open state to the Closed state, disconnecting the transmission selection function of the forwarding
10 process from the queue and preventing it from selecting frames from that queue.

11 **3.109 gate-open event:** An event that occurs when the transmission gate associated with a queue transitions
12 from the Closed state to the Open state, connecting the transmission selection function of the forwarding
13 process to a queue and allowing it to select frames from that queue.

14 **3.110 gating cycle:** The period of time over which the sequence of operations in a gate control list repeats.

15 **3.111 Generalized Almost Directed Acyclic Graph (GADAG):** A directed graph that has only Almost
16 Directed Acyclic Graphs (ADAGs) as all of its topology blocks.

17 **3.112 Group:** An association of

- 18 a) A group Media Access Control (MAC) address
- 19 b) A set of properties that define membership characteristics
- 20 c) A set of properties that define the forwarding/filtering behavior of a Bridge with respect to frames
21 destined for members of that group MAC address
22 with a set of end stations that all wish to receive information destined for that group MAC address.
- 23 NOTE—An example of the information that Group members might wish to receive is a multicast video data
24 stream.

25 **3.113 GroupID:** A service instance identifier used in Virtual Station Interface (VSI) Discovery and
26 Configuration Protocol (VDP).

27 **3.114 I-component (I-Comp):** A Service Virtual Local Area Network (S-VLAN) component with one or
28 more Provider Instance Ports (PIPs).

29 **3.115 I-DEI field:** A field of the Backbone Service Instance tag (I-TAG) that indicates the drop eligibility of
30 a frame in a backbone service instance.

31 **3.116 I-PCP field:** A field of the Backbone Service Instance tag (I-TAG) that indicates the priority of a
32 frame in a backbone service instance.

33 **3.117 I-SID field:** A field of the Backbone Service Instance tag (I-TAG) that identifies the backbone service
34 instance of a frame.

35 **3.118 I-tagged frame:** A frame that contains a Backbone Service Instance tag (I-TAG) immediately
36 following the Source MAC Address field.

1 **3.119 IEEE 802 Local Area Network [LAN] (IEEE 802 LAN):** LAN technologies that provide a Media
2 Access Control (MAC) Service equivalent to the MAC Service defined in IEEE Std 802.1AC. IEEE 802
3 LANs include IEEE 802.3 (Ethernet) and IEEE 802.11 (Wireless) LANs.

4 NOTE—The term “Local Area Network” and the abbreviation LAN are used exclusively to refer to an individual LAN
5 specified by a MAC technology, without the inclusion of Bridges. This precise use of terminology within this
6 specification allows a Bridged Network to be distinguished from an individual LAN that has been bridged to other LANs
7 in the network (a bridged LAN). In more general usage, such precise terminology is not required, as it is an explicit goal
8 of IEEE Std 802.1Q that Bridges are transparent to the users of the MAC Service.

9 **3.120 Independent Virtual Local Area Network [VLAN] Learning (IVL):** Configuration and operation
10 of the Learning Process and the Filtering Database (FDB) such that, for a given set of VLAN Identifiers
11 (VIDs), if a given individual Media Access Control (MAC) Address is learned in association with one VID,
12 that learned information is not used in forwarding decisions taken for that address relative to any other VID
13 in the given set.

14 NOTE—In a Bridge that supports only IVL operation, the “given set of VIDs” is the set of all VIDs.

15 **3.121 Independent Virtual Local Area Network [VLAN] Learning [IVL] Bridge (IVL Bridge):** A
16 Bridge that supports only IVL.

17 **3.122 Infrastructure Protection Group (IPG):** In the case of 1:1 Infrastructure Protection Switching (IPS),
18 a set comprising a Working Segment and a Protection Segment or, in the case of M:1 IPS, a set comprising a
19 Working Segment and one or more Protection Segments.

20 **3.123 Infrastructure Protection Switching (IPS):** The switching of a selected group of Traffic
21 Engineering service instances (TESIs) from a Working Segment to a Protection Segment or from a
22 Protection Segment to a Working Segment due to failure of connectivity, restoration of connectivity, or
23 administrative command.

24 **3.124 Infrastructure Segment:** A sequence of Provider Network Ports (PNPs) and the intervening Local
25 Area Networks (LANs) and Bridge relay entities.

26 **3.125 Intermediate Service Access Point (ISAP):** A Service Access Point (SAP), interior to a Maintenance
27 Domain, through which frames can pass in transit from Domain Service Access Point (DoSAP) to DoSAP.

28 **3.126 Intermediate System to Intermediate System (IS-IS) Path Control and Reservation (ISIS-PCR):**
29 IS-IS with the Path Control and Reservation extensions specified by Clause 45 of IEEE Std 802.1Q-2022.

30 **3.127 Internal Spanning Tree (IST):** The connectivity provided by the Common and Internal Spanning
31 Tree (CIST) within a Multiple Spanning Tree (MST) Region.

32 **3.128 Internal Sublayer Service (ISS) Service Access Point (ISS-SAP):** An instance of the ISS.

33 NOTE—See IEEE Std 802.1AC.

34 **3.129 latency:** The delay experienced by a frame in the course of its propagation between two points in a
35 network, measured from the time that a known reference point in the frame passes the first point to the time
36 that the reference point in the frame passes the second point.

37 NOTE—Latency is sometimes referred to as *frame delay*.

38 **3.130 Layer Management Interface (LMI):** An interface used for carrying management controls,
39 counters, status parameters, or event indications that are not communicated to an entity’s peers.

40 NOTE—This is not one of the Service Access Points (SAPs) defined in this standard.

1 **3.131 Legacy region:** A set of Local Area Networks (LANs) connected such that there is physical
2 connectivity between any pair of LANs using only Media Access Control (MAC) Bridges.

3 NOTE—If, in a Bridged Network containing both MAC Bridges and Virtual Local Area Network (VLAN) Bridges, all
4 VLAN Bridges were to be removed, the result would be one or more Bridged Networks, each with its own distinct
5 spanning tree. Each of those networks is a legacy region.

6 **3.132 Link status notification:** Notification of the status of a Local Area Network (LAN) by transmitting a
7 frame conveying information about the MAC_Operational parameter associated with the LAN.

8 **3.133 Linktrace Message (LTM):** A Connectivity Fault Management (CFM) protocol data unit (PDU)
9 initiated by a Maintenance Association (MA) Endpoint (MEP) to trace a path to a target Media Access
10 Control (MAC) address, forwarded from Maintenance Domain (MD) Intermediate Point (MIP) to MIP, up to
11 the point at which the LTM reaches its target, a MEP, or can no longer be forwarded.

12 **3.134 Linktrace Output Multiplexer (LOM):** A Connectivity Fault Management (CFM) shim that enables
13 the Linktrace Responder to forward Linktrace Messages (LTMs) out of a specific Bridge Port without
14 passing the LTMs through the Frame filtering entity.

15 **3.135 Linktrace Reply (LTR):** A unicast Connectivity Fault Management (CFM) protocol data unit (PDU)
16 sent by an Maintenance Point (MP) to a Maintenance Association (MA) Endpoint (MEP), in response to
17 receiving a Linktrace Message (LTM) from that MEP.

18 **3.136 Listener:** The end station that is the destination, receiver, or consumer of a stream.

19 **3.137 Loop-Free Alternate (LFA):** A loop-free backup path for local repair after a failure event.

20 **3.138 Loopback Message (LBM):** A unicast Connectivity Fault Management (CFM) protocol data unit
21 (PDU) transmitted by a Maintenance Association (MA) Endpoint (MEP), addressed to a specific
22 Maintenance Point (MP), in the expectation of receiving a Loopback Reply (LBR).

23 **3.139 Loopback Reply (LBR):** A unicast Connectivity Fault Management (CFM) protocol data unit (PDU)
24 transmitted by an Maintenance Point (MP) to a Maintenance Association (MA) Endpoint (MEP), in
25 response to a Loopback Message (LBM) received from that MEP.

26 **3.140 Maintenance Association Identifier (MAID):** An identifier for a Maintenance Association (MA),
27 unique over the domain that Connectivity Fault Management (CFM) is to protect against the accidental
28 concatenation of service instances. The MAID has two parts: the Maintenance Domain Name and the Short
29 MA Name.

30 **3.141 Maintenance Association (MA):** A set of MA Endpoints (MEPs), each configured with the same
31 Maintenance Association Identifier (MAID) and Maintenance Domain (MD) Level, established to verify the
32 integrity of a single service instance. Equivalently, an MA is the closure of the Maintenance Entities among
33 a set of MEPs so configured.

34 **3.142 Maintenance Association [MA] Endpoint [MEP] Identifier [ID] (MEPID):** A small integer,
35 unique over a given MA, identifying a specific MEP.

36 **3.143 Maintenance Association [MA] Endpoint (MEP):** An actively managed Connectivity Fault
37 Management (CFM) entity, associated with a specific Domain Service Access Point (DoSAP) of a service
38 instance, which can generate and receive CFM protocol data units (PDUs) and track any responses. It is an
39 endpoint of a single MA and is an endpoint of a separate Maintenance Entity for each of the other MEPs in
40 the same MA.

1 **3.144 Maintenance Association [MA] Endpoint [MEP] Continuity Check Message [CCM] Database**
2 **(MEP CCM Database):** A database, maintained by every MEP, that maintains received information about
3 other MEPs in the MA.

4 **3.145 Maintenance Domain:** The network or the part of the network for which faults in connectivity can be
5 managed. The boundary of a Maintenance Domain is defined by a set of Domain Service Access Points
6 (DoSAPs), each of which can become a point of connectivity to a service instance.

7 **3.146 Maintenance Domain Intermediate Point (MIP):** A Connectivity Fault Management (CFM) entity
8 consisting of two MIP Half Functions (MHFs).

9 **3.147 Maintenance Domain Intermediate Point [MIP] Continuity Check Message [CCM] Database**
10 **(MIP CCM Database):** A database, maintained optionally by a MIP or Maintenance Association (MA)
11 Endpoint (MEP), that maintains received information about the MEPs in the Maintenance Domain.

12 **3.148 Maintenance Domain Intermediate Point [MIP] Half Function (MHF):** A Connectivity Fault
13 Management (CFM) entity, associated with a single Maintenance Domain, and thus with a single
14 Maintenance Domain Level (MD Level) and a set of Virtual Local Area Network (VLAN) Identifiers
15 (VIDs), that can generate CFM protocol data units (PDUs), but only in response to received CFM PDUs.

16 **3.149 Maintenance Domain Level (MD Level):** A small integer in a field in a Connectivity Fault
17 Management (CFM) protocol data unit (PDU) that is used, along with the Virtual Local Area Network
18 (VLAN) Identifier (VID) in the VLAN tag, to identify to which Maintenance Domain among those
19 associated with the CFM frame's VID, and thus to which Maintenance Association (MA), a CFM PDU
20 belongs.

21 **3.150 Maintenance Domain Name:** The identifier, unique over the domain that Connectivity Fault
22 Management (CFM) is to protect against accidental concatenation of service instances, of a particular
23 Maintenance Domain.

24 **3.151 Maintenance Entity:** A point-to-point relationship between two Maintenance Association (MA)
25 Endpoints (MEPs) within a single MA.

26 **3.152 Maintenance Point (MP):** One of either a Maintenance Association (MA) Endpoint (MEP) or a
27 Maintenance Domain Intermediate Point (MIP).

28 **3.153 Maintenance Point (MP) Level Demultiplexer:** The component of an MP that separates
29 Connectivity Fault Management (CFM) protocol data units (PDUs) at different Maintenance Domain Levels
30 (MD Levels).

31 **3.154 Maintenance Point (MP) Multiplexer:** A component of an MP that merges frames from more than
32 one input Service Access Point (SAP), sending them to a single output SAP.

33 **3.155 Maintenance Point (MP) OpCode Demultiplexer:** A component of an MP that identifies and
34 separates Connectivity Fault Management (CFM) protocol data units (PDUs) with different values of the
35 OpCode field.

36 **3.156 Maintenance Point (MP) Type Demultiplexer:** A component of an MP that identifies and separates
37 Connectivity Fault Management (CFM) protocol data units (PDUs) based on the Length/Type field.

38 **3.157 manual switch:** An administrative command to move the backbone service instances assigned to a
39 Traffic Engineering (TE) protection group to a specific TE service instance (TESI) associated with this TE
40 protection group in absence of signal failures on both of the entities in the group.

1 **3.158 Maximally Redundant Trees (MRTs):** A pair of trees with a common MRT Root where the path
2 from any leaf Bridge to the MRT Root along the first tree (MRT-Blue) and the path from the same leaf
3 Bridge along the second tree (MRT-Red) share the minimum number of Bridges and the minimum number
4 of links.

5 **3.159 Media Access Control (MAC) Bridge:** A Bridge that does not recognize Virtual Local Area Network
6 (VLAN) tagged frames.

7 **3.160 Media Access Control (MAC) Bridge component:** The media access method-independent
8 functionality supporting an instance of the Internal Sublayer Service (ISS) at each Bridge Port and the
9 functionality that relays frames between Bridge Ports.

10 **3.161 Media Access Control (MAC) Bridged Network:** A concatenation of individual IEEE 802 Local
11 Area Networks (LANs) interconnected by MAC Bridges.

12 **3.162 Media Access Control (MAC) status notification:** A layer management interaction with an
13 underlying MAC Service to change the status of the MAC_Operational parameter in a Bridge that is a peer
14 user of that service.

15 **3.163 Media Access Control (MAC) status propagation:** The process of communicating a
16 MAC_Operational parameter value through one or more Two-Part MAC Relays (TPMRs), using link status
17 notification, MAC status notification, or both.

18 **3.164 MRT-Blue:** One of the two Maximally Redundant Trees (MRTs), specifically, the increasing MRT
19 where links in the Generalized Almost Directed Acyclic Graph (GADAG) are taken in the direction from a
20 lower topologically ordered Bridge to a higher one.

21 **3.165 MRT-Red:** One of the two Maximally Redundant Trees (MRTs), specifically, the decreasing MRT
22 where links in the Generalized Almost Directed Acyclic Graph (GADAG) are taken in the direction from a
23 higher topologically ordered Bridge to a lower one.

24 **3.166 MRT Root:** The common root of the two Maximally Redundant Trees (MRTs), i.e., of MRT-Blue and
25 MRT-Red.

26 **3.167 Multi-Topology (MT):** One of a set of independent topologies that the Intermediate System to
27 Intermediate System (IS-IS) Protocol supports within an IS-IS domain.

28 **3.168 Multi-Topology Identifier (MTID):** The identifier of an Intermediate System to Intermediate System
29 (IS-IS) Protocol Multi-Topology (MT).

30 **3.169 Multicast:**

- 31 a) A group Media Access Control (MAC) address (noun).
- 32 b) To transmit a frame using a group MAC address as the destination address (verb).
- 33 c) Containing a group MAC address as the destination address (adjective).

34 **3.170 Multiple Spanning Tree Instance (MSTI):** One of a number of spanning trees calculated by the
35 Multiple Spanning Tree Algorithm and Protocol (MSTP) within a Multiple Spanning Tree (MST) Region.

36 **3.171 Multiple Spanning Tree (MST) Bridge:** A Bridge capable of supporting the Common Spanning Tree
37 (CST), and one or more Multiple Spanning Tree Instances (MSTIs), and of selectively mapping frames
38 classified in any given Virtual Local Area Network (VLAN) to the CST or a given MSTI.

3.172 Multiple Spanning Tree [MST] Configuration Identifier (MCID): A name for, revision level, and a summary of a given allocation of Virtual Local Area Networks (VLANs) to spanning trees.

NOTE—Each MST Bridge uses a single MST Configuration Table and Configuration Identifier.

3.173 Multiple Spanning Tree (MST) Configuration Table: A configurable table that allocates each and every possible Virtual Local Area Network (VLAN) Identifier (VID) to the Common Spanning Tree (CST) or a specific Multiple Spanning Tree Instance (MSTI).

3.174 Multiple Spanning Tree (MST) Region: One or more MST Bridges with the same MST Configuration Identifiers (MCIDs), interconnected by and including Local Area Networks (LANs) for which one of those Bridges is the Designated Bridge for the Common and Internal Spanning Tree (CIST) and that have no Bridges attached that cannot receive and transmit Rapid Spanning Tree (RST) Bridge Protocol Data Units (BPDUs).

3.175 Multiple Stream Registration Protocol (MSRP): A protocol designed to provide quality of service (QoS) for streams in bridged networks by reserving resources within each Bridge along the streams' paths.

3.176 Network Virtualization Edge (NVE): A term as defined in IETF RFC 7365.

3.177 Network Virtualization Overlays over Layer 3 (NVO3): A framework conforming to IETF RFC 7365.

3.178 nNVE: Network-side VNE as defined in IETF RFC 8394.

3.179 Operations, Administration, and Maintenance (OAM): A group of network management functions that provide network fault indication, performance information, and data and diagnosis functions.

NOTE—Examples are ATM OAM ITU-T I.610 (02/1999) [B51]¹⁹ and IEEE 802.3™ Clause 57 OAM.

3.180 operator: (A) A provider of a single network of Provider Bridges or a single Layer 2 or Layer 3 backbone network to the service provider. (B) An enumeration of a state machine event. (C) A symbol denoting a mathematical operation.

NOTE 1—(A) An operator can, in fact, be identical to, or a part of the same organization as, the service provider, but for the purposes of IEEE Std 802.1Q, the operator and service provider are presumed to be separate organizations.

NOTE 2—(A) Certain terms in this standard, including “customer,” “service provider,” and “operator,” reflect common business relationships that often exist among organizations and individuals that use equipment implemented in accordance with this standard. Terms such as “customer VID” or “operator MD Level” are no more than convenient labels for entities defined here; their use does not imply any requirement upon the business relationships among vendors or users of devices compliant with this standard.

3.181 Owner: A user of a system (and in particular, a Bridge) who has full access to the System Group of the SNMPv2-MIB (IETF RFC 3418).

3.182 packet: A protocol data unit, comprising data, addressing, and protocol identification information, sent by an instance of an identified class of protocol entities and transmitted in one or more frames (e.g., an IPv6 packet).

3.183 Passive Service Access Point [SAP] (Passive SAP): The SAP, bounding an Maintenance Point (MP), that is on the side of the MP away from the monitored Maintenance Association (MA).

3.184 Path Computation Element (PCE): An entity that is capable of computing a path through a network based on a representation of the network's topology (obtained by undefined means external to the PCE). A PCE is a higher layer entity in a Bridge or an end station.

¹⁹ Numbers in brackets correspond to the numbers in the bibliography in Annex X.

1 **3.185 Path Control Agent (PCA):** The agent that is part of the Intermediate System to Intermediate System
2 (IS-IS) Domain and thus can perform IS-IS operations on behalf of a Path Computation Element (PCE), e.g.,
3 maintain Link State Database (LSDB) and send Link State protocol data units (PDUs) (LSPs).

4 **3.186 Paused state:** A state of a queue in which the transmission selection entity does not select frames
5 from the queue.

6 **3.187 Point of Local Repair (PLR):** A Bridge that locally redirects the data traffic to a backup path, e.g., to
7 a loop-free alternate path after a failure event.

8 **3.188 point-to-multipoint Ethernet Switched Path [ESP] (point-to-multipoint ESP):** An ESP among
9 one root Customer Backbone Port (CBP) and n leaf CBPs.

10 **3.189 point-to-multipoint Traffic Engineering service instance [TESI] (point-to-multipoint TESI):** A
11 TESI supported by a set of Ethernet Switched Paths (ESPs) that comprises one point-to-multipoint ESP from
12 a root to n leaves plus a point-to-point ESP from each of the leaves to the root.

13 **3.190 point-to-point Ethernet Switched Path [ESP] (point-to-point ESP):** An ESP between two
14 Customer Backbone Ports (CBPs). The ESP-DA and the ESP-SA in the ESP's 3-tuple identifier are the
15 individual Media Access Control (MAC) addresses of the two CBPs.

16 **3.191 point-to-point Traffic Engineering service instance [TESI] (point-to-point TESI):** A TESI
17 supported by two point-to-point Ethernet Switched Paths (ESPs) where the ESPs' endpoints have the same
18 Customer Backbone Port (CBP) Media Access Control (MAC) addresses.

19 **3.192 policing:** A process of monitoring network traffic and deliberately dropping frames that are in excess
20 of previously defined criteria.

21 **3.193 Port:** A Service Access Point (SAP) and the interface stack supporting that SAP.

22 **3.194 Port State:** One of a number of possible combinations of forwarding and learning variables, used by a
23 spanning tree protocol or Shortest Path Tree (SPT) protocol for each tree for each Bridge Port, to enforce the
24 active topology of that tree.

25 **3.195 Port-based service interface:** The interface provided by a Provider Bridge that associates all
26 customer frames with a single service instance.

27 **3.196 Port-mapping Service Virtual Local Area Network [S-VLAN] component (Primary S-VLAN
28 component):** An S-VLAN component with one externally accessible port and one or more additional ports,
29 each connected to another Bridge component via an internal Local Area Network (LAN), and for which
30 each registered S-VLAN Identifier's (S-VID's) member set includes the externally accessible port and
31 exactly one other port.

32 NOTE—See 6.14.

33 **3.197 preemption:** The suspension of the transmission of a preemptable frame to allow one or more express
34 frames to be transmitted before transmission of the preemptable frame is resumed.

35 **3.198 Primary Virtual Local Area Network [VLAN] Identifier [VID] (Primary VID):** The VID, among
36 a list of VIDs associated with a service instance, on which all Connectivity Fault Management (CFM)
37 protocol data units (PDUs) generated by Maintenance Points (MPs) except for forwarded Linktrace
38 Messages (LTMs) are to be transmitted.

39 **3.199 Priority-tagged frame:** A tagged frame whose tag header carries priority information but carries no
40 Virtual Local Area Network (VLAN) identification information.

1 **3.200 Protection Segment:** An Infrastructure Segment associated with an Infrastructure Protection Group
2 (IPG) that can carry Traffic Engineering service instance (TESI) traffic when the Working Segment
3 associated with that IPG has failed or when directed by administrative command.

4 **3.201 protocol group database:** Specifies a group of protocols by assigning a unique protocol group
5 identifier to all protocols of the same group.

6 **3.202 protocol group identifier:** Designates a group of protocols that are associated together when
7 assigning a Virtual Local Area Network (VLAN) Identifier (VID) to a frame.

8 **3.203 protocol template:** A tuple of values that specify a data-link encapsulation format and an
9 identification of the protocol layer above the data-link layer.

10 **3.204 Provider Access Port (PAP):** A Port-mapping Service Virtual Local Area Network (S-VLAN)
11 component port within a Provider Edge Bridge (PEB) that receives and transmits frames for a single
12 customer [single S-VLAN Identifier (S-VID)].

13 **3.205 Provider Backbone Bridge (PBB):** A Backbone Core Bridge (BCB) or a Backbone Edge Bridge
14 (BEB).

15 **3.206 Provider Backbone Bridge Traffic Engineering (PBB-TE) Region:** A contiguous set of combined
16 I type and B type Backbone Edge Bridges (IB-BEBs) and Backbone Core Bridges (BCBs) that implement
17 traffic engineering capabilities.

18 **3.207 Provider Backbone Bridged Network (PBBN):** A network using Backbone Edge Bridges (BEBs)
19 and Backbone Core Bridges (BCBs) to interconnect Provider Bridged Networks (PBNs) and other networks.

20 **3.208 Provider Bridge:** A Provider Edge Bridge (PEB) or a Service Virtual Local Area Network (S-VLAN)
21 Bridge.

22 **3.209 Provider Bridged Network (PBN):** A network using Provider Bridges to offer customer protocol
23 transparent connectivity.

24 **3.210 Provider Edge Bridge (PEB):** A system comprising one or more Service Virtual Local Area Network
25 (S-VLAN) components and one or more Customer Virtual Local Area Network (C-VLAN) components.

26 **3.211 Provider Edge Port (PEP):** A Customer Virtual Local Area Network (C-VLAN) component Port
27 within a Provider Edge Bridge (PEB) that connects to a Customer Network Port (CNP) and receives and
28 transmits frames for a single customer.

29 **3.212 Provider Instance Port (PIP):** The set of Virtual Instance Ports (VIPs) that are supported by a single
30 instance of the Internal Sublayer Service (ISS).

31 **3.213 Provider Network Port (PNP):** A Service Virtual Local Area Network (S-VLAN) component Port
32 on a Provider Bridge that can transmit and receive frames for multiple customers.

33 **3.214 Rapid Spanning Tree (RST) Bridge:** A Bridge that uses the Rapid Spanning Tree Algorithm and
34 Protocol (RSTP) to calculate a single spanning tree and is only capable of forwarding frames on that tree.

35 **3.215 Reaction Point (RP):** An end station port function that controls the transmission rate of frames for
36 one or more Congestion Controlled Flows (CCFs).

37 NOTE—See 31.2.2.2.

1 **3.216 redundant trees:** A pair of trees with a common root where the paths from any leaf Bridge to the root
2 along the first tree and the second tree are disjoint.

3 **3.217 Reflected Frame Message (RFM):** A Connectivity Fault Management (CFM) protocol data unit
4 (PDU) transmitted by a Reflection Responder (RR) in order to perform forward path test (FPT).

5 **3.218 Reflection Responder (RR):** An entity that encapsulates and reflects selected data frames to a target
6 location and allows a copy of each reflected frame to continue to its destination without encapsulation.

7 **3.219 Reflective relay:** A mode of operation of the active topology enforcement function in which a
8 received frame on a port that supports reflective operation can be forwarded on the same port on which it
9 was received.

10 **3.220 Remote Customer Access Port (RCAP):** A Port-mapping Service Virtual Local Area Network
11 (S-VLAN) component port on a Provider Edge Bridge (PEB) that is intended to be directly connected to
12 another Provider Bridged Network (PBN) under a different administration, and receives and transmits
13 frames for one or more customers.

14 **3.221 return path test (RPT):** A test to determine if a data frame can be sent without error from a specific
15 location within a network to a station specified by the destination_address field of the data frame.

16 **3.222 reverse path congruent:** A phrase to describe two paths where a unicast frame traverses exactly the
17 same Bridges and Local Area Networks (LANs) along the path from the source to the destination, but in the
18 reverse order as a frame transmitted from that destination to the source.

19 **3.223 rooted-multipoint:** A service instance in which each Service Access Point (SAP) is either a Root or a
20 Leaf, and a Data.Request at a Leaf does not result in a Data.Indication at any other Leaf.

21 NOTE—See 6.2.1 and F.1.3.2.

22 **3.224 S-channel:** A point-to-point Service Virtual Local Area Network (S-VLAN) established between a
23 Port-mapping S-VLAN component in an Edge Virtual Bridging (EVB) Bridge and a Port-mapping S-VLAN
24 component in an EVB station.

25 **3.225 S-channel Access Port (CAP):** The Port that terminates an S-channel.

26 **3.226 S-channel Discovery and Configuration Protocol (CDCP):** A protocol that is used to configure
27 Service Virtual Local Area Network (S-VLAN) components to create S-channels.

28 **3.227 S-tagged service interface:** The interface provided by a Provider Bridge to allow the attached
29 customer to select between and identify service instances using the Service Virtual Local Area Network
30 (S-VLAN) Identifier (S-VID) associated with transmitted and received frames.

31 **3.228 Segment Endpoint Bridge (SEB):** A Provider Backbone Bridge (PBB) at the endpoint of an
32 Infrastructure Segment.

33 **3.229 Segment Endpoint Port (SEP):** A Provider Network Port (PNP) at the endpoint of an Infrastructure
34 Segment.

35 **3.230 Segment Identifier (SEG-ID):** A pair of Segment Monitoring Path Identifiers (SMP-IDs) specifying
36 the identity of an Infrastructure Segment.

37 **3.231 Segment Intermediate Bridge (SIB):** A Provider Backbone Bridge (PBB) having an intermediate
38 position within an Infrastructure Segment.

1 **3.232 Segment Intermediate Port (SIP):** A Provider Network Port (PNP) having an intermediate position
2 within an Infrastructure Segment.

3 **3.233 Segment Monitoring Path (SMP):** A unidirectional path carrying Connectivity Fault Management
4 (CFM) traffic associated with the monitoring of an Infrastructure Segment.

5 **3.234 Segment Monitoring Path Identifier (SMP-ID):** A 3-tuple <SMP-DA, SMP-SA, SMP-VID> where
6 the SMP-DA is the Media Access Control (MAC) address of the destination Provider Network Port (PNP)
7 of the Segment Monitoring Path (SMP), the SMP-SA is the MAC address of the origin PNP of the SMP, and
8 SMP-VID is a Virtual Local Area Network (VLAN) Identifier (VID) allocated to the Traffic Engineering
9 Multiple Spanning Tree Instance Identifier (TE-MSTID) and associated with the SMP.

10 NOTE—See 8.9.

11 **3.235 Send Frame Message (SFM):** A Connectivity Fault Management (CFM) protocol data unit (PDU)
12 destined to a Decapsulator Responder (DR) in order to perform return path test (RPT).

13 **3.236 Send Frame Message (SFM) Originator:** A function on a Bridge interface, an end station, or a test
14 equipment to send SFM encapsulated data frames.

15 **3.237 Service Access Point (SAP):** The point at which a service is offered.

16 NOTE—In this standard, an instance of either the Internal Sublayer Service (ISS) (IEEE Std 802.1AC) or the Enhanced
17 Internal Sublayer Service (EISS) (6.8). This term is used in place of the awkward “ISS Service Access Point” or “EISS
18 Service Access Point,” where no loss of clarity is introduced.

19 **3.238 service frame:** A Local Area Network (LAN) frame exchanged between a provider and a customer.

20 **3.239 service instance:** A set of Service Access Points (SAPs) such that a Data.Request primitive presented
21 to one SAP can result in a Data.Indication primitive occurring at one or more of the other SAPs in that set.

22 NOTE—In general, a service provider can use a number of technologies to support a service instance; within this
23 standard, a service instance is supported by a single Service Virtual Local Area Network (S-VLAN).

24 **3.240 service provider:** An organization that contracts to provide one or more service instances to a
25 customer.

26 **3.241 Service Virtual Local Area Network (S-VLAN):** A Virtual Local Area Network (VLAN) identified
27 by an S-VLAN Identifier (S-VID).

28 **3.242 Service Virtual Local Area Network (S-VLAN) Bridge:** A system comprising a single S-VLAN
29 component.

30 **3.243 Service Virtual Local Area Network (S-VLAN) component:** A Virtual Local Area Network
31 (VLAN) Bridge component with each Port supported by an instance of the Enhanced Internal Sublayer
32 Service (EISS) that can recognize, insert, and remove S-VLAN tags (S-TAGs).

33 **3.244 Service Virtual Local Area Network [S-VLAN] Identifier (S-VID):** A Virtual Local Area Network
34 (VLAN) Identifier (VID) conveyed in an S-VLAN tag (S-TAG).

35 **3.245 Service Virtual Local Area Network [S-VLAN] tag (S-TAG):** A Virtual Local Area Network
36 (VLAN) tag with a Tag Protocol Identification value allocated for “802.1Q Service Tag EtherType.”

37 **3.246 Shared Risk Link Group (SRLG):** A set of links that share a resource whose failure affects each
38 link.

1 **3.247 Shared Virtual Local Area Network [VLAN] Learning (SVL):** Configuration and operation of the
2 Learning Process and the Filtering Database (FDB) such that, for a given set of VLANs, if an individual
3 Media Access Control (MAC) Address is learned in one VLAN, that learned information is used in
4 forwarding decisions taken for that address relative to all other VLANs in the given set.

5 NOTE—In a Bridge that supports only SVL operation, the “given set of VLANs” is the set of all VLANs.

6 **3.248 Shared Virtual Local Area Network [VLAN] Learning [SVL] Bridge (SVL Bridge):** A type of
7 Bridge that supports only SVL.

8 **3.249 Shared Virtual Local Area Network [VLAN] Learning [SVL]/Independent VLAN Learning**
9 **[IVL] Bridge (SVL/IVL Bridge):** A type of Bridge that simultaneously supports both SVL and IVL.

10 **3.250 shim:** A protocol entity that uses the same service as it provides.

11 NOTE—Within this standard, shims make use of the Internal Sublayer Service (ISS) or the Enhanced Internal Sublayer
12 Service (EISS).

13 **3.251 Short Maintenance Association [MA] Name (Short MA Name):** The part of the Maintenance
14 Association Identifier (MAID) that is unique within the Maintenance Domain and is appended to the
15 Maintenance Domain Name to form the MAID.

16 **3.252 Shortest Path Bridging (SPB):** The method of bridging using Shortest Path Trees (SPTs) for
17 forwarding from each Bridge. SPB is used for references that are applicable to all modes of SPB: Shortest
18 Path Bridging Media Access Control (MAC) (SPBM) mode and Shortest Path Bridging Virtual Local Area
19 Network (VLAN) Identifier (VID) (SPBV) mode.

20 **3.253 Shortest Path Bridging (SPB) System Identifier:** A unique 6-byte value for each Intermediate
21 System to Intermediate System Protocol for Shortest Path Bridging (ISIS-SPB) capable Bridge and is
22 equivalent to the Intermediate System to Intermediate System (IS-IS) Protocol System Identifier (ID).

23 NOTE—See 8.13.8.

24 **3.254 Shortest Path Bridging Media Access Control [MAC] [SPBM] mode (SPBM mode):** The Shortest
25 Path Bridging (SPB) method where all Shortest Path Trees (SPTs) are rooted at a Backbone Edge Bridge
26 (BEB). In frames with a group destination address, the SPT is identified by a destination MAC address
27 derived from the SPSrcID; in frames with an individual address, the SPT is identified by a MAC address
28 of the source BEB. This method does not use learning.

29 **3.255 Shortest Path Bridging Media Access Control [MAC] [SPBM] mode Maintenance Association**
30 **[MA] (SPBM MA):** An MA for monitoring and diagnosing connectivity associated with a Virtual Local
31 Area Network (VLAN) Identifier (VID) assigned to the SPBM Multiple Spanning Tree Instance Identifier
32 (MSTID). There are three types of SPBM MA—SPBM VID MAs, SPBM path MAs, and SPBM group
33 MAs.

34 NOTE—See 27.18.1.

35 **3.256 Shortest Path Bridging Media Access Control [MAC] [SPBM] mode Virtual Local Area**
36 **Network [VLAN] Identifier [VID] (SPBM VID):** A VID associated with the SPBM-MSTID value of the
37 Multiple Spanning Tree Instance Identifier (MSTID) in the Multiple Spanning Tree (MST) Configuration
38 Table.

39 **3.257 Shortest Path Bridging Virtual Local Area Network [VLAN] Identifier [VID] [SPBV] mode**
40 **(SPBV mode):** The Shortest Path Bridging (SPB) method where Shortest Path VLAN Identifiers (SPVIDs)
41 are used to identify the Shortest Path Trees (SPTs) for individual and group addresses. This method is
42 learning capable.

1 **3.258 Shortest Path First (SPF):** Abbreviation for the Dijkstra Shortest Path First Algorithm that is used in
2 link state protocols such as Intermediate System to Intermediate System (IS-IS) Protocol.

3 **3.259 Shortest Path Source Identifier (SPSourceID):** A 20-bit identifier of a Bridge that is unique within
4 a Shortest Path Tree (SPT) Domain. SPSourceID is used to auto-create SPT Domain significant group
5 Media Access Control (MAC) addresses utilizing the local address bit in Shortest Path Bridging MAC
6 (SPBM) mode.

7 NOTE—See 27.10.

8 **3.260 Shortest Path Tree (SPT) Bridge:** A Virtual Local Area Network (VLAN) Bridge capable of
9 operating Shortest Path Bridging (SPB) protocols.

10 **3.261 Shortest Path Tree (SPT) Domain:** A set of SPT Bridges that are interconnected by Local Area
11 Networks (LANs) and have the same configured Intermediate System to Intermediate System (IS-IS)
12 Protocol area address. An SPT Domain contains one or more SPT Regions, each comprising some or all of
13 the Bridges in the Domain.

14 **3.262 Shortest Path Tree (SPT) Region:** A set of Local Area Networks (LANs) and SPT Bridges
15 interconnected via Ports on those SPT Bridges, where each LAN's Common and Internal Spanning Tree
16 (CIST) Designated Bridge is an SPT Bridge, and each Port is either the Designated Port on one of the LANs
17 or else a non-Designated Port of an SPT Bridge that is connected to one of the LANs, whose Multiple
18 Spanning Tree (MST) Configuration Identifier (MCID) matches exactly the MCID or the Auxiliary MCID
19 of the Designated Bridge of that LAN.

20 NOTE—See 27.4.1.

21 **3.263 Shortest Path Tree (SPT) Set:** A set of SPTs determined by one Equal Cost Tree (ECT) Algorithm
22 and capable of supporting one or more Virtual Local Area Networks (VLANs) within an SPT Region. The
23 set is the union of all the SPTs from all sources to all other Bridges using that single ECT Algorithm. A
24 single SPT Set is created for each ECT Algorithm. Active SPT Sets are identified by their {ECT Algorithm,
25 FID} tuples.

26 NOTE—See 8.9.4 and 13.6.

27 **3.264 Shortest Path Virtual Local Area Network [VLAN] Identifier (SPVID):** The identifier of both the
28 VLAN and the Shortest Path Tree (SPT) that is used for the transmission of a tagged frame in Shortest Path
29 Bridging VLAN Identifier (VID) (SPBV) mode.

30 **3.265 Single Spanning Tree (SST) Bridge:** A Bridge capable of supporting only a single spanning tree.

31 **3.266 spanning tree:** A simply and fully connected active topology formed from the arbitrary physical
32 topology of connected Bridged Network components by relaying frames through selected Bridge Ports and
33 not through others. The protocol parameters and states used and exchanged to facilitate the calculation of
34 that active topology and to control the Media Access Control (MAC) relay function.

35 **3.267 Station-facing Bridge Port (SBP):** A Bridge Port that supports the Edge Virtual Bridging (EVB)
36 status parameters with an EVBMode parameter value of “EVB Bridge”. For NVO3 a port of an nNVE that
37 supports VDP.

38 NOTE—See 40.4.

39 **3.268 Stream:** A unidirectional flow of data (e.g., audio and/or video) from a Talker to one or more
40 Listeners.

1 **3.269 Stream Reservation Protocol (SRP) domain boundary port:** A Port of a station that is a member of
2 an SRP domain for a given stream reservation (SR) class and that is connected via an individual Local Area
3 Network (LAN) to a Port of a station that either is within a different SRP domain for that SR class or is not
4 part of any SRP domain.

5 NOTE—The term *SRP domain* is defined in 35.1.4.

6 **3.270 Stream Reservation Protocol (SRP) domain core port:** A Port of a station that is a member of an
7 SRP domain for a given stream reservation (SR) class and that is connected, via an individual Local Area
8 Network (LAN) that is part of the active topology, to a Port of a station that is a member of the same SRP
9 domain for that SR class.

10 **3.271 stream reservation (SR) class:** A traffic class whose bandwidth can be reserved for time-sensitive
11 streams using the Stream Reservation Protocol (SRP). A priority value is associated with each SR class. SR
12 classes are denoted by consecutive letters of the alphabet, starting with A and continuing for up to seven
13 classes.

14 **3.272 StreamID:** A 64-bit field that uniquely identifies a stream.

15 **3.273 symmetric:** An adjective to describe paths, trees, or sets of trees. Symmetric algorithms are used to
16 create a set of trees that are *reverse path congruent* with respect to sources and destinations. Symmetric
17 algorithms are used to create a set of trees that are *unicast multicast congruent* with respect to a given
18 source. *See also:* **reverse path congruent** and **unicast multicast congruent**.

19 **3.274 T-component:** A Two-Port Media Access Control (MAC) Relay (TPMR) component with one
20 Provider Instance Port (PIP).

21 **3.275 Tag header:** A header that allows priority information, and optionally, Virtual Local Area Network
22 (VLAN) identification information, to be associated with a frame.

23 **3.276 Tagged frame:** A frame that contains a tag header immediately following the Source MAC Address
24 field of the frame.

25 **3.277 Talker:** The end station that is the source or producer of a stream.

26 **3.278 time-aware:** An adjective to describe use of time that is synchronized with other stations using a
27 protocol (e.g., IEEE Std 802.1AS).

28 **3.279 time-sensitive stream:** A stream of data frames that are required to be delivered with a bounded
29 latency.

30 **3.280 tNVE:** Terminal-side NVE as defined in IETF RFC 8394.

31 **3.281 topology block:** A maximally two-connected (induced) subgraph, a cut-link with the two Bridges at
32 either end, or an isolated Bridge.

33 **3.282 traffic class:** A classification used to expedite transmission of frames generated by critical or
34 time-sensitive services.

35 **3.283 Traffic Engineering Database (TED):** A database storing the traffic engineering information
36 propagated by a link state protocol, e.g., Intermediate System to Intermediate System (IS-IS).

1 **3.284 Traffic Engineering service instance (TESI):** An instance of the Media Access Control (MAC)
2 Service supported by a set of Ethernet Switched Paths (ESPs) and identified by Traffic Engineering service
3 instance Identifier (TE-SID), forming a bidirectional service. A TESI is point-to-point or
4 point-to-multipoint.

5 **3.285 Traffic Engineering service instance Identifier (TE-SID):** An identifier of the Traffic Engineering
6 service instance (TESI) that corresponds to a series of 3-tuples <ESP-DA, ESP-SA, ESP-VID>, each one
7 identifying one of the TESI's Ethernet Switched Paths (ESPs).

8 NOTE—The TE-SID is not used as a tag parameter.

9 **3.286 Traffic Engineering (TE):** The process that controls the placement of traffic demands in a network in
10 order to optimize the resource utilization and to ensure that the quality of service (QoS) objectives for each
11 defined class of service are met.

12 **3.287 Traffic Engineering (TE) protection group:** A group of two point-to-point Traffic Engineering
13 service instances (TESIs) between a pair of Customer Backbone Ports (CBPs), which carries an assigned set
14 of backbone service instances and continues to carry these backbone service instances if any one of the
15 TESIs in the group is failed or disabled.

16 **3.288 traffic specification (TSpec):** A specification that characterizes the bandwidth that a stream can
17 consume.

18 **3.289 transmission gate:** A gate that connects or disconnects the transmission selection function of the
19 forwarding process from the queue, allowing or preventing it from selecting frames from that queue. The
20 gate has two states, Open and Closed.

21 **3.290 TSN features:** The protocols and mechanisms that constitute the set of tools available for building a
22 time-sensitive network.

23 **3.291 Two-connected:** A topology that has no cut-bridges. This is a topology that requires at least two
24 Bridges to be removed before it is partitioned.

25 **3.292 Two-Port Media Access Control [MAC] Relay (TPMR):** A Bridge that has exactly two externally
26 accessible Ports and supports a subset of the functionality of a MAC Bridge as specified in 5.15 of
27 IEEE Std 802.1Q-2022.

28 **3.293 Two-Port Media Access Control [MAC] Relay (TPMR) component:** A MAC Bridge component
29 with two Bridge Ports.

30 **3.294 unicast:**

- 31 a) An individual Media Access Control (MAC) address (noun).
- 32 b) To transmit a frame using an individual MAC address as the destination address (verb).
- 33 c) Containing an individual MAC address as the destination address (adjective).

34 **3.295 unicast multicast congruent:** A phrase to describe the path taken by a unicast frame to a given
35 destination when that path is the same as the path taken by any multicast frame [assigned to the same Virtual
36 Local Area Network (VLAN)] from that source to the same destination.

37 **3.296 Untagged frame:** A frame that does not contain a tag header immediately following the Source MAC
38 Address field of the frame.

1 **3.297 Up Maintenance Association [MA] Endpoint [MEP] (Up MEP):** A MEP residing in a Bridge that
2 transmits Connectivity Fault Management (CFM) protocol data units (PDUs) toward, and receives them
3 from, the direction of the Media Access Control (MAC) Relay Entity.

4 **3.298 Up Maintenance Point [MP] (Up MP):** A Maintenance Association (MA) Endpoint (MEP) or a
5 Maintenance Domain Intermediate Point (MIP) Half Function (MHF) residing in a Bridge that transmits
6 Connectivity Fault Management (CFM) protocol data units (PDUs) toward, and receives them from, the
7 direction of the Media Access Control (MAC) Relay Entity.

8 **3.299 Uplink Access Port (UAP):** A Port on a Port-mapping Service Virtual Local Area Network
9 (S-VLAN) component that connects an Edge Virtual Bridging (EVB) Bridge with an EVB station.

10 **3.300 uplink relay port (URP):** A port of an edge relay (ER) that supports the Edge Virtual Bridging (EVB)
11 status parameters with an EVBMode parameter value of “EVB station.”
12 NOTE—See 40.4.

13 **3.301 Virtual Bridged Network:** A concatenation of individual IEEE 802 Local Area Networks (LANs)
14 interconnected by Bridges, including Virtual Local Area Network (VLAN) Bridges.

15 **3.302 virtual edge bridge (VEB):** An edge relay (ER) that requires reflective relay service to be disabled on
16 the Station-facing Bridge Port (SBP) of the attached Bridge.

17 **3.303 virtual edge port aggregator (VEPA):** An edge relay (ER) that always forwards frames through its
18 uplink relay port (URP) and that can make use of reflective relay service provided by the Station-facing
19 Bridge Port (SBP) of the attached Bridge.

20 **3.304 Virtual Instance Port (VIP):** A Bridge Port on an I-component or a T-component in a Backbone
21 Edge Bridge (BEB) that provides access to a single backbone service instance.

22 **3.305 Virtual Local Area Network (VLAN):** The closure of a set of Media Access Control (MAC) Service
23 Access Points (MSAPs) such that a data request in one MSAP in the set is expected to result in a data
24 indication in another MSAP in the set.

25 NOTE—See also 6.2.

26 **3.306 Virtual Local Area Network (VLAN) Bridge:** A system comprising a single VLAN Bridge
27 component implemented in accordance with Clause 5 of IEEE Std 802.1Q-2022.

28 **3.307 Virtual Local Area Network (VLAN) Bridge component:** The media access method-independent
29 functionality supporting an instance of the Enhanced Internal Sublayer Service (EISS) at each Bridge Port
30 that can recognize, insert, and remove VLAN tags, and the functionality that relays frames between Bridge
31 Ports.

32 **3.308 Virtual Local Area Network (VLAN) Bridged Network:** A Virtual Bridged Network.

33 **3.309 Virtual Local Area Network (VLAN) tagged frame:** A tagged frame whose tag header carries both
34 VLAN identification and priority information.

35 **3.310 Virtual Network Instance (VNI):** A term as defined in IETF RFC 7365.

36 **3.311 Virtual Network Instance Identifier (VNI ID):** A 3-octet identifier for a VNI.

37 **3.312 virtual station:** An end station instantiated within an Edge Virtual Bridging (EVB) station.

1 **3.313 Virtual Station Interface (VSI):** An interface to a virtual station that is attached to a downlink relay
2 port (DRP) of an edge relay (ER). For NVO3 a VSI is equivalent to an NVO3 Tenant Station Interface (TSI)
3 (see IETF RFC 8394).

4 **3.314 Virtual Station Interface [VSI] Discovery and Configuration Protocol (VDP):** A protocol that
5 supports the association of a VSI with a Bridge Port.

6 **3.315 Working Segment:** The Infrastructure Segment, among the Infrastructure Segments associated with
7 an Infrastructure Protection Group (IPG), on which Traffic Engineering service instance (TESI) traffic is
8 carried when no connectivity failure of that segment has been detected and no administrative command is
9 active.

10

4. Abbreviations

The following abbreviations are used in this standard:

AAB	Auto Attach BEB
AAD	Auto Attach Device
AAP	Auto Attach Protocol
AAS	Auto Attach System
ACK	acknowledgment
ADAG	Almost Directed Acyclic Graph
AP	Agreement Protocol
ATS	Asynchronous Traffic Shaping
AV	audio/video
AVB	audio/video bridging
B-BEB	B type Backbone Edge Bridge
B-Comp	B-component
B-DA	Backbone Destination MAC address
B-DEI	B-VLAN Drop Eligible Indicator
B-MAC	Backbone Media Access Control
B-PCP	B-VLAN Priority Code Point
B-SA	Backbone Source MAC address
B-TAG	B-VLAN tag
B-VID	Backbone VLAN Identifier
B-VLAN	Backbone Virtual Local Area Network
BCB	Backbone Core Bridge
BEB	Backbone Edge Bridge
BLCE	Bridge Local Computation Engine
BNF	Backus-Naur Form
BPDU	Bridge Protocol Data Unit
BSI	backbone service instance
CBS	committed burst size
C-DA	Customer Destination MAC address
C-MAC	Customer Media Access Control
C-SA	Customer Source MAC address
C-TAG	C-VLAN tag
C-VID	Customer VLAN Identifier
C-VLAN	Customer Virtual Local Area Network
CAP	S-channel Access Port
CBN	Customer Bridged Network
CBP	Customer Backbone Port
CCF	Congestion Controlled Flow
CCM	Continuity Check Message
CCP	Current Congestion Point
CDCP	S-channel Discovery and Configuration Protocol
CE	customer equipment
CEP	Customer Edge Port
CFM	Connectivity Fault Management
CI	Congestion Isolation
CID	Company ID ²⁰ . (See also OUI)
CIM	Congestion Isolation Message

1	CIP	Congestion Isolation Point
3	CIST	Common and Internal Spanning Tree
5	CIST-MSTID	Common and Internal Spanning Tree Multiple Spanning Tree Instance Identifier
7	CN	congestion notification
9	CNC	Centralized Network Configuration
12	CN-TAG	Congestion Notification tag
14	CND	Congestion Notification Domain
16	CNM	Congestion Notification Message
18	CNP	Customer Network Port
20	CNPV	Congestion Notification Priority Value
22	CP	Congestion Point
24	CPID	Congestion Point Identifier
26	CQF	cyclic queuing and forwarding
28	CR	constrained routing
30	CRC	Cyclic Redundancy Check
32	CSN	Coordinated Shared Network
34	CSPF	Constrained Shortest Path First
36	CST	Common Spanning Tree
38	CUC	Centralized User Configuration
40	DAG	Directed Acyclic Graph
42	DCB	Data Center Bridging
44	DCBX	Data Center Bridging eXchange protocol
46	DCCP	Datagram Congestion Control Protocol
48	DCN	data center network
50	DDCFM	data-driven and data-dependent connectivity fault management
52	DDF	data-driven and data-dependent fault
54	DEI	Drop Eligible Indicator
56	DLSDU	Data Link Service Data Unit
58	DMN	Designated MSRP Node
60	DoSAP	Domain Service Access Point
62	DR	Decapsulator Responder
64	DRP	downlink relay port
66	ECMP	Equal Cost Multiple Paths
68	ECP	Edge Control Protocol
70	ECPDU	Edge Control Protocol Data Unit
72	ECT	Equal Cost Tree
74	EISS	Enhanced Internal Sublayer Service (6.8)
76	eMAC	express Media Access Control
78	ER	edge relay
80	ESP	Ethernet Switched Path
82	ET	Explicit Tree
84	ETDB	Explicit Tree Database
86	ETS	Enhanced Transmission Selection
88	EUI-48	48-bit Extended Unique Identifier ²¹
90	EVB	Edge Virtual Bridging
92	F-TAG	flow filtering tag
94	F_b	Quantized Feedback

²⁰ See <https://standards.ieee.org/develop/regauth/tut/eui.pdf>.

²¹ A tutorial on the structure and use of EUI-48 identifiers can be found at <https://standards.ieee.org/develop/regauth/tut/eui.pdf>.

1	FCS	Frame Check Sequence
3	FDB	Filtering Database
5	FID	Filtering Identifier (8.8.8, 8.9.3)
7	FPT	forward path test
9	FQTSS	Forwarding and Queuing Enhancements for time-sensitive streams
11	FS	Forced Switch
13	GADAG	Generalized Almost Directed Acyclic Graph
15	I-BEB	I type Backbone Edge Bridge
17	I-Comp	I-component
19	I-DEI	Backbone Service Instance Drop Eligible Indicator
21	I-PCP	Backbone Service Instance Priority Code Point
23	I-TAG	Backbone Service Instance tag
25	I-SID	Backbone Service Instance Identifier
27	IB-BEB	combined I type and B type Backbone Edge Bridge
29	ID	identifier
31	IP	Internet Protocol
33	IPG	Infrastructure Protection Group
35	IPS	Infrastructure Protection Switching
37	IPV	internal priority value specification
39	IPv6	Internet Protocol version 6
41	ISAP	Intermediate Service Access Point
43	ISIS-PCR	Intermediate System to Intermediate System with Path Control and Reservation extensions
45	ISIS-SPB	Intermediate System to Intermediate System Protocol for Shortest Path Bridging
47	ISS	Internal Sublayer Service (IEEE Std 802.1AC)
49	IST	Internal Spanning Tree
51	ITB-BEB	any valid combination of multi-component Backbone Edge Bridge, e.g., IB-BEB
53	IVL	Independent VLAN Learning (3.120)
55	LACP	Link Aggregation Control Protocol
57	LAG	Link Aggregation Group
59	LAN	Local Area Network (IEEE Std 802)
61	LBM	Loopback Message
63	LBR	Loopback Reply
65	LFA	Loop-Free Alternate
67	LLC	Logical Link Control (ISO/IEC 8802-2)
69	LLDP	Link Layer Discovery Protocol
71	LMI	Layer Management Interface
73	Local-SID	Local Service Instance Identifier
75	LOM	Linktrace Output Multiplexer
77	LoP	Lockout of Protection
79	LSB	least significant bit
81	LSDB	Link State Database
83	LSP	Link State PDU
85	LT	Loose Tree
87	LTM	Linktrace Message
89	LTR	Linktrace Reply
91	LTS	Loose Tree Set
93	MA	Maintenance Association
95	MAC	Medium Access Control (IEEE Std 802)
97	MAD	MRP Attribute Declaration
99	MAID	Maintenance Association Identifier

1	MAP	MRP Attribute Propagation
3	MCID	MST Configuration Identifier
5	MD Level	Maintenance Domain Level
7	MEP	MA Endpoint
9	MEPID	MA Endpoint Identifier
11	MHF	MIP Half Function
13	MIB	Management Information Base (Clause 17)
15	MIP	Maintenance domain Intermediate Point
17	MIRP	Multiple I-SID Registration Protocol
19	MIRPDU	Multiple I-SID Registration Protocol Data Unit
21	MMRP	Multiple MAC Registration Protocol
23	MMRPDU	Multiple MAC Registration Protocol Data Unit
25	MO	Managed Object
27	MoCA	Multimedia over Coax Alliance
29	MP	Maintenance Point
31	MPDU	MAC Protocol Data Unit
33	MRP	Multiple Registration Protocol
35	MRPDU	Multiple Registration Protocol Data Unit
37	MRTs	Maximally Redundant Trees
39	MS	MAC Service
41	MSAP	MAC Service Access Point
43	MSB	most significant bit
45	MSDU	MAC Service Data Unit (IEEE Std 802.1AC)
47	MSP	MAC Status Protocol or MAC status propagation
49	MSPDU	MAC Status Protocol Data Unit
51	MSPE	MAC Status Propagation Entity
53	MSRP	Multiple Stream Registration Protocol
55	MSRPDU	Multiple Stream Registration Protocol Data Unit
57	MSS	MAC Status Shim
59	MST	Multiple Spanning Tree
61	MST BPDU	Multiple Spanning Tree Bridge Protocol Data Unit
63	MSTI	Multiple Spanning Tree Instance
65	MSTID	Multiple Spanning Tree Instance Identifier
67	MSTP	Multiple Spanning Tree Algorithm and Protocol
69	MT	Multi-Topology (Note that “MT state” is used in Clause 10 for the “Empty state.”)
71	MTID	Multi-Topology Identifier
73	MVRP	Multiple VLAN Registration Protocol
75	MVRPDU	Multiple VLAN Registration Protocol Data Unit
77	NETCONF	Network Configuration Protocol
79	NLPID	Network Layer Protocol Identifier
81	NVE	Network Virtualization Edge
83	NVO3	Network Virtualization Overlays over Layer 3
85	OAM	Operations, Administration, and Maintenance
87	OUI	organizationally unique Identifier
89	PAE	Port Access Entity
91	PAP	Provider Access Port
93	PATHID	Path Identifier
95	PB	Provider Bridge
97	PBB	Provider Backbone Bridge
99	PBB-TE	Provider Backbone Bridge Traffic Engineering

1	PBBN	Provider Backbone Bridged Network
3	PBN	Provider Bridged Network
5	PCA	Path Control Agent
7	PCE	Path Computation Element
9	PCP	Priority Code Point
11	PCR	Path Control and Reservation
13	PDU	protocol data unit
15	PFC	Priority-based Flow Control
17	PICS	Protocol Implementation Conformance Statement (Annex A, Annex B)
19	PID	Protocol Identifier
21	PIP	Provider Instance Port
23	PLR	Point of Local Repair
25	pMAC	preemptable Media Access Control
27	PNP	Provider Network Port
29	PPVID	port and protocol VLAN Identifier
31	PQoS	Parameterized Quality of Service
33	PSFP	Per-Stream Filtering and Policing
35	PTP	IEEE 1588 precision time protocol
37	PVID	port VLAN Identifier
39	QCN	Quantized Congestion Notification protocol
41	QoS	quality of service
43	RCAP	Remote Customer Access Port
45	RCSI	Remote Customer Service Interface
47	RFM	Reflected Frame Message
49	RP	Reaction Point
51	RPT	return path test
53	RR	Reflection Responder
55	RST	Rapid Spanning Tree
57	RST BPDU	Rapid Spanning Tree Bridge Protocol Data Unit
59	RSTP	Rapid Spanning Tree Algorithm and Protocol
61	RSVP	Resource Reservation Protocol
63	S-TAG	S-VLAN tag
65	S-VID	Service VLAN Identifier
67	S-VLAN	Service Virtual Local Area Network
69	SAP	Service Access Point
71	SBP	Station-facing Bridge Port
73	SCID	S-channel Identifier
75	SCTP	Stream Control Transmission Protocol
77	SDU	service data unit
79	SEB	Segment Endpoint Bridge
81	SEG-ID	Segment Identifier
83	SEP	Segment Endpoint Port
85	SF	Signal Fail
87	SFM	Send Frame Message
89	SIB	Segment Intermediate Bridge
91	SIP	Segment Intermediate Port
93	SMP	Segment Monitoring Path
95	SMP-ID	Segment Monitoring Path Identifier
97	SNAP	Subnetwork Access Protocol
99	SNM	Status Notification state machine

1	SNMP	Simple Network Management Protocol
3	SPB	Shortest Path Bridging
5	SPBM	Shortest Path Bridging MAC
7	SPBV	Shortest Path Bridging VID
9	SPF	Shortest Path First
11	SPT	Shortest Path Tree
13	SPVID	Shortest Path VLAN Identifier
15	SR	stream reservation
17	SR_PVID	Stream Reservation Port VLAN Identifier
19	SRLG	Shared Risk Link Group
21	SRP	Stream Reservation Protocol
23	SST	Single Spanning Tree
25	ST	Strict Tree
27	ST BPDU	Spanning Tree Bridge Protocol Data Unit
29	STM	Status Transition state machine
31	STP	Spanning Tree Algorithm and Protocol
33	SVL	Shared VLAN Learning (3.247)
35	TAI	Temps Atomic International—International Atomic Time
37	TC	textual convention
39	TCI	Tag Control Information (9.3)
41	TCP	Transmission Control Protocol
43	TE	Traffic Engineering
45	TE-MSTID	Traffic Engineering Multiple Spanning Tree Instance Identifier
47	TE-SID	Traffic Engineering service instance Identifier
49	TED	Traffic Engineering Database
51	TESI	Traffic Engineering service instance
53	TLV	Type, Length, Value
55	TPID	Tag Protocol Identifier (9.3)
57	TPMR	Two-Port MAC Relay
59	TSpec	traffic specification
61	TSN	Time-Sensitive Networking
63	TTL	time-to-live
65	UAP	Uplink Access Port
67	UDP	User Datagram Protocol
69	ULP	upper layer protocol
71	ULPDU	upper layer protocol data unit
73	UNI	User Network Interface
75	URP	uplink relay port
77	UUID	Universally Unique Identifier
79	VDP	VSI Discovery and Configuration Protocol
81	VEB	virtual edge bridge
83	VEPA	virtual edge port aggregator
85	VID	VLAN Identifier (7.2, 9.3)
87	VIP	Virtual Instance Port
89	VLAN	Virtual Local Area Network
91	VNI	Virtual Network Instance
93	VNI ID	Virtual Network Instance Identifier
95	VS	Virtual Station Interface
97	VSID	VSI Instance Identifier
99	VTID	VSI Type Identifier

1 YANG Yet Another Next Generation²²

3

²²YANG is best viewed as a name, not an acronym.

5. Conformance

This clause specifies the mandatory and optional capabilities provided by conformant implementations of this standard. An implementation can:

- a) Compose all or part of the functionality of a system.
- b) Provide, as specified by this standard, one or more instances of the MAC Service to other functional entities whose specification is outside the scope of this standard.
- c) Provide, as specified by this standard, one or more instances of the MAC Internal Sublayer Service (ISS) to other implementations or instances of the same implementation that conform to this standard.

Accordingly, and as detailed in 5.4, this clause specifies conformance requirements for common systems and for functional components within systems, possibly connected to other system components with interfaces that are not otherwise accessible.

5.1 Requirements terminology

For consistency with existing IEEE and IEEE 802.1™ standards, requirements placed upon conformant implementations of this standard are expressed using the following terminology:

- a) **Shall** is used for mandatory requirements.
- b) **May** is used to describe implementation or administrative choices (“may” means “is permitted to,” and hence, “may” and “may not” mean precisely the same thing).
- c) **Should** is used for recommended choices (the behaviors described by “should” and “should not” are both permissible but not equally desirable choices).

NOTE—The usage in 17.7 follows the IETF model.

The Protocol Implementation Conformance Statement (PICS) proformas (see Annex A for Bridges and Annex B for end station implementations) reflect the occurrences of the words “shall,” “may,” and “should” within the standard.

The standard avoids needless repetition and apparent duplication of its formal requirements by using **is**, **is not**, **are**, and **are not** for definitions and the logical consequences of conformant behavior. Behavior that is permitted but is neither always required nor directly controlled by an implementer or administrator, or whose conformance requirement is detailed elsewhere, is described by **can**. Behavior that never occurs in a conformant implementation or system of conformant implementations is described by **cannot**. The word **allow** is used as a replacement for the phrase “Support the ability for,” and the word **capability** means “can be configured to.”

5.2 Conformant components and equipment

This subclause specifies requirements and options for the following core components:

- a) VLAN Bridge component (5.4)
- b) MAC Bridge component (5.13)

for the following components that use that core functionality:

- c) Customer VLAN (C-VLAN) component (5.5)
- d) Service VLAN (S-VLAN) component (5.6)
- e) I-component (5.7)
- f) B-component (5.8)

- 1 g) TPMR component (5.15)
- 2 h) T-component (5.17)
- 3 i) Edge relay (ER) (5.24.1)

4 and for the following systems that include instances of the above components:

- 5 j) C-VLAN Bridge (5.9)
- 6 k) S-VLAN Bridge (5.10.1)
- 7 l) Provider Edge Bridge (5.10.2)
- 8 m) Backbone Edge Bridge (BEB) (5.12)
- 9 n) MAC Bridge (5.14)
- 10 o) TPMR (5.16)
- 11 p) Edge Virtual Bridging (EVB) Bridge (5.23)
- 12 q) EVB station (5.24)
- 13 r) TSN CNC station (5.29)
- 14 s) VDP-NVO3 (5.30)

15 NOTE—Both S-VLAN Bridges and Provider Edge Bridges are examples of Provider Bridges.

16 5.3 Protocol Implementation Conformance Statement (PICS)

17 A claim of conformance specifies implementation of a C-VLAN component, an S-VLAN component, an
18 I-component, a B-component, a VLAN-unaware MAC Bridge component, or a specific system. A
19 component or system can support multiple claims for a range of possible behaviors.

20 The supplier of an implementation that is claimed to conform to this standard shall provide the information
21 necessary to identify both the supplier and the implementation, and shall complete a copy of the PICS
22 proforma provided in Annex A for that specific Bridge component or system, or Annex B for an end station
23 implementation, together with any further information and completed PICS(s) required to identify
24 subcomponents.

25 NOTE 1—C-VLAN component and S-VLAN component PICS both require completion of a PICS for a VLAN Bridge
26 component; the VLAN Bridge PICS requires a claim of conformance for a single C-VLAN component; the Provider
27 Edge Bridge requires a claim of conformance for a single S-VLAN component and one or more claims for C-VLAN
28 components.

29 NOTE 2—The claim of conformance that could be made to this standard for an implementation of a VLAN-aware
30 Bridge is replaced by a claim of conformance to a VLAN Bridge. Although the current and subsequent amendment(s) or
31 revision(s) of this standard have changed the presentation of the information, the technical requirements of conformance
32 remain unchanged.

33 NOTE 3—I-component and B-component PICS both require completion of a PICS for an S-VLAN component.

34 5.4 VLAN Bridge component requirements

35 An implementation of a VLAN Bridge component shall:

- 36 a) Conform to the relevant standard for the MAC technology implemented at each Port in support of
37 the MAC ISS, as specified in IEEE Std 802.1AC.
- 38 b) Support the MAC Enhanced Internal Sublayer Service (EISS) at each Port, as specified in 6.8 and
39 6.9.
- 40 c) Implement an ISO/IEC 8802-2 conformant Logical Link Control (LLC) class with Type 1 operation
41 as required by 8.2.
- 42 d) Relay and filter frames as described in 8.1 and specified in 8.5, 8.6, 8.7, and 8.8.
- 43 e) Maintain the information required to support Basic Filtering Services, as described in 6.16 and in 8.1
44 and specified in 8.5, 8.7, and 8.8.

- 1 f) Conform to the provisions for addressing specified in 8.13.
- 2 g) Implement Rapid Spanning Tree Algorithm and Protocol (RSTP), as specified in Clause 13.
- 3 h) Encode transmitted BPDUs and validate received BPDUs as specified in Clause 14.
- 4 i) Specify the following parameters of the implementation:
 - 5 1) Filtering Database Size, the maximum number of entries
 - 6 2) Permanent Database Size, the maximum number of entries
- 7 j) Specify the following performance characteristics of the implementation:
 - 8 1) A Guaranteed Port Filtering Rate for each Bridge Port
 - 9 2) A Guaranteed Bridge Relaying Rate for the Bridge
 - 10 3) The related time intervals T_F and T_R for these parameters as specified in Clause 24
- 11 k) Operation of the Bridge within the specified parameters shall not violate any of the other
- 12 conformance provisions of this standard.
- 13 l) On each Port, support at least one of the permissible values for the Acceptable Frame Types
- 14 parameter, as defined in 6.9.
- 15 m) Support the following on each Port that supports untagged and priority-tagged frames:
 - 16 1) A Port VLAN Identifier (PVID) value (6.9)
 - 17 2) Configuration of at least one VID whose untagged set includes that Port (8.8.2)
 - 18 3) Configuration of the PVID value via management operations (12.10)
 - 19 4) Configuration of Static Filtering Entries via management operations (12.7)
- 20 n) Allow tag headers to be inserted, modified, and removed from relayed frames, as specified in 8.1
- 21 and Clause 9, as required by the value(s) of the Acceptable Frame Types parameter supported on
- 22 each Port, and by the ability of each Port to transmit VLAN-tagged and/or untagged frames.
- 23 o) Allow automatic configuration and management of VLAN topology using the Multiple VLAN
- 24 Registration Protocol (MVRP) (5.4.2, Clause 11) on all Ports.
- 25 p) Allow static and dynamic configuration information for at least one VID, by means of Static and
- 26 Dynamic VLAN Registration Entries in the FDB (8.8).
- 27 q) Support at least one Filtering Identifier (FID) (8.8.3, 8.8.8, 8.8.9, and IEEE Std 802.1AC).
- 28 r) Allow allocation of at least one VID to each FID that is supported (8.8.3, 8.8.8, 8.8.9, and
- 29 IEEE Std 802.1AC).
- 30 NOTE—Under some circumstances, the ability for VLAN Bridges to successfully interoperate depends on the
- 31 number of FIDs supported and on the number of VIDs that can be allocated to each FID. These circumstances
- 32 are discussed in Annex B, along with interoperability implications.

33 5.4.1 VLAN Bridge component options

34 An implementation of a VLAN Bridge component may:

- 35 a) Support MST operation (5.4.1.1).
- 36 b) Support Port-and-Protocol-based VLAN classification (5.4.1.2), including multiple VID values per
- 37 port, administrative control of the values of the multiple VIDs, and a Protocol Group Database.
- 38 c) Support CFM operation (5.4.1.4).
- 39 d) Support Shortest Path Bridging (SPB) operation (5.4.5).
- 40 e) Support congestion notification (5.4.3).
- 41 f) Support Priority-based Flow Control (PFC) (5.11).
- 42 g) Support Extended Filtering Services (6.16.5) and the operation of Multiple MAC Registration
- 43 Protocol (MMRP, 10.9) to support automatic configuration and management of MAC address
- 44 information on all Ports (5.4.1.3).
- 45 h) Allow the FDB to contain Static and Dynamic VLAN Registration Entries (8.8) for more than one
- 46 VID, up to a maximum of 4094 VIDs.

- 1 i) Allow translation of VIDs through support of the VID Translation Table or through support of both
2 the VID Translation Table and Egress VID translation table on one or more Bridge Ports (6.9).
- 3 NOTE 1—The maximum number of VIDs that can be supported is 4094 rather than 4096, as the VID values 0
4 and FFF are reserved, as indicated in Table 9-2. As conformance to this standard is only with regard to
5 externally visible protocol behavior, this limit on the number of VIDs that can be supported does not imply any
6 such limitation with regard to the internal architecture of a Bridge.
- 7 j) On each Port, support all of the permissible values for the Acceptable Frame Types parameter, as
8 defined in 6.9, and support configuration of the parameter value via management.
- 9 k) Support enabling and disabling of Ingress Filtering (8.6.2).
- 10 l) Allow configuration of more than one VID whose untagged set includes that Port (8.8.2).
- 11 m) Support the management functionality defined in Clause 12.
- 12 n) Support the use of a remote management protocol. Bridges claiming to support remote management
13 shall:
- 14 1) State which remote management protocol standard(s) or specification(s) are supported.
- 15 2) State which standard(s) or specification(s) for managed object definitions and encodings are
16 supported for use by the remote management protocol.
- 17 o) Support multiple Traffic Classes for relaying and filtering frames through one or more outbound
18 Ports, controlling the mapping of the priority of forwarded frames as specified in 6.9.4 and
19 IEEE Std 802.1AC.
- 20 p) Support more than one FID (8.8).
- 21 q) Allow allocation of more than one VID to each supported FID (8.8, 8.8.8).
- 22 r) Allow configuration of fixed VID to FID allocations (8.8.8, 12.10.3).
- 23 s) Allow configuration of the Restricted_MAC_Address_Registration parameter (10.12.2.3) for each
24 Port of the Bridge.
- 25 t) Support the ability to configure the value of the Restricted_VLAN_Registration parameter
26 (11.2.3.2.3) for each Port of the Bridge.
- 27 u) Support Forwarding and Queuing Enhancements for time-sensitive streams (FQTSS) (5.4.1.5).
- 28 v) Support SMIv2 MIB modules for the management of VLAN Bridge capabilities (Clause 17).
- 29 NOTE 2—In previous versions of this standard, the MIB modules were maintained by IETF (e.g., IETF RFC
30 4363 for the Q-BRIDGE MIB). For this version of the standard, the relevant set of MIB modules required to
31 manage a Bridge is contained in Clause 17.
- 32 w) Support YANG modules for the management of VLAN Bridge capabilities (Clause 48).
- 33 x) Support Multiple Stream Registration Protocol (MSRP) (Clause 35).
- 34 y) Support the operation of MVRP (5.4.2, 11.2) as a New-Only Participant.
- 35 z) Support the MVRP extension MIB module defined in 17.7.15.
- 36 aa) Support Enhanced Transmission Selection (ETS) (5.4.1.6).
- 37 ab) Support Data Center Bridging eXchange protocol (DCBX) (5.4.1.7).
- 38 ac) Support the enhancements for scheduled traffic as specified in 8.6.8.4.
- 39 ad) Support the management entities for scheduled traffic as specified in 12.29.
- 40 ae) Support frame preemption as specified in 6.7.1, 6.7.2, and 8.6.8.
- 41 af) Support Stream reservation remote management (5.4.1.11).
- 42 ag) Support for Congestion Isolation (CI) operation (5.4.7).

1 5.4.1.1 Multiple Spanning Tree (MST) operation (optional)

2 A VLAN Bridge implementation in conformance to the provisions of this standard for an MST Bridge
3 (5.4.1, 8.3, 8.4, 8.6.1, 8.9, 8.10, 11.2, 11.2.3.1, Clause 13, and Clause 14) shall:

- 4 a) Support MSTP as specified in Clause 13.
- 5 b) Support the Common and Internal Spanning Tree (CIST) plus a stated maximum number of
6 Multiple Spanning Tree Instances (MSTIs), where that number is at least 2 (8.9) and at most 64
7 (13.14).
8 NOTE—In other words, a conformant MST Bridge supports a minimum of three spanning tree instances—the
9 CIST and at least two additional MSTIs.
- 10 c) Support a stated maximum number of FIDs not less than the number of MSTIs (8.9).
- 11 d) Support the ability to associate each FID to a spanning tree (8.9.3).
- 12 e) Support the transmission and reception of MST Configuration Identifier (MCID) information
13 (8.9.2).
- 14 f) Support a Port State for each Port for each spanning tree instance supported (8.4, 13.38).
- 15 g) Support operation of a spanning tree protocol for each spanning tree instance and Port (8.10,
16 Clause 13).
- 17 h) Use the Bridge Group Address as specified in 8.13.3.
- 18 i) Support the default values for Bridge Forward Delay and Bridge Priority parameters specified in
19 13.26.
- 20 j) Support the operation of MVRP in each supported spanning tree context (11.2.3.1.1, 11.2.3.1.2).
- 21 k) Support the Bridge management functions for the Bridge protocol entity for each supported
22 spanning tree, independently (12.8).
- 23 l) Support, in particular, management of the Bridge priority parameters, and of the port priority and
24 path cost parameters for every port, independently for each supported spanning tree (12.8.1.1,
25 12.8.1.3, 13.27).
- 26 m) Support VLAN management functions for each supported spanning tree (12.10.1 and 12.11.1).
- 27 n) Support management of the MSTI configuration (12.12).

28 A VLAN Bridge implementation in conformance to the provisions of this standard for an MST Bridge
29 (5.4.1, Clause 13) may:

- 30 o) Support a greater number of FIDs than spanning trees (8.8.8).
- 31 p) Support SMIv2 MIB modules for the management of VLAN Bridge capabilities (Clause 17).
- 32 q) Support YANG modules for the management of MSTP (48.6.26, 48.6.27, 48.6.28, 48.6.29).

33 5.4.1.2 Port-and-Protocol-based VLAN classification (optional)

34 A VLAN Bridge component implementation in conformance to the provisions of this standard for
35 Port-and-Protocol-based VLAN classification (5.4.1) shall:

- 36 a) Support one or more of the following Protocol Classifications and Protocol Template formats:
37 Ethernet, RFC_1042, SNAP_8021H, SNAP_Other, or LLC_Other (6.12).

38 A VLAN Bridge component implementation in conformance to the provisions of this standard for
39 Port-and-Protocol-based VLAN classification (5.4.1) may:

- 40 b) Support configuration of the contents of the Protocol Group Database.

1 5.4.1.3 Multiple MAC Registration Protocol (MMRP) operation (optional)

2 A Bridge implementation in conformance to the provisions of this standard for the support of MMRP shall:

- 3 a) Conform to the operation of the MRP Applicant, Registrar, LeaveAll, and Periodic Transmission
4 state machines, as defined in 10.7.7, 10.7.8, 10.7.9, and 10.7.10, as required for one or the other of
5 the following variants of the MRP Participant, as defined in 10.6 and 10.7:
 - 6 1) The Full Participant
 - 7 2) The Full Participant, point-to-point subset
- 8 b) Exchange Multiple Registration Protocol Data Units (MRPDUs) as required by those state
9 machines, formatted in accordance with the generic protocol data unit (PDU) format described in
10 10.8, and able to carry application-specific information as defined in 10.12, using the group MAC
11 addresses reserved for use by MRP applications, as defined in Table 10-1.
- 12 c) Implement the MMRP Application component as defined in 10.12.
- 13 d) Propagate registration information in accordance with the operation of MAP for the Base Spanning
14 Tree Context, as specified in 10.3.1.
- 15 e) Forward, filter, or discard MAC frames carrying any MRP Application address as the destination
16 MAC address in accordance with the requirements of 8.13.6.

17 5.4.1.4 Connectivity Fault Management (CFM) (optional)

18 A VLAN Bridge component implementation that conforms to the provisions of this standard for CFM
19 (Clause 18 through Clause 22) shall:

- 20 a) Support the creation of Maintenance Domains at eight Maintenance Domain Levels (MD Levels),
21 with multiple nonoverlapping Maintenance Domains at each MD Level, using the managed objects
22 specified in 12.14.
- 23 b) Support the creation of a Maintenance Association (MA) for each VID supported by the Bridge for
24 each MD Level.
- 25 c) Support the creation of a single Maintenance Domain Intermediate Point (MIP) for each
26 Maintenance Domain on each Port, all MIPs being at the same MD Level.
- 27 d) Support the creation of eight Up MA Endpoints (MEPs) for each VID on each Port, each MEP at a
28 different MD Level.
- 29 e) Support the creation of eight Down MEPs for each VID on each Port, each MEP at a different MD
30 Level.
- 31 f) Support the creation of eight Down MEPs attached to no VID on each Port, each MEP at a different
32 MD Level.
- 33 g) Support the maintenance of a MEP CCM Database.
- 34 h) Provide control via all of the required managed objects specified in 12.14.
- 35 i) Conform to the state machines and procedures in Clause 20.
- 36 j) Transmit and receive required CFM PDUs in the formats specified in Clause 21.

37 A VLAN Bridge component implementation that conforms to the provisions of this standard for CFM may:

- 38 k) Support the creation of MIPs at different MD Levels on a single Port.
- 39 l) Support the creation of MIPs at MD Levels lower than or equal to MEPs on the same Port.
- 40 m) Support the maintenance of a MIP CCM Database in a MIP or MEP.
- 41 n) Support the CFM MIB module defined in 17.5 and/or the CFM YANG modules defined in
42 Clause 48.
- 43 o) Support the creation of MAs that are associated with more than one VID (22.1.5).
- 44 p) Support the creation and operation of Reflection Responder (RR) and Reflected Frame Message
45 (RFM) Receiver for the forward path test (FPT).
- 46 q) Support the creation and operation of Decapsulator Responder (DR) and Send Frame Message
47 (SFM) Originator for the return path test (RPT).

1 5.4.1.5 Forwarding and Queuing Enhancements for time-sensitive streams 2 (FQTSS)—requirements

3 A VLAN Bridge component implementation that conforms to the provisions of this standard for FQTSS
4 shall:

- 5 a) Support a minimum of two traffic classes on all Ports, of which:
 - 6 1) A minimum of one traffic class supports the strict priority algorithm for transmission selection
7 (8.6.8.1), and
 - 8 2) One traffic class is a stream reservation (SR) class.
- 9 b) Support the operation of the credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission
10 selection algorithm used for the SR class.
- 11 c) Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the
12 default priority regeneration override value defined in Table 6-5, for SR class “B.”
- 13 d) Support the tables and procedures for mapping priorities to traffic classes as defined in 34.5.

14 A VLAN Bridge component implementation that conforms to the provisions of this standard for FQTSS
15 may:

- 16 e) Support the management entities defined in 12.20 by means of the SNMPv2 MIB module defined in
17 17.7.12.
- 18 f) Support two or more SR classes (a maximum of seven), and support the operation of the
19 credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission selection algorithm used for
20 those SR classes. The number of SR classes supported shall be stated in the PICS.
- 21 g) Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the
22 default priority regeneration override value defined in Table 6-5, for SR class “A.” If more than two
23 SR classes are supported, the default priority regeneration override values used for the additional SR
24 classes shall be stated in the PICS.
- 25 h) Support the classMeasurementInterval attribute (12.20.1).
- 26 i) Support the SR Class to Priority Mapping Table (12.20.4).

27 5.4.1.6 ETS Bridge requirements

28 A device supporting ETS shall:

- 29 a) Support the ETS algorithm (8.6.8.3).
- 30 b) Support at least 3 traffic classes (37.3).
31 NOTE—A minimum of 3 traffic classes allows a minimum configuration such that one traffic class contains
32 priorities with PFC enabled, one traffic class contains priorities with PFC disabled, and one traffic class using
33 strict priority.
- 34 c) Support bandwidth configuration with a granularity of 1% or finer (37.3).
- 35 d) Support bandwidth allocation with a precision of 10% (37.3).
- 36 e) Support a transmission selection policy such that if one of the traffic classes does not consume its
37 allocated bandwidth, then any unused bandwidth is available to other traffic classes (37.3).
- 38 f) Support DCBX (Clause 38).

39 5.4.1.7 DCBX Bridge requirements

40 A device supporting DCBX shall:

- 41 a) Support Link Layer Discovery Protocol (LLDP) transmit and receive mode (IEEE Std 802.1AB).
- 42 b) Support the DCBX ETS Configuration Type, Length, Value (TLV) (D.2.8).
- 43 c) Support the ETS Recommendation TLV (D.2.9).

- 1 d) Support the Priority-based Flow Control Configuration TLV (D.2.10).
- 2 e) Support the Application Priority TLV (D.2.11).
- 3 f) Support the asymmetric and symmetric DCBX state machines (38.4).
- 4 g) Support the Application VLAN TLV (D.2.14).

5 5.4.1.8 Per-Stream Filtering and Policing (PSFP) requirements

6 A VLAN Bridge component implementation that conforms to the provisions of this standard for PSFP shall:

- 7 a) Support PSFP as specified in 8.6.5.2.1 and 8.6.6 items d) and e).
- 8 b) Support the state machines for stream gate control as specified in 8.6.10.
- 9 c) Support the management entities for PSFP as specified in 12.31.

10 5.4.1.9 Cyclic queuing and forwarding (CQF) requirements

11 A VLAN Bridge component implementation that conforms to the provisions of this standard for CQF (see
12 Annex T) shall:

- 13 a) Support the enhancements for scheduled traffic as specified in 8.6.8.4.
- 14 b) Support the state machines for scheduled traffic as specified in 8.6.9.
- 15 c) Support the state machines for stream gate control as specified in 8.6.10.
- 16 d) Support the management entities for scheduled traffic as specified in 12.29.
- 17 e) Support the requirements for Per-Stream Filtering and Policing (PSFP) as stated in 5.4.1.8.
- 18 f) Support the management entities for PSFP as specified in 12.31.

19 5.4.1.10 Asynchronous Traffic Shaping (ATS) requirements

20 A VLAN Bridge component implementation that conforms to the provisions of this standard for ATS shall:

- 21 a) Support per-stream classification and metering for ATS as specified in 8.6.5.2.2.
- 22 b) Support queuing with support for stream gates as specified in 8.6.6 items d) and e).
- 23 c) Support the ATS transmission selection algorithm as specified in 8.6.8.5.
- 24 d) Support the ATS scheduler state machines as specified in 8.6.11.
- 25 e) Support the management entities for ATS as specified in 12.31.

26 5.4.1.11 Stream reservation remote management (optional)

27 A VLAN-aware Bridge component implementation that conforms to the provisions of this standard for
28 Stream reservation remote management shall:

- 29 a) Support the management functionality defined in Clause 12.
- 30 b) Support the use of a remote management protocol. Bridges claiming to support remote management
31 shall state the following:
 - 32 1) Which remote management protocol standard(s) or specification(s) are supported for the server
33 implementation.
 - 34 2) Which standard(s) or specification(s) for managed object definitions and encodings are
35 supported for use by the remote management protocol.
- 36 c) Support TE-MSTID (5.5.2).
- 37 d) Support the entities for Stream reservation remote management (12.32).
- 38 e) Support the adminIdleSlope attribute (12.20.1).

1 5.4.2 Multiple VLAN Registration Protocol (MVRP) requirements

2 A VLAN Bridge implementation in conformance to the provisions of this standard for the support of MVRP
3 shall:

- 4 a) Conform to the operation of the MRP Applicant, Registrar, LeaveAll, and Periodic Transmission
5 state machines, as defined in 10.7.7, 10.7.8, 10.7.9, and 10.7.10, as required for one or the other of
6 the following variants of the MRP Participant, as defined in 10.6 and 10.7:
 - 7 1) The Full Participant
 - 8 2) The Full Participant, point-to-point subset
- 9 b) Exchange MRPDUs as required by those state machines, formatted in accordance with the generic
10 PDU format described in 10.8, and able to carry application-specific information as defined in 11.2,
11 using the group MAC addresses reserved for use by MVRP.
- 12 c) Implement the MVRP Application component as defined in 11.2.
- 13 d) Propagate registration information:
 - 14 1) In a Single Spanning Tree (SST) Bridge, in accordance with the operation of MAP for the Base
15 Spanning Tree Context, as specified in 10.3 and 10.3.1; or
 - 16 2) In an MST Bridge, in accordance with the operation of MAP for MST contexts as specified in
17 11.2.3.1.2 of this standard.
- 18 e) Forward, filter, or discard MAC frames carrying any MRP Application address as the destination
19 MAC address in accordance with the requirements of 8.13.6.
- 20 f) If a VID Translation Table (6.9) is in use for a Bridge Port, translate VIDs in Multiple VLAN
21 Registration Protocol Data Units (MRPDUs) as specified in 11.2.4.

22 5.4.3 VLAN Bridge requirements for congestion notification

23 A VLAN Bridge implementation that conforms to the provisions of this standard for congestion notification
24 (Clause 30 through Clause 33) shall:

- 25 a) Support, on one or more Ports, the creation of at least one Congestion Point (CP) (32.1.1).
- 26 b) Support, at each CP, the generation of Congestion Notification Messages (CNMs) (32.7) and the
27 removal of Congestion Notification tags (CN-TAGs) (32.1.1).
- 28 c) Support the ability to configure the variables controlling the operation of each CP (12.21.1, 12.21.2,
29 12.21.3, 12.21.4).
- 30 d) Support the operation of each Port in each of the Congestion Notification Domain (CND) defense
31 modes separately for each Congestion Notification Priority Value (CNPV) (32.1.1).
- 32 e) Support the LLDP capabilities required for conformance to IEEE Std 802.1AB.
- 33 f) Support the use of the Congestion Notification TLV in LLDP (32.1.1).

34 A Provider Instance Port (PIP, 6.10) that conforms to the provisions of this standard for congestion
35 notification shall:

- 36 g) Support CNM translation (32.16).

37 A Provider Edge Port (PEP) (15.4) that conforms to the provisions of this standard for congestion
38 notification shall:

- 39 h) Support CNM translation (32.16).

40 A VLAN Bridge implementation that conforms to the provisions of this standard for congestion notification
41 may:

- 42 i) Support the creation of up to seven CPs on a Bridge Port (31.1.1).
- 43 j) Support the IEEE8021-CN-MIB (17.7.13).

1 5.4.4 Multiple Stream Registration Protocol (MSRP) requirements

2 A VLAN Bridge implementation in conformance to the provisions of this standard for the support of MSRP
3 shall:

- 4 a) Conform to the operation of the MRP Applicant, Registrar, and LeaveAll state machines, as defined
5 in 10.7.7, 10.7.8, and 10.7.9, as required for one or more of the following variants of the MRP
6 Participant, as defined in 10.6 and 10.7:
 - 7 1) The Full Participant
 - 8 2) The Full Participant, point-to-point subset
 - 9 b) Exchange MRPDUs as required by those state machines, formatted in accordance with the generic
10 PDU format described in 10.8, and able to carry application-specific information as defined in
11 Clause 35, using the “Individual LAN Scope group address, Nearest Bridge group address” as
12 defined in Table 8-1, Table 8-2, and Table 8-3 (C-VLAN, S-VLAN, and TPMR component
13 Reserved addresses, respectively).
 - 14 c) Implement the MSRP Application component as defined in Clause 35.
 - 15 d) Propagate registration information in accordance with the operation of MAP for the Base Spanning
16 Tree Context, as specified in 10.3.1.
 - 17 e) Forward, filter, or discard MAC frames carrying any MRP Application address as the destination
18 MAC address in accordance with the requirements of 8.13.6.
 - 19 f) Not redeclare its attributes unless responding to a LeaveAll event.
- 20 NOTE—In order to meet the requirement specified in 5.4.4(f), the Periodic Transmission state machine
21 (10.7.10) is specifically excluded from MSRP. A Bridge is allowed to generate an MSRP LeaveAll event to
22 force an immediate redeclaration of all MSRP attributes from its neighbor(s).

23 5.4.5 Shortest Path Bridging (SPB) operation (optional)

24 A VLAN Bridge implementation that conforms to the provisions of this standard for SPB in Clause 27 shall:

- 25 a) Support SPB, in either VID mode (SPBV) or MAC mode (SPBM), as specified in Clause 27.
- 26 b) Support the IS-IS Link State Protocol with procedures to ensure Loop Prevention as specified in
27 Clause 28 to provide Shortest Path Tree (SPT) computation for SPB.
- 28 c) Encode, decode, and validate SPT BPDUs for the Agreement Protocol (13.17) and support
29 Agreement Protocol logic in Intermediate System to Intermediate System (IS-IS) Protocol as
30 specified in Clause 28.
- 31 d) Support at least three FIDs.
- 32 e) Support the management functionality specified in Clause 12.

33 A VLAN Bridge implementation in conformance to the provisions of this standard for SPB in Clause 27
34 may:

- 35 f) Support both SPB VID mode (SPBV) and SPB MAC mode (SPBM).
- 36 g) Support the SPB MIB module defined in 17.7.19.
- 37 h) Support both the VID Translation Table and Egress VID translation table (6.9) on SPBV Boundary
38 Ports.
- 39 i) Support the CFM modifications for SPBM as specified in Clause 20 and summarized in 27.18.
- 40 j) Support Equal Cost Multiple Paths (ECMP) operation of SPBM (5.4.5.1).
- 41 k) Support ECMP operation of SPBM with flow filtering (5.4.5.2).

1 5.4.5.1 SPBM ECMP operation (optional)

2 An SPBM-capable VLAN-aware bridge implementation that conforms to the provisions of this standard for
3 ECMP in 44.1 shall:

- 4 a) Support SPBM ECMP as specified in 44.1.
- 5 b) Support the management functionality specified in Clause 12.

6 An SPBM-capable VLAN-aware bridge implementation that conforms to the provisions of this standard for
7 ECMP may:

- 8 c) Support the IEEE8021-ECMP-MIB module objects `ieee8021EcmpEctStaticTable` and
9 `ieee8021EcmpTopSrvTable` defined in 17.7.21.

10 5.4.5.2 SPBM ECMP operation with flow filtering (optional)

11 An SPBM-capable VLAN-aware bridge implementation that conforms to the provisions of this standard for
12 ECMP with flow filtering shall:

- 13 a) Support SPBM ECMP as specified in 44.1.
- 14 b) Support flow filtering as specified in 44.2, including inserting, modifying, and removing flow
15 filtering tags (F-TAGs) from relayed frames as required by the applicable values in the Flow
16 Filtering Control Table for each port.
- 17 c) Support the management functionality specified in Clause 12.

18 An SPBM-capable VLAN-aware bridge implementation that conforms to the provisions of this standard for
19 ECMP with flow filtering may:

- 20 d) Support the CFM modifications for ECMP using flow filtering as specified in Clause 20 and
21 summarized in 44.2.5.
- 22 e) Support the IEEE8021-ECMP-MIB module defined in 17.7.21.

23 5.4.6 Path Control and Reservation (PCR) (optional)

24 A VLAN Bridge implementation that conforms to the provisions of this standard for PCR in Clause 45 shall:

- 25 a) Support the IS-IS link state protocol with the extensions for explicit path control (ISIS-PCR) as
26 specified in 45.1.
- 27 b) Support the SPB Link Metric sub-TLV as specified in 28.12.7.
- 28 c) Support the SPB Base VLAN-Identifiers sub-TLV as specified in 28.12.4.
- 29 d) Support the SPB Instance sub-TLV as specified in 28.12.5.
- 30 e) Support the SPBV MAC address sub-TLV as specified in 28.12.9.
- 31 f) Support the SPBM Service Identifier and Unicast Address sub-TLV as specified in 28.12.10.
- 32 g) Support the ST ECT Algorithm as specified in 45.1.2 and the use of the ST ECT Algorithm as
33 specified in 45.3.2.
- 34 h) Support the Topology sub-TLV, the Hop sub-TLV, and the corresponding ISIS-PCR operations for
35 explicit trees as specified in 45.1.
- 36 i) Support the management functionality specified in 12.26.

37 A VLAN Bridge implementation that conforms to the provisions of this standard for PCR in Clause 45 may:

- 38 j) Support the LT ECT Algorithm as specified in 45.1.2.
- 39 k) Support the LTS ECT Algorithm as specified in 45.1.2.
- 40 l) Support the MRT ECT Algorithm as specified in 45.3.3.
- 41 m) Support the MRTG ECT Algorithm as specified in 45.3.4.

- 1 n) Support the Administrative Group sub-TLV and the corresponding ISIS-PCR operations as specified
- 2 in 45.1.11.
- 3 o) Support the Bandwidth Constraint sub-TLV and the corresponding ISIS-PCR operations as specified
- 4 in 45.1.12.
- 5 p) Support the Bandwidth Assignment sub-TLV and the corresponding ISIS-PCR operations as
- 6 specified in 45.2.1.
- 7 q) Support the Timestamp sub-TLV as specified in 45.2.2.
- 8 r) Support LFA for unicast data flows as specified in 45.3.1.
- 9 s) Support the PCR Management Information Base (MIB) objects defined in 17.7.19.

10 **5.4.7 VLAN Bridge requirements for congestion isolation (optional)**

11 A VLAN-aware Bridge implementation that conforms to the provisions of this standard for congestion
12 isolation in Clause 49 shall:

- 13 a) Support, on one or more Ports, the creation of at least one Congestion Isolation Point (49.3.1).
- 14 b) Support Explicit Congestion Notification (ECN) as defined by IETF RFC 3168 and associated
- 15 updates along with Active Queue Management (AQM) as described in 49.2.1.
- 16 c) Support per-stream classification and metering for CI as specified in 8.6.5.2.3.
- 17 d) Support, at each Congestion Isolation Point, the variables and procedures of the Congestion
- 18 Isolation Protocol (49.4).
- 19 e) Support the ability to configure the variables controlling the operation of Congestion Isolation
- 20 (12.33.1), the CI Peer Table (12.33.2), the CI Stream Table (12.33.3), and each CIP (12.33.4).
- 21 f) Conform to the required capabilities of IEEE Std 802.1AB.
- 22 g) Support the use of the Congestion Isolation TLV in LLDP (D.2.15).

23 A VLAN Bridge implementation that conforms to the provisions of this standard for congestion isolation
24 may:

- 25 h) Support the monitoring of more than one queue on a Bridge Port (49.4.1.2.5).
- 26 i) Support transmission selection algorithms other than strict priority.
- 27 j) Support the de-isolation of congesting flows by means other than the empty status of a congesting
- 28 queue (49.2.6).
- 29 k) Support the Congestion Isolation YANG model (48.3.8).
- 30 l) Support Topology Recognition (49.5).

31 **5.5 C-VLAN component conformance**

32 A C-VLAN component comprises a VLAN Bridge component with the EISS on all Ports supported by the
33 use of a C-VLAN tag (C-TAG) (6.8, 9.5).

34 A conformant implementation of a C-VLAN component shall:

- 35 a) Comprise a single conformant VLAN Bridge component.
- 36 b) Recognize and use C-TAGs.
- 37 c) Filter the reserved MAC addresses specified in Table 8-1.
- 38 d) Use the Customer Bridge MVRP Address specified in Table 10-1.

39 A conformant implementation of a C-VLAN component shall not:

- 40 e) Use S-VLAN tags (S-TAGs) except in support of the functionality specified in 6.13.

1 5.5.1 C-VLAN component options

2 A conformant C-VLAN component may implement the options specified for a VLAN Bridge component
3 whose use is not specifically prohibited by 5.4.1 and may:

- 4 a) Support encoding the drop_eligible parameter in the PCP field of the tag header (6.9.3).
- 5 b) Allow support of the ISS as specified in 6.13 to facilitate priority selection on a Provider Bridged
6 Network (PBN).
- 7 c) Implement RSTP, with the enhancements to support Customer Edge Ports (CEPs), as specified in
8 13.41.
- 9 d) Support TE-MSTID (5.5.2).

10 5.5.2 TE-MSTID (optional)

11 A C-VLAN component implementation that conforms to the provisions of this standard for TE-MSTID
12 shall:

- 13 a) Disable learning for a set of Customer VLAN Identifiers (C-VIDs) allocated to the Traffic
14 Engineering Multiple Spanning Tree Instance Identifier (TE-MSTID) as specified in 8.4 and 8.9,
15 with the exception that support for PBB-TE is not required; and
- 16 b) Discard frames with unregistered destination addresses for C-VIDs allocated to the TE-MSTID
17 (8.8.1).

18 A C-VLAN component implementation that conforms to the provisions of this standard for TE-MSTID may

- 19 c) Allow control of all C-VIDs by an external agent, conforming to the Reserved VID values of
20 Table 9-2.

21 5.6 S-VLAN component conformance

22 An S-VLAN component comprises a VLAN Bridge component with the EISS on all Ports supported by the
23 use of an S-TAG (6.8, 9.5).

24 A conformant implementation of an S-VLAN component shall:

- 25 a) Comprise a single conformant VLAN Bridge component.
- 26 b) Recognize and use S-TAGs.
- 27 c) Support encoding the drop_eligible parameter in the PCP field of the tag header (6.9.3).
- 28 d) Filter the reserved MAC addresses specified in Table 8-2.
- 29 e) Use the Provider Bridge MVRP Address specified in Table 8-1.
- 30 f) Allow the Acceptable Frame Types parameter (6.9) to be set to *Admit All Frames* for each Port.
- 31 g) Allow the Enable Ingress Filtering parameter (8.6.2) to be set for each Port.

32 A conformant implementation of an S-VLAN component shall not:

- 33 h) Recognize or use C-TAGs.
- 34 i) Allow support of the ISS as specified in 6.13 for any of its Ports.
- 35 j) Configure the Customer Bridge MVRP Addresses specified in Table 10-1 in the FDB (8.8) or
36 Permanent Database (8.8.11).
- 37 k) Use the Customer Bridge MVRP Address specified in Table 10-1.

1 5.6.1 S-VLAN component options

2 A conformant S-VLAN component may implement any of the options specified for a VLAN Bridge
3 component (5.4.1).

4 5.6.2 S-VLAN component requirements for Provider Backbone Bridge Traffic Engineering 5 (PBB-TE)

6 An S-VLAN component implementation that conforms to the provisions of this standard for PBB-TE
7 (25.10) shall:

- 8 a) Disable learning for a set of Backbone VLAN Identifiers (B-VIDs) allocated to Traffic Engineering
9 Multiple Spanning Tree Instance Identifier (TE-MSTID) as specified in 8.4 and in 8.9.
- 10 b) Discard frames with unregistered destination addresses for B-VIDs allocated to TE-MSTID (8.8.1).

11 An S-VLAN component implementation that conforms to the provisions of this standard for PBB-TE
12 (25.10) may:

- 13 c) Allow control of all B-VIDs by an external agent, conforming to the Reserved VID values of
14 Table 9-2.
- 15 d) Support MIP creation and the corresponding CFM extensions for Traffic Engineering service
16 instances (TESIs) as specified in 26.9.
- 17 e) Support Infrastructure Protection Switching (IPS) as specified in 26.11.

18 5.6.3 S-VLAN component requirements for PBB-TE IPS

19 An S-VLAN component implementation that conforms to the provisions of this standard for PBB-TE IPS
20 (26.11) shall:

- 21 a) Support 1:1 IPS as specified in 26.11.2.

22 An S-VLAN component implementation that conforms to the provisions of this standard for PBB-TE IPS
23 (26.11) may:

- 24 b) Support M:1 IPS as specified in 26.11.5.

25 5.6.4 S-VLAN component requirements for ECMP with flow filtering

26 An S-VLAN component implementation that conforms to the provisions of this standard for ECMP with
27 flow filtering shall:

- 28 a) Support F-TAG processing as specified in 44.2.2 on each Provider Network Port (PNP).

29 5.7 I-component conformance

30 An I-component comprises an S-VLAN component (5.6) with the EISS on each Customer Network Port
31 (CNP) supported by the use of an S-TAG (6.9, 9.5), and the EISS for each Virtual Instance Port (VIP)
32 configured on a PIP supported by the use of both an S-TAG (6.9, 9.5) and a Backbone Service Instance tag
33 (I-TAG) (6.10, 9.5).

34 A conformant implementation of an I-component shall:

- 35 a) Comprise a single conformant S-VLAN component.
- 36 b) Recognize and use I-TAGs on one or more PIPs (6.10).
- 37 c) Support 1:1 mapping between Service VLAN Identifier (S-VID) values and Backbone Service
38 Instance Identifier (I-SID) values.

- 1 d) Support the `connection_identifier` parameter on the EISS of each VIP (6.10) and in the FDB
2 (8.8.12).
- 3 e) Support the termination of PBN spanning trees by inhibiting transmission of PBN BPDUs at a PIP.

4 **5.7.1 I-component options**

5 A conformant I-component may implement any of the options specified for an S-VLAN component (5.6.1),
6 and may:

- 7 a) Support many-to-one mapping from S-VID values to I-SID values.
- 8 b) Support a Common Spanning Tree (CST) among PBNs interconnected with one or more backbone
9 service instances by permitting transmission of PBN BPDUs through VIPs at a PIP.
- 10 c) Filter frames received at PIPs that have the Backbone Source MAC address (B-SA) used by the PIP,
11 to prevent such frames from circulating if backbone service instances in two or more attached
12 PBBNs are configured in a loop.
- 13 d) Use back-to-back Backbone Service Instance Multiplex Entities (6.18) on a PIP to enable
14 monitoring of backbone service instances with CFM.
- 15 e) Support the Multiple I-SID Registration Protocol (MIRP) defined in 39.1.1.
- 16 f) Support the MVRP Extension MIB module defined in 17.7.15.
- 17 g) Support the MIRP MIB module defined in 17.7.16.

18 **5.8 B-component conformance**

19 A B-component comprises an S-VLAN component with the EISS on each PNP supported by the use of an
20 S-TAG (6.9, 9.5), and the EISS on each Customer Backbone Port (CBP) supported by the use of both an
21 S-TAG (6.9, 9.5) and an I-TAG (6.11, 9.5). A conformant implementation of a B-component shall:

- 22 a) Comprise a single conformant S-VLAN component.
- 23 b) Recognize and use I-TAGs on one or more CBPs (6.11).
- 24 c) Terminate PBBN spanning tree by inhibiting transmission of PBBN BPDUs at a CBP.

25 **5.8.1 B-component options**

26 A conformant B-component may implement any of the options specified for an S-VLAN component (5.6.1),
27 and may:

- 28 a) Translate I-SID values by supporting the Local-SID field in the Backbone Service Instance table in
29 CBPs (6.11).
- 30 b) Assign B-VID values based on I-SID values by supporting the B-VID field in the Backbone Service
31 Instance table in CBPs (6.11).
- 32 c) Use the Default Backbone Destination field in the Backbone Service Instance table in CBPs (6.11)
33 as the destination MAC address for any frames received with the Backbone Service Instance Group
34 address (26.4).
- 35 d) Use back-to-back Backbone Service Instance Multiplex Entities (6.18) on a CBP to enable
36 monitoring of backbone service instances with CFM.
- 37 e) Support the MIRP defined in 39.1.2.
- 38 f) Support the MVRP Extension MIB module defined in 17.7.15.
- 39 g) Support the MIRP MIB module defined in 17.7.16.

1 5.8.2 B-component requirements for PBB-TE

2 A B-component implementation that conforms to the provisions of this standard for PBB-TE (25.10) shall:

- 3 a) Disable learning for a set of B-VIDs allocated to TE-MSTID as specified in 8.4 and in 8.9.
- 4 b) Discard frames with unregistered destination addresses for B-VIDs allocated to TE-MSTID (8.8.1).
- 5 c) Support the Default Backbone Destination field in the Backbone Service Instance tables on the
- 6 CBPs providing TESI as specified in 6.11.
- 7 d) Support the B-VID field in the Backbone Service Instance tables on the CBPs providing TESI as
- 8 specified in 6.11.
- 9 e) Support the creation of MEPs for the TESI on CBPs as specified in 26.9.
- 10 f) Support 1:1 protection switching as specified in 26.10.3.
- 11 g) Provide control via all of the PBB-TE required managed objects specified in Clause 12.

12 A B-component implementation that conforms to the provisions of this standard for PBB-TE (25.10) may:

- 13 h) Allow control of all B-VIDs by an external agent, conforming to the Reserved VID values of
- 14 Table 9-2.
- 15 i) Support MIP creation and the corresponding CFM extensions for TESI as specified in 26.9.
- 16 j) Support the Hold-Off timer used by the protection switching protocol as specified in 26.10.3.
- 17 k) Support sharing of TESI among TE protection groups in order to provide 1:1 protection switching
- 18 that is capable load sharing as specified in 12.14.1.2 and 25.10.2.
- 19 l) Support the Mismatch defect identification as specified in 26.11.
- 20 m) Support the PBB-TE MIB module defined in Clause 17.
- 21 n) Support IPS as specified in 26.11.

22 5.8.3 B-component requirements for PBB-TE IPS

23 A B-component implementation that conforms to the provisions of this standard for PBB-TE IPS (26.11)

24 shall:

- 25 a) Support 1:1 IPS as specified in 26.11.2.

26 A B-component implementation that conforms to the provisions of this standard for PBB-TE IPS (26.11)

27 may:

- 28 b) Support M:1 IPS as specified in 26.11.5.

29 5.8.4 B-component requirements for ECMP with flow filtering

30 A B-component implementation that conforms to the provisions of this standard for ECMP with flow

31 filtering shall:

- 32 a) Support F-TAG processing as specified in 44.2.2 on each CBP and PNP.

33 5.9 C-VLAN Bridge conformance

34 A C-VLAN Bridge shall comprise a single conformant C-VLAN component (5.5).

35 Each C-VLAN Bridge Port shall be capable of attaching directly to an IEEE 802 LAN.

36 5.9.1 C-VLAN Bridge options

37 A C-VLAN Bridge may implement any C-VLAN component option (5.5).

1 5.9.2 C-VLAN Bridge requirements for PBBN Auto Attach (optional)

2 A C-VLAN Bridge that conforms to the provisions for PBBN Auto Attach (Clause 50) shall:

- 3 a) Support Link Layer Discovery Protocol (LLDP) transmit and receive mode (IEEE Std 802.1AB™).
- 4 b) Support the PBBN Auto Attach System TLV (D.2.17).
- 5 c) Support the PBBN Auto Attach Assignment TLV (D.2.18).
- 6 d) Support the AAD state machine (50.6).
- 7 e) Support management entities for PBBN Auto Attach as specified in 12.34.

8 A C-VLAN Bridge that conforms to the provisions for PBBN Auto Attach may:

- 9 f) Support SNMP MIB extensions for AAP (17.7.26).
- 10 g) Support SNMP MIB extensions for AAP LLDP TLVs (D.5.6).
- 11 h) Support YANG for AAP LLDP TLVs (D.6.6.7).

12 5.10 Provider Bridge conformance

13 A Provider Bridge shall comprise one S-VLAN (5.6) component, zero or more C-VLAN components (5.5),
14 and zero or more Port-mapping S-VLAN components (5.6).

15 NOTE—The one mandatory S-VLAN component in a Provider Bridge is generally referred to simply as the S-VLAN
16 component; however, when this component needs to be distinguished from other components in a Provider Bridge
17 (particularly Port-mapping S-VLAN components), it is sometimes referred to as the “primary S-VLAN component.”

18 Each Port shall be capable of being configured as one of, and may be capable of being configured as any of:

- 19 a) A Provider Network Port (PNP)
- 20 b) A Customer Network Port (CNP)
- 21 c) A Customer Edge Port (CEP)
- 22 d) A Remote Customer Access Port (RCAP)

23 as specified in Clause 15. Each Port configured as a PNP or CNP shall be capable of attaching the S-VLAN
24 component of the Provider Bridge directly to an IEEE 802 LAN. Each Port configured as a CEP shall be
25 capable of attaching a C-VLAN component within the Provider Bridge directly to an IEEE 802 LAN. Each
26 Port configured as an RCAP shall be capable of attaching a Port-mapping S-VLAN component (15.6) within
27 the Provider Bridge directly to an IEEE 802 LAN.

28 5.10.1 S-VLAN Bridge conformance

29 An S-VLAN Bridge shall comprise a single conformant S-VLAN component (5.6). An S-VLAN Bridge
30 does not have any physical interfaces configured as a CEP or RCAP, nor does it include any C-VLAN
31 components.

32 5.10.2 Provider Edge Bridge conformance

33 A Provider Edge Bridge is a conformant Provider Bridge with a primary S-VLAN component, the capability
34 to include one or more C-VLAN components as specified in 15.4 and the capability to include zero or more
35 Port-mapping S-VLAN components as specified in 15.6.

36 Each C-VLAN component shall comprise a single CEP and a single distinct PEP for each service instance
37 that can be provided through that CEP. Each PEP shall be connected within the Provider Edge Bridge, as
38 specified in 6.14, to a distinct CNP on the S-VLAN component. Each C-VLAN component shall implement
39 RSTP, with the enhancements to support CEPs, as specified in 13.41.

Each Port-mapping S-VLAN component shall comprise a single RCAP and one or more Provider Access Ports (PAPs) each associated with one Remote Customer Service Interface (RCSI) provided through that RCAP. Each PAP shall be connected within the Provider Edge Bridge, as specified in 6.14, to a distinct CNP on the primary S-VLAN component or CEP on a C-VLAN component. A PNP shall also be provided, connected internally to a PNP on the primary S-VLAN component as specified in 15.6.

NOTE—The single CEP supported by a C-VLAN component and the single RCAP supported by a Port-mapping S-VLAN component can be supported by two or more independent instances of a MAC, aggregated as specified by Link Aggregation (IEEE Std 802.1AX).

5.10.3 S-VLAN Bridge requirements for PBBN Auto Attach (optional)

An S-VLAN Bridge that conforms to the provisions for PBBN Auto Attach (Clause 50) shall:

- a) Support Link Layer Discovery Protocol (LLDP) transmit and receive mode (IEEE Std 802.1AB).
- b) Support the PBBN Auto Attach System TLV (D.2.17).
- c) Support the PBBN Auto Attach Assignment TLV (D.2.18).
- d) Support the AAD state machine (50.6).
- e) Support management entities for PBBN Auto Attach as specified in 12.34.

An S-VLAN Bridge that conforms to the provisions for PBBN Auto Attach may:

- f) Support SNMP MIB extensions for AAP (17.7.26).
- g) Support SNMP MIB extensions for AAP LLDP TLVs (D.5.6).
- h) Support YANG for AAP LLDP TLVs (D.6.6.7).

5.11 System requirements for Priority-based Flow Control (PFC)

A system that conforms to the provisions of this standard for PFC (see Clause 36) shall:

- a) Support, on one or more ports, enabling PFC on at least one priority (36.1.2).
- b) Support, for each PFC Priority, processing PFC M_CONTROL.requests (36.1.3.1).
- c) Support, for each PFC Priority, processing PFC M_CONTROL.indications (36.1.3.3).
- d) Abide by the PFC delay constraints (36.1.3.3).
- e) Provide PFC-aware system queue functions (36.2).
- f) Enable use of PFC only in a domain controlled by DCBX (Clause 38).

A system that conforms to the provisions of this standard for PFC may:

- g) Support enabling PFC on up to eight priorities per port.
- h) Support the IEEE8021-PFC-MIB (17.7.17).

5.12 Backbone Edge Bridge (BEB) conformance

A BEB system shall comprise zero or more I-components and zero or one B-component and zero or more T-components, but at least one I-component or one B-component or one T-component, supporting externally accessible ports as specified in Clause 25. Each externally accessible port shall be designated as one of, and may be capable of being configured as any of the following:

- a) A PNP
- b) A PIP
- c) A CBP
- d) A CNP
- e) A CEP
- f) An RCAP

1 5.12.1 BEB requirements for PBB-TE

2 A BEB that conforms to the provisions of this standard for PBB-TE (25.10) shall comprise one
3 B-component capable of providing TESIs (5.8.2) and one or more I-components (5.7) or T-components
4 (5.17)1

5 5.12.2 Backbone Edge Bridge requirements for PBBN Auto Attach (optional)

6 An Auto Attach BEB (AAB) shall be a conformant Backbone Edge Bridge (BEB, 5.12) with at least one
7 I-component (5.7) and shall have one or more C-VLAN components providing externally visible Customer
8 Edge Ports (CEP) as specified in 50.2.

9 Each externally accessible AAB port that supports PBBN Auto Attach shall:

- 10 a) Support Link Layer Discovery Protocol (LLDP) transmit and receive mode (IEEE Std 802.1AB).
- 11 b) Support the PBBN Auto Attach System TLV (D.2.17).
- 12 c) Support the PBBN Auto Attach Assignment TLV (D.2.18).
- 13 d) Support the C-tagging service interface (50.2.2) and S-tagging service interface (50.2.1).
- 14 e) Support the AAB state machine (50.7).
- 15 f) Support the management entities for PBBN Auto Attach as specified in 12.34.

16 Each externally accessible AAB port that supports PBBN Auto Attach may:

- 17 g) Support the IEEE Std 802.1Q Organizationally Specific Management VID TLV (D.2.6)
- 18 h) Support SNMP MIB extensions for AAP (17.7.26).
- 19 i) Support SNMP MIB extensions for AAP LLDP TLVs (D.5.6).
- 20 j) Support YANG for AAP LLDP TLVs (D.6.6.7).

21 5.13 MAC Bridge component requirements

22 An implementation of a VLAN-unaware MAC Bridge component shall:

- 23 a) Conform to the relevant standard for the MAC technology implemented at each Port in support of
24 the MAC ISS, as specified in IEEE Std 802.1AC.
- 25 b) Support the VLAN-unaware MAC Relay Entity by means of an ISS interface (not an EISS
26 interface).
- 27 c) Implement an ISO/IEC 8802-2 conformant LLC class with Type 1 operation as required by 8.2.
- 28 d) Maintain the information required to support Basic Filtering Services, as described in
29 IEEE Std 802.1AC and in 8.1 and specified in 8.5, 8.7, and 8.8.
- 30 e) Conform to the provisions for addressing specified in 8.13.
- 31 f) Implement RSTP, as specified in Clause 13.
- 32 g) Encode transmitted BPDUs and validate received BPDUs as specified in Clause 14.
- 33 h) Specify the following parameters of the implementation:
 - 34 1) Filtering Database Size, the maximum number of entries
 - 35 2) Permanent Database Size, the maximum number of entries
- 36 i) Specify the following performance characteristics of the implementation:
 - 37 1) A Guaranteed Port Filtering Rate for each Bridge Port
 - 38 2) A Guaranteed Bridge Relaying Rate for the Bridge
 - 39 3) The related time intervals T_F and T_R for these parameters as specified in Clause 24
- 40 j) Operation of the Bridge within the specified parameters shall not violate any of the other
41 conformance provisions of this standard.

1 5.13.1 MAC Bridge component options

2 An implementation of a MAC Bridge component may:

- 3 a) Support Extended Filtering Services (6.16.5) and the operation of MMRP (10.9) to support
4 automatic configuration and management of MAC address information on all Ports (5.4.1.3).
- 5 b) Support the management functionality defined in Clause 12.
- 6 c) Support the use of a remote management protocol. Bridges claiming to support remote management
7 shall:
 - 8 1) State which remote management protocol standard(s) or specification(s) are supported.
 - 9 2) State which standard(s) or specification(s) for managed object definitions and encodings are
10 supported for use by the remote management protocol.
- 11 d) Support multiple Traffic Classes for relaying and filtering frames through one or more outbound
12 Ports, controlling the mapping of the priority of forwarded frames as specified in IEEE Std 802.1AC
13 and 6.9.4.
- 14 e) Allow configuration of the Restricted_MAC_Address_Registration parameter (10.12.2.3) for each
15 Port of the Bridge.
- 16 f) Support the ISS with signaled priority on each Bridge Port (6.20).
- 17 g) Support SMIv2 MIB modules for the management of MAC Bridge capabilities (Clause 17).
- 18 h) Support YANG modules for the management of MAC Bridge capabilities (Clause 48);
- 19 i) Support CFM operation (5.4.1.4).
- 20 j) Support ETS (5.4.1.6).
- 21 k) Support PFC (5.11).
- 22 l) Support DCBX (5.4.1.7).
- 23 m) Support the enhancements for scheduled traffic as specified in 8.6.8.4.
- 24 n) Support the state machines for scheduled traffic as specified in 8.6.9.
- 25 o) Support frame preemption as specified in 6.7.1, 6.7.2, and 8.6.8.

26 5.13.1.1 Per-Stream Filtering and Policing (PSFP) requirements

27 A MAC Bridge component implementation that conforms to the provisions of this standard for PSFP shall:

- 28 a) Support PSFP as specified in 8.6.5.2 and 8.6.6.
- 29 b) Support the state machines for stream gate control as specified in 8.6.10.
- 30 c) Support the management entities for PSFP as specified in 12.31.

31 5.13.1.2 Cyclic queuing and forwarding requirements

32 A MAC Bridge component implementation that conforms to the provisions of this standard for CQF (see
33 Annex T) shall:

- 34 a) Support the enhancements for scheduled traffic as specified in 8.6.8.4.
- 35 b) Support the state machines for scheduled traffic as specified in 8.6.9.
- 36 c) Support the state machines for stream gate control as specified in 8.6.10.
- 37 d) Support the management entities for scheduled traffic as specified in 12.29.
- 38 e) Support the requirements for PSFP as stated in 5.13.1.1.
- 39 f) Support the management entities for PSFP as specified in 12.31.

1 5.13.1.3 Asynchronous Traffic Shaping (ATS) requirements

2 A MAC Bridge component implementation that conforms to the provisions of this standard for ATS shall:

- 3 a) Support per-stream classification and metering for ATS as specified in 8.6.5.2.2.
- 4 b) Support queuing with support for stream gates as specified in 8.6.6 items d) and e).
- 5 c) Support the ATS transmission selection algorithm as specified in 8.6.8.5.
- 6 d) Support the ATS scheduler state machines as specified in 8.6.11.
- 7 e) Support the management entities for ATS as specified in 12.31.

8 5.14 MAC Bridge conformance

9 A MAC Bridge shall comprise a single conformant MAC Bridge component (5.13).

10 Each Bridge Port shall be capable of attaching directly to an IEEE 802 LAN.

11 5.14.1 MAC Bridge options

12 A MAC Bridge may implement any MAC Bridge component option (5.13.1).

13 5.14.2 MAC Bridge requirements for PBBN Auto Attach (optional)

14 A MAC Bridge that conforms to the provisions for PBBN Auto Attach (Clause 50) shall:

- 15 a) Support Link Layer Discovery Protocol (LLDP) transmit and receive mode (IEEE Std 802.1AB).
- 16 b) Support the PBBN Auto Attach System TLV (D.2.17).
- 17 c) Support the PBBN Auto Attach Assignment TLV (D.2.18).
- 18 d) Support the AAD state machine (50.6).
- 19 e) Support the management entities for PBBN Auto Attach as specified in 12.34.

20 A MAC Bridge that conforms to the provisions for PBBN Auto Attach may:

- 21 f) Support SNMP MIB extensions for AAP (17.7.26).
- 22 g) Support SNMP MIB extensions for AAP LLDP TLVs (D.5.6).
- 23 h) Support YANG for AAP LLDP TLVs (D.6.6.7).

24 5.15 TPMR component conformance

25 A TPMR component comprises a MAC Bridge component. A conformant implementation of a TPMR
26 component shall:

- 27 a) Support exactly two Bridge Ports other than a Management Port.
- 28 b) Support the operation of the following functions of the Forwarding Process:
 - 29 1) Frame filtering (8.6.3)
 - 30 2) Queuing frames (8.6.6)
 - 31 3) Transmission selection (8.6.8)
- 32 c) Support the operation of the MAC Status Propagation Entity (MSPE), by means of the functions and
33 protocol specified in Clause 23.
- 34 d) Support at least one traffic class on each Bridge Port.
- 35 e) Support the management functionality specified in 12.19.

- 1 f) Support only the following entries in the FDB:
- 2 1) Static Filtering Entries, permanently configured in the FDB and that cannot be modified by
- 3 management, that specify filtering on both of the TPMR's Bridge Ports for the group MAC
- 4 addresses defined in Table 8-3.
- 5 2) A Static Filtering Entry, permanently configured in the FDB and that cannot be modified by
- 6 management, that specifies forwarding or filtering on each of the TPMR's Bridge Ports for the
- 7 MAC address associated with the management entity of the TPMR. The values that are
- 8 configured into this filtering entry are determined by the implementer, and shall be stated in the
- 9 PICS.
- 10 g) Support a MIP on an MA that is not associated with any VLAN on each of the TPMR's Bridge
- 11 Ports. The default MD level shall be zero.
- 12 NOTE—As a TPMR is a non-VLAN-aware Bridge, these MPs are not associated with a particular VID
- 13 (22.1.7). This default level 0 MIP allows detection and identification of the TPMR without configuration. Such
- 14 a MIP can of course be disabled by management.

15 A conformant implementation of a TPMR component shall not:

- 16 h) Implement RSTP defined in Clause 13.
- 17 i) Encode transmitted BPDUs and validate received BPDUs (Clause 14).
- 18 j) Support Extended Filtering Services for relaying and filtering frames (6.16.5).

19 **5.15.1 TPMR component options**

20 An implementation of a TPMR component may:

- 21 a) Support multiple traffic classes on each Bridge Port.
- 22 b) Support the ISS with signaled priority on each Bridge Port (6.20).

23 **5.16 TPMR conformance**

24 An implementation of a TPMR shall:

- 25 a) Comprise a single conformant TPMR component (5.15).
- 26 b) Support exactly two externally accessible Ports, each capable of attaching directly to an IEEE 802
- 27 LAN.
- 28 c) Be capable of supporting a management entity using the TPMR Port Connectivity (8.5.2) on one of
- 29 its externally accessible ports.
- 30 d) Support remote management of the TPMR managed objects specified in 12.19.

31 **5.16.1 TPMR options**

32 A TPMR may implement any TPMR component option (5.15.1).

33 An implementation of a TPMR may:

- 34 a) Support a management entity using the Bridge Port Connectivity (8.5.1) on a Management Port (8.3)
- 35 or the externally accessible ports.
- 36 b) Support remote management using Simple Network Management Protocol (SNMP) over IEEE 802
- 37 Networks (IETF RFC 4789, 8.13.7).
- 38 NOTE—While IETF RFC 4789 is specifically referenced, the standard does not preclude the support of other
- 39 transports for SNMP.
- 40 c) Support remote management using the TPMR MIB module defined in 17.7.11.
- 41 d) Support remote management using the TPMR YANG modules defined in 48.3.2.

1 5.17 T-component conformance

2 A T-component comprises a TPMR component (5.15) with one PIP (6.11).

3 A conformant implementation of an T-component shall:

- 4 a) Comprise a single conformant TPMR component.
- 5 b) Recognize and use I-TAGs on one PIP (6.10).
- 6 c) Support a single VIP at the PIP.

7 5.17.1 T-component options

8 A conformant T-component may implement any of the options specified for a TPMR component (5.15.1).

9 5.18 End station requirements for MMRP, MVRP, and MSRP

10 This subclause defines the conformance requirements for end station implementations claiming
11 conformance to MVRP, MMRP, and MSRP. Although this standard is principally concerned with defining
12 the requirements for VLAN Bridges, the conformance requirements for end station implementations of
13 MVRP, MMRP, and MSRP are included in order to give guidance to such implementations. The PICS
14 proforma defined in Annex B is concerned with conformance claims with respect to end station
15 implementations.

16 For the reasons stated in 10.6, it is recommended that end stations that do not require the ability to perform
17 Source Pruning implement the Applicant-Only Participant, in preference to the Simple-Applicant
18 Participant.

19 5.18.1 MMRP requirements and options

20 An end station for which conformance to MMRP is claimed shall:

- 21 a) Conform to the operation of the MRP Applicant, Registrar, LeaveAll, and Periodic Transmission
22 state machines, as defined in 10.7.7, 10.7.8, 10.7.9, and 10.7.10, as required for one or more of the
23 following variants of the MRP Participant, as defined in 10.6 and 10.7:
 - 24 1) The Full Participant
 - 25 2) The Full Participant, point-to-point subset
 - 26 3) The Applicant-Only Participant
 - 27 4) The Simple-Applicant Participant
- 28 b) Exchange MPDUs as required by the MRP state machine(s) implemented, formatted in accordance
29 with the generic PDU format described in 10.8, and able to carry application-specific information as
30 defined in 10.12.1, using the “Individual LAN Scope group address, Nearest Bridge group address”
31 as defined in Table 8-1, Table 8-2, and Table 8-3 (C-VLAN and MAC Bridge, S-VLAN, and TPMR
32 component Reserved addresses, respectively).

33 An end station for which conformance to the operation of the Applicant state machine (10.7.7) is claimed
34 may:

- 35 c) Perform source pruning, as defined in 10.10.3 and 10.12.

36 It is recommended that only those end stations that require the ability to perform Source Pruning (10.10.3)
37 conform to the operation of the Full Participant or the Full Participant, point-to-point subset.

38 NOTE—If an end station does not require the ability to perform Source Pruning, then there is no need for it to
39 implement Registrar functionality.

1 5.18.2 MVRP requirements and options

2 An end station for which conformance to MVRP is claimed shall:

- 3 a) Conform to the operation of the MRP Applicant, Registrar, LeaveAll, and Periodic Transmission
4 state machines, as defined in 10.7.7, 10.7.8, 10.7.9, and 10.7.10, as required for one or more of the
5 following variants of the MRP Participant, as defined in 10.6 and 10.7:
 - 6 1) The Full Participant
 - 7 2) The Full Participant, point-to-point subset
 - 8 3) The Applicant-Only Participant
 - 9 4) The Simple-Applicant Participant
- 10 b) Exchange MPDUs as required by the MRP state machine(s) implemented, formatted in accordance
11 with the generic PDU format described in 10.8, and able to carry application-specific information as
12 defined in 11.2.3, using the MVRP Application address as defined in Table 10-1.

13 An end station for which conformance to MVRP is claimed may:

- 14 c) Perform source pruning, as defined in 10.10.3 and 11.2.1.1.

15 It is recommended that only those end stations that require the ability to perform Source Pruning (11.2.1.1)
16 conform to the operation of the Full Participant or the Full Participant, point-to-point subset.

17 NOTE—If an end station does not require the ability to perform Source Pruning, then there is no need for it to
18 implement Registrar functionality.

- 19 d) Support the operation of MVRP (5.4.2, 11.2) as a New-Only Participant.
- 20 e) Support the MVRP Extension MIB module defined in 17.7.15.

21 5.18.3 MSRP requirements and options

22 An end station for which conformance to MSRP is claimed shall:

- 23 a) Conform to the operation of the MRP Applicant, Registrar, and LeaveAll state machines, as defined
24 in 10.7.7, 10.7.8, and 10.7.9, as required for one or more of the following variants of the MRP
25 Participant, as defined in 10.6 and 10.7:
 - 26 1) The Full Participant
 - 27 2) The Full Participant, point-to-point subset
 - 28 3) The Applicant-Only Participant
 - 29 4) The Simple-Applicant Participant
- 30 b) Exchange MPDUs as required by the MRP state machine(s) implemented, formatted in accordance
31 with the generic PDU format described in 10.8, and able to carry application-specific information as
32 defined in 35.2.2, using the “Individual LAN Scope group address, Nearest Bridge group address”
33 as defined in Table 8-1, Table 8-2, and Table 8-3 (C-VLAN, S-VLAN, and TPMR component
34 Reserved addresses, respectively).
- 35 c) Not redeclare its attributes unless responding to a LeaveAll event.

36 An end station for which conformance to the operation of the Applicant state machine (10.7.7) is claimed
37 may:

- 38 d) Perform Talker pruning, as defined in 35.2.1.4(b), 35.2.3.1, and 35.2.4.3.2.
- 39 e) Perform Listener pruning as described in 35.2.3.1.

40 NOTE—In order to meet the requirement specified in 5.18.3(c), the Periodic Transmission state machine (10.7.10) is
41 specifically excluded from MSRP. An end station is allowed to generate an MSRP LeaveAll event to force an immediate
42 redeclaration of all MSRP attributes from its neighbor(s).

1 5.19 VLAN-aware end station requirements for CFM

2 A VLAN-aware Station implementation that conforms to the provisions of this standard for CFM (Clause 18
3 through Clause 22) shall:

- 4 a) Support the creation of Maintenance Domains at eight MD Levels, with multiple nonoverlapping
5 Maintenance Domains at each MD Level, using the managed objects specified in 12.14.
- 6 b) Support the creation of an MA on each VID supported by the Station for each MD Level.
- 7 c) Support the creation of eight Down MEPs on each VID on each Port, each MEP at a different MD
8 Level.
- 9 d) Support the creation of eight Down MEPs attached to no VID on each Port, each MEP at a different
10 MD Level.
- 11 e) Support the maintenance of a MEP CCM Database.
- 12 f) Provide control via all of the required managed objects specified in 12.14.
- 13 g) Conform to the state machines and procedures in Clause 20.
- 14 h) Transmit and receive required CFM PDUs in the formats specified in Clause 21.

15 A VLAN-aware Station implementation that conforms to the provisions of this standard for CFM may:

- 16 i) Support the CFM MIB module defined in 17.5.
- 17 j) Support the CFM YANG modules defined in Clause 48.
- 18 k) Support the creation of MAs that are associated with more than one VID (22.1.5).

19 5.20 End station requirements—FQTSS

20 An end station implementation that conforms to the provisions of this standard for FQTSS shall:

- 21 a) Support a minimum of two traffic classes on all Ports, of which:
 - 22 1) A minimum of one traffic class supports the strict priority algorithm for transmission selection
23 (8.6.8.1), and
 - 24 2) One traffic class is an SR class.
- 25 b) Support the operation of the credit-based shaper algorithm (8.6.8.2) as the transmission selection
26 algorithm used for frames transmitted for each stream associated with the SR class.
- 27 c) Support the operation of the credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission
28 selection algorithm used for the SR class.
- 29 d) Use the default priority associated with SR class “B” as shown in Table 6-5 as the priority value
30 carried in transmitted SR class “B” data frames.

31 An end station implementation that conforms to the provisions of this standard for FQTSS may:

- 32 e) Support two or more SR classes (a maximum of seven), and support the operation of the
33 credit-based shaper algorithm (8.6.8.2) on all Ports as the transmission selection algorithm used for
34 those SR classes. The number of SR classes supported shall be stated in the PICS.
- 35 f) Use the default priority associated with SR class “A” as shown in Table 6-5 as the priority value
36 carried in transmitted SR class “A” data frames. If more than two SR classes are supported, the
37 priority value carried in transmitted data frames for the additional SR classes shall be stated in the
38 PICS.

1 5.21 End station requirements for congestion notification

2 A VLAN-aware end station implementation that conforms to the provisions of this standard for congestion
3 notification (Clause 30 through Clause 33) shall:

- 4 a) Support the creation of at least one Reaction Point (RP) (31.2.2.2).
- 5 b) Support, if CPs are supported, the ability to configure the variables controlling the operation of each
6 CP, if any (12.21.1, 12.21.2, 12.21.3, 12.21.4).
- 7 c) Support, at each supported CP, if any, the generation of CNMs (32.7).
- 8 d) Support the ability to configure the variables controlling the operation of each RP (12.21.1, 12.21.5,
9 12.21.6).
- 10 e) Support the operation of its Port in at least the `cptDisabled` and `cptInterior` CND defense modes
11 separately for each CNPV (32.1.1).
- 12 f) Conform to the required specifications of the LLDP of IEEE Std 802.1AB.
- 13 g) Support the use of the Congestion Notification TLV in LLDP (32.1.1).
- 14 h) Support, in each RP, the limiting of frames output in response to the reception of CNMs (31.2.2.2).
- 15 i) Support, if multiple RPs are supported, the ability to distinguish among CNMs pertaining to
16 different RPs, and direct each such message to the correct RP (31.2.5).

17 A VLAN-aware end station implementation that conforms to the provisions of this standard for congestion
18 notification (Clause 30 through Clause 33) may:

- 19 j) Support the creation of one or more CPs (31.1.1).
- 20 k) Support the creation of more than one RP (31.2.2.2).
- 21 l) Support more than one CNPV per RP (31.2.1).
- 22 m) Support the operation of its Port in both the `cptInterior` and `cptInteriorReady` CND defense modes
23 separately for each CNPV (31.1.1).
- 24 n) Support the IEEE8021-MIRP-MIB (17.7.13).

25 5.22 MAC-specific bridging methods

26 MAC-specific bridging methods may exist. Use of a MAC-specific bridging method and the method
27 specified in this standard on the same LAN shall:

- 28 a) Not prevent communication between stations in a network.
- 29 b) Preserve the MAC Service.
- 30 c) Preserve the characteristics of each bridging method within its own domain.
- 31 d) Provide for the ability of both bridging techniques to coexist simultaneously on a LAN without
32 adverse interaction.

33 5.23 EVB Bridge requirements

34 An EVB Bridge shall comprise a single conformant C-VLAN component (5.5) and zero or one
35 Port-mapping S-VLAN component (5.6) per externally accessible port.

36 Each externally accessible port shall be capable of being configured as one of, and may be capable of being
37 configured as any of the following:

- 38 a) A C-VLAN Bridge Port
- 39 b) A Station-facing Bridge Port (SBP)

1 c) An Uplink Access Port (UAP)

2 as specified in Clause 40.

3 A conformant EVB Bridge implementation shall:

4 d) Support the functionality of a C-VLAN component (5.5).

5 e) Support at least one SBP on the C-VLAN component (Clause 40).

6 f) Support the EVB status parameters for EVBMode = EVB Bridge (40.4).

7 g) Support an LLDP nearest Customer Bridge database (Clause 40).

8 h) Support the EVB TLV on each SBP (D.2.12).

9 i) Support Edge Control Protocol (ECP) on each SBP (Clause 43).

10 j) Support the Bridge role of Virtual Station Interface (VSI) Discovery and Configuration Protocol
11 (VDP) on each SBP (Clause 41).

12 A conformant EVB Bridge may support S-channels. A conformant EVB Bridge with S-channel support
13 shall:

14 k) Support at least one Port-mapping S-VLAN component (22.6.4) and associated UAP, configured as
15 specified in 40.2 a)–d).

16 l) Support S-channel Discovery and Configuration Protocol (CDCP), as specified in Clause 42,
17 operating in Bridge mode.

18 m) Support the enhanced filtering utility criteria (8.7.2) and not support the default filtering utility
19 criteria (8.7.1).

20 A conformant EVB Bridge implementation may:

21 n) Support configuration of reflective relay on each SBP of the C-VLAN component (8.6.1).

22 o) Support management for the EVB components (12.4–12.12, 12.26).

23 p) Support the MIB module defined in 17.7.20.

24 q) Support assignment of VIDs to GroupIDs (41.2.9).

25 r) Support the use of the M and S bits in VDP (41.2.3).

26 5.24 EVB station requirements

27 An EVB station shall comprise one or more conformant ERs (5.24.1) and zero or one Port-mapping
28 S-VLAN component (5.6) per externally accessible port.

29 Each externally accessible port shall be capable of being configured as one of, and may be capable of being
30 configured as any of the following:

31 a) An Uplink Access Port (UAP)

32 b) An uplink relay port (URP)

33 as specified in Clause 40.

34 Each downlink relay port (DRP) shall be capable of attaching to one or more VSIs.

35 Each URP shall be capable of attaching its ER to the LAN connecting to an EVB Bridge, or in the
36 case where a Port-mapping S-VLAN component is present, to an internal LAN (6.14) connecting the URP
37 to a S-channel Access Port (CAP).

38 A conformant EVB station implementation shall:

39 c) Support at least one ER (5.24.1, Clause 40).

40 d) Support the EVB status parameters for EVBMode = EVB station on each URP (40.4).

- 1 e) Support an LLDP nearest Customer Bridge database (Clause 40).
- 2 f) Support the EVB TLV on each URP of each ER (D.2.12).
- 3 g) Support ECP on each URP of each ER (Clause 43).
- 4 h) Support the station role of VDP for each URP of each ER (Clause 41).

5 In addition, a conformant EVB station implementation that supports a Port-mapping S-VLAN components
6 shall:

- 7 i) Support a Port-mapping S-VLAN component (22.6.4) on each port configured as a UAP (40.2)
8 configured as specified in 40.2 (a)–(d).
- 9 j) Support CDCP, as specified in Clause 42, operating in Station mode.
- 10 k) Support the enhanced filtering utility criteria (8.7.2) and not support the default filtering utility
11 criteria (8.7.1).

12 A conformant EVB station implementation may:

- 13 l) Support multiple ERs (Clause 40).
- 14 m) Support management for the EVB components (12.4–12.12, 12.26).
- 15 n) Support the MIB module defined in 17.7.20.
- 16 o) Support assignment of VIDs to GroupIDs (41.2.9).
- 17 p) Support the use of the M and S bits in VDP (41.2.3).

18 5.24.1 Edge relay (ER) requirements

19 A conformant implementation of an ER shall:

- 20 a) Conform to the relevant standard for the MAC technology implemented at each Port in support of
21 the MAC ISS, as specified in IEEE Std 802.1AC and 6.14.
- 22 b) Support the MAC EISS at each Port, as specified in 6.8 and 6.9.
- 23 c) Recognize and use C-TAGs (6.9).
- 24 d) Relay and filter frames as described in 8.1 and specified in 8.5, 8.6, and 8.8.
- 25 e) Support the following on each DRP Port that supports untagged and priority-tagged frames:
26 1) A PVID value (6.9)
27 2) Configuration of at least one VID whose untagged set includes that Port (8.8.2)
- 28 f) Support setting the Acceptable Frame Types parameter (6.9) to *Admit Only VLAN-tagged Frames* on
29 the URP.
- 30 g) Allow tag headers to be inserted, modified, and removed from relayed frames, as specified in 8.1
31 and Clause 9, as required by the value(s) of the Acceptable Frame Types parameter supported on
32 each Port, and by the ability of each Port to transmit VLAN-tagged and/or untagged frames.
- 33 h) Support at least one FID (IEEE Std 802.1AC, 8.8.3, 8.8.8, and 8.8.9).
- 34 i) Allow allocation of at least one VID to each FID that is supported (IEEE Std 802.1AC, 8.8.3, 8.8.8,
35 and 8.8.9).
- 36 j) Support exactly one URP (Clause 40) supporting the parameters of 40.4 for EVBMode = EVB
37 station.
- 38 k) Support one or more DRPs each supporting access to VSIs (Clause 40).
- 39 l) Filter the reserved MAC addresses specified in Table 8-1.
- 40 m) If more than one DRP is supported, support setting the Enable Ingress Filtering parameter (8.6.2) on
41 each DRP and the URP.
- 42 n) Support the requirements of either a virtual edge bridge (VEB) ER (5.24.1.1) or a virtual edge port
43 aggregator (VEPA) ER (5.24.1.2).

1 A conformant implementation of an ER may:

- 2 o) Support the following if the URP supports untagged and priority-tagged frames:
 - 3 1) A PVID value (6.9)
 - 4 2) Configuration of at least one VID whose untagged set includes that Port (8.8.2)
- 5 p) Comprise a single conformant C-VLAN component (5.4).
- 6 q) Support disabling of learning on each DRP (8.6.1).
- 7 r) Support the ability to discard frames received at each DRP if there is no entry in the FDB that
- 8 specifies forwarding on that Port for the frame's source MAC address and VLAN.
- 9 s) Support the operation of the learning process as described in 8.7.

10 **5.24.1.1 VEB ER requirements**

11 In addition to the requirements stated in 5.24.1, a conformant VEB ER implementation shall:

- 12 a) Request that reflective relay service not be provided by setting adminReflectiveRelayRequest to
- 13 FALSE (40.4).

14 **5.24.1.2 VEPA ER requirements**

15 In addition to the requirements stated in 5.24.1, a conformant VEPA ER implementation shall:

- 16 a) Disable learning on the URP (8.6.1).
- 17 b) Filter frames as specified in 8.6.3.1.

18 A conformant VEPA ER implementation may:

- 19 c) Filter frames received at each DRP that are destined for the URP until reflective relay is enabled
- 20 (40.4).

21 A conformant VEPA ER implementation should:

- 22 d) Request the provision of reflective relay service by setting adminReflectiveRelayRequest to TRUE
- 23 (40.4).

24 NOTE—This item is optional because there can be cases where an EVB station is configured to prohibit VSIs

25 from communicating with each other in VEPA mode.

26 **5.25 End station requirements—enhancements for scheduled traffic**

27 An end station implementation that conforms to the provisions of this standard for enhancements for

28 scheduled traffic shall:

- 29 a) Support the enhancements for scheduled traffic as specified in 8.6.8.4.
- 30 b) Support the state machines for scheduled traffic as specified in 8.6.9.

1 5.26 End station requirements—enhancements for frame preemption

2 An end station implementation that conforms to the provisions of this standard for frame preemption shall:

- 3 a) Support the provisions of 6.7.1, 6.7.2, and 8.6.8.

4 5.27 End station requirements—PSFP

5 An end-station implementation that conforms to the provisions of this standard for PSFP shall:

- 6 a) Support PSFP as specified in 8.6.5.2.1 and 8.6.6.
7 b) Support the state machines for stream gate control as specified in 8.6.10.
8 c) Support the management entities for PSFP as specified in 12.31.

9 5.28 End station requirements—Cyclic queuing and forwarding

10 An end station implementation that conforms to the provisions of this standard for CQF (see Annex T) shall:

- 11 a) Support the enhancements for scheduled traffic as specified in 8.6.8.4.
12 b) Support the state machines for scheduled traffic as specified in .
13 c) Support the state machines for stream gate control as specified in 8.6.10.
14 d) Support the management entities for scheduled traffic as specified in 12.29.
15 e) Support the requirements for PSFP as stated in 5.27.
16 f) Support the management entities for PSFP as specified in 12.31.

17 5.29 TSN CNC station requirements

18 This subclause (5.29) defines the conformance requirements for a station that supports the Time-Sensitive
19 Networking (TSN) Centralized Network Configuration (CNC) requirements (46.1.3). The TSN CNC station
20 component is implemented within an end station or Bridge.

21 A TSN CNC station implementation that conforms to the provisions of this standard shall:

- 22 a) Support the use of a remote management protocol. The TSN CNC claiming to support remote
23 management shall state the following:
24 1) Which remote management protocol standard(s) or specification(s) are supported for the client
25 implementation.
26 2) Which standard(s) or specification(s) for managed object definitions and encodings are
27 supported for use by the remote management protocol.
28 b) Support the managed object definitions and encodings for Stream reservation remote management
29 (12.32).
30 c) Support the use of at least one protocol for User/network configuration information that complies
31 with the requirements for protocol integration defined in 46.2. The TSN CNC shall state which
32 User/network configuration information protocol standard(s) or specification(s) are supported.
33 d) If SRP (Clause 35) is supported by the TSN CNC for the User/network configuration information,
34 the TSN CNC shall support MRP External Control (12.32.4).

35 A TSN CNC station implementation that conforms to the provisions of this standard may:

- 36 e) Use the YANG modules identified in 46.3.

1 5.30 VDP-NVO3 requirements

2 In the Split-NVE scenario, the nNVE implements the bridge role VDP and tNVE implements the station role
3 VDP. While the nNVE and tNVE share the VDP functionality of an EVB Bridge and an EVB Station, their
4 conformance requirements are different. This clause lists the conformance requirements for nNVE and
5 tNVE to operate VDP in the Split-NVE scenario.

6 5.30.1 VDP-NVO3 nNVE requirements

7 A conformant VDP-NVO3 nNVE implementation shall:

- 8 a) Support the Bridge role of VDP on each SBP (Clause 41).
- 9 b) Support assignment of VIDs to GroupIDs (41.2.9).

10 A conformant VDP-NVO3 nNVE implementation may:

- 11 c) Support the functionality of a C-VLAN component (5.5).
- 12 d) Support at least one SBP on the C-VLAN component (Clause 40).
- 13 e) Support an LLDP nearest Customer Bridge database (Clause 40).
- 14 f) Support the EVB status parameters for EVBMode = NVO3 Mode (40.4.4) and NVERole = nNVE
15 (40.5.1).
- 16 g) Support the EVB Bridge status parameters about IPv4 and IPv6 address capability (D.2.12.3).
- 17 h) Support the EVB TLV on each SBP (D.2.12).
- 18 i) Support ECP on each SBP (Clause 43).
- 19 j) Support the use of the M, S, and N bits in VDP (41.2.3).
- 20 k) Support the use of IP addresses in VDP filter info format (41.2.9).

21 5.30.2 VDP-NVO3 tNVE requirements

22 A conformant VDP-NVO3 tNVE implementation shall:

- 23 a) Support the station role of VDP for each URP (Clause 41).
- 24 b) Support assignment of VIDs to GroupIDs (41.2.9).

25 A conformant VDP-NVO3 tNVE implementation may:

- 26 c) Support one ER (5.24.1, Clause 40).
- 27 d) Support an LLDP nearest Customer Bridge database (Clause 40).
- 28 e) Support the EVB status parameters for EVBMode = NVO3 Mode (40.4.4) and NVERole = tNVE
29 (40.5.2).
- 30 f) Support the EVB station status parameters about IPv4 and IPv6 address capability (D.2.12.4).
- 31 g) Support the EVB TLV on each URP (D.2.12).
- 32 h) Support ECP on each URP (Clause 43).
- 33 i) Support the use of the M, S, and N bits in VDP (41.2.3).
- 34 j) Support the use of IP addresses in VDP filter info format (41.2.9).

35 5.31 End station requirements—ATS

36 An end station implementation that conforms to the provisions of this standard for ATS shall:

- 37 a) Support a minimum of two traffic classes, of which:
 - 38 1) One traffic class supports the strict priority transmission selection algorithm (8.6.8.1) and
 - 39 2) One traffic class supports the ATS transmission selection algorithm (8.6.8.5).
- 40 b) Support the ATS talker transmission behavior, as specified in 47.1.

1 5.32 End station requirements for congestion isolation

2 An end station implementation that conforms to the provisions of this standard for congestion isolation in
3 Clause 49 shall

- 4 a) Support Explicit Congestion Notification (ECN) as defined by IETF RFC 3168 and associated
5 updates along with Active Queue Management (AQM) as described in 49.2.1.
- 6 b) Support, at each Congestion Isolation Point, the variables and procedures of the Congestion
7 Isolation Protocol (49.4).
- 8 c) Conform to the required capabilities of IEEE Std 802.1AB.
- 9 d) Support the use of the Congestion Isolation TLV in LLDP (D.2.15).

10 An end station implementation that conforms to the provisions of this standard for congestion isolation in
11 Clause 49 may

- 12 e) Support transmission selection algorithms other than strict priority.
- 13 f) Support the de-isolation of congesting flows by means other than the empty status of a congesting
14 queue (49.2.6).
- 15 g) Support the Congestion Isolation YANG model (48.3.8).
- 16 h) Support non-relay end station or server functionality of Topology Recognition (49.5).

17 5.33 End station requirements for PBBN Auto Attach (optional)

18 An end station that conforms to the provisions for PBBN Auto Attach (Clause 50) shall:

- 19 a) Support Link Layer Discovery Protocol (LLDP) transmit and receive mode (IEEE Std 802.1AB).
- 20 b) Support the PBBN Auto Attach System TLV (D.2.17).
- 21 c) Support the PBBN Auto Attach Assignment TLV (D.2.18).
- 22 d) Support the AAD state machine (50.6).
- 23 e) Support the management entities for PBBN Auto Attach as specified in 12.34.

24 An end station implementation that conforms to the provisions for PBBN Auto Attach may:

- 25 f) Support SNMP MIB extensions for AAP (17.7.26).
- 26 g) Support SNMP MIB extensions for AAP LLDP TLVs (D.5.6).
- 27 h) Support YANG for AAP LLDP TLVs (D.6.6.7).

28

2 VLAN Bridges interconnect the separate IEEE 802 LANs that compose a Virtual Bridged Network by
3 relaying and filtering frames between the separate MACs of the bridged LANs.

Figure 6-1—Internal organization of the MAC sublayer

- 7 a) Media access method-specific functions²³ that realize transmission and reception of MAC Protocol
8 Data Units (MPDUs).
- 9 b) Media access method-dependent convergence functions that use item a) to provide a media access
10 method-independent service.
- 11 c) Media access method-independent functions that use a media-independent service to provide the
12 same or another media-independent service.

15 A VLAN Bridge's MAC Relay Entity forwards frames between the instances of the media-independent
16 EISS (6.8). The EISS is provided by the functions specified in 6.9 using the media-independent ISS
17 (IEEE Std 802.1AC). The convergence functions that provide the ISS using the media-specific functions for
18 each IEEE 802 LAN MAC type are specified in IEEE Std 802.1AC.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards draft, subject to change.

1 This clause:

- 2 a) Summarizes basic architectural concepts and terms used throughout this standard, introduces the
3 primitives and parameters of the MAC Service, and defines VLANs in terms of the connectivity
4 provided to service users (IEEE Std 802.1AC, 6.2, 6.3).
- 5 b) Describes how Bridges preserve and maintain the quality of the MAC Service (6.4, 6.5).
- 6 c) Specifies the MAC ISS and MAC status parameters, their support by specific media access methods
7 (IEEE Std 802.1AC), and support for signaled priority (IEEE Std 802.1AC, 6.14, 6.15, 6.20).
- 8 d) Specifies the EISS used by VLAN-aware stations, the encoding of VID and priority parameters in
9 transmitted frames, and the classification of received frames that do not explicitly convey those
10 parameters (6.8, 6.9, 6.12).
- 11 e) Specifies how entities defined in terms of the ISS can support the EISS (6.17).

12 6.1 Basic architectural concepts and terms

13 The architectural concepts used in this and other IEEE 802.1 standards are based on the layered protocol
14 model introduced by the OSI Reference Model (ISO/IEC 7498-1) and used in the MAC Service definition
15 (IEEE Std 802.1AC), in IEEE Std 802, in other IEEE 802 standards, and elsewhere in networking.
16 IEEE 802.1 standards in particular have developed terms and distinctions useful in describing the MAC
17 Service and its support by protocol entities within the MAC Sublayer.²⁴ For further detailed discussion of
18 these concepts and terms, see IEEE Std 802.1AC.

19 6.2 Provision of the MAC Service

20 The MAC Service provided in end stations attached to MAC Bridged Networks and Virtual Bridged
21 Networks is the (unconfirmed) connectionless mode MAC Service defined in IEEE Std 802.1AC. The MAC
22 Service is defined as an abstraction of the features common to a number of specific MAC Services; it
23 describes the transfer of user data between source and destination end stations, via MA-UNITDATA request
24 primitives and corresponding MA-UNITDATA indication primitives issued at MAC Service Access Points
25 (MSAPs). Each MA-UNITDATA request and indication primitive has four parameters: Destination Address,
26 Source Address, MAC Service data unit (MSDU), and Priority.

27 NOTE 1—The primitives of the MAC Service are closely aligned with those of the ISS (IEEE Std 802.1AC).

28 In a MAC Bridged Network, MAC Bridges provide a single instance of the MAC Service for the network as
29 a whole—a single LAN, as defined in IEEE Std 802.1AC—by forwarding between the individual LANs that
30 compose the network, though frames with specified group destination addresses are confined to individual
31 LANs.

32 NOTE 2—MAC Bridges can be configured to completely partition a bridged network, though that is rarely intended.

33 In a Virtual Bridged Network, VLANs are defined in terms of the connectivity associations
34 (IEEE Std 802.1AC) supported by the network: if service requests made at two different MSAPs both result
35 in service indications at any third MSAP, then the two MSAPs are transmitting on the same VLAN; and if
36 the service requests made at any given MSAP can result in service indications at either (or both) of two
37 MSAPs, then those two MSAPs are receiving on the same VLAN. When only two MSAPs are in
38 bidirectional communication, they are using the same VLAN for both transmit and receive if a third station
39 could be added in such a way that it would receive transmissions from both.

40 NOTE 3—This definition of a VLAN accommodates an unavoidable and sometimes useful consequence of
41 configuration possibilities: use of a single MSAP can result in transmission on one VLAN and reception on another.

42 NOTE 4—Protocol VLAN classification can be used to classify frames transmitted from a single MSAP into separate
43 VLANs. Where protocol classification is used, the preceding rules apply to frames of a single classification.

²⁴ Drawing on prior network layer standards, including ISO/IEC 7498-1, wherever possible.

1 6.2.1 Point-to-point, multipoint-to-multipoint, and rooted-multipoint connectivity

2 The types of connectivity supported by associations of Bridge Ports in Virtual Bridged Networks are often
3 classified into three categories: point-to-point, multipoint-to-multipoint, and rooted-multipoint.
4 Point-to-point connectivity is an association of exactly two Bridge Ports such that an ingress frame (for a
5 given VLAN) at one Bridge Port may result in an egress frame at the other Bridge Port.
6 Multipoint-to-multipoint connectivity is an association of two or more Bridge Ports such that an ingress
7 frame at one Bridge Port may result in an egress frame at any or all of the other Bridge Ports.
8 Rooted-multipoint connectivity is an association of two or more Bridge Ports where each Bridge Port is
9 designated as either a Root or a Leaf Port, and an ingress frame at a Root Port may result in an egress frame
10 at any or all Leaf Ports and other Root Ports, while an ingress frame at a Leaf Port may result in an egress
11 frame at any or all Root Ports but never results in an egress frame at another Leaf Port. Note that
12 point-to-point connectivity is a special case of the more general multipoint-to-multipoint connectivity, and
13 that both point-to-point connectivity and multipoint-to-multipoint connectivity are special cases of
14 rooted-multipoint connectivity where all Bridge Ports are designated as Root Ports.

15 6.3 Support of the MAC Service

16 In a MAC Bridged Network at most one LAN can be supported on each of the individual LANs that, bridged
17 together, compose the network. On the individual LANs of a Virtual Bridged Network, frames for different
18 VLANs can be distinguished by the addition of a VLAN tag as the initial octets of a frame's MSDU.

19 When a VLAN-unaware end station is attached to an individual LAN, it will transmit and receive only
20 untagged frames, and thus uses the VID(s) assigned by the Bridge Ports that classify those frames.

21 NOTE 1—Where there are only two stations attached to an individual LAN, each can assign received untagged frames
22 to a different VID. However, such assignments can defeat the operation of network configuration protocols and can only
23 be recommended at the edge of a network to facilitate initial classification of untagged frames transmitted by
24 VLAN-unaware end stations. Otherwise, a change in a frame's VID can severely impact network operation by
25 introducing frame forwarding loops. The same considerations apply to the use of protocol VLAN classification. The use
26 of different VID assignments by Bridge Ports attached to the same LAN is usually a network configuration error.

27 NOTE 2—The LLC Entity in a VLAN-unaware end station can receive MAC Service indications that correspond to the
28 receipt of VLAN-tagged frames but, being unable to recognize the tag's EtherType value, will discard the frame and not
29 deliver it to any LLC User.

30 A VLAN-aware end station can use the EISS Multiplex Entity (6.17) to provide multiple SAPs, one per VID
31 of interest, to separate MAC Clients.

32 The objective of Bridge operation is to maximize the availability of the MAC Service to end stations and to
33 assist in the maintenance of the network. Bridges can be configured to provide redundant paths between end
34 stations, enabling the network to continue providing service in the event of component failure (of Bridge or
35 LAN). The paths supported between end stations are predictable and configurable. Group addressed
36 (multicast and broadcast) frames for any given VLAN take the same path as frames addressed to an
37 individual end station (unicast), thus permitting the use of either individually addressed or group addressed
38 frames to configure or diagnose paths for frames with either group or individual destination addresses. The
39 paths taken for frames for any given VLAN between any two stations are symmetric, i.e., frames from
40 station A (say) to station B traverse the same Bridges and LANs as those from B to A, only in the reverse
41 order. This symmetry reduces the probability of one-way connectivity between stations and permits the use
42 of station location learning to localize traffic (6.5.10).

43 The VLAN Identifier (VID) conveyed in the VLAN tag associates each frame with the following:

- 44 a) A VLAN
- 45 b) A loop-free active topology

1 6.4 Preservation of the MAC Service

2 The MAC Service offered by a network consisting of LANs interconnected by Bridges is similar to that
3 offered by a single LAN (see 6.5).

- 4 a) Frames transmitted between end stations carry the MAC addresses of the peer-end stations in their
5 destination and source address fields, not an address of a Bridge. The MAC Relay is not directly
6 addressed by communicating end stations.
- 7 b) The MAC addresses of end stations are not restricted by the network's topology or configuration.
- 8 c) All MAC addresses need to be unique within a VLAN, and within any set of VLANs for which
9 filtering information is shared by a Bridge.

10 6.5 Quality of service (QoS) maintenance

11 6.5.1 Service availability

12 Service availability is measured as that fraction of some total time during which the MAC Service is
13 provided. The operation of a Bridge can increase or lower the service availability.

14 The service availability can be increased by automatic reconfiguration of the network in order to avoid the
15 use of a failed component (e.g., repeater, cable, or connector) in the data path. The service availability can be
16 lowered by failure of a Bridge, through denial of service by the Bridge, or through frame filtering by the
17 Bridge. Changes in topology, caused by component failures, the addition or removal of components, or
18 administrative changes, are detected and signaled by the following means:

- 19 a) Physical detection of component failure and signaling of that failure by the EISS (6.8 and 6.9).
- 20 b) Detection of component failure through the operation of a spanning tree algorithm and protocol.
- 21 c) Explicit signaling of reconfiguration events through the operation of a spanning tree algorithm and
22 protocol.

23 NOTE 1—Each reference to spanning tree algorithms and protocols throughout Clause 6 includes any whose purpose is
24 to verify that each transmitted frame gives rise to at most one service indication at any of the MSAPs for which frames
25 for that LAN or VLAN are to be received. Such protocols include RSTP (13.4), MSTP (13.5), and SPB (Clause 27).

26 Automatic reconfiguration can be achieved rapidly on the detection of a physical topology change (see
27 Clause 13), thus minimizing any service denial that is caused by the reconfiguration.

28 A Bridge may deny service and discard frames (6.5.2) in order to preserve other aspects of the MAC Service
29 (6.5.3 and 6.5.4) when automatic reconfiguration takes place. Service may be denied to end stations that do
30 not benefit from the reconfiguration; hence, the service availability is lowered for those end stations. Bridges
31 may filter frames in order to localize traffic in the network. Should an end station move, it may then be
32 unable to receive frames from other end stations until the filtering information held by the Bridges is
33 updated.

34 To minimize the effects of service denial caused by reconfiguration events, filtering information that has
35 been dynamically learned can be modified when automatic reconfiguration takes place, or in preparation for
36 future reconfiguration events (Clause 13 and 13.16). However, filtering information that is statically
37 configured cannot be modified in this way.

38 A Bridge may deny service and discard frames in order to prevent access to the network by devices that are
39 not authorized for such access.

40 To maximize the service availability, no loss of service or delay in service provision should be caused by
41 Bridges, except as a consequence of a failure, removal, or insertion of a network component; or as a
42 consequence of the movement of an end station; or as a consequence of an attempt to perform unauthorized

1 access. These events are regarded as extraordinary. The operation of any additional protocol necessary to
2 maintain the quality of the MAC Service is thus limited to the configuration of the network and is
3 independent of individual instances of service provision.

4 NOTE 2—This is true only in circumstances where admission control mechanisms are not present, i.e., where the
5 Bridges provide a “best effort” service.

6 NOTE 3—The operation of management on the Bridge can result in the Bridge being reset, either as a result of a specific
7 Bridge reset operation or as a consequence of manipulating the Bridge’s configuration. From the point of view of service
8 availability, resetting the Bridge is an extraordinary event that has a similar effect to physical removal of the Bridge from
9 the network, followed by reinsertion of the Bridge into the network.

10 6.5.2 Frame loss

11 The MAC Service does not guarantee the delivery of Service Data Units (SDUs). Frames transmitted by a
12 source station arrive, uncorrupted, at the destination station with high probability. The operation of a Bridge
13 introduces minimal additional frame loss.

14 A frame transmitted by a source station can fail to reach its destination station as a result of:

- 15 a) Frame corruption during physical layer transmission or reception.
- 16 b) Frame discard by a Bridge because:
 - 17 1) It is unable to transmit the frame within some maximum period of time and, hence, must
 - 18 discard the frame to prevent the maximum frame lifetime (6.5.6) from being exceeded.
 - 19 2) It is unable to continue to store the frame due to exhaustion of internal buffering capacity as
 - 20 frames continue to arrive at a rate in excess of that at which they can be transmitted.
 - 21 3) The size of the SDU carried by the frame exceeds the maximum supported by the MAC
 - 22 procedures employed on the LAN to which the frame is to be relayed.
 - 23 4) Changes in the connected topology of the network necessitate frame discard for a limited
 - 24 period of time to maintain other aspects of QoS (13.16).
 - 25 5) The device attached to the Bridge is not authorized for access to the network.
 - 26 6) The configuration of Static Filtering Entries or Static VLAN Registration Entries in the FDB
 - 27 (8.8.1, 8.8.2) disallows the forwarding of frames with particular destination addresses or
 - 28 VLAN classifications on specific Ports.
 - 29 7) A flow metering algorithm (8.6.5) determines that discard is necessary.
 - 30 8) The Bridge supports enhancements for scheduled traffic (8.6.8.4) and the size of the service
 - 31 data unit exceeds the value of queueMaxSDU (8.6.8.4) for the traffic class queue on which the
 - 32 frame is to be queued.

33 NOTE—As Static Filtering Entries and Static VLAN Registration Entries are associated with particular Ports or
34 combinations of Ports, there is a possibility that misconfiguration of such entries will lead to unintended frame discard
35 during or following automatic reconfiguration of the network.

36 6.5.3 Frame misordering

37 The MAC Service (IEEE Std 802.1AC) permits a negligible rate of reordering of frames with a given
38 priority for a given combination of destination address, source address, and flow hash, if present, transmitted
39 on a given VLAN. MA_UNITDATA.indication service primitives corresponding to
40 MA_UNITDATA.request primitives, with the same requested priority and for the same combination of
41 VLAN classification, destination address, source address, and flow hash, if present, are received in the same
42 order as the request primitives were processed.

43 NOTE 1—The operation of the Forwarding Process in Bridges (8.6) is such that the frame-ordering characteristics of the
44 MAC Service are preserved.

1 Where Bridges in a network are capable of connecting the individual MACs in such a way that multiple
2 paths between any source station–destination station pairs exist, the operation of a protocol is required to
3 ensure that a single path is used for every sequence of frames whose order must be maintained.

4 NOTE 2—Where STP is in use (see Clause 8 of IEEE Std 802.1D, 1998 Edition [B13]), frame misordering cannot occur
5 during normal operation. When a protocol that is capable of rapid reconfiguration after network component failure (such
6 as RSTP, MSTP, or SPB) is used, there is an increased probability that frames that are in transit through the network will
7 be misordered, because a Bridge can buffer frames awaiting transmission through its Ports. The probability of
8 misordering occurring as a result of such an event is dependent on implementation choices and is associated with
9 reconfiguration events. Some known LAN protocols, for example, LLC Type 2 (ISO/IEC 8802-2), are sensitive to frame
10 misordering and duplication; in order to allow Bridges that support RSTP to be used in environments where sensitive
11 protocols are in use, the forceVersion parameter (13.7.2) can be used to force a Bridge that supports RSTP to operate in
12 an STP-compatible manner. A more detailed discussion of misordering in RSTP can be found in K.3 of
13 IEEE Std 802.1D-2004 [B14].

14 **6.5.4 Frame duplication**

15 The MAC Service (IEEE Std 802.1AC) permits a negligible rate of duplication of frames. The operation of
16 Bridges introduces a negligible rate of duplication of user data frames.

17 The potential for frame duplication in a bridged network arises through the possibility of the following:

- 18 a) Repeated transmission, through a given Bridge Port, of a frame received on another Bridge Port
- 19 b) Multiple paths between source and destination end stations
- 20 c) A loop in a path between source and destination stations

21 A Bridge does not transmit any received frame more than once through any Port (8.6.7).

22 When Bridges in a network connect individual LANs in such a way that physical topology is capable of
23 providing multiple paths between any source and destination, a protocol is required to ensure that the active
24 topology comprises a single path. SST Bridges assign all frames to a single active topology, other Bridges
25 assign frames to an active topology using the VID or VID and destination address. For frames assigned to
26 Ethernet Switched Paths (ESPs) by PBB-TE, the protocol comprises the operation of an external agent that
27 configures Static Filtering Entries in the FDB (see 25.10); for all other frames RSTP, MSTP, or Intermediate
28 System to Intermediate System Protocol for Shortest Path Bridging (ISIS-SPB) is used to configure the
29 controls used for active topology enforcement (8.6.1). Reception of duplicate copies of a single transmitted
30 frame can be avoided by loop prevention (6.5.4.1) or loop mitigation (6.5.4.2).

31 NOTE—Where RSTP, MSTP, or SPB is used (see Clause 13), there is a probability that a reconfiguration event can
32 cause frames that are in transit through the network to be duplicated, because a Bridge can buffer frames awaiting
33 transmission through its Ports. As the probability of duplication occurring as a result of such an event is small, and the
34 frequency of reconfiguration events is also small, the degradation of the properties of the MAC Service caused by this
35 source of frame duplication is considered to be negligible. A more detailed discussion of frame duplication in RSTP can
36 be found in K.2 of IEEE Std 802.1D-2004 [B14].

37 **6.5.4.1 Loop prevention**

38 Each Bridge that participates in RSTP, MSTP, or ISIS-SPB calculates a spanning tree for each active
39 topology, and prevents loops by exchanging information between neighboring Bridges to ensure either that
40 their topology calculations are consistent and each frame is consistently allocated to the same active
41 topology, or that frames are not relayed.

6.5.4.2 Loop mitigation

Each Bridge that participates in SPBM may mitigate loops. The properties of loop mitigation are listed below. Allowing temporary loops to form for unicast traffic and using loop mitigation provides for faster response to topology changes. Loop mitigation is accomplished if a unicast frame can only be forwarded by the Bridges as follows:

- a) From at most one reception Port to at most one transmission Port; and
- b) To a different transmission Port from the reception Port;
and
- c) Each Bridge does not relay, from any reception Port, a frame whose source address and VID are the same as that used by any entity within the Bridge; and
- d) Each Bridge does not transmit, through any Port, a frame whose destination address and VID are the same as that used by any entity within the Bridge;
and
- e) All LANs in the network provide only point-to-point connectivity.

Loop mitigation limits the effect of loops since any loop in the active topology cannot include either the source or destination of the frame, nor can the loop cause multiple copies of the initial frame to be delivered to the destination, or to or from any LAN in the network that is not part of the loop.

Loop mitigation does not require that neighboring Bridges assign a given frame to the same active topology; or even that the reception and transmission Ports for that frame have been derived from the same topology calculation within a given Bridge. While this standard specifies the calculation of paths that are reverse path congruent (Clause 3) to support bidirectional communication, that property has no bearing on loop mitigation. As a consequence, the Dynamic Filtering Entries that are used for active topology enforcement (8.6.1) and loop mitigation with SPBM (27.4.1) can be updated one at a time after ISIS-SPB has completed a link state calculation or during the course of that calculation.

6.5.5 Transit delay

The MAC Service introduces a frame transit delay that is dependent on the particular media and MAC method employed. Frame transit delay is the elapsed time between an MA_UNITDATA.request primitive and the corresponding MA_UNITDATA.indication primitive. Elapsed time values are calculated only on SDUs that are successfully transferred.

Since the MAC Service is provided at an abstract interface within an end station, it is not possible to specify precisely the total frame transit delay. It is, however, possible to measure those components of delay associated with media access and with transmission and reception; and the transit delay introduced by an intermediate system, in this case a Bridge, can be measured.

The minimum additional transit delay introduced by a Bridge is the time taken to receive a frame plus that taken to access the media onto which the frame is to be relayed. Note that the frame is completely received before it is relayed as the Frame Check Sequence (FCS) is to be calculated and the frame discarded if in error.

6.5.6 Frame lifetime

The MAC Service ensures that an upper bound to the transit delay is experienced for a particular instance of communication. This maximum frame lifetime is necessary to ensure the correct operation of higher layer protocols. The additional transit delay introduced by a Bridge is discussed in 6.5.5.

1 To enforce the maximum frame lifetime, a Bridge may be required to discard frames. Since the information
2 provided by the MAC Sublayer to a Bridge does not include the transit delay already experienced by any
3 particular frame, Bridges discard frames to enforce a maximum delay in each Bridge.

4 The value of the maximum Bridge transit delay is based on both the maximum delays imposed by all
5 Bridges in the network and the desired maximum frame lifetime. A recommended and an absolute
6 maximum value are specified in Table 6-1.

Table 6-1—Bridge transit delay

Parameter	Recommended value	Absolute maximum
Maximum Bridge transit delay	1.0s	4.0s

7 6.5.7 Undetected frame error rate

8 The MAC Service introduces a very low undetected frame error rate in transmitted frames. Undetected
9 errors are protected against by the use of an FCS that is appended to the frame by the MAC Sublayer of the
10 source station prior to transmission, and checked by the destination station on reception.

11 The FCS calculated for a given SDU is dependent on the MAC method employed. It is therefore necessary
12 to recalculate the FCS within a Bridge providing a relay function between IEEE 802 LAN MACs of
13 dissimilar types if differences in the method of calculation and/or the coverage of the FCS, or changes to the
14 data that is within the coverage of the FCS, would lead to a different FCS being calculated for the SDU by
15 the two MAC methods. This introduces the possibility of additional undetected errors arising from the
16 operation of a Bridge. For frames relayed between LANs of the same MAC type, the Bridge shall not
17 introduce an undetected frame error rate greater than that which would have been achieved by preserving the
18 FCS.

19 NOTE—Application of the techniques described in Annex O allows an implementation to achieve an arbitrarily small
20 increase in undetected frame error rate, even in cases where the data that is within the coverage of the FCS is changed.

21 6.5.8 Maximum Service Data Unit Size

22 The Maximum Service Data Unit Size that can be supported by an IEEE 802 LAN varies with the MAC
23 method and its associated parameters (e.g., speed, electrical characteristics). It may be constrained by the
24 owner of the LAN. The Maximum Service Data Unit Size supported by a Bridge between two LANs is the
25 smaller of that supported by the LANs. No attempt is made by a Bridge to relay a frame to a LAN that does
26 not support the size of SDU conveyed by that frame.

27 6.5.9 Priority

28 The MAC Service includes priority as a QoS parameter. MA_UNITDATA.request primitives with a high
29 priority may be given precedence over other request primitives made at the same station, or at other stations
30 attached to the same LAN and can give rise to earlier MA_UNITDATA.indication primitives.

31 The MAC Sublayer maps the requested priority onto the priorities supported by the individual MAC
32 method. The requested priority may be conveyed to the destination station.

33 The transmission delay experienced by a frame in a Bridge can be managed by associating a priority with the
34 frame.

1 The transmission delay comprises:

- 2 a) A queuing delay until the frame becomes first in line for transmission on the Port, in accordance
3 with the procedure for selecting frames for transmission described in 8.6.8.
- 4 b) The access delay for transmission of the frame.

5 Queuing delays can be managed using priority. Access delays can be managed using priority in MAC
6 methods that support more than one priority.

7 The priority associated with a frame can be signaled by means of the priority signaling mechanisms inherent
8 in some IEEE 802 LAN MAC types. Since not all IEEE 802 LAN MAC types are able to signal the priority
9 associated with a frame, VLAN Bridges regenerate priority based on a combination of signaled information
10 and configuration information held in the Bridge.

11 The Bridge maps priority values onto one or more traffic classes; Bridges that support more than one traffic
12 class are able to support expedited classes of traffic. The Forwarding Process, 8.6, describes the use of
13 priority and traffic classes in Bridges. Given the constraints placed on frame misordering in a Bridge, as
14 expressed in 6.5.3, the mappings of priority and traffic class are static.

15 NOTE 1—The term “traffic class,” as used in this standard, is used only in the context of the operation of the priority
16 handling and queuing functions of the Forwarding Process, as described in 8.6. Any other meanings attached to this term
17 in other contexts do not apply to the use of the term in this standard.

18 The ability to signal priority in IEEE 802 LANs, coupled with a consistent approach to the mapping of
19 priority to traffic classes, and of priority to the priority requested from the individual LAN or supporting
20 service, allows consistent use of priority information to be made, according to the capabilities of the Bridges
21 and MAC methods that are involved in the transmission path.

22 NOTE 2—This standard defines a frame format and associated procedures that can be used to carry priority information
23 across LAN MAC types that are not able to signal priority.

24 Under normal circumstances, priority is not modified in transit through the relay function of a Bridge;
25 however, there may be some circumstances where it is desirable for management purposes to control how
26 priority is propagated. The Priority Regeneration Table (Table 6-4) provides the ability to map incoming
27 priority values on a per-Port basis, under management control. In its default state, this table provides an
28 identity mapping from priority values to Regenerated priority values; i.e., by default, the Regenerated
29 priority is identical to the incoming priority.

30 **6.5.10 Throughput**

31 The total throughput provided by a network can be significantly greater than that provided by an equivalent
32 single LAN. Bridges may localize traffic within the network by filtering frames. Filtering services available
33 in bridged networks are described in 6.16.

34 The throughput between end stations on individual LANs, communicating through a Bridge, can be lowered
35 by frame discard in the Bridge due to the inability to transmit at the required rate on the LAN forming the
36 path to the destination for an extended period.

1 6.6 Internal Sublayer Service (ISS)

2 The service and control primitives and parameters, the status parameters, and the point-to-point parameters
3 of the ISS are specified in IEEE Std 802.1AC.

4 6.7 Support of the ISS by specific MAC procedures

5 Support of the ISS by MAC Entities that use specific IEEE 802 media access methods is specified in
6 IEEE Std 802.1AC.

7 6.7.1 Support of the ISS by IEEE Std 802.3 (Ethernet)

8 Mapping between M_CONTROL.requests/indications and IEEE 802.3
9 MA_CONTROL.requests/indications is performed as specified in IEEE Std 802.1AC. If the MAC supports
10 the MAC Merge sublayer specified in IEEE Std 802.3, then PFC M_CONTROL.requests are mapped onto
11 the MAC control interface associated with the express MAC (eMAC).

12 If frame preemption (6.7.2) is supported on a Port, then the IEEE 802.3 MAC provides the following two
13 MAC service interfaces:

- 14 a) A preemptable MAC (pMAC) service interface
- 15 b) An express MAC (eMAC) service interface

16 For priority values that are identified in the frame preemption status table (6.7.2) as *preemptable*, frames that
17 are selected for transmission shall be transmitted using the pMAC service instance, and for priority values
18 that are identified in the frame preemption status table as *express*, frames that are selected for transmission
19 shall be transmitted using the eMAC service instance.

20 In all other respects, the Port behaves as if it is supported by a single MAC service interface. In particular, all
21 frames received by the Port are treated as if they were received on a single MAC service interface regardless
22 of whether they were received on the eMAC service interface or the pMAC service interface, except with
23 respect to frame preemption.

24 If the value of the holdRequest managed object (12.30.1.5) transitions from *release (2)* to *hold (1)*, a
25 MM_CTL.request(hold_req) primitive is issued to the underlying IEEE 802.3 MAC, with a hold_req
26 parameter value of HOLD, as described in IEEE Std 802.3. If the value of the holdRequest managed object
27 (12.30.1.5) transitions from *hold (1)* to *release (2)*, a MM_CTL.request(hold_req) primitive is issued to the
28 underlying IEEE 802.3 MAC, with a hold_req parameter value of RELEASE.

29 NOTE—This additional material will be moved to IEEE Std 802.1AC in a future revision.

30 6.7.2 Frame preemption

31 If the Port supports frame preemption, then a value of frame preemption status is assigned to each value of
32 priority via a *frame preemption status table*. The possible values of frame preemption status are *express* or
33 *preemptable*.

34 The frame preemption status table can be changed by management as described in 12.30.1.1. The default
35 value of frame preemption status is *express* for all priority values.

6.8 Enhanced Internal Sublayer Service (EISS)

The EISS is derived from the ISS (IEEE Std 802.1AC) by augmenting that specification with elements necessary to the operation of the tagging and untagging functions of a VLAN Bridge (3.306). Within the attached end station, these elements can be considered either to be below the MAC Service boundary, and pertinent only to the operation of the service provider, or to be local matters not forming part of the peer-to-peer nature of the MAC Service.

The EISS provides the same service status and point-to-point parameters as the ISS (IEEE Std 802.1AC).

6.8.1 Service primitives

The unit-data primitives that define this service are:

```
EM_UNITDATA.indication    (  
    destination_address,  
    source_address,  
    mac_service_data_unit,  
    priority,  
    drop_eligible,  
    vlan_identifier,  
    frame_check_sequence,  
    service_access_point_identifier,  
    connection_identifier,  
    flow_hash,  
    time_to_live  
)  
  
EM_UNITDATA.request      (  
    destination_address,  
    source_address,  
    mac_service_data_unit,  
    priority,  
    drop_eligible,  
    vlan_identifier,  
    frame_check_sequence,  
    service_access_point_identifier,  
    connection_identifier,  
    flow_hash,  
    time_to_live  
)
```

The **destination_address**, **source_address**, **mac_service_data_unit**, **priority**, **drop_eligible**, **service_access_point_identifier**, **connection_identifier**, and **frame_check_sequence** parameters are as defined for the ISS.

NOTE—Some of the functions supporting the EISS may result in changes to the **mac_service_data_unit** or other parameters used to construct a frame. The original FCS associated with a frame is invalidated if there are changes to any fields of the frame, if fields are added or removed, or if bit ordering or other aspects of the frame encoding have changed. An invalid FCS is signaled in the EISS by an unspecified value in the **frame_check_sequence** parameter. This signals the need for the FCS to be regenerated according to the normal procedures for the transmitting MAC. Two options for regenerating the FCS under these circumstances are discussed in Annex O.

The **vlan_identifier** parameter carries the VID.

The **flow_hash** parameter either is null or carries a value that may be used by the forwarding process in selecting a single forwarding port from a set of available forwarding ports (8.6.3). Transmission order is maintained for all frames from a source to a destination with the same flow hash value. A flow hash value may be retrieved from an F-TAG or otherwise derived from the contents of a service primitive (44.2.2). A flow hash is not required by the MAC Relay Entity, and successful filtering can be provided without this additional information; however, it is required for flow filtering services supported by SPBM with ECMP. Any protocol entity in the interface stack that does not specify use of the flow_hash assigns the flow_hash value (if any) supplied with a request from the user of the protocol entity (or with an indication from the provider of the service used by the protocol entity) to the flow_hash on associated requests made (or indications generated) by the protocol entity.

The **time_to_live** parameter either is null or carries a value that may be used to filter frames that have reached their limit of permitted network hops. A time_to_live value may be retrieved from an F-TAG in the mac_service_data_unit and processed as described in 44.2.2. Any protocol entity in the interface stack that does not specify use of the time_to_live assigns the time_to_live value (if any) supplied with a request from the user of the protocol entity (or with an indication from the provider of the service used by the protocol entity) to the time_to_live on associated requests made (or indications generated) by the protocol entity.

6.8.2 Status parameters

The EISS also makes available the **MAC_Enabled** and **MAC_Operational** status parameters that reflect the operational state and administrative controls over each instance of the service provided. The values of these parameters are mapped directly from the values made available by the ISS (IEEE Std 802.1AC).

6.8.3 Point-to-point parameters

The EISS also makes available the **operPointToPointMAC** and **adminPointToPointMAC** status parameters that reflect the point-to-point status of each instance of the service provided and provide administrative control over the use of that information. The values of these parameters are mapped directly from the values made available by the ISS (IEEE Std 802.1AC).

6.8.4 Control primitives and parameters

The EISS provides two control primitives, an EM_CONTROL.request and an EM_CONTROL.indication, and their associated parameters. The values of these parameters are mapped directly to/from the values made available by the corresponding ISS primitives (11.4 of IEEE Std 802.1AC-2016 [B10]).

The EM_CONTROL.request primitive has the form:

```
EM_CONTROL.request      (
destination_address
opcode
request_operand_list
)
```

The EM_CONTROL.indication primitive has the form:

```
EM_CONTROL.indication   (
opcode
indication_operand_list
)
```

6.9 Support of the EISS

The EISS is supported by tagging and detagging functions that in turn use the ISS (IEEE Std 802.1AC). Any given instance of the EISS shall be supported by using one but not both of the following VLAN tag types:

- a) C-VLAN tag (C-TAG) or
- b) S-VLAN tag (S-TAG)

selected as specified in 9.5.

Each Bridge Port shall support the following parameters for use by these functions:

- c) An Acceptable Frame Types parameter with at least one of the following values:
 - 1) *Admit Only VLAN-tagged frames*
 - 2) *Admit Only Untagged and Priority-tagged frames*
 - 3) *Admit All frames*
- d) A PVID parameter for Port-based VLAN classification

Each Bridge Port may support the following parameters:

- e) A VID Set for Port-and-Protocol-based classification (6.12)
- f) A VID translation table
- g) An Egress VID translation table

All three values for Acceptable Frame Types may be supported; if so, they shall be configurable using the management operations defined in Clause 12, and the default shall be *Admit All Frames*. A frame is treated as Untagged if the initial octets of the `mac_service_data_unit` parameter do not contain a VLAN tag of the type used to support the EISS (9.5), as Priority-tagged if the value contained in the VID field of the VLAN tag is zero, and as VLAN-tagged if the value is nonzero.

NOTE 1—The VID translation tables are for use only at the edge of administrative or control protocol domains, e.g., Network-Network Interfaces, SPB domains, or Root and Leaf interfaces to a rooted-multipoint service, as described in this standard. Control protocols that are designed for operation through ports where VID translation is permitted (e.g., MSTP, MVRP, CFM) explicitly compensate for the translation of VID values carried within their PDUs. Successful operation and maintenance of Bridged Networks depends on the fact that VID values are not translated at other ports or at ports within the operational span of control protocols that do not recognize translation.

The PVID and VID Set shall contain valid VID values (Table 9-2) and may be configured by management. If they have not been explicitly configured, the PVID shall assume the value of the default PVID defined in Table 9-2 and the VID Set shall be empty.

NOTE 2—The default behavior of a Bridge that supports Port-and-Protocol-based classification is the same as that of a Bridge that supports only Port-based classification, since all the Protocol Group Identifiers in the VID Set for each Port assign the same VID as the PVID.

The VID translation table and Egress VID translation table provide a mapping between “local VID” values and “relay VID” values. The local VID is the value encoded in a VLAN tag header in the initial octets of the `mac_service_data_unit` parameter of an ISS service primitive (`M_UNITDATA.indication` or `M_UNITDATA.request`). The relay VID is the value contained in the `vlan_identifier` parameter of an EISS service primitive (`EM_UNITDATA.indication` or `EM_UNITDATA.request`). The VID translation table may be supported by itself, or in combination with the Egress VID translation table. When only the VID translation table is supported it is used for bidirectional mapping between the local VID and the relay VID in both indication and request service primitives. This enforces a symmetric one-to-one translation (i.e., if local VID A maps to relay VID B then relay VID B will map to local VID A). When both tables are supported the VID translation table is used only for mapping the local VID to a relay VID in indication service primitives, and the Egress VID translation table is used for mapping the relay VID to a local VID in request service primitives. This allows asymmetric and many-to-one translations (e.g., for translating to and from SPVIDs at the edge of an SPT Region). Both tables are configurable by management, and the default configuration maps each value of the local VID to the same value for the relay VID, and vice versa. For Bridge Ports at the boundary of an SPBV SPT Region the tables are dynamically modified by the ISIS-SPB.

6.9.1 Data indications

On receipt of an M_UNITDATA.indication primitive from the ISS, the received frame is discarded if:

- a) The initial octets of the mac_service_data_unit do not contain a valid VLAN tag header (9.3) of the type used to support the EISS (9.5), and the Acceptable Frame Types is *Admit Only VLAN-tagged frames*; or
- b) The initial octets of the mac_service_data_unit contain a valid VLAN tag header (9.3) of the type used to support the EISS (9.5), and:
 - 1) The VID value is FFF, reserved in Table 9-2 for implementation use; or
 - 2) The VID translation table is supported and the translated VID value (relay VID) is FFF (indicating local discard); or
 - 3) The VID value is 000 (indicating Priority-tagged), and the Acceptable Frame Types is *Admit Only VLAN-tagged frames*; or
 - 4) The VID value is in the range 001-FFE, and the Acceptable Frame Types is *Admit Only Untagged and Priority-tagged frames*.

Otherwise, an EM_UNITDATA.indication primitive is invoked, with parameter values determined as follows:

The **destination_address**, **source_address**, and **frame_check_sequence** parameters carry values equal to the corresponding parameters in the received data indication. The values of the remaining parameters are affected by the contents of the tag header if the frame contains a VLAN tag of the type supported by this instance of the EISS.

The value of the **mac_service_data_unit** parameter is as follows:

- c) If the frame is tagged, then the value used is equal to the value of the received mac_service_data_unit following removal of the tag header.
- d) Otherwise, the value used is equal to the value of the received mac_service_data_unit.

The value of the **vlan_identifier** parameter is as follows:

- e) The value contained in the VID field, optionally translated using the VID translation table, if the frame is VLAN-tagged.
- f) The value of the PVID for the Port, if Port-and-Protocol-based VLAN classification is not supported and the frame is untagged, Priority-tagged, or the VID translation table is supported and the translated VID value (relay VID) is zero.
- g) As determined by Port-and-Protocol-based VLAN classification (6.12) if that capability is implemented and the frame is untagged or Priority-tagged or the VID translation table is supported and the translated VID value (relay VID) is zero.

The value of the **drop_eligible** and **priority** parameters are determined as follows:

- h) If the frame is tagged, the value of the drop_eligible parameter and the received priority value are decoded from the tag header as described in 6.9.3. Otherwise,
- i) The received priority value and the drop_eligible parameter value are the values in the M_UNITDATA.indication.
- j) The value of the priority parameter is then regenerated from the received priority, as specified in 6.9.4.

6.9.2 Data requests

On invocation of an EM_UNITDATA.request primitive by a user of the EISS, the frame is discarded if the Egress VID translation table is supported and the translated VID value (local VID) is FFF (reserved in Table 9-2). Otherwise, an M-UNITDATA.request primitive is invoked, with parameter values as follows:

The **destination_address**, **source_address**, **drop_eligible**, and **priority** parameters carry values equal to the corresponding parameters in the received data request.

Unless the Bridge Port is a member of the untagged set (8.8.2) for the VID identified by the **vlan_identifier** parameter, then a tag header, formatted as necessary for the destination MAC type, is inserted as the initial octets of the **mac_service_data_unit** parameter. The values of the **vlan_identifier** (optionally translated using the VID translation table or Egress VID translation table), **priority**, and **drop_eligible** parameters are used to determine the contents of the tag header, in accordance with Clause 9.

If the Bridge Port is a member of the untagged set (8.8.2) for the VID identified by the **vlan_identifier** parameter, no tag header is inserted.

The remaining octets of the **mac_service_data_unit** parameter are those accompanying the EISS-request. If the data request is a consequence of relaying a frame and the MAC type of the Port differs from that used to receive the frame, they are modified, if necessary, in accordance with the procedures described in ISO/IEC 11802-5, IETF RFC 1042, and IETF RFC 1390.

The value of the **frame_check_sequence** parameter is determined as follows:

- a) If the **frame_check_sequence** parameter received in the data request either is unspecified or still carries a valid value, then that value is used.
- b) Otherwise, the value used is either unspecified or derived from the received FCS information by modification to take account of the conditions that have caused it to become invalid.

6.9.3 Priority Code Point encoding

The **priority** and **drop_eligible** parameters are encoded in the PCP field of the VLAN tag using the Priority Code Point Encoding Table for the Port, and they are decoded from the PCP using the Priority Code Point Decoding Table. For each Port, the Priority Code Point Encoding Table has 16 entries, corresponding to each of the possible combinations of the eight possible values of priority (0 through 7) with the two possible values of **drop_eligible** (True or False). For each Port, the Priority Code Point Decoding Table has 8 entries, corresponding to each of the possible PCP values.

Alternative values for each table are specified as rows in Table 6-2 and Table 6-3, with each alternative labeled by the number of distinct priorities that can be communicated, and the number of these for which drop precedence can be communicated. For example, the table entries 6P2D allow six distinct priorities with drop precedence for two. The default values (8P0D) specify a PCP value equal to the priority value and do not support communication of drop precedence in the PCP field. The combination of the **priority** and **drop_eligible** parameters is shown by the priority alone if **drop_eligible** is False, and by the priority followed by the letters “DE” if **drop_eligible** is True.

If the VLAN tag is an S-TAG, then Table 6-2 and Table 6-3 shall be supported. If the VLAN tag is a C-TAG, then the 8P0D row of each table shall be supported, and the remaining rows may be supported.

NOTE—To avoid potential misordering of frames, all Ports on a LAN should select the same row of Table 6-2 and Table 6-3. Interoperability considerations with Bridges supporting only the 8P0D row are discussed in I.8.

Table 6-2—Priority Code Point encoding

priority drop_eligible		7	7DE	6	6DE	5	5DE	4	4DE	3	3DE	2	2DE	1	1DE	0	0DE
PCP	8P0D (default)	7	7	6	6	5	5	4	4	3	3	2	2	1	1	0	0
	7P1D	7	7	6	6	5	4	5	4	3	3	2	2	1	1	0	0
	6P2D	7	7	6	6	5	4	5	4	3	2	3	2	1	1	0	0
	5P3D	7	7	6	6	5	4	5	4	3	2	3	2	1	0	1	0

Table 6-3—Priority Code Point decoding

PCP		7	6	5	4	3	2	1	0
priority drop_eligible	8P0D (default)	7	6	5	4	3	2	1	0
	7P1D	7	6	4	4DE	3	2	1	0
	6P2D	7	6	4	4DE	2	2DE	1	0
	5P3D	7	6	4	4DE	2	2DE	0	0DE

1 The values in the Priority Code Point Encoding Table and Priority Code Point Decoding Table may be
2 modified by management, as described in Clause 12. If this capability is provided, the value of the table
3 entries shall be independently settable for each Port. The values shall be constrained as follows:

- 4 a) For any two priority values with the same drop_eligible value, the lower priority shall not map to a
5 higher PCP than the higher priority.
- 6 b) The lower of any two PCP values shall not map to a higher priority than the higher PCP.
- 7 c) With the exception of values 0 and 1, any priority value combined with drop_eligible True shall not
8 map to a higher PCP than the same priority value combined with drop_eligible False.
- 9 d) With the exception of values 0 and 1, any PCP value shall not map to a priority value combined with
10 drop_eligible True if a lower PCP maps to the same priority value combined with drop_eligible
11 False.

12 The values may be further constrained, but if the tables can be modified, the default values shown in
13 Table 6-2 and Table 6-3, and the alternative sets of values in Table 6-2 and Table 6-3 shall be settable.

14 The drop_eligible parameter may also be encoded in and decoded from the Drop Eligible Indicator (DEI) in
15 the VLAN tag. Use of the DEI allows the VLAN tag to convey eight distinct priorities, each with a DEI. If
16 this capability is provided, it shall be independently manageable for each Port by means of a Use_DEI
17 parameter. If the Use_DEI is True for the Port, the drop_eligible parameter is encoded in the DEI of
18 transmitted frames, and the drop_eligible parameter shall be True for a received frame if the DEI is set in the
19 VLAN tag or the Priority Code Point Decoding Table indicates drop_eligible True for the received PCP
20 value. If the Use_DEI parameter is False, the DEI shall be transmitted as zero and ignored on receipt. The
21 default value of the Use_DEI parameter is False.

6.9.4 Regenerating priority

The priority of each received frame is regenerated using priority information contained in the frame and the Priority Regeneration Table for the reception Port. For each reception Port, the Priority Regeneration Table has eight entries, corresponding to the eight possible values of priority (0 through 7). Each entry specifies, for the given value of received priority, the corresponding regenerated value.

For untagged frames, the priority parameter signaled in the corresponding M_UNITDATA.indication is taken to be the received priority. For tagged frames, the priority signaled in the PCP field of the tag header is taken to be the received priority.

NOTE 1—IEEE 802 LAN technologies signal a maximum of eight priority values. Annex I further explains the use of priority values and how they map to traffic classes.

Table 6-4 specifies default regenerated priority values for each of the eight possible values of the received priority. These default values shall be used as the initial values of the corresponding entries of the Priority Regeneration Table for each Port.

Table 6-4—Priority regeneration

Received priority	Default regenerated priority	Range
0	0	0–7
1	1	0–7
2	2	0–7
3	3	0–7
4	4	0–7
5	5	0–7
6	6	0–7
7	7	0–7

The values in the Priority Regeneration Table may be modified by management, as described in Clause 12. If this capability is provided, the value of the table entries shall be independently settable for each reception Port and for each value of received priority, and the Bridge shall have the capability to use the full range of values in the parameter ranges specified in Table 6-4.

NOTE 2—The regeneration and mapping of priority within the Bridge should be consistent with the end-to-end significance of that priority across the rest of the Bridged Network. The regenerated priority value is used:

— Via the traffic class table (8.6.6) to determine the traffic class for a given outbound Port, and

— Via fixed, MAC type-specific mappings (IEEE Std 802.1AC) to determine the access priority that will be used for certain media access methods.

In Bridges that support SRP, an SRP domain boundary port priority regeneration override table is maintained for each reception Port. This table associates a priority value with each SR class supported by the Bridge, and specifies the priority value to be used as the regenerated priority, in preference to the value in the Priority Regeneration table, when the SRPdomainBoundaryPort parameter (35.1.4) for that SR class has the value TRUE. Table 6-5 specifies the default values for priority and regenerated priority for two SR classes, SR class A and SR class B. In cases where only SR class B is supported, only the default value shown in the table for SR Class B is used.

Table 6-5—Default SRP domain boundary port priority regeneration override values

SR class	Default priority	Default regenerated priority for SRP domain boundary Ports	Range
A	3	0	0–7
B	2	0	0–7

The priority values contained in the SRP domain boundary port priority regeneration override table may be modified by management, as described in Clause 12. If this capability is provided, the value of the table entries shall be independently settable for each reception Port and for each supported SR class, and the Bridge shall have the capability to use the full range of values in the parameter ranges specified in Table 6-5.

NOTE 3—The default values specified for the SRP domain boundary port Priority Regeneration Table map inbound PCP values onto the next lowest priority that is available, leaving the remaining priority values unchanged. The consequence of this remapping is that traffic entering an SRP domain that carries the priority value associated with that domain is forwarded by the Bridge at the domain boundary using a lower priority. There is no means of mapping these priority values back to their original values as frames exit from an SRP domain. The default mappings are based on the assumption that normally, two SR classes (A and B) are used to support traffic for which bandwidth has been reserved; if only the lower class (B) is supported, then the default override value for the unsupported A class does not apply.

NOTE 4—The ability for the priority regeneration table to be freely modified by management means that it is possible to create mappings that result in reserved and nonreserved traffic sharing the same priority. This will have unpredictable effects upon the behavior of the network with respect to the worst-case latency that can be assumed for reserved traffic.

6.10 Support of the ISS/EISS by PIPs

Each PIP on a BEB uses a single ISS-SAP (IEEE Std 802.1AC) and the I-TAG (9.5) to provide a separate Bridge Port (the VIP) for each backbone service instance configured for the PIP. Each VIP supports either the VLAN-unaware MAC Relay Entity of a T-component or the VLAN-aware MAC Relay Entity of an I-component. A single VIP ISS-SAP is provided to a T-component. An I-component uses an EISS-SAP, supported by its own instance of the functions specified in 6.9 with the S-TAG (9.5), for each VIP ISS-SAP. See Figure 6-2.

The EISS of each Virtual Instance Port (VIP-EISS) shall be supported by an instance of the functions of 6.9 using the S-TAG type (9.5). The ISS of all of the Virtual Instance Ports (VIP-ISS) shall be supported by a single instance of the functions specified in this subclause using the I-TAG type (9.5).

NOTE 1—Higher Layer Entities may interface to any VIP and/or the PIP at the Service Access Points (SAPs) represented by any VIP-ISS and/or the PIP-ISS using the Bridge Port Connectivity specified in 8.5.1. Protocol shims such as CFM MPs may interface at any VIP-EISS using the EISS Multiplex Entity specified in 6.17, or at the PIP-ISS using the Backbone Service Instance Multiplex Entity specified in 6.18. Figure 26-2 shows a PIP with both Bridge Port Connectivity and CFM shims.

Each PIP shall have an individual Backbone Media Access Control (B-MAC) address referred to as the PIP MAC address (26.4) for use by the functions specified in this subclause.

Each VIP shall support the following parameters for use by these functions:

- a) A VIP-ISID parameter for the I-SID of the backbone service instance associated with this VIP
- b) A Default Backbone Destination
- c) An adminPointToPointMAC
- d) An enableConnectionIdentifier

The VIP-ISID shall contain a valid I-SID value (9.7), which is configured by management.

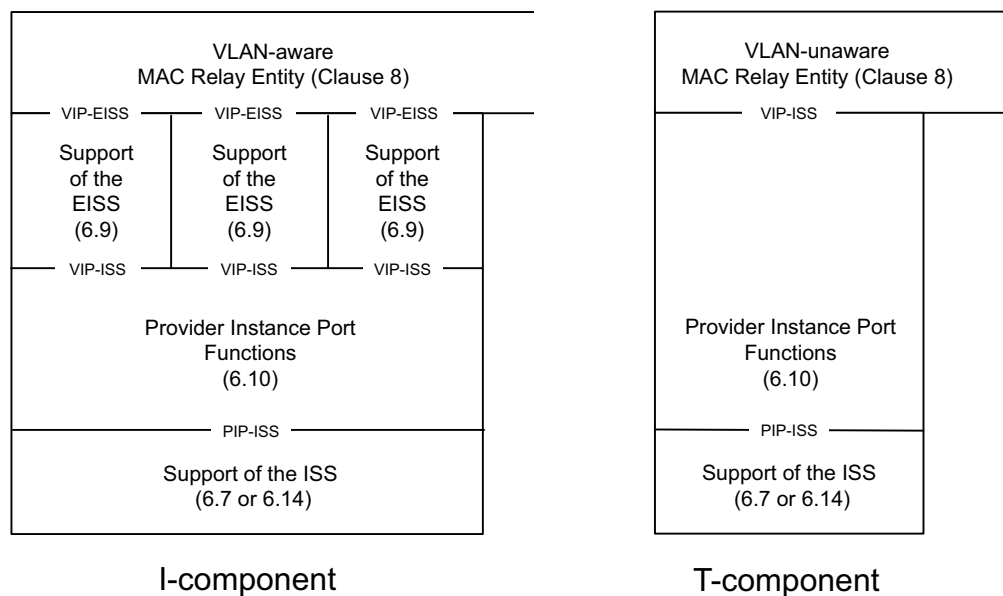


Figure 6-2—Provider Instance Ports (PIPs)

1 The Default Backbone Destination parameter contains a MAC address to be used in the destination_address
2 parameter of an M_UNITDATA.request at the PIP-ISS when a backbone destination address cannot be
3 derived from the connection_identifier parameter of the M_UNITDATA.request from the VIP-ISS. The
4 default value of the Default Backbone Destination is the Backbone Service Instance Group address for the
5 VIP-ISID (generated by concatenating the Provider Backbone Bridge (PBB) organizationally unique
6 identifier (OUI) with the VIP-ISID value as specified in 26.4).

7 The adminPointToPointMAC parameter of the VIP-ISS reflects the point-to-point status of the backbone
8 service instance associated with the VIP. The default value of the adminPointToPointMAC parameter is
9 ForceFalse. The value may be configured by management to ForceTrue when instantiating a VIP for a
10 point-to-point backbone service instance. A value of ForceFalse or Auto results in a operPointToPointMAC
11 value of FALSE; a value of ForceTrue results in a operPointToPointMAC value of TRUE. Whenever the
12 operPointToPointMAC parameter transitions to FALSE, the value for the Default Backbone Destination
13 parameter is set to the Backbone Service Instance Group address for the VIP-ISID. When the
14 operPointToPointMAC parameter is TRUE, the Default Backbone Destination parameter is modified by
15 subsequent M_UNITDATA.indications as specified in 6.10.1 (and described in 26.4.1).

16 The enableConnectionIdentifier parameter allows the VIP to use the connection_identifier parameter to
17 learn associations between a B-MAC address and a Customer Media Access Control (C-MAC) address. The
18 default value is TRUE. This parameter should be configured to FALSE at the root node of a
19 point-to-multipoint TESI.

20 NOTE 2—There is a 1:1 relationship between a given value of the connection_identifier and a B-MAC address. This
21 level of indirection is provided to allow the use of the connection_identifier parameter for other purposes by other types
22 of Bridge Ports. The relationship between a given connection_identifier value and a B-MAC address is maintained as
23 long as any FDB entry contains this value for the connection_identifier. No ageing mechanism other than that specified
24 for Dynamic Filtering Entries is implied.

6.10.1 Data indications

On receipt of an M_UNITDATA.indication primitive from the PIP-ISS, if the PIP is congestion aware (5.4.1.4) and the initial octets of the mac_service_data_unit contain a valid CNM encapsulation, the received frame is processed according to 32.16. If the PIP is congestion isolation aware (5.4.7) and the initial octets of the mac_service_data_unit contain a valid CIM encapsulation (49.4.3), the received frame is processed according to 49.4.2.6. Otherwise, the received frame shall be discarded if:

- a) The initial octets of the mac_service_data_unit do not contain a valid I-TAG header; or
- b) The Use Customer addresses (UCA) bit in the I-TAG is not zero; or
- c) The Res2 field in the I-TAG is not zero; or
- d) The I-SID value in the I-TAG does not match the VIP-ISID parameter for one of the VIPs supported by this PIP; or
- e) The destination_address parameter of the received M_UNITDATA.indication primitive does not match one of the following:
 - 1) The PIP MAC address; or
 - 2) A Backbone Service Instance Group address; or
 - 3) The broadcast address.

NOTE 1—The generation of I-tagged frames specified in 6.10.2 and the allowed manipulation of backbone addresses specified in 6.11 never result in an I-tagged frame with a broadcast destination address. The broadcast address is included in the previous list on the principle that the broadcast address is a valid address at any reception port by definition.

The received frame may be discarded if:

- f) The source_address parameter of the received M_UNITDATA.indication primitive is the PIP MAC address.

Otherwise, an M_UNITDATA.indication primitive is invoked at the VIP-ISS with a VIP-ISID value matching the I-SID in the I-TAG. The parameter values of the indication are determined as follows:

The **destination_address** parameter contains the value of the encapsulated C-DA field of the I-TAG.

The **source_address** parameter contains the value of the encapsulated C-SA field of the I-TAG.

If enableConnectionIdentifier is TRUE, then the **connection_identifier** parameter references the value of the source_address parameter (B-SA) of the M_UNITDATA.indication primitive received from the PIP-ISS; otherwise, the connection_identifier parameter is null. The connection_identifier parameter in the M_UNITDATA.indication primitive received from the PIP-ISS is ignored.

NOTE 2—The connection_identifier has only local significance meaning that it is used only at a specific PIP. The only requirement imposed by this subclause is that the connection_identifier uniquely identifies an individual B-MAC address. Whether the connection_identifier takes the value of that address, or a value from which that address can be determined, is an implementation decision that is not externally observable.

If the value of the operPointToPointMAC parameter of the VIP-ISS is TRUE, then the Default Backbone Destination parameter is set to the value of the source_address parameter of the M_UNITDATA.indication primitive.

The **service_access_point_identifier** parameter contains a value that uniquely identifies the VIP within the I-component. This is a local value that is used only within the I-component, and is used by the relay functions (Clause 8) as the port number of the VIP. The service_access_point_identifier parameter received in the M_UNITDATA.indication primitive is ignored.

The value of the **drop_eligible** and **priority** parameters are decoded from the I-PCP and I-DEI fields of the I-TAG as described in 6.10.3.

1 The value of the **mac_service_data_unit** parameter is the value of the received **mac_service_data_unit**
2 following the removal of the I-TAG.

3 The value of the **frame_check_sequence** parameter is either unspecified or derived from the received FCS
4 information modified to take into account the changes to the frame.

5 **6.10.2 Data requests**

6 On invocation of an **M_UNITDATA.request** primitive at any VIP-ISS, an **M_UNITDATA.request** primitive
7 is invoked at the PIP-ISS, with parameter values as follows:

8 If **enableConnectionIdentifier** is **TRUE**, the **connection_identifier** is not null, and the **connection_identifier**
9 references an address retained by the PIP, then the value for the **destination_address** is the address
10 referenced by the **connection_identifier**. Otherwise, the value for the **destination_address** is the contents of
11 the Default Backbone Destination parameter of the VIP.

12 The value for the **source_address** is the PIP MAC address.

13 The **priority** and **drop_eligible** parameters carry the same value as the corresponding parameter in the
14 VIP-ISS request.

15 The **connection_identifier** parameter is **NULL**.

16 The **mac_service_data_unit** parameter is constructed from the **mac_service_data_unit** of the
17 **M_UNITDATA.request** primitive from the VIP-ISS by prepending an I-TAG as follows:

- 18 a) The encapsulated C-SA field is the value of the **source_address** parameter of the VIP-ISS request.
- 19 b) The encapsulated C-DA field is the value of the **destination_address** parameter of the VIP-ISS
20 request.
- 21 c) The UCA, Res1, and Res2 fields are all zero.
- 22 d) The I-SID field is the value contained in the VIP-ISID parameter.
- 23 e) The I-PCP and I-DEI fields are encoded from the **priority** and **drop_eligible** parameters of the
24 VIP-ISS request as specified in 6.10.3.

25 The value of the **frame_check_sequence** parameter is either unspecified or derived from the received FCS
26 information modified to take into account the changes to the frame.

27 **6.10.3 Priority Code Point encoding**

28 The encoding (and decoding) of the I-PCP and I-DEI fields from (to) the **priority** and **drop_eligible**
29 parameters is accomplished using a Priority Code Point Encoding Table (and Priority Code Point Decoding
30 Table) as described in 6.9.3.

6.11 Support of the EISS by CBPs

The functions specified in this subclause replace the functions specified in 6.9 when the EISS is supported by a CBP in a BEB (Clause 25). See Figure 6-3. The functions specified in this subclause use a single instance of the ISS (IEEE Std 802.1AC) and provide a single instance of the EISS. The EISS shall be supported by functions using both the S-TAG and the I-TAG as specified in 9.5.

NOTE 1—Although the CBP does not transmit or receive S-tagged frames, it supports the S-TAG type in the sense that it is a port on an S-VLAN component and the `vlan_identifier` parameter contains an S-VID.

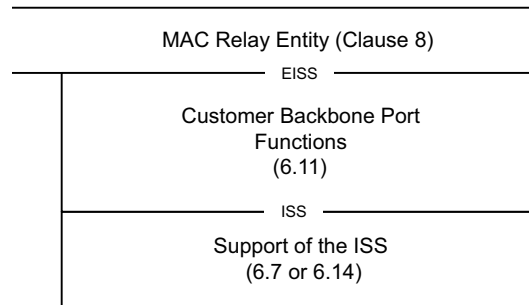


Figure 6-3—B-Component CBP

NOTE 2—Higher Layer Entities may interface to the CBP at the SAP represented by the ISS using the Bridge Port Connectivity specified in 8.5.1. Protocol shims such as CFM MPs may interface at the EISS using the EISS Multiplex Entity specified in 6.17, or at the ISS using the Backbone Service Instance Multiplex Entity specified in 6.18. Figure 26-2 shows a CBP with both Bridge Port Connectivity and CFM shims.

The CBP shall support the following parameters for use by these functions:

- a) A PVID parameter for Port-based B-VLAN classification.
The PVID shall contain a valid VID value (Table 9-2) and may be configured by management. If it has not been explicitly configured, the PVID shall assume the value of the default PVID defined in Table 9-2.
- b) A Backbone Service Instance table.
The Backbone Service Instance table is configurable by management and has one entry for each backbone service instance supported on the CBP. The Backbone Service Instance table is used to filter frames containing an I-SID value that is not configured on this CBP. It may also be used to provide backbone service instance-specific assignment of a B-VID, to translate I-SID values, and/or to translate backbone destination addresses. Entries in the Backbone Service Instance table may be set automatically by the protection switching procedures described in 26.10.3.4. The Backbone Service Instance table shall contain the following field for each entry:
- c) Backbone-SID (I-SID). This contains the identifier used in the backbone network for the backbone service instance.

The Backbone Service Instance table may also contain one or more of the following additional fields for each entry:

- d) B-VID (Backbone VLAN Identifier). If the B-VID field is supported, it shall contain a valid VID value (Table 9-2) for that backbone service instance. The default value is the same as the PVID.
- e) Local-SID (Local Service Instance Identifier). This field is used to perform a bidirectional 1:1 mapping between the Backbone-SID (contained in the I-TAG in the `mac_service_data_unit` at the EISS) and the Local-SID (contained in the I-TAG in the `mac_service_data_unit` at the ISS). The default value of the Local-SID field is the value of the Backbone-SID field.

- 1 f) Default Backbone Destination. If this field is present, it is used as the `destination_address` in the
2 EISS indication when the `destination_address` parameter of the ISS indication is the Backbone
3 Service Instance Group address. It allows the backbone provider to replace the Backbone Service
4 Instance Group address with an individual or group address to limit the distribution of the frame
5 within a B-VLAN. The default value is the Backbone Service Instance Group address.
6 NOTE 3—If a CBP is associated with a TESI the Default Backbone Destination field and the B-VID field is
7 supported.

8 6.11.1 Data indications

9 On receipt of an `M_UNITDATA.indication` primitive from the ISS, the received frame shall be discarded if:

- 10 a) The initial octets of the `mac_service_data_unit` do not contain a valid I-TAG; or
11 b) The `Res2` field in the I-TAG is not zero; or
12 c) The Local-SID field of the Backbone Service Instance table is supported and the I-SID value in the
13 I-TAG does not match the Local-SID value in any entry in the Backbone Service Instance table; or
14 d) The Local-SID field of the Backbone Service Instance table is not supported and the I-SID value in
15 the I-TAG does not match the Backbone-SID value in any entry in the Backbone Service Instance
16 table.

17 Otherwise, an `EM_UNITDATA.indication` primitive is invoked at the EISS with parameter values
18 determined using one entry in the Backbone Service Instance table. If the Local-SID field of the Backbone
19 Service Instance table is supported then the entry with a Local-SID value that matches the I-SID value in the
20 I-TAG is selected. Otherwise, the entry with a Backbone-SID value that matches the I-SID value in the
21 I-TAG is selected. The parameter values are then determined as follows:

22 The `mac_service_data_unit` parameter is determined as follows:

- 23 e) If the Local-SID field of the Backbone Service Instance table is supported, then value of the
24 `mac_service_data_unit` parameter is the value of the `mac_service_data_unit` parameter of the ISS
25 indication modified by replacing the value of the I-SID field in the I-TAG with the value of
26 Backbone-SID field in the selected Backbone Service Instance table entry.
27 f) Otherwise, the `mac_service_data_unit` parameter contains the same value as the
28 `mac_service_data_unit` parameter of the ISS indication.

29 The `destination_address` parameter is determined as follows:

- 30 g) If the `destination_address` in the ISS indication is the Backbone Service Instance Group address and
31 the Default Backbone Destination field of the Backbone Service Instance table is supported, then the
32 `destination_address` parameter is the Default Backbone Destination value in the selected Backbone
33 Service Instance table entry.
34 h) Otherwise, if the `destination_address` in the ISS indication is the Backbone Service Instance Group
35 address and the Local-SID field of the Backbone Service Instance table is supported, then the
36 `destination_address` parameter is the Backbone Service Instance Group address constructed using
37 the Backbone-SID value in the selected Backbone Service Instance table entry.
38 i) Otherwise, the `destination_address` parameter contains the same value as the `destination_address`
39 parameter of the ISS indication.

40 The `source_address` parameter contains the same value as the `source_address` parameter of the ISS
41 indication.

42 The value of the `vlan_identifier` parameter is determined as follows:

- 43 j) If the B-VID field of the Backbone Service Instance table is supported, then the `vlan_identifier`
44 parameter contains the B-VID value in the selected Backbone Service Instance table entry.
45 k) Otherwise, the `vlan_identifier` parameter contains the value of the PVID parameter.

1 The value of the **drop_eligible** and **priority** parameters are determined as follows:

- 2 l) The value of the **drop_eligible** parameter and the received_priority value are decoded from the
- 3 I-DEI and the I-PCP fields of the I-TAG in the mac_service_data_unit as described in 6.11.3.
- 4 m) The value of the **priority** parameter is then regenerated from the received_priority value, as
- 5 specified for the received_priority in 6.11.4.

6 The value of the **frame_check_sequence** parameter is either unspecified or derived from the received FCS

7 information modified as necessary to take into account changes to the frame.

8 6.11.2 Data requests

9 On invocation of an EM_UNITDATA.request primitive by a user of the EISS, the frame shall be discarded

10 if:

- 11 a) The initial octets of the mac_service_data_unit do not contain a valid I-TAG.
- 12 b) The Res2 field in the I-TAG is not zero.
- 13 c) The I-SID value in the I-TAG does not match the Backbone-SID value in any of the entries in the
- 14 Backbone Service Instance table.

15 Otherwise, an M_UNITDATA.request primitive is invoked, with parameter values determined using the

16 Backbone Service Instance table entry with a Backbone-SID value that matches the I-SID value in the

17 I-TAG. The parameter values are determined as follows:

18 The value of the **mac_service_data_unit** parameter is determined as follows:

- 19 d) If the Local-SID field of the Backbone Service Instance table is supported, then
- 20 **mac_service_data_unit** parameter contains the value of the mac_service_data_unit parameter of
- 21 the EISS request modified by replacing the I-SID in the I-TAG with the Local-SID value in the
- 22 selected Backbone Service Instance table entry.
- 23 e) Otherwise, the **mac_service_data_unit** parameter contains the same value as the
- 24 mac_service_data_unit parameter of the EISS request.

25 The **destination_address** parameter is determined as follows:

- 26 f) If the Local-SID field of the Backbone Service Instance table is supported and the
- 27 destination_address in the EISS request is the address of this CBP, or is a group address, then the
- 28 destination_address parameter is the Backbone Service Instance Group address constructed from the
- 29 Local-SID value in the selected Backbone Service Instance table entry.
- 30 g) If the Local-SID field of the Backbone Service Instance table is not supported and the
- 31 destination_address in the EISS request is the address of this CBP, or is a group address, then the
- 32 destination_address parameter is the Backbone Service Instance Group address constructed from the
- 33 Backbone-SID value in the selected Backbone Service Instance table entry.
- 34 h) Otherwise, the **destination_address** parameter contains the same value as the destination_address
- 35 parameter of the EISS request.

36 The **source_address** parameter contains the same value as the source_address parameter of the EISS

37 request.

38 The **priority** and **drop_eligible** parameters carry the same value as the corresponding parameter in the EISS

39 request.

40 The value of the **frame_check_sequence** parameter is either unspecified or derived from the received FCS

41 information modified to take into account any changes to the frame.

6.11.3 Priority Code Point decoding

The decoding of the I-PCP and I-DEI fields to the **priority** and **drop_eligible** parameters is accomplished using a Priority Code Point Decoding Table as described in 6.9.3.

6.11.4 Regenerating priority

The priority of received frames is regenerated using priority information contained in the I-PCP and the Priority Regeneration table for the reception Port. For each reception Port, the Priority Regeneration table has eight entries, corresponding to the eight possible values of priority (0 through 7). Each entry specifies, for the given value of received_priority, the corresponding regenerated value. The priority decoded from the I-PCP field of the I-TAG is taken to be the received_priority.

NOTE—IEEE 802 LAN technologies signal a maximum of eight priority values. Annex I further explains the use of priority values and how they map to traffic classes.

Table 6-4 specifies default regenerated priority values for each of the eight possible values of the received_priority. These default values shall be used as the initial values of the corresponding entries of the Priority Regeneration table for each Port. The table may be configured as described in 6.9.4.

6.12 Protocol VLAN classification

Each instance of the tagging and detagging functions that supports the EISS (6.9), and implements the optional Port-and-Protocol-based VLAN classification, shall implement a VID Set, each member of which associates values of a Protocol Group Identifier (6.12.2) with a VID. Each Untagged and Priority-tagged frame received is assigned a **vlan_identifier** equal to the VID Set value for the reception Port and the Protocol Group Identifier selected by matching the received frame with a Protocol Template. See Figure 6-4.

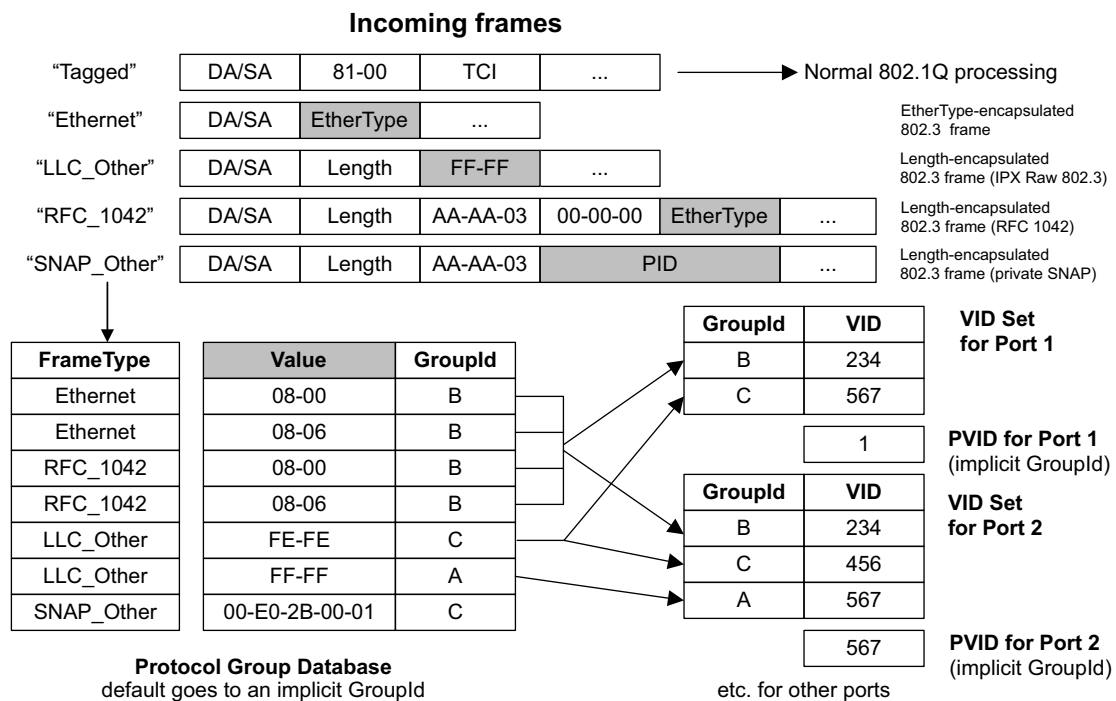


Figure 6-4—Example of operation of Port-and-Protocol-based classification

1 The **detagged_frame_type** parameter indicates the frame format. Its value is determined as follows:

- 2 a) If the frame is Untagged or Priority-tagged, this parameter is present and indicates the link-layer
3 encapsulation format of the *Detagged Frame*. The Detagged Frame of an Untagged Frame is the
4 frame itself. The Detagged Frame of a Tagged Frame or Priority-tagged Frame is the frame that
5 results from untagging the frame in accordance with the frame format described in Clause 9. The
6 value of **detagged_frame_type** is as follows:
 - 7 1) Ethernet, if the Detagged Frame uses EtherType-encapsulated IEEE 802.3 format.
 - 8 2) RFC_1042, if the Detagged Frame is of the format specified by 9.4 in IEEE Std 802™-2014
9 [B7] for the encoding of an IEEE 802.3 EtherType Field in an ISO/IEC 8802-2/SNAP header
10 (this supersedes the original definition, which appeared in IETF RFC 1042 (1988)).
 - 11 3) SNAP_Other, if the Detagged Frame contains an LLC UI PDU with DSAP and SSAP fields
12 equal to the LLC address reserved for SNAP and the 5-octet Subnetwork Access Protocol
13 (SNAP) Protocol Identifier (PID) value is not in either of the ranges used for RFC_1042 above.
 - 14 4) LLC_Other, if the Detagged Frame contains both a DSAP and an SSAP address field in the
15 positions specified by ISO/IEC 8802-2 Logical Link Control, but is not any of the formats
16 described for LLC frames above.
- 17 b) Else the parameter is not present.

18 The **EtherType** value is present if the **detagged_frame_type** parameter is present and has the value Ethernet
19 or RFC_1042. Its value is the value of the IEEE 802.3 Length/Type Field present in the Detagged Frame.
20 The value is determined as follows:

- 21 c) If the **detagged_frame_type** parameter is present and has the value Ethernet or RFC_1042, then this
22 parameter is present and has the value of the IEEE 802.3 EtherType Field present in the Detagged
23 Frame.
- 24 d) Else the parameter is not present.

25 The **llc_saps** parameter is present if the **detagged_frame_type** parameter is present and has the value
26 LLC_Other. Its value is determined as follows:

- 27 e) If the **detagged_frame_type** parameter is present and has the value LLC_Other, then this parameter
28 is present and its value is the pair of LLC ISO/IEC 8802-2 DSAP and SSAP address field values.
- 29 f) Else the parameter is not present.

30 NOTE 1—A frame that is encapsulated using values of hex FF/FF in the position where an LLC header is to be expected
31 (as defined by ISO/IEC 8802-2) is known as a “Novell IPX Raw” encapsulation. Such frames do not conform to
32 ISO/IEC 8802-2, in that they do not include some of the other required LLC fields. For the purposes of this standard,
33 they are treated as LLC_Other, regardless of whether they are legal LLC frames.

34 NOTE 2—Bridges are not required, for the purposes of this standard, to completely verify whether the format of frames
35 meets ISO/IEC 8802-2. They are required only to recognize the DSAP and SSAP fields of such frames.

36 The **snap_pid** parameter is present if the **detagged_frame_type** parameter is present and has the value
37 SNAP_Other. Its value is determined as follows:

- 38 g) If the **detagged_frame_type** parameter is present and has the value SNAP_Other, then the parameter
39 is present and its value is the contents of the 5 octets following the LLC header, i.e., the PID field.
- 40 h) Else the parameter is not present.

1 6.12.1 Protocol Templates

2 In a Bridge that supports Port-and-Protocol-based VLAN classification, a Protocol Template is a tuple that
3 specifies a protocol to be identified in received frames. A Protocol Template has one of the following
4 formats:

- 5 a) A value “Ethernet” and a 16-bit IEEE 802.3 EtherType Field value
- 6 b) A value “RFC_1042” and a 16-bit IEEE 802.3 EtherType Field value
- 7 c) A value “SNAP_Other” and a 40-bit PID value
- 8 d) A value “LLC_Other” and a pair of ISO/IEC 8802-2 LSAP values: DSAP and SSAP

9 A Protocol Template matches a frame if:

- 10 e) The frame’s `detagged_frame_type` is Ethernet, the Protocol Template is of type Ethernet, and the
11 frame’s IEEE 802.3 EtherType Field is equal to the value of the IEEE 802.3 EtherType Field of the
12 Protocol Template, or
- 13 f) The frame’s `detagged_frame_type` is RFC_1042, the Protocol Template is of type RFC_1042 and
14 the frame’s IEEE 802.3 EtherType Field is equal to the IEEE 802.3 EtherType Field of the Protocol
15 Template, or
- 16 g) The frame’s `detagged_frame_type` is SNAP_Other, the Protocol Template is of type SNAP_Other,
17 and the frame’s `snap_pid` is equal to the PID of the Protocol Template, or
- 18 h) The frame’s `detagged_frame_type` is LLC_Other, the Protocol Template is of type LLC_Other, and
19 the frame’s `llc_saps` matches the value of the DSAP and SSAP of the Protocol Template.

20 NOTE—If a port does not support Protocol Templates of the frame’s `detagged_frame_type`, then no match will occur.

21 6.12.2 Protocol Group Identifiers

22 A Bridge that supports Port-and-Protocol-based VLAN classification shall support Protocol Group
23 Identifiers.

24 A Protocol Group Identifier, shown as “Group Id” in Figure 6-4, designates a group of protocols that will be
25 associated with one member of the VID Set of a Port. The association of protocols into groups is established
26 by the contents of the Protocol Group Database, as described in 6.12.3. The identifier has scope only within
27 a single Bridge.

28 An implicit Protocol Group Identifier is assigned to frames that match none of the entries in the Protocol
29 Group Database. Therefore, every incoming frame can be assigned to a Protocol Group Identifier.

30 6.12.3 Protocol Group Database

31 A Bridge that supports Port-and-Protocol-based VLAN classification shall support a single Protocol Group
32 Database. The Protocol Group Database groups together a set of one or more Protocols by assigning them
33 the same Protocol Group Identifier (6.12.2). Each entry of the Protocol Group Database comprises the
34 following:

- 35 a) A Protocol Template
- 36 b) A Protocol Group Identifier

37 The Protocol Group Database specifies a mapping from Protocol Templates to Protocol Group Identifiers. If
38 two entries of the Protocol Group Database contain different Protocol Group Identifiers, then their Protocol
39 Templates must also be different.

40 The entries of the Protocol Group Database may be configured by management. A Bridge that supports
41 Port-and-Protocol-based VLAN classification shall support at least one of the Protocol Template formats.

1 An implicit Protocol Group Database entry exists that matches all frames. This entry is invoked for frames
2 that do not match the template of any of the other entries. It references an implicit Protocol Group Identifier
3 that selects the PVID on each port. In this way, it is ensured that all incoming frames are matched by a
4 Protocol Group Identifier and, hence, are assigned to a VID.

5 NOTE—If there are no entries in the Protocol Group Database, then the frame relay behavior of this Bridge is identical
6 to the frame relay behavior of a Bridge having the same number of Ports that supports only Port-based VLAN
7 classification.

8 6.13 Support of the ISS for attachment to a PBN

9 This standard specifies both Customer Bridges (3.59) and Provider Bridges (3.208). The operation of
10 Provider Bridges and PBNs is, by design, largely transparent to Customer Bridges and Customer Bridged
11 Networks (CBNs). Figure 15-1 illustrates the relationship of the service provided by the MAC Sublayer
12 functionality of Provider Bridges to that used by a Customer Bridge. This subclause enables a mechanism
13 for a Customer Bridge attached to a PBN to request priority handling of frames.

14 The functions specified in this subclause provide the ISS to the MAC Relay Entity of a Customer Bridge by
15 making use of the underlying ISS provided by the Media Access Method Convergence function as shown in
16 Figure 6-5. These functions shall be one of the following:

- 17 a) When Service Access Priority Selection is disabled, the functions shall be null; i.e., each
18 M_UNITDATA.request received from the provided ISS results in an M_UNITDATA.request with
19 identical parameters presented to the underlying ISS, and each M_UNITDATA.indication received
20 from the underlying ISS results in an M_UNITDATA.indication with identical parameters presented
21 to the provided ISS.
- 22 b) When Service Access Priority Selection is enabled, the mac_service_data_unit in each
23 M_UNITDATA.request is Priority-tagged with an S-TAG header when presented to the underlying
24 ISS, and if an S-TAG header is present in the mac_service_data_unit parameter in an
25 M_UNITDATA.indication received from the underlying ISS, the S-TAG header is removed before
26 the M_UNITDATA.indication is presented to the provided ISS.

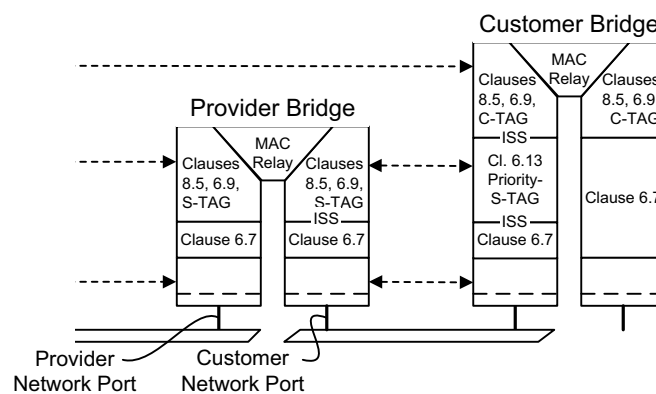


Figure 6-5—Service access priority selection

27 Specification of a null function allows a Customer Bridge without additional functionality to connect to a
28 PBN.

29 Specification of the Service Access Priority Selection function allows the Customer Bridge Port to request
30 priority handling of the frame from a Port-based service interface (15.3) to a PBN on the basis of the priority
31 assigned within the CBN, while allowing for differences in cost and service level ascribed to individual
32 values of priority within the service provider and customer networks.

6.13.1 Data requests

For each M_UNITDATA.request received from the provided ISS, an M_UNITDATA.request is presented to the underlying ISS with the following parameters:

The destination_address, source_address, and priority parameters are unchanged.

A tag header, formatted as necessary for the destination MAC type, is inserted as the initial octets of the mac_service_data_unit parameter. The Tag Protocol Identification is the Service VLAN Tag EtherType (9.5). The Tag Control Information (TCI) contains a null VID field, and PCP and DEI fields constructed as specified in 6.9.3 using a drop_eligible value of FALSE and a service_access_priority value derived from the priority parameter using Table 6-6.

Table 6-6—Service Access Priority

Received priority	Default service access priority	Range
0	0	0–7
1	1	0–7
2	2	0–7
3	3	0–7
4	4	0–7
5	5	0–7
6	6	0–7
7	7	0–7

Table 6-6 specifies default service_access_priority values for each of the eight possible values of the priority parameter in the M_UNITDATA.request received from the provided ISS. These default values shall be used as the initial values of the corresponding entries of the Service Access Priority Table for each Port supporting Service Access Priority Selection. The values in the table may be modified by management. If this capability is provided, the value of the table entries shall be independently settable for each Port and for each value of priority, and the Bridge shall have the capability to use the full range of the values in the parameter ranges specified in the table.

The frame_check_sequence parameter is either unspecified or contains a value modified from the received frame_check_sequence value taking into account the changes in the mac_service_data_unit parameter.

NOTE—The priority of the original service request made of the customer network can be carried transparently and unchanged across the PBN in the C-TAG.

6.13.2 Data indications

For each M_UNITDATA.indication received from the underlying ISS, an M_UNITDATA.request is presented to the supported ISS with the following parameters:

The destination_address and source_address parameters are unchanged.

1 If the initial octets of the `mac_service_data_unit` do not contain a tag header with the Service VLAN Tag
2 EtherType value (9.5), then the `mac_service_data_unit` and `frame_check_sequence` parameters are
3 unchanged. Otherwise,

- 4 a) The `mac_service_data_unit` is detagged by removing the octets of the S-TAG header,
- 5 b) The priority parameter is decoded from the PCP field (6.9.3),
- 6 c) The `drop_eligible` parameter is decoded from the Drop Eligible Indicator field (6.9.3), and
- 7 d) The `frame_check_sequence` parameter is either unspecified or contains a value modified from the
8 received `frame_check_sequence` value taking into account the changes in the `mac_service_data_unit`
9 parameter.

10 6.14 Support of the ISS within a system

11 A single system can comprise two or more instances of VLAN Bridge components connected by one or
12 more of their Ports. If those Ports are not otherwise accessible, the ISS may be provided by means outside
13 the scope of this specification. Figure 15-4 provides an example. Each instance of such an implementation
14 of the ISS shall support the MAC status and Point-to-point parameters. An `M_UNITDATA.request` at one of
15 the Ports connected to such an instance of the ISS shall result in `M_UNITDATA.indications` with identical
16 parameters at all other Ports connected to that instance.

17 For convenience of management, such instances of ISS provision are assigned the LAN MAC Type 802.1.
18 Management parameters common to MACs of this type are specified in Clause 12.

19 NOTE—The ISS can also be supported at Bridge Ports wholly contained within a system by any IEEE 802 LAN
20 technology, subject to the system requirements of the particular media access method.

21 6.15 Support of the ISS by additional technologies

22 The ISS may be supported by other technologies that provide either an IEEE 802 MAC Service or an
23 emulated IEEE 802 MAC Service. The technology is responsible for invoking an `M_UNITDATA.indication`
24 with appropriate parameters (IEEE Std 802.1AC) for each received frame, and transmitting a frame in
25 response to each `M_UNITDATA.request`. The `mac_service_data_unit` parameter in an
26 `M_UNITDATA.indication` may include a tag header of a type recognized by the functions using the EISS
27 (6.8). If multiplexing of multiple virtual instances of the MAC Service is provided, each virtual service must
28 be:

- 29 a) Identified by the VID field in a tag header of a type recognized by the functions using the EISS, or
- 30 b) Attached to separate instance of the ISS with no tag header in the `mac_service_data_unit`

31 (i.e., the only multiplexing recognized by the functions of the MAC Relay are those identified by a VID).

32 6.16 Filtering services in Bridged Networks

33 Bridges provide filtering services that support aspects of the maintenance of QoS—in particular, transit
34 delay, priority, and throughput. In addition, these services provide a degree of administrative control over the
35 propagation of particular MAC addresses in the network.

36 The services described are services in the most general sense; i.e., descriptions of functionality available to a
37 MAC Service user or an administrator to control and access filtering capabilities. The descriptions make no
38 assumptions about how the service might be realized. There are at least the following possibilities:

- 39 a) Use of existing protocols and mechanisms, defined in IEEE 802 standards and elsewhere.
- 40 b) Use of management functionality, either locally defined or via remote management protocols.
- 41 c) Other means, standardized or otherwise.

1 **6.16.1 Purpose(s) of filtering service provision**

2 Filtering services are provided for the purposes described in 6.16.1.1 and 6.16.1.2.

3 **6.16.1.1 Administrative control**

4 Filtering services provide administrative control over the use of particular source and destination addresses
5 in designated parts of the network. Such control allows network managers and administrators to limit the
6 extent of operation of network layer and other protocols that use individual and group MAC addresses by
7 establishing administrative boundaries across which specific MAC addresses are not forwarded.

8 **6.16.1.2 Throughput and end station load**

9 Filtering services increase the overall throughput of the network, and reduce the load placed on end stations
10 caused by the reception of frames that are destined for other end stations, by:

- 11 a) Limiting frames destined for specific MAC addresses to parts of the network that, to a high
12 probability, lie along a path between the source MAC address and the destination MAC address.
- 13 b) Reducing the extent of group addressed frames to those parts of the network that contain end
14 stations that are legitimate recipients of that traffic.
- 15 c) In Virtual Bridged Networks, reducing the extent of frames on specific VLANs to those parts of the
16 network that contain stations that are legitimate recipients of traffic on that VLAN.

17 NOTE—Some aspects of the filtering services described in this standard are dependent upon the active participation of
18 end stations. Where such participation is not possible, those aspects of the filtering services will be unavailable.

19 **6.16.2 Goals of filtering service provision**

20 The filtering services provided can be used to:

- 21 a) Allow the MAC Service provider to dynamically learn where the recipients of frames addressed to
22 individual MAC addresses are located.
- 23 b) Allow end stations that are the potential recipients of MAC frames destined for group MAC
24 addresses to dynamically indicate to the MAC Service provider which destination MAC address(es)
25 they wish to receive.
- 26 c) Exercise administrative control over the extent of propagation of specific MAC addresses.

27 **6.16.3 Users of filtering services**

28 The filtering services provided are available to the following users:

- 29 a) Network management and administration, for the purposes of applying administrative control.
30 Interactions between administrators of the network and the filtering service provider may be
31 achieved by local means or by means of explicit management mechanisms.
- 32 b) End stations, for the purposes of controlling the destination addresses that they will receive.
33 Interactions between end stations and the filtering service provider may be implicit, as is the case
34 with the Learning Process (8.7), or by explicit use of filtering service primitives.

35 **6.16.4 Basis of service**

36 Filtering in Bridged Networks relies on the establishment of filtering rules, and subsequent filtering
37 decisions, that are based on the value(s) contained in the Source MAC Address, Destination MAC Address,
38 or VID fields in MAC frames.

39 NOTE—The filtering services defined by this standard use source address learning, destination address, and VID
40 filtering.

1 6.16.5 Categories of service

2 Filtering services in Bridged Networks fall into the following categories:

- 3 a) *Basic Filtering Services.* These services are supported by the Forwarding Process (8.6) and by Static
4 Filtering Entries (8.8.1), Static VLAN Registration Entries (8.8.2), Dynamic Filtering Entries
5 (8.8.3), and Dynamic VLAN Registration Entries (8.8.5) in the FDB. The information contained in
6 the Dynamic Filtering Entries is maintained through the operation of the Learning Process (8.7),
7 while the information contained in the Dynamic VLAN Registration Entries is maintained through
8 the operation of MVRP (11.2) and MIRP (Clause 39).
- 9 b) *Extended Filtering Services.* These services are supported by the Forwarding Process (8.6), and the
10 Static Filtering Entries (8.8.1) and MAC Address Registration Entries (8.8.4) in the FDB. The
11 information contained in the MAC Address Registration Entries is maintained through the operation
12 of MMRP (10.9).

13 All Bridges shall support Basic Filtering Services and may support Extended Filtering Services.

14 6.16.6 Service configuration

15 In the absence of explicit information in the FDB, the behavior of the Forwarding Process with respect to the
16 forwarding or filtering of frames destined for group MAC addresses depends upon the categories of service
17 supported by the Bridge.

18 Basic Filtering Services support the filtering behavior required for regions of a Bridged Network in which
19 potential recipients of multicast frames exist, but where either the recipients or the Bridges are unable to
20 support the dynamic configuration of filtering information for those group MAC addresses, or the recipients
21 have a requirement to receive all traffic destined for group MAC addresses.

22 Extended Filtering Services support the filtering behavior required for regions of a network in which
23 potential recipients of multicast frames exist, and where both the potential recipients of frames and the
24 Bridges are able to support dynamic configuration of filtering information for group MAC addresses. In
25 order to integrate this extended filtering behavior with the needs of regions of the network that support only
26 Basic Filtering Services, Bridges that support Extended Filtering Services can be statically and dynamically
27 configured to modify their filtering behavior on a per-group MAC address basis, and also on the basis of the
28 overall filtering service provided by each outbound Port with regard to multicast frames. The latter
29 capability permits configuration of the Port's default forwarding or filtering behavior with regard to group
30 MAC addresses for which no specific static or dynamic filtering information has been configured.

31 Service configuration provides the ability to configure the overall filtering for the following cases:

- 32 a) Bridges that only implement Basic Filtering Services.
- 33 b) Bridges that support Extended Filtering Services in a heterogeneous environment, where some
34 equipment is unable to participate in Dynamic Multicast Filtering, or where some equipment (e.g.,
35 routers) has specific needs to see unfiltered traffic.
- 36 c) Bridges that support Extended Filtering Services in a homogeneous environment, where all
37 equipment is able to participate in Dynamic Multicast Filtering.

38 6.16.7 Service definition for Extended Filtering Services

39 The Filtering Services are described by means of service primitives that define particular types of interaction
40 between MAC Service users and the MAC Service provider across the MAC Service boundary. As these
41 interactions are not defined between peer entities, they are described simply in terms of service requests sent
42 from the MAC Service user to the MAC Service provider.

6.16.7.1 Dynamic registration and deregistration services

These services allow MAC Service users dynamic control over the set of destination MAC addresses that they will receive from the MAC Service provider, by:

- a) Registering/deregistering membership of specific Groups associated with those addresses.
- b) Registering/deregistering individual MAC address information.
- c) Registering/deregistering service requirements with regard to the overall forwarding/filtering behavior for Groups.

Provision of these services is achieved by means of MMRP and its associated procedures, as described in 10.9.

NOTE 1—These services can provide the MAC Service user with dynamic control over access to multicast data streams, for example, multiple video channels made available by a server using a different group MAC address for each channel. The ability to both register and deregister Group membership, coupled with the filtering action associated with the Group membership, limits the impact of such services on the bandwidth available in the network. These services can be used to control the reception of other categories of multicast traffic, for similar reasons.

REGISTER_MAC_ADDRESS (MAC_ADDRESS)

Indicates to the MAC Service provider that the MAC Service user wishes to receive frames containing the MAC address indicated in the MAC_ADDRESS parameter as the destination address. The MAC addresses that can be carried by this parameter do not include the following:

- d) Any of the Reserved Addresses identified in Table 8-1, Table 8-2, or Table 8-3.
- e) Any of the MRP Application addresses, as defined in Table 10-1.

DEREGISTER_MAC_ADDRESS (MAC_ADDRESS)

Indicates to the MAC Service provider that the MAC Service user no longer wishes to receive frames containing the MAC address indicated in the MAC_ADDRESS parameter as the destination address.

REGISTER_SERVICE_REQUIREMENT (REQUIREMENT_SPECIFICATION)

Indicates to the MAC Service provider that the MAC Service user has a requirement for any devices that support Extended Filtering Services to forward frames in the direction of the MAC Service user in accordance with the definition of the service requirement defined by the REQUIREMENT_SPECIFICATION parameter. The values that can be carried by this parameter are as follows:

- f) Forward All Groups.
- g) Forward Unregistered Groups.

DEREGISTER_SERVICE_REQUIREMENT (REQUIREMENT_SPECIFICATION)

Indicates to the MAC Service provider that the MAC Service user no longer has a requirement for any devices that support Extended Filtering Services to forward frames in the direction of the MAC Service user in accordance with the definition of the service requirement defined by the REQUIREMENT_SPECIFICATION parameter. The values that can be carried by this parameter are as follows:

- h) Forward All Groups.
- i) Forward Unregistered Groups.

The use of these services can result in the propagation of group MAC address and service requirement information across the spanning tree, affecting the contents of MAC Address Registration Entries (8.8.4) in Bridges and end stations, and thereby affecting the frame forwarding behavior of the Bridges and end stations with regard to multicast frames.

NOTE 2—If the EISS is supported, the Extended Filtering Service primitives issued by the MAC Service user should also include a VID parameter in order to identify the VLAN associated with the MAC_ADDRESS or REQUIREMENT_SPECIFICATION specified.

6.17 EISS Multiplex Entity

The EISS Multiplex Entity enables shims defined for the ISS to use the EISS. Figure 6-6 illustrates two EISS Multiplex Entities placed back-to-back.

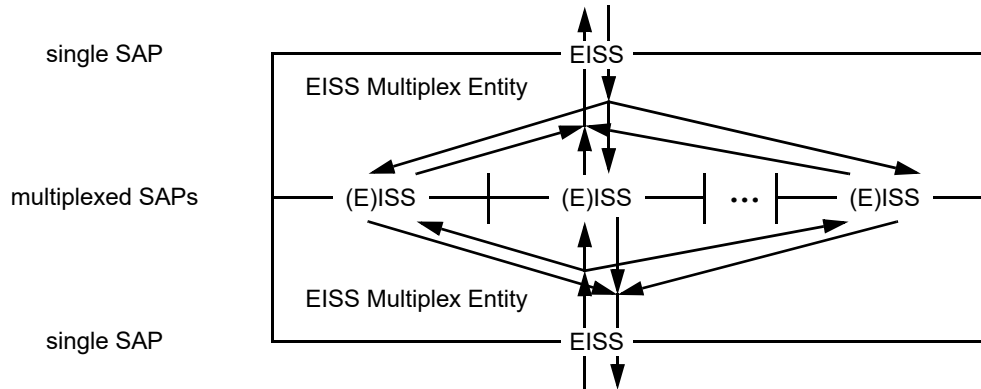


Figure 6-6—Two back-to-back EISS Multiplex Entities

An EISS Multiplex Entity has one EISS SAP, and a number of multiplexed SAPs, each either an ISS SAP or an EISS SAP. Each multiplexed ISS SAP is assigned to a single `vlan_identifier` value. Each multiplexed EISS SAP is assigned to one or more `vlan_identifier` values. Every `vlan_identifier` is assigned to some multiplexed SAP. Upon receiving a Request or Indication from its single EISS SAP, the EISS Multiplex Entity uses the `vlan_identifier` to select the corresponding one of its multiplexed SAPs to present the Request or Indication. Similarly, any Request or Indication received from a multiplexed SAP is presented to the single EISS SAP; the `vlan_identifier` parameter presented on the single EISS SAP is the one associated with the multiplexed ISS SAP, or the one obtained from the multiplexed EISS SAP.

Shims can be multiplexed in this fashion to separately serve multiple `vlan_identifiers`. Figure 22-1 illustrates CFM shims deployed in this manner.

The `MAC_Operational` status parameter (IEEE Std 802.1AC) presented to the uppermost EISS-SAP in Figure 6-6 is TRUE if and only if:

- a) The uppermost EISS-SAP's `MAC_Enabled` parameter is TRUE; and
- b) At least one of its multiplexed SAPs' `MAC_Operational` status parameters is TRUE.

The `MAC_Operational` status parameter of each of the lower EISS Multiplex Entity's multiplexed SAPs in Figure 6-6 is computed separately and is TRUE if and only if:

- c) The lowermost EISS-SAP's `MAC_Operational` parameter is TRUE; and
- d) That multiplexed SAP's `MAC_Enabled` parameter is TRUE.

6.18 Backbone Service Instance Multiplex Entity

The Backbone Service Instance Multiplex Entity allows shims defined for the ISS to be instantiated per backbone service instance at a SAP that supports multiple backbone service instances. Figure 6-7 illustrates two Backbone Service Instance Multiplex Entities placed back-to-back. A set of back-to-back Backbone Service Instance Multiplex Entities may be used at the ISS of a PIP or CBP to provide per Backbone Service Instance SAPs that support CFM shims at the Backbone Service Instance MD Level as shown in Figure 6-8 and the figures of 26.8.

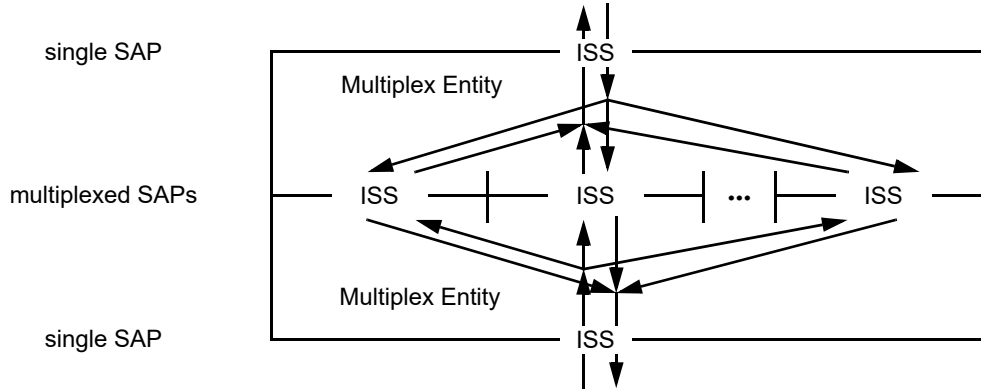


Figure 6-7—Two back-to-back Backbone Service Instance Multiplex Entities

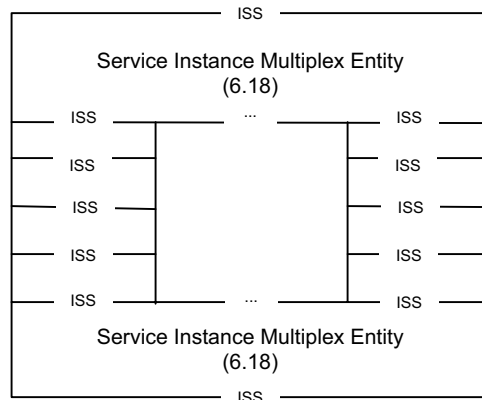


Figure 6-8—Backbone Service Instance Multiplex Entities with example CFM shims

NOTE 1—Figure 6-8 shows a representative placement of CFM shims between back-to-back Backbone Service Instance Multiplex Entities. This is not intended to imply any restriction that this is the only possible configuration of MEPs and MIPs.

A Backbone Service Instance Multiplex Entity has one ISS SAP that supports multiple backbone service instances, and a number of multiplexed ISS SAPs each supporting a single backbone service instance. Each multiplexed SAP has a single assigned I-SID value. Every I-SID value is assigned to a multiplexed SAP, and no I-SID value is assigned to more than one multiplexed SAP.

NOTE 2—There are implicitly 2^{24} multiplexed SAPs—one for each potential I-SID value. This does not require any implementation to support 2^{24} SAPs. The combined structure of back-to-back Backbone Service Instance Multiplex Entities is transparent to any frames with an I-SID value corresponding to a multiplexed SAP that does not have a protocol entity (e.g., CFM) instantiated at that SAP.

1 In the multiplexing direction, any Request or Indication received from a multiplexed SAP results in the
2 generation of a corresponding Request or Indication to the single ISS SAP, with an I-TAG header added to
3 the `mac_service_data_unit` as specified in 6.18.2. Similarly, in the demultiplexing direction any Request or
4 Indication received from the single ISS SAP results in the generation of a corresponding Request or
5 Indication at one of the multiplexed SAPs (determined by the I-SID contained in a I-TAG in the
6 `mac_service_data_unit`), with the I-TAG removed from the `mac_service_data_unit` as specified in 6.18.1.
7 When an I-TAG header with the UCA field containing a value of zero is removed, the C_DA and C_SA
8 fields of the I-TAG remain in the `mac_service_data_unit`. In this case, a new EtherType field is prepended so
9 the `mac_service_data_unit` still begins with a valid EtherType value. The Encapsulated Addresses EtherType
10 value in Table 6-7 is allocated for this purpose.

Table 6-7—Encapsulated Addresses EtherType

Name	Value
IEEE 802.1Q Encapsulated Addresses EtherType	89-10

11 NOTE 3—This EtherType value is used only internally at the Backbone Service Multiplex Entity and does not appear in
12 any frame on a LAN in the PBBN.

13 6.18.1 Demultiplexing direction

14 On receipt of an `M_UNITDATA.request` or `M_UNITDATA.indication` primitive from the single ISS SAP,
15 the received frame shall be discarded if:

- 16 a) The initial octets of the `mac_service_data_unit` do not contain a valid I-TAG; or
- 17 b) The Res2 field in the I-TAG is not zero.

18 Otherwise, an `M_UNITDATA.request` or `M_UNITDATA.indication` primitive is invoked at the multiplexed
19 ISS SAP corresponding to the I-SID value in the I-TAG. The parameter values are determined as follows:

20 The **connection_identifier** parameter contains the same value as in the received primitive.

21 The **destination_address** and **source_address** parameters are determined as follows:

- 22 c) If the UCA bit is zero, the **destination_address** and **source_address** parameters contain the
23 respective values in the received primitive.
- 24 d) Otherwise, the **destination_address** and **source_address** parameters contain the respective values
25 in the C-DA and C-SA fields of the I-TAG.

26 The value of the **drop_eligible** and **priority** parameters are decoded from the I-DEI and the I-PCP fields of
27 the I-TAG header in the `mac_service_data_unit` as described in 6.18.3.

28 The value of the **mac_service_data_unit** parameter is constructed from the value of the received
29 `mac_service_data_unit` modified as follows:

- 30 e) If the UCA field in the I-TAG is zero, then the I-PCP, I-DEI, UCA, Res1, Res2, and I-SID fields of
31 the I-TAG are removed and the Backbone Service Instance Tag EtherType value is replaced with the
32 Encapsulated Addresses EtherType value (Table 6-7).
- 33 f) Otherwise, the entire I-TAG is removed.

34 The value of the **frame_check_sequence** parameter is either unspecified or derived from the received FCS
35 information modified as necessary to take into account changes to the frame.

6.18.2 Multiplexing direction

On receipt of an M_UNITDATA.request or M_UNITDATA.indication primitive from any of the multiplexed ISS SAPs, an M_UNITDATA.request or M_UNITDATA.indication primitive is invoked at the single ISS SAP. The parameter values are determined as follows:

The **source_address**, **priority**, **drop_eligible**, and **connection_identifier** parameters contain the same value as in the received primitive.

The **destination_address** parameter is determined as follows:

- a) If the **destination_address** in the received primitive is a group address and the UCA bit is set, then the **destination_address** parameter contains the Backbone Service Instance Group address constructed from the I-SID corresponding to the multiplexed ISS SAP at which the primitive was received.
- b) Otherwise the **destination_address** parameter contains the same value as in the received primitive.

The value of the **mac_service_data_unit** parameter is equal to the value of the received **mac_service_data_unit** modified as follows:

- c) If the initial two octets of the **mac_service_data_unit** match the Encapsulated Addresses EtherType value, then these octets are replaced with the Backbone Service Instance Tag EtherType value, and the I-PCP, I-DEI, UCA, Res1, Res2, and I-SID fields of an I-TAG are inserted following this EtherType value. The UCA, Res1, and Res2 fields all contain a value of zero. The I-SID field contains the value of the I-SID corresponding to the multiplexed ISS SAP at which the primitive was received. The values of the I-PCP and I-DEI fields are encoded from the **drop_eligible** and **priority** parameters as specified in 6.18.3.
- d) If the initial two octets of the **mac_service_data_unit** do not match the Encapsulated Addresses EtherType value, then a complete I-TAG is prepended to the **mac_service_data_unit**. The UCA field contains a value of one. The Res1 and Res2 fields contain a value of zero. The C-DA and C-SA fields contain the **destination_address** and **source_address** from the received primitive, respectively. The I-SID field contains the value of the I-SID corresponding to the multiplexed ISS SAP at which the primitive was received. The values of the I-PCP and I-DEI fields are encoded from the **drop_eligible** and **priority** parameters as specified in 6.18.3.

The value of the **frame_check_sequence** parameter is either unspecified or derived from the received FCS information modified as necessary to take into account changes to the frame.

6.18.3 Priority Code Point encoding

The encoding (and decoding) of the I-PCP and I-DEI fields from (to) the **priority** and **drop_eligible** parameters is accomplished using a Priority Code Point Encoding Table (and Priority Code Point Decoding Table) as described in 6.9.3.

6.18.4 Status parameters

The MAC_Operational status parameter (IEEE Std 802.1AC) presented to the uppermost single ISS SAP in Figure 6-7 is TRUE if and only if:

- a) The uppermost single ISS SAP's MAC_Enabled parameter is TRUE; and
- b) At least one of its multiplexed SAPs' MAC_Operational status parameters is TRUE.

The MAC_Operational status parameter of each of the multiplexed SAPs in Figure 6-7 is computed separately, and is TRUE if and only if:

- c) The lowermost single ISS SAP's MAC_Operational parameter is TRUE; and
- d) That multiplexed SAP's MAC_Enabled parameter is TRUE.

6.19 TESI Multiplex Entity

The TESI Multiplex Entity allows shims defined for PBB-TE to be instantiated per TESI at a SAP that supports multiple TESI. Figure 6-9 illustrates two TESI Multiplex Entities, placed back-to-back. A set of back-to-back TESI Multiplex Entities may be used at the multiplexed (E)ISS SAPs that are associated with ESP-VIDs on a CBP to provide per TESI SAPs that support CFM shims for PBB-TE MAs as shown in Figure 26-8.

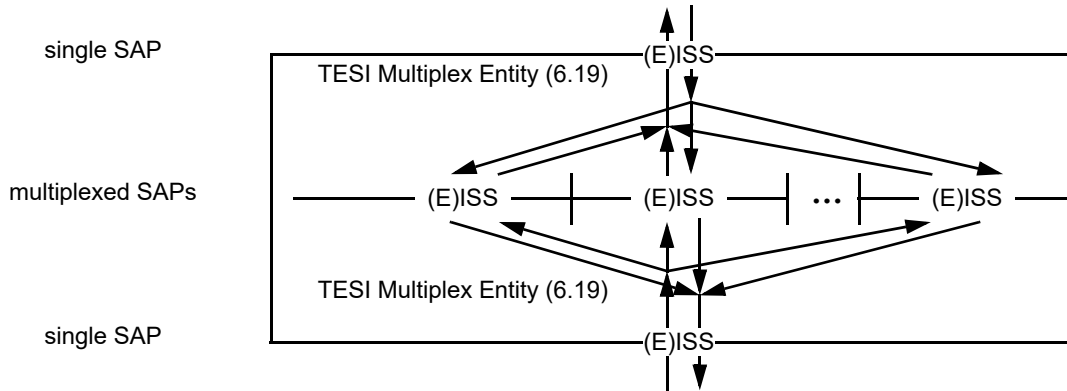


Figure 6-9—Two back-to-back Up and Down TESI Multiplex Entities

A TESI Multiplex Entity has one (E)ISS SAP that supports multiple TESI, and a number of multiplexed (E)ISS SAPs each supporting a single TESI. Each multiplexed SAP has destination_address, source_address, and vlan_identifier combinations assigned by the TESI Multiplex Entity. Every destination_address, source_address, and vlan_identifier combination can be assigned to a multiplexed SAP, and no destination_address, source_address, and vlan_identifier combination is assigned to more than one multiplexed SAP.

Upon receiving a Request or Indication from its single (E)ISS SAP, the TESI Multiplex Entity uses the destination_address, source_address, and vlan_identifier to select the corresponding one of its multiplexed SAPs to present the Request or Indication. The Request or Indication presented at the multiplexed SAPs has the same parameters as the original Request or Indication at the Single SAP. Similarly, any Request or Indication received from a multiplexed SAP is transparently presented to the single (E)ISS SAP.

The MAC_Operational status parameter (6.8.2) presented to the uppermost single (E)ISS SAP in Figure 6-9 is TRUE if and only if:

- a) The uppermost single (E)ISS SAP's MAC_Enabled parameter is TRUE; and
- b) At least one of its multiplexed SAPs' MAC_Operational status parameters is TRUE.

The MAC_Operational status parameter of each of the multiplexed SAPs in Figure 6-9 is computed separately, and is TRUE if and only if:

- c) The lowermost single (E)ISS SAP's MAC_Operational parameter is TRUE; and
- d) That multiplexed SAP's MAC_Enabled parameter is TRUE.

The MAC_Operational parameter of the passive SAP on a PBB-TE MEP is set to FALSE when the PBB-TE MEP declares an errorCCMdefect (20.21.3) or an xconCCMdefect (20.23.3), setting the MAC_Operational parameter of the associated multiplexed SAP on the TESI Multiplex Entity to FALSE. The MAC_Operational parameter of the passive SAP on a PBB-TE MEP is set back to TRUE when both defects are cleared, setting back to TRUE the MAC_Operational parameter of the associated multiplexed SAP.

6.20 Support of the ISS with signaled priority

The functions specified in this subclause comprise a shim (3.243, IEEE Std 802.1AC) that uses an ISS SAP supported by a MAC entity (IEEE Std 802.1AC) to provide explicitly signaled priority information to an ISS SAP that supports a VLAN-unaware MAC Relay (Figure 6-10). Signaled priority information is derived from a VLAN tag header (9.3), if such header is present in the `mac_service_data_unit`. Any given instance of the ISS shall be supported by recognizing one but not both of the following VLAN Tag EtherTypes:

- a) C-VLAN tag (C-TAG) or
- b) S-VLAN tag (S-TAG)

The VLAN Tag EtherType is selected as specified in 9.5.

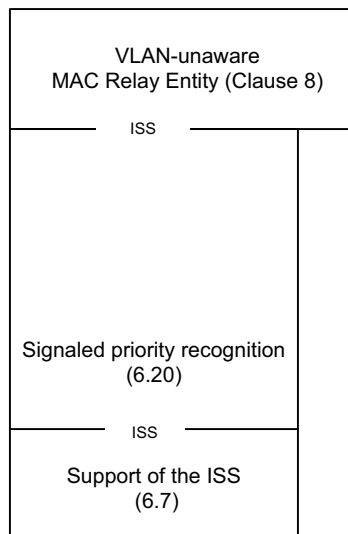


Figure 6-10—Supporting the ISS with signaled priority

6.20.1 Data indications

On receipt of an `M_UNITDATA.indication` primitive from the lower ISS SAP (Figure 6-10), an `M_UNITDATA.indication` primitive is invoked at the upper ISS SAP, with parameter values determined as follows:

The **`destination_address`**, **`source_address`**, **`connection_identifier`**, **`mac_service_data_unit`**, and **`frame_check_sequence`** parameters carry values equal to the corresponding parameters in the received data indication.

The value of the **`drop_eligible`** and **`priority`** parameters are determined as follows:

- a) If the initial octets of the `mac_service_data_unit` contain a valid VLAN tag header (9.3) of the type used to support the EISS (9.5), the value of the `drop_eligible` parameter and the received priority value are decoded from the tag header as described in 6.9.3. Otherwise,
- b) The received priority value and the `drop_eligible` parameter value are the values in the received data indication.
- c) The value of the `priority` parameter is then regenerated from the received priority, as specified in 6.9.4.

6.20.2 Data requests

On receipt of an M_UNITDATA.request primitive from the upper ISS SAP (Figure 6-10), an M-UNITDATA.request primitive with identical parameter values is invoked at the lower ISS SAP.

6.21 Infrastructure Segment Multiplex Entity

The Infrastructure Segment Multiplex Entity allows shims defined for Segment Endpoint Ports (SEPs) and Segment Intermediate Ports (SIPs) to be instantiated per Infrastructure Segment at a SAP that supports multiple Infrastructure Segments. Figure 6-11 illustrates two Infrastructure Segment Multiplex Entities, placed back-to-back. A set of back-to-back Infrastructure Segment Multiplex Entities may be used at the multiplexed (E)ISS SAPs that are associated with SMP-VIDs on a PNP to provide per Infrastructure Segment SAPs that support CFM shims for Infrastructure Segment MAs as shown in Figure 26-10.

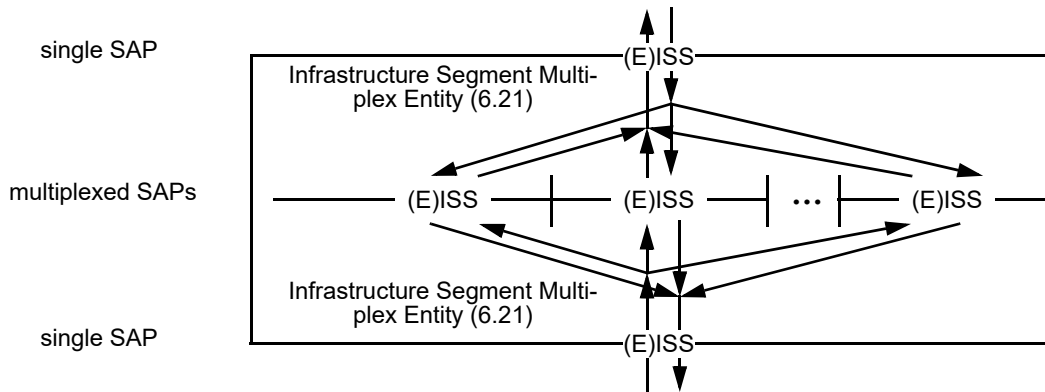


Figure 6-11—Two back-to-back Up and Down Infrastructure Segment Multiplex Entities

An Infrastructure Segment Multiplex Entity has one (E)ISS SAP that supports multiple Infrastructure Segment instances, and a number of multiplexed (E)ISS SAPs each supporting a single Infrastructure Segment instance. Each multiplexed SAP has destination_address, source_address, and vlan_identifier combinations assigned by the Infrastructure Segment Multiplex Entity. Every destination_address, source_address, and vlan_identifier combination can be assigned to a multiplexed SAP, and no destination_address, source_address, and vlan_identifier combination is assigned to more than one multiplexed SAP.

Upon receiving a Request or Indication from its single (E)ISS SAP, the Infrastructure Segment Multiplex Entity uses the destination_address, source_address, and vlan_identifier to select the corresponding one of its multiplexed SAPs to present the Request or Indication. The Request or Indication presented at the multiplexed SAPs has the same parameters as the original Request or Indication at the Single SAP. Similarly, any Request or Indication received from a multiplexed SAP is transparently presented to the single (E)ISS SAP.

The MAC_Operational status parameter (IEEE Std 802.1AC) presented to the uppermost single (E)ISS SAP in Figure 6-9 is TRUE if and only if:

- a) The uppermost single (E) ISS SAP's MAC_Enabled parameter is TRUE; and
- b) At least one of its multiplexed SAPs' MAC_Operational status parameters is TRUE.

The MAC_Operational status parameter of each of the multiplexed SAPs in Figure 6-9 is computed separately, and is TRUE if and only if:

- c) The lowermost single (E)ISS SAP's MAC_Operational parameter is TRUE; and
- d) That multiplexed SAP's MAC_Enabled parameter is TRUE.

1 The MAC_Operational parameter of the passive SAP on an Infrastructure Segment MEP is set to FALSE
2 when the Infrastructure Segment MEP declares an errorCCMdefect (20.21.3) or an xconCCMdefect
3 (20.23.3), setting the MAC_Operational parameter of the associated multiplexed SAP on the Infrastructure
4 Segment Multiplex Entity to FALSE. The MAC_Operational parameter of the passive SAP on an
5 Infrastructure Segment MEP is set back to TRUE when both defects are cleared, setting back to TRUE the
6 MAC_Operational parameter of the associated multiplexed SAP.

7 **6.22 PDU and protocol discrimination and media**

8 As described more fully in Clause 9 of IEEE Std 802-2014 [B7], the format of the data parameter passed to
9 or from a specific MAC procedure (see Figure 6-1 and 6.7) is identified using the first few octets of the
10 MSDU, by either source and destination LLC addresses or a two-octet EtherType.

11 Length/Type networks have a Length/Type field in the first two octets of the MSDU (e.g., IEEE Std 802.3).
12 Depending on the value of the Length/Type field (see IEEE Std 802.1AC), the Length/Type field is either an
13 EtherType or is a Length followed immediately by the LSAPs and Control octets.

14 Networks that do not use a Length/Type field in the MSDU (e.g., some applications of IEEE Std 802.11) can
15 implement a Media Access Method Dependent Convergence Function to perform any necessary translations
16 from Length/Type-encoded protocol identifiers. See Clause 13 of IEEE Std 802.1AC-2016 [B10].

17 NOTE—The encoding of tagged frames is clarified in G.3.

7. Principles of Virtual Bridged Network operation

This clause establishes the principles and a model of Virtual Bridged Network operation. It defines the context necessary for:

- a) The operation of individual VLAN Bridges (Clause 8);
- b) Their participation in the Spanning Tree Algorithm and Protocol (STP) (Clause 8 of IEEE Std 802.1D, 1998 Edition [B13]), Rapid Spanning Tree Algorithm and Protocol (RSTP) (Clause 13), Multiple Spanning Tree Algorithm and Protocol (MSTP) (Clause 13), Shortest Path Bridging (SPB) (Clause 27), and Path Control and Reservation (PCR) (Clause 45) protocols;
- c) The management of individual Bridges (Clause 12); and
- d) The management of VLAN Topology (Clause 11)

to support, preserve, and maintain the quality of the MAC Service as discussed in Clause 6.

7.1 Network overview

The operation of a Virtual Bridged Network, the Bridges, and the LANs that compose that network comprises:

- a) A physical topology comprising LANs, Bridges, and Bridge Ports. Each Bridge Port attaches a LAN to a Bridge and is capable of providing bidirectional connectivity for MAC user data frames. Each LAN is connected to every other LAN by a Bridge and zero or more other LANs and Bridges.
- b) Calculation of one or more active topologies, each a loop-free subset of the physical topology.
- c) Rules for the classification of MAC user data frames that allow each Bridge to allocate, directly or indirectly, each frame to one and only one active topology.
- d) Management control of the connectivity provided for differently classified data frames by the selected active topology.
- e) Implicit or explicit configuration of end station location information, identifying LANs with attached end stations that need to receive user data frames with a given destination address.
- f) Communication of end station location information to allow Bridges to restrict user data frames to LANs in the path provided to their destination(s) by the chosen active topology.

These elements and their interrelationships are illustrated in Figure 7-1.

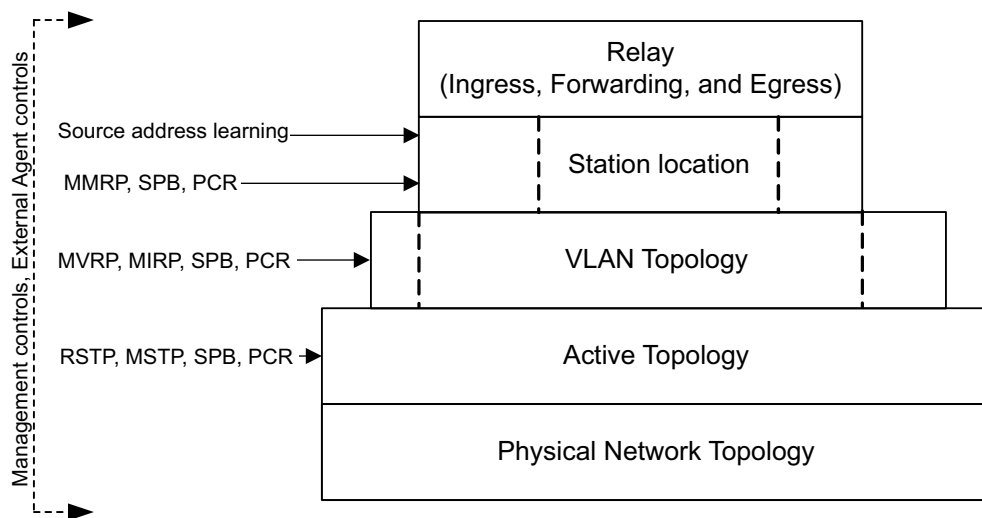


Figure 7-1—VLAN Bridging overview

1 NOTE 1—The physical and active topologies can be represented as bipartite graphs. Bridges and LANs are nodes in
2 these graphs (including LANs that are point-to-point links) and the Bridge Ports are edges.

3 NOTE 2—This standard applies the notion of a physical topology to media access control methods, like wireless, where
4 there is no tangible physical connection between a Bridge and an attached LAN. A Bridge Port models this association
5 just as for wired connectivity.

6 NOTE 3—Each of the active topologies [see item b)] does not necessarily span the entire network, but does span all
7 those Bridges and LANs between which data frame connectivity is desired for frames allocated to that active topology.

8 NOTE 4—The destination addressing information associated with a MAC user data frame includes the VLAN
9 classification of the frame [see item d)].

10 NOTE 5—Implicit configuration [see item d)], or recognition, of end station location includes observation of the source
11 address of the user data frame transmitted by that station. The user data frame communicates that location information
12 [see item e)] along the portion of the chosen active topology leading to the frame's destination(s).

13 7.2 Use of VLANs

14 VLANs and their VIDs provide a convenient and consistent network-wide reference for VLAN Bridges to:

- 15 a) Identify rules for the classification of user data frames into VLANs.
- 16 b) Effectively extend the source and destination MAC addresses, by treating frames and addressing
17 information for different VLANs independently.
- 18 c) Identify and select from different active topologies.
- 19 d) Identify configuration parameters that partition a physical network.

20 Taken together these capabilities allow VLAN Bridges to emulate a number of separately manageable, or
21 virtual, Bridged Networks. A LAN that has been selected by network management to receive frames
22 assigned to a given VLAN is said to form part of, belong to, or be a member of the VLAN. Similarly, end
23 stations that are attached to those LANs and that can receive frames assigned to the VLAN are said to be
24 attached to that VLAN.

25 NOTE—Separate control over the transmission and over the reception of frames to and from LANs and end stations is
26 possible. To avoid ambiguity, VLAN membership is defined by reception.

27 Inclusion of the VID in VLAN-tagged frames guards against connectivity loops arising from differing
28 classifications by different Bridges, and permits enhanced classification rules to be used by some Bridges,
29 while others simply forward the previously classified frames.

30 The VLAN tag allows frames to carry priority information, even if the frame has not been classified as
31 belonging to a particular VLAN.

32 7.3 Active topology

33 Bridges cooperate to calculate one or more loop-free and fully connected active topologies, i.e., spanning
34 trees. The algorithms and protocols—RSTP (13.4), MSTP (13.5), SPB (Clause 27), PCR (Clause 45)—that
35 support that calculation provide rapid recovery from network component failure (6.5.1) by using alternate
36 physical connectivity, without requiring management intervention.

37 The Bridge's forwarding processes constrain the potential path for each user data frame to a single spanning
38 tree. A Bridge that uses RSTP or STP allocates all frames to a single CST. An MST or SPT Region (13.8,
39 27.4.1) comprises the transitive closure of Bridges that use MSTP or SPB and that agree on the allocation of
40 frames with a given VID to a given subtree within the region. That subtree can be an MSTI, a SPT, an
41 Explicit Tree (ET), or the IST that is the logical continuation of the CST through the region. Interoperability

1 between MST and SPT Bridges and SST Bridges, and between differently configured MST/SPT Bridges, is
2 achieved by allocating all frames to the CST at region boundaries. Thus the spanning tree for a given frame
3 can comprise subtrees of the CST together with an MSTI or SPT within each region. Frames with different
4 VIDs can be assigned to the same or to different trees and subtrees within a region.

5 NOTE—In a stable network, the MSTP and SPB algorithms verify that each MST/SPT Region is fully connected
6 internally. However, if the CST Root lies outside the region, the region can be partitioned temporarily as necessary to
7 prevent external connectivity between Bridge Ports at the region boundary from creating a loop. A given region of the
8 network can be an MST Region, an SPT Region, or both, but the protocol mechanisms that verify connectivity internal
9 to the region require that regions do not overlap or subset other regions.

10 7.4 VLAN topology

11 Typically each of the VIDs for frames assigned to the CST, and to the IST or an MSTI within an MST
12 Region, supports a different VLAN, and each VLAN is supported by only one VID and only one subtree
13 within the region. However, frames for VLANs supported by SPB are assigned to the SPT rooted at the
14 Bridge where they enter an SPT Region, and SPBV requires that each source Bridge for a given VLAN uses
15 a unique SPVID. To provide interoperability with other regions and end systems that support a VLAN with
16 a single VID, the VIDs for SPBV VLANs are translated at the SPT Region boundary (6.9, 27.12). SPBM
17 uses a single VID for all the SPTs for a given VLAN.

18 NOTE 1—Support of a network comprising a single SPT Region with directly attached VLAN-unaware or priority
19 tagging end stations does not require VID translation. This includes, for example, a PBN or PBBN where the edge of the
20 network corresponds with the boundary of the SPT Region.

21 Within a given region, the SPTs that support any given shortest path VLAN provide full and symmetric
22 connectivity (6.3). To allow frames in different VLANs to make use of equal cost shortest paths, multiple
23 SPT Sets can be used, each providing full symmetric connectivity between all Bridges in the region.

24 Each VLAN may occupy the full extent of the active topology of its associated spanning tree(s) or a
25 continuously connected subset of that active topology. The connected subset must be continuously
26 connected; otherwise, the VLAN is split. The maximum extent of the connected subset may be bounded by
27 management by explicitly excluding certain Bridge Ports from a VLAN's connectivity.

28 At any time the current extent of a VLAN can be further reduced from the maximum to include only those
29 LANs that provide communication between attached devices, by the use of protocol that allows end stations
30 to request and release services that use the VLAN. Such a protocol is specified in Clause 11. The dynamic
31 determination of VLAN extent provides flexibility and bandwidth conservation, at the cost of network
32 management complexity.

33 NOTE 2—Dynamic determination of VLAN extent is generally preferable to static configuration for bandwidth
34 conservation, as the latter is error prone and can defeat potential alternate connectivity—requiring active management
35 intervention to recover from network component failure.

36 NOTE 3—To accommodate end stations that do not participate in the MVRP specified in Clause 11, management
37 controls associated with each Bridge Port allow the Port to identify the attached LAN as connecting end stations that
38 require services using specified VLANs.

39 NOTE 4—Where a given VLAN is supported by multiple VIDs, as for SPB, management controls identify the VLAN
40 by a Base VID (that used when frames are allocated to the CST). Use of this single VID reduces the configuration
41 burden, particularly when Bridges are added to the SPT Region and further SPVIDs are allocated.

1 7.5 Locating end stations

2 Functioning as a distributed system, Bridges within the current extent of a VLAN can, through explicit and
3 or implicit cooperation, locate those LANs where an attached end station or end stations are intended to
4 receive frames addressed to a specified individual address or group address. Bridges can thus reduce traffic
5 by confining frames to the LANs where their transmission is necessary.

6 NOTE 1—Individual Bridges do not determine the precise location of end stations but merely determine which of their
7 Bridge Ports need to forward frames toward the destination(s). For the system of Bridges this is sufficient to restrict
8 frames to the paths necessary to reach the destination LANs.

9 As specified in 13.6.2 of IEEE Std 802.1AC-2016 [B10], a Bridge with a Point-to-Multipoint Network port
10 (PMPN, G.4.2, G.4.3) can detect when an attached PMPN node establishes or breaks a connection to another
11 PMPN node.

12 The Multiple MAC Registration Protocol (MMRP), specified in Clause 10, allows end stations to advertise
13 their presence and their desire to join (or leave) a multicast group, or to register an individual MAC address,
14 in the context of a VLAN. The protocol communicates this information to other Bridges, using the VLAN
15 and its active topology.

16 NOTE 2—To accommodate end stations that do not participate in MMRP, management controls associated with each
17 Bridge Port allow the port to identify the attached LAN as connecting end stations that are intended to receive specified
18 group addresses. The continuous operation of MMRP and the propagation of location information through Bridges using
19 the current active topology for the VLAN support multicast traffic reduction, while ensuring rapid restoration of
20 multicast connectivity without management intervention if alternate connectivity is selected following network
21 component failure.

22 Each end station implicitly advertises its attachment to an individual LAN and its individual MAC address
23 whenever it transmits a frame. Bridges learn from the source address as they forward the frame along the
24 active topology to its destination or destinations — or throughout the Bridged Network or VLAN if the
25 location of the destination or destinations is unknown. The learned information is stored in the FDB (8.8)
26 and used to filter frames on the basis of their destination addresses.

27 Address information learned from a frame with a given VID can be used to filter frames with other VIDs.
28 For example, information learned from a frame with any of the VIDs that support SPB for a given VLAN
29 (i.e., VIDs associated with the same Base VID) is used to filter frames with any of the VIDs for that VLAN.
30 In some network configurations, it is also necessary to use address information learned in any one of a set of
31 VLANs to filter frames with VIDs that identify any VLAN in the set (Shared VLAN Learning (SVL),
32 Clause 3). In other configurations it is necessary or desirable not to share learned information between
33 certain VLANs (Independent VLAN Learning (IVL), Clause 3) while in some cases it is immaterial whether
34 address information is shared between VLANs. Annex F discusses the network configuration requirements,
35 and some of the related interoperability issues.

36 Clause 15 and Clause 16 describe how the mapping of C-VLANs into S-VLANs is accomplished within
37 PBNs and PBBNs. These networks provide additional mapping facilities to support hierarchies of VLANs
38 allowing a provider a separate FDB from customers. The conformance definitions of 5.3 have been extended
39 to support S-VLANs. Clause 25 and Clause 26 describe how the mapping of S-VLANs into backbone
40 service instances is accomplished within PBBNs.

41 7.6 Ingress, forwarding, and egress rules

42 The relay function provided by each Bridge controls the following:

- 1 a) Classification of each received frame as belonging to one and only one VLAN, and discard or
2 acceptance of the frame for further processing on the basis of that classification and the received
3 frame format, which can be one of three possible types:
 - 4 1) Untagged, and not explicitly identifying the frame as being associated with a particular VID.
 - 5 2) Priority-tagged, i.e., including a tag header conveying explicit priority information but not
6 identifying the frames as being associated with a specific VID.
 - 7 3) VLAN-tagged, i.e., explicitly associating the frames with a particular VID.
- 8 This aspect of relay implements the *ingress* rules.
- 9 b) Implementation of the decisions governing where each frame is to be forwarded as determined by
10 the current extent of the VLAN topology (7.4), station location information (7.5), and the additional
11 management controls specified in Clause 8. This aspect of relay implements the *forwarding* rules.
- 12 c) Queuing of frames for transmission through the selected Bridge Ports, management of the queued
13 frames, selection of frames for transmission, and determination of the appropriate frame format
14 type, VLAN-tagged or untagged. This aspect of relay implements the *egress* rules.

15 The structuring of the relay functionality into the implementation of ingress, forwarding, and egress rules
16 constitutes a generic approach to the provision of VLAN functionality. All VLAN Bridges can correctly
17 forward received frames that are already VLAN-tagged. These are classified as belonging to the VLAN
18 identified by the VID in the tag header. All VLAN Bridges can also classify untagged and priority-tagged
19 frames received on any given port as belonging to a specified VLAN. In addition to this default Port-based
20 ingress classification, this standard specifies an optional Port-and-Protocol-based classification.

21 The classification of untagged and priority-tagged frames, and the addition or removal of tag headers, is
22 performed in support of the EISS (6.9).

23 Frames that carry control information to determine the active topology and current extent of each VLAN,
24 i.e., spanning tree or SPB and MVRPDUs, and frames from other link constrained protocols, such as
25 EAPOL and LLDP, are not forwarded. Permanently configured static entries in the FDB (8.2, 8.3, and 8.12)
26 ensure that such frames are discarded by the Forwarding Process (8.6).

27 NOTE—MRPDUs destined for any MRP application are forwarded or filtered depending on whether the application
28 concerned is supported by the Bridge, as specified in 8.1.2.

29 The forwarding rules specified for VLAN-tagged frames facilitate the interoperation of VLAN Bridges
30 conformant to this standard with end stations that directly support attachment of MAC Service users to
31 VLANs by transmitting VLAN-tagged frames, and with Bridges that are capable of additional proprietary
32 ingress classification methods.

33 Frames transmitted on a given LAN by a VLAN Bridge for a given VLAN shall be one of the following:

- 34 d) All untagged or
- 35 e) All VLAN-tagged with the same VID or
- 36 f) All VLAN-tagged, each with an SPVID allocated to the Base VID for that VLAN

1 8. Principles of Bridge operation

2 This clause:

- 3 a) Explains the principal elements of the operation of Bridges, and the operation of Bridge components
- 4 within systems, and lists the supporting functions.
- 5 b) Establishes a Bridge architecture that governs the provision of these functions.
- 6 c) Provides a model of Bridge operation in terms of processes and entities that support the functions.
- 7 d) Details the addressing requirements in a bridged network.
- 8 e) Specifies the addressing of entities in a Bridge.

9 8.1 Bridge operation

10 The principal elements of Bridge operation are:

- 11 a) Relay and filtering of frames (8.1.1).
- 12 b) Maintenance of the information required to make frame filtering and relaying decisions (8.1.2).
- 13 c) Management of the above (Clause 12).

14 8.1.1 Relay

15 A Bridge relays individual MAC user data frames between the separate MACs of the individual LANs
16 connected to its Ports. The functions that support relaying of frames and maintain the QoS are:

- 17 a) Frame reception (8.5).
- 18 b) Discard on received frame in error (6.5.2).
- 19 c) Discard of frames that do not carry user data (IEEE Std 802.1AC).
- 20 d) Priority and drop eligibility decoding from a VLAN tag, if present, and regeneration of priority, if
- 21 required (6.9).
- 22 e) Assignment of a VID to each received frame (6.9).
- 23 f) Ensure that each frame is confined to an active topology (8.6.1).
- 24 g) Frame discard to support management control over the active topology of each VLAN (6.9, 8.6.2,
- 25 8.6.4).
- 26 h) Frame discard following the application of filtering information (8.6.3).
- 27 i) Metering of frames, potentially marking as drop eligible or discarding frames exceeding bandwidth
- 28 limits (8.6.5).
- 29 j) Forwarding of received frames to other Bridge Ports (8.6.6).
- 30 k) Selection of traffic class and queuing of frames by traffic class (8.6.6).
- 31 l) Frame discard to ensure that a maximum Bridge transit delay is not exceeded (6.5.6, 8.6.7).
- 32 m) Preferential discard of drop eligible frames to preserve QoS for other frames (8.6.7).
- 33 n) Selection of queued frames for transmission (8.6.8).
- 34 o) Mapping of SDUs and FCS recalculation, if required (6.5.7).
- 35 p) Frame discard on transmittable SDU size exceeded (6.5.8).
- 36 q) Selection of outbound access priority (6.5.9).
- 37 r) Frame transmission (8.5).

38 A VLAN-unaware MAC Relay Entity is supported by the ISS, and not the EISS, and correspondingly the
39 functions that support relaying of frames in a MAC Bridge cannot make use of any parameters explicit to
40 EISS, like the VIDs. This means in particular that for a MAC Bridge, item e) in the list above is not
41 supported, item d) and item g) would be supported in the form of:

- 42 s) Regeneration of priority, if required (6.9.4)
 - 43 t) Frame discard to support management control over the physical topology of the network
- 44 and, every other item on the list is supported but without making any use of, or qualification upon, VIDs.

1 Figure 8-1 gives an example of the physical topology of a Bridged Network.

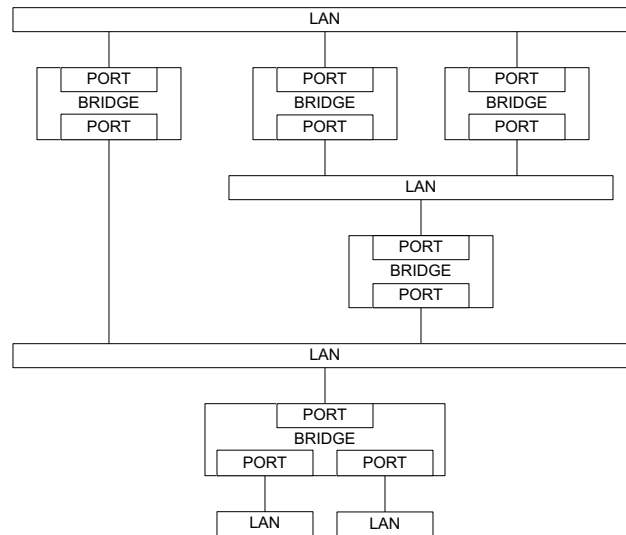


Figure 8-1—A Bridged Network

2 8.1.2 Filtering and relaying information

3 A Bridge maintains filtering and relaying information for the following purposes:

- 4 a) Duplicate frame prevention: to maintain a loop-free active topology for every frame
- 5 b) Traffic segregation: to separate communication by different sets of network users
- 6 c) Traffic reduction: to confine frames to the path(s) between their source and destination(s)
- 7 d) Traffic expediting: to classify frames in order to expedite time-critical traffic
- 8 e) Frame format conversion: to tag or untag as appropriate for the destination LAN and stations

9 8.1.3 Duplicate frame prevention

10 A Bridge filters frames, i.e., does not relay frames received by a Bridge Port to other Bridge Port on that
11 Bridge, in order to prevent the duplication of frames (6.5.4). The functions that support the use and
12 maintenance of information for this purpose are:

- 13 a) Configuration and calculation of one or more spanning tree active topologies.
- 14 b) In MST Bridges, explicit configuration to assign VIDs to MSTIs (8.9).
- 15 c) In SPT Bridges: explicit or default selection of shortest path, CIST, or MSTI support for specified
16 VLANs; and automatic assignment of VIDs to SPTs for shortest path VLANs.

17 8.1.4 Traffic segregation

18 A VLAN Bridge can filter frames to confine them to LANs that belong to the VLAN to which they are
19 assigned and, thus, define the VLAN's maximum extent (7.4). The functions that support the use and
20 maintenance of information for this purpose are:

- 21 a) Configuration of a PVID for each Port, to associate a VID with untagged and priority-tagged
22 received frames (6.9.1), and parameters for Protocol VLAN Classification (6.12) if implemented.
- 23 b) Configuration of Static VLAN Registration Entries (8.8.2).
- 24 c) Configuration of the Enable Ingress Filtering parameter to enable or disable application of Static
25 VLAN Registration Entries to received frames.

1 A VLAN Bridge can filter frames to partially partition a Virtual Bridged Network. Frames assigned to any
2 given VID and addressed to specific end stations or groups of end stations can be excluded from relay to
3 certain Bridge Ports. The functions that support the use and maintenance of information for this purpose are:

- 4 d) Permanent configuration of Reserved Addresses (Table 8-1, Table 8-2, and Table 8-3).
- 5 e) Configuration of Static Filtering Entries (8.8.1) and MAC Address Registration Entries (8.8.4).

6 NOTE—The use of VLANs is generally less error prone and is preferred to filtering using destination addresses if a
7 Bridged Network is to be partitioned for reasons of scale, efficiency, management, or security. Destination address
8 filtering is the only mechanism available to MAC Bridges.

9 8.1.5 Traffic reduction

10 A Bridge can filter frames so that LANs not attaching to or forming part of the path between the source and
11 the potential destination(s) of any given communication do not have to support the transmission of related
12 frames, potentially improving the quality of the MAC Service for other communications. The functions that
13 support the use and maintenance of information for this purpose are:

- 14 a) Automatic learning of dynamic filtering information for unicast destination addresses through
15 observation of source addresses of frames.
- 16 b) Ageing out or flushing of dynamic filtering information that has been learned to support the
17 movement of end stations and changes in active topology.
- 18 c) Automatic configuration of MAC Address Registration Entries by means of MMRP exchanges.
- 19 d) Explicit configuration of the management controls associated with the operation of MMRP by
20 means of MAC Address Registration Entries.

21 A VLAN Bridge can also filter frames to confine them to LANs either that have end stations attached to
22 their assigned VLAN or that connect those LANs and, thus, define the current practical extent of the VLAN
23 (7.5).

- 24 e) Automatic inclusion and removal of Bridge Ports in the VLAN, through configuration of Dynamic
25 VLAN Registration Entries by means of MVRP (8.8.5 and 11.2) or MIRP (Clause 39).
- 26 f) Explicit configuration of management controls associated with the operation of MVRP and MIRP
27 by means of Static VLAN Registration Entries (8.8.2 and 11.2).

28 Item e) and item f) are only supported by VLAN Bridges, while all other items in the list above are
29 supported by every Bridge.

30 8.1.6 Traffic expediting

31 A Bridge classifies frames into traffic classes in order to expedite transmission of frames generated by
32 critical or time-sensitive services. The function that supports the use and maintenance of information for this
33 purpose is:

- 34 a) Explicit configuration of traffic class information associated with the Ports of the Bridge.

35 8.1.7 Conversion of frame formats

36 A VLAN Bridge adds and removes tag headers (9.3) from frames and performs the associated frame
37 translations that may be required. The function that supports the use and maintenance of information for this
38 purpose is:

- 39 a) Explicit configuration of tagging requirements on transmission for each Port (8.8.2, 6.9.2).

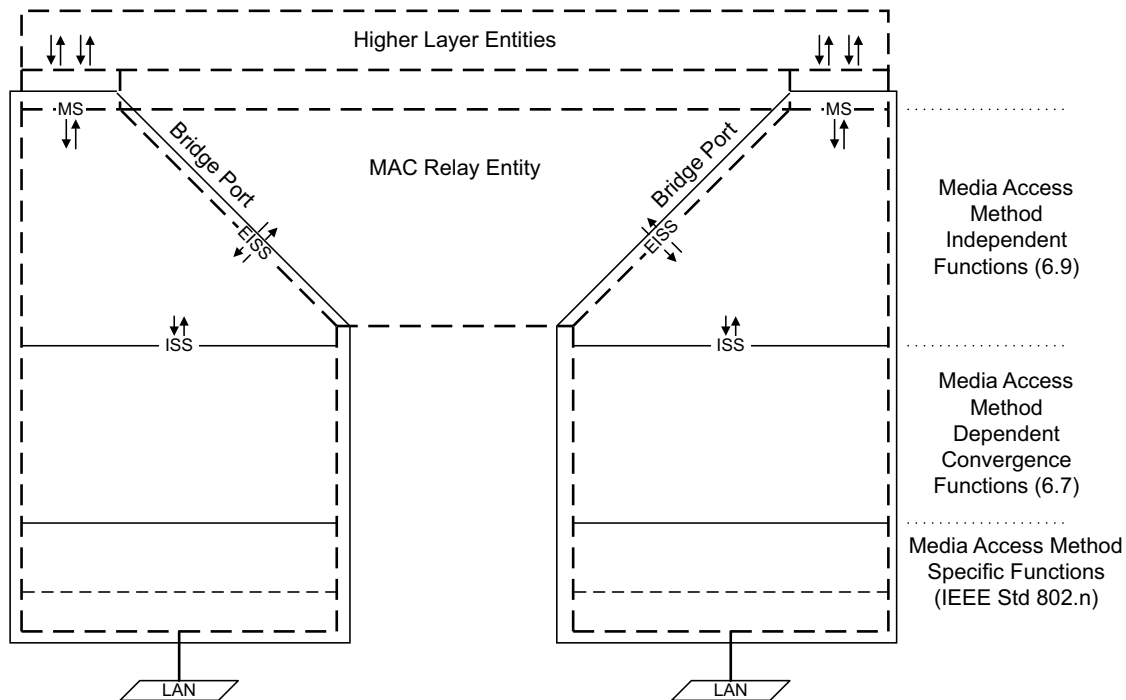
40 NOTE—As all incoming frames, including priority-tagged frames, are classified as belonging to a VLAN, the
41 transmitting Port transmits VLAN-tagged frames or untagged frames. Hence, a station sending a priority-tagged frame
42 via a Bridge will receive a response that is either VLAN-tagged or untagged, as described in 8.5.

8.2 Bridge architecture

A Bridge comprises at least one Bridge component. A Bridge component comprises the following:

- a) A MAC Relay Entity that interconnects the Bridge's Ports
- b) At least two Ports
- c) Higher layer entities, including at least a Spanning Tree Protocol Entity

The VLAN Bridge architecture is illustrated in Figure 8-2. The MAC Relay Entity handles the media access method-independent functions of relaying frames among Bridge Ports, filtering frames, and learning filtering information. It uses the EISS (6.8, 6.9) provided by each Bridge Port.



NOTE—The notation “IEEE Std 802.n” in this figure indicates that the specifications for these functions can be found in the relevant standard for the media access method concerned; for example, n would be 3 (IEEE Std 802.3) in the case of Ethernet.

Figure 8-2—VLAN Bridge architecture

1 The MAC Bridge architecture is illustrated in Figure 8-3. The main difference with the VLAN
2 Bridge architecture is that it does not use the EISS and the VLAN-unaware MAC relay entity is supported
3 by the ISS.

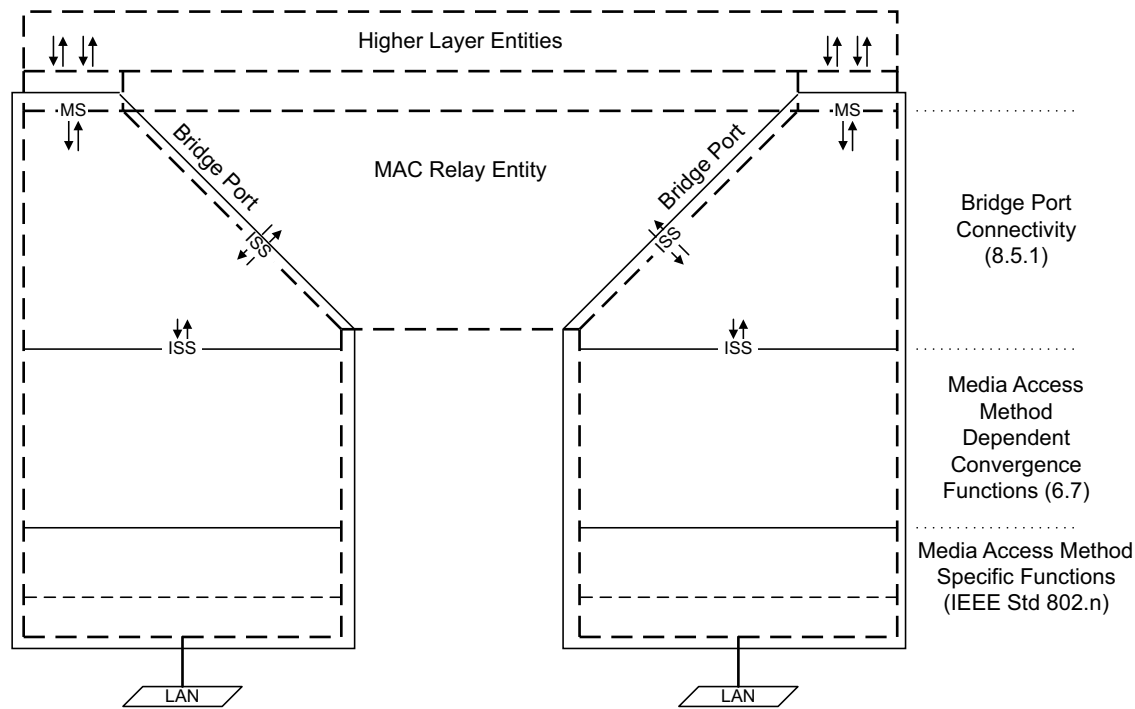


Figure 8-3—MAC Bridge architecture

4 Each Bridge Port also functions as an end station and shall provide the MAC Service to an LLC Entity that
5 operates LLC Type 1 procedures to support protocol identification, multiplexing, and demultiplexing, for
6 PDU transmission and reception by the Spanning Tree Protocol Entity and other higher layer entities. Other
7 types of LLC procedures can also be supported for use by other protocols. The Bridge Port provides
8 additional MSAPs if those are explicitly required by the specifications of the attached higher layer entities.
9 Each instance of the MAC Service is provided to a distinct LLC Entity that supports protocol identification,
10 multiplexing, and demultiplexing, for PDU transmission and reception by one or more higher layer entities.

11 NOTE—For simplicity of specification, this standard refers to a single LLC Entity that can provide both the procedures
12 specified by ISO/IEC 8802-2 and EtherType protocol discrimination in the cases where the media access method for the
13 attached LAN supports the latter.

14 If the Bridge Port can be directly attached to an IEEE 802 LAN, an instance of the MAC for that LAN type
15 is permanently associated with the Port, handles the media access method-specific functions (MAC protocol
16 and procedures), and provides an instance of the ISS as specified in IEEE Std 802.1AC to support frame
17 transmission and reception by the other processes and entities that compose the Port.

18 Figure 8-2 illustrates a VLAN Bridge, and Figure 8-3 illustrates a MAC Bridge, with two Ports, each
19 directly connected to a LAN.

1 8.3 Model of operation

2 The model of operation is simply a basis for describing the functionality of the Bridge. It is in no way
3 intended to constrain real implementations of a Bridge; these may adopt any internal model of operation
4 compatible with the externally visible behavior that this standard specifies. Conformance of equipment to
5 this standard is purely in respect of observable protocol.

6 The processes and entities that model the operation of a Bridge Port include the following:

- 7 a) A Bridge Port Transmit and Receive Process (8.5) that:
 - 8 1) Receives and transmits frames from and to the attached LAN (8.5, 6.8, 6.9,
9 IEEE Std 802.1AC).
 - 10 2) Can filter received frames if the VLAN tag is absent, or present, or conveys a null VID.
 - 11 3) Classifies received frames into VLANs, assigning each a VID value.
 - 12 4) Determines the format, VLAN-tagged or untagged, of transmitted frames.
 - 13 5) Delivers and accepts frames to and from the MAC Relay Entity and LLC Entities.
- 14 b) The LLC Entity or Entities that support Higher Layer Entities such as the following:
 - 15 1) Spanning Tree Protocol
 - 16 2) Multiple Registration Protocol (MRP)
 - 17 3) Bridge Management
 - 18 4) Linktrace response
- 19 c) Zero or more CFM entities, each of which can be one of the following:
 - 20 1) An MA Endpoint (MEP) or
 - 21 2) An MA Intermediate Point (MIP) or
 - 22 3) A Linktrace Output Multiplexer (LOM)

23 As Bridge Ports on MAC Bridges provide only the ISS, item a2), item a3), and item a4) in the list above are
24 not supported by MAC Bridges.

- 25 d) The Forwarding Process (8.6) that:
 - 26 1) Enforces loop-free active topologies for all frames (8.1.3, 8.4, 8.6.1);
 - 27 2) Filters frames using their VID and destination MAC addresses (8.1.4, 8.6.2, 8.6.3, 8.6.4);
 - 28 3) Optionally, classifies and meters received frames that are to be relayed to other Bridge Ports
29 (8.6.5);
 - 30 4) Forwards received frames that are to be relayed to other Bridge Ports (8.6.6, 8.6.7, 8.6.8).
- 31 e) The Learning Process (8.7) that observes the source addresses of frames received on each Port and
32 updates the FDB (8.1.5, 8.4).
- 33 f) The FDB (8.8) that holds filtering information and supports queries by the Forwarding Process
34 about whether frames with given values of VID and destination MAC address field can be
35 forwarded to a given Port.

36 NOTE 1—The processes and entities that model the operation of the VLAN-unaware MAC Relays on MAC Bridges are
37 the same as the ones described above but excluding any use or qualification based on VID values.

38 NOTE 2—In Figure 8-4, Figure 8-5, Figure 8-6, and Figure 8-7, the EISS is used by the VLAN-aware MAC Relay
39 entities and the ISS is used by the VLAN-unaware MAC Relay entities.

1 Figure 8-4 illustrates a single instance of frame relay between the Ports of a Bridge with two Ports.

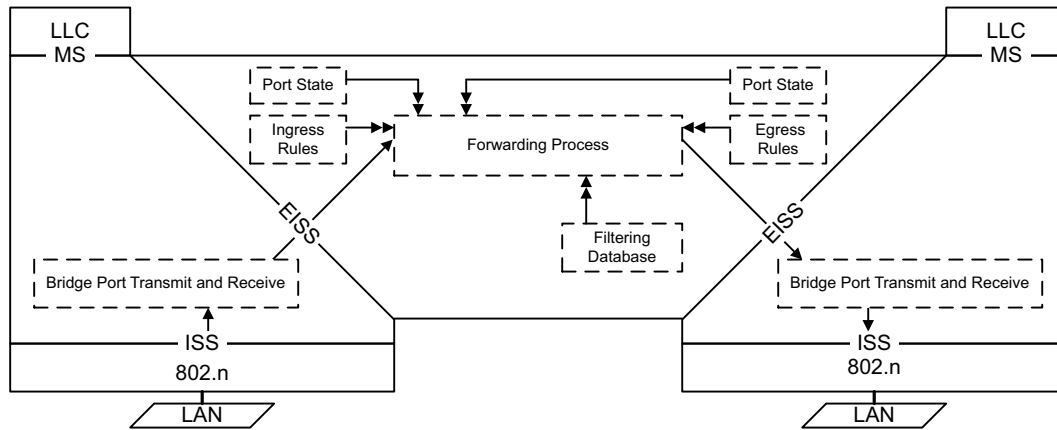


Figure 8-4—Relaying MAC frames

2 Figure 8-5 illustrates the inclusion of information carried by a single frame, received on one of the Ports of a
3 Bridge with two Ports, in the FDB.

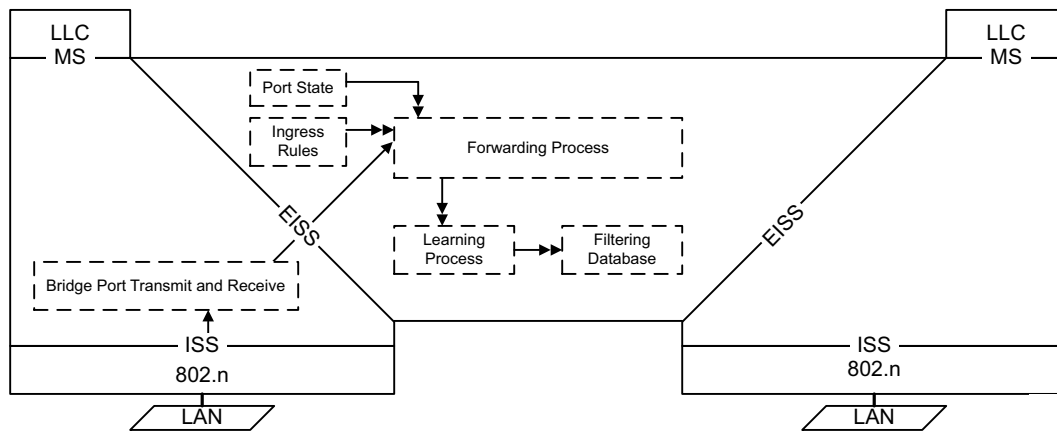


Figure 8-5—Observation of network traffic

1 Figure 8-6 illustrates the operation of the Spanning Tree Protocol Entity.

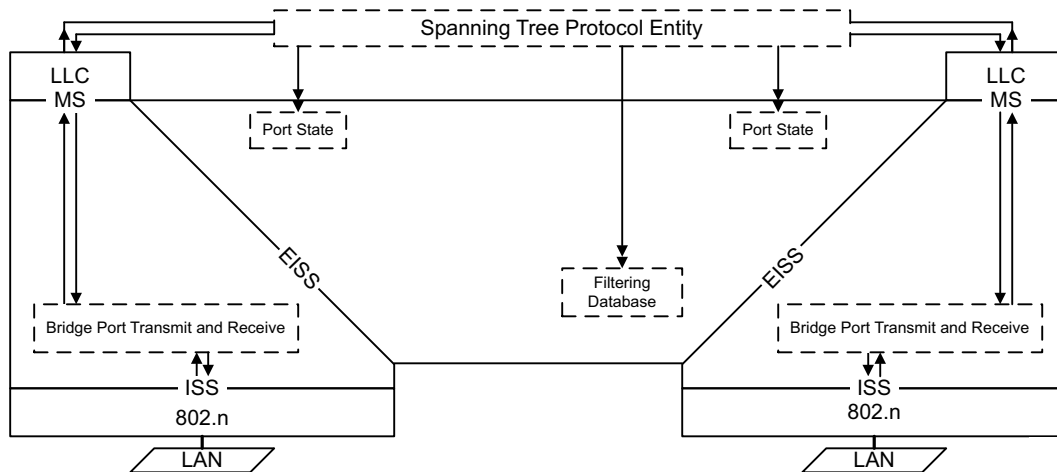


Figure 8-6—Operation of Spanning Tree Protocol Entity

2 Figure 8-7 illustrates the operation of the MRP Entity (8.11).

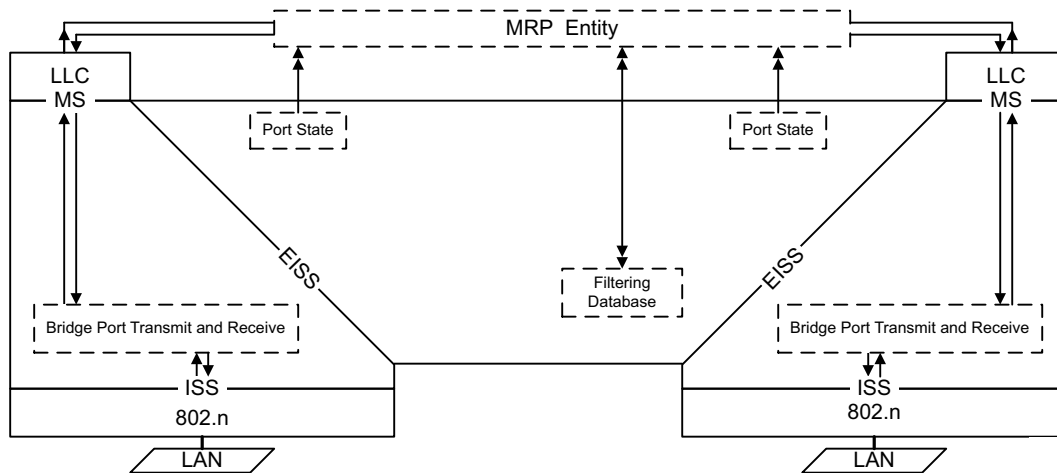


Figure 8-7—Operation of MRP

3 Higher Layer Entities that require only one point of attachment for the Bridge as a whole may attach to an
4 LLC Entity that uses an instance of the MAC Service provided by a Management Port. A Management Port
5 does not use an instance of the ISS to attach to a network but uses the Bridge Port Transmit and Receive
6 Process and the MAC Relay Entity to provide connectivity to other Bridge Ports and the attached LANs.

7 NOTE 3—Management port functionality may also be provided by an end station connected to an IEEE 802 LAN that is
8 wholly contained within the system that incorporates the Bridge. The absence of external connectivity to the LAN means
9 that the management port is always forwarding.

1 Figure 8-8 illustrates the reception and transmission by an entity attached to a Management Port.

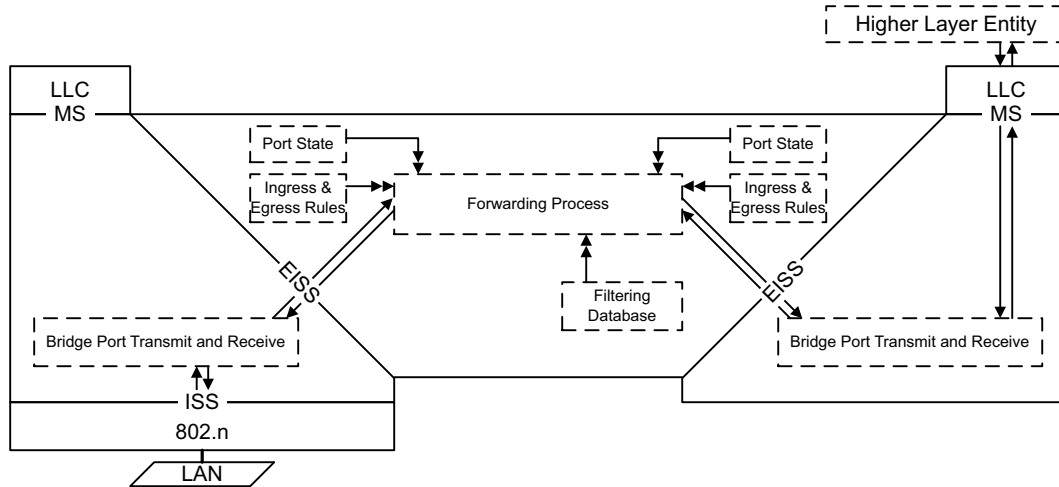


Figure 8-8—Management Port transmission and reception

2 Figure 8-9 illustrates the operation of the IPS Control entity. The IPS Control entity functions as an
3 intermediary between the Bridge Management entity and the FDB for operations related to TESIs associated
4 with an IPG. Infrastructure Segment MEPs associated with an IPG communicate the operational state of the
5 associated Infrastructure Segment to the IPS control entity (26.11.3).

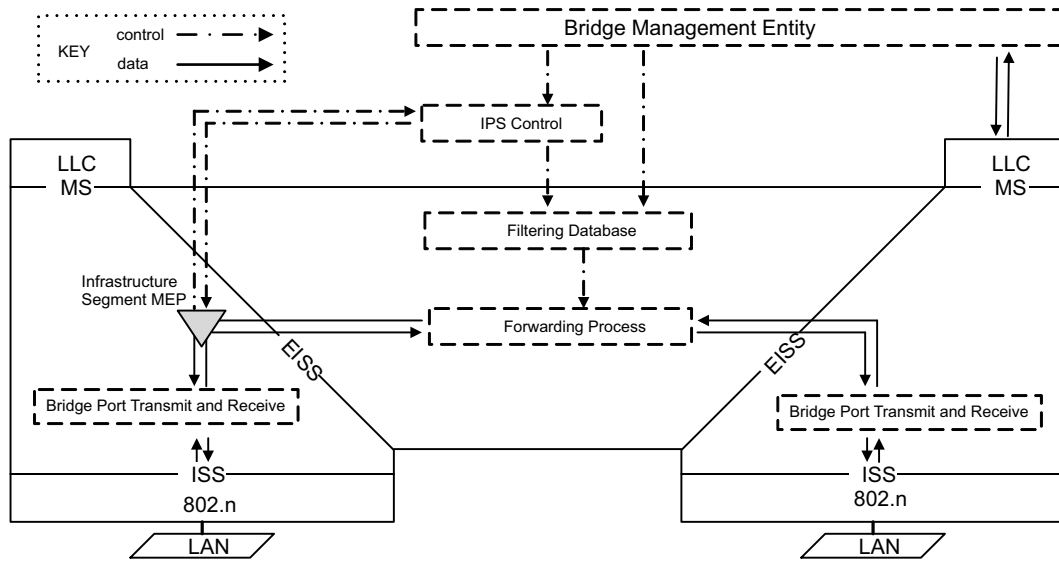


Figure 8-9—Infrastructure Segment MEP placement in a PNP

6 8.4 Active topologies, learning, and forwarding

7 An *active topology* is the connectivity provided, for frames with a specified set of VID, destination address,
8 and source address values, by interconnecting the LANs and Bridges in a bridged network with *forwarding*
9 Bridge Ports.

1 The distributed spanning tree algorithms and protocols, i.e., RSTP (Clause 13), MSTP (Clause 13), and
2 ISIS-SPB (Clause 27), construct one or more active topologies, each simply and fully connected for frames
3 being associated with any given VLAN and transmitted from any end station to any other. The *forwarding*
4 and *learning* performed by each Bridge Port for each spanning tree is dynamically managed by RSTP,
5 MSTP, or ISIS-SPB to prevent temporary loops and reduce excessive traffic in the network while
6 minimizing denial of service following any change in the *physical topology* of the network.

7 PBB-TE enables construction of active topologies by the external agent that is responsible for setting up
8 Ethernet Switched Paths (ESPs).

9 Any port that is not enabled, i.e., has MAC_Operational (IEEE Std 802.1AC) False or has been excluded
10 from the active topology by management setting of the Administrative Bridge Port State to Disabled, has
11 both forwarding and learning disabled for all spanning trees and ESPs.

12 If the Bridge Port is enabled, PBB-TE disables learning and enables forwarding for all frames allocated to
13 each ESP-VID. An external agent manages the FDB to control the forwarding of frames with particular
14 values of ESP-VID and destination MAC address.

15 The term Port State summarizes per-tree combinations of forwarding and learning, and any additional
16 per-tree variables used by a given spanning tree protocol to enforce the active topologies it has calculated,
17 and is used by RSTP and MSTP as follows. Any port that has learning and forwarding disabled is assigned
18 the Port State *Discarding*. A Port that has learning enabled but forwarding disabled has the Port State
19 *Learning*, and a Port that both learns and forwards frames has the Port State *Forwarding*. However, the
20 RSTP and MSTP state machines (Clause 13) do not control the Port State directly but use independent
21 forwarding and learning variables for each tree.

22 ISIS-SPB controls both forwarding and learning variables for each SPT and Bridge Port directly when used
23 to support Shortest Path Bridging VID (SPBV, Clause 27) just like RSTP and MSTP. ISIS-SPB disables
24 learning and sets forwarding to TRUE when supporting Shortest Path Bridging MAC (SPBM, Clause 27).

25 RSTP constructs a single spanning tree, the CST, and maintains a single Port State for each Bridge Port. SST
26 Bridges allocate all frames to that single spanning tree irrespective of their VLAN classification or source
27 and destination MAC addresses.

28 MSTP constructs multiple spanning trees, the CIST and additional MSTIs, and maintains a Port State for
29 each spanning tree for each Port. An MST Bridge allocates all frames classified as belonging to a given
30 VLAN to the CIST or to one of the MSTIs using the MST Configuration Table. An MSTID of 0 identifies
31 the CIST. A reserved MSTID value (TE-MSTID) is used to identify VIDs for use by PBB-TE with ESPs.
32 Further MSTID values are used to identify the use of SPBV or SPBM. A single VID is used to identify
33 frames assigned to any given VLAN that is supported by the CIST or an MSTI. That VID is used by the
34 Forwarding Process (8.6) to identify the spanning tree for the relayed frame, and thus identifies the
35 applicable Port State.

36 ISIS-SPB constructs symmetric (Clause 3) SPTs rooted at each Bridge within an SPT Region, and maintains
37 independent forwarding and learning variables for each SPT for each Bridge Port.

38 Each frame classified as belonging to a VLAN supported by SPBV, is allocated to an SPT rooted at the
39 Bridge that first relays the frame into the Region. If a C-VLAN, S-VLAN, or B-VLAN is supported by
40 SPBV (Clause 27) the frame is assigned a Shortest Path VLAN Identifier (SPVID) that is used by the
41 Forwarding Process (8.6) to identify both the VLAN and the SPT, and thus identifies the applicable learning
42 and forwarding values.

1 If a B-VLAN is supported by SPBM, (Clause 27) the frame is assigned a B-VID value that identifies the
2 frame as subject to SPBM and identifies the B-VLAN and the SPT Set (Clause 3). The SPT is identified
3 (from among those in that set) by the source address of the frame (B-SA), which identifies the SPT root.
4 When ISIS-SPB sets forwarding True for that SPT, a Dynamic Filtering Entry for that B-VID, source
5 address tuple is included in the FDB so that it passes the active topology enforcement check (8.6.1). An
6 individual destination address identifies the SPT that is rooted at that destination and belongs to the SPT Set
7 identified by the VID. If relaxed ingress checking (45.3.1) is supported, then a MAC Address Registration
8 Entry is included in the FDB for that VID, individual destination address tuple if there is any source whose
9 path to the destination in that SPT is via the given Bridge. The Port Map of those MAC Address Registration
10 entries comprises only the ports that are upstream for a given individual destination address (45.3.1). If a
11 B-VLAN is supported by SPBM, learning is disabled for all frames for that B-VID.

12 Figure 8-6 illustrates the operation of the Spanning Tree Protocol Entity, which operates the spanning tree
13 algorithm and its related protocols, and its modification of Port state information as part of determining the
14 active topology of the network.

15 Figure 8-4 illustrates the Forwarding Process's use of the Port State: first, for a Port receiving a frame, to
16 determine whether the received frame is to be relayed through any other Ports; and second, for another Port
17 in order to determine whether the relayed frame is to be forwarded through that particular Port.

18 Figure 8-5 illustrates the use of the Port state information for a Port receiving a frame, in order to determine
19 whether the station location information is to be incorporated in the FDB.

20 8.5 Bridge Port Transmit and Receive

21 The Bridge Port Transmit and Receive process supports the attachment of the Bridge Port to a network. It
22 comprises the following functions:

- 23 a) In a VLAN Bridge component, mapping between the EISS (6.8) provided to the MAC Relay Entity,
24 and the ISS (IEEE Std 802.1AC), adding, recognizing, interpreting, and removing VLAN tags as
25 specified in 6.9 and illustrated in Figure 8-10.
- 26 b) In a MAC Bridge component, an optional shim (IEEE Std 802.1AC) that uses and provides the ISS
27 (IEEE Std 802.1AC), recognizing and interpreting VLAN tags as specified in 6.20 and illustrated in
28 Figure 8-11.
- 29 c) Connectivity, as specified in 8.5.1 (for non-TPMR Bridges) or 8.5.2 (for TPMRs), between the
30 following ISS access points:
31 1) That provided by the MAC Entity for the LAN attached to the Port, as specified in
32 IEEE Std 802.1AC.
33 2) That supporting the MAC Relay Entity.
34 3) One or more that support Higher Layer Entities attached to the Port (8.5.3).

35 A single Port for a Bridge, known as the Management Port, may support Higher Layer Entities without
36 providing a point of attachment to a LAN (see Figure 8-8).

37 8.5.1 Bridge Port connectivity

38 Each M_UNITDATA.indication provided by the ISS access point for an attached LAN [(1) in Figure 8-10]
39 shall result in a corresponding M_UNITDATA.indication with identical parameters at each of the access
40 points supporting the MAC Relay and Higher Layer Entities [(2), (3a) and (3b)]. Each
41 M_UNITDATA.request from the ISS access point supporting the MAC Relay Entity shall result in a
42 corresponding M_UNITDATA.indication with identical parameters at each of the access points for the
43 Higher Layer Entities [(3a) and (3b)], and a corresponding M_UNITDATA.request with identical parameters
44 at the access point for the LAN [(1)]. Each M_UNITDATA.request from an ISS access point supporting a

1 Higher Layer Entity shall result in a corresponding M_UNITDATA.indication with identical parameters at
2 the access points for the MAC Relay Entity, and at other access points for Higher Layer Entities, and a
3 corresponding M_UNITDATA.request with identical parameters at the access point for the LAN.

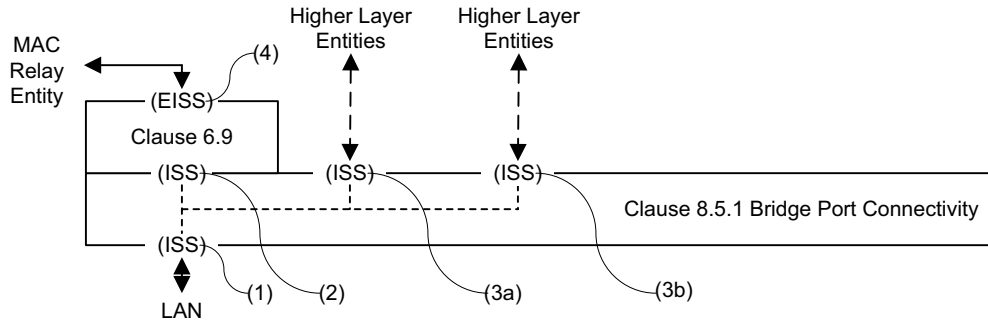


Figure 8-10—Bridge Port Transmit and Receive

4 NOTE—Figure 8-10 shows the relative position of the Bridge Port Transmit and Receive process and the VLAN
5 tagging function (6.9) for a Customer Bridge or Provider Bridge that does not support CFM. For the relative positioning
6 of these functions in a Bridge supporting CFM, see Figure 8-9. For the relative positioning of these functions in a BEB,
7 see Figure 26-2.

8 The MAC_Enabled, MAC_Operational, and operPointToPointMAC status parameters for the ISS access
9 point for the MAC Relay Entity and Higher Layer Entities [(2), (3a), and (3b) in Figure 8-10] shall take the
10 same value as that for the LAN (1) if that is present, and shall be True otherwise (i.e., if the Port is a
11 Management Port).

12 The consequence of the above connectivity is that frames relayed to a Bridge Port are both submitted to that
13 Port's MAC Service users and transmitted on the attached LAN (see 8.13.9).

14 8.5.2 TPMR Port connectivity

15 Each M_UNITDATA.indication provided by the ISS access point for an attached LAN [(1) in Figure 8-11]
16 shall result in a corresponding M_UNITDATA.indication with identical parameters at each of the access
17 points supporting the MAC Relay and Higher Layer Entities [(2), (3a), and (3b)].

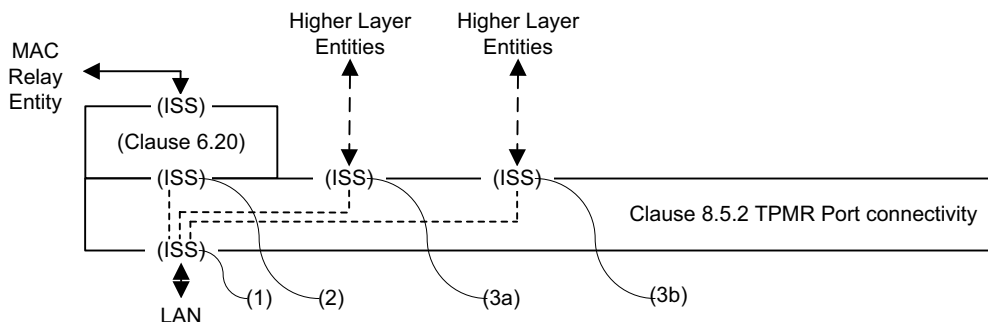


Figure 8-11—TPMR Port Transmit and Receive

18 Each M_UNITDATA.request from the ISS access point supporting the MAC Relay Entity (2) or a Higher
19 Layer Entity [(3a) or (3b)] shall result in a corresponding M_UNITDATA.request with identical parameters
20 at the access point for the LAN (1).

1 The MAC_Enabled, MAC_Operational, and operPointToPointMAC status parameters for the ISS access
2 point for the MAC Relay Entity and Higher Layer Entities [(2), (3a), and (3b) in Figure 8-11] shall take the
3 same value as for the LAN (1).

4 NOTE—These connectivity rules allow a higher layer entity to “listen” on both TPMR Ports to be able to determine the
5 LAN on which a particular frame was received, and ensure that any frames sent by a higher layer entity are propagated
6 only on the LAN associated with the Port on which the frame is transmitted. This differs from the propagation rules for
7 other Bridges, where frames transmitted by a higher layer entity are also submitted to the forwarding process, and frames
8 transmitted by the Forwarding Process on a given Port are received by any higher layer entity associated with that Port.

9 8.5.3 Support of Higher Layer Entities

10 The MAC Service is provided to a Higher Layer Entity using one of ISS access points provided for that
11 purpose by the Bridge Port Connectivity function (8.5.1) or the TPMR Port Connectivity function (8.5.2).

12 Each ISS M_UNITDATA.indication with a destination MAC address that is either the individual address of
13 an MSAP provided by the Bridge Port or a group address used by the attached LLC Entity shall cause an
14 MA_UNITDATA.indication at that MSAP with destination address, source address, MSDU, and priority
15 parameters identical to those in the M_UNITDATA.indication. No other indications or frames give rise to
16 indications to the MAC Service user.

17 Each MA_UNITDATA.request at the MSAP shall result in an M_UNITDATA.request at the ISS access
18 point with identical destination address, source address, MSDU, and priority parameters.

19 NOTE 1—Appropriate selection of the PVID for a Management Port facilitates attachment of an Internet Protocol (IP)
20 stack supporting an SNMP Management Agent to any selected VID relayed by the Bridge. This function is not
21 supported in Bridges that do not support the EISS, such as TPMRs.

22 NOTE 2—Higher Layer Entities attached to a Bridge Port can use the VLAN tag of received frames to achieve
23 VLAN-awareness.

24 8.6 The Forwarding Process

25 Each frame submitted to the MAC Relay Entity shall be forwarded subject to the constituent functions of the
26 Forwarding Process (Figure 8-12). Each function is described in terms of the action taken for a given frame
27 received on a given Port (termed “the reception Port”). The frame can be forwarded for transmission on
28 some Ports (termed “transmission Ports”) and discarded without being transmitted at the other Ports.

29 NOTE 1—IEEE Std 802.1Q, 2003 Edition, included frame formatting and FCS recalculation functions within the
30 Forwarding Process. This standard now places those functions below the EISS interface, to allow the specification of
31 additional methods for Bridge Port support of the EISS.

32 NOTE 2—IEEE Std 802.1D-2004 [B14], included priority mapping functions within the Forwarding Process. This
33 standard now places this function below the ISS interface.

34 NOTE 3—The Forwarding Process models the MAC relay function and does not take into consideration what may occur
35 once frames are passed to the Bridge Port for transmission. Conformant implementations of some media access methods
36 can vary the transmission order in apparent violation of the transmission selection rules when observing frames on the
37 medium. It may therefore not be possible to test conformance to this standard for some implementations simply by
38 relating observed LAN traffic to the functionality of the forwarding process; tests also have to allow for the (conformant)
39 behavior of the MAC.

40 Figure 8-4 illustrates the operation of the Forwarding Process in a single instance of frame relay between the
41 Ports of a Bridge with two Ports.

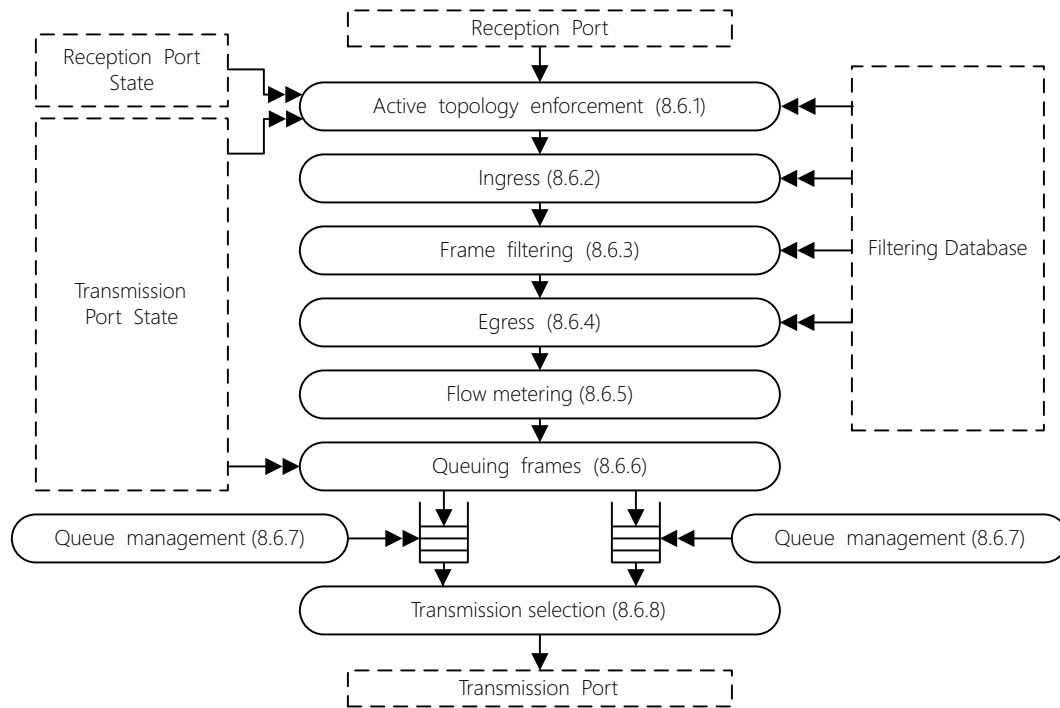


Figure 8-12—Forwarding process functions

1 8.6.1 Active topology enforcement

2 To prevent data loops and unwanted learning of source MAC addresses, the Forwarding Process determines
3 the values (TRUE or FALSE) of the learning and forwarding controls (8.4) appropriate to each received
4 frame and Bridge Port. If learning is TRUE for the reception Port and ingress filtering (8.6.2) would not
5 cause the received frame to be discarded, the source address and VID are submitted to the Learning Process.
6 NOTE 1—VIDs are not applicable to MAC Bridges and only the source address is submitted to the Learning Process.

7 If forwarding is TRUE for the reception Port, and either the EVBMode parameter value (40.4) for the Port is
8 not “EVB Bridge” or the value of the operReflectiveRelayControl parameter for the Port is FALSE, each
9 Bridge Port, other than the reception Port, with forwarding TRUE is identified as a potential transmission
10 Port. If forwarding is TRUE for the receiving Port and the EVBMode parameter value (40.4) for the Port is
11 “EVB Bridge” and the operReflectiveRelayControl parameter value for the Port is TRUE, each Bridge Port,
12 including the reception Port, with forwarding TRUE is identified as a potential transmission Port.

13 In an edge relay (ER), the forwarding process may set learning FALSE for all frames.

1 An SST Bridge supports a single active topology, the CST. For each Bridge Port, RSTP determines a single
2 value for forwarding and a single value for learning (8.4) for all frames.

3 Bridges with MST, PBB-TE, or SPB capabilities use the VID of the received frame to determine forwarding
4 and learning for that frame for each Bridge Port that is enabled, i.e., has MAC_Operational True and an
5 Administrative Bridge Port State of Enabled, as follows:

6 If the Bridge supports PBB-TE, and the VID is an ESP-VID, forwarding is TRUE and learning is FALSE.
7 Control over the active topology of each ESP is provided by configuration of Static Filtering Entries in the
8 FDB (8.8.1, 25.10.1).

9 If the Bridge uses MSTP to configure the CIST, and the VID identifies a VLAN assigned to the CIST,
10 forwarding and learning are as determined by MSTP (Clause 13) for the CIST.

11 If the Bridge uses MSTP to configure MSTIs, and the VID identifies a VLAN assigned to an MSTI
12 calculated by MSTP, forwarding and learning are as determined by MSTP for that MSTI.

13 If the Bridge uses ISIS-SPB to configure the CIST, and the VID identifies a VLAN assigned to the CIST,
14 forwarding and learning take the value of forwarding determined by ISIS-SPB (Clause 28) for the CIST.

15 If the Bridge uses ISIS-SPB, and the VID is used by SPBV as an SPVID, both forwarding and learning take
16 the value of forwarding determined by ISIS-SPB for the SPT identified by that SPVID.

17 If the Bridge uses ISIS-SPB and the VID identifies a VLAN supported by SPBM, then for that VID:

- 18 — Learning is False for all Bridge Ports.
- 19 — Forwarding is True for the reception Port if and only if a Dynamic Filtering Entry for the frame's
20 source address exists and specifies Forward for that Port or a MAC Address Registration Entry for
21 the frame's individual destination address exists and specifies Forward for that Port.
- 22 — Forwarding is True for all other Bridge Ports.

23 NOTE 2—Effective control over the egress component of an SPT for VLANs supported by SPBM is provided by
24 ISIS-SPB configuration of (a) a Dynamic Filtering Entry for the VLAN's VID and Root MAC address (for unicast
25 frames); and (b) MAC Address Registration Entries for the VID- and SPT-specific group MAC addresses (for multicast
26 frames); and (c) a MAC Address Registration Entry for the VID to filter "All Unregistered Group Addresses" at all
27 Ports.

28 All Bridges other than SST Bridges implement the MST Configuration Table (8.9.1). The use of VIDs to
29 determine learning and forwarding, as required by this clause, shall be consistent with that table as follows.
30 VIDs allocated by the MST Configuration Table (8.9.1) of value:

- 31 — CIST-MSTID (0x000) are CIST VIDs.
- 32 — SPBM-MSTID (0xFFC) are Base VIDs supported by SPBM.
- 33 — SPBV-MSTID (0xFFD) are Base VIDs supported by SPBV.
- 34 — TE-MSTID (0xFFE) are ESP-VIDs supported by PBB-TE.
- 35 — SPVID-Pool-MSTID (0xFFF) are VIDs reserved for use by ISIS-SPB as SPVIDs.
- 36 — All VIDs allocated to other values of MSTID are assigned to the MSTI identified by that MSTID.

37 NOTE 3—While the MST Configuration Table is capable of detailing the configuration of each SPVID, it captures the
38 allocation of Base VIDs only. Individual allocation of SPVIDs are captured in the ISIS-SPB database, the SPB ECT
39 Static Entry (12.25.4), and in the port VID translation tables. By specifying only the type of VID and its allocation to
40 SPB, backwards compatibility with STP, RSTP, and MSTP is maintained.

1 8.6.1.1 Requirements for the use of reflective relay

2 VEPA ERs (8.6.3.1) used in EVB (Clause 40) require reflective relay (40.4, 8.6.1) to be enabled on an
3 attached EVB Bridge SBP in order to ensure that all VSIs connected to the VEPA ER are able to receive
4 frames transmitted on one of the other VSIs. The following requirements ensure that a device operates
5 correctly when using reflective relay:

- 6 a) The operation of the device shall be such that it prevents the establishment of loops, i.e., the device
7 shall not both request and provide reflective relay on the same port.
- 8 b) The device shall ensure that any frame that it transmits on a given port, and that is reflected back to
9 the device through the attached Bridge Port, is filtered by the device in order to prevent it being
10 delivered to the originating port.
- 11 c) Any device requesting reflective relay is responsible for performing frame replication as necessary
12 for delivery to multiple ports.

13 NOTE—Information that can be used to assist in meeting these requirements is the source MAC address and the FID
14 derived from the VID.

15 8.6.2 Ingress filtering

16 Each Port may support an Enable Ingress Filtering parameter. A frame received on a Port that is not in the
17 member set (8.8.10) associated with the frame's VID shall be discarded if this parameter is set. The default
18 value for this parameter is reset, i.e., Disable Ingress Filtering, for all Ports. Any Port that supports setting
19 this parameter shall also support resetting it. The parameter may be configured by the management
20 operations defined in Clause 12.

21 This function is not applicable to VLAN-unaware MAC Relays.

22 8.6.3 Frame filtering

23 The Forwarding Process takes filtering decisions, i.e., reduces the set of potential transmission Ports (8.6.1),
24 for each received frame on the basis of:

- 25 a) Destination MAC address
- 26 b) VID
- 27 c) Flow hash
- 28 d) The information contained in the FDB for that MAC address and VID
- 29 e) The default Group filtering behavior for the potential transmission Port (8.8.6)

30 in accordance with the definition of the FDB entry types (8.8.1, 8.8.3, and 8.8.4). The required behavior is
31 summarized in 8.8.6, 8.8.9, Table 8-10, Table 8-11, and Table 8-12. Item b), specifying a VID value, is not
32 applicable to VLAN-unaware MAC Relays. Flow hash is applicable only on Bridges supporting flow
33 filtering (44.2).

34 Each of the reserved MAC addresses specified in Table 8-1 shall be permanently configured in the FDB in
35 C-VLAN components and ERs. Each of the reserved MAC addresses specified in Table 8-2 shall be
36 permanently configured in the FDB in S-VLAN components. Each of the reserved MAC addresses specified
37 in Table 8-3 shall be permanently configured in the FDB in TPMR components. The FDB entries for
38 reserved MAC addresses shall specify filtering for all Bridge Ports and all VIDs. Management shall not
39 provide the capability to modify or remove entries for reserved MAC addresses.

40 The addresses in Table 8-1, Table 8-2, and Table 8-3 determine the scope of propagation of PDUs within a
41 Bridged Network, as follows:

- 42 f) The *Nearest Bridge* group address is an address that no conformant TPMR component, S-VLAN
43 component, C-VLAN component, or MAC Bridge can forward. PDUs transmitted using this

destination address, or any of the other addresses that appear in all three tables, can therefore travel no further than those stations that can be reached via a single individual LAN from the originating station. Hence the Nearest Bridge group address is also known as the Individual LAN Scope group address.

g) The ***Nearest non-TPMR*** Bridge group address is an address that no conformant S-VLAN component, C-VLAN component, or MAC Bridge can forward; however, this address is relayed by a TPMR component. PDUs using this destination address, or any of the other addresses that appear in both Table 8-1 and Table 8-2 but not in Table 8-3, will be relayed by any TPMRs but will propagate no further than the nearest S-VLAN component, C-VLAN component, or MAC Bridge.

h) The ***Nearest Customer Bridge*** group address is an address that no conformant C-VLAN component or MAC Bridge can forward; however, it is relayed by TPMR components and S-VLAN components. PDUs using this destination address, or any of the other addresses that appear in Table 8-1 but not in either Table 8-2 or Table 8-3, will be relayed by TPMR components and S-VLAN components but will propagate no further than the nearest C-VLAN component or MAC Bridge.

NOTE 1—The reserved MAC addresses specified in Table 8-1, Table 8-2, and Table 8-3 determine the span of connectivity for the protocol frames using the address; they do not identify the protocol. The protocols are identified by the EtherType field following the address fields.

NOTE 2—An instance of IEEE 802.1AX™ Link Aggregation Control Protocol (LACP) intended to operate between non-TPMR Bridges, even through a TPMR, can use the Nearest non-TPMR Bridge group address. See K.2 for discussion of the use of TPMRs in a link using LACP.

1

Table 8-1—C-VLAN and MAC Bridge component Reserved addresses

Assignment	Value
Bridge Group Address, Nearest Customer Bridge group address ^a	01-80-C2-00-00-00
IEEE MAC-specific Control Protocols group address	01-80-C2-00-00-01
IEEE 802.3 Slow_Protocols_Multicast address	01-80-C2-00-00-02
Nearest non-TPMR Bridge group address	01-80-C2-00-00-03
IEEE MAC-specific Control Protocols group address	01-80-C2-00-00-04
Reserved for future standardization	01-80-C2-00-00-05
Reserved for future standardization	01-80-C2-00-00-06
MEF Forum ELMI protocol group address ^b	01-80-C2-00-00-07
Provider Bridge Group Address	01-80-C2-00-00-08
Reserved for future standardization	01-80-C2-00-00-09
Reserved for future standardization	01-80-C2-00-00-0A
EDE-SS PEP Address (IEEE Std 802.1AE)	01-80-C2-00-00-0B
Reserved for future standardization	01-80-C2-00-00-0C
Provider Bridge MVRP Address	01-80-C2-00-00-0D
Individual LAN Scope group address ^c , Nearest Bridge group address	01-80-C2-00-00-0E
Reserved for future standardization	01-80-C2-00-00-0F

^a As stated in 13.41, in a Provider Edge Bridge, a C-VLAN component supporting a single service instance (i.e., with a single PEP) may forward (not filter) frames with the Nearest Customer Bridge Address as the destination address.

^b This address is not exclusively reserved for this purpose; other uses are reserved for future standardization.

^c It is intended that no IEEE 802.1 relay device will be defined that will forward frames that carry this destination address.

Table 8-2—S-VLAN component Reserved addresses

Assignment	Value
IEEE MAC-specific Control Protocols group address	01-80-C2-00-00-01
IEEE 802.3 Slow_Protocols_Multicast address	01-80-C2-00-00-02
Nearest non-TPMR Bridge group address	01-80-C2-00-00-03
IEEE MAC-specific Control Protocols group address	01-80-C2-00-00-04
Reserved for future standardization	01-80-C2-00-00-05
Reserved for future standardization	01-80-C2-00-00-06
MEF Forum ELMI protocol group address ^a	01-80-C2-00-00-07
Provider Bridge Group Address ^b	01-80-C2-00-00-08
Reserved for future standardization	01-80-C2-00-00-09
Reserved for future standardization	01-80-C2-00-00-0A
Individual LAN Scope group address ^c , Nearest Bridge group address	01-80-C2-00-00-0E

^a This address is not exclusively reserved for this purpose; other uses are reserved for future standardization.

^b It is intended that no IEEE 802.1 protocol or device will be defined that will require the inclusion of this address in the FDB to prevent or restrict its use for communication between a port of a Provider Bridge's S-VLAN component and the S-VLAN component of the nearest Provider Bridge.

^c It is intended that no IEEE 802.1 relay device will be defined that will forward frames that carry this destination address.

Table 8-3—TPMR component Reserved addresses

Assignment	Value
IEEE MAC-specific Control Protocols group address	01-80-C2-00-00-01
IEEE 802.3 Slow_Protocols_Multicast address	01-80-C2-00-00-02
IEEE MAC-specific Control Protocols group address	01-80-C2-00-00-04
Individual LAN Scope group address ^a , Nearest Bridge group address	01-80-C2-00-00-0E

^a It is intended that no IEEE 802.1 relay device will be defined that will forward frames that carry this destination address.

1 8.6.3.1 Virtual edge port aggregator (VEPA) filtering

2 A VEPA ER filters frames as follows.

3 If the receiving port is a DRP, then the URP shall be selected as the only transmission Port.

4 If the receiving port is a URP, then in addition to the filtering specified in 8.6.3, if there is a filtering entry
5 that specifies forwarding for the source MAC address and VLAN of the frame for any DRP, then that DRP
6 shall be removed from the list of potential transmission Ports.

8.6.4 Egress filtering

This function is not applicable to VLAN-unaware MAC Relays.

Any Port that is not in the member set (8.8.10) for the frame's VID is removed from the set of potential egress Ports.

8.6.5 Flow classification and metering

The Forwarding Process can apply flow classification and metering to frames that are received on a Bridge Port and have one or more potential transmission ports. Bridge Ports and end stations may support Per-Stream Filtering and Policing (PSFP), Asynchronous Traffic Shaping (ATS) filtering and eligibility time assignment, Congestion Isolation (CI), or the general flow classification rules specified in 8.6.5.1.

NOTE—The general flow classification and metering specification was added to this standard by IEEE Std 802.1Q-2005, PSFP by IEEE Std 802.1Qci-2017, ATS by IEEE Std 802.1Qcr-2020, and CI by IEEE Std 802.1Qcz-2023.

PSFP, ATS, and CI share common per-stream classification and metering elements, as shown in Figure 8-13. The Stream identification function specified in IEEE Std 802.1CB can be used to associate received frames with these elements.

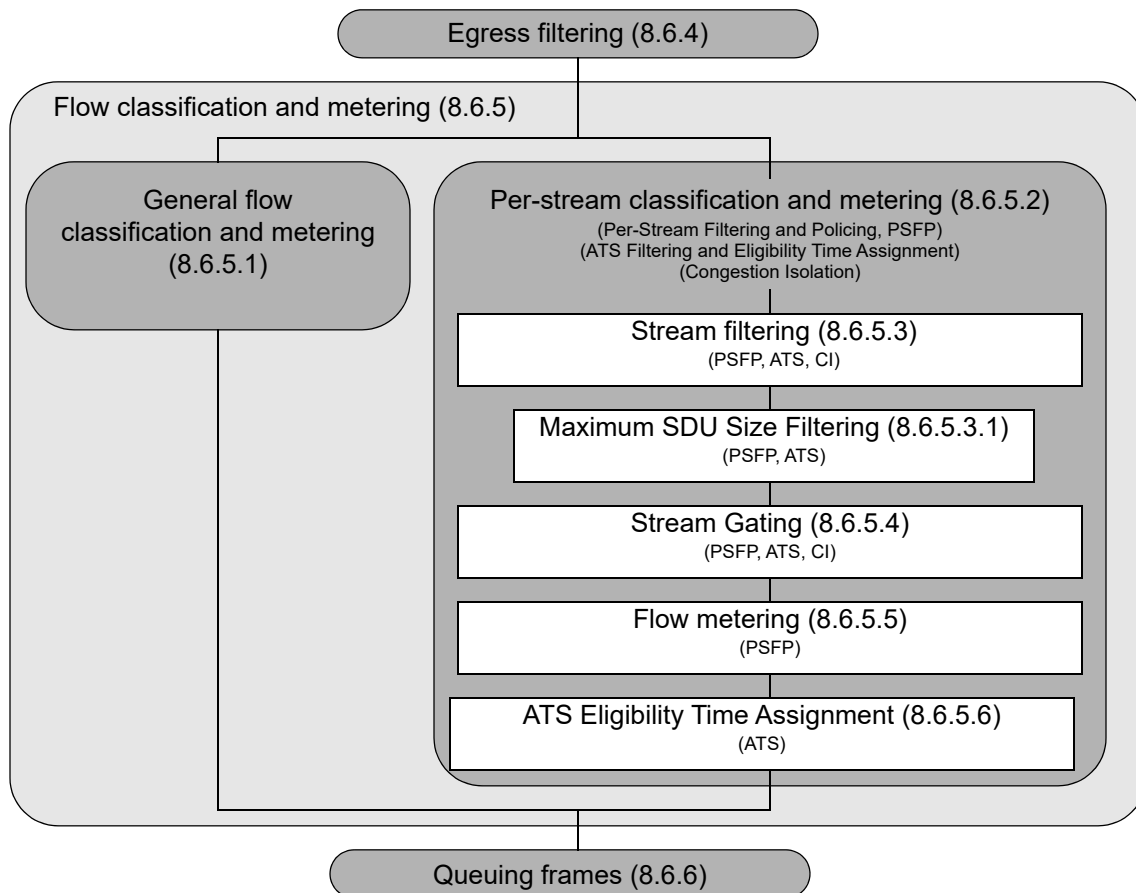


Figure 8-13—Flow classification and metering

1 8.6.5.1 General flow classification and metering

2 Bridges that implement general flow classification and metering can identify subsets of traffic (frames), each
3 subject to the same flow metering and forwarding. Flow classification rules may be based on the following:

- 4 a) Destination MAC address
- 5 b) Source MAC address
- 6 c) VID
- 7 d) Priority

8 Item c), specifying a VID value, is not applicable to VLAN-unaware MAC Relays.

9 Frames classified using the same set of classification rules are subject to the same flow meter. The flow
10 meter can change the drop_eligible parameter associated with each frame and can discard frames on the
11 basis of the following parameters for each received frame and previously received frames, and the time
12 elapsed since those frames were received:

- 13 e) The received value of the drop_eligible parameter
- 14 f) The mac_service_data_unit size

15 The flow meter shall not base its decision on the parameters of frames received on other Bridge Ports, or on
16 any other parameters of those Ports. The bandwidth profile algorithm described in the MEF Forum
17 Technical Specification 10.3 (MEF 10.3) should be used.

18 NOTE 1—The flow meter described here can encompass a number of meters, each with a simpler specification.
19 However, given the breadth of implementation choice permitted, further structuring to specify, for example, that frames
20 can bypass a meter or are subject only to one of a number of meters provides no additional information.

21 NOTE 2—Although flow metering is applied after egress (Figure 8-12), the meter(s) operate per reception Port, not per
22 potential transmission Port(s).

23 8.6.5.2 Per-stream classification and metering

24 When Per-Stream Filtering and Policing (PSFP), Asynchronous Traffic Shaping (ATS), or Congestion
25 Isolation (CI) is used, filtering and policing decisions for received frames are made, and subsequent queuing
26 (8.6.6) and transmission selection decisions (8.6.8) supported, as follows:

- 27 a) Each received frame can be associated with a stream filter, as specified in 8.6.5.3. If a matching
28 stream filter is specified (8.6.5.3), that is used to process the frame. Wildcard stream filters can be
29 configured to match and discard frames not associated with a specified stream. If no matching
30 stream filter is found, the frame is queued for transmission as specified in 8.6.6.
- 31 b) If the stream filter specifies maximum SDU size filtering (8.6.5.3.1), that is used to process the
32 frame. The frame can be discarded if a maximum SDU size is exceeded. The ATS scheduler state
33 machine operation (8.6.11) assumes that the sizes of frames that it processes are less than or equal to
34 the associated CommittedBurstSize parameter (8.6.11.3.5).
- 35 c) The stream filter specifies a stream gate (8.6.5.4), that is used to process the frame. The frame can be
36 discarded if it is received outside of permitted reception intervals or a given data limit within a
37 reception interval is exceeded. The frame's priority can be mapped to an internal priority value
38 (IPV) that can influence subsequent queuing decisions (8.6.6).
- 39 d) If the stream filter specifies a flow meter (8.6.5.5), that is used to process the frame. The frame can
40 be discarded or marked as drop eligible if a traffic limit of a flow meter is exceeded. A given stream
41 filter can be configured with flow meters and an ATS scheduler if both PSFP and ATS are supported.
- 42 e) If the stream filter specifies an ATS scheduler (8.6.5.6), that is used to process the frame. It
43 computes an eligibility time for the frame for subsequent use by the ATS transmission selection
44 algorithm (8.6.8.5). The frame can be discarded if a maximum eligibility time is exceeded
45 (8.6.11.3.13).

8.6.5.2.1 PSFP support

Each Bridge component or end station that implements PSFP shall support stream filtering, maximum SDU size filtering, stream gating, and flow metering, with the following:

- a) A single *Stream Filter Instance Table* (8.6.5.3).
- b) A single *Stream Gate Instance Table* (8.6.5.4).
- c) A single *Flow Meter Instance Table* (8.6.5.5).

The relationship between stream filters, stream gates, flow meters for streams subject to PSFP processing (as identified by the stream filter) is illustrated by Figure 8-14 for a number of streams.

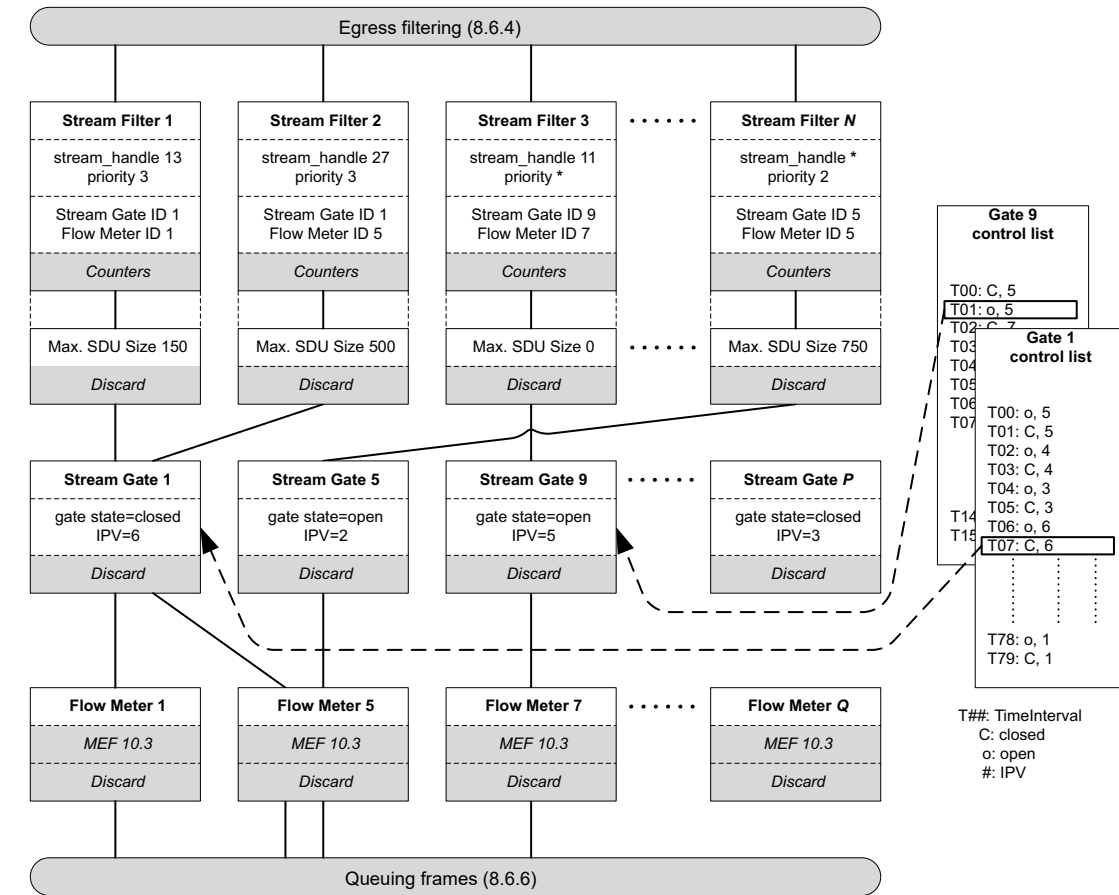


Figure 8-14—Per-stream classification for PSFP

1 8.6.5.2.2 ATS support

2 Each Bridge component that implements ATS shall support stream filtering, maximum SDU size filtering,
3 stream gates supporting IPV assignment, ATS schedulers, and ATS scheduler groups, with the following:

- 4 a) A single *Stream Filter Instance Table* (8.6.5.3).
- 5 b) A single *Stream Gate Instance Table* (8.6.5.4).
- 6 If the Bridge component does not support PSFP in addition to ATS, each stream gate only supports
7 IPV assignment (8.6.5.4). IPV can be used as part of adjusting per-hop delay bounds to meet specific
8 networks' end-to-end delay requirements.
- 9 c) A single *ATS Scheduler Instance Table* (8.6.5.6).
- 10 d) A single *ATS Scheduler Group Instance Table* (8.6.5.6).
- 11 e) An *ATS Port Parameter Table* for each Bridge Port (8.6.5.6).

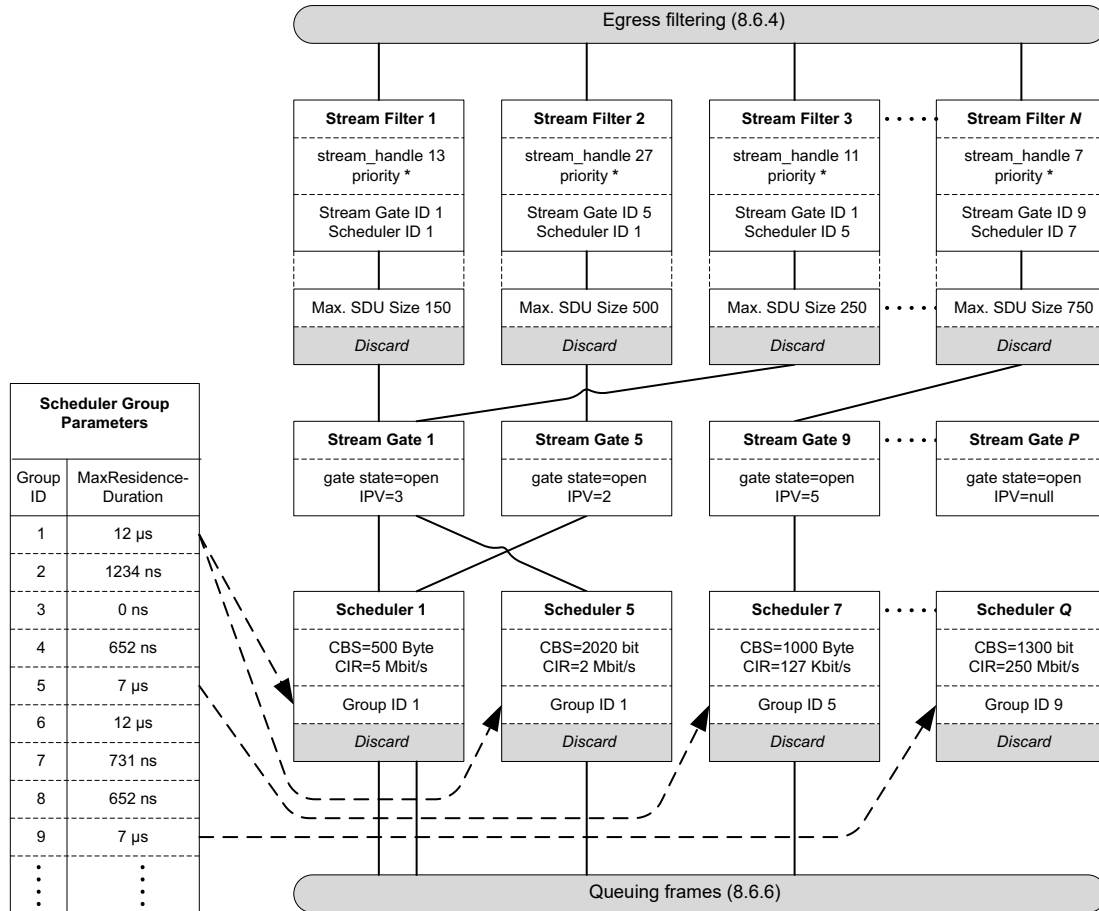
12 NOTE 1—The operation of ATS schedulers (8.6.11) provides policing capabilities similar to those provided by flow
13 meters (8.6.5.5). This handling of improper traffic can be sufficient, such that no additional flow meters are required.

14 NOTE 2—For bridges with support for ATS, and without support for PSFP, stream gates of ATS traffic will never close.
15 In this case, stream gates are only used for IPV assignment.

16 NOTE 3—In presence of state changes of stream gates associated with ATS traffic, the asynchronous nature of ATS
17 traffic can require that state changes of multiple of these associated gates are executed simultaneously.

18 The relationship between stream filters, stream gates, and ATS schedulers for streams subject to ATS
19 processing is illustrated by Figure 8-15.

20 ATS support in end stations is provided by a modified variant of per-stream classification and metering for
21 ATS, as specified in 47.1.



KEY

Discard: Frame discarding abilities and parameters (Clause 8.6.5.3.1, Clause 8.6.5.4, 8.6.5.6).

CBS: CommittedBurstSize parameter (8.6.5.6, 8.6.11.3.5).

CIR: CommittedInformationRate parameter (8.6.5.6, 8.6.11.3.6).

Figure 8-15—Per-stream classification and metering for ATS

8.6.5.2.3 CI Support

Each Bridge component or an end station that implements Congestion Isolation supports stream identification and stream gates to allow traffic class modification, with the following:

- a) A single Stream Filter Instance Table (8.6.5.3).
- b) A single Stream Gate Instance Table (8.6.5.4).

The relationship between stream filters and stream gates for streams subject to CI processing (as identified by the stream filter) is illustrated by Figure 8-16 for a number of streams.

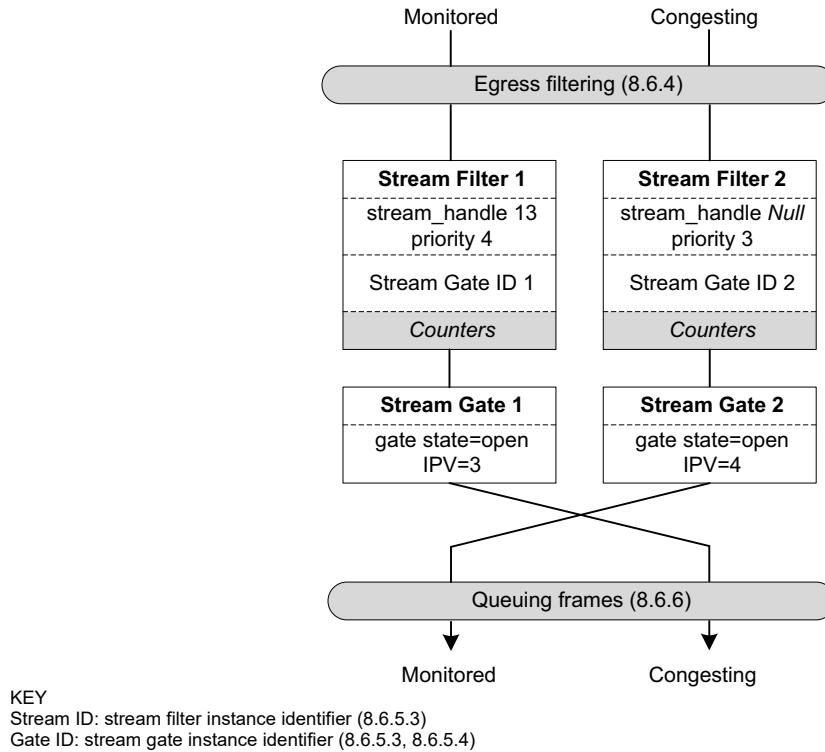


Figure 8-16—Per-stream classification and assignment for CI

Congestion isolation uses two stream filter instances from the stream filter instance table (8.6.5.3) to select stream gate instances that will modify the priority of flows. One stream filter instance is used to modify the priority of congesting flows from the monitored queue to the congesting queue. The other stream filter instance is used to modify the priority of congesting flows from the congesting queue back to the monitored queue. There are no *filter specifications* used by congestion isolation.

The *stream_handle* and *priority* parameters associated with a received frame select the stream filter instance of congestion isolation for a particular monitored traffic class. The purpose of the stream filter is to select the stream gate instance that will modify the priority of congesting flow frames so they will traverse the congesting queue. The absence of a *stream_handle* and the *priority* parameter associated with a received frame select the stream gate instance that will modify the priority of received frames back to the priority of the monitored queue.

NOTE—Changes to the frame's priority are intended to persist (see 49.2.4 and 49.2.6) by modifying the Priority Code Point (see 6.9.3) or other indicators used to select the traffic class, such as the Differentiated Services Code Point (see IETF RFC 2474). As noted in 6.8.1, modification to a frame will require the FCS to be regenerated. Options for regenerating the FCS are discussed in Annex O.

1 Congestion isolation uses two stream gate instances from the stream gate instance table (8.6.5.4) to modify
2 the priority of flows. The operational and an administrative *stream gate state* (8.6.10.4, 8.6.10.5, 12.31.3) is
3 set to Open for Congestion Isolation and the *GateClosedDueToInvalidRxEnable* and
4 *GateClosedDueToOctetsExceededEnable* parameters are set to FALSE. The internal priority value (IPV) is
5 used in place of the priority value associated with the frame to determine the frame's traffic class, using the
6 Traffic Class Table as specified in 8.6.6. The *stream gate control list* feature is not used by Congestion
7 Isolation and is set to null.

8 8.6.5.3 Stream filtering

9 A received frame can be associated with a stream filter using the frame's *stream_handle* and *priority*
10 parameters. The *stream_handle* is a sub-parameter of the *connection_identifier* parameter of the ISS (6.6),
11 provided by the stream identification function specified in IEEE Std 802.1CB.

12 Each stream filter comprises the following:

- 13 a) An integer *stream filter identifier*.
- 14 b) A *stream_handle* specification, either:
 - 15 1) A single value, as specified in IEEE Std 802.1CB.
 - 16 2) A wildcard, that matches any *stream_handle*.
 - 17 3) If congestion isolation is supported, a null-handle, that matches when no *stream_handle* is
18 provided.
- 19 c) A *priority* specification, either:
 - 20 1) A single priority value.
 - 21 2) A wildcard value that matches any priority value.
- 22 d) Maximum SDU size filtering (8.6.5.3.1) information, comprising:
 - 23 1) An integer *Maximum SDU size*, in octets. A value of 0 disables maximum SDU size filtering
24 for this stream filter.
 - 25 2) A boolean *StreamBlockedDueToOversizeFrameEnable* parameter.
 - 26 3) A boolean *StreamBlockedDueToOversizeFrame* parameter.
- 27 e) An integer *stream gate identifier* (8.6.5.4).
- 28 f) An integer *flow meter instance identifier* (8.6.5.5).
29 If this parameter is absent, frames associated with the stream filter are not subject to flow metering.
- 30 g) An integer *ATS scheduler instance identifier* (8.6.5.6).
31 If this parameter is absent, frames associated with the stream filter are not subject to ATS scheduling
32 and transmission selection.

33 Each stream filter also comprises the following counters for frames associated with the stream filter:

- 34 h) *MatchingFramesCount*: all frames associated with that stream filter.
- 35 i) *PassingSDUCount*: frames passing the maximum SDU size filter (8.6.5.3.1).
- 36 j) *NotPassingSDUCount*: frames not passing the maximum SDU size filter (8.6.5.3.1).
- 37 k) *PassingFrameCount*: frames passing the associated stream gate (8.6.5.4).
- 38 l) *NotPassingFrameCount*: frames not passing the stream gate (8.6.5.4).
- 39 m) *RedFramesCount*: frames discarded by the flow meter (8.6.5.5).

1 The *stream filter identifier* uniquely identifies the stream filter, indexing a *Stream Filter Instance Table* of up
2 to *MaxStreamFilterInstances* stream filters. Each received frame is associated with the stream filter with the
3 lowest *stream filter identifier* whose *stream_handle* and *priority* specification match the frame's parameters,
4 and the *MatchingFramesCount* is incremented for that filter.

5 NOTE 1—The use of *stream_handle* and *priority*, along with the wild-carding rules previously stated, allow
6 configuration possibilities that go beyond the selection of individual streams, for example, per-priority filtering and
7 policing, or per-priority per-reception Port filtering and policing can be configured using these rules.

8 NOTE 2—If it is desired to discard frames that do not match any other stream filter, rather than such frames being
9 processed without filtering, a stream filter can be placed at the end of the table with *stream_handle* and *priority*
10 specifications both wild-carded and the stream gate identifier of a permanently closed stream gate.

11 NOTE 3—A discarded frame is not processed or counted by subsequent elements of the forwarding process. As a
12 consequence, implementations can internally represent all counters [items h) through m) above] by the four counters
13 *MatchingFramesCount* [item h) above], *NotPassingSDUCount* [item j) above], *NotPassingFrameCount* [item l) above],
14 and *RedFramesCount* [item m) above].

15 8.6.5.3.1 Maximum SDU Size Filtering

16 If the SDU size of a frame exceeds the value of the associated stream filter's Maximum SDU size parameter,
17 the frame is discarded and that stream filter's *NotPassingSDUCount* is incremented. If the stream filter's
18 *StreamBlockedDueToOversizeFrameEnable* parameter is configured to be TRUE, the
19 *StreamBlockedDueToOversizeFrame* parameter is set to TRUE and all subsequent frames will be discarded
20 until *StreamBlockedDueToOversizeFrame* is administratively reset to FALSE.

21 Otherwise, the stream filter's *PassingSDUCount* is incremented (see 8.6.5.3). The default configuration of
22 both *StreamBlockedDueToOversizeFrameEnable* and *StreamBlockedDueToOversizeFrame* is FALSE.

23 NOTE—The Maximum SDU size is defined per stream filter and can therefore differ from the *queueMaxSDU* specified
24 in 8.6.8.4. As *queueMaxSDU* is applied after the flow classification and metering, it is possible that a frame that passes
25 the Maximum SDU size filter will later be discarded because its SDU size exceeds *queueMaxSDU*.

26 8.6.5.4 Stream gating

27 Stream gates can discard frames whose reception times contradict a given time schedule. Stream gates can
28 also map the frame's priority to an internal priority value (IPV) that is used to make subsequent queuing
29 decisions (8.6.6), while retaining the frame's original priority for transmission.

30 NOTE 1—The IPV facilitates ATS per-hop delay bound adjustment to satisfy specific networks' end-to-end delay
31 requirements. Annex T (CQF) describes another IPV use case. Another use case is the ability to isolate congesting flows
32 to a congesting flow queue as specified by Congestion Isolation (Clause 49).

33 Each stream gate comprises the following:

- 34 a) An integer *stream gate instance identifier*.
- 35 b) An administrative and an operational *stream gate state* parameter. These parameters take the
36 following values:
 - 37 1) Open: Frames are permitted to pass through the stream gate.
 - 38 2) Closed: Frames are not permitted to pass through the stream gate.
- 39 c) An administrative and an operational *internal priority value specification* parameter. These
40 parameters take the following values:
 - 41 1) Null, in this case the received frame's priority parameter is used as the IPV.
 - 42 2) A specific IPV for the frame.

43 The *stream gate instance identifier* uniquely identifies the stream gate, indexing a *Stream Gate Instance*
44 *Table* of up to *MaxStreamGateInstances* stream gates.

45 If PSFP are supported, each stream gate also includes the following:

- 46 d) An administrative and an operational *stream gate control list*.

- e) A boolean *GateClosedDueToInvalidRxEnable* parameter.
- f) A boolean *GateClosedDueToInvalidRx* parameter.
- g) A boolean *GateClosedDueToOctetsExceededEnable* parameter.
- h) A boolean *GateClosedDueToOctetsExceeded* parameter.

An instance of the stream gate control state machine (8.6.10) determines the operational values of the *stream gate state* and the *internal priority value specification* [items b) and c) above] by the cyclical execution of the control operations (see Table 8-4) specified in the stream gate's *stream gate control list* [item d) above]. The administrative *stream gate state* and *internal priority value specification* parameters are used to determine the initial values of the corresponding operational parameters, and the administrative *stream gate control list* parameter allows configuration of a new control list prior to enabling its use in a running system.

If a frame is passed by a stream gate, the *PassingFrameCount* of the stream filter (8.6.5.3) associated with that frame is incremented. The *NotPassingFrameCount* is incremented if the frame is discarded.

Table 8-4—Stream gate control operations

Operation name	Parameter(s)	Action
SetGateAndIPV	StreamGateState, IPV, TimeInterval, IntervalOctetMax	The StreamGateState parameter specifies a desired state, <i>open</i> or <i>closed</i> , for the stream gate, and the IPV parameter specifies a desired value of the IPV associated with the stream. On execution, the StreamGateState and IPV parameter values are used to set the operational values of the stream gate state and internal priority value specification parameters for the stream. After <i>TimeInterval</i> ticks (8.6.9.4.16) has elapsed since the completion of the previous stream gate control operation in the stream gate control list, control passes to the next stream gate control operation. The optional IntervalOctetMax parameter specifies the maximum number of MSDU octets that are permitted to pass the gate during the specified TimeInterval. If the IntervalOctetMax parameter is omitted, there is no limit on the number of octets that can pass the gate.

Stream gates are able to permanently discard frames and thus effectively override the operational gate state (i.e., the stream gate behaves as if the operational stream gate state is Closed). This capability is provided in case a frame is discarded due to a closed gate state [8.6.5.4 item b)], and in case a frame is discarded because there are insufficient octets left (i.e., the value of the IntervalOctetsLeft parameter, as specified in 8.6.10.8, is lower than the frame length).

- i) Permanent frame discarding due to a frame received during a closed gate state is enabled if the *GateClosedDueToInvalidRxEnable* parameter is TRUE, and disabled if this parameter is FALSE. If enabled and any frame is discarded, then the *GateClosedDueToInvalidRx* parameter is set to TRUE, and all subsequent frames are discarded as long as the *GateClosedDueToInvalidRxEnable* and *GateClosedDueToInvalidRx* parameters are TRUE. Changes of the *GateClosedDueToInvalidRxEnable* parameter and transitions from TRUE to FALSE of the *GateClosedDueToInvalidRx* parameter are administrative actions.
- j) Permanent frame discarding due to insufficient left octets is enabled if the *GateClosedDueToOctetsExceededEnable* parameter is TRUE, and disabled if this parameter is FALSE. If enabled and any frame is discarded because there are insufficient octets left, then the *GateClosedDueToOctetsExceeded* parameter is set to TRUE, and all subsequent frames are discarded as long as the *GateClosedDueToOctetsExceededEnable* and *GateClosedDueToOctetsExceededEnable* parameters are TRUE. Changes of the *GateClosedDueToOctetsExceededEnable* parameter and transitions from TRUE to FALSE of the *GateClosedDueToOctetsExceeded* parameter are administrative actions.

1 Per default, permanent frame discarding is disabled and all associated parameters have the default value
2 FALSE.

3 NOTE 2—Permanent frame discarding allows the detection of incoming frames during time periods when the stream
4 gate is in the closed state and exceptionally large ingress bursts to result in the stream gate behaving as it is in a
5 permanently closed state, until such a time as management action is taken to reset the condition. The intent is to support
6 applications where the transmission and reception of frames across the network is coordinated such that frames are
7 received only when the stream gate is open with a limited overall amount of ingress octets. Hence, frames received by
8 the stream gate when it is in the closed state and unexpected amounts of ingress octets represent invalid receive
9 conditions.

10 **8.6.5.5 Flow metering**

11 The flow meters specified by this subclause (8.6.5.5) implement the parameters and algorithm specified in
12 *Bandwidth Profile Parameters and Algorithm* in MEF 10.3 with the additions described in this subclause.

13 Each flow meter comprises the following:

- 14 a) An integer *flow meter identifier*.
- 15 b) An integer *Committed information rate (CIR)*, in bits per second (MEF 10.3).
- 16 c) An integer *Committed burst size (CBS)*, in octets (MEF 10.3).
- 17 d) An integer *Excess Information Rate (EIR)*, in bits per second (MEF 10.3).
- 18 e) An integer *Excess burst size (EBS) per bandwidth profile flow*, in octets (MEF 10.3).
- 19 f) A *Coupling flag (CF)*, which takes the value 0 or 1 (MEF 10.3).
- 20 g) A *Color mode (CM)*, which takes the value *color-blind* or *color-aware* (MEF 10.3).
- 21 h) A boolean *DropOnYellow* parameter.
- 22 i) A boolean *MarkAllFramesRedEnable* parameter.
- 23 j) A boolean *MarkAllFramesRed* parameter.

24 Each frame has a predetermined color for subsequent processing by a flow meter. If this color is determined
25 by the *drop_eligible* parameter (6.9.1), the color is determined as follows:

- 26 k) If the *drop_eligible* parameter is FALSE, the color is green.
- 27 l) If the *drop_eligible* parameter is TRUE, the color is yellow.

28 NOTE 1—This standard specifies the relationship among frame length, media-dependent overhead, ATS scheduler
29 operation, and the associated parameters in 8.6.11. In contrast, the operation and parameters of flow meters in 8.6.5.5 are
30 as specified by MEF 10.3.

31 NOTE 2—Envelope and Rank, as defined in MEF 10.3, are not used by the flow meters described in this subclause; i.e.,
32 the reduced functionality algorithm described in 12.2 of MEF 10.3 is used.

33 The *flow meter identifier* uniquely identifies the flow meter instance, indexing a *Flow Meter Instance Table*
34 of up to *MaxFlowMeterInstances* flow meters.

35 The *DropOnYellow* parameter indicates whether frames marked yellow by the MEF 10.3 algorithm are
36 discarded or marked as drop eligible:

- 37 m) A value of TRUE indicates that yellow frames are discarded.
- 38 n) A value of FALSE indicates that the *drop_eligible* parameter of yellow frames is set to TRUE.

39 NOTE 3—Changing the value of the *drop_eligible* parameter may change the contents of the frame, depending on how
40 the frame is tagged when transmitted, which may then require updating the *frame_check_sequence*. Mechanisms for
41 conveying information from ingress to egress that the *frame_check_sequence* may require updating are implementation
42 dependent.

43 Flow meters can permanently discard all frames after an initial frame has been discarded. This capability is
44 enabled if the *MarkAllFramesRedEnable* parameter is TRUE, and disabled if this parameter is FALSE. If
45 enabled and the flow meter discards a frame, then the *MarkAllFramesRed* parameter is set to TRUE, and all

1 subsequent frames are discarded as long as the `MarkAllFramesRedEnable` and `MarkAllFramesRed`
2 parameters are TRUE.

3 Changes of the `MarkAllFramesRedEnable` parameter and transitions from TRUE to FALSE of the
4 `MarkAllFramesRed` parameter are administrative actions. Per default, permanent frame discarding is
5 disabled and both associated parameters have the default value FALSE.

6 Each time a flow meter discards a frame, the `RedFramesCount` counter of the originating stream filter
7 (8.6.5.3) is increased.

8 **8.6.5.6 ATS eligibility time assignment**

9 Asynchronous Traffic Shaping (ATS) schedulers assign eligibility times to frames which are then used for
10 traffic regulation by the ATS transmission selection algorithm (8.6.8.5).

11 NOTE 1—ATS schedulers, as defined in this subclause (8.6.5.6), realize the computational part of the overall traffic
12 shaping operation of ATS. The complete operation is provided in combination with the ATS transmission selection
13 algorithm (8.6.8.5), which uses the assigned eligibility times to regulate the traffic for transmission.

14 Each ATS scheduler comprises the following:

- 15 a) An integer *scheduler identifier*.
- 16 b) An integer *scheduler group identifier*.
- 17 c) An integer *CommittedBurstSizeParameter* parameter, in bits (8.6.11.3.5).
- 18 d) An integer *CommittedInformationRate* parameter, in bits per second (8.6.11.3.6).
- 19 e) An internal *bucket empty time* state variable, in seconds (8.6.11.3.3).

20 ATS schedulers are organized in *ATS scheduler groups*. There is one ATS scheduler group per reception Port
21 per upstream traffic class, where the latter refers to the transmitting traffic class in the device connected to
22 the given reception Port. All ATS schedulers that process frames from a particular reception Port and a
23 particular upstream traffic class are in the respective ATS scheduler group.

24 Each ATS scheduler group comprises the following:

- 25 f) An integer *scheduler group identifier*.
- 26 g) An integer *MaximumResidenceTime* parameter, shared by all ATS schedulers in a scheduler group,
27 in nanoseconds (8.6.11.3.13).
- 28 h) An internal *group eligibility time* state variable, shared by all ATS schedulers in a scheduler group,
29 in seconds (8.6.11.3.10).

30 NOTE 2—The organization of ATS schedulers into groups results in a non-decreasing ordering of eligibility times of
31 successive frames associated with a single group. This permits frames of one group to be queued in FIFO order.

32 Each ATS scheduler assigns eligibility times to the associated frames, and discards frames in exceptional
33 situations. The underlying operations are performed by an ATS scheduler state machine (8.6.11) associated
34 with an ATS scheduler. This state machine updates the associated bucket empty time and group eligibility
35 time state variables based on the `CommittedBurstSize`, the `CommittedInformationRate`, the
36 `MaxResidenceTime`, the frame arrival times, and the frame lengths (including media-specific overhead).

37 Each Port is associated with the following variable for ATS schedulers:

- 38 i) An integer *DiscardedFramesCount* of frames discarded by the ATS schedulers associated with that
39 reception Port.

Each Bridge component provides an *ATS Scheduler Instance Table* of up to *MaxSchedulerInstances* ATS schedulers, an *ATS Scheduler Group Instance Table* of up to *MaxSchedulerGroupInstances* ATS scheduler groups, and a *ATS Scheduler Port Parameter Table* shared by all ATS schedulers associated with a reception Port.

NOTE 3—Whether ATS scheduler instances, ATS scheduler group instances, the scheduler instance table, and the scheduler group instance table are located in reception ports or in transmission ports is implementation specific.

8.6.6 Queuing frames

The Forwarding Process shall queue each received frame to each of the potential transmission Ports (8.6.1, 8.6.3, 8.6.4).

NOTE 1—The Forwarding Process is modeled as receiving a frame as the parameters of a data indication and transmitting through supplying the parameters of a data request. Queuing a frame awaiting transmission amounts to placing the parameters of a data request on an outbound queue.

The Forwarding Process provides storage for queued frames, awaiting an opportunity to submit these for transmission. The order of frames received on the same Bridge Port shall be preserved for:

- a) Unicast frames with a given VID, priority, flow hash, and destination address and source address combination.
- b) Multicast frames with a given VID, priority, flow hash, and destination address.

Flow hash is applicable only on Bridges supporting flow filtering (44.2).

The Forwarding Process provides one or more queues for a given Bridge Port, each corresponding to a distinct traffic class. Each frame is mapped to a traffic class using the Traffic Class Table for the Port. The priority value used for this mapping is determined as follows:

- c) If stream gates are unsupported (8.6.5.4), the frame's priority is used.
- d) If stream gates are supported and the IPV specification assigned to the frame is the null value, the frame's priority is used.
- e) If stream gates are supported and the IPV specification assigned to the frame is an IPV, this IPV is used.

Traffic class tables may be managed. Table 8-5 shows the recommended mapping for the number of classes implemented, in implementations that do not support the credit-based shaper transmission selection algorithm (8.6.8.2). The requirements for priority to traffic class mappings in implementations that support the credit-based shaper transmission selection algorithm are defined in 34.5. Up to eight traffic classes may be supported, allowing separate queues for each priority.

NOTE 2—Different Ports can use different numbers of traffic classes. Ports with media access methods that support a single transmission priority, such as Ethernet, can support more than one traffic class.

NOTE 3—A queue in this context is not necessarily a single FIFO data structure. A queue is a record of all frames of a given traffic class awaiting transmission on a given Bridge Port. The transmission selection algorithm (8.6.8) determines which traffic class, among those classes with frames available for transmission, provides the next frame for transmission. A variety of queue structures such as a single FIFO, or a set of FIFOs with one for each VLAN or priority, is allowed.

In a congestion-aware Bridge (Clause 30) or a congestion isolation aware Bridge (Clause 49), the act of queuing a frame for transmission on a Bridge Port can result in the Forwarding Process generating a CNM or a CIM. The CNM and CIM are injected back into the Forwarding Process (8.6.1) as if they had been received on that Bridge Port.

If PSFP is supported (8.6.5.2), and the IPV associated with the stream filter that passed the frame is anything other than the null value, then that IPV is used to determine the traffic class of the frame, in place of the frame's priority, via the Traffic Class Table specified in 8.6.6. In all other respects, the queuing actions specified in 8.6.6 are unchanged.

Table 8-5—Recommended priority to traffic class mappings

		Number of available traffic classes							
		1	2	3	4	5	6	7	8
Priority	0 (Default)	0	0	0	0	0	1	1	1
	1	0	0	0	0	0	0	0	0
	2	0	0	0	1	1	2	2	2
	3	0	0	0	1	1	2	3	3
	4	0	1	1	2	2	3	4	4
	5	0	1	1	2	2	3	4	5
	6	0	1	2	3	3	4	5	6
	7	0	1	2	3	4	5	6	7
NOTE—The rationale for these mappings is discussed in Annex I.									

¹ The IPV is used only to determine the traffic class associated with a frame, and hence select an outbound queue; for all other purposes, the received priority is used.

8.6.7 Queue management

A frame queued for transmission on a Port shall be removed from that queue:

- a) Following a transmit data request. No further attempt shall be made to transmit the frame on that Port even if the transmission is known to have failed.
- b) If that is necessary to ensure that the maximum Bridge transit delay (6.5.6) will not be exceeded at the time at which the frame would subsequently be transmitted.
- c) If the transmission Port is no longer forwarding for that frame (8.4, 8.6.1). Frames entering the queue subsequent to the transition out of the Forwarding state (e.g., CFM frames from Down MPs, as described in 22.1, Figure 22-4, and Figure 22-7) are not discarded.

NOTE—CFM frames may be excepted, and not removed from the queue, on a transition out of the Forwarding state (see 22.1.3).

The frame can be removed from the queue, and not subsequently transmitted:

- d) If the time for which buffering is guaranteed has been exceeded for that frame
- e) By a queue management algorithm that attempts to improve the QoS provided by deterministically or probabilistically managing the queue depth based on the current and past queue depths.

Removal of a frame from a queue for any particular Port does not affect queuing of that frame for transmission on any other Port.

The probability of removing a frame with drop_eligible set shall not be less than that of removing a frame with drop_eligible False, all other conditions being equal. If a queue management algorithm is implemented, it should preferentially discard frames with drop_eligible True. If a Bridge supports encoding or decoding of drop_eligible from the PCP field of a VLAN tag (6.9.3) on any of its Ports, then it shall implement a boolean parameter Require Drop Encoding on each of its Ports with default value FALSE. If Require Drop Encoding is True and the Bridge Port cannot encode particular priorities with drop_eligible, then frames queued with those priorities and drop_eligible True shall be discarded and not transmitted.

1 8.6.8 Transmission selection

2 For each Port, frames are selected for transmission on the basis of the traffic classes that the Port supports
3 and the operation of the transmission selection algorithms supported by the corresponding queues on that
4 Port. For a given Port and traffic class, frames are selected from the corresponding queue for transmission if
5 and only if:

- 6 a) The operation of the transmission selection algorithm supported by that queue determines that there
7 is a frame available for transmission; and
- 8 b) For each queue corresponding to a numerically higher value of traffic class supported by the Port,
9 the operation of the transmission selection algorithm supported by that queue determines that there
10 is no frame available for transmission.

11 In a port of a Bridge or station that supports PFC, a frame of priority n is not available for transmission if
12 that priority is paused, i.e., if `Priority_Paused[n]` is TRUE (see 36.1.3.2) on that port. When Transmission
13 Selection is running above Link Aggregation, a frame of priority n is not available for transmission if that
14 priority is paused on the physical port to which the frame is to be distributed.

15 NOTE 1—Two or more priorities can be combined in a single queue. In this case if one or more of the priorities in the
16 queue are paused, it is possible for frames in that queue not belonging to the paused priority to not be scheduled for
17 transmission.

18 NOTE 2—Mixing PFC and non-PFC priorities in the same queue results in non-PFC traffic being paused causing
19 congestion spreading, and therefore is not recommended.

20 In a port of a Bridge or station that supports frame preemption, a frame of priority n is not available for
21 transmission if that priority is identified in the frame preemption status table (6.7.2) as preemptable and
22 either the `holdRequest` object (12.30.1.5) is set to the value *hold*, or the transmission of a prior preemptable
23 frame has yet to complete because it has been interrupted to allow the transmission of an express frame.

24 The strict priority transmission selection algorithm defined in 8.6.8.1 shall be supported by all Bridges as the
25 default algorithm for selecting frames for transmission. The credit-based shaper transmission selection
26 algorithm defined in 8.6.8.2, the ETS algorithm defined in 8.6.8.3, and the ATS transmission selection
27 algorithm defined in 8.6.8.5 may be supported in addition to the strict priority algorithm. Further
28 transmission selection algorithms, selectable by management means, may be supported as an
29 implementation option so long as the requirements of 8.6.6 are met.

30 The Transmission Selection Algorithm Table for a given Port assigns, for each traffic class that the Port
31 supports, the transmission selection algorithm that is to be used to select frames for transmission from the
32 corresponding queue. Transmission Selection Algorithm Tables may be managed, and allow the
33 identification of vendor-specific transmission selection algorithms. The transmission selection algorithms
34 are identified in the Transmission Selection Algorithm Table by means of integer identifiers, as defined in
35 Table 8-6.

36 A recommended configuration of the Transmission Selection Algorithm Tables for use with the credit-based
37 shaper transmission selection algorithms is specified in 34.5. When the enhancements for scheduled traffic
38 are supported, additional requirements to determine whether a frame is available for transmission apply, as
39 specified in 8.6.8.4.

Table 8-6—Transmission selection algorithm identifiers

Transmission selection algorithm	Identifier
Strict priority (8.6.8.1)	0
Credit-based shaper (8.6.8.2)	1
Enhanced Transmission Selection (ETS) (8.6.8.3)	2
ATS Transmission Selection (8.6.8.5)	3
Reserved for future standardization	4–254
Vendor-specific Transmission Selection algorithm value for use with DCBX (D.2.8.8)	255
Vendor-specific	A four-octet integer, where the most significant 3 octets hold an OUI or CID value, and the least significant octet holds an integer value in the range 0–255 assigned by the owner of the OUI or CID.

1 8.6.8.1 Strict priority algorithm

2 For a given queue that supports strict priority transmission selection, the algorithm determines that there is a
3 frame available for transmission if the queue contains one or more frames. The order in which frames are
4 selected for transmission from the queue shall maintain the ordering requirement specified in 8.6.6.

5 8.6.8.2 Credit-based shaper algorithm

6 For a given queue that supports credit-based shaper transmission selection, the algorithm determines that a
7 frame is available for transmission if the following conditions are all true:

- 8 a) The queue contains one or more frames.
- 9 b) The *transmitAllowed* signal for the queue is TRUE.

10 The order in which frames are selected for transmission from the queue shall maintain the ordering
11 requirement specified in 8.6.6.

12 The following external parameters are associated with each queue that supports the operation of the
13 credit-based shaper algorithm:

- 14 c) ***portTransmitRate***. The transmission rate, in bits per second, that the underlying MAC Service that
15 supports transmission through the Port provides. The value of this parameter is determined by the
16 operation of the MAC.
- 17 d) ***idleSlope***. The rate of change of *credit*, in bits per second, when the value of *credit* is increasing (i.e.,
18 while *transmit* is FALSE and the transmission gate for the queue is open [8.6.8.4]). The value of
19 *idleSlope* can never exceed *portTransmitRate*. If the enhancements for scheduled traffic (8.6.8.4) are
20 not supported, or if GateEnabled is FALSE (8.6.9.4.14), the value of *idleSlope* for a given queue is
21 equal to the value of the *operIdleSlope(N)* parameter for that queue, as defined in 34.3. If the
22 enhancements for scheduled traffic (8.6.8.4) are supported, and GateEnabled is TRUE (8.6.9.4.14),
23 then:

$$24 \quad \textit{idleSlope} = (\textit{operIdleSlope}(N) \times \textit{OperCycleTime} / \textit{GateOpenTime})$$

1 where *OperCycleTime* is as defined in 8.6.9.4.19 and *GateOpenTime* is equal to the total amount of
2 time during the gating cycle that the gate state for the queue is Open.

3 NOTE 1—When scheduled traffic operation is enabled, *credit* is accumulated only while the gate is open; therefore, the
4 effective data rate of the *idleSlope* is increased to reflect the duty cycle for the transmission gate associated with the
5 queue; however, the value of *operIdleSlope(N)* for the queue remains unchanged.

6 The following internal parameters are associated with each queue that supports the credit-based shaper
7 algorithm:

8 e) ***transmit***. Takes the value TRUE for the duration of a frame transmission from the queue; FALSE
9 when any frame transmission from the queue has completed. If the credit-based shaper algorithm is
10 used in combination with frame preemption (6.7.2), *transmit* only takes the value TRUE while the
11 frame is actually being transmitted by the MAC. If the frame transmission is delayed or interrupted
12 (e.g., the frame is a preemptable frame and its transmission is interrupted to allow the transmission
13 of an express frame from a different queue, or the frame is an express frame and there is a delay
14 before transmission can start because a preemptable frame was being transmitted) *transmit* takes the
15 value FALSE until transmission of the frame commences or is resumed. *Transmit* also takes the
16 value FALSE during the transmission of any overhead that is a consequence of frame preemption;
17 i.e., any additional frame overhead that is added to the preemptable frame when preemption occurs.

18 NOTE 2—The consequence of this is that any overhead associated with preemption does not come out of the reserved
19 bandwidth for the credit-based shaper.

20 f) ***credit***. The transmission credit, in bits, that is currently available to the queue. If, at any time, there
21 are no frames in the queue, and the *transmit* parameter is FALSE, and *credit* is positive, and the
22 transmission gate for the queue is open (8.6.8.4), and there is no preemptable frame from this queue
23 for which transmission is in progress but has been interrupted, then *credit* is set to zero.

24 NOTE 3—The condition that the *transmit* parameter is FALSE and a preemptable frame or part-frame is waiting in the
25 MAC for transmission can only occur if the credit-based shaper algorithm is used in combination with Preemption.

26 g) ***sendSlope***. The rate of change of *credit*, in bits per second, when the value of *credit* is decreasing
27 (i.e., while *transmit* is TRUE). The value of *sendSlope* is defined as follows:

$$28 \quad \textit{sendSlope} = (\textit{idleSlope} - \textit{portTransmitRate})$$

29 h) ***transmitAllowed***. Takes the value TRUE when the *credit* parameter is zero or positive; FALSE when
30 the *credit* parameter is negative.

31 NOTE 4—The value of *credit* is naturally constrained by the operating parameters for the algorithm, and also by the
32 operation of any higher priority queues that use this algorithm. The lowest value that *credit* can reach depends on
33 *sendSlope* and the largest frame that will be transmitted from the queue. This largest frame size is a consequence of the
34 type of traffic that is being transmitted using the queue, rather than a limit imposed by the algorithm or by management.
35 The highest value that can be accumulated in *credit* depends on *idleSlope* and the length of time that the algorithm may
36 have to wait when the queue is not empty and there is other traffic being transmitted through the Port; for any given
37 queue, that length of time is bounded, as discussed in Annex L, which also illustrates how the algorithm operates under
38 varying conditions, and how its operation affects the maximum latency that can be experienced by SR traffic.

39 NOTE 5—In order for the credit-based shaper algorithm to operate as intended, it is necessary for all traffic classes that
40 support the algorithm to be numerically higher than any traffic classes that support the strict priority algorithm defined in
41 8.6.8.1. The mapping of priorities onto traffic classes, and the choice of traffic classes that support particular
42 transmission selection algorithms, is defined in 34.5.

43 Traffic classes using the credit-based shaper algorithm shall not use PFC and shall ignore the setting of the
44 bits related to such classes in the PFC Enable bit vector (see D.2.10.6)

8.6.8.3 ETS algorithm

If ETS is enabled for a traffic class, transmission selection is performed based on the allocation of bandwidth to that traffic class. Bandwidth is distributed among ETS traffic classes that support ETS algorithm such that each traffic class is allocated available bandwidth in proportion to its TCBandwidth (see Clause 37).

For a given queue that supports ETS, the algorithm determines that there is a frame available for transmission if the following conditions are all true:

- The queue contains one or more frames.
- The ETS algorithm (37.3) determines that a frame should be transmitted from the queue.
- There are no frames available for transmission for any queues running strict priority or credit-based shaper algorithms.

The order in which frames are selected for transmission from the queue shall maintain the ordering requirement specified in 8.6.6.

8.6.8.4 Enhancements for scheduled traffic

A Bridge or an end station may support enhancements that allow transmission from each queue to be scheduled relative to a known timescale. In order to achieve this, a transmission gate is associated with each queue; the state of the transmission gate determines whether or not queued frames can be selected for transmission (see Figure 8-17). For a given queue, the transmission gate can be in one of two states:

- Open*: Queued frames are selected for transmission, in accordance with the definition of the transmission selection algorithm associated with the queue.
- Closed*: Queued frames are not selected for transmission.

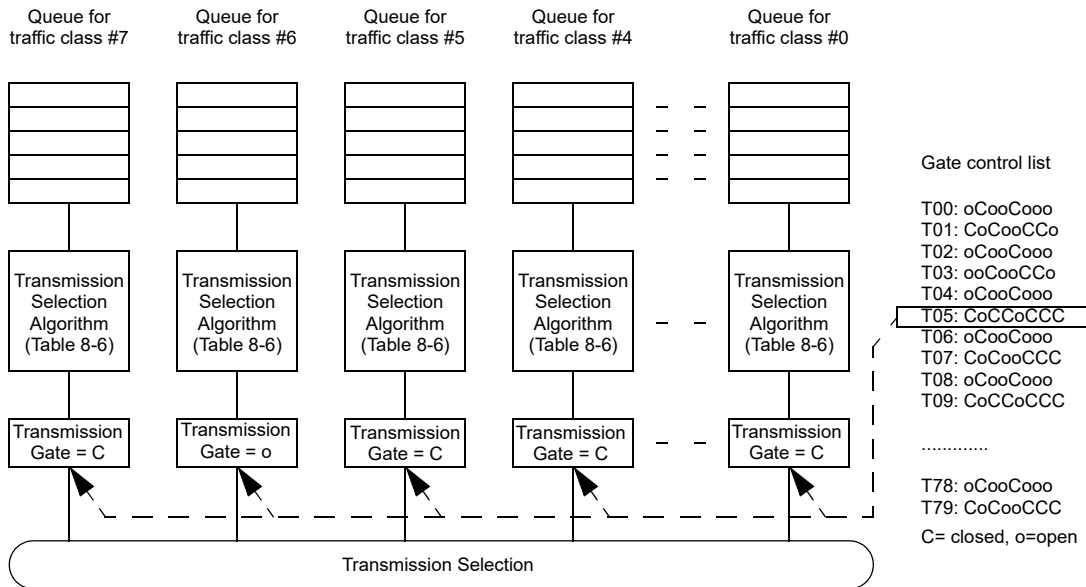


Figure 8-17—Transmission selection with gates

A *gate control list* associated with each Port contains an ordered list of gate operations. Each gate operation changes the transmission gate state for the gate associated with each of the Port's traffic class queues and allows associated control operations to be scheduled. In an implementation that does not support enhancements for scheduled traffic, all gates are assumed to be permanently in the *open* state. Table 8-7 identifies the gate operation types, their parameters, and the actions that result from their execution. The state machines that control the execution of the gate control list, along with their variables and procedures, are specified in 8.6.9.

Table 8-7—Gate operations

Operation name	Parameter(s)	Action
SetGateStates	GateState, TimeInterval	The <i>GateState</i> parameter indicates a value, <i>open</i> or <i>closed</i> , for each of the Port's queues. The gates are immediately set to the states indicated in the <i>GateState</i> parameter. This causes gate-close events (3.108) and/or gate-open events (3.109) to occur for any queue where the new <i>GateState</i> represents a change of state relative to the current state of the gate. After <i>TimeInterval</i> ticks (8.6.9.4.16) have elapsed since the completion of the previous gate operation in the gate control list, control passes to the next gate operation.
Set-And-Hold-MAC	GateState, TimeInterval	Performs all of the actions defined for the SetGateStates operation; in addition, the start of this operation marks the point in the sequence of gate operations at which the MAC associated with the port is to have stopped transmitting preemptable frames. This is achieved by setting the holdRequest managed object to the value <i>hold (1)</i> , at holdAdvance (Table 12-33) nanoseconds in advance of this point for the hold to have taken effect at this point. If frame preemption is not supported or not enabled (preemptionActive is FALSE), this operation behaves the same as SetGateStates.
Set-And-Release-MAC	GateState, TimeInterval	Performs all of the actions defined for the SetGateStates operation; in addition, the start of this operation marks the point in the sequence of gate operations at which the MAC associated with the port is permitted to resume transmitting preemptable frames; if an express frame is currently being transmitted by the MAC, the release takes effect at the end of that transmission. This is achieved by setting the holdRequest managed object to the value <i>release (2)</i> , at releaseAdvance (Table 12-33) nanoseconds in advance of this point for the release to have taken effect at this point. ^a If frame preemption is not supported or not enabled (preemptionActive is FALSE), this operation behaves the same as SetGateStates.

^a The releaseAdvance parameter allows the timing of when the release command is issued to vary depending upon the constraints of a particular implementation. Its value should be less than the minimum frame size so that release does not occur too early and interfere with transmission of the last express frame.

1 In addition to the other checks carried out by the transmission selection algorithm, a frame on a traffic class
2 queue is not available for transmission [as required for tests (a) and (b) in 8.6.8] if the transmission gate is in
3 the closed state or if there is insufficient time available to transmit the entirety of that frame, plus the
4 media-dependent overhead specified in 12.4.2.2, before the next gate-close event (3.108) associated with
5 that queue (Figure 8-18).

6 NOTE 1—For example, in the case of IEEE 802.3 media, the media-dependent overhead prior to the gate-close event
7 includes the preamble of a potentially following frame from a different traffic class.

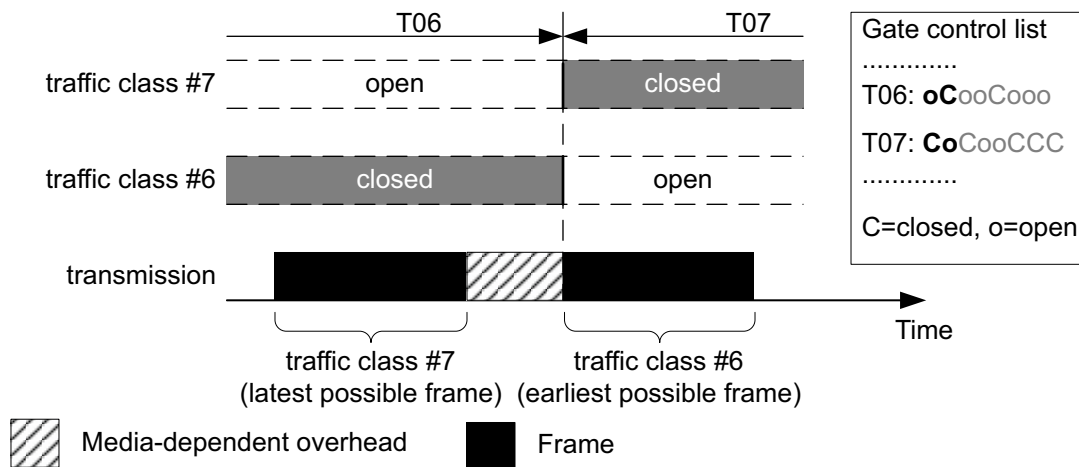


Figure 8-18—Frame timing at gate-close events

1 A per-traffic class counter, TransmissionOverrun (12.29.1.1.2), is incremented if the implementation detects
2 that a frame from a given queue is still being transmitted by the MAC when the gate-close event for that
3 queue occurs.

4 NOTE 2—There can be reasons why the frame size, and therefore the length of time needed for its transmission, is
5 unknown. For example, where frame preemption is supported and there is no way of telling in advance how many times
6 a given frame will be preempted before its transmission is complete. It is desirable that the schedule for such traffic is
7 designed to accommodate the intended pattern of transmission without overrunning the next gate-close event for the
8 traffic classes concerned.

9 NOTE 3— It is assumed that the implementation can determine the time at which the next gate-close event will occur
10 from the sequence of gate events. In the normal case, this can be achieved by inspecting the sequence of gate operations
11 in OperControlList (8.6.9.4.18) and associated variables. However, when a new configuration is about to be installed, it
12 would be necessary to inspect the contents of both the OperControlList and the AdminControlList (8.6.9.4.2) and
13 associated variables in order to determine the time of the next gate-close event, as the gating cycle for the new
14 configuration may differ in size and phasing from the old cycle.

15 A per-traffic class queue queueMaxSDU parameter defines the maximum service data unit size for each
16 queue; frames that exceed queueMaxSDU are discarded [item b8) in 6.5.2]. The value of queueMaxSDU for
17 each queue is configurable by management (12.29); its default value is the maximum SDU size supported by
18 the MAC procedures employed on the LAN to which the frame is to be relayed [item b3) in 6.5.2].

19 NOTE 4—The use of PFC is likely to interfere with a traffic schedule, because PFC is transmitted by a higher layer
20 entity (see Clause 36).

21 In order for an end station to support the scheduled transmission of frames, it is necessary for its behavior to
22 be compatible with the operation of the forwarding and queuing mechanisms employed in the Bridges to
23 which it connects. In effect, the requirements for an end station are for its transmission selection to operate
24 as if it is a single outbound Port of a Bridge that supports scheduled traffic. There are no particular
25 requirements for end station support for the reception of scheduled traffic; only for the transmission of
26 scheduled traffic.

27 8.6.8.5 ATS transmission selection algorithm

28 For a given queue that supports ATS transmission selection, a frame is eligible for transmission if the
29 assigned eligibility time (8.6.5.6, 8.6.11.3.2) is earlier than or at the current time.

30 The current time is determined by the TransmissionSelection Clock, which is an implementation specific
31 local system clock function.

1 Frames are selected for transmission in ascending order of the assigned eligibility times. Transmission
2 selection of frames with identical assigned eligibility times shall maintain the ordering requirement specified
3 in 8.6.6.

4 NOTE—In the case of frames with non-identical eligibility times, the ordering requirement from 8.6.6 is automatically
5 satisfied due the operation of the ATS scheduler state machines (8.6.11), which assign eligibility times in a
6 non-decreasing order.

7 **8.6.8.6 Enhancements for congestion isolation**

8 A Bridge component or an end station may support enhancements to isolate the frames of congesting flows
9 to a designated congesting traffic class. During the isolation of a congesting flow it may be possible for
10 frames of the same flow to queue in both the congesting and non-congesting traffic class. The transmission
11 gate associated with each monitored non-congesting queue and each congesting queue can be used to meet
12 the ordering requirements of 8.6.6. The state of the transmission gate determines whether or not queued
13 frames can be selected for transmission. For a given queue, the transmission gate can be either *Open* or
14 *Closed* as described in 8.6.8.4.

15 The need to control the transmission gate depends upon the transmission selection algorithm used and the
16 technique for determining when to return a congesting flow to non-congesting status. Congestion isolation
17 specifies that congesting and non-congesting queues use the same Active Queue Management (AQM)
18 approach and the same transmission selection algorithm. Additionally, the congesting queues have lower
19 priority than the monitored non-congesting queues.

20 When the transmission selection algorithm is strict priority and the technique for returning a congesting flow
21 to non-congesting status is only when the congesting queue is empty, the state of the transmission gate can
22 be set to *Open*. When the transmission selection algorithm is anything other than strict priority or the
23 technique for returning a congesting flow to non-congesting status allows for congesting frames to exist in
24 the congesting queue during the transition, the state of the transmission gate is controlled by the
25 *cipGateControl* (49.4.1.4.1) variable. An example implementation of setting *cipGateControl* to provide
26 transmission gate control for the purposes of ensuring ordering can be seen in Annex X.

27 Congestion isolation controls the transmission gate of the congesting and non-congesting queues to maintain
28 the order of congesting flow frames. As a consequence, transmission gates used for congestion isolation
29 cannot be used for other purposes.

30 **8.6.9 Scheduled traffic state machines**

31 The execution of the gate operations in a Port's gate control list (8.6.8.4) is controlled by three state
32 machines:

- 33 a) The Cycle Timer state machine (8.6.9.1)
- 34 b) The List Execute state machine (8.6.9.2)
- 35 c) The List Config state machine (8.6.9.3)

36 One instance of each state machine is instantiated for each Port that supports the enhancements for
37 scheduled traffic.

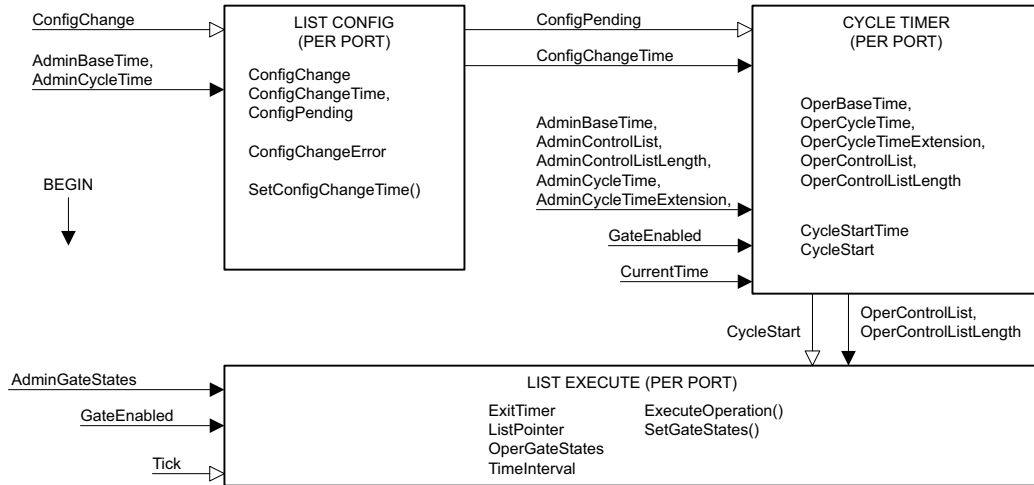
38 The Cycle Timer state machine initiates the execution of the gate control list and ensures that the gating
39 cycle time defined for the Port is maintained.

40 The List Execute state machine executes the gate operations in the gate control list, in sequence, and
41 establishes the appropriate time delay between each operation.

1 The List Config state machine manages the process of updating the current active schedule, interrupting the
2 operation of the other two state machines while the update process is performed, and restarting them once
3 the new schedule has been installed.

4 The state machine notation is specified in Annex E.

5 An overview of the state machines, showing their relationships and the variables that are used by them, can
6 be seen in Figure 8-19.



NOTATION:

Variables are shown both within the machine where they are initialized and between machines where they are used to communicate information. In the latter case the arrow styles, running from one machine to another, provide an overview of how the variables are used:

- ▶ Not changed by the target machine, this variable communicates between state machines for the same port.
- ▷ Set (or cleared) by the originating machine, cleared (or set) by the target machine, communicates between machines for the same port.

Figure 8-19—Scheduled traffic state machines—overview and relationships

8.6.9.1 Cycle Timer state machine

The Cycle Timer state machine shall implement the function specified by the state diagram in Figure 8-20 and the attendant definitions in 8.6.9.4.

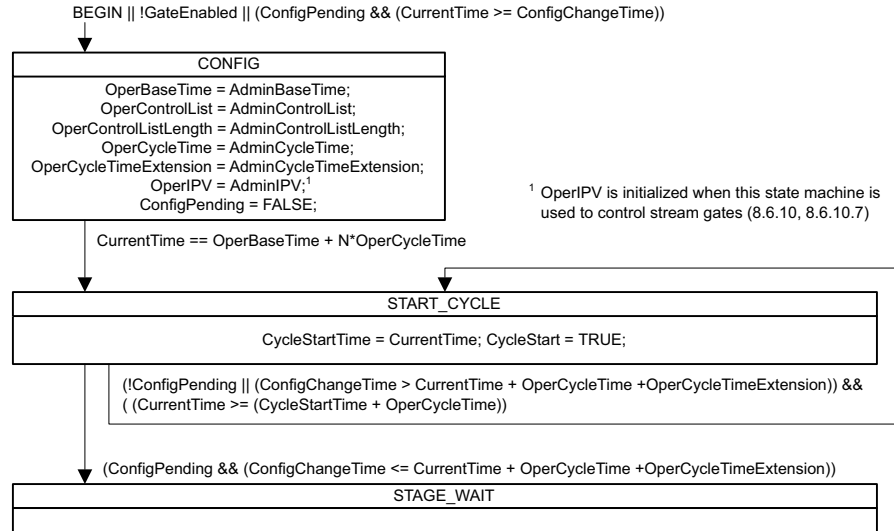


Figure 8-20—Cycle Timer state machine

NOTE 1—Since the origin of the PTP timescale is 1 January 1970 00:00:00 TAI, CycleStartTime will be larger than 1.3×10^{18} ns. If sufficient precision is not maintained when computing N, CycleStartTime will not be an integer multiple of OperCycleTime, which could result in misalignment of the cycles at ports on different bridges.

NOTE 2—If AdminBaseTime is set to the same time in the past in all bridges and end stations, OperBaseTime is always in the past, and all cycles start synchronized. Using AdminBaseTime in the past is appropriate when you can start schedules prior to starting the application that uses the schedules. Use of AdminBaseTime in the future is intended to change a currently running schedule in all bridges and end stations to a new schedule at a future time. Using AdminBaseTime in the future is appropriate when schedules must be changed without stopping the application.

1 8.6.9.2 List Execute state machine

2 The List Execute state machine shall implement the function specified by the state diagram in Figure 8-21
3 and the attendant definitions in 8.6.9.2.1, 8.6.9.2.2, and 8.6.9.4.

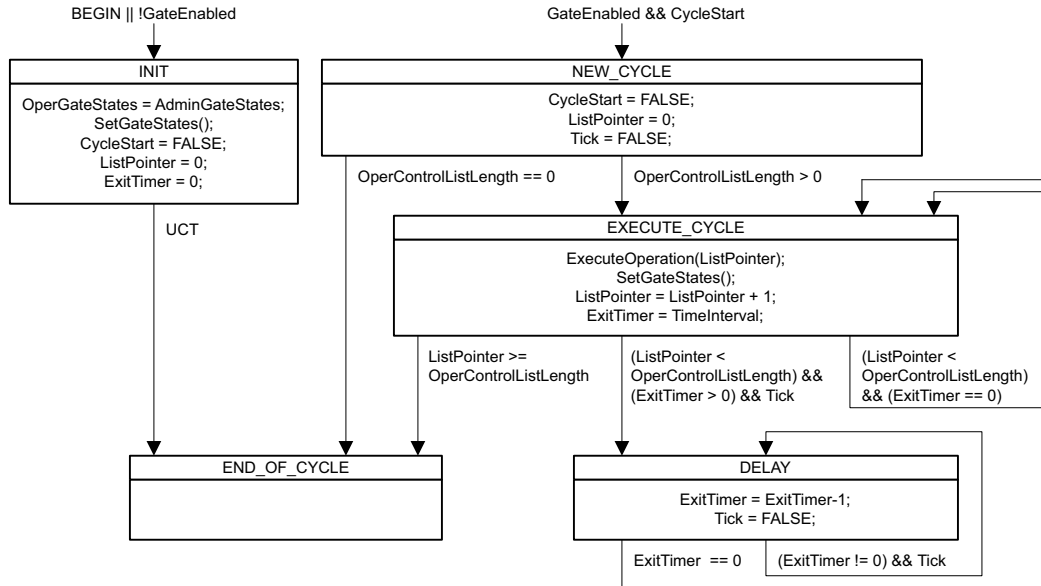


Figure 8-21—List Execute state machine

4 8.6.9.2.1 ExecuteOperation()

5 The ExecuteOperation() procedure is responsible for fetching the next gate operation from the
6 OperControlList, along with any parameters associated with it, and performing actions based upon the gate
7 operation that has been fetched. The value of the ListPointer variable (8.6.9.4.15) is used as an index into
8 OperControlList. The procedure processes the operation according to its operation name (Table 8-7) as
9 follows:

- 10 a) If the operation name is SetGateStates, then the GateState parameter value associated with the
11 operation is assigned to the OperGateStates variable (8.6.9.4.21), and the TimeInterval parameter
12 value associated with the operation is assigned to the TimeInterval variable (8.6.9.4.23). If the
13 TimeInterval parameter value associated with the operation was 0, the TimeInterval variable is
14 assigned the value 1.
- 15 b) If the operation name is unrecognized, then the ListPointer variable (8.6.9.4.15) is assigned the
16 value of the OperControlListLength variable (8.6.9.4.22) and the TimeInterval variable (8.6.9.4.23)
17 is assigned the value 0.
- 18 c) If there is no TimeInterval parameter associated with the operation, then the TimeInterval variable is
19 assigned the value 0.

20 8.6.9.2.2 SetGateStates()

21 This procedure sets the gate state for each of the Port's queues as specified by the value of the
22 OperGateStates variable (8.6.9.4.21).

23 NOTE—If the OperGateStates value differs from the previous gate states, the SetGateStates() procedure causes
24 gate-open and/or gate-close events to occur (3.108, 3.109). It is possible that, on a given queue, the maximum time
25 interval between any gate-open event and a subsequent gate-close event is smaller than the value of queueMaxSDU, in
26 which case, frames that are too long to transmit but are shorter than queueMaxSDU could be queued on that queue. Such
27 frames would never be transmitted, but would eventually be discarded because they exceed the maximum frame lifetime
28 (6.5.2, 6.5.6). It should also be noted that a Bridge is allowed to discard frames that could never be transmitted [item b8)
29 in 6.5.2].

1 8.6.9.3 List Config state machine

2 The List Config state machine shall implement the function specified by the state diagram in Figure 8-22
3 and the attendant definitions in 8.6.9.3.1 and 8.6.9.4.

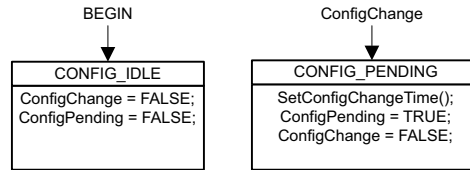


Figure 8-22—List Config state machine

4 8.6.9.3.1 SetConfigChangeTime()

5 The SetConfigChangeTime() procedure determines the time at which the administrative values of the cycle
6 configuration variables are to be copied across to the operational variables. The procedure sets the
7 ConfigChangeTime variable (8.6.9.4.9) to the start time, based upon the values of AdminBaseTime
8 (8.6.9.4.1) and AdminCycleTime (8.6.9.4.3), according to the following rules:

- 9 a) If:
 - 10 AdminBaseTime $\geq$ CurrentTime (8.6.9.4.10)
 - 11 (i.e., AdminBaseTime specifies the current time or a future time)
 - 12 Then:
 - 13 ConfigChangeTime = AdminBaseTime
- 14 b) If:
 - 15 (AdminBaseTime < CurrentTime) and (GateEnabled = FALSE)
 - 16 (i.e., AdminBaseTime specifies a time in the past, and the current schedule is stopped)
 - 17 Then:
 - 18 ConfigChangeTime = (AdminBaseTime + N*AdminCycleTime)
 - 19 where N is the smallest integer for which the relation
 - 20 ConfigChangeTime $\geq$ CurrentTime
 - 21 would be TRUE.
- 22 c) If:
 - 23 (AdminBaseTime < CurrentTime) and (GateEnabled = TRUE)
 - 24 (i.e., AdminBaseTime specifies a time in the past, and the current schedule is running)
 - 25 Then:
 - 26 Increment ConfigChangeError counter (12.29.1)
 - 27 ConfigChangeTime = (AdminBaseTime + N*AdminCycleTime)
 - 28 where N is the smallest integer for which the relation
 - 29 ConfigChangeTime $\geq$ CurrentTime
 - 30 would be TRUE.

1 8.6.9.4 State machine variables

2 8.6.9.4.1 AdminBaseTime

3 The administrative value of base time, expressed as an IEEE 1588 precision time protocol (PTP) [B17]
4 timescale (see IEEE Std 802.1AS™). This value can be changed by management, and is used by the List
5 Config state machine (8.6.9.3) to set the value of OperBaseTime (8.6.9.4.17).

6 NOTE—Time is expressed in the PTP timescale as the number of seconds, nanoseconds, and fractional nanoseconds that
7 have elapsed since 1 January 1970 00:00:00 TAI.

8 8.6.9.4.2 AdminControlList

9 The administrative version of the gate control list for the Port. This value can be changed by management,
10 and is used by the List Config state machine (8.6.9.3) to set the value of OperControlList (8.6.9.4.18).

11 8.6.9.4.3 AdminCycleTime

12 The administrative value of the duration of the gating cycle for the Port (3.110). This value can be changed
13 by management, and is used by the List Config state machine (8.6.9.3) to set the value of OperCycleTime
14 (8.6.9.4.19). The AdminCycleTime variable is a rational number of seconds, defined by an integer
15 numerator and an integer denominator.

16 8.6.9.4.4 AdminCycleTimeExtension

17 An integer number of nanoseconds, defining the maximum amount of time by which the gating cycle for the
18 Port (3.110) is permitted to be extended when a new cycle configuration is being installed. This
19 administrative value can be changed by management, and is used by the List Config state machine (8.6.9.3)
20 to set the value of OperCycleTimeExtension (8.6.9.4.20).

21 8.6.9.4.5 AdminGateStates

22 The initial state of the gate associated with each queue for the Port is set by the List Execute state machine
23 (8.6.9.2), and is determined by the value of the AdminGateStates variable. The default value of
24 AdminGateStates is *open* for all queues. The value of AdminGateStates can be changed by management.

25 8.6.9.4.6 AdminControlListLength

26 This value can be changed by management, and is used by the List Config state machine (8.6.9.3) to set the
27 value of OperControlListLength (8.6.9.4.22).

28 8.6.9.4.7 ConfigChange

29 A Boolean variable that acts as a start signal to the List Config state machine (8.6.9.3) that the administrative
30 variable values for the Port are ready to be copied into their corresponding operational variables. This
31 variable is set TRUE by management and is set FALSE by the List Config state machine.

32 8.6.9.4.8 ConfigPending

33 A Boolean variable, set TRUE by the List Config state machine (8.6.9.3) to signal that there is a new cycle
34 configuration awaiting installation. The variable is set FALSE when the List Config state machine has
35 installed the new configuration. The variable is used by the Cycle Timer state machine (8.6.9.1) to control
36 the length of the cycle that immediately precedes the first cycle that uses the new configuration values. This
37 value can be read by management, as specified in 12.29.1.

1 **8.6.9.4.9 ConfigChangeTime**

2 The time at which the administrative variables that determine the cycle are to be copied across to the
3 corresponding operational variables, expressed as a PTP timescale. The value of this variable is set by the
4 SetConfigChangeTime() procedure (8.6.9.3.1) in the List Config state machine (8.6.9.3).

5 **8.6.9.4.10 CurrentTime**

6 The current time maintained by the local system, expressed as a PTP timescale (see IEEE Std 802.1AS).
7 NOTE 1—Other sources of times can be used to provide a value for the time. The implementation has to ensure that the
8 same view of time is shared by the implementation and whatever entity is responsible for establishing the time schedule.
9 The described mechanisms will work also with other formats of time.
10 NOTE 2—A discontinuity in time will have a predictable effect upon the operation of the state machines, as a
11 consequence of their definitions. The implementer and/or user is advised to understand what those effects could be, and
12 either limit such discontinuities or ensure that the results will not adversely affect the application.

13 **8.6.9.4.11 CycleStart**

14 A Boolean variable used as a signal from the Cycle Timer state machine (8.6.9.1) to the List Execute state
15 machine to start executing the gate control list. Set TRUE by the Cycle Timer state machine and set FALSE
16 by the List Execute state machine (8.6.9.2).

17 **8.6.9.4.12 CycleStartTime**

18 The time at which the next gate control list execution cycle is to start, expressed as a PTP timescale. The
19 value of this parameter is set by the Cycle Timer state machine (8.6.9.1).

20 **8.6.9.4.13 ExitTimer**

21 A timer that implements the delay associated with the currently executing gate operation, expressed as an
22 integer number of nanoseconds. The value is set by the operation of the List Execute state machine (8.6.9.2).

23 **8.6.9.4.14 GateEnabled**

24 A Boolean variable that indicates whether the operation of the state machines is enabled (TRUE) or disabled
25 (FALSE). This variable is set by management. The default value of this variable is FALSE.

26 **8.6.9.4.15 ListPointer**

27 An integer used as a pointer to entries in the OperControlList (8.6.9.4.18), each entry consisting of a gate
28 operation with its associated parameters (Table 8-7). A value of zero points at the first entry in the list; a
29 value of (OperControlListLength – 1) points at the last entry.

30 **8.6.9.4.16 Tick**

31 A Boolean variable, set to TRUE by an implementation-specific system clock function at one nanosecond
32 intervals, that controls the decrementing of the ExitTimer variable (8.6.9.4.13). This variable is set FALSE
33 by the operation of the List Execute state machine (8.6.9.2).

34 NOTE—While the state machine is documented on the basis of a nanosecond clock “tick,” it is anticipated that real
35 implementations will use a wide variety of clocks that differ in frequency accuracy and granularity. Hence, the
36 management parameters specified in 12.29 allow a management station to discover the characteristics of an
37 implementation’s cycle timer clock (TickGranularity) and to set the parameters for the gating cycle accordingly.

1 **8.6.9.4.17 OperBaseTime**

2 The operational value of base time, expressed as a PTP timescale (see IEEE Std 802.1AS). This variable is
3 used by the List Config state machine (8.6.9.3).

4 **8.6.9.4.18 OperControlList**

5 The active Gate Control List for the Port; the List Execute state machine executes this list. This list is
6 populated dynamically from the AdminControlList (8.6.9.4.2) under the control of the List Config state
7 machine (8.6.9.3).

8 **8.6.9.4.19 OperCycleTime**

9 The operational value of the duration of the gating cycle for the Port (3.110). This variable is set
10 dynamically from the AdminCycleTime variable (8.6.9.4.3) under the control of the List Config state
11 machine (8.6.9.3). OperCycleTime is used by the Cycle Timer state machine (8.6.9.1) to enforce the cycle
12 time for the Port. The OperCycleTime variable is a rational number of seconds, defined by an integer
13 numerator and an integer denominator.

14 **8.6.9.4.20 OperCycleTimeExtension**

15 An integer number of nanoseconds, defining the maximum amount of time by which the gating cycle for the
16 Port (3.110) is permitted to be extended when a new cycle configuration is installed. This operational value
17 is set by the List Config state machine (8.6.9.3) to the value of AdminCycleTimeExtension (8.6.9.4.4). The
18 value of OperCycleTimeExtension is used by the Cycle Timer state machine (8.6.9.1).

19 **8.6.9.4.21 OperGateStates**

20 The current state of the gate associated with each queue for the Port. OperGateStates is set by the List
21 Execute state machine (8.6.9.2), and its initial value is determined by the value of the AdminGateStates
22 variable (8.6.9.4.5).

23 **8.6.9.4.22 OperControlListLength**

24 This variable is set dynamically from the AdminControlListLength parameter (8.6.9.4.6) under the control
25 of the List Config state machine (8.6.9.3). OperControlListLength indicates the number of entries in the
26 OperControlList and is used by the List Execute state machine (8.6.9.2) to determine when the end of the list
27 has been reached.

28 **8.6.9.4.23 TimeInterval**

29 Set to the value of the TimeInterval parameter associated with the currently executing gate operation. Used
30 in the operation of the List Execute state machine (8.6.9.2).

31 **8.6.10 Stream gate control state machines**

32 The execution of the gate operations in a stream gate control list (8.6.5.4) is controlled by the three state
33 machines specified in 8.6.9:

- 34 a) The Cycle Timer state machine (8.6.9.1)
- 35 b) The List Execute state machine (8.6.9.2)
- 36 c) The List Config state machine (8.6.9.3)

One instance of each state machine is instantiated for each stream gate control list associated with instances of stream gates in a Bridge component that supports stream gates. An overview of the operation of these state machines can be found in Figure 8-19.

The operation of these state machines is as defined in 8.6.9, with the exception of the definitions of the `ExecuteOperation()` procedure, the `SetGateStates()` procedure, the `ListPointer` variable, the `AdminGateStates` variable, and the `OperGateStates` variable; amended versions of these definitions appear in 8.6.10.1 through 8.6.10.5. Table 8-8 shows the correspondence between the procedures/variables used in 8.6.9 and the stream gate versions of these procedures/variables.

Three additional variables needed by the `ExecuteStreamGateOperation` procedure are defined in 8.6.10.6 and 8.6.10.7.

Table 8-8—Scheduled Traffic and Stream Gate procedures/variables

Procedure/variable name in 8.6.9	StreamGate procedure/variable name
<code>ExecuteOperation()</code> (8.6.9.2.1)	<code>ExecuteStreamGateOperation()</code> (8.6.10.1)
<code>SetGateStates()</code> (8.6.9.2.2)	<code>SetStreamGateStates()</code> (8.6.10.2)
<code>ListPointer</code> (8.6.9.4.15)	<code>StreamGateListPointer</code> (8.6.10.3)
<code>AdminGateStates</code> (8.6.9.4.5)	<code>StreamGateAdminGateStates</code> (8.6.10.4)
<code>OperGateStates</code> (8.6.9.4.21)	<code>StreamGateOperGateStates</code> (8.6.10.5)

8.6.10.1 `ExecuteStreamGateOperation()`

The `ExecuteStreamGateOperation()` procedure is responsible for fetching the next gate operation from the `OperControlList`, along with any parameters associated with it, and performing actions based upon the gate operation that has been fetched. The value of the `StreamGateListPointer` variable (8.6.9.2.2) is used as an index into `OperControlList`. The procedure processes the operation according to its operation name (Table 8-4) as follows:

- a) If the operation name is `SetGateAndIPV`, then the `StreamGateState` parameter value associated with the operation is assigned to the `StreamGateOperGateStates` variable (8.6.10.5), the `IPV` parameter value is assigned to the `OperIPV` variable (8.6.10.7), and the `TimeInterval` parameter value associated with the operation is assigned to the `TimeInterval` variable (8.6.9.4.23). If the `TimeInterval` parameter value associated with the operation was 0, the `TimeInterval` variable is assigned the value 1. If there is an `IntervalOctetMax` parameter associated with the gate operation, then that parameter value is used to set the value of the `IntervalOctetsLeft` variable (8.6.10.8); otherwise, the `IntervalOctetsLeft` variable is set to a value greater than the maximum possible number of octets that the gate could pass during `TimeInterval`.
- b) If the operation name is unrecognized, then the `StreamGateListPointer` variable (8.6.9.4.15) is assigned the value of the `OperControlListLength` variable (8.6.9.4.22) and the `TimeInterval` variable (8.6.9.4.23) is assigned the value 0.
- c) If there is no `TimeInterval` parameter associated with the operation, then the `TimeInterval` variable is assigned the value 0.

8.6.10.2 `SetStreamGateStates()`

This procedure sets the stream gate state as specified by the value of the `StreamGateOperGateStates` variable (8.6.9.4.21).

1 8.6.10.3 StreamGateListPointer

2 An integer used as a pointer to entries in the OperControlList (8.6.9.4.18), each entry consisting of a stream
3 gate control operation with its associated parameters (Table 8-4). A value of zero points at the first entry in
4 the list; a value of (OperControlListLength – 1) points at the last entry.

5 8.6.10.4 StreamGateAdminGateStates

6 The initial state of the gate associated with the stream gate is set by the List Execute state machine (8.6.9.2)
7 and is determined by the value of the StreamGateAdminGateStates variable. The default value of
8 StreamGateAdminGateStates is open. The value of StreamGateAdminGateStates can be changed by
9 management.

10 8.6.10.5 StreamGateOperGateStates

11 The current state of the gate associated with the stream gate. StreamGateOperGateStates is set by the List
12 Execute state machine (8.6.9.2), and its initial value is determined by the value of the
13 StreamGateAdminGateStates variable (8.6.10.4).

14 8.6.10.6 AdminIPV

15 The initial value of the OperIPV variable (8.6.10.7) associated with the stream gate is determined by the
16 value of the AdminIPV variable. The default value of AdminIPV variable is the null value. The value of the
17 AdminIPV variable can be changed by management.

18 8.6.10.7 OperIPV

19 The current value of the IPV associated with the stream gate. The initial value of OperIPV is set equal to the
20 value of the AdminIPV variable (8.6.10.6) by the Cycle Timer state machine (8.6.9.1) with the operational
21 cycle configuration variables when the state machines are initialized, when GateEnabled (8.6.9.4.14) is
22 FALSE, and at ConfigChange Time (8.6.9.4.9). Subsequently, if there is a stream gate control list associated
23 with the stream gate instance, its value is controlled by the contents of the operational stream gate control
24 list and the operation of the List Execute state machine (8.6.9.2).

25 8.6.10.8 IntervalOctetsLeft

26 The current value of the IntervalOctetsLeft parameter indicates how many more MSDU octets can be passed
27 by the stream gate during the current TimeInterval. This variable is initialized by the
28 ExecuteStreamGateOperation() procedure (8.6.10.1). If a frame that would otherwise pass the gate is larger
29 than the current value of IntervalOctetsLeft, it is treated as if the gate is in the *closed* state; i.e., it is
30 discarded. If a frame that would otherwise pass the gate is smaller than the current value of
31 IntervalOctetsLeft, the number of MSDU octets is subtracted from the value of IntervalOctetsLeft.

32 8.6.11 ATS Scheduler state machines

33 The ATS scheduler state machine operation is based on the ATS scheduler clocks (8.6.11.1), which is in
34 relationship to transmission selection clocks (8.6.8.5). The state machine operation is specified by the
35 ProcessFrame(frame) procedure (8.6.11.3) and the therein used state variables (8.6.11.3.3, 8.6.11.3.10).
36 Parameters used by the ProcessFrame(frame) procedure are as defined in 8.6.5.6. Each arriving frame causes
37 invocation of the ProcessFrame(frame) procedure.

1 8.6.11.1 ATS Scheduler Clocks

2 An ATS scheduler clock is an implementation specific local system clock function. It is used to determine
3 the arrival time of frames (8.6.11.3.1). A Bridge component may utilize one or more ATS scheduler clock
4 instances. In the case of multiple scheduler clock instances, all ATS scheduler instances associated with the
5 same reception Port share the same ATS scheduler clock instance (i.e., the arrival time of all frames received
6 from a particular reception port is determined by the same ATS scheduler clock instance).

7 8.6.11.2 Relationship between ATS Scheduler Clocks and Transmission Selection Clocks

8 ATS scheduler clocks and transmission selection clocks (8.6.8.5) run at the same long term rate with
9 bounded offset variation. A difference between an arbitrary instant of time t_{FA} , as recognized by an ATS
10 scheduler clock instance, and the same instant of time t_{TS} , as recognized by a transmission selection clock
11 instance, may be observed during the processing of a frame. The time difference may vary over a sequence
12 of frames, which is characterized by the following equation:

$$13 \quad \text{ClockOffsetMin} \leq t_{TS} - t_{FA} \leq \text{ClockOffsetMax}$$

14 where ClockOffsetMin and ClockOffsetMax are implementation specific constants that limit the variation.
15 The range is characterized by ClockOffsetVariationMax (12.31.8.3) as follows:

$$16 \quad \text{ClockOffsetVariationMax} = \text{ClockOffsetMax} - \text{ClockOffsetMin}$$

17 NOTE 1—ClockOffsetMin and ClockOffsetMax capture implementation specific properties such as the resolution of the
18 associated clocks, associated rounding errors, constant offsets between clocks, Bridge-internal synchronization
19 inaccuracies in presence of different underlying oscillators, and similar.

20 A pair of a scheduler clock instance and a transmission selection clock instance has an implementation
21 specific nominal rate, and a maximal absolute deviation from this nominal rate during operation, as
22 characterized by the ClockRateDeviationMax parameter (Clause 12.31.8.4).

23 NOTE 2—ClockRateDeviationMax captures implementation specific properties such as oscillator rate deviation,
24 numeric resolution for the operations specified in 8.6.11.3, and similar.

25 NOTE 3—ATS scheduler clocks and transmission selection clocks provide a model to express different sources of delay,
26 delay variation, and inaccuracy. It is not required to implement different multiple physical oscillators/clocks (i.e., ATS
27 scheduler clocks and transmission selection clocks can actually be the same physical clock or can be generated from the
28 same oscillator), but the model captures the properties of implementations with and without different physical
29 oscillators/clocks in a unified manner.

1 8.6.11.3 ProcessFrame(frame)

2 This procedure computes eligibility time, assigns the eligibility times to frames, and updates the ATS
3 scheduler state machine variables.

4 The procedure is described by the following pseudo-code in a neutral manner: The arithmetic precision, and
5 the resolution of variables, are implementation specific, unless externally visible by management (12.31).
6 The impact of the associated inaccuracies is discussed in Annex V.

```

7 ProcessFrame(frame) {
8     lengthRecoveryDuration    = length(frame) /
9                               CommittedInformationRate;
10    emptyToFullDuration        = CommittedBurstSize /
11                                CommittedInformationRate;
12    schedulerEligibilityTime    = BucketEmptyTime +
13                                lengthRecoveryDuration;
14    bucketFullTime              = BucketEmptyTime +
15                                emptyToFullDuration;
16    eligibilityTime              = max(arrivalTime(frame),
17                                     GroupEligibilityTime,
18                                     schedulerEligibilityTime);
19
20    if (eligibilityTime <= (arrivalTime(frame) + MaxResidenceTime/1.0e9)){
21        // The frame is valid
22        GroupEligibilityTime = eligibilityTime;
23        BucketEmptyTime     = (eligibilityTime < bucketFullTime) ?
24                               schedulerEligibilityTime :
25                               schedulerEligibilityTime + eligibilityTime - bucketFullTime;
26        AssignAndProceed(frame, eligibilityTime);
27    } else {
28        // The frame is invalid
29        Discard(frame);
30    }
31 }
```

32 8.6.11.3.1 arrivalTime(frame)

33 The arrival time of the frame, in seconds. The arrival time refers to the instant of time at which the
34 associated ATS scheduler clock instance recognizes the arrival of the entire frame.

35 The point at which this recognition happens on the path between reception Ports and transmission Ports is
36 implementation dependent. The earliest option is when the frame passes the boundary between the network
37 physical medium and a reception Port, the latest option is during invocation of the ProcessFrame procedure.

38 For all frames arriving at all reception Ports, the arrival time is determined relative to the frame end. The
39 maximum delay between the time a frame passes the boundary between the network physical medium and
40 its subsequent recognition by the associated ATS scheduler clock instance is implementation specific, and is
41 characterized by the ArrivalRecognitionDelayMax parameter (12.31.8.5).

42 The order of all frames associated with one ATS scheduler group at the boundary between the network
43 physical medium and a reception Port is the same as the order of the arrival times determined for these
44 frames.

45 NOTE—For example, the arrival time of frames may be determined based on invocation of the
46 M_UNITDATA.indication service primitive of the ISS (6.6), as specified in IEEE Std 802.1AC.

47 8.6.11.3.2 AssignAndProceed(frame, eligibilityTime)

48 This procedure calculates an assigned eligibility time parameter (assignedEligibilityTime) that is used by the
49 ATS transmission selection algorithm (8.6.8.5).

1 The assignedEligibilityTime is derived from the eligibilityTime parameter. Its calculation accounts for
2 variations between the associated ATS scheduler clock and transmission selection clock instances, and for
3 processing delays through the forwarding process, as detailed in the following.

4 Any frame may experience an additional, non-negative, processing delay, between its arrival time
5 recognition by the associated ATS scheduler clock instance and the time at which it is queued and becomes
6 available for transmission selection (8.6.8.5). This delay may vary per frame, thus that there is a delay
7 variation over a sequence of frames. The processing delay is characterized by the following equation:

$$8 \quad \text{ProcessingDelayMin} \leq \text{processingDelay}(\text{frame}) \leq \text{ProcessingDelayMax}$$

9 where ProcessingDelayMin (12.31.8.6) and ProcessingDelayMax (12.31.8.7) are implementation specific
10 parameters, and processingDelay(frame) denotes the processing delay of an actual frame, including any
11 potential delay introduced by the operation of the associated ATS scheduler state machine instance.

12 The assigned eligibility time is calculated as follows:

$$13 \quad \text{assignedEligibilityTime} = \text{eligibilityTime} + \text{ClockOffsetMin} + \text{ProcessingDelayMax}$$

14 **8.6.11.3.3 BucketEmptyTime**

15 A variable that embodies the current state of the ATS scheduler instance. This variable is initialized with a
16 time earlier than CommittedBurstSize/CommittedInformationRate in the past, as perceived by the ATS
17 Scheduler Clock.

18 **8.6.11.3.4 bucketFullTime**

19 A temporary variable used when updating BucketEmptyTime after determining a frame's
20 schedulerEligibilityTime. The prior value of BucketEmptyTime plus the emptyToFullDuration.

21 **8.6.11.3.5 CommittedBurstSize**

22 The committed burst size of the ATS scheduler instance, in bits (8.6.5.6).

23 **8.6.11.3.6 CommittedInformationRate**

24 The committed information rate of the ATS scheduler instance, in bits per second (8.6.5.6).

25 **8.6.11.3.7 Discard(frame)**

26 This procedure discards the frame and increases the DiscardedFramesCount counter of the associated
27 reception port (8.6.5.6). The procedure is called in exceptional situations only (e.g., misbehavior of the
28 connected upstream system).

29 **8.6.11.3.8 eligibilityTime**

30 The eligibility time of the frame, without taking the implementation specific device-internal timing
31 properties of the forwarding process into account. These timing properties are taken into account by the
32 AssignAndProceed(frame) procedure (8.6.11.3.2).

33 **8.6.11.3.9 emptyToFullDuration**

34 The duration CommittedBurstSize/CommittedInformationRate.

1 **8.6.11.3.10 GroupEligibilityTime**

2 A state variable that contains the most recent value of the eligibilityTime variable from the previous frame,
3 as processed by any ATS scheduler instance in the same ATS scheduler group, in seconds.

4 The GroupEligibilityTime variable is initialized with a time earlier or equal to the current time, as perceived
5 by the ATS scheduler clock.

6 **8.6.11.3.11 length(frame)**

7 The length of the frame, including all media-dependent overhead (12.4.2.2), in bits.

8 **8.6.11.3.12 lengthRecoveryDuration**

9 The duration $\text{length}(\text{frame})/\text{CommittedInformationRate}$.

10 **8.6.11.3.13 MaxResidenceTime**

11 The MaximumResidenceTime parameter of the ATS scheduler group instance associated with the ATS
12 scheduler instance, in nanoseconds (8.6.5.6). The parameter limits the duration for which frames can reside
13 in a Bridge.

14 NOTE—A consistent setup of MaxResidenceTime parameter can be determined by the per hop delay bound, which is a
15 result of the timing analysis (Annex V).

16 **8.6.11.3.14 schedulerEligibilityTime**

17 A temporary variable used in the calculation of a frame's eligibilityTime. The prior value of
18 BucketEmptyTime plus the lengthRecoveryDuration for the frame.

1 8.7 The Learning Process

2 The Learning Process receives the source MAC addresses and VIDs, or only the source MAC addresses in
3 the case of VLAN-unaware MAC Relays, of received frames from the Forwarding Process, subject to active
4 topology enforcement (8.6.1) and the application of ingress filtering (8.6.2). The Learning Process is not
5 invoked (8.6.1) for frames whose VID is an ESP-VID or identifies a VLAN supported by SPBM.

6 When invoked, the Learning Process shall create or update a Dynamic Filtering Entry (8.8.3) that specifies
7 the reception Port for the frame's source address and, in the case of VLAN Bridge components, the frame's
8 VID, if and only if the source address is an individual address, i.e., is not a group address, the resulting
9 number of entries would not exceed the capacity of the FDB, and the filtering utility criteria for the receiving
10 Bridge Port are met, as specified below.

11 If the FDB is already filled to capacity, but a new entry would otherwise be made, then an existing entry may
12 be removed to make room for the new entry.

13 The purpose of filtering utility criteria is to reduce the capacity requirements of the FDB and to reduce the
14 time for which service can be denied (6.5.1) by retaining filtering information learned prior to a change in
15 the physical topology of the network. Filtering utility criteria shall be applied to the learning and retention of
16 information for each FID (8.8.8). In Bridges other than EVB Bridges (5.23), enhanced filtering utility
17 criteria may be implemented for any Bridge Port as specified below (8.7.2); if implemented, both the default
18 (8.7.1) and the enhanced criteria shall be selectable by management. In EVB Bridges, the enhanced filtering
19 utility criteria shall be implemented for all Bridge Ports, and the default filtering utility criteria shall not be
20 implemented.

21 Figure 8-5 illustrates the operation of the Learning Process in the inclusion of station location information
22 carried by a single frame, received on one of the Ports of a Bridge, in the FDB.

23 8.7.1 Default filtering utility criteria

24 The default for a VLAN Bridge is that the member set (8.8.10) for the frame's VID includes at least one
25 Port.

26 NOTE—If the member set for a given VID is empty, that VID is not currently active, and the Bridge will filter all frames
27 destined for that VID, regardless of their destination address.

28 The default filtering utility criteria are always met for MAC Bridges.

29 8.7.2 Enhanced filtering utility criteria

30 For a VLAN Bridge, the enhanced criteria are satisfied if at least one VID that uses the FID includes the
31 reception Port and at least one other Port with a Port State of Learning or Forwarding in its member set, and:

- 32 a) The operPointToPointMAC parameter is FALSE for the reception Port, or
- 33 b) Ingress for the VID is permitted through a third Port, or
- 34 c) The reception Port has reflective relay enabled (40.4.2).

35 NOTE—The third port can, but is not required to, be in the member set.

36 For a MAC Bridge, the enhanced criteria are satisfied if there exists at least one Port different than the
37 reception Port with a Port State of Learning or Forwarding, and:

- 38 d) The operPointToPointMAC parameter is FALSE for the reception Port, or
- 39 e) There exists third Port with a Port State of Learning or Forward, or
- 40 f) The reception Port has reflective relay enabled (40.4.2).

8.7.3 Ageing of Dynamic Filtering Entries

Dynamic Filtering Entries shall be automatically removed after a specified time, the Ageing Time, has elapsed since the entry was created or last updated by the Learning Process.

The ageing out of Dynamic Filtering Entries ensures that end stations that have been moved to a different part of the network will not be permanently prevented from receiving frames. It also takes account of changes in the active topology of the network that can cause end stations to appear to move from the point of view of the Bridge; i.e., the path to those end stations subsequently lies through a different Bridge Port.

The Ageing Time may be set by management (Clause 12). A range of applicable values and a recommended default is specified in Table 8-9; this is suggested to remove the need for explicit configuration in most cases. If the value of Ageing Time can be set by management, the Bridge shall have the capability to use values in the range specified, with a granularity of 1 s.

Table 8-9—Ageing time parameter value

Parameter	Recommended default value	Range
Ageing time	300.0 s	10.0–1 000 000.0 s

NOTE—The granularity is specified in order to establish a common basis for the granularity expressed in the management operations defined in Clause 12, not to constrain the granularity of the actual timer supported by a conformant implementation. If the implementation supports a granularity other than 1 s, then it is possible that the value read back by management following a Set operation will not match the actual value expressed in the Set.

The STP specified in Clause 8 of IEEE Std 802.1D, 1998 Edition [B13], includes a procedure for notifying all Bridges in the network of topology change. It specifies a short value for the Ageing Timer, to be enforced for a period after any topology change (8.3.5 of IEEE Std 802.1D, 1998 Edition [B13]). While the topology is not changing, this procedure allows normal ageing to accommodate extended periods during which addressed end stations do not generate frames themselves, perhaps through being powered down, without sacrificing the ability of the network to continue to provide service after automatic configuration.

8.8 The Filtering Database (FDB)

The FDB supports queries by the Forwarding Process to determine whether received frames, with given values of, destination MAC address, and for VLAN Bridge components, VID, are to be forwarded through a given potential transmission Port (8.6.1, 8.6.3, 8.6.4). It can also be used, by SPBM (8.6.1), to determine whether received frames with given values of source B-MAC address and B-VID, are to be forwarded from a given reception Port. The FDB contains filtering information in the form of filtering entries that are either:

- a) Static, and explicitly configured by management action; or
- b) Dynamic, and automatically entered into the FDB by the normal operation of the Bridge and the protocols it supports.

Two entry types are used to represent static filtering information. The Static Filtering Entry represents static information in the FDB for individual and for group MAC addresses. It allows administrative control of:

- c) Forwarding of frames with particular destination addresses.
- d) The inclusion in the FDB of dynamic filtering information associated with Extended Filtering Services, and use of this information.

1 The FDB shall contain entries of the Static Filtering Entry type.

2 The Static VLAN Registration Entry represents all static information in the FDB for VLANs. It allows
3 administrative control of:

- 4 e) Forwarding of frames with particular VIDs.
- 5 f) The inclusion/removal of tag headers in forwarded frames.
- 6 g) The inclusion in the FDB of dynamic VLAN membership information, and use of this information.

7 The FDB associated with VLAN Bridge components may contain entries of the Static VLAN Registration
8 Entry type. The Static VLAN Registration Entry type is not applicable to MAC Bridge components.

9 Static filtering information is added to, modified, and removed from the FDB only under explicit
10 management control except when PBB-TE IPS is deployed, in which case the static filtering information for
11 traffic associated with IPGs configured on the Bridge is added to, modified, and removed from the FDB only
12 under the direction of IPS Control (26.11.3). It shall not be automatically removed by any ageing
13 mechanism. Management of static filtering information may be carried out by use of the remote
14 management capability provided by Bridge Management (8.12) using the operations specified in Clause 12.

15 Four entry types are used to represent dynamic filtering information:

- 16 h) Each Dynamic Filtering Entry specifies the Port through which a frame with a given individual
17 MAC address, and in the case of VLAN Bridge components, VID, has been or can be received.
18 Dynamic Filtering Entries can be created and updated by the Learning Process (8.7) and by
19 ISIS-SPB. Entries that are updated by the Learning Process are subject to ageing and removal by the
20 FDB.

21 NOTE 1—ISIS-SPB has to create the Dynamic Filtering Entries for VLANs supported by SPBM, since learning is not
22 used. Initial creation of these entries does no harm for VLANs supported by SPBV (when learning is used) as it can
23 reduce flooding of frames to unknown destinations.

- 24 i) MAC Address Registration Entries support the registration of MAC addresses. They are created,
25 updated, and removed by MMRP in support of Extended Filtering Services (8.8.4 and Clause 10)
26 and by ISIS-SPB. If the `Restricted_MAC_Address_Registration` management control (10.12.2.3) is
27 TRUE, then the creation of a MAC Address Registration Entry by MMRP is not permitted unless a
28 Static Filtering Entry exists that permits dynamic registration for the Group concerned.
- 29 j) Dynamic VLAN Registration Entries are used to specify the Ports on which VLAN membership has
30 been dynamically registered. They are not applicable to MAC Bridge components. They are created,
31 updated, and removed by MVRP, MIRP, and ISIS-SPB, in support of automatic VLAN membership
32 configuration (Clause 11 and Clause 39). The MVRP controlled Dynamic VLAN Registration
33 Entries are subject to the state of the `Restricted_VLAN_Registration` management control
34 (11.2.3.2.3). If the value of this control is TRUE, then the creation of a Dynamic VLAN Registration
35 Entry is not permitted unless a Static VLAN Registration Entry exists that permits dynamic
36 registration for the VID concerned.
- 37 k) Dynamic Reservation Entries are used to specify the Ports on which stream reservations have been
38 made. They are created, updated, and removed by the operation of SRP defined in Clause 35.

39 Static Filtering Entries and MAC Address Registration Entries comprise:

- 40 l) A MAC address specification.
- 41 m) A VID.
- 42 n) A Port Map, with a control element for each outbound Port to specify filtering for, that MAC address
43 specification, and in the case of VLAN Bridge components, that VID.

44 Item m), and any reference to VIDs, are not applicable to MAC Bridge components.

1 Dynamic Filtering Entries comprise:

- 2 o) A MAC address specification.
- 3 p) A locally significant FID (see 8.8.8).
- 4 q) A Port Map, with a control element for each outbound Port to specify filtering for, that MAC address
- 5 specification, and in the case of VLAN Bridge components, for the VID(s) allocated to that FID.

6 Item p) and any reference to VIDs and FIDs are not applicable to MAC Bridge components.

7 The Port Map in Static and Dynamic Filtering Entries may contain a **connection_identifier** (8.8.12) for each
8 outbound Port.

9 Static and Dynamic VLAN Registration Entries comprise:

- 10 r) A VID.
- 11 s) A Port Map, with a control element for each outbound Port to specify filtering for the VID.

12 Static and Dynamic VLAN Registration Entries are not applicable to MAC Bridge components.

13 Dynamic Reservation Entries comprise:

- 14 t) A MAC address specification.
- 15 u) A VID.
- 16 v) A Port Map, with a control element for each outbound Port to specify filtering for that MAC address
- 17 specification, and in the case of VLAN Bridge components, in the VLAN specified by the VID.

18 Dynamic filtering information may be read by use of the remote management capability provided by Bridge
19 Management (8.12) using the operations specified in Clause 12.

20 The Filtering Services supported by a Bridge (Basic and Extended Filtering Services) determine the default
21 behavior of the Bridge with respect to the forwarding of frames destined for group MAC addresses. In
22 Bridges that support Extended Filtering Services, the default forwarding behavior for group MAC addresses,
23 for each Port, and for each VID, can be configured both statically and dynamically by means of Static
24 Filtering Entries and/or MAC Address Registration Entries that can carry the following MAC address
25 specifications:

- 26 w) All Group Addresses, for which no more specific Static Filtering Entry exists.
- 27 x) All Unregistered Group Addresses (i.e., all group MAC addresses for which no MAC Address
- 28 Registration Entry exists), for which no more specific Static Filtering Entry exists.

29 NOTE 2—The All Group Addresses specification in item w), when used in a Static Filtering Entry with an appropriate
30 control specification, provides the ability to configure a Bridge that supports Extended Filtering Services to behave as a
31 Bridge that supports only Basic Filtering Services on some or all of its Ports. This might be done for the following
32 reasons:

- 33 — The Ports concerned serve “legacy” devices that wish to receive multicast traffic, but are unable to
- 34 register Group membership.
- 35 — The Ports concerned serve devices that need to receive all multicast traffic, such as routers or
- 36 diagnostic devices.

37 The FDB shall support the creation, updating, and removal of Dynamic Filtering Entries by the Learning
38 Process (8.7). In Bridges that support Extended Filtering Services, the FDB shall support the creation,
39 updating, and removal of MAC Address Registration Entries by MMRP (Clause 10). In Bridges that support
40 SPBM, Dynamic Filtering Entries are populated by ISIS-SPB (Clause 28). In Bridges that support SPBV the
41 Dynamic Filtering Entries are created and updated by the Learning Process (8.7) and the use of ISIS-SPB for
42 this operation is optional (28.12.9).

1 SPB (27.10) allocates SPVIDs dynamically to support shortest path bridging of frames assigned to one or
2 more VLANs, each identified by a Base VID (Clause 3). A Static Filtering Entry should not be created for
3 an SPVID, irrespective of whether that SPVID is currently used, and SPVIDs should not be used in
4 registration protocols that create MAC Address Registration Entries or Dynamic VLAN Registration
5 Entries. In an SPT Bridge, configuration or registration of an FDB entry with the Base VID of a VLAN
6 supported by SPBV shall cause the Bridge to filter frames as if that entry was also present for every one of
7 the SPVIDs for that Base VID.

8 In Bridges that support SRP, the FDB shall support the creation, updating, and removal of Dynamic
9 Reservation Entries by SRP (Clause 35).

10 Figure 8-4 illustrates use of the FDB by the Forwarding Process in a single instance of frame relay between
11 the Ports of a Bridge with two Ports.

12 Figure 8-5 illustrates the creation or update of a dynamic entry in the FDB by the Learning Process. The
13 entries in the FDB, associated with VLAN Bridge components, allow MAC address information to be
14 learned independently for each VID or set of VIDs, by relating a MAC address to the VID or set of VIDs on
15 which that address was learned.

16 NOTE 3—The ability to create VLAN-dependent FDB entries allows a VLAN Bridge to support multiple end stations
17 with the same individual MAC address residing on different VLANs. It also supports end stations with multiple
18 interfaces, each using the same individual MAC address, as long as not more than one end station or interface that uses a
19 given MAC address resides in a given VLAN.

20 Figure 8-6 illustrates the operation of the Spanning Tree Protocol Entity (8.10), which operates STP, and its
21 notification of the FDB of changes in active topology signaled by that protocol.

22 There are no standardized constraints on the size of the FDB in an implementation for which conformance to
23 this standard is claimed. The PICS proforma in Annex A requires the following to be specified for a given
24 implementation:

- 25 y) The total number of entries (Static Filtering Entries, Dynamic Filtering Entries, MAC Address
26 Registration Entries, Static VLAN Registration Entries, Dynamic VLAN Registration Entries, and
27 Dynamic Reservation Entries) that the implementation of the FDB can support.
- 28 z) Of that total number, the total number of VLAN Registration Entries (static and dynamic) that the
29 FDB can support.

30 8.8.1 Static Filtering Entries

31 A Static Filtering Entry specifies:

- 32 a) A MAC address specification, comprising
 - 33 1) An individual MAC address, or
 - 34 2) A group MAC address, or
 - 35 3) All Individual Addresses, for which no more specific Filtering Entry exists, or
 - 36 4) All Group Addresses, for which no more specific Static Filtering Entry exists, or
 - 37 5) All Unregistered Group Addresses, for which no more specific Static Filtering Entry exists.
- 38 b) A VID specification, comprising:
 - 39 1) The VID of a specific VLAN to which the static filtering information applies, or
 - 40 2) The wildcard VID (Table 9-2), indicating that the static filtering information applies to all VIDs
41 for which no specific Static Filtering Entry exists.
- 42 c) A Port Map, containing a control element for each outbound Port, specifying that a frame with a
43 destination MAC address, and in the case of VLAN Bridge components, VID that meets this
44 specification is to be:
 - 45 1) Forwarded, independently of any dynamic filtering information held by the FDB, or

- 1 2) Filtered, independently of any dynamic filtering information, or
- 2 3) Forwarded or filtered on the basis of dynamic filtering information, or on the basis of the
- 3 default Group filtering behavior for the outbound Port (8.8.6) if no dynamic filtering
- 4 information is present specifically for the MAC address.

5 The Port Map may contain a connection_identifier (8.8.12) for each outbound Port.

6 MAC Bridges shall have the capability to support the first two values for the MAC address specification,
7 and all three values for each control element for all Static Filtering Entries [i.e., shall have the capability to
8 support item a1), item a2), item c1), item c2), and item c3)]. All VLAN Bridges shall have the capability to
9 support the first two values for the MAC address specification, both values of the VID specification, and all
10 three values for each control element for all Static Filtering Entries [i.e., shall have the capability to support
11 item a1), item a2), item b1), item b2), item c1), item c2), and item c3)]. All Bridges supporting PBB-TE or
12 SPT Bridges shall, in addition, support item a3) and item a4) for ESP-VIDs (8.9) or VIDs identifying
13 VLANs supported by SPBM, respectively, and shall always have in their FDBs Static Filtering Entries
14 specifying item a3) and item a4), and a Port Map specifying filtering for each outbound port, for every
15 ESP-VID or SPBM VID, respectively.

16 A Bridge that supports Extended Filtering Services shall have the capability to support item a1), item a2),
17 item a4), and item a5) for the MAC address specification and all three control element values for all Static
18 Filtering Entries.

19 For a given MAC address specification, a separate Static Filtering Entry with a distinct Port Map may be
20 created for each VID from which frames are received by the Forwarding Process.

21 In addition to controlling the forwarding of frames, Static Filtering Entries provide the Registrar
22 Administrative Control values for MMRP (Clause 10). Static configuration of forwarding of a specific
23 destination MAC address to an outbound port indicates Registration Fixed on that port, which indicates a
24 desire to receive the specific frames even in the absence of dynamic information. Static configuration of
25 filtering of frames that might otherwise be sent to an outbound port indicates Registration Forbidden. The
26 absence of a Static Filtering Entry for a MAC address, or the configuration of forwarding or filtering on the
27 basis of dynamic filtering information, indicates Normal Registration.

28 8.8.2 Static VLAN Registration Entries

29 A Static VLAN Registration Entry specifies:

- 30 a) The VID to which the static filtering information applies.
- 31 b) A Port Map, consisting of a control element for each outbound Port, specifying:
 - 32 1) The Registrar Administrative Control values for MVRP (Clause 11) and MIRP (Clause 39) for
 - 33 the VID. In addition to providing control over the operation of MVRP and MIRP, these values
 - 34 can also directly affect the forwarding behavior of the Bridge, as described in 8.8.10. The
 - 35 values that can be represented are:
 - 36 i) Registration Fixed (New ignored), or
 - 37 ii) Registration Fixed (New propagated), or
 - 38 iii) Registration Forbidden, or
 - 39 iv) Normal Registration.
 - 40 2) Whether frames are to be VLAN-tagged or untagged when transmitted. The entries in the Port
 - 41 Map that specify untagged transmission compose the untagged set for the VID. The untagged
 - 42 set is empty if no Static VLAN Registration Entry exists for the VID.

43 Static VLAN Registration Entries are not applicable to MAC Bridge components.

1 A separate Static VLAN Registration Entry with a distinct Port Map may be created for each VID. All
2 Bridges shall be capable of supporting all values for each control element for all Static VLAN Registration
3 Entries; however, the ability to support more than one untagged VID on egress on any given Port is optional
4 (see 5.4).

5 NOTE 1—In other words, it is possible to configure any VID as untagged on egress, but it is not necessarily possible to
6 configure more than one VID as untagged on egress. It is an implementation option about whether only a single
7 untagged VID per Port on egress is supported or whether multiple untagged VIDs per Port on egress are supported.
8 When a single untagged VID per Port is supported, that VID is not necessarily the same as the PVID.

9 The initial state of the Permanent Database contains a Static VLAN Registration Entry for the VID
10 corresponding to the Default PVID (Table 9-2). The Port Map in this entry specifies Registration Fixed and
11 forwarding untagged for all Ports of the Bridge. This entry may be modified or removed from the Permanent
12 Database by means of the management operations defined in Clause 12 if the implementation supports these
13 operations.

14 NOTE 2—This initial state causes the default tagging state for the PVID to be untagged, and for all other VIDs to be
15 tagged, unless otherwise configured; however, the management configuration mechanisms allow any VID (including the
16 PVID) to be specified as VLAN-tagged or untagged on any Port.

17 NOTE 3—For VLANs supported by SPBV the Filtering specified by a static VLAN registration entry for the Base VID
18 is applied for all SPVIDs associated with that Base VID.

19 8.8.3 Dynamic Filtering Entries

20 A Dynamic Filtering Entry specifies:

- 21 a) An individual MAC address.
- 22 b) The FID, an identifier assigned by the MAC Bridge (8.8.8) to identify a set of VIDs for which no
23 more than one Dynamic Filtering Entry can exist for any individual MAC address.
24 NOTE 1—An FID identifies a set of VIDs among which Shared VLAN Learning (SVL) (3.247) takes place.
25 Any pair of FIDs identifies two sets of VIDs between which Independent VLAN Learning (IVL) (3.120) takes
26 place. The allocation of FIDs by a Bridge is described in 8.8.8.
- 27 c) A Port Map specifying forwarding for the destination MAC address and FID to a single Port or, in
28 the case of ECMP with flow filtering (44.1, 44.2), to a set of Ports from which one must be selected.

29 Item b), and any reference to FIDs and VIDs in the list above, are not applicable to MAC Bridge
30 components.

31 The Port Map may contain a **connection_identifier** (8.8.12) for each outbound Port.

32 Dynamic Filtering Entries can be created, updated, and removed by the Learning Process (8.7). They shall
33 be automatically removed after a specified time, the Ageing Time (8.7.3), has elapsed since the entry was
34 created or last updated.

35 Dynamic Filtering Entries can also be created and updated by SPB (Clause 27). ISIS-SPB (Clause 28)
36 refreshes these entries, so they are not aged out by the normal operation of the Learning Process. These
37 entries are used by SPBM (Clause 27) to support both unicast loop mitigation (6.5.4.2) and multicast loop
38 prevention (6.5.4.1) for active topology enforcement (8.6.1) as well as to locate end stations. Entries
39 indicating a set of potential forwarding ports from which one must be selected are used by ECMP with flow
40 filtering. SPBV updates Dynamic Filtering Entries by the normal Learning Process, but they may also be
41 controlled by ISIS-SPB. This allows entries for the higher layer entities in a Bridge (see Figure 8-12) to be
42 populated rapidly after any topology change.

43 No more than one Dynamic Filtering Entry shall be created in the FDB for a given combination of MAC
44 address and FID. Dynamic Filtering Entries cannot be created or updated by management.

45 NOTE 2—Dynamic Filtering Entries may be read by management (Clause 12). In the case of VLAN Bridge
46 components, the FID is represented in the management Read operation by any one of the VIDs that it represents. For a
47 given VID, the set of VIDs that share the same FID may also be determined by management.

1 8.8.4 MAC Address Registration Entries

2 A MAC Address Registration Entry specifies:

- 3 a) A MAC address specification, comprising:
 - 4 1) An individual MAC address, or
 - 5 2) A group MAC address, or
 - 6 3) All Group Addresses, for which no more specific Static Filtering Entry exists, or
 - 7 4) All Unregistered Group Addresses, for which no more specific Static Filtering Entry exists.
- 8 b) The VID for which the dynamic filtering information was registered.
- 9 c) A Port Map, consisting of a control element for each outbound Port, which specifies forwarding
10 (Registered) or filtering (Not registered) of frames destined to the MAC address and, in the case of
11 VLAN Bridge components, the VID.

12 Item b) and any reference to VIDs, in the list above, are not applicable to MAC Bridge components.

13 MAC Address Registration Entries are created, modified, and deleted by the operation of MMRP
14 (Clause 10) or by ISIS-SPB. No more than one MAC Address Registration Entry shall be created in the FDB
15 for a given combination of MAC address specification and VID.

16 NOTE—It is possible to have a Static Filtering Entry that has values of Forward or Filter on some or all Ports that mask
17 the dynamic values held in a corresponding MAC Address Registration Entry. The values in the MAC Address
18 Registration Entry will continue to be updated by MMRP; hence, subsequent modification of that entry to allow the use
19 of dynamic filtering information on one or more Ports immediately activates the true MMRP registration state that was
20 hitherto masked by the static information.

21 The creation of MAC Address Registration Entries by MMRP is subject to the
22 Restricted_MAC_Address_Registration management control (10.12.2.3). If the value of this control is
23 TRUE, a dynamic entry for a given MAC address may only be created if a Static Filtering Entry already
24 exists for that MAC address, in which the Registrar Administrative Control value is Normal Registration.

25 8.8.5 Dynamic VLAN Registration Entries

26 A Dynamic VLAN Registration Entry specifies:

- 27 a) The VID to which the dynamic filtering information applies.
- 28 b) A Port Map with a control element for each outbound Port specifying whether the VID is registered
29 on that Port.

30 NOTE—Dynamic VLAN Registration Entries can be accessed by ingress ports or egress ports.

31 Dynamic VLAN Registration Entries are not applicable to MAC Bridge components.

32 A separate Dynamic VLAN Registration Entry with a distinct Port Map may be created for each VID from
33 which frames are received by the Forwarding Process.

34 The creation of Dynamic VLAN Registration Entries is subject to the Restricted_VLAN_Registration
35 management control (11.2.3.2.3). If the value of this control is TRUE, a dynamic entry for a given VID may
36 only be created if a Static VLAN Registration Entry already exists for that VID, in which the Registrar
37 Administrative Control value is Normal Registration.

1 8.8.6 Default Group filtering behavior

2 Forwarding and filtering of group-addressed frames may be managed by specifying defaults for each VID
3 and outbound Port, or just outbound Ports in the case of MAC Bridges. The behavior of each of these
4 defaults, as modified by the control elements of more explicit FDB entries applicable to a given frame's
5 MAC address, VID, and outbound Port, is as follows:

6 NOTE 1—As stated in 8.8.1 for VLAN Bridge components, for a given MAC address, there may be separate Static
7 Filtering Entries with a distinct Port Map for each VID.

- 8 a) *Forward All Groups*. The frame is forwarded, unless an explicit Static Filtering Entry specifies
9 filtering independent of any dynamic filtering information.
- 10 b) *Forward Unregistered Groups*. The frame is forwarded, unless:
 - 11 1) An explicit Static Filtering Entry specifies filtering independent of any dynamic filtering
12 information; or
 - 13 2) An explicit Static Filtering Entry specifies forwarding or filtering on the basis of dynamic
14 filtering information, and an applicable explicit MAC Address Registration Entry exists
15 specifying filtering; or
 - 16 3) An applicable explicit Static Filtering Entry does not exist, but an applicable MAC Address
17 Registration Entry specifies filtering.
- 18 c) *Filter Unregistered Groups*. The frame is filtered unless:
 - 19 1) An explicit Static Filtering Entry specifies forwarding independent of any dynamic filtering
20 information; or
 - 21 2) An explicit Static Filtering Entry specifies forwarding or filtering on the basis of dynamic
22 filtering information, and an applicable explicit MAC Address Registration Entry exists
23 specifying forwarding; or
 - 24 3) An applicable explicit Static Filtering Entry does not exist, but an applicable MAC Address
25 Registration Entry specifies forwarding.

26 In Bridges that support only Basic Filtering Services, the default Group filtering behavior is Forward All
27 Groups for all Ports of the Bridge, for all VIDs.

28 NOTE 2—Forward All Groups corresponds directly to the behavior specified in IEEE Std 802.1D, 1993 Edition [B12]
29 when forwarding group MAC addressed frames for which no static filtering information exists in the FDB. Forward All
30 Groups makes use of information contained in Static Filtering Entries for specific group MAC addresses, but overrides
31 any information contained in MAC Address Registration Entries. Forward Unregistered Groups is analogous to the
32 forwarding behavior of a Bridge with respect to individual MAC addresses. If there is no static or dynamic information
33 for a specific group MAC address, then the frame is forwarded; otherwise, the frame is forwarded in accordance with the
34 statically configured or dynamically learned information.

35 In Bridges that support Extended Filtering Services, the default Group filtering behavior for each outbound
36 Port, and in the case of VLAN Bridge components, for each VID, is determined by the following
37 information contained in the FDB:

- 38 d) Any Static Filtering Entries that are applicable to that VID in the case of VLAN Bridge components,
39 with a MAC address specification of All Group Addresses or All Unregistered Group Addresses.
- 40 e) Any MAC Address Registration Entries that are applicable to that VID in the case of VLAN Bridge
41 components, with a MAC address specification of All Group Addresses or All Unregistered Group
42 Addresses.

43 The means whereby this information determines the default Group filtering behavior is specified in 8.8.9,
44 Table 8-11, and Table 8-12.

45 NOTE 3—The result is that the default Group filtering behavior, applicable in the case of VLAN Bridge components for
46 each VID, can be configured for each Port of the Bridge via Static Filtering Entries, determined dynamically via MAC
47 Address Registration Entries created/updated by MMRP (Clause 10), or both. For example, in the absence of any static
48 or dynamic information in the FDB for All Group Addresses or All Unregistered Group Addresses, the default Group
49 filtering behavior will be Filter Unregistered Groups on all Ports, for all VIDs. Subsequently, the creation of a Dynamic

1 MAC Address Registration Entry for All Unregistered Group Addresses indicating “Registered”, for a given VID in the
2 case of VLAN Bridge components, on a given Port would cause that Port to exhibit Forward Unregistered Groups
3 behavior, for that VID in the case of VLAN Bridge components. Similarly, creating a Static Filtering Entry for All Group
4 Addresses indicating “Registration Fixed” on a given Port, in the case of VLAN Bridge components for that VID, would
5 cause that Port to exhibit Forward All Groups behavior.

6 Hence, by using appropriate combinations of “Registration Fixed,” “Registration Forbidden,” and “Normal
7 Registration” in the Port Maps of Static Filtering Entries for the All Group Addresses and All Unregistered Group
8 Addresses address specifications, it is possible, for a given Port, and in the case of VLAN Bridge components, VID, to:

- 9 — Fix the default Group filtering behavior to be just one of the three behaviors described above; or
- 10 — Restrict the choice of behaviors to a subset of the three, and allow MMRP registrations (or their absence) to
11 determine the final choice; or
- 12 — Allow any one of the three behaviors to be adopted, in accordance with any registrations received via MMRP.

13 8.8.7 Dynamic Reservation Entries

14 A Dynamic Reservation Entry specifies the following:

- 15 a) A group MAC address or an individual MAC address.
- 16 b) The VID for which the dynamic reservation information was registered.
- 17 c) A Reservation Port Map, consisting of a control element for each outbound Port that specifies
18 forwarding (reservation information exists for this Port) or filtering (reservation information does
19 not exist for this Port) of frames for this Port. The initial value shall be filtering.

20 Dynamic Reservation Entries are created, modified, and deleted by the operation of SRP. No more than one
21 Dynamic Reservation Entry shall be created in the FDB for a given combination of MAC address and VID.

22 8.8.8 Allocation of VIDs to FIDs

23 The allocation of VIDs to FIDs within a VLAN Bridge determines how learned individual MAC address
24 information is used in forwarding and filtering decisions. If two VIDs are allocated to the same FID learned
25 information is shared, i.e., an individual MAC address learned following receipt of a frame affects the
26 forwarding or filtering of a subsequent frame with the same address as its destination address if either of the
27 two VIDs have been assigned to each of the two frames. If two VIDs are allocated to different FIDs learned
28 information is independent, i.e., an individual MAC address for frames that have one of the VIDs assigned
29 does not affect forwarding or filtering of frames with the other VID. IVL can allow two separate stations to
30 use the same MAC address, provided they use different VLANs, and also allows frames transmitted by a
31 single station to be assigned to independent active topologies. SVL allows support of a single VLAN by
32 multiple VIDs, enabling further control over the filtering of frames for that VLAN, and allowing SPB to
33 support the VLAN with a set of symmetric trees while still using learning. Further considerations for the use
34 of IVL and/or SVL are described in Annex F (informative).

35 A VLAN Bridge shall support either SVL or IVL, and may support both. If IVL is not supported the Bridge
36 shall support a single FID. If IVL is supported, the number of FIDs shall be the same as the maximum
37 number of VLANs supported by the implementation. For the purposes of the management operations, FIDs
38 are numbered from 1 through N, where N is the maximum number of FIDs supported by the
39 implementation. A Bridge that supports both SVL and IVL shall be capable of allocating any VID to any
40 FID, through configuration of the VID to FID allocation table. Following any set of management operations
41 designed to change the allocation of VIDs to FIDs, whether initiated by an administrative request or as a
42 consequence of other aspects of the Bridge’s operation, the value of any FID that is in use should be the
43 same as that of one of the VIDs allocated to that FID. One exception to this rule is the allocation of SPVIDs
44 to FIDs for SPBV (Clause 27). SPVIDs that are allocated are associated with a Base VID’s FID. The
45 learning behavior of SPVIDs assigned to a FID is determined by the learning behavior of the associated
46 Base VID’s FID (27.11).

1 The VID to FID allocation table may be read and modified by means of the management operations defined
2 in Clause 12. For each VID supported by the implementation, the allocation table indicates one of the
3 following:

- 4 a) The VID is currently not allocated to any FID, or
- 5 b) A fixed allocation has been defined (via management), allocating this VID to a specific FID, or
- 6 c) A dynamic allocation has been defined, allocating this VID to a specific FID.

7 An SPT Bridge can dynamically allocate VIDs, that have been reserved for use as SPVIDs, to FIDs.

8 An MST Bridge shall support IVL and may support SVL. An MST Bridge shall be capable of allocating any
9 FID to any MSTID through configuration of the FID to MSTID Allocation Table (12.12.2). An SPT Bridge
10 supports both IVL and SVL.

11 8.8.9 Querying the FDB

12 If a frame is assigned to a VLAN supported by SPBM, then the Filtering Database is queried upon reception
13 of the frame by the ingress port for ingress checking and the frame is forwarded or filtered. If single port
14 admittance ingress checking (6.5.4.2) is applied, then the frame is forwarded only if a Dynamic Filtering
15 Entry for that VID, source address tuple specifies forward for the ingress port; otherwise, the frame is
16 filtered. If relaxed ingress checking (45.3.1) is applied, then the frame is forwarded only if the ingress port is
17 in the Port Map of the MAC Address Registration Entry for that VID, individual destination address tuple;
18 otherwise, the frame is filtered.

19 If the VID assigned to a relayed frame identifies a VID or an ESP with a given outbound Bridge Port in its
20 member set (8.8.10), the FDB entries that are applicable to the frame's destination MAC address and VID
21 determine whether the frame is filtered or forwarded through that Bridge Port. More than one entry can
22 apply to a given frame. This subclause (8.8.9) specifies the effect of combining applicable entries, and of the
23 absence of certain types of entries (not all possible entries are necessarily present). For an overview of the
24 different types of entries, see the introductory text of 8.8.

25 The FDB entries associated with a VLAN Bridge component, applicable to a frame whose destination MAC
26 address is a specific individual MAC address are the Dynamic Filtering Entry (if present) with that MAC
27 address and the FID to which the frame's VID is allocated (8.8.8), and all Static Filtering Entries (if any)
28 with that MAC address, or with "All Individual MAC Address," as their MAC address specification and a
29 VID that is also allocated to that FID, as their VID specification. An entry with a wildcard VID applies only
30 if there is no applicable Static Filtering Entry for a specific VID. Table 8-10 specifies the result of
31 combining this information for an individual MAC address and a given outbound Bridge Port, and can be
32 summarized as follows:

- 33 — IF any static entry for a VID allocated to the FID specifies Forward, THEN Forward.
- 34 — ELSE IF any static entry for a VID allocated to the FID specifies Filter, THEN Filter.
- 35 — ELSE IF a static entry for the wildcard VID specifies Forward, THEN Forward.
- 36 — ELSE IF a static entry for the wildcard VID specifies Filter, THEN Filter.
- 37 — ELSE IF dynamic (learned filtering information) entry for the FID specifies Filter, THEN Filter
- 38 — ELSE Forward.

39 Table 8-10 is also applicable to MAC Bridge components but in this case there is no specific qualification on
40 FIDs in the FDB entries and the result of combining this information for an individual MAC address and a
41 given outbound Bridge Port can be summarized as follows:

- 42 — IF any static entry specifies Forward, THEN Forward.
- 43 — ELSE IF any static entry specifies Filter, THEN Filter.
- 44 — ELSE IF dynamic (learned filtering information) entry specifies Filter, THEN Filter.

Table 8-10—Combining Static and Dynamic Filtering Entries for an individual MAC address

Filtering Information	Control Elements in any Static Filtering Entry or Entries for this individual MAC address, FID, and outbound Port specify:				
	Forward (Any Static Filtering Entry for this Address/FID/Port specifies Forward)	Filter (No Static Filtering Entry for this Address/FID/Port specifies Forward)	Use Dynamic Filtering Information (No Static Filtering Entry for this Address/FID/Port specifies Forward or Filter), or no Static Filtering Entry present. Dynamic Filtering Entry Control Element for this individual MAC address, FID, and outbound Port specifies:		
			Forward	Filter	No Dynamic Filtering Entry present
Result	Forward	Filter	Forward	Filter	Forward

1 — ELSE Forward.

2 The FDB entries applicable (if present) to a frame whose destination MAC address is a specific group MAC
3 address are the Static Filtering Entries and MAC Address Registration Entries (for the frame's VID and for
4 the wildcard VID) whose address specification is that group MAC address or "All Group Addresses" or "All
5 Unregistered Group Addresses." A Static Filtering Entry for a wildcard VID applies only if there is no
6 applicable Static Filtering Entry for a specific VID, MAC Address Registration Entries do not use wildcard
7 VID. Table 8-11 specifies the results, Registered or Not Registered for a given outbound Bridge Port, of
8 combining the entries for the "All Group Addresses" and for combining the entries for "All Unregistered
9 Group Addresses." Table 8-12 combines these results with the entries for the specific group MAC address.
10 The result of Table 8-11 for "All Group Addresses" can be summarized as follows:

- 11 — IF a static entry for "All Group Addresses" and the frame's VID specifies Forward (Registration
12 Fixed), THEN "All Group Addresses" is Registered.
- 13 — ELSE IF a static entry for "All Group Addresses" and the frame's VID specifies Filter (Registration
14 Forbidden), THEN "All Group Addresses" is Not Registered.
- 15 — ELSE IF a dynamic (MAC Address Registration) entry for "All Group Addresses" and the frame's
16 VID specifies Forward (Registered), THEN "All Group Addresses" is Registered.
- 17 — ELSE "All Group Addresses" is NOT Registered.

18 NOTE 1—The result for "All Unregistered Group Addresses" is given by substituting that address specification for "All
19 Group Address" throughout the summary.

20 The result of Table 8-12 can be summarized as follows:

- 21 — IF a static entry for the specific group address and the frame's VID specifies Forward, THEN
22 Forward.
- 23 — ELSE IF a static entry for the specific group address and the frame's VID specifies Filter, THEN
24 Filter.
- 25 — ELSE IF a static entry for the specific group address and the wildcard VID specifies Forward,
26 THEN Forward.
- 27 — ELSE IF a static entry for the specific group address and the wildcard VID specifies Filter, THEN
28 Filter.
- 29 — ELSE IF the Table 8-11 result for "All Group Addresses" is Registered, THEN Forward.
- 30 — ELSE IF the Table 8-11 result for "All Unregistered Group Addresses" is Registered, THEN
31 Forward.

Table 8-11—Combining Static Filtering Entry and MAC Address Registration Entry for “All Group Addresses” and “All Unregistered Group Addresses”

Filtering Information	Static Filtering Entry Control Element for this group MAC address, VID, and outbound Port specifies:				
	Registration Fixed (Forward)	Registration Forbidden (Filter)	Use MAC Address Registration Information, or no Static Filtering Entry present. MAC Address Registration Entry Control Element for this group MAC address, VID, and outbound Port specifies:		
			Registered (Forward)	Not Registered (Filter)	No MAC Address Registration Entry present
Result	Registered	Not Registered	Registered	Not Registered	Not Registered

- 1 — ELSE IF a dynamic (MAC Address Registration) entry for the specific group address and the
- 2 frame’s VID specifies Forward, THEN Forward.
- 3 — ELSE Filter.

4 Table 8-11 and Table 8-12 are also applicable to MAC Bridge components but in this case there is no
5 specific qualification on VIDs in the FDB entries and the result of combining this information for an
6 individual MAC address and a given outbound Bridge Port can be summarized as follows:

- 7 — IF a static entry for the specific group address specifies Forward, THEN Forward.
- 8 — ELSE IF a static entry for the specific group address specifies Filter, THEN Filter.
- 9 — ELSE IF a static entry for the specific group address specifies Forward, THEN Forward.
- 10 — ELSE IF a static entry for the specific group address specifies Filter, THEN Filter.
- 11 — ELSE IF the Table 8-11 result for “All Group Addresses” is Registered, THEN Forward.
- 12 — ELSE IF the Table 8-11 result for “All Unregistered Group Addresses” is Registered, THEN
- 13 Forward.
- 14 — ELSE IF a dynamic (MAC Address Registration) entry for the specific group address specifies
- 15 Forward, THEN Forward.
- 16 — ELSE Filter.

17 When forwarding or filtering a frame with a destination group MAC address, a VLAN Bridge may:

- 18 a) Ignore the allocation of VIDs to FID, and use Table 8-12 directly for the frame’s VID; or
- 19 b) Take the same decision for all VIDs allocated to any given FID, forwarding if Table 8-12 specifies
- 20 Forward for any VID allocated to the same FID as the frame’s VID, and filtering otherwise.

21 For a given destination MAC address and VID, a Dynamic Reservation Entry (8.8.7) may be present in the
22 FDB, indicating, by means of its port map, which Ports have valid bandwidth reservations. If a Dynamic
23 Reservation Entry exists in the FDB for that MAC address and VID, the port map in the
24 Dynamic Reservation Entry is used to prevent frames being forwarded on Ports where no reservation exists.

Table 8-12—Forwarding or Filtering for specific group MAC addresses

				Static Filtering Entry Control Element for this group MAC address, VID, and outbound Port specifies:				
				Registration Fixed (Forward)	Registration Forbidden (Filter)	Use MAC Address Registration Information, or no Static Filtering Entry present. MAC Address Registration Entry Control Element for this group MAC address, VID, and outbound Port specifies:		
						Registered (Forward)	Not Registered (Filter)	No MAC Address Registration Entry present
All Group Addresses control elements for this VID and Port specify (Table 8-11):	Not Registered	All Unregistered Group Addresses control elements for this VID and Port specify (Table 8-11):	Not Registered	Forward	Filter	Forward	Filter (Filter Unregistered Groups)	Filter (Filter Unregistered Groups)
			Registered	Forward	Filter	Forward	Forward (Forward Unregistered Groups)	Forward (Forward Unregistered Groups)
	Registered				Forward	Filter	Forward (Forward All Groups)	Forward (Forward All Groups)

1 Table 8-13 combines the output of Table 8-10 (for an individual MAC address) or Table 8-12 (for a group
2 MAC address) with a Dynamic Reservation Entry to specify forwarding, or filtering, of a frame with that
3 destination MAC address and VID through an outbound Port.

Table 8-13—Forwarding or Filtering with Dynamic Reservation Entries

		Dynamic Reservation Entry Control Element for this individual or group MAC address, VID, and outbound Port specifies:	
		Forward	Filter
Output of Table 8-10 (individual MAC address) or Table 8-12 (group MAC address) for this group or individual MAC address, VID, and outbound Port specifies:	Forward	Forward	Filter
	Filter	Filter	Filter

4 NOTE 2—Where a group MAC address is used as the destination address for reserved traffic, the reservation
5 mechanisms used by SRP will prevent reservations being made for multiple streams using the same combination of
6 MAC address and VID. The Dynamic Reservation Entry can then control which Port(s) the stream associated with the
7 MAC address and VID is permitted to be transmitted through. The restriction preventing multiple streams from using the
8 same group MAC address/VID is necessary because those different streams may be destined for different destination
9 Ports; if the restriction was not imposed, it would not be possible to distinguish which frames belong to which streams
10 and therefore through which Ports they should be transmitted.

11 8.8.10 Determination of the member set for a VID

12 The VLAN configuration information contained in the FDB of a VLAN Bridge component for a given VID
13 may include a Static VLAN Registration Entry (8.8.2) and/or a Dynamic VLAN Registration Entry (8.8.5).
14 This information defines the member set, the Ports through which members of the VLAN identified by that
15 VID can be reached.

16 For a given VID, the FDB can contain a Static VLAN Registration Entry, a Dynamic VLAN Registration
17 Entry, both, or neither. Table 8-14 combines Static VLAN Registration Entry and Dynamic VLAN
18 Registration Entry information for a VID and Port to give a result *member*, or *not member*, for the Port. The
19 member set for a given VID consists of all Ports for which the result is member.

Table 8-14—Determination of whether a Port is in a VID's member set

Filtering Information	Static VLAN Registration Entry Control Element for this VID and Port specifies:				
	Registration Fixed	Registration Forbidden	Normal Registration, or no Static VLAN Registration Entry present. Dynamic VLAN Registration Entry Control Element for this VID and Port specifies:		
			Registered	Not Registered	No Dynamic VLAN Registration Entry present
Result	Member	Not member	Member	Not member	Not member

20 The Forwarding Process uses the member set to apply ingress (8.6.2) and egress rules (8.6.4) for the VID.

1 8.8.11 Permanent Database

2 The Permanent Database provides fixed storage for a number of Static Filtering Entries and Static VLAN
3 Registration Entries. The FDB shall be initialized with the FDB entries contained in this fixed data store.

4 Entries may be added to and removed from the Permanent Database under explicit management control,
5 using the management functionality defined in Clause 12. Changes to the contents of Static Filtering Entries
6 or Static VLAN Registration Entries in the Permanent Database do not affect forwarding and filtering
7 decisions taken by the Forwarding Process or the egress rules until such a time as the FDB is reinitialized.

8 NOTE 1—This aspect of the Permanent Database can be viewed as providing a “boot image” for the FDB, defining the
9 contents of all initial entries, before any dynamic filtering information is added.

10 NOTE 2—Subclause 10.12.2.3 defines an initial state for the contents of the Permanent Database, required for the
11 purposes of MMRP operation.

12 8.8.12 Connection_Identifier

13 A connection_identifier may be associated with the Bridge Port value maintained in a Dynamic Filtering
14 Entry of the FDB for Bridge Ports that generate EM_UNITDATA.indications with a non-null
15 connection_identifier parameter (e.g., a VIP as specified in 6.10). The connection_identifier has no effect on
16 the Bridge operation specified in this clause other than being stored in the FDB and conveyed across the
17 EISS as a parameter in an EM_UNITDATA.request or EM_UNITDATA.indication (6.8.1).

18 When the result of an FDB query (8.8.9) for a particular frame is that the frame will be forwarded, and the
19 FDB entry determining that result has a connection_identifier, then that connection_identifier is included in
20 the EM_UNITDATA.request. For a given VID and MAC address, a connection_identifier can be included in
21 a Dynamic Filtering Entry of the FDB. The FDB entry that determines a filtering or forwarding result is
22 specified in Table 8-10. If there are no entries for that MAC address and VID in the FDB, then a null value
23 for the connection_identifier is used in the EM_UNITDATA.request.

24 When an EM_UNITDATA.indication with a non-null connection_identifier is received from an ingress port,
25 and the Learning Process identifies or creates a Dynamic Filtering Entry for that frame’s source address and
26 VID, the connection_identifier is included in the entry created for that port in the FDB.

27 8.9 MST, SPB, and ESP configuration information

28 In order to support multiple spanning trees, and SPB, all the Bridges in an MST or SPT Region have to be
29 configured with assignments of VIDs to spanning trees.

30 The combination of the VID to FID allocations (8.8.8) and the FID to MSTI allocations (8.9.3) defines a
31 mapping of VIDs to spanning trees, MSTIDs, represented by the MST Configuration Table (8.9.1). If an
32 MSTID is included in the MSTI List (12.12.1), frames with VIDs allocated to that MSTID will be supported
33 by an MSTI calculated by MSTP (Clause 13). If the MSTID is not included in the MSTI List, and is not a
34 reserved MSTID, those frames will be supported by the IST.

35 A Bridge supporting PBB-TE can assign specific VID values to ESPs but can also support any of the
36 spanning tree protocols. This is achieved by allocating the VIDs associated with the PBB-TE ESPs to a
37 special MSTID, the TE-MSTID.

38 An SPT Bridge can allocate any given VLAN to the CIST, or to the reserved MSTIs, as well as supporting
39 SPB for other VLANs within an SPT Region. The assignment of VLANs and VIDs to MSTIs is subject to
40 similar considerations and constraints as for MST Bridges. In addition to the Base VID used to identify

1 frames for the VLAN when transmitted on the CST outside the region, each VLAN supported by SPBV uses
2 a number of dynamically allocated shortest path VIDs (SPVIDs), one for each SPT Bridge participating in
3 the VLAN within the region.

4 The network administrator allocates enough SPVIDs for SPBV to avoid unintentionally excluding Bridges
5 from the topology. The details of SPVID allocation, including mitigation of the consequences of not
6 allocating enough SPVIDs, are specified in 27.10.

7 Administratively, by not translating from Base VID to SPVID, an SPBV VLAN can be forced to the IST.
8 When there is no shortage of SPVIDs and the MST Configuration Identifiers (MCIDs) match but the Base
9 VID algorithms disagree the algorithm defaults to the spanning tree. This behavior allows SPBV to change
10 the Equal Cost Tree (ECT) Algorithms.

11 Dynamic SPVID allocation is supported by ISIS-SPB and allows the addition of Bridges to existing SPT
12 Regions without disruptive configuration changes, as well as allowing supporting auto configuration of
13 SPB in simple networks where all user data is assigned to a VLAN identified by the default PVID of 1
14 (Table 9-2). However, prior to SPVID allocation, all Bridges in an SPT Region have to agree which VLANs
15 are to be shortest path bridged, and which SPT Set (ECT Algorithm) is to be used for each of those VLANs.
16 By default, all VLANs whose Base VID is allocated to MSTID 0xFFD will be supported by SPBV using the
17 SPB default ECT Algorithm. The SPT configuration information can also specify that frames for SPBV and
18 SPBM be supported by another specified SPB ECT Algorithm as specified by configuration (12.25.4,
19 Table 28-1).

20 NOTE—The use of MSTIDs to support allocation of Base VIDs to SPB allows the per VLAN components of the
21 necessary information to be conveyed in the MST Configuration Digest, thus avoiding any need for a separate digest.
22 However, the digest is not specific about how the VIDs are mapped to specific SPTs, just that they are allocated to SPBV,
23 SPBM, or SPVIDs.

24 8.9.1 MST Configuration Table

25 The MST Configuration Table specifies an MSTID for each possible VID.

26 In an MST Bridge, each MSTID identifies the MSTI to which the VID is allocated and an MSTID of zero
27 identifies the CIST.

28 The combination of the VID to FID allocations (8.8.8) and the FID to MSTID allocations (8.9.3) define a
29 mapping of VIDs to MSTIDs summarized in the MST Configuration Table (8.9.1).

30 8.9.2 MST configuration identification

31 For two or more MST Bridges to be members of the same MST Region (3.174), it is necessary for those
32 Bridges to support the same MST Region configuration and to be interconnected only by means of LANs,
33 with all intervening Bridges being members of the region or being transparent to the BPDUs of the members
34 of the region. Two MST Region configurations are considered to be the same if the correspondence between
35 VIDs and MSTIDs is identical in both configurations and they use the same information to identify the
36 configuration, including the same Configuration Name.

37 NOTE 1—If two adjacent MST Bridges consider themselves to be in the same MST Region despite having different
38 mappings of VIDs to MSTIDs, then the possibility exists of undetectable loops arising within the MST Region.

39 In order to ensure that adjacent MST Bridges are able to determine whether they are part of the same MST
40 Region, the MST BPDU supports the communication of an MCID (13.8).

41 NOTE 2—As the MCID is smaller than the mapping information that it summarizes, there is a small but finite
42 possibility that two MST Bridges will assume that they have the same MST Region Configuration when this is not
43 actually the case. However, given the size of the identifier, this standard assumes that this possibility is sufficiently small
44 that it can safely be ignored. Appropriate use of the Configuration Name and Revision Level portions of the identifier
45 can remove the possibility of an accidental match between MCIDs that are derived from different configurations within
46 a single administrative domain (see 13.8).

1 8.9.3 FID to MSTI Allocation Table

2 The FID to MSTI Allocation Table defines, for all FIDs that the Bridge supports, the MSTID to which the
3 FID is allocated.

- 4 a) The IST is identified by the reserved MSTID value 0.
- 5 b) The use of PBB-TE is identified by the reserved MSTID value TE-MSTID (0xFFE).
- 6 c) Each MSTID in the MSTI List identifies an MSTI.
 - 7 The reserved MSTID values 0 and TE-MSTID, SPBV-MSTID, SPBM-MSTID and
 - 8 SPVID-Pool-MSTID are never used in the MSTI List.
- 9 d) The following MSTID values identify the method used to support the VLANs identified by the Base
10 VIDs allocated to those MSTIDs:
 - 11 1) 0xFFC—SPBM-MSTID.
 - 12 2) 0xFFD—SPBV-MSTID.
 - 13 3) 0xFFF—SPVID-Pool-MSTID: Allocated to FIDs that are not used to filter frames including
 - 14 SPVIDs for SPBV.

15 NOTE—MSTIDs that are present in the MSTI List (12.12) identify spanning tree instances supported by MSTP.
16 MSTIDs identify (indirectly) VLANs that are supported by SPB.

17 8.9.4 SPT Configuration Identification

18 For two or more SPT Bridges to be members of the same SPT Region (Clause 3), it is necessary for those
19 Bridges to support the same SPT Region configuration and to be interconnected only by means of LANs,
20 with all intervening Bridges being members of the region or being transparent to the BPDUs and/or SPB
21 Hello PDUs of the members of the region. This requirement is identical to that for MST Regions: a given
22 SPT Region can support selected MSTIs as well as SPTs. MCIDs only specify the VIDs that are configured.

23 Two SPT Region configurations are considered to be the same if they have the same MST Region
24 configuration or alternate MST Region configuration, they agree on the VLANs that are to be shortest path
25 bridged and they use the same ISIS-SPB. Furthermore by using ISIS-SPB they agree on which SPT Set is to
26 be used for each of these VLANs (Clause 27).

27 The MST Region information captures the VIDs for SPB but it does not capture the operational state of the
28 VIDs for SPB. This allows allocation of SPVIDs and upgrade of multiple ECT Algorithms without change
29 to the MST Region when sufficient pre-allocation of VIDs has been configured. However, in situations
30 where the network grows beyond the current allocation of VIDs, the options are to temporarily create a
31 region boundary as the new configuration is deployed or allow in-service upgrades by making changes that
32 are compatible with the current MST region configuration.

33 To support in-service upgrade from one set of VID to MSTID allocations to another, SPT Bridges support a
34 second (auxiliary) MCID (28.9). Two such Bridges consider themselves to be in the same MST Region
35 provided that at least one of their MCIDs matches with one of their neighbors. In this way, a new
36 configuration can be incrementally introduced without time constraints, and put into operation, all the while
37 maintaining operation as a single SPT Region.

38 8.10 Spanning Tree Protocol Entity

39 Figure 8-6 illustrates the operation of the Spanning Tree Protocol Entity, including the reception and
40 transmission of frames containing BPDUs, the modification of the state information associated with
41 individual Bridge Ports, and the notification of the FDB of changes in active topology.

1 A given MST Bridge is not required to support all of the spanning trees that exist within the MST bridged
2 network. That is, the number of spanning trees operated by the Spanning Tree Protocol Entity in a given
3 Bridge may be different from the number operated by that in another Bridge. However, as a direct
4 consequence of the conditions stated in 8.9.2, the number of instances of the spanning tree protocol operated
5 by a given MST Bridge is the same for all Bridges that are members of a given MST Region.

6 The ISIS-SPB Entity comprises an instance of IS-IS with SPB extensions as specified in Clause 28. The
7 ISIS-SPB Entity is described, for the purposes of management, as distinct from the Spanning Tree Protocol
8 Entity. Nonetheless, the ISIS-SPB and Spanning Tree Protocol Entities that calculate active topologies for
9 the same Bridge Port use the same ISS SAP (ISS-SAP, 8.5, 27.2), and the same individual MAC address.
10 However, ISIS-SPB uses one of the group MAC addresses in Table 8-15 and does not use the group MAC
11 addresses used by the spanning tree protocol specified in Table 8-1. The ISIS-SPB Entity uses the state
12 machines and state machine variables supported by the Spanning Tree Protocol Entity (Clause 27, 13.6) and
13 calculates the Agreement Digest that the Spanning Tree Protocol Entity transmits in SPT BPDUs and/or
14 SPB Hello PDUs.

15 **8.11 MRP entities**

16 The MRP entities operate the algorithms and protocols associated with the MRP Applications supported by
17 the Bridge and consist of the set of MRP Participants for those MRP Applications (Clause 10).

18 Figure 8-7 illustrates the operation of an MRP entity, including the reception and transmission of frames
19 containing MRPDUs, the use of control information contained in the FDB, and the notification of the FDB
20 of changes in filtering information.

21 **8.12 Bridge Management Entity**

22 Remote management capabilities (as opposed to management via a console or other device attached directly
23 to a Bridge) are modeled as performed by a Bridge Management Entity supported by a protocol stack as
24 described in 8.13.7. To facilitate interoperable remote management, the Bridge Management Entity should
25 use either the SMIPv2 MIB modules specified in Clause 17 with SNMP or the YANG modules specified in
26 Clause 48 with a network configuration protocol, e.g., NETCONF, RESTCONF.

27 **8.13 Addressing**

28 All MAC Entities communicating across a bridged network use 48-bit MAC addresses (IEEE Std 802).
29 These can be universally administered addresses or a combination of universally administered and locally
30 administered addresses.

31 **8.13.1 End stations**

32 Frames transmitted between end stations using the MAC Service carry the MAC address of the source and
33 destination peer end stations in the source and destination address fields of the frames, respectively. The
34 address, or other means of identification, of a Bridge is not carried in frames transmitted between peer users
35 for the purpose of frame relay in the network.

36 In the absence of explicit filters configured via management as Static Filtering Entries, or via MMRP as
37 MAC Address Registration Entries (8.8, Clause 10), frames with a destination address of the broadcast
38 address or any other group address that is not a Reserved Address (8.6.3) are assigned by a VLAN Bridge
39 component a VID and relayed throughout the VLAN identified by that VID.

8.13.2 Bridge Ports

A separate individual MAC address is associated with each instance of the MAC Service provided to an LLC Entity. That MAC address is used as the source address of frames transmitted by the LLC Entity.

Media access method-specific procedures can require the transmission and reception of frames that use an individual MAC address associated with the Bridge Port, but neither originate from nor are delivered to a MAC Service user. Where an individual MAC address is associated with the provision of an instance of the MAC Service by the Port, that address can be used as the source and/or the destination address of such frames, unless the specification of the media access method-specific procedures requires otherwise.

8.13.3 Use of LLC by Spanning Tree Protocol Entities

Spanning Tree Protocol Entities use the DL_UNITDATA.request and DL_UNITDATA.indication primitives (ISO/IEC 8802-2) provided by individual LLC Entities associated with each Bridge Port to transmit and receive BPDUs (Clause 14). The source_address and destination_address parameters of the DL_UNITDATA.request shall both denote the standard LLC address assigned to the Bridge spanning tree protocol (Table 8-15). Each DL_UNITDATA request primitive gives rise to the transmission of an LLC UI command PDU, which conveys the BPDU in its information field.²⁵

Table 8-15—Standard LLC address assignment

Assignment	Value
Bridge spanning tree protocol	01000010
IS-IS PDUs	11111110
NOTE—Code Representation: The least significant bit (LSB) of the value shown is the right-most. The bits increase in significance from right to left. It should be noted that the code representation used here has been chosen in order to maintain consistency with the representation used elsewhere in this standard.	

A Protocol Identifier field, present in all BPDUs (Clause 14), serves to identify different protocols supported within the scope of the LLC address assignment. Further values of this field are reserved for future standardization. A Spanning Tree Protocol Entity that receives a BPDU with an unknown PID shall discard that PDU.

IS-IS PDUs are LLC (encoded with a DSAP and SSAP of 0xFE (section 7.5 of ISO/IEC 10589:2002), then a single byte indicating Control 0x03 (see ISO/IEC/IEEE 8802-2) immediately followed by a IS-IS PDU starting with an Interdomain Routing Protocol Discriminator equal to 0x83 (section 5.3 of ISO/IEC 9577:1999).

8.13.4 Reserved MAC addresses

Any frame with a destination address that is a reserved MAC address shall not be forwarded by a Bridge. Reserved MAC addresses for MAC Bridge components and C-VLAN components, for S-VLAN components, and for TPMR components are specified in Table 8-1, Table 8-2, and Table 8-3, respectively. These group MAC addresses are reserved for assignment to standard protocols, according to the criteria for such assignments.²⁶

²⁵ Administration of LLC address assignments is the responsibility of the IEEE Registration Authority (IEEE RA). A full list of standard LLC address assignments, and the criteria for assignment, can be found on the IEEE RA Website at <https://standards.ieee.org/regauth/index.html>.

1 8.13.5 Group MAC addresses for spanning tree entity

2 A Spanning Tree Protocol Entity uses an instance of the MAC Service provided by each of the Bridge's
3 Ports to transmit frames addressed to the Spanning Tree Protocol Entities of all the other Bridges attached to
4 the same LAN as that Port.

5 8.13.5.1 Group MAC addresses for spanning tree protocols

6 Spanning tree protocols use a reserved address for peer to peer communication between entities of the same
7 type.

8 A universally administered standard group MAC address, known as the Bridge Group Address, has been
9 assigned (Table 8-1) for use by C-VLAN components for this purpose.

10 It is essential to the operation of the spanning tree protocols that frames with this destination address both
11 reach all peer protocol entities attached to the same LAN and do not reach protocol entities attached to other
12 LANs. The Bridge Group Address is therefore included in the block of MAC Bridge and C-VLAN
13 components reserved MAC addresses and is always filtered by Customer Bridges and C-VLAN components
14 of Provider Edge Bridges (8.6.3).

15 As a network of Provider Bridges needs to appear to attached Customer Bridges as if it was a single LAN,
16 frames addressed to the Bridge Group Address are forwarded by S-VLAN components. Provider Bridges
17 also use a spanning tree protocol to provide one or more loop-free active topologies, so a distinct universally
18 administered standard group MAC address, the Provider Bridge Group Address, which can be confined to
19 the LANs that their Bridge Ports attach, has been assigned (Table 8-2). The Provider Bridge Group Address
20 is included in both the C-VLAN component and the S-VLAN component reserved MAC addresses and is
21 always filtered by all Bridges (8.6.3).

22 The source MAC address field of frames transmitted by Spanning Tree Protocol Entities contains the
23 individual MAC address for the Bridge Port used to transmit the frame.

24 8.13.5.2 Group MAC addresses for SPB

25 ISIS-SPB use a group address for ISIS-SPB peer to peer communication. ISIS-SPB is configured to use only
26 one of the addresses listed in Table 8-16. All other addresses in this table will be ignored by IS-IS and will
27 not be filtered by the MAC Relay. The choice of which address is used depends on the environment.

28 Table 8-16 lists the various addresses that may be configured for ISIS-SPB. Table 8-17 lists the deployment
29 environments for SPB. The usage of ISIS-SPB is described in Clause 28. These addresses are from a
30 reserved address space. These addresses have the property that:

- 31 a) Where a given address in the set is used by a Bridge component to support ISIS-SPB, frames
32 destined for that address shall not be forwarded by that Bridge component; i.e., a Static Filtering
33 Entry for that address is maintained for that group MAC address in order to prevent the forwarding
34 of frames destined for that address.
- 35 b) Where a given address in the set is not used by a Bridge component to support ISIS-SPB, frames
36 destined for that address received on any Port that is part of a given active topology shall be
37 forwarded by that Bridge component on all other Ports that are part of that active topology.

²⁶ Administration of standard group MAC address assignments is the responsibility of the IEEE Registration Authority (IEEE RA). A full list of standard group MAC address assignments, and the criteria for assignment, can be found on the IEEE RA Website at <https://standards.ieee.org/regauth/index.html>.

Table 8-16—ISIS-SPB reserved addresses

Assignment	Value
All Level 1 Intermediate Systems	01-80-C2-00-00-14 ^a
All Level 2 Intermediate Systems	01-80-C2-00-00-15 ^a
All Intermediate Systems	09-00-2B-00-00-05 ^a
All Provider Bridge Intermediate Systems	01-80-C2-00-00-2E
All Customer Bridge Intermediate Systems	01-80-C2-00-00-2F

^a The first three addresses are existing IS-IS addresses (Table 9 of ISO/IEC 10589:2002).

¹ An SPT Bridge, whether it is running SPBV, or SPBM, or both, uses a single group MAC address.

² Table 8-17 provides the reference for the possible environments.

³ In Provider Bridge only environments, for SPBV the address should be the allocated All Provider Bridge Intermediate Systems address.

⁵ In Customer Bridge only environments where only SPBV is supported the address should be the All Customer Bridge Intermediate Systems address.

⁷ In SPBM or SPBV deployments where only Provider backbone components are configured the use of addresses is flexible as indicated in Table 8-17. To allow peering with other systems capable of using IS-IS for IP, the existing IS-IS addresses may be used.

Table 8-17—ISIS-SPB Recommended Address Usage

Network Deployment	Level 1	Level 2	All Levels
SPT Bridges Peering with other IS-IS-capable systems	01-80-C2-00-00-14	01-80-C2-00-00-15	09-00-2B-00-00-05
SPT Bridges PBBs only	01-80-C2-00-00-14	01-80-C2-00-00-15	09-00-2B-00-00-05 or 01-80-C2-00-00-2E
SPT Bridges in SPBV mode for S-VLANs Provider Bridge	01-80-C2-00-00-2E		
SPT Bridges in SPBV mode for C-VLANs Customer Bridge	01-80-C2-00-00-2F		

10 8.13.6 Group MAC addresses for MRP Applications

¹¹ An MRP Entity that supports a given MRP Application transmits frames addressed to all other MRP Entities that implement the same MRP Application. The peers of each such entity bound a region of the network that contains no peers, commonly a single LAN in the case where all Bridges attached to the LAN implement the application.

¹⁵ Universally administered standard group MAC addresses are assigned to MRP applications, from a set of universally administered group MAC addresses, known as MRP application addresses, as shown in Table 10-1; these addresses are for use by VLAN components that support the MRP applications concerned.

1 FDB entries for each MRP application address assigned to an application that is supported by a VLAN
2 component should be configured in the FDB so frames for that application are confined to the peer region,
3 while addresses for applications that are not supported should not be included. These group MAC addresses
4 are reserved for assignment to standard protocols, according to the criteria for such assignments (see 8.13.4).

5 When an MRP Application operates among both Customer and Provider Bridges, a distinct universally
6 administered group MAC address is assigned for use by both C-VLAN components and S-VLAN
7 components. MMRP is such an application, and uses the Customer and Provider Bridge MMRP Address
8 assigned in Table 10-1. When an MRP application operates separately among Customer Bridges or among
9 Provider Bridges, different distinct universally administered group MAC addresses are assigned for use by
10 C-VLAN components and for use by S-VLAN components. MVRP is such an application; C-VLAN
11 components use the Customer Bridge MVRP address assigned in Table 10-1, while S-VLAN components
12 use the Provider Bridge MVRP address assigned in Table 8-1. Note that because the Provider Bridge MVRP
13 address is selected from the set of addresses that appear in Table 8-1 but not Table 8-2, it will always be
14 filtered by C-VLAN components. Although the Provider Bridge MVRP Address is not selected from
15 Table 10-1, an S-VLAN component should treat this as an MRP Application Address and configure an FDB
16 entry to filter frames with this address.

17 An MRP Entity that supports a given MRP Application uses the individual MAC address for the Bridge Port
18 through which the PDU is transmitted (8.13.2) as the source_address field of MAC frames conveying
19 MRPDUs for that application.

20 **8.13.7 Bridge Management Entities**

21 Network configuration and management protocols typically use IP as a network layer protocol. SNMP can
22 also be supported directly over an IEEE 802 LAN, as specified in IETF RFC 4789. The management
23 protocol stack and address used shall be supported by a single LLC Entity attached to a Bridge Port. The
24 Port should be a Management Port for the Bridge, as described in 8.3 and Figure 8-8, but may be a Port
25 attached to a LAN, as described in 8.13.9, Figure 8-25, and Figure 8-26.

26 NOTE—A universally administered standard group MAC address, known as the All LANs Bridge Management Group
27 Address with a value of 01-80-C2-00-00-10 was assigned and recorded in the 1990 Edition of this standard. That address
28 should not be used for Bridge management or for any other purpose.

29 **8.13.8 Unique identification of a Bridge**

30 A unique EUI-48 Universally Administered MAC address, termed the Bridge Address, shall be assigned to
31 each Bridge. The Bridge Address may be the individual MAC address of a Bridge Port; in which case, use of
32 the address of the lowest numbered Bridge Port (Port 1) is recommended.

33 NOTE—RSTP (Clause 13) and MSTP (Clause 13) require that a single unique identifier be associated with each Bridge.
34 That identifier is derived from the Bridge Address as specified in 14.2.5.

35 **8.13.9 Points of attachment and connectivity for Higher Layer Entities**

36 The Higher Layer Entities in a Bridge, such as the Spanning Tree Protocol Entity (8.10), MRP entities
37 (8.11), the ISIS-SPB Entity (Clause 28), and Bridge Management (8.12), are modeled as attaching directly
38 to one or more individual LANs connected by the Bridge's Ports, in the same way that any distinct end
39 station is attached to the network. While these entities and the relay function of the Bridge use the same
40 individual MAC entities to transmit and receive frames, the addressing and connectivity to and from these
41 entities is the same as if they were attached as separate end stations "outside" the Port or Ports where they
42 are actually attached. Figure 8-23 is functionally equivalent to Figure 8-2 but illustrates this logical
43 separation between the points of attachment used by the Higher Layer Entities and those used by the MAC
44 Relay Entity.

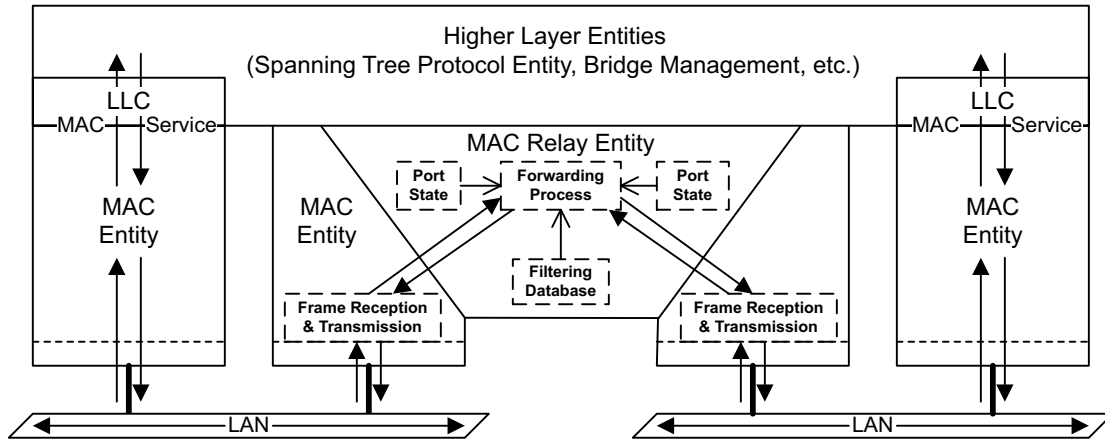


Figure 8-23—Logical points of attachment of the Higher Layer and Relay Entities

1 Figure 8-24 depicts the information used to control the forwarding of frames from one Bridge Port to
2 another (the Port States and the content of the FDB) as a series of switches (shown in the open, disconnected
3 state) inserted in the path provided by the MAC Relay Entity. For the Bridge to forward a given frame
4 between two Ports, all three switches must be in the closed state. While showing Higher Layer Entities
5 sharing the point of attachment to each LAN used by each Bridge Port to forward frames, this figure further
6 illustrates a point made by Figure 8-23. Controls placed in the forwarding path have no effect on the ability
7 of a Higher Layer Entity to transmit and receive frames to or from a given LAN using a direct attachment to
8 that LAN (e.g., from entity A to LAN A); they only affect the path taken by any indirect transmission or
9 reception (e.g., from entity A to or from LAN B).

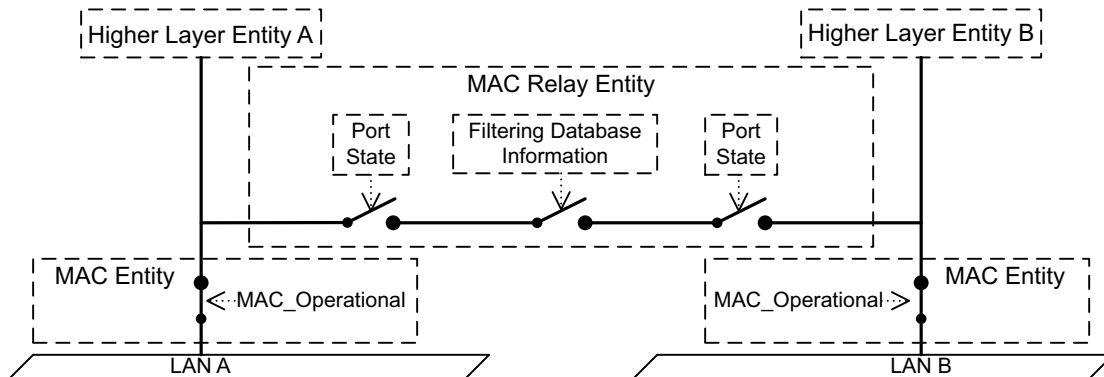


Figure 8-24—Effect of control information on the forwarding path

10 The functions provided by Higher Layer Entities can be categorized as requiring either:

- 11 a) A single point of attachment to the Bridged Network, providing connectivity to stations attached to
12 the network at any point (subject to administrative control), as does Bridge Management; or
- 13 b) A distinct point of attachment to each individual LAN attached by a Bridge Port, providing
14 connectivity only to peer entities connected directly to that LAN, as do the Spanning Tree Protocol
15 Entity and the MRP entity.

1 In the latter case, it is essential that the function associate each received and transmitted frame with a point
2 of attachment. Frames transmitted or received via one point of attachment are not to be relayed to and from
3 other Ports and attached LANs, so the MAC addresses (8.13.4) used to reach these functions are
4 permanently configured in the FDB.

5 NOTE 1—Addresses used to reach functions with distinct points of attachment are generally group MAC addresses.

6 NOTE 2—A single higher layer entity can incorporate both a function requiring a single point of attachment and a
7 function requiring distinct points of attachment. The two functions are reached using different MAC addresses.

8 Figure 8-25 illustrates forwarding path connectivity for frames destined for Higher Layer Entities requiring
9 per-Port points of attachment. Configuration of the Permanent Database in all Bridges to prevent relay of
10 frames addressed to these entities means that they receive frames only via their direct points of attachment
11 (i.e., from LAN A to entity A, and from LAN B to entity B), regardless of Port states.

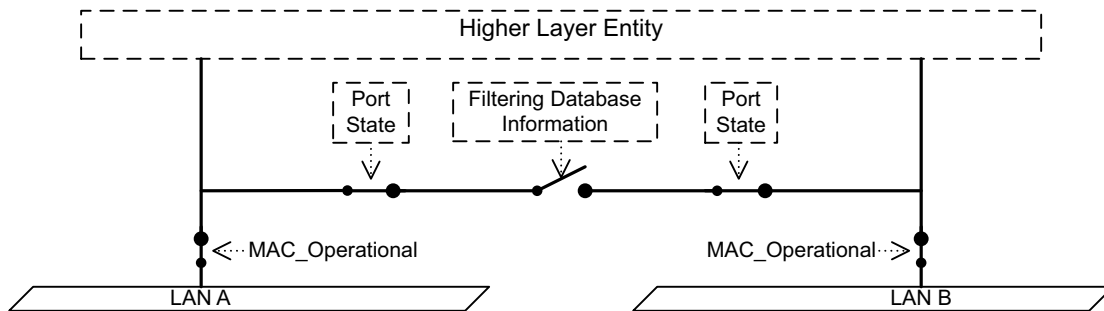


Figure 8-25—Per-Port points of attachment

12 Figure 8-26 and Figure 8-27 illustrate forwarding path connectivity for frames destined for a Higher Layer
13 Entity requiring a single point of attachment. In both figures, the FDB permits relay of frames, as do the Port
14 states in Figure 8-26 where frames received from LAN B are relayed by the Bridge to the entity and to
15 LAN A.

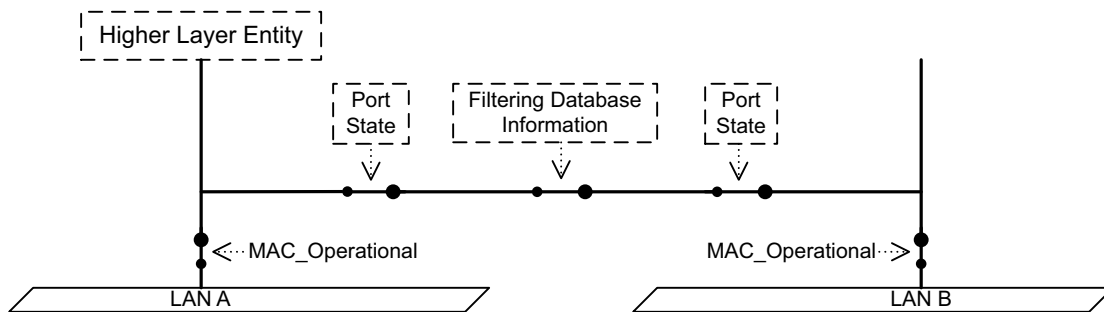


Figure 8-26—Single point of attachment—relay permitted

16 In Figure 8-27, frames received from LAN A are received by the entity directly, but frames received from
17 LAN B are not relayed by the Bridge and will only be received by the entity if another forwarding path is
18 provided between LANs A and B. If the Discarding Port state shown resulted from spanning tree
19 computation (and not from disabling the Administrative Bridge Port State), such a path will exist via one or
20 more Bridges. If there is no active spanning tree path from B to A, the network has partitioned into two
21 separate Bridged Networks, and the Higher Layer Entity shown is reachable only via LAN A.

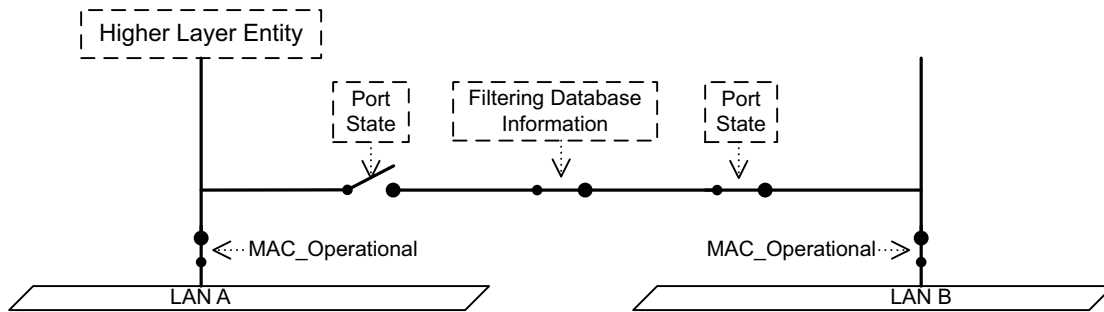


Figure 8-27—Single point of attachment—relay not permitted

1 Specific Higher Layer Entities can take notice of the Administrative Bridge Port State, as required by their
2 specification. The Spanning Tree Protocol Entity is one such example—BPDUs are never transmitted or
3 received on Ports with an Administrative Bridge Port State of Disabled.

4 If a Bridge Port's MAC Entity is not operational, a Higher Layer Entity directly attached at the Port will not
5 be reachable, as Figure 8-28 illustrates. The Spanning Tree Protocol Entity ensures that the Port State is
6 Discarding if the MAC_Operational (IEEE Std 802.1AC) is FALSE even if the Administrative Bridge Port
7 State is Enabled.

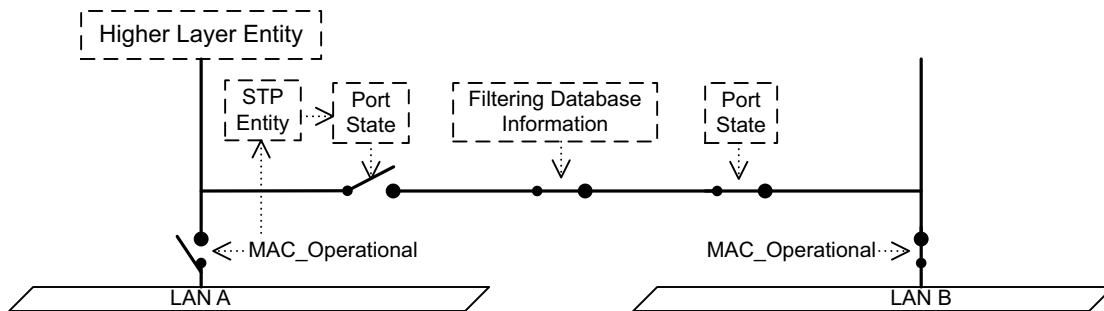


Figure 8-28—Effect of Port State

8 Port-based network access control (IEEE Std 802.1X) and MAC Security (IEEE Std 802.1AE) can be used
9 to provide a further level of control over the connectivity provided by a Bridge Port to the MAC Relay
10 Entity and the Higher Layer Entities within a Bridge. Port-based network access control creates two distinct
11 SAPs for the LAN: the Controlled Port and the Uncontrolled Port. MAC_Operational is TRUE for the
12 Controlled Port only when criteria for authentication, authorization, and secure connectivity have been
13 satisfied; while MAC_Operational for the Uncontrolled Port is the same as that for direct access to the LAN.
14 Both the Spanning Tree Protocol Entity and the MAC Relay Entity attach to the Controlled Port, thus
15 ensuring that the connectivity seen by each of those entities is the same. If MAC_Operational for the
16 Controlled Port is FALSE, the Spanning Tree Protocol Entity will ensure that both forwarding and learning
17 (8.4) will be FALSE for that Port (i.e., the Port State will be Discarding). The Uncontrolled Port supports the
18 operation of protocol entities such as the Port Access Entity (PAE) specified by IEEE Std 802.1X, that
19 participate in the authentication exchanges necessary before Controlled Port connectivity is permitted or that
20 distribute other unsecured information.

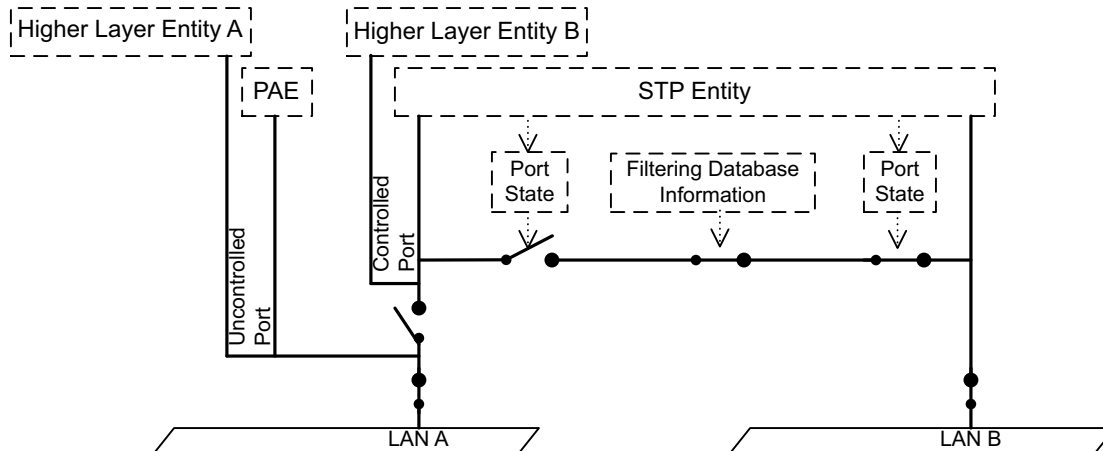


Figure 8-29—Controlled and Uncontrolled Port connectivity

1 Figure 8-29 illustrates the connectivity provided to Higher Layer Entities if the MAC entity is physically
2 capable of transmitting and receiving frames but MAC_Operational is FALSE for the Controlled Port.
3 Higher Layer Entity A and the PAE are connected to an Uncontrolled Port and can transmit and receive
4 frames using the MAC entity associated with the Port for LAN A, which Higher Layer Entity B cannot.
5 None of the three entities can transmit or receive to or from LAN B.

6 NOTE 3—Reference should be made to IEEE Std 802.1X-2010 or later.

7 NOTE 4—The administrative and operational state values associated with the MAC, the Port's authorization state, and
8 the Bridge Port State equate to the ifAdminStatus and ifOperStatus parameters associated with the corresponding
9 interface definitions; see IETF RFC 2863.

10

11 8.13.10 VLAN attachment and connectivity for Higher Layer Entities

12 In VLAN Bridges, two more switches appear in the forwarding path, corresponding to the actions taken by
13 the Forwarding Process (8.6) in applying the ingress and egress rules (8.6.2 and 8.6.4), as illustrated in
14 Figure 8-30.

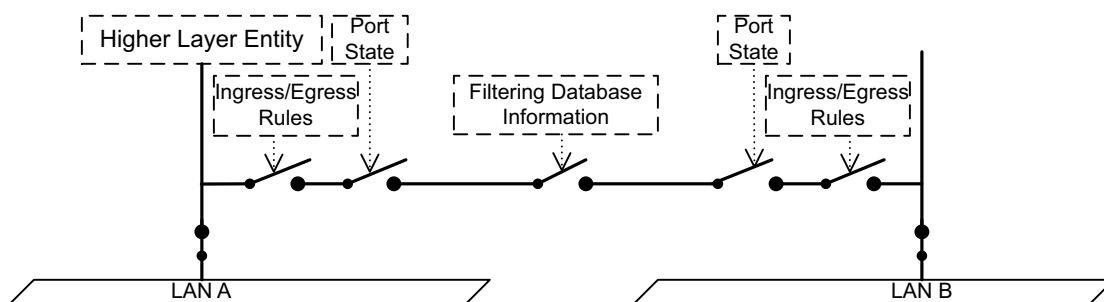


Figure 8-30—Ingress/egress control information in the forwarding path

15 As with Port state information, the configuration of the ingress and egress rules does not affect the reception
16 of frames received on the same LAN as a Higher Layer Entity's point of attachment. For example, the
17 reception of a frame by Higher Layer Entity A that was transmitted on LAN A is unaffected by the ingress or

1 egress configuration of either Port. However, for Higher Layer Entities that require only a single point of
2 attachment, the ingress and egress configuration affects the forwarding path. For example, frames destined
3 for Higher Layer Entity A that are transmitted on LAN B would be subjected to the ingress rules that apply
4 to Port B and the egress rules that apply to Port A.

5 The decision about whether frames transmitted by Higher Layer Entities are VLAN-tagged or untagged
6 depends on the Higher Layer Entity concerned and the connectivity that it requires:

7 a) Spanning Tree Bridge Protocol Data Units (ST BPDUs) transmitted by the Spanning Tree Protocol
8 Entity are not forwarded by Bridges and must be visible to all other Spanning Tree Protocol Entities
9 attached to the same LAN. Such frames shall be transmitted untagged.

10 NOTE—Any BPDUs or MVRPDUs that carry a tag header are not recognized as well-formed BPDUs or
11 MVRPDUs and are not forwarded by the Bridge.

12 b) The definition of the MVRP application (11.2.3) calls for all MVRP frames to be transmitted
13 untagged for similar reasons.

14 c) The definition of the MMRP application (Clause 10) calls for all MMRP frames originating from
15 VLAN devices to be transmitted VLAN-tagged, in order for the VID in the tag to be used to identify
16 the VLAN context in which the registration applies.

17 d) It may be necessary for PDUs transmitted for Bridge Management (8.12) to be VLAN-tagged in
18 order to achieve the necessary connectivity for management in a Virtual Bridged Network. This is
19 normally achieved by routing a packet containing the PDU to the routed subnet associated with the
20 VLAN. Transmission of the packet through the router interface to that VLAN and subsequent
21 forwarding of the resulting frame by VLAN Bridges ensures that the frame is correctly
22 VLAN-tagged, as required.

23 8.13.11 CFM entities

24 CFM entities reside in shims. The entities in a CFM shim inspect every frame that passes through either of
25 the shim's two SAPs, and decides whether to pass that frame through to the other SAP, process it, or both.
26 The decision whether pass it through is not dependent on the destination_address, but the decision whether
27 to process the frame is dependent on the destination_address (see Clause 19). A CFM entity shall discard
28 any frame, directed to it by reason of its VID and MD Level, whose destination_address parameter does not
29 correspond to a MAC address recognized by that CFM entity.

30 Any given CFM entity is configured to recognize one or more individual MAC addresses, and one or more
31 group MAC addresses, in received frames, and to use one or more individual MAC address or one or more
32 group MAC addresses for transmitted frames. The individual MAC address is unique, within a bridged
33 network, to a specific Bridge, though not necessarily to a single Bridge Port (J.6). For VLAN-based and
34 backbone service instances, the group MAC addresses are selected from Table 8-18 and Table 8-19
35 according to the MD Level and OpCode fields in the CFM PDU header. For these service instances,
36 Loopback Messages (LBMs, 20.2), Loopback Replies (LBRs, 20.2), and Linktrace Replies (LTRs, 20.3) are
37 carried in unicast frames while Continuity Check Messages (CCMs, 20.1) use addresses from Table 8-18,
38 and Linktrace Messages (LTMs, 20.3) use addresses from Table 8-19. In the case of TESI, all CFM
39 messages use the individual MAC addresses or the group MAC addresses that are associated with the
40 monitored service (20.1, 20.2, 20.3).

41 NOTE—The group destination MAC addresses in Table 8-18 are there primarily to make it possible for Bridges built
42 prior to this standard to process CFM. This restriction on the choice of group destination MAC addresses to be used on a
43 CFM PDU is relaxed for ESP-VIDs. See 22.8.

Table 8-18—CCM group destination MAC addresses

01-80-C2-00-00-3y	
MD Level of CCM	Four address bits “y”
7	7
6	6
5	5
4	4
3	3
2	2
1	1
0	0

Table 8-19—LTM group destination MAC addresses

01-80-C2-00-00-3y	
MD Level of LTM	Four address bits “y”
7	F
6	E
5	D
4	C
3	B
2	A
1	9
0	8

9. Tagged frame format

This clause specifies the format of the tags added to and removed from user data frames by the tag encoding and decoding functions that support the EISS (6.8, 6.9, 6.10, 6.11) and the Backbone Service Instance Multiplex Entity (6.18). It:

- a) Reviews the purpose of tagging, and the functionality provided.
- b) Specifies generic rules for the representation of tag fields and their encoding in the octets of an MSDU.
- c) Specifies a general tag format, comprising a Tag Protocol Identifier (TPID), Tag Control Information (TCI), with additional information as signaled in the TCI.
- d) Specifies the format of the TPID for each IEEE 802 media access control method.
- e) Describes the types of tags that can be used, including the following:
 - 1) A C-TAG used at ports on a C-VLAN component
 - 2) An S-TAG used at ports on an S-VLAN component
 - 3) An I-TAG used at PIPs on an I-component and CBPs on a B-component
- f) Documents the allocation of EtherType values to identify the types of tag specified in this standard.
- g) Specifies the format of the TCI and additional information for each tag type.

Further analysis of the frame formats and the format translations that can occur when frames are tagged or untagged when relayed between different media access control methods can be found in Annex G.

9.1 Purpose of tagging

Tagging a frame with a VLAN tag:

- a) Allows a VID to be conveyed, facilitating consistent VLAN classification of the frame throughout the network and enabling segregation of frames assigned to different VIDs.
- b) Allows priority (IEEE Std 802.1AC, 6.8) to be conveyed with the frame when using IEEE 802 LAN media access control methods that provide no inherent capability to signal priority.

9.2 Representation and encoding of tag fields

In this subclause, octets are numbered starting from 1 and increasing in the order in which they are encoded in the sequence of octets that constitute an MSDU.

Where bits in consecutive octets are used to encode a binary number in a single field, the lower octet number encodes the more significant bits of the field, and the LSB of the lower octet number and the most significant bit (MSB) of the next octet both form part of the field.

Where the value of a field comprising a sequence of octets is represented as a sequence of two-digit hexadecimal values separated by hyphens (e.g., A1-5B-03), the leftmost hexadecimal value (A1 in this example) appears in the lowest numbered octet of the field and the rightmost hexadecimal value (03 in this example) appears in the highest numbered octet of the field.

The bits in an octet are numbered from 1 to 8, where 1 is the LSB.

When the terms set and reset are used in the text to indicate the values of single-bit fields, set is encoded as a binary 1 and reset as a binary 0 (zero).

- 1 When the encoding of a field or a number of fields is represented using a diagram,
- 2 a) Octets are shown with the lowest numbered octet nearest the top of the page, the octet numbering
3 increasing from the top to bottom; or
 - 4 b) Octets are shown with the lowest numbered octet nearest the left of the page, the octet numbering
5 increasing from left to right;
 - 6 c) Within an octet, bits are shown with bit 8 to the left and bit 1 to the right.

7 **9.3 Tag format**

8 Each tag comprises the following sequential information elements:

- 9 a) A Tag Protocol Identifier (TPID) (9.4)
- 10 b) Tag Control Information (TCI) that is dependent on the tag type (9.5, 9.6)
- 11 c) Additional information, if and as required by the tag type and TCI

12 The tag encoding function supports each EISS (6.8) instance by using an instance of the ISS to transmit and
13 receive frames and encodes the above information in the first and subsequent octets of the MSDU that will
14 accompany an ISS M_UNITDATA.request, immediately prior to encoding the sequence of octets that
15 constitute the corresponding EISS M_UNITDATA.request's MSDU. On reception the tag decoding function
16 is selected by the TPID and decodes the TCI and additional information octets (if present) prior to issuing an
17 EISS M_UNITDATA.indication with an MSDU that comprises the subsequent octets.

18 **9.4 TPID formats**

19 The TPID includes an EtherType value that is used to identify the frame as a tagged frame and to select the
20 correct tag decoding functions. Such an EtherType is known as the Tag EtherType. The TPID is EtherType
21 encoded, i.e., is two octets in length and contains only the assigned EtherType value.

22 **9.5 Tag Protocol identification**

23 The following types of tags are specified:

- 24 a) A C-TAG, for general use by C-VLAN Bridges (5.9) and C-VLAN components of Provider Edge
25 Bridges (5.10.1).
- 26 b) An S-TAG, reserved for use by S-VLAN Bridges (5.10), the S-VLAN component of Provider Edge
27 Bridges (5.10.1) and BEBs (5.12), and C-VLAN Bridges signaling priority to a PBN or a PBBN
28 (6.13).
- 29 c) An I-TAG, reserved for use by BEBs (5.7, 5.8, 5.12).

30 NOTE—The S-TAG is identical to the B-TAG (see 25.2).

31 A distinct EtherType has been allocated (Table 9-1) for use in the TPID field (9.4) of each tag type so they
32 can be distinguished from each other, and from other protocols.

Table 9-1—IEEE 802.1Q™ EtherType allocations

Tag Type	Name	Value
Customer VLAN Tag	IEEE 802.1Q Tag Protocol EtherType (802.1QTagType)	81-00
Service VLAN Tag or Backbone VLAN Tag	IEEE 802.1Q Service Tag EtherType (802.1QSTagType)	88-A8
Backbone Service Instance Tag	IEEE 802.1Q Backbone Service Instance Tag EtherType (802.1QITagType)	88-E7

9.6 VLAN Tag Control Information (TCI)

The VLAN TCI field (Figure 9-1) is two octets in length and encodes the vlan_identifier, drop_eligible, and priority parameters of the corresponding EISS M_UNITDATA.request as unsigned binary numbers.



Figure 9-1—VLAN TCI format

The VID is encoded in a 12-bit field. A VLAN Bridge may not support the full range of VID values but shall support the use of all VID values in the range 0 through a maximum N, less than or equal to 4094 and specified for that implementation. Table 9-2 identifies VID values that have specific meanings or uses.

Table 9-2—Reserved VID values

VID value (hexadecimal)	Meaning/Use
0	The null VID. Indicates that the tag header contains only priority information; no VID is present in the frame. This VID value shall not be configured as a PVID or a member of a VID Set, or configured in any FDB entry, or used in any Management operation.
1	The default PVID value used for classifying frames on ingress through a Bridge Port. The PVID value of a Port can be changed by management.
2	The default SR_PVID value used for SRP [35.2.1.4 item i)] Stream related traffic. The SR_PVID value of a Port can be changed by management.
FFF	Reserved for implementation use. This VID value shall not be configured as a PVID or a member of a VID Set, or transmitted in a tag header. This VID value may be used to indicate a wildcard match for the VID in management operations or FDB entries.

NOTE 1—There is a distinction between the range of VIDs that an implementation can support and the maximum number of active VIDs supported at any one time. An implementation supports only 16 active VIDs, for example, may use VIDs chosen from anywhere in the identifier space, or from a limited range. The latter can result in difficulties where different Bridges in the same network support different maximums. It is recommended that new implementations of this standard support the full range of VIDs, even if the number of active VIDs is limited.

The priority and drop_eligible parameters are conveyed in the 3-bit PCP field and the DEI field as specified in 6.9.3.

NOTE 2—Previous versions of this standard used bit 5 of octet 1 of the TCI in C-TAGs as a Canonical Format Indicator (CFI). Bridges conformant to this version of the standard will not interoperate with Bridges that set the CFI as a result of inserting C-TAGs in frames received from an IEEE 802.5™ Token Ring LAN. Bridges conformant to this version of the standard will interoperate with Bridges that forward bit 5 of octet 1 as a CFI but do not have 802.5 Token Ring interfaces.

9.7 Backbone Service Instance Tag Control Information (I-TAG TCI)

The I-TAG TCI field (Figure 9-2) is 16 octets in length and encodes the priority, drop_eligible, destination_address, and source_address parameters of the corresponding service request primitive as unsigned binary numbers. I-TAGs are encoded and decoded at PIPs and CBPs of BEBs as specified in 6.10, 6.11, and 6.18.

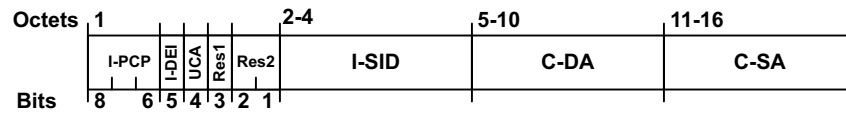


Figure 9-2—I-TAG TCI format

The I-TAG TCI contains the following fields:

- I-PCP (Priority Code Point)*—This 3-bit field encodes the priority and drop_eligible parameters of the service request primitive associated with this frame using the same encoding as specified for VLAN tags in 6.9.3. The PBBN operates on the priority associated with the B-TAG.
- I-DEI (Drop Eligible Indicator)*—This 1-bit field carries the drop_eligible parameter of the service request primitive associated with this frame. The PBBN operates on the drop eligibility associated with the B-TAG.
- UCA (Use Customer Addresses)*—A single-bit flag that, when containing a value of one, signals a Backbone Service Instance Multiplex Entity (6.18) to use the addresses contained in the C-DA and C-SA fields.
- Res1 (Reserved 1)*—This 1-bit field is reserved for future standardization. The Res1 field contains a value of zero when the tag is encoded, and is ignored when the tag is decoded.
- Res2 (Reserved 2)*—This 2-bit field is reserved for future standardization. The Res2 field contains a value of zero when the tag is encoded. The frame will be discarded if this field contains a nonzero value when the tag is decoded.
- I-SID (Backbone Service Instance Identifier)*—This 24-bit field carries the identifier of the backbone service instance.
- C-DA (Encapsulated Customer Destination MAC address)*—Contains the address in the destination_address parameter of the service request primitive associated with this frame. The address is represented using the Hexadecimal Representation as specified in IEEE Std 802 with the octet containing the I/G bit in the lowest numbered octet of the field.
- C-SA (Encapsulated Customer Source MAC address)*—Contains the address in the source_address parameter of the service request primitive associated with this frame. The address is represented using the Hexadecimal Representation as specified in IEEE Std 802 with the octet containing the I/G bit in the lowest numbered octet of the field.

¹ The I-SID is encoded in a 24-bit field. Table 9-3 identifies I-SID values that have specific meanings or uses.

Table 9-3—Reserved I-SID values

I-SID value (hexadecimal)	Meaning/Use
0	Reserved for implementation use. This I-SID value shall not be configured as an identifier for a backbone service instance or transmitted in an I-TAG header. This I-SID value may be used in management and control operations, e.g., to indicate a deleted or null I-SID in ISIS-SPB TLVs (see Clause 28).
1	Default value—Unassigned ISID on a VIP.
2 through FE	Reserved for future standardization.
FF	SPBM default I-SID.
FFFFFF	Reserved for implementation use. This I-SID value shall not be configured as an identifier for a backbone service instance or transmitted in an I-TAG header. This I-SID value may be used to indicate a wildcard match for the I-SID in management operations.

² NOTE—There is a distinction between the range of I-SIDs that an implementation can support, and the maximum
³ number of active backbone service instances supported at any one time. An implementation that supports only 2^{16} active
⁴ backbone service instances, for example, may use I-SIDs chosen from anywhere in the identifier space. Implementations
⁵ of this standard support the full range of I-SIDs, even if the number of active backbone service instances is limited.

10. Multiple Registration Protocol (MRP) and Multiple MAC Registration Protocol (MMRP)

The Multiple Registration Protocol allows participants in an MRP application to register attributes with other participants in a Bridged Network. The definition of attribute types, their values, and the semantics associated with values when registered, are specific to each MRP application.

This clause:

- a) Provides an overview of the use of MRP within a bridged network (10.1).
- b) Describes the architecture of MRP participants for end stations and Bridge Ports (10.2).
- c) Specifies registration propagation between the per-Port Participants in a Bridge (10.3).
- d) Details requirements to be met by the MRP design (10.4).
- e) Details requirements for interoperability between MRP participants (10.5).
- f) Provides an overview of protocol operation (10.6).
- g) Provides a detailed specification of the protocol (10.7).
- h) Describes the structure of PDUs exchanged between MRP participants (10.8).

Subclauses 10.9 through 10.12 define an MRP application, the Multiple MAC Registration Protocol (MMRP), that registers attributes of two types—MAC addresses and Group service requirements. Values of these attributes control MAC address filtering by MMRP participants.

Clause 11 defines a second MRP application, the Multiple VLAN Registration Protocol (MVRP), that registers VLAN membership information.

Clause 35 defines a third MRP application, the Multiple Stream Registration Protocol (MSRP), that registers data Stream characteristics and reserves Bridge resources as appropriate to provide QoS guarantees.

Clause 39 defines a fourth MRP application, the Multiple I-SID Registration Protocol (MIRP), that serves as an alternative to MVRP for per-VLAN flushing of learned MAC address information in I-components.

10.1 MRP overview

MRP allows a participant in a given MRP application to make or withdraw *declarations* of *attributes*, and for those declarations (or withdrawals) to result in the *registration* (or removal of registrations) of those attributes with the other MRP Participants for that application.

A declaration by an MRP Participant for an end station or Bridge Port is recorded by an Applicant state machine for the declared attribute and Port. Changes in the Applicant state machine's variables trigger the transmission of MRPDUs to communicate the declaration (or withdrawal).

A registration is recorded by a Registrar state machine for the attribute at each participating end station and Bridge Port that receives the MRPDU. Removal of a given attribute registration occurs only if all the other participants connected to the same LAN withdraw the declaration.

1 Attributes registered on Bridge Ports that are part of the applicable *active topology* (8.4, 10.3.1) are declared
2 on all the other Bridge Ports that are also part of that active topology. Hence, a given declaration is
3 propagated to all application participants, and registered in each Bridge on those Ports that are “nearest” to
4 the source or sources of the declaration within the active topology.

5 NOTE—Unless otherwise stated, the following description assumes operation within the Base Spanning Tree Context
6 (see 10.3.1). While registration can occur on any Bridge Port, regardless of Port State (8.4), propagation follows the
7 spanning tree active topology. All the Bridge Ports shown in Figure 10-1, Figure 10-2, and Figure 10-3 are in the
8 Forwarding Port State.

9 Figure 10-1 illustrates the result of a single end station making a declaration, and shows the Bridge Ports that
10 also make declarations to propagate the attribute. The attribute is propagated to all LANs in the Bridged
11 Network, but the directional nature of the propagation results in registration only on Bridge Ports that
12 receive a declaration of that attribute from the attached LAN.

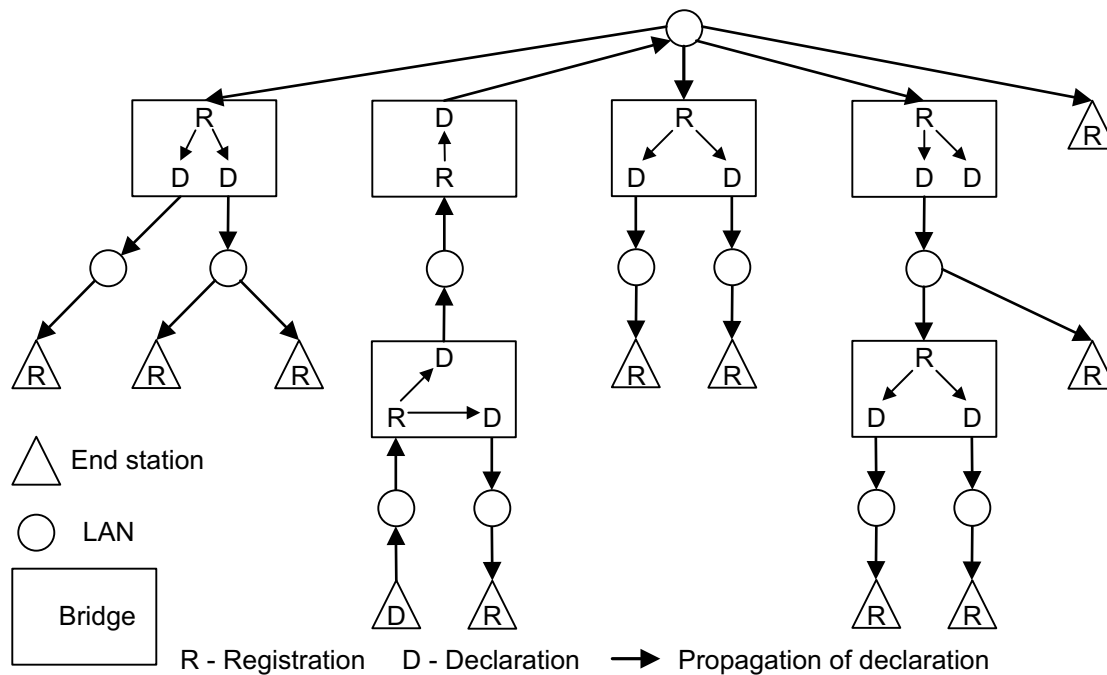


Figure 10-1—Example—Attribute value propagation from one station

13 Figure 10-2 illustrates the result of different end stations declaring the same attribute on different LANs. All
14 end stations register the attribute, and some Bridges register it on more than one Port.

15 The set of Bridge Ports and end stations that both declare and register a given attribute defines the subset of
16 the active topology that connects all the participants interested in that attribute. A registration can be
17 regarded as a pointer to participants that have declared that attribute, as illustrated in Figure 10-3 (using the
18 same set of declarations and registrations that were illustrated in Figure 10-2).

19 Situations where it is desirable to form “reachability trees” are generally good candidates for the use of
20 MRP. For example, if the attribute in Figure 10-3 is a group MAC address that is used as a destination
21 address for a video data stream, and it is deemed desirable for that video data to be sent only to the subset of
22 the active topology that contains end stations that have declared that attribute, then an end station that is the
23 source of that video stream could use the presence or absence of a registration as an indication of whether to
24 send the data on the LAN to which it is attached. Any Bridge receiving the data could determine on which
25 Ports the data should be forwarded.

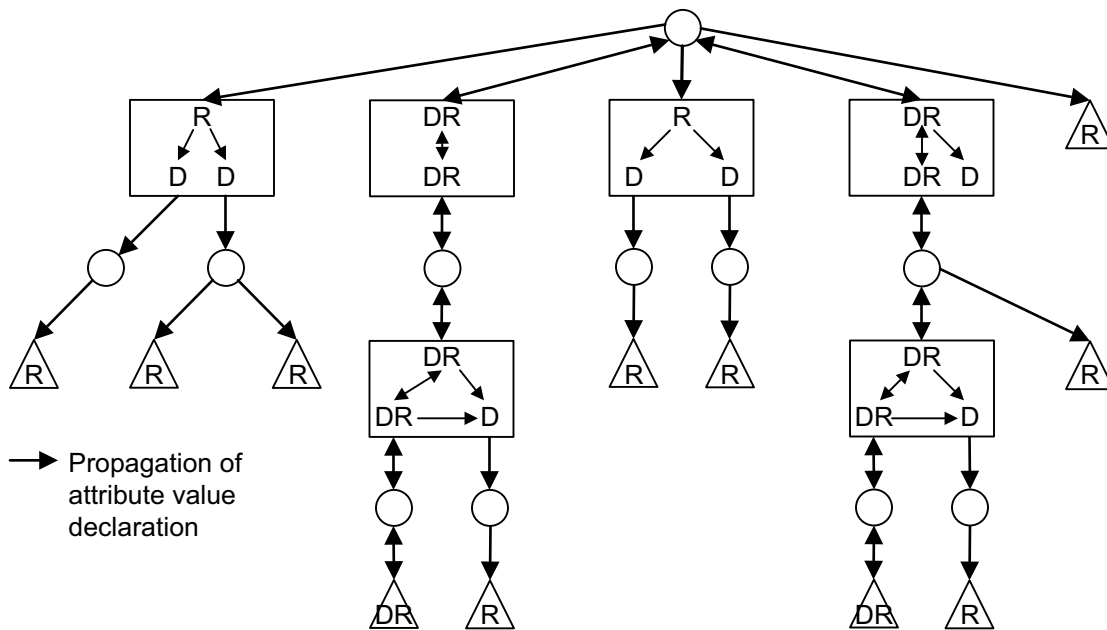


Figure 10-2—Example—Attribute value propagation from two stations

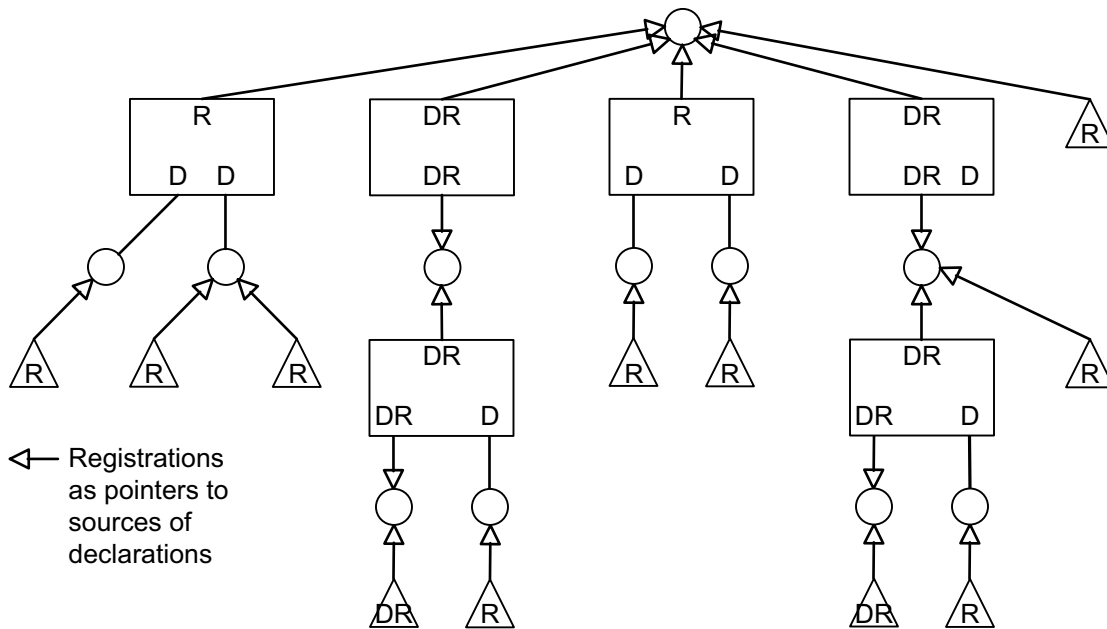


Figure 10-3—Example—Registrations as pointers to the sources of declarations

- 1 VLAN Bridges that do not support MRP are transparent to MRP exchanges, and forward received MRPDUs
- 2 on all Ports in the active topology. Similarly, VLAN Bridges that do not implement a given MRP application
- 3 are transparent to MRP exchanges destined for that application, and forward any such received MRPDUs on
- 4 all Ports that are participating in the active topology for the VLAN concerned.

1 MRP operates only on Ports that are MAC_Operational (IEEE Std 802.1AC). If the Port is operating as a
2 network access port (IEEE Std 802.1X), MRP uses the controlled port (8.13.9). On any Port whose
3 MAC_Operational parameter is FALSE, any MRP entity shall not transmit MRPDUs, and shall discard,
4 without processing, any received MRPDUs.

5 MRP provides a means to mark initial attribute declarations and propagated attribute declarations as “new,”
6 signaling to the recipient of the declaration that the attribute value is being newly declared, or is being
7 redeclared following a change in the underlying topology. The rules applied to the marking and propagation
8 of newly declared values in this way are common to all MRP Applications; however, the action taken on
9 receipt of an attribute declaration marked as “new” is specific to each MRP Application. For example,
10 MMRP (10.9) makes no use of the “new” marking, whereas MVRP (Clause 11) uses the “new” marking to
11 permit FDB entries to be flushed on a per-VID basis following a topology change.

12 10.2 MRP architecture

13 An MRP Participant consists of an application component and an MRP Attribute Declaration (MAD)
14 component. The application component is responsible for the semantics associated with attribute values and
15 their registration, including the use of explicitly signaled new declarations, and uses the following two
16 primitives to request MAD to make or withdraw attribute declarations:

17 MAD_Join.request (attribute_type, attribute_value, new)

18 MAD_Leave.request (attribute_type, attribute_value)

19 where *attribute_type* specifies the type of the attribute declaration, and *attribute_value* specifies the instance
20 of that type, and the Boolean *new* parameter indicates an explicit new declaration.

21 Within a Bridge, a per application MRP Attribute Propagation (MAP) component (10.3) propagates
22 information between the per-Port Participants, using the same request and indication primitives. If the value
23 of *tcDetected* (13.25) for the Port and MAP Context (10.3.1) associated with the MRP Participant is
24 nonzero, then the value of the *new* parameter in the propagated MAD_Join.request is set TRUE.

25 The MAD component executes MRP (10.6, 10.7), generating MRP messages for transmission and
26 processing messages received from other Participants, and uses the following two primitives to notify its
27 application component of a change in Attribute registration:

28 MAD_Join.indication (attribute_type, attribute_value, new)

29 MAD_Leave.indication (attribute_type, attribute_value)

30 One such Participant, per MRP application, exists for each point of attachment to a LAN where Attributes
31 for that application are to be declared or registered, i.e., one Participant per application in an end station, and
32 one per application per Port in a Bridge. The encoding of Attribute values in MRPDUs and their subsequent
33 decoding is application specific, within the general structure specified in 10.8, and each MRPDU conveys
34 messages generated by the MAD component for a single application.

35 Figure 10-4 illustrates the components of MRP Participants in a two-Port Bridge and an end station.

36 For each MRP application, the following are defined:

- 37 a) A set of Attribute types used by the application.
- 38 b) The Attribute values permitted for each Attribute type.
- 39 c) The semantics associated with each Attribute type and value.
- 40 d) The use made of MAP Contexts by the application.
- 41 e) The group MAC address and EtherType for protocol exchanges between application Participants.

- f) The structure and encoding of the Attribute types and values in MRPDUs.
 - g) The requirements for MRP state machine support in end stations and Bridges.
 - h) The circumstances, if any, in which the application makes use of the “new” declaration capability.
- NOTE 1—Not all applications of MRP will make use of the “new” declaration capability.
- i) If the application makes use of the “new” declaration capability, the semantics that new registrations carry in that application.
 - j) The number of attribute values out of the possible range of values for which the application is expected to be capable of maintaining current state information.
- NOTE 2—In some applications, such as MVRP, the failure of one Bridge in a network to maintain state for all possible attribute values would have major consequences for the integrity of the network. For such applications, the application definition will mandate the ability to track the state of all possible attribute values.

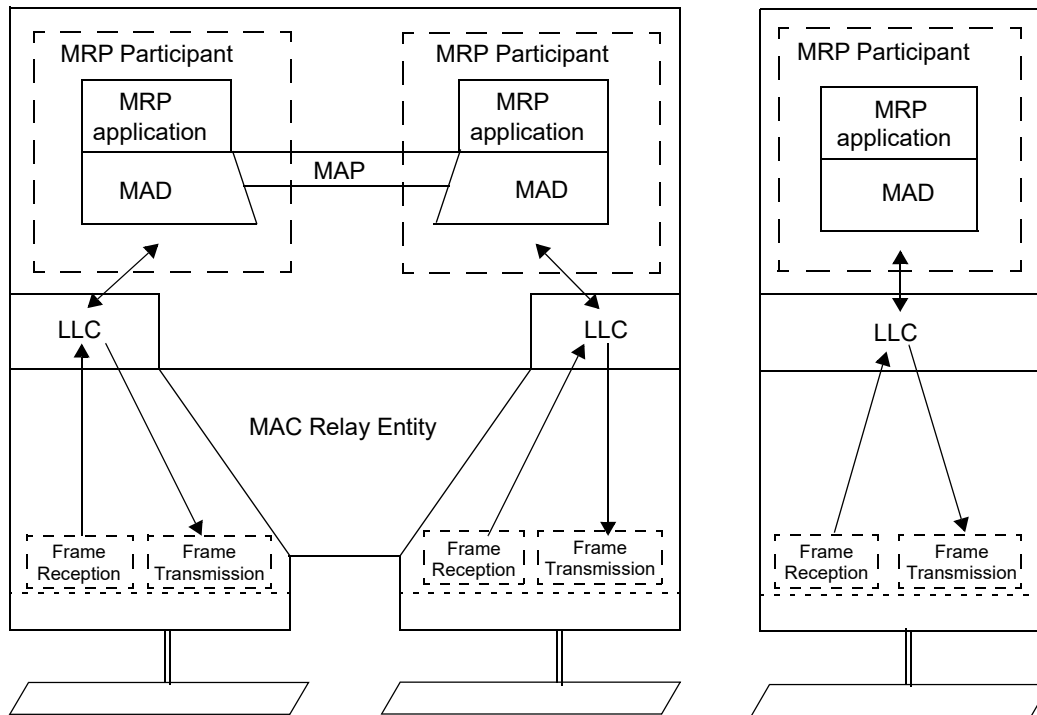


Figure 10-4—MRP architecture

10.3 MRP Attribute Propagation (MAP)

The MAP function enables propagation of attributes registered on Bridge Ports across the network to other participants. Each MRP application specifies the operation of the MAP function. This subclause specifies the operation of the MAP function for the MMRP application, the MVRP application (11.2.1), and the MSRP application (35.2). In addition, 35.2.4 specifies additional MSRP attribute processing rules that modify the MAP function defined below.

For a given MRP application and MAP Context (10.3.1), and for the set of Ports that have a Port State (8.4) of Forwarding for that MAP Context, and for the set of attributes associated with that MAP Context:

- a) Any MAD_Join.indication, or any MAD_Join.request issued by the MRP application, received by MAP from a given Port in the set is propagated as a MAD_Join.request to the instance(s) of MAD associated with each other Port in the set. If the value of tcDetected (13.25) for the given Port and MAP Context is nonzero, then the value of the *new* parameter in the propagated MAD_Join.request

1 is set TRUE, regardless of the value of this parameter in the indication or request that is being
2 propagated.

3 b) Any MAD_Leave.indication, or any MAD_Leave.request issued by the MRP application, received
4 by MAP from a given Port in the set is propagated as a MAD_Leave.request to the instance(s) of
5 MAD associated with each other Port in the set (Port P, say) if and only if no registration now exists
6 for that Attribute on any other Port in the set excluding P.

7 These rules propagate attribute registrations through any given Port if any other Port has seen a registration
8 for that attribute, and propagate deregistrations if all other Ports are now deregistered.

9 As the set of Ports with a Port State of Forwarding for a given MAP Context can change dynamically, for
10 example as a result of spanning tree reconfiguration, MAP operates as follows after such a change:

11 c) If a Port is added to the set, and that Port has registered an attribute (i.e., a MAD_Join.indication or
12 MAD_Join.request has occurred more recently than any MAD_Leave.indication or
13 MAD_Leave.request for the attribute), then MAD_Join.requests are propagated to the MAD
14 instances for each of the other Ports in the set.

15 d) If a Port is added to the set, but that Port has not declared an attribute that other Ports in the set have
16 registered, then MAD_Join.requests are propagated by those other Ports to the MAD instance for
17 that Port.

18 e) If a Port is removed from the set, and that Port has registered an attribute and no other Port has, then
19 MAD_Leave.requests are propagated to the MAD instances for each of the other Ports in the set.

20 NOTE—If a Port is removed from the set, and that Port has declared one or more attributes, then this Port
21 transmits a Leave message (see 10.6) for every attribute that it has declared.

22 f) If a Port is removed from the set, and that Port has registered an attribute that another Port has also
23 registered, then a MAD_Leave.request is propagated to the MAD instance for that other Port.

24 10.3.1 MAP Context

25 For a given Port of an MRP-aware Bridge and MRP application supported by that Bridge, an instance of an
26 MRP Participant can exist for each *MRP Attribute Propagation Context (MAP Context)* understood by the
27 Bridge. A MAP Context identifies the set of Bridge Ports that form the applicable *active topology* (8.4).

28 Examples of MAP Contexts are as follows:

29 a) The active topology formed by the operation of RSTP (Clause 13). This MAP Context provides the
30 same connectivity as the spanning tree in each Bridge and is known as the *Base Spanning Tree*
31 *Context*.

32 b) The active topology formed by the subset of Bridge Ports, and the underlying spanning tree, that
33 support a given VID. This MAP Context is known as a *VLAN Context*.

34 NOTE—This standard uses the Base Spanning Tree Context and VLAN Contexts to define the operation of MMRP and
35 MVRP; however, other MAP Contexts may be used for other applications, or for extending MMRP functionality.

36 MRP exchanges can occur on all of the Ports of a Bridge; however propagation across a Bridged Network of
37 attribute registrations for a given application uses only those Bridge Ports that are part of the active topology
38 identified by the MAP Context. Each MRP application specification identifies the contexts it can operate
39 within and assigns MAP Context identifiers for use in conjunction with the operation of MRP and its
40 administrative controls (10.7.2, 10.7.3).

41 A MAP Context identifier of 0 always identifies the *Base Spanning Tree Context*. The MRP application
42 specifies how the MAP Context of each MRPD is identified if any other context is used.

10.3.1.1 MAD and Port role changes

A spanning tree provides a loop-free active topology connecting all the Bridges and all the LANs in a network. Hence, any Bridge Port that forwards frames connects two otherwise disconnected parts of the topology. Each Alternate Port on a Bridge A (say) connects to a LAN that has a Port of another Bridge with a better priority vector than A can supply; hence, each of A's Alternate Ports is a potential connection to part of the network that is not connected to any of A's Designated Ports, all LANs and Bridges in any such part having an equal or worse priority vector than A. Therefore, an Alternate Port on A potentially connects to the part of the network connected to by A's Root Port. Since any part of the network is fully connected, in the absence of registration controls, A's Alternate Port will have registered the same attributes as its Root Port.

For implementations running over RSTP or MSTP, this gives rise to the risk of information loops when Port roles change; because of the store and forward nature of attribute propagation and the potentially rapid transitions of Port roles (compared to the relatively slow transitions that occurred with STP), these can arise even when there are no data loops.

To prevent such information loops from occurring, the information held by MAD's Registrars for a Port (i.e., information registered on a Port as a result of protocol activity on the LAN to which that Port is connected) is discarded whenever the Port transitions from an Alternate port or Root Port role to become a Designated Port. No such discard is needed for changes in the other direction, i.e., changes from Designated Port to Root Port or Alternate Port.

10.4 Requirements to be met by MRP

MRP establishes, maintains, withdraws, and disseminates attribute declarations and registrations among the MRP Participants attached to a Bridged Network. The protocol meets the following requirements for Applicant and Registrar behavior, error recovery, performance, scalability, compatibility with non-MRP-aware devices, and the load imposed on Bridges, end stations, and the network:

- a) Participants can issue declarations for MRP application attributes (10.2, 10.3, 10.7.3, and 10.7.7).
- b) Participants can withdraw declarations for attributes (10.2, 10.3, 10.7.3, and 10.7.7).
- c) Each Bridge propagates declarations to MRP Participants (10.3).
- d) MRP Participants can track the current state of declaration and registration of attributes on each Port of the participant device (10.7.7 and 10.7.8).
- e) MRP Participants can remove state information relating to attributes that are no longer active within port or all of the network, e.g., as a result of the failure of a participant (10.7.8 and 10.7.9).
- f) The latency involved in issuing, propagating, or revoking attribute declarations, is small (i.e., comparable to the frame propagation delay) and increases linearly as a function of the diameter of the network (10.7.7, 10.7.8, and 10.7.9).
- g) MRP is resilient in the face of the failure of MRP Participants.
- h) MRP is resilient in the face of single packet loss.
- i) MRP will operate correctly in networks where:
 - 1) All Bridges support both Basic and Extended Filtering Services; or where
 - 2) Some Bridges support only Basic Filtering Services and some both Basic and Extended Filtering Services (10.5, 10.7.7, 10.7.8, and 10.7.9).
- j) The communications bandwidth consumed on any particular LAN by Applicants and Registrars in exchanging MRPDUs will be a small percentage of the total available bandwidth, and independent of the total traffic supported by the network. The bandwidth consumed will be a function of the number of attributes registered.

10.5 Requirements for interoperability between MRP Participants

To ensure the interoperability of MRP, the following are required:

- a) All MRP applications use a group MAC address as the destination address of MRPDUs, selected in accordance with the stated requirements of the MRP application concerned for the type of Bridge component in which the application is implemented. The addresses used may be taken from the set of addresses specified in Table 10-1, or taken from the set of reserved addresses specified in Table 8-1, Table 8-2, and Table 8-3, or other group MAC addresses, chosen according to the particular properties required by the application concerned.

NOTE 1—The addresses in Table 8-1, Table 8-2, Table 8-3, and Table 10-1 differ from other group MAC addresses in terms of the scope of transmission within a network, as a consequence of the forwarding/filtering decisions that are taken relative to them by different types of Bridge component.
- b) Table 10-1 specifies a set of group MAC addresses, some that have been assigned for use by existing applications defined in this standard, and others reserved for future standardization. Addresses in this set have the property that:
 - 1) Where a given address in the set is used by a Bridge component to support an MRP application, frames destined for that address shall not be forwarded by that Bridge component; i.e., a Static Filtering Entry for that address is maintained for that group MAC address in order to prevent the forwarding of frames destined for that address.
 - 2) Where a given address in the set is not used by a Bridge component to support any MRP application, frames destined for that address received on any Port that is part of a given active topology shall be forwarded by that Bridge component on all other Ports that are part of that active topology.

Table 10-1—MRP application addresses

Assignment ^a	Value
Customer and Provider Bridge MMRP address	01-80-C2-00-00-20
Customer Bridge MVRP address	01-80-C2-00-00-21
Reserved	01-80-C2-00-00-22
Reserved	01-80-C2-00-00-23
Reserved	01-80-C2-00-00-24
Reserved	01-80-C2-00-00-25
Reserved	01-80-C2-00-00-26
Reserved	01-80-C2-00-00-27
Reserved	01-80-C2-00-00-28
Reserved	01-80-C2-00-00-29
Reserved	01-80-C2-00-00-2A
Reserved	01-80-C2-00-00-2B
Reserved	01-80-C2-00-00-2C
Reserved	01-80-C2-00-00-2D
All Provider Bridge Intermediate Systems	01-80-C2-00-00-2E
All Customer Bridge Intermediate Systems	01-80-C2-00-00-2F

^a MIRP is an MRP application, but is not assigned an address from this table (39.2.1.5, 39.2.1.6).

- c) The transmission and reception of MRPDUs between MRP Participants, formatted as defined for the application using the generic PDU format defined in 10.8, shall use LLC procedures. Each MRP application uses a unique EtherType value in order to identify the application protocol. Table 10-2 specifies the EtherType values assigned to existing applications.

Table 10-2—MRP EtherType values

Assignment	Value
MMRP EtherType	88-F6
MVRP EtherType	88-F5
MSRP EtherType	22-EA
MIRP EtherType	89-29

NOTE 2—For the purposes of this standard, the expression “use LLC procedures” includes making use of the service provided by link layer protocol entities that support protocol discrimination by means of an EtherType value.

- d) MRPDUs, i.e., frames with the destination MAC addresses selected as specified in item a) and the EtherType values specified in item c), that are destined for applications supported by a Bridge component, and that are not well formed (i.e., are not structured and encoded as defined in 10.8 and with attribute types and values encoded as defined by the MRP application), shall be discarded on receipt.

10.6 Protocol operation

This subclause provides an informal introduction. The definitive specification of MRP is contained in 10.7 and 10.8.

MRP is a simple, fully distributed, many-to-many protocol, that supports efficient, reliable, and rapid declaration and registration of attributes by multiple participants on shared and virtual shared media. MRP also incorporates optimizations to speed attribute declarations and withdrawals on point-to-point media. Correctness of MRP operation is independent of the relative values of protocol timers, and the protocol design is based primarily on the exchange of idempotent protocol state rather than commands.

A full MRP Participant maintains Registrar and Applicant state machines for each Attribute of interest, and a LeaveAll state machine and PeriodicTransmission state machine for the participant as a whole.

The job of the Registrar is to record declarations of the attribute made by other Participants on the LAN. It does not send any protocol messages, as the Applicant looks after the interests of all would-be Participants.

The job of the Applicant is twofold:

- a) To ensure that this Participant’s declaration is correctly registered by other Participants’ Registrars.
- b) To prompt other Participants to reregister after one withdraws a declaration.

NOTE 1—Applicant-Only implementations are concerned only with item a).

The basic design of MRP is oriented to LAN environments, where there is generally a low probability of frame loss, and ensures that timely registration of attributes is unaffected unless at least two out of a set of immediately related frames are lost. The LeaveAll state machine periodically ensures that Participants reregister attributes, thus guarding against an extended failure to register or deregister. In potentially lossy

1 environments such as service instances supported by PBNs, where frame losses are temporally correlated,
2 the PeriodicTransmission machine ensures successful registration without immediately contributing to
3 congestive loss.

4 If one Applicant has both declared and registered an Attribute, other Applicants do not need to reiterate the
5 declaration. MRP messages convey both Applicant and Registrar state to allow suppression of such repeated
6 declarations. The Applicant state machine distinguishes between Active Participants (that have sent a
7 message or messages to make a declaration), Passive Participants (that require registration, but have not had
8 to declare the attribute so far to register it, and will not have to explicitly deregister), and Observers (that do
9 not require registration at present, but track the attribute's registration in case they do and become Passive
10 Participants). The following four distinct messages communicate the transmitting participant's state for an
11 Attribute:

- 12 — Empty—Not declared, and not registered.
- 13 — In—Not declared, but registered.
- 14 — JoinEmpty—Declared, but not registered.
- 15 — JoinIn—Declared and registered.

16 One message communicates a withdrawal of a prior declaration, so that Registrars do not have to track
17 individual Applicants:

- 18 — Leave—Previously registered, but now withdrawn.

19 A further message conveys a declaration from a new Participant, or a Participant newly added to the active
20 topology reached through a Bridge Port. This information is propagated by the MAP component, and so
21 includes additional registrations elsewhere in the bridged network:

- 22 — New—Newly declared, and possibly not previously registered.

23 The LeaveAll state machine uses the following single message that applies to all Attributes.

- 24 — LeaveAll—All registrations will shortly be deregistered; Participants need to reregister.

25 The Registrar for each Attribute actually implements the following three states for the Attribute:

- 26 — IN—Registered.
- 27 — LV—Previously registered, but now being timed out.
- 28 — MT—Not registered.

29 A single timer, the leavetimer, is associated with each Registrar and operates in the LV state. In that state,
30 MRP messages for the Attribute report it as unregistered (using Empty or Join Empty) but the MAD Leave
31 indication is delayed until the leavetimer expires and the state transitions to MT.

32 NOTE 2—The accuracy required for the leavetimer is sufficiently coarse to permit the use of a single operating system
33 timer per Participant with 2 bits of state for each Registrar.

34 The Applicant for each Attribute implements states that record whether it wishes to make a new declaration,
35 to maintain or withdraw an existing declaration, or has no declaration to make. It also records whether it has
36 actively made a declaration, or has been passive, taking advantage of or simply observing the declarations of
37 others. It counts the New, JoinIn, and JoinEmpty messages it has sent, and JoinIn messages sent by others, to
38 ensure that at least two such messages have been sent since it last received a LeaveAll or Leave message,
39 and at least one since it last received a JoinEmpty or Empty message. This ensures that each of the other
40 Participant's Registrars for the Attribute either have received (assuming no packet loss) two Join or New
41 messages or have reported the Attribute as registered. The Applicant state machine (Table 10-3) uses the
42 following states:

- 43 — VO—Very anxious Observer. The applicant is not declaring the attribute, and has not received a
44 JoinIn message since the state machine was initialized, or since last receiving a Leave or LeaveAll.

- 1 — VP—Very anxious Passive. The applicant is declaring the attribute, but has neither sent a Join nor
2 received a JoinIn since the state machine was initialized, or since last receiving a LeaveAll or Leave.
- 3 — VN—Very anxious New. The applicant is declaring the attribute, but has not sent a message since
4 receiving a MAD Join request for a new declaration.
- 5 — AN—Anxious New. The applicant is declaring the attribute, and has sent a single New message
6 since receiving the MAD Join request for the new declaration.
- 7 — AA—Anxious Active. The applicant is declaring the attribute, and has sent a Join message, since the
8 last Leave or LeaveAll, but either has not received another JoinIn or In, or has received a subsequent
9 message specifying an Empty registrar state.
- 10 — QA—Quiet Active. The applicant is declaring the attribute and has sent at least one of the required
11 Join or New messages since the last Leave or LeaveAll, has seen or sent the other, and has received
12 no subsequent messages specifying an Empty registrar state.
- 13 — LA—Leaving Active. The applicant has sent a Join or New message since last receipt of a Leave or
14 LeaveAll, but has subsequently received a MAD Leave request and has not yet sent a Leave
15 message.
- 16 — AO—Anxious Observer. The applicant is not declaring the attribute, but has received a JoinIn since
17 last receiving a Leave or LeaveAll.
- 18 — QO—Quiet Observer. The applicant is not declaring the attribute, but has received two JoinIns since
19 last receiving a Leave or LeaveAll, and at least one since last receiving a message specifying an
20 Empty registrar state.
- 21 — AP—Anxious Passive. The applicant is declaring the attribute, and has not sent a Join or a New
22 since last receiving a Leave or a LeaveAll but has received messages as for the Anxious Observer
23 state.
- 24 — QP—Quiet Passive. The applicant is declaring the attribute, and has not sent a Join or a New since
25 last receiving a Leave or a LeaveAll but has received messages as for the Quiet Observer state.
- 26 — LO—Leaving Observer. The applicant is not declaring the attribute, and has received a Leave or
27 LeaveAll message.

28 If an Applicant receives a Leave or LeaveAll message it will send a JoinEmpty or Empty message, and that
29 message will prompt any other Applicants that are declaring the attribute to send a JoinIn. An Applicant that
30 sends a LeaveAll message will also ensure that an Empty message is sent to prompt other Applicants to
31 repeat their declaration. Thus, if any Participant's Registrar deregisters an Attribute, at least two messages
32 will be lost before another Participant's Applicant fails to make a required redeclaration.

33 To facilitate and encourage the transmission of timely protocol information and the encoding of messages
34 for multiple Attributes within the same MRPDU, rather than the use and subsequent queuing of multiple
35 PDUs prior to transmission, PDU transmission is specified in terms of requests for a transmission
36 opportunity, and the Applicant and LeaveAll state machines specify the necessary addition of messages to an
37 MRPDU when that opportunity occurs. Whenever a state machine transitions to a state that requires
38 transmission of a message, a transmit opportunity is requested if one is not already pending. The message
39 actually transmitted (if any) is that appropriate to the state of the machine when the opportunity is presented.

40 NOTE 3—Specifying transmit opportunity requests and their subsequent use is also intended to aid correct
41 implementation in systems where a transmission is not possible immediately, e.g., shortly after whole or part of the
42 system is initialized when other claims on system resources take precedence or interfaces are not yet available.

43 To support the efficient encoding of many messages in a single MRPDU, the number of message types has
44 been kept to the minimum consistent with protocol goals and requirements, and the state machines specify
45 both whether it is necessary to send a message and a message type that can always be sent—thus removing
46 the need for a further “no message” encoding. The MRPDU structure and compact encoding (10.8) allows
47 all potential attributes for certain MRP applications, e.g., MVRP (Clause 11), to be encoded in a single
48 IEEE 802.3 frame and this further promotes protocol efficiency through ready detection of missing
49 registrations. Protocol efficiency and correct registration are further supported by encoding a LeaveAll

1 message (if required) at the beginning of the MRPDU, and by using an Applicant state machine that
2 minimizes state changes for those Attributes whose redeclaration can be encoded in the same PDU—if the
3 LeaveAll is received by another Participant so also will be the redeclaration, and sequential processing of
4 the encoded messages by the recipient will ensure that the registration is uninterrupted. Applicant states for
5 Attributes that cannot be redeclared in the same PDU allow for receipt of the LeaveAll but loss of the
6 redeclaration.

7 LeaveAll messages are transmitted to ensure that any failure to withdraw a declaration does not result in an
8 unwanted permanent registration—perhaps the system or process responsible for the registration has been
9 removed from the LAN or has ceased to operate. Attributes registered by the Participant transmitting the
10 LeaveAll as well as those receiving it are therefore timed out. The LeaveAll state machine (10.7.9) for the
11 Participant operates a single timer, the leaveAllTimer, that causes a transmit opportunity to be requested
12 when it expires, and transmission of a LeaveAll at the next opportunity. Reception of a LeaveAll message
13 from another Participant causes the timer to be restarted without generating a message, thus suppressing
14 multiple LeaveAll messages from Participants connected to the same LAN.

15 NOTE 4—In the face of changing inputs, arbitrary loss, delay, reordering, and unsignaled interruption in participation,
16 no protocol will meet its objectives. What can be said is that the objectives will be met after a known period of operation
17 within specified limits, and what failures are most likely otherwise. MRP favors ensuring that registrations are made, at
18 the expense of tolerating prolonged registrations. Thus, the LeaveAll mechanism, when it has an effect, most often
19 implements “garbage collection.” At the same time it will also ensure that failed registrations are (re-)established. These
20 objectives cannot be ensured if the LeaveAllTime is set to zero; therefore, it is recommended to set the LeaveAllTime to
21 a nonzero value as soon as is practicable.

22 When MRP Participants are connected by a shared or virtual shared medium, protocol performance is
23 improved and the effect on other protocols using the LAN minimized if the Participants transmit at different
24 times, avoiding a multicast storm and allowing some to optimize their transmissions based on messages
25 received. A request for a transmit opportunity starts a randomized join timer, with a maximum value chosen
26 to ensure successful reregistration(s) within a Leave time period (10.7.4), and the transmit opportunity is
27 offered when the timer expires. If more messages are to be sent than can fit in a single PDU, a further
28 transmit opportunity is requested.

29 When two MRP Participants are connected by a point-to-point medium or service instance delaying
30 MRPDU transmission provides no benefit. In bridged networks it is desirable to transmit without delay,
31 minimizing the denial of service that might occur while registration changes propagate after reconfiguration,
32 and maximizing the benefit from using protocols such as RSTP and MSTP. When operPointToPointMAC
33 (IEEE Std 802.1AC) is TRUE, transmit opportunities are scheduled immediately on request, subject to rate
34 limiting (10.7.4).

35 Use of point-to-point service connecting at most two Participants allows further protocol optimization, e.g.,
36 receipt of an In message does acknowledge registration. However, not all LAN media support reliable
37 determination of point-to-point status, particularly if nonstandard bridge-like devices are present. When
38 operating in point-to-point mode, MRP avoids behavior that could cause failure, potentially continuous
39 rapid exchanges of messages, or flapping registrations if there are more than two Participants. This standard
40 permits simple point-to-point subset implementations of MRP, but these will successfully operate on shared
41 media—albeit at reduced efficiency.

42 NOTE 5—IEEE Std 802.1AE (MAC Security) can ensure that there are at most two communicating Participants.

43 In certain contexts, often encountered in PBNs or PBBNs, it is necessary to control the VLAN entries in the
44 FDB using Static VLAN Registration Entries, and retain only the ability to signal the need to flush learned
45 MAC Address Entries via MRP. A New-Only Participant supports this need by implementing only a part of
46 the Applicant and Registrar state machines, but not implementing the LeaveAll state machine, and by
47 transmitting and receiving only New messages. It is also possible to simplify an MRP Participant that only
48 wishes to make declarations; for example, for an end station that uses MMRP (10.9) to declare a need to
49 receive group addressed frames, but is not a source of such frames, and therefore does not need to support
50 source pruning by registering declarations from other Participants. Such an Applicant-Only Participant does

1 not implement the Registrar or LeaveAll state machines, never sends LeaveAll, Empty, or JoinEmpty
2 messages (which would elicit unnecessary Joins from its peer Participants), and does not implement the
3 administrative controls defined in 10.7.2 and 10.7.3. The following five types of MRP implementation
4 conform to this standard:

- 5 — Full Participant
- 6 — Full Participant, point-to-point subset
- 7 — New-Only Participant
- 8 — Applicant-Only Participant
- 9 — Applicant-Only point-to-point subset, also referred to as the Simple-Applicant Participant

10 While Simple-Applicant and Full Participant point-to-point subset implementations can operate on shared
11 media, their initial Join and Leave messages are not suppressed. Significant additional, and unnecessary,
12 traffic can result from attaching several such implementations to the same shared medium. Devices that do
13 not perform registration should use an Applicant-Only Participant rather than a Simple-Applicant
14 Participant.

15 NOTE 6—At the time of the development of this standard the LAN MACs most commonly used were point-to-point but
16 the use of virtual shared media was increasing.

17 10.7 Protocol specification

18 The operation of MRP as executed by the MRP Attribute Declaration (MAD, 10.2) component of an MRP
19 Participant is represented by the following state machines:

- 20 a) A per-Attribute Applicant state machine (10.7.7)
- 21 b) A per-Attribute Registrar state machine (10.7.8)
- 22 c) A LeaveAll state machine for the Participant as a whole (10.7.9)
- 23 d) A PeriodicTransmission state machine for the Participant as a whole (10.7.10)

24 These state machines are specified as compact state tables, and make use of the following:

- 25 e) Notational conventions and abbreviations for protocol events, actions, and timer operations (10.7.1)
- 26 f) Registrar Administrative Controls (10.7.2)
- 27 g) Applicant Administrative Controls (10.7.3)
- 28 h) Protocol timers (10.7.4)
- 29 i) Protocol event definitions (10.7.5)
- 30 j) Protocol action definitions (10.7.6)

31 A Full Participant implements the complete Applicant state machine (Table 10-3) and the Registrar state
32 machine (Table 10-4) for each Attribute declared, registered, or tracked, together with a single instance of
33 the LeaveAll state machine (Table 10-5) and the PeriodicTransmission state machine (Table 10-6).

34 The point-to-point subset of the Full Participant implements the same state machines, but omits certain
35 Applicant state machine states and actions as specified by Table 10-3.

36 A New-Only Participant implements the Applicant state machine, with the omission of certain states and
37 actions, as specified in Table 10-3, and the Registrar state machine (Table 10-4), for each Attribute declared,
38 registered, or tracked, but does not implement the LeaveAll state machine (Table 10-5) or the
39 PeriodicTransmission state machine (Table 10-6). Applications that permit both Full Participants and New-
40 Only Participants define managed objects (12.9) to make the selection on a per-Port basis.

1 An Applicant-Only Participant implements the Applicant state machine, with the omission of certain states
2 and actions as specified by Table 10-3, for each Attribute declared, registered, or tracked, together with a
3 single instance of the PeriodicTransmission state machine (Table 10-6).

4 The point-to-point subset of the Applicant-Only Participant (the Simple-Applicant Participant) implements
5 the same state machines, but omits certain Applicant state machine states and actions as specified by
6 Table 10-3.

7 NOTE—Conceptually, per-Attribute state is maintained for all possible values of all Attribute types that are defined for
8 a given application; however, in real implementations of MRP, it is likely that the range of possible Attribute values in
9 some applications will preclude this, and the implementation will limit the state to those Attribute values in which the
10 Participant has an immediate interest, either as a Member or as a likely future Member.

11 Timer values, their relationships, and default values are described in 10.7.11 and Table 10-7, protocol
12 management statistics accumulated by each MAD component in 10.7.12, and interoperability considerations
13 for potentially misordering networks in 10.7.13.

14 The encoding of MRP messages in MRPUs is specified in 10.8, which also specifies the parsing and checks
15 applied to received PDUs.

16 10.7.1 Notational conventions and abbreviations

17 The following conventions are used in the abbreviations used in this subclause:

18 rXXX	receive PDU XXX
19 sXXX	send PDU XXX
20 txXXX	transmit opportunity
21 XXX!	state machine event
22 !XXX	“Not XXX”; i.e., logical NOT applied to the condition XXX

23 The following abbreviations are used in the state machine descriptions. For their meaning, see 10.7.5 and
24 10.7.6.

25 Protocol events:

26 Begin!	Initialize state machine (10.7.5.1)
27 New!	A new declaration (10.7.5.4)
28 Join!	Declaration without signaling new registration (10.7.5.5)
29 Lv!	Withdraw a declaration (10.7.5.6)
30 tx!	Transmission opportunity without a LeaveAll (10.7.5.7)
31 txLA!	Transmission opportunity with a LeaveAll (10.7.5.8)
32 txLAF!	Transmission opportunity with a LeaveAll, and with no room (Full) (10.7.5.9)
33 rNew!	receive New message (10.7.5.14)
34 rJoinIn!	receive JoinIn message (10.7.5.15)
35 rIn!	receive In message (10.7.5.18)
36 rJoinMt!	receive JoinEmpty message (10.7.5.16)
37 rMt!	receive Empty message (10.7.5.19)
38 rLv!	receive Leave message (10.7.5.17)
39 rLA!	receive a LeaveAll message (10.7.5.20)
40 Flush!	Port role changes from Root Port or Alternate Port to Designated Port (10.7.5.2)
41 Re-Declare!	Port role changes from Designated to Root Port or Alternate Port (10.7.5.3)
42 periodic!	A periodic transmission event occurs (10.7.5.10)
43 leavetimer!	leavetimer has expired (10.7.5.21)
44 leavealltimer!	leavealltimer has expired (10.7.5.22)
45 periodictimer!	periodictimer has expired (10.7.5.23)

1 Protocol actions:

2 New	send a New indication to MAP and the MRP application (10.7.6.12)
3 Join	send a Join indication to MAP and the MRP application (10.7.6.13)
4 Lv	send a Lv indication to MAP and the MRP application (10.7.6.14)
5 sN	send a New message (10.7.6.2)
6 sJ	send a JoinIn or JoinMT message (10.7.6.3)
7 sL	send a Lv message (10.7.6.4)
8 s	send an In or an Empty message (10.7.6.5)
9 [s]	send an In or an Empty message, if required for optimization of the encoding (10.7.6.5)
10 [sL]	send a Lv message, if required for optimization of the encoding (10.7.6.4)
11 [sJ]	send a Join message, if required for optimization of the encoding (10.7.6.3)
12 sLA	send a Leave All message (10.7.6.6)
13 periodic	Periodic transmission event (10.7.6.7).
14 leavetimer	Leave period timer (10.7.4.2)
15 leavealltimer	Leave All period timer (10.7.4.3)
16 periodictimer	Periodic Transmission timer (10.7.4.4)
17 -x-	Inapplicable event/state combination. No action or state transition occurs in this case.

18 Timers are used in the state machine descriptions in order to cause actions to be taken after defined time
19 periods have elapsed. The following terminology is used in the state machine descriptions to define timer
20 states and the actions that can be performed upon them:

- 21 a) A timer is said to be *running* if the most recent action to be performed upon it was a *start*.
- 22 b) A running timer is said to have *expired* when the time period associated with the timer has elapsed
23 since the most recent start action took place.
- 24 c) A timer is said to be *stopped* if it has expired or if the most recent action to be performed upon it was
25 a *stop* action.
- 26 d) A *start* action sets a stopped timer to the running state, and associates a time period with the timer.
27 This time period supersedes any periods that might have been associated with the timer by previous
28 start events.
- 29 e) A *stop* action sets a timer to the stopped state.

30 The following abbreviations are used for the state names in the state tables and state diagrams:

31 **Registrar states** (see 10.6):

32 IN	In
33 LV	Leaving
34 MT	Empty

35 **Applicant and Simple-Applicant states** (see 10.6):

36 VO	Very anxious Observer
37 VP	Very anxious Passive
38 VN	Very anxious New
39 AN	Anxious New
40 AA	Anxious Active
41 QA	Quiet Active
42 LA	Leaving Active
43 AO	Anxious Observer
44 QO	Quiet Observer
45 AP	Anxious Passive
46 QP	Quiet Passive
47 LO	Leaving Observer

1 10.7.2 Registrar Administrative Controls

2 Associated with each instance of the Registrar state machines are *Registrar Administrative Control*
3 parameters. These parameters allow administrative control to be exercised over the registration state of each
4 Attribute value, and hence, via the propagation mechanism provided by MAP, allow control to be exercised
5 over the propagation of declarations.

- 6 a) *Normal Registration*. The Registrar responds to incoming MRP messages as specified by
7 Table 10-4.
- 8 b) *Registration Fixed (New ignored)*. The Registrar ignores all MRP messages, and remains IN
9 (registered).
- 10 c) *Registration Fixed (New propagated)*. The Registrar ignores all MRP messages except New, and
11 remains IN (registered).
- 12 d) *Registration Forbidden*. The Registrar ignores all MRP messages, and remains MT (unregistered).

13 The default value of this parameter is *Normal Registration*.

14 If the value of this parameter is *Registration Fixed (New ignored)* or *Registration Fixed (New propagated)*,
15 In and JoinIn messages are sent. If the value of this parameter is *Registration Forbidden*, Empty or
16 JoinEmpty messages are sent. If the value of this parameter is *Registration Fixed (New propagated)*, only
17 New messages are accepted.

18 NOTE—The Registrar Administrative Controls are realized by means of the contents of the Port Map parameters of
19 static entries in the FDB for all MRP applications. In the case of MMRP, the static entries concerned are Static Filtering
20 Entries (8.8.1); in the case of MVRP and MIRP, the static entries concerned are Static VLAN Registration Entries
21 (8.8.2). The contents of the Port Map parameters in static entries can be modified by means of the management
22 operations defined in Clause 12. In the absence of such control information for a given attribute, the default value
23 “Normal Registration” is assumed.

24 10.7.3 Applicant Administrative Controls

25 An overall control parameter for each Applicant state machine, the *Applicant Administrative Control*,
26 determines whether the Applicant state machine participates in MRP exchanges.

- 27 a) *Normal Participant*. The state machine participates normally in MRP exchanges.
- 28 b) *New-Only Participant*. The state machine sends only New MRP messages.
- 29 c) *Non-Participant*. The state machine does not send any MRP messages.

30 The default value of this parameter is Normal Participant.

31 NOTE 1—The Applicant Administrative Control parameters can be modified for any MRP application by means of the
32 management operations defined in Clause 12. In the absence of such information for a given attribute, the default value
33 “Normal Participant” is assumed.

34 NOTE 2—The Applicant Administrative Control parameters can be set per attribute type (see 12.9.2).

35 10.7.4 Protocol timers

36 10.7.4.1 jointimer

37 The Join Period Timer, jointimer, controls the interval between transmit opportunities that are applied to the
38 Applicant state machine. An instance of this timer is required on a per-Port, per-MRP Participant basis. The
39 value of JoinTime used to initialize this timer is determined in accordance with 10.7.11.

40 10.7.4.2 leavetimer

41 The Leave Period Timer, leavetimer, controls the period of time that the Registrar state machine will wait in
42 the LV state before transiting to the MT state. An instance of the timer is required for each state machine that
43 is in the LV state. The Leave Period Timer is set to the value LeaveTime when it is started; LeaveTime is
44 defined in Table 10-7.

1 **10.7.4.3 leavealltimer**

2 The Leave All Period Timer, `leavealltimer`, controls the frequency with which the LeaveAll state machine
3 generates LeaveAll PDUs. The timer is required on a per-Port, per-MRP Participant basis. If LeaveAllTime
4 is zero, the Leave All Period Timer is not started; otherwise, the Leave All Period Timer is set to a random
5 value, T , in the range $\text{LeaveAllTime} < T < 1.5 \times \text{LeaveAllTime}$ when it is started. LeaveAllTime is defined
6 in Table 10-7.

7 **10.7.4.4 periodictimer**

8 The Periodic Transmission timer, `periodictimer`, controls the frequency with which the PeriodicTransmission
9 state machine generates periodic! events. The timer is required on a per-Port basis. The Periodic
10 Transmission timer is set to one second when it is started.

11 **10.7.5 Protocol event definitions**

12 Unless stated otherwise in these event definitions, MRDPDU reception in a Bridge can occur through all Ports
13 of a Bridge, and events generated as a result of such reception affect only those state machines that are
14 associated with the Port through which the PDU was received.

15 **10.7.5.1 Begin!**

16 The state machine is initialized or reinitialized.

17 **10.7.5.2 Flush!**

18 A Flush! event signals to the Registrar state machine that there is a need to rapidly deregister information on
19 the Port associated with the state machine as a result of a topology change that has occurred in the network
20 topology that supports the propagation of MRP information. If the network topology is maintained by means
21 of the spanning tree protocol state machines, then, for the set of Registrar state machines associated with a
22 given Port and spanning tree instance, this event is generated when the Port Role changes from either Root
23 Port or Alternate Port to Designated Port.

24 When a Flush! event occurs for a given Port and spanning tree instance, a `leavealltimer!` event (10.7.5.22) is
25 also signaled to the LeaveAll state machine for that Port and spanning tree instance.

26 **10.7.5.3 Re-declare!**

27 A Re-declare! event signals to the Applicant and Registrar state machines that there is a need to rapidly
28 redeclare registered information on the Port associated with the state machines as a result of a topology
29 change that has occurred in the network topology that supports the propagation of MRP information. If the
30 network topology is maintained by means of the spanning tree protocol state machines, then, for the set of
31 Applicant and Registrar state machines associated with a given Port and spanning tree instance, this event is
32 generated when the Port Role changes from Designated Port to either Root Port or Alternate Port.

33 **10.7.5.4 New!**

34 A new declaration is made. The event is deemed to have occurred if the MAD Service User issues a
35 `MAD_Join.request` service primitive for the Attribute instance associated with that state machine, indicating
36 a new declaration.

37 **10.7.5.5 Join!**

38 A declaration is made. The event is deemed to have occurred if the MAD Service User issues a
39 `MAD_Join.request` service primitive for the Attribute instance associated with that state machine.

1 **10.7.5.6 Lv!**

2 A declaration is withdrawn. The event is deemed to have occurred if the MAD Service User issues a
3 MAD_Leave.request service primitive for the Attribute instance associated with that state machine.

4 **10.7.5.7 tx!**

5 A transmission opportunity occurs, without a LeaveAll being signaled by the LeaveAll state machine.

6 NOTE—The tx! event is modified by the behavior of the LeaveAll state machine. If the LeaveAll state machine has
7 signaled LeaveAll, then tx! is modified to txLA! (see 10.7.5.8).

8 **10.7.5.8 txLA!**

9 A transmission opportunity occurs, with a LeaveAll being signaled by the LeaveAll state machine.

10 **10.7.5.9 txLAF!**

11 A transmission opportunity occurs, with a LeaveAll being signaled by the LeaveAll state machine, and with
12 no room available in the PDU.

13 **10.7.5.10 periodic!**

14 This event indicates to the Applicant state machine that the timer used to stimulate periodic transmission has
15 expired.

16 **10.7.5.11 periodicEnabled!**

17 This event indicates to the Periodic Transmission state machine that it has been enabled by management
18 action (12.9.3).

19 **10.7.5.12 periodicDisabled!**

20 This event indicates to the Periodic Transmission state machine that it has been disabled by management
21 action (12.9.3).

22 **10.7.5.13 Message reception events**

23 For an instance of the Applicant state machine or the Registrar state machine, a message reception event is
24 deemed to have occurred if an MRPDU (10.8) is received, and the following conditions are true:

- 25 a) The PDU was addressed to an MRP application (the MRP application address, Table 10-1, or MRP
26 address, 39.2.1.6) and had an EtherType (Table 10-2) in accordance with that of the MRP
27 application associated with the state machine.
- 28 b) The PDU contains a Message (10.8.1) in which the Attribute Type is the type associated with the
29 state machine.
- 30 c) The Message contains a VectorAttribute (10.8.1.2) where the range defined by the FirstValue and
31 NumberOfValues includes the attribute value associated with the state machine.

32 The specific type of message reception event is determined by the event value (10.8.2.5) associated with the
33 state machine. The possible message reception event types are specified in 10.7.5.14 through 10.7.5.20.

34 **10.7.5.14 rNew!**

35 For an instance of the Applicant or Registrar state machine, a message reception event (10.7.5.13) occurs,
36 and the event value (10.8.2.5) associated with the state machine specifies the New message.

1 **10.7.5.15 rJoinIn!**

2 For an instance of the Applicant or Registrar state machine, a message reception event (10.7.5.13) occurs,
3 and the event value (10.8.2.5) associated with the state machine specifies the JoinIn message.

4 **10.7.5.16 rJoinMt!**

5 For an instance of the Applicant or Registrar state machine, a message reception event (10.7.5.13) occurs,
6 and the event value (10.8.2.5) associated with the state machine specifies the JoinMt message.

7 **10.7.5.17 rLv!**

8 For an instance of the Applicant or Registrar state machine, a message reception event (10.7.5.13) occurs,
9 and the event value (10.8.2.5) associated with the state machine specifies the Lv message.

10 **10.7.5.18 rIn!**

11 For an instance of the Applicant or Registrar state machine, a message reception event (10.7.5.13) occurs,
12 and the event value (10.8.2.5) associated with the state machine specifies the In message.

13 **10.7.5.19 rMt!**

14 For an instance of the Applicant or Registrar state machine, a message reception event (10.7.5.13) occurs,
15 and the event value (10.8.2.5) associated with the state machine specifies the Mt message.

16 **10.7.5.20 rLA!**

17 For an instance of the Applicant state machine, the Registrar state machine, or the LeaveAll state machine,
18 the rLA! event is deemed to have occurred if either:

- 19 a) The LeaveAll state machine associated with that instance of the Applicant or Registrar state
20 machine performs the sLA action (10.7.6.6); or
- 21 b) An MRPD (10.8) is received, and the following conditions are true:
 - 22 1) The PDU's destination MAC address and EtherType are in accordance with the definition of
23 the MRP application associated with the state machine.
 - 24 2) The PDU contains a Message (10.8.1) in which the Attribute Type is the type associated with
25 the state machine.
 - 26 3) The Message contains a LeaveAllEvent (10.8.2.6) that carries the value LeaveAll.

27 NOTE—The LeaveAll state machine operates on a per-application (not per-Attribute Type) basis, but the LeaveAll
28 message operates on a per-Attribute Type basis. Hence, when the LeaveAll state machine issues a LeaveAll, it must
29 generate a LeaveAll Attribute for each Attribute Type supported by the application concerned.

30 **10.7.5.21 leavetimer!**

31 For an instance of the Registrar state machine, the leavetimer! event is deemed to have occurred when the
32 leavetimer associated with that state machine expires.

33 **10.7.5.22 leavealltimer!**

34 For an instance of the LeaveAll state machine, the leavealltimer! event is deemed to have occurred either
35 when the leavealltimer associated with that state machine expires or when a Flush! event (10.7.5.2) occurs
36 for the Port and spanning tree instance associated with that state machine.

37 **10.7.5.23 periodictimer!**

38 For an instance of the PeriodicTransmission state machine, the periodictimer! event is deemed to have
39 occurred when the periodictimer associated with that state machine expires.

1 10.7.6 Protocol Action definitions

2 10.7.6.1 MRPDU transmission actions

3 Unless stated otherwise in these action definitions, MRPDU transmission as a result of the operation of a
4 state machine in a Bridge occurs only through the Port associated with that state machine. MRPDUs shall be
5 transmitted using the destination MAC address and EtherType value defined by the MRP application
6 associated with the state machine.

7 When the action specifies the transmission of attribute state information, an MRPDU, formatted as defined
8 in 10.8.1, is transmitted so that:

- 9 a) The PDU contains a Message (10.8.1.2) that carries an Attribute Type (10.8.2.2) that corresponds to
10 the type of attribute associated with the state machine concerned.
- 11 b) The Message contains a VectorAttribute (10.8.2.10) in which the FirstValue (10.8.2.7) and
12 NumberOfValues (10.8.2.8) defines a range of attribute values that includes the attribute value
13 associated with the state machine.
- 14 c) The Vector (10.8.2.10) encodes an AttributeEvent value in the vector position corresponding to the
15 attribute value for this state machine. The choice of AttributeEvent value is determined by the
16 specific transmission action, as defined below.

17 NOTE—As explained in 10.8.2.10.1 each attribute in a vector is associated with its own individual state machine. A
18 naive implementation of MRP can therefore be quite inefficient.

19 In the case of LeaveAll transmission actions, the LeaveAll event value specified by the transmission action
20 is encoded in the NumberOfValues field of the VectorAttribute.

21 10.7.6.2 sN

22 The AttributeEvent value New is encoded in the Vector as specified in 10.7.6.1.

23 10.7.6.3 sJ, [sJ]

24 If the Registrar state is IN, then the AttributeEvent value JoinIn is encoded in the Vector as specified in
25 10.7.6.1.

26 If the Registrar state is MT or LV, then the AttributeEvent value JoinMt is encoded in the Vector as specified
27 in 10.7.6.1.

28 NOTE—The [sJ] variant indicates that the action is only necessary in cases where transmitting the value, rather than
29 terminating a vector and starting a new one, makes for more optimal encoding; i.e., transmitting the value is not
30 necessary for correct protocol operation.

31 10.7.6.4 sL

32 The AttributeEvent value Lv is encoded in the Vector as specified in 10.7.6.1.

33 10.7.6.5 s, [s]

34 If the Registrar state is IN, then the AttributeEvent value In is encoded in the Vector as specified in 10.7.6.1.

35 If the Registrar state is MT or LV, then the AttributeEvent value Mt is encoded in the Vector as specified in
36 10.7.6.1.

37 NOTE—In the [s] variant, this value is transmitted only if its inclusion makes for more optimal encoding; i.e.,
38 transmitting the value is not necessary for correct protocol operation.

39 10.7.6.6 sLA

40 The LeaveAll event value specified by the transmission action is encoded in the NumberOfValues field of
41 the VectorAttribute as specified in 10.7.6.1.

1 The sLA action also gives rise to a rLA! event (10.7.5.20) against all instances of the Applicant state
2 machine, or the Registrar state machine, associated with the MRP participant.

3 **10.7.6.7 periodic**

4 Causes a periodic! event (10.7.5.10) against all Applicant state machines associated with the participant.

5 **10.7.6.8 Start leavetimer**

6 Causes leavetimer to be started, in accordance with the definition of the timer in 10.7.4.2.

7 **10.7.6.9 Stop leavetimer**

8 Causes leavetimer to be stopped.

9 **10.7.6.10 Start leavealltimer**

10 When LeaveAllTime is nonzero, this causes leavealltimer to be started, in accordance with the definition of
11 the timer in 10.7.4.3.

12 **10.7.6.11 Start periodictimer**

13 Causes periodictimer to be started, in accordance with the definition of the timer in 10.7.4.4.

14 **10.7.6.12 New**

15 This action causes a MAD_Join.indication primitive to be issued to the MAD Service User, indicating the
16 Attribute instance corresponding to the state machine concerned, with the *new* parameter set TRUE.

17 **10.7.6.13 Join**

18 This action causes a MAD_Join.indication primitive to be issued to the MAD Service User, indicating the
19 Attribute instance corresponding to the state machine concerned, with the *new* parameter set FALSE.

20 **10.7.6.14 Lv**

21 This action causes a MAD_Leave.indication primitive to be issued to the MAD Service User, indicating the
22 Attribute instance corresponding to the state machine concerned.

23 **10.7.7 Applicant state machine**

24 A full MRP Participant maintains a single instance of the Applicant state machine (Table 10-3) for each
25 Attribute value for which the Participant needs to maintain state information.

26 **10.7.8 Registrar state machine**

27 A full MRP Participant maintains a single instance of this state machine for each Attribute value that is
28 currently registered, or that the Registrar state machine is in the process of deregistering.

29 NOTE—As with the Applicant, state information is conceptually maintained for all possible values of all Attribute types
30 that are defined for a given application; however, in real implementations of MRP, it is likely that the range of possible
31 Attribute values in some applications will preclude this, and the implementation will limit the state to those Attribute
32 values in which the Participant has an immediate interest. In the case of simple devices that have no interest in what
33 other Participants have registered, it may be appropriate for that device to ignore Registrar operation altogether.

34 The detailed operation of this state machine is described in Table 10-4.

Table 10-3—Applicant state table

		STATE											
		VO ^{11,12}	VP ⁶	VN ^{6, 12}	AN ^{6, 12}	AA ⁶	QA ¹²	LA ⁶	AO ^{3,11}	QO ^{3,11}	AP ^{3,6}	QP ³	LO ⁶
EVENT	Begin! ¹²	—	VO	VO	VO	VO	VO	VO	VO	VO	VO	VO	VO
	New! ¹²	VN	VN	—	—	VN	VN	VN	VN	VN	VN	VN	VN
	Join!	VP	—	—	—	—	—	AA	AP	QP	—	—	VP
	Lv!	—	VO	LA	LA	LA	LA	—	—	—	AO	QO	—
	rNew!	—	—	—	—	—	—	—	—	—	—	—	—
	rJoinIn!	AO ⁴	AP ⁴	—	—	QA	—	—	QO	—	QP	—	—
	rIn!	—	—	—	—	QA ⁵	—	—	—	—	—	—	—
	rJoinMt! rMt!	—	—	—	—	—	AA	—	—	AO	—	AP	VO
	rLv! rLA! Re-declare!	LO ¹	—	—	VN	VP ⁹	VP ⁹	— ¹⁰	LO ¹	LO ¹	VP	VP	—
	periodic!	—	—	—	—	—	AA	—	—	—	—	AP	—
	tx! ^{7, 12}	[s] —	sJ AA	sN AN	sN QA ⁸	sJ QA	[sJ] —	sL VO	[s] —	[s] —	sJ QA	[s] —	s VO
	txLA! ²	[s] LO	s AA	sN AN	sN QA	sJ QA	sJ —	[s] LO	[s] LO	[s] LO	sJ QA	sJ QA	[s] —
	txLAF! ²	LO	VP	VN	VN	VP	VP	LO	LO	LO	VP	VP	—

Notes to the table:

- ¹ Applicant-Only participants exclude the LO state, and transition to VO.
- ² These events do not occur for Applicant-Only participants.
- ³ Point-to-point subset participants exclude the AO, QO, AP, and QP states.
- ⁴ Ignored (no transition) if point-to-point subset or if operPointToPointMAC is TRUE.
- ⁵ Ignored (no transition) if operPointToPointMAC is FALSE. See MRP Design Notes below.
- ⁶ Request opportunity to transmit on entry to VN, AN, AA, LA, VP, AP, and LO states.
- ⁷ If the MRDPDU is full and cannot convey a required message there is no change of state and an additional transmit opportunity is requested if that has not been done already.
- ⁸ QA if the Registrar is IN, and AA otherwise. See MRP Design Notes below.

MRP design notes:

- ⁵ On shared media the receipt of In does not confirm registration by all Participants, and the In could have been sent by an Applicant-Only participant.
- ⁸ Since New messages do not convey registrar state, a Leave could have been received without an Empty or JoinEmpty prompt being sent, the transition to AA guards against loss of that Leave by another Applicant.
- ⁹ The design accepts a small possibility of a continued registration (after rLv! if a Lv! occurs before a further Join is sent) in return for not accumulating many Active participants when Join!s and Lv!s are frequent. rLv! processing is deliberately not optimized for point-to-point.
- ¹⁰ If a Leave has been received, the Registrar for the transmitting participant is very probably IN, as this Applicant has not yet sent a Leave, so the pending Leave is required. The small savings from avoiding transmission of Leaves pending on receipt of LeaveAlls does not merit distinguishing the rLv! and rLA! cases.
- ¹¹ The VO, AO, and QO states represent states where the attribute is neither being declared by the Participant nor being registered by any other station on the LAN. In implementations where dynamic creation and discarding of state machines is desirable, the state machine can be discarded when in any of these states, pending a future requirement to declare or register that attribute value.
- ¹² A New-Only Participant recognizes only the Begin!, New!, and tx! events, and ignores all others, so can only reach the VO, VN, AN, and QA states.

Table 10-4—Registrar state table

		STATE		
		IN	LV	MT
EVENT	Begin!	MT	MT	MT
	rNew!	New IN	New Stop leavetimer IN	New IN
	rJoinIn! rJoinMt!	IN	Stop leavetimer IN	Join IN
	rLv! rLA! txLA! Re-declare!	Start leavetimer LV	-x-	-x-
	Flush!	Lv MT	Lv MT	MT
	leavetimer!	-x-	Lv MT	MT ^a

^aIf operPointToPointMAC is TRUE, proceed directly to the Lv/MT transition as defined in the LV State for the leavetimer! Event. This avoids the delay resulting from the leavetimer. If operPointToPointMAC is FALSE, the state machine is executed as shown.

1 10.7.9 LeaveAll state machine

2 A single LeaveAll state machine exists for each full MRP Participant. Leave All messages generated by this
3 state machine also generate LeaveAll events against all the Applicant and Registrar state machines
4 associated with that Participant and Port; hence, LeaveAll generation is treated by those state machines in
5 the same way as reception of a LeaveAll message from an external source.

6 If the LeaveAllTime is changed from a value of zero to a nonzero value, a Begin! event shall be signaled to
7 the LeaveAll state machine to cause it to reinitialize.

8 The detailed operation of this state machine is described in Table 10-5.

9 10.7.10 PeriodicTransmission state machine

10 A single PeriodicTransmission state machine exists for each Port. Periodic Transmission events are
11 generated on a regular basis, against all Applicant state machines that are associated with that Port.

12 The detailed operation of this state machine is described in Table 10-6.

Table 10-5—LeaveAll state table

		STATE	
		Active ^a	Passive
EVENT	Begin!	Start leavealltimer Passive	Start leavealltimer Passive
	tx!	sLA Passive	-x-
	rLA!	Start leavealltimer Passive	Start leavealltimer Passive
	leavealltimer!	Start leavealltimer Active	Start leavealltimer Active

^a Request opportunity to transmit on entry to the Active state

Table 10-6—PeriodicTransmission state table

		STATE	
		Active	Passive
EVENT	Begin!	Start periodictimer Active	Start periodictimer Active
	periodicEnabled!	-x-	Start periodictimer Active
	periodicDisabled!	Passive	-x-
	periodictimer!	Start periodictimer periodic Active	-x-

1 10.7.11 Timer values

2 MRP *correctness* is not critically dependent on the values of protocol timers. However, the protocol operates
3 more efficiently, and with less likelihood of unwanted deregistrations, if the following relationships are
4 maintained between timer values used by peer Participants:

- 5 a) LeaveTime should be at least twice the maximum JoinTime, plus six times the timer resolution, to
6 allow reregistration after a Leave or LeaveAll even if a message is lost.
- 7 b) To minimize the volume of rejoining traffic generated following a LeaveAll, the value chosen for
8 LeaveAllTime should be large relative to LeaveTime.

9 The default timer values (Table 10-7) maintain these relationships. They may be modified on a per-Port
10 basis by means of the management functionality defined in Clause 12.

11 NOTE—The default values for the MRP timers are independent of media access method or data rate. This is a deliberate
12 choice, made in the interests of maximizing the “plug and play” characteristics of the protocol.

13 Implementation of the timers for MRP shall be based on a timer resolution of 1 centisecond or less.

Table 10-7—MRP timer parameter default values

Parameter	Value (centiseconds)
JoinTime	20
LeaveTime	60–100
LeaveAllTime	1000

1 If operPointToPointMAC (IEEE Std 802.1AC) is TRUE, a request for a transmit opportunity should result
2 in such an opportunity as soon as is practicable, given other system constraints, and shall occur within the
3 value specified for JoinTime subject to not more than three such transmission opportunities occurring in any
4 period of $1.5 \times \text{JoinTime}$.

5 If operPointToPointMAC is FALSE, and there is no pending request, a transmit opportunity shall occur at a
6 time value randomized between 0 and JoinTime seconds. These provisions shall apply even if a point-to-
7 point subset Applicant has been implemented.

8 **10.7.12 Operational reporting and statistics**

9 **10.7.12.1 Failure to register**

10 Each MRP Participant maintains a count of the number of times that it has received a registration request,
11 but has failed to register the attribute concerned due to lack of space in the FDB to record the registration.
12 The value of this count may be examined by management (see 12.9.2.1.3).

13 NOTE—Further action to be taken on such events is a matter for implementation choice.

14 **10.7.12.2 Peer tracking**

15 An implementation may support the ability to record against each Registrar state machine the MAC address
16 of the originator of the MRPDU that caused the most recent state change for that state machine (see
17 12.9.3.1.3).

18 **10.7.13 Interoperability considerations**

19 Correct operation of MRP for a given MRP application requires that protocol exchanges among a given set
20 of communicating MRP Participants maintain sequentiality; i.e., that Participant A cannot receive
21 MRPDU B (generated as a consequence of Participant B receiving MRPDU A) before Participant A has
22 received MRPDU A. In circumstances where the Participants concerned are all attached to the same LAN,
23 such sequentiality is ensured. However, if a set of MRP Participants communicates via an intervening
24 Bridge that does not implement that MRP application (or does not implement MRP at all), the sequentiality
25 constraints expressed in 8.6.6, 8.6.7, and 8.6.8 are insufficient to guarantee the correct operation of MRP. In
26 order for the correct sequencing of PDUs to be maintained through such a Bridge, the following constraint
27 needs to be met:

28 If MRPDU A is received on Port X, and is due to be forwarded on Ports Y and Z, and subsequent to
29 being forwarded on Y, MRPDU B is received on Port Y for forwarding on Port Z, then forwarding
30 of B cannot precede A on Port Z.

31 NOTE—This expresses a stronger sequencing constraint for multicast frames than is stated in 8.6, but a weaker
32 constraint than was required for conformance to IEEE Std 802.1D, 1993 Edition [B12].

33 The consequence of failure to meet this constraint is that the users of a given MRP application may
34 experience an increased incidence of loss of registration. Therefore, it is inadvisable to construct LAN
35 configurations involving forwarding of MRPDU through intervening Bridges if those Bridges do not meet
36 the constraint expressed previously.

10.7.14 External control

The MRP External Control feature (12.32.4) specifies a managed object that the network manager uses to:

- a) Disable MRP attribute propagation (MAP) for the MRP Participant of a Bridge Port.
- b) Read MRP attribute registrations that the MRP Participant receives.
- c) Write MRP attribute values for the MRP Participant to declare.

There is one MRP External Control managed object for each MRP Participant (per Port per MRP application per MAP Context, as specified in 10.2 and 10.3.1).

Under default operation, the MRP External Control feature (externalControl attribute FALSE) is ignored by the operation of MRP and has no effect on MRP Attribute Propagation (MAP) specified in 10.3.

When the MRP External Control feature is enabled by network management (externalControl attribute TRUE), the operation of MAP in the Bridge performs as specified in 12.32.4, but all other aspects of MRP operation are performed as specified in Clause 10.

10.8 Structure and encoding of Multiple Registration Protocol Data Units (MRPDUs)

This subclause describes the generic structure and encoding of the MRPDUs exchanged between all MRP Participants. The structure and encoding of elements that are specific to the operation of the MRP applications are defined by the applications themselves.

Each MRPDU identifies the MRP application by which it was generated, and to which it is being transmitted. Bridges that receive MRPDUs identified as belonging to an MRP application that they do not support shall forward such PDUs on all other Ports with a Port State (8.4) of Forwarding.

NOTE 1—If MRP is used to support an application that can operate in any MAP Context other than 0 (the Base Spanning Tree Context), the application specification describes how that context is identified in protocol exchanges.

Each MRPDU carries one or more MRP messages, each of which identify one or more MRP events (e.g., Join, Leave, LeaveAll) and the attribute class(es) and value(s) to which each event applies. A given MRP Participant shall process MRPDUs in the order in which they are received, and shall process the MRP Messages in a PDU in the order in which they were put into the Data Link Service Data Unit (DLSDU).

NOTE 2—Any messages generated as a consequence of state machine responses to an sLA action and its associated LeaveAll events will be put into the DLSDU after the LeaveAll message(s), or into a later DLSDU.

10.8.1 Structure

10.8.1.1 Transmission and representation of octets

All MRPDUs consist of an integral number of octets, numbered starting from 1 and increasing in the order that they are put into a DLSDU. The bits in each octet are numbered from 1 to 8, where 1 is the low-order bit.

When consecutive octets are used to represent a binary number, the lower octet number has the most significant value.

When the encoding of (an element of) an MRPDU is represented using a diagram in this clause, the following representations are used:

- a) Octet 1 is shown toward the top of the page, higher numbered octets being toward the bottom.
- b) Where more than one octet appears on a given line, octets are shown with the lowest numbered octet to the left, higher numbered octets being to the right.
- c) Within an octet, bits are shown with bit 8 to the left and bit 1 to the right.

10.8.1.2 Structure definition

MRP makes use of an EtherType value as the means of identifying the MRP application that has transmitted, and that will receive, a given MRPDU. Table 10-2 lists the set of MRP applications that are defined, and the EtherType values that correspond to them.

A protocol version field is included in the structure of the MRPDU, in order to provide the ability to identify enhancements to MRP applications.

MRPDUs exchanged according to the protocol specified in this clause shall have the following structure:

- a) The first octet contains the *ProtocolVersion*.
- b) Following the Protocol Version are one or more *Messages*. The last element in the PDU is an *EndMark*.
- c) Each Message consists of an *AttributeType*, an *AttributeLength*, and an *AttributeList*, in that order.
- d) An Attribute List consists of one or more *VectorAttributes*. The last element in the AttributeList is an *EndMark*.
- e) A VectorAttribute consists of a *VectorHeader*, a *FirstValue*, and a *Vector*, in that order. The VectorHeader is able to encode both a *LeaveAllEvent* and the number of attribute events encoded in the vector.
- f) If the end of an MRPDU is encountered before an EndMark is reached, then processing of the PDU is terminated as if an EndMark had been reached.

The following Backus-Naur Form (BNF) productions give the formal description of the MRPDU structure:

```
MRPDU ::= ProtocolVersion, Message {, Message}, EndMark
ProtocolVersion BYTE ::= Defined by the specific MRP application
Message ::= AttributeType, AttributeLength[, AttributeListLength], AttributeList
AttributeType BYTE ::= Nonzero integer defined by the specific MRP application
AttributeLength BYTE ::= Nonzero integer defined by the specific MRP application
AttributeListLength SHORT ::= Nonzero integer defined by the specific MRP application
AttributeList ::= VectorAttribute {, VectorAttribute}, EndMark
VectorAttribute ::= VectorHeader, FirstValue {, Vector}
VectorHeader SHORT ::= (LeaveAllEvent * 8192) + NumberOfValues
FirstValue ::= Defined by the specific MRP application
Vector ::= ThreePackedEvents {, ThreePackedEvents}
           [, FourPackedEvents {, FourPackedEvents}]
ThreePackedEvents BYTE ::= (((((AttributeEvent) * 6) + AttributeEvent) * 6) + AttributeEvent) * 6) + AttributeEvent
AttributeEvent BYTE ::= New | JoinIn | In | JoinMt | Mt | Lv
FourPackedEvents BYTE ::= ((FourPackedType * 64) + (FourPackedType * 16)
                           + (FourPackedType * 4) + FourPackedType)
FourPackedType BYTE ::= Integer defined by the specific MRP application
LeaveAllEvent BYTE ::= NullLeaveAllEvent | LeaveAll
NumberOfValues SHORT ::= Number of events encoded in the vector
EndMark SHORT ::= 0x0000 | End of PDU
NullLeaveAllEvent ::= 0
LeaveAll ::= 1
New ::= 0
JoinIn ::= 1
In ::= 2
JoinMt ::= 3
Mt ::= 4
Lv ::= 5
```

1 The parameters carried in MRPDUs, as identified in this structure definition, shall be encoded as specified in
2 10.8.2.

3 Figure 10-5 illustrates the structure of the MRPDU and its components.

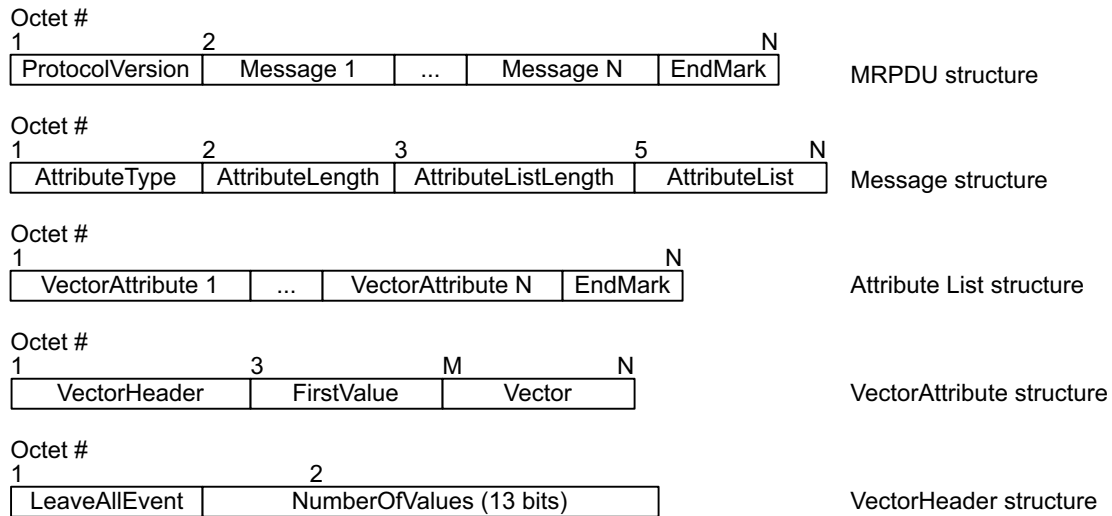


Figure 10-5—Format of the major components of an MRPDU

4 10.8.2 Encoding of MRPDU parameters

5 10.8.2.1 Encoding of ProtocolVersion

6 A ProtocolVersion shall be encoded in a single octet, taken to represent an unsigned binary number. It takes
7 a value that is determined by the application concerned.

8 NOTE—The handling of protocol version information is defined in 10.8.3.5.

9 10.8.2.2 Encoding of AttributeType

10 An AttributeType shall be encoded as a single octet, taken to represent an unsigned binary number. The
11 AttributeType identifies the type of Attribute to which the message applies. The range of values that can be
12 taken by the AttributeType, and the meanings of those values, are defined by the application concerned. The
13 value 0 is reserved, and shall not be used as an AttributeType by any MRP application. MRP applications
14 may otherwise allocate meanings to any set of values of AttributeType in the range 1 through 255.

15 The definition of AttributeType values in the application associates the following with each AttributeType
16 value defined:

- 17 a) The size, in octets, of attribute values, and how they are encoded.
- 18 b) Any restrictions that are placed upon the range of attribute values that the vector is permitted to
19 represent.

20 10.8.2.3 Encoding of AttributeLength

21 An AttributeLength shall be encoded as a single octet, taken to represent an unsigned binary number. The
22 AttributeLength indicates the length, in octets, of the FirstValue field (10.8.2.7) for the Attribute to which
23 the message applies.

1 10.8.2.4 Encoding of AttributeListLength

2 An AttributeListLength shall be encoded as two octets, taken to represent an unsigned binary number. The
3 AttributeListLength indicates the length, in octets, of the AttributeList field for the Attribute to which the
4 message applies. This field is not present in all MRPDUs. Specifically, Multiple MAC Registration Protocol
5 Data Units (MMRPDUs) and Multiple VLAN Registration Protocol Data Units (MVRPDUs) do not use this
6 field, whereas Multiple Stream Protocol Data Units (MSRPDUs) do use this field.

7 10.8.2.5 Encoding of AttributeEvent

8 An AttributeEvent shall be encoded as an unsigned decimal number in the range 0 through 5. The permitted
9 values and meanings of the AttributeEvent are as follows:

10	0:	New operator
11	1:	JoinIn operator
12	2:	In operator
13	3:	JoinMt operator
14	4:	Mt operator
15	5:	Lv operator

16 Further values of AttributeEvent are reserved.

17 The AttributeEvent is interpreted on receipt as a MAD event to be applied to the state machine for the
18 Attribute defined by the AttributeType and AttributeValue to which the AttributeEvent relates.

19 10.8.2.6 Encoding of LeaveAllEvent

20 A LeaveAllEvent shall be encoded as an unsigned binary number. The permitted values and meanings of
21 LeaveAllEvent are as follows:

22	0:	NullLeaveAllEvent operator
23	1:	LeaveAll operator

24 Further values of LeaveAllEvent are reserved.

25 The LeaveAllEvent is interpreted on receipt as a MAD Leave All event to be applied to the state machines
26 for all Attributes of the type defined by the AttributeType field.

27 The value NullLeaveAllEvent signifies that there is no Leave All event to process, and is included purely for
28 encoding efficiency in the vector attribute structures. Receipt of this value does not cause any event to be
29 applied to any state machine.

30 10.8.2.7 Encoding of FirstValue

31 A FirstValue is encoded in N octets, taken to be an unsigned binary number, in accordance with the
32 specification for the AttributeType defined by the MRP application concerned. The length, in octets, of the
33 FirstValue field is specified by the value contained in the AttributeLength field (10.8.2.3). When
34 NumberOfValues is greater than one (1) FirstValue is incremented in a way that is defined by each MRP
35 application. For example MMRP simply increments FirstValue by adding the number one (1) to it, whereas
36 MSRP adds one (1) to multiple fields within the FirstValue for each increment. Throughout this
37 specification FirstValue incremented by one is denoted as FirstValue + 1, incrementing by two is denoted as
38 FirstValue + 2, etc.

10.8.2.8 Encoding of VectorHeader

The VectorHeader is used to encode both the value of the LeaveAllEvent (10.8.2.6) and the NumberOfValues, the number of AttributeEvent values encoded in a Vector (10.8.2.10). The VectorHeader is taken to be an unsigned binary number, encoded in two octets, as follows:

- a) The value of the LeaveAllEvent is multiplied by 8192.
- b) The resulting number is added to NumberOfValues.

The range of values that NumberOfValues can take is restricted, such that the following are true:

- c) The size of the Vector that is defined by this number will fit in the available space in the PDU.
- d) Incrementing FirstValue the number of times specified in NumberOfValues does not exceed the permitted numeric range of FirstValue as defined for the application concerned.
- e) The number of AttributeEvent values does not exceed 8191.
- f) If the number of AttributeEvent values is zero, FirstValue is ignored and the value is skipped. However, FirstValue is still present, and of the correct length, in octets, as specified by the AttributeLength field (10.8.2.3) for the Attribute to which the message applies.
- g) If LeaveAllEvent is a NullLeaveAllEvent, the number of AttributeEvent values is nonzero.

10.8.2.9 Encoding of EndMark

An EndMark shall be encoded as two octets, taken to represent an unsigned binary number. It takes the numeric value 0.

Further values of EndMark are reserved and shall not be used.

NOTE—As defined by the MRPDU structure definition in 10.8.1, if the end of the MRPDU is encountered, this is taken to be an End Mark from the point of view of processing the PDU contents.

10.8.2.10 Encoding of Vector

10.8.2.10.1 Encoding of Vector ThreePackedEvents

The Vector is encoded as zero or more 8-bit values, each containing a numeric value, ThreePackedEvents, derived from three packed numeric values, each of which represents an AttributeEvent, in the range 0 through 5.

As can be seen from the BNF definition of ThreePackedEvents, each 8-bit value is derived by successively adding an event value and multiplying the result by 6. In order to facilitate the subsequent description, the event values are numbered from *first* to *third*, as follows:

ThreePackedEvents BYTE ::=
$$((((firstAttributeEvent) \times 6) + secondAttributeEvent) \times 6) + thirdAttributeEvent$$

The NumberOfValues field in the VectorHeader of the VectorAttribute determines the number of 8-bit ThreePackedEvents values, *E*, that will be present in the vector; hence *E* is determined by dividing NumberOfValues by 3 and rounding any noninteger answer up to the nearest larger integer.

The FirstValue field of the VectorAttribute determines which of the originator's state machines the *first* AttributeEvent value in the first ThreePackedEvents value relates to. The *second* AttributeEvent value in the first ThreePackedEvents value corresponds to the state machine identified by (FirstValue + 1), and the *third* AttributeEvent value in the first ThreePackedEvents value corresponds to the state machine identified by (FirstValue + 2), and so on, through subsequent packed values.

1 Where the NumberOfValues field carries a value that is not a multiple of 3, there will be either one or two
2 AttributeEvent values packed in the final ThreePackedEvents that are ignored; these values are encoded as
3 the numeric value 0 on transmission and are ignored on receipt.

4 NOTE—If NumberOfValues is zero, there will be no ThreePackedEvents encoded in the vector.

5 10.8.2.10.2 Encoding of Vector FourPackedEvents

6 The Vector is encoded as zero or more 8-bit values, each containing a numeric value, FourPackedEvents,
7 derived from four packed numeric values, each of which represent a FourPackedType, in the range 0 through
8 3. The FourPackedTypes are defined by each MRP application that uses FourPackedEvents. Note that not all
9 MRP applications use FourPackedEvents.

10 As can be seen from the BNF definition of FourPackedEvents, each 8-bit value is derived by successively
11 adding a FourPackedType value and multiplying the result by 4. In order to facilitate the subsequent
12 description, the event values are numbered from first to fourth, as follows:

13 FourPackedEvents BYTE ::= ((firstFourPackedType × 64) + (secondFourPackedType × 16) +
14 (thirdFourPackedType × 4) + (fourthFourPackedType))

15 The NumberOfValues field in the VectorHeader of the VectorAttribute determines the number of 8-bit
16 FourPackedEvents values, E, that will be present in the vector; hence E is determined by dividing
17 NumberOfValues by 4 and rounding any noninteger answer up to the nearest larger integer.

18 The FirstValue field of the VectorAttribute determines which of the originator's state machines the first
19 FourPackedType value in the first FourPackedEvents value relates to. The second FourPackedType value in
20 the first FourPackedEvents value corresponds to the state machine identified by (FirstValue + 1). The third
21 FourPackedType value in the first FourPackedEvents value corresponds to the state machine identified by
22 (FirstValue + 2), and the fourth FourPackedType value in the first FourPackedEvents value corresponds to
23 the state machine identified by (FirstValue + 3), and so on, through subsequent packed values.

24 Where the NumberOfValues field carries a value that is not a multiple of 4, there will be one, two, or three
25 FourPackedType values packed in the final FourPackedEvent that are ignored; these values are encoded as
26 the numeric value 0 on transmission and are ignored on receipt.

27 NOTE—If NumberOfValues is zero, there will be no FourPackedEvents encoded in the vector.

28 10.8.3 Packing and parsing MRPDUs

29 The use of the End Mark (10.8.2.9) to signal the end of an AttributeList and the end of an MRPDu, and the
30 fact that the (physical) end of the PDU is interpreted as an End Mark, simplifies the requirements both for
31 packing information into MRPDUs and for correctly interpreting that information on receipt.

32 10.8.3.1 Packing

33 Successive Messages are packed into the MRPDu, and within each Message, successive VectorAttributes
34 are packed into each Message, until the end of the PDU is encountered or there are no more VectorAttributes
35 to pack at that time. The following cases can occur:

- 36 a) The PDU has sufficient room for all the VectorAttributes that require to be transmitted at that time to
37 be packed. In this case, the PDU is transmitted, and subsequent PDUs are transmitted when there are
38 further VectorAttributes to transmit.
- 39 b) The PDU has enough room for the first N VectorAttributes that require to be transmitted at that time
40 to be packed. In this case, the PDU is transmitted, and the next N VectorAttributes are encoded in a
41 subsequent PDU.

10.8.3.2 Handling of received MRPDUs

Received MRPDUs, i.e., PDUs that are constructed in accordance with the MRPDU format defined in 10.8, that carry a destination MAC address and MRP EtherType value as specified for use by a defined MRP application, are processed as follows:

- a) In Bridges and end stations that support the operation of the MRP application concerned, all such PDUs shall be submitted to the MRP Participant associated with the reception Port for further processing.
- b) In Bridges that do not support the operation of the MRP application concerned, all such PDUs shall be submitted to the Forwarding Process.

10.8.3.3 Discarding badly formed MRPDUs

An MRP Participant that receives an MRPDU shall discard that PDU if the PDU is not formatted according to the MRPDU format defined in 10.8.

10.8.3.4 Parsing

Successive Messages, and within each Message, successive VectorAttributes, are unpacked from the PDU. If this process terminates because the end of the PDU is reached, then the end of the PDU is taken to signal termination both of the current AttributeList and the overall PDU. The following two cases can occur:

- a) The last VectorAttribute to be unpacked was complete. In this case, the VectorAttribute is processed normally, and processing of the PDU terminates.
- b) The last VectorAttribute to be unpacked was incomplete. In this case, the entire PDU is discarded, and processing of the PDU terminates.

10.8.3.5 Handling of protocol versions

In order to ensure compatibility with previous protocol versions, while allowing the protocol to be extended, the following requirements shall be met by the implementation:

- a) MRPDUs that carry a protocol version lower than the protocol version implemented shall be interpreted according to the definition corresponding to the protocol version carried in the PDU.
- b) MRPDUs that carry a protocol version equal to or higher than the protocol version implemented shall be interpreted according to the definition corresponding to the protocol version implemented.
- c) Where the MRPDU carries a higher protocol version than the version implemented, then:
 - 1) If a Message is encountered in which the AttributeType is not recognized, then that Message is discarded. This is achieved by discarding the successive VectorAttributes in the AttributeList until either an EndMark or the end of the PDU is reached. If an EndMark is reached, processing continues with the next Message.
 - 2) If a VectorAttribute is encountered in which the AttributeEvent is not recognized for the AttributeType concerned, then the VectorAttribute is discarded and processing continues with the next VectorAttribute, if the end of the PDU has not been reached.
 - 3) For the MSRP application (Clause 35), if a FirstValue is encountered in which a TLV is not recognized for the Type concerned, then the TLV is discarded and processing continues with the next TLV, if the end of the PDU has not been reached.

10.9 Multiple MAC Registration Protocol (MMRP)—Purpose

MMRP provides a mechanism that allows end stations and MAC Bridges to dynamically register (and subsequently, deregister) Group membership or individual MAC address information with the Bridges attached to the same LAN, and disseminates that information across all the Bridges that support Extended Filtering Services in the Bridged Network. The operation of MMRP relies upon the services provided by MRP.

The information registered, deregistered, and disseminated via MMRP can appear in any of the following forms:

- a) *Group membership information.* This indicates the presence of MMRP participants that are members of a particular Group (or Groups), and carries the group MAC address(es) associated with the Group(s). The exchange of specific Group membership information can result in the creation or updating of MAC Address Registration Entries in the FDB to indicate the Port(s) and VID(s) of the VLAN(s) on which members of the Group(s) have been registered. The structure of these entries is described in 8.8.4.
- b) *Group service requirement information.* This indicates that one or more MMRP participants require Forward All Groups or Forward Unregistered Groups to be the default Group filtering behavior (see 6.16.7 and 8.8.6).
- c) *Individual MAC address information.*

Registration of Group membership information makes Bridges aware that frames destined for the group MAC address concerned should only be forwarded in the direction of the registered members of the Group. Therefore, forwarding of frames destined for the address associated with that Group occurs only on Ports on which such membership registration has been received.

Registration of Group service requirement information makes the Bridges aware that Ports that can forward frames in the direction from which the information has been received should modify their default Group forwarding behavior in accordance with the service requirement expressed.

NOTE 1—Modification of default Group forwarding behavior allows Bridge Ports to accommodate MMRP-unaware devices in the Bridged Network by forwarding frames destined for unregistered group MAC addresses.

The operation of MMRP can result in:

- d) The propagation of Group membership information and Group service requirement information, and consequent creation, updating, or deletion of MAC Address Registration Entries in the FDBs of all Bridges in the network that support Extended Filtering Services.
- e) Consequent changes to the Group filtering behavior of such Bridges.

In VLAN Bridges, MMRP operates only when the Bridge Filtering Mode is set to Extended Filtering Mode. Bridges that are unable to operate in Extended Filtering Mode, or have been set to operate in Basic Filtering Mode, are transparent with respect to MMRP exchanges, and forward any MRPDUs destined for the MMRP application through all Ports that are in Forwarding.

NOTE 2—An MMRP user is not prevented from registering an individual MAC address associated with a location different from its actual location. Such registration could result in denial of service to the station associated with that MAC address and, possibly, the interception of associated traffic by an unintended recipient. Such consequences of the misuse of individual MAC address registration can be prevented by using MMRP individual MAC address registration only in secure network environments (i.e., only in networks providing perimeter security).

10.10 MMRP Model of operation

MMRP defines an *MRP application* that provides the extended filtering services defined in 6.16.5 and 6.16.7. To this end, MMRP makes use of:

- a) The declaration and propagation services offered by *MRP Attribute Declaration* (MAD; 10.2 and 10.3,) and *MRP Attribute Propagation* (MAP; 10.2 and 10.3) to declare and propagate Group membership, Group service requirement, and individual MAC address information within the Bridged Network.
- b) The registration services offered by MAD (10.2 and 10.3) to allow Group membership, Group service requirement, and individual MAC address information to control the frame filtering behavior of participating devices.

Figure 10-6 illustrates the architecture of MMRP in the case of a two-Port Bridge and an end station, for a given VLAN Context. Where MMRP is used in multiple VLAN Contexts, an instance of the MMRP Participant exists for each VLAN context.

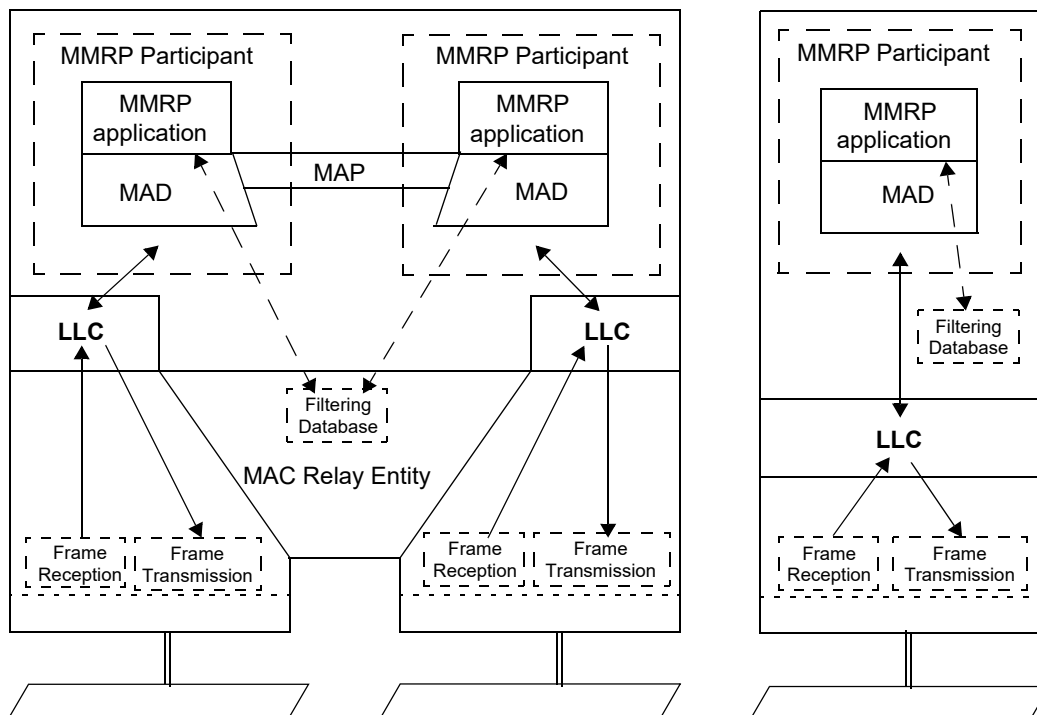


Figure 10-6—Operation of MMRP for a single VLAN Context

As shown in the diagram, the MMRP Participant consists of the following components:

- c) The MMRP application, described in 10.12
- d) MRP Attribute Propagation (MAP), described in 10.3
- e) MRP Attribute Declaration (MAD), described in 10.2

10.10.1 Propagation of Group Membership information

2 The Forwarding Process uses the MAC Address Registration entries in the FDBs to ensure that frames are
3 transmitted only through those Bridge Ports necessary to reach LANs to which Group members are attached.
4 Figure 10-7 illustrates the MAC Address Registration Entries created by MMRP for a single Group.

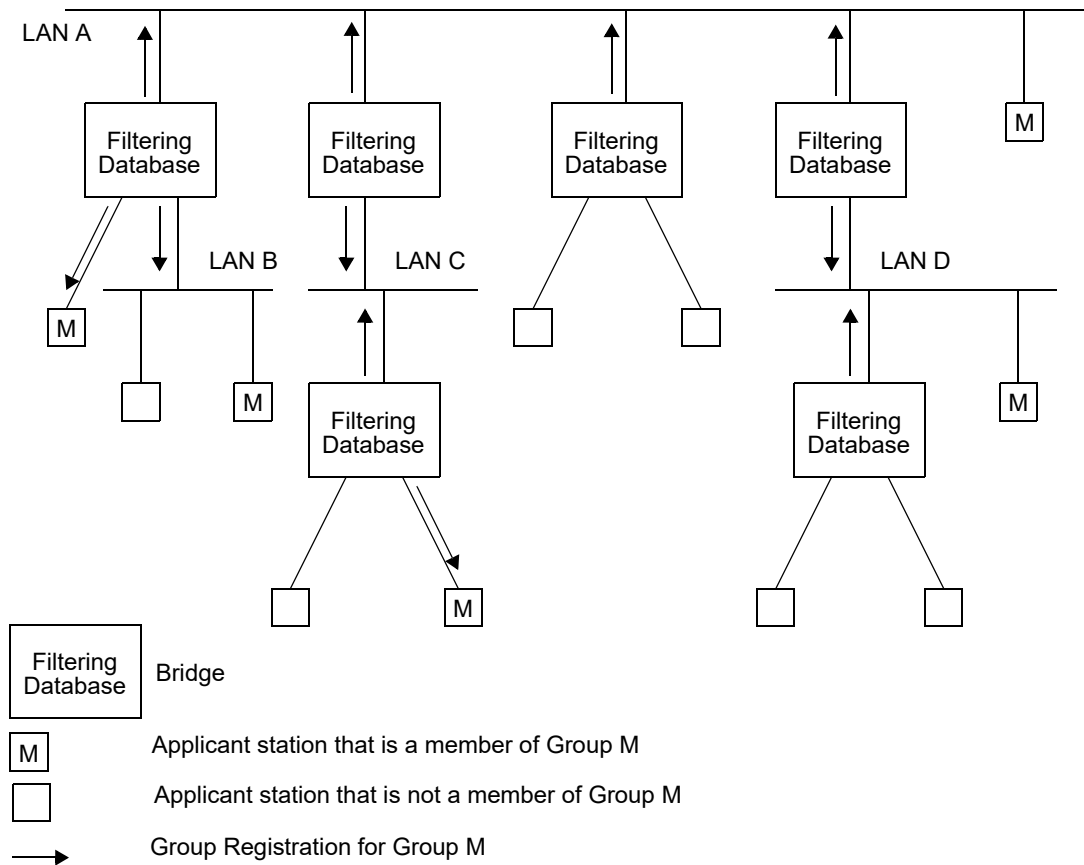


Figure 10-7—Example Directed Graph

5 By receiving frames from all Ports and forwarding only through Ports for which MMRP has created MAC
6 Address Registration Entries, Bridges facilitate Group distribution mechanisms based on the concept of an
7 Open Host Group. Any MMRP Participants (10.2) that wish to receive frames transmitted to a particular
8 Group or Groups request membership of the Group(s) concerned. Any MAC Service user that wishes to
9 send frames to a particular Group can do so from any point of attachment to the Bridged Network. These
10 frames can be received on all LANs to which registered MMRP Participants are attached, but the filtering
11 applied by Bridges ensures that frames are not transmitted on LANs that are not part of the active topology
12 between the sources of the frames and the registered Group members. MMRP and the MAC Address
13 Registration Entries thus restrict the frames to pruned subsets of the overall loop-free active topology.

14 NOTE—The term “Open Host Group” comes from the terminology introduced in the definition of the Internet Group
15 Membership Protocol (IGMP) defined by the IETF.

16 MAC Service users that are sources of MAC frames destined for the Group do not have to register as
17 members of the Group themselves unless they also wish to receive frames transmitted to the Group address
18 by other sources.

10.10.2 Propagation of Group service requirement information

MMRP propagates Group service requirement information in the same manner as for Group registration information. If any Port in a given Bridge has a registered Group service requirement of All Groups or All Unregistered Groups (expressed in terms of the control information in Static Filtering Entries and/or MAC Address Registration Entries with a MAC address specification of All Groups or All Unregistered Groups), this fact is propagated on all other Ports of the Bridge, resulting in the registration of that information on Ports of adjacent Bridges. As a consequence of that registration, the default Group filtering behavior of those Ports can change in order to maintain compatibility with the service requirements expressed by the registered information, as defined in 8.8.6. This ensures that connectivity can be maintained in LANs where the service requirements of different regions of the Bridged Network differ.

NOTE—In a Bridged Network where the default Group filtering behavior is not the same for all “edge” Ports, service requirement propagation will tend to result in all “backbone” Ports switching to the highest precedence Group filtering behavior in use in the network. The precedence rules are defined in 8.8.6.

10.10.3 Source pruning

As described in 10.10.1, the operation of MMRP defines a subtree of the spanning tree as a result of the creation of MAC Address Registration Entries in the FDBs of the Bridges. End stations are also able to make use of the Group Membership information registered via MMRP to allow them to keep track of the set of Groups for which active members currently exist and the service requirements of upstream devices. This allows end stations that are sources of frames destined for a Group to suppress the transmission of such frames, if their registered Group membership and Group service requirement information indicates that there are no valid recipients of those frames reachable via the LANs to which they are attached.

NOTE—In effect, for the purposes of frame transmission, the end station can be viewed as if it operates as a single Port Bridge, with its own default Group filtering behavior and FDB entries updated via MMRP that tell it whether multicast frames that it has generated should be forwarded onto the attached LAN. In order to achieve this, it is necessary for the end station to implement both the Registrar and the Applicant functionality of MRP, as described in 10.6, 10.7.7, and 10.7.8. The Applicant-Only and Simple-Applicant Participants described in 10.6 do not contain the Registrar functionality that would be required for source pruning.

This end system behavior is known as *source pruning*. Source pruning allows MAC Service users that are sources of MAC frames destined for a number of Groups, such as server stations or routers, to avoid unnecessary flooding of traffic on their local LANs in circumstances where there are no current Group members in the network that wish to receive such traffic.

10.10.4 Use of Group service requirement registration by end stations

The ability to propagate Group service requirement information is described in this standard primarily as a means of propagating the requirements of the Bridges themselves. However, this mechanism can also be used by end stations that have requirements involving some aspect of promiscuous reception, such as Routers or network monitors. A MMRP-aware end station wishing to receive all multicast traffic can declare membership of All Groups on the LAN to which it is attached; similarly, an end station that wishes to receive unregistered multicast traffic can do so by declaring membership of All Unregistered Groups.

10.11 Default Group filtering behavior and MMRP propagation

The propagation of MMRP registrations within a VLAN Context has implications with respect to the choice of default Group filtering behavior within a Bridged Network. As MMRP frames are transmitted only on outbound Ports that are in the Member set (8.8.10) for the VID concerned, propagation of Group registrations by a given Bridge occurs only toward regions of the Bridged Network where that VID has been (statically or dynamically) registered. This is illustrated in Figure 10-8; dotted lines in the diagram show those regions of the Bridged Network where propagation of registrations for Group M for VID V does not occur. Consequently, the FDBs of the lower two Bridges will not contain any Dynamic MAC Address Registration Entry for Group M for VID V.

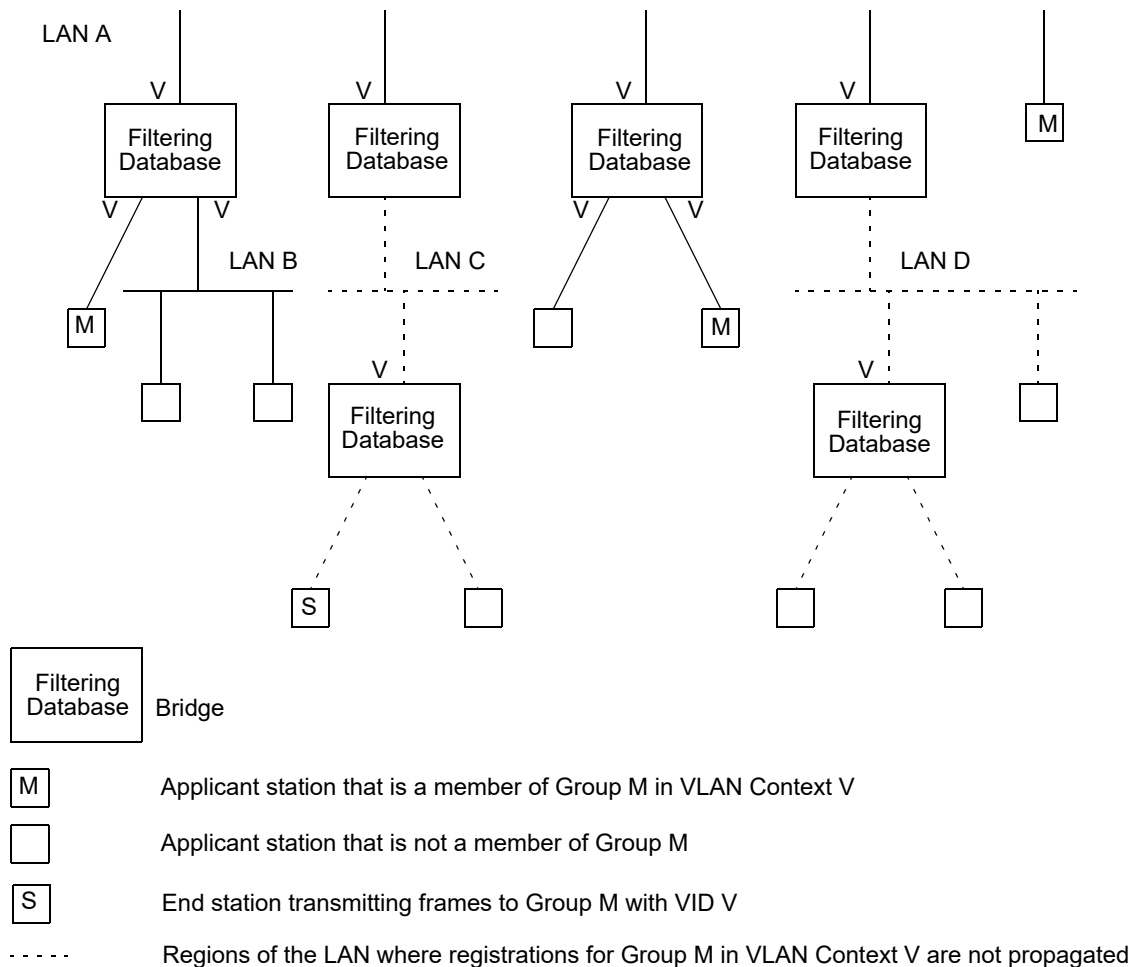


Figure 10-8—Example of MMRP propagation in a VLAN Context

1 The action of these two Bridges on receipt of frames, on either of their lower Ports, destined for Group M
2 and VID V, will depend upon the Default Group Filtering Behavior adopted by their upper Ports, which are
3 the Ports that are in the Member set for VID V. If the Default Group Filtering Behavior is either Forward All
4 Groups or Forward Unregistered Groups, then these Bridges will forward the frames. If the Default Group
5 Filtering Behavior is Filter Unregistered Groups, then these Bridges will filter the frames. In the scenario
6 shown, the choice of Default Group Filtering Behavior is therefore crucial with respect to whether end
7 station S, or any other station that is “outside” the VLAN, is able to send frames to members of the Group.
8 The choice between Filter Unregistered Groups and the other default behaviors therefore has the effect of
9 defining VLANs that are closed to external unregistered traffic (Filter Unregistered Groups) or open to
10 external unregistered traffic (either of the other default behaviors).

1 10.12 Definition of the MMRP application

2 10.12.1 Definition of MRP elements

3 10.12.1.1 Use of MAP Contexts by MMRP

4 MMRP operates within a MAP Context, or set of MAP Contexts, that is determined by the type of Bridge
5 component supporting the MMRP application. In a MAC Bridge component MMRP operates in the
6 Base Spanning Tree Context. In a C-VLAN component or S-VLAN component MMRP operates in the set
7 of C-VLAN Contexts or S-VLAN Contexts, respectively, that corresponds to the VIDs that are supported by
8 the Virtual Bridged Network.

9 The MAP Context Identifier used to identify a VLAN Context shall be equal to the VID used to identify the
10 corresponding VLAN.

11 The set of Ports of a Bridge defined to be part of the active topology for a given VLAN Context shall be
12 equal to the set of Ports of a Bridge for which the following are true:

- 13 a) The Port is a member of the *Member set* (8.8.10) for that VID.
- 14 b) The Port is one of the Ports of the Bridge that are part of the active topology for the spanning tree
15 that supports that VID.

16 NOTE—The above definition applies equally to SST and MST environments. It ensures that MMRP operates in either
17 environment, without the MMRP implementations needing to be aware of whether the VLAN Contexts that apply are all
18 supported by the same spanning tree, as in the SST environment, or are potentially distributed across two or more
19 spanning trees, as in the MST environment.

20 VLAN Contexts can be hierarchical. For example, in a PBN that forwards C-VLAN-tagged frames between
21 CBNs, there can be multiple C-VLAN Contexts within each S-VLAN context. Likewise, there can be
22 multiple C-VLAN Contexts and/or S-VLAN Contexts within the Base Spanning Tree Context. While it is
23 theoretically possible for an MMRP application to support hierarchical MAP Contexts by maintaining an
24 MMRP Participant for each possible (or at least anticipated) combination of C-VLAN, S-VLAN and Base
25 Spanning Tree MAP Contexts, an MMRP application is not required to support hierarchical MAP Contexts.
26 The identification of MMRPDUs belonging to a hierarchical MAP Context, and the treatment of such
27 MMRPDUs, is discussed in 10.12.1.2.

28 10.12.1.2 Context identification in MMRP

29 In a MAC Bridge component, received MMRPDUs that are untagged are identified with the Base Spanning
30 Tree Context and are directed to the MRP Participant for the Base Spanning Tree. Received MMRPDUs that
31 are VLAN-tagged identify hierarchical MAP Contexts. If the MMRP application does not support
32 hierarchical MAP Contexts (i.e., maintain an MRP Participant for each possible S-VLAN and/or C-VLAN
33 Context within the Base Spanning Tree Context), then the received MMRPDUs are not directed to any MRP
34 Participant and are transmitted, including the original VLAN tag(s), on each port in the Base Spanning Tree
35 Context except the reception port.

36 Implementations of MMRP in VLAN Bridges apply the same ingress rules (8.6.2) to received MMRPDUs
37 that are defined for the reception Port. Therefore,

- 38 a) MMRP frames with no VLAN classification (i.e., untagged or priority-tagged MMRPDUs) are
39 discarded if the Acceptable Frame Types parameter (6.9) for the Port is set to *Admit Only VLAN-*
40 *tagged frames*. Otherwise, they are classified either according to the PVID for the Port, or as
41 determined by protocol-based VLAN classification (6.12) if that capability is implemented.
- 42 b) VLAN-tagged MMRP frames are classified according to the VID carried in the tag header,
43 optionally translated using the VID translation table (6.9).
- 44 c) If Ingress Filtering (8.6.2) is enabled, and if the Port is not in the Member set (8.8.10) for the MMRP
45 frame's VLAN classification, then the frame is discarded.

1 The VLAN classification thus associated with a received MMRP frame establishes the C-VLAN Context or
2 S-VLAN Context for the received PDU, depending on whether the reception port is on a C-VLAN
3 component or S-VLAN component, and identifies the MRP Participant instance to which the PDU is
4 directed. MMRPDUs received at an S-VLAN component and containing a C-TAG, either alone or within an
5 S-TAG, identify hierarchical MAP Contexts. If the MMRP application does not support hierarchical MAP
6 Contexts (i.e., maintain an MRP Participant for each possible C-VLAN Context within each S-VLAN
7 Context), then the received MMRPDUs are not directed to any MRP Participant and are transmitted,
8 including the original C-TAG, on each port in the S-VLAN Context except the reception port.

9 MMRPDUs transmitted by MMRP Participants are VLAN classified according to the VLAN Context
10 associated with that Participant. MMRP Participants in VLAN Bridges apply the same egress rules that are
11 defined for the transmission Port (8.6.4). Therefore,

12 d) MMRPDUs are transmitted through a given Port only if the Member Set for the VID concerned
13 indicates that the VID is registered on that Port.

14 e) MMRPDUs are transmitted through a given Port as VLAN-tagged frames or as untagged frames in
15 accordance with the state of the Untagged Set (8.8.10) for that Port and for the VID concerned.

16 Where VLAN-tagged frames are transmitted, the VID field of the tag header carries the VLAN
17 Context Identifier value, optionally translated using the VID translation table (6.9).

18 **10.12.1.3 MMRP application address**

19 The group MAC address used as the destination address for MRPDUs destined for MMRP Participants shall
20 be the group MAC address identified in Table 10-1 as “Customer and Provider Bridge MMRP address.”

21 **10.12.1.4 MMRP application EtherType**

22 The EtherType used for MRPDUs destined for MMRP Participants shall be the MMRP EtherType identified
23 in Table 10-2.

24 **10.12.1.5 MMRP ProtocolVersion**

25 The ProtocolVersion for the version of MMRP defined in this standard takes the hexadecimal value 0x00.

26 **10.12.1.6 MMRP AttributeType definitions**

27 MMRP defines two AttributeTypes (10.8.2.2) that are carried in MRP exchanges, as follows:

- 28 a) The Service Requirement Vector Attribute Type
- 29 b) The MAC Vector Attribute Type

30 Attributes identified by the Service Requirement Vector Attribute Type are instances of VectorAttributes
31 (10.8.1), used to identify values of Group service requirements. The value of AttributeType used to identify
32 the Service Requirement Vector Attribute Type in MRPDUs (10.8.2.2) shall be 1.

33 Attributes identified by the MAC Vector Attribute Type are instances of VectorAttributes (10.8.1), used to
34 identify a sequence of values of MAC addresses. The value of AttributeType used to identify the MAC
35 Vector Attribute Type in MRPDUs (10.8.2.2) shall be 2.

10.12.1.7 MMRP FirstValue definitions

The FirstValue field (10.8.2.7) in instances of the MAC Vector Attribute Type shall be encoded in MRPDUs as six octets, each taken to represent an unsigned binary number. The octets are derived from the Hexadecimal Representation of a MAC Address (defined in IEEE Std 802) as follows:

- a) Each two-digit hexadecimal numeral in the Hexadecimal Representation is taken to represent an unsigned hexadecimal value, in the normal way, i.e., the rightmost digit of each numeral represents the least significant digit of the value, the leftmost digit is the most significant.
- b) The first octet of the attribute value encoding is derived from the left-most hexadecimal value in the Hexadecimal Representation of the MAC address. The LSB of the octet (bit 1) is assigned the LSB of the hexadecimal value, the second significant bit is assigned the value of the second significant bit of the hexadecimal value, and so on.
- c) The second through sixth octets of the encoding are similarly assigned the value of the second through sixth hexadecimal values in the Hexadecimal Representation of the MAC address.

FirstValue + 1 is defined as adding 1 to FirstValue. There are no restrictions on the range of values that can be represented in this data type.

The FirstValue field in instances of the Service Requirement Attribute Type shall be encoded in MRPDUs (10.8.2.7) as a single octet, taken to represent an unsigned binary number. Only two values of this type are defined:

- d) All Groups shall be encoded as the value 0.
- e) All Unregistered Groups shall be encoded as the value 1.

The remaining possible values (2 through 255) are reserved.

10.12.1.8 MMRP AttributeLength definitions

The AttributeLength field (10.8.2.3) in instances of the MAC Vector Attribute Type shall be encoded in MRPDUs (10.8) as an unsigned binary number, equal to the value 0x06.

The AttributeLength field (10.8.2.3) in instances of the Service Requirement Vector Attribute Type shall be encoded in MRPDUs (10.8) as an unsigned binary number, equal to the value 0x01.

10.12.1.9 MMRP AttributeListLength definitions

The AttributeListLength field (10.8.2.4) is not present in the MMRPDUs.

10.12.1.10 MMRP Vector definitions

The ThreePackedEvent vectors are encoded as defined in 10.8.2.10.1.

The FourPackedEvent vectors (10.8.2.10.2) are not present in the MMRPDUs.

10.12.2 Provision and support of Extended Filtering Services

10.12.2.1 Initiating MMRP registration and deregistration

The MMRP application element of an MRP Participant provides the dynamic registration and deregistration services defined in 6.16.7.1, as follows:

On receipt of a REGISTER_MAC_ADDRESS service primitive, the MMRP Participant issues a MAD_Join.request. The attribute_type parameter of the request carries the value of the MAC Vector Attribute Type (10.12.1.6) and the attribute_value parameter carries the value of the MAC_ADDRESS parameter of the service primitive.

1 On receipt of a DEREGISTER_MAC_ADDRESS service primitive, the MMRP Participant issues a
2 MAD_Leave.request. The attribute_type parameter of the request carries the value of the MAC Vector
3 Attribute Type (10.12.1.6) and the attribute_value parameter carries the value of the MAC_ADDRESS
4 parameter of the service primitive.

5 On receipt of a REGISTER_SERVICE_REQUIREMENT service primitive, the MMRP Participant issues a
6 MAD_Join.request. The attribute_type parameter of the request carries the value of the Service Requirement
7 Vector Attribute Type (10.12.1.6) and the attribute_value parameter carries the value of the
8 REQUIREMENT_SPECIFICATION parameter of the service primitive.

9 On receipt of a DEREGISTER_SERVICE_REQUIREMENT service primitive, the MMRP Participant
10 issues a MAD_Leave.request. The attribute_type parameter of the request carries the value of the Service
11 Requirement Vector Attribute Type (10.12.1.6) and the attribute_value parameter carries the value of the
12 REQUIREMENT_SPECIFICATION parameter of the service primitive.

13 NOTE—If the EISS is supported, the Extended Filtering service primitives issued by the MAC Service user should
14 include a VID parameter in addition to the MAC_ADDRESS or REQUIREMENT_SPECIFICATION parameters, in
15 order to identify the MMRP Participant associated with the primitive.

16 10.12.2.2 Registration and deregistration events

17 The MMRP application element of an MRP Participant responds to registration and deregistration events
18 signaled by MAD as follows:

19 On receipt of a MAD_Join.indication, the MMRP application element specifies the Port associated with the
20 MMRP Participant as Forwarding in the Port Map field of the MAC Address Registration Entry (8.8.4) for
21 the MAC address specification carried in the attribute_value parameter and the VID associated with the
22 MAP Context. If such a MAC Address Registration Entry does not exist in the FDB, a new MAC Address
23 Registration Entry is created.

24 Creation of new MAC Address Registration Entries may be restricted by the
25 Restricted_MAC_Address_Registration control (10.12.2.3). If this control is TRUE, creation of a new
26 dynamic entry is permitted only if there is a Static Filtering Entry for the MAC address with a Registrar
27 Administrative Control value of Normal Registration.

28 NOTE—Group Membership and Group service requirement information can be recorded in the FDB by means of MAC
29 Address Registration Entries (see 8.8.4). In the case of Group Membership Information, the MAC address specification
30 in the MAC Address Registration Entry is a group MAC address. In the case of Group service requirement information,
31 the MAC address specification is either “All Group Addresses” or “All Unregistered Group Addresses.”

32 On receipt of a MAD_Leave.indication, the MMRP application element specifies the Port associated with
33 the MMRP Participant as Filtering in the Port Map field of the MAC Address Registration Entry (8.8.4) for
34 the MAC address specification carried in the attribute_value parameter and the VID associated with the
35 MAP Context. If such an FDB entry does not exist in the FDB, then the indication is ignored. If setting that
36 Port to Filtering results in there being no Ports in the Port Map specified as Forwarding (i.e., all MMRP
37 members are deregistered), then that MAC Address Registration Entry is removed from the FDB.

38 10.12.2.3 Administrative controls

39 The provision of static control over the registration state of the state machines associated with the MMRP
40 application is achieved by means of the Registrar Administrative Control parameters associated with the
41 operation of MRP (10.7.2). These parameters are represented in the FDB by the information held in Static
42 Filtering Entries (8.8.1). If no Static Filtering Entry exists for a given MAC address specification, the value
43 of the Registrar Administrative Control parameter for the corresponding attribute value is Normal
44 Registration for all Ports of the Bridge.

1 The initial state of the Permanent Database (i.e., the state of the Permanent Database in a Bridge that has not
2 been otherwise configured by management action) includes a Static Filtering Entry with a MAC address
3 specification of All Groups, in which the Port Map indicates Registration Fixed (New ignored). This Static
4 Filtering Entry will have the effect of determining the default Group filtering behavior of all Ports of the
5 Bridge to be Forward All Groups. This Permanent Database entry may be deleted or updated by
6 management action.

7 NOTE—This specification of an initial Static Filtering Entry means that operation using Forward Unregistered Groups
8 or Filter Unregistered Groups requires a conscious action on the part of the network manager or administrator.

9 Where management capability is implemented, the Registrar Administrative Control parameters can be
10 applied and modified by means of the management functionality defined in 12.7.

11 The provision of static control over the ability of Applicant state machines to participate in protocol
12 exchanges is achieved by means of the Applicant Administrative Control parameters associated with the
13 operation of MRP (10.7.3). Where management capability is implemented, the Applicant Administrative
14 Control parameters can be applied and modified by means of the management functionality defined in
15 Clause 12.

16 Further administrative control over dynamic MAC address registration may be achieved, if supported, by
17 means of a per-Port Restricted_MAC_Address_Registration control parameter. If the value of this control is
18 TRUE for a given Port, the creation or modification of Dynamic MAC Address Registration Entries as a
19 result of MMRP exchanges on that Port shall be restricted only to those MAC addresses for which Static
20 Filtering Entries exist in which the Registrar Administrative Control value is Normal Registration. If the
21 value of the Restricted_MAC_Address_Registration control is FALSE, dynamic MAC address registration
22 is not so restricted. Where management capability is implemented, the value of the
23 Restricted_MAC_Address_Registration control can be manipulated by means of the management
24 functionality defined in Clause 12. If management of this parameter is not supported, the value of this
25 parameter shall be FALSE for all Ports.

26 **10.12.3 Use of “new” declaration capability**

27 MMRP does not make use of the “new” declaration capability.

28 **10.12.4 Attribute value support requirements**

29 Implementations of MMRP shall maintain state information for all attribute values that support the Group
30 service requirement registration (10.12.1.7).

31 Implementations of MMRP shall be capable of supporting any attribute value in the range of possible values
32 that can be registered using Group membership and individual MAC address registration (10.12.1.7);
33 however, the maximum number of attribute values for which the implementation is able to maintain current
34 state information is an implementation decision. The number of values that the implementation can support
35 shall be stated in the PICS.

36 **10.12.5 Registrar Administrative Controls**

37 MMRP supports the Registration Fixed (New ignored) value, but not the Registration Fixed (New
38 propagated) value, of the Registrar Administrative Controls (10.7.2).

11. VLAN topology management

The egress rules (8.6.4) defined for the Forwarding Process in VLAN Bridges rely on the existence of configuration information for each VID that defines the set of Ports of the Bridge through which one or more members are reachable. This set of Ports is known as the Member Set (8.8.10), and its membership is determined by the presence or absence of configuration information in the FDB, in the form of Static and Dynamic VLAN Registration Entries (8.8.2, 8.8.5).

Reliable operation of the VLAN infrastructure requires VLAN membership information held in the FDB to be maintained in a consistent manner across all VLAN Bridges in the Virtual Bridged Network, in order to ensure that frames destined for end station(s) on a given VLAN can be correctly delivered, regardless of where in the Bridged Network the frame is generated. Maintenance of this information by end stations that are sources of VLAN-tagged frames can allow such stations to suppress transmission of such frames if no members exist for the VLAN concerned.

This standard defines the following mechanisms that allow VLAN membership information to be configured:

- a) Dynamic configuration and distribution of VLAN membership information by means of the MVRP, as described in 11.2.
- b) Static configuration of VLAN membership information via Management mechanisms, as described in Clause 12, which allow configuration of Static VLAN Registration Entries.

These mechanisms provide for the configuration of VLAN membership information as a result of the following:

- c) Dynamic registration actions taken by end stations or Bridges in the Bridged Network.
- d) Administrative actions.

11.1 Static and dynamic VLAN configuration

The combined functionality provided by the ability to configure Static VLAN Registration Entries in the FDB, coupled with the use of the Restricted_VLAN_Registration control (11.2.3.2.3) and the ability of MVRP to dynamically create and update Dynamic VLAN Registration Entries, offers the following possibilities with respect to how VIDs are configured on a given Port:

- a) *Static configuration only.* The management facilities described in Clause 12 are used to establish precisely which VIDs have this Port in their Member set, and the MVRP management controls are used to disable the operation of MVRP on that Port. Hence, any use of MVRP by devices reachable via that Port is ignored, and the Member set for all VIDs can therefore only be determined by means of static entries in the FDB.
- b) *Static configuration with topology changes.* The management facilities described in Clause 12 are used to establish precisely which VLANs have this Port in their Member set, and the MVRP or MIRP management controls are used to permit the signaling, via MVRP or MIRP, of network topology changes that require flushing learned address information from the FDB. MVRPDUs and MIRPDUs received from devices reachable via that Port are ignored for the purposes of creating or deleting Dynamic VLAN Registration Entries in the FDB on that Port, and the Member set for all VLANs can therefore only be determined by means of static entries in the FDB.
- c) *Dynamic configuration only.* The operation of MVRP is relied upon to establish Dynamic VLAN Registration Entries that will dynamically reflect which VIDs are registered on the Port, their contents changing as the configuration of the network changes. The MVRP management controls are set to enable the operation of MVRP on that Port.
- d) *Combined static and dynamic configuration.* The static configuration mechanisms are used in order to configure some VLAN membership information; for other VIDs, MVRP is relied upon to establish the configuration. The MVRP management controls are set to enable the operation of MVRP on that Port.

1 All of the previous approaches are supported by the mechanisms defined in this standard, and each approach
2 is applicable in different circumstances. For example:

- 3 e) Use of static configuration may be appropriate on Ports where the configuration of the attached
4 devices is fixed, or where the network administrator wishes to establish an administrative boundary
5 outside of which any MVRP registration information is to be ignored. For example, it might be
6 desirable for all Ports serving end user devices to be statically configured in order to ensure that
7 particular end users have access only to VLANs identified by particular VIDs.
- 8 f) Use of static configuration with topology changes can be appropriate where the configuration of
9 VLANs and ports is static in some part of the network, but dynamic in another part, so that changes
10 in the topology of the dynamic part can necessitate the flushing of learned MAC address information
11 in Bridges in the static part. For example, in a PBBN, a topology change in a customer network with
12 dual attachments to the provider can require a provider's I-component to signal a MAC address flush
13 across the backbone, even though the VLANs on the VIPs involved are statically configured.
- 14 g) Use of dynamic configuration may be appropriate on Ports where the VLAN configuration is
15 inherently dynamic; where users of particular VIDs can connect to the network via different Ports on
16 an ad hoc basis, or where it is desirable to allow dynamic reconfiguration in the face of spanning tree
17 topology changes. In particular, if the "core" of the Virtual Bridged Network contains redundant
18 paths that are pruned by the operation of the spanning tree protocol, then it is desirable for Bridge
19 Ports that form the core network to be dynamically configured.
- 20 h) Use of both static and dynamic configuration can be appropriate on Ports where it is desirable to
21 place restrictions on the configuration of some VIDs, while maintaining the flexibility of dynamic
22 registration for others. For example, on Ports serving mobile end user devices, this would maintain
23 the benefits of dynamic VLAN registration from the point of view of traffic reduction, while still
24 allowing administrative control over some VIDs via that Port.

25 11.2 Multiple VLAN Registration Protocol (MVRP)

26 MVRP defines an *MRP application* that provides the VLAN registration service defined in 11.2.2. MVRP
27 makes use of MRP Attribute Declaration (MAD) and MRP Attribute Propagation (MAP), which provide the
28 common state machine descriptions and the common attribute propagation mechanisms defined for use in
29 MRP-based applications. The MRP architecture, MAD, and MAP are defined in Clause 10.

30 MVRP provides a mechanism for dynamic maintenance of the contents of Dynamic VLAN Registration
31 Entries for each VID, and for propagating the information they contain to other Bridges. This information
32 allows MVRP-aware devices to dynamically establish and update their knowledge of the set of VIDs
33 associated with VLANs that currently have active members, and through which Ports those members can be
34 reached.

35 11.2.1 MVRP overview

36 The operation of MVRP is closely similar to the operation of MMRP (10.9), which is used for registering
37 Group membership and individual MAC address information. The primary differences are as follows:

- 38 a) The attribute values carried by the protocol are 12-bit VID values, rather than Group service
39 requirement information and MAC Addresses.
- 40 b) The act of registering/deregistering a VID affects the contents of Dynamic Registration Entries
41 (8.8.5), rather than the contents of MAC Address Registration Entries (8.8.4).
- 42 c) In an SST environment, there is a single MVRP Participant per port, as opposed to one MMRP
43 Participant per VID per port, and the MVRP Participants all operate in a single MAP Context.
- 44 d) In an MST environment, there is again a single MVRP Participant per port, but each MVRP
45 Participant operates in multiple MAP Contexts.

1 MVRP allows both end stations and Bridges in a Bridged Network to issue and revoke declarations relating
2 to membership of VLANs. The effect of issuing such a declaration is that each MVRP Participant that
3 receives the declaration will create or update a Dynamic VLAN Registration Entry in the FDB to indicate
4 that VID is registered on the reception Port. Subsequently, if all Participants on a LAN that had an interest in
5 a given VID revoke their declarations, the Port attached to that LAN is set to Unregistered in the Dynamic
6 VLAN Registration Entry for that VID by each MVRP Participant attached to that LAN.

7 Figure 11-1 illustrates the architecture of MVRP in the case of a two-Port Bridge and an end station.

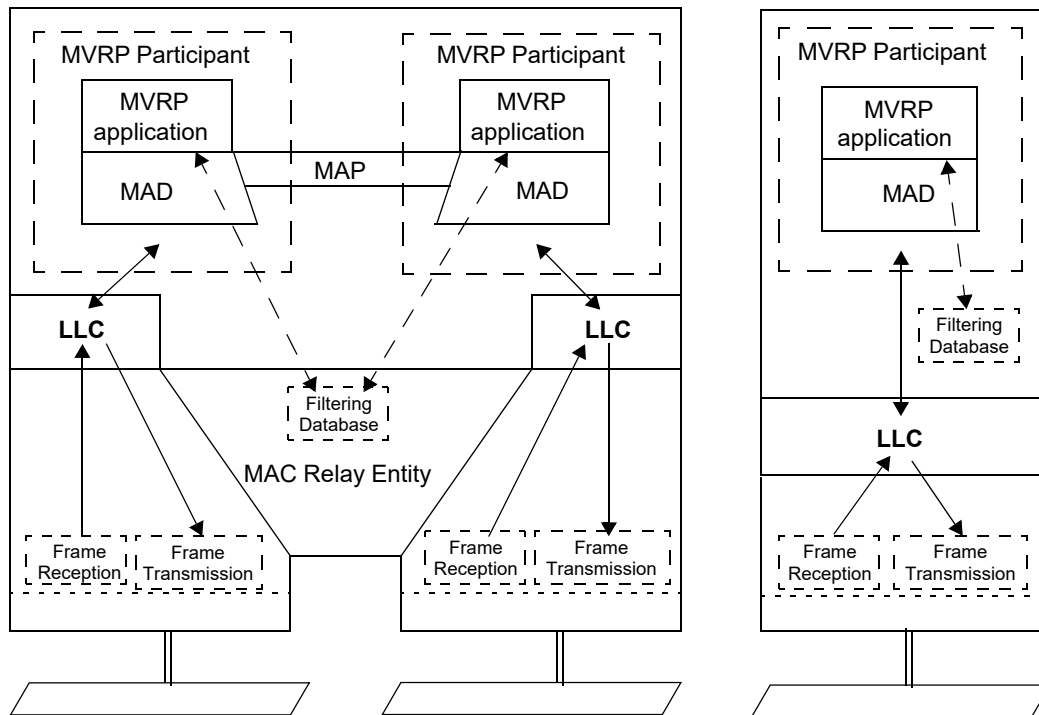


Figure 11-1—Operation of MVRP

8 As shown in the diagram, the MVRP Participant consists of the following components:

- 9 e) The MVRP application, described in 11.2.3
- 10 f) MRP Attribute Propagation (MAP), described in 10.3
- 11 g) MRP Attribute Declaration (MAD), described in 10.2

12 11.2.1.1 Behavior of end stations

13 VLAN-aware end stations participate in MVRP activity, as appropriate for the set of VIDs of VLANs of
14 which they are currently members. MVRP provides a way for such an end station to ensure that the VIDs of
15 VLAN(s) of which it is a member are registered for each Port on any LAN to which the end station is
16 attached. MVRP also provides for that VID information to be propagated across the spanning tree to other
17 VLAN-aware devices, as described in 11.2.1.2.

1 Incoming VLAN membership information (from all other devices on the same LAN) allows such end
2 stations to “source prune” (i.e., discard at source; see 10.10.3) any traffic destined for VLANs that currently
3 have no other members in the Bridged Network, thus avoiding the generation of unnecessary traffic on their
4 local LANs. This is illustrated in Figure 11-1 by an FDB shown as being present in the end station.

5 NOTE—Non-VLAN-aware end stations have no need to register VLAN membership via MVRP; indeed, this would be
6 impossible for them to achieve if truly VLAN-unaware, as they would have no knowledge of the set of VLANs in which
7 they participate. Their VLAN registration requirements are taken care of by means of the configuration of PVIDs (and
8 possibly other VLAN classification mechanisms) and the propagation of registered VIDs by the Bridges.

9 11.2.1.2 Behavior of Bridges

10 VLAN Bridges register and propagate VLAN memberships on all Bridge Ports that are part of the active
11 topology of the underlying spanning tree(s). Incoming VID registration and deregistration information is
12 used to update the Dynamic VLAN Registration Entries associated with each VID. Any changes in the state
13 of registration of a given VID on a given Port are propagated on Ports that are part of the active topology of
14 the underlying spanning tree, in order to ensure that other MVRP-aware devices in the Bridged Network
15 update their FDBs appropriately. In Bridges that support multiple spanning tree instances, the MST
16 Configuration Table (12.12, 13.8) is used to determine which spanning tree instance is to be used to
17 propagate registration information for each supported VID.

18 The FDBs in all MVRP-aware devices are thus automatically configured such that the Port Map in the
19 Dynamic VLAN Registration Entry for a given VID indicates that a given Port is registered if one or more
20 members of the corresponding VLAN are reachable through the Port.

21 NOTE—The information that determines whether frames destined for each VID are transmitted VLAN-tagged or
22 untagged is carried in Static VLAN Registration Entries (8.8.2); if no such entry exists for a VID, then it is assumed that
23 frames for that VID are transmitted VLAN-tagged on all Ports. Therefore, if the configuration information held in the
24 FDB for a given VID consists only of information configured by the operation of MVRP (i.e., only a Dynamic VLAN
25 Registration Entry), then all traffic for that VID will be VLAN-tagged on transmission.

26 11.2.1.3 Use of the PVID and VID Set

27 The initial state of the Permanent Database contains a Static VLAN Registration Entry for the Default PVID,
28 in which the Port Map indicates Registration Fixed (New ignored) on all Ports. This ensures that in the
29 default state, where the value of every PVID of each Port is the Default PVID and where the VID Set of each
30 Port is empty, membership of the Default PVID is propagated across the Bridged Network to all other
31 MVRP-aware devices. Subsequent management action may change both the Permanent Database and the
32 FDB in order to modify or remove this initial setting, and may change the PVID and/or VID Set value(s) on
33 any Port of the Bridge.

34 NOTE—In the absence of any modification of these initial settings, this ensures that connectivity is established across
35 the Bridged Network for the VLAN corresponding to the Default PVID.

36 11.2.2 VLAN registration service definition

37 The VLAN registration service allows MAC Service users to indicate to the MAC Service provider the set
38 of VLANs in which they wish to participate; i.e., that the MAC Service user wishes to receive traffic
39 destined for members of that set of VLANs. The service primitives allow the service user to:

- 40 a) Register membership of a VLAN.
- 41 b) Deregister membership of a VLAN.

42 Provision of these services is achieved by means of MVRP and its associated procedures, as described in
43 11.2.3.

44 ES_REGISTER_VLAN_MEMBER (VID)

45 Indicates to the MAC Service provider that the MAC Service user wishes to receive frames destined for the
46 VLAN identified by the VID parameter.

1 ES_DEREGISTER_VLAN_MEMBER (VID)

2 Indicates to the MAC Service provider that the MAC Service user no longer wishes to receive frames
3 destined for the VLAN identified by the VID parameter.

4 The use of these services can result in the propagation of VID information across the spanning tree, affecting
5 the contents of Dynamic VLAN Registration Entries (8.8.5) in Bridges and end stations in the Bridged
6 Network, and thereby affecting the frame forwarding behavior of those Bridges and end stations.

7 **11.2.3 Definition of the MVRP application**

8 **11.2.3.1 Definition of MRP elements**

9 **11.2.3.1.1 MAP Context for MVRP in SST environments**

10 In an SST environment, MVRP operates in the Base Spanning Tree Context (10.3.1); i.e., MVRP operates
11 only on the CIST. Consequently, all MVRPDUs sent and received by MVRP Participants in SST Bridges are
12 transmitted as untagged frames.

13 **11.2.3.1.2 MAP Contexts for MVRP in MST environments**

14 In an MST environment, MVRP operates in multiple spanning tree contexts, one for each of the spanning
15 trees. Each spanning tree context consists of the ports that are in the forwarding state for the corresponding
16 spanning tree. The MVRP Participants associated with a given spanning tree operate in the MAP context
17 identified by that spanning tree instance. MST Bridges can identify the MAP Contexts using the mappings
18 of VID values to MSTID values (see 8.9.1). All MVRPDUs sent and received by MVRP Participants in
19 MST Bridges are transmitted as untagged frames.

20 **11.2.3.1.3 MVRP application address**

21 The group MAC address used as the destination address for MRPDUs destined for MVRP Participants shall
22 be either:

- 23 a) The group MAC address identified in Table 10-1 as “Customer Bridge MVRP address” if the
24 MVRP Participant is implemented in a C-VLAN component; or
- 25 b) The group MAC address identified in Table 8-1 as “Provider Bridge MVRP Address” if the MVRP
26 Participant is implemented in an S-VLAN component.

27 **11.2.3.1.4 MVRP application EtherType**

28 The EtherType used for MRPDUs destined for MVRP Participants shall be the MVRP EtherType identified
29 in Table 10-2.

30 **11.2.3.1.5 MVRP ProtocolVersion**

31 The ProtocolVersion for the version of MVRP defined in this standard takes the hexadecimal value 0x00.

32 **11.2.3.1.6 MVRP AttributeType definitions**

33 MVRP defines a single Attribute Type (10.8.2.2) that is carried in MRP exchanges, as follows:

34 Attributes identified by the VID Vector Attribute Type are instances of VectorAttributes (10.8.1), used to
35 identify a sequence of values of VIDs. The value of AttributeType used to identify the VID Vector Attribute
36 Type in MRPDUs (10.8.2.2) shall be 1.

11.2.3.1.7 MVRP FirstValue definitions

The FirstValue field in instances of the VID Vector Attribute Type shall be encoded in MRPDUs (10.4) as two octets, taken to represent an unsigned binary number, and equal to the value of the VID that is to be encoded.

The range of permitted values that can be encoded in the FirstValue fields in MVRP is restricted to the range 0 through 4094. The value 0 represents the VID of the untagged frame on which the MRPDUs are transmitted or received, and 1 through 4094 identify specific VIDs. If enabled by management [item e) in 12.9.2.2.2], and if there is only one VID whose untagged set includes the Port of an MVRP Participant, the Participant shall transmit the value 0 in place of that VID. All MVRP Participants shall translate the value 0 in a received message to the PVID of the receiving Port.

NOTE—Previous revisions of this Standard did not provide for transmission or reception of 0 as an MVRP Attribute value. No automatic means for determining whether the other MVRP Participants on a LAN are capable of receiving the value 0 are provided. Therefore, extreme care is in order when configuring an MVRP Participant to transmit Attribute value 0, as compliant implementations of previous revisions of this standard could drop such MRPDUs as badly formed.

11.2.3.1.8 MVRP AttributeLength definitions

The AttributeLength field (10.12.1.8) in instances of the VID Vector Attribute Type shall be encoded in MRPDUs (10.8) as an unsigned binary number, equal to the value 0x02.

11.2.3.1.9 MVRP AttributeListLength definitions

The AttributeListLength field (10.8.2.4) is not present in the MRPDUs.

11.2.3.1.10 MVRP Vector definitions

The ThreePackedEvent vectors are encoded as defined in 10.8.2.10.1.

The FourPackedEvent vectors (10.8.2.10.2) are not present in the MRPDUs.

11.2.3.2 Provision and support of the VLAN registration service

11.2.3.2.1 Initiating VLAN membership declaration

The MVRP application element of a MVRP Participant provides the dynamic registration and deregistration services defined in 11.2.2, as follows:

On receipt of an ES_REGISTER_VLAN_MEMBER service primitive, the MVRP Participant issues a MAD_Join.request service primitive (10.2, 10.3). The attribute_type parameter of the request carries the value of the VID Vector Attribute Type (11.2.3.1.6) and the attribute_value parameter carries the value of the VID parameter carried in the ES_REGISTER_VLAN_MEMBER primitive.

On receipt of an ES_DEREGISTER_VLAN_MEMBER service primitive, the MVRP Participant issues a MAD_Leave.request service primitive (10.2, 10.3). The attribute_type parameter of the request carries the value of the VID Vector Attribute Type (11.2.3.1.6) and the attribute_value parameter carries the value of the VID parameter carried in the ES_DEREGISTER_VLAN_MEMBER primitive.

11.2.3.2.2 VLAN membership registration

The MVRP application element of a MVRP Participant responds to registration and deregistration events signaled by MAD as follows:

On receipt of a MAD_Join.indication whose attribute_type is equal to the value of the VID Vector Attribute Type (11.2.3.1.6), the MVRP application element indicates the reception Port as Registered in the Port Map of the Dynamic VLAN Registration Entry for the VID indicated by the attribute_value parameter. If no such entry exists, there is sufficient room in the FDB, and the VID is within the range of values supported by the implementation (see 9.6), then an entry is created. If not, then the indication is not propagated and the registration fails.

The creation of new Dynamic VLAN Registration Entries can be restricted by use of the Restricted_VLAN_Registration control (11.2.3.2.3). If the value of this control is TRUE, then creation of a new dynamic entry is permitted only if there is a Static VLAN Registration Entry for the VID concerned, in which the Registrar Administrative Control value is Normal Registration.

On receipt of a MAD_Leave.indication whose attribute_type is equal to the value of the VID Vector Attribute Type (11.2.3.1.6), the MVRP application element indicates the reception Port as not Registered in the Port Map of the Dynamic VLAN Registration Entry for the VID indicated by the attribute_value parameter. If no such entry exists, the indication is ignored.

11.2.3.2.3 Administrative controls

The provision of static control over the declaration or registration state of the state machines associated with the MVRP application is achieved by means of the Registrar Administrative Control parameters provided by MRP (10.7.2). These parameters are represented as Static VLAN Registration Entries in the FDB (8.8.2). Where management capability is implemented, these parameters can be manipulated by means of the management functionality defined in 12.7.

The provision of static control over the ability of Applicant and Registrar state machines to participate in protocol exchanges is achieved by means of the Applicant Administrative Control parameters and Registrar Administrative Control parameters associated with the operation of MRP (10.7.2, 10.7.3). Where management capability is implemented, the Applicant and Registrar Administrative Control parameters can be applied and modified by means of the management functionality defined in 12.9. A separate managed variable [item c) in 12.9.2.1.3] controls whether the Applicant state machine transmits 0 or the VID as the Attribute for a VLAN on a Port in that VLAN's untagged set.

Further administrative control over dynamic VLAN registration can be achieved, if supported, by means of a per-Port Restricted_VLAN_Registration control parameter. If the value of this control is TRUE for a given Port, the creation or modification of Dynamic VLAN Registration Entries as a result of MVRP exchanges on that Port shall be restricted only to those VIDs for which Static VLAN Registration Entries exist in which the Registrar Administrative Control value is Normal Registration. If the value of the Restricted_VLAN_Registration control is FALSE, dynamic VLAN registration is not so restricted. The recommended default value of this parameter is FALSE. Where management capability is implemented, the value of the Restricted_VLAN_Registration control can be manipulated by means of the management functionality defined in 12.10. If management of this parameter is not supported, the value of this parameter shall be FALSE for all Ports.

11.2.4 VID translation table

If a VID Translation Table (6.9) is in use for a Bridge Port, the VID values received in MVRP Attributes are translated on reception of MVRPDUs prior to MVRP processing as specified in this subclause (11.2), and translated after processing for encoding MVRP Attributes in transmitted MVRPDUs. The interpretation of the MVRP Attribute value 0 is not affected by the VID Translation Table in either direction.

NOTE—In the case that the Port on which an MVRPDU is transmitted is in the untagged set of the VID matching the PVID of the Port receiving the MVRPDU, the use of the value 0 in the FirstValue field effectively performs a VID translation.

11.2.5 Use of “new” declaration capability

MVRP makes use of the “new” declaration capability in order to allow FDB entries to be removed or rapidly aged out on a per-VID basis following a topology change in the network connectivity supporting the VID concerned. The rules for propagation of “new” declarations as defined in Clause 10 ensure that all MADs that originate from, or are propagated through, a Bridge Port that has recently transitioned to Forwarding are marked as “new” declarations.

When any MVRP declaration marked as “new” is received on a given Port, either as a result of receiving an MVRPDU from the attached LAN (MAD_Join.indication), or as a result of receiving a request from MAP or the MVRP Application (MAD_Join.request), any Dynamic Filtering Entries in the FDB for that Port and for the VID corresponding to the attribute value in the MAD_Join primitive are removed.

11.2.6 New-Only Participant and Registrar Administrative Controls

A Bridge may support an MVRP New-Only Participant (10.6). This can be useful in an I-component that has statically configured S-VLAN registrations, but must still accommodate signaling the flushing of learned MAC address information. The choice of whether a given Port’s MVRP Participant operates as a Full Participant or a New-Only Participant is controlled by a managed object (12.9.2). This managed object is separate from the one controlling whether a given Port uses Registration Fixed (New ignored) or Registration Fixed (New propagated) Static VLAN Registration Entries [item d) in 12.7.7.3.3].

NOTE 1—Configuring an MVRP Participant as a New-Only Participant makes the assumption that all of the MVRP Participants on the LAN are configured completely via Static VLAN Registration Entries, and that no Dynamic VLAN Registration Entries are needed. If this assumption is violated, then the Bridged Network can fail to provide connectivity for the VLANs that need Dynamic VLAN Registration Entries. Therefore, extreme care is in order when configuring a New-Only MVRP Participant.

If a MAD_Join.indication is received by the MVRP MAD with the *new* parameter set, e.g., because a “new” MVRP declaration is received on a Port configured for New propagated [item d) in 12.7.7.3.3] as a result of receiving an MVRPDU from the attached LAN, and that MAD_Join.indication does not correspond to a Static VLAN Registration Entry of Registration Fixed (New propagated) in the FDB, that MAD_Join.indication shall be discarded without being propagated.

NOTE 2—If only one S-VLAN is configured per VIP, then using a MVRP Participant (Clause 39) can provide bandwidth and processing savings over using an MVRP New-Only Participant on a VIP. See 39.2.1.1 and 39.2.1.6.

11.2.7 Attribute value support requirements

Implementations of MVRP shall be capable of supporting all attribute values in the range of possible values that can be registered using MVRP, and shall be capable of maintaining current state information for all attributes in the range of possible values.

12. Bridge management

This clause defines the set of managed objects, and their functionality, that allow administrative configuration of VLANs.

This clause:

- a) Introduces the functions of management to assist in the identification of the requirements placed on Bridges for the support of management facilities.
- b) Establishes the correspondence between the Processes used to model the operation of the Bridge (8.3) and the managed objects of the Bridge.
- c) Specifies the management operations supported by each managed object.

12.1 Management functions

Management functions relate to the users' needs for facilities that support the planning, organization, supervision, control, protection, and security of communications resources, and account for their use. These facilities may be categorized as supporting the functional areas of Configuration, Fault, Performance, Security, and Accounting Management. Each functional area is summarized in 12.1.1 through 12.1.5, together with the facilities commonly required for the management of communication resources, and the particular facilities provided in that functional area by Bridge Management.

It can be necessary, particularly in PBNs, for multiple organizations to manage a single Bridge, with each organization having different abilities to manage, or even detect the existence of, Bridge Ports, Bridges, or entire networks. Access to all of the managed objects in Clause 12 for reading, writing, or detection can be restricted. Access by certain organizations can be limited to only a small number of managed objects, particularly to those specified in 12.14.

12.1.1 Configuration Management

Configuration Management provides for the identification of communications resources, initialization, reset and close-down, the supply of operational parameters, and the establishment and discovery of the relationship between resources. The facilities provided by Bridge Management in this functional area are:

- a) The identification of all Bridges that together make up the network and their respective locations and, as a consequence of that identification, the location of specific end stations to particular individual LANs.
- b) The ability to remotely reset, i.e., reinitialize, specified Bridges.
- c) The ability to control the priority with which a Bridge Port transmits frames.
- d) The ability to force a specific configuration of a spanning tree.
- e) The ability to control the propagation of frames with specific group MAC addresses to certain parts of the configured network.
- f) The ability to identify the VIDs in use, and through which Ports of the Bridge and for which protocols frames destined for a given VID may be received and/or forwarded.
- g) The ability to create and delete the functional elements of CFM and to control their operation.
- h) The ability to create and delete the functional elements of DDCFM and to control their operations.
- i) The ability to create and delete the functional elements of congestion notification and to control their operation.
- j) The ability to control the operation of MIRP.
- k) The ability to control the operation of SPB.
- l) The ability to configure the functional elements of EVB and to control their operation.

- 1 m) The ability to create and delete the functional elements of congestion isolation and to control their
2 operation.

3 **12.1.2 Fault Management**

4 Fault Management provides for fault prevention, detection, diagnosis, and correction. The facilities provided
5 by Bridge Management in this functional area are:

- 6 a) The ability to identify and correct Bridge malfunctions, including error logging and reporting.
7 b) Within the context of multiple management organizations, the ability to:
8 1) Actively monitor the connectivity of a set of managed endpoints on individual VLANs,
9 simultaneously over multiple physical extents.
10 2) Actively monitor and ensure the segregation of data among different VLANs.
11 3) Issue trains of point-to-point query-response messages to selected network components in a
12 VLAN.
13 4) Issue multicast query-relay-response messages to determine the path taken by frames addressed
14 to specific individual MAC addresses through a VLAN.
15 5) Determine whether a flow of specified data frames (e.g., frames associated with a service
16 instance or selected data frames with certain destination address) can reach a particular
17 destination.
18 6) Determine whether a flow of data frames can be sent without error from a specific location
19 within a network to a station or stations specified by the DA field of each frame.

20 **12.1.3 Performance Management**

21 Performance Management provides for evaluation of the behavior of communications resources and of the
22 effectiveness of communication activities. The facilities provided by Bridge Management in this functional
23 area are as follows:

- 24 a) The ability to gather statistics relating to performance and traffic analysis. Specific metrics include
25 network utilization, frame forward, and frame discard counts for individual Ports within a Bridge.

26 **12.1.4 Security Management**

27 Security Management provides for the protection of resources. Bridge Management does not provide any
28 specific facilities in this functional area.

29 **12.1.5 Accounting Management**

30 Accounting Management provides for the identification and distribution of costs and the setting of charges.
31 Bridge Management does not provide any specific facilities in this functional area.

32 **12.2 VLAN Bridge objects**

33 Managed objects model the semantics of management operations. Operations on an object supply
34 information concerning, or facilitate control over, the process or entity associated with that object.

35 The managed resources of a VLAN Bridge or Bridge Component are those of the processes and entities in
36 8.3. Specifically,

- 37 a) The Bridge Management Entity (12.4 and 8.12)
38 b) The individual MAC Entities associated with each Bridge Port (12.5, 8.2, and 8.5)
39 c) The Forwarding Process of the MAC Relay Entity (12.6, 8.2, and 8.6)

- 1 d) The FDB of the MAC Relay Entity (12.7 and 8.8)
- 2 e) The Bridge Protocol Entity (12.8 and 8.10)
- 3 f) MRP participants (Clause 10)
- 4 g) MVRP participants (12.10, Clause 11)
- 5 h) MMRP participants (12.11, Clause 10)
- 6 i) The MST Configuration Table (12.12)
- 7 j) Additional objects to support Provider Bridge management (12.13)
- 8 k) The CFM Entities (12.14)
- 9 l) The DDCFM Entities (12.17)
- 10 m) The PBB-TE protection switching Entities (12.18)
- 11 n) The TPMR Entities (12.19)
- 12 o) The management entities for FQTSS (12.20)
- 13 p) The congestion notification entities (12.20.4)
- 14 q) The Shortest Path Bridging managed objects
- 15 r) The EVB entities (12.26)
- 16 s) The congestion isolation entities (12.33)

17 The management of each of these resources is described in terms of managed objects and operations in 12.4
18 through 12.12.

19 NOTE—The values specified in this clause, as inputs and outputs of management operations, are abstract information
20 elements. Questions of formats or encodings are a matter for particular protocols that convey or otherwise represent this
21 information.

22 Some data elements that are represented by the managed objects specified in this clause are required to be
23 persistent across reinitializations or rebooting of the system within which the objects are implemented. For
24 example, it may be desirable for changes in configuration parameters to persist across such events. Where
25 such a requirement exists, this is indicated in the specification by marking the object concerned with the
26 keyword “*Persistent*.”

27 12.3 Data types

28 This subclause specifies the semantics of operations independent of their encoding in management protocol.
29 The data types of the parameters of operations are defined only as required for that specification.

30 The following data types are used:

- 31 a) Boolean.
- 32 b) Enumerated, for a collection of named values.
- 33 c) Unsigned, for all parameters specified as “the number of” some quantity, and for spanning tree
34 priority values that are numerically compared. When comparing spanning tree priority values, the
35 lower number represents the higher priority value.
- 36 d) MAC address.
- 37 e) Latin1 String, as defined by ANSI X3.159, for all text strings.
- 38 f) Time Interval, an Unsigned value representing a positive integral number of seconds, for all
39 spanning tree protocol timeout parameters.
- 40 g) Counter, for all parameters specified as a “count” of some quantity. A counter increments and wraps
41 with a modulus of 2 to the power of 64.
- 42 h) MRP Time Interval, an Unsigned value representing a positive integral number of centiseconds, for
43 all MRP timeout parameters.

- 1 i) Port Number, an Unsigned value assigned to a Port as part of a Port Identifier. Valid Port Numbers
2 are in the range 1 through 4095.
- 3 j) Port Priority, an Unsigned value used to represent the priority component of a Port Identifier. Valid
4 Port Priorities are in the range 0 through 240, in steps of 16.
- 5 k) Bridge Priority, an Unsigned value used to represent the priority component of a Bridge Identifier.
6 Valid Bridge Priorities are in the range 0 through 61440, in steps of 4096.
- 7 l) ComponentID, an unsigned value used to uniquely identify the management objects for a particular
8 VLAN Bridge Component (12.2, Clause 8, 5.4) within a system (such as a BEB) comprising
9 multiple such components. ComponentIDs start at 1 and go through 4294967295. If the system has a
10 single component it will have a ComponentID equal to 1.
- 11 m) ComponentType, an enumerated list used to classify a particular VLAN Bridge Component within a
12 system comprising multiple components.
- 13 n) Port Index, a handle, unique within a system, that identifies a port.
- 14 o) PIP Index, a Port Index for a PIP.
- 15 p) Percentage.
- 16 q) ECT-ALGORITHM. A 4-byte unsigned identifier. Used as a worldwide unique definition of an
17 Equal Cost Tree (ECT) Algorithm, the first 3 bytes are expected to be taken from the OUI or CID
18 space for the organization that has defined the algorithm. The last byte is allocated by that
19 organization.
- 20 r) SPSourceID. A 20-bit Unsigned identifier. Used to represent a node uniquely within an SPT Domain
21 (27.10).
- 22 s) Timer exp, an unsigned value from 0–31 representing a positive integer for the exponent of 2, which
23 forms the multiplier of 10 μ s, used for EVB protocol timeout parameters.
24 NOTE—For example, a value of 4 represents $2^4 \times 10 \mu$ s, or 160 μ s.
- 25 t) Boolean array, an array of Boolean values.
- 26 u) IP Address, an IPv4 address, IPv6 address, or null for no IP address.

27 12.4 Bridge Management Entity

28 The Bridge Management Entity is described in 8.12.

29 This managed resource comprises the following objects:

- 30 a) The Bridge Configuration (12.4.1)
- 31 b) The Port Configuration for each Port (12.4.2)

32 12.4.1 Bridge Configuration

33 The Bridge Configuration object models the operations that modify, or inquire about, the configuration of
34 the Bridge's resources. There is a single Bridge Configuration object per Bridge.

35 The management operations that can be performed on the Bridge Configuration are as follows:

- 36 a) Discover Bridge (12.4.1.1)
- 37 b) Read Bridge (12.4.1.2)
- 38 c) Set Bridge Name (12.4.1.3)
- 39 d) Reset Bridge (12.4.1.4)
- 40 e) Read component table entry (12.4.1.5)
- 41 f) Update component table entry (12.4.1.5)

1 12.4.1.1 Discover Bridge

2 12.4.1.1.1 Purpose

3 To solicit configuration information regarding the Bridge(s) in the network.

4 12.4.1.1.2 Inputs

5 a) Inclusion Range, a set of ordered pairs of specific MAC addresses. Each pair specifies a range of
6 MAC addresses. A Bridge shall respond if and only if:

- 7 1) For one of the pairs, the numerical comparison of its Bridge Address with each MAC address
8 of the pair shows it to be greater than or equal to the first, and
- 9 2) Less than or equal to the second, and
- 10 3) Its Bridge Address does not appear in the Exclusion List parameter below.

11 The numerical comparison of one MAC address with another, for the purpose of this operation, is
12 achieved by deriving a number from the MAC address according to the following procedure. The
13 consecutive octets of the MAC address are taken to represent a binary number; the first octet that
14 would be transmitted on a LAN medium when the MAC address is used in the source or destination
15 fields of a MAC frame has the most significant value, the next octet the next most significant value.
16 Within each octet, the first bit of each octet is the LSB.

17 b) Exclusion List, a list of specific MAC addresses.

18 12.4.1.1.3 Outputs

19 a) Bridge Address—the MAC address for the Bridge from which the Bridge Identifiers used by STP,
20 RSTP, and MSTP are derived (8.13.8, 13.26).

21 b) Bridge Name—a text string of up to 32 characters, of locally determined significance.

22 c) Number of Ports—the number of Bridge Ports (MAC Entities).

23 d) Port Addresses—a list specifying the following for each Port:

- 24 1) Port Number—the number of the Bridge Port (13.27).
- 25 2) Port Address—the specific MAC address of the individual MAC Entity associated with the
26 Port (8.13.2).

27 e) Uptime—count in seconds of the time elapsed since the Bridge was last reset or initialized.

28 NOTE—Events that are considered to reset or initialize the Bridge include changing the MCID.

29 12.4.1.2 Read Bridge

30 12.4.1.2.1 Purpose

31 To obtain general information regarding the Bridge.

32 12.4.1.2.2 Inputs

33 None.

34 12.4.1.2.3 Outputs

35 a) Bridge Address—the MAC address for the Bridge from which the Bridge Identifiers used by RSTP
36 and MSTP are derived (8.13.8, 13.26).

37 b) Bridge Name—a text string of up to 32 characters, of locally determined significance.

38 c) Number of Ports—the number of Bridge Ports (MAC Entities).

39 d) Port Addresses—a list specifying the following for each Port:

- 40 1) Port Number (13.27).

- 1 2) Port Address—the specific MAC address of the individual MAC Entity associated with the
2 Port (8.13.2).
3 e) Uptime—count in seconds of the time elapsed since the Bridge was last reset or initialized.

4 **12.4.1.3 Set Bridge Name**

5 **12.4.1.3.1 Purpose**

6 To associate a text string, readable by the Read Bridge operation, with a Bridge.

7 **12.4.1.3.2 Inputs**

- 8 a) Bridge Name—a text string of up to 32 characters.

9 **12.4.1.3.3 Outputs**

10 None.

11 **12.4.1.4 Reset Bridge**

12 **12.4.1.4.1 Purpose**

13 To reset the specified Bridge. The FDB is cleared and initialized with the entries specified in the Permanent
14 Database, and the Bridge Protocol Entity is initialized.

15 **12.4.1.4.2 Inputs**

16 None.

17 **12.4.1.4.3 Outputs**

18 None.

19 **12.4.1.5 Bridge Component configuration**

20 There is a single Bridge Component table per system. Each entry in the component table represents a
21 component of the system (Table 12-1). The entries hold the parameters for each component including the
22 component type and capabilities.

Table 12-1—Component table entry managed object

Name	Data type	Operations supported ^a	References
compComponentId	ComponentID	R	12.3 l)
compMACAddress	MAC address	R	8.13.8, 13.24
compNumberPorts	unsigned (1..4095)	R	12.4.1.1.3 c)
compComponentType	ComponentType	R	12.3 m)
compDeviceCapabilities	Boolean array (0..7)	R	12.10.1.1.3 b)
compTrafficClassesEnabled	Boolean	RW	—
compMmrpEnabledStatus	Boolean	RW	—

^a R = Read-only access; RW = Read/Write access.

1 The operations that can be implemented on component table entries are as follows:

- 2 a) Read component table entry
- 3 b) Update component table entry

4 NOTE—The Bridge Component table is implemented in Clause 17 as the BridgeBaseTable (see Table 17-3).

5 **12.4.1.5.1 Component type enumeration**

6 The compComponentType parameter can be assigned the following values:

- 7 a) iComponent—An I-Component (5.7)
- 8 b) bComponent—A B-Component (5.8)
- 9 c) cVlanComponent—A C-VLAN Component (5.5)
- 10 d) sVlanComponent—An S-VLAN Component (5.6)
- 11 e) dBridgeComponent—A MAC Bridge Component (5.13)
- 12 f) edgeRelay—An EVB station ER (5.24.1)

13 **12.4.1.5.2 Component device capabilities**

14 The compDeviceCapabilities parameter contains an array of Boolean values for the following:

- 15 a) ExtendedFilteringServices
- 16 b) TrafficClasses
- 17 c) StaticEntryIndividualPort
- 18 d) IVLCapable
- 19 e) SVLCapable
- 20 f) HybridCapable
- 21 g) ConfigurablePvidTagging
- 22 h) LocalVlanCapable

23 **12.4.2 Port configuration**

24 The Port Configuration object models the operations that modify, or inquire about, the configuration of the
25 Ports of a Bridge. Unless the system explicitly supports the ability to dynamically create and/or delete ports,
26 there is a fixed set of Bridge Ports per Bridge (one for each MAC interface), and each is identified by a
27 permanently allocated Port Number.

28 The allocated Port Numbers are not required to be consecutive. Also, some Port Numbers may be dummy
29 entries, with no actual LAN Port (for example, to allow for expansion of the Bridge by addition of further
30 MAC interfaces). Such dummy Ports shall support the Port Configuration management operations and other
31 Port-related management operations in a manner consistent with the Port being permanently disabled.

32 The information provided by the Port Configuration consists of summary data indicating its name and type.
33 Specific counter information pertaining to the number of frames forwarded, filtered, and in error is
34 maintained by the Forwarding Process resource. The management operations supported by the Bridge
35 Protocol Entity allow for controlling the states of each Port.

36 A port table entry can be implemented by a Bridge for each Port of each component (Table 12-2). It
37 comprises the parameters for each Port including the port type, capabilities, and statistics.

Table 12-2—Port table entry

Name	Data type	Operations supported ^a	References
portComponentId	ComponentID	R	12.4.1.5
portPortNumber	Port Number	R	13.25
portMACAddress	MAC address	R	12.4.1.1.3 a)
portDelayExceededDiscards	counter	R	—
portMtuExceededDiscards	counter	R	—
portCapabilities	unsigned	R	—
portTypeCapabilities	unsigned	R	—
portType	enumerated	R	12.4.2.1
portExternal	Boolean	R	—
portAdminPointToPoint	unsigned	RW	IEEE Std 802.1AC
portOperPointToPoint	Boolean	R	IEEE Std 802.1AC
portName	Latin1 String (SIZE(0..32))	RW	—
portMediaDependentOverhead	unsigned	R	12.4.2.2

^a R = Read-only access; RW = Read/Write access.

1 The management operations that can be performed on the port table are as follows:

- 2 a) Read port table entry (12.4.2.1)
- 3 b) Update port table entry (12.4.2.1)

4 **12.4.2.1 Port type capabilities and enumeration**

5 The portTypeCapabilities array has a bit for each port type the Port can take, while the portType is an
6 enumeration. The portTypeCapabilities can take any combination of port types while the portType can take
7 exactly one of the following values:

- 8 a) C-VLAN Bridge Port
- 9 b) Provider Network Port (PNP)
- 10 c) Customer Network Port (CNP)
- 11 d) Customer Edge Port (CEP)
- 12 e) Customer Backbone Port (CBP)
- 13 f) Virtual Instance Port (VIP)
- 14 g) D-Bridge Port
- 15 h) Remote Customer Access Port (RCAP) (12.13.3)
- 16 i) Station-facing Bridge Port (SBP) (12.26.2)
- 17 j) Uplink Access Port (UAP) (12.26.4)
- 18 k) Uplink relay port (URP) (12.26.5)

19 NOTE—A portType is not required for a downlink relay port (DRP) or an S-channel Access Port (CAP) as no special
20 EVB objects are necessary. A DRP is type C-VLAN Bridge Port while a CAP is type CNP.

12.4.2.2 Media-dependent overhead

The portMediaDependentOverhead parameter provides the number of additional octets for media-dependent framing. The overhead includes:

- a) All octets prior to the first octet of the Destination Address field and
- b) The minimum number of octets after the last octet of the frame check sequence before a subsequent frame, including its media-dependent overhead, can be transmitted.

NOTE—An example of media-dependent overhead is 20 octets to account preamble, start of frame delimiter, and a minimal inter-frame gap (IFG) of 12 octets for IEEE 802.3 point-to-point media.

12.5 MAC entities

The Management Operations and Facilities provided by the MAC Entities are those specified in the Layer Management standards of the individual MACs. A MAC Entity is associated with each Bridge Port.

12.5.1 ISS Port Number table managed object (optional)

An instance of the ISS Port Number table can be implemented by a Bridge system to identify the ISS interfaces that can be assigned to Bridge Ports. The ISS table is required when the Bridge Port assigned to an ISS and the ISS itself are referenced using different Port Numbers. The ISS table is keyed on the ISS Port Number. Each ISS table entry identifies a mapping from the ISS Port Number to a Bridge Port's ComponentID and Port Number. An issToComponentID value of 0 indicates the ISS is not bound to a Bridge Port. The issToComponentID and issToPortNumber parameters are updated indirectly as a result of creating or updating other system-specific Port objects.

The operation that can be implemented on an ISS Port Number table is as follows:

- a) Read ISS Port Number table entry (see Table 12-3)

Table 12-3—ISS Port Number table entry

Name	Data type	Operations supported ^a	References
issPortNumber	Port Number	R	12.3 i)
issMACAddress	MAC address	R	8.13.2
issToComponentID	ComponentID, 0	R	12.4.2
issToPortNumber	Port Number, 0	R	12.4.2

^a R = Read-only access; RW = Read/Write access.

12.6 Forwarding process

The Forwarding Process contains information relating to the forwarding of frames. Counters are maintained that provide information on the number of frames forwarded, filtered, and dropped due to error. Configuration data, defining how frame priority is handled, is maintained by the Forwarding Process.

This managed resource comprises the following objects:

- a) The Port Counters (12.6.1)
- b) The Priority Handling objects for each Port (12.6.2)
- c) The Traffic Class Table for each Port (12.6.3)

12.6.1 The Port Counters

The Port Counters object models the operations that can be performed on the Port counters of the Forwarding Process resource. There are multiple instances (one for each VID for each MAC Entity) of the Port Counters object per Bridge.

The management operation that can be performed on the Port Counters is Read Forwarding Port Counters (12.6.1.1).

12.6.1.1 Read forwarding port counters

12.6.1.1.1 Purpose

To read the forwarding counters associated with a specific Bridge Port.

12.6.1.1.2 Inputs

- a) Port Number (13.27)
- b) Optionally, VID (9.6)

If the VID parameter is supported, then the forwarding Port counters are maintained per VID per Port. If the parameter is not supported, then the forwarding Port counters are maintained per Port only.

12.6.1.1.3 Outputs

- a) Frames Received—count of all valid frames received (including BPDUs, frames addressed to the Bridge as an end station, and frames that were submitted to the Forwarding Process, 8.5).
- b) Optionally, Octets Received—count of the total number of octets in all valid frames received (including BPDUs, frames addressed to the Bridge as an end station, and frames that were submitted to the Forwarding Process).
- c) Discard Inbound—count of valid frames received that were discarded by the Forwarding Process (8.6).
- d) Forward Outbound—count of frames forwarded to the associated MAC Entity (8.5).
- e) Discard Lack of Buffers—count of frames that were to be transmitted through the associated Port but were discarded due to lack of buffers (8.6.6).
- f) Discard Transit Delay Exceeded—count of frames that were to be transmitted but were discarded due to the maximum Bridge transit delay being exceeded (buffering may have been available, 8.6.6).
- g) Discard on Error—count of frames that were to be forwarded on the associated MAC but could not be transmitted (e.g., frame would be too large, 6.5.8).
- h) If Ingress Filtering is supported (8.6.2), Discard on Ingress Filtering—count of frames that were discarded as a result of Ingress Filtering being enabled.
- i) If flow filtering is supported (44.2), Discard TTL Expired—count of frames discarded because they were received with the TTL field expired.

12.6.2 Priority handling

The Priority Handling object models the operations that can be performed on, or inquire about, the Default Priority parameter, the Priority Regeneration Table parameter, the Outbound Access Priority Table parameter, the Priority Code Point parameters, and the Service Access Priority parameters for each Port. The operations that can be performed on this object are as follows:

- a) Read Port Default Priority (12.6.2.1)
- b) Set Port Default Priority (12.6.2.2)
- c) Read Port Priority Regeneration Table (12.6.2.3)
- d) Set Port Priority Regeneration Table (12.6.2.4)
- e) Read Port Priority Code Point Selection (12.6.2.5)
- f) Set Port Priority Code Point Selection (12.6.2.6)
- g) Read Port Priority Code Point Decoding Table (12.6.2.7)
- h) Optionally, Set Port Priority Code Point Decoding Table (12.6.2.8)
- i) Read Port Priority Code Point Encoding Table (12.6.2.9)
- j) Optionally, Set Port Priority Code Point Encoding Table (12.6.2.10)
- k) Read Use_DEI parameter (12.6.2.11)
- l) Optionally, Set Use_DEI parameter (12.6.2.12)
- m) Read Require Drop Encoding parameter (12.6.2.13)
- n) Optionally, Set Require Drop Encoding parameter (12.6.2.14)
- o) Read Service Access Priority Selection (12.6.2.15)
- p) Optionally, Set Service Access Priority Selection (12.6.2.16)
- q) Read Service Access Priority Table (12.6.2.17)
- r) Optionally, Set Service Access Priority Table (12.6.2.18)

12.6.2.1 Read Port Default Priority

12.6.2.1.1 Purpose

To read the current state of the Default Priority parameter (IEEE Std 802.1AC) for a specific Bridge Port.

12.6.2.1.2 Inputs

- a) Port number.

12.6.2.1.3 Outputs

- a) Default Priority value—Integer in range 0–7.

12.6.2.2 Set Port Default Priority

12.6.2.2.1 Purpose

To set the current state of the Default Priority parameter (IEEE Std 802.1AC) for a specific Bridge Port.

12.6.2.2.2 Inputs

- a) Port number.
- b) Default Priority value—Integer in range 0–7.

12.6.2.2.3 Outputs

None.

12.6.2.3 Read Port Priority Regeneration Table

12.6.2.3.1 Purpose

To read the current state of the Priority Regeneration Table parameter (6.9.4) for a specific Bridge Port.

12.6.2.3.2 Inputs

- a) Port number.

12.6.2.3.3 Outputs

- a) Regenerated Priority value for Received Priority 0—Integer in range 0–7.
- b) Regenerated Priority value for Received Priority 1—Integer in range 0–7.
- c) Regenerated Priority value for Received Priority 2—Integer in range 0–7.
- d) Regenerated Priority value for Received Priority 3—Integer in range 0–7.
- e) Regenerated Priority value for Received Priority 4—Integer in range 0–7.
- f) Regenerated Priority value for Received Priority 5—Integer in range 0–7.
- g) Regenerated Priority value for Received Priority 6—Integer in range 0–7.
- h) Regenerated Priority value for Received Priority 7—Integer in range 0–7.

12.6.2.4 Set Port Priority Regeneration Table

12.6.2.4.1 Purpose

To set the current state of the Priority Regeneration Table parameter (6.9.4) for a specific Bridge Port.

12.6.2.4.2 Inputs

- a) Port number.
- b) Regenerated Priority value for Received Priority 0—Integer in range 0–7.
- c) Regenerated Priority value for Received Priority 1—Integer in range 0–7.
- d) Regenerated Priority value for Received Priority 2—Integer in range 0–7.
- e) Regenerated Priority value for Received Priority 3—Integer in range 0–7.
- f) Regenerated Priority value for Received Priority 4—Integer in range 0–7.
- g) Regenerated Priority value for Received Priority 5—Integer in range 0–7.
- h) Regenerated Priority value for Received Priority 6—Integer in range 0–7.
- i) Regenerated Priority value for Received Priority 7—Integer in range 0–7.

12.6.2.4.3 Outputs

None.

12.6.2.5 Read Port Priority Code Point Selection

12.6.2.5.1 Purpose

To read which row of the Priority Code Point Encoding Table and Priority Code Point Decoding Table (6.9.3) is currently selected for use on this Port.

1 12.6.2.5.2 Inputs

- 2 a) Port Number: the number of the Bridge Port.

3 12.6.2.5.3 Outputs

- 4 a) Priority Code Point Selection. This takes one of the following values:

- 5 1) 8P0D
6 2) 7P1D
7 3) 6P2D
8 4) 5P3D

9 12.6.2.6 Set Port Priority Code Point Selection

10 12.6.2.6.1 Purpose

11 To set which row of the Priority Code Point Encoding Table and Priority Code Point Decoding Table (6.9.3)
12 will be selected for use on this Port.

13 12.6.2.6.2 Inputs

- 14 a) Port Number: the number of the Bridge Port.
15 b) Priority Code Point Selection. This takes one of the following values:
16 1) 8P0D
17 2) 7P1D
18 3) 6P2D
19 4) 5P3D

20 12.6.2.6.3 Outputs

21 None.

22 12.6.2.7 Read Priority Code Point Decoding Table

23 12.6.2.7.1 Purpose

24 To read the current contents of a row in the Priority Code Point Decoding Table (6.9.3) for a Port.

25 12.6.2.7.2 Inputs

- 26 a) Port Number: the number of the Bridge Port.
27 b) Priority Code Point Row. This takes one of the following values:
28 1) 8P0D
29 2) 7P1D
30 3) 6P2D
31 4) 5P3D

32 12.6.2.7.3 Outputs

- 33 a) Priority value for Priority Code Point 0: Integer in range 0–7.
34 b) Drop_eligible value for Priority Code Point 0: Boolean.
35 c) Priority value for Priority Code Point 1: Integer in range 0–7.
36 d) Drop_eligible value for Priority Code Point 1: Boolean.
37 e) Priority value for Priority Code Point 2: Integer in range 0–7.
38 f) Drop_eligible value for Priority Code Point 2: Boolean.
39 g) Priority value for Priority Code Point 3: Integer in range 0–7.

- 1 h) Drop_eligible value for Priority Code Point 3: Boolean.
- 2 i) Priority value for Priority Code Point 4: Integer in range 0–7.
- 3 j) Drop_eligible value for Priority Code Point 4: Boolean.
- 4 k) Priority value for Priority Code Point 5: Integer in range 0–7.
- 5 l) Drop_eligible value for Priority Code Point 5: Boolean.
- 6 m) Priority value for Priority Code Point 6: Integer in range 0–7.
- 7 n) Drop_eligible value for Priority Code Point 6: Boolean.
- 8 o) Priority value for Priority Code Point 7: Integer in range 0–7.
- 9 p) Drop_eligible value for Priority Code Point 7: Boolean.

10 **12.6.2.8 Set Priority Code Point Decoding Table**

11 **12.6.2.8.1 Purpose**

12 To modify the contents of a row in the Priority Code Point Decoding Table (6.9.3) for a Port.

13 **12.6.2.8.2 Inputs**

- 14 a) Port Number: the number of the Bridge Port.
- 15 b) Priority Code Point Row. This takes one of the following values:
 - 16 1) 8P0D
 - 17 2) 7P1D
 - 18 3) 6P2D
 - 19 4) 5P3D
- 20 c) Priority value for Priority Code Point 0: Integer in range 0–7.
- 21 d) Drop_eligible value for Priority Code Point 0: Boolean.
- 22 e) Priority value for Priority Code Point 1: Integer in range 0–7.
- 23 f) Drop_eligible value for Priority Code Point 1: Boolean.
- 24 g) Priority value for Priority Code Point 2: Integer in range 0–7.
- 25 h) Drop_eligible value for Priority Code Point 2: Boolean.
- 26 i) Priority value for Priority Code Point 3: Integer in range 0–7.
- 27 j) Drop_eligible value for Priority Code Point 3: Boolean.
- 28 k) Priority value for Priority Code Point 4: Integer in range 0–7.
- 29 l) Drop_eligible value for Priority Code Point 4: Boolean.
- 30 m) Priority value for Priority Code Point 5: Integer in range 0–7.
- 31 n) Drop_eligible value for Priority Code Point 5: Boolean.
- 32 o) Priority value for Priority Code Point 6: Integer in range 0–7.
- 33 p) Drop_eligible value for Priority Code Point 6: Boolean.
- 34 q) Priority value for Priority Code Point 7: Integer in range 0–7.
- 35 r) Drop_eligible value for Priority Code Point 7: Boolean.

36 **12.6.2.8.3 Outputs**

37 None.

1 12.6.2.9 Read Priority Code Point Encoding Table

2 12.6.2.9.1 Purpose

3 To read the current contents of a row in the Priority Code Point Encoding Table (6.9.3) for a Port.

4 12.6.2.9.2 Inputs

- 5 a) Port Number: the number of the Bridge Port.
- 6 b) Priority Code Point Row. This takes one of the following values:
 - 7 1) 8P0D
 - 8 2) 7P1D
 - 9 3) 6P2D
 - 10 4) 5P3D

11 12.6.2.9.3 Outputs

- 12 a) Priority Code Point value for priority 0 with drop_eligible False: Integer in range 0–7.
- 13 b) Priority Code Point value for priority 0 with drop_eligible True: Integer in range 0–7.
- 14 c) Priority Code Point value for priority 1 with drop_eligible False: Integer in range 0–7.
- 15 d) Priority Code Point value for priority 1 with drop_eligible True: Integer in range 0–7.
- 16 e) Priority Code Point value for priority 2 with drop_eligible False: Integer in range 0–7.
- 17 f) Priority Code Point value for priority 2 with drop_eligible True: Integer in range 0–7.
- 18 g) Priority Code Point value for priority 3 with drop_eligible False: Integer in range 0–7.
- 19 h) Priority Code Point value for priority 3 with drop_eligible True: Integer in range 0–7.
- 20 i) Priority Code Point value for priority 4 with drop_eligible False: Integer in range 0–7.
- 21 j) Priority Code Point value for priority 4 with drop_eligible True: Integer in range 0–7.
- 22 k) Priority Code Point value for priority 5 with drop_eligible False: Integer in range 0–7.
- 23 l) Priority Code Point value for priority 5 with drop_eligible True: Integer in range 0–7.
- 24 m) Priority Code Point value for priority 6 with drop_eligible False: Integer in range 0–7.
- 25 n) Priority Code Point value for priority 6 with drop_eligible True: Integer in range 0–7.
- 26 o) Priority Code Point value for priority 7 with drop_eligible False: Integer in range 0–7.
- 27 p) Priority Code Point value for priority 7 with drop_eligible True: Integer in range 0–7.

28 12.6.2.10 Set Priority Code Point Encoding Table

29 12.6.2.10.1 Purpose

30 To modify the contents of a row in the Priority Code Point Encoding Table (6.9.3) for a Port.

31 12.6.2.10.2 Inputs

- 32 a) Port Number: the number of the Bridge Port.
- 33 b) Priority Code Point Row. This takes one of the following values:
 - 34 1) 8P0D
 - 35 2) 7P1D
 - 36 3) 6P2D
 - 37 4) 5P3D
- 38 c) Priority Code Point value for priority 0 with drop_eligible False: Integer in range 0–7.
- 39 d) Priority Code Point value for priority 0 with drop_eligible True: Integer in range 0–7.
- 40 e) Priority Code Point value for priority 1 with drop_eligible False: Integer in range 0–7.
- 41 f) Priority Code Point value for priority 1 with drop_eligible True: Integer in range 0–7.
- 42 g) Priority Code Point value for priority 2 with drop_eligible False: Integer in range 0–7.
- 43 h) Priority Code Point value for priority 2 with drop_eligible True: Integer in range 0–7.
- 44 i) Priority Code Point value for priority 3 with drop_eligible False: Integer in range 0–7.

- 1 j) Priority Code Point value for priority 3 with drop_eligible True: Integer in range 0–7.
- 2 k) Priority Code Point value for priority 4 with drop_eligible False: Integer in range 0–7.
- 3 l) Priority Code Point value for priority 4 with drop_eligible True: Integer in range 0–7.
- 4 m) Priority Code Point value for priority 5 with drop_eligible False: Integer in range 0–7.
- 5 n) Priority Code Point value for priority 5 with drop_eligible True: Integer in range 0–7.
- 6 o) Priority Code Point value for priority 6 with drop_eligible False: Integer in range 0–7.
- 7 p) Priority Code Point value for priority 6 with drop_eligible True: Integer in range 0–7.
- 8 q) Priority Code Point value for priority 7 with drop_eligible False: Integer in range 0–7.
- 9 r) Priority Code Point value for priority 7 with drop_eligible True: Integer in range 0–7.

10 **12.6.2.10.3 Outputs**

11 None.

12 **12.6.2.11 Read Use_DEI Parameter**

13 **12.6.2.11.1 Purpose**

14 To read the current state of the Use_DEI parameter (6.9.3) for the Port.

15 **12.6.2.11.2 Inputs**

- 16 a) Port Number: the number of the Bridge Port.

17 **12.6.2.11.3 Outputs**

- 18 a) Use_DEI parameter: Boolean.

19 **12.6.2.12 Set Use_DEI Parameter**

20 **12.6.2.12.1 Purpose**

21 To set the current state of the Use_DEI parameter (6.9.3) for the Port.

22 **12.6.2.12.2 Inputs**

- 23 a) Port Number: the number of the Bridge Port.
- 24 b) Use_DEI parameter: Boolean.

25 **12.6.2.12.3 Outputs**

26 None.

27 **12.6.2.13 Read Require Drop Encoding Parameter**

28 **12.6.2.13.1 Purpose**

29 To read the current state of the Require Drop Encoding parameter (8.6.6) for the Port.

30 **12.6.2.13.2 Inputs**

- 31 a) Port Number: the number of the Bridge Port.

32 **12.6.2.13.3 Outputs**

- 33 a) Require Drop Encoding parameter: Boolean.

1 **12.6.2.14 Set Require Drop Encoding Parameter**

2 **12.6.2.14.1 Purpose**

3 To set the current state of the Require Drop Encoding parameter (8.6.6) for the Port.

4 **12.6.2.14.2 Inputs**

- 5 a) Port Number: the number of the Bridge Port.
- 6 b) Require Drop Encoding parameter: Boolean.

7 **12.6.2.14.3 Outputs**

8 None.

9 **12.6.2.15 Read Service Access Priority Selection**

10 **12.6.2.15.1 Purpose**

11 To read the current state of whether Service Access Priority Selection is enabled (6.13) for the Port.

12 **12.6.2.15.2 Inputs**

- 13 a) Port Number: the number of the Bridge Port.

14 **12.6.2.15.3 Outputs**

- 15 a) Enable Service Access Priority Selection: the permissible values are as follows:
 - 16 1) Enabled
 - 17 2) Disabled

18 **12.6.2.16 Set Service Access Priority Selection**

19 **12.6.2.16.1 Purpose**

20 To enable or disable Service Access Priority Selection (6.13) for the Port.

21 **12.6.2.16.2 Inputs**

- 22 a) Port Number: the number of the Bridge Port.
- 23 b) Service Access Priority Selection: the permissible values are as follows:
 - 24 1) Enabled
 - 25 2) Disabled

26 **12.6.2.16.3 Outputs**

27 None.

28 **12.6.2.17 Read Service Access Priority Table**

29 **12.6.2.17.1 Purpose**

30 To read the current contents of the Service Access Priority Table (6.13.1) for the Port.

1 12.6.2.17.2 Inputs

- 2 a) Port Number: the number of the Bridge Port.

3 12.6.2.17.3 Outputs

- 4 a) Service Access Priority value for Received Priority 0: Integer in range 0–7.
5 b) Service Access Priority value for Received Priority 1: Integer in range 0–7.
6 c) Service Access Priority value for Received Priority 2: Integer in range 0–7.
7 d) Service Access Priority value for Received Priority 3: Integer in range 0–7.
8 e) Service Access Priority value for Received Priority 4: Integer in range 0–7.
9 f) Service Access Priority value for Received Priority 5: Integer in range 0–7.
10 g) Service Access Priority value for Received Priority 6: Integer in range 0–7.
11 h) Service Access Priority value for Received Priority 7: Integer in range 0–7.

12 12.6.2.18 Set Service Access Priority Table

13 12.6.2.18.1 Purpose

14 To modify the contents of the Service Access Priority Table (6.13.1) for the Port.

15 12.6.2.18.2 Inputs

- 16 a) Port Number: the number of the Bridge Port.
17 b) Service Access Priority value for Received Priority 0: Integer in range 0–7.
18 c) Service Access Priority value for Received Priority 1: Integer in range 0–7.
19 d) Service Access Priority value for Received Priority 2: Integer in range 0–7.
20 e) Service Access Priority value for Received Priority 3: Integer in range 0–7.
21 f) Service Access Priority value for Received Priority 4: Integer in range 0–7.
22 g) Service Access Priority value for Received Priority 5: Integer in range 0–7.
23 h) Service Access Priority value for Received Priority 6: Integer in range 0–7.
24 i) Service Access Priority value for Received Priority 7: Integer in range 0–7.

25 12.6.2.18.3 Outputs

26 None.

27 12.6.3 Traffic Class Table

28 The Traffic Class Table object models the operations that can be performed on, or inquire about, the current
29 contents of the Traffic Class Table (8.6.6) for a given Port. The operations that can be performed on this
30 object are Read Port Traffic Class Table and Set Port Traffic Class Table.

31 12.6.3.1 Read Port Traffic Class Table

32 12.6.3.1.1 Purpose

33 To read the contents of the Traffic Class Table (8.6.6) for a given Port.

34 12.6.3.1.2 Inputs

- 35 a) Port Number.

36 12.6.3.1.3 Outputs

- 37 a) The number of traffic classes, in the range 1 through 8, supported on the Port.
38 b) For each value of traffic class supported on the Port, the value of the traffic class in the range 0
39 through 7, and the set of priority values assigned to that traffic class.

1 12.6.3.2 Set Port Traffic Class Table

2 12.6.3.2.1 Purpose

3 To set the contents of the Traffic Class Table (8.6.6) for a given Port.

4 12.6.3.2.2 Inputs

5 a) Port number.

6 b) For each value of traffic class supported on the Port, the value of the traffic class in the range 0
7 through 7, and the set of priority values assigned to that traffic class.

8 NOTE—If a traffic class value greater than the largest traffic class available on the Port is specified, then the value
9 applied to the Traffic Class Table is the largest available traffic class.

10 12.6.3.2.3 Outputs

11 None.

12 12.7 Filtering Database (FDB)

13 The FDB is described in 8.8. It contains filtering information used by the Forwarding Process (8.6) in
14 deciding through which Ports of the Bridge frames should be forwarded.

15 This managed resource comprises the following objects:

- 16 a) The Filtering Database (12.7.1)
- 17 b) The Static Filtering Entries (12.7.2)
- 18 c) The Dynamic Filtering Entries (12.7.3)
- 19 d) The MAC Address Registration Entries (12.7.4)
- 20 e) The Static VLAN Registration Entries (12.7.5)
- 21 f) The Dynamic VLAN Registration Entries (12.7.5)
- 22 g) The Permanent Database (12.7.6)

23 12.7.1 The Filtering Database object

24 The Filtering Database object models the operations that can be performed on, or affect, the FDB as a whole.

25 There is a single Filtering Database object per Bridge.

26 The management operations that can be performed on the Database are as follows:

- 27 a) Read Filtering Database (12.7.1.1)
- 28 b) Set Filtering Database Ageing Time (12.7.1.2)
- 29 c) Read Permanent Database (12.7.6.1)
- 30 d) Create Filtering Entry (12.7.7.1)
- 31 e) Delete Filtering Entry (12.7.7.2)
- 32 f) Read Filtering Entry (12.7.7.3)
- 33 g) Read Filtering Entry Range (12.7.7.4)

34 12.7.1.1 Read Filtering Database

35 12.7.1.1.1 Purpose

36 To obtain general information regarding the Bridge's FDB.

37 12.7.1.1.2 Inputs

38 None.

12.7.1.1.3 Outputs

- a) Filtering Database Size—the maximum number of entries that can be held in the FDB.
- b) Number of Static Filtering Entries—the number of Static Filtering Entries (8.8.1) currently in the FDB.
- c) Number of Dynamic Filtering Entries—the number of Dynamic Filtering Entries (8.8.3) currently in the FDB.
- d) Number of Static VLAN Registration Entries—the number of Static VLAN Registration Entries (8.8.2) currently in the FDB.
- e) Number of Dynamic VLAN Registration Entries—the number of Dynamic VLAN Registration Entries (8.8.5) currently in the FDB.
- f) Ageing Time—for ageing out Dynamic Filtering Entries when the Port associated with the entry is in the Forwarding state (8.8.3).
- g) If Extended Filtering Services are supported, Number of MAC Address Registration Entries—the number of MAC Address Registration Entries (8.8.4) currently in the FDB.

12.7.1.2 Set Filtering Database Ageing Time

12.7.1.2.1 Purpose

To set the ageing time for Dynamic Filtering Entries (8.8.3).

12.7.1.2.2 Inputs

- a) Ageing Time.

12.7.1.2.3 Outputs

None.

12.7.2 A Static Filtering Entry object

A Static Filtering Entry object models the operations that can be performed on a single Static Filtering Entry in the FDB. The set of Static Filtering Entry objects within the FDB changes only under management control.

A Static Filtering Entry object supports the following operations:

- a) Create Filtering Entry (12.7.7.1)
- b) Delete Filtering Entry (12.7.7.2)
- c) Read Filtering Entry (12.7.7.3)
- d) Read Filtering Entry Range (12.7.7.4)

12.7.3 A Dynamic Filtering Entry object

A Dynamic Filtering Entry object models the operations that can be performed on a single Dynamic Filtering Entry (i.e., one that is created by the Learning Process as a result of the observation of network traffic) in the FDB.

A Dynamic Filtering Entry object supports the following operations:

- a) Delete Filtering Entry (12.7.7.2)
- b) Read Filtering Entry (12.7.7.3)
- c) Read Filtering Entry Range (12.7.7.4)

1 **12.7.4 A MAC Address Registration Entry object**

2 A MAC Address Registration Entry object models the operations that can be performed on a single MAC
3 Address Registration Entry in the FDB. The set of MAC Address Registration Entry objects within the FDB
4 changes only as a result of MMRP exchanges.

5 A MAC Address Registration Entry object supports the following operations:

- 6 a) Read Filtering Entry (12.7.7.3)
- 7 b) Read Filtering Entry Range (12.7.7.4)

8 **12.7.5 A VLAN Registration Entry object**

9 A VLAN Registration Entry object models the operations that can be performed on a single VLAN
10 Registration Entry in the FDB. The set of VLAN Registration Entry objects within the FDB changes under
11 management control and also as a result of MVRP exchanges.

12 **12.7.5.1 Static VLAN Registration Entry object**

13 A Static VLAN Registration Entry object supports the following operations:

- 14 a) Create Filtering Entry (12.7.7.1)
- 15 b) Delete Filtering Entry (12.7.7.2)
- 16 c) Read Filtering Entry (12.7.7.3)
- 17 d) Read Filtering Entry Range (12.7.7.4)

18 **12.7.5.2 Dynamic VLAN Registration Entry object**

19 A Dynamic VLAN Registration Entry object supports the following operations:

- 20 a) Read Filtering Entry (12.7.7.3)
- 21 b) Read Filtering Entry Range (12.7.7.4)

22 **12.7.6 Permanent Database object**

23 The Permanent Database object models the operations that can be performed on, or affect, the Permanent
24 Database. There is a single Permanent Database per FDB.

25 The management operations that can be performed on the Permanent Database are as follows:

- 26 a) Read Permanent Database (12.7.6.1)
- 27 b) Create Filtering Entry (12.7.7.1)
- 28 c) Delete Filtering Entry (12.7.7.2)
- 29 d) Read Filtering Entry (12.7.7.3)
- 30 e) Read Filtering Entry Range (12.7.7.4)

31 **12.7.6.1 Read Permanent Database**

32 **12.7.6.1.1 Purpose**

33 To obtain general information regarding the Permanent Database (8.8.11).

34 **12.7.6.1.2 Inputs**

35 None.

12.7.6.1.3 Outputs

- a) Permanent Database Size—maximum number of entries that can be held in the Permanent Database.
- b) Number of Static Filtering Entries—number of Static Filtering Entries (8.8.1) currently in the Permanent Database.
- c) Number of Static VLAN Registration Entries—number of Static VLAN Registration Entries (8.8.2) currently in the Permanent Database.

12.7.7 General FDB operations

In these operations on the FDB, the operation parameters make use of VID values, even when operating on a Dynamic Filtering Entry (8.8.3) whose structure carries an FID rather than a VID. In this case, the value used in the VID parameter can be any VID that has been allocated to the FID concerned (8.8.8).

12.7.7.1 Create Filtering Entry

12.7.7.1.1 Purpose

To create or update a Static Filtering Entry (8.8.1) or Static VLAN Registration Entry (8.8.2) in the FDB or Permanent Database. Only static entries may be created in the FDB or Permanent Database.

12.7.7.1.2 Inputs

- a) Identifier—FDB or Permanent Database.
- b) Address—MAC address of the entry (not present in VLAN Registration Entries).
- c) VID—VLAN Identifier of the entry.
- d) Port Map—a set of control indicators, one for each Port, as specified in 8.8.1 and 8.8.2.

12.7.7.1.3 Outputs

- a) Operation rejected because a Port identified by the Port Map includes a port already in the member set of a VID of a different type than the currently registered VID.
- b) Operation rejected because the specified Static Filtering Entry is associated with and Infrastructure Protection Group (IPG) and is administered by IPS Control.
- c) Operation rejected because the VID identifies an SPVID.
- d) Operation accepted.

12.7.7.2 Delete Filtering Entry

12.7.7.2.1 Purpose

To delete a Filtering Entry or VLAN Registration Entry from the FDB or Permanent Database.

12.7.7.2.2 Inputs

- a) Identifier—FDB or Permanent Database.
- b) Address—MAC address of the desired entry (not present in VLAN Registration Entries).
- c) VID—VLAN Identifier of the entry.

12.7.7.2.3 Outputs

None.

12.7.7.3 Read Filtering Entry

12.7.7.3.1 Purpose

To read a Filtering Entry, MAC Address Registration Entry, or VLAN Registration Entry from the Filtering or Permanent Databases.

12.7.7.3.2 Inputs

- a) Identifier—FDB or Permanent Database.
- b) Address—MAC address of the desired entry (not present in VLAN Registration Entries).
- c) VID—VLAN Identifier of the entry.
- d) Type—Static or Dynamic entry.

12.7.7.3.3 Outputs

- a) Address—MAC address of the desired entry (not present in VLAN Registration Entries).
- b) VID—VLAN Identifier of the entry.
- c) Type—Static or Dynamic entry.
- d) Port Map—a set of control indicators as appropriate for the entry (that may include a Connection Identifier), as specified in 8.8.1 through 8.8.5.

12.7.7.4 Read Filtering Entry range

12.7.7.4.1 Purpose

To read a range of FDB entries (of any type) from the Filtering or Permanent Databases.

Since the number of values to be returned in the requested range may have exceeded the capacity of the SDU conveying the management response, the returned entry range is identified. The indices that define the range take on values from zero up to Filtering Database Size minus one.

12.7.7.4.2 Inputs

- a) Identifier—FDB or Permanent Database.
- b) Start Index—inclusive starting index of the desired entry range.
- c) Stop Index—inclusive ending index of the desired range.

12.7.7.4.3 Outputs

- a) Start Index—inclusive starting index of the returned entry range.
- b) Stop Index—inclusive ending index of the returned entry range.
- c) For each index returned:
 - 1) Address—MAC address of the desired entry (not present in VLAN Registration Entries).
 - 2) VID—VLAN Identifier of the entry.
 - 3) Type—Static or Dynamic entry.
 - 4) Port Map—a set of control indicators as appropriate for the entry (that may include a Connection Identifier), as specified in 8.8.1 through 8.8.5.

12.8 Bridge Protocol Entity

The Bridge Protocol Entity is described in 8.10 and Clause 13.

This managed resource comprises the following objects:

- a) The Protocol Entity
- b) The Ports under its control

12.8.1 The Protocol Entity

The Protocol Entity object models the operations that can be performed on, or inquire about, the operation of STP. There is a single Protocol Entity per Bridge; it can therefore be identified as a single fixed component of the Protocol Entity resource.

The management operations that can be performed on the Protocol Entity are as follows:

- a) Read CIST Bridge Protocol Parameters (12.8.1.1)
- b) Read MSTI Bridge Protocol Parameters (12.8.1.2)
- c) Set CIST Bridge Protocol Parameters (12.8.1.3)
- d) Set MSTI Bridge Protocol Parameters (12.8.1.4)

12.8.1.1 Read CIST Bridge Protocol Parameters

12.8.1.1.1 Purpose

To obtain information regarding the Bridge's Bridge Protocol Entity for the CIST.

12.8.1.1.2 Inputs

None.

12.8.1.1.3 Outputs

- a) Bridge Identifier—as defined in 13.26.2. The Bridge Identifier for the CIST.
- b) Time Since Topology Change—in an STP Bridge, the count in seconds of the time elapsed since the Topology Change flag parameter for the Bridge (8.5.3.12 of IEEE Std 802.1D, 1998 Edition [B13]) was last True, or in an RSTP or MSTP Bridge, the count in seconds since tcWhile timer (13.25) for any Port was nonzero.
- c) Topology Change Count—in an STP Bridge, the count of the times the Topology Change flag parameter for the Bridge has been set (i.e., transitioned from False to True) since the Bridge was powered on or initialized, or in an RSTP or MSTP Bridge, the count of times that there has been at least one nonzero tcWhile timer (13.25).
- d) CIST Root Identifier (13.26.10).
- e) CIST External Root Path Cost (13.26.10).
- f) Root Port (13.26.9).
- g) Max Age (13.26.11).
- h) Forward Delay (13.26.11).
- i) Bridge Max Age (13.26.4).
- j) Bridge Hello Time (13.25). This parameter is present only if the Bridge supports STP or RSTP.
- k) Bridge Forward Delay (13.26.4).
- l) Hold Time (8.5.3.14 of IEEE Std 802.1D, 1998 Edition [B13]) or Transmission Limit (TxHoldCount in 13.26.12).

1 The following parameter is present only if the Bridge supports RSTP or MSTP:

- 2 m) forceVersion—the value of the Force Protocol Version parameter for the Bridge (13.7.2).

3 The following additional parameters are present only if the Bridge supports MSTP:

- 4 n) CIST Regional Root Identifier (13.26.10). The Bridge Identifier of the current CIST Regional Root.
5 o) CIST Path Cost (CIST Internal Root Path Cost, in 13.26.10). The CIST path cost from the
6 transmitting Bridge to the CIST Regional Root.
7 p) MaxHops (13.26.4).

8 12.8.1.2 Read MSTI Bridge Protocol Parameters

9 12.8.1.2.1 Purpose

10 In an MST Bridge, to obtain information regarding the Bridge's Bridge Protocol Entity for the specified
11 spanning tree instance.

12 12.8.1.2.2 Inputs

- 13 a) MSTID—Identifies the set of parameters that will be returned. For Bridges that support MSTP, this
14 parameter is the identifier of the spanning tree for which the operation is being performed. This
15 parameter takes a value in the range 1 through 4094.

16 12.8.1.2.3 Outputs

- 17 a) MSTID—identifies the set of parameters that are being returned. This parameter is the identifier of
18 the MST Instance for which the operation is being performed.
19 b) Bridge Identifier—as defined in 13.26.2. The Bridge Identifier for the spanning tree instance
20 identified by the MSTID.
21 c) Time Since Topology Change—count in seconds of the time elapsed since tcWhile (13.25) was last
22 nonzero for any Port for the given MSTI.
23 d) Topology Change Count—count of the times tcWhile has been nonzero for any Port for the given
24 MSTI since the Bridge was powered on or initialized.
25 e) Topology Change (tcWhile, 13.25). True if tcWhile is nonzero for any Port for the given MST.
26 f) Designated Root (MSTI Regional Root Identifier, 13.26.10). The Bridge Identifier of the Root
27 Bridge for the spanning tree instance identified by the MSTID.
28 g) Root Path Cost (MSTI Internal Root Path Cost, 13.26.10). The path cost from the transmitting
29 Bridge to the Root Bridge for the spanning tree instance identified by the MSTID.
30 h) Root Port (13.26.9). The Root Port for the spanning tree instance identified by the MSTID.

31 12.8.1.3 Set CIST Bridge Protocol Parameters

32 12.8.1.3.1 Purpose

33 To modify parameters in the Bridge's Bridge Protocol Entity for the CIST, in order to force a configuration
34 of the spanning tree and/or tune the reconfiguration time to suit a specific topology. In RSTP and MSTP
35 implementations, this operation causes these values to be set for all Ports of the Bridge.

36 12.8.1.3.2 Inputs

- 37 a) Bridge Max Age—the new value (13.26.4).
38 b) Bridge Hello Time—the new value (13.25) This parameter is present only if the Bridge supports
39 STP or RSTP.
40 c) Bridge Forward Delay—the new value (13.26.4).
41 d) Bridge Priority—the new value of the priority part of the Bridge Identifier (13.26.2) for the CIST.

1 The following parameters are present only if the Bridge supports RSTP or MSTP:

- 2 e) forceVersion—the new value of the Force Protocol Version parameter for the Bridge (13.7.2).
- 3 f) TxHoldCount—the new value of TxHoldCount (13.26.12).

4 The following parameter is present only if the Bridge supports MSTP:

- 5 g) MaxHops—the new value of MaxHops (13.26.4).

6 **12.8.1.3.3 Outputs**

7 a) Operation status. This takes one of the following values:

- 8 1) Operation rejected due to invalid Bridge Priority value (12.3).
- 9 2) Operation rejected due to the specified Max Age, Hello Time, or Forward Delay values being
- 10 outside the range specified for the parameter (see 12.8.1.3.4).
- 11 3) Operation rejected due to the specified Max Age, Hello Time, or Forward Delay values not
- 12 being in compliance with the requirements of this standard (see 12.8.1.3.4).
- 13 4) Operation rejected due to the specified MaxHops value not being within the permitted range
- 14 specified in Table 13-5.
- 15 5) Operation accepted.

16 **12.8.1.3.4 Procedure**

17 In the following description, the references to Bridge Hello Time apply only to Bridges that support STP or
18 RSTP.

19 The input parameter values are checked for compliance with their definitions in Clause 13. If they do not
20 comply, or the value of Bridge Max Age or Bridge Forward Delay is less than the lower limit of the
21 range specified in Table 13-5, then no action shall be taken for any of the supplied parameters. If the value of
22 any of Bridge Max Age, Bridge Forward Delay, or Bridge Hello Time is outside the range specified in
23 Table 13-5, then the Bridge need not take action.

24 Otherwise,

- 25 a) The Bridge's Bridge Max Age, Bridge Hello Time, and Bridge Forward Delay parameters are set to
- 26 the supplied values.
- 27 b) In STP Bridges, the Set Bridge Priority procedure (8.8.4 of IEEE Std 802.1D, 1998 Edition [B13]) is
- 28 used to set the priority part of the Bridge Identifier to the supplied value.
- 29 c) In RSTP and MSTP Bridges, the priority component of the Bridge Identifier (13.26.2) is updated
- 30 using the supplied value. For all Ports of the Bridge, the reselect for the CIST parameter (13.27) is
- 31 set TRUE, and the selected parameter for the CIST is set FALSE.

32 **12.8.1.4 Set MSTI Bridge Protocol Parameters**

33 **12.8.1.4.1 Purpose**

34 To modify parameters in the Bridge's Bridge Protocol Entity for the specified spanning tree instance, in
35 order to force a configuration of the spanning tree and/or tune the reconfiguration time to suit a specific
36 topology.

37 **12.8.1.4.2 Inputs**

- 38 a) MSTID—identifies the set of parameters upon which the operation will be performed.
- 39 b) Bridge Priority—the new value of the priority part of the Bridge Identifier (13.26.2) for the spanning
- 40 tree instance identified by the MSTID.

1 **12.8.1.4.3 Outputs**

- 2 a) Operation status. This takes one of the following values:
- 3 1) Operation rejected due to invalid Bridge Priority value (12.3).
 - 4 2) Operation rejected due to invalid MSTID (i.e, there is currently no MST Instance with that
 - 5 value of MSTID supported by the Bridge).
 - 6 3) Operation accepted.

7 **12.8.1.4.4 Procedure**

8 The Bridge Priority parameter value is checked for compliance with its definition in Clause 13. If it does not
9 comply, then no action shall be taken.

10 Otherwise, the priority part of the Bridge Identifier is set to the supplied value for the specified spanning tree
11 instance.

12 **12.8.2 Bridge Port**

13 A Bridge Port object models the operations related to an individual Bridge Port in relation to the operation of
14 STP. There are a fixed set of Bridge Ports per Bridge; each can therefore be identified by a permanently
15 allocated Port Number, as a fixed component of the Protocol Entity resource.

16 The management operations that can be performed on a Bridge Port are as follows:

- 17 a) Read CIST Port Parameters (12.8.2.1)
- 18 b) Read MSTI Port Parameters (12.8.2.2)
- 19 c) Set CIST Port Parameters (12.8.2.3)
- 20 d) Set MSTI Port Parameters (12.8.2.4)
- 21 e) Force BPDU Migration Check (12.8.2.5)

22 **12.8.2.1 Read CIST Port Parameters**

23 **12.8.2.1.1 Purpose**

24 To obtain information regarding a specific Port within the Bridge's Bridge Protocol Entity, for the CIST.

25 **12.8.2.1.2 Inputs**

- 26 a) Port Number—the number of the Bridge Port.

27 **12.8.2.1.3 Outputs**

- 28 a) Uptime—count in seconds of the time elapsed since the Port was last reset or initialized (BEGIN,
29 13.26).
- 30 b) Port State—Discarding, Listening, Learning, or Forwarding (8.4).
- 31 c) Port Identifier—the unique Port identifier comprising two parts, the Port Number and the Port
32 Priority field (13.27.46).
- 33 d) Path Cost (ExternalPortPathCost, 13.27.25).
- 34 e) Designated Root (CIST Root Identifier, 13.27.20).
- 35 f) Designated Cost (External Root Path Cost, 13.27.20).
- 36 g) Designated Bridge (13.27.20).
- 37 h) Designated Port (13.27.20).
- 38 i) Topology Change Acknowledge (13.27.72).

- 1 j) Hello Time (13.27.48).
- 2 k) adminEdgePort (13.27.1). Present in implementations that support the identification of edge ports.
- 3 l) operEdgePort (13.27.44). Present in implementations that support the identification of edge ports.
- 4 m) autoEdgePort (13.27.6). Optional and provided only in implementations that support the automatic
5 identification of edge ports.
- 6 n) autoIsolatePort (13.27.19). Present in implementations that support detection of fragile Bridges.
- 7 o) isolatePort (13.27.27). Present in implementations that support detection of fragile Bridges.
- 8 p) MAC Enabled—the current state of the MAC Enabled parameter (IEEE Std 802.1AC). Present if
9 the implementation supports the MAC Enabled parameter.
- 10 q) MAC Operational—the current state of the MAC Operational parameter (IEEE Std 802.1AC).
11 Present if the implementation supports the MAC Operational parameter.
- 12 r) adminPointToPointMAC—the current state of the adminPointToPointMAC parameter
13 (IEEE Std 802.1AC). Present if the implementation supports the adminPointToPointMAC
14 parameter.
- 15 s) operPointToPointMAC—the current state of the operPointToPointMAC parameter
16 (IEEE Std 802.1AC). Present if the implementation supports the operPointToPointMAC parameter.
- 17 t) restrictedRole—the current state of the restrictedRole parameter for the Port (13.27.63).
- 18 u) restrictedTcn—the current state of the restrictedTcn parameter for the Port (13.27.65).
- 19 v) Port Role—the current Port Role for the Port (i.e., Root, Alternate, Designated, or Backup)
- 20 w) Disputed—the current value of the disputed variable for the CIST for the Port (13.27.22).
- 21 x) enableBPDURx—the value of enableBPDURx (13.27.23).
- 22 y) enableBPDUTx—the value of enableBPDUTx (13.27.24).
- 23 z) pseudoRootId—the value of Bridge Identifier used by the L2 gateway protocol (13.27.51).
- 24 aa) isL2gp—the value of isL2gp (13.27.26).

25 The following additional parameters are present only if the Bridge supports MSTP:

- 26 ab) CIST Regional Root Identifier (13.26.10). The Bridge Identifier of the current CIST Regional Root.
- 27 ac) CIST Path Cost (CIST Internal Root Path Cost, in 13.26.10). The CIST path cost from the
28 transmitting Bridge to the CIST Regional Root.
- 29 ad) Port Hello Time. The administrative value of Hello Time for the Port (13.27.48).

30 12.8.2.2 Read MSTI Port Parameters

31 12.8.2.2.1 Purpose

32 To obtain information regarding a specific Port within the Bridge's Bridge Protocol Entity, for a given
33 MSTI.

34 12.8.2.2.2 Inputs

- 35 a) Port Number—the number of the Bridge Port.
- 36 b) MSTID—identifies the set of parameters that will be returned, in the range 1 through 4094.

37 12.8.2.2.3 Outputs

- 38 a) MSTID—identifies the set of parameters that are being returned. This parameter is the identifier of
39 the spanning tree for which the operation is being performed.
- 40 b) Uptime—count in seconds of the time elapsed since the Port was last reset or initialized (BEGIN).
- 41 c) State—the current state of the Port (i.e., Disabled, Listening, Learning, Forwarding, or Blocking)
42 (8.4, 13.38).

- 1 d) Port Identifier—the unique Port identifier comprising two parts, the Port Number and the Port
- 2 Priority field (13.27.46).
- 3 e) Path Cost (13.27.20).
- 4 f) Designated Root (13.27.20).
- 5 g) Designated Cost (13.27.20).
- 6 h) Designated Bridge (13.27.20).
- 7 i) Designated Port (13.27.20).
- 8 j) Port Role—the current Port Role for the Port (i.e., Root, Alternate, Designated, or Backup,
- 9 13.27.66).
- 10 k) Disputed—the current value of the disputed variable (13.27.22).
- 11 l) pseudoRootId—the new value of Bridge Identifier used by the L2 gateway protocol (13.27.51).

12 12.8.2.3 Set CIST port parameters

13 12.8.2.3.1 Purpose

14 To modify parameters for a Port in the Bridge's Bridge Protocol Entity in order to force a configuration of
15 the spanning tree for the CIST.

16 12.8.2.3.2 Inputs

- 17 a) Port Number—the number of the Bridge Port.
- 18 b) Path Cost—the new ExternalPortPathCost (13.27.25).
- 19 c) Port Priority—the new value of the priority field for the Port Identifier (13.27.46).
- 20 d) adminEdgePort—the new value of the adminEdgePort parameter (13.27.1). Present in
- 21 implementations that support the identification of edge ports.
- 22 e) autoEdgePort—the new value of the autoEdgePort parameter (13.27.6). Optional and provided only
- 23 in implementations that support the automatic identification of edge ports.
- 24 f) autoIsolatePort—the new value of the autoIsolatePort parameter (13.27.19). Present in
- 25 implementations that support detection of fragile Bridges.
- 26 g) MAC Enabled—the new value of the MAC Enabled parameter (IEEE Std 802.1AC). May be
- 27 present if the implementation supports the MAC Enabled parameter.
- 28 h) adminPointToPointMAC—the new value of the adminPointToPointMAC parameter
- 29 (IEEE Std 802.1AC). May be present if the implementation supports the adminPointToPointMAC
- 30 parameter.
- 31 i) restrictedRole—the new value of the restrictedRole parameter for the Port (13.27.63).
- 32 j) restrictedTcn—the new value of the restrictedTcn parameter for the Port (13.27.65).
- 33 k) enableBPDURx—the new value of enableBPDURx (13.27.23).
- 34 l) enableBPDUTx—the new value of enableBPDUTx (13.27.24).
- 35 m) pseudoRootId—the new value of Bridge Identifier used by the L2 gateway protocol (13.27.51).
- 36 n) isL2gp—the new value of isL2gp (13.27.26).

37 12.8.2.3.3 Outputs

- 38 a) Operation status. This takes one of the following values:
- 39 1) Operation rejected due to invalid Port Priority value (12.3)
- 40 2) Operation accepted

12.8.2.3.4 Procedure

In STP Bridges, the Set Path Cost procedure (8.8.6 of IEEE Std 802.1D, 1998 Edition [B13]) is used to set the Path Cost parameter for the specified Port for the specified spanning tree instance. The Set Port Priority procedure (8.8.5 of IEEE Std 802.1D, 1998 Edition [B13]) is used to set the priority part of the Port Identifier (8.5.5.1 of IEEE Std 802.1D, 1998 Edition [B13]) for the CIST to the supplied value.

In RSTP and MSTP Bridges, the Path Cost (13.27.25) and Port Priority (13.27.47) parameters for the Port are updated using the supplied values. The reselect parameter value for the CIST for the Port (13.27.62) is set TRUE, and the selected parameter for the CIST for the Port (13.27.67) is set FALSE.

12.8.2.4 Set MSTI port parameters

12.8.2.4.1 Purpose

To modify parameters for a Port in the Bridge's Bridge Protocol Entity in order to force a configuration of the spanning tree for the specified spanning tree instance.

12.8.2.4.2 Inputs

- a) MSTID—identifies the set of parameters upon which the operation will be performed. This parameter is the identifier of the spanning tree for which the operation is being performed.
- b) Port Number—the number of the Bridge Port.
- c) Path Cost—the new value (13.27.25).
- d) Port Priority—the new value of the priority field for the Port Identifier (13.27.46).
- e) pseudoRootId—the new value of Bridge Identifier used by the L2 gateway protocol (13.27.51).

12.8.2.4.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to invalid Port Priority value (12.3).
 - 2) Operation rejected due to invalid MSTID (i.e., there is currently no spanning tree instance with that value of MSTID supported by the Bridge).
 - 3) Operation accepted.

12.8.2.4.4 Procedure

The Path Cost (13.27.25) and Port Priority (13.27.47) parameters for the specified MSTI and Port are updated using the supplied values. The reselect parameter value for the MSTI for the Port (13.27.62) is set TRUE, and the selected parameter for the MSTI for the Port (13.27.67) is set FALSE.

12.8.2.5 Force BPDU Migration Check

This operation is available only in Bridges that support RSTP or MSTP, as specified in Clause 13.

12.8.2.5.1 Purpose

To force the specified Port to transmit Rapid Spanning Tree (RST) or MST BPDUs (see 13.32).

12.8.2.5.2 Inputs

- a) Port Number—the number of the Bridge Port.

12.8.2.5.3 Outputs

None.

1 12.8.2.5.4 Procedure

2 The mcheck variable (13.27.38) for the specified Port is set to the value TRUE if the value of the
3 forceVersion variable (13.7.2) is greater than or equal to 2.

4 12.9 MRP Entities

5 The operation of MRP is described in Clause 10.

6 This managed resource comprises the following objects:

- 7 a) The MRP Timer objects (12.9.1)
- 8 b) The MRP Attribute Type objects (12.9.2)
- 9 c) The Periodic State Machine objects (12.9.3)

10 12.9.1 The MRP Timer object

11 The MRP Timer object models the operations that can be performed on, or inquire about, the current settings
12 of the timers used by MRP on a given Port. The management operations that can be performed on the MRP
13 Participant are as follows:

- 14 a) Read MRP Timers (12.9.1.1)
- 15 b) Set MRP Timers (12.9.1.2)

16 NOTE—The MRP timer values modeled by this object are the values used to initialize timer instances that are used
17 within the MRP state machines, not the timer instances themselves. Hence, there is a single MRP Timer object per Port,
18 regardless of whether the Bridge supports single or multiple spanning trees.

19 12.9.1.1 Read MRP Timers

20 12.9.1.1.1 Purpose

21 To read the current MRP Timers for a given Port.

22 12.9.1.1.2 Inputs

- 23 a) The Port identifier.

24 12.9.1.1.3 Outputs

- 25 a) Current value of JoinTime—Centiseconds, *Persistent*. (10.7.4.1 and 10.7.11).
- 26 b) Current value of LeaveTime—Centiseconds, *Persistent*. (10.7.4.2 and 10.7.11).
- 27 c) Current value of LeaveAllTime—Centiseconds, *Persistent*. (10.7.4.3 and 10.7.11).

28 12.9.1.2 Set MRP Timers

29 12.9.1.2.1 Purpose

30 To set new values for the MRP Timers for a given Port.

31 12.9.1.2.2 Inputs

- 32 a) The Port identifier.
- 33 b) New value of JoinTime—Centiseconds (10.7.4.1 and 10.7.11).
- 34 c) New value of LeaveTime—Centiseconds (10.7.4.2 and 10.7.11).
- 35 d) New value of LeaveAllTime—Centiseconds (10.7.4.3 and 10.7.11).

1 12.9.1.2.3 Outputs

2 None.

3 12.9.2 The MRP Attribute Type object

4 The MRP Attribute Type object models the operations that can be performed on, or inquire about, the
5 operation of MRP for a given Attribute Type (10.8.2.2). The management operations that can be performed
6 on an MRP Attribute Type are as follows:

- 7 a) Read MRP Applicant Controls (12.9.2.1)
- 8 b) Set MRP Applicant Controls (12.9.2.2)

9 12.9.2.1 Read MRP Applicant Controls

10 12.9.2.1.1 Purpose

11 To read the current values of the MRP Applicant Administrative control parameters (10.7.3) and Transmit
12 Zero parameters (11.2.3.1.7) associated with all MRP Participants for a given Port, MRP Application, and
13 Attribute Type.

14 12.9.2.1.2 Inputs

- 15 a) The Port identifier.
- 16 b) The MRP Application.
- 17 c) The Attribute Type (10.8.2.2).

18 12.9.2.1.3 Outputs

- 19 a) The current Applicant Administrative Control Value, *Persistent*. (10.7.3);
- 20 b) Failed Registrations—count of the number of times that this MRP Application has failed to register
21 an attribute of this type due to lack of space in the FDB, *Persistent*. (12.10.1.6).
- 22 c) Transmit zero enable—whether the transmission of 0 as an Attribute value is currently enabled
23 (MVRP only). (11.2.3.1.7).

24 12.9.2.2 Set MRP Applicant Controls

25 12.9.2.2.1 Purpose

26 To set new values for the MRP Applicant Administrative control parameters (10.7.3) and Transmit Zero
27 parameters (11.2.3.1.7) associated with all MRP Participants for a given Port, MRP Application, and
28 Attribute Type.

29 12.9.2.2.2 Inputs

- 30 a) The Port identifier
- 31 b) The MRP Application
- 32 c) The Attribute Type (10.8.2.2) associated with the state machine
- 33 d) The desired Applicant Administrative Control Value (10.7.3)
- 34 e) The desired Transmit zero enable value (MVRP only) (11.2.3.1.7)

35 12.9.2.2.3 Outputs

36 None.

1 **12.9.2.2.4 Procedures**

2 A Bridge shall not allow both MIRP and MVRP to be enabled on the same VIP.

3 **12.9.3 Periodic state machine objects**

4 The Periodic state machine object models the operations that can be performed upon, or inquire about, the
5 operation of the Periodic state machine for a given Port.

6 The management operations that can be performed on a Periodic state machine are Read Periodic state
7 machine state and Set Periodic state machine state.

8 **12.9.3.1 Read Periodic state machine state**

9 **12.9.3.1.1 Purpose**

10 To inquire whether a particular Periodic state machine is enabled or disabled.

11 **12.9.3.1.2 Inputs**

12 a) The Port identifier.

13 **12.9.3.1.3 Outputs**

14 a) The Port identifier.

15 b) The state of the Periodic state machine. This can take either of the values “enabled” or “disabled.”

16 **12.9.3.2 Set Periodic state machine state**

17 **12.9.3.2.1 Purpose**

18 To enable or disable a particular Periodic state machine.

19 **12.9.3.2.2 Inputs**

20 a) The Port identifier.

21 b) The desired state of the Periodic state machine. This can take either of the values “enabled” or
22 “disabled.”

23 **12.9.3.2.3 Outputs**

24 None.

25 **12.10 Bridge VLAN managed objects**

26 The following managed objects define the semantics of the management operations that can be performed
27 on the VLAN aspects of a Bridge:

28 a) The Bridge VLAN Configuration managed object (12.10.1)

29 b) The VLAN Configuration managed object (12.10.2)

30 c) The VID to FID allocation managed object (12.10.3)

12.10.1 Bridge VLAN Configuration managed object

The Bridge VLAN Configuration managed object models operations that modify, or inquire about, the overall configuration of the Bridge's VLAN resources. There is a single Bridge VLAN Configuration managed object per Bridge Component.

The management operations that can be performed on the Bridge VLAN Configuration managed object are as follows:

- a) Read Bridge VLAN Configuration (12.10.1.1)
- b) Configure PVID and VID Set values (12.10.1.2)
- c) Configure Acceptable Frame Types parameters (12.10.1.3)
- d) Configure Enable Ingress Filtering parameters (12.10.1.4)
- e) Reset Bridge (12.10.1.5)
- f) Configure Restricted_VLAN_Registration parameters (12.10.1.6)
- g) Configure Protocol Group Database (12.10.1.7)
- h) Configure VID to FID allocation (12.10.3)
- i) Configure VID Translation Table (12.10.1.8)
- j) Configure Egress VID Translation Table (12.10.1.9)

12.10.1.1 Read Bridge VLAN Configuration

12.10.1.1.1 Purpose

To obtain general VLAN information from a Bridge.

12.10.1.1.2 Inputs

None.

12.10.1.1.3 Outputs

- a) The IEEE 802.1Q VLAN Version number. Reported as "1" by Bridges that support only SST operation, and reported as "2" by Bridges that support MST operation.

NOTE—No IEEE 802.1Q VLAN version numbers other than 1 and 2 are specified.

- b) The optional VLAN features supported by the implementation:
 - 1) The maximum number of VIDs supported.
 - 2) Whether the implementation supports the ability to override the default PVID setting, and its egress status (VLAN-tagged or untagged) on each Port.
 - 3) For a Bridge that supports Port-and-Protocol-based VLAN classification, which of the Protocol Template formats (6.12.1) are supported by the implementation.
 - 4) For MST Bridges, the maximum number of MSTIs supported within an MST Region (i.e., the number of spanning tree instances that can be supported in addition to the CIST). For SST Bridges, this parameter may be either omitted or reported as "0."
- c) For each Port:
 - 1) The Port number.
 - 2) The PVID value (6.9) currently assigned to that Port.
 - 3) For a Bridge that supports Port-and-Protocol-based VLAN classification, whether the implementation supports Port-and-Protocol-based VLAN classification on that Port.
 - 4) For a Bridge that supports Port-and-Protocol-based VLAN classification on that Port, the maximum number of entries supported in the VID Set on that Port; the VID value and Protocol Group Identifier currently assigned to each entry in the VID Set (8.6.2) on that Port.

- 1 5) The state of the Acceptable Frame Types parameter (6.9). The permissible values for this
2 parameter are as follows:
 - 3 i) *Admit Only VLAN-tagged frames*
 - 4 ii) *Admit Only Untagged and Priority Tagged frames*
 - 5 iii) *Admit All frames*
- 6 6) The state of the Enable Ingress Filtering parameter (6.9); Enabled or Disabled;
- 7 7) The state of the Restricted_VLAN_Registration parameter (11.2.3.2.3), TRUE or FALSE.
- 8 8) Whether the Bridge supports a VID Translation Table on that Port; whether the Bridge supports
9 an Egress VID Translation Table on that Port;
- 10 9) For a Bridge that supports a VID Translation Table on that Port, a set of (Local VID, Relay
11 VID) bindings (6.9);
- 12 10) For a Bridge that supports an Egress VID Translation Table on that Port, a set of (Relay VID,
13 Local VID) bindings (6.9).
- 14 d) For a Bridge that supports Port-and-Protocol-based VLAN classification: the contents of the
15 Protocol Group Database comprising a set of {Protocol Template, Protocol Group Identifier}
16 bindings (6.12.1, 6.12.2, and 6.12.3); the maximum number of entries supported in the Protocol
17 Group Database.

18 12.10.1.2 Configure PVID and VID Set values

19 12.10.1.2.1 Purpose

20 To configure the PVID and VID Set value(s) (6.9) associated with one or more Ports.

21 12.10.1.2.2 Inputs

- 22 a) For each Port to be configured, a Port number and the PVID value to be associated with that Port.
- 23 b) In addition, for a Bridge that supports Port-and-Protocol-based VLAN classification: for each Port
24 to be configured, a Port number, a Protocol Group Identifier, and a VID value for the member of the
25 Port's VID Set that is to be configured.

26 12.10.1.2.3 Outputs

- 27 a) Operation status for each Port to be configured. This takes one of the following values:
 - 28 1) Operation rejected due to there being no spare VID Set entries on this Port
 - 29 2) Operation rejected due to the PVID or VID being out of the supported range for this Port
 - 30 3) Operation accepted

31 12.10.1.3 Configure Acceptable Frame Types parameters

32 12.10.1.3.1 Purpose

33 To configure the Acceptable Frame Types parameter (6.9) associated with one or more Ports.

34 12.10.1.3.2 Inputs

- 35 a) For each Port to be configured, a Port number and the value of the Acceptable Frame Types
36 parameter to be associated with that Port. The permissible values of this parameter are (as defined in
37 6.9) as follows:
 - 38 1) *Admit Only VLAN-tagged frames*
 - 39 2) *Admit Only Untagged and Priority Tagged frames*
 - 40 3) *Admit All frames*

1 **12.10.1.3.3 Outputs**

2 None.

3 **12.10.1.4 Configure Enable Ingress Filtering parameters**

4 **12.10.1.4.1 Purpose**

5 To configure the Enable Ingress Filtering parameter(s) (8.6.2) associated with one or more Ports.

6 **12.10.1.4.2 Inputs**

7 a) For each Port to be configured, a Port number and the value of the Enable Ingress Filtering
8 parameter to be associated with that Port. The permissible values for the parameter are as follows:

- 9 1) Enabled
- 10 2) Disabled

11 **12.10.1.4.3 Outputs**

12 None.

13 **12.10.1.5 Reset Bridge**

14 **12.10.1.5.1 Purpose**

15 To reset all statically configured VLAN-related information in the Bridge to its default state. This operation:

- 16 a) Deletes all VLAN Configuration managed objects.
- 17 b) Resets the PVID associated with each Bridge Port to the Default PVID value (Table 9-2).
- 18 c) Removes all entries in the Protocol Group Database and removes all members of the VID Set on
19 each port, for a Bridge that supports Port-and-Protocol-based VLAN classification.
- 20 d) Resets the Acceptable Frame Types parameter value associated with each Port to the default value
21 (6.9).

22 **12.10.1.5.2 Inputs**

23 None.

24 **12.10.1.5.3 Outputs**

25 None.

26 **12.10.1.6 Configure Restricted_VLAN_Registration parameters**

27 **12.10.1.6.1 Purpose**

28 To configure the Restricted_VLAN_Registration parameter (11.2.3.2.3) associated with one or more Ports.

29 **12.10.1.6.2 Inputs**

30 a) For each Port to be configured, a Port number and the value of the Restricted_VLAN_Registration
31 parameter. The permissible values of this parameter are (as defined in 11.2.3.2.3) as follows:

- 32 1) TRUE
- 33 2) FALSE

34 **12.10.1.6.3 Outputs**

35 None.

1 **12.10.1.7 Configure Protocol Group Database**

2 To configure a Protocol Group Database (6.12.3) entry. This operation is not applicable to a Bridge that does
3 not support Port-and-Protocol-based VLAN classification.

4 NOTE—Implementation of the Configure Protocol Group Database operation is not mandatory; conformant
5 implementations may implement a fixed set of Protocol Group Database entries.

6 **12.10.1.7.1 Inputs**

- 7 a) A value representing the frame format to be matched: Ethernet, RFC_1042, SNAP_8021H,
8 SNAP_Other, or LLC_Other (6.12.1).
- 9 b) One of:
 - 10 1) An IEEE 802.3 EtherType value, for matching frame formats of Ethernet, RFC_1042, or
 - 11 SNAP_8021H.
 - 12 2) A 40-bit Protocol Identifier (PID), for matching frame formats of SNAP_Other.
 - 13 3) A pair of ISO/IEC 8802-2 DSAP and SSAP address field values, for matching frame formats of
 - 14 LLC_Other.
- 15 c) A Protocol Group Identifier (6.12.2).

16 NOTE—While the intent is to identify VID traffic, an ID of 0 may be used to indicate untagged traffic that needs
17 prioritization.

18 **12.10.1.7.2 Outputs**

- 19 a) Operation status. This takes one of the following values:
 - 20 1) Operation rejected due to there being no spare Protocol Group Database entries.
 - 21 2) Operation rejected due to an unsupported frame format.
 - 22 3) Operation rejected due to an unsupported value for an IEEE 802.3 EtherType value, PID,
 - 23 DSAP, or SSAP.
 - 24 4) Operation accepted.

25 **12.10.1.8 VID Translation Configuration managed object**

26 To configure the VID Translation Table (6.9) associated with a Port. This object is not applicable to Ports
27 that do not support a VID Translation Table. The default configuration of the table has the value of the Relay
28 VID equal to the value of the Local VID for all Local VID values. If the port supports an Egress VID
29 translation table, the VID Translation Configuration object configures the Local VID to Relay VID mapping
30 on ingress only. If an Egress VID translation is not supported, the VID Translation Configuration object
31 defines a single bidirectional mapping.

32 The management operations that can be performed are as follows:

- 33 a) Read VID Translation Table Entry (12.10.1.8.1)
- 34 b) Configure VID Translation Table Entry (12.10.1.8.2)

35 **12.10.1.8.1 Read VID Translation Table Entry**

36 **12.10.1.8.1.1 Purpose**

37 To read the relay VID associated with the local VID.

38 **12.10.1.8.1.2 Inputs**

- 39 a) Port Number: the number of the port that is capable of performing VID translation.
- 40 b) Local VLAN Identifier: a 12-bit VID.

1 **12.10.1.8.1.3 Outputs**

- 2 a) Relay VLAN Identifier: a 12-bit VID.

3 **12.10.1.8.2 Configure VID Translation Table Entry**

4 **12.10.1.8.2.1 Purpose**

5 To modify an entry in the VID Translation Table.

6 **12.10.1.8.2.2 Inputs**

- 7 a) Port Number: the number of the port that is capable of performing VID translation. This port should
8 be at the edge of an administrative or a control protocol domain (6.9).
9 b) Local VLAN Identifier: a 12-bit VID.
10 c) Relay VLAN Identifier: a 12-bit VID.

11 **12.10.1.8.2.3 Outputs**

- 12 a) Operation status. This takes one of the following values:
13 1) Operation rejected because the port does not have Egress VID translation capabilities
14 2) Operation accepted

15 **12.10.1.9 Egress VID Translation Configuration managed object**

16 To configure the Egress VID Translation Table (6.9) associated with a Port. This object is not applicable to
17 Ports that do not support an Egress VID Translation Table. The default configuration of the table has the
18 value of the Local VID equal to the value of the Relay VID for all Relay VID values.

19 The management operations that can be performed are as follows:

- 20 a) Read Egress VID Translation Table Entry (12.10.1.9.1)
21 b) Configure Egress VID Translation Table Entry (12.10.1.9.2)

22 **12.10.1.9.1 Read Egress VID Translation Table Entry**

23 **12.10.1.9.1.1 Purpose**

24 To read the local VID associated with the relay VID.

25 **12.10.1.9.1.2 Inputs**

- 26 a) Port Number: the number of the port that is capable of performing Egress VID translation.
27 b) Relay VLAN Identifier: a 12-bit VID.

28 **12.10.1.9.1.3 Outputs**

- 29 a) Local VLAN Identifier: a 12-bit VID.

30 **12.10.1.9.2 Configure Egress VID Translation Table Entry**

31 **12.10.1.9.2.1 Purpose**

32 To modify an entry in the Egress VID Translation Table.

33 **12.10.1.9.2.2 Inputs**

- 34 a) Port Number: the number of the port that is capable of performing Egress VID translation. This port
35 should be at the edge of an administrative or a control protocol domain (6.9).
36 b) Relay VLAN Identifier: a 12-bit VID.
37 c) Local VLAN Identifier: a 12-bit VID.

1 **12.10.1.9.2.3 Outputs**

- 2 a) Operation rejected because the port does not have Egress VID translation capabilities.
- 3 b) Operation accepted.

4 **12.10.2 VLAN Configuration managed object**

5 The VLAN Configuration object models operations that modify, or inquire about, the configuration of a
6 particular VID within a Bridge. There are multiple VLAN Configuration objects per Bridge; only one such
7 object can exist for a given VID.

8 The management operations that can be performed on the VLAN Configuration are as follows:

- 9 a) Read VLAN Configuration (12.10.2.1)
- 10 b) Create VLAN Configuration (12.10.2.2)
- 11 c) Delete VLAN Configuration (12.10.2.3)

12 **12.10.2.1 Read VLAN Configuration**

13 **12.10.2.1.1 Purpose**

14 To obtain general information regarding a specific VLAN Configuration.

15 **12.10.2.1.2 Inputs**

- 16 a) VLAN Identifier: a 12-bit VID.

17 **12.10.2.1.3 Outputs**

- 18 a) VLAN Name: A text string of up to 32 characters of locally determined significance.
- 19 b) List of Untagged Ports: The set of Port numbers in the untagged set (8.8.2) for this VID.
- 20 c) List of Egress Ports: The set of Port numbers in the member set (8.8.10) for this VID.

21 NOTE—The values of the member set and the untagged set are determined by the values held in VLAN Registration
22 Entries in the FDB (8.8.2, 8.8.5, and 8.8.10).

23 **12.10.2.2 Create VLAN Configuration**

24 **12.10.2.2.1 Purpose**

25 To create or update a VLAN Configuration managed object.

26 **12.10.2.2.2 Inputs**

- 27 a) VLAN Identifier: a 12-bit VID.
- 28 b) VLAN Name: a text string of up to 32 characters of locally determined significance.

29 NOTE—Static configuration of the member set and the Untagged set is achieved by means of the management
30 operations for manipulation of VLAN Registration Entries (12.7.5).

31 **12.10.2.2.3 Outputs**

32 None.

33 **12.10.2.3 Delete VLAN Configuration**

34 **12.10.2.3.1 Purpose**

35 To delete a VLAN Configuration managed object.

1 **12.10.2.3.2 Inputs**

- 2 a) VLAN Identifier: a 12-bit VID.

3 **12.10.2.3.3 Outputs**

- 4 None.

5 **12.10.3 The VID to FID allocation managed object**

6 The VID to FID allocations managed object models operations that modify, or inquire about VID to FID
7 allocations (8.8.8) that apply to the operation of the Learning Process and the FDB. The object is modeled as
8 a fixed-length table, as follows:

- 9 a) A VID to FID allocation table (8.8.8) with an entry per VID supported by the implementation. Each
10 table entry indicates, for that VID, that there is currently:
11 1) No allocation defined or
12 2) A fixed allocation to FID X or
13 3) A dynamic allocation to FID X.

14 The management operations that can be performed on the VID to FID allocations managed object are as
15 follows:

- 16 b) Read VID to FID allocations (12.10.3.1)
17 c) Read FID allocation for VID (12.10.3.2)
18 d) Read VIDs allocated to FID (12.10.3.3)
19 e) Set VID to FID allocation (12.10.3.4)
20 f) Delete VID to FID allocation (12.10.3.5)

21 **12.10.3.1 Read VID to FID allocations**

22 **12.10.3.1.1 Purpose**

23 To read the contents of a range of one or more entries in the VID to FID allocation table.

24 **12.10.3.1.2 Inputs**

- 25 a) First Entry—VID of first entry to be read.
26 b) Last Entry—VID of last entry to be read.

27 **12.10.3.1.3 Outputs**

- 28 a) List of Entries—For each entry that was read:
29 1) VID—the VLAN Identifier for this entry.
30 2) Allocation Type—the type of the allocation: Undefined, Fixed, or Dynamic.
31 3) FID—the FID to which the VID is allocated (if not of type Undefined).

32 NOTE 1—Where this operation is implemented using a remote management protocol, PDU size constraints may restrict
33 the number of entries that are actually read to fewer than was requested in the input parameters. In such cases, retrieving
34 the remainder of the desired entry range can be achieved by repeating the operation with a modified entry range
35 specification.

36 NOTE 2—The Allocation Type of Dynamic is applicable only for SPT Bridges and VIDs that have been reserved for use
37 as SPVIDs.

38 **12.10.3.2 Read FID allocation for VID**

39 **12.10.3.2.1 Purpose**

40 To read the FID to which a specified VID is currently allocated.

1 12.10.3.2.2 Inputs

- 2 a) VID—the VLAN Identifier to which the read operation applies.

3 12.10.3.2.3 Outputs

- 4 a) VID—the VLAN Identifier to which the read operation applies.
5 b) Allocation Type—the type of the allocation: Undefined, Fixed, or Dynamic.
6 c) FID—the FID to which the VID is allocated (if not of type Undefined).

7 NOTE—The Allocation Type of Dynamic is applicable only for SPT Bridges and VIDs that have been reserved for use
8 as SPVIDs.

9 12.10.3.3 Read VIDs allocated to FID

10 12.10.3.3.1 Purpose

11 To read all VIDs currently allocated to a given FID.

12 12.10.3.3.2 Inputs

- 13 a) FID—the Filtering Identifier to which the read operation applies.

14 12.10.3.3.3 Outputs

- 15 a) FID—the Filtering Identifier to which the read operation applies.
16 b) Allocation List—a list of allocations for this FID. For each element in the list:
17 1) Allocation Type—the type of the allocation: Fixed or Dynamic.
18 2) VID—the VID that is allocated.

19 NOTE—The Allocation Type of Dynamic is applicable only for SPT Bridges and VIDs that have been reserved for use
20 as SPVIDs.

21 12.10.3.4 Set VID to FID allocation

22 12.10.3.4.1 Purpose

23 To establish a fixed allocation of a VID to an FID.

24 12.10.3.4.2 Inputs

- 25 a) VID—the VID of the entry to be set.
26 b) FID—the FID to which the VID is to be allocated.

27 12.10.3.4.3 Outputs

- 28 a) Operation status. This takes one of the following values:
29 1) Operation rejected due to VID exceeding the maximum VID supported by the allocation table
30 2) Operation rejected due to FID exceeding the maximum ID supported by the implementation
31 3) Operation accepted

32 12.10.3.4.4 Procedure

33 In MST Bridges, the Configuration Digest element of the MCID is recalculated, in accordance with the
34 definition in 13.8, following any change in the allocation of VIDs to FIDs that results in a change in the
35 allocation of VIDs to spanning trees.

1 **12.10.3.5 Delete VID to FID allocation**

2 **12.10.3.5.1 Purpose**

3 To remove a fixed VID to FID allocation from the VID to FID allocation table. This operation has the effect
4 of setting the value of the specified table entry to “Undefined.”

5 NOTE—If the VID concerned represents a currently active VID, then removal of a fixed allocation may result in the
6 “Undefined” value in the table immediately being replaced by a dynamic allocation to an FID.

7 **12.10.3.5.2 Inputs**

- 8 a) VID—VID of the allocation to be deleted.

9 **12.10.3.5.3 Outputs**

- 10 a) Operation status. This takes one of the following values:
11 1) Operation rejected due to VID exceeding the maximum value supported by the allocation table
12 2) Operation accepted

13 **12.10.3.5.4 Procedure**

14 In MST Bridges, the Configuration Digest element of the MCID is recalculated, in accordance with the
15 definition in 13.8, and the MSTP state machine variables are reinitialized by asserting BEGIN, following
16 any change in the allocation of VID to FIDs that results in a change in the allocation of VID to spanning
17 trees.

18 **12.11 MMRP entities**

19 The following managed object defines the semantics of the management operations that can be performed
20 on the operation of MMRP in a Bridge:

- 21 a) The MMRP Configuration managed object (12.11.1)

22 **12.11.1 MMRP Configuration managed object**

23 The MMRP Configuration managed object models operations that modify, or inquire about, the overall
24 configuration of the operation of MMRP. There is a single MMRP Configuration managed object per
25 Bridge.

26 The management operations that can be performed on the MMRP Configuration managed object are as
27 follows:

- 28 a) Read MMRP Configuration (12.11.1.1)
29 b) Notify MAC address registration failure (12.11.1.2)
30 c) Configure Restricted_MAC_Address_Registration parameters (12.11.1.3)

31 **12.11.1.1 Read MMRP Configuration**

32 **12.11.1.1.1 Purpose**

33 To obtain general MMRP configuration information from a Bridge.

34 **12.11.1.1.2 Inputs**

35 None.

1 12.11.1.1.3 Outputs

- 2 a) For each Port:
 - 3 1) The Port number.
 - 4 2) The state of the Restricted_MAC_Address_Registration parameter, *Persistent*. (10.12.2.3),
 - 5 TRUE or FALSE.

6 12.11.1.2 Notify MAC address registration failure

7 12.11.1.2.1 Purpose

8 To notify a manager that MMRP has failed to register a given MAC address owing to lack of
9 resources in the FDB for the creation of a MAC Address Registration Entry (8.8.4) or to the
10 Restricted_MAC_Address_Registration parameter.

11 12.11.1.2.2 Inputs

12 None.

13 12.11.1.2.3 Outputs

- 14 a) The MAC address that MMRP failed to register
- 15 b) The Port number of the Port on which the registration request was received
- 16 c) The reason for the failure:
 - 17 1) Lack of Resources or
 - 18 2) Registration Restricted

19 12.11.1.3 Configure Restricted_MAC_Address_Registration parameters

20 12.11.1.3.1 Purpose

21 To configure the Restricted_MAC_Address_Registration parameter (10.12.2.3) associated with one or more
22 Ports.

23 12.11.1.3.2 Inputs

- 24 a) For each Port to be configured, a Port number and the value of the
25 Restricted_MAC_Address_Registration parameter. The permissible values of this parameter are as
26 follows (as defined in 10.12.2.3):
 - 27 1) TRUE
 - 28 2) FALSE

29 12.11.1.3.3 Outputs

30 None.

31 12.12 MST configuration entities

32 The following managed objects define the semantics of the management operations that can be performed
33 on the MST configuration in a Bridge:

- 34 a) The MSTI List object (12.12.1)
- 35 b) The FID to MSTID Allocation Table object (12.12.2)
- 36 c) The MST Configuration Table object (12.12.3)

1 **12.12.1 The MSTI List**

2 For MST Bridges, the MSTI List object models the operations that modify, or inquire about, the list of MST
3 spanning tree instances supported by the Bridge. The object is modeled as a list of MSTIDs corresponding to
4 the MSTIs supported by the Bridge.

5 The MSTID List object supports the following operations:

- 6 a) Read MSTI List (12.12.1.1)
- 7 b) Create MSTI (12.12.1.2)
- 8 c) Delete MSTI (12.12.1.3)

9 **12.12.1.1 Read MSTI List**

10 **12.12.1.1.1 Purpose**

11 To read the list of MSTIDs that are currently supported by the Bridge.

12 **12.12.1.1.2 Inputs**

13 None.

14 **12.12.1.1.3 Outputs**

- 15 a) MSTID list. The list of MSTID values that are currently supported by the Bridge.

16 **12.12.1.2 Create MSTI**

17 **12.12.1.2.1 Purpose**

18 To create a new MSTI and its associated state machines and parameters, and to add its MSTID to the MSTI
19 List.

20 **12.12.1.2.2 Inputs**

- 21 a) The MSTID of the MSTI to be created.

22 **12.12.1.2.3 Outputs**

- 23 a) Operation status. This takes one of the following values:
 - 24 1) Operation rejected due to the number of MSTIs currently supported by the Bridge being equal
25 to the maximum number of MSTIs that the Bridge is able to support.
 - 26 2) Operation rejected as the MSTID value supplied in the input parameters is already present in
27 the MSTI List.
 - 28 3) Operation rejected as the MSTID value supplied in the input parameters is one of the reserved
29 MSTID values (8.9.3).
 - 30 4) Operation accepted.

31 **12.12.1.3 Delete MSTI**

32 **12.12.1.3.1 Purpose**

33 To delete an existing MSTI and its associated state machines and parameters, and to remove its MSTID from
34 the MSTI List.

1 12.12.1.3.2 Inputs

- 2 a) The MSTID of the MSTI to be deleted.

3 12.12.1.3.3 Outputs

- 4 a) Operation status. This takes one of the following values:
- 5 1) Operation rejected as the MSTID value supplied in the input parameters is not present in the
- 6 MSTI List.
- 7 2) Operation rejected as the MSTID value supplied in the input parameter currently has one or
- 8 more FIDs allocated to it in the FID to MSTID Allocation Table.
- 9 3) Operation accepted.

10 12.12.2 The FID to MSTID Allocation Table

11 For MST Bridges, the FID to MSTID Allocation Table object models the operations that modify, or inquire
12 about, the assignment of FIDs to spanning tree instances currently supported by the Bridge (8.9.3). The
13 object is modeled as a fixed-length table in which each entry in the table corresponds to a FID, and the value
14 of the entry specifies the MSTID of the spanning tree to which the set of VIDs supported by that FID are
15 assigned. A value of zero in an entry specifies that the set of VIDs supported by that FID are assigned to the
16 CST.

17 The MSTID Allocation Table object supports the following operations:

- 18 a) Read FID to MSTID allocations (12.12.2.1).
- 19 b) Set FID to MSTID allocation (12.12.2.2).

20 12.12.2.1 Read FID to MSTID allocations

21 12.12.2.1.1 Purpose

22 To read a range of one or more entries in the FID to MSTID Allocation Table.

23 12.12.2.1.2 Inputs

- 24 a) First FID—the FID of the first entry to be read.
- 25 b) Last FID—the FID of the last entry to be read.

26 If the value of Last FID is numerically equal to, or smaller than, the value of First FID, then a single table
27 entry is read, corresponding to the value of First FID.

28 12.12.2.1.3 Outputs

- 29 a) List of entries—for each entry that was read:
- 30 1) The FID of the entry
- 31 2) The MSTID to which that FID is allocated

32 12.12.2.2 Set FID to MSTID allocation

33 12.12.2.2.1 Purpose

34 To change the contents of an entry in the FID to MSTID Allocation Table.

35 12.12.2.2.2 Inputs

- 36 a) FID—the FID of the entry to be changed.
- 37 b) MSTID—the MSTID to which the FID is to be allocated.

12.12.2.2.3 Outputs

- a) Operation status. This takes one of the following values:
- 1) Operation rejected as the MSTID value supplied in the input parameters is not present in the MSTI List.
 - 2) Operation rejected as the FID value supplied in the input parameters is invalid or is not supported.
 - 3) Operation accepted.

12.12.2.2.4 Procedure

The Configuration Digest element of the MCID is recalculated, in accordance with the definition in 13.8, following any change in the allocations of FIDs to MSTIDs.

12.12.3 The MST Configuration Table

The MST Configuration Table managed object models the operations that can be performed on the MST Configuration Table for the Bridge (3.173, 8.9.1, and 13.8). Associated with the table is the MCID for the Bridge (8.9.2, 13.8).

The MST Configuration Table is a read-only table, its elements derived from other configuration information held by the Bridge; specifically, the current state of the VID to FID allocation table (8.8.8, 12.10.3), and the FID to MSTID allocation table (8.9.3, 12.12.2). Hence, changes made to either of these tables can in turn affect the contents of the MST Configuration Table, and also affect the value of the “digest” element of the MCID. The MST Configuration Table is modeled as a fixed table of 4096 elements, as described in 13.8.

The MST Configuration Table managed object supports the following operations:

- a) Read MST Configuration Table Element (12.12.3.1).
- b) Read VIDs assigned to MSTID (12.12.3.2).
- c) Read MST Configuration Identifier (12.12.3.3).
- d) Set MST Configuration Identifier Elements (12.12.3.4).

12.12.3.1 Read MST Configuration Table Element

12.12.3.1.1 Purpose

To read a single element of the current MST Configuration Table for the Bridge (13.8).

12.12.3.1.2 Inputs

- a) A VID value, in the range 0 through 4094.

12.12.3.1.3 Outputs

- a) A VID value, in the range 0 through 4094.
- b) The MSTID value corresponding to that VID.

12.12.3.2 Read VIDs assigned to MSTID

12.12.3.2.1 Purpose

To read the list of VIDs that are currently assigned to a given MSTID in the MST Configuration Table for the Bridge (13.8).

1 **12.12.3.2.2 Inputs**

- 2 a) An MSTID value, in the range 0 through 4095.

3 **12.12.3.2.3 Outputs**

- 4 a) A MSTID value, in the range 0 through 4095.
5 b) A 4096-bit vector in which bit N is set TRUE if VID N is assigned to the given MSTID and is
6 otherwise set FALSE.

7 **12.12.3.3 Read MST Configuration Identifier**

8 **12.12.3.3.1 Purpose**

9 To read the current value of the MCID for the Bridge (13.8).

10 **12.12.3.3.2 Inputs**

11 None.

12 **12.12.3.3.3 Outputs**

- 13 a) The MCID (13.8), consisting of:
14 1) The Configuration Identifier Format Selector in use by the Bridge. This has a value of 0 to
15 indicate the format specified in this standard.
16 2) The Configuration Name.
17 3) The Revision Level.
18 4) The Configuration Digest.

19 **12.12.3.4 Set MST Configuration Identifier Elements**

20 **12.12.3.4.1 Purpose**

21 To change the current values of the modifiable elements of the MCID for the Bridge (13.8).

22 NOTE—The Configuration Digest element of the MCID is read-only; its value is recalculated whenever configuration
23 changes occur that result in a change in the allocation of VIDs to MSTIs.

24 **12.12.3.4.2 Inputs**

- 25 a) The MCID (13.8) Format Selector in use by the Bridge. This has a value of 0 to indicate the format
26 specified in this standard.
27 b) The Configuration Name.
28 c) The Revision Level.

29 **12.12.3.4.3 Outputs**

- 30 a) Operation Status. This can take the following values:
31 1) Operation rejected due to unsupported Configuration Identifier Format Selector value
32 2) Operation accepted

12.13 Provider Bridge management

The conformance requirements of Provider Bridges are specified in 5.10. The S-VLAN Component and the externally accessible ports of all Provider Bridges, including Provider Edge Bridges, are managed using the managed objects defined in 12.4 through 12.12. This subclause defines additional managed objects specific to the operation of Provider Bridges.

The internal ports, LANs, C-VLAN Components, and Port-mapping S-VLAN Components of a Provider Edge Bridge are not managed directly using the managed objects defined in 12.4 through 12.12. Their operation is controlled and monitored through managed objects defined in this subclause.

Each externally accessible Bridge Port on a Provider Bridge is designated as a PNP, CNP, CEP, or RCAP. Designating a port as a CEP implies Provider Edge Bridge functionality and, specifically, the existence of a C-VLAN Component associated with that port. This C-VLAN Component is uniquely identified within the Bridge by the port number of the associated CEP. The management of the forwarding process, filtering data base, and C-VLANs of the C-VLAN Component and the internal connections are achieved through the Customer Edge Port Configuration managed object defined here (12.13.2).

Designating a port as a RCAP implies the existence of a Port-mapping S-VLAN Component associated with that port. The Port-mapping S-VLAN Component is uniquely identified within the Bridge by the port number of the associated RCAP. Designating an internal port on the Port-mapping S-VLAN Component as a C-tagged RCSI implies a C-VLAN Component with a CEP connected via an internal LAN to that internal port (PAP). This C-VLAN Component and internal connection are uniquely identified by the port number of the RCAP and the S-VID associated with the PAP or by the port number of the internal CEP. Designating an internal port on the Port-mapping S-VLAN Component as a Port-based RCSI or a PNP implies an internal LAN connecting this port with a CNP or PNP, respectively. These ports and internal connections are uniquely identified by the port number of the RCAP and the S-VID associated with the PAP or by the port number of the CNP or PNP. The management of the forwarding process, filtering data base, and S-VLANs of Port-mapping S-VLAN Components and their internal connections are achieved through the Remote Customer Access Port Configuration managed object defined here (12.13.3).

An internal connection between a CNP on the S-VLAN Component and a PEP on the C-VLAN Component is instantiated for each service instance. The CNP is identified by the CEP identifier and the S-VID value used for the PVID of the CNP. The PEP is identified by the CEP identifier and the C-VLAN Identifier (C-VID) value used for the PVID of the PEP. These PVID values and the connection between the CNP and PEP are established by reciprocal entries in the C-VID Registration Table (12.13.2.2) and the PEP Configuration Table (12.13.2.4) as follows:

- a) An entry in the C-VID Registration Table is created for each C-VLAN supported in the C-VLAN Component associated with a CEP. The CEP identifier and C-VID combination identify a PEP. The entry contains (among other parameters) an S-VID value corresponding to the PVID of the CNP, which associates the PEP with a specific CNP. Note that the CEP/C-VID combination identifies a single PEP; however, multiple CEP/C-VID combinations can identify the same PEP if the entries for those CEP/C-VID combinations contain the same S-VID value. This many-to-one mapping of CEP/C-VID to PEP permits “bundling”, i.e., mapping multiple C-VLANs to the same service instance.
- b) An entry in the PEP Configuration Table is created for each S-VID corresponding to a service instance accessed by the C-VLAN Component. The CEP identifier and S-VID combination identify a CNP. The entry contains (among other parameters) a C-VID value corresponding to the PVID of the PEP, which associates the CNP with a specific PEP. Note that the CEP/S-VID combination identifies a single CNP; however, multiple CEP/S-VID combinations can identify the same CNP if the entries for those CEP/S-VID combinations contain the same C-VID value. This many-to-one mapping of CEP/S-VID to CNP permits configuration of asymmetric VLANs and can be used to establish rooted-multipoint connectivity (F.1.3.2).

1 Management control of the member sets and untagged sets for C-VIDs in the C-VLAN Component is
2 provided in C-VID Registration Table entries rather than through Static VLAN Registration Entries (thus
3 eliminating the need for Filtering Database managed objects for the C-VLAN Component). Management
4 control of the member sets and untagged sets for S-VIDs in the S-VLAN Component is provided through
5 Static VLAN Registration Entries in the FDB. Creating an entry for an S-VID in the PEP Configuration
6 Table does not automatically modify the Static VLAN Registration Entries for the corresponding S-VLAN.
7 A CNP is added to the member set and untagged set of an S-VLAN by including the CEP identifier (since
8 the CEP identifier and S-VID combination identifies the CNP) in the Port Map of a Static VLAN
9 Registration Entry for that S-VLAN (12.7.7.1, 8.8.2).

10 A C-VLAN Component with more than one PEP (i.e., supporting more than one service instance)
11 participates in the customer network spanning tree protocol by running an instance of RSTP with the
12 enhancements specified in 13.41. This protocol instance is managed using the managed objects defined in
13 12.8 and 12.12. All BPDUs generated by this protocol instance use the MAC address of the CEP as a source
14 address and as the Bridge address portion of Bridge Identifier. For each PEP, the protocol uses the S-VID
15 value that is the PVID of the associated CNP as the port number. For the CEP, the protocol uses the value
16 0xFFFF as the port number.

17 The following managed objects define the semantics of the management operations specific to Provider
18 Bridges:

- 19 c) The Provider Bridge Port Type managed object (12.13.1)
- 20 d) The Customer Edge Port Configuration managed object (12.13.2)
- 21 e) The Remote Customer Access Port Configuration managed object (12.13.3)

22 12.13.1 Provider Bridge Port Type managed object

23 The management operations that can be performed on the Provider Bridge Port Type managed object are as
24 follows:

- 25 a) Read Provider Bridge Port Type (12.13.1.1)
- 26 b) Configure Provider Bridge Port Type (12.13.1.2)

27 12.13.1.1 Read Provider Bridge Port Type

28 12.13.1.1.1 Purpose

29 To obtain information regarding the designated type of an externally accessible port on a Provider Bridge.

30 12.13.1.1.2 Inputs

- 31 a) Port Number: the number of the Bridge Port.

32 12.13.1.1.3 Outputs

- 33 a) Port Type. This takes one of the following values:
 - 34 1) PNP
 - 35 2) CNP
 - 36 3) CEP
 - 37 4) RCAP

38 12.13.1.2 Configure Provider Bridge Port Type

39 12.13.1.2.1 Purpose

40 To designate the type of an externally accessible port on a Provider Bridge.

1 12.13.1.2.2 Inputs

- 2 a) Port Number: the number of the Bridge Port.
- 3 b) Port Type. This takes one of the following values:
 - 4 1) PNP
 - 5 2) CNP
 - 6 3) CEP
 - 7 4) RCAP

8 12.13.1.2.3 Outputs

9 None.

10 12.13.2 Customer Edge Port Configuration managed object

11 The Customer Edge Port Configuration managed object applies to each externally accessible CEP on a
12 Provider Edge Bridge. It includes:

- 13 a) The C-VID Registration Table, which provides a mapping between a C-VID and the service instance
14 represented by an S-VID selected for that C-VLAN. This table provides the equivalent functionality
15 of :
 - 16 1) configuring the PVID of the internal CNP on the S-VLAN Component;
 - 17 2) adding the corresponding PEP on the C-VLAN Component to the member set of the C-VLAN;
 - 18 3) adding the PEP and/or CEP to the untagged set of the C-VLAN (if it is desired that frames
19 forwarded to that port are transmitted untagged for this C-VLAN).
- 20 b) The PEP configuration parameters, which provide the subset of the Bridge VLAN Configuration
21 managed object (12.10.1) that is relevant for the internal ports of the C-VLAN Component
22 associated with the CEP.
- 23 c) The Service Priority Regeneration Table, which provides the Priority Regeneration Table (12.6.2)
24 for each internal CNP connected to the C-VLAN Component associated with the CEP.

25 The management operations that can be performed are as follows:

- 26 d) Read C-VID Registration Table Entry (12.13.2.1)
- 27 e) Configure C-VID Registration Table Entry (12.13.2.2)
- 28 f) Read Provider Edge Port Configuration (12.13.2.3)
- 29 g) Set Provider Edge Port Configuration (12.13.2.4)
- 30 h) Read Service Priority Regeneration Table (12.13.2.5)
- 31 i) Set Service Priority Regeneration Table (12.13.2.6)

32 12.13.2.1 Read C-VID Registration Table Entry

33 12.13.2.1.1 Purpose

34 To read the VID of the service associated with a specific Customer VLAN in the C-VLAN Component of a
35 Provider Edge Bridge.

36 12.13.2.1.2 Inputs

- 37 a) Port Number: the number of the CEP.
- 38 b) Customer VLAN Identifier: a 12-bit C-VID.

12.13.2.1.3 Outputs

- a) Service VLAN Identifier: a 12-bit S-VID.
- b) Untagged PEP: a boolean indicating frames for this C-VLAN should be forwarded untagged through the Provider Edge Port.
- c) Untagged CEP: a boolean indicating frames for this C-VLAN should be forwarded untagged through the Customer Edge Port.

12.13.2.2 Configure C-VID Registration Table Entry

12.13.2.2.1 Purpose

To modify an entry in the C-VID Registration Table.

12.13.2.2.2 Inputs

- a) Port Number: the number of the CEP.
- b) Customer VLAN Identifier: a 12-bit C-VID.
- c) Service VLAN Identifier: a 12-bit S-VID.
- d) Untagged PEP: a boolean indicating frames for this C-VLAN should be forwarded untagged through the Provider Edge Port.
- e) Untagged CEP: a boolean indicating frames for this C-VLAN should be forwarded untagged through the Customer Edge Port.

12.13.2.2.3 Outputs

None.

12.13.2.3 Read Provider Edge Port Configuration

12.13.2.3.1 Purpose

To read the current configuration of an internal PEP in the C-VLAN Component of a Provider Edge Bridge.

12.13.2.3.2 Inputs

- a) Port Number: the number of the CEP.
- b) Service VLAN Identifier: a 12-bit S-VID.

12.13.2.3.3 Outputs

- a) PVID: a 12-bit C-VID to be used for untagged frames received at the PEP.
- b) Default priority: an integer range 0–7 to be used for untagged frames received at the PEP.
- c) Acceptable Frame Types: the Acceptable Frame Types (8.6.2) for frames received at the PEP. The permissible values for the parameter are as follows:
 - 1) *Admit only VLAN-tagged frames*
 - 2) *Admit only Untagged and Priority-tagged frames*
 - 3) *Admit all frames*
- d) Enable Ingress Filtering: the Enable Ingress Filtering parameter for frames received at the PEP. The permissible values for the parameter are as follows:
 - 1) Enabled
 - 2) Disabled

1 12.13.2.4 Set Provider Edge Port Configuration

2 12.13.2.4.1 Purpose

3 To modify the configuration of a PEP in the C-VLAN Component of a Provider Edge Bridge.

4 12.13.2.4.2 Inputs

- 5 a) Port Number: the number of the CEP.
- 6 b) Service VLAN Identifier: a 12-bit S-VID.
- 7 c) PVID: a 12-bit C-VID to be used for untagged frames received at the PEP.
- 8 d) Default priority: an integer range 0–7 to be used for untagged frames received at the PEP.
- 9 e) Acceptable Frame Types: the Acceptable Frame Types (8.6.2) for frames received at the PEP. The
10 permissible values for the parameter are as follows:
 - 11 1) *Admit only VLAN-tagged frames*
 - 12 2) *Admit only Untagged and Priority-tagged frames*
 - 13 3) *Admit all frames*
- 14 f) Enable Ingress Filtering: the Enable Ingress Filtering parameter for frames received at the PEP. The
15 permissible values for the parameter are as follows:
 - 16 1) Enabled
 - 17 2) Disabled

18 12.13.2.4.3 Outputs

19 None.

20 12.13.2.5 Read Service Priority Regeneration Table

21 12.13.2.5.1 Purpose

22 To read the current contents of the Priority Regeneration Table (6.9.3) for an internal CNP connected to the
23 C-VLAN Component associated with a CEP.

24 12.13.2.5.2 Inputs

- 25 a) Port Number: the number of the CEP.
- 26 b) Service VLAN Identifier: a 12-bit S-VID.

27 12.13.2.5.3 Outputs

- 28 a) Regenerated Priority value for Received Priority 0: Integer in range 0–7.
- 29 b) Regenerated Priority value for Received Priority 1: Integer in range 0–7.
- 30 c) Regenerated Priority value for Received Priority 2: Integer in range 0–7.
- 31 d) Regenerated Priority value for Received Priority 3: Integer in range 0–7.
- 32 e) Regenerated Priority value for Received Priority 4: Integer in range 0–7.
- 33 f) Regenerated Priority value for Received Priority 5: Integer in range 0–7.
- 34 g) Regenerated Priority value for Received Priority 6: Integer in range 0–7.
- 35 h) Regenerated Priority value for Received Priority 7: Integer in range 0–7.

36 12.13.2.6 Set Service Priority Regeneration Table

37 12.13.2.6.1 Purpose

38 To modify the contents of the Priority Regeneration Table (6.9.3) for an internal CNP connected to the
39 C-VLAN Component associated with a CEP.

12.13.2.6.2 Inputs

- a) Port Number: the number of the CEP.
- b) Service VLAN Identifier: a 12-bit S-VID.
- c) Regenerated Priority value for Received Priority 0: Integer in range 0–7.
- d) Regenerated Priority value for Received Priority 1: Integer in range 0–7.
- e) Regenerated Priority value for Received Priority 2: Integer in range 0–7.
- f) Regenerated Priority value for Received Priority 3: Integer in range 0–7.
- g) Regenerated Priority value for Received Priority 4: Integer in range 0–7.
- h) Regenerated Priority value for Received Priority 5: Integer in range 0–7.
- i) Regenerated Priority value for Received Priority 6: Integer in range 0–7.
- j) Regenerated Priority value for Received Priority 7: Integer in range 0–7.

12.13.2.6.3 Outputs

None.

12.13.3 Remote Customer Access managed object

Designating an external port as a RCAP automatically creates a Port-mapping S-VLAN Component associated with that port. This Port-mapping S-VLAN Component includes one internal PNP.

The Remote Customer Access managed object applies to each externally accessible RCAP on a Provider Edge Bridge. It includes the following:

- a) The Internal Interface Table, which configures the internal interfaces between the Port-mapping S-VLAN Component and the S-VLAN or C-VLAN Components. This table provides the following equivalent functionality:
 - 1) For an RCSI supporting a Port-based service interface:
 - i) Creating an internal PAP connected to a CNP on the S-VLAN Component.
 - ii) Configuring both the PAP and CNP to accept only untagged frames.
 - iii) Setting the member set for the external S-VID to include the PAP and RCAP.
 - iv) Setting the PAP's PVID to the external S-VID and adding that port to that S-VID's untagged set.
 - v) Setting the CNP's PVID to the internal S-VID for the Port-based service instance and adding the port to that S-VID's untagged set.
 - 2) For an RCSI supporting a C-tagged service interface:
 - i) Creating a PAP connected to a CEP on a C-VLAN Component.
 - ii) Setting the member set for the external S-VID to include the PAP and RCAP.
 - iii) Configuring the PAP's PVID to the external S-VID and adding the port to that S-VID's untagged set.
 - 3) For a PNP:
 - i) Setting the member set for the external S-VID to include the PNP and RCAP.
 - ii) Adding the associated primary S-VLAN Component PNP to the member set of the internal S-VLAN.
 - iii) Configuring the associated primary S-VLAN Component PNP's VID translation table to translate between the external S-VID and the internal S-VID for the selected service instance.

Note that in the case of an RCSI supporting a C-tagged service interface the C-VLAN Component can be further configured using the Customer Edge managed object. This configuration includes the following:

- The parameters to set the C-VID Registration Table entries (12.13.2.2)
- The PEP configuration parameters (12.13.2.4)
- The Service Priority Regeneration Table entries (12.13.2.6)

all associated with the configuration of the internal CEP, which is directly connected to the internal PAP identified by the external S-VID and the RCAP.

The management operations that can be performed on the Remote Customer Access managed object are as follows:

- b) Read Internal Interface Table Entry (12.13.3.1)
- c) Configure Internal Interface Table Entry (12.13.3.2)

Further configuration related to a C-VLAN Component associated with an RCSI supporting a C-tagged service interface is accomplished using the Customer Edge managed object (12.13.2).

12.13.3.1 Read Internal Interface Table Entry

12.13.3.1.1 Purpose

To read the Internal Interface Table entry associated with a specific external S-VID in a Port-mapping S-VLAN Component of a RCAP.

12.13.3.1.2 Inputs

- a) External Port Number: the number of the RCAP.
- b) External S-VLAN Identifier: a 12-bit S-VID.

12.13.3.1.3 Outputs

- a) Internal Port Number: the port number used to reference the internal interface.
- b) A value indicating the type of internal interface associated with the external S-VID, one of the following:
 - 1) Port-based RCSI
 - 2) C-tagged RCSI
 - 3) PNP
 - 4) Discard (external S-VID is not associated with an internal port)
- c) Internal S-VLAN Identifier: a 12-bit S-VID (not applicable for a C-tagged RCSI).

12.13.3.2 Configure Internal Interface Table Entry

12.13.3.2.1 Purpose

To add or delete an entry in the Internal Interface Table.

12.13.3.2.2 Inputs

- a) External Port Number: the number of the RCAP.
- b) External S-VLAN Identifier: a 12-bit S-VID.
- c) Internal Port Number (optional): the port number used to reference the internal interface;
- d) A value indicating the type of internal interface associated with the external S-VID, one of the following:
 - 1) Port-based RCSI
 - 2) C-tagged RCSI
 - 3) PNP
 - 4) Discard (delete entry for the external S-VID)
- e) Internal S-VLAN Identifier: a 12-bit S-VID (not applicable for a C-tagged RCSI).

1 12.13.3.2.3 Outputs

2 None

3 12.14 CFM entities

4 The CFM managed objects model operations that modify, or inquire about, the configuration and the
5 operation of CFM. Figure 12-1 illustrates the simple hierarchical relationships among the various CFM
6 managed objects, including the control of access to those objects. See 17.4.7 for an explanation of methods
7 available to distinguish between an Owner and a Maintenance Domain administrator. See Clause 18 for an
8 explanation of the use of Maintenance Domains and Maintenance Associations (MAs). The following are
9 the CFM managed objects in a Bridge:

- 10 a) Maintenance Domain list managed object (12.14.1)
- 11 b) CFM Stack managed object (12.14.2)
- 12 c) Default MD Level managed object (12.14.3)
- 13 d) Configuration Error List managed object (12.14.4)
- 14 e) Maintenance Domain managed objects (12.14.5)
- 15 f) Maintenance Association managed objects (12.14.6)
- 16 g) Maintenance association Endpoint managed objects (12.14.7)

17 12.14.1 Maintenance Domain list managed object

18 There is one Maintenance Domain list managed object per Bridge. It contains a list of the Maintenance
19 Domains that have been configured on the Bridge.

20 The management operations that can be performed on the Maintenance Domain list managed object are as
21 follows:

- 22 a) Read Maintenance Domain list (12.14.1.1)
- 23 b) Create Maintenance Domain managed object (12.14.1.2)
- 24 c) Delete Maintenance Domain managed object (12.14.1.3)

25 NOTE—In Provider Bridge applications, each Maintenance Domain, and thus each Maintenance Domain managed
26 object, can be controlled by a different administrative organization. Some or all of these administrative organizations can
27 be different from that which controls the Maintenance Domain list managed object or the managed objects in Clause 12
28 other than those in 12.14. See 17.3.7 for a discussion of the means by which control of and access to different instances
29 of the Maintenance Domain managed object are allocated to these different administrative organizations.

30 12.14.1.1 Read Maintenance Domain list

31 12.14.1.1.1 Purpose

32 To obtain information about all of the Maintenance Domains in a Bridge.

33 12.14.1.1.2 Inputs

34 None.

35 12.14.1.1.3 Outputs

- 36 a) A list, perhaps empty, of the Maintenance Domain managed objects configured on the Bridge. For
37 each item in the list, the Read Maintenance Domain list command returns the following:
 - 38 1) A Maintenance Domain Name, including the format specifier from Table 21-18 (21.6.5.1)
 - 39 2) A reference to that Maintenance Domain's Maintenance Domain managed object (12.14.5)

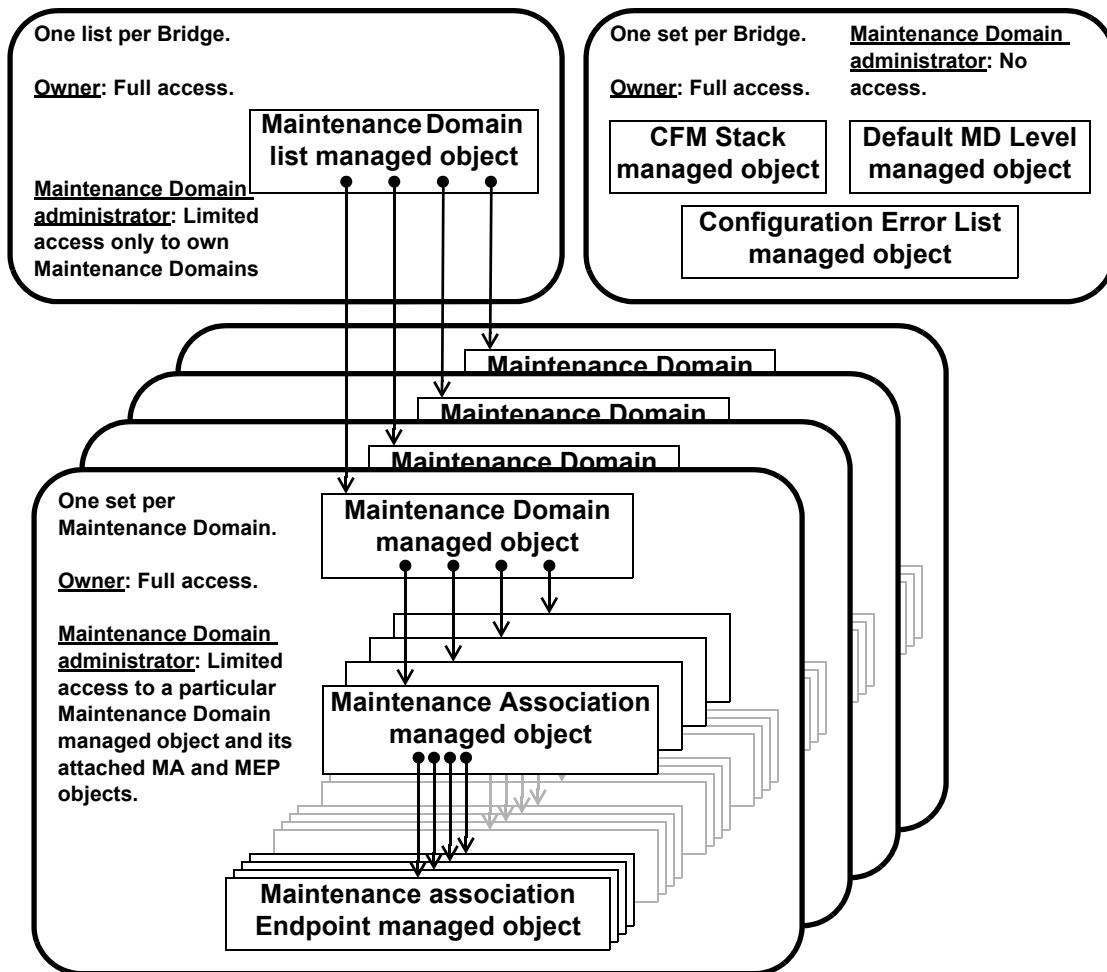


Figure 12-1—Relationships among CFM managed objects

12.14.1.2 Create Maintenance Domain managed object

12.14.1.2.1 Purpose

To create a new Maintenance Domain managed object in a Bridge, and add it to the Bridge's Maintenance Domain list managed object.

12.14.1.2.2 Inputs

- a) A Maintenance Domain Name, including the format specifier from Table 21-18 (21.6.5.1); default value is the character string (format 4) "DEFAULT."
- b) The MD Level at which the Maintenance Domain is to be created; default value is 0.

12.14.1.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to invalid Maintenance Domain Name (21.6.5.1)
 - 2) Operation rejected because Maintenance Domain Name already exists

- 1 3) Operation rejected due to invalid MD Level
- 2 4) Operation accepted

3 **12.14.1.3 Delete Maintenance Domain managed object**

4 **12.14.1.3.1 Purpose**

5 To delete a specific Maintenance Domain managed object from a Bridge and from the Bridge's Maintenance
6 Domain list managed object.

7 **12.14.1.3.2 Inputs**

- 8 a) A reference to a particular Maintenance Domain managed object (12.14.5).

9 **12.14.1.3.3 Outputs**

- 10 a) Operation status. This takes one of the following values:
 - 11 1) Operation rejected due to invalid or nonexistent Maintenance Domain
 - 12 2) Operation accepted

13 **12.14.2 CFM Stack managed object**

14 There is one CFM Stack managed object per Bridge. It permits retrieval of information about the
15 MPs configured on a particular Bridge Port or aggregated port. The management operation that can be
16 performed is as follows:

- 17 a) Read CFM Stack managed object (12.14.2.1)

18 **12.14.2.1 Read CFM Stack managed object**

19 **12.14.2.1.1 Purpose**

20 To obtain the contents of the CFM Stack managed object, which allows a network administrator to discover
21 the relationships among MEPs and MHFs configured on a particular Bridge Port or aggregated port.

22 **12.14.2.1.2 Inputs**

- 23 a) An interface, either a Bridge Port, or an aggregated IEEE 802.3 port within a Bridge Port, on which
24 MPs might be configured.
- 25 b) An MD Level.
- 26 c) A value indicating the direction in which any reported MP faces, either:
 - 27 1) Down (Active SAP is further away from the Frame filtering entity) or
 - 28 2) Up (Active SAP is closer to the Frame filtering entity).
- 29 d) A specific VID, I-SID, Traffic Engineering service instance Identifier (TE-SID), or Segment
30 Identifier (SEG-ID) associated with an MP, or 0, in the case that the MP is associated with no VID,
31 I-SID, TE-SID, or SEG-ID.

32 **12.14.2.1.3 Outputs**

- 33 a) Operation status. This takes one of the following values:
 - 34 1) Operation rejected due to illegal inputs.
 - 35 2) No MP is configured on the indicated Bridge Port or aggregated port at the indicated MD Level
36 and direction.
 - 37 3) A MEP is configured.
 - 38 4) An MHF is configured.

- 1 b) Either the Maintenance Domain managed object (12.14.5) to which the MP's MA is associated, or
- 2 an indication that there is no Maintenance Domain associated with the MP.
- 3 c) Either the Maintenance Association managed object (12.14.6) to which the MP is associated, or an
- 4 indication that there is no MA associated with the MP.
- 5 d) If a MEP is configured, its MA Endpoint Identifier (MEPID), else 0.
- 6 e) The MAC address of the MP.

7 **12.14.3 Default MD Level managed object**

8 There is a single Default MD Level managed object per Bridge Component. It controls MIP Half Function
9 (MHF) creation for VIDs or I-SIDs of I- or B-Components that are not contained in the list of VIDs attached
10 to any specific Maintenance Association managed object [item c) in 12.14.5.3.2], and the transmission of the
11 Sender ID TLV (21.5.3) by those MHFs.

12 The management commands that can be performed on the Default MD Level managed object are as follows:

- 13 a) Read Default MD Level managed object (12.14.3.1).
- 14 b) Write Default MD Level managed object (12.14.3.2).

15 **12.14.3.1 Read Default MD Level managed object**

16 **12.14.3.1.1 Purpose**

17 To read the table of default parameters for controlling MHFs not associated with any MA configured on the
18 Bridge Component.

19 **12.14.3.1.2 Input**

- 20 a) A VID or I-SID in an I- or B-Component.

21 **12.14.3.1.3 Output**

- 22 a) A list of VIDs associated with any MHF on the VID named in item a) in 12.14.3.1.2, always
- 23 including that VID, or the Backbone-SID of the B-Component or VIP-SID of the I-Component
- 24 associated with any MHF on the I-SID named in item a) in 12.14.3.1.2. The first VID is the MAs'
25 Primary VID. List is empty if no VID specified in 12.14.3.1.2.
- 26 b) A Boolean value indicating whether this entry is in effect or has been overridden by the existence of
27 a Maintenance Association managed object associated with the same VID or I-SID of I- or
28 B-Components and MD Level, and on which is configured an Up MEP. True if this Maintenance
29 Domain managed object is in effect.
- 30 c) (writable) The MD Level at which MHFs are to be created.
- 31 d) (writable) An enumerated value indicating whether the management entity can create MHFs for this
32 VID(s) or I-SID(s) of I- or B-Components: defMHFnone: [item n) in 22.2.3, the default],
33 defMHFdefault [item o)], or defMHFexplicit [item q)].
- 34 e) (writable) An enumerated value indicating what, if anything, is to be included in the Sender ID TLV
35 (21.5.3) transmitted by MPs configured in the Default Maintenance Domain: sendIdNone (1, the
36 default value), sendIdChassis (2), sendIdManage (3), or sendIdChassisManage (4).

37 **12.14.3.2 Write Default MD Level managed object**

38 **12.14.3.2.1 Purpose**

39 To write entries in the table of default parameters for controlling MHFs on VIDs or on I-SIDs used in an I-
40 or B-Component not associated with any MA.

1 12.14.3.2.2 Inputs

- 2 a) A list of VIDs or the VIP-SIDs used in the I-Component or Backbone-SID used in the
- 3 B-Component, or 0, indicating no VID or I-SID. The first VID in the list is the Primary VID.
- 4 b) One of the writable managed object in 12.14.3.1.3.

5 12.14.3.2.3 Outputs

- 6 a) Operation status. This takes one of the following values:
 - 7 1) Operation rejected because creating or assigning these VIDs or I-SIDs to this MD Level would
 - 8 exceed the number of MD Levels supported by some Bridge Port [item c) in 22.2.4].
 - 9 2) Operation rejected because the VID list provided [item a) in 12.14.3.2.2] specifies a different
 - 10 Primary VID, or contains one or more but not all of the VIDs, in the definition of an MA or
 - 11 some other entry in the Default MD Level managed object.
 - 12 3) Operation accepted.

13 12.14.4 Configuration Error List managed object

14 The Configuration Error List managed object is a list of {identifier, port} pairs configured in error together
15 with the identity of the configuration error. The identifier is a VID, I-SID, TE-SID, or SEG-ID and the port
16 is a simple Bridge Port or aggregated Bridge Port.

17 12.14.4.1 Read Configuration Error List managed object

18 12.14.4.1.1 Purpose

19 To list the entries in the Configuration Error List where each entry describes the identifier and port pair that
20 identifies the entry, together with a description of the particular configuration error.

21 12.14.4.1.2 Inputs

- 22 a) A VID, I-SID, or TE-SID, specifying which service instance to check for ports in error or a SEG-ID
- 23 specifying which Infrastructure Segment to check for ports in error.
- 24 b) An interface, either a Bridge Port or an aggregated IEEE 802.3 port within a Bridge Port.

25 12.14.4.1.3 Outputs

- 26 a) An indication of the result of the operation:
 - 27 1) Operation rejected due to illegal vlan_identifier
 - 28 2) Operation rejected due to illegal or nonexistent Bridge Port or aggregated port
 - 29 3) Operation accepted
- 30 b) If the operation was accepted, a vector of Boolean error conditions from 22.2.4, any of which may
31 be true:
 - 32 1) CFMleak
 - 33 2) ConflictingVIDs
 - 34 3) ExcessiveLevels and/or
 - 35 4) OverlappedLevels

36 12.14.5 Maintenance Domain managed object

37 There can be any number of Maintenance Domain managed objects per Bridge. A Maintenance Domain
38 managed object is required in order to create an MA with a Maintenance Association Identifier (MAID) that
39 includes that Maintenance Domain's Name. From this Maintenance Domain managed object, all
40 Maintenance Association managed objects associated with that Maintenance Domain managed object can be
41 accessed, and thus controlled.

1 The management operations that can be performed on the Maintenance Domain managed object are as
2 follows:

- 3 a) Read Maintenance Domain managed object (12.14.5.1)
- 4 b) Write Maintenance Domain managed object (12.14.5.2)
- 5 c) Create Maintenance Association managed object (12.14.5.3)
- 6 d) Delete Maintenance Association managed object (12.14.5.4)

7 NOTE 1—The Maintenance Domain managed object is defined in such a manner that it can be configured identically in
8 all of the Bridges in that Bridged Network. All of the information local to a particular MA or a particular Bridge is
9 contained in other managed objects, e.g., the Maintenance Association managed object (12.14.6) or the Maintenance
10 association Endpoint managed object (12.14.7).

11 NOTE 2—A single Maintenance Domain managed object can be used to manage any number of separate Maintenance
12 Domains, as Maintenance Domains are defined in 18.1. For example, the Maintenance Association managed object for
13 all of the physical links in a Bridged Network that are protected by MAs can be grouped together under a single
14 Maintenance Domain managed object for the convenience of the network administrator.

15 NOTE 3—Multiple Maintenance Domain managed objects can be used to manage a single Maintenance Domain, as
16 Maintenance Domains are defined in 18.1. For example, a PBN's Domain Service Access Points (DoSAPs) could be
17 split among two Maintenance Domain managed objects, one for point-to-point services and one for multipoint services.

18 **12.14.5.1 Read Maintenance Domain managed object**

19 **12.14.5.1.1 Purpose**

20 To obtain information from a Bridge about a specific Maintenance Domain managed object.

21 **12.14.5.1.2 Inputs**

- 22 a) A reference to a particular Maintenance Domain managed object (12.14.5).

23 **12.14.5.1.3 Outputs**

- 24 a) Operation status. This takes one of the following values:
 - 25 1) Operation rejected due to nonexistent Maintenance Domain
 - 26 2) Operation accepted
- 27 b) The MD Level of the specific Maintenance Domain identified by the Input.
- 28 c) (writable) An enumerated value indicating whether the management entity can create MHFs for this
29 Maintenance Domain: defMHFnone: [item n) in 22.2.3, the default], defMHFdefault [item o)], or
30 defMHFexplicit [item p)].
- 31 d) (writable) An enumerated value indicating what, if anything, is to be included in the Sender ID TLV
32 (21.5.3) transmitted by MPs configured in this Maintenance Domain: sendIdNone (1, the default
33 value), sendIdChassis (2), sendIdManage (3), or sendIdChassisManage (4) [item d) in 12.14.6.1.3].
- 34 e) (writable) The network address to which Fault Alarms (12.14.7.7) are to be transmitted, or a value
35 indicating "Fault Alarms are not to be transmitted." Default value is "not transmitted."

36 NOTE—The variables item c) through item e) could be writable by the organization operating the Maintenance Domain.

- 37 f) A list of references to the Maintenance Association managed objects (12.14.6) attached to the
38 indicated Maintenance Domain managed object.

39 **12.14.5.2 Write Maintenance Domain managed object**

40 **12.14.5.2.1 Purpose**

41 To alter a value of a specific Maintenance Domain managed object.

1 12.14.5.2.2 Inputs

- 2 a) A reference to a particular Maintenance Domain managed object (12.14.5).
- 3 b) A reference to a writable managed object in 12.14.5.1.3 to be altered.
- 4 c) A new value for the managed object.

5 12.14.5.2.3 Outputs

- 6 a) Operation status. This takes one of the following values:
 - 7 1) Operation rejected due to nonexistent Maintenance Domain
 - 8 2) Operation rejected due to the selected managed object being Read-Only
 - 9 3) Operation rejected due to invalid value for the selected managed object
 - 10 4) Operation accepted

11 12.14.5.3 Create Maintenance Association managed object

12 12.14.5.3.1 Purpose

13 To create a new Maintenance Association managed object within a Maintenance Domain managed object.

14 12.14.5.3.2 Inputs

- 15 a) A reference to a particular Maintenance Domain managed object (12.14.5).
- 16 b) The Short MA Name of an MA within that Maintenance Domain, including the format specifier
17 from Table 21-19 (21.6.5.4). In the case of a PBB-TE, VID-based, or Infrastructure Segment MA,
18 the default value is a 2-octet integer (format 3) containing the primary VID, or 0, if the MA is not
19 attached to a VID.
- 20 c) The list of VIDs, the I-SID, the TE-SID, or the SEG-ID monitored by this MA, or 0, if the MA is not
21 attached to any VID, I-SID, TE-SID, or SEG-ID. In the case that a list of VIDs is specified, the first
22 VID in the list is the MA's Primary VID (default none). The specification of I-SID is allowed only in
23 the case of I- or B-Components. The TE-SID is allowed only in the case that PBB-TE or SPBM is
24 supported. The SEG-ID is allowed only in the case that IPS is supported.

25 12.14.5.3.3 Outputs

- 26 a) Operation status. This takes one of the following values:
 - 27 1) Operation rejected due to nonexistent Maintenance Domain.
 - 28 2) Operation rejected due to invalid Short MA Name (21.6.5.4).
 - 29 3) Operation rejected due to the existence of another MA in the same Maintenance Domain with
30 the same Short MA Name.
 - 31 4) Operation rejected because creation of this MA would exceed the number of MD Levels
32 supported by some Bridge Port [item c) in 22.2.4].
 - 33 5) Operation rejected because the VID list provided [item c) in 12.14.5.3.2] specifies a different
34 Primary VID, or contains one or more, but not all, of the VIDs, in the definition of another MA
35 or an entry in the Default MD Level managed object.
 - 36 6) Operation rejected because the I-SID value specified does not correspond to a valid backbone
37 service instance.
 - 38 7) Operation rejected because the TE-SID value specified does not correspond to a valid TESI or
39 set of SPBM path endpoints.
 - 40 8) Operation rejected because the SEG-ID value specified does not correspond to a valid
41 Infrastructure Segment.
 - 42 9) Operation accepted.

1 **12.14.5.4 Delete Maintenance Association managed object**

2 **12.14.5.4.1 Purpose**

3 To delete a specific Maintenance Association managed object from a Maintenance Domain managed object.

4 **12.14.5.4.2 Inputs**

- 5 a) A reference to a particular Maintenance Association managed object (12.14.6).

6 **12.14.5.4.3 Outputs**

- 7 a) Operation status. This takes one of the following values:

- 8 1) Operation rejected due to nonexistent MA
9 2) Operation accepted

10 **12.14.6 Maintenance Association managed object**

11 There can be any number of Maintenance Association managed objects per Bridge, one for each service
12 instance for which an MP is defined on that Bridge. From this Maintenance Association managed object, all
13 Maintenance association Endpoint managed objects associated with that Maintenance Association managed
14 object can be accessed, and thus controlled.

15 The management operations that can be performed on the Maintenance Association managed object are as
16 follows:

- 17 a) Read Maintenance Association managed object (12.14.6.1)
18 b) Write Maintenance Association managed object (12.14.6.2)
19 c) Create Maintenance association Endpoint managed object (12.14.6.3)
20 d) Delete Maintenance association Endpoint managed object (12.14.6.4)

21 NOTE 1—The Maintenance Association managed object is defined in such a manner that it can be configured
22 identically in all of the Bridges in its Maintenance Domain. All information local to a particular Bridge is contained in
23 other managed objects, e.g., the Maintenance association Endpoint managed object (12.14.7).

24 NOTE 2—Although MHFs are created via the Maintenance Domain managed object, the Default MD Level managed
25 object, and the Maintenance Association managed object, and although there is a Maintenance association Endpoint
26 managed object, there is no “MIP Half Function managed object.” One reason for this omission is that no controls over
27 the behavior of the MHF beyond its existence is required. Another is that the burden of implementing, providing access
28 to, and providing the maintenance capabilities to access the counters and similar managed objects appropriate to an MHF
29 Managed Object seemed to the developers of this standard to outweigh the benefits of having that information available.

30 **12.14.6.1 Read Maintenance Association managed object**

31 **12.14.6.1.1 Purpose**

32 To obtain information from a Bridge about a specific Maintenance Association managed object.

33 **12.14.6.1.2 Inputs**

- 34 a) A reference to a particular Maintenance Association managed object (12.14.6).

35 **12.14.6.1.3 Outputs**

- 36 a) Operation status. This takes one of the following values:

- 37 1) Operation rejected due to nonexistent MA
38 2) Operation accepted

- 1 b) The VID(s), I-SID, TE-SID, or SEG-ID monitored by this MA, or 0, if the MA is not attached to any
2 VID, I-SID, TE-SID, or SEG-ID. In the case of a list of VIDs, the first VID returned is the MA's
3 Primary VID.
- 4 c) (writable) An enumerated value indicating whether the management entity can create MHFs for this
5 MA: defMHFnone: [item n] in 22.2.3], defMHFdefault [item o)], or defMHFexplicit [item p)] or
6 defMHFdefer [item q)], default value is defMHFdefer.
- 7 d) (writable) An enumerated value indicating what, if anything, is to be included in the Sender ID TLV
8 (21.5.3) transmitted by MPs configured in this MA:
- 9 1) **sendIdNone**: The Sender ID TLV is not to be sent.
- 11 2) **sendIdChassis**: The Chassis ID Length, Chassis ID Subtype, and Chassis ID fields of the
13 Sender ID TLV are to be sent, but not the Management Address Length or Management
14 Address fields.
- 15 3) **sendIdManage**: The Management Address Length and Management Address of the Sender ID
17 TLV are to be sent, but the Chassis ID Length is to be transmitted with a 0 value, and the
18 Chassis ID Subtype and Chassis ID fields not sent.
- 19 4) **sendIdChassisManage**: The Chassis ID Length, Chassis ID Subtype, Chassis ID,
21 Management Address Length, and Management Address fields are all to be sent.
- 22 5) **sendIdDefer**: (the default value) The contents of the Sender ID TLV are determined by the
24 Maintenance Domain managed object, item d) in 12.14.5.1.3.
- 25 e) (writable) An enumerated value, other than 0, indicating the interval between CCM transmissions to
26 be used by all MEPs in the MA (20.8.1, 21.6.1.3) (default 1 s).
- 27 f) (writable) The network address to which Fault Alarms (12.14.7.7) are to be transmitted, a value
28 indicating "not specified," or a value indicating "Fault Alarms are not to be transmitted." If "not
29 specified," the address used is that from the Maintenance Domain managed object [item e) in
30 12.14.5.1.3].
- 31 g) (writable) A list of the MEPIDs of the MEPs in the MA.

32 12.14.6.2 Write Maintenance Association managed object

33 12.14.6.2.1 Purpose

34 To alter a value of a specific Maintenance Association managed object.

35 12.14.6.2.2 Inputs

- 36 a) A reference to a particular Maintenance Association managed object (12.14.6).
- 37 b) A reference to one of the writable managed objects in 12.14.6.1.3 is to be altered.
- 38 c) A new value for the managed object.

39 12.14.6.2.3 Outputs

- 40 a) Operation status. This takes one of the following values:
- 41 1) Operation rejected due to nonexistent MA
- 42 2) Operation rejected due to the selected managed object being Read-Only
- 43 3) Operation rejected due to lack of authority to set this variable
- 44 4) Operation rejected due to invalid value for the selected managed object
- 45 5) Operation accepted

46 12.14.6.3 Create Maintenance association Endpoint managed object

47 12.14.6.3.1 Purpose

48 To create a new Maintenance association Endpoint managed object within a Maintenance Association
49 managed object on a specific Bridge Port or aggregated port.

12.14.6.3.2 Inputs

- a) A reference to a particular Maintenance Association managed object (12.14.6).
- b) A MEPID for the created MEP.
- c) A value indicating the direction in which the MEP faces on the Bridge Port or aggregated port, either:
 - 1) Down (Active SAP is further away from the Frame filtering entity) (the default) or
 - 2) Up (Active SAP is closer to the Frame filtering entity).
- d) An interface, either a Bridge Port or an aggregated IEEE 802.3 port within a Bridge Port.

12.14.6.3.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to nonexistent MA.
 - 2) Operation rejected because the specified MEPID already exists in this MA in this Bridge.
 - 3) Operation rejected because MEPID is not in the MA's list of MEPIDs [item g) in 12.14.6.1.3].
 - 4) Operation rejected due to the existence of a MEP in the same direction (Up or Down) at that same MD Level, for the same VID(s), I-SID, TE-SID, or SEG-ID as this MEP (or for no VID, I-SID, TE-SID, or SEG-ID, if this MEP's MA has no VID, I-SID, TE-SID, or SEG-ID), on that Bridge Port or aggregated port.
 - 5) Operation failed because one or more of the listed VIDs, I-SID, or TE-SID in this MA are assigned to some other MA on which Up MEPs are configured on any Bridge Port.
 - 6) Operation rejected because the Bridge is incapable of instantiating a MEP on that Bridge Port or aggregated IEEE 802.3 port.
 - 7) Operation rejected because an Up MEP cannot be configured for an MA that has no VID, I-SID, or TE-SID.
 - 8) Operation rejected because a MEP exists on this MA that has a different value for its Up/Down parameter [item c) in 12.14.6.3.2].
 - 9) Operation accepted.

12.14.6.4 Delete Maintenance association Endpoint managed object

12.14.6.4.1 Purpose

To delete an existing Maintenance association Endpoint managed object from a Maintenance Association managed object.

12.14.6.4.2 Inputs

- a) A reference to a particular Maintenance association Endpoint managed object (12.14.7).

12.14.6.4.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to nonexistent MEP
 - 2) Operation accepted

12.14.7 Maintenance association Endpoint managed object

There can be any number of Maintenance association Endpoint managed objects per Bridge, one for each MEP defined within that Bridge. From this Maintenance association Endpoint managed object, all management objects related to that MEP can be controlled.

1 The management operations that can be performed on the Maintenance association Endpoint managed
2 object are as follows:

- 3 a) Read Maintenance association Endpoint managed object (12.14.7.1)
- 4 b) Write Maintenance association Endpoint managed object (12.14.7.2)
- 5 c) Transmit Loopback Messages (12.14.7.3)
- 6 d) Transmit Linktrace Message (12.14.7.4)
- 7 e) Read Linktrace Reply (12.14.7.5)
- 8 f) Read MEP Database (12.14.7.6)
- 9 g) Transmit MEP Fault Alarm (12.14.7.7)

10 **12.14.7.1 Read Maintenance association Endpoint managed object**

11 **12.14.7.1.1 Purpose**

12 To obtain information from a Bridge about a specific Maintenance association Endpoint managed object.

13 **12.14.7.1.2 Inputs**

- 14 a) A reference to a particular Maintenance association Endpoint managed object (12.14.7).

15 **12.14.7.1.3 Outputs**

- 16 a) Operation status. This takes one of the following values:
 - 17 1) Operation rejected due to nonexistent MEP
 - 18 2) Operation accepted
- 19 b) An interface, either a Bridge Port or an aggregated IEEE 802.3 port within a Bridge Port, to which
20 the MEP is attached.
- 21 c) A value indicating the direction in which the MEP faces on the interface, either:
 - 22 1) Down (Active SAP is further away from Frame filtering entity) or
 - 23 2) Up (Active SAP is closer to the Frame filtering entity).
- 24 d) (writable) An integer indicating the Primary VID of the MEP, always one of the VIDs assigned to
25 the MEP's MA. The value 0 indicates either that the Primary VID is that of the MEP's MA or that
26 the MEP's MA is associated with no VID. In the case of a PBB-TE associated MEP, the Primary
27 VID is not writable but is always associated with the value of the ESP-VID parameter identifying
28 the MA's ESP that has the MEP's MAC address in its ESP-SA field (MEPprimaryVID, 20.9.7). In
29 the case of an Infrastructure Segment associated MEP, the Primary VID (MEPprimaryVID, 20.9.7)
30 is not writable but is always associated with the value of the SMP-VID parameter identifying the
31 MA's SMP that has the MEP's MAC address in its SMP-SA field. In the case of an SPB associated
32 MEP, the Primary VID is not writable but is always associated with the value of the SPVID for the
33 node on which the MEP is configured (for SPBV) or the Base VID (for SPBM).
- 34 e) (writable) A Boolean flag indicating the administrative state of the MEP (20.9.1) (default FALSE).
- 35 f) A value indicating the current state of the MEP Fault Notification Generator state machine (20.37).
- 36 g) (writable) A Boolean flag indicating whether the MEP is or is not to generate CCMs (CCIenabled,
37 20.10.1) (default FALSE, no CCMs).
- 38 h) (writable) The priority parameter for CCMs and LTMs transmitted by the MEP (default value: the
39 highest priority, i.e., that with the highest numerical value, allowed to pass through the Bridge Port
40 for any of this MEP's VIDs, I-SID, TE-SID, or SEG-ID).

41 NOTE—The management entity can obtain the default value for this variable from the Priority regeneration
42 table in 6.9.3 by extracting the highest priority value in the table on this MEP's Bridge Port (1 is lowest, then 2,
43 then 0, then 3–7), appropriate to the direction in which frames are emitted from the MEP's Active SAP.

- 1 i) The MAC address of the MEP (19.4). In the case of a MEP that is associated with a TESI, the MAC
2 address of the CBP upon which the MEP is operating. In the case of a MEP that is associated with an
3 Infrastructure Segment, the MAC address of the PNP upon which the MEP is operating.
- 4 j) (writable) The network address to which Fault Alarms (12.14.7.7) are to be transmitted, a value
5 indicating “not specified,” or a value indicating “Fault Alarms are not to be transmitted.” If “not
6 specified,” the address used is that from the Maintenance Association managed object [item f]
7 in 12.14.6.1.3].
- 8 k) (writable) An integer value specifying the lowest priority defect that is allowed to generate a Fault
9 Alarm (20.9.5):
 - 10 1) All defects: DefRDICCM, DefMACstatus, DefRemoteCCM, DefErrorCCM, and
11 DefXconCCM.
 - 12 2) Only DefMACstatus, DefRemoteCCM, DefErrorCCM, and DefXconCCM (the default).
 - 13 3) Only DefRemoteCCM, DefErrorCCM, and DefXconCCM.
 - 14 4) Only DefErrorCCM and DefXconCCM.
 - 15 5) Only DefXconCCM.
 - 16 6) No defects are to be reported.
- 17 l) (writable) The time that defects must be present before a Fault Alarm is issued (20.35.3)
18 (default 2.5 s).
- 19 m) (writable) The time that defects must be absent before resetting a Fault Alarm (20.35.4)
20 (default 10 s).
- 21 n) An enumerated value indicating the highest priority defect that has been present since the MEP Fault
22 Notification Generator state machine was last in the FNG_RESET state (20.35.9):
 - 23 1) **DefNone**: No defect has been present since the last FNG_RESET state.
 - 24 2) **DefRDICCM**: The last CCM received by this MEP from some remote MEP contained the RDI
25 bit, so variable item o) in 12.14.7.1.3 is set.
 - 26 3) **DefMACstatus**: The last CCM received by this MEP from some remote MEP indicated that
27 the transmitting MEP’s associated MAC is reporting an error status via the Port Status TLV
28 (21.5.4) or Interface Status TLV (21.5.5), so variable item p) in 12.14.7.1.3 is set.
 - 29 4) **DefRemoteCCM**: This MEP is not receiving CCMs from some other MEP in its configured
30 list [item g) in 12.14.6.1.3], so variable item q) in 12.14.7.1.3 is set.
 - 31 5) **DefErrorCCM**: This MEP is receiving invalid CCMs, so variable item r) in 12.14.7.1.3, is set.
 - 32 6) **DefXconCCM**: This MEP is receiving CCMs that could be from some other MA, so variable
33 item s) in 12.14.7.1.3 is set.
- 34 o) A Boolean flag (20.35.7) indicating that some other MEP in this MEP’s MA is transmitting the RDI
35 bit (can trigger DefRDICCM).
- 36 p) A Boolean flag (20.35.6) indicating that a Port Status TLV (21.5.4) or Interface Status TLV (21.5.5)
37 is indicating an error condition (can trigger DefMACstatus).
- 38 q) A Boolean flag (20.35.5) indicating that CCMs are not being received from at least one of the
39 configured remote MEPs (can trigger DefRemoteCCM).
- 40 r) A Boolean flag (20.21.3) indicating that erroneous CCMs are being received from some MEP in this
41 MEP’s MA (can trigger DefErrorCCM).
- 42 s) A Boolean flag (20.23.3) indicating that CCMs are being received from a MEP that could be in
43 some other MA (can trigger DefXconCCM).
- 44 t) The last-received CCM (20.21.2) that triggered a DefErrorCCM fault (20.21.3).
- 45 u) The last-received CCM (20.23.2) that triggered a DefXconCCM fault (20.23.3).
- 46 v) (optional) The total number of out-of-sequence CCMs received (20.16.12).
- 47 w) The total number of CCMs transmitted (20.10.2).
- 48 x) The next Loopback Transaction Identifier to be sent in an LBM (20.30.2).
- 49 y) The total number of valid, in-order LBRs received (20.33.1).
- 50 z) The total number of valid, out-of-order LBRs received (20.33.1).

- 1 aa) (optional) The total number of LBRs received whose mac_service_data_unit did not match (except
2 for the OpCode) that of the corresponding LBM (20.2.3).
- 3 ab) The next LTM Transaction Identifier to be sent in an LTM (20.41.1).
- 4 ac) The total number of unexpected LTRs received (20.44.1).
- 5 ad) The total number of LBRs transmitted (20.28.2).
- 6 ae) (writable) A list indicating which of the remote MEPs in the same MA are active. The Remote MEP
7 state machines (20.20) are instantiated only for the active remote MEPs. By default, all configured
8 remote MEPs in the same MA are active.
- 9 af) (writable) Only applicable for PBB-TE MEPs supporting the Traffic field (21.6.1.4), a Boolean flag
10 indicating that the capability to report the presence of traffic is enabled. The default value is FALSE.
- 11 ag) (writable) Only applicable for PBB-TE MEPs supporting the Traffic field (21.6.1.4), a Boolean flag
12 indicating that the mismatch defect is allowed to generate a Fault Alarm. The default value is
13 FALSE.
- 14 ah) Only applicable for PBB-TE MEPs supporting the Traffic field (21.6.1.4), a Boolean flag indicating
15 the presence of a traffic field mismatch defect (20.25.2) (can trigger DefMmCCM).
- 16 ai) Only applicable for PBB-TE MEPs supporting the Traffic field (21.6.1.4), a Boolean flag indicating
17 the presence of a local mismatch defect (20.25.5) (can trigger DefMmCCM).
- 18 aj) Only applicable for PBB-TE MEPs supporting the Traffic field (21.6.1.4), an enumerated value
19 indicating if the mismatch defect that has been present since the MEP Mismatch Fault Notification
20 Generator state machine (20.40) was last in the MFNG_RESET state, either:
 - 21 1) **DefNone**: No defect has been present since the last MFNG_RESET state; or
 - 22 2) **DefMmCCM**: This MEP is receiving CCMs that have different value in their Traffic field than
23 the CCMs transmitted by this MEP or is transmitting CCMs with both the RDI and the Traffic
24 field bit set to 1 and the disableLocdefect (20.25.4) variable is cleared, so variable item ah) in
25 12.14.7.1.3 is set.
- 26 ak) Only applicable for PBB-TE MEPs supporting the Traffic field (21.6.1.4), a value indicating the
27 current state of the MEP Mismatch Fault Notification Generator state machine (20.40).
- 28 al) (writable) Only applicable for ECMP path MEPs, a list of flow hash values to be used in a path
29 testing cycle (20.10.5).

30 12.14.7.2 Write Maintenance association Endpoint managed object

31 12.14.7.2.1 Purpose

32 To alter a value in a specific managed object within a specific Maintenance association Endpoint managed
33 object.

34 12.14.7.2.2 Inputs

- 35 a) A reference to a particular Maintenance association Endpoint managed object (12.14.7).
- 36 b) A reference to one of the writable managed objects in 12.14.7.1.3 is to be altered.
- 37 c) A new value for the managed object.

38 12.14.7.2.3 Outputs

- 39 a) Operation status. This takes one of the following values:
 - 40 1) Operation rejected due to nonexistent MEP
 - 41 2) Operation rejected due to the selected managed object being Read-Only
 - 42 3) Operation rejected due to invalid value for the selected managed object
 - 43 4) Operation accepted

12.14.7.3 Transmit Loopback Messages

12.14.7.3.1 Purpose

To signal to the MEP to transmit some number of LBMs.

NOTE—The Maintenance association Endpoint managed object (12.14.7) can be examined to determine whether the corresponding LBRs have been received by the MEP.

12.14.7.3.2 Inputs

- a) A reference to a particular Maintenance association Endpoint managed object (12.14.7).
- b) An indication of the target MAC address for LBMs transmitted by the MEP (20.31.1):
 - 1) The MEPID of a MEP in that MA or
 - 2) An Individual destination MAC address or
 - 3) For PBB-TE and SPBM VID MEPs, an individual MAC address to be used as the MIP MAC address in a PBB-TE MIP TLV.
- c) The number of LBMs to be transmitted (default 1).
- d) An arbitrary amount of data to be included in a Data TLV, along with an indication whether the Data TLV is to be included (20.31.1) (default no Data TLV).
- e) The priority and drop_eligible parameters to be used in the transmitted LBMs [default is CCM priority item h) in 12.14.7.1.3, drop_eligible FALSE].
- f) In the case of a PBB-TE MEP, the Reverse VID to be used in the associated PBB-TE MIP TLV 21.7.5. The parameter is not required in the case of PBB-TE MEPs associated with point-to-point TESIs or if the b3) entry is a MAC address corresponding to an entry in the ESP-DA field of any of the MA's ESPs.
- g) In the case of an SPBM VID MEP, an individual or group MAC address to be used as the destination_address for the LBM primitive. If no value is provided the group MAC address for the SPBM default I-SID (Table 9-3, Figure 27-3) is used.
- h) In the case of ECMP with flow filtering, a flow hash value if the destination address is an individual address.

12.14.7.3.3 Outputs

- a) An indication of whether the command was accepted by the MEP:
 - 1) Operation rejected due to nonexistent MEP;
 - 2) Operation rejected due to invalid number of LBMs to be transmitted;
 - 3) Operation rejected due to too much data to be transmitted;
 - 4) Operation rejected due to invalid priority or drop_eligible parameters to be transmitted;
 - 5) Operation rejected due to invalid ESP-VID;
 - 6) The LBM(s) will be (or have been) sent; or
 - 7) The LBM(s) will not be sent (e.g., because the Bridge is busy with another LBM transmit operation).
- b) The Loopback Transaction Identifier (20.30.2) of the first LBM (to be) sent. The value returned is undefined if the Transmit Loopback Messages command was not accepted.

12.14.7.4 Transmit Linktrace Message

12.14.7.4.1 Purpose

To signal to the MEP to transmit an LTM and to create an LTM entry in the MEP's Linktrace Database.

NOTE—The Maintenance association Endpoint managed object (12.14.7) can be examined to determine whether the corresponding LTRs have been received by the MEP.

12.14.7.4.2 Inputs

- a) A reference to a particular Maintenance association Endpoint managed object (12.14.7).
- b) The Flags field for LTMs transmitted by the MEP (20.42.1).
- c) An indication of the Target MAC Address field to be transmitted, either:
 - 1) The MEPID of another MEP in the same MA or
 - 2) An individual MAC address.
- d) An initial value for the LTM TTL field (21.8.4). Default value, if not specified, is 64.
- e) In the case of a PBB-TE MEP, the Reverse VID to be used in the associated PBB-TE MIP TLV (21.7.5). The parameter is not required in the case of PBB-TE MEPs associated with point-to-point TESI.
- f) In the case of an SPBM VID MEP, an individual or group MAC address to be used as the destination_address for the LTM primitive. If no value is provided the group MAC address for the SPBM default I-SID (Table 9-3, Figure 27-3) is used.
- g) In the case of ECMP with flow filtering, a flow hash value if the destination address is an Individual address.

12.14.7.4.3 Outputs

- a) An indication of whether the command was accepted by the MEP:
 - 1) Operation rejected due to nonexistent MEP;
 - 2) Operation rejected due to invalid Flags field to be transmitted;
 - 3) Operation rejected due to invalid Target MAC Address to be transmitted;
 - 4) Operation rejected due to invalid LTM TTL field to be transmitted;
 - 5) Operation rejected due to invalid ESP-VID;
 - 6) The command was accepted and the LTM has (or will be) transmitted; or
 - 7) The command was rejected and no LTM will be sent (e.g., because the Bridge is busy with another LTM transmit operation).
- b) The LTM Transaction Identifier (20.41.1) of the LTM sent. The value returned is undefined if the Transmit Linktrace Message command was not accepted.
- c) The LTM Egress Identifier TLV value transmitted in the LTM. The value returned is undefined if the Transmit Linktrace Message command was not accepted.

12.14.7.5 Read Linktrace Reply

12.14.7.5.1 Purpose

To obtain from the MEP's Linktrace database (20.41.2) the LTM entry for a previously transmitted LTM with a specific LTM Transaction Identifier. An LTM entry consists of a list of LTR entries, each corresponding to a Linktrace Reply (LTR) PDU received in response to that LTM. The LTM Transaction Identifier returned by successful Transmit Linktrace Message command [item b) in 12.14.7.5.2] is used in the Read Linktrace Reply command [item b) in 12.14.7.5.2] to select the LTM entry.

See J.5 for a discussion of how to interpret the output from the Read Linktrace Reply command.

12.14.7.5.2 Inputs

- a) A reference to a particular Maintenance association Endpoint managed object (12.14.7).
- b) The LTM Transaction Identifier returned from a previous Transmit Linktrace Message command, indicating the LTM entry to which the LTR entries will be attached.
- c) An index to distinguish among multiple LTRs with the same LTR Transaction Identifier field value.

12.14.7.5.3 Outputs

A list of LTR entries returned corresponding to the selected LTM Transaction Identifier (20.41.2). Each LTR entry in the list returns:

- a) An indication of whether the command was accepted by the MEP:
 - 1) Operation rejected due to nonexistent MEP;
 - 2) Operation rejected due to LTM Transaction Identifier not found;
 - 3) Operation rejected due to LTR entry has been discarded because too many LTRs have been returned corresponding to that LTM Transaction Identifier;
 - 4) Operation rejected due to LTR entry with that index has not yet been returned; or
 - 5) Operation accepted.
- b) The integer Reply TTL field value returned in the LTR (20.41.2.2).
- c) A Boolean value stating whether an LTM was forwarded by the responding MP, as returned in the FwdYes flag of the Flags field (20.41.2.1).
- d) A Boolean value stating whether the forwarded LTM reached a MEP enclosing its MA, as returned in the TerminalMEP flag of the Flags field (20.41.2.1).
- e) An octet string holding the Last Egress Identifier field returned in the LTR Egress Identifier TLV of the LTR (20.41.2.3).
- f) An octet string holding the Next Egress Identifier field returned in the LTR Egress Identifier TLV of the LTR (20.41.2.4).
- g) An enumerated value indicating the value returned in the Relay Action field (20.41.2.5): RlyHit, RlyFDB, or RlyMPDB.
- h) (if returned in the LTR) An enumerated value indicating the format of the Chassis ID (21.5.3.2).
- i) (if returned in the LTR) The Chassis ID (21.5.3.3).
- j) (if returned in the LTR) The Management Address information of the Bridge transmitting the LTR (21.5.3.7).
- k) (if returned in the LTR) An enumerated value indicating the value returned in the Ingress Action field (20.41.2.6) of the LTR: IngOK, IngDown, IngBlocked, or IngVID.
- l) (if returned in the LTR) The MAC address returned in the Ingress MAC Address field (20.41.2.7).
- m) (if returned in the LTR) An enumerated value indicating the format of the Ingress Port ID field (20.41.2.8).
- n) (if returned in the LTR) The Ingress Port ID (20.41.2.9).
- o) (if returned in the LTR) An enumerated value indicating the value returned in the Egress Action field (20.41.2.10) of the LTR: EgrOK, EgrDown, EgrBlocked, or EgrVID.
- p) (if returned in the LTR) The MAC address returned in the Egress MAC Address field (20.41.2.11).
- q) (if returned in the LTR) An enumerated value indicating the format of the Egress Port ID (20.41.2.12).
- r) (if returned in the LTR) The Egress Port ID (20.41.2.13).
- s) (if returned in the LTR) The OUI or CID and contents of any Organization-Specific TLVs (21.5.2) returned in the LTR.

12.14.7.6 Read MEP Database

12.14.7.6.1 Purpose

To obtain from the MEP the contents of one element in the MEP CCM Database.

1 12.14.7.6.2 Inputs

- 2 a) A reference to a particular Maintenance association Endpoint managed object (12.14.7).
- 3 b) The MEPID of a remote MEP whose information from the selected database is to be returned.

4 12.14.7.6.3 Outputs

- 5 a) An indication of whether the command was accepted by the MEP:
 - 6 1) Operation rejected due to nonexistent MEP;
 - 7 2) Operation rejected due remote MEPID not configured in MA; or
 - 8 3) Operation accepted.
- 9 b) An enumerated value indicating the operational state of the Remote MEP state machine (20.20) for
10 this remote MEP:
 - 11 1) RMEP_IDLE;
 - 12 2) RMEP_START;
 - 13 3) RMEP_FAILED; or
 - 14 4) RMEP_OK.
- 15 c) The time (SysUpTime, IETF RFC 3418) at which the Remote MEP state machine last entered either
16 the RMEP_FAILED or RMEP_OK state, or 0 if it has not yet entered either of those states.
- 17 d) The MAC address of the remote MEP (the source_address of the last received CCM) or 0 if no CCM
18 has been received (20.19.7).
- 19 e) A Boolean value indicating the state of the RDI bit in the last received CCM (TRUE for RDI = 1), or
20 FALSE, if none has been received (20.19.2). In the case of an ECMP path MEP, an array of Boolean
21 values indicating the value of rMEPlastRDI[i] for i=0..(Npaths – 1) indicating the RDI values in the
22 last CCMs received in each path test cycle period. These values are true for any RDI = 1 or false if
23 all RDI = 0 or if no CCM has ever been received (20.19.2).
- 24 f) (optional) The enumerated value from the Port Status TLV (20.19.3) from the last CCM received
25 from the remote MEP, or the default value:
 - 26 1) **psNoPortStateTLV**: Indicates either that no CCM has been received, or that no Port Status
27 TLV was present in the last CCM received.
- 28 g) (optional) The enumerated value from the Interface Status TLV (20.19.4) from the last CCM
29 received from the remote MEP, or the default value:
 - 30 1) **isNoInterfaceStatusTLV**: Indicates either that no CCM has been received or that no Interface
31 Status TLV was present in the last CCM received.
- 32 h) (optional) The Sender ID TLV (20.19.5) from the last CCM received from the remote MEP or the
33 default value:
 - 34 1) **isNoSenderIdTLV**: Indicates either that no CCM has been received, or that no Sender ID TLV
35 was present in the last CCM received.

36 12.14.7.7 Transmit MEP Fault Alarm

37 12.14.7.7.1 Purpose

38 To alert the Manager to the existence of a fault in a monitored MA by issuing a Fault Alarm. Transmit MEP
39 Fault Alarm is not a command; it is an unsolicited notification from the xmitFaultAlarm() procedure, issued
40 upon detection of the Fault Alarm condition by the MEP Fault Notification Generator state machine (20.37),
41 or by the MEP Mismatch Fault Notification Generator state machine (20.40). The Fault Alarm is sent to the
42 address specified in item j) in 12.14.7.1.3.

1 12.14.7.7.2 Outputs

- 2 a) An indication of which Bridge is reporting the Fault Alarm.
- 3 b) A reference, in a suitable format for the method of delivering the Fault Alarm, to the reporting MEP.
- 4 c) An enumerated value, equal to the contents of the variable highestDefect (20.35.9), and to the
- 5 corresponding MEP managed object [item n) in 12.14.7.1.3] or indicating the presence of a
- 6 mismatch defect [item aj) in 12.14.7.1.3].

7 12.15 Backbone Core Bridge (BCB) management

8 Each BCB is a conformant S-VLAN Bridge (5.10.1) that is managed by the managed objects summarized in
9 12.2 and specified in detail in 12.4 through 12.14.

10 12.16 Backbone Edge Bridge (BEB) management

11 The conformance requirements for BEBs are specified in 5.12. Each of the I-Components and (or) the
12 B-Component of a BEB are managed using the managed objects defined in 12.2, while this subclause adds
13 managed objects used in conjunction with those of 12.2 to manage the overall configuration and the VIPs,
14 PIPs, and CBPs of a BEB. This subclause specifies the additional managed objects specific to the operation
15 of BEBs, which:

- 16 a) Support a BEB configuration managed object identifying all the BEB Components (12.16.1).
- 17 b) Identify BEB port types (12.16.2).
- 18 c) Provide additional managed objects for management of the VIPs, PIPs, and CBPs (12.16.3, 12.16.4,
- 19 12.16.5).

20 Only a single view of the BEB managed objects exists. This view may only contain the objects needed for
21 the type of BEB under management. If the BEB contains only a single I-Component then the view will have
22 only the BEB configuration, BEB, VIP configuration, and PIP configuration managed objects. If the BEB
23 has only a B-Component, then the view will have only the BEB configuration, BEB, and CBP configuration
24 managed objects.

25 An instance of a managed object associated with a BEB is referenced by a unique identifier. Rules
26 associated with the selection of this identifier are specified within this subclause. Each BEB has a single
27 BEB configuration managed object, which is described in this subclause and illustrated in Figure 12-2. This
28 managed object is accessed at only one of the components within a given BEB, the B-Component if that is
29 present, and the single I-Component otherwise, and is referenced by that component's MAC address.

1

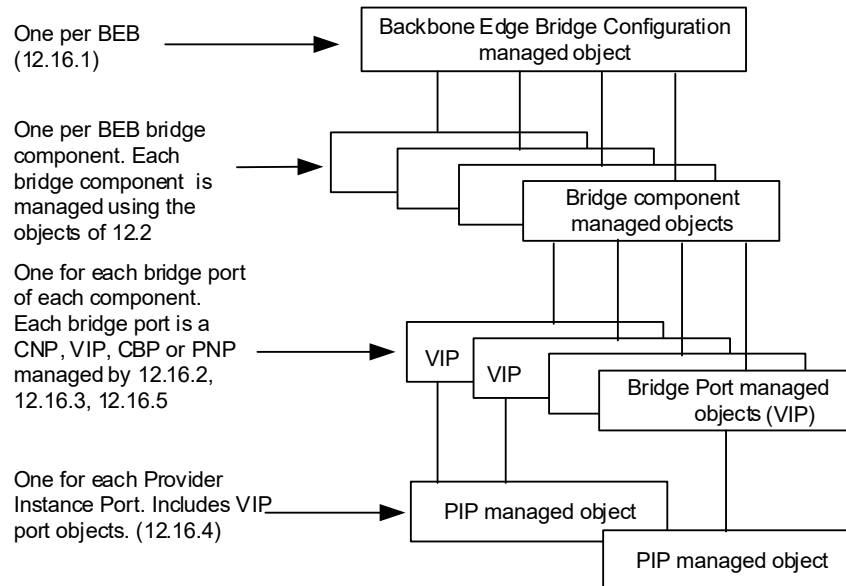


Figure 12-2—Relationship among BEB managed objects

2 Each VLAN Bridge Component relays frames between Bridge Ports. In a BEB, these Bridge Ports can
3 provide external connectivity directly, or can connect to other Bridge Components through internal LANs.
4 Each component contained within a BEB is associated with the BEB configuration managed object
5 representing that BEB. The component is uniquely identified within the PBBN by the combination of a
6 componentID unique within the BEB and by the MAC address of the BEB within that the component is
7 contained. Each component may also be identified by the MAC address of the component that is unique
8 within the PBBN. Within a BEB, a B-Component uses two types of Bridge Ports—PNPs and CBPs—and an
9 I-Component uses two types of Bridge Ports—VIPs and CNPs. Each Bridge Port is represented by a Bridge
10 Port managed object (12.4.2), a Provider Bridge Port Type managed object (12.13.1), and a BEB port
11 configuration managed object (12.16.2). Each such managed object is uniquely identified, within the BEB,
12 by the combination of the ComponentID and Port Number (12.3) of the associated Bridge Port.

13 Each VIP associated with an I-Component is mapped to a PIP associated with that I-Component. In general,
14 multiple VIPs are associated with a single PIP. PIPs are not Bridge Ports, though they are ports on BEBs, as
15 they are not used directly by the MAC Relay Entity (see Clause 8) of a Bridge Component. Each VIP is
16 supported by one and only one PIP. Each PIP is therefore identified uniquely by the ComponentID and Port
17 Number of any of the VIPs that it supports and by convention is referenced by the smallest Port Number
18 supported by the PIP. Each PIP is managed using the PIP configuration managed object (12.16.4), and each
19 VIP is managed using the VIP configuration managed object of (12.16.3).

20 PIPs and CBPs of a BEB may be internal or external. Each externally accessible port on a BEB is designated
21 as a PNP, CBP, PIP, or CNP.

22 The following managed objects define the semantics of the management operations specific to BEBs:

- 23 d) The BEB configuration managed object (12.16.1)
- 24 e) The BEB/PB/VLAN Bridge Port configuration managed object (12.16.2)
- 25 f) The VIP configuration managed object (12.16.3)
- 26 g) The PIP configuration managed object (12.16.4)

1 h) The CBP configuration managed object (12.16.5)

2 NOTE—All objects specified in 12.16 are persistent over power-up/down and system reboot unless otherwise specified.

3 **12.16.1 BEB configuration managed object**

4 The BEB configuration managed object identifies a BEB and all the managed objects that manage the BEB.

5 The management operations that can be performed on the BEB configuration managed object are as follows:

- 6 a) Read BEB configuration (12.16.1.1)
- 7 b) Set BEB configuration (12.16.1.2)
- 8 c) Create BEB Component (12.16.1.3)
- 9 d) Delete BEB Component (12.16.1.4)
- 10 e) Create BEB Bridge Port (12.16.1.5)
- 11 f) Create BEB PIP (12.16.1.6)
- 12 g) Delete BEB Bridge Port (12.16.1.7)
- 13 h) Delete BEB PIP (12.16.1.8)

14 Items marked “(MIRP only)” shall be supported if the B-Component supports the Multiple I-SID
15 Registration Protocol (MIRP) (Clause 39).

16 **12.16.1.1 Read BEB configuration**

17 **12.16.1.1.1 Purpose**

18 All BEBs shall implement the read BEB configuration function to obtain information regarding the type of
19 components and ports within a BEB. Items marked “(MIRP only)” shall be supported if the B-Component
20 supports the Multiple I-SID Registration Protocol (MIRP) (Clause 39).

21 **12.16.1.1.2 Inputs**

22 None.

23 **12.16.1.1.3 Outputs**

- 24 a) BEB address—the MAC address for the BEB used to access the BEB configuration managed object
25 (8.13.8). Each BEB has a single BEB address even if the BEB has many components.
- 26 b) BEB name—a text string of up to 32 characters, of locally determined significance.
- 27 c) Number of I-Components—in the BEB, which may range from 0 to many.
- 28 d) I-Component addresses—a list specifying the following for each I-Component or a NULL list value
29 for no I-Components:
 - 30 1) ComponentID—the number of the I-Component.
 - 31 2) A Bridge address that is the MAC address for the I-Component (8.13.8).
- 32 e) Number of B-Components—in the BEB, which may be 0 or 1.
- 33 f) B-Component address—a list specifying the following for the B-Component or a NULL list value
34 for no B-Component:
 - 35 1) ComponentID—the number of the B-Component.
 - 36 2) A Bridge address that is the MAC address for the B-Component (8.13.8).
- 37 g) Number of BEB ports (MAC Entities).
- 38 h) Port addresses—a list specifying the following for each port:
 - 39 1) ComponentID—the number identifying the Bridge Component associated with this port.
 - 40 2) An interface—either that component’s Port Number for the Bridge Port, or in the case of a PIP,
41 the PIP Index.

- 1 3) Port address—the specific MAC address associated with the port (8.13.2).
2 4) External port—a Boolean indicating that the port is an external port.
- 3 i) (MIRP only) A Boolean value specifying whether MIRP is (TRUE) or is not (FALSE) enabled in
4 this B-Component (Clause 39).
- 5 j) (MIRP only) The MIRP B-VID to which MIRPDUs transmitted from a PNP are to be assigned if
6 item k) contains the value cbpMirpGroup (1). The value 0 (the default value) indicates that the CBP
7 PVID is to be used (39.2.1.6).
- 8 k) (MIRP only) An enumerated value specifying what destination_address and vlan_identifier are to be
9 used when the MIRP Participant transmits an MIRPDU from a PNP:
- 10 1) cbpMirpGroup: Use the Nearest Customer Bridge group address from Table 8-1 with the MIRP
12 B-VID (this is the default value) [item a) in 39.2.1.6];
- 13 2) cbpMirpVlan: Use the Nearest Customer Bridge group address from Table 8-1 with the B-VID
15 field from the Backbone Service Instance table [item b) in 39.2.1.6]; or
- 16 3) cbpMirpTable: Use the Default Backbone Destination and B-VID fields from the Backbone
18 Service Instance table [item c) in 39.2.1.6].
- 19 l) (MIRP only) A Boolean value specifying the administrative status requested by management for
20 attaching a MIRP Participant to a PNP or management Port. If TRUE, the BEB is to attach a MIRP
21 Participant to exactly one PNP or management Port, and if FALSE, no MIRP Participant is to be
22 present on any PNP or management Port. Default value is TRUE (39.1.2).
- 23 m) (MIRP only) The read-only Bridge Port Number (or 0, if none) of the PNP to which the MIRP
24 Participant is attached (39.1.2).

25 12.16.1.2 Set BEB configuration

26 12.16.1.2.1 Purpose

27 All BEBs shall implement the set BEB configuration function to set the BEB Name. Items marked “(MIRP
28 only)” shall be supported if the B-Component supports the Multiple I-SID Registration Protocol (MIRP)
29 (Clause 39). All items are persistent over power up or reboot.

30 12.16.1.2.2 Inputs

- 31 a) BEB Name—a text string of up to 32 characters, of locally determined significance.
- 32 b) (MIRP only) A Boolean value specifying whether MIRP is (TRUE) or is not (FALSE) enabled in
33 this B-Component (Clause 39).
- 34 c) (MIRP only) The MIRP B-VID to which received MIRPDUs are to be assigned if item d) contains
35 the value cbpMirpGroup (1). The value 0 (the default) indicates that the CBP PVID is to be used
36 (39.2.1.6).
- 37 d) (MIRP only) An enumerated value specifying what destination_address and vlan_identifier are to be
38 used when the MIRP Participant transmits an MIRPDU towards the MAC relay entity, either
39 cbpMirpGroup (1), cbpMirpVlan (2), or cbpMirpTable (3) (39.2.1.6).
- 40 e) (MIRP only) A Boolean value specifying the administrative status requested by management for
41 attaching a MIRP Participant to a PNP or management Port. If TRUE, the BEB is to attach a MIRP
42 Participant to exactly one PNP or management Port, and if FALSE, no MIRP Participant is to be
43 present on any PNP or management Port (39.1.2).

44 12.16.1.2.3 Outputs

45 None.

1 **12.16.1.3 Create BEB component (optional)**

2 **12.16.1.3.1 Purpose**

3 A BEB may implement the create BEB component function to allow the dynamic creation of new
4 components. Components created by this command are persistent over power-up or reboot.

5 **12.16.1.3.2 Inputs**

6 a) ComponentType. This takes one of the following values:

7 1) I-Comp

8 2) B-Comp

9 **12.16.1.3.3 Outputs**

10 None.

11 **12.16.1.4 Delete BEB component (optional)**

12 **12.16.1.4.1 Purpose**

13 A BEB may implement the delete BEB component function to allow deletion of components. Component
14 deletion using this command is persistent over power-up or reboot.

15 **12.16.1.4.2 Inputs**

16 a) ComponentID—indicates the component being deleted.

17 **12.16.1.4.3 Outputs**

18 None.

19 **12.16.1.5 Create BEB Bridge Port (optional)**

20 **12.16.1.5.1 Purpose**

21 A BEB may implement the create BEB Bridge Port function to allow dynamic creation of Bridge Ports.
22 Bridge Ports created by this command are persistent over power-up or reboot.

23 **12.16.1.5.2 Inputs**

24 a) Port Index—indicates the port where the new Bridge Port will be created.

25 b) ComponentID—indicates the component where the new Bridge Port will be created.

26 c) Port type. This takes one of the following values:

27 1) CNP

28 2) PNP

29 3) CBP

30 4) VIP

31 **12.16.1.5.3 Outputs**

32 None.

1 **12.16.1.6 Create BEB PIP (optional)**

2 **12.16.1.6.1 Purpose**

3 A BEB may implement the create BEB PIP function to allow dynamic creation of PIPs. PIPs created by this
4 command are persistent over power-up or reboot.

5 **12.16.1.6.2 Inputs**

- 6 a) Port Index—indicates the port where the new PIP will be created.
- 7 b) ComponentID—indicates the component where the new PIP will be created.

8 **12.16.1.6.3 Outputs**

9 None.

10 **12.16.1.7 Delete BEB Bridge Port (optional)**

11 **12.16.1.7.1 Purpose**

12 A BEB may implement the delete BEB Bridge Port function to allow deletion of Bridge Ports. Bridge Port
13 deletion using this command is persistent over power-up or reboot.

14 **12.16.1.7.2 Inputs**

- 15 a) ComponentID—the component for the Bridge Port.
- 16 b) Port Number—the Port Number for the Bridge Port.

17 **12.16.1.7.3 Outputs**

18 None.

19 **12.16.1.8 Delete BEB PIP (optional)**

20 **12.16.1.8.1 Purpose**

21 A BEB may implement the delete BEB PIP function to allow deletion of PIPs. PIP deletion using this
22 command is persistent over power-up or reboot.

23 **12.16.1.8.2 Inputs**

- 24 a) PIP Index—identifies the PIP within the BEB.

25 **12.16.1.8.3 Outputs**

26 None.

12.16.2 BEB/PB/VLAN Bridge Port configuration managed object

The management operation that can be performed on the BEB/PB/VLAN Bridge Port configuration managed object is as follows:

- a) Read BEB/PB/VLAN Bridge Port configuration (12.16.2.1)

12.16.2.1 Read BEB/PB/VLAN Bridge Port configuration

12.16.2.1.1 Purpose

All BEBs shall implement the read BEB/PB/VLAN Bridge Port configuration function to obtain information regarding the port type of a Bridge Port on the BEB.

12.16.2.1.2 Inputs

- a) ComponentID—the number identifying the Bridge Component associated with this port.
- b) Port Number—that component's Port Number for the Bridge Port.

12.16.2.1.3 Outputs

- a) Port Type. This takes one of the following values:
 - 1) C-VLAN Port
 - 2) PNP (identical to the PNP Bridge Port type specified by 12.13.1)
 - 3) CNP (identical to the CNP Bridge Port type specified by 12.13.1)
 - 4) CEP (identical to the CEP Bridge Port type specified by 12.13.1)
 - 5) CBP (5.12, 6.11, 12.13.1 will indicate PNP)
 - 6) VIP (5.12, 6.10, 12.13.1 will indicate PNP)

NOTE—Support for C-VLAN ports requires equipment with VLAN Bridge (5.9) functions in addition to BEB (5.12) functions. Support for CEP ports requires equipment with Provider Bridge (5.10) functions in addition to BEB functions.

12.16.3 VIP configuration managed object

The VIP configuration managed object applies to each VIP (6.10) on a BEB whether its associated PIP is externally or internally accessible.

It includes:

- a) The VIP configuration parameters, which provide the subset of the Bridge VLAN Configuration managed object (12.10.1) that is relevant for the configuration of the backbone service instance.

The management operations that can be performed are as follows:

- b) Read VIP configuration (12.16.3.1)
- c) Set VIP configuration (12.16.3.2)

12.16.3.1 Read VIP configuration

12.16.3.1.1 Purpose

All BEBs with I-Components shall implement the read VIP configuration function to allow reading the current configuration of the VIPs (6.10) in an I-Component of a BEB. Each VIP is configured by 12.8, 12.10, 12.16.3.2. The adminPointToPointMAC parameter is configured by 12.8.2.3.2 while the PVID and Acceptable Frame Types parameters are supplied by the Bridge Port configuration of 12.10.1.2 and 12.10.1.3.

1 12.16.3.1.2 Inputs

- 2 a) ComponentID—the number identifying the Bridge Component associated with this port.
- 3 b) Port Number—that component's Port Number for the Bridge Port.

4 12.16.3.1.3 Outputs

- 5 a) ComponentID—the number identifying the Bridge Component associated with this port.
- 6 b) Port Number—that component's Port Number for the Bridge Port.
- 7 c) PIP Name—a text string of up to 32 characters that identifies the PIP within a BEB with which this
- 8 VIP is associated.
- 9 d) VIP-ISID—the I-SID associated with this VIP (6.10).
- 10 e) Default Backbone Destination—an EUI-48 DA (6.10.2).
- 11 f) enableConnectionIdentifier—A Boolean to indicate if the connection_identifier parameter is
- 12 allowed to learn associations between a B-MAC address and a C-MAC address (6.10).

13 12.16.3.2 Set VIP configuration

14 12.16.3.2.1 Purpose

15 All BEBs with I-Components shall implement the set VIP configuration function to allow setting the
16 configuration of the internal VIP in the I-Component of a BEB.

17 12.16.3.2.2 Inputs

- 18 a) ComponentID—the number identifying the Bridge Component associated with this port.
- 19 b) Port Number—that component's Port Number for the Bridge Port.
- 20 c) VIP-ISID—the I-SID associated with this VIP (6.10).
- 21 d) enableConnectionIdentifier—A Boolean that determines if the connection_identifier parameter is
- 22 allowed to learn associations between a B-MAC address and a C-MAC address (6.10). The default
- 23 value is TRUE. This parameter should be configured to FALSE at the root node of a
- 24 point-to-multipoint TESI.

25 12.16.3.2.3 Outputs

- 26 a) Operation status. This takes one of the following values:
 - 27 1) Operation rejected because the Bridge Port identified by the ComponentID and Port Number is
 - 28 not a VIP.
 - 29 2) Operation accepted.

30 12.16.4 PIP configuration managed object

31 The PIP Configuration managed object applies to each PIP (6.10) on a BEB.

32 It includes the following:

- 33 a) The VIP mapping, which provides the set of VIPs multiplexed on this PIP
- 34 b) The PIP MAC address, which is used as the B-SA for all encapsulated frames

35 The management operations that can be performed are as follows:

- 36 c) Read PIP Configuration (12.16.4.1)
- 37 d) Optionally, Set PIP Configuration (12.16.4.2)
- 38 e) Read VIP to PIP mapping (12.16.4.3)
- 39 f) Optionally, Set VIP to PIP mapping (12.16.4.4)

- 1 g) Read PIP Priority Code Point Selection (12.16.4.5)
- 2 h) Set PIP Priority Code Point Selection (12.16.4.6)
- 3 i) Read PIP Priority Code Point Decoding Table (12.16.4.7)
- 4 j) Optionally, Set PIP Priority Code Point Decoding Table (12.16.4.8)
- 5 k) Read PIP Priority Code Point Encoding Table (12.16.4.9)
- 6 l) Optionally, Set PIP Priority Code Point Encoding Table (12.16.4.10)
- 7 m) Read PIP Use_DEI parameter (12.16.4.11)
- 8 n) Optionally, Set PIP Use_DEI parameter (12.16.4.12)
- 9 o) Read PIP Require Drop Encoding parameter (12.16.4.13)
- 10 p) Optionally, Set PIP Require Drop Encoding parameter (12.16.4.14)

11 **12.16.4.1 Read PIP configuration**

12 **12.16.4.1.1 Purpose**

13 All BEBs with I-Components shall implement the read PIP configuration function to allow reading the
14 current configuration of the PIP in the I-Component of a BEB.

15 **12.16.4.1.2 Inputs**

- 16 a) PIP Index—an identifier for the PIP within this BEB.

17 **12.16.4.1.3 Outputs**

- 18 a) PIP B-MAC address—identifies the PIP (6.10.2).
- 19 b) PIP Name—a text string of up to 32 characters used to identify a PIP within a BEB.
- 20 c) ComponentID—the number identifying the I-Component with this PIP.
- 21 d) VIP map—a bit array of 4094 bits indicating all the VIPs associated with this port.

22 **12.16.4.2 Set PIP configuration (optional)**

23 **12.16.4.2.1 Purpose**

24 All BEBs with I-Components may implement the set PIP configuration function to set the current
25 configuration of the PIP in the I-Component of a BEB.

26 **12.16.4.2.2 Inputs**

- 27 a) PIP Index—an identifier for the PIP within this BEB.
- 28 b) PIP B-MAC address—the B-MAC address used by this PIP in the B-SA field of transmitted frames.
- 29 c) PIP Name—a text string of up to 32 characters used to identify the PIP within a BEB.
- 30 d) VIP map—a bit array of 4094 bits indicating all the VIPs associated with this PIP.

31 **12.16.4.2.3 Outputs**

- 32 a) Operation status. This takes one of the following values:
 - 33 1) Operation rejected because the PIP Index is invalid
 - 34 2) Operation rejected because the VIP map identifies a port that is not a VIP
 - 35 3) Operation accepted

12.16.4.3 Read VIP to PIP mapping

12.16.4.3.1 Purpose

All BEBs with I-Components shall implement the read VIP to PIP mapping function to allow reading the VIP to PIP map information from the PIP configuration managed object.

12.16.4.3.2 Inputs

- a) ComponentID—the number identifying a Bridge Component with a VIP.
- b) Port Number—that component's Port Number for this VIP.

12.16.4.3.3 Outputs

- a) ComponentID—the number identifying a Bridge Component with a VIP.
- b) Port Number—that component's Port Number for this VIP.
- c) PIP Index—identifies the PIP supporting this VIP.
- d) PIP B-MAC address—the B-MAC address used by this PIP supporting this VIP (6.10.2).
- e) PIP Name—a text string of up to 32 characters identifying the PIP supporting this VIP.

12.16.4.4 Set VIP to PIP mapping (optional)

12.16.4.4.1 Purpose

All BEBs with I-Components may implement the set VIP to PIP mapping function to allow binding individual VIPs to PIPs. The set VIP to PIP mapping function allows setting map entries for individual VIPs while the set PIP configuration function sets the entire VIP to PIP map.

12.16.4.4.2 Inputs

- a) ComponentID—the number identifying a Bridge Component with a VIP.
- b) Port Number—that component's Port Number for this VIP.
- c) PIP Index—an identifier for this PIP within this component of the BEB.

12.16.4.4.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because the ComponentID and Port Number do not identify a VIP.
 - 2) Operation rejected because the PIP Index does not identify a PIP within the specified component.
 - 3) Operation accepted.

12.16.4.5 Read PIP Priority Code Point Selection

12.16.4.5.1 Purpose

To read which row of the Priority Code Point Encoding Table and Priority Code Point Decoding Table (6.10.3) is currently selected for use on this PIP.

1 12.16.4.5.2 Inputs

- 2 a) PIP Index—an identifier for this PIP within this component of the BEB.

3 12.16.4.5.3 Outputs

- 4 a) Priority Code Point Selection. This takes one of the following values:

- 5 1) 8P0D
6 2) 7P1D
7 3) 6P2D
8 4) 5P3D

9 12.16.4.6 Set PIP Priority Code Point Selection

10 12.16.4.6.1 Purpose

11 To set which row of the Priority Code Point Encoding Table and Priority Code Point Decoding Table
12 (6.10.3) will be selected for use on this PIP.

13 12.16.4.6.2 Inputs

- 14 a) PIP Index—an identifier for this PIP within this component of the BEB.

- 15 b) Priority Code Point Selection. This takes one of the following values:

- 16 1) 8P0D
17 2) 7P1D
18 3) 6P2D
19 4) 5P3D

20 12.16.4.6.3 Outputs

21 None.

22 12.16.4.7 Read PIP Priority Code Point Decoding Table

23 12.16.4.7.1 Purpose

24 To read the current contents of a row in the Priority Code Point Decoding Table (6.10.3) for a PIP.

25 12.16.4.7.2 Inputs

- 26 a) PIP Index—an identifier for this PIP.

- 27 b) Priority Code Point Row. This takes one of the following values:

- 28 1) 8P0D
29 2) 7P1D
30 3) 6P2D
31 4) 5P3D

32 12.16.4.7.3 Outputs

- 33 a) Priority value for Priority Code Point 0: Integer in range 0–7.
34 b) Drop_eligible value for Priority Code Point 0: Boolean.
35 c) Priority value for Priority Code Point 1: Integer in range 0–7.
36 d) Drop_eligible value for Priority Code Point 1: Boolean.
37 e) Priority value for Priority Code Point 2: Integer in range 0–7.
38 f) Drop_eligible value for Priority Code Point 2: Boolean.

- 1 g) Priority value for Priority Code Point 3: Integer in range 0–7.
- 2 h) Drop_eligible value for Priority Code Point 3: Boolean.
- 3 i) Priority value for Priority Code Point 4: Integer in range 0–7.
- 4 j) Drop_eligible value for Priority Code Point 4: Boolean.
- 5 k) Priority value for Priority Code Point 5: Integer in range 0–7.
- 6 l) Drop_eligible value for Priority Code Point 5: Boolean.
- 7 m) Priority value for Priority Code Point 6: Integer in range 0–7.
- 8 n) Drop_eligible value for Priority Code Point 6: Boolean.
- 9 o) Priority value for Priority Code Point 7: Integer in range 0–7.
- 10 p) Drop_eligible value for Priority Code Point 7: Boolean.

11 **12.16.4.8 Set PIP Priority Code Point Decoding Table (optional)**

12 **12.16.4.8.1 Purpose**

13 To modify the contents of a row in the Priority Code Point Decoding Table (6.10.3) for a PIP.

14 **12.16.4.8.2 Inputs**

- 15 a) PIP Index—an identifier for this PIP.
- 16 b) b) Priority Code Point Row. This takes one of the following values:
 - 17 1) 8P0D
 - 18 2) 7P1D
 - 19 3) 6P2D
 - 20 4) 5P3D
- 21 c) Priority value for Priority Code Point 0: Integer in range 0–7.
- 22 d) Drop_eligible value for Priority Code Point 0: Boolean.
- 23 e) Priority value for Priority Code Point 1: Integer in range 0–7.
- 24 f) Drop_eligible value for Priority Code Point 1: Boolean.
- 25 g) Priority value for Priority Code Point 2: Integer in range 0–7.
- 26 h) Drop_eligible value for Priority Code Point 2: Boolean.
- 27 i) Priority value for Priority Code Point 3: Integer in range 0–7.
- 28 j) Drop_eligible value for Priority Code Point 3: Boolean.
- 29 k) Priority value for Priority Code Point 4: Integer in range 0–7.
- 30 l) Drop_eligible value for Priority Code Point 4: Boolean.
- 31 m) Priority value for Priority Code Point 5: Integer in range 0–7.
- 32 n) Drop_eligible value for Priority Code Point 5: Boolean.
- 33 o) Priority value for Priority Code Point 6: Integer in range 0–7.
- 34 p) Drop_eligible value for Priority Code Point 6: Boolean.
- 35 q) Priority value for Priority Code Point 7: Integer in range 0–7.
- 36 r) Drop_eligible value for Priority Code Point 7: Boolean.

37 **12.16.4.8.3 Outputs**

38 None.

39 **12.16.4.9 Read PIP Priority Code Point Encoding Table**

40 **12.16.4.9.1 Purpose**

41 To read the current contents of a row in the Priority Code Point Encoding Table (6.10.3) for a Port.

1 12.16.4.9.2 Inputs

- 2 a) PIP Index—an identifier for this PIP.
- 3 b) Priority Code Point Row. This takes one of the following values:
 - 4 1) 8P0D
 - 5 2) 7P1D
 - 6 3) 6P2D
 - 7 4) 5P3D

8 12.16.4.9.3 Outputs

- 9 a) Priority Code Point value for priority 0 with drop_eligible False: Integer in range 0–7.
- 10 b) Priority Code Point value for priority 0 with drop_eligible True: Integer in range 0–7.
- 11 c) Priority Code Point value for priority 1 with drop_eligible False: Integer in range 0–7.
- 12 d) Priority Code Point value for priority 1 with drop_eligible True: Integer in range 0–7.
- 13 e) Priority Code Point value for priority 2 with drop_eligible False: Integer in range 0–7.
- 14 f) Priority Code Point value for priority 2 with drop_eligible True: Integer in range 0–7.
- 15 g) Priority Code Point value for priority 3 with drop_eligible False: Integer in range 0–7.
- 16 h) Priority Code Point value for priority 3 with drop_eligible True: Integer in range 0–7.
- 17 i) Priority Code Point value for priority 4 with drop_eligible False: Integer in range 0–7.
- 18 j) Priority Code Point value for priority 4 with drop_eligible True: Integer in range 0–7.
- 19 k) Priority Code Point value for priority 5 with drop_eligible False: Integer in range 0–7.
- 20 l) Priority Code Point value for priority 5 with drop_eligible True: Integer in range 0–7.
- 21 m) Priority Code Point value for priority 6 with drop_eligible False: Integer in range 0–7.
- 22 n) Priority Code Point value for priority 6 with drop_eligible True: Integer in range 0–7.
- 23 o) Priority Code Point value for priority 7 with drop_eligible False: Integer in range 0–7.
- 24 p) Priority Code Point value for priority 7 with drop_eligible True: Integer in range 0–7.

25 12.16.4.10 Set PIP Priority Code Point Encoding Table (optional)

26 12.16.4.10.1 Purpose

27 To modify the contents of a row in the Priority Code Point Encoding Table (6.10.3) for a PIP.

28 12.16.4.10.2 Inputs

- 29 a) PIP Index—an identifier for this PIP.
- 30 b) Priority Code Point Row. This takes one of the following values:
 - 31 1) 8P0D
 - 32 2) 7P1D
 - 33 3) 6P2D
 - 34 4) 5P3D
- 35 c) Priority Code Point value for priority 0 with drop_eligible False: Integer in range 0–7.
- 36 d) Priority Code Point value for priority 0 with drop_eligible True: Integer in range 0–7.
- 37 e) Priority Code Point value for priority 1 with drop_eligible False: Integer in range 0–7.
- 38 f) Priority Code Point value for priority 1 with drop_eligible True: Integer in range 0–7.
- 39 g) Priority Code Point value for priority 2 with drop_eligible False: Integer in range 0–7.
- 40 h) Priority Code Point value for priority 2 with drop_eligible True: Integer in range 0–7.
- 41 i) Priority Code Point value for priority 3 with drop_eligible False: Integer in range 0–7.
- 42 j) Priority Code Point value for priority 3 with drop_eligible True: Integer in range 0–7.
- 43 k) Priority Code Point value for priority 4 with drop_eligible False: Integer in range 0–7.
- 44 l) Priority Code Point value for priority 4 with drop_eligible True: Integer in range 0–7.
- 45 m) Priority Code Point value for priority 5 with drop_eligible False: Integer in range 0–7.
- 46 n) Priority Code Point value for priority 5 with drop_eligible True: Integer in range 0–7.

- 1 o) Priority Code Point value for priority 6 with drop_eligible False: Integer in range 0–7.
- 2 p) Priority Code Point value for priority 6 with drop_eligible True: Integer in range 0–7.
- 3 q) Priority Code Point value for priority 7 with drop_eligible False: Integer in range 0–7.
- 4 r) Priority Code Point value for priority 7 with drop_eligible True: Integer in range 0–7.

5 **12.16.4.10.3 Outputs**

6 None.

7 **12.16.4.11 Read PIP Use_DEI parameter**

8 **12.16.4.11.1 Purpose**

9 To read the current state of the Use_DEI parameter (6.10.3) for the PIP.

10 **12.16.4.11.2 Inputs**

- 11 a) PIP Index—an identifier for this PIP.

12 **12.16.4.11.3 Outputs**

- 13 a) Use_DEI parameter: Boolean.

14 **12.16.4.12 Set PIP Use_DEI parameter (optional)**

15 **12.16.4.12.1 Purpose**

16 To set the current state of the Use_DEI parameter (6.10.3) for the PIP.

17 **12.16.4.12.2 Inputs**

- 18 a) PIP Index: an identifier for this PIP.
- 19 b) Use_DEI parameter: Boolean.

20 **12.16.4.12.3 Outputs**

21 None.

22 **12.16.4.13 Read PIP Require Drop Encoding parameter**

23 **12.16.4.13.1 Purpose**

24 To read the current state of the Require Drop Encoding parameter (8.6.6) for the PIP.

25 **12.16.4.13.2 Inputs**

- 26 a) PIP Index—an identifier for this PIP.

27 **12.16.4.13.3 Outputs**

- 28 a) Require Drop Encoding parameter—Boolean.

29 **12.16.4.14 Set PIP Require Drop Encoding parameter (optional)**

30 **12.16.4.14.1 Purpose**

31 To set the current state of the Require Drop Encoding parameter (8.6.6) for the PIP.

1 **12.16.4.14.2 Inputs**

- 2 a) PIP Index—an identifier for this PIP.
- 3 b) Require Drop Encoding parameter—Boolean.

4 **12.16.4.14.3 Outputs**

5 None.

6 **12.16.5 CBP Configuration managed object**

7 The CBP Configuration managed object includes the following:

- 8 a) A Backbone Service Instance table (6.11)
- 9 b) A Flow Filtering Control Table (44.2.2)

10 Item b) is applicable only on BEBs supporting ECMP with flow filtering.

11 The management operations that can be performed are as follows:

- 12 c) Read Backbone Service Instance table entry (12.16.5.1)
- 13 d) Set Backbone Service Instance table entry (12.16.5.2)
- 14 e) TESI assignment managed object (12.16.5.3)
- 15 f) Read Flow Filtering Control Table entry (12.16.5.4)
- 16 g) Set Flow Filtering Control Table entry (12.16.5.5)

17 **12.16.5.1 Read Backbone Service Instance table entry**

18 **12.16.5.1.1 Purpose**

19 All BEBs shall implement the read Backbone Service Instance table entry function to allow reading the
20 Backbone Service Instance table entries in the CBP configuration managed object.

21 **12.16.5.1.2 Inputs**

- 22 a) ComponentID—the number identifying the Bridge Component associated with this port.
- 23 b) Port Number—that component's Port Number for the Bridge Port.
- 24 c) Backbone-SID—identifying the backbone service instance (6.11).

25 **12.16.5.1.3 Outputs**

- 26 a) ComponentID—the number identifying the Bridge Component associated with this port.
- 27 b) Port Number—that component's Port Number for the Bridge Port.
- 28 c) Backbone-SID—the identifier used in the PBBN for the backbone service instance (6.11).
- 29 d) B-VID (optional)—12-bit VID (6.11).
- 30 e) Default Backbone Destination (optional)—EUI-48 DA (6.11).
- 31 f) Local-SID (optional)—24-bit I-SID value (6.11).

12.16.5.2 Set Backbone Service Instance table entry

12.16.5.2.1 Purpose

All BEBs shall implement the set Backbone Service Instance table entry function to allow configuration of the backbone service instances available on each CBP. This information is persistent over power-up or reboot.

12.16.5.2.2 Inputs

- a) ComponentID—the number identifying the Bridge Component associated with this port.
- b) Port Number—that component's Port Number for the Bridge Port.
- c) Backbone-SID—24-bit I-SID value (6.11).
- d) B-VID (optional)—12-bit VID (6.11).
- e) Default Backbone Destination (optional)—EUI-48 DA (6.11).
- f) Local-SID (optional)—24-bit I-SID value (6.11).

12.16.5.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because the ComponentID and Port Number do not identify a valid CBP.
 - 2) Operation rejected because the B-VID is assigned to ECMP with flow filtering (12.25.4.1, Table 44-1) and the Bridge does not support flow filtering.
 - 3) Operation accepted.

12.16.5.3 TESI assignment managed object

12.16.5.3.1 Purpose

All CBPs providing TESIs shall implement the TESI assignment managed object to assign a particular TESI on this CBP.

12.16.5.3.2 Inputs

- a) ComponentID—the number identifying the Bridge Component associated with this port.
- b) Port Number—that component's Port Number for the Bridge Port.
- c) A series of ordered 3-tuples of the form <ESP-DA, ESP-SA, ESP-VID>.

12.16.5.3.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because the ComponentID and Port Number do not identify a valid CBP.
 - 2) Operation rejected because a 3-tuple in the provided series has a VID value in its ESP-VID field that is not allocated to TE-MSTID.
 - 3) Operation rejected because no 3-tuple in the provided series has the CBP's MAC address in its ESP-SA field.
 - 4) Operation rejected because there are only two 3-tuples provided and the values in their ESP-DA and ESP-SA fields are not reversed.
 - 5) Operation rejected because there are more than two different values specified in the ESP-DA fields of the provided series of 3-tuples.
 - 6) Operation accepted.

1 12.16.5.4 Read Flow Filtering Control Table entry

2 12.16.5.4.1 Purpose

3 BEBs supporting ECMP with flow filtering shall implement the read Flow Filtering Control Table entry
4 function to allow reading the Flow Filtering Control Table entries in the CBP configuration managed object.

5 12.16.5.4.2 Inputs

- 6 a) ComponentID—the number identifying the Bridge Component associated with this port.
- 7 b) Port Number—that component's Port Number for the Bridge Port.
- 8 c) VID—identifying a B-VID.

9 12.16.5.4.3 Outputs

- 10 a) Operation status. This takes one of the following values:
 - 11 1) Operation rejected because the ComponentID and Port Number do not identify a valid CBP
 - 12 2) Operation rejected because the CBP does not support F-TAG processing
 - 13 3) Operation accepted
- 14 b) ComponentID—the number identifying the Bridge Component associated with this port.
- 15 c) Port Number—that component's Port Number for the Bridge Port.
- 16 d) VID—the identifier of a B-VLAN.
- 17 e) Flow Filtering—indicates whether flow filtering behavior is enabled on the port for the VID
18 (enabled/disabled).
- 19 f) Flow Hash Generation—indicates whether flow hash generation is enabled on the port for the VID
20 (enabled/disabled).
- 21 g) TTL Value—the initial TTL value for frames entering the flow filtering SPT Domain (1..63).

22 12.16.5.5 Set Flow Filtering Control Table entry

23 12.16.5.5.1 Purpose

24 BEBs supporting ECMP with flow filtering shall implement the set Flow Filtering Control Table entry
25 function to allow configuration of the initial TTL value for frames entering an SPT Domain. This
26 information is persistent over power-up or reboot.

27 12.16.5.5.2 Inputs

- 28 a) ComponentID—the number identifying the Bridge Component associated with this port.
- 29 b) Port Number—that component's Port Number for the Bridge Port.
- 30 c) VID—12-bit VID.
- 31 d) TTL Value—the initial TTL value for frames entering the SPT Domain, Integer in range 1–63. The
32 default value is 8.

33 NOTE—The Flow Filtering and Flow Hash Generation fields in the Flow Filtering Control Table are controlled by
34 ISIS-SPB (28.8) and therefore are not settable by management.

35 12.16.5.5.3 Outputs

- 36 a) Operation status. This takes one of the following values:
 - 37 1) Operation rejected because the ComponentID and Port Number do not identify a valid CBP
 - 38 2) Operation rejected because the CBP does not support F-TAG processing
 - 39 3) Operation accepted

12.17 DDCFM entities

The following are the DDCFM managed objects in a Bridge:

- a) DDCFM Stack managed object (12.17.1)
- b) Reflection Responder managed object (12.17.2)
- c) RFM Receiver managed object (12.17.3)
- d) Decapsulator Responder managed object (12.17.4)
- e) SFM Originator managed object (12.17.5)

12.17.1 DDCFM Stack managed object

There is one DDCFM Stack managed object per Bridge. It allows the network administrator to retrieve DDCFM entities configured on a particular Bridge Port or an aggregated port within a Bridge Port. The management operation that can be performed is as follows:

- a) Read DDCFM Stack managed object (12.17.1.1)

12.17.1.1 Read DDCFM Stack managed object

12.17.1.1.1 Purpose

To retrieve DDCFM entities configured on a particular Bridge Port or an aggregated port within a Bridge Port.

12.17.1.1.2 Input

- a) An interface, either a Bridge Port, or an aggregated port within a Bridge Port.

12.17.1.1.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because of invalid reference to an interface;
 - 2) Operation accepted, but there is no DDCFM entity configured on the specified interface; or
 - 3) Operation accepted, there are (is a) DDCFM entities (entity) configured on the specified interface, and
- b) If there are DDCFM entities configured on the specified interface, the following attributes should be returned:
 - 1) An RR is configured, its associated maintenance domain, and the value indicating its direction; and/or
 - 2) An RFM Receiver is configured, its associated maintenance domain; and/or
 - 3) A DR is configured, its associated maintenance domain, and its associated MA; and/or
 - 4) An SFM Originator is configured, its associated maintenance domain, its associated MA, and the value indicating its direction.

12.17.2 Reflection Responder managed object

The management operations that can be performed on the Reflection Responder managed object are as follows:

- a) Create Reflection Responder managed object (12.17.2.1)
- b) Write Reflection Responder managed object's attributes (12.17.2.2)
- c) Read Reflection Responder managed object's attributes (12.17.2.3)
- d) Delete Reflection Responder managed object (12.17.2.4)

- 1 e) Activate Reflection Responder (12.17.2.5)
- 2 f) Deactivate Reflection Responder (12.17.2.6)

3 **12.17.2.1 Create Reflection Responder managed object**

4 **12.17.2.1.1 Purpose**

5 To create a Reflection Responder (RR). After the creation, all the writable attributes have their default
6 values unless being configured differently by the Write Reflection Responder managed object operation
7 (12.17.2.2). The RR is in the inactive state after the creation and remains so until activated by a subsequent
8 Activate Reflection Responder operation (12.17.2.5).

9 **12.17.2.1.2 Inputs**

- 10 a) A reference to a particular Reflection Responder managed object, which includes the following:
 - 11 1) An interface (either a Bridge Port or an aggregated port within a Bridge Port) on which the RR
 - 12 is defined. The interface identifier could be MAC address or port identifier.
 - 13 2) A reference to a particular Maintenance Domain managed object (12.14.5).
 - 14 3) A value indicating the direction in which the RR faces, which can be either Down [Active SAP
 - 15 is further away from the Frame filtering entity (default)], or Up (Active SAP is closer to the
 - 16 Frame filtering entity).

17 **12.17.2.1.3 Outputs**

- 18 a) Operation status. This takes one of the following values:
 - 19 1) Operation rejected because of invalid interface reference or invalid Maintenance Domain level
 - 20 2) Operation accepted

21 **12.17.2.2 Write Reflection Responder managed object's attributes**

22 **12.17.2.2.1 Purpose**

23 To configure and alter the value of one or more writable attribute(s) of an Reflection Responder managed
24 object. The attributes of an Reflection Responder managed object cannot be modified when the RR is active.

25 **12.17.2.2.2 Inputs**

- 26 a) Reference to a Reflection Responder managed object [item a) in 12.17.2.1.2].
- 27 b) Writable attributes are as follows:
 - 28 1) Reference to a Maintenance Association managed object (12.14.6) within the RR's
 - 29 Maintenance Domain. If set to 0, the MA associated with the filtered frame shall be used for the
 - 30 Reflected Frame Message (RFM) emitted by the RR.
 - 31 2) Reflection Filter definition(s), which is to specify what data frames are selected to be reflected.
 - 32 Multiple filters can be combined together by "&& (and)", "|| (or)", or "!(negation)" operations
 - 33 with precedence association being determined by "(". DDCFM-capable Bridges shall support
 - 34 each of the following filter primitives. Implementers may support additional filters, such as
 - 35 length of data frame being equal, larger, or smaller than a particular value.
 - 36 —All.
 - 37 —VID== vid; This primitive uses SVID if the RR is defined on a S Component, and CVID if
 - 38 the RR is defined on a C Component. Untagged frame shall not be selected by this filter
 - 39 primitive.
 - 40 —DA == xx.xx.xx.xx.xx.xx.
 - 41 —SA == xx.xx.xx.xx.xx.xx.
 - 42 —The first 2 bytes of the EtherType field ==xx.

- 1 3) Sampling Interval (29.2.3.4).
- 2 4) Reflection Target Address, which is a MAC address to which the reflected frames are targeted.
- 3 Only individual address is allowed for the Reflection Target Address. The default is the
- 4 source_address of the selected data frame being used for Reflection Target Address.
- 5 5) Continue option, indicating whether the selected data frames are to be continued toward the
- 6 destination_address specified in the data frame. The default is TRUE to allow data frames to be
- 7 forwarded to their destinations.
- 8 6) Boolean variable indicating if the duration is in seconds or in number of data frames.
- 9 7) Duration in seconds for the RR to remain active after being activated. Minimum 1 and
- 10 maximum is implementation dependent.
- 11 8) The number data frames for the RR to reflect after being activated. Once this frame count is
- 12 reached, the RR shall be deactivated automatically. Default is 0, which means the duration is
- 13 limited by RR's timer.
- 14 9) The priority and drop_eligible parameters to be used in the transmitted encapsulated frames
- 15 (default value: the highest priority, i.e., that with the highest numerical value, allowed to pass
- 16 through the Bridge Port for any of the VID. Default value for drop_eligible is FALSE).
- 17 10) FloodingEnabled Flag indicating whether flooding is allowed if Egress port cannot be
- 18 identified for RFM by the FDB.
- 19 11) Truncation flag indicating if the received data frame should be truncated to keep the length of
- 20 RFM encapsulated frame not exceeding the Maximum Service Data Unit Size (6.5.8). If the
- 21 Truncation flag is not set and the RFM encapsulated frame exceeds the Maximum Service Data
- 22 Unit Size, the filtered data frame is fragmented to two smaller frames and encapsulated in two
- 23 separate RFMs.

24 **12.17.2.2.3 Outputs**

- 25 a) Operation status. This takes one of the following values:
- 26 1) Operation rejected because the referenced Reflection Responder managed object does not exist
- 27 2) Operation rejected because of one or more invalid writable attribute(s)
- 28 3) Operation rejected because the referenced RR is active
- 29 4) Operation accepted

30 **12.17.2.3 Read Reflection Responder managed object's attributes**

31 **12.17.2.3.1 Purpose**

32 To retrieve values of attributes in a Reflection Responder managed object.

33 **12.17.2.3.2 Input**

- 34 a) Reference to a particular Reflection Responder managed object [item a) in 12.17.2.1.2].

35 **12.17.2.3.3 Outputs**

- 36 a) Operation status. This takes one of the following values:
- 37 1) Operation rejected because of invalid reference to the Reflection Responder managed object
- 38 2) Operation rejected because there is no RR with the given reference
- 39 3) Operation accepted
- 40 b) If the operation is accepted, the following attributes for the RR should be returned:
- 41 1) Reference to the Reflection Responder managed object.
- 42 2) Reference to a Maintenance Association managed object (12.14.6) for the RFMs generated by
- 43 the RR (writable).
- 44 3) Reflection Filter definition(s) (writable).
- 45 4) Sampling Interval (writable).

- 1 5) The Reflection Target Address (writable).
- 2 6) Continue option (writable).
- 3 7) Duration for RR to stay active after being activated (writable).
- 4 8) DurationInTimeFlag (writable).
- 5 9) The priority and drop_eligible parameters to be used in the transmitted RFMs (writable).
- 6 10) FloodingFlag (writable).
- 7 11) Truncation flag (writable).
- 8 12) Activation status (TRUE or FALSE).
- 9 13) Remaining time or count left for the RR to be active.
- 10 14) nextRFMtransID (29.3.1.16) value.

11 **12.17.2.4 Delete Reflection Responder managed object**

12 **12.17.2.4.1 Purpose**

13 To delete a Reflection Responder managed object. If the corresponding RR is still active, the deletion will
14 deactivate the RR and then delete the corresponding managed object.

15 **12.17.2.4.2 Input**

- 16 a) Reference to a Reflection Responder managed object [item a) in 12.17.2.1.2].

17 **12.17.2.4.3 Outputs**

- 18 a) Operation status. This takes one of the following values:
- 19 1) Operation rejected because of invalid reference to RR
 - 20 2) Operation rejected because the referenced RR managed does not exist
 - 21 3) Operation accepted

22 **12.17.2.5 Activate Reflection Responder**

23 **12.17.2.5.1 Purpose**

24 To activate an RR.

25 Depending on the filter conditions specified for each RR, some RRs can potentially reflect large amount of
26 data frames into the network, which could cause excessive extra traffic injected into the network and impact
27 network performance. For those RRs, it is recommended that no more than one instance of the Reflection
28 Responder managed object be activated per MD for an RR associated with an MP and that no more than one
29 instance of the Reflection Responder managed object be activated per VID for an RR not associated with an
30 MP (e.g., for an RR created on a Bridge Port).

31 **12.17.2.5.2 Inputs**

- 32 a) Reference to a Reflection Responder managed object [item a) in 12.17.2.1.2].

33 **12.17.2.5.3 Outputs**

- 34 a) Operation status. This takes one of the following values:
- 35 1) Operation rejected because invalid reference or the referenced RR does not exist
 - 36 2) Operation rejected because of the referenced RR already being activated
 - 37 3) Operation accepted

1 **12.17.2.6 Deactivate Reflection Responder**

2 **12.17.2.6.1 Purpose**

3 To deactivate an active RR before its duration expires.

4 **12.17.2.6.2 Input**

- 5 a) Reference to a Reflection Responder managed object [item a) in 12.17.2.1.2].

6 **12.17.2.6.3 Outputs**

- 7 a) Operation status. This takes one of the following values:
- 8 1) Operation rejected because invalid reference or the referenced RR does not exist
- 9 2) Operation rejected because of the referenced RR not being activated
- 10 3) Operation accepted

11 **12.17.3 RFM Receiver managed object**

12 The management operations that can be performed on the RFM Receiver managed object are as follows:

- 13 a) Create RFM Receiver managed object (12.17.3.1)
- 14 b) Delete RFM Receiver managed object (12.17.3.2)

15 An RFM Receiver is active when the corresponding RFM Receiver managed object is created. Therefore,
16 there is no “Activate and Deactivate” operations for the RFM Receiver managed object.

17 **12.17.3.1 Create RFM Receiver managed object**

18 **12.17.3.1.1 Purpose**

19 To create an RFM Receiver managed object. Once an RFM Receiver managed object is created, the
20 corresponding RFM Receiver is able to receive RFMs.

21 **12.17.3.1.2 Inputs**

- 22 a) Reference to an RFM Receiver managed object, which could be either:
- 23 1) Reference to an MP or
- 24 2) An interface (either a Bridge Port or an aggregated port within a Bridge Port) on which RFM
- 25 Receiver is created and a reference to the corresponding RR’s Maintenance Domain managed
- 26 object.

27 **12.17.3.1.3 Outputs**

- 28 a) Operation status. This takes one of the following values:
- 29 1) Operation rejected because of invalid reference to the RFM Receiver managed object
- 30 2) Operation rejected because the referenced RFM Receiver managed object already exists
- 31 3) Operation accepted

32 **12.17.3.2 Delete RFM Receiver managed object**

33 **12.17.3.2.1 Purpose**

34 To delete an RFM Receiver managed object.

1 **12.17.3.2.2 Input**

- 2 a) Reference to an RFM Receiver managed object [item a) in 12.17.3.1.2].

3 **12.17.3.2.3 Outputs**

- 4 a) Operation status. This takes one of the following values:
- 5 1) Operation rejected because of invalid reference to the RFM Receiver managed object
 - 6 2) Operation rejected because the referenced RFM Receiver managed object does not exist
 - 7 3) Operation accepted

8 **12.17.4 Decapsulator Responder managed object**

9 The management operations that can be performed on the Decapsulator Responder managed object are as
10 follows:

- 11 a) Create Decapsulator Responder managed object (12.17.4.1)
- 12 b) Read Decapsulator Responder managed object (12.17.4.2)
- 13 c) Write Decapsulator Responder managed object's attributes (12.17.4.3)
- 14 d) Delete Decapsulator Responder managed object (12.17.4.4)
- 15 e) Activate Decapsulator Responder (12.17.4.5)
- 16 f) Deactivate Decapsulator Responder (12.17.4.6)

17 **12.17.4.1 Create Decapsulator Responder managed object**

18 **12.17.4.1.1 Purpose**

19 To create a Decapsulator Responder managed object and corresponding Decapsulator Responder (DR).

20 **12.17.4.1.2 Inputs**

- 21 a) Reference to a DR, which could be either:
- 22 1) Reference to an MP or
 - 23 2) A triple consisting of an interface (either a Bridge Port or an aggregated port within a Bridge
 - 24 Port), a reference to a Maintenance Domain managed object, and a reference to a Maintenance
 - 25 Association managed object.

26 **12.17.4.1.3 Outputs**

- 27 a) Operation status. This takes one of the following values:
- 28 1) Operation rejected because of invalid reference to a Decapsulator Responder managed object.
 - 29 2) Operation rejected because the referenced Decapsulator Responder managed object already
 - 30 exists.
 - 31 3) Operation accepted.

32 **12.17.4.2 Read Decapsulator Responder managed object**

33 **12.17.4.2.1 Purpose**

34 To retrieve the attributes of a Decapsulator Responder managed object.

35 **12.17.4.2.2 Input**

- 36 a) Reference to a Decapsulator Responder managed object [item a) in 12.17.4.1.2].

12.17.4.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because of invalid reference to a Decapsulator Responder managed object.
 - 2) Operation rejected because the referenced Decapsulator Responder managed object does not exist.
 - 3) Operation accepted.
- b) If the operation is accepted, the following attributes for the DR should be returned:
 - 1) Reference to the Decapsulator Responder managed object.
 - 2) MAC address of the SFM Originator (writable).
 - 3) The value of SourceAddressStayFlag (writable).
 - 4) The value of FloodingEnabled flag (writable).
 - 5) The specified duration in seconds, for DR to remain active once activated (writable).
 - 6) Activation status.
 - 7) Remaining time left for DR to be active.
 - 8) The total number of out-of-sequence SFMs received (29.3.8.7).

12.17.4.3 Write Decapsulator Responder managed object's attributes

12.17.4.3.1 Purpose

To alter the value of one or more attributes of a Decapsulator Responder managed object when the corresponding DR is not active.

12.17.4.3.2 Inputs

- a) Reference to a Decapsulator Responder managed object [item a) in 12.17.4.1.2].
- b) Reference to the attribute whose value is to be changed:
 - 1) Boolean flag, SourceAddressStayFlag, to enforce the DR not to replace the source_address field of the decapsulated frame with the DR's own MAC address. Default is FALSE.
 - 2) MAC address of the SFM Originator, which is optional.
 - 3) Boolean FloodingEnabled flag indicating if flooding is allowed (TRUE) or not (FALSE) if the Egress port cannot be identified by the FDB.
 - 4) Duration in seconds, for DR to remain active once activated.

12.17.4.3.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because of invalid reference to a Decapsulator Responder managed object.
 - 2) Operation rejected because the referenced Decapsulator Responder managed object does not exist.
 - 3) Operation rejected because of invalid attribute reference or invalid value.
 - 4) Operation rejected because the referenced DR is active.
 - 5) Operation accepted.

12.17.4.4 Delete Decapsulator Responder managed object

12.17.4.4.1 Purpose

To delete a Decapsulator Responder managed object. If the DR is still active, the deletion will deactivate the DR first and then delete the corresponding managed object.

1 **12.17.4.4.2 Input**

- 2 a) Reference to a Decapsulator Responder managed object [item a) in 12.17.4.1.2].

3 **12.17.4.4.3 Outputs**

- 4 a) Operation status. This takes one of the following values:
- 5 1) Operation rejected because of invalid reference to a Decapsulator Responder managed object.
 - 6 2) Operation rejected because the referenced Decapsulator Responder managed object does not
 - 7 exist.
 - 8 3) Operation accepted.

9 **12.17.4.5 Activate Decapsulator Responder**

10 **12.17.4.5.1 Purpose**

11 To activate a DR.

12 **12.17.4.5.2 Input**

- 13 a) Reference to the Decapsulator Responder managed object [item a) in 12.17.4.1.2].

14 **12.17.4.5.3 Outputs**

- 15 a) Operation status. This takes one of the following values:
- 16 1) Operation rejected because of invalid reference.
 - 17 2) Operation rejected because the referenced Decapsulator Responder managed object does not
 - 18 exist.
 - 19 3) Operation rejected because the referenced DR has already been activated.
 - 20 4) Operation accepted.

21 **12.17.4.6 Deactivate Decapsulator Responder**

22 **12.17.4.6.1 Purpose**

23 To deactivate a DR.

24 **12.17.4.6.2 Input**

- 25 a) Reference to the Decapsulator Responder managed object [item a) in 12.17.4.1.2].

26 **12.17.4.6.3 Outputs**

- 27 a) Operation status. This takes one of the following values:
- 28 1) Operation rejected because of invalid reference.
 - 29 2) Operation rejected because the referenced Decapsulator Responder managed object does not
 - 30 exist.
 - 31 3) Operation rejected because the referenced DR is not active.
 - 32 4) Operation accepted.

33 **12.17.5 SFM Originator managed object**

34 The management operations that can be performed on the SFM Originator managed object are as follows:

- 35 a) Create SFM Originator managed object (12.17.5.1)
- 36 b) Read SFM Originator managed object (12.17.5.2)
- 37 c) Delete SFM Originator managed object (12.17.5.3)

- 1 d) Write SFM Originator managed object's attributes (12.17.5.4)
- 2 e) Activate SFM Originator (12.17.5.5)
- 3 f) Deactivate SFM Originator (12.17.5.6)

4 **12.17.5.1 Create SFM Originator managed object**

5 **12.17.5.1.1 Purpose**

6 To create a SFM Originator managed object.

7 **12.17.5.1.2 Inputs**

- 8 a) Reference to a SFM Originator managed object, which could be either:
 - 9 1) Reference to an MP or
 - 10 2) An interface (either a Bridge Port, or an aggregated port within a Bridge Port), a reference to a
 - 11 Maintenance Domain managed object, a reference to a Maintenance Association managed
 - 12 object, and a value indicating the direction in which the SFM Originator faces on the interface.

13 **12.17.5.1.3 Outputs**

- 14 a) Operation status. This takes one of the following values:
 - 15 1) Operation rejected because of invalid reference
 - 16 2) Operation rejected because the referenced SFM Originator managed object already exists
 - 17 3) Operation accepted

18 **12.17.5.2 Read SFM Originator managed object**

19 **12.17.5.2.1 Purpose**

20 To retrieve attributes of SFM Originator managed object.

21 **12.17.5.2.2 Inputs**

- 22 a) Reference to a SFM Originator managed object [item a) in 12.17.5.1.2].

23 **12.17.5.2.3 Outputs**

- 24 a) Operation status. This takes one of the following values:
 - 25 1) Operation rejected because of invalid reference
 - 26 2) Operation rejected because the referenced SFM Originator managed object does not exist
 - 27 3) Operation accepted
- 28 b) If the operation is accepted, following attributes for the SFM Originator should be returned:
 - 29 1) MAC address of the SFM Originator
 - 30 2) MAC address of the corresponding DR (writable)
 - 31 3) Duration (writable)
 - 32 4) Activation status: TRUE/FALSE
 - 33 5) The remaining time left for the SFM Originator to remain active

34 **12.17.5.3 Delete SFM Originator managed object**

35 **12.17.5.3.1 Purpose**

36 To delete a SFM Originator managed object, and the corresponding SFM Originator.

1 **12.17.5.3.2 Inputs**

- 2 a) The reference to a SFM Originator managed object [item a) in 12.17.5.1.2].

3 **12.17.5.3.3 Outputs**

- 4 a) Operation status. This takes one of the following values:
- 5 1) Operation rejected because of invalid reference
- 6 2) Operation rejected because the referenced SFM Originator managed object does not exist
- 7 3) Operation accepted

8 **12.17.5.4 Write SFM Originator managed object's attribute**

9 **12.17.5.4.1 Purpose**

10 To change an attribute of a SFM Originator managed object.

11 **12.17.5.4.2 Inputs**

- 12 a) Reference to a SFM Originator managed object [item a) in 12.17.5.1.2].
- 13 b) MAC address of the corresponding DR.
- 14 c) Duration in seconds, for SFM Originator to stay active once activated.

15 **12.17.5.4.3 Outputs**

- 16 a) Operation status. This takes one of the following values:
- 17 1) Operation rejected because of invalid reference of SFM Originator managed object or the
- 18 attributes.
- 19 2) Operation rejected because the referenced SFM Originator managed object does not exist.
- 20 3) Operation accepted.

21 **12.17.5.5 Activate SFM Originator**

22 **12.17.5.5.1 Purpose**

23 To activate a SFM Originator.

24 **12.17.5.5.2 Input**

- 25 a) Reference to the SFM Originator managed object [item a) in 12.17.5.1.2].

26 **12.17.5.5.3 Outputs**

- 27 a) Operation status. This takes one of the following values:
- 28 1) Operation rejected because of invalid reference
- 29 2) Operation rejected because the referenced SFM Originator managed object does not exist
- 30 3) Operation rejected because the referenced SFM Originator has already been activated
- 31 4) Operation accepted

32 **12.17.5.6 Deactivate SFM Originator**

33 **12.17.5.6.1 Purpose**

34 To deactivate a SFM Originator.

1 12.17.5.6.2 Input

- 2 a) Reference to a SFM Originator managed object [item a) in 12.17.5.1.2].

3 12.17.5.6.3 Output

- 4 a) Operation status. This takes one of the following values:
- 5 1) Operation rejected because of invalid reference
 - 6 2) Operation rejected because the referenced SFM Originator managed object does not exist
 - 7 3) Operation rejected because the referenced SFM Originator is not active
 - 8 4) Operation accepted

9 12.18 PBB-TE Protection Switching managed objects

10 The PBB-TE Protection Switching managed objects model operations that create, modify, delete, or inquire
11 about, the configuration and the operation of protection switching. The following are the PBB-TE Protection
12 Switching managed objects in a B-Component of an IB-BEB capable of supporting TESIs:

- 13 a) The TE protection group list managed object
- 14 b) The TE protection group managed object

15 12.18.1 TE protection group list managed object

16 There is one TE protection group list managed object per IB-BEB. It contains a list of the TE protection
17 groups that have been configured on the IB-BEB.

18 The management operations that can be performed on the TE protection group list managed object are as
19 follows:

- 20 a) Read TE protection group list
- 21 b) Create TE protection group managed object
- 22 c) Delete TE protection group managed object

23 12.18.1.1 Read TE protection group list

24 12.18.1.1.1 Purpose

25 To obtain information about all of the TE protection groups in an IB-BEB.

26 12.18.1.1.2 Inputs

27 None.

28 12.18.1.1.3 Outputs

29 A list, perhaps empty, of the TE protection group managed objects configured on the IB-BEB. For each item
30 in the list, the Read TE protection group list command returns:

- 31 a) A reference to a particular Maintenance Association managed object (12.14.6) identifying a
32 PBB-TE MA that corresponds to the group's working entity.
- 33 b) A reference to a particular Maintenance Association managed object (12.14.6) identifying a
34 PBB-TE MA with the same endpoints as the one provided in item b) that corresponds to the group's
35 protection entity.
- 36 c) A reference to all TE protection group managed objects (12.18.2) that share a working or protection
37 entity with the TE protection group in the list.

12.18.1.2 Create TE protection group managed object

12.18.1.2.1 Purpose

To create a new TE protection group managed object in an IB-BEB and add it to the IB-BEB's TE protection group list managed object.

12.18.1.2.2 Inputs

- a) A reference to a particular Maintenance Association managed object (12.14.6) identifying a PBB-TE MA that corresponds to the working entity.
- b) A reference to a particular Maintenance Association managed object identifying a PBB-TE MA with the same endpoints as the one provided in item a) that will monitor the protection entity.

12.18.1.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to nonexistent Maintenance Domain.
 - 2) Operation rejected due to nonexistent MAs.
 - 3) Operation rejected because the referred MAs do not correspond to PBB-TE services.
 - 4) Operation rejected because the two referred MAs do not have the same endpoints.
 - 5) Operation rejected because the two referred MAs do not belong to the same MD.
 - 6) Operation rejected due to duplicate MAs.
 - 7) Operation accepted. In addition, the list, perhaps empty, of the TE protection group managed objects (12.18.2) that share a working or protection entity with the created TE protection group is provided.

12.18.1.3 Delete TE protection group managed object

12.18.1.3.1 Purpose

To delete a specific TE protection group managed object from an IB-BEB and from the IB-BEB's TE protection group list managed object.

12.18.1.3.2 Inputs

- a) A reference to a particular TE protection group managed object (12.18.2).

12.18.1.3.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to nonexistent TE protection group
 - 2) Operation rejected because the TE protection group is enabled
 - 3) Operation accepted

12.18.2 TE protection group managed object

There can be any number of TE protection group managed objects per IB-BEB, one for each working and protection pair of TESIs with common endpoints.

The management operations that can be performed on the TE protection group managed object are as follows:

- a) Read TE protection group managed object
- b) Write TE protection group managed object
- c) TE protection group administrative commands managed object

1 12.18.2.1 Read TE protection group managed object

2 12.18.2.1.1 Purpose

3 To obtain information from an IB-BEB about a specific TE protection group managed object.

4 12.18.2.1.2 Inputs

- 5 a) A reference to a particular TE protection group managed object.

6 12.18.2.1.3 Outputs

- 7 a) Operation status. This takes one of the following values:
- 8 1) Operation rejected due to nonexistent TE protection group
- 9 2) Operation accepted
- 10 b) (writable) A list of Backbone-SIDs, identifying the protected backbone service instances, or NULL
- 11 indicating that no backbone service instances are assigned to the TE protection group and that the
- 12 TE protection group is disabled.
- 13 c) An enumerated value indicating the operational state of the Service Mapping state machine
- 14 (Figure 26-17) for each TE protection group:
- 15 1) WORKING_PATH
- 16 2) PROTECTION_PATH
- 17 3) WTR
- 18 4) PROT_ADMIN
- 19 d) An enumerated value indicating the status of active requests within the TE protection group:
- 20 1) NoRequest: No administrative command is in effect.
- 22 2) LoP: Set if LoP (26.10.3.3.4) is TRUE, indicating that an administrative command to prohibit
- 24 the use of the protection entity is in effect.
- 25 3) FS: Set if FS (26.10.3.3.5) is TRUE, indicating that an administrative command to perform
- 27 forced switching to the protection path(s) is in effect.
- 28 4) p.SFH: Set if SFH (26.10.3.3.3) is TRUE on the protection entity.
- 30 5) w.SFH: Set if SFH is TRUE on the working entity.
- 32 6) MStoProtection: Set if MStoProtection (26.10.3.3.6) is TRUE, indicating that an administrative
- 34 command to perform manual switching to the protection path(s) is in effect.
- 35 7) MStoWorking: Set if MStoWorking (26.10.3.3.7) is TRUE, indicating that an administrative
- 37 command to perform manual switching to the working path is in effect.
- 38 e) (writable) (optional) The Wait-to-restore (WTR) period (26.10.3.3.8). In revertive operation it may
- 39 be configured in steps of 1 min between 5 min and 12 min; the default value is 5 min. The value 0
- 40 indicates nonrevertive operation.
- 41 f) (writable) (optional) The Hold-Off period (26.10.3.3.9). The range of the Hold-Off period is 0 s to
- 42 10 s in steps of 100 ms; the default value is 0.

43 12.18.2.2 Write TE protection group managed object

44 12.18.2.2.1 Purpose

45 To alter a value of a specific TE protection group managed object.

46 12.18.2.2.2 Inputs

- 47 a) A reference to a particular TE protection group managed object.
- 48 b) A reference to one of the writable managed objects in 12.18.2.1.3 that is to be altered.
- 49 c) A new value for the managed object.

12.18.2.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to nonexistent TE protection group
 - 2) Operation rejected due to the selected managed object being Read-Only
 - 3) Operation rejected due to lack of authority to set this variable
 - 4) Operation rejected due to invalid value for the selected managed object
 - 5) Operation accepted

12.18.2.3 TE protection group administrative commands managed object

12.18.2.3.1 Purpose

To set an administrative command on a TE protection group managed object.

12.18.2.3.2 Inputs

- a) A reference to a particular TE protection group managed object.
- b) An enumerated value indicating the exercised administrative command:
 - 1) Clear: An indication to clear all other administrative commands.
 - 2) Lockout of Protection: An administrative command to prohibit the use of the protection entity.
 - 3) Forced Switch: An administrative command to perform forced switching to the protection path(s).
 - 4) Manual Switch To Protection: An administrative command to perform manual switching to the protection path(s) if it is operational.
 - 5) Manual Switch To Working: An administrative command to perform manual switching to the working path if it is operational.

12.18.2.3.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to nonexistent TE protection group
 - 2) Operation rejected due to lack of authority to set this variable
 - 3) Operation rejected due to an active higher priority request
 - 4) Operation accepted

12.19 TPMR managed objects

The managed resources of a TPMR, accessed via the TPMR management entity, are those of the processes and entities defined in this clause, and listed as follows:

- a) The TPMR management entity (12.19.1)
- b) The individual MAC and PHY entities associated with each TPMR Port
- c) The Forwarding Process of the MAC Relay Entity (8.6, 12.19.3)
- d) The MAC Status Propagation Entity (MSPE) (23.3, 12.19.4)

The management of each of these resources is described in terms of managed objects and operations in the following subclauses.

12.19.1 TPMR management entity

This managed resource comprises the following objects:

- a) The TPMR Configuration (12.19.1.1)
- b) The Port Configuration for each Port (12.19.1.2)

12.19.1.1 TPMR configuration

The TPMR configuration object models the operations that modify, or inquire about, the configuration of the TPMR's resources. There is a single TPMR configuration object per TPMR.

The management operations that can be performed on the TPMR configuration are as follows:

- a) Read TPMR (12.19.1.1.1)
- b) Set TPMR Name (12.19.1.1.2)

12.19.1.1.1 Read TPMR

12.19.1.1.1.1 Purpose

To obtain general information regarding the TPMR.

12.19.1.1.1.2 Inputs

None.

12.19.1.1.1.3 Outputs

- a) TPMR Name—a text string of up to 32 characters, of locally determined significance.
- b) TPMR MAC addresses—the MAC addresses of the two externally accessible TPMR Ports. The following information is returned for each Port:
 - 1) The Port number.
 - 2) The MAC address of the Port.
 - 3) A Boolean value, which is TRUE if the MAC address is the management address for the TPMR, and is otherwise FALSE.

NOTE 1—The TPMR management entity may make use of one or both Ports of a TPMR to transmit and receive management frames. However, the MAC address used by the TPMR management entity as the source MAC address in transmitted management frames (the management MAC address) is the individual MAC address associated with one of the Ports of the TPMR (a management Port, see 8.3, 8.5).

NOTE 2—As the transmission and reception rules for a TPMR (8.5.2) mean that frames forwarded by a TPMR are not received by higher layer entities on the forwarding Port, by receiving management frames on either Port the TPMR management entity can determine which LAN incoming management frames were transmitted on, as incoming frames are only presented to higher layer entities attached to the reception Port.

- c) Uptime—count in seconds of the time elapsed since the TPMR was last reset or initialized.

12.19.1.1.2 Set TPMR Name

12.19.1.1.2.1 Purpose

To associate a text string, readable by the Read TPMR operation, with a TPMR.

12.19.1.1.2.2 Inputs

- a) TPMR Name—a text string of up to 32 characters.

12.19.1.1.2.3 Outputs

None.

1 **12.19.1.2 Port configuration**

2 The Port Configuration object models the operations that modify, or inquire about, the configuration of the
3 Ports of a TPMR. By definition there are two Ports per TPMR (one for each MAC interface), and each is
4 identified by a permanently allocated Port number.

5 The management operations that can be performed on the Port Configuration are as follows:

- 6 a) Read Port (12.19.1.2.1)
- 7 b) Set Port Name (12.19.1.2.2)

8 **12.19.1.2.1 Read Port**

9 **12.19.1.2.1.1 Purpose**

10 To obtain general information regarding a specific TPMR Port.

11 **12.19.1.2.1.2 Inputs**

- 12 a) Port Number—the number of the Port. This parameter can take the integer values 1 and 2 only.

13 **12.19.1.2.1.3 Outputs**

- 14 a) Port Name—a text string of up to 32 characters, of locally determined significance.
- 15 b) Port Type—the MAC Entity type of the Port (IEEE Std 802.3; ISO/IEC 8802-11; ISO 9314; other).
- 16 c) Management address forwarding status. This takes the value TRUE if forwarding is enabled for the
17 management address on the Port or FALSE if forwarding is disabled for the management address on
18 the Port (see 5.15).

19 **12.19.1.2.2 Set Port Name**

20 **12.19.1.2.2.1 Purpose**

21 To associate a text string, readable by the Read Port operation, with a TPMR Port.

22 **12.19.1.2.2.2 Inputs**

- 23 a) Port Number—the number of the Port. This parameter can take the integer values 1 and 2 only.
- 24 b) Port Name—a text string of up to 32 characters.

25 **12.19.1.2.2.3 Outputs**

26 None.

27 **12.19.2 MAC and PHY entities**

28 The management operations and facilities provided by the MAC and PHY entities are those specified in the
29 layer management standards of the individual MACs and PHYs. A MAC entity, and its underlying PHY
30 entity, is associated with each TPMR Port.

31 **12.19.3 Forwarding Process**

32 The Forwarding Process contains information relating to the forwarding of frames. Counters are maintained
33 that provide information on the number of frames forwarded, filtered, and dropped due to error.

34 This managed resource comprises the following objects:

- 35 a) The Port Counters (12.19.3.1)
- 36 b) Optionally, the Priority Handling objects for each Port (12.19.3.2)

- 1 c) Optionally, The Traffic Class Table for each Port (12.6.3)

2 **12.19.3.1 The Port Counters**

3 The Port Counters object models the operations that can be performed on the Port counters of the
4 Forwarding Process resource. There is one instance of the Port Counters object for each MAC Entity per
5 TPMR (i.e., two instances total).

6 The management operation that can be performed on the Port Counters is as follows:

- 7 a) Read Forwarding Port Counters (12.19.3.1.1)

8 **12.19.3.1.1 Read forwarding port counters**

9 **12.19.3.1.1.1 Purpose**

10 To read the forwarding counters associated with a specific TPMR Port.

11 **12.19.3.1.1.2 Inputs**

- 12 a) Port Number—the number of the Port. This parameter can take the integer values 1 and 2 only.

13 **12.19.3.1.1.3 Outputs**

- 14 a) Frames Received—count of all valid frames received on this Port (including BPDUs, frames
15 addressed to the TPMR as an end station, and frames that were submitted to the Forwarding Process,
16 8.6).
- 17 b) Octets Received—count of the total number of octets in all valid frames received on this Port
18 (including BPDUs, frames addressed to the TPMR as an end station, and frames that were submitted
19 to the Forwarding Process).
- 20 c) Frames Discarded by Forwarding Process—count of all frames that were received on this Port but
21 were discarded by the Forwarding Process for any reason.
- 22 d) Frames Forwarded to Transmission Port—count of all frames that were received on this Port and
23 were forwarded to the transmission Port (8.6.8).
- 24 e) Frames Discarded due to Queue Full—count of all frames received on this Port that were to be
25 transmitted through the transmission Port but were discarded due to lack of available queue space
26 (8.6.6, 8.6.7).
- 27 f) Frames Discarded Lifetime Exceeded—count of all frames received on this Port that were to be
28 transmitted through the transmission Port but were discarded due to their frame lifetime having been
29 exceeded (8.6.7).
- 30 g) Frames Discarded on Error—count of all frames received on this Port that were to be transmitted
31 through the transmission Port but could not be transmitted (e.g., frame would be too large, 6.5.8).
- 32 h) Frame Discard on Error Details—a list of 16 elements, each containing the source address of a frame
33 and the reason why the frame was discarded (e.g., frame too large). The list is maintained as a
34 circular buffer. The reason for discard on error, at present, is as follows:
- 35 1) Transmissible SDU size exceeded

36 **12.19.3.2 Priority handling**

37 The Priority Handling object models the operations that can be performed on, or inquire about, the Default
38 Priority parameter, the Priority Regeneration Table parameter, the Outbound Access Priority Table
39 parameter, the Priority Code Point parameters, and the Service Access Priority parameters for each Port. The
40 operations that can be performed on this object are as follows:

- 41 a) Read Port Default Priority (12.6.2.1)
42 b) Set Port Default Priority (12.6.2.2)

- 1 c) Read Port Priority Regeneration Table (12.6.2.3)
- 2 d) Set Port Priority Regeneration Table (12.6.2.4)
- 3 e) Read Port Priority Code Point Selection (12.6.2.5)
- 4 f) Set Port Priority Code Point Selection (12.6.2.6)
- 5 g) Optionally, Read Port Priority Code Point Decoding Table (12.6.2.7)
- 6 h) Read Use_DEI parameter (12.6.2.11)
- 7 i) Optionally, Set Use_DEI parameter (12.6.2.12)

8 **12.19.3.2.1 Read Port Default Priority**

9 **12.19.3.2.1.1 Purpose**

10 To read the current state of the Default Priority parameter (IEEE Std 802.1ACI) for a specific TPMR Port.

11 **12.19.3.2.1.2 Inputs**

- 12 a) Port number.

13 **12.19.3.2.1.3 Outputs**

- 14 a) Default Priority value—Integer in range 0–7.

15 **12.19.3.2.2 Set Port Default Priority**

16 **12.19.3.2.2.1 Purpose**

17 To set the current state of the Default Priority parameter (IEEE Std 802.1AC) for a specific TPMR Port.

18 **12.19.3.2.2.2 Inputs**

- 19 a) Port number.
- 20 b) Default Priority value—Integer in range 0–7.

21 **12.19.3.2.2.3 Outputs**

22 None.

23 **12.19.3.2.3 Read Port Priority Regeneration Table**

24 **12.19.3.2.3.1 Purpose**

25 To read the current state of the Priority Regeneration Table parameter (6.9.4) for a specific TPMR Port.

26 **12.19.3.2.3.2 Inputs**

- 27 a) Port number.

28 **12.19.3.2.3.3 Outputs**

- 29 a) Regenerated Priority value for Received Priority 0—Integer in range 0–7.
- 30 b) Regenerated Priority value for Received Priority 1—Integer in range 0–7.
- 31 c) Regenerated Priority value for Received Priority 2—Integer in range 0–7.
- 32 d) Regenerated Priority value for Received Priority 3—Integer in range 0–7.
- 33 e) Regenerated Priority value for Received Priority 4—Integer in range 0–7.
- 34 f) Regenerated Priority value for Received Priority 5—Integer in range 0–7.
- 35 g) Regenerated Priority value for Received Priority 6—Integer in range 0–7.
- 36 h) Regenerated Priority value for Received Priority 7—Integer in range 0–7.

37 **12.19.3.2.4 Set Port Priority Regeneration Table**

38 **12.19.3.2.4.1 Purpose**

39 To set the current state of the Priority Regeneration Table parameter (6.9.4) for a specific TPMR Port.

1 **12.19.3.2.4.2 Inputs**

- 2 a) Port number;
- 3 b) Regenerated Priority value for Received Priority 0—Integer in range 0–7.
- 4 c) Regenerated Priority value for Received Priority 1—Integer in range 0–7.
- 5 d) Regenerated Priority value for Received Priority 2—Integer in range 0–7.
- 6 e) Regenerated Priority value for Received Priority 3—Integer in range 0–7.
- 7 f) Regenerated Priority value for Received Priority 4—Integer in range 0–7.
- 8 g) Regenerated Priority value for Received Priority 5—Integer in range 0–7.
- 9 h) Regenerated Priority value for Received Priority 6—Integer in range 0–7.
- 10 i) Regenerated Priority value for Received Priority 7—Integer in range 0–7.

11 **12.19.3.2.4.3 Outputs**

12 None.

13 **12.19.3.2.5 Read Port Priority Code Point Selection**

14 **12.19.3.2.5.1 Purpose**

15 To read which row of the Priority Code Point Decoding Table (6.9.3) is currently selected for use on this
16 Port.

17 **12.19.3.2.5.2 Inputs**

- 18 a) Port Number: the number of the TPMR Port.

19 **12.19.3.2.5.3 Outputs**

- 20 a) Priority Code Point Selection. This takes one of the following values:
 - 21 1) 8P0D
 - 22 2) 7P1D
 - 23 3) 6P2D
 - 24 4) 5P3D

25 **12.19.3.2.6 Set Port Priority Code Point Selection**

26 **12.19.3.2.6.1 Purpose**

27 To set which row of the Priority Code Point Decoding Table (6.9.3) will be selected for use on this Port.

28 **12.19.3.2.6.2 Inputs**

- 29 a) Port Number: the number of the TPMR Port.
- 30 b) Priority Code Point Selection. This takes one of the following values:
 - 31 1) 8P0D
 - 32 2) 7P1D
 - 33 3) 6P2D
 - 34 4) 5P3D

35 **12.19.3.2.6.3 Outputs**

36 None.

37 **12.19.3.2.7 Read Priority Code Point Decoding Table**

38 **12.19.3.2.7.1 Purpose**

39 To read the current contents of a row in the Priority Code Point Decoding Table (6.9.3) for a Port.

1 **12.19.3.2.7.2 Inputs**

- 2 a) Port Number: the number of the Bridge Port.
- 3 b) Priority Code Point Row. This takes one of the following values:
 - 4 1) 8P0D
 - 5 2) 7P1D
 - 6 3) 6P2D
 - 7 4) 5P3D

8 **12.19.3.2.7.3 Outputs**

- 9 a) Priority value for Priority Code Point 0: Integer in range 0–7.
- 10 b) Drop_eligible value for Priority Code Point 0: Boolean.
- 11 c) Priority value for Priority Code Point 1: Integer in range 0–7.
- 12 d) Drop_eligible value for Priority Code Point 1: Boolean.
- 13 e) Priority value for Priority Code Point 2: Integer in range 0–7.
- 14 f) Drop_eligible value for Priority Code Point 2: Boolean.
- 15 g) Priority value for Priority Code Point 3: Integer in range 0–7.
- 16 h) Drop_eligible value for Priority Code Point 3: Boolean.
- 17 i) Priority value for Priority Code Point 4: Integer in range 0–7.
- 18 j) Drop_eligible value for Priority Code Point 4: Boolean.
- 19 k) Priority value for Priority Code Point 5: Integer in range 0–7.
- 20 l) Drop_eligible value for Priority Code Point 5: Boolean.
- 21 m) Priority value for Priority Code Point 6: Integer in range 0–7.
- 22 n) Drop_eligible value for Priority Code Point 6: Boolean.
- 23 o) Priority value for Priority Code Point 7: Integer in range 0–7.
- 24 p) Drop_eligible value for Priority Code Point 7: Boolean.

25 **12.19.3.2.8 Read Use_DEI Parameter**

26 **12.19.3.2.8.1 Purpose**

27 To read the current state of the Use_DEI parameter (6.9.3) for the Port.

28 **12.19.3.2.8.2 Inputs**

- 29 a) Port Number: the number of the TPMR Port.

30 **12.19.3.2.8.3 Outputs**

- 31 a) Use_DEI parameter: Boolean.

32 **12.19.3.2.9 Set Use_DEI Parameter**

33 **12.19.3.2.9.1 Purpose**

34 To set the current state of the Use_DEI parameter (6.9.3) for the Port.

35 **12.19.3.2.9.2 Inputs**

- 36 a) Port Number: the number of the TPMR Port.
- 37 b) Use_DEI parameter: Boolean.

38 **12.19.3.2.9.3 Outputs**

39 None.

1 **12.19.3.3 Traffic Class Table**

2 The Traffic Class Table object models the operations that can be performed on, or inquire about, the current
3 contents of the Traffic Class Table (8.6.6) for a given Port. The operations that can be performed on this
4 object are Read Port Traffic Class Table and Set Port Traffic Class Table.

5 **12.19.3.3.1 Read Port Traffic Class Table**

6 **12.19.3.3.1.1 Purpose**

7 To read the contents of the Traffic Class Table (8.6.6) for a given Port.

8 **12.19.3.3.1.2 Inputs**

- 9 a) Port Number.

10 **12.19.3.3.1.3 Outputs**

- 11 a) The number of traffic classes, in the range 1 through 8, supported on the Port.
12 b) For each value of traffic class supported on the Port, the value of the traffic class in the range 0
13 through 7, and the set of priority values assigned to that traffic class.

14 **12.19.3.3.2 Set Port Traffic Class Table**

15 **12.19.3.3.2.1 Purpose**

16 To set the contents of the Traffic Class Table (8.6.6) for a given Port.

17 **12.19.3.3.2.2 Inputs**

- 18 a) Port number.
19 b) For each value of traffic class supported on the Port, the value of the traffic class in the range 0
20 through 7, and the set of priority values assigned to that traffic class.

21 NOTE—If a traffic class value greater than the largest traffic class available on the Port is specified, then the value
22 applied to the traffic class Table is the largest available traffic class.

23 **12.19.3.3.2.3 Outputs**

24 None.

25 **12.19.4 MAC Status Propagation Entity (MSPE)**

26 This managed resource comprises the following object:

- 27 a) The MAC status propagation configuration for each Port of the TPMR (12.19.4.1)

28 **12.19.4.1 MAC status propagation (MSP)**

29 The MAC status propagation object models the operations that modify, or inquire about, the configuration
30 and statistics of the MSPE for a particular Port of the TPMR.

31 The management operations that can be performed on the MAC status propagation configuration are as
32 follows:

- 33 a) Read MSP performance parameters (12.19.4.1.1)
34 b) Set MSP performance parameters (12.19.4.1.2)
35 c) Read MSP statistics (12.19.4.1.3)

1 **12.19.4.1.1 Read MSP performance parameters**

2 **12.19.4.1.1.1 Purpose**

3 To solicit information regarding the current configuration of the MSP performance parameters with respect
4 to a particular Port of the TPMR.

5 **12.19.4.1.1.2 Inputs**

6 a) Port number. The number of the Port of the TPMR. This can take the values 1 or 2 only.

7 **12.19.4.1.1.3 Outputs**

8 a) LinkNotify. The current value (Boolean) of LinkNotify (23.5.1) being used by the MSP state
9 machines.

10 b) LinkNotifyWait. The current value, in centiseconds, of LinkNotifyWait (23.5.2) being used by the
11 MSP state machines.

12 c) LinkNotifyRetry. The current value, in centiseconds, of LinkNotifyRetry (23.5.3) being used by the
13 MSP state machines.

14 d) MACNotify. The current value (Boolean) of MACNotify (23.5.4) being used by the MSP state
15 machines.

16 e) MACNotifyTime. The current value, in centiseconds, of MACNotifyTime (23.5.5) being used by
17 the MSP state machines.

18 f) MACRecoverTime. The current value, in centiseconds, of MACRecoverTime (23.5.6) being used
19 by the MSP state machines.

20 **12.19.4.1.2 Set MSP performance parameters**

21 **12.19.4.1.2.1 Purpose**

22 To change the MSP performance parameters to be used for a particular Port of the TPMR.

23 **12.19.4.1.2.2 Inputs**

24 a) Port number. The number of the Port of the TPMR. This can take the values 1 or 2 only.

25 b) LinkNotify. The desired value (Boolean) of LinkNotify (23.5.1) to be used by the MSP state
26 machines.

27 c) LinkNotifyWait. The desired value, in centiseconds, of LinkNotifyWait (23.5.2) to be used by the
28 MSP state machines.

29 d) LinkNotifyRetry. The desired value, in centiseconds, of LinkNotifyRetry (23.5.3) to be used by the
30 MSP state machines.

31 e) MACNotify. The desired value (Boolean) of MACNotify (23.5.4) to be used by the MSP state
32 machines.

33 f) MACNotifyTime. The desired value, in centiseconds, of MACNotifyTime (23.5.5) to be used by the
34 MSP state machines.

35 g) MACRecoverTime. The desired value, in centiseconds, of MACRecoverTime (23.5.6) to be used by
36 the MSP state machines.

37 **12.19.4.1.2.3 Outputs**

38 None.

39 **12.19.4.1.3 Read MSP statistics**

40 **12.19.4.1.3.1 Purpose**

41 To read the MSP statistics (23.12) for a particular Port of the TPMR.

1 12.19.4.1.3.2 Inputs

- 2 a) Port number. The number of the Port of the TPMR. This can take the values 1 or 2 only.

3 12.19.4.1.3.3 Outputs

- 4 a) `acksTransmitted`: The number of *acks* transmitted (23.6.15) by the Port's Transmit Process as a
5 consequence of `txAck` being set.
- 6 b) `addNotificationsTransmitted`: The number of *adds* transmitted (23.6.16) by the Port's Transmit
7 Process as a consequence of `txAdd` being set.
- 8 c) `addConfirmationsTransmitted`: The number of *add confirms* transmitted (23.6.17) by the Port's
9 Transmit Process as a consequence of `txAddConfirm` being set.
- 10 d) `lossNotificationsTransmitted`: The number of *losses* transmitted (23.6.18) by the Port's Transmit
11 Process as a consequence of `txLoss` being set.
- 12 e) `lossConfirmationsTransmitted`: The number of *loss confirms* transmitted (23.6.19) by the Port's
13 Transmit Process as a consequence of `txLossConfirm` being set.
- 14 f) `acksReceived`: The number of *acks* received (23.6.10) by the Port's Receive Process.
- 15 g) `addNotificationsReceived`: The number of *adds* received (23.6.11) by the Port's Receive Process.
- 16 h) `addConfirmationsReceived`: The number of *add confirms* received (23.6.12) by the Port's Receive
17 Process.
- 18 i) `lossNotificationsReceived`: The number of *losses* received (23.6.13) by the Port's Receive Process.
- 19 j) `lossConfirmationsReceived`: The number of *loss confirms* received (23.6.14) by the Port's Receive
20 Process.
- 21 k) `addEvents`: The number of transitions to `STM:ADD` directly from `STM:DOWN` or `STM:LOSS`
22 (23.8).
- 23 l) `lossEvents`: The number of transitions to `STM:LOSS` directly from `STM:UP` or `STM:ADD` (23.8).
- 24 m) `macStatusNotifications`: The number of transitions to `SNM:MAC_NOTIFICATION` (23.9).

25 12.20 Management entities for FQTSS

26 The Bridge enhancements for support of FQTSS are defined in Clause 34.

27 This managed resource comprises the following objects:

- 28 a) The Bandwidth Availability Parameter Table (12.20.1)
- 29 b) The Transmission Selection Algorithm Table (12.20.2)
- 30 c) The Priority Regeneration Override Table (12.20.3)
- 31 d) The SR Class to Priority Mapping Table (12.20.4)

32 12.20.1 The Bandwidth Availability Parameter Table

33 There is one Bandwidth Availability Parameter Table per Port of a Bridge Component. Each table row
34 contains a set of parameters for each traffic class configured for use with time-sensitive streams, as detailed
35 in Table 12-4. Rows in the table can be created or removed dynamically in implementations that support
36 dynamic configuration of Ports and components.

Table 12-4—Bandwidth Availability Parameter Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
trafficClass ^c	unsigned integer [0..7]	R	BE	3.280, 8.6.8
deltaBandwidth	percentage	RW	BE	34.3
adminIdleSlope	unsigned integer	RW	BEC	34.3
operIdleSlope	unsigned integer	R	BE	34.3
classMeasurementInterval	unsigned integer	RW	be	34.3, 34.4, 34.6
lockClassBandwidth	Boolean	RW	be	34.3

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of FQTSS; E = Required for end station support of FQTSS; b = Optional for Bridge or Bridge Component support of FQTSS; e = Optional for end station support of FQTSS; C = Required for Bridge or Bridge Component support of Stream reservation remote management,

^c The term “trafficClass” in this table is the same as the “traffic class” definition of this standard.

The classMeasurementInterval attribute provides management control of the classMeasurementInterval parameter specified in 34.3. The classMeasurementInterval attribute uses units of nanoseconds, converted to/from units of seconds for use in 34.3. If management of classMeasurementInterval is not supported, the default values (34.5) are used as the fixed Port configuration.

If management of lockClassBandwidth is not supported, the default value of false is used as the fixed Port configuration.

1 12.20.2 The Transmission Selection Algorithm Table

2 There is one Transmission Selection Algorithm Table per Port of a Bridge Component. Each table row
3 contains a set of parameters for each traffic class that the Port supports, as detailed in Table 12-5.

4 NOTE—To manage the mapping of traffic class to priority, refer to the Traffic Class Table managed object (12.6.3).

Table 12-5—Transmission Selection Algorithm Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
Traffic class	unsigned integer [0..7]	R	B	8.6.8
Transmission selection algorithm	enumerated (see Table 8-6)	RW	B	8.6.8, Table 8-6

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of FQTSS.

5 12.20.3 The Priority Regeneration Override Table

6 There is one Priority Regeneration Override Table per Port of a Bridge Component. This is in addition to the
7 Priority Handling managed object (12.6.2) that would also exist per Port of a Bridge Component. Each table
8 row contains a set of parameters for each received priority value that is associated with an SR class (3.271,
9 6.9.4), as detailed in Table 12-6.

Table 12-6—Priority Regeneration Override Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
Received priority	integer [0..7]	R	B	6.9.4
Regenerated priority	integer [0..7]	RW	B	6.9.4
SRPdomainBoundaryPort	boolean	R	B	35.1.4

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of FQTSS.

12.20.4 SR Class to Priority Mapping Table

There is one SR Class to Priority Mapping Table per Bridge Component. Each table row corresponds to an SR class, as detailed in Table 12-7. Rows in the table can be created or removed dynamically in implementations that support dynamic configuration of Ports and components, using Priority to specify the row to create/delete.

Table 12-7—SR Class to Priority Mapping Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
Priority	unsigned integer [0..7]	R	b	6.9.3
SRclassID	unsigned integer [0..6]	RW	b	35.2.2.9.2

^a R = Read only access; RW = Read/Write access.

^b b = Optional for Bridge or Bridge Component support of FQTSS.

The SRclassID attribute specifies the SR class ID from Table 35-6 of 35.2.2.9.2, mapped to the associated Priority (SRclassPriority of 35.2.2.9.3). The default values for this managed object use the default values specified in 34.5 (i.e., Priority 3 for SRclassID 6 and Priority 2 for SRclassID 5).

If this managed object is not supported, the default values specified in 34.5 are used as the fixed configuration.

NOTE 1—This managed object is not needed for an end station, since according to the requirements of 35.2.2.9.3, an end station uses the SR class to priority mapping that the neighboring Bridge provides in SRP's Domain attribute.

NOTE 2—To manage the mapping of traffic class to priority, refer to the Traffic Class Table managed object (12.6.3).

NOTE 3—The maximum number of rows supported for this table is provided in the maxSRclasses parameter of 12.22.1.

12.21 Congestion Notification managed objects

A number of the variables that implement congestion notification, including Congestion Points (CPs), Reaction Points (RPs), and Congestion Notification Domain (CND) defense, are manageable objects. There are a number of managed objects, each including a number of variables. The managed objects are as follows:

- CN component managed object (12.21.1)
- CN component priority managed object (12.21.2)
- CN Port priority managed object (12.21.3)
- Congestion Point managed object (12.21.4)

1 e) Reaction Point port priority managed object (12.21.5)

2 f) Reaction Point group managed object (12.21.6)

3 NOTE—If multiple managed objects are altered over a period of time, then between the time the first and last object has
4 been altered, operation of the state machines could produce unexpected results. This standard assumes that any number
5 of managed objects can be altered as an atomic operation, so that no inconsistent intermediate states can occur. See
6 17.7.13 for one mechanism to ensure consistency.

7 12.21.1 CN component managed object

8 A single instance of the CN component managed object shall be implemented by a Bridge Component or
9 end station that is congestion aware. It comprises all of the variables included in the CN component
10 variables (32.2), as illustrated in Table 12-8. An end station may omit the managed objects noted “C” in the
11 Conformance column of Table 12-8 if it does not support CPs, and may in any case omit the managed object
12 marked “e.”

Table 12-8—CN component managed object row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
cngMasterEnable	Boolean	RW	BE	32.2.1
cngCnmTransmitPriority	unsigned integer [0..7]	R	BC	32.2.2
cngDiscardedFrames	counter	R	BC	32.2.3
cngErroredPortList	list	R	Be	32.2.4

^a R = Read only access; RW = Read/Write access.

^b B = Required for a Bridge or Bridge Component that is congestion aware; C = Required for an end station that implements one or more CPs; E = Required for an end station that is congestion aware; e = Optional for an end station that is congestion aware.

13 12.21.2 CN component priority managed object

14 The CN component priority managed object contains the managed objects that control a single CNPV for all
15 Ports in an end station or Bridge Component. It comprises all of the variables included in the congestion
16 notification per-CNPV variables (32.3), as illustrated in Table 12-9. In one common use case for congestion
17 notification, every Port of a Bridge or end station has the same number of CPs, each configured for the same
18 set of priority values. This managed object facilitates configuring that case. These objects can override, and
19 can be overridden by, the CN Port priority managed objects (12.21.3) as described in 32.1.3.

20 A CN component priority managed object shall be implemented by a Bridge Component or end station that
21 is congestion aware for each priority value that can be a CNPV. An end station may omit the managed
22 objects noted “C” in the Conformance column in Table 12-9 if it does not support CPs. The operations that
23 can be performed on a congestion-aware system’s CN component priority managed object are as follows:

24 a) Create CN component priority managed object (12.21.2.1)

25 b) Delete CN component priority managed object (12.21.2.2)

Table 12-9—CN component priority managed object row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
cncpDefModeChoice	enum{cpcAdmin, cpcAuto}	RW	BE	32.3.1
cncpAlternatePriority	integer [0..7]	RW	BC	32.3.2
cncpAutoAltPri	integer [0..7]	R	BC	32.3.3
cncpAdminDefenseMode	enum{cptDisabled, cptInterior, cptInteriorReady, cptEdge}	RW	BE	32.3.4
cncpCreation	enum{cncpAutoEnable, cncpAutoDisable}	RW	BE	32.3.5
cncpLldpInstanceChoice	enum{cnlNone, cnlAdmin}	RW	BE	32.3.6
cncpLldpInstanceSelector	IEEE 802.1AB™ LLDP instance selector	RW	BE	32.3.7

^a R = Read only access; RW = Read/Write access.

^b B = Required for a Bridge or Bridge Component that is congestion aware; C = Required for an end station that implements one or more CPs; E = Required for an end station that is congestion aware.

12.21.2.1 Create CN component priority managed object

Creating a CN component priority managed object creates an instance of each of the congestion notification per-CNPV variables (32.3), and also creates the corresponding CN Port priority managed objects (12.21.3) and all their dependent managed objects and variables, on every Port in the Bridge Component or end station, as illustrated in CN Port priority managed object. Depending on the value of cncpCreation (32.3.5), creating a CN component priority managed object can make the selected priority a CNPV throughout the Bridge Component or end station.

12.21.2.2 Delete CN component priority managed object

Deleting a CN component priority managed object deletes all of the CN component variables (32.2), and also deletes the corresponding CN Port priority managed objects (12.21.3) and all their dependent managed objects and variables, on all Ports in the Bridge Component or end station, thus making the priority not a CNPV throughout the Bridge Component or end station.

12.21.3 CN Port priority managed object

There is one CN Port priority managed object per Port per priority in a congestion-aware end station or Bridge Component. It comprises some of the variables included in the CND defense per-Port per-CNPV variables (32.4), as illustrated in Table 12-10. These objects can override, and can be overridden by, the CN component managed objects (12.21.1) and CN component priority managed objects (12.21.2) as described in 32.1.3.

A CN Port priority managed object is created or deleted when the corresponding Port or CN component priority managed object is created or deleted, and its initial state on creation is determined by cncpCreation (32.3.5).

Table 12-10—CN Port priority managed object row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
cnpdDefModeChoice	enum{cpcAdmin, cpcAuto, cpcComp}	RW	BE	32.4.1
cnpdAdminDefenseMode	enum{cptDisabled, cptInterior, cptInteriorReady, cptEdge}	RW	BE	32.4.2
cnpdAutoDefenseMode	enum{cptDisabled, cptInterior, cptInteriorReady, cptEdge}	R	BE	32.4.3
cnpdLdpInstanceChoice	enum{cnlNone, cnlAdmin}	RW	BE	32.4.4
cnpdLdpInstanceSelector	IEEE 802.1AB LLDP instance selector	RW	BE	32.4.5
cnpdAlternatePriority	integer [0..7]	RW	BC	32.4.6

^a R = Read only access; RW = Read/Write access.

^b B = Required for a Bridge or Bridge Component that is congestion aware; C = Required for an end station that implements one or more CPs; E = Required for an end station that is congestion aware.

¹ The CN Port priority managed object shall be implemented by a Bridge Component or end station that is
² congestion aware. An end station may omit the managed objects noted “C” in the Conformance column in
³ Table 12-10 if it does not support CPs.

⁴ 12.21.4 Congestion Point managed object

⁵ There is one Congestion Point managed object for each CP in a Bridge Component or end station. It
⁶ comprises some of the variables included in the CP variables (32.8), as illustrated in Table 12-11.

Table 12-11—Congestion Point managed object row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
cpMacAddress	MAC address	R	BC	32.8.1
cpId	octet string (size = 8)	R	BC	32.8.2
cpQSp	unsigned integer	R	BC	32.8.3
cpW	real number	RW	BC	32.8.6
cpSampleBase	unsigned integer	RW	BC	32.8.11
cpDiscardedFrames	counter	R	BC	32.8.12
cpTransmittedFrames	counter	R	BC	32.8.13
cpTransmittedCnms	counter	R	BC	32.8.14
cpMinHeaderOctets	unsigned integer	RW	BC	32.8.15

^a R = Read only access; RW = Read/Write access.

^b B = Required for a Bridge or Bridge Component that is congestion aware; C = Required for an end station that implements one or more CPs.

¹ The Congestion Point managed object shall be implemented by a congestion-aware Bridge Component. It
² shall be implemented by an end station that supports CPs.

³ NOTE—The Recommended priority to traffic class mappings in Table 8-5 can assign more than one CNPV to the same
⁴ traffic class, the same queue, and hence the same CP. There can be only one CP controlling a given queue, and that CP
⁵ has one set of controlling managed objects, not one set per CNPV. That set of managed objects can be accessed using
⁶ any of the CNPV values assigned to the CP's queue. Thus, changing a managed object for one CNPV changes the
⁷ managed object for all CNPVs assigned to the same queue.

⁸ 12.21.5 Reaction Point port priority managed object

⁹ A congestion-aware end station shall implement one Reaction Point port priority managed object for each
¹⁰ CNPV on each port. The Reaction Point port priority managed object controls the creation of RPs on the
¹¹ port and CNPV, as illustrated in Table 12-12.

Table 12-12—Reaction Point port priority managed object row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
rpppMaxRps	unsigned integer	RW	E	32.10.1
rpppCreatedRps	counter	R	E	32.10.2
rpppRpCentiseconds	unsigned integer	R	E	32.10.3

^a R = Read only access; RW = Read/Write access.

^b E = Required for an end station that is congestion aware.

¹² 12.21.6 Reaction Point group managed object

¹³ There is one Reaction Point group managed object for each set of RP group variables (32.11), as illustrated
¹⁴ in Table 12-13. The Reaction Point group managed object shall be implemented by an end station that is
¹⁵ congestion aware.

Table 12-13—Reaction Point group managed object row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
rpgEnable	Boolean	RW	E	32.11.1
rpgTimeReset	unsigned integer	RW	E	32.11.2
rpgByteReset	unsigned integer	RW	E	32.11.3
rpgThreshold	unsigned integer	RW	E	32.11.4
rpgMaxRate	unsigned integer	RW	E	32.11.5
rpgAiRate	unsigned integer	RW	E	32.11.6
rpgHaiRate	unsigned integer	RW	E	32.11.7

Table 12-13—Reaction Point group managed object row elements (*continued*)

Name	Data type	Operations supported ^a	Conformance ^b	References
rpgGd	real number	RW	E	32.11.8
rpgMinDecFac	real number	RW	E	32.11.9
rpgMinRate	unsigned integer	RW	E	32.11.10

^a RW = Read/Write access.

^b E = Required for an end station that is congestion aware.

1 12.22 Stream Reservation Protocol (SRP) entities

2 The Bridge enhancements for support of SRP are defined in Clause 35.

3 This managed resource comprises the following objects:

- 4 a) SRP Bridge Base Table (12.22.1)
- 5 b) SRP Bridge Port Table (12.22.2)
- 6 c) SRP Latency Parameter Table (12.22.3)
- 7 d) SRP Stream Table (12.22.4)
- 8 e) SRP Reservations Table (12.22.5)

9 12.22.1 SRP Bridge Base Table

10 There is a set of parameters that configure SRP operation for the entire device. Those parameters are shown
11 in Table 12-14.

Table 12-14—SRP Bridge Base Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
msrpEnabledStatus	Boolean	RW	B	35.2.1.4(d)
talkerPruning	Boolean	RW	B	35.2.1.4(b)
msrpMaxFanInPorts	unsigned integer	RW	B	35.2.1.4(f)
msrpLatencyMaxFrameSize	unsigned integer	RW	B	35.2.1.4(g)
talkerVlanPruning	Boolean	RW	b	35.2.1.4 item l)
maxSRclasses	unsigned integer	R	b	35.2.1.4 item m)

^a R = Read only access; RW = Read/Write access.

^b B = required for Bridge or Bridge Component support of SRP; E = required for end station support of SRP; b = Optional for Bridge or Bridge Component support of SRP.

1 12.22.2 SRP Bridge Port Table

2 There is one SRP Configuration Parameter Table per Port of a Bridge Component. Each table row contains a
3 set of parameters for each MSRP entity per Port, as detailed in Table 12-15.

Table 12-15—SRP Bridge Port Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
msrpPortEnabledStatus	Boolean	RW	B	35.2.1.4(e)
MSRP Failed Registrations	counter	R	B	10.7.12.1
MSRP Last PDU Origin	MAC address	R	B	10.7.12.2
SR_PVID	unsigned integer[1..4094]	RW	B	Table 9-2 35.2.1.4(i)
talkerPruningPerPort	Boolean	RW	b	35.2.1.4 item k)

^a R = Read only access; RW = Read/Write access.

^b B = required for Bridge or Bridge Component support of SRP; E = required for end station support of SRP; b = Optional for Bridge or Bridge Component support of SRP.

4 12.22.3 SRP Latency Parameter Table

5 There is one SRP Latency Parameter Table per Port of a Bridge Component. Each table row contains a set of
6 parameters for each traffic class supported on a port, as detailed in Table 12-16. Rows in the table can be
7 created or removed dynamically in implementations that support dynamic configuration of ports and
8 components.

Table 12-16—SRP Latency Parameter Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
Traffic class	unsigned integer [0..7]	R	BE	34.3
portTcMaxLatency	unsigned integer	R	BE	35.2.1.4(a), 35.2.2.8.6

^a R = Read only access; RW = Read/Write access.

^b B = required for Bridge or Bridge Component support of SRP; E = required for end station support of SRP.

9 12.22.4 SRP Stream Table

10 There is one SRP Stream Table per Bridge Component. Each table contains a set of parameters for each
11 StreamID that is registered on the Bridge, as detailed in Table 12-17. Rows in the table are created and
12 removed dynamically as StreamIDs are registered and deregistered on the Bridge.

Table 12-17—SRP Stream Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamID	octet string(size(8))	RW	BE	35.2.2.8.2
Stream Destination Address	MAC address	R	BE	35.2.2.8.3(a)
Stream VID	unsigned integer [0..4094]	R	BE	35.2.2.8.3(b)
MaxFrameSize	unsigned integer [0..65 535]	R	BE	35.2.2.8.4(a)
MaxIntervalFrames	unsigned integer [0..65 535]	R	BE	35.2.2.8.4(b)
Data Frame Priority	unsigned integer [0..7]	R	BE	35.2.2.8.5(a)
Rank	unsigned integer [0..1]	R	BE	35.2.2.8.5(b)

^a R = Read only access; RW = Read/Write access.

^b B = required for Bridge or Bridge Component support of SRP; E = required for end station support of SRP.

1 12.22.5 SRP Reservations Table

2 There is one SRP Reservations Table per reservation direction per port of a Bridge Component. Each table
3 contains a set of parameters for each Talker or Listener Reservation that is registered on a port of the Bridge,
4 as detailed in Table 12-18. Rows in the table can be created or removed dynamically as Talker and Listener
5 declarations are registered and deregistered on a port of the Bridge.

6

Table 12-18—SRP Reservations Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamID	octet string(size(8))	RW	BE	35.2.2.8.2
Direction	unsigned integer[0..1]	R	BE	35.2.1.2
Declaration Type	unsigned integer [0..4]	R	BE	35.2.1.3
Accumulated Latency	unsigned integer	R	BE	35.2.2.8.6
Failed system ID	BridgeId	R	BE	35.2.2.8.7(a)
Failure Code	unsigned integer [0..16]	R	BE	35.2.2.8.7(b)
Dropped Frames	counter	R	BE	35.2.5.1
Stream Age	unsigned integer	R	BE	35.2.1.4(c)

^aR = Read only access; RW = Read/Write access.

^bB = required for Bridge or Bridge Component support of SRP; E = required for end station support of SRP.

1 12.22.6 SRP Stream Preload Table

2 There is one SRP Stream Preload Table per Bridge Component. Each table contains a set of parameters for
3 each StreamID that is preloaded on the Bridge as it initializes (i.e., powers up/reboots), as detailed in
4 Table 12-17. The preloading of Table 12-17 on each Port of the Bridge is specified in 12.22.5. Rows in the
5 table can be created or removed dynamically in implementations that support dynamic configuration of
6 Ports and components.

Table 12-19—SRP Stream Preload Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamID	octet string(size(8))	RW	b	35.2.2.8.2
Stream Destination Address	MAC Address	RW	b	35.2.2.8.3 item a)
Stream VLAN ID	unsigned integer [0..4094]	RW	b	35.2.2.8.3 item b)
MaxFrameSize	unsigned integer [0..65535]	RW	b	35.2.2.8.4 item a)
MaxIntervalFrames	unsigned integer [0..65535]	RW	b	35.2.2.8.4 item b)
Data Frame Priority	unsigned integer [0..7]	RW	b	35.2.2.8.4 item a)
Rank	unsigned integer [0..1]	RW	b	35.2.2.8.5 item b)

^a R = Read only access; RW = Read/Write access.

^b b = Optional for Bridge or Bridge Component support of SRP.

7 12.22.7 SRP Reservations Preload Table

8 There is one SRP Reservations Preload Table per reservation direction per Port of a Bridge Component.
9 Each table contains a set of parameters for each Talker or Listener registration that is preloaded on the
10 Bridge as it initializes (i.e., powers up/reboots), as detailed in Table 12-18. Rows in the table can be created
11 or removed dynamically in implementations that support dynamic configuration of Ports and
12 components.

Table 12-20—SRP Reservations Preload Table row elements

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamID	octet string(size(8))	RW	b	35.2.2.8.2
Direction	unsigned integer[0..1]	RW	b	35.2.1.2
Accumulated Latency	unsigned integer	RW	b	35.2.2.8.6

^aR = Read only access; RW = Read/Write access.

^bb = optional for Bridge or Bridge Component support of SRP.

13 The SRP preload tables (Table 12-17 and Table 12-18) are used to initialize Streams within each Bridge as it
14 powers up, to preload the Stream registrations that will later be provided by operation of SRP. When the
15 Bridge initializes, the SRP preload tables shall be used for a two phase operation as follows:

1 In the first phase, the Bridge iterates Table 12-18 to find rows for which the Direction is Talker. For each
2 Talker row, if the StreamID in Table 12-18 is also found in Table 12-17, the Bridge shall:

- 3 a) Invoke the MVRP ES_REGISTER_VLAN_MEMBER service primitive (11.2.3.2.1) on the Port.
4 The attribute_type parameter carries the VID Vector Attribute Type (11.2.3.1.6). The attribute_value
5 parameter carries the Stream VLAN ID for the associated StreamID in Table 12-17.
- 6 b) Invoke MSRP REGISTER_STREAM.request (35.2.3.1.1) on the Port. The attribute_type parameter
7 carries the Talker Advertise Vector Attribute Type [item a) in 35.2.2.4]. The Declaration Type in the
8 attribute_value parameter is Talker Advertise. The attribute_value parameter carries the following
9 parameters from Table 12-17: StreamID, DataFrameParameters.destination_address from Stream
10 Destination Address, DataFrameParameters.vlan_identifier from Stream VLAN ID,
11 TSpec.MaxFrameSize, TSpec.MaxIntervalFrames, PriorityAndRank.Data Frame Priority, and
12 PriorityAndRank.Rank. The attribute_value parameter carries the following per-Port parameters
13 from Table 12-18: Accumulated Latency. The Failure Information in the attribute_value parameter
14 is not applicable to Talker Advertise.

15 In the second phase, the Bridge iterates Table 12-18 to find rows for which the Direction is Listener. For
16 each Listener row, if the StreamID was processed as a Talker during the first phase, then the Bridge shall:

- 17 c) Invoke MSRP REGISTER_ATTACH.request (35.2.3.1.5) on the Port. The attribute_type parameter
18 carries the Listener Vector Attribute Type [item c) in 35.2.2.4]. The Declaration Type in the
19 attribute_value parameter is Listener Ready. The attribute_value parameter carries StreamID from
20 Table 12-18.
- 21 d) Invoke the MVRP ES_REGISTER_VLAN_MEMBER service primitive (11.2.3.2.1) on the Port.
22 The attribute_type parameter carries the VID Vector Attribute Type (11.2.3.1.6). The attribute_value
23 parameter carries the Stream VLAN ID for the associated StreamID in Table 12-17.
- 24 e) Update the Dynamic Reservation Entries (8.8.7) for the Port to Forwarding, using the Stream
25 Destination Address for the associated StreamID in Table 12-17.

26 For traffic class N that is mapped to the Data Frame Priority of Table 12-17, increase the *operIdleSlope(N)*
27 variable [item d) in 34.3] using the bandwidth requirements of the Stream (MaxFrameSize and
28 MaxIntervalFrames of Table 12-17).

29 12.23 Priority-based Flow Control objects

30 The following Priority-based Flow Control objects exist for each port that support PFC:

- 31 a) **PFCLinkDelayAllowance:** the allowance made for round-trip propagation delay of the link in bits
- 32 b) **PFCRequests:** a count of the invoked PFC M_CONTROL.request primitives
- 33 c) **PFCIndications:** a count of the received PFC M_CONTROL.indication primitives

Table 12-21 shows the format and applicability of these objects.

Table 12-21—Priority-based Flow Control objects

Name	Data type	Operations supported ^a	Conformance ^b
PFCLinkDelayAllowance	unsigned integer	RW	BE
PFCRequests	unsigned integer	R	BE
PFCIndications	unsigned integer	R	BE

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of PFC; E = Required for end station support of PFC.

NOTE—The PFC Initiator (see 36.2.1) can use the PFCLinkDelayAllowance parameter as one of the factors to determine when to issue a PFC M_CONTROL.request in order to not discard frames. The parameter can be written to adjust to different link characteristics that affect the link delay (e.g., link length or link technology). See Annex N for an example of how to compute this parameter.

12.24 1:1 PBB-TE IPS managed objects

The IPS managed objects model operations that create, modify, delete, or inquire about the configuration and the operation of IPS (26.11). To this end, they describe objects related to an Infrastructure Protection Group (IPG) (26.11.2.1). The following are the IPS managed objects associated with a Segment Endpoint Bridge (SEB):

- a) TE protection group list managed object (12.18.1)
- b) IPG managed object (12.24.2)

12.24.1 IPG list managed object

There is one TE protection group list managed object per SEB. The IPG list managed object contains a list of the IPGs that have been configured on the SEB.

The management operations that can be performed on the TE protection group list managed object are as follows:

- a) Read Maintenance Domain list (12.14.1.1)
- b) Create Maintenance Domain managed object (12.14.1.2)
- c) Delete Maintenance Association managed object (12.14.5.4)

12.24.1.1 Read IPG list

12.24.1.1.1 Purpose

To obtain information about the set of IPGs associated with a SEB.

12.24.1.1.2 Inputs

None.

12.24.1.1.3 Outputs

A list, perhaps empty, of the IPG managed objects configured on the SEB. For each item in the list, the Read Maintenance Domain list command returns the following:

- a) A value identifying the IPG.
- b) A reference to the particular Maintenance Association managed object (12.14.6) identifying the Infrastructure Segment MA that corresponds to the working entity of the IPG.
- c) A reference to the particular Maintenance Association managed object identifying the Infrastructure Segment MA that corresponds to the protection entity of the IPG.
- d) In the case of M:1 IPS, a prioritized list of references to Maintenance Association managed object, each of which identifies a candidate Protection Segment associated with the IPG.

12.24.1.2 Create IPG managed object

12.24.1.2.1 Purpose

To create a new IPG managed object in a SEB and add it to the TE protection group list managed object associated with the SEB.

12.24.1.2.2 Inputs

- a) A reference to a particular Maintenance Association managed object identifying the Infrastructure Segment MA that corresponds to the working entity of the IPG.
- b) In the case of 1:1 IPS, a reference to a particular Maintenance Association managed object identifying the Infrastructure Segment MA that corresponds to the protection entity of the IPG or, in the case of M:1 IPS, a prioritized list of references to Maintenance Association managed object, each of which identifies an Infrastructure Segment MA associated with an Infrastructure Segment that is a candidate Protection Segment for the IPG.

12.24.1.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected due to nonexistent MD.
 - 2) Operation rejected due to a nonexistent MA.
 - 3) Operation rejected because a referenced MA is not associated with an Infrastructure Segment.
 - 4) Operation rejected because the referenced MAs do not belong to the same MD.
 - 5) Operation rejected because two or more managed objects referenced are associated with the same MA.
 - 6) Operation rejected because neither Segment Monitoring Path Identifier (SMP-ID) associated with a SEG-ID identifying one of the referenced Infrastructure MAs contains in its SMP-SA field the MAC address of a PNP associated with the SEB.
 - 7) Operation accepted.

12.24.1.3 Delete IPG managed object

12.24.1.3.1 Purpose

To remove a specific IPG managed object from the TE protection group list managed object associated with the SEB and to delete that IPG managed object.

1 12.24.1.3.2 Inputs

- 2 a) A reference to a particular IPG managed object (12.24.2).

3 12.24.1.3.3 Outputs

- 4 a) Operation status. This takes one of the following values:
- 5 1) Operation rejected due to nonexistent IPG
- 6 2) Operation rejected because the IPG is enabled
- 7 3) Operation accepted

8 12.24.2 IPG managed object

9 There can be any number of IPG managed objects per SEB.

10 The management operations that can be performed on the IPG managed object are as follows:

- 11 a) Read Maintenance Association managed object (12.14.6.1)
- 12 b) Write Maintenance Association managed object (12.14.6.2)
- 13 c) TE protection group administrative commands managed object (12.18.2.3)

14 12.24.2.1 Read IPG managed object

15 12.24.2.1.1 Purpose

16 To obtain information from a SEB regarding a specified IPG managed object.

17 12.24.2.1.2 Inputs

- 18 a) A reference to an IPG managed object.

19 12.24.2.1.3 Outputs

- 20 a) Operation status. This takes one of the following values:
- 21 1) Operation rejected due to nonexistent IPG
- 22 2) Operation accepted
- 23 b) The MAID and Port Number associated with the Working Segment and an indication (TRUE or
- 24 FALSE) about whether the Working Segment is operational.
- 25 c) The MAID and Port Number associated with the Protection Segment and an indication (TRUE or
- 26 FALSE) about whether the Protection Segment is operational.
- 27 d) (optional) In the case of M:1 IPS, the list of MAIDs and Port Numbers associated with Candidate
- 28 Protection Segments ordered from highest to lowest priority and an indication (TRUE or FALSE)
- 29 about whether each candidate Protection Segment is operational.
- 30 e) (writable) A list of TE-SIDs, possibly NULL, identifying TESIs protected by the IPG. If not NULL,
- 31 the IPG is considered to be enabled.
- 32 f) An enumerated value indicating the operational state of the Service Mapping state machine
- 33 (Figure 26-22) for each IPG:
- 34 1) WORKING_SEGMENT
- 35 2) PROTECTION_SEGMENT
- 36 3) WTR
- 37 4) PROT_ADMIN
- 38 g) An enumerated value indicating the status of active requests within the IPG:
- 39 1) **NoRequest:** No administrative command is in effect.
- 40 2) **LoP:** Set if LoP (26.10.3.3.4) is TRUE, indicating that an administrative command to prohibit
- 41 the use of the Protection Segment is in effect.

- 1 3) **FS:** Set if FS (26.10.3.3.5) is TRUE, indicating that an administrative command to perform
2 forced switching to the Protection Segment is in effect.
- 3 4) **p.SFH:** Set if SFH (26.10.3.3.3) is TRUE on the Protection Segment.
- 4 5) **w.SFH:** Set if SFH is TRUE on the Working Segment.
- 5 6) **MStoProtection:** Set if MStoProtection (26.10.3.3.6) is TRUE, indicating that an
6 administrative command to perform manual switching to the Protection Segment is in effect.
- 7 7) **MStoWorking:** Set if MStoWorking (26.10.3.3.7) is TRUE, indicating that an administrative
8 command to perform manual switching to the Working Segment is in effect.
- 9 h) (writable) (optional) The wait-to-restore (WTR) period (26.10.3.3.8.) In revertive operation it may
10 be configured in steps of 1 min between 5 and 12 min; the default value is 5 min. The value 0
11 indicates nonrevertive operation. The value 0 is not permitted when the IPG is associated with M:1
12 IPS.
- 13 i) (writable) (optional) The hold-off period (26.10.3.3.9). The range of the hold-off period is 0 to 10 s
14 in steps of 100 ms; the default value is 0.
- 15 j) (optional) An enumerated value indicating the operational state of the Protection Segment Selection
16 state machine (Figure 26-26) for each IPG:
 - 17 1) PS_ASSIGNED
 - 18 2) SEGMENT_OK
 - 19 3) SEGMENT_FAILED
 - 20 4) ASSIGN_NEW_PS
 - 21 5) REVERT_TO_BETTER_PS
- 22 k) (writable) (optional) The M:1 wait-to-restore (MWTR) period (26.11.5.4.7.). In revertive operation
23 it may be configured in steps of 1 min between 5 and 12 min; the default value is 5 min. The value 0
24 indicates nonrevertive operation.

25 **12.24.2.2 Write IPG managed object**

26 **12.24.2.2.1 Purpose**

27 To update a writable item associated with a specified IPG managed object.

28 **12.24.2.2.2 Inputs**

- 29 a) A reference to an IPG managed object.
- 30 b) The identity of a writable item in 12.18.2.1.3 that is to be updated.
- 31 c) The updated value of the specified item.

32 **12.24.2.2.3 Outputs**

- 33 a) Operation status. This takes one of the following values:
 - 34 1) Operation rejected due to specification of nonexistent IPG
 - 35 2) Operation rejected due to the selection of an item that is not writable
 - 36 3) Operation rejected due to lack of authority to set the value of this item
 - 37 4) Operation rejected due to the specification of an invalid value for the selected item
 - 38 5) Operation accepted

39 **12.24.2.3 Apply administrative command to IPG managed object**

40 **12.24.2.3.1 Purpose**

41 To apply a specified administrative command to an IPG managed object.

1 12.24.2.3.2 Inputs

- 2 a) A reference to an IPG managed object.
- 3 b) An enumerated value indicating the exercised administrative command:
 - 4 1) **Clear:** An indication to clear all other administrative commands.
 - 5 2) **Lockout of Protection:** An administrative command to prohibit the use of the Protection
 - 6 Segment.
 - 7 3) **Forced Switch:** An administrative command to perform forced switching to the Protection
 - 8 Segment.
 - 9 4) **Manual Switch To Protection:** An administrative command to perform manual switching to
 - 10 the Protection Segment if that Infrastructure Segment is operational.
 - 11 5) **Manual Switch To Working:** An administrative command to perform manual switching to the
 - 12 Working Segment if that Infrastructure Segment is operational.

13 12.24.2.3.3 Outputs

- 14 a) Operation status. This takes one of the following values:
 - 15 1) Operation rejected due to specification of a nonexistent IPG.
 - 16 2) Operation rejected due to lack of authority to apply the specified command to the identified
 - 17 IPG managed object.
 - 18 3) Operation rejected due to an active higher priority request.
 - 19 4) Operation accepted.

20 12.25 Shortest Path Bridging managed objects

21 The Shortest Path Bridging managed objects model operations that create, modify, delete, or inquire about
22 the configuration and the operation of an SPT Bridge and the region in which it is operating. These managed
23 objects model only behaviors unique to SPB. Figure 12-3 illustrates the hierarchical relationships among the
24 various SPB managed objects.

25 IS-IS provides a logical topology instance and within each instance the SPT Domain defines a set of
26 managed objects. The following are the managed objects in an SPT Bridge:

- 27 a) The SPB System managed object (12.25.1)
- 28 b) The SPB MTID Static managed object (12.25.2) (see Multi-Topology Identifier (MTID) in
- 29 Clause 3)
- 30 c) The SPB Topology Instance Dynamic managed object (12.25.3)
- 31 d) The SPB ECT Static Entry managed object (12.25.4)
- 32 e) The SPB ECT Dynamic Entry managed object (12.25.5)
- 33 f) The SPB Adjacency Static Entry managed object (12.25.6)
- 34 g) The SPB Adjacency Dynamic Entry managed object (12.25.7)
- 35 h) The SPBM BSI Static Entry managed object (12.25.8)
- 36 i) The SPB Topology Node Table managed object (12.25.9)
- 37 j) The SPB Topology ECT Table managed object (12.25.10)
- 38 k) The SPB Topology Edge Table managed object (12.25.11)
- 39 l) The SPBM Topology Service Table managed object (12.25.12)
- 40 m) The SPBV Topology Service Table managed object (12.25.13)
- 41 n) The ECMP ECT Static Entry managed object (12.25.14)
- 42 o) The PCR ECT Static Entry managed object (12.28.1)
- 43 p) The PCR Topology ECT Table managed object (12.28.2)

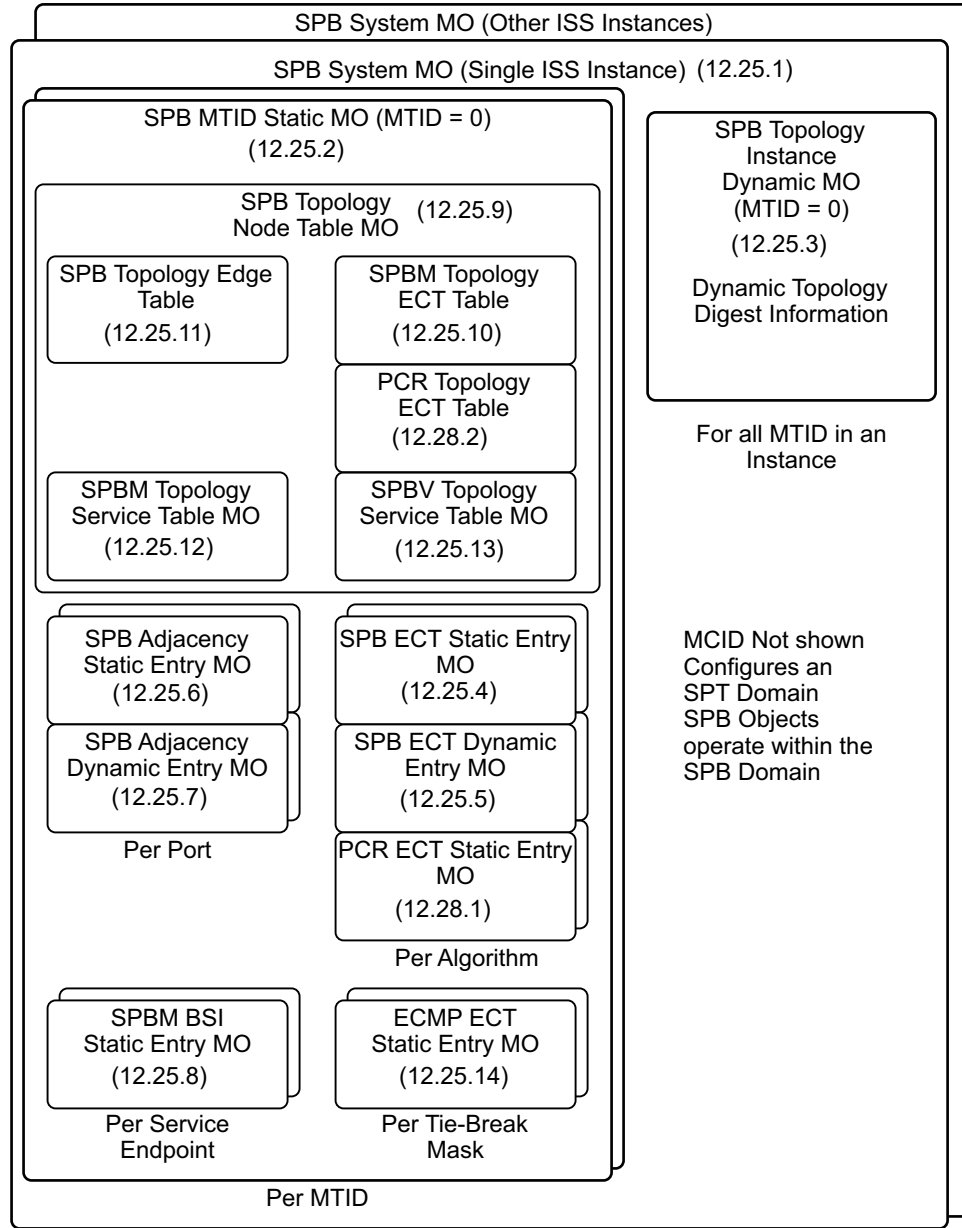


Figure 12-3—SPB managed objects (MOs)

12.25.1 The SPB System managed object

There is one SPB System managed object per ISIS-SPB area supported by an SPT Bridge. A single area only is supported in this version of the specification. It contains the fundamental SPB configuration parameters for the Bridge. It is persistent over reboot. The management operations that can be performed on the SPB System managed object are as follows:

- Create SPB System managed object (12.25.1.1)
- Write SPB System managed object (12.25.1.2)
- Read SPB System managed object (12.25.1.3)
- Delete SPB System managed object (12.25.1.4)

1 12.25.1.1 Create SPB System Managed object

2 12.25.1.1.1 Purpose

3 To create and configure the SPB System managed object on a Bridge.

4 12.25.1.1.2 Inputs

- 5 a) The part of the IS-IS Area Address that is domain specific (see Figure 4, ISO/IEC 10589:2002). This
6 shall be set to zero in this version of this specification whenever SPB is the only protocol being
7 supported by this IS-IS instance.
- 8 b) The MAC address to be used as the SPB System Identifier by all instances of ISIS-SPB on this
9 Bridge. The System ID defaults to the Bridge Address if this input is not provided. The Bridge
10 Address is the value returned by item a) of Discover Bridge (12.4.1.1.3) and Read Bridge
11 (12.4.1.2.3).
- 12 c) The Group address used to address this ISIS-SPB instance (see Table 8-16). This address is
13 configured to manage the transparency to ISIS-SPB of neighboring network elements, and also to
14 specify the IS-IS Level at which this SPB instance is operating, which shall be Level 1 in this
15 version of this specification.

16 12.25.1.1.3 Outputs

- 17 a) Operation status. This takes one of the following values:
 - 18 1) Operation rejected because the specified IS-IS Area Address is already in use.
 - 19 2) Operation rejected because the provided Group address does not correspond to an allowed
20 address (Table 8-16).
 - 21 3) Operation rejected because the specified SPB System Identifier value is not valid.
 - 22 4) Operation rejected due to the Bridge's inability to support the requested IS-IS Level.
 - 23 5) Operation accepted.

24 12.25.1.2 Write SPB System managed object

25 12.25.1.2.1 Purpose

26 To configure an SPB System managed object on a Bridge.

27 12.25.1.2.2 Inputs

- 28 a) The part of the IS-IS Area Address that is domain specific (see Figure 4, ISO/IEC 10589:2002). This
29 shall be zero in this version of this specification.
- 30 b) The SPB System Shortest Path Source Identifier (SPSourceID) (27.10).
- 31 c) If SPBV mode is used, whether the SPVIDs associated with this Bridge are assigned, or are to be
32 auto-allocated. The use of SPBV mode is inferred from the assignment of at least one VID to the
33 reserved MSTID value 0xFFD (27.4) in the FID to MSTID Allocation Managed Object (12.12.2.2).
- 34 d) If SPBM mode is used, whether the SPsourceID associated with this Bridge is assigned, or is to be
35 auto-allocated. The use of SPBM mode is inferred from the assignment of at least one VID to the
36 reserved MSTID value 0xFFC (27.4) in the FID to MSTID Allocation Managed Object (12.12.2.2).
- 37 e) The agreement digest Convention identifier (28.4.3), specifying the agreement rules being used. It
38 can take the values 1 to 3 as specified in 28.4.3.

39 12.25.1.2.3 Outputs

- 40 a) Operation status. This takes one of the following values:
 - 41 1) Operation rejected due to the Bridge's inability to support the requested SPB capability
 - 42 2) Operation accepted

12.25.1.3 Read SPB System managed object

12.25.1.3.1 Purpose

To obtain information about SPB System managed object on a Bridge.

12.25.1.3.2 Inputs

- a) The part of the IS-IS Area Address that is domain specific (see Figure 4, ISO/IEC 10589:2002).

12.25.1.3.3 Outputs

- a) The part of the IS-IS Area Address that is domain specific (see Figure 4, ISO/IEC 10589:2002).
- b) The SPB System Identifier used by all instances of ISIS-SPB on this Bridge.
- c) The Group address used to address this ISIS-SPB instance (Table 8-16). This address also identifies the IS-IS Level at which this SPB instance is operating.
- d) The SPB System Name used by all instances of ISIS-SPB. This takes the value configured by Set Bridge Name (12.4.1.3) and which can also be read as item b) of Read Bridge (12.4.1.2).
- e) The Bridge Priority (13.26.2). The value used by SPB is set in the CIST Bridge Protocol Parameters (12.8.1.3).
- f) The SPB System SPSourceID (27.10).
- g) The SPBV Mode, identifying whether, if SPBV mode is used, the SPVIDs associated with this Bridge are assigned, or are to be auto-allocated.
- h) The SPBM Mode, identifying whether, if SPBM mode is used, the SPsourceID associated with this Bridge is assigned, or is to be auto-allocated.
- i) The agreement digest Convention identifier (28.4.3), specifying the agreement rules being used.

12.25.1.4 Delete SPB System managed object

12.25.1.4.1 Purpose

To delete the SPB System managed object on a Bridge, so disabling it as an SPT Bridge.

12.25.1.4.2 Inputs

- a) The part of the IS-IS Area Address that is domain specific (see Figure 4, ISO/IEC 10589:2002).

12.25.1.5 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because one or more SPB MTID Static managed objects still exist and have not been deleted for this SPB instance.
 - 2) Operation accepted.

12.25.2 The SPB MTID Static managed object

There is one SPB MTID Static managed object per IS-IS Multi-Topology Identifier (MTID) used for SPB operation. It contains the SPB MTID value and the ISIS-SPB Overload flag for this MTID. It is persistent over reboot. The management operations that can be performed on the SPB MTID Static managed object are as follows:

- a) Create SPB MTID Static managed object (12.25.2.1)
- b) Write SPB MTID Static managed object (12.25.2.2)
- c) Read SPB MTID Static managed object (12.25.2.3)
- d) Delete SPB MTID Static managed object (12.25.2.4)

1 **12.25.2.1 Create SPB MTID Static managed object**

2 **12.25.2.1.1 Purpose**

3 To create and configure an SPB MTID Static managed object on a Bridge.

4 **12.25.2.1.2 Inputs**

- 5 a) An MTID to support IS-IS MT operation.

6 **12.25.2.1.3 Outputs**

- 7 a) Operation status. This takes one of the following values:
8 1) Operation rejected because the MTID is already in use
9 2) Operation accepted
10 b) A Topology Index value that uniquely identifies this MTID Static managed object.

11 **12.25.2.2 Write SPB MTID Static managed object**

12 **12.25.2.2.1 Purpose**

13 To configure an SPB MTID Static managed object.

14 **12.25.2.2.2 Inputs**

- 15 a) An MTID to support IS-IS MT operation.
16 b) The Overload flag value, which has ISIS “no transit” semantics, applied only in this SPB topology.

17 **12.25.2.2.3 Outputs**

- 18 a) The MTID value.
19 b) The Overload flag value, which has ISIS “no transit” semantics, but applied only in this SPB
20 topology.

21 **12.25.2.3 Read SPB MTID Static managed object**

22 **12.25.2.3.1 Purpose**

23 To obtain information about an SPB MTID Static managed object.

24 **12.25.2.3.2 Inputs**

- 25 a) An MTID to support IS-IS MT operation.

26 **12.25.2.3.3 Outputs**

- 27 a) The MTID value.
28 b) The Overload flag value, which has ISIS “no transit” semantics, but applied only in this SPB
29 topology.
30 c) A Topology Index value that uniquely identifies this MTID Static managed object.

31 **12.25.2.4 Delete SPB MTID Static managed object**

32 **12.25.2.4.1 Purpose**

33 To delete an SPB MTID Static managed object.

1 **12.25.2.4.2 Inputs**

- 2 a) An MTID to support IS-IS MT operation.

3 **12.25.2.4.3 Outputs**

- 4 a) Operation status. This takes one of the following values:
- 5 1) Operation rejected because not all peer links in the SPB Adjacency Static Entry managed object
- 6 for this MTID have an Administrative State of Disabled.
- 7 2) Operation accepted.

8 **12.25.3 The SPB Topology Instance Dynamic managed object**

9 There is one SPB Topology Instance Dynamic managed object for each SPB MTID Static managed object. It
10 contains the dynamic parameters computed for this Bridge as a consequence of its participation in SPB in
11 the ISIS-SPB topology defined by the specified MTID within this instance of SPB. This object is
12 automatically created as a consequence of the creation of the SPB MTID Static managed object associated
13 with the topology. It is deleted with the deletion of the SPB MTID Static managed object associated with the
14 topology. The management operation that can be performed on the SPB Topology Instance Dynamic
15 managed object is as follows:

- 16 a) Read SPB Topology Instance Dynamic managed object

17 **12.25.3.1 Read SPB Topology Instance Dynamic managed object**

18 **12.25.3.1.1 Purpose**

19 To obtain information about the dynamic parameters computed by this Bridge as a consequence of its
20 participation in SPB in the ISIS-SPB topology defined by the specified MTID within this instance of SPB.

21 **12.25.3.1.2 Inputs**

- 22 a) A Topology Index value that uniquely identifies a topology instance.

23 **12.25.3.1.3 Outputs**

- 24 a) The Topology Index value that uniquely identifies this topology instance.
- 25 b) The value of the topology Agreement Digest (13.17.1) being advertised by this Bridge to its peers in
- 26 this ISIS-SPB Topology. This includes all parameters defined in the Agreement Digest (28.4).
- 27 c) The value of the MCID (8.9, 13.5, 13.8) being advertised by this Bridge to its peers.
- 28 d) The value of the Auxiliary MCID (27.4.1, 28.12.2) being advertised by this Bridge to its peers.

29 **12.25.4 The SPB ECT Static Entry managed object**

30 There is one SPB ECT Static Entry managed object per Base VID allocated to SPB operation. A Base VID
31 exists in the context of a single ISIS-SPB MTID. The static Entry contains the SPB configuration parameters
32 for the forwarding plane associated with the Base VID. It is persistent over reboot. The management
33 operations that can be performed on the ECT Static Entry managed object are as follows:

- 34 a) Create an SPB ECT Static Entry managed object (12.25.4.1)
- 35 b) Read an SPB ECT Static Entry managed object (12.25.4.2)
- 36 c) Delete an SPB ECT Static Entry managed object (12.25.4.3)

1 12.25.4.1 Create SPB ECT Static Entry managed object

2 12.25.4.1.1 Purpose

3 To create and configure and interrogate an SPB ECT Static Entry managed object on a Bridge.

4 12.25.4.1.2 Inputs

- 5 a) A Topology Index value that uniquely identifies a topology instance.
- 6 b) A Base VID. This VID must have been assigned either to the reserved MSTID value 0xFFC for SPT
7 Bridges in SPBM operation, or to the reserved MSTID value 0xFFD for SPBV (27.4) in the FID to
8 MSTI Allocation table (12.12.2.2).
- 9 c) The identifier of the ECT Algorithm (ECT-ALGORITHM) to be used for this Base VID
10 (Table 28-1, Table 44-1, Table 45-1). The Default is the LowPATHID 00-80-C2-01 (28.8).
- 11 d) The pre-configured value of the SPVID assigned to this Bridge if operating in SPBV mode. This
12 input is ignored if auto-allocated is selected [item c) in 12.25.1.2.2]. This VID value must have been
13 assigned to the SPVID-Pool-MSTID (27.4) in the FID to MSTI Allocation Managed Object
14 (12.12.2.2).

15 12.25.4.1.3 Outputs

- 16 a) Operation status. This takes one of the following values:
 - 17 1) Operation rejected because the Topology Index value is not defined
 - 18 2) Operation rejected because the Base VID is already in use
 - 19 3) Operation rejected due to requested Base VID not assigned to specified SPB use
 - 20 4) Operation rejected due to requested SPVID not assigned to the SPBV pool
 - 21 5) Operation accepted

22 12.25.4.2 Read SPB ECT Static Entry managed object

23 12.25.4.2.1 Purpose

24 To obtain information about the SPB ECT Static Entry managed object for a Base VID.

25 12.25.4.2.2 Inputs

- 26 a) A Topology Index value that uniquely identifies a topology instance.
- 27 b) A Base VID. The value 4095 is a wildcard indicating that information, from any Base VID assigned
28 to SPB operation, is requested.

29 12.25.4.2.3 Outputs

- 30 a) Operation status. This takes one of the following values:
 - 31 1) Operation rejected because the Base VID is not assigned to SPB use
 - 32 2) Operation accepted
- 33 b) A Topology Index value that uniquely identifies this topology instance.
- 34 c) The Base VID.
- 35 d) The identifier of the ECT algorithm (ECT-ALGORITHM) to be used for this Base VID.
- 36 e) The value of the SPVID to be used by this Bridge for this Base VID if operating in SPBV mode.

1 **12.25.4.3 Delete SPB ECT Static Entry managed object**

2 **12.25.4.3.1 Purpose**

3 To delete the SPB ECT Static Entry managed object for a Base VID.

4 **12.25.4.3.2 Inputs**

- 5 a) A Topology Index value that uniquely identifies a topology instance.
- 6 b) A Base VID.

7 **12.25.4.3.3 Outputs**

- 8 a) Operation status. This takes one of the following values:
 - 9 1) Operation rejected because the Bridge has services currently assigned to this Base VID
 - 10 2) Operation rejected because the SPT Region has services currently using this Base VID
 - 11 3) Operation accepted

12 **12.25.5 The SPB ECT Dynamic Entry managed object**

13 There is one SPB ECT Dynamic Entry managed object per Base VID allocated to SPB operation. It contains
14 the SPB dynamic parameters associated with the Base VID. This object is automatically created as a
15 consequence of the creation of the SPB ECT Static Entry managed object. It is deleted with the deletion of
16 the SPB ECT Static Entry managed object. The management operation that can be performed on the ECT
17 Dynamic Entry managed object is as follows:

- 18 a) Read SPB ECT Dynamic Entry managed object

19 **12.25.5.1 Read SPB ECT Dynamic Entry managed object**

20 **12.25.5.1.1 Purpose**

21 To obtain information about the SPB ECT Dynamic Entry managed object for a Base VID.

22 **12.25.5.1.2 Inputs**

- 23 a) A Topology Index value that uniquely identifies a topology instance.
- 24 b) A Base VID. The value 4095 is a wildcard indicating that information, from any Base VID assigned
25 to SPB operation, is requested.

26 **12.25.5.1.3 Outputs**

- 27 a) The Topology Index value that uniquely identifies this topology instance.
- 28 b) The Base VID.
- 29 c) A flag identifying whether the operational mode of this Base VID is SPBV or SPBM.
- 30 d) A flag indicating that this Bridge has services currently assigned to this Base VID.
- 31 e) A flag indicating that the SPT Region has services currently using this Base VID.
- 32 f) A count of the number of ingress check failures (an error condition) on this Base VID since creation
33 of this SPB ECT Dynamic Entry managed object.

12.25.6 The SPB Adjacency Static Entry managed object

There is one SPB Adjacency Static Entry managed object per port per ISIS-SPB topology. It contains the SPB configuration parameters for the port in a topology. It is persistent over reboot. The management operations that can be performed on the SPB Adjacency Static Entry managed object are as follows:

- a) Write SPB Adjacency Static Entry managed object (12.25.6.1)
- b) Read SPB Adjacency Static Entry managed object (12.25.6.2)

12.25.6.1 Write SPB Adjacency Static Entry managed object

12.25.6.1.1 Purpose

To configure an SPB Adjacency Static Entry managed object on a port for an ISIS-SPB topology.

12.25.6.1.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) The system interface index of a port.
- c) The value of the ieee802.1 SPB Link State metric for the link on this port. For MTID = 0, or when a single MTID only is in use, this value should be set equal to the Path Cost set in the CIST Port Parameters Managed Object (see 12.8.2.1 and 12.8.2.3).
- d) The SPB administrative state of a port.

NOTE—Item c) – The primary function of IS-IS MT is to allow different link metrics to be associated with each topology, so that different routes are preferentially favored in different topologies. However, care must be taken when configuring IS-IS MT that CIST path costs are assigned to one topology, and that this topology is the one used to compute the CIST, in order that SPB calculates its part of the CIST consistently.

12.25.6.1.3 Outputs

- a) The Topology Index value that uniquely identifies this topology instance.
- b) The system interface index of the port.
- c) The value of the ieee802.1 SPB Link State metric for the link on this port.
- d) The SPB administrative state of this port.

12.25.6.2 Read SPB Adjacency Static Entry managed object

12.25.6.2.1 Purpose

To read back the parameters of an SPB Adjacency Static Entry managed object on a port for an ISIS-SPB topology.

12.25.6.2.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) The system interface index of a port. The value of zero is a wildcard indicating information, for any interface on which SPB operation is enabled, is requested.

12.25.6.2.3 Outputs

- a) The Topology Index value that uniquely identifies this topology instance.
- b) The system interface index of the port.
- c) The value of the ieee802.1 SPB Link State metric for the link on this port.
- d) The SPB administrative state of this port.

12.25.7 The SPB Adjacency Dynamic Entry managed object

There is one SPB Adjacency Dynamic Entry managed object per port per ISIS-SPB topology. It contains the dynamically discovered SPB parameters for the port in that topology. The management operation that can be performed on the SPB Adjacency Dynamic Entry managed object is as follows:

- a) Read SPB Adjacency Dynamic Entry managed object in a topology

12.25.7.1 Read SPB Adjacency Dynamic Entry managed object

12.25.7.1.1 Purpose

To obtain information about an SPB Adjacency Dynamic Entry managed object on a port.

12.25.7.1.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) The system interface index of a port. The value of zero is a wildcard indicating information, for any interface on which SPB operation is enabled, is requested.
- c) The SPB System Identifier of a peer attached to the port.

12.25.7.1.3 Outputs

- a) The Topology Index value that uniquely identifies this topology instance.
- b) The system interface index of the port.
- c) The SPB System Identifier of the peer attached to the port.
- d) The Bridge Port Number (same as item d1) in 12.4.1.2.3).
- e) The SPB operational state of this port.
- f) The IS-IS System name of the peer attached to the port.
- g) The value of the topology Agreement Digest (13.17.1) being advertised by the peer. This includes all parameters defined in the Agreement Digest (28.4).
- h) The value of the MCID (8.9, 13.5, 13.8) being advertised by the peer.
- i) The value of the Auxiliary MCID (27.4.1, 28.12.2) being advertised by the peer.
- j) The value of the Local Circuit ID of this node, used by ISIS to select a single adjacency if more than one link connects a pair of SPT Bridges.
- k) The value of the Local Circuit ID of the peer node, used by ISIS to select a single adjacency if more than one link connects a pair of SPT Bridges.
- l) The value of the Port Identifier on this node for the link selected by ISIS to form the adjacency, equal to the value read as item c) from the CIST Port Parameters Managed Object (see 12.8.2.1) on this node.
- m) The value of the Port Identifier on the peer node for the link selected by ISIS to form the adjacency with this node.
- n) An index pointing into the isisCircTable in the IS-IS MIB for cross-referencing.

12.25.8 The SPBM BSI Static Entry managed object

There is one SPBM BSI Static Entry managed object for each I-SID assigned to an SPBM VID on a CBP. It contains the SPBM configuration parameters for an I-SID. It is persistent over reboot. The management operations that can be performed on the SPBM BSI Static Entry managed object are as follows:

- a) Write SPBM BSI Static Entry managed object (12.25.8.1)
- b) Read SPBM BSI Static Entry managed object (12.25.8.2)

1 12.25.8.1 Write SPBM BSI Static Entry managed object

2 12.25.8.1.1 Purpose

3 To configure an SPBM BSI Static Entry managed object for an I-SID assigned to an SPBM VID on a CBP.

4 12.25.8.1.2 Inputs

- 5 a) A Topology Index value that uniquely identifies a topology instance.
- 6 b) The system interface index of a CBP.
- 7 c) The I-SID identifying the backbone service instance.
- 8 d) A B-VID associated with this I-SID. There may be more than one object for a given I-SID, allowing
9 it to be associated with multiple B-VIDs. This B-VID may differ from the B-VID specified in
10 12.16.5.2 that governs the VID assigned to the I-SID at the CBP; for example, to allow changing
11 ISIS-SPB parameters when moving an I-SID from one B-VID to another. The B-VID to I-SID
12 association in this managed object is advertised in ISIS-SPB so that forwarding state for the
13 association is installed throughout the SPT domain, in addition to CBP state installed by 12.16.5.2. It
14 is only the B-VID defined in 12.16.5.2 on which a CBP transmits; however, a CBP must receive
15 from any B-VID with which its I-SID has an association.
- 16 e) Multicast transmit indication bit [item e) in 28.12.10].
- 17 f) Multicast receive indication bit [item f) in 28.12.10].
- 18 g) Shared tree transmit indication bit [item g) in 28.12.10].
- 19 h) Tie-Break Mask value [item h) in 28.12.10].

20 12.25.8.1.3 Outputs

- 21 a) Operation status. This takes one of the following values:
 - 22 1) Operation rejected because the Topology Index value is not defined
 - 23 2) Operation rejected because the identified port does not exist or is not a CBP
 - 24 3) Operation rejected because the I-SID is not assigned to an SPBM VID
 - 25 4) Operation accepted

26 12.25.8.2 Read SPBM BSI Static Entry managed object

27 12.25.8.2.1 Purpose

28 To read back the parameters of an SPBM BSI Static Entry managed object for an I-SID assigned to an
29 SPBM VID on a CBP.

30 12.25.8.2.2 Inputs

- 31 a) A Topology Index value that uniquely identifies a topology instance.
- 32 b) The system interface index of a CBP. The value of zero is a wildcard indicating information, for any
33 interface on which SPB operation is enabled, is requested.
- 34 c) An I-SID identifying a backbone service instance. The value 0xFFFFFFFF is a wildcard indicating that
35 information on all I-SIDs on the specified port(s) is requested.
- 36 d) A B-VID associated with this I-SID. The value 4095 is a wildcard indicating that information for
37 any B-VID associated with the specified I-SID(s) is requested.

12.25.8.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because the Topology Index value is not defined
 - 2) Operation rejected because the identified port does not exist or is not a CBP
 - 3) Operation rejected because the I-SID is not assigned to an SPBM VID
 - 4) Operation accepted
- b) The Topology Index value that uniquely identifies this topology instance.
- c) The system interface index of the port.
- d) The I-SID identifying a backbone service instance.
- e) The B-VID associated with this I-SID.
- f) Multicast transmit indication bit [item e) in 28.12.10].
- g) Multicast receive indication bit [item f) in 28.12.10].
- h) Shared tree transmit indication bit [item g) in 28.12.10].
- i) Tie-Break Mask value [item h) in 28.12.10].

If item b) in 12.25.8.2.2 is configured with the value 0, a list indexed by the CBP ports on the system containing item a) to item i) [with the exception of item a2)].

12.25.9 The SPB Topology Node Table managed object

There is one SPB Topology Node Table managed object per Bridge per ISIS-SPB topology. It is populated by ISIS-SPB with the nodal SPB parameters discovered for all SPT Bridges participating in that topology within the SPT Region. This object is automatically created as a consequence of the creation of the SPB MTID Static managed object for that topology. It is deleted with the deletion of that SPB MTID Static managed object. The management operation that can be performed on the SPB Topology Node Table managed object is as follows:

- a) Read SPB Topology Node Table managed object

12.25.9.1 Read SPB Topology Node Table managed object

12.25.9.1.1 Purpose

To obtain nodal information about all SPT Bridges in an SPT Region for an ISIS-SPB topology.

12.25.9.1.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) The SPB System Identifier of ISIS-SPB on a remote Bridge. The value zero is a wildcard indicating that information, from all remote Bridges in the SPT region, is requested.

12.25.9.1.3 Outputs

A list, which may be empty, comprising one entry for each Bridge within the SPT Region of which the local IS-IS instance is aware. Each entry contains the following information about a remote Bridge:

- a) The Topology Index value that uniquely identifies this topology instance
- b) The SPB System Identifier of ISIS-SPB on the remote Bridge
- c) The Bridge Priority [the value set in CIST Protocol Parameters (12.8.1.3) for the remote Bridge]
- d) The SPSourceID (27.10) of the remote Bridge
- e) The SPB System Name of the remote Bridge

If b) identifies a specific remote Bridge then the list above will comprise one entry per Base VID, as specified in item c), containing only information about the specific remote Bridge.

12.25.10 The SPB Topology ECT Table managed object

There is one SPB Topology ECT Table managed object per Bridge per ISIS-SPB topology. It is populated by ISIS-SPB with the nodal ECT parameters discovered for all SPT Bridges participating in that topology within the SPT Region. This object is automatically created as a consequence of the creation of the SPB MTID Static managed object. It is deleted with the deletion of that SPB MTID Static managed object. The management operation that can be performed on the SPB Topology ECT Table managed object is as follows:

- a) Read SPB Topology ECT Table managed object

12.25.10.1 Read SPB Topology ECT Table managed object

12.25.10.1.1 Purpose

To obtain nodal ECT information, including the binding of ECT Algorithms to Base VIDs, about all SPT Bridges in an SPT Region for an ISIS-SPB topology.

12.25.10.1.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) The SPB System Identifier of ISIS-SPB on a remote Bridge. The value zero is a wildcard indicating information, for all System identifiers, is requested.
- c) The Base VID. The value 4095 is a wildcard indicating that information, from any Base VID assigned for SPB operation, is requested.

12.25.10.1.3 Outputs

A list, which may be empty, comprising one entry for each Bridge within the SPT Region of which the local ISIS-SPB instance is aware. Each entry contains the following information about a remote Bridge:

- a) The Topology Index value that uniquely identifies this topology instance
- b) The SPB System Identifier of ISIS-SPB on the remote Bridge
- c) The Base VID
- d) The identifier of the ECT Algorithm being used
- e) The SPB operational mode (SPBV or SPBM)
- f) The SPVID allocation mode (manual or automatic)
- g) The value of the SPVID when operating in SPBV mode
- h) A flag indicating that this Bridge has services currently assigned to this Base VID

If item b) identifies a specific remote Bridge then the list above will comprise one entry per Base VID as specified in item c), containing only information about the specific remote Bridge.

12.25.11 The SPB Topology Edge Table managed object

There is one SPB Topology Edge Table managed object per Bridge per ISIS-SPB topology. It is populated by ISIS-SPB with the discovered Edge (link) parameters for that topology within the SPT Region, referenced by the System Identifiers of the pair of SPT Bridges forming the end-points. This object is automatically created as a consequence of the creation of the SPB MTID Static managed object for that topology. It is deleted with the deletion of that SPB MTID Static managed object. The management operation that can be performed on the SPB Topology Edge Table managed object is as follows:

- a) Read SPB Topology Edge Table managed object

12.25.11.1 Read SPB Topology Edge Table managed object

12.25.11.1.1 Purpose

To obtain information about all Edges interconnecting SPT Bridges within an SPT Region for an ISIS-SPB topology.

NOTE—The use of the ISIS Local Circuit ID for tie breaking enables a single Edge (adjacency) per node pair and a unique Port Identifier on each node for spanning tree compatibility; refer to 28.11 and the Read operation on SPB Adjacency Dynamic Entry managed object.

12.25.11.1.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) The SPB System Identifier used by the near Bridge on an Edge. The value zero is a wildcard indicating information, for all System identifiers, is requested.
- c) The SPB System Identifier used by the far Bridge on an Edge. The value zero is a wildcard indicating information, for all System identifiers, is requested.

12.25.11.1.3 Outputs

A list, which may be empty, comprising one entry for each Edge within the SPT Region of which the local IS-IS instance is aware. Each entry contains the following information about an Edge:

- a) The Topology Index value that uniquely identifies this topology instance.
- b) The SPB System Identifier used by the near Bridge on the Edge.
- c) The SPB System Identifier used by the far Bridge on the Edge.
- d) The value of the ieee802.1 SPB Link State metric for this Edge, which is being advertised by the near Bridge.
- e) The value of the ieee802.1 SPB Link State metric for this Edge, which is being advertised by the far Bridge.

NOTE—The terms “near” and “far” are used locally solely to distinguish the two ends of the Edge and associated metric values, and do not signify any relationship between the SPT Bridges advertising the Edge and the local Bridge supporting the SPB Topology Edge Table managed object. As a consequence of this definition, in a stable topology two Edges are created per physical link, one advertised in a Link State PDU (LSP) by each Bridge.

12.25.12 The SPBM Topology Service Table managed object

There is one SPBM Topology Service Table managed object per Bridge per ISIS-SPB topology. It is populated by ISIS-SPB with the mapping of Service instances to SPT Bridges operating in SPBM mode participating in that topology discovered within the SPT Region. This object is automatically created as a consequence of the creation of the SPB MTID Static managed object for that topology. It is deleted with the deletion of that SPB MTID Static managed object. The SPBM Topology Service Table managed object only exists to provide Region-wide visibility of I-SID instance locations. Configuration of local I-SID instances is performed using the BEB Management (12.16) and the mapping of VIPs to PIPs (and hence to the B-MAC address associated with an I-SID) by PIP Management (12.16.4).

The management operation that can be performed on the SPBM Topology Service Table managed object is as follows:

- a) Read SPBM Topology Service Table managed object

12.25.12.1 Read SPBM Topology Service Table managed object

12.25.12.1.1 Purpose

To obtain information about the location of all I-SID instances within an SPT Region in an ISIS-SPB topology.

12.25.12.1.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) The SPB System Identifier of an SPT Bridges operating in SPBM mode hosting an I-SID. The value zero is a wildcard indicating information, from all remote Bridges in the SPT region hosting any I-SIDs, is requested.
- c) An I-SID value. The value 0xFFFFFFFF is a wildcard indicating that information, from the remote Bridge(s) specified in item b) above hosting any I-SID, is requested.
- d) The Base VID associated with this I-SID instance. The value 4095 is a wildcard indicating that information, from any Base VID assigned to SPB operation, is requested.

12.25.12.1.3 Outputs

A list, which may be empty, comprising one entry for each I-SID assigned to a Base VID within the SPT Region of which the local IS-IS instance is aware. Each entry contains the following information about the I-SID:

- a) The Topology Index value that uniquely identifies this topology instance.
- b) The SPB System Identifier of the SPT Bridges operating in SPBM mode hosting an I-SID instance.
- c) The I-SID value.
- d) The Base VID associated with this I-SID (and hence the ECT-ALGORITHM applied to this service).
- e) The B-MAC address associated with this I-SID. If multiple I-SID instances are configured on a single Bridge but are reachable by different B-MAC addresses, a separate record is returned for each B-MAC address associated with that I-SID.
- f) The I-SID instance service characteristic; transmit-only, receive-only, or both [item e) and item f) in 28.12.10; item e) and item f) in 12.25.8.1].
- g) If ECMP is supported, the shared tree transmit indication [item g) in 28.12.10; item g) in 12.25.8.1.2].
- h) If ECMP is supported, the Tie-Break Mask value [item h) in 28.12.10; item h) in 12.25.8.1.2].

NOTE—In general, an I-SID will be assigned to a single Base VID. During the administratively mediated migration of traffic from one ECT-ALGORITHM to a new one, an I-SID may be assigned to two Base VIDs for the duration of the transition in order to achieve loss-less migration (28.9.2).

12.25.13 The SPBV Topology Service Table managed object

There is one SPBV Topology Service Table managed object per Bridge per ISIS-SPB topology. It is populated by ISIS-SPB with the mapping of group MAC addresses to SPT Bridges operating in SPBV mode participating in that topology discovered within the SPT Region. This object is automatically created as a consequence of the creation of the SPB MTID Static managed object. It is deleted with the deletion of that SPB MTID Static managed object. The SPBV Topology Service Table managed object only exists to provide Region-wide visibility of group MAC address instance locations. Local configuration of group MAC address instances is performed using the procedures for management of Static Filtering Entries (12.7.2), or MMRP procedures (10.9).

The management operation that can be performed on the SPBV Topology Service Table managed object is as follows:

- a) Read SPBV Topology Service Table managed object

1 **12.25.13.1 Read SPBV Topology Service Table managed object**

2 **12.25.13.1.1 Purpose**

3 To obtain information about all group MAC address instances within an SPT Region in an ISIS-SPB
4 topology.

5 **12.25.13.1.2 Inputs**

- 6 a) A Topology Index value that uniquely identifies a topology instance.
- 7 b) The SPB System Identifier of an SPT Bridges operating in SPBV mode hosting a group MAC
8 address. The value zero is a wildcard indicating information, from all remote Bridges in the SPT
9 region hosting group MAC addresses, is requested.
- 10 c) The group MAC address value. The value zero is a wildcard indicating information, about all Group
11 MACs for the specific Bridge(s) with the System Identifier as specified by item b), is requested.

12 **12.25.13.1.3 Outputs**

13 A list, which may be empty, and comprising one entry for each group MAC address instance within the SPT
14 Region of which the local IS-IS instance is aware. Each entry contains the following information about the
15 group MAC address:

- 16 a) The Topology Index value that uniquely identifies this topology instance.
- 17 b) The SPB System Identifier of the SPT Bridges operating in SPBV mode hosting a group MAC
18 address instance.
- 19 c) The group MAC address value.
- 20 d) The Base VID (and hence ECT-ALGORITHM applied to this service).
- 21 e) The group MAC address instance service characteristic; transmit-only, receive-only, or both.

22 If item b) identifies a specific remote Bridge hosting an group address, then the list above will comprise only
23 information about the specific remote Bridge.

24 **12.25.14 The ECMP ECT Static Entry managed object**

25 There is one ECMP ECT Static Entry managed object per Tie-Break Mask. It contains the Bridge Priority to
26 be used for shared tree root selection for that Tie-Break Mask value. It is persistent over reboot. The
27 management operations that can be performed on the ECMP ECT Static Entry managed object are as
28 follows:

- 29 a) Write ECMP ECT Static Entry managed object (12.25.14.1)
- 30 b) Read ECMP ECT Static Entry managed object (12.25.14.2)

31 **12.25.14.1 Write ECMP ECT Static Entry managed object**

32 **12.25.14.1.1 Purpose**

33 To configure a different Bridge Priority to be used solely for shared tree root selection for a given Tie-Break
34 Mask (28.12.6.1, 28.8).

35 **12.25.14.1.2 Inputs**

- 36 a) A Topology Index value that uniquely identifies a topology instance.
- 37 b) A Tie-Break Mask value [item d) in 28.12.6.1].
- 38 c) Bridge Priority [item e) in 28.12.6.1].

12.25.14.1.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because the Topology Index value is not defined
 - 2) Operation accepted

12.25.14.2 Read ECMP ECT Static Entry managed object

12.25.14.2.1 Purpose

To read back the Bridge Priority for a given Tie-Break Mask.

12.25.14.2.2 Inputs

- a) A Topology Index value that uniquely identifies a topology instance.
- b) A Tie-Break Mask value [item d) in 28.12.6.1].

12.25.14.2.3 Outputs

- a) Operation status. This takes one of the following values:
 - 1) Operation rejected because the Topology Index value is not defined
 - 2) Operation accepted
- b) The Topology Index value that uniquely identifies this topology instance.
- c) The Tie-Break Mask value [item d) in 28.12.6.1].
- d) The Bridge Priority [item e) in 28.12.6.1].

12.26 Edge Virtual Bridging (EVB) management

The conformance requirements for EVB Bridges and EVB stations are defined in 5.23 and 5.24, respectively. Each C-VLAN Component, ER (5.24.1), and Port-mapping S-VLAN Component can be managed using the managed objects of 12.4 through 12.12 along with the EVB managed objects specified in this subclause.

An EVB Bridge system (Figure 12-4) supports the EVB managed objects defined in 12.26.1 through 12.26.3 and 12.27. Optionally, an EVB Bridge supports the managed objects for S-channels defined in 12.26.4.

An EVB station system (Figure 12-5) supports the EVB managed objects defined in 12.26.1, 12.26.3, 12.26.5, and 12.27. Optionally, an EVB station supports the managed objects for S-channels defined in 12.26.4.

The EVB-specific managed objects defined here:

- a) Provide managed objects for identifying and configuring an EVB system and its system-wide default parameters for LLDP, Virtual Station Interface (VSI) Discovery and Configuration Protocol (VDP), and S-channel (12.26.1).
- b) Provide managed objects for configuring the Station-facing Bridge Ports (SBPs) of EVB Bridges (12.26.2).
- c) Provide a VSI table that contains the current VSI and VDP state for each VSI that is active in the EVB system (12.26.3).
- d) Provide managed objects for configuring the Port-mapping S-VLAN Components and S-channels (12.26.4).
- e) Provide managed objects for configuring the URPs of the ERs (12.26.5).

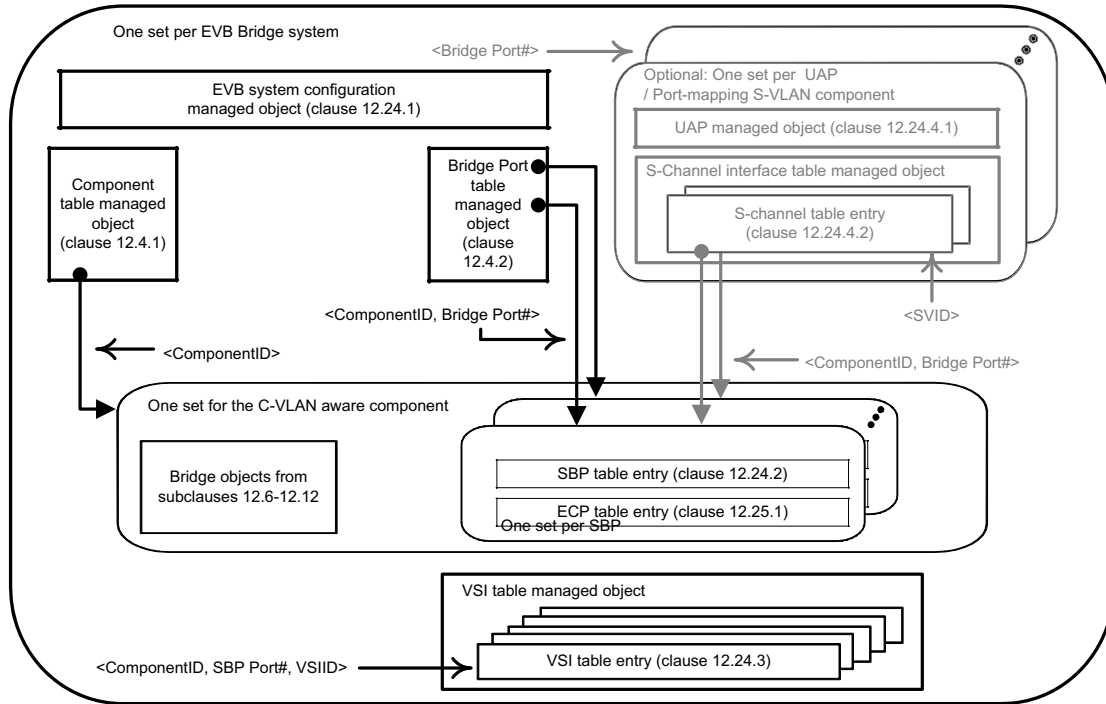
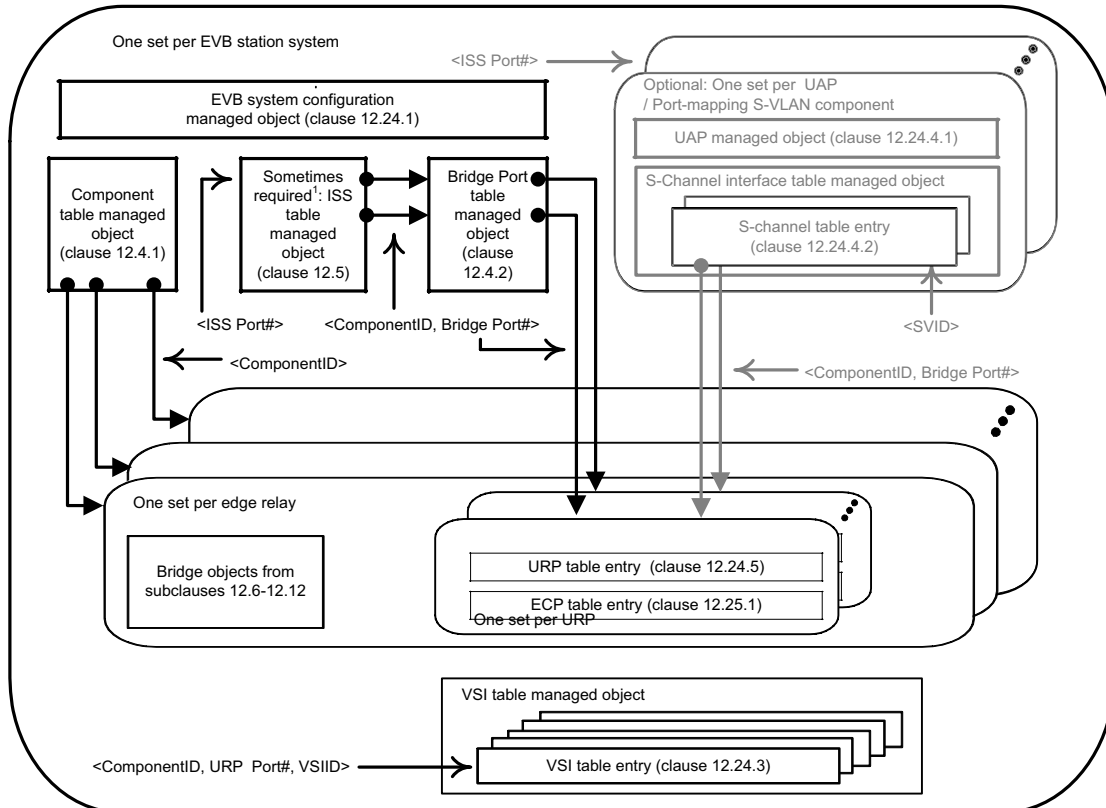


Figure 12-4—Relationships among EVB Bridge managed objects



Note 1: ISS table managed object is required when URP Port# != ISS Port#

Figure 12-5—Relationship among EVB station managed objects

1 For each EVB system, an EVB system configuration managed object exists containing the system-wide
2 defaults used to initialize the other EVB objects.

3 Every Port of the EVB system (40.1, 40.2, 40.3) is uniquely identified by a ComponentID (12.3) and Port
4 Number (12.3) [which together identify a Port Index (12.3)]. In EVB systems where the ISS and bound
5 Bridge Port have different Port Numbers, an ISS table (12.5.1) allows determining the associated Bridge
6 Port's ComponentID and Port Number from the scalar ISS Port Number. An example where the ISS table
7 would be needed is an EVB station where the station has multiple LANs each attaching to different ERs and
8 where the ERs each use the same Port Number (however different ComponentIDs) to identify their URPs
9 (Figure 40-3). The ISS table is normally not necessary in an EVB Bridge since the single C-VLAN-aware
10 Component provides a Bridge Port Number that is unique within the EVB Bridge for every ISS, allowing
11 the ISS and the Bridge Port to use the same Port Number independent from the ComponentID.

12 The ERs of an EVB station have two types of ports. Each ER DRP provides attachment to one or more VSIs.
13 No special EVB objects are used to manage the DRPs. URPs can attach to a LAN that can be externally
14 accessible or attach through an internal LAN to a CAP of an S-channel. URPs are managed using the URP
15 table. Each URP table entry identifies a URP within the system and allows configuration and monitoring of
16 LLDP and VDP for the URP (12.26.5). For each URP there is also a ECP table entry that allows
17 configuration and monitoring of ECP (12.27).

18 NOTE 1—The most common case is that a single VSI is associated with each DRP.

19 The C-VLAN Component of an EVB Bridge has two types of Ports. Each C-VLAN Bridge Port is
20 externally accessible and does not provide any EVB capabilities. No special EVB objects are used to
21 manage the C-VLAN Bridge Ports. Each SBP attaches to a LAN that can be externally accessible or attach
22 though an internal LAN to a CAP of an S-channel. SBPs are managed using the SBP table. Each SBP table
23 entry identifies a single SBP within the system and allows configuration and monitoring of LLDP and VDP
24 for the SBP (12.26.2). For each SBP there is also a ECP table entry that allows configuration and monitoring
25 of ECP (12.27).

26 Within each EVB station and EVB Bridge, a table holding the VSIs allows monitoring of the active VSIs
27 (12.26.3). VSI table entries are keyed on the VSI Instance Identifier (VSIID) and the URP's or SBP's
28 ComponentID and Port Number. During the movement of VSIs within the data center network (DCN) it is
29 possible to have a transient condition where the same VSIID exists at two locations. When these two
30 instances are at the same URP or SBP, they are indistinguishable; therefore, the table holds only the state for
31 the most recent VSI command. However, when the movement is between different URPs or SBPs, the table
32 will hold two copies of the same VSIID differentiated by the ComponentID and Port Number of the URPs or
33 SBPs.

34 NOTE 2—In the case where a VSI is migrated between DRPs of the same ER, there is no reason to de-associate, as the
35 table does not maintain any state that identifies the DRP of the ER to which the VSI is attached.

36 An association is identified by its URP and VSIID. If an association is established for a VSI, and a new
37 association is established on the same URP for the same VSIID, then the new association replaces the old
38 one. A subsequent de-association for that VSIID will remove the current association, regardless of the
39 parameters.

40 In EVB systems supporting S-channels, the optional UAP table and S-channel interface table are used to
41 configure and control the Port-mapping S-VLAN Components and CDCP (12.26.4).

42 Each ISS that is bound to an Uplink Access Port (UAP) automatically has a Port-mapping S-VLAN
43 Component with a default S-channel (using S-VID 1). Each S-channel is identified by the ISS Port Number
44 of the UAP and the S-VID, for SBPs, or S-channel Identifier (SCID), for URPs, identifying the S-channel.
45 The default S-channel carries all the un-S-tagged traffic that traverses the Port-mapping S-VLAN
46 Component.

1 In an EVB Bridge system, S-channels are automatically connected to SBPs when the S-Channel is created.
2 If the S-channel is the default S-channel, it is connected to the SBP with the same Port Number as the ISS
3 bound to the UAP. If the S-channel is not the default S-channel, a new SBP is allocated to the S-channel
4 (which has a Port Number different from any ISS Port Number where a UAP or C-VLAN Bridge Port could
5 be bound) and connected to the CAP through an internal LAN (6.14). With CDCP enabled and operating in
6 the ‘B’ role, the S-channel interface table entries are automatically created (deleted) by CDCP to fill
7 requests from the peer CDCP operating the ‘S’ role.

8 In an EVB station, S-channels are connected to URPs by the operating system. The URP Port Numbers are
9 not reserved, so the UAP’s ISS Port Number may not be the same as the URP Port Number. The operating
10 system is not required to automatically attach each S-channel to a URP; however, it can if it chooses. The
11 URP Port Numbers have no restrictions beyond the normal requirement that every relay port be uniquely
12 identified in the EVB system by a ComponentID and Port Number pair. With CDCP enabled and operating
13 in the ‘S’ role, CDCP uses the S-channel interface table to build requests for creating (deleting) S-channels
14 from the peer CDCP operating in the ‘B’ role and enables these S-channels after the peer grants an S-VID
15 for use by the S-channel.

16 Management of the UAP, Port-mapping S-VLAN Component and CDCP is achieved through the UAP table
17 (12.26.4.1). The management of CAPs and their LAN attachments is accomplished by the S-channel
18 interface table entries (12.26.4.2). A UAP table entry along with the Port-mapping S-VLAN creation rules
19 specified in 12.26.4 are sufficient to manage the Port-mapping S-VLAN Component. The Bridge managed
20 objects in 12.4 through 12.12 can optionally be used, in addition to the objects of 12.26.4, to manage the
21 Port-mapping S-VLAN Components.

22 The following managed objects, illustrated in Figure 12-4 and Figure 12-5, define the semantics of the
23 management operations specific to EVB Bridges and stations:

- 24 — The EVB system base managed object (12.26.1)
- 25 — The SBP table entry managed object (12.26.2)
- 26 — The VSI table entry managed object (12.26.3)
- 27 — The MAC/VID pair table entry managed object (12.26.3)
- 28 — The UAP table entry managed object (12.26.4)
- 29 — The S-channels interface table entry managed object (12.26.4)
- 30 — The URP table entry managed object (12.26.5)

31 **12.26.1 EVB system base table**

32 An instance of the EVB system base table (Table 12-22) can be implemented by an EVB Bridge or EVB
33 station. It comprises the identifiers for an EVB system (Clause 40) and system-wide default parameters used
34 to support EVB services (D.2.12, 40.4), VSI discovery (Clause 41), and ECP (43.3).

35 The management operations that can be performed on an EVB system table entry are as follows:

- 36 a) Read EVB system table entry
- 37 b) Update EVB system table entry

38 **12.26.1.1 System identifiers**

39 The evbSysType identifies the system type. The enumerated types for evbSysType are as follows:

- 40 a) sysB—EVB Bridge
- 41 b) sysS—EVB station

Table 12-22—EVB system base table

Name	Data type	Operations supported ^a	Conformance ^b	References
evbSysType	enumerated {sysB, sysS}	R	BE	5.23, 5.24
evbSysNumExternalPorts	unsigned [1...4095]	R	BE	12.4.2, 12.5.1
evbSysEvpLldpTxEnable	Boolean	RW	BE	D.2.12
evbSysEvpLldpManual	Boolean	RW	BE	D.2.12
evbSysEvpLldpGidCapable	Boolean	RW	BE	D.2.12
evbSysEcpDfltAckTimerInit	timer exp	RW	BE	D.2, 43.3.6.1
evbSysEcpDfltMaxTries	unsigned [0...7]	RW	BE	D.2, 43.3.7.4
evbSysVdpDfltRsrcWaitDelay	timer exp	RW	BE	D.2.12, 41.5.5.7
evbSysVdpDfltReinitKeepAlive	timer exp	RW	BE	D.2.12, 41.5.5.5

^a R = Read-only access; RW = Read/Write access.

^b B = Required for an EVB Bridge system; E = Required for an EVB station system.

1 12.26.1.2 System defaults for EVB

2 The parameters evbSysEvpLldpTxEnable and evbSysEvpLldpManual are used to initialize the LLDP EVB
3 objects for new SBPs and URPs. When evbSysLldpTxEnable is TRUE, a new SBP or URP will place the
4 local EVB objects in the LLDP nearest Customer database; when FALSE, a new SBP or URP will not place
5 the local EVB objects in the LLDP database. When evbSysLldpManual is FALSE, the operating
6 configuration will be determined by the comparison between the local and remote LLDP EVB objects
7 (automatic), regardless of the setting of evbSysLldpTxEnable. When evbSysLldpManual is TRUE, the
8 configuration will be determined by the setting of the local EVB objects only (manual).

9 The evbSysLldpGidCapable parameter indicates if the port is capable of processing GroupIDs. GroupIDs
10 can be used if both the EVB Bridge and EVB station indicate they are capable by setting the EVB LLDP
11 object for GroupID capable to TRUE. On an EVB Station this means the station can provide GroupIDs
12 rather than VIDs in VDP requests and accept VIDs in response from the EVB Bridge. On an EVB Bridge
13 this means the Bridge can provide a VID corresponding to the GroupID provided by the EVB station.

14 The default value for evbSysEcpDfltAckTimerInit is 14, which provides a time of 164 ms. Systems are not
15 required to implement smaller times and can reject requests to set the timers to small times; however,
16 systems are required to implement 14 and above to allow fall back to the longest time proposed by the EVB
17 Bridge or EVB station.

18 The default value for evbSysVdpDfltRsrcWaitDelay and evbSysVdpDfltReinitKeepAlive is system
19 dependent. All systems support the values of 20 and above, which provide times of 10.5 seconds and greater.
20 Systems are not required to implement smaller times and can reject requests to set the timers to small times;
21 however, systems are required to implement 20 and above to allow fall back to the longest time proposed by
22 the EVB Bridge or EVB station.

23 Table 12-23 shows how the system defaults are used to initialize the parameters in the SBP, URP, and ECP
24 table entries.

Table 12-23—EVB system parameter defaults

System parameter	Default value	LLDP, SBP, URP, ECP entry parameter
evbSysEvpLldpTxEnable	TRUE	LLDP Transmit Enable
evbSysEvpLldpManual	FALSE	sbpLldpManual, urpLldpManual
evbSysEvpLldpGidCapable	system dependent	LLDP GID Capable
evbSysEcpDfltAckTimerInit	14, for 164 milliseconds	ecpAdminAckTimerInit
evbSysEcpDfltMaxTries	3	ecpAdminMaxTries
evbSysVdpDfltRsrcWaitDelay	system dependent	sbpVdpAdminRsrcWaitDelay, urpVdpAdminRsrcWaitDelay
evbSysVdpDfltReinitKeepAlive	system dependent	sbpVdpAdminReinitKeepAlive, urpVdpAdminReinitKeepAlive

1 12.26.2 SBP table entry

2 SBP table entries (Table 12-24) may be created explicitly or implicitly as a result of creating an entry in the
3 S-channel interface table. When an SBP table entry is created, the port type in the Port table entry
4 (Table 12-2) changes to type ‘Station-facing Bridge Port’. When an SBP table entry is deleted, the Port table
5 entry returns to type C-VLAN Bridge Port.

Table 12-24—SBP table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
sbpComponentID	ComponentID	R	B	12.4.1.5
sbpPortNumber	Port Number	R	B	12.4.2
sbpLldpManual	Boolean	RW	B	—
sbpVdpOperRsrcWaitDelay	timer exp	R	B	D.2.12, 41.5.5.7
sbpVdpOperReinitKeepAlive	timer exp	R	B	D.2.12, 41.5.5.5
sbpVdpOperToutKeepAlive	unsigned	R	B	D.2.12, 41.5.5.13

^a R = Read-only access; RW = Read/Write access.

^b B = Required for an EVB Bridge system.

6 Whenever a new SBP table entry is created, a new entry is also created in the ECP table (12.27) keyed under
7 the ComponentID and Port Number of the SBP. Whenever an SBP table entry is deleted, the corresponding
8 entry in the ECP table is deleted.

9 The management operations that can be performed on an SBP table entry are as follows:

- 10 a) Read SBP table entry
- 11 b) Update SBP table entry
- 12 c) Create SBP table entry
- 13 d) Delete SBP table entry

12.26.3 VSI table entry

Each EVB system maintains a table of the active VSIs. The structure of a VSI table entry is shown in Table 12-25. This read-only table provides the current operation parameters of each VSI along with the VDP state associated with the VSI. The table is keyed on the SBP's or URP's ComponentID and Port Number and on the VSIID. The operation that can be performed on the VSI table is as follows:

- a) Read entry for a ComponentID, Port Number, and VSIID

Table 12-25—VSI table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
evbVsiComponentID	ComponentID	R	BE	12.4.1.5
evbVsiPortNumber	Port Number	R	BE	12.4.2
evbVsiIDType	enumerated	R	BE	41.2.6, Table 41-5
evbVsiID	Latin1 String (SIZE(16))	R	BE	41.2.7
evbVsiTimeSinceCreate	time interval	R	BE	Clause 41
evbVsiVdpOperCmd	enumerated	R	BE	41.2.1, Table 41-1
evbVsiOperRevert	Boolean	R	BE	41.2.3
evbVsiOperHard	Boolean	R	BE	41.2.3
evbVsiOperReason	unsigned (0..15)	R	BE	41.2.3
evbVsiMgrID	Latin1 String (SIZE(1))	R	BE	41.1.3
evbVsiType	Latin1 String (SIZE(3))	R	BE	41.2.4
evbVsiTypeVersion	Latin1 String (SIZE(1))	R	BE	41.2.5
evbVsiMvFormat	Latin1 String (SIZE(1))	R	BE	41.2.8
evbVsiNumMACs	unsigned	R	BE	41.2.9
evbVdpMachineState	enumerated	R	BE	41.5.5.14
evbVdpCmdsSucceeded	counter	R	BE	41.5
evbVdpCmdsFailed	counter	R	BE	41.5
evbVdpCmdsReverts	counter	R	BE	41.5

^a R = Read-only access; RW = Read/Write access.

^b B = Required for an EVB Bridge system; E = Required for an EVB station system.

Each EVB Bridge or EVB station maintains a table of the VID/MACs on each VSI. The structure of a VSI MAC/VLAN table entry is shown in Table 12-26. This read-only table provides the current GroupID/VID/MAC assignments for each VSI. The operations that can be performed on the VSI table are as follows:

- b) Read entries for a ComponentID, Port Number, and VSIID
- c) Read entries for a ComponentID and Port Number

Table 12-26—VSI MAC/VLAN table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
evbMvComponentID	ComponentID	R	BE	12.4.1.5
evbMvPortNumber	Port Number	R	BE	12.4.2
evbMvVsiIDType	enumerated	R	BE	41.2.6, Table 41-5
evbMvVsiID	Latin1 String (SIZE(16))	R	BE	41.2.7
evbMvVsiGroupID	unsigned	R	BE	41.2.9
evbMvVsiVID	unsigned (1..4094)	R	BE	41.2.9
evbMvVsiMAC	MAC address	R	BE	41.2.9
evbMvVsiIpAddress	IP address	R		41.2.9

^a R = Read-only access; RW = Read/Write access.

^b B = Required for an EVB Bridge system; E = Required for an EVB station system.

1 12.26.4 S-channel configuration and management

2 The S-channel managed objects are not required unless the system implements S-channels.

3 Creating a UAP table entry (40.2) causes a Port-mapping S-VLAN Component (15.6) to be instantiated and
4 sets the portType parameter of the Port table entry (12.4.2) to type ‘Uplink Access Port’. The Port-mapping
5 S-VLAN Component automatically includes a default S-channel with one CAP that can attach to an SBP or
6 a URP through an internal LAN (6.14). Each UAP is identified by the ISS Port Number where the UAP is
7 attached. The default S-channel and CAP shall be identified by the S-VID 1 and SCID 1.

Table 12-27—UAP table entry parameters

UAP table name	Default values
uapSchCdcAdminEnable	TRUE
uapSchCdcAdminRole	schS if EVB station and schB if EVB Bridge
uapSchCdcAdminChnCap	1
uapSchAdminCdcSvidPoolLow	0
uapSchAdminCdcSvidPoolHigh	0

8 12.26.4.1 UAP table entry

9 The management operations that can be performed on the UAP table entry (Table 12-28) managed object are
10 as follows:

- 11 a) Read UAP table entry
- 12 b) Update UAP table entry
- 13 c) Create UAP table entry

- 1 d) Delete UAP table entry

Table 12-28—UAP table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
uapISSPortNumber	Port Number	R	BE	12.4.2, 12.5.1
uapComponentID	ComponentID	R	be	12.4.1.5
uapPortNumber	Port Number	R	be	12.4.2
uapSchCdcPAdminEnable	Boolean	RW	BE	42.4.2
uapSchCdcPAdminRole	enumerated	RW	BE	42.4.2
uapSchCdcPAdminChnCap	unsigned [1...167]	RW	BE	42.4.1
uapSchCdcPOperChnCap	unsigned [1...167]	R	BE	42.4.8
uapSchAdminCdcPSvidPoolLow	unsigned [0,2...4094]	RW	BE	42.4.7
uapSchAdminCdcPSvidPoolHigh	unsigned [0,2...4094]	RW	BE	42.4.7
uapSchOperState	enumerated	R	BE	42.4.14
uapSchCdcPRemoteEnabled	Boolean	R	BE	42.4.14
uapSchCdcPRemoteRole	enumerated	R	BE	42.4.12

^a R = Read-only access; RW = Read/Write access.

^b B = Required for an EVB Bridge system; E = Required for an EVB station system; b = Optional for an EVB Bridge system; e = Optional for an EVB station system.

2 Table 12-27 shows the default values for the UAP table entry parameters.

3 The available SVIDs determined by the range of uapSchAdminCdcPSvidPoolLow and
4 uapSchAdminCdcPSvidPoolHigh limit the S-channel capacity indicated by uapSchCdcPAdminChnCap to
5 the available SVIDs plus one for the default S-channel. If the capacity is greater than the VID range plus
6 one, then the VID range overrides the capacity (i.e., the actual capacity is never bigger than the available
7 VID range plus one).

8 **12.26.4.2 S-channel interface table entry**

9 The S-channel interface table entry applies to each internal S-channel configured on an EVB Bridge or EVB
10 station, as shown in Table 12-29. The management operations that can be performed on an S-channel
11 interface table entry are as follows:

- 12 a) Read S-channel interface table entry
13 b) Update S-channel interface table entry
14 c) Create S-channel interface table entry
15 d) Delete S-channel interface table entry

16 If the S-channel interface table is being used with a UAP operating in the cdcP role, then the table is keyed
17 on schUapISSPortNumber and schScId. If the S-channel interface table is being used with a UAP operating
18 in the cdcPB role, then the table is keyed on the schUapISSPortNumber and schSvid.

19 The schComponentID and schCapPortNumber refer to the Port-mapping S-VLAN Component and the Port,
20 which may also be identified by the schISSPortNumber and S-VID (or SCID).

Table 12-29—S-channel interface table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
schUapISSPortNumber	Port Number	R	BE	12.4.2, 12.5.1
schScid	unsigned [1...4094]	R	bE	42.4.3
schSvid	unsigned [0...4094]	R	BE	42.4.3
schComponentID	ComponentID	R	be	42.1, 12.4.1.5
schCapPortNumber	Port Number	R	be	42.1, 12.4.2
schSbpOrUrpComponentID	ComponentID	RW	BE	12.4.1.5
schSbpOrUrpPortNumber	Port Number	RW	BE	12.4.2

^a R = Read-only access; RW = Read/Write access.

^b B = Required for an EVB Bridge system; E = Required for an EVB station system; b = Optional for an EVB Bridge system; e = Optional for an EVB station system.

1 The schSbpOrUrpComponentID and schSbpOrUrpPortNumber are the ComponentID and Port Number of
2 the attached SBP or URP.

3 **12.26.5 ER management**

4 ERs can be built dynamically or statically within an EVB station. Each ER is assigned a ComponentID that
5 is unique for the EVB station system. An ER always has a single URP that exists as long as the ER exists.
6 The DRPs of an ER may be built either along with the ER or on demand.

7 **12.26.5.1 URP table entry**

8 When a URP table entry is created, a corresponding Component table entry is created for the ER along with
9 a Port table entry for the URP. When a URP table entry is deleted, the corresponding Component table entry
10 is deleted along with the corresponding Port table entry.

11 When each URP table entry is created, a corresponding entry is created in the ECP table (12.27) and keyed
12 under the ComponentID and Port Number of the URP. Whenever a URP table entry is deleted, the
13 corresponding entry in the ECP table is deleted.

14 The management operations that can be performed on the URP table entry (Table 12-30) managed object are
15 as follows:

- 16 a) Read URP table entry
- 17 b) Update URP table entry
- 18 c) Create URP table entry
- 19 d) Delete URP table entry

Table 12-30—URP table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
urpComponentID	ComponentID	R	E	12.4.1.5
urpPortNumber	Port Number	R	E	12.4.2
urpBindToISSPortNumber	unsigned [0...4095]	RW	e	12.5.1
urpLdpManual	Boolean	RW	E	
urpVdpOperRsrcWaitDelay	timer exp	R	E	D.2.12, 41.5.5.7
urpVdpOperRespWaitDelay	unsigned	R	E	D.2.12, 41.5.5.9
urpVdpOperReinitKeepAlive	timer exp	R	E	D.2.12, 41.5.5.5

^a R = Read-only access; RW = Read/Write access.

^b E = Required for an EVB station system; e = Optional for an EVB station system.

1 12.27 Edge Control Protocol (ECP) management

2 12.27.1 ECP table entry

3 The management operation that can be performed on the ECP table entry managed object is as follows:

4 a) Read ECP table entry

5 ECP table entries (Table 12-31) are created or deleted implicitly as a result of the creation or deletion of
6 other port objects.

Table 12-31—ECP table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
ecpComponentID	ComponentID	R	BE	12.4.1.5
ecpPortNumber	Port Number	R	BE	12.4.2
ecpOperAckTimerInit	timer exp	R	BE	D.2.12, 43.3.7.1
ecpOperMaxRetries	unsigned [0...7]	R	BE	D.2.12, 43.3.7.4
ecpTxFrameCount	counter	R	BE	Clause 43
ecpTxRetryCount	counter	R	BE	Clause 43
ecpTxFailures	counter	R	BE	Clause 43
ecpRxFrameCount	counter	R	BE	Clause 43

^a R = Read-only access; RW = Read/Write access.

^b B = Required for an EVB Bridge system; E = Required for an EVB station system.

12.28 Path Control and Reservation (PCR) management

The VLAN Bridges that support IS-IS implement the following SPB managed objects because they are required for IS-IS operations in a Bridged Network. Therefore, these SPB managed objects are required for explicit path control via IS-IS, i.e., for PCR operations:

- SPB System managed object (12.25.1)
- SPB MTID Static managed object (12.25.2)
- SPB Topology Instance Dynamic managed object (12.25.3)
- SPB ECT Static Entry managed object (12.25.4)
- SPB ECT Dynamic Entry managed object (12.25.5)
- SPB Adjacency Static Entry managed object (12.25.6)
- SPB Adjacency Dynamic Entry managed object (12.25.7)
- SPBM BSI Static Entry managed object (12.25.8)
- SPB Topology Node Table managed object (12.25.9)
- SPB Topology ECT Table managed object (12.25.10)
- SPB Topology Edge Table managed object (12.25.11)
- SPBM Topology Service Table managed object (12.25.12)
- SPBV Topology Service Table managed object (12.25.13)

The above listed SPB managed objects shall be used to configure PCR operations in SPT Bridges as specified in Clause 45. For instance, if a VLAN is under explicit path control via IS-IS, then the corresponding explicit ECT Algorithm value of Table 45-1 has to be used for the VLAN's Base VID when the SPB ECT Static Entry managed object (12.25.4) is created.

NOTE—The SPBM BSI Static Entry managed object (12.25.8) and the SPBM Topology Service Table managed object (12.25.12) are not required if no backbone Service Instance is present in the SPT Region.

Managed objects dedicated to PCR are needed only for Maximally Redundant Trees (MRTs). Due to the dependence on SPB managed objects, the PCR managed objects are specified as part of the SPB managed object hierarchy as illustrated in Figure 12-3. The following are the PCR managed objects in an SPT Bridge:

- a) The PCR ECT Static Entry managed object (12.28.1)
- b) The PCR Topology ECT Table managed object (12.28.2)

The SPVID parameter of an ISIS-SPB sub-TLV and an SPB managed object does not convey Shortest Path VID if it refers to an explicit tree (Clause 45), but the SPVID parameter is used to carry another type of VID as specified by 45.1.3, e.g., an MRT VID. These VIDs are always configured and never auto-allocated.

If the ECT Algorithm takes the value of the LTS ECT Algorithm (Table 45-1) and it is a learning VLAN (45.1.3), i.e., the VLAN's Base VID is allocated to the SPBV-MSTID (0xFFD), then the Bridge's VID [also allocated to the SPBV-MSTID, and not to the SPVID pool (SPVID-Pool-MSTID: 0xFFFF) as is done for Shortest Path Bridging operation] has to be configured in place of the SPVID parameter of the SPB ECT Static Entry managed object [item d) in 12.25.4.1.2].

If the ECT Algorithm is one of the MRT algorithms (Table 45-1), then MRT VID values are domain-wide for a non-learning VLAN, whereas they are Bridge-local for a learning VLAN. If it is a learning VLAN (45.1.3), i.e., the VLAN's Base VID is allocated to the SPBV-MSTID, then the Bridge's MRT VIDs (also allocated to the SPBV-MSTID) have to be configured as the MRT-Blue VID and MRT-Red VID parameters of the PCR ECT Static Entry managed object (12.28.1). MRT VIDs are configured; they cannot be auto-allocated. The Base VID's SPVID entry in the SPB ECT Static managed object (12.25.2) is ignored for learning VLANs if the MRTs protect each other. If the MRTs protect an SPT for a learning VLAN, then a Shortest Path VID (a real SPVID allocated to the SPVID-Pool-MSTID) has to be associated with the SPT Bridge, in addition to its MRT VIDs, which is provided as specified by 12.25. The Shortest Path VID is

1 configured via the SPB ECT Static Entry managed object (see item d) in 12.25.4.1.2] and its allocation
2 method (automatic or manual) is configured via the SPB System managed object (see item c) in
3 12.25.1.2.2].

4 **12.28.1 The PCR ECT Static Entry managed object**

5 Each Bridge has one PCR ECT Static Entry managed object per Base VID assigned with MRT operation
6 (45.3.3, 45.3.4). The static Entry contains the MRT VID configuration parameters for the Base VID. It is
7 persistent over reboot. The management operations that can be performed on the ECT Static Entry managed
8 object are as follows:

- 9 a) Create a PCR ECT Static Entry managed object (12.28.1.1)
- 10 b) Read a PCR ECT Static Entry managed object (12.28.1.2)
- 11 c) Write a PCR ECT Static Entry managed object (12.28.1.3)
- 12 d) Delete a PCR ECT Static Entry managed object (12.28.1.4)

13 **12.28.1.1 Create PCR ECT Static Entry managed object**

14 **12.28.1.1.1 Purpose**

15 To create, configure, and interrogate a PCR ECT Static Entry managed object on a Bridge.

16 **12.28.1.1.2 Inputs**

- 17 a) A Topology Index value that uniquely identifies a topology instance.
- 18 b) A Base VID. This VID has been allocated either to the SPBM-MSTID (0xFFC) for non-learning
19 VLANs or to the SPBV-MSTID (0xFFD) for learning VLANs in the FID to MSTI Allocation table
20 (12.12.2.2). An SPB ECT Static Entry managed object (12.25.4) has to be created for the Base VID
21 and Topology Index prior to the creation of the PCR ECT Static Entry managed object.
- 22 c) The MRT-Blue VID assigned with the Base VID of the corresponding SPB ECT Static Entry
23 managed object. The MRT-Blue VID has to be allocated to the same MSTID as the Base VID. If it is
24 a non-learning VLAN, i.e., the Base VID is allocated to the SPBM-MSTID (0xFFC), and the MRTs
25 are used to protect each other, then the value of the MRT-Blue VID has to take the value of the Base
26 VID. An MRT-Blue VID is always assigned by configuration. In the case of a learning VLAN or if
27 the MRTs protect SPTs of a non-learning VLAN, the values of Base VID, MRT-Blue VID, and
28 MRT-Red VID are all different.
- 29 d) The MRT-Red VID assigned with the Base VID of the corresponding SPB ECT Static Entry
30 managed object. The MRT-Red VID has to be allocated to the same MSTID as the Base VID. An
31 MRT-Red VID is always assigned by configuration.

32 **12.28.1.1.3 Outputs**

- 33 a) Operation status. This takes one of the following values:
 - 34 1) Operation rejected because the Topology Index value is not defined.
 - 35 2) Operation rejected because the SPB ECT Static Entry managed object does not exist for the
36 Base VID and Topology Index.
 - 37 3) Operation rejected because the requested Base VID was not assigned with MRT operation.
 - 38 4) Operation rejected due to MRT-Blue VID not allocated to the same MSTID as the Base VID.
 - 39 5) Operation rejected due to MRT-Red VID not allocated to the same MSTID as the Base VID.
 - 40 6) Operation accepted.

1 12.28.1.2 Read PCR ECT Static Entry managed object

2 12.28.1.2.1 Purpose

3 To obtain information about the PCR ECT Static Entry managed object for a Base VID.

4 12.28.1.2.2 Inputs

- 5 a) A Topology Index value that uniquely identifies a topology instance.
- 6 b) A Base VID. The value 4095 is a wildcard indicating that information on any Base VID assigned
7 with MRT operation is requested.

8 12.28.1.2.3 Outputs

- 9 a) Operation status. This takes one of the following values:
 - 10 1) Operation rejected
 - 11 2) Operation accepted
- 12 b) A Topology Index value that uniquely identifies this topology instance.
- 13 c) A list of <Base VID, MRT-Blue VID, MRT-Red VID> 3-tuples assigned with MRT operation. The
14 list is null if no VLAN is assigned with MRT operation. The list comprises a single 3-tuple if the
15 Base VID parameter is not the wildcard (12.28.1.2.2).

16 12.28.1.3 Write PCR ECT Static Entry managed object

17 12.28.1.3.1 Purpose

18 To configure a PCR ECT Static Entry managed object for a Base VID.

19 12.28.1.3.2 Inputs

- 20 a) A Topology Index value that uniquely identifies a topology instance.
- 21 b) A Base VID. This VID has been allocated either to the SPBM-MSTID (0xFFC) for non-learning
22 VLANs, or to the SPBV-MSTID (0xFFD) for learning VLANs in the FID to MSTI Allocation table
23 (12.12.2.2). An SPB ECT Static Entry managed object (12.25.4) has to be created for the Base VID
24 and Topology Index prior to the creation of the PCR ECT Static Entry managed object.
- 25 c) The MRT-Blue VID assigned with the Base VID of the corresponding SPB ECT Static Entry
26 managed object. The MRT-Blue VID has to be allocated to the same MSTID as the Base VID. If it is
27 a non-learning VLAN, i.e., the Base VID is allocated to the SPBM-MSTID (0xFFC), and the MRTs
28 are used to protect each other, then the value of the MRT-Blue VID has to take the value of the Base
29 VID. An MRT-Blue VID is always assigned by configuration. In the case of a learning VLAN or if
30 the MRTs protect SPTs of a non-learning VLAN, the values of Base VID, MRT-Blue VID, and
31 MRT-Red VID are all different.
- 32 d) The MRT-Red VID assigned with the Base VID of the corresponding SPB ECT Static Entry
33 managed object. The MRT-Red VID has to be allocated to the same MSTID as the Base VID. An
34 MRT-Red VID is always assigned by configuration.

35 12.28.1.3.3 Outputs

- 36 a) Operation status. This takes one of the following values:
 - 37 1) Operation rejected because the Topology Index value is not defined.
 - 38 2) Operation rejected because the SPB ECT Static Entry managed object does not exist for the
39 Base VID and Topology Index.
 - 40 3) Operation rejected because the requested Base VID was not assigned with MRT operation.
 - 41 4) Operation rejected due to MRT-Blue VID not allocated to the same MSTID as the Base VID.
 - 42 5) Operation rejected due to MRT-Red VID not allocated to the same MSTID as the Base VID.

1 6) Operation accepted.

2 **12.28.1.4 Delete PCR ECT Static Entry managed object**

3 **12.28.1.4.1 Purpose**

4 To delete the PCR ECT Static Entry managed object for a Base VID.

5 **12.28.1.4.2 Inputs**

- 6 a) A Topology Index value that uniquely identifies a topology instance.
- 7 b) A Base VID.

8 **12.28.1.4.3 Outputs**

- 9 a) Operation status. This takes one of the following values:
 - 10 1) Operation rejected because the Bridge has services currently assigned with this Base VID.
 - 11 2) Operation rejected because the SPT Region has services currently using this Base VID.
 - 12 3) Operation rejected because PCR ECT Static Entry managed object does not exist for the Base
 - 13 VID.
 - 14 4) Operation accepted.

15 **12.28.2 The PCR Topology ECT Table managed object**

16 There is one PCR Topology ECT Table managed object per Bridge per IS-IS topology. It is populated by
17 IS-IS with the nodal ECT parameters discovered for all SPT Bridges participating in that topology within the
18 SPT Region. This object is automatically created as a consequence of the creation of a PCR ECT Static
19 Entry managed object for the given IS-IS topology. It is automatically deleted with the deletion of the last
20 PCR ECT Static Entry managed object for the given IS-IS topology. The management operation that can be
21 performed on the PCR Topology ECT Table managed object is as follows:

- 22 a) Read PCR Topology ECT Table managed object

23 **12.28.2.1 Read PCR Topology ECT Table managed object**

24 **12.28.2.1.1 Purpose**

25 To obtain nodal information from all SPT Bridges of an SPT Region on the binding of Base VIDs to the
26 MRT algorithms in an IS-IS topology.

27 **12.28.2.1.2 Inputs**

- 28 a) A Topology Index value that uniquely identifies a topology instance.
- 29 b) The System Identifier of a remote Bridge. The value zero is a wildcard indicating the information on
30 all System identifiers is requested.
- 31 c) The Base VID. The value 4095 is a wildcard indicating that information on any Base VID assigned
32 with MRT operation is requested.

33 **12.28.2.1.3 Outputs**

34 A list comprising one entry for each Base VID assigned to MRT operation at a Bridge within the SPT
35 Region that the local IS-IS instance is aware. The list can be empty. Each entry contains the following
36 information about a remote Bridge:

- 37 a) The Topology Index value that uniquely identifies this topology instance
- 38 b) The System Identifier of the remote Bridge

- 1 c) The Base VID
- 2 d) The VLAN's operational mode (learning or non-learning)
- 3 e) MRT-Blue VID
- 4 f) MRT-Red VID

5 If item b) identifies a specific remote Bridge, then the above list comprises one entry per Base VID as
6 specified in item c), containing only information about the specific remote Bridge.

7 **12.29 Managed objects for scheduled traffic**

8 The Bridge enhancements for support of scheduled traffic are defined in 8.6.8 and 8.6.8.4.

9 This managed resource comprises the following object:

- 10 a) The Gate Parameter Table (12.29.1)

11 **12.29.1 The Gate Parameter Table**

12 There is one Gate Parameter Table per Port of a Bridge Component. Each table row contains a set of
13 parameters that supports the enhancements for scheduled traffic (8.6.8.4), as detailed in Table 12-32. Rows
14 in the table can be created or removed dynamically in implementations that support dynamic configuration
15 of ports and components.

Table 12-32—The Gate Parameter Table

Name	Data type	Operations supported ^a	Conformance ^b	References
queueMaxSDUTable ^c	sequence of queueMaxSDU	RW	B, E	8.6.8.4, 12.29.1.1, 12.29.1.1.1
GateEnabled	Boolean	RW	B, E	8.6.9.4.14
AdminGateStates	gateStatesValue	RW	B, E	8.6.9.4.5, 12.29.1.2, 12.29.1.2.2
OperGateStates	gateStatesValue	R	B, E	8.6.9.4.21, 12.29.1.2, 12.29.1.2.2
AdminControlListLength	unsigned integer	RW	B, E	8.6.9.4.6, 12.29.1.2
OperControlListLength	unsigned integer	R	B, E	8.6.9.4.22, 12.29.1.2
AdminControlList	sequence of GateControlEntry	RW	B, E	8.6.9.4.2, 12.29.1.2, 12.29.1.2.1
OperControlList	sequence of GateControlEntry	R	B, E	8.6.9.4.18, 12.29.1.2, 12.29.1.2.1
AdminCycleTime	RationalNumber (seconds)	RW	B, E	8.6.9.4.3, 12.29.1.3
OperCycleTime	RationalNumber (seconds)	R	B, E	8.6.9.4.19, 12.29.1.3
AdminCycleTimeExtension	Integer (nanoseconds)	RW	B, E	8.6.9.4.4
OperCycleTimeExtension	Integer (nanoseconds)	R	B, E	8.6.9.4.20
AdminBaseTime	PTPtime	RW	B, E	8.6.9.4.1, 12.29.1.4

Table 12-32—The Gate Parameter Table (continued)

Name	Data type	Operations supported ^a	Conformance ^b	References
OperBaseTime	PTPtime	R	B, E	8.6.9.4.17, 12.29.1.4
ConfigChange	Boolean	RW	B, E	8.6.9.4.7
ConfigChangeTime	PTPtime	R	B, E	8.6.9.4.9, 12.29.1.4
TickGranularity	Integer (tenths of nanoseconds)	R	B, E	8.6.9.4.16
CurrentTime	PTPtime	R	B, E	8.6.9.4.10, 12.29.1.4
ConfigPending	Boolean	R	B, E	8.6.9.3, 8.6.9.4.8
ConfigChangeError	Integer	R	B, E	8.6.9.3.1
SupportedListMax	Integer	R	B, E	12.29.1.5
SupportedCycleMax	RationalNumber (seconds)	R	B, E	8.6.8.4, 12.29.1.3, 12.29.1.6
SupportedIntervalMax	Integer	R	B, E	8.6.8.4, 12.29.1.7

^a R= Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of enhancements for scheduled traffic; E = Required for end station support of enhancements for scheduled traffic.

^c The number of queues supported by a Port is available from the traffic class table (12.6.3)

1 12.29.1.1 The queueMaxSDUTable structure and data types

2 The queueMaxSDUTable consists of up to 8 entries, one per traffic class supported by the implementation,
3 each entry consisting of a queueMaxSDU value (12.29.1.1.1) and a TransmissionOverrun counter.

4 12.29.1.1.1 queueMaxSDU

5 An unsigned integer value, denoting the maximum SDU size supported by the queue.

6 12.29.1.1.2 TransmissionOverrun

7 A counter that is incremented when the implementation detects that the transmission gate associated with a
8 queue has closed and a frame that originated from the queue is still being transmitted by the MAC.

9 12.29.1.2 The gate control list structure and data types

10 The AdminControlList and OperControlList are ordered lists containing AdminControlListLength or
11 OperControlListLength entries, respectively. Each entry represents a gate operation as defined in Table 8-7.
12 Each entry in the list is structured as a GateControlEntry (12.29.1.2.1).

13 12.29.1.2.1 GateControlEntry

14 A GateControlEntry consists of an operation name, followed by two parameters associated with the
15 operation, as detailed in Table 8-7. The first parameter is a gateStatesValue (12.29.1.2.2); the second
16 parameter is a timeIntervalValue (12.29.1.2.3).

1 **12.29.1.2.2 gateStatesValue**

2 A list of up to 8 tuples, one for each traffic class supported by the Port, each of which consists of a traffic
3 class number in the range 0–7 and a gate state that is an enumerated value representing the gate state, *open*
4 or *closed*, for that traffic class (see GateState in Table 8-7).

5 **12.29.1.2.3 timeIntervalValue**

6 An unsigned integer, denoting a TimeInterval in nanoseconds (see TimeInterval in Table 8-7).

7 **12.29.1.3 RationalNumber**

8 A rational number represented by an integer numerator and an integer denominator.

9 **12.29.1.4 PTPtime**

10 A time value, expressed as a PTP timescale (see IEEE Std 802.1AS).

11 **12.29.1.5 SupportedListMax**

12 The maximum value supported by this Port of the AdminControlListLength and OperControlListLength
13 parameters. It is available for use by schedule computation software to determine the port's control list
14 capacity prior to computation.

15 **12.29.1.6 SupportedCycleMax**

16 The maximum value supported by this Port of the AdminCycleTime (8.6.9.4.3) and OperCycleTime
17 (8.6.9.4.19) parameters.

18 **12.29.1.7 SupportedIntervalMax**

19 The maximum value supported by this Port of the timeIntervalValue (12.29.1.2.3) parameter.

20 **12.29.2 Timing points for scheduled traffic**

21 Table 12-32 specifies times (e.g., AdminBaseTime) with reference to the on-the-wire timing point at which
22 the start of a frame crosses the boundary between the physical network media and the PHY. This is the
23 message timestamp point specified by IEEE Std 802.1AS for various media.

24 Figure 12-6 shows both the on-the-wire timing point and the transmission selection timing point within the
25 interface stack (above MAC and PHY) that is used for gate open/close as described in 8.6.8.4. Each timing
26 point will have variance. A delay exists between the transmission selection timing point and the on-the-wire
27 timing point. This delay will have variance that is bounded (minimum/maximum).

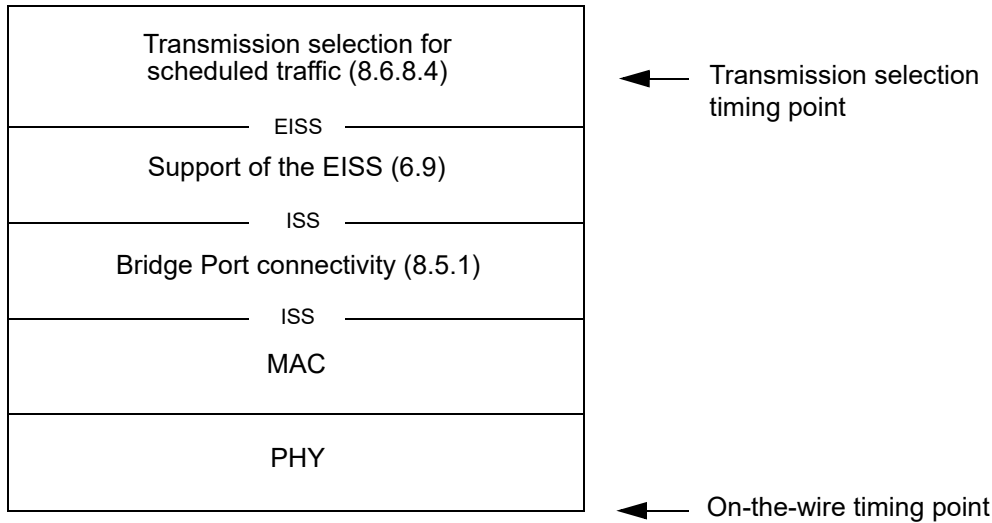


Figure 12-6—Timing points for scheduled traffic

1 The Bridge contains the information needed in order to compute the minimum/maximum delay from the
2 transmission selection timing point to the on-the-wire timing point. The Bridge shall adjust using the
3 minimum delay, such that a frame's on-the-wire timing point occurs no earlier than the gate events
4 represented in managed objects.

5 NOTE 1—As an example, consider an IEEE 802.3 port configured to transmit a single traffic class with known frame
6 lengths, where the managed objects are configured for a cycle of open for two milliseconds, then closed for two
7 milliseconds. If a one millisecond burst of frames is transmitted through the Bridge to the port, the preamble and SFD
8 can be transmitted during the closed window, but no other symbol of a frame can be transmitted from the PHY during
9 the closed window (including the FCS).

10 NOTE 2—Although the managed objects apply to a Bridge, the preceding specification of on-the-wire timing point can
11 be applied to an end station.

12 12.30 Managed objects for frame preemption

13 The Bridge enhancements for support of frame preemption are defined in 8.6.8, 8.6.8.4, and 6.7.2.

14 This managed resource comprises the following object:

- 15 a) Frame Preemption Parameter Table (12.30.1)

16 12.30.1 Frame Preemption Parameter Table

17 There is one Frame Preemption Parameter table per Port of a Bridge Component or end station. Each table
18 row contains a set of parameters that supports the enhancements for frame preemption (6.7.2), as detailed in
19 Table 12-33. Rows in the table can be created or removed dynamically in implementations that support
20 dynamic configuration of ports and components.

21 12.30.1.1 framePreemptionStatusTable structure and data types

22 The framePreemptionStatusTable (6.7.2) consists of 8 framePreemptionAdminStatus values (12.30.1.1.1),
23 one per priority.

Table 12-33—Frame Preemption Parameter table

Name	Data type	Operations supported ^a	Conformance ^b	References
framePreemptionStatusTable	sequence of framePreemptionAdminStatus values	RW	B, E	6.7.2, 12.30.1.1, 12.30.1.1.1.
holdAdvance	Integer, nanoseconds	R	B, E	Table 8-7, 12.30.1.2
releaseAdvance	Integer, nanoseconds	R	B, E	Table 8-7, 12.30.1.3
preemptionActive	Boolean	R	B, E	12.30.1.4
holdRequest	Integer {hold (1), release (2)}	R	B, E	Table 8-7, 12.30.1.5

^a R = Read only access; RW = Read/Write access

^b B = Required for Bridge or Bridge Component support of enhancements for frame preemption; E = Required for end station support of enhancements for frame preemption.

1 12.30.1.1.1 framePreemptionAdminStatus

2 This parameter is the administrative value of the preemption status for the priority. It takes value *express* if
3 frames queued for the priority are to be transmitted using the express service for the Port, or *preemptable* if
4 frames queued for the priority are to be transmitted using the preemptable service for the Port and
5 preemption is enabled for the Port.

6 Priorities that all map to the same traffic class should be constrained to use the same value of preemption
7 status.

8 12.30.1.2 holdAdvance object

9 The holdAdvance object contains an integer value representing the maximum number of nanoseconds that
10 can elapse between issuing a HOLD (12.30.1.5) to the MAC and the MAC ceasing to transmit any
11 preemptable frame that is in the process of transmission or any preemptable frames that are queued for
12 transmission, including any MAC-specific delay before transmission of an express frame could start once
13 preemptable frame transmission has ceased. This object exists per Port, and is a characteristic of the
14 underlying MAC.

15 12.30.1.3 releaseAdvance object

16 The releaseAdvance object contains an integer value representing the maximum number of nanoseconds that
17 can elapse between issuing a RELEASE (12.30.1.5) to the MAC and the MAC being ready to resume
18 transmission of preemptable frames, in the absence of there being any express frames available for
19 transmission. This object exists per Port, and is a characteristic of the underlying MAC.

20 12.30.1.4 preemptionActive object

21 TRUE if preemption is both supported by the MAC and currently active.

1 **12.30.1.5 holdRequest object**

2 The holdRequest object contains an enumerated integer value, with *hold (1)* indicating that a
3 MM_CTL.request(HOLD) has been issued, and release (2) indicating that a MM_CTL.request(RELEASE)
4 has been issued. This object exists per Port.

5 NOTE 1—In order that the operations enumerated in the table occur at the specified times (8.6.8.4), an implementation
6 needs to trigger the MM_CTL.request in advance, as appropriate.

7 NOTE 2—In order to determine support of frame preemption by the Bridge, a network management
8 application could attempt to read the preemptionActive object. If the read operation returns an error, frame
9 preemption is not supported. Determining whether the MAC supports preemption could similarly be achieved
10 by examining the objects in the MAC MIB.

12.31 Managed objects for per-stream classification and metering

The Bridge enhancements for support of per-stream classification and metering are defined in 8.6.5.2, 8.6.5.3, 8.6.5.4, 8.6.5.5, and 8.6.5.6. The associated state machines are defined in 8.6.10 and 8.6.11.

This managed resource comprises the following objects:

- a) The Stream Parameter Table (12.31.1)
- b) The Stream Filter Instance Table (12.31.2)
- c) The Stream Gate Instance Table (12.31.3)
- d) The Flow Meter Instance Table (12.31.4)
- e) The Scheduler Instance Table (12.31.5)
- f) The Scheduler Group Instance Table (12.31.6)
- g) The Scheduler Port Parameter Table (12.31.7)
- h) The Scheduler Timing Characteristics Table (12.31.8)

12.31.1 The Stream Parameter Table

There is one Stream Parameter Table per Bridge Component. The table contains a set of parameters that supports PSFP (8.6.5.2.1), ATS (8.6.5.2.2), and CI (8.6.5.2.3), as detailed in Table 12-34. Tables can be created or removed dynamically in implementations that support dynamic configuration of Bridge components.

Table 12-34—The Stream Parameter Table

Name	Data type	Operations supported ^a	Conformance ^b	References
MaxStreamFilterInstances	integer	R	PSFP, ATS, CI	8.6.5.3, 12.31.2
MaxStreamGateInstances	integer	R	PSFP, ATS, CI	8.6.5.4, 12.31.3
MaxFlowMeterInstances	integer	R	PSFP, ats	8.6.5.5, 12.31.4
SupportedListMax	integer	R	PSFP, ats	8.6.5.4, 12.31.4
MaxSchedulerInstances	integer	R	psfp, ATS	8.6.5.4, 12.31.4
MaxSchedulerGroupInstances	integer	R	psfp, ATS	8.6.5.4, 12.31.4
SupportedCycleMax	RationalNumber (seconds)	R	PSFP, ATS	8.6.5.4, 12.29.1.3, 12.29.1.6
SupportedIntervalMax	Integer	R	PSFP, ATS	8.6.5.4, 12.29.1.7

^a R= Read only access; RW = Read/Write access.

^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.

psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.

ATS = Required for Bridge or Bridge Component support of ATS.

ats = Optional for Bridge or Bridge Component support of ATS.

CI = Required for Bridge or Bridge Component support of CI.

12.31.1.1 MaxStreamFilterInstances

The maximum number of Stream Filter instances supported by this Bridge Component (8.6.5.3).

1 **12.31.1.2 MaxStreamGateInstances**

2 The maximum number of Stream Gate instances supported by this Bridge Component (8.6.5.4).

3 **12.31.1.3 MaxFlowMeterInstances**

4 The maximum number of Flow Meter instances supported by this Bridge Component (8.6.5.5).

5 **12.31.1.4 SupportedListMax**

6 The maximum value supported by this Bridge Component of the AdminControlListLength (8.6.9.4.6) and
7 OperControlListLength (8.6.9.4.22) parameters. It is available for use by schedule computation software to
8 determine the Bridge Component's control list capacity prior to computation.

9 **12.31.1.5 MaxSchedulerInstances**

10 The maximum number of ATS scheduler instances supported by this Bridge Component (8.6.5.6).

11 **12.31.1.6 MaxSchedulerGroupInstances**

12 The maximum number of ATS scheduler group instances supported by this Bridge Component (8.6.5.6).

13 **12.31.1.7 SupportedCycleMax**

14 The maximum value supported by this Port of the AdminCycleTime (8.6.9.4.3) and OperCycleTime
15 (8.6.9.4.19) parameters.

16 **12.31.1.8 SupportedIntervalMax**

17 The maximum value supported by this Bridge Component of the timeIntervalValue (12.31.3.2.4) parameter.

18 **12.31.2 The Stream Filter Instance Table**

19 There is one Stream Filter Instance Table per Bridge Component. Each table row contains a set of
20 parameters that defines a single Stream Filter (8.6.5.3), as detailed in Table 12-35. The table rows form an
21 ordered list of filter instances, the order being determined by the StreamFilterInstance parameter. Tables can
22 be created or removed dynamically in implementations that support dynamic configuration of Bridge
23 Components. Rows in the table can be created or removed dynamically in implementations that support
24 dynamic configuration of stream filters.

25 **12.31.2.1 StreamFilterInstance**

26 An integer index value that determines the place of the stream filter in the ordered list of stream filter
27 instances. The values of StreamFilterInstance are ordered according to their integer value; smaller values
28 appear earlier in the ordered list.

29 **12.31.2.2 stream_handle specification data type**

30 The stream_handle specification data type allows either of the following to be represented:

- 31 a) A stream_handle value, represented as an integer
- 32 b) The wild card value
- 33 c) If congestion isolation is supported, the null-handle value.

Table 12-35—Stream Filter Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamFilterInstance	integer	R	PSFP, ATS, CI	8.6.5.3
StreamHandleSpec	stream_handle specification	RW	PSFP, ATS, CI	8.6.5.3
PrioritySpec	priority specification	RW	PSFP, ATS, CI	8.6.5.3
MaximumSDUSize	integer	RW	PSFP, ATS	8.6.5.3.1, 12.31.2.5
StreamGateInstanceID	integer	RW	PSFP, ATS, CI	8.6.5.2, 8.6.5.4
FlowMeterInstanceID	integer	RW	PSFP, ats	8.6.5.5, 12.31.2.5
FlowMeterEnable	Boolean	RW	PSFP, ats	
SchedulerInstanceID	integer	RW	psfp, ATS	
SchedulerEnable	Boolean	RW	psfp, ATS	
MatchingFramesCount	counter	R	PSFP, ats, ci	8.6.5.3
PassingFramesCount	counter	R	PSFP, ats	8.6.5.3, 8.6.5.4
NotPassingFramesCount	counter	R	PSFP, ats	8.6.5.3, 8.6.5.4
PassingSDUCount	counter	R	PSFP, ats	8.6.5.3, 8.6.5.3.1
NotPassingSDUCount	counter	R	PSFP, ats	8.6.5.3, 8.6.5.3.1
REDFramesCount	counter	R	PSFP, ats	8.6.5.3
StreamBlockedDueToOversizeFrameEnable	Boolean	RW	PSFP, ATS	8.6.5.3, 8.6.5.3.1
StreamBlockedDueToOversizeFrame	Boolean	RW	PSFP, ATS	8.6.5.3, 8.6.5.3.1

^a R= Read only access; RW = Read/Write access.

^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.

psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.

ATS = Required for Bridge or Bridge Component support of ATS.

ats = Optional for Bridge or Bridge Component support of ATS.

CI = Required for Bridge Component or end station support of CI.

ci = Optional for Bridge Component or end station support of CI.

1 12.31.2.3 priority specification data type

2 The priority specification data type allows either of the following to be represented:

- 3 a) A priority value, represented as an integer
- 4 b) The wild card value

12.31.2.4 MaxSDUSize

The MaxSDUSize parameter defines the maximum SDU size to be accepted by the stream filter (8.6.5.3). A value of 0 disables the maximum SDU size filtering for frames associated with the stream filter.

12.31.2.5 StreamGateInstanceID

The StreamGateInstanceID parameter identifies the stream gate instance (12.31.3) that is associated with the stream filter. The relationship between stream filters and stream gates is many to one; a given stream filter can be associated with only one stream gate, but there can be multiple stream filters associated with a given stream gate.

12.31.2.6 FlowMeterInstanceID and FlowMeterEnable

If FlowMeterEnable is set to TRUE, the FlowMeterInstanceID parameter identifies the flow meter instance (12.31.4) that is associated with the stream filter. The relationship between stream filters and flow meters is many to one; a given stream filter can be associated with only one flow meter, but there can be multiple stream filters associated with a given flow meter. If FlowMeterEnable is set to FALSE, no flow meter instance is associated with the stream filter.

12.31.2.7 SchedulerInstanceID and SchedulerEnable

If SchedulerEnable is set to TRUE, the SchedulerInstanceID parameter identifies the ATS scheduler instance (12.31.5) that is associated with the stream filter. The relationship between stream filters and ATS schedulers is many to one; a given stream filter can be associated with only one ATS scheduler, but there can be multiple stream filters associated with a given ATS scheduler instance. If SchedulerEnable is set to FALSE, no ATS scheduler instance is associated with the stream filter.

12.31.3 The Stream Gate Instance Table

There is one Stream Gate Instance Table per Bridge Component. Each table row contains a set of parameters that defines a single Stream Gate (8.6.5.4), as detailed in Table 12-36. Tables can be created or removed dynamically in implementations that support dynamic configuration of Bridge Components. Rows in the table can be created or removed dynamically in implementations that support dynamic configuration of stream gates.

Table 12-36—The Stream Gate Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamGateInstance	integer	R	PSFP, ATS, CI	8.6.5.4
StreamGateEnabled	Boolean	RW	PSFP, ATS, CI	8.6.9.4.14
StreamGateAdminGateStates	StreamGateStatesValue	RW	PSFP, ATS, CI	8.6.10.4, 12.29.1.2.2
StreamGateOperGateStates	StreamGateStatesValue	R	PSFP, ATS, CI	8.6.10.5, 12.29.1.2.2
StreamGateAdminControlListLength	unsigned integer	RW	PSFP, ats	8.6.9.4.6, 12.31.3.2
StreamGateOperControlListLength	unsigned integer	R	PSFP, ats	8.6.9.4.22, 12.31.3.2

Table 12-36—The Stream Gate Instance Table (*continued*)

Name	Data type	Operations supported ^a	Conformance ^b	References
StreamGateAdminControlList	sequence of StreamGateGateControl Entry	RW	PSFP, ats	8.6.9.4.2, 12.31.3.2, 12.31.3.2.2
StreamGateOperControlList	sequence of StreamGateGateControl Entry	R	PSFP, ats	8.6.9.4.18, 12.31.3.2, 12.31.3.2.2
StreamGateAdminCycleTime	RationalNumber	RW	PSFP, ats	8.6.9.4.3, 12.29.1.3
StreamGateOperCycleTime	RationalNumber (seconds)	R	PSFP, ats	8.6.9.4.19, 12.29.1.3
StreamGateAdminCycleTime Extension	Integer (nanoseconds)	RW	PSFP, ats	8.6.9.4.4
StreamGateOperCycleTimeExtension	Integer (nanoseconds)	R	PSFP, ats	8.6.9.4.20
StreamGateAdminBaseTime	PTPtime	RW	PSFP, ats	8.6.9.4.1, 12.29.1.4
StreamGateOperBaseTime	PTPtime	R	PSFP, ats	8.6.9.4.17, 12.29.1.4
StreamGateConfigChange	Boolean	RW	PSFP, ats	8.6.9.4.7
StreamGateConfigChangeTime	PTPtime	R	PSFP, ats	8.6.9.4.9, 12.29.1.4
StreamGateTickGranularity	Integer (tenths of nanoseconds)	R	PSFP, ats	8.6.9.4.16
StreamGateCurrentTime	PTPtime	R	PSFP, ats	8.6.9.4.10, 12.29.1.4
StreamGateConfigPending	Boolean	R	PSFP, ats	8.6.9.3, 8.6.9.4.8
StreamGateConfigChangeError	Integer	R	PSFP, ats	8.6.9.3.1
StreamGateAdminIPV	IPV	RW	PSFP, ATS, CI	8.6.5.4, 8.6.10.6, 12.31.3.3
StreamGateOperIPV	IPV	RW	PSFP, ats, CI	8.6.5.4, 8.6.10.7, 12.31.3.3
StreamGateGateClosedDueToInvalid Rx-Enable	Boolean	RW	PSFP, ats	8.6.5.4
StreamGateGateClosedDueToInvalid Rx	Boolean	RW	PSFP, ats	8.6.5.4
StreamGateGateClosedDueToOctets ExceededEnable	Boolean	RW	PSFP, ats	8.6.5.4
StreamGateGateClosedDueToOctets Exceeded	Boolean	RW	PSFP, ats	8.6.5.4

^a R= Read only access; RW = Read/Write access.

- ^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.
psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.
ATS = Required for Bridge or Bridge Component support of ATS.
ats = Optional for Bridge or Bridge Component support of ATS.
CI = Required for Bridge Component or end station support of CI.

1 **12.31.3.1 StreamGateInstance**

- 2 An integer table index that allows the stream gate to be referenced from Stream Filter Instance Table entries.

3 **12.31.3.2 The gate control list structure and data types**

- 4 The AdminControlList and OperControlList are ordered lists containing AdminControlListLength or
5 OperControlListLength entries, respectively. Each entry represents a gate operation as defined in Table 8-4.
6 Each entry in the list is structured as a GateControlEntry (12.31.3.2.2).

7 **12.31.3.2.1 StreamGateStatesValue**

- 8 The StreamGateStatesValue indicates the desired gate state, *open* or *closed*, for the stream gate.

9 **12.31.3.2.2 StreamGateControlEntry**

- 10 A StreamGateControlEntry consists of an operation name, followed by three mandatory parameters and one
11 optional parameter associated with the operation, as detailed in Table 8-4. The first parameter is a
12 StreamGateStatesValue (8.6.10.5, 12.31.3.2.1); the second parameter is an IPV value (8.6.10.7, 12.31.3.2.3);
13 the third parameter is a timeIntervalValue (8.6.9.4.23, 12.31.3.2.4); and the fourth parameter is an
14 IntervalOctetMaxValue (8.6.10.1, 12.31.3.2.5). IntervalOctetMaxValue is optional.

15 **12.31.3.2.3 IPV value**

- 16 The IPV value indicates the IPV (12.31.3.3) to be associated with frames that pass the gate (8.6.10.7).

17 **12.31.3.2.4 timeIntervalValue**

- 18 An unsigned integer, denoting a TimeInterval in nanoseconds (see TimeInterval in Table 8-4).

19 **12.31.3.2.5 IntervalOctetMaxValue**

- 20 An unsigned integer, denoting an IntervalOctetMax in MSDU octets (see IntervalOctetMax in Table 8-4).

21 **12.31.3.3 The internal priority value specification (IPV) data type**

- 22 The IPV data type represents an IPV value (8.6.5.4); this is either the null value or an internal priority value.

23 **12.31.3.4 Representation of times**

- 24 Table 12-36 specifies times (e.g., StreamGateAdminBaseTime) with reference to the on-the-wire timing
25 point at which the start of a frame crosses the boundary between the physical network media and PHY. This
26 is the message timestamp point specified by IEEE Std 802.1AS for various media.

- 27 Figure 12-7 shows both the on-the-wire timing point and the Stream Gate timing point within the interface
28 stack (above MAC and PHY) that is used for gate open/close as described in 8.6.5.4. Each timing point will
29 have variance. A delay exists between the on-the-wire timing point and the Stream Gate timing point. This
30 delay will have variance that is bounded (minimum/maximum).

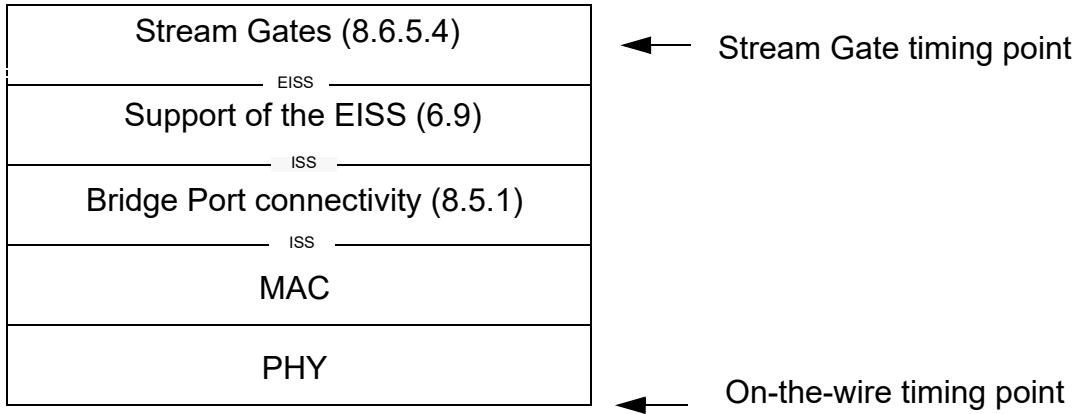


Figure 12-7—Timing points for PSFP

1 The Bridge contains the information needed in order to compute the minimum/maximum delay from the
2 on-the-wire timing point to the Stream Gate timing point. The Bridge shall adjust using the maximum delay,
3 such that a frame's on-the-wire timing point occurs no later than the gate events represented in managed
4 objects.

5 NOTE—Although the managed objects apply to a Bridge, the preceding specification of on-the-wire timing point can be
6 applied to an end station.

7 12.31.4 The Flow Meter Instance Table

8 There is one Flow Meter Instance Table per Bridge Component. Each table row contains a set of parameters
9 that defines a single Flow Meter Instance (8.6.5.5), as detailed in Table 12-37. Tables can be created or
10 removed dynamically in implementations that support dynamic configuration of Bridge Components. Rows
11 in the table can be created or removed dynamically in implementations that support dynamic configuration
12 of flow meters.

13 12.31.5 The Scheduler Instance Table

14 There is one Scheduler Instance Table per Bridge Component. Each table row in the Scheduler Instance
15 Table comprises a set of parameters that defines a single ATS scheduler instance, as detailed in Table 12-38.

16 NOTE—ATS scheduler groups establish the relationship between ATS scheduler instances and the per port management
17 variables (12.31.7), as described in 8.6.5.6. As a result, the scheduler instance table does not contain references to ports.

18 12.31.5.1 SchedulerInstanceID

19 An integer table index that allows the ATS scheduler instance to be referenced from Stream Filter Instance
20 Table entries.

21 12.31.5.2 CommittedBurstSize

22 As specified in 8.6.11.3.5.

23 12.31.5.3 CommittedInformationRate

24 As specified in 8.6.11.3.6.

Table 12-37—The Flow Meter Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
FlowMeterInstanceID	integer	R	PSFP, ats	8.6.5.2, 8.6.5.5
CIR	integer, bit/second	RW	PSFP, ats	8.6.5.5
CBS	integer, octets	RW	PSFP, ats	8.6.5.5
EIR	integer, bit/second	RW	PSFP, ats	8.6.5.5
EBS	integer, octets	RW	PSFP, ats	8.6.5.5
CF	integer, 0 or 1	RW	PSFP, ats	8.6.5.5
CM	enumerated, color-blind or color-aware	RW	PSFP, ats	8.6.5.5
DropOnYellow	Boolean	RW	PSFP, ats	8.6.5.5
MarkAllFramesRedEnable	Boolean	RW	PSFP, ats	8.6.5.5
MarkAllFramesRed	Boolean	RW	PSFP, ats	8.6.5.5

^a R= Read only access; RW = Read/Write access.

^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.
psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.
ATS = Required for Bridge or Bridge Component support of ATS.
ats = Optional for Bridge or Bridge Component support of ATS.

Table 12-38—The Scheduler Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
SchedulerInstanceID	integer	R	psfp, ATS	8.6.5.6, 12.31.5.1
CommittedBurstSize	integer, bits	RW	psfp, ATS	8.6.5.6, 8.6.11.3.5, 12.31.5.2
CommittedInformationRate	integer, bits/s	RW	psfp, ATS	8.6.5.6, 8.6.11.3.6, 12.31.5.3
SchedulerGroupInstanceID	integer	RW	psfp, ATS	8.6.5.6, 12.31.5.4

^a R= Read only access; RW = Read/Write access.

^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.
psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.
ATS = Required for Bridge or Bridge Component support of ATS.
ats = Optional for Bridge or Bridge Component support of ATS.

12.31.5.4 SchedulerGroupInstanceID

The SchedulerGroupInstanceID parameter identifies the ATS scheduler group (12.31.6) that is associated with the ATS scheduler instance. Multiple scheduler instance can be associated to one ATS scheduler group, as detailed in 8.6.5.6.

12.31.6 The Scheduler Group Instance Table

There is one Scheduler Group Instance Table per Bridge Component. Each table row in the Scheduler Group Instance Table comprises a set of parameters that defines a single ATS scheduler group instance (8.6.5.6), as detailed in Table 12-39.

Table 12-39—The Scheduler Group Instance Table

Name	Data type	Operations supported ^a	Conformance ^b	References
SchedulerGroupInstanceID	integer	R	psfp, ATS	8.6.5.6, 12.31.6.1
MaxResidenceTime	integer, nanoseconds	RW	psfp, ATS	8.6.5.6, 12.31.6.2

^a R = Read only access; RW = Read/Write access.

^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.

psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.

ATS = Required for Bridge or Bridge Component support of ATS.

ats = Optional for Bridge or Bridge Component support of ATS.

12.31.6.1 SchedulerGroupInstanceID

An integer table index that allows the ATS scheduler group instance to be referenced from Scheduler Instance Table entries.

12.31.6.2 MaxResidenceTime

As specified in 8.6.11.3.13.

12.31.7 The Scheduler Port Parameter Table

There is one Scheduler Port Parameter Table per Bridge. Each table row in the Scheduler Port Parameter Table comprises a set of parameters shared by all ATS scheduler instance associated with a reception Port, as detailed in Table 12-40.

Table 12-40—The Scheduler Port Parameter Table

Name	Data type	Operations supported ^a	Conformance ^b	References
PortNumber	integer	R	psfp, ATS	12.31.7.1
DiscardedFramesCount	integer	R	psfp, ATS	8.6.5.6, 8.6.11.3.7, 12.31.7.2

^a R = Read only access; RW = Read/Write access.

^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.

psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.

ATS = Required for Bridge or Bridge Component support of ATS.

ats = Optional for Bridge or Bridge Component support of ATS.

12.31.7.1 PortNumber

An unique index of the associated Bridge Port (12.4.2).

12.31.7.2 DiscardedFramesCount

As specified in 8.6.5.6 and 8.6.11.3.7.

12.31.8 The Scheduler Timing Characteristics Table

There is one Scheduler Timing Characteristics Table per Bridge Component. Each row in this table comprises the timing characteristics of a reception Port transmission Port pair, as detailed in Table 12-41.

Table 12-41—The Timing Characteristics Table

Name	Data type	Operations supported ^a	Conformance ^b	References
ReceptionPortNumber	integer	R	psfp, ATS	12.31.8.1
TransmissionPortNumber	integer	R	psfp, ATS	12.31.8.2
ClockOffsetVariationMax	integer	R	psfp, ATS	8.6.11.2, 12.31.8.3
ClockRateDeviationMax	integer	R	psfp, ATS	8.6.11.2, 12.31.8.4
ArrivalRecognitionDelayMax	integer	R	psfp, ATS	8.6.11.3.1, 12.31.8.5
ProcessingDelayMin	integer	R	psfp, ATS	8.6.11.3.2, 12.31.8.6
ProcessingDelayMax	integer	R	psfp, ATS	8.6.11.3.2, 12.31.8.7

^a R = Read only access; RW = Read/Write access.

^b PSFP = Required for Bridge, Bridge Component, or end station support of PSFP.

psfp = Optional for Bridge, Bridge Component, or end station support of PSFP.

ATS = Required for Bridge or Bridge Component support of ATS.

ats = Optional for Bridge or Bridge Component support of ATS.

12.31.8.1 ReceptionPortNumber

An unique index of the associated reception Port of the Bridge (12.4.2).

12.31.8.2 TransmissionPortNumber

An unique index of the associated transmission Port of the Bridge (12.4.2).

1 **12.31.8.3 ClockOffsetVariationMax**

2 As specified in 8.6.11.2, in nanoseconds, rounded to the next numerically higher representable value.

3 **12.31.8.4 ClockRateDeviationMax**

4 As specified in 8.6.11.2, in ppm, rounded to the next numerically higher representable value.

5 **12.31.8.5 ArrivalRecognitionDelayMax**

6 As specified in 8.6.11.3.1, in nanoseconds, rounded to the next numerically higher representable value.

7 **12.31.8.6 ProcessingDelayMin**

8 As specified in 8.6.11.3.2, in nanoseconds, rounded to the next numerically lower representable value.

9 **12.31.8.7 ProcessingDelayMax**

10 As specified in 8.6.11.3.2, in nanoseconds, rounded to the next numerically higher representable value.

12.32 Stream reservation remote management

Clause 46 defines Time-Sensitive Networking (TSN) configuration, including a Centralized Network Configuration (CNC) station that uses managed objects for configuration of each Bridge. This subclause specifies managed objects within the Bridge that can be used by the TSN CNC station or any other management component.

This managed resource comprises the following objects:

- c) Bridge Delay (12.32.1)
- d) Propagation Delay (12.32.2)
- e) Static Trees (12.32.3)
- f) MRP External Control (12.32.4)

12.32.1 Bridge Delay

This subclause specifies attributes of a managed object that is used to determine the delay of frames as they pass through the Bridge's relay. There is one Bridge Delay managed object per Port pair per traffic class of a Bridge Component. Each set of Bridge Delay attributes is accessed using three indices: ingress Port, egress Port, and traffic class. The set of managed object attributes is shown in Table 12-42.

Table 12-42—Bridge Delay attributes

Name	Data type	Operations supported ^a	Conformance ^b	References
independentDelayMin	unsigned integer	R	B	12.32.1.1
independentDelayMax	unsigned integer	R	B	12.32.1.1
dependentDelayMin	unsigned integer	R	B	12.32.1.2
dependentDelayMax	unsigned integer	R	B	12.32.1.2

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of Stream reservation remote management; b = Optional for Bridge or Bridge Component support of Stream reservation remote management.

For each set of Bridge Delay attributes, the total delay through the Bridge consists of the delay that is independent of frame length, plus the delay that is dependent on frame length.

Each delay is provided as a range, with both minimum attribute and maximum attribute.

The delays represent the worst-case range per the design of the Bridge, and are not measured. If the Bridge design varies based on the configuration of features in the Bridge, the delays are returned according to the current configuration of the Bridge.

NOTE 1—In order to obtain the most accurate delays, the TSN CNC typically configures the Stream from Talker to Listeners, and then reads the delays for the Ports that ingress/egress that Stream. For example, use of IEEE Std 802.1CB could change the implementation in the Bridge, and therefore it is best to configure those features in the Bridge prior to reading the delays.

Each delay is provided as if the corresponding frame encounters zero delay for transmission selection (8.6.8). This occurs when the queue for the frame's traffic class is empty, the frame's traffic class has permission to transmit, and the egress Port is idle (not transmitting).

1 NOTE 2—Computation of delay factors due to use of multiple traffic classes, frame preemption, traffic shaping, and
2 traffic scheduling is the responsibility of the TSN CNC, and therefore these factors are not included in the delay
3 attributes of this managed object. For example, assume that a frame of a TSN Stream encounters 12 μ s of delay due to
4 traffic scheduling (i.e., waiting for gate to open), 5 μ s due to a previous frame of the same traffic class, and 700 ns to
5 800 ns of hardware delay (e.g., PHYs and relay). For this example, independentDelayMin is 700, and
6 independentDelayMax is 800 (i.e., the extra 17 μ s for queuing/scheduling is not included in this managed object).

7 **12.32.1.1 independentDelayMin/Max**

8 These attributes provide the portion of delay that is independent of frame length.

9 Each length-independent delay attribute is an unsigned integer in units of nanoseconds.

10 The independentDelayMin attribute provides the minimum length-independent delay for a frame to forward
11 from ingress Port to egress Port for this traffic class. If the actual delay is a fraction of a nanosecond,
12 independentDelayMin shall be the greatest integer number of nanoseconds that is less than or equal to the
13 actual delay.

14 The independentDelayMax attribute provides the maximum length-independent delay for a frame to forward
15 from ingress Port to egress Port for this traffic class. If the actual delay is a fraction of a nanosecond,
16 independentDelayMax shall be the least integer number of nanoseconds that is greater than or equal to the
17 actual delay.

18 NOTE 1—Since the minimum to maximum range is intended to be worst-case, the rules for handling a fraction of a
19 nanosecond are intended to result in the widest range.

20 The delay begins when the message timestamp point of the ingress frame passes the reference plane marking
21 the boundary between the network media and PHY. The delay ends when the message timestamp point of
22 the egress frame passes the reference plane marking the boundary between the network media and PHY. The
23 message timestamp point is specified by IEEE Std 802.1AS for various media, near the start of the frame.

24 NOTE 2—This delay includes all aspects of length-independent delay for a frame that is forwarded, including handling
25 of error conditions.

26 **12.32.1.2 dependentDelayMin/Max**

27 These attributes provide the portion of delay that is dependent on frame length, where frame length is the
28 number of octets that transfer across the MAC Service interfaces. Each length-dependent delay attribute
29 specifies the time for a single octet of the frame to transfer from ingress to egress.

30 Each length-dependent delay attribute is an unsigned integer in units of picoseconds.

31 The dependentDelayMin attribute provides the minimum length-dependent delay from ingress Port to egress
32 Port for this traffic class. If the actual delay is a fraction of a picosecond, dependentDelayMin shall be the
33 greatest integer number of picoseconds that is less than or equal to the actual delay.

34 The dependentDelayMax attribute provides the maximum length-dependent delay from ingress Port to
35 egress Port for this traffic class. If the actual delay is a fraction of a picosecond, dependentDelayMax shall
36 be the least integer number of picoseconds that is greater than or equal to the actual delay.

37 NOTE—The length-dependent delay typically includes the time to receive and store each octet of the frame, and this
38 time depends on the link speed of the ingress Port. For example, if the ingress Port is operating at 1 Gb/s, both
39 dependentDelayMin and dependentDelayMax can return 8000. If an internal communication mechanism exists within
40 the Bridge's architecture (i.e., in the relay), the time for that mechanism is added to the frame-dependent delay. For
41 example, if a 100 Mb/s serial link connects the ingress and egress Ports, the previous example with 1 Gb/s link speed can
42 return 88000 for both dependentDelayMin and dependentDelayMax.

12.32.2 Propagation Delay

This subclause specifies attributes of a managed object used to determine the delay along the network media (e.g., cable) for a frame transmitted from the specified Port of this station to the neighboring Port on a different station. There is one Propagation Delay managed object per Port of a station (i.e., Bridge or end station). The set of managed object attributes is shown in Table 12-43.

Table 12-43—Propagation Delay attributes

Name	Data type	Operations supported ^a	Conformance ^b	References
txPropagationDelay	unsigned integer	R	BE	12.32.2.1

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of Stream reservation remote management; E = Required for end station support of Stream reservation remote management.

12.32.2.1 txPropagationDelay

The txPropagationDelay begins when the message timestamp point of an egress frame passes the reference plane marking the boundary between the network media and PHY of this station's Port. The txPropagationDelay ends when the message timestamp point of an ingress frame on the neighboring station's Port passes the reference plane marking the boundary between the network media and PHY. The message timestamp point is specified by IEEE Std 802.1AS for various media, near the start of the frame.

The txPropagationDelay attribute is an unsigned integer in units of nanoseconds. If the actual propagation delay is measured to a fraction of a nanosecond, the txPropagationDelay value shall be least integer number of nanoseconds that is greater than or equal to the actual delay.

NOTE 1—Propagation delay in one direction of transmit is not necessarily the same as propagation delay of transmit in the reverse direction. In order to determine if a delay asymmetry exists, management can read the txPropagationDelay attribute on the neighboring station's Port. The txPropagationDelay attribute of the neighboring station's Port effectively provides the propagation delay for receiving frames on this station's Port.

NOTE 2—The txPropagationDelay attribute is typically measured using a time synchronization protocol. For example, when IEEE Std 802.1AS is in use, the value of this attribute can be obtained using the managed objects neighborPropDelay and delayAsymmetry.

NOTE 3—One of the primary purposes of the txPropagationDelay attribute is to add to Bridge Delay in order to compute total latency for a Stream. Although time synchronization protocols based on IEEE Std 1588 typically measure propagation delay to units of 2–16 ns, that degree of precision is not required for txPropagationDelay, since Bridge Delay is expressed as nanoseconds.

12.32.3 Static Trees

This subclause specifies attributes of a managed object used to determine if the static trees feature is supported by the Bridge. There is one Static Trees managed object that applies to the Bridge. The set of managed object attributes is shown in Table 12-44.

Table 12-44—Static Trees attributes

Name	Data type	Operations supported ^a	Conformance ^b	References
staticTreesSupported	Boolean	R	B	12.32.3.1

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component support of Stream reservation remote management.

12.32.3.1 staticTreesSupported

For some use cases, the TSN CNC needs to configure static (persistent) trees through Bridges from Talker to Listeners (e.g., for use of IEEE Std 802.1CB). The Traffic Engineering Multiple Spanning Tree Instance Identifier (TE-MSTID) can be leveraged for these use cases. A VLAN ID is allocated to the TE-MSTID in the MST Configuration Table if it is not under control of either a spanning tree protocol or IS-IS. Static Filtering Entries (8.8.1) specify frame forwarding and filtering for VLAN IDs allocated to the TE-MSTID.

The TE-MSTID feature is used as the basis of a set of features called Provider Backbone Bridge Traffic Engineering (PBB-TE) as specified in 25.10. For many TSN CNCs, use of the TE-MSTID feature is essential, but the additional features of PBB-TE are not necessary. The TSN CNC uses the staticTreesSupported attribute to determine that TE-MSTID is supported by the Bridge. A Bridge that supports PBB-TE without support of Stream reservation remote management is not required to implement the staticTreesSupported attribute.

A Bridge that supports Stream reservation remote management shall return the value TRUE from the staticTreesSupported attribute, and shall support the TE-MSTID as specified in 5.5.2.

NOTE—In order to determine support for TE-MSTID by the Bridge, the TSN CNC attempts to read the staticTreesSupported attribute. If the read operation returns an error, or if the read returns FALSE, then the TSN CNC concludes that TE-MSTID is not supported.

12.32.4 MRP External Control

In the Centralized network/distributed user model shown in Figure 46-2, user configuration data is exchanged in a distributed manner by each Talker/Listener end station, and the network is configured in a centralized manner (using a CNC). A Bridge near the Talker/Listener acts as a proxy with the CNC. One protocol solution for this model is to use the MRP protocol for user data (i.e., MSRP, MVRP, and/or MMRP applications), and a network management protocol as the proxy with the network manager (i.e., CNC). This enables the end stations to use the same MRP protocol as the Fully distributed model of Figure 46-1.

This subclause specifies attributes of a managed object that the network manager uses to:

- a) Disable MRP attribute propagation (MAP) for the MRP Participant of a Bridge Port.
- b) Read MRP attribute registrations that the MRP Participant receives.
- c) Write MRP attribute values for the MRP Participant to declare.

There is one MRP External Control managed object for each MRP Participant (per Port per MRP application per MAP Context, as specified in 10.2 and 10.3.1).

For the MSRP application, although 35.2.4.5 implies that a single MAP Context exists, the specifications of 35.1.3.1 ensure that every distinct value of vlan_identifier propagates along the corresponding VLAN context (10.3.1). Therefore, for the purposes of indexing this managed object, MSRP uses the vlan_identifier as the MAP Context identifier.

The set of managed object attributes is shown in Table 12-45.

12.32.4.1 externalControl

When the externalControl attribute is FALSE, MRP attributes propagate on the MRP Participant according to the specifications for MAP in 10.3 and specifications of the MRP Application (e.g., 35.2.4). Ports with the externalControl attribute FALSE are considered as candidates for the MRP Application's MAP Context. The remaining attributes of this managed object (12.32.4.2 through 12.32.4.8) are ignored by Ports with the externalControl attribute FALSE.

Table 12-45—MRP External Control attributes

Name	Data type	Operations supported ^a	Conformance ^b	References
externalControl	Boolean	RW	B	12.32.4.1
indicationList	octet string	R	B	12.32.4.2
indicationListLength	unsigned integer	R	B	12.32.4.3
indicationChangeCounter	counter	R	B	12.32.4.4
adminRequestList	octet string	RW	B	12.32.4.5
adminRequestListLength	unsigned integer	RW	B	12.32.4.6
operRequestList	octet string	R	B	12.32.4.7
operRequestListLength	unsigned integer	R	B	12.32.4.8

^a R = Read only access; RW = Read/Write access.

^b B = Required for Bridge or Bridge Component when both Stream reservation remote management and the corresponding MRP application are supported.

1 When the externalControl attribute is TRUE, the MRP Participant is removed from the MRP Application's
2 MAP Context. The MRP Participant performs all other aspects of MRP, including MRP operation (10.6),
3 MRP specifications (10.7), and MRPDUs encodings (10.8). The remaining attributes of this managed object
4 (12.32.4.2 through 12.32.4.8) act as an application component as specified in 10.2. The application
5 component stores MAD indications for registration received on the Port, and invokes MAD requests for
6 declarations on the Port.

7 The default value of the externalControl attribute is FALSE.

8 **12.32.4.2 indicationList**

9 As specified in 10.2, the MAD_Join.indication primitive is invoked in the application component when
10 MRP detects that an MRP attribute has joined (i.e., registered), and the MAD_Leave.indication primitive is
11 invoked in the application component when MRP detects that the MRP attribute has left (i.e., no longer
12 registered). When the externalControl attribute is TRUE, the indicationList attribute stores the list of all
13 joined MRP attributes for the MRP Participant.

14 When the MAD_Join.indication primitive is invoked, the MRP Participant forms a sequence of octets,
15 *temporaryIndication*, from the MAD_Join.indication parameters as follows:

- 16 d) The first octet of *temporaryIndication* contains the *attribute_type* parameter.
- 17 e) The second octet of *temporaryIndication* contains the length of the *attribute_value* parameter as a
18 single octet.
- 19 f) Subsequent octets of *temporaryIndication* contain the sequence of octets for the *attribute_value*
20 parameter.

21 NOTE 1—The *new* parameter is not used for indicationList.

22 The MRP Participant searches the indicationList to determine if the same octets as *temporaryIndication*
23 (i.e., same *attribute_type* and *attribute_value*) exist. If no match is found, the contents of
24 *temporaryIndication* are appended to the end of indicationList. The result of this specification is that the
25 indicationList consists of an octet string that represents multiple MAD_Join.indication primitives in the
26 order initially invoked.

1 When the MAD_Leave.indication primitive is invoked, the MRP Participant forms a sequence of octets,
2 *temporaryIndication*, as specified for MAD_Join.indication. The MRP Participant searches the
3 indicationList to determine if the same octets as *temporaryIndication* exist. If a match is found, the
4 attribute's octets (i.e., matching *temporaryIndication*) are removed from the indicationList.

5 NOTE 2—When the indication's *attribute_type* is Listener Vector Attribute Type, the Listener's Declaration Type
6 (35.2.1.3) is encoded as the last octet of *attribute_value*, even though the Declaration Type is not encoded in the MRP
7 FirstValue (10.8.2.7), but the MRP FourPackedEvent (10.8.2.10.2). For the Listener Enhanced Vector Attribute Type,
8 the Listener's Declaration Type is not encoded as the last octet, since the equivalent information is available in the
9 ListenerStatus element (46.2.5.1).

10 The default value of the indicationList attribute is the empty octet string.

11 NOTE 3—The attribute encoding of indicationList does not correspond directly to the MRPDU. For example, if the
12 received MRPDU uses NumberOfValues of three, then the MRPDU encodes only one attribute value (FirstValue),
13 whereas indicationList encodes three distinct attribute type/length/value triplets.

14 When the externalControl attribute is FALSE, the indicationList attribute is ignored by the MRP Participant,
15 and returns the empty octet string.

16 **12.32.4.3 indicationListLength**

17 The indicationListLength attribute returns the number of octets in the indicationList attribute.

18 When the externalControl attribute is FALSE, the indicationListLength attribute returns zero.

19 **12.32.4.4 indicationChangeCounter**

20 The indicationChangeCounter attribute increments by one for each change to indicationList as specified in
21 12.32.4.2.

22 The default value of the indicationChangeCounter attribute is zero.

23 When the externalControl attribute is FALSE, the indicationChangeCounter attribute returns zero.

24 **12.32.4.5 adminRequestList**

25 The adminRequestList attribute provides the administrative value of the current list of MAD requests for the
26 MRP Participant (operRequestList). Changes to adminRequestList result in invocation of
27 MAD_Join.request and MAD_Leave.request primitives (12.32.4.7). Each attribute in adminRequestList is
28 encoded as the *attribute_type* parameter as a single octet, followed by the length of the *attribute_value*
29 parameter as a single octet, followed by a sequence of octets for the *attribute_value* parameter.

30 NOTE—The *new* parameter is not used for adminRequestList.

31 The default value of the adminRequestList attribute is the empty octet string.

32 When the externalControl attribute is TRUE, the adminRequestList attribute is copied to the
33 operRequestList attribute as soon as possible according to the implementation. For example, the
34 implementation could copy the list between each MRPDU transmit, or it could copy the list after all
35 attributes in the previous operRequestList complete transmission as MRPDUs.

36 When the externalControl attribute is FALSE, the adminRequestList attribute is ignored by the MRP
37 Participant, but it retains its value.

1 **12.32.4.6 adminRequestListLength**

2 The adminRequestListLength attribute provides the administrative value for the number of octets in
3 adminRequestList.

4 The default value of the adminRequestListLength attribute is zero.

5 When the externalControl attribute is TRUE, the adminRequestListLength attribute is copied to the
6 operRequestListLength attribute at the same time that the adminRequestList attribute is copied to the
7 operRequestList attribute.

8 When the externalControl attribute is FALSE, the adminRequestListLength attribute is ignored by the MRP
9 Participant, but it retains its value.

10 **12.32.4.7 operRequestList**

11 The operRequestList attribute is the operational value of the current list of MAD requests for the MRP
12 Participant. Prior to copying adminRequestList to operRequestList, the lists are compared as follows:

- 13 g) If an attribute exists in adminRequestList that does not exist in operRequestList, invoke
14 MAD_Join.request for that attribute, with the *new* parameter set TRUE.
- 15 h) If an attribute exists in operRequestList that does not exist in adminRequestList, invoke
16 MAD_Leave.request for that attribute.

17 When the comparison is complete, adminRequestList is copied to operRequestList.

18 The default value of the operRequestList attribute is the empty octet string.

19 **12.32.4.8 operRequestListLength**

20 The operRequestListLength attribute is the operational value of the adminRequestListLength attribute, and
21 it is copied at the same time that adminRequestList attribute is copied to operRequestList.

22 The default value of the operRequestListLength attribute is zero.

23 **12.33 Congestion Isolation managed objects**

24 Several variables control the operation of Congestion Isolation in a congestion isolation aware Bridge. The
25 managed objects are as follows:

- 26 a) CI entity managed object (12.33.1)
- 27 b) CI Peer Table (12.33.2)
- 28 c) CI Stream Table (12.33.3)
- 29 d) CIP entity managed object (12.33.4)

30 **12.33.1 CI entity managed object**

31 A single instance of the CI entity managed object shall be implemented by a Bridge Component or end
32 station that is congestion isolation aware. It comprises all the variables included in the CI entity variables
33 (49.4.1.1) as illustrated in Table 12-42.

Table 12-42—CI entity managed object

Name	Data type	Operations supported ^a	Conformance ^b	References
ciMasterEnable	Boolean	RW	CI	49.4.1.1.1
ciCIMTransmitPriority	unsigned integer [0..7]	R	CI	49.4.1.1.2
ciMaxFlowLife	unsigned integer	R	CI	49.4.1.1.3

^a R= Read only access; RW = Read/Write access.

^b CI = Required for Bridge Component or end station support of CI.

1 12.33.2 CI Peer Table

2 A single instance of the CI Peer Table shall be implemented by a Bridge Component or end station that is
3 congestion isolation aware. Each table row contains the parameters needed to assist in the creation of a CIM
4 as illustrated in Table 12-43. There is a row in the table for each port of the Bridge or router that is
5 supporting congestion isolation.

Table 12-43—CI Peer Table entry

Name	Data type	Operations supported ^a	Conformance	References
ciCIMtype	enum {12, ipv4, ipv6}	R		49.3.6
ciPeerMacAddress	MAC address	R		49.3.6
ciPeerIPv4Address	IPv4 address	R	IETF RFC 791	49.3.6
ciPeerIPv6Address	IPv6 address	R	IETF RFC 8200	49.3.6
ciPeerUDPPort	UDP Port Number	R	IETF RFC 768	49.3.6
ciPeerCIMEncapLen	integer [0..512]	R		49.3.6

^a R= Read only access; RW = Read/Write access.

6 12.33.3 CI Stream Table

7 A single instance of the CI Stream Table (49.3.7) shall be implemented by a Bridge Component or end
8 station that is congestion isolation aware. Each table row contains stream management variables associated
9 with a tsnStreamIdEntry object created in the IEEE Std 802.1CB Stream identity table. The variables assist
10 in the processes associated with the isolating and de-isolating of congesting flows. There is a row in the table
11 for each congesting flow as illustrated in Table 12-44.

12 12.33.4 CIP entity managed object

13 There is one Congestion Isolation Point (CIP) managed object for each CIP in a Bridge Component or end
14 station that is congestion isolation aware. The CIP managed object comprises some of the variables included
15 in the CIP variables (49.4.1.2) as illustrated in Table 12-45.

Table 12-44—CI Stream Table entry

Name	Data type	Operations supported ^a	Conformance ^b	References
ciStreamIdHandle	stream_handle value	R	IEEE Std 802.1CB	49.4.1.5.1
ciCIMCount	integer	RW	CI	49.4.1.5.2
ciCreateTime	sysUpTime (IETF RFC 3418)	R	IETF RFC 3418	49.4.1.5.3
ciStreamCreateMask	2-bit mask	RW	CI	49.4.1.5.4
ciQueueKey	integer	R	CI	49.4.1.5.5
ciDestination_address	MAC address	R	CI	49.4.1.5.6
ciSource_address	MAC address	R	CI	49.4.1.5.7
ciVlan_identifier	12-bit VID	R	CI	49.4.1.5.8
ciMsdu	octet string (size 64..512)	R	CI	49.4.1.5.9

^a R= Read only access; RW = Read/Write access.

^b CI = Required for Bridge Component or end station support of CI.

Table 12-45—CIP entity managed object

Name	Data type	Operations supported ^a	Conformance ^b	References
cipMacAddress	MAC address	R	CI	49.4.1.2.1
cipIPv4Address	IPv4 address	R	IETF RFC 791	49.4.1.2.2
cipIPv6Address	IPv6 address	R	IETF RFC 8200	49.4.1.2.3
cipCIMUDPPort	UDP Port Number	R	IETF RFC 768	49.4.1.2.4
cipQueueMap[]	array of integers [−8..8]	RW	CI	49.4.1.2.5
cipMinHeaderOctets	integer	RW	CI	49.4.1.2.6
cipMaxCIM	integer	RW	CI	49.4.1.2.7

^a R= Read only access; RW = Read/Write access.

^b CI = Required for Bridge Component or end station support of CI.

12.34 Managed objects for PBBN Auto Attach

The conformance requirements for PBBN Auto Attach support are specified in 5.9.2, 5.10.3, 5.12.2, 5.14.2, and 5.33. PBBN Auto Attach is specified in Clause 50.

The PBBN Auto Attach specific managed objects defined here:

- a) Provide managed objects for identifying and configuring an Auto Attach System (12.34.1).
- b) Provide managed objects for configuring Bridge Ports, LAGs, and DRNIs for Auto Attach (12.34.2).
- c) Provide managed objects to catalog discovered remote ports supporting Auto Attach (12.34.3).
- d) Provide managed objects that indicate the desired VID to I-SID assignments for AADs and the operating assignments for AABs for each Bridge Port, LAG, or DRNI (12.34.4).
- e) Provide per port PBBN Auto Attach statistics (12.34.5).

12.34.1 PBBN Auto Attach System objects

The PBBN Auto Attach System objects (Table 12-46) identify the AAS within the administrative domain, indicate the type of AAS, either AAB or one of the AAD types, and enable Auto Attach within the AAS. The aaSystemMAC uniquely identifies the system including all associated components.

Table 12-46—PBBN Auto Attach System objects

Name	Data type	Operations supported ^a	Conformance ^b	References
aaSystemEnable	Boolean	RW	B, D	50.4.2.1
aaSystemMAC	macAddress	R	B, D	50.4.2.4
aaSystemType	enumerated {AAB, CVLAN-AWARE-AAD, VLAN-UNAWARE-AAD, SVLAN-AWARE-AAD}	R	B, D	50.4.2.5
aaSystemResetTime	unsigned (1...1200)	RW	B, D	50.4.2.6

^a R = Read only access; RW = Read/Write access.

^b B = required for AAB; D= required for AAD.

In the case of an AAB a single aaSystemMAC identifies all I-components and T-components as well as all embedded S-components and C-components used to support external C-VLAN and S-VLAN interfaces. In the case where a pair of BEBs form a DRNI, the single aaSystemMAC identifies all components of both BEBs coupled to form a DRNI.

For an AAD the single aaSystemMAC identifies all end station and Bridge components within the AAD.

The object aaSystemEnable defaults to FALSE allowing configuration of the system before Auto Attach operation. Changing aaSystemEnable from FALSE to TRUE starts Auto Attach on all configured and enabled ports. Setting aaSystemEnable to FALSE breaks all attachments and disables the Auto Attach feature on all ports of the AAS.

12.34.2 PBBN Auto Attach port table

Each AAS has a port table that lists the ports that can be advertised by this system using the AAP. Each entry in the PBBN Auto Attach port table (Table 12-47) can be created, deleted, and updated by the system administrator to configure ports for Auto Attach.

Table 12-47—PBBN Auto Attach port table

Name	Data type	Operations supported ^a	Conformance ^b	References
aaPortId	If-ref	R	B, D	50.4.3.13
aaPortNetId	String	R	B, D	50.4.3.9, 50.4.1.3 portNetId ^c
aaPortEnable	Boolean	RW	B, D	50.4.3.1

^a R = Read only access; RW = Read/Write access.

^b B = required for AAB; D= required for AAD.

^c This is the object within the referenced structure.

Each local port is uniquely identified within the AAS by its aaPortId and uniquely identified over the administrative domain by aaPortNetId. The aaPortNetId is composed of the aaSystemMAC and the aaPortID. The management object aaPortNetId reflects the content of the state machine variable aaOperLocSysTlv.portNetId.

The object aaPortEnable controls the state machine variable aaAdminPortEnable that enables/disables Auto Attach operation on this port. Setting aaPortEnable = FALSE forces the AAP state machines for this port to restart, detaching all active associations and clearing the current statistics.

12.34.3 PBBN Discovered Auto Attach Systems table

The PBBN Discovered AAS table (Table 12-48) object provides information about active remote Auto Attach ports identified by the Auto Attach Protocol (AAP) through the Auto Attach LLDP database system objects. The discovered Auto Attach Systems table is read only.

Table 12-48—PBBN Discovered Auto Attach Systems table

Name	Data type	Operations supported ^a	Conformance ^b	References
aaDiscLocPortId	If-ref	R	B, D	50.4.3.13
aaDiscRemPortNetId	String	R	B, D	50.4.3.11, 50.4.1.3, D.2.17.10 portNetId ^c
aaDiscRemSystemType	Enumeration {AAB, CVLAN-AWARE-AAD, VLAN-UNAWARE-AAD, SVLAN-AWARE-AAD}	R	B, D	50.4.3.11, 50.4.1.3, D.2.17.7 sysType ^c
aaDiscRemPortTagging	Enumeration {TAG-ALL, TAG-OR-UNTAG, UNTAG-ONLY}	R	B, D	50.4.3.11, 50.4.1.3, D.2.17.8 portTagging ^c
aaDiscRemPortAssocState	Enumeration {NOT_READY, READY_TO_ASSOC, READY_TO_ATTACH, ASSOC_FAILED_TYPES, ASSOC_FAILED_TAGS, ASSOC_FAILED_TOPO, ASSOC_FAILED_OTHER, ASSOC_ATTACHED, ASSOC_STANDBY, ASSOC_INVALID}	R	B, D	50.4.3.11, 50.4.1.3, D.2.17.5 state ^c
aaDiscLocPortAssocState	Enumeration {NOT_READY, READY_TO_ASSOC, READY_TO_ATTACH, ASSOC_FAILED_TYPES, ASSOC_FAILED_TAGS, ASSOC_FAILED_TOPO, ASSOC_FAILED_OTHER, ASSOC_ATTACHED, ASSOC_STANDBY, ASSOC_INVALID}	R	B, D	50.4.3.9, 50.4.1.3, D.2.17.5 state ^c

^a Operations Supported: R = Read only access; RW = Read/Write access.

^b B = required for AAB; D= required for AAD.

^c This is the object within the referenced structure

The aaDiscLocPortId provides a reference to the local port, LAG, or DRNI where the remote attach port was discovered. The aaDiscRemPortNetId uniquely identifies the remote port, LAG, or DRNI discovered at the local port.

12.34.4 PBBN Auto Attach assignment table

There is one PBBN Auto Attach assignment table for each port configured in the Auto Attach port table (12.34.2). The Auto Attach assignment table is used by management to set the VLAN ID/I-SID assignments desired by the AAD and to monitor the current Auto Attach assignment requests received at the AAB. Entries in the Auto Attach assignment table can be created, updated, and deleted for the AAD, but are read only for the AAB.

Each entry of the Auto Attach assignment table (Table 12-49) contains three variables specifying the VID, I-SID, and status of each assignment. In an AAD the desired assignments are stored and read from the per port state machine variable aaAdminVidIsidAsgns[] (50.4.3.2). One aaAdminVidIsidAsgns[] exists for each local port identified by the Auto Attach port table.

Table 12-49—PBBN Auto Attach assignment table

Name	Data type	Operations supported ^a	Conformance ^b	References
aaVidIsidAsgnsVid	unsigned[1...4094]	R for AAB, RW for AAD	B, D	50.4.3.2 vid ^c
aaVidIsidAsgnsIsid	unsigned [1, 256...16777214]	R for AAB, RW for AAD	B, D	50.4.3.2 isid ^c
aaVidIsidAsgnsStatus	Enumerated { PENDING, ACCEPTED, REJECTED, REJ-RES-UNAVAIL, REJ-INVALID-VID, REJ-VLAN-UNAVAIL, REJ-INVALID-I-SID, REJ-I-SID-UNAVAIL, REJ-APP-ISSUE, REJ-NOT-ALLOWED	R	B, D	AAD 50.4.3.10, AAB 50.4.3.8, 50.4.1.1 asgnsArray[] status ^c

^a R = Read only access; RW = Read/Write access.

^b B = required for AAB; D= required for AAD.

^c This is the object within the referenced structure

The status indicated in the Auto Attach assignment table for each VID to I-SID assignment is determined from the state machine variable aaOperRemAsgnsTlv.asgnsArray.status (50.4.3.10) in AADs or from the state machine variable aaOperLocAsgnsTlv.asgns.status (50.4.3.8) in AABs. The status at the AAD is the status from the VID/ I-SID assignments reported in the remote LLDP assignment objects received from the AAB. The status at the AAB is the status determined by the AAB for the VID/ I-SID assignment requests received from the AAD. At the AAD, if no remote LLDP assignment objects exist in the LLDP database or if the VID/I-SID assignment is not listed in the remote AAB LLDP database objects, then the status is set to PENDING in the Auto Attach assignment table.

The AAD administrator can create new entries, delete entries, and update entries in the Auto Attach assignment table. When the administrator deletes, adds, or updates an entry in the Auto Attach assignment table the AAP state machine variable aaAdminVidIsidAsgn[] associated with the affected port is updated to match the Auto Attach assignment table. When a new entry is created or updated, the new VID and I-SID is provided by management and the status begins as PENDING and then transitions to reflect the current status indicated in the remote AAB LLDP objects.

¹ In an AAB the Auto Attach assignment table is read only and reflects the current assignments within the
² local LLDP assignment database. The Auto Attach assignment table contains the status, VID, and I-SID
³ (D.2.18.6, D.2.18.7, D.2.18.8) from the local LLDP assignment database objects.

⁴ **12.34.5 PBBN Auto Attach statistics table**

⁵ There is one PBBN Auto Attach statistics table for each port configured in the Auto Attach port table
⁶ (Table 12-47). Each Auto Attach statistics table (Table 12-50) provides per port statistics. All objects of the
⁷ Auto Attach statistics table are read only. The contents of the table is maintained by the Auto Attach state
⁸ machines (50.6, 50.7). All statistic counters on all ports are cleared on system reset.

Table 12-50—PBBN Auto Attach statistics table

Name	Data type	Operations supported ^a	Conformance ^b	References
aaStatsAssocAttached	Counter32	R	B, D	50.4.3.18
aaStatsAssocFailed	Counter32	R	B, D	50.4.3.19
aaStatsAssocReset	Counter32	R	B, D	50.4.3.20
aaStatsAssocStandby	Counter32	R	B, D	50.4.3.21
aaStatsVidIsidAsgnsAccepted	Counter32	R	B, D	50.4.3.22
aaStatsVidIsidAsgnsRejected	Counter32	R	B, D	50.4.3.23
aaStatsVidIsidAsgnsRequested	Counter32	R	B, D	50.4.3.24
aaStatsVidIsidAsgnsWithdrawn	Counter32	R	B, D	50.4.3.25

^a R = Read only access; RW = Read/Write access.

^b B = required for AAB; D= required for AAD.

⁹ The counters provided are as follows:

- ¹⁰ a) aaStatsAssocAttached: Counts the number of associations that were attached.
- ¹¹ b) aaStatsAssocFailed: Counts the number of associations that failed.
- ¹² c) aaStatsAssocReset: Counts the number of association resets.
- ¹³ d) aaStatsAssocStandby: Counts the number of associations that were standby.
- ¹⁴ e) aaStatsVidIsidAsgnsRequested: Counts the number of new VID to I-SID assignments requested.
- ¹⁵ f) aaStatsVidIsidAsgnsAccepted: Counts the number of VID to I-SID assignment requests accepted.
- ¹⁶ g) aaStatsVidIsidAsgnsRejected: Counts the number of VID to I-SID assignment requests rejected.
- ¹⁷ h) aaStatsVidIsidAsgnsWithdrawn: Counts the number of VID to I-SID assignment requests
¹⁸ withdrawn.

13. Spanning tree protocols

The spanning tree algorithms and protocols specified by this standard provide simple and full connectivity throughout a Bridged Network comprising arbitrarily interconnected Bridges. Each Bridge can use Rapid Spanning Tree Algorithm and Protocol (RSTP), Multiple Spanning Tree Algorithm and Protocol (MSTP), or Shortest Path Bridging (SPB) protocols.

NOTE 1—The spanning tree protocols specified by this standard supersede the STP specified in IEEE Std 802.1D revisions prior to 2004 ([B12], [B13]), but facilitate migration by interoperating with the latter without configuration restrictions beyond those previously imposed by STP. However, networks that include Bridges using STP can reconfigure slowly and constrain active topologies.

NOTE 2—Although the active topologies determined by the spanning tree protocols connect all the components of a Bridged Network, filtering (MVRP, etc.) can restrict frames to a subset of each active topology.

RSTP assigns all frames to a Common Spanning Tree (CST), without being aware of the active topology assignments made by MSTP or SPB that allow frames to follow separate paths within Multiple Spanning Tree (MST) or Shortest Path Tree (SPT) Regions. Each of these regions comprises MST or SPT Bridges that consistently assign any given frame to the same active topology (see 8.4) and the LANs that interconnect those Bridges. These regions and other Bridges and LANs are connected into the CST, to provide loop-free network wide connectivity even if active topology assignments or spanning tree protocols differ locally.

MSTP and SPB connect all Bridges and LANs with a single Common and Internal Spanning Tree (CIST) that supports the automatic determination of each region, choosing its maximum possible extent. The connectivity calculated for the CIST provides the CST for interconnecting the regions, and an Internal Spanning Tree (IST) within each region. MSTP calculates a number of independent Multiple Spanning Tree Instances (MSTIs) within each region, and ensures that frames with a given VID are assigned to one and only one of the MSTIs or the IST within the region (or reserved for use by SPB or PBB-TE), that the assignment is consistent among all Bridges within the region, and that the stable connectivity of each MSTI and the IST at the boundary of the region matches that of the CST. SPB calculates symmetric sets of SPTs, each rooted at a Bridge within a region, and ensures that frames for any given VLAN are assigned to the same symmetric SPT set within the region (or to the IST, an MSTI, or PBB-TE). Within an SPT Region, all Bridges that receive a given shortest path bridged frame assign that frame to the same SPT. The assignment process is specified in Clause 27, and can be done in one of two ways: by using additional VIDs (SPBV, Clause 3), or by using the frame's MAC addresses (SPBM, Clause 3). SPT Bridges use the VID of each received frame to decide whether to use the IST, MSTI, PBB-TE, SPBV, or SPBM. ISIS-SPB can dynamically allocate the additional VIDs (SPVIDs) used by SPBV mode, and ensure that their allocation is consistent within a region.

Spanning Tree Protocol Entities transmit and receive BPDUs (Clause 14) to convey parameters used by RSTP and MSTP to calculate CST, CIST, and MSTI spanning trees. BPDUs also convey parameters that all the spanning tree protocols use to interoperate with each other, that determine the extent of MST and SPT Regions, and that ensure that temporary loops are not created when neighboring Bridges are acting on different topology information. SPB uses ISIS-SPB (see Clause 28) to share information used to calculate the IST and SPTs, and to perform that calculation. SPT BPDUs that ensure loop-free connectivity even if neighboring Bridges' views of the topology differ (13.17) is also carried in SPB Hello PDUs (28.12.2).

This clause:

- a) Specifies protocol design and support requirements (13.1, 13.2) and design goals (13.3).
- b) Provides an overview of RSTP (13.4), MSTP (13.5), SPBV, and SPBM (13.6) operation.
- c) Describes how the spanning tree protocols interoperate and coexist (13.7).
- d) Specifies how spanning tree priority vectors (13.9) are calculated (13.10, 13.11) and used to assign the Port Roles (13.12) that determine the Port States, i.e., forwarding and learning (8.4), for each tree.

- 1 e) Shows that RSTP, MSTP, and ISIS-SPB provide stable connectivity (13.13).
- 2 f) Describes how spanning tree priority vectors are communicated (13.14) and changed (13.15).
- 3 g) Describes how Port Roles are used to change Port States without introducing loops (13.16).
- 4 h) Recommends defaults and ranges for the parameters that determine each tree's topology (13.18).
- 5 i) Describes the updating of learned station location information when a tree reconfigures (13.19).
- 6 j) Specifies additional controls that can speed reconfiguration or prevent unwanted outcomes (13.20).
- 7 k) Describes how loops are prevented when a LAN is only providing one-way connectivity (13.21),
8 and can be prevented when the network includes Bridges whose protocol operation can fail (13.23).
- 9 l) Describes how a Bridge's protocol processing can be 'hot upgraded' in an active network (13.22).
- 10 m) Specifies RSTP, MSTP, and support for SPB using state machines (13.24–13.40).
- 11 n) Specifies the use and configuration of the spanning tree protocols for the special cases of a Provider
12 Edge Bridge's CEPs (13.41), a BEB's VIPs (13.42), and an L2 Gateway Port connecting a customer
13 to a provider (13.20, 13.40).

14 NOTE 3—Readers of this specification are urged to begin by familiarizing themselves with RSTP.

15 Clause 14 specifies the format of BPDUs. Clause 27 describes the uses of SPB. Clause 28 specifies
16 ISIS-SPB. The text of this clause (Clause 13) takes precedence should any conflict be apparent between it
17 and the text in other parts of this standard (in particular, Clause 12, Clause 14, and Annex A). Within this
18 clause (Clause 13) the state machine specifications (13.24–13.40) take precedence over the general
19 description (13.1–13.23). A distinctive font is used to highlight references to state machine variables,
20 procedures, and STATES in the general description.

21 13.1 Protocol design requirements

22 The spanning tree algorithm and its associated protocols operate in Bridged Networks of arbitrary physical
23 topology comprising MST or SST Bridges connecting shared media or point-to-point LANs, to support,
24 preserve, and maintain the quality of the MAC Service in all its aspects as specified by Clause 6.

25 RSTP configures the Port State (8.4) of each Bridge Port. MSTP configures the Port State for the CIST and
26 each MSTI, and verifies the allocation of VIDs to FIDs and FIDs to trees. ISIS-SPB configures the Port
27 State for the CIST and each SPT, configures the VID Translation Table for each Bridge Port, allocates FIDs
28 to SPT sets, and assigns frames to SPTs using SPVIDs or MAC addresses (for SPBV and SPBM,
29 respectively, see Clause 27). Operating both independently and together RSTP, MSTP, and ISIS-SPB meet
30 the following requirements:

- 31 a) They configure one or more active topologies that fully connect all physically connected LANs and
32 Bridges, and stabilize (with high probability) within a short, known bounded interval after any
33 change in the physical topology, maximising service availability (6.5.1).
- 34 b) The active topology for any given frame remains simply connected at all times (6.5.3, 6.5.4), and
35 will (with high probability) continue to provide simple and full connectivity for frames even in the
36 presence of administrative errors (e.g., in the allocation of VIDs to MSTIs).
- 37 c) The configured stable active topologies are unicast multicast congruent, downstream congruent, and
38 reverse path congruent (symmetric) (3.295, 3.83, 6.3).
- 39 d) The same symmetric active topology is used, in a stable network, for all frames using the same FID,
40 i.e., between any two LANs all such frames are forwarded through the same Bridge Ports (6.3).
- 41 e) The active topology for a given VID can be chosen by the network administrator to be a common
42 spanning tree, one of multiple spanning trees (if MSTP is implemented), or shortest path (if
43 ISIS-SPB is implemented).
- 44 f) Each active topology is predictable, reproducible, and manageable, allowing Configuration
45 Management (following traffic analysis) to meet Performance Management goals (6.5 and 6.5.10).

- 1 g) The configured network can support VLAN-unaware end stations, such that they are unaware of
2 their attachment to a single LAN or a Bridged Network, or their use of a VID (6.2).
- 3 h) The communications bandwidth on any particular LAN is always a small fraction of the total
4 available bandwidth (6.5.10).

5 NOTE—The spanning tree protocols cannot protect against temporary loops caused by the interconnection of LANs by
6 devices other than Bridges (e.g., LAN repeaters) that operate invisibly with respect to support of the MAC ISS and the
7 MAC_Operational status parameter (IEEE Std 802.1AC).

8 13.2 Protocol support requirements

9 In order for the spanning tree protocols to operate, the following are required:

- 10 a) A unique group MAC address used by the Spanning Tree Protocol Entities (8.10) of participating
11 Bridges or Bridge components (5.2), and recognized by all the Bridges attached to a LAN.
- 12 b) An identifier for each Bridge or Bridge component, unique within the Bridged Network.
- 13 c) An identifier for each Bridge Port, unique within a Bridge or Bridge component.

14 Values for each of these parameters shall be provided by each Bridge. The unique MAC address that
15 identifies the Spanning Tree Protocol Entities of MAC Bridges, VLAN Bridges (5.9), and C-VLAN
16 components (5.5) is the Bridge Group Address (Table 8-1). The unique MAC address that identifies the
17 Spanning Tree Protocol Entities of S-VLAN components is the Provider Bridge Group Address (Table 8-2).

18 To allow management of active topology (for RSTP, MSTP, or SPB) means of assigning values to the
19 following are required:

- 20 d) The relative Bridge Priority of each Bridge in the network
- 21 e) A Port Path Cost for each Bridge Port
- 22 f) The relative Port Priority of each Bridge Port

23 13.2.1 MSTP support requirements

24 MSTP does not require any additional configuration, provided that communication between end stations is
25 supported by a number of VLANs. However, to realize the improved throughput and associated frame loss
26 and transit delay performance improvements made possible by the use of multiple spanning trees, the
27 following are required:

- 28 a) Assessment of the probable distribution of traffic between VLANs and between sets of
29 communicating end stations using those VLANs.
- 30 b) Per MSTI assignment of Bridge Priority and Internal Port Path Costs to configure the MSTIs.
- 31 c) Consistent assignment of VIDs to MSTIDs within each potential MST Region (3.174).
- 32 d) Administrative agreement on the Configuration Name and Revision Level used to represent the
33 assignments of VIDs to MSTIDs.

34 13.2.2 SPB support requirements

35 In order for the ISIS-SPB protocol to operate, the following are required:

- 36 a) An MTID indicating the topology information to be used for route calculation.
- 37 b) An IS-IS Area Address for the IS-IS routing area.
- 38 c) An IS-IS system ID for the SPT Bridge, normally the Bridge Identifier.
- 39 d) A group MAC address for use by IS-IS SPB (Table 8-17).
- 40 e) A Base VID for each SPB VLAN.
- 41 f) An ECT-ALGORITHM identifying the algorithm for calculating SPTs and selecting from among
42 Equal Cost Trees.

1 SPBV can provide both CST and SPT support for VLANs without further configuration. To support default
2 SPB operation this standard specifies a default configuration for SPT Bridges (13.8).

3 **13.3 Protocol design goals**

4 All the spanning tree protocols meet the following goal, which simplifies operational practice:

- 5 a) Bridges do not have to be individually configured before being added to the network, other than
6 having their MAC addresses assigned through normal procedures.
- 7 b) In normal operation, the time taken to configure the active topology of a network comprising
8 point-to-point LANs is independent of the timer values of the protocol.

9 RSTP and MSTP meet the following goal, which limits the complexity of Bridges and their configuration:

- 10 c) The memory requirements associated with each Bridge Port are independent of the number of
11 Bridges and LANs in the network.

12 It is highly desirable that the operation of ISIS-SPB supports updating of the SPB configuration so that:

- 13 d) SPT Bridges can be added to a region without disrupting communication for existing VLANs.
- 14 e) Additional VLANs can be supported by SPB without disrupting communication for the VLANs that
15 are already supported by SPB, or for those VLANs that are supported by RSTP or MSTP.
- 16 f) SPB support can be enabled or disabled for individual VLANs that are being used to support user
17 communication, with the minimum of frame loss on those VLANs.

18 **13.4 RSTP overview**

19 RSTP configures the Port State (8.4) of each Bridge Port in the Bridge Local Area Network. RSTP ensures
20 that the stable connectivity provided by each Bridge between its ports and by the individual LANs attached
21 to those ports is predictable, manageable, full, simple, and symmetric. RSTP further ensures that temporary
22 loops in the active topology do not occur if the network has to reconfigure in response to the failure,
23 removal, or addition of a network component, and that erroneous station location information is removed
24 from the FDB after reconfiguration.

25 Each of the Bridges in the network transmits Configuration Messages (13.14). Each message contains
26 spanning tree priority vector (13.9) information that identifies one Bridge as the Root Bridge of the network,
27 and allows each Bridge to compute its own lowest path cost to that Root Bridge before transmitting its own
28 Configuration Messages. A Port Role (13.12) of Root Port is assigned to the one port on each Bridge that
29 provides that lowest cost path to the Root Bridge, and a Port Role of Designated Port to the one Bridge Port
30 that provides the lowest cost path from the attached LAN to the Root Bridge. Alternate Port and Backup Port
31 roles are assigned to Bridge Ports that can provide connectivity if other network components fail.

32 State machines associated with the Port Roles maintain and change the Port States that control forwarding
33 (8.6) and learning (8.7) of frames. In a stable network, Root Ports and Designated Ports are Forwarding,
34 while Alternate, Backup, and Disabled Ports are Discarding. Each Port's role can change if a Bridge, Bridge
35 Port, or LAN fails, is added to, or removed from network. Port state transitions to Learning and Forwarding
36 are delayed, and ports can temporarily transition to the Discarding state to prevent loops and to ensure that
37 misordering (6.5.3) and duplication (6.5.4) rates remain negligible.

38 RSTP provides rapid recovery of connectivity to minimize frame loss (6.5.2). A new Root Port, and
39 Designated Ports attached to point-to-point LANs, can transition to Forwarding without waiting for protocol
40 timers to expire. A Root Port can transition to Forwarding without transmitting or receiving messages from
41 other Bridges, while a Designated Port attached to a point-to-point LAN can transition when it receives an
42 explicit agreement transmitted by the other Bridge attached to that LAN. The forwarding transition delay

1 used by a Designated Port attached to a shared media LAN is long enough for other Bridges attached to that
2 LAN to receive and act on transmitted messages, but is independent of the overall network size. If all the
3 LANs in a network are point-to-point, RSTP timers define worst-case delays that only occur if protocol
4 messages are lost or rate transmission limits are exceeded.

5 A Bridge Port attached to a LAN that has no other Bridges attached to it may be administratively configured
6 as an Edge Port. RSTP monitors the LAN to ensure that no other Bridges are connected, and may be
7 configured to automatically detect an Edge Port. Each Edge Port transitions directly to the Forwarding Port
8 State, since there is no possibility of it participating in a loop.

9 **13.4.1 Computation of the active topology**

10 The Bridge with the best Bridge Identifier is selected as the Root Bridge. The unique Bridge Identifier for
11 each Bridge is derived, in part, from the Bridge Address (8.13.8) and, in part, from a manageable priority
12 component (13.26). The relative priority of Bridges is determined by the numerical comparison of the
13 unique identifiers, with the lower numerical value indicating the better identifier.

14 Every Bridge has a Root Path Cost associated with it. For the Root Bridge this is zero. For all other Bridges,
15 it is the sum of the Port Path Costs on the least cost path to the Root Bridge. Each port's Path Cost may be
16 managed, 13.18 recommends default values for ports attached to LANs of various speeds.

17 The Bridge Port on each Bridge with the lowest Root Path Cost is assigned the role of Root Port for that
18 Bridge (the Root Bridge does not have a Root Port). If a Bridge has two or more ports with the same Root
19 Path Cost, then the port with the best Port Identifier is selected as the Root Port. Part of the Port Identifier is
20 fixed and different for each port on a Bridge, and part is a manageable priority component (13.26). The
21 relative priority of Bridge Ports is determined by the numerical comparison of the unique identifiers, with
22 the lower numerical value indicating the better identifier.

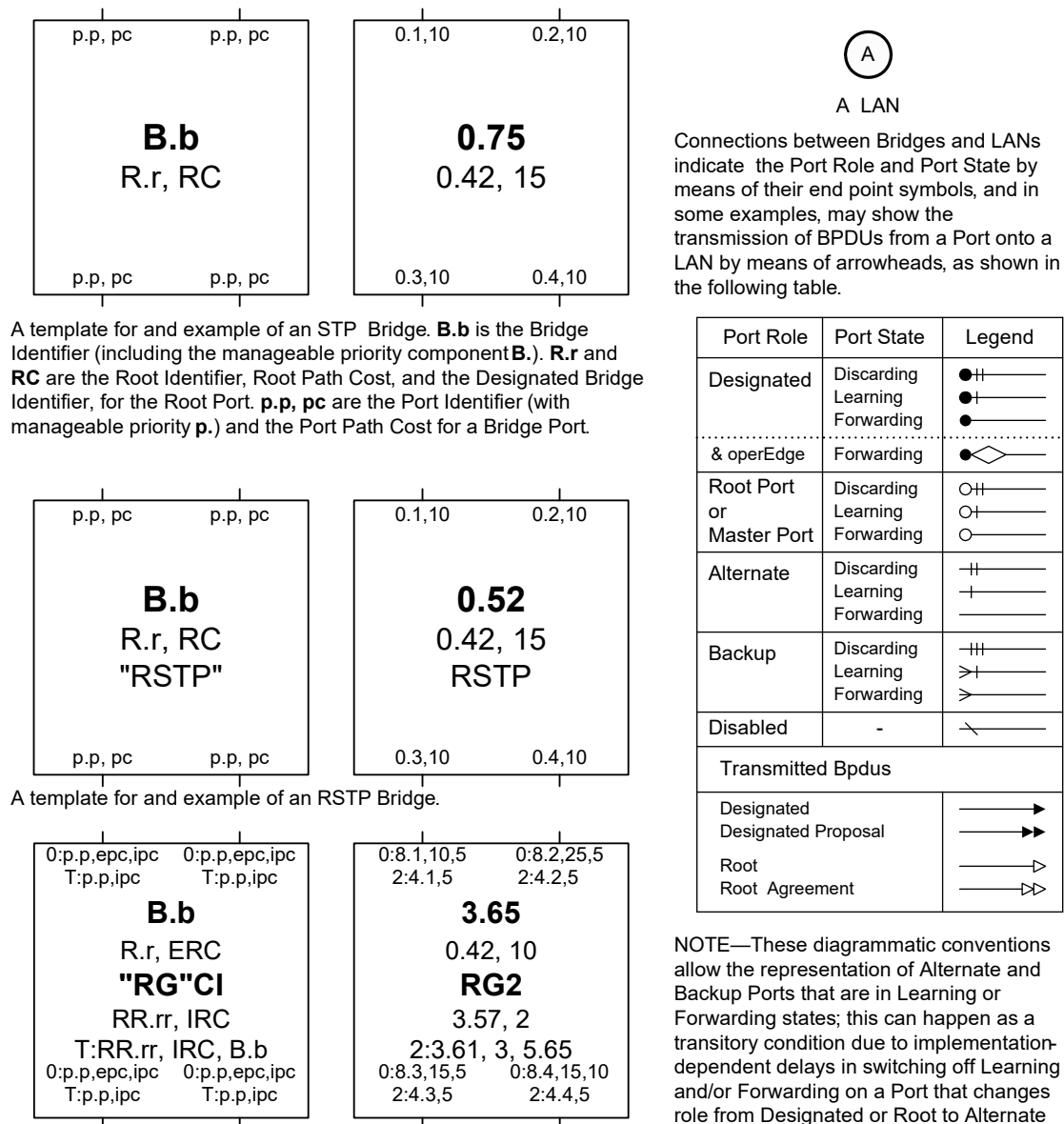
23 Each LAN in the Bridged Network also has an associated Root Path Cost. This is the Root Path Cost of the
24 lowest cost Bridge with a Bridge Port connected to that LAN. This Bridge is selected as the Designated
25 Bridge for that LAN. If there are two or more Bridges with the same Root Path Cost, then the Bridge with
26 the best priority (least numerical value) is selected as the Designated Bridge. The Bridge Port on the
27 Designated Bridge that is connected to the LAN is assigned the role of Designated Port for that LAN. If the
28 Designated Bridge has two or more ports connected to the LAN, then the Bridge Port with the best priority
29 Port Identifier (least numerical value) is selected as the Designated Port.

30 In a Bridged Network whose physical topology is stable, i.e., RSTP has communicated consistent
31 information throughout the network, every LAN has one and only one Designated Port, and every Bridge
32 with the exception of the Root Bridge has a single Root Port connected to a LAN. Since each Bridge
33 provides connectivity between its Root Port and its Designated Ports, the resulting active topology connects
34 all LANs (is "spanning") and will be loop-free (is a "tree").

35 Any Bridge Port that is enabled, but not a Root or Designated Port, is a Backup Port if that Bridge is the
36 Designated Bridge for the attached LAN, and an Alternate Port otherwise. An Alternate Port offers an
37 alternate path in the direction of the Root Bridge to that provided by the Bridge's own Root Port, whereas a
38 Backup Port acts as a backup for the path provided by a Designated Port in the direction of the leaves of the
39 spanning tree. Backup Ports exist only where there are two or more connections from a given Bridge to a
40 given LAN; hence, they (and the Designated Ports that they back up) can only exist where the Bridge has
41 two or more ports attached to a shared media LAN, or directly connected by a point-to-point LAN.

1 13.4.2 Example topologies

2 The spanning tree examples in this clause use the conventions of Figure 13-1.



A template for and an example of an MSTP Bridge. **B.b** is the CIST Bridge Identifier. **R.r**, **ERC**, **RR.rr** are the CIST Root Identifier, External Root Path Cost, and Regional Root Identifier. **CI** identifies the Configuration Identifier for the Bridge. **RR.rr**, **IRC** the CIST Regional Root Identifier and the Internal Root Path Cost **T:RR.rr**, **IRC**, **B.b** is the Regional Root Identifier, Internal Root Path Cost IRC, and Bridge Identifier for the MSTI with MSTID **T**.

0:p.p, **epc**, **ipc** are the CIST Port Identifier, External Port Path Cost, and Internal Port Path Cost for a Bridge Port. **T:p.p**, **ipc** are the Port Identifiers and their Regional Costs for MSTI **T**.

Any of the above information may be selectively omitted if deemed irrelevant for the purposes of a diagram.

Figure 13-1—Diagrammatic conventions for spanning tree topologies

1 Figure 13-2 shows a simple, redundantly connected, structured wiring configuration, with Bridges
2 connected by point-to-point LANs A through N, and a possible spanning tree active topology of the same
3 network. Bridge 111 has been selected as the Root (though one cannot tell simply by looking at the active
4 topology which Bridge is the Root).

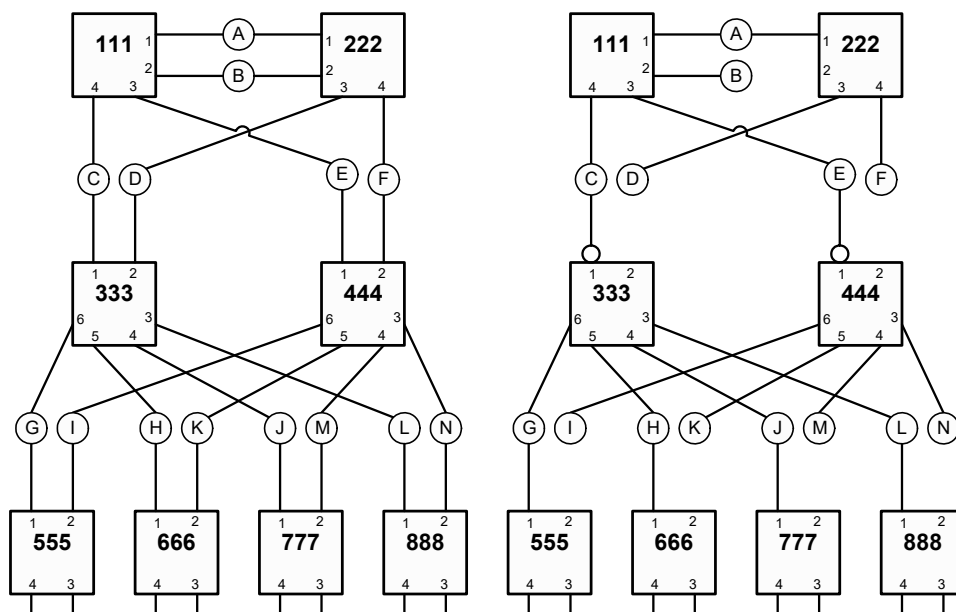


Figure 13-2—Physical topology and active topology

5 Figure 13-3 shows the Port Roles and Port States of each Bridge Port. It can be seen that Bridge 111 is the
6 Root, as its Ports are all Designated Ports, each of the remaining Bridges have one Root Port.

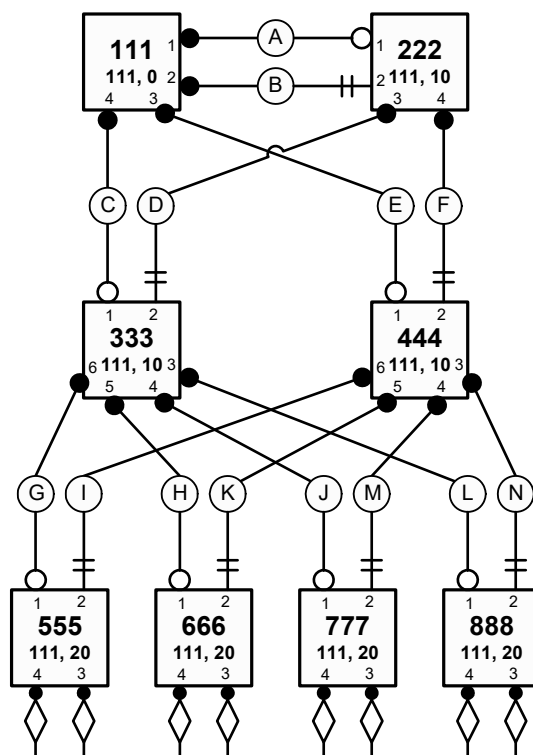


Figure 13-3—Port Roles and Port States

1 Figure 13-4 shows the result of connecting two of the ports of Bridge 888 to the same LAN. As port 4 of
2 Bridge 888 has worse priority than port 3 and both offer the same Root Path Cost, port 4 will be assigned the
3 Backup Port Role and will therefore be in the Discarding Port State. Should port 3 or its connection to LAN
4 O fail, port 4 will be assigned the Designated Port Role and will transition to the Forwarding Port State.

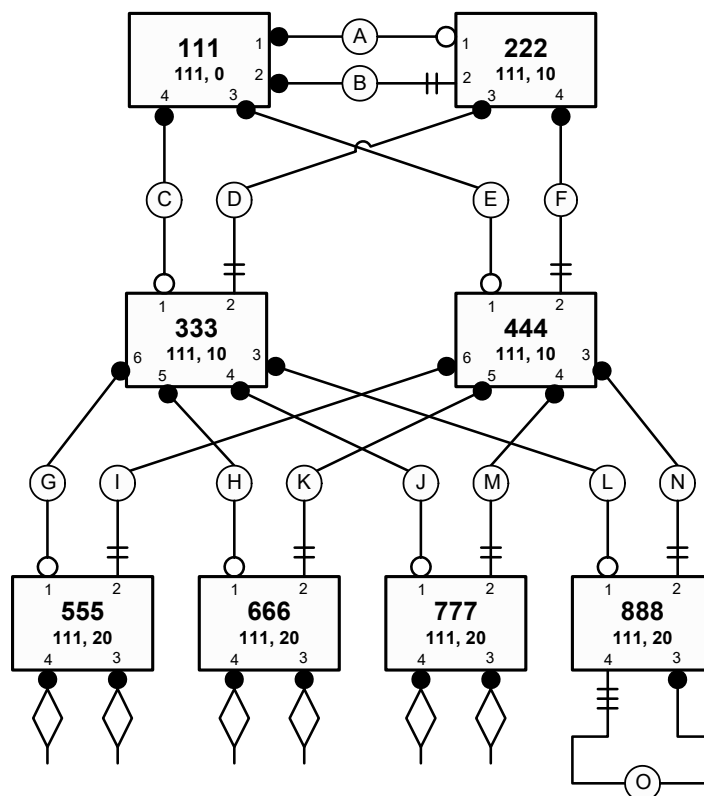


Figure 13-4—A Backup Port

5 Figure 13-5 shows a “ring” topology constructed from point-to-point links, as in some resilient backbone
6 configurations. Bridge 111 is the Root, as in previous examples.

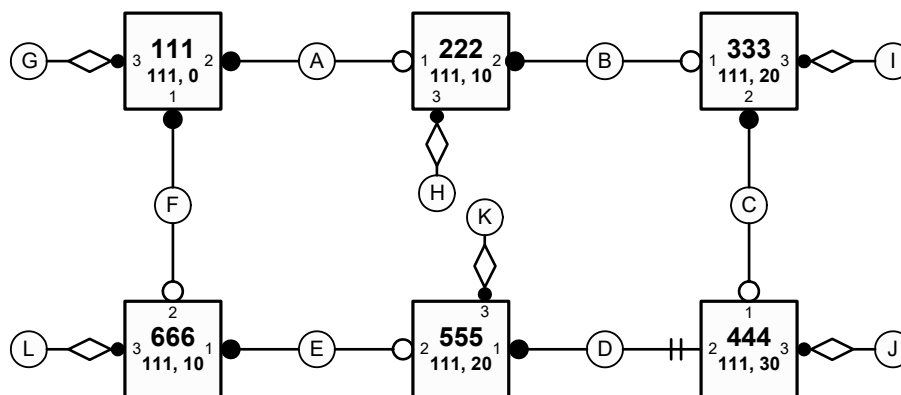


Figure 13-5—“Ring Backbone” example

13.5 MSTP overview

MSTP specifies the following:

- a) An MST Configuration Identifier (MCID) (13.8) that allows each Bridge to advertise its assignment, to a specified MSTI or to the IST, of frames with any given VID.
- b) A priority vector (13.9) that comprises Bridge identifier and path cost information for constructing a deterministic and manageable single spanning tree active topology, the CIST, that:
 - 1) Fully and simply connects all Bridges and LANs in a Bridged Network.
 - 2) Permits the construction and identification of MST Regions of Bridges and LANs that are guaranteed fully connected by the Bridges and LANs within each region.
 - 3) Ensures that paths within each MST Region are always preferred to paths outside the region.
- c) An MSTI priority vector (13.9), comprising information for constructing a deterministic and independently manageable active topology for any given MSTI within each region.
- d) Comparisons and calculations performed by each Bridge in support of the distributed spanning tree algorithm (13.10). These select a CIST priority vector for each Bridge Port, based on the priority vectors and MCIDs received from other Bridges and on an incremental Path Cost associated with each reception port. The resulting priority vectors are such that in a stable network:
 - 1) One Bridge is selected to be the CIST Root of the Bridged Network as a whole.
 - 2) A minimum cost path to the CIST Root is selected for each Bridge and LAN, thus preventing loops while ensuring full connectivity.
 - 3) The one Bridge in each MST Region whose minimum cost path to the Root is not through another Bridge using the same MCID is identified as its region's CIST Regional Root.
 - 4) Conversely, each Bridge whose minimum cost path to the Root is through a Bridge using the same MCID is identified as being in the same region as that Bridge.
- e) Priority vector comparisons and calculations performed by each Bridge for each MSTI (13.11). In a stable network:
 - 1) One Bridge is independently selected for each MSTI to be the MSTI Regional Root.
 - 2) A minimum cost path to the MSTI Regional Root that lies wholly within the region is selected for each Bridge and LAN.
- f) CIST Port Roles (13.12) that identify the role in the CIST active topology played by each port on a Bridge:
 - 1) The Root Port provides the minimum cost path from the Bridge to the CIST Root (if the Bridge is not the CIST Root) through the Regional Root (if the Bridge is not a Regional Root).
 - 2) A Designated Port provides the least cost path from the attached LAN through the Bridge to the CIST Root.
 - 3) Alternate or Backup Ports provide connectivity if other Bridges, Bridge Ports, or LANs fail or are removed.
- g) MSTI and SPT Port Roles (13.12) that identify the role played by each port on a Bridge for each MSTI's or SPT's active topology within and at the boundaries of a region.
 - 1) The Root Port provides the minimum cost path from the Bridge to the Regional Root (if the Bridge is not the Regional Root for the tree).
 - 2) A Designated Port provides the least cost path from the attached LAN through the Bridge to the Regional Root.
 - 3) A Master Port provides connectivity from the region to a CIST Root that lies outside the region. The Bridge Port that is the CIST Root Port for the CIST Regional Root is the Master Port for all MSTIs and SPTs.
 - 4) Alternate or Backup Ports provide connectivity if other Bridges, Bridge Ports, or LANs fail or are removed.

- h) State machines and state variables associated with each spanning tree (CIST, MSTI, or SPT), port, and port role, to select and change the Port State (8.4, 13.24) that controls the processing and forwarding of frames assigned to that tree by a MAC Relay Entity (8.3).

4 13.5.1 Example topologies

5 Figure 13-6 is an example Bridged Network, using the conventions of Figure 13-1, and chosen to illustrate
6 MSTP calculations rather than as an example of a common or desirable physical topology.

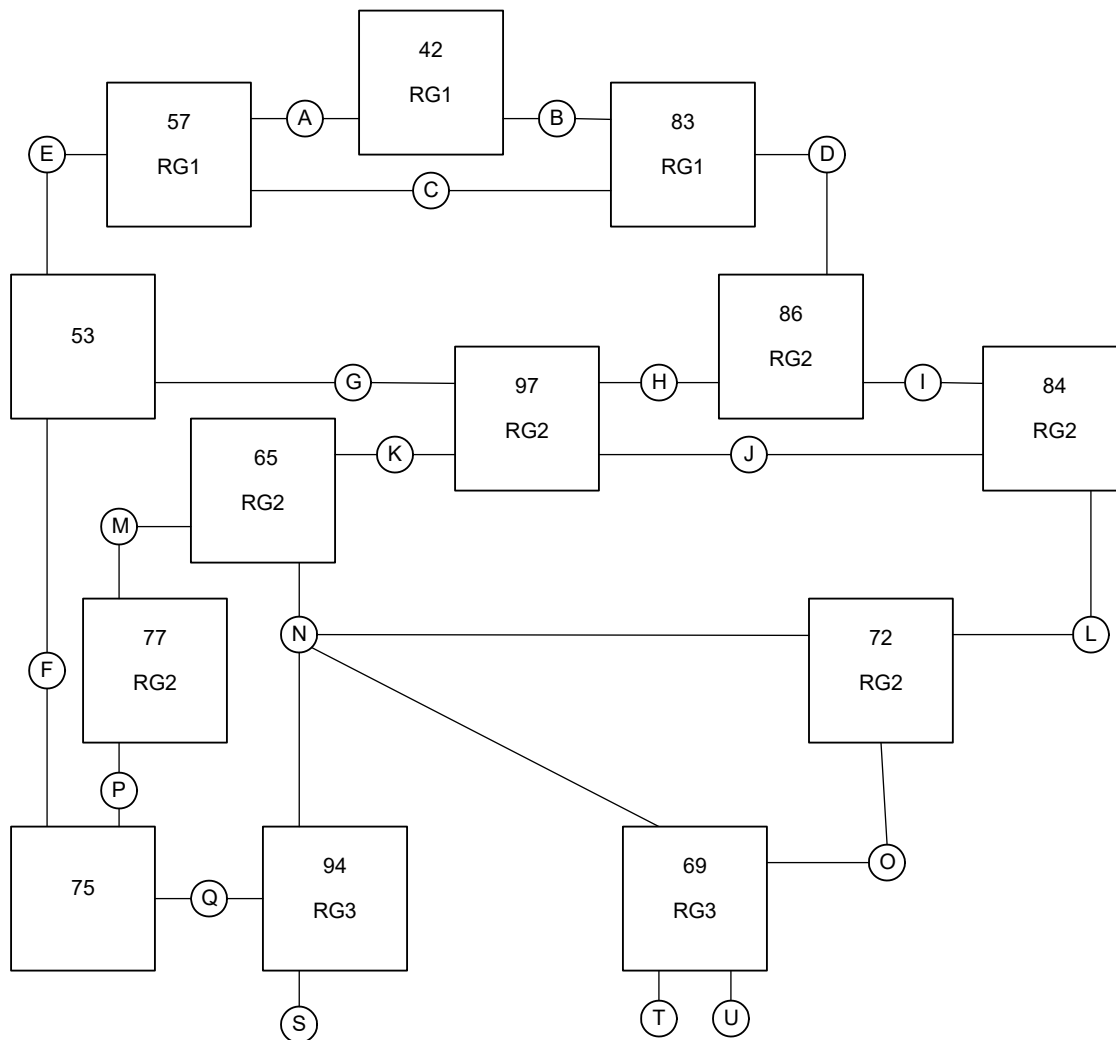


Figure 13-6—An MST Bridge network

7 Figure 13-7 is the same network showing Bridges and LANs with better CIST spanning tree priorities higher
8 on the page, and including CIST priority vectors, port roles, and MST Regions. In this example:

- 9 a) Bridge 0.42 is the CIST Root because it has the best (numerically lowest) Bridge Identifier.
- 10 b) Bridges 0.57 and 2.83 are in the same MST Region (1) as 0.42, because they have the same MCID
- 11 as the latter. Because they are in the same MST Region as the CIST Root, their External Root Path
- 12 Cost is 0, and their CIST Regional Root is the CIST Root.

- 1 c) LANs A, B, C, and D are in Region 1 because their CIST Designated Bridge is a Region 1 MST
- 2 Bridge, and no STP Bridges are attached to those LANs. LAN E is not in an MST Region (or in its
- 3 own region—an equivalent view) because it is attached to Bridge 0.53, which is not an MST Bridge.
- 4 d) Bridges 0.77, 0.65, 0.97, 0.86, 3.84, and 3.72 are in the same MST Region (2) since they have the
- 5 same MCID and are interconnected by LANs for which one of them is the CIST Designated Bridge.
- 6 e) Bridge 0.86 is the CIST Regional Root for Region 2 because it is has the lowest External Root Path
- 7 Cost through a Boundary Port.
- 8 f) LAN N is in Region 2 because its CIST Designated Bridge is in Region 2. Frames assigned to
- 9 different MSTIDs may reach N from Bridge 0.86 (for example) by either Bridge 0.65 or Bridge
- 10 3.72, even though Bridges 0.94 and 0.69 with MCIDs that differ from those for Bridges in Region 2
- 11 are attached to this shared LAN.
- 12 g) Bridges 0.94 and 0.69 are in different regions, even though they have the same MCID, because the
- 13 LAN that connects them (N) is in a different region.

14 Figure 13-8 shows a possible active topology of MSTI 2 within Region 2.

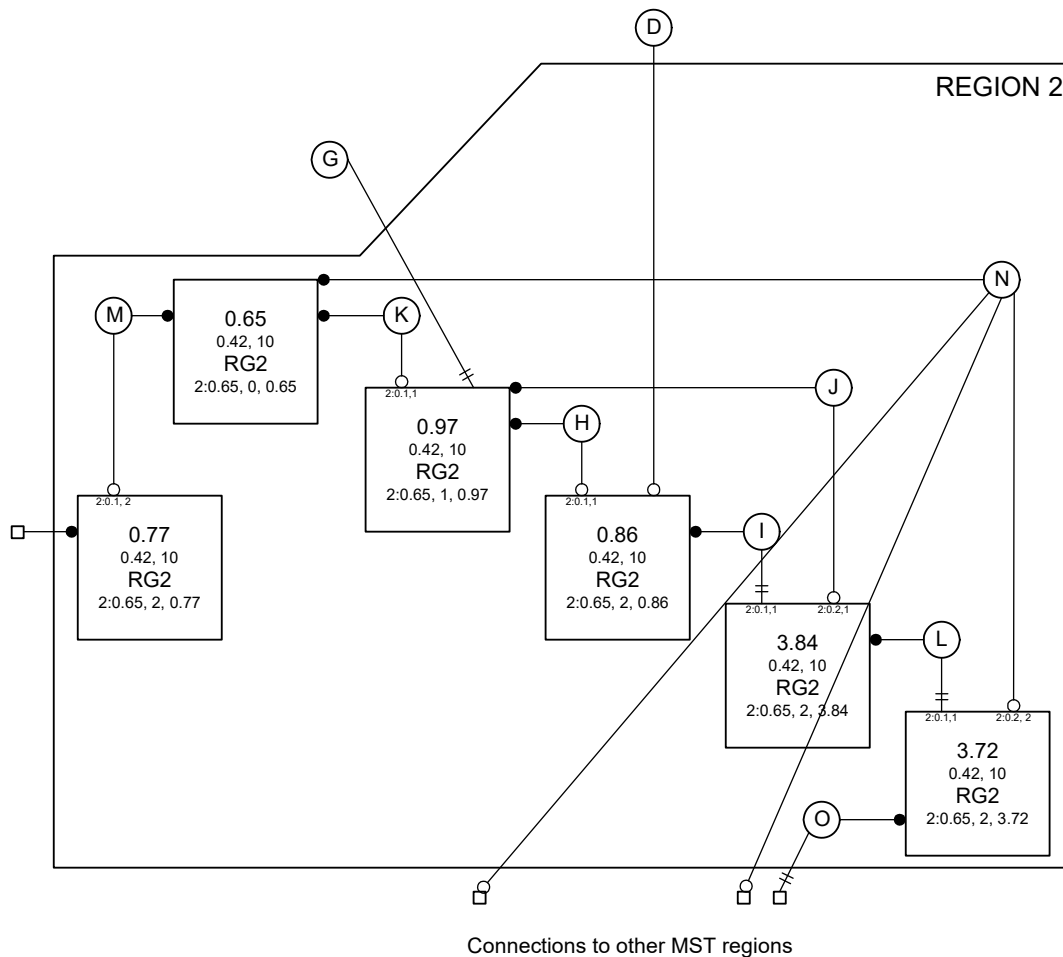


Figure 13-8—MSTI Active Topology in Region 2

- 15 h) Bridge 0.65 has been chosen as the MSTI Regional Root because it has the best (numerically the
- 16 lowest) Bridge Identifier of all Bridges in the region for this MSTI.

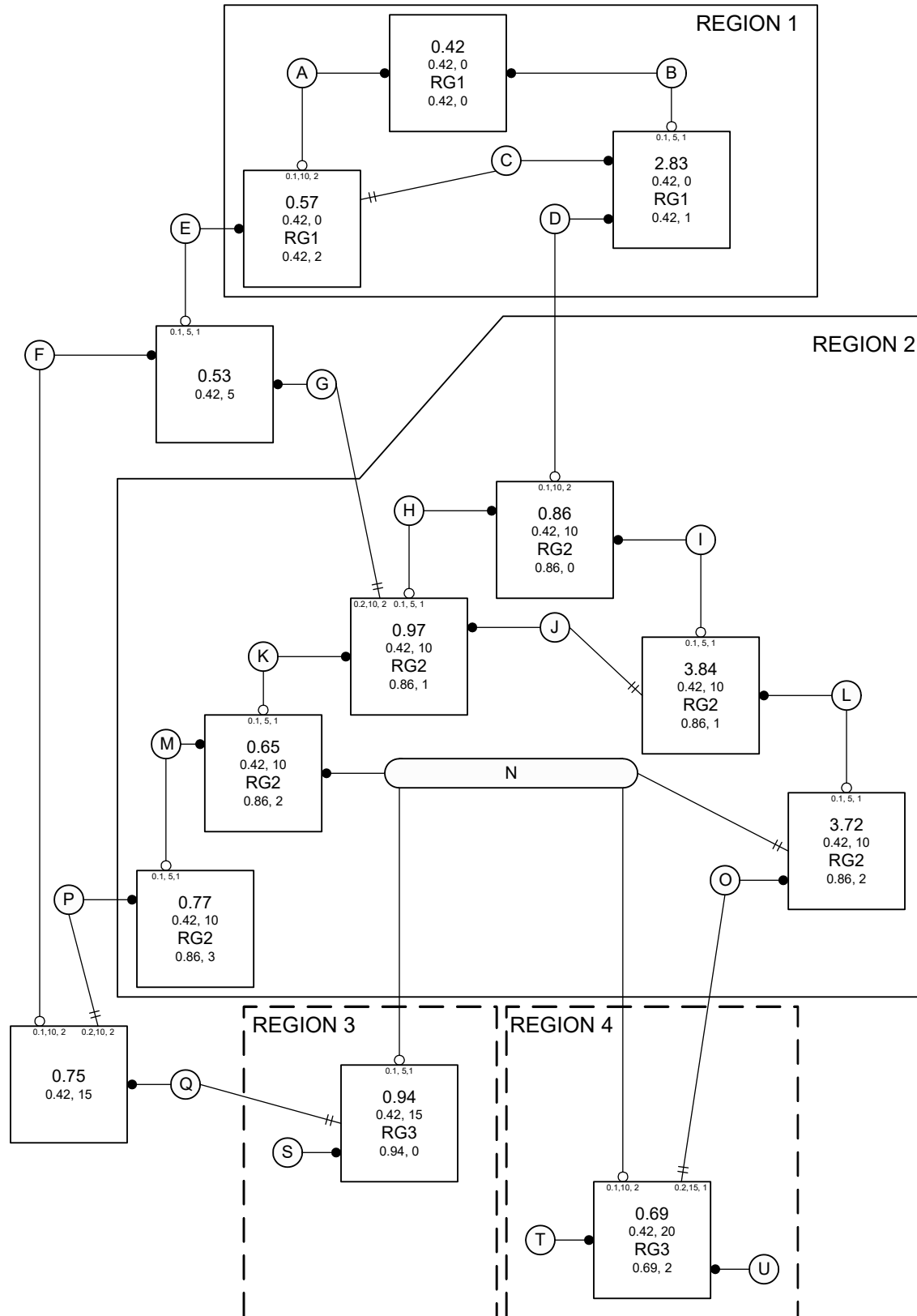


Figure 13-7—CIST Priority Vectors, Port Roles, and MST Regions

- 1 i) The connectivity between the whole of Region 2 and Region 1 is provided through a single Bridge
2 Port, the Master Port on Bridge 0.86. This port was selected for this role because it is the CIST Root
3 Port on the CIST Regional Root for the region (see Figure 13-7).
- 4 j) The connectivity between the whole of Region 2 and LANs and Bridges outside the region for the
5 MSTI is the same as that for the CIST. This connectivity is similar to that which might result by
6 replacing the entire region by a single SST Bridge. The region has a single Root Port (this port is the
7 Master Port for each MSTI) and a number of Designated Ports.

8 **13.5.2 Relationship of MSTP to RSTP**

9 MSTP is based on RSTP, extended so frames for different VLANs can follow different trees within regions.

- 10 a) The same fundamental spanning tree algorithm selects the CIST Root Bridge and Port Roles, but
11 extended priority vector components are used within (13.9, 13.10) in each region. As a result each
12 region resembles a single Bridge from the point of view of the CST as calculated by RSTP.
- 13 b) Each MSTI's Regional Root Bridge and Port Roles are also computed using the same fundamental
14 spanning tree algorithm with modified priority vector components (13.11).
- 15 c) Different Bridges may be selected as the Regional Root for different MSTIs by modifying the
16 manageable priority component of the Bridge Identifier differently for the MSTIs.
- 17 d) MST Configuration Identification is not applicable to RSTP.
- 18 e) The Port Roles used by the CIST (Root, Designated, Alternate, Backup, or Disabled Port) are the
19 same as those of RSTP. The MSTIs use the additional port role Master Port. The Port States
20 associated with each spanning tree and port are the same as those of RSTP.
- 21 f) The state variables for each Bridge Port for each tree and for the Bridge itself are those specified for
22 RSTP as per port and per Bridge with a few exceptions, additions, and enhancements.
- 23 g) The performance parameters specified for RSTP apply to the CIST, with a few exceptions,
24 additions, and enhancements. A simplified set of performance parameters apply to the MSTIs.
- 25 h) This standard specifies RSTP state machines and procedures as a subset of MSTP.

26 **13.5.3 Modeling an MST or SPT Region as a single Bridge**

27 The nominal replacement of an entire region by a single RST Bridge leads to little impact on the remainder
28 of the Bridged Network. This design is intended to assist those familiar with RSTP to comprehend and
29 verify MSTP, and to administer networks using MSTP. Treating the MST Regions as single Bridges
30 provides the network administrator with a natural hierarchy. The internal management of MST Regions can
31 be largely separated from the management of the active topology of the network as a whole.

32 The portion of the active topology of the network that connects any two Bridges in the same MST Region
33 traverses only MST Bridges and LANs in that region and never Bridges of any kind outside the region; in
34 other words, connectivity within the region is independent of external connectivity. This is because the
35 protocol parameters that determine the active topology of the network as a whole, the Root Identifier and
36 Root Path Cost (known in the MSTP specification as the CIST Root Identifier and CIST External Root Path
37 Cost) are carried unchanged throughout and across the MST Region, so Bridges within the region will
38 always prefer spanning tree information that has been propagated within the region to information that has
39 exited the region and is attempting to reenter it.

40 NOTE 1—No LAN can be in more than one MST Region at a time, so two Bridges (0.11 and 0.22 say) that would
41 otherwise be in the same region by virtue of having the same MST Configuration and of being directly connected by a
42 LAN, may be in distinct regions if that is a shared LAN with other Bridges attached (having a different MST
43 Configuration) and no other connectivity between 0.11 and 0.22 and lying wholly within their region is available. The
44 region that the shared LAN belongs to may be dynamically determined. No such dynamic partitioning concerns arise
45 with single Bridges. Obviously the sharing of LANs between administrative regions militates against the partitioning of
46 concerns and should only be done following careful analysis.

1 The Port Path Cost (MSTP's External Port Path Cost) is added to the Root Path Cost just once at the Root
2 Port of the CIST Regional Root, the closest Bridge in the region to the Root Bridge of the entire network.
3 The Message Age used by STP and RSTP is also only incremented at this port. If the CIST Root is within a
4 region, it also acts as the Regional Root, and the Root Path Cost and Message Age advertised are zero, just
5 as for a single Bridge.

6 Within an MST Region, each MSTI operates in much the same way as an independent instance of RSTP
7 with dedicated Regional Root Identifier, Internal Root Path Cost, and Internal Port Path Cost parameters.
8 Moreover, the overall spanning tree (the CIST) includes a fragment (the IST) within each MST Region that
9 can be viewed as operating in the same way as an MSTI with the Regional Root as its root.

10 NOTE 2—Since an MST Region behaves like a single Bridge and does not partition (except in the unusual configuration
11 involving shared LANs noted above), it has a single Root Port in the CST active topology. Partitioning a network into
12 two or more regions can therefore force nonoptimal blocking of Bridge Ports at the boundaries of those regions.

13 13.6 SPB overview

14 Clause 27 provides a comprehensive overview of SPT Bridges using SPBV and SPBM mode operation.
15 This subclause (13.6) summarizes aspects of SPB operation that relate to the transmission and reception of
16 BPDUs, and their role in providing interoperability with RSTP and MSTP, and in carrying the agreements
17 that ensure that each SPT provides loop-free connectivity throughout an SPT Region. The details of
18 agreements are considered in 13.17, the assignment of frames to SPTs in 8.4 and Clause 27, and protocols
19 and procedures to allow plug-and-play generation of SPVIDs in Clause 28.

20 Considerations of backward and forward compatibility and interoperability figure largely in this clause
21 (Clause 13) and to some extent these consideration revolve around the notion of MST or SPT Region, with
22 each region capable of using different protocols or different configurations. These considerations should not
23 be allowed to obscure the fact that the ideal network configuration (from the point of view of connectivity
24 and bandwidth efficiency) comprises a single region, or possibly one core region with RST Bridges
25 attached, and that separate regions are most likely to arise when continued connectivity is being provided by
26 the CST as configuration changes are made to Bridges in the network. If an SPT Region is bounded by
27 BEBs or LAN with no other Bridges attached, B-VLANs can be supported by SPBM mode, with frames
28 assigned to SPTs using their MAC addresses, while other B-VLANs and S-VLANs can be supported by
29 SPBV mode, with frames assigned to each SPT by SPVID. In both cases, however, SPTs are calculated in
30 the same way and the effects on this clause (Clause 13) are limited to allowing loop mitigation (6.5.4.2) for
31 unicast frames supported by SPBM mode as well as loop prevention (6.5.4.1). The need for unicast multicast
32 congruence (Clause 3), the simplification possible by not introducing differences between SPBM mode
33 unicast forwarding and the other uses of SPTs, and the need for source address lookup to support loop
34 mitigation means that all shortest path bridged frames are assigned to an SPT rooted at the source Bridge and
35 follow the standard bridging paradigm of multicast distribution with filtering, even when the frame has a
36 single destination and the path to the destination is known through the use of routing protocol (ISIS-SPB).

37 NOTE—When loop mitigation is used for unicast frames, the Port State used to prevent loops (as expressed by the state
38 machine variables forwarding and learning) need only apply to frames with group destination addresses.

39 SPT Bridges send and receive BPDUs in order to advertise their presence to, and recognize the presence of,
40 other Bridges. The requirement for loop-free SPT connectivity also means that SPT Bridges have to
41 communicate with their nearest neighbors, when calculations that follow the reception of ISIS-SPB
42 messages have been completed rather than as a part of the IS-IS protocol itself. The agreements (13.17) that
43 satisfy this requirement could be separated conceptually from other uses of BPDUs, but the opportunity to
44 share common information elements, synchronization of neighbor state, and an overall reduction in the
45 number of protocol frames sent make it convenient to use BPDUs to integrate its specification with the other
46 protocol variables, procedures, and state machines in this clause (13.24–13.40). Agreement Protocol

1 variables with exactly the same semantics are also carried in SPB Hello PDUs. The protocol handles
2 duplicated and misordered agreement information, so that updated information can be received from SPB
3 Hello PDUs or both SPB Hello PDUs and BPDUs.

4 SPT Bridges use the configuration and active topology management parameters (Bridge Identifiers, Port
5 Path Costs) already required for the CIST. They also retain MSTP capabilities as a subset of their operation,
6 though it is always possible to configure zero MSTIs.

7 **13.7 Compatibility and interoperability**

8 RSTP, MSTP, and the ISIS-SPB are designed to interoperate with each other and with STP. This subclause
9 (13.7) reviews aspects of their design that are important to meeting that requirement. ISIS-SPB and SPT
10 BPDUs include the functionality provided by MSTP and MST BPDUs, so the compatibility with RSTP and
11 STP provided by the latter extends to SPB.

12 **13.7.1 Designated Port selection**

13 Correct operation of the spanning tree protocols requires that all Bridge Ports attached to any given LAN
14 agree on a single CIST Designated Port after a short interval sufficient for any Bridge Port to receive a
15 configuration message from that Designated Port.

16 A unique spanning tree priority (13.9) is required for each Bridge Port for STP, which has no other way of
17 communicating port roles. Since port numbers on different Bridges are not guaranteed to be unique, this
18 necessitates the inclusion of the transmitting Bridge's Bridge Identifier in the STP BPDU. RSTP and
19 MSTP's Port Protocol Migration state machines (13.32) ensure that all Bridges attached to any LAN with an
20 attached STP Bridge send and receive STP BPDUs exclusively.

21 NOTE 1—This behavior satisfies the requirement for unique, agreed Designated Port for LANs with attached STP
22 Bridges, but means that an MST Region cannot completely emulate a single Bridge since the transmitted Designated
23 Bridge Identifier can differ on Bridge Ports at the region's boundary.

24 MSTP transmits and receives the Regional Root Identifier and not the Designated Bridge Identifier in the
25 BPDU fields recognized by RSTP (14.4) to allow both the MST and the RST Bridges potentially connected
26 to a single LAN to perform comparisons (13.9, 13.10) between all spanning tree priority vectors transmitted
27 that yield a single conclusion about which RST Bridge or MST Region includes the Designated Port. MST
28 and RST BPDUs convey the transmitting port's CIST Port Role. This is checked on receipt by RSTP when
29 receiving messages from a Designated Bridge, thus ensuring that an RST Bridge does not incorrectly
30 identify one MST Bridge Port as being Designated rather than another, even while omitting the competing
31 Bridge Ports' Designated Bridge Identifiers from comparisons.

32 NOTE 2—This ability of MST Bridges to communicate the full set of MSTP information on shared LANs to which RST
33 Bridges are attached avoids the need for the Port Protocol Migration machines to detect RST Bridges. Two or more MST
34 and one or more RST Bridges may be connected to a shared LAN, with full MSTP operation. This includes the
35 possibility of different MSTI Designated Ports (see 13.5.1).

36 **13.7.2 Force Protocol Version**

37 A Force Protocol Version parameter, controlled by management, permits emulation of aspects of the
38 behavior of earlier versions of spanning tree protocol that are not strictly required for interoperability. The
39 value of this parameter applies to all Bridge Ports.

- 40 a) STP BPDUs, rather than MST BPDUs, are transmitted if Force Protocol Version is 0.
- 41 b) RST BPDUs, rather than MST BPDUs, are transmitted if Force Protocol Version is 2. RST BPDUs
42 omit the MCID and all MSTI Information.
- 43 c) All received BPDUs are treated as being from a different MST Region if Force Protocol Version is 0
44 or 2.

- 1 d) Rapid transitions are disabled if Force Protocol Version is 0. This allows MST Bridges to support
- 2 applications and protocols that can be sensitive to the increased rates of frame duplication and
- 3 misordering that can arise under some circumstances, as discussed in Annex P.
- 4 e) The MSTP state machines allow full MSTP behavior if Force Protocol Version is 3 or more.
- 5 f) SPT BPDUs are transmitted if Force Protocol Version is 4 or more.
- 6 NOTE—Force Protocol Version does not support multiple spanning trees with rapid transitions disabled.

7 **13.8 MST Configuration Identifier (MCID)**

8 It is essential that all Bridges within an MST or SPT Region agree on the allocation of VIDs to spanning
9 trees. If the allocation differs, frames for some VIDs may be duplicated or not delivered to some LANs at all.
10 MST and SPT Bridges check that they are allocating VIDs to the same spanning trees as their neighbors in
11 the same region by transmitting and receiving MCIDs in BPDUs. Each MCID includes a Configuration
12 Digest that is compact but designed so that two matching identifiers have a very high probability of denoting
13 the same allocation of VIDs to MSTIDs (Clause 3, 8.4) even if the identifiers are not explicitly managed.
14 Suitable management practices for equipment deployment and for choosing Configuration Names and
15 Revision Levels (see below) can guarantee that the identifiers will differ if the VID to tree allocation differs
16 within a single administrative domain.

17 An MST or SPT Region comprises one or more MST or SPT Bridges with the same MCIDs, interconnected
18 by and including LANs for which one of those Bridges is the Designated Bridge for the CIST and which
19 have no Bridges attached that cannot receive and transmit RST BPDUs.

20 SPT BPDUs are a superset of MST BPDUs received and validated by MST Bridges as if they were MST
21 BPDUs, so MSTP operates within an SPT Region just as if it were an MST Region. However, each shortest
22 path VLAN is represented by a reserved MSTID value that is included in the Configuration Digest, so an
23 SPT Region contains only SPT Bridges.

24 Each MCID contains the following components:

- 25 a) A Configuration Identifier Format Selector, the value 0 encoded in a fixed field of one octet to
- 26 indicate the use of the following components as specified in this standard.
- 27 b) The Configuration Name, a variable length text string encoded within a fixed field of 32 octets,
- 28 conforming to IETF RFC 2271's definition of SnmpAdminString. If the Configuration Name is less
- 29 than 32 characters, the text string should be terminated by the NUL character, with the remainder of
- 30 the 32-octet field filled with NUL characters. Otherwise, the text string is encoded with no
- 31 terminating NUL character.
- 32 c) The Revision Level, an unsigned integer encoded within a fixed field of 2 octets.
- 33 d) The Configuration Digest, a 16-octet signature of type HMAC-MD5 (see IETF RFC 2104) created
- 34 from the MST Configuration Table (Clause 3, 8.9). To calculate the digest, the table is considered to
- 35 contain 4096 consecutive two octet elements, where each element of the table (with the exception of
- 36 the first and last) contains an MSTID value encoded as a binary number, with the first octet being
- 37 most significant. The first element of the table contains the value 0, the second element the MSTID
- 38 value corresponding to VID 1, the third element the MSTID value corresponding to VID 2, and so
- 39 on, with the next to last element of the table containing the MSTID value corresponding to VID
- 40 4094, and the last element containing the value 0. The key used to generate the signature consists of
- 41 the 16-octet string specified in Table 13-1.

42 NOTE—The formulation of the signature as described above does not imply that a separate VID to MSTID translation
43 table has to be maintained by the implementation; rather that it should be possible for the implementation to derive the
44 logical contents of such a table, and the signature value as specified above, from the other configuration information
45 maintained by the implementation, as described in Clause 12.

Table 13-1—Configuration Digest Signature Key

Parameter	Mandatory value
Configuration Digest Signature Key	0x13AC06A62E47FD51F95D2BA243CD0346

¹ The Configuration Digests of some VID to MSTID translations are shown in Table 13-2 to help verify
² implementations of this specification.

Table 13-2—Sample Configuration Digest Signature Keys

VID to MSTID translation	Configuration Digest
All VIDs map to the CIST, no VID mapped to any MSTI	0xAC36177F50283CD4B83821D8AB26DE62
All VIDs map to MSTID 1	0xE13A80F11ED0856ACD4EE3476941C73B
Every VID maps to the MSTID equal to (VID modulo 32) + 1	0x9D145C267DBE9FB5D893441BE3BA08CE

³ It is recommended that MST Bridge implementations provide an easily selectable or default configuration
⁴ comprising a Configuration Name of the Bridge Address as a text string using the Hexadecimal
⁵ Representation specified in IEEE Std 802, a Revision Level of 0, and a Configuration Digest representing a
⁶ VID to MSTID translation table containing the value 0 for every element. Such a table represents the
⁷ mapping of all VIDs to the CIST. Since the Bridge Address is unique to each Bridge, no two Bridges using
⁸ this default configuration will be identified as belonging to the same region.

⁹ It is recommended that SPT Bridge implementations provide an easily selectable or default configuration
¹⁰ comprising a Configuration Name of 'IEEE802.1 SPB Default', a Revision Level of 0, and a Configuration
¹¹ Digest representing an MST Configuration Table containing the value 0xFFD for VID 1, the value 0xFFFF
¹² for VIDs 3600 to 3999, and the value 0 for every other element. Such a table represents the mapping of the
¹³ default PVID to SPBV, an SPVID pool containing VIDs 3600 to 3999, and mapping all other VLANs to the
¹⁴ CIST. This will allow SPT Bridges using the default configuration to form an SPT Region and operate the
¹⁵ default VLAN in SPBV mode.

¹⁶ An Auxiliary MCID (14.4, 27.4.1, 28.12.2) is specified for SPB configuration and is conveyed in SPT
¹⁷ BPDUs and SPB Hello PDUs. When a configuration change that cannot introduce loops is made, the
¹⁸ previous MCID can be retained in the Auxiliary MCID, allowing a neighboring Bridge whose configuration
¹⁹ has not yet been changed to maintain communication without creating a region boundary. This allows an
²⁰ operator to increase the number of SPVIDs available to SPBV mode or the number of Base VIDs for SPBV
²¹ or SPBM modes (for example) without disrupting communication. To avoid creating a region boundary all
²² the Bridges have to be configured so they are all using the same new MCID before any other change can be
²³ made.

²⁴ 13.9 Spanning tree priority vectors

²⁵ Priority vectors permit concise specification of each protocol's computation of the active topology, both in
²⁶ terms of the entire network and of the operation of individual Bridges in support of the distributed algorithm.
²⁷ MST, RST, and STP Bridges use *spanning tree priority vector* information in Configuration Messages
²⁸ (13.14), sent and received from neighboring Bridges, to assign Port Roles that determine each port's
²⁹ participation in a fully and simply connected active topology based on one or more spanning trees. SPT

1 Bridges use ISIS-SPB to disseminate the information necessary to calculate Port Roles throughout SPT
2 Regions, and to perform those calculations, but also use this Configuration Message information for the
3 CIST to ensure that neighboring Bridges agree on that active topology, and to receive the information if the
4 CIST Root lies outside their SPT Region.

5 CIST priority vectors comprise the following components:

- 6 a) CIST Root Identifier, the Bridge Identifier of the CIST Root.
- 7 b) CIST External Root Path Cost, the inter-regional cost from the transmitting Bridge to the CIST
8 Root.
- 9 c) CIST Regional Root Identifier, the Bridge Identifier of the single Bridge in a region whose CIST
10 Root Port connects to a LAN in a different region, or of the CIST Root if that is within the region.
- 11 d) CIST Internal Root Path Cost, the cost to the CIST Regional Root.
- 12 e) CIST Designated Bridge Identifier, the Bridge Identifier for the transmitting Bridge for the CIST.
- 13 f) CIST Designated Port Identifier, the Port Identifier for the transmitting port for the CIST.
- 14 g) CIST Receiving Port Identifier (not conveyed in Configuration Messages, used as tie-breaker
15 between otherwise equal priority vectors within a receiving Bridge).

16 The first two components of the CIST priority vector are significant throughout the network. The CIST
17 External Root Path Cost transmitted by a Bridge is propagated along each path from the CIST Root, is added
18 to at Bridge Ports that receive the priority vector from a Bridge in a different region, and thus accumulates
19 costs at the Root Ports of Bridges that are not MST or SPT Bridges or are CIST Regional Roots and is
20 constant within a region. The CIST Internal Root Path Cost is only significant and defined within a region.
21 The last three components are used as locally significant tie breakers, not propagated within or between
22 regions. The set of all CIST spanning tree priority vectors is thus totally ordered.

23 Since RSTP is not aware of regions, RSTP specifications also refer to the CIST Root Identifier and CIST
24 External Root Path Cost simply as the Root Bridge Identifier and Root Path Cost, respectively, and omit the
25 CIST Internal Root Path Cost (as does STP). MSTP encodes the CIST Regional Root Identifier in the BPDU
26 field used by RSTP to convey the Designated Bridge Identifier (14.1.1), so an entire region appears to an
27 RSTP-capable Bridge as a single Bridge. RSTP's CST use of CIST priority vectors can be conveniently
28 specified by the use of the zero for the Internal Root Path Cost and the same values for both the Regional
29 Root Identifier and Designated Bridge Identifier.

30 NOTE 1—The path to the CIST Root from a Bridge with a CIST Root Port within a region always goes to or through the
31 CIST Regional Root.

32 NOTE 2—STP lacks the fields necessary for MST Bridges to communicate the Designated Bridge Identifier to resolve a
33 potential priority vector tie, and MSTP BPDUs are not sent on a LAN to which an STP Bridge is attached.

34 Each MSTI priority vector comprises the following components for the particular MSTI in a given region:

- 35 h) MSTI Regional Root Identifier, the Bridge Identifier of the MSTI Regional Root
- 36 i) MSTI Internal Root Path Cost, the path cost to the MSTI Regional Root
- 37 j) MSTI Designated Bridge Identifier, the Bridge Identifier for the transmitting Bridge for this MSTI
- 38 k) MSTI Designated Port Identifier, the Port Identifier for the transmitting port for this MSTI
- 39 l) MSTI Receiving Port Identifier (not conveyed in Configuration Messages)

40 The set of priority vectors for a given MSTI is only defined within a region. Within each region they are
41 totally and uniquely ordered. A CIST Root Identifier, CIST External Root Path Cost, and CIST Regional
42 Root Identifier tuple defines the connection of the region to the external CST and is required to be associated
43 with the source of the MSTI priority vector information when assessing the agreement of information for
44 rapid transitions to forwarding, but plays no part in priority vector calculations.

1 As each Bridge and Bridge Port receives priority vector information from Bridges and ports closer to the
2 Root, calculations and comparisons are made to decide which priority vectors to record, and what
3 information to pass on. Decisions about a given port's role are made by comparing the priority vector
4 components that could be transmitted with that received by the port. For all components, a lesser numerical
5 value is better, and earlier components in the above lists are more significant. As each Bridge Port receives
6 information from ports closer to the Root, additions are made to one or more priority vector components to
7 yield a worse priority vector for potential transmission through other ports of the same Bridge.

8 NOTE 3—The consistent use of lower numerical values to indicate better information is deliberate as the Designated
9 Port that is closest to the Root Bridge, i.e., has a numerically lowest path cost component, is selected from among
10 potential alternatives for any given LAN (13.9). Adopting the conventions that lower numerical values indicate better
11 information, that where possible more significant priority components are encoded earlier in the octet sequence of a
12 BPDU (14.1), and that earlier octets in the encoding of individual components are more significant (14.2) allow
13 concatenated octets that compose a priority vector to be compared as if they were a multiple octet encoding of a single
14 number, without regard to the boundaries between the encoded components. To reduce the confusion that naturally arises
15 from having the lesser of two numerical values represent the better of the two, i.e., that chosen all other factors being
16 equal, this clause uses the following consistent terminology. Relative numeric values are described as “least,” “lesser,”
17 “equal,” and “greater,” and their comparisons as “less than,” “equal to,” or “greater than,” while relative spanning tree
18 priorities are described as “best,” “better,” “the same,” “different,” and “worse” and their comparisons as “better than,”
19 “the same as,” “different from,” and “worse than.” The operators “<” and “=” represent less than and equal to,
20 respectively. The terms “superior” and “inferior” are used for comparisons that are not simply based on priority but
21 include the fact that a priority vector can replace an earlier vector transmitted by the same Bridge Port. All of these terms
22 are defined for priority vectors in terms of the numeric comparison of components below (13.10, 13.11).

23 NOTE 4—To ensure that the CIST and each MSTI's view of the boundaries of each region remain in synchronization at
24 all times, each BPDU carries priority vector information for the CIST as well as for MSTIs. Associating the CIST Root
25 Identifier, External Path Cost, and Regional Root Identifier with the priority vector information for each MSTI does not
26 therefore raise a requirement to transmit these components separately. A single bit per MSTI vector, the Agreement flag,
27 satisfies the requirement to indicate that the vector beginning with the MSTI Regional Root Identifier for that specific
28 MSTI has always been associated with the single CIST Root Identifier, etc. transmitted in the BPDU.

29 To allow the active topology to be managed for each tree through adjusting the relative priority of different
30 Bridges and Bridge Ports for selection as the CIST Root, a CIST or MSTI Regional Root, Designated
31 Bridge, or Designated Port, the priority component of the Bridge's Bridge Identifier can be independently
32 chosen for the CIST and for each MSTI. The priority component used by the CIST for its CIST Regional
33 Root Identifier can also be chosen independently of that used for the CIST Root Identifier. Independent
34 configuration of Port Path Cost and Port Priority values for the CIST and for each MSTI can also be used to
35 control selection of the various roles for the CIST and for each MSTI.

36 In principle an SPT priority vector could be defined within an SPT Region, comprising a Regional Root
37 Identifier and Internal Root Path Cost, and would reflect the construction of each SPT (lowest Internal Root
38 Path Cost from each Bridge and LAN to the Regional Root). However, the set of such vectors cannot be
39 totally ordered by the addition of purely local tie-breaker components: as each SPT Set has to be symmetric.
40 Clause 28 specifies ISIS-SPB's calculation of SPTs.

41 13.10 CIST Priority Vector calculations

42 The *port priority vector* is the priority vector held for the port when the reception of BPDUs and any
43 pending update of information has been completed:

44
$$\text{port priority vector} = \{ \text{RootID} : \text{ExtRootPathCost} : \\$$

45
$$\text{RRootID} : \text{IntRootPathCost} : \\$$

46
$$\text{DesignatedBridgeID} : \text{DesignatedPortID} : \text{RcvPortID} \}$$

47 The *message priority vector* is the priority vector conveyed in a received Configuration Message. For a
48 Bridge with Bridge Identifier B receiving a Configuration Message on a port P_B from a Designated Port P_D
49 on Bridge D claiming a CIST Root Identifier of R_D , a CIST External Root Path Cost of ERC_D , a CIST
50 Regional Root Identifier of RR_D , and a CIST Internal Root Path Cost of IRC_D :

51
$$\text{message priority vector} = \{ R_D : ERC_D : RR_D : IRC_D : D : P_D : P_B \}$$

1 If B is not in the same region as D , the Internal Root Path Cost has no meaning to B and is set to 0.

2 NOTE 1—If a Configuration Message is received in an RST or STP BPDU, both the Regional Root Identifier and the
3 Designated Bridge Identifier are decoded from the single BPDU field used for the Designated Bridge Parameter (the
4 MST BPDU field in this position encodes the CIST Regional Root Identifier). An STP or RST Bridge is always treated
5 by MSTP as being in an region of its own, so the Internal Root Path Cost is decoded as zero.

6 The received CIST message priority vector is the same as B 's port priority vector if:

7 $(R_D == RootID) \ \&\& \ (ERC_D == ExtRootPathCost) \ \&\& \ (RR_D == RRootID) \ \&\&$
8 $(IRC_D == IntRootPathCost) \ \&\& \ (D == DesignatedBridgeID) \ \&\& \ (P_D == DesignatedPortID)$

9 and is better if:

10 $((R_D < RootID)) \ ||$
11 $((R_D == RootID) \ \&\& \ (ERC_D < ExtRootPathCost)) \ ||$
12 $((R_D == RootID) \ \&\& \ (ERC_D == ExtRootPathCost) \ \&\& \ (RR_D < RRootID)) \ ||$
13 $((R_D == RootID) \ \&\& \ (ERC_D == ExtRootPathCost) \ \&\& \ (RR_D == RRootID)$
14 $\ \&\& \ (IRC_D < IntRootPathCost)) \ ||$
15 $((R_D == RootID) \ \&\& \ (ERC_D == ExtRootPathCost) \ \&\& \ (RR_D == RRootID)$
16 $\ \&\& \ (IRC_D == IntRootPathCost) \ \&\& \ (D < DesignatedBridgeID)) \ ||$
17 $((R_D == RootID) \ \&\& \ (ERC_D == ExtRootPathCost) \ \&\& \ (RR_D == RRootID)$
18 $\ \&\& \ (IRC_D == IntRootPathCost) \ \&\& \ (D == DesignatedBridgeID)$
19 $\ \&\& \ (P_D < DesignatedPortID))$

20 A received CIST message priority vector is superior to the port priority vector if, and only if, the message
21 priority vector is better than the port priority vector, or the Designated Bridge Identifier Bridge Address and
22 Designated Port Identifier Port Number components are the same; in which case, the message has been
23 transmitted from the same Designated Port as a previously received superior message, i.e., if:

24 $\{R_D : ERC_D : RR_D : IRC_D : D : P_D : P_B\}$
25 *is better than*
26 $\{RootID : ExtRootPathCost : RRootID : IntRootPathCost :$
27 $DesignatedBridgeID : DesignatedPortID : RcvPortID\}$
28 $\} \ || \ ((D.BridgeAddress == DesignatedBridgeID.BridgeAddress) \ \&\&$
29 $(P_D.PortNumber == DesignatedPortID.PortNumber))$

30 If the message priority vector received in a Configuration Message from a Designated Port is superior, it will
31 replace the current port priority vector.

32 A *root path priority vector* for a port can be calculated from a port priority vector that contains information
33 from a message priority vector, as follows:

34 If the port priority vector was received from a Bridge in a different region (13.29.8), the External Port Path
35 Cost EPC_{PB} is added to the External Root Path Cost component, and the Regional Root Identifier is set to
36 the value of the Bridge Identifier for the receiving Bridge. The Internal Root Path Cost component will have
37 been set to zero on reception.

38 $root \ path \ priority \ vector = \{R_D : ERC_D + EPC_{PB} : B : 0 : D : P_D : P_B\}$

39 If the port priority vector was received from a Bridge in the same region (13.29.8), the Internal Port Path
40 Cost IPC_{PB} is added to the Internal Root Path Cost component.

41 $root \ path \ priority \ vector = \{R_D : ERC_D : RR_D : IRC_D + IPC_{PB} : D : P_D : P_B\}$

1 The *Bridge priority vector* for a Bridge B is the priority vector that would, with the Designated Port
2 Identifier set equal to the transmitting Port Identifier, be used as the message priority vector in Configuration
3 Messages transmitted on Bridge B 's Designated Ports if B was selected as the Root Bridge of the CIST.

4
$$\text{Bridge priority vector} = \{B : 0 : B : 0 : B : 0 : 0\}$$

5 The *root priority vector* for Bridge B is the best priority vector of the set of priority vectors comprising:

- 6 a) The Bridge priority vector; plus
- 7 b) All root path priority vectors that:
 - 8 1) Have a Designated Bridge Identifier D that is not equal to B , and
 - 9 2) Were received from a Bridge Port attached to a LAN that is not in the same SPT Region as B ;
- 10 plus
- 11 c) The root path priority vector calculated by ISIS-SPB (if SPB is enabled, and the attached LAN is
12 within the Bridge's SPT Region).

13 NOTE 2—The BPDUs sent and received by all Bridges attached to a LAN allow MST and SPT Bridges to determine
14 whether each attached LAN is within their region independently of priority vector values. SPT Bridges take advantage of
15 this fact by using ISIS-SPB to communicate the CST priority vector for each of SPT Region's potential Master Ports
16 throughout the region, at the same time as ISIS-SPB calculates the Port Roles for each SPT (see Clause 28).

17 If the Bridge priority vector is the best of this set of priority vectors, Bridge B has been selected as the CIST
18 Root. Otherwise, the root priority vector will only be that calculated by ISIS-SPB if the Root Bridge or
19 Regional Root is within the Bridge's SPT Region.

20 The *designated priority vector* for a port Q on Bridge B is the root priority vector with B 's Bridge Identifier
21 B substituted for the *DesignatedBridgeID* and Q 's Port Identifier Q_B substituted for the *DesignatedPortID*
22 and *RcvPortID* components. If Q is attached to a LAN that has one or more STP Bridges attached (as
23 determined by the Port Protocol Migration state machine), B 's Bridge Identifier B is also substituted for the
24 *RRootID* component.

25 If the designated priority vector is better than the port priority vector and the LAN attached to the port is not
26 within the Bridge's SPT Region (possibly because SPB is not enabled), the port will be the Designated Port
27 for that LAN and the current port priority vector will be updated. If the attached LAN is within the Bridge's
28 SPT Region, then the Port Role is as calculated by ISIS-SPB and the port priority vector will be updated
29 with the designated priority vector if and only if the port is a Designated Port. The message priority vector in
30 Configuration Messages transmitted by a port always comprises the components of the designated priority
31 vector for the port, even if the port is a Root Port.

32 13.11 MST Priority Vector calculations

33 The *port priority vector* for a given MSTI is the priority vector held for the port per MSTI when the
34 reception of BPDUs and any pending update of information has been completed:

35
$$\text{port priority vector} = \{RRootID : IntRootPathCost : \\ 36 \text{DesignatedBridgeID} : \text{DesignatedPortID} : RcvPortID\}$$

37 The *message priority vector* for a given MSTI is the MSTI priority vector conveyed in a received
38 Configuration Message. For a Bridge with Bridge Identifier B receiving a Configuration Message on a
39 Regional Port P_B from a Designated Port P_D on Bridge D belonging to the same MST Region and claiming
40 an Internal Root Path Cost of IRC_D :

41
$$\text{message priority vector} = \{RR_D : IRC_D : D : P_D : P_B\}$$

42 An MSTI message priority vector received from a Bridge not in the same MST Region is discarded.

1 An MSTI message priority vector received from a Bridge Port internal to the region is the same as the port
2 priority vector if:

3 $((RR_D == RRootID) \&\& (IRC_D == IntRootPathCost) \&\& (D == DesignatedBridgeID)$
4 $\&\& (P_D == DesignatedPortID))$

5 and is better if:

6 $((RR_D < RRootID)) ||$
7 $((RR_D == RRootID) \&\& (IRC_D < IntRootPathCost)) ||$
8 $((RR_D == RRootID) \&\& (IRC_D == IntRootPathCost) \&\& (D < DesignatedBridgeID)) ||$
9 $((RR_D == RRootID) \&\& (IRC_D == IntRootPathCost) \&\& (D == DesignatedBridgeID)$
10 $\&\& (P_D < DesignatedPortID))$

11 An MSTI message priority vector is superior to the port priority vector if, and only if, the message priority
12 vector is better than the port priority vector, or the Designated Bridge Identifier Bridge Address and
13 Designated Port Identifier Port Number components are the same; in which case, the message has been
14 transmitted from the same Designated Port as a previously received superior message, i.e., if:

15 $\{RR_D : IRC_D : D : P_D : P_B\}$
16 *is better than*
17 $\{RRootID : IntRootPathCost : DesignatedBridgeID : DesignatedPortID : RcvPortID\}$
18 $) || ((D.BridgeAddress == DesignatedBridgeID.BridgeAddress) \&\&$
19 $(P_D.PortNumber == DesignatedPortID.PortNumber))$

20 If the message priority vector received in a Configuration Message from a Designated Port for the MSTI is
21 superior, it will replace the current port priority vector.

22 NOTE 1—The agree flag (13.27.3) for the port and this MSTI will be cleared if the CIST Root Identifier, CIST External
23 Root Path Cost, and CIST Regional Root Identifier in the received BPDU are not better than or the same as those for the
24 CIST designated priority vector for the port following processing of the received BPDU.

25 A *root path priority vector* for a given MSTI can be calculated for a port that has received a port priority
26 vector from a Bridge in the same region by adding the Internal Port Path Cost IPC_{PB} to the Internal Root
27 Path Cost component.

28 $root\ path\ priority\ vector = \{RR_D : IRC_D + IPC_{PB} : D : P_D : P_B\}$

29 NOTE 2—Internal Port Path Costs are independently manageable for each MSTI, as are the priority components of the
30 Bridge and Port Identifiers. The ability to independently manage the topology of each MSTI without transmitting
31 individual Port Path Costs is a key reason for retaining the use of a Distance Vector protocol for constructing MSTIs. A
32 simple link state protocol requires transmission (or *a priori* sharing) of all Port Costs for all links.

33 The *Bridge priority vector* for a Bridge B for a given MSTI is the priority vector that would, with the
34 Designated Port Identifier set equal to the transmitting Port Identifier, be used as the message priority vector
35 in Configuration Messages transmitted on Bridge B 's Designated Ports if B was selected as the Root Bridge
36 of a given tree.

37 $Bridge\ priority\ vector = \{B : 0 : B : 0\}$

38 The *root priority vector* for Bridge B is the best priority vector of the set of priority vectors comprising the
39 Bridge priority vector plus all root path priority vectors whose Designated Bridge Identifier D is not equal to
40 B . If the Bridge priority vector is the best of this set of priority vectors, Bridge B has been selected as the
41 Root of the tree.

42 The *designated priority vector* for a port Q on Bridge B is the root priority vector with B 's Bridge Identifier
43 B substituted for the *DesignatedBridgeID* and Q 's Port Identifier Q_B substituted for the *DesignatedPortID*
44 and *RcvPortID* components.

1 If the designated priority vector is better than the port priority vector, the port will be the Designated Port for
2 the attached LAN and the current port priority vector will be updated. The message priority vector in MSTP
3 BPDUs transmitted by a port always comprises the components of the designated priority vector of the port,
4 even if the port is a Root Port.

5 Figure 13-8 shows the priority vectors and the active topology calculated for an MSTI in a region of the
6 example network of Figure 13-6.

7 **13.12 Port Role assignments**

8 Each Bridge assigns CIST Port Roles (when new information becomes available as specified in this
9 subclause, 13.12) before assigning MSTI or SPT Port Roles. The calculations specified in 13.10 are used to
10 assign a role to each Bridge Port that is enabled as follows:

- 11 a) If the Bridge is not the CIST Root, the source of the root priority vector is the Root Port.
- 12 b) Each port whose port priority vector is the designated priority vector is a Designated Port.
- 13 c) Each port, other than the Root Port, with a port priority vector received from another Bridge is a
14 Alternate Port.
- 15 d) Each port with a port priority vector received from another port on this Bridge is a Backup Port.

16 If the port is not enabled, i.e., its MAC_Operational status is FALSE or its Administrative Bridge Port state
17 is Disabled (8.4), it is assigned the Disabled Port role for the CIST, all MSTIs, and all SPTs, to identify it as
18 having no part in the operation of any of the spanning trees or the active topology of the network.

19 If the Bridge is an MST or SPT Bridge, the calculations specified in 13.11 are used to assign a role to each
20 enabled Bridge Port for each MSTI as follows:

- 21 e) If the port is the CIST Root Port and the CIST port priority vector was received from a Bridge in
22 another MST or SPT Region, the port is the Master Port.
- 23 f) If the Bridge is not the MSTI Regional Root, the port that is the source of the MSTI root priority
24 vector is the Root Port.
- 25 g) Each port whose port priority vector is the designated priority vector derived from the root priority
26 vector is a Designated Port.
- 27 h) Each port, other than the Master Port or the Root Port, with a port priority vector received from
28 another Bridge or a CIST port priority vector from a Bridge in another region, is an Alternate Port.
- 29 i) Each port that has a port priority vector that has been received from another port on this Bridge is a
30 Backup Port.

31 Independently of priority vector values and active topology calculations, each SPT Bridge Port determines
32 from received BPDUs whether all the Bridges attached to its LAN are in the same SPT Region. If not, the
33 port is a Boundary Port, and its role for each SPT is determined by its CIST Port Role as follows:

- 34 j) If the port is the CIST Root Port, the port is the Master Port for all SPTs.
- 35 k) If the port is not the CIST Root Port, the port's role is the same as that for the CIST.

36 By excluding Boundary Ports from the physical topology used to calculate SPTs, and adopting CIST
37 connectivity at those ports, ISIS-SPB ensures that the use of SPVIDs (see Clause 27) is not required on
38 shared media LANs attached to Bridges in other regions.

39 SPT Bridges use ISIS-SPB to assign Port Roles for each SPT to non-Boundary Ports as follows:

- 40 l) If the Bridge is not the SPT Root Bridge, the port that ISIS-SPB has calculated as providing the path
41 for frames assigned to the SPT and forwarded to the Bridge from that SPT Root is the Root Port.
- 42 m) Each port, other than the Root Port, that ISIS-SPB has calculated as providing a path for frames
43 forwarded from the SPT Root to the attached LAN is a Designated Port.
- 44 n) Each port that is attached to the same LAN as another port on that same Bridge that is a Designated
45 Port for the SPT, is a Backup Port.
- 46 o) Each port not assigned a Root, Designated, or Backup Port role is an Alternate Port.

13.13 Stable connectivity

This subclause provides an analysis to show that RSTP, MSTP, and ISIS-SPB meet the goal of providing full and simple connectivity for frames assigned to any given VID in a stable network, i.e., where the physical topology has remained constant for long enough that the spanning tree information communicated and processed by Bridges is not changing.

NOTE 1—The FDB can be configured to prevent connectivity, in particular this analysis assumes that every Bridge Port is a member of every VID's Member Set (8.8.10). Spanning tree protocol controls can also be used to prevent new connectivity (to allow for upgrades), or to disallow certain topologies (restricting the location of the CIST Root, for example). This analysis assumes that those controls are not being used, that all the Bridges are using conformant protocol implementations and that the LANs are providing omnidirectional connectivity.

Every LAN provides connectivity for all frames between all attached Bridge Ports. Every Bridge provides connectivity between and only between its CIST Root and Designated Ports for frames assigned to the CIST; between the Root, Designated, and Master Ports for a given MSTI for frames assigned to that MSTI; and between Root and Designated Ports for a given SPT for frames assigned to that SPT. Any given Bridge does not assign frames to more than one tree and has one Root Port per tree, unless it is the Root of that tree.

Every LAN has one and only one CIST Designated Port, and every Bridge apart from the CIST Root has one and only one CIST Root Port. The CIST spanning tree priority vector of the Designated Port attached to the LAN that is connected to a Bridge's Root Port is better than of any Designated Port of that Bridge. The CIST thus connects all Bridges and LANs (is "spanning") and loop-free (is a "tree").

Each MST or SPT Region is bounded by CST Root and Alternate Ports. At the CST Root Ports connectivity for frames assigned to MSTIs or SPTs within the connected regions is the same as that for the CIST. Every region apart from that containing the CIST Root has a single CST Root Port, identified as the Master Port for each MSTI. The CIST spanning tree priority vector of the LAN attached to the region's CST Root Port is better than that of any CST Designated Port of a Bridge in the region attached to a LAN also attached to the CST Root Port of another region. The CST thus provides loop-free connectivity between all regions.

NOTE 2—The term "Common Spanning Tree (CST)" refers to the CIST connectivity between regions, and the term "Internal Spanning Tree (IST)" to the CIST connectivity within each region. An RST Bridge and the LANs for which it is the Designated Bridge are conveniently considered as forming an MST region of limited extent.

Within each region each frame is consistently assigned to the CIST, an MSTI, or an SPT, and each of these spanning trees provides full loop-free connectivity to each of the Bridges within the region, just as the CIST does for the network as a whole, including connectivity between the CST Root Port (Master Port) and the CST Designated Ports. Since each Bridge or LAN is in one and only one region, and it has already been shown that loop-free connectivity is provided between regions, loop-free connectivity is thus provided between all the Bridges and LANs in the network.

Figure 13-9 illustrates the above connectivity with the simple example of Region 1 from the example network of Figure 13-6 and Figure 13-8. Bridge 0.42 has been selected as the CIST Root and Regional Root, Bridge 0.57 as the Regional Root for MSTI 1, and Bridge 2.83 for MSTI 2 by management of the per MSTI Bridge Identifier priority component. The potential loop through the three Bridges in the region is blocked at different Bridge Ports for the CIST, and each MSTI, but the connectivity across the region and from each LAN and Bridge in the region through the boundaries of the region is the same in all cases.

13.14 Communicating spanning tree information

A Spanning Tree Protocol Entity transmits and receives group addressed BPDUs (Clause 14, 8.13.4) through each of its Bridge Ports to communicate with the Spanning Tree Protocol Entities of the other Bridges attached to the same LAN. The group address used is one of a small number of addresses that identify frames that are not directly forwarded by Bridges (8.6.3), but the information in the BPDU can be used by a Bridge in calculating its own BPDUs to transmit and can stimulate that transmission.

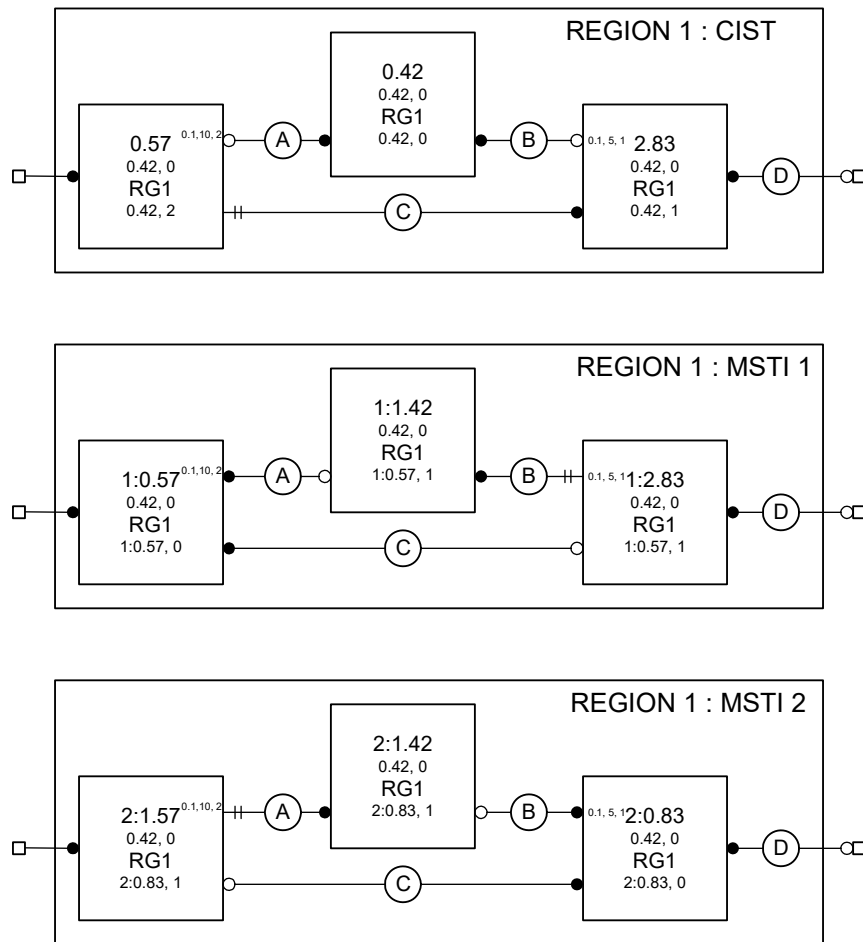


Figure 13-9—CIST and MSTI active topologies in Region 1 of the example network

1 BPDUs are used to convey the following:

- 2 a) Configuration Messages
- 3 b) Topology Change Notification (TCN) Messages
- 4 c) MCIDs
- 5 d) Agreement parameters to support SPB

6 Designated Ports also transmit BPDUs at intervals to guard against loss and to assist in the detection of
7 failed components (LANs, Bridges, or Bridge Ports), so all messages are designed to be idempotent.

8 A Configuration Message for the CIST can be encoded in an STP Configuration BPDU, an RST BPDU, an
9 MST BPDU, or an SPT BPDU (14.1, 14.5). A TCN Message for the CIST can be encoded in an STP
10 Topology Change Notification BPDU (14.2), or an RST, MST, or SPT BPDU with the TC flag set.
11 Configuration and TCN Messages for the CIST and for all MSTIs in an MST Region are encoded in a single
12 MST or SPT BPDU, as is the MCID. No more than 64 MSTI Configuration Messages shall be encoded in an
13 MST BPDU, and no more than 64 MSTIs shall be supported by an MST Bridge.

1 When SPB is enabled, ISIS-SPB is used to communicate CST priority vectors, IST topology information,
2 and CST topology change information to and from the other Bridges in the SPT Region and to calculate IST
3 and SPT Port Roles and designated priority vectors. However, the full CIST priority information is still
4 conveyed in SPT BPDUs, partly to encode agreement information for the IST, but principally to support
5 interoperability at the boundaries of each region without having to assess whether the transmitting port is at
6 a boundary before deciding what to encode in each BPDU.

7 Configuration and Topology Change Notification BPDUs are distinguished from each other and from RST
8 and MST BPDUs by their BPDU Type (Clause 14). RST and MST BPDUs share the same BPDU Type and
9 are distinguished by their version identifiers.

10 Bridges implementing STP (Clause 8 of IEEE Std 802.1D, 1998 Edition [B13]) transmit and decode
11 Configuration and Topology Change Notification BPDUs, and ignore RST and MST BPDUs on receipt.
12 This ensures that connection of a Bridge Port of such a Bridge to a LAN that is also attached to a Bridge
13 implementing RSTP or MSTP is detected, as transmission of RSTP or MSTP BPDUs does not suppress
14 regular transmissions by the STP Bridge. This functionality is provided by the Port Protocol Migration state
15 machine for RSTP (13.32). The Port Protocol Migration state machines select the BPDU types used to
16 encode spanning tree messages so that all Bridges attached to the same LAN participate in a spanning tree
17 protocol, while maximizing the available functionality. If one or more attached Bridges only implement STP,
18 only Configuration and Topology Change Notification BPDUs will be used and the functionality provided
19 by the protocol will be constrained.

20 **13.15 Changing spanning tree information**

21 Addition, removal, failure, or management of the parameters of Bridges and LAN connectivity can change
22 spanning tree information and require Port Role changes in all or part of the network (for the CIST) or all or
23 part of an MST or SPT Region (for an MSTI or SPT Set). A CIST or MSTI configuration message received
24 in a BPDU is considered superior to, and will replace, that recorded in the reception Port's port priority
25 vector if its message priority vector is better, or if it was transmitted by the same Designated Bridge and
26 Designated Port and the message priority vector, timer, or hop count information differ from those recorded.

27 RSTP and MSTP propagate new information rapidly from Bridge to Bridge, superseding prior information
28 and stimulating further transmissions until it reaches either Designated Ports that have already received the
29 new information through redundant paths in the network or the leaves of the spanning tree, as defined by the
30 new configuration. Configuration Message transmissions will then once more occur at regular intervals from
31 ports selected as Designated Ports.

32 To ensure that old information does not endlessly circulate through redundant paths in the network,
33 preventing the effective propagation of the new information, MSTP associates a hop count with the
34 information for each spanning tree. The hop count is assigned by the CIST Regional Root or the MSTI
35 Regional Root and decremented by each reception Port. Received information is discarded and the port
36 made a Designated Port if the hop count reaches zero.

37 RSTP and MSTP's CST processing do not use an explicit hop count (for reasons of STP compatibility), but
38 detect circulating aged information by treating the BPDU Message Age parameter as an incrementing hop
39 count with Max Age as its maximum value. MSTP increments Message Age for information received at the
40 boundary of an MST Region, discarding the information if necessary.

1 If a Bridge Port's MAC_Operational parameter becomes FALSE, the port becomes a Disabled Port and
2 received information is discarded. Spanning tree information for the tree can be recomputed, the Bridge's
3 Port Roles changed, and new spanning tree information transmitted if necessary. Not all component failure
4 conditions can be detected in this way, so each Designated Port transmits BPDUs at regular intervals and a
5 reception Port will discard information and become a Designated Port if two transmissions are missed.

6 NOTE—Use of a separate hop count and message loss detection timer provides superior reconfiguration performance
7 compared with the original use of Message Age and Max Age by STP. Connectivity loss detection is not compromised
8 by the need to allow for the overall diameter of the network, nor does the time allowed extend the number of hops
9 permitted to aged recirculating information. Management calculation of the necessary parameters for custom topologies
10 is also facilitated, as no allowance needs to be made for relative timer jitter and accuracy in different Bridges.

11 ISIS-SPB communicates CST, IST, and SPT information throughout an SPT Region using link state
12 procedures specified in Clause 28. In addition to the normal hop-by-hop distribution of this information,
13 which is essential to guarantee its dissemination, each link state PDU (LSP) is also broadcast on the SPT
14 rooted at the originating Bridge, so new information can reach Bridges in the region with the same delay as
15 data.

16 13.16 Changing Port States with RSTP or MSTP

17 The Port State for the CIST and each MSTI for each Bridge Port is controlled by state machines whose goal
18 is to maximize connectivity without introducing temporary loops in each of these active topologies. Root
19 Ports, Master Ports, and Designated Ports are transitioned to the Forwarding Port State, and Alternate Ports
20 and Backup Ports to the Discarding Port State, as rapidly as possible. Transitions to the Discarding Port State
21 can be simply effected without the risk of data loops. This subclause (13.16) describes the conditions that
22 RSTP and MSTP use to transition a Port State for a given spanning tree to Forwarding.

23 Starting with the assumption that any connected fragment of a network is composed of Bridges, Bridge
24 Ports, and connected LANs that form a subtree of a spanning tree, ports with Root Port, Master Port, or
25 Designated Port roles are transitioned using conditions that ensure that the newly enlarged fragment
26 continues to form either a subtree or the whole of the spanning tree. Since the conditions are used every time
27 a fragment is enlarged, it is possible to trace the growth of a fragment from a single Bridge—a consistent, if
28 small, subtree of a spanning tree—to any sized fragment, thus justifying the initial assumption.

29 Port States in two subtrees, each bounded by ports that are not forwarding or are attached to LANs not
30 attached to any other Bridge, can be made consistent by waiting for any changes in the priority vector
31 information used to assign Port Roles to reach all Bridges in the network, thus ensuring that the subtrees are
32 not, and are not about to be, joined by other Forwarding Ports. However, it can be shown that a newly
33 selected Root Port can forward frames as soon as prior recent root ports on the same Bridge cease to do so,
34 without further communication from other Bridges. Rapid transitions of Designated Ports and Master Ports
35 do require an explicit Agreement from the Bridges in the subtrees to be connected. The Agreement
36 mechanism is described, together with a Proposal mechanism that forces satisfaction of the conditions if
37 they have not already been met by blocking Designated Ports connecting lower subtrees that are not yet in
38 agreement. The same Agreement mechanism is then used to transition the newly blocked ports back to
39 forwarding, advancing any temporary cut in the active topology toward the edge of the network.

40 13.16.1 Subtree connectivity and priority vectors

41 Any given Bridge B , the LANs connected through its Forwarding Designated Ports, the further Bridges
42 connected to those LANs through their Root Ports, the LANs connected to their Forwarding Designated
43 Ports, and so on, recursively, constitute a subtree S_B . Any LAN L that is part of S_B will be connected to B
44 through a Forwarding Designated Port P_{CL} on a Bridge C also in S_B . L cannot be directly connected to any
45 port P_B on Bridge B unless B and C are one and the same, since the message priority vector for P_B is better
46 than that of any port of any other Bridge in S_B , and prior to Forwarding P_{CL} will have advertised its spanning

1 port priority vector for long enough for it to receive any better message priority vector (within the design
2 probabilities of protocol failure due to repeated BPDU loss) or will have engaged in an explicit confirmed
3 exchange (see below) with all other Bridge Ports attached to that LAN.

4 NOTE—The analysis for the distance vector-based RSTP and MSTP differs from that for the link state-based ISIS-SPB
5 (see 13.17). In the latter C's priority vector can become better than B's while C remains in S_B , without B being aware of
6 the improvement first.

7 13.16.2 Root Port transition to Forwarding

8 It follows from the above that B's Root Port can be transitioned to Forwarding immediately whether it is
9 attached to a LAN in S_B or in the rest of the network, provided that all prior recent Root Ports on B (that
10 might be similarly arbitrarily attached) have been transitioned to Discarding and the Root Port was not a
11 Backup Port recently (B and C the same as above).

12 13.16.3 Designated Port transition to Forwarding

13 On any given Bridge A, the Designated Port P_{AM} connected to a LAN M can be transitioned to Forwarding
14 provided that the message priority advertised by the Designated Port P_{CL} on any LAN L in any subtree S_{M1} ,
15 S_{M2} , etc. connected to M is worse than that advertised by P_{AM} ; that any Bridge D attached to L has agreed
16 that P_{CL} is the Designated Port; and that only the Root Port and Designated Ports on D are Forwarding. A
17 sufficient condition for P_{AM} to transition to Forwarding is that M is a point-to-point link attached to the Root
18 Port P_{BM} of a Bridge B, that the port priority of P_{BM} is the same as or worse than that of P_{AM} , and any port
19 P_{BN} on B is Discarding or similarly attached to a Bridge C. P_{BM} signals this condition to P_{AM} by setting the
20 Agreement flag in a Configuration Message carrying P_{BM} 's designated priority and Port Role.

21 NOTE 1—RSTP and MSTP use adminPointToPointMAC and operPointToPointMAC (IEEE Std 802.1AC) to allow the
22 point-to-point status of LANs to be managed and used by the Port Role Transition state machines for Designated Ports.
23 A newly selected Root Port can be transitioned to Forwarding rapidly, even if attached to a shared media LAN.

24 Figure 13-10 illustrates the generation of an Agreement at a Bridge's Root Port from an Agreement received
25 or a Port State of Discarding at each of its Designated Ports, and a Port State of Discarding at each of its
26 Alternate and Backup Ports. A Bridge receiving a Proposal transitions any Designated Port not already
27 synchronized to Discarding so it can send the Agreement, and that port solicits an Agreement by sending a
28 Proposal in its turn.

29 NOTE 2—Agreements can be generated without prior receipt of a Proposal as soon as the necessary conditions are met.
30 Subsequent receipt of a Proposal serves to elicit a further Agreement. If all other ports have already been synchronized
31 (allSynced in Figure 13-10) and the Proposal's priority vector does not convey worse information, synchronization is
32 maintained and there is no need to transition Designated Ports to Discarding once more, or to transmit further Proposals.

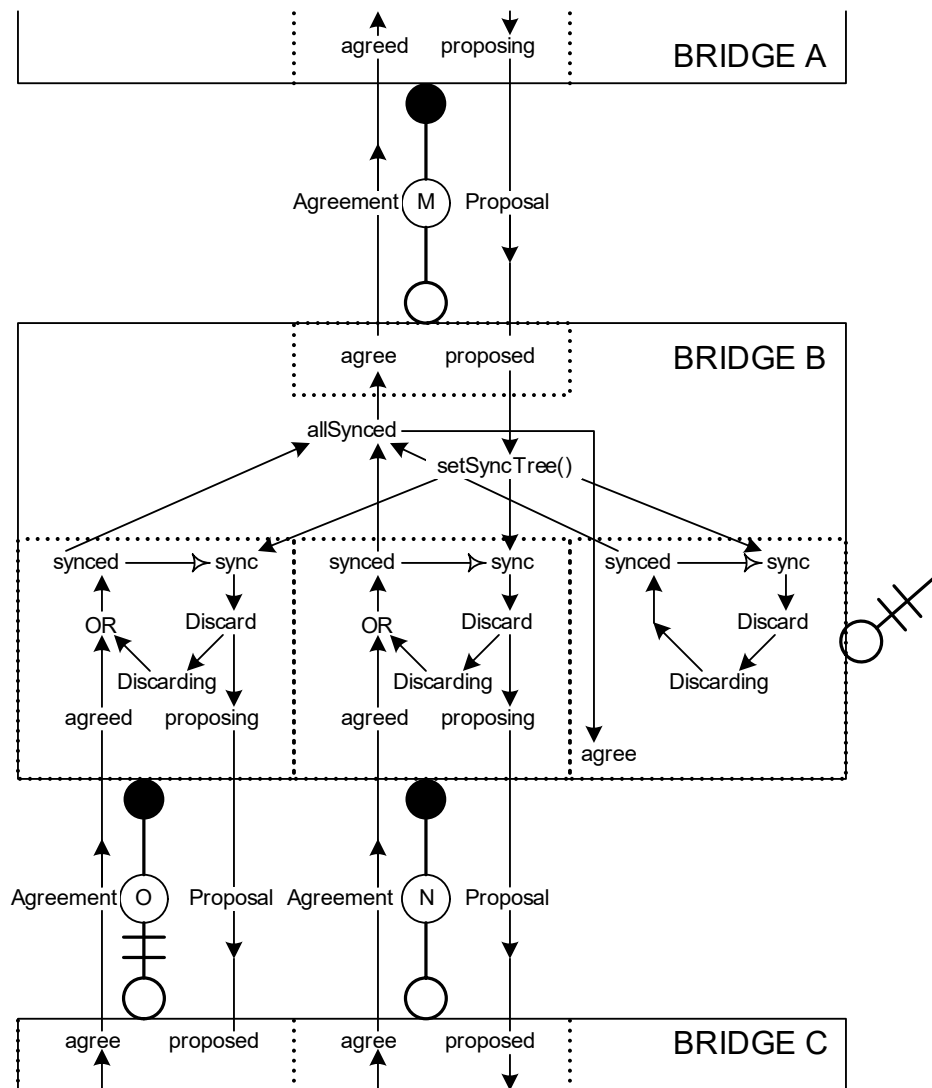


Figure 13-10—Agreements and Proposals

13.16.4 Master Port transition to Forwarding

While the connectivity of the CIST from the CIST Regional Root through an MST Region to the rest of the CST comprises a subtree rooted in the CIST Regional Root, the connectivity of the MSTI from the Master Port includes both a subtree below the CIST Regional Root and a subtree rooted in the MSTI Regional Root and connected to the CIST Regional Root by an MSTI Root Port. In the example network of Figure 13-6, this latter subtree continues CST connectivity, from the Master Port on Bridge 86 through to LAN N, for frames allocated to the MSTI within Region 2 (see Figure 13-11). In general either MSTI subtree could be providing CST connectivity through a prior Master Port: the connectivity of both subtrees has to agree with the new CIST Regional Root, before a new Master Port transitions to Forwarding.

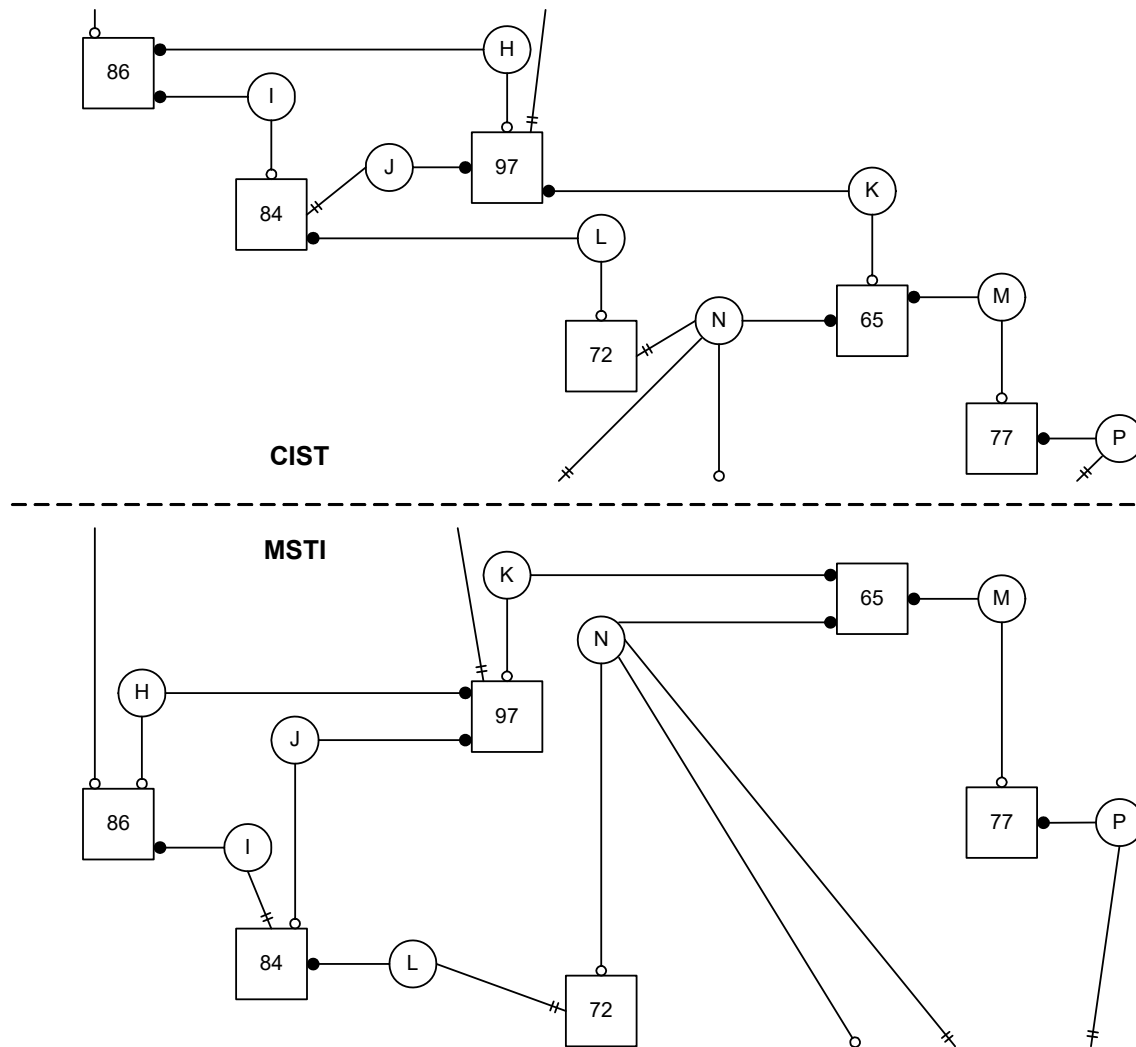


Figure 13-11—CIST and MSTI Active Topologies in Region 2 of Figure 13-6

NOTE 1—The physical layout shown in the two halves of Figure 13-11 differs in order to reflect the different priorities and logical topologies for the two spanning tree instances. The layout convention is that Designated Ports are shown as horizontal lines, Root Ports as vertical lines, and Alternate Ports as diagonal lines.

Figure 13-12 illustrates the extension of the Agreement mechanism to signal from Designated Ports to Root Ports as well as vice versa. To ensure that an MSTI does not connect alternate Master Ports, an Agreement is only recognized at an MSTI Port when the associated CIST Regional Root information matches that selected

1 by the reception Port. Proposals, eliciting Agreements, necessarily flow from Designated Ports to Root Ports
2 with the propagation of spanning tree information, so a new CIST Regional Root cannot transmit a Proposal
3 directly on its MSTI Root Ports. However, updating a CIST Designated Port's port priority vector with a
4 new Regional Root Identifier forces the port to discard frames for all MSTIs, thus initiating the Proposal
5 from the first Bridge nearer the MSTI Regional Root that learns of the new Regional Root.

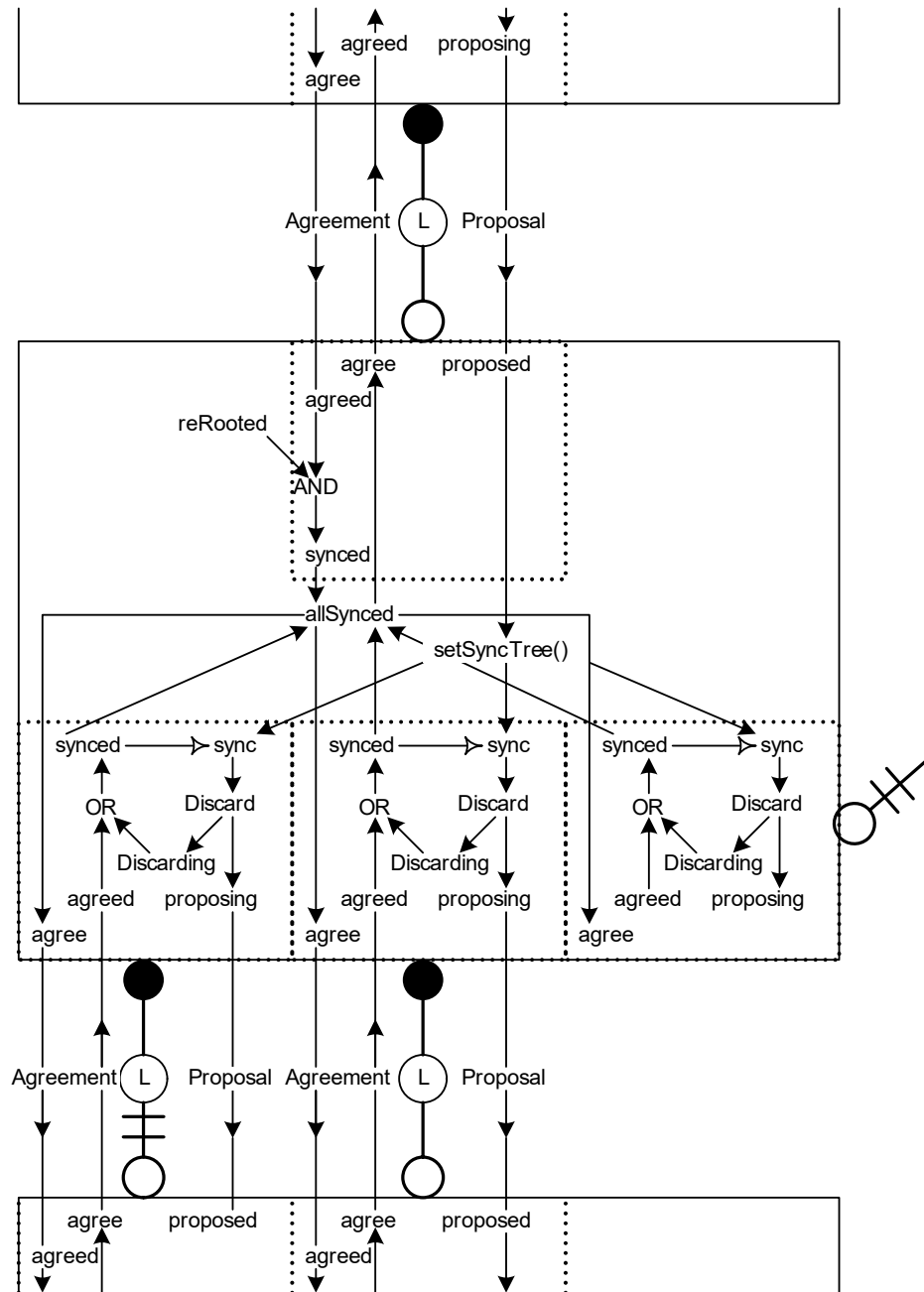


Figure 13-12—Enhanced Agreements

1 When an Agreement A_{MR} is sent by a Root Port P_{MR} on a Regional Root M , it attests that the CIST Root
2 Identifier and External Root Path Cost components of the message priority advertised on all LANs
3 connected to the CIST by P_{MR} through M are the same as or worse than those accompanying A_{MR} . The
4 connectivity provided by each MSTI can be independent of that provided by the CIST within the MST
5 Region and can therefore connect P_{MR} and one or more CIST Root Ports external to but attached at the
6 boundary of the region even as CIST connectivity within the region is interrupted in order to satisfy the
7 conditions for generating A_{MR} . The Agreement cannot therefore be generated unless all MSTI subtrees as
8 well as the CIST subtree internal to the region are in Agreement. To ensure that an MSTI does not connect to
9 a CIST subtree external to the region that does not meet the constraints on the CST priority vector
10 components, an Agreement received at an MSTI Designated Port from a Bridge Port not internal to the
11 region is only recognized if the CIST Root Identifier and External Root Path Cost of the CIST root priority
12 vector selected by the transmitting Bridge Port are equal to or worse than those selected by the receiver.
13 Updating of a CIST Designated Port's port priority vector with a worse CIST Root Identifier and External
14 Root Path Cost forces the port to discard frames for all MSTIs, thus initiating a Proposal that will elicit
15 agreement.

16 NOTE 2—MSTI Designated Ports are prompted to discard frames, as required above, as follows. The CIST Port
17 Information state machine sets sync for all MSTIs on a transition into the UPDATE state if updating the port priority with
18 the designated priority changes the Regional Root Identifier or replaces the CIST Root Identifier or External Path Cost
19 with a worse tuple. The MSTI's Port Role Transition machine acts on the sync, instructing the port to discard frames, and
20 setting synced and canceling sync when the port is discarding or an agreement is received.

21 NOTE 3—A "cut" in an MSTI can be transferred to the CST, either at a Designated Port attached to the same LAN as an
22 STP Bridge or at the Root Port of a Bridge in an adjacent region. If the CST priority components have already been
23 synced, as is likely if the original cut was caused by changes in physical topology within the region, the cut will terminate
24 there. Otherwise, the transferred cut precedes a cut in the CIST, and the synced port may terminate the latter. In that way,
25 cuts in the CST will proceed through an MST Region by the quickest tree that will carry them.

26 NOTE 4—In the important topology where the CIST Root Bridge is within an MST Region, cuts are not transferred
27 from the region's IST to any MSTI. CIST cuts propagating in the region will not disrupt MSTI connectivity.

28 13.17 Changing Port States with SPB

29 While the link state routing protocols are less likely to create temporary loops than distance vector-based
30 protocols, loops are still possible if (for example) two (or more) links fail about the same time. Any potential
31 loop is a serious problem for bridged networks, as a frame that is multicast or whose destination has not been
32 learned and is traveling around a loop is copied to each of the connected subtrees on every circuit,
33 consuming bandwidth throughout the network. Each SPT Bridge Port uses SPT BPDUs and/or SPB Hello
34 PDUs (28.2) to ensure that additions to each active topology do not occur unless there is sufficient
35 agreement between neighboring Bridges (with some ports temporarily transitioned to Discarding if
36 necessary) to ensure that a temporary loop will not be created.

37 Like RSTP and MSTP, agreements (see 13.16, Figure 13-12) are used to ensure that ports that transition to
38 forwarding do not create loops. However, IS-IS distributes information in parallel with computation, rather
39 than computing a new active topology hop-by-hop as priority vector information is passed from the Root
40 Bridge to members of a potential subtree. Moreover the distribution path is not restricted to the potential
41 subtree. Arbitrary distribution and parallel computation means that different Bridges can complete topology
42 calculations at quite different times, with two major consequences as follows.

43 First, each Bridge is not necessarily aware of priority vector changes passed to any subtree connected
44 through one of its Designated Ports before any other Bridge in that subtree. A forwarding Designated Port
45 on a given Bridge can only provide connectivity to another Bridge through the latter's Root Port; and new
46 connectivity is only allowed if the Designated Port has received an agreement showing that the Root Port's
47 designated priority is worse than its own. When RSTP (or MSTP) is used priority vectors propagate down
48 the tree: so each Bridge knows that its forwarding parents (connected through its Root Port) either have a
49 better *designated priority* or have discarded any agreement previously received. None of those parents can
50 therefore provide an agreement allowing connectivity through their Root Port to one of the given Bridge's

1 Designated Ports (potentially creating a loop), unless one of the Designated Ports between the parent and the
2 given Bridge becomes Discarding—allowing an agreement to be sent but interrupting the potential loop.
3 When the CIST or an SPT is supported by ISIS-SPB, a Bridge Port needs to receive an explicit message
4 from its neighbors discarding any outstanding agreement before it can start forwarding as a Designated Port
5 or (if it is the Root Port) before the Bridge's Designated Ports can use an improved designated priority to
6 transition to forwarding. A given port cannot discard outstanding agreements until all the other ports of the
7 Bridge have either received suitable agreements or stopped forwarding, or until the given port itself is
8 Discarding.

9 The second major consequence of arbitrary distribution and parallel computation is that received
10 Agreements that appear to contradict the results of the last local link state computation can become useful
11 after a future computation completes. Agreements received after the last computation can be retained until
12 they prove useful (following a further link state update) or are superseded by the receipt of a further
13 agreement or a changed Port Role. The Bridge transmitting the agreement will not create any connectivity
14 that is inconsistent with the agreement until it has been explicitly discarded. For example, two Designated
15 Ports attached to the same LAN cannot both transition to forwarding, each using an old Agreement with a
16 Root Port role from the other. To minimize the number of messages needed to create new connectivity,
17 unsatisfactory received Agreements are voluntarily discarded when a link state topology calculation
18 completes. In addition to the Agreement Digest, the Agreement Protocol (13.16, 13.17) uses a two-bit
19 Agreement Number and a Discarded Agreement Number (in SPT BPDUs and SPB Hello PDUs) to allow for
20 message delay, loss, duplication, and misordering. Once an Agreement has been sent it is considered
21 outstanding until the Agreement Protocol declares a new topology match.

22 Agreements for the IST use CIST information present in MST and SPT BPDUs. This minimizes active
23 topology disruption while allowing parallel adoption of a new topology since neighboring Bridges can
24 communicate IST priority vectors, propagate agreements, and signal the need for temporary cuts to speed
25 transitions to forwarding: even while those neighbors' views of the entire active topology differ as a
26 consequence of differences in their knowledge of the latest physical topology. A single BPDUs or SPB Hello
27 PDU cannot convey the same per-tree state for all possible SPTs, so the input to their link state calculation is
28 summarized in a Agreement Digest (28.4). If a received digest matches the Bridge's own digest, the Bridge
29 can compute the implied received Agreements. Using a digest means neighboring Bridges cannot
30 synchronize their local views of any SPT's active topology until they have synchronized their views of the
31 physical topology of the entire SPT Region, but reducing the number of messages is worthwhile.

32 NOTE 1—Management VLANs with modest bandwidth needs should be allocated to the IST in SPT Regions.

33 The per-port per-tree state machine variable agreed (see Figure 13-12) is recomputed after ISIS-SPB
34 completes a link state calculation, including the calculation of each port's designated priority vector, and
35 after each BPDUs is received: agreed is TRUE if and only if the port is a Root or Alternate Port with no
36 outstanding agreements worse than its designated priority vector, or a Designated Port with no outstanding
37 agreements and a received agreement that is worse than its designated priority vector and any outstanding
38 agreement for the Bridge's Root Port. As in Figure 13-12 each port is synced if it is either agreed or
39 Discarding for the tree, and agree is TRUE for a given port when all other ports are synced. Unused
40 outstanding agreements are discarded, by their recipient, only when agree is TRUE, since any continued
41 connectivity has to be supported by continued agreement if loops are to be prevented.

42 To avoid delays that might otherwise occur waiting for Agreements to percolate up the tree, sync is set for
43 all Designated Ports to force those that are not agreed to Discarding, so that the Root Port can transmit an
44 Agreement to its neighbor. However, it is an implementation issue about how long sync requires to take
45 effect: the likelihood of receiving an Agreement for the CIST from the Designated Port's neighbor before
46 the port can be transitioned to Discarding and then to Forwarding again can be taken into account. Each
47 BPDUs CIST Proposal flag should be set when a Designated Port is Discarding, thus providing a prompt
48 for the return of an Agreement just as described above (13.16) for RSTP and MSTP.

1 The Agreement Digest conveys fresh Agreements for all SPTs (all identified by the same Agreement
2 Number used for the CIST) and discards those received for prior topologies (identified by the Discarded
3 Agreement Number). Since different trees will have different Root Ports (in the same Bridge), if all Bridges
4 waited for agreed to be set for all other ports for all other trees before transmitting a fresh digest the protocol
5 would deadlock. If the Root Port is not agreed but all other ports are synced, then agree is also set for any
6 port that is Discarding, thus avoiding the potential deadlock. When a link state computation completes, each
7 Bridge transitions any Designated Port that is not agreed for an SPT to Discarding as rapidly as possible so
8 that it can send the fresh digest to its neighbors, and does not attempt to optimize Port State transitions by
9 waiting to receive a matching digest.

10 NOTE 2—In all comparisons between SPT priority vectors, the specified comparisons take place within the CST
11 context, and agreed is only set if the CST Root and External Path Cost match exactly. This allow the Port Role Transition
12 state machine for the Master Port at the SPT Region boundary, where the IST and each SPT are connected into the CST,
13 to function just as for MSTP.

14 The considerations and state machine conditions detailed in this subclause (13.17) can be summarized (for
15 an SPT Bridge with all ports providing connectivity within an SPT Region) as follows:

16 A Root Port satisfies the conditions for Forwarding if and only if:

- 17 a) The Agreement Protocol has declared a topology match; and
- 18 b) Since (and including) that topology match, no outstanding Agreement Digest sent through the port is
19 for a topology where:
 - 20 1) This Bridge is or was farther from the SPT's Root Bridge than in the current topology; or
 - 21 2) The port was a Designated Port or Backup Port.

22 A Designated Port satisfies the conditions for Forwarding if and only if:

- 23 c) The Agreement Protocol has declared a topology match; and
- 24 d) Since (and including) that topology match, no outstanding Agreement Digest sent through the port is
25 for a topology where:
 - 26 1) Any other Bridge attached to the same LAN is or was as close or closer to the SPT's Root
27 Bridge than this Bridge is in the current topology; or
 - 28 2) The port was a Root Port or Alternate Port.

29 Note that a Root Port can only become Forwarding following receipt of a fresh Agreement Digest if all
30 Designated Ports either satisfy the conditions for Forwarding or are made Discarding. A Designated Port can
31 only become Forwarding if the Root Port also satisfies the conditions for Forwarding, or is made Discarding.

32 An Agreement Digest is sent on all ports as soon as possible after ISIS-SPB completes a link state
33 calculation, but is only sent if (for every SPT)

- 34 e) The port sending the Agreement Digest is Discarding; or
- 35 f) The port is a Root Port or a Designated Port, that satisfies the necessary conditions for Forwarding
36 [item a) to item d) above], and every other port of the Bridge is either Discarding or is a Root or
37 Designated Port that also satisfies those conditions.

38 13.17.1 Agreement Digest

39 The Agreement Digest is calculated as specified in 28.4.

40 The purpose of the Agreement Digest is to prevent temporary loops and its use as an input to the Port Role
41 Transition and Port State Transition state machines is specified in 13.29, 13.37, and 13.38. Successful
42 operation of ISIS-SPB relies on the fact that SPT Bridges attached to the same LAN will, in the absence of
43 continual physical topology changes in the network or continual loss of ISIS-SPB frames, advertise the same
44 Agreement Digest in transmitted BPDUs and SPB Hello PDUs. Conditions that would result in permanent
45 disagreement, and consequent failure to provide SPB, are captured in the MCID (13.8). Differences in the
46 MCID result in both MST and SPT Bridge Ports attached to the LAN being identified as bounding a region,
47 so connectivity can be provided using the CST.

13.18 Managing spanning tree topologies

The active topology of the CIST, and the topologies that can result after the failure or addition of network components, may be managed by assigning values to some or all of the following:

- The Bridge Priority component of the CIST Bridge Identifier for one or more Bridges
- The External Port Path Cost (also referred to as the Port Path Cost for RSTP) for some Bridge Ports
- Components of the MCID for Bridges with the same Configuration Digest
- The CIST Internal Port Path Cost Port (for MSTP and ISIS-SPB) for some Bridge Ports
- The Port Priority component of the Port Identifier for some Bridge Ports

Within an MST Region, the active topology of each MSTI may be managed by assigning values to some or all of the following:

- The Bridge Priority component of the MSTI Regional Root Identifier
- The Internal Port Path Cost for the MSTI for some Bridge Ports
- The Port Priority component of the MSTI's Port Identifier for some Bridge Ports

In general topology management objectives can be met by modifying only a few parameter values in a few Bridges in the network. Table 13-3 specifies default values and ranges for Bridge Priorities and Port Priorities. If these parameters can be updated by management, the Bridge shall have the capability to use the full range of values with the granularity specified.

Table 13-3—Bridge and Port Priority values

Parameter	Recommended or default value	Range
Bridge Priority	32 768	0–61 440 in steps of 4096
Port Priority	128	0–240 in steps of 16

NOTE 1—The stated ranges and granularities for Bridge Priority and Port Priority differ from those in IEEE Std 802.1D, 1998 Edition [B13], and earlier revisions of that standard. Expressing these values in steps of 4096 and 16 allows consistent management of old and new implementations of this standard; the steps chosen ensure that bits that have been re-assigned are not modified, but priority values can be directly compared.

Table 13-4 recommends defaults and ranges for Port Path Cost and Internal Port Path Cost values, chosen according to the speed of the attached LAN, to minimize the administrative effort required to provide reasonable active topologies. If these values can be set by management, the Bridge shall be able to use the full range of values in the parameter ranges specified, with a granularity of 1.

When two or more links are aggregated (see IEEE Std 802.1AX), Port Path Cost and Internal Port Path Cost values can be modified to reflect the actual throughput. However, as the primary purpose of Path Cost is to select active topologies, it can be inappropriate to track throughput too closely, as the resultant active topology could fluctuate or differ from that intended by the network administrator. For example, if the network administrator had chosen aggregated links for resilience (rather than for increased data rate), it would be inappropriate to change topology as a result of one of the links in an aggregation failing. Similarly, with links that can autonegotiate their data rate, reflecting such changes of data rate in changes to Path Cost is not necessarily appropriate. As a default behavior, dynamic changes of data rate should not automatically cause changes in Port Path Cost.

Table 13-4—Port Path Cost values

Link Speed	Recommended value	Recommended range	Range
<=100 kb/s	200 000 000	20 000 000–200 000 000	1–200 000 000
1 Mb/s	20 000 000	2 000 000–200 000 000	1–200 000 000
10 Mb/s	2 000 000	200 000–20 000 000	1–200 000 000
100 Mb/s	200 000	20 000–2 000 000	1–200 000 000
1 Gb/s	20 000	2000–200 000	1–200 000 000
10 Gb/s	2000	200–20 000	1–200 000 000
100 Gb/s	200	20–2000	1–200 000 000
1 Tb/s	20	2–200	1–200 000 000
10 Tb/s	2	1–20	1–200 000 000

1 NOTE 2—BPDUs are capable of carrying 32 bits of Root Path Cost information, though IEEE Std 802.1D, 1998 Edition
2 [B13], and its earlier revisions limited the range of the Port Path Cost parameter to a 16-bit unsigned integer value.
3 Table 13-4 uses the full 32-bit range to extend the range of supported link speeds. Additional recommended values can
4 be calculated as 20 000 000 000/(Link Speed in kb/s). Limiting the range of the Path Cost parameter to 1–200 000 000
5 ensures that the accumulated Path Cost cannot exceed 32 bits over a concatenation of 20 hops. Where Bridges using the
6 IEEE Std 802.1D, 1998 Edition [B13], recommendations and others using Table 13-4 are mixed in the same Bridged
7 Network, explicit configuration is likely to be necessary to obtain reasonable CST topologies.

8 There exist media for which it is difficult to define a Link Speed. This includes, for example,
9 Ethernet-over-packet-network technologies, where the frames are carried over a layer 3 best-effort network.
10 Another example is IEEE 802.11 wireless media, where the effective bandwidth is shared among stations
11 operating at different speeds, all varying on short timescales. Individual media specifications provide values
12 for Link Speed suitable for use with Table 13-4 to determine a Port Path Cost.

13 13.19 Updating learned station location information

14 In normal stable operation, learned station location information held in the FDB need only change as a
15 consequence of the physical relocation of stations. It is therefore desirable to employ a long ageing time for
16 Dynamic Filtering Entries (8.8.3), especially as many end stations transmit frames following power-up
17 causing the information to be relearned.

18 However, when the active topology reconfigures, stations can appear to move from the point of view of any
19 given Bridge even if that Bridge's Port States have not changed. If a Bridge Port is no longer part of an
20 active topology, stations are no longer reachable through that port, and its Dynamic Filtering Entries are
21 removed from that Bridge's FDB. Conversely, stations formerly reachable through other ports might be
22 reachable through a newly active port. Dynamic Filtering Entries for the other ports are removed, and RSTP
23 and MSTP transmit Topology Change Notification Messages both through the newly active Port and
24 through the other active Ports on that Bridge. TCNs signal additional connectivity, not just changes in
25 connectivity, as relearning a station's location is only possible if it can be reached, and if that is possible
26 when a port is removed from the active topology another port will be added. A TCN is sent when a Bridge
27 Port joins the active topology, and not before, so that Bridges can relearn removed station location
28 information and minimize unnecessary flooding of frames. A Bridge that receives a TCN on an active port
29 removes Dynamic Filtering Entries for their other active ports and propagates the TCN through those ports.

1 NOTE 1—STP allowed for the presence of LAN repeaters that could partition a shared media LAN, thus causing
2 stations to appear to move when the partition was repaired later—with the only Bridge Port changing Port Role or Port
3 State transitioning to Discarding at that time. This scenario does not occur with current technology, and its future
4 likelihood does not justify the use of TCNs to signal connectivity loss. Bridge Ports that participate in the MAC status
5 propagation protocol (Clause 23) should be capable of originating TCNs when that protocol signals additional
6 connectivity.

7 The Topology Change state machine (13.39) avoids removing learned information when ports temporarily
8 revert to Discarding to suppress loops. It treats a port as joining the active topology when it becomes
9 forwarding, and no longer active when it becomes an Alternate, Backup, or Disabled Port and stops
10 forwarding and learning. TCNs are not generated following Edge Port (operEdge, 13.27.44) Port State
11 changes, as these do not affect connectivity or station location information in the rest of the network, nor are
12 Dynamic Filtering Entries for Edge Ports removed when TCNs are received.

13 Dynamic Filtering Entries for MAC addresses previously learned on a Root Port may be modified to move
14 those addresses to an Alternate Port that becomes the new Root Port and a TCN sent only through the new
15 Root Port (and not through other active ports), reducing the need to flood frames. This optimization is
16 possible because a retiring Root Port that becomes Discarding temporarily partitions the active topology into
17 two subtrees, one including all Bridges and LANs hitherto reachable through the retiring Root Port, and the
18 other including all the others. If the new Root Port simply provides a new path to the first of these subtrees,
19 its stations will not appear to move from the point of view of Bridges in the other subtree. Alternatively if a
20 the tree reconfiguration is more complex one or more newly Designated Port will become active and will
21 transmit the necessary TCNs.

22 NOTE 2—The rules described require removal of potentially invalid learned information for a minimum set of ports on
23 each Bridge. A Bridge implementation can flush information from more ports than strictly necessary, removing (for
24 example) all Dynamic Filtering Entries rather than just those for the specified ports. This does not result in incorrect
25 operation, but will result in more flooding of frames with unknown destination addresses.

26 Changes in the active topology of any given MSTI or SPT do not change Dynamic Filtering Entries for the
27 CIST or any other MSTI or SPT, unless the underlying changes in the physical topology that gave rise to the
28 reconfiguration also cause those trees to reconfigure. Changes to the CST, i.e., the connectivity provided
29 between regions, can cause end station location changes for all trees. Changes to an IST can cause CST end
30 station location changes but do not affect MSTIs in that region unless those trees also reconfigure.

31 On receipt of a CIST TCN Message from a Bridge Port not internal to the region, or on a change in Port Role
32 for a Bridge Port at the region boundary, TCN Messages are transmitted through each of the other ports of
33 the receiving Bridge for each MSTI and the Dynamic Filtering Entries for those ports are removed.

34 NOTE 3—The port receiving the CIST TCN Message can be a Master Port, a Designated Port attached to the same LAN
35 as an STP Bridge, or a Designated Port attached to the same LAN as the Root Ports of Bridges in other regions.

36 TCN Messages for the CIST are always encoded in the same way, irrespective of whether they are perceived
37 to have originated from topology changes internal to the region or outside it. This allows RST Bridges
38 whose Root Ports attach to a LAN within an MST Region to receive these TCN Messages correctly.

39 Since each SPT is rooted at the source of the frames assigned to that SPT, or at the point where those frames
40 enter the SPT Region, entries are only learned for the Root Port of an SPT. Any apparent changes in station
41 location due to changes in an SPT's active topology only occur when a Bridge changes the Root Port for that
42 SPT, and are accommodated by removing the Dynamic Filtering Entries for the prior Root Port and FID
43 associated with the SPT Set. If the same new Root Port is chosen for all SPTs, in the SPT set, with that prior
44 Root Port, the entries can be moved to the new Root Port rather than simply being deleted.

45 NOTE 4—Dynamic Filtering Entries created by ISIS-SPB as a result of that protocol's direct knowledge of the location
46 of some stations are not removed when a TCN is received, but can be changed by ISIS-SPB as result of its calculations.

47 NOTE 5—Topology changes for the IST are always propagated by TCNs received and transmitted in BPDUs, and are
48 not injected as a result of ISIS-SPB calculations as the latter complete prior to new connectivity being established.

1 13.20 Managing reconfiguration

2 The priority component of the Bridge Identifier can be managed for the CIST, and independently for each
3 MSTI, to allow preferential selection of the first choice Root Bridge and of alternate Roots that will be used
4 if the better choices have failed or lack connectivity. The Port Path Cost of each Bridge Port can also be
5 managed to allow preferential selection of the path from each Bridge to the Root, with the priority
6 component of each Bridge's Bridge Identifier providing a manageable tie-breaker between equal cost paths.

7 In the event of LAN or Bridge failure, fastest reconfiguration (and thus highest service availability) can
8 usually be provided by a Bridge that can substitute a prior Alternate Port for a Root Port that now lacks or
9 provides inferior connectivity to the Root. Figure 13-2 illustrates a simple topology where most failures can
10 be handled in this way. Configuration to place a backup Root Bridge adjacent to the primary (Bridges 222
11 and 111 in the figure, respectively) enables Alternate Port to Root Port failover in the other Bridges. If the
12 original Root Port and its replacement Alternate Port have the same Root Port Path Cost, Bridges further
13 from the Root will not see a topology change. Figure 13-5 illustrates a ring topology, where simple failover
14 copes with few failures. However, locating a backup Root Bridge adjacent to the preferred Root remains
15 effective in reducing the effect of any change.

16 RSTP and MSTP are distance vector protocols: following the failure of a Root Bridge (or selection of a
17 costlier path to the Root) spanning tree priority vectors conveying the prior Root can circulate in the network
18 until that information ages out (i.e., until Message Age exceeds Max Age, or the remainingHops parameter
19 is decremented to zero). To accommodate a wide range of networks, the default setting of these parameters
20 permit 20 hops. However, if the preferred and alternate Roots are located at the center of the network, many
21 networks can be configured with minimal values (6 hops, Table 13-5) so each message can only be received
22 by any node at most twice. In larger networks, for example ring backbones with other Bridges redundantly
23 attached to adjacent Bridges in the ring, the unwanted effects of recirculating information can be prevented
24 by configuring the restrictedRole parameter (13.27.64). This parameter prevents information from being
25 propagated through ports that should never provide connectivity toward the Root. In Figure 13-3, for
26 example, restrictedRole could be configured for any or all of the Designated Ports.

27 Other potentially disruptive effects of reconfiguration can be prevented or mitigated by setting the
28 restrictedTcn parameter (13.27.65) for a port.

29 The configuration of a subtree of a spanning tree defined by the connectivity provided by a given Bridge *B*
30 (say) can be made independent of the rest of the tree by configuring *B*'s Root Port as a Layer 2 Gateway Port
31 (L2GP). An L2GP Port behaves as if it is in continual receipt of a fixed, manageable, spanning tree priority
32 vector. Provided that this fixed priority vector is worse than the actual priority vector transmitted by the
33 Designated Port for the LAN attached to *B*'s Root Port, there is no potential for a loop. Each port in the
34 subtree will transmit a worse priority vector. If a received message priority vector is worse than the fixed
35 value the disputed flag is set, causing the L2GP port to transition to Discarding. Thus stability of the
36 spanning tree priority vector information in the subtree is maintained, at the possible expense of the
37 connectivity between the subtree and the rest of the tree. Alternatively, if another Bridge within *B*'s subtree
38 becomes the source of better priority information, causing *B*'s L2GP port to assume a Designated Port Role,
39 a dispute will exist between the newly propagated spanning tree information and the fixed information for
40 the L2GP port, and will also cause that port to become Discarding.

41 The L2GP capability is optional, and is primarily intended to be used for service access protection (25.9)
42 when a single customer's Bridged Network is connected to a PBBN or a PBN. Two or more Bridge Ports on
43 one or more of the Bridges within the customer's network, are connected to a single service instance
44 supported by the provider's network, and are configured as L2GP ports. The L2GP ports' fixed priority
45 vectors are each defined by a different, configured pseudoRootId (13.27.51). Only the L2GP port with the
46 best pseudoRootId can provide connectivity to the provider's network. Spanning tree information from that
47 port will be disseminated throughout the customer's network and will be disputed by the other L2GP port,
48 causing the latter to become Discarding. If all the customer network's ports that are bridged to other

networks (such as PBNs or PBBNs) are configured as L2GP ports, those ports neither have to transmit or receive BPDUs to prevent loops—though connectivity through providers' networks to other networks will not be protected. Enabling BPDU reception for each L2GP port prevents loops that might otherwise be caused by attaching another of the customer's network's ports to the provider network.

13.21 Partial and disputed connectivity

It is possible for the connectivity between Bridge Ports attached to the same LAN to fail, in system or media access method-dependent ways, so that BPDUs and data frames are transmitted in one direction only. As a result it is possible for more than one port attached to the same LAN to believe itself to be the Designated Port. To ensure that the active topology remains loop-free, a Designated Port will recognize that a dispute is in progress and stop learning from or forwarding frames, if it receives a BPDU with a worse message priority and the Learning flag set from another port that claims to be Designated.

If two (Designated) ports attached to the same shared media LAN cannot communicate with each other at all, but can each communicate with a third (Root) port, a potential loop exists if one of the Designated Ports has a priority vector that is worse than that of the Root Port. To ensure that such loop does not occur, a Root Port that receives an inferior message from a Designated Port detects a dispute if the Learning flag is set, and transitions to Discarding.

13.22 In-service upgrades

It can be desirable to upgrade the control plane software of a bridging system, without interrupting existing data connectivity, while the Spanning Tree Protocol Entity is not operating. This can be done, without the risk of creating data loops, providing that the other Bridges in the network are suitably configured. However, the failure of a network component (Bridge or LAN) while the upgrade is in progress can result in a loss of connectivity, as the ways in which the network can reconfigure are restricted. This subclause (13.22) describes the necessary configuration of the other Bridges in the network; the behavior of the upgrading system is assumed to be system dependent and is not specified in detail, except as follows:

- a) Frames can be received from and transmitted to ports that were forwarding prior to beginning the upgrade, and are not forwarded to or from any other port.
- b) Frames received on ports that were learning prior to beginning the upgrade can be submitted to the Learning Process, while frames received on other ports are not.
- c) BPDUs are not transmitted, and received BPDUs are discarded, while the upgrade is in progress.
- d) If there is to be no interruption in connectivity when the upgrade is complete, the parameters of BPDUs received prior to beginning the upgrade are retained, or recovered using ISIS-SPB.

NOTE 1—If received BPDU information is not retained, the spanning tree protocol variables and state machines are reinitialized by asserting BEGIN, and there will be a brief interruption in connectivity.

The need for a control plane software upgrade can result from the need to change any component of that software, and rarely from upgrades to the Spanning Tree Protocol Entity itself. The in-service upgrade procedures described here depend on the operation of specifically identified features of the RSTP, MSTP, and ISIS-SPB specified in this standard (Clause 13), and do not ensure loop-free operation if the necessary criteria are not met by all Bridges in the network. In particular, safe upgrades are not supported in networks including Bridges operating STP as specified in the IEEE Std 802.1D, 1998 Edition [B13], and earlier revisions. It is essential for the network administrator to base-line the network, i.e., verify its configuration and the configuration of its components, prior to attempting an in-service upgrade. It is recommended that a single system be upgraded at a time, and its subsequent correct operation verified prior to making further changes.

1 If all the LANs that connect the Bridges in the network provide point-to-point connectivity, connecting at
2 most two Bridges, and each Bridge Port thus connecting to another Bridge meets the following conditions:

- 3 e) `operPointToPointMAC` (13.28.16) is TRUE and
- 4 f) `operEdge` (13.27.44) is FALSE,

5 NOTE 2—Apart from the requirement for a point-to-point physical topology, these conditions can be ensured by setting
6 `adminPointToPointMAC` (IEEE Std 802.1AC) TRUE, and both `AdminEdge` and `AutoEdge` FALSE.

7 then a Designated Port will not transition from Discarding to Learning or Forwarding without receiving an
8 Agreement from its immediate neighbor (see 13.37). This prevents the introduction of data loops while one
9 or more Bridges are being upgraded, even if other Bridges or LANs fail or are added to the network. To
10 prevent existing connectivity being disrupted (provided no other network additions or failures occur) the
11 following are also necessary:

- 12 g) Prior to the upgrade, each of the upgrading Bridge's immediately neighboring ports that is a Root
13 Port (for a given tree) has to be configured as an Layer 2 Gateway Port (13.20) (for that tree), with
14 the same information that it was previously receiving from the upgrading Bridge.
- 15 h) After the upgrade, each of the ports thus temporarily configured as an Layer 2 Gateway Port needs
16 to have its normal configuration restored.

17 NOTE 3—If the upgrading Bridge is capable of sending periodic 'canned BPDUs' containing the same information as
18 immediately prior to the upgrade, there is no requirement for neighbor L2GP configuration, or for the point-to-point and
19 `operEdge` conditions. Attempts by neighboring Bridges to operate outside the parameters dictated by the 'canned
20 BPDUs' will result in disputes, preventing new connectivity.

21 On the SPB ports within an SPT Region, the functions of the Layer 2 Gateway Port described in item g) and
22 item h) above are replaced by those of "Restart Signaling for IS-IS" (IETF RFC 5306 [B37]), which allows
23 the forwarding state across an adjacency to be maintained while an IS-IS element is restarted. When the
24 neighbor (running) shortest path Bridge receives a restart request, it performs the procedures of IETF
25 RFC 5306, and for as long as it maintains the state of the IS-IS adjacency as "UP" it also maintains its
26 received Agreement Digest at the last value received from the restarting Bridge before it requested restart. If
27 the adjacency is successfully reacquired by the procedures of IETF RFC 5306, as signaled by the restarting
28 Bridge clearing the Restart Request bit in an SPB Hello PDU, its running neighbor shall include its current
29 Agreement Digest value in its SPB Hello response [and update it as required thereafter according to the
30 Agreement Protocol (13.17)].

31 Once the restarting Bridge has achieved LSP Database Synchronization according to the procedures of IETF
32 RFC 5306 [B37], it shall compute its local value of the Agreement Digest prior to making any changes to its
33 FDB. On ports for which a Digest match is achieved with a neighbor, full installation of unicast and
34 multicast state may be performed immediately. On ports that do not have a Digest match (caused for
35 example by a propagating topology change), the restarting Bridge shall assume when computing allowed
36 forwarding entries that Digest Agreement has never been achieved on the affected adjacency (where
37 "allowed" is determined by the value of the Agreement Digest Convention Identifier; see 28.4.3). In either
38 event, a Digest Agreement shall be sent to the neighbor as soon as possible, as specified by the Agreement
39 Protocol.

40 SPB ports on the boundary of an SPT Region use the procedures of a) to h) unaltered.

41 13.23 Fragile Bridges

42 Some, nonconformant, bridging systems are known to be "fragile," i.e., they can suffer from unpredictable
43 interruptions to Spanning Tree Protocol Entity operation and will forward data frames while no longer
44 sending or receiving BPDUs, on some or all ports. If the spanning tree protocol implementations of the other
45 Bridges in the network conform to prior revisions of this standard and IEEE Std 802.1D-2004 [B14], these
46 interruptions can result data loops even in networks of point-to-point LANs. This revision of this standard
47 allows a Bridge Port attached to point-to-point LAN to ensure that its neighbor remains capable of receiving

1 and transmitting BPDUs, even if that neighbor's RSTP or MSTP implementation conforms to a prior
2 revision of this standard, and to block connectivity (by transitioning to Discarding) otherwise. The CIST
3 Proposal flag is set in all BPDUs transmitted by a Designated Port, and used to solicit an Agreement from
4 the neighbor (which might otherwise not transmit). This capability is controlled by the AutoIsolate
5 (13.27.19) variable, is disabled by default to allow for in-service upgrades, and is only effective if the
6 neighboring Bridge is capable of RSTP or MSTP operation.

7 **13.24 Spanning tree protocol state machines**

8 Each Spanning Tree Protocol Entity's operation of the protocols specified in this clause (Clause 13) is
9 specified by the following state machines:

10 a) Port Role Selection (PRS, 13.36)

11 for the CIST and for each MSTI, with the following state machines for each Bridge Port:

12 b) Port Timers (PTI, 13.30)

13 c) Port Protocol Migration (PPM, 13.32)

14 d) Port Receive (PRX, 13.31)

15 e) Port Transmit (PRT, 13.34)

16 f) Bridge Detection (BDM, 13.33)

17 and the following optional state machine for each Bridge Port:

18 g) Layer 2 Gateway Port Receive (L2GPRX, 13.40)

19 and the following state machines for each Bridge Port for the CIST and for each MSTI:

20 h) Port Information (PIM, 13.35)

21 i) Topology Change (TCM, 13.39)

22 and the following state machines for each Bridge Port for the CIST, for each MSTI, and for each SPT:

23 j) Port Role Transitions (PRT, 13.37)

24 k) Port State Transition (PST, 13.38)

25 Each state machine and its associated variable and procedural definitions are specified in detail in 13.25
26 through 13.40. The state machine notation is specified in Annex E. Figure 13-13 is not itself a state machine
27 but provides an overview of the state machines, their state variables, and communication between machines.
28 Figure 13-14 describes its notation.

29 **13.25 State machine timers**

30 The timer variables declared in this subclause are part of the specification. The accompanying descriptions
31 are provided to aid in the comprehension of the protocol only, and are not part of the specification. Each
32 timer variable represents an integral number of seconds before timer expiry.

33 One instance of the following shall be implemented per port:

34 a) edgeDelayWhile (13.25.1)

35 b) helloWhen (13.25.3)

36 c) mdelayWhile (13.25.4)

37 One instance of the following shall be implemented per port when L2GP functionality is provided:

38 d) pseudoInfoHelloWhen (13.25.10)

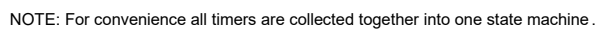


Figure 13-13—Spanning tree protocol state machines—overview and relationships

NOTATION:

Variables are shown both within the machine where they are principally used and between machines where they are used to communicate information. In the latter case they are shown with a variety of arrow styles, running from one machine to another, that provide an overview of how the variables are typically used:

→ Not changed by the target machine. Where the state machines are both per Port, this variable communicates between machine instances for the same port.

⇨ Set (or cleared) by the originating machine, cleared (or set) by the target machine. Where the state machines are both per Port, this variable communicates between machine instances for the same port.

⇨⇨ As above except that the originating per port machine instance communicates with multiple port machine instances (by setting or clearing variables owned by those Ports).

⇨⇨⇨ As above except that multiple per Port instances communicate with (an) other instance(s) (by setting or clearing variables owned by the originating Ports).

ABBREVIATIONS:

BDM: Bridge Detection Machine
PIM: Port Information Machine
PPM: Port Protocol Migration Machine
PRS: Port Role Selection Machine
PRT: Port Role Transitions Machine
PRX: Port Receive Machine
PST: Port State Transitions Machine
PTI: Port Timers Machine
PTX: Port Transmit Machine
TCM: Topology Change Machine

Figure 13-14—MSTP overview notation

1 One instance per port of the following shall be implemented for the CIST and one per port for each MSTI:

- 2 e) fdWhile (13.25.2)
- 3 f) rrWhile (13.25.7)
- 4 g) rbWhile (13.25.5)
- 5 h) tcWhile (13.25.9)
- 6 i) rcvdInfoWhile (13.25.6)
- 7 j) tcDetected (13.25.8)

8 Table 13-5 specifies values and ranges for the initial values of timers and for transmission rate limiting
9 performance parameters. Default values are specified to avoid the need to set values prior to operation in
10 most cases, and are widely applicable to networks using the spanning tree protocols specified in this
11 standard. The table recommends Bridge Hello Time, Bridge Max Age, and Bridge Forward Delay values
12 that maximize interoperability in networks that include Bridges using STP as specified in IEEE Std 802.1D,
13 1998 Edition [B13]. Ranges are specified to ensure that the protocols operate correctly.

Table 13-5—Timer and related parameter values

Parameter	Default	Permitted range	Interoperability recommendations
Migrate Time	3.0	— ^a	— ^a
(Bridge) Hello Time	2.0	— ^a	— ^a
Bridge Max Age	20.0	6.0–40.0	20.0
Bridge Forward Delay	15.0	4.0–30.0	15.0
Transmit Hold Count	6	1–10	6
Max Hops	20	6–100	—

All times are in seconds. —^a Not applicable, value is fixed.

14 NOTE—Changes to Bridge Forward Delay do not affect reconfiguration times, unless the network includes Bridges that
15 do not conform to this revision of this standard. Changes to Bridge Max Age can have an effect, as it is possible for old
16 information to persist in loops in the physical topology for a number of “hops” equal to the value of Max Age in seconds,
17 and thus exhaust the Transmit Hold Count in small loops. The IEEE Std 802.1Qdy amendment to this standard increased
18 the permitted range of Max Hops to allow a spanning tree to provide initial and remedial connectivity in extended ring
19 topologies whenever physical connectivity is possible. Once configured such networks are not expected to depend on
20 spanning tree for failure protection.

1 Bridge Max Age, Bridge Forward Delay, and Transmit Hold Count may be set by management, if this
2 capability is provided the Bridge shall have the capability to use the full range of values in the parameter
3 ranges specified in the “Permitted range” column of Table 13-5, with a timer resolution of r seconds, where
4 $0 < r \leq 1$. To support interoperability with previous revisions of this standard and IEEE Std 802.1D-2004
5 [B14], a Bridge shall enforce the following relationships:

$$6 \quad 2 \times (\text{Bridge_Forward_Delay} - 1.0 \text{ seconds}) \geq \text{Bridge_Max_Age}$$

$$7 \quad \text{Bridge_Max_Age} \geq 2 \times (\text{Bridge_Hello_Time} + 1.0 \text{ seconds})$$

8 **13.25.1 edgeDelayWhile**

9 The Edge Delay timer. The time remaining, in the absence of a received BPDU, before this port is identified
10 as an operEdgePort.

11 **13.25.2 fdWhile**

12 The Forward Delay timer. Used to delay Port State transitions until other Bridges have received spanning
13 tree information.

14 **13.25.3 helloWhen**

15 The Hello timer. Used to ensure that at least one BPDU is transmitted by a Designated Port in each
16 HelloTime period.

17 **13.25.4 mdelayWhile**

18 The Migration Delay timer. Used by the Port Protocol Migration state machine to allow time for another
19 RST Bridge on the same LAN to synchronize its migration state with this port before the receipt of a BPDU
20 can cause this port to change the BPDU types it transmits. Initialized to MigrateTime (13.26.6).

21 **13.25.5 rbWhile**

22 The Recent Backup timer. Maintained at its initial value, twice HelloTime, while the port is a Backup Port.

23 **13.25.6 rcvdInfoWhile**

24 The Received Info timer. The time remaining before information, i.e., portPriority (13.27.47) and portTimes
25 (13.27.48), received in a Configuration Message is aged out if a further message is not received.

26 **13.25.7 rrWhile**

27 The Recent Root timer.

28 **13.25.8 tcDetected**

29 The Topology Change timer for MRP application usage. ‘New’ messages are sent while this timer is running
30 (see 10.2).

31 **13.25.9 tcWhile**

32 The Topology Change timer. TCN Messages are sent while this timer is running.

33 **13.25.10 pseudoInfoHelloWhen**

34 The Pseudo Info Hello When timer. Used by the Layer 2 Gateway Port Receive (L2GPRX, 13.40) state
35 machine to prompt the simulated reception at HelloTime intervals of a BPDU (see 13.29.11).

1 **13.26 Per Bridge variables**

2 The variables declared in this subclause (13.26) are part of the specification. The accompanying descriptions
3 are provided to aid in the comprehension of the protocol only, and are not part of the specification.

4 There is one instance per Bridge component of the following variable(s):

- 5 a) ForceProtocolVersion (13.26.5)
- 6 b) TxHoldCount (13.26.12)
- 7 c) MigrateTime (13.26.6)

8 One instance of the following shall be implemented per Bridge component if MSTP or ISIS-SPB is
9 implemented:

- 10 d) MstConfigId (13.26.7)

11 The above parameters [item a) through item d)] are not modified by the operation of the spanning tree
12 protocols, but are treated as constants by the state machines. If ForceProtocolVersion or MstConfigId are
13 modified by management, BEGIN shall be asserted for all state machines.

14 There is one instance per Bridge of each of the following for the CIST and one for each MSTI.

- 15 e) BridgeIdentifier (13.26.2)
- 16 f) BridgePriority (13.26.3)
- 17 g) BridgeTimes (13.26.4)
- 18 h) rootPortId (13.26.9)
- 19 i) rootPriority (13.26.10)
- 20 j) rootTimes (13.26.11)

21 BridgeIdentifier, BridgePriority, and BridgeTimes are not modified by the operation of the spanning tree
22 protocols but are treated as constants by the state machines. If they are modified by management, spanning
23 tree priority vectors and Port Role assignments for all trees shall be recomputed, as specified by the
24 operation of the Port Role Selection state machine (13.36) by clearing selected (13.27.67) and setting
25 reselect (13.27.62) for all Bridge Ports for the relevant MSTI and for all trees if the CIST parameter is
26 changed.

27 If the ISIS-SPB is implemented there is one instance per Bridge component, of the following variable(s),
28 with that single instance supporting all SPTs:

- 29 k) agreementDigest (13.26.1)
- 30 l) AuxMstConfigId (13.26.8)

31 **13.26.1 agreementDigest**

32 The Agreement Digest calculated by ISIS-SPB (28.4), and updated with other calculated parameters (see
33 13.36).

34 **13.26.2 BridgeIdentifier**

35 The unique Bridge Identifier assigned to this Bridge for this tree (CIST or MSTI).

36 The 12-bit system ID extension component of a Bridge Identifier (14.2.5) shall be set to zero for the CIST
37 and to the value of the MSTID for an MSTI, thus allocating distinct Bridge Identifiers to the CIST and each
38 MSTI—all based on the use of a single Bridge Address component value for the MST Bridge as a whole.

39 NOTE—This convention is used to convey the MSTID for each MSTI Configuration Message in an MST BPDU.

1 The most significant 4 bits of the Bridge Identifier (the settable Priority component) for the CIST and for
2 each MSTI can be modified independently of the setting of those bits for all other trees, as a part of allowing
3 full and independent configuration control to be exerted over each spanning tree instance.

4 **13.26.3 BridgePriority**

5 For the CIST, the value of the CIST Bridge priority vector, as defined in 13.10. The CIST Root Identifier,
6 CIST Regional Root Identifier, and Designated Bridge Identifier components are all equal to the value of the
7 CIST Bridge Identifier. The remaining components (External Root Path Cost, Internal Root Path Cost, and
8 Designated Port Identifier) are set to zero.

9 For a given MSTI, the value of the MSTI Bridge priority vector, as defined in 13.11. The MSTI Regional
10 Root Identifier and Designated Bridge Identifier components are equal to the value of the MSTI Bridge
11 Identifier (13.26.2). The remaining components (MSTI Internal Root Path Cost, Designated Port Identifier)
12 are set to zero.

13 BridgePriority is used by updtRolesTree() in determining the value of the rootPriority variable (see 13.26.10).

14 **13.26.4 BridgeTimes**

15 For the CIST, BridgeTimes comprises:

- 16 a) The current values of Bridge Forward Delay and Bridge Max Age (13.25, Table 13-5).
- 17 b) A Message Age value of zero.
- 18 c) The current value of Max Hops (13.25, Table 13-5). This parameter value is determined only by
19 management.

20 For a given MSTI, BridgeTimes comprises:

- 21 d) The current value of MaxHops (Max Hops in Table 13-5), the initial value of remainingHops for
22 MSTI information generated at the boundary of an MSTI region (see 13.26.11).

23 BridgeTimes is used by updtRolesTree() in determining the value of the rootTimes variable (13.26.11).

24 **13.26.5 ForceProtocolVersion**

25 The Force Protocol Version parameter for the Bridge (13.7.2).

26 **13.26.6 MigrateTime**

27 The value of the Migrate Time parameter as specified in Table 13-5. This value shall not be changed.

28 **13.26.7 MstConfigId**

29 The current value of the Bridge's MCID (13.8). Changes in this parameter cause BEGIN to be asserted for
30 the state machines for the Bridge, for all trees, and for each port.

31 **13.26.8 AuxMstConfigId**

32 The previous value of the Bridge's MCID (13.8). This value is saved in nonvolatile memory when certain
33 non-critical configuration is made if the Bridge supports the ISIS-SPB non critical changes (27.4.1).

34 **13.26.9 rootPortId**

35 For the CIST, the Port Identifier of the Root Port, and a component of the CIST root priority vector (13.10).

36 For a given MSTI, the Port Identifier of the Root Port, and a component of the MSTI root priority vector
37 (13.11).

1 **13.26.10 rootPriority**

2 For the CIST: the Root Identifier, External Root Path Cost, Regional Root Identifier, Internal Root Path
3 Cost, Designated Bridge Identifier, and Designated Port Identifier components of the Bridge's CIST root
4 priority vector (13.10).

5 For a given MSTI: the MSTI Regional Root Identifier, Internal Root Path Cost, Designated Bridge
6 Identifier, and Designated Port Identifier components of the Bridge's MSTI root priority vector (13.11).

7 **13.26.11 rootTimes**

8 For the CIST, the Bridge's timer parameter values (Message Age, Max Age, Forward Delay, and
9 remainingHops). The values of these timers are derived (see 13.29.34) from the values stored in the CIST's
10 portTimes parameter (13.27.48) for the Root Port or from BridgeTimes (13.26.4).

11 For a given MSTI, the value of remainingHops derived (13.29.34) from the value stored in the MSTI's
12 portTimes parameter (13.27.48) for the Root Port or from BridgeTimes (13.26.4).

13 **13.26.12 TxHoldCount**

14 The value of Transmit Hold Count (Table 13-5) for the Bridge. If this is modified, the value of txCount
15 (13.27) for all ports shall be set to zero.

16 **13.27 Per port variables**

17 The variables declared in this subclause (13.27) are part of the specification. The accompanying descriptions
18 are provided to aid in the comprehension of the protocol only, and are not part of the specification.

19 There is one instance per port of each of the following variables:

- 20 a) AdminEdge (13.27.1)
- 21 b) ageingTime (13.27.2)
- 22 c) AutoEdge (13.27.18)
- 23 d) Autololate (13.27.19)
- 24 e) enableBPDURx (13.27.23)
- 25 f) enableBPDUTx (13.27.24)
- 26 g) ExternalPortPathCost (13.27.25)
- 27 h) isL2gp (13.27.26)
- 28 i) isolate (13.27.27)
- 29 j) mcheck (13.27.38)
- 30 k) newInfo (13.27.42)
- 31 l) operEdge (13.27.44)
- 32 m) portEnabled (13.27.45)
- 33 n) rcvdBpdu (13.27.52)
- 34 o) rcvdRSTP (13.27.56)
- 35 p) rcvdSTP (13.27.57)
- 36 q) rcvdTcAck (13.27.59)
- 37 r) rcvdTcn (13.27.60)
- 38 s) restrictedRole (13.27.64)
- 39 t) restrictedTcn (13.27.65)
- 40 u) sendRSTP (13.27.69)
- 41 v) tcAck (13.27.72)
- 42 w) tick (13.27.74)
- 43 x) txCount (13.27.75)

1 If MSTP or the ISIS-SPB is implemented, there is one instance per port, applicable to the CIST and to all
2 MSTIs and SPTs, of the following variable(s):

- 3 y) rcvdInternal (13.27.54)
- 4 z) restrictedDomainRole (13.27.63)

5 If MSTP or the ISIS-SPB is implemented, there is one instance per port of each of the following variables
6 for the CIST:

- 7 aa) infoInternal (13.27.31)
- 8 ab) master (13.27.36)
- 9 ac) mastered (13.27.37)

10 A single per port instance of the following variable(s) applies to all MSTIs:

- 11 ad) newInfoMsti (13.27.43)

12 If the Layer 2 Gateway Port Receive state machine (13.20, 13.40) is implemented for a port, there is one
13 instance for the CIST, and one instance for each MSTI, for that port of the following variable:

- 14 ae) pseudoRootId (13.27.51)

15 If the ISIS-SPB is implemented, there is one instance per port of the following variable(s):

- 16 af) agreedMisorder (13.27.10)
- 17 ag) agreedN (13.27.11)
- 18 ah) agreedND (13.27.12)
- 19 ai) agreeN (13.27.16)
- 20 aj) agreeND (13.27.17)

21 If the ISIS-SPB is implemented, there is one instance per port of the following variable(s), with that single
22 instance supporting all SPTs:

- 23 ak) agreedDigest (13.27.6)
- 24 al) agreedDigestValid (13.27.7)
- 25 am) agreeDigest (13.27.8)
- 26 an) agreeDigestValid (13.27.9)
- 27 ao) agreedTopology (13.27.14)

28 There is one instance per port of each of the following variables for the CIST, one per port for each MSTI
29 and one per port for each SPT:

- 30 ap) agree (13.27.3)
- 31 aq) agreed (13.27.4)
- 32 ar) designatedPriority (13.27.20)
- 33 as) designatedTimes (13.27.21)
- 34 at) disputed (13.27.22)
- 35 au) fdbFlush (13.27.28)
- 36 av) forward (13.27.29)
- 37 aw) forwarding (13.27.30)
- 38 ax) infols (13.27.32)
- 39 ay) InternalPortPathCost (13.27.33)
- 40 az) learn (13.27.34)
- 41 ba) learning (13.27.35)
- 42 bb) msgPriority (13.27.39)
- 43 bc) msgTimes (13.27.40)
- 44 bd) portId (13.27.46)
- 45 be) portPriority (13.27.47)

- 1 bf) portTimes (13.27.48)
- 2 bg) proposed (13.27.49)
- 3 bh) proposing (13.27.50)
- 4 bi) rcvdInfo (13.27.53)
- 5 bj) rcvdMsg (13.27.55)
- 6 bk) rcvdTc (13.27.58)
- 7 bl) reRoot (13.27.61)
- 8 bm) reselect (13.27.62)
- 9 bn) role (13.27.66)
- 10 bo) selected (13.27.67)
- 11 bp) selectedRole (13.27.68)
- 12 bq) sync (13.27.70)
- 13 br) synced (13.27.71)
- 14 bs) tcProp (13.27.73)
- 15 bt) updtInfo (13.27.76)

16 If the ISIS-SPB is implemented, there is one instance per port of the following variable(s) for the CIST and
17 one per port for each SPT:

- 18 bu) agreedPriority (13.27.13)
- 19 bv) agreementOutstanding (13.27.15)

20 If the ISIS-SPB is implemented, there is one instance per port of the following variables for each SPT:

- 21 bw) agreedAbove (13.27.5)
- 22 bx) neighbourPriority (13.27.41)

23 **13.27.1 AdminEdge**

24 A Boolean. Set by management if the port is to be identified as operEdge immediately on initialization,
25 without a delay to detect other Bridges attached to the LAN. The recommended default value is FALSE.

26 **13.27.2 ageingTime**

27 FDB entries for this port are aged out after ageingTime has elapsed since they were first created or refreshed
28 by the Learning Process. The value of this parameter is normally Ageing Time (8.8.3, Table 8-9), and is
29 changed to FwdDelay (13.28.10) for a period of FwdDelay after fdbFlush (13.27.28) is set by the topology
30 change state machine if stpVersion (13.28.23) is TRUE.

31 **13.27.3 agree**

32 A Boolean. See 13.16 and Figure 13-12.

33 **13.27.4 agreed**

34 A Boolean. Set, for the CIST or an MSTI, if an Agreement has been received indicating that the Port States
35 and transmitted priority vectors for the other Bridge attached to this LAN are (and, in the absence of further
36 communication with this Bridge and within the design probabilities of protocol failure due to repeated
37 BPDU loss, will remain) compatible with a loop-free active topology determined by this port's priority
38 vectors (13.16, 13.24).

39 **13.27.5 agreedAbove**

40 A Boolean controlled by the updtAgreement() procedure to indicate agreement on the neighbor(s) superior
41 priority vector.

1 **13.27.6 agreedDigest**

2 The Agreement Digest conveyed in the last BPDU received, if the port is internal to an SPT Region.

3 **13.27.7 agreedDigestValid**

4 A Boolean. TRUE if the Agreement Valid flag was set in the last BPDU or SPB Hello PDU that conveyed an
5 Agreement Digest.

6 **13.27.8 agreeDigest**

7 The Agreement Digest currently encoded in SPT BPDUs transmitted from the port.

8 **13.27.9 agreeDigestValid**

9 A Boolean. TRUE if the Agreement Valid flag is to be set in transmitted SPT BPDUs and SPB Hello PDUs.

10 **13.27.10 agreedMisorder**

11 A Boolean. TRUE if the Agreement Protocol (13.16, 13.17) has received an out of order message as
12 indicated by the received Agreement Number (13.27.16), reset when an in order message is received.

13 **13.27.11 agreedN**

14 The Agreement Number associated with the last BPDU received, if the port is within an SPT Region.
15 It takes values 0, 1, 2, or 3, and all arithmetic using this variable is modulo 4, e.g., $1 + 1 = 2$, $3 + 1 = 0$, $0 - 1$
16 $= 3$.

17 **13.27.12 agreedND**

18 The Discarded Agreement Number (see 13.17) to be included in SPT BPDUs transmitted from the port. The
19 variable values 0, 1, 2, or 3, are encoded in a two-bit field and all arithmetic using this variable is modulo 4.

20 **13.27.13 agreedPriority**

21 The priority vector associated with the worst priority received by a CIST or SPT Designated Port from an
22 SPT Region in SPT BPDUs that are sent by Root Ports, zero if there is no such agreement.

23 **13.27.14 agreedTopology**

24 A Boolean. TRUE if the Agreement Protocol has declared a topology match for agreeDigest (13.27.8), and
25 agreeDigest has the same value as the last calculated agreementDigest (13.26.1). Reset to FALSE whenever
26 agreementDigest is recalculated by ISIS-SPB, and set TRUE by updtDigest().

27 **13.27.15 agreementOutstanding**

28 The priority vector for the worst priority associated with outstanding Agreements made by a CIST or SPT
29 Root Port, Alternate Port, or Backup Port within an SPT Region. in SPT BPDUs. When an SPT BPDU, from
30 the same SPT Region, is received with a Discarded Agreement Number that is the immediate precursor of
31 the current agreeN (13.27.16), the CIST's agreementOutstanding is set to the port's designatedPriority. On
32 reception of a BPDU that is either not an SPT BPDU or is from a different SPT Region, the agreeND
33 (13.27.17) is set to the value of the agreeN (13.27.16), so there is no outstanding Agreement.

34 **13.27.16 agreeN**

35 The Agreement Number (see 13.17) transmitted in SPT BPDUs. The variable values 0, 1, 2, or 3, are
36 encoded in a two-bit field and arithmetic using this variable is modulo 4.

1 **13.27.17 agreeND**

2 The last Discarded Agreement Number (see 13.17) received in an SPT BPDU. It takes values 0, 1, 2, or 3,
3 and arithmetic using this variable is modulo 4. If agreeND matches agreeN, there is no outstanding
4 agreement.

5 **13.27.18 AutoEdge**

6 A Boolean. Set by management if the Bridge detection state machine (BDM, 13.33) is to detect other
7 Bridges attached to the LAN, and set operEdge automatically. The recommended default is TRUE.

8 **13.27.19 Autolsolate**

9 A Boolean. Set by management if isolate (13.27.27) is to be set, causing a Designated Port to transition to
10 Discarding if both AdminEdge (13.27.1) and AutoEdge (13.27.18) are FALSE and the other Bridge
11 presumed to be attached to the same point-to-point LAN does not transmit periodic BPDUs—either as a
12 Designated Port or in response to BPDUs with the Proposal flag set (see 13.23). The recommended default
13 of this parameter is FALSE. AdminEdge and AutoEdge are both reset only on ports that are known to
14 connect to other Bridges.

15 **13.27.20 designatedPriority**

16 For the CIST and a given port, the CIST Root Identifier, External Root Path Cost, Regional Root Identifier,
17 Internal Root Path Cost, Designated Bridge Identifier, and Designated Port Identifier components of the
18 port's CIST designated priority vector, as defined in 13.10.

19 For a given MSTI and port, the Regional Root Identifier, Internal Root Path Cost, Designated Bridge
20 Identifier, and Designated Port Identifier components of the designated priority vector, as defined in 13.11.

21 **13.27.21 designatedTimes**

22 For the CIST and a given port, the set of timer parameter values (Message Age, Max Age, Forward Delay,
23 and remainingHops) that are used to update Port Times when updtnfo is set. These timer parameter values
24 are used in BPDUs transmitted from the port. The value of designatedTimes is copied from the CIST
25 rootTimes Parameter (13.26.11) by the operation of the updRolesTree() procedure.

26 For a given MSTI and port, the value of remainingHops used to update this MSTI's portTimes parameter
27 when updtnfo is set. This timer parameter value is used in BPDUs transmitted from the port. The
28 updRolesTree() procedure (13.29.34) copies designatedTimes from the MSTI's rootTimes (13.26.11).

29 **13.27.22 disputed**

30 A Boolean. See 13.21, 13.20, Figure 13-20, 13.29.16, Figure 13-24, and Figure 13-25.

31 **13.27.23 enableBPDURx**

32 A Boolean. This per port management parameter is set by default, and should not be clear unless the port is
33 configured as a Layer Two Gateway Port (i.e., isL2gp is set). When clear it can allow loops to be created, or
34 can result in no connectivity. When cleared, BPDUs received on the port are discarded and not processed.

35 **13.27.24 enableBPDUTx**

36 A Boolean. This per port management parameter is set by default, and should not be clear unless the port is
37 configured as a Layer Two Gateway Port (i.e., isL2gp is set). When clear it can allow loops to be created, or
38 can result in no connectivity. When cleared, no BPDUs are transmitted by this port.

1 **13.27.25 ExternalPortPathCost**

2 The port's contribution, when it is the CIST Master Port (for MSTP and ISIS-SPB) or the CST Root Port
3 (for RSTP and MSTP), to the External Root Path Cost (13.9) for the Bridge.

4 **13.27.26 isL2gp**

5 A Boolean. Set by management to identify a port configured to be a Layer Two Gateway Port. This
6 parameter is set to FALSE by default. When set, enableBPDUTx should be cleared.

7 **13.27.27 isolate**

8 A Boolean. Set by the Bridge detection state machine (BDM, 13.33) when the Spanning Tree Protocol
9 Entity of a neighboring Bridge has apparently failed (see 13.23, 13.27.19).

10 **13.27.28 fdbFlush**

11 A Boolean. Set by the topology change state machine to instruct the FDB to remove entries for this port,
12 immediately if rstpVersion (13.28.21) is TRUE, or by rapid ageing (13.27.2) if stpVersion (13.28.23) is
13 TRUE. Reset by the FDB once the entries are removed if rstpVersion is TRUE, and immediately if
14 stpVersion is TRUE. Setting the fdbFlush variable does not result in removal of FDB entries in the case that
15 the port is an Edge Port (i.e., operEdge is TRUE). The FDB removes entries only for those VIDs that have a
16 fixed registration (see 10.7.2) on any port of the Bridge that is not an Edge Port.

17 NOTE—If MVRP or MIRP is in use, the topology change notification and flushing mechanisms defined in MRP
18 (Clause 10) and MVRP (11.2.5) are responsible for filtering entries in the FDB for VIDs that are dynamically registered
19 using MVRP or MIRP (i.e., for which there is no fixed registration in the Bridge on non-Edge Ports).

20 **13.27.29 forward**

21 A Boolean. Set or cleared by the Port Role Transitions state machine (13.37) to instruct the Port State
22 Transitions state machine (13.38) to enable or disable forwarding.

23 **13.27.30 forwarding**

24 A Boolean. Set or cleared by the Port State Transitions state machine (13.38) to indicate that forwarding has
25 been enabled or disabled.

26 **13.27.31 infoInternal**

27 If infoIs Received, indicating that the port has received current information from the Designated Bridge for
28 the attached LAN, infoInternal is set if that Designated Bridge is in the same MST Region as the receiving
29 Bridge and reset otherwise.

30 **13.27.32 infoIs**

31 A variable that takes the values Mine, Aged, Received, or Disabled, to indicate the origin/state of the port's
32 Spanning Tree information (portInfo) held for the port, as follows:

- 33 a) If infoIs Received, the port has received current (not aged out) information from the Designated
34 Bridge for the attached LAN (a point-to-point Bridge link being a special case of a LAN).
- 35 b) If infoIs Mine, information for the port has been derived from the Root Port for the Bridge (with
36 the addition of root port cost information). This includes the possibility that the Root Port is
37 "Port 0," i.e., the Bridge is the Root Bridge for the Bridged Network.
- 38 c) If infoIs Aged, information from the Root Bridge has been aged out. Just as for reselect (13.27.62),
39 the state machine does not formally allow the Aged state to persist. However, if there is a delay in
40 recomputing the new root port, correct processing of a received BPDUs is specified.
- 41 d) Finally if the port is disabled, infoIs is Disabled.

1 **13.27.33 InternalPortPathCost**

2 The port's contribution, when it is an IST or SPT Root Port (for MSTP and ISIS-SPB) to the Internal Root
3 Path Cost (13.9) for the Bridge.

4 **13.27.34 learn**

5 A Boolean. Set or cleared by the Port Role Transitions state machine (13.37) to instruct the Port State
6 Transitions state machine (13.38) to enable or disable learning.

7 **13.27.35 learning**

8 A Boolean. Set or cleared by the Port State Transitions state machine (13.38) to indicate that learning has
9 been enabled or disabled.

10 **13.27.36 master**

11 A Boolean. Used to determine the value of the Master flag for a given MSTI and port in transmitted MST
12 BPDUs.

13 Set TRUE if the Port Role for the MSTI and Port is Root Port or Designated Port, and the Bridge has
14 selected one of its ports as the Master Port for this MSTI or the mastered flag is set for this MSTI for any
15 other Bridge Port with a Root Port or Designated Port Role. Set FALSE otherwise.

16 **13.27.37 mastered**

17 A Boolean. Used to record the value of the Master flag for a given MSTI and port in MST BPDUs received
18 from the attached LAN.

19 NOTE—master and mastered signal the connection of the MSTI to the CST via the Master Port throughout the MSTI.
20 These variables and their supporting procedures do not affect the connectivity provided by this standard but permit
21 future enhancements to MSTP providing increased flexibility in the choice of Master Port without abandoning
22 plug-and-play network migration. They are therefore omitted from the overviews of protocol operation, including
23 Figure 13-13.

24 **13.27.38 mcheck**

25 A Boolean. May be set by management to force the Port Protocol Migration state machine to transmit RST
26 (or MST or SPT) BPDUs for a MigrateTime (13.26.6, Table 13-5) period, to test whether all STP Bridges on
27 the attached LAN have been removed and the port can continue to transmit RST BPDUs. Setting mcheck
28 has no effect if stpVersion (13.28.23) is TRUE.

29 **13.27.39 msgPriority**

30 For the CIST and a given port, the CIST Root Identifier, External Root Path Cost, Regional Root Identifier,
31 Internal Root Path Cost, Designated Bridge Identifier, and Designated Port Identifier components of the
32 CIST message priority vector conveyed in a received BPDU, as defined in 13.10.

33 For a given MSTI and port, the Regional Root Identifier, Internal Root Path Cost, Designated Bridge
34 Identifier, and Designated Port Identifier components of the MSTI message priority vector, as defined in
35 13.11 and conveyed in a received BPDU for this MSTI.

36 **13.27.40 msgTimes**

37 For the CIST and a given port, the timer parameter values (Message Age, Max Age, Forward Delay, Hello
38 Time, and remainingHops) conveyed in a received BPDU. If the BPDU is an STP or RST BPDU without
39 MSTP parameters, remainingHops is set to the value of the MaxHops component of BridgeTimes (13.26.4).

40 For a given MSTI and port, the value of remainingHops received in the same BPDU as the message priority
41 components of this MSTI's msgPriority parameter.

1 **13.27.41 neighbourPriority**

2 For a port attached to a point-to-point LAN within the same SPT Region, the designated priority of the other
3 Bridge attached to that LAN as calculated by ISIS-SPB for a given SPT. Receipt of an Agreement Digest
4 matching the Bridge component's agreementDigest implies receipt of an Agreement, with agreedPriority ==
5 neighbourPriority, for each SPT for which the port's selectedRole is DesignatedPort.

6 If the port is attached to a shared media LAN that is within the same SPT Region, i.e., all Bridges attached to
7 that LAN are within the Region, then neighbourPriority is the best (13.9) of all the neighbor's calculated
8 designated priority vectors, and an SPT BPDUs with a matching digest has to be received from all the
9 neighbors before agreedPriority is updated.

10 **13.27.42 newInfo**

11 A Boolean. Set TRUE if a BPDUs conveying changed CIST information is to be transmitted. It is set FALSE
12 by the Port Transmit state machine.

13 **13.27.43 newInfoMsti**

14 A Boolean. Set TRUE if a BPDUs conveying changed MSTI information is to be transmitted. It is set FALSE
15 by the Port Transmit state machine.

16 **13.27.44 operEdge**

17 A Boolean. The value of the operEdgePort parameter [item l) in 12.18.2.1.3], as determined by the operation
18 of the Bridge Detection state machine (13.33).

19 **13.27.45 portEnabled**

20 A Boolean. Set if the Bridge's MAC Relay Entity and Spanning Tree Protocol Entity can use the MAC
21 Service provided by the Bridge Port's MAC entity to transmit and receive frames to and from the attached
22 LAN, i.e., portEnabled is TRUE if and only if:

- 23 a) MAC_Operational (IEEE Std 802.1AC) is TRUE; and
- 24 b) Administrative Bridge Port State (8.4, 13.12) for the port is Enabled.

25 **13.27.46 portId**

26 The Port Identifier for this port. This variable forms a component of the port priority and designated priority
27 vectors (13.10, 13.11).

28 The most significant 4 bits of the Port Identifier (the settable Priority component) for the CIST and for each
29 MSTI can be modified independently of the setting of those bits for all other trees, as a part of allowing full
30 and independent configuration control to be exerted over each spanning tree instance.

31 **13.27.47 portPriority**

32 For the CIST and a given port, the CIST Root Identifier, External Root Path Cost, Regional Root Identifier,
33 Internal Root Path Cost, Designated Bridge Identifier, and Designated Port Identifier components of the
34 port's port priority vector (13.10).

35 For a given MSTI and port, the Regional Root Identifier, Internal Root Path Cost, Designated Bridge
36 Identifier, and Designated Port Identifier components of the port's MSTI port priority vector (13.11).

37 **13.27.48 portTimes**

38 For the CIST and a given port, the port's timer parameter values (Message Age, Max Age, Forward Delay,
39 Hello Time, and remainingHops). The Hello Time timer parameter value is used in transmitted BPDUs.

40 For a given MSTI and port, the value of remainingHops for this MSTI in transmitted BPDUs.

1 **13.27.49 proposed**

2 A Boolean. See 13.16, Figure 13-10, Figure 13-12, 13.29.14, 13.29.19, 13.37, and Figure 13-24.

3 **13.27.50 proposing**

4 A Boolean. See 13.16, Figure 13-10, Figure 13-12, 13.29.14, 13.29.15, 13.33, and Figure 13-25.

5 **13.27.51 pseudoRootId**

6 A Bridge Identifier configured by management for L2GP operation. By default, it is set to the
7 BridgeIdentifier (13.26.2).

8 **13.27.52 rcvdBPDU**

9 A Boolean. Set by system-dependent processes, this variable notifies the Port Receive state machine (13.31)
10 when a valid (Clause 14) Configuration, TCN, RST, MST, or SPT BPDU (14.3) is received on the port.
11 Reset by the Port Receive state machine.

12 **13.27.53 rcvdInfo**

13 Set to the result of the rcvInfo() procedure (13.29.12).

14 **13.27.54 rcvdInternal**

15 A Boolean. Set TRUE by the Receive Machine if the BPDU received was transmitted by a Bridge in the
16 same MST Region as the receiving Bridge.

17 **13.27.55 rcvdMsg**

18 A Boolean. See 13.29.13 and 13.31.

19 **13.27.56 rcvdRSTP**

20 A Boolean. See 13.31, 13.29.31, and 13.32.

21 **13.27.57 rcvdSTP**

22 A Boolean. See 13.31, 13.29.31, and 13.32.

23 **13.27.58 rcvdTc**

24 A Boolean. See 13.29.13, 13.29.24, and 13.39.

25 **13.27.59 rcvdTcAck**

26 A Boolean. See 13.29.24 and 13.39.

27 **13.27.60 rcvdTcn**

28 A Boolean. See 13.29.13, 13.29.24, and 13.39.

29 **13.27.61 reRoot**

30 A Boolean. Set by a newly selected Root Port to force prior Root Ports to Discarding, before it becomes
31 forwarding. See Figure 13-24, 13.29.21, Figure 13-23, Figure 13-25, and Figure 13-26.

32 **13.27.62 reselect**

33 A Boolean. Set to prompt recomputation of the CIST or an MSTI. See 13.35 and 13.36.

1 **13.27.63 restrictedDomainRole**

2 A Boolean. Set by management. If TRUE, causes a port that is a Boundary Port of an SPT Region not to be
3 selected as Root Port for the CIST, any MSTI, or any SPT, even it has the best spanning tree priority vector.
4 See 13.20, 27.7, and restrictedRole (13.27.64).

5 **13.27.64 restrictedRole**

6 A Boolean. Set by management. If TRUE causes the port not to be selected as Root Port for the CIST or any
7 MSTI, even it has the best spanning tree priority vector. Such a port will be selected as an Alternate Port
8 after the Root Port has been selected. This parameter should be FALSE by default. If set, it can cause lack of
9 spanning tree connectivity. It is set by a network administrator to prevent Bridges external to a core region of
10 the network influencing the spanning tree active topology, possibly because those Bridges are not under the
11 full control of the administrator.

12 **13.27.65 restrictedTcn**

13 A Boolean. Set by management. If TRUE causes the port not to propagate received topology change
14 notifications and topology changes to other ports. This parameter should be FALSE by default. If set it can
15 cause temporary loss of connectivity after changes in a spanning trees active topology as a result of
16 persistent incorrectly learned station location information. It is set by a network administrator to prevent
17 Bridges external to a core region of the network, causing address flushing in that region, possibly because
18 those Bridges are not under the full control of the administrator or MAC_Operational for the attached LANs
19 transitions frequently.

20 **13.27.66 role**

21 The current Port Role. DisabledPort, RootPort, DesignatedPort, AlternatePort, BackupPort, or MasterPort.
22 NOTE—The MasterPort role applies to each MSTI when the CIST Port Role is RootPort and connects to another MST
23 Region. An MSTI Master Port is part of the stable active topology for frames assigned to that MSTI, just as the CIST
24 Root Port forwards frames for the IST. The Port State for each MSTI may differ as required to suppress temporary loops.

25 **13.27.67 selected**

26 A Boolean. See 13.36 and 13.29.22.

27 **13.27.68 selectedRole**

28 A newly computed role for the port.

29 **13.27.69 sendRSTP**

30 A Boolean. See 13.32 and 13.34.

31 **13.27.70 sync**

32 A Boolean. Set to force the Port State to be compatible with the loop-free active topology determined by the
33 priority vectors held by this Bridge (13.16, 13.24) for this tree (CIST or MSTI), transitioning the Port State
34 to Discarding, and soliciting an Agreement if possible, if the port is not already synchronized (13.27.71).

35 **13.27.71 synced**

36 A Boolean. TRUE only if the Port State is compatible with the loop-free active topology determined by the
37 priority vectors held by this Bridge for this tree (13.16, 13.24).

38 **13.27.72 tcAck**

39 A Boolean. Set to transmit a Configuration Message with a topology change acknowledge flag set.

1 **13.27.73 tcProp**

2 A Boolean. Set by the Topology Change state machine of any other port, to indicate that a topology change
3 should be propagated through this port.

4 **13.27.74 tick**

5 A Boolean. See the Port Timers state machine (13.30).

6 **13.27.75 txCount**

7 A counter. Incremented by the Port Transmission (13.34) state machine on every BPDU transmission, and
8 decremented used by the Port Timers state machine (13.30) once a second. Transmissions are delayed if
9 txCount reaches TxHoldCount (13.26.12).

10 **13.27.76 updtInfo**

11 A Boolean. Set by the Port Role Selection state machine (13.36, 13.29.34) to tell the Port Information state
12 machine that it should copy designatedPriority to portPriority and designatedTimes to portTimes.

13 **13.28 State machine conditions and parameters**

14 The following variable evaluations are defined for notational convenience in the state machines:

- 15 a) allSptAgree (13.28.1)
- 16 b) allSynced (13.28.2)
- 17 c) allTransmitReady (13.28.3)
- 18 d) BestAgreementPriority (13.28.4)
- 19 e) cist (13.28.5)
- 20 f) cistRootPort (13.28.6)
- 21 g) cistDesignatedPort (13.28.7)
- 22 h) EdgeDelay (13.28.8)
- 23 i) forwardDelay (13.28.9)
- 24 j) FwdDelay (13.28.10)
- 25 k) HelloTime (13.28.11)
- 26 l) MaxAge (13.28.12)
- 27 m) msti (13.28.13)
- 28 n) mstiDesignatedOrTCpropagatingRootPort (13.28.14)
- 29 o) mstiMasterPort (13.28.15)
- 30 p) operPointToPointMAC (13.28.16)
- 31 q) rcvdAnyMsg (13.28.17)
- 32 r) rcvdCistMsg (13.28.18)
- 33 s) rcvdMstiMsg (13.28.19)
- 34 t) reRooted (13.28.20)
- 35 u) rstpVersion (13.28.21)
- 36 v) spt (13.28.22)
- 37 w) stpVersion (13.28.23)
- 38 x) updtCistInfo (13.28.24)
- 39 y) updtMstiInfo (13.28.25)

40 **13.28.1 allSptAgree**

41 TRUE, if and only if, agree is TRUE for the given port for all SPTs.

1 **13.28.2 allSynced**

2 The condition allSynced is TRUE for a given port, for a given tree, if and only if:

- 3 a) For all ports for the given tree, selected is TRUE, the port's role is the same as its selectedRole, and
4 updtInfo is FALSE; and
- 5 b) The role of the given port is:
 - 6 1) Root Port or Alternate Port and synced is TRUE for all ports for the given tree other than the
7 Root Port; or
 - 8 2) Designated Port and synced is TRUE for all ports for the given tree other than the given port; or
 - 9 3) Designated Port, and the tree is an SPT or the IST, and the Root Port of the tree and the given
10 port are both within the Bridge's SPT Region, and both learning and forwarding are FALSE for
11 the given port; or
 - 12 4) Master Port and synced is TRUE for all ports for the given tree other than the given port.

13 **13.28.3 allTransmitReady**

14 TRUE, if and only if, for the given port for all trees:

- 15 a) selected is TRUE; and
- 16 b) updtInfo is FALSE.

17 **13.28.4 BestAgreementPriority**

18 The best of all possible priority vectors, i.e., numerically the lowest.

19 **13.28.5 cist**

20 TRUE only for CIST state machines; i.e., FALSE for MSTI or SPT state machine instances.

21 **13.28.6 cistRootPort**

22 TRUE if the CIST role for the given port is RootPort.

23 **13.28.7 cistDesignatedPort**

24 TRUE if the CIST role for the given port is DesignatedPort.

25 **13.28.8 EdgeDelay**

26 Returns the value of MigrateTime if operPointToPointMAC is TRUE, and the value of MaxAge otherwise.

27 **13.28.9 forwardDelay**

28 Returns the value of HelloTime if sendRSTP is TRUE, and the value of FwdDelay otherwise.

29 **13.28.10 FwdDelay**

30 The Forward Delay component of the CIST's designatedTimes parameter (13.27.21).

31 **13.28.11 HelloTime**

32 The Hello Time component of the CIST's portTimes parameter (13.27.48) with the recommended default
33 value given in Table 13-5.

34 **13.28.12 MaxAge**

35 The Max Age component of the CIST's designatedTimes parameter (13.27.21).

1 **13.28.13 msti**

2 TRUE only for MSTI state machines; i.e., FALSE for CIST or SPT state machine instances.

3 **13.28.14 mstiDesignatedOrTCpropagatingRootPort**

4 TRUE if the role for any MSTI for the given port is either:

- 5 a) DesignatedPort or
- 6 b) RootPort, and the instance for the given MSTI and port of the tcWhile timer is not zero.

7 **13.28.15 mstiMasterPort**

8 TRUE if the role for any MSTI for the given port is MasterPort.

9 **13.28.16 operPointToPoint**

10 TRUE if operPointToPointMAC (IEEE Std 802.1AC) is TRUE for the Bridge Port.

11 **13.28.17 rcvdAnyMsg**

12 TRUE for a given port if rcvdMsg is TRUE for the CIST or any MSTI for that port.

13 **13.28.18 rcvdCistMsg**

14 TRUE for a given port if and only if rcvdMsg is TRUE for the CIST for that port.

15 **13.28.19 rcvdMstiMsg**

16 TRUE for a given port and MSTI if and only if rcvdMsg is FALSE for the CIST for that port and rcvdMsg is

17 TRUE for the MSTI for that port.

18 **13.28.20 reRooted**

19 TRUE if the rrWhile timer is clear (zero) for all Ports for the given tree other than the given Port.

20 **13.28.21 rstpVersion**

21 TRUE if ForceProtocolVersion (13.7.2) is greater than or equal to 2.

22 **13.28.22 spt**

23 TRUE only for SPT state machines, in an SPT Bridge; i.e., FALSE for CIST and MSTI state machine
24 instances.

25 **13.28.23 stpVersion**

26 TRUE if Force Protocol Version (13.7.2) is less than 2.

27 **13.28.24 updtCistInfo**

28 TRUE for a given port if and only if updtInfo is TRUE for the CIST for that port.

29 **13.28.25 updtMstiInfo**

30 TRUE for a given port and MSTI if and only if updtInfo is TRUE for the MSTI for that port or updtInfo is

31 TRUE for the CIST for that port.

32 NOTE—The dependency of rcvdMstiMsg and updtMstiInfo on CIST variables for the port reflects the fact that MSTIs
33 exist in a context of CST parameters. The state machines ensure that the CIST parameters from received BPDUs are
34 processed and updated prior to processing MSTI information.

1 13.29 State machine procedures

2 The following procedures perform the functions specified for both the state machines for all trees, or
3 specifically for the CIST, a given MSTI, or a given SPT:

- 4 a) betterorsameInfo(newInfol) (13.29.1)
- 5 b) clearAllRcvdMsgs() (13.29.2)
- 6 c) clearReselectTree() (13.29.3)
- 7 d) disableForwarding() (13.29.4)
- 8 e) disableLearning() (13.29.5)
- 9 f) enableForwarding() (13.29.6)
- 10 g) enableLearning() (13.29.7)
- 11 h) fromSameRegion() (13.29.8)
- 12 i) newTcDetected() (13.29.9)
- 13 j) newTcWhile() (13.29.10)
- 14 k) pseudoRcvMsgs() (13.29.11)
- 15 l) rcvInfo() (13.29.12)
- 16 m) rcvMsgs() (13.29.13)
- 17 n) recordAgreement() (13.29.15)
- 18 o) recordDispute() (13.29.16)
- 19 p) recordMastered() (13.29.17)
- 20 q) recordPriority() (13.29.18)
- 21 r) recordProposal() (13.29.19)
- 22 s) recordTimes() (13.29.20)
- 23 t) setReRootTree() (13.29.21)
- 24 u) setSelectedTree() (13.29.22)
- 25 v) setSyncTree() (13.29.23)
- 26 w) setTcFlags() (13.29.24)
- 27 x) setTcPropTree() (13.29.25)
- 28 y) syncMaster() (13.29.26)
- 29 z) txConfig() (13.29.27)
- 30 aa) txRstp() (13.29.28)
- 31 ab) txTcn() (13.29.29)
- 32 ac) updtBPDUVersion() (13.29.31)
- 33 ad) updtRcvdInfoWhile() (13.29.33)
- 34 ae) updtRolesTree() (13.29.34)
- 35 af) updtRolesDisabledTree() (13.29.35)

36 The following procedures perform the functions specified for all SPTs, or for a given SPT or the CIST when
37 ISIS-SPB is implemented:

- 38 ag) rcvAgreements() (13.29.14)
- 39 ah) updtAgreement() (13.29.30)
- 40 ai) updtDigest() (13.29.32)

41 All references to named variables in the specification of procedures are to instances of the variables
42 corresponding to the instance of the state machine using the function, i.e., to the CIST or the given MSTI or
43 the given SPT as appropriate. References to forwarding and learning apply to frames assigned to the
44 specified tree.

1 **13.29.1 betterorsameInfo(newInfoIs)**

2 Returns TRUE if, for a given port and tree (CIST or MSTI), either:

- 3 a) The procedure's parameter newInfoIs is Received, and infoIs is Received and the msgPriority vector
4 is better than or the same as (13.10) the portPriority vector; or
- 5 b) The procedure's parameter newInfoIs is Mine, and infoIs is Mine and the designatedPriority vector is
6 better than or the same as (13.10) the portPriority vector.

7 Returns False otherwise.

8 NOTE—This procedure is not invoked (in the case of a MSTI) if the received BPDU carrying the MSTI information was
9 received from another MST Region. In that event, the Port Receive Machine (using rcvMsgs()) does not set rcvdMsg
10 for any MSTI, and the Port Information Machine's SUPERIOR_DESIGNATED state is not entered.

11 **13.29.2 clearAllRcvdMsgs()**

12 Clears rcvdMsg for the CIST and all MSTIs, for this port.

13 **13.29.3 clearReselectTree()**

14 Clears reselect for the tree (the CIST or a given MSTI) for all ports of the Bridge.

15 **13.29.4 disableForwarding()**

16 An implementation-dependent procedure that causes the Forwarding Process (8.6) to stop forwarding frames
17 through the port. The procedure does not complete until forwarding has stopped.

18 **13.29.5 disableLearning()**

19 An implementation-dependent procedure that causes the Learning Process (8.7) to stop learning from the
20 source address of frames received on the port. The procedure does not complete until learning has stopped.

21 **13.29.6 enableForwarding()**

22 An implementation-dependent procedure that causes the Forwarding Process (8.6) to start forwarding
23 frames through the port. The procedure does not complete until forwarding has been enabled.

24 **13.29.7 enableLearning()**

25 An implementation-dependent procedure that causes the Learning Process (8.7) to start learning from frames
26 received on the port. The procedure does not complete until learning has been enabled.

27 **13.29.8 fromSameRegion()**

28 Returns TRUE if rcvdRSTP is TRUE, and the received BPDU conveys a MCID that matches that held for
29 the Bridge. Returns FALSE otherwise. If SPB protocols are implemented, fromSameRegion() returns TRUE
30 if the Bridge detects that either its MCID (13.26.7) or its Auxiliary MCID (13.26.8) match either the MCID
31 or Auxiliary MCID carried by the received SPT BPDU.

32 **13.29.9 newTcDetected()**

33 If the value of tcDetected is zero and sendRSTP is TRUE, this procedure sets the value of tcDetected to
34 HelloTime plus one second. The value of HelloTime is taken from the CIST's portTimes parameter (13.27.48)
35 for this port.

36 If the value of tcDetected is zero and sendRSTP is FALSE, this procedure sets the value of tcDetected to the
37 sum of the Max Age and Forward Delay components of rootTimes.

38 Otherwise, the procedure takes no action.

1 **13.29.10 newTcWhile()**

2 If the value of tcWhile is zero and sendRSTP is TRUE, this procedure sets the value of tcWhile to HelloTime
3 plus one second and sets either newInfo TRUE for the CIST or newInfoMsti TRUE for a given MSTI. The
4 value of HelloTime is taken from the CIST's portTimes parameter (13.27.48) for this port.

5 If the value of tcWhile is zero and sendRSTP is FALSE, this procedure sets the value of tcWhile to the sum
6 of the Max Age and Forward Delay components of rootTimes and does not change the value of either
7 newInfo or newInfoMsti.

8 Otherwise, the procedure takes no action.

9 **13.29.11 pseudoRcvMsgs()**

10 Using local parameters, this procedure simulates the processing that would be applied by rcvInfo() and
11 rcvMsgs() to a BPDU received on the port, from the same region and with the following parameters:

- 12 a) Message Age, Max Age, Hello Time, and Forward Delay are derived from BridgeTimes (13.26.4).
- 13 b) The CIST information carries the message priority vector (13.10) with a value of {pseudoRootId, 0,
14 pseudoRootId, 0, 0, 0}.
- 15 c) A CIST Port Role of Designated Port, with the Learning and Forwarding flags set.
- 16 d) The Version 1 Length is 0 and Version 3 Length calculated appropriately.
- 17 e) For each MSTI configured on the Bridge, the corresponding MSTI Configuration Message carries
18 the following:
 - 19 1) A message priority vector with a value of {pseudoRootId, 0, 0, 0}
 - 20 2) A Port Role of Designated Port, with the Learning and Forwarding flags set
 - 21 3) MSTI Remaining Hops set to the value of the MaxHops component of BridgeTimes (13.26.4)

22 NOTE—If two L2GP ports are configured with the same CIST pseudoRootId then the IST may partition within the MST
23 Region, but either of the L2GP ports can be selected to provide connectivity from the Region/customer network to a
24 provider's network on an MSTI by MSTI basis.

25 **13.29.12 rcvInfo()**

26 Returns SuperiorDesignatedInfo if, for a given port and tree (CIST or MSTI),

- 27 a) The received CIST or MSTI message conveys a Designated Port Role and
 - 28 1) The message priority (msgPriority—13.27.39) is superior (13.10 or 13.11) to the port's port
29 priority vector; or
 - 30 2) The message priority is the same as the port's port priority vector, and any of the received timer
31 parameter values (msgTimes—13.27.40) differ from those already held for the port
32 (portTimes—13.27.48).

33 Otherwise, returns RepeatedDesignatedInfo if, for a given port and tree (CIST or MSTI),

- 34 b) The received CIST or MSTI message conveys a Designated Port Role and
 - 35 1) A message priority vector and timer parameters that are the same as the port's port priority
36 vector and timer values and
 - 37 2) infols is Received.

38 Otherwise, returns InferiorDesignatedInfo if, for a given port and tree (CIST or MSTI),

- 39 c) The received CIST or MSTI message conveys a Designated Port Role.

40 Otherwise, returns InferiorRootAlternateInfo if, for a given port and tree (CIST or MSTI),

- 41 d) The received CIST or MSTI message conveys a Root Port, Alternate Port, or Backup Port Role and
42 a CIST or MSTI message priority that is the same as or worse than the CIST or MSTI port priority
43 vector.

44 Otherwise, returns OtherInfo.

45 NOTE—A Configuration BPDU implicitly conveys a Designated Port Role.

1 **13.29.13 rcvMsgs()**

2 This procedure is invoked by the Port Receive state machine (13.31) to decode a received BPDU. Sets
3 rcvdTcn and rcvdTc for each and every MSTI if a TCN BPDU has been received, and extracts the message
4 priority and timer values from the received BPDU storing them in the msgPriority and msgTimes variables.

5 If ISIS-SPB is implemented, ForceProtocolVersion is 4 (or greater), the BPDU is an SPT BPDU, and has
6 been received on a Bridge Port that is internal to the SPT Region (i.e., is not a Boundary Port, see 13.12),
7 then the rcvAgreements() procedure processes the CIST and SPT information conveyed by the BPDU.

8 Otherwise (i.e., if rcvAgreements() is not used), this procedure sets rcvdMsg for the CIST and makes the
9 received CST or CIST message available to the CIST Port Information state machines.

10 If and only if rcvdInternal is set, this procedure sets rcvdMsg for each and every MSTI for which a MSTI
11 message is conveyed in the BPDU, and makes available each MSTI message and the common parts of the
12 CIST message priority (the CIST Root Identifier, External Root Path Cost, and Regional Root Identifier) to
13 the Port Information state machine for that MSTI.

14 **13.29.14 rcvAgreements()**

15 This procedure is used to process agreements received in SPT BPDUs and in SPB Hello PDUs. The same
16 variables are used in both cases. In the case of SPT BPDUs, this procedure is invoked by the rcvMsgs()
17 procedure that decodes received BPDUs.

18 If the received Agreement Number is equal to the value of agreedN plus three, agreedMisorder is set TRUE.

19 The variables agreedN and agreeND are set to the values of the received Agreement Number and Discard
20 Agreement Number, respectively. The variables agreedDigest and agreedDigestValid are updated with the
21 values of the received Agreement Digest and Agreement Digest flag, respectively.

22 The CIST proposed flag is set to the value of the Proposal flag for the CIST in the received BPDU.

23 The updtDigest() procedure is invoked. The updtAgreement() procedure is invoked for the port, for the
24 CIST and for each SPT.

25 **13.29.15 recordAgreement()**

26 For the CIST and a given port, if rstpVersion is TRUE, operPointToPointMAC (IEEE Std 802.1AC) is
27 TRUE, and the received CIST Message has the Agreement flag set, then the CIST agreed flag is set and the
28 CIST proposing flag is cleared. Otherwise, the CIST agreed flag is cleared. Additionally, if the CIST
29 message was received from a Bridge in a different MST Region, i.e., the rcvdInternal flag is clear, the agreed
30 and proposing flags for this port for all MSTIs are set or cleared to the same value as the CIST agreed and
31 proposing flags. If the CIST message was received from a Bridge in the same MST Region, the MSTI
32 agreed and proposing flags are not changed.

33 For a given MSTI and port, if operPointToPointMAC (IEEE Std 802.1AC) is TRUE, and:

- 34 a) The message priority vector of the CIST Message accompanying the received MSTI Message (i.e.,
35 received in the same BPDU) has the same CIST Root Identifier, CIST External Root Path Cost, and
36 Regional Root Identifier as the CIST port priority vector, and
- 37 b) The received MSTI Message has the Agreement flag set,

38 the MSTI agreed flag is set and the MSTI proposing flag is cleared. Otherwise, the MSTI agreed flag is
39 cleared.

40 NOTE—MSTI Messages received from Bridges external to the MST Region are discarded and not processed by
41 recordAgreement() or recordProposal().

1 **13.29.16 recordDispute()**

2 For the CIST and a given port, if the CIST message has the learning flag set:

- 3 a) The disputed variable is set, and
- 4 b) The agreed variable is cleared.

5 Additionally, if the CIST message was received from a Bridge in a different MST region (i.e., if the
6 rcvdInternal flag is clear), then for all the MSTIs:

- 7 c) The disputed variable is set, and
- 8 d) The agreed variable is cleared.

9 For a given MSTI and port, if the received MSTI message has the learning flag set:

- 10 e) The disputed variable is set, and
- 11 f) The agreed variable is cleared.

12 **13.29.17 recordMastered()**

13 For the CIST and a given port, if the CIST message was received from a Bridge in a different MST Region,
14 i.e., the rcvdInternal flag is clear, the mastered variable for this port is cleared for all MSTIs.

15 For a given MSTI and port, if the MSTI message was received on a point-to-point link and the MSTI
16 Message has the Master flag set, set the mastered variable for this MSTI. Otherwise, reset the mastered
17 variable.

18 **13.29.18 recordPriority()**

19 Sets the components of the portPriority variable to the values of the corresponding msgPriority components.

20 **13.29.19 recordProposal()**

21 For the CIST and a given port, if the received CIST Message conveys a Designated Port Role, and has the
22 Proposal flag set, the CIST proposed flag is set. Otherwise, the CIST proposed flag is not changed.
23 Additionally, if the CIST Message was received from a Bridge in a different MST Region, i.e., the
24 rcvdInternal flag is clear, the proposed flags for this port for all MSTIs are set or cleared to the same value as
25 the CIST proposed flag. If the CIST message was received from a Bridge in the same MST Region, the
26 MSTI proposed flags are not changed.

27 For a given MSTI and port, if the received MSTI Message conveys a Designated Port Role, and has the
28 Proposal flag set, the MSTI proposed flag is set. Otherwise, the MSTI proposed flag is not changed.

29 **13.29.20 recordTimes()**

30 For the CIST and a given port, sets portTimes' Message Age, Max Age, Forward Delay, and remainingHops
31 to the received values held in msgTimes and portTimes' Hello Time to the default specified in Table 13-5.

32 For a given MSTI and port, sets portTime's remainingHops to the received value held in msgTimes.

33 **13.29.21 setReRootTree()**

34 Sets reRoot TRUE for this tree (the CIST or a given MSTI) for all ports of the Bridge.

35 **13.29.22 setSelectedTree()**

36 Sets selected TRUE for this tree (the CIST or a given MSTI) for all ports of the Bridge if reselect is FALSE
37 for all ports in this tree. If reselect is TRUE for any port in this tree, this procedure takes no action.

38 **13.29.23 setSyncTree()**

39 Sets sync TRUE for this tree (the CIST or a given MSTI) for all ports of the Bridge.

1 **13.29.24 setTcFlags()**

2 For the CIST and a given port:

- 3 a) If the Topology Change Acknowledgment flag is set for the CIST in the received BPDU, sets
4 rcvdTcAck TRUE.
- 5 b) If rcvdInternal is clear and the Topology Change flag is set for the CIST in the received BPDU, sets
6 rcvdTc TRUE for the CIST and for each and every MSTI.
- 7 c) If rcvdInternal is set, sets rcvdTc for the CIST if the Topology Change flag is set for the CIST in the
8 received BPDU.

9 For a given MSTI and port, sets rcvdTc for this MSTI if the Topology Change flag is set in the corresponding
10 MSTI message.

11 **13.29.25 setTcPropTree()**

12 If and only if restrictedTcn is FALSE for the port that invoked the procedure, sets tcProp TRUE for the given
13 tree (the CIST or a given MSTI) for all other ports.

14 **13.29.26 syncMaster()**

15 For all MSTIs, for each port that has infoInternal set:

- 16 a) Clears the agree, agreed, and synced variables; and
- 17 b) Sets the sync variable.

18 **13.29.27 txConfig()**

19 Transmits a Configuration BPDU. The first four components of the message priority vector (13.27.39)
20 conveyed in the BPDU are set to the value of the CIST Root Identifier, External Root Path Cost, Bridge
21 Identifier, and Port Identifier components of the CIST's designatedPriority parameter (13.27.20) for this
22 port. The topology change flag is set if (tcWhile != 0) for the port. The topology change acknowledgment flag
23 is set to the value of tcAck for the port. The remaining flags are set to zero. The value of the Message Age,
24 Max Age, and Fwd Delay parameters conveyed in the BPDU are set to the values held in the CIST's
25 designatedTimes parameter (13.27.21) for the port. The value of the Hello Time parameter conveyed in the
26 BPDU is set to the value held in the CIST's portTimes parameter (13.27.48) for the port.

27 **13.29.28 txRstp()**

28 Transmits a RST BPDU, MST BPDU, or SPT BPDU as determined by the value of ForceProtocolVersion
29 (13.7.2), and encoded as specified by Clause 14. All per port variables referenced in this subclause
30 (13.29.28) are those for the transmitting port.

31 The first six components of the CIST message priority vector (13.27.39) conveyed in the BPDU are set to
32 the value of the CIST's designatedPriority parameter (13.27.20). The Port Role in the BPDU (14.2.9) is set to
33 the current value of the CIST's role (13.27.66). The Agreement and Proposal flags in the BPDU are set to
34 the values of agree (13.27.3) and proposing (13.27.50), respectively. The CIST topology change flag is set if
35 (tcWhile != 0) for the port. The topology change acknowledge flag in the BPDU is never used and is set to
36 zero. The Learning and Forwarding flags in the BPDU are set to the values of learning (13.27.35) and
37 forwarding (13.27.30) for the CIST, respectively. The value of the Message Age, Max Age, and Fwd Delay
38 parameters conveyed in the BPDU are set to the values held in the CIST's designatedTimes parameter
39 (13.27.21). The value of the Hello Time parameter conveyed in the BPDU is set to the value held in the
40 CIST's portTimes parameter (13.27.48).

1 If the value of the Force Protocol Version parameter is less than 3, no further parameters are encoded in the
2 BPDU and the protocol version parameter is set to 2 (denoting a RST BPDU). Otherwise, MST BPDU
3 parameters are encoded as follows:

- 4 a) The version 3 length.
- 5 b) The MST Configuration Identifier parameter of the BPDU is the value of MstConfigId (13.26.7).
- 6 c) The CIST Internal Root Path Cost (13.27.20).
- 7 d) The CIST Bridge Identifier (CIST Designated Bridge Identifier—13.27.20).
- 8 e) The CIST Remaining Hops (13.27.21).
- 9 f) The parameters of each MSTI message, encoded in MSTID order.

10 NOTE—No more than 64 MSTIs may be supported. The parameter sets for all of these can be encoded in a
11 standard-sized Ethernet frame. The number of MSTIs supported can be zero: an SPT Bridge, for example, is not obliged
12 to have MSTIs configured in order to support SPB.

13 If the value of the Force Protocol Version parameter is less than 3, no further parameters are encoded in the
14 BPDU and the protocol version parameter is set to 3 (denoting a MST BPDU). Otherwise, SPT BPDU
15 parameters are encoded, and the protocol version parameter is set to 4 (denoting an SPT BPDU):

- 16 g) agreeN (13.27.16) is encoded in the Agreement Number field.
- 17 h) agreedND (13.27.17) is encoded in the Discarded Agreement Number field.
- 18 i) agreeDigest (13.27.8) is encoded in the Agreement Digest field.

19 **13.29.29 txTcn()**

20 Transmits a TCN BPDU.

21 **13.29.30 updtAgreement()**

22 If ISIS-SPB is implemented, this procedure is invoked for a given port, for the CIST or a given SPT, by the
23 updtRolesTree() procedure used by the Port Role Selection machine (13.36), and by the rcvAgreements()
24 procedure. The Port Role Selection machine itself could have been invoked following receipt of a BPDU on
25 a port that is at the boundary of the SPT Region, possibly changing the context for the link state calculation,
26 or following a link state update computed by ISIS-SPB following receipt of an IS-IS PDU.

27 If the procedure has been invoked for the CIST:

- 28 a) If the port's selectedRole is DesignatedPort, agreed is TRUE if and only if:
 - 29 1) agreedTopology is TRUE; and
 - 30 2) designatedPriority is the same or better (13.9) than agreedPriority.
- 31 b) If the port's selectedRole is not DesignatedPort, agreed is TRUE if and only if:
 - 32 1) agreementOutstanding is the same or better than designatedPriority.

33 If the procedure has been invoked for an SPT, and the port's selectedRole is DesignatedPort:

- 34 c) agreedAbove is reset to FALSE;
- 35 d) if agreedTopology is TRUE:
 - 36 1) agreementOutstanding (13.27.15) is set to BestAgreementPriority (13.28.4); and
 - 37 2) agreedPriority (13.27.13) is set to the sum of neighbourPriority (13.27.41) and
38 InternalPortPathCost.
- 39 e) if agreedTopology is FALSE and agreedPriority is less than the sum of neighbourPriority and
40 InternalPortPathCost:
 - 41 1) agreedPriority (13.27.13) is set to the sum of neighbourPriority (13.27.41) and
42 InternalPortPathCost.
- 43 f) agreed (13.27.4) is TRUE if and only if:
 - 44 1) designatedPriority is better than agreedPriority.

1 If the procedure has been invoked for an SPT and the port's selectedRole is not DesignatedPort:

- 2 g) agreedPriority is set to BestAgreementPriority;
- 3 h) if agreedTopology is TRUE:
 - 4 1) agreedAbove is set TRUE, and
 - 5 2) agreementOutstanding (13.27.15) is set to the sum of neighbourPriority (13.27.41) and
 - 6 InternalPortPathCost.
- 7 i) if agreedTopology is FALSE and agreementOutstanding is better than the sum of neighbourPriority
- 8 and InternalPortPathCost:
 - 9 1) agreementOutstanding is set to the sum of neighbourPriority (13.27.41) and
 - 10 InternalPortPathCost.
- 11 j) agreed is TRUE if and only if:
 - 12 1) agreedAbove is TRUE, and
 - 13 2) agreementOutstanding is the same as or better than designatedPriority.

14 Independently of whether the procedure was invoked for the CIST or an SPT:

- 15 k) if agreed is FALSE, then the sync variable is set TRUE.

16 **13.29.31 updtBPDUVersion()**

17 Sets rcvdSTP TRUE if the BPDU received is a version 0 or version 1 TCN or a Config BPDU. Sets
18 rcvdRSTP TRUE if the received BPDU is a RST BPDU or a MST BPDU.

19 **13.29.32 updtDigest()**

20 Updates agreeDigest and agreeN, following calculation of a new topology or topologies by ISIS-SPB, and
21 checks for a topology match with the updated values of those variables, as follows.

22 If agreeDigest is not equal to agreementDigest and:

- 23 a) agreeN is equal to agreeND, or
- 24 b) agreeN plus one is equal to agreeND:

25 then

- 26 c) agreeDigest is set equal to agreementDigest, and
- 27 d) agreeDigestValid is reset to FALSE, and
- 28 e) agreeN is set equal to agreeN plus one.

29 Additionally, if agreeDigest is now equal to agreementDigest and:

- 30 f) agreeDigest equals agreedDigest, and
- 31 g) agreedDigestValid is TRUE:

32 then

- 33 h) if agreedND is not equal to agreedN plus one, then
 - 34 1) agreedND is set equal to agreedN plus one, and
 - 35 2) newInfoMsti is set.

36 and

- 37 i) if agreeND is equal to agreeN and agreedMisorder is FALSE, or
- 38 j) if agreeND is set equal to agreeN plus one, then
 - 39 1) agreedTopology is set TRUE, and
 - 40 2) agreedMisorder is set FALSE;

41 otherwise, i.e., if (f) and (g) above are not both TRUE:

- 1 k) if `agreedND` is not equal to `agreedN`, then:
2 1) `agreedND` is set equal to `agreedN`, and
3 2) `newInfoMsti` is set.

4 The `agreeDigest`, `agreeN`, and `agreedND` variables determine the values of the Agreement Digest,
5 Agreement Number (AN), and Discarded Agreement Number (DAN), transmitted in SPT BPDU and SPB
6 Hello PDUs. This procedure is used by `updtRolesTree()` before using `updtAgreement()` for each SPT, and
7 also by `rcvAgreements()`, since the latter can rotate the AN sequence window number and thus allow
8 `agreeDigest` to be updated with `agreementDigest`.

9 Wherever `newInfoMsti` is set in this procedure, transmission of an SPB Hello PDU to convey the updated
10 digest and sequence number information is also requested. SPT BPDUs and SPB Hello PDUs perform
11 additional functions and are subject to different rate limiters, but both convey the Agreement Digest and
12 related information.

13 13.29.33 `updtRcvdInfoWhile()`

14 Updates `rcvdInfoWhile` (13.25). The value assigned to `rcvdInfoWhile` is three times the Hello Time, if either:

- 15 a) Message Age, incremented by 1 second and rounded to the nearest whole second, does not exceed
16 Max Age and the information was received from a Bridge external to the MST Region (`rcvdInternal`
17 `FALSE`); or
18 b) `remainingHops`, decremented by one, is greater than zero and the information was received from a
19 Bridge internal to the MST Region (`rcvdInternal TRUE`);

20 and is zero otherwise.

21 The values of Message Age, Max Age, `remainingHops`, and Hello Time used in these calculations are taken
22 from the CIST's `portTimes` parameter (13.27.48) and are not changed by this procedure.

23 13.29.34 `updtRolesTree()`

24 This procedure calculates the following priority vectors (13.9, 13.10 for the CIST, 13.11 for a MSTI) and
25 timer values, for the CIST or a given MSTI:

- 26 a) The *root path priority* vector for each Bridge Port that is not Disabled and has a *port priority* vector
27 (`portPriority` plus `portId`—see 13.27.47 and 13.27.46) that has been recorded from a received message
28 and not aged out (`infoIs == Received`).
29 b) The *root path priority* vector propagated and calculated by ISIS-SPB (if ISIS-SPB is implemented
30 and `ForceProtocolVersion` is 4).
31 c) The Bridge's *root priority vector* (`rootPortId`, `rootPriority`—13.26.9, 13.26.10), chosen as the best of
32 the set of priority vectors comprising the Bridge's own *Bridge priority vector*
33 (`BridgePriority`—13.26.3) plus all calculated root path priority vectors whose:
34 1) DesignatedBridgeID Bridge Address component is not equal to that component of the Bridge's
35 own Bridge priority vector (13.10) and
36 2) Port's `restrictedRole` parameter is `FALSE` and
37 3) Port's `restrictedDomainRole` parameter is `FALSE` or the port is not a Boundary Port of the SPT
38 Region (13.12).

39 NOTE 1—If ISIS-SPB is being used but did not provide the selected *root priority vector* for the CIST, that priority
40 vector will be associated with the Master Port of the SPT Region (a Boundary Port of, and not internal to, the Region)
41 and will be used by ISIS-SPB to propagate the CST component of the priority vector throughout the Region.

- 1 d) The Bridge's *root times*, (*rootTimes*—13.26.11), set equal to:
- 2 1) *BridgeTimes* (13.26.4), if the chosen root priority vector is the Bridge priority vector, or was
3 calculated by ISIS-SPB; otherwise,
- 4 2) *portTimes* (13.27.48) for the port associated with the selected root priority vector, with the
5 Message Age component incremented by 1 second and rounded to the nearest whole second if
6 the information was received from a Bridge external to the MST Region (*rcvdInternal* FALSE),
7 and with *remainingHops* decremented by one if the information was received from a Bridge
8 internal to the MST Region (*rcvdInternal* TRUE).
- 9 e) The *designated priority vector* (*designatedPriority*—13.27.20) for each port that is not internal an
10 SPT Region.
- 11 f) The *designated times* (*designatedTimes*—13.27.21) for each port set equal to the value of *root times*.

12 If the root priority vector for the CIST is recalculated, and has a different Regional Root Identifier than that
13 previously selected, and has or had a nonzero CIST External Root Path Cost, the *syncMaster()* procedure
14 (13.29.26) is invoked.

15 NOTE 2—Changes in Regional Root Identifier will not cause loops if the Regional Root is within an MST Region, as is
16 the case if and only if the MST Region is the Root of the CST. This important optimization allows the MSTIs to be fully
17 independent of each other in the case where they compose the core of a network.

18 The CIST, MSTI, or SPT Port Role for each port is assigned, and its port priority vector and timer
19 information are updated as specified in the remainder of this subclause (13.41.2).

20 If the port is Disabled (*infol* == Disabled), *selectedRole* is set to DisabledPort.

21 Otherwise, if this procedure was invoked for an MSTI or an SPT, for a port that is not Disabled, and that has
22 CIST port priority information that was received from a Bridge external to its Bridge's Region
23 (*infol* == Received and *infolInternal* == FALSE), then:

- 24 g) If the selected CIST Port Role (calculated for the CIST prior to invoking this procedure for an MSTI
25 or SPT) is RootPort, *selectedRole* is set to MasterPort.
- 26 h) If selected CIST Port Role is AlternatePort, *selectedRole* is set to AlternatePort.
- 27 i) Additionally, *updtInfo* is set if the port priority vector differs from the designated priority vector or
28 the port's associated timer parameter differs from the one for the Root Port.

29 Otherwise, for the CIST for a port that is not Disabled and not internal to an SPT Region, or for an MSTI for
30 a port of that is not Disabled and whose CIST port priority information was not received from a Bridge
31 external to the Region (*infol* != Received or *infolInternal* == TRUE), the CIST or MSTI port role is assigned,
32 and the port priority vector and timer information updated as follows:

- 33 j) If the port priority vector information was aged (*infol* = Aged), *updtInfo* is set and *selectedRole* is
34 set to DesignatedPort.
- 35 k) If the port priority vector was derived from another port on the Bridge or from the Bridge itself as
36 the Root Bridge (*infol* = Mine), *selectedRole* is set to DesignatedPort. Additionally, *updtInfo* is set if
37 the port priority vector differs from the designated priority vector or the port's associated timer
38 parameter(s) differ(s) from the Root Port's associated timer parameters.
- 39 l) If the port priority vector was received in a Configuration Message and is not aged
40 (*infol* == Received), and the root priority vector is now derived from it, *selectedRole* is set to
41 RootPort, and *updtInfo* is reset.
- 42 m) If the port priority vector was received in a Configuration Message and is not aged
43 (*infol* == Received), the root priority vector is not now derived from it, the designated priority
44 vector is not better than the port priority vector, and the designated Bridge and designated port
45 components of the port priority vector do not reflect another port on this Bridge, *selectedRole* is set
46 to AlternatePort, and *updtInfo* is reset.

- 1 n) If the port priority vector was received in a Configuration Message and is not aged
2 (infols == Received), the root priority vector is not now derived from it, the designated priority
3 vector is not better than the port priority vector, and the designated Bridge and designated port
4 components of the port priority vector reflect another port on this Bridge, selectedRole is set to
5 BackupPort, and updtInfo is reset.
- 6 o) If the port priority vector was received in a Configuration Message and is not aged
7 (infols == Received), the root priority vector is not now derived from it, the designated priority
8 vector is better than the port priority vector, selectedRole is set to DesignatedPort, and updtInfo is
9 set.

10 Otherwise, for the CIST or an SPT, for a port that is not Disabled and is internal to an SPT Region, ISIS-SPB
11 determines the selectedRole (and other parameters, see 13.36), and uses updtAgreement() (13.29.30).

12 13.29.35 updtRolesDisabledTree()

13 This procedure sets selectedRole to DisabledPort for all ports of the Bridge for a given tree (CIST, or MSTI,
14 or SPT).

15 13.30 The Port Timers state machine

16 The Port Timers state machine shall implement the function specified by the state diagram in Figure 13-15
17 and the attendant definitions in 13.25 through 13.29.

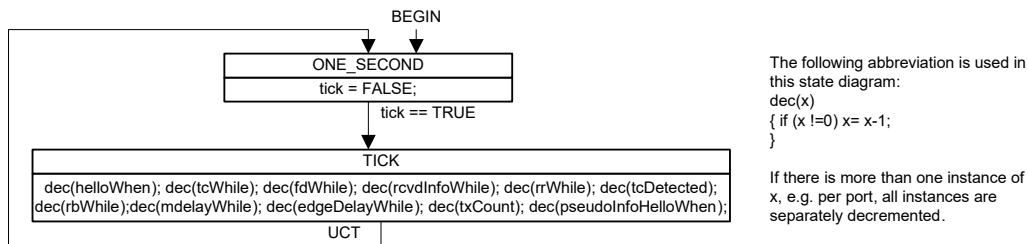


Figure 13-15—Port Timers state machine

18 The state machine uses tick (13.27), a signal set by an implementation-specific system clock function at one
19 second intervals, to decrement the timer variables for the CIST and all MSTIs for the port. The state machine
20 that uses a given timer variable is responsible for setting its initial value.

13.31 Port Receive state machine

The Port Receive state machine shall implement the function specified by the state diagram in Figure 13-16 and the attendant definitions in 13.25 through 13.29.

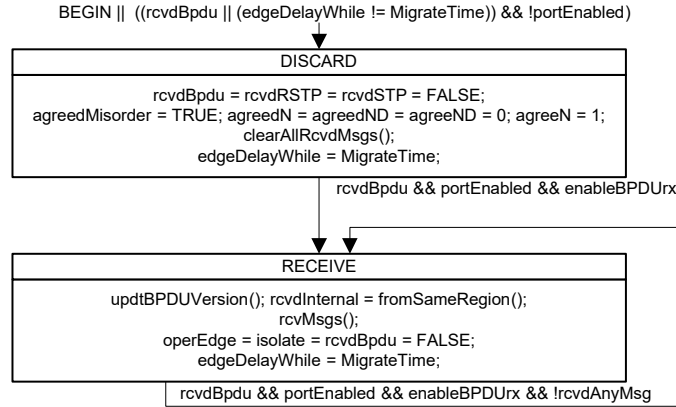


Figure 13-16—Port Receive state machine

This state machine receives and decodes validated BPDUs. The rcvdMsg flag is set for the CIST and for each MSTI supported by the receiving Bridge if the received BPDU is from the same MST or SPT Region. The next BPDU is not processed until all rcvdMsg flags have been cleared by the per-tree state machines.

NOTE—This standard does not specify or constrain the means used by the rcvMsgs() procedure to identify or extract per-tree information from a BPDU for processing by rcvInfo() in the Port Information Machine RECEIVE state.

1 13.32 Port Protocol Migration state machine

2 The Port Protocol Migration state machine shall implement the function specified by the state diagram in
3 Figure 13-17 and the attendant definitions in 13.25 through 13.29.

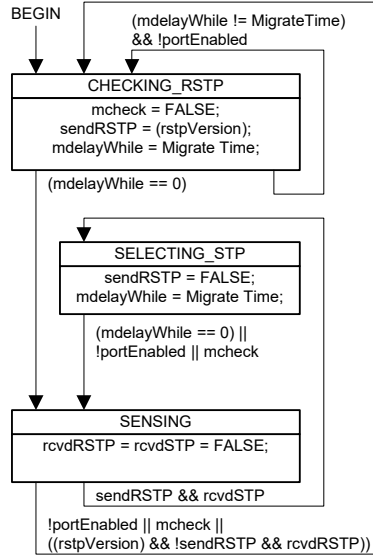


Figure 13-17—Port Protocol Migration state machine

4 13.33 Bridge Detection state machine

5 The Bridge Detection state machine shall implement the function specified by the state diagram in
6 Figure 13-18 and the attendant definitions in 13.25 through 13.29.

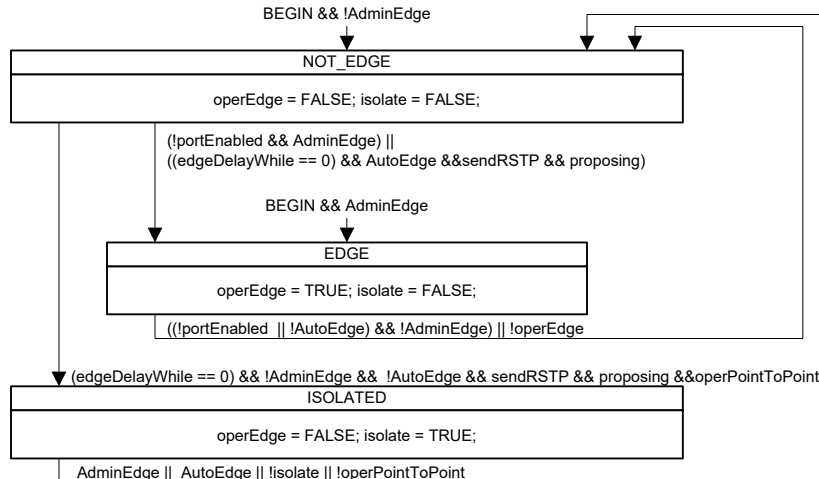


Figure 13-18—Bridge Detection state machine

1 13.34 Port Transmit state machine

2 The Port Transmit state machine shall implement the function specified by the state diagram in Figure 13-19
3 and the attendant definitions in 13.25 through 13.29.

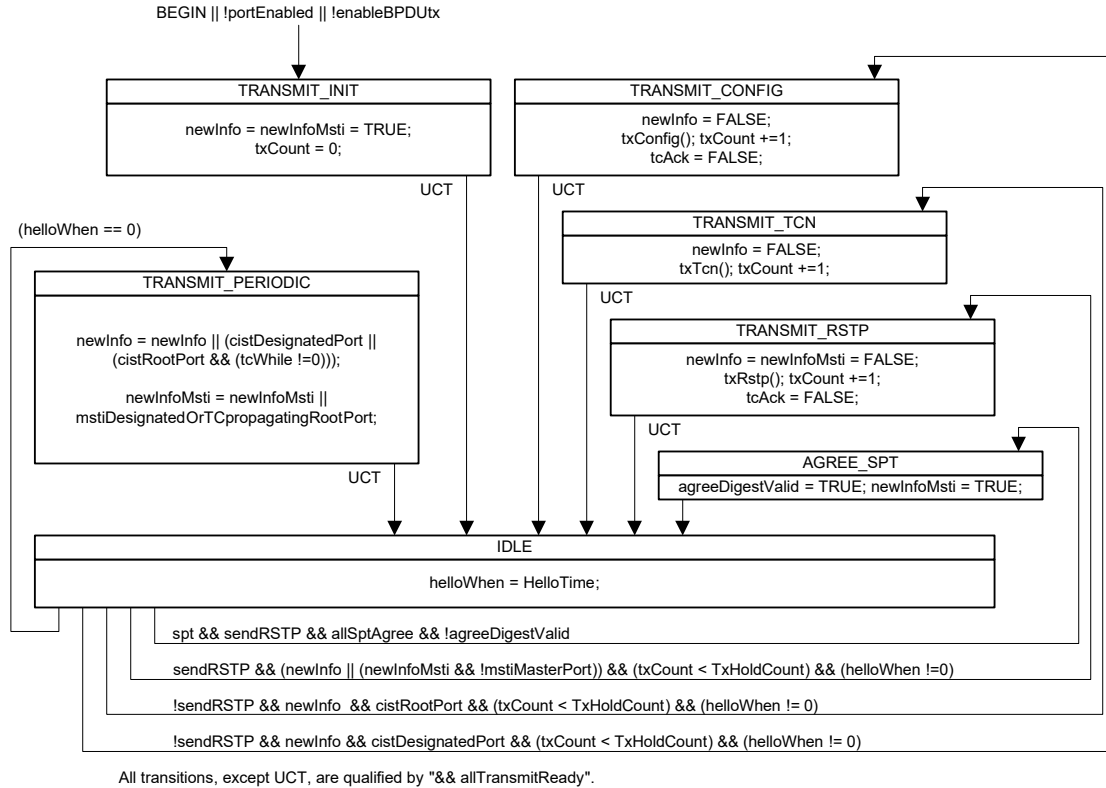


Figure 13-19—Port Transmit state machine

4 This state machine is responsible for transmitting BPDUs. It also determines when the Agreement Digest in
5 SPT BPDUs can be updated, and prompts transmission when that has been done.

6 NOTE 1—Any single received BPDU that changes the CIST Root Identifier, CIST External Root Path Cost, or CIST
7 Regional Root associated with MSTIs should be processed entirely, or not at all, before encoding BPDUs for
8 transmission. This recommendation minimizes the number of BPDUs to be transmitted following receipt of a BPDU with
9 new information. It is not required for correctness and has not therefore been incorporated into the state machines.

10 NOTE 2—If a CIST state machine sets **newInfo**, this machine will ensure that a BPDU is transmitted conveying the new
11 CIST information. If MST BPDUs can be transmitted through the port, this BPDU will also convey new MSTI
12 information for all MSTIs. If a MSTI state machine sets **newInfoMsti**, and MST BPDUs can be transmitted through the
13 port, this machine will ensure that a BPDU is transmitted conveying information for the CIST and all MSTIs. Separate
14 **newInfo** and **newInfoMsti** variables are provided to avoid requiring useless transmission of a BPDU through a port that
15 can only transmit STP BPDUs (as required by the Force Protocol Version parameter or Port Protocol Migration
16 machine) following a change in MSTI information without any change to the CIST.

13.35 Port Information state machine

The Port Information state machine for each tree shall implement the function specified by the state diagram in Figure 13-20 and the attendant definitions in 13.25 through 13.29.

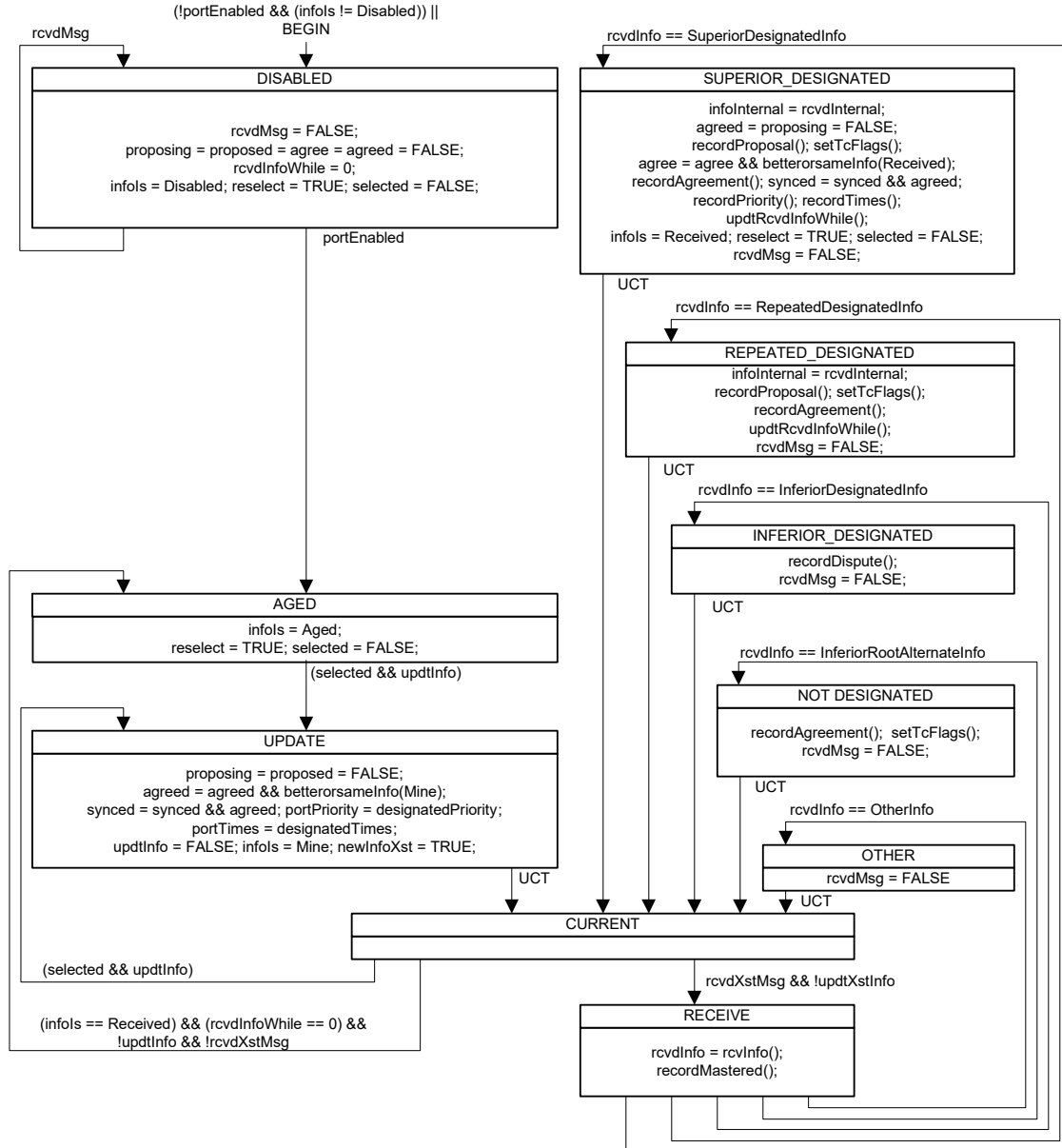


Figure 13-20—Port Information state machine

This state machine is responsible for recording the spanning tree information currently in use by the CIST or a given MSTI for a given port, ageing that information out if it was derived from an incoming BPDU, and recording the origin of the information in the infols variable. The selected variable is cleared and reselect set to signal to the Port Role Selection machine that port roles need to be recomputed. The infols and portPriority variables from all ports are used in that computation and, together with portTimes, determine new values of designatedPriority and designatedTimes. The selected variable is set by the Port Role Selection machine once the computation is complete.

13.36 Port Role Selection state machine

The Port Role Selection state machine shall implement the function specified by the state diagram in Figure 13-21 and the attendant definitions in 13.25 through 13.29.

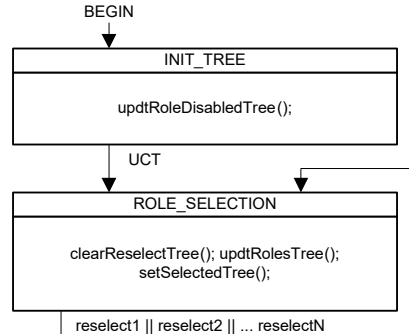


Figure 13-21—Port Role Selection state machine

When ISIS-SPB finishes a link state calculation, it clears the selected variable for each port for each SPT before updating agreementDigest (13.26.1) for the Bridge component, the root priority vector (rootPortId, rootPriority—13.26.9, 13.26.10) for each SPT and the selectedRole (13.12, 13.27.68), designated priority vector (designatedPriority—13.27.20), and neighbourPriority for each port for each SPT. ISIS-SPB then sets reselect for each SPT.

13.37 Port Role Transitions state machine

The Port Role Transitions state machine shall implement the function specified by the state diagram in the following figures:

- Part 1: Figure 13-22 for both the initialization of this state machine and the states associated with the DisabledPort role
- Part 2: Figure 13-23 for the states associated with the MasterPort role
- Part 3: Figure 13-24 for the states associated with the RootPort role
- Part 4: Figure 13-25 for the states associated with the DesignatedPort role
- Part 5: Figure 13-26 for the states associated with the AlternatePort and BackupPort roles

and the attendant definitions in 13.25 through 13.29.

1 As Figure 13-22, Figure 13-23, Figure 13-24, Figure 13-25, and Figure 13-26 are component parts of the
2 same state machine, the global transitions associated with these diagrams are possible exit transitions from
3 the states shown in any of the diagrams.

4 Figure 13-22 and Figure 13-26 show the Port Roles for ports that do not form part of the active topology of
5 the given tree.

6 Figure 13-23, Figure 13-24, and Figure 13-25 show the Port Roles that form part of the active topology.

7

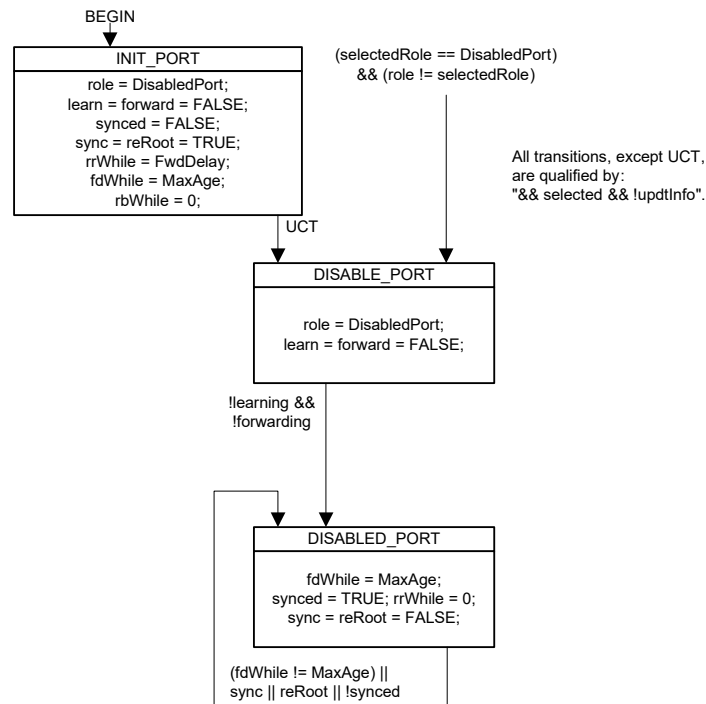


Figure 13-22—Disabled Port role transitions

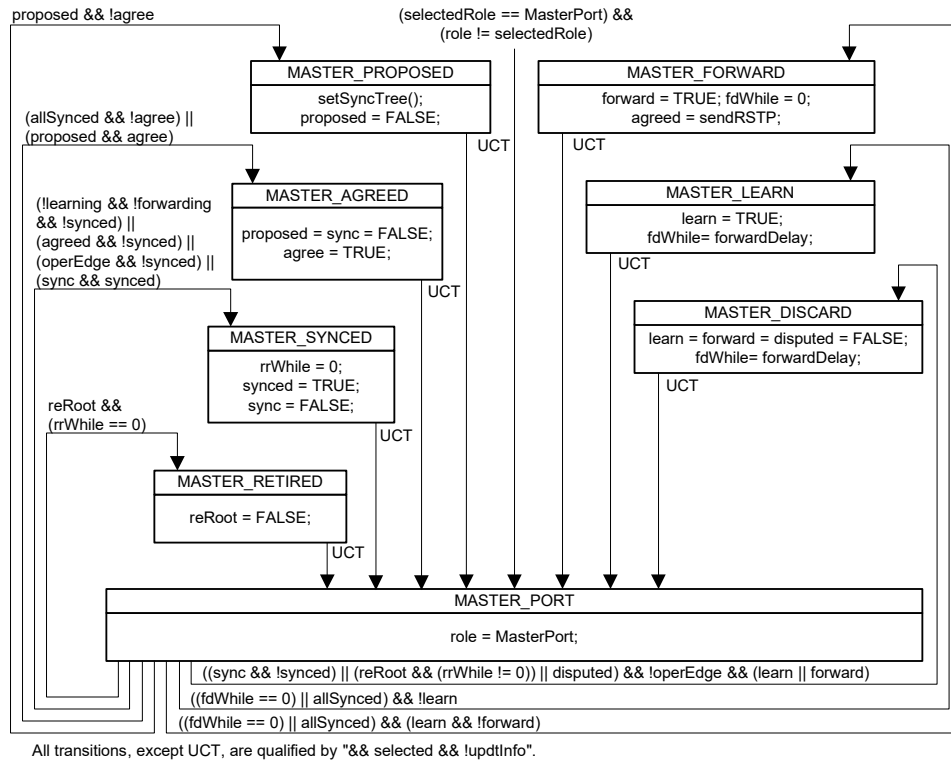


Figure 13-23—Port Role Transitions state machine—MasterPort

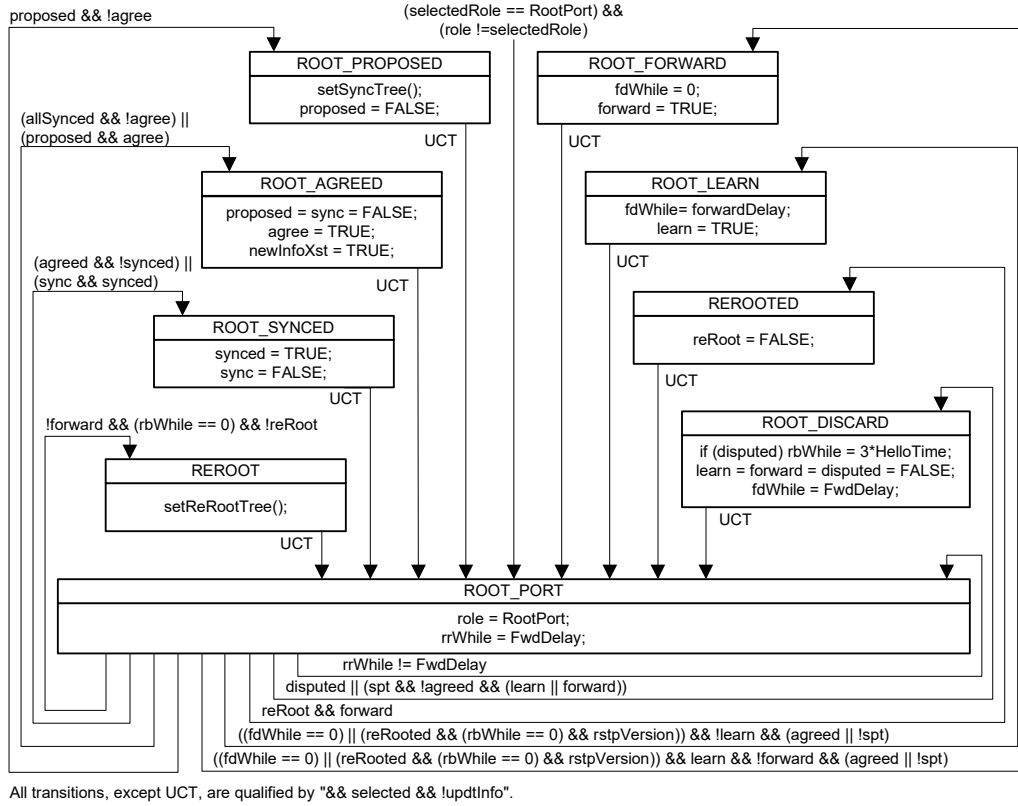
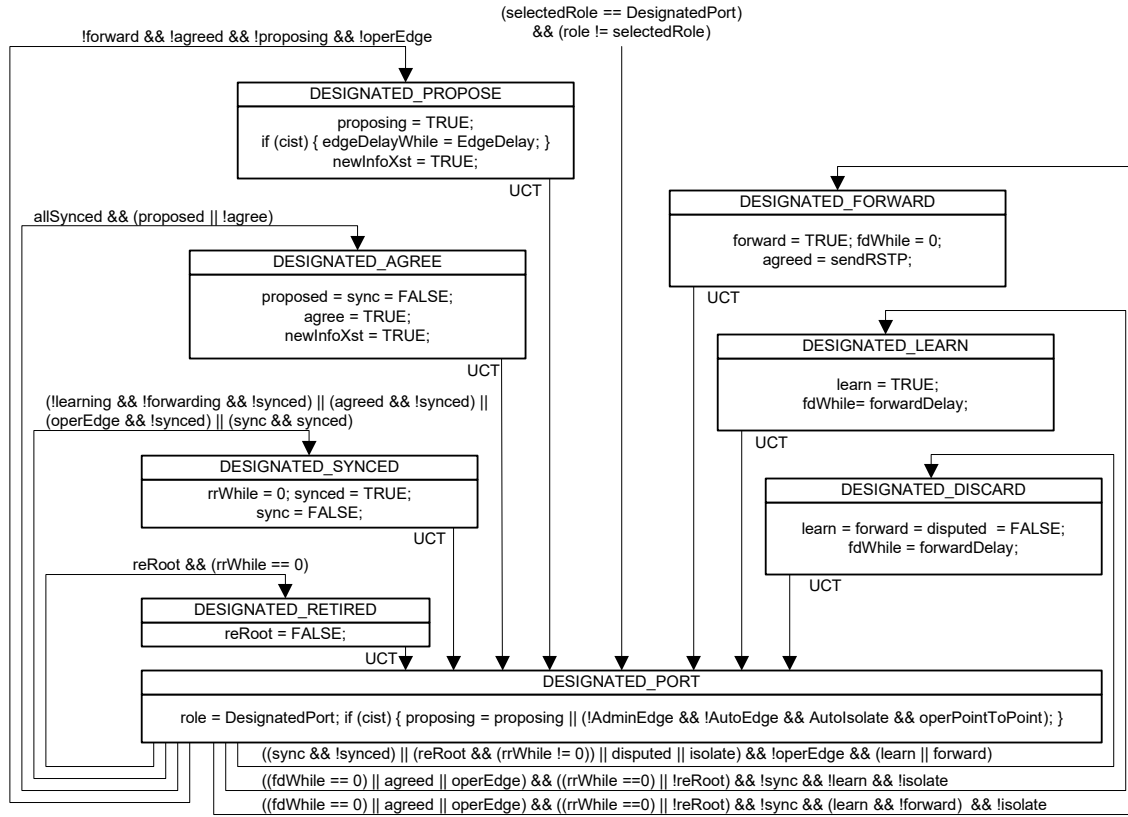


Figure 13-24—Port Role Transitions state machine—RootPort



All transitions, except UCT, are qualified by "&& selected && !updtInfo".

Figure 13-25—Port Role Transitions state machine—DesignatedPort

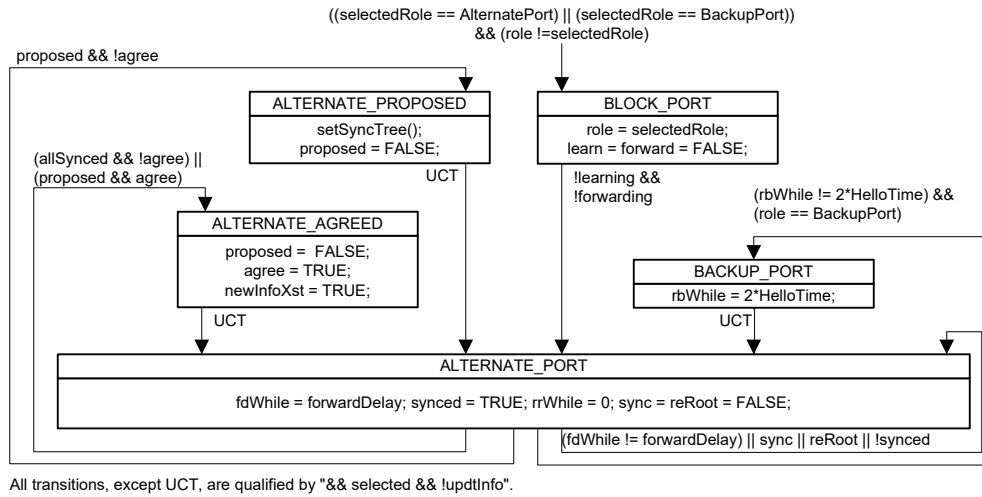


Figure 13-26—Port Role Transitions state machine—AlternatePort and BackupPort

1 13.38 Port State Transition state machine

- 2 The Port State Transition state machine shall implement the function specified by the state diagram in
3 Figure 13-27 and the attendant definitions in 13.25 through 13.29.

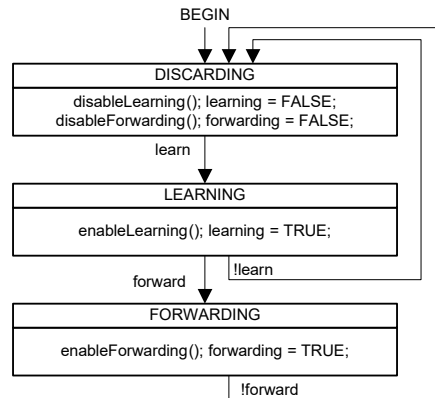


Figure 13-27—Port State Transition state machine

- 4 NOTE—A small system-dependent delay may occur on each of the transitions shown in the referenced state machine.

5 This state machine operates independently of the type of tree (CIST, MSTI, or SPT) and whether backbone
6 bridging is being supported. However, the way in which the Bridge supports the learn and forward variables
7 and the disableForwarding(), disableLearning(), enableForwarding(), and enableLearning() procedures are
8 supported does vary (see 8.4, 8.6, 8.6.1). The forwarding and learning variables provide
9 implementation-independent reporting of the current state.

1 **13.38.1 Port State transitions for the CIST and MSTIs**

2 The CIST and each MSTI are always supported by an explicit Port State, enforced by Bridge's
3 implementation of the Forwarding Process, and the procedures prompt that implementation to take the
4 necessary action to forward and/or learn from received frames (as requested).

5 **13.38.2 Port State transitions for SPTs**

6 The `enableLearning()` and `disableLearning()` procedures return without taking an action: for SPBV mode
7 learning does not occur until forwarding is enabled (8.6.1); for SPBM mode the Learning Process (8.7)
8 neither creates or deletes Dynamic Filtering Entries.

9 When a VLAN is supported by an SPT Bridge using SPBV mode, a Dynamic VLAN Registration Entry is
10 created for each SPVID and Enable Ingress Filtering (8.6.2) enabled on each port. Connectivity is provided,
11 and loops prevented, by adding and removing ports from the Dynamic VLAN Registration Entry (27.13).

12 The `enableForwarding()` procedure adds the port to the Dynamic VLAN Registration Entry for each SPVID
13 that is associated with the SPT (provided the VLAN topology extends through the port, see 27.13). Similarly
14 `disableForwarding()` removes the port from the registration entry.

15 When frames with a given VID are supported by SPBM mode, MAC address-based ingress filtering is used
16 to discard a received frame if there is no corresponding Dynamic Filtering Entry (8.8, 8.6.1, 27.14) for that
17 VID and the frame's source MAC address with the receiving port in the Port Map.

18 ISIS-SPB identifies the port that provides the shortest path to a given Bridge: this is the Root Port for the
19 SPT rooted at that Bridge. The individual MAC addresses (there can be more than one) known to ISIS-SPB
20 as identifying sources for which that given Bridge first transmits frames within the SPT Region (entering the
21 region from an attached Bridge or station, or from a protocol entity within the Bridge) or as identifying
22 destinations for which that Bridge is the last recipient within the region (delivering frames to an attached
23 Bridge or station, or to a protocol entity within the Bridge) are associated with that Root Port. If loop
24 mitigation (6.5.4.2) is used for unicast frames, a Dynamic Filtering Entry is created for those addresses and
25 VID, permitting forwarding through that port and no others, without reference to the state machines
26 specified in this clause (Clause 13), and the setting or clearing of the forward variable for any port. The
27 `enableForwarding()` procedure creates or modifies these Dynamic Filtering Entries, so that forwarding to
28 and from a given port is permitted, if and only if loop prevention (6.5.4.1) is used for both unicast and
29 multicast frames and the port's role is `RootPort`, and has no effect otherwise. Similarly if loop prevention is
30 being used, `disableForwarding()` modifies or removes any existing Dynamic Filtering Entry so that
31 forwarding to or from the port is not permitted.

32 NOTE—The Port Role Transition machine allows an SPT to transition a Root Port to Discarding in support of
33 `enableForwarding()` and `disableForwarding()` as specified in this subclause (13.38.2). The need to specify that
34 transition could have been avoided by specifying that the Dynamic Filtering Entry would only permit forwarding
35 through a given port if `forward` was `TRUE` for the port for every other SPT for which it was a Designated Port. While
36 equivalent, such a specification approach would have been obscure and many implementors would have failed to spot
37 the way to avoid checking variables for all SPTs when processing a change for a single tree.

38 An SPT Bridge using SPBM mode uses source-specific multicast addresses, so that the destination address
39 alone can be used to identify the SPT (Clause 27) on egress, and a Static Filtering Entry (8.8.1) is created for
40 that multicast address and the B-VIDs supported by the SPT. If `enableForwarding()` is invoked for a
41 Designated Port, the Static Filtering Entry's control element for that port is updated to specify that frames
42 should be Forwarded through the port. The control element is modified to specify Filtered if
43 `enableForwarding()` is invoked and the port's role is not Designated Port, or if `disabledForwarding()` is used.

13.39 Topology Change state machine

2 The Topology Change state machine for each tree shall implement the function specified by the state
3 diagram in Figure 13-28 and the attendant definitions in 13.25 through 13.29.

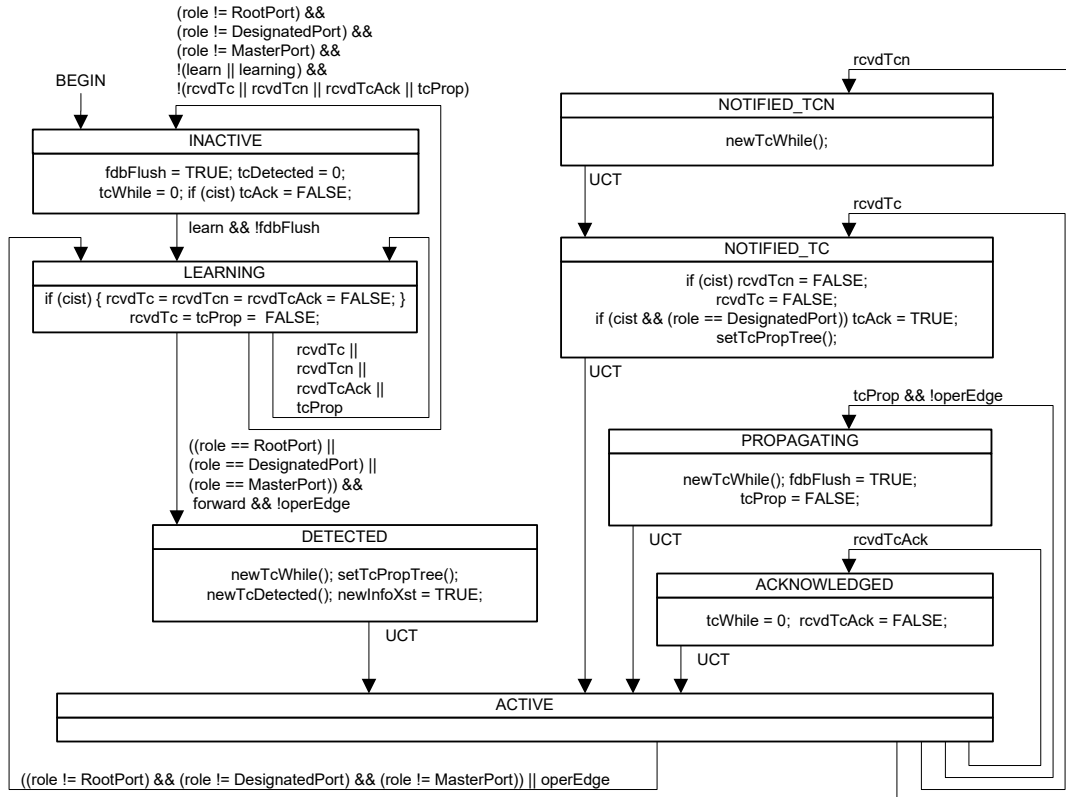


Figure 13-28—Topology Change state machine

⁴ NOTE—MRP (Clause 10) uses the `tcDetected` variable maintained by this state machine.

1 13.40 Layer 2 Gateway Port Receive state machine

2 If implemented, the Layer 2 Gateway Port state machine for each port shall implement the function specified
3 by the state diagram in Figure 13-29 and the attendant definitions in 13.25 through 13.29.

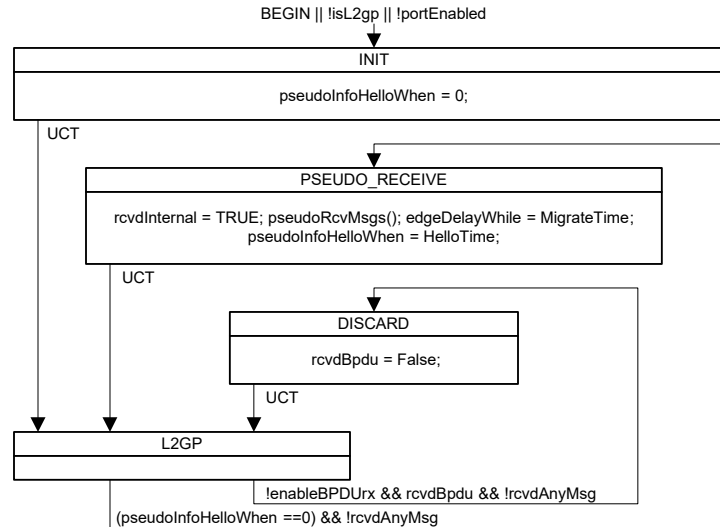


Figure 13-29—L2 Gateway Port Receive state machine

4 NOTE—The functionality provided by this state machine is discussed in 13.20, 25.9.2, 26.4.3, and 13.29.11.

5 13.41 CEP spanning tree operation

6 This clause specifies the operation of the Spanning Tree Protocol Entity within a C-VLAN component that
7 supports a CEP (Figure 15-4) of a Provider Edge Bridge. The CEP and each PEP are treated as separate
8 Bridge Ports by the spanning tree protocol.

9 If the C-VLAN component connects to the S-VLAN component with a single PEP, and the associated
10 service instance supports no more than two customer interfaces, then all frames (including ST BPDUs)
11 addressed to the Bridge Group Address may be relayed between the two ports of the C-VLAN component
12 without modification. Otherwise, the Spanning Tree Protocol Entity shall execute RSTP, as modified by the
13 provisions of this subclause (13.41).

14 The RSTP enhancements specified do not reduce PBN connectivity between CEPs to a single spanning tree
15 of service instances but ensure that connectivity for frames assigned to any given C-VLAN is loop-free. In
16 this respect, the C-VLAN component's spanning tree protocol operation is equivalent to, but simpler to
17 manage than, the operation of MSTP.

18 13.41.1 PEP operPointToPointMAC and operEdge

19 The value of the adminPointToPointMAC parameter for a PEP is always Auto, and no management control
20 over its setting is provided. The value of the operPointToPointMAC parameter, used by the RSTP state
21 machines, shall be TRUE if the service instance corresponding to the PEP connects at most two customer
22 interfaces, and FALSE otherwise.

1 The value of the AdminEdge, AutoEdge, and operEdge parameters for a PEP are always FALSE, TRUE, and
2 FALSE, respectively. No management control over their setting is provided.

3 **13.41.2 updtRolesTree()**

4 The spanning tree priority vectors timer values are calculated, and the port role for each port, its port priority
5 vector, and timer information updated as specified in 13.29.34, with one exception. If selectedRole was to be
6 set to AlternatePort, the port is a PEP, and the root priority vector was derived from another PEP, then the
7 selectedRole shall be set to RootPort.

8 NOTE—The effect of this enhancement is to allow the C-VLAN component to have multiple Root Ports (just as if
9 separate per S-VLAN trees were being provided), if they are all PEPs. As the C-VLAN component assigns each frame to
10 a single C-VLAN and maps any given C-VLAN to and from at most one PEP, no loop is created.

11 **13.41.3 setReRootTree(), setSyncTree(), setTcPropTree()**

12 The setReRootTree() and setSyncTree() procedures specified in 13.29.21 and 13.29.23 set the reRoot and
13 sync variables for all ports of the Bridge, and the setTcPropTree() as specified in 13.29.25 sets the tcProp
14 variable for all ports other than the port that invoked the procedure. If the port invoking the procedure is a
15 CEP, then this behavior is unchanged; if it is a PEP, then the behavior of each procedure shall be as follows:

- 16 — The setReRootTree() procedure sets reRoot for the port invoking the procedure and for the CEP.
- 17 — The setSyncTree() procedure sets sync for the port invoking the procedure and for the CEP.
- 18 — The setTcPropTree() procedure sets tcProp for the CEP.

19 **13.41.4 allSynced, reRooted**

20 RSTP specifies a single value of the allSynced and reRooted state machine conditions for all Bridge Ports.
21 This specification requires an independent value of each of these conditions for each port of the C-VLAN
22 component. If that port is the CEP, then allSynced shall be TRUE if and only if synced is TRUE for all PEPs,
23 and reRooted shall be TRUE if and only if rrWhile is zero for all PEPs. If the port for which the condition is
24 being evaluated is a PEP, then allSynced shall take the value of synced for the CEP, and reRooted shall be
25 TRUE if and only if rrWhile is zero for the CEP.

26 **13.41.5 Configuration parameters**

27 All configuration parameters for RSTP should be set to their recommended defaults, with the exception of
28 the following, which are chosen to minimize the chance of interfering with the customer's configuration
29 (e.g., by the C-VLAN component becoming the root of the customer spanning tree), as follows:

- 30 a) The Bridge Priority (13.18, Table 13-3, 13.26.2) should be set to 61 440. This sets the priority part
31 of the Bridge Identifier (the most significant 4 bits) to hex F.
- 32 b) The following 12 bits (the Bridge Identifier system ID extension) should be set to hex FFF.
- 33 c) The Port Priority (13.18, Table 13-3, 13.27.47) should be set to 32. This sets the priority part of the
34 Port Identifier (the most significant 4 bits) to hex 2, a higher priority than the default (128, or hex 8).
- 35 d) The Port Path Cost values for PEPs should be set to 128.

36 All BPDUs generated by the Spanning Tree Protocol Entity within a C-VLAN component use the MAC
37 address of the CEP as a source address and as the Bridge address portion of the Bridge Identifier. For each
38 internal PEP, the protocol uses the S-VID associated with the corresponding internal CNP on the S-VLAN
39 component as a port number. For the CEP, the value 0xFFFF is used as the port number.

1 **13.42 Virtual Instance Port (VIP) spanning tree operation**

2 This subclause specifies the operation of the Spanning Tree Protocol Entity within an I-component in a
3 BEB. The CNPs and VIPs are treated as separate Bridge Ports by the spanning tree protocol.

4 If the I-component has a single CNP and a single VIP supported by a point-to-point backbone service
5 instance, then all frames (including ST BPDUs) addressed to the Provider Bridge Group address may be
6 relayed between the two ports of the I-component without modification. Otherwise, the Spanning Tree
7 Protocol Entity shall execute RSTP, as modified by the provisions of this subclause.

8 The RSTP enhancements specified ensure that connectivity for frames assigned to any given S-VLAN is
9 loop-free.

10 The parameters and functions of the RSTP used on the VIPs get the same values and functionality as defined
11 for PEPs of a C-VLAN component as defined in 13.41. The Bridge Identifier Priority and system ID
12 extension get the values specified in 13.41.5. These changes in RSTP ensure that no VIP is blocked due to
13 the operation of RSTP and the I-component will never be elected as root.

14 NOTE—The effect of not blocking any VIP in the I-component (never set the port role alternate to a VIP) will not cause
15 a loop since the I-component maps any given S-VID to at most one VIP.

16

14. Encoding of Bridge Protocol Data Units (BPDUs)

This clause specifies formats, and the encoding, decoding, and validation of BPDUs exchanged by protocol entities operating RSTP, MSTP, and other protocols enhancing or designed to interoperate with STP, RSTP, or MSTP (Clause 13).

Interoperability with STP, as specified in IEEE Std 802.1D, 1998 Edition [B13], and prior revisions, is provided as specified by the state machines and procedures in Clause 13 of this standard. Parameter type encodings for STP BPDUs are a subset of those used for RSTP, MSTP, and SPT BPDUs, and the fields encoded in STP Configuration BPDUs are a subset of those used in RST BPDUs.

The format of MST BPDUs is compatible with that specified for RST BPDUs, with the addition of fields to convey information for the IST and each MSTI, and is shown in Figure 14-1.

The format of SPT BPDUs comprises the fields specified for MST BPDUs, with the addition of fields that communicate the Agreement Number, Discarded Agreement Number, and Agreement Digest (13.17).

The Protocol Version Identifier encoded in all BPDUs serves to distinguish RST BPDUs, MST BPDUs, and SPT BPDUs. A BPDU Type is also encoded in all BPDUs, and distinguishes the Configuration and TCN BPDUs used by STP and by the spanning tree protocols specified in this standard when interoperating with STP implementations.

Figure 14-1 shows the overall format of RST, MST, SPT, and STP Configuration BPDUs. Figure 14-2 shows the format of STP TCN BPDUs.

14.1 BPDU Structure

14.1.1 Transmission and representation of octets

All BPDUs shall contain an integral number of octets. The octets in a BPDU are numbered starting from 1 and increasing in the order they are put into a Data Link Service Data Unit (DLSDU). When bit positions in an octet or a sequence of octets encode a number, the number is encoded as an unsigned binary numeral with bit positions in lower octet numbers having more significance. Within an octet the bits are numbered from 8 to 1, where 1 is the low-order bit. Where sequences of bits are represented, high order bits are shown to the left of lower order bits in the same octet, and bits in lower octet numbers are shown to the left of bits in higher octet numbers.

14.1.2 Common BPDU fields

A Protocol Identifier is encoded in the initial octets of all BPDUs. The single Protocol Identifier value of 0000 0000 0000 0000 identifies the spanning tree family of protocols (STP, RSTP, and MSTP) and BPDUs whose formats are compatible with:

- a) STP BPDUs, as specified in this clause and in IEEE Std 802.1D, 1998 Edition [B13]
- b) RST BPDUs, as specified in this clause and Clause 9 of IEEE Std 802.1D-2004 [B14]
- c) The BPDUs specified in IEEE Std 802.1G, now withdrawn
- d) MST BPDUs, as specified in this clause
- e) SPT BPDUs, as specified in this clause

NOTE—ISIS-SPB (Clause 27, Clause 28) sends and receives its own PDUs to support the distributed computation of symmetric SPTs.

A Protocol Version Identifier and BPDU Type are also encoded in all BPDUs.

	Octet
Protocol Identifier	1–2
Protocol Version Identifier	3
BPDU Type	4
CIST Flags	5
CIST Root Identifier	6–13
CIST External Path Cost	14–17
CIST Regional Root Identifier	18–25
CIST Port Identifier	26–27
Message Age	28–29
Max Age	30–31
Hello Time	32–33
Forward Delay	34–35
Version 1 Length = 0 (RST, MST, SPT BPDUs only)	36
Version 3 Length (MST, SPT BPDUs only)	37–38
MST Configuration Identifier	39–89
CIST Internal Root Path Cost	90–93
CIST Bridge Identifier	94–101
CIST Remaining Hops	102
MSTI Configuration Messages (may be absent)	103–39 + <i>Version 3 Length</i>
Version 4 Length (SPT BPDUs only)	(40 + <i>Version 3 Length</i>) – (41 + <i>Version 3 Length</i>)
Auxiliary MCID (SPT BPDUs only)	(42 + <i>Version 3 Length</i>) – (92 + <i>Version 3 Length</i>)
SPT Agreement Number, Discarded Agreement Number, Agreement Digest (SPT BPDUs only)	(93 + <i>Version 3 Length</i>) – (41 + <i>Version 3 Length</i> + <i>Version 4 Length</i>)

NOTE—BPDUs are encoded in LLC Type 1 frames following the DSAP, LSAP, and UI fields, so if that frame is received from directly an IEEE 802.3 MAC with the MAC addresses aligned on an even octet boundaries then the BPDU octet pairs 6 and 7, 14 and 15, 18 and 19, etc. will also be aligned on those boundaries.

Figure 14-1—RST, MST, SPT, and STP Configuration BPDU format

	Octet
Protocol Identifier	1–2
Protocol Version Identifier	3
BPDU Type	4

Figure 14-2—STP TCN BPDU format

1 14.2 Encoding of parameter types

2 14.2.1 Encoding of Protocol Identifiers

3 A Protocol Identifier shall be encoded in two octets.

4 14.2.2 Encoding of Protocol Version Identifiers

5 A Protocol Version Identifier shall be encoded in one octet. If two Protocol Version Identifiers are
6 interpreted as unsigned binary numbers, the greater identifies the more recently defined Protocol Version.

7 14.2.3 Encoding of BPDUs types

8 The type of the BPDUs shall be encoded as a single octet. The bit pattern contained in the octet merely serves
9 to distinguish the type; no ordering relationship between BPDUs of different types is implied.

10 14.2.4 Encoding of flags

11 A flag shall be encoded as a bit in a single octet. A flag is set if the bit takes the value 1. A number of flags
12 may be encoded in a single octet. Bits in the octet that do not correspond to flags defined for the BPDUs
13 type are reset, i.e., shall take the value 0. No additional flags will be defined for a BPDUs of given protocol
14 version and type.

15 14.2.5 Encoding of Bridge Identifiers

16 A Bridge Identifier is a 64-bit unsigned integer. If two Bridge Identifiers are numerically compared, the
17 lesser number denotes the Bridge of the better priority.

18 NOTE 1—Use of the terms “higher” and “lower” to describe both the relative numerical values and the relative priority
19 of Spanning Tree information can cause confusion, as lesser numbers convey better priorities. In this clause and in
20 Clause 13, relative numeric values are described as “least,” “lesser,” “equal,” and “greater,” and their comparisons as
21 “less than,” “equal to,” or “greater than,” while relative spanning tree priorities are described as “best,” “better,” “the
22 same,” “different,” and “worse” and their comparisons as “better than,” “the same as,” “different from,” and “worse
23 than.” The terms “superior” and “inferior” describe comparisons not simply based on strict ordered comparison of
24 priority components.

25 The most significant 4 bits of a Bridge Identifier comprise a settable priority component (13.20, 13.26.2),
26 and can be modified for the CIST or for an MSTI independently of those for other MSTIs (or for the CIST),
27 allowing the Bridge Identifier priority to be used to provide full and independent control over the
28 configuration of each of those trees. Each SPT uses the same Bridge Identifier as the CIST, so SPT Bridge
29 Identifiers are not separately encoded.

30 NOTE 2—To maintain management compatibility with implementations of revisions of this standard prior to the
31 IEEE Std 802.1s-2002 amendment, and of IEEE Std 802.1D, 1998 Edition [B13] and prior revisions, the priority
32 component is considered to be a 16-bit value for management purposes, but the values that it can be set to are restricted
33 to only those values where the least significant 12 bits are zero (i.e., only the most significant 4 bits are settable).

34 The next most significant 12 bits of a Bridge Identifier (when encoded, the least significant 4 bits of the most
35 significant octet plus the second most significant octet) comprise a locally assigned system ID extension,
36 that provides a distinct Bridge Identifier for each MSTI. The CIST is identified by the system ID extension
37 of zero. Each MSTI has a system ID extension value equal to its MSTID, a convention used to convey the
38 MSTID for each MSTI parameter set in an MST BPDUs.

39 The least significant 48 bits of the Bridge Identifier ensure its uniqueness and are derived from the globally
40 unique Bridge Address (8.13.8) and are encoded in the third through eighth most significant octets of the
41 encoded identifier according to the following procedure.

1 The third most significant octet is derived from the initial octet of the MAC address; the LSB of the octet
2 (Bit 1) is assigned the value of the first bit of the Bridge Address, the next most significant bit is assigned the
3 value of the second bit of the Bridge Address, and so on. The fourth through eighth octets are similarly
4 assigned the values of the second to the sixth octets of the Bridge Address.

5 **14.2.6 Encoding of External Root Path Cost and Internal Root Path Cost**

6 The External Root Path Cost shall be encoded in four octets as a number of arbitrary cost units. The Internal
7 Root Path Cost is similarly encoded in four octets. Subclause 13.18 recommends Port Path Cost values so
8 that a common interpretation can be placed on path cost values, and reasonable spanning tree configurations
9 obtained without explicit management.

10 NOTE—IEEE Std 802.1D-2004 [B14] refers to the External Root Path Cost as Root Path Cost and does not use Internal
11 Root Path Cost.

12 **14.2.7 Encoding of Port Identifiers**

13 A Port Identifier is a 16-bit unsigned integer. If two Port Identifiers are numerically compared, the lesser
14 number denotes the port of better priority. The most significant 4 bits of a Port Identifier is a settable priority
15 component that permits the relative priority of ports on the same Bridge to be managed (13.27.46), and can
16 be modified independently for the CIST and each MSTI. The less significant 12 bits is the Port Number
17 expressed as an unsigned binary number, and is same for the CIST and all MSTIs. The value 0 is not used as
18 a Port Number.

19 The Port Identifier for the CIST is encoded in two octets, comprising both the CIST's priority component
20 and the Port Number used for all spanning trees. The 4-bit priority component for each MSTI is encoded as
21 a binary number, in the MSTI Configuration Message for that MSTI. Each SPT uses the same Port Identifier
22 as the CIST, so SPT Port Identifiers are not separately encoded.

23 NOTE—IEEE Std 802.1D, 1998 Edition [B13], and prior revisions specified a priority component of 8 bits and a Port
24 Number of 8 bits. To maintain management compatibility with prior implementations the priority component is still
25 considered to be an 8-bit value, but its values are restricted to those where the least significant 4 bits are zero (and hence
26 ignored in encoding).

27 **14.2.8 Encoding of Timer Values**

28 Timer Values shall be encoded in two octets, taken to represent an unsigned binary number multiplied by a
29 unit of time of 1/256 of a second. This permits times in the range 0 to, but not including, 256 s to be
30 represented.

31 **14.2.9 Encoding of Port Role values**

32 Port Role values shall be encoded in two consecutive flag bits, taken to represent an unsigned integer, as
33 follows:

- 34 a) A value of 0 indicates Master Port
- 35 b) A value of 1 indicates Alternate or Backup
- 36 c) A value of 2 indicates Root
- 37 d) A value of 3 indicates Designated

38 NOTE—IEEE Std 802.1D-2004 [B14] identified the Port Role value of 0 as Unknown, as it not used as a CIST Port
39 Role in transmitted BPDUs. A received BPDUs with a CIST Port Role value of 0 is identified as a Configuration BPDUs.

40 **14.2.10 Encoding of Length Values**

41 Version 1 Length Values are encoded in one octet, taken to represent an unsigned binary number. No further
42 length values are encoded for Version 2. Length Values for Version 3 and 4 are encoded in two octets.

43 **14.2.11 Encoding of Hop Counts**

44 The number of remaining Hops parameter shall be encoded in a single octet.

14.3 Transmission of BPDUs

A Bridge Protocol Entity shall encode 0000 0000 0000 0000 in Octets 1 and 2 (conveying the Protocol Identifier), the remaining fields shall be encoded to convey an STP Configuration BPDU, an STP TCN BPDU, an RST BPDU, or an MST BDU as required by the Force Protocol Version parameter, the Port Protocol Migration state machine, and other protocol parameters, all as specified in Clause 13.

- a) If transmission of an STP Configuration BPDU is required, the Protocol Version Identifier shall be 0, and the BPDU Type shall be 0000 0000.
- b) If transmission of an STP TCN BPDU is required, the Protocol Version Identifier shall be 0, and the BPDU Type shall be 1000 0000.
- c) If transmission of an RST BPDU is required, the Protocol Version Identifier shall be 2, and the BPDU Type shall be 0000 0010.
- d) If transmission of an MST BPDU is required, the Protocol Version Identifier shall be 3, and the BPDU Type shall be 0000 0010.
- e) If transmission of an SPT BPDU is required, the Protocol Version Identifier shall be 4, and the BPDU Type shall be 0000 0010.

The remaining parameters for STP Configuration, RST, MST, and SPT BPDUs shall be encoded as specified below (14.4).

14.4 Encoding and decoding of STP Configuration, RST, MST, and SPT BPDUs

STP Configuration, RST, MST, and SPT BPDU protocol parameters are encoded for transmission, and decoded, checked, or ignored on receipt as follows:

- a) Bit 1 of Octet 5 conveys the CIST Topology Change flag.
- b) Bit 2 of Octet 5 conveys the CIST Proposal flag in RST, MST, and SPT BPDUs. It is unused in STP Configuration BPDUs, and shall be transmitted as 0 and ignored on receipt.
- c) Bits 3 and 4 of Octet 5 conveys the CIST Port Role in RST, MST, and SPT BPDUs. It is unused in STP Configuration BPDUs, and shall be transmitted as 0 and ignored on receipt.
- d) Bit 5 of Octet 5 conveys the CIST Learning flag in RST, MST, and SPT BPDUs. It is unused in STP Configuration BPDUs, and shall be transmitted as 0 and ignored on receipt.
- e) Bit 6 of Octet 5 conveys the CIST Forwarding flag in RST, MST, and SPT BPDUs. It is unused in STP Configuration BPDUs, and shall be transmitted as 0 and ignored on receipt.
- f) Bit 7 of Octet 5 conveys the CIST Agreement flag in RST, MST, and SPT BPDUs. It is unused in STP Configuration BPDUs, and shall be transmitted as 0 and ignored on receipt.
- g) Bit 8 of Octet 5 conveys the Topology Change Acknowledge Flag in STP Configuration BPDUs. It is unused in RST, MST, and SPT BPDUs, and shall be transmitted as 0 and ignored on receipt.
- h) Octets 6 through 13 convey the CIST Root Identifier.
NOTE 1—The 12-bit system ID extension component of the CIST Root Identifier can be received and subsequently transmitted as an arbitrary value, even in MST BPDUs, since the CIST Root may be an STP Bridge.
- i) Octets 14 through 17 convey the CIST External Root Path Cost.
- j) Octets 18 through 25 shall take the value of the CIST Regional Root Identifier when transmitted in RST and MST BPDUs, and the value of the CIST Bridge Identifier of the transmitting Bridge when transmitted in STP Configuration BPDUs. On receipt of an STP Configuration or RST BPDU both the CIST Regional Root Identifier and the CIST Designated Bridge Identifier shall be decoded from this field. On receipt of an MST BPDU the CIST Regional Root Identifier shall be decoded from this field.
- k) Octets 26 and 27 convey the CIST Port Identifier of the transmitting Bridge Port.

- 1 l) Octets 28 and 29 convey the Message Age timer value.
- 2 m) Octets 30 and 31 convey the Max Age timer value.
- 3 n) Octets 32 and 33 convey the Hello Time timer value used by the transmitting Bridge Port.
- 4 o) Octets 34 and 35 convey the Forward Delay timer value.

5 No further octets shall be encoded in STP Configuration BPDUs. Additional octets in received BPDUs
6 identified by the validation procedure (14.5) as STP Configuration BPDUs shall be ignored. The
7 specification of encoding or decoding of further octets in this subclause refers only to RST, MST, and SPT
8 BPDUs.

- 9 p) Octet 36 conveys the Version 1 Length. This shall be transmitted as 0. It is checked on receipt by the
10 validation procedure (14.5).

11 No further octets shall be encoded in RST BPDUs. Additional octets in received BPDUs identified by the
12 validation procedure (14.5) as RST BPDUs shall be ignored. The specification of encoding or decoding of
13 further octets in this subclause refers only to MST and SPT BPDUs.

14 NOTE 2—As Version 2 does not specify any additional fields beyond the end of the Version 0 information, there is no
15 Version 2 Length field specified in Version 2 of the protocol and therefore no need for a Version 2 length field here.

- 16 q) Octets 37 and 38 convey the Version 3 Length. Its value is the number of octets taken by the
17 parameters that follow in the BPDU. It is checked on receipt by the validation procedure (14.5).

- 18 r) Octets 39 through 89 convey the elements of the MCID (13.8):

- 19 1) The Configuration Identifier Format Selector is encoded in octet 39 and shall take the value
20 0000 0000.
- 21 2) The Configuration Name is encoded in octets 40 through 71.
- 22 3) The Revision Level is encoded as a number in octets 72 through 73.
- 23 4) The Configuration Digest is encoded in octets 74 through 89.

- 24 s) Octets 90 through 93 convey the CIST Internal Root Path Cost.

- 25 t) Octets 94 through 101 convey the CIST Bridge Identifier of the transmitting Bridge. The 12-bit
26 system ID extension component of the CIST Bridge Identifier shall be transmitted as 0. The
27 behavior on receipt is unspecified if it is nonzero.

28 NOTE 3—The most significant 4 bits of the Bridge Identifier constitute the manageable priority component for
29 each MSTI and are separately encoded in MSTI Configuration Messages in the BPDU.

- 30 u) Octet 102 encodes the value of remaining Hops for the CIST.

- 31 v) A sequence of zero or more, up to a maximum of 64, MSTI Configuration Messages follows, each
32 encoded as specified in 14.4.1.

33 No further octets shall be encoded in MST Configuration BPDUs. Additional octets in received BPDUs
34 identified by the validation procedure (14.5) as MST Configuration BPDUs shall be ignored. The
35 specification of encoding or decoding of further octets in this subclause refers only to SPT BPDUs.

- 36 w) Octets 1 and 2 following the Version 3 information convey the Version 4 Length. Its value is the
37 number of octets that follow the Version 4 Length in the BPDU, up to but not including octets added
38 by subsequent revisions of this standard and associated with Versions 5 or greater. It is checked on
39 receipt by the validation procedure (14.5).

- 40 x) Octets 1 through 51 following the Version 4 Length encode the Auxiliary MCID (if present) as
41 follows:

- 42 1) The Configuration Identifier Format Selector is encoded in octet 1 and shall take the value
43 0000 0000.
- 44 2) The Configuration Name is encoded in octets 2 through 33.
- 45 3) The Revision Level is encoded as a number in octets 34 through 35.
- 46 4) The Configuration Digest is encoded in octets 36 through 51.

- 1 y) Octets 1 and 2 following the Auxiliary MCID are encoded (if present) as follows:
- 2 1) Bits 1 and 2 of Octet 1 convey the Agreement Number (13.27.11, 13.27.16).
- 3 2) Bits 3 and 4 of Octet 1 convey the Discarded Agreement Number (13.27.12, 13.27.17).
- 4 3) Bit 5 of Octet 1 conveys the Agreement Valid flag (13.27.7, 13.27.9).
- 5 4) Bit 6 of Octet 1 conveys the Restricted Role flag (13.27.63, 27.20).
- 6 5) Bits 7 through 8 of Octet 1 are unused, and shall be transmitted as 0 and ignored on receipt.
- 7 6) Octet 2 is unused, and shall be transmitted as 0 and ignored on receipt.
- 8 z) The remaining Octets following the Version 4 Length and covered by that length comprise the
- 9 Agreement Digest (13.17.1, 28.4).

10 No further octets shall be encoded in BPDUs. Additional octets in received BPDUs shall be ignored.

11 14.4.1 MSTI Configuration Messages

12 A single instance of the following set of parameters is encoded for each MSTI supported by the transmitting

13 Bridge.

- 14 a) Bits 1, 2, 3 and 4, 5, 6, 7, and 8, respectively, of Octet 1 convey the Topology Change flag, Proposal
- 15 flag, Port Role, Learning flag, Forwarding flag, Agreement flag, and Master flag for this MSTI.
- 16 b) Octets 2 through 9 convey the Regional Root Identifier (13.27.20) as illustrated in Figure 14-3. This
- 17 includes the value of the MSTID for this Configuration Message encoded in bits 4 through 1 of
- 18 Octet 1, and bits 8 through 1 of Octet 2.
- 19 NOTE—The most significant 4 bits of each MSTI's Regional Root Identifier constitute a manageable priority
- 20 component.
- 21 c) Octets 10 through 13 convey the Internal Root Path Cost.
- 22 d) Bits 5 through 8 of Octet 14 convey the value of the Bridge Identifier Priority for this MSTI. Bits 1
- 23 through 4 of Octet 14 shall be transmitted as 0, and ignored on receipt.
- 24 e) Bits 5 through 8 of Octet 15 convey the value of the Port Identifier Priority for this MSTI. Bits 1
- 25 through 4 of Octet 15 shall be transmitted as 0, and ignored on receipt.
- 26 f) Octet 16 conveys the value of remainingHops for this MSTI (13.27.21).

	Octet
MSTI Flags	1
MSTI Regional Root Identifier	2–9
MSTI Internal Root Path Cost	10–13
MSTI Bridge Priority	14
MSTI Port Priority	15
MSTI Remaining Hops	16

Figure 14-3—MSTI Configuration Message parameters and format

1 14.5 Validation of received BPDUs

2 The receiving protocol entity shall examine Octets 1 and 2 (conveying the Protocol Identifier), Octet 3
3 (conveying the Protocol Version Identifier encoded as a number), Octet 4 (conveying the BPDU Type) and
4 the total length of the received BPDU (including the preceding fields, but none prior to the Protocol
5 Identifier) to determine the further processing required as follows:

- 6 a) If the Protocol Identifier is 0000 0000 0000 0000, the BPDU Type is 0000 0000, and the BPDU
7 contains 35 or more octets, it shall be decoded as an STP Configuration BPDU.
- 8 b) If the Protocol Identifier is 0000 0000 0000 0000, the BPDU Type is 1000 0000 (where bit 8 is
9 shown at the left of the sequence), and the BPDU contains 4 or more octets, it shall be decoded as an
10 STP TCN BPDU (9.3.2 of IEEE Std 802.1D-2004 [B14]).
- 11 c) If the Protocol Identifier is 0000 0000 0000 0000, the Protocol Version Identifier is 2, and the BPDU
12 Type is 0000 0010 (where bit 8 is shown at the left of the sequence), and the BPDU contains 36 or
13 more octets, it shall be decoded as an RST BPDU.
- 14 d) If the Protocol Identifier is 0000 0000 0000 0000, the Protocol Version Identifier is 3 or greater, the
15 BPDU Type is 0000 0010, the Bridge is configured as an MST Bridge or an SPT Bridge or
16 according to a future revision of this standard that intends to provide interoperability with prior
17 revisions, and the BPDU:
 - 18 1) Contains 35 or more but less than 103 octets, or
 - 19 2) Contains a Version 1 Length that is not 0, or
 - 20 3) Contains a Version 3 length that does not represent an integral number, from 0 to 64 inclusive,
21 of MSTI Configuration Messages,22 it shall be decoded as an RST BPDU.
- 23 e) If the Protocol Identifier is 0000 0000 0000 0000, the Protocol Version Identifier is 3 or greater and
24 the Bridge is configured as an MST Bridge or the Protocol Version Identifier is 3 and the Bridge is
25 configured as an SPT Bridge or according to a future revision of this standard that intends to provide
26 interoperability with prior revisions, the BPDU Type is 0000 0010, and the BPDU:
 - 27 1) Contains 102 or more octets, and
 - 28 2) Contains a Version 1 Length of 0, and
 - 29 3) Contains a Version 3 length representing an integral number, from 0 to 64 inclusive, of MSTI
30 Configuration Messages,31 it shall be decoded as an MST BPDU.
- 32 f) If the Protocol Identifier is 0000 0000 0000 0000, the Protocol Version Identifier is 3 or greater, the
33 BPDU Type is 0000 0010, the Bridge is configured as an SPT Bridge or according to a future
34 revision of this standard that intends to provide interoperability with prior revisions, and the BPDU:
 - 35 1) Contains 102 or more octets, and
 - 36 2) Contains a Version 1 Length of 0, and
 - 37 3) Contains a Version 3 length representing an integral number, from 0 to 64 inclusive, of MSTI
38 Configuration Messages, and
 - 39 4) Is not a well-formed an SPT BPDU, i.e.,
 - 40 i) Contains less than 6 octets following the octets specified by the Version 3 length,
 - 41 ii) Has a Version 4 length that is less than 55,
 - 42 iii) Does not contain an MCID Format Selector of 1,43 it shall be decoded as an MST BPDU.
- 44 g) If the Protocol Identifier is 0000 0000 0000 0000, the Protocol Version Identifier is 4 or greater, the
45 BPDU Type is 0000 0010, the Bridge is configured as an MST Bridge or an SPT Bridge or
46 according to a future revision of this standard that intends to provide interoperability with prior
47 revisions, and the BPDU:
 - 48 1) Contains 106 or more octets, and
 - 49 2) Contains a Version 1 Length of 0, and

- 1) Contains a Version 3 length representing an integral number, from 0 to 64 inclusive, of MSTI Configuration Messages, and
- 2) Is a well-formed an SPT BPDUs, i.e., contains
 - i) At least 6 octets following the octets specified by the Version 3 length,
 - ii) A Version 4 length of 55 or greater,
 - iii) An MCID Format Selector of 1,it shall be decoded as an SPT BPDUs.
- h) Otherwise, the BPDUs shall be discarded and not processed.

NOTE 1—The LLC LSAP that identifies BPDUs is reserved for standard protocols, no other protocols using that LSAP have been standardized though they may be at some future time. At that time, BPDUs with different PIDs may be processed according to the rules of those protocols but will still be discarded from the point of view of MSTP.

NOTE 2—These validation rules do not contain a loopback check of the form specified for use with STP Configuration BPDUs in 9.3.4 of IEEE Std 802.1D-2004 [B14].

14.6 Validation and interoperability

The validation rules above (14.5) follow a consistent general approach that allows future version enhancements to be made while retaining backwards compatibility. In particular, tests (a) and (b) in 14.5 above do not check the Protocol Version Identifier.

In general, for an implementation that supports version A of a given protocol, a received PDU of a given type that carries a protocol version number B is interpreted as follows:

- a) Where B is greater than or equal to A, the PDU shall be interpreted as if it carried the supported version number, A. Specifically,
 - 1) All PDU types, parameters, and flags that are defined in version A shall be interpreted in the manner specified for version A of the protocol for the given BPDUs type.
 - 2) All PDU types, parameters, and flags that are undefined in version A for the given BPDUs type shall be ignored.
 - 3) All octets that appear in the PDU beyond the largest numbered octet defined for version A for the given BPDUs type shall be ignored.
- b) Where B is less than A, the PDU shall be interpreted as specified for the version number, B, carried in the BPDUs. Specifically,
 - 1) All PDU parameters and flags shall be interpreted in the manner specified for version B of the protocol for the given PDU type.
 - 2) All PDU parameters and flags that are undefined in version B for the given BPDUs type shall be ignored.
 - 3) All octets that appear in the PDU beyond the largest numbered octet defined for version B for the given BPDUs type shall be ignored.

NOTE—In other words, if the protocol version implemented differs from the protocol version number carried in the PDU, then only those PDU types, parameters, and flags that are specified within the lesser numbered protocol version are interpreted by the implementation (in accordance with the lesser numbered protocol version's specification), and no attempt is made to interpret any additional PDU types, parameters, and flags that may be specified within the greater numbered protocol version. In the specific case of STP (version 0) and RSTP (version 2), as there is only a single RST BPDUs type defined in version 2, and as the RST BPDUs type is undefined in version 0, a version 0 implementation will ignore all RST BPDUs. Version 2 implementations, however, recognize and process both version 0 and version 2 BPDUs. As version 2 makes no changes to the BPDUs types defined for version 0 (and always transmits such BPDUs types with 0 as the version identifier), version 0 BPDUs are always interpreted by version 2 implementations according to their version 0 definition.

15. Support of the MAC Service by PBNs

Provider Bridges interconnect the separate MACs of the IEEE 802 LANs that compose a PBN, relaying frames to provide connectivity between all LANs that provide customer interfaces for each service instance. The position of the Provider Bridge S-VLAN component bridging function within the MAC Sublayer is shown in Figure 15-1.

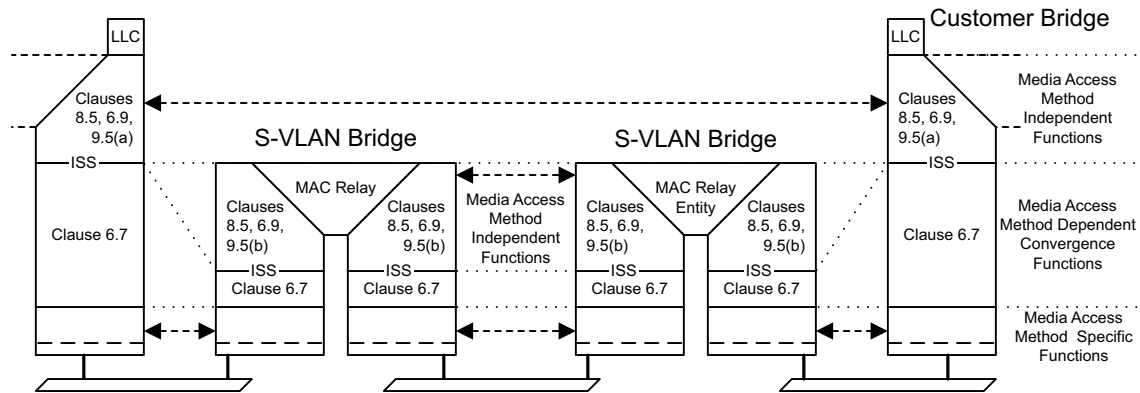


Figure 15-1—Internal organization of the MAC sublayer in a PBN

This clause discusses the following aspects of provisioning service instances on a PBN:

- a) Service transparency
- b) Customer service interfaces
- c) Service instance segregation
- d) Service instance selection and identification
- e) Service priority selection
- f) Service access protection

NOTE—In describing the MAC Service, this standard makes use of term “service” as defined by the OSI Reference Model (ISO/IEC 7498-1). In this sense, a service comprises a set of primitives and associated parameters, provided by one protocol layer in the architectural model to the protocol layer above, and the causal relationships between the primitives invoked by an upper layer protocol (ULP) entity in one system with those resulting indications to a peer entity in another system. The term “service” used by service providers, while including layering concepts, goes far beyond this formal definition, and commonly specifies some or all of the following: interfacing considerations across multiple protocol layers (including physical connectors, for example); selection of interface points; interfacing equipment; QoS guarantees and measurement methods; charging methods and responsibilities; connectivity verification and other management tools; and regulatory issues. Many of these aspects lie outside the scope of this standard; the reader is referred to the bibliography in Annex X, which includes references to completed and ongoing work in the MEF Forum and the ITU.

15.1 Service transparency

The operation of Provider Bridges and the networks they compose is, by design, largely transparent to Customer Bridges and CBNs as illustrated by Figure 15-1.

The service provided by Provider Bridges is transparent to the use of the MAC Service by end stations attached to the CBNs and transparent to the operation of media access method-independent functions by Customer Bridges.

The service is not transparent to the operation of media access method-dependent convergence functions, specified in IEEE Std 802.1AC, or to the operation of the media access method-specific functions specified by standards for each media access method. Media access method-dependent and -specific functions operate

1 between Bridges, whether Customer Bridges or Provider Bridges, attached to the same LAN. Where these
2 functions make use of standard group MAC addresses, those addresses are included in the Reserved
3 Addresses that are always filtered by Customer Bridges (Table 8-1) and by Provider Bridges (Table 8-2).

4 Frames transmitted and received by media access method-independent functions particular to PBN
5 operation are not forwarded by Customer Bridges between provider networks. Where these frames are
6 addressed using standard group MAC addresses, those addresses are included in the Reserved Addresses
7 that are always filtered by Customer Bridges (Table 8-1). In addition, such frames may be filtered from
8 PEPs.

9 15.2 Customer service interfaces

10 A service provider can offer a customer one or more types of service interfaces, each providing different
11 capabilities for service selection, priority selection, and service access protection (15.8, 15.9, and 15.10).
12 Some service interfaces are provided by the service provider operating systems that include C-VLAN
13 components, or by customer operating systems that include S-VLAN components. In all cases, segregation
14 of different service instances is achieved at an interface wholly under the control of the service provider by
15 authentication and authorization of the attached customer systems, and by verification of customer provided
16 parameters that provide service instance selection.

17 NOTE—The term “service access protection” describes provision of service access over multiple access LANs with
18 redundancy and rapid failover in case of failure of an access LAN or attached equipment.

19 Access to a given service instance can be provided through different types of customer interface.

20 15.3 Port-based service interface

21 The customer service interfaces that can be provided by a provider network are specified by reference to a
22 CNP provided by the S-VLAN component of a Provider Bridge. The CNP provides a single service
23 instance, as illustrated in Figure 15-2 and Figure 15-3. The attached customer system can be a Bridge, a
24 router, or an end station.

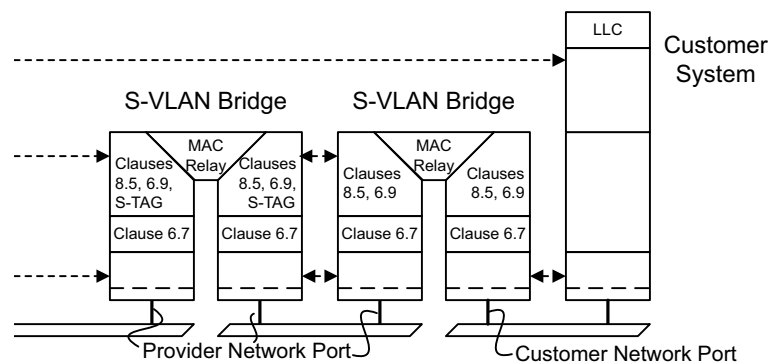


Figure 15-2—Port-based service interface to a PBN

1 This interface is Port-based; i.e., customers select and identify different service instances by associating each
2 with a different CNP. Frames transmitted to a CNP by a C-VLAN-aware customer system do not include an
3 S-VID but can be priority-tagged with an S-TAG (6.13).

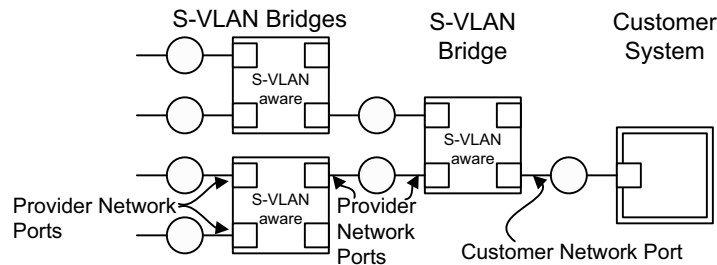


Figure 15-3—Port-based service interface to a PBN

4 NOTE—The terms “Customer Network Port,” “Customer Edge Port,” “Provider Network Port,” and “Provider Edge
5 Port,” do not refer to the ownership of equipment, or necessarily to differently implemented Ports, but to Ports that are
6 configured to fulfill the requirements of precise roles within a structured provider network design. These requirements
7 are described in 15.6 through 15.10 and in Clause 16. All “Network” Ports are part of S-VLAN components, and all
8 “Edge” Ports are part of C-VLAN components, whereas all “Customer” Ports receive data from a single customer
9 inbound to the network and transmit data outbound from the network to a single customer. See Clause 3 for definitions.

10 15.4 C-tagged service interface

11 A C-tagged service interface can be provided by a Provider Edge Bridge comprising one or more C-VLAN
12 components attached to Port-based service interfaces provided by a single S-VLAN component, as
13 illustrated by Figure 15-4 and Figure 15-5.

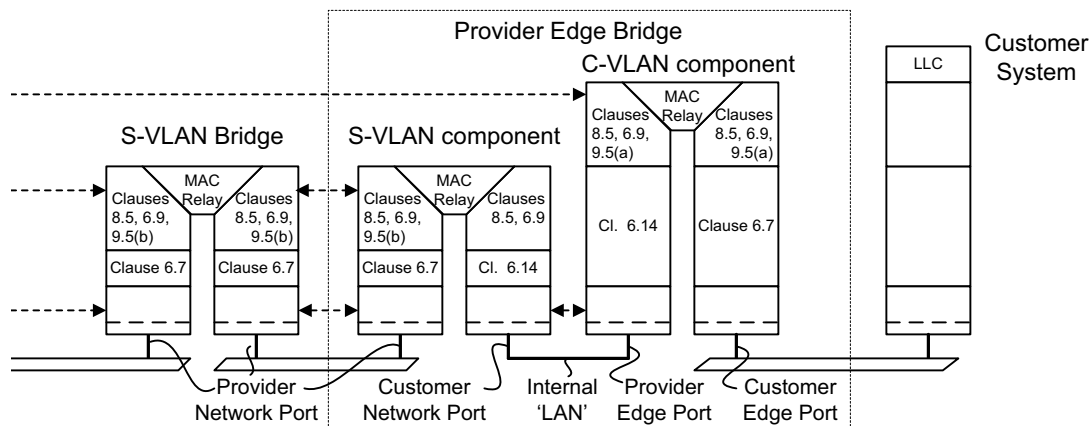


Figure 15-4—C-tagged service interface to a PBN

14 The C-tagged service interface allows service instance selection and identification by C-VID. Each frame
15 from the customer system is assigned to a C-VLAN and presented at zero or one internal Port-based service
16 interfaces, each supporting a single service instance that the customer desires to carry that C-VLAN.

17 NOTE—The restriction that each C-VLAN map to a single service instance allows the customer equipment receiving
18 frames to correctly identify the service instance used, supports mechanisms that guard against accidental creation of data
19 loops, and prevents configuration of the C-VLAN component to create a multi-point service from point-to-point service
20 instances. The service provider can offer a multi-point service through appropriate configuration of the S-VLAN
21 component.

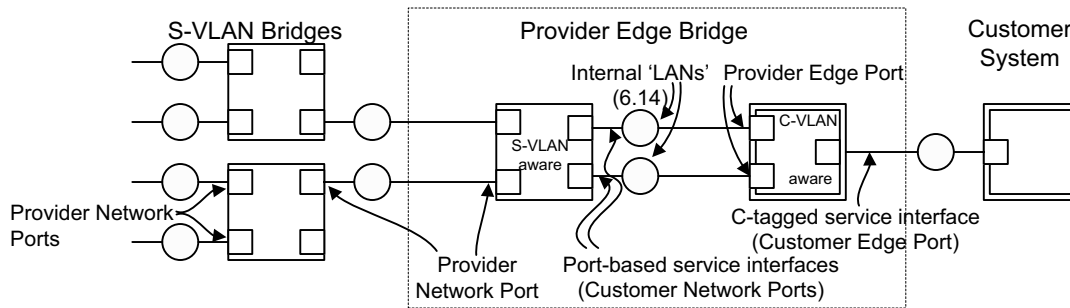


Figure 15-5—C-tagged service interface to a PBN

1 Similarly frames from the provider network are assigned to an internal interface or “LAN” on the basis of
2 the S-VID. As each internal interface supports a single service instance, no S-TAG is used at this interface. If
3 multiple C-VLANs are supported by this service instance, the frames will have C-TAGs with the possible
4 exception of frames for a single C-VLAN. The C-VLAN component applies a PVID to untagged frames
5 received on each internal “LAN,” allowing full control over the delivery of frames for each C-VLAN
6 through the CEP.

7 Each Provider Edge Bridge can support multiple CEPs for the same customer or for multiple customers.
8 Each CEP is supported by a dedicated C-VLAN component as illustrated in Figure 15-6.

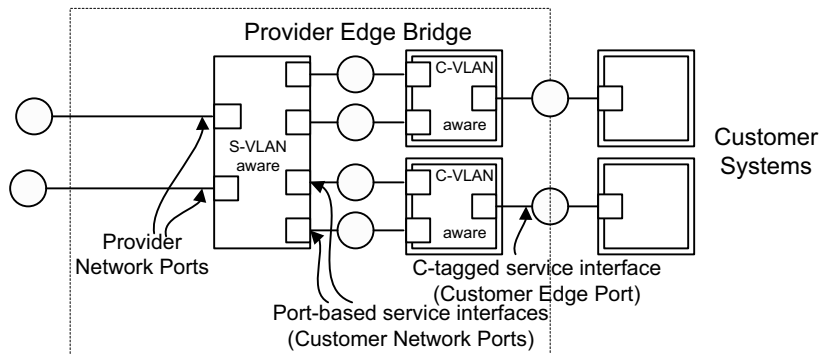


Figure 15-6—Customer Edge Ports (CEPs)

9 15.5 S-tagged service interface

10 An S-tagged service interface can be provided to an S-VLAN Bridge operated by a customer as illustrated
11 by Figure 15-7 and Figure 15-8, or to a customer-operated Provider Edge Bridge that in turn provides
12 C-tagged service interfaces within the customer’s own network as described in 15.4.

13 15.6 Remote customer service interfaces (RCSIs)

14 An RCSI provides a C-tagged or Port-based service interface to a customer attached via an access PBN as
15 depicted in Figure 16-2 (16.2). Multiple RCSIs can be provided over a LAN interconnecting two PBNs
16 through the use of a Port-mapping S-VLAN component as shown in Figure 15-2. The Port-mapping

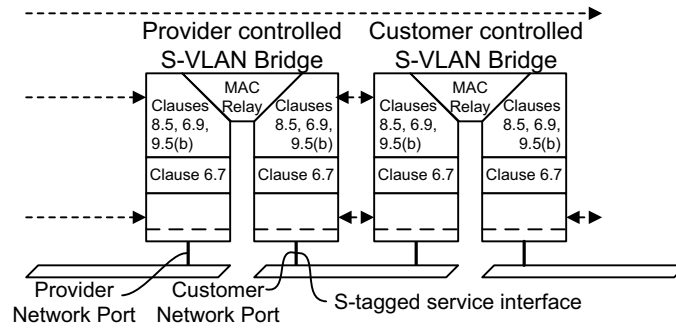


Figure 15-7—S-tagged service interface to a PBN

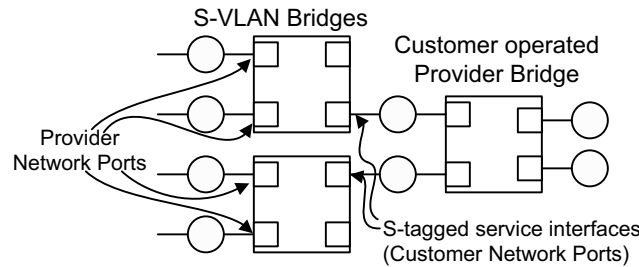


Figure 15-8—S-tagged interface to a PBN

1 S-VLAN component has one external port (RCAP) and one or more internal Ports (PAPs) that each support
2 one RCSI. The Port-mapping S-VLAN component also can have an internal PNP for service instances that
3 do not require an RCSI.

4 The Port-mapping S-VLAN component is configured to associate a unique S-VID with each PAP.
5 Furthermore, each S-VID configured in the Port-mapping S-VLAN component has the RCAP and exactly
6 one other Port in its member set. This ensures that each external service instance is mapped to only one
7 internal port. See Figure 15-9.

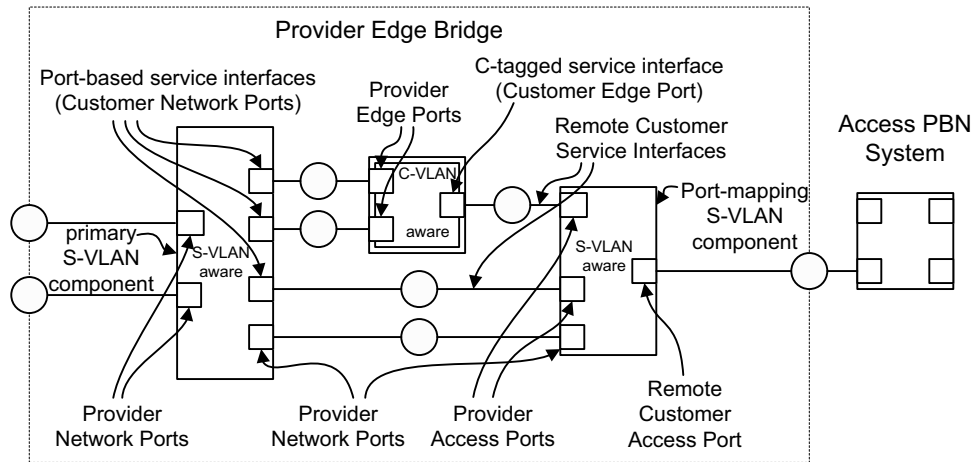


Figure 15-9—RCSIs to a PBN

8 Each RCSI supported by a RCAP is associated with a PAP connected via an internal LAN to either a CNP,
9 providing a Port-based service interface (15.3), or a CEP, providing a C-tagged service interface (15.4). The
10 external S-VID mapped to an RCSI identifies a service instance originating at a remote customer interface
11 LAN attached to an access PBN (16.2). To provide separation between the external S-VID space used by the

1 access PBN and the internal S-VID space for this PBN, frames on an RCSI do not carry an S-TAG. When
2 frames transit an RCSI the internal S-VID is set by the PVID of the CNP and the external S-VID by the
3 PVID of the PAP.

4 NOTE—If two CNPs associated with different RCSIs on the same RCAP belong to the same internal S-VLAN, frames
5 received on the RCAP with one S-VID can be transmitted on that same external port with a different S-VID. This is
6 sometimes referred to as “Hairpin Switching” and is described in 16.2.

7 In addition to the RCSIs, a PNP on the Port-mapping S-VLAN component is connected via an internal LAN
8 to a PNP on the primary S-VLAN component. External service instances that are not associated with an
9 RCSI can be mapped to this Port. This allows multiple service instances to be passed between the PBNs
10 without requiring a separate Port for each service instance. On this interface service frames carry S-TAGs
11 and the VID translation table of the PNP on the S-VLAN component is used to provide independence
12 between internal and external S-VID spaces.

13 A Provider Edge Bridge can support multiple RCAPs connected to other PBNs. Each RCAP can support
14 multiple RCSIs for one or more customers. Each RCAP is supported by a Port-mapping S-VLAN
15 component as illustrated in Figure 15-10. As shown in the figure, each Port-mapping S-VLAN component
16 with a RCAP can have one PNP and multiple PAPs providing either Port-based or C-tagged RCSIs.

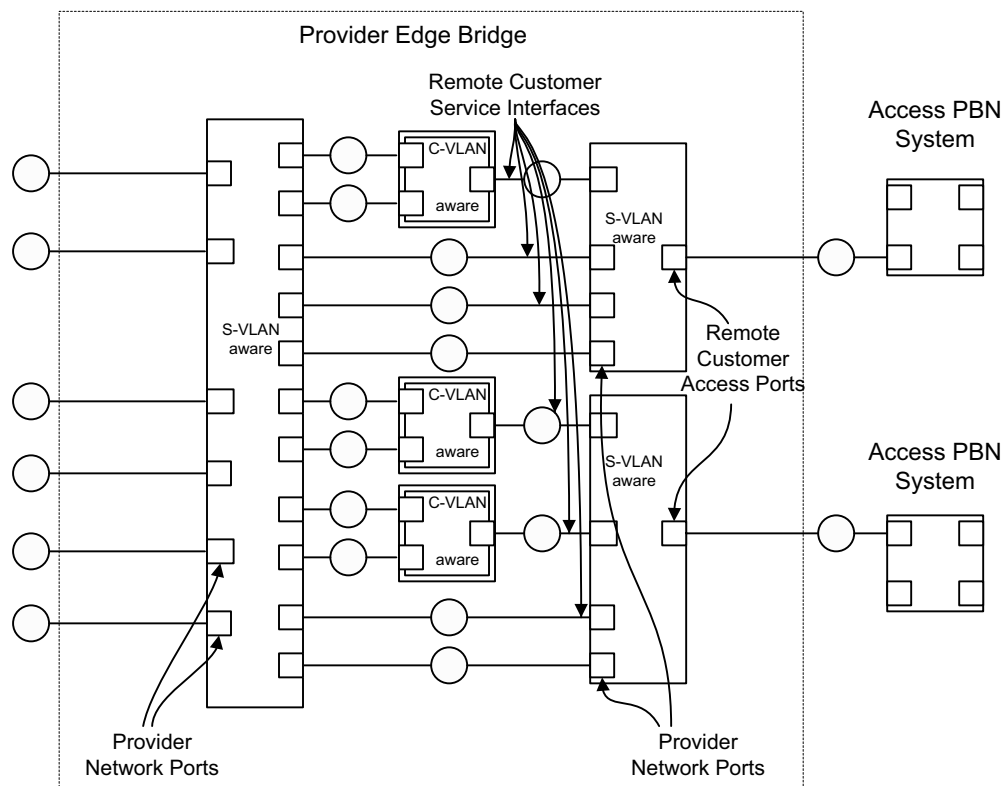
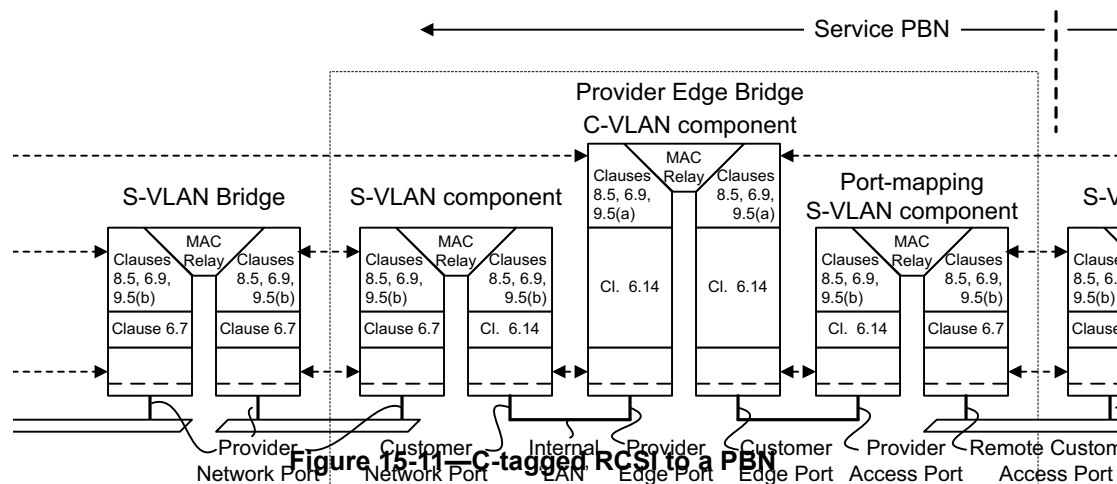


Figure 15-10—Remote Customer Access Ports (RCAPs)

1 Figure 15-11 shows a C-tagged RCSI. Since S-VLAN components are transparent to C-VLAN control
2 PDUs, i.e., PDUs using addresses in Table 8-1 but not in Table 8-2, these control PDUs exchanged via the
3 CEP reach the next (e.g., customer controlled) C-VLAN component across the access PBN. S-VLAN
4 control PDUs, using addresses in Table 8-2, exchanged on the RCAP will normally reach the neighboring
5 Provider Bridge in the adjacent PBN.



6 Frames traversing the internal LAN between the Port-mapping S-VLAN component and the C-VLAN
7 component do not contain an S-TAG. The S-VID for frames received on this interface by the Port-mapping
8 S-VLAN component is provided by the PAP's PVID.

9 Figure 15-12 shows a Port-based RCSI. In this case the PAP is connected to a CNP on the S-VLAN
10 component. Frames carried over the internal LAN between the PAP and CNP do not contain an S-TAG. An
11 S-VID is provided at each Port by the respective Port's PVID. These S-VIDs are not required to have the
12 same value.

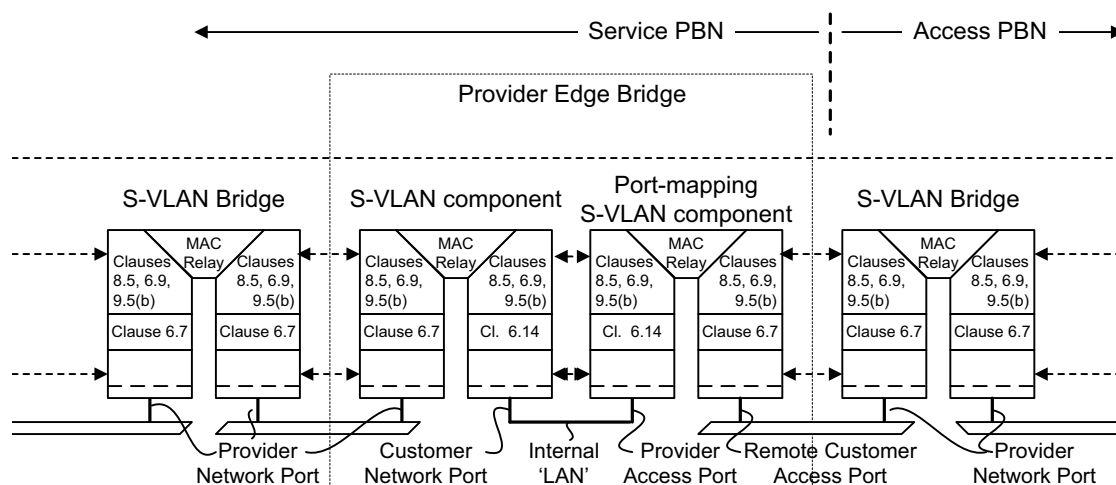


Figure 15-12—Port-based RCSI to a PBN

1 Figure 15-13 shows the PNP on the Port-mapping S-VLAN component connected to a PNP on the S-VLAN
2 component. Service frames carried over this interface have S-TAGs and S-VID translation can be
3 performed.

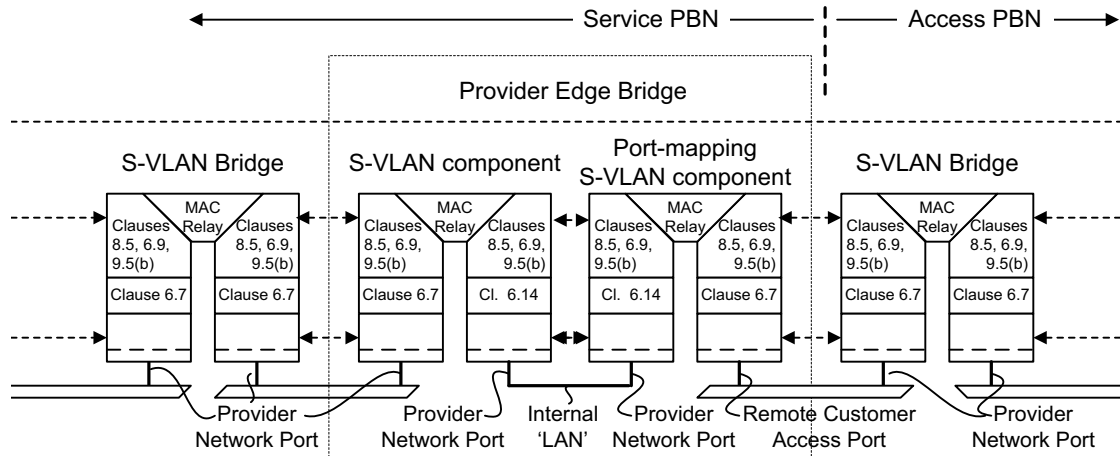


Figure 15-13—Provider Network Port (PNP) interface

4 15.7 Service instance segregation

5 Segregation of data frames associated with different MAC Service instances is achieved by supporting each
6 service instance with a separate S-VLAN and ensuring that:

- 7 a) Provider Bridges are configured such that no customer data frames are transmitted through a PNP
8 untagged, i.e., without an S-TAG.
- 9 b) No frames are accepted, i.e., received and relayed, from any customer system without first being
10 subject to service instance selection.
- 11 c) No frames are delivered to any customer system without explicit service instance identification.
- 12 d) Prior to transmission through a PNP, customer data frames are received through a CNP within the
13 provider network that is exclusively accessed by a single customer. The S-VIDs of all frames
14 received through that CNP correspond to service instances that the customer is permitted to access.
- 15 e) Provider Bridges and the S-VLAN component of each Provider Edge Bridge within the provider
16 network can only be directly controlled by the service provider. Customer equipment, including
17 customer-owned Provider Bridges, are not within the provider network and are controlled by the
18 customer.
- 19 f) Only frames that have been transmitted through a PNP can be received through other PNPs within
20 the provider network.

21 15.8 Service instance selection and identification

22 Service instance selection is provided for Port-based service interfaces by configuring a CNP with a PVID
23 value corresponding to the S-VID used to identify the service instance and an Acceptable Frame Types value
24 of *Admit Only Untagged and Priority-tagged frames*.

25 Service instance selection is provided for C-tagged service interfaces by a C-VLAN component internal to a
26 Provider Edge Bridge. The C-VLAN component uses the C-VID to direct frames to an internal PEP
27 supporting a specific service instance. Frames for at most one C-VLAN can be conveyed untagged over a
28 single service instance. Management control of associating C-VIDs with PEPs is accomplished using the
29 C-VID Registration Table (12.13.2), which provides equivalent functionality to configuring the PVID of the
30 internal CNP with the S-VID of the service instance, and adding the PEP to the Member Set, and possibly

1 Untagged Set, of the C-VLAN. No management control is provided for Protocol-based VID assignment on
2 internal CNPs.

3 NOTE 1—A Provider Edge Bridge can configure the C-VLAN component associated with a CEP to select the same
4 service instance for all frames. This creates a service interface similar, but not identical, to a Port-based service interface.
5 The C-VLAN component allows modification of the C-TAG (insertion of a C-TAG with the PVID value in untagged
6 frames, assigning the PVID value to priority-tagged frames, or stripping the C-TAG from frames forwarded through the
7 PEP) but never forwards a frame with a null C-VID. A Port-based service interface does not modify the C-TAG of
8 received frames in any way. A Provider Edge Bridge may offer a Port-based service interface by configuring the Port to
9 be a CNP rather than a CEP; in which case, there is no associated C-VLAN component.

10 Service instance selection is provided by the attached customer system for S-tagged service interfaces. The
11 CNP is configured with Enable Ingress Filtering (8.6.2), and the Port is only included in the Member Set for
12 S-VLANs corresponding to service instances that the customer is permitted to use.

13 For all service interfaces described, the CNP determines the S-VID for each customer data frame as
14 specified in 6.9.1 for an EISS instance using an S-TAG type.

15 The VID Translation Table for the Port (6.9.1) allows a service provider to assign S-VIDs independently
16 from those used by a customer (or other service provider) to identify service instances on an S-tagged
17 service interface (15.5). The table also allows customers to identify the same service instance by different
18 VIDs at different interfaces.

19 NOTE 2—The means used by a service provider and a customer to determine the VIDs used by the customer to select
20 and identify a given service instance are outside the scope of this standard.

21 The service instance for each frame received by the attached customer system is identified in the same way
22 as frames transmitted using the same interface, but not necessarily in the same way that the service instance
23 is selected or identified at other interfaces. A single service instance can support Port-based, C-tagged, and
24 S-tagged service interfaces.

25 **15.9 Service priority selection**

26 For all service interface types, the service priority is selected using the received priority for each frame,
27 possible regenerated using the Priority Regeneration Table (6.9.4). The mechanism for determining the
28 received priority varies with the type of service interface.

29 Service priority selection is provided for Port-based service interfaces using the received priority signaled
30 from the media access method of the port. If the media access method used to attach to the interface does not
31 directly support priority, this will result in the selection of a single value for all frames. A customer system
32 may also signal priority to a Port-based service interface on a per-frame basis by priority-tagging each frame
33 with an S-TAG with a null VID. Subclause 6.13 specifies a function to support priority-tagging with an
34 S-TAG on Customer Bridges.

35 Service priority selection is provided by C-tagged service interfaces using the priority conveyed in the
36 C-TAG of each frame. A C-tagged service interface can provide a single service instance for all C-VIDs
37 received and in this way function much as a Port-based service interface with the addition of the capability
38 of the customer to independently signal priority with each frame.

39 Service priority selection is provided by S-tagged interfaces using the received priority decoded from the
40 PCP field in the S-TAG.

41 **15.10 Service access protection**

42 A customer system or systems at a single location can attach to two or more service interfaces using separate
43 LANs for attachment, thus providing fault tolerance through redundancy of the interface components.

16. Principles of Provider Bridged Network (PBN) operation

This clause establishes the principles and a model of PBN operation. It provides the context necessary to understand how the:

- a) Operation of individual Provider Bridges (Clause 8),
- b) Configuration and management of individual Provider Bridges (Clause 12), and
- c) Management of spanning tree and VLAN topologies within a provider network (Clause 7, Clause 11, Clause 13)

support, preserve, and maintain the quality of each instance of the MAC Service offered to the customers of the provider network (Clause 6, Clause 15), including:

- d) Independence of each service instance supported by a service provider from other service instances (Clause 15).
- e) Identification of service instances within the provider network (Clause 15, 8.8).
- f) Maintenance of service availability in the event of the failure, restoration, removal, or insertion of LAN components connecting a customer network to a provider network (Clause 6, Clause 11, Clause 13, 16.2).

A PBN is a Virtual Bridged Network that comprises Provider Bridges (S-VLAN Bridges and Provider Edge Bridges) and attached LANs, under the administrative control of a single service provider. The principal elements of provider network operation are those specified in Clause 7 for Virtual Bridged Networks in general, as amended by this clause.

NOTE 1—Unless explicitly stated, the term “provider network” in this standard refers to a PBN. The term “Provider Bridged Network (PBN)” is used exclusively to refer to networks configured and managed as specified by this clause and comprising only (a) Provider Bridges and Provider Edge Bridges and (b) communications media and equipment providing the ISS (IEEE Std 802.1AC). Although the requirements of Clause 15 are generally applicable to similar services, a generalized framework for all network designs that could support these requirements, while useful in the context of other equipment and services, is outside the scope of this standard. This clause describes specific best practice for PBNs to ensure that the requirements for Bridge functionality are clear. Conformance of a Provider Bridge implementation to this standard does not require that the implementation be used as specified in this clause, merely that it is capable of being so used.

NOTE 2—Within a provider network, an instance or instances of the MAC Service are reserved for the service provider’s own use to configure and manage the network. All frames associated with such service instances, and that are not confined to an individual LAN, are subject to service instance selection, segregation, and identification as specified in 16.1.

16.1 PBN overview

The principal elements of PBN operation comprise:

- a) Service instance segregation within the provider network for customer frames (15.6).
- b) Service instance selection on ingress, and service instance identification on egress, for each customer frame (15.8).
- c) Resource allocation and configuration to provide service instance connectivity (16.3).

The principal elements of PBN operation may also include:

- d) Management of customer end station address learning (16.4).
- e) Prevention of connectivity loops formed through attached networks (16.5).

16.2 Provider Bridged Network (PBN)

An example PBN is illustrated by Figure 16-1.

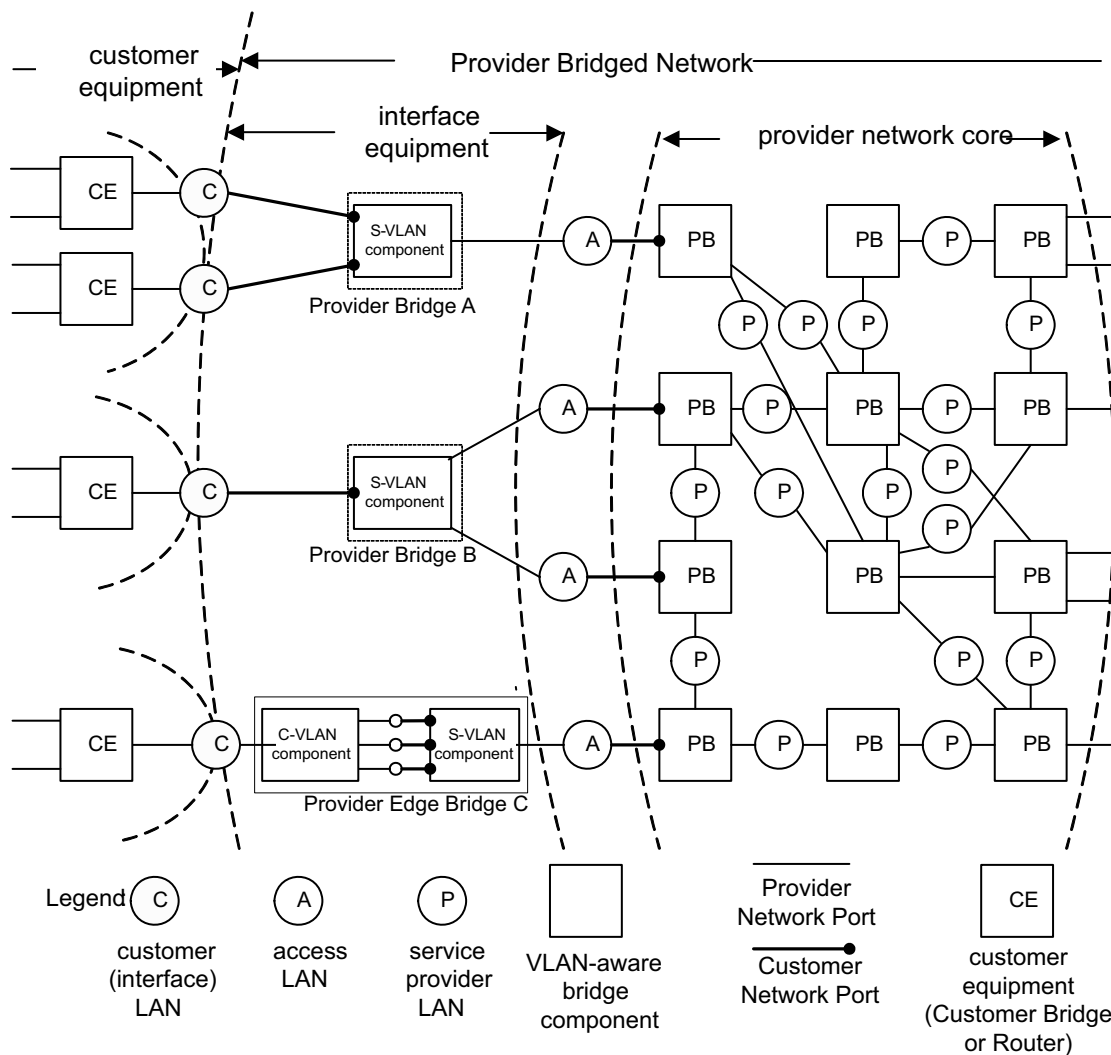


Figure 16-1—PBN with interface examples

1 Customer equipment attaches, via one or more customer interface LANs, to equipment that provides the
2 service interfaces specified in Clause 15. That interface equipment can be located in the core of the provider
3 network, such that both the equipment and the attached LANs are secure against direct addition of frames
4 either by the customer or by others. More commonly the interface equipment connects to the network core
5 using one or more access LANs, which can be subject to external interference. Figure 16-1 provides
6 examples. Within the network core, Provider Bridges and LANs are secured so that only the service provider
7 can manage the reception, transmission, and relay of frames between Provider Bridges.

8 The arbitrary physical network topology of the network core and the connectivity that it provides to support
9 segregated instances (15.6) of the MAC Service is designed and managed (16.3) by the service provider to
10 meet bandwidth and service availability requirements at the PNP. Application of the S-VLAN ingress and
11 egress rules at these Ports in support of service instance selection and identification (15.8) ensures that
12 frames cannot be transmitted or received on any service instance by any customer's equipment without prior
13 agreement with the service provider.

1 Although the application of the ingress and egress rules, together with the use of the MSTP restrictedRole
2 and restrictedTcn (13.27.64, 13.27.65) parameters and MVRP registration controls (16.3), permit service
3 providers to allow direct attachment of customer-operated equipment to access LANs, there are commonly
4 other reasons, such as Operations, Administration, and Maintenance (OAM) support of access LANs, why
5 interface equipment is mandated. The interface equipment, as illustrated by, but not limited to, the examples
6 in Figure 16-1 can be used to partition and enhance network access functionality to provide:

- 7 a) Service instance multiplexing on a single access LAN
- 8 b) Provision of resilient, and optionally physically route diverse, access
- 9 c) Reduced management of customer use of multiple service instances
- 10 d) Selective multiplexing of C-VLANs onto service instances
- 11 e) Reliable identification of the customer point of attachment

12 without requiring customer systems to understand the internal details of the provider network.

13 Provider Bridge A in Figure 16-1 uses physically separate customer interface LANs to provide separate
14 Port-based service interfaces to the customer equipment and uses S-TAGs to multiplex the corresponding
15 service instances over a single access LAN. The interface equipment can be managed by the service
16 provider to use S-VIDs that do not require use of a VID Translation Table at the edge of the core network.
17 Alternatively, the latter can translate S-VIDs to remove the need for such management, with all interface
18 equipment using the same S-VIDs in the same way.

19 Provider Bridge B in Figure 16-1 uses a single customer interface LAN to provide a Port-based service
20 interface to the customer equipment and uses two access LANs to provide resilient connectivity to distinct
21 Provider Bridges in the core network. Receipt of Provider Bridge BPDUs by Bridge B protects against the
22 failure of an access LAN, of one of the core Bridges, or of some internal physical connectivity within the
23 provider network. Use of the MSTP restrictedRole and restrictedTcn parameters by PNP ensures that
24 receipt of frames from the access LANs cannot disrupt the active topology or address learning within the
25 core network. Where multiple service instances are provided at a single customer point of attachment, both
26 access links can be used.

27 Provider Edge Bridge C in Figure 16-1 provides a C-tagged service interface to the customer equipment and
28 uses C-TAGs on the customer interface LAN to select between multiple S-VLAN-tagged service instances
29 on the access LAN. Individual C-VLANs can be conveyed on any service instance, and this distribution of
30 C-VLANs can be changed in response to changes in the connectivity offered by the service instances
31 through customer systems at other points of customer attachment. The traffic for a particular C-VLAN can
32 be the majority of that carried on a specific service instance; specifying that frames for that C-VLAN are
33 untagged on the internal LAN that corresponds to that service instance within the Provider Edge Bridge
34 allows those frames to be carried through the provider network without a C-VID following the S-VID if the
35 overhead of conveying the additional octets is a concern.

36 NOTE 1—In the scenario where a customer network has multiple service interfaces to a provider network, and the
37 customer network VLAN topology is such that frames sent from a single MAC source on different C-VLANs will enter
38 the provider network on different service interfaces and be mapped to the same service instance (same S-VLAN), then
39 the C-MAC address will appear to be duplicated and potentially cause oscillation in the learning process within the
40 Provider Network. This scenario can be avoided by restricting a MAC address to enter the provider network at the same
41 service interface for all C-VLANs, or by mapping the different C-VLANs to different service instances (different
42 S-VLANs), or by restricting the service instance to be point-to-point so the provider Bridges do not need to learn
43 the C-MAC addresses.

44 In some cases a service provider needs to reach a customer interface LAN that is not directly attached to the
45 service provider's network but is attached instead to another PBN. Figure 16-2 shows some examples in
46 which a customer interface LAN is connected to an access PBN that is not the PBN providing service to the
47 customer.

48 NOTE 2—The PBN providing access to the customer interface LANs in these examples can provide other services to its
49 own customers as well.

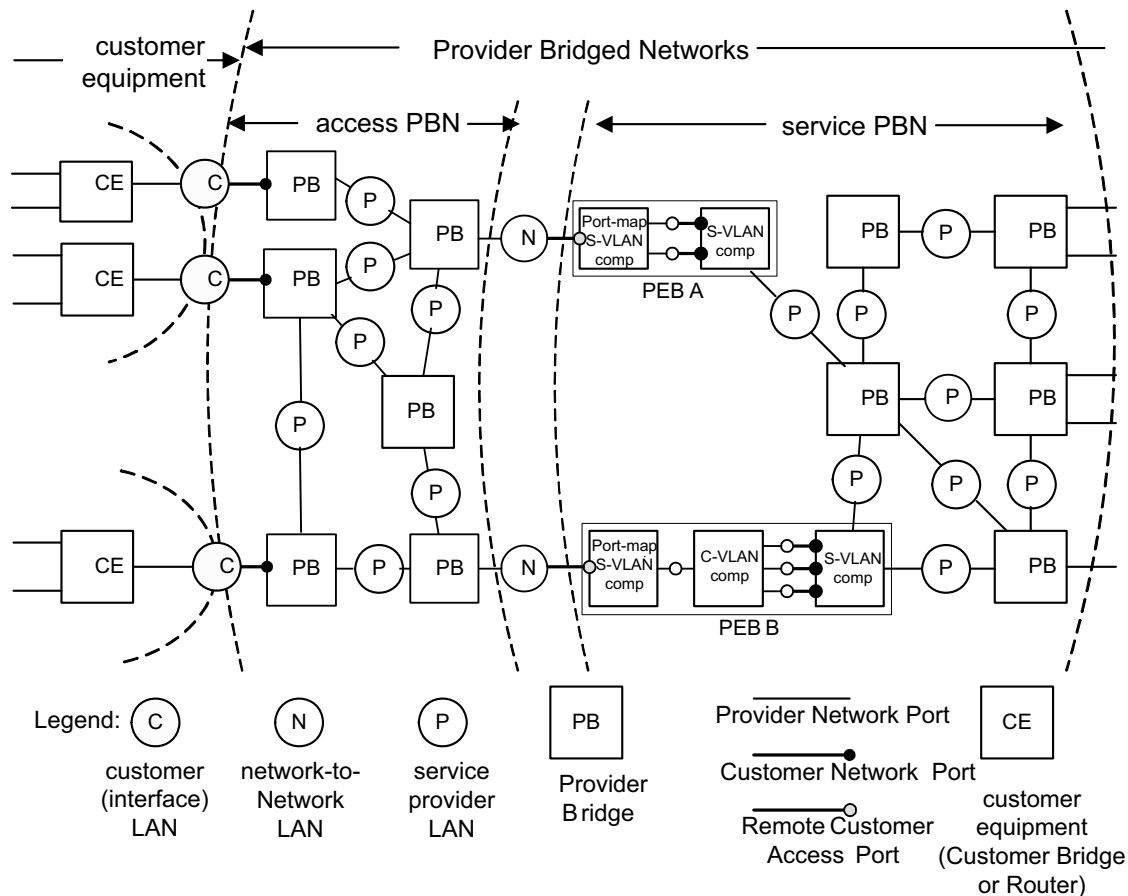


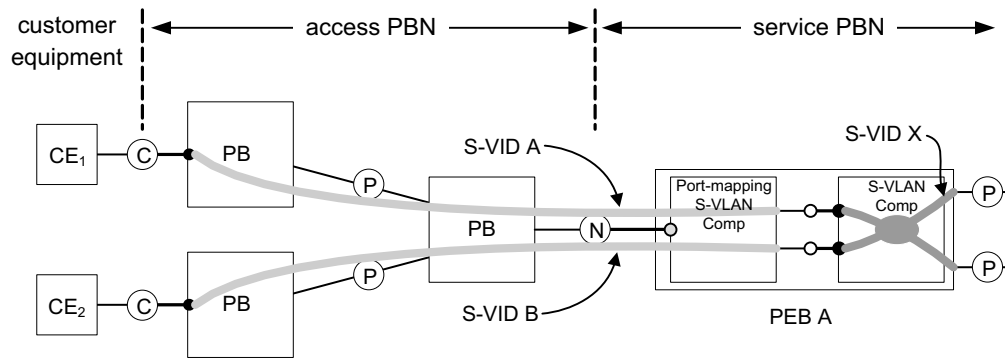
Figure 16-2—Examples of remote customer service access via a second PBN

1 In these examples the service provider uses a Port-based service provided by the access PBN to reach the
2 customer interface LANs. Each such access service is identified by a distinct S-VID at the network-to-
3 network interface LAN. The service PBN uses this S-VID to select an RCSI that provides either a
4 Port-based or C-tagged service interface as specified in Clause 15. An RCSI provides service instance
5 selection and identification (15.3) and service management for each customer interface LAN. Each RCSI
6 carries frames to/from a single customer interface LAN.

7 PEB A in Figure 16-2 receives traffic from two customer interface LANs attached to the access PBN. The
8 traffic from each customer interface LAN is carried over a separate RCSI. In the example shown each RCSI
9 is connected to a Port-based service interface (CNP). If the two customer interface LANs participate in the
10 same service instance (i.e., their RCSI CNPs belong to the same S-VLAN) a frame received from one RCSI
11 CNP can be transmitted on the other RCSI CNP. Thus a frame received on the network-to-network LAN
12 with one S-VID can be subsequently transmitted on that same external LAN with a different S-VID. This
13 behavior is sometimes called “Hairpin Switching”.²⁷

14 The structure allowing a frame to be received and subsequently transmitted on the same external interface
15 while avoiding loops and misdelivery of multicast frames is illustrated in Figure 16-3. Frames from/to CE₁
16 have S-VID A and frames from/to CE₂ have S-VID B. Thus the two customer interface LANs cannot
17 exchange traffic within the access PBN. In PEB A external S-VIDs A and B select or identify different RCSI
18 PAPs.

²⁷ For example, this behavior is called “Hairpin Switching” in MEF 26 [B55].



1 In this example, both RCSI CNPs belong to the same internal S-VLAN in the service provider's network
2 identified by S-VID X. Frames can be relayed between all Ports belonging to S-VID X, including between
3 the two RCSIs. When a frame is received on one RCSI and subsequently transmitted on the other RCSI the
4 frame is received and transmitted on the same external RCAP. However, in this case the external S-VIDs in
5 the received and transmitted frames are different. If the sets of endpoints in the access PBN associated with
6 each external S-VID are disjoint there is no possibility of looping or misdelivery of multicast frames.

⁸ The capabilities of a C-tagged RCSI are the same as those provided by an external CEP as described above.

9 16.3 Service instance connectivity

10 The VLAN Topology of each S-VLAN is established by the mechanisms introduced in 7.1 and Figure 7-1.
11 The service provider can use and configure MSTP to provide a number of independent spanning tree active
12 topologies and can assign each S-VLAN independently to one of these to best use the resources in the
13 network. MVRP running in the context of each spanning tree active topology configures the extent of each
14 S-VLAN to the subset of that active topology necessary to support connectivity between the customer points
15 of attachment to the MAC Service instance provided, and it can reconfigure that connectivity as required if
16 the spanning tree active topology changes.

NOTE 1—Autoconfiguration of the extent of each S-VLAN is accomplished by the service provider configuring the MVRP Administrative Control “Registration Fixed (New ignored)” for the S-VLAN on each CNP where the corresponding MAC Service Instance can be selected. The Enable Ingress Filtering parameter is not typically used within an individual provider network, as it limits the ability of the network to carry service instances following changes in the active topology. However, it can be used to limit the reachability of service instances used by the service provider for network management and to restrict service instances carried from one provider network domain to another.

The operation of MSTP within a provider network is independent of the operation of any spanning tree protocol within attached customer networks. This independence is achieved by using the Provider Bridge Group Address (Table 8-1) as the destination address of all MSTP BPDUs transmitted on all Provider Bridge Ports and by setting the restrictedRole and restrictedTen parameters (13.27.64, 13.27.65) for CNPs. Frames received by CNPs and addressed to the Bridge Group Address are subject to service instance selection and relay in the same way as customer data frames.

29 NOTE 2—Customer BPDUs addressed to the Bridge Group Address are conveyed transparently, allowing the customer
30 to use an instance of MSTP or RSTP that is completely independent of the provider network to establish and maintain
31 full and loop-free connectivity of the customer connected networks and services. A customer can also use MVRP to
32 limit the transmission of frames assigned to C-VLANs to the service instances required for C-VLAN connectivity.

1 The operation of MVRP within a provider network is independent of the operation of any configuration
2 protocol within attached customer networks. The Provider Bridge MVRP Address (Table 8-1) is used as the
3 destination address of all MVRPDUs transmitted in support of the MVRP Application. Frames received by
4 CNPs and addressed to an MRP Application Address (Table 10-1) not in use by the S-VLAN component are
5 subject to service instance selection and relay in the same way as customer data frames. The MVRP
6 Administrative Control for each S-VLAN is either “Registration Fixed (New ignored)” or “Registration
7 Forbidden” on all CNPs, so no information is received from any Provider Bridge MVRPDU that has been
8 erroneously transmitted by a customer system.

9 **16.4 Service provider learning of customer end station addresses**

10 Customer data frames for any given MAC Service instance are restricted to that part of the provider network
11 that supports the VLAN topology of the associated S-VLAN as described in 16.3 and are further restricted
12 by learning the source addresses of frames as described in Clause 7 and Clause 8.

13 In a PBN that commonly provides interfaces to each customer at a small fraction of the total number of
14 customer interfaces provided, the requirement for learning customer end station addresses can be much
15 reduced by applying enhanced filtering utility criteria (8.7). In particular, learning can be restricted to the
16 ingress and egress Provider Bridge Ports of each S-VLAN that connects only two customer points of
17 attachment, or to the customer systems attached to those Ports.

18 The “new” flag in MVRP can be used to flush learned addresses on a specific service instance in a PBN in
19 response to a topology change in the Customer Network.

20 **16.5 Detection of connectivity loops through attached networks**

21 The transmission and reception of MSTP BPDUs through CNPs will detect accidental direct connection of
22 those ports, or their interconnection by a network that is transparent to frames with the Provider Bridge
23 Group Address as the destination MAC address. However, a service provider cannot rely on any customer
24 network relaying such frames and should develop a policy and mechanisms to deal with potential data loops
25 that can arise if the attached customer systems do not correctly operate their own instance or instances of a
26 spanning tree protocol.

27 NOTE 1—Use of the restrictedRole parameter at ingress ports ensures that receipt of BPDUs addressed to the Provider
28 Bridge Group Address cannot disrupt internal connectivity within the provider network.

29 NOTE 2—Specification of service provider policies, mechanisms, and heuristics used to detect or minimize the impact
30 of data loops created by customer systems is not addressed by this standard. They can include, but are not limited to,
31 bandwidth limitation, charging policies, detection of the repetitive movement of the apparent location of customer
32 stations, and customer agreement to allow the use of service provider loop detection protocols by not filtering the
33 associated frames.

34 NOTE 3—A data loop is not the only possible cause of excess bandwidth consumption by a given customer of a
35 provider network, and the service provider is usually required to meet service guarantees to other customers irrespective
36 of the cause of the excess bandwidth demand. Data loops are not a unique threat to satisfactory overall network
37 performance. Their distinct characteristic is consumption of discretionary bandwidth without benefiting any customer.
38 The customer that creates the loop can suffer particularly serious network degradation or excess cost as the service
39 provider limits the total bandwidth consumed by that customer. It is therefore in the interests of each customer and the
40 service provider to raise service satisfaction by preventing and detecting loops.

41 Each C-VLAN component within a Provider Edge Bridge implements RSTP, with the enhancements to
42 support loop-free connectivity between CEPs, as specified in 13.41, thus interoperating with customer
43 equipment and spanning tree configurations to support reliable and deterministic prevention of loops, and
44 supporting provision of redundant connectivity.

1 16.6 Network management

2 Management of a Provider Bridge is directly under the control of the service provider. Provider network
3 customers shall not have access to managed objects related to elements of Provider Bridges within the
4 provider network.

17. Management Information Base (MIB)

This clause (Clause 17) contains a complete SMIV2 MIB set for all features of this standard.

17.1 Internet Standard Management Framework

For a detailed overview of the documents that describe the Internet Standard Management Framework, refer to section 7 of IETF RFC 3410 (2002).

Managed objects are accessed via a virtual information store, termed the *Management Information Base* or *MIB*. MIB objects are generally accessed through the Simple Network Management Protocol (SNMP). Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information (SMI). This clause specifies a MIB module that is compliant to the SMIV2, which is described in IETF STD 58, IETF RFC 2578, IETF RFC 2579, and IETF RFC 2580.

Control of the transmission of Fault Alarms, which are Notifications in SNMP, are described by IETF STD 62, IETF RFC 3413, IETF RFC 3417, and IETF RFC 4789. As a consequence of the concentration of control of the reporting of SNMP Notifications in the SNMP-TARGET-MIB and SNMP-NOTIFICATION-MIB of IETF RFC 3413, the variables controlling Fault Alarm notifications require no specific objects in the MIB modules defined in this standard.

17.2 Structure of the MIB

The IEEE 802.1Q MIB set comprises a number of modules. These are summarized in Table 17-1 with the name of each module and references to its specification in this clause (Clause 17) and to the clause that specifies the managed functionality. The table also identifies the amendment or amendment project that introduced that functionality²⁸ (and unless otherwise noted) the MIB module for its management. The definitive specification of both functionality and MIB module is provided by this standard, not by any prior amendment.

Prior revisions of this standard relied on MIB modules defined by the IETF Bridge Working Group. IETF RFC 4663 [B35] describes more details of the IETF transition of responsibility for bridging-related MIB modules from the IETF Bridge MIB Working Group to the IEEE 802.1 Working Group.

The REVISION statements in each MIB module describe its history and development.

Table 17-1—IEEE 802.1Q MIB modules

Module	Structure, relationships, security considerations, module definition	Managed functionality	Initial functionality and MIB specification
IEEE8021-TC-MIB	17.2.1, 17.3.1, 17.4.1, 17.7.1	Textual conventions	IEEE Std 802.1ap
IEEE8021-BRIDGE-MIB	17.2.2, 17.3.2, 17.4.2, 17.7.2	Clause 8	IEEE Std 802.1D, 802.1p, IEEE Std 802.1t. MIB adapted from IETF RFC 4188, IETF RFC 4363
IEEE8021-SPANNING-TREE-MIB	17.2.3, 17.3.3, 17.4.3, 17.7.3	Clause 13	IEEE Std 802.1w. MIB adapted from IETF RFC 4318

²⁸ Functionality added to an IEEE standard is often referred to using an amendment or amendment project designation long after it has been incorporated in a revision of the standard, even if the amendment was not independently published and the functionality has since been revised. This table column is provided as an aid to identifying this standard's specification of that functionality and MIB module.

Table 17-1—IEEE 802.1Q MIB modules (*continued*)

Module	Structure, relationships, security considerations, module definition	Managed functionality	Initial functionality and MIB specification
IEEE8021-Q-BRIDGE-MIB	17.2.4, 17.3.4, 17.4.4, 17.7.4	Clause 8	IEEE Std 802.1Q, IEEE Std 802.1u, IEEE Std 802.1v. MIB adapted from IETF RFC 4363.
IEEE8021-PB-MIB	17.2.5, 17.3.5, 17.4.5, 17.7.5	Clause 16	IEEE Std 802.1ad. MIB in IEEE Std 802.1ap
IEEE8021-MSTP-MIB	17.2.6, 17.3.6, 17.4.6, 17.7.6	Clause 13	IEEE Std 802.1s. MIB in IEEE Std 802.1ap
IEEE8021-CFM-MIB	17.2.7, 17.3.7, 17.4.7, 17.7.7.1	Clause 20	IEEE Std 802.1ag
IEEE8021-CFM-V2-MIB	17.2.7, 17.3.7, 17.4.7, 17.7.7.2	Clause 20	IEEE Std 802.1ag. MIB updated in IEEE Std 802.1ap
IEEE8021-PBB-MIB	17.2.8, 17.3.8, 17.4.8, 17.7.8	Clause 26	IEEE Std 802.1ah. MIB updated in IEEE Std 802.1ap
IEEE8021-DDCFM-MIB	17.2.9, 17.3.9, 17.4.9, 17.7.9	Clause 29	IEEE Std 802.1Qaw
IEEE8021-PBBTE-MIB	17.2.10, 17.3.10, 17.4.10, 17.7.10	25.10	IEEE Std 802.1Qay
IEEE8021-TPMR-MIB	17.2.11, 17.3.11, 17.4.11, 17.7.11	5.13, 5.15	IEEE Std 802.1aj
IEEE8021-FQTSS-MIB	17.2.12, 17.3.12, 17.4.1, 17.7.12	Clause 34	IEEE Std 802.1Qav
IEEE8021-CN-MIB	17.2.13, 17.3.13, 17.4.13, 17.7.13	Clause 30	IEEE Std 802.1Qau
IEEE8021-SRP-MIB	17.2.14, 17.3.14, 17.4.14, 17.7.14	Clause 35	IEEE Std 802.1Qat
IEEE8021-MVRPX-MIB	17.2.15, 17.3.15, 17.4.15, 17.7.15	11.2	IEEE Std 802.1Qbe
IEEE8021-MIRP-MIB	17.2.16, 17.3.16, 17.4.16, 17.7.16	Clause 39	IEEE Std 802.1Qbe
IEEE8021-PFC-MIB	17.2.17, 17.3.17, 17.4.17, 17.7.17	Clause 36	IEEE Std 802.1Qbb
IEEE8021-TEIPS-MIB	17.2.18, 17.3.18, 17.4.18, 17.7.18	26.11	IEEE Std 802.1Qbf
IEEE8021-SPB-MIB	17.2.19, 17.3.19, 17.4.19, 17.7.19	Clause 27, Clause 28, Clause 45	IEEE Std 802.1aq. IEEE Std 802.1Qca added PCR and MIB objects.
IEEE8021-EVB-MIB	17.2.20, 17.3.20, 17.4.20, 17.7.20	5.23, 5.24	IEEE Std 802.1Qbg
IEEE8021-ECMP-MIB	17.2.21, 17.3.21, 17.4.21, 17.7.21	Clause 44, Clause 28	IEEE Std 802.1Qbp
IEEE8021-ST-MIB	17.2.22, 17.3.22, 17.4.22, 17.7.22	8.6, 12.29	IEEE Std 802.1Qbv
IEEE8021-Preemption-MIB	17.2.23, 17.3.23, 17.4.23, 17.7.23	8.6, 12.30	IEEE Std 802.1Qbu
IEEE8021-PSFP-MIB	17.2.24, 17.3.24, 17.4.24, 17.7.24	8.6, 12.31	IEEE Std 802.1Qci
IEEE8021-TSN-REMOTE-MANAGEMENT-MIB	17.2.25, 17.3.25, 17.4.25, 17.7.25	12.32	IEEE Std 802.1Qcc
IEEE8021-PBBN-AA-MIB	17.2.26, 17.3.26, 17.4.26, 17.7.26	Clause 50	IEEE Std 802.1Qcj

1 17.2.1 Structure of the IEEE8021-TC-MIB

2 The IEEE8021-TC-MIB defines textual conventions used throughout the IEEE 802.1Q MIB modules and
3 summarized in Table 17-2. Textual conventions (TCs) originally appeared in each of the MIB modules for
4 IEEE Std 802.1Q produced by the IETF. Many of the original IETF TCs are still used in the various MIB
5 modules for VLAN-Bridge management and have not been imported into this module.

Table 17-2—IEEE8021-TC-MIB structure

IEEE8021-TC-MIB object	Reference
IEEE8021PbbComponentIdentifier	12.3 l)
IEEE8021PbbComponentIdentifierOrZero	12.3 l)
IEEE8021PbbServiceIdentifier	12.16.3, 12.16.5
IEEE8021PbbServiceIdentifierOrUnassigned	12.16.3, 12.16.5
IEEE8021PbbIngressEgress	12.16.3, 12.16.5
IEEE8021PriorityCodePoint	12.6.2
IEEE8021BridgePortNumber	12.3 i), 17.3.2.2
IEEE8021BridgePortNumberOrZero	12.3 i), 17.3.2.2
IEEE8021BridgePortType	12.16.1.1.3 h4), 12.16.2.1, 12.13.1.1, 12.13.1.2
IEEE8021VlanIndex	9.6
IEEE8021VlanIndexOrWildcard	9.6
IEEE8021MstIdentifier	13.8
IEEE8021ServiceSelectorType	—
IEEE8021ServiceSelectorValueOrNone	—
IEEE8021ServiceSelectorValue	—
IEEE8021PortAcceptableFrameTypes	12.10.1.3, 12.13.2.3, 12.13.2.4
IEEE8021PriorityValue	12.13.2.3
IEEE8021PbbTeProtectionGroupId	12.19.2
IEEE8021PbbTeEsp	3.95
IEEE8021PbbTeTSidId	3.287
IEEE8021PbbTeProtectionGroupConfigAdmin	26.10.3.3.4, 26.10.3.3.5, 26.10.3.3.6, 26.10.3.3.7, 12.18.2.1.3d)
IEEE8021PbbTeProtectionGroupActiveRequests	12.18.2.3.2
ieee8021TeipsV2Ipgid	12.24.2
ieee8021TeipsV2IpgConfigAdmin	26.10.3.3.4, 26.10.3.3.5, 26.10.3.3.6, 26.10.3.3.7, 12.14.2.1.2 f) and j)
ieee8021TeipsV2IpgActiveRequests	12.24.2.2.1 b)
ieee8021TeipsV2Segid	26.11.1, 3.124
ieee8021TeipsV2Smpid	26.11.1, 3.234

The IEEE8021-FQTSS-MIB module in 17.7.12 defines additional TEXTUAL-CONVENTIONS that are utilized by the managed objects that support use of the Credit-based shaper algorithm (8.6.8.2, Clause 34).

The textual conventions used by IEEE 802.1Q MIB modules not contained in this TC module, but contained in IETF RFC 4318 and IETF RFC 4363 are as follows:

BridgeId
Timeout
PortList
VlanIdOrAny
VlanIdOrNone
VlanIdOrAnyOrNone

17.2.2 Structure of the IEEE8021-BRIDGE-MIB

This MIB module is based upon the BRIDGE MIB version in IETF RFC 4188 and the P-BRIDGE MIB originally in IETF RFC 4363. While many tables and objects are similar, the IEEE8021-BRIDGE-MIB has been reindexed and rerooted.

Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects. The overall structure and assignment of objects to their subtrees is shown below. Where appropriate, the corresponding IETF RFC 4188 object name, IEEE 802.1D™-2004 [B14] management object name, and reference are also included. If the IETF RFC mapping is missing, it means there is no equivalent in the original IETF MIB modules. If the Managed Object is missing, it means that no managed object is specified in the referenced clause.

Table 17-3 indicates the structure of the IEEE8021-BRIDGE-MIB module as well as showing its relationship to the IETF MIB that is being obsoleted.

Table 17-3—IEEE8021-BRIDGE-MIB structure

IEEE8021-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021BridgeBaseTable	dot1dBase	12.4.1
ieee8021BridgeBaseComponentId*	—	—
ieee8021BridgeBaseBridgeAddress	dot1dBaseBridgeAddress	12.4.1.1.3 a)
ieee8021BridgeBaseNumPorts	dot1dBaseNumPorts	12.4.1.1.3 c)
ieee8021BridgeBaseComponentType	—	12.3 m)
ieee8021BridgeBaseDeviceCapabilities	dot1dDeviceCapabilities	12.10.1.1.3 b)
ieee8021BridgeBaseTrafficClassesEnabled	dot1dTrafficClassesEnabled	—
ieee8021BridgeBaseMmrpEnabledStatus	dot1dGmrpStatus	—
ieee8021BridgeBaseRowStatus	—	—

Table 17-3—IEEE8021-BRIDGE-MIB structure (continued)

IEEE8021-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021BridgeBasePortTable	dot1dBasePortTable	12.4.2
ieee8021BridgeBasePortComponentId*	—	—
ieee8021BridgeBasePort*	dot1dBasePort	12.4.2.1
ieee8021BridgeBasePortIfIndex	dot1dBasePortIfIndex	—
ieee8021BridgeBasePortDelayExceededDiscards	dot1dBasePortDelayExceededDiscards	12.6.1.1.3 f)
ieee8021BridgeBasePortMtuExceededDiscards	dot1dBasePortMtuExceededDiscards	12.6.1.1.3 g)
ieee8021BridgePortCapabilities	dot1dPortCapabilities	12.10.1.1.3 c)
ieee8021BridgeBasePortTypeCapabilities	—	—
ieee8021BridgeBasePortType	—	—
ieee8021BridgeBasePortExternal	—	—
ieee8021BridgeBasePortAdminPointToPoint	dot1dStpPortAdminPointToPoint	12.8.2.1.3 o)
ieee8021BridgeBasePortOperPointToPoint	dot1dStpPortOperPointToPoint	12.8.2.1.3 p)
ieee8021BridgeBasePortName	ifDescr	12.4.2.1
ieee8021BridgeTpPortTable	dot1dTpPortTable	12.4.2
ieee8021BridgeTpPortComponentId*	—	—
ieee8021BridgeTpPort*	dot1dTpPort	—
ieee8021BridgeTpPortMaxInfo	dot1dTpPortMaxInfo	—
ieee8021BridgeTpPortInFrames	dot1dTpPortInFrames	12.6.1.1.3 a)
ieee8021BridgeTpPortOutFrames	dot1dTpPortOutFrames	12.6.1.1.3 d)
ieee8021BridgeTpPortInDiscards	dot1dTpPortInDiscards	12.6.1.1.3 c)
ieee8021BridgePortPriorityTable	dot1dPortPriorityTable	12.6.2
(AUGMENTS ieee8021BridgeBasePortEntry)	(AUGMENTS dot1dBasePortEntry)	—
ieee8021BridgePortDefaultUserPriority	dot1dPortDefaultUserPriority	—
ieee8021BridgePortNumTrafficClasses	dot1dPortNumTrafficClasses	—
ieee8021BridgePortPriorityCodePointSelection	—	12.6.2.6, 12.6.2.7
ieee8021BridgePortUseDEI	—	12.6.2.11, 12.6.2.12
ieee8021BridgePortRequireDropEncoding	—	12.6.2.13, 12.6.2.14
ieee8021BridgePortServiceAccessPrioritySelection	—	12.6.2.15, 12.6.2.16

Table 17-3—IEEE8021-BRIDGE-MIB structure (continued)

IEEE8021-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021BridgeUserPriorityRegenTable	dot1dUserPriorityRegenTable	IEEE Std 802.1AC
ieee8021BridgeUserPriorityRegenComponentId*	—	—
ieee8021BridgeBasePort*	dot1dBasePort	—
ieee8021BridgeUserPriority*	dot1dUserPriority	—
ieee8021BridgeRegenUserPriority	dot1dRegenUserPriority	—
ieee8021BridgeTrafficClassTable	dot1dTrafficClassTable	Table 8-5
ieee8021BridgeTrafficClassComponentId*	—	—
ieee8021BridgeBasePort*	dot1dBasePort	—
ieee8021BridgeTrafficClassPriority*	dot1dTrafficClassPriority	—
ieee8021BridgeTrafficClass	dot1dTrafficClass	—
ieee8021BridgePortOutboundAccessPriorityTable	dot1dPortOutboundAccessPriority Table	Table 8-5
ieee8021BridgePortOutboundAccessPriorityComponentId*	—	—
ieee8021BridgeBasePort*	dot1dBasePort	—
ieee8021BridgePortOutboundAccessPriority*	dot1dPortOutboundAccessPriority	—
ieee8021BridgePortDecodingTable	—	12.6.2
ieee8021BridgePortDecodingComponentId*	—	—
ieee8021BridgePortDecodingPortNum*	—	—
ieee8021BridgePortDecodingPriorityCodePointRow*	—	—
ieee8021BridgePortDecodingPriorityCodePoint*	—	—
ieee8021BridgePortDecodingPriority	—	12.6.2.7, 12.6.2.8
ieee8021BridgePortDecodingDropEligible	—	12.6.2.11, 12.6.2.12
ieee8021BridgePortEncodingTable	—	12.6.2
ieee8021BridgePortEncodingComponentId*	—	—
ieee8021BridgePortEncodingPortNum*	—	—
ieee8021BridgePortEncodingPriorityCodePointRow*	—	—
ieee8021BridgePortEncodingPriorityCodePoint*	—	—
ieee8021BridgePortEncodingDropEligible*	—	—
ieee8021BridgePortEncodingPriority	—	12.6.2.9, 12.6.2.10

Table 17-3—IEEE8021-BRIDGE-MIB structure (continued)

IEEE8021-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021BridgeServiceAccessPriorityTable	—	—
ieee8021BridgeServiceAccessPriorityComponentId*	—	—
ieee8021BridgeServiceAccessPriorityPortNum*	—	—
ieee8021BridgeServiceAccessPriorityReceived*	—	—
ieee8021BridgeServiceAccessPriorityValue	—	12.6.2.15, 12.6.2.16
ieee8021BridgePortMrpTable	dot1dPortGarpTable	12.9
(AUGMENTS ieee8021BridgeBasePortEntry)	(AUGMENTS dot1dBasePortEntry)	12.9.1.1, 12.9.1.2
ieee8021BridgePortMrpJoinTime	dot1dPortGarpJoinTime	—
ieee8021BridgePortMrpLeaveTime	dot1dPortGarpLeaveTime	—
ieee8021BridgePortMrpLeaveAllTime	dot1dPortGarpLeaveAllTime	—
ieee8021BridgePortMmrpTable	dot1dPortGmrpTable	12.11
(AUGMENTS ieee8021BridgeBasePortEntry)	(AUGMENTS dot1dBasePortEntry)	—
ieee8021BridgePortMmrpEnabledStatus	dot1dPortGmrpStatus	—
ieee8021BridgePortMmrpFailedRegistrations	dot1dPortGmrpFailedRegistrations	—
ieee8021BridgePortMmrpLastPduOrigin	dot1dPortGmrpLastPduOrigin	—
ieee8021BridgePortRestrictedGroupRegistration	dot1dPortRestrictedGroupRegistration	12.11.1.3
ieee8021BridgeILanIfTable	—	17.3.2.2
ieee8021BridgeILanIfRowStatus	—	—
ieee8021BridgeDot1dPortTable	—	17.5.3
ieee8021BridgeBasePortComponentId*	—	—
ieee8021BridgeBasePort*	—	—
ieee8021BridgeDot1dPortRowStatus	—	—
ieee8021BridgePortTable	—	12.5.1
ieee8021BridgePhyPort	—	—
ieee8021BridgePhyIfIndex	—	—
8021BridgePhyMacAddress	—	—
ieee8021BridgePhyPortToComponentId	—	—
ieee8021BridgePhyPortToInternalPort	—	—
ieee8021BridgeBaseIfToPortTable	—	17.3.2
ieee8021BridgeBaseIfToPortComponentID	—	—
ieee8021BridgeBaseIfIndexPort	—	—

*This object is an INDEX of the table in which it resides.

1 The IEEE8021-BRIDGE-MIB contains the following object subtrees:

- 2 a) The ieee8021BridgeBase Subtree
3 This subtree contains the objects that are applicable to all types of Bridges.
- 4 b) The ieee8021BridgeTp Subtree
5 This subtree contains objects that describe the entity's state with respect to transparent bridging. If
6 transparent bridging is not supported, this subtree will not be implemented. This subtree is
7 applicable to transparent-only Bridges.
- 8 c) The ieee8021BridgePriority Subtree
9 This subtree contains the objects for configuring and reporting status of Priority-based queuing
10 mechanisms in a Bridge. This includes per Bridge Port user_priority treatment, mapping of
11 user_priority in frames into internal traffic classes, and outbound user_priority and access_priority.
- 12 d) The ieee8021BridgeMrp Subtree
13 This subtree contains the objects for configuring and reporting on operation of the Multiple
14 Registration Protocol (MRP).
- 15 e) The ieee8021BridgeMmrp Subtree
16 This subtree contains the objects for configuring and reporting on operation of the Multiple MAC
17 Registration Protocol (MMRP).
- 18 f) The ieee8021BridgeInternal LAN Subtree
19 This subtree contains the objects for configuring and reporting on operation of the Internal LAN
20 interface. An I-LAN Interface is used to create internal connections between Bridge Ports in an
21 IEEE 802.1 device.

22 The IEEE 802.1D-2004 [B14] management objects in Table 17-4 have not been included in the
23 IEEE8021-BRIDGE-MIB module for the indicated reasons. Note that there is a conformance requirement
24 that the base modules SNMPv2-MIB [RFC3418] and IF-MIB [RFC2863] will be implemented as a default
25 on all Bridges.

Table 17-4—IEEE 802.1D objects not in the IEEE8021-BRIDGE-MIB

IEEE 802.1D object	Disposition
Bridge.BridgeName	Same as sysDescr (SNMPv2-MIB)
Bridge.BridgeUpTime	Same as sysUpTime (SNMPv2-MIB)
Bridge.PortAddresses	Same as ifPhysAddress (IF-MIB)

26 17.2.3 Structure of the IEEE8021-SPANNING-TREE MIB

27 A RSTP MIB originally appeared in IETF RFC 4318. While many tables and objects are similar, the
28 IEEE8021-SPANNING-TREE-MIB has been merged with spanning tree objects from the BRIDGE MIB in
29 IETF RFC 4188 and also reindexed and rerooted with its inclusion in this subclause.

30 This subclause defines an SMIV2 MIB module for managing RSTP capability originally defined by the
31 IEEE Std 802.1t™ and IEEE Std 802.1w™ amendments to IEEE Std 802.1D, 1998 Edition [B13], for

- 1 bridging between LANs. The objects in this MIB are defined to apply both to transparent bridging and to
2 Bridges connected by subnetworks other than LANs.
- 3 The MIB module in this clause is based on IEEE Std 802.1D-2004 [B14], though the references are to this
4 standard.
- 5 Table 17-5 indicates the structure of the IEEE8021-SPANNING_TREE-MIB module as well as showing its
6 relationship to the IETF MIB that is being obsoleted.

Table 17-5—IEEE8021-SPANNING-TREE MIB structure

IEEE8021-SPANNING-TREE MIB table/object	IETF MIB table/object	Reference
ieee8021SpanningTreeTable	dot1dStp	12.8.1
ieee8021SpanningTreeComponentId*	—	—
ieee8021SpanningTreeProtocolSpecification	dot1dStpProtocolSpecification	—
ieee8021SpanningTreePriority	dot1dStpPriority	12.8.1.1.3 a)
ieee8021SpanningTreeTimeSinceTopologyChange	dot1dStpTimeSinceTopologyChange	12.8.1.1.3 b)
ieee8021SpanningTreeTopChanges	dot1dStpTopChanges	12.8.1.1.3 c)
ieee8021SpanningTreeDesignatedRoot	dot1dStpDesignatedRoot	12.8.1.1.3 e)
ieee8021SpanningTreeRootCost	dot1dStpRootCost	12.8.1.1.3 f)
ieee8021SpanningTreeRootPort	dot1dStpRootPort	12.8.1.1.3 g)
ieee8021SpanningTreeMaxAge	dot1dStpMaxAge	12.8.1.1.3 h)
ieee8021SpanningTreeHelloTime	dot1dStpHelloTime	12.8.1.1.3 k)
ieee8021SpanningTreeHoldTime	dot1dStpHoldTime	12.8.1.1.3 m)
ieee8021SpanningTreeForwardDelay	dot1dStpForwardDelay	12.8.1.1.3 i)
ieee8021SpanningTreeBridgeMaxAge	dot1dStpBridgeMaxAge	12.8.1.1.3 j)
ieee8021SpanningTreeBridgeHelloTime	dot1dStpBridgeHelloTime	12.8.1.1.3 k)
ieee8021SpanningTreeBridgeForwardDelay	dot1dStpBridgeForwardDelay	12.8.1.1.3 l)
ieee8021SpanningTreeVersion	dot1dStpVersion	12.8.1.1.3 n)
ieee8021SpanningTreeRstpTxHoldCount	dot1dStpTxHoldCount	12.8.1.1.3 m)
ieee8021SpanningTreePortTable	dot1dStpPortTable	12.8.2
ieee8021SpanningTreePortComponentId*	—	—
ieee8021SpanningTreePort*	dot1dStpPort	12.8.2.1.2 a)
ieee8021SpanningTreePortPriority	dot1dStpPortPriority	12.8.2.1.3 c)
ieee8021SpanningTreePortState	dot1dStpPortState	12.8.2.1.3 b)
ieee8021SpanningTreePortEnable	dot1dStpPortEnable	12.8.2.1.3 m)
ieee8021SpanningTreePortPathCost	dot1dStpPortPathCost	12.8.2.1.3 d)
ieee8021SpanningTreePortDesignatedRoot	dot1dStpPortDesignatedRoot	12.8.2.1.3 e)
ieee8021SpanningTreePortDesignatedCost	dot1dStpPortDesignatedCost	12.8.2.1.3 f)
ieee8021SpanningTreePortDesignatedBridge	dot1dStpPortDesignatedBridge	12.8.2.1.3 g)

Table 17-5—IEEE8021-SPANNING-TREE MIB structure (continued)

IEEE8021-SPANNING-TREE MIB table/object	IETF MIB table/object	Reference
ieee8021SpanningTreePortDesignatedPort	dot1dStpPortDesignatedPort	12.8.2.1.3 h)
ieee8021SpanningTreePortForwardTransitions	dot1dStpPortForwardTransitions	—
ieee8021SpanningTreeRstpPortProtocolMigration	dot1dStpPortProtocolMigration	12.8.2.5
ieee8021SpanningTreeRstpPortAdminEdgePort	dot1dStpPortAdminEdgePort	12.8.2.1.3 k)
ieee8021SpanningTreeRstpPortOperEdgePort	dot1dStpPortOperEdgePort	12.8.2.1.3 l)
ieee8021SpanningTreeRstpPortAdminPathCost	dot1dStpPortAdminPathCost	12.8.2.1.3 d)
ieee8021SpanningTreePortExtensionTable AUGMENTS ieee8021SpanningTreePortEntry	—	12.8.2
ieee8021SpanningTreeRstpPortAutoEdgePort	—	12.8.2.1.3
ieee8021SpanningTreeRstpPortAutoIsolatePort	—	12.8.2.1.3
ieee8021SpanningTreeRstpPortIsolatePort	—	12.8.2.1.3

*This object is an INDEX of the table in which it resides.

1 The IEEE8021-SPANNING-TREE MIB contains the following object subtrees:

- 2 a) The ieee8021SpanningTree Subtree
- 3 This subtree contains the objects that denote the Bridge's state with respect to the spanning tree
- 4 protocol.
- 5 b) The ieee8021SpanningTreePort Subtree
- 6 This subtree contains the objects that denote the Bridge Port's state with respect to the spanning tree
- 7 protocol.

8 For compatibility with the original MIB defined in IETF RFC 4318, the IEEE8021-SPANNING-TREE-MIB
9 continues to use disabled, blocking, listening, learning, forwarding, and broken
10 ieee8021SpanningTreePortStates. The learning and forwarding states correspond exactly to the Learning
11 and Forwarding Port States specified in Clause 12 of this standard. Disabled, blocking, listening, and broken
12 all correspond to the Discarding Port State—while those ieee8021SpanningTreePortStates serve to
13 distinguish reasons for discarding frames, the operation of the Forwarding and Learning processes is the
14 same for all of them. The ieee8021SpanningTreePortState broken represents the failure or unavailability of
15 the Port's MAC as indicated by MAC_Operational FALSE; disabled represents exclusion of the Port from
16 the active topology by management setting of the Administrative Port State to Disabled; blocking represents
17 exclusion of the Port from the active topology by the spanning tree algorithm [computing an Alternate or
18 Backup Port Role (17.7 of IEEE Std 802.1D-2004 [B14])]; listening represents a Port that the spanning tree
19 algorithm has selected to be part of the active topology (computing a Root Port or Designated Port role) but
20 is temporarily discarding frames to guard against loops or incorrect learning.

1 The IEEE 802.1D management objects in Table 17-6 have not been included for the indicated reasons.

Table 17-6—Clause 12 objects not in the IEEE8021-SPANNING-TREE MIB

IEEE 802.1D object	Disposition
SpanningTreeProtocol	
.BridgeIdentifier	Combination of ieee8021SpanningTreePriority and ieee8021BridgeBaseBridgeAddress
SpanningTreeProtocolPort	
.Uptime	Same as ifLastChange (IF-MIB)
.PortIdentifier	Combination of ieee8021SpanningTreePort and ieee8021SpanningTreePortPriority
.DiscardLackOfBuffers	Redundant

2 17.2.4 Structure of the IEEE8021-Q-BRIDGE-MIB

3 A Q-BRIDGE MIB originally appeared in IETF RFC 4363. While many tables and objects are similar, it has
4 been reindexed and rerooted with its inclusion in this subclause.

5 Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects.
6 Where appropriate, the corresponding IETF RFC 4363 object name and IEEE Std 802.1Q management
7 reference is also included.

8 Unlike IEEE Std 802.1D, 1998 Edition [B13], this standard does not define exact syntax for a set of
9 managed objects. The following cross-references indicate only the subclause numbering of the descriptions
10 of management operations from Clause 12.

11 Table 17-7 indicates the structure of the IEEE8021-Q-BRIDGE-MIB module as well as showing its
12 relationship to the IETF MIB that is being obsoleted.

Table 17-7—IEEE8021-Q-BRIDGE MIB structure

IEEE8021-Q-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021QBridgeTable	dot1qBase	12.4
ieee8021QBridgeComponentId*	—	—
ieee8021QBridgeVlanVersionNumber	dot1qVlanVersionNumber	12.10.1.1
ieee8021QBridgeMaxVlanId	dot1qMaxVlanId	9.6
ieee8021QBridgeMaxSupportedVlans	dot1qMaxSupportedVlans	12.10.1.1
ieee8021QBridgeNumVlans	dot1qNumVlans	12.7.1.1
ieee8021QBridgeMvrpEnabledStatus	dot1qGvrpStatus	—
ieee8021QBridgeCVlanPortTable	—	—
ieee8021QBridgeCVlanPortComponentId*	—	—
ieee8021QBridgeCVlanPortNumber	—	—
ieee8021QBridgeCVlanPortRowStatus	—	—
ieee8021QBridgeFdbTable	dot1qFdbTable	12.7.1

Table 17-7—IEEE8021-Q-BRIDGE MIB structure (continued)

IEEE8021-Q-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021QBridgeFdbComponentId*	—	—
ieee8021QBridgeFdbId*	dot1qFdbId	—
ieee8021QBridgeFdbDynamicCount	dot1qFdbDynamicCount	12.7.1.1.3
ieee8021QBridgeFdbLearnedEntryDiscards	—	—
ieee8021QBridgeFdbAgingTime	—	—
ieee8021QBridgeTpFdbTable	dot1qTpFdbTable	12.7.1
ieee8021QBridgeFdbComponentId*	—	—
ieee8021QBridgeFdbId*	—	—
ieee8021QBridgeTpFdbAddress*	dot1qTpFdbAddress	—
ieee8021QBridgeTpFdbPort	dot1qTpFdbPort	—
ieee8021QBridgeTpFdbStatus	dot1qTpFdbStatus	—
ieee8021QBridgeTpGroupTable	dot1qTpGroupTable	12.7.4
ieee8021QBridgeVlanCurrentComponentId*	—	—
ieee8021QBridgeVlanIndex*	—	—
ieee8021QBridgeTpGroupAddress*	dot1qTpGroupAddress	—
ieee8021QBridgeTpGroupEgressPorts	dot1qTpGroupEgressPorts	—
ieee8021QBridgeTpGroupLearnt	dot1qTpGroupLearnt	—
ieee8021QBridgeForwardAllTable	dot1qForwardAllTable	12.7.2, 12.7.7
ieee8021QBridgeForwardAllComponentId*	—	—
ieee8021QBridgeVlanIndex*	—	—
ieee8021QBridgeForwardAllPorts	dot1qForwardAllPorts	—
ieee8021QBridgeForwardAllStaticPorts	dot1qForwardAllStaticPorts	—
ieee8021QBridgeForwardAllForbiddenPorts	dot1qForwardAllForbiddenPorts	—
ieee8021QBridgeForwardUnregisteredTable	dot1qForwardUnregisteredTable	12.7.2, 12.7.7
ieee8021QBridgeForwardUnregisteredComponentId*	—	—
ieee8021QBridgeVlanIndex*	—	—
ieee8021QBridgeForwardUnregisteredPorts	dot1qForwardUnregisteredPorts	—
ieee8021QBridgeForwardUnregisteredStaticPorts	dot1qForwardUnregisteredStaticPorts	—
ieee8021QBridgeForwardUnregisteredForbiddenPorts	dot1qForwardUnregisteredForbiddenPorts	—
ieee8021QBridgeStaticUnicastTable	dot1qStaticUnicastTable	12.7.7, 8.8.1
ieee8021QBridgeFdbComponentId*	—	—
ieee8021QBridgeFdbId*	—	—
ieee8021QBridgeStaticUnicastAddress*	dot1qStaticUnicastAddress	—

Table 17-7—IEEE8021-Q-BRIDGE MIB structure (continued)

IEEE8021-Q-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021QBridgeStaticUnicastReceivePort*	dot1qStaticUnicastReceivePort	—
ieee8021QBridgeStaticUnicastStaticEgressPorts	dot1qStaticUnicastAllowedToGoTo	8.5
ieee8021QBridgeStaticUnicastForbiddenEgressPorts	dot1qStaticUnicastAllowedToGoTo	8.5
ieee8021QBridgeStaticUnicastStorageType	—	—
ieee8021QBridgeStaticUnicastRowStatus	dot1qStaticUnicastStatus	—
ieee8021QBridgeStaticMulticastTable	dot1qStaticMulticastTable	12.7.7, 8.8.1
ieee8021QBridgeVlanCurrentComponentId*	—	—
ieee8021QBridgeVlanIndex*	—	—
ieee8021QBridgeStaticMulticastAddress*	dot1qStaticMulticastAddress	—
ieee8021QBridgeStaticMulticastReceivePort*	dot1qStaticMulticastReceivePort	—
ieee8021QBridgeStaticMulticastStaticEgressPorts	dot1qStaticMulticastStaticEgressPorts	—
ieee8021QBridgeStaticMulticastForbiddenEgressPorts	dot1qStaticMulticastForbiddenEgressPorts	—
ieee8021QBridgeStaticMulticastStorageType	—	—
ieee8021QBridgeStaticMulticastRowStatus	dot1qStaticMulticastStatus	—
ieee8021QBridgeVlan	dot1qVlan	—
ieee8021QBridgeVlanNumDeletes	dot1qVlanNumDeletes	—
ieee8021QBridgeVlanCurrentTable	dot1qVlanCurrentTable	12.10.2
ieee8021QBridgeVlanCurrentComponentId*	—	—
ieee8021QBridgeVlanTimeMark*	dot1qVlanTimeMark	—
ieee8021QBridgeVlanIndex*	dot1qVlanIndex	—
ieee8021QBridgeVlanFdbId	dot1qVlanFdbId	—
ieee8021QBridgeVlanCurrentEgressPorts	dot1qVlanCurrentEgressPorts	12.10.2.1
ieee8021QBridgeVlanCurrentUntaggedPorts	dot1qVlanCurrentUntaggedPorts	12.10.2.1
ieee8021QBridgeVlanStatus	dot1qVlanStatus	—
ieee8021QBridgeVlanCreationTime	dot1qVlanCreationTime	—
ieee8021QBridgeVlanStaticTable	dot1qVlanStaticTable	12.7.5
ieee8021QBridgeVlanStaticComponentId*	—	—
ieee8021QBridgeVlanStaticVlanIndex*	—	—
ieee8021QBridgeVlanStaticName	dot1qVlanStaticName	12.10.2.1
ieee8021QBridgeVlanStaticEgressPorts	dot1qVlanStaticEgressPorts	12.7.7.3, 11.2.3.2.3
ieee8021QBridgeVlanForbiddenEgressPorts	dot1qVlanForbiddenEgressPorts	12.7.7.3, 11.2.3.2.3

Table 17-7—IEEE8021-Q-BRIDGE MIB structure (continued)

IEEE8021-Q-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021QBridgeVlanStaticUntaggedPorts	dot1qVlanStaticUntaggedPorts	12.10.2.1
ieee8021QBridgeVlanStaticRowStatus	dot1qVlanStaticRowStatus	—
ieee8021QBridgeNextFreeLocalVlanTable	dot1qVlan	—
ieee8021QBridgeNextFreeLocalVlanComponentId*	—	—
ieee8021QBridgeNextFreeLocalVlanIndex	dot1qNextFreeLocalVlanIndex	—
ieee8021QBridgePortVlanTable	dot1qPortVlanTable	12.10.1
(AUGMENTS ieee8021BridgeBasePortEntry)	(AUGMENTS dot1dBasePortEntry)	—
ieee8021QBridgePvid	dot1qPvid	12.10.1.1
ieee8021QBridgePortAcceptableFrameTypes	dot1qPortAcceptableFrameTypes	12.10.1.3
ieee8021QBridgePortIngressFiltering	dot1qPortIngressFiltering	12.10.1.4
ieee8021QBridgePortmvrpEnabledStatus	dot1qPortGvrpStatus	—
ieee8021QBridgePortmvrpFailedRegistrations	dot1qPortGvrpFailedRegistrations	—
ieee8021QBridgePortmvrpLastPduOrigin	dot1qPortGvrpLastPduOrigin	—
ieee8021QBridgePortRestrictedVlanRegistration	dot1qPortRestrictedVlanRegistration	11.2.3.2.3, 12.10.1.7
ieee8021QBridgePortVlanStatisticsTable	dot1qPortVlanStatisticsTable	12.1.3
ieee8021BridgeBasePortComponentId*	—	—
ieee8021BridgeBasePort*	dot1dBasePort	—
ieee8021QBridgeVlanIndex*	dot1qVlanIndex	—
ieee8021QBridgeTpVlanPortInFrames	dot1qTpVlanPortInFrames dot1qTpVlanPortHCInFrames	12.6.1.1.3 a)
ieee8021QBridgeTpVlanPortOutFrames	dot1qTpVlanPortOutFrames dot1qTpVlanPortHCOutFrames	12.6.1.1.3 d)
ieee8021QBridgeTpVlanPortInDiscards	dot1qTpVlanPortInDiscards dot1qTpVlanPortHCInDiscards	12.6.1.1.3
ieee8021QBridgeProtocolGroupTable	dot1vProtocolGroupTable	12.10.1
ieee8021QBridgeProtocolGroupComponentId*	—	—
ieee8021QBridgeProtocolTemplateFrameType*	dot1vProtocolTemplateFrameType	12.10.1.7
ieee8021QBridgeProtocolTemplateProtocolValue*	dot1vProtocolTemplateProtocolValue	12.10.1.7
ieee8021QBridgeProtocolGroupId	dot1vProtocolGroupId	12.10.1.7
ieee8021QBridgeProtocolGroupRowStatus	dot1vProtocolGroupRowStatus	—
ieee8021QBridgeProtocolPortTable	dot1vProtocolPortTable	12.10.1
ieee8021BridgeBasePortComponentId*	—	—
ieee8021BridgeBasePort*	dot1dBasePort	—
ieee8021QBridgeProtocolPortGroupId*	dot1vProtocolPortGroupId	12.10.1.7

Table 17-7—IEEE8021-Q-BRIDGE MIB structure (continued)

IEEE8021-Q-BRIDGE-MIB table/object	IETF MIB table/object	Reference
ieee8021QBridgeProtocolPortGroupVid	dot1vProtocolPortGroupVid	12.10.1.7
ieee8021QBridgeProtocolPortRowStatus	dot1vProtocolPortRowStatus	—
ieee8021QBridgeVidXTable	—	12.10.1.8, 12.13.2.1
ieee8021QBridgeBasePortComponentId*	—	—
ieee8021QBridgeBasePort*	—	—
ieee8021QBridgeVidXLocalVid*	—	—
ieee8021QBridgeVidXRelayVid	—	—
ieee8021QBridgeVidXRowStatus	—	—
ieee8021QBridgeEgressVidXV2Table	—	12.10.1.9, 12.13.2.1
ieee8021BridgeBasePortComponentId*	—	—
ieee8021QBridgeBasePort*	—	—
ieee8021QBridgeEgressVidXV2RelayVid*	—	—
ieee8021QBridgeEgressVidXV2LocalVid	—	—
ieee8021QBridgeEgressVidXV2RowStatus	—	—

*This object is an INDEX of the table in which it resides.

1 The IEEE8021-Q-BRIDGE-MIB contains the following object subtrees:

- 2 a) The ieee8021QBridgeBase Subtree
3 This subtree contains the objects that are applicable to all Bridges implementing IEEE Std 802.1Q
4 VLANs.
- 5 b) The ieee8021QBridgeTp Subtree
6 This subtree contains objects that control the operation and report the status of transparent bridging.
7 This includes management of the dynamic FDBs for both unicast and multicast forwarding. This
8 subtree will be implemented by all Bridges that perform destination-address filtering.
- 9 c) The ieee8021QBridgeStatic Subtree
10 This subtree contains objects that control static configuration information for transparent bridging.
11 This includes management of the static entries in the FDBs for both unicast and multicast
12 forwarding.
- 13 d) The ieee8021QBridgeVlan Subtree
14 This subtree contains objects that control configuration and report status of the VLANs known to a
15 Bridge. This includes management of the statically configured VIDs as well as reporting VIDs
16 discovered by other means [e.g., Multiple VLAN Registration Protocol (MVRP)]. It also controls
17 configuration and reports status of per-Port objects relating to VIDs and reports traffic statistics. It
18 also provides for management of the VID to FID allocation.
- 19 e) The ieee8021QBridgeProtocol Subtree
20 This subtree contains objects that control configuration and report status of the mappings from
21 Protocol Templates to Protocol Group Identifiers used for Port-and-Protocol-based VLAN
22 classification.
- 23 f) The ieee8021QBridgeVIDX Subtree
24 This subtree contains objects that control configuration of VID translation for VLAN ports. A single

1 symmetric translation value or an individual ingress and egress translation is allowed depending on
2 the management table supported.

3 The Clause 12 management objects in Table 17-8 have not been included in the
4 IEEE8021-Q-BRIDGE-MIB for the indicated reasons. Note that the assumption is made the base modules
5 SNMPv2-MIB and IF-MIB will be implemented as a default on all Bridges.

Table 17-8—Clause 12 management not in IEEE8021-Q-BRIDGE-MIB

Clause 12 operation	Disposition
Reset Bridge (12.4.1.4)	Not appropriate for this MIB module
Reset VLAN Bridge (12.10.1.5)	Not appropriate for this MIB module
Read Permanent Database (12.7.6.1)	Count rows
Permanent Database Size	Count rows
Number of Static Filtering Entries	Count rows in ieee8021QBridgeStaticUnicastTable + ieee8021QBridgeStaticMulticastTable
Number of Static VLAN Registration Entries	Count rows in ieee8021QBridgeVlanStaticTable
Read Filtering Entry Range (12.7.7.4)	Use GetNext operation
Read Filtering Database (12.7.1.1)	Count rows
Filtering Database Size	Count rows
Number of Dynamic Group Address Entries	Count rows applicable to each FDB in ieee8021QBridgeTpGroupTable

6 17.2.5 Structure of the IEEE8021-PB-MIB

7 The IEEE8021-PB-MIB provides objects to configure and manage Provider Bridges.

8 Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects.

9 Where appropriate, the corresponding Clause 12 management reference is also included.

¹ Table 17-9 indicates the structure of the IEEE8021-PB-MIB module.

Table 17-9—IEEE8021-PB-MIB structure

IEEE8021-PB-MIB table/object	Reference
ieee8021PbCVidRegistrationTable	12.13.2.1, 12.13.2.2
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbCVidRegistrationCVid*	—
ieee8021PbCVidRegistrationSVid	—
ieee8021PbCVidRegistrationUntaggedPep	—
ieee8021PbCVidRegistrationUntaggedCep	—
ieee8021PbCVidRegistrationRowStatus	—
ieee8021PbEdgePortTable	12.13.2.3, 12.13.2.4
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbEdgePortSVid*	—
ieee8021PbEdgePortPVID	—
ieee8021PbEdgePortDefaultUserPriority	—
ieee8021PbEdgePortAcceptableFrameTypes	—
ieee8021PbEdgePortEnableIngressFiltering	—
ieee8021PbServicePriorityRegenerationTable	12.13.2.5, 12.13.2.6
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbServicePriorityRegenerationSVid*	—
ieee8021PbServicePriorityRegenerationReceivedPriority*	—
ieee8021PbServicePriorityRegenerationRegeneratedPriority	—
ieee8021PbCnpTable	12.13.2
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbCnpCComponentId	—
ieee8021PbCnpSVid	—
ieee8021PbCnpRowStatus	—

Table 17-9—IEEE8021-PB-MIB structure (continued)

IEEE8021-PB-MIB table/object	Reference
ieee8021PbPnpTable	12.13.2
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbPnpRowStatus	—
ieee8021PbCepTable	12.13.2
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbCepCComponentId	—
ieee8021PbCepCepPortNumber	—
ieee8021PbCepRowStatus	—
ieee8021PbRcapTable	12.13.3
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbRcapSComponentId	—
ieee8021PbRcapRcapPortNumber	—
ieee8021PbRcapRowStatus	—
ieee8021PbInternalInterfaceTable	12.13.3
ieee8021BridgeBasePortComponentId*	—
ieee8021BridgeBasePort*	—
ieee8021PbIiExternalSVid*	—
ieee8021PbIiInternalPortNumber	—
ieee8021PbIiInternalPortType	—
ieee8021PbIiInternalSVid	—
ieee8021PbIiRowStatus	—

*This object is an INDEX of the table in which it resides.

1 17.2.6 Structure of the IEEE8021-MSTP-MIB

² The IEEE8021-MSTP-MIB provides objects to configure and manage multiple spanning trees.

³ Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects.

⁴ Where appropriate, the corresponding Clause 12 management reference is also included.

¹ Table 17-10 indicates the structure of the IEEE8021-MSTP-MIB module.

Table 17-10—IEEE8021-MSTP-MIB structure

IEEE8021-MSTP-MIB table/object	Reference
ieee8021MstpCistTable	12.8.1.1, 12.8.1.3
ieee8021MstpCistComponentId*	—
ieee8021MstpCistBridgeIdentifier	—
ieee8021MstpCistTopologyChange	—
ieee8021MstpCistRegionalRootIdentifier	13.26.3
ieee8021MstpCistPathCost	13.9 d), 13.11
ieee8021MstpCistMaxHops	13.26.4
ieee8021MstpTable	12.8.1.2, 12.8.1.4, 12.12.3.2, 12.12.1
ieee8021MstpComponentId*	—
ieee8021MstpId*	—
ieee8021MstpBridgeId	13.26.2
ieee8021MstpTimeSinceTopologyChange	13.23
ieee8021MstpTopologyChanges	13.23
ieee8021MstpTopologyChange	—
ieee8021MstpDesignatedRoot	13.26.3
ieee8021MstpRootPathCost	13.26.3
ieee8021MstpRootPort	13.26.9
ieee8021MstpBridgePriority	13.26.3
ieee8021MstpVids0	—
ieee8021MstpVids1	—
ieee8021MstpVids2	—
ieee8021MstpVids3	—
ieee8021MstpRowStatus	—
ieee8021MstpCistPortTable	12.8.2.1, 12.8.2.3, 12.8.2.5
ieee8021MstpCistPortComponentId*	—
ieee8021MstpCistPortNum*	—
ieee8021MstpCistPortUptime	—
ieee8021MstpCistPortAdminPathCost	13.27.25
ieee8021MstpCistPortDesignatedRoot	13.27.20
ieee8021MstpCistPortTopologyChangeAck	13.27.72
ieee8021MstpCistPortAdminHelloTime	12.8.2.1

Table 17-10—IEEE8021-MSTP-MIB structure (continued)

IEEE8021-MSTP-MIB table/object	Reference
ieee8021MstpCistPortAdminEdgePort	13.27.1
ieee8021MstpCistPortOperEdgePort	13.27.44
ieee8021MstpCistPortOperHelloTime	12.8.2.1
ieee8021MstpCistPortMacEnabled	12.8.2.1.3 m)
ieee8021MstpCistPortMacOperational	12.8.2.1.3 n)
ieee8021MstpCistPortRestrictedRole	13.27.63
ieee8021MstpCistPortRestrictedTcn	13.27.65
ieee8021MstpCistPortRole	—
ieee8021MstpCistPortDisputed	13.27.22
ieee8021MstpCistPortCistRegionalRootId	13.9 c), 13.11
ieee8021MstpCistPortCistPathCost	13.27.25
ieee8021MstpCistPortProtocolMigration	13.27.22
ieee8021MstpCistPortEnableBPDURx	13.27.31
ieee8021MstpCistPortEnableBPDUTx	13.27.32
ieee8021MstpCistPortPseudoRootId	13.27.33
ieee8021MstpCistPortIsL2Gp	13.27.31
ieee8021MstpPortTable	12.8.2.2, 12.8.2.4
ieee8021MstpPortComponentId*	—
ieee8021MstpPortMstpId*	—
ieee8021MstpPortNum*	—
ieee8021MstpPortUptime	—
ieee8021MstpPortState	13.38
ieee8021MstpPortPriority	13.27.47
ieee8021MstpPortPathCost	13.27.25
ieee8021MstpPortDesignatedRoot	13.27.20
ieee8021MstpPortDesignatedCost	13.27.20
ieee8021MstpPortDesignatedBridge	13.27.20
ieee8021MstpPortDesignatedPort	13.27.20
ieee8021MstpPortRole	—
ieee8021MstpPortDisputed	13.27.22
ieee8021MstpFidToMstiV2Table	12.12.2.2
ieee8021MstpFidToMstiV2ComponentId*	—
ieee8021MstpFidToMstiV2Fid*	—

Table 17-10—IEEE8021-MSTP-MIB structure (continued)

IEEE8021-MSTP-MIB table/object	Reference
ieee8021MstpFidToMstiV2MstpId*	—
ieee8021MstpVlanV2Table	12.12.3.1
ieee8021MstpVlanV2ComponentId*	—
ieee8021MstpVlanV2Id*	—
ieee8021MstpVlanV2MstpId*	—
ieee8021MstpConfigIdTable	12.12.3.3, 12.12.3.4
ieee8021MstpConfigIdComponentId*	—
ieee8021MstpConfigIdFormatSelector	13.8:1
ieee8021MstpConfigurationName	13.8:2
ieee8021MstpRevisionLevel	13.8:3
ieee8021MstpConfigurationDigest	13.8:4
ieee8021MstpCistPortExtensionTable AUGMENTS ieee8021MstpCistPortEntry	12.8.2
ieee8021MstpCistPortAutoEdgePort	12.8.2.1.3
ieee8021MstpCistPortAutoIsolatePort	12.8.2.1.3

*This object is an INDEX of the table in which it resides.

1 17.2.7 Structure of the IEEE8021-CFM-MIB

2 Subclause 12.14 of this document defines the information model associated with this standard in a
3 protocol-independent manner. Table 17-11 and Table 17-12 describe the relationship between the SMIV2
4 objects defined in the MIB module in 17.4.9, the variables defined in Clause 20, and the
5 protocol-independent objects defined in 12.14.

6 Note that there are actually two MIB modules—IEEE8021-CFM-MIB and IEEE8021-CFM-V2-MIB. The
7 former contains all original and current tables as well as deprecated tables. The latter contains reindexed
8 tables (to support PBB per Clause 26) and a complete conformance clause that lists all the current tables.

1 Table 17-11 lists only the current tables (i.e., the deprecated tables are not listed) in the
2 IEEE8021-CFM-MIB module, and Table 17-12 lists the reindexed tables in the IEEE8021-CFM-V2-MIB
3 module (that replace the deprecated tables of IEEE8021-CFM-MIB).

Table 17-11—IEEE8021-CFM-MIB structure

Variable	IEEE8021-CFM-MIB table/object	Reference
Maintenance Domain managed object (12.14.5)	dot1agCfmMdTable	—
	dot1agCfmMdIndex *	12.14.1.1.3 a2)
	dot1agCfmMdFormat, dot1agCfmMdName	12.14.1.2.2 a)
mdLevel (20.7.1)	dot1agCfmMdMdLevel	12.14.5.1.3 b)
	dot1agCfmMdMhfCreation	12.14.5.1.3 c)
	dot1agCfmMdMhfIdPermission	12.14.5.1.3 d)
	fault alarm controlled by IETF RFC 3413	12.14.5.1.3 e)
	dot1agCfmMdIndex	12.14.5.1.3 f)
Maintenance Association managed object (12.14.6)	dot1agCfmMaNetTable, dot1agCfmMaMepListTable	—
	dot1agCfmMaIndex *	12.14.6.1.2 a)
	dot1agCfmMaNetFormat, dot1agCfmMaNetName	12.14.5.3.2 b)
	dot1agCfmVlanVid, dot1agCfmMaCompNumberOfVids	12.14.6.1.3 b)
	dot1agCfmMaCompMhfCreation	12.14.6.1.3 c)
	dot1agCfmMaCompIdPermission	12.14.6.1.3 d)
CCMinterval (20.8.1)	dot1agCfmMaNetCcmInterval	12.14.6.1.3 e)
	fault alarm controlled by IETF RFC 3413	12.14.6.1.3 f)
	dot1agCfmMaMepListIdentifier	12.14.6.1.3 g)

Table 17-11—IEEE8021-CFM-MIB structure (*continued*)

Variable	IEEE8021-CFM-MIB table/object	Reference
MEP managed object (12.14.7)	dot1agCfmMepTable	—
	dot1agCfmMdIndex* dot1agCfmMaIndex* dot1agCfmMepIdentifier*	12.14.7.1.2 a)
	dot1agCfmMepIfIndex	12.14.7.1.3 b)
	dot1agCfmMepDirection	12.14.7.1.3 c)
	dot1agCfmMepPrimaryVid	12.14.7.1.3 d)
MEPactive (20.9.1)	dot1agCfmMepActive	12.14.7.1.3 e)
20.37	dot1agCfmMepFngState	12.14.7.1.3 f)
CCIenabled (20.10.1)	dot1agCfmMepCciEnabled	12.14.7.1.3 g)
	dot1agCfmMepCcmLtmPriority	12.14.7.1.3 h)
	dot1agCfmMepMacAddress	12.14.7.1.3 i)
	fault alarm controlled by IETF RFC 3413	12.14.7.1.3 j)
lowestAlarmPri (20.9.5)	dot1agCfmMepLowPrDef	12.14.7.1.3 k)
fngAlarmTime (20.35.3)	dot1agCfmMepFngAlarmTime	12.14.7.1.3 l)
fngResetTime (20.35.4)	dot1agCfmMepFngResetTime	12.14.7.1.3 m)
highestDefect (20.35.9)	dot1agCfmMepHighestPrDefect	12.14.7.1.3 n)
someRDId defect (20.35.7)	dot1agCfmMepDefects	12.14.7.1.3 o)
someMACstatusDefect (20.35.6)		12.14.7.1.3 p)
someRMEPCCMdefect (20.35.5)		12.14.7.1.3 q)
errorCCMdefect (20.21.3)		12.14.7.1.3 r)
xconCCMdefect (20.23.3)		12.14.7.1.3 s)
errorCCMlastFailure (20.21.2)	dot1agCfmMepErrorCcmLastFailure	12.14.7.1.3 t)
xconCCMlastFailure (20.23.2)	dot1agCfmMepXconCcmLastFailure	12.14.7.1.3 u)
CCMsequenceErrors (20.16.12)	dot1agCfmMepCcmSequenceErrors	12.14.7.1.3 v)
CCIscntCCMs (20.10.2)	dot1agCfmMepCciScntCcms	12.14.7.1.3 w)
nextLBMtransID (20.30.2)	dot1agCfmMepNextLbmTransId	12.14.7.1.3 x)
	dot1agCfmMepLbrIn	12.14.7.1.3 y)
	dot1agCfmMepLbrInOutOfOrder	12.14.7.1.3 z)
	dot1agCfmMepLbrBadMsdu	12.14.7.1.3 aa)
nextLTMtransID (20.41.1)	dot1agCfmMepLtmNextSeqNumber	12.14.7.1.3 ab)
	dot1agCfmMepUnexpLtrIn	12.14.7.1.3 ac)
	dot1agCfmMepLbrOut	12.14.7.1.3 ad)
	dot1agCfmMepPbbTeCanReportPbbTePresence	12.14.7.1.3 af)

Table 17-11—IEEE8021-CFM-MIB structure (continued)

Variable	IEEE8021-CFM-MIB table/object	Reference
	dot1agCfmMepPbbTeTrafficMismatchDefect	12.14.7.1.3 ah)
	dot1agCfmMepPbbTeLbmReverseVid	12.14.7.3.2 f)
	dot1agCfmMepPbbTeLtmReverseVid	12.14.7.4.2 e)
	dot1agCfmMepPbbTeMismatchAlarm	12.14.7.1.3 ag)
	dot1agCfmMepPbbTeMismatchDefect	12.14.7.1.3 ai)
	dot1agCfmMepPbbTeMismatchSinceReset	12.14.7.1.3 aj)
Transmit Loopback Messages (12.14.7.3)	dot1agCfmMepTable, dot1agCfmMepTransmitLbmStatus	—
	dot1agCfmMdIndex* dot1agCfmMaIndex* dot1agCfmMepIdentifier*	12.14.7.3.2 a)
	dot1agCfmMepTransmitLbmDestMacAddress, dot1agCfmMepTransmitLbmDestMepId, dot1agCfmMepTransmitLbmDestIsMepId	12.14.7.3.2 b)
	dot1agCfmMepTransmitLbmMessages	12.14.7.3.2 c)
	dot1agCfmMepTransmitLbmDataTlv	12.14.7.3.2 d)
	dot1agCfmMepTransmitLbmVlanPriority, dot1agCfmMepTransmitLbmVlanDropEnable	12.14.7.3.2 e)
	dot1agCfmMepTransmitLbmResultOK	12.14.7.3.3 a)
	dot1agCfmMepTransmitLbmSeqNumber	12.14.7.3.3 b)
Transmit Linktrace Message (12.14.7.4)	dot1agCfmMepTable	—
	dot1agCfmMdIndex* dot1agCfmMaIndex* dot1agCfmMepIdentifier*	12.14.7.4.2 a)
	dot1agCfmMepTransmitLtmFlags	12.14.7.4.2 b)
	dot1agCfmMepTransmitLtmTargetMacAddress, dot1agCfmMepTransmitLtmTargetMepId, dot1agCfmMepTransmitLtmTargetIsMepId	12.14.7.4.2 c)
	dot1agCfmMepTransmitLtmTtl	12.14.7.4.2 d)
	dot1agCfmMepTransmitLtmResult	12.14.7.4.3 a)
	dot1agCfmMepTransmitLtmSeqNumber	12.14.7.4.3 b)
	dot1agCfmMepTransmitLtmEgressIdentifier	12.14.7.4.3 c)

Table 17-11—IEEE8021-CFM-MIB structure (*continued*)

Variable	IEEE8021-CFM-MIB table/object	Reference
Read Linktrace Reply (12.14.7.5)	dot1agCfmLtrTable	—
	dot1agCfmMdIndex* dot1agCfmMaIndex* dot1agCfmMepIdentifier*	12.14.7.5.2 a)
	dot1agCfmLtrSeqNumber*	12.14.7.5.2 b)
	dot1agCfmLtrReceiveOrder*	12.14.7.5.2 c)
ltrReplyTTL (20.41.2.2)	dot1agCfmLtrTtl	12.14.7.5.3 b)
ltrFlags (20.41.2.1)	dot1agCfmLtrForwarded	12.14.7.5.3 c)
	dot1agCfmLtrTerminalMep	12.14.7.5.3 d)
ltrLastEgressId (20.41.2.3)	dot1agCfmLtrLastEgressIdentifier	12.14.7.5.3 e)
ltrNextEgressId (20.41.2.4)	dot1agCfmLtrNextEgressIdentifier	12.14.7.5.3 f)
ltrRelayAction (20.41.2.5)	dot1agCfmLtrRelay	12.14.7.5.3 g)
ltrIngressAction (20.41.2.6)	dot1agCfmLtrIngress	12.14.7.5.3 k)
ltrIngressAddress (20.41.2.7)	dot1agCfmLtrIngressMac	12.14.7.5.3 l)
ltrIngressPortIdSubtype (20.41.2.8)	dot1agCfmLtrIngressPortIdSubtype	12.14.7.5.3 m)
ltrIngressPortId (20.41.2.9)	dot1agCfmLtrIngressPortId	12.14.7.5.3 n)
ltrEgressAction (20.41.2.10)	dot1agCfmLtrEgress	12.14.7.5.3 o)
ltrEgressAddress (20.41.2.11)	dot1agCfmLtrEgressMac	12.14.7.5.3 p)
ltrEgressPortIdSubtype (20.41.2.12)	dot1agCfmLtrEgressPortIdSubtype	12.14.7.5.3 q)
ltrEgressPortId (20.41.2.13)	dot1agCfmLtrEgressPortId	12.14.7.5.3 r)
ltrOrgSpecTlv (20.41.2.15)	dot1agCfmLtrOrganizationSpecificTlv	12.14.7.5.3 s)
ltrSenderIdTlv (20.41.2.14)	dot1agCfmLtrChassisIdSubtype	12.14.7.5.3 h)
	dot1agCfmLtrChassisId	12.14.7.5.3 i)
	dot1agCfmLtrManAddressDomain dot1agCfmLtrManAddress	12.14.7.5.3 j)

Table 17-11—IEEE8021-CFM-MIB structure (continued)

Variable	IEEE8021-CFM-MIB table/object	Reference
Read MEP Database (12.14.7.6)	dot1agCfmMepDbTable	—
	dot1agCfmMdIndex* dot1agCfmMaIndex* dot1agCfmMepIdentifier*	12.14.7.6.2 a)
	dot1agCfmMepDbRMepIdentifier*	12.14.7.6.2 b)
20.20	dot1agCfmMepDbRMepState	12.14.7.6.3 b)
	dot1agCfmMepDbRMepFailedOkTime	12.14.7.6.3 c)
rMEPmacAddress (20.19.7)	dot1agCfmMepDbMacAddress	12.14.7.6.3 d)
rMEPlastRDI and rMEPlastRDI[i] (20.19.2)	dot1agCfmMepDbRdi	12.14.7.6.3 e)
rMEPlastPortState (20.19.3)	dot1agCfmMepDbPortStatusTlv	12.14.7.6.3 f)
rMEPlastInterfaceStatus (20.19.4)	dot1agCfmMepDbInterfaceStatusTlv	12.14.7.6.3 g)
rMEPlastSenderId (20.19.5)	dot1agCfmMepDbChassisIdSubtype	12.14.7.6.3 h)
	dot1agCfmMepDbChassisId	
	dot1agCfmMepDbManAddressDomain dot1agCfmMepDbManAddress	
	dot1agCfmMepDbRMepIsActive	12.14.7.1.3 ae)
Transmit MEP Fault Alarm (12.14.7.7)	dot1agCfmFaultAlarm	—
	dot1agCfmMepHighestPrDefect	12.14.7.7.2 b), 12.14.7.7.2 c)

*This object is an INDEX of the table in which it resides.

Table 17-12—IEEE8021-CFM-V2-MIB structure

Variable	IEEE-CFM-V2-MIB table/object	Reference
CFM Stack managed object (12.14.2)	ieee8021CfmStackTable	—
	ieee8021CfmStackIfIndex*	12.14.2.1.2 a)
	ieee8021CfmStackServiceSelectorType*	12.14.2.1.2 d)
	ieee8021CfmStackServiceSelectorOrNone*	12.14.2.1.2 d)
	ieee8021CfmStackMdLevel*	12.14.2.1.2 b)
	ieee8021CfmStackDirection*	12.14.2.1.2 c)
	ieee8021CfmStackMdIndex	12.14.2.1.3 b)
	ieee8021CfmStackMaIndex	12.14.2.1.2 c)
	ieee8021CfmStackMepId	12.14.2.1.3 d)
	ieee8021CfmStackMacAddress	12.14.2.1.3 e)

Table 17-12—IEEE8021-CFM-V2-MIB structure (continued)

Variable	IEEE-CFM-V2-MIB table/object	Reference
Default MD Level managed object (12.14.3)	ieee8021CfmDefaultMdTable	—
	ieee8021CfmDefaultMdComponentId* ieee8021CfmDefaultMdPrimarySelectorType* ieee8021CfmDefaultMdPrimarySelector*	12.14.3.1.3 a), 12.14.3.2.2 a)
	ieee8021CfmDefaultMdStatus	12.14.3.1.3 b)
	ieee8021CfmDefaultMdLevel	12.14.3.2.2 b)
	ieee8021CfmDefaultMdDefMhfCreation ieee8021CfmDefaultMdMhfCreation	12.14.3.1.3 d)
	ieee8021CfmDefaultMdDefIdPermission ieee8021CfmDefaultMdIdPermission	12.14.3.1.3 e)
Configuration Error List managed object (12.14.4)	ieee8021CfmConfigErrorListTable	—
	ieee8021CfmConfigErrorListSelectorType* ieee8021CfmConfigErrorListSelector*	12.14.4.1.2 a)
	ieee8021CfmConfigErrorListIfIndex*	12.14.4.1.2 b)
	ieee8021CfmConfigErrorListErrorType	12.14.4.1.3 b)
Maintenance Association managed object (12.14.6)	ieee8021CfmVlanTable	12.14.3.1.3 a), 12.14.3.2.2 a), 12.14.5.3.2 c), 12.14.6.1.3 b)
	ieee8021CfmVlanComponentId*	—
	ieee8021CfmVlanSelector*	—
	ieee8021CfmVlanRowStatus	—
Maintenance Association managed object (12.14.6)	ieee8021CfmMaCompTable	—
	ieee8021CfmMaComponentId* dot1agCfmMdIndex* dot1agCfmMaIndex*	12.14.6.1.2 a)
	ieee8021CfmMaCompPrimarySelectorType ieee8021CfmMaCompPrimarySelectorOrNone	—
	ieee8021CfmMaNetFormat ieee8021CfmMaNetName	12.14.5.3.2 b)
	ieee8021CfmMaCompNumberOfVids	12.14.6.1.3 b)
	ieee8021CfmMaCompMhfCreation	12.14.6.1.3 c)
	ieee8021CfmMaCompIdPermission	12.14.6.1.3 d)
CCMinterval (20.8.1)	ieee8021CfmMaNetCcmInterval	12.14.6.1.3 e)
	controlled by IETF RFC 3413	12.14.6.1.3 f)

*This object is an INDEX of the table in which it resides.

17.2.8 Structure of the IEEE8021-PBB-MIB

The IEEE8021-PBB-MIB provides objects to configure and manage PBBs.

Objects in this MIB module are arranged based on their description in Clause 12. Where appropriate in the table, the corresponding Clause 12 management reference is included.

Table 17-13 indicates the structure of the IEEE8021-PBB-MIB module.

Table 17-13—IEEE8021-PBB-MIB structure

Variable	IEEE8021-PBB-MIB table/group	Reference
Backbone Edge Bridge Configuration	ieee8021PbbBackboneEdgeBridgeObjects	12.16.1.1, 12.16.1.2
	ieee8021PbbBackboneEdgeBridgeAddress	—
	ieee8021PbbBackboneEdgeBridgeName	—
	ieee8021PbbNumberOfComponents	—
	ieee8021PbbNumberOfBComponents	—
	ieee8021PbbNumberOfBebPorts	—
	ieee8021PbbNextAvailablePipIfIndex	—
Virtual Instance Port Configuration	ieee8021PbbVipTable	12.16.2.1
	ieee8021BridgeBasePortComponentId*	—
	ieee8021BridgeBasePort*	—
	ieee8021PbbVipPipIfIndex	—
	ieee8021PbbVipISid	—
	ieee8021PbbVipDefaultDstBMAC	—
	ieee8021PbbVipRowStatus	—
I-SID to VIP Cross Reference	ieee8021IsidToVipTable	12.16.3.1, 12.16.3.2
	ieee8021PbbISidToVipISid*	—
	ieee8021PbbISidToVipComponentId	—
	ieee8021PbbISidToVipPort	—

Table 17-13—IEEE8021-PBB-MIB structure (*continued*)

Variable	IEEE8021-PBB-MIB table/group	Reference
Provider Instance Port Configuration	ieee8021PbbPipTable	12.16.4.1, 12.16.4.2
	ieee8021PbbPipIfIndex *	—
	ieee8021PbbPipBMACAddress	—
	ieee8021PbbPipName	—
	ieee8021PbbPipIComponentId	—
	ieee8021PbbPipVipMap	—
	ieee8021PbbPipVipMap1	—
	ieee8021PbbPipVipMap2	—
	ieee8021PbbPipVipMap3	—
	ieee8021PbbPipVipMap4	—
	ieee8021PbbPipRowStatus	—
Provider Instance Port Priority	ieee8021PbbPipPriorityTable	12.16.4.1, 12.16.4.2
	(AUGMENTS ieee8021PbbPipEntry)	—
	ieee8021PbbPipPriorityCodePointSelection	12.16.4.5, 12.16.4.6
	ieee8021PbbPipUseDEI	12.16.4.11, 12.16.4.12
	ieee8021PbbPipRequireDropEncoding	12.16.4.13, 12.16.4.14
Provider Instance Port Traffic Class	ieee8021PbbPipTrafficClassTable	—
	ieee8021PbbPipIfIndex *	—
	ieee8021PbbPipTrafficClassPriority *	—
	ieee8021PbbPipTrafficClass	—
PIP Priority Code Point Decoding Table	ieee8021PbbPipDecodingTable	12.16.4.7, 12.16.4.8
	ieee8021PbbPipIfIndex *	—
	ieee8021PbbPipDecodingPriorityCodePointRow *	—
	ieee8021PbbPipDecodingPriorityCodePoint *	—
	ieee8021PbbPipDecodingPriority	—
	ieee8021PbbPipDecodingDropEligible	—

Table 17-13—IEEE8021-PBB-MIB structure (continued)

Variable	IEEE8021-PBB-MIB table/group	Reference
PIP Priority Code Point Encoding Table	ieee8021PbbPipEncodingTable	12.16.4.9, 12.16.4.10
	ieee8021PbbPipIfIndex*	—
	ieee8021PbbPipEncodingPriorityCodePointRow*	—
	ieee8021PbbPipEncodingPriorityCodePoint*	—
	ieee8021PbbPipEncodingDropEligible*	—
	ieee8021PbbPipEncodingPriority	—
VIP to PIP Cross Reference	ieee8021PbbVipToPipMappingTable	12.16.4.3, 12.16.4.4
	ieee8021BridgeBasePortComponentId*	—
	ieee8021BridgeBasePort*	—
	ieee8021PbbVipToPipMappingPipIfIndex	—
	ieee8021PbbVipToPipMappingStorageType	—
	ieee8021PbbVipToPipMappingRowStatus	—
Customer Backbone Port Configuration	ieee8021PbbCBPServiceMappingTable	12.16.5.1, 12.16.5.2
	ieee8021BridgeBasePortComponentId*	—
	ieee8021BridgeBasePort*	—
	ieee8021PbbCBPServiceMappingBackboneSid*	—
	ieee8021PbbCBPServiceMappingBVid	—
	ieee8021PbbCBPServiceMappingDefaultBackboneDest	—
	ieee8021PbbCBPServiceMappingLocalSid	—
	ieee8021PbbCBPServiceMappingRowStatus	—
Customer Backbone Port creation/deletion	ieee8021PbbCbpTable	17.5.3.3.4
	ieee8021BridgeBasePortComponentId*	—
	ieee8021BridgeBasePort*	—
	ieee8021PbbCbpRowStatus	—

*This object is an INDEX of the table in which it resides.

1 17.2.9 Structure of the IEEE8021-DDCFM-MIBs

2 The IEEE8021-DDCFM-MIB provides objects to configure and manage the DDCFM capabilities.

1 Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects
2 as follows:

- 3 a) RR subtree: this subtree contains the objects that are applicable to Bridges that support DDCFM's
4 RR function.
- 5 b) RFM Receiver subtree: this subtree contains the objects that are applicable to Bridges or end stations
6 that support DDCFM's RFM Receiver function.
- 7 c) DR subtree: this subtree contains the objects that are applicable to Bridges that support DDCFM's
8 DR function.
- 9 d) SFM Originator subtree: this subtree contains the objects that are applicable to Bridges or end
10 stations that support DDCFM's SFM Originator function.

11 Table 17-14 indicates the structure of the IEEE8021-DDCFM-MIB module.

Table 17-14—IEEE8021-DDCFM-MIB structure

Variable	IEEE8021-DDCFM-MIB Table/Group	Reference
	ieee8021DdcfmStackTable	DDCFM Stack managed object (12.17.1)
	ieee8021DdcfmStackIfIndex	12.17.1.1.2 a)
	ieee8021DdcfmStackRrMdLevel	12.17.1.1.3 b1)
	ieee8021DdcfmStackRrDirection	
	ieee8021DdcfmStackRFMreceiverMdLevel	12.17.1.1.3 b2)
	ieee8021DdcfmStackDrMdLevel	12.17.1.1.3 b3)
	ieee8021DdcfmStackDrVlanIdOrNone	
	ieee8021DdcfmStackSFMOriginatorMdLevel	12.17.1.1.3 b4)
	ieee8021DdcfmStackSFMOriginatorVlanIdOrNone	
	ieee8021DdcfmStackSFMOriginatorDirection	
	ieee8021DdcfmRrTable	Reflection Responder managed object (12.17.2)
	ieee8021DdcfmRr	—
	ieee8021DdcfmRrIfIndex	12.17.2.1.2 a1)
	ieee8021DdcfmRrMdIndex	12.17.2.1.2 a2)
	ieee8021DdcfmRrDirection	12.17.2.1.2 a3)
	ieee8021DdcfmRrPrimaryVlanIdOrNone	12.17.2.2.2 b1)
	ieee8021DdcfmRrFilter	12.17.2.2.2 b2)
	ieee8021DdcfmRrSamplingInterval	12.17.2.2.2 b3)
	ieee8021DdcfmRrTargetAddress	12.17.2.2.2 b4)
	ieee8021DdcfmRrContinueFlag	12.17.2.2.2 b5)
	ieee8021DdcfmRrDuration	12.17.2.2.2 b7)

Table 17-14—IEEE8021-DDCFM-MIB structure (continued)

Variable	IEEE8021-DDCFM-MIB Table/Group	Reference
	ieee8021DdcfmRrDurationInTimeFlag	12.17.2.2.2 b6)
	ieee8021DdcfmRrVlanPriority ieee8021DdcfmRrVlanDropEligible	12.17.2.2.2 b9)
	ieee8021DdcfmRrFloodingFlag	12.17.2.2.2 b10)
	ieee8021DdcfmRrTruncationFlag	12.17.2.2.2 b.11
ReflectionRequest (29.3.1.15)	ieee8021DdcfmRrActivationStatus	12.18.1.2.3 b12)
RRwhile (29.3.1.9)	ieee8021DdcfmRrRemainDuration	12.17.2.3.3 b13)
nextRFMtransID (29.3.1.16)	ieee8021DdcfmRrNextRfmTransID	12.17.2.3.3 b14)
	ieee8021DdcfmRFMReceiverTable	RFM Receiver managed object (12.17.3)
	ieee8021DdcfmRFMReceiver	—
	ieee8021DdcfmRfmReceiverIfIndex ieee8021DdcfmRfmReceiverMdIndex	12.18.2.1.2 a2)
	ieee8021DdcfmDrTable	Decapsulator Responder managed object (12.17.4)
	ieee8021DdcfmDr	—
	ieee8021DdcfmDrIfIndex ieee8021DdcfmDrMdIndex ieee8021DdcfmDrVlanIdOrNone	12.17.3.1.2 a2)
	ieee8021DdcfmDrSourceAddressStayFlag	12.17.4.3.2 b1)
	ieee8021DdcfmDrSfmOriginator	12.17.4.3.2 b2)
	ieee8021DdcfmDrFloodingFlag	12.17.4.3.2 b3)
DRtime (29.3.8.2)	ieee8021DdcfmDrDuration	12.17.4.3.2 b4)

Table 17-14—IEEE8021-DDCFM-MIB structure (continued)

Variable	IEEE8021-DDCFM-MIB Table/Group	Reference
DRactive (29.3.8.4)	ieee8021DdcfmDrActivationStatus	12.17.3.2.3 b6)
DRwhile (29.3.8.1)	ieee8021DdcfmDrRemainDuration	12.17.3.2.3 b7)
SFMsequenceErrors (29.3.8.7)	ieee8021DdcfmDrSFMsequenceErrors	12.17.4.2.3 b8)
	ieee8021DdcfmSoTable	SFM Originator managed object (12.17.5)
	ieee8021DdcfmSFMOrganizer	—
	ieee8021DdcfmSfmIfIndex ieee8021DdcfmSoMdIndex ieee8021DdcfmSoVlanIdOrNone ieee8021DdcfmSoDirection	12.17.4.1.2 a2)
	ieee8021DdcfmSoDrMacAddress	12.17.5.4.3 b2)
	ieee8021DdcfmSoDuration	12.17.5.4.3 b3)
	ieee8021DdcfmSoActivationStatus	12.17.4.2.3 b4)
	ieee8021DdcfmSoRemainDuration	12.17.4.2.3 b5)

1 17.2.10 Structure of the IEEE8021-PBBTE-MIB

2 The IEEE8021-PBBTE-MIB provides objects to configure and manage PBBs that support Traffic
3 Engineering.

4 Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects.
5 Where appropriate, the corresponding Clause 12 management reference is also included.

6 Table 17-15 shows the mapping of the PBB-TE Clause 12 managed objects to the tables and columns of the
7 PBB-TE MIB that model those managed objects.

Table 17-15—IEEE8021-PBBTE-MIB structure

Variable	IEEE8021- PBBTE-MIB table/object	Reference
—	ieee8021PbbTeProtectionGroupListTable	—
—	ieee8021PbbTeProtectionGroupListGroupId	12.18.1
—	ieee8021PbbTeProtectionGroupListWorkingMA	12.18.1.1.3 a)
—	ieee8021PbbTeProtectionGroupListProtectionMA	12.18.1.1.3 b)
—	ieee8021PbbTeMASHaredGroupTable	—
—	ieee8021PbbTeMASHaredGroupId	12.18.1.1.3 c)
—	ieee8021PbbTeTesiTable	—
—	ieee8021PbbTeTesiComponent	12.16.5.3.2 a)
—	ieee8021PbbTeTesiBridgePort	12.16.5.3.2 b)
—	ieee8021PbbTeTeSiEspTable	—

Table 17-15—IEEE8021-PBBTE-MIB structure (continued)

Variable	IEEE8021- PBBTE-MIB table/object	Reference
—	ieee8021PbbTeTeSiEspEspIndex	12.16.5.3.2 c)
—	ieee8021PbbTeTeSiEspEsp	12.16.5.3.2 c)
—	ieee8021PbbTeProtectionGroupConfigTable	—
—	ieee8021PbbTeProtectionGroupConfigState	12.18.2.1.3 d)
—	ieee8021PbbTeProtectionGroupConfigCommandStatus	12.18.2.1.3 d)
—	ieee8021PbbTeProtectionGroupConfigCommandLast	12.18.2.1.3 b)
—	ieee8021PbbTeProtectionGroupConfigAdmin	12.18.2.3.2 b)
—	ieee8021PbbTeProtectionGroupConfigActiveRequests	12.18.2.1.3 d)
WTRwhile (26.10.3.2.1)	ieee8021PbbTeProtectionGroupConfigWTR	12.18.2.1.3 e)
HoldOffWhile (26.10.3.2.2)	ieee8021PbbTeProtectionGroupConfigHoldOff	12.18.2.1.3 f)
—	ieee8021PbbTeProtectionGroupISidTable	—
—	ieee8021PbbTeProtectionGroupISidGroupId	12.18.2.1.3 b)
—	ieee8021PbbTeBridgeStaticForwardAnyUnicastTable	—
—	ieee8021PbbTeBridgeStaticForwardAnyUnicastVlanIndex	8.8.1
—	ieee8021PbbTeBridgeStaticForwardAnyUnicastEgressPorts	8.8.1
—	ieee8021pbbTeBridgeStaticForwardAnyUnicastForbiddenPorts	8.8.1

1 This standard defines a TE-SID as an implementation-dependent identifier used to refer to a TESI. However,
2 TE-SIDs are used externally by management to refer to these TESIs; in particular they are used to create
3 MAs that monitor the health of a TESI. Thus, while the identifier itself is implementation dependent, a
4 canonical external representation of the identifiers is needed for the purpose of MIB definition. The textual
5 convention `IEEE8021PbbTeTSidId` is used to represent TE-SID values externally. This convention
6 defines TE-SIDs to be a simple integer of local significance to a particular Bridge component for the
7 purpose of MIB management.

8 A TESI consists of a set of ESPs. Each ESP is identified by a 3 tuple consisting of an
9 <ESP-DA,ESP-SA,ESP-VID>. These tuples are represented by the textual convention
10 `IEEE8021PbbTeEsp`, which is a fixed length octet string containing the two MAC addresses along with
11 the VID value.

12 TESIs themselves are represented by the `ieee8021PbbTeTeSiEspTable`. This table has five columns.
13 Two are indices and consist of the TE-SID and an index into the list of ESPs. The remaining columns consist
14 of the ESP itself along with a storage type and row status. The storage type indicates if the value defined by
15 this row of the MIB persists across system restarts. The row status column is used by the management
16 station to add, delete, activate, and deactivate rows from this table. Consider Figure 26-7 of this standard.

The `ieee8021PbbTeTeSiEspTable` for that switch would have the information shown in Table 17-16, assuming no other TESIs were configured. In the example shown in Table 17-16, two TESIs are present in the system. The first has a single ESP, while the second has two ESPs.

Table 17-16—Example of `ieee8021PbbTeTeSiEspTable`

ID	EspIndex	Esp	StorageType	RowStatus
1	1	<DA3, SA1, VID-1>	nonVolatile	Active
2	1	<DA3, SA2, VID-1>	nonVolatile	Active
2	2	<DA3, SA3, VID-2>	nonVolatile	Active

17.2.10.1 Using the MIB to create MAs associated with TESIs

In order to support protection switching, PBB-TE defines a mechanism to allow redundant TESIs to carry the same traffic. The implementation of protection switching requires a mechanism to monitor the health of this TESI. The CFM MA is used as this mechanism.

Configuring an MA requires adding a row to the CFM MIB's `StackTable`. Two columns are used in the `StackTable` to define the data stream upon which the MA operates. These are the `IEEE8021ServiceSelectorType` column and the `IEEE8021ServiceSelectorValue` column. The first column defines how the value in the second column is to be interpreted. If the value of the first column is set to `tesid(3)`, then the second column is interpreted as the external representation of a TE-SID.

Thus, TESIs, and their corresponding identifiers, configured in the `ieee8021PbbTeTeSiEspTable` provide the TE-SID values that are used as parameters to rowcreate operations for the `CFMStackTable` to create MAs that monitor the health of TESIs in a PBB-TE network. Creating entries in the `CFMStackTable`, in turn, defines MAID values, represented as unsigned integers, that are used subsequently to configure the protected service.

17.2.10.2 Using the MIB to create protection groups

A TE protection group is a group of two TESIs referred to as WORKING and PROTECTION between a pair of CBPs. The TE protection group carries a set of backbone service instances between these CBPs and in the event of failure of one of the TESIs, the other TESI is used to transport the backbone service instances.

The TE protection groups are created by adding rows to the `ieee8021PbbTeProtectionGroupListTable`. Each entry in the table contains two columns indicating the MAs associated with the TESIs used to carry the traffic and a `RowStatus` column used to managed the creation and deletion of the TE protection group. The MAs are referred to by MAID and can be found in the `StackTable` of the CFM MIB managed objects.

17.2.10.3 Using the MIB to query and configure protection groups

Each configuration group has status and configuration information that can be managed. This information is located in the `ieee8021PbbTeProtectionGroupConfigTable`. This table allows management stations to determine which TESI is being used to transmit traffic and it also allows for management commands to switch the traffic between the two TESIs. There are also optional parameters that configure the time to trigger a switchover and the time to restore after a failure condition has cleared.

17.2.10.4 Using the MIB to prevent erroneous forwarding in the PBB-TE network

Normally in a bridged network, a frame with an unknown unicast destination address is flooded to an appropriate set of ports determined by the frame's VID. PBB-TE networks do not operate this way. A frame with an unknown unicast destination MAC address whose VID is an ESP-VID is discarded. Item a) 3) of 8.8.1 takes care of this. This clause allows for a new kind of filtering entry. This filtering entry applies to all frames with individual, as opposed to group or broadcast, destination addresses to which no other filtering entry applies. By associating a drop action with this filtering action, the relay function will cause all frames that arrive with a VID belonging to the set of PBB-TE VIDs with an unknown DA to be dropped. The table that sets up these filtering entries is called the `ieee8021PbbTeBridgeStaticForwardAnyUnicastTable`.

17.2.11 Structure of the TPMR MIB

The TPMR MIB is organized into the following MIB groups:

- TPMR Configuration
- TPMR Port Configuration
- TPMR Port Counters
- TPMR Port Discard Details
- TPMR MSP Configuration
- TPMR MSP Statistics

Clause 12 defines the information model associated with this standard in a protocol-independent manner. Table 17-17 describes the relationship between the SMIV2 objects defined in this MIB module and the protocol-independent objects defined in Clause 12.

Table 17-17—IEEE8021-TPMR-MIB structure

Managed object definition	IEEE8021-TPMR-MIB table/object	Reference
TPMR name TPMR uptime	system (SNMPv2-MIB) sysName sysUpTime	12.19.1.1.1.3:a and 12.19.1.1.2.2:a 12.19.1.1.1.3:c
TPMR port name TPMR port type TPMR port MAC address	ifTable (IF-MIB) ifName ifType ifPhysAddress	12.19.1.2.1.3:a and 12.19.1.2.2.2:b 12.19.1.2.1.3:b 12.19.1.1.1.3:b,2
TPMR port number TPMR port management MAC address TPMR port management MAC address forwarding	ieee8021TpmrPortTable ieee8021TpmrPortNumber ieee8021TpmrPortMgmtAddr ieee8021TpmrPortMgmtAddrForwarding	12.19.1.1.1.3:b,1 12.19.1.1.1.3:b,3 12.19.1.2.1.3:c

Table 17-17—IEEE8021-TPMR-MIB structure (continued)

Managed object definition	IEEE8021-TPMR-MIB table/object	Reference
TPMR port Rx Frames TPMR port Rx Octets TPMR port Forwarded Frames TPMR port Discarded Frames TPMR port Discarded Frames Queue Full TPMR port Discarded Frames Lifetime TPMR port Discarded Frames Error	ieee8021TpmrPortStatsTable AUGMENTS ieee8021TpmrPortTable ieee8021TpmrPortStatsRxFrames ieee8021TpmrPortStatsRxOctets ieee8021TpmrPortStatsFramesForwarded ieee8021TpmrPortStatsFramesDiscarded ieee8021TpmrPortStatsFramesDiscardedQueueFull ieee8021TpmrPortStatsFramesDiscardedLifetime ieee8021TpmrPortStatsFramesDiscardedError	12.19.3.1.1.3:a 12.19.3.1.1.3:b 12.19.3.1.1.3:d 12.19.3.1.1.3:c 12.19.3.1.1.3:e 12.19.3.1.1.3:f 12.19.3.1.1.3:g
TPMR port discard details: source address TPMR port discard details: error reason	ieee8021TpmrPortDiscardDetailsTable ieee8021BridgeBasePortComponentId* ieee8021TpmrPortNumber* ieee8021TpmrPortDiscardDetailsIndex* ieee8021TpmrPortDiscardDetailsSource ieee8021TpmrPortDiscardDetailsReason	12.19.3.1.1.3:h 12.19.3.1.1.3:h
TPMR MSP link notify TPMR MSP link notify wait TPMR MSP link notify retry TPMR MSP MAC notify TPMR MSP MAC notify time TPMR MSP MAC notify recover	ieee8021TpmrMspTable AUGMENTS ieee8021TpmrPortTable ieee8021TpmrMspLinkNotify ieee8021TpmrMspLinkNotifyWait ieee8021TpmrMspLinkNotifyRetry ieee8021TpmrMspMacNotify ieee8021TpmrMspMacNotifyTime ieee8021TpmrMspMacRecoverTime	12.19.4.1.1.3:a and 12.19.4.1.2.2:b 12.19.4.1.1.3:b and 12.19.4.1.2.2:c 12.19.4.1.1.3:c and 12.19.4.1.2.2:d 12.19.4.1.1.3:d and 12.19.4.1.2.2:e 12.19.4.1.1.3:e and 12.19.4.1.2.2:f 12.19.4.1.1.3:f and 12.19.4.1.2.2:g
TPMR MSP tx acks TPMR MSP tx add notifications TPMR MSP tx add conformations TPMR MSP tx loss notifications TPMR MSP tx loss conformations TPMR MSP rx acks TPMR MSP rx add notifications TPMR MSP rx add conformations TPMR MSP rx loss notifications TPMR MSP rx loss conformations TPMR MSP add events TPMR MSP loss events TPMR MSP MAC status notifications	ieee8021TpmrMspStatsTable AUGMENTS ieee8021TpmrPortTable ieee8021TpmrMspStatsTxAcks ieee8021TpmrMspStatsTxAddNotifications ieee8021TpmrMspStatsTxAddConformations ieee8021TpmrMspStatsTxLossNotifications ieee8021TpmrMspStatsTxLossConformations ieee8021TpmrMspStatsRxAcks ieee8021TpmrMspStatsRxAddNotifications ieee8021TpmrMspStatsRxAddConformations ieee8021TpmrMspStatsRxLossNotifications ieee8021TpmrMspStatsRxLossConformations ieee8021TpmrMspStatsAddEvents ieee8021TpmrMspStatsLossEvents ieee8021TpmrMspStatsMacStatusNotifications	12.19.4.1.3.3:a 12.19.4.1.3.3:b 12.19.4.1.3.3:c 12.19.4.1.3.3:d 12.19.4.1.3.3:e 12.19.4.1.3.3:f 12.19.4.1.3.3:g 12.19.4.1.3.3:h 12.19.4.1.3.3:i 12.19.4.1.3.3:j 12.19.4.1.3.3:k 12.19.4.1.3.3:l 12.19.4.1.3.3:m

*This object is an INDEX of the table in which it resides.

17.2.12 Structure of the IEEE8021-FQTSS-MIB

The IEEE8021-FQTSS-MIB provides objects to configure and manage those aspects of a VLAN Bridge that are related to Forwarding and Queuing Enhancements for time-sensitive streams (FQTSS) (see Clause 34).

Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects. Where appropriate, the corresponding Clause 12 management reference is also included.

Table 17-18 indicates the structure of the IEEE8021-FQTSS-MIB module.

Table 17-18—IEEE8021-FQTSS-MIB structure

IEEE8021-FQTSS-MIB table/object	References
<i>ieee8021FqtssBap subtree</i>	
ieee8021FqtssBapTable	Bandwidth Availability Parameter Table, 12.20.1, 34.3
ieee8021FqtssBAPTrafficClass *	Traffic class
ieee8021FqtssDeltaBandwidth	deltaBandwidth, 12.20.1, 34.3
ieee8021FqtssOperIdleSlopeMs	operIdleSlope, 12.20.1, 34.3
ieee8021FqtssOperIdleSlopeLs	operIdleSlope, 12.20.1, 34.3
ieee8021FqtssAdminIdleSlopeMs	adminIdleSlope, 12.20.1, 34.3
ieee8021FqtssAdminIdleSlopeLs	adminIdleSlope, 12.20.1, 34.3
<i>ieee8021FqtssMappings subtree</i>	
ieee8021FqtssTxSelectionAlgorithmTable	Transmission Selection Algorithm Table, 12.20.2, 8.6.8
ieee8021FqtssTrafficClass *	Traffic class
ieee8021FqtssTxSelectionAlgorithmID	Transmission selection algorithm, 12.20.2, 8.6.8
ieee8021FqtssSrpRegenOverrideTable	SRP domain boundary port priority regeneration override table, 12.20.3, 35.1.4, 6.9.4
ieee8021FqtssSrClassPriority *	Received priority
ieee8021FqtssPriorityRegenOverride	Regenerated priority, 12.20.3, 6.9.4
ieee8021FqtssSrpBoundaryPort	SRPdomainBoundaryPort, 12.20.3, 35.1.4

*This object is an INDEX of the table in which it resides.

17.2.13 Structure of the IEEE8021-CN-MIB

Table 17-19 describes the relationship between the SMIV2 objects defined in the IEEE8021-CN-MIB module (17.7.13) and the Congestion Notification variables and managed objects defined in Clause 12 and Clause 32.

Table 17-19—IEEE8021-CN-MIB structure

Clause 32 table or variable	Clause 12/ Clause 32 reference	IEEE8021-CN-MIB table/object
CN component managed object, CN component variables	12.21.1 32.2	ieee8021CnGlobalTable
cngMasterEnable	32.2.1	ieee8021CnGlobalMasterEnable
cngCnmTransmitPriority	32.2.2	ieee8021CnGlobalCnmTransmitPriority
cngDiscardedFrames	32.2.3	ieee8021CnGlobalDiscardedFrames
cngErroredPortList	32.2.4	ieee8021CnErroredPortTable
CN component priority managed object, congestion notification per-CNPV variables	12.21.2 32.3	ieee8021CnCompntPriTable
cncpDefModeChoice	32.3.1	ieee8021CnComPriDefModeChoice
cncpAlternatePriority	32.3.2	ieee8021CnComPriAlternatePriority
cncpAutoAltPri	32.3.3	ieee8021CnComPriAutoAltPri
cncpAdminDefenseMode	32.3.4	ieee8021CnComPriAdminDefenseMode
cncpCreation	32.3.5	ieee8021CnComPriCreation
cncpLldpInstanceChoice	32.3.6	ieee8021CnComPriLldpInstanceChoice
cncpLldpInstanceSelector	32.3.7	ieee8021CnComPriLldpInstanceSelector
CN Port priority managed object, CND defense per-Port per-CNPV variables	12.21.2 32.4	ieee8021CnPortPriTable
cnpdDefModeChoice	32.4.1	ieee8021CnPortPriDefModeChoice
cnpdAdminDefenseMode	32.4.2	ieee8021CnPortPriAdminDefenseMode
cnpdAutoDefenseMode	32.4.3	ieee8021CnPortPriAutoDefenseMode
cnpdLldpInstanceChoice	32.4.4	ieee8021CnPortPriLldpInstanceChoice
cnpdLldpInstanceSelector	32.4.5	ieee8021CnPortPriLldpInstanceSelector
cnpdAlternatePriority	32.4.6	ieee8021CnPortPriAlternatePriority

Table 17-19—IEEE8021-CN-MIB structure (continued)

Clause 32 table or variable	Clause 12/ Clause 32 reference	IEEE8021-CN-MIB table/object
Congestion Point managed object, CP variables	12.21.4 32.8	ieee8021CnCpTable
cpMacAddress	32.8.1	ieee8021CnCpMacAddress
cpId	32.8.2	ieee8021CnCpIdentifier
cpQSp	32.8.3	ieee8021CnCpQueueSizeSetPoint
cpW	32.8.6	ieee8021CnCpFeedbackWeight
cpSampleBase	32.8.11	ieee8021CnCpMinSampleBase
cpDiscardedFrames	32.8.12	ieee8021CnCpDiscardedFrames
cpTransmittedFrames	32.8.13	ieee8021CnCpTransmittedFrames
cpTransmittedCnms	32.8.14	ieee8021CnCpTransmittedCnms
cpMinHeaderOctets	32.8.15	ieee8021CnCpMinHeaderOctets
(None)^a	(none)	ieee8021CnCpidToInterfaceTable
Reaction Point port priority managed object, RP per-Port per-CNPV variables	12.21.5 32.10	ieee8021CnRpPortPriTable
rpppMaxRps	32.10.1	ieee8021CnRpPortPriMaxRps
rpppCreatedRps	32.10.2	ieee8021CnRpPortPriCreatedRps
rpppRpCentiseconds	32.10.3	ieee8021CnRpPortPriCentiseconds
Reaction Point group managed object, RP group variables	12.21.6 32.11	ieee8021CnRpGroupTable
rpgEnable	32.11.1	ieee8021CnRpgEnable
rpgTimeReset	32.11.2	ieee8021CnRpgTimeReset
rpgByteReset	32.11.3	ieee8021CnRpgByteReset
rpgThreshold	32.11.4	ieee8021CnRpgThreshold
rpgMaxRate	32.11.5	ieee8021CnRpgMaxRate
rpgAiRate	32.11.6	ieee8021CnRpgAiRate
rpgHaiRate	32.11.7	ieee8021CnRpgHaiRate
rpgGd	32.11.8	ieee8021CnRpgGd
rpgMinDecFac	32.11.9	ieee8021CnRpgMinDecFac
rpgMinRate	32.11.10	ieee8021CnRpgMinRate

^a This table is an artifact of SNMP, required to find an entry in the ieee8021CnCpTable, given a CPID (32.8.2, 33.4.4).

17.2.14 Structure of the IEEE8021-SRP-MIB

The IEEE8021-SRP-MIB provides objects to configure and manage those aspects of a VLAN Bridge that are related to the Stream Reservation Protocol (SRP).

Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects. Where appropriate, the corresponding Clause 12 management reference is also included.

Table 17-20 indicates the structure of the IEEE8021-SRP-MIB module.

Table 17-20—IEEE8021-SRP-MIB structure

IEEE8021-SRP-MIB table/object	References
<i>ieee8021SrpConfiguration subtree</i>	
ieee8021SrpBridgeBaseTable	SRP control and status information for a Bridge. AUGMENTS ieee8021BridgeBaseEntry.
ieee8021SrpBridgeBaseMsrpEnabledStatus	Is MSRP enabled on this device? True or False. 12.22.1, 35.2.1.4 d).
ieee8021SrpBridgeBaseMsrpTalkerPruning	talkerPruning, 12.22.1, 35.2.1.4 b).
ieee8021SrpBridgeBaseMsrpMaxFanInPorts	msrpMaxFanInPorts, 12.22.1, 35.2.1.4 f).
ieee8021SrpBridgeBaseMsrpLatencyMaxFrameSize	msrpLatencyMaxFrameSize, 12.22.1, 35.2.1.4 g).
ieee8021SrpBridgePortTable	SRP Control and Status information for each port on the Bridge. AUGMENTS ieee8021BridgeBasePortEntry.
ieee8021SrpBridgePortMsrpEnabledStatus	Is MSRP enabled on this port? True or False. 12.22.2, 35.2.1.4 e).
ieee8021SrpBridgePortMsrpFailedRegistrations	How many failed registrations have there been on this port, 10.7.12.1, 12.22.2.
ieee8021SrpBridgePortMsrpLastPduOrigin	Source MAC address of last MSRPDU received on this port, 10.7.12.2, 12.22.2.
ieee8021SrpBridgePortSrPvid	Default VID for Streams on this port, Table 9-2, Table 12-15, 12.22.2, 35.2.1.4 i).
<i>ieee8021SrpLatency subtree</i>	
ieee8021SrpLatencyTable	Maximum port latency per traffic class, 12.22.3, 35.2.2.8.6.
ieee8021SrpTrafficClass	Traffic class (Table index).
ieee8021SrpPortTcLatency	Maximum port latency for the associated traffic class, 12.22.3, 35.2.1.4 a), 35.2.2.8.6.
<i>ieee8021SrpStreams subtree</i>	
ieee8021SrpStreamTable	Components that define the characteristics of a Stream., 12.22.4, 35.2.2.8.
ieee8021SrpStreamID	StreamID (Table index), 12.22.4, 35.2.2.8.2.
ieee8021SrpStreamDestinationAddress	Stream destination MAC address, 12.22.4, 35.2.2.8.3 a).
ieee8021SrpStreamVlanID	VID for Stream (0=default), 12.22.4, 35.2.2.8.3 b).

Table 17-20—IEEE8021-SRP-MIB structure (*continued*)

IEEE8021-SRP-MIB table/object	References
ieee8021SrpStreamTspecMaxFrameSize	Maximum frame size sent by Talker, 12.22.4, 35.2.2.8.4 a).
ieee8021SrpStreamTspecMaxIntervalFrames	Maximum number of frames sent per class measurement interval, 12.22.4, 35.2.2.8.4 b), 34.4.
ieee8021SrpStreamDataFramePriority	The Priority Code Point (PCP) value that the data Stream will be tagged with, 12.22.4, 35.2.2.8.5 a).
ieee8021SrpStreamRank	Emergency/nonemergency Rank associated with the Stream, 12.22.4, 35.2.2.8.5 b).
<i>ieee8021SrpReservations subtree</i>	
ieee8021SrpReservationsTable	A table containing Stream attribute registrations per port, 12.22.5, 35.2.4.
ieee8021SrpReservationStreamId	StreamID (Table index), 12.22.5, 35.2.2.8.2.
ieee8021SrpReservationDirection	Talker or Listener (Table index), 12.22.5, 35.2.1.2.
ieee8021SrpReservationDeclarationType	Advertise or Failed for Talkers. Asking Failed, Ready, or Ready Failed for Listeners. 12.22.5, 35.2.1.3.
ieee8021SrpReservationAccumulatedLatency	Latency at ingress port for Talker registrations, or latency at end of egress media for Listener Declarations, 12.22.5, 35.2.2.8.6.
ieee8021SrpReservationFailureSystemId	system ID of system that changed Talker Advertise to Talker Failed, 12.22.5, 35.2.2.8.7 a).
ieee8021SrpReservationFailureCode	Failure Code associated with system that changed Talker Advertise to Talker Failed, 12.22.5, 35.2.2.8.7 b).
ieee8021SrpReservationDroppedStreamFrames	Number of stream data frames (not MSRPDU) that have been dropped for this stream on this port, 12.22.5, 35.2.5.1.
ieee8021SrpReservationStreamAge	Number of seconds since reservation was established, 12.22.5, 35.2.1.4 c).

1 17.2.15 Structure of the IEEE8021-MVRPX-MIB

2 The IEEE8021-MVRPX-MIB extends the Bridge Port VLAN table to add three variables to control whether
3 the MVRP Participant on each Port operates as a Full Participant or a New-Only Participant (10.6), how
4 New messages are treated, and whether to transmit the 0 VID. Table 17-21 describes the relationship
5 between the SMIV2 objects defined in the MIB module in 17.7.15 and the variables and managed objects
6 defined in Clause 12.

Table 17-21—IEEE8021-MVRPX-MIB structure

IEEE8021-MVRPX-MIB table/object	Reference
ieee8021MvrpxPortTable	12.9.2
(AUGMENTS ieee8021BridgeBasePortEntry)	—
ieee8021MvrpxPortNewOnly	12.9.2.1.3, 12.9.2.2.2
ieee8021MvrpxPortMvrpNewPropagated	12.7.7.1.2 d), 12.7.7.3.3 d)
ieee8021MvrpxPortXmitZero	12.9.2.1.3 c), 12.9.2.2.2 e)

7 17.2.16 Structure of the IEEE8021-MIRP-MIB

8 The IEEE8021-MIRP-MIB provides objects to configure the Multiple I-SID Registration Protocol defined
9 in Clause 39. Table 17-22 describes the relationship between the SMIV2 objects defined in the MIB module
10 in 17.7.13 and the variables and managed objects defined in Clause 12.

Table 17-22—IEEE8021-MIRP-MIB structure

IEEE8021-MIRP-MIB table/object	Reference
ieee8021MirpV2PortTable	12.7.7
(AUGMENTS ieee8021BridgeBasePortEntry)	—
ieee8021MirpV2PortEnabledStatus	12.7.7.1, 12.7.7.2
ieee8021PbbBackboneEdgeBridgeObjects	12.6.1.1, 12.16.1.2
ieee8021PbbMirpEnableStatus	12.16.1.1.3 i)
ieee8021PbbMirpBvid	12.16.1.1.3 j), 12.16.1.2.2 c)
ieee8021PbbMirpDestSelector	Table 8-1, 12.16.1.1.3 k), 12.16.1.2.2 d)
ieee8021PbbMirpPnpEnable	12.16.1.1.3 j), 12.16.1.2.2 c)
ieee8021PbbMirpPnpPortNumber	12.16.1.1.3 j), 12.16.1.2.2 c)

1 17.2.17 Structure of the IEEE8021-PFC-MIB

2 Table 17-23 describes the relationship between the SMIV2 objects defined in the PFC-MIB module (17.7.13)
3 and the variables and managed objects defined in Clause 12 and Clause 36.

Table 17-23—PFC-MIB structure

Variable	Reference	IEEE8021-PFC-MIB table/object
PFC Interface Table	17.7.13	ieee8021PfcIfTable
(AUGMENTS ifEntry)	—	—
PFCLinkDelayAllowance	12.22.6	ieee8021PfcLinkDelayAllowance
PFCRequests	12.22.6	ieee8021PfcRequests
PFCIndications	12.22.6	ieee8021PfcIndications

4 17.2.18 Structure of the IEEE8021-TEIPS-MIB

5 The IEEE8021-TEIPS-MIB provides objects to configure and manage IPS in PBB-TE Region (26.11).

6 Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of related objects.
7 Where appropriate, the corresponding Clause 12 management reference is also included.

8 Table 17-24 shows the mapping of the IPS Clause 12 managed objects to the tables and columns of the
9 TEIPS MIB that model those managed objects.

Table 17-24—IEEE8021-TEIPS MIB structure

Variable	IEEE8021-TEIPS-MIB table/object	Reference
	ieee8021TeipsV2IpgTable	
	ieee8021TeipsV2Ipgid	12.24.1.1.3 a)
	ieee8021TeipsV2IpgWorkingMA	12.24.1.1.3 b)
	ieee8021TeipsV2IpgProtectionMA	12.24.1.1.3 c)
	ieee8021TeipsV2IpgWorkingPortNumber	12.24.2.2.1 b)
	ieee8021TeipsV2IpgProtectionPortNumber	12.24.2.1.1 c)
	ieee8021TeipsV2TesiTable	
	ieee8021TeipsV2TesiId	12.24.2.1.3 f)
	ieee8021TeipsV2CandidatePsTable	
	ieee8021TeipsV2CandidatePsMA	12.24.2.1.3 e)
	ieee8021TeipsV2CandidatePsPortNumber	12.24.2.1.3 e)
	ieee8021TeipsV2CandidatePsOperational	12.24.2.1.3 e)

Table 17-24—IEEE8021-TEIPS MIB structure (continued)

Variable	IEEE8021-TEIPS-MIB table/object	Reference
	ieee8021TeipsV2IpgConfigTable	
	ieee8021TeipsV2IpgConfigState	12.24.2.1.3 g)
	ieee8021TeipsV2IpgConfigCommandStatus	12.24.2.1.3 h)
	ieee8021TeipsV2IpgConfigCommandLast	12.24.2.3.2 b)
	ieee8021TeipsV2IpgConfigAdmin	12.24.2.2.1 b)
	ieee8021TeipsV2IpgConfigActiveRequests	12.24.2.1.3 h)
WTRwhile (26.10.3.2.1)	ieee8021TeipsV2IpgConfigWTR	12.24.2.1.3 i)
HoldOffWhile (26.10.3.2.2)	ieee8021TeipsV2IpgConfigHoldOff	12.24.2.1.3 j)
	ieee8021TeipsV2IpgM1ConfigState	12.24.2.1.3 k)
WTRwhile (26.10.3.2.1)	ieee8021TeipsV2IpgConfigMWTR	12.24.2.1.3 l)

1 An IPG can be referenced externally. For this reason, a canonical external IPG identifier is provided via the
2 textual convention `IEEE8021TeipsIpgid`. The convention defines a simple integer of local significance
3 to a particular Bridge component for the purpose of MIB management.

4 An Infrastructure Segment can be referenced externally. For this reason, a canonical external Infrastructure
5 Segment Identifier (SEG-ID) is provided via the textual convention `IEEE8021TeipsSegid`. The
6 convention defines a simple integer of local significance to a particular Bridge component for the purpose of
7 MIB management.

8 In the case of 1:1 IPS, an IPG specifies a Working Segment and a Protection Segment. In the case of M:1
9 IPS, the IPG specifies a Working Segment and a list of candidate Protection Segments from which the
10 Protection Segment is selected. Each Infrastructure Segment is associated with a pair of Segment
11 Monitoring Paths (SMPs). Each SMP is identified by a 3-tuple consisting of a
12 <SMP-DA,SMP-SA,SMP-VID> tuple. Each such tuple is represented by the textual convention
13 `IEEE8021TeipsSmpid`, which is a fixed-length octet string containing the two MAC addresses and the
14 VID value.

15 17.2.18.1 Using the TEIPS MIB to create MAs for Infrastructure Segments

16 IPS requires a mechanism to monitor the health of each Infrastructure Segment associated with an IPG. The
17 CFM MA provides this mechanism.

18 Configuring an MA requires adding a row to the CFM MIB's `StackTable`. Two columns are used in the
19 `StackTable` to define the Infrastructure Segment upon which the MA operates. These are the
20 `IEEE8021ServiceSelectorType` column and the `IEEE8021ServiceSelectorValue`
21 column. The first column defines how the value in the second column is to be interpreted. If the value of the
22 first column is set to `segid(4)`, then the second column is interpreted as the external representation of a
23 SEG-ID.

24 Thus, Infrastructure Segments, and their corresponding identifiers, configured in the
25 `ieee8021TeipsV2SegTable` provide the SEG-ID values that are used as parameters to rowcreate
26 operations for the `CFMStackTable` to create MAs that monitor the health of Infrastructure Segments.
27 Creating entries in the `CFMStackTable`, in turn, defines MAID values, represented as unsigned integers,
28 that are used subsequently to configure the protected services.

1 17.2.18.2 Using the TEIPS MIB to create IPGs

2 IPGs are created by adding rows to the `ieee8021TeipsV2IpgTable`. Each entry in the table contains
3 columns representing the Working and Protection Segment MAs associated with the IPG, the Port Number
4 associated with each Infrastructure Segment MA, and a RowStatus column used to manage the creation and
5 deletion of the IPG. The MAs are referred to by MAID and can be found in the `StackTable` of the CFM
6 MIB managed objects. The value of the Port Number column is derived from the corresponding
7 Infrastructure Segment MA.

8 17.2.18.3 Using the TEIPS MIB to query and configure IPGs

9 Each IPG has status and configuration information that can be managed. This information is located in the
10 `ieee8021TeipsV2IpgConfigTable`. This table allows management stations to determine whether
11 the Working Segment or Protection Segment is active, and it allows traffic to be switched by management
12 command. It provides optional parameters for configuration of the waiting time before triggering a
13 switchover and the time to restore after a failure condition has cleared.

14 17.2.19 Structure of the IEEE8021-SPB-MIB

15 The IEEE8021-SPB-MIB provides objects to configure both SPBV and SPBM as defined in Clause 27 and
16 Clause 28. Table 17-25 describes the relationship between the SMIV2 objects defined in the MIB module in
17 (17.7.19) and the variables and managed objects defined in Clause 12.

Table 17-25—IEEE8021-SPB-MIB structure

IEEE8021-SPB-MIB table/object	Reference
ieee8021SpbSys	12.25.1
<code>ieee8021SpbSysAreaAddress</code>	12.25.1.3.2, 12.25.1.3.3
<code>ieee8021SpbSysId</code>	12.25.1.3.3, Clause 3
<code>ieee8021SpbSysControlAddr</code>	12.25.1.3.3
<code>ieee8021SpbSysName</code>	12.25.1.3.3
<code>ieee8021SpbSysBridgePriority</code>	12.25.1.3.3, 13.26.3
<code>ieee8021SpbmSysSPSourceId</code>	12.25.1.3.3, Clause 3
<code>ieee8021SpbvSysMode</code>	12.25.1.3.3, Clause 3
<code>ieee8021SpbmSysMode</code>	12.25.1.3.3, Clause 3
<code>ieee8021SpbSysDigestConvention</code>	12.25.1.3.3, 28.4.3
ieee8021SpbMtidStaticTable	12.25.2
<code>ieee8021SpbMtidStaticTableEntry</code>	12.25.2.3.3
<code>ieee8021SpbMtidStaticEntryMtid</code>	12.25.1.3.2, 12.25.2.3.3, 28.12
<code>ieee8021SpbMTidStaticEntryMtidOverload</code>	12.25.2.3.3, 27.8.1
<code>ieee8021SpbMtidStaticEntryRowStatus</code>	12.25.2.3.3
<code>ieee8021SpbTopIx</code>	12.25.2.3.3
ieee8021SpbTopIxDynamicTable	12.25.3
<code>ieee8021SpbTopIxDynamicTableEntry</code>	12.25.3

Table 17-25—IEEE8021-SPB-MIB structure (*continued*)

IEEE8021-SPB-MIB table/object	Reference
ieee8021SpbTopIxDynamicEntryTopIx	12.25.3.1.2, 28.12
ieee8021SpbTopIxDynamicEntryAgreeDigest	12.25.3.1.3, 28.4
ieee8021SpbTopIxDynamicEntryMCID	12.25.3.1.3, 13.8
ieee8021SpbTopIxDynamicEntryAuxMCID	12.25.3.1.3, 28.9
ieee8021SpbEctStaticTable	12.25.4
ieee8021SpbEctStaticTableEntry	12.25.4
ieee8021SpbEctStaticEntryTopIx	12.25.4.1.2, 28.12
ieee8021SpbEctStaticEntryBaseVid	12.25.4.1.2, Clause 3
ieee8021SpbEctStaticEntryEctAlgorithm	12.25.4.1.2, Clause 3
ieee8021SpbvEctStaticEntrySpvid	12.25.4.1.2, Clause 3
ieee8021SpbEctStaticEntryRowStatus	12.25.4.1.2
ieee8021SpbEctDynamicTable	12.25.5
ieee8021SpbEctDynamicTableEntry	12.25.5
ieee8021SpbEctDynamicEntryTopIx	12.25.5.1.2, 12.25.5.1.3, 28.12
ieee8021SpbEctDynamicEntryBaseVid	12.25.5.1.2, 12.25.5.1.3, Clause 3
ieee8021SpbEctDynamicEntryMode	12.25.5.1.3, 28.12.4
ieee8021SpbEctDynamicEntryLocalUse	12.25.5.1.3, 28.12.4
ieee8021SpbEctDynamicEntryRemoteUse	12.25.5.1.3, 28.12.4
ieee8021SpbEctDynamicEntryIngressCheckDiscards	12.25.5.1.3
ieee8021SpbAdjStaticTable	12.25.6
ieee8021SpbAdjStaticTableEntry	12.25.6
ieee8021SpbAdjStaticEntryTopIx	12.25.6.1.2, 12.25.6.1.3, 28.12
ieee8021SpbAdjStaticEntryIfIndex	12.25.6.1.2, 12.25.6.1.3
ieee8021SpbAdjStaticEntryMetric	12.25.6.1.2, 12.25.6.1.3, 28.12.7
ieee8021SpbAdjStaticEntryIfAdminState	12.25.6.1.2, 12.25.6.1.3
ieee8021SpbAdjStaticEntryRowStatus	12.25.6.1.3
ieee8021SpbAdjDynamicTable	12.25.7
ieee8021SpbAdjDynamicTableEntry	12.25.7
ieee8021SpbAdjDynamicEntryTopIx	12.25.7.1.2, 12.25.7.1.3, 28.12
ieee8021SpbAdjDynamicEntryIfIndex	12.25.7.1.2, 12.25.7.1.3
ieee8021SpbAdjDynamicEntryPeerSysId	12.25.7.1.3, Clause 3
ieee8021SpbAdjDynamicEntryPort	12.25.7.1.3
ieee8021SpbAdjDynamicEntryIfOperState	12.25.7.1.3

Table 17-25—IEEE8021-SPB-MIB structure (continued)

IEEE8021-SPB-MIB table/object	Reference
ieee8021SpbAdjDynamicEntryPeerSysName	12.25.7.1.3
ieee8021SpbAdjDynamicEntryPeerAgreeDigest	12.25.7.1.3, 28.4
ieee8021SpbAdjDynamicEntryPeerMCID	12.25.7.1.3, 28.12.2, 27.4.1
ieee8021SpbAdjDynamicEntryPeerAuxMCID	12.25.7.1.3, 28.12.2
ieee8021SpbAdjDynamicEntryLocalCircuitID	12.25.7.1.3, 28.11
ieee8021SpbAdjDynamicEntryPeerLocalCircuitID	12.25.7.1.3, 28.11
ieee8021SpbAdjDynamicEntryPortIdentifier	12.25.7.1.3, 28.11, 14.2.7
ieee8021SpbAdjDynamicEntryPeerPortIdentifier	12.25.7.1.3, 28.11, 14.2.7
ieee8021SpbAdjDynamicEntryIshCircIndex	12.25.7.1.3, 28.11
ieee8021SpbTopNodeTable	12.25.9
ieee8021SpbTopNodeTableEntry	12.25.9
ieee8021SpbTopNodeEntryTopIx	12.25.9.1.2, 12.25.9.1.3, 28.12
ieee8021SpbTopNodeEntrySysId	12.25.9.1.2, 12.25.9.1.3, Clause 3
ieee8021SpbTopNodeEntryBridgePriority	12.25.9.1.3, 13.26.3
ieee8021SpbmTopNodeEntrySPsourceID	12.25.9.1.3, Clause 3
ieee8021SpbTopNodeEntrySysName	12.25.9.1.3
ieee8021SpbTopEctTable	12.25.10
ieee8021SpbTopEctTableEntry	12.25.10
ieee8021SpbTopEctEntryTopIx	12.25.10.1.2, 12.25.10.1.3
ieee8021SpbTopEctEntrySysId	12.25.10.1.2, 12.25.10.1.3, Clause 3
ieee8021SpbTopEctEntryBaseVid	12.25.10.1.2, 12.25.10.1.3, Clause 3
ieee8021SpbTopEctEntryEctAlgorithm	12.25.10.1.3, Clause 3
ieee8021SpbTopEctEntryMode	12.25.10.1.3
ieee8021SpbvTopEctSysMode	12.25.10.1.3
ieee8021SpbvTopEctEntrySpvid	12.25.10.1.3
ieee8021SpbTopEctEntryLocalUse	12.25.10.1.3, 28.12.5
ieee8021SpbTopEdgeTable	12.25.11
ieee8021SpbTopEdgeTableEntry	12.25.11
ieee8021SpbTopEdgeEntryTopIx	12.25.11.1.2, 12.25.11.1.3, 28.12
ieee8021SpbTopEdgeEntrySysIdNear	12.25.11.1.2, 12.25.11.1.3, Clause 3
ieee8021SpbTopEdgeEntrySysIdFar	12.25.11.1.2, 12.25.11.1.3, Clause 3
ieee8021SpbTopEdgeEntryMetricNear2Far	12.25.11.1.3, 28.12.7
ieee8021SpbTopEdgeEntryMetricFar2Near	12.25.11.1.3, 28.12.7

Table 17-25—IEEE8021-SPB-MIB structure (*continued*)

IEEE8021-SPB-MIB table/object	Reference
ieee8021SpbmTopSrvTable	12.25.12
ieee8021SpbmTopSrvTableEntry	12.25.12
ieee8021SpbmTopSrvEntryTopIx	12.25.12.1.2, 12.25.11.1.3, 28.12
ieee8021SpbmTopSrvEntrySysId	12.25.12.1.2, 12.25.11.1.3, Clause 3
ieee8021SpbmTopSrvEntryIsid	12.25.12.1.2, 12.25.11.1.3, 28.12.10
ieee8021SpbmTopSrvEntryBaseVid	12.25.12.1.2, 12.25.11.1.3, 28.12.10
ieee8021SpbmTopSrvEntryIsidFlags	12.25.12.1.2, 12.25.11.1.3, 28.12.10
ieee8021SpbmTopSrvEntryMac	12.25.11.1.3, 28.12.10
ieee8021SpbvTopSrvTable	12.25.13
ieee8021SpbvTopSrvTableEntry	12.25.13
ieee8021SpbvTopSrvEntryTopIx	12.25.13.1.2, 12.25.13.1.3, 28.12
ieee8021SpbvTopSrvEntrySysId	12.25.13.1.2, 12.25.13.1.3, Clause 3
ieee8021SpbvTopSrvEntryMMac	12.25.13.1.2, 12.25.13.1.3, 28.12.9
ieee8021SpbvTopSrvEntryMMacFlags	12.25.13.1.3, 28.12.9
ieee8021SpbvTopSrvEntryBaseVid	12.25.13.1.3, Clause 3
ieee8021SpbmBsiStaticTable	12.25.8, 28.12.10
ieee8021SpbmBsiStaticTableEntry	12.25.8
ieee8021SpbmBsiStaticEntryIsid	12.25.8.1, 12.25.8.2
ieee8021SpbmBsiStaticEntryBaseVid	12.25.8.1, 12.25.8.2
ieee8021SpbmBsiStaticEntryTBit	12.25.8.1, 12.25.8.2
ieee8021SpbmBsiStaticEntryRBit	12.25.8.1, 12.25.8.2
ieee8021SpbmBsiStaticEntryTsBit	12.25.8.1, 12.25.8.2
ieee8021SpbmBsiStaticEntryTieBreakMask	12.25.8.1, 12.25.8.2
ieee8021SpbmBsiStaticEntryRowStatus	—
dot1agCfmMepSpbmTable	12.14.7, 27.19
dot1agCfmMepSpbmEntry	12.14.7
dot1agCfmMepTransmitLbmSpbmDA	12.14.7.3
dot1agCfmMepTransmitLtmSpbmDA	12.14.7.4
dot1agCfmMepSpbmEspTable	12.14.5.3.2, 27.19.1
dot1agCfmMepSpbmEspEntry	12.14.5.3.2
dot1agCfmMepSpbmEspIndex	12.14.5.3.2
dot1agCfmMepSpbmEspEsp	12.14.5.3.2
dot1agCfmMepSpbmEspRowStatus	—

Table 17-25—IEEE8021-SPB-MIB structure (continued)

IEEE8021-SPB-MIB table/object	Reference
ieee8021PcrEctStaticTable	12.28.1
ieee8021PcrEctStaticTableEntry	12.28.1
ieee8021PcrEctStaticEntryTopIx	12.28.1.1.2
ieee8021PcrEctStaticEntryBaseVid	12.28.1.1.2
ieee8021PcrEctStaticEntryMrtBlueVid	12.28.1.1.2, 45.3.3, 45.3.4
ieee8021PcrEctStaticEntryMrtRedVid	12.28.1.1.2, 45.3.3, 45.3.4
ieee8021PcrEctStaticEntryRowStatus	12.28.2.1.3
ieee8021PcrTopEctTable	12.28.2
ieee8021PcrTopEctTableEntry	12.28.2
ieee8021PcrTopEctEntryTopIx	12.28.2.1.2, 12.28.2.1.3
ieee8021PcrTopEctEntrySysId	12.28.2.1.2, 12.28.2.1.3
ieee8021PcrTopEctEntryBaseVid	12.28.2.1.2, 12.28.2.1.3
ieee8021PcrTopEctEntryMode	12.28.2.1.2, 12.28.2.1.3
ieee8021PcrTopEctEntryMrtBlueVid	12.28.2.1.3, 45.3.3, 45.3.4
ieee8021PcrTopEctEntryMrtRedVid	12.28.2.1.3, 45.3.3, 45.3.4

1 17.2.20 Structure of the IEEE8021-EVB-MIB

2 The IEEE8021-EVB-MIB provides objects to configure and manage an EVB station system or EVB Bridge
3 system. Objects in this MIB module are arranged into subtrees. Each subtree is organized as a set of
4 related objects. Where appropriate, the corresponding Clause 12 management reference is also included.
5 Table 17-26 indicates the structure of the IEEE8021-EVB-MIB module.

Table 17-26—IEEE8021-EVB-MIB structure

IEEE8021-EVB-MIB table/object	References
<i>ieee8021BridgeEvbNotifications subtree</i>	
—	—
<i>ieee8021BridgeEvbObjects subtree</i>	
ieee8021BridgeEvbSys	12.26.1
ieee8021BridgeEvbSysType	—
ieee8021BridgeEvbSysNumExternalPorts	—
ieee8021BridgeEvbSysEvbLdpTxEnable	—
ieee8021BridgeEvbSysEvbLdpManual	—
ieee8021BridgeEvbSysEvbLdpGidCapable	—
ieee8021BridgeEvbSysEcpAckTimerExp	—

Table 17-26—IEEE8021-EVB-MIB structure (continued)

IEEE8021-EVB-MIB table/object	References
ieee8021BridgeEvbSysEcpMaxRetries	—
ieee8021BridgeEvbSysVdpDfltRsrcWaitDelayExp	—
ieee8021BridgeEvbSysVdpDfltReinitKeepAliveExp	—
ieee8021BridgeEvbSbpTable	12.26.2
ieee8021BridgeEvbSbpComponentID	—
ieee8021BridgeEvbSbpPortNumber	—
ieee8021BridgeEvbSbpLdpManual	—
ieee8021BridgeEvbSbpVdpOperRsrcWaitDelayExp	—
ieee8021BridgeEvbSbpVdpOperReinitKeepAliveExp	—
ieee8021BridgeEvbSbpVdpOperToutKeepAlive	—
ieee8021BridgeEvbSbpRowStatus	—
ieee8021BridgeEvbVsiDbTable	12.26.3
ieee8021BridgeEvbVsiComponentID	—
ieee8021BridgeEvbVsiPortNumber	—
ieee8021BridgeEvbVsiID	—
ieee8021BridgeEvbVsiTimeSinceCreate	—
ieee8021BridgeEvbVsiVdpOperCmd	—
ieee8021BridgeEvbVsiOperRevert	—
ieee8021BridgeEvbVsiOperHard	—
ieee8021BridgeEvbVsiOperReason	—
ieee8021BridgeEvbVsiMgrID (Deprecated)	—
ieee8021BridgeEvbVsiType	—
ieee8021BridgeEvbVsiTypeVersion	—
ieee8021BridgeEvbVsiMvFormat	—
ieee8021BridgeEvbVsiNumMACs	—
ieee8021BridgeEvbVDPMachineState	—
ieee8021BridgeEvbVDPCommandsSucceeded	—
ieee8021BridgeEvbVDPCommandsFailed	—
ieee8021BridgeEvbVDPCommandReverts	—
ieee8021BridgeEvbVsiMgrID16	—
ieee8021BridgeEvbVsiMvFormat	—
ieee8021BridgeEvbVsiDbMacTable	12.26.3
ieee8021BridgeEvbVsiComponentID	—

Table 17-26—IEEE8021-EVB-MIB structure (continued)

IEEE8021-EVB-MIB table/object	References
ieee8021BridgeEvbVsiPortNumber	—
ieee8021BridgeEvbVsiID	—
ieee8021BridgeEvbGroupID	—
ieee8021BridgeEvbVsiMac	—
ieee8021BridgeEvbVsiVlanId	—
ieee8021BridgeEvbUapConfigTable	12.26.4.1
ieee8021BridgePort	—
ieee8021BridgeEvbUapComponentId	—
ieee8021BridgeEvbUapPort	—
ieee8021BridgeEvbUapConfigIfIndex	—
ieee8021BridgeEvbUapCdcAdminEnable	—
ieee8021BridgeEvbUapAdminCdcRole	—
ieee8021BridgeEvbUapAdminCdcChanCap	—
ieee8021BridgeEvbUapOperCdcChanCap	—
ieee8021BridgeEvbUapAdminCdcSVIDPoolLow	—
ieee8021BridgeEvbUapAdminCdcSVIDPoolHigh	—
ieee8021BridgeEvbUapOperState	—
ieee8021BridgeEvbUapCdcRemoteEnabled	—
ieee8021BridgeEvbUapCdcRemoteRole	—
ieee8021BridgeEvbUapConfigRowStatus	—
ieee8021BridgeEvbCapConfigTable	12.26.4.2
ieee8021BridgeBridgePort	—
ieee8021BridgeEvbCapSchID	—
ieee8021BridgeEvbCapComponentId	—
ieee8021BridgeEvbCapIfIndex	—
ieee8021BridgeEvbCapPortNumber	—
ieee8021BridgeEvbCapSChannelID	—
ieee8021BridgeEvbCapAssociateSbpOrUrpCompID	—
ieee8021BridgeEvbCapAssociateSbpOrUrpPort	—
ieee8021BridgeEvbCapRowStatus	—
ieee8021BridgeEvbUrpTable	12.26.5
ieee8021BridgeEvbUrpComponentID	—
ieee8021BridgeEvbUrpPort	—

Table 17-26—IEEE8021-EVB-MIB structure (continued)

IEEE8021-EVB-MIB table/object	References
ieee8021BridgeEvpUrpIfIndex	—
ieee8021BridgeEvpUrpBindToIssPort	—
ieee8021BridgeEvpUrpLldpManual	—
ieee8021BridgeEvpUrpVdpOperRsrcWaitDelayExp	—
ieee8021BridgeEvpUrpVdpOperRespWaitDelay	—
ieee8021BridgeEvpUrpVdpOperReinitKeepAliveExp	—
ieee8021BridgeEvpUrpRowStatus	—
ieee8021BridgeEvpEcpTable	12.27.1
ieee8021BridgeEvpEcpComponentID	—
ieee8021BridgeEvpEcpPort	—
ieee8021BridgeEvpEcpOperAckTimerInitExp	—
ieee8021BridgeEvpEcpOperMaxRetries	—
ieee8021BridgeEvpEcpTxFrameCount	—
ieee8021BridgeEvpEcpTxRetryCount	—
ieee8021BridgeEvpEcpFailures	—
ieee8021BridgeEvpEcpRxFrameCount	—
<i>ieee8021BridgeEvpConformance subtree</i>	
ieee8021BridgeEvpGroups	—
ieee8021BridgeEvpSysGroup	—
ieee8021BridgeEvpSbpConfigGroup	—
ieee8021BridgeEvpVsiDbGroup	—
ieee8021BridgeEvpUapConfigGroup	—
ieee8021BridgeEvpCapConfigGroup	—
ieee8021BridgeEvpUrpConfigGroup	—
ieee8021BridgeEvbsEcpConfigGroup	—
ieee8021BridgeEvpCompliances	—
ieee8021BridgeEvbbCompliance	—
ieee8021BridgeEvbsCompliance	—

1 17.2.21 Structure of the IEEE8021-ECMP-MIB

2 The IEEE8021-ECMP-MIB provides for configuration of flow filtering (44.2) on ports (CBPs and PNPs)
3 and the configuration of ISIS-SPB advertisements specific to the ECMP ECT Algorithm (44.1.2). It also
4 allows reading of FDB state and statistics related to ECMP and flow filtering. Table 17-27 indicates the
5 relationship between the SMIV2 objects defined in the MIB module (17.7.21) and managed objects defined
6 in Clause 12.

Table 17-27—IEEE8021-ECMP-MIB structure

IEEE8021-ECMP-MIB table/object	Reference
ieee8021QBridgeEcmpFdbTable	12.7.7.3, 8.8.3
(AUGMENTS ieee8021QBridgeTpFdbEntry)	—
ieee8021QBridgeEcmpFdbPortList	—
ieee8021EcmpFlowFilterCtlTable	12.16.5.4, 12.16.5.5, 44.2.2
ieee8021EcmpFlowFilterCtlVid	—
ieee8021EcmpFlowFilterCtlEnabled	—
ieee8021EcmpFlowFilterCtlHashGen	—
ieee8021EcmpFlowFilterCtlTtl	—
ieee8021EcmpEctStaticTable	12.25.14, 28.12.6.1
ieee8021EcmpEctStaticEntryTieBreakMask	—
ieee8021EcmpEctStaticEntryBridgePriority	—
ieee8021EcmpEctStaticEntryRowStatus	—
ieee8021EcmpTopSrvTable	12.25.12.1.3, 28.12.10
(AUGMENTS ieee8021SpbmTopSrvTableEntry)	—
ieee8021EcmpTopSrvEntryTsBit	—
ieee8021EcmpTopSrvEntryTieBreakMask	—
ieee8021QBridgePortVlanTtlStatisticsTable	12.6.1.1.3, 44.2.2.1
(AUGMENTS ieee8021QBridgePortVlanStatisticsEntry)	—
ieee8021QBridgeTpVlanPortTtlDiscards	—

1 17.2.22 Structure of the IEEE8021-ST-MIB

2 The IEEE8021-ST-MIB provides for configuration of scheduled traffic (8.6.8, 8.6.8.4) on ports. Table 17-28
3 indicates the relationship between the SMIV2 objects defined in the MIB module (17.7.22) and managed
4 objects defined in 12.29.

Table 17-28—IEEE8021-ST-MIB structure

IEEE8021-ST-MIB table/object	Reference
<i>ieee8021STMaxSDU subtree</i>	
ieee8021STMaxSDUTable	Max SDU table, 8.6.8.4, 8.6.9, 12.29.1
ieee8021STTrafficClass*	Traffic Class
ieee8021STMaxSDU	queueMaxSDU, 8.6.8.4, 8.6.9, 12.29.1, 12.29.1.1.1
ieee8021TransmissionOverrun	TransmissionOverrun, 8.6.8.4, 8.6.9, 12.29.1, 12.29.1.1.2
<i>ieee8021STParameters</i>	
ieee8021STParametersTable	Scheduled Traffic parameter table, 8.6.8.4, 8.6.9, 12.29.1
ieee8021STGateEnabled	GateEnabled, 8.6.8.4, 8.6.9, 8.6.9.4.14, 12.29.1
ieee8021STAdminGateStates	AdminGateStates, 8.6.8.4, 8.6.9, 8.6.9.4.5, 12.29.1
ieee8021STOperGateStates	OperGateStates, 8.6.8.4, 8.6.9, 8.6.9.4.21, 12.29.1
ieee8021STAdminControlListLength	AdminControlListLength, 8.6.8.4, 8.6.9, 8.6.9.4.6, 12.29.1
ieee8021STOperControlListLength	OperControlListLength, 8.6.8.4, 8.6.9, 8.6.9.4.22, 12.29.1
ieee8021STAdminControlList	AdminControlList, 8.6.8.4, 8.6.9, 8.6.9.4.2, 12.29.1
ieee8021STOperControlList	OperControlList, 8.6.8.4, 8.6.9, 8.6.9.4.18, 12.29.1
ieee8021STAdminCycleTimeNumerator	Numerator—AdminCycleTime, 8.6.8.4, 8.6.9, 8.6.9.4.3, 12.29.1
ieee8021STAdminCycleTimeDenominator	Denominator—AdminCycleTime, 8.6.8.4, 8.6.9, 8.6.9.4.3, 12.29.1
ieee8021STOperCycleTimeNumerator	Numerator—OperCycleTime, 8.6.8.4, 8.6.9, 8.6.9.4.19, 12.29.1
ieee8021STOperCycleTimeDenominator	Denominator—OperCycleTime, 8.6.8.4, 8.6.9, 8.6.9.4.19, 12.29.1
ieee8021STAdminCycleTimeExtension	AdminCycleTimeExtension, 8.6.8.4, 8.6.9, 8.6.9.4.4, 12.29.1
ieee8021STOperCycleTimeExtension	OperCycleTimeExtension, 8.6.8.4, 8.6.9, 8.6.9.4.20, 12.29.1
ieee8021STAdminBaseTime	AdminBaseTime, 8.6.8.4, 8.6.9, 8.6.9.4.1, 12.29.1
ieee8021STOperBaseTime	OperBaseTime, 8.6.8.4, 8.6.9, 8.6.9.4.17, 12.29.1

Table 17-28—IEEE8021-ST-MIB structure (*continued*)

IEEE8021-ST-MIB table/object	Reference
ieee8021STConfigChange	ConfigChange, 8.6.8.4, 8.6.9, 8.6.9.4.7, 12.29.1
ieee8021STConfigChangeTime	ConfigChangeTime, 8.6.8.4, 8.6.9, 8.6.9.4.9, 12.29.1
ieee8021STTickGranularity	TickGranularity, 8.6.8.4, 8.6.9, 12.29.1
ieee8021STCurrentTime	CurrentTime, 8.6.8.4, 8.6.9, 8.6.9.4.10, 12.29.1
ieee802STConfigPending	ConfigPending, 8.6.9.4.8
ieee8021STSupportedListMax	SupportedListMax, 12.29.1.5
ieee8021STSupportedCycleMaxNumerator	SupportedCycleMax, 12.29.1.6
ieee8021STSupportedCycleMaxDenominator	SupportedCycleMax, 12.29.1.6
ieee8021STSupportedIntervalMax	SupportedIntervalMax, 12.29.1.7

*This object is an INDEX of the table in which it resides.

1 17.2.23 Structure of the IEEE8021-Preemption-MIB

- 2 The IEEE8021-Preemption-MIB provides for configuration of frame preemption (6.7.2, 8.6.8) on ports.
3 Table 17-29 indicates the relationship between the SMIPv2 objects defined in the MIB module (17.7.23) and
4 managed objects defined in 12.30.

Table 17-29—IEEE8021-Preemption-MIB structure

IEEE8021-Preemption-MIB table/object	Reference
<i>ieee8021PreemptionParameterTable subtree</i>	
ieee8021PreemptionParameterTable	Frame Preemption parameter table, 6.7.2, 12.30.1
ieee8021PreemptionPriority	Priority (Table index)
ieee8021FramePreemptionAdminStatus	framePreemptionAdminStatus, 6.7.2, 12.30.1
ieee8021PreemptionConfigTable	Frame Preemption configuration table, 6.7.2, 12.30.1
ieee8021FramePreemptionHoldAdvance	framePreemptionHoldAdvance, 6.7.2, 12.30.1, 12.30.1.2
ieee8021FramePreemptionReleaseAdvance	framePreemptionReleaseAdvance, 6.7.2, 12.30.1, 12.30.1.3
ieee8021FramePreemptionActive	framePreemptionActive, 6.7.2, 12.30.1, 12.30.1.4
ieee8021FramePreemptionHoldRequest	framePreemptionHoldRequest, 6.7.2, 12.30.1, 12.30.1.5

1 17.2.24 Structure of the IEEE8021-PSFP-MIB

2 The IEEE8021-PSFP-MIB provides for configuration of PSFP (8.6.5, 8.6.5.2, 8.6.10) on reception Ports.

3 Table 17-30 indicates the relationship between the SMIPv2 objects defined in the MIB module (17.7.24) and

4 managed objects defined in 12.31.

Table 17-30—IEEE8021-PSFP-MIB structure

IEEE8021-PSFP-MIB table/object	Reference
<i>ieee8021PSFPStreamFilterParameters subtree</i>	
ieee8021PSFPStreamFilterTable	Stream Filter Instance Table, 8.6.5.2.1, 8.6.5.3, 12.31.2
ieee8021PSFPStreamFilterInstance	StreamFilterInstance, 8.6.5.2.1, 8.6.5.3, 12.31.2
ieee8021PSFPStreamHandleSpec	StreamHandleSpec, 8.6.5.2.1, 8.6.5.3, 12.31.2
ieee8021PSFPPrioritySpec	PrioritySpec, 8.6.5.2.1, 8.6.5.3, 12.31.2
ieee8021PSFPStreamGateInstanceID	StreamGateInstanceID, 8.6.5.2.1, 8.6.5.3, 12.31.2
ieee8021PSFPFilterSpecificationList ^a	MaxSDUSize, FlowMeterInstanceID, FlowMeterEnable, 8.6.5.2.1, 8.6.5.3, 12.31.2, 12.31.2.4, 12.31.2.6
ieee8021PSFPMatchingFramesCount	MatchingFramesCount, 8.6.5.2.1, 8.6.5.3, 12.31.2
ieee8021PSFPPassingFramesCount	PassingFramesCount, 8.6.5.2.1, 8.6.5.3, 8.6.5.4, 12.31.2
ieee8021PSFPNotPassingFramesCount	NotPassingFramesCount, 8.6.5.2.1, 8.6.5.3, 8.6.5.4, 12.31.2
ieee8021PSFPPassingSDUCount	PassingSDUCount, 8.6.5.2.1, 8.6.5.3, 8.6.5.3.1, 12.31.2
ieee8021PSFPNotPassingSDUCount	NotPassingSDUCount, 8.6.5.2.1, 8.6.5.3, 8.6.5.3.1, 12.31.2
ieee8021PSFPREDFramesCount	REDFramesCount, 8.6.5.2.1, 8.6.5.3, 8.6.5.5, 12.31.2
ieee8021PSFPStreamBlockedDueToOversizeFrameEnable	StreamBlockedDueToOversizeFrameEnable, 8.6.5.2.1, 8.6.5.3.1, 12.31.2
ieee8021PSFPStreamBlockedDueToOversizeFrame	StreamBlockedDueToOversizeFrame, 8.6.5.2.1, 8.6.5.3.1, 12.31.2

Table 17-30—IEEE8021-PSFP-MIB structure (continued)

IEEE8021-PSFP-MIB table/object	Reference
<i>ieee8021PSFPStreamGateParameters</i>	
ieee8021FPStPSreamGateTable	Stream Gate Instance Table, 8.6.5.2.1, 8.6.5.4, 12.31.3
ieee8021PSFPStreamGateInstance	StreamGateInstance, 8.6.5.2.1, 8.6.5.4, 12.31.3
ieee8021PSFPGateEnabled	StreamGateEnabled, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminGateStates	StreamGateAdminGateStates, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPOperGateStates	StreamGateOperGateStates, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminControlListLength	StreamGateAdminControlListLength, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPOperControlListLength	StreamGateOperControlListLength, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminControlList	StreamGateAdminControlList, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPOperControlList	StreamGateOperControlList, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminCycleTimeNumerator	StreamGateAdminCycleTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminCycleTimeDenominator	StreamGateAdminCycleTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPOperCycleTimeNumerator	StreamGateOperCycleTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPOperCycleTimeDenominator	StreamGateOperCycleTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminCycleTimeExtension	StreamGateAdminCycleTimeExtension, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPOperCycleTimeExtension	StreamGateOperCycleTimeExtension, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminBaseTime	StreamGateAdminBaseTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPOperBaseTime	StreamGateOperBaseTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPConfigChange	StreamGateConfigChange, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPConfigChangeTime	StreamGateConfigChangeTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPTickGranularity	StreamGateTickGranularity, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPCurrentTime	StreamGateCurrentTime, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPConfigPending	StreamGateConfigPending, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3

Table 17-30—IEEE8021-PSFP-MIB structure (continued)

IEEE8021-PSFP-MIB table/object	Reference
ieee8021PSFPConfigChangeError	StreamGateConfigChangeError, 8.6.5.2.1, 8.6.5.4, 8.6.10, 12.31.3
ieee8021PSFPAdminIPV	StreamGateAdminIPV, 8.6.5.2.1, 8.6.5.4, 12.31.3
ieee8021PSFPOperIPV	StreamGateOperIPV, 8.6.5.2.1, 8.6.5.4, 12.31.3
ieee8021PSFPGateClosedDueToInvalidRxEnable	StreamGateGateClosedDueToInvalidRxEnable, 8.6.5.2.1, 8.6.5.4
ieee8021PSFPGateClosedDueToInvalidRx	StreamGateGateClosedDueToInvalidRx, 8.6.5.2.1, 8.6.5.4
ieee8021PSFPGateClosedDueToOctetsExceededEnable	StreamGateGateClosedDueToOctetsExceededEnable, 8.6.5.2.1, 8.6.5.4
ieee8021PSFPGateClosedDueToOctetsExceeded	StreamGateGateClosedDueToOctetsExceeded, 8.6.5.2.1, 8.6.5.4
<i>ieee8021PSFPFlowMeterParameters</i>	
ieee8021PSFPFlowMeterTable	Flow Meter Instance Table, 8.6.5.2.1, 8.6.5.4, 12.31.4
ieee8021PSFPFlowMeterInstance	FlowMeterInstanceID, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterCIR	CIR, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterCBS	CBS, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterEIR	EIR, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterCF	CF, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterCM	CM, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterDropOnYellow	DropOnYellow, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterMarkAllFramesRedEnable	MarkAllFramesRedEnable, 8.6.5.2.1, 8.6.5.5, 12.31.4
ieee8021PSFPFlowMeterMarkAllFramesRed	MarkAllFramesRed, 8.6.5.2.1, 8.6.5.5, 12.31.4
<i>ieee8021PSFPStreamParameters</i>	
ieee8021PSFPStreamParameterTable	StreamParameterTable, 8.6.5.2.1, 12.31.1
ieee8021PSFPMaxStreamFilterInstances	MaxStreamFilterInstances, 8.6.5.2.1, 8.6.5.3, 12.31.1
ieee8021PSFPMaxStreamGateInstances	MaxStreamGateInstances, 8.6.5.2.1, 8.6.5.4, 12.31.1
ieee8021PSFPMaxFlowMeterInstances	MaxFlowMeterInstances, 8.6.5.2.1, 8.6.5.5, 12.31.1
ieee8021PSFPSupportedListMax	SupportedListMax, 8.6.5.2.1, 8.6.5.4, 12.31.1
ieee8021PSFPSupportedCycleMaxNumerator	SupportedCycleMax, 12.31.1.7
ieee8021PSFPSupportedCycleMaxDenominator	SupportedCycleMax, 12.31.1.7
ieee8021PSFPSupportedIntervalMax	SupportedIntervalMax, 12.31.1.8

^a To allow the PSFP MIB originally specified in IEEE Std 802.1Q-2018 to manage systems conformant to this standard, the encoding of a Maximum SDU size and a flow meter identifier in an ieee8021PSFPFilterSpecificationList has been retained.

1 17.2.25 Structure of the IEEE8021-TSN-REMOTE-MANAGEMENT-MIB

2 The IEEE8021-TSN-REMOTE-MANAGEMENT-MIB module defines managed objects for remote
3 management of TSN stream reservations (12.32).

Table 17-31—IEEE8021-TSN-REMOTE-MANAGEMENT-MIB structure

IEEE8021-TSN-REMOTE-MANAGEMENT-MIB table/object	Reference
ieee8021TsnRemoteMgmtBridgeDelayTable ieee8021BridgeTrafficClass	Bridge Delay, 12.32.1 traffic class (index, from IEEE8021-TC-MIB)
ieee8021TsnRemoteMgmtBridgeIngressPort	ingress Port (Table index)
ieee8021TsnRemoteMgmtBridgeEgressPort	egress Port (Table index)
ieee8021TsnRemoteMgmtBridgeIndependentDelayMin	independentDelayMin, 12.32.1.1
ieee8021TsnRemoteMgmtBridgeIndependentDelayMax	independentDelayMax, 12.32.1.1
ieee8021TsnRemoteMgmtBridgeDependentDelayMin	dependentDelayMin, 12.32.1.2
ieee8021TsnRemoteMgmtBridgeDependentDelayMax	dependentDelayMax, 12.32.1.2
ieee8021TsnRemoteMgmtPropagationDelayTable ieee8021BridgeBase Port	Propagation Delay, 12.32.2 Bridge Port (index, from IEEE8021-TC-MIB)
ieee8021TsnRemoteMgmtTxPropagationDelay	txPropagationDelay, 12.32.2.1
ieee8021TsnRemoteMgmtStaticTreesSupported	staticTreesSupported, 12.32.3.1
ieee8021TsnRemoteMgmtMsrpMrpExternalControlTable	12.32.4, Table 12-41
ieee8021TsnRemoteMgmtMsrpMrpExternalControl	externalControl, 12.32.4.1
ieee8021TsnRemoteMgmtMrpIndicationList	indicationList, 12.32.4.2
ieee8021TsnRemoteMgmtMrpIndicationListLength	indicationListLength, 12.32.4.3
ieee8021TsnRemoteMgmtMrpIndicationChangeCounter	indicationListLength, 12.32.4.4
ieee8021TsnRemoteMgmtMrpAdminRequestList	adminRequestList, 12.32.4.5
ieee8021TsnRemoteMgmtMrpAdminRequestListLength	adminRequestListLength, 12.32.4.6
ieee8021TsnRemoteMgmtMrpOperRequestList	operRequestList, 12.32.4.7
ieee8021TsnRemoteMgmtMrpOperRequestListLength	operRequestListLength, 12.32.4.8

1 17.2.26 Structure of the IEEE8021-PBBN-AA-MIB

2 The IEEE8021-PBBN-AA-MIB module provides objects to configure and manage the Auto Attach Protocol

3 (AAP, Clause 50).

Table 17-32—IEEE8021-PBBN-AA-MIB Structure

IEEE8021-PBBN-AA-MIB table/objects	Reference
<i>ieee8021AaConfig</i>	
ieee8021AaSystemEnable	aaSystemEnable, 12.34.1, 50.4.2.1
ieee8021AaSystemType	aaSystemType, 12.34.1, 50.4.2.5
ieee8021AaSystemMAC	aaSystemMAC, 12.34.1, 50.4.2.4
ieee8021AaSystemResetTime	aaSystemResetTime, 12.34.1, 50.4.2.6
ieee8021AaPortTable	PBBN Auto Attach port table, 12.34.2
ieee8021AaPortEntry	12.34.2
ieee8021AaPortIfIndex*	aaPortId, 12.34.2, 50.4.3.13
ieee8021AaPortNetId*	aaPortNetId, 12.34.2, 50.4.3.9, 50.4.1.3
ieee8021AaPortEnable	aaPortEnable, 12.34.2, 50.4.3.1
ieee8021AaPortRowStatus	—
ieee8021AaDiscSystemsTable	PBBN Discovered Auto Attach Systems table, 12.34.3
ieee8021AaDiscSystemsEntry	12.34.3
ieee8021AaDiscLocPortIfIndex*	aaDiscLocPortId, 12.34.2, 50.4.3.13
ieee8021AaDiscRemSystemType	aaDiscRemSystemType, 12.34.3, 50.4.3.11, 50.4.1.3
ieee8021AaDiscRemPortNetId*	aaDiscRemPortNetId, 12.34.3, 50.4.3.11, 50.4.1.3
ieee8021AaDiscRemPortTagging	aaDiscRemPortTagging, 12.34.3, 50.4.3.11, 50.4.1.3
ieee8021AaDiscRemPortAssocState	aaDiscRemPortAssocState, 12.34.3, 50.4.3.11, 50.4.1.3
ieee8021AaDiscLocPortAssocState	aaDiscLocPortAssocState, 12.34.3, 50.4.3.9, 50.4.1.3
ieee8021AaIsidVidAsgnsTable	Auto Attach assignment table, 12.34.4
ieee8021AaIsidVidAsgnsEntry	12.34.4
ieee8021AaIsidVidAsgnsIfIndex*	aaPortId, 12.34.2, 50.4.3.13
ieee8021AaIsidVidAsgnsIsid*	aaVidIsidAsgnsIsid, 12.34.4, 50.4.3.2, 50.4.1.4
ieee8021AaIsidVidAsgnsVlan*	aaVidIsidAsgnsVid, 12.34.4, 50.4.3.2, 50.4.1.4
ieee8021AaIsidVidAsgnsStatus	aaVidIsidAsgnsStatus, 12.34.4, 50.4.3.10, 50.4.3.8
ieee8021AaIsidVidAsgnsRowStatus	—

Table 17-32—IEEE8021-PBBN-AA-MIB Structure (*continued*)

IEEE8021-PBBN-AA-MIB table/objects	Reference
<i>ieee8021AaStats</i>	
ieee8021AaStatsTable	Auto Attach statistics table, 12.34.5
ieee8021AaStatsEntry	12.34.5
ieee8021AaStatsPortIfIndex*	aaPortId, 12.34.2, 50.4.3.13
ieee8021AaStatsAssocAttached	aaStatsAssocAttached, 12.34.5, 50.4.3.18
ieee8021AaStatsAssocFailed	aaStatsAssocFailed, 12.34.5, 50.4.3.19
ieee8021AaStatsAssocReset	aaStatsAssocReset, 12.34.5, 50.4.3.20
ieee8021AaStatsAssocStandby	aaStatsAssocStandby, 12.34.5, 50.4.3.21
ieee8021AaStatsAsgnsRequested	aaStatsVidIsidAsgnsRequested, 12.34.5, 50.4.3.24
ieee8021AaStatsAsgnsAccepted	aaStatsVidIsidAsgnsAccepted, 12.34.5, 50.4.3.22
ieee8021AaStatsAsgnsRejected	aaStatsVidIsidAsgnsRejected, 12.34.5, 50.4.3.23
ieee8021AaStatsAsgnsWithdrawn	aaStatsVidIsidAsgnsWithdrawn, 12.34.5, 50.4.3.25
<i>ieee8021AaNotifyObjects</i>	
ieee8021AaNotifications	—
ieee8021AaDiscoveredSystem	—
ieee8021AaDiscRemSystemType	aaDiscRemSystemType, 12.34.3, 50.4.3.11, 50.4.1.3
ieee8021AaDiscRemPortNetId	aaDiscRemPortNetId, 12.34.3, 50.4.3.11, 50.4.1.3
ieee8021AaDiscSystemsDescr	IEEE Std 802.1AB-2016, 8.5.7.2
ieee8021AaDiscSystemsMgmtOid	IEEE Std 802.1AB-2016, 8.5.9.8
ieee8021AaNotifyObjects	—
ieee8021AaDiscSystemsDescr	IEEE Std 802.1AB-2016, 8.5.7.2
ieee8021AaDiscSystemsMgmtOid	IEEE Std 802.1AB-2016, 8.5.9.8

*This object is an INDEX of the table in which it resides.

17.3 MIB module relationships

In order to facilitate interoperable management of VLAN Bridges by remote means, 17.7 contains a complete set of SMIPv2 MIB modules.

Some of these MIB modules have been derived from, and now replace, previous IETF MIB modules, as noted in Table 17-1. This was necessary because significant changes had to be made in these MIB modules, in particular the table indexing schemes, to support the evolution of VLAN Bridges. For example, Provider Backbone Bridging, described in Clause 26, supports multiple Bridge instances within a BEB. To avoid duplication of the base Bridge MIB modules within a PBB MIB module, the base Bridge modules have been reindexed to allow their use within a BEB. A BEB may have multiple I-components and zero or one B-components. A component identifier index, unique within a particular BEB, is used in the various MIB tables of these modules to distinguish between instances. In Bridges supporting multiple component or Bridge instances, the component identifier index is still present but is set to a default value of 1.

17.3.1 Relationship of the IEEE8021-TC-MIB to other MIB modules

Textual conventions (TCs) originally appeared in each of the MIB modules for IEEE Std 802.1Q produced by the IETF. However, with the transition of all modules for IEEE Std 802.1Q to this clause, it made sense to have a single module with all the TCs contained within it. Note that many of the original IETF TCs are still used in the various MIB modules for VLAN-Bridge management since they can be used unchanged.

These TCs in the IEEE8021-TC-MIB module, as well as many TCs from IETF MIB modules, are used by all of the MIB modules presented in this clause.

The component identifier (IEEE8021PbbComponentIdentifierTC) is defined in this module and is used as the syntax for component ID table indices in subsequent MIB modules.

17.3.2 Relationship of the IEEE8021-BRIDGE-MIB to other MIB modules

As described in Table 17-4 of 17.2.2, some IEEE 802.1D management objects have not been included in this MIB module because they overlap with objects in other MIB modules that are applicable to a Bridge implementing this MIB module.

17.3.2.1 Relationship to the SNMPv2-MIB

The SNMPv2-MIB [RFC3418] defines objects that are generally applicable to managed devices. These objects apply to the device as a whole, irrespective of whether the device's sole functionality is bridging, or whether bridging is only a subset of the device's functionality.

As explained in Table 17-4 of 17.2.2, full support for the IEEE 802.1D management objects requires that the SNMPv2-MIB objects sysDescr and sysUpTime be implemented. Note that compliance with the current SNMPv2-MIB module requires additional objects and notifications to be implemented, as specified in IETF RFC 3418.

17.3.2.2 Relationship to the IF-MIB

The IF-MIB [RFC2863], requires that any MIB that is an adjunct of the IF-MIB clarify specific areas within the IF-MIB. These areas were intentionally left vague in the IF-MIB in order to avoid over-constraining the MIB, thereby precluding management of certain media types.

The IF-MIB enumerates several areas that a media-specific MIB must clarify. Each of these areas is addressed in a following subclause. The implementor is referred to the IF-MIB in order to understand the general intent of these areas.

1 The IF-MIB [RFC2863] defines managed objects for managing network interfaces. A network interface is
2 thought of as being attached to a *subnetwork*. Note that this term is not to be confused with *subnet*, which
3 refers to an addressing partitioning scheme used in the Internet suite of protocols. The term *segment* is used
4 in this clause to refer to such a subnetwork, whether it be an Ethernet LAN, a *ring*, a WAN link, or even an
5 SDH virtual circuit.

6 As explained in 17.2.2, full support for the IEEE 802.1D management objects requires that the IF-MIB
7 objects ifIndex, ifType, ifDescr, ifPhysAddress, and ifLastChange are implemented. Note that compliance to
8 the current IF-MIB module requires additional objects and notifications to be implemented as specified in
9 IETF RFC 2863.

10 Implicit in this BRIDGE-MIB is the notion of Bridge Ports (3.30). Each Bridge Port is associated with one
11 interface of the *interfaces* subtree (one row in ifTable), and in most situations, each Bridge Port is associated
12 with a different interface. However, there are situations in which multiple Bridge Ports are associated with
13 the same interface. That is, for a given IF-MIB, interface refers to one of the interface points in the bridging
14 architecture (in Figure 8-1), and that zero or more multiple interface table entries can thus be instantiated for
15 a given Bridge Port. An example of such a situation would be several Bridge Ports each corresponding
16 one-to-one with several Ethernet private lines (or SDH virtual circuits) but all on the same interface.
17 Alternatively, there is the Link Aggregation (IEEE Std 802.1AX) case where there are many physical Ports
18 for one Bridge Port.

19 Each Bridge Port is uniquely identified by a Port Number. A Port Number has no mandatory relationship to
20 an interface number, but in the simple case, a Port Number will have the same value as the corresponding
21 interface's interface number. As a result of limitations in the BPDU for STP, this standard [see item i) in
22 12.3] limits the maximum number of Bridge Ports to 4095. However, in the absence of spanning tree there is
23 no restriction. As a result, Port Numbers are in the range (1..65 535) to allow correspondence to interface
24 numbers—but the STP restriction must be adhered to if it is supported on a Bridge.

25 Discontinuities in the value of the counter on the Bridge Port can occur at reinitialization of the management
26 system, and at other times as indicated by the value of ifCounterDiscontinuityTime object of the associated
27 interface (if any).

28 Some entities perform other functionalities as well as bridging through the sending and receiving of data on
29 their interfaces. In such situations, only a subset of the data sent/received on an interface is within the
30 domain of the entity's bridging functionality. This subset is considered to be delineated according to a set of
31 protocols, with some protocols being bridged, and other protocols not being bridged. For example, in an
32 entity that exclusively performs bridging, all protocols would be considered as bridged, whereas in an entity
33 that performs IP routing on IP datagrams and only Bridges other protocols, only the non-IP data would be
34 considered as having been bridged.

35 Thus, this BRIDGE-MIB (and in particular, its counters) are applicable only to that subset of the data on an
36 entity's interfaces that is sent/received for a protocol being bridged. All such data is sent/received via the
37 Ports of the Bridge.

38 The BRIDGE MIB also defines a new interface type called an internal LAN (I-LAN). This interface is
39 identified by an ifIndex and is intended to help manage internal associations between bridging components.
40 The BRIDGE-MIB provides the ability to create and destroy I-LAN interfaces. When an I-LAN interface is
41 created, a corresponding row is created in the ifTable. The ifType of this interface is 247.²⁹

²⁹ IANA ifType registry is <https://www.iana.org/assignments/ianaiftype-mib/ianaiftype-mib>.

1 I-LAN interfaces support stacking through the ifStackTable. Through this mechanism, the I-LAN interface
2 can create internal connections between components. For example, an I-LAN interface can connect two
3 C-VLAN components in a system. To complete this process, the following steps are required:

- 4 a) Create C-VLAN component 1.
- 5 b) Create C-VLAN Port 1 on component 1.
- 6 c) Create a Bridge Port interface (ifType 209) identified by ifIndex 1 using a external mechanism.
- 7 d) Associate C-VLAN Port 1 on component 1 to Bridge Port interface 1.
- 8 e) Create an I-LAN interface identified with ifIndex 2.
- 9 f) Stack Bridge Port interface 1 on the I-LAN interface 2.
- 10 g) Create C-VLAN component 2.
- 11 h) Create C-VLAN Port 1 on component 2.
- 12 i) Create a Bridge Port interface (ifType 209) identified by ifIndex 3 using a external mechanism.
- 13 j) Associate C-VLAN Port 1 on component 2 to Bridge Port interface 3.
- 14 k) Stack Bridge Port interface 3 on the I-LAN interface 2.

15 The result of the previous configuration steps are displayed in the Figure 17-1.

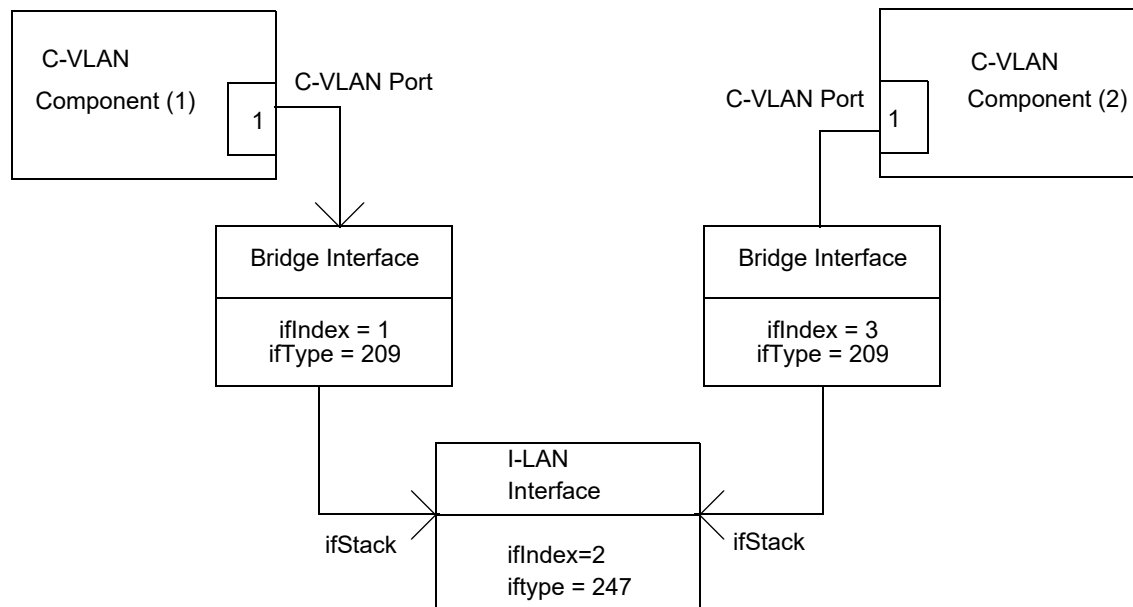


Figure 17-1—C-VLAN component internal LAN managed system

16 As a result, traffic can now be forwarded between C-VLAN component 1 and C-VLAN component 2
17 through the internal LAN.

18 17.3.2.3 FDB for IEEE 802.1D Bridges

19 The tables needed to explicitly manage the FDB for IEEE 802.1D Bridges have been removed from the
20 IEEE8021-BRIDGE-MIB. There were separate tables for this function in the original IETF MIBs, but these
21 duplicate tables in the IEEE8021-QBRIDGE MIB and are not needed.

22 As a result, single FDB implementations, like IEEE Std 802.1D, should just implement the
23 ieee8021QBridgeTpFdbTable with an ieee8021QBridgeFdbId of 1, and setting any VID indices to 0.

17.3.3 Relationship of the IEEE8021-RSTP MIB to other MIB modules

The objects in the RSTP MIB supplement those defined in the BRIDGE MIB.

The original IETF BRIDGE-MIB [RFC1493] and its SMIPv2-compliant version [RFC4188] have been replaced by 17.7.2. Besides using different prefix names for modules, an additional index was added to allow use of the RSTP MIB module in PBB.

17.3.4 Relationship of the IEEE8021-Q-BRIDGE-MIB to other MIB modules

17.3.4.1 Relationship to the IF-MIB

This standard assumes the interpretation of the Interfaces Subtree to be in accordance with the IF-MIB [RFC2863], which states that the interfaces table (ifTable) contains information on the managed resource's interfaces and that each sublayer below the internetwork layer of a network interface is considered an interface.

This standard does not make any assumption that within an entity, VIDs that are instantiated as an entry in dot1qVlanCurrentTable—either by management configuration through dot1qVlanStaticTable or by dynamic means (e.g., through MVRP)—are also represented by an entry in ifTable.

Where an entity contains higher-layer protocol entities (e.g., IP-layer interfaces that transmit and receive traffic to/from a VLAN), these should be represented by an interface that represents the protocol entity and an interface of ifType l2vlan (135) with the ifStackTable indicating the stacking relationship between the two entities.

17.3.4.1.1 ifStackTable

In addition, the IF-MIB [RFC2863] defines a table 'ifStackTable' for describing the relationship between logical interfaces within an entity. It is anticipated that implementors will use this table to describe the binding of (for example) Link Aggregation Group (LAG) interfaces to physical Ports, although the presence of VLANs makes the representation less than perfect for showing connectivity. The ifStackTable cannot represent the full capability of this standard, since this standard makes a distinction between VLAN bindings on *ingress* to and *egress* from a Port. These relationships may or may not be symmetrical; whereas, Interface MIB Evolution assumes a symmetrical binding for transmit and receive. This makes it necessary to define other manageable objects for configuring which Ports are in the member set for which VIDs.

17.3.4.1.2 ifRcvAddressTable

This table contains all MAC addresses, unicast, multicast, and broadcast, for which an interface will receive frames and forward them up to a higher-layer entity for local consumption. Note that this does not include addresses for data-link layer control protocols such as STP, RSTP, MSTP, MMRP, or MVRP. The format of the address, contained in ifRcvAddressAddress, is the same as for ifPhysAddress.

This table does not include unicast or multicast addresses that are accepted for possible forwarding out some other Port. This table is explicitly not intended to provide a Bridge address filtering mechanism.

17.3.4.2 Relationship to IEEE8021-BRIDGE-MIB

This subclause defines how objects in the IEEE8021-BRIDGE-MIB module should be represented for devices that implement the extensions. Some of the old objects are less useful in such devices, but must still be implemented for reasons of backwards compatibility.

1 **17.3.4.2.1 ieee8021BridgeBase subtree**

2 This subtree contains objects that are applicable to all types of Bridges. Interpretation of this subtree is
3 unchanged.

4 **17.3.4.2.2 ieee8021BridgeTp subtree**

5 This subtree contains objects that describe the entity's state with respect to transparent bridging.

6 In a device operating with a single FDB, interpretation of this subtree is unchanged.

7 In a device supporting multiple FDBs, this subtree is interpreted as follows:

- 8 a) ieee8021BridgeTpLearnedEntryDiscards
9 The number of times that any of the FDBs became full.
- 10 b) ieee8021BridgeTpAgingTime
11 This applies to all FDBs.
- 12 c) ieee8021BridgeTpFdbTable
13 Report MAC addresses learned on each Port, regardless of which FDB they have been learned in. If
14 an address has been learned in multiple databases on a single Port, report it only once. If an address
15 has been learned in multiple databases on more than one Port, report the entry on any one of the
16 valid Ports.
- 17 d) ieee8021BridgeTpPortTable
18 This table is Port-based and is not affected by multiple FDBs or multiple VIDs. The counters should
19 include frames received or transmitted for all VIDs.

20 **17.3.4.2.3 ieee8021BridgeStatic subtree**

21 This optional subtree contains objects that describe the configuration of destination-address filtering.

22 In a device operating with a single FDB, interpretation of this subtree is unchanged.

23 In a device supporting multiple FDBs, this subtree is interpreted as follows:

- 24 a) ieee8021BridgeStaticTable
25 Entries read from this table include all static entries from all of the FDBs. Entries for the same MAC
26 address and receive Port in more than one FDB must appear only once, since these are the indices of
27 this table. This table should be implemented as read-only in devices that support multiple FDBs.
28 Instead, write access should be provided through dot1qStaticUnicastTable and
29 dot1qStaticMulticastTable, as defined in this standard.

30 **17.3.4.2.4 Additions to the IEEE8021-BRIDGE-MIB**

31 To supplement the BRIDGE-MIB, this module contains the following:

- 32 a) Support for multiple traffic classes and dynamic multicast filtering as per IEEE Std 802.1D.
- 33 b) Support for bridged VLANs as per IEEE Std 802.1Q.
- 34 c) Support for 64-bit versions of BRIDGE-MIB Port counters.

35 **17.3.5 Relationship of the IEEE8021-PB-BRIDGE MIB to other MIB modules**

36 To supplement the Q-BRIDGE-MIB, this module contains the following:

- 37 a) The Q-Bridge MIB directly supports VID translation. The original bidirectional VID translation of
38 PB MIB is supported by the Q-Bridge MIB as well as separate ingress and egress VID translation.
- 39 b) Support for Provider Edge Bridges.

1 17.3.6 Relationship of the IEEE8021-MSTP-MIB to other MIB modules

2 The objects in the IEEE8021-MSTP-MIB supplement those defined in the IEEE8021-RSTP MIB.

3 17.3.7 Relationship of the IEEE8021-CFM-MIB to other MIB modules

4 17.3.7.1 Relationship to Interface MIB

5 Subclause 17.7.7 defines a CFM MIB module that supports 12.14 with the textual conventions imported
6 from the TC MIB in 17.7.1. A system implementing the CFM MIB module in 17.7.7 shall also implement at
7 least the System Group of the SNMPv2-MIB defined in IETF RFC 3418 and the Interfaces Group (the
8 Interfaces MIB module, or IF-MIB) defined in IETF RFC 2863. The Interfaces Group has one conceptual
9 row in a table for every interface in a system. 3.3 of IETF RFC 2863 defines hierarchical relationships
10 among interfaces.

11 IETF RFC 2863 also requires that any MIB module that is an adjunct of the Interface Group clarify specific
12 areas within the Interface MIB module. These areas were intentionally left vague in IETF RFC 2863 to
13 avoid over-constraining the MIB, thereby precluding management of certain media types. These areas are
14 clarified in other clauses that define the Bridge MIB modules. Even if a Bridge supports none of these, if it
15 supports the CFM MIB module, and hence, the Interfaces Group, the clarifications from the other clauses
16 shall be applied to the Interfaces Group. The relationship between IETF RFC 2863 and IETF RFC 3418
17 interfaces and Bridge Ports is also described in previous subclauses of 17.3. In addition, if both the CFM
18 MIB module (17.7.7) and IEEE Std 802.1AX are supported, a Bridge shall

- 19 a) Assign each IEEE 802.1AX Port in the aggregation its own conceptual row in the IETF RFC 2863
20 IF-MIB.
- 21 b) Assign one conceptual row in the IETF RFC 2863 IF-MIB that references the aggregated Port. This
22 may be the same as the interface identifying the Bridge Port. It shall not be the same as any of the
23 IEEE 802.1AX Ports being aggregated.

24 17.3.7.2 IEEE8021-CFM-MIB and IEEE8021-CFM-V2-MIB

25 The inclusion of PBB (Clause 26) required the addition of additional indices to numerous tables in the CFM
26 MIB. As a result, these tables are deprecated in the existing IEEE8021-CFM-MIB module and new
27 reindexed tables are created in a second IEEE8021-CFM-V2-MIB module. Unchanged tables remain in the
28 IEEE8021-CFM-MIB module. Specifically, groups with tables left unchanged are imported in the
29 IEEE8021-CFM-V2-MIB module to appear in the compliance statement and new groups containing the new
30 reindexed tables are added to the compliance statement.

31 The following tables are renamed:

32 dot1agPbbCfmStackTable
33 dot1agPbbCfmVlanTable
34 dot1agPbbCfmDefaultMdTable
35 dot1agPbbCfmConfigErrorListTable
36 dot1agPbbCfmMaCompTable

37 The new names are as follows:

38 ieee8021CfmStackTable
39 ieee8021CfmVlanTable
40 ieee8021CfmDefaultMdTable
41 ieee8021CfmConfigErrorListTable
42 ieee8021CfmMaCompTable

17.3.8 Relationship of the IEEE8021-PBB-MIB to other MIB modules

The IF-MIB, [RFC2863], requires that any MIB that is an adjunct of the IF-MIB clarify specific areas within the IF-MIB. These areas were intentionally left vague in the IF-MIB in order to avoid over-constraining the MIB, thereby precluding management of certain media types. The IF-MIB enumerates several areas that a media-specific MIB must clarify. Each of these areas is addressed in a following subclause. The implementor is referred to the IF-MIB in order to understand the general intent of these areas. The IF-MIB [RFC2863] defines managed objects for managing network interfaces.

A network interface is considered attached to a *subnetwork*. The term *segment* is used to refer to such a subnetwork, whether it be an Ethernet LAN, a *ring*, a WAN link, or even an SDH virtual circuit. Full support for PBB managed objects requires that the IF-MIB objects `ifIndex`, `ifType`, `ifDescr`, `ifPhysAddress`, and `ifLastChange` be implemented. Note that compliance to the current IF-MIB module requires additional objects and notifications to be implemented as specified in IETF RFC 2863.

The interpretation of the Interfaces Subtree is assumed to be in accordance with the IF-MIB, which states that the interfaces table (`ifTable`) contains information on the managed resource's interfaces and that each sublayer below the internetwork layer of a network interface is considered an interface. Specifically, in the PBB MIB module, the PIP index has been equated to an interface index (`ifIndex`). Therefore, when a PIP is created, a row is also created in the `ifTable` identified by the PIP's `ifIndex`. The `ifType` of this interface is 248.³⁰

In addition, the IF-MIB defines a table 'ifStackTable' for describing the relationship between logical interfaces within an entity. The `ifStackTable` cannot represent the full capability of this standard, since this standard makes a distinction between VLAN bindings on *ingress* to and *egress* from a Port. These relationships may or may not be symmetrical; whereas, Interface MIB Evolution assumes a symmetrical binding for transmit and receive. This makes it necessary to define other manageable objects for configuring which Ports are in the member set for which VIDs.

A PIP supports stacking through the `ifStackTable`. For external PIP interfaces, a PIP can be stacked on an Ethernet interface (`ifType` 6). If the PIP is internal, it can be stacked on an I-LAN interface to create a connection between an I-component and a B-component. The following is an example of how this process is completed:

- a) Create I-component 1.
- b) Create VIP 1 on component 1.
- c) Create a PIP identified by `ifIndex` 1.
- d) Associate VIP 1 on component 1 to the PIP.
- e) Create an I-LAN interface identified with `ifIndex` 2.
- f) Stack PIP 1 on the I-LAN interface 2.
- g) Create B-component 2.
- h) Create CBP 1 on component 2.
- i) Associate CBP 1 on component 2 to I-LAN interface 2.

The result of the previous configuration steps are displayed in Figure 17-2.

As a result, traffic can now be forwarded between I-component 1 and B-component 2 once a service is provisioned.

³⁰ IANA `ifType` registry is <https://www.iana.org/assignments/ianaiftype-mib/ianaiftype-mib>.

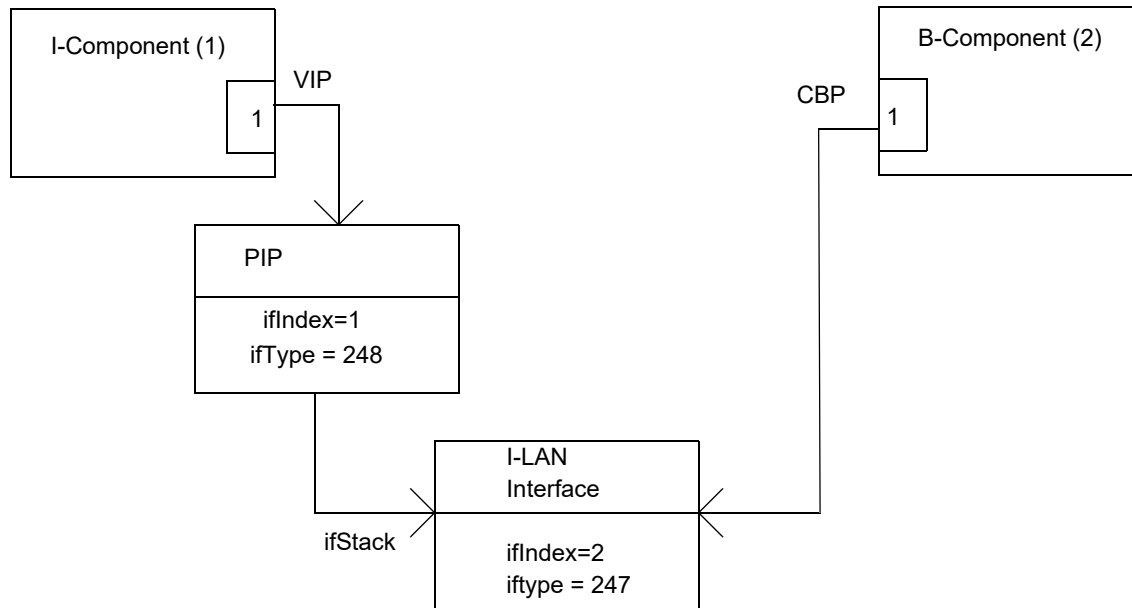


Figure 17-2—I/B-component internal LAN managed system

17.3.9 Relationship of the IEEE8021-DDCFM to other MIB modules

Subclause 17.7.9 defines the DDCFM MIB module that supports 12.17. A system implementing the DDCFM MIB module in 17.7.9 shall also implement at least the System Group of the SNMPv2-MIB defined in IETF RFC 3418 and Interfaces Group (the Interfaces MIB module, or IF-MIB) defined in IETF RFC 2863.

17.3.10 Relationship of the IEEE8021-PBBTE-MIB to other MIB modules

The IEEE8021-PBBTE-MIB is used to manage IB-BEBs that support additional PBB-TE functionality for the configuration of TESIs across provider backbone networks. The IEEE8021-PBBTE-MIB uses textual conventions and objects from the following MIB modules:

- IEEE8021-TC-MIB
- IEEE8021-BRIDGE-MIB
- Q-BRIDGE-MIB
- IEEE8021-Q-BRIDGE-MIB

The IEEE8021-PBBTE-MIB has the purpose of managing PBB-TE-specific functionality. As a PBB-TE IB-BEB is a specialized PBB IB-BEB with additional functionality, management of the system will also require that the system support the following Clause 17 MIB modules:

- IEEE8021-BRIDGE-MIB
- IEEE8021-CFM-MIB
- IEEE8021-CFM-V2-MIB
- IEEE8021-Q-BRIDGE-MIB
- IEEE8021-MSTP-MIB
- IEEE8021-PB-MIB
- IEEE8021-PBB-MIB

1 Specifically, support of the IEEE-PBBTE-MIB requires that the system support the MIB and compliances in
2 Table 17-1.

Table 17-1—PBB-TE required MIB compliances

MIB	Compliance
IEEE8021-PBB-MIB	ieee8021PbbWithPbbTeCompliance
IEEE8021-CFM-MIB	dot1agCfmWithPbbTeCompliance
IEEE8021-CFM-V2-MIB	

3 17.3.11 Relationship of the IEEE8021-TPMR MIB to other MIB modules

4 Subclause 17.7.11 defines a TPMR MIB module that supports the managed objects defined in 12.19. A
5 system implementing the TPMR MIB module in 17.7.11 shall also implement at least the System Group of
6 the SNMPv2-MIB defined in IETF RFC 3418 and the Interfaces Group (the Interfaces MIB module, or
7 IF-MIB) defined in IETF RFC 2863. The Interfaces Group has one conceptual row in a table for every
8 interface in a system. See Table 17-17 for the key objects from the System and Interfaces groups.

9 17.3.12 Relationship of the IEEE8021-FQTSS-MIB to other MIB modules

10 The IEEE8021-FQTSS-MIB provides objects that extend the core management functionality of a Bridge, as
11 defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the additional management
12 functionality needed when the FQTSS, as defined in Clause 34, is supported by the Bridge. As support of the
13 objects defined in the IEEE8021-FQTSS-MIB also requires support of the IEEE8021-BRIDGE-MIB, the
14 provisions of 17.3.2 apply to implementations claiming support of the IEEE8021-FQTSS-MIB.

15 17.3.13 Relationship of the IEEE802-CN-MIB to other MIB modules

16 17.3.13.1 Interface MIB

17 Subclause 17.7.13 defines a Congestion Notification MIB (IEEE802-CN-MIB) module that supports
18 congestion notification with the textual conventions imported from the TC MIB in 17.7.13. A system
19 implementing the IEEE802-CN-MIB module in 17.7.13 shall also implement at least the System Group of
20 the SNMPv2-MIB defined in IETF RFC 3418 and the Interfaces Group (the Interfaces MIB module, or
21 IF-MIB) defined in IETF RFC 2863. The Interfaces Group has one conceptual row in a table for every
22 interface in a system. 3.3 of IETF RFC 2863 defines hierarchical relationships among interfaces. IETF RFC
23 2863 also requires that any MIB module that is an adjunct of the Interface Group clarify specific areas
24 within the Interface MIB module. These areas were intentionally left vague in IETF RFC 2863 to avoid
25 over-constraining the MIB, thereby precluding management of certain media types. These areas are clarified
26 in other clauses that define the MIB modules in this standard. Even if a system supports none of these, if it
27 supports the IEEE802-CN-MIB module, and hence, the Interfaces Group, the clarifications from the other
28 clauses shall be applied to the Interfaces Group. The relationship between IETF RFC 2863 and IETF RFC
29 3418 interfaces and ports is also described in previous subclauses of 17.3. In addition, if both the
30 IEEE802-CN-MIB module (17.7.13) and IEEE Std 802.1AX are supported, a Bridge component or end
31 station may

- 32 a) Assign each port in the aggregation its own conceptual row in the IETF RFC 2863 IF-MIB.
- 33 b) Assign one conceptual row in the IETF RFC 2863 IF-MIB that references the aggregated port. This
34 can be the same as the interface identifying the port. It shall not be the same as any of the ports being
35 aggregated.

17.3.13.2 IEEE8021-CFM-MIB and IEEE8021-CFM-V2-MIBs

Subclause 31.1 allows a system that supports either the dot1agCfmVlanTable from the IEEE8021-CFM-MIB module or the ieee8021CfmVlanTable from the IEEE8021-CFM-V2-MIB module to use those tables to select the vlan_identifier for a transmitted CNM.

17.3.14 Relationship of the IEEE8021-SRP-MIB to other MIB modules

The IEEE8021-SRP-MIB provides objects that extend the core management functionality of a Bridge, as defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the management functionality needed when the SRP extensions, as defined in Clause 35, are supported by the Bridge. As support of these objects defined in the IEEE8021-SRP-MIB also requires support of the IEEE8021-TC-MIB, IEEE8021-BRIDGE-MIB, and IEEE8021-FQTSS-MIB, the provisions of 17.3.2 apply to implementations claiming support of the IEEE8021-SRP-MIB.

17.3.15 Relationship of the IEEE8021-MVRPX-MIB to other MIB modules

The IEEE8021-MVRPX-MIB, because it adds variables to the ieee8021BridgeBasePortEntry in the IEEE8021-BRIDGE-MIB, depends upon that MIB. It also imports items from the SNMPv2-SMI, SNMPv2-TC, and SNMPv2-CONF MIBs, and requires the systemGroup of the SNMPv2-MIB for conformance.

17.3.16 Relationship of the IEEE8021-MIRP-MIB to other MIB modules

The IEEE8021-MIRP-MIB, because it extends the variables in the IEEE8021-PBB-MIB (17.7.8), depends upon that MIB. Because it adds a variable to the ieee8021BridgeBasePortEntry in the IEEE8021-BRIDGE-MIB, it also depends upon that MIB. It also imports items from the SNMPv2-SMI, SNMPv2-TC, SNMPv2-CONF, and Q-BRIDGE-MIB MIBs, and requires the systemGroup of the SNMPv2-MIB for conformance.

17.3.17 Relationship of the IEEE8021-PFC-MIB to other MIB modules

A system implementing the IEEE8021-PFC-MIB module (17.7.17) shall also implement at least the System Group of the SNMPv2-MIB defined in IETF RFC 3418 and the Interfaces Group (the Interfaces MIB module, or IF-MIB) defined in IETF RFC 2863. The Interfaces Group has one conceptual row in a table for every interface in a system. 3.3 of IETF RFC 2863 defines hierarchical relationships among interfaces. IETF RFC 2863 also requires that any MIB module that is an adjunct of the Interface Group clarify specific areas within the Interface MIB module. These areas were intentionally left vague in IETF RFC 2863 to avoid over constraining the MIB, thereby precluding management of certain media types. These areas are clarified in other clauses which define the MIB modules in this standard. Even if a system supports none of these, if it supports the PFC MIB module, and hence, the Interfaces Group, the clarifications from the other clauses shall be applied to the Interfaces Group. The relationship between IETF RFC 2863 and IETF RFC 3418 interfaces and ports is also described in previous subclauses of 17.3.

17.3.18 Relationship of the IEEE8021-TEIPS-MIB to other MIB modules

The IEEE8021-TEIPS-MIB is used to manage IB-BEBs and BCBs that support IPS functionality. The IEEE8021-TEIPS-MIB uses textual conventions and objects from the following MIB modules:

- IEEE8021-TC-MIB
- IEEE8021-BRIDGE-MIB

The IEEE8021-TEIPS-MIB has the purpose of managing IPS in PBB-TE Region (26.11). Thus, use of the IEEE8021-TEIPS-MIB requires that the system support the same set of MIB modules specified within 17.3.10 and requires the compliances described by Table 17-21.

1 **17.3.19 Relationship of the IEEE8021-SPB-MIB to other MIB modules**

2 The IEEE8021-SPB-MIB, because it provides mainly control plane functions and reuses VLANs in the
3 Q-BRIDGE-MIB depends on that MIB. It also imports items from the SNMPv2-SMI, SNMPv2-TC,
4 IF-MIB, and IEEE8021-TC-MIB, and requires SNMPv2-CONF for conformance.

5 PCR operations are based on ISIS-SPB sub-TLVs, with additional objects that support MRT operations.

6 **17.3.20 Relationship of the IEEE8021-EVB-MIB to other MIB modules**

7 The IEEE8021-EVB-MIB provides objects that extend the core management functionality of a Bridge, as
8 defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the management functionality needed
9 for EVB (5.23, 5.24), as defined in Clause 40, Clause 41, Clause 42, and Clause 43. As support of the
10 objects defined in the IEEE8021-EVB-MIB also requires support of the IEEE8021-TC-MIB and
11 IEEE8021-BRIDGE-MIB, the provisions of 17.3.2 apply to implementations claiming support of the
12 IEEE8021-EVB-MIB.

13 **17.3.21 Relationship of the IEEE8021-ECMP-MIB to other MIB modules**

14 The IEEE8021-ECMP-MIB extends SPBM functionality and therefore is closely related to the
15 IEEE8021-SPB-MIB and IEEE8021-PBB-MIB. For example, it includes configuration for an optional
16 ISIS-SPB sub-TLV for the ECMP ECT Algorithm (28.12.6.1) and configuration for flow filtering on PNPs
17 and CBPs (44.2.2).

18 Because it provides control functions related to VIDs, it depends on the Q-BRIDGE-MIB. It also augments
19 tables defined in the IEEE8021-Q-BRIDGE-MIB (adding the ability to read the FDB Port Map for an
20 individual address and adding TTL discard statistics). It also imports items from the SNMPv2-SMI,
21 SNMPv2-TC, and IEEE8021-TC-MIB and requires SNMPv2-CONF for conformance.

22 **17.3.22 Relationship of the IEEE8021-ST-MIB to other MIB modules**

23 The IEEE8021-ST-MIB provides objects that extend the core management functionality of a Bridge, as
24 defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the additional management
25 functionality needed when the scheduled traffic extensions, as defined in 8.6.8.4, are supported by
26 the Bridge. As support of the objects defined in the IEEE8021-ST-MIB also requires support of the
27 IEEE8021-BRIDGE-MIB, the provisions of 17.3.2 apply to implementations claiming support of
28 the IEEE8021-ST-MIB.

29 **17.3.23 Relationship of the IEEE8021-Preemption-MIB to other MIB modules**

30 The IEEE8021-Preemption-MIB provides objects that extend the core management functionality of a
31 Bridge, as defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the additional management
32 functionality needed when the frame preemption extensions are supported by the Bridge. As support of the
33 objects defined in the IEEE8021-Preemption-MIB also requires support of the IEEE8021-BRIDGE-MIB,
34 the provisions of 17.3.2 apply to implementations claiming support of the IEEE8021-Preemption-MIB.

35 **17.3.24 Relationship of the IEEE8021-PSFP-MIB to other MIB modules**

36 The IEEE8021-PSFP-MIB provides objects that extend the core management functionality of a Bridge, as
37 defined by the IEEE8021-BRIDGE-MIB (17.7.2), in order to support the additional management
38 functionality needed when the PSFP extensions, as defined in 8.6.5.2.1 and 8.6.10, are supported by the
39 Bridge. As support of the objects defined in the IEEE8021-PSFP-MIB also requires support of the
40 IEEE8021-BRIDGE-MIB, the provisions of 17.3.2 apply to implementations claiming support of the
41 IEEE8021-PSFP-MIB.

1 17.3.25 Relationship of the IEEE8021-TSN-REMOTE-MANAGEMENT-MIB to other MIB 2 modules

3 The IEEE8021-TSN-REMOTE-MANAGEMENT-MIB provides objects that support management of TSN
4 stream reservations by providing Bridge transit delay and Bridge Port propagation delay information, by
5 allowing a remote manager to determine whether static trees are supported (12.32.3), and provides
6 additional control over MSRP attribute propagation. As support of the objects defined in the
7 IEEE8021-TSN-REMOTE-MANAGEMENT-MIB also requires support of the IEEE8021-BRIDGE-MIB,
8 the provisions of 17.3.2 apply to implementations claiming support of the
9 IEEE8021-TSN-REMOTE-MANAGEMENT-MIB.

10 17.3.26 Relationship of the IEEE8021-PBBN-AA-MIB to other MIB modules

11 The IEEE8021-PBBN-AA-MIB provides objects that extend the functionality of IEEE802.1AB LLDP.

12 17.4 Security considerations

13 SNMP versions prior to SNMPv3 did not include adequate security. Even if the network itself is secure (for
14 example, by using IPsec), there is no control about who on the secure network is allowed to access and
15 GET/SET (read/change/create/delete) the objects in these MIB modules.

16 It is recommended that implementers consider the security features as provided by the SNMPv3 framework
17 (see IETF RFC 3410 (2002), section 8), including full support for the SNMPv3 cryptographic mechanisms
18 (for authentication and privacy).

19 Further, deployment of SNMP versions prior to SNMPv3 is not recommended. Instead, it is recommended
20 to deploy SNMPv3 and to enable cryptographic security. It is then a customer/operator responsibility to
21 ensure that the SNMP entity giving access to an instance of these MIB modules is properly configured to
22 give access to the objects only to those principals (users) that have legitimate rights to indeed GET or SET
23 (change/create/delete) them.

24 17.4.1 Security considerations of the IEEE8021-TC-MIB

25 This module does not define any management objects. Instead, it defines a set of textual conventions that
26 may be used by other MIB modules to define management objects. Meaningful security considerations are
27 contained in the following subclauses that describe security considerations for other MIB modules that
28 define management objects with a SYNTAX of any of these textual conventions.

29 17.4.2 Security considerations of the IEEE8021-BRIDGE-MIB

30 There are a number of management objects defined in the IEEE8021-BRIDGE-MIB module that have a
31 MAX-ACCESS clause of read-write and/or read-create. Such objects may be considered sensitive or
32 vulnerable in some network environments. The support for SET operations in a nonsecure environment
33 without proper protection can have a negative effect on network operations.

34 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
35 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
36 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
37 even encrypt the values of these objects when sending them over the network via SNMP.

1 The following tables and objects in the IEEE8021-BRIDGE-MIB can be manipulated to interfere with the
2 operation of priority classes. This could, for example, be used to force a reinitialization of state machines,
3 thus causing network instability. Another possibility would be for an attacker to override established policy
4 on Port priorities, thus giving a user (or an attacker) unauthorized preferential treatment.

5 ieee8021TrafficClassesEnabled
6 ieee8021GmrpStatus
7 ieee8021PortPriorityTable
8 ieee8021UserPriorityRegenTable
9 ieee8021TrafficClassTable
10 ieee8021PortGarpTable
11 ieee8021PortGmrpTable

- 12 a) The writable object ieee8021BridgeTpAgingTime controls how fast dynamically-learned
13 forwarding information is aged out. Setting this object to a large value may simplify FDB overflow
14 attacks. Setting this object to too small a value may compromise the throughput of the network by
15 causing excessive flooding.
- 16 b) The writable ieee8021BridgeStaticTable provides a filtering mechanism controlling to which Ports
17 frames originating from a specific source may be forwarded. Write access to this table can be used to
18 turn provisioned filtering off or to add filters to prevent rightful use of the network.

19 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
20 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
21 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
22 even encrypt the values of these objects when sending them over the network via SNMP. These tables and
23 objects and their sensitivity/vulnerability are described as follows:

- 24 — The objects ieee8021DeviceCapabilities and ieee8021PortCapabilitiesTable in the
25 IEEE8021-P-BRIDGE-MIB could be used by an attacker to determine which attacks might be
26 useful to attempt against a given device.
- 27 — The readable objects defined in the IEEE8021-BRIDGE-MIB module provide information about the
28 topology of a bridged network and the attached active stations. The addresses listed in the
29 ieee8021BridgeTpFdbTable usually reveal information about the manufacturer of the MAC
30 hardware, which can be useful information for mounting other specific attacks.
- 31 — The two notifications, newRoot and topologyChange, are emitted during spanning tree computation
32 and may trigger management systems to inspect the status of Bridges and to recompute internal
33 topology information. Hence, forged notifications may cause management systems to perform
34 unnecessary computations and to generate additional SNMP traffic directed to the Bridges in a
35 network. Therefore, forged notifications may be part of a denial of service attack.

36 17.4.3 Security considerations of the IEEE8021-SPANNING-TREE MIB

37 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
38 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
39 environments. The support for SET operations in a nonsecure environment without proper protection can
40 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
41 described below.

42 The following writable objects could be misused to cause network delays and spanning tree instabilities:

43 ieee8021SpanningTreeVersion
44 ieee8021SpanningTreeRstpTxHoldCount
45 ieee8021SpanningTreeRstpPortProtocolMigration
46 ieee8021SpanningTreeRstpPortAdminEdgePort

1 ieee8021SpanningTreeRstpPortAdminPathCost
2 ieee8021SpanningTreeRstpPortAdminPointToPoint

3 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
4 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
5 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
6 even encrypt the values of these objects when sending them over the network via SNMP. These tables and
7 objects and their sensitivity/vulnerability are described below:

8 ieee8021SpanningTreeVersion could be read by an attacker to identify environments containing applications
9 or protocols that are potentially sensitive to RSTP mode.

10 The writable objects:

11 ieee8021SpanningTreePriority
12 ieee8021SpanningTreeBridgeMaxAge
13 ieee8021SpanningTreeBridgeHelloTime
14 ieee8021SpanningTreeBridgeForwardDelay
15 ieee8021SpanningTreePortPriority
16 ieee8021SpanningTreePortEnable
17 ieee8021SpanningTreePortPathCost

18 influence the spanning tree protocol. Unauthorized write access to these objects can cause the spanning tree
19 protocol to compute other default topologies or it can change the speed in which the spanning tree protocol
20 reacts to failures.

21 17.4.4 Security considerations of the IEEE8021-Q-BRIDGE-MIB

22 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
23 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
24 environments. The support for SET operations in a nonsecure environment without proper protection can
25 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
26 described below.

27 The following tables and objects in the IEEE8021-Q-BRIDGE-MIB could be manipulated to interfere with
28 the operation of VLANs. This could, for example, be used to force a reinitialization of state machines to
29 cause network instability, or to change the forwarding and filtering policies.

30 ieee8021GvrpStatus
31 ieee8021ForwardAllTable
32 ieee8021StaticUnicastTable
33 ieee8021StaticMulticastTable
34 ieee8021VlanStaticTable
35 ieee8021PortVlanTable
36 ieee8021LearningConstraintsTable
37 ieee8021ProtocolGroupTable
38 ieee8021ProtocolPortTable

39 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
40 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
41 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
42 even encrypt the values of these objects when sending them over the network via SNMP. These tables and
43 objects and their sensitivity/vulnerability are described below.

1 The following read-only tables and objects in the IEEE8021-Q-BRIDGE-MIB could be used by an attacker
2 to determine which attacks might be useful to attempt against a given device, could be used by an attacker to
3 detect whether their attacks are being blocked or filtered, or could be used to understand the logical topology
4 of the network.

5 ieee8021MaxVlanID
6 ieee8021MaxSupportedVlans
7 ieee8021NumVlans
8 ieee8021FdbTable
9 ieee8021TpFdbTable
10 ieee8021TpPortGroupTable
11 ieee8021VlanCurrentTable
12 ieee8021PortVlanStatisticsTable

13 **17.4.5 Security considerations of the IEEE8021-PB-MIB**

14 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
15 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
16 environments. The support for SET operations in a non-secure environment without proper protection can
17 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
18 described below.

19 The following tables and objects in the PB-MIB could be manipulated to interfere with the operation of
20 VLANs. This could, for example, be used to force a reinitialization of state machines to cause network
21 instability, or to change the forwarding and filtering policies.

22 ieee8021PbProviderBridgePortTable
23 ieee8021PbVidTranslationTable
24 ieee8021PbCVidRegistrationTable
25 ieee8021PbEdgePortTable
26 ieee8021PbInternalInterfaceTable

27 **17.4.6 Security considerations of the IEEE8021-MSTP-MIB**

28 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
29 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
30 environments. The support for SET operations in a nonsecure environment without proper protection can
31 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
32 described below.

33 Writable objects that could be misused to cause network delays and spanning tree instabilities include the
34 following:

35 ieee8021CistEnableBPDURx
36 ieee8021CistEnableBPDUTx
37 ieee8021CistPortAdminEdgePort
38 ieee8021CistPortMacEnabled
39 ieee8021MstpPortPriority
40 ieee8021MstpPortPathCost

1 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
2 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
3 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
4 even encrypt the values of these objects when sending them over the network via SNMP. These tables and
5 objects and their sensitivity/vulnerability are described below:

6 ieee8021SpanningTreeVersion could be read by an attacker to identify environments containing applications
7 or protocols that are potentially sensitive to RSTP mode.

8 **17.4.7 Security considerations of the IEEE8021-CFM-MIB**

9 It is expected that, in some provider networks, management access to a Bridge can be granted, by an
10 organization that owns and is responsible for the normal operations of the Bridge, to another organization
11 that needs to configure and manage CFM entities in that Bridge. For example, a Provider could sell a service
12 to a customer that connects sites in cities thousands of kilometers apart. That Provider might not have a
13 physical presence in all of those cities (or any of them, for that matter) and therefore would need to contract
14 with other Providers to supply a number of interconnected PBNs that together are able to offer the service to
15 the customer. The relationship between the Provider and Owner is not necessarily cordial; there can be
16 personal or financial incentives for misbehavior.

17 To support this scenario, the MIB module in 17.7.7 defines the Maintenance Domain table—
18 dot1agCfmMdTable. Each row (dot1agCfmMdEntry) in this table corresponds to one Maintenance Domain
19 managed object (12.14.5). A Maintenance Domain managed object can be created, read, and written by the
20 Bridge's Owner. More limited access to that object can be granted to another Provider, who thereby becomes
21 that Maintenance Domain's administrator. Associated with each row in the Maintenance Domain table are
22 any number of rows (dot1agCfmMaNetEntry and dot1agCfmMaCompEntry) in the MA
23 tables—dot1agCfmMaNetTable and dot1agCfmMaCompTable. Each of the paired rows in these two tables
24 corresponds to a Maintenance Association managed object (12.14.6). A Maintenance Domain administrator
25 has the ability to create and alter these rows. Additional MIB tables are defined in 17.7.7 that correspond to
26 the Maintenance Domain list managed object (12.14.1), the CFM Stack managed object (12.14.2), and the
27 Default MD Level managed object (12.14.3), all of which can be used by an Owner of the Bridge, but not by
28 a Maintenance Domain administrator.

29 The SNMPv3 framework (IETF STD 62) defines a securityName, which provides for secure identification
30 of a user (or group of users) via an SNMPv3 Security Model. For ease of reference in this present standard,
31 the "Owner" of a system (in particular, a Bridge) is defined as a user whose "MIB view" includes
32 READ-WRITE access to the System Group of the SNMPv2-MIB (IETF RFC 3418). A Management
33 Domain administrator is then a user whose MIB view includes more limited access to the Maintenance
34 Domain managed object, and full READ-WRITE access to its subordinate Maintenance Association
35 managed objects (12.14.6), each of which is a pair of rows (dot1agCfmMaNetEntry and
36 dot1agCfmMaCompEntry) in the two tables—dot1agCfmMaNetTable and dot1agCfmMaCompTable.

37 A Bridge can support IETF STD 62 (SNMPv3) in order to provide a secure means for confining access by
38 Management Domain administrators to their corresponding Maintenance Association managed objects, and
39 to prevent inappropriate access to the rest of the Bridge's management objects. If the SNMPv3 framework is
40 not implemented in a Bridge, an Owner is defined, imprecisely, as an administrator of the Bridge, and a
41 Management Domain administrator (even less precisely) as an administrator of one Management Domain.

42 Therefore, it is recommended that the implementors consider the security features as provided by the
43 SNMPv3 framework, including full support for the SNMPv3 cryptographic mechanisms (for authentication
44 and privacy). Specifically, a Bridge can use the User-Based Security Model, IETF RFC 3414, and/or the
45 View-Based Access Control Model, IETF RFC 3415, for controlling access to the CFM managed objects. It

1 is then a customer/user responsibility to ensure that the SNMP agent giving access to an instance of this MIB
2 is properly configured to give access to the objects only to those principals (users) that have legitimate rights
3 to GET or SET (change/create/delete) them.

4 The control of access by Providers to other MIB modules, for example, the Interface Group, is not specified
5 by this standard.

6 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
7 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
8 environments. The support for SET operations in a nonsecure environment without proper protection can
9 have a negative effect on network operations. Table 17-2 lists the MIB tables and objects and their
10 sensitivity/vulnerability.

Table 17-2—Sensitive managed objects: tables and notifications

Table or object	Reason for sensitivity to security considerations
dot1agCfmDefaultMdTable dot1agCfmVlanTable	The integrity of the spanning tree(s) running in a network is dependent on certain configuration parameters, especially Static VLAN Registration Entries (8.8.2). The assignment of VIDs to particular service instances, and hence to MAs, is controlled by the administrator responsible for the integrity of the spanning tree, namely the Owner of the Bridge.
dot1agCfmMdTable dot1agCfmMaNetTable dot1agCfmMaCompTable dot1agCfmVlanTable dot1agCfmMaMepListTable dot1agCfmMepTable dot1agCfmMepDbTable dot1agCfmLtrTable dot1agCfmFaultAlarm	The business relationships among the various Maintenance Domain administrators, and between them and the Owner of the Bridge, are not defined by this standard. The ability of one Maintenance Domain administrator to identify another, or even detect the presence of another Maintenance Domain administrator, in the same Bridge, may jeopardize those business relationships. The CFM MIB module separates objects by Maintenance Domain so that the Bridge Owner can ensure that management by one Maintenance Domain administrator does not conflict with management by another.

11 There is no detailed list in this standard of all vulnerable objects with a MAX-ACCESS clause of read-write
12 or read-create and of the security threats associated to their intentional or unintentional manipulation by
13 write actions. The reason is that this standard takes the approach that objects within a maintenance domain
14 are protected as if in a “walled garden,” accessible only by administrators authorized for the respective
15 Maintenance Domains. See Figure 12-1 for an illustration of this concept.

16 The only exceptions to this approach are the read-create MIB objects that define the maintenance
17 domains themselves and their capabilities. The vulnerabilities associated with these objects are shown in
18 Table 17-3.

19 **17.4.8 Security considerations of the IEEE8021-PBB-MIB**

20 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
21 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
22 environments. The support for SET operations in a nonsecure environment without proper protection can
23 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
24 described next.

25 The following tables and objects in the PBB MIB could be manipulated to interfere with the operation of
26 PBBs. This could, for example, be used to force a reinitialization of state machines to cause network
27 instability, or changing the forwarding and filtering policies. The following are vulnerable writable objects
28 from the IEEE8021-PBB-MIB:

Table 17-3—Sensitive managed objects: variables in dot1agCfmMdTable

Object	Reason for sensitivity to security considerations
dot1agCfmMdFormat dot1agCfmMdName	An intentional or accidental write operation on these objects may lead to the modification of the identifiers of the Maintenance Domains and would impact the capacity of existing administrators to identify and manage the entities within those Maintenance Domains.
dot1agCfmMdMdLevel	An intentional or accidental write operation on this object may impact the CFM operations on the Maintenance Domain by altering the MD Level associated with the Maintenance Domain.
dot1agCfmMdMhfCreation	An intentional or accidental write operation on this object may impact the capabilities for creating MIP Half Functions (MHFs) in the Maintenance Domain.

1 ieee8021PbbVipISid

2 ieee8021PbbPipVipMap

3 In addition, the following are vulnerable tables from the IEEE8021-PBB-MIB:

4 ieee8021PbbCBPServiceMappingTable

5 ieee8021PbbPipPriorityTable

6 ieee8021PbbVipToPipMappingTable

7 ieee8021PbbCbpTable

8 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
9 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
10 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
11 even encrypt the values of these objects when sending them over the network via SNMP.

12 The following read-only tables and objects in this MIB could be used by an attacker to determine which
13 attacks might be useful to attempt against a given device, or could be used to understand the logical topology
14 of the network:

15 ieee8021PbbBackboneEdgeBridgeAddress

16 ieee8021PbbVipPipIfIndex

17 ieee8021PbbISidToVipComponentId

18 ieee8021PbbISidToVipPort

19 17.4.9 Security considerations of the IEEE8021-DDCFM-MIB

20 There are a number of management objects defined in DDCFM MIB module with a MAX-ACCESS clause
21 of read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
22 environments. The support for SET operations in a nonsecure environment without proper protection can
23 have a negative effect on network operations. Table 17-4 lists the MIB tables and objects and their
24 sensitivity/vulnerability.

25 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
26 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
27 important to control even GET and/or NOTIFY access to these objects and possibly to even encrypt the
28 values of these objects when sending them over the network via SNMP. Table 17-5 lists the MIB tables and
29 objects that are sensitive to read.

Table 17-4—Sensitive managed objects (of DDCFM): tables and notifications

Table or object	Reason for sensitivity to security considerations
ieee8021DdcfmRr	Once created and activated, ieee8021DdcfmRr can intercept selected data frames traversing through a Bridge interface, and send its copy encapsulated in RFM header to a specific destination within the maintenance domain. The action can create extra traffic within the maintenance domain.
ieee8021DdcfmDr	Once created and activated, ieee8021DdcfmDr will decapsulate received SFM and send the decapsulated data frame to location specified by the data frame's destination_address. This action practically injects specific traffic into the network.
ieee8021DdcfmSFMSOriginator	Once created and activated, ieee8021DdcfmSFMSOriginator injects SFMs into the maintenance domain. However, if the corresponding ieee8021DdcfmDr is not activated, the received SFMs are dropped. Therefore, ieee8021DdcfmSFMSOriginator is less sensitive than the corresponding ieee8021DdcfmDr.

1

Table 17-5—Sensitive managed objects (of DDCFM) for read

Table or object	Reason for sensitivity to security considerations
ieee8021DdcfmRFMReceiver	Once created, ieee8021DdcfmRFMReceiver can receive the RFM, which encapsulate actual data frame that could be sensitive and need to be protected. It is very important to control the GET operation on this object.

2 17.4.10 Security considerations of the IEEE8021-PBBTE-MIB

3 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
4 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
5 environments. The support for SET operations in a nonsecure environment without proper protection can
6 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
7 described below.

8 The following tables and objects in the PBB MIB could be manipulated to interfere with the operation of
9 IB PBBs. This could, for example, be used to force a reinitialization of state machines to cause network
10 instability, or changing the forwarding and filtering policies. The following are all the writable objects from
11 the IEEE8021-PBB-MIB:

- 12 ieee8021PbbTeProtectionGroupListWorkingMA
- 13 ieee8021PbbTeProtectionGroupListProtectionMA
- 14 ieee8021PbbTeProtectionGroupListRowStatus
- 15 ieee8021PbbTeTesiComponent
- 16 ieee8021PbbTeTesiBridgePort
- 17 ieee8021PbbTeProtectionGroupConfigCommandAdmin
- 18 ieee8021PbbTeProtectionGroupConfigWTR
- 19 ieee8021PbbTeProtectionGroupConfigHoldOff
- 20 ieee8021PbbTeProtetcionGroupConfigNotifyEnable
- 21 ieee8021PbbTeProtectionGroupISidGroupId
- 22 ieee8021PbbTeProtectionGroupISidRowStatus
- 23 ieee8021PbbTeBridgeStaticForwardAnyUnicastTable
- 24 ieee8021PbbTeBridgeStaticForwardAnyUnicastEgressPorts
- 25 ieee8021PbbTeBridgeStaticForwardAnyUnicastForbiddenPorts

17.4.11 Security considerations of the IEEE8021-TPMR-MIB

There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a nonsecure environment without proper protection can have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are described below.

The following object in the SNMPv2-MIB could be manipulated to interfere with the operation of a management system and its ability to recognize and manage a TPMR device.

sysName

The following object in the SNMPv2-MIB could be manipulated to interfere with the operation of a management system and its ability to recognize and manage a TPMR Port.

ifName

The following objects in the TPMR-MIB could be manipulated to interfere with the operation of MSP on a TPMR Port and, for example, be used to cause network instability.

ieee8021TpmrMspLinkNotify
ieee8021TpmrMspLinkNotifyWait
ieee8021TpmrMspLinkNotifyRetry
ieee8021TpmrMspMacNotify
ieee8021TpmrMspMacNotifyTime
ieee8021TpmrMspMacRecoverTime

17.4.12 Security considerations of the IEEE8021-FQTSS-MIB

There are a number of management objects defined in the IEEE8021-FQTSS-MIB module that have a MAX-ACCESS clause of read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a nonsecure environment without proper protection can have a negative effect on network operations.

Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than notaccessible) may be considered sensitive or vulnerable in some network environments. It is thus important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to even encrypt the values of these objects when sending them over the network via SNMP.

The following tables and objects in the IEEE8021-FQTSS-MIB can be manipulated to interfere with the operation of the forwarding and queuing mechanisms in a manner that would be detrimental to the transmission of time-sensitive streams:

ieee8021FqtssDeltaBandwidth
ieee8021FqtssAdminIdleSlopeMs
ieee8021FqtssAdminIdleSlopeLs
ieee8021FqtssClassMeasurementInterval
ieee8021FqtssSrClassId
ieee8021FqtssTxSelectionAlgorithmID
ieee8021FqtssPriorityRegenOverride

- a) ieee8021FqtssDeltaBandwidth can be manipulated to reduce the amount of bandwidth available to a given SR class.
- b) ieee8021FqtssAdminIdleSlopeMs and ieee8021FqtssAdminIdleSlopeLs can be manipulated to change the overall amount of bandwidth available to stream traffic on a Port, in a network where SRP is not used for stream reservations.

- c) ieee8021FqtssClassMeasurementInterval can be manipulated to change the measurement interval that is used, together with TSpec, to calculate reserved bandwidth.
- d) ieee8021FqtssSrClassId can be manipulated to change the priority mapping of a SRclassID to a priority. This can impact the behavior of Streams.
- e) ieee8021FqtssTxSelectionAlgorithmID can be manipulated in order to apply the wrong transmission selection algorithm to a traffic class that was being used by stream traffic.
- f) ieee8021FqtssPriorityRegenOverride can be manipulated to disrupt stream traffic within an SRP domain with non-stream traffic entering from outside the SRP domain boundary.

17.4.13 Security considerations of the IEEE8021-CN-MIB

There are a number of management objects defined in the IEEE8021-MIRP-MIB module with a MAX-ACCESS clause of read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a nonsecure environment without proper protection can have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are described in the paragraphs that follow.

Improper manipulation of certain tables and objects can result in the misidentification of the boundaries of CNDs (30.6). Any such errors can result in the following problems:

- a) Frames not originated through an RP (31.2.2.2) can pass through CPs (31.1.1). The inability to throttle these frames via CNMs (33.3) can cause legitimate, RP-originated frames to experience unacceptably high loss rates.
- b) Frames not originating through an RP (31.2.2.2) can pass through queues in the bridged network that are not CPs (31.1.1), where they can encounter uncontrolled congestion. This can cause those frames to experience unacceptably high loss rates.
- c) The CN-TAG (30.5) in a data frame or CNM can be improperly discarded. This can cause that CNM, or a CNM resulting from the data frame, to be discarded by the end station that sourced the frame. As a result, that flow, and all flows passing through the same CPs, can experience unacceptably high loss rates.
- d) The CN-TAG in a data frame or CNM can be improperly retained. This can result in a failure of two end stations to communicate, because the receiving end station is unable to process frames with CN-TAGs.

The following tables and objects in the IEEE8021-MIRP-MIB can cause such CND boundary identification errors:

- ieee8021CnGlobalMasterEnable
- ieee8021CnComPriDefModeChoice
- ieee8021CnComPriAdminDefenseMode
- ieee8021CnComPriCreation
- ieee8021CnComPriLldpInstanceChoice
- ieee8021CnComPriLldpInstanceSelector
- ieee8021CnComPriRowStatus
- ieee8021CnPortPriDefModeChoice
- ieee8021CnPortPriAdminDefenseMode
- ieee8021CnPortPriLldpInstanceChoice
- ieee8021CnPortPriLldpInstanceSelector

Improper manipulation of certain other objects can result in assigning a data frame or CNM to the incorrect priority. This can result in the affected data streams experiencing loss rates that are either higher or lower than expected by the network administrator. Lower than expected loss rates for one data stream can cause

1 higher than expected loss rates for other streams. These affects are typically, but not universally, less severe
2 than the loss rate anomalies caused by CND boundary misidentification. The following variables can cause
3 improper priority assignment:

4 ieee8021CnGlobalCnmTransmitPriority
5 ieee8021CnComPriAlternatePriority
6 ieee8021CnPortPriAlternatePriority

7 Improper manipulation of the remainder of the read-write and read-create objects can cause the congestion
8 notification algorithm to operate in ways not intended by the network administrator. This can result in
9 excessively high data frame loss rates and/or low link utilization rates. These variables are the following:

10 ieee8021CnCpQueueSizeSetPoint
11 ieee8021CnCpFeedbackWeight
12 ieee8021CnCpMinSampleBase
13 ieee8021CnCpMinHeaderOctets
14 ieee8021CnRpPortPriMaxRps
15 ieee8021CnRpgEnable
16 ieee8021CnRpgTimeReset
17 ieee8021CnRpgByteReset
18 ieee8021CnRpgThreshold
19 ieee8021CnRpgMaxRate
20 ieee8021CnRpgAiRate
21 ieee8021CnRpgHaiRate
22 ieee8021CnRpgGd
23 ieee8021CnRpgMinDecFac
24 ieee8021CnRpgMinRate

25 Unintended access to any of the readable tables or variables in the IEEE8021-MIRP-MIB alerts the reader
26 that congestion notification is configured, and (for all tables and variables except those in the
27 ieee8021CnGlobalTable and ieee8021CnCpidToInterfaceTable) on which priority value or values
28 congestion notification is configured. This information can suggest to an attacker what applications are
29 being run, and thus suggest application-specific attacks, or to enable the attacker to detect whether their
30 attacks are successful. It is thus important to control even GET and/or NOTIFY access to these objects and
31 possibly to even encrypt the values of these objects when sending them over the network via SNMP.

32 **17.4.14 Security considerations of the IEEE8021-SRP-MIB**

33 The purpose of MSRP is to create reservations for various types of data streams, including audio/video (AV)
34 content. Access to the objects within the IEEE8021-SRP-MIB module, whether they have MAX-ACCESS
35 of read-write, read-create, or read-only, may reveal sensitive information in some network environments.
36 Very serious health and safety situations could arise if MSRP was involved in configuring network resources
37 for an emergency public safety announcement and the MSRP behavior of the bridged network was allowed
38 to be modified unexpectedly.

39 With these considerations in mind it is thus important to control all types of access (including GET and/or
40 NOTIFY) to these objects and possibly even encrypt their values when sending them over the network via
41 SNMP.

42 The following tables and objects in the IEEE8021-SRP-MIB can be manipulated to interfere with the
43 operation of the stream reservation mechanisms in a manner that would be detrimental to the transmission of
44 the associated stream data:

45 ieee8021SrpBridgeBaseMsrpEnabledStatus
46 ieee8021SrpBridgeBaseMsrpTalkerPruning

- 1 ieee8021SrpBridgeBaseMsrpMaxFanInPorts
- 2 ieee8021SrpBridgeBaseMsrpLatencyMaxFrameSize
- 3 ieee8021SrpBridgeBaseMsrpTalkerVlanPruning
- 4 ieee8021SrpBridgePortMsrpEnabledStatus
- 5 ieee8021SrpBridgePortSrPvid
- 6 ieee8021SrpBridgeBaseMsrpTalkerPruningPerPort
- 7 a) ieee8021SrpBridgeBaseMsrpEnabledStatus can be manipulated to enable or disable MSRP
- 8 operations for the entire Bridge.
- 9 b) ieee8021SrpBridgeBaseMsrpTalkerPruning can be manipulated to stop the propagation of Talker
- 10 attributes if Listeners are not configured to support Talker Pruning.
- 11 c) ieee8021SrpBridgeBaseMsrpMaxFanInPorts can be manipulated to set the number of ingress Ports
- 12 supporting streaming down to one, which would stop Talkers streams from coming in on any other
- 13 Port.
- 14 d) ieee8021SrpBridgeBaseMsrpLatencyMaxFrameSize can be manipulated to set a frame size that is
- 15 so large or so small that it causes the Bridge to calculate unreasonable maximum latency.
- 16 e) ieee8021SrpBridgeBaseMsrpTalkerVlanPruning can be manipulated to change the behavior of a
- 17 Bridge regarding Talker declarations. This can either restrict Talker declarations in an unintended
- 18 way or cause Talker declarations to be forwarded on unintended Ports.
- 19 f) ieee8021SrpBridgePortMsrpEnabledStatus can be manipulated to enable or disable MSRP on a
- 20 particular Port, perhaps allowing sensitive stream data to be sent to unacceptable devices.
- 21 g) ieee8021SrpBridgePortSrPvid can be manipulated to move Streams to a VLAN that has been
- 22 blocked by management, thus disabling reception of the Stream by one or more Listeners.
- 23 h) ieee8021SrpBridgeBaseMsrpTalkerPruningPerPort can be manipulated to disable MAC address
- 24 pruning per Port, causing Talker declarations to be forwarded on unintended Ports.

25 ieee8021SrpBridgePortSrPvid can be manipulated to move Streams to a VLAN that has been blocked by
26 management, thus disabling reception of the Stream by one or more Listeners.

27 **17.4.15 Security considerations of the IEEE8021-MVRPX-MIB**

28 The objects in IEEE8021-MVRPX-MIB could be manipulated to interfere with the
29 operation of PBBs. Setting ieee8021MvrpxPortNewOnly, ieee8021MvrpxPortMvrpNewPropagated, or
30 ieee8021MvrpxPortXmitZero to the wrong value could cause New messages to be transmitted when they
31 should not be or suppressed when they should be transmitted. This can result in the FDB being flushed either
32 too often or not at all, or in lost or spurious VLAN registrations. Extra New messages waste resources in the
33 Bridge's supervisory processor and can cause unnecessary flooding of data frames to unknown destinations,
34 and lost New messages can prevent connectivity until the entries in the FDB time out. Lost or spurious
35 VLAN registrations can cause the loss of connectivity for particular VLANs, or waste system resources
36 transporting data frames to parts of the network where they are not needed.

37 **17.4.16 Security considerations of the IEEE8021-MIRP-MIB**

38 The following tables and objects in the IEEE8021-MIRP-MIB could be manipulated to interfere with the
39 operation of PBBs. They could, for example, prevent Multiple I-SID Registration Protocol Data Units
40 (MIRPDUs) from reaching their intended destinations, or cause them to reach unintended destinations. The
41 former could result in temporary loss of service due to MAC address entries being timed out, instead of
42 being flushed. The latter could result in excessive computation time taken by the PIPs or I-components
43 unnecessarily receiving the MIRPDUs. The following are vulnerable writable objects from the
44 IEEE8021-MIRP-MIB:

- 45 ieee8021MirpV2PortTable
- 46 ieee8021MirpV2PortEnabledStatus

1 ieee8021PbbMirpEnableStatus
2 ieee8021PbbMirpBvid
3 ieee8021PbbMirpDestSelector
4 ieee8021PbbMirpPnpEnable
5 ieee8021PbbMirpPnpPortNumber

6 **17.4.17 Security considerations of the IEEE8021-PFC-MIB**

7 One management object defined in the IEEE8021-PFC-MIB module has a MAXACCESS clause of
8 read-write. Such object can be considered sensitive or vulnerable in some network environments. The
9 support for SET operations in a nonsecure environment without proper protection can have a negative effect
10 on network operations. The management object is:

11 PFCLinkDelayAllowance

12 Improper setting of this management object can result in improper network operations. If the value of this
13 management object is too high, then PFC can be invoked excessively, negatively impacting the link
14 bandwidth. If the value of this management object is too low, then PFC can be invoked too late and frame
15 loss can occur.

16 **17.4.18 Security considerations of the IEEE8021-TEIPS-MIB**

17 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
18 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
19 environments. The support for SET operations in a nonsecure environment without proper protection can
20 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
21 described below.

22 The following tables and objects in the TEIPS-MIB could be manipulated to interfere with the operation of
23 PBBs. This could, for example, be used to force a reinitialization of state machines to cause network
24 instability or to change the forwarding and filtering policies. The following are all the writable objects from
25 the IEEE8021-TEIPS-MIB:

26 ieee8021TeipsV2Ipgid
27 ieee8021TeipsV2IpgWorkingMA
28 ieee8021TeipsV2IpgProtectionMA
29 ieee8021TeipsV2TesiId
30 ieee8021TeipsV2CandidatePsMA
31 ieee8021TeipsV2IpgConfigWTR
32 ieee8021TeipsV2IpgConfigHoldOff
33 ieee8021TeipsV2IpgConfigMWTR

34 **17.4.19 Security considerations of the IEEE8021-SPB-MIB**

35 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
36 read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network
37 environments. The support for SET operations in a non-secure environment without proper protection can
38 have a negative effect on network operations. These tables and objects and their sensitivity/vulnerability are
39 described next.

40 ieee8021SpbMtidStaticEntryMtid

1 The following tables and objects in the SPB-MIB could be manipulated to interfere with the operation of
2 Shortest Path Bridges. This could, for example, be used to misconfigure the network to cause loss of
3 connectivity, or misconnect traffic. The following are vulnerable writable objects from the
4 IEEE8021-SPB-MIB:

5 ieee8021SpbSys
6 ieee8021SpbSysAreaAddress
7 ieee8021SpbSysId
8 ieee8021SpbSysControlAddr
9 ieee8021SpbSysName
10 ieee8021SpbSysBridgePriority
11 ieee8021SpbMtidStaticTable
12 ieee8021SpbMtidStaticTableEntry
13 ieee8021SpbMTidStaticEntryMtidOverload
14 ieee8021SpbmBsiStaticTable
15 dot1agCfmMepSpbmEspTable
16 ieee8021PcrEctStaticTable

17 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than not
18 accessible) may be considered sensitive or vulnerable in some network environments. It is thus important to
19 control all types of access (including GET and/or NOTIFY) to these objects and possibly to even encrypt the
20 values of these objects when sending them over the network via SNMP. The following read-only tables with
21 their respective objects in this MIB could be used by an attacker to understand the logical topology of the
22 network:

23 ieee8021SpbEctDynamicTable
24 ieee8021SpbAdjDynamicTable
25 ieee8021SpbTopNodeTable
26 ieee8021SpbTopEctTable
27 ieee8021SpbTopEdgeTable
28 ieee8021PcrTopEctTable

29 **17.4.20 Security considerations of the IEEE8021-EVB-MIB**

30 The purpose of EVB is to coordinate VSIs within an EVB station with a data center network (DCN). In this
31 environment the EVB station and the EVB Bridge may be under different management authorities. Access
32 to the objects within the IEEE8021-EVB-MIB module of the EVB Bridge by the EVB station and access to
33 objects within the IEEE8021-EVB-MIB module of the EVB station by the EVB Bridge may therefore need
34 to be restricted.

35 Access to the objects within the IEEE8021-EVB-MIB module, whether they have MAX-ACCESS of
36 read-write, read-create, or read-only, can reveal sensitive information in some network environments. Very
37 serious health and safety situations could arise if EVB systems were involved in configuring network
38 resources for an emergency public safety announcement and the EVB Bridge system behavior of the bridged
39 network was allowed to be modified unexpectedly.

40 With these considerations in mind, it is thus important to control all types of access (including GET and/or
41 NOTIFY) to these objects. SNMP versions prior to SNMPv3 did not include adequate security. Even if the
42 network itself is secure (for example, by using IPsec), there is no control about who on the secure network is
43 allowed to access and GET/SET (read/change/create/delete) the objects in this MIB module.

1 It is recommended that implementers consider the security features as provided by the SNMPv3 framework
2 (see IETF RFC 3410 (2002), section 8), including full support for the SNMPv3 cryptographic mechanisms
3 (for authentication and privacy).

4 Further, deployment of SNMP versions prior to SNMPv3 is not recommended. Instead, it is recommended
5 to deploy SNMPv3 and to enable cryptographic security. It is then a customer/operator responsibility to
6 ensure that the SNMP entity giving access to an instance of this MIB module is properly configured to give
7 access to the objects only to the principals (users) that have legitimate rights to indeed GET or SET
8 (change/create/delete) them.

9 There are a number of management objects defined in IEEE8021-EVB-MIB module with a MAX-ACCESS
10 clause of read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some
11 network environments. The support for SET operations in a non-secure environment without proper
12 protection can have a negative effect on network operations. Table 17-6 describes the tables and objects and
13 their sensitivities/vulnerabilities.

Table 17-6—Sensitive managed objects (of EVB): tables and notifications

Table or object	Reason for sensitivity to security considerations
evbSysEvbLldpTxEnable evbSysEvbLldpManual evbSysEvbLldpGidCapable	The EVB TLV exchange controls how the VDP, ECP, and reflective relay parameters are set. These parameters allow manual configurations of the EVB parameters, which can be used to disable the operation of these protocols.
ieee8021BridgeEvbSysEcpAckTimer ieee8021BridgeEvbSysEcpMaxRetires	The ECP timer and retry count are set to provide reliable delivery of control frames. Incorrect settings can cause failures of the control protocols.
ieee8021BridgeEvbSysVdpDfltRsrcWaitDelay	The VDP resource timer determines the time required by the system to locate a profile. Setting this time too short may make VDP requests always fail.
ieee8021BridgeEvbSbpRowStatus ieee8021BridgeEvbUapRowStatus ieee8021BridgeEvbCapRowStatus ieee8021BridgeEvbUrpRowStatus	These variables allow creations of new ports on an EVB Bridge or EVB station. Inappropriate use of these may allow unauthorized interception of data.

14 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
15 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
16 important to control even GET and/or NOTIFY access to these objects and possibly to even encrypt the
17 values of these objects when sending them over the network via SNMP. Table 17-7 provides the identity of
18 VSIs and the addresses used to access them that could be used to identify the user application and what
19 addresses can be used to interfere with or intercept their traffic.

Table 17-7—Sensitive managed objects (of EVB) for read

Table	Reason for sensitivity to security considerations
ieee8021BridgeEvbVsiDbTable	The VSI database provides a list of all operating VSIs along with their primary system keys. This information may be used to spy on the operating applications and the activity associated with each application.
ieee8021BridgeEvbVsiDbMacTable	The VSI MAC table provides the network addresses associated with operating applications. This information may be used in system attacks.

17.4.21 Security considerations of the IEEE8021-ECMP-MIB

There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of read-write and/or read-create. Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations. These objects and their sensitivity/vulnerability are described next.

The following objects in the IEEE8021-ECMP-MIB could be manipulated to interfere with the operation of ECMP networks. These could, for example, be used to misconfigure the network to cause loss of connectivity or undesirable path selection. The following are vulnerable writable objects from the IEEE8021-ECMP-MIB:

- ieee8021EcmpFlowFilterCtlTtl
- ieee8021EcmpEctStaticEntryBridgePriority
- ieee8021EcmpEctStaticEntryRowStatus

17.4.22 Security considerations of the IEEE8021-ST-MIB

There are a number of management objects defined in the IEEE8021-ST-MIB module that have a MAX-ACCESS clause of read-write. Such objects may be considered sensitive or vulnerable in some network environments. The support for SET operations in a nonsecure environment without proper protection can have a negative effect on network operations.

The following tables and objects in the IEEE8021-ST-MIB can be misconfigured to interfere with the operation of the forwarding and queuing mechanisms in a manner that would be detrimental to the transmission of scheduled traffic:

- ieee8021STMaxSDU
 - ieee8021STGateEnabled
 - ieee8021STAdminGateStates
 - ieee8021STAdminControlListLength
 - ieee8021STAdminControlList
 - ieee8021STAdminCycleTimeNumerator
 - ieee8021STAdminCycleTimeDenominator
 - ieee8021STAdminCycleTimeExtension
 - ieee8021STAdminBaseTime
 - ieee8021STConfigChange
 - ieee8021STConfigChangeTime
- a) ieee8021STMaxSDU can be misconfigured to affect the ability of a Port to transmit frames of greater than a given SDU size.
 - b) ieee8021STGateEnabled can be misconfigured to enable/disable scheduled traffic processing.
 - c) ieee8021STAdminGateStates can be misconfigured to affect the gate states of a Port on startup.
 - d) ieee8021STAdminControlListLength, ieee8021STAdminControlList, ieee8021STAdminCycleTimeNumerator, ieee8021STAdminCycleTimeDenominator, ieee8021STAdminCycleTimeExtension, ieee8021STAdminBaseTime, ieee8021STConfigChange, and ieee8021STConfigChangeTime can be misconfigured to affect the traffic schedule for the Port.

Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than not accessible) can be considered sensitive or vulnerable in some network environments. Thus it is important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to encrypt the values of these objects when sending them over the network via SNMP.

1 17.4.23 Security considerations of the IEEE8021-Preemption-MIB

2 There is one management object defined in the IEEE8021-Preemption-MIB module that has a
3 MAX-ACCESS clause of read-write. Such objects may be considered sensitive or vulnerable in some
4 network environments. The support for SET operations in a nonsecure environment without proper
5 protection can have a negative effect on network operations.

6 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than
7 not-accessible) may be considered sensitive or vulnerable in some network environments. It is thus
8 important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to
9 even encrypt the values of these objects when sending them over the network via SNMP.

10 The following tables and objects in the IEEE8021-Preemption-MIB can be manipulated to interfere with the
11 operation of the forwarding and queuing mechanisms in a manner that would be detrimental to the
12 transmission of frames:

13 ieee8021FramePreemptionAdminStatus

- 14 a) Misconfiguration of the ieee8021FramePreemptionAdminStatus object can lead to the degradation
15 of the quality of service for the application on the port at the respective traffic class that is wrongly
16 preempted, or to the saturation of the priority queues on the port because of preemptable traffic
17 classes receiving express processing.

18 17.4.24 Security considerations of the IEEE8021-PSFP-MIB

19 There are a number of management objects defined in the IEEE8021-PSFP-MIB module that have a
20 MAX-ACCESS clause of read-write or read-create. Such objects may be considered sensitive or vulnerable
21 in some network environments. The support for SET operations in a nonsecure environment without proper
22 protection can have a negative effect on network operations.

23 The following tables and objects in the IEEE8021-PSFP-MIB can be misconfigured to interfere with the
24 operation of the forwarding and queuing mechanisms in a manner that would be detrimental to the operation
25 of PSFP:

26 ieee8021PSFPStreamHandleSpec
27 ieee8021PSFPPrioritySpec
28 ieee8021PSFPStreamGateInstanceID
29 ieee8021PSFPFilterSpecificationList
30 ieee8021PSFPStreamBlockedDueToOversizeFrameEnable
31 ieee8021PSFPStreamBlockedDueToOversizeFrame
32 ieee8021PSFPGateEnabled
33 ieee8021PSFPAdminGateStates
34 ieee8021PSFPAdminControlListLength
35 ieee8021PSFPAdminControlList
36 ieee8021PSFPAdminCycleTimeNumerator
37 ieee8021PSFPAdminCycleTimeDenominator
38 ieee8021PSFPAdminCycleTimeExtension
39 ieee8021PSFPAdminBaseTime
40 ieee8021PSFPConfigChange
41 ieee8021PSFPConfigChangeTime
42 ieee8021PSFPAdminIPV
43 ieee8021PSFPOperIPV
44 ieee8021PSFPGateClosedDueToInvalidRxEnable

- 1 ieee8021PSFPGateClosedDueToInvalidRx
2 ieee8021PSFPGateClosedDueToOctetsExceededEnable
3 ieee8021PSFPGateClosedDueToOctetsExceeded
4 ieee8021PSFPFlowMeterInstance
5 ieee8021PSFPFlowMeterCIR
6 ieee8021PSFPFlowMeterCBS
7 ieee8021PSFPFlowMeterEIR
8 ieee8021PSFPFlowMeterCF
9 ieee8021PSFPFlowMeterCM
10 ieee8021PSFPFlowMeterDropOnYellow
11 ieee8021PSFPFlowMeterMarkAllFramesRedEnable
12 ieee8021PSFPFlowMeterMarkAllFramesRed
- 13 a) ieee8021PSFPStreamHandleSpec, ieee8021PSFPPrioritySpec,
14 ieee8021PSFPStreamGateInstanceID, and ieee8021PSFPFilterSpecificationList can be
15 misconfigured to adversely affect the policing functions that are applied to received frames.
- 16 b) ieee8021PSFPGateEnabled can be misconfigured to enable/disable scheduled traffic processing.
- 17 c) ieee8021PSFPAdminGateStates can be misconfigured to affect the gate state of a stream filter on
18 startup.
- 19 d) ieee8021PSFPAdminControlListLength, ieee8021PSFPAdminControlList,
20 ieee8021PSFPAdminCycleTimeNumerator, ieee8021PSFPAdminCycleTimeDenominator,
21 ieee8021PSFPAdminCycleTimeExtension, ieee8021PSFPAdminBaseTime,
22 ieee8021PSFPConfigChange, and ieee8021PSFPConfigChangeTime can be misconfigured to affect
23 the filter schedule for the Port.
- 24 ieee8021PSFPFlowMeterInstance, ieee8021PSFPFlowMeterCIR, ieee8021PSFPFlowMeterCBS,
25 ieee8021PSFPFlowMeterEIR, ieee8021PSFPFlowMeterCF, ieee8021PSFPFlowMeterCM,
26 ieee8021PSFPFlowMeterDropOnYellow, ieee8021PSFPFlowMeterMarkAllFramesRedEnable,
27 ieee8021PSFPFlowMeterMarkAllFramesRed, ieee8021PSFPGateClosedDueToInvalidRxEnable,
28 ieee8021PSFPGateClosedDueToInvalidRx,
29 ieee8021PSFPGateClosedDueToOctetsExceededEnable,
30 ieee8021PSFPGateClosedDueToOctetsExceeded,
31 ieee8021PSFPStreamBlockedDueToOversizeFrameEnable, and
32 ieee8021PSFPStreamBlockedDueToOversizeFrame can be misconfigured to adversely affect the
33 way that flow metering operates.

34 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than not
35 accessible) can be considered sensitive or vulnerable in some network environments. It is thus important to
36 control all types of access (including GET and/or NOTIFY) to these objects and possibly to encrypt the
37 values of these objects when sending them over the network via SNMP.

38 17.4.25 Security considerations of the 39 IEEE8021-TSN-REMOTE-MANAGEMENT-MIB

40 There are a number of management objects defined in the
41 IEEE8021-TSN-REMOTE-MANAGEMENT-MIB module that have a MAX-ACCESS clause of read-write
42 or read-create. Such objects may be considered sensitive or vulnerable in some network environments. The
43 support for SET operations in a nonsecure environment without proper protection can have a negative effect
44 on network operations.

45 The following tables and objects in the IEEE8021-TSN-REMOTE-MANAGEMENT-MIB can be
46 misconfigured to interfere with the operation of the forwarding and queuing mechanisms in a manner that
47 would be detrimental to the connectivity, delay, and jitter provided to TSN streams and other network traffic:

1 ieee8021TsnRemoteMgmtMsrpMrpExternalControl

2 **17.4.26 Security considerations of the IEEE8021-PBBN-AA-MIB**

3 There are a number of management objects defined in the IEEE8021-PBBN-AA-MIB module that have a
4 MAX-ACCESS clause of read-write. Such objects can be considered sensitive or vulnerable in some
5 network environments. The support for SET operations in a non-secure environment without proper
6 protection can have a negative effect on network operations.

7 The following tables and objects in the IEEE8021-PBBN-AA-MIB can be misconfigured to interfere with
8 the operation of the forwarding and queuing mechanisms in a manner that would be detrimental to the
9 transmission of scheduled traffic:

10	ieee8021AaSystemEnable
11	ieee8021AaPortRowStatus
12	ieee8021AaPortEnable
13	ieee8021AaIsidVidAsgnsRowStatus
14	ieee8021AaIsidVidAsgnsIsid
15	ieee8021AaIsidVidAsgnsVlan

16 Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than not
17 accessible) can be considered sensitive or vulnerable in some network environments. It is thus important to
18 control all types of access (including GET and/or NOTIFY) to these objects and possibly to encrypt the
19 values of these objects when sending them over the network via SNMP.

20

17.5 Dynamic component and Port creation

17.5.1 Overview of the dynamically created Bridge entities

The IEEE Bridge MIBs allow for the possibility of Bridge components and Bridge Ports to be created under the control of management operations. This functionality is optional. Compliant Bridges need not support this functionality or may tie the creation of components and Ports to the insertion or extraction of system hardware. This subclause outlines the rules and table interactions used by management stations to create soft configurable dynamic components and Ports on those systems that support this functionality.

A Bridge component contains, at a minimum, a collection of Bridge Ports, a relay unit that forwards and filters frames that travel between Ports, and managed objects to control the operation of the component. Certain types of components may contain other objects. These will be addressed later in this subclause in component-specific text.

Bridge components are the Owners of Bridge Ports, so components must be created before components can be populated with Ports.

The tables that are indexed by a single component ID index are as follows:

ieee8021BridgeBaseTable
ieee8021QBridgeTable
ieee8021QBridgeNextFreeLocalVlanTable
ieee8021QBridgeLearningConstraintDefaultTable
ieee8021CistTable
ieee8021MstConfigIdTable
ieee8021CfmVlanTable

17.5.1.1 Components

A component contains a relay function whose purpose is to move frames between interfaces to the relay called a Bridge Port or an Edge Relay Port. Different types of components are provided in this standard that are used to construct different types of Bridges.

17.5.1.2 Bridge Ports

A Bridge Port or Edge Relay Port is a frame source or sink directly attached to the relay function of a Bridge component.

17.5.1.3 Internal LAN connections

A BEB is composed of zero or one B-component and zero or more I-components. When the BEB has both a B-component and some I-components, the PIPs and CBPs of these I-components are connected by internal LAN connections. There needs to be a way to specify the interconnections between the PIPs on the I-component and the CBP on the B-component. This is done via the ieee8021BridgeILanIfTable defined in the IEEE8021-BRIDGE-MIB.

Essentially, this table allows for the creation of a new “interface” to represent the connection and then the ifStackTable is used to specify the interconnection.

1 These interconnections are used in multi-component Bridges, such as:

- 2 Provider Edge Bridges
- 3 BEBs
- 4 EVB stations
- 5 EVB Bridges

6 to specify the relationship between two interfaces in the following manner:

7 Two Port interfaces are interconnected if the invocation of a request operation at one of the interfaces causes
8 an indication operation with the same parameters to happen at the other interface.

9 **17.5.1.4 Provider Instance Ports (PIPs)**

10 Even though it is not a Bridge Port, the creation of PIPs on I-components are discussed as part of the Bridge
11 Port creation logic for I-components.

12 **17.5.2 Component creation**

13 A component is created by making an entry in the `ieee8021BridgeBaseTable` with the
14 `ieee8021BridgeBaseComponentType` set to the proper value. Components can also be created indirectly by
15 making entries in other system-specific tables, which then automatically create entries in the
16 `ieee8021BridgeBaseTable`.

17 A Bridge component consists of a relay function and related Bridge Ports or Edge Relay Ports. The
18 component type determines if the relay operates on untagged, C-tagged, or S-tagged frames. It also
19 determines which specific type(s) of Bridge Ports or Edge Relay Ports may be created on the component.

20 **17.5.2.1 MAC Bridge component creation**

21 The MAC Bridge component has no specific component creation rules.

22 **17.5.2.2 C-VLAN component creation**

23 C-VLAN components are used in the following different types of Bridges:

- 24 a) C-VLAN Bridges
- 25 b) Provider Edge Bridges
- 26 c) EVB Bridges

27 Provider Edge Bridge C-VLAN components are created implicitly by the creation of a CEP on the S-VLAN
28 component of the Provider Edge Bridge. C-VLAN components that belong to customer Bridges or to EVB
29 Bridges are created by a management station performing a row-create on the component table or by implicit
30 action such as the insertion of blades into a system.

31 **17.5.2.3 S-VLAN component creation**

32 S-VLAN components are used in three different ways in Bridges. The first is as the S-VLAN component of
33 an S-VLAN Bridge or Provider Edge Bridge or the foundation for an I-component or B-component. The
34 second is as a Port-mapping S-VLAN component in a Provider Edge Bridge. The third is as a Port-mapping
35 S-VLAN component in an EVB station or an EVB Bridge.

36 Provider Edge Bridge Port-mapping S-VLAN components are created implicitly by the creation of a RCAP
37 on the primary S-VLAN component of the Provider Edge Bridge.

1 EVB Bridge and EVB station Port-mapping S-VLAN components are created implicitly by the creation of
2 an Uplink Access Port (UAP) on an ISS of an EVB Bridge or an EVB station. For an EVB Bridge the ISS is
3 allocated to a Bridge Port of the primary C-VLAN component. On an EVB station the ISS has no permanent
4 Bridge Port or Edge Relay Port assignment.

5 **17.5.2.4 B-component creation**

6 The B-component has no specific component creation rules.

7 **17.5.2.5 I-component creation**

8 The I-component has no specific component creation rules.

9 **17.5.2.6 ER creation**

10 ERs of an EVB station are created implicitly by the creation of a URP, or actions such as the insertion of
11 blades into a system or the installation of a software driver on the EVB station.

12 **17.5.2.7 T-component creation**

13 The T-component has no specific component creation rules.

14 **17.5.3 Port creation**

15 This subclause discusses how Ports of each relevant Port type are created on each relevant component type.

16 The general procedure is for the network administrator to perform an SNMP row-create operation on a table
17 specific to the type of Port being created. If the operation succeeds, an entry will be implicitly created in the
18 ieee8021BridgeBasePortTable by the agent.

19 The specific details are outlined in the 17.5.3.1 through 17.5.3.7.

20 **17.5.3.1 Port creation on MAC Bridge components**

21 MAC Bridge components are used in MAC Bridges. Creation of a Port on a MAC Bridge component
22 requires specifying the component and Port index in the ieee8021BridgeCreatableBaseBridgeGroup table.

23 The type of the component referred to by the component ID parameter must be dBridgeComponent.

24 The implicitly constructed ieee8021BridgeBasePortTable entry will have the following fields filled in:

25	ieee8021BridgeBaseComponentId	- Implementation Specific
26	ieee8021BridgeBasePort	- Implementation Specific
27	ieee8021BridgeBasePortIfIndex	- Implementation Specific Action
28	ieee8021BridgeBasePortDelayExceededDiscards	- Statistic, reset to 0 by creation
29	ieee8021BridgeBasePortMtuExceededDiscards	- Statistic, reset to 0 by creation
30	ieee8021BridgeBasePortCapabilities	- Implementation Specific
31	ieee8021BridgeBasePortTypeCapabilities	- Implementation Specific
32		bit customerVlanPort(0) must be set
33	ieee8021BridgeBasePortType	- dBridgePort(8)
34	ieee8021BridgeBasePortExternal	- Implementation Specific

17.5.3.2 Port creation on C-VLAN components

C-VLAN components support several different types of Bridge Ports. These are C-VLAN Ports, CEPs, PEPs, and SBPs.

A C-VLAN component that is not part of a Provider Edge Bridge may have C-VLAN Ports. A C-VLAN component that is part of a Provider Edge Bridge must have exactly 1 CEP and any number of PEPs.

The only type of Ports that may be created by operating on the C-VLAN component that is not part of an EVB Bridge are the C-VLAN Ports. On an EVB Bridge it is possible to create both C-VLAN Bridge Ports and SBPs.

CEPs are created by a management action on the S-VLAN component of a Provider Edge Bridge. From a management perspective, these entities are managed through the S-VLAN component or via management operations specific to the Provider Edge Bridge.

PEPs, together with the associated CNP and the internal connection between them, are created by performing row-create operations for a C-VID in the `ieee8021PbCvidRegistrationTable` and the corresponding S-VID in the `ieee8021PbEdgePortTable`.

17.5.3.2.1 Creating C-VLAN Ports

C-VLAN Ports are created by performing a row-create operation on the `ieee8021QBridgeCVlanPortTable` for a C-VLAN component that is configured to act as a VLAN Bridge. The required columns are the component ID and the Port Number to use for the newly created Port.

The type of the component referred to by the component ID parameter must be `cVlanComponent` configured for Q-Bridge operation.

The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>QBridgeCVlanPortTable</code>
<code>ieee8021BridgeBasePort</code>	- As per <code>QBridgeCVlanPortTable</code>
<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific
	bit <code>customerVlanPort(0)</code> must be set
<code>ieee8021BridgeBasePortType</code>	- <code>customerVlanPort(2)</code>
<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17.5.3.2.2 Creating SBPs

SBPs are created by performing a row-create operation on the `ieee8021BridgeEvbSbpConfigTable` for a C-VLAN component that is configured to act as an EVB Bridge. The required columns are the component ID and the Port Number to use for the newly created Port.

The type of the component referred to by the component ID parameter is a `cVlanComponent` configured for Q-Bridge operation.

1 The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

2	<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>ieee8021BridgeEvbSbpConfigTable</code>
3	<code>ieee8021BridgeBasePort</code>	- As per <code>ieee8021BridgeEvbSbpConfigTable</code>
4	<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
5	<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset at creation
6	<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset at creation
7	<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
8	<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific:
9		bit <i>Station-facing Bridge Port</i> (8) is set
10	<code>ieee8021BridgeBasePortType</code>	- Station-facing Bridge Port (8)
11	<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

12 17.5.3.2.3 Creating CEPs

13 CEPs are created by doing a RowStatus create operation on the CEP table belonging to the PEB.

14 17.5.3.2.4 Creating PEPs

15 PEPs, together with the associated CNP and the internal connection between them, are created by
16 performing row-create operations for a C-VID in the `ieee8021PbCvidRegistrationTable` and the
17 corresponding S-VID in the `ieee8021PbEdgePortTable`. See the discussion on provisioning a C-tagged
18 service interface in 17.6.1.2.

19 17.5.3.3 Port creation on S-VLAN components

20 S-VLAN components are the S-VLAN component of an S-VLAN Bridge or Provider Bridges. There are
21 several different types of Ports that may be created on S-VLAN components. These are PNPs, CNPs, CEPs,
22 RCAPs, and UAPs.

23 17.5.3.3.1 Creating PNPs

24 A PNP on an S-VLAN component works pretty much the same way as a `customerVlanPort` on a C-VLAN
25 component except that the VID value used for forwarding and filtering decisions must come from an S-TAG
26 and not a C-TAG.

27 Creating a PNP on a Provider Bridge is done by adding an entry to the `ieee8021PbPnpTable` specifying the
28 `componentId` and Port Number.

29	<code>ieee8021BridgeBasePortComponentId</code>	- Component to which the new Port belongs
30	<code>ieee8021BrdigeBasePort</code>	- Port Number for the new Port

31 The type of the component referred to by the component ID parameter must be `sVlanComponent`.

32 This will cause an implicit entry in the `ieee8021QBridgePortVlan` table associated with the S-VLAN
33 component.

34 The Component ID must refer to the S-VLAN component of a Provider Edge Bridge and the Port Number
35 must refer to the Port Number of the PNP associated with that S-VLAN component.

1 The implicitly constructed ieee8021BridgeBasePortTable entry will have the following fields filled in:

2	ieee8021BridgeBasePortComponentId	- As per PbPnpTable
3	ieee8021BridgeBasePort	- As per PbPnpTable
4	ieee8021BridgeBasePortIfIndex	- Implementation Specific Action
5	ieee8021BridgeBasePortDelayExceededDiscards	- Statistic, reset to 0 by creation
6	ieee8021BridgeBasePortMtuExceededDiscards	- Statistic, reset to 0 by creation
7	ieee8021BridgeBasePortCapabilities	- Implementation Specific
8	ieee8021BridgeBasePortTypeCapabilities	- Implementation Specific
9		bit providerNetworkPort(1) must be set
10	ieee8021BridgeBasePortType	- providerNetworkPort(3)
11	ieee8021BridgeBasePortExternal	- Implementation Specific

12 17.5.3.3.2 Creating CNPs

13 There are two variants on the creation of CNPs. CNPs are either internal or external. Internal CNPs are
14 directly connected to either a PEP on a C-VLAN component or a PAP on a Port-mapping S-VLAN
15 component of a PEB and provide a C-tagged or Port-based service interface, respectively. External CNPs are
16 connected to an external customer system and provide either a Port-based or S-tagged service interface.

17 a) Creating an external CNP.

18 The external CNP is used to provide a Port-based or S-tagged service interface to customer of the
19 PBN. Creating an external CNP requires specifying the component ID and Bridge Port Number of
20 the Port to be created. This is done by creating an entry in the ieee8021PbCnpTable. This creates a
21 new entry in the ieee8021QBridgePortVlanTable for the provider Bridge's S-VLAN
22 component. This operation also created a new entry in the ieee8021BridgeBasePortTable.

23 The implicitly constructed ieee8021BridgeBasePortTable entry will have the following fields filled
24 in:

25	ieee8021BridgeBasePortComponentId	- As per PbCnpTable
26	ieee8021BridgeBasePort	- As per PbCnpTable
27	ieee8021BridgeBasePortIfIndex	- Implementation Specific Action
28	ieee8021BridgeBasePortDelayExceededDiscards	- Statistic, reset to 0 by creation
29	ieee8021BridgeBasePortMtuExceededDiscards	- Statistic, reset to 0 by creation
30	ieee8021BridgeBasePortCapabilities	- Implementation Specific
31	ieee8021BridgeBasePortTypeCapabilities	- Implementation Specific
32		bit customerNetworkPort(2) must be set
33	ieee8021BridgeBasePortType	- customerNetworkPort(4)
34	ieee8021BridgeBasePortExternal	- Implementation Specific

35 b) Creating an internal CNP.

36 The internal CNP is used to provide a C-tagged service interface or a Port-based RCSI to the
37 customer of a PBN.

38 An internal CNP, together with the associated PEP and the internal connection between them, is
39 created by performing row-create operations for an S-VID in the ieee8021PbEdgePortTable and the
40 corresponding C-VID in the ieee8021PbCvidRegistrationTable. See the discussion on
41 provisioning a C-tagged service interface in 17.6.1.2. An internal CNP together with the associated
42 PAP and the internal connection between them are created by configuring a Port-based RCSI,
43 that is, performing a row-create operation on the ieee8021PbInternalInterfaceTable with an
44 ieee8021PbIInternalPortType of customerNetworkPort.

45 Internal CNPs are not directly manageable.

17.5.3.3.3 Creating a CEP

CEPs are created either by doing a row-create operation on the Provider Edge Bridge's CEP table or as a side effect of mapping an external S-VID on a RCAP to a C-tagged RCSI. The `ieee8021PbCepTable` contains the following columns:

<code>ieee8021BridgeBasePortComponentId</code>	- S-VLAN component ID that "owns" the CEP
<code>ieee8021BridgeBasePort</code>	- The Port Number of the CEP on the S-VLAN component.
<code>ieee8021PbCepCComponentId</code>	- C-VLAN component read only index cross-ref for entity MIB
<code>ieee8021PbCepCepPortNumber</code>	- C-VLAN component Port read-only index cross-ref for entity MIB
<code>ieee8021PbCepRowStatus</code>	- Controls the creation and deletion of the Port

Note that the C-VLAN component containing the newly created CEP does not appear in the IEEE 802.1 Bridge MIB's list of components. So `ieee8021PbCepCComponentId` and the `ieee8021PbCepCepPortNumber`, are index values that are not further interpreted by any IEEE 802.1 MIB. These index values, if present, can be used in an implementation-dependent manner to allow management stations to cross reference entries in other MIBs, such as the IETF's entity MIB, back to the information managed by the IEEE MIBs.

The newly created Port will be added to the Port list of the S-VLAN component of the Provider Edge Bridge.

The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>CustomerEdgePortTable</code>
<code>ieee8021BridgeBasePort</code>	- As per <code>CustomerEdgePortTable</code>
<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific
	bit <code>customerEdgePort(3)</code> must be set
<code>ieee8021BridgeBasePortType</code>	- <code>customerEdgePort(5)</code>
<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17.5.3.3.4 Creating an RCAP

RCAPs are created by doing a row-create operation on the Provider Edge Bridge's RCAP table. The `ieee8021PbRcapTable` contains the following columns:

<code>ieee8021BridgeBasePortComponentId</code>	- primary S-VLAN component ID that "owns" the RCAP
<code>ieee8021BridgeBasePort</code>	- The Port Number of the RCAP on the primary S-VLAN component.
<code>ieee8021PbRcapSComponentId</code>	- Port-mapping S-VLAN component read only index cross-ref for entity MIB
<code>ieee8021PbRcapRcapPortNumber</code>	- Port-mapping S-VLAN component Port read-only index cross-ref for entity MIB
<code>ieee8021PbRcapRowStatus</code>	- Controls the creation and deletion of the Port

Note that the Port-mapping S-VLAN component containing the newly created RCAP does not appear in the IEEE 802.1 Bridge MIB's list of components. So `ieee8021PbRcapSComponentId` and the `ieee8021PbRcapRcapPortNumber`, are index values that are not further interpreted by any IEEE 802.1 MIB.

1 These index values, if present, can be used in an implementation-dependent manner to allow management
2 stations to cross reference entries in other MIBs, such as the IETF's entity MIB, back to the information
3 managed by the IEEE MIBs.

4 The newly created Port will be added to the Port list of the primary S-VLAN component of the Provider
5 Edge Bridge.

6 The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

7	<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>RcapTable</code>
8	<code>ieee8021BridgeBasePort</code>	- As per <code>RcapTable</code>
9	<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
10	<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
11	<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
12	<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
13	<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific
14		bit <code>remoteCustomerAccessPort(7)</code> must be set
15	<code>ieee8021BridgeBasePortType</code>	- <code>remoteCustomerAccessPort(9)</code>
16	<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17 17.5.3.3.5 Creating an Uplink Access Port (UAP)

18 UAPs are created by doing a row-create operation on the EVB Bridge's or EVB station's
19 `ieee8021BridgeEvbUapConfigTable`. The required column is the ISS Port Number to use for the newly
20 created Port. The `ComponentID` and `PortNumber` are specified if the system chooses to build an implicit
21 `ieee8021BridgeBasePortTable` entry. The `ieee8021BridgeEvbUapConfigTable` contains the following
22 columns:

23	<code>ieee8021BridgeEvbUapIssPortNumber</code>
24	<code>ieee8021BridgeEvbUapComponentId</code>
25	<code>ieee8021BridgeEvbUapPortNumber</code>
26	<code>ieee8021BridgeEvbUapCdcAdminEnable</code>
27	<code>ieee8021BridgeEvbUapAdminCDCPRole</code>
28	<code>ieee8021BridgeEvbUapAdminCDCPChanCap</code>
29	<code>ieee8021BridgeEvbUapOperCDCPChanCap</code>
30	<code>ieee8021BridgeEvbUapAdminCDCPSVIDPoolLow</code>
31	<code>ieee8021BridgeEvbUapAdminCDCPSVIDPoolHigh</code>
32	<code>ieee8021BridgeEvbUapOperState</code>
33	<code>ieee8021BridgeEvbUapCdcRemoteEnabled</code>
34	<code>ieee8021BridgeEvbUapCdcRemoteRole</code>
35	<code>ieee8021BridgeEvbUapConfigRowStatus</code>

36 The optional implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled
37 in:

38	<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>ieee8021BridgeEvbUapConfigTable</code>
39	<code>ieee8021BridgeBasePort</code>	- As per <code>ieee8021BridgeEvbUapConfigTable</code>
40		<code>UapPortNumber</code>
41	<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
42	<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
43	<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
44	<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
45	<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific:
46		bit <i>Uplink Access Port</i> (9) is set
47	<code>ieee8021BridgeBasePortType</code>	- Uplink Access Port
48	<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17.5.3.4 Port creation on B-components

B- and I-components are the two components that constitute a BEB (see Clause 26). These components provide what is generally known as “MAC in MAC” transport of Ethernet frames.

Ports on these BEB components are created using 12.16 managed objects.

17.5.3.4.1 Creating PNPs

A B-component is an S-VLAN component with one or more CBPs. A B-component may be a component of a BEB. PNP creation on a B-component follows the same logic used to create a PNP on an S-VLAN component.

Creating a PNP on a BEB is done by adding an entry to the `ieee8021PbPnpTable` specifying the `componentId` and Port Number.

11 <code>ieee8021BridgeBasePortComponentId</code>	- Component to which the new Port belongs
12 <code>ieee8021BridgeBasePort</code>	- Port Number for the new Port

The type of the component referred to by the component ID parameter must be `sVlanComponent`.

As a PNP is a Port on a B-component, which is a type of S-VLAN component, the act of creating a PNP causes entries to be made in the `ieee8021QBridgePortVlanTable` and the `ieee8021BridgeBasePortTable`.

The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

17 <code>ieee8021BridgeBasePortComponentId</code>	- As per <code>PbPnpTable</code>
18 <code>ieee8021BridgeBasePort</code>	- As per <code>PbPnpTable</code>
19 <code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
20 <code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
21 <code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
22 <code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
23 <code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific
24	bit <code>providerNetworkPort(1)</code> must be set
25 <code>ieee8021BridgeBasePortType</code>	- <code>providerNetworkPort(3)</code>
26 <code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17.5.3.4.2 Creating CBPs

Creating a CBP on a B-component is done by creating an entry in the `ieee8021PbbCbpTable` with the following column values:

<code>ieee8021BridgeBasePortComponentId</code>	- Component number of the B-component
<code>ieee8021BridgeBasePort</code>	- Port Number for the newly created Port

As a CBP is a Port on a B-component, which is a type of S-VLAN component, the act of creating a PNP causes entries to be made in the `ieee8021QBridgePortVlanTable` and the `ieee8021BridgeBasePortTable`.

The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>PbbCbpTable</code>
<code>ieee8021BridgeBasePort</code>	- As per <code>PbbCbpTable</code>
<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific
	bit <code>customerBackbonePort(4)</code> must be set
<code>ieee8021BridgeBasePortType</code>	- <code>customerBackbonePort(6)</code>
<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17.5.3.5 Port creation on I-components

Creating a Bridge Port on an I-component works the same way as creating a Port on a B-component with the exception that the rules for determining the legality of the Port type are different.

Creating PIPs works in a manner analogous to the creation of a Bridge Port, but as PIPs are not Bridge Ports the details are somewhat different.

17.5.3.5.1 Creating CNPs

The CNP is used to provide a Port-based or S-tagged service interface to customer of the PBBN.

Creating a CNP of requires specifying the I-component ID and Bridge Port Number of the Port to be created. This is done by creating an entry in the `ieee8021PbCnpTable`. This creates a new entry in the `ieee8021QBridgePortVlanTable` for the provider Bridge's I-component. This operation also created a new entry in the `ieee8021BridgeBasePortTable`.

The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>QBridgePortVlanTable</code>
<code>ieee8021BridgeBasePort</code>	- As per <code>QBridgePortVlanTable</code>
<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific
	bit (2) must be set
<code>ieee8021BridgeBasePortType</code>	- <code>customerNetworkPort(4)</code>
<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17.5.3.5.2 Creating VIPs

The creation of VIP starts with the creation of an entry in the `ieee8021PbbVipTable` with the following columns set:

<code>ieee8021BridgeBasePortComponentId</code>	- Component Number of I-component
<code>ieee8021BridgeBasePort</code>	- As appropriate
<code>ieee8021PbbVipRowStatus</code>	- Probably either <code>createAndGo</code> or <code>createAndWait</code>

An entry will be created in the `ieee8021PbbVipTable` with the following columns set:

<code>ieee8021PbbVipPipIfIndex</code>	- 0
<code>ieee8021PbbVipISid</code>	- 1
<code>ieee8021PbbVipDefaultDstBMAC</code>	- 00-1E-83-00-00-01
<code>ieee8021PbbVipRowStatus</code>	- “Active”

As a VIP is a Port on a I-component, which is a type of S-VLAN component, the act of creating a VIP causes entries to be made in the `ieee8021QBridgePortVlanTable` and the `ieee8021BridgeBasePortTable`.

The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>QBridgePortVlanTable</code>
<code>ieee8021BridgeBasePort</code>	- As per <code>QBridgePortVlanTable</code>
<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific
	bit <code>virtualInstancePort</code> must be set
<code>ieee8021BridgeBasePortType</code>	- <code>virtualInstancePort</code>
<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

Note that before becoming operational, the VIP must be assigned to a PIP and must be assigned to a service by setting an I-SID value. Furthermore, the mapping tables must be set up in a consistent basis.

17.5.3.5.3 Creating PIPs

PIPs are not Bridge Ports, but they are necessary parts of the internal operation of a BEB. They share many characteristics of Bridge Ports, simply because both PIPs and Bridge Ports are interfaces that transfer MAC frames, but are not connected to a MAC relay function.

PIPs are created using a BEB managed object. This object maps to the `ieee8021PbbPipTable` in the PBB MIB module. An entry needs to be created in the table with the following attributes:

<code>ieee8021PbbPipIfIndex</code>	- A previously unallocated <code>ifIndex</code> value
<code>ieee8021PbbPipBMACAddress</code>	- Implementation Default or specified value
<code>ieee8021PbbPipName</code>	- Implementation Default or specified value
<code>ieee8021PbbPipIComponentId</code>	- I-component index to which this PIP belongs
<code>ieee8021PbbPipVipMap</code>	- Empty Mapping

Note that before the operation can occur on this PIP, it may be necessary to modify the B-MAC address value of the PIP, especially in the case where local MAC addresses are used in the PBBN.

17.5.3.6 Required post creation operations

Before a Bridge Port is ready for operational use, the Bridge Port must be associated with an entity that acts as a frame source and frame sink. For a Bridge with or without the dynamic Port creation capability, this will likely be an ISS. For a Bridge with dynamic Port creation, it will be an ISS.

Most Bridge Ports manage this association by using the `ieee8021BridgeBasePortIfIndex` column in the `ieee8021BridgeBasePort` table. It is system dependent if the `ieee8021BridgeBasePortIfIndex` column in the `ieee8021BridgeBasePort` table needs to be filled in with the appropriate `ifIndex` for the newly created Bridge Port. Systems that enforce a specific mapping between Bridge Port Numbers and physical interfaces may have the agent automatically fill in this field and the association may not be modifiable.

The association for CEPs is managed by the `ieee8021PbCepTable`.

The association between PEPs and CNPs for PEBs is managed implicitly by the creation of the PEP/CNP pair by adding a CEP to the member set of an S-VID.

The association between PAPs and CNPs for PEBs is managed implicitly by the creation of a PAP/CNP pair by mapping an external S-VID on a RCAP to a Port-based RCSI. The association between PAPs and CEPs for PEBs is managed implicitly by the creation of a PAP/CEP pair by mapping an external S-VID on a RCAP to a C-tagged RCSI.

The association between PIPs and CBPs is managed by the internal LAN table (`ieee8021BridgeILanIfTable`).

Additionally, for Ports on elements of Provider Bridges and PBBs, instances in service tables must be created before the Ports are capable of passing frames according to the relevant clauses of this standard and subsequent amendments.

17.5.3.7 Port creation on ERs

Creating a URP implicitly creates the ER itself. When a URP is created it is not necessarily bound to an ISS. The operating environment of the EVB system may choose to bind the URP to either an S-channel or to an ISS when it chooses. The ER may be created with a DRP or it may dynamically create them on demand from the operating environment.

Creating a DRP on an ER works the same way as creating a Port on a C-VLAN-aware component (17.5.3.2) with the exception that the rules for determining the legality of the Port type are different.

17.5.3.7.1 Creating DRPs

DRPs are created by performing a row-create operation on the `ieee8021EvbPortTable` for an ER. The required columns are the component ID and the Port Number to use for the newly created DRP.

17.5.3.7.2 Creating URPs

URPs are created by performing a row-create operation on the `ieee8021BridgeEvpUrpConfigTable` of an EVB station. The required columns are the component ID and the Port Number to use for the newly created Port.

The implicitly constructed `ieee8021BridgeBaseTable` entry will have the following fields filled in:

<code>ieee8021BridgeBaseComponentId</code>	- As per <code>ieee8021BridgeEvpUrpConfigTable</code>
<code>ieee8021BridgeBaseBridgeAddress</code>	- As per <code>ieee8021BridgeEvpUrpConfigTable</code>
<code>ieee8021BridgeBaseNumPorts</code>	- Implementation Specific Action
<code>ieee8021BridgeBaseComponentType</code>	- Type edge relay
<code>ieee8021BridgeBaseDeviceCapabilities</code> -	- Implementation Specific
<code>ieee8021BridgeBaseTrafficClassesEnabled</code>	- Implementation Specific
<code>ieee8021BridgeBaseMmrpEnabledStatus</code>	- Implementation Specific
<code>ieee8021BridgeBaseRowStatus</code>	

The implicitly constructed `ieee8021BridgeBasePortTable` entry will have the following fields filled in:

<code>ieee8021BridgeBasePortComponentId</code>	- As per <code>ieee8021BridgeEvpUrpConfigTable</code>
<code>ieee8021BridgeBasePort</code>	- As per <code>ieee8021BridgeEvpUrpConfigTable</code>
<code>ieee8021BridgeBasePortIfIndex</code>	- Implementation Specific Action
<code>ieee8021BridgeBasePortDelayExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortMtuExceededDiscards</code>	- Statistic, reset to 0 by creation
<code>ieee8021BridgeBasePortCapabilities</code>	- Implementation Specific
<code>ieee8021BridgeBasePortTypeCapabilities</code>	- Implementation Specific: bit Uplink Relay Port(10) is set
<code>ieee8021BridgeBasePortType</code>	- Uplink Relay Port (10)
<code>ieee8021BridgeBasePortExternal</code>	- Implementation Specific

17.6 MIB operations for service interface configuration

This subclause outlines, on a per-service interface basis, the MIB operations that a Provider must perform to configure each type of service interface. The clause is organized into two broad subclauses. The first subclause explains how service interfaces for Provider networks (Clause 15, Clause 16, etc.), are configured. The second subclause explains how service interfaces for Provider Backbone Networks (Clause 25, Clause 26, etc.), are configured. Both of these subclauses have the same substructure. First, a description of the MIB operations that are service interface type independent, and secondly a description of the service type MIB operations organized in a per-service type manner.

17.6.1 Provisioning PBN service interfaces

This clause assumes that the components and Ports, if dynamic, have been created, although there may be some notes on the parameters used for Port creation on a per-service basis, if there are restrictions. This subclause describes the MIB operations needed to configure service interfaces for PBN equipment.

Table 17-1 lists the parameters required for each of the service interface types that can be used to interface to a provider network. These parameter names are used in subsequent clauses as values to be entered in the appropriate MIB tables.

Table 17-1—Provider Bridge service interface parameters

Parameter	Description	Customer service interface type		
		Port-based	C-tagged	S-tagged
ComponentID	ID of the S-VLAN component	Mandatory	Mandatory	Mandatory
Port Number	Bridge Port Number that Receives Customer Frames	CNP Port Number	CEP Port Number	CNP Port Number
RelaySVid	S-VID value used within the provider network	Mandatory	Mandatory	Mandatory
CVid	C-VID value used to identify frames to be transported by the specified service	Not needed	Mandatory	Not needed
LocalSVid	S-VID value used to identify the frames to be transported by the service	Not needed	Not Needed	Optional: if not specified, the RelaySVid value is used
Default CVID value	C-VID value to be used for frames received from the Provider Network without C-TAGs	Not needed	Mandatory	Not needed
Default Priority Value	Priority value to be used for frames received from the Provider Network without C-TAGs	Not needed	Mandatory	Not needed

17.6.1.1 Provisioning a Port-based service interface

This subclause describes the MIB operations needed to provision a Port-based service interface.

Creating a Port-based service on a CNP requires manipulation of the Port member sets of the `ieee8021QBridgeVlanStaticTable`. One must manipulate the Port member sets for the row with the following index values:

<code>ieee8021QBridgeVlanStaticComponentId</code>	- ComponentId
<code>ieee8021QBridgeVlanStaticVlanIndex</code>	- RelaySVid

The CNP must be added to the PortList specified by the `ieee8021QBridgeVlanStaticEgressPorts` column of this row and must be added to the PortList specified by the `ieee8021QBridgeVlanStaticUntaggedPorts` column of this row.

One may, or may not, wish to allow the customer to specify priorities for the arriving customer traffic. In either case one must set the `ieee8021QBridgePortAcceptableFrameTypes` column of the entry in the `ieee8021QBridgePortVlanTable` indexed by:

<code>ieee8021BridgeBasePortComponentId</code>	- ComponentId
<code>ieee8021BridgeBasePort</code>	- PortNumber

to `admitUntaggedAndPriority` and set the `ieee8021QBridgePortIngressFiltering` column to TRUE to ensure that customers may not “spoof” the provider network by sending S-tagged frames to the CNP.

17.6.1.2 Provisioning a C-tagged service interface

This subclause describes the MIB operations needed to provision a C-tagged service interface.

The CEP must be added to the Port member set of each S-VID on the S-VLAN component of the Provider Edge Bridge that is to provide service for the CEP. This is done by modifying the `ieee8021QBridgeVlanStaticTable` `ieee8021QBridgeStaticEgressPorts` PortList whose matching index column values are as follows:

<code>ieee8021QBridgeVlanStaticComponentId</code>	- ComponentId
<code>ieee8021QBridgeVlanStaticVlanIndex</code>	- RelaySVid

Adding the CEP to the member set of an S-VID implies the existence of a CNP on the S-VLAN component that is internally connected to a PEP on the C-VLAN component. Enabling the CEP to send and receive frames from the service instance identified by the RelaySVid requires creating a linkage between the S-VID, the CNP, the PEP, and a C-VID. This is accomplished by configuring the `ieee8021PbEdgePortTable` and `ieee8021PbCVidRegistrationTable`.

Each S-VID value that is associated with a provider service instance accessed through the CEP requires that an entry be made in the `ieee8021PbEdgePortTable`. An entry in this table maps an S-VID to a CNP and associates that CNP with a specific PEP. The CNP is identified by the CEP port number and the S-VID that form the index values for the table. The contents of the table entry identify the PEP by specifying the C-VID value used as the PVID for that PEP (as well as providing other configuration parameters for that PEP). A CNP is implicitly created by the first entry identifying a given PEP (i.e., the first entry that contains a given C-VID value). In some cases it is useful to create multiple entries to map multiple S-VIDs to the same CNP. Two or more entries map to the same CNP if the C-VID value identifying the PVID of the PEP is the same (the other values of the entry should be the same as well). This many-to-one mapping of S-VIDs to CNP/PEP pairs can be used to configure asymmetric VLANs for supporting rooted-multipoint connectivity (F.1.3.2).

1 The table entry with the following index values:

2	ieee8021BridgeBasePortComponentId	- ComponentId
3	ieee8021BridgeBasePort	- PortNumber
4	ieee8021PAEdgePortSVid	- RelaySVid

5 needs to have the following columns set:

6	ieee8021PbEdgePortPVID	- DefaultCVid
7	ieee8021PbEdgePortDefaultUserPriority	- DefaultPriority
8	ieee8021PbEdgePortAcceptableFrameType	- As Desired
9	ieee8021PbEdgePortEnableIngressFiltering	- As Desired

10 Each C-VID value that is to be mapped to a provider service instance requires that an entry be made in the
11 ieee8021PbCVidRegistrationTable. An entry in this table maps a C-VID to a PEP and associates that PEP
12 with a specific CNP. The PEP is identified by the CEP port number and the C-VID that form the index
13 values for the table. The contents of the table entry identify the CNP by specifying the S-VID value used as
14 the PVID for that CNP (as well as providing member set and untagged set information for the C-VID). A
15 PEP is implicitly created by the first entry identifying a given CNP (i.e., the first entry that contains a given
16 S-VID value). In some cases it is useful to create multiple entries to map multiple C-VIDs to the same PEP.
17 Two or more entries map to the same PEP if the S-VID value identifying the PVID of the CNP is the same.
18 This many-to-one mapping of C-VIDs to PEP/CNP pairs can be used to support bundling multiple
19 C-VLANs into a single service instance. The table entry with the following index values:

20	ieee8021BridgeBasePortComponentId	- ComponentId
21	ieee8021BridgeBasePort	- PortNumber
22	ieee8021PbCVidRegistrationCVid	- CVid

23 needs to have the following columns set:

24	ieee8021PbCVidRegistrationSVid	- RelaySVid
25	ieee8021PbCVidRegistrationUntaggedPep	- As Desired
26	ieee8021PbCVidRegistrationUntaggedCep	- As Desired

27 17.6.1.3 Provisioning an S-tagged service interface

28 This subclause describes the MIB operations needed to provision an S-tagged service interface.

29 Creating an S-tagged-based service interface requires manipulation of the Port member sets of the CNP
30 entry in the ieee8021QBridgeVlanStaticTable. One must manipulate the Port member sets for the row with
31 the following index values:

32	ieee8021QBridgeVlanStaticComponentId	- ComponentId
33	ieee8021QBridgeVlanStaticVlanIndex	- RelaySVid

34 The CNP must be added to the PortList specified by the ieee8021QBridgeVlanStaticEgressPorts column of
35 this row. The CNP must not be in the PortList specified by the ieee8021QBridgeVlanStaticUntaggedPorts
36 column of this row as this service requires S-TAGS in all frames.

37 If the Provider wishes to use separate S-TAG values for the customer and provider S-VLAN spaces, then the
38 entry in the ieee8021PbVidTranslationTable indexed by the following:

39	ieee8021BridgeBasePortComponentId	- ComponentId
40	ieee8021BridgeBasePort	- PortNumber
41	ieee8021PbVidTranslationLocalVid	- LocalSVid

42 must be created or changed so that the ieee8021PbVidTranslationRelayVid column is set to the RelaySVid.

17.6.2 Provisioning Backbone Bridged Network service interfaces

This subclause describes the MIB operations needed to provision service interfaces on PBBNs. In particular, this subclause assumes that the provider backbone network service instance exists and that the following information is available:

The I-SID value that identifies the correct backbone service instance is known.

At any BEB that is to have a service interface map customer data to a backbone service instance, given the I-SID, the following information can be determined:

- The Component ID of the B-component that contains the appropriate PNP used to access the service instance.
- The Port number of the PNP.
- The B-VID value used to transport that service over the Backbone Core.
- The DA used to transport the frame over the Backbone Core. Either a default MAC address, or the service MAC address.

Furthermore, the core of the network is assumed to be operationally correct. That is, if a frame is presented to the Backbone Core by a BEB with the proper B-VID value and proper Backbone Destination MAC address (B-DA) at the proper PNP, then the frame will arrive at the appropriate VIP(s). This can be achieved by statically configuring the member sets of the BCBs or by configuring the network such that dynamic address and VLAN registration protocols ensure that the Core Bridges Port member sets and FDBs are set appropriately.

Provisioning a service interface in a BEB to access a backbone service interface requires configuring two components. The first is the I-component. The I-component determines the backbone service instance to which customer data is mapped. The second component is the B-component. The B-component determines how the backbone service instance is forwarded over the PBBN core. Thus, the I-component determines the I-SID value, and thus the backbone service instance, to which the customer data belongs and the B-component determines the B-VID and B-DA values that will be used by the relay functions of the BCBs to determine the forwarding path through the core network used by the customer's data.

In the case of an I-tagged service interface, the service interface used by the customer is configured on the CBP of a BEB and the physical interconnection between the backbone network and the customer network is between the CBP of a provider managed B-component and the PIP of a customer managed I-component.

To simplify the discussion of service interface provisioning, this subclause first discusses interface type-independent provisioning of the B-component, and then service interface-independent provisioning of the I-component. Following that discussion, interface type-specific configuration is discussed. This type-specific configuration consists of instructions on how to determine the parameters needed by the service independent configuration of both the I- and B-components along with the provisioning of service-specific tables in the I- and B-components.

Table 17-2 lists the parameters required to provision each type of service interface per Clause 25. The parameters are used in the descriptions of the MIB table operations needed to configure each service in the following subclauses.

Table 17-2—PBB service interface parameters

Parameter	Description	Service interface type			
		Port-based	Unbundled one-to-one S-tagged	Bundled many-to-one or all-to-one S-tagged	I-tagged service interface
IComponentID	Component ID of the I-component of a BEB	Mandatory	Mandatory	Mandatory	Not needed
BComponentID	Component ID of the B-component of a BEB	Mandatory	Mandatory	Mandatory	Mandatory
CBPPortNumber	Port number of a Customer Backbone Port	Mandatory	Mandatory	Mandatory	Mandatory
BackboneSid	I-SID value used between the PIP and CBP	Mandatory	Mandatory	Mandatory	Mandatory
MappingType	Used for point-to-multipoint services where ingress or egress limiting is needed	Optional	Optional	Optional	Optional
DefaultDest Address	MAC address to use as the backbone DA	Optional: if not specified, the service default address is used	Optional: if not specified, the service default address is used	Optional: if not specified, the service default address is used	Optional: if not specified, the service default address is used
B-Vid	Backbone VID that identifies the VLAN to transport the service over the backbone	Optional: if not specified, the default B-VLAN is used	Optional: if not specified, the default B-VLAN is used	Optional: if not specified, the default B-VLAN is used	Optional: if not specified, the default B-VLAN is used
LocalSID	I-SID value to be used in the backbone network.	Optional: if not specified, the BackboneSID value is used	Optional: if not specified, the BackboneSID value is used	Optional: if not specified, the BackboneSID value is used	Optional: if not specified, the BackboneSID value is used
PCPSpec	Priority Code Point encoding and decoding specifications	Optional: note that separate values may be required for the I- and B-component s	Optional: note that separate values may be required for the I- and B-component s	Optional: note that separate values may be required for the I- and B-component s	Not needed
LocalSVid	S-TAG value that designates traffic at the CNP, if different from the RelaySVid	Not needed	Optional: RelaySVid used if not specified	Optional: RelaySVid used if not specified	Not needed

Table 17-2—PBB service interface parameters (continued)

Parameter	Description	Service interface type			
		Port-based	Unbundled one-to-one S-tagged	Bundled many-to-one or all-to-one S-tagged	I-tagged service interface
RelaySVid	S-TAG value used within the I-component to identify traffic belonging to a particular service instance	Mandatory	Mandatory	Mandatory	Not needed
VIPPortNumber	Port number of the VIP	Mandatory	Mandatory	Mandatory	Not needed
PIPIIndex	Identifier of the PIP to which the VIP belongs	Mandatory	Mandatory	Mandatory	Not needed
VIP-SID	ISID value used by a VIP	Mandatory	Mandatory	Mandatory	Not needed
adminPointToPointMAC	Indicates if the service is a point-to-point service	Optional	Optional	Optional	Optional
CNPPortNumber	The Bridge Port number of a CNP on an I-component	Mandatory	Mandatory	Mandatory	Not needed

17.6.2.1 Service type-independent provisioning of the B-component

All frames that are transmitted over a PBBN service instance ultimately start at a CBP. Thus part of provisioning any service interface requires configuration of the CBP. This is done with the `ieee8021PbbCBPServiceMappingTable`.

Provisioning this portion of the service interface is straightforward. One needs to add a row to the `ieee8021PbbCBPServiceMappingTable` whose columns are set to the values specified in the required parameters. Furthermore, one may need to ensure that both the CBP and PNP belong to the member set of the B-VID if dynamic configuration of VIDs is not supported.

17.6.2.2 Service type-independent provisioning of the I-component

The I-component needs to be configured when either Port-based or S-tagged service interfaces are configured. Traffic belonging to an I-tagged service interface does not transit the I-component; hence there is no configuration of the I-component for that service. The I-component's provisioning is more service interface dependent than the provisioning of the B-component. However, some of the logic needed to provision the I-component does not depend upon the service being provisioned. In particular the configuration of the `ieee8021PbbVIPTable` and the `ieee8021PbbPIPTable` is provisioned almost identically for both Port-based and S-tagged service instances.

1 17.6.2.2.1 Configuring the ieee8021PbbVipTable

2 Traffic for both Port-based and S-tagged services interfaces transitions a VIP on the I-component. The
3 particular VIP over which the traffic transits determines the backbone service instance, and thus the I-SID
4 value for the traffic. A VIP must be created for each service instance for which the I-component forwards
5 traffic. Bundled services interfaces map multiple customer S-VIDs to a single service instance. Generally,
6 when creating a service the first step is to create the VIP that will carry this service. This is done by creating
7 a row in the ieee8021PbbVipTable. The following columns must be set in the table:

8	ieee8021BridgeBasePortComponentId	- set to IComponentId
9	ieee8021BridgeBasePort	- set to VIPPortNum
10	ieee8021PbbVipISid	- set to the VIP-SID (which may be localSID or
11		backboneSID depending on the use of I-SID
12		translation at the B-component)

13 As a side effect of this row creation, a row will be created in the ieee8021PbbISidToVipTable. Essentially,
14 this table implements the inverse mapping, taking an I-SID value to the VIP that is the local endpoint of the
15 service instance.

16 17.6.2.2.2 Configuring the ieee8021PbbPipTable

17 For Port-based, unbundled service interfaces, and for the first S-VID allocated to a bundled service, then the
18 VIP used for the service must be configured to use the appropriate PIP. Use the IComponentID and PIPIndex
19 to select a row in this table. Set the bit corresponding to the VIPPortNum in the appropriate MAP column of
20 that row to 1. This adds the VIP to the PIP and, as a side effect, sets the ieee8021PbbVipPipIndex column
21 in the ieee8021PbbVipTable.

22 17.6.2.3 Service-dependent provisioning for an I-tagged service interface

23 The I-tagged service interface of a PBBN allows a customer to attach directly to a CBP of a B-component of
24 a BEB allowing the customer to transport frames that already possess an I-TAG. A scenario for this is when
25 the customer is also a backbone provider and is purchasing transport across another Provider's network.
26 Another scenario is when an access network provides an I-tagged interface. The I-tagged service interface is
27 defined in 25.5.

28 The provisioning is accomplished by using the parameter values of Table 17-7 and applying the procedures
29 outlined in 17.6.2.1 to provision the B-component.

30 17.6.2.4 Service-dependent provisioning a Port-based service interface

31 The Port-based service interface of a PBBN causes all of the traffic arriving at the CNP of the I-component
32 of a BEB to be mapped to a single backbone service instance. The Port-based interface of a PBBN provides the
33 same customer service as a Port-based service interface of a PBN. The implementation of this service
34 interface in a PBBN is defined in 25.3.

35 17.6.2.4.1 Configuring the B-component

36 Given the PIP, one uses the I-LAN table to find the B-component and CBP that the traffic for this service
37 will transit. This gives us the following two parameters:

38	BComponentId—B-component ID for the CBP
39	CBPPortNumber—Port number of the CBP on this B-component

40 These two service parameters found from the I-LAN table along with the remainder of the parameters
41 defined in Table 17-5 are used to configure the B-component as outlined in 17.6.2.1. As a side effect of
42 configuring the CBP via the ServiceMapping table, the ieee8021PbbVipDefaultDstBMAC column is set in
43 the ieee8021PbbVipTable.

17.6.2.4.2 Configuring the Port operating modes and I-component VID

The first step is to configure the I-component VID used to interconnect the VIP with the CNP. This is done by creating an entry in the `ieee8021QBridgeVlanStaticTable` with the columns set to the following values:

<code>ieee8021QBridgeVlanStaticComponentId</code>	- IcomponentId
<code>ieee8021QBridgeVlanStaticVlanIndex</code>	- IcompPVid
<code>ieee8021QBridgeVlanStaticName</code>	- Something administratively convenient
<code>ieee8021QBridgeVlanStaticEgressPorts</code>	- Port List containing just the VIP and the CNP
<code>ieee8021QBridgeVlanStaticUntaggedPorts</code>	- PortList containing just the VIP and the CNP

This sets the VIP and the CNP to be the only Ports present on the new I-component PVID. Furthermore, they are both set for untagged traffic. Thus, traffic arriving at one Port will be relayed to the other Port without any S-TAGs.

The final step is to configure the ingress filtering on both the VIP and CNP. This is done by modifying their entries in the `ieee8021QBridgePortVlanTable`. For both of these Ports, the following columns in the table must be set:

<code>ieee8021QBridgePortAcceptableFrameTypes</code>	- acceptUntaggedAndPriority
<code>ieee8021QBridgePortIngressFiltering</code>	- TRUE
<code>ieee8021QBridgePVid</code>	- IcompPVid

17.6.2.5 Service-dependent provisioning for an S-tagged service interfaces

This subclause describes the MIB operations specific to S-tagged service interface provisioning. Configuration is done in a manner similar to Port-based services. The configuration differences are largely confined to different configurations of the CNP and VIP Port member sets, tagging, and ingress filtering.

The parameters for an S-tagged service interface are similar to those for a Port-based service interface.

17.6.2.5.1 Configuring the B-component

If the service interface to be created is unbundled, or if the service interface is bundled and the newly configured service interface is the first S-VID value to be transported on this Backbone I-SID, then the B-component will need to be configured.

The first step is to find the B-component ID and CBP for the traffic. Given the PIP, one uses the I-LAN table to find the B-component and CBP that the traffic for this service interface will transit. This gives us the following two parameters:

<code>BComponentId</code>	—B-component ID for the CBP
<code>CBPPortNumber</code>	—Port number of the CBP on this B-component

These, together with the `BackboneSid`, `DefaultDestAddress`, and `BVid` value can be used to configure the B-component as per the previous instructions. As a side effect of configuring the CBP via the `ServiceMapping` table, the `ieee8021PbbVipDefaultDstBMAC` column is set in the `ieee8021PbbVipTable`.

1 17.6.2.5.2 Configuring the I-component for S-tagged service interfaces

2 These service interfaces are configured by creating one or more static VLANs on the I-component whose
3 S-VIDs are the values specified as the S-VID parameters to the service. For a one-to-one service, exactly
4 one S-VID value is specified. For a many-to-one or all-to-one service, multiple S-VID values are specified.
5 This is done by creating an entry in the ieee8021QBridgeVlanStaticTable with the columns set to the
6 following values:

7	ieee8021QBridgeVlanStaticComponentId	- IcomponentId
8	ieee8021QBridgeVlanStaticVlanIndex	- S-VID
9	ieee8021QBridgeVlanStaticName	- Something administratively convenient
10	ieee8021QBridgeVlanStaticEgressPorts	- Port List containing just the VIP and the CNP

11 For the one-to-one service set:

12	ieee8021QBridgeVlanStaticUntaggedPorts	- PortList containing just the VIP
----	--	------------------------------------

13 For the many-to-one or all-to-one service:

14	ieee8021QBridgeVlanStaticUntaggedPorts	- Null set
----	--	------------

15 For the unbundled one-to-one service, the VIP and CNP are set to be the only Ports that are members of the
16 I-component's VLAN identified by S-VID. Adding the VIP to the set of untagged Ports means that the data
17 will be sent to the CBP on the B-component untagged.

18 For the bundled many-to-one or all-to-one service interfaces, the VIP may be in the member set of more than
19 one S-VID. This is the mechanism by which multiple CNP services, identified by S-VID, are mapped to the
20 same I-SID value. Furthermore, unlike the unbundled service, the VIP is not a member of the untagged
21 egress set of the S-VID.

1 17.7 MIB modules^{31 32}

2 In this subclause, certain terms (e.g., “SHOULD” and “MUST”) denote requirements terminology according
3 to the usage specified in IETF RFC 2119, as is customary for IETF MIB module definitions. In the MIB
4 modules definitions below, if any discrepancy between the DESCRIPTION text and the corresponding
5 definition in any other part of this standard occur, the definitions outside this subclause take precedence.

6 17.7.1 Definitions for the IEEE8021-TC-MIB module

```

7 IEEE8021-TC-MIB DEFINITIONS ::= BEGIN
8
9 -- =====
10 -- TEXTUAL-CONVENTIONS MIB for IEEE 802.1
11 -- =====
12
13 IMPORTS
14     MODULE-IDENTITY, Unsigned32, org
15     FROM SNMPv2-SMI -- RFC 2578
16     TEXTUAL-CONVENTION
17     FROM SNMPv2-TC; -- RFC 2579
18
19 ieee8021TcMib MODULE-IDENTITY
20     LAST-UPDATED "202211080000Z" -- November 8, 2022
21     ORGANIZATION "IEEE 802.1 Working Group"
22     CONTACT-INFO
23         " WG-URL: http://www.ieee802.org/1/
24         WG-EMail: stds-802-1-1@ieee.org
25         Contact: IEEE 802.1 Working Group Chair
26         Postal: C/O IEEE 802.1 Working Group
27                 IEEE Standards Association
28                 445 Hoes Lane
29                 Piscataway, NJ 08854
30                 USA
31         E-mail: stds-802-1-chairs@ieee.org"
32     DESCRIPTION
33         "Textual conventions used throughout the various IEEE 802.1 MIB
34         modules.
35
36         Unless otherwise indicated, the references in this MIB
37         module are to IEEE Std 802.1Q-2022.
38
39         Copyright (C) IEEE (2022).
40         This version of this MIB module is part of IEEE Std 802.1Q;
41         see that standard for full legal notices."
42
43     REVISION "202211080000Z" -- November 8, 2022
44     DESCRIPTION
45         "Published as part of IEEE Std 802.1Q-2022.
46         Cross references and contact information updated."
47
48     REVISION "201807010000Z" -- July 1, 2018
49     DESCRIPTION
50         "Published as part of IEEE Std 802.1Q 2018 revision.
51         Cross references updated and corrected."
52
53     REVISION "201412150000Z" -- December 15, 2014
54     DESCRIPTION
55         "Published as part of IEEE Std 802.1Q 2014 revision.
56         Cross references updated and corrected.
57         Updating of definition of IEEE8021PbbIngressEgress
58         New identifier types for new SPBM MA types"
59
60     REVISION "201202150000Z" -- February 15, 2012
```

³¹ Copyright release for MIBs: Users of this standard may freely reproduce the MIB modules in this standard so that they can be used for their intended purpose.

³² An ASCII version of this MIB module is attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/mib-modules/>.

```
1  DESCRIPTION
2      "Modified IEEE8021BridgePortType textual convention to
3      include stationFacingBridgePort,
4      uplinkAccessPort, and uplinkRelayPort types."
5
6  REVISION      "201108230000Z" -- August 23, 2011
7  DESCRIPTION
8      "Modified textual conventions to support the IEEE 802.1
9      MIBs for PBB-TE Infrastructure Protection Switching."
10
11 REVISION      "201104060000Z" -- April 6, 2011
12 DESCRIPTION
13     "Modified textual conventions to support Remote Customer
14     Service Interfaces."
15
16 REVISION      "201102270000Z" -- February 27, 2011
17 DESCRIPTION
18     "Minor edits to contact information etc. as part of
19     2011 revision of IEEE Std 802.1Q."
20
21
22 REVISION      "200811180000Z" -- November 18, 2008
23 DESCRIPTION
24     "Added textual conventions needed to support the IEEE 802.1
25     MIBs for PBB-TE. Additionally, some textual conventions were
26     modified for the same reason."
27
28 REVISION      "200810150000Z" -- October 15, 2008
29 DESCRIPTION
30     "Initial version."
31 ::= { org ieee(111) standards-association-numbers-series-standards(2)
32     lan-man-stds(802) ieee802dot1(1) 1 1 }
33
34 ieee802dot1mibs OBJECT IDENTIFIER
35 ::= { org ieee(111) standards-association-numbers-series-standards(2)
36     lan-man-stds(802) ieee802dot1(1) 1 }
37
38 -- =====
39 -- Textual Conventions
40 -- =====
41
42 IEEE8021PbbComponentIdentifier ::= TEXTUAL-CONVENTION
43     DISPLAY-HINT "d"
44     STATUS      current
45     DESCRIPTION
46         "The component identifier is used to distinguish between the
47         multiple virtual Bridge instances within a PB or PBB. Each
48         virtual Bridge instance is called a component. In simple
49         situations where there is only a single component the default
50         value is 1. The component is identified by a component
51         identifier unique within the BEB and by a MAC address unique
52         within the PBBN. Each component is associated with a Backbone
53         Edge Bridge (BEB) Configuration managed object."
54     REFERENCE "12.3 1)"
55     SYNTAX      Unsigned32 (1..4294967295)
56
57 IEEE8021PbbComponentIdentifierOrZero ::= TEXTUAL-CONVENTION
58     DISPLAY-HINT "d"
59     STATUS      current
60     DESCRIPTION
61         "The component identifier is used to distinguish between the
62         multiple virtual Bridge instances within a PB or PBB. In simple
63         situations where there is only a single component the default
64         value is 1. The component is identified by a component
65         identifier unique within the BEB and by a MAC address unique
66         within the PBBN. Each component is associated with a Backbone
67         Edge Bridge (BEB) Configuration managed object.
68
69         The special value '0' means 'no component identifier'. When
70         this TC is used as the SYNTAX of an object, that object must
71         specify the exact meaning for this value."
72     REFERENCE "12.3 1)"
```

```
1 SYNTAX      Unsigned32 (0 | 1..4294967295)
2
3 IEEE8021PbbServiceIdentifier ::= TEXTUAL-CONVENTION
4   DISPLAY-HINT "d"
5   STATUS      current
6   DESCRIPTION
7       "The service instance identifier is used at the Customer
8       Backbone Port of a PBB to distinguish a service instance
9       (Local-SID). If the Local-SID field is supported, it is
10      used to perform a bidirectional 1:1 mapping between the
11      Backbone I-SID and the Local-SID. If the Local-SID field
12      is not supported, the Local-SID value is the same as the
13      Backbone I-SID value."
14  REFERENCE "12.16.3, 12.16.5"
15  SYNTAX      Unsigned32 (256..16777214)
16
17 IEEE8021PbbServiceIdentifierOrUnassigned ::= TEXTUAL-CONVENTION
18   DISPLAY-HINT "d"
19   STATUS      current
20   DESCRIPTION
21       "The service instance identifier is used at the Customer
22       Backbone Port of a PBB to distinguish a service instance
23       (Local-SID). If the Local-SID field is supported, it is
24       used to perform a bidirectional 1:1 mapping between the
25       Backbone I-SID and the Local-SID. If the Local-SID field
26       is not supported, the Local-SID value is the same as the
27       Backbone I-SID value.
28
29       The special value of 1 indicates an unassigned I-SID."
30  REFERENCE "12.16.3, 12.16.5"
31  SYNTAX      Unsigned32 (1|256..16777214)
32
33 IEEE8021PbbIngressEgress ::= TEXTUAL-CONVENTION
34   STATUS      current
35   DESCRIPTION
36       "A 2 bit selector that determines if frames on this VIP may
37       ingress to the PBBN but not egress the PBBN, egress to the
38       PBBN but not ingress the PBBN, or both ingress and egress
39       the PBBN."
40  REFERENCE "12.16.3, 12.16.5"
41  SYNTAX      BITS {
42              ingress(0),
43              egress(1)
44          }
45
46 IEEE8021PriorityCodePoint ::= TEXTUAL-CONVENTION
47   STATUS      current
48   DESCRIPTION
49       "Bridge ports may encode or decode the PCP value of the
50       frames that traverse the port. This textual convention
51       names the possible encoding and decoding schemes that
52       the port may use. The priority and drop_eligible
53       parameters are encoded in the Priority Code Point (PCP)
54       field of the VLAN tag using the Priority Code Point
55       Encoding Table for the Port, and they are decoded from
56       the PCP using the Priority Code Point Decoding Table."
57  REFERENCE "12.6.2.6"
58  SYNTAX      INTEGER {
59              codePoint8p0d(1),
60              codePoint7p1d(2),
61              codePoint6p2d(3),
62              codePoint5p3d(4)
63          }
64
65 IEEE8021BridgePortNumber ::= TEXTUAL-CONVENTION
66   DISPLAY-HINT "d"
67   STATUS      current
68   DESCRIPTION
69       "An integer that uniquely identifies a Bridge Port, as
70       specified in 17.3.2.2.
71       This value is used within the spanning tree
72       protocol to identify this port to neighbor Bridges."
```

```

1  REFERENCE "17.3.2.2"
2  SYNTAX      Unsigned32 (1..65535)
3
4  IEEE8021BridgePortNumberOrZero ::= TEXTUAL-CONVENTION
5  DISPLAY-HINT "d"
6  STATUS      current
7  DESCRIPTION
8      "An integer that uniquely identifies a Bridge Port. The value
9      0 means no port number, and this must be clarified in the
10     DESCRIPTION clause of any object defined using this
11     TEXTUAL-CONVENTION."
12  REFERENCE "17.3.2.2"
13  SYNTAX      Unsigned32 (0..65535)
14
15  IEEE8021BridgePortType ::= TEXTUAL-CONVENTION
16  STATUS      current
17  DESCRIPTION
18      "A port type. The possible port types are:
19
20      customerVlanPort(2) - Indicates a port is a
21      C-tag-aware port of an enterprise VLAN-aware Bridge.
22
23      providerNetworkPort(3) - Indicates a port is an S-tag-
24      aware port of a Provider Bridge or Backbone Edge
25      Bridge used for connections within a PBN or PBBN.
26
27      customerNetworkPort(4) - Indicates a port is an S-tag-
28      aware port of a Provider Bridge or Backbone Edge
29      Bridge used for connections to the exterior of a
30      PBN or PBBN.
31
32      customerEdgePort(5) - Indicates a port is a C-tag-
33      aware port of a Provider Bridge used for connections
34      to the exterior of a PBN or PBBN.
35
36      customerBackbonePort(6) - Indicates a port is a I-tag-
37      aware port of a Backbone Edge Bridge's B-component.
38
39      virtualInstancePort(7) - Indicates a port is a virtual
40      S-tag-aware port within a Backbone Edge Bridge's
41      I-component that is responsible for handling
42      S-tagged traffic for a specific backbone service
43      instance.
44
45      dBridgePort(8) - Indicates a port is a VLAN-unaware
46      member of an IEEE 802.1D Bridge.
47
48      remoteCustomerAccessPort (9) - Indicates a port is an
49      S-tag-aware port of a Provider Bridge used for
50      connections to remote customer interface LANs
51      through another PBN.
52
53      stationFacingBridgePort (10) - Indicates a port of a
54      Bridge that supports the EVB status parameters
55      (40.4) with an EVBMode parameter value of
56      EVB Bridge.
57
58      uplinkAccessPort (11) - Indicates a port on a
59      Port-mapping S-VLAN component that connects an EVB
60      Bridge with an EVB station.
61
62      uplinkRelayPort (12) - Indicates a port of an edge relay
63      that supports the EVB status parameters (40.4)
64      with an EVBMode parameter value of EVB station."
65  REFERENCE "40.4, 12.13.1.1, 12.13.1.2, 12.16, 12.16.1.1.3
66          12.16.2.1, 12.26"
67  SYNTAX      INTEGER {
68      none(1),
69      customerVlanPort(2),
70      providerNetworkPort(3),
71      customerNetworkPort(4),
72      customerEdgePort(5),

```

```

1         customerBackbonePort(6),
2         virtualInstancePort(7),
3         dBridgePort(8),
4         remoteCustomerAccessPort(9),
5         stationFacingBridgePort(10),
6         uplinkAccessPort(11),
7         uplinkRelayPort(12)
8     }
9
10 IEEE8021VlanIndex ::= TEXTUAL-CONVENTION
11     DISPLAY-HINT "d"
12     STATUS current
13     DESCRIPTION
14         "A value used to index per-VLAN tables: values of 0 and
15         4095 are not permitted. If the value is between 1 and
16         4094 inclusive, it represents an IEEE 802.1Q VLAN-ID with
17         global scope within a given bridged domain (see VlanId
18         textual convention). If the value is greater than 4095,
19         then it represents a VLAN with scope local to the
20         particular agent, i.e., one without a global VLAN-ID
21         assigned to it. Such VLANs are outside the scope of
22         IEEE 802.1Q, but it is convenient to be able to manage them
23         in the same way using this MIB."
24     REFERENCE "9.6"
25     SYNTAX      Unsigned32 (1..4094|4096..4294967295)
26
27 IEEE8021VlanIndexOrWildcard ::= TEXTUAL-CONVENTION
28     DISPLAY-HINT "d"
29     STATUS current
30     DESCRIPTION
31         "A value used to index per-VLAN tables. The value 0 is not
32         permitted, while the value 4095 represents a 'wildcard'
33         value. An object whose SYNTAX is IEEE8021VlanIndexOrWildcard
34         must specify in its DESCRIPTION the specific meaning of the
35         wildcard value. If the value is between 1 and
36         4094 inclusive, it represents an IEEE 802.1Q VLAN-ID with
37         global scope within a given bridged domain (see VlanId
38         textual convention). If the value is greater than 4095,
39         then it represents a VLAN with scope local to the
40         particular agent, i.e., one without a global VLAN-ID
41         assigned to it. Such VLANs are outside the scope of
42         IEEE 802.1Q, but it is convenient to be able to manage them
43         in the same way using this MIB."
44     REFERENCE "9.6"
45     SYNTAX      Unsigned32 (1..4294967295)
46
47 IEEE8021MstIdentifier ::= TEXTUAL-CONVENTION
48     DISPLAY-HINT "d"
49     STATUS current
50     DESCRIPTION
51         "In an MSTP Bridge, an MSTID, i.e., a value used to identify
52         a spanning tree (or MST) instance. In the PBB-TE environment
53         the value 4094 is used to identify VIDs managed by the PBB-TE
54         procedures."
55     SYNTAX      Unsigned32 (1..4094)
56
57 IEEE8021ServiceSelectorType ::= TEXTUAL-CONVENTION
58     STATUS current
59     DESCRIPTION
60         "A value that represents a type (and thereby the format)
61         of a IEEE8021ServiceSelectorValue. The value can be one of
62         the following:
63
64         ieeeReserved(0)    Reserved for definition by IEEE 802.1
65                           recommend to not use zero unless
66                           absolutely needed.
67         vlanId(1)          12-Bit identifier as described in IEEE Std 802.1Q.
68         isid(2)            24-Bit identifier as described in IEEE Std 802.1ah.
69         tesid(3)           32 Bit identifier as described below.
70         segid(4)           32 Bit identifier as described below.
71         path-tesid(5)      32 Bit identifier as described below.
72         group-isid(6)      24 Bit identifier as described below.

```

1 ieeeReserved(7) Reserved for definition by IEEE Std 802.1
2
3 To support future extensions, the IEEE8021ServiceSelectorType
4 textual convention SHOULD NOT be subtyped in object type
5 definitions. It MAY be subtyped in compliance statements in
6 order to require only a subset of these address types for a
7 compliant implementation.
8
9 The tesid is used as a service selector for MAs that are present
10 in Bridges that implement PBB-TE functionality. A selector of
11 this type is interpreted as a 32 bit unsigned value of type
12 IEEE8021PbbTeTSidId. This type is used to index the
13 ieee8021PbbTeTeSiEspTable to find the ESPs that comprise the TE
14 Service Instance named by this TE-SID value.
15
16 The segid is used as a service selector for MAs that are present
17 in Bridges that implement IPS functionality. A selector of
18 this type is interpreted as a 32 bit unsigned value of type
19 IEEE8021TeipsSegid. This type is used to index the
20 Ieee8021TeipsSegTable to find the SMPs that comprise the
21 Infrastructure Segment named by this segid value.
22
23 The path-tesid is used as a service selector for SPBM path MAs.
24 A selector of this type is interpreted as a 32 bit unsigned value
25 corresponding to the MA index dotlagCfmMaIndex. This type is used
26 to index the dotlagCfmMepSpbmEspTable to find the ESPs that
27 comprise the SPBM path associated with an SPBM path MA.
28
29 The group-isid is used as a service selector for SPBM group MAs.
30 A selector of this type is interpreted as a 24 bit unsigned value
31 corresponding to the I-SID associated with an SPBM group MA.
32
33 Implementations MUST ensure that IEEE8021ServiceSelectorType
34 objects and any dependent objects (e.g.,
35 IEEE8021ServiceSelectorValue objects) are consistent. An
36 inconsistentValue error MUST be generated if an attempt to
37 change an IEEE8021ServiceSelectorType object would, for
38 example, lead to an undefined IEEE8021ServiceSelectorValue value."
39 SYNTAX INTEGER {
40 vlanId(1),
41 isid(2),
42 tesid(3),
43 segid(4),
44 path-tesid(5),
45 group-isid(6),
46 ieeeReserved(7)
47 }
48
49 IEEE8021ServiceSelectorValueOrNone ::= TEXTUAL-CONVENTION
50 DISPLAY-HINT "d"
51 STATUS current
52 DESCRIPTION
53 "An integer that uniquely identifies a generic MAC Service,
54 or none. Examples of service selectors are a VLAN-ID
55 (IEEE 802.1Q) and an I-SID (IEEE Std 802.1ah).
56
57 An IEEE8021ServiceSelectorValueOrNone value is always
58 interpreted within the context of an
59 IEEE8021ServiceSelectorType value. Every usage of the
60 IEEE8021ServiceSelectorValueOrNone textual convention is
61 required to specify the IEEE8021ServiceSelectorType object
62 that provides the context. It is suggested that the
63 IEEE8021ServiceSelectorType object be logically registered
64 before the object(s) that use the
65 IEEE8021ServiceSelectorValueOrNone textual convention, if
66 they appear in the same logical row.
67
68 The value of an IEEE8021ServiceSelectorValueOrNone object
69 must always be consistent with the value of the associated
70 IEEE8021ServiceSelectorType object. Attempts to set an
71 IEEE8021ServiceSelectorValueOrNone object to a value
72 inconsistent with the associated

1 IEEE8021ServiceSelectorType must fail with an
2 inconsistentValue error.
3
4 The special value of zero is used to indicate that no
5 service selector is present or used. This can be used in
6 any situation where an object or a table entry MUST either
7 refer to a specific service, or not make a selection.
8
9 Note that a MIB object that is defined using this
10 TEXTUAL-CONVENTION SHOULD clarify the meaning of
11 'no service' (i.e., the special value 0), as well as the
12 maximum value (i.e., 4094, for a VLAN ID)."
13 SYNTAX Unsigned32 (0 | 1..4294967295)
14
15 IEEE8021ServiceSelectorValue ::= TEXTUAL-CONVENTION
16 DISPLAY-HINT "d"
17 STATUS current
18 DESCRIPTION
19 "An integer that uniquely identifies a generic MAC Service.
20 Examples of service selectors are a VLAN-ID (IEEE 802.1Q)
21 and an I-SID (IEEE Std 802.1ah).
22
23 An IEEE8021ServiceSelectorValue value is always interpreted
24 within the context of an IEEE8021ServiceSelectorType value.
25 Every usage of the IEEE8021ServiceSelectorValue textual
26 convention is required to specify the
27 IEEE8021ServiceSelectorType object that provides the context.
28 It is suggested that the IEEE8021ServiceSelectorType object
29 be logically registered before the object(s) that use the
30 IEEE8021ServiceSelectorValue textual convention, if they
31 appear in the same logical row.
32
33 The value of an IEEE8021ServiceSelectorValue object must
34 always be consistent with the value of the associated
35 IEEE8021ServiceSelectorType object. Attempts to set an
36 IEEE8021ServiceSelectorValue object to a value inconsistent
37 with the associated IEEE8021ServiceSelectorType must fail
38 with an inconsistentValue error.
39
40 Note that a MIB object that is defined using this
41 TEXTUAL-CONVENTION SHOULD clarify the
42 maximum value (i.e., 4094, for a VLAN ID)."
43 SYNTAX Unsigned32 (1..4294967295)
44
45 IEEE8021PortAcceptableFrameTypes ::= TEXTUAL-CONVENTION
46 STATUS current
47 DESCRIPTION
48 "Acceptable frame types on a port."
49 REFERENCE "12.10.1.3, 12.13.2.3, 12.13.2.4"
50 SYNTAX INTEGER {
51 admitAll(1),
52 admitUntaggedAndPriority(2),
53 admitTagged(3)
54 }
55
56 IEEE8021PriorityValue ::= TEXTUAL-CONVENTION
57 DISPLAY-HINT "d"
58 STATUS current
59 DESCRIPTION
60 "An IEEE 802.1Q user priority value."
61 REFERENCE "12.13.2.3"
62 SYNTAX Unsigned32 (0..7)
63
64 IEEE8021PbbTeProtectionGroupId ::= TEXTUAL-CONVENTION
65 DISPLAY-HINT "d"
66 STATUS current
67 DESCRIPTION
68 "The PbbTeProtectionGroupId identifier is used to distinguish
69 protection group instances present in the B Component of
70 an IB-BEB."
71 REFERENCE "12.18.2"
72 SYNTAX Unsigned32 (1..4294967295)


```
1
2 IEEE8021PbbTeEsp ::= TEXTUAL-CONVENTION
3   STATUS current
4   DESCRIPTION
5       "This textual convention is used to represent the logical
6       components that constitute the 3-tuple that identifies an
7       Ethernet Switched Path. The 3-tuple consists of a
8       destination MAC address, a source MAC address and a VID.
9       Bytes (1..6) of this textual convention contain the
10      ESP-MAC-DA, bytes (7..12) contain the ESP-MAC-SA, and bytes
11      (13..14) contain the ESP-VID."
12  REFERENCE "3.95"
13  SYNTAX OCTET STRING ( SIZE(14))
14
15 IEEE8021PbbTeTSidId ::= TEXTUAL-CONVENTION
16  DISPLAY-HINT "d"
17  STATUS current
18  DESCRIPTION
19      "This textual convention is used to represent an identifier
20      that refers to a TE Service Instance. Note that, internally
21      a TE-SID is implementation dependent. This textual convention
22      defines the external representation of TE-SID values."
23  REFERENCE
24      "3.284"
25  SYNTAX Unsigned32 (1..42947295)
26
27 IEEE8021PbbTeProtectionGroupConfigAdmin ::= TEXTUAL-CONVENTION
28  STATUS current
29  DESCRIPTION
30      "This textual convention is used to represent administrative
31      commands that can be issued to a protection group. The value
32      noAdmin(1) is used to indicate that no administrative action
33      is to be performed."
34  REFERENCE "26.10.3.3.5
35            26.10.3.3.6
36            26.10.3.3.7
37            12.18.2.3.2"
38  SYNTAX INTEGER {
39      clear(1),
40      lockOutProtection(2),
41      forceSwitch(3),
42      manualSwitchToProtection(4),
43      manualSwitchToWorking(5)
44  }
45
46 IEEE8021PbbTeProtectionGroupActiveRequests ::= TEXTUAL-CONVENTION
47  STATUS current
48  DESCRIPTION
49      "This textual convention is used to represent the status of
50      active requests within a protection group."
51  REFERENCE
52      "12.18.2.1.3 d)"
53  SYNTAX INTEGER {
54      noRequest(1),
55      loP(2),
56      fs(3),
57      pSFH(4),
58      wSFH(5),
59      manualSwitchToProtection(6),
60      manualSwitchToWorking(7)
61  }
62
63 IEEE8021TeipsIpgid ::= TEXTUAL-CONVENTION
64  DISPLAY-HINT "d"
65  STATUS current
66  DESCRIPTION
67      "The TEIPS IPG identifier is used to distinguish
68      IPG instances present in a PBB."
69  REFERENCE "12.24.1.1.3 a)"
70  SYNTAX Unsigned32 (1..429467295)
71
72 IEEE8021TeipsSegid ::= TEXTUAL-CONVENTION
```

```
1  DISPLAY-HINT "d"
2  STATUS current
3  DESCRIPTION
4      "This textual convention is used to represent an
5      identifier that refers to an Infrastructure Segment.
6      Note that, internally a SEG-ID implementation
7      dependent. This textual convention defines the
8      external representation of SEG-ID values."
9  REFERENCE
10     "26.11.1"
11  SYNTAX  Unsigned32 (1..42947295)
12
13 IEEE8021TeipsSmpid ::= TEXTUAL-CONVENTION
14     STATUS current
15     DESCRIPTION
16         "This textual convention is used to represent the logical
17         components that constitute the 3-tuple that identifies a
18         Segment Monitoring Path (SMP). The 3-tuple consists of a
19         destination MAC address, a source MAC address and a VID.
20         Bytes (1..6) of this textual convention contain the
21         SMP-MAC-DA, bytes (7..12) contain the SMP-MAC-SA, and bytes
22         (13..14) contain the SMP-VID."
23     REFERENCE "26.11.1"
24     SYNTAX OCTET STRING ( SIZE(14))
25
26 IEEE8021TeipsIpgConfigAdmin ::= TEXTUAL-CONVENTION
27     STATUS current
28     DESCRIPTION
29         "This textual convention is used to represent administrative
30         commands that can be issued to an IPG. The value
31         clear(1) is used to indicate that no administrative action
32         is to be performed."
33     REFERENCE "12.24.2.1.3 h)"
34     SYNTAX  INTEGER {
35         clear(1),
36         lockOutProtection(2),
37         forceSwitch(3),
38         manualSwitchToProtection(4),
39         manualSwitchToWorking(5)
40     }
41
42 IEEE8021TeipsIpgConfigActiveRequests ::= TEXTUAL-CONVENTION
43     STATUS current
44     DESCRIPTION
45         "This textual convention is used to represent the status of
46         active requests within an IPG."
47     REFERENCE
48         "12.24.2.1.3 d)"
49     SYNTAX  INTEGER {
50         noRequest(1),
51         loP(2),
52         fs(3),
53         pSFH(4),
54         wSFH(5),
55         manualSwitchToProtection(6),
56         manualSwitchToWorking(7)
57     }
58
59 END
```

1 17.7.2 Definitions for the IEEE8021-BRIDGE-MIB module

```
2 IEEE8021-BRIDGE-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- Bridge MIB for IEEE 802.1Q devices
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE,
10     Integer32, Counter64
11     FROM SNMPv2-SMI
12     RowStatus, MacAddress, TruthValue, TimeInterval
13     FROM SNMPv2-TC
14     MODULE-COMPLIANCE, OBJECT-GROUP
15     FROM SNMPv2-CONF
16     ifIndex, InterfaceIndexOrZero, ifGeneralInformationGroup
17     FROM IF-MIB
18     ieee802dot1mibs, IEEE8021PbbComponentIdentifier,
19     IEEE8021BridgePortNumber, IEEE8021PriorityCodePoint,
20     IEEE8021BridgePortType, IEEE8021PriorityValue,
21     IEEE8021PbbComponentIdentifierOrZero,
22     IEEE8021BridgePortNumberOrZero
23     FROM IEEE8021-TC-MIB
24     SnmpAdminString
25     FROM SNMP-FRAMEWORK-MIB
26     systemGroup
27     FROM SNMPv2-MIB
28 ;
29
30 ieee8021BridgeMib MODULE-IDENTITY
31     LAST-UPDATED "202211080000Z" -- November 8, 2022
32     ORGANIZATION "IEEE 802.1 Working Group"
33     CONTACT-INFO
34         " WG-URL: http://www.ieee802.org/1/
35         WG-EMail: stds-802-1-1@ieee.org
36         Contact: IEEE 802.1 Working Group Chair
37         Postal: C/O IEEE 802.1 Working Group
38             IEEE Standards Association
39             445 Hoes Lane
40             Piscataway, NJ 08854
41             USA
42         E-mail: stds-802-1-chairs@ieee.org"
43     DESCRIPTION
44         "The Bridge MIB module for managing devices that support
45         IEEE Std 802.1Q.
46
47         Unless otherwise indicated, the references in this MIB
48         module are to IEEE Std 802.1Q-2022.
49
50         Copyright (C) IEEE (2022).
51         This version of this MIB module is part of IEEE Std 802.1Q;
52         see that standard for full legal notices."
53
54     REVISION "202211080000Z" -- November 8, 2022
55     DESCRIPTION
56         "Published as part of IEEE Std 802.1Q-2022.
57         Cross references and contact information updated."
58
59     REVISION "201807010000Z" -- July 1, 2018
60     DESCRIPTION
61         "Published as part of IEEE Std 802.1Q 2018 revision.
62         Cross references updated and corrected."
63
64     REVISION "201412150000Z" -- December 15, 2014
65     DESCRIPTION "Published as part of IEEE Std 802.1Q 2014 revision.
66         Cross references updated and corrected."
67
68     REVISION "201208100000Z" -- August 10, 2012
69     DESCRIPTION "Added an enumeration for tComponent in
70         ieee8021BridgeBaseComponentType
71         as part of IEEE Std 802.1Q Cor-2."
```

```
1
2 REVISION      "201202150000Z" -- February 15, 2012
3 DESCRIPTION   "Extended ieee8021BridgeBaseComponentType to
4               include erComponent and
5               ieee8021BridgeBasePortTypeCapabilities to include
6               stationFacingBridgePort, uplinkAccessPort and
7               uplinkRelayPort.
8               Added tables ieee8021BridgeBaseIfToPortTable and
9               ieee8021BridgePortTable
10              as part of IEEE Std 802.1Qbg."
11
12 REVISION      "201104060000Z" -- April 6, 2011
13 DESCRIPTION   "Modifications to support Remote Customer Service
14               Interfaces."
15
16 REVISION      "201102270000Z" -- February 27, 2011
17 DESCRIPTION   "Minor edits to contact information etc. as part of
18               2011 revision of IEEE Std 802.1Q."
19
20 REVISION      "200810150000Z" -- October 15, 2008
21 DESCRIPTION   "Initial revision, derived from RFC 4188."
22 ::= { ieee802dot1mibs 2 }
23
24 -- =====
25 -- subtrees in the Bridge MIB
26 -- =====
27
28 ieee8021BridgeNotifications
29   OBJECT IDENTIFIER ::= { ieee8021BridgeMib 0 }
30
31 ieee8021BridgeObjects
32   OBJECT IDENTIFIER ::= { ieee8021BridgeMib 1 }
33
34 ieee8021BridgeConformance
35   OBJECT IDENTIFIER ::= { ieee8021BridgeMib 2 }
36
37 ieee8021BridgeBase
38   OBJECT IDENTIFIER ::= { ieee8021BridgeObjects 1 }
39 ieee8021BridgeTp
40   OBJECT IDENTIFIER ::= { ieee8021BridgeObjects 2 }
41 ieee8021BridgePriority
42   OBJECT IDENTIFIER ::= { ieee8021BridgeObjects 3 }
43 ieee8021BridgeMrp
44   OBJECT IDENTIFIER ::= { ieee8021BridgeObjects 4 }
45 ieee8021BridgeMmrp
46   OBJECT IDENTIFIER ::= { ieee8021BridgeObjects 5 }
47 ieee8021BridgeInternalLan
48   OBJECT IDENTIFIER ::= { ieee8021BridgeObjects 6 }
49 ieee8021BridgeDot1d
50   OBJECT IDENTIFIER ::= { ieee8021BridgeObjects 7 }
51
52 -- =====
53 -- the ieee8021BridgeBase subtree
54 -- =====
55 -- Implementation of the ieee8021BridgeBase subtree is mandatory
56 -- for all Bridges.
57 -- =====
58
59 -- =====
60 -- the ieee8021BridgeBaseTable
61 -- =====
62 ieee8021BridgeBaseTable OBJECT-TYPE
63   SYNTAX      SEQUENCE OF Ieee8021BridgeBaseEntry
64   MAX-ACCESS  not-accessible
65   STATUS      current
66   DESCRIPTION
67     "A table that contains generic information about every
68     Bridge component. All writable objects in this table
69     MUST be persistent over power up restart/reboot."
70   REFERENCE   "12.4.1"
71   ::= { ieee8021BridgeBase 1 }
72
```

```
1 ieee8021BridgeBaseEntry OBJECT-TYPE
2   SYNTAX      Ieee8021BridgeBaseEntry
3   MAX-ACCESS  not-accessible
4   STATUS      current
5   DESCRIPTION
6       "A list of objects containing information for each Bridge
7       component."
8   INDEX { ieee8021BridgeBaseComponentId }
9   ::= { ieee8021BridgeBaseTable 1 }
10
11 Ieee8021BridgeBaseEntry ::=
12   SEQUENCE {
13       ieee8021BridgeBaseComponentId
14           IEEE8021PbbComponentIdentifier,
15       ieee8021BridgeBaseBridgeAddress
16           MacAddress,
17       ieee8021BridgeBaseNumPorts
18           Integer32,
19       ieee8021BridgeBaseComponentType
20           INTEGER,
21       ieee8021BridgeBaseDeviceCapabilities
22           BITS,
23       ieee8021BridgeBaseTrafficClassesEnabled
24           TruthValue,
25       ieee8021BridgeBaseMmrpEnabledStatus
26           TruthValue,
27       ieee8021BridgeBaseRowStatus
28           RowStatus
29   }
30
31 ieee8021BridgeBaseComponentId OBJECT-TYPE
32   SYNTAX      IEEE8021PbbComponentIdentifier
33   MAX-ACCESS  not-accessible
34   STATUS      current
35   DESCRIPTION
36       "The component identifier is used to distinguish between the
37       multiple virtual Bridge instances within a PBB. In simple
38       situations where there is only a single component the default
39       value is 1."
40   ::= { ieee8021BridgeBaseEntry 1 }
41
42 ieee8021BridgeBaseBridgeAddress OBJECT-TYPE
43   SYNTAX      MacAddress
44   MAX-ACCESS  read-create
45   STATUS      current
46   DESCRIPTION
47       "The MAC address used by this Bridge when it is
48       referred to in a unique fashion. It is recommended
49       that this be the numerically smallest MAC address of
50       all ports that belong to this Bridge. However, it is
51       only required to be unique. When concatenated with
52       ieee8021SpanningTreePriority, a unique BridgeIdentifier
53       is formed, which is used in the Spanning Tree Protocol.
54
55       This object may not be modified while the corresponding
56       instance of ieee8021BridgeBaseRowStatus is active(1).
57
58       The value of this object MUST be retained across
59       reinitializations of the management system."
60   REFERENCE  "12.4.1.1.3 a)"
61   ::= { ieee8021BridgeBaseEntry 2 }
62
63 ieee8021BridgeBaseNumPorts OBJECT-TYPE
64   SYNTAX      Integer32
65   UNITS       "ports"
66   MAX-ACCESS  read-only
67   STATUS      current
68   DESCRIPTION
69       "The number of ports controlled by this bridging
70       entity."
71   REFERENCE  "12.4.1.1.3 c)"
72   ::= { ieee8021BridgeBaseEntry 3 }
```

```

1
2 ieee8021BridgeBaseComponentType OBJECT-TYPE
3     SYNTAX      INTEGER {
4         iComponent(1),
5         bComponent(2),
6         cVlanComponent(3),
7         sVlanComponent(4),
8         dBridgeComponent(5),
9         erComponent (6),
10        tComponent (7)
11    }
12    MAX-ACCESS   read-create
13    STATUS       current
14    DESCRIPTION
15        "Indicates the component type(s) of this Bridge. The
16        following component types are possible:
17
18        iComponent(1) - An S-VLAN component of a Backbone
19        Edge Bridge that performs encapsulation of customer
20        frames.
21
22        bComponent(2) - An S-VLAN component of a Backbone
23        Edge Bridge that bundles backbone service instances
24        into B-VLANs.
25
26        cVlanComponent(3) - A C-VLAN component of an
27        enterprise VLAN Bridge or of a Provider Bridge used
28        to process C-tagged frames.
29
30        sVlanComponent(4) - An S-VLAN component of a
31        Provider Bridge.
32
33        dBridgeComponent(5) - A VLAN unaware component of an
34        IEEE 802.1Q Bridge.
35
36        erComponent (6) - An Edge Relay component of an EVB Station.
37
38        tComponent (7) - A TPMR component in a Backbone Edge Bridge.
39
40        This object may not be modified while the corresponding
41        instance of ieee8021BridgeBaseRowStatus is active(1).
42
43        The value of this object MUST be retained across
44        reinitializations of the management system."
45    REFERENCE    "12.3 m)"
46    ::= { ieee8021BridgeBaseEntry 4 }
47
48 ieee8021BridgeBaseDeviceCapabilities OBJECT-TYPE
49     SYNTAX      BITS {
50         dotldExtendedFilteringServices(0),
51         dotldTrafficClasses(1),
52         dotlqStaticEntryIndividualPort(2),
53         dotlqIVLCapable(3),
54         dotlqSVLCapable(4),
55         dotlqHybridCapable(5),
56         dotlqConfigurablePvidTagging(6),
57         dotldLocalVlanCapable(7)
58    }
59    MAX-ACCESS   read-create
60    STATUS       current
61    DESCRIPTION
62        "Indicates the optional parts of IEEE Std 802.1Q
63        that are implemented by this device and are manageable
64        through this MIB. Capabilities that are allowed on a
65        per-port basis are indicated in
66        ieee8021BridgeBasePortCapabilities.
67
68        dotldExtendedFilteringServices(0),
69            -- can perform filtering of
70            -- individual multicast addresses
71            -- controlled by MMRP.
72        dotldTrafficClasses(1),

```

```

1          -- can map priority to
2          -- multiple traffic classes.
3      dot1qStaticEntryIndividualPort(2),
4          -- dot1qStaticUnicastReceivePort &
5          -- dot1qStaticMulticastReceivePort
6          -- can represent non-zero entries.
7      dot1qIVLCapable(3),      -- Independent VLAN Learning (IVL).
8      dot1qSVLCapable(4),    -- Shared VLAN Learning (SVL).
9      dot1qHybridCapable(5),
10         -- both IVL & SVL simultaneously.
11      dot1qConfigurablePvidTagging(6),
12         -- whether the implementation
13         -- supports the ability to
14         -- override the default PVID
15         -- setting and its egress status
16         -- (VLAN-Tagged or Untagged) on
17         -- each port.
18      dot1dLocalVlanCapable(7)
19         -- can support multiple local
20         -- Bridges, outside of the scope
21         -- of IEEE 802.1Q defined VLANs.
22
23      This object may not be modified while the corresponding
24      instance of ieee8021BridgeBaseRowStatus is active(1).
25
26      The value of this object MUST be retained across
27      reinitializations of the management system."
28  REFERENCE    "12.10.1.1.3 b)"
29  ::= { ieee8021BridgeBaseEntry 5 }
30
31  ieee8021BridgeBaseTrafficClassesEnabled OBJECT-TYPE
32      SYNTAX      TruthValue
33      MAX-ACCESS  read-create
34      STATUS      current
35      DESCRIPTION
36          "The value true(1) indicates that Traffic Classes are
37          enabled on this Bridge. When false(2), the Bridge
38          operates with a single priority level for all traffic.
39
40          This object may be modified while the corresponding
41          instance of ieee8021BridgeBaseRowStatus is active(1).
42
43          The value of this object MUST be retained across
44          reinitializations of the management system."
45      DEFVAL      { true }
46      ::= { ieee8021BridgeBaseEntry 6 }
47
48  ieee8021BridgeBaseMmrpEnabledStatus OBJECT-TYPE
49      SYNTAX      TruthValue
50      MAX-ACCESS  read-create
51      STATUS      current
52      DESCRIPTION
53          "The administrative status requested by management for
54          MMRP. The value true(1) indicates that MMRP should
55          be enabled on this device, in all VLANs, on all ports
56          for which it has not been specifically disabled. When
57          false(2), MMRP is disabled, in all VLANs and on all
58          ports, and all MMRP packets will be forwarded
59          transparently. This object affects both Applicant and
60          Registrar state machines. A transition from false(2)
61          to true(1) will cause a reset of all MMRP state
62          machines on all ports.
63
64          This object may be modified while the corresponding
65          instance of ieee8021BridgeBaseRowStatus is active(1).
66
67          The value of this object MUST be retained across
68          reinitializations of the management system."
69      DEFVAL      { true }
70      ::= { ieee8021BridgeBaseEntry 7 }
71
72  ieee8021BridgeBaseRowStatus OBJECT-TYPE

```

```

1  SYNTAX      RowStatus
2  MAX-ACCESS  read-create
3  STATUS      current
4  DESCRIPTION
5      "The object indicates the status of an entry, and is used
6      to create/delete entries.
7
8      The following objects MUST be set prior to making a new
9      entry active:
10         ieee8021BridgeBaseBridgeAddress
11         ieee8021BridgeBaseComponentType
12         ieee8021BridgeBaseDeviceCapabilities
13      It is recommended that these three objects not be allowed
14      to be modified while the corresponding instance of
15      ieee8021BridgeBaseRowStatus object is active(1).
16
17      The following objects are not required to be set before
18      making a new entry active (they will take their defaults),
19      and they also may be modified while the corresponding
20      instance of this object is active(1):
21         ieee8021BridgeBaseTrafficClassesEnabled
22         ieee8021BridgeBaseMmrpEnabledStatus
23
24      The value of this object and all corresponding instances
25      of other objects in this table MUST be retained across
26      reinitializations of the management system."
27 ::= { ieee8021BridgeBaseEntry 8 }
28
29 -- =====
30 -- The Generic Bridge Port Table
31 -- =====
32 ieee8021BridgeBasePortTable OBJECT-TYPE
33     SYNTAX      SEQUENCE OF Ieee8021BridgeBasePortEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "A table that contains generic information about every
38         port that is associated with this Bridge.  Transparent,
39         and source-route ports are included."
40     REFERENCE   "12.4.2"
41     ::= { ieee8021BridgeBase 4 }
42
43 ieee8021BridgeBasePortEntry OBJECT-TYPE
44     SYNTAX      Ieee8021BridgeBasePortEntry
45     MAX-ACCESS  not-accessible
46     STATUS      current
47
48     DESCRIPTION
49         "A list of objects containing information for each port
50         of the Bridge."
51     INDEX      { ieee8021BridgeBasePortComponentId,
52                 ieee8021BridgeBasePort }
53     ::= { ieee8021BridgeBasePortTable 1 }
54
55 Ieee8021BridgeBasePortEntry ::=
56     SEQUENCE {
57         ieee8021BridgeBasePortComponentId
58         IEEE8021PbbComponentIdentifier,
59         ieee8021BridgeBasePort
60         IEEE8021BridgePortNumber,
61         ieee8021BridgeBasePortIfIndex
62         InterfaceIndexOrZero,
63         ieee8021BridgeBasePortDelayExceededDiscards
64         Counter64,
65         ieee8021BridgeBasePortMtuExceededDiscards
66         Counter64,
67         ieee8021BridgeBasePortCapabilities
68         BITS,
69         ieee8021BridgeBasePortTypeCapabilities
70         BITS,
71         ieee8021BridgeBasePortType
72         IEEE8021BridgePortType,

```



```

1      ieee8021BridgeBasePortExternal
2      TruthValue,
3      ieee8021BridgeBasePortAdminPointToPoint
4      INTEGER,
5      ieee8021BridgeBasePortOperPointToPoint
6      TruthValue,
7      ieee8021BridgeBasePortName
8      SnmpAdminString
9  }
10
11 ieee8021BridgeBasePortComponentId OBJECT-TYPE
12     SYNTAX      IEEE8021PbbComponentIdentifier
13     MAX-ACCESS  not-accessible
14     STATUS      current
15     DESCRIPTION
16         "The component identifier is used to distinguish between the
17         multiple virtual Bridge instances within a PBB. In simple
18         situations where there is only a single component the default
19         value is 1."
20     ::= { ieee8021BridgeBasePortEntry 1 }
21
22 ieee8021BridgeBasePort OBJECT-TYPE
23     SYNTAX      IEEE8021BridgePortNumber
24     MAX-ACCESS  not-accessible
25     STATUS      current
26     DESCRIPTION
27         "The port number of the port for which this entry
28         contains Bridge management information."
29     REFERENCE   "12.4.2.1"
30     ::= { ieee8021BridgeBasePortEntry 2 }
31
32 ieee8021BridgeBasePortIfIndex OBJECT-TYPE
33     SYNTAX      InterfaceIndexOrZero
34     MAX-ACCESS  read-write
35     STATUS      current
36     DESCRIPTION
37         "The value of the instance of the IfIndex object,
38         defined in the IF-MIB, for the interface corresponding
39         to this port, or the value 0 if the port has not been
40         bound to an underlying frame source and sink.
41
42         It is an implementation specific decision as to whether this object
43         may be modified if it has been created or if 0 is a legal value.
44
45         The underlying IfEntry indexed by this column MUST be persistent
46         across reinitializations of the management system."
47     ::= { ieee8021BridgeBasePortEntry 3 }
48
49 ieee8021BridgeBasePortDelayExceededDiscards OBJECT-TYPE
50     SYNTAX      Counter64
51     UNITS        "frames"
52     MAX-ACCESS  read-only
53     STATUS      current
54     DESCRIPTION
55         "The number of frames discarded by this port due
56         to excessive transit delay through the Bridge. It
57         is incremented by both transparent and source
58         route Bridges.
59
60         Discontinuities in the value of the counter can occur
61         at re-initialization of the management system, and at
62         other times as indicated by the value of
63         ifCounterDiscontinuityTime object of the associated
64         interface (if any)."
65     REFERENCE   "12.6.1.1.3 f)"
66     ::= { ieee8021BridgeBasePortEntry 4 }
67
68 ieee8021BridgeBasePortMtuExceededDiscards OBJECT-TYPE
69     SYNTAX      Counter64
70     UNITS        "frames"
71     MAX-ACCESS  read-only
72     STATUS      current

```

```

1  DESCRIPTION
2      "The number of frames discarded by this port due
3      to an excessive size. It is incremented by both
4      transparent and source route Bridges.
5
6      Discontinuities in the value of the counter can occur
7      at re-initialization of the management system, and at
8      other times as indicated by the value of
9      ifCounterDiscontinuityTime object of the associated
10     interface (if any)."
```

11 REFERENCE "12.6.1.1.3 g)"

12 ::= { ieee8021BridgeBasePortEntry 5 }

13

14 ieee8021BridgeBasePortCapabilities OBJECT-TYPE

15 SYNTAX BITS {

16 dot1qDot1qTagging(0),

17 dot1qConfigurableAcceptableFrameTypes(1),

18 dot1qIngressFiltering(2)

19 }

20 MAX-ACCESS read-only

21 STATUS current

22 DESCRIPTION

23 "Indicates the parts of IEEE 802.1Q that are

24 optional on a per-port basis, that are implemented by

25 this device, and that are manageable through this MIB.

26

27 dot1qDot1qTagging(0), -- supports IEEE 802.1Q VLAN tagging of

28 -- frames and MVRP.

29 dot1qConfigurableAcceptableFrameTypes(1),

30 -- allows modified values of

31 -- dot1qPortAcceptableFrameTypes.

32 dot1qIngressFiltering(2)

33 -- supports the discarding of any

34 -- frame received on a Port whose

35 -- VLAN classification does not

36 -- include that Port in its Member

37 -- set."

38 REFERENCE "12.10.1.1.3 c)"

39 ::= { ieee8021BridgeBasePortEntry 6 }

40

41 ieee8021BridgeBasePortTypeCapabilities OBJECT-TYPE

42 SYNTAX BITS {

43 customerVlanPort(0),

44 providerNetworkPort(1),

45 customerNetworkPort(2),

46 customerEdgePort(3),

47 customerBackbonePort(4),

48 virtualInstancePort(5),

49 dBridgePort(6),

50 remoteCustomerAccessPort(7),

51 stationFacingBridgePort(8),

52 uplinkAccessPort(9),

53 uplinkRelayPort(10)

54 }

55 MAX-ACCESS read-only

56 STATUS current

57 DESCRIPTION

58 "Indicates the capabilities of this port. The corresponding

59 instance of ieee8021BridgeBasePortType can potentially take

60 any of the values for which the corresponding bit in this

61 object is 1. The possible port types are as follows:

62

63 customerVlanPort(0) - Indicates the port can be a C-TAG-

64 aware port of an enterprise VLAN-aware Bridge.

65

66 providerNetworkPort(1) - Indicates the port can be an

67 S-TAG-aware port of a Provider Bridge or Backbone

68 Edge Bridge used for connections within a PBN or

69 PBBN.

70

71 customerNetworkPort(2) - Indicates the port can be an

72 S-TAG-aware port of a Provider Bridge or Backbone

1 Edge Bridge used for connections to the exterior of
2 a PBN or PBBN.
3
4 customerEdgePort(3) - Indicates the port can be a C-TAG-
5 aware port of a Provider Bridge used for connections
6 to the exterior of a PBN or PBBN.
7
8 customerBackbonePort(4) - Indicates the port can be a
9 I-TAG-aware port of a Backbone Edge Bridge's
10 B-component.
11
12 virtualInstancePort(5) - Indicates the port can be a
13 virtual S-TAG-aware port within a Backbone Edge
14 Bridge's I-component that is responsible for
15 handling S-tagged traffic for a specific backbone
16 service instance.
17
18 dBridgePort(6) - Indicates the port can be a VLAN-unaware
19 member of an IEEE 802.1Q Bridge.
20
21 remoteCustomerAccessPort(7) - Indicates the port can be an
22 S-TAG-aware port of a Provider Bridge capable of providing
23 Remote Customer Service Interfaces.
24
25 stationFacingBridgePort(8) - Indicates the station-facing
26 Bridge Port in a EVB Bridge.
27
28 uplinkAccessPort(9) - Indicates the uplink access port
29 in an EVB Bridge or EVB station.
30
31 uplinkRelayPort(10) - Indicates the uplink relay port
32 in an EVB station."
33
34 REFERENCE "40.4, 12.13.1.1, 12.13.1.2, 12.16,
35 12.16.2.1, 12.26"
36 ::= { ieee8021BridgeBasePortEntry 7 }
37
38 ieee8021BridgeBasePortType OBJECT-TYPE
39 SYNTAX IEEE8021BridgePortType
40 MAX-ACCESS read-only
41 STATUS current
42 DESCRIPTION
43 "The port type. This value MUST be persistent over power up
44 restart/reboot."
45 REFERENCE "40.4, 12.13.1.1, 12.13.1.2, 12.16,
46 12.16.2.1, 12.26"
47 ::= { ieee8021BridgeBasePortEntry 8 }
48
49 ieee8021BridgeBasePortExternal OBJECT-TYPE
50 SYNTAX TruthValue
51 MAX-ACCESS read-only
52 STATUS current
53 DESCRIPTION
54 "A boolean indicating whether the port is external. A value of
55 true(1) means the port is external. A value of false(2) means
56 the port is internal."
57 REFERENCE "12.4.2.1"
58 ::= { ieee8021BridgeBasePortEntry 9 }
59
60 ieee8021BridgeBasePortAdminPointToPoint OBJECT-TYPE
61 SYNTAX INTEGER {
62 forceTrue(1),
63 forceFalse(2),
64 auto(3)
65 }
66 MAX-ACCESS read-write
67 STATUS current
68 DESCRIPTION
69 "For a port running spanning tree, this object represents the
70 administrative point-to-point status of the LAN segment
71 attached to this port, using the enumeration values of
72 IEEE Std 802.1AC. A value of forceTrue(1) indicates

that this port should always be treated as if it is connected to a point-to-point link. A value of forceFalse(2) indicates that this port should be treated as having a shared media connection. A value of auto(3) indicates that this port is considered to have a point-to-point link if it is an Aggregator and all of its members are aggregatable, or if the MAC entity is configured for full duplex operation, either through auto-negotiation or by management means. Manipulating this object changes the underlying adminPointToPointMAC.

For a VIP, the adminPointToPointMAC parameter controls the mechanism by which the Default Backbone Destination parameter for the VIP is determined. For a backbone service instance that includes only 2 VIPs, the value may be set to forceTrue(1) which permits dynamic learning of the Default Backbone Destination parameter. For a backbone service instance that includes more than 2 VIPs, the value MUST be set to ForceFalse(2) or auto(3).

When this object is set to forceTrue(1) for a VIP, the Default Backbone Destination parameter is modified by the subsequent M_UNITDATA.indications as specified in 6.10.1 (and described in 26.4.1). Whenever the parameter is set to forceFalse(2) or auto(3), the value for the Default Backbone Destination parameter is set to the Backbone Service Instance Group Address for the VIP-ISID.

The value of this object MUST be retained across reinitializations of the management system."

REFERENCE "IEEE Std 802.1AC, 6.10, 12.8.2.1.3 o), 12.8.2.3.2 f), 26.4.1"

DEFVAL { forceFalse }

::= { ieee8021BridgeBasePortEntry 10 }

ieee8021BridgeBasePortOperPointToPoint OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"For a port running spanning tree, this object represents the operational point-to-point status of the LAN segment attached to this port. It indicates whether a port is considered to have a point-to-point connection. If adminPointToPointMAC is set to auto(2), then the value of operPointToPointMAC is determined in accordance with the specific procedures defined for the MAC entity concerned, as defined in IEEE Std 802.1AC. The value is determined dynamically; that is, it is re-evaluated whenever the value of adminPointToPointMAC changes, and whenever the specific procedures defined for the MAC entity evaluate a change in its point-to-point status.

For a VIP, this object simply reflects the value of the corresponding instance of ieee8021BridgeBasePortAdminPointToPoint. The value will be true(1) if that object is forceTrue(1), and the value will be false(2) if the value of that object is either forceFalse(2) or auto(3)."

REFERENCE "IEEE Std 802.1AC, 6.10, 12.8.2.1.3 p), 12.8.2.3.2 f), 26.4.1"

::= { ieee8021BridgeBasePortEntry 11 }

ieee8021BridgeBasePortName OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A text string of up to 32 characters, of locally determined significance."

REFERENCE "12.4.2.1"

::= { ieee8021BridgeBasePortEntry 12 }

-- =====

```

1 -- The Generic Bridge ifIndex to Port Table
2 -- =====
3 ieee8021BridgeBaseIfToPortTable OBJECT-TYPE
4     SYNTAX      SEQUENCE OF Ieee8021BridgeBaseIfToPortEntry
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "A table that contains generic information about every
9         ifIndex that is associated with this Bridge."
10    REFERENCE   "17.2.2"
11    ::= { ieee8021BridgeBase 5 }
12
13 ieee8021BridgeBaseIfToPortEntry OBJECT-TYPE
14     SYNTAX      Ieee8021BridgeBaseIfToPortEntry
15     MAX-ACCESS  not-accessible
16     STATUS      current
17
18     DESCRIPTION
19         "A list of objects containing information for each ifIndex
20         of the Bridge."
21     INDEX { ifIndex }
22     ::= { ieee8021BridgeBaseIfToPortTable 1 }
23
24 Ieee8021BridgeBaseIfToPortEntry ::=
25     SEQUENCE {
26         ieee8021BridgeBaseIfIndexComponentId
27         IEEE8021PbbComponentIdentifier,
28         ieee8021BridgeBaseIfIndexPort
29         IEEE8021BridgePortNumber
30     }
31
32 ieee8021BridgeBaseIfIndexComponentId OBJECT-TYPE
33     SYNTAX      IEEE8021PbbComponentIdentifier
34     MAX-ACCESS  read-only
35     STATUS      current
36     DESCRIPTION
37         "The component ID for this ifIndex."
38     ::= { ieee8021BridgeBaseIfToPortEntry 1 }
39
40 ieee8021BridgeBaseIfIndexPort OBJECT-TYPE
41     SYNTAX      IEEE8021BridgePortNumber
42     MAX-ACCESS  read-only
43     STATUS      current
44     DESCRIPTION
45         "The port for this ifIndex."
46     ::= { ieee8021BridgeBaseIfToPortEntry 2 }
47
48
49 -- =====
50 -- port number table section 12.5.1
51 -- =====
52
53 ieee8021BridgePhyPortTable OBJECT-TYPE
54     SYNTAX      SEQUENCE OF Ieee8021BridgePhyPortEntry
55     MAX-ACCESS  not-accessible
56     STATUS      current
57     DESCRIPTION
58         "A table that contains ISS port number to Bridge
59         componentID and port number mapping."
60     REFERENCE   "12.5.1"
61     ::= { ieee8021BridgeBase 6}
62
63 ieee8021BridgePhyPortEntry OBJECT-TYPE
64     SYNTAX      Ieee8021BridgePhyPortEntry
65     MAX-ACCESS  not-accessible
66     STATUS      current
67     DESCRIPTION
68         "A list of objects containing mapping for ISS port
69         numbers to Bridge componentID and port numbers "
70     INDEX { ieee8021BridgePhyPort }
71     ::= { ieee8021BridgePhyPortTable 1 }
72

```

```

1  Ieee8021BridgePhyPortEntry ::=
2  SEQUENCE {
3      ieee8021BridgePhyPort
4          IEEE8021BridgePortNumber,
5      ieee8021BridgePhyPortIfIndex
6          InterfaceIndexOrZero,
7      ieee8021BridgePhyMacAddress
8          MacAddress,
9      ieee8021BridgePhyPortToComponentId
10         IEEE8021PbbComponentIdentifierOrZero,
11      ieee8021BridgePhyPortToInternalPort
12         IEEE8021BridgePortNumberOrZero
13  }
14
15
16 ieee8021BridgePhyPort OBJECT-TYPE
17     SYNTAX      IEEE8021BridgePortNumber
18     MAX-ACCESS  not-accessible
19     STATUS      current
20     DESCRIPTION
21         "The ISS port."
22     REFERENCE   "12.26"
23     ::= { ieee8021BridgePhyPortEntry 1 }
24
25
26 ieee8021BridgePhyPortIfIndex OBJECT-TYPE
27     SYNTAX      InterfaceIndexOrZero
28     MAX-ACCESS  read-only
29     STATUS      current
30     DESCRIPTION
31         "The value of the instance of the IfIndex object,
32         defined in the IF-MIB, for the interface corresponding
33         to this port, or the value 0 if the port has not been
34         bound to an underlying frame source and sink.
35
36         The underlying IfEntry indexed by this column MUST
37         be persistent across reinitializations of the
38         management system."
39     ::= { ieee8021BridgePhyPortEntry 2 }
40
41 ieee8021BridgePhyMacAddress OBJECT-TYPE
42     SYNTAX      MacAddress
43     MAX-ACCESS  read-only
44     STATUS      current
45     DESCRIPTION "The MAC address"
46     ::= { ieee8021BridgePhyPortEntry 3 }
47
48 ieee8021BridgePhyPortToComponentId OBJECT-TYPE
49     SYNTAX      IEEE8021PbbComponentIdentifierOrZero
50     MAX-ACCESS  read-only
51     STATUS      current
52     DESCRIPTION
53         "The component ID that this ISS port belongs to."
54
55     ::= { ieee8021BridgePhyPortEntry 4 }
56
57
58 ieee8021BridgePhyPortToInternalPort OBJECT-TYPE
59     SYNTAX      IEEE8021BridgePortNumberOrZero
60     MAX-ACCESS  read-only
61     STATUS      current
62     DESCRIPTION
63         "The port number to which this ISS port maps to."
64
65     ::= { ieee8021BridgePhyPortEntry 5 }
66
67 -- =====
68 -- the ieee8021BridgeTp subtree
69 -- =====
70 -- This is implemented by those Bridges that support the
71 -- transparent bridging mode. A transparent Bridge will
72 -- implement this subtree.

```

```

1 -- =====
2
3 -- =====
4 -- Port Table for Transparent Bridges
5 -- =====
6
7 ieee8021BridgeTpPortTable OBJECT-TYPE
8     SYNTAX      SEQUENCE OF Ieee8021BridgeTpPortEntry
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "A table that contains information about every port that
13         is associated with this transparent Bridge."
14    REFERENCE   "12.4.2"
15    ::= { ieee8021BridgeTp 1 }
16
17 ieee8021BridgeTpPortEntry OBJECT-TYPE
18     SYNTAX      Ieee8021BridgeTpPortEntry
19     MAX-ACCESS  not-accessible
20     STATUS      current
21     DESCRIPTION
22         "A list of objects containing information for each port of
23          a transparent Bridge."
24     INDEX      { ieee8021BridgeTpPortComponentId,
25                  ieee8021BridgeTpPort }
26     ::= { ieee8021BridgeTpPortTable 1 }
27
28 Ieee8021BridgeTpPortEntry ::=
29     SEQUENCE {
30         ieee8021BridgeTpPortComponentId
31             IEEE8021PbbComponentIdentifier,
32         ieee8021BridgeTpPort
33             IEEE8021BridgePortNumber,
34         ieee8021BridgeTpPortMaxInfo
35             Integer32,
36         ieee8021BridgeTpPortInFrames
37             Counter64,
38         ieee8021BridgeTpPortOutFrames
39             Counter64,
40         ieee8021BridgeTpPortInDiscards
41             Counter64
42     }
43
44 ieee8021BridgeTpPortComponentId OBJECT-TYPE
45     SYNTAX      IEEE8021PbbComponentIdentifier
46     MAX-ACCESS  not-accessible
47     STATUS      current
48     DESCRIPTION
49         "The component identifier is used to distinguish between the
50         multiple virtual Bridge instances within a PBB. In simple
51         situations where there is only a single component the default
52         value is 1."
53     ::= { ieee8021BridgeTpPortEntry 1 }
54
55 ieee8021BridgeTpPort OBJECT-TYPE
56     SYNTAX      IEEE8021BridgePortNumber
57     MAX-ACCESS  not-accessible
58     STATUS      current
59     DESCRIPTION
60         "The port number of the port for which this entry
61         contains Transparent bridging management information."
62     ::= { ieee8021BridgeTpPortEntry 2 }
63
64 ieee8021BridgeTpPortMaxInfo OBJECT-TYPE
65     SYNTAX      Integer32
66     UNITS       "bytes"
67     MAX-ACCESS  read-only
68     STATUS      current
69     DESCRIPTION
70         "The maximum size of the INFO (non-MAC) field that
71         this port will receive or transmit."
72     ::= { ieee8021BridgeTpPortEntry 3 }

```

```

1
2 ieee8021BridgeTpPortInFrames OBJECT-TYPE
3     SYNTAX      Counter64
4     UNITS       "frames"
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "The number of frames that have been received by this
9         port from its segment. Note that a frame received on the
10        interface corresponding to this port is only counted by
11        this object if and only if it is for a protocol being
12        processed by the local bridging function, including
13        Bridge management frames.
14
15        Discontinuities in the value of the counter can occur
16        at re-initialization of the management system, and at
17        other times as indicated by the value of
18        ifCounterDiscontinuityTime object of the associated
19        interface (if any)."
```

REFERENCE "12.6.1.1.3 a)"

```

21 ::= { ieee8021BridgeTpPortEntry 4 }
22
23 ieee8021BridgeTpPortOutFrames OBJECT-TYPE
24     SYNTAX      Counter64
25     UNITS       "frames"
26     MAX-ACCESS  read-only
27     STATUS      current
28     DESCRIPTION
29         "The number of frames that have been transmitted by this
30         port to its segment. Note that a frame transmitted on
31         the interface corresponding to this port is only counted
32         by this object if and only if it is for a protocol being
33         processed by the local bridging function, including
34         Bridge management frames.
35
36         Discontinuities in the value of the counter can occur
37         at re-initialization of the management system, and at
38         other times as indicated by the value of
39         ifCounterDiscontinuityTime object of the associated
40         interface (if any)."
```

REFERENCE "12.6.1.1.3 d)"

```

42 ::= { ieee8021BridgeTpPortEntry 5 }
43
44 ieee8021BridgeTpPortInDiscards OBJECT-TYPE
45     SYNTAX      Counter64
46     UNITS       "frames"
47     MAX-ACCESS  read-only
48     STATUS      current
49     DESCRIPTION
50         "Count of received valid frames that were discarded
51         (i.e., filtered) by the Forwarding Process.
52
53         Discontinuities in the value of the counter can occur
54         at re-initialization of the management system, and at
55         other times as indicated by the value of
56         ifCounterDiscontinuityTime object of the associated
57         interface (if any)."
```

REFERENCE "12.6.1.1.3 c)"

```

59 ::= { ieee8021BridgeTpPortEntry 6 }
60
61 -- =====
62 -- the ieee8021BridgePriority subtree
63 -- =====
64
65 -- =====
66 -- Port Priority Table
67 -- =====
68
69 ieee8021BridgePortPriorityTable OBJECT-TYPE
70     SYNTAX      SEQUENCE OF Ieee8021BridgePortPriorityEntry
71     MAX-ACCESS  not-accessible
72     STATUS      current
```



```

1  DESCRIPTION
2      "A table that contains information about every port that
3      is associated with this transparent Bridge."
4      ::= { ieee8021BridgePriority 1 }
5
6  ieee8021BridgePortPriorityEntry OBJECT-TYPE
7      SYNTAX      Ieee8021BridgePortPriorityEntry
8      MAX-ACCESS  not-accessible
9      STATUS      current
10     DESCRIPTION
11         "A list of Default User Priorities for each port of a
12         transparent Bridge. This is indexed by
13         ieee8021BridgeBasePortComponentId and
14         ieee8021BridgeBasePort."
15     AUGMENTS { ieee8021BridgeBasePortEntry }
16     ::= { ieee8021BridgePortPriorityTable 1 }
17
18 Ieee8021BridgePortPriorityEntry ::=
19     SEQUENCE {
20         ieee8021BridgePortDefaultUserPriority
21             IEEE8021PriorityValue,
22         ieee8021BridgePortNumTrafficClasses
23             Integer32,
24         ieee8021BridgePortPriorityCodePointSelection
25             IEEE8021PriorityCodePoint,
26         ieee8021BridgePortUseDEI
27             TruthValue,
28         ieee8021BridgePortRequireDropEncoding
29             TruthValue,
30         ieee8021BridgePortServiceAccessPrioritySelection
31             TruthValue
32     }
33
34 ieee8021BridgePortDefaultUserPriority OBJECT-TYPE
35     SYNTAX      IEEE8021PriorityValue
36     MAX-ACCESS  read-write
37     STATUS      current
38     DESCRIPTION
39         "The default ingress priority for this port. This
40         only has effect on media, such as Ethernet, that do not
41         support native priority.
42
43         The value of this object MUST be retained across
44         reinitializations of the management system."
45     ::= { ieee8021BridgePortPriorityEntry 1 }
46
47 ieee8021BridgePortNumTrafficClasses OBJECT-TYPE
48     SYNTAX      Integer32 (1..8)
49     MAX-ACCESS  read-write
50     STATUS      current
51     DESCRIPTION
52         "The number of egress traffic classes supported on this
53         port. This object may optionally be read-only.
54
55         The value of this object MUST be retained across
56         reinitializations of the management system."
57     ::= { ieee8021BridgePortPriorityEntry 2 }
58
59 ieee8021BridgePortPriorityCodePointSelection OBJECT-TYPE
60     SYNTAX      IEEE8021PriorityCodePoint
61     MAX-ACCESS  read-write
62     STATUS      current
63     DESCRIPTION
64         " This object identifies the rows in the PCP encoding and
65         decoding tables that are used to remark frames on this
66         port if this remarking is enabled."
67     REFERENCE   "12.6.2.6, 12.6.2.7"
68     ::= { ieee8021BridgePortPriorityEntry 3 }
69
70 ieee8021BridgePortUseDEI OBJECT-TYPE
71     SYNTAX      TruthValue
72     MAX-ACCESS  read-write

```

```

1  STATUS      current
2  DESCRIPTION
3      "If the Use_DEI is set to true(1) for the Port then the
4      drop_eligible parameter is encoded in the DEI of transmitted
5      frames, and the drop_eligible parameter shall be true(1) for a
6      received frame if the DEI is set in the VLAN tag or the Priority
7      Code Point Decoding Table indicates drop_eligible True for
8      the received PCP value. If the Use_DEI parameter is false(2),
9      the DEI shall be transmitted as zero and ignored on receipt.
10     The default value of the Use_DEI parameter is false(2)."
```

11 REFERENCE "12.6.2.11, 12.6.2.12"

12 ::= { ieee8021BridgePortPriorityEntry 4 }

13

14 ieee8021BridgePortRequireDropEncoding OBJECT-TYPE

15 SYNTAX TruthValue

16 MAX-ACCESS read-write

17 STATUS current

18 DESCRIPTION

19 "If a Bridge supports encoding or decoding of drop_eligible

20 from the PCP field of a VLAN tag (6.9.3) on any of its Ports,

21 then it shall implement a Boolean parameter Require Drop

22 Encoding on each of its Ports with default value false(2). If

23 Require Drop Encoding is True and the Bridge Port cannot

24 encode particular priorities with drop_eligible, then frames

25 queued with those priorities and drop_eligible true(1) shall

26 be discarded and not transmitted."

27 REFERENCE "12.6.2.13, 12.6.2.14"

28 DEFVAL { false }

29 ::= { ieee8021BridgePortPriorityEntry 5 }

30

31 ieee8021BridgePortServiceAccessPrioritySelection OBJECT-TYPE

32 SYNTAX TruthValue

33 MAX-ACCESS read-write

34 STATUS current

35 DESCRIPTION

36 "Indication of whether the Service Access Priority Selection

37 function is supported on the Customer Bridge Port to request

38 priority handling of the frame from a Port-based service

39 interface."

40 REFERENCE "12.6.2.15, 12.6.2.16"

41 ::= { ieee8021BridgePortPriorityEntry 6 }

42

43 -- =====

44 -- Priority Regeneration Table

45 -- =====

46

47 ieee8021BridgeUserPriorityRegenTable OBJECT-TYPE

48 SYNTAX SEQUENCE OF Ieee8021BridgeUserPriorityRegenEntry

49 MAX-ACCESS not-accessible

50 STATUS current

51 DESCRIPTION

52 "A list of Regenerated User Priorities for each received

53 priority on each port of a Bridge. The regenerated

54 priority value may be used to index the Traffic

55 Class Table for each input port. This only has effect

56 on media that support native priority. The default

57 values for Regenerated User Priorities are the same as

58 the User Priorities."

59 REFERENCE "6.5.9, 6.9.4"

60 ::= { ieee8021BridgePriority 2 }

61

62 ieee8021BridgeUserPriorityRegenEntry OBJECT-TYPE

63 SYNTAX Ieee8021BridgeUserPriorityRegenEntry

64 MAX-ACCESS not-accessible

65 STATUS current

66 DESCRIPTION

67 "A mapping of incoming priority to a regenerated

68 priority."

69 INDEX { ieee8021BridgeBasePortComponentId,

70 ieee8021BridgeBasePort,

71 ieee8021BridgeUserPriority }

72 ::= { ieee8021BridgeUserPriorityRegenTable 1 }

```

1
2 Ieee8021BridgeUserPriorityRegenEntry ::=
3     SEQUENCE {
4         ieee8021BridgeUserPriority
5             IEEE8021PriorityValue,
6         ieee8021BridgeRegenUserPriority
7             IEEE8021PriorityValue
8     }
9
10 ieee8021BridgeUserPriority OBJECT-TYPE
11     SYNTAX      IEEE8021PriorityValue
12     MAX-ACCESS  not-accessible
13     STATUS      current
14     DESCRIPTION
15         "The priority for a frame received on this port."
16     ::= { ieee8021BridgeUserPriorityRegenEntry 1 }
17
18 ieee8021BridgeRegenUserPriority OBJECT-TYPE
19     SYNTAX      IEEE8021PriorityValue
20     MAX-ACCESS  read-write
21     STATUS      current
22     DESCRIPTION
23         "The regenerated priority that the incoming User
24         Priority is mapped to for this port.
25
26         The value of this object MUST be retained across
27         reinitializations of the management system."
28     ::= { ieee8021BridgeUserPriorityRegenEntry 2 }
29
30 -- =====
31 -- Traffic Class Table
32 -- =====
33
34 ieee8021BridgeTrafficClassTable OBJECT-TYPE
35     SYNTAX      SEQUENCE OF Ieee8021BridgeTrafficClassEntry
36     MAX-ACCESS  not-accessible
37     STATUS      current
38     DESCRIPTION
39         "A table mapping evaluated priority to Traffic
40         Class, for forwarding by the Bridge. Traffic class is a
41         number in the range (0..(ieee8021BridgePortNumTrafficClasses-1))."
42     REFERENCE   "8.6.6, Table 8-5"
43     ::= { ieee8021BridgePriority 3 }
44
45 ieee8021BridgeTrafficClassEntry OBJECT-TYPE
46     SYNTAX      Ieee8021BridgeTrafficClassEntry
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "Priority to Traffic Class mapping."
51     INDEX       { ieee8021BridgeBasePortComponentId,
52                 ieee8021BridgeBasePort,
53                 ieee8021BridgeTrafficClassPriority }
54     ::= { ieee8021BridgeTrafficClassTable 1 }
55
56 Ieee8021BridgeTrafficClassEntry ::=
57     SEQUENCE {
58         ieee8021BridgeTrafficClassPriority
59             IEEE8021PriorityValue,
60         ieee8021BridgeTrafficClass
61             Integer32
62     }
63
64 ieee8021BridgeTrafficClassPriority OBJECT-TYPE
65     SYNTAX      IEEE8021PriorityValue
66     MAX-ACCESS  not-accessible
67     STATUS      current
68     DESCRIPTION
69         "The Priority value determined for the received frame.
70         This value is equivalent to the priority indicated in
71         the tagged frame received, or one of the evaluated
72         priorities, determined according to the media-type.

```

```

1      For untagged frames received from Ethernet media, this
2      value is equal to the ieee8021BridgePortDefaultUserPriority value
3      for the ingress port.
4
5      For untagged frames received from non-Ethernet media,
6      this value is equal to the ieee8021BridgeRegenUserPriority value
7      for the ingress port and media-specific priority."
8      ::= { ieee8021BridgeTrafficClassEntry 1 }
9
10 ieee8021BridgeTrafficClass OBJECT-TYPE
11     SYNTAX      Integer32 (0..7)
12     MAX-ACCESS  read-write
13     STATUS      current
14     DESCRIPTION
15         "The Traffic Class the received frame is mapped to.
16
17         The value of this object MUST be retained across
18         reinitializations of the management system."
19     ::= { ieee8021BridgeTrafficClassEntry 2 }
20
21 -- =====
22 -- Outbound Access Priority Table
23 -- =====
24
25 ieee8021BridgePortOutboundAccessPriorityTable OBJECT-TYPE
26     SYNTAX      SEQUENCE OF Ieee8021BridgePortOutboundAccessPriorityEntry
27     MAX-ACCESS  not-accessible
28     STATUS      current
29     DESCRIPTION
30         "A table mapping regenerated priority to Outbound
31         Access Priority. This is a fixed mapping for all port
32         types."
33     REFERENCE   "IEEE Std 802.1AC"
34     ::= { ieee8021BridgePriority 4 }
35
36 ieee8021BridgePortOutboundAccessPriorityEntry OBJECT-TYPE
37     SYNTAX      Ieee8021BridgePortOutboundAccessPriorityEntry
38     MAX-ACCESS  not-accessible
39     STATUS      current
40     DESCRIPTION
41         "Regenerated priority to Outbound Access Priority
42         mapping."
43     INDEX       { ieee8021BridgeBasePortComponentId,
44                  ieee8021BridgeBasePort,
45                  ieee8021BridgeRegenUserPriority }
46     ::= { ieee8021BridgePortOutboundAccessPriorityTable 1 }
47
48 Ieee8021BridgePortOutboundAccessPriorityEntry ::=
49     SEQUENCE {
50         ieee8021BridgePortOutboundAccessPriority
51         IEEE8021PriorityValue
52     }
53
54 ieee8021BridgePortOutboundAccessPriority OBJECT-TYPE
55     SYNTAX      IEEE8021PriorityValue
56     MAX-ACCESS  read-only
57     STATUS      current
58     DESCRIPTION
59         "The Outbound Access Priority the received frame is
60         mapped to."
61     ::= { ieee8021BridgePortOutboundAccessPriorityEntry 1 }
62
63 -- =====
64 -- ieee8021BridgePortDecodingTable:
65 -- =====
66
67 ieee8021BridgePortDecodingTable OBJECT-TYPE
68     SYNTAX      SEQUENCE OF Ieee8021BridgePortDecodingEntry
69     MAX-ACCESS  not-accessible
70     STATUS      current
71     DESCRIPTION
72         "A table that contains information about Priority Code

```

```

1      Point Decoding Table for a Port of a provider Bridge.
2      Alternative values for each table are specified as rows
3      in Table 6-3 (6.9.3), with each alternative labeled by
4      the number of distinct priorities that can be communicated,
5      and the number of these for which drop precedence can
6      be communicated. All writable objects in this table MUST
7      be persistent over power up restart/reboot."
8      ::= { ieee8021BridgePriority 5 }
9
10 ieee8021BridgePortDecodingEntry OBJECT-TYPE
11     SYNTAX      Ieee8021BridgePortDecodingEntry
12     MAX-ACCESS  not-accessible
13     STATUS      current
14     DESCRIPTION
15         "A list of objects containing Priority Code Point Decoding
16         information for a port of a provider Bridge."
17     INDEX { ieee8021BridgePortDecodingComponentId,
18             ieee8021BridgePortDecodingPortNum,
19             ieee8021BridgePortDecodingPriorityCodePointRow,
20             ieee8021BridgePortDecodingPriorityCodePoint }
21     ::= { ieee8021BridgePortDecodingTable 1 }
22
23 Ieee8021BridgePortDecodingEntry ::= SEQUENCE {
24     ieee8021BridgePortDecodingComponentId
25         IEEE8021PbbComponentIdentifier,
26     ieee8021BridgePortDecodingPortNum
27         IEEE8021BridgePortNumber,
28     ieee8021BridgePortDecodingPriorityCodePointRow
29         IEEE8021PriorityCodePoint,
30     ieee8021BridgePortDecodingPriorityCodePoint
31         Integer32,
32     ieee8021BridgePortDecodingPriority
33         IEEE8021PriorityValue,
34     ieee8021BridgePortDecodingDropEligible
35         TruthValue
36 }
37
38 ieee8021BridgePortDecodingComponentId OBJECT-TYPE
39     SYNTAX      IEEE8021PbbComponentIdentifier
40     MAX-ACCESS  not-accessible
41     STATUS      current
42     DESCRIPTION
43         "The component identifier is used to distinguish between the
44         multiple virtual Bridge instances within a PBB. In simple
45         situations where there is only a single component the default
46         value is 1."
47     ::= { ieee8021BridgePortDecodingEntry 1 }
48
49 ieee8021BridgePortDecodingPortNum OBJECT-TYPE
50     SYNTAX      IEEE8021BridgePortNumber
51     MAX-ACCESS  not-accessible
52     STATUS      current
53     DESCRIPTION
54         "A unique identifier of a port controlled by this VLAN
55         bridging entity."
56     ::= { ieee8021BridgePortDecodingEntry 2 }
57
58 ieee8021BridgePortDecodingPriorityCodePointRow OBJECT-TYPE
59     SYNTAX      IEEE8021PriorityCodePoint
60     MAX-ACCESS  not-accessible
61     STATUS      current
62     DESCRIPTION
63         "The specific row in Table 6-2 (6.9.3) indicating the PCP."
64     ::= { ieee8021BridgePortDecodingEntry 3 }
65
66 ieee8021BridgePortDecodingPriorityCodePoint OBJECT-TYPE
67     SYNTAX      Integer32 (0..7)
68     MAX-ACCESS  not-accessible
69     STATUS      current
70     DESCRIPTION
71         "The specific PCP value in Table 6-2 (6.9.3)."
72     ::= { ieee8021BridgePortDecodingEntry 4 }

```

```

1
2 ieee8021BridgePortDecodingPriority OBJECT-TYPE
3   SYNTAX      IEEE8021PriorityValue
4   MAX-ACCESS  read-write
5   STATUS      current
6   DESCRIPTION
7       "The specific priority value in Table 6-2 (6.9.3)."

```

```

1      "The component identifier is used to distinguish between the
2      multiple virtual Bridge instances within a PBB. In simple
3      situations where there is only a single component the default
4      value is 1."
5      ::= { ieee8021BridgePortEncodingEntry 1 }
6
7 ieee8021BridgePortEncodingPortNum OBJECT-TYPE
8     SYNTAX      IEEE8021BridgePortNumber
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "A unique identifier of a port controlled by this VLAN bridging
13        entity."
14    ::= { ieee8021BridgePortEncodingEntry 2 }
15
16 ieee8021BridgePortEncodingPriorityCodePointRow OBJECT-TYPE
17    SYNTAX      IEEE8021PriorityCodePoint
18    MAX-ACCESS  not-accessible
19    STATUS      current
20    DESCRIPTION
21        "The specific row in Table 6-2 (6.9.3) indicating the PCP row.
22        (i.e., 8P0D, 7P1D, 6P2D, 5P3D)"
23    ::= { ieee8021BridgePortEncodingEntry 3 }
24
25 ieee8021BridgePortEncodingPriorityCodePoint OBJECT-TYPE
26    SYNTAX      Integer32 (0..7)
27    MAX-ACCESS  not-accessible
28    STATUS      current
29    DESCRIPTION
30        "The specific row in Table 6-2 (6.9.3) indicating the PCP.
31        (i.e., 0,1,2,3,4,5,6,7)."
32    ::= { ieee8021BridgePortEncodingEntry 4 }
33
34 ieee8021BridgePortEncodingDropEligible OBJECT-TYPE
35    SYNTAX      TruthValue
36    MAX-ACCESS  not-accessible
37    STATUS      current
38    DESCRIPTION
39        "The specific row in Table 6-2 (6.9.3) indicating the drop
40        eligibility. A value of true(1) means eligible for drop."
41    ::= { ieee8021BridgePortEncodingEntry 5 }
42
43 ieee8021BridgePortEncodingPriority OBJECT-TYPE
44    SYNTAX      IEEE8021PriorityValue
45    MAX-ACCESS  read-write
46    STATUS      current
47    DESCRIPTION
48        "The encoding priority in Table 6-2 (6.9.3)."
49    REFERENCE   "6.9.3, 12.6.2.9, 12.6.2.10"
50    ::= { ieee8021BridgePortEncodingEntry 6 }
51
52 -- =====
53 -- ieee8021BridgeServiceAccessPriorityTable:
54 -- =====
55
56 ieee8021BridgeServiceAccessPriorityTable OBJECT-TYPE
57    SYNTAX      SEQUENCE OF Ieee8021BridgeServiceAccessPriorityEntry
58    MAX-ACCESS  not-accessible
59    STATUS      current
60    DESCRIPTION
61        "A table that contains information about the Service Access
62        Priority Selection function for a provider Bridge. The use
63        of this table enables a mechanism for a Customer Bridge
64        attached to a Provider Bridged Network to request priority
65        handling of frames. All writable objects in this table MUST
66        be persistent over power up restart/reboot."
67    ::= { ieee8021BridgePriority 7 }
68
69 ieee8021BridgeServiceAccessPriorityEntry OBJECT-TYPE
70    SYNTAX      Ieee8021BridgeServiceAccessPriorityEntry
71    MAX-ACCESS  not-accessible
72    STATUS      current

```

```
1  DESCRIPTION
2      "A list of objects containing information about the Service
3      Access Priority Selection function for a provider Bridge."
4  INDEX { ieee8021BridgeServiceAccessPriorityComponentId,
5          ieee8021BridgeServiceAccessPriorityPortNum,
6          ieee8021BridgeServiceAccessPriorityReceived }
7  ::= { ieee8021BridgeServiceAccessPriorityTable 1 }
8
9  Ieee8021BridgeServiceAccessPriorityEntry ::= SEQUENCE {
10     ieee8021BridgeServiceAccessPriorityComponentId
11     IEEE8021PbbComponentIdentifier,
12     ieee8021BridgeServiceAccessPriorityPortNum
13     IEEE8021BridgePortNumber,
14     ieee8021BridgeServiceAccessPriorityReceived
15     IEEE8021PriorityValue,
16     ieee8021BridgeServiceAccessPriorityValue
17     IEEE8021PriorityValue
18 }
19
20 ieee8021BridgeServiceAccessPriorityComponentId OBJECT-TYPE
21     SYNTAX      IEEE8021PbbComponentIdentifier
22     MAX-ACCESS  not-accessible
23     STATUS      current
24     DESCRIPTION
25         "The component identifier is used to distinguish between the
26         multiple virtual Bridge instances within a PBB. In simple
27         situations where there is only a single component the default
28         value is 1."
29     ::= { ieee8021BridgeServiceAccessPriorityEntry 1 }
30
31 ieee8021BridgeServiceAccessPriorityPortNum OBJECT-TYPE
32     SYNTAX      IEEE8021BridgePortNumber
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "A unique identifier of a port controlled by this VLAN bridging
37         entity."
38     ::= { ieee8021BridgeServiceAccessPriorityEntry 2 }
39
40 ieee8021BridgeServiceAccessPriorityReceived OBJECT-TYPE
41     SYNTAX      IEEE8021PriorityValue
42     MAX-ACCESS  not-accessible
43     STATUS      current
44     DESCRIPTION
45         "The default received priority value in Table 6-4.
46         (i.e., 0,1,2,3,4,5,6,7)"
47     ::= { ieee8021BridgeServiceAccessPriorityEntry 3 }
48
49 ieee8021BridgeServiceAccessPriorityValue OBJECT-TYPE
50     SYNTAX      IEEE8021PriorityValue
51     MAX-ACCESS  read-write
52     STATUS      current
53     DESCRIPTION
54         "The regenerated priority value in Table 6-4.
55         (i.e., 0,1,2,3,4,5,6,7)"
56     REFERENCE   "12.6.2.17, 112.6.2.18"
57     ::= { ieee8021BridgeServiceAccessPriorityEntry 4 }
58
59 -- =====
60 -- the ieee8021BridgeMrp subtree
61 -- =====
62
63 -- =====
64 -- The MRP Port Table
65 -- =====
66
67 ieee8021BridgePortMrpTable OBJECT-TYPE
68     SYNTAX      SEQUENCE OF Ieee8021BridgePortMrpEntry
69     MAX-ACCESS  not-accessible
70     STATUS      current
71     DESCRIPTION
72         "A table of MRP control information about every Bridge
```



```

1      port. This is indexed by ieee8021BridgeBasePortComponentId
2      and ieee8021BridgeBasePort."
3      ::= { ieee8021BridgeMrp 1 }
4
5 ieee8021BridgePortMrpEntry OBJECT-TYPE
6     SYNTAX      Ieee8021BridgePortMrpEntry
7     MAX-ACCESS  not-accessible
8     STATUS      current
9     DESCRIPTION
10        "MRP control information for a Bridge Port."
11     AUGMENTS { ieee8021BridgeBasePortEntry }
12     ::= { ieee8021BridgePortMrpTable 1 }
13
14 Ieee8021BridgePortMrpEntry ::=
15     SEQUENCE {
16         ieee8021BridgePortMrpJoinTime
17         TimeInterval,
18         ieee8021BridgePortMrpLeaveTime
19         TimeInterval,
20         ieee8021BridgePortMrpLeaveAllTime
21         TimeInterval
22     }
23
24 ieee8021BridgePortMrpJoinTime OBJECT-TYPE
25     SYNTAX      TimeInterval
26     UNITS       "centi-seconds"
27     MAX-ACCESS  read-write
28     STATUS      current
29     DESCRIPTION
30        "The MRP Join time, in centiseconds."
31
32        The value of this object MUST be retained across
33        reinitializations of the management system."
34     DEFVAL     { 20 }
35     ::= { ieee8021BridgePortMrpEntry 1 }
36
37 ieee8021BridgePortMrpLeaveTime OBJECT-TYPE
38     SYNTAX      TimeInterval
39     UNITS       "centi-seconds"
40     MAX-ACCESS  read-write
41     STATUS      current
42     DESCRIPTION
43        "The MRP Leave time, in centiseconds."
44
45        The value of this object MUST be retained across
46        reinitializations of the management system."
47     DEFVAL     { 60 }
48     ::= { ieee8021BridgePortMrpEntry 2 }
49
50 ieee8021BridgePortMrpLeaveAllTime OBJECT-TYPE
51     SYNTAX      TimeInterval
52     UNITS       "centi-seconds"
53     MAX-ACCESS  read-write
54     STATUS      current
55     DESCRIPTION
56        "The MRP LeaveAll time, in centiseconds."
57
58        The value of this object MUST be retained across
59        reinitializations of the management system."
60     DEFVAL     { 1000 }
61     ::= { ieee8021BridgePortMrpEntry 3 }
62
63 -- =====
64 -- The MMRP Port Configuration and Status Table
65 -- =====
66
67 ieee8021BridgePortMmrpTable OBJECT-TYPE
68     SYNTAX      SEQUENCE OF Ieee8021BridgePortMmrpEntry
69     MAX-ACCESS  not-accessible
70     STATUS      current
71     DESCRIPTION
72        "A table of MMRP control and status information about

```

```

1      every Bridge Port. Augments the ieee8021BridgeBasePortTable."
2      ::= { ieee8021BridgeMmrp 1 }
3
4 ieee8021BridgePortMmrpEntry OBJECT-TYPE
5     SYNTAX      Ieee8021BridgePortMmrpEntry
6     MAX-ACCESS  not-accessible
7     STATUS      current
8     DESCRIPTION
9         "MMRP control and status information for a Bridge Port."
10    AUGMENTS { ieee8021BridgeBasePortEntry }
11    ::= { ieee8021BridgePortMmrpTable 1 }
12
13 Ieee8021BridgePortMmrpEntry ::=
14     SEQUENCE {
15         ieee8021BridgePortMmrpEnabledStatus
16         TruthValue,
17         ieee8021BridgePortMmrpFailedRegistrations
18         Counter64,
19         ieee8021BridgePortMmrpLastPduOrigin
20         MacAddress,
21         ieee8021BridgePortRestrictedGroupRegistration
22         TruthValue
23     }
24
25 Ieee8021BridgePortMmrpEnabledStatus OBJECT-TYPE
26     SYNTAX      TruthValue
27     MAX-ACCESS  read-write
28     STATUS      current
29     DESCRIPTION
30         "The administrative state of MMRP operation on this port. The
31         value true(1) indicates that MMRP is enabled on this port
32         in all VLANs as long as ieee8021BridgeMmrpEnabledStatus is
33         also true(1). A value of false(2) indicates that MMRP is
34         disabled on this port in all VLANs: any MMRP packets received
35         will be silently discarded, and no MMRP registrations will be
36         propagated from other ports. Setting this to a value of
37         true(1) will be stored by the agent but will only take
38         effect on the MMRP protocol operation if
39         ieee8021BridgeMmrpEnabledStatus
40         also indicates the value true(1). This object affects
41         all MMRP Applicant and Registrar state machines on this
42         port. A transition from false(2) to true(1) will
43         cause a reset of all MMRP state machines on this port.
44
45         The value of this object MUST be retained across
46         reinitializations of the management system."
47     DEFVAL      { true }
48     ::= { ieee8021BridgePortMmrpEntry 1 }
49
50 Ieee8021BridgePortMmrpFailedRegistrations OBJECT-TYPE
51     SYNTAX      Counter64
52     UNITS       "failed MMRP registrations"
53     MAX-ACCESS  read-only
54     STATUS      current
55     DESCRIPTION
56         "The total number of failed MMRP registrations, for any
57         reason, in all VLANs, on this port."
58     ::= { ieee8021BridgePortMmrpEntry 2 }
59
60 Ieee8021BridgePortMmrpLastPduOrigin OBJECT-TYPE
61     SYNTAX      MacAddress
62     MAX-ACCESS  read-only
63     STATUS      current
64     DESCRIPTION
65         "The Source MAC Address of the last MMRP message
66         received on this port."
67     ::= { ieee8021BridgePortMmrpEntry 3 }
68
69 Ieee8021BridgePortRestrictedGroupRegistration OBJECT-TYPE
70     SYNTAX      TruthValue
71     MAX-ACCESS  read-write
72     STATUS      current

```

```

1  DESCRIPTION
2      "The state of Restricted Group Registration on this port.
3      If the value of this control is true(1), then creation
4      of a new dynamic entry is permitted only if there is a
5      Static Filtering Entry for the VLAN concerned, in which
6      the Registrar Administrative Control value is Normal
7      Registration.
8
9      The value of this object MUST be retained across
10     reinitializations of the management system."
11  REFERENCE    "11.2.3.2.3, 12.11.1.3"
12  DEFVAL      { false }
13  ::= { ieee8021BridgePortMmrpEntry 4 }
14
15 -- =====
16 -- I-LAN Interface configuration table
17 -- =====
18
19 ieee8021BridgeILanIfTable OBJECT-TYPE
20     SYNTAX      SEQUENCE OF Ieee8021BridgeILanIfEntry
21     MAX-ACCESS  not-accessible
22     STATUS      current
23     DESCRIPTION
24         "This table is a sparse augmentation of ifTable and controls
25         the creation of the I-LAN Interface. An I-LAN Interface is
26         used to create internal connections between Bridge Ports in a
27         802.1 device. An I-LAN Interfaces can be directly associated
28         with a set of Bridge Ports. An I-LAN Interfaces can also be
29         used as a stacking interface to relate other interfaces before
30         association to Bridge Ports.
31
32         For example, an I-LAN interface can be created to link traffic
33         between a PIP and a CBP. In this case a CBP is created on the
34         B-Component and the CBP's related IfEntry is stacked upon the
35         IfEntry of the I-LAN. The PIP is stacked upon the I-LAN using
36         the IfStackTable. Finally, a VIP is created on the I-Component
37         and is associated with the PIP, thus completing the path from
38         the I-Component's MAC relay to the CBP on the B-Component.
39
40         Entries in this table MUST be persistent over power up
41         restart/reboot."
42     REFERENCE    "17.3.2.2"
43     ::= { ieee8021BridgeInternalLan 1 }
44
45 ieee8021BridgeILanIfEntry OBJECT-TYPE
46     SYNTAX      Ieee8021BridgeILanIfEntry
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "Each entry consists of a Row Status to control creation."
51     INDEX       { ifIndex }
52     ::= { ieee8021BridgeILanIfTable 1 }
53
54 Ieee8021BridgeILanIfEntry ::=
55     SEQUENCE {
56         ieee8021BridgeILanIfRowStatus
57         RowStatus
58     }
59
60 ieee8021BridgeILanIfRowStatus OBJECT-TYPE
61     SYNTAX      RowStatus
62     MAX-ACCESS  read-create
63     STATUS      current
64     DESCRIPTION
65         "This object is used to create and delete entries in this
66         table and the Interface table."
67     ::= { ieee8021BridgeILanIfEntry 1 }
68
69 -- =====
70 -- Dynamic Port Creation table
71 -- =====
72

```

```

1 ieee8021BridgeDot1dPortTable OBJECT-TYPE
2     SYNTAX      SEQUENCE OF Ieee8021BridgeDot1dPortEntry
3     MAX-ACCESS  not-accessible
4     STATUS      current
5     DESCRIPTION
6         "This table provides the capability to dynamically create and
7         delete MAC Bridge Ports. Each entry in this table MUST
8         have a corresponding entry in the ieee8021BridgeBasePortTable.
9
10        Entries in this table MUST be persistent over power up
11        restart/reboot."
12     REFERENCE   "17.5.3"
13     ::= { ieee8021BridgeDot1d 1 }
14
15 ieee8021BridgeDot1dPortEntry OBJECT-TYPE
16     SYNTAX      Ieee8021BridgeDot1dPortEntry
17     MAX-ACCESS  not-accessible
18     STATUS      current
19     DESCRIPTION
20         "Each entry consists of a Row Status to control creation."
21     INDEX { ieee8021BridgeBasePortComponentId,
22             ieee8021BridgeBasePort }
23     ::= { ieee8021BridgeDot1dPortTable 1 }
24
25 Ieee8021BridgeDot1dPortEntry ::=
26     SEQUENCE {
27         ieee8021BridgeDot1dPortRowStatus
28         RowStatus
29     }
30
31 ieee8021BridgeDot1dPortRowStatus OBJECT-TYPE
32     SYNTAX      RowStatus
33     MAX-ACCESS  read-create
34     STATUS      current
35     DESCRIPTION
36         "This object is used to create and delete entries in this
37         table and the ieee8021BridgeBasePortTable."
38     ::= { ieee8021BridgeDot1dPortEntry 1 }
39
40
41 -- =====
42 -- IEEE 802.1Q Bridge MIB - Conformance Information
43 -- =====
44
45 ieee8021BridgeCompliances
46     OBJECT IDENTIFIER ::= { ieee8021BridgeConformance 1 }
47 ieee8021BridgeGroups
48     OBJECT IDENTIFIER ::= { ieee8021BridgeConformance 2 }
49
50 -- =====
51 -- units of conformance
52 -- =====
53
54 -- =====
55 -- the ieee8021BridgeBase group
56 -- =====
57
58 ieee8021BridgeBaseBridgeGroup OBJECT-GROUP
59     OBJECTS {
60         ieee8021BridgeBaseBridgeAddress,
61         ieee8021BridgeBaseNumPorts,
62         ieee8021BridgeBaseComponentType
63     }
64     STATUS      current
65     DESCRIPTION
66         "Bridge level information for this device."
67     ::= { ieee8021BridgeGroups 1 }
68
69 ieee8021BridgeBasePortGroup OBJECT-GROUP
70     OBJECTS {
71         ieee8021BridgeBasePortIfIndex,
72         ieee8021BridgeBasePortDelayExceededDiscards,

```

```

1      ieee8021BridgeBasePortMtuExceededDiscards,
2      ieee8021BridgeBasePortType,
3      ieee8021BridgeBasePortExternal,
4      ieee8021BridgeBasePortAdminPointToPoint,
5      ieee8021BridgeBasePortOperPointToPoint,
6      ieee8021BridgeBasePortName
7  }
8  STATUS      current
9  DESCRIPTION
10     "Information for each port on this device."
11     ::= { ieee8021BridgeGroups 2 }
12
13 ieee8021BridgeCapGroup OBJECT-GROUP
14     OBJECTS {
15         ieee8021BridgeBaseDeviceCapabilities,
16         ieee8021BridgeBasePortCapabilities,
17         ieee8021BridgeBasePortTypeCapabilities
18     }
19     STATUS      current
20     DESCRIPTION
21         "A collection of objects indicating the optional
22         capabilities of the device."
23     ::= { ieee8021BridgeGroups 3 }
24
25 ieee8021BridgeDeviceMmrpGroup OBJECT-GROUP
26     OBJECTS {
27         ieee8021BridgeBaseMmrpEnabledStatus
28     }
29     STATUS      current
30     DESCRIPTION
31         "A collection of objects providing device-level control
32         for the Multicast Filtering extended Bridge services."
33     ::= { ieee8021BridgeGroups 4 }
34
35 -- =====
36 -- the ieee8021BridgeTp group
37 -- =====
38
39 ieee8021BridgeTpPortGroup OBJECT-GROUP
40     OBJECTS {
41         ieee8021BridgeTpPortMaxInfo,
42         ieee8021BridgeTpPortInFrames,
43         ieee8021BridgeTpPortOutFrames,
44         ieee8021BridgeTpPortInDiscards
45     }
46     STATUS      current
47     DESCRIPTION
48         "Dynamic Filtering Database information for each port of
49         the Bridge."
50     ::= { ieee8021BridgeGroups 6 }
51
52 -- =====
53 -- Bridge Priority groups
54 -- =====
55
56 ieee8021BridgeDevicePriorityGroup OBJECT-GROUP
57     OBJECTS {
58         ieee8021BridgeBaseTrafficClassesEnabled
59     }
60     STATUS      current
61     DESCRIPTION
62         "A collection of objects providing device-level control
63         for the Priority services."
64     ::= { ieee8021BridgeGroups 7 }
65
66 ieee8021BridgeDefaultPriorityGroup OBJECT-GROUP
67     OBJECTS {
68         ieee8021BridgePortDefaultUserPriority,
69         ieee8021BridgePortPriorityCodePointSelection,
70         ieee8021BridgePortUseDEI,
71         ieee8021BridgePortRequireDropEncoding,
72         ieee8021BridgePortServiceAccessPrioritySelection

```

```

1      }
2      STATUS      current
3      DESCRIPTION
4          "A collection of objects defining the priority
5          applicable to each port for media that do not support
6          native priority."
7      ::= { ieee8021BridgeGroups 8 }
8
9  ieee8021BridgeRegenPriorityGroup OBJECT-GROUP
10     OBJECTS {
11         ieee8021BridgeRegenUserPriority
12     }
13     STATUS      current
14     DESCRIPTION
15         "A collection of objects defining the User Priorities
16         applicable to each port for media that support native
17         priority."
18     ::= { ieee8021BridgeGroups 9 }
19
20  ieee8021BridgePriorityGroup OBJECT-GROUP
21     OBJECTS {
22         ieee8021BridgePortNumTrafficClasses,
23         ieee8021BridgeTrafficClass
24     }
25     STATUS      current
26     DESCRIPTION
27         "A collection of objects defining the traffic classes
28         within a Bridge for each evaluated priority."
29     ::= { ieee8021BridgeGroups 10 }
30
31  ieee8021BridgeAccessPriorityGroup OBJECT-GROUP
32     OBJECTS {
33         ieee8021BridgePortOutboundAccessPriority
34     }
35     STATUS      current
36     DESCRIPTION
37         "A collection of objects defining the media-dependent
38         outbound access level for each priority."
39     ::= { ieee8021BridgeGroups 11 }
40
41  ieee8021BridgePortMrpGroup OBJECT-GROUP
42     OBJECTS {
43         ieee8021BridgePortMrpJoinTime,
44         ieee8021BridgePortMrpLeaveTime,
45         ieee8021BridgePortMrpLeaveAllTime
46     }
47     STATUS      current
48     DESCRIPTION
49         "A collection of objects providing port level control
50         and status information for MRP operation."
51     ::= { ieee8021BridgeGroups 12 }
52
53  ieee8021BridgePortMmrpGroup OBJECT-GROUP
54     OBJECTS {
55         ieee8021BridgePortMmrpEnabledStatus,
56         ieee8021BridgePortMmrpFailedRegistrations,
57         ieee8021BridgePortMmrpLastPduOrigin,
58         ieee8021BridgePortRestrictedGroupRegistration
59     }
60     STATUS      deprecated
61     DESCRIPTION
62         "A collection of objects providing port level control
63         and status information for MMRP operation."
64     ::= { ieee8021BridgeGroups 13 }
65
66  ieee8021BridgePortDecodingGroup OBJECT-GROUP
67     OBJECTS {
68         ieee8021BridgePortDecodingPriority,
69         ieee8021BridgePortDecodingDropEligible
70     }
71     STATUS      current
72     DESCRIPTION

```

```

1      "A collection of objects providing statistics counters for
2      decoding priority and drop eligibility for Bridge Ports."
3      ::= { ieee8021BridgeGroups 14 }
4
5 ieee8021BridgePortEncodingGroup OBJECT-GROUP
6     OBJECTS {
7         ieee8021BridgePortEncodingPriority
8     }
9     STATUS      current
10    DESCRIPTION
11    "A collection of objects providing statistics counters for
12    encoding priority and drop eligibility for Bridge Ports."
13    ::= { ieee8021BridgeGroups 15 }
14
15 ieee8021BridgeServiceAccessPriorityGroup OBJECT-GROUP
16     OBJECTS {
17         ieee8021BridgeServiceAccessPriorityValue
18     }
19     STATUS      current
20     DESCRIPTION
21     "A collection of objects providing statistics
22     counters for service access priority."
23     ::= { ieee8021BridgeGroups 16 }
24
25 -- =====
26 -- Internal LAN group
27 -- =====
28
29 ieee8021BridgeInternalLANGroup OBJECT-GROUP
30     OBJECTS {
31         ieee8021BridgeILanIfRowStatus
32     }
33     STATUS      current
34     DESCRIPTION
35     "A collection of objects providing control of internal
36     LAN configuration."
37     ::= { ieee8021BridgeGroups 17 }
38
39 -- =====
40 -- Bridge Creation Group
41 -- =====
42
43 ieee8021BridgeCreatableBaseBridgeGroup OBJECT-GROUP
44     OBJECTS {
45         ieee8021BridgeBaseRowStatus
46     }
47     STATUS      current
48     DESCRIPTION
49     "Controls the management system directed creation of
50     Bridge Components."
51     ::= { ieee8021BridgeGroups 18 }
52
53 -- =====
54 -- Dot1d Dynamic Port Creation group
55 -- =====
56
57 ieee8021BridgeDot1dDynamicPortCreationGroup OBJECT-GROUP
58     OBJECTS {
59         ieee8021BridgeDot1dPortRowStatus
60     }
61     STATUS      current
62     DESCRIPTION
63     "A collection of objects providing dynamic creation and
64     deletion of MAC Bridge Ports."
65     ::= { ieee8021BridgeGroups 19 }
66
67
68 -- =====
69 -- Bridge interface index to port table group
70 -- =====
71
72 ieee8021BridgeBaseIfToPortGroup OBJECT-GROUP

```

```

1  OBJECTS {
2  ieee8021BridgeBaseIfIndexComponentId,
3  ieee8021BridgeBaseIfIndexPort
4
5  }
6  STATUS      current
7  DESCRIPTION
8      "A collection of objects providing a map between interface
9      index and component ID and Bridge Ports."
10 ::= { ieee8021BridgeGroups 20 }
11
12
13
14 -- =====
15 -- Bridge interface index to component group
16 -- =====
17 ieee8021BridgePhyPortGroup  OBJECT-GROUP
18 OBJECTS {
19     ieee8021BridgePhyPortIfIndex,
20     ieee8021BridgePhyMacAddress,
21     ieee8021BridgePhyPortToComponentId,
22     ieee8021BridgePhyPortToInternalPort
23 }
24 STATUS      current
25 DESCRIPTION
26 "The collection of objects used to represent a ISS port management objects."
27 ::= { ieee8021BridgeGroups 21 }
28
29
30 -- =====
31 -- compliance statements
32 -- =====
33
34 ieee8021BridgeCompliance1 MODULE-COMPLIANCE
35 STATUS      current
36 DESCRIPTION
37     "The compliance statement for devices supporting
38     VLAN-unaware bridging services as defined in
39     IEEE Std 802.1Q. Such devices support
40     path cost values of 32-bits, and Bridge and port priority
41     values are more restricted than in IEEE Std 802.1D-1995.
42
43     Full support for the IEEE 802.1Q management objects requires
44     implementation of the objects listed in the systemGroup
45     from the SNMPv2-MIB [RFC3418], as well as the objects
46     listed in the ifGeneralInformationGroup from the
47     IF-MIB [RFC2863]."
48
49 MODULE SNMPv2-MIB -- The SNMPv2-MIB, RFC 3418
50     MANDATORY-GROUPS {
51         systemGroup
52     }
53
54 MODULE IF-MIB -- The interfaces MIB, RFC 2863
55     MANDATORY-GROUPS {
56         ifGeneralInformationGroup
57     }
58
59 MODULE
60     MANDATORY-GROUPS {
61         ieee8021BridgeBaseBridgeGroup,
62         ieee8021BridgeBasePortGroup
63     }
64
65 GROUP ieee8021BridgeCreatableBaseBridgeGroup
66 DESCRIPTION
67     "Implementation of this group is mandatory for
68     Bridges that allow management systems to add and delete
69     Bridge components. Provider Backbone Edge Bridges would
70     typically fall in this category."
71
72 GROUP ieee8021BridgeTpPortGroup

```



```
1 DESCRIPTION
2     "Implementation of this group is mandatory for
3     Bridges that support the transparent bridging
4     mode. A transparent Bridge will implement
5     this group."
6
7 GROUP ieee8021BridgeInternalLanGroup
8 DESCRIPTION
9     "Implementation of this group is optional. It can be supported
10    to provide control over the relationship between interfaces and
11    Bridge Ports where such relationships are more complex than a
12    simple 1-to-1 mapping."
13
14 GROUP ieee8021BridgeDot1dDynamicPortCreationGroup
15 DESCRIPTION
16     "Implementation of this group is optional. It can be supported
17    to provide the ability to dynamically create and delete
18    MAC Bridge Ports."
19
20 GROUP ieee8021BridgeBaseIfToPortGroup
21 DESCRIPTION
22     "A collection of objects providing a map between interface
23    index and component ID and Bridge Ports."
24 GROUP ieee8021BridgePhyPortGroup
25 DESCRIPTION
26     "A collection of objects providing a map between port numbers
27    to the component id, interface index."
28
29 ::= { ieee8021BridgeCompliances 3 }
30
31
32 ieee8021BridgeCompliance MODULE-COMPLIANCE
33     STATUS current
34     DESCRIPTION
35         "The compliance statement for devices supporting
36         VLAN-unaware bridging services as defined in
37         IEEE Std 802.1Q. Such devices support
38         path cost values of 32-bits, and Bridge and port priority
39         values are more restricted than in IEEE Std 802.1D-1995.
40
41         Full support for the IEEE 802.1Q management objects requires
42         implementation of the objects listed in the systemGroup
43         from the SNMPv2-MIB [RFC3418], as well as the objects
44         listed in the ifGeneralInformationGroup from the
45         IF-MIB [RFC2863]."
46
47 MODULE SNMPv2-MIB -- The SNMPv2-MIB, RFC 3418
48     MANDATORY-GROUPS {
49         systemGroup
50     }
51
52 MODULE IF-MIB -- The interfaces MIB, RFC 2863
53     MANDATORY-GROUPS {
54         ifGeneralInformationGroup
55     }
56
57 MODULE
58     MANDATORY-GROUPS {
59         ieee8021BridgeBaseBridgeGroup,
60         ieee8021BridgeBasePortGroup
61     }
62
63 GROUP ieee8021BridgeCreatableBaseBridgeGroup
64 DESCRIPTION
65     "Implementation of this group is mandatory for
66     Bridges that allow management systems to add and delete
67     Bridge components. Provider Backbone Edge Bridges would
68     typically fall in this category."
69
70 GROUP ieee8021BridgeTpPortGroup
71 DESCRIPTION
72     "Implementation of this group is mandatory for
```

```
1      Bridges that support the transparent bridging
2      mode. A transparent Bridge will implement
3      this group."
4
5      GROUP      ieee8021BridgeInternalLANGroup
6      DESCRIPTION
7          "Implementation of this group is optional. It can be supported
8          to provide control over the relationship between interfaces and
9          Bridge Ports where such relationships are more complex than a
10         simple 1-to-1 mapping."
11
12     GROUP      ieee8021BridgeDot1dDynamicPortCreationGroup
13     DESCRIPTION
14         "Implementation of this group is optional. It can be supported
15         to provide the ability to dynamically create and delete
16         Bridge Ports."
17
18     ::= { ieee8021BridgeCompliances 1 }
19
20 ieee8021BridgePriorityAndMulticastFilteringCompliance MODULE-COMPLIANCE
21     STATUS deprecated
22     DESCRIPTION
23         "The compliance statement for device support of Priority
24         and Multicast Filtering extended bridging services."
25
26     MODULE
27         MANDATORY-GROUPS { ieee8021BridgeCapGroup }
28
29         GROUP      ieee8021BridgeDeviceMmrpGroup
30         DESCRIPTION
31             "This group is mandatory for devices supporting the MMRP
32             application, defined by IEEE 802.1Q Extended Filtering
33             Services."
34
35         GROUP      ieee8021BridgeDevicePriorityGroup
36         DESCRIPTION
37             "This group is mandatory only for devices supporting
38             the priority forwarding operations defined by IEEE
39             802.1Q."
40
41         GROUP      ieee8021BridgeDefaultPriorityGroup
42         DESCRIPTION
43             "This group is mandatory only for devices supporting
44             the priority forwarding operations defined by the
45             extended Bridge services with media types, such as
46             Ethernet, that do not support native priority."
47
48         GROUP      ieee8021BridgeRegenPriorityGroup
49         DESCRIPTION
50             "This group is mandatory only for devices supporting
51             the priority forwarding operations defined by IEEE Std
52             802.1Q and that have interface media types that support
53             native priority."
54
55         GROUP      ieee8021BridgePriorityGroup
56         DESCRIPTION
57             "This group is mandatory only for devices supporting
58             the priority forwarding operations defined by
59             IEEE Std 802.1Q."
60
61         GROUP      ieee8021BridgeAccessPriorityGroup
62         DESCRIPTION
63             "This group is optional and is relevant only for devices
64             supporting the priority forwarding operations defined by
65             IEEE Std 802.1Q and that have interface media types that
66             support native Access Priority."
67
68         GROUP      ieee8021BridgePortMrpGroup
69         DESCRIPTION
70             "This group is mandatory for devices supporting any
71             of the MRP applications: e.g., MMRP, defined by the
72             extended filtering services; or MVRP.
```

```
1       Refer to the Q-BRIDGE-MIB for
2       conformance statements for MVRP."
3
4       GROUP      ieee8021BridgePortMmrpGroup
5       DESCRIPTION
6       "This group is mandatory for devices supporting the
7       MMRP application, as defined by IEEE 802.1Q Extended
8       Filtering Services."
9
10      GROUP      ieee8021BridgePortDecodingGroup
11      DESCRIPTION
12      "This group is optional and supports Priority Code Point
13      Decoding Table for a Port of a provider Bridge."
14
15      GROUP      ieee8021BridgePortEncodingGroup
16      DESCRIPTION
17      "This group is optional and supports Priority Code Point
18      Encoding Table for a Port of a provider Bridge."
19
20      GROUP      ieee8021BridgeServiceAccessPriorityGroup
21      DESCRIPTION
22      "This group is optional and supports Priority Code Point
23      Encoding Table for a Port of a provider Bridge."
24
25      OBJECT      ieee8021BridgePortNumTrafficClasses
26      MIN-ACCESS  read-only
27      DESCRIPTION
28      "Write access is not required."
29
30      OBJECT      ieee8021BridgeTrafficClass
31      MIN-ACCESS  read-only
32      DESCRIPTION
33      "Write access is not required."
34
35      OBJECT      ieee8021BridgeRegenUserPriority
36      MIN-ACCESS  read-only
37      DESCRIPTION
38      "Write access is not required."
39
40      ::= { ieee8021BridgeCompliances 2 }
41
42 END
```

1 17.7.3 Definitions for the IEEE8021-SPANNING-TREE-MIB module

```
2 IEEE8021-SPANNING-TREE-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for IEEE 802.1Q spanning tree devices
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE, NOTIFICATION-TYPE,
10     Counter64, Integer32, TimeTicks
11     FROM SNMPv2-SMI
12     MODULE-COMPLIANCE, OBJECT-GROUP, NOTIFICATION-GROUP
13     FROM SNMPv2-CONF
14     TruthValue
15     FROM SNMPv2-TC
16     ieee802dot1mibs, IEEE8021PbbComponentIdentifier,
17     IEEE8021BridgePortNumber
18     FROM IEEE8021-TC-MIB
19     BridgeId, Timeout
20     FROM BRIDGE-MIB
21 ;
22
23 ieee8021SpanningTreeMib MODULE-IDENTITY
24     LAST-UPDATED "202211080000Z" -- November 8, 2022
25     ORGANIZATION "IEEE 802.1 Working Group"
26     CONTACT-INFO
27         " WG-URL: http://www.ieee802.org/1/
28         WG-EMail: stds-802-1-1@ieee.org
29         Contact: IEEE 802.1 Working Group Chair
30         Postal: C/O IEEE 802.1 Working Group
31               IEEE Standards Association
32               445 Hoes Lane
33               Piscataway, NJ 08854
34               USA
35         E-mail: stds-802-1-chairs@ieee.org"
36     DESCRIPTION
37         "The Spanning-Tree MIB module for managing devices that
38         support IEEE 802.1Q.
39
40         Unless otherwise indicated, the references in this MIB
41         module are to IEEE Std 802.1Q-2022.
42
43         Copyright (C) IEEE (2022).
44         This version of this MIB module is part of IEEE Std 802.1Q;
45         see that standard for full legal notices."
46
47     REVISION "202211080000Z" -- November 8, 2022
48     DESCRIPTION
49         "Published as part of IEEE Std 802.1Q-2022.
50         Cross references and contact information updated."
51
52     REVISION "201807010000Z" -- July 1, 2018
53     DESCRIPTION
54         "Published as part of IEEE Std 802.1Q 2018 revision.
55         Cross references updated and corrected."
56
57     REVISION "201412150000Z" -- December 15, 2014
58     DESCRIPTION
59         "Published as part of IEEE Std 802.1Q 2014 revision.
60         Cross references updated and corrected.
61         ieee8021SpanningTreeRstpTxHoldCount default value
62         changed to 6 to match Table 13-5."
63
64     REVISION "201103240000Z" -- March 24, 2011
65     DESCRIPTION
66         "Minor edits to contact information and addition of
67         fragile Bridge as part of 2011 revision of
68         IEEE Std 802.1Q."
69
70     REVISION "200810150000Z" -- October 15, 2008
71     DESCRIPTION
```

```

1      "Initial revision, derived from RFC 4188."
2      ::= { ieee802dot1mibs 3 }
3
4  -- =====
5  -- subtrees in the Spanning-Tree MIB
6  -- =====
7
8  ieee8021SpanningTreeNotifications
9      OBJECT IDENTIFIER ::= { ieee8021SpanningTreeMib 0 }
10
11 ieee8021SpanningTreeObjects
12     OBJECT IDENTIFIER ::= { ieee8021SpanningTreeMib 1 }
13
14 ieee8021SpanningTreeConformance
15     OBJECT IDENTIFIER ::= { ieee8021SpanningTreeMib 2 }
16
17
18  -- =====
19  -- the ieee8021SpanningTreeTable
20  -- =====
21 ieee8021SpanningTreeTable OBJECT-TYPE
22     SYNTAX      SEQUENCE OF Ieee8021SpanningTreeEntry
23     MAX-ACCESS  not-accessible
24     STATUS      current
25     DESCRIPTION
26         "A table that contains information related to STP about
27         every Bridge."
28     REFERENCE   "12.8.1"
29     ::= { ieee8021SpanningTreeObjects 1 }
30
31 ieee8021SpanningTreeEntry OBJECT-TYPE
32     SYNTAX      Ieee8021SpanningTreeEntry
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "A list of objects containing information for each Bridge
37         about the Spanning Tree Protocol for that Bridge."
38     INDEX      { ieee8021SpanningTreeComponentId }
39     ::= { ieee8021SpanningTreeTable 1 }
40
41 Ieee8021SpanningTreeEntry ::=
42     SEQUENCE {
43         ieee8021SpanningTreeComponentId
44         IEEE8021PbbComponentIdentifier,
45         ieee8021SpanningTreeProtocolSpecification
46         INTEGER,
47         ieee8021SpanningTreePriority
48         Integer32,
49         ieee8021SpanningTreeTimeSinceTopologyChange
50         TimeTicks,
51         ieee8021SpanningTreeTopChanges
52         Counter64,
53         ieee8021SpanningTreeDesignatedRoot
54         BridgeId,
55         ieee8021SpanningTreeRootCost
56         Integer32,
57         ieee8021SpanningTreeRootPort
58         IEEE8021BridgePortNumber,
59         ieee8021SpanningTreeMaxAge
60         Timeout,
61         ieee8021SpanningTreeHelloTime
62         Timeout,
63         ieee8021SpanningTreeHoldTime
64         Integer32,
65         ieee8021SpanningTreeForwardDelay
66         Timeout,
67         ieee8021SpanningTreeBridgeMaxAge
68         Timeout,
69         ieee8021SpanningTreeBridgeHelloTime
70         Timeout,
71         ieee8021SpanningTreeBridgeForwardDelay
72         Timeout,

```

```

1      ieee8021SpanningTreeVersion
2          INTEGER,
3      ieee8021SpanningTreeRstpTxHoldCount
4          Integer32
5  }
6
7 ieee8021SpanningTreeComponentId OBJECT-TYPE
8     SYNTAX      IEEE8021PbbComponentIdentifier
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "The component identifier is used to distinguish between the
13        multiple virtual Bridge instances within a PBB. In simple
14        situations where there is only a single component the default
15        value is 1."
16    ::= { ieee8021SpanningTreeEntry 1 }
17
18 ieee8021SpanningTreeProtocolSpecification OBJECT-TYPE
19     SYNTAX      INTEGER {
20         unknown(1),
21         decLb100(2),
22         ieee8021d(3),
23         ieee8021q(4)
24     }
25     MAX-ACCESS  read-only
26     STATUS      current
27     DESCRIPTION
28         "An indication of what version of the Spanning Tree Protocol is
29         being run. The value 'decLb100(2)' indicates the DEC LANbridge
30         100 Spanning Tree protocol. IEEE 802.1D implementations will
31         return 'ieee8021d(3)'. New enumerated values may be added in
32         the future to the definition of this object to reflect future
33         versions of the IEEE Spanning Tree protocol."
34     ::= { ieee8021SpanningTreeEntry 2 }
35
36 ieee8021SpanningTreePriority OBJECT-TYPE
37     SYNTAX      Integer32 (0..65535)
38     MAX-ACCESS  read-write
39     STATUS      current
40     DESCRIPTION
41         "The value of the write-able portion of the Bridge ID
42         (i.e., the first two octets of the (8 octet long) Bridge
43         ID). The other (last) 6 octets of the Bridge ID are
44         given by the value of ieee8021BridgeBaseBridgeAddress.
45         On Bridges supporting IEEE 802.1t or IEEE 802.1w,
46         permissible values are 0-61440, in steps of 4096.
47
48         The value of this object MUST be retained across
49         reinitializations of the management system."
50     REFERENCE  "12.8.1.1.3 a)"
51     ::= { ieee8021SpanningTreeEntry 3 }
52
53 ieee8021SpanningTreeTimeSinceTopologyChange OBJECT-TYPE
54     SYNTAX      TimeTicks
55     UNITS       "centi-seconds"
56     MAX-ACCESS  read-only
57     STATUS      current
58     DESCRIPTION
59         "The time (in hundredths of a second) since the
60         last time a topology change was detected by the
61         Bridge entity.
62         For RSTP, this reports the time since the tcWhile
63         timer for any port on this Bridge was nonzero."
64     REFERENCE  "12.8.1.1.3 b)"
65     ::= { ieee8021SpanningTreeEntry 4 }
66
67 ieee8021SpanningTreeTopChanges OBJECT-TYPE
68     SYNTAX      Counter64
69     UNITS       "topology changes"
70     MAX-ACCESS  read-only
71     STATUS      current
72     DESCRIPTION

```

```

1      "The total number of topology changes detected by
2      this Bridge since the management entity was last
3      reset or initialized.
4
5      Discontinuities in the value of the counter can occur
6      at re-initialization of the management system."
7  REFERENCE   "12.8.1.1.3 c)"
8  ::= { ieee8021SpanningTreeEntry 5 }
9
10 ieee8021SpanningTreeDesignatedRoot OBJECT-TYPE
11     SYNTAX      BridgeId
12     MAX-ACCESS  read-only
13     STATUS      current
14     DESCRIPTION
15         "The Bridge identifier of the root of the spanning
16         tree, as determined by the Spanning Tree Protocol,
17         as executed by this node. This value is used as
18         the Root Identifier parameter in all Configuration
19         Bridge PDUs originated by this node."
20     REFERENCE   "12.8.1.1.3 e)"
21     ::= { ieee8021SpanningTreeEntry 6 }
22
23 ieee8021SpanningTreeRootCost OBJECT-TYPE
24     SYNTAX      Integer32
25     MAX-ACCESS  read-only
26     STATUS      current
27     DESCRIPTION
28         "The cost of the path to the root as seen from
29         this Bridge."
30     REFERENCE   "12.8.1.1.3 f)"
31     ::= { ieee8021SpanningTreeEntry 7 }
32
33 ieee8021SpanningTreeRootPort OBJECT-TYPE
34     SYNTAX      IEEE8021BridgePortNumber
35     MAX-ACCESS  read-only
36     STATUS      current
37     DESCRIPTION
38         "The port number of the port that offers the lowest
39         cost path from this Bridge to the root Bridge."
40     REFERENCE   "12.8.1.1.3 g)"
41     ::= { ieee8021SpanningTreeEntry 8 }
42
43 ieee8021SpanningTreeMaxAge OBJECT-TYPE
44     SYNTAX      Timeout
45     UNITS        "centi-seconds"
46     MAX-ACCESS  read-only
47     STATUS      current
48     DESCRIPTION
49         "The maximum age of Spanning Tree Protocol information
50         learned from the network on any port before it is
51         discarded, in units of hundredths of a second. This is
52         the actual value that this Bridge is currently using."
53     REFERENCE   "12.8.1.1.3 h)"
54     ::= { ieee8021SpanningTreeEntry 9 }
55
56 ieee8021SpanningTreeHelloTime OBJECT-TYPE
57     SYNTAX      Timeout
58     UNITS        "centi-seconds"
59     MAX-ACCESS  read-only
60     STATUS      current
61     DESCRIPTION
62         "The amount of time between the transmission of
63         Configuration Bridge PDUs by this node on any port when
64         it is the root of the spanning tree, or trying to become
65         so, in units of hundredths of a second. This is the
66         actual value that this Bridge is currently using."
67     REFERENCE   "12.8.1.1.3 k)"
68     ::= { ieee8021SpanningTreeEntry 10 }
69
70 ieee8021SpanningTreeHoldTime OBJECT-TYPE
71     SYNTAX      Integer32
72     UNITS        "centi-seconds"

```

```

1  MAX-ACCESS    read-only
2  STATUS        current
3  DESCRIPTION
4      "This time value determines the interval length
5      during which no more than two Configuration Bridge
6      PDUs shall be transmitted by this node, in units
7      of hundredths of a second."
8  REFERENCE     "12.8.1.1.3 m)"
9  ::= { ieee8021SpanningTreeEntry 11 }
10
11 ieee8021SpanningTreeForwardDelay OBJECT-TYPE
12     SYNTAX      Timeout
13     UNITS        "centi-seconds"
14     MAX-ACCESS   read-only
15     STATUS        current
16     DESCRIPTION
17         "This time value, measured in units of hundredths of a
18         second, controls how fast a port changes its spanning
19         state when moving towards the Forwarding state. The
20         value determines how long the port stays in each of the
21         Listening and Learning states, which precede the
22         Forwarding state. This value is also used when a
23         topology change has been detected and is underway, to
24         age all dynamic entries in the Filtering Database.
25         [Note that this value is the one that this Bridge is
26         currently using, in contrast to
27         ieee8021SpanningTreeBridgeForwardDelay, which is the
28         value that this Bridge and all others would start using
29         if/when this Bridge were to become the root.]"
30     REFERENCE     "12.8.1.1.3 i)"
31     ::= { ieee8021SpanningTreeEntry 12 }
32
33 ieee8021SpanningTreeBridgeMaxAge OBJECT-TYPE
34     SYNTAX      Timeout (600..4000)
35     UNITS        "centi-seconds"
36     MAX-ACCESS   read-write
37     STATUS        current
38     DESCRIPTION
39         "The value that all Bridges use for MaxAge when this
40         Bridge is acting as the root. Note that IEEE Std 802.1D-1998
41         specifies that the range for this parameter is related
42         to the value of ieee8021SpanningTreeBridgeHelloTime. The
43         granularity of this timer is specified by IEEE Std 802.1D-1998
44         to be 1 second. An agent may return an SNMP badValue error
45         (or its equivalent if another protocol is used) if a set
46         is attempted to a value that is not a whole number of
47         seconds.
48
49         The value of this object MUST be retained across
50         reinitializations of the management system."
51     REFERENCE     "12.8.1.1.3 j)"
52     ::= { ieee8021SpanningTreeEntry 13 }
53
54 ieee8021SpanningTreeBridgeHelloTime OBJECT-TYPE
55     SYNTAX      Timeout (100..1000)
56     UNITS        "centi-seconds"
57     MAX-ACCESS   read-write
58     STATUS        current
59     DESCRIPTION
60         "The value that all Bridges use for HelloTime when this
61         Bridge is acting as the root. The granularity of this
62         timer is specified by IEEE Std 802.1D-1998 to be 1 second.
63         An agent may return an SNMP badValue error (or its equivalent
64         if another protocol is used) if a set is attempted to a
65         value that is not a whole number of seconds.
66
67         The value of this object MUST be retained across
68         reinitializations of the management system."
69     REFERENCE     "12.8.1.1.3 k)"
70     ::= { ieee8021SpanningTreeEntry 14 }
71
72 ieee8021SpanningTreeBridgeForwardDelay OBJECT-TYPE

```



```

1  SYNTAX      Timeout (400..3000)
2  UNITS       "centi-seconds"
3  MAX-ACCESS  read-write
4  STATUS      current
5  DESCRIPTION
6      "The value that all Bridges use for ForwardDelay when
7      this Bridge is acting as the root. Note that IEEE Std
8      802.1D-1998 specifies that the range for this parameter
9      is related to the value of ieee8021SpanningTreeBridgeMaxAge.
10     The granularity of this timer is specified by IEEE Std 802.1D-1998
11     to be 1 second. An agent may return an SNMP badValue error
12     (or its equivalent if another protocol is used) if a set is
13     attempted to a value that is not a whole number of seconds.
14
15     The value of this object MUST be retained across
16     reinitializations of the management system."
17  REFERENCE   "12.8.1.1.3 l)"
18  ::= { ieee8021SpanningTreeEntry 15 }
19
20 ieee8021SpanningTreeVersion OBJECT-TYPE
21  SYNTAX      INTEGER {
22              stp(0),
23              rstp(2),
24              mstp(3)
25          }
26  MAX-ACCESS  read-write
27  STATUS      current
28  DESCRIPTION
29      "The version of Spanning Tree Protocol the Bridge is
30      currently running. The values are directly from
31      the IEEE standard. New values may be defined as future
32      versions of the protocol become available.
33
34      The value 'stp(0)' indicates the Bridge is running the
35      Spanning Tree Protocol specified in IEEE Std 802.1D-1998.
36
37      The value 'rstp(2)' indicates the Bridge is running RSTP
38      specified in IEEE 802.1Q.
39
40      The value 'mstp(3)' indicates the Bridge is running
41      MSTP specified in Clause 13 of IEEE Std 802.1Q.
42
43      The value of this object MUST be retained across
44      reinitializations of the management system."
45  REFERENCE   "12.8.1.1.3 n)"
46  ::= { ieee8021SpanningTreeEntry 16 }
47
48 ieee8021SpanningTreeRstpTxHoldCount OBJECT-TYPE
49  SYNTAX      Integer32 (1..10)
50  MAX-ACCESS  read-write
51  STATUS      current
52  DESCRIPTION
53      "The value used by the Port Transmit state machine to limit
54      the maximum transmission rate. This is used by Bridges
55      that are running RSTP.
56
57      The value of this object MUST be retained across
58      reinitializations of the management system."
59  REFERENCE   "12.8.1.1.3 m), 13.26.12"
60  DEFVAL      { 6 }
61  ::= { ieee8021SpanningTreeEntry 17 }
62
63 -- =====
64 -- The Spanning Tree Port Table
65 -- =====
66
67 ieee8021SpanningTreePortTable OBJECT-TYPE
68  SYNTAX      SEQUENCE OF Ieee8021SpanningTreePortEntry
69  MAX-ACCESS  not-accessible
70  STATUS      current
71  DESCRIPTION
72      "A table that contains port-specific information

```

```

1      for the Spanning Tree Protocol."
2  REFERENCE    "12.8.2"
3  ::= { ieee8021SpanningTreeObjects 2 }
4
5  ieee8021SpanningTreePortEntry OBJECT-TYPE
6  SYNTAX      Ieee8021SpanningTreePortEntry
7  MAX-ACCESS  not-accessible
8  STATUS      current
9  DESCRIPTION
10     "A list of objects containing information maintained by
11     every port about the Spanning Tree Protocol state for
12     that port."
13  INDEX      { ieee8021SpanningTreePortComponentId,
14              ieee8021SpanningTreePort }
15  ::= { ieee8021SpanningTreePortTable 1 }
16
17 Ieee8021SpanningTreePortEntry ::=
18     SEQUENCE {
19         ieee8021SpanningTreePortComponentId
20             IEEE8021PbbComponentIdentifier,
21         ieee8021SpanningTreePort
22             IEEE8021BridgePortNumber,
23         ieee8021SpanningTreePortPriority
24             Integer32,
25         ieee8021SpanningTreePortState
26             INTEGER,
27         ieee8021SpanningTreePortEnabled
28             TruthValue,
29         ieee8021SpanningTreePortPathCost
30             Integer32,
31         ieee8021SpanningTreePortDesignatedRoot
32             BridgeId,
33         ieee8021SpanningTreePortDesignatedCost
34             Integer32,
35         ieee8021SpanningTreePortDesignatedBridge
36             BridgeId,
37         ieee8021SpanningTreePortDesignatedPort
38             OCTET STRING,
39         ieee8021SpanningTreePortForwardTransitions
40             Counter64,
41         ieee8021SpanningTreeRstpPortProtocolMigration
42             TruthValue,
43         ieee8021SpanningTreeRstpPortAdminEdgePort
44             TruthValue,
45         ieee8021SpanningTreeRstpPortOperEdgePort
46             TruthValue,
47         ieee8021SpanningTreeRstpPortAdminPathCost
48             Integer32
49     }
50
51 ieee8021SpanningTreePortComponentId OBJECT-TYPE
52 SYNTAX      IEEE8021PbbComponentIdentifier
53 MAX-ACCESS  not-accessible
54 STATUS      current
55 DESCRIPTION
56     "The component identifier is used to distinguish between the
57     multiple virtual Bridge instances within a PBB. In simple
58     situations where there is only a single component the default
59     value is 1."
60 ::= { ieee8021SpanningTreePortEntry 1 }
61
62 ieee8021SpanningTreePort OBJECT-TYPE
63 SYNTAX      IEEE8021BridgePortNumber
64 MAX-ACCESS  not-accessible
65 STATUS      current
66 DESCRIPTION
67     "The port number of the port for which this entry
68     contains Spanning Tree Protocol management information."
69 REFERENCE  "12.8.2.1.2 a)"
70 ::= { ieee8021SpanningTreePortEntry 2 }
71
72 ieee8021SpanningTreePortPriority OBJECT-TYPE

```

```
1 SYNTAX Integer32 (0..255)
2 MAX-ACCESS read-write
3 STATUS current
4 DESCRIPTION
5     "The value of the priority field that is contained in
6     the first (in network byte order) octet of the (2 octet
7     long) Port ID. The other octet of the Port ID is given
8     by the value of ieee8021SpanningTreePort.
9     On Bridges supporting IEEE 802.1t or IEEE 802.1w,
10    permissible values are 0-240, in steps of 16.
11
12    The value of this object MUST be retained across
13    reinitializations of the management system."
14 REFERENCE "12.8.2.1.3 c)"
15 ::= { ieee8021SpanningTreePortEntry 3 }
16
17 ieee8021SpanningTreePortState OBJECT-TYPE
18 SYNTAX INTEGER {
19     disabled(1),
20     blocking(2),
21     listening(3),
22     learning(4),
23     forwarding(5),
24     broken(6)
25 }
26 MAX-ACCESS read-only
27 STATUS current
28 DESCRIPTION
29     "The port's current state, as defined by application of
30     the Spanning Tree Protocol. This state controls what
31     action a port takes on reception of a frame. If the
32     Bridge has detected a port that is malfunctioning, it
33     will place that port into the broken(6) state. For
34     ports that are disabled (see
35     ieee8021SpanningTreePortEnabled), this object will have a
36     value of disabled(1). The values disabled, blocking,
37     listening, and broken correspond to the Clause 12 port
38     state of 'Discarding'. The value learning corresponds to
39     the Clause 12 port state of 'Learning'. The value forwarding
40     corresponds to the Clause 12 port state of 'Forwarding'."
41
42 REFERENCE "12.8.2.1.3 b)"
43 ::= { ieee8021SpanningTreePortEntry 4 }
44
45 ieee8021SpanningTreePortEnabled OBJECT-TYPE
46 SYNTAX TruthValue
47 MAX-ACCESS read-write
48 STATUS current
49 DESCRIPTION
50     "The enabled/disabled status of the port. A value of true(1)
51     means the spanning-tree protocol is enabled for this port.
52
53     The value of this object MUST be retained across
54     reinitializations of the management system."
55 REFERENCE "12.8.2.1.3 m)"
56 ::= { ieee8021SpanningTreePortEntry 5 }
57
58 ieee8021SpanningTreePortPathCost OBJECT-TYPE
59 SYNTAX Integer32 (1..2000000000)
60 MAX-ACCESS read-write
61 STATUS current
62 DESCRIPTION
63     "The contribution of this port to the path cost of
64     paths towards the spanning tree root that include
65     this port. Table 13-4 recommends defaults and ranges
66     for Port Path Cost values, in inverse proportion
67     to the speed of the attached LAN. If this object is used
68     to set the Path Cost it is possible to restore the
69     default setting using the
70     ieee8021SpanningTreeRstpPortAdminPathCost object.
71
72     Table 13-4 recommends defaults and ranges for Port
```

```

1      Path Cost values, in inverse proportion to the speed of the
2      attached LAN. If this object is used to set the Path
3      Cost it is possible to restore the default
4      setting using the ieee8021MstpPortAdminPathCost object.
5
6      The value of this object MUST be retained across
7      reinitializations of the management system."
8  REFERENCE  "12.8.2.1.3 d)"
9  ::= { ieee8021SpanningTreePortEntry 6 }
10
11 ieee8021SpanningTreePortDesignatedRoot OBJECT-TYPE
12     SYNTAX      BridgeId
13     MAX-ACCESS  read-only
14     STATUS      current
15     DESCRIPTION
16         "The unique Bridge Identifier of the Bridge
17         recorded as the Root in the Configuration BPDUs
18         transmitted by the Designated Bridge for the
19         segment to which the port is attached."
20     REFERENCE  "12.8.2.1.3 e)"
21     ::= { ieee8021SpanningTreePortEntry 7 }
22
23 ieee8021SpanningTreePortDesignatedCost OBJECT-TYPE
24     SYNTAX      Integer32
25     MAX-ACCESS  read-only
26     STATUS      current
27     DESCRIPTION
28         "The path cost of the Designated Port of the segment
29         connected to this port. This value is compared to the
30         Root Path Cost field in received Bridge PDUs."
31     REFERENCE  "12.8.2.1.3 f)"
32     ::= { ieee8021SpanningTreePortEntry 8 }
33
34 ieee8021SpanningTreePortDesignatedBridge OBJECT-TYPE
35     SYNTAX      BridgeId
36     MAX-ACCESS  read-only
37     STATUS      current
38     DESCRIPTION
39         "The Bridge Identifier of the Bridge that this
40         port considers to be the Designated Bridge for
41         this port's segment."
42     REFERENCE  "12.8.2.1.3 g)"
43     ::= { ieee8021SpanningTreePortEntry 9 }
44
45 ieee8021SpanningTreePortDesignatedPort OBJECT-TYPE
46     SYNTAX      OCTET STRING (SIZE (2))
47     MAX-ACCESS  read-only
48     STATUS      current
49     DESCRIPTION
50         "The Port Identifier of the port on the Designated
51         Bridge for this port's segment."
52     REFERENCE  "12.8.2.1.3 h)"
53     ::= { ieee8021SpanningTreePortEntry 10 }
54
55 ieee8021SpanningTreePortForwardTransitions OBJECT-TYPE
56     SYNTAX      Counter64
57     UNITS       "forwarding transitions"
58     MAX-ACCESS  read-only
59     STATUS      current
60     DESCRIPTION
61         "The number of times this port has transitioned
62         from the Learning state to the Forwarding state.
63
64         Discontinuities in the value of the counter can occur
65         at re-initialization of the management system, and at
66         other times as indicated by the value of
67         ifCounterDiscontinuityTime object of the associated
68         interface (if any)."
69     ::= { ieee8021SpanningTreePortEntry 11 }
70
71 ieee8021SpanningTreeRstpPortProtocolMigration OBJECT-TYPE
72     SYNTAX      TruthValue

```

```

1  MAX-ACCESS  read-write
2  STATUS      current
3  DESCRIPTION
4      "When operating in RSTP (version 2) mode, writing true(1)
5      to this object forces this port to transmit RSTP BPDUs.
6      Any other operation on this object has no effect and
7      it always returns false(2) when read."
8  REFERENCE   "12.8.2.5"
9  ::= { ieee8021SpanningTreePortEntry 12 }
10
11 ieee8021SpanningTreeRstpPortAdminEdgePort OBJECT-TYPE
12     SYNTAX      TruthValue
13     MAX-ACCESS  read-write
14     STATUS      current
15     DESCRIPTION
16         "The administrative value of the Edge Port parameter.
17         A value of true(1) indicates that this port should be
18         assumed as an edge-port, and a value of false(2) indicates
19         that this port should be assumed as a non-edge-port.
20
21         Setting this object will also cause the corresponding
22         instance of ieee8021SpanningTreeRstpPortOperEdgePort to
23         change to the same value. Note that even when this
24         object's value is true(1), the value of the corresponding
25         instance of ieee8021SpanningTreeRstpPortOperEdgePort can
26         be false(2) if a BPDU has been received.
27
28         The value of this object MUST be retained across
29         reinitializations of the management system."
30     REFERENCE   "12.8.2.1.3 k)"
31     ::= { ieee8021SpanningTreePortEntry 13 }
32
33 ieee8021SpanningTreeRstpPortOperEdgePort OBJECT-TYPE
34     SYNTAX      TruthValue
35     MAX-ACCESS  read-only
36     STATUS      current
37     DESCRIPTION
38         "The operational value of the Edge Port parameter. The
39         object is initialized to the value of the corresponding
40         instance of ieee8021SpanningTreeRstpPortAdminEdgePort.
41         When the corresponding instance of
42         ieee8021SpanningTreeRstpPortAdminEdgePort is set, this
43         object will be changed as well. This object will also be
44         changed to false(2) on reception of a BPDU."
45     REFERENCE   "12.8.2.1.3 l)"
46     ::= { ieee8021SpanningTreePortEntry 14 }
47
48 ieee8021SpanningTreeRstpPortAdminPathCost OBJECT-TYPE
49     SYNTAX      Integer32 (0..2000000000)
50     MAX-ACCESS  read-write
51     STATUS      current
52     DESCRIPTION
53         "The administratively assigned value for the contribution
54         of this port to the path cost of paths toward the spanning
55         tree root.
56
57         Writing a value of '0' assigns the automatically calculated
58         default Path Cost value to the port. If the default Path
59         Cost is being used, this object returns '0' when read.
60
61         This complements the object ieee8021SpanningTreePortPathCost,
62         which returns the operational value of the path cost.
63
64         The value of this object MUST be retained across
65         reinitializations of the management system."
66     REFERENCE   "12.8.2.1.3 d)"
67     ::= { ieee8021SpanningTreePortEntry 15 }
68
69
70
71 -- =====
72 -- The Spanning Tree Port Extension Table

```

```

1 -- =====
2
3 ieee8021SpanningTreePortExtensionTable OBJECT-TYPE
4     SYNTAX      SEQUENCE OF Ieee8021SpanningTreePortExtensionEntry
5     MAX-ACCESS   not-accessible
6     STATUS      current
7     DESCRIPTION
8         "A table that contains port-specific information
9         for the Spanning Tree Protocol."
10    REFERENCE    "12.8.2"
11    ::= { ieee8021SpanningTreeObjects 3 }
12
13 ieee8021SpanningTreePortExtensionEntry OBJECT-TYPE
14     SYNTAX      Ieee8021SpanningTreePortExtensionEntry
15     MAX-ACCESS   not-accessible
16     STATUS      current
17     DESCRIPTION
18         "A list of additional objects containing information
19         maintained by every port about the Spanning Tree
20         Protocol state for that port."
21     AUGMENTS { ieee8021SpanningTreePortEntry }
22     ::= { ieee8021SpanningTreePortExtensionTable 1 }
23
24 Ieee8021SpanningTreePortExtensionEntry ::=
25     SEQUENCE {
26
27         ieee8021SpanningTreeRstpPortAutoEdgePort
28         TruthValue,
29         ieee8021SpanningTreeRstpPortAutoIsolatePort
30         TruthValue,
31         ieee8021SpanningTreeRstpPortIsolatePort
32         TruthValue
33     }
34
35 ieee8021SpanningTreeRstpPortAutoEdgePort OBJECT-TYPE
36     SYNTAX      TruthValue
37     MAX-ACCESS   read-write
38     STATUS      current
39     DESCRIPTION
40         "The administrative value of the Auto Edge Port parameter.
41         A value of true(1) indicates if the Bridge detection state
42         machine (BDM, 13.33) is to detect other Bridges
43         attached to the LAN, and set
44         ieee8021SpanningTreeRstpPortOperEdgePort automatically.
45         The default value is true(1)
46
47         This is optional and provided only by implementations
48         that support the automatic identification of edge ports.
49
50         The value of this object MUST be retained across
51         reinitializations of the management system."
52     REFERENCE    "12.8.2.1.3 m)"
53     ::= { ieee8021SpanningTreePortExtensionEntry 1 }
54
55 ieee8021SpanningTreeRstpPortAutoIsolatePort OBJECT-TYPE
56     SYNTAX      TruthValue
57     MAX-ACCESS   read-only
58     STATUS      current
59     DESCRIPTION
60         "The operational value of the Isolate Port parameter.
61
62         A value of true(1) indicates a Designated Port will
63         transition to discarding if both
64         ieee8021SpanningTreeRstpPortAdminEdgePort and
65         ieee8021SpanningTreeRstpPortAutoEdgePort are FALSE and
66         the other Bridge presumed to be attached to the same
67         point-to-point LAN does not transmit periodic BPDUs.
68
69         This is optional and provided only by implementations
70         that support the automatic identification of edge ports."
71     REFERENCE    "12.8.2.1.3 n)"
72     ::= { ieee8021SpanningTreePortExtensionEntry 2 }

```

```

1
2 ieee8021SpanningTreeRstpPortIsolatePort OBJECT-TYPE
3     SYNTAX      TruthValue
4     MAX-ACCESS  read-only
5     STATUS      current
6     DESCRIPTION
7         "The operational value of the Isolate Port parameter.
8
9     A value of true(1), Set by the Bridge detection state
10        machine (BDM, 13.33), indicates when the Spanning Tree
11        Protocol Entity of a neighboring Bridge has apparently
12        failed ."
13     REFERENCE   "12.8.2.1.3 o)"
14     ::= { ieee8021SpanningTreePortExtensionEntry 3 }
15
16
17
18 -- =====
19 -- Notifications for use by Bridges
20 -- =====
21 -- Notifications for the Spanning Tree Protocol
22 -- =====
23
24 ieee8021SpanningTreeNewRoot NOTIFICATION-TYPE
25     -- OBJECTS      { }
26     STATUS          current
27     DESCRIPTION
28         "The ieee8021SpanningTreeNewRoot notification indicates that
29         the sending agent has become the new root of the Spanning Tree;
30         the notification is sent by a Bridge soon after its election
31         as the new root, e.g., upon expiration of the Topology Change
32         Timer, immediately subsequent to its election."
33     ::= { ieee8021SpanningTreeNotifications 1 }
34
35 ieee8021SpanningTreeTopologyChange NOTIFICATION-TYPE
36     -- OBJECTS      { }
37     STATUS          current
38     DESCRIPTION
39         "A ieee8021SpanningTreeTopologyChange notification is sent
40         by a Bridge when any of its configured ports transitions from
41         the Learning state to the Forwarding state, or from the
42         Forwarding state to the Blocking state. The notification
43         is not sent if a ieee8021SpanningTreeNewRoot notification
44         is sent for the same transition."
45     ::= { ieee8021SpanningTreeNotifications 2 }
46
47 -- =====
48 -- IEEE 802.1D MIB - Conformance Information
49 -- =====
50
51 ieee8021SpanningTreeCompliances
52     OBJECT IDENTIFIER ::= { ieee8021SpanningTreeConformance 1 }
53 ieee8021SpanningTreeGroups
54     OBJECT IDENTIFIER ::= { ieee8021SpanningTreeConformance 2 }
55
56 -- =====
57 -- units of conformance
58 -- =====
59
60 -- =====
61 -- the ieee8021SpanningTree group
62 -- =====
63
64 ieee8021SpanningTreeGroup OBJECT-GROUP
65     OBJECTS {
66         ieee8021SpanningTreeProtocolSpecification,
67         ieee8021SpanningTreePriority,
68         ieee8021SpanningTreeTimeSinceTopologyChange,
69         ieee8021SpanningTreeTopChanges,
70         ieee8021SpanningTreeDesignatedRoot,
71         ieee8021SpanningTreeRootCost,
72         ieee8021SpanningTreeRootPort,

```

```

1      ieee8021SpanningTreeMaxAge,
2      ieee8021SpanningTreeHelloTime,
3      ieee8021SpanningTreeHoldTime,
4      ieee8021SpanningTreeForwardDelay,
5      ieee8021SpanningTreeBridgeMaxAge,
6      ieee8021SpanningTreeBridgeHelloTime,
7      ieee8021SpanningTreeBridgeForwardDelay,
8      ieee8021SpanningTreeVersion
9  }
10 STATUS      current
11 DESCRIPTION
12     "Bridge level Spanning Tree data for this device."
13 ::= { ieee8021SpanningTreeGroups 1 }
14
15 ieee8021SpanningTreeRstpGroup OBJECT-GROUP
16 OBJECTS {
17     ieee8021SpanningTreeRstpTxHoldCount
18 }
19 STATUS      current
20 DESCRIPTION
21     "Bridge level Rstp data for this device."
22 ::= { ieee8021SpanningTreeGroups 2 }
23
24 ieee8021SpanningTreePortGroup OBJECT-GROUP
25 OBJECTS {
26     ieee8021SpanningTreePortPriority,
27     ieee8021SpanningTreePortState,
28     ieee8021SpanningTreePortEnabled,
29     ieee8021SpanningTreePortPathCost,
30     ieee8021SpanningTreePortDesignatedRoot,
31     ieee8021SpanningTreePortDesignatedCost,
32     ieee8021SpanningTreePortDesignatedBridge,
33     ieee8021SpanningTreePortDesignatedPort,
34     ieee8021SpanningTreePortForwardTransitions
35 }
36 STATUS      current
37 DESCRIPTION
38     "Spanning Tree data for each port on this device."
39 ::= { ieee8021SpanningTreeGroups 3 }
40
41 ieee8021SpanningTreeRstpPortGroup OBJECT-GROUP
42 OBJECTS {
43     ieee8021SpanningTreeRstpPortProtocolMigration,
44     ieee8021SpanningTreeRstpPortAdminEdgePort,
45     ieee8021SpanningTreeRstpPortOperEdgePort,
46     ieee8021SpanningTreeRstpPortAdminPathCost
47 }
48 STATUS      current
49 DESCRIPTION
50     "Rstp data for each port on this device."
51 ::= { ieee8021SpanningTreeGroups 4 }
52
53 ieee8021SpanningTreeRstpFragileGroup OBJECT-GROUP
54 OBJECTS {
55     ieee8021SpanningTreeRstpPortAutoEdgePort,
56     ieee8021SpanningTreeRstpPortAutoIsolatePort,
57     ieee8021SpanningTreeRstpPortIsolatePort
58 }
59 STATUS      current
60 DESCRIPTION
61     "Rstp fragile Bridge data for each port
62     on this device."
63 ::= { ieee8021SpanningTreeGroups 6 }
64
65 -- =====
66 -- The Notification Group
67 -- =====
68
69 ieee8021SpanningTreeNotificationGroup NOTIFICATION-GROUP
70 NOTIFICATIONS {
71     ieee8021SpanningTreeNewRoot,
72     ieee8021SpanningTreeTopologyChange

```



```

1      }
2      STATUS      current
3      DESCRIPTION
4          "Group of notifications."
5      ::= { ieee8021SpanningTreeGroups 5 }
6
7  -- =====
8  -- compliance statements
9  -- =====
10
11 ieee8021SpanningTreeCompliance MODULE-COMPLIANCE
12     STATUS      current
13     DESCRIPTION
14         "The compliance statement for devices supporting the
15         Spanning Tree Protocol."
16
17     MODULE
18         MANDATORY-GROUPS {
19             ieee8021SpanningTreeGroup,
20             ieee8021SpanningTreePortGroup
21         }
22
23     OBJECT ieee8021SpanningTreePriority
24     SYNTAX Integer32 (0|4096|8192|12288|16384|20480|24576
25                     |28672|32768|36864|40960|45056|49152
26                     |53248|57344|61440)
27     DESCRIPTION
28         "The possible values defined by IEEE 802.1t."
29
30     OBJECT ieee8021SpanningTreePortPriority
31     SYNTAX Integer32 (0|16|32|48|64|80|96|112|128
32                     |144|160|176|192|208|224|240)
33     DESCRIPTION
34         "The possible values defined by IEEE 802.1t."
35
36     GROUP ieee8021SpanningTreeNotificationGroup
37     DESCRIPTION
38         "Implementation of this group is optional."
39
40     ::= { ieee8021SpanningTreeCompliances 1 }
41
42 ieee8021SpanningTreeRstpCompliance MODULE-COMPLIANCE
43     STATUS      current
44     DESCRIPTION
45         "The compliance statement for devices supporting RSTP."
46
47     MODULE
48         MANDATORY-GROUPS {
49             ieee8021SpanningTreeGroup,
50             ieee8021SpanningTreeRstpGroup,
51             ieee8021SpanningTreePortGroup,
52             ieee8021SpanningTreeRstpPortGroup
53         }
54
55     OBJECT ieee8021SpanningTreePriority
56     SYNTAX Integer32 (0|4096|8192|12288|16384|20480|24576
57                     |28672|32768|36864|40960|45056|49152
58                     |53248|57344|61440)
59     DESCRIPTION
60         "The possible values defined by IEEE 802.1t."
61
62     OBJECT ieee8021SpanningTreePortPriority
63     SYNTAX Integer32 (0|16|32|48|64|80|96|112|128
64                     |144|160|176|192|208|224|240)
65     DESCRIPTION
66         "The possible values defined by IEEE 802.1t."
67
68     GROUP ieee8021SpanningTreeNotificationGroup
69     DESCRIPTION
70         "Implementation of this group is optional."
71
72     GROUP ieee8021SpanningTreeRstpFragileGroup

```

```
1  DESCRIPTION
2      "Implementation of this group is optional."
3
4  ::= { ieee8021SpanningTreeCompliances 2 }
5
6  END
```

1 17.7.4 Definitions for the IEEE8021-Q-BRIDGE-MIB module

```
2 IEEE8021-Q-BRIDGE-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for IEEE 802.1Q Devices
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE, Gauge32,
10     Counter64, Unsigned32, TimeTicks, Integer32
11     FROM SNMPv2-SMI
12     RowStatus, StorageType, TruthValue, MacAddress
13     FROM SNMPv2-TC
14     SnmpAdminString
15     FROM SNMP-FRAMEWORK-MIB
16     MODULE-COMPLIANCE, OBJECT-GROUP
17     FROM SNMPv2-CONF
18     ieee8021BridgeBasePortComponentId, ieee8021BridgeBasePort,
19     ieee8021BridgeBasePortEntry
20     FROM IEEE8021-BRIDGE-MIB
21     ieee802dot1mibs, IEEE8021PbbComponentIdentifier,
22     IEEE8021BridgePortNumber, IEEE8021BridgePortNumberOrZero,
23     IEEE8021VlanIndex, IEEE8021VlanIndexOrWildcard,
24     IEEE8021PortAcceptableFrameTypes
25     FROM IEEE8021-TC-MIB
26     PortList, VlanId
27     FROM Q-BRIDGE-MIB
28     TimeFilter
29     FROM RMON2-MIB;
30
31 ieee8021QBridgeMib MODULE-IDENTITY
32     LAST-UPDATED "202211080000Z" -- November 8, 2022
33     ORGANIZATION "IEEE 802.1 Working Group"
34     CONTACT-INFO
35         " WG-URL: http://www.ieee802.org/1/
36         WG-EMail: stds-802-1-1@ieee.org
37         Contact: IEEE 802.1 Working Group Chair
38         Postal: C/O IEEE 802.1 Working Group
39             IEEE Standards Association
40             445 Hoes Lane
41             Piscataway, NJ 08854
42             USA
43         E-mail: stds-802-1-chairs@ieee.org"
44     DESCRIPTION
45         "The VLAN Bridge MIB module for managing Virtual Bridged
46         Local Area Networks, as defined by IEEE Std 802.1Q.
47
48         Unless otherwise indicated, the references in this MIB
49         module are to IEEE Std 802.1Q-2022.
50
51         Copyright (C) IEEE (2022).
52         This version of this MIB module is part of IEEE Std 802.1Q;
53         see that standard for full legal notices."
54
55     REVISION "202211080000Z" -- November 8, 2022
56     DESCRIPTION
57         "Published as part of IEEE Std 802.1Q-2022.
58         Cross references and contact information updated."
59
60     REVISION "201807010000Z" -- July 1, 2018
61     DESCRIPTION
62         "Published as part of IEEE Std 802.1Q 2018 revision.
63         Cross references updated and corrected."
64
65     REVISION "201412150000Z" -- December 15, 2014
66     DESCRIPTION
67         "Published as part of IEEE Std 802.1Q 2014 revision.
68         Cross references updated and corrected.
69         VLAN Learning Constraints objects deprecated
70         and removed from compliances.
71         ieee8021QbridgeEgressVidXTable properly reindexed;
```

```

1      old version deprecated and new version named
2      ieee8021QbridgeEgressVidXV2Table."
3
4      REVISION      "201112120000Z" -- December 12, 2011
5      DESCRIPTION
6          "Addition of the VID Translation MIB Subtree for IEEE Std 802.1aq"
7
8      REVISION      "201102270000Z" -- February 27, 2011
9      DESCRIPTION
10         "Minor edits to contact information etc. as part of
11         2011 revision of IEEE Std 802.1Q."
12
13      REVISION      "200810150000Z" -- October 15, 2008
14      DESCRIPTION
15         "Initial version, derived from RFC 4363."
16      ::= { ieee802dot1mibs 4 }
17
18 ieee8021QBridgeMibObjects OBJECT IDENTIFIER ::= { ieee8021QBridgeMib 1 }
19
20 -- =====
21 -- subtrees in the Q-BRIDGE MIB
22 -- =====
23
24 ieee8021QBridgeBase      OBJECT IDENTIFIER ::= { ieee8021QBridgeMibObjects 1 }
25 ieee8021QBridgeTp        OBJECT IDENTIFIER ::= { ieee8021QBridgeMibObjects 2 }
26 ieee8021QBridgeStatic    OBJECT IDENTIFIER ::= { ieee8021QBridgeMibObjects 3 }
27 ieee8021QBridgeVlan      OBJECT IDENTIFIER ::= { ieee8021QBridgeMibObjects 4 }
28 ieee8021QBridgeProtocol  OBJECT IDENTIFIER ::= { ieee8021QBridgeMibObjects 5 }
29 ieee8021QBridgeVIDX      OBJECT IDENTIFIER ::= { ieee8021QBridgeMibObjects 6 }
30
31 -- =====
32 -- ieee8021QBridgeBase subtree
33 -- =====
34
35 -- =====
36 -- ieee8021QBridgeTable - Table of VLAN Bridges
37 -- =====
38
39 ieee8021QBridgeTable OBJECT-TYPE
40     SYNTAX      SEQUENCE OF Ieee8021QBridgeEntry
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44         "A table that contains generic information about every
45         VLAN Bridge."
46     REFERENCE   "12.4"
47     ::= { ieee8021QBridgeBase 1 }
48
49 ieee8021QBridgeEntry OBJECT-TYPE
50     SYNTAX      Ieee8021QBridgeEntry
51     MAX-ACCESS  not-accessible
52     STATUS      current
53     DESCRIPTION
54         "A list of objects containing information for each VLAN Bridge."
55     INDEX       { ieee8021QBridgeComponentId }
56     ::= { ieee8021QBridgeTable 1 }
57
58 Ieee8021QBridgeEntry ::=
59     SEQUENCE {
60         ieee8021QBridgeComponentId      IEEE8021PbbComponentIdentifier,
61         ieee8021QBridgeVlanVersionNumber  INTEGER,
62         ieee8021QBridgeMaxVlanId          VlanId,
63         ieee8021QBridgeMaxSupportedVlans  Unsigned32,
64         ieee8021QBridgeNumVlans           Gauge32,
65         ieee8021QBridgeMvrpEnabledStatus  TruthValue
66     }
67
68 ieee8021QBridgeComponentId OBJECT-TYPE
69     SYNTAX      IEEE8021PbbComponentIdentifier
70     MAX-ACCESS  not-accessible
71     STATUS      current
72     DESCRIPTION

```

```

1      "The component identifier is used to distinguish between the
2      multiple virtual Bridge instances within a PBB. In simple
3      situations where there is only a single component the default
4      value is 1."
5      ::= { ieee8021QBridgeEntry 1 }
6
7 ieee8021QBridgeVlanVersionNumber OBJECT-TYPE
8     SYNTAX      INTEGER {
9         version1(1),
10        version2(2)
11    }
12     MAX-ACCESS  read-only
13     STATUS      current
14     DESCRIPTION
15         "The version number of IEEE 802.1Q that this device
16         supports. Reported as 1 by VLAN Bridges that support
17         only SST operation, and reported as 2 by VLAN Bridges
18         that support MST operation."
19     REFERENCE   "12.10.1.1"
20     ::= { ieee8021QBridgeEntry 2 }
21
22 ieee8021QBridgeMaxVlanId OBJECT-TYPE
23     SYNTAX      VlanId
24     MAX-ACCESS  read-only
25     STATUS      current
26     DESCRIPTION
27         "The maximum IEEE 802.1Q VLAN-ID that this device
28         supports."
29     REFERENCE   "9.6"
30     ::= { ieee8021QBridgeEntry 3 }
31
32 ieee8021QBridgeMaxSupportedVlans OBJECT-TYPE
33     SYNTAX      Unsigned32
34     UNITS       "vlans"
35     MAX-ACCESS  read-only
36     STATUS      current
37     DESCRIPTION
38         "The maximum number of IEEE 802.1Q VLANs that this
39         device supports."
40     REFERENCE   "12.10.1.1"
41     ::= { ieee8021QBridgeEntry 4 }
42
43 ieee8021QBridgeNumVlans OBJECT-TYPE
44     SYNTAX      Gauge32
45     UNITS       "vlans"
46     MAX-ACCESS  read-only
47     STATUS      current
48     DESCRIPTION
49         "The current number of IEEE 802.1Q VLANs that are
50         configured in this device."
51     REFERENCE   "12.7.1.1"
52     ::= { ieee8021QBridgeEntry 5 }
53
54 ieee8021QBridgeMvrpEnabledStatus OBJECT-TYPE
55     SYNTAX      TruthValue
56     MAX-ACCESS  read-write
57     STATUS      current
58     DESCRIPTION
59         "The administrative status requested by management for
60         MVRP. The value true(1) indicates that MVRP should
61         be enabled on this device, on all ports for which it has
62         not been specifically disabled. When false(2), MVRP
63         is disabled on all ports, and all MVRP packets will be
64         forwarded transparently. This object affects all MVRP
65         Applicant and Registrar state machines. A transition
66         from false(2) to true(1) will cause a reset of all
67         MVRP state machines on all ports.
68
69         The value of this object MUST be retained across
70         reinitializations of the management system."
71     DEFVAL      { true }
72     ::= { ieee8021QBridgeEntry 6 }

```

```

1
2 -- =====
3 -- ieee8021QBridgeCVlanPortTable - Table of C-VLAN ports
4 -- =====
5
6 ieee8021QBridgeCVlanPortTable OBJECT-TYPE
7     SYNTAX      SEQUENCE OF Ieee8021QBridgeCVlanPortEntry
8     MAX-ACCESS  not-accessible
9     STATUS      current
10    DESCRIPTION
11        "This table provides the capability to create and delete
12         customer VLAN ports. Entries in this table must be
13         persistent over power up restart/reboot."
14    REFERENCE    "12.16.1.1.3 h4), 12.16.2.1,
15                 12.13.1.1, 12.13.1.2, 12.15"
16    ::= { ieee8021QBridgeBase 2 }
17
18 ieee8021QBridgeCVlanPortEntry OBJECT-TYPE
19     SYNTAX      Ieee8021QBridgeCVlanPortEntry
20     MAX-ACCESS  not-accessible
21     STATUS      current
22     DESCRIPTION
23         "A list of objects containing information for each VLAN Bridge."
24     INDEX       { ieee8021QBridgeCVlanPortComponentId,
25                  ieee8021QBridgeCVlanPortNumber }
26     ::= { ieee8021QBridgeCVlanPortTable 1 }
27
28 Ieee8021QBridgeCVlanPortEntry ::=
29     SEQUENCE {
30         ieee8021QBridgeCVlanPortComponentId  IEEE8021PbbComponentIdentifier,
31         ieee8021QBridgeCVlanPortNumber       IEEE8021BridgePortNumber,
32         ieee8021QBridgeCVlanPortRowStatus    RowStatus
33     }
34
35 ieee8021QBridgeCVlanPortComponentId OBJECT-TYPE
36     SYNTAX      IEEE8021PbbComponentIdentifier
37     MAX-ACCESS  not-accessible
38     STATUS      current
39     DESCRIPTION
40         "The component containing the customer VLAN port represented
41         by this row."
42     ::= { ieee8021QBridgeCVlanPortEntry 1 }
43
44 ieee8021QBridgeCVlanPortNumber OBJECT-TYPE
45     SYNTAX      IEEE8021BridgePortNumber
46     MAX-ACCESS  not-accessible
47     STATUS      current
48     DESCRIPTION
49         "The customer VLAN port number represented by this row."
50     ::= { ieee8021QBridgeCVlanPortEntry 2 }
51
52 ieee8021QBridgeCVlanPortRowStatus OBJECT-TYPE
53     SYNTAX      RowStatus
54     MAX-ACCESS  read-create
55     STATUS      current
56     DESCRIPTION
57         "This indicates the status of the entry, and is used to create
58         and delete entries in this table. Each entry in this table that
59         is valid will have a corresponding entry in the
60         ieee8021BridgeBasePortTable whose value for
61         ieee8021BridgeBasePortType is customerVlanPort(2). The
62         corresponding value of ieee8021BridgeBasePortIfIndex must
63         be set at the time the value of this object transitions
64         to valid(1).
65
66         Entries in this table must be persistent across
67         reinitializations of the management system."
68     ::= { ieee8021QBridgeCVlanPortEntry 3 }
69
70 -- =====
71 -- the ieee8021QBridgeTp subtree
72 -- =====

```

```

1
2 -- =====
3 -- the current Filtering Database Table
4 -- =====
5
6 ieee8021QBridgeFdbTable OBJECT-TYPE
7     SYNTAX      SEQUENCE OF Ieee8021QBridgeFdbEntry
8     MAX-ACCESS  not-accessible
9     STATUS      current
10    DESCRIPTION
11        "A table that contains configuration and control
12         information for each Filtering Database currently
13         operating on this device. Entries in this table appear
14         automatically when VLANs are assigned FDB IDs in the
15         ieee8021QBridgeVlanCurrentTable."
16    REFERENCE   "12.7.1"
17    ::= { ieee8021QBridgeTp 1 }
18
19 ieee8021QBridgeFdbEntry OBJECT-TYPE
20     SYNTAX      Ieee8021QBridgeFdbEntry
21     MAX-ACCESS  not-accessible
22     STATUS      current
23     DESCRIPTION
24        "Information about a specific Filtering Database."
25     INDEX       { ieee8021QBridgeFdbComponentId,
26                  ieee8021QBridgeFdbId }
27     ::= { ieee8021QBridgeFdbTable 1 }
28
29 Ieee8021QBridgeFdbEntry ::=
30     SEQUENCE {
31         ieee8021QBridgeFdbComponentId
32         IEEE8021PbbComponentIdentifier,
33         ieee8021QBridgeFdbId
34         Unsigned32,
35         ieee8021QBridgeFdbDynamicCount
36         Gauge32,
37         ieee8021QBridgeFdbLearnedEntryDiscards
38         Counter64,
39         ieee8021QBridgeFdbAgingTime
40         Integer32
41     }
42
43 ieee8021QBridgeFdbComponentId OBJECT-TYPE
44     SYNTAX      IEEE8021PbbComponentIdentifier
45     MAX-ACCESS  not-accessible
46     STATUS      current
47     DESCRIPTION
48        "The component identifier is used to distinguish between the
49         multiple virtual Bridge instances within a PBB. In simple
50         situations where there is only a single component the default
51         value is 1."
52     ::= { ieee8021QBridgeFdbEntry 1 }
53
54 ieee8021QBridgeFdbId OBJECT-TYPE
55     SYNTAX      Unsigned32 (0..4294967295)
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59        "The identity of this Filtering Database."
60     ::= { ieee8021QBridgeFdbEntry 2 }
61
62 ieee8021QBridgeFdbDynamicCount OBJECT-TYPE
63     SYNTAX      Gauge32
64     UNITS       "database entries"
65     MAX-ACCESS  read-only
66     STATUS      current
67     DESCRIPTION
68        "The current number of dynamic entries in this
69         Filtering Database."
70     REFERENCE   "12.7.1.1.3"
71     ::= { ieee8021QBridgeFdbEntry 3 }
72

```

```

1 ieee8021QBridgeFdbLearnedEntryDiscards OBJECT-TYPE
2     SYNTAX      Counter64
3     UNITS       "database entries"
4     MAX-ACCESS  read-only
5     STATUS      current
6     DESCRIPTION
7         "The total number of Filtering Database entries that
8         have been or would have been learned, but have been
9         discarded due to a lack of storage space in the
10        Filtering Database. If this counter is increasing, it
11        indicates that the Filtering Database is regularly
12        becoming full (a condition that has unpleasant
13        performance effects on the subnetwork). If this counter
14        has a significant value but is not presently increasing,
15        it indicates that the problem has been occurring but is
16        not persistent.
17
18        Discontinuities in the value of the counter can occur
19        at re-initialization of the management system."
20 ::= { ieee8021QBridgeFdbEntry 4 }
21
22 ieee8021QBridgeFdbAgingTime OBJECT-TYPE
23     SYNTAX      Integer32 (10..1000000)
24     UNITS       "seconds"
25     MAX-ACCESS  read-write
26     STATUS      current
27     DESCRIPTION
28         "The timeout period in seconds for aging out
29         dynamically-learned forwarding information.
30         IEEE Std 802.1D-1998 recommends a default of 300 seconds.
31
32         The value of this object MUST be retained across
33         reinitializations of the management system."
34     REFERENCE   "12.7.1.2"
35 ::= { ieee8021QBridgeFdbEntry 5 }
36
37 -- =====
38 -- Multiple Filtering Databases for IEEE 802.1Q Transparent Devices
39 -- This table is an alternative to the ieee8021QBridgeTpFdbTable,
40 -- previously defined for IEEE 802.1D devices that only support a
41 -- single Filtering Database.
42 -- =====
43
44 ieee8021QBridgeTpFdbTable OBJECT-TYPE
45     SYNTAX      SEQUENCE OF Ieee8021QBridgeTpFdbEntry
46     MAX-ACCESS  not-accessible
47     STATUS      current
48     DESCRIPTION
49         "A table that contains information about unicast entries
50         for which the device has forwarding and/or filtering
51         information. This information is used by the
52         transparent bridging function in determining how to
53         propagate a received frame."
54     REFERENCE   "12.7.1"
55 ::= { ieee8021QBridgeTp 2 }
56
57 ieee8021QBridgeTpFdbEntry OBJECT-TYPE
58     SYNTAX      Ieee8021QBridgeTpFdbEntry
59     MAX-ACCESS  not-accessible
60     STATUS      current
61     DESCRIPTION
62         "Information about a specific unicast MAC address for
63         which the device has some forwarding and/or filtering
64         information."
65     INDEX       { ieee8021QBridgeFdbComponentId,
66                  ieee8021QBridgeFdbId,
67                  ieee8021QBridgeTpFdbAddress }
68 ::= { ieee8021QBridgeTpFdbTable 1 }
69
70 Ieee8021QBridgeTpFdbEntry ::=
71     SEQUENCE {
72         ieee8021QBridgeTpFdbAddress

```



```

1      MacAddress,
2      ieee8021QBridgeTpFdbPort
3      IEEE8021BridgePortNumberOrZero,
4      ieee8021QBridgeTpFdbStatus
5      INTEGER
6  }
7
8 ieee8021QBridgeTpFdbAddress OBJECT-TYPE
9     SYNTAX      MacAddress
10    MAX-ACCESS  not-accessible
11    STATUS      current
12    DESCRIPTION
13       "A unicast MAC address for which the device has
14       forwarding and/or filtering information."
15    ::= { ieee8021QBridgeTpFdbEntry 1 }
16
17 ieee8021QBridgeTpFdbPort OBJECT-TYPE
18     SYNTAX      IEEE8021BridgePortNumberOrZero
19     MAX-ACCESS  read-only
20     STATUS      current
21     DESCRIPTION
22        "Either the value '0', or the port number of the port on
23        which a frame having a source address equal to the value
24        of the corresponding instance of ieee8021QBridgeTpFdbAddress has
25        been seen. A value of '0' indicates that the port
26        number has not been learned but that the device does
27        have some forwarding/filtering information about this
28        address (e.g., in the ieee8021QBridgeStaticUnicastTable).
29        Implementors are encouraged to assign the port value to
30        this object whenever it is learned, even for addresses
31        for which the corresponding value of ieee8021QBridgeTpFdbStatus is
32        not learned(3)."
```

```

33    ::= { ieee8021QBridgeTpFdbEntry 2 }
34
35 ieee8021QBridgeTpFdbStatus OBJECT-TYPE
36     SYNTAX      INTEGER {
37                 other(1),
38                 invalid(2),
39                 learned(3),
40                 self(4),
41                 mgmt(5)
42             }
43     MAX-ACCESS  read-only
44     STATUS      current
45     DESCRIPTION
46        "The status of this entry. The meanings of the values
47        are:
48        other(1) - none of the following. This may include
49        the case where some other MIB object (not the
50        corresponding instance of ieee8021QBridgeTpFdbPort, nor an
51        entry in the ieee8021QBridgeStaticUnicastTable) is being
52        used to determine if and how frames addressed to
53        the value of the corresponding instance of
54        ieee8021QBridgeTpFdbAddress are being forwarded.
55        invalid(2) - this entry is no longer valid (e.g., it
56        was learned but has since aged out), but has not
57        yet been flushed from the table.
58        learned(3) - the value of the corresponding instance
59        of ieee8021QBridgeTpFdbPort was learned and is being used.
60        self(4) - the value of the corresponding instance of
61        ieee8021QBridgeTpFdbAddress represents one of the device's
62        addresses. The corresponding instance of
63        ieee8021QBridgeTpFdbPort indicates which of the device's
64        ports has this address.
65        mgmt(5) - the value of the corresponding instance of
66        ieee8021QBridgeTpFdbAddress is also the value of an
67        existing instance of ieee8021QBridgeStaticUnicastAddress."
68    ::= { ieee8021QBridgeTpFdbEntry 3 }
69
70 -- =====
71 -- Dynamic Group Registration Table
72 -- =====

```

```

1
2 ieee8021QBridgeTpGroupTable OBJECT-TYPE
3   SYNTAX      SEQUENCE OF Ieee8021QBridgeTpGroupEntry
4   MAX-ACCESS  not-accessible
5   STATUS      current
6   DESCRIPTION
7       "A table containing filtering information for VLANs
8       configured into the Bridge by (local or network)
9       management, or learned dynamically, specifying the set of
10      ports to which frames received on a VLAN for this FDB
11      and containing a specific Group destination address are
12      allowed to be forwarded."
13   REFERENCE   "12.7.4"
14   ::= { ieee8021QBridgeTp 3 }
15
16 ieee8021QBridgeTpGroupEntry OBJECT-TYPE
17   SYNTAX      Ieee8021QBridgeTpGroupEntry
18   MAX-ACCESS  not-accessible
19   STATUS      current
20   DESCRIPTION
21       "Filtering information configured into the Bridge by
22       management, or learned dynamically, specifying the set of
23       ports to which frames received on a VLAN and containing
24       a specific Group destination address are allowed to be
25       forwarded. The subset of these ports learned dynamically
26       is also provided."
27   INDEX       { ieee8021QBridgeVlanCurrentComponentId,
28                 ieee8021QBridgeVlanIndex,
29                 ieee8021QBridgeTpGroupAddress }
30   ::= { ieee8021QBridgeTpGroupTable 1 }
31
32 Ieee8021QBridgeTpGroupEntry ::=
33   SEQUENCE {
34       ieee8021QBridgeTpGroupAddress
35       MacAddress,
36       ieee8021QBridgeTpGroupEgressPorts
37       PortList,
38       ieee8021QBridgeTpGroupLearnt
39       PortList
40   }
41
42 ieee8021QBridgeTpGroupAddress OBJECT-TYPE
43   SYNTAX      MacAddress
44   MAX-ACCESS  not-accessible
45   STATUS      current
46   DESCRIPTION
47       "The destination Group MAC address in a frame to which
48       this entry's filtering information applies."
49   ::= { ieee8021QBridgeTpGroupEntry 1 }
50
51 ieee8021QBridgeTpGroupEgressPorts OBJECT-TYPE
52   SYNTAX      PortList
53   MAX-ACCESS  read-only
54   STATUS      current
55   DESCRIPTION
56       "The complete set of ports, in this VLAN, to which
57       frames destined for this Group MAC address are currently
58       being explicitly forwarded. This does not include ports
59       for which this address is only implicitly forwarded, in
60       the ieee8021QBridgeForwardAllPorts list."
61   ::= { ieee8021QBridgeTpGroupEntry 2 }
62
63 ieee8021QBridgeTpGroupLearnt OBJECT-TYPE
64   SYNTAX      PortList
65   MAX-ACCESS  read-only
66   STATUS      current
67   DESCRIPTION
68       "The subset of ports in ieee8021QBridgeTpGroupEgressPorts that
69       were learned by MMRP or some other dynamic mechanism, in
70       this Filtering database."
71   ::= { ieee8021QBridgeTpGroupEntry 3 }
72

```

```

1 -- =====
2 -- Service Requirements subtree
3 -- =====
4
5 ieee8021QBridgeForwardAllTable OBJECT-TYPE
6     SYNTAX      SEQUENCE OF Ieee8021QBridgeForwardAllEntry
7     MAX-ACCESS  not-accessible
8     STATUS      current
9     DESCRIPTION
10        "A table containing forwarding information for each
11        VLAN, specifying the set of ports to which forwarding of
12        all multicasts applies, configured statically by
13        management or dynamically by MMRP. An entry appears in
14        this table for all VLANs that are currently
15        instantiated."
16     REFERENCE   "12.7.2, 12.7.7"
17     ::= { ieee8021QBridgeTp 4 }
18
19 ieee8021QBridgeForwardAllEntry OBJECT-TYPE
20     SYNTAX      Ieee8021QBridgeForwardAllEntry
21     MAX-ACCESS  not-accessible
22     STATUS      current
23     DESCRIPTION
24        "Forwarding information for a VLAN, specifying the set
25        of ports to which all multicasts should be forwarded,
26        configured statically by management or dynamically by
27        MMRP."
28     INDEX       { ieee8021QBridgeVlanCurrentComponentId,
29                   ieee8021QBridgeForwardAllVlanIndex }
30     ::= { ieee8021QBridgeForwardAllTable 1 }
31
32 Ieee8021QBridgeForwardAllEntry ::=
33     SEQUENCE {
34         ieee8021QBridgeForwardAllVlanIndex
35             IEEE8021VlanIndexOrWildcard,
36         ieee8021QBridgeForwardAllPorts
37             PortList,
38         ieee8021QBridgeForwardAllStaticPorts
39             PortList,
40         ieee8021QBridgeForwardAllForbiddenPorts
41             PortList
42     }
43
44 ieee8021QBridgeForwardAllVlanIndex OBJECT-TYPE
45     SYNTAX      IEEE8021VlanIndexOrWildcard
46     MAX-ACCESS  not-accessible
47     STATUS      current
48     DESCRIPTION
49        "The VLAN-ID or other identifier referring to this VLAN."
50     ::= { ieee8021QBridgeForwardAllEntry 1 }
51
52 ieee8021QBridgeForwardAllPorts OBJECT-TYPE
53     SYNTAX      PortList
54     MAX-ACCESS  read-only
55     STATUS      current
56     DESCRIPTION
57        "The complete set of ports in this VLAN to which all
58        multicast group-addressed frames are to be forwarded.
59        This includes ports for which this need has been
60        determined dynamically by MMRP, or configured statically
61        by management."
62     ::= { ieee8021QBridgeForwardAllEntry 2 }
63
64 ieee8021QBridgeForwardAllStaticPorts OBJECT-TYPE
65     SYNTAX      PortList
66     MAX-ACCESS  read-write
67     STATUS      current
68     DESCRIPTION
69        "The set of ports configured by management in this VLAN
70        to which all multicast group-addressed frames are to be
71        forwarded. Ports entered in this list will also appear
72        in the complete set shown by ieee8021QBridgeForwardAllPorts. This

```

```

1      value will be restored after the device is reset. This
2      only applies to ports that are members of the VLAN,
3      defined by ieee8021QBridgeVlanCurrentEgressPorts. A port may not
4      be added in this set if it is already a member of the
5      set of ports in ieee8021QBridgeForwardAllForbiddenPorts. The
6      default value is a string of ones of appropriate length,
7      to indicate the standard behavior of using basic
8      filtering services, i.e., forward all multicasts to all
9      ports.
10
11      The value of this object MUST be retained across
12      reinitializations of the management system."
13      ::= { ieee8021QBridgeForwardAllEntry 3 }
14
15 ieee8021QBridgeForwardAllForbiddenPorts OBJECT-TYPE
16     SYNTAX      PortList
17     MAX-ACCESS  read-write
18     STATUS      current
19     DESCRIPTION
20         "The set of ports configured by management in this VLAN
21         for which the Service Requirement attribute Forward All
22         Multicast Groups may not be dynamically registered by
23         MMRP. This value will be restored after the device is
24         reset. A port may not be added in this set if it is
25         already a member of the set of ports in
26         ieee8021QBridgeForwardAllStaticPorts. The default value is a
27         string of zeros of appropriate length.
28
29         The value of this object MUST be retained across
30         reinitializations of the management system."
31     ::= { ieee8021QBridgeForwardAllEntry 4 }
32
33 ieee8021QBridgeForwardUnregisteredTable OBJECT-TYPE
34     SYNTAX      SEQUENCE OF Ieee8021QBridgeForwardUnregisteredEntry
35     MAX-ACCESS  not-accessible
36     STATUS      current
37     DESCRIPTION
38         "A table containing forwarding information for each
39         VLAN, specifying the set of ports to which forwarding of
40         multicast group-addressed frames for which no
41         more specific forwarding information applies. This is
42         configured statically by management and determined
43         dynamically by MMRP. An entry appears in this table for
44         all VLANs that are currently instantiated."
45     REFERENCE   "12.7.2, 12.7.7"
46     ::= { ieee8021QBridgeTp 5 }
47
48 ieee8021QBridgeForwardUnregisteredEntry OBJECT-TYPE
49     SYNTAX      Ieee8021QBridgeForwardUnregisteredEntry
50     MAX-ACCESS  not-accessible
51     STATUS      current
52     DESCRIPTION
53         "Forwarding information for a VLAN, specifying the set
54         of ports to which all multicasts for which there is no
55         more specific forwarding information shall be forwarded.
56         This is configured statically by management or
57         dynamically by MMRP."
58     INDEX      { ieee8021QBridgeVlanCurrentComponentId,
59                  ieee8021QBridgeForwardUnregisteredVlanIndex }
60     ::= { ieee8021QBridgeForwardUnregisteredTable 1 }
61
62 Ieee8021QBridgeForwardUnregisteredEntry ::=
63     SEQUENCE {
64         ieee8021QBridgeForwardUnregisteredVlanIndex
65         IEEE8021VlanIndexOrWildcard,
66         ieee8021QBridgeForwardUnregisteredPorts
67         PortList,
68         ieee8021QBridgeForwardUnregisteredStaticPorts
69         PortList,
70         ieee8021QBridgeForwardUnregisteredForbiddenPorts
71         PortList
72     }

```

```

1
2 ieee8021QBridgeForwardUnregisteredVlanIndex OBJECT-TYPE
3   SYNTAX      IEEE8021VlanIndexOrWildcard
4   MAX-ACCESS  not-accessible
5   STATUS      current
6   DESCRIPTION
7       "The VLAN-ID or other identifier referring to this VLAN."
8   ::= { ieee8021QBridgeForwardUnregisteredEntry 1 }
9
10 ieee8021QBridgeForwardUnregisteredPorts OBJECT-TYPE
11   SYNTAX      PortList
12   MAX-ACCESS  read-only
13   STATUS      current
14   DESCRIPTION
15       "The complete set of ports in this VLAN to which
16       multicast group-addressed frames for which there is no
17       more specific forwarding information will be forwarded.
18       This includes ports for which this need has been
19       determined dynamically by MMRP, or configured statically
20       by management."
21   ::= { ieee8021QBridgeForwardUnregisteredEntry 2 }
22
23 ieee8021QBridgeForwardUnregisteredStaticPorts OBJECT-TYPE
24   SYNTAX      PortList
25   MAX-ACCESS  read-write
26   STATUS      current
27   DESCRIPTION
28       "The set of ports configured by management, in this
29       VLAN, to which multicast group-addressed frames for
30       which there is no more specific forwarding information
31       are to be forwarded. Ports entered in this list will
32       also appear in the complete set shown by
33       ieee8021QBridgeForwardUnregisteredPorts. This value will be
34       restored after the device is reset. A port may not be
35       added in this set if it is already a member of the set
36       of ports in ieee8021QBridgeForwardUnregisteredForbiddenPorts. The
37       default value is a string of zeros of appropriate
38       length, although this has no effect with the default
39       value of ieee8021QBridgeForwardAllStaticPorts.
40
41       The value of this object MUST be retained across
42       reinitializations of the management system."
43   ::= { ieee8021QBridgeForwardUnregisteredEntry 3 }
44
45 ieee8021QBridgeForwardUnregisteredForbiddenPorts OBJECT-TYPE
46   SYNTAX      PortList
47   MAX-ACCESS  read-write
48   STATUS      current
49   DESCRIPTION
50       "The set of ports configured by management in this VLAN
51       for which the Service Requirement attribute Forward
52       Unregistered Multicast Groups may not be dynamically
53       registered by MMRP. This value will be restored after
54       the device is reset. A port may not be added in this
55       set if it is already a member of the set of ports in
56       ieee8021QBridgeForwardUnregisteredStaticPorts. The default value
57       is a string of zeros of appropriate length.
58
59       The value of this object MUST be retained across
60       reinitializations of the management system."
61   ::= { ieee8021QBridgeForwardUnregisteredEntry 4 }
62
63 -- =====
64 -- The Static (Destination-Address Filtering) Database
65 -- =====
66
67 ieee8021QBridgeStaticUnicastTable OBJECT-TYPE
68   SYNTAX      SEQUENCE OF Ieee8021QBridgeStaticUnicastEntry
69   MAX-ACCESS  not-accessible
70   STATUS      current
71   DESCRIPTION
72       "A table containing filtering information for Unicast

```

```

1      MAC addresses for each Filtering Database, configured
2      into the device by (local or network) management
3      specifying the set of ports to which frames received
4      from specific ports and containing specific unicast
5      destination addresses are allowed to be forwarded.
6      Entries are valid for unicast addresses only.
7
8      Two modes of operation are supported by this table. When
9      the receive port index is non-zero, this table is
10     supporting an IEEE 802.1D filtering database as specified in
11     14.7.6.1 of IEEE Std 802.1D. If the receive port is zero,
12     the table is operating as specified in IEEE Std 802.1Q
13     8.8.1 and 12.7.7. An agent must at least
14     support the IEEE Std 802.1Q mode of operation."
15     REFERENCE      "8.8.1, 12.7.7;
16                   IEEE Std 802.1D 7.9.1, 14.7.6.1"
17     ::= { ieee8021QBridgeStatic 1 }
18
19 ieee8021QBridgeStaticUnicastEntry OBJECT-TYPE
20     SYNTAX          Ieee8021QBridgeStaticUnicastEntry
21     MAX-ACCESS      not-accessible
22     STATUS          current
23     DESCRIPTION
24         "Filtering information configured into the device by
25         (local or network) management specifying the set of
26         ports to which frames received from a specific port and
27         containing a specific unicast destination address are
28         allowed to be forwarded."
29     INDEX           {
30         ieee8021QBridgeStaticUnicastComponentId,
31         ieee8021QBridgeStaticUnicastVlanIndex,
32         ieee8021QBridgeStaticUnicastAddress,
33         ieee8021QBridgeStaticUnicastReceivePort
34     }
35     ::= { ieee8021QBridgeStaticUnicastTable 1 }
36
37 Ieee8021QBridgeStaticUnicastEntry ::=
38     SEQUENCE {
39         ieee8021QBridgeStaticUnicastComponentId
40             IEEE8021PbbComponentIdentifier,
41         ieee8021QBridgeStaticUnicastVlanIndex
42             IEEE8021VlanIndexOrWildcard,
43         ieee8021QBridgeStaticUnicastAddress
44             MacAddress,
45         ieee8021QBridgeStaticUnicastReceivePort
46             IEEE8021BridgePortNumberOrZero,
47         ieee8021QBridgeStaticUnicastStaticEgressPorts
48             PortList,
49         ieee8021QBridgeStaticUnicastForbiddenEgressPorts
50             PortList,
51         ieee8021QBridgeStaticUnicastStorageType
52             StorageType,
53         ieee8021QBridgeStaticUnicastRowStatus
54             RowStatus
55     }
56
57 ieee8021QBridgeStaticUnicastComponentId OBJECT-TYPE
58     SYNTAX          IEEE8021PbbComponentIdentifier
59     MAX-ACCESS      not-accessible
60     STATUS          current
61     DESCRIPTION
62         "The component identifier is used to distinguish between the
63         multiple virtual Bridge instances within a PBB. In simple
64         situations where there is only a single component the default
65         value is 1."
66     ::= { ieee8021QBridgeStaticUnicastEntry 1 }
67
68 ieee8021QBridgeStaticUnicastVlanIndex OBJECT-TYPE
69     SYNTAX          IEEE8021VlanIndexOrWildcard
70     MAX-ACCESS      not-accessible
71     STATUS          current
72     DESCRIPTION

```

```

1      "The Vlan to which this entry applies."
2      ::= { ieee8021QBridgeStaticUnicastEntry 2 }
3
4 ieee8021QBridgeStaticUnicastAddress OBJECT-TYPE
5     SYNTAX      MacAddress
6     MAX-ACCESS  not-accessible
7     STATUS      current
8     DESCRIPTION
9         "The destination MAC address in a frame to which this
10        entry's filtering information applies. This object must
11        take the value of a unicast address."
12     ::= { ieee8021QBridgeStaticUnicastEntry 3 }
13
14 ieee8021QBridgeStaticUnicastReceivePort OBJECT-TYPE
15     SYNTAX      IEEE8021BridgePortNumberOrZero
16     MAX-ACCESS  not-accessible
17     STATUS      current
18     DESCRIPTION
19         "Either the value '0' or the port number of the port
20        from which a frame must be received in order for this
21        entry's filtering information to apply. A value of zero
22        indicates that this entry applies on all ports of the
23        device for which there is no other applicable entry. An
24        implementation is required to support the '0' value and
25        may optionally support non-zero values for this column."
26     ::= { ieee8021QBridgeStaticUnicastEntry 4 }
27
28 ieee8021QBridgeStaticUnicastStaticEgressPorts OBJECT-TYPE
29     SYNTAX      PortList
30     MAX-ACCESS  read-create
31     STATUS      current
32     DESCRIPTION
33         "The set of ports to which frames received from a
34        specific port and destined for a specific unicast address
35        must be forwarded, regardless of
36        any dynamic information, e.g., from MMRP. A port may not
37        be added in this set if it is already a member of the
38        set of ports in ieee8021QBridgeStaticUnicastForbiddenEgressPorts.
39        The default value of this object is a string of ones of
40        appropriate length."
41     DEFVAL      { 'H' }
42     ::= { ieee8021QBridgeStaticUnicastEntry 5 }
43
44
45 ieee8021QBridgeStaticUnicastForbiddenEgressPorts OBJECT-TYPE
46     SYNTAX      PortList
47     MAX-ACCESS  read-create
48     STATUS      current
49     DESCRIPTION
50         "The set of ports to which frames received from a
51        specific port and destined for a specific unicast
52        MAC address must not be forwarded, regardless
53        of any dynamic information, e.g., from MMRP. A port may
54        not be added in this set if it is already a member of the
55        set of ports in ieee8021QBridgeStaticUnicastStaticEgressPorts.
56        The default value of this object is a string of zeros of
57        appropriate length."
58     DEFVAL      { 'H' }
59     ::= { ieee8021QBridgeStaticUnicastEntry 6 }
60
61 ieee8021QBridgeStaticUnicastStorageType OBJECT-TYPE
62     SYNTAX      StorageType
63     MAX-ACCESS  read-create
64     STATUS      current
65     DESCRIPTION
66         "The storage type for this conceptual row. If this object
67        has a value of permanent(4), then no other objects are
68        required to be able to be modified."
69     DEFVAL      { nonVolatile }
70     ::= { ieee8021QBridgeStaticUnicastEntry 7 }
71
72 ieee8021QBridgeStaticUnicastRowStatus OBJECT-TYPE

```

```

1  SYNTAX      RowStatus
2  MAX-ACCESS  read-create
3  STATUS      current
4  DESCRIPTION
5      "This object indicates the status of this entry, and is used
6      to create/delete entries in the table.
7
8      An entry of this table may be set to active without setting
9      any other columns of the table. Also, other columns of this
10     table may be set while the value of this object is active(1)."
11 ::= { ieee8021QBridgeStaticUnicastEntry 8 }
12
13 ieee8021QBridgeStaticMulticastTable OBJECT-TYPE
14     SYNTAX      SEQUENCE OF Ieee8021QBridgeStaticMulticastEntry
15     MAX-ACCESS  not-accessible
16     STATUS      current
17     DESCRIPTION
18         "A table containing filtering information for Multicast
19         and Broadcast MAC addresses for each VLAN, configured
20         into the device by (local or network) management
21         specifying the set of ports to which frames received
22         from specific ports and containing specific Multicast
23         and Broadcast destination addresses are allowed to be
24         forwarded. A value of zero in this table (as the port
25         number from which frames with a specific destination
26         address are received) is used to specify all ports for
27         which there is no specific entry in this table for that
28         particular destination address. Entries are valid for
29         Multicast and Broadcast addresses only."
30     REFERENCE   "12.7.7, 8.8.1"
31     ::= { ieee8021QBridgeStatic 2 }
32
33 ieee8021QBridgeStaticMulticastEntry OBJECT-TYPE
34     SYNTAX      Ieee8021QBridgeStaticMulticastEntry
35     MAX-ACCESS  not-accessible
36     STATUS      current
37     DESCRIPTION
38         "Filtering information configured into the device by
39         (local or network) management specifying the set of
40         ports to which frames received from this specific port
41         for this VLAN and containing this Multicast or Broadcast
42         destination address are allowed to be forwarded."
43     INDEX      {
44         ieee8021QBridgeVlanCurrentComponentId,
45         ieee8021QBridgeVlanIndex,
46         ieee8021QBridgeStaticMulticastAddress,
47         ieee8021QBridgeStaticMulticastReceivePort
48     }
49     ::= { ieee8021QBridgeStaticMulticastTable 1 }
50
51 Ieee8021QBridgeStaticMulticastEntry ::=
52     SEQUENCE {
53         ieee8021QBridgeStaticMulticastAddress
54             MacAddress,
55         ieee8021QBridgeStaticMulticastReceivePort
56             IEEE8021BridgePortNumberOrZero,
57         ieee8021QBridgeStaticMulticastStaticEgressPorts
58             PortList,
59         ieee8021QBridgeStaticMulticastForbiddenEgressPorts
60             PortList,
61         ieee8021QBridgeStaticMulticastStorageType
62             StorageType,
63         ieee8021QBridgeStaticMulticastRowStatus
64             RowStatus
65     }
66
67 ieee8021QBridgeStaticMulticastAddress OBJECT-TYPE
68     SYNTAX      MacAddress
69     MAX-ACCESS  not-accessible
70     STATUS      current
71     DESCRIPTION
72         "The destination MAC address in a frame to which this

```



```

1      entry's filtering information applies. This object must
2      take the value of a Multicast or Broadcast address."
3      ::= { ieee8021QBridgeStaticMulticastEntry 1 }
4
5 ieee8021QBridgeStaticMulticastReceivePort OBJECT-TYPE
6     SYNTAX      IEEE8021BridgePortNumberOrZero
7     MAX-ACCESS  not-accessible
8     STATUS      current
9     DESCRIPTION
10        "Either the value '0' or the port number of the port
11        from which a frame must be received in order for this
12        entry's filtering information to apply. A value of zero
13        indicates that this entry applies on all ports of the
14        device for which there is no other applicable entry. An
15        implementation is required to support the '0' value and
16        may optionally support non-zero values for this column."
17        ::= { ieee8021QBridgeStaticMulticastEntry 2 }
18
19 ieee8021QBridgeStaticMulticastStaticEgressPorts OBJECT-TYPE
20     SYNTAX      PortList
21     MAX-ACCESS  read-create
22     STATUS      current
23     DESCRIPTION
24        "The set of ports to which frames received from a
25        specific port and destined for a specific Multicast or
26        Broadcast MAC address must be forwarded, regardless of
27        any dynamic information, e.g., from MMRP. A port may not
28        be added in this set if it is already a member of the
29        set of ports in ieee8021QBridgeStaticMulticastForbiddenEgressPorts.
30        The default value of this object is a string of ones of
31        appropriate length."
32     DEFVAL      { 'H' }
33     ::= { ieee8021QBridgeStaticMulticastEntry 3 }
34
35 ieee8021QBridgeStaticMulticastForbiddenEgressPorts OBJECT-TYPE
36     SYNTAX      PortList
37     MAX-ACCESS  read-create
38     STATUS      current
39     DESCRIPTION
40        "The set of ports to which frames received from a
41        specific port and destined for a specific Multicast or
42        Broadcast MAC address must not be forwarded, regardless
43        of any dynamic information, e.g., from MMRP. A port may
44        not be added in this set if it is already a member of the
45        set of ports in ieee8021QBridgeStaticMulticastStaticEgressPorts.
46        The default value of this object is a string of zeros of
47        appropriate length."
48     DEFVAL      { 'H' }
49     ::= { ieee8021QBridgeStaticMulticastEntry 4 }
50
51 ieee8021QBridgeStaticMulticastStorageType OBJECT-TYPE
52     SYNTAX      StorageType
53     MAX-ACCESS  read-create
54     STATUS      current
55     DESCRIPTION
56        "The storage type for this conceptual row. If this object
57        has a value of permanent(4), then no other objects are
58        required to be able to be modified."
59     DEFVAL      { nonVolatile }
60     ::= { ieee8021QBridgeStaticMulticastEntry 5 }
61
62 ieee8021QBridgeStaticMulticastRowStatus OBJECT-TYPE
63     SYNTAX      RowStatus
64     MAX-ACCESS  read-create
65     STATUS      current
66     DESCRIPTION
67        "This object indicates the status of this entry, and is used
68        to create/delete entries in the table.
69
70        An entry of this table may be set to active without setting
71        any other columns of the table. Also, other columns of this
72        table may be set while the value of this object is active(1)."
```

```

1 ::= { ieee8021QBridgeStaticMulticastEntry 6 }
2
3 -- =====
4 -- The Current VLAN Database
5 -- =====
6
7 ieee8021QBridgeVlanNumDeletes OBJECT-TYPE
8     SYNTAX      Counter64
9     UNITS        "vlan deletions"
10    MAX-ACCESS   read-only
11    STATUS        current
12    DESCRIPTION
13        "The number of times a VLAN entry has been deleted from
14        the ieee8021QBridgeVlanCurrentTable (for any reason).
15        If an entry is deleted, then inserted, and then deleted,
16        this counter will be incremented by 2. Discontinuities
17        in this value can only occur at a reboot."
18 ::= { ieee8021QBridgeVlan 1 }
19
20 ieee8021QBridgeVlanCurrentTable OBJECT-TYPE
21     SYNTAX      SEQUENCE OF Ieee8021QBridgeVlanCurrentEntry
22     MAX-ACCESS   not-accessible
23     STATUS        current
24     DESCRIPTION
25         "A table containing current configuration information
26         for each VLAN currently configured into the device by
27         (local or network) management, or dynamically created
28         as a result of MVRP requests received."
29     REFERENCE    "12.10.2"
30 ::= { ieee8021QBridgeVlan 2 }
31
32 ieee8021QBridgeVlanCurrentEntry OBJECT-TYPE
33     SYNTAX      Ieee8021QBridgeVlanCurrentEntry
34     MAX-ACCESS   not-accessible
35     STATUS        current
36     DESCRIPTION
37         "Information for a VLAN configured into the device by
38         (local or network) management, or dynamically created
39         as a result of MVRP requests received."
40     INDEX        { ieee8021QBridgeVlanTimeMark,
41                    ieee8021QBridgeVlanCurrentComponentId,
42                    ieee8021QBridgeVlanIndex }
43 ::= { ieee8021QBridgeVlanCurrentTable 1 }
44
45 Ieee8021QBridgeVlanCurrentEntry ::=
46     SEQUENCE {
47         ieee8021QBridgeVlanTimeMark
48         TimeFilter,
49         ieee8021QBridgeVlanCurrentComponentId
50         IEEE8021PbbComponentIdentifier,
51         ieee8021QBridgeVlanIndex
52         IEEE8021VlanIndex,
53         ieee8021QBridgeVlanFdbId
54         Unsigned32,
55         ieee8021QBridgeVlanCurrentEgressPorts
56         PortList,
57         ieee8021QBridgeVlanCurrentUntaggedPorts
58         PortList,
59         ieee8021QBridgeVlanStatus
60         INTEGER,
61         ieee8021QBridgeVlanCreationTime
62         TimeTicks
63     }
64
65 ieee8021QBridgeVlanTimeMark OBJECT-TYPE
66     SYNTAX      TimeFilter
67     MAX-ACCESS   not-accessible
68     STATUS        current
69     DESCRIPTION
70         "A TimeFilter for this entry. See the TimeFilter
71         textual convention to see how this works."
72 ::= { ieee8021QBridgeVlanCurrentEntry 1 }

```

```

1
2 ieee8021QBridgeVlanCurrentComponentId OBJECT-TYPE
3     SYNTAX      IEEE8021PbbComponentIdentifier
4     MAX-ACCESS  not-accessible
5     STATUS      current
6     DESCRIPTION
7         "The component identifier is used to distinguish between the
8         multiple virtual Bridge instances within a PBB. In simple
9         situations where there is only a single component the default
10        value is 1."
11     ::= { ieee8021QBridgeVlanCurrentEntry 2 }
12
13 ieee8021QBridgeVlanIndex OBJECT-TYPE
14     SYNTAX      IEEE8021VlanIndex
15     MAX-ACCESS  not-accessible
16     STATUS      current
17     DESCRIPTION
18         "The VLAN-ID or other identifier referring to this VLAN."
19     ::= { ieee8021QBridgeVlanCurrentEntry 3 }
20
21 ieee8021QBridgeVlanFdbId OBJECT-TYPE
22     SYNTAX      Unsigned32
23     MAX-ACCESS  read-only
24     STATUS      current
25     DESCRIPTION
26         "The Filtering Database used by this VLAN. This is one
27         of the ieee8021QBridgeFdbId values in the ieee8021QBridgeFdbTable.
28         This value is allocated automatically by the device whenever
29         the VLAN is created: either dynamically by MVRP, or by
30         management, in ieee8021QBridgeVlanStaticTable. Allocation of this
31         value follows the learning constraints defined for this
32         VLAN in ieee8021QBridgeLearningConstraintsTable."
33     ::= { ieee8021QBridgeVlanCurrentEntry 4 }
34
35 ieee8021QBridgeVlanCurrentEgressPorts OBJECT-TYPE
36     SYNTAX      PortList
37     MAX-ACCESS  read-only
38     STATUS      current
39     DESCRIPTION
40         "The set of ports that are transmitting traffic for
41         this VLAN as either tagged or untagged frames."
42     REFERENCE   "12.10.2.1"
43     ::= { ieee8021QBridgeVlanCurrentEntry 5 }
44
45 ieee8021QBridgeVlanCurrentUntaggedPorts OBJECT-TYPE
46     SYNTAX      PortList
47     MAX-ACCESS  read-only
48     STATUS      current
49     DESCRIPTION
50         "The set of ports that are transmitting traffic for
51         this VLAN as untagged frames."
52     REFERENCE   "12.10.2.1"
53     ::= { ieee8021QBridgeVlanCurrentEntry 6 }
54
55 ieee8021QBridgeVlanStatus OBJECT-TYPE
56     SYNTAX      INTEGER {
57         other(1),
58         permanent(2),
59         dynamicMvrp(3)
60     }
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64         "This object indicates the status of this entry.
65         other(1) - this entry is currently in use, but the
66         conditions under which it will remain so differ
67         from the following values.
68         permanent(2) - this entry, corresponding to an entry
69         in ieee8021QBridgeVlanStaticTable, is currently in use and
70         will remain so after the next reset of the
71         device. The port lists for this entry include
72         ports from the equivalent ieee8021QBridgeVlanStaticTable

```

```

1         entry and ports learned dynamically.
2     dynamicMvrp(3) - this entry is currently in use
3         and will remain so until removed by MVRP. There
4         is no static entry for this VLAN, and it will be
5         removed when the last port leaves the VLAN."
6     ::= { ieee8021QBridgeVlanCurrentEntry 7 }
7
8 ieee8021QBridgeVlanCreationTime OBJECT-TYPE
9     SYNTAX      TimeTicks
10    MAX-ACCESS   read-only
11    STATUS       current
12    DESCRIPTION
13        "The value of sysUpTime when this VLAN was created."
14    ::= { ieee8021QBridgeVlanCurrentEntry 8 }
15
16 -- =====
17 -- The Static VLAN Database
18 -- =====
19
20 ieee8021QBridgeVlanStaticTable OBJECT-TYPE
21     SYNTAX      SEQUENCE OF Ieee8021QBridgeVlanStaticEntry
22     MAX-ACCESS   not-accessible
23     STATUS       current
24     DESCRIPTION
25         "A table containing static configuration information for
26         each VLAN configured into the device by (local or
27         network) management. All entries are persistent and will
28         be restored after the device is reset."
29     REFERENCE    "12.7.5"
30     ::= { ieee8021QBridgeVlan 3 }
31
32 ieee8021QBridgeVlanStaticEntry OBJECT-TYPE
33     SYNTAX      Ieee8021QBridgeVlanStaticEntry
34     MAX-ACCESS   not-accessible
35     STATUS       current
36     DESCRIPTION
37         "Static information for a VLAN configured into the
38         device by (local or network) management."
39     INDEX        { ieee8021QBridgeVlanStaticComponentId,
40                   ieee8021QBridgeVlanStaticVlanIndex }
41     ::= { ieee8021QBridgeVlanStaticTable 1 }
42
43 Ieee8021QBridgeVlanStaticEntry ::=
44     SEQUENCE {
45         ieee8021QBridgeVlanStaticComponentId
46             IEEE8021PbbComponentIdentifier,
47         ieee8021QBridgeVlanStaticVlanIndex
48             IEEE8021VlanIndex,
49         ieee8021QBridgeVlanStaticName
50             SnmpAdminString,
51         ieee8021QBridgeVlanStaticEgressPorts
52             PortList,
53         ieee8021QBridgeVlanForbiddenEgressPorts
54             PortList,
55         ieee8021QBridgeVlanStaticUntaggedPorts
56             PortList,
57         ieee8021QBridgeVlanStaticRowStatus
58             RowStatus
59     }
60
61 ieee8021QBridgeVlanStaticComponentId OBJECT-TYPE
62     SYNTAX      IEEE8021PbbComponentIdentifier
63     MAX-ACCESS   not-accessible
64     STATUS       current
65     DESCRIPTION
66         "The component identifier is used to distinguish between the
67         multiple virtual Bridge instances within a PBB. In simple
68         situations where there is only a single component the default
69         value is 1."
70     ::= { ieee8021QBridgeVlanStaticEntry 1 }
71
72 ieee8021QBridgeVlanStaticVlanIndex OBJECT-TYPE

```

```

1  SYNTAX      IEEE8021VlanIndex
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "The VLAN-ID or other identifier referring to this VLAN."
6  ::= { ieee8021QBridgeVlanStaticEntry 2 }
7
8  ieee8021QBridgeVlanStaticName OBJECT-TYPE
9  SYNTAX      SnmpAdminString (SIZE (0..32))
10 MAX-ACCESS  read-create
11 STATUS      current
12 DESCRIPTION
13     "An administratively assigned string, which may be used
14     to identify the VLAN."
15 REFERENCE   "12.10.2.11"
16 ::= { ieee8021QBridgeVlanStaticEntry 3 }
17
18 ieee8021QBridgeVlanStaticEgressPorts OBJECT-TYPE
19 SYNTAX      PortList
20 MAX-ACCESS  read-create
21 STATUS      current
22 DESCRIPTION
23     "The set of ports that are permanently assigned to the
24     egress list for this VLAN by management. Changes to a
25     bit in this object affect the per-port, per-VLAN
26     Registrar control for Registration Fixed for the
27     relevant MVRP state machine on each port. A port may
28     not be added in this set if it is already a member of
29     the set of ports in ieee8021QBridgeVlanForbiddenEgressPorts. The
30     default value of this object is a string of zeros of
31     appropriate length, indicating not fixed."
32 REFERENCE   "12.7.7.3, 11.2.3.2.3"
33 ::= { ieee8021QBridgeVlanStaticEntry 4 }
34
35 ieee8021QBridgeVlanForbiddenEgressPorts OBJECT-TYPE
36 SYNTAX      PortList
37 MAX-ACCESS  read-create
38 STATUS      current
39 DESCRIPTION
40     "The set of ports that are prohibited by management
41     from being included in the egress list for this VLAN.
42     Changes to this object that cause a port to be included
43     or excluded affect the per-port, per-VLAN Registrar
44     control for Registration Forbidden for the relevant MVRP
45     state machine on each port. A port may not be added in
46     this set if it is already a member of the set of ports
47     in ieee8021QBridgeVlanStaticEgressPorts. The default value of
48     this object is a string of zeros of appropriate length,
49     excluding all ports from the forbidden set."
50 REFERENCE   "12.7.7.3, 11.2.3.2.3"
51 ::= { ieee8021QBridgeVlanStaticEntry 5 }
52
53 ieee8021QBridgeVlanStaticUntaggedPorts OBJECT-TYPE
54 SYNTAX      PortList
55 MAX-ACCESS  read-create
56 STATUS      current
57 DESCRIPTION
58     "The set of ports that should transmit egress frames
59     for this VLAN as untagged. The default value of this
60     object for the default VLAN (ieee8021QBridgeVlanIndex = 1) is a string
61     of appropriate length including all ports. There is no
62     specified default for other VLANs. If a device agent cannot
63     support the set of ports being set, then it will reject the
64     set operation with an error. For example, a
65     manager might attempt to set more than one VLAN to be untagged
66     on egress where the device does not support this IEEE 802.1Q
67     option."
68 REFERENCE   "12.10.2.1"
69 ::= { ieee8021QBridgeVlanStaticEntry 6 }
70
71 ieee8021QBridgeVlanStaticRowStatus OBJECT-TYPE
72 SYNTAX      RowStatus

```

```

1  MAX-ACCESS  read-create
2  STATUS      current
3  DESCRIPTION
4      "This object indicates the status of this entry, and is used
5      to create/delete entries. Any object in an entry of this table
6      may be modified while the value of the corresponding instance
7      of this object is active(1)."
```

8 ::= { ieee8021QBridgeVlanStaticEntry 7 }

```

9
10 ieee8021QBridgeNextFreeLocalVlanTable OBJECT-TYPE
11 SYNTAX      SEQUENCE OF Ieee8021QBridgeNextFreeLocalVlanEntry
12 MAX-ACCESS  not-accessible
13 STATUS      current
14 DESCRIPTION
15     "A table that contains information about the next free VLAN
16     value for a statically configured VLAN Bridge."
17 ::= { ieee8021QBridgeVlan 4 }
```

```

18
19 ieee8021QBridgeNextFreeLocalVlanEntry OBJECT-TYPE
20 SYNTAX      Ieee8021QBridgeNextFreeLocalVlanEntry
21 MAX-ACCESS  not-accessible
22 STATUS      current
23 DESCRIPTION
24     "The next free VLAN value for a statically configured VLAN Bridge"
```

25 INDEX { ieee8021QBridgeNextFreeLocalVlanComponentId }

```

26 ::= { ieee8021QBridgeNextFreeLocalVlanTable 1 }
```

```

27
28 Ieee8021QBridgeNextFreeLocalVlanEntry ::=
29     SEQUENCE {
30         ieee8021QBridgeNextFreeLocalVlanComponentId
31         IEEE8021PbbComponentIdentifier,
32         ieee8021QBridgeNextFreeLocalVlanIndex
33         Unsigned32
34     }
35
36 ieee8021QBridgeNextFreeLocalVlanComponentId OBJECT-TYPE
37 SYNTAX      IEEE8021PbbComponentIdentifier
38 MAX-ACCESS  not-accessible
39 STATUS      current
40 DESCRIPTION
41     "The component identifier is used to distinguish between the
42     multiple virtual Bridge instances within a PBB. In simple
43     situations where there is only a single component the default
44     value is 1."
```

45 ::= { ieee8021QBridgeNextFreeLocalVlanEntry 1 }

```

46
47 ieee8021QBridgeNextFreeLocalVlanIndex OBJECT-TYPE
48 SYNTAX      Unsigned32 (0|4096..4294967295)
49 MAX-ACCESS  read-only
50 STATUS      current
51 DESCRIPTION
52     "The next available value for ieee8021QBridgeVlanIndex of a local
53     VLAN entry in ieee8021QBridgeVlanStaticTable. This will report
54     values >=4096 if a new Local VLAN may be created or else
55     the value 0 if this is not possible."
56
57     A row creation operation in this table for an entry with a local
58     VlanIndex value may fail if the current value of this object
59     is not used as the index. Even if the value read is used,
60     there is no guarantee that it will still be the valid index
61     when the create operation is attempted; another manager may
62     have already got in during the intervening time interval.
63     In this case, ieee8021QBridgeNextFreeLocalVlanIndex should be re-read
64     and the creation re-tried with the new value."
65
66     This value will automatically change when the current value is
67     used to create a new row."
```

68 ::= { ieee8021QBridgeNextFreeLocalVlanEntry 2 }

```

69
70 -- =====
71 -- The VLAN Port Configuration Table
72 -- =====
```

```

1
2 ieee8021QBridgePortVlanTable OBJECT-TYPE
3   SYNTAX      SEQUENCE OF Ieee8021QBridgePortVlanEntry
4   MAX-ACCESS  not-accessible
5   STATUS      current
6   DESCRIPTION
7       "A table containing per-port control and status
8       information for VLAN configuration in the device."
9   REFERENCE   "12.10.1"
10  ::= { ieee8021QBridgeVlan 5 }
11
12 ieee8021QBridgePortVlanEntry OBJECT-TYPE
13   SYNTAX      Ieee8021QBridgePortVlanEntry
14   MAX-ACCESS  not-accessible
15   STATUS      current
16   DESCRIPTION
17       "Information controlling VLAN configuration for a port
18       on the device. This is indexed by ieee8021BridgeBasePort."
19   AUGMENTS { ieee8021BridgeBasePortEntry }
20  ::= { ieee8021QBridgePortVlanTable 1 }
21
22 Ieee8021QBridgePortVlanEntry ::=
23   SEQUENCE {
24       ieee8021QBridgePvid
25       IEEE8021VlanIndex,
26       ieee8021QBridgePortAcceptableFrameTypes
27       IEEE8021PortAcceptableFrameTypes,
28       ieee8021QBridgePortIngressFiltering
29       TruthValue,
30       ieee8021QBridgePortMvrpEnabledStatus
31       TruthValue,
32       ieee8021QBridgePortMvrpFailedRegistrations
33       Counter64,
34       ieee8021QBridgePortMvrpLastPduOrigin
35       MacAddress,
36       ieee8021QBridgePortRestrictedVlanRegistration
37       TruthValue
38   }
39
40 ieee8021QBridgePvid OBJECT-TYPE
41   SYNTAX      IEEE8021VlanIndex
42   MAX-ACCESS  read-write
43   STATUS      current
44   DESCRIPTION
45       "The PVID, the VLAN-ID assigned to untagged frames or
46       Priority-tagged frames received on this port.
47
48       The value of this object MUST be retained across
49       reinitializations of the management system."
50   REFERENCE   "12.10.1.1"
51   DEFVAL      { 1 }
52  ::= { ieee8021QBridgePortVlanEntry 1 }
53
54 ieee8021QBridgePortAcceptableFrameTypes OBJECT-TYPE
55   SYNTAX      IEEE8021PortAcceptableFrameTypes
56   MAX-ACCESS  read-write
57   STATUS      current
58   DESCRIPTION
59       "When this is admitTagged(3), the device will
60       discard untagged frames or Priority-tagged frames
61       received on this port. When admitAll(1), untagged
62       frames or Priority-tagged frames received on this port
63       will be accepted and assigned to a VID based on the
64       PVID and VID Set for this port.
65
66       This control does not affect VLAN-independent Bridge
67       Protocol Data Unit (BPDU) frames, such as MVRP and
68       Spanning Tree Protocol (STP). It does affect VLAN-
69       dependent BPDU frames, such as MMRP.
70
71       The value of this object MUST be retained across
72       reinitializations of the management system."

```

```

1  REFERENCE      "12.10.1.3"
2  DEFVAL         { admitAll }
3  ::= { ieee8021QBridgePortVlanEntry 2 }
4
5  ieee8021QBridgePortIngressFiltering OBJECT-TYPE
6      SYNTAX      TruthValue
7      MAX-ACCESS  read-write
8      STATUS      current
9      DESCRIPTION
10         "When this is true(1), the device will discard incoming
11         frames for VLANs that do not include this Port in its
12         Member set.  When false(2), the port will accept all
13         incoming frames.
14
15         This control does not affect VLAN-independent BPDU
16         frames, such as MVRP and STP.  It does affect VLAN-
17         dependent BPDU frames, such as MMRP.
18
19         The value of this object MUST be retained across
20         reinitializations of the management system."
21  REFERENCE      "12.10.1.4"
22  DEFVAL         { false }
23  ::= { ieee8021QBridgePortVlanEntry 3 }
24
25  ieee8021QBridgePortMvrpEnabledStatus OBJECT-TYPE
26      SYNTAX      TruthValue
27      MAX-ACCESS  read-write
28      STATUS      current
29      DESCRIPTION
30         "The state of MVRP operation on this port.  The value
31         true(1) indicates that MVRP is enabled on this port,
32         as long as ieee8021QBridgeMvrpEnabledStatus is also enabled
33         for this device.  When false(2) but
34         ieee8021QBridgeMvrpEnabledStatus is still
35         enabled for the device, MVRP is disabled on this port:
36         any MVRP packets received will be silently discarded, and
37         no MVRP registrations will be propagated from other
38         ports.  This object affects all MVRP Applicant and
39         Registrar state machines on this port.  A transition
40         from false(2) to true(1) will cause a reset of all
41         MVRP state machines on this port.
42
43         The value of this object MUST be retained across
44         reinitializations of the management system."
45  DEFVAL         { true }
46  ::= { ieee8021QBridgePortVlanEntry 4 }
47
48  ieee8021QBridgePortMvrpFailedRegistrations OBJECT-TYPE
49      SYNTAX      Counter64
50      UNITS       "failed MVRP registrations"
51      MAX-ACCESS  read-only
52      STATUS      current
53      DESCRIPTION
54         "The total number of failed MVRP registrations, for any
55         reason, on this port.
56
57         Discontinuities in the value of the counter can occur
58         at re-initialization of the management system, and at
59         other times as indicated by the value of
60         ifCounterDiscontinuityTime object of the associated
61         interface (if any)."
```

```

62  ::= { ieee8021QBridgePortVlanEntry 5 }
63
64  ieee8021QBridgePortMvrpLastPduOrigin OBJECT-TYPE
65      SYNTAX      MacAddress
66      MAX-ACCESS  read-only
67      STATUS      current
68      DESCRIPTION
69         "The Source MAC Address of the last MVRP message
70         received on this port."
71  ::= { ieee8021QBridgePortVlanEntry 6 }
72

```



```

1 ieee8021QBridgePortRestrictedVlanRegistration OBJECT-TYPE
2     SYNTAX      TruthValue
3     MAX-ACCESS  read-write
4     STATUS      current
5     DESCRIPTION
6         "The state of Restricted VLAN Registration on this port.
7         If the value of this control is true(1), then creation
8         of a new dynamic VLAN entry is permitted only if there
9         is a Static VLAN Registration Entry for the VLAN concerned,
10        in which the Registrar Administrative Control value for
11        this port is Normal Registration.
12
13        The value of this object MUST be retained across
14        reinitializations of the management system."
15     REFERENCE   "11.2.3.2.3, 12.10.1.6."
16     DEFVAL      { false }
17     ::= { ieee8021QBridgePortVlanEntry 7 }
18
19 -- =====
20 -- Per port VLAN Statistics Table
21 -- =====
22
23 ieee8021QBridgePortVlanStatisticsTable OBJECT-TYPE
24     SYNTAX      SEQUENCE OF Ieee8021QBridgePortVlanStatisticsEntry
25     MAX-ACCESS  not-accessible
26     STATUS      current
27     DESCRIPTION
28         "A table containing per-port, per-VLAN statistics for
29         traffic received."
30     ::= { ieee8021QBridgeVlan 6 }
31
32 ieee8021QBridgePortVlanStatisticsEntry OBJECT-TYPE
33     SYNTAX      Ieee8021QBridgePortVlanStatisticsEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "Traffic statistics for a VLAN on an interface."
38     INDEX      { ieee8021BridgeBasePortComponentId,
39                 ieee8021BridgeBasePort,
40                 ieee8021QBridgeVlanIndex }
41     ::= { ieee8021QBridgePortVlanStatisticsTable 1 }
42
43 Ieee8021QBridgePortVlanStatisticsEntry ::=
44     SEQUENCE {
45         ieee8021QBridgeTpVlanPortInFrames
46             Counter64,
47         ieee8021QBridgeTpVlanPortOutFrames
48             Counter64,
49         ieee8021QBridgeTpVlanPortInDiscards
50             Counter64
51     }
52
53 ieee8021QBridgeTpVlanPortInFrames OBJECT-TYPE
54     SYNTAX      Counter64
55     UNITS        "frames"
56     MAX-ACCESS  read-only
57     STATUS      current
58     DESCRIPTION
59         "The number of valid frames received by this port from
60         its segment that were classified as belonging to this
61         VLAN. Note that a frame received on this port is
62         counted by this object if and only if it is for a
63         protocol being processed by the local forwarding process
64         for this VLAN. This object includes received Bridge
65         management frames classified as belonging to this VLAN
66         (e.g., MMRP, but not MVRP or STP.
67
68         Discontinuities in the value of the counter can occur
69         at re-initialization of the management system, and at
70         other times as indicated by the value of
71         ifCounterDiscontinuityTime object of the associated
72         interface (if any)."
```

```

1  REFERENCE    "12.6.1.1.3(a)"
2  ::= { ieee8021QBridgePortVlanStatisticsEntry 1 }
3
4  ieee8021QBridgeTpVlanPortOutFrames OBJECT-TYPE
5  SYNTAX      Counter64
6  UNITS       "frames"
7  MAX-ACCESS  read-only
8  STATUS      current
9  DESCRIPTION
10     "The number of valid frames transmitted by this port to
11     its segment from the local forwarding process for this
12     VLAN. This includes Bridge management frames originated
13     by this device that are classified as belonging to this
14     VLAN (e.g., MMRP, but not MVRP or STP).
15
16     Discontinuities in the value of the counter can occur
17     at re-initialization of the management system, and at
18     other times as indicated by the value of
19     ifCounterDiscontinuityTime object of the associated
20     interface (if any)."
```

```

21  REFERENCE    "12.6.1.1.3(d)"
22  ::= { ieee8021QBridgePortVlanStatisticsEntry 2 }
23
24  ieee8021QBridgeTpVlanPortInDiscards OBJECT-TYPE
25  SYNTAX      Counter64
26  UNITS       "frames"
27  MAX-ACCESS  read-only
28  STATUS      current
29  DESCRIPTION
30     "The number of valid frames received by this port from
31     its segment that were classified as belonging to this
32     VLAN and that were discarded due to VLAN-related reasons.
33     Specifically, the IEEE 802.1Q counters for Discard
34     Inbound and Discard on Ingress Filtering.
35
36     Discontinuities in the value of the counter can occur
37     at re-initialization of the management system, and at
38     other times as indicated by the value of
39     ifCounterDiscontinuityTime object of the associated
40     interface (if any)."
```

```

41  REFERENCE    "12.6.1.1.3"
42  ::= { ieee8021QBridgePortVlanStatisticsEntry 3 }
43
44  -- =====
45  -- The VLAN Learning Constraints Table (now deprecated)
46  -- =====
47
48  ieee8021QBridgeLearningConstraintsTable OBJECT-TYPE
49  SYNTAX      SEQUENCE OF Ieee8021QBridgeLearningConstraintsEntry
50  MAX-ACCESS  not-accessible
51  STATUS      deprecated
52  DESCRIPTION
53     "A table containing learning constraints for sets of
54     Shared and Independent VLANs. Entries in this table are
55     persistent and are preserved across reboots."
56  REFERENCE    "12.10.3.1 of IEEE Std 802.1Q-2012"
57  ::= { ieee8021QBridgeVlan 8 }
58
59  ieee8021QBridgeLearningConstraintsEntry OBJECT-TYPE
60  SYNTAX      Ieee8021QBridgeLearningConstraintsEntry
61  MAX-ACCESS  not-accessible
62  STATUS      deprecated
63  DESCRIPTION
64     "A learning constraint defined for a VLAN."
65  INDEX      { ieee8021QBridgeLearningConstraintsComponentId,
66              ieee8021QBridgeLearningConstraintsVlan,
67              ieee8021QBridgeLearningConstraintsSet }
68  ::= { ieee8021QBridgeLearningConstraintsTable 1 }
69
70  Ieee8021QBridgeLearningConstraintsEntry ::=
71  SEQUENCE {
72      ieee8021QBridgeLearningConstraintsComponentId
```

```

1         IEEE8021PbbComponentIdentifier,
2         ieee8021QBridgeLearningConstraintsVlan
3         IEEE8021VlanIndex,
4         ieee8021QBridgeLearningConstraintsSet
5         Integer32,
6         ieee8021QBridgeLearningConstraintsType
7         INTEGER,
8         ieee8021QBridgeLearningConstraintsStatus
9         RowStatus
10    }
11
12 ieee8021QBridgeLearningConstraintsComponentId OBJECT-TYPE
13     SYNTAX      IEEE8021PbbComponentIdentifier
14     MAX-ACCESS  not-accessible
15     STATUS      deprecated
16     DESCRIPTION
17         "The component identifier is used to distinguish between the
18         multiple virtual Bridge instances within a PBB. In simple
19         situations where there is only a single component the default
20         value is 1."
21     ::= { ieee8021QBridgeLearningConstraintsEntry 1 }
22
23 ieee8021QBridgeLearningConstraintsVlan OBJECT-TYPE
24     SYNTAX      IEEE8021VlanIndex
25     MAX-ACCESS  not-accessible
26     STATUS      deprecated
27     DESCRIPTION
28         "The index of the row in ieee8021QBridgeVlanCurrentTable for the
29         VLAN constrained by this entry."
30     ::= { ieee8021QBridgeLearningConstraintsEntry 2 }
31
32 ieee8021QBridgeLearningConstraintsSet OBJECT-TYPE
33     SYNTAX      Integer32 (0..65535)
34     MAX-ACCESS  not-accessible
35     STATUS      deprecated
36     DESCRIPTION
37         "The identity of the constraint set to which
38         ieee8021QBridgeLearningConstraintsVlan belongs. These values may
39         be chosen by the management station."
40     ::= { ieee8021QBridgeLearningConstraintsEntry 3 }
41
42 ieee8021QBridgeLearningConstraintsType OBJECT-TYPE
43     SYNTAX      INTEGER {
44         independent(1),
45         shared(2)
46     }
47     MAX-ACCESS  read-create
48     STATUS      deprecated
49     DESCRIPTION
50         "The type of constraint this entry defines.
51         independent(1) - the VLAN, ieee8021QBridgeLearningConstraintsVlan,
52         uses a filtering database independent from all
53         other VLANs in the same set, defined by
54         ieee8021QBridgeLearningConstraintsSet.
55         shared(2) - the VLAN, ieee8021QBridgeLearningConstraintsVlan,
56         shares the same filtering database as all other VLANs
57         in the same set, defined by
58         ieee8021QBridgeLearningConstraintsSet."
59     ::= { ieee8021QBridgeLearningConstraintsEntry 4 }
60
61 ieee8021QBridgeLearningConstraintsStatus OBJECT-TYPE
62     SYNTAX      RowStatus
63     MAX-ACCESS  read-create
64     STATUS      deprecated
65     DESCRIPTION
66         "The status of this entry. Any object in an entry of this table
67         may be modified while the value of the corresponding instance
68         of this object is active(1)."
69     ::= { ieee8021QBridgeLearningConstraintsEntry 5 }
70
71 ieee8021QBridgeLearningConstraintDefaultsTable OBJECT-TYPE
72     SYNTAX      SEQUENCE OF Ieee8021QBridgeLearningConstraintDefaultsEntry

```

```

1  MAX-ACCESS    not-accessible
2  STATUS        deprecated
3  DESCRIPTION
4      "A table containing learning constraints for sets of
5      Shared and Independent VLANs."
6  REFERENCE     "12.10.3.1 of IEEE Std 802.1Q-2012"
7  ::= { ieee8021QBridgeVlan 9 }
8
9  ieee8021QBridgeLearningConstraintDefaultsEntry OBJECT-TYPE
10  SYNTAX         Ieee8021QBridgeLearningConstraintDefaultsEntry
11  MAX-ACCESS     not-accessible
12  STATUS         deprecated
13  DESCRIPTION
14      "A learning constraint defined for a VLAN."
15  INDEX          { ieee8021QBridgeLearningConstraintDefaultsComponentId }
16  ::= { ieee8021QBridgeLearningConstraintDefaultsTable 1 }
17
18  Ieee8021QBridgeLearningConstraintDefaultsEntry ::=
19      SEQUENCE {
20          ieee8021QBridgeLearningConstraintDefaultsComponentId
21          IEEE8021PbbComponentIdentifier,
22          ieee8021QBridgeLearningConstraintDefaultsSet
23          Integer32,
24          ieee8021QBridgeLearningConstraintDefaultsType
25          INTEGER
26      }
27
28  ieee8021QBridgeLearningConstraintDefaultsComponentId OBJECT-TYPE
29  SYNTAX         IEEE8021PbbComponentIdentifier
30  MAX-ACCESS     not-accessible
31  STATUS         deprecated
32  DESCRIPTION
33      "The component identifier is used to distinguish between the
34      multiple virtual Bridge instances within a PBB. In simple
35      situations where there is only a single component the default
36      value is 1."
37  ::= { ieee8021QBridgeLearningConstraintDefaultsEntry 1 }
38
39  ieee8021QBridgeLearningConstraintDefaultsSet OBJECT-TYPE
40  SYNTAX         Integer32 (0..65535)
41  MAX-ACCESS     read-write
42  STATUS         deprecated
43  DESCRIPTION
44      "The identity of the constraint set to which a VLAN
45      belongs, if there is not an explicit entry for that VLAN
46      in ieee8021QBridgeLearningConstraintsTable.
47
48      The value of this object MUST be retained across
49      reinitializations of the management system."
50  ::= { ieee8021QBridgeLearningConstraintDefaultsEntry 2 }
51
52  ieee8021QBridgeLearningConstraintDefaultsType OBJECT-TYPE
53  SYNTAX         INTEGER {
54              independent(1),
55              shared(2)
56          }
57  MAX-ACCESS     read-write
58  STATUS         deprecated
59  DESCRIPTION
60      "The type of constraint set to which a VLAN belongs, if
61      there is not an explicit entry for that VLAN in
62      ieee8021QBridgeLearningConstraintsTable. The types are as defined
63      for ieee8021QBridgeLearningConstraintsType.
64
65      The value of this object MUST be retained across
66      reinitializations of the management system."
67  ::= { ieee8021QBridgeLearningConstraintDefaultsEntry 3 }
68
69  -- =====
70  -- ieee8021QBridgeProtocol subtree
71  -- =====
72

```

```

1 ieee8021QBridgeProtocolGroupTable OBJECT-TYPE
2   SYNTAX      SEQUENCE OF Ieee8021QBridgeProtocolGroupEntry
3   MAX-ACCESS  not-accessible
4   STATUS      current
5   DESCRIPTION
6       "A table that contains mappings from Protocol
7       Templates to Protocol Group Identifiers used for
8       Port-and-Protocol-based VLAN Classification.
9
10      Entries in this table must be persistent over power
11      up restart/reboot."
12   REFERENCE  "12.10.1"
13   ::= { ieee8021QBridgeProtocol 1 }
14
15 ieee8021QBridgeProtocolGroupEntry OBJECT-TYPE
16   SYNTAX      Ieee8021QBridgeProtocolGroupEntry
17   MAX-ACCESS  not-accessible
18   STATUS      current
19   DESCRIPTION
20       "A mapping from a Protocol Template to a Protocol
21       Group Identifier."
22   REFERENCE  "12.10.1.1.3 d)"
23   INDEX      { ieee8021QBridgeProtocolGroupComponentId,
24               ieee8021QBridgeProtocolTemplateFrameType,
25               ieee8021QBridgeProtocolTemplateProtocolValue }
26   ::= { ieee8021QBridgeProtocolGroupTable 1 }
27
28 Ieee8021QBridgeProtocolGroupEntry ::=
29   SEQUENCE {
30       ieee8021QBridgeProtocolGroupComponentId
31       IEEE8021PbbComponentIdentifier,
32       ieee8021QBridgeProtocolTemplateFrameType
33       INTEGER,
34       ieee8021QBridgeProtocolTemplateProtocolValue
35       OCTET STRING,
36       ieee8021QBridgeProtocolGroupId
37       Integer32,
38       ieee8021QBridgeProtocolGroupRowStatus
39       RowStatus
40   }
41
42 ieee8021QBridgeProtocolGroupComponentId OBJECT-TYPE
43   SYNTAX      IEEE8021PbbComponentIdentifier
44   MAX-ACCESS  not-accessible
45   STATUS      current
46   DESCRIPTION
47       "The component identifier is used to distinguish between the
48       multiple virtual Bridge instances within a PBB. In simple
49       situations where there is only a single component the default
50       value is 1."
51   ::= { ieee8021QBridgeProtocolGroupEntry 1 }
52
53 ieee8021QBridgeProtocolTemplateFrameType OBJECT-TYPE
54   SYNTAX      INTEGER {
55       ethernet (1),
56       rfc1042 (2),
57       snap8021H (3),
58       snapOther (4),
59       llcOther (5)
60   }
61   MAX-ACCESS  not-accessible
62   STATUS      current
63   DESCRIPTION
64       "The data-link encapsulation format or the
65       'detagged_frame_type' in a Protocol Template."
66   REFERENCE  "12.10.1.8"
67   ::= { ieee8021QBridgeProtocolGroupEntry 2 }
68
69 ieee8021QBridgeProtocolTemplateProtocolValue OBJECT-TYPE
70   SYNTAX      OCTET STRING (SIZE (2 | 5))
71   MAX-ACCESS  not-accessible
72   STATUS      current

```

```

1  DESCRIPTION
2      "The identification of the protocol above the data-link
3      layer in a Protocol Template. Depending on the
4      frame type, the octet string will have one of the
5      following values:
6
7      For 'ethernet', 'rfc1042' and 'snap8021H',
8          this is the 16-bit (2-octet) IEEE 802.3 Type Field.
9      For 'snapOther',
10         this is the 40-bit (5-octet) PID.
11     For 'llcOther',
12         this is the 2-octet IEEE 802.2 Link Service Access
13         Point (LSAP) pair: first octet for Destination Service
14         Access Point (DSAP) and second octet for Source Service
15         Access Point (SSAP)."
```

16 REFERENCE "12.10.1.8"

17 ::= { ieee8021QBridgeProtocolGroupEntry 3 }

18

19 ieee8021QBridgeProtocolGroupId OBJECT-TYPE

20 SYNTAX Integer32 (0..2147483647)

21 MAX-ACCESS read-create

22 STATUS current

23 DESCRIPTION

24 "Represents a group of protocols that are associated

25 together when assigning a VID to a frame."

26 REFERENCE "12.10.1.8"

27 ::= { ieee8021QBridgeProtocolGroupEntry 4 }

28

29 ieee8021QBridgeProtocolGroupRowStatus OBJECT-TYPE

30 SYNTAX RowStatus

31 MAX-ACCESS read-create

32 STATUS current

33 DESCRIPTION

34 "This object indicates the status of this entry."

35 ::= { ieee8021QBridgeProtocolGroupEntry 5 }

36

37 ieee8021QBridgeProtocolPortTable OBJECT-TYPE

38 SYNTAX SEQUENCE OF Ieee8021QBridgeProtocolPortEntry

39 MAX-ACCESS not-accessible

40 STATUS current

41 DESCRIPTION

42 "A table that contains VID sets used for

43 Port-and-Protocol-based VLAN Classification."

44 REFERENCE "12.10.1"

45 ::= { ieee8021QBridgeProtocol 2 }

46

47 ieee8021QBridgeProtocolPortEntry OBJECT-TYPE

48 SYNTAX Ieee8021QBridgeProtocolPortEntry

49 MAX-ACCESS not-accessible

50 STATUS current

51 DESCRIPTION

52 "A VID set for a port."

53 REFERENCE "12.10.1.1.3 c)"

54 INDEX { ieee8021BridgeBasePortComponentId,

55 ieee8021BridgeBasePort,

56 ieee8021QBridgeProtocolPortGroupId }

57 ::= { ieee8021QBridgeProtocolPortTable 1 }

58

59 Ieee8021QBridgeProtocolPortEntry ::=

60 SEQUENCE {

61 ieee8021QBridgeProtocolPortGroupId

62 Integer32,

63 ieee8021QBridgeProtocolPortGroupVid

64 VlanId,

65 ieee8021QBridgeProtocolPortRowStatus

66 RowStatus

67 }

68

69 ieee8021QBridgeProtocolPortGroupId OBJECT-TYPE

70 SYNTAX Integer32 (1..2147483647)

71 MAX-ACCESS not-accessible

72 STATUS current

```

1  DESCRIPTION
2      "Designates a group of protocols in the Protocol
3      Group Database."
4  REFERENCE  "12.10.1.2"
5  ::= { ieee8021QBridgeProtocolPortEntry 1 }
6
7  ieee8021QBridgeProtocolPortGroupVid OBJECT-TYPE
8      SYNTAX      VlanId
9      MAX-ACCESS  read-create
10     STATUS      current
11     DESCRIPTION
12         "The VID associated with a group of protocols for
13         each port."
14     REFERENCE  "12.10.1.2"
15     ::= { ieee8021QBridgeProtocolPortEntry 2 }
16
17  ieee8021QBridgeProtocolPortRowStatus OBJECT-TYPE
18      SYNTAX      RowStatus
19      MAX-ACCESS  read-create
20      STATUS      current
21      DESCRIPTION
22          "This object indicates the status of this entry."
23      ::= { ieee8021QBridgeProtocolPortEntry 3 }
24
25  -- =====
26  -- ieee8021QBridgeVIDX subtree
27  --
28  -- =====
29
30  ieee8021QBridgeVIDXTable OBJECT-TYPE
31      SYNTAX      SEQUENCE OF Ieee8021QBridgeVIDXEntry
32      MAX-ACCESS  not-accessible
33      STATUS      current
34      DESCRIPTION
35          "This table is used to configure the VID Translation
36          Table defined in 12.10.1.8 and 6.9. The Bridge VID
37          Translation Table is used to implement a mapping between a
38          local VID, and a relay VID, used by the filtering and
39          forwarding process. Each row in this table is indexed by
40          component, port, and local VID value and a value to be used
41          for the specified VID as specified in (6.9). Entries in
42          this table must be persistent over power up restart/reboot."
43  REFERENCE  "12.10.1.8, 6.9"
44      ::= { ieee8021QBridgeVIDX 1 }
45
46  ieee8021QBridgeVIDXEntry OBJECT-TYPE
47      SYNTAX      Ieee8021QBridgeVIDXEntry
48      MAX-ACCESS  not-accessible
49      STATUS      current
50      DESCRIPTION
51          "An entry for the S-VID translation table, which includes
52          both the Local and Relay S-VIDs."
53
54      INDEX { ieee8021BridgeBasePortComponentId,
55              ieee8021BridgeBasePort,
56              ieee8021QBridgeVIDXLocalVid }
57      ::= { ieee8021QBridgeVIDXTable 1 }
58
59  Ieee8021QBridgeVIDXEntry ::=
60      SEQUENCE {
61          ieee8021QBridgeVIDXLocalVid VlanId,
62          ieee8021QBridgeVIDXRelayVid VlanId,
63          ieee8021QBridgeVIDXRowStatus RowStatus
64      }
65
66  ieee8021QBridgeVIDXLocalVid OBJECT-TYPE
67      SYNTAX      VlanId
68      MAX-ACCESS  not-accessible
69      STATUS      current
70      DESCRIPTION
71          "The Local VID after translation received at the ISS or EISS."
72      REFERENCE  "12.10.1.8.1, 12.10.1.8.2 "

```

```

1 ::= { ieee8021QBridgeVIDXEntry 1 }
2
3 ieee8021QBridgeVIDXRelayVid OBJECT-TYPE
4     SYNTAX      VlanId
5     MAX-ACCESS  read-create
6     STATUS      current
7     DESCRIPTION
8         "The Relay VID received before translation received at ISS or EISS."
9     REFERENCE   "12.10.1.8.1, 12.10.1.8.2"
10    ::= { ieee8021QBridgeVIDXEntry 2 }
11
12 ieee8021QBridgeVIDXRowStatus OBJECT-TYPE
13     SYNTAX      RowStatus
14     MAX-ACCESS  read-create
15     STATUS      current
16     DESCRIPTION
17         "This indicates the status of an entry in this table,
18         and is used to create/delete entries. It is an
19         implementation specific decision as to whether
20         any column in this table may be set while the
21         corresponding instance of this object is valid(1)."

```



```

1  REFERENCE    "12.10.1.9.1, 12.10.1.9.2"
2  ::= { ieee8021QBridgeEgressVidXEntry 1 }
3
4  ieee8021QBridgeEgressVidXLocalVid OBJECT-TYPE
5  SYNTAX      VlanId
6  MAX-ACCESS  read-create
7  STATUS      deprecated
8  DESCRIPTION
9      "The Local VID before translation transmitted to the
10     ISS or EISS."
11  REFERENCE    "12.10.1.9.1, 12.10.1.9.2"
12  ::= { ieee8021QBridgeEgressVidXEntry 2 }
13
14
15  ieee8021QBridgeEgressVidXRowStatus OBJECT-TYPE
16  SYNTAX      RowStatus
17  MAX-ACCESS  read-create
18  STATUS      deprecated
19  DESCRIPTION
20      "This indicates the status of an entry in this table,
21      and is used to create/delete entries. It is an
22      implementation specific decision as to whether
23      any column in this table may be set while the
24      corresponding instance of this object is valid(1)."
```

25 REFERENCE "12.10.1.9.1, 12.10.1.9.2"

```

26  ::= { ieee8021QBridgeEgressVidXEntry 3 }
27
28  -- =====
29  -- ieee8021QBridgeEgressVidXV2Table:
30  -- =====
31
32  ieee8021QBridgeEgressVidXV2Table OBJECT-TYPE
33  SYNTAX      SEQUENCE OF Ieee8021QBridgeEgressVidXV2Entry
34  MAX-ACCESS  not-accessible
35  STATUS      current
36  DESCRIPTION
37      "This table is used to configure the VID Translation
38      Table defined in 12.10.1.9 and 6.9. The Bridge VID
39      Egress Translation Table is used to implement a mapping between a
40      relay VID, and a local VID, used by the filtering and
41      forwarding process. Each row in this table is indexed by
42      component, port, and relay VID value and a value to be used
43      for the specified local VID as specified in (6.9). Entries in
44      this table must be persistent over power up restart/reboot."
45  REFERENCE    "12.10.1.9, 6.9"
46  ::= { ieee8021QBridgeVIDX 3 }
47
48  ieee8021QBridgeEgressVidXV2Entry OBJECT-TYPE
49  SYNTAX      Ieee8021QBridgeEgressVidXV2Entry
50  MAX-ACCESS  not-accessible
51  STATUS      current
52  DESCRIPTION
53      "An entry for the Egress VID translation table, which includes
54      both the relay and local IDs between which the PNP or CNP
55      translates."
56
57  INDEX { ieee8021BridgeBasePortComponentId,
58          ieee8021BridgeBasePort,
59          ieee8021QBridgeEgressVidXV2RelayVid }
60  ::= { ieee8021QBridgeEgressVidXV2Table 1 }
61
62  Ieee8021QBridgeEgressVidXV2Entry ::=
63      SEQUENCE {
64          ieee8021QBridgeEgressVidXV2RelayVid VlanId,
65          ieee8021QBridgeEgressVidXV2LocalVid VlanId,
66          ieee8021QBridgeEgressVidXV2RowStatus RowStatus
67      }
68
69  ieee8021QBridgeEgressVidXV2RelayVid OBJECT-TYPE
70  SYNTAX      VlanId
71  MAX-ACCESS  not-accessible
72  STATUS      current

```

```

1  DESCRIPTION
2      "The Relay VID after translation transmitted to the
3      ISS or EISS."
4  REFERENCE  "12.10.1.9.1, 12.10.1.9.2"
5      ::= { ieee8021QBridgeEgressVidXV2Entry 1 }
6
7  ieee8021QBridgeEgressVidXV2LocalVid OBJECT-TYPE
8      SYNTAX      VlanId
9      MAX-ACCESS  read-create
10     STATUS      current
11     DESCRIPTION
12         "The Local VID before translation transmitted to the
13         ISS or EISS."
14     REFERENCE  "12.10.1.9.1, 12.10.1.9.2"
15     ::= { ieee8021QBridgeEgressVidXV2Entry 2 }
16
17
18  ieee8021QBridgeEgressVidXV2RowStatus OBJECT-TYPE
19      SYNTAX      RowStatus
20      MAX-ACCESS  read-create
21      STATUS      current
22      DESCRIPTION
23          "This indicates the status of an entry in this table,
24          and is used to create/delete entries. It is an
25          implementation specific decision as to whether
26          any column in this table may be set while the
27          corresponding instance of this object is valid(1)."
28      REFERENCE  "12.10.1.9.1, 12.10.1.9.2"
29      ::= { ieee8021QBridgeEgressVidXV2Entry 3 }
30
31
32  -- =====
33  -- IEEE 802.1Q MIB - Conformance Information
34  -- =====
35
36  ieee8021QBridgeConformance
37      OBJECT IDENTIFIER ::= { ieee8021QBridgeMib 2 }
38
39  ieee8021QBridgeGroups
40      OBJECT IDENTIFIER ::= { ieee8021QBridgeConformance 1 }
41
42  ieee8021QBridgeCompliances
43      OBJECT IDENTIFIER ::= { ieee8021QBridgeConformance 2 }
44
45  -- =====
46  -- units of conformance
47  -- =====
48
49  ieee8021QBridgeBaseGroup OBJECT-GROUP
50      OBJECTS {
51          ieee8021QBridgeVlanVersionNumber,
52          ieee8021QBridgeMaxVlanId,
53          ieee8021QBridgeMaxSupportedVlans,
54          ieee8021QBridgeNumVlans,
55          ieee8021QBridgeMvrpEnabledStatus
56      }
57      STATUS      current
58      DESCRIPTION
59          "A collection of objects providing device-level control
60          and status information for the VLAN Bridge
61          services."
62      ::= { ieee8021QBridgeGroups 1 }
63
64  ieee8021QBridgeFdbUnicastGroup OBJECT-GROUP
65      OBJECTS {
66          ieee8021QBridgeFdbDynamicCount,
67          ieee8021QBridgeFdbLearnedEntryDiscards,
68          ieee8021QBridgeFdbAgingTime,
69          ieee8021QBridgeTpFdbPort,
70          ieee8021QBridgeTpFdbStatus
71      }
72      STATUS      current

```

```

1  DESCRIPTION
2      "A collection of objects providing information about all
3      unicast addresses, learned dynamically or statically
4      configured by management, in each Filtering Database."
5      ::= { ieee8021QBridgeGroups 2 }
6
7  ieee8021QBridgeFdbMulticastGroup OBJECT-GROUP
8      OBJECTS {
9          ieee8021QBridgeTpGroupEgressPorts,
10         ieee8021QBridgeTpGroupLearnt
11     }
12     STATUS      current
13     DESCRIPTION
14         "A collection of objects providing information about all
15         multicast addresses, learned dynamically or statically
16         configured by management, in each Filtering Database."
17         ::= { ieee8021QBridgeGroups 3 }
18
19  ieee8021QBridgeServiceRequirementsGroup OBJECT-GROUP
20      OBJECTS {
21         ieee8021QBridgeForwardAllPorts,
22         ieee8021QBridgeForwardAllStaticPorts,
23         ieee8021QBridgeForwardAllForbiddenPorts,
24         ieee8021QBridgeForwardUnregisteredPorts,
25         ieee8021QBridgeForwardUnregisteredStaticPorts,
26         ieee8021QBridgeForwardUnregisteredForbiddenPorts
27     }
28     STATUS      current
29     DESCRIPTION
30         "A collection of objects providing information about
31         service requirements, learned dynamically or statically
32         configured by management, in each Filtering Database."
33         ::= { ieee8021QBridgeGroups 4 }
34
35  ieee8021QBridgeFdbStaticGroup OBJECT-GROUP
36      OBJECTS {
37         ieee8021QBridgeStaticUnicastStaticEgressPorts,
38         ieee8021QBridgeStaticUnicastForbiddenEgressPorts,
39         ieee8021QBridgeStaticUnicastStorageType,
40         ieee8021QBridgeStaticUnicastRowStatus,
41         ieee8021QBridgeStaticMulticastStaticEgressPorts,
42         ieee8021QBridgeStaticMulticastForbiddenEgressPorts,
43         ieee8021QBridgeStaticMulticastStorageType,
44         ieee8021QBridgeStaticMulticastRowStatus
45     }
46     STATUS      current
47     DESCRIPTION
48         "A collection of objects providing information about
49         unicast and multicast addresses statically configured by
50         management, in each Filtering Database or VLAN."
51         ::= { ieee8021QBridgeGroups 5 }
52
53  ieee8021QBridgeVlanGroup OBJECT-GROUP
54      OBJECTS {
55         ieee8021QBridgeVlanNumDeletes,
56         ieee8021QBridgeVlanFdbId,
57         ieee8021QBridgeVlanCurrentEgressPorts,
58         ieee8021QBridgeVlanCurrentUntaggedPorts,
59         ieee8021QBridgeVlanStatus,
60         ieee8021QBridgeVlanCreationTime
61     }
62     STATUS      current
63     DESCRIPTION
64         "A collection of objects providing information about
65         all VLANs currently configured on this device."
66         ::= { ieee8021QBridgeGroups 6 }
67
68  ieee8021QBridgeVlanStaticGroup OBJECT-GROUP
69      OBJECTS {
70         ieee8021QBridgeVlanStaticName,
71         ieee8021QBridgeVlanStaticEgressPorts,
72         ieee8021QBridgeVlanForbiddenEgressPorts,

```

```

1         ieee8021QBridgeVlanStaticUntaggedPorts,
2         ieee8021QBridgeVlanStaticRowStatus,
3         ieee8021QBridgeNextFreeLocalVlanIndex
4     }
5     STATUS          current
6     DESCRIPTION
7         "A collection of objects providing information about
8         VLANs statically configured by management."
9     ::= { ieee8021QBridgeGroups 7 }
10
11 ieee8021QBridgeVlanStatisticsGroup OBJECT-GROUP
12     OBJECTS {
13         ieee8021QBridgeTpVlanPortInFrames,
14         ieee8021QBridgeTpVlanPortOutFrames,
15         ieee8021QBridgeTpVlanPortInDiscards
16     }
17     STATUS          current
18     DESCRIPTION
19         "A collection of objects providing per-port frame
20         statistics for all VLANs currently configured on this
21         device."
22     ::= { ieee8021QBridgeGroups 8 }
23
24 ieee8021QBridgeLearningConstraintsGroup OBJECT-GROUP
25     OBJECTS {
26         ieee8021QBridgeLearningConstraintsType,
27         ieee8021QBridgeLearningConstraintsStatus
28     }
29     STATUS          deprecated
30     DESCRIPTION
31         "A collection of objects defining the Filtering Database
32         constraints all VLANs have with each other."
33     ::= { ieee8021QBridgeGroups 9 }
34
35 ieee8021QBridgeLearningConstraintDefaultGroup OBJECT-GROUP
36     OBJECTS {
37         ieee8021QBridgeLearningConstraintDefaultsSet,
38         ieee8021QBridgeLearningConstraintDefaultsType
39     }
40     STATUS          deprecated
41     DESCRIPTION
42         "A collection of objects defining the default Filtering
43         Database constraints for VLANs that have no specific
44         constraints defined."
45     ::= { ieee8021QBridgeGroups 10 }
46
47 ieee8021QBridgeClassificationDeviceGroup OBJECT-GROUP
48     OBJECTS {
49         ieee8021QBridgeProtocolGroupId,
50         ieee8021QBridgeProtocolGroupRowStatus
51     }
52     STATUS          current
53     DESCRIPTION
54         "VLAN classification information for the Bridge."
55     ::= { ieee8021QBridgeGroups 11 }
56
57 ieee8021QBridgeClassificationPortGroup OBJECT-GROUP
58     OBJECTS {
59         ieee8021QBridgeProtocolPortGroupVid,
60         ieee8021QBridgeProtocolPortRowStatus
61     }
62     STATUS          current
63     DESCRIPTION
64         "VLAN classification information for individual ports."
65     ::= { ieee8021QBridgeGroups 12 }
66
67 ieee8021QBridgePortGroup2 OBJECT-GROUP
68     OBJECTS {
69         ieee8021QBridgePvid,
70         ieee8021QBridgePortAcceptableFrameTypes,
71         ieee8021QBridgePortIngressFiltering,
72         ieee8021QBridgePortMvrpEnabledStatus,

```

```

1         ieee8021QBridgePortMvrpFailedRegistrations,
2         ieee8021QBridgePortMvrpLastPduOrigin,
3         ieee8021QBridgePortRestrictedVlanRegistration
4     }
5     STATUS          current
6     DESCRIPTION
7         "A collection of objects providing port-level VLAN
8         control and status information for all ports."
9     ::= { ieee8021QBridgeGroups 13 }
10
11 ieee8021QBridgeCVlanPortGroup OBJECT-GROUP
12     OBJECTS {
13         ieee8021QBridgeCVlanPortRowStatus
14     }
15     STATUS          current
16     DESCRIPTION
17         "Objects used to create/delete customer VLAN ports."
18     ::= { ieee8021QBridgeGroups 14 }
19
20 ieee8021QBridgeVIDXGroup OBJECT-GROUP
21     OBJECTS {
22         ieee8021QBridgeVIDXRelayVid,
23         ieee8021QBridgeVIDXRowStatus
24     }
25     STATUS          current
26     DESCRIPTION
27         "Ingress or Ingress/Egress VID translation for
28         individual ports."
29     ::= { ieee8021QBridgeGroups 15 }
30
31 ieee8021QBridgeEgressVIDXGroup OBJECT-GROUP
32     OBJECTS {
33         ieee8021QBridgeEgressVidXLocalVid,
34         ieee8021QBridgeEgressVidXRowStatus
35     }
36     STATUS          deprecated
37     DESCRIPTION
38         "Egress VID translation for individual ports."
39     ::= { ieee8021QBridgeGroups 16 }
40
41 ieee8021QBridgeEgressVIDXV2Group OBJECT-GROUP
42     OBJECTS {
43         ieee8021QBridgeEgressVidXV2LocalVid,
44         ieee8021QBridgeEgressVidXV2RowStatus
45     }
46     STATUS          current
47     DESCRIPTION
48         "Egress VID translation for individual ports."
49     ::= { ieee8021QBridgeGroups 17 }
50
51
52 -- =====
53 -- compliance statements
54 -- =====
55
56 ieee8021QBridgeCompliance MODULE-COMPLIANCE
57     STATUS deprecated
58     DESCRIPTION
59         "The compliance statement for device support of Virtual
60         LAN Bridge services."
61
62     MODULE
63         MANDATORY-GROUPS {
64             ieee8021QBridgeBaseGroup,
65             ieee8021QBridgeVlanGroup,
66             ieee8021QBridgeVlanStaticGroup,
67             ieee8021QBridgePortGroup2
68         }
69
70         GROUP          ieee8021QBridgeFdbUnicastGroup
71         DESCRIPTION
72             "This group is mandatory for Bridges that implement

```

```
1      IEEE 802.1Q transparent bridging."
2
3      GROUP      ieee8021QBridgeFdbMulticastGroup
4      DESCRIPTION
5          "This group is mandatory for Bridges that implement
6          IEEE 802.1Q transparent bridging."
7      GROUP      ieee8021QBridgeServiceRequirementsGroup
8      DESCRIPTION
9          "This group is mandatory for Bridges that implement
10         extended filtering services. All objects must be
11         read-write if extended-filtering services are
12         enabled."
13
14      GROUP      ieee8021QBridgeFdbStaticGroup
15      DESCRIPTION
16          "This group is optional."
17
18      GROUP      ieee8021QBridgeVlanStatisticsGroup
19      DESCRIPTION
20          "This group is optional as there may be significant
21         implementation cost associated with its support."
22
23      GROUP      ieee8021QBridgeClassificationDeviceGroup
24      DESCRIPTION
25          "This group is mandatory ONLY for devices implementing
26         VLAN Classification as specified in IEEE 802.1v."
27
28      GROUP      ieee8021QBridgeClassificationPortGroup
29      DESCRIPTION
30          "This group is mandatory ONLY for devices implementing
31         VLAN Classification as specified in IEEE 802.1v."
32
33      GROUP      ieee8021QBridgeCVlanPortGroup
34      DESCRIPTION
35          "This group is mandatory ONLY for devices supporting
36         creation/deletion of customer VLAN ports."
37
38      GROUP      ieee8021QBridgeVIDXGroup
39      DESCRIPTION
40          "This group is mandatory ONLY for devices supporting
41         VID translation of customer and/or provider VLAN ports."
42
43      GROUP      ieee8021QBridgeEgressVIDXGroup
44      DESCRIPTION
45          "This group is mandatory ONLY for devices supporting
46         separate Ingress and Egress VID translation of
47         of customer and provider VLAN ports."
48
49      OBJECT      ieee8021QBridgePortAcceptableFrameTypes
50      MIN-ACCESS  read-only
51      DESCRIPTION
52          "Write access is not required as this is an optional
53         capability in IEEE 802.1Q."
54
55      OBJECT      ieee8021QBridgePortIngressFiltering
56      MIN-ACCESS  read-only
57      DESCRIPTION
58          "Write access is not required as this is an optional
59         capability in IEEE 802.1Q."
60
61      OBJECT      ieee8021QBridgeLearningConstraintDefaultsSet
62      MIN-ACCESS  read-only
63      DESCRIPTION
64          "Write access is not required as this is an optional
65         capability in IEEE 802.1Q-2012."
66
67      OBJECT      ieee8021QBridgeLearningConstraintDefaultsType
68      MIN-ACCESS  read-only
69      DESCRIPTION
70          "Write access is not required as this is an optional
71         capability in IEEE 802.1Q-2012."
72
```

```
1      OBJECT      ieee8021QBridgeProtocolGroupId
2      MIN-ACCESS  read-only
3      DESCRIPTION
4          "Write access is not required as this is an optional
5          capability in IEEE 802.1v."
6
7      OBJECT      ieee8021QBridgeProtocolGroupRowStatus
8      MIN-ACCESS  read-only
9      DESCRIPTION
10         "Write access is not required as this is an optional
11         capability in IEEE 802.1v."
12
13         ::= { ieee8021QBridgeCompliances 1 }
14
15 ieee8021QBridgeComplianceV2 MODULE-COMPLIANCE
16     STATUS current
17     DESCRIPTION
18         "The compliance statement for device support of Virtual
19         LAN Bridge services."
20
21     MODULE
22     MANDATORY-GROUPS {
23         ieee8021QBridgeBaseGroup,
24         ieee8021QBridgeVlanGroup,
25         ieee8021QBridgeVlanStaticGroup,
26         ieee8021QBridgePortGroup2
27     }
28
29     GROUP      ieee8021QBridgeFdbUnicastGroup
30     DESCRIPTION
31         "This group is mandatory for Bridges that implement
32         IEEE 802.1Q transparent bridging."
33
34     GROUP      ieee8021QBridgeFdbMulticastGroup
35     DESCRIPTION
36         "This group is mandatory for Bridges that implement
37         IEEE 802.1Q transparent bridging."
38     GROUP      ieee8021QBridgeServiceRequirementsGroup
39     DESCRIPTION
40         "This group is mandatory for Bridges that implement
41         extended filtering services. All objects must be
42         read-write if extended-filtering services are
43         enabled."
44
45     GROUP      ieee8021QBridgeFdbStaticGroup
46     DESCRIPTION
47         "This group is optional."
48
49     GROUP      ieee8021QBridgeVlanStatisticsGroup
50     DESCRIPTION
51         "This group is optional as there may be significant
52         implementation cost associated with its support."
53
54     GROUP      ieee8021QBridgeClassificationDeviceGroup
55     DESCRIPTION
56         "This group is mandatory ONLY for devices implementing
57         VLAN Classification as specified in IEEE 802.1v."
58
59     GROUP      ieee8021QBridgeClassificationPortGroup
60     DESCRIPTION
61         "This group is mandatory ONLY for devices implementing
62         VLAN Classification as specified in IEEE 802.1v."
63
64     GROUP      ieee8021QBridgeCVlanPortGroup
65     DESCRIPTION
66         "This group is mandatory ONLY for devices supporting
67         creation/deletion of customer VLAN ports."
68
69     GROUP      ieee8021QBridgeVIDXGroup
70     DESCRIPTION
71         "This group is mandatory ONLY for devices supporting
72         VID translation of customer and/or provider VLAN ports."
```

```
1
2  GROUP          ieee8021QBridgeEgressVIDXV2Group
3  DESCRIPTION
4      "This group is mandatory ONLY for devices supporting
5      separate Ingress and Egress VID translation of
6      of customer and provider VLAN ports."
7
8  OBJECT          ieee8021QBridgePortAcceptableFrameTypes
9  MIN-ACCESS      read-only
10 DESCRIPTION
11     "Write access is not required as this is an optional
12     capability in IEEE 802.1Q."
13
14 OBJECT          ieee8021QBridgePortIngressFiltering
15 MIN-ACCESS      read-only
16 DESCRIPTION
17     "Write access is not required as this is an optional
18     capability in IEEE 802.1Q."
19
20 OBJECT          ieee8021QBridgeProtocolGroupId
21 MIN-ACCESS      read-only
22 DESCRIPTION
23     "Write access is not required as this is an optional
24     capability in IEEE 802.1v."
25
26 OBJECT          ieee8021QBridgeProtocolGroupRowStatus
27 MIN-ACCESS      read-only
28 DESCRIPTION
29     "Write access is not required as this is an optional
30     capability in IEEE 802.1v."
31
32 ::= { ieee8021QBridgeCompliances 2 }
33
34 END
```


1 17.7.5 Definitions for the IEEE8021-PB-MIB module

```
2 IEEE8021-PB-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for IEEE Std 802.1Q Provider Bridge Devices
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE
10     FROM SNMPv2-SMI
11     TruthValue, RowStatus
12     FROM SNMPv2-TC
13     ieee802dot1mibs, IEEE8021PbbComponentIdentifierOrZero,
14     IEEE8021PortAcceptableFrameTypes, IEEE8021PriorityValue,
15     IEEE8021BridgePortNumberOrZero, IEEE8021BridgePortType
16     FROM IEEE8021-TC-MIB
17     ieee8021BridgeBasePortComponentId, ieee8021BridgeBasePort
18     FROM IEEE8021-BRIDGE-MIB
19     VlanId, VlanIdOrNone
20     FROM Q-BRIDGE-MIB
21     MODULE-COMPLIANCE, OBJECT-GROUP
22     FROM SNMPv2-CONF;
23
24 ieee8021PbMib MODULE-IDENTITY
25     LAST-UPDATED "202211080000Z" -- November 8, 2022
26     ORGANIZATION "IEEE 802.1 Working Group"
27     CONTACT-INFO
28         " WG-URL: http://www.ieee802.org/1/
29         WG-EMail: stds-802-1-1@ieee.org
30         Contact: IEEE 802.1 Working Group Chair
31         Postal: C/O IEEE 802.1 Working Group
32             IEEE Standards Association
33             445 Hoes Lane
34             Piscataway, NJ 08854
35             USA
36         E-mail: stds-802-1-chairs@ieee.org"
37     DESCRIPTION
38         "Provider Bridge MIB module.
39
40         Unless otherwise indicated, the references in this MIB
41         module are to IEEE Std 802.1Q-2022.
42
43         Copyright (C) IEEE (2022).
44         This version of this MIB module is part of IEEE Std 802.1Q;
45         see that standard for full legal notices."
46
47     REVISION "202211080000Z" -- November 8, 2022
48     DESCRIPTION
49         "Published as part of IEEE Std 802.1Q-2022.
50         Cross references and contact information updated."
51
52     REVISION "201807010000Z" -- July 1, 2018
53     DESCRIPTION
54         "Published as part of IEEE Std 802.1Q 2018 revision.
55         Cross references updated and corrected."
56
57     REVISION "201412150000Z" -- December 15, 2014
58     DESCRIPTION
59         "Published as part of IEEE Std 802.1Q 2014 revision.
60         Cross references updated and corrected.
61         Bug fixes to conformance section."
62
63
64     REVISION "201202100000Z" -- February 10, 2012"
65     DESCRIPTION
66         "Deprecated ieee8021PbVidTranslationTable
67         moved the new object to the Q Bridge as part
68         of VID translation for IEEE Std 802.1aq."
69
70     REVISION "201104060000Z" -- April 6, 2011
71     DESCRIPTION
```

```

1      "Additions to support Remote Customer Service Interfaces."
2
3      REVISION      "201102270000Z" -- February 27, 2011
4      DESCRIPTION
5          "Change to ieee8021PbEdgePortAcceptableFrameTypes
6           permissible values, addition of
7           IEEE8021BridgePortNumberOrZero to IMPORTS,
8           as part of 2011 revision of IEEE Std 802.1Q."
9
10     REVISION      "201008260000Z" -- August 26, 2010
11     DESCRIPTION
12         "Minor edits to contact information etc. as part of
13         revision of Std 802.1Q."
14
15     REVISION      "200810150000Z" -- October 15, 2008
16     DESCRIPTION
17         "Initial version."
18     ::= { ieee802dot1mibs 5 }
19
20 ieee8021PbNotifications OBJECT IDENTIFIER ::= { ieee8021PbMib 0 }
21 ieee8021PbObjects       OBJECT IDENTIFIER ::= { ieee8021PbMib 1 }
22 ieee8021PbConformance  OBJECT IDENTIFIER ::= { ieee8021PbMib 2 }
23
24 -- =====
25 -- ieee8021PbVidTranslationTable:
26 -- Deprecated and moved VID translation to Q Bridge MIB.
27 -- =====
28
29 ieee8021PbVidTranslationTable OBJECT-TYPE
30     SYNTAX      SEQUENCE OF Ieee8021PbVidTranslationEntry
31     MAX-ACCESS  not-accessible
32     STATUS      deprecated
33     DESCRIPTION
34         "This table is used to configure the VID Translation Table
35         defined in 12.10.1.8. The VID
36         Translation Table is used to implement a bidirectional
37         mapping between a local S-VID, used in data and protocol
38         frames transmitted and received through a CNP or PNP,
39         and a relay S-VID, used by the filtering and forwarding
40         process. Each row in this table is indexed by component,
41         port, and local S-VID value and indicates the relay S-VID
42         value to be used for the specified S-VID. If no entry for
43         a component, port, and local-svid is present in this table
44         is present then the relay S-VID used for a frame received
45         on that port with the local S-VID value will be the S-VID
46         that has the same numeric value as the local S-VID of the
47         received frame.
48
49         Entries in this table must be persistent over power up
50         restart/reboot."
51     REFERENCE   "12.10.1.8"
52     ::= { ieee8021PbObjects 1 }
53
54 ieee8021PbVidTranslationEntry OBJECT-TYPE
55     SYNTAX      Ieee8021PbVidTranslationEntry
56     MAX-ACCESS  not-accessible
57     STATUS      deprecated
58     DESCRIPTION
59         "An entry for the S-VID translation table, which includes
60         both the Local and Relay S-VIDs between which the PNP or CNP
61         translates.
62
63         Note that the component ID of entries in this table must refer
64         to the S-VLAN Component of a Provider Bridge."
65     INDEX { ieee8021BridgeBasePortComponentId,
66             ieee8021BridgeBasePort,
67             ieee8021PbVidTranslationLocalVid }
68     ::= { ieee8021PbVidTranslationTable 1 }
69
70 Ieee8021PbVidTranslationEntry ::= SEQUENCE {
71     ieee8021PbVidTranslationLocalVid
72     VlanId,

```

```

1      ieee8021PbVidTranslationRelayVid
2          VlanId,
3      ieee8021PbVidTranslationRowStatus
4          RowStatus
5 }
6
7 ieee8021PbVidTranslationLocalVid OBJECT-TYPE
8     SYNTAX      VlanId
9     MAX-ACCESS  not-accessible
10    STATUS      deprecated
11    DESCRIPTION
12        "The S-VID on received (transmitted) at the ISS of a CNP or PNP."
13    ::= { ieee8021PbVidTranslationEntry 1 }
14
15 ieee8021PbVidTranslationRelayVid OBJECT-TYPE
16     SYNTAX      VlanId
17     MAX-ACCESS  read-create
18     STATUS      deprecated
19     DESCRIPTION
20        "The translated S-VID delivered (received) over the EISS from a
21        CNP or PNP. The default value of this object on creation will
22        be the value of the corresponding instance of
23        ieee8021PbVidTranslationLocalVid."
24    ::= { ieee8021PbVidTranslationEntry 2 }
25
26 ieee8021PbVidTranslationRowStatus OBJECT-TYPE
27     SYNTAX      RowStatus
28     MAX-ACCESS  read-create
29     STATUS      deprecated
30     DESCRIPTION
31        "This indicates the status of an entry in this table, and is
32        used to create/delete entries.
33
34        It is an implementation specific decision as to whether any
35        column in this table may be set while the corresponding
36        instance of this object is valid(1)."
37    ::= { ieee8021PbVidTranslationEntry 3 }
38
39 -- =====
40 -- ieee8021PbCvidRegistrationTable:
41 -- =====
42
43 ieee8021PbCvidRegistrationTable OBJECT-TYPE
44     SYNTAX      SEQUENCE OF Ieee8021PbCvidRegistrationEntry
45     MAX-ACCESS  not-accessible
46     STATUS      current
47     DESCRIPTION
48        "A table used in a CEP to create the mapping between a C-VID
49        and a service represented by an S-VID.
50
51        Note that the component ID of entries in this table must refer
52        to the S-VLAN component of a Provider Edge Bridge and the Port
53        Number must refer to the port number of the Customer Edge Port
54        associated with that Provider Edge Bridge.
55
56        Entries in this table must be persistent over power up
57        restart/reboot."
58     REFERENCE   "12.13.2.1, 12.13.2.2"
59     ::= { ieee8021PbObjects 2 }
60
61 ieee8021PbCvidRegistrationEntry OBJECT-TYPE
62     SYNTAX      Ieee8021PbCvidRegistrationEntry
63     MAX-ACCESS  not-accessible
64     STATUS      current
65     DESCRIPTION
66        "An element of the C-VID registration table accessed by PB
67        C-VLAN component, Customer Edge Port Bridge Port number, and
68        C-VID. Each element contains the mapping between a C-VID and
69        the S-VID that carries the service and booleans for handling
70        untagged frames at the PEP and CEP."
71     INDEX { ieee8021BridgeBasePortComponentId,
72            ieee8021BridgeBasePort,

```

```

1         ieee8021PbCvidRegistrationCvid }
2 ::= { ieee8021PbCvidRegistrationTable 1 }
3
4 Ieee8021PbCvidRegistrationEntry ::= SEQUENCE {
5     ieee8021PbCvidRegistrationCvid
6     VlanId,
7     ieee8021PbCvidRegistrationSvid
8     VlanId,
9     ieee8021PbCvidRegistrationUntaggedPep
10    TruthValue,
11    ieee8021PbCvidRegistrationUntaggedCep
12    TruthValue,
13    ieee8021PbCvidRegistrationRowStatus
14    RowStatus
15 }
16
17 ieee8021PbCvidRegistrationCvid OBJECT-TYPE
18     SYNTAX      VlanId
19     MAX-ACCESS  not-accessible
20     STATUS      current
21     DESCRIPTION
22         "C-VID of this C-VID registration entry."
23     ::= { ieee8021PbCvidRegistrationEntry 1 }
24
25 ieee8021PbCvidRegistrationSvid OBJECT-TYPE
26     SYNTAX      VlanId
27     MAX-ACCESS  read-create
28     STATUS      current
29     DESCRIPTION
30         "S-VID for this C-VID registration entry."
31     ::= { ieee8021PbCvidRegistrationEntry 2 }
32
33 ieee8021PbCvidRegistrationUntaggedPep OBJECT-TYPE
34     SYNTAX      TruthValue
35     MAX-ACCESS  read-create
36     STATUS      current
37     DESCRIPTION
38         "A flag indicating if this C-VID should be carried untagged
39         at the PEP. A value of true(1) means untagged."
40     DEFVAL { true }
41     ::= { ieee8021PbCvidRegistrationEntry 3 }
42
43 ieee8021PbCvidRegistrationUntaggedCep OBJECT-TYPE
44     SYNTAX      TruthValue
45     MAX-ACCESS  read-create
46     STATUS      current
47     DESCRIPTION
48         "A flag indicating if this C-VID should be carried untagged
49         at the CEP. A value of true(1) means untagged."
50     DEFVAL { true }
51     ::= { ieee8021PbCvidRegistrationEntry 4 }
52
53 ieee8021PbCvidRegistrationRowStatus OBJECT-TYPE
54     SYNTAX      RowStatus
55     MAX-ACCESS  read-create
56     STATUS      current
57     DESCRIPTION
58         "This indicates the status of an entry in this table, and is
59         used to create/delete entries.
60
61         The value of ieee8021PbCvidRegistrationSvid must be set before
62         an entry in this table can be made valid.
63
64         It is an implementation specific decision as to whether any
65         column in this table may be set while the corresponding
66         instance of this object is valid(1)."
67     ::= { ieee8021PbCvidRegistrationEntry 5 }
68
69 -- =====
70 -- ieee8021PbEdgePortTable:
71 -- =====
72

```

```

1 ieee8021PbEdgePortTable OBJECT-TYPE
2   SYNTAX      SEQUENCE OF Ieee8021PbEdgePortEntry
3   MAX-ACCESS  not-accessible
4   STATUS      current
5   DESCRIPTION
6       "A Provider Edge Port (PEP) table that indicates the subset of
7       parameters needed for each PEP."
8   REFERENCE   "12.13.2.3, 12.13.2.4"
9   ::= { ieee8021PbObjects 3 }
10
11 ieee8021PbEdgePortEntry OBJECT-TYPE
12   SYNTAX      Ieee8021PbEdgePortEntry
13   MAX-ACCESS  not-accessible
14   STATUS      current
15   DESCRIPTION
16       "An entry in the PEP table indexed by ComponentID and S-VID and
17       containing parameters used to configure ingress filtering on
18       the PEP, thus affecting traffic transiting from the provider
19       network to the customer edge port. The columns allow the
20       default C-VID value and default priority to be specified
21       and PEP's ingress filtering operation to be controlled.
22
23       Note that the component ID of entries in this table must refer
24       to an S-VLAN component of a provider edge Bridge and the Bridge
25       Port number must refer to the port number of a Customer Edge
26       Port associated with that Provider Edge Bridge."
27   INDEX { ieee8021BridgeBasePortComponentId,
28           ieee8021BridgeBasePort,
29           ieee8021PbEdgePortSVid }
30   ::= { ieee8021PbEdgePortTable 1 }
31
32 Ieee8021PbEdgePortEntry ::= SEQUENCE {
33     ieee8021PbEdgePortSVid
34         VlanId,
35     ieee8021PbEdgePortPVID
36         VlanId,
37     ieee8021PbEdgePortDefaultUserPriority
38         IEEE8021PriorityValue,
39     ieee8021PbEdgePortAcceptableFrameTypes
40         IEEE8021PortAcceptableFrameTypes,
41     ieee8021PbEdgePortEnableIngressFiltering
42         TruthValue
43 }
44
45 ieee8021PbEdgePortSVid OBJECT-TYPE
46   SYNTAX      VlanId
47   MAX-ACCESS  not-accessible
48   STATUS      current
49   DESCRIPTION
50       "The 12 bit S-VID associated with the PEP."
51   ::= { ieee8021PbEdgePortEntry 1 }
52
53 ieee8021PbEdgePortPVID OBJECT-TYPE
54   SYNTAX      VlanId
55   MAX-ACCESS  read-write
56   STATUS      current
57   DESCRIPTION
58       "A 12-bit C-VID to be used for untagged frames received at
59       the Provider Edge Port."
60   ::= { ieee8021PbEdgePortEntry 2 }
61
62 ieee8021PbEdgePortDefaultUserPriority OBJECT-TYPE
63   SYNTAX      IEEE8021PriorityValue
64   MAX-ACCESS  read-write
65   STATUS      current
66   DESCRIPTION
67       "An integer range 0-7 to be used for untagged frames received
68       at the Provider Edge Port."
69   ::= { ieee8021PbEdgePortEntry 3 }
70
71 ieee8021PbEdgePortAcceptableFrameTypes OBJECT-TYPE
72   SYNTAX      IEEE8021PortAcceptableFrameTypes

```

```

1  MAX-ACCESS    read-write
2  STATUS       current
3  DESCRIPTION
4      "the Acceptable Frame Types for frames received at the PEP.
5      The permissible values for the parameter are:
6          1) Admit all frames;
7          2) Admit only untagged and Priority-tagged frames;
8          3) Admit only VLAN-tagged frames."
9  DEFVAL { admitAll }
10 ::= { ieee8021PbEdgePortEntry 4 }
11
12 ieee8021PbEdgePortEnableIngressFiltering OBJECT-TYPE
13     SYNTAX      TruthValue
14     MAX-ACCESS  read-write
15     STATUS      current
16     DESCRIPTION
17         "Filtering parameter for frames received at the PEP.
18         The permissible values for the parameter are:
19             true(1) Enabled;
20             false(2) Disabled."
21     DEFVAL { true }
22     ::= { ieee8021PbEdgePortEntry 5 }
23
24 -- =====
25 -- ieee8021PbServicePriorityRegenerationTable:
26 -- =====
27
28 ieee8021PbServicePriorityRegenerationTable OBJECT-TYPE
29     SYNTAX      SEQUENCE OF Ieee8021PbServicePriorityRegenerationEntry
30     MAX-ACCESS  not-accessible
31     STATUS      current
32     DESCRIPTION
33         "The priority regeneration table for this PEP."
34     REFERENCE   "12.13.2.5, 12.13.2.6"
35     ::= { ieee8021PbObjects 4 }
36
37 ieee8021PbServicePriorityRegenerationEntry OBJECT-TYPE
38     SYNTAX      Ieee8021PbServicePriorityRegenerationEntry
39     MAX-ACCESS  not-accessible
40     STATUS      current
41     DESCRIPTION
42         "An element of the PEP priority regeneration table indexed
43         by Component ID, Bridge Port number, S-VID, and received
44         priority. Each element contains the regenerated priority.
45
46         Note that the component ID of entries in this table must refer
47         to the S-VLAN component of a Provider Edge Bridge and the Port
48         Number must refer to the port number of the Customer Edge Port
49         associated with that S-VLAN component."
50     INDEX { ieee8021BridgeBasePortComponentId,
51             ieee8021BridgeBasePort,
52             ieee8021PbServicePriorityRegenerationSvid,
53             ieee8021PbServicePriorityRegenerationReceivedPriority }
54     ::= { ieee8021PbServicePriorityRegenerationTable 1 }
55
56 Ieee8021PbServicePriorityRegenerationEntry ::= SEQUENCE {
57     ieee8021PbServicePriorityRegenerationSvid
58     VlanId,
59     ieee8021PbServicePriorityRegenerationReceivedPriority
60     IEEE8021PriorityValue,
61     ieee8021PbServicePriorityRegenerationRegeneratedPriority
62     IEEE8021PriorityValue
63 }
64
65 ieee8021PbServicePriorityRegenerationSvid OBJECT-TYPE
66     SYNTAX      VlanId
67     MAX-ACCESS  not-accessible
68     STATUS      current
69     DESCRIPTION
70         "S-VID for this regeneration table entry."
71     ::= { ieee8021PbServicePriorityRegenerationEntry 1 }
72

```

```

1 ieee8021PbServicePriorityRegenerationReceivedPriority OBJECT-TYPE
2     SYNTAX      IEEE8021PriorityValue
3     MAX-ACCESS  not-accessible
4     STATUS      current
5     DESCRIPTION
6         "Received priority for this regeneration table entry."
7     ::= { ieee8021PbServicePriorityRegenerationEntry 2 }
8
9 ieee8021PbServicePriorityRegenerationRegeneratedPriority OBJECT-TYPE
10    SYNTAX      IEEE8021PriorityValue
11    MAX-ACCESS  read-write
12    STATUS      current
13    DESCRIPTION
14        "The regenerated priority contained in this regeneration table
15         entry."
16    ::= { ieee8021PbServicePriorityRegenerationEntry 3 }
17
18 -- =====
19 -- ieee8021PbCnpTable
20 -- =====
21
22 ieee8021PbCnpTable OBJECT-TYPE
23     SYNTAX      SEQUENCE OF Ieee8021PbCnpEntry
24     MAX-ACCESS  not-accessible
25     STATUS      current
26     DESCRIPTION
27         "This table is used for dynamic creation and deletion of
28         Customer Network Ports on S-VLAN components or I-components.
29         Creation of an entry in this table will implicitly also
30         create a corresponding entry in the ieee8021BridgeBasePortTable.
31
32         Entries in this table must be persistent across reinitializations
33         of the management system."
34     REFERENCE   "12.13.2"
35     ::= { ieee8021PbObjects 5 }
36
37 ieee8021PbCnpEntry OBJECT-TYPE
38     SYNTAX      Ieee8021PbCnpEntry
39     MAX-ACCESS  not-accessible
40     STATUS      current
41     DESCRIPTION
42         "Represents a dynamically created Customer Network Port."
43     INDEX { ieee8021BridgeBasePortComponentId,
44             ieee8021BridgeBasePort }
45     ::= { ieee8021PbCnpTable 1 }
46
47 Ieee8021PbCnpEntry ::= SEQUENCE {
48     ieee8021PbCnpCComponentId
49         IEEE8021PbbComponentIdentifierOrZero,
50     ieee8021PbCnpSVid
51         VlanIdOrNone,
52     ieee8021PbCnpRowStatus
53         RowStatus
54 }
55
56 ieee8021PbCnpCComponentId OBJECT-TYPE
57     SYNTAX      IEEE8021PbbComponentIdentifierOrZero
58     MAX-ACCESS  read-create
59     STATUS      current
60     DESCRIPTION
61         "The component ID of the C-Vlan component if this is an
62         internal customer network port. The value must be 0 for
63         an external customer network port.
64
65         This value must be consistent with the value of the
66         corresponding instance of ieee8021PbCnpSVid.
67         Both must be non-zero, or both must be zero."
68     ::= { ieee8021PbCnpEntry 1 }
69
70 ieee8021PbCnpSVid OBJECT-TYPE
71     SYNTAX      VlanIdOrNone
72     MAX-ACCESS  read-create

```

```

1  STATUS      current
2  DESCRIPTION
3      "The S-VID for service for an internal customer network port.
4      For an external port, this value must be 0.
5
6      This value must be consistent with the value of the
7      corresponding instance of ieee8021PbCnpCComponentId.
8      Both must be non-zero, or both must be zero."
9  ::= { ieee8021PbCnpEntry 2 }
10
11 ieee8021PbCnpRowStatus OBJECT-TYPE
12     SYNTAX      RowStatus
13     MAX-ACCESS  read-create
14     STATUS      current
15     DESCRIPTION
16         "This object is used for creation/deletion of entries in
17         this table.
18
19         All columns in this table must have valid values before
20         this object can be set to active(1).
21
22         While the value of this object is active(1), the values
23         of other columns in the same entry may not be modified."
24     ::= { ieee8021PbCnpEntry 3 }
25
26 -- =====
27 -- ieee8021PbPnpTable
28 -- =====
29
30 ieee8021PbPnpTable OBJECT-TYPE
31     SYNTAX      SEQUENCE OF Ieee8021PbPnpEntry
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35         "This table is used for dynamic creation and deletion of
36         Provider Network Ports on S-VLAN components or B-components.
37         Creation of an entry in this table will implicitly also
38         create a corresponding entry in the ieee8021BridgeBasePortTable.
39
40         Entries in this table must be persistent across reinitializations
41         of the management system."
42     REFERENCE   "12.13.1"
43     ::= { ieee8021PbObjects 6 }
44
45 ieee8021PbPnpEntry OBJECT-TYPE
46     SYNTAX      Ieee8021PbPnpEntry
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "Represents a dynamically created Provider Network Port."
51     INDEX { ieee8021BridgeBasePortComponentId,
52             ieee8021BridgeBasePort }
53     ::= { ieee8021PbPnpTable 1 }
54
55 Ieee8021PbPnpEntry ::= SEQUENCE {
56     ieee8021PbPnpRowStatus
57     RowStatus
58 }
59
60 ieee8021PbPnpRowStatus OBJECT-TYPE
61     SYNTAX      RowStatus
62     MAX-ACCESS  read-create
63     STATUS      current
64     DESCRIPTION
65         "This object is used for creation/deletion of entries in
66         this table."
67     ::= { ieee8021PbPnpEntry 1 }
68
69 -- =====
70 -- ieee8021PbCepTable
71 -- =====
72

```



```

1 ieee8021PbCepTable OBJECT-TYPE
2     SYNTAX      SEQUENCE OF Ieee8021PbCepEntry
3     MAX-ACCESS  not-accessible
4     STATUS      current
5     DESCRIPTION
6         "This table is used to create Customer Edge Ports, CEPs, on a
7         provider edge Bridge. It is indexed by the ComponentId of the
8         PEB's S-VLAN component and by the port number for the CEP. Note that
9         the CEP's port number belongs to the set of port numbers
10        associated with the PEB's S-VLAN component.
11
12        Entries in this table must be persistent across reinitializations
13        of the management system. However, note that some column values,
14        as noted below, may change across system reinitializations."
15    ::= { ieee8021PbObjects 7 }
16
17 ieee8021PbCepEntry OBJECT-TYPE
18     SYNTAX      Ieee8021PbCepEntry
19     MAX-ACCESS  not-accessible
20     STATUS      current
21     DESCRIPTION
22         "The entry of the ieee8021PbCepTable. Note that the component
23         index must refer to the S-VLAN component of a PEB, and that the port
24         number for the CEP is allocated from the port number space of
25         that S-VLAN component."
26     INDEX { ieee8021BridgeBasePortComponentId,
27             ieee8021BridgeBasePort }
28     ::= { ieee8021PbCepTable 1 }
29
30 Ieee8021PbCepEntry ::=
31     SEQUENCE {
32         ieee8021PbCepCComponentId  IEEE8021PbbComponentIdentifierOrZero,
33         ieee8021PbCepCepPortNumber IEEE8021BridgePortNumberOrZero,
34         ieee8021PbCepRowStatus     RowStatus
35     }
36
37 ieee8021PbCepCComponentId OBJECT-TYPE
38     SYNTAX      IEEE8021PbbComponentIdentifierOrZero
39     MAX-ACCESS  read-only
40     STATUS      current
41     DESCRIPTION
42         "This column is an implementation specific column that may be
43         used to map the C component associated with this CEP to other
44         tables within the system, such as the Entity MIB. This
45         column may not be created or modified by management station
46         action. A value of 0 is always legal, and non-zero values
47         will be interpreted in an implementation specific manner.
48         The value of this column may or may not persist across system
49         restarts."
50     ::= { ieee8021PbCepEntry 1 }
51
52 ieee8021PbCepCepPortNumber OBJECT-TYPE
53     SYNTAX      IEEE8021BridgePortNumberOrZero
54     MAX-ACCESS  read-only
55     STATUS      current
56     DESCRIPTION
57         "This column is an implementation specific column that may be
58         used to map the CEP to other tables within the system, for
59         example the Entity MIB. This column may not be created or
60         modified by management station action. A value of 0 is
61         always legal, and non-zero values will be interpreted in an
62         implementation specific manner. The value of this column
63         may or may not persist across system restarts."
64     ::= { ieee8021PbCepEntry 2 }
65
66 ieee8021PbCepRowStatus OBJECT-TYPE
67     SYNTAX      RowStatus
68     MAX-ACCESS  read-create
69     STATUS      current
70     DESCRIPTION
71         "This indicates the status of the entry, and is used to create
72         and delete entries in this table."

```

```

1      ::= { ieee8021PbCepEntry 3 }
2
3 -- =====
4 -- ieee8021PbRcapTable
5 -- =====
6
7 ieee8021PbRcapTable OBJECT-TYPE
8     SYNTAX      SEQUENCE OF Ieee8021PbRcapEntry
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "This table is used to create Remote Customer Access Ports, on a
13        provider edge Bridge. It is indexed by the ComponentId of the
14        PEB's S-VLAN component and by the port number for the RCAP. Note that
15        the index port number belongs to the set of port numbers
16        associated with the PEB's primary S-VLAN component.
17
18        Entries in this table must be persistent across reinitializations
19        of the management agent. However, note that some column values,
20        as noted below, may change across system reinitializations."
21    ::= { ieee8021PbObjects 8 }
22
23 ieee8021PbRcapEntry OBJECT-TYPE
24     SYNTAX      Ieee8021PbRcapEntry
25     MAX-ACCESS  not-accessible
26     STATUS      current
27     DESCRIPTION
28         "The entry of the ieee8021PbRcapTable. Note that the component
29         index must refer to the primary S-VLAN component of a PEB, and that
30         the port number index for the RCAP is allocated from the port
31         number space of that S-VLAN component."
32     INDEX { ieee8021BridgeBasePortComponentId,
33             ieee8021BridgeBasePort }
34     ::= { ieee8021PbRcapTable 1 }
35
36 Ieee8021PbRcapEntry ::=
37     SEQUENCE {
38         ieee8021PbRcapSComponentId  IEEE8021PbbComponentIdentifierOrZero,
39         ieee8021PbRcapRcapPortNumber IEEE8021BridgePortNumberOrZero,
40         ieee8021PbRcapRowStatus      RowStatus
41     }
42
43 ieee8021PbRcapSComponentId OBJECT-TYPE
44     SYNTAX      IEEE8021PbbComponentIdentifierOrZero
45     MAX-ACCESS  read-only
46     STATUS      current
47     DESCRIPTION
48         "This column is an implementation specific column that may be
49         used to map the Port-mapping S-VLAN component associated with
50         this RCAP to other tables within the system, such as the
51         Entity MIB. This column may not be created or modified
52         by management station action. A value of 0 is always legal,
53         and non-zero values will be interpreted in an implementation
54         specific manner. The value of this column may or may not
55         persist across system restarts."
56     ::= { ieee8021PbRcapEntry 1 }
57
58 ieee8021PbRcapRcapPortNumber OBJECT-TYPE
59     SYNTAX      IEEE8021BridgePortNumberOrZero
60     MAX-ACCESS  read-only
61     STATUS      current
62     DESCRIPTION
63         "This column is an implementation specific column that may be
64         used to map the RCAP to other tables within the system, for
65         example the Entity MIB. This column may not be created or
66         modified by management station action. A value of 0 is
67         always legal, and non-zero values will be interpreted in an
68         implementation specific manner. The value of this column
69         may or may not persist across system restarts."
70     ::= { ieee8021PbRcapEntry 2 }
71
72 ieee8021PbRcapRowStatus OBJECT-TYPE

```

```

1  SYNTAX      RowStatus
2  MAX-ACCESS  read-create
3  STATUS      current
4  DESCRIPTION
5      "This indicates the status of the entry, and is used to create
6      and delete entries in this table."
7      ::= { ieee8021PbRcapEntry 3 }
8
9  -- =====
10 -- ieee8021PbInternalInterfaceTable:
11 -- =====
12
13 ieee8021PbInternalInterfaceTable OBJECT-TYPE
14     SYNTAX      SEQUENCE OF Ieee8021PbIiEntry
15     MAX-ACCESS  not-accessible
16     STATUS      current
17     DESCRIPTION
18         "A table used in a Port-mapping S-VLAN component associated with
19         a Remote Customer Access Port to manage the mapping between
20         external S-VIDs and internal interfaces/S-VIDs.
21
22         Note that the component ID of entries in this table must refer
23         to the primary S-VLAN component of a Provider Edge Bridge and
24         the Port Number must refer to the port number of a Remote
25         Customer Access Port associated with that S-VLAN component.
26
27         Entries in this table must be persistent over power up
28         restart/reboot."
29     REFERENCE   "12.13.3.1, 12.13.3.2"
30     ::= { ieee8021PbObjects 9 }
31
32 ieee8021PbIiEntry OBJECT-TYPE
33     SYNTAX      Ieee8021PbIiEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "An element of the internal interface table accessed by PB
38         S-VLAN component ID, Remote Customer Access Port Bridge Port
39         number, and external S-VID. Each element contains the mapping
40         between an external S-VID and the internal port it selects and,
41         except in the case of a C-tagged service interface the
42         internal S-VID that carries the service ."
43     INDEX { ieee8021BridgeBasePortComponentId,
44             ieee8021BridgeBasePort,
45             ieee8021PbIiExternalSvid }
46     ::= { ieee8021PbInternalInterfaceTable 1 }
47
48 Ieee8021PbIiEntry ::= SEQUENCE {
49     ieee8021PbIiExternalSvid      VlanId,
50     ieee8021PbIiInternalPortNumber IEEE8021BridgePortNumberOrZero,
51     ieee8021PbIiInternalPortType  IEEE8021BridgePortType,
52     ieee8021PbIiInternalSvid      VlanIdOrNone,
53     ieee8021PbIiRowStatus         RowStatus
54 }
55
56 ieee8021PbIiExternalsvid OBJECT-TYPE
57     SYNTAX      VlanId
58     MAX-ACCESS  not-accessible
59     STATUS      current
60     DESCRIPTION
61         "External S-VID for this internal interface table entry."
62     ::= { ieee8021PbIiEntry 1 }
63
64 ieee8021PbIiInternalPortNumber OBJECT-TYPE
65     SYNTAX      IEEE8021BridgePortNumberOrZero
66     MAX-ACCESS  read-only
67     STATUS      current
68     DESCRIPTION
69         "The port number for the internal port on the primary
70         S-VLAN component. This port number is used in FDB entries
71         that reference an RCSI.
72

```

```

1      The port number of the Remote Customer Access Port can
2      be used to identify a PNP on the primary S-VLAN component
3      connected to a PNP on the Port-mapping S-VLAN component."
4      ::= { ieee8021PbIiEntry 2 }
5
6 ieee8021PbIiInternalPortType OBJECT-TYPE
7     SYNTAX      IEEE8021BridgePortType
8     MAX-ACCESS  read-create
9     STATUS      current
10    DESCRIPTION
11        "The port type of the internal port on the primary
12        S-VLAN component. This indicates the type of RCSI as follows:
13
14            providerNetworkPort(3) - Indicates a PNP (not an RCSI)
15            customerNetworkPort(4) - Indicates a Port-based RCSI
16            customerEdgePort(5) - Indicates a C-tagged RCSI
17
18        Other port type values are not valid for this field."
19    ::= { ieee8021PbIiEntry 3 }
20
21 ieee8021PbIiInternalSvid OBJECT-TYPE
22     SYNTAX      VlanIdOrNone
23     MAX-ACCESS  read-create
24     STATUS      current
25     DESCRIPTION
26        "Internal S-VID for this external S-VID entry."
27    ::= { ieee8021PbIiEntry 4 }
28
29 ieee8021PbIiRowStatus OBJECT-TYPE
30     SYNTAX      RowStatus
31     MAX-ACCESS  read-create
32     STATUS      current
33     DESCRIPTION
34        "This indicates the status of an entry in this table, and is
35        used to create/delete entries.
36
37        The value of ieee8021PbIiExternalSvid must be set before
38        an entry in this table can be made valid.deprecated
39
40        It is an implementation specific decision as to whether any
41        column in this table may be set while the corresponding
42        instance of this object is valid(1)."
43    ::= { ieee8021PbIiEntry 5 }
44
45 -- =====
46 -- Conformance Information
47 -- =====
48
49 ieee8021PbGroups
50     OBJECT IDENTIFIER ::= { ieee8021PbConformance 1 }
51 ieee8021PbCompliances
52     OBJECT IDENTIFIER ::= { ieee8021PbConformance 2 }
53
54 -- =====
55 -- Units of conformance
56 -- =====
57
58 ieee8021PbVidTranslationGroup OBJECT-GROUP
59     OBJECTS {
60         ieee8021PbVidTranslationRelayVid,
61         ieee8021PbVidTranslationRowStatus
62     }
63     STATUS      deprecated
64     DESCRIPTION
65        "The collection of objects used to represent a PB
66        C-VID/S-VID translation."
67    ::= { ieee8021PbGroups 1 }
68
69 ieee8021PbCVidRegistrationGroup OBJECT-GROUP
70     OBJECTS {
71         ieee8021PbCVidRegistrationSvid,
72         ieee8021PbCVidRegistrationUntaggedPep,

```

```

1         ieee8021PbCVidRegistrationUntaggedCep,
2         ieee8021PbCVidRegistrationRowStatus
3     }
4     STATUS          current
5     DESCRIPTION
6         "The collection of objects used to represent a CEP translation."
7     ::= { ieee8021PbGroups 2 }
8
9 ieee8021PbEdgePortGroup OBJECT-GROUP
10    OBJECTS {
11        ieee8021PbEdgePortPVID,
12        ieee8021PbEdgePortDefaultUserPriority,
13        ieee8021PbEdgePortAcceptableFrameTypes,
14        ieee8021PbEdgePortEnableIngressFiltering
15    }
16    STATUS          current
17    DESCRIPTION
18        "The collection of objects user to represent a PEP."
19    ::= { ieee8021PbGroups 3 }
20
21 ieee8021PbServicePriorityRegenerationGroup OBJECT-GROUP
22    OBJECTS {
23        ieee8021PbServicePriorityRegenerationRegeneratedPriority
24    }
25    STATUS          current
26    DESCRIPTION
27        "A regenerated priority value for a PEP."
28    ::= { ieee8021PbGroups 4 }
29
30 ieee8021PbDynamicCnpGroup OBJECT-GROUP
31    OBJECTS {
32        ieee8021PbCnpCComponentId,
33        ieee8021PbCnpSVid,
34        ieee8021PbCnpRowStatus
35    }
36    STATUS          current
37    DESCRIPTION
38        "A set of objects used for dynamic creation and deletion
39        of customer network ports."
40    ::= { ieee8021PbGroups 5 }
41
42 ieee8021PbDynamicPnpGroup OBJECT-GROUP
43    OBJECTS {
44        ieee8021PbPnpRowStatus
45    }
46    STATUS          current
47    DESCRIPTION
48        "A set of objects used for dynamic creation and deletion
49        of provider network ports."
50    ::= { ieee8021PbGroups 6 }
51
52 ieee8021PbDynamicCepGroup OBJECT-GROUP
53    OBJECTS {
54        ieee8021PbCepCComponentId,
55        ieee8021PbCepCepPortNumber,
56        ieee8021PbCepRowStatus
57    }
58    STATUS          current
59    DESCRIPTION
60        "A set of objects used for dynamic creation and deletion
61        of customer edge ports."
62    ::= { ieee8021PbGroups 7 }
63
64 ieee8021PbDynamicRcapGroup OBJECT-GROUP
65    OBJECTS {
66        ieee8021PbRcapSComponentId,
67        ieee8021PbRcapRcapPortNumber,
68        ieee8021PbCepRowStatus
69    }
70    STATUS          deprecated
71    DESCRIPTION
72        "A set of objects used for dynamic creation and deletion

```

```

1      of remote customer access ports."
2      ::= { ieee8021PbGroups 8 }
3
4 ieee8021PbInternalInterfaceGroup OBJECT-GROUP
5     OBJECTS {
6         ieee8021PbIiInternalPortNumber,
7         ieee8021PbIiInternalPortType,
8         ieee8021PbIiInternalSVid,
9         ieee8021PbIiRowStatus
10    }
11    STATUS      current
12    DESCRIPTION
13        "A set of objects used for dynamic creation and deletion
14        of internal interfaces on a Port-mapping S-VLAN component."
15    ::= { ieee8021PbGroups 9 }
16
17 ieee8021PbDynamicRcapV2Group OBJECT-GROUP
18     OBJECTS {
19         ieee8021PbRcapSComponentId,
20         ieee8021PbRcapRcapPortNumber,
21         ieee8021PbRcapRowStatus
22     }
23     STATUS      current
24     DESCRIPTION
25         "A set of objects used for dynamic creation and deletion
26         of remote customer access ports."
27     ::= { ieee8021PbGroups 10 }
28
29
30 -- =====
31 -- Compliance statements
32 -- =====
33
34 ieee8021PbCompliance MODULE-COMPLIANCE
35     STATUS      deprecated
36     DESCRIPTION
37         "The compliance statement for device support of Provider
38         Bridge services."
39
40     MODULE
41         MANDATORY-GROUPS {
42             ieee8021PbVidTranslationGroup,
43             ieee8021PbCVidRegistrationGroup,
44             ieee8021PbEdgePortGroup,
45             ieee8021PbServicePriorityRegenerationGroup
46         }
47
48     GROUP      ieee8021PbDynamicCnpGroup
49     DESCRIPTION
50         "This group is optional and supports dynamic creation
51         and deletion of customer network ports."
52
53     GROUP      ieee8021PbDynamicPnpGroup
54     DESCRIPTION
55         "This group is optional and supports dynamic creation
56         and deletion of provider network ports."
57
58     GROUP      ieee8021PbDynamicCepGroup
59     DESCRIPTION
60         "This group is optional and supports dynamic creation
61         and deletion of customer edge ports."
62
63     GROUP      ieee8021PbDynamicRcapGroup
64     DESCRIPTION
65         "This group is optional and supports dynamic creation
66         and deletion of remote customer access ports."
67
68     GROUP      ieee8021PbInternalInterfaceGroup
69     DESCRIPTION
70         "This group is optional and supports dynamic creation
71         and deletion of internal interfaces on Port-mapping
72         S-VLAN components."

```

```

1
2 ::= { ieee8021PbCompliances 1 }
3
4 ieee8021PbComplianceV2 MODULE-COMPLIANCE
5     STATUS      current
6     DESCRIPTION
7         "The compliance statement for device support of Provider
8         Bridge services."
9
10    MODULE
11        MANDATORY-GROUPS {
12            ieee8021PbCvidRegistrationGroup,
13            ieee8021PbEdgePortGroup,
14            ieee8021PbServicePriorityRegenerationGroup
15        }
16
17        GROUP      ieee8021PbDynamicCnpGroup
18        DESCRIPTION
19            "This group is optional and supports dynamic creation
20            and deletion of customer network ports."
21
22        GROUP      ieee8021PbDynamicPnpGroup
23        DESCRIPTION
24            "This group is optional and supports dynamic creation
25            and deletion of provider network ports."
26
27        GROUP      ieee8021PbDynamicCepGroup
28        DESCRIPTION
29            "This group is optional and supports dynamic creation
30            and deletion of customer edge ports."
31
32        GROUP      ieee8021PbDynamicRcapV2Group
33        DESCRIPTION
34            "This group is optional and supports dynamic creation
35            and deletion of remote customer access ports."
36
37        GROUP      ieee8021PbInternalInterfaceGroup
38        DESCRIPTION
39            "This group is optional and supports dynamic creation
40            and deletion of internal interfaces on Port-mapping
41            S-VLAN components."
42
43 ::= { ieee8021PbCompliances 2 }
44
45 END

```

1 17.7.6 Definitions for the IEEE8021-MSTP-MIB module

```
2 IEEE8021-MSTP-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for IEEE 802.1Q Multiple Spanning Tree Bridge Devices
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE, Integer32, Counter64,
10     Unsigned32, TimeTicks
11     FROM SNMPv2-SMI
12     TruthValue, RowStatus
13     FROM SNMPv2-TC
14     ieee802dot1mibs, IEEE8021PbbComponentIdentifier,
15     IEEE8021BridgePortNumber, IEEE8021VlanIndex,
16     IEEE8021MstIdentifier
17     FROM IEEE8021-TC-MIB
18     BridgeId
19     FROM BRIDGE-MIB
20     SnmpAdminString
21     FROM SNMP-FRAMEWORK-MIB
22     MODULE-COMPLIANCE, OBJECT-GROUP
23     FROM SNMPv2-CONF;
24
25 ieee8021MstpMib MODULE-IDENTITY
26     LAST-UPDATED "202211080000Z" -- November 8, 2022
27     ORGANIZATION "IEEE 802.1 Working Group"
28     CONTACT-INFO
29         " WG-URL: http://www.ieee802.org/1/
30         WG-EMail: stds-802-1-1@ieee.org
31         Contact: IEEE 802.1 Working Group Chair
32         Postal: C/O IEEE 802.1 Working Group
33             IEEE Standards Association
34             445 Hoes Lane
35             Piscataway, NJ 08854
36             USA
37         E-mail: stds-802-1-chairs@ieee.org"
38     DESCRIPTION
39         "The Bridge MIB modules for managing devices that support
40         IEEE Std 802.1Q multiple spanning tree groups.
41
42         Unless otherwise indicated, the references in this MIB
43         module are to IEEE Std 802.1Q-2022.
44
45         Copyright (C) IEEE (2022).
46         This version of this MIB module is part of IEEE Std 802.1Q;
47         see that standard for full legal notices."
48
49     REVISION "202211080000Z" -- November 8, 2022
50     DESCRIPTION
51         "Published as part of IEEE Std 802.1Q-2022.
52         Cross references and contact information updated."
53
54     REVISION "201807010000Z" -- July 1, 2018
55     DESCRIPTION
56         "Published as part of IEEE Std 802.1Q 2018 revision.
57         Cross references updated and corrected."
58
59     REVISION "201412150000Z" -- December 15, 2014
60     DESCRIPTION
61         "Published as part of IEEE Std 802.1Q 2014 revision.
62         Cross references updated and corrected.
63         Instances of ...value of mstp(2)... changed to
64         ...value of mstp(3).
65         Defval for ieee8021MstpCistPortAdminEdgePort
66         changed to false.
67         ieee8021MstpVlanV2Table DESCRIPTION updated
68         (4094 not 4096).
69         Bug fixes to conformance section."
70
71     REVISION "201208100000Z" -- August 10, 2012
```


1 DESCRIPTION
2 "Updated cross references to other clauses, particularly
3 Clause 13, as part of IEEE 802.1Q Cor-2."
4
5 REVISION "201112120000Z" -- December 12, 2011
6 DESCRIPTION
7 "Deprecated ieee8021MstpFidToMstiTable for an identical
8 ieee8021MstpFidToMstiV2Table to add 4095 to the range
9 of ieee8021MstpFidToMstiV2Fid and to add 0 and 4095 to
10 the range of ieee8021MstpFidToMstiV2MstId for IEEE Std 802.1aq.
11 Deprecated ieee8021MstpVlanTable for an identical
12 ieee8021MstpVlanV2Table to add 0 & 4095 to the range
13 of ieee8021MstpVlanV2MstId for IEEE Std 802.1aq"
14
15 REVISION "201103230000Z" -- March 23, 2011
16 DESCRIPTION
17 "Minor edits to contact information, correction to range of
18 ieee8021MstpCistMaxHops and addition of fragile Bridge
19 as part of 2011 revision of IEEE Std 802.1Q."
20
21 REVISION "200810150000Z" -- October 15, 2008
22 DESCRIPTION
23 "Initial version."
24 ::= { ieee802dot1mibs 6 }
25
26 ieee8021MstpNotifications OBJECT IDENTIFIER ::= { ieee8021MstpMib 0 }
27 ieee8021MstpObjects OBJECT IDENTIFIER ::= { ieee8021MstpMib 1 }
28 ieee8021MstpConformance OBJECT IDENTIFIER ::= { ieee8021MstpMib 2 }
29
30 -- =====
31 -- MSTP CIST Table
32 -- =====
33
34 ieee8021MstpCistTable OBJECT-TYPE
35 SYNTAX SEQUENCE OF Ieee8021MstpCistEntry
36 MAX-ACCESS not-accessible
37 STATUS current
38 DESCRIPTION
39 "The Common and Internal Spanning Tree (CIST) Table. Each row in
40 the table represents information regarding a Bridge's Bridge
41 Protocol Entity for the CIST.
42
43 Note that entries will exist in this table only for Bridge
44 components for which the corresponding instance of
45 ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
46 has a value of mstp(3).
47
48 This table contains objects corresponding to the following items
49 from 12.8.1.1 and 12.8.1.3. Some of those items are provided
50 in the IEEE8021-SPANNING-TREE-MIB as noted below.
51
52 From 12.8.1.1:
53 Items a), c), o), p), and q) are defined in this table
54 The remaining items are covered in the
55 IEEE8021-SPANNING-TREE-MIB:
56 b) ieee8021SpanningTreeTimeSinceTopologyChange
57 c) ieee8021SpanningTreeTopChanges
58 e) ieee8021SpanningTreeDesignatedRoot
59 f) ieee8021SpanningTreeRootCost
60 g) ieee8021SpanningTreeRootPort
61 h) ieee8021SpanningTreeMaxAge
62 i) ieee8021SpanningTreeForwardDelay
63 j) ieee8021SpanningTreeBridgeMaxAge
64 k) ieee8021SpanningTreeBridgeHelloTime
65 l) ieee8021SpanningTreeBridgeForwardDelay
66 m) ieee8021SpanningTreeHoldTime
67 n) ieee8021SpanningTreeVersion
68 From 12.8.1.3:
69 Item g) is defined in this table
70 The remaining items are covered in the
71 IEEE8021-SPANNING-TREE-MIB:
72 a) ieee8021SpanningTreeBridgeMaxAge

```

1          b) ieee8021SpanningTreeBridgeHelloTime
2          c) ieee8021SpanningTreeBridgeForwardDelay
3          d) ieee8021SpanningTreePriority
4          e) ieee8021SpanningTreeVersion
5          f) ieee8021RstpStpExtTxHoldCount"
6  REFERENCE "12.8.1.1, 12.8.1.3"
7  ::= { ieee8021MstpObjects 1 }
8
9  ieee8021MstpCistEntry OBJECT-TYPE
10   SYNTAX      Ieee8021MstpCistEntry
11   MAX-ACCESS  not-accessible
12   STATUS      current
13   DESCRIPTION
14       "A CIST Table entry."
15   INDEX { ieee8021MstpCistComponentId }
16   ::= { ieee8021MstpCistTable 1 }
17
18  Ieee8021MstpCistEntry ::= SEQUENCE {
19      ieee8021MstpCistComponentId      IEEE8021PbbComponentIdentifier,
20      ieee8021MstpCistBridgeIdentifier  BridgeId,
21      ieee8021MstpCistTopologyChange    TruthValue,
22      ieee8021MstpCistRegionalRootIdentifier BridgeId,
23      ieee8021MstpCistPathCost          Unsigned32,
24      ieee8021MstpCistMaxHops           Integer32
25  }
26
27  ieee8021MstpCistComponentId OBJECT-TYPE
28   SYNTAX      IEEE8021PbbComponentIdentifier
29   MAX-ACCESS  not-accessible
30   STATUS      current
31   DESCRIPTION
32       "The component identifier is used to distinguish between the
33       multiple virtual Bridge instances within a PBB. In simple
34       situations where there is only a single component the default
35       value is 1."
36   ::= { ieee8021MstpCistEntry 1 }
37
38  ieee8021MstpCistBridgeIdentifier OBJECT-TYPE
39   SYNTAX      BridgeId
40   MAX-ACCESS  read-only
41   STATUS      current
42   DESCRIPTION
43       "The Bridge Identifier for the CIST."
44   REFERENCE "12.8.1.1"
45   ::= { ieee8021MstpCistEntry 2 }
46
47  ieee8021MstpCistTopologyChange OBJECT-TYPE
48   SYNTAX      TruthValue
49   MAX-ACCESS  read-only
50   STATUS      current
51   DESCRIPTION
52       "In an STP Bridge, the value of the Topology Change parameter
53       (14.8.1.1.3, item d of IEEE Std 802.1D, 2004 Edition), or in
54       an RSTP or MSTP Bridge, asserted if the tcWhile timer for any
55       Port for the CIST is non-zero."
56   REFERENCE "13.25.9, 14.8.1.1.3:d of IEEE Std 802.1D-2004"
57   ::= { ieee8021MstpCistEntry 3 }
58
59  ieee8021MstpCistRegionalRootIdentifier OBJECT-TYPE
60   SYNTAX      BridgeId
61   MAX-ACCESS  read-only
62   STATUS      current
63   DESCRIPTION
64       "In an MSTP Bridge, the CIST Regional Root Identifier parameter,
65       i.e., the Bridge Identifier of the current CIST Regional Root."
66   REFERENCE "13.16.4, 13.26.3"
67   ::= { ieee8021MstpCistEntry 4 }
68
69  ieee8021MstpCistPathCost OBJECT-TYPE
70   SYNTAX      Unsigned32 (0..2147483647)
71   MAX-ACCESS  read-only
72   STATUS      current

```

```

1  DESCRIPTION
2      "In an MSTP Bridge, the CIST Path Cost parameter, i.e., the CIST
3      path cost from the transmitting Bridge to the CIST Regional Root.
4      The sum (about 20 possible out of the given range) of multiple
5      port path costs. Also, if the 'transmitting Bridge' is
6      the 'CIST Regional Root', then this value could be zero."
7  REFERENCE    "13.9:d, 13.10"
8  ::= { ieee8021MstpCistEntry 5 }
9
10 ieee8021MstpCistMaxHops OBJECT-TYPE
11     SYNTAX      Integer32 (6..40)
12     MAX-ACCESS  read-write
13     STATUS      current
14     DESCRIPTION
15         "In an MSTP Bridge, the MaxHops parameter.
16
17         The value of this object MUST be retained across
18         reinitializations of the management system."
19     REFERENCE    "13.26.4"
20     ::= { ieee8021MstpCistEntry 6 }
21
22 -- =====
23 -- ieee8021MstpTable:
24 -- =====
25
26 ieee8021MstpTable OBJECT-TYPE
27     SYNTAX      SEQUENCE OF Ieee8021MstpEntry
28     MAX-ACCESS  not-accessible
29     STATUS      current
30     DESCRIPTION
31         "In an MSTP Bridge, the MSTP Table. Each row in the Table
32         represents information regarding a Bridge's Bridge Protocol
33         Entity for the specified Spanning Tree instance.
34
35         Entries in this table MUST be retained across
36         reinitializations of the management system.
37
38         Note that entries can be created in this table only for Bridge
39         components for which the corresponding instance of
40         ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
41         has a value of mstp(3)."
42     REFERENCE    "12.8.1.2, 12.8.1.4, 12.12.3.2, 12.12.1"
43     ::= { ieee8021MstpObjects 2 }
44
45 ieee8021MstpEntry OBJECT-TYPE
46     SYNTAX      Ieee8021MstpEntry
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "A MSTP Table entry."
51     INDEX { ieee8021MstpComponentId, ieee8021MstpId }
52     ::= { ieee8021MstpTable 1 }
53
54 Ieee8021MstpEntry ::= SEQUENCE {
55     ieee8021MstpComponentId      IEEE8021PbbComponentIdentifier,
56     ieee8021MstpId               IEEE8021MstIdentifier,
57     ieee8021MstpBridgeId         BridgeId,
58     ieee8021MstpTimeSinceTopologyChange TimeTicks,
59     ieee8021MstpTopologyChanges Counter64,
60     ieee8021MstpTopologyChange   TruthValue,
61     ieee8021MstpDesignatedRoot   BridgeId,
62     ieee8021MstpRootPathCost     Integer32,
63     ieee8021MstpRootPort         IEEE8021BridgePortNumber,
64     ieee8021MstpBridgePriority    Integer32,
65     ieee8021MstpVids0            OCTET STRING,
66     ieee8021MstpVids1            OCTET STRING,
67     ieee8021MstpVids2            OCTET STRING,
68     ieee8021MstpVids3            OCTET STRING,
69     ieee8021MstpRowStatus        RowStatus
70 }
71
72 ieee8021MstpComponentId OBJECT-TYPE

```

```

1  SYNTAX      IEEE8021PbbComponentIdentifier
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "The component identifier is used to distinguish between the
6      multiple virtual Bridge instances within a PBB. In simple
7      situations where there is only a single component the default
8      value is 1."
9      ::= { ieee8021MstpEntry 1 }
10
11 ieee8021MstpId OBJECT-TYPE
12     SYNTAX      IEEE8021MstIdentifier
13     MAX-ACCESS  not-accessible
14     STATUS      current
15     DESCRIPTION
16         "In an MSTP Bridge, this parameter is the MSTID, i.e., the
17         identifier of a Spanning Tree (or MST) Instance."
18     ::= { ieee8021MstpEntry 2 }
19
20 ieee8021MstpBridgeId OBJECT-TYPE
21     SYNTAX      BridgeId
22     MAX-ACCESS  read-only
23     STATUS      current
24     DESCRIPTION
25         "In an MSTP Bridge, the Bridge Identifier for the MSTI."
26     REFERENCE   "13.26.2"
27     ::= { ieee8021MstpEntry 3 }
28
29 ieee8021MstpTimeSinceTopologyChange OBJECT-TYPE
30     SYNTAX      TimeTicks
31     UNITS       "centi-seconds"
32     MAX-ACCESS  read-only
33     STATUS      current
34     DESCRIPTION
35         "In an MSTP Bridge, count in seconds of the time elapsed since
36         tcWhile was last non-zero for any Port for the MSTI."
37     REFERENCE   "13.25.9"
38     ::= { ieee8021MstpEntry 4 }
39
40 ieee8021MstpTopologyChanges OBJECT-TYPE
41     SYNTAX      Counter64
42     UNITS       "topology changes"
43     MAX-ACCESS  read-only
44     STATUS      current
45     DESCRIPTION
46         "In an MSTP Bridge, count of the times tcWhile has been
47         non-zero for any Port for the MSTI since the Bridge was powered
48         on or initialized."
49     REFERENCE   "13.25.9"
50     ::= { ieee8021MstpEntry 5 }
51
52 ieee8021MstpTopologyChange OBJECT-TYPE
53     SYNTAX      TruthValue
54     MAX-ACCESS  read-only
55     STATUS      current
56     DESCRIPTION
57         "In an MSTP Bridge, the Topology Change parameter value: true(1)
58         if tcWhile is non-zero for any Port for the MSTI."
59     REFERENCE   "13.25.9"
60     ::= { ieee8021MstpEntry 6 }
61
62 ieee8021MstpDesignatedRoot OBJECT-TYPE
63     SYNTAX      BridgeId
64     MAX-ACCESS  read-only
65     STATUS      current
66     DESCRIPTION
67         "In an MSTP Bridge, the Designated Root parameter value, i.e., the
68         Bridge Identifier of the Root Bridge for the MSTI."
69     REFERENCE   "13.27.20"
70     ::= { ieee8021MstpEntry 7 }
71
72 ieee8021MstpRootPathCost OBJECT-TYPE

```

```

1  SYNTAX      Integer32
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "In an MSTP Bridge, the Root Path Cost parameter value, i.e., the
6      path cost from the transmitting Bridge to the Root Bridge for
7      the MSTI."
8  REFERENCE   "13.27.20"
9  ::= { ieee8021MstpEntry 8 }
10
11 ieee8021MstpRootPort OBJECT-TYPE
12     SYNTAX      IEEE8021BridgePortNumber
13     MAX-ACCESS  read-only
14     STATUS      current
15     DESCRIPTION
16         "In an MSTP Bridge, the Root Port parameter value, i.e., the Root
17         Port for the MSTI."
18     REFERENCE   "13.26.9"
19     ::= { ieee8021MstpEntry 9 }
20
21 ieee8021MstpBridgePriority OBJECT-TYPE
22     SYNTAX      Integer32 (0..61440)
23     MAX-ACCESS  read-create
24     STATUS      current
25     DESCRIPTION
26         "In an MSTP Bridge, the Bridge Priority parameter value for the
27         MSTI, i.e., the most significant 4 bits of the Bridge Identifier
28         for the MSTI."
29     REFERENCE   "13.26.3"
30     ::= { ieee8021MstpEntry 10 }
31
32 ieee8021MstpVids0 OBJECT-TYPE
33     SYNTAX      OCTET STRING (SIZE(128))
34     MAX-ACCESS  read-only
35     STATUS      current
36     DESCRIPTION
37         "This object contains the first 1024 bits of the 4096 bit vector
38         indicating which VIDs are assigned to this MSTID. The high order
39         bit of the first octet corresponds to the first bit of the vector,
40         while the low order bit of the last octet corresponds to the last
41         bit of this portion of the vector. A bit that is on (equal to 1)
42         indicates that the corresponding VID is assigned to this MSTID."
43     ::= { ieee8021MstpEntry 11 }
44
45 ieee8021MstpVids1 OBJECT-TYPE
46     SYNTAX      OCTET STRING (SIZE(128))
47     MAX-ACCESS  read-only
48     STATUS      current
49     DESCRIPTION
50         "This object contains the second 1024 bits of the 4096 bit vector
51         indicating which VIDs are assigned to this MSTID. The high order
52         bit of the first octet corresponds to the first bit of this
53         portion of the vector, while the low order bit of the last octet
54         corresponds to the last bit of this portion of the vector. A bit
55         that is on (equal to 1) indicates that the corresponding VID is
56         assigned to this MSTID."
57     ::= { ieee8021MstpEntry 12 }
58
59 ieee8021MstpVids2 OBJECT-TYPE
60     SYNTAX      OCTET STRING (SIZE(128))
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64         "This object contains the third 1024 bits of the 4096 bit vector
65         indicating which VIDs are assigned to this MSTID. The high order
66         bit of the first octet corresponds to the first bit of this
67         portion of the vector, while the low order bit of the last octet
68         corresponds to the last bit of this portion of the vector. A bit
69         that is on (equal to 1) indicates that the corresponding VID is
70         assigned to this MSTID."
71     ::= { ieee8021MstpEntry 13 }
72

```

```

1 ieee8021MstpVids3 OBJECT-TYPE
2     SYNTAX      OCTET STRING (SIZE(128))
3     MAX-ACCESS  read-only
4     STATUS      current
5     DESCRIPTION
6         "This object contains the fourth 1024 bits of the 4096 bit vector
7         indicating which VIDs are assigned to this MSTID. The high order
8         bit of the first octet corresponds to the first bit of this
9         portion of the vector, while the low order bit of the last octet
10        corresponds to the last bit of this portion of the vector. A bit
11        that is on (equal to 1) indicates that the corresponding VID is
12        assigned to this MSTID."
13    ::= { ieee8021MstpEntry 14 }
14
15 ieee8021MstpRowStatus OBJECT-TYPE
16     SYNTAX      RowStatus
17     MAX-ACCESS  read-create
18     STATUS      current
19     DESCRIPTION
20         "The status of the row.
21
22         Read SNMPv2-TC (RFC2579) for an
23         explanation of the possible values this object can take.
24
25         The writable columns in a row cannot be changed if the row
26         is active. All columns must have a valid value before a row
27         can be activated."
28    ::= { ieee8021MstpEntry 15 }
29
30 -- =====
31 -- ieee8021MstpCistPortTable:
32 -- =====
33
34 ieee8021MstpCistPortTable OBJECT-TYPE
35     SYNTAX      SEQUENCE OF Ieee8021MstpCistPortEntry
36     MAX-ACCESS  not-accessible
37     STATUS      current
38     DESCRIPTION
39         "The CIST Port Table. Each row in the Table represents information
40         regarding a specific Port within the Bridge's Bridge Protocol
41         Entity, for the CIST.
42
43         The values of all writable objects in this table MUST be
44         retained across reinitializations of the management system.
45
46         Note that entries will exist in this table only for Bridge
47         components for which the corresponding instance of
48         ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
49         has a value of mstp(3).
50
51         This table contains objects corresponding to the following items
52         from 12.8.2.1, 12.8.2.3, and 12.8.2.5. Some of those items are
53         provided in the IEEE8021-SPANNING-TREE-MIB as noted below.
54
55         From 12.8.2.1:
56             Items a), d), e), and i) through w) are defined in this table
57             The remaining items are covered in the
58             IEEE8021-SPANNING-TREE-MIB:
59                 b) ieee8021SpanningTreePortState
60                 c) ieee8021SpanningTreePortPriority
61                 d) ieee8021SpanningTreePortPathCost
62                 f) ieee8021SpanningTreePortDesignatedCost
63                 g) ieee8021SpanningTreePortDesignatedBridge
64                 h) ieee8021SpanningTreePortDesignatedPort
65
66         From 12.8.2.3:
67             Items a), b), and d) through h) are defined in this table
68             (item a is the index)
69             The remaining items are covered in the
70             IEEE8021-SPANNING-TREE-MIB:
71                 b) ieee8021SpanningTreePortPathCost,
72                 c) ieee8021SpanningTreePortPriority
73
74         From 12.8.2.5:

```

```

1      All items are defined in this table
2      Also from 12.8.2.1:
3          Items u), v), w), and x) are defined in this table
4      Also from 12.8.2.3:
5          Items i), j), k), and l) are defined in this table"
6  REFERENCE "12.8.2.1, 12.8.2.3, 12.8.2.5"
7  ::= { ieee8021MstpObjects 3 }
8
9  ieee8021MstpCistPortEntry OBJECT-TYPE
10     SYNTAX      Ieee8021MstpCistPortEntry
11     MAX-ACCESS  not-accessible
12     STATUS      current
13     DESCRIPTION
14         "A CIST Port Table entry."
15     INDEX { ieee8021MstpCistPortComponentId, ieee8021MstpCistPortNum }
16     ::= { ieee8021MstpCistPortTable 1 }
17
18  Ieee8021MstpCistPortEntry ::= SEQUENCE {
19      ieee8021MstpCistPortComponentId      IEEE8021PbbComponentIdentifier,
20      ieee8021MstpCistPortNum              IEEE8021BridgePortNumber,
21      ieee8021MstpCistPortUptime           TimeTicks,
22      ieee8021MstpCistPortAdminPathCost    Integer32,
23      ieee8021MstpCistPortDesignatedRoot  BridgeId,
24      ieee8021MstpCistPortTopologyChangeAck TruthValue,
25      ieee8021MstpCistPortHelloTime        Integer32,
26      ieee8021MstpCistPortAdminEdgePort    TruthValue,
27      ieee8021MstpCistPortOperEdgePort     TruthValue,
28      ieee8021MstpCistPortMacEnabled        TruthValue,
29      ieee8021MstpCistPortMacOperational   TruthValue,
30      ieee8021MstpCistPortRestrictedRole    TruthValue,
31      ieee8021MstpCistPortRestrictedTcn     TruthValue,
32      ieee8021MstpCistPortRole              INTEGER,
33      ieee8021MstpCistPortDisputed          TruthValue,
34      ieee8021MstpCistPortCistRegionalRootId BridgeId,
35      ieee8021MstpCistPortCistPathCost      Unsigned32,
36      ieee8021MstpCistPortProtocolMigration TruthValue,
37      ieee8021MstpCistPortEnableBPDURx      TruthValue,
38      ieee8021MstpCistPortEnableBPDUTx      TruthValue,
39      ieee8021MstpCistPortPseudoRootId      BridgeId,
40      ieee8021MstpCistPortIsL2Gp            TruthValue
41  }
42
43  ieee8021MstpCistPortComponentId OBJECT-TYPE
44     SYNTAX      IEEE8021PbbComponentIdentifier
45     MAX-ACCESS  not-accessible
46     STATUS      current
47     DESCRIPTION
48         "The component identifier is used to distinguish between the
49         multiple virtual Bridge instances within a PBB. In simple
50         situations where there is only a single component the default
51         value is 1."
52     ::= { ieee8021MstpCistPortEntry 1 }
53
54  ieee8021MstpCistPortNum OBJECT-TYPE
55     SYNTAX      IEEE8021BridgePortNumber
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59         "The Port's Port Number parameter value for the CIST, i.e., the
60         number of the Bridge Port for the CIST."
61     ::= { ieee8021MstpCistPortEntry 2 }
62
63  ieee8021MstpCistPortUptime OBJECT-TYPE
64     SYNTAX      TimeTicks
65     UNITS        "centi-seconds"
66     MAX-ACCESS  read-only
67     STATUS      current
68     DESCRIPTION
69         "The Port's Uptime parameter value for the CIST, i.e., the count
70         in seconds of the time elapsed since the Port was last reset or
71         initialized (BEGIN, Annex E)."
72     ::= { ieee8021MstpCistPortEntry 3 }

```

```

1
2 ieee8021MstpCistPortAdminPathCost OBJECT-TYPE
3   SYNTAX      Integer32 (0..2000000000)
4   MAX-ACCESS  read-write
5   STATUS      current
6   DESCRIPTION
7       "The administratively assigned value for the contribution
8       of this port to the path cost of paths toward the spanning
9       tree root.
10
11       Writing a value of '0' assigns the automatically calculated
12       default Path Cost value to the port.  If the default Path
13       Cost is being used, this object returns '0' when read.
14
15       This complements the object ieee8021MstpCistPortCistPathCost,
16       which returns the operational value of the port path cost.
17
18       The value of this object MUST be retained across
19       reinitializations of the management system."
20   REFERENCE   "13.27.25, 17.13.11 of IEEE Std 802.1D"
21   ::= { ieee8021MstpCistPortEntry 4 }
22
23 ieee8021MstpCistPortDesignatedRoot OBJECT-TYPE
24   SYNTAX      BridgeId
25   MAX-ACCESS  read-only
26   STATUS      current
27   DESCRIPTION
28       "The CIST Regional Root Identifier component of the Port's port
29       priority vector, as defined in 13.10, for the CIST."
30   REFERENCE   "13.27.47"
31   ::= { ieee8021MstpCistPortEntry 5 }
32
33 ieee8021MstpCistPortTopologyChangeAck OBJECT-TYPE
34   SYNTAX      TruthValue
35   MAX-ACCESS  read-only
36   STATUS      current
37   DESCRIPTION
38       "The Port's Topology Change Acknowledge parameter value.
39       True(1) if a Configuration Message with a topology change
40       acknowledge flag set is to be transmitted. "
41   REFERENCE   "13.27.72, 17.19.41 of IEEE Std 802.1D"
42   ::= { ieee8021MstpCistPortEntry 6 }
43
44 ieee8021MstpCistPortHelloTime OBJECT-TYPE
45   SYNTAX      Integer32 (100..1000)
46   UNITS       "centi-seconds"
47   MAX-ACCESS  read-only
48   STATUS      current
49   DESCRIPTION
50       "The Port's Hello Time timer parameter value, for the CIST.
51       In centi-seconds"
52   REFERENCE   "13.27.48"
53   ::= { ieee8021MstpCistPortEntry 7 }
54
55 ieee8021MstpCistPortAdminEdgePort OBJECT-TYPE
56   SYNTAX      TruthValue
57   MAX-ACCESS  read-write
58   STATUS      current
59   DESCRIPTION
60       "In a Bridge that supports the identification of edge ports, the
61       Port's Admin Edge Port parameter value, for the CIST."
62   REFERENCE   "13.27.1"
63   DEFVAL      { false }
64   ::= { ieee8021MstpCistPortEntry 8 }
65
66 ieee8021MstpCistPortOperEdgePort OBJECT-TYPE
67   SYNTAX      TruthValue
68   MAX-ACCESS  read-only
69   STATUS      current
70   DESCRIPTION
71       "In a Bridge that supports the identification of edge ports, the
72       Port's operational Edge Port parameter value, for the CIST."

```



```

1      True(1) if it is an operEdge Port."
2      REFERENCE    "13.27.44"
3      ::= { ieee8021MstpCistPortEntry 9 }
4
5 ieee8021MstpCistPortMacEnabled OBJECT-TYPE
6     SYNTAX      TruthValue
7     MAX-ACCESS  read-write
8     STATUS      current
9     DESCRIPTION
10      "In a Bridge that supports the MAC Enabled parameter, the current
11      state of the MAC Enabled parameter.
12      True(1) indicates that administratively the MAC is set as if it
13      was connected to a point-to-point LAN."
14     REFERENCE    "12.8.2.1.3 p)"
15     ::= { ieee8021MstpCistPortEntry 10 }
16
17 ieee8021MstpCistPortMacOperational OBJECT-TYPE
18     SYNTAX      TruthValue
19     MAX-ACCESS  read-only
20     STATUS      current
21     DESCRIPTION
22      "In a Bridge that supports the MAC Operational parameter, the
23      current state of the MAC Operational parameter.
24      True(1) indicates the MAC is operational."
25     REFERENCE    "12.8.2.1.3 q)"
26     ::= { ieee8021MstpCistPortEntry 11 }
27
28 ieee8021MstpCistPortRestrictedRole OBJECT-TYPE
29     SYNTAX      TruthValue
30     MAX-ACCESS  read-write
31     STATUS      current
32     DESCRIPTION
33      "The current state of the restrictedRole parameter for the Port.
34      True(1) causes the Port not to be selected as Root Port for the
35      CIST or any MSTI. "
36     REFERENCE    "13.27.64"
37     ::= { ieee8021MstpCistPortEntry 12 }
38
39 ieee8021MstpCistPortRestrictedTcn OBJECT-TYPE
40     SYNTAX      TruthValue
41     MAX-ACCESS  read-write
42     STATUS      current
43     DESCRIPTION
44      "The current state of the restrictedTcn parameter for the Port.
45      True(1) causes the Port not to propagate topology changes to
46      other Ports."
47     REFERENCE    "13.27.65"
48     ::= { ieee8021MstpCistPortEntry 13 }
49
50 ieee8021MstpCistPortRole OBJECT-TYPE
51     SYNTAX      INTEGER {
52         root(1),
53         alternate(2),
54         designated(3),
55         backup(4)
56     }
57     MAX-ACCESS  read-only
58     STATUS      current
59     DESCRIPTION
60      "The current Port Role for the Port (i.e., Root, Alternate,
61      Designated, or Backup), for the CIST."
62     REFERENCE    "12.8.2.1.3 v)"
63     ::= { ieee8021MstpCistPortEntry 14 }
64
65 ieee8021MstpCistPortDisputed OBJECT-TYPE
66     SYNTAX      TruthValue
67     MAX-ACCESS  read-only
68     STATUS      current
69     DESCRIPTION
70      "The current value of the disputed variable for the CIST for
71      the Port. A value of true(1) indicates that the disputed
72      variable is set. A value of false(2) indicates that the

```

```

1         agreed variable is cleared."
2     REFERENCE    "13.27.22"
3     ::= { ieee8021MstpCistPortEntry 15 }
4
5 ieee8021MstpCistPortCistRegionalRootId OBJECT-TYPE
6     SYNTAX      BridgeId
7     MAX-ACCESS  read-only
8     STATUS      current
9     DESCRIPTION
10        "In an MSTP Bridge, the CIST Regional Root Identifier, i.e., the
11        Bridge Identifier of the current CIST Regional Root, for the CIST."
12     REFERENCE    "13.9:c, 13.10, 13.27.47"
13     ::= { ieee8021MstpCistPortEntry 16 }
14
15 ieee8021MstpCistPortCistPathCost OBJECT-TYPE
16     SYNTAX      Unsigned32 (0..2147483647)
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20        "In an MSTP Bridge, the Port's Port Path Cost
21        parameter value for the CIST."
22     REFERENCE    "13.27.25, 17.13.11 of IEEE Std 802.1D"
23     ::= { ieee8021MstpCistPortEntry 17 }
24
25 ieee8021MstpCistPortProtocolMigration OBJECT-TYPE
26     SYNTAX      TruthValue
27     MAX-ACCESS  read-write
28     STATUS      current
29     DESCRIPTION
30        "In an MSTP Bridge, the current value of the mcheck variable for
31        the Port. A value of true(1) forces the state machine to
32        perform functions as per 13.27.38."
33     REFERENCE    "13.27.38"
34     ::= { ieee8021MstpCistPortEntry 18 }
35
36 ieee8021MstpCistPortEnableBPDURx OBJECT-TYPE
37     SYNTAX      TruthValue
38     MAX-ACCESS  read-write
39     STATUS      current
40     DESCRIPTION
41        "In an MSTP Bridge, the enableBPDURx parameter value. A value
42        of false(2) indicates that BPDUs are ignored."
43     REFERENCE    "13.27.38"
44     DEFVAL { true }
45     ::= { ieee8021MstpCistPortEntry 19 }
46
47 ieee8021MstpCistPortEnableBPDUTx OBJECT-TYPE
48     SYNTAX      TruthValue
49     MAX-ACCESS  read-write
50     STATUS      current
51     DESCRIPTION
52        "In an MSTP Bridge, the enableBPDUTx parameter value. A value
53        of false(2) indicates that BPDUs are not transmitted."
54     REFERENCE    "13.27.24"
55     DEFVAL { true }
56     ::= { ieee8021MstpCistPortEntry 20 }
57
58 ieee8021MstpCistPortPseudoRootId OBJECT-TYPE
59     SYNTAX      BridgeId
60     MAX-ACCESS  read-write
61     STATUS      current
62     DESCRIPTION
63        "In an MSTP Bridge, the pseudoRootId parameter value."
64     REFERENCE    "13.27.51"
65     ::= { ieee8021MstpCistPortEntry 21 }
66
67 ieee8021MstpCistPortIsL2Gp OBJECT-TYPE
68     SYNTAX      TruthValue
69     MAX-ACCESS  read-write
70     STATUS      current
71     DESCRIPTION
72        "In an MSTP Bridge, the isL2gp parameter value. A value of

```

```

1      true(1) indicates this is an L2GP port."
2      REFERENCE      "13.27.26"
3      DEFVAL { false }
4      ::= { ieee8021MstpCistPortEntry 22 }
5
6  -- =====
7  -- ieee8021MstpPortTable:
8  -- =====
9
10 ieee8021MstpPortTable OBJECT-TYPE
11     SYNTAX          SEQUENCE OF Ieee8021MstpPortEntry
12     MAX-ACCESS      not-accessible
13     STATUS          current
14     DESCRIPTION
15         "The MSTP Port Table. Each row in the Table represents information
16         regarding a specific Port within the Bridge's Bridge Protocol
17         Entity, for a given MSTI.
18
19         The values of all writable objects in this table MUST be
20         retained across reinitializations of the management system.
21
22         Note that entries will exist in this table only for Bridge
23         components for which the corresponding instance of
24         ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
25         has a value of mstp(3)."
26     REFERENCE      "12.8.2.2, 12.8.2.4"
27     ::= { ieee8021MstpObjects 4 }
28
29 ieee8021MstpPortEntry OBJECT-TYPE
30     SYNTAX          Ieee8021MstpPortEntry
31     MAX-ACCESS      not-accessible
32     STATUS          current
33     DESCRIPTION
34         "A MSTP Port Table entry."
35     INDEX { ieee8021MstpPortComponentId,
36             ieee8021MstpPortMstId,
37             ieee8021MstpPortNum }
38     ::= { ieee8021MstpPortTable 1 }
39
40 Ieee8021MstpPortEntry ::= SEQUENCE {
41     ieee8021MstpPortComponentId      IEEE8021PbbComponentIdentifier,
42     ieee8021MstpPortMstId            IEEE8021MstIdentifier,
43     ieee8021MstpPortNum              IEEE8021BridgePortNumber,
44     ieee8021MstpPortUptime           TimeTicks,
45     ieee8021MstpPortState            INTEGER,
46     ieee8021MstpPortPriority          Integer32,
47     ieee8021MstpPortPathCost         Integer32,
48     ieee8021MstpPortDesignatedRoot   BridgeId,
49     ieee8021MstpPortDesignatedCost   Integer32,
50     ieee8021MstpPortDesignatedBridge BridgeId,
51     ieee8021MstpPortDesignatedPort   IEEE8021BridgePortNumber,
52     ieee8021MstpPortRole              INTEGER,
53     ieee8021MstpPortDisputed         TruthValue,
54     ieee8021MstpPortAdminPathCost    Integer32
55 }
56
57 ieee8021MstpPortComponentId OBJECT-TYPE
58     SYNTAX          IEEE8021PbbComponentIdentifier
59     MAX-ACCESS      not-accessible
60     STATUS          current
61     DESCRIPTION
62         "The component identifier is used to distinguish between the
63         multiple virtual Bridge instances within a PBB. In simple
64         situations where there is only a single component the default
65         value is 1."
66     ::= { ieee8021MstpPortEntry 1 }
67
68 ieee8021MstpPortMstId OBJECT-TYPE
69     SYNTAX          IEEE8021MstIdentifier
70     MAX-ACCESS      not-accessible
71     STATUS          current
72     DESCRIPTION

```

```

1      "In an MSTP Bridge, this parameter is the MSTID, i.e., the
2      identifier of a Spanning Tree (or MST) Instance."
3      ::= { ieee8021MstpPortEntry 2 }
4
5 ieee8021MstpPortNum OBJECT-TYPE
6     SYNTAX      IEEE8021BridgePortNumber
7     MAX-ACCESS  not-accessible
8     STATUS      current
9     DESCRIPTION
10      "In an MSTP Bridge, the Port's Port Number parameter value for
11      the MSTI, i.e., the number of the Bridge Port for the MSTI."
12      ::= { ieee8021MstpPortEntry 3 }
13
14 ieee8021MstpPortUptime OBJECT-TYPE
15     SYNTAX      TimeTicks
16     UNITS       "centi-seconds"
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20      "In an MSTP Bridge, the Port's Uptime parameter value for the
21      MSTI, i.e., the count in seconds of the time elapsed since the
22      Port was last reset or initialized (BEGIN, Annex E)."
23      ::= { ieee8021MstpPortEntry 4 }
24
25 ieee8021MstpPortState OBJECT-TYPE
26     SYNTAX      INTEGER {
27         disabled(1),
28         listening(2),
29         learning(3),
30         forwarding(4),
31         blocking(5)
32     }
33     MAX-ACCESS  read-only
34     STATUS      current
35     DESCRIPTION
36      "In an MSTP Bridge, the current state of the Port (i.e., Disabled,
37      Listening, Learning, Forwarding, or Blocking), for the MSTI."
38     REFERENCE   "13.38"
39     ::= { ieee8021MstpPortEntry 5 }
40
41 ieee8021MstpPortPriority OBJECT-TYPE
42     SYNTAX      Integer32 (0..240)
43     MAX-ACCESS  read-write
44     STATUS      current
45     DESCRIPTION
46      "In an MSTP Bridge, the Port's Port Priority parameter value for
47      the MSTI, i.e., the priority field for the Port Identifier for the
48      Port for the MSTI."
49     REFERENCE   "13.27.47"
50     ::= { ieee8021MstpPortEntry 6 }
51
52 ieee8021MstpPortPathCost OBJECT-TYPE
53     SYNTAX      Integer32 (1..200000000)
54     MAX-ACCESS  read-write
55     STATUS      current
56     DESCRIPTION
57      "In an MSTP Bridge, the Port's Port Path Cost parameter value for
58      the MSTI."
59     REFERENCE   "13.27.33"
60     ::= { ieee8021MstpPortEntry 7 }
61
62 ieee8021MstpPortDesignatedRoot OBJECT-TYPE
63     SYNTAX      BridgeId
64     MAX-ACCESS  read-only
65     STATUS      current
66     DESCRIPTION
67      "In an MSTP Bridge, the Regional Root Identifier component of the
68      Port's MSTI port priority vector, as defined in 13.11, for the MSTI."
69     REFERENCE   "13.27.47"
70     ::= { ieee8021MstpPortEntry 8 }
71
72 ieee8021MstpPortDesignatedCost OBJECT-TYPE

```

```

1  SYNTAX      Integer32
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "In an MSTP Bridge, the Internal Root Path Cost component of the
6      Port's MSTI port priority vector, as defined in 13.11, for the MSTI."
7  REFERENCE   "13.27.47"
8  ::= { ieee8021MstpPortEntry 9 }
9
10 ieee8021MstpPortDesignatedBridge OBJECT-TYPE
11     SYNTAX      BridgeId
12     MAX-ACCESS  read-only
13     STATUS      current
14     DESCRIPTION
15         "In an MSTP Bridge, the Designated Bridge Identifier component of
16         the Port's MSTI port priority vector, as defined in 13.11, for
17         the MSTI."
18     REFERENCE   "13.27.47"
19     ::= { ieee8021MstpPortEntry 10 }
20
21 ieee8021MstpPortDesignatedPort OBJECT-TYPE
22     SYNTAX      IEEE8021BridgePortNumber
23     MAX-ACCESS  read-only
24     STATUS      current
25     DESCRIPTION
26         "In an MSTP Bridge, the Designated Port Identifier component of the
27         Port's MSTI port priority vector, as defined in 13.11, for the MSTI."
28     REFERENCE   "13.27.47"
29     ::= { ieee8021MstpPortEntry 11 }
30
31 ieee8021MstpPortRole OBJECT-TYPE
32     SYNTAX      INTEGER {
33         root(1),
34         alternate(2),
35         designated(3),
36         backup(4)
37     }
38     MAX-ACCESS  read-only
39     STATUS      current
40     DESCRIPTION
41         "In an MSTP Bridge, the current Port Role for the Port (i.e., Root,
42         Alternate, Designated, or Backup), for the MSTI."
43     ::= { ieee8021MstpPortEntry 12 }
44
45 ieee8021MstpPortDisputed OBJECT-TYPE
46     SYNTAX      TruthValue
47     MAX-ACCESS  read-only
48     STATUS      current
49     DESCRIPTION
50         "In an MSTP Bridge, the current value of the disputed variable for
51         the MSTI for the Port."
52     REFERENCE   "13.27.22"
53     ::= { ieee8021MstpPortEntry 13 }
54
55
56 ieee8021MstpPortAdminPathCost OBJECT-TYPE
57     SYNTAX      Integer32 (1..2000000000)
58     MAX-ACCESS  read-write
59     STATUS      current
60     DESCRIPTION
61         "In an MSTP Bridge, the administrative value of the Port's
62         Port Path Cost parameter value for the MSTI.
63
64         Writing a value of '0' assigns the automatically calculated
65         default Path Cost value to the Port. If the default Path
66         Cost is being used, this object returns '0' when read.
67
68         This complements the object ieee8021MstpPortPathCost,
69         which returns the operational value of the path cost.
70
71         The value of this object MUST be retained across
72         reinitializations of the management system."

```

```

1  REFERENCE    "13.27.33"
2  ::= { ieee8021MstpPortEntry 14 }
3
4  -- =====
5  -- ieee8021MstpFidToMstiTable deprecated
6  -- see ieee8021MstpFidToMstiV2Table below
7  -- =====
8
9  ieee8021MstpFidToMstiTable OBJECT-TYPE
10   SYNTAX      SEQUENCE OF Ieee8021MstpFidToMstiEntry
11   MAX-ACCESS  not-accessible
12   STATUS      deprecated
13   DESCRIPTION
14       "In an MSTP Bridge, the fixed-length FID to MSTID Allocation Table
15       entry. Each entry in the Table corresponds to a FID, and the value
16       of the entry specifies the MSTID of the spanning tree to which the
17       set of VLANs supported by that FID are assigned. A value of zero
18       in an entry specifies that the set of VLANs supported by that FID
19       are assigned to the CST.
20
21       The values of all writable objects in this table MUST be
22       retained across reinitializations of the management system.
23
24       Note that entries will exist in this table only for Bridge
25       components for which the corresponding instance of
26       ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
27       has a value of mstp(3)."
```

REFERENCE "12.12.2"

```

29  ::= { ieee8021MstpObjects 5 }
30
31  ieee8021MstpFidToMstiEntry OBJECT-TYPE
32   SYNTAX      Ieee8021MstpFidToMstiEntry
33   MAX-ACCESS  not-accessible
34   STATUS      deprecated
35   DESCRIPTION
36       "In an MSTP Bridge, a FID to MSTID Allocation Table entry."
37   INDEX { ieee8021MstpFidToMstiComponentId, ieee8021MstpFidToMstiFid }
38   ::= { ieee8021MstpFidToMstiTable 1 }
39
40  Ieee8021MstpFidToMstiEntry ::= SEQUENCE {
41      ieee8021MstpFidToMstiComponentId  IEEE8021PbbComponentIdentifier,
42      ieee8021MstpFidToMstiFid          Unsigned32,
43      ieee8021MstpFidToMstiMstId        IEEE8021MstIdentifier
44  }
45
46  ieee8021MstpFidToMstiComponentId OBJECT-TYPE
47   SYNTAX      IEEE8021PbbComponentIdentifier
48   MAX-ACCESS  not-accessible
49   STATUS      deprecated
50   DESCRIPTION
51       "The component identifier is used to distinguish between the
52       multiple virtual Bridge instances within a PBB. In simple
53       situations where there is only a single component the default
54       value is 1."
55   ::= { ieee8021MstpFidToMstiEntry 1 }
56
57  ieee8021MstpFidToMstiFid OBJECT-TYPE
58   SYNTAX      Unsigned32 (1..4094)
59   MAX-ACCESS  not-accessible
60   STATUS      deprecated
61   DESCRIPTION
62       "In an MSTP Bridge, the FID of the entry in the FID to MSTID
63       Allocation Table."
64   ::= { ieee8021MstpFidToMstiEntry 2 }
65
66  ieee8021MstpFidToMstiMstId OBJECT-TYPE
67   SYNTAX      IEEE8021MstIdentifier
68   MAX-ACCESS  read-write
69   STATUS      deprecated
70   DESCRIPTION
71       "In an MSTP Bridge, the MSTID to which the FID (of the entry in
72       the FID to MSTID Allocation Table) is to be allocated."
```

```

1 ::= { ieee8021MstpFidToMstiEntry 3 }
2
3 -- =====
4 -- ieee8021MstpFidToMstiV2Table
5 -- =====
6
7 ieee8021MstpFidToMstiV2Table OBJECT-TYPE
8     SYNTAX      SEQUENCE OF Ieee8021MstpFidToMstiV2Entry
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "In an MSTP Bridge, the fixed-length FID to MSTID Allocation Table
13        entry. Each entry in the Table corresponds to a FID, and the value
14        of the entry specifies the MSTID of the spanning tree to which the
15        set of VLANs supported by that FID are assigned. A value of zero
16        in an entry specifies that the set of VLANs supported by that FID
17        are assigned to the CST.
18
19        The values of all writable objects in this table MUST be
20        retained across reinitializations of the management system.
21
22        Note that entries will exist in this table only for Bridge
23        components for which the corresponding instance of
24        ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
25        has a value of mstp(3)."
```

REFERENCE "12.12.2"

```

27 ::= { ieee8021MstpObjects 9 }
28
29 ieee8021MstpFidToMstiV2Entry OBJECT-TYPE
30     SYNTAX      Ieee8021MstpFidToMstiV2Entry
31     MAX-ACCESS  not-accessible
32     STATUS      current
33     DESCRIPTION
34         "In an MSTP Bridge, a FID to MSTID Allocation Table entry."
35     INDEX { ieee8021MstpFidToMstiV2ComponentId, ieee8021MstpFidToMstiV2Fid }
36     ::= { ieee8021MstpFidToMstiV2Table 1 }
37
38 Ieee8021MstpFidToMstiV2Entry ::= SEQUENCE {
39     ieee8021MstpFidToMstiV2ComponentId  IEEE8021PbbComponentIdentifier,
40     ieee8021MstpFidToMstiV2Fid          Unsigned32,
41     ieee8021MstpFidToMstiV2MstId       Unsigned32
42 }
43
44 ieee8021MstpFidToMstiV2ComponentId OBJECT-TYPE
45     SYNTAX      IEEE8021PbbComponentIdentifier
46     MAX-ACCESS  not-accessible
47     STATUS      current
48     DESCRIPTION
49         "The component identifier is used to distinguish between the
50         multiple virtual Bridge instances within a PBB. In simple
51         situations where there is only a single component the default
52         value is 1."
53     ::= { ieee8021MstpFidToMstiV2Entry 1 }
54
55 ieee8021MstpFidToMstiV2Fid OBJECT-TYPE
56     SYNTAX      Unsigned32 (1..4095)
57     MAX-ACCESS  not-accessible
58     STATUS      current
59     DESCRIPTION
60         "In an MSTP Bridge, the FID of the entry in the FID to MSTID
61         Allocation Table."
62     ::= { ieee8021MstpFidToMstiV2Entry 2 }
63
64 ieee8021MstpFidToMstiV2MstId OBJECT-TYPE
65     SYNTAX      Unsigned32 (0..4095)
66     MAX-ACCESS  read-write
67     STATUS      current
68     DESCRIPTION
69         "In an MSTP Bridge, the MSTID to which the FID (of the entry in
70         the FID to MSTID Allocation Table) is to be allocated.
71         In an SPT Bridge, the value 4095 is used to indicate unused
72         (non-filtering) FIDs."
```

```

1 ::= { ieee8021MstpFidToMstiV2Entry 3 }
2
3 -- =====
4 -- ieee8021MstpVlanTable deprecated
5 -- see ieee8021MstpVlanV2Table below
6 -- =====
7
8 ieee8021MstpVlanTable OBJECT-TYPE
9     SYNTAX      SEQUENCE OF Ieee8021MstpVlanEntry
10    MAX-ACCESS  not-accessible
11    STATUS      deprecated
12    DESCRIPTION
13        "In an MSTP Bridge, the fixed-length (4094 elements), read-only,
14        MST Configuration Table. Its elements are derived from other
15        configuration information held by the Bridge; specifically, the
16        current state of the VID to FID Allocation Table (8.8.8,
17        12.10.1), and the FID to MSTID Allocation Table (8.9.3, 12.12.2).
18        Hence, changes made to either of these Tables can in turn affect
19        the contents of the MST Configuration Table, and also affect the
20        value of the digest element of the MST Configuration Identifier.
21
22        The values of all writable objects in this table MUST be
23        retained across reinitializations of the management system.
24
25        Note that entries will exist in this table only for Bridge
26        components for which the corresponding instance of
27        ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
28        has a value of mstp(3)."
```

REFERENCE "12.12.3.1"

```

29 ::= { ieee8021MstpObjects 6 }
30
31
32 ieee8021MstpVlanEntry OBJECT-TYPE
33     SYNTAX      Ieee8021MstpVlanEntry
34     MAX-ACCESS  not-accessible
35     STATUS      deprecated
36     DESCRIPTION
37         "In an MSTP Bridge, a MST Configuration Table entry."
38     INDEX { ieee8021MstpVlanComponentId, ieee8021MstpVlanId }
39     ::= { ieee8021MstpVlanTable 1 }
40
41 Ieee8021MstpVlanEntry ::= SEQUENCE {
42     ieee8021MstpVlanComponentId  IEEE8021PbbComponentIdentifier,
43     ieee8021MstpVlanId          IEEE8021VlanIndex,
44     ieee8021MstpVlanMstId       IEEE8021MstIdentifier
45 }
46
47 ieee8021MstpVlanComponentId OBJECT-TYPE
48     SYNTAX      IEEE8021PbbComponentIdentifier
49     MAX-ACCESS  not-accessible
50     STATUS      deprecated
51     DESCRIPTION
52         "The component identifier is used to distinguish between the
53         multiple virtual Bridge instances within a PBB. In simple
54         situations where there is only a single component the default
55         value is 1."
56     ::= { ieee8021MstpVlanEntry 1 }
57
58 ieee8021MstpVlanId OBJECT-TYPE
59     SYNTAX      IEEE8021VlanIndex
60     MAX-ACCESS  not-accessible
61     STATUS      deprecated
62     DESCRIPTION
63         "In an MSTP Bridge, the VID of the entry in the MST
64         Configuration Table."
65     ::= { ieee8021MstpVlanEntry 2 }
66
67 ieee8021MstpVlanMstId OBJECT-TYPE
68     SYNTAX      IEEE8021MstIdentifier
69     MAX-ACCESS  read-only
70     STATUS      deprecated
71     DESCRIPTION
72         "In an MSTP Bridge, the MSTID value corresponding to the VID
```



```

1      of the entry in the MST Configuration Table."
2      ::= { ieee8021MstpVlanEntry 3 }
3
4 -- =====
5 -- ieee8021MstpVlanV2Table
6 -- =====
7
8 ieee8021MstpVlanV2Table OBJECT-TYPE
9     SYNTAX      SEQUENCE OF Ieee8021MstpVlanV2Entry
10    MAX-ACCESS  not-accessible
11    STATUS      current
12    DESCRIPTION
13        "In an MSTP Bridge, the fixed-length (4094 elements), read-only,
14        MST Configuration Table. Its elements are derived from other
15        configuration information held by the Bridge; specifically, the
16        current state of the VID to FID Allocation Table (8.8.8,
17        12.10.1), and the FID to MSTID Allocation Table (8.9.3, 12.12.2).
18        Hence, changes made to either of these Tables can in turn affect
19        the contents of the MST Configuration Table, and also affect the
20        value of the digest element of the MST Configuration Identifier.
21
22        The values of all writable objects in this table MUST be
23        retained across reinitializations of the management system.
24
25        Note that entries will exist in this table only for Bridge
26        components for which the corresponding instance of
27        ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
28        has a value of mstp(3)."
29    REFERENCE    "12.12.3.1"
30    ::= { ieee8021MstpObjects 10 }
31
32 ieee8021MstpVlanV2Entry OBJECT-TYPE
33     SYNTAX      Ieee8021MstpVlanV2Entry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "In an MSTP Bridge, a MST Configuration Table entry."
38     INDEX { ieee8021MstpVlanV2ComponentId, ieee8021MstpVlanV2Id }
39     ::= { ieee8021MstpVlanV2Table 1 }
40
41 Ieee8021MstpVlanV2Entry ::= SEQUENCE {
42     ieee8021MstpVlanV2ComponentId  IEEE8021PbbComponentIdentifier,
43     ieee8021MstpVlanV2Id           IEEE8021VlanIndex,
44     ieee8021MstpVlanV2MstId        Unsigned32
45 }
46
47 ieee8021MstpVlanV2ComponentId OBJECT-TYPE
48     SYNTAX      IEEE8021PbbComponentIdentifier
49     MAX-ACCESS  not-accessible
50     STATUS      current
51     DESCRIPTION
52         "The component identifier is used to distinguish between the
53         multiple virtual Bridge instances within a PBB. In simple
54         situations where there is only a single component the default
55         value is 1."
56     ::= { ieee8021MstpVlanV2Entry 1 }
57
58 ieee8021MstpVlanV2Id OBJECT-TYPE
59     SYNTAX      IEEE8021VlanIndex
60     MAX-ACCESS  not-accessible
61     STATUS      current
62     DESCRIPTION
63         "In an MSTP Bridge, the VID of the entry in the MST
64         Configuration Table."
65     ::= { ieee8021MstpVlanV2Entry 2 }
66
67 ieee8021MstpVlanV2MstId OBJECT-TYPE
68     SYNTAX      Unsigned32 (0..4095)
69     MAX-ACCESS  read-only
70     STATUS      current
71     DESCRIPTION
72         "In an MSTP Bridge, the MSTID value corresponding to the VID

```

```

1      of the entry in the MST Configuration Table.
2      In an SPT Bridge, a value of 4095 is used to indicate
3      SPVIDs."
4      ::= { ieee8021MstpVlanV2Entry 3 }
5
6  -- =====
7  -- MST Configuration Identifier Table
8  -- =====
9
10 ieee8021MstpConfigIdTable OBJECT-TYPE
11     SYNTAX      SEQUENCE OF Ieee8021MstpConfigIdEntry
12     MAX-ACCESS  not-accessible
13     STATUS      current
14     DESCRIPTION
15         "A table containing the MST Configuration Identifier for each
16         virtual Bridge. In simple situations where there is only
17         a single component, there will only be a single entry in
18         this table (i.e., only a single MST Configuration Identifier).
19
20         The values of all writable objects in this table MUST be
21         retained across reinitializations of the management system.
22
23         Note that entries will exist in this table only for Bridge
24         components for which the corresponding instance of
25         ieee8021SpanningTreeVersion (from the IEEE8021-SPANNING-TREE-MIB)
26         has a value of mstp(3)."
27     REFERENCE   "12.12.3.3, 12.12.3.4"
28     ::= { ieee8021MstpObjects 7 }
29
30 ieee8021MstpConfigIdEntry OBJECT-TYPE
31     SYNTAX      Ieee8021MstpConfigIdEntry
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35         "An entry containing the MST Configuration Identifier of a Bridge."
36     INDEX { ieee8021MstpConfigIdComponentId }
37     ::= { ieee8021MstpConfigIdTable 1 }
38
39 Ieee8021MstpConfigIdEntry ::= SEQUENCE {
40     ieee8021MstpConfigIdComponentId  IEEE8021PbbComponentIdentifier,
41     ieee8021MstpConfigIdFormatSelector  Integer32,
42     ieee8021MstpConfigurationName      SnmpAdminString,
43     ieee8021MstpRevisionLevel          Unsigned32,
44     ieee8021MstpConfigurationDigest    OCTET STRING
45 }
46
47 ieee8021MstpConfigIdComponentId OBJECT-TYPE
48     SYNTAX      IEEE8021PbbComponentIdentifier
49     MAX-ACCESS  not-accessible
50     STATUS      current
51     DESCRIPTION
52         "The component identifier is used to distinguish between the
53         multiple virtual Bridge instances within a PBB. In simple
54         situations where there is only a single component the default
55         value is 1."
56     ::= { ieee8021MstpConfigIdEntry 1 }
57
58 ieee8021MstpConfigIdFormatSelector OBJECT-TYPE
59     SYNTAX      Integer32 (0..0)
60     MAX-ACCESS  read-write
61     STATUS      current
62     DESCRIPTION
63         "In an MSTP Bridge, the Configuration Identifier Format Selector
64         in use by the Bridge, in the MST Configuration Identifier. This
65         has a value of 0 to indicate the format specified in IEEE Std 802.1Q."
66     REFERENCE   "13.8:1"
67     ::= { ieee8021MstpConfigIdEntry 2 }
68
69 ieee8021MstpConfigurationName OBJECT-TYPE
70     SYNTAX      SnmpAdminString (SIZE(32))
71     MAX-ACCESS  read-write
72     STATUS      current

```

```

1  DESCRIPTION
2      "In an MSTP Bridge, the Configuration Name in the MST
3      Configuration Identifier."
4  REFERENCE  "13.8:2"
5  ::= { ieee8021MstpConfigIdEntry 3 }
6
7  ieee8021MstpRevisionLevel OBJECT-TYPE
8      SYNTAX      Unsigned32 (0..65535)
9      MAX-ACCESS  read-write
10     STATUS      current
11     DESCRIPTION
12         "In an MSTP Bridge, the Revision Level in the MST
13         Configuration Identifier."
14     REFERENCE  "13.8:3"
15     ::= { ieee8021MstpConfigIdEntry 4 }
16
17  ieee8021MstpConfigurationDigest OBJECT-TYPE
18      SYNTAX      OCTET STRING (SIZE(16))
19      MAX-ACCESS  read-only
20      STATUS      current
21      DESCRIPTION
22          "In an MSTP Bridge, the Configuration Digest in the MST
23          Configuration Identifier."
24      REFERENCE  "13.8:4"
25      ::= { ieee8021MstpConfigIdEntry 5 }
26
27
28  -- =====
29  -- Ieee8021MstpCistPortExtensionTable:
30  -- =====
31
32  ieee8021MstpCistPortExtensionTable OBJECT-TYPE
33      SYNTAX      SEQUENCE OF Ieee8021MstpCistPortExtensionEntry
34      MAX-ACCESS  not-accessible
35      STATUS      current
36      DESCRIPTION
37          "The CIST Port Extensions Table. Each row in the Table represents information
38          regarding a specific Port within the Bridge's Bridge Protocol
39          Entity, for the CIST."
40      REFERENCE  "12.8.2"
41      ::= { ieee8021MstpObjects 8 }
42
43  ieee8021MstpCistPortExtensionEntry OBJECT-TYPE
44      SYNTAX      Ieee8021MstpCistPortExtensionEntry
45      MAX-ACCESS  not-accessible
46      STATUS      current
47      DESCRIPTION
48          "A list of additional objects containing information
49          maintained by every port about the CIST
50          state for that port."
51      AUGMENTS { ieee8021MstpCistPortEntry }
52      ::= { ieee8021MstpCistPortExtensionTable 1 }
53
54  Ieee8021MstpCistPortExtensionEntry ::=
55      SEQUENCE {
56          ieee8021MstpCistPortAutoEdgePort
57              TruthValue,
58          ieee8021MstpCistPortAutoIsolatePort
59              TruthValue
60      }
61
62  ieee8021MstpCistPortAutoEdgePort OBJECT-TYPE
63      SYNTAX      TruthValue
64      MAX-ACCESS  read-write
65      STATUS      current
66      DESCRIPTION
67          "The administrative value of the Auto Edge Port parameter.
68          A value of true(1) indicates if the Bridge detection state
69          machine (BDM, 13.33) is to detect other Bridges
70          attached to the LAN, and set
71          ieee8021SpanningTreeRstpPortOperEdgePort automatically.
72          The default value is true(1)

```

```

1
2     This is optional and provided only by implementations
3     that support the automatic identification of edge ports.
4
5     The value of this object MUST be retained across
6     reinitializations of the management system."
7     REFERENCE    "12.8.2.1.3 )"
8     ::= { ieee8021MstpCistPortExtensionEntry 1 }
9
10 ieee8021MstpCistPortAutoIsolatePort OBJECT-TYPE
11     SYNTAX      TruthValue
12     MAX-ACCESS  read-only
13     STATUS      current
14     DESCRIPTION
15         "The operational value of the Isolate Port parameter.
16
17         A value of true(1) indicates a Designated Port will
18         transition to discarding if both
19         ieee8021SpanningTreeRstpPortAdminEdgePort and
20         ieee8021SpanningTreeRstpPortAutoEdgePort are FALSE and
21         the other Bridge presumed to be attached to the same
22         point-to-point LAN does not transmit periodic BPDUs.
23
24         This is optional and provided only by implementations
25         that support the automatic identification of fragile
26         Bridges."
27     REFERENCE    "12.8.2.1.3"
28     ::= { ieee8021MstpCistPortExtensionEntry 2 }
29
30
31 -- =====
32 -- Conformance Information
33 -- =====
34
35 ieee8021MstpGroups
36     OBJECT IDENTIFIER ::= { ieee8021MstpConformance 1 }
37 ieee8021MstpCompliances
38     OBJECT IDENTIFIER ::= { ieee8021MstpConformance 2 }
39
40 -- =====
41 -- Units of conformance
42 -- =====
43
44 ieee8021MstpCistGroup OBJECT-GROUP
45     OBJECTS {
46         ieee8021MstpCistBridgeIdentifier,
47         ieee8021MstpCistTopologyChange,
48         ieee8021MstpCistRegionalRootIdentifier,
49         ieee8021MstpCistPathCost,
50         ieee8021MstpCistMaxHops
51     }
52     STATUS      current
53     DESCRIPTION
54         "Objects for the CIST group"
55     ::= { ieee8021MstpGroups 1 }
56
57 ieee8021MstpGroup OBJECT-GROUP
58     OBJECTS {
59         ieee8021MstpBridgeId,
60         ieee8021MstpTimeSinceTopologyChange,
61         ieee8021MstpTopologyChanges,
62         ieee8021MstpTopologyChange,
63         ieee8021MstpDesignatedRoot,
64         ieee8021MstpRootPathCost,
65         ieee8021MstpRootPort,
66         ieee8021MstpBridgePriority,
67         ieee8021MstpVids0,
68         ieee8021MstpVids1,
69         ieee8021MstpVids2,
70         ieee8021MstpVids3,
71         ieee8021MstpRowStatus
72     }

```

```

1  STATUS      current
2  DESCRIPTION
3      "Objects for the MST group"
4  ::= { ieee8021MstpGroups 2 }
5
6  ieee8021MstpCistPortGroup OBJECT-GROUP
7      OBJECTS {
8          ieee8021MstpCistPortUptime,
9          ieee8021MstpCistPortAdminPathCost,
10         ieee8021MstpCistPortDesignatedRoot,
11         ieee8021MstpCistPortTopologyChangeAck,
12         ieee8021MstpCistPortHelloTime,
13         ieee8021MstpCistPortAdminEdgePort,
14         ieee8021MstpCistPortOperEdgePort,
15         ieee8021MstpCistPortMacEnabled,
16         ieee8021MstpCistPortMacOperational,
17         ieee8021MstpCistPortRestrictedRole,
18         ieee8021MstpCistPortRestrictedTcn,
19         ieee8021MstpCistPortRole,
20         ieee8021MstpCistPortDisputed,
21         ieee8021MstpCistPortCistRegionalRootId,
22         ieee8021MstpCistPortCistPathCost,
23         ieee8021MstpCistPortProtocolMigration,
24         ieee8021MstpCistPortEnableBPDURx,
25         ieee8021MstpCistPortEnableBPDUTx,
26         ieee8021MstpCistPortPseudoRootId,
27         ieee8021MstpCistPortIsL2Gp
28     }
29  STATUS      current
30  DESCRIPTION
31      "Objects for the CIST Port group"
32  ::= { ieee8021MstpGroups 3 }
33
34  ieee8021MstpPortGroup OBJECT-GROUP
35      OBJECTS {
36          ieee8021MstpPortUptime,
37          ieee8021MstpPortState,
38          ieee8021MstpPortPriority,
39          ieee8021MstpPortPathCost,
40          ieee8021MstpPortDesignatedRoot,
41          ieee8021MstpPortDesignatedCost,
42          ieee8021MstpPortDesignatedBridge,
43          ieee8021MstpPortDesignatedPort,
44          ieee8021MstpPortRole,
45          ieee8021MstpPortDisputed,
46          ieee8021MstpPortAdminPathCost
47      }
48  STATUS      current
49  DESCRIPTION
50      "Objects for the MST Port group"
51  ::= { ieee8021MstpGroups 4 }
52
53  ieee8021MstpFidToMstiGroup OBJECT-GROUP
54      OBJECTS {
55          ieee8021MstpFidToMstiMstId
56      }
57  STATUS      deprecated
58  DESCRIPTION
59      "Objects for the MST FID to MSTID Allocation Table group"
60  ::= { ieee8021MstpGroups 5 }
61
62  ieee8021MstpVlanGroup OBJECT-GROUP
63      OBJECTS {
64          ieee8021MstpVlanMstId
65      }
66  STATUS      deprecated
67  DESCRIPTION
68      "Objects for the MST Configuration Table group"
69  ::= { ieee8021MstpGroups 6 }
70
71
72  ieee8021MstpConfigIdGroup OBJECT-GROUP

```

```

1  OBJECTS {
2      ieee8021MstpConfigIdFormatSelector,
3      ieee8021MstpConfigurationName,
4      ieee8021MstpRevisionLevel,
5      ieee8021MstpConfigurationDigest
6  }
7  STATUS      current
8  DESCRIPTION
9      "Objects for the MST Configuration Identifier group"
10 ::= { ieee8021MstpGroups 7 }
11
12
13 ieee8021MstpCistPortExtensionGroup OBJECT-GROUP
14     OBJECTS {
15         ieee8021MstpCistPortAutoEdgePort,
16         ieee8021MstpCistPortAutoIsolatePort
17     }
18     STATUS      current
19     DESCRIPTION
20         "Objects for the CIST Port Extension group
21         for fragile Bridges"
22     ::= { ieee8021MstpGroups 8 }
23
24 ieee8021MstpFidToMstiV2Group OBJECT-GROUP
25     OBJECTS {
26         ieee8021MstpFidToMstiV2MstId
27     }
28     STATUS      current
29     DESCRIPTION
30         "Objects for the MST FID to MSTID Allocation Table group
31         for SPB"
32     ::= { ieee8021MstpGroups 9 }
33
34 ieee8021MstpVlanV2Group OBJECT-GROUP
35     OBJECTS {
36         ieee8021MstpVlanV2MstId
37     }
38     STATUS      current
39     DESCRIPTION
40         "Objects for the MST Configuration Table group for SPB"
41     ::= { ieee8021MstpGroups 10 }
42
43 -- =====
44 -- Compliance statements
45 -- =====
46
47 ieee8021MstpCompliance MODULE-COMPLIANCE
48     STATUS      deprecated
49     DESCRIPTION
50         "The compliance statement for devices supporting Multiple
51         Spanning Tree as defined in 13 of IEEE Std 802.1Q."
52
53     MODULE
54         MANDATORY-GROUPS {
55             ieee8021MstpCistGroup,
56             ieee8021MstpGroup,
57             ieee8021MstpCistPortGroup,
58             ieee8021MstpPortGroup,
59             ieee8021MstpFidToMstiGroup,
60             ieee8021MstpVlanGroup,
61             ieee8021MstpConfigIdGroup
62         }
63
64     GROUP ieee8021MstpCistPortExtensionGroup
65     DESCRIPTION
66         "Implementation of this group is optional."
67
68     ::= { ieee8021MstpCompliances 1 }
69
70 ieee8021MstpComplianceV2 MODULE-COMPLIANCE
71     STATUS      current
72     DESCRIPTION

```

```
1      "The compliance statement for devices supporting Multiple
2      Spanning Tree as defined in 13 of IEEE Std 802.1Q."
3
4  MODULE
5      MANDATORY-GROUPS {
6          ieee8021MstpCistGroup,
7          ieee8021MstpGroup,
8          ieee8021MstpCistPortGroup,
9          ieee8021MstpPortGroup,
10         ieee8021MstpFidToMstiV2Group,
11         ieee8021MstpVlanV2Group,
12         ieee8021MstpConfigIdGroup
13     }
14
15     GROUP ieee8021MstpCistPortExtensionGroup
16     DESCRIPTION
17         "Implementation of this group is optional."
18
19     ::= { ieee8021MstpCompliances 2 }
20
21 END
```

1 17.7.7 Definitions for the CFM MIB modules

2 There are two modules for CFM as a result of the reindexing required with the addition of PBB (see 17.3.2).
3 Since the IEEE8021-CFM-MIB module contains deprecated tables replaced by reindexed tables in
4 IEEE8021-CFM-V2 MIB module, both modules are needed for CFM implementations.

5 17.7.7.1 Definitions for the IEEE8021-CFM-MIB module

```
6 IEEE8021-CFM-MIB DEFINITIONS ::= BEGIN
7
8 -- *****
9 -- IEEE 802.1Q(TM) CFM MIB
10 -- *****
11
12 IMPORTS
13     MODULE-IDENTITY,
14     OBJECT-TYPE,
15     NOTIFICATION-TYPE,
16     Integer32, Counter32,
17     Unsigned32          FROM SNMPv2-SMI    -- [RFC2578]
18     TEXTUAL-CONVENTION,
19     TimeInterval,
20     TimeStamp, RowStatus,
21     TruthValue, MacAddress,
22     TDomain, TAddress   FROM SNMPv2-TC    -- [RFC2579]
23     MODULE-COMPLIANCE,
24     OBJECT-GROUP,
25     NOTIFICATION-GROUP FROM SNMPv2-CONF   -- [RFC2580]
26     InterfaceIndex,
27     InterfaceIndexOrZero FROM IF-MIB      -- [RFC2863]
28     LldpChassisId,
29     LldpChassisIdSubtype,
30     LldpPortId,
31     LldpPortIdSubtype   FROM LLDP-MIB     -- [IEEE Std 802.1AB]
32     VlanIdOrNone, VlanId FROM Q-BRIDGE-MIB -- [RFC4363]
33     ieee802dot1mibs,
34     IEEE8021VlanIndex   FROM IEEE8021-TC-MIB
35 ;
36
37 ieee8021CfmMib MODULE-IDENTITY
38     LAST-UPDATED "202211080000Z" -- November 8, 2022
39     ORGANIZATION "IEEE 802.1 Working Group"
40     CONTACT-INFO
41         " WG-URL: http://www.ieee802.org/1/
42           WG-EMail: stds-802-1-1@ieee.org
43           Contact: IEEE 802.1 Working Group Chair
44           Postal: C/O IEEE 802.1 Working Group
45                 IEEE Standards Association
46                 445 Hoes Lane
47                 Piscataway, NJ 08854
48                 USA
49           E-mail: stds-802-1-chairs@ieee.org"
50     DESCRIPTION
51         "Connectivity Fault Management module.
52
53         Unless otherwise indicated, the references in this MIB
54         module are to IEEE Std 802.1Q-2022.
55
56         Copyright (C) IEEE (2022).
57         This version of this MIB module is part of IEEE Std 802.1Q-2022;
58         see that standard for full legal notices."
59
60     REVISION "202211080000Z" -- November 8, 2022
61     DESCRIPTION
62         "Published as part of IEEE Std 802.1Q-2022.
63         Cross references and contact information updated."
64
65     REVISION "201412150000Z" -- December 15, 2014
66     DESCRIPTION
```



```
1      "Published as part of IEEE Std 802.1Q 2014 revision.
2      Cross references updated and corrected."
3
4      REVISION      "201102270000Z" -- February 27, 2011
5      DESCRIPTION
6          "Addition of support for ICC format and minor edits
7          as part of 2011 revision of IEEE Std 802.1Q."
8
9      REVISION      "200811180000Z" -- November 18, 2008
10     DESCRIPTION
11
12     "Added new columns to the dotlagCfmMepTable to support new
13     MEP functionality required for PBB-TE support. Modified
14     dotlagCfmMepDbTable to support new functionality required
15     for PBB-TE. Modified conformance clauses to indicate objects
16     needed for PBB-TE support."
17
18     REVISION      "200810150000Z" -- October 15, 2008
19     DESCRIPTION
20     "The IEEE8021-CFM-MIB Module was originally included in IEEE
21     Std 802.1ag-2007. Some objects in this module are deprecated
22     and replaced by objects in the IEEE8021-CFM-V2-MIB module
23     defined in IEEE Std 802.1ap.
24
25     This revision is included in IEEE Std 802.1ap."
26
27     REVISION      "200706100000Z"      -- 06/10/2007 00:00GMT
28     DESCRIPTION
29     "Included in IEEE Std 802.1ag-2007."
30
31     ::= { ieee802dot1mibs 8 }
32
33     dotlagNotifications      OBJECT IDENTIFIER ::= { ieee8021CfmMib 0 }
34     dotlagMIBObjects        OBJECT IDENTIFIER ::= { ieee8021CfmMib 1 }
35     dotlagCfmConformance    OBJECT IDENTIFIER ::= { ieee8021CfmMib 2 }
36
37 -- *****
38 -- Groups in the CFM MIB Module
39 -- *****
40     dotlagCfmStack          OBJECT IDENTIFIER ::= { dotlagMIBObjects 1 }
41     dotlagCfmDefaultMd      OBJECT IDENTIFIER ::= { dotlagMIBObjects 2 }
42     dotlagCfmVlan           OBJECT IDENTIFIER ::= { dotlagMIBObjects 3 }
43     dotlagCfmConfigErrorList OBJECT IDENTIFIER ::= { dotlagMIBObjects 4 }
44     dotlagCfmMd             OBJECT IDENTIFIER ::= { dotlagMIBObjects 5 }
45     dotlagCfmMa             OBJECT IDENTIFIER ::= { dotlagMIBObjects 6 }
46     dotlagCfmMep           OBJECT IDENTIFIER ::= { dotlagMIBObjects 7 }
47
48 -- *****
49 -- Textual conventions
50 -- *****
51
52     DotlagCfmMaintDomainNameType ::= TEXTUAL-CONVENTION
53         STATUS      current
54         DESCRIPTION
55             "A value that represents a type (and thereby the format)
56             of a DotlagCfmMaintDomainName. The value can be one of
57             the following:
58
59
60             ieeeReserved(0)    Reserved for definition by IEEE 802.1
61                               recommend to not use zero unless
62                               absolutely needed.
63             none(1)           No format specified, usually because
64                               there is not (yet) a Maintenance
65                               Domain Name. In this case, a zero
66                               length OCTET STRING for the Domain
67                               Name field is acceptable.
68             dnsLikeName(2)     Domain Name like string, globally unique
69                               text string derived from a DNS name.
70             macAddrAndUint(3)  MAC address + 2-octet (unsigned) integer.
71             charString(4)      RFC2579 DisplayString, except that the
72                               character codes 0-31 (decimal) are not
```

1 used.

2 ieeeReserved(xx) Reserved for definition by IEEE 802.1

3 xx values can be [5..31] and [96..255]

4 ituReserved(xx) Reserved for definition by ITU-T G.8013/Y.1731

5 xx values range from [32..63]

6 ietfReserved(xx) Reserved for definition by IETF. xx values range

7 from [64..95].

8

9 To support future extensions, the DotlagCfmMaintDomainNameType

10 textual convention SHOULD NOT be subtyped in object type

11 definitions. It MAY be subtyped in compliance statements in

12 order to require only a subset of these address types for a

13 compliant implementation.

14

15 Implementations MUST ensure that DotlagCfmMaintDomainNameType

16 objects and any dependent objects (e.g.,

17 DotlagCfmMaintDomainName objects) are consistent. An

18 inconsistentValue error MUST be generated if an attempt to

19 change an DotlagCfmMaintDomainNameType object would, for

20 example, lead to an undefined DotlagCfmMaintDomainName value.

21 In particular,

22 DotlagCfmMaintDomainNameType/DotlagCfmMaintDomainName pairs

23 MUST be changed together if the nameType changes.

24 "

25 REFERENCE

26 "21.6.5, Table 21-18"

27 SYNTAX INTEGER {

28 none (1),

29 dnsLikeName (2),

30 macAddressAndUint (3),

31 charString (4)

32 }

33

34 DotlagCfmMaintDomainName ::= TEXTUAL-CONVENTION

35 STATUS current

36 DESCRIPTION

37 "Denotes a generic Maintenance Domain Name.

38

39 A DotlagCfmMaintDomainName value is always interpreted within

40 the context of a DotlagCfmMaintDomainNameType value. Every

41 usage of the DotlagCfmMaintDomainName textual convention is

42 required to specify the DotlagCfmMaintDomainNameType object

43 that provides the context. It is suggested that the

44 DotlagCfmMaintDomainNameType object be logically registered

45 before the object(s) that use the DotlagCfmMaintDomainName

46 textual convention, if they appear in the same logical row.

47

48 The value of a DotlagCfmMaintDomainName object MUST always

49 be consistent with the value of the associated

50 DotlagCfmMaintDomainNameType object. Attempts to set

51 an DotlagCfmMaintDomainName object to a value inconsistent

52 with the associated DotlagCfmMaintDomainNameType MUST fail

53 with an inconsistentValue error.

54

55 When this textual convention is used as the syntax of an

56 index object, there may be issues with the limit of 128

57 sub-identifiers specified in SMIV2, IETF STD 58. In this

58 case, the object definition MUST include a 'SIZE' clause

59 to limit the number of potential instance sub-identifiers;

60 otherwise the applicable constraints MUST be stated in

61 the appropriate conceptual row DESCRIPTION clauses, or

62 in the surrounding documentation if there is no single

63 DESCRIPTION clause that is appropriate.

64

65 A value of none(1) in the associated

66 DotlagCfmMaintDomainNameType object means that no Maintenance

67 Domain name is present, and the contents of the

68 DotlagCfmMaintDomainName object are meaningless.

69

70 See the DESCRIPTION of the DotlagCfmMaintAssocNameType

71 TEXTUAL-CONVENTION for a discussion of the length limits on

72 the Maintenance Domain name and Maintenance Association name.

```

1      "
2      REFERENCE
3      "21.6.5"
4      SYNTAX      OCTET STRING (SIZE(1..43))
5
6      DotlagCfmMaintAssocNameType ::= TEXTUAL-CONVENTION
7          STATUS      current
8          DESCRIPTION
9              "A value that represents a type (and thereby the format)
10             of a DotlagCfmMaintAssocName. The value can be one of
11             the following:
12
13             ieeeReserved(0)    Reserved for definition by IEEE 802.1
14                                recommend to not use zero unless
15                                absolutely needed.
16             primaryVid(1)      Primary VLAN ID.
17                                12 bits represented in a 2-octet integer:
18                                - least significant 4 bits of the first
19                                  byte contains the most significant
20                                  4 bits of the 12 bits primary VID
21                                - second byte contains the least
22                                  significant 8 bits of the primary VID
23
24                                0 1 2 3 4 5 6 7 8
25                                +-+---+---+---+---+
26                                |0 0 0 0| (MSB) |
27                                +-+---+---+---+---+
28                                |  VID   LSB   |
29                                +-+---+---+---+---+
30
31             charString(2)      RFC2579 DisplayString, except that the
32                                character codes 0-31 (decimal) are not
33                                used. (1..45) octets
34             unsignedInt16 (3)  2-octet integer/big endian
35             rfc2685VpnId(4)    RFC 2685 VPN ID
36                                3 octet VPN authority Organizationally
37                                Unique Identifier (OUI) or Company ID (CID)
38                                followed by 4 octet VPN index identifying VPN
39                                according to the OUI or CID:
40
41                                0 1 2 3 4 5 6 7 8
42                                +-+---+---+---+---+
43                                |VPN OUI/CID MSB|
44                                +-+---+---+---+---+
45                                |VPN OUI/CID   |
46                                +-+---+---+---+---+
47                                |VPN OUI/CID LSB|
48                                +-+---+---+---+---+
49                                |VPN Index (MSB)|
50                                +-+---+---+---+---+
51                                |  VPN Index   |
52                                +-+---+---+---+---+
53                                |  VPN Index   |
54                                +-+---+---+---+---+
55                                |VPN Index (LSB)|
56                                +-+---+---+---+---+
57
58             ieeeReserved(xx)   Reserved for definition by IEEE 802.1
59                                xx values can be [5..31] and [96..255]
60             iccFormat(32)      ICC-based format as specified in ITU-T G.8013/Y.1731
61             ituReserved(xx)    Reserved for definition by ITU-T G.8013/Y.1731
62                                xx values range from [33..63]
63             ietfReserved(xx)   Reserved for definition by IETF
64                                xx values range from [64..95]
65
66             To support future extensions, the DotlagCfmMaintAssocNameType
67             textual convention SHOULD NOT be subtyped in object type
68             definitions. It MAY be subtyped in compliance statements in
69             order to require only a subset of these address types for a
70             compliant implementation.
71
72             Implementations MUST ensure that DotlagCfmMaintAssocNameType

```

objects and any dependent objects (e.g., DotlagCfmMaintAssocName objects) are consistent. An inconsistentValue error MUST be generated if an attempt to change an DotlagCfmMaintAssocNameType object would, for example, lead to an undefined DotlagCfmMaintAssocName value. In particular, DotlagCfmMaintAssocNameType/DotlagCfmMaintAssocName pairs MUST be changed together if the nameType changes.

The Maintenance Domain name and Maintenance Association name, when put together into the CCM PDU, MUST total 48 octets or less. If the DotlagCfmMaintDomainNameType object contains none(1), then the DotlagCfmMaintAssocName object MUST be 45 octets or less in length. Otherwise, the length of the DotlagCfmMaintDomainName object plus the length of the DotlagCfmMaintAssocName object, added together, MUST total less than or equal to 44 octets.

"

REFERENCE

"21.6.5.4, Table 21-19"

SYNTAX INTEGER {

primaryVid (1),

charString (2),

unsignedInt16 (3),

rfc2865VpnId (4),

iccFormat (32)

}

DotlagCfmMaintAssocName ::= TEXTUAL-CONVENTION

STATUS current

DESCRIPTION

"Denotes a generic Maintenance Association Name. It is the part of the Maintenance Association Identifier that is unique within the Maintenance Domain Name and is appended to the Maintenance Domain Name to form the Maintenance Association Identifier (MAID).

A DotlagCfmMaintAssocName value is always interpreted within the context of a DotlagCfmMaintAssocNameType value. Every usage of the DotlagCfmMaintAssocName textual convention is required to specify the DotlagCfmMaintAssocNameType object that provides the context. It is suggested that the DotlagCfmMaintAssocNameType object be logically registered before the object(s) that use the DotlagCfmMaintAssocName textual convention, if they appear in the same logical row.

The value of a DotlagCfmMaintAssocName object MUST always be consistent with the value of the associated DotlagCfmMaintAssocNameType object. Attempts to set an DotlagCfmMaintAssocName object to a value inconsistent with the associated DotlagCfmMaintAssocNameType MUST fail with an inconsistentValue error.

When this textual convention is used as the syntax of an index object, there may be issues with the limit of 128 sub-identifiers specified in SMIV2, IETF STD 58. In this case, the object definition MUST include a 'SIZE' clause to limit the number of potential instance sub-identifiers; otherwise the applicable constraints MUST be stated in the appropriate conceptual row DESCRIPTION clauses, or in the surrounding documentation if there is no single DESCRIPTION clause that is appropriate.

"

REFERENCE

"21.6.5.4, 21.6.5.5, 21.6.5.6"

SYNTAX OCTET STRING (SIZE(1..45))

DotlagCfmMDLevel ::= TEXTUAL-CONVENTION

DISPLAY-HINT "d"

STATUS current

DESCRIPTION

```

1      "Integer identifying the Maintenance Domain Level (MD Level).
2      Higher numbers correspond to higher Maintenance Domains,
3      those with the greatest physical reach, with the highest
4      values for customers' CFM PDUs. Lower numbers correspond
5      to lower Maintenance Domains, those with more limited
6      physical reach, with the lowest values for CFM PDUs
7      protecting single Bridges or physical links.
8      "
9  REFERENCE
10     "18.3, 21.4.1"
11  SYNTAX      Integer32 (0..7)
12
13 DotlagCfmMDLevelOrNone ::= TEXTUAL-CONVENTION
14     DISPLAY-HINT "d"
15     STATUS      current
16     DESCRIPTION
17         "Integer identifying the Maintenance Domain Level (MD Level).
18         Higher numbers correspond to higher Maintenance Domains,
19         those with the greatest physical reach, with the highest
20         values for customers' CFM frames. Lower numbers correspond
21         to lower Maintenance Domains, those with more limited
22         physical reach, with the lowest values for CFM PDUs
23         protecting single Bridges or physical links.
24
25         The value (-1) is reserved to indicate that no MA Level has
26         been assigned.
27     "
28  REFERENCE
29     "18.3, 12.14.3.1.3:c"
30  SYNTAX      Integer32 (-1 | 0..7)
31
32 DotlagCfmMpDirection ::= TEXTUAL-CONVENTION
33     STATUS      current
34     DESCRIPTION
35         "Indicates the direction in which the Maintenance
36         association (MEP or MIP) faces on the Bridge Port:
37
38         down(1)      Sends Continuity Check Messages away from the
39                       MAC Relay Entity.
40         up(2)         Sends Continuity Check Messages towards the
41                       MAC Relay Entity.
42     "
43  REFERENCE
44     "12.14.6.3.2:c"
45  SYNTAX      INTEGER {
46              down  (1),
47              up    (2)
48          }
49
50 DotlagCfmPortStatus ::= TEXTUAL-CONVENTION
51     STATUS      current
52     DESCRIPTION
53         "An enumerated value from the Port Status TLV from the last CCM
54         received from the last MEP. It indicates the ability of the
55         Bridge Port on which the transmitting MEP resides to pass
56         ordinary data, regardless of the status of the MAC
57         (Table 21-9).
58
59         psNoPortStateTLV(0) Indicates either that no CCM has been
60                             received or that no port status TLV was
61                             present in the last CCM received.
62
63         psBlocked(1)       Ordinary data cannot pass freely through
64                             the port on which the remote MEP resides.
65                             Value of enableRmepDefect is equal to
66                             false.
67
68         psUp(2):           Ordinary data can pass freely through
69                             the port on which the remote MEP resides.
70                             Value of enableRmepDefect is equal to
71                             true.
72

```

```

1      NOTE: A 0 value is used for psNoPortStateTLV, so that
2      additional code points can be added in a manner
3      consistent with the DotlagCfmInterfaceStatus textual
4      convention.
5      "
6      REFERENCE
7      "12.14.7.6.3:f, 20.19.3, 21.5.4"
8      SYNTAX      INTEGER {
9                  psNoPortStateTLV    (0),
10                 psBlocked             (1),
11                 psUp                  (2)
12             }
13
14 DotlagCfmInterfaceStatus ::= TEXTUAL-CONVENTION
15     STATUS      current
16     DESCRIPTION
17         "An enumerated value from the Interface Status TLV from the
18         last CCM received from the last MEP. It indicates the status
19         of the Interface within which the MEP transmitting the CCM
20         is configured, or the next lower Interface in the Interface
21         Stack, if the MEP is not configured within an Interface.
22
23     isNoInterfaceStatusTLV(0)  Indicates either that no CCM has been
24                               received or that no interface status TLV
25                               was present in the last CCM received.
26
27     isUp(1)                    The interface is ready to pass frames.
28
29     isDown(2)                  The interface cannot pass frames.
30
31     isTesting(3)               The interface is in some test mode.
32
33     isUnknown(4)               The interface status cannot be determined
34                               for some reason.
35
36     isDormant(5)               The interface is not in a state to pass
37                               frames but is in a pending state, waiting
38                               for some external event.
39
40     isNotPresent(6)             Some component of the interface is missing
41
42     isLowerLayerDown(7)        The interface is down due to state of the
43                               lower layer interfaces
44
45     NOTE: A 0 value is used for isNoInterfaceStatusTLV, so that
46     these code points can be kept consistent with new code
47     points added to ifOperStatus in the IF-MIB.
48     "
49
50     REFERENCE
51     "12.14.7.6.3:g, 20.19.4, 21.5.5"
52     SYNTAX      INTEGER {
53                 isNoInterfaceStatusTLV (0),
54                 isUp                    (1),
55                 isDown                  (2),
56                 isTesting                (3),
57                 isUnknown                (4),
58                 isDormant                (5),
59                 isNotPresent             (6),
60                 isLowerLayerDown        (7)
61             }
62
63 DotlagCfmHighestDefectPri ::= TEXTUAL-CONVENTION
64     STATUS      current
65     DESCRIPTION
66         "An enumerated value, equal to the contents of the variable
67         highestDefect (20.35.9 and Table 20-1), indicating the
68         highest-priority defect that has been present since the MEP
69         Fault Notification Generator State Machine was last in the
70         FNG_RESET state, either:
71
72     none(0)                    no defects since FNG_RESET

```

```

1      defRDICCM(1)      DefRDICCM
2      defMACstatus(2)   DefMACstatus
3      defRemoteCCM(3)   DefRemoteCCM
4      defErrorCCM(4)    DefErrorCCM
5      defXconCCM(5)     DefXconCCM
6
7      The value 0 is used for no defects so that additional higher
8      priority values can be added, if needed, at a later time, and
9      so that these values correspond with those in
10     DotlagCfmLowestAlarmPri.
11     "
12     REFERENCE
13     "12.14.7.7.2, 20.1.2, 20.35.9 "
14     SYNTAX      INTEGER {
15         none                (0),
16         defRDICCM           (1),
17         defMACstatus        (2),
18         defRemoteCCM        (3),
19         defErrorCCM         (4),
20         defXconCCM          (5)
21     }
22
23 DotlagCfmLowestAlarmPri ::= TEXTUAL-CONVENTION
24     STATUS      current
25     DESCRIPTION
26     "An integer value specifying the lowest priority defect
27     that is allowed to generate a Fault Alarm (20.9.5), either:
28
29     allDef(1)      DefRDICCM, DefMACstatus, DefRemoteCCM,
30                    DefErrorCCM, and DefXconCCM;
31     macRemErrXcon(2) Only DefMACstatus, DefRemoteCCM,
32                    DefErrorCCM, and DefXconCCM (default);
33     remErrXcon(3)   Only DefRemoteCCM, DefErrorCCM,
34                    and DefXconCCM;
35     errXcon(4)      Only DefErrorCCM and DefXconCCM;
36     xcon(5)         Only DefXconCCM; or
37     noXcon(6)       No defects DefXcon or lower are to be
38                    reported;
39     "
40     REFERENCE
41     "12.14.7.1.3:k, 20.9.5"
42     SYNTAX      INTEGER {
43         allDef              (1),
44         macRemErrXcon       (2),
45         remErrXcon          (3),
46         errXcon             (4),
47         xcon                (5),
48         noXcon              (6)
49     }
50
51 DotlagCfmMepId ::= TEXTUAL-CONVENTION
52     DISPLAY-HINT "d"
53     STATUS      current
54     DESCRIPTION
55     "Maintenance association Endpoint Identifier (MEPID): A small
56     integer, unique over a given Maintenance Association,
57     identifying a specific MEP.
58     "
59     REFERENCE
60     "3.142, 19.2.1"
61     SYNTAX      Unsigned32 (1..8191)
62
63 DotlagCfmMepIdOrZero ::= TEXTUAL-CONVENTION
64     DISPLAY-HINT "d"
65     STATUS      current
66     DESCRIPTION
67     "Maintenance association Endpoint Identifier (MEPID): A small
68     integer, unique over a given Maintenance Association,
69     identifying a specific MEP.
70
71     The special value 0 is allowed to indicate special cases, for
72     example that no MEPID is configured.

```

```
1
2     Whenever an object is defined with this SYNTAX, then the
3     DESCRIPTION clause of such an object MUST specify what the
4     special value of 0 means.
5     "
6     REFERENCE
7         "19.2.1"
8     SYNTAX      Unsigned32 (0 | 1..8191)
9
10 DotlagCfmMhfCreation ::= TEXTUAL-CONVENTION
11     STATUS      current
12     DESCRIPTION
13         "Indicates if the Management Entity can create MHFs.
14         The valid values are:
15
16         defMHFnone(1)      No MHFs can be created for this VID.
17         defMHFdefault(2)   MHFs can be created on this VID on any
18                             Bridge port through which this VID can
19                             pass.
20         defMHFexplicit(3)  MHFs can be created for this VID only on
21                             Bridge ports through which this VID can
22                             pass, and only if a MEP is created at some
23                             lower MD Level.
24         defMHFdefer(4)     The creation of MHFs is determined by the
25                             corresponding Maintenance Domain variable
26                             (dotlagCfmMaCompMhfCreation).
27     "
28     REFERENCE
29         "12.14.5.1.3:c, 22.2.3"
30     SYNTAX      INTEGER {
31         defMHFnone      (1),
32         defMHFdefault   (2),
33         defMHFexplicit  (3),
34         defMHFdefer     (4)
35     }
36
37 DotlagCfmIdPermission ::= TEXTUAL-CONVENTION
38     STATUS      current
39     DESCRIPTION
40         "Indicates what, if anything, is to be included in the Sender
41         ID TLV transmitted in CCMs, LBMs, LTM, and LTRs. The valid
42         values are:
43
44         sendIdNone(1)      The Sender ID TLV is not to be sent.
45         sendIdChassis(2)   The Chassis ID Length, Chassis ID
46                             Subtype, and Chassis ID fields of the
47                             Sender ID TLV are to be sent.
48         sendIdManage(3)    The Management Address Length and
49                             Management Address of the Sender ID TLV
50                             are to be sent.
51         sendIdChassisManage(4) The Chassis ID Length, Chassis ID
52                             Subtype, Chassis ID, Management Address
53                             Length and Management Address fields are
54                             all to be sent.
55         sendIdDefer(5)     The contents of the Sender ID TLV are
56                             determined by the corresponding
57                             Maintenance Domain variable
58                             (dotlagCfmMaCompIdPermission).
59     "
60     REFERENCE
61         "12.14.6.1.3:d, 21.5.3"
62     SYNTAX      INTEGER {
63         sendIdNone      (1),
64         sendIdChassis    (2),
65         sendIdManage     (3),
66         sendIdChassisManage (4),
67         sendIdDefer      (5)
68     }
69
70 DotlagCfmCcmInterval ::= TEXTUAL-CONVENTION
71     STATUS      current
72     DESCRIPTION
```



```

1      "Indicates the interval at which CCMs are sent by a MEP.
2      The possible values are:
3      intervalInvalid(0) No CCMs are sent (disabled).
4      interval300Hz(1)   CCMs are sent every 3 1/3 milliseconds
5                          (300Hz).
6      interval10ms(2)   CCMs are sent every 10 milliseconds.
7      interval100ms(3)  CCMs are sent every 100 milliseconds.
8      interval1s(4)     CCMs are sent every 1 second.
9      interval10s(5)    CCMs are sent every 10 seconds.
10     interval1min(6)   CCMs are sent every minute.
11     interval10min(7)  CCMs are sent every 10 minutes.
12
13     Note: enumerations start at zero to match the 'CCM Interval
14           field' protocol field.
15
16     REFERENCE
17     "12.14.6.1.3:e, 20.8.1, 21.6.1.3"
18     SYNTAX      INTEGER {
19         intervalInvalid  (0),
20         interval300Hz    (1),
21         interval10ms     (2),
22         interval100ms    (3),
23         interval1s       (4),
24         interval10s      (5),
25         interval1min     (6),
26         interval10min    (7)
27     }
28
29 DotlagCfmFngState ::= TEXTUAL-CONVENTION
30     STATUS      current
31     DESCRIPTION
32     "Indicates the different states of the MEP Fault Notification
33     Generator State Machine.
34
35     fngReset(1)      No defect has been present since the
36                      dotlagCfmMepFngResetTime timer
37                      expired, or since the state machine
38                      was last reset.
39
40     fngDefect(2)      A defect is present, but not for a
41                      long enough time to be reported
42                      (dotlagCfmMepFngAlarmTime).
43
44     fngReportDefect(3) A momentary state during which the
45                      defect is reported by sending a
46                      dotlagCfmFaultAlarm notification,
47                      if that action is enabled.
48
49     fngDefectReported(4) A defect is present, and some defect
50                      has been reported.
51
52     fngDefectClearing(5) No defect is present, but the
53                      dotlagCfmMepFngResetTime timer has
54                      not yet expired.
55
56     "
57     REFERENCE
58     "12.14.7.1.3:f, 20.35"
59     SYNTAX      INTEGER {
60         fngReset          (1),
61         fngDefect          (2),
62         fngReportDefect    (3),
63         fngDefectReported  (4),
64         fngDefectClearing  (5)
65     }
66 DotlagCfmRelayActionFieldValue ::= TEXTUAL-CONVENTION
67     STATUS      current
68     DESCRIPTION
69     "Possible values the Relay action field can take."
70     REFERENCE
71     "12.14.7.5.3:g, 20.41.2.5, 21.9.5, Table 21-26"
72     SYNTAX      INTEGER {

```

```

1         rlyHit      (1),
2         rlyFdb      (2),
3         rlyMpdb     (3)
4     }
5
6 DotlagCfmIngressActionFieldValue ::= TEXTUAL-CONVENTION
7     STATUS      current
8     DESCRIPTION
9         "Possible values returned in the ingress action field."
10    REFERENCE
11        "12.14.7.5.3:g, 20.41.2.6, 21.9.8.1, Table 21-29"
12    SYNTAX      INTEGER {
13        ingNoTlv    (0),
14        ingOk       (1),
15        ingDown     (2),
16        ingBlocked  (3),
17        ingVid      (4)
18    }
19
20 DotlagCfmEgressActionFieldValue ::= TEXTUAL-CONVENTION
21     STATUS      current
22     DESCRIPTION
23         "Possible values returned in the egress action field"
24     REFERENCE
25        "12.14.7.5.3:o, 20.41.2.10, 21.9.9.1, Table 21-31"
26     SYNTAX      INTEGER {
27        egrNoTlv    (0),
28        egrOK       (1),
29        egrDown     (2),
30        egrBlocked  (3),
31        egrVid      (4)
32    }
33
34 DotlagCfmRemoteMepState ::= TEXTUAL-CONVENTION
35     STATUS      current
36     DESCRIPTION
37         "Operational state of the remote MEP state machine. This
38         state machine monitors the reception of valid CCMs from a
39         remote MEP with a specific MEPID. It uses a timer that
40         expires in 3.5 times the length of time indicated by the
41         dotlagCfmMaNetCcmInterval object.
42
43         rMepIdle(1)           Momentary state during reset.
44
45         rMepStart(2)          The timer has not expired since the
46                               state machine was reset, and no valid
47                               CCM has yet been received.
48
49         rMepFailed(3)         The timer has expired, both since the
50                               state machine was reset, and since a
51                               valid CCM was received.
52
53         rMepOk(4)             The timer has not expired since a
54                               valid CCM was received.
55 "
56     REFERENCE
57        "12.14.7.6.3:b, 20.22"
58     SYNTAX      INTEGER {
59        rMepIdle     (1),
60        rMepStart    (2),
61        rMepFailed   (3),
62        rMepOk       (4)
63    }
64
65 DotlafCfmIndexIntegerNextFree ::= TEXTUAL-CONVENTION
66     DISPLAY-HINT "d"
67     STATUS      current
68     DESCRIPTION
69         "An integer that may be used as a new Index in a table.
70
71         The special value of 0 indicates that no more new entries can
72         be created in the relevant table.

```

```

1
2   When a MIB is used for configuration, an object with this
3   SYNTAX always contains a legal value (if non-zero) for an
4   index that is not currently used in the relevant table. The
5   Command Generator (Network Management Application) reads this
6   variable and uses the (non-zero) value read when creating a
7   new row with an SNMP SET. When the SET is performed, the
8   Command Responder (agent) MUST determine whether the value is
9   indeed still unused; Two Network Management Applications may
10  attempt to create a row (configuration entry) simultaneously
11  and use the same value. If it is currently unused, the SET
12  succeeds and the Command Responder (agent) changes the value
13  of this object, according to an implementation-specific
14  algorithm. If the value is in use, however, the SET fails.
15  The Network Management Application MUST then re-read this
16  variable to obtain a new usable value.
17
18  An OBJECT-TYPE definition using this SYNTAX MUST specify the
19  relevant table for which the object is providing this
20  functionality.
21  "
22  SYNTAX      Unsigned32 (0..4294967295)
23
24  Dot1agCfmMepDefects ::= TEXTUAL-CONVENTION
25      STATUS      current
26      DESCRIPTION
27          "A MEP can detect and report a number of defects, and multiple
28          defects can be present at the same time. These defects are:
29
30          bDefRDICCM(0) A remote MEP is reported the RDI bit in its
31                      last CCM.
32          bDefMACstatus(1) Either some remote MEP is reporting its
33                      Interface Status TLV as not isUp, or all remote
34                      MEPs are reporting a Port Status TLV that
35                      contains some value other than psUp.
36          bDefRemoteCCM(2) The MEP is not receiving valid CCMs from at
37                      least one of the remote MEPs.
38          bDefErrorCCM(3) The MEP has received at least one invalid CCM
39                      whose CCM Interval has not yet timed out.
40          bDefXconCCM(4) The MEP has received at least one CCM from
41                      either another MAID or a lower MD Level whose
42                      CCM Interval has not yet timed out.
43  "
44  REFERENCE
45      "12.14.7.1.3:o, 12.14.7.1.3:p, 12.14.7.1.3:q,
46      12.14.7.1.3:r, 12.14.7.1.3:s."
47  SYNTAX BITS {
48      bDefRDICCM(0),
49      bDefMACstatus(1),
50      bDefRemoteCCM(2),
51      bDefErrorCCM(3),
52      bDefXconCCM(4)
53  }
54
55  Dot1agCfmConfigErrors ::= TEXTUAL-CONVENTION
56      STATUS      current
57      DESCRIPTION
58          "While making the MIP creation evaluation described in
59          22.2.3, the management entity can encounter errors in
60          the configuration. These are possible errors that can be
61          encountered:
62
63          CFMleak(0)   MA x is associated with a specific VID list,
64                      one or more of the VIDs in MA x can pass through
65                      the Bridge Port, no Down MEP is configured on
66                      any Bridge Port for MA x, and some other MA y,
67                      at a higher MD Level than MA x, and associated
68                      with at least one of the VID(s) also in MA x,
69                      does have a MEP configured on the Bridge Port.
70
71          conflictingVids(1) MA x is associated with a specific VID
72                      list, an Up MEP is configured on MA x on the

```

```

1         Bridge Port, and some other MA y, associated
2         with at least one of the VID(s) also in MA x,
3         also has an Up MEP configured on some Bridge
4         Port.
5
6         ExcessiveLevels(2) The number of different MD Levels at
7         which MIPs are to be created on this port
8         exceeds the Bridge's capabilities (22.3).
9
10        OverlappedLevels(3) A MEP is created for one VID at one MD
11        Level, but a MEP is configured on another
12        VID at that MD Level or higher, exceeding
13        the Bridge's capabilities.
14    "
15    REFERENCE
16        "12.14.4.1.3:b, 22.2.3, 22.2.4"
17    SYNTAX BITS {
18        cfmLeak(0),
19        conflictingVids(1),
20        excessiveLevels(2),
21        overlappedLevels(3)
22    }
23
24    DotlagCfmPbbComponentIdentifier
25 ::= TEXTUAL-CONVENTION
26     DISPLAY-HINT "d"
27     STATUS      current
28     DESCRIPTION
29         "A Provider Backbone Bridge (PBB) can comprise a number of
30         components, each of which can be managed in a manner
31         essentially equivalent to an IEEE 802.1Q Bridge. In order to
32         access these components easily, an index is used in a number of
33         tables. If any two tables are indexed by
34         DotlagCfmPbbComponentIdentifier, then entries in those tables
35         indexed by the same value of DotlagCfmPbbComponentIdentifier
36         correspond to the same component.
37     "
38     REFERENCE
39         "12.3 1)"
40     SYNTAX      Unsigned32 (1..4294967295)
41
42 -- *****
43 -- The Stack Object. This group will contain all the MIBs objects
44 -- needed to access the Stack managed object.
45 -- *****
46
47 -- *****
48 -- The CFM Stack Table
49 -- *****
50
51    dotlagCfmStackTable OBJECT-TYPE
52     SYNTAX      SEQUENCE OF DotlagCfmStackEntry
53     MAX-ACCESS  not-accessible
54     STATUS      deprecated
55     DESCRIPTION
56         "There is one CFM Stack table per Bridge. It permits
57         the retrieval of information about the Maintenance Points
58         configured on any given interface.
59         **NOTE: this object is deprecated due to re-indexing of the
60         table.
61     "
62     REFERENCE
63         "12.14.2"
64     ::= { dotlagCfmStack 1 }
65
66    dotlagCfmStackEntry OBJECT-TYPE
67     SYNTAX      DotlagCfmStackEntry
68     MAX-ACCESS  not-accessible
69     STATUS      deprecated
70     DESCRIPTION
71         "The Stack table entry
72         **NOTE: this object is deprecated due to re-indexing of the

```

```

1  table.
2  "
3  INDEX { dotlagCfmStackifIndex, dotlagCfmStackVlanIdOrNone,
4          dotlagCfmStackMdLevel, dotlagCfmStackDirection
5          }
6  ::= { dotlagCfmStackTable 1 }
7
8  DotlagCfmStackEntry ::= SEQUENCE {
9      dotlagCfmStackifIndex      InterfaceIndex,
10     dotlagCfmStackVlanIdOrNone  VlanIdOrNone,
11     dotlagCfmStackMdLevel       DotlagCfmMDLevel,
12     dotlagCfmStackDirection     DotlagCfmMpDirection,
13     dotlagCfmStackMdIndex       Unsigned32,
14     dotlagCfmStackMaIndex       Unsigned32,
15     dotlagCfmStackMepId         DotlagCfmMepIdOrZero,
16     dotlagCfmStackMacAddress     MacAddress
17 }
18
19 dotlagCfmStackifIndex OBJECT-TYPE
20     SYNTAX      InterfaceIndex
21     MAX-ACCESS  not-accessible
22     STATUS      deprecated
23     DESCRIPTION
24         "This object represents the Bridge Port or aggregated port
25         on which MEPS or MHFS might be configured.
26
27         Upon a restart of the system, the system SHALL, if necessary,
28         change the value of this variable, and rearrange the
29         dotlagCfmStackTable, so that it indexes the entry in the
30         interface table with the same value of ifAlias that it
31         indexed before the system restart. If no such entry exists,
32         then the system SHALL delete all entries in the
33         dotlagCfmStackTable with the interface index.
34         **NOTE: this object is deprecated due to re-indexing of the
35         table.
36         "
37     REFERENCE
38         "12.14.2.1.2:a"
39     ::= { dotlagCfmStackEntry 1 }
40
41 dotlagCfmStackVlanIdOrNone OBJECT-TYPE
42     SYNTAX      VlanIdOrNone
43     MAX-ACCESS  not-accessible
44     STATUS      deprecated
45     DESCRIPTION
46         "VLAN ID to which the MP is attached, or 0, if none.
47         **NOTE: this object is deprecated due to re-indexing of the
48         table.
49         "
50     REFERENCE
51         "12.14.2.1.2:d, 22.1.7"
52     ::= { dotlagCfmStackEntry 2 }
53
54 dotlagCfmStackMdLevel OBJECT-TYPE
55     SYNTAX      DotlagCfmMDLevel
56     MAX-ACCESS  not-accessible
57     STATUS      deprecated
58     DESCRIPTION
59         "MD Level of the Maintenance Point.
60         **NOTE: this object is deprecated due to re-indexing of the
61         table.
62         "
63     REFERENCE
64         "12.14.2.1.2:b"
65     ::= { dotlagCfmStackEntry 3 }
66
67 dotlagCfmStackDirection OBJECT-TYPE
68     SYNTAX      DotlagCfmMpDirection
69     MAX-ACCESS  not-accessible
70     STATUS      deprecated
71     DESCRIPTION
72         "Direction in which the MP faces on the Bridge Port

```

```

1      **NOTE: this object is deprecated due to re-indexing of the
2      table.
3      "
4      REFERENCE
5      "12.14.2.1.2:c"
6      ::= { dotlagCfmStackEntry 4 }
7
8      dotlagCfmStackMdIndex OBJECT-TYPE
9          SYNTAX      Unsigned32
10         MAX-ACCESS  read-only
11         STATUS      deprecated
12         DESCRIPTION
13             "The index of the Maintenance Domain in the dotlagCfmMdTable
14             to which the MP is associated, or 0, if none.
15         "
16         REFERENCE
17             "12.14.2.1.3:b"
18         ::= { dotlagCfmStackEntry 5 }
19
20     dotlagCfmStackMaIndex OBJECT-TYPE
21         SYNTAX      Unsigned32
22         MAX-ACCESS  read-only
23         STATUS      deprecated
24         DESCRIPTION
25             "The index of the MA in the dotlagCfmMaNetTable and
26             dotlagCfmMaCompTable to which the MP is associated, or 0, if
27             none.
28             **NOTE: this object is deprecated due to re-indexing of the
29             table.
30         "
31         REFERENCE
32             "12.14.2.1.3:c"
33         ::= { dotlagCfmStackEntry 6 }
34
35     dotlagCfmStackMepId OBJECT-TYPE
36         SYNTAX      DotlagCfmMepIdOrZero
37         MAX-ACCESS  read-only
38         STATUS      deprecated
39         DESCRIPTION
40             "If an MEP is configured, the MEPID, else 0"
41         REFERENCE
42             "12.14.2.1.3:d"
43             **NOTE: this object is deprecated due to re-indexing of the
44             table.
45         "
46         ::= { dotlagCfmStackEntry 7 }
47
48     dotlagCfmStackMacAddress OBJECT-TYPE
49         SYNTAX      MacAddress
50         MAX-ACCESS  read-only
51         STATUS      deprecated
52         DESCRIPTION
53             "MAC address of the MP.
54             **NOTE: this object is deprecated due to re-indexing of the
55             table.
56         "
57         REFERENCE
58             "12.14.2.1.3:e"
59         ::= { dotlagCfmStackEntry 8 }
60
61 -- *****
62 -- The VLAN Table
63 -- *****
64
65     dotlagCfmVlanTable OBJECT-TYPE
66         SYNTAX      SEQUENCE OF DotlagCfmVlanEntry
67         MAX-ACCESS  not-accessible
68         STATUS      deprecated
69         DESCRIPTION
70             "This table defines the association of VIDs into VLANs. There
71             is an entry in this table, for each component of the Bridge,
72             for each VID that is:

```

```

1         a) a VID belonging to a VLAN associated with more than
2           one VID; and
3         b) not the Primary VLAN of that VID.
4     The entry in this table contains the Primary VID of the VLAN.
5
6     By default, this table is empty, meaning that every VID is
7     the Primary VID of a single-VID VLAN.
8
9     VLANs that are associated with only one VID SHOULD NOT have
10    an entry in this table.
11
12    The writable objects in this table need to be persistent
13    upon reboot or restart of a device.
14    **NOTE: this object is deprecated due to re-indexing of the
15    table.
16    "
17    REFERENCE
18        "12.14.3.1.3:a, 12.14.3.2.2:a, 12.14.5.3.2:c,
19        12.14.6.1.3:b, 22.1.5."
20    ::= { dotlagCfmVlan 1 }
21
22 dotlagCfmVlanEntry OBJECT-TYPE
23     SYNTAX      DotlagCfmVlanEntry
24     MAX-ACCESS  not-accessible
25     STATUS      deprecated
26     DESCRIPTION
27         "The VLAN table entry.
28         **NOTE: this object is deprecated due to re-indexing of the
29         table.
30         "
31     INDEX { dotlagCfmVlanComponentId, dotlagCfmVlanVid }
32     ::= { dotlagCfmVlanTable 1 }
33
34 DotlagCfmVlanEntry ::= SEQUENCE {
35     dotlagCfmVlanComponentId DotlagCfmPbbComponentIdentifier,
36     dotlagCfmVlanVid         VlanId,
37     dotlagCfmVlanPrimaryVid  VlanId,
38     dotlagCfmVlanRowStatus   RowStatus
39 }
40
41 dotlagCfmVlanComponentId OBJECT-TYPE
42     SYNTAX      DotlagCfmPbbComponentIdentifier
43     MAX-ACCESS  not-accessible
44     STATUS      deprecated
45     DESCRIPTION
46         "The Bridge component within the system to which the information
47         in this dotlagCfmVlanEntry applies. If the system is not a
48         Bridge, or if only one component is present in the Bridge, then
49         this variable (index) MUST be equal to 1.
50         **NOTE: this object is deprecated due to re-indexing of the
51         table.
52         "
53     REFERENCE
54         "12.3 1)"
55     ::= { dotlagCfmVlanEntry 1 }
56
57 dotlagCfmVlanVid OBJECT-TYPE
58     SYNTAX      VlanId
59     MAX-ACCESS  not-accessible
60     STATUS      deprecated
61     DESCRIPTION
62         "This is a VLAN ID belonging to a VLAN that is associated with
63         more than one VLAN ID, and this is not the Primary VID of the
64         VLAN.
65         **NOTE: this object is deprecated due to re-indexing of the
66         table.
67         "
68     ::= { dotlagCfmVlanEntry 2 }
69
70 dotlagCfmVlanPrimaryVid OBJECT-TYPE
71     SYNTAX      VlanId
72     MAX-ACCESS  read-create

```

```

1  STATUS      deprecated
2  DESCRIPTION
3  "This is the Primary VLAN ID of the VLAN with which this
4  entry's dotlagCfmVlanVid is associated. This value MUST not
5  equal the value of dotlagCfmVlanVid.
6  **NOTE: this object is deprecated due to re-indexing of the
7  table.
8  "
9  ::= { dotlagCfmVlanEntry 3 }
10
11 dotlagCfmVlanRowStatus OBJECT-TYPE
12     SYNTAX      RowStatus
13     MAX-ACCESS  read-create
14     STATUS      deprecated
15     DESCRIPTION
16         "The status of the row.
17
18         The writable columns in a row cannot be changed if the row
19         is active. All columns MUST have a valid value before a row
20         can be activated.
21         **NOTE: this object is deprecated due to re-indexing of the
22         table.
23         "
24     ::= { dotlagCfmVlanEntry 4 }
25
26 -- *****
27 -- The Default MD Level object. This group will contain all the
28 -- MIB objects needed to access and modify default MD level
29 -- managed objects.
30 -- *****
31
32 dotlagCfmDefaultMdDefLevel OBJECT-TYPE
33     SYNTAX      DotlagCfmMDLevel
34     MAX-ACCESS  read-write
35     STATUS      current
36     DESCRIPTION
37         "A value indicating the MD Level at which MHFs are to be
38         created, and Sender ID TLV transmission by those MHFs is to
39         be controlled, for each dotlagCfmDefaultMdEntry whose
40         dotlagCfmDefaultMdLevel object contains the value -1.
41
42         After this initialization, this object needs to be persistent
43         upon reboot or restart of a device.
44         "
45     REFERENCE
46         "12.14.3.1.3:c, 12.14.3.2.2:b"
47     DEFVAL {0}
48     ::= { dotlagCfmDefaultMd 1 }
49
50 dotlagCfmDefaultMdDefMhfCreation OBJECT-TYPE
51     SYNTAX      DotlagCfmMhfCreation {
52         defMHFnone      (1),
53         defMHFdefault    (2),
54         defMHFexplicit  (3)
55     }
56     MAX-ACCESS  read-write
57     STATUS      current
58     DESCRIPTION
59         "A value indicating if the Management entity can create MHFs
60         (MIP Half Function) for the VID, for each
61         dotlagCfmDefaultMdEntry whose dotlagCfmDefaultMdMhfCreation
62         object contains the value defMHFdefer. Since, in this
63         variable, there is no encompassing Maintenance Domain, the
64         value defMHFdefer is not allowed.
65
66         After this initialization, this object needs to be persistent
67         upon reboot or restart of a device.
68         "
69     REFERENCE
70         "12.14.3.1.3:d"
71     DEFVAL {defMHFnone}
72     ::= { dotlagCfmDefaultMd 2 }

```



```

1
2 dotlagCfmDefaultMdDefIdPermission OBJECT-TYPE
3     SYNTAX      DotlagCfmIdPermission {
4         sendIdNone      (1),
5         sendIdChassis   (2),
6         sendIdManage    (3),
7         sendIdChassisManage (4)
8     }
9     MAX-ACCESS  read-write
10    STATUS      current
11    DESCRIPTION
12        "Enumerated value indicating what, if anything, is to be
13        included in the Sender ID TLV (21.5.3) transmitted by MHFs
14        created by the Default Maintenance Domain, for each
15        dotlagCfmDefaultMdEntry whose dotlagCfmDefaultMdIdPermission
16        object contains the value sendIdDefer. Since, in this
17        variable, there is no encompassing Maintenance Domain, the
18        value sendIdDefer is not allowed.
19
20        After this initialization, this object needs to be persistent
21        upon reboot or restart of a device.
22    "
23    REFERENCE
24        "12.14.3.1.3:e"
25    DEFVAL { sendIdNone }
26    ::= { dotlagCfmDefaultMd 3 }
27
28 -- *****
29 -- The Default MD Level Table
30 -- *****
31
32 dotlagCfmDefaultMdTable OBJECT-TYPE
33     SYNTAX      SEQUENCE OF DotlagCfmDefaultMdEntry
34     MAX-ACCESS  not-accessible
35     STATUS      deprecated
36     DESCRIPTION
37         "For each Bridge component, the Default MD Level Managed Object
38         controls MHF creation for VIDs that are not attached to a
39         specific Maintenance Association Managed Object, and Sender ID
40         TLV transmission by those MHFs.
41
42         For each Bridge Port, and for each VLAN ID whose data can
43         pass through that Bridge Port, an entry in this table is
44         used by the algorithm in 22.2.3 only if there is no
45         entry in the Maintenance Association table defining an MA
46         for the same VLAN ID and MD Level as this table's entry, and
47         on which MA an Up MEP is defined. If there exists such an
48         MA, that MA's objects are used by the algorithm in
49         22.2.3 in place of this table entry's objects. The
50         agent maintains the value of dotlagCfmDefaultMdStatus to
51         indicate whether this entry is overridden by an MA.
52
53         When first initialized, the agent creates this table
54         automatically with entries for all VLAN IDs,
55         with the default values specified for each object.
56
57         After this initialization, the writable objects in this
58         table need to be persistent upon reboot or restart of a
59         device.
60         **NOTE: this object is deprecated due to re-indexing of the
61         table.
62    "
63    REFERENCE
64        "12.14.3"
65    ::= { dotlagCfmDefaultMd 4 }
66
67 dotlagCfmDefaultMdEntry OBJECT-TYPE
68     SYNTAX      DotlagCfmDefaultMdEntry
69     MAX-ACCESS  not-accessible
70     STATUS      deprecated
71     DESCRIPTION
72        "The Default MD Level table entry.

```

```

1      **NOTE: this object is deprecated due to re-indexing of the
2      table.
3      "
4      INDEX { dotlagCfmDefaultMdComponentId,
5              dotlagCfmDefaultMdPrimaryVid }
6      ::= { dotlagCfmDefaultMdTable 1 }
7
8      DotlagCfmDefaultMdEntry ::= SEQUENCE {
9          dotlagCfmDefaultMdComponentId DotlagCfmPbbComponentIdentifier,
10         dotlagCfmDefaultMdPrimaryVid   VlanId,
11         dotlagCfmDefaultMdStatus       TruthValue,
12         dotlagCfmDefaultMdLevel        DotlagCfmMDLevelOrNone,
13         dotlagCfmDefaultMdMhfCreation  DotlagCfmMhfCreation,
14         dotlagCfmDefaultMdIdPermission DotlagCfmIdPermission
15     }
16
17 dotlagCfmDefaultMdComponentId OBJECT-TYPE
18     SYNTAX      DotlagCfmPbbComponentIdentifier
19     MAX-ACCESS  not-accessible
20     STATUS      deprecated
21     DESCRIPTION
22         "The Bridge component within the system to which the information
23         in this dotlagCfmDefaultMdEntry applies. If the system is not
24         a Bridge, or if only one component is present in the Bridge,
25         then this variable (index) MUST be equal to 1.
26         **NOTE: this object is deprecated due to re-indexing of the
27         table.
28         "
29     REFERENCE
30         "12.3 1)"
31     ::= { dotlagCfmDefaultMdEntry 1 }
32
33 dotlagCfmDefaultMdPrimaryVid OBJECT-TYPE
34     SYNTAX      VlanId
35     MAX-ACCESS  not-accessible
36     STATUS      deprecated
37     DESCRIPTION
38         "The Primary VID of the VLAN to which this entry's objects
39         apply.
40         **NOTE: this object is deprecated due to re-indexing of the
41         table.
42         "
43     ::= { dotlagCfmDefaultMdEntry 2 }
44
45 dotlagCfmDefaultMdStatus OBJECT-TYPE
46     SYNTAX      TruthValue
47     MAX-ACCESS  read-only
48     STATUS      deprecated
49     DESCRIPTION
50         "State of this Default MD Level table entry. True if there is
51         no entry in the Maintenance Association table defining an MA
52         for the same VLAN ID and MD Level as this table's entry, and
53         on which MA an Up MEP is defined, else false.
54         **NOTE: this object is deprecated due to re-indexing of the
55         table.
56         "
57     REFERENCE
58         "12.14.3.1.3:b"
59     ::= { dotlagCfmDefaultMdEntry 3 }
60
61 dotlagCfmDefaultMdLevel OBJECT-TYPE
62     SYNTAX      DotlagCfmMDLevelOrNone
63     MAX-ACCESS  read-write
64     STATUS      deprecated
65     DESCRIPTION
66         "A value indicating the MD Level at which MHFs are to be
67         created, and Sender ID TLV transmission by those MHFs is to
68         be controlled, for the VLAN to which this entry's objects
69         apply. If this object has the value -1, the MD Level for MHF
70         creation for this VLAN is controlled by
71         dotlagCfmDefaultMdDefLevel.
72         **NOTE: this object is deprecated due to re-indexing of the

```

```

1  table.
2  "
3  REFERENCE
4  "12.14.3.1.3:c, 12.14.3.2.2:b"
5  DEFVAL {-1}
6  ::= { dotlagCfmDefaultMdEntry 4 }
7
8 dotlagCfmDefaultMdMhfCreation OBJECT-TYPE
9  SYNTAX      DotlagCfmMhfCreation
10 MAX-ACCESS  read-write
11 STATUS      deprecated
12 DESCRIPTION
13     "A value indicating if the Management entity can create MHFs
14     (MIP Half Function) for this VID at this MD Level.  If this
15     object has the value defMHFdefer, MHF creation for this VLAN
16     is controlled by dotlagCfmDefaultMdDefMhfCreation.
17
18     The value of this variable is meaningless if the values of
19     dotlagCfmDefaultMdStatus is false.
20     **NOTE: this object is deprecated due to re-indexing of the
21     table.
22     "
23 REFERENCE
24 "12.14.3.1.3:d"
25 DEFVAL {defMHFdefer}
26 ::= { dotlagCfmDefaultMdEntry 5 }
27
28 dotlagCfmDefaultMdIdPermission OBJECT-TYPE
29 SYNTAX      DotlagCfmIdPermission
30 MAX-ACCESS  read-write
31 STATUS      deprecated
32 DESCRIPTION
33     "Enumerated value indicating what, if anything, is to be
34     included in the Sender ID TLV (21.5.3) transmitted by MHFs
35     created by the Default Maintenance Domain.  If this object
36     has the value sendIdDefer, Sender ID TLV transmission for
37     this VLAN is controlled by dotlagCfmDefaultMdDefIdPermission.
38
39     The value of this variable is meaningless if the values of
40     dotlagCfmDefaultMdStatus is false.
41     **NOTE: this object is deprecated due to re-indexing of the
42     table.
43     "
44 REFERENCE
45 "12.14.3.1.3:e"
46 DEFVAL { sendIdDefer }
47 ::= { dotlagCfmDefaultMdEntry 6 }
48
49 -- *****
50 -- The CFM configuration error list managed object. This group will
51 -- contain all the MIB objects used to read the interfaces and VIDs
52 -- configured incorrectly.
53 -- *****
54
55 -- *****
56 -- The CFM Configuration Error List Table
57 -- *****
58
59 dotlagCfmConfigErrorListTable OBJECT-TYPE
60 SYNTAX      SEQUENCE OF DotlagCfmConfigErrorListEntry
61 MAX-ACCESS  not-accessible
62 STATUS      deprecated
63 DESCRIPTION
64     "The CFM Configuration Error List table provides a list of
65     Interfaces and VIDs that are incorrectly configured.
66     **NOTE: this object is deprecated due to re-indexing of the
67     table.
68     "
69 REFERENCE
70 "12.14.4"
71 ::= { dotlagCfmConfigErrorList 1}
72

```

```

1 dotlagCfmConfigErrorListEntry OBJECT-TYPE
2   SYNTAX      DotlagCfmConfigErrorListEntry
3   MAX-ACCESS  not-accessible
4   STATUS      deprecated
5   DESCRIPTION
6     "The Config Error List Table  entry
7     **NOTE: this object is deprecated due to re-indexing of the
8     table.
9     "
10  INDEX { dotlagCfmConfigErrorListVid,
11          dotlagCfmConfigErrorListIfIndex
12          }
13  ::= { dotlagCfmConfigErrorListTable 1 }
14
15 DotlagCfmConfigErrorListEntry ::= SEQUENCE {
16   dotlagCfmConfigErrorListVid      VlanId,
17   dotlagCfmConfigErrorListIfIndex  InterfaceIndex,
18   dotlagCfmConfigErrorListErrorType DotlagCfmConfigErrors
19 }
20
21
22 dotlagCfmConfigErrorListVid OBJECT-TYPE
23   SYNTAX      VlanId
24   MAX-ACCESS  not-accessible
25   STATUS      deprecated
26   DESCRIPTION
27     "The VLAN ID of the VLAN with interfaces in error.
28     **NOTE: this object is deprecated due to re-indexing of the
29     table.
30     "
31   REFERENCE
32     "12.14.4.1.2:a"
33   ::= { dotlagCfmConfigErrorListEntry 1 }
34
35 dotlagCfmConfigErrorListIfIndex OBJECT-TYPE
36   SYNTAX      InterfaceIndex
37   MAX-ACCESS  not-accessible
38   STATUS      deprecated
39   DESCRIPTION
40     "This object is the IfIndex of the interface.
41
42     Upon a restart of the system, the system SHALL, if necessary,
43     change the value of this variable so that it indexes the
44     entry in the interface table with the same value of ifAlias
45     that it indexed before the system restart.  If no such
46     entry exists, then the system SHALL delete any entries in
47     dotlagCfmConfigErrorListTable indexed by that
48     InterfaceIndex value.
49     **NOTE: this object is deprecated due to re-indexing of the
50     table.
51     "
52   REFERENCE
53     "12.14.4.1.2:b"
54   ::= { dotlagCfmConfigErrorListEntry 2 }
55
56 dotlagCfmConfigErrorListErrorType OBJECT-TYPE
57   SYNTAX      DotlagCfmConfigErrors
58   MAX-ACCESS  read-only
59   STATUS      deprecated
60   DESCRIPTION
61     "A vector of Boolean error conditions from 22.2.4, any of
62     which may be true:
63
64     0) CFMleak;
65     1) ConflictingVids;
66     2) ExcessiveLevels;
67     3) OverlappedLevels.
68     **NOTE: this object is deprecated due to re-indexing of the
69     table.
70     "
71   REFERENCE
72     "12.14.4.1.3:b"

```

```

1 ::= { dotlagCfmConfigErrorListEntry 3 }
2
3 -- *****
4 -- The Maintenance Domain Managed Object. This group contains all
5 -- the MIB objects used to maintain Maintenance Domains.
6 -- *****
7
8 dotlagCfmMdTableNextIndex OBJECT-TYPE
9     SYNTAX      DotlafCfmIndexIntegerNextFree
10    MAX-ACCESS  read-only
11    STATUS      current
12    DESCRIPTION
13        "This object contains an unused value for dotlagCfmMdIndex in
14        the dotlagCfmMdTable, or a zero to indicate that none exist.
15        "
16    ::= { dotlagCfmMd 1 }
17
18 -- *****
19 -- The Maintenance Domain Table
20 -- *****
21
22 dotlagCfmMdTable OBJECT-TYPE
23     SYNTAX      SEQUENCE OF DotlagCfmMdEntry
24     MAX-ACCESS  not-accessible
25     STATUS      current
26     DESCRIPTION
27         "The Maintenance Domain table. Each row in the table
28         represents a different Maintenance Domain.
29
30         A Maintenance Domain is described 3.145 as the
31         network or the part of the network for which faults in
32         connectivity are to be managed. The boundary of a Maintenance
33         Domain is defined by a set of DSAPs, each of which can become
34         a point of connectivity to a service instance.
35         "
36     REFERENCE
37         "3.145, 18.1"
38     ::= { dotlagCfmMd 2 }
39
40 dotlagCfmMdEntry OBJECT-TYPE
41     SYNTAX      DotlagCfmMdEntry
42     MAX-ACCESS  not-accessible
43     STATUS      current
44     DESCRIPTION
45         "The Maintenance Domain table entry. This entry is not lost
46         upon reboot. It is backed up by stable storage.
47         "
48     INDEX {dotlagCfmMdIndex }
49     ::= { dotlagCfmMdTable 1 }
50
51 DotlagCfmMdEntry ::= SEQUENCE {
52     dotlagCfmMdIndex      Unsigned32,
53     dotlagCfmMdFormat     DotlagCfmMaintDomainNameType,
54     dotlagCfmMdName       DotlagCfmMaintDomainName,
55     dotlagCfmMdMdLevel    DotlagCfmMDLevel,
56     dotlagCfmMdMhfCreation DotlagCfmMhfCreation,
57     dotlagCfmMdMhfIdPermission DotlagCfmIdPermission,
58     dotlagCfmMdMaNextIndex DotlafCfmIndexIntegerNextFree,
59     dotlagCfmMdRowStatus  RowStatus
60 }
61
62 dotlagCfmMdIndex OBJECT-TYPE
63     SYNTAX      Unsigned32(1..4294967295)
64     MAX-ACCESS  not-accessible
65     STATUS      current
66     DESCRIPTION
67         "The index to the Maintenance Domain table.
68
69         dotlagCfmMdTableNextIndex needs to be inspected to find an
70         available index for row-creation.
71
72         Referential integrity is required, i.e., the index needs to be

```

```

1      persistent upon a reboot or restart of a device. The index
2      can never be reused for other Maintenance Domain. The index
3      value SHOULD keep increasing up to the time that they wrap
4      around. This is to facilitate access control based on OID.
5      "
6      ::= { dotlagCfmMdEntry 1 }
7
8 dotlagCfmMdFormat OBJECT-TYPE
9     SYNTAX      DotlagCfmMaintDomainNameType
10    MAX-ACCESS   read-create
11    STATUS       current
12    DESCRIPTION
13        "The type (and thereby format) of the Maintenance Domain Name."
14    REFERENCE
15        "21.6.5.1"
16    DEFVAL { charString }
17    ::= { dotlagCfmMdEntry 2 }
18
19 dotlagCfmMdName OBJECT-TYPE
20    SYNTAX      DotlagCfmMaintDomainName
21    MAX-ACCESS   read-create
22    STATUS       current
23    DESCRIPTION
24        "The Maintenance Domain name. The type/format of this object
25         is determined by the value of the dotlagCfmMdNameType object.
26
27         Each Maintenance Domain has unique name among all those
28         used or available to a service provider or operator. It
29         facilitates easy identification of administrative
30         responsibility for each Maintenance Domain.
31
32         3.150 defines a Maintenance Domain name as the
33         identifier, unique over the domain for which CFM is to
34         protect against accidental concatenation of Service
35         Instances, of a particular Maintenance Domain.
36        "
37    REFERENCE
38        "3.150, 12.14.5, 21.6.5.3"
39    DEFVAL { "DEFAULT" }
40    ::= { dotlagCfmMdEntry 3 }
41
42 dotlagCfmMdMdLevel OBJECT-TYPE
43    SYNTAX      DotlagCfmMDLevel
44    MAX-ACCESS   read-create
45    STATUS       current
46    DESCRIPTION
47        "The Maintenance Domain Level."
48    REFERENCE
49        "12.14.5.1.3:b"
50    DEFVAL { 0 }
51    ::= { dotlagCfmMdEntry 4 }
52
53 dotlagCfmMdMhfCreation OBJECT-TYPE
54    SYNTAX      DotlagCfmMhfCreation {
55        defMHFnone (1),
56        defMHFdefault (2),
57        defMHFexplicit (3)
58    }
59    MAX-ACCESS   read-create
60    STATUS       current
61    DESCRIPTION
62        "Enumerated value indicating whether the management entity can
63         create MHFs (MIP Half Function) for this Maintenance Domain.
64         Since, in this variable, there is no encompassing Maintenance
65         Domain, the value defMHFdefer is not allowed.
66        "
67    REFERENCE
68        "12.14.5.1.3:c"
69    DEFVAL { defMHFnone }
70    ::= { dotlagCfmMdEntry 5 }
71
72 dotlagCfmMdMhfIdPermission OBJECT-TYPE

```

```

1  SYNTAX      DotlagCfmIdPermission {
2              sendIdNone          (1),
3              sendIdChassis       (2),
4              sendIdManage        (3),
5              sendIdChassisManage (4)
6          }
7  MAX-ACCESS  read-create
8  STATUS      current
9  DESCRIPTION
10     "Enumerated value indicating what, if anything, is to be
11     included in the Sender ID TLV (21.5.3) transmitted by MPs
12     configured in this Maintenance Domain. Since, in this
13     variable, there is no encompassing Maintenance Domain, the
14     value sendIdDefer is not allowed.
15     "
16  REFERENCE
17     "12.14.5.1.3:d"
18  DEFVAL { sendIdNone }
19  ::= { dotlagCfmMdEntry 6 }
20
21 dotlagCfmMdMaNextIndex OBJECT-TYPE
22  SYNTAX      DotlagCfmIndexIntegerNextFree
23  MAX-ACCESS  read-only
24  STATUS      current
25  DESCRIPTION
26     "Value to be used as the index of the MA table entries, both
27     the dotlagCfmMaNetTable and the dotlagCfmMaCompTable, for
28     this Maintenance Domain when the management entity wants to
29     create a new row in those tables.
30     "
31  ::= { dotlagCfmMdEntry 7 }
32
33 dotlagCfmMdRowStatus OBJECT-TYPE
34  SYNTAX      RowStatus
35  MAX-ACCESS  read-create
36  STATUS      current
37  DESCRIPTION
38     "The status of the row.
39
40     The writable columns in a row cannot be changed if the row
41     is active. All columns MUST have a valid value before a row
42     can be activated.
43     "
44  ::= { dotlagCfmMdEntry 8 }
45
46 -- *****
47 -- The Maintenance Association Object. This group contains all the
48 -- MIB objects used to read, create, modify, and delete Maintenance
49 -- Associations in the MIB.
50 -- *****
51
52 -- *****
53 -- The Maintenance Association (MA) Network Table
54 -- *****
55
56 dotlagCfmMaNetTable OBJECT-TYPE
57  SYNTAX      SEQUENCE OF DotlagCfmMaNetEntry
58  MAX-ACCESS  not-accessible
59  STATUS      current
60  DESCRIPTION
61     "The Maintenance Association table. Each row in the table
62     represents an MA. An MA is a set of MEPs, each configured
63     with a single service instance.
64
65     This is the part of the complete MA table that is constant
66     across all Bridges in a Maintenance Domain, and across all
67     components of a single Bridge. That part of the MA table that
68     can vary from Bridge component to Bridge component is contained
69     in the dotlagCfmMaCompTable.
70
71     Creation of a Service Instance establishes a connectionless
72     association among the selected DSAPs. Configuring a

```

1 Maintenance association Endpoint (MEP) at each of the
2 DSAPs creates a Maintenance Association (MA) to monitor
3 that connectionless connectivity. The MA is identified by a
4 Short MA Name that is unique within the Maintenance Domain
5 and chosen to facilitate easy identification of the Service
6 Instance. Together, the Maintenance Domain Name and the
7 Short MA Name form the Maintenance Association Identifier
8 (MAID) that is carried in CFM Messages to identify
9 incorrect connectivity among Service Instances. A small
10 integer, the Maintenance association Endpoint Identifier
11 (MEPID), identifies each MEP among those configured on a
12 single MA (3.142, 18.2).
13
14 This table uses two indices, first index is the index of the
15 Maintenance Domain table. The second index is the same as the
16 index of the dotlagCfmMaCompEntry for the same MA.
17
18 The writable objects in this table need to be persistent
19 upon reboot or restart of a device.
20
21 "
22 REFERENCE
23 "18.2"
24 ::= { dotlagCfmMa 1 }
25
26 dotlagCfmMaNetEntry OBJECT-TYPE
27 SYNTAX DotlagCfmMaNetEntry
28 MAX-ACCESS not-accessible
29 STATUS current
30 DESCRIPTION
31 "The MA table entry."
32 INDEX {dotlagCfmMdIndex, dotlagCfmMaIndex }
33 ::= { dotlagCfmMaNetTable 1 }
34
35 DotlagCfmMaNetEntry ::= SEQUENCE {
36 dotlagCfmMaIndex Unsigned32,
37 dotlagCfmMaNetFormat DotlagCfmMaintAssocNameType,
38 dotlagCfmMaNetName DotlagCfmMaintAssocName,
39 dotlagCfmMaNetCcmInterval DotlagCfmCcmInterval,
40 dotlagCfmMaNetRowStatus RowStatus
41 }
42
43 dotlagCfmMaIndex OBJECT-TYPE
44 SYNTAX Unsigned32(1..4294967295)
45 MAX-ACCESS not-accessible
46 STATUS current
47 DESCRIPTION
48
49 "Index of the MA table dotlagCfmMdMaNextIndex needs to
50 be inspected to find an available index for row-creation.
51 "
52 ::= { dotlagCfmMaNetEntry 1 }
53
54 dotlagCfmMaNetFormat OBJECT-TYPE
55 SYNTAX DotlagCfmMaintAssocNameType
56 MAX-ACCESS read-create
57 STATUS current
58 DESCRIPTION
59 "The type (and thereby format) of the Maintenance Association
60 Name.
61 "
62 REFERENCE
63 "21.6.5.4"
64 ::= { dotlagCfmMaNetEntry 2 }
65
66 dotlagCfmMaNetName OBJECT-TYPE
67 SYNTAX DotlagCfmMaintAssocName
68 MAX-ACCESS read-create
69 STATUS current
70 DESCRIPTION
71 "The Short Maintenance Association name. The type/format of
72 this object is determined by the value of the


```

1      dotlagCfmMaNetNameType object. This name MUST be unique within
2      a maintenance domain.
3      "
4      REFERENCE
5      "21.6.5.6, Table 21-19"
6      ::= { dotlagCfmMaNetEntry 3 }
7
8 dotlagCfmMaNetCcmInterval OBJECT-TYPE
9     SYNTAX      DotlagCfmCcmInterval
10    MAX-ACCESS   read-create
11    STATUS       current
12    DESCRIPTION
13        "Interval between CCM transmissions to be used by all MEPS
14        in the MA.
15        "
16    REFERENCE
17        "12.14.6.1.3:e"
18    DEFVAL { intervalls }
19    ::= { dotlagCfmMaNetEntry 4 }
20
21 dotlagCfmMaNetRowStatus OBJECT-TYPE
22    SYNTAX      RowStatus
23    MAX-ACCESS   read-create
24    STATUS       current
25    DESCRIPTION
26        "The status of the row.
27
28        The writable columns in a row cannot be changed if the row
29        is active. All columns MUST have a valid value before a row
30        can be activated.
31        "
32    ::= { dotlagCfmMaNetEntry 5 }
33
34 -- *****
35 -- The Maintenance Association (MA) Component Table
36 -- *****
37
38 dotlagCfmMaCompTable OBJECT-TYPE
39    SYNTAX      SEQUENCE OF DotlagCfmMaCompEntry
40    MAX-ACCESS   not-accessible
41    STATUS       deprecated
42    DESCRIPTION
43        "The Maintenance Association table. Each row in the table
44        represents an MA. An MA is a set of MEPS, each configured
45        with a single service instance.
46
47        This is the part of the complete MA table that is variable
48        across the Bridges in a Maintenance Domain, or across the
49        components of a single Bridge. That part of the MA table that
50        is constant across the Bridges and their components in a
51        Maintenance Domain is contained in the dotlagCfmMaNetTable.
52
53        This table uses three indices, first index is the
54        DotlagCfmPbbComponentIdentifier that identifies the component
55        within the Bridge for which the information in the
56        dotlagCfmMaCompEntry applies. The second is the index of the
57        Maintenance Domain table. The third index is the same as the
58        index of the dotlagCfmMaNetEntry for the same MA.
59
60        The writable objects in this table need to be persistent
61        upon reboot or restart of a device.
62
63        **NOTE: this object is deprecated due to re-indexing of the
64        table.
65        "
66    REFERENCE
67        "18.2"
68    ::= { dotlagCfmMa 2 }
69
70 dotlagCfmMaCompEntry OBJECT-TYPE
71    SYNTAX      DotlagCfmMaCompEntry
72    MAX-ACCESS   not-accessible

```

```

1  STATUS      deprecated
2  DESCRIPTION
3      "The MA table entry.
4      **NOTE: this object is deprecated due to re-indexing of the
5  table.
6  "
7  INDEX {dotlagCfmMaComponentId,
8          dotlagCfmMdIndex, dotlagCfmMaIndex }
9  ::= { dotlagCfmMaCompTable 1 }
10
11 DotlagCfmMaCompEntry ::= SEQUENCE {
12     dotlagCfmMaComponentId      DotlagCfmPbbComponentIdentifier,
13     dotlagCfmMaCompPrimaryVlanId VlanIdOrNone,
14     dotlagCfmMaCompMhfCreation   DotlagCfmMhfCreation,
15     dotlagCfmMaCompIdPermission  DotlagCfmIdPermission,
16     dotlagCfmMaCompNumberOfVids  Unsigned32,
17     dotlagCfmMaCompRowStatus     RowStatus
18 }
19
20 dotlagCfmMaComponentId OBJECT-TYPE
21     SYNTAX      DotlagCfmPbbComponentIdentifier
22     MAX-ACCESS  not-accessible
23     STATUS      deprecated
24     DESCRIPTION
25         "The Bridge component within the system to which the information
26         in this dotlagCfmMaCompEntry applies. If the system is not a
27         Bridge, or if only one component is present in the Bridge, then
28         this variable (index) MUST be equal to 1.
29     **NOTE: this object is deprecated due to re-indexing of the
30     table.
31     "
32     REFERENCE
33         "12.3 1)"
34     ::= { dotlagCfmMaCompEntry 1 }
35
36 dotlagCfmMaCompPrimaryVlanId OBJECT-TYPE
37     SYNTAX      VlanIdOrNone
38     MAX-ACCESS  read-create
39     STATUS      deprecated
40     DESCRIPTION
41         "The Primary VLAN ID with which the Maintenance Association is
42         associated, or 0 if the MA is not attached to any VID. If
43         the MA is associated with more than one VID, the
44         dotlagCfmVlanTable lists them.
45     **NOTE: this object is deprecated due to re-indexing of the
46     table.
47     "
48     REFERENCE
49         "12.14.6.1.3:b"
50     ::= { dotlagCfmMaCompEntry 2 }
51
52 dotlagCfmMaCompMhfCreation OBJECT-TYPE
53     SYNTAX      DotlagCfmMhfCreation
54     MAX-ACCESS  read-create
55     STATUS      deprecated
56     DESCRIPTION
57         "Indicates if the Management entity can create MHFs (MIP Half
58         Function) for this MA.
59     **NOTE: this object is deprecated due to re-indexing of the
60     table.
61     "
62     REFERENCE
63         "12.14.6.1.3:c"
64     DEFVAL { defMHFdefer }
65     ::= { dotlagCfmMaCompEntry 3 }
66
67 dotlagCfmMaCompIdPermission OBJECT-TYPE
68     SYNTAX      DotlagCfmIdPermission
69     MAX-ACCESS  read-create
70     STATUS      deprecated
71     DESCRIPTION
72         "Enumerated value indicating what, if anything, is to be

```

```

1      included in the Sender ID TLV (21.5.3) transmitted by MPs
2      configured in this MA.
3      **NOTE: this object is deprecated due to re-indexing of the
4      table.
5      "
6      REFERENCE
7      "12.14.6.1.3:d"
8      DEFVAL { sendIdDefer }
9      ::= { dotlagCfmMaCompEntry 4 }
10
11 dotlagCfmMaCompNumberOfVids OBJECT-TYPE
12     SYNTAX      Unsigned32
13     MAX-ACCESS   read-create
14     STATUS      deprecated
15     DESCRIPTION
16         "The number of VIDs associated with the MA.
17         **NOTE: this object is deprecated due to re-indexing of the
18         table.
19         "
20     REFERENCE
21     "12.14.6.1.3:b"
22     ::= { dotlagCfmMaCompEntry 5 }
23
24 dotlagCfmMaCompRowStatus OBJECT-TYPE
25     SYNTAX      RowStatus
26     MAX-ACCESS   read-create
27     STATUS      deprecated
28     DESCRIPTION
29         "The status of the row.
30
31         The writable columns in a row cannot be changed if the row
32         is active. All columns MUST have a valid value before a row
33         can be activated.
34         **NOTE: this object is deprecated due to re-indexing of the
35         table.
36         "
37     ::= { dotlagCfmMaCompEntry 6 }
38
39 -- *****
40 -- The list of known MEPs for a given MA
41 -- *****
42
43 dotlagCfmMaMepListTable OBJECT-TYPE
44     SYNTAX      SEQUENCE OF DotlagCfmMaMepListEntry
45     MAX-ACCESS   not-accessible
46     STATUS      current
47     DESCRIPTION
48         "List of MEPIDs that belong to this MA.
49
50         12.14.6.1.3 specifies that a list of MEPIDs in all
51         Bridges in that MA, but since SNMP SMI does not allow to
52         state in a MIB that an object in a table is an array, the
53         information has to be stored in another table with two
54         indices, being the first index, the index of the table that
55         contains the list or array.
56
57         For all Bridges in which the same MAID {dotlagCfmMdFormat,
58         dotlagCfmMdName, dotlagCfmMaNetFormat, and dotlagCfmMaNetName}
59         is configured, the same set of dotlagCfmMaMepListIdentifiers
60         MUST be configured in the Bridges' dotlagCfmMaMepListTables.
61         This allows each MEP to determine whether or not it is
62         receiving CCMs from all of the other MEPs in the MA.
63
64         For example, if one were creating a new MA whose MAID were
65         {charString, 'Dom1', charString, 'MA1'}, that had 2 MEPs, whose
66         MEPIDs were 1 and 3, one could, in Bridge A:
67         1. Get a new MD index d from dotlagCfmMdTableNextIndex.
68         2. Create the Maintenance Domain {charString, 'Dom1'}.
69         3. Get a new MA index a from dotlagCfmMdMaNextIndex [d].
70         4. Create the Maintenance Association {charString, 'MA1'}.
71         5. Create a new dotlagCfmMaMepListEntry for each of the MEPs
72         in the MA: [d, a, 1] and [d, a, 3].

```

```

1      6. Create one of the new MEPS, say [d, a, 1].
2      Then, in Bridge B:
3      7. Do all of these steps 1-6, except for using the other MEPID
4         for the new MEP in Step 6, in this example, MEPID 3.
5      Note that, when creating the MA, MEP List Table, and MEP
6      entries in the second Bridge, the indices 'd' and 'a'
7      identifying the MAID {charString, 'Dom1', charString, 'MA1'}
8      may have different values than those in the first Bridge.
9      "
10     REFERENCE
11     "12.14.6.1.3:g"
12     ::= { dotlagCfmMa 3 }
13
14 dotlagCfmMaMepListEntry OBJECT-TYPE
15     SYNTAX      DotlagCfmMaMepListEntry
16     MAX-ACCESS  not-accessible
17     STATUS      current
18     DESCRIPTION
19         "The known MEPS table entry."
20     INDEX { dotlagCfmMdIndex,
21             dotlagCfmMaIndex,
22             dotlagCfmMaMepListIdentifier
23         }
24     ::= { dotlagCfmMaMepListTable 1 }
25
26 DotlagCfmMaMepListEntry ::= SEQUENCE {
27     dotlagCfmMaMepListIdentifier DotlagCfmMepId,
28     dotlagCfmMaMepListRowStatus RowStatus
29 }
30
31 dotlagCfmMaMepListIdentifier OBJECT-TYPE
32     SYNTAX      DotlagCfmMepId
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "MEPID"
37     REFERENCE
38     "12.14.6.1.3:g"
39     ::= { dotlagCfmMaMepListEntry 1 }
40
41 dotlagCfmMaMepListRowStatus OBJECT-TYPE
42     SYNTAX      RowStatus
43     MAX-ACCESS  read-create
44     STATUS      current
45     DESCRIPTION
46         "The status of the row. Read SNMPv2-TC (RFC1903) for an
47         explanation of the possible values this object can take.
48     "
49     ::= { dotlagCfmMaMepListEntry 2 }
50
51 -- *****
52 -- The MEP Object. This object represents a Maintenance End
53 -- Point as described in IEEE Std 802.1Q.
54 -- *****
55
56 -- *****
57 -- The MEP Table
58 -- *****
59
60 dotlagCfmMepTable OBJECT-TYPE
61     SYNTAX      SEQUENCE OF DotlagCfmMepEntry
62     MAX-ACCESS  not-accessible
63     STATUS      current
64     DESCRIPTION
65         "The Maintenance Association Endpoint (MEP) table.
66
67         Each row in the table represents a different MEP. A MEP is
68         an actively managed CFM entity, associated with a specific
69         DSAP of a Service Instance, which can generate and receive
70         CFM PDUs and track any responses. It is an endpoint of a
71         single Maintenance Association, and is an endpoint of a
72         separate Maintenance Entity for each of the other MEPS in

```

the same Maintenance Association (3.142).

This table uses three indices. The first two indices are the indices of the Maintenance Domain and MA tables, the reason being that a MEP is always related to an MA and Maintenance Domain.

The MEP table also stores all the managed objects for sending LBM and LTM.

*LBM Managed objects

LBM Managed objects in the MEP table enables the management entity to initiate transmission of Loopback messages. It will signal the MEP that it SHOULD transmit some number of Loopback messages and detect the detection (or lack thereof) of the corresponding Loopback messages.

Steps to use entries in this table:

- 1) Wait for dotlagCfmMepTransmitLbmStatus value to be false. To do this do this sequence:
 - a. an SNMP GET for both SnmpSetSerialNo and dotlagCfmMepTransmitLbmStatus objects (in same SNMP PDU).
 - b. Check if value for dotlagCfmMepTransmitLbmStatus is false.
 - if not, wait x seconds, go to step a above.
 - if yes, save the value of SnmpSetSerialNo and go to step 2) below
- 2) Change dotlagCfmMepTransmitLbmStatus value from false to true to ensure no other management entity will use the service. In order to not disturb a possible other NMS do this by sending an SNMP SET for both SnmpSetSerialNo and dotlagCfmMepTransmitLbmStatus objects (in same SNMP PDU, and make sure SnmpSetSerialNo is the first varBind). For the SnmpSetSerialNo varBind, use the value that you obtained in step 1)a.. This ensures that two cooperating NMSes will not step on each others toes. Setting this MIB object does not set the corresponding LBIActive state machine variable.
- 3) Setup the different data to be sent (number of messages, optional TLVs,...), except do not set dotlagCfmMepTransmitLbmMessages.
- 4) Record the current values of dotlagCfmMepLbrIn, dotlagCfmMepLbrInOutOfOrder, and dotlagCfmMepLbrBadMsdu.
- 6) Set dotlagCfmMepTransmitLbmMessages to a non-zero value to initiate transmission of Loopback messages. The dotlagCfmMepTransmitLbmMessages indicates the number of LBMs to be sent and is not decremented as loopbacks are actually sent. dotlagCfmMepTransmitLbmMessages is not equivalent to the LBMTToSend state machine variable.
- 7) Check the value of dotlagCfmMepTransmitLbmResultOK to find out if the operation was successfully initiated or not.
- 8) Monitor the value of dotlagCfmMepTransmitLbmStatus. When it is reset to false, the last LBM has been transmitted. Wait an additional 5 seconds to ensure that all LBRs have been returned.
- 9) Compare dotlagCfmMepLbrIn, dotlagCfmMepLbrInOutOfOrder, and dotlagCfmMepLbrBadMsdu to their old values from step 4, above, to get the results of the test.

*LTM Managed objects

The LTM Managed objects in the MEP table are used in a manner similar to that described for LBM transmission, above. A SET operation to the variable dotlagCfmMepTransmitLtmFlags triggers the transmission of an LTM. Then, the variables dotlagCfmMepTransmitLtmSeqNumber and dotlagCfmMepTransmitLtmEgressIdentifier return the information required to recover the results of the LTM from the dotlagCfmLtrTable.

```

1      "
2      REFERENCE
3      "12.14.7, 19.2"
4      ::= { dotlagCfmMep 1 }
5
6      dotlagCfmMepEntry OBJECT-TYPE
7          SYNTAX      DotlagCfmMepEntry
8          MAX-ACCESS  not-accessible
9          STATUS      current
10         DESCRIPTION
11             "The MEP table entry"
12         INDEX { dotlagCfmMdIndex,
13                 dotlagCfmMaIndex,
14                 dotlagCfmMepIdentifier
15             }
16         ::= { dotlagCfmMepTable 1 }
17
18     DotlagCfmMepEntry ::= SEQUENCE {
19         dotlagCfmMepIdentifier          DotlagCfmMepId,
20         dotlagCfmMepIfIndex             InterfaceIndexOrZero,
21         dotlagCfmMepDirection           DotlagCfmMpDirection,
22         dotlagCfmMepPrimaryVid          Unsigned32,
23         dotlagCfmMepActive              TruthValue,
24         dotlagCfmMepFngState            DotlagCfmFngState,
25         dotlagCfmMepCciEnabled          TruthValue,
26         dotlagCfmMepCcmLtmPriority      Unsigned32,
27         dotlagCfmMepMacAddress          MacAddress,
28         dotlagCfmMepLowPrDef            DotlagCfmLowestAlarmPri,
29         dotlagCfmMepFngAlarmTime        TimeInterval,
30         dotlagCfmMepFngResetTime        TimeInterval,
31         dotlagCfmMepHighestPrDefect     DotlagCfmHighestDefectPri,
32         dotlagCfmMepDefects             DotlagCfmMepDefects,
33         dotlagCfmMepErrorCcmLastFailure OCTET STRING,
34         dotlagCfmMepXconCcmLastFailure  OCTET STRING,
35         dotlagCfmMepCcmSequenceErrors   Counter32,
36         dotlagCfmMepCciSentCcms         Counter32,
37         dotlagCfmMepNextLbmTransId      Unsigned32,
38         dotlagCfmMepLbrIn               Counter32,
39         dotlagCfmMepLbrInOutOfOrder     Counter32,
40         dotlagCfmMepLbrBadMsdu          Counter32,
41         dotlagCfmMepLtmNextSeqNumber    Unsigned32,
42         dotlagCfmMepUnexpLtrIn          Counter32,
43         dotlagCfmMepLbrOut              Counter32,
44         dotlagCfmMepTransmitLbmStatus    TruthValue,
45         dotlagCfmMepTransmitLbmDestMacAddress MacAddress,
46         dotlagCfmMepTransmitLbmDestMepId DotlagCfmMepIdOrZero,
47         dotlagCfmMepTransmitLbmDestIsMepId TruthValue,
48         dotlagCfmMepTransmitLbmMessages Integer32,
49         dotlagCfmMepTransmitLbmDataTlv  OCTET STRING,
50         dotlagCfmMepTransmitLbmVlanPriority Integer32,
51         dotlagCfmMepTransmitLbmVlanDropEnable TruthValue,
52         dotlagCfmMepTransmitLbmResultOK  TruthValue,
53         dotlagCfmMepTransmitLbmSeqNumber Unsigned32,
54         dotlagCfmMepTransmitLtmStatus    TruthValue,
55         dotlagCfmMepTransmitLtmFlags     BITS,
56         dotlagCfmMepTransmitLtmTargetMacAddress MacAddress,
57         dotlagCfmMepTransmitLtmTargetMepId DotlagCfmMepIdOrZero,
58         dotlagCfmMepTransmitLtmTargetIsMepId TruthValue,
59         dotlagCfmMepTransmitLtmTtl       Unsigned32,
60         dotlagCfmMepTransmitLtmResult    TruthValue,
61         dotlagCfmMepTransmitLtmSeqNumber Unsigned32,
62         dotlagCfmMepTransmitLtmEgressIdentifier OCTET STRING,
63         dotlagCfmMepRowStatus            RowStatus,
64         dotlagCfmMepPbbTeCanReportPbbTePresence TruthValue,
65         dotlagCfmMepPbbTeTrafficMismatchDefect TruthValue,
66         dotlagCfmMepPbbTeTransmitLbmLtmReverseVid IEEE8021VlanIndex,
67         dotlagCfmMepPbbTeMismatchAlarm        TruthValue,
68         dotlagCfmMepPbbTeLocalMismatchDefect  TruthValue,
69         dotlagCfmMepPbbTeMismatchSinceReset   TruthValue
70     }
71
72     dotlagCfmMepIdentifier OBJECT-TYPE

```

```

1  SYNTAX      DotlagCfmMepId
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "Integer that is unique among all the MEPs in the same MA.
6      Other definition is: a small integer, unique over a given
7      Maintenance Association, identifying a specific Maintenance
8      association Endpoint (3.142).
9
10     MEP Identifier is also known as the MEPID.
11     "
12  REFERENCE
13      "3.142, 19.2, 12.14.7"
14      ::= { dotlagCfmMepEntry 1 }
15
16 dotlagCfmMepIfIndex OBJECT-TYPE
17     SYNTAX      InterfaceIndexOrZero
18     MAX-ACCESS  read-create
19     STATUS      current
20     DESCRIPTION
21         "This object is the interface index of the interface either a
22         Bridge Port, or an aggregated IEEE 802.1 link within a Bridge
23         port, to which the MEP is attached.
24
25         Upon a restart of the system, the system SHALL, if necessary,
26         change the value of this variable so that it indexes the
27         entry in the interface table with the same value of ifAlias
28         that it indexed before the system restart. If no such
29         entry exists, then the system SHALL set this variable to 0.
30         "
31     REFERENCE
32         "12.14.7.1.3:b"
33     ::= { dotlagCfmMepEntry 2 }
34
35 dotlagCfmMepDirection OBJECT-TYPE
36     SYNTAX      DotlagCfmMpDirection
37     MAX-ACCESS  read-create
38     STATUS      current
39     DESCRIPTION
40         "The direction in which the MEP faces on the Bridge port."
41     REFERENCE
42         "12.14.7.1.3:c, 19.2"
43     ::= { dotlagCfmMepEntry 3 }
44
45 dotlagCfmMepPrimaryVid OBJECT-TYPE
46     SYNTAX      Unsigned32(0..16777215)
47     MAX-ACCESS  read-create
48     STATUS      current
49     DESCRIPTION
50         "An integer indicating the Primary VID of the MEP, always
51         one of the VIDs assigned to the MEP's MA. The value 0
52         indicates that either the Primary VID is that of the
53         MEP's MA, or that the MEP's MA is associated with no VID."
54     REFERENCE
55         "12.14.7.1.3:d"
56     DEFVAL { 0 }
57     ::= { dotlagCfmMepEntry 4 }
58
59 dotlagCfmMepActive OBJECT-TYPE
60     SYNTAX      TruthValue
61     MAX-ACCESS  read-create
62     STATUS      current
63     DESCRIPTION
64         "Administrative state of the MEP
65
66         A Boolean indicating the administrative state of the MEP.
67
68         True indicates that the MEP is to function normally, and
69         false that it is to cease functioning."
70     REFERENCE
71         "12.14.7.1.3:e, 20.9.1"
72     DEFVAL { false }

```

```

1 ::= { dotlagCfmMepEntry 5 }
2
3 dotlagCfmMepFngState OBJECT-TYPE
4     SYNTAX      DotlagCfmFngState
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "Current state of the MEP Fault Notification Generator
9         State Machine.
10        "
11     REFERENCE
12         "12.14.7.1.3:f, 20.35"
13     DEFVAL { fngReset }
14     ::= { dotlagCfmMepEntry 6 }
15
16 dotlagCfmMepCciEnabled OBJECT-TYPE
17     SYNTAX      TruthValue
18     MAX-ACCESS  read-create
19     STATUS      current
20     DESCRIPTION
21         "If set to true, the MEP will generate CCM messages."
22     REFERENCE
23         "12.14.7.1.3:g, 20.10.1"
24     DEFVAL { false }
25     ::= { dotlagCfmMepEntry 7 }
26
27 dotlagCfmMepCcmLtmPriority OBJECT-TYPE
28     SYNTAX      Unsigned32 (0..7)
29     MAX-ACCESS  read-create
30     STATUS      current
31     DESCRIPTION
32         "The priority value for CCMs and LTMs transmitted by the MEP.
33         Default Value is the highest priority value allowed to pass
34         through the Bridge Port for any of this MEPs VIDs.
35         The management entity can obtain the default value for this
36         variable from the priority regeneration table by extracting the
37         highest priority value in this table on this MEPs Bridge Port.
38         (1 is lowest, then 2, then 0, then 3-7).
39        "
40     REFERENCE
41         "12.14.7.1.3:h"
42     ::= { dotlagCfmMepEntry 8 }
43
44 dotlagCfmMepMacAddress OBJECT-TYPE
45     SYNTAX      MacAddress
46     MAX-ACCESS  read-only
47     STATUS      current
48     DESCRIPTION
49         "MAC address of the MEP."
50     REFERENCE
51         "12.14.7.1.3:i, 19.4"
52     ::= { dotlagCfmMepEntry 9 }
53
54 dotlagCfmMepLowPrDef OBJECT-TYPE
55     SYNTAX      DotlagCfmLowestAlarmPri
56     MAX-ACCESS  read-create
57     STATUS      current
58     DESCRIPTION
59         "An integer value specifying the lowest priority defect
60         that is allowed to generate fault alarm.
61        "
62     REFERENCE
63         "12.14.7.1.3:k, 20.9.5, Table 20-1"
64     DEFVAL { macRemErrXcon }
65     ::= { dotlagCfmMepEntry 10}
66
67 dotlagCfmMepFngAlarmTime OBJECT-TYPE
68     SYNTAX      TimeInterval (250..1000)
69     MAX-ACCESS  read-create
70     STATUS      current
71     DESCRIPTION
72         "The time that defects MUST be present before a Fault Alarm is

```



```

1      issued (fngAlarmTime, 20.3.3) (default 2.5s).
2      "
3      REFERENCE
4      "12.14.7.1.3:l, 20.3.3"
5      DEFVAL { 250 }
6      ::= { dotlagCfmMepEntry 11 }
7
8      dotlagCfmMepFngResetTime OBJECT-TYPE
9      SYNTAX      TimeInterval (250..1000)
10     MAX-ACCESS   read-create
11     STATUS       current
12     DESCRIPTION
13     "The time that defects MUST be absent before resetting a
14     Fault Alarm (fngResetTime, 20.35.4) (default 10s).
15     "
16     REFERENCE
17     "12.14.7.1.3:m, 20.35.4"
18     DEFVAL { 1000 }
19     ::= { dotlagCfmMepEntry 12 }
20
21     dotlagCfmMepHighestPrDefect OBJECT-TYPE
22     SYNTAX      DotlagCfmHighestDefectPri
23     MAX-ACCESS   read-only
24     STATUS       current
25     DESCRIPTION
26     "The highest priority defect that has been present since the
27     MEPS Fault Notification Generator State Machine was last in
28     the FNG_RESET state.
29     "
30     REFERENCE
31     "12.14.7.1.3:n, 20.35.9, Table 20-1"
32     ::= { dotlagCfmMepEntry 13 }
33
34     dotlagCfmMepDefects OBJECT-TYPE
35     SYNTAX      DotlagCfmMepDefects
36     MAX-ACCESS   read-only
37     STATUS       current
38     DESCRIPTION
39     "A vector of Boolean error conditions from Table 20-1, any of
40     which may be true:
41
42     DefRDICCM(0)
43     DefMACstatus(1)
44     DefRemoteCCM(2)
45     DefErrorCCM(3)
46     DefXconCCM(4)
47     "
48     REFERENCE
49     "12.14.7.1.3:o, 12.14.7.1.3:p, 12.14.7.1.3:q,
50     12.14.7.1.3:r, 12.14.7.1.3:s, 20.21.3, 20.23.3, 20.35.5,
51     20.35.6, 20.35.7."
52     ::= { dotlagCfmMepEntry 14 }
53
54     dotlagCfmMepErrorCcmLastFailure OBJECT-TYPE
55     SYNTAX      OCTET STRING (SIZE(1..1522))
56     MAX-ACCESS   read-only
57     STATUS       current
58     DESCRIPTION
59     "The last-received CCM that triggered an DefErrorCCM fault."
60     REFERENCE
61     "12.14.7.1.3:t, 20.21.2"
62     ::= { dotlagCfmMepEntry 15 }
63
64     dotlagCfmMepXconCcmLastFailure OBJECT-TYPE
65     SYNTAX      OCTET STRING (SIZE(1..1522))
66     MAX-ACCESS   read-only
67     STATUS       current
68     DESCRIPTION
69     "The last-received CCM that triggered a DefXconCCM fault."
70     REFERENCE
71     "12.14.7.1.3:u, 20.23.2"
72     ::= { dotlagCfmMepEntry 16 }

```

```

1
2 dotlagCfmMepCcmSequenceErrors OBJECT-TYPE
3     SYNTAX      Counter32
4     MAX-ACCESS  read-only
5     STATUS      current
6     DESCRIPTION
7         "The total number of out-of-sequence CCMS received from all
8         remote MEPS.
9     "
10    REFERENCE
11        "12.14.7.1.3:v, 20.16.12"
12    ::= { dotlagCfmMepEntry 17 }
13
14 dotlagCfmMepCciSentCcms OBJECT-TYPE
15     SYNTAX      Counter32
16     MAX-ACCESS  read-only
17     STATUS      current
18     DESCRIPTION
19         "Total number of Continuity Check messages transmitted."
20    REFERENCE
21        "12.14.7.1.3:w, 20.10.2"
22    ::= { dotlagCfmMepEntry 18 }
23
24 dotlagCfmMepNextLbmTransId OBJECT-TYPE
25     SYNTAX      Unsigned32
26     MAX-ACCESS  read-only
27     STATUS      current
28     DESCRIPTION
29         "Next sequence number/transaction identifier to be sent in a
30         Loopback message. This sequence number can be zero because
31         it wraps around.
32     "
33    REFERENCE
34        "12.14.7.1.3:x, 20.28.2"
35    ::= { dotlagCfmMepEntry 19 }
36
37 dotlagCfmMepLbrIn OBJECT-TYPE
38     SYNTAX      Counter32
39     MAX-ACCESS  read-only
40     STATUS      current
41     DESCRIPTION
42         "Total number of valid, in-order Loopback Replies received."
43    REFERENCE
44        "12.14.7.1.3:y, 20.31.1"
45    ::= { dotlagCfmMepEntry 20 }
46
47 dotlagCfmMepLbrInOutOfOrder OBJECT-TYPE
48     SYNTAX      Counter32
49     MAX-ACCESS  read-only
50     STATUS      current
51     DESCRIPTION
52         "The total number of valid, out-of-order Loopback Replies
53         received.
54     "
55    REFERENCE
56        "12.14.7.1.3:z, 20.31.1"
57    ::= { dotlagCfmMepEntry 21 }
58
59 dotlagCfmMepLbrBadMsdu OBJECT-TYPE
60     SYNTAX      Counter32
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64         "The total number of LBRs received whose
65         mac_service_data_unit did not match (except for the OpCode)
66         that of the corresponding LBM (20.2.3).
67     "
68    REFERENCE
69        "12.14.7.1.3:aa, 20.2.3"
70    ::= { dotlagCfmMepEntry 22 }
71
72 dotlagCfmMepLtmNextSeqNumber OBJECT-TYPE

```

```

1  SYNTAX      Unsigned32
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "Next transaction identifier/sequence number to be sent in a
6      Linktrace message. This sequence number can be zero because
7      it wraps around.
8      "
9  REFERENCE
10     "12.14.7.1.3:ab, 20.41.1"
11     ::= { dotlagCfmMepEntry 23 }
12
13 dotlagCfmMepUnexplTrIn OBJECT-TYPE
14     SYNTAX      Counter32
15     MAX-ACCESS  read-only
16     STATUS      current
17     DESCRIPTION
18         "The total number of unexpected LTRs received (20.39.1).
19         "
20     REFERENCE
21         "12.14.7.1.3:ac, 20.44.1"
22     ::= { dotlagCfmMepEntry 24 }
23
24 dotlagCfmMepLbrOut OBJECT-TYPE
25     SYNTAX      Counter32
26     MAX-ACCESS  read-only
27     STATUS      current
28     DESCRIPTION
29         "Total number of Loopback Replies transmitted."
30     REFERENCE
31         "12.14.7.1.3:ad, 20.28.2"
32     ::= { dotlagCfmMepEntry 25 }
33
34 dotlagCfmMepTransmitLbmStatus OBJECT-TYPE
35     SYNTAX      TruthValue
36     MAX-ACCESS  read-create
37     STATUS      current
38     DESCRIPTION
39         "A Boolean flag set to true by the MEP Loopback Initiator State
40         Machine or an MIB manager to indicate
41         that another LBM is being transmitted.
42         Reset to false by the MEP Loopback Initiator State Machine."
43     DEFVAL { false }
44     ::= { dotlagCfmMepEntry 26 }
45
46 dotlagCfmMepTransmitLbmDestMacAddress OBJECT-TYPE
47     SYNTAX      MacAddress
48     MAX-ACCESS  read-create
49     STATUS      current
50     DESCRIPTION
51         "The Target MAC Address Field to be transmitted: A unicast
52         destination MAC address.
53         This address will be used if the value of the column
54         dotlagCfmMepTransmitLbmDestIsMepId is 'false'.
55         "
56     REFERENCE
57         "12.14.7.3.2:b"
58     ::= { dotlagCfmMepEntry 27 }
59
60 dotlagCfmMepTransmitLbmDestMepId OBJECT-TYPE
61     SYNTAX      DotlagCfmMepIdOrZero
62     MAX-ACCESS  read-create
63     STATUS      current
64     DESCRIPTION
65         "The Maintenance association Endpoint Identifier of another
66         MEP in the same Maintenance Association to which the LBM is
67         to be sent.
68         This address will be used if the value of the column
69         dotlagCfmMepTransmitLbmDestIsMepId is 'true'.
70         "
71     REFERENCE
72         "12.14.7.3.2:b"

```

```
1 ::= { dotlagCfmMepEntry 28 }
2
3 dotlagCfmMepTransmitLbmDestIsMepId OBJECT-TYPE
4     SYNTAX      TruthValue
5     MAX-ACCESS  read-create
6     STATUS      current
7     DESCRIPTION
8         "True indicates that MEPID of the target MEP is used for
9         Loopback transmission.
10        False indicates that unicast destination MAC address of the
11        target MEP is used for Loopback transmission.
12        "
13     REFERENCE
14         "12.14.7.3.2:b"
15     ::= { dotlagCfmMepEntry 29 }
16
17 dotlagCfmMepTransmitLbmMessages OBJECT-TYPE
18     SYNTAX      Integer32(1..1024)
19     MAX-ACCESS  read-create
20     STATUS      current
21     DESCRIPTION
22         "The number of Loopback messages to be transmitted."
23     REFERENCE
24         "12.14.7.3.2:c"
25     DEFVAL { 1 }
26     ::= { dotlagCfmMepEntry 30 }
27
28 dotlagCfmMepTransmitLbmDataTlv OBJECT-TYPE
29     SYNTAX      OCTET STRING
30     MAX-ACCESS  read-create
31     STATUS      current
32     DESCRIPTION
33         "An arbitrary amount of data to be included in the Data TLV,
34         if the Data TLV is selected to be sent. The intent is to be able
35         to fill the frame carrying the CFM PDU to its maximum length.
36         This may lead to fragmentation in some cases.
37         "
38     REFERENCE
39         "12.14.7.3.2:d"
40     ::= { dotlagCfmMepEntry 31 }
41
42 dotlagCfmMepTransmitLbmVlanPriority OBJECT-TYPE
43     SYNTAX      Integer32(0..7)
44     MAX-ACCESS  read-create
45     STATUS      current
46     DESCRIPTION
47         "Priority. 3 bit value to be used in the VLAN tag, if present
48         in the transmitted frame.
49
50         The default value is CCM priority.
51         "
52     REFERENCE
53         "12.14.7.3.2:e"
54     ::= { dotlagCfmMepEntry 32 }
55
56 dotlagCfmMepTransmitLbmVlanDropEnable OBJECT-TYPE
57     SYNTAX      TruthValue
58     MAX-ACCESS  read-create
59     STATUS      current
60     DESCRIPTION
61         "Drop Enable bit value to be used in the VLAN tag, if present
62         in the transmitted frame.
63
64         For more information about VLAN Drop Enable, check
65         IEEE Std 802.1ad.
66         "
67     REFERENCE
68         "12.14.7.3.2:e"
69     DEFVAL { false }
70     ::= { dotlagCfmMepEntry 33 }
71
72 dotlagCfmMepTransmitLbmResultOK OBJECT-TYPE
```

```

1  SYNTAX      TruthValue
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "Indicates the result of the operation:
6
7      - true          The Loopback Message(s) will be
8                      (or has been) sent.
9      - false         The Loopback Message(s) will not
10                     be sent.
11
12  REFERENCE
13      "12.14.7.3.3:a"
14  DEFVAL { true }
15  ::= { dotlagCfmMepEntry 34 }
16
17 dotlagCfmMepTransmitLbmSeqNumber OBJECT-TYPE
18     SYNTAX      Unsigned32
19     MAX-ACCESS  read-only
20     STATUS      current
21     DESCRIPTION
22         "The Loopback Transaction Identifier
23         (dotlagCfmMepNextLbmTransId) of the first LBM (to be) sent.
24         The value returned is undefined if
25         dotlagCfmMepTransmitLbmResultOK is false.
26
27     REFERENCE
28         "12.14.7.3.3:a"
29     ::= { dotlagCfmMepEntry 35 }
30
31 dotlagCfmMepTransmitLtmStatus OBJECT-TYPE
32     SYNTAX      TruthValue
33     MAX-ACCESS  read-create
34     STATUS      current
35     DESCRIPTION
36         "A Boolean flag set to true by the Bridge Port to indicate
37         that another LTM may be transmitted.
38         Reset to false by the MEP Linktrace Initiator State Machine."
39     DEFVAL { true }
40     ::= { dotlagCfmMepEntry 36 }
41
42 dotlagCfmMepTransmitLtmFlags OBJECT-TYPE
43     SYNTAX      BITS {
44         useFDBOnly    (0)
45     }
46     MAX-ACCESS  read-create
47     STATUS      current
48     DESCRIPTION
49         "The flags field for LTMs transmitted by the MEP."
50     REFERENCE
51         "12.14.7.4.2:b, 20.42.1"
52     DEFVAL { {useFDBOnly} }
53     ::= { dotlagCfmMepEntry 37 }
54
55 dotlagCfmMepTransmitLtmTargetMacAddress OBJECT-TYPE
56     SYNTAX      MacAddress
57     MAX-ACCESS  read-create
58     STATUS      current
59     DESCRIPTION
60         "The Target MAC Address Field to be transmitted: A unicast
61         destination MAC address.
62         This address will be used if the value of the column
63         dotlagCfmMepTransmitLtmTargetIsMepId is 'false'.
64
65     REFERENCE
66         "12.14.7.4.2:c"
67     ::= { dotlagCfmMepEntry 38 }
68
69 dotlagCfmMepTransmitLtmTargetMepId OBJECT-TYPE
70     SYNTAX      DotlagCfmMepIdOrZero
71     MAX-ACCESS  read-create
72     STATUS      current

```

```

1  DESCRIPTION
2      "An indication of the Target MAC Address Field to be
3      transmitted:
4      The Maintenance association Endpoint Identifier of
5      another MEP in the same Maintenance Association
6      This address will be used if the value of the column
7      dotlagCfmMepTransmitLtmTargetIsMepId is 'true'.
8      "
9  REFERENCE
10     "12.14.7.4.2:c"
11     ::= { dotlagCfmMepEntry 39 }
12
13 dotlagCfmMepTransmitLtmTargetIsMepId OBJECT-TYPE
14     SYNTAX      TruthValue
15     MAX-ACCESS  read-create
16     STATUS      current
17     DESCRIPTION
18         "True indicates that MEPID of the target MEP is used for
19         Linktrace transmission.
20         False indicates that unicast destination MAC address of the
21         target MEP is used for Loopback transmission.
22         "
23     REFERENCE
24         "12.14.7.4.2:c"
25     ::= { dotlagCfmMepEntry 40 }
26
27 dotlagCfmMepTransmitLtmTtl OBJECT-TYPE
28     SYNTAX      Unsigned32 (0..255)
29     MAX-ACCESS  read-create
30     STATUS      current
31     DESCRIPTION
32         "The LTM TTL field. Default value, if not specified, is 64.
33         The TTL field indicates the number of hops remaining to the
34         LTM. Decrementd by 1 by each Linktrace Responder that
35         handles the LTM. The value returned in the LTR is one less
36         than that received in the LTM. If the LTM TTL is 0 or 1, the
37         LTM is not forwarded to the next hop, and if 0, no LTR is
38         generated.
39         "
40     REFERENCE
41         "12.14.7.4.2:d, 21.8.4"
42     DEFVAL { 64 }
43     ::= { dotlagCfmMepEntry 41 }
44
45 dotlagCfmMepTransmitLtmResult OBJECT-TYPE
46     SYNTAX      TruthValue
47     MAX-ACCESS  read-only
48     STATUS      current
49     DESCRIPTION
50         "Indicates the result of the operation:
51
52         - true    The Linktrace Message will be (or has been) sent.
53         - false   The Linktrace Message will not be sent"
54     REFERENCE
55         "12.14.7.4.3:a"
56     DEFVAL { true }
57     ::= { dotlagCfmMepEntry 42 }
58
59 dotlagCfmMepTransmitLtmSeqNumber OBJECT-TYPE
60     SYNTAX      Unsigned32
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64         "The LTM Transaction Identifier
65         (dotlagCfmMepLtmNextSeqNumber) of the LTM sent.
66         The value returned is undefined if
67         dotlagCfmMepTransmitLtmResult is false.
68         "
69     REFERENCE
70         "12.14.7.4.3:a"
71     ::= { dotlagCfmMepEntry 43 }
72

```

```

1 dotlagCfmMepTransmitLtmEgressIdentifier OBJECT-TYPE
2     SYNTAX      OCTET STRING (SIZE(8))
3     MAX-ACCESS  read-create
4     STATUS      current
5     DESCRIPTION
6         "Identifies the MEP Linktrace Initiator that is originating,
7         or the Linktrace Responder that is forwarding, this LTM.
8         The low-order six octets contain a 48-bit IEEE MAC address
9         unique to the system in which the MEP Linktrace Initiator
10        or Linktrace Responder resides. The high-order two octets
11        contain a value sufficient to uniquely identify the MEP
12        Linktrace Initiator or Linktrace Responder within that system.
13
14        For most Bridges, the address of any MAC attached to the
15        Bridge will suffice for the low-order six octets, and 0 for
16        the high-order octets. In some situations, e.g., if multiple
17        virtual Bridges utilizing emulated LANs are implemented in a
18        single physical system, the high-order two octets can be used
19        to differentiate among the transmitting entities.
20
21        The value returned is undefined if
22        dotlagCfmMepTransmitLtmResult is false.
23    "
24    REFERENCE
25        "12.14.7.4.3:b, 21.8.8"
26    ::= { dotlagCfmMepEntry 44 }
27
28 dotlagCfmMepRowStatus OBJECT-TYPE
29     SYNTAX      RowStatus
30     MAX-ACCESS  read-create
31     STATUS      current
32     DESCRIPTION
33         "The status of the row.
34
35         The writable columns in a row cannot be changed if the row
36         is active. All columns MUST have a valid value before a row
37         can be activated.
38    "
39    ::= { dotlagCfmMepEntry 45 }
40
41 dotlagCfmMepPbbTeCanReportPbbTePresence OBJECT-TYPE
42     SYNTAX      TruthValue
43     MAX-ACCESS  read-create
44     STATUS      current
45     DESCRIPTION
46         "A Boolean valued parameter that is set to true if the system
47         has the capability to report the presence of traffic and that
48         the capability is enabled. Traffic presence reporting is an
49         optional PBB-TE feature."
50     REFERENCE
51         "12.14.7.1.3:af, 21.6.1.4"
52     DEFVAL { false }
53     ::= { dotlagCfmMepEntry 46 }
54
55 dotlagCfmMepPbbTeTrafficMismatchDefect OBJECT-TYPE
56     SYNTAX      TruthValue
57     MAX-ACCESS  read-only
58     STATUS      current
59     DESCRIPTION
60         "A Boolean valued parameter that is set to true if the system
61         has detected a traffic field mismatch defect. Mismatch detection
62         is an optional PBB-TE feature."
63     REFERENCE
64         "12.14.7.1.3:ah, 21.6.1.4"
65     ::= { dotlagCfmMepEntry 47 }
66
67 dotlagCfmMepPbbTransmitLbmLtmReverseVid OBJECT-TYPE
68     SYNTAX      IEEE8021VlanIndex
69     MAX-ACCESS  read-create
70     STATUS      current
71     DESCRIPTION
72         "This column specifies the value to use in the Reverse VID value

```

```

1      field of PBB-TE MIP TLVs contained within TransmitLTM pdus."
2  REFERENCE
3      "12.14.7.4.2"
4  ::= { dotlagCfmMepEntry 48 }
5
6  dotlagCfmMepPbbTeMismatchAlarm OBJECT-TYPE
7      SYNTAX      TruthValue
8      MAX-ACCESS   read-create
9      STATUS      current
10     DESCRIPTION
11         "A Boolean valued parameter that is set to true if the system
12         is to allow a mismatch defect to generate a fault alarm."
13     REFERENCE
14         "12.14.7.1.3:ag, 21.6.1.4"
15     DEFVAL { false }
16     ::= { dotlagCfmMepEntry 49 }
17
18 dotlagCfmMepPbbTeLocalMismatchDefect OBJECT-TYPE
19     SYNTAX      TruthValue
20     MAX-ACCESS   read-only
21     STATUS      current
22     DESCRIPTION
23         "A Boolean valued parameter that is set to true if the system
24         has detected a local mismatch defect. Mismatch detection
25         is an optional PBB-TE feature."
26     REFERENCE
27         "12.14.7.1.3:ai, 21.6.1.4"
28     ::= { dotlagCfmMepEntry 50 }
29
30 dotlagCfmMepPbbTeMismatchSinceReset OBJECT-TYPE
31     SYNTAX      TruthValue
32     MAX-ACCESS   read-only
33     STATUS      current
34     DESCRIPTION
35         "A Boolean valued parameter indicating if the mismatch defect
36         has been present since the MEP Mismatch Fault Notification
37         Generator was last in the MFNG_RESET state."
38     REFERENCE
39         "12.14.7.1.3:aj"
40     ::= { dotlagCfmMepEntry 51 }
41
42
43
44 -- *****
45 -- The Linktrace Reply Table
46 -- *****
47
48 dotlagCfmLtrTable OBJECT-TYPE
49     SYNTAX      SEQUENCE OF DotlagCfmLtrEntry
50     MAX-ACCESS   not-accessible
51     STATUS      current
52     DESCRIPTION
53         "This table extends the MEP table and contains a list of
54         Linktrace replies received by a specific MEP in response to
55         a linktrace message.
56
57         SNMP SMI does not allow to state in a MIB that an object in
58         a table is an array. The solution is to take the index (or
59         indices) of the first table and add one or more indices.
60         "
61     REFERENCE
62         "12.14.7.5"
63     ::= { dotlagCfmMep 2 }
64
65 dotlagCfmLtrEntry OBJECT-TYPE
66     SYNTAX      DotlagCfmLtrEntry
67     MAX-ACCESS   not-accessible
68     STATUS      current
69     DESCRIPTION
70         "The Linktrace Reply table entry."
71     INDEX { dotlagCfmMdIndex,
72            dotlagCfmMaIndex,

```



```

1      dotlagCfmMepIdentifier,
2      dotlagCfmLtrSeqNumber,
3      dotlagCfmLtrReceiveOrder
4  }
5  ::= { dotlagCfmLtrTable 1 }
6
7  dotlagCfmLtrEntry ::= SEQUENCE {
8      dotlagCfmLtrSeqNumber      Unsigned32,
9      dotlagCfmLtrReceiveOrder   Unsigned32,
10     dotlagCfmLtrTtl             Unsigned32,
11     dotlagCfmLtrForwarded       TruthValue,
12     dotlagCfmLtrTerminalMep     TruthValue,
13     dotlagCfmLtrLastEgressIdentifier OCTET STRING,
14     dotlagCfmLtrNextEgressIdentifier OCTET STRING,
15     dotlagCfmLtrRelay           DotlagCfmRelayActionFieldValue,
16     dotlagCfmLtrChassisIdSubtype LldpChassisIdSubtype,
17     dotlagCfmLtrChassisId       LldpChassisId,
18     dotlagCfmLtrManAddressDomain TDomain,
19     dotlagCfmLtrManAddress      TAddress,
20     dotlagCfmLtrIngress         DotlagCfmIngressActionFieldValue,
21     dotlagCfmLtrIngressMac      MacAddress,
22     dotlagCfmLtrIngressPortIdSubtype LldpPortIdSubtype,
23     dotlagCfmLtrIngressPortId   LldpPortId,
24     dotlagCfmLtrEgress         DotlagCfmEgressActionFieldValue,
25     dotlagCfmLtrEgressMac      MacAddress,
26     dotlagCfmLtrEgressPortIdSubtype LldpPortIdSubtype,
27     dotlagCfmLtrEgressPortId   LldpPortId,
28     dotlagCfmLtrOrganizationSpecificTlv OCTET STRING
29  }
30
31  dotlagCfmLtrSeqNumber OBJECT-TYPE
32      SYNTAX      Unsigned32 (0..4294967295)
33      MAX-ACCESS  not-accessible
34      STATUS      current
35      DESCRIPTION
36          "Transaction identifier/Sequence number returned by a previous
37          transmit linktrace message command, indicating which LTM's
38          response is going to be returned.
39          "
40      REFERENCE
41          "12.14.7.5.2:b"
42      ::= { dotlagCfmLtrEntry 1}
43
44  dotlagCfmLtrReceiveOrder OBJECT-TYPE
45      SYNTAX      Unsigned32(1..4294967295)
46      MAX-ACCESS  not-accessible
47      STATUS      current
48      DESCRIPTION
49          "An index to distinguish among multiple LTRs with the same LTR
50          Transaction Identifier field value. dotlagCfmLtrReceiveOrder
51          are assigned sequentially from 1, in the order that the
52          Linktrace Initiator received the LTRs.
53          "
54      REFERENCE
55          "12.14.7.5.2:c"
56      ::= { dotlagCfmLtrEntry 2 }
57
58  dotlagCfmLtrTtl OBJECT-TYPE
59      SYNTAX      Unsigned32 (0..255)
60      MAX-ACCESS  read-only
61      STATUS      current
62      DESCRIPTION
63          "TTL field value for a returned LTR."
64      REFERENCE
65          "12.14.7.5, 20.41.2.2"
66      ::= { dotlagCfmLtrEntry 3 }
67
68  dotlagCfmLtrForwarded OBJECT-TYPE
69      SYNTAX      TruthValue
70      MAX-ACCESS  read-only
71      STATUS      current
72      DESCRIPTION

```

```

1      "Indicates if a LTM was forwarded by the responding MP, as
2      returned in the 'FwdYes' flag of the flags field.
3      "
4      REFERENCE
5      "12.14.7.5.3:c, 20.41.2.1"
6      ::= { dotlagCfmLtrEntry 4 }
7
8      dotlagCfmLtrTerminalMep OBJECT-TYPE
9          SYNTAX      TruthValue
10         MAX-ACCESS  read-only
11         STATUS      current
12         DESCRIPTION
13             "A Boolean value stating whether the forwarded LTM reached a
14             MEP enclosing its MA, as returned in the Terminal MEP flag of
15             the Flags field.
16             "
17         REFERENCE
18             "12.14.7.5.3:d, 20.41.2.1"
19         ::= { dotlagCfmLtrEntry 5 }
20
21     dotlagCfmLtrLastEgressIdentifier OBJECT-TYPE
22         SYNTAX      OCTET STRING (SIZE(8))
23         MAX-ACCESS  read-only
24         STATUS      current
25         DESCRIPTION
26             "An octet field holding the Last Egress Identifier returned
27             in the LTR Egress Identifier TLV of the LTR.
28             The Last Egress Identifier identifies the MEP Linktrace
29             Initiator that originated, or the Linktrace Responder that
30             forwarded, the LTM to which this LTR is the response. This
31             is the same value as the Egress Identifier TLV of that LTM.
32             "
33         REFERENCE
34             "12.14.7.5.3:e, 20.41.2.3"
35         ::= { dotlagCfmLtrEntry 6 }
36
37     dotlagCfmLtrNextEgressIdentifier OBJECT-TYPE
38         SYNTAX      OCTET STRING (SIZE(8))
39         MAX-ACCESS  read-only
40         STATUS      current
41         DESCRIPTION
42             "An octet field holding the Next Egress Identifier returned
43             in the LTR Egress Identifier TLV of the LTR. The Next Egress
44             Identifier Identifies the Linktrace Responder that
45             transmitted this LTR, and can forward the LTM to the next
46             hop. This is the same value as the Egress Identifier TLV of
47             the forwarded LTM, if any. If the FwdYes bit of the Flags
48             field is false, the contents of this field are undefined,
49             i.e., any value can be transmitted, and the field is ignored
50             by the receiver.
51             "
52         REFERENCE
53             "12.14.7.5.3:f, 20.41.2.4"
54         ::= { dotlagCfmLtrEntry 7 }
55
56     dotlagCfmLtrRelay OBJECT-TYPE
57         SYNTAX      DotlagCfmRelayActionFieldValue
58         MAX-ACCESS  read-only
59         STATUS      current
60         DESCRIPTION
61             "Value returned in the Relay Action field."
62         REFERENCE
63             "12.14.7.5.3:g, 20.41.2.5"
64         ::= { dotlagCfmLtrEntry 8 }
65
66     dotlagCfmLtrChassisIdSubtype OBJECT-TYPE
67         SYNTAX      LldpChassisIdSubtype
68         MAX-ACCESS  read-only
69         STATUS      current
70         DESCRIPTION
71             "This object specifies the format of the Chassis ID returned
72             in the Sender ID TLV of the LTR, if any. This value is

```

```

1      meaningless if the dotlagCfmLtrChassisId has a length of 0."
2  REFERENCE
3      "12.14.7.5.3:h, 21.5.3.2"
4      ::= { dotlagCfmLtrEntry 9 }
5
6  dotlagCfmLtrChassisId OBJECT-TYPE
7      SYNTAX      LldpChassisId
8      MAX-ACCESS  read-only
9      STATUS      current
10     DESCRIPTION
11         "The Chassis ID returned in the Sender ID TLV of the LTR, if
12         any. The format of this object is determined by the
13         value of the dotlagCfmLtrChassisIdSubtype object.
14         "
15     REFERENCE
16         "12.14.7.5.3:i, 21.5.3.2"
17         ::= { dotlagCfmLtrEntry 10 }
18
19  dotlagCfmLtrManAddressDomain OBJECT-TYPE
20     SYNTAX      TDomain
21     MAX-ACCESS  read-only
22     STATUS      current
23     DESCRIPTION
24         "The TDomain that identifies the type and format of
25         the related dotlagCfmMepDbManAddress object, used to access
26         the SNMP agent of the system transmitting the LTR. Received
27         in the LTR Sender ID TLV from that system.
28
29         Typical values will be one of (not all inclusive) list:
30
31
32         snmpUDPDomain          (from SNMPv2-TM, RFC3417)
33         snmpIeee802Domain      (from SNMP-IEEE802-TM-MIB, RFC4789)
34
35         The value 'zeroDotZero' (from RFC2578) indicates 'no management
36         address was present in the LTR', in which case the related
37         object dotlagCfmMepDbManAddress MUST have a zero-length OCTET
38         STRING as a value.
39         "
40     REFERENCE
41         "12.14.7.5.3:j, 21.5.3.5, 21.9.6"
42         ::= { dotlagCfmLtrEntry 11 }
43
44  dotlagCfmLtrManAddress OBJECT-TYPE
45     SYNTAX      TAddress
46     MAX-ACCESS  read-only
47     STATUS      current
48     DESCRIPTION
49         "The TAddress that can be used to access the SNMP
50         agent of the system transmitting the CCM, received in the CCM
51         Sender ID TLV from that system.
52
53         If the related object dotlagCfmLtrManAddressDomain contains
54         the value 'zeroDotZero', this object dotlagCfmLtrManAddress
55         MUST have a zero-length OCTET STRING as a value.
56         "
57     REFERENCE
58         "12.14.7.5.3:j, 21.5.3.7, 21.9.6"
59         ::= { dotlagCfmLtrEntry 12 }
60
61  dotlagCfmLtrIngress OBJECT-TYPE
62     SYNTAX      DotlagCfmIngressActionFieldValue
63     MAX-ACCESS  read-only
64     STATUS      current
65     DESCRIPTION
66         "The value returned in the Ingress Action Field of the LTM.
67         The value ingNoTlv(0) indicates that no Reply Ingress TLV was
68         returned in the LTM."
69     REFERENCE
70         "12.14.7.5.3:k, 20.41.2.6"
71         ::= { dotlagCfmLtrEntry 13 }
72

```

```

1 dotlagCfmLtrIngressMac OBJECT-TYPE
2   SYNTAX      MacAddress
3   MAX-ACCESS  read-only
4   STATUS      current
5   DESCRIPTION
6     "MAC address returned in the ingress MAC address field.
7     If the dotlagCfmLtrIngress object contains the value
8     ingNoTlv(0), then the contents of this object are meaningless."
9   REFERENCE
10    "12.14.7.5.3:1, 20.41.2.7"
11    ::= { dotlagCfmLtrEntry 14 }
12
13 dotlagCfmLtrIngressPortIdSubtype OBJECT-TYPE
14   SYNTAX      LldpPortIdSubtype
15   MAX-ACCESS  read-only
16   STATUS      current
17   DESCRIPTION
18     "Format of the Ingress Port ID.
19     If the dotlagCfmLtrIngress object contains the value
20     ingNoTlv(0), then the contents of this object are meaningless."
21   REFERENCE
22    "12.14.7.5.3:m, 20.41.2.8"
23    ::= { dotlagCfmLtrEntry 15 }
24
25 dotlagCfmLtrIngressPortId OBJECT-TYPE
26   SYNTAX      LldpPortId
27   MAX-ACCESS  read-only
28   STATUS      current
29   DESCRIPTION
30     "Ingress Port ID. The format of this object is determined by
31     the value of the dotlagCfmLtrIngressPortIdSubtype object.
32     If the dotlagCfmLtrIngress object contains the value
33     ingNoTlv(0), then the contents of this object are meaningless."
34   REFERENCE
35    "12.14.7.5.3:n, 20.41.2.9"
36    ::= { dotlagCfmLtrEntry 16 }
37
38 dotlagCfmLtrEgress OBJECT-TYPE
39   SYNTAX      DotlagCfmEgressActionFieldValue
40   MAX-ACCESS  read-only
41   STATUS      current
42   DESCRIPTION
43     "The value returned in the Egress Action Field of the LTM.
44     The value egrNoTlv(0) indicates that no Reply Egress TLV was
45     returned in the LTM."
46   REFERENCE
47    "12.14.7.5.3:o, 20.41.2.10"
48    ::= { dotlagCfmLtrEntry 17 }
49
50 dotlagCfmLtrEgressMac OBJECT-TYPE
51   SYNTAX      MacAddress
52   MAX-ACCESS  read-only
53   STATUS      current
54   DESCRIPTION
55     "MAC address returned in the egress MAC address field.
56     If the dotlagCfmLtrEgress object contains the value
57     egrNoTlv(0), then the contents of this object are meaningless."
58   REFERENCE
59    "12.14.7.5.3:p, 20.41.2.11"
60    ::= { dotlagCfmLtrEntry 18 }
61
62 dotlagCfmLtrEgressPortIdSubtype OBJECT-TYPE
63   SYNTAX      LldpPortIdSubtype
64   MAX-ACCESS  read-only
65   STATUS      current
66   DESCRIPTION
67     "Format of the egress Port ID.
68     If the dotlagCfmLtrEgress object contains the value
69     egrNoTlv(0), then the contents of this object are meaningless."
70   REFERENCE
71    "12.14.7.5.3:q, 20.41.2.12"
72    ::= { dotlagCfmLtrEntry 19 }

```

```

1
2 dotlagCfmLtrEgressPortId OBJECT-TYPE
3     SYNTAX      LldpPortId
4     MAX-ACCESS  read-only
5     STATUS      current
6     DESCRIPTION
7         "Egress Port ID. The format of this object is determined by
8         the value of the dotlagCfmLtrEgressPortIdSubtype object.
9         If the dotlagCfmLtrEgress object contains the value
10        egrNoTlv(0), then the contents of this object are meaningless."
11    REFERENCE
12        "12.14.7.5.3:r, 20.41.2.13"
13    ::= { dotlagCfmLtrEntry 20 }
14
15 dotlagCfmLtrOrganizationSpecificTlv OBJECT-TYPE
16     SYNTAX      OCTET STRING (SIZE(0|4..1500))
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20         "All Organization specific TLVs returned in the LTR, if
21         any. Includes all octets including and following the TLV
22         Length field of each TLV, concatenated together."
23    REFERENCE
24        "12.14.7.5.3:s, 21.5.2"
25    ::= { dotlagCfmLtrEntry 21 }
26
27 -- *****
28 -- The MEP Database Table
29 -- *****
30
31 dotlagCfmMepDbTable OBJECT-TYPE
32     SYNTAX      SEQUENCE OF DotlagCfmMepDbEntry
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "The MEP Database. A database, maintained by every MEP, that
37         maintains received information about other MEPs in the
38         Maintenance Domain.
39
40         The SMI does not allow to state in a MIB that an object in
41         a table is an array. The solution is to take the index (or
42         indices) of the first table and add one or more indices.
43         "
44     REFERENCE
45        "19.2.15"
46    ::= { dotlagCfmMep 3 }
47
48 dotlagCfmMepDbEntry OBJECT-TYPE
49     SYNTAX      DotlagCfmMepDbEntry
50     MAX-ACCESS  not-accessible
51     STATUS      current
52     DESCRIPTION
53         "The MEP Database table entry."
54     INDEX { dotlagCfmMdIndex,
55             dotlagCfmMaIndex,
56             dotlagCfmMepIdentifier,
57             dotlagCfmMepDbRMepIdentifier
58           }
59    ::= { dotlagCfmMepDbTable 1 }
60
61 DotlagCfmMepDbEntry ::= SEQUENCE {
62     dotlagCfmMepDbRMepIdentifier      DotlagCfmMepId,
63     dotlagCfmMepDbRMepState           DotlagCfmRemoteMepState,
64     dotlagCfmMepDbRMepFailedOkTime    TimeStamp,
65     dotlagCfmMepDbMacAddress           MacAddress,
66     dotlagCfmMepDbRdi                 TruthValue,
67     dotlagCfmMepDbPortStatusTlv       DotlagCfmPortMepDbPortStatus,
68     dotlagCfmMepDbInterfaceStatusTlv  DotlagCfmInterfaceStatus,
69     dotlagCfmMepDbChassisIdSubtype    LldpChassisIdSubtype,
70     dotlagCfmMepDbChassisId           LldpChassisId,
71     dotlagCfmMepDbManAddressDomain     TDomain,
72     dotlagCfmMepDbManAddress           TAddress,

```

```

1      dotlagCfmMepDbRMepIsActive          TruthValue
2      }
3
4 dotlagCfmMepDbRMepIdentifier OBJECT-TYPE
5     SYNTAX          DotlagCfmMepId
6     MAX-ACCESS      not-accessible
7     STATUS          current
8     DESCRIPTION
9         "Maintenance association Endpoint Identifier of a remote MEP
10        whose information from the MEP Database is to be returned.
11        "
12     REFERENCE
13         "12.14.7.6.2:b"
14     ::= { dotlagCfmMepDbEntry 1 }
15
16 dotlagCfmMepDbRMepState OBJECT-TYPE
17     SYNTAX          DotlagCfmRemoteMepState
18     MAX-ACCESS      read-only
19     STATUS          current
20     DESCRIPTION
21         "The operational state of the remote MEP IFF State machines."
22     REFERENCE
23         "12.14.7.6.3:b, 20.22"
24     ::= { dotlagCfmMepDbEntry 2 }
25
26 dotlagCfmMepDbRMepFailedOkTime OBJECT-TYPE
27     SYNTAX          TimeStamp
28     MAX-ACCESS      read-only
29     STATUS          current
30     DESCRIPTION
31         "The time (SysUpTime) at which the IFF Remote MEP state machine
32         last entered either the RMEP_FAILED or RMEP_OK state.
33         "
34     REFERENCE
35         "12.14.7.6.3:c"
36     ::= { dotlagCfmMepDbEntry 3 }
37
38 dotlagCfmMepDbMacAddress OBJECT-TYPE
39     SYNTAX          MacAddress
40     MAX-ACCESS      read-only
41     STATUS          current
42     DESCRIPTION
43         "The MAC address of the remote MEP."
44     REFERENCE
45         "12.14.7.6.3:d, 20.19.7"
46     ::= { dotlagCfmMepDbEntry 4 }
47
48 dotlagCfmMepDbRdi OBJECT-TYPE
49     SYNTAX          TruthValue
50     MAX-ACCESS      read-only
51     STATUS          current
52     DESCRIPTION
53         "State of the RDI bit in the last received CCM (true for
54         RDI=1), or false if none has been received.
55         "
56     REFERENCE
57         "12.14.7.6.3:e, 20.19.2"
58     ::= { dotlagCfmMepDbEntry 5 }
59
60 dotlagCfmMepDbPortStatusTlv OBJECT-TYPE
61     SYNTAX          DotlagCfmPortStatus
62     MAX-ACCESS      read-only
63     STATUS          current
64     DESCRIPTION
65         "An enumerated value of the Port status TLV received in the
66         last CCM from the remote MEP or the default value
67         psNoPortStateTLV indicating either no CCM has been received,
68         or that nor port status TLV was received in the last CCM.
69         "
70     REFERENCE
71         "12.14.7.6.3:f, 20.19.3"
72     DEFVAL { psNoPortStateTLV }

```

```

1      ::= { dotlagCfmMepDbEntry 6}
2
3 dotlagCfmMepDbInterfaceStatusTlv OBJECT-TYPE
4     SYNTAX      DotlagCfmInterfaceStatus
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "An enumerated value of the Interface status TLV received
9         in the last CCM from the remote MEP or the default value
10        isNoInterfaceStatus TLV indicating either no CCM has been
11        received, or that no interface status TLV was received in
12        the last CCM.
13        "
14     REFERENCE
15         "12.14.7.6.3:g, 20.19.4"
16     DEFVAL { isNoInterfaceStatusTLV }
17     ::= { dotlagCfmMepDbEntry 7}
18
19 dotlagCfmMepDbChassisIdSubtype OBJECT-TYPE
20     SYNTAX      LldpChassisIdSubtype
21     MAX-ACCESS  read-only
22     STATUS      current
23     DESCRIPTION
24         "This object specifies the format of the Chassis ID received
25         in the last CCM."
26     REFERENCE
27         "12.14.7.6.3:h, 21.5.3.2"
28     ::= { dotlagCfmMepDbEntry 8 }
29
30 dotlagCfmMepDbChassisId OBJECT-TYPE
31     SYNTAX      LldpChassisId
32     MAX-ACCESS  read-only
33     STATUS      current
34     DESCRIPTION
35         "The Chassis ID. The format of this object is determined by the
36         value of the dotlagCfmLtrChassisIdSubtype object.
37         "
38     REFERENCE
39         "12.14.7.6.3:h, 21.5.3.3"
40     ::= { dotlagCfmMepDbEntry 9 }
41
42 dotlagCfmMepDbManAddressDomain OBJECT-TYPE
43     SYNTAX      TDomain
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "The TDomain that identifies the type and format of
48         the related dotlagCfmMepDbManAddress object, used to access
49         the SNMP agent of the system transmitting the CCM. Received
50         in the CCM Sender ID TLV from that system.
51
52         Typical values will be one of (not all inclusive) list:
53
54         snmpUDPDDomain          (from SNMPv2-TM, RFC3417)
55         snmpIeee802Domain       (from SNMP-IEEE802-TM-MIB, RFC4789)
56
57         The value 'zeroDotZero' (from RFC2578) indicates 'no management
58         address was present in the LTR', in which case the related
59         object dotlagCfmMepDbManAddress MUST have a zero-length OCTET
60         STRING as a value.
61         "
62     REFERENCE
63         "12.14.7.6.3:h, 21.5.3.5, 21.6.7"
64     ::= { dotlagCfmMepDbEntry 10 }
65
66
67 dotlagCfmMepDbManAddress OBJECT-TYPE
68     SYNTAX      TAddress
69     MAX-ACCESS  read-only
70     STATUS      current
71     DESCRIPTION
72         "The TAddress that can be used to access the SNMP

```

```

1      agent of the system transmitting the CCM, received in the CCM
2      Sender ID TLV from that system.
3
4      If the related object dotlagCfmMepDbManAddressDomain contains
5      the value 'zeroDotZero', this object dotlagCfmMepDbManAddress
6      MUST have a zero-length OCTET STRING as a value.
7      "
8      REFERENCE
9      "12.14.7.6.3:h, 21.5.3.7, 21.6.7"
10     ::= { dotlagCfmMepDbEntry 11 }
11
12 dotlagCfmMepDbRMepIsActive OBJECT-TYPE
13     SYNTAX      TruthValue
14     MAX-ACCESS  read-write
15     STATUS      current
16     DESCRIPTION
17         "A Boolean value stating if the remote MEP is active."
18     REFERENCE
19         "12.14.7.1.3:ae"
20     ::= { dotlagCfmMepDbEntry 12 }
21
22 -- *****
23 -- NOTIFICATIONS (TRAPS)
24 -- These notifications will be sent to the management entity
25 -- whenever a MEP loses/restores contact with one or more other MEPs.
26 -- *****
27
28 dotlagCfmFaultAlarm NOTIFICATION-TYPE
29     OBJECTS      { dotlagCfmMepHighestPrDefect
30                     }
31     STATUS      current
32     DESCRIPTION
33         "A MEP has a persistent defect condition. A notification
34         (fault alarm) is sent to the management entity with the OID
35         of the MEP that has detected the fault.
36
37         Whenever a MEP has a persistent defect,
38         it may or may not generate a Fault Alarm to warn the system
39         administrator of the problem, as controlled by the MEP
40         Fault Notification Generator State Machine and associated
41         Managed Objects. Only the highest-priority defect, as shown
42         in Table 20-1, is reported in the Fault Alarm.
43
44         If a defect with a higher priority is raised after a Fault
45         Alarm has been issued, another Fault Alarm is issued.
46
47         The management entity receiving the notification can identify
48         the system from the network source address of the
49         notification, and can identify the MEP reporting the defect
50         by the indices in the OID of the dotlagCfmMepHighestPrDefect
51         variable in the notification:
52
53         dotlagCfmMdIndex - Also the index of the MEP's
54                           Maintenance Domain table entry
55                           (dotlagCfmMdTable).
56         dotlagCfmMaIndex - Also an index (with the MD table index)
57                           of the MEP's Maintenance Association
58                           network table entry
59                           (dotlagCfmMaNetTable), and (with the MD
60                           table index and component ID) of the
61                           MEP's MA component table entry
62                           (dotlagCfmMaCompTable).
63         dotlagCfmMepIdentifier - MEP Identifier and final index
64                                into the MEP table (dotlagCfmMepTable).
65     "
66     REFERENCE
67         "12.14.7.7"
68     ::= { dotlagNotifications 1 }
69
70 -- *****
71 -- CFM MIB Module - Conformance Information
72 -- *****

```



```

1
2 dotlagCfmCompliances OBJECT IDENTIFIER ::= { dotlagCfmConformance 1 }
3 dotlagCfmGroups      OBJECT IDENTIFIER ::= { dotlagCfmConformance 2 }
4
5 -- *****
6 -- Units of conformance
7 -- *****
8 dotlagCfmStackGroup OBJECT-GROUP
9     OBJECTS {
10         dotlagCfmStackMdIndex,
11         dotlagCfmStackMaIndex,
12         dotlagCfmStackMepId,
13         dotlagCfmStackMacAddress
14     }
15     STATUS      deprecated
16     DESCRIPTION
17         "Objects for the Stack group."
18     ::= { dotlagCfmGroups 1 }
19
20 dotlagCfmDefaultMdGroup OBJECT-GROUP
21     OBJECTS {
22         dotlagCfmDefaultMdDefLevel,
23         dotlagCfmDefaultMdDefMhfCreation,
24         dotlagCfmDefaultMdDefIdPermission,
25         dotlagCfmDefaultMdStatus,
26         dotlagCfmDefaultMdLevel,
27         dotlagCfmDefaultMdMhfCreation,
28         dotlagCfmDefaultMdIdPermission
29     }
30     STATUS      deprecated
31     DESCRIPTION
32         "Objects for the Default MD Level group."
33     ::= { dotlagCfmGroups 2 }
34
35 dotlagCfmVlanIdGroup OBJECT-GROUP
36     OBJECTS {
37         dotlagCfmVlanPrimaryVid,
38         dotlagCfmVlanRowStatus
39     }
40     STATUS      deprecated
41     DESCRIPTION
42         "Objects for the VLAN ID group."
43     ::= { dotlagCfmGroups 3 }
44
45 dotlagCfmConfigErrorListGroup OBJECT-GROUP
46     OBJECTS {
47         dotlagCfmConfigErrorListErrorType
48     }
49     STATUS deprecated
50     DESCRIPTION
51         "Objects for the CFM Configuration Error List Group."
52     ::= { dotlagCfmGroups 4 }
53
54 dotlagCfmMdGroup OBJECT-GROUP
55     OBJECTS {
56         dotlagCfmMdTableNextIndex,
57         dotlagCfmMdName,
58         dotlagCfmMdFormat,
59         dotlagCfmMdMdLevel,
60         dotlagCfmMdMhfCreation,
61         dotlagCfmMdMhfIdPermission,
62         dotlagCfmMdMaNextIndex,
63         dotlagCfmMdRowStatus
64     }
65     STATUS      current
66     DESCRIPTION
67         "Objects for the Maintenance Domain Group."
68     ::= { dotlagCfmGroups 5 }
69
70 dotlagCfmMaGroup OBJECT-GROUP
71     OBJECTS {
72         dotlagCfmMaNetFormat,

```

```

1      dotlagCfmMaNetName,
2      dotlagCfmMaNetCcmInterval,
3      dotlagCfmMaNetRowStatus,
4      dotlagCfmMaCompPrimaryVlanId,
5      dotlagCfmMaCompMhfCreation,
6      dotlagCfmMaCompIdPermission,
7      dotlagCfmMaCompRowStatus,
8      dotlagCfmMaCompNumberOfVids,
9      dotlagCfmMaMepListRowStatus
10     }
11     STATUS      deprecated
12     DESCRIPTION
13         "Objects for the MA group."
14     ::= { dotlagCfmGroups 6 }
15
16 dotlagCfmMepGroup OBJECT-GROUP
17     OBJECTS {
18         dotlagCfmMepIfIndex,
19         dotlagCfmMepDirection,
20         dotlagCfmMepPrimaryVid,
21         dotlagCfmMepActive,
22         dotlagCfmMepFngState,
23         dotlagCfmMepCciEnabled,
24         dotlagCfmMepCcmLtmPriority,
25         dotlagCfmMepMacAddress,
26         dotlagCfmMepLowPrDef,
27         dotlagCfmMepFngAlarmTime,
28         dotlagCfmMepFngResetTime,
29         dotlagCfmMepHighestPrDefect,
30         dotlagCfmMepDefects,
31         dotlagCfmMepErrorCcmLastFailure,
32         dotlagCfmMepXconCcmLastFailure,
33         dotlagCfmMepCcmSequenceErrors,
34         dotlagCfmMepCciSentCcms,
35         dotlagCfmMepNextLbmTransId,
36         dotlagCfmMepLbrIn,
37         dotlagCfmMepLbrInOutOfOrder,
38         dotlagCfmMepLbrBadMsdu,
39         dotlagCfmMepLtmNextSeqNumber,
40         dotlagCfmMepUnexpLtrIn,
41         dotlagCfmMepLbrOut,
42         dotlagCfmMepTransmitLbmStatus,
43         dotlagCfmMepTransmitLbmDestMacAddress,
44         dotlagCfmMepTransmitLbmDestMepId,
45         dotlagCfmMepTransmitLbmDestIsMepId,
46         dotlagCfmMepTransmitLbmMessages,
47         dotlagCfmMepTransmitLbmDataTlv,
48         dotlagCfmMepTransmitLbmVlanPriority,
49         dotlagCfmMepTransmitLbmVlanDropEnable,
50         dotlagCfmMepTransmitLbmResultOK,
51         dotlagCfmMepTransmitLbmSeqNumber,
52         dotlagCfmMepTransmitLtmStatus,
53         dotlagCfmMepTransmitLtmFlags,
54         dotlagCfmMepTransmitLtmTargetMacAddress,
55         dotlagCfmMepTransmitLtmTargetMepId,
56         dotlagCfmMepTransmitLtmTargetIsMepId,
57         dotlagCfmMepTransmitLtmTtl,
58         dotlagCfmMepTransmitLtmResult,
59         dotlagCfmMepTransmitLtmSeqNumber,
60         dotlagCfmMepTransmitLtmEgressIdentifier,
61         dotlagCfmMepRowStatus,
62         dotlagCfmLtrForwarded,
63         dotlagCfmLtrRelay,
64         dotlagCfmLtrChassisIdSubtype,
65         dotlagCfmLtrChassisId,
66         dotlagCfmLtrManAddress,
67         dotlagCfmLtrManAddressDomain,
68         dotlagCfmLtrIngress,
69         dotlagCfmLtrIngressMac,
70         dotlagCfmLtrIngressPortIdSubtype,
71         dotlagCfmLtrIngressPortId,
72         dotlagCfmLtrEgress,

```

```

1      dotlagCfmLtrEgressMac,
2      dotlagCfmLtrEgressPortIdSubtype,
3      dotlagCfmLtrEgressPortId,
4      dotlagCfmLtrTerminalMep,
5      dotlagCfmLtrLastEgressIdentifier,
6      dotlagCfmLtrNextEgressIdentifier,
7      dotlagCfmLtrTtl,
8      dotlagCfmLtrOrganizationSpecificTlv
9  }
10     STATUS      current
11     DESCRIPTION
12         "Objects for the MEP group."
13     ::= { dotlagCfmGroups 7 }
14
15 dotlagCfmMepDbGroup OBJECT-GROUP
16     OBJECTS {
17         dotlagCfmMepDbRMepState,
18         dotlagCfmMepDbRMepFailedOkTime,
19         dotlagCfmMepDbMacAddress,
20         dotlagCfmMepDbRdi,
21         dotlagCfmMepDbPortStatusTlv,
22         dotlagCfmMepDbInterfaceStatusTlv,
23         dotlagCfmMepDbChassisIdSubtype,
24         dotlagCfmMepDbChassisId,
25         dotlagCfmMepDbManAddressDomain,
26         dotlagCfmMepDbManAddress
27     }
28     STATUS      current
29     DESCRIPTION
30         "Objects for the MEP group."
31     ::= { dotlagCfmGroups 8 }
32
33 dotlagCfmNotificationsGroup NOTIFICATION-GROUP
34     NOTIFICATIONS {
35         dotlagCfmFaultAlarm
36     }
37     STATUS      current
38     DESCRIPTION
39         "Objects for the Notifications group."
40     ::= { dotlagCfmGroups 9 }
41
42
43 ieee8021CfmMaNetGroup OBJECT-GROUP
44     OBJECTS {
45         dotlagCfmMaNetFormat,
46         dotlagCfmMaNetName,
47         dotlagCfmMaNetCcmInterval,
48         dotlagCfmMaNetRowStatus,
49         dotlagCfmMaMepListRowStatus
50     }
51     STATUS      current
52     DESCRIPTION
53         "Objects for the MA Net group."
54     ::= { dotlagCfmGroups 10 }
55
56
57 ieee8021CfmDefaultMdDefGroup OBJECT-GROUP
58     OBJECTS {
59         dotlagCfmDefaultMdDefLevel,
60         dotlagCfmDefaultMdDefMhfCreation,
61         dotlagCfmDefaultMdDefIdPermission
62     }
63     STATUS      current
64     DESCRIPTION
65         "Objects for the Default MD default Level group."
66     ::= { dotlagCfmGroups 11 }
67
68 ieee8021CfmPbbTeExtensionGroup OBJECT-GROUP
69     OBJECTS {
70         dotlagCfmMepDbRMepIsActive,
71         dotlagCfmMepPbbTransmitLbmLtmReverseVid
72     }

```

```

1  STATUS      current
2  DESCRIPTION
3      "Objects needed for systems that support PBB-TE CFM functionality."
4  ::= { dotlagCfmGroups 12 }
5
6  ieee8021CfmPbbTeTrafficBitGroup OBJECT-GROUP
7      OBJECTS {
8          dotlagCfmMepDbManAddress,
9          dotlagCfmMepPbbTeCanReportPbbTePresence,
10         dotlagCfmMepPbbTeMismatchAlarm,
11         dotlagCfmMepPbbTeTrafficMismatchDefect,
12         dotlagCfmMepPbbTeLocalMismatchDefect,
13         dotlagCfmMepPbbTeMismatchSinceReset
14     }
15     STATUS      current
16     DESCRIPTION
17         "Objects needed for PBB-TE supporting systems that support the
18         optional traffic bit."
19     ::= { dotlagCfmGroups 13 }
20
21 -- *****
22 -- MIB Module Compliance statements
23 -- *****
24
25 dotlagCfmCompliance MODULE-COMPLIANCE
26     STATUS      deprecated
27     DESCRIPTION
28         "The compliance statement for support of the CFM MIB module."
29     MODULE
30         MANDATORY-GROUPS {
31             dotlagCfmStackGroup,
32             dotlagCfmDefaultMdGroup,
33             dotlagCfmConfigErrorListGroup,
34             dotlagCfmMdGroup,
35             dotlagCfmMaGroup,
36             dotlagCfmMepGroup,
37             dotlagCfmMepDbGroup,
38             dotlagCfmNotificationsGroup
39         }
40
41     GROUP dotlagCfmVlanIdGroup
42     DESCRIPTION "The VLAN ID group is optional."
43
44     OBJECT dotlagCfmMepLbrBadMsdu
45     MIN-ACCESS not-accessible
46     DESCRIPTION "The dotlagCfmMepLbrBadMsdu variable is optional. It
47         MUST not be present if the system cannot compare a
48         received LBR to the corresponding LBM."
49
50     OBJECT dotlagCfmMdRowStatus
51     SYNTAX      RowStatus { active(1), notInService(2) }
52     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
53         destroy(6) }
54     DESCRIPTION "Support for createAndWait is not required."
55
56     OBJECT dotlagCfmMaNetRowStatus
57     SYNTAX      RowStatus { active(1), notInService(2) }
58     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
59         destroy(6) }
60     DESCRIPTION "Support for createAndWait is not required."
61
62     OBJECT dotlagCfmMaCompRowStatus
63     SYNTAX      RowStatus { active(1), notInService(2) }
64     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
65         destroy(6) }
66     DESCRIPTION "Support for createAndWait is not required."
67
68     OBJECT dotlagCfmVlanRowStatus
69     SYNTAX      RowStatus { active(1), notInService(2) }
70     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
71         destroy(6) }
72     DESCRIPTION "Support for createAndWait is not required."

```

```
1
2 OBJECT dotlagCfmMaMepListRowStatus
3   SYNTAX      RowStatus { active(1), notInService(2) }
4   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
5                        destroy(6) }
6   DESCRIPTION "Support for createAndWait is not required."
7
8 OBJECT dotlagCfmMepRowStatus
9   SYNTAX      RowStatus { active(1), notInService(2) }
10  WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
11                        destroy(6) }
12  DESCRIPTION "Support for createAndWait is not required."
13
14 ::= { dotlagCfmCompliances 1 }
15 END
16
```

1 17.7.7.2 Definitions for the IEEE8021-CFM-V2-MIB module

```
2 IEEE8021-CFM-V2-MIB DEFINITIONS ::= BEGIN
3
4 -- *****
5 -- IEEE 802.1Q(TM) CFM MIB - V2
6 -- *****
7
8 IMPORTS
9     MODULE-IDENTITY,
10     OBJECT-TYPE,
11     Unsigned32          FROM SNMPv2-SMI    -- [RFC2578]
12     RowStatus,
13     TruthValue, MacAddress
14     FROM SNMPv2-TC      -- [RFC2579]
15     MODULE-COMPLIANCE,
16     OBJECT-GROUP
17     FROM SNMPv2-CONF    -- [RFC2580]
18     InterfaceIndex
19     FROM IF-MIB         -- [RFC2863]
20     IEEE8021ServiceSelectorType,
21     IEEE8021ServiceSelectorValue,
22     IEEE8021ServiceSelectorValueOrNone,
23     IEEE8021PbbComponentIdentifier,
24     ieee802dot1mibs     FROM IEEE8021-TC-MIB
25 --cfm types
26     DotlagCfmMhfCreation,
27     DotlagCfmIdPermission,
28     DotlagCfmMDLevel,
29     DotlagCfmMpDirection,
30     DotlagCfmMepIdOrZero,
31     DotlagCfmMDLevelOrNone,
32     DotlagCfmConfigErrors,
33 -- cfm indexes
34     dotlagCfmMdIndex,
35     dotlagCfmMaIndex,
36 --cfm groups
37     dotlagCfmStack,
38     dotlagCfmDefaultMd,
39     dotlagCfmVlan,
40     dotlagCfmConfigErrorList,
41     dotlagCfmMa,
42 -- cfm row items
43     dotlagCfmMepLbrBadMsdu,
44     dotlagCfmMdRowStatus,
45     dotlagCfmMaNetRowStatus,
46     dotlagCfmMaMepListRowStatus,
47     dotlagCfmMepRowStatus,
48 --cfm conformance groups
49     dotlagCfmCompliances,
50     dotlagCfmGroups,
51     dotlagCfmMdGroup,
52     dotlagCfmMepGroup,
53     dotlagCfmMepDbGroup,
54     dotlagCfmNotificationsGroup,
55     ieee8021CfmDefaultMdDefGroup,
56     ieee8021CfmMaNetGroup,
57     ieee8021CfmPbbTeExtensionGroup,
58     ieee8021CfmPbbTeTrafficBitGroup      FROM IEEE8021-CFM-MIB
59 ;
60
61 ieee8021CfmV2Mib MODULE-IDENTITY
62     LAST-UPDATED "202211080000Z" -- November 8, 2022
63     ORGANIZATION "IEEE 802.1 Working Group"
64     CONTACT-INFO
65         " WG-URL: http://www.ieee802.org/1/
66         WG-EMail: stds-802-1-1@ieee.org
67         Contact: IEEE 802.1 Working Group Chair
68         Postal: C/O IEEE 802.1 Working Group
69             IEEE Standards Association
70             445 Hoes Lane
71             Piscataway, NJ 08854
```

```
1          USA
2          E-mail: stds-802-1-chairs@ieee.org"
3 DESCRIPTION
4     "Connectivity Fault Management V2 module.
5
6     Unless otherwise indicated, the references in this MIB
7     module are to IEEE Std 802.1Q-2022.
8
9     Copyright (C) IEEE (2022).
10    This version of this MIB module is part of IEEE Std 802.1Q-2022;
11    see that standard for full legal notices."
12
13 REVISION "202211080000Z" -- November 8, 2022
14 DESCRIPTION
15     "Published as part of IEEE Std 802.1Q-2022.
16     Cross references and contact information updated."
17
18 REVISION "201412150000Z" -- December 15, 2014
19
20 DESCRIPTION
21     "Published as part of IEEE Std 802.1Q 2014 revision.
22     Cross references updated and corrected."
23
24 REVISION "201102270000Z" -- February 27, 2011
25 DESCRIPTION
26     "Minor edits to contact information etc. as part of
27     2011 revision of IEEE Std 802.1Q."
28
29 REVISION "200811180000Z" -- November 18, 2008
30 DESCRIPTION
31     "Added a new compliance clause for utilization by systems
32     that support CFM and PBB-TE."
33
34 REVISION "200810150000Z" -- October 15, 2008
35 DESCRIPTION
36     "The IEEE8021-CFM-V2-MIB Module contains objects that
37     replace those deprecated in the IEEE8021-CFM-MIB module.
38
39     This version is included in IEEE Std 802.1ap."
40
41 ::= { ieee802dot1mibs 7 }
42
43
44 -- *****
45 -- Note: Re-indexed IEEE 802.1ag tables
46 -- *****
47 -- This section contains new tables replacing deprecated tables in
48 -- this version of the MIB
49
50 -- *****
51 -- The CFM Stack Table
52 -- *****
53
54 ieee8021CfmStackTable OBJECT-TYPE
55     SYNTAX      SEQUENCE OF Ieee8021CfmStackEntry
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59         "There is one CFM Stack table per Bridge. It permits
60         the retrieval of information about the Maintenance Points
61         configured on any given interface.
62         "
63     REFERENCE
64         "12.14.2"
65     ::= { dot1agCfmStack 2 }
66
67 ieee8021CfmStackEntry OBJECT-TYPE
68     SYNTAX      Ieee8021CfmStackEntry
69     MAX-ACCESS  not-accessible
70     STATUS      current
71     DESCRIPTION
72         "The Stack table entry"
```

```

1  INDEX { ieee8021CfmStackifIndex, ieee8021CfmStackServiceSelectorType,
2          ieee8021CfmStackServiceSelectorOrNone,
3          ieee8021CfmStackMdLevel, ieee8021CfmStackDirection
4      }
5  ::= { ieee8021CfmStackTable 1 }
6
7  ieee8021CfmStackEntry ::= SEQUENCE {
8      ieee8021CfmStackifIndex          InterfaceIndex,
9      ieee8021CfmStackServiceSelectorType IEEE8021ServiceSelectorType,
10     ieee8021CfmStackServiceSelectorOrNone IEEE8021ServiceSelectorValueOrNone,
11     ieee8021CfmStackMdLevel          DotlagCfmMDLevel,
12     ieee8021CfmStackDirection        DotlagCfmMpDirection,
13     ieee8021CfmStackMdIndex          Unsigned32,
14     ieee8021CfmStackMaIndex          Unsigned32,
15     ieee8021CfmStackMepId            DotlagCfmMepIdOrZero,
16     ieee8021CfmStackMacAddress        MacAddress
17 }
18
19 ieee8021CfmStackifIndex OBJECT-TYPE
20     SYNTAX      InterfaceIndex
21     MAX-ACCESS  not-accessible
22     STATUS      current
23     DESCRIPTION
24         "This object represents the Bridge Port or aggregated port
25         on which MEPs or MHFs might be configured.
26
27         Upon a restart of the system, the system SHALL, if necessary,
28         change the value of this variable, and rearrange the
29         ieee8021CfmStackTable, so that it indexes the entry in the
30         interface table with the same value of ifAlias that it
31         indexed before the system restart. If no such entry exists,
32         then the system SHALL delete all entries in the
33         ieee8021CfmStackTable with the interface index.
34     "
35     REFERENCE
36         "12.14.2.1.2:a"
37     ::= { ieee8021CfmStackEntry 1 }
38
39 ieee8021CfmStackServiceSelectorType OBJECT-TYPE
40     SYNTAX      IEEE8021ServiceSelectorType
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44         "Type of the Service Selector identifier indicated by
45     ieee8021CfmStackServiceSelectorOrNone.
46         See textual convention IEEE8021ServiceSelectorType for details.
47     "
48     REFERENCE
49         "12.14.2.1.2:d, 22.1.7"
50     ::= { ieee8021CfmStackEntry 2 }
51
52 ieee8021CfmStackServiceSelectorOrNone OBJECT-TYPE
53     SYNTAX      IEEE8021ServiceSelectorValueOrNone
54     MAX-ACCESS  not-accessible
55     STATUS      current
56     DESCRIPTION
57         "Service Selector identifier to which the MP is attached, or 0, if none.
58         See textual convention IEEE8021ServiceSelectorValue for details.
59     "
60     REFERENCE
61         "12.14.2.1.2:d, 22.1.7"
62     ::= { ieee8021CfmStackEntry 3 }
63
64 ieee8021CfmStackMdLevel OBJECT-TYPE
65     SYNTAX      DotlagCfmMDLevel
66     MAX-ACCESS  not-accessible
67     STATUS      current
68     DESCRIPTION
69         "MD Level of the Maintenance Point."
70     REFERENCE
71         "12.14.2.1.2:b"
72     ::= { ieee8021CfmStackEntry 4 }

```



```
1
2 ieee8021CfmStackDirection OBJECT-TYPE
3   SYNTAX      DotlagCfmMpDirection
4   MAX-ACCESS  not-accessible
5   STATUS      current
6   DESCRIPTION
7       "Direction in which the MP faces on the Bridge Port"
8   REFERENCE
9       "12.14.2.1.2:c"
10  ::= { ieee8021CfmStackEntry 5 }
11
12 ieee8021CfmStackMdIndex OBJECT-TYPE
13   SYNTAX      Unsigned32
14   MAX-ACCESS  read-only
15   STATUS      current
16   DESCRIPTION
17       "The index of the Maintenance Domain in the ieee8021CfmMdTable
18       to which the MP is associated, or 0, if none."
19   REFERENCE
20       "12.14.2.1.3:b"
21  ::= { ieee8021CfmStackEntry 6 }
22
23 ieee8021CfmStackMaIndex OBJECT-TYPE
24   SYNTAX      Unsigned32
25   MAX-ACCESS  read-only
26   STATUS      current
27   DESCRIPTION
28       "The index of the MA in the dotlagCfmMaNetTable and
29       ieee8021CfmMaCompTable to which the MP is associated, or 0, if
30       none."
31   REFERENCE
32       "12.14.2.1.3:c"
33  ::= { ieee8021CfmStackEntry 7 }
34
35 ieee8021CfmStackMepId OBJECT-TYPE
36   SYNTAX      DotlagCfmMepIdOrZero
37   MAX-ACCESS  read-only
38   STATUS      current
39   DESCRIPTION
40       "If an MEP is configured, the MEPID, else 0"
41   REFERENCE
42       "12.14.2.1.3:d"
43  ::= { ieee8021CfmStackEntry 8 }
44
45 ieee8021CfmStackMacAddress OBJECT-TYPE
46   SYNTAX      MacAddress
47   MAX-ACCESS  read-only
48   STATUS      current
49   DESCRIPTION
50       "MAC address of the MP."
51   REFERENCE
52       "12.14.2.1.3:e"
53  ::= { ieee8021CfmStackEntry 9 }
54
55 -- *****
56 -- The CFM VLAN Table
57 -- *****
58
59 ieee8021CfmVlanTable OBJECT-TYPE
60   SYNTAX      SEQUENCE OF Ieee8021CfmVlanEntry
61   MAX-ACCESS  not-accessible
62   STATUS      current
63   DESCRIPTION
64       "This table defines the association of VIDs into VLANs. There
65       is an entry in this table, for each component of the Bridge,
66       for each VID that is:
67           a) a VID belonging to a VLAN associated with more than
68             one VID; and
69           b) not the Primary VID of that VLAN.
70       The entry in this table contains the Primary VID of the VLAN.
71
72       By default, this table is empty, meaning that every VID is
```

```

1      the Primary VID of a single-VID VLAN.
2
3      VLANs that are associated with only one VID SHOULD NOT have
4      an entry in this table.
5
6      The writable objects in this table need to be persistent
7      upon reboot or restart of a device.
8      "
9      REFERENCE
10     "12.14.3.1.3:a, 12.14.3.2.2:a, 12.14.5.3.2:c,
11     12.14.6.1.3:b, 22.1.5"
12     ::= { dotlagCfmVlan 2 }
13
14 ieee8021CfmVlanEntry OBJECT-TYPE
15     SYNTAX      Ieee8021CfmVlanEntry
16     MAX-ACCESS  not-accessible
17     STATUS      current
18     DESCRIPTION
19         "The VLAN table entry."
20     INDEX { ieee8021CfmVlanComponentId,
21            ieee8021CfmVlanSelector}
22     ::= { ieee8021CfmVlanTable 1 }
23
24 Ieee8021CfmVlanEntry ::= SEQUENCE {
25     ieee8021CfmVlanComponentId      IEEE8021PbbComponentIdentifier,
26     ieee8021CfmVlanSelector          IEEE8021ServiceSelectorValue,
27     ieee8021CfmVlanPrimarySelector  IEEE8021ServiceSelectorValue,
28     ieee8021CfmVlanRowStatus         RowStatus
29 }
30
31 ieee8021CfmVlanComponentId OBJECT-TYPE
32     SYNTAX      IEEE8021PbbComponentIdentifier
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "The Bridge component within the system to which the information
37         in this ieee8021CfmVlanEntry applies. If the system is not a
38         Bridge, or if only one component is present in the Bridge, then
39         this variable (index) MUST be equal to 1.
40     "
41     REFERENCE
42         "12.3 1)"
43     ::= { ieee8021CfmVlanEntry 1 }
44
45 ieee8021CfmVlanSelector OBJECT-TYPE
46     SYNTAX      IEEE8021ServiceSelectorValue
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "This is a service ID belonging to a service that is associated
51         with more than one Service Selector identifiers, and this is not the Primary
52         Service ID of the service. The type of this Service Selector is the same
53         as the primary Service Selector's type defined by
54 ieee8021CfmMaCompPrimarySelectorType
55         in the ieee8021CfmMaCompTable.
56     "
57     ::= { ieee8021CfmVlanEntry 3 }
58
59 ieee8021CfmVlanPrimarySelector OBJECT-TYPE
60     SYNTAX      IEEE8021ServiceSelectorValue
61     MAX-ACCESS  read-create
62     STATUS      current
63     DESCRIPTION
64         "This is the Primary Service selector for a Service that is associated
65         with more than one Service Selector identifiers. This value MUST not
66         equal the value of ieee8021CfmVlanSelector. The type of this Service Selector is the
67 same
68         as the primary Service Selector's type defined by
69 ieee8021CfmMaCompPrimarySelectorType
70         in the ieee8021CfmMaCompTable.
71     "
72     ::= { ieee8021CfmVlanEntry 5 }

```

```
1
2 ieee8021CfmVlanRowStatus OBJECT-TYPE
3     SYNTAX      RowStatus
4     MAX-ACCESS   read-create
5     STATUS      current
6     DESCRIPTION
7         "The status of the row.
8
9         The writable columns in a row cannot be changed if the row
10        is active. All columns MUST have a valid value before a row
11        can be activated.
12        "
13    ::= { ieee8021CfmVlanEntry 6 }
14
15
16 -- *****
17 -- The CFM Default MD Level Table
18 -- *****
19
20 ieee8021CfmDefaultMdTable OBJECT-TYPE
21     SYNTAX      SEQUENCE OF Ieee8021CfmDefaultMdEntry
22     MAX-ACCESS   not-accessible
23     STATUS      current
24     DESCRIPTION
25         "For each Bridge component, the Default MD Level Managed Object
26         controls MHF creation for VIDs that are not attached to a
27         specific Maintenance Association Managed Object, and Sender ID
28         TLV transmission by those MHFs.
29
30         For each Bridge Port, and for each VLAN ID whose data can
31         pass through that Bridge Port, an entry in this table is
32         used by the algorithm in 22.2.3 only if there is no
33         entry in the Maintenance Association table defining an MA
34         for the same VLAN ID and MD Level as this table's entry, and
35         on which MA an Up MEP is defined. If there exists such an
36         MA, that MA's objects are used by the algorithm in
37         22.2.3 in place of this table entry's objects. The
38         agent maintains the value of ieee8021CfmDefaultMdStatus to
39         indicate whether this entry is overridden by an MA.
40
41         When first initialized, the agent creates this table
42         automatically with entries for all VLAN IDs,
43         with the default values specified for each object.
44
45         After this initialization, the writable objects in this
46         table need to be persistent upon reboot or restart of a
47         device.
48        "
49     REFERENCE
50         "12.14.3"
51    ::= { dot1agCfmDefaultMd 5 }
52
53 ieee8021CfmDefaultMdEntry OBJECT-TYPE
54     SYNTAX      Ieee8021CfmDefaultMdEntry
55     MAX-ACCESS   not-accessible
56     STATUS      current
57     DESCRIPTION
58         "The Default MD Level table entry."
59     INDEX { ieee8021CfmDefaultMdComponentId,
60            ieee8021CfmDefaultMdPrimarySelectorType,
61            ieee8021CfmDefaultMdPrimarySelector }
62    ::= { ieee8021CfmDefaultMdTable 1 }
63
64 Ieee8021CfmDefaultMdEntry ::= SEQUENCE {
65     ieee8021CfmDefaultMdComponentId      IEEE8021PbbComponentIdentifier,
66     ieee8021CfmDefaultMdPrimarySelectorType IEEE8021ServiceSelectorType,
67     ieee8021CfmDefaultMdPrimarySelector  IEEE8021ServiceSelectorValue,
68     ieee8021CfmDefaultMdStatus            TruthValue,
69     ieee8021CfmDefaultMdLevel             Dot1agCfmMDLevelOrNone,
70     ieee8021CfmDefaultMdMhfCreation       Dot1agCfmMhfCreation,
71     ieee8021CfmDefaultMdIdPermission      Dot1agCfmIdPermission
72 }
```

```

1
2 ieee8021CfmDefaultMdComponentId OBJECT-TYPE
3     SYNTAX      IEEE8021PbbComponentIdentifier
4     MAX-ACCESS  not-accessible
5     STATUS      current
6     DESCRIPTION
7         "The Bridge component within the system to which the information
8         in this ieee8021CfmDefaultMdEntry applies. If the system is not
9         a Bridge, or if only one component is present in the Bridge,
10        then this variable (index) MUST be equal to 1.
11        "
12     REFERENCE
13         "12.3 1)"
14     ::= { ieee8021CfmDefaultMdEntry 1 }
15
16 ieee8021CfmDefaultMdPrimarySelectorType OBJECT-TYPE
17     SYNTAX      IEEE8021ServiceSelectorType
18     MAX-ACCESS  not-accessible
19     STATUS      current
20     DESCRIPTION
21         "Type of the Primary Service Selector identifier indicated by
22         ieee8021CfmDefaultMdPrimarySelector. See textual
23         convention IEEE8021ServiceSelectorType for details.
24         "
25     ::= { ieee8021CfmDefaultMdEntry 2 }
26
27 ieee8021CfmDefaultMdPrimarySelector OBJECT-TYPE
28     SYNTAX      IEEE8021ServiceSelectorValue
29     MAX-ACCESS  not-accessible
30     STATUS      current
31     DESCRIPTION
32         "Primary Service Selector identifier of a Service Instance with
33         no MA configured. See IEEE8021ServiceSelectorValue for details.
34         "
35     ::= { ieee8021CfmDefaultMdEntry 3 }
36
37 ieee8021CfmDefaultMdStatus OBJECT-TYPE
38     SYNTAX      TruthValue
39     MAX-ACCESS  read-only
40     STATUS      current
41     DESCRIPTION
42         "State of this Default MD Level table entry. True if there is
43         no entry in the Maintenance Association table defining an MA
44         for the same VLAN ID and MD Level as this table's entry, and
45         on which MA an Up MEP is defined, else false.
46         "
47     REFERENCE
48         "12.14.3.1.3:b"
49     ::= { ieee8021CfmDefaultMdEntry 4 }
50
51 ieee8021CfmDefaultMdLevel OBJECT-TYPE
52     SYNTAX      DotlagCfmMDLevelOrNone
53     MAX-ACCESS  read-write
54     STATUS      current
55     DESCRIPTION
56         "A value indicating the MD Level at which MHFs are to be
57         created, and Sender ID TLV transmission by those MHFs is to
58         be controlled, for the VLAN to which this entry's objects
59         apply.
60         "
61     REFERENCE
62         "12.14.3.1.3:c, 12.14.3.2.2:b"
63     DEFVAL {-1}
64     ::= { ieee8021CfmDefaultMdEntry 5 }
65
66 ieee8021CfmDefaultMdMhfCreation OBJECT-TYPE
67     SYNTAX      DotlagCfmMhfCreation
68     MAX-ACCESS  read-write
69     STATUS      current
70     DESCRIPTION
71         "A value indicating if the Management entity can create MHFs
72         (MIP Half Function) for this VID at this MD Level. If this

```

```

1      object has the value defMHFdefer, MHF creation for this VLAN
2      is controlled by ieee8021CfmDefaultMdDefMhfCreation.
3
4      The value of this variable is meaningless if the values of
5      ieee8021CfmDefaultMdStatus is false.
6      "
7  REFERENCE
8      "12.14.3.1.3:d"
9  DEFVAL {defMHFdefer}
10 ::= { ieee8021CfmDefaultMdEntry 6 }
11
12 ieee8021CfmDefaultMdIdPermission OBJECT-TYPE
13     SYNTAX      DotlagCfmIdPermission
14     MAX-ACCESS  read-write
15     STATUS      current
16     DESCRIPTION
17         "Enumerated value indicating what, if anything, is to be
18         included in the Sender ID TLV (21.5.3) transmitted by MHFs
19         created by the Default Maintenance Domain. If this object
20         has the value sendIdDefer, Sender ID TLV transmission for
21         this VLAN is controlled by ieee8021CfmDefaultMdDefIdPermission.
22
23         The value of this variable is meaningless if the values of
24         ieee8021CfmDefaultMdStatus is false.
25     "
26     REFERENCE
27         "12.14.3.1.3:e"
28     DEFVAL { sendIdDefer }
29     ::= { ieee8021CfmDefaultMdEntry 7 }
30
31
32 -- *****
33 -- The CFM Configuration Error List Table
34 -- *****
35
36 ieee8021CfmConfigErrorListTable OBJECT-TYPE
37     SYNTAX      SEQUENCE OF Ieee8021CfmConfigErrorListEntry
38     MAX-ACCESS  not-accessible
39     STATUS      current
40     DESCRIPTION
41         "The CFM Configuration Error List table provides a list of
42         Interfaces and VIDs that are incorrectly configured.
43     "
44     REFERENCE
45         "12.14.4"
46     ::= { dotlagCfmConfigErrorList 2 }
47
48 ieee8021CfmConfigErrorListEntry OBJECT-TYPE
49     SYNTAX      Ieee8021CfmConfigErrorListEntry
50     MAX-ACCESS  not-accessible
51     STATUS      current
52     DESCRIPTION
53         "The Config Error List Table entry"
54     INDEX { ieee8021CfmConfigErrorListSelectorType,
55             ieee8021CfmConfigErrorListSelector,
56             ieee8021CfmConfigErrorListIfIndex
57         }
58     ::= { ieee8021CfmConfigErrorListTable 1 }
59
60 Ieee8021CfmConfigErrorListEntry ::= SEQUENCE {
61     ieee8021CfmConfigErrorListSelectorType  IEEE8021ServiceSelectorType,
62     ieee8021CfmConfigErrorListSelector      IEEE8021ServiceSelectorValue,
63     ieee8021CfmConfigErrorListIfIndex       InterfaceIndex,
64     ieee8021CfmConfigErrorListErrorType     DotlagCfmConfigErrors
65 }
66
67 ieee8021CfmConfigErrorListSelectorType OBJECT-TYPE
68     SYNTAX      IEEE8021ServiceSelectorType
69     MAX-ACCESS  not-accessible
70     STATUS      current
71     DESCRIPTION
72         "Type of the Service Selector identifier indicated by

```

```

1         ieee8021CfmConfigErrorListSelector. See textual
2         convention IEEE8021ServiceSelectorType for details.
3         "
4     REFERENCE
5         "12.14.4.1.2:a"
6     ::= { ieee8021CfmConfigErrorListEntry 1 }
7
8 ieee8021CfmConfigErrorListSelector OBJECT-TYPE
9     SYNTAX      IEEE8021ServiceSelectorValue
10    MAX-ACCESS  not-accessible
11    STATUS      current
12    DESCRIPTION
13        "The Service Selector Identifier of the Service with interfaces
14        in error. See IEEE8021ServiceSelectorValue for details.
15        "
16    REFERENCE
17        "12.14.4.1.2:a"
18    ::= { ieee8021CfmConfigErrorListEntry 2 }
19
20 ieee8021CfmConfigErrorListIfIndex OBJECT-TYPE
21    SYNTAX      InterfaceIndex
22    MAX-ACCESS  not-accessible
23    STATUS      current
24    DESCRIPTION
25        "This object is the IfIndex of the interface.
26
27        Upon a restart of the system, the system SHALL, if necessary,
28        change the value of this variable so that it indexes the
29        entry in the interface table with the same value of ifAlias
30        that it indexed before the system restart. If no such
31        entry exists, then the system SHALL delete any entries in
32        ieee8021CfmConfigErrorListTable indexed by that
33        InterfaceIndex value.
34        "
35    REFERENCE
36        "12.14.4.1.2:b"
37    ::= { ieee8021CfmConfigErrorListEntry 3 }
38
39 ieee8021CfmConfigErrorListErrorType OBJECT-TYPE
40    SYNTAX      DotlagCfmConfigErrors
41    MAX-ACCESS  read-only
42    STATUS      current
43    DESCRIPTION
44        "A vector of Boolean error conditions from 22.2.4, any of
45        which may be true:
46
47        0) CFMleak;
48        1) ConflictingVids;
49        2) ExcessiveLevels;
50        3) OverlappedLevels.
51        "
52    REFERENCE
53        "12.14.4.1.3:b"
54    ::= { ieee8021CfmConfigErrorListEntry 4 }
55
56 -- *****
57 -- The CFM Maintenance Association (MA) Component Table
58 -- *****
59
60 ieee8021CfmMaCompTable OBJECT-TYPE
61    SYNTAX      SEQUENCE OF Ieee8021CfmMaCompEntry
62    MAX-ACCESS  not-accessible
63    STATUS      current
64    DESCRIPTION
65        "The Maintenance Association table. Each row in the table
66        represents an MA. An MA is a set of MEPs, each configured
67        with a single service instance.
68
69        This is the part of the complete MA table that is variable
70        across the Bridges in a Maintenance Domain, or across the
71        components of a single Bridge. That part of the MA table that
72        is constant across the Bridges and their components in a

```

```

1      Maintenance Domain is contained in the dotlagCfmMaNetTable.
2
3      This table uses three indices, first index is the
4      IEEE8021PbbComponentIdentifier that identifies the component
5      within the Bridge for which the information in the
6      ieee8021CfmMaCompEntry applies. The second is the index of the
7      Maintenance Domain table. The third index is the same as the
8      index of the ieee8021CfmMaNetEntry for the same MA.
9
10     The writable objects in this table need to be persistent
11     upon reboot or restart of a device.
12
13     "
14     REFERENCE
15     "18.2"
16     ::= { dotlagCfmMa 4 }
17
18 ieee8021CfmMaCompEntry OBJECT-TYPE
19     SYNTAX      Ieee8021CfmMaCompEntry
20     MAX-ACCESS  not-accessible
21     STATUS      current
22     DESCRIPTION
23         "The MA table entry."
24     INDEX {ieee8021CfmMaComponentId,
25           dotlagCfmMdIndex, dotlagCfmMaIndex }
26     ::= { ieee8021CfmMaCompTable 1 }
27
28 Ieee8021CfmMaCompEntry ::= SEQUENCE {
29     ieee8021CfmMaComponentId      IEEE8021PbbComponentIdentifier,
30     ieee8021CfmMaCompPrimarySelectorType  IEEE8021ServiceSelectorType,
31     ieee8021CfmMaCompPrimarySelectorOrNone IEEE8021ServiceSelectorValueOrNone,
32     ieee8021CfmMaCompMhfCreation      DotlagCfmMhfCreation,
33     ieee8021CfmMaCompIdPermission      DotlagCfmIdPermission,
34     ieee8021CfmMaCompNumberOfVids      Unsigned32,
35     ieee8021CfmMaCompRowStatus          RowStatus
36 }
37
38 ieee8021CfmMaComponentId OBJECT-TYPE
39     SYNTAX      IEEE8021PbbComponentIdentifier
40     MAX-ACCESS  not-accessible
41     STATUS      current
42     DESCRIPTION
43         "The Bridge component within the system to which the information
44         in this ieee8021CfmMaCompEntry applies. If the system is not a
45         Bridge, or if only one component is present in the Bridge, then
46         this variable (index) MUST be equal to 1."
47     "
48     REFERENCE
49     "12.3 1)"
50     ::= { ieee8021CfmMaCompEntry 1 }
51
52 ieee8021CfmMaCompPrimarySelectorType OBJECT-TYPE
53     SYNTAX      IEEE8021ServiceSelectorType
54     MAX-ACCESS  read-create
55     STATUS      current
56     DESCRIPTION
57         "Type of the Service Selector identifiers indicated by
58         ieee8021CfmMaCompPrimarySelectorOrNone. If the service
59         instance is defined by more than one Service Selector, this
60         parameter also indicates the type of the
61         ieee8021CfmVlanPrimarySelector and ieee8021CfmVlanSelector
62         in the ieee8021CfmVlanTable.
63         In Services instances made of multiple Service Selector
64         identifiers, ensures that the type of the Service selector
65         identifiers is the same. See textual convention
66         IEEE8021ServiceSelectorType for details."
67     "
68     REFERENCE
69     "12.14.6.1.3:b"
70     ::= { ieee8021CfmMaCompEntry 2 }
71
72 ieee8021CfmMaCompPrimarySelectorOrNone OBJECT-TYPE

```

```
1 SYNTAX IEEE8021ServiceSelectorValueOrNone
2 MAX-ACCESS read-create
3 STATUS current
4 DESCRIPTION
5     "Service Selector identifier to which the MP is attached,
6     or 0, if none. If the MA is associated with more than one
7     Service Selectors Identifiers, the ieee8021CfmVlanTable
8     lists them.
9     "
10 REFERENCE
11     "12.14.6.1.3:b"
12 ::= { ieee8021CfmMaCompEntry 3 }
13
14
15 ieee8021CfmMaCompMhfCreation OBJECT-TYPE
16 SYNTAX DotlagCfmMhfCreation
17 MAX-ACCESS read-create
18 STATUS current
19 DESCRIPTION
20     "Indicates if the Management entity can create MHFs (MIP Half
21     Function) for this MA.
22     "
23 REFERENCE
24     "12.14.6.1.3:c"
25 DEFVAL { defMHFdefer }
26 ::= { ieee8021CfmMaCompEntry 4 }
27
28 ieee8021CfmMaCompIdPermission OBJECT-TYPE
29 SYNTAX DotlagCfmIdPermission
30 MAX-ACCESS read-create
31 STATUS current
32 DESCRIPTION
33     "Enumerated value indicating what, if anything, is to be
34     included in the Sender ID TLV (21.5.3) transmitted by MPs
35     configured in this MA.
36     "
37 REFERENCE
38     "12.14.6.1.3:d"
39 DEFVAL { sendIdDefer }
40 ::= { ieee8021CfmMaCompEntry 5 }
41
42 ieee8021CfmMaCompNumberOfVids OBJECT-TYPE
43 SYNTAX Unsigned32
44 MAX-ACCESS read-create
45 STATUS current
46 DESCRIPTION
47     "The number of VIDs associated with the MA.
48     "
49 REFERENCE
50     "12.14.6.1.3:b"
51 ::= { ieee8021CfmMaCompEntry 6 }
52
53 ieee8021CfmMaCompRowStatus OBJECT-TYPE
54 SYNTAX RowStatus
55 MAX-ACCESS read-create
56 STATUS current
57 DESCRIPTION
58     "The status of the row.
59
60     The writable columns in a row cannot be changed if the row
61     is active. All columns MUST have a valid value before a row
62     can be activated.
63     "
64 ::= { ieee8021CfmMaCompEntry 7 }
65
66 -- *****
67 -- Units of conformance
68 -- *****
69
70 ieee8021CfmStackGroup OBJECT-GROUP
71 OBJECTS {
72     ieee8021CfmStackMdIndex,
```



```

1      ieee8021CfmStackMaIndex,
2      ieee8021CfmStackMepId,
3      ieee8021CfmStackMacAddress
4  }
5  STATUS      current
6  DESCRIPTION
7      "Objects for the Stack group."
8  ::= { dotlagCfmGroups 12 }
9
10 ieee8021CfmMaGroup OBJECT-GROUP
11     OBJECTS {
12         ieee8021CfmMaCompPrimarySelectorType,
13         ieee8021CfmMaCompPrimarySelectorOrNone,
14         ieee8021CfmMaCompMhfCreation,
15         ieee8021CfmMaCompIdPermission,
16         ieee8021CfmMaCompRowStatus,
17         ieee8021CfmMaCompNumberOfVids
18     }
19     STATUS      current
20     DESCRIPTION
21         "Objects for the MA group."
22     ::= { dotlagCfmGroups 13 }
23
24 ieee8021CfmDefaultMdGroup OBJECT-GROUP
25     OBJECTS {
26         ieee8021CfmDefaultMdStatus,
27         ieee8021CfmDefaultMdLevel,
28         ieee8021CfmDefaultMdMhfCreation,
29         ieee8021CfmDefaultMdIdPermission
30     }
31     STATUS      current
32     DESCRIPTION
33         "Objects for the Default MD Level group."
34     ::= { dotlagCfmGroups 14 }
35
36 ieee8021CfmVlanIdGroup OBJECT-GROUP
37     OBJECTS {
38         ieee8021CfmVlanPrimarySelector,
39         ieee8021CfmVlanRowStatus
40     }
41     STATUS      current
42     DESCRIPTION
43         "Objects for the VLAN ID group."
44     ::= { dotlagCfmGroups 15 }
45
46 ieee8021CfmConfigErrorListGroup OBJECT-GROUP
47     OBJECTS {
48         ieee8021CfmConfigErrorListErrorType
49     }
50     STATUS      current
51     DESCRIPTION
52         "Objects for the CFM Configuration Error List Group."
53     ::= {dotlagCfmGroups 16 }
54
55 -- *****
56 -- MIB Module Compliance statements
57 -- *****
58
59 ieee8021CfmComplianceV2 MODULE-COMPLIANCE
60     STATUS      current
61     DESCRIPTION
62         "The compliance statement for support of the CFM MIB module."
63
64     MODULE
65         MANDATORY-GROUPS {
66             ieee8021CfmStackGroup,
67             ieee8021CfmMaGroup,
68             ieee8021CfmDefaultMdGroup,
69             ieee8021CfmConfigErrorListGroup
70         }
71
72     GROUP ieee8021CfmVlanIdGroup

```

```

1  DESCRIPTION "The VLAN ID group is optional."
2
3  OBJECT ieee8021CfmMaCompRowStatus
4      SYNTAX      RowStatus { active(1), notInService(2) }
5      WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
6                          destroy(6) }
7      DESCRIPTION "Support for createAndWait is not required."
8
9  OBJECT ieee8021CfmVlanRowStatus
10     SYNTAX      RowStatus { active(1), notInService(2) }
11     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
12                         destroy(6) }
13     DESCRIPTION "Support for createAndWait is not required."
14
15  MODULE IEEE8021-CFM-MIB
16     MANDATORY-GROUPS {
17         dotlagCfmMdGroup,
18         dotlagCfmMepGroup,
19         dotlagCfmMepDbGroup,
20         dotlagCfmNotificationsGroup,
21         ieee8021CfmDefaultMdDefGroup,
22         ieee8021CfmMaNetGroup
23     }
24
25  OBJECT dotlagCfmMepLbrBadMsdu
26  MIN-ACCESS not-accessible
27  DESCRIPTION "The dotlagCfmMepLbrBadMsdu variable is optional. It
28              MUST not be present if the system cannot compare a
29              received LBR to the corresponding LBM."
30
31  OBJECT dotlagCfmMdRowStatus
32     SYNTAX      RowStatus { active(1), notInService(2) }
33     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
34                         destroy(6) }
35     DESCRIPTION "Support for createAndWait is not required."
36
37  OBJECT dotlagCfmMaNetRowStatus
38     SYNTAX      RowStatus { active(1), notInService(2) }
39     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
40                         destroy(6) }
41     DESCRIPTION "Support for createAndWait is not required."
42
43  OBJECT dotlagCfmMaMepListRowStatus
44     SYNTAX      RowStatus { active(1), notInService(2) }
45     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
46                         destroy(6) }
47     DESCRIPTION "Support for createAndWait is not required."
48
49  OBJECT dotlagCfmMepRowStatus
50     SYNTAX      RowStatus { active(1), notInService(2) }
51     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
52                         destroy(6) }
53     DESCRIPTION "Support for createAndWait is not required."
54
55  ::= { dotlagCfmCompliances 2 }
56
57 dotlagCfmWithPbbTeCompliance MODULE-COMPLIANCE
58     STATUS      current
59     DESCRIPTION
60         "The compliance statement for support of the CFM MIB for
61         systems that support PBB-TE."
62
63  MODULE
64     MANDATORY-GROUPS {
65         ieee8021CfmStackGroup,
66         ieee8021CfmMaGroup,
67         ieee8021CfmDefaultMdGroup,
68         ieee8021CfmConfigErrorListGroup
69     }
70
71  GROUP ieee8021CfmVlanIdGroup
72  DESCRIPTION "The VLAN ID group is optional."

```

```
1
2 OBJECT ieee8021CfmMaCompRowStatus
3   SYNTAX      RowStatus { active(1), notInService(2) }
4   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
5                       destroy(6) }
6   DESCRIPTION "Support for createAndWait is not required."
7
8 OBJECT ieee8021CfmVlanRowStatus
9   SYNTAX      RowStatus { active(1), notInService(2) }
10  WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
11                      destroy(6) }
12  DESCRIPTION "Support for createAndWait is not required."
13
14 MODULE IEEE8021-CFM-MIB
15   MANDATORY-GROUPS {
16     dotlagCfmMdGroup,
17     dotlagCfmMepGroup,
18     dotlagCfmMepDbGroup,
19     dotlagCfmNotificationsGroup,
20     ieee8021CfmDefaultMdDefGroup,
21     ieee8021CfmMaNetGroup,
22     ieee8021CfmPbbTeExtensionGroup
23   }
24
25 GROUP ieee8021CfmPbbTeTrafficBitGroup
26 DESCRIPTION "The objects needed to support the traffic bit are optional
27             as traffic bit support, itself, is optional."
28
29 OBJECT dotlagCfmMepLbrBadMsdu
30 MIN-ACCESS not-accessible
31 DESCRIPTION "The dotlagCfmMepLbrBadMsdu variable is optional. It
32             MUST not be present if the system cannot compare a
33             received LBR to the corresponding LBM."
34
35 OBJECT dotlagCfmMdRowStatus
36   SYNTAX      RowStatus { active(1), notInService(2) }
37   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
38                       destroy(6) }
39   DESCRIPTION "Support for createAndWait is not required."
40
41 OBJECT dotlagCfmMaNetRowStatus
42   SYNTAX      RowStatus { active(1), notInService(2) }
43   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
44                       destroy(6) }
45   DESCRIPTION "Support for createAndWait is not required."
46
47 OBJECT dotlagCfmMaMepListRowStatus
48   SYNTAX      RowStatus { active(1), notInService(2) }
49   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
50                       destroy(6) }
51   DESCRIPTION "Support for createAndWait is not required."
52
53 OBJECT dotlagCfmMepRowStatus
54   SYNTAX      RowStatus { active(1), notInService(2) }
55   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
56                       destroy(6) }
57   DESCRIPTION "Support for createAndWait is not required."
58
59 ::= { dotlagCfmCompliances 3 }
60
61 END
```

1 17.7.8 Definitions for the IEEE8021-PBB-MIB module

```
2 IEEE8021-PBB-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for IEEE 802.1Q Provider Backbone Bridge Devices
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE, Unsigned32, Integer32
10    FROM SNMPv2-SMI
11    RowStatus, StorageType, MacAddress, TruthValue
12    FROM SNMPv2-TC
13    SnmpAdminString
14    FROM SNMP-FRAMEWORK-MIB
15    ieee802dot1mibs, IEEE8021PbbComponentIdentifier,
16    IEEE8021BridgePortNumber, IEEE8021PbbServiceIdentifier,
17    IEEE8021PbbServiceIdentifierOrUnassigned, IEEE8021PbbIngressEgress,
18    IEEE8021PriorityValue, IEEE8021PriorityCodePoint
19    FROM IEEE8021-TC-MIB
20    ieee8021BridgeBasePortComponentId, ieee8021BridgeBasePort
21    FROM IEEE8021-BRIDGE-MIB
22    VlanId
23    FROM Q-BRIDGE-MIB
24    InterfaceIndex, InterfaceIndexOrZero
25    FROM IF-MIB
26    MODULE-COMPLIANCE, OBJECT-GROUP
27    FROM SNMPv2-CONF;
28
29 ieee8021PbbMib MODULE-IDENTITY
30     LAST-UPDATED "202211080000Z" -- November 8, 2022
31     ORGANIZATION "IEEE 802.1 Working Group"
32     CONTACT-INFO
33         " WG-URL: http://www.ieee802.org/1/
34         WG-EMail: stds-802-1-1@ieee.org
35         Contact: IEEE 802.1 Working Group Chair
36         Postal: C/O IEEE 802.1 Working Group
37             IEEE Standards Association
38             445 Hoes Lane
39             Piscataway, NJ 08854
40             USA
41         E-mail: stds-802-1-chairs@ieee.org"
42     DESCRIPTION
43         "The Provider Backbone Bridge (PBB) MIB module for managing
44         devices that support PBB.
45
46         Unless otherwise indicated, the references in this MIB
47         module are to IEEE Std 802.1Q-2022.
48
49         Copyright (C) IEEE (2022).
50         This version of this MIB module is part of IEEE Std 802.1Q;
51         see that standard for full legal notices."
52
53     REVISION "202211080000Z" -- November 8, 2022
54     DESCRIPTION
55         "Published as part of IEEE Std 802.1Q-2022.
56         Cross references and contact information updated."
57
58     REVISION "201807010000Z" -- July 1, 2018
59     DESCRIPTION
60         "Published as part of IEEE Std 802.1Q 2018 revision.
61         Cross references updated and corrected."
62
63     REVISION "201412150000Z" -- December 15, 2014
64     DESCRIPTION
65         "Published as part of IEEE Std 802.1Q 2014 revision.
66         Cross references updated and corrected.
67         ieee8021PbbVipType and ieee8021PbbCBPServiceMappingType
68         deprecated."
69
70     REVISION "201102270000Z" -- February 27, 2011
71     DESCRIPTION
```

```

1      "Minor edits to contact information etc. as part of
2      2011 revision of IEEE Std 802.1Q."
3
4      REVISION      "200811180000Z" -- November 18, 2008
5      DESCRIPTION
6      "Modified VIP table to support the enableConnectionIdentifier
7      configuration option."
8
9      REVISION      "200810150000Z" -- October 15, 2008
10     DESCRIPTION
11     "Initial version published in IEEE Std 802.1ap."
12     ::= { ieee802dot1mibs 9 }
13
14     ieee8021PbbNotifications OBJECT IDENTIFIER ::= { ieee8021PbbMib 0 }
15     ieee8021PbbObjects      OBJECT IDENTIFIER ::= { ieee8021PbbMib 1 }
16     ieee8021PbbConformance OBJECT IDENTIFIER ::= { ieee8021PbbMib 2 }
17
18 --
19 -- IEEE 802.1ah MIB Objects
20 --
21
22     ieee8021PbbProviderBackboneBridge
23     OBJECT IDENTIFIER ::= { ieee8021PbbObjects 1 }
24
25 -- =====
26 -- 12.16.1.1/12.16.1.2 Backbone Edge Bridge (BEB) configuration
27 -- =====
28 -- items a), b), c), e), and g), see below
29 -- d) ieee8021BridgeBaseBridgeAddress from IEEE8021-BRIDGE-MIB
30 -- f) ieee8021BridgeBaseBridgeAddress from IEEE8021-BRIDGE-MIB
31 -- i) and j) ifPhysAddress from the IF-MIB, the correct instance
32 -- can be found using ieee8021BridgeBasePortIfIndex
33 -- from the IEEE8021-BRIDGE-MIB
34 -- =====
35
36     ieee8021PbbBackboneEdgeBridgeObjects
37     OBJECT IDENTIFIER ::= { ieee8021PbbProviderBackboneBridge 1 }
38
39     ieee8021PbbBackboneEdgeBridgeAddress OBJECT-TYPE
40     SYNTAX      MacAddress
41     MAX-ACCESS  read-only
42     STATUS      current
43     DESCRIPTION
44     "The MAC Address used by the BEB when it must be referred
45     to in a unique fashion."
46     REFERENCE   "12.16.1.1.3 a)"
47     ::= { ieee8021PbbBackboneEdgeBridgeObjects 1 }
48
49     ieee8021PbbBackboneEdgeBridgeName OBJECT-TYPE
50     SYNTAX      SnmpAdminString (SIZE (0..32))
51     MAX-ACCESS  read-write
52     STATUS      current
53     DESCRIPTION
54     "A text string of locally determined significance. This value
55     must be persistent over power up restart/reboot."
56     REFERENCE   "12.16.1.1.3 b), 12.16.1.2.2"
57     ::= { ieee8021PbbBackboneEdgeBridgeObjects 2 }
58
59     ieee8021PbbNumberOfIComponents OBJECT-TYPE
60     SYNTAX      Unsigned32
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64     "The number of I-components in this BEB."
65     REFERENCE   "12.16.1.1.3 c)"
66     ::= { ieee8021PbbBackboneEdgeBridgeObjects 3 }
67
68     ieee8021PbbNumberOfBComponents OBJECT-TYPE
69     SYNTAX      Unsigned32 (0..1)
70     MAX-ACCESS  read-only
71     STATUS      current
72     DESCRIPTION

```

```

1      "The number of B-components in this BEB."
2      REFERENCE "12.16.1.1.3 e)"
3      ::= { ieee8021PbbBackboneEdgeBridgeObjects 4 }
4
5 ieee8021PbbNumberOfBebPorts OBJECT-TYPE
6     SYNTAX      Unsigned32
7     MAX-ACCESS  read-only
8     STATUS      current
9     DESCRIPTION
10      "The number of CNPs, PIPs, CBPs, and PNPs in this BEB."
11     REFERENCE "12.16.1.1.3 g)"
12     ::= { ieee8021PbbBackboneEdgeBridgeObjects 5 }
13
14 ieee8021PbbNextAvailablePipIfIndex OBJECT-TYPE
15     SYNTAX      InterfaceIndex
16     MAX-ACCESS  read-only
17     STATUS      current
18     DESCRIPTION
19      "This object provides an available ifIndex value that can
20      be used for creation of a PIP."
21     REFERENCE "12.16.4.1, 12.16.4.2"
22     ::= { ieee8021PbbBackboneEdgeBridgeObjects 6 }
23
24 -- =====
25 -- 12.16.3.1/2 Virtual Instance Port (VIP) configuration table
26 -- =====
27
28 ieee8021PbbVipTable OBJECT-TYPE
29     SYNTAX      SEQUENCE OF Ieee8021PbbVipEntry
30     MAX-ACCESS  not-accessible
31     STATUS      current
32     DESCRIPTION
33      "This table lists the additional PBB parameters for each
34      VIP. Entries in this table must be persistent over power
35      up restart/reboot."
36     REFERENCE "12.16.3.1, 12.16.3.2"
37     ::= { ieee8021PbbProviderBackboneBridge 2 }
38
39 ieee8021PbbVipEntry OBJECT-TYPE
40     SYNTAX      Ieee8021PbbVipEntry
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44      "Each entry consists of the name string, I-SID, Default
45      Destination MAC, Service Type, and possible B-MAC."
46     INDEX      { ieee8021BridgeBasePortComponentId,
47                  ieee8021BridgeBasePort }
48     ::= { ieee8021PbbVipTable 1 }
49
50 Ieee8021PbbVipEntry ::=
51     SEQUENCE {
52         ieee8021PbbVipPipIfIndex
53             InterfaceIndexOrZero,
54         ieee8021PbbVipISid
55             IEEE8021PbbServiceIdentifierOrUnassigned,
56         ieee8021PbbVipDefaultDstBMAC
57             MacAddress,
58         ieee8021PbbVipType
59             IEEE8021PbbIngressEgress,
60         ieee8021PbbVipRowStatus
61             RowStatus,
62         ieee8021PbbVipEnableConnectionId
63             TruthValue
64     }
65
66 ieee8021PbbVipPipIfIndex OBJECT-TYPE
67     SYNTAX      InterfaceIndexOrZero
68     MAX-ACCESS  read-only
69     STATUS      current
70     DESCRIPTION
71      "Identifies the PIP associated with this VIP within the BEB.
72      A value of zero indicates the VIP is not currently associated

```

```

1         with any PIP.
2
3         The value of this object must be persistent across
4         reinitializations of the management system."
5     DEFVAL      { 0 }
6     ::= { ieee8021PbbVipEntry 1 }
7
8 ieee8021PbbVipISid OBJECT-TYPE
9     SYNTAX      IEEE8021PbbServiceIdentifierOrUnassigned
10    MAX-ACCESS   read-create
11    STATUS       current
12    DESCRIPTION
13        "The I-SID for this VIP.
14
15        Within an I-Component, an VIP can only be serviced
16        by one I-SID. And the ISID is a configurable parameter
17        of the VIP.
18
19        The value of this object must be persistent across
20        reinitializations of the management system."
21    DEFVAL { 1 }
22    ::= { ieee8021PbbVipEntry 2 }
23
24 ieee8021PbbVipDefaultDstBMAC OBJECT-TYPE
25     SYNTAX      MacAddress
26     MAX-ACCESS   read-only
27     STATUS       current
28     DESCRIPTION
29         "The Default Destination B-MAC for this VIP."
30     DEFVAL { '001e83000001'h }
31     ::= { ieee8021PbbVipEntry 3 }
32
33 ieee8021PbbVipType OBJECT-TYPE
34     SYNTAX      IEEE8021PbbIngressEgress
35     MAX-ACCESS   read-create
36     STATUS       deprecated
37     DESCRIPTION
38         "This feature is used to support asymmetric VLANs.
39
40         The value of this object must be persistent across
41         reinitializations of the management system."
42     DEFVAL { { egress, ingress } }
43     ::= { ieee8021PbbVipEntry 4 }
44
45 ieee8021PbbVipRowStatus OBJECT-TYPE
46     SYNTAX      RowStatus
47     MAX-ACCESS   read-create
48     STATUS       current
49     DESCRIPTION
50         "This indicates the status of an entry in this table, and is
51         used to create/delete entries.
52
53         It is an implementation specific decision as to whether any
54         column in this table may be set while the corresponding
55         instance of this object is valid(1).
56
57         The value of this object must be persistent across
58         reinitializations of the management system."
59     ::= { ieee8021PbbVipEntry 5 }
60
61 ieee8021PbbVipEnableConnectionId OBJECT-TYPE
62     SYNTAX      TruthValue
63     MAX-ACCESS   read-write
64     STATUS       current
65     DESCRIPTION
66         "This indicates if the connection_identifier parameter is allowed
67         to learn associations between a backbone MAC address and a customer
68         MAC address. The default value is true, indicating that such
69         learning is allowed. This parameter should be configured to false
70         at the root node of a Point-to-multipoint TE service instance."
71     DEFVAL      { true }
72     ::= { ieee8021PbbVipEntry 6 }

```

```

1
2 -- =====
3 -- 12.16.3.1/12.16.3.2 I-SID to VIP cross-reference table
4 -- =====
5
6 ieee8021PbbISidToVipTable OBJECT-TYPE
7     SYNTAX      SEQUENCE OF Ieee8021PbbISidToVipEntry
8     MAX-ACCESS  not-accessible
9     STATUS      current
10    DESCRIPTION
11        "This table contains a cross-reference of I-SID values
12         to the VIPs with which they are associated. This allows
13         VIPs to be located easily by their associated I-SID."
14    REFERENCE   "12.16.3.1, 12.16.3.2"
15    ::= { ieee8021PbbProviderBackboneBridge 3 }
16
17 ieee8021PbbISidToVipEntry OBJECT-TYPE
18     SYNTAX      Ieee8021PbbISidToVipEntry
19     MAX-ACCESS  not-accessible
20     STATUS      current
21     DESCRIPTION
22        "A mapping from an I-SID to the VIP with which it is
23         associated. An entry will exist for each entry in
24         the ieee8021PbbVipTable."
25     INDEX       { ieee8021PbbISidToVipISid }
26     ::= { ieee8021PbbISidToVipTable 1 }
27
28 Ieee8021PbbISidToVipEntry ::=
29     SEQUENCE {
30         ieee8021PbbISidToVipISid
31             IEEE8021PbbServiceIdentifier,
32         ieee8021PbbISidToVipComponentId
33             IEEE8021PbbComponentIdentifier,
34         ieee8021PbbISidToVipPort
35             IEEE8021BridgePortNumber
36     }
37
38 ieee8021PbbISidToVipISid OBJECT-TYPE
39     SYNTAX      IEEE8021PbbServiceIdentifier
40     MAX-ACCESS  not-accessible
41     STATUS      current
42     DESCRIPTION
43        "The I-SID of a VIP."
44     ::= { ieee8021PbbISidToVipEntry 1 }
45
46 ieee8021PbbISidToVipComponentId OBJECT-TYPE
47     SYNTAX      IEEE8021PbbComponentIdentifier
48     MAX-ACCESS  read-only
49     STATUS      current
50     DESCRIPTION
51        "The component identifier of the VIP to which this I-SID
52         is associated."
53     ::= { ieee8021PbbISidToVipEntry 2 }
54
55 ieee8021PbbISidToVipPort OBJECT-TYPE
56     SYNTAX      IEEE8021BridgePortNumber
57     MAX-ACCESS  read-only
58     STATUS      current
59     DESCRIPTION
60        "The port number of the VIP to which this I-SID is associated."
61     ::= { ieee8021PbbISidToVipEntry 3 }
62
63 -- =====
64 -- 12.16.4.1/12.16.4.2 Provider Instance Port (PIP) configuration
65 --      table
66 -- =====
67
68 ieee8021PbbPipTable OBJECT-TYPE
69     SYNTAX      SEQUENCE OF Ieee8021PbbPipEntry
70     MAX-ACCESS  not-accessible
71     STATUS      current
72     DESCRIPTION

```



```

1      "This table contains the parameters for each PIP, and
2      can be used to configure the PIP port names. Entries
3      in this table must be persistent over power up
4      restart/reboot."
5  REFERENCE      "12.16.4.1, 12.16.4.2"
6  ::= { ieee8021PbbProviderBackboneBridge 4 }
7
8  ieee8021PbbPipEntry OBJECT-TYPE
9      SYNTAX      Ieee8021PbbPipEntry
10     MAX-ACCESS  not-accessible
11     STATUS      current
12     DESCRIPTION
13         "The parameters for a PIP.  "
14     INDEX        { ieee8021PbbPipIfIndex }
15     ::= { ieee8021PbbPipTable 1 }
16
17 Ieee8021PbbPipEntry ::=
18     SEQUENCE {
19         ieee8021PbbPipIfIndex
20         InterfaceIndex,
21         ieee8021PbbPipBMACAddress
22         MacAddress,
23         ieee8021PbbPipName
24         SnmpAdminString,
25         ieee8021PbbPipIComponentId
26         IEEE8021PbbComponentIdentifier,
27         ieee8021PbbPipVipMap
28         OCTET STRING,
29         ieee8021PbbPipVipMap1
30         OCTET STRING,
31         ieee8021PbbPipVipMap2
32         OCTET STRING,
33         ieee8021PbbPipVipMap3
34         OCTET STRING,
35         ieee8021PbbPipVipMap4
36         OCTET STRING,
37         ieee8021PbbPipRowStatus
38         RowStatus
39     }
40
41 ieee8021PbbPipIfIndex OBJECT-TYPE
42     SYNTAX      InterfaceIndex
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46         "The number identifying the PIP."
47     ::= { ieee8021PbbPipEntry 1 }
48
49 ieee8021PbbPipBMACAddress OBJECT-TYPE
50     SYNTAX      MacAddress
51     MAX-ACCESS  read-create
52     STATUS      current
53     DESCRIPTION
54         "The B-MAC used by this PIP for the B-SA."
55     ::= { ieee8021PbbPipEntry 2 }
56
57 ieee8021PbbPipName OBJECT-TYPE
58     SYNTAX      SnmpAdminString (SIZE(0..32))
59     MAX-ACCESS  read-create
60     STATUS      current
61     DESCRIPTION
62         "A text string of local significance that identifies the
63         PIP within a BEB."
64     DEFVAL { 'H' }
65     ::= { ieee8021PbbPipEntry 3 }
66
67 ieee8021PbbPipIComponentId OBJECT-TYPE
68     SYNTAX      IEEE8021PbbComponentIdentifier
69     MAX-ACCESS  read-only
70     STATUS      current
71     DESCRIPTION
72         "Identifies the I-component associated with this PIP."

```

```

1 ::= { ieee8021PbbPipEntry 4 }
2
3 ieee8021PbbPipVipMap OBJECT-TYPE
4     SYNTAX      OCTET STRING (SIZE(0..512))
5     MAX-ACCESS  read-create
6     STATUS      current
7     DESCRIPTION
8         "This object contains a bitmap indicating all the VIPs
9         in the range 1 through 4094 that are associated with this
10        PIP. The bits correspond to Bridge Port numbers in the
11        range 1 through 4094. The high-order bit of the first
12        octet corresponds to port number 1, and subsequent bits
13        of the octet string correspond to subsequent port numbers.
14        The following formula can be used to find the bit
15        corresponding to a particular port number B:
16            octet[(B-1)/8] & (1 >> ((B-1)%8))
17        If the bit for a particular port number is 1, that VIP is
18        associated with this PIP.
19
20        The value of this object may be truncated to remove
21        trailing octets of all zeros."
22     DEFVAL { ''H }
23 ::= { ieee8021PbbPipEntry 5 }
24
25 ieee8021PbbPipVipMap1 OBJECT-TYPE
26     SYNTAX      OCTET STRING (SIZE(0..2048))
27     MAX-ACCESS  read-create
28     STATUS      current
29     DESCRIPTION
30         "This object contains a bitmap indicating all the VIPs
31         in the range 4095 through 20478 that are associated with
32         this PIP. The bits correspond to Bridge Port numbers in
33         the range 4095 through 20478. The high-order bit of the first
34         octet corresponds to port number 1, and subsequent bits
35         of the octet string correspond to subsequent port numbers.
36         The following formula can be used to find the bit
37         corresponding to a particular port number B:
38            octet[(B-4095)/8] & (1 >> ((B-4095)%8))
39        If the bit for a particular port number is 1, that VIP is
40        associated with this PIP.
41
42        Note that ports numbers greater than 4094 cannot be used
43        with xSTP.
44
45        The value of this object may be truncated to remove
46        trailing octets of all zeros."
47     DEFVAL { ''H }
48 ::= { ieee8021PbbPipEntry 6 }
49
50 ieee8021PbbPipVipMap2 OBJECT-TYPE
51     SYNTAX      OCTET STRING (SIZE(0..2048))
52     MAX-ACCESS  read-create
53     STATUS      current
54     DESCRIPTION
55         "This object contains a bitmap indicating all the VIPs
56         in the range 20479 through 36861 that are associated with
57         this PIP. The bits correspond to Bridge Port numbers in
58         the range 20479 through 36861. The high-order bit of the first
59         octet corresponds to port number 1, and subsequent bits
60         of the octet string correspond to subsequent port numbers.
61         The following formula can be used to find the bit
62         corresponding to a particular port number B:
63            octet[(B-20479)/8] & (1 >> ((B-20479)%8))
64        If the bit for a particular port number is 1, that VIP is
65        associated with this PIP.
66
67        Note that ports numbers greater than 4094 cannot be used
68        with xSTP.
69
70        The value of this object may be truncated to remove
71        trailing octets of all zeros."
72     DEFVAL { ''H }

```

```

1      ::= { ieee8021PbbPipEntry 7 }
2
3 ieee8021PbbPipVipMap3 OBJECT-TYPE
4     SYNTAX      OCTET STRING (SIZE(0..2048))
5     MAX-ACCESS  read-create
6     STATUS      current
7     DESCRIPTION
8         "This object contains a bitmap indicating all the VIPs
9         in the range 36862 through 53245 that are associated with
10        this PIP. The bits correspond to Bridge Port numbers in
11        the range 36862 through 53245. The high-order bit of the first
12        octet corresponds to port number 1, and subsequent bits
13        of the octet string correspond to subsequent port numbers.
14        The following formula can be used to find the bit
15        corresponding to a particular port number B:
16            octet[(B-36862)/8] & (1 >> ((B-36862)%8))
17        If the bit for a particular port number is 1, that VIP is
18        associated with this PIP.
19
20        Note that ports numbers greater than 4094 cannot be used
21        with xSTP.
22
23        The value of this object may be truncated to remove
24        trailing octets of all zeros."
25     DEFVAL { 'H' }
26     ::= { ieee8021PbbPipEntry 8 }
27
28 ieee8021PbbPipVipMap4 OBJECT-TYPE
29     SYNTAX      OCTET STRING (SIZE(0..1537))
30     MAX-ACCESS  read-create
31     STATUS      current
32     DESCRIPTION
33         "This object contains a bitmap indicating all the VIPs
34         in the range 53246 through 65535 that are associated with
35         this PIP. The bits correspond to Bridge Port numbers in
36         the range 53246 through 65535. The high-order bit of the first
37         octet corresponds to port number 1, and subsequent bits
38         of the octet string correspond to subsequent port numbers.
39         The following formula can be used to find the bit
40         corresponding to a particular port number B:
41            octet[(B-53246)/8] & (1 >> ((B-53246)%8))
42        If the bit for a particular port number is 1, that VIP is
43        associated with this PIP.
44
45        Note that ports numbers greater than 4094 cannot be used
46        with xSTP.
47
48        The value of this object may be truncated to remove
49        trailing octets of all zeros."
50     DEFVAL { 'H' }
51     ::= { ieee8021PbbPipEntry 9 }
52
53 ieee8021PbbPipRowStatus OBJECT-TYPE
54     SYNTAX      RowStatus
55     MAX-ACCESS  read-create
56     STATUS      current
57     DESCRIPTION
58         "Indicates the status of an entry in this table, and is used
59         to create/delete entries.
60
61         The object ieee8021PbbPipBMACAddress must be set before this
62         object can be made active(1).
63
64         The value of ieee8021PbbPipBMACAddress cannot be modified
65         while this object is active(1)."
66     ::= { ieee8021PbbPipEntry 10 }
67
68 -- =====
69 -- 12.16.4.1    Provider Instance Port (PIP)
70 --              Priority Table
71 -- =====
72

```

```

1 ieee8021PbbPipPriorityTable OBJECT-TYPE
2   SYNTAX      SEQUENCE OF Ieee8021PbbPipPriorityEntry
3   MAX-ACCESS  not-accessible
4   STATUS      current
5   DESCRIPTION
6       "A table that contains information about every PIP that
7       is associated with this PBB."
8   REFERENCE   "12.16.4.1"
9   ::= { ieee8021PbbProviderBackboneBridge 5 }
10
11 ieee8021PbbPipPriorityEntry OBJECT-TYPE
12   SYNTAX      Ieee8021PbbPipPriorityEntry
13   MAX-ACCESS  not-accessible
14   STATUS      current
15   DESCRIPTION
16       "A list of Default User Priorities for each PIP of a
17       PBB. This is indexed by ieee8021PbbPipIfIndex."
18   AUGMENTS { ieee8021PbbPipEntry }
19   ::= { ieee8021PbbPipPriorityTable 1 }
20
21 Ieee8021PbbPipPriorityEntry ::=
22   SEQUENCE {
23       ieee8021PbbPipPriorityCodePointSelection
24       IEEE8021PriorityCodePoint,
25       ieee8021PbbPipUseDEI
26       TruthValue,
27       ieee8021PbbPipRequireDropEncoding
28       TruthValue
29   }
30
31 ieee8021PbbPipPriorityCodePointSelection OBJECT-TYPE
32   SYNTAX      IEEE8021PriorityCodePoint
33   MAX-ACCESS  read-write
34   STATUS      current
35   DESCRIPTION
36       " This object identifies the rows in the PCP encoding and
37       decoding tables that are used to remark frames on this
38       PIP if this remarking is enabled."
39   REFERENCE   "12.16.4.5, 12.16.4.6"
40   ::= { ieee8021PbbPipPriorityEntry 1 }
41
42 ieee8021PbbPipUseDEI OBJECT-TYPE
43   SYNTAX      TruthValue
44   MAX-ACCESS  read-write
45   STATUS      current
46   DESCRIPTION
47       "If the Use_DEI is set to true(1) for the PIP then the
48       drop_eligible parameter is encoded in the DEI of transmitted
49       frames, and the drop_eligible parameter shall be true(1) for a
50       received frame if the DEI is set in the VLAN tag or the Priority
51       Code Point Decoding Table indicates drop_eligible True for
52       the received PCP value. If the Use_DEI parameter is false(2),
53       the DEI shall be transmitted as zero and ignored on receipt.
54       The default value of the Use_DEI parameter is false(2)."

```

```

1 ::= { ieee8021PbbPipPriorityEntry 3 }
2
3 -- =====
4 -- 12.16.4.7/12.16.4.8 Provider Instance Port (PIP)
5 -- Priority Code Point (PCP) Decoding Table
6 -- =====
7
8 ieee8021PbbPipDecodingTable OBJECT-TYPE
9     SYNTAX      SEQUENCE OF Ieee8021PbbPipDecodingEntry
10    MAX-ACCESS  not-accessible
11    STATUS      current
12    DESCRIPTION
13        "A table that contains information about Priority Code
14        Point Decoding Table for a PIP of a provider Bridge.
15        Alternative values for each table are specified as rows
16        in Table 6-3 (6.9.3), with each alternative labeled by
17        the number of distinct priorities that can be communicated,
18        and the number of these for which drop precedence can
19        be communicated. All writable objects in this table must
20        be persistent over power up restart/reboot."
21    REFERENCE    "12.16.4.7, 12.16.4.8"
22    ::= { ieee8021PbbProviderBackboneBridge 6 }
23
24 ieee8021PbbPipDecodingEntry OBJECT-TYPE
25     SYNTAX      Ieee8021PbbPipDecodingEntry
26     MAX-ACCESS  not-accessible
27     STATUS      current
28     DESCRIPTION
29         "A list of objects containing Priority Code Point Decoding
30         information for a PIP of a provider Bridge."
31     INDEX { ieee8021PbbPipIfIndex,
32             ieee8021PbbPipDecodingPriorityCodePointRow,
33             ieee8021PbbPipDecodingPriorityCodePoint }
34     ::= { ieee8021PbbPipDecodingTable 1 }
35
36 Ieee8021PbbPipDecodingEntry ::= SEQUENCE {
37     ieee8021PbbPipDecodingPriorityCodePointRow
38         IEEE8021PriorityCodePoint,
39     ieee8021PbbPipDecodingPriorityCodePoint
40         Integer32,
41     ieee8021PbbPipDecodingPriority
42         IEEE8021PriorityValue,
43     ieee8021PbbPipDecodingDropEligible
44         TruthValue
45 }
46
47 ieee8021PbbPipDecodingPriorityCodePointRow OBJECT-TYPE
48     SYNTAX      IEEE8021PriorityCodePoint
49     MAX-ACCESS  not-accessible
50     STATUS      current
51     DESCRIPTION
52         "The specific row in Table 6-3 (6.9.3) indicating the PCP."
53     ::= { ieee8021PbbPipDecodingEntry 1 }
54
55 ieee8021PbbPipDecodingPriorityCodePoint OBJECT-TYPE
56     SYNTAX      Integer32 (0..7)
57     MAX-ACCESS  not-accessible
58     STATUS      current
59     DESCRIPTION
60         "The specific PCP value in Table 6-3 (6.9.3)."
61     ::= { ieee8021PbbPipDecodingEntry 2 }
62
63 ieee8021PbbPipDecodingPriority OBJECT-TYPE
64     SYNTAX      IEEE8021PriorityValue
65     MAX-ACCESS  read-write
66     STATUS      current
67     DESCRIPTION
68         "The specific priority value in Table 6-3 (6.9.3)."
69     REFERENCE    "12.6.2.8, 12.6.2.9"
70     ::= { ieee8021PbbPipDecodingEntry 3 }
71
72 ieee8021PbbPipDecodingDropEligible OBJECT-TYPE

```

```

1  SYNTAX      TruthValue
2  MAX-ACCESS  read-write
3  STATUS      current
4  DESCRIPTION
5      "The drop eligibility value in 12.6.2.8."
6  REFERENCE   "12.6.2.8, 12.6.2.9"
7  ::= { ieee8021PbbPipDecodingEntry 4 }
8
9  -- =====
10 -- 12.16.4.9/10 Provider Instance Port (PIP)
11 --      Priority Code Point (PCP) Encoding Table
12 -- =====
13
14 ieee8021PbbPipEncodingTable OBJECT-TYPE
15     SYNTAX      SEQUENCE OF Ieee8021PbbPipEncodingEntry
16     MAX-ACCESS  not-accessible
17     STATUS      current
18     DESCRIPTION
19         "A table that contains information about Priority Code
20         Point Decoding Table for a PIP of a provider Bridge.
21         Alternative values for each table are specified as rows
22         in Table 6-3 (6.9.3), with each alternative labeled by
23         the number of distinct priorities that can be communicated,
24         and the number of these for which drop precedence can be
25         communicated. All writable objects in this table must be
26         persistent over power up restart/reboot."
27     REFERENCE   "12.16.4.9, 12.16.4.10"
28     ::= { ieee8021PbbProviderBackboneBridge 7 }
29
30 ieee8021PbbPipEncodingEntry OBJECT-TYPE
31     SYNTAX      Ieee8021PbbPipEncodingEntry
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35         "A list of objects containing Priority Code Point Encoding
36         information for a PIP of a provider Bridge."
37     INDEX { ieee8021PbbPipIfIndex,
38             ieee8021PbbPipEncodingPriorityCodePointRow,
39             ieee8021PbbPipEncodingPriorityCodePoint,
40             ieee8021PbbPipEncodingDropEligible }
41     ::= { ieee8021PbbPipEncodingTable 1 }
42
43 Ieee8021PbbPipEncodingEntry ::= SEQUENCE {
44     ieee8021PbbPipEncodingPriorityCodePointRow
45     IEEE8021PriorityCodePoint,
46     ieee8021PbbPipEncodingPriorityCodePoint
47     Integer32,
48     ieee8021PbbPipEncodingDropEligible
49     TruthValue,
50     ieee8021PbbPipEncodingPriority
51     IEEE8021PriorityValue
52 }
53
54 ieee8021PbbPipEncodingPriorityCodePointRow OBJECT-TYPE
55     SYNTAX      IEEE8021PriorityCodePoint
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59         "The specific row in Table 6-3 (6.9.3) indicating the PCP row.
60         (i.e., 8P0D, 7P1D, 6P2D, 5P3D)"
61     ::= { ieee8021PbbPipEncodingEntry 1 }
62
63 ieee8021PbbPipEncodingPriorityCodePoint OBJECT-TYPE
64     SYNTAX      Integer32 (0..7)
65     MAX-ACCESS  not-accessible
66     STATUS      current
67     DESCRIPTION
68         "The specific row in Table 6-3 (6.9.3) indicating the PCP.
69         (i.e., 0,1,2,3,4,5,6,7)."
70     ::= { ieee8021PbbPipEncodingEntry 2 }
71
72 ieee8021PbbPipEncodingDropEligible OBJECT-TYPE

```

```

1  SYNTAX      TruthValue
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "The specific row in Table 6-3 (6.9.3) indicating the drop
6      eligibility. A value of true(1) means eligible for drop."
7  ::= { ieee8021PbbPipEncodingEntry 3 }
8
9  ieee8021PbbPipEncodingPriority OBJECT-TYPE
10 SYNTAX      IEEE8021PriorityValue
11 MAX-ACCESS  read-write
12 STATUS      current
13 DESCRIPTION
14      "The encoding priority in Table 6-3 (6.9.3)."
```

REFERENCE "12.6.2.10, 12.6.2.11"

```

16 ::= { ieee8021PbbPipEncodingEntry 4 }
17
18 -- =====
19 -- 12.16.4.3/12.16.4.4 Virtual Instance Port (VIP) to Provider
20 -- Instance Port (PIP) mapping table
21 -- =====
22
23 ieee8021PbbVipToPipMappingTable OBJECT-TYPE
24 SYNTAX      SEQUENCE OF Ieee8021PbbVipToPipMappingEntry
25 MAX-ACCESS  not-accessible
26 STATUS      current
27 DESCRIPTION
28      "This table lists VIPs and the PIP to which each is
29      associated, and allows the PIP associated with each
30      VIP to be configured. Entries in this table must be
31      persistent over power up restart/reboot."
32 REFERENCE  "12.16.4.3, 12.16.4.4"
33 ::= { ieee8021PbbProviderBackboneBridge 8 }
34
35 ieee8021PbbVipToPipMappingEntry OBJECT-TYPE
36 SYNTAX      Ieee8021PbbVipToPipMappingEntry
37 MAX-ACCESS  not-accessible
38 STATUS      current
39 DESCRIPTION
40      " The PIP is identified by the value of the
41      ieee8021PbbVipToPipNumber. This value may be used to
42      index the ieee8021PbbPipTable to set or retrieve the
43      PIP's configuration information"
44 INDEX      { ieee8021BridgeBasePortComponentId,
45              ieee8021BridgeBasePort }
46 ::= { ieee8021PbbVipToPipMappingTable 1 }
47
48 Ieee8021PbbVipToPipMappingEntry ::=
49     SEQUENCE {
50         ieee8021PbbVipToPipMappingPipIfIndex
51             InterfaceIndex,
52         ieee8021PbbVipToPipMappingStorageType
53             StorageType,
54         ieee8021PbbVipToPipMappingRowStatus
55             RowStatus
56     }
57
58 ieee8021PbbVipToPipMappingPipIfIndex OBJECT-TYPE
59 SYNTAX      InterfaceIndex
60 MAX-ACCESS  read-create
61 STATUS      current
62 DESCRIPTION
63      "The PIP's interface number."
64 ::= { ieee8021PbbVipToPipMappingEntry 1 }
65
66 ieee8021PbbVipToPipMappingStorageType OBJECT-TYPE
67 SYNTAX      StorageType
68 MAX-ACCESS  read-create
69 STATUS      current
70 DESCRIPTION
71      "Indicates the storage type of this entry. An entry whose
72      storage type is permanent(4) need not allow write access to
```

```

1      other columns in that entry."
2      ::= { ieee8021PbbVipToPipMappingEntry 2 }
3
4 ieee8021PbbVipToPipMappingRowStatus OBJECT-TYPE
5     SYNTAX      RowStatus
6     MAX-ACCESS  read-create
7     STATUS      current
8     DESCRIPTION
9         "Indicates the status of an entry in this table, and is used
10        to create/delete entries.
11
12        The corresponding instance of ieee8021PbbVipToPipMappingPipIfIndex
13        must be set before this object can be made active(1).
14
15        The corresponding instance of ieee8021PbbVipToPipMappingPipIfIndex
16        may not be changed while this object is active(1)."
```

-- 12.16.5.1/12.16.5.2 Service Mapping configuration table

```

22
23 ieee8021PbbCBPServiceMappingTable OBJECT-TYPE
24     SYNTAX      SEQUENCE OF Ieee8021PbbCBPServiceMappingEntry
25     MAX-ACCESS  not-accessible
26     STATUS      current
27     DESCRIPTION
28         "The CBP table of I-SID values (6.11). The contents of this
29         table are not persistent over power up restart/reboot."
30     REFERENCE   "12.16.5.1, 12.16.5.2"
31     ::= { ieee8021PbbProviderBackboneBridge 9 }
32
33 ieee8021PbbCBPServiceMappingEntry OBJECT-TYPE
34     SYNTAX      Ieee8021PbbCBPServiceMappingEntry
35     MAX-ACCESS  not-accessible
36     STATUS      current
37     DESCRIPTION
38         "Each entry includes the B-VID to carry and optionally an
39         I-SID for mapping I-SIDs normally used at a Peer E-NNI
40         (6.11, 26.6.2). The table is indexed by the component ID
41         of the relevant B-Component of the PBB, the Bridge port
42         number of the CBP on that Bcomponent, and the I-SID for
43         the service. "
```

```

44     INDEX      { ieee8021BridgeBasePortComponentId,
45                  ieee8021BridgeBasePort,
46                  ieee8021PbbCBPServiceMappingBackboneSid }
47     ::= { ieee8021PbbCBPServiceMappingTable 1 }
48
49 Ieee8021PbbCBPServiceMappingEntry ::=
50     SEQUENCE {
51         ieee8021PbbCBPServiceMappingBackboneSid
52         IEEE8021PbbServiceIdentifier,
53         ieee8021PbbCBPServiceMappingBVID
54         VlanId,
55         ieee8021PbbCBPServiceMappingDefaultBackboneDest
56         MacAddress,
57         ieee8021PbbCBPServiceMappingType
58         IEEE8021PbbIngressEgress,
59         ieee8021PbbCBPServiceMappingLocalSid
60         IEEE8021PbbServiceIdentifierOrUnassigned,
61         ieee8021PbbCBPServiceMappingRowStatus
62         RowStatus
63     }
64
65 ieee8021PbbCBPServiceMappingBackboneSid OBJECT-TYPE
66     SYNTAX      IEEE8021PbbServiceIdentifier
67     MAX-ACCESS  not-accessible
68     STATUS      current
69     DESCRIPTION
70         "The I-SID that will be transmitted over the PBBN."
71     ::= { ieee8021PbbCBPServiceMappingEntry 1 }
72

```



```

1 ieee8021PbbCBPServiceMappingBVID OBJECT-TYPE
2     SYNTAX      VlanId
3     MAX-ACCESS  read-create
4     STATUS      current
5     DESCRIPTION
6         "The B-VID that will carry this service instance."
7     ::= { ieee8021PbbCBPServiceMappingEntry 2 }
8
9 ieee8021PbbCBPServiceMappingDefaultBackboneDest OBJECT-TYPE
10    SYNTAX      MacAddress
11    MAX-ACCESS  read-create
12    STATUS      current
13    DESCRIPTION
14        "A default destination B-MAC for the CBP to use."
15    ::= { ieee8021PbbCBPServiceMappingEntry 3 }
16
17 ieee8021PbbCBPServiceMappingType OBJECT-TYPE
18    SYNTAX      IEEE8021PbbIngressEgress
19    MAX-ACCESS  read-create
20    STATUS      deprecated
21    DESCRIPTION
22        "Used for Pt-MPT service where ingress or egress limiting
23         is desired."
24    ::= { ieee8021PbbCBPServiceMappingEntry 4 }
25
26 ieee8021PbbCBPServiceMappingLocalSid OBJECT-TYPE
27    SYNTAX      IEEE8021PbbServiceIdentifierOrUnassigned
28    MAX-ACCESS  read-create
29    STATUS      current
30    DESCRIPTION
31        "The I-SID value used in frames transmitted and received through
32         this CustomerBackbonePort."
33    DEFVAL { 1 }
34    ::= { ieee8021PbbCBPServiceMappingEntry 5 }
35
36 ieee8021PbbCBPServiceMappingRowStatus OBJECT-TYPE
37    SYNTAX      RowStatus
38    MAX-ACCESS  read-create
39    STATUS      current
40    DESCRIPTION
41        "Indicates the status of an entry in this table, and is used
42         to create/delete entries.
43
44         The corresponding instances of the following objects
45         must be set before this object can be made active(1):
46             ieee8021PbbCBPServiceMappingBVID
47             ieee8021PbbCBPServiceMappingDefaultBackboneDest
48             ieee8021PbbCBPServiceMappingType
49
50         The corresponding instances of the following objects
51         may not be changed while this object is active(1):
52             ieee8021PbbCBPServiceMappingBVID
53             ieee8021PbbCBPServiceMappingDefaultBackboneDest
54             ieee8021PbbCBPServiceMappingType
55             ieee8021PbbCBPServiceMappingLocalSid"
56    ::= { ieee8021PbbCBPServiceMappingEntry 6 }
57
58 -- =====
59 -- CBP port creation/deletion table
60 -- =====
61
62 ieee8021PbbCbpTable OBJECT-TYPE
63    SYNTAX      SEQUENCE OF Ieee8021PbbCbpEntry
64    MAX-ACCESS  not-accessible
65    STATUS      current
66    DESCRIPTION
67        "This table is used to dynamically create/delete Customer
68         Backbone Ports in a PBB.
69
70         Entries in this table must be persistent across reinitializations
71         of the management system."
72    REFERENCE   "17.5.3.4"

```

```

1 ::= { ieee8021PbbProviderBackboneBridge 10 }
2
3 ieee8021PbbCbpEntry OBJECT-TYPE
4     SYNTAX      Ieee8021PbbCbpEntry
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "An entry representing a dynamically created CBP in a PBB."
9     INDEX       { ieee8021BridgeBasePortComponentId,
10                ieee8021BridgeBasePort }
11 ::= { ieee8021PbbCbpTable 1 }
12
13 Ieee8021PbbCbpEntry ::=
14     SEQUENCE {
15         ieee8021PbbCbpRowStatus
16         RowStatus
17     }
18
19 ieee8021PbbCbpRowStatus OBJECT-TYPE
20     SYNTAX      RowStatus
21     MAX-ACCESS  read-create
22     STATUS      current
23     DESCRIPTION
24         "This object is used for creation/deletion of entries in
25         this table."
26 ::= { ieee8021PbbCbpEntry 1 }
27
28 -- =====
29 -- Conformance Information
30 -- =====
31
32 ieee8021PbbGroups
33     OBJECT IDENTIFIER ::= { ieee8021PbbConformance 1 }
34 ieee8021PbbCompliances
35     OBJECT IDENTIFIER ::= { ieee8021PbbConformance 2 }
36
37 -- =====
38 -- Units of conformance
39 -- =====
40
41 ieee8021PbbBackboneEdgeBridgeGroup OBJECT-GROUP
42     OBJECTS {
43         ieee8021PbbBackboneEdgeBridgeAddress,
44         ieee8021PbbBackboneEdgeBridgeName,
45         ieee8021PbbNumberOfIComponents,
46         ieee8021PbbNumberOfBComponents,
47         ieee8021PbbNumberOfBebPorts
48     }
49     STATUS      current
50     DESCRIPTION
51         "The collection of objects used to represent a Backbone
52         Edge Bridge."
53 ::= { ieee8021PbbGroups 1 }
54
55 ieee8021PbbVipGroup OBJECT-GROUP
56     OBJECTS {
57         ieee8021PbbVipPipIfIndex,
58         ieee8021PbbVipISid,
59         ieee8021PbbVipDefaultDstBMAC,
60         ieee8021PbbVipType,
61         ieee8021PbbVipRowStatus,
62         ieee8021PbbISidToVipComponentId,
63         ieee8021PbbISidToVipPort
64     }
65     STATUS      current
66     DESCRIPTION
67         "The collection of objects used to represent a Virtual
68         Instance Port (VIP)."
69 ::= { ieee8021PbbGroups 2 }
70
71 ieee8021PbbPipGroup OBJECT-GROUP
72     OBJECTS {

```

```

1      ieee8021PbbNextAvailablePipIfIndex,
2      ieee8021PbbPipBMACAddress,
3      ieee8021PbbPipName,
4      ieee8021PbbPipIComponentId,
5      ieee8021PbbPipVipMap,
6      ieee8021PbbPipVipMap1,
7      ieee8021PbbPipVipMap2,
8      ieee8021PbbPipVipMap3,
9      ieee8021PbbPipVipMap4,
10     ieee8021PbbPipRowStatus
11   }
12   STATUS      current
13   DESCRIPTION
14     "The collection of objects used to represent a Provider
15     Instance Port (PIP)."
```

::= { ieee8021PbbGroups 3 }

```

17
18 ieee8021PbbDefaultPriorityGroup OBJECT-GROUP
19   OBJECTS {
20     ieee8021PbbPipPriorityCodePointSelection,
21     ieee8021PbbPipUseDEI,
22     ieee8021PbbPipRequireDropEncoding
23   }
24   STATUS      current
25   DESCRIPTION
26     "A collection of objects defining the priority
27     applicable to each port for media that do not support
28     native priority."
29   ::= { ieee8021PbbGroups 4 }
```

30

```

31 ieee8021PbbPipDecodingGroup OBJECT-GROUP
32   OBJECTS {
33     ieee8021PbbPipDecodingPriority,
34     ieee8021PbbPipDecodingDropEligible
35   }
36   STATUS      current
37   DESCRIPTION
38     "A collection of objects providing statistics counters for
39     decoding priority and drop eligibility for Bridge Ports."
40   ::= { ieee8021PbbGroups 5 }
```

41

```

42 ieee8021PbbPipEncodingGroup OBJECT-GROUP
43   OBJECTS {
44     ieee8021PbbPipEncodingPriority
45   }
46   STATUS      current
47   DESCRIPTION
48     "A collection of objects providing statistics counters for
49     encoding priority and drop eligibility for Bridge Ports."
50   ::= { ieee8021PbbGroups 6 }
```

51

```

52 ieee8021PbbVipToPipMappingGroup OBJECT-GROUP
53   OBJECTS {
54     ieee8021PbbVipToPipMappingPipIfIndex,
55     ieee8021PbbVipToPipMappingStorageType,
56     ieee8021PbbVipToPipMappingRowStatus
57   }
58   STATUS      current
59   DESCRIPTION
60     "The collection of objects used to represent the mapping
61     of a VIP to a PIP."
62   ::= { ieee8021PbbGroups 7 }
```

63

```

64 ieee8021PbbCBPServiceMappingGroup OBJECT-GROUP
65   OBJECTS {
66     ieee8021PbbCBPServiceMappingBvid,
67     ieee8021PbbCBPServiceMappingDefaultBackboneDest,
68     ieee8021PbbCBPServiceMappingType,
69     ieee8021PbbCBPServiceMappingLocalSid,
70     ieee8021PbbCBPServiceMappingRowStatus
71   }
72   STATUS      current
```

```

1  DESCRIPTION
2      "The collection of objects used to represent a service instance."
3  ::= { ieee8021PbbGroups 8 }
4
5  ieee8021PbbDynamicCbpGroup OBJECT-GROUP
6      OBJECTS {
7          ieee8021PbbCbpRowStatus
8      }
9      STATUS      current
10     DESCRIPTION
11         "The collection of objects used to dynamically create/delete
12         CBPs in a PBB."
13     ::= { ieee8021PbbGroups 9 }
14
15  ieee8021PbbVipPbbTeGroup OBJECT-GROUP
16      OBJECTS {
17          ieee8021PbbVipEnableConnectionId
18      }
19      STATUS      current
20     DESCRIPTION
21         "The collection of objects specific to PBB Bridges operating
22         in a PBB-TE-aware manner."
23     ::= { ieee8021PbbGroups 10 }
24
25 -- =====
26 -- Compliance statements
27 -- =====
28
29  ieee8021PbbCompliance MODULE-COMPLIANCE
30      STATUS      current
31      DESCRIPTION
32          "The compliance statement for devices supporting Provider
33          Backbone Bridging as defined in IEEE Std 802.1ah."
34      MODULE
35          MANDATORY-GROUPS {
36              ieee8021PbbBackboneEdgeBridgeGroup,
37              ieee8021PbbVipGroup,
38              ieee8021PbbPipGroup,
39              ieee8021PbbVipToPipMappingGroup,
40              ieee8021PbbCBPServiceMappingGroup,
41              ieee8021PbbDynamicCbpGroup
42          }
43
44      GROUP          ieee8021PbbDefaultPriorityGroup
45      DESCRIPTION
46          "This group is mandatory only for devices supporting
47          the priority forwarding operations defined by the
48          extended Bridge services with media types, such as
49          Ethernet, that do not support native priority."
50
51      GROUP          ieee8021PbbPipDecodingGroup
52      DESCRIPTION
53          "This group is optional and supports Priority Code Point
54          Decoding Table for a PIP of a provider Bridge."
55
56      GROUP          ieee8021PbbPipEncodingGroup
57      DESCRIPTION
58          "This group is optional and supports Priority Code Point
59          Encoding Table for a PIP of a provider Bridge."
60
61      OBJECT ieee8021PbbPipName
62      MIN-ACCESS read-only
63      DESCRIPTION
64          "Read-write access for this object is not required."
65
66      OBJECT ieee8021PbbPipVipMap
67      MIN-ACCESS read-only
68      DESCRIPTION
69          "Read-write access for this object is not required."
70
71      OBJECT ieee8021PbbVipToPipMappingPipIfIndex
72      MIN-ACCESS read-only

```

```

1  DESCRIPTION
2      "Read-write access for this object is not required."
3
4  OBJECT ieee8021PbbCBPServiceMappingBVid
5  MIN-ACCESS not-accessible
6  DESCRIPTION
7      "Read-only and read-write access for this object are optional."
8
9  OBJECT ieee8021PbbCBPServiceMappingDefaultBackboneDest
10 MIN-ACCESS not-accessible
11 DESCRIPTION
12     "Read-only and read-write access for this object are optional."
13
14 OBJECT ieee8021PbbCBPServiceMappingLocalSid
15 MIN-ACCESS not-accessible
16 DESCRIPTION
17     "Read-only and read-write access for this object are optional."
18
19 ::= { ieee8021PbbCompliances 1 }
20
21 ieee8021PbbWithPbbTeCompliance MODULE-COMPLIANCE
22 STATUS current
23 DESCRIPTION
24     "The compliance statement for devices supporting Provider
25     Backbone Bridging with traffic engineering as defined in IEEE Std 802.1ah
26     and IEEE Std 802.1Qay."
27 MODULE
28     MANDATORY-GROUPS {
29         ieee8021PbbBackboneEdgeBridgeGroup,
30         ieee8021PbbVipGroup,
31         ieee8021PbbPipGroup,
32         ieee8021PbbVipToPipMappingGroup,
33         ieee8021PbbCBPServiceMappingGroup,
34         ieee8021PbbDynamicCbpGroup,
35         ieee8021PbbVipPbbTeGroup
36     }
37
38 GROUP ieee8021PbbDefaultPriorityGroup
39 DESCRIPTION
40     "This group is mandatory only for devices supporting
41     the priority forwarding operations defined by the
42     extended Bridge services with media types, such as
43     Ethernet, that do not support native priority."
44
45 GROUP ieee8021PbbPipDecodingGroup
46 DESCRIPTION
47     "This group is optional and supports Priority Code Point
48     Decoding Table for a PIP of a provider Bridge."
49
50 GROUP ieee8021PbbPipEncodingGroup
51 DESCRIPTION
52     "This group is optional and supports Priority Code Point
53     Encoding Table for a PIP of a provider Bridge."
54
55 OBJECT ieee8021PbbPipName
56 MIN-ACCESS read-only
57 DESCRIPTION
58     "Read-write access for this object is not required."
59
60 OBJECT ieee8021PbbPipVipMap
61 MIN-ACCESS read-only
62 DESCRIPTION
63     "Read-write access for this object is not required."
64
65 OBJECT ieee8021PbbVipToPipMappingPipIfIndex
66 MIN-ACCESS read-only
67 DESCRIPTION
68     "Read-write access for this object is not required."
69
70 OBJECT ieee8021PbbCBPServiceMappingBVid
71 MIN-ACCESS not-accessible
72 DESCRIPTION

```

```
1      "Read-only and read-write access for this object are optional."  
2  
3      OBJECT ieee8021PbbCBPServiceMappingDefaultBackboneDest  
4      MIN-ACCESS not-accessible  
5      DESCRIPTION  
6      "Read-only and read-write access for this object are optional."  
7  
8      OBJECT ieee8021PbbCBPServiceMappingLocalSid  
9      MIN-ACCESS not-accessible  
10     DESCRIPTION  
11     "Read-only and read-write access for this object are optional."  
12  
13     ::= { ieee8021PbbCompliances 2 }  
14  
15 END
```

1 17.7.9 Definitions for the IEEE8021-DDCFM-MIB module

```
2
3 IEEE8021-DDCFM-MIB DEFINITIONS ::= BEGIN
4
5 -- *****
6 -- IEEE802.1Q DDCFM MIB
7 -- *****
8
9 IMPORTS
10     MODULE-IDENTITY, OBJECT-TYPE, Integer32, Unsigned32
11     FROM SNMPv2-SMI -- [RFC2578]
12     TruthValue, RowStatus, MacAddress
13     FROM SNMPv2-TC -- [RFC2579]
14     ieee802dot1mibs
15     FROM IEEE8021-TC-MIB
16     MODULE-COMPLIANCE, OBJECT-GROUP
17     FROM SNMPv2-CONF -- [RFC2580]
18     InterfaceIndex
19     FROM IF-MIB -- [RFC2863]
20     VlanIdOrNone
21     FROM Q-BRIDGE-MIB -- [RFC4363]
22     Dot1agCfmMDLevel, Dot1agCfmMpDirection
23     FROM IEEE8021-CFM-MIB;
24
25 ieee8021DdcfmMIB MODULE-IDENTITY
26     LAST-UPDATED "202211080000Z" -- November 8, 2022
27     ORGANIZATION "IEEE 802.1 Working Group"
28     CONTACT-INFO
29         "WG-URL: http://www.ieee802.org/1/
30         WG-Email: stds-802-1-l@ieee.org
31
32         Contact: IEEE 802.1 Working Group Chair
33         Postal: C/O IEEE 802.1 Working Group
34             IEEE Standards Association
35             445 Hoes Lane
36             Piscataway
37             NJ 08854
38             USA
39         E-mail: STDS-802-1-L@IEEE.ORG"
40     DESCRIPTION
41         "module for managing Data Dependent and Data Driven
42         Connectivity Fault Management.
43
44         Unless otherwise indicated, the references in this MIB
45         module are to IEEE Std 802.1Q-2022.
46
47         Copyright (C) IEEE (2022).
48         This version of this MIB module is part of IEEE Std 802.1Q;
49         see that standard for full legal notices."
50
51     REVISION "202211080000Z" -- November 8, 2022
52     DESCRIPTION
53         "Published as part of IEEE Std 802.1Q-2022.
54         Cross references and contact information updated."
55
56     REVISION "201807010000Z" -- July 1, 2018
57     DESCRIPTION
58         "Published as part of IEEE Std 802.1Q 2018 revision.
59         Cross references updated and corrected."
60
61     REVISION "201412150000Z" -- December 15, 2014
62     DESCRIPTION
63         "Published as part of IEEE Std 802.1Q 2014 revision.
64         Cross references updated and corrected."
65
66     REVISION "201102270000Z" -- February 27, 2011
67     DESCRIPTION
68         "Minor edits to contact information etc. as part of
69         2011 revision of IEEE Std 802.1Q."
70
71     REVISION "200904060000Z" -- 04/06/2009 00:00GMT
```

```

1  DESCRIPTION
2      "Included in IEEE P802.1Qaw D5.0
3      Copyright (c) IEEE"
4      ::= { ieee802dot1mibs 11}
5
6  ieee8021MIBObjects OBJECT IDENTIFIER ::= { ieee8021DdcfmMIB 1 }
7  ieee8021DdcfmConformance OBJECT IDENTIFIER ::= { ieee8021DdcfmMIB 2 }
8
9  -- *****
10 -- Groups in the DDCFM MIB Module
11 -- *****
12
13  ieee8021DdcfmStack OBJECT IDENTIFIER ::= { ieee8021MIBObjects 1}
14  ieee8021DdcfmRr OBJECT IDENTIFIER ::= { ieee8021MIBObjects 2}
15  ieee8021DdcfmRFMReceiver OBJECT IDENTIFIER ::= { ieee8021MIBObjects 3}
16  ieee8021DdcfmDr OBJECT IDENTIFIER ::= { ieee8021MIBObjects 4}
17  ieee8021DdcfmSFMOrganizer OBJECT IDENTIFIER ::= { ieee8021MIBObjects 5}
18
19  -- *****
20 -- The DDCFM Stack Table
21 -- *****
22  ieee8021DdcfmStackTable OBJECT-TYPE
23      SYNTAX SEQUENCE OF Ieee8021DdcfmStackEntry
24      MAX-ACCESS not-accessible
25      STATUS current
26      DESCRIPTION
27          "The DDCFM Stack MIB object table. This table is for operator to
28          retrieve all the DDCFM entities defined on a specified interface.
29          This table is created by default."
30      REFERENCE
31          "12.17.1"
32      ::= { ieee8021DdcfmStack 1 }
33
34  ieee8021DdcfmStackEntry OBJECT-TYPE
35      SYNTAX Ieee8021DdcfmStackEntry
36      MAX-ACCESS not-accessible
37      STATUS current
38      DESCRIPTION
39          "The DDCFM Stack Table. "
40      INDEX { ieee8021DdcfmStackIfIndex}
41      ::= { ieee8021DdcfmStackTable 1 }
42
43  Ieee8021DdcfmStackEntry ::= SEQUENCE {
44      ieee8021DdcfmStackIfIndex InterfaceIndex,
45      ieee8021DdcfmStackRrMdLevel DotlagCfmMDLevel,
46      ieee8021DdcfmStackRrDirection DotlagCfmMpDirection,
47      ieee8021DdcfmStackRFMreceiverMdLevel DotlagCfmMDLevel,
48      ieee8021DdcfmStackDrMdLevel DotlagCfmMDLevel,
49      ieee8021DdcfmStackDrVlanIdOrNone VlanIdOrNone,
50      ieee8021DdcfmStackSFMOrganizerMdLevel DotlagCfmMDLevel,
51      ieee8021DdcfmStackSFMOrganizerVlanIdOrNone VlanIdOrNone,
52      ieee8021DdcfmStackSFMOrganizerDirection DotlagCfmMpDirection
53  }
54
55  ieee8021DdcfmStackIfIndex OBJECT-TYPE
56      SYNTAX InterfaceIndex
57      MAX-ACCESS not-accessible
58      STATUS current
59      DESCRIPTION
60          "This object is the interface index of the interface either a
61          Bridge Port, or an aggregated port within a Bridge
62          Port. When the ifIndex value corresponds to the ifIndex of a
63          Bridge Port, the value in this column must match the value in the
64          ieee8021BridgeBasePortIfIndex column for the Bridge Port."
65      REFERENCE
66          "12.17.1.1.2 a.1"
67      ::= { ieee8021DdcfmStackEntry 1 }
68
69  ieee8021DdcfmStackRrMdLevel OBJECT-TYPE
70      SYNTAX DotlagCfmMDLevel
71      MAX-ACCESS read-only
72      STATUS current

```



```

1  DESCRIPTION
2      "MD level of the Reflection Responder managed object."
3  REFERENCE
4      "12.17.1.1.3 b.1"
5      ::= { ieee8021DdcfmStackEntry 2 }
6
7  ieee8021DdcfmStackRrDirection OBJECT-TYPE
8      SYNTAX DotlagCfmMpDirection
9      MAX-ACCESS read-only
10     STATUS current
11     DESCRIPTION
12         "The direction in which the RR faces."
13     REFERENCE
14         "12.17.1.1.3 b1"
15     ::= { ieee8021DdcfmStackEntry 3 }
16
17  ieee8021DdcfmStackRFMreceiverMdLevel OBJECT-TYPE
18      SYNTAX DotlagCfmMDLevel
19      MAX-ACCESS read-only
20      STATUS current
21      DESCRIPTION
22          "MD level of the RFM Receiver MO configured on the interface."
23      REFERENCE
24          "12.17.1.1.3 b.2"
25      ::= { ieee8021DdcfmStackEntry 4 }
26
27  ieee8021DdcfmStackDrMdLevel OBJECT-TYPE
28      SYNTAX DotlagCfmMDLevel
29      MAX-ACCESS read-only
30      STATUS current
31      DESCRIPTION
32          "MD level of the Deflection Responder managed object."
33      REFERENCE
34          "12.17.1.1.3 b.3"
35      ::= { ieee8021DdcfmStackEntry 5 }
36
37  ieee8021DdcfmStackDrVlanIdOrNone OBJECT-TYPE
38      SYNTAX VlanIdOrNone
39      MAX-ACCESS read-only
40      STATUS current
41      DESCRIPTION
42          "The MA of the DR configured on the interface."
43      REFERENCE
44          "12.17.1.1.3 b.3"
45      ::= { ieee8021DdcfmStackEntry 6 }
46
47  ieee8021DdcfmStackSFMOoriginatorMdLevel OBJECT-TYPE
48      SYNTAX DotlagCfmMDLevel
49      MAX-ACCESS read-only
50      STATUS current
51      DESCRIPTION
52          "MD level of the SFM Originator MO configured on the interface."
53      REFERENCE
54          "12.17.1.1.3 b.4"
55      ::= { ieee8021DdcfmStackEntry 7 }
56
57  ieee8021DdcfmStackSFMOoriginatorVlanIdOrNone OBJECT-TYPE
58      SYNTAX VlanIdOrNone
59      MAX-ACCESS read-only
60      STATUS current
61      DESCRIPTION
62          "The MA of the SFM Originator configured on the interface."
63      REFERENCE
64          "12.17.1.1.3 b.4"
65      ::= { ieee8021DdcfmStackEntry 8 }
66
67  ieee8021DdcfmStackSFMOoriginatorDirection OBJECT-TYPE
68      SYNTAX DotlagCfmMpDirection
69      MAX-ACCESS read-only
70      STATUS current
71      DESCRIPTION
72          "The direction of which the SFM Originator is facing."

```

```

1  REFERENCE
2  "12.17.1.1.3 b.4"
3  ::= { ieee8021DdcfmStackEntry 9 }
4
5  -- *****
6  -- The Reflection Responder Table
7  -- *****
8  ieee8021DdcfmRrTable OBJECT-TYPE
9      SYNTAX SEQUENCE OF Ieee8021DdcfmRrEntry
10     MAX-ACCESS not-accessible
11     STATUS current
12     DESCRIPTION
13         "The Reflection Responder MIB object table. Each row
14         in the table represents a different Reflection Responder.
15         All rows in this table persist across a system restart,
16         however after such a restart, the value of the
17         ActivationStatus column will be false."
18     REFERENCE
19         "12.17.2"
20     ::= { ieee8021DdcfmRr 1 }
21
22  ieee8021DdcfmRrEntry OBJECT-TYPE
23      SYNTAX Ieee8021DdcfmRrEntry
24      MAX-ACCESS not-accessible
25      STATUS current
26      DESCRIPTION
27          "The Reflection Responder. Each entry associated with a Reflection
28          Responder."
29      INDEX { ieee8021DdcfmRrIfIndex,
30              ieee8021DdcfmRrMdIndex,
31              ieee8021DdcfmRrDirection
32            }
33      ::= { ieee8021DdcfmRrTable 1 }
34
35  Ieee8021DdcfmRrEntry ::= SEQUENCE {
36      ieee8021DdcfmRrIfIndex InterfaceIndex,
37      ieee8021DdcfmRrMdIndex DotlagCfmMDLevel,
38      ieee8021DdcfmRrDirection DotlagCfmMpDirection,
39      ieee8021DdcfmRrPrimaryVlanIdOrNone VlanIdOrNone,
40      ieee8021DdcfmRrFilter OCTET STRING,
41      ieee8021DdcfmRrSamplingInterval Unsigned32,
42      ieee8021DdcfmRrTargetAddress MacAddress,
43      ieee8021DdcfmRrContinueFlag TruthValue,
44      ieee8021DdcfmRrDuration Unsigned32,
45      ieee8021DdcfmRrDurationInTimeFlag TruthValue,
46      ieee8021DdcfmRrVlanPriority Integer32,
47      ieee8021DdcfmRrVlanDropEligible TruthValue,
48      ieee8021DdcfmRrFloodingFlag TruthValue,
49      ieee8021DdcfmRrTruncationFlag TruthValue,
50      ieee8021DdcfmRrActivationStatus TruthValue,
51      ieee8021DdcfmRrRemainDuration Unsigned32,
52      ieee8021DdcfmRrNextRfmTransID Unsigned32,
53      ieee8021DdcfmRrRowStatus RowStatus
54  }
55
56  ieee8021DdcfmRrIfIndex OBJECT-TYPE
57      SYNTAX InterfaceIndex
58      MAX-ACCESS not-accessible
59      STATUS current
60      DESCRIPTION
61          "This object is the interface index of the interface either a
62          Bridge Port, or an aggregated port within a Bridge
63          Port, on which Reflection Responder is defined.
64          When the ifIndex value corresponds to the ifIndex of a
65          Bridge Port, the value in this column must match the value in the
66          ieee8021BridgeBasePortIfIndex column for the Bridge Port."
67      REFERENCE
68          "12.17.2.1.2 a.1"
69      ::= { ieee8021DdcfmRrEntry 1 }
70
71  ieee8021DdcfmRrMdIndex OBJECT-TYPE
72      SYNTAX DotlagCfmMDLevel

```

```

1  MAX-ACCESS not-accessible
2  STATUS current
3  DESCRIPTION
4      "MD level of the Reflection Responder managed object."
5  REFERENCE
6      "12.17.2.1.2 a.2"
7      ::= { ieee8021DdcfmRrEntry 2 }
8
9  ieee8021DdcfmRrDirection OBJECT-TYPE
10     SYNTAX DotlagCfmMpDirection
11     MAX-ACCESS not-accessible
12     STATUS current
13     DESCRIPTION
14         "The direction in which the RR faces."
15     REFERENCE
16         "12.17.2.1.2 a.3"
17     ::= { ieee8021DdcfmRrEntry 3 }
18
19  ieee8021DdcfmRrPrimaryVlanIdOrNone OBJECT-TYPE
20     SYNTAX VlanIdOrNone
21     MAX-ACCESS read-create
22     STATUS current
23     DESCRIPTION
24         "The VID to be used on RFM frames."
25     REFERENCE
26         "12.17.2.2.2 b.1"
27     DEFVAL { 0 }
28     ::= { ieee8021DdcfmRrEntry 4 }
29
30  ieee8021DdcfmRrFilter OBJECT-TYPE
31     SYNTAX OCTET STRING (SIZE(0..1500))
32     MAX-ACCESS read-create
33     STATUS current
34     DESCRIPTION
35         "A pattern string specifies what data frames are selected to be
36         reflected. Below are the primary Reflection Filters all
37         Implementers should support. Multiple primary filters can be
38         combined together by
39         && (and), || (or), or !(negation).
40         1) All;
41         2) VID= vid;
42         3) I-SID = x;
43         4) DA = xx.xx.xx.xx.xx.xx;
44         5) SA = xx.xx.xx.xx.xx.xx;
45         6) EtherType =xx.
46         For the reason that this management object allows a max size of
47         1500, messages carrying this object may be fragmented on some
48         segments."
49     REFERENCE
50         "12.17.2.2.2 b.2"
51     ::= { ieee8021DdcfmRrEntry 5 }
52
53  ieee8021DdcfmRrSamplingInterval OBJECT-TYPE
54     SYNTAX Unsigned32
55     UNITS "milliseconds"
56     MAX-ACCESS read-create
57     STATUS current
58     DESCRIPTION
59         "Indicates a time interval in which only the first frame matching
60         the filter conditions is reflected."
61     REFERENCE
62         "12.17.2.2.2 b.3"
63     ::= { ieee8021DdcfmRrEntry 6 }
64
65  ieee8021DdcfmRrTargetAddress OBJECT-TYPE
66     SYNTAX MacAddress
67     MAX-ACCESS read-create
68     STATUS current
69     DESCRIPTION
70         "The Reflection Target Address, which is a MAC address to which the
71         reflected frames are targeted. Only individual address is allowed
72         for the Reflection Target Address.
```

```

1      If not specified, the source_address of the selected data frame is
2      used for Reflection Target Address."
3  REFERENCE
4      "12.17.2.2.2 b.4"
5      ::= { ieee8021DdcfmRrEntry 7 }
6
7  ieee8021DdcfmRrContinueFlag OBJECT-TYPE
8      SYNTAX TruthValue
9      MAX-ACCESS read-create
10     STATUS current
11     DESCRIPTION
12         "True indicates that the selected data frames are to be continued
13         towards the DA specified in the frame header.
14         False indicates that the selected data frames are terminated."
15     REFERENCE
16         "12.17.2.2.2 b.5"
17     DEFVAL { true }
18     ::= { ieee8021DdcfmRrEntry 8 }
19
20  ieee8021DdcfmRrDuration OBJECT-TYPE
21     SYNTAX Unsigned32
22     MAX-ACCESS read-create
23     STATUS current
24     DESCRIPTION
25         "Duration of time in the unit of seconds or the number
26         of frames to be reflected, for Reflection Responder to
27         remain active after activation; Minimum 2 octets (65536
28         seconds) are needed for the duration of time;"
29     REFERENCE
30         "12.17.2.2.2 b.7"
31     ::= { ieee8021DdcfmRrEntry 9 }
32
33  ieee8021DdcfmRrDurationInTimeFlag OBJECT-TYPE
34     SYNTAX TruthValue
35     MAX-ACCESS read-create
36     STATUS current
37     DESCRIPTION
38         "True indicates that duration is in seconds;
39         False indicates that duration is by the total number of frames reflected."
40     REFERENCE
41         "12.17.2.2.2 b.6"
42     DEFVAL { true }
43     ::= { ieee8021DdcfmRrEntry 10 }
44
45  ieee8021DdcfmRrVlanPriority OBJECT-TYPE
46     SYNTAX Integer32(0..7)
47     MAX-ACCESS read-create
48     STATUS current
49     DESCRIPTION
50         "Priority, 3 bit value to be used in the VLAN tag, to be used in the
51         transmitted encapsulated frames. The default value is the highest
52         priority."
53     REFERENCE
54         "12.17.2.2.2 b.9"
55     DEFVAL { 7 }
56     ::= { ieee8021DdcfmRrEntry 11 }
57
58  ieee8021DdcfmRrVlanDropEligible OBJECT-TYPE
59     SYNTAX TruthValue
60     MAX-ACCESS read-create
61     STATUS current
62     DESCRIPTION
63         "It indicates that the drop_eligible bit value to be used in
64         the VLAN tag to be used in the transmitted encapsulated
65         frames is set as True or False accordingly."
66     REFERENCE
67         "12.17.2.2.2 b.9"
68     DEFVAL { false }
69     ::= { ieee8021DdcfmRrEntry 12 }
70
71  ieee8021DdcfmRrFloodingFlag OBJECT-TYPE
72     SYNTAX TruthValue

```

```

1  MAX-ACCESS read-create
2  STATUS current
3  DESCRIPTION
4      "True indicates that flooding is allowed if Egress port cannot be
5      identified for RFM by the Filtering Database, False otherwise."
6  REFERENCE
7      "12.17.2.2.2 b.10"
8  DEFVAL { true }
9  ::= { ieee8021DdcfmRrEntry 13 }
10
11
12 ieee8021DdcfmRrTruncationFlag OBJECT-TYPE
13     SYNTAX TruthValue
14     MAX-ACCESS read-create
15     STATUS current
16     DESCRIPTION
17         "True indicates that the received data frame will be truncated to
18         keep the constructed RFM size not exceeding the egress port's
19         Maximum Service Data Unit Size, False otherwise."
20     REFERENCE
21         "12.17.2.2.2 b.11"
22     DEFVAL { true }
23     ::= { ieee8021DdcfmRrEntry 14 }
24
25 ieee8021DdcfmRrActivationStatus OBJECT-TYPE
26     SYNTAX TruthValue
27     MAX-ACCESS read-only
28     STATUS current
29     DESCRIPTION
30         "True When receiving a request to activate a Reflection Responder,
31         False When receiving a request to stop Reflection Responder or
32         its timer expires."
33     REFERENCE
34         "12.17.2.2.2 b.12"
35     DEFVAL { false }
36     ::= { ieee8021DdcfmRrEntry 15 }
37
38 ieee8021DdcfmRrRemainDuration OBJECT-TYPE
39     SYNTAX Unsigned32
40     UNITS "seconds"
41     MAX-ACCESS read-only
42     STATUS current
43     DESCRIPTION
44         "The value indicates the time, in the unit of seconds, or count
45         left for Reflection Responder to be active."
46     REFERENCE
47         "12.17.2.2.2 b.13"
48     ::= { ieee8021DdcfmRrEntry 16 }
49
50 ieee8021DdcfmRrNextRfmTransID OBJECT-TYPE
51     SYNTAX Unsigned32
52     MAX-ACCESS read-only
53     STATUS current
54     DESCRIPTION
55         "It indicates the value of RFM Transaction Identifier field of the
56         next RFM transmitted. It is incremented by 1 with each
57         transmission of RFM."
58     REFERENCE
59         "12.17.2.2.2 b.14"
60     ::= { ieee8021DdcfmRrEntry 17 }
61
62 ieee8021DdcfmRrRowStatus OBJECT-TYPE
63     SYNTAX RowStatus
64     MAX-ACCESS read-create
65     STATUS current
66     DESCRIPTION
67         "The status of the row.
68         The writable columns in a row cannot be changed if the row is
69         active."
70     ::= { ieee8021DdcfmRrEntry 18 }
71
72 -- *****

```

```

1 -- The RFM Receiver Table
2 -- *****
3
4 ieee8021DdcfmRFMReceiverTable OBJECT-TYPE
5     SYNTAX SEQUENCE OF Ieee8021DdcfmRFMReceiverEntry
6     MAX-ACCESS not-accessible
7     STATUS current
8     DESCRIPTION
9         "The RFM Receiver MIB object table. Each row in the table
10        represents a different RFM Receiver.
11        All rows associated with this table persist across system restart."
12    REFERENCE
13        "12.17.3"
14    ::= { ieee8021DdcfmRFMReceiver 1 }
15
16 ieee8021DdcfmRFMReceiverEntry OBJECT-TYPE
17     SYNTAX Ieee8021DdcfmRFMReceiverEntry
18     MAX-ACCESS not-accessible
19     STATUS current
20     DESCRIPTION
21         "The DDCFM RFM Receiver. Each entry associated with a DDCFM RFM
22        Receiver that reference to a MP."
23     INDEX { ieee8021DdcfmRfmReceiverIfIndex,
24            ieee8021DdcfmRfmReceiverMdIndex }
25     ::= { ieee8021DdcfmRFMReceiverTable 1 }
26
27 Ieee8021DdcfmRFMReceiverEntry ::= SEQUENCE {
28     ieee8021DdcfmRfmReceiverIfIndex InterfaceIndex,
29     ieee8021DdcfmRfmReceiverMdIndex DotlagCfmMDLevel,
30     ieee8021DdcfmRFMRowStatus RowStatus
31 }
32
33 ieee8021DdcfmRfmReceiverIfIndex OBJECT-TYPE
34     SYNTAX InterfaceIndex
35     MAX-ACCESS not-accessible
36     STATUS current
37     DESCRIPTION
38         "The interface index of the interface either a
39        Bridge Port, or an aggregated port within a Bridge
40        Port, on which the RFM Receiver is created.
41        When the ifIndex value corresponds to the ifIndex of a
42        Bridge Port, the value in this column must match the value in the
43        ieee8021BridgeBasePortIfIndex column for the Bridge Port."
44     REFERENCE
45         "12.17.3.1.2 a.2 "
46     ::= { ieee8021DdcfmRFMReceiverEntry 1 }
47
48 ieee8021DdcfmRfmReceiverMdIndex OBJECT-TYPE
49     SYNTAX DotlagCfmMDLevel
50     MAX-ACCESS not-accessible
51     STATUS current
52     DESCRIPTION
53         "MD level of the RFM Receiver managed object."
54     REFERENCE
55         "12.17.3.1.2 a.2"
56     ::= { ieee8021DdcfmRFMReceiverEntry 2 }
57
58 ieee8021DdcfmRFMRowStatus OBJECT-TYPE
59     SYNTAX RowStatus
60     MAX-ACCESS read-create
61     STATUS current
62     DESCRIPTION
63         "The status of the row.
64        The writable columns in a row cannot be changed if the row is
65        active."
66     ::= { ieee8021DdcfmRFMReceiverEntry 3 }
67
68 -- *****
69 -- The Decapsulator Responder Table
70 -- *****
71 ieee8021DdcfmDrTable OBJECT-TYPE
72     SYNTAX SEQUENCE OF Ieee8021DdcfmDrEntry

```

```

1  MAX-ACCESS not-accessible
2  STATUS current
3  DESCRIPTION
4      "The DDCFM Decapsulator Responder MIB object table. Each row in the
5      table represents a different DDCFM Decapsulator Responder. All rows
6      in this table persist across a system restart; however after such
7      a restart, the value of the ActivationStatus column will be false."
8  REFERENCE
9      "12.17.4"
10 ::= { ieee8021DdcfmDr 1 }
11
12 ieee8021DdcfmDrEntry OBJECT-TYPE
13     SYNTAX Ieee8021DdcfmDrEntry
14     MAX-ACCESS not-accessible
15     STATUS current
16     DESCRIPTION
17         "The DDCFM Decapsulator Responder. Each entry associated with a
18         DDCFM RFM Receiver which reference to a MP."
19     INDEX { ieee8021DdcfmDrIfIndex,
20             ieee8021DdcfmDrMdIndex,
21             ieee8021DdcfmDrVlanIdOrNone }
22     ::= { ieee8021DdcfmDrTable 1 }
23
24 Ieee8021DdcfmDrEntry ::= SEQUENCE {
25     ieee8021DdcfmDrIfIndex InterfaceIndex,
26     ieee8021DdcfmDrMdIndex DotlagCfmMDLevel,
27     ieee8021DdcfmDrVlanIdOrNone VlanIdOrNone,
28     ieee8021DdcfmDrSfmOriginator MacAddress,
29     ieee8021DdcfmDrSourceAddressStayFlag TruthValue,
30     ieee8021DdcfmDrFloodingFlag TruthValue,
31     ieee8021DdcfmDrDuration Unsigned32,
32     ieee8021DdcfmDrActivationStatus TruthValue,
33     ieee8021DdcfmDrRemainDuration Unsigned32,
34     ieee8021DdcfmDrSFMsequenceErrors Unsigned32,
35     ieee8021DdcfmDrRowStatus RowStatus
36 }
37
38 ieee8021DdcfmDrIfIndex OBJECT-TYPE
39     SYNTAX InterfaceIndex
40     MAX-ACCESS not-accessible
41     STATUS current
42     DESCRIPTION
43         "The interface index of the interface either a Bridge
44         Port, or an aggregated port within a Bridge
45         Port, on which the Decapsulator Responder is created.
46         When the ifIndex value corresponds to the ifIndex of a
47         Bridge Port, the value in this column must match the value in the
48         ieee8021BridgeBasePortIfIndex column for the Bridge Port."
49     REFERENCE
50         "12.17.4.1.2 a.2 "
51     ::= { ieee8021DdcfmDrEntry 1 }
52
53 ieee8021DdcfmDrMdIndex OBJECT-TYPE
54     SYNTAX DotlagCfmMDLevel
55     MAX-ACCESS not-accessible
56     STATUS current
57     DESCRIPTION
58         "MD level of the Decapsulator Responder managed object."
59     REFERENCE
60         "12.17.4.1.2 a.2"
61     ::= { ieee8021DdcfmDrEntry 2 }
62
63 ieee8021DdcfmDrVlanIdOrNone OBJECT-TYPE
64     SYNTAX VlanIdOrNone
65     MAX-ACCESS not-accessible
66     STATUS current
67     DESCRIPTION
68         "An integer indicating the VID expected from the Send Frame Message
69         frames."
70     REFERENCE
71         "12.17.4.1.2 a.2"
72     ::= { ieee8021DdcfmDrEntry 3 }

```

```

1
2 ieee8021DdcfmDrSfmOriginator OBJECT-TYPE
3     SYNTAX MacAddress
4     MAX-ACCESS read-create
5     STATUS current
6     DESCRIPTION
7         "MAC address reference to the corresponding Send Frame Message
8         Originator."
9     REFERENCE
10        "12.17.4.2.3 b.2"
11    ::= { ieee8021DdcfmDrEntry 4 }
12
13 ieee8021DdcfmDrSourceAddressStayFlag OBJECT-TYPE
14     SYNTAX TruthValue
15     MAX-ACCESS read-create
16     STATUS current
17     DESCRIPTION
18         "True indicates that Decapsulator Responder does not replace the
19         source_address field of the decapsulated frame with the
20         Decapsulator Responder's own MAC address, False otherwise."
21     REFERENCE
22        "12.17.4.2.3 b.3"
23     DEFVAL { true }
24    ::= { ieee8021DdcfmDrEntry 5 }
25
26 ieee8021DdcfmDrFloodingFlag OBJECT-TYPE
27     SYNTAX TruthValue
28     MAX-ACCESS read-create
29     STATUS current
30     DESCRIPTION
31         "True indicates that broadcast is allowed if Egress port cannot be
32         identified by the Filtering Database, False otherwise."
33     REFERENCE
34        "12.17.4.3.2 b.3"
35     DEFVAL { true }
36    ::= { ieee8021DdcfmDrEntry 6 }
37
38 ieee8021DdcfmDrDuration OBJECT-TYPE
39     SYNTAX Unsigned32
40     UNITS "seconds"
41     MAX-ACCESS read-create
42     STATUS current
43     DESCRIPTION
44         "The time that the Decapsulator Responder can stay active after
45         its activation in the unit of seconds."
46     REFERENCE
47        "12.17.4.3.2 b.4"
48    ::= { ieee8021DdcfmDrEntry 7 }
49
50 ieee8021DdcfmDrActivationStatus OBJECT-TYPE
51     SYNTAX TruthValue
52     MAX-ACCESS read-only
53     STATUS current
54     DESCRIPTION
55         "True When receiving a request to activate a Decapsulator
56         Responder, false When receiving a request to stop the Decapsulator
57         Responder or its timer expires."
58     REFERENCE
59        "12.17.4.2.3 b.6"
60     DEFVAL { false }
61    ::= { ieee8021DdcfmDrEntry 8 }
62
63 ieee8021DdcfmDrRemainDuration OBJECT-TYPE
64     SYNTAX Unsigned32
65     UNITS "seconds"
66     MAX-ACCESS read-only
67     STATUS current
68     DESCRIPTION
69         "The value indicates the time left for Decapsulator Responder keep
70         active. Its granularity is in seconds."
71     REFERENCE
72        "12.17.4.2.3 b.7"

```



```

1 ::= { ieee8021DdcfmDrEntry 9 }
2
3 ieee8021DdcfmDrSFMsequenceErrors OBJECT-TYPE
4     SYNTAX Unsigned32
5     UNITS "integer"
6     MAX-ACCESS read-only
7     STATUS current
8     DESCRIPTION
9         "The value indicates the total number of out-of-sequence SFMs."
10    REFERENCE
11        "12.17.4.2.3 b.8"
12    ::= { ieee8021DdcfmDrEntry 10 }
13
14 ieee8021DdcfmDrRowStatus OBJECT-TYPE
15     SYNTAX RowStatus
16     MAX-ACCESS read-create
17     STATUS current
18     DESCRIPTION
19         "The status of the row.
20          The writable columns in a row cannot be changed if the row is
21          active."
22    ::= { ieee8021DdcfmDrEntry 11 }
23
24 -- *****
25 -- The SFM Originator Table
26 -- *****
27 ieee8021DdcfmSoTable OBJECT-TYPE
28     SYNTAX SEQUENCE OF Ieee8021DdcfmSoEntry
29     MAX-ACCESS not-accessible
30     STATUS current
31     DESCRIPTION
32         "The DDCFM Send Frame Message Originator MIB object table. Each row
33          in the table represents a different DDCFM Send Frame Message
34          Originator. All rows in this table persist across a system restart;
35          however after such a restart, the value of the ActivationStatus
36          column will be false."
37     REFERENCE
38         "12.17.5"
39     ::= { ieee8021DdcfmSFMOriginator 1 }
40
41 ieee8021DdcfmSoEntry OBJECT-TYPE
42     SYNTAX Ieee8021DdcfmSoEntry
43     MAX-ACCESS not-accessible
44     STATUS current
45     DESCRIPTION
46         "Each entry associated with a Send Frame Message Originator."
47     INDEX { ieee8021DdcfmSfmIfIndex,
48             ieee8021DdcfmSoMdIndex,
49             ieee8021DdcfmSoVlanIdOrNone,
50             ieee8021DdcfmSoDirection }
51     ::= { ieee8021DdcfmSoTable 1 }
52
53 Ieee8021DdcfmSoEntry ::= SEQUENCE {
54     ieee8021DdcfmSfmIfIndex InterfaceIndex,
55     ieee8021DdcfmSoMdIndex DotlagCfmMDLevel,
56     ieee8021DdcfmSoVlanIdOrNone VlanIdOrNone,
57     ieee8021DdcfmSoDirection DotlagCfmMpDirection,
58     ieee8021DdcfmSoDrMacAddress MacAddress,
59     ieee8021DdcfmSoDuration Unsigned32,
60     ieee8021DdcfmSoActivationStatus TruthValue,
61     ieee8021DdcfmSoRemainDuration Unsigned32,
62     ieee8021DdcfmSoRowStatus RowStatus
63 }
64
65 ieee8021DdcfmSfmIfIndex OBJECT-TYPE
66     SYNTAX InterfaceIndex
67     MAX-ACCESS not-accessible
68     STATUS current
69     DESCRIPTION
70         "The interface index of the interface either a Bridge
71          port, or an aggregated port within a Bridge
72          port, on which Send Frame Message Originator is created.

```

```

1      When the ifIndex value corresponds to the ifIndex of a
2      Bridge Port, the value in this column must match the value in the
3      ieee8021dBridgeBasePortIfIndex column for the Bridge Port."
4  REFERENCE
5      "12.17.5.1.2 a.2"
6      ::= { ieee8021dDdcfmSoEntry 1 }
7
8  ieee8021dDdcfmSoMdIndex OBJECT-TYPE
9      SYNTAX DotlagCfmMDLevel
10     MAX-ACCESS not-accessible
11     STATUS current
12     DESCRIPTION
13         "MD level of the Send Frame Message Originator managed object."
14     REFERENCE
15         "12.17.5.1.2 a.2"
16     ::= { ieee8021dDdcfmSoEntry 2 }
17
18  ieee8021dDdcfmSoVlanIdOrNone OBJECT-TYPE
19     SYNTAX VlanIdOrNone
20     MAX-ACCESS not-accessible
21     STATUS current
22     DESCRIPTION
23         "An integer indicating the VID to be used on Send Frame Message
24         frames."
25     REFERENCE
26         "12.17.5.1.2 a.2"
27     ::= { ieee8021dDdcfmSoEntry 3 }
28
29  ieee8021dDdcfmSoDirection OBJECT-TYPE
30     SYNTAX DotlagCfmMpDirection
31     MAX-ACCESS not-accessible
32     STATUS current
33     DESCRIPTION
34         "The direction in which the SFM Originator faces."
35     REFERENCE
36         "12.17.5.1.2 a.2"
37     ::= { ieee8021dDdcfmSoEntry 4 }
38
39  ieee8021dDdcfmSoDrMacAddress OBJECT-TYPE
40     SYNTAX MacAddress
41     MAX-ACCESS read-create
42     STATUS current
43     DESCRIPTION
44         "MAC Address of the corresponding Decapsulator Responder."
45     REFERENCE
46         "12.17.5.4.2 b"
47     ::= { ieee8021dDdcfmSoEntry 5 }
48
49  ieee8021dDdcfmSoDuration OBJECT-TYPE
50     SYNTAX Unsigned32
51     UNITS "seconds"
52     MAX-ACCESS read-create
53     STATUS current
54     DESCRIPTION
55         "Duration, in the unit of seconds, of Send Frame Message Originator
56         staying active once activated."
57     REFERENCE
58         "12.17.5.4.2 c"
59     ::= { ieee8021dDdcfmSoEntry 6 }
60
61  ieee8021dDdcfmSoActivationStatus OBJECT-TYPE
62     SYNTAX TruthValue
63     MAX-ACCESS read-only
64     STATUS current
65     DESCRIPTION
66         "True When receiving a request to activate a Send Frame Message
67         Originator, false When receiving a request to stop the Send Frame
68         Message Originator or its timer expires."
69     REFERENCE
70         "12.17.5.2.3 b.4"
71     DEFVAL { false }
72     ::= { ieee8021dDdcfmSoEntry 7 }

```

```

1
2 ieee8021DdcfmSoRemainDuration OBJECT-TYPE
3     SYNTAX Unsigned32
4     UNITS "seconds"
5     MAX-ACCESS read-only
6     STATUS current
7     DESCRIPTION
8         "The value indicates the time left for Send Frame Message
9         Originator keep active. Its granularity is in seconds."
10    REFERENCE
11        "12.17.5.2.3 b.5"
12    ::= { ieee8021DdcfmSoEntry 8 }
13
14 ieee8021DdcfmSoRowStatus OBJECT-TYPE
15     SYNTAX RowStatus
16     MAX-ACCESS read-create
17     STATUS current
18     DESCRIPTION
19         "The status of the row.
20         The writable columns in a row cannot be changed if the row is
21         active."
22    ::= { ieee8021DdcfmSoEntry 9 }
23
24
25 -- *****
26 -- IEEE802.1Qaw MIB Module - Conformance Information
27 -- *****
28 ieee8021DdcfmCompliances OBJECT IDENTIFIER ::= { ieee8021DdcfmConformance 1 }
29 ieee8021DdcfmGroups OBJECT IDENTIFIER ::= { ieee8021DdcfmConformance 2 }
30
31 -- *****
32 -- Units of conformance
33 -- *****
34 ieee8021DdcfmStackGroup OBJECT-GROUP
35     OBJECTS {
36         ieee8021DdcfmStackRrMdLevel,
37         ieee8021DdcfmStackRrDirection,
38         ieee8021DdcfmStackRFMreceiverMdLevel,
39         ieee8021DdcfmStackDrMdLevel,
40         ieee8021DdcfmStackDrVlanIdOrNone,
41         ieee8021DdcfmStackSFMOriginatorMdLevel,
42         ieee8021DdcfmStackSFMOriginatorVlanIdOrNone,
43         ieee8021DdcfmStackSFMOriginatorDirection
44     }
45     STATUS current
46     DESCRIPTION
47         "Objects for the DDCFM Stack group."
48    ::= { ieee8021DdcfmGroups 1 }
49
50 ieee8021DdcfmRrGroup OBJECT-GROUP
51     OBJECTS {
52         ieee8021DdcfmRrPrimaryVlanIdOrNone,
53         ieee8021DdcfmRrFilter,
54         ieee8021DdcfmRrSamplingInterval,
55         ieee8021DdcfmRrTargetAddress,
56         ieee8021DdcfmRrContinueFlag,
57         ieee8021DdcfmRrDuration,
58         ieee8021DdcfmRrDurationInTimeFlag,
59         ieee8021DdcfmRrVlanPriority,
60         ieee8021DdcfmRrVlanDropEligible,
61         ieee8021DdcfmRrFloodingFlag,
62         ieee8021DdcfmRrTruncationFlag,
63         ieee8021DdcfmRrActivationStatus,
64         ieee8021DdcfmRrRemainDuration,
65         ieee8021DdcfmRrNextRfmTransID,
66         ieee8021DdcfmRrRowStatus
67     }
68     STATUS current
69     DESCRIPTION
70         "Objects for the Reflection Responder group."
71    ::= { ieee8021DdcfmGroups 2 }
72

```

```

1 ieee8021DdcfmRFMReceiverGroup OBJECT-GROUP
2   OBJECTS {
3     ieee8021DdcfmRFMRowStatus
4   }
5   STATUS current
6   DESCRIPTION
7     "Objects for the RFM Receiver group."
8   ::= { ieee8021DdcfmGroups 3 }
9
10 ieee8021DdcfmDrGroup OBJECT-GROUP
11   OBJECTS {
12     ieee8021DdcfmDrSourceAddressStayFlag,
13     ieee8021DdcfmDrSfmOriginator,
14     ieee8021DdcfmDrFloodingFlag,
15     ieee8021DdcfmDrDuration,
16     ieee8021DdcfmDrActivationStatus,
17     ieee8021DdcfmDrRemainDuration,
18     ieee8021DdcfmDrSFMsequenceErrors,
19     ieee8021DdcfmDrRowStatus
20   }
21   STATUS current
22   DESCRIPTION
23     "Objects for the Decapsulator Responder group."
24   ::= { ieee8021DdcfmGroups 4 }
25
26 ieee8021DdcfmSoGroup OBJECT-GROUP
27   OBJECTS {
28     ieee8021DdcfmSoDrMacAddress,
29     ieee8021DdcfmSoDuration,
30     ieee8021DdcfmSoActivationStatus,
31     ieee8021DdcfmSoRemainDuration,
32     ieee8021DdcfmSoRowStatus
33   }
34   STATUS current
35   DESCRIPTION
36     "Objects for the Send Frame Message Originator group."
37   ::= { ieee8021DdcfmGroups 5 }
38
39
40 --*****
41 -- MIB Module Compliance statements
42 --*****
43 ieee8021DdcfmCompliance MODULE-COMPLIANCE
44   STATUS current
45   DESCRIPTION
46     "The compliance statement for support of the DDCFM MIB module."
47   MODULE
48   MANDATORY-GROUPS {
49     ieee8021DdcfmStackGroup,
50     ieee8021DdcfmRrGroup,
51     ieee8021DdcfmRFMReceiverGroup,
52     ieee8021DdcfmDrGroup,
53     ieee8021DdcfmSoGroup
54   }
55
56   OBJECT ieee8021DdcfmRrRowStatus
57   SYNTAX RowStatus { active(1), notInService(2) }
58   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
59     destroy(6) }
60   DESCRIPTION "Support for createAndWait is not required."
61
62   OBJECT ieee8021DdcfmRFMRowStatus
63   SYNTAX RowStatus { active(1), notInService(2) }
64   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
65     destroy(6) }
66   DESCRIPTION "Support for createAndWait is not required."
67
68   OBJECT ieee8021DdcfmDrRowStatus
69   SYNTAX RowStatus { active(1), notInService(2) }
70   WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
71     destroy(6) }
72   DESCRIPTION "Support for createAndWait is not required."

```

```
1
2 OBJECT ieee8021DdcfmSoRowStatus
3 SYNTAX RowStatus { active(1), notInService(2) }
4 WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
5     destroy(6) }
6 DESCRIPTION "Support for createAndWait is not required."
7 ::= { ieee8021DdcfmCompliances 1 }
8
9 END
```

1 17.7.10 Definitions for the IEEE8021-PBBTE-MIB module

```
2
3 IEEE8021-PBBTE-MIB DEFINITIONS ::= BEGIN
4
5 -- =====
6 -- MIB for IEEE 802.1Q PBB-TE Devices
7 -- =====
8
9 IMPORTS
10     MODULE-IDENTITY,
11     NOTIFICATION-TYPE,
12     OBJECT-TYPE,
13     Unsigned32
14         FROM SNMPv2-SMI
15     RowStatus,
16     StorageType,
17     TruthValue
18         FROM SNMPv2-TC
19     ieee802dot1mibs,
20     IEEE8021BridgePortNumber,
21     IEEE8021PbbComponentIdentifier,
22     IEEE8021PbbServiceIdentifier,
23     IEEE8021PbbTeEsp,
24     IEEE8021PbbTeProtectionGroupActiveRequests,
25     IEEE8021PbbTeProtectionGroupId,
26     IEEE8021PbbTeProtectionGroupConfigAdmin,
27     IEEE8021PbbTeTSidId,
28     IEEE8021VlanIndexOrWildcard
29         FROM IEEE8021-TC-MIB
30     ieee8021BridgeBaseComponentId
31         FROM IEEE8021-BRIDGE-MIB
32     PortList
33         FROM Q-BRIDGE-MIB
34     ieee8021QBridgeVlanCurrentComponentId
35         FROM IEEE8021-Q-BRIDGE-MIB
36     MODULE-COMPLIANCE,
37     NOTIFICATION-GROUP,
38     OBJECT-GROUP
39         FROM SNMPv2-CONF;
40
41 ieee8021PbbTeMib MODULE-IDENTITY
42     LAST-UPDATED "202211080000Z" -- November 8, 2022
43     ORGANIZATION "IEEE 802.1 Working Group"
44     CONTACT-INFO
45         " WG-URL: http://www.ieee802.org/1/
46         WG-EMail: stds-802-1-1@ieee.org
47         Contact: IEEE 802.1 Working Group Chair
48         Postal: C/O IEEE 802.1 Working Group
49                 IEEE Standards Association
50                 445 Hoes Lane
51                 Piscataway, NJ 08854
52                 USA
53         E-mail: stds-802-1-chairs@ieee.org"
54
55     DESCRIPTION
56         "Unless otherwise indicated, the references in this MIB
57         module are to IEEE Std 802.1Q-2022.
58
59         Copyright (C) IEEE (2022).
60         This version of this MIB module is part of IEEE Std 802.1Q;
61         see that standard for full legal notices."
62
63     REVISION "202211080000Z" -- November 8, 2022
64     DESCRIPTION
65         "Published as part of IEEE Std 802.1Q-2022.
66         Cross references and contact information updated."
67
68     REVISION "201807010000Z" -- July 1, 2018
69     DESCRIPTION
70         "Published as part of IEEE Std 802.1Q 2018 revision.
71         Cross references updated and corrected."
```

```

1
2 REVISION "201412150000Z" -- December 15, 2014
3 DESCRIPTION
4     "Published as part of IEEE Std 802.1Q 2014 revision.
5     Cross references updated and corrected."
6
7 REVISION "201102270000Z" -- February 27, 2011
8 DESCRIPTION
9     "Minor edits to contact information etc. as part of
10    2011 revision of IEEE Std 802.1Q."
11
12 REVISION "200811180000Z" -- November 18, 2008
13 DESCRIPTION
14     "Initial version of the PBB-TE MIB module based upon draft 3.2
15     of the MIB modules defined in IEEE Std 802.1ap and IEEE Std 802.1Qay"
16
17 ::= { ieee802dot1mibs 10 }
18
19 ieee8021PbbTeNotifications OBJECT IDENTIFIER ::= { ieee8021PbbTeMib 0 }
20 ieee8021PbbTeObjects OBJECT IDENTIFIER ::= { ieee8021PbbTeMib 1 }
21 ieee8021PbbTeConformance OBJECT IDENTIFIER ::= { ieee8021PbbTeMib 2 }
22
23 --
24 -- 802.1Qay MIB Objects
25 --
26
27 ieee8021PbbTeProtectionGroupListTable OBJECT-TYPE
28     SYNTAX SEQUENCE OF Ieee8021PbbTeProtectionGroupListEntry
29     MAX-ACCESS not-accessible
30     STATUS current
31     DESCRIPTION
32         "The PBB-TE Protection group list table. Each entry in this table
33         corresponds to a configured PBB-TE Protection Group configured on
34         the B-Component of an IB-BEB."
35     REFERENCE
36         "12.18.1"
37     ::= { ieee8021PbbTeObjects 1 }
38
39 ieee8021PbbTeProtectionGroupListEntry OBJECT-TYPE
40     SYNTAX Ieee8021PbbTeProtectionGroupListEntry
41     MAX-ACCESS not-accessible
42     STATUS current
43     DESCRIPTION
44         "The PBB-TE protection group list table entry.
45         Note that the component ID must refer to an B component"
46     INDEX { ieee8021BridgeBaseComponentId,
47             ieee8021PbbTeProtectionGroupListGroupId }
48     ::= { ieee8021PbbTeProtectionGroupListTable 1 }
49
50 Ieee8021PbbTeProtectionGroupListEntry ::=
51     SEQUENCE {
52         ieee8021PbbTeProtectionGroupListGroupId IEEE8021PbbTeProtectionGroupId,
53         ieee8021PbbTeProtectionGroupListMD Unsigned32,
54         ieee8021PbbTeProtectionGroupListWorkingMA Unsigned32,
55         ieee8021PbbTeProtectionGroupListProtectionMA Unsigned32,
56         ieee8021PbbTeProtectionGroupListStorageType StorageType,
57         ieee8021PbbTeProtectionGroupListRowStatus RowStatus
58     }
59
60 ieee8021PbbTeProtectionGroupListGroupId OBJECT-TYPE
61     SYNTAX IEEE8021PbbTeProtectionGroupId
62     MAX-ACCESS not-accessible
63     STATUS current
64     DESCRIPTION
65         "The unique identifier for the protection group."
66     REFERENCE
67         "12.18.2"
68     ::= { ieee8021PbbTeProtectionGroupListEntry 1 }
69
70 ieee8021PbbTeProtectionGroupListMD OBJECT-TYPE
71     SYNTAX Unsigned32
72     MAX-ACCESS read-create

```

```

1  STATUS      current
2  DESCRIPTION
3      "This refers to the Maintenance Domain that qualifies the
4      WorkingMA and ProtectionMA columns of this table. The MD
5      index in this column must hold a value that matches the
6      in the dotlagCfmStackMdIndex in the dotlagCfmStackTable
7      for the corresponding WorkingMA and ProtectionMA columns
8      of this table. This correspondence must hold for the RowStatus
9      of this row to be set to Active. Furthermore, this column may
10     not be modified while the RowStatus for this row is Active"
11     REFERENCE
12     "12.18.1.1.3 b)"
13 ::= { ieee8021PbbTeProtectionGroupListEntry 2 }
14
15
16 ieee8021PbbTeProtectionGroupListWorkingMA OBJECT-TYPE
17     SYNTAX      Unsigned32
18     MAX-ACCESS  read-create
19     STATUS      current
20     DESCRIPTION
21         "This refers to the Maintenance Association that refers
22         to the PBB-TE MA that corresponds to the group's working
23         entity. The MA index in this column must hold a value
24         that is the value of dotlagCfmStackMaIndex column for
25         some entry in the dotlagCfmStackTable before the RowStatus
26         for this row can be set to Active. Furthermore, this column
27         may not be modified when the RowStatus for this row is Active."
28     REFERENCE
29     "12.18.1.1.3 b)"
30 ::= { ieee8021PbbTeProtectionGroupListEntry 3 }
31
32 ieee8021PbbTeProtectionGroupListProtectionMA OBJECT-TYPE
33     SYNTAX      Unsigned32
34     MAX-ACCESS  read-create
35     STATUS      current
36     DESCRIPTION
37         "This refers to the Maintenance Association that refers
38         to the PBB-TE MA that corresponds to the group's protection
39         entity. The MA index in this column must hold a value
40         that is the value of dotlagCfmStackMaIndex column for
41         some entry in the dotlagCfmStackTable before the RowStatus
42         for this row can be set to Active. Furthermore, this column
43         may not be modified when the RowStatus for this row is Active."
44     REFERENCE
45     "12.18.1.1.3 c)"
46 ::= { ieee8021PbbTeProtectionGroupListEntry 4 }
47
48 ieee8021PbbTeProtectionGroupListStorageType OBJECT-TYPE
49     SYNTAX      StorageType
50     MAX-ACCESS  read-create
51     STATUS      current
52     DESCRIPTION
53         "This object indicates the persistence of this entry. All read-create
54         columns must be writable if this column is set to permanent."
55     DEFVAL { nonVolatile }
56 ::= { ieee8021PbbTeProtectionGroupListEntry 5 }
57
58 ieee8021PbbTeProtectionGroupListRowStatus OBJECT-TYPE
59     SYNTAX      RowStatus
60     MAX-ACCESS  read-create
61     STATUS      current
62     DESCRIPTION
63         "The status of this row.
64
65         The writable columns in a row cannot be changed if the row is
66         active. The PbbTeProtectionGroupListWorkingMA, and
67         PbbTeProtectionGroupListProtectionMA columns must be specified
68         before the row can be activated."
69     REFERENCE
70     "12.18.1.2"
71 ::= { ieee8021PbbTeProtectionGroupListEntry 6 }
72

```



```

1 ieee8021PbbTeMASharedGroupTable OBJECT-TYPE
2   SYNTAX SEQUENCE OF Ieee8021PbbTeMASharedGroupEntry
3   MAX-ACCESS not-accessible
4   STATUS current
5   DESCRIPTION
6     "This table contains references to all protection groups that
7     share a working or protection entity with a given protection
8     group."
9   REFERENCE
10    "12.18.1.1.3 d)"
11    ::= { ieee8021PbbTeObjects 2 }
12
13 ieee8021PbbTeMASharedGroupEntry OBJECT-TYPE
14   SYNTAX Ieee8021PbbTeMASharedGroupEntry
15   MAX-ACCESS not-accessible
16   STATUS current
17   DESCRIPTION
18     "The ieee801PbbTeMASharedGroupEntry table. The table is
19     indexed by protection group and by a simple integer. The
20     table lists all protection groups that load share with that
21     group."
22   INDEX { ieee8021BridgeBaseComponentId,
23           ieee8021PbbTeProtectionGroupListGroupId,
24           ieee8021PbbTeMASharedGroupSubIndex }
25   ::= { ieee8021PbbTeMASharedGroupTable 1 }
26
27 Ieee8021PbbTeMASharedGroupEntry ::=
28   SEQUENCE {
29     ieee8021PbbTeMASharedGroupSubIndex Unsigned32,
30     ieee8021PbbTeMASharedGroupId IEEE8021PbbTeProtectionGroupId
31   }
32
33 ieee8021PbbTeMASharedGroupSubIndex OBJECT-TYPE
34   SYNTAX Unsigned32(1..4294967295)
35   MAX-ACCESS not-accessible
36   STATUS current
37   DESCRIPTION
38     "A simple integer to distinguish the members of the set
39     of MAs that comprise the set of load sharing MAs for the
40     specified protection group."
41   ::= { ieee8021PbbTeMASharedGroupEntry 1 }
42
43 ieee8021PbbTeMASharedGroupId OBJECT-TYPE
44   SYNTAX IEEE8021PbbTeProtectionGroupId
45   MAX-ACCESS read-only
46   STATUS current
47   DESCRIPTION
48     "This column holds the group id of a protection group that shares
49     a working or protection group with the group whose index is the
50     first index of this row."
51   ::= { ieee8021PbbTeMASharedGroupEntry 2 }
52
53
54 ieee8021PbbTeTesiTable OBJECT-TYPE
55   SYNTAX SEQUENCE OF Ieee8021PbbTeTesiEntry
56   MAX-ACCESS not-accessible
57   STATUS current
58   DESCRIPTION
59     "The PBB-TE TESI table contains information for each
60     TE Service Instance known to a system."
61   REFERENCE
62     "12.16.5.3.1"
63   ::= { ieee8021PbbTeObjects 3 }
64
65 ieee8021PbbTeTesiEntry OBJECT-TYPE
66   SYNTAX Ieee8021PbbTeTesiEntry
67   MAX-ACCESS not-accessible
68   STATUS current
69   DESCRIPTION
70     "The PBB-TE TESI entry. Each entry maps a TESI to
71     a component and CBP."
72   INDEX { ieee8021PbbTeTesiId }

```

```

1 ::= { ieee8021PbbTeTesiTable 1 }
2
3 Ieee8021PbbTeTesiEntry ::=
4     SEQUENCE {
5         ieee8021PbbTeTesiId          IEEE8021PbbTeTSidId,
6         ieee8021PbbTeTesiComponent  IEEE8021PbbComponentIdentifier,
7         ieee8021PbbTeTesiBridgePort IEEE8021BridgePortNumber,
8         ieee8021PbbTeTesiStorageType StorageType,
9         ieee8021PbbTeTesiRowStatus  RowStatus
10    }
11
12 ieee8021PbbTeTesiId OBJECT-TYPE
13     SYNTAX      IEEE8021PbbTeTSidId
14     MAX-ACCESS  not-accessible
15     STATUS      current
16     DESCRIPTION
17         "This is the unique identifier for a PBB-TE TE-SID."
18     REFERENCE
19         "3.285"
20     ::= { ieee8021PbbTeTesiEntry 1 }
21
22 ieee8021PbbTeTesiComponent OBJECT-TYPE
23     SYNTAX      IEEE8021PbbComponentIdentifier
24     MAX-ACCESS  read-create
25     STATUS      current
26     DESCRIPTION
27         "This is the component upon which the Bridge Port of the
28         TESI is located."
29     REFERENCE
30         "12.16.5.3.2 a)"
31     ::= { ieee8021PbbTeTesiEntry 2 }
32
33 ieee8021PbbTeTesiBridgePort OBJECT-TYPE
34     SYNTAX      IEEE8021BridgePortNumber
35     MAX-ACCESS  read-create
36     STATUS      current
37     DESCRIPTION
38         "This is the Bridge Port of the TESI."
39     REFERENCE
40         "12.16.5.3.2 b)"
41     ::= { ieee8021PbbTeTesiEntry 3 }
42
43 ieee8021PbbTeTesiStorageType OBJECT-TYPE
44     SYNTAX      StorageType
45     MAX-ACCESS  read-create
46     STATUS      current
47     DESCRIPTION
48         "This object indicates the persistence of this entry. All read-create
49         columns must be writable for rows whose StorageType is permanent."
50     DEFVAL { nonVolatile }
51     ::= { ieee8021PbbTeTesiEntry 4 }
52
53 ieee8021PbbTeTesiRowStatus OBJECT-TYPE
54     SYNTAX      RowStatus
55     MAX-ACCESS  read-create
56     STATUS      current
57     DESCRIPTION
58         "This column holds the status for this row.
59
60         When the status is active, no columns of this table may be
61         modified. All columns must have a valid value before the row
62         can be activated."
63     ::= { ieee8021PbbTeTesiEntry 5 }
64
65 ieee8021PbbTeTesiEspTable OBJECT-TYPE
66     SYNTAX      SEQUENCE OF Ieee8021PbbTeTesiEspEntry
67     MAX-ACCESS  not-accessible
68     STATUS      current
69     DESCRIPTION
70         "The PBB-TE TE-SID table contains information for each TE
71         service instance known to a system."
72     REFERENCE

```

```

1      "12.16.5.3.2 c)"
2      ::= { ieee8021PbbTeObjects 4 }
3
4 ieee8021PbbTeTeSiEspEntry OBJECT-TYPE
5     SYNTAX      Ieee8021PbbTeTeSiEspEntry
6     MAX-ACCESS  not-accessible
7     STATUS      current
8     DESCRIPTION
9         "The PBB-TE TE-SID entry. Each entry refers to a
10        TE Service Instance by identifier and contains information about
11        one of the ESPs that comprise this TE Service Instance."
12     INDEX { ieee8021PbbTeTesiId,
13             ieee8021PbbTeTeSiEspEspIndex }
14     ::= { ieee8021PbbTeTeSiEspTable 1 }
15
16 Ieee8021PbbTeTeSiEspEntry ::=
17     SEQUENCE {
18         ieee8021PbbTeTeSiEspEspIndex      Unsigned32,
19         ieee8021PbbTeTeSiEspEsp           IEEE8021PbbTeEsp,
20         ieee8021PbbTeTeSiEspStorageType   StorageType,
21         ieee8021PbbTeTeSiEspRowStatus     RowStatus
22     }
23
24 ieee8021PbbTeTeSiEspEspIndex OBJECT-TYPE
25     SYNTAX      Unsigned32 (1..4294967295)
26     MAX-ACCESS  not-accessible
27     STATUS      current
28     DESCRIPTION
29         "This is an identifier, of local significance to a particular
30         PBB-TE TE-SID that is used to index all of the ESPs associated
31         with the TE-SID."
32     REFERENCE
33         "12.16.5.3.2 c)"
34     ::= { ieee8021PbbTeTeSiEspEntry 1 }
35
36 ieee8021PbbTeTeSiEspEsp OBJECT-TYPE
37     SYNTAX      IEEE8021PbbTeEsp
38     MAX-ACCESS  read-create
39     STATUS      current
40     DESCRIPTION
41         "This column holds the ESP identifier for one of the Ethernet
42         Switched Paths that define the TE service instance."
43     REFERENCE
44         "12.16.5.3.2 c)"
45     ::= { ieee8021PbbTeTeSiEspEntry 2 }
46
47 ieee8021PbbTeTeSiEspStorageType OBJECT-TYPE
48     SYNTAX      StorageType
49     MAX-ACCESS  read-create
50     STATUS      current
51     DESCRIPTION
52         "This object indicates the persistence of this entry. All read-create
53         columns must be writable for permanent rows."
54     DEFVAL { nonVolatile }
55     ::= { ieee8021PbbTeTeSiEspEntry 3 }
56
57 ieee8021PbbTeTeSiEspRowStatus OBJECT-TYPE
58     SYNTAX      RowStatus
59     MAX-ACCESS  read-create
60     STATUS      current
61     DESCRIPTION
62         "This column holds the status for this row.
63
64         When the status is active, no columns of this table may be
65         modified. All columns must have a valid value before the row
66         can be activated."
67     ::= { ieee8021PbbTeTeSiEspEntry 4 }
68
69 ieee8021PbbTeProtectionGroupConfigTable OBJECT-TYPE
70     SYNTAX      SEQUENCE OF Ieee8021PbbTeProtectionGroupConfigEntry
71     MAX-ACCESS  not-accessible
72     STATUS      current

```

```

1  DESCRIPTION
2
3      "The PBB-TE Protection group config table contains
4      configuration and status information for each configuration
5      group configured in the system. Entries in this table are
6      created implicitly by the creation of entries in the
7      ieee8021PbbTeProtectionGroupListTable table."
8  REFERENCE
9      "12.18.2"
10 ::= { ieee8021PbbTeObjects 5 }
11
12 ieee8021PbbTeProtectionGroupConfigEntry OBJECT-TYPE
13     SYNTAX      Ieee8021PbbTeProtectionGroupConfigEntry
14     MAX-ACCESS  not-accessible
15     STATUS      current
16     DESCRIPTION
17         "The protection group configuration table entry. Rows are
18         created in this table implicitly when a row is added to the
19         ieee8021PbbTeProtectionGroupListTable."
20     INDEX { ieee8021BridgeBaseComponentId,
21             ieee8021PbbTeProtectionGroupListGroupId }
22     ::= { ieee8021PbbTeProtectionGroupConfigTable 1 }
23
24 Ieee8021PbbTeProtectionGroupConfigEntry ::=
25     SEQUENCE {
26         ieee8021PbbTeProtectionGroupConfigState
27         INTEGER,
28         ieee8021PbbTeProtectionGroupConfigCommandStatus
29         IEEE8021PbbTeProtectionGroupConfigAdmin,
30         ieee8021PbbTeProtectionGroupConfigCommandLast
31         IEEE8021PbbTeProtectionGroupConfigAdmin,
32         ieee8021PbbTeProtectionGroupConfigCommandAdmin
33         IEEE8021PbbTeProtectionGroupConfigAdmin,
34         ieee8021PbbTeProtectionGroupConfigActiveRequests
35         IEEE8021PbbTeProtectionGroupActiveRequests,
36         ieee8021PbbTeProtectionGroupConfigWTR
37         Unsigned32,
38         ieee8021PbbTeProtectionGroupConfigHoldOff
39         Unsigned32,
40         ieee8021PbbTeProtectionGroupConfigNotifyEnable
41         TruthValue,
42         ieee8021PbbTeProtectionGroupConfigStorageType
43         StorageType
44     }
45
46 ieee8021PbbTeProtectionGroupConfigState OBJECT-TYPE
47     SYNTAX      INTEGER {
48         workingPath(1),
49         protectionPat(2),
50         waitToRestore(3),
51         protAdmin(4)
52     }
53     MAX-ACCESS  read-only
54     STATUS      current
55     DESCRIPTION
56         "This column indicates the current state of the protection
57         switching state machine for a protection group."
58     REFERENCE  "26.10.3.5 12.18.2.1.3 c)"
59     ::= { ieee8021PbbTeProtectionGroupConfigEntry 1 }
60
61 ieee8021PbbTeProtectionGroupConfigCommandStatus OBJECT-TYPE
62     SYNTAX      IEEE8021PbbTeProtectionGroupConfigAdmin
63     MAX-ACCESS  read-only
64     STATUS      current
65     DESCRIPTION
66         "This column indicates the status of administrative commands
67         within the protection group. It reflects the current
68         operational administrative command being acted upon by the
69         protection group."
70     REFERENCE  "12.18.2.1.3 d)"
71     ::= { ieee8021PbbTeProtectionGroupConfigEntry 2 }
72

```

```

1 ieee8021PbbTeProtectionGroupConfigCommandLast OBJECT-TYPE
2   SYNTAX      IEEE8021PbbTeProtectionGroupConfigAdmin
3   MAX-ACCESS  read-only
4   STATUS      current
5   DESCRIPTION
6       "This column indicates the last attempted administrative
7       command applied to the protection group. It is changed
8       whenever a write is made to the CommandAdmin column of
9       this table and is a record of the last attempted
10      administrative operation."
11      REFERENCE "12.18.2.1.3 d)"
12      ::= { ieee8021PbbTeProtectionGroupConfigEntry 3 }
13
14 ieee8021PbbTeProtectionGroupConfigCommandAdmin OBJECT-TYPE
15   SYNTAX      IEEE8021PbbTeProtectionGroupConfigAdmin
16   MAX-ACCESS  read-create
17   STATUS      current
18   DESCRIPTION
19       "This column is used by the operator to request that the
20       protection group state machine perform some administrative
21       operation. The operator requests a command by writing the
22       command value to this column. The state machine indicates the
23       command that it is performing by setting the value of the
24       CommandStatus column of this table. This column always reads
25       back as clear(1)."
26       REFERENCE "12.18.2.3.2"
27       DEFVAL { clear }
28       ::= { ieee8021PbbTeProtectionGroupConfigEntry 4 }
29
30 ieee8021PbbTeProtectionGroupConfigActiveRequests OBJECT-TYPE
31   SYNTAX      IEEE8021PbbTeProtectionGroupActiveRequests
32   MAX-ACCESS  read-only
33   STATUS      current
34   DESCRIPTION
35       "This column shows the status of active requests within
36       the TE protection group."
37       REFERENCE "12.18.2.1.3 d)"
38       ::= { ieee8021PbbTeProtectionGroupConfigEntry 5 }
39
40 ieee8021PbbTeProtectionGroupConfigWTR OBJECT-TYPE
41   SYNTAX      Unsigned32 ( 0 | 5..12 )
42   MAX-ACCESS  read-create
43   STATUS      current
44   DESCRIPTION
45       "This column is used to configure the wait-to-restore timer
46       for the protection group operation. The timer may be
47       configured in steps of 1 minute between 5 and 12 minutes, the
48       default being 5. Additionally, the value 0 is used to
49       indicate that the protection group is to operate
50       non-revertively."
51       REFERENCE "26.10.3.3.8 12.18.2.1.3 e)"
52       DEFVAL { 5 }
53       ::= { ieee8021PbbTeProtectionGroupConfigEntry 6 }
54
55 ieee8021PbbTeProtectionGroupConfigHoldOff OBJECT-TYPE
56   SYNTAX      Unsigned32 ( 0..100 )
57   MAX-ACCESS  read-create
58   STATUS      current
59   DESCRIPTION
60       "This column is used to configure the hold off
61       timer. The purpose is to allow a service layer protection
62       mechanism to have a chance to fix the problem before
63       switching at the client layer, or to allow an upstream
64       protected domain to switch before a downstream domain. The
65       hold off timer has a period of from 0 to 10 seconds, the
66       default being 0, with a 100ms granularity."
67       REFERENCE "12.18.2.1.3 f)"
68       DEFVAL { 0 }
69       ::= { ieee8021PbbTeProtectionGroupConfigEntry 7 }
70
71 ieee8021PbbTeProtectionGroupConfigNotifyEnable OBJECT-TYPE
72   SYNTAX      TruthValue

```

```

1  MAX-ACCESS read-create
2  STATUS current
3  DESCRIPTION
4      "This column is used to enable or disable transmission
5      of ieee8021PbbTeProtectionGroupAdminFailure
6      notifications. These notifications are generated
7      whenever an administrative command cannot be performed
8      by the protection group."
9  DEFVAL { false }
10 ::= { ieee8021PbbTeProtectionGroupConfigEntry 8 }
11
12 ieee8021PbbTeProtectionGroupConfigStorageType OBJECT-TYPE
13     SYNTAX StorageType
14     MAX-ACCESS read-create
15     STATUS current
16     DESCRIPTION
17         "This object indicates the persistence of this entry. For
18         permanent objects the
19         ieee8021PbbTeProtectionGroupConfigCommandAdmin column
20         must be writable."
21
22     DEFVAL { nonVolatile }
23     ::= { ieee8021PbbTeProtectionGroupConfigEntry 9 }
24
25 ieee8021PbbTeProtectionGroupISidTable OBJECT-TYPE
26     SYNTAX SEQUENCE OF Ieee8021PbbTeProtectionGroupISidEntry
27     MAX-ACCESS not-accessible
28     STATUS current
29     DESCRIPTION
30         "This table contains entries for each I-SID that is
31         transported over TE-SIDs that belong to protection groups.
32         Each I-SID maps to a single protection group."
33     REFERENCE "12.18.2.1.3 b)"
34     ::= { ieee8021PbbTeObjects 6 }
35
36 ieee8021PbbTeProtectionGroupISidEntry OBJECT-TYPE
37     SYNTAX Ieee8021PbbTeProtectionGroupISidEntry
38     MAX-ACCESS not-accessible
39     STATUS current
40     DESCRIPTION
41         "The ieee8021PbbTeProtectionGroupISidTable entry."
42     INDEX { ieee8021PbbTeProtectionGroupISidIndex }
43     ::= { ieee8021PbbTeProtectionGroupISidTable 1 }
44
45 Ieee8021PbbTeProtectionGroupISidEntry ::=
46     SEQUENCE {
47         ieee8021PbbTeProtectionGroupISidIndex IEEE8021PbbServiceIdentifier,
48         ieee8021PbbTeProtectionGroupISidComponentId IEEE8021PbbComponentIdentifier,
49         ieee8021PbbTeProtectionGroupISidGroupId IEEE8021PbbTeProtectionGroupId,
50         ieee8021PbbTeProtectionGroupISidStorageType StorageType,
51         ieee8021PbbTeProtectionGroupISidRowStatus RowStatus
52     }
53
54 ieee8021PbbTeProtectionGroupISidIndex OBJECT-TYPE
55     SYNTAX IEEE8021PbbServiceIdentifier
56     MAX-ACCESS not-accessible
57     STATUS current
58     DESCRIPTION
59         "This is the I-Sid that is to be mapped to a protection
60         group."
61     ::= { ieee8021PbbTeProtectionGroupISidEntry 1 }
62
63 ieee8021PbbTeProtectionGroupISidComponentId OBJECT-TYPE
64     SYNTAX IEEE8021PbbComponentIdentifier
65     MAX-ACCESS read-create
66     STATUS current
67     DESCRIPTION
68         "This column qualifies the GroupId column to a particular
69         component."
70     ::= { ieee8021PbbTeProtectionGroupISidEntry 2 }
71
72 ieee8021PbbTeProtectionGroupISidGroupId OBJECT-TYPE

```

```

1      SYNTAX      IEEE8021PbbTeProtectionGroupId
2      MAX-ACCESS  read-create
3      STATUS      current
4      DESCRIPTION
5          "This column contains the Id of the protection group used
6          to transport the data belonging to the service identified
7          by the I-SID value specified in the ISidIndex column of this
8          table."
9      ::= { ieee8021PbbTeProtectionGroupISidEntry 3 }
10
11 ieee8021PbbTeProtectionGroupISidStorageType OBJECT-TYPE
12     SYNTAX      StorageType
13     MAX-ACCESS  read-create
14     STATUS      current
15     DESCRIPTION
16         "This object indicates the persistence of this entry."
17     DEFVAL { nonVolatile }
18     ::= { ieee8021PbbTeProtectionGroupISidEntry 4 }
19
20 ieee8021PbbTeProtectionGroupISidRowStatus OBJECT-TYPE
21     SYNTAX      RowStatus
22     MAX-ACCESS  read-create
23     STATUS      current
24     DESCRIPTION
25         "This column contains the status for this row. Once active
26         none of the columns in the row may be modified. All columns
27         must be specified when creating the row."
28     ::= { ieee8021PbbTeProtectionGroupISidEntry 5 }
29
30 ieee8021PbbTeBridgeStaticForwardAnyUnicastTable OBJECT-TYPE
31     SYNTAX SEQUENCE OF Ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry
32     MAX-ACCESS not-accessible
33     STATUS      current
34     DESCRIPTION
35         "A table containing forwarding information for each vlan
36         specifying the set of ports to which forwarding of unicast
37         addressed frames for which no more specific forwarding information
38         applies. This is configured statically by management."
39     REFERENCE "8.8.1"
40     ::= { ieee8021PbbTeObjects 7 }
41
42 ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry OBJECT-TYPE
43     SYNTAX Ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry
44     MAX-ACCESS not-accessible
45     STATUS      current
46     DESCRIPTION
47         "Forwarding information for a VLAN, specifying the
48         set of ports to which forwarding of unicast addressed
49         frames for which no more specific forwarding information
50         applies.
51
52         The EgressPorts and ForbiddenPorts PortList objects, together,
53         implement the PortMap control element listed in IEEE Std 802.1Q 8.8.1.c."
54     INDEX { ieee8021QBridgeVlanCurrentComponentId,
55             ieee8021PbbTeBridgeStaticForwardAnyUnicastVlanIndex }
56     ::= { ieee8021PbbTeBridgeStaticForwardAnyUnicastTable 1 }
57
58
59 Ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry ::=
60     SEQUENCE {
61         ieee8021PbbTeBridgeStaticForwardAnyUnicastVlanIndex
62         IEEE8021VlanIndexOrWildcard,
63         ieee8021PbbTeBridgeStaticForwardAnyUnicastEgressPorts
64         PortList,
65         ieee8021PbbTeBridgeStaticForwardAnyUnicastForbiddenPorts
66         PortList,
67         ieee8021PbbTeBridgeStaticForwardAnyUnicastStorageType
68         StorageType,
69         ieee8021PbbTeBridgeStaticForwardAnyUnicastRowStatus
70         RowStatus
71     }
72

```

```
1 ieee8021PbbTeBridgeStaticForwardAnyUnicastVlanIndex OBJECT-TYPE
2   SYNTAX IEEE8021VlanIndexOrWildcard
3   MAX-ACCESS not-accessible
4   STATUS current
5   DESCRIPTION
6     "The VLAN-ID or other identifier referring to the VLAN to
7     which this static filtering entry applies."
8   ::= { ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry 1 }
9
10 ieee8021PbbTeBridgeStaticForwardAnyUnicastEgressPorts OBJECT-TYPE
11   SYNTAX PortList
12   MAX-ACCESS read-create
13   STATUS current
14   DESCRIPTION
15     "The complete set of ports to which a unicast addressed frame
16     is to be forwarded. This value is persistent and will be restored
17     upon system reboot. A port may not be added to this set if it
18     is already a member of
19     ieee8021PbbTeBridgeStaticForwardAnyUnicastForbiddenPorts. The default
20     value is a string of zeros of appropriate length.
21
22     The value of this object MUST be retained across
23     reinitialization of the management system."
24   REFERENCE "8.8.1 c)"
25   ::= { ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry 2 }
26
27 ieee8021PbbTeBridgeStaticForwardAnyUnicastForbiddenPorts OBJECT-TYPE
28   SYNTAX PortList
29   MAX-ACCESS read-create
30   STATUS current
31   DESCRIPTION
32     "The complete set of ports to which a unicast addressed frame
33     is to be filtered. This value is persistent and will be restored
34     upon system reboot. A port may not be added to this set if it
35     is already a member of
36     ieee8021PbbTeBridgeStaticForwardAnyUnicastEgress. The default
37     value is a string of zeros of appropriate length.
38
39     The value of this object MUST be retained across
40     reinitialization of the management system."
41   REFERENCE "8.8.1 c)"
42   ::= { ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry 3 }
43
44 ieee8021PbbTeBridgeStaticForwardAnyUnicastStorageType OBJECT-TYPE
45   SYNTAX StorageType
46   MAX-ACCESS read-create
47   STATUS current
48   DESCRIPTION
49     "The storage type for this row. All read-create columns must be
50     writable for permanent entries."
51   DEFVAL { nonVolatile }
52   ::= { ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry 4 }
53
54 ieee8021PbbTeBridgeStaticForwardAnyUnicastRowStatus OBJECT-TYPE
55   SYNTAX RowStatus
56   MAX-ACCESS read-create
57   STATUS current
58   DESCRIPTION
59     "This column contains the status for this row."
60   ::= { ieee8021PbbTeBridgeStaticForwardAnyUnicastEntry 5 }
61
62 -- *****
63 -- NOTIFICATIONS (TRAPS)
64 -- These notifications will be sent to the management entity
65 -- whenever a ProtectionGroup admin command cannot be performed
66 -- *****
67
68 ieee8021PbbTeProtectionGroupAdminFailure NOTIFICATION-TYPE
69   OBJECTS {
70     ieee8021PbbTeProtectionGroupConfigState,
71     ieee8021PbbTeProtectionGroupConfigCommandStatus,
72     ieee8021PbbTeProtectionGroupConfigCommandLast
```



```

1      }
2      STATUS current
3      DESCRIPTION
4          "A protection group generates this notification whenever
5          an administrative command cannot be executed by the
6          protection state machine. For example, a requested
7          manual switch cannot be performed because of a signal
8          failure condition.
9
10         The management entity receiving the notification can
11         identify the system from the network source address of
12         the notification and can identify the protection group
13         by the indices of the OID
14         of the ieee8021PbbTeProtectionGroupConfigState variable in the
15         notification:
16
17         ieee8021BridgeBaseComponentId - Identifies the
18         component on the Bridge where the protection group
19         is configured.
20
21         ieee8021PbbTeProtectionGroupListGroupId - The id
22         of the protection group.
23     "
24     ::= { ieee8021PbbTeNotifications 1 }
25
26 --
27 -- MIB Module Compliance Statements
28 --
29
30 ieee8021PbbTeCompliances OBJECT IDENTIFIER ::= { ieee8021PbbTeConformance 1 }
31 ieee8021PbbTeGroups      OBJECT IDENTIFIER ::= { ieee8021PbbTeConformance 2 }
32
33 --
34 -- Units of Conformance
35
36 ieee8021PbbTeListGroup OBJECT-GROUP
37     OBJECTS {
38         ieee8021PbbTeProtectionGroupListMD,
39         ieee8021PbbTeProtectionGroupListWorkingMA,
40         ieee8021PbbTeProtectionGroupListProtectionMA,
41         ieee8021PbbTeProtectionGroupListStorageType,
42         ieee8021PbbTeProtectionGroupListRowStatus
43     }
44     STATUS current
45     DESCRIPTION
46         "Objects for the GroupList group."
47     ::= { ieee8021PbbTeGroups 1 }
48
49 ieee8021PbbTeMASharedGroup OBJECT-GROUP
50     OBJECTS {
51         ieee8021PbbTeMASharedGroupId
52     }
53     STATUS current
54     DESCRIPTION
55         "Objects for the MA Load Sharing Table Group."
56     ::= { ieee8021PbbTeGroups 2 }
57
58
59 ieee8021PbbTeTesiGroup OBJECT-GROUP
60     OBJECTS {
61         ieee8021PbbTeTesiComponent,
62         ieee8021PbbTeTesiBridgePort,
63         ieee8021PbbTeTesiStorageType,
64         ieee8021PbbTeTesiRowStatus
65     }
66     STATUS current
67     DESCRIPTION
68         "Objects for the TE SI group "
69     ::= { ieee8021PbbTeGroups 3 }
70
71 ieee8021PbbTeSiEspGroup OBJECT-GROUP
72     OBJECTS {

```

```
1         ieee8021PbbTeTeSiEspEsp,  
2         ieee8021PbbTeTeSiEspStorageType,  
3         ieee8021PbbTeTeSiEspRowStatus  
4     }  
5     STATUS current  
6     DESCRIPTION  
7         "Objects for the TESI ESP group."  
8     ::= { ieee8021PbbTeGroups 4 }  
9  
10 ieee8021PbbTeProtectionConfigManGroup OBJECT-GROUP  
11     OBJECTS {  
12         ieee8021PbbTeProtectionGroupConfigState,  
13         ieee8021PbbTeProtectionGroupConfigCommandStatus,  
14         ieee8021PbbTeProtectionGroupConfigCommandLast,  
15         ieee8021PbbTeProtectionGroupConfigCommandAdmin,  
16         ieee8021PbbTeProtectionGroupConfigActiveRequests,  
17         ieee8021PbbTeProtectionGroupConfigNotifyEnable,  
18         ieee8021PbbTeProtectionGroupConfigStorageType  
19     }  
20     STATUS current  
21     DESCRIPTION  
22         "Objects for the PbbTeConfiguration group."  
23     ::= { ieee8021PbbTeGroups 5 }  
24  
25 ieee8021PbbTeProtectionConfigOptGroup OBJECT-GROUP  
26     OBJECTS {  
27         ieee8021PbbTeProtectionGroupConfigWTR,  
28         ieee8021PbbTeProtectionGroupConfigHoldOff  
29     }  
30     STATUS current  
31     DESCRIPTION  
32         "Objects for the PbbTeConfiguration group."  
33     ::= { ieee8021PbbTeGroups 6 }  
34  
35 ieee8021PbbTeProtectionGroupISidGroup OBJECT-GROUP  
36     OBJECTS {  
37         ieee8021PbbTeProtectionGroupISidComponentId,  
38         ieee8021PbbTeProtectionGroupISidGroupId,  
39         ieee8021PbbTeProtectionGroupISidStorageType,  
40         ieee8021PbbTeProtectionGroupISidRowStatus  
41     }  
42     STATUS current  
43     DESCRIPTION  
44         "Objects for the ieee8021PbbTeProtectionGroupISidGroup group."  
45     ::= { ieee8021PbbTeGroups 7 }  
46  
47 ieee8021PbbTeFdbGroup OBJECT-GROUP  
48     OBJECTS {  
49         ieee8021PbbTeBridgeStaticForwardAnyUnicastEgressPorts,  
50         ieee8021PbbTeBridgeStaticForwardAnyUnicastForbiddenPorts,  
51         ieee8021PbbTeBridgeStaticForwardAnyUnicastStorageType,  
52         ieee8021PbbTeBridgeStaticForwardAnyUnicastRowStatus  
53     }  
54     STATUS current  
55     DESCRIPTION  
56         "Fdb extension objects group"  
57     ::= { ieee8021PbbTeGroups 8 }  
58  
59 ieee8021PbbTeNotificationsGroup NOTIFICATION-GROUP  
60     NOTIFICATIONS {  
61         ieee8021PbbTeProtectionGroupAdminFailure  
62     }  
63     STATUS current  
64     DESCRIPTION  
65         "Objects for the notifications group."  
66     ::= { ieee8021PbbTeGroups 9 }  
67  
68 ieee8021PbbTeCompliance MODULE-COMPLIANCE  
69     STATUS current  
70     DESCRIPTION  
71         "The compliance statement for support of the PBB-TE MIB module."  
72     MODULE
```

```
1      MANDATORY-GROUPS {
2          ieee8021PbbTeGroupListGroup,
3          ieee8021PbbTeMASharedGroup,
4          ieee8021PbbTeTesiGroup,
5          ieee8021PbbTeSiEspGroup,
6          ieee8021PbbTeProtectionConfigManGroup,
7          ieee8021PbbTeProtectionGroupISidGroup,
8          ieee8021PbbTeFdbGroup,
9          ieee8021PbbTeNotificationsGroup
10     }
11     GROUP ieee8021PbbTeProtectionConfigOptGroup
12     DESCRIPTION
13         "This group allows implementation to optionally change the
14         WaitToRestore and HoldOff timers for protection groups."
15
16     OBJECT ieee8021PbbTeProtectionGroupConfigWTR
17     MIN-ACCESS not-accessible
18     DESCRIPTION "This object is optional."
19
20     OBJECT ieee8021PbbTeProtectionGroupConfigHoldOff
21     MIN-ACCESS not-accessible
22     DESCRIPTION "This object is optional."
23
24     OBJECT ieee8021PbbTeProtectionGroupListRowStatus
25     SYNTAX      RowStatus { active(1), notInService(2) }
26     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
27                             destroy(6) }
28     DESCRIPTION "Support for createAndWait is not required."
29
30     OBJECT ieee8021PbbTeTeSiEspRowStatus
31     SYNTAX      RowStatus { active(1), notInService(2) }
32     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
33                             destroy(6) }
34     DESCRIPTION "Support for createAndWait is not required."
35
36     OBJECT ieee8021PbbTeProtectionGroupISidRowStatus
37     SYNTAX      RowStatus { active(1), notInService(2) }
38     WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
39                             destroy(6) }
40     DESCRIPTION "Support for createAndWait is not required."
41
42     ::= { ieee8021PbbTeCompliances 1 }
43
44 END
45
```

1 17.7.11 Definitions for the IEEE8021-TPMR-MIB module

```
2
3 IEEE8021-TPMR-MIB DEFINITIONS ::= BEGIN
4
5 -- =====
6 -- MIB for IEEE 802.1Q TPMR Devices
7 -- =====
8
9 IMPORTS
10  MODULE-IDENTITY, OBJECT-TYPE, Counter32, Counter64,
11  Unsigned32
12    FROM SNMPv2-SMI
13  TEXTUAL-CONVENTION, TruthValue, MacAddress, TimeInterval,
14  StorageType
15    FROM SNMPv2-TC
16  MODULE-COMPLIANCE, OBJECT-GROUP
17    FROM SNMPv2-CONF
18  ifCounterDiscontinuityGroup
19    FROM IF-MIB
20  IEEE8021BridgePortNumber, ieee802dot1mibs
21    FROM IEEE8021-TC-MIB
22  ieee8021BridgeBasePortComponentId
23    FROM IEEE8021-BRIDGE-MIB;
24
25 ieee8021TpmrMib MODULE-IDENTITY
26   LAST-UPDATED "202211080000Z" -- November 8, 2022
27   ORGANIZATION "IEEE 802.1 Working Group"
28   CONTACT-INFO
29     " WG-URL: http://www.ieee802.org/1/
30     WG-EMail: stds-802-1-1@ieee.org
31     Contact: IEEE 802.1 Working Group Chair
32     Postal: C/O IEEE 802.1 Working Group
33             IEEE Standards Association
34             445 Hoes Lane
35             Piscataway, NJ 08854
36             USA
37     E-mail: stds-802-1-chairs@ieee.org"
38   DESCRIPTION
39     "Two-Port MAC Relay (TPMR) MIB module.
40
41     Unless otherwise indicated, the references in this MIB
42     module are to IEEE Std 802.1Q-2022.
43
44     Copyright (C) IEEE (2022).
45     This version of this MIB module is part of IEEE Std 802.1Q;
46     see that standard for full legal notices."
47
48   REVISION "202211080000Z" -- November 8, 2022
49   DESCRIPTION
50     "Published as part of IEEE Std 802.1Q-2022.
51     Cross references and contact information updated."
52
53   REVISION "201807010000Z" -- July 1, 2018
54   DESCRIPTION
55     "Published as part of IEEE Std 802.1Q 2018 revision.
56     Cross references updated and corrected."
57
58   REVISION "201412150000Z" -- December 15, 2014
59   DESCRIPTION
60     "Published as part of IEEE Std 802.1Q 2014 revision.
61     Cross references updated and corrected."
62
63   REVISION "201102270000Z" -- February 27, 2011
64   DESCRIPTION
65     "Minor edits to contact information etc. as part of
66     2011 revision of IEEE Std 802.1Q."
67
68   REVISION "200909040000Z" -- September 4, 2009
69   DESCRIPTION
70     "Initial version as published in IEEE Std 802.1aj"
71 ::= { ieee802dot1mibs 14 }
```

```

1
2 ieee8021TpmrNotifications OBJECT IDENTIFIER ::= { ieee8021TpmrMib 0 }
3 ieee8021TpmrObjects       OBJECT IDENTIFIER ::= { ieee8021TpmrMib 1 }
4 ieee8021TpmrConformance   OBJECT IDENTIFIER ::= { ieee8021TpmrMib 2 }
5
6 -- -----
7 -- Textual conventions
8 -- -----
9
10 IEEE8021TpmrFrameDiscardErrorReason ::= TEXTUAL-CONVENTION
11     STATUS current
12     DESCRIPTION
13         "A reason code for a frame discard error."
14     REFERENCE
15         "12.19.3.1.1.3:h"
16     SYNTAX INTEGER {
17         txSduSizeExceeded (1) -- transmissible SDU size exceeded
18     }
19
20 -- -----
21 -- ieee8021TpmrPort objects
22 -- -----
23
24 ieee8021TpmrPortTable OBJECT-TYPE
25     SYNTAX SEQUENCE OF Ieee8021TpmrPortEntry
26     MAX-ACCESS not-accessible
27     STATUS current
28     DESCRIPTION
29         "The TPMR port table. Each row in the table represents a
30         TPMR port. By definition there are two ports per TPMR.
31
32         Note that the indices of this table are equivalent to
33         those of the ieee8021BridgeBasePortTable in the
34         IEEE8021-BRIDGE-MIB, with ieee8021TpmrPortNumber having
35         a more limited value range than ieee8021BridgeBasePort."
36     REFERENCE
37         "12.19.1.2.1, 12.19.1.2.2"
38     ::= { ieee8021TpmrObjects 1 }
39
40 ieee8021TpmrPortEntry OBJECT-TYPE
41     SYNTAX Ieee8021TpmrPortEntry
42     MAX-ACCESS not-accessible
43     STATUS current
44     DESCRIPTION
45         "A TPMR port table entry."
46     INDEX { ieee8021BridgeBasePortComponentId,
47             ieee8021TpmrPortNumber }
48     ::= { ieee8021TpmrPortTable 1 }
49
50 Ieee8021TpmrPortEntry ::= SEQUENCE {
51     ieee8021TpmrPortNumber      IEEE8021BridgePortNumber,
52     ieee8021TpmrPortMgmtAddr    TruthValue,
53     ieee8021TpmrPortMgmtAddrForwarding TruthValue
54 }
55
56 ieee8021TpmrPortNumber OBJECT-TYPE
57     SYNTAX IEEE8021BridgePortNumber (1..2)
58     MAX-ACCESS not-accessible
59     STATUS current
60     DESCRIPTION
61         "The number of this TPMR port."
62     REFERENCE
63         "12.19.1.1.1.3:b,1"
64     ::= { ieee8021TpmrPortEntry 1 }
65
66 ieee8021TpmrPortMgmtAddr OBJECT-TYPE
67     SYNTAX TruthValue
68     MAX-ACCESS read-only
69     STATUS current
70     DESCRIPTION
71         "Is 'true' if the TPMR port MAC address is the management
72         address of the TPMR, otherwise 'false'."

```

```

1  REFERENCE
2  "12.19.1.1.1.3:b,3"
3  ::= { ieee8021TpmrPortEntry 2 }
4
5  ieee8021TpmrPortMgmtAddrForwarding OBJECT-TYPE
6  SYNTAX      TruthValue
7  MAX-ACCESS  read-only
8  STATUS      current
9  DESCRIPTION
10     "Is 'true' if forwarding is enabled for frames destined to the
11     management address of the TPMR, otherwise 'false'."
12  REFERENCE
13     "12.19.1.2.1.3:c"
14  ::= { ieee8021TpmrPortEntry 3 }
15
16  -- -----
17  -- ieee8021TpmrPortStats objects
18  -- -----
19
20  ieee8021TpmrPortStatsTable OBJECT-TYPE
21  SYNTAX SEQUENCE OF Ieee8021TpmrPortStatsEntry
22  MAX-ACCESS not-accessible
23  STATUS      current
24  DESCRIPTION
25     "The TPMR port statistics table. Each row in the table
26     represents a TPMR port. By definition there are two
27     ports per TPMR.
28
29     Discontinuities in the value of counters in this table
30     can occur at re-initialization of the management system,
31     and at other times as indicated by the value of IF-MIB
32     ifCounterDiscontinuityTime."
33  REFERENCE
34     "12.19.3.1"
35  ::= { ieee8021TpmrObjects 2 }
36
37  ieee8021TpmrPortStatsEntry OBJECT-TYPE
38  SYNTAX      Ieee8021TpmrPortStatsEntry
39  MAX-ACCESS  not-accessible
40  STATUS      current
41  DESCRIPTION
42     "A TPMR port counters table entry."
43  AUGMENTS { ieee8021TpmrPortEntry }
44  ::= { ieee8021TpmrPortStatsTable 1 }
45
46  Ieee8021TpmrPortStatsEntry ::= SEQUENCE {
47      ieee8021TpmrPortStatsRxFrames      Counter64,
48      ieee8021TpmrPortStatsRxOctets      Counter64,
49      ieee8021TpmrPortStatsFramesForwarded Counter64,
50      ieee8021TpmrPortStatsFramesDiscarded Counter64,
51      ieee8021TpmrPortStatsFramesDiscardedQueueFull Counter64,
52      ieee8021TpmrPortStatsFramesDiscardedLifetime Counter64,
53      ieee8021TpmrPortStatsFramesDiscardedError Counter64
54  }
55
56  ieee8021TpmrPortStatsRxFrames OBJECT-TYPE
57  SYNTAX      Counter64
58  UNITS       "frames"
59  MAX-ACCESS  read-only
60  STATUS      current
61  DESCRIPTION
62     "Count of all valid frames received on this port (including
63     BPDUs, frames addressed to the TPMR as an end station, and
64     frames that were submitted to the Forwarding Process)."
65  REFERENCE
66     "12.19.3.1.1.3:a"
67  ::= { ieee8021TpmrPortStatsEntry 1 }
68
69  ieee8021TpmrPortStatsRxOctets OBJECT-TYPE
70  SYNTAX      Counter64
71  UNITS       "octets"
72  MAX-ACCESS  read-only

```

```
1  STATUS      current
2  DESCRIPTION
3      "Count of the total number of octets in all valid frames
4      received on this port (including BPDUs, frames addressed
5      to the TPMR as an end station, and frames that were
6      submitted to the Forwarding Process)."
```

7 REFERENCE

```
8      "12.19.3.1.1.3:b"
9      ::= { ieee8021TpmrPortStatsEntry 2 }
```

10

11 ieee8021TpmrPortStatsFramesForwarded OBJECT-TYPE

```
12  SYNTAX      Counter64
13  UNITS       "frames"
14  MAX-ACCESS  read-only
15  STATUS      current
16  DESCRIPTION
17      "Count of all frames that were received on this port and
18      were forwarded to the transmission port."
```

19 REFERENCE

```
20      "12.19.3.1.1.3:d"
21      ::= { ieee8021TpmrPortStatsEntry 3 }
```

22

23 ieee8021TpmrPortStatsFramesDiscarded OBJECT-TYPE

```
24  SYNTAX      Counter64
25  UNITS       "frames"
26  MAX-ACCESS  read-only
27  STATUS      current
28  DESCRIPTION
29      "Count of all frames that were received on this port but
30      were discarded by the Forwarding Process for any reason."
```

31 REFERENCE

```
32      "12.19.3.1.1.3:c"
33      ::= { ieee8021TpmrPortStatsEntry 4 }
```

34

35 ieee8021TpmrPortStatsFramesDiscardedQueueFull OBJECT-TYPE

```
36  SYNTAX      Counter64
37  UNITS       "frames"
38  MAX-ACCESS  read-only
39  STATUS      current
40  DESCRIPTION
41      "Count of all frames received on this port that were to
42      be transmitted through the transmission port but were
43      discarded due to lack of available queue space."
```

44 REFERENCE

```
45      "12.19.3.1.1.3:e"
46      ::= { ieee8021TpmrPortStatsEntry 5 }
```

47

48 ieee8021TpmrPortStatsFramesDiscardedLifetime OBJECT-TYPE

```
49  SYNTAX      Counter64
50  UNITS       "frames"
51  MAX-ACCESS  read-only
52  STATUS      current
53  DESCRIPTION
54      "Count of all frames received on this port that were to
55      be transmitted through the transmission port but were
56      discarded due to their frame lifetime having been
57      exceeded."
```

58 REFERENCE

```
59      "12.19.3.1.1.3:f"
60      ::= { ieee8021TpmrPortStatsEntry 6 }
```

61

62 ieee8021TpmrPortStatsFramesDiscardedError OBJECT-TYPE

```
63  SYNTAX      Counter64
64  UNITS       "frames"
65  MAX-ACCESS  read-only
66  STATUS      current
67  DESCRIPTION
68      "Count of all frames received on this port that were to
69      be transmitted through the transmission port but could
70      not be transmitted (e.g., frame too large)."
```

71 REFERENCE

```
72      "12.19.3.1.1.3:g"
```

```

1  ::= { ieee8021TpmrPortStatsEntry 7 }
2
3  -- -----
4  -- ieee8021TpmrPortDiscardDetails objects
5  -- -----
6
7  ieee8021TpmrPortDiscardDetailsTable OBJECT-TYPE
8      SYNTAX SEQUENCE OF Ieee8021TpmrPortDiscardDetailsEntry
9      MAX-ACCESS not-accessible
10     STATUS current
11     DESCRIPTION
12         "The TPMR frames discard details table. Each row in
13         the table represents a discarded frame on a TPMR port.
14         By definition there are two ports per TPMR.
15
16         This table is maintained as a FIFO. A new entry is
17         inserted in the first row, and existing entries are
18         shuffled down, with the last entry being discarded.
19
20         Because of the FIFO behavior, the relationship between
21         the index and contents will change when an entry is
22         added to the table. This may result in apparent
23         duplication of row content during a table traversal."
24     REFERENCE
25         "12.19.3.1.1.3:h"
26     ::= { ieee8021TpmrObjects 3 }
27
28  ieee8021TpmrPortDiscardDetailsEntry OBJECT-TYPE
29      SYNTAX Ieee8021TpmrPortDiscardDetailsEntry
30      MAX-ACCESS not-accessible
31      STATUS current
32      DESCRIPTION
33         "A TPMR frames discarded error details table entry."
34      INDEX { ieee8021BridgeBasePortComponentId,
35              ieee8021TpmrPortNumber,
36              ieee8021TpmrPortDiscardDetailsIndex }
37      ::= { ieee8021TpmrPortDiscardDetailsTable 1 }
38
39  Ieee8021TpmrPortDiscardDetailsEntry ::= SEQUENCE {
40      ieee8021TpmrPortDiscardDetailsIndex Unsigned32,
41      ieee8021TpmrPortDiscardDetailsSource MacAddress,
42      ieee8021TpmrPortDiscardDetailsReason IEEE8021TpmrFrameDiscardErrorReason
43  }
44
45  ieee8021TpmrPortDiscardDetailsIndex OBJECT-TYPE
46      SYNTAX Unsigned32 (1..16)
47      MAX-ACCESS not-accessible
48      STATUS current
49      DESCRIPTION
50         "The second index of a TPMR frames discard details
51         table entry."
52      ::= { ieee8021TpmrPortDiscardDetailsEntry 1 }
53
54  ieee8021TpmrPortDiscardDetailsSource OBJECT-TYPE
55      SYNTAX MacAddress
56      MAX-ACCESS read-only
57      STATUS current
58      DESCRIPTION
59         "The source MAC address of the discarded frame."
60      REFERENCE
61         "12.19.3.1.1.3:h"
62      ::= { ieee8021TpmrPortDiscardDetailsEntry 2 }
63
64  ieee8021TpmrPortDiscardDetailsReason OBJECT-TYPE
65      SYNTAX IEEE8021TpmrFrameDiscardErrorReason
66      MAX-ACCESS read-only
67      STATUS current
68      DESCRIPTION
69         "The reason why the frame was discarded."
70      REFERENCE
71         "12.19.3.1.1.3:h"
72      ::= { ieee8021TpmrPortDiscardDetailsEntry 3 }

```



```
1
2 -- -----
3 -- ieee8021TpmrMsp objects
4 -- -----
5
6 ieee8021TpmrMspTable OBJECT-TYPE
7   SYNTAX SEQUENCE OF Ieee8021TpmrMspEntry
8   MAX-ACCESS not-accessible
9   STATUS current
10  DESCRIPTION
11    "The TPMR MAC status propagation performance table. Each
12     row in the table represents a TPMR port. By definition
13     there are two ports per TPMR.
14
15     The persistence of writable objects in a conceptual row
16     of this table is determined by the value of the
17     ieee8021TpmrMspStorageType object."
18  REFERENCE
19    "12.19.4.1.1, 12.19.4.1.2"
20  ::= { ieee8021TpmrObjects 4 }
21
22 ieee8021TpmrMspEntry OBJECT-TYPE
23   SYNTAX Ieee8021TpmrMspEntry
24   MAX-ACCESS not-accessible
25   STATUS current
26   DESCRIPTION
27     "A TPMR MAC status propagation performance table entry."
28   AUGMENTS { ieee8021TpmrPortEntry }
29   ::= { ieee8021TpmrMspTable 1 }
30
31 Ieee8021TpmrMspEntry ::= SEQUENCE {
32   ieee8021TpmrMspLinkNotify TruthValue,
33   ieee8021TpmrMspLinkNotifyWait TimeInterval,
34   ieee8021TpmrMspLinkNotifyRetry TimeInterval,
35   ieee8021TpmrMspMacNotify TruthValue,
36   ieee8021TpmrMspMacNotifyTime TimeInterval,
37   ieee8021TpmrMspMacRecoverTime TimeInterval,
38   ieee8021TpmrMspStorageType StorageType
39 }
40
41 ieee8021TpmrMspLinkNotify OBJECT-TYPE
42   SYNTAX TruthValue
43   MAX-ACCESS read-write
44   STATUS current
45   DESCRIPTION
46     "The value of LinkNotify used by the MSP state machines."
47   REFERENCE
48     "12.19.4.1.1.3:a, 12.19.4.1.2.2:b"
49   DEFVAL { true }
50   ::= { ieee8021TpmrMspEntry 1 }
51
52 ieee8021TpmrMspLinkNotifyWait OBJECT-TYPE
53   SYNTAX TimeInterval (20..100)
54   UNITS "centiseconds"
55   MAX-ACCESS read-write
56   STATUS current
57   DESCRIPTION
58     "The value of LinkNotifyWait used by the MSP state machines."
59   REFERENCE
60     "12.19.4.1.1.3:b, 12.19.4.1.2.2:c"
61   DEFVAL { 40 }
62   ::= { ieee8021TpmrMspEntry 2 }
63
64 ieee8021TpmrMspLinkNotifyRetry OBJECT-TYPE
65   SYNTAX TimeInterval (10..100)
66   UNITS "centiseconds"
67   MAX-ACCESS read-write
68   STATUS current
69   DESCRIPTION
70     "The value of LinkNotifyRetry used by the MSP state machines."
71   REFERENCE
72     "12.19.4.1.1.3:c, 12.19.4.1.2.2:d"
```

```

1  DEFVAL { 100 }
2  ::= { ieee8021TpmrMspEntry 3 }
3
4  ieee8021TpmrMspMacNotify OBJECT-TYPE
5      SYNTAX      TruthValue
6      MAX-ACCESS  read-write
7      STATUS      current
8      DESCRIPTION
9          "The value of MacNotify used by the MSP state machines."
10     REFERENCE
11         "12.19.4.1.1.3:d, 12.19.4.1.2.2:e"
12     DEFVAL { true }
13     ::= { ieee8021TpmrMspEntry 4 }
14
15  ieee8021TpmrMspMacNotifyTime OBJECT-TYPE
16      SYNTAX      TimeInterval (1..50)
17      UNITS       "centiseconds"
18      MAX-ACCESS  read-write
19      STATUS      current
20      DESCRIPTION
21          "The value of MacNotifyTime used by the MSP state machines."
22     REFERENCE
23         "12.19.4.1.1.3:e, 12.19.4.1.2.2:f"
24     DEFVAL { 20 }
25     ::= { ieee8021TpmrMspEntry 5 }
26
27  ieee8021TpmrMspMacRecoverTime OBJECT-TYPE
28      SYNTAX      TimeInterval (2..50)
29      UNITS       "centiseconds"
30      MAX-ACCESS  read-write
31      STATUS      current
32      DESCRIPTION
33          "The value of MacRecoverTime used by the MSP state machines."
34     REFERENCE
35         "12.19.4.1.1.3:f, 12.19.4.1.2.2:g"
36     DEFVAL { 10 }
37     ::= { ieee8021TpmrMspEntry 6 }
38
39  ieee8021TpmrMspStorageType OBJECT-TYPE
40      SYNTAX      StorageType
41      MAX-ACCESS  read-write
42      STATUS      current
43      DESCRIPTION
44          "The storage type for all read-write objects within this
45          row. Conceptual rows having the value 'permanent' need
46          not allow write access to any columnar objects in the row.
47
48          If this object has the value 'volatile', modifications
49          to read-write objects in this row are not persistent
50          across reboots or restarts. If this object has the value
51          'nonVolatile', modifications to objects in this row
52          are persistent."
53     DEFVAL { nonVolatile }
54     ::= { ieee8021TpmrMspEntry 7 }
55
56  -- -----
57  -- ieee8021TpmrMspStats objects
58  -- -----
59
60  ieee8021TpmrMspStatsTable OBJECT-TYPE
61      SYNTAX SEQUENCE OF Ieee8021TpmrMspStatsEntry
62      MAX-ACCESS not-accessible
63      STATUS      current
64      DESCRIPTION
65          "The TPMR MAC status propagation statistics table. Each
66          row in the table represents a TPMR port. By definition
67          there are two ports per TPMR.
68
69          Discontinuities in the value of counters in this table
70          can occur at re-initialization of the management system,
71          and at other times as indicated by the value of IF-MIB
72          ifCounterDiscontinuityTime."

```

```

1  REFERENCE
2      "12.19.4.1.3"
3  ::= { ieee8021TpmrObjects 5 }
4
5  ieee8021TpmrMspStatsEntry OBJECT-TYPE
6      SYNTAX      Ieee8021TpmrMspStatsEntry
7      MAX-ACCESS  not-accessible
8      STATUS      current
9      DESCRIPTION
10         "A TPMR MAC status propagation statistics table entry."
11     AUGMENTS { ieee8021TpmrPortEntry }
12     ::= { ieee8021TpmrMspStatsTable 1 }
13
14 Ieee8021TpmrMspStatsEntry ::= SEQUENCE {
15     ieee8021TpmrMspStatsTxAcks          Counter32,
16     ieee8021TpmrMspStatsTxAddNotifications Counter32,
17     ieee8021TpmrMspStatsTxAddConfirmations Counter32,
18     ieee8021TpmrMspStatsTxLossNotifications Counter32,
19     ieee8021TpmrMspStatsTxLossConfirmations Counter32,
20     ieee8021TpmrMspStatsRxAcks          Counter32,
21     ieee8021TpmrMspStatsRxAddNotifications Counter32,
22     ieee8021TpmrMspStatsRxAddConfirmations Counter32,
23     ieee8021TpmrMspStatsRxLossNotifications Counter32,
24     ieee8021TpmrMspStatsRxLossConfirmations Counter32,
25     ieee8021TpmrMspStatsAddEvents       Counter32,
26     ieee8021TpmrMspStatsLossEvents      Counter32,
27     ieee8021TpmrMspStatsMacStatusNotifications Counter32
28 }
29
30 ieee8021TpmrMspStatsTxAcks OBJECT-TYPE
31     SYNTAX      Counter32
32     UNITS        "MSPDUs"
33     MAX-ACCESS  read-only
34     STATUS      current
35     DESCRIPTION
36         "The number of acks transmitted by the port's Transmit
37         Process as a consequence of txAck being set."
38     REFERENCE
39         "12.19.4.1.3.3:a"
40     ::= { ieee8021TpmrMspStatsEntry 1 }
41
42 ieee8021TpmrMspStatsTxAddNotifications OBJECT-TYPE
43     SYNTAX      Counter32
44     UNITS        "MSPDUs"
45     MAX-ACCESS  read-only
46     STATUS      current
47     DESCRIPTION
48         "The number of adds transmitted by the port's Transmit
49         Process as a consequence of txAdd being set."
50     REFERENCE
51         "12.19.4.1.3.3:b"
52     ::= { ieee8021TpmrMspStatsEntry 2 }
53
54 ieee8021TpmrMspStatsTxAddConfirmations OBJECT-TYPE
55     SYNTAX      Counter32
56     UNITS        "MSPDUs"
57     MAX-ACCESS  read-only
58     STATUS      current
59     DESCRIPTION
60         "The number of add confirms transmitted by the port's
61         Transmit Process as a consequence of txAddConfirm
62         being set."
63     REFERENCE
64         "12.19.4.1.3.3:c"
65     ::= { ieee8021TpmrMspStatsEntry 3 }
66
67 ieee8021TpmrMspStatsTxLossNotifications OBJECT-TYPE
68     SYNTAX      Counter32
69     UNITS        "MSPDUs"
70     MAX-ACCESS  read-only
71     STATUS      current
72     DESCRIPTION

```

```
1      "The number of losses transmitted by the port's Transmit
2      Process as a consequence of txLoss being set."
3  REFERENCE
4      "12.19.4.1.3.3:d"
5  ::= { ieee8021TpmrMspStatsEntry 4 }
6
7  ieee8021TpmrMspStatsTxLossConfirmations OBJECT-TYPE
8      SYNTAX      Counter32
9      UNITS       "MSPDUs"
10     MAX-ACCESS   read-only
11     STATUS       current
12     DESCRIPTION
13         "The number of loss confirms transmitted by the port's
14         Transmit Process as a consequence of txLossConfirm
15         being set."
16     REFERENCE
17         "12.19.4.1.3.3:e"
18     ::= { ieee8021TpmrMspStatsEntry 5 }
19
20  ieee8021TpmrMspStatsRxAcks OBJECT-TYPE
21     SYNTAX      Counter32
22     UNITS       "MSPDUs"
23     MAX-ACCESS   read-only
24     STATUS       current
25     DESCRIPTION
26         "The number of acks received by the port's Receive
27         Process."
28     REFERENCE
29         "12.19.4.1.3.3:f"
30     ::= { ieee8021TpmrMspStatsEntry 6 }
31
32  ieee8021TpmrMspStatsRxAddNotifications OBJECT-TYPE
33     SYNTAX      Counter32
34     UNITS       "MSPDUs"
35     MAX-ACCESS   read-only
36     STATUS       current
37     DESCRIPTION
38         "The number of adds received by the port's Receive
39         Process."
40     REFERENCE
41         "12.19.4.1.3.3:g"
42     ::= { ieee8021TpmrMspStatsEntry 7 }
43
44  ieee8021TpmrMspStatsRxAddConfirmations OBJECT-TYPE
45     SYNTAX      Counter32
46     UNITS       "MSPDUs"
47     MAX-ACCESS   read-only
48     STATUS       current
49     DESCRIPTION
50         "The number of add confirms received by the port's
51         Receive Process."
52     REFERENCE
53         "12.19.4.1.3.3:h"
54     ::= { ieee8021TpmrMspStatsEntry 8 }
55
56  ieee8021TpmrMspStatsRxLossNotifications OBJECT-TYPE
57     SYNTAX      Counter32
58     UNITS       "MSPDUs"
59     MAX-ACCESS   read-only
60     STATUS       current
61     DESCRIPTION
62         "The number of losses received by the port's Receive
63         Process."
64     REFERENCE
65         "12.19.4.1.3.3:i"
66     ::= { ieee8021TpmrMspStatsEntry 9 }
67
68  ieee8021TpmrMspStatsRxLossConfirmations OBJECT-TYPE
69     SYNTAX      Counter32
70     UNITS       "MSPDUs"
71     MAX-ACCESS   read-only
72     STATUS       current
```

```
1  DESCRIPTION
2      "The number of loss confirms received by the port's
3      Receive Process."
4  REFERENCE
5      "12.19.4.1.3.3:j"
6      ::= { ieee8021TpmrMspStatsEntry 10 }
7
8  ieee8021TpmrMspStatsAddEvents OBJECT-TYPE
9      SYNTAX      Counter32
10     UNITS        "MSP transitions"
11     MAX-ACCESS   read-only
12     STATUS       current
13     DESCRIPTION
14         "The number of transitions to STM:ADD directly from
15         STM:DOWN or STM:LOSS."
16     REFERENCE
17         "12.19.4.1.3.3:k"
18     ::= { ieee8021TpmrMspStatsEntry 11 }
19
20  ieee8021TpmrMspStatsLossEvents OBJECT-TYPE
21     SYNTAX      Counter32
22     UNITS        "MSP transitions"
23     MAX-ACCESS   read-only
24     STATUS       current
25     DESCRIPTION
26         "The number of transitions to STM:LOSS directly from
27         STM:UP or STM:ADD."
28     REFERENCE
29         "12.19.4.1.3.3:l"
30     ::= { ieee8021TpmrMspStatsEntry 12 }
31
32  ieee8021TpmrMspStatsMacStatusNotifications OBJECT-TYPE
33     SYNTAX      Counter32
34     UNITS        "MSP transitions"
35     MAX-ACCESS   read-only
36     STATUS       current
37     DESCRIPTION
38         "The number of transitions to SNM:MAC_NOTIFICATION."
39     REFERENCE
40         "12.19.4.1.3.3:m"
41     ::= { ieee8021TpmrMspStatsEntry 13 }
42
43 -- -----
44 -- IEEE 802.1aj MIB - Conformance Information
45 -- -----
46
47  ieee8021TpmrCompliances OBJECT IDENTIFIER ::= { ieee8021TpmrConformance 1 }
48  ieee8021TpmrGroups      OBJECT IDENTIFIER ::= { ieee8021TpmrConformance 2 }
49
50 -- -----
51 -- Units of conformance
52 -- -----
53
54  ieee8021TpmrPortGroup OBJECT-GROUP
55      OBJECTS {
56          ieee8021TpmrPortMgmtAddr,
57          ieee8021TpmrPortMgmtAddrForwarding
58      }
59      STATUS current
60      DESCRIPTION
61          "TPMR port objects."
62      ::= { ieee8021TpmrGroups 1 }
63
64  ieee8021TpmrPortStatsGroup OBJECT-GROUP
65      OBJECTS {
66          ieee8021TpmrPortStatsRxFrames,
67          ieee8021TpmrPortStatsRxOctets,
68          ieee8021TpmrPortStatsFramesForwarded,
69          ieee8021TpmrPortStatsFramesDiscarded,
70          ieee8021TpmrPortStatsFramesDiscardedQueueFull,
71          ieee8021TpmrPortStatsFramesDiscardedLifetime,
72          ieee8021TpmrPortStatsFramesDiscardedError
```

```
1     }
2     STATUS current
3     DESCRIPTION
4         "TPMR port statistics objects."
5     ::= { ieee8021TpmrGroups 2 }
6
7 ieee8021TpmrPortDiscardDetailsGroup OBJECT-GROUP
8     OBJECTS {
9         ieee8021TpmrPortDiscardDetailsSource,
10        ieee8021TpmrPortDiscardDetailsReason
11    }
12    STATUS current
13    DESCRIPTION
14        "TPMR port discard details objects."
15    ::= { ieee8021TpmrGroups 3 }
16
17 ieee8021TpmrMspGroup OBJECT-GROUP
18     OBJECTS {
19         ieee8021TpmrMspLinkNotify,
20         ieee8021TpmrMspLinkNotifyWait,
21         ieee8021TpmrMspLinkNotifyRetry,
22         ieee8021TpmrMspMacNotify,
23         ieee8021TpmrMspMacNotifyTime,
24         ieee8021TpmrMspMacRecoverTime,
25         ieee8021TpmrMspStorageType
26     }
27     STATUS current
28     DESCRIPTION
29         "TPMR port MSP objects."
30     ::= { ieee8021TpmrGroups 4 }
31
32 ieee8021TpmrMspStatsGroup OBJECT-GROUP
33     OBJECTS {
34         ieee8021TpmrMspStatsTxAcks,
35         ieee8021TpmrMspStatsTxAddNotifications,
36         ieee8021TpmrMspStatsTxAddConfirmations,
37         ieee8021TpmrMspStatsTxLossNotifications,
38         ieee8021TpmrMspStatsTxLossConfirmations,
39         ieee8021TpmrMspStatsRxAcks,
40         ieee8021TpmrMspStatsRxAddNotifications,
41         ieee8021TpmrMspStatsRxAddConfirmations,
42         ieee8021TpmrMspStatsRxLossNotifications,
43         ieee8021TpmrMspStatsRxLossConfirmations,
44         ieee8021TpmrMspStatsAddEvents,
45         ieee8021TpmrMspStatsLossEvents,
46         ieee8021TpmrMspStatsMacStatusNotifications
47     }
48     STATUS current
49     DESCRIPTION
50         "TPMR port MSP statistics objects."
51     ::= { ieee8021TpmrGroups 5 }
52
53 -- -----
54 -- Compliance statements
55 -- -----
56
57 ieee8021TpmrCompliance MODULE-COMPLIANCE
58     STATUS current
59     DESCRIPTION
60         "The compliance statement for device support of TPMR."
61     MODULE IF-MIB
62         MANDATORY-GROUPS {
63             ifCounterDiscontinuityGroup
64         }
65     MODULE
66         MANDATORY-GROUPS {
67             ieee8021TpmrPortGroup,
68             ieee8021TpmrPortStatsGroup,
69             ieee8021TpmrPortDiscardDetailsGroup,
70             ieee8021TpmrMspGroup,
71             ieee8021TpmrMspStatsGroup
72         }
```

```
1 ::= { ieee8021TpmrCompliances 1 }  
2  
3 END  
4  
5
```

1 17.7.12 Definitions for the IEEE8021-FQTSS-MIB module

```
2 IEEE8021-FQTSS-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for support of the Forwarding & Queuing Enhancements
6 -- for Time-Sensitive Streams (FQTSS) in IEEE 802.1Q Bridges.
7 -- =====
8
9 IMPORTS
10     MODULE-IDENTITY,
11     OBJECT-TYPE,
12     Unsigned32
13         FROM SNMPv2-SMI
14     TEXTUAL-CONVENTION,
15     TruthValue,
16     RowStatus
17         FROM SNMPv2-TC
18     MODULE-COMPLIANCE,
19     OBJECT-GROUP
20         FROM SNMPv2-CONF
21     ieee802dot1mibs,
22     IEEE8021PriorityValue
23         FROM IEEE8021-TC-MIB
24     ieee8021BridgeBaseComponentId,
25     ieee8021BridgeBaseEntry,
26     ieee8021BridgeBasePort,
27     ieee8021BridgeBasePortEntry
28         FROM IEEE8021-BRIDGE-MIB
29     BridgeId
30         FROM BRIDGE-MIB
31 ;
32
33 ieee8021FqtssMib MODULE-IDENTITY
34     LAST-UPDATED "202211080000Z" -- November 8, 2022
35     ORGANIZATION "IEEE 802.1 Working Group"
36     CONTACT-INFO
37         " WG-URL: http://www.ieee802.org/1/
38         WG-EMail: stds-802-1-1@ieee.org
39         Contact: IEEE 802.1 Working Group Chair
40         Postal: C/O IEEE 802.1 Working Group
41             IEEE Standards Association
42             445 Hoes Lane
43             Piscataway, NJ 08854
44             USA
45         E-mail: stds-802-1-chairs@ieee.org"
46     DESCRIPTION
47         "The Bridge MIB module for managing devices that support
48         the Forwarding and Queuing Enhancements
49         for Time-Sensitive Streams.
50
51         Unless otherwise indicated, the references in this MIB
52         module are to IEEE Std 802.1Q-2022.
53
54         Copyright (C) IEEE (2022).
55         This version of this MIB module is part of IEEE Std 802.1Q;
56         see that standard for full legal notices."
57
58     REVISION "202211080000Z" -- November 8, 2022
59     DESCRIPTION
60         "Published as part of IEEE Std 802.1Q-2022.
61         Cross references and contact information updated."
62
63     REVISION "201810040000Z" -- October 4, 2018
64     DESCRIPTION
65         "Published as part of IEEE 802.1Qcc-2018.
66         Added managed objects for Stream Reservation
67         Protocol (SRP) Enhancements and Performance
68         Improvements"
69
70     REVISION "201806280000Z" -- June 28, 2018
71     DESCRIPTION
```



```
1      "Published as part of IEEE Std 802.1Q 2018.
2      Cross references corrected and updated. "
3
4      REVISION "201512020000Z" -- December 2, 2015
5      DESCRIPTION
6          "Published as part of IEEE Std 802.1Q 2014 Cor-1.
7          ETS code point added to the textual convention
8          IEEE8021FqtssTxSelectionAlgorithmIDValue "
9
10     REVISION "201412150000Z" -- December 15, 2014
11     DESCRIPTION
12         "Published as part of IEEE Std 802.1Q 2014 revision.
13         Cross references updated and corrected."
14
15     REVISION      "201102270000Z" -- February 27, 2011
16     DESCRIPTION
17         "Minor edits to contact information etc. as part of
18         2011 revision of IEEE Std 802.1Q."
19
20     REVISION      "200910010000Z" -- October 1, 2009
21     DESCRIPTION
22         "Initial revision, included in IEEE 802.1Qav."
23     ::= { ieee802dot1mibs 16 }
24
25 -- =====
26 -- Textual Conventions
27 -- =====
28
29 IEEE8021FqtssTrafficClassValue ::= TEXTUAL-CONVENTION
30     DISPLAY-HINT "d"
31     STATUS      current
32     DESCRIPTION
33         "An 802.1 FQTSS traffic class value.
34         This is the numerical value associated with a traffic
35         class in a Bridge. Larger values are associated with
36         higher priority traffic classes."
37     REFERENCE   "12.20.1"
38     SYNTAX      Unsigned32 (0..7)
39
40 IEEE8021FqtssDeltaBandwidthValue ::= TEXTUAL-CONVENTION
41     DISPLAY-HINT "d"
42     STATUS      current
43     DESCRIPTION
44         "An 802.1 FQTSS delta bandwidth percentage,
45         represented as a fixed point number scaled by
46         1 000 000."
47     REFERENCE   "12.20.1, 34.4"
48     SYNTAX      Unsigned32 (0..100000000)
49
50 IEEE8021FqtssTxSelectionAlgorithmIDValue ::= TEXTUAL-CONVENTION
51     DISPLAY-HINT "d"
52     STATUS      current
53     DESCRIPTION
54         "An 802.1 transmission selection algorithm identifier
55         value. This is an integer, with the following
56         interpretation placed on the value:
57
58         0: Strict priority algorithm,
59         1: Credit-based shaper algorithm,
60         2: Enhanced Transmission Selection algorithm,
61         3-255: Reserved for future standardization,
62         256-4294967295: Vendor-specific transmission selection
63                       algorithm identifiers, consisting of a
64                       four-octet integer, where the most
65                       significant 3 octets hold an OUI or CID value,
66                       and the least significant octet holds
67                       an integer value in the range 0-255
68                       assigned by the owner of the OUI or CID."
69     REFERENCE   "8.6.4, 12.20.2"
70     SYNTAX      Unsigned32
71
72 -- =====
```

```

1 -- subtrees in the FQTSS MIB
2 -- =====
3
4 ieee8021FqtssNotifications
5   OBJECT IDENTIFIER ::= { ieee8021FqtssMib 0 }
6
7 ieee8021FqtssObjects
8   OBJECT IDENTIFIER ::= { ieee8021FqtssMib 1 }
9
10 ieee8021FqtssConformance
11   OBJECT IDENTIFIER ::= { ieee8021FqtssMib 2 }
12
13 ieee8021FqtssBap
14   OBJECT IDENTIFIER ::= { ieee8021FqtssObjects 1 }
15
16 ieee8021FqtssMappings
17   OBJECT IDENTIFIER ::= { ieee8021FqtssObjects 2 }
18
19 ieee8021FqtssBapX
20   OBJECT IDENTIFIER ::= { ieee8021FqtssObjects 3 }
21
22 -- =====
23 -- The ieee8021FqtssBap subtree
24 -- This subtree defines the objects necessary for the management
25 -- of bandwidth allocation for queues that support FQTSS.
26 -- =====
27
28 -- =====
29 -- the ieee8021FqtssBapTable
30 -- =====
31
32 ieee8021FqtssBapTable OBJECT-TYPE
33   SYNTAX      SEQUENCE OF Ieee8021FqtssBapEntry
34   MAX-ACCESS  not-accessible
35   STATUS      current
36   DESCRIPTION
37     "A table containing a set of bandwidth availability
38     parameters for each traffic class that supports the
39     credit-based shaper algorithm.
40     All writable objects in this table must be
41     persistent over power up restart/reboot."
42   REFERENCE   "12.20.1"
43   ::= { ieee8021FqtssBap 1 }
44
45 ieee8021FqtssBapEntry OBJECT-TYPE
46   SYNTAX      Ieee8021FqtssBapEntry
47   MAX-ACCESS  not-accessible
48   STATUS      current
49   DESCRIPTION
50     "A list of objects containing bandwidth allocation
51     information for each traffic class that supports the
52     credit-based shaper algorithm. Rows in the table are
53     automatically created and deleted as a result of the
54     operation of the algorithm described in 34.5. "
55   INDEX       { ieee8021BridgeBaseComponentId,
56                 ieee8021BridgeBasePort,
57                 ieee8021FqtssBAPTrafficClass }
58   ::= { ieee8021FqtssBapTable 1 }
59
60 Ieee8021FqtssBapEntry ::=
61   SEQUENCE {
62     ieee8021FqtssBAPTrafficClass
63       IEEE8021FqtssTrafficClassValue,
64     ieee8021FqtssDeltaBandwidth
65       IEEE8021FqtssDeltaBandwidthValue,
66     ieee8021FqtssOperIdleSlopeMs
67       Unsigned32,
68     ieee8021FqtssOperIdleSlopeLs
69       Unsigned32,
70     ieee8021FqtssAdminIdleSlopeMs
71       Unsigned32,
72     ieee8021FqtssAdminIdleSlopeLs

```

```
1         Unsigned32,
2         ieee8021FqtssBapRowStatus
3         RowStatus
4     }
5
6 ieee8021FqtssBAPTrafficClass OBJECT-TYPE
7     SYNTAX      IEEE8021FqtssTrafficClassValue
8     MAX-ACCESS  not-accessible
9     STATUS      current
10    DESCRIPTION
11        "The traffic class number associated with the row of
12        the table.
13
14        A row in this table is created for each traffic class
15        that supports the credit-based shaper algorithm. The
16        recommended mappings of priorities to traffic classes
17        for support of the credit-based shaper algorithm are
18        described in 34.5."
19    REFERENCE    "12.20.2, 34.3, 34.5"
20    ::= { ieee8021FqtssBapEntry 1 }
21
22
23 ieee8021FqtssDeltaBandwidth OBJECT-TYPE
24     SYNTAX      IEEE8021FqtssDeltaBandwidthValue
25     UNITS       "percent"
26     MAX-ACCESS  read-write
27     STATUS      current
28     DESCRIPTION
29        "The value of the deltaBandwidth parameter
30        for the traffic class.
31        This value is represented as a fixed point number
32        scaled by a factor of 1 000 000; i.e., 100 000 000
33        (the maximum value) represents 100%.
34
35        The default value of the deltaBandwidth parameter
36        for the highest numbered traffic class that supports
37        the credit-based shaper algorithm is 75%; for all
38        lower numbered traffic classes that support the
39        credit-based shaper algorithm the default value is 0%.
40
41        The value of this object MUST be retained across
42        reinitializations of the management system."
43    REFERENCE    "12.20.1, 34.3"
44    ::= { ieee8021FqtssBapEntry 2 }
45
46 ieee8021FqtssOperIdleSlopeMs OBJECT-TYPE
47     SYNTAX      Unsigned32
48     UNITS       "bits per second"
49     MAX-ACCESS  read-only
50     STATUS      current
51     DESCRIPTION
52        "The most significant 32 bits of the bandwidth,
53        in bits per second, that is currently allocated to the
54        traffic class (idleSlope(N)). This object MUST be read
55        at the same time as ieee8021FqtssOperIdleSlopeLs,
56        which represents the LS 32 bits of the value, in order
57        for the read operation to succeed.
58
59        If SRP is supported and in operation, then the reserved
60        bandwidth is determined by the operation of SRP; otherwise,
61        the value of ieee8021FqtssOperIdleSlopeMs is equal to
62        the value of ieee8021FqtssAdminIdleSlopeMs.
63
64        The value of this object MUST be retained across
65        reinitializations of the management system."
66    REFERENCE    "12.20.1, 34.3"
67    ::= { ieee8021FqtssBapEntry 3 }
68
69 ieee8021FqtssOperIdleSlopeLs OBJECT-TYPE
70     SYNTAX      Unsigned32
71     UNITS       "bits per second"
72     MAX-ACCESS  read-only
```

```
1  STATUS      current
2  DESCRIPTION
3      "The least significant 32 bits of the bandwidth,
4      in bits per second, that is currently allocated to the
5      traffic class (idleSlope(N)). This object MUST be read
6      at the same time as ieee8021FqtssOperIdleSlopeMs,
7      which represents the LS 32 bits of the value, in order
8      for the read operation to succeed.
9
10     If SRP is supported and in operation, then the reserved
11     bandwidth is determined by the operation of SRP; otherwise,
12     the value of ieee8021FqtssOperIdleSlopeLs is equal to
13     the value of ieee8021FqtssAdminIdleSlopeMs.
14
15     The value of this object MUST be retained across
16     reinitializations of the management system."
17  REFERENCE   "12.20.1, 34.3"
18  ::= { ieee8021FqtssBapEntry 4 }
19
20 ieee8021FqtssAdminIdleSlopeMs OBJECT-TYPE
21     SYNTAX      Unsigned32
22     UNITS        "bits per second"
23     MAX-ACCESS   read-write
24     STATUS      current
25     DESCRIPTION
26         "The most significant 32 bits of the bandwidth,
27         in bits per second, that the manager desires to allocate
28         to the traffic class as idleSlope(N). This object MUST be
29         read or written at the same time as
30         ieee8021FqtssAdminIdleSlopeLs,
31         which represents the LS 32 bits of the value, in order
32         for the read or write operation to succeed.
33
34         If SRP is supported and in operation, then the reserved
35         bandwidth is determined by the operation of SRP, and any
36         changes to the value of this object have no effect on the
37         operational value of idleSlope(N).
38
39         The value of this object MUST be retained across
40         reinitializations of the management system."
41     REFERENCE   "12.20.1, 34.3"
42     DEFVAL { 0 }
43     ::= { ieee8021FqtssBapEntry 5 }
44
45 ieee8021FqtssAdminIdleSlopeLs OBJECT-TYPE
46     SYNTAX      Unsigned32
47     UNITS        "bits per second"
48     MAX-ACCESS   read-write
49     STATUS      current
50     DESCRIPTION
51         "The least significant 32 bits of the bandwidth,
52         in bits per second, that the manager desires to allocate
53         to the traffic class as idleSlope(N). This object MUST be
54         read or written at the same time as
55         ieee8021FqtssAdminIdleSlopeMs,
56         which represents the LS 32 bits of the value, in order
57         for the read or write operation to succeed.
58
59         If SRP is supported and in operation, then the reserved
60         bandwidth is determined by the operation of SRP, and any
61         changes to the value of this object have no effect on the
62         operational value of idleSlope(N).
63
64         The value of this object MUST be retained across
65         reinitializations of the management system."
66     REFERENCE   "12.20.1, 34.3"
67     DEFVAL { 0 }
68     ::= { ieee8021FqtssBapEntry 6 }
69
70 ieee8021FqtssBapRowStatus OBJECT-TYPE
71     SYNTAX      RowStatus
72     MAX-ACCESS   read-create
```

```

1  STATUS current
2  DESCRIPTION
3      "Indicates the status of an entry (row) in this table, and is
4      used to create/delete entries.
5
6      The corresponding instances of the following objects
7      must be set before this object can be made active(1):
8          ieee8021FqtssBAPTrafficClass
9          ieee8021FqtssDeltaBandwidth
10         ieee8021FqtssOperIdleSlopeMs
11         ieee8021FqtssOperIdleSlopeLs
12         ieee8021FqtssAdminIdleSlopeMs
13         ieee8021FqtssAdminIdleSlopeLs
14
15         The corresponding instances of the following objects
16         may not be changed while this object is active(1):
17         ieee8021FqtssBAPTrafficClass"
18 ::= { ieee8021FqtssBapEntry 7 }
19
20 -- =====
21 -- The ieee8021FqtssMappings subtree
22 -- This subtree defines the objects necessary for the assignment
23 -- of transmission selection algorithms to traffic classes,
24 -- and definition of regeneration table override values.
25 -- =====
26
27 -- =====
28 -- the ieee8021FqtssTxSelectionAlgorithmTable
29 -- =====
30
31 ieee8021FqtssTxSelectionAlgorithmTable OBJECT-TYPE
32     SYNTAX      SEQUENCE OF Ieee8021FqtssTxSelectionAlgorithmEntry
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "A table containing the assignment of transmission
37         selection algorithms to traffic classes for the Port.
38         This table provides management of the Transmission
39         Selection Algorithm Table defined in 8.6.8.
40
41         For a given Port, a row in the table exists for each
42         traffic class that is supported by the Port.
43
44         The default assignments of transmission selection
45         algorithms to traffic classes in the table are made
46         on instantiation of the table, in accordance
47         with the defaults defined in 8.6.8 and 34.5.
48
49         All writable objects in this table must be
50         persistent over power up restart/reboot."
51     REFERENCE   "8.6.8, 12.20.2, 34.5"
52     ::= { ieee8021FqtssMappings 1 }
53
54 ieee8021FqtssTxSelectionAlgorithmEntry OBJECT-TYPE
55     SYNTAX      Ieee8021FqtssTxSelectionAlgorithmEntry
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59         "A list of objects that contain the mapping of a
60         traffic class value to a transmission selection algorithm
61         value."
62     INDEX { ieee8021BridgeBaseComponentId,
63            ieee8021BridgeBasePort,
64            ieee8021FqtssTrafficClass }
65     ::= { ieee8021FqtssTxSelectionAlgorithmTable 1 }
66
67 Ieee8021FqtssTxSelectionAlgorithmEntry ::=
68     SEQUENCE {
69         ieee8021FqtssTrafficClass
70         IEEE8021FqtssTrafficClassValue,
71         ieee8021FqtssTxSelectionAlgorithmID
72         IEEE8021FqtssTxSelectionAlgorithmIDValue

```

```

1      }
2
3 ieee8021FqtssTrafficClass OBJECT-TYPE
4     SYNTAX      IEEE8021FqtssTrafficClassValue
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "The traffic class to which the transmission selection
9         algorithm is assigned.
10
11         The value of this object MUST be retained across
12         reinitializations of the management system."
13     REFERENCE   "8.6.8, 12.20.2, 34.5"
14     ::= { ieee8021FqtssTxSelectionAlgorithmEntry 1 }
15
16 ieee8021FqtssTxSelectionAlgorithmID OBJECT-TYPE
17     SYNTAX      IEEE8021FqtssTxSelectionAlgorithmIDValue
18     MAX-ACCESS  read-write
19     STATUS      current
20     DESCRIPTION
21         "The identifier of the transmission selection algorithm
22         assigned to the traffic class.
23
24         The value of this object MUST be retained across
25         reinitializations of the management system."
26     REFERENCE   "8.6.8, 12.20.2, 34.5"
27     ::= { ieee8021FqtssTxSelectionAlgorithmEntry 2 }
28
29 -- =====
30 -- the ieee8021FqtssSrpRegenOverrideTable
31 -- =====
32
33 ieee8021FqtssSrpRegenOverrideTable OBJECT-TYPE
34     SYNTAX      SEQUENCE OF Ieee8021FqtssSrpRegenOverrideEntry
35     MAX-ACCESS  not-accessible
36     STATUS      current
37     DESCRIPTION
38         "A table containing the set of priority regeneration
39         table override values for the Port.
40
41         The recommended default values of priorities
42         associated with SR classes, and the corresponding
43         override values, are defined in 6.9.4.
44
45         All writable objects in this table must be
46         persistent over power up restart/reboot."
47     REFERENCE   "35.1.4, 6.9.4, 12.20.3"
48     ::= { ieee8021FqtssMappings 2 }
49
50 ieee8021FqtssSrpRegenOverrideEntry OBJECT-TYPE
51     SYNTAX      Ieee8021FqtssSrpRegenOverrideEntry
52     MAX-ACCESS  not-accessible
53     STATUS      current
54     DESCRIPTION
55         "A list of objects that contain the mapping of a
56         priority value to a priority regeneration override
57         value, and a boundary port indication.
58         Rows in the table exist for all priorities that are
59         associated with SR classes."
60     INDEX { ieee8021BridgeBaseComponentId,
61             ieee8021BridgeBasePort,
62             ieee8021FqtssSrClassPriority }
63     ::= { ieee8021FqtssSrpRegenOverrideTable 1 }
64
65 Ieee8021FqtssSrpRegenOverrideEntry ::=
66     SEQUENCE {
67         ieee8021FqtssSrClassPriority
68         IEEE8021PriorityValue,
69         ieee8021FqtssPriorityRegenOverride
70         IEEE8021PriorityValue,
71         ieee8021FqtssSrpBoundaryPort
72         TruthValue

```

```

1      }
2
3 ieee8021FqtssSrClassPriority OBJECT-TYPE
4     SYNTAX      IEEE8021PriorityValue
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "The priority value that is overridden at the
9         SRP domain boundary. "
10    REFERENCE   "35.1.4, 6.9.4, 12.20.3"
11    ::= { ieee8021FqtssSrpRegenOverrideEntry 1 }
12
13 ieee8021FqtssPriorityRegenOverride OBJECT-TYPE
14    SYNTAX      IEEE8021PriorityValue
15    MAX-ACCESS  read-write
16    STATUS      current
17    DESCRIPTION
18        "The priority value that is used to override the
19        priority regeneration table entry at the SRP
20        domain boundary.
21
22        The value of this object MUST be retained across
23        reinitializations of the management system."
24    REFERENCE   "35.1.4, 6.9.4, 12.20.3"
25    ::= { ieee8021FqtssSrpRegenOverrideEntry 2 }
26
27 ieee8021FqtssSrpBoundaryPort OBJECT-TYPE
28    SYNTAX      TruthValue
29    MAX-ACCESS  read-only
30    STATUS      current
31    DESCRIPTION
32        "The value of the SRPdomainBoundaryPort parameter
33        (35.1.4) for the priority. "
34    REFERENCE   "35.1.4, 6.9.4, 12.20.3"
35    ::= { ieee8021FqtssSrpRegenOverrideEntry 3 }
36
37    -- =====
38    -- the ieee8021FqtssSRClassToPriorityTable
39    -- =====
40
41 ieee8021FqtssSRClassToPriorityTable OBJECT-TYPE
42    SYNTAX      SEQUENCE OF Ieee8021FqtssSRClassToPriorityEntry
43    MAX-ACCESS  not-accessible
44    STATUS      current
45    DESCRIPTION
46        "A table containing the mapping of the SR Class to
47        the associated priority.
48
49        The default values for the entries of this table are
50        specified in 34.5"
51    REFERENCE   "12.20.4, 35.2.2.9.2, 6.9.3"
52    ::= { ieee8021FqtssMappings 3 }
53
54 ieee8021FqtssSRClassToPriorityEntry OBJECT-TYPE
55    SYNTAX      Ieee8021FqtssSRClassToPriorityEntry
56    MAX-ACCESS  not-accessible
57    STATUS      current
58    DESCRIPTION
59        "This attribute holds the Data Frame Priority
60        (35.2.2.8.5(a)) value that will be used for streams
61        that belong to the associated SR class."
62    INDEX { ieee8021BridgeBaseComponentId,
63            ieee8021BridgeBasePort,
64            ieee8021FqtssSrClassPriority }
65    ::= { ieee8021FqtssSRClassToPriorityTable 1 }
66
67 Ieee8021FqtssSRClassToPriorityEntry ::=
68     SEQUENCE {
69         ieee8021FqtssSRClassToPrioritySrClassID
70         IEEE8021FqtssTrafficClassValue,
71         ieee8021FqtssSRClassToPriorityRowStatus
72         RowStatus

```

```
1      }
2
3      ieee8021FqtssSRClassToPrioritySrClassID OBJECT-TYPE
4          SYNTAX      IEEE8021FqtssTrafficClassValue
5          MAX-ACCESS   read-create
6          STATUS       current
7          DESCRIPTION
8              "The srClassId attribute provides the SR class ID
9              from Table 35-6 of 35.2.2.9.2, so that management
10             software can associate the traffic class to the
11             corresponding SR class A or B used by protocols
12             such as SRP.
13
14             The default values for this attribute use the
15             default values specified in 34.5 (i.e. Priority 3
16             for SRclassID 6 and Priority 2 for SRclassID 5).
17
18             If this managed object is not supported, the default
19             values specified in 34.5 are used as the fixed
20             configuration."
21          REFERENCE    "12.20.4, 35.2.2.9.2"
22          ::= { ieee8021FqtssSRClassToPriorityEntry 1 }
23
24      ieee8021FqtssSRClassToPriorityRowStatus OBJECT-TYPE
25          SYNTAX RowStatus
26          MAX-ACCESS   read-create
27          STATUS       current
28          DESCRIPTION
29              "Indicates the status of an entry (row) in this table, and is
30              used to create/delete entries."
31          ::= { ieee8021FqtssSRClassToPriorityEntry 2 }
32
33      ieee8021FqtssBapXTable OBJECT-TYPE
34          SYNTAX      SEQUENCE OF Ieee8021FqtssBapXEntry
35          MAX-ACCESS   not-accessible
36          STATUS       current
37          DESCRIPTION
38              "A table containing a set of bandwidth availability
39              parameters for each traffic class configured for use with time-sensitive streams.
40              All writable objects in this table must be
41              persistent over power up restart/reboot."
42          REFERENCE    "12.20.2"
43          ::= { ieee8021FqtssBapX 1 }
44
45      ieee8021FqtssBapXEntry OBJECT-TYPE
46          SYNTAX      Ieee8021FqtssBapXEntry
47          MAX-ACCESS   not-accessible
48          STATUS       current
49          DESCRIPTION
50              "A list of objects containing bandwidth allocation
51              information for each traffic class configured for use with time-sensitive streams.
52              Rows in the table are
53              automatically created and deleted as a result of the
54              operation of the algorithm described in 34.5."
55          AUGMENTS    { ieee8021FqtssBapEntry }
56          ::= { ieee8021FqtssBapXTable 1 }
57
58      Ieee8021FqtssBapXEntry ::=
59          SEQUENCE {
60              ieee8021FqtssBAPClassMeasurementInterval
61              Unsigned32,
62              ieee8021FqtssBAPLockClassBandwidth
63              TruthValue
64          }
65
66      ieee8021FqtssBAPClassMeasurementInterval OBJECT-TYPE
67          SYNTAX      Unsigned32
68          MAX-ACCESS   read-create
69          STATUS       current
70          DESCRIPTION
71              "The value of the ClassMeasurementInterval
72              parameter for the traffic class.
```



```
1      This attribute uses units of nanoseconds,
2      converted to/from units of seconds for use in
3      34.3.
4
5      If management of classMeasurementInterval is
6      not supported, the default values (34.5) are
7      used as the fixed Port configuration.
8
9      The value of this object MUST be retained across
10     reinitializations of the management system."
11     REFERENCE      "12.20.1, 34.3, 34.4, 34.6"
12     ::= { ieee8021FqtssBapXEntry 1 }
13
14 ieee8021FqtssBAPLockClassBandwidth OBJECT-TYPE
15     SYNTAX          TruthValue
16     MAX-ACCESS      read-create
17     STATUS          current
18     DESCRIPTION
19         "This attribute determines the interpretation of
20         deltaBandwidth. For the value false(2), deltaBandwidth
21         is specified in 34.3.1. For true(1), deltaBandwidth is
22         specified in 34.3.2"
23     REFERENCE      "12.20.1, 34.3"
24     DEFVAL { false }
25     ::= { ieee8021FqtssBapXEntry 2 }
26
27 -- =====
28 -- IEEE8021 FQTSS MIB - Conformance Information
29 -- =====
30
31 ieee8021FqtssCompliances
32     OBJECT IDENTIFIER ::= { ieee8021FqtssConformance 1 }
33 ieee8021FqtssGroups
34     OBJECT IDENTIFIER ::= { ieee8021FqtssConformance 2 }
35
36 -- =====
37 -- units of conformance
38 -- =====
39
40 -- =====
41 -- the ieee8021FqtssBap group
42 -- =====
43
44 ieee8021FqtssBapGroup OBJECT-GROUP
45     OBJECTS {
46         ieee8021FqtssDeltaBandwidth,
47         ieee8021FqtssOperIdleSlopeMs,
48         ieee8021FqtssOperIdleSlopeLs,
49         ieee8021FqtssAdminIdleSlopeMs,
50         ieee8021FqtssAdminIdleSlopeLs,
51         ieee8021FqtssBapRowStatus
52     }
53     STATUS          current
54     DESCRIPTION
55         "Objects that define bandwidth allocation for FQTSS."
56     ::= { ieee8021FqtssGroups 1 }
57
58 -- =====
59 -- the ieee8021FqtssTxSelectionAlgorithm group
60 -- =====
61
62 ieee8021FqtssTxSelectionAlgorithmGroup OBJECT-GROUP
63     OBJECTS {
64         ieee8021FqtssTxSelectionAlgorithmID
65     }
66     STATUS          current
67     DESCRIPTION
68         "Objects that define transmission selection
69         mappings for FQTSS."
70     ::= { ieee8021FqtssGroups 2 }
71
72 -- =====
```

```

1 -- the ieee8021FqtssBoundaryPort group
2 -- =====
3
4 ieee8021FqtssBoundaryPortGroup OBJECT-GROUP
5     OBJECTS {
6         ieee8021FqtssPriorityRegenOverride,
7         ieee8021FqtssSrpBoundaryPort
8     }
9     STATUS      current
10    DESCRIPTION
11        "Objects that define boundary port priority override
12        mappings for FQTSS."
13    ::= { ieee8021FqtssGroups 3 }
14
15 -- =====
16 -- the ieee8021FqtssBapMeasurement group
17 -- =====
18
19 ieee8021FqtssBapMeasurementGroup OBJECT-GROUP
20     OBJECTS {
21         ieee8021FqtssBAPClassMeasurementInterval,
22         ieee8021FqtssBAPLockClassBandwidth
23     }
24     STATUS      current
25     DESCRIPTION
26        "Objects that define the SRP TSpec measurement interval
27        and deltaBandwidth interpretation for FQTSS."
28    ::= { ieee8021FqtssGroups 4 }
29
30 -- =====
31 -- the ieee8021FqtssSRClassPriority group
32 -- =====
33
34 ieee8021FqtssSRClassPriorityGroup OBJECT-GROUP
35     OBJECTS {
36         ieee8021FqtssSRClassToPrioritySrClassID,
37         ieee8021FqtssSRClassToPriorityRowStatus
38     }
39     STATUS      current
40     DESCRIPTION
41        "Objects that define mappings of the SR class ID to
42        the associated priority for FQTSS."
43    ::= { ieee8021FqtssGroups 5 }
44
45 -- =====
46 -- compliance statements
47 -- =====
48
49 ieee8021FqtssCompliance MODULE-COMPLIANCE
50     STATUS      current
51     DESCRIPTION
52        "The compliance statement for devices supporting
53        forwarding and queuing for time sensitive streams.
54
55        Support of the objects defined in the IEEE8021-FQTSS MIB
56        also requires support of the IEEE8021-BRIDGE-MIB; the
57        provisions of 17.3.2 apply to implementations claiming
58        support of the IEEE8021-FQTSS MIB. "
59
60     MODULE -- this module
61         MANDATORY-GROUPS {
62             ieee8021FqtssBapGroup,
63             ieee8021FqtssTxSelectionAlgorithmGroup,
64             ieee8021FqtssBoundaryPortGroup
65         }
66
67     GROUP    ieee8021FqtssBapMeasurementGroup
68     DESCRIPTION
69        "Implementation of this group is optional. Implementation
70        will allow management of the TSpec measurement interval
71        and deltaBandwidth interpretation."
72

```

```
1  GROUP    ieee8021FqtssSRClassPriorityGroup
2  DESCRIPTION
3      "Implementation of this group is optional. Implementation
4      will allow management of the mapping of SR class IDs to the
5      associated Priority."
6
7      ::= { ieee8021FqtssCompliances 1 }
8
9  END
10
```

1 17.7.13 Definitions for the IEEE8021-CN-MIB module

```
2
3 IEEE8021-CN-MIB
4 DEFINITIONS ::= BEGIN
5
6 -- *****
7 -- IEEE Std 802.1Q(TM) Congestion Management MIB
8 -- *****
9
10 IMPORTS
11     MODULE-IDENTITY,
12     OBJECT-TYPE,
13     Integer32
14         FROM SNMPv2-SMI -- [RFC2578]
15     Counter32
16         FROM SNMPv2-SMI -- [RFC2578]
17     Counter64
18         FROM SNMPv2-SMI -- [RFC2578]
19     Unsigned32
20         FROM SNMPv2-SMI -- [RFC2578]
21     TEXTUAL-CONVENTION,
22     TimeInterval
23         FROM SNMPv2-TC -- [RFC2579]
24     RowStatus
25         FROM SNMPv2-TC -- [RFC2579]
26     TruthValue
27         FROM SNMPv2-TC -- [RFC2579]
28     MacAddress
29         FROM SNMPv2-TC -- [RFC2579]
30     MODULE-COMPLIANCE,
31     OBJECT-GROUP
32         FROM SNMPv2-CONF -- [RFC2580]
33     InterfaceIndex
34         FROM IF-MIB -- [RFC2863]
35     ifGeneralInformationGroup
36         FROM IF-MIB -- [RFC2863]
37     ieee802dot1mibs
38         FROM IEEE8021-TC-MIB
39     IEEE8021PriorityValue
40         FROM IEEE8021-TC-MIB
41     IEEE8021PbbComponentIdentifier
42         FROM IEEE8021-TC-MIB -- [IEEE Std 802.1ap]
43     LldpV2DestAddressTableIndex
44         FROM LLDP-V2-TC-MIB
45     systemGroup
46         FROM SNMPv2-MIB -- [RFC3418]
47 ;
48 ieee8021CnMib
49 MODULE-IDENTITY
50     LAST-UPDATED "202211080000Z" -- November 8, 2022
51     ORGANIZATION "IEEE 802.1 Working Group"
52     CONTACT-INFO
53         " WG-URL: http://www.ieee802.org/1/
54         WG-EMail: stds-802-1-1@ieee.org
55         Contact: IEEE 802.1 Working Group Chair
56         Postal: C/O IEEE 802.1 Working Group
57             IEEE Standards Association
58             445 Hoes Lane
59             Piscataway, NJ 08854
60             USA
61         E-mail: stds-802-1-chairs@ieee.org"
62 DESCRIPTION
63     "Congestion notification module.
64
65     Unless otherwise indicated, the references in this MIB
66     module are to IEEE Std 802.1Q-2022.
67
68     Copyright (C) IEEE (2022).
69     This version of this MIB module is part of IEEE Std 802.1Q;
70     see that standard for full legal notices."
71
```

```
1 REVISION "202211080000Z" -- November 8, 2022
2 DESCRIPTION
3     "Published as part of IEEE Std 802.1Q-2022.
4     Cross references and contact information updated."
5
6 REVISION "201807010000Z" -- July 1, 2018
7 DESCRIPTION
8     "Published as part of IEEE Std 802.1Q 2018 revision.
9     Cross references updated and corrected."
10
11 REVISION "201412150000Z" -- December 15, 2014
12 DESCRIPTION
13     "Published as part of IEEE Std 802.1Q 2014 revision.
14     Cross references updated and corrected.
15     Imports tidied up to remove orphaned commas."
16
17 REVISION "201102270000Z" -- February 27, 2011
18 DESCRIPTION
19     "Minor edits to contact information etc. as part of
20     2011 revision of IEEE Std 802.1Q."
21
22 REVISION "200912180000Z" -- 12/18/2009 00:00GMT
23 DESCRIPTION
24     "Included in IEEE 802.1Qau-2010
25
26     Copyright (C) IEEE."
27     ::= { ieee802dot1mibs 18 }
28
29 ieee8021CnMIBObjects
30     OBJECT IDENTIFIER ::= { ieee8021CnMib 1 }
31 ieee8021CnConformance
32     OBJECT IDENTIFIER ::= { ieee8021CnMib 2 }
33
34 -- *****
35 -- Textual conventions
36 -- *****
37
38 Ieee8021CnControlChoice
39 ::= TEXTUAL-CONVENTION
40     STATUS current
41     DESCRIPTION
42         "This controls what other object selects the CND defense mode and
43         the Congestion Notification Priority Value (CNPV) alternate
44         priority for a CNPV in an end station or Bridge component, or
45         for a CNPV on a particular Port in an end station or Bridge
46         component. It can take the following values:
47
48         cpcAdmin(1) The CND defense mode and alternate priority are
49                     controlled by the administrative variables in the
50                     same table entry as this object.
51         cpcAuto(2) This Port or all Ports' CND defense modes are
52                     controlled automatically, as indicated by
53                     ieee8021CnPortPriAutoDefenseMode, and the
54                     alternate priority by ieee8021CnComPriAutoAltPri.
55         cpcComp(3) This CND defense mode and alternate priority are
56                     both controlled by ieee8021CnPortPriTable.
57     "
58     REFERENCE
59         "32.3.1, 32.4.1, Table 32-2"
60     SYNTAX INTEGER {
61         cpcAdmin(1),
62         cpcAuto(2),
63         cpcComp(3)
64     }
65
66 Ieee8021CnDefenseMode
67 ::= TEXTUAL-CONVENTION
68     STATUS current
69     DESCRIPTION
70         "For a given Congestion Notification Priority Value (CNPV), a
71         port can operate in one of four CND defense modes. The CND
72         defense mode determines whether congestion notification is
```

1 enabled or not, and if enabled, whether the port translates the
2 CNPV to a non-CNPV value on input, and whether the port removes
3 CN-TAGs on output.
4

5 cptDisabled(1) The congestion notification capability is
6 administratively disabled for this priority
7 value and port. This priority is not a CNPV.
8 The priority regeneration table controls the
9 remapping of input frames on this port for
10 this priority. CN-TAGs are neither added by
11 an end station nor stripped by a Bridge.
12 cptInterior(2) On this port and for this CNPV, the priority
13 parameters of input frames are not remapped,
14 regardless of the priority regeneration
15 table. CN-TAGs are not added by an end
16 station, and are removed from frames before
17 being output by a Bridge.
18 cptInteriorReady(3) On this port and for this CNPV, the priority
19 parameters of input frames are not remapped,
20 regardless of the priority regeneration
21 table. CN-TAGs can be added by an end
22 station, and are not removed from frames by
23 a Bridge.
24 cptEdge(4) On this port and for this CNPV, the priority
25 parameters of input frames are remapped to
26 an alternate (non-CNPV) value, regardless of
27 the priority regeneration table. CN-TAGs are
28 not added by an end station, and are removed
29 from frames before being output by a Bridge.
30 This mode is optional for an end station.
31 "

32 REFERENCE
33 "32.1.1, 32.3.4, 32.4.2, 32.4.3, Table 32-2"
34 SYNTAX INTEGER {
35 cptDisabled(1),
36 cptInterior(2),
37 cptInteriorReady(3),
38 cptEdge(4)
39 }

40

41 Ieee8021CnLldpChoice
42 ::= TEXTUAL-CONVENTION
43 STATUS current
44 DESCRIPTION
45 "Specifies how to determine what index value is to be used as
46 the index to lldpDestAddressTable, the table of destination MAC
47 addresses, for both the destination addresses on transmitted
48 LLDPDUs and on received LLDPDUs, found in the LLDP-MIB, either:
49

50 cnlNone(1) No LLDP Congestion Notification TLV is to
51 carry Per-priority CNPV indicators or
52 Per-priority Ready indicators on this Port
53 for this priority (or all Ports and all
54 priorities, as appropriate to the managed
55 object).
56 cnlAdmin(2) The administrative LLDP instance selector
57 in the same table entry as this object
58 governs which LLDP instance will carry the
59 Per-priority CNPV indicators and
60 Per-priority Ready indicators for this
61 priority in its Congestion Notification TLV
62 on this Port (or all Ports and all
63 priorities, as appropriate to the managed
64 object).
65 cnlComponent(3) ieee8021CnComPriLldpInstanceSelector governs
66 LLDP instance selection for this Port and
67 priority.
68 "

69 REFERENCE
70 "32.3.6, 32.4.4, Table 32-1"
71 SYNTAX INTEGER {
72 cnlNone(1),

```

1      cnlAdmin(2),
2      cnlComponent(3)
3  }
4
5
6  -- *****
7  -- The Global Table. This group will contain the MIB objects that
8  -- apply to the whole Bridge component or end station.
9  -- *****
10
11 ieee8021CnGlobalTable
12 OBJECT-TYPE
13     SYNTAX      SEQUENCE OF Ieee8021CnGlobalEntry
14     MAX-ACCESS  not-accessible
15     STATUS      current
16     DESCRIPTION
17         "A table containing the global variables for each component of
18         a Bridge or for an end station. A value of 1 is used in a
19         Bridge or end station that does not have multiple components.
20
21         The contents of this table SHALL be maintained across a restart
22         of the system.
23         "
24     REFERENCE
25         "12.21.1"
26         ::= { ieee8021CnMIBObjects 1 }
27
28 ieee8021CnGlobalEntry
29 OBJECT-TYPE
30     SYNTAX      Ieee8021CnGlobalEntry
31     MAX-ACCESS  not-accessible
32     STATUS      current
33     DESCRIPTION
34         "A list of objects pertaining to a whole Bridge component or
35         end station.
36         "
37     INDEX { ieee8021CnGlobalComponentId }
38     ::= { ieee8021CnGlobalTable 1 }
39
40 Ieee8021CnGlobalEntry
41 ::= SEQUENCE {
42     ieee8021CnGlobalComponentId  IEEE8021PbbComponentIdentifier,
43     ieee8021CnGlobalMasterEnable  TruthValue,
44     ieee8021CnGlobalCnmTransmitPriority  IEEE8021PriorityValue,
45     ieee8021CnGlobalDiscardedFrames  Counter64
46 }
47
48 ieee8021CnGlobalComponentId
49 OBJECT-TYPE
50     SYNTAX      IEEE8021PbbComponentIdentifier
51     MAX-ACCESS  not-accessible
52     STATUS      current
53     DESCRIPTION
54         "The Bridge component within the system to which the information
55         in this ieee8021CnGlobalEntry applies. If the system is not
56         a Bridge, or if only one component is present in the Bridge,
57         then this variable (index) MUST be equal to 1.
58         "
59     ::= { ieee8021CnGlobalEntry 1 }
60
61 ieee8021CnGlobalMasterEnable
62 OBJECT-TYPE
63     SYNTAX      TruthValue
64     MAX-ACCESS  read-write
65     STATUS      current
66     DESCRIPTION
67         "The state of the congestion notification feature on this Bridge
68         component or system. If true, Congestion notification is
69         enabled, and if false, congestion notification is disabled.
70         "
71     REFERENCE
72         "32.2.1"

```

```

1 ::= { ieee8021CnGlobalEntry 2 }
2
3 ieee8021CnGlobalCnmTransmitPriority
4 OBJECT-TYPE
5     SYNTAX      IEEE8021PriorityValue
6     MAX-ACCESS  read-write
7     STATUS      current
8     DESCRIPTION
9         "The priority to use for all Congestion Notification Messages
10        transmitted by this Bridge component or end station.
11        "
12     REFERENCE
13         "32.2.2"
14 ::= { ieee8021CnGlobalEntry 3 }
15
16 ieee8021CnGlobalDiscardedFrames
17 OBJECT-TYPE
18     SYNTAX      Counter64
19     UNITS       "frames"
20     MAX-ACCESS  read-only
21     STATUS      current
22     DESCRIPTION
23         "The number of frames discarded from full CP queues, in spite
24         of the efforts of congestion notification to avoid discards.
25
26         This object is incremented whenever any of the
27         ieee8021CnCpDiscardedFrames objects on any Port or priority in
28         this same component are incremented.
29
30         Discontinuities in the value of this counter can occur
31         at re-initialization of the management system, and at
32         other times as indicated by the value of
33         ifCounterDiscontinuityTime object of the associated
34         interface (if any).
35         "
36     REFERENCE
37         "32.2.3"
38 ::= { ieee8021CnGlobalEntry 4 }
39
40
41
42 -- *****
43 -- The CCF Errored Port Table. One per Bridge component or end station.
44 -- Scanning this table reveals which Interfaces at which priorities
45 -- have an Alternate Priority value that is a
46 -- Congestion Notification Priority Value.
47 -- *****
48
49 ieee8021CnErroredPortTable
50 OBJECT-TYPE
51     SYNTAX      SEQUENCE OF Ieee8021CnErroredPortEntry
52     MAX-ACCESS  not-accessible
53     STATUS      current
54     DESCRIPTION
55         "There is one Errored Port Table per Bridge component or end
56         station. It permits the retrieval of information about which
57         interfaces have congestion notification configuration errors,
58         namely, those specifying an alternate priority that is a CNPV.
59         "
60     REFERENCE
61         "32.2.4"
62 ::= { ieee8021CnMIBObjects 2 }
63
64 ieee8021CnErroredPortEntry
65 OBJECT-TYPE
66     SYNTAX      Ieee8021CnErroredPortEntry
67     MAX-ACCESS  not-accessible
68     STATUS      current
69     DESCRIPTION
70         "A list of interfaces whose ieee8021CnComPriAlternatePriority
71         and/or ieee8021CnPortPriAlternatePriority specify a priority
72         value that is a Congestion Notification Priority Value.

```



```

1      "
2      REFERENCE
3      "32.2.4"
4      INDEX { ieee8021CnEpComponentId,
5               ieee8021CnEpPriority,
6               ieee8021CnEpIfIndex}
7      ::= { ieee8021CnErroredPortTable 1 }
8
9      Ieee8021CnErroredPortEntry
10     ::= SEQUENCE {
11         ieee8021CnEpComponentId      IEEE8021PbbComponentIdentifier,
12         ieee8021CnEpPriority          IEEE8021PriorityValue,
13         ieee8021CnEpIfIndex          InterfaceIndex
14     }
15
16     ieee8021CnEpComponentId
17     OBJECT-TYPE
18         SYNTAX      IEEE8021PbbComponentIdentifier
19         MAX-ACCESS  not-accessible
20         STATUS      current
21         DESCRIPTION
22             "The Bridge component within the system to which the information
23             in this ieee8021CnErroredPortEntry applies. If the system is
24             not a Bridge, or if only one component is present in the
25             Bridge, then this variable (index) MUST be equal to 1.
26             "
27         REFERENCE
28             "32.2.4"
29         ::= { ieee8021CnErroredPortEntry 1 }
30
31     ieee8021CnEpPriority
32     OBJECT-TYPE
33         SYNTAX      IEEE8021PriorityValue
34         MAX-ACCESS  not-accessible
35         STATUS      current
36         DESCRIPTION
37             "The priority value whose alternate priority is misconfigured.
38             "
39         REFERENCE
40             "32.2.4"
41         ::= { ieee8021CnErroredPortEntry 2 }
42
43     ieee8021CnEpIfIndex
44     OBJECT-TYPE
45         SYNTAX      InterfaceIndex
46         MAX-ACCESS  read-only
47         STATUS      current
48         DESCRIPTION
49             "This object represents the Bridge Port or aggregated port
50             on which the congestion notification alternate priority is
51             misconfigured.
52             Upon a restart of the system, the system SHALL, if necessary,
53             change the value of this variable so that it references the row
54             in the ifXTable with the same value of ifAlias that it
55             referenced before the system restart. If no such row exists,
56             then the system SHALL delete this row in the
57             ieee8021CnErroredPortTable.
58             "
59         REFERENCE
60             "32.2.4"
61         ::= { ieee8021CnErroredPortEntry 3 }
62
63
64 -- *****
65 -- The CCF Per-component per-priority table. One table per Bridge
66 -- component or end station, one entry per
67 -- Congestion Notification Priority Value
68 -- *****
69
70     ieee8021CnCompntPriTable
71     OBJECT-TYPE
72         SYNTAX      SEQUENCE OF Ieee8021CnCompntPriEntry

```

```

1  MAX-ACCESS    not-accessible
2  STATUS        current
3  DESCRIPTION
4      "Each row in this table supplies default values for one
5      Congestion Notification Priority Value for a whole Bridge
6      component or end station.
7
8      Creating a row in this table makes the priority value of
9      ieee8021CnComPriPriority a
10     Congestion Notification Priority Value.
11     Deleting a row in this table makes the value in the deleted
12     ieee8021CnComPriPriority no longer a
13     Congestion Notification Priority Value.
14
15     A system SHALL NOT allow eight rows in this table
16     to be created with the same value of
17     ieee8021CnComPriComponentId; see the description of
18     ieee8021CnComPriRowStatus.
19
20     The contents of this table SHALL be maintained across a restart
21     of the system.
22     "
23  REFERENCE
24      "12.21.2, 12.21.2.1, 12.21.2.2"
25  ::= { ieee8021CnMIBObjects 3 }
26
27 ieee8021CnCompntPriEntry
28  OBJECT-TYPE
29      SYNTAX      Ieee8021CnCompntPriEntry
30      MAX-ACCESS  not-accessible
31      STATUS      current
32      DESCRIPTION
33          "One entry per Congestion Notification Priority Value per
34          Bridge component or end station.
35          "
36      REFERENCE
37          "12.21.2, 12.21.2.1, 12.21.2.2"
38      INDEX { ieee8021CnComPriComponentId,
39              ieee8021CnComPriPriority }
40      ::= { ieee8021CnCompntPriTable 1 }
41
42 Ieee8021CnCompntPriEntry
43 ::= SEQUENCE {
44     ieee8021CnComPriComponentId      IEEE8021PbbComponentIdentifier,
45     ieee8021CnComPriPriority          IEEE8021PriorityValue,
46     ieee8021CnComPriDefModeChoice    Ieee8021CnControlChoice,
47     ieee8021CnComPriAlternatePriority IEEE8021PriorityValue,
48     ieee8021CnComPriAutoAltPri       IEEE8021PriorityValue,
49     ieee8021CnComPriAdminDefenseMode Ieee8021CnDefenseMode,
50     ieee8021CnComPriCreation         INTEGER,
51     ieee8021CnComPriLldpInstanceChoice Ieee8021CnLldpChoice,
52     ieee8021CnComPriLldpInstanceSelector LldpV2DestAddressTableIndex,
53     ieee8021CnComPriRowStatus        RowStatus
54 }
55
56 ieee8021CnComPriComponentId
57  OBJECT-TYPE
58      SYNTAX      IEEE8021PbbComponentIdentifier
59      MAX-ACCESS  not-accessible
60      STATUS      current
61      DESCRIPTION
62          "The Bridge component within the system to which the information
63          in this ieee8021CnCompntPriEntry applies. If the system is
64          not a Bridge, or if only one component is present in the
65          Bridge, then this variable (index) MUST be equal to 1.
66          "
67      REFERENCE
68          "12.21.2, 12.21.2.1, 12.21.2.2"
69      ::= { ieee8021CnCompntPriEntry 1 }
70
71 ieee8021CnComPriPriority
72  OBJECT-TYPE

```

```

1  SYNTAX      IEEE8021PriorityValue
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "The Congestion Notification Priority Value for which this
6      row supplies default values.
7      "
8  REFERENCE
9      "802.1Qau clauses 12.21.2"
10 ::= { ieee8021CnCompntPriEntry 2 }
11
12 ieee8021CnComPriDefModeChoice
13 OBJECT-TYPE
14     SYNTAX      Ieee8021CnControlChoice {
15                 cpcAdmin(1),
16                 cpcAuto(2)
17             }
18     MAX-ACCESS  read-create
19     STATUS      current
20     DESCRIPTION
21         "Specifies how the default CND defense mode and alternate
22         priority for this Congestion Notification Priority Value on all
23         ports on this Bridge component or end station are to be chosen,
24         either:
25
26             cpcAdmin(1) Default CND defense mode is chosen by
27                         ieee8021CnComPriAdminDefenseMode, and alternate
28                         priority by ieee8021CnComPriAlternatePriority.
29             cpcAuto(2)  Default CND defense mode is chosen by
30                         ieee8021CnPortPriAutoDefenseMode, and alternate
31                         priority by ieee8021CnComPriAutoAltPri.
32
33         This variable can be overridden by
34         ieee8021CnPortPriDefModeChoice.
35         "
36     REFERENCE
37         "32.1.3, 32.3.1"
38     DEFVAL { cpcAuto }
39     ::= { ieee8021CnCompntPriEntry 3 }
40
41 ieee8021CnComPriAlternatePriority
42 OBJECT-TYPE
43     SYNTAX      IEEE8021PriorityValue
44     MAX-ACCESS  read-create
45     STATUS      current
46     DESCRIPTION
47         "The Congestion Notification Priority Value to which an
48         incoming frame is to be mapped, in spite of what the
49         Priority Regeneration Table says, if 1) Congestion
50         Notification is enabled and 2) the CND defense mode of the
51         port is cptEdge.
52
53         Deleting a row in this table does not alter the value of any
54         other row's ieee8021CnComPriAlternatePriority.
55         "
56     REFERENCE
57         "802.1Qau clauses 32.3.2"
58     DEFVAL { 0 }
59     ::= { ieee8021CnCompntPriEntry 4 }
60
61 ieee8021CnComPriAutoAltPri
62 OBJECT-TYPE
63     SYNTAX      IEEE8021PriorityValue
64     MAX-ACCESS  read-only
65     STATUS      current
66     DESCRIPTION
67         "The Congestion Notification Priority Value to which an
68         incoming frame can be mapped, in spite of what the
69         Priority Regeneration Table says, if 1) Congestion
70         Notification is enabled, 2) the CND defense mode of the
71         port is cptEdge, and 3) ieee8021CnComPriDefModeChoice
72         contains the value cpcAuto(2).

```

1
2 The value of this object is the next lower priority value
3 than this row's ieee8021CnComPriPriority that is not a CNPV,
4 or the next higher non-CNPV, if all lower values are CNPVs.
5
6 The value of this object, and any consequent priority
7 regeneration, is automatically updated by the managed system
8 whenever a row in the ieee8021CnCompntPriTable is created or
9 deleted. The value of this object is not dependent upon
10 whether congestion notification is enabled or disabled for any
11 priority or for the whole Bridge component or end station; it
12 depends only upon whether the ieee8021CnCompntPriTable row
13 exists.
14
15 REFERENCE
16 "802.1Qau clauses 32.3.3"
17 ::= { ieee8021CnCompntPriEntry 5 }
18
19 ieee8021CnComPriAdminDefenseMode
20 OBJECT-TYPE
21 SYNTAX Ieee8021CnDefenseMode
22 MAX-ACCESS read-create
23 STATUS current
24 DESCRIPTION
25 "The default CND defense mode for this
26 Congestion Notification Priority Value on all ports on this
27 Bridge component or end station.
28
29 This variable can be overridden by
30 ieee8021CnPortPriAdminDefenseMode.
31
32 REFERENCE
33 "32.1.3, 32.3.4"
34 DEFVAL { cptInterior }
35 ::= { ieee8021CnCompntPriEntry 6 }
36
37 ieee8021CnComPriCreation
38 OBJECT-TYPE
39 SYNTAX INTEGER {
40 cncpAutoEnable(1),
41 cncpAutoDisable(2)
42 }
43 MAX-ACCESS read-create
44 STATUS current
45 DESCRIPTION
46 "The default value for ieee8021CnComPriDefModeChoice for
47 newly-created entries in the ieee8021CnPortPriTable:
48
49 cncpAutoEnable (1) Newly-created
50 ieee8021CnPortPriDefModeChoice
51 objects take the value cpcComp(3).
52 cncpAutoDisable(2) Newly-created
53 ieee8021CnPortPriDefModeChoice
54 objects take the value cpcAdmin (1).
55
56 REFERENCE
57 "32.3.5"
58 DEFVAL { cncpAutoEnable }
59 ::= { ieee8021CnCompntPriEntry 7 }
60
61 ieee8021CnComPriLldpInstanceChoice
62 OBJECT-TYPE
63 SYNTAX Ieee8021CnLldpChoice {
64 cnlNone (1),
65 cnlAdmin (2)
66 }
67 MAX-ACCESS read-create
68 STATUS current
69 DESCRIPTION
70 "Specifies whether or not the default value for all Ports is to
71 send and receive the Congestion Notification TLV in LLDPDUs,
72 either:

```

1          cnlNone(1) Do not send Congestion Notification TLVs, and
2                      ignore them on receipt.
3          cnlAdmin(2) Use the LLDP instance selected by
4                      ieee8021CnComPriLldpInstanceSelector to send and
5                      receive the Congestion Notification TLV.
6
7          This object can be overridden by
8          ieee8021CnPortPriLldpInstanceChoice.
9          "
10         REFERENCE
11         "32.1.3, 32.3.6"
12         DEFVAL { cnlAdmin }
13         ::= { ieee8021CnCompntPriEntry 8 }
14
15 ieee8021CnComPriLldpInstanceSelector
16 OBJECT-TYPE
17     SYNTAX      LldpV2DestAddressTableIndex
18     MAX-ACCESS  read-create
19     STATUS      current
20     DESCRIPTION
21         "Specifies a default value for which LLDP instance is to be
22         used to provide the information for automatic configuration
23         of ports' CND defense modes (ieee8021CnPortPriAutoDefenseMode).
24
25         This object is ignored by the managed system if
26         ieee8021CnComPriLldpInstanceChoice contains the value cnlNone
27         (1).
28
29         This object can be overridden by
30         ieee8021CnPortPriLldpInstanceChoice and
31         ieee8021CnPortPriLldpInstanceChoice.
32         "
33     REFERENCE
34     "32.1.3, 32.3.7"
35     DEFVAL { 1 }
36     ::= { ieee8021CnCompntPriEntry 9 }
37
38 ieee8021CnComPriRowStatus
39 OBJECT-TYPE
40     SYNTAX      RowStatus
41     MAX-ACCESS  read-create
42     STATUS      current
43     DESCRIPTION
44         "This object indicates the status of an entry, and is used
45         to create/delete entries.
46
47         A system SHALL NOT permit eight ieee8021CnComPriRowStatus
48         objects, all with the same value of ieee8021CnComPriComponentId,
49         to have the value active(1). An attempt to create or activate
50         a row when there are already seven active rows SHALL result in
51         that eighth row's ieee8021CnComPriRowStatus having the value
52         notReady(3), and the return of an inconsistentValue error.
53         "
54     REFERENCE
55     "30.4"
56     ::= { ieee8021CnCompntPriEntry 10 }
57
58
59 -- *****
60 -- The CCF Per-component per-interface per-priority table. One table
61 -- per end station or Bridge component. One entry per interface per
62 -- priority value, but only for priority values that are
63 -- Congestion Notification Priority Values (CNPVs). Controls
64 -- a CNPV on a specific port when the default values in
65 -- ieee8021CnCompntPriTable need to be overridden.
66 -- *****
67
68 ieee8021CnPortPriTable
69 OBJECT-TYPE
70     SYNTAX      SEQUENCE OF Ieee8021CnPortPriEntry
71     MAX-ACCESS  not-accessible
72     STATUS      current

```

```
1 DESCRIPTION
2 "Each row in this table supplies values for one port's
3 Congestion Notification Priority Value (CNPV).
4
5 Creating an entry in ieee8021CnCompntPriTable creates this
6 entry, with the default values, on all ports in the Bridge
7 component or end station. Deleting an entry in
8 ieee8021CnCompntPriTable deletes this ieee8021CnCompntPriEntry
9 on all ports in the Bridge component or end station.
10
11 The contents of this table SHALL be maintained across a restart
12 of the system, except as noted in the description of
13 ieee8021CnPortPriIfIndex.
14 "
15 REFERENCE
16 "12.21.3"
17 ::= { ieee8021CnMIBObjects 4 }
18
19 ieee8021CnPortPriEntry
20 OBJECT-TYPE
21 SYNTAX Ieee8021CnPortPriEntry
22 MAX-ACCESS not-accessible
23 STATUS current
24 DESCRIPTION
25 "One entry per port per Congestion Notification Priority Value
26 per Bridge component or end station.
27 "
28 REFERENCE
29 "12.21.3"
30 INDEX { ieee8021CnPortPriComponentId,
31 ieee8021CnPortPriority,
32 ieee8021CnPortPriIfIndex }
33 ::= { ieee8021CnPortPriTable 1 }
34
35 Ieee8021CnPortPriEntry
36 ::= SEQUENCE {
37 ieee8021CnPortPriComponentId IEEE8021PbbComponentIdentifier,
38 ieee8021CnPortPriority IEEE8021PriorityValue,
39 ieee8021CnPortPriIfIndex InterfaceIndex,
40 ieee8021CnPortPriDefModeChoice Ieee8021CnControlChoice,
41 ieee8021CnPortPriAdminDefenseMode Ieee8021CnDefenseMode,
42 ieee8021CnPortPriAutoDefenseMode Ieee8021CnDefenseMode,
43 ieee8021CnPortPriLldpInstanceChoice Ieee8021CnLldpChoice,
44 ieee8021CnPortPriLldpInstanceSelector
45 LldpV2DestAddressTableIndex,
46 ieee8021CnPortPriAlternatePriority IEEE8021PriorityValue
47 }
48
49 ieee8021CnPortPriComponentId
50 OBJECT-TYPE
51 SYNTAX IEEE8021PbbComponentIdentifier
52 MAX-ACCESS not-accessible
53 STATUS current
54 DESCRIPTION
55 "The Bridge component within the system to which the information
56 in this ieee8021CnPortPriEntry applies. If the system is
57 not a Bridge, or if only one component is present in the
58 Bridge, then this variable (index) MUST be equal to 1.
59 "
60 REFERENCE
61 "12.21.3"
62 ::= { ieee8021CnPortPriEntry 1 }
63
64 ieee8021CnPortPriority
65 OBJECT-TYPE
66 SYNTAX IEEE8021PriorityValue
67 MAX-ACCESS not-accessible
68 STATUS current
69 DESCRIPTION
70 "The Congestion Notification Priority Value for which
71 this row supplies default values.
72 "
```

```

1  REFERENCE
2  "802.1Qau clauses 12.21.3"
3  ::= { ieee8021CnPortPriEntry 2 }
4
5  ieee8021CnPortPriIfIndex
6  OBJECT-TYPE
7      SYNTAX      InterfaceIndex
8      MAX-ACCESS  not-accessible
9      STATUS      current
10     DESCRIPTION
11         "This object represents the port or aggregated port
12         to which the entry applies.
13
14         Upon a restart of the system, the system SHALL, if necessary,
15         change the value of this object, and rearrange the order of the
16         ieee8021CnPortPriTable, so that the value in
17         ieee8021CnPortPriIfIndex references the row in the ifXTable
18         with the same value for ifAlias that it referenced before the
19         system restart. If no such entry exists in the ifXTable, then
20         the system SHALL delete the row in the ieee8021CnPortPriTable.
21     "
22     REFERENCE
23         "12.21.3"
24     ::= { ieee8021CnPortPriEntry 3 }
25
26  ieee8021CnPortPriDefModeChoice
27  OBJECT-TYPE
28      SYNTAX      Ieee8021CnControlChoice
29      MAX-ACCESS  read-write
30      STATUS      current
31      DESCRIPTION
32         "This object determines how the CND defense mode and alternate
33         priority value of this port for this CNPV is to be selected,
34         either:
35
36         cpcAdmin(1) CND defense mode is controlled by
37                     ieee8021CnPortPriAdminDefenseMode, and alternate
38                     priority by ieee8021CnPortPriAlternatePriority.
39         cpcAuto(2)  CND defense mode is controlled by
40                     ieee8021CnPortPriAutoDefenseMode and alternate
41                     priority by ieee8021CnComPriAlternatePriority.
42         cpcComp(3)  CND defense mode and alternate priority are
43                     controlled by
44                     ieee8021CnComPriDefModeChoice.
45
46         This variable can override ieee8021CnComPriDefModeChoice.
47     "
48     REFERENCE
49         "IEEE 32.1.3, 32.4.1"
50     DEFVAL { cpcComp }
51     ::= { ieee8021CnPortPriEntry 4 }
52
53  ieee8021CnPortPriAdminDefenseMode
54  OBJECT-TYPE
55      SYNTAX      Ieee8021CnDefenseMode
56      MAX-ACCESS  read-write
57      STATUS      current
58      DESCRIPTION
59         "This object indicates the operator's choice for the CND defense
60         mode in which this port is to operate for this CNPV whenever
61         ieee8021CnPortPriDefModeChoice has the value cpcAdmin(1).
62
63         This variable can override ieee8021CnComPriDefModeChoice.
64     "
65     REFERENCE
66         "IEEE 32.1.3, 32.4.1"
67     DEFVAL { cptDisabled }
68     ::= { ieee8021CnPortPriEntry 5 }
69
70  ieee8021CnPortPriAutoDefenseMode
71  OBJECT-TYPE
72      SYNTAX      Ieee8021CnDefenseMode {

```

```

1          cptInterior      (2),
2          cptInteriorReady (3),
3          cptEdge          (4)
4      }
5      MAX-ACCESS read-only
6      STATUS current
7      DESCRIPTION
8          "This object indicates in which the CND defense mode this port
9          would operate for this CNPV as determined by the LLDP
10         Congestion Notification TLV.
11         "
12     REFERENCE
13         "IEEE 32.4.3"
14     ::= { ieee8021CnPortPriEntry 6 }
15
16
17 ieee8021CnPortPriLldpInstanceChoice
18 OBJECT-TYPE
19     SYNTAX      Ieee8021CnLldpChoice
20     MAX-ACCESS  read-write
21     STATUS      current
22     DESCRIPTION
23         "Specifies how to determine the LLDP instance to be used for the
24         Congestion Notification TLV, either:
25             cnlNone(1)      No LLDP Congestion Notification TLV is to
26                             carry Per-priority CNPV indicators or
27                             Per-priority Ready indicators on this Port
28                             for this priority.
29             cnlAdmin(2)     ieee8021CnPortPriLldpInstanceSelector
30                             governs which LLDP instance is to carry
31                             Per-priority CNPV indicators and
32                             Per-priority Ready indicators for this
33                             priority in its Congestion Notification TLV
34                             on this Port
35             cnlComponent(3) ieee8021CnComPriLldpInstanceChoice
36                             governs LLDP instance selection for this
37                             Port and priority.
38
39         This object can override ieee8021CnComPriLldpInstanceChoice.
40         "
41     REFERENCE
42         "32.1.3, 32.4.4"
43     DEFVAL { cnlComponent }
44     ::= { ieee8021CnPortPriEntry 7 }
45
46 ieee8021CnPortPriLldpInstanceSelector
47 OBJECT-TYPE
48     SYNTAX      LldpV2DestAddressTableIndex
49     MAX-ACCESS  read-write
50     STATUS      current
51     DESCRIPTION
52         "This object determines which LLDP instance selector, if any,
53         is used for automatic determination of the CND defense mode for
54         this port and CNPV.
55
56         This object can override ieee8021CnComPriLldpInstanceSelector.
57         "
58     REFERENCE
59         "32.1.3, 32.4.5"
60     DEFVAL { 3 }
61     ::= { ieee8021CnPortPriEntry 8 }
62
63 ieee8021CnPortPriAlternatePriority
64 OBJECT-TYPE
65     SYNTAX      IEEE8021PriorityValue
66     MAX-ACCESS  read-write
67     STATUS      current
68     DESCRIPTION
69         "The Congestion Notification Priority Value to which an
70         incoming frame is to be mapped, in spite of what the
71         Priority Regeneration Table says, if 1) Congestion
72         Notification is enabled and 2) the port is acting in the

```



```

1      cptEdge (4) CND defense mode.
2
3      This object is ignored unless ieee8021CnPortPriDefModeChoice
4      contains the value cpcAdmin (1).
5      "
6      REFERENCE
7      "32.4.6"
8      DEFVAL { 0 }
9      ::= { ieee8021CnPortPriEntry 9 }
10
11
12
13 -- *****
14 -- The CCF per-CP table. One table per end station or Bridge component.
15 -- One entry per Congestion Point. An entry in this table controls one
16 -- Congestion Point (CP).
17 -- *****
18
19 ieee8021CnCpTable
20 OBJECT-TYPE
21     SYNTAX      SEQUENCE OF Ieee8021CnCpEntry
22     MAX-ACCESS  not-accessible
23     STATUS      current
24     DESCRIPTION
25         "Each row in this table supplies values for one
26         Congestion Point (CP).
27
28         This table is indexed by component, port (interface), and
29         an arbitrary CP index. This arbitrary CP index is not
30         necessarily the Congestion Point Identifier (CPID) carried in
31         Congestion Notification Messages (CNMs).
32
33         Creating an entry in ieee8021CnCpTable can create an
34         entry in this table, with the default values, on all ports in
35         the Bridge component or end station. Because more than one
36         Congestion Notification Priority Value (CNPV) can flow
37         through a single CP, the creation of an entry in
38         ieee8021CnCpTable does not necessarily create a new
39         entry in this table. An end station can have more than one
40         CP for the same CNPV, so creating an entry in
41         ieee8021CnCpTable can create multiple entries in this
42         table.
43
44         Because each port in a Bridge component or end station can have
45         a different relationship between CNPVs and CPs, the entries
46         created or deleted on each port can be different.
47
48         Deleting the last entry in ieee8021CnCpTable for a
49         CNPV passing through the CP controlled by this entry deletes
50         the entry on some or all of the ports in the Bridge component
51         or end station.
52
53         Because each port in a Bridge component or end station can have
54         a different relationship between CNPVs and CPs, the entries
55         created or deleted on each port can be different.
56
57         The relationship between ieee8021CnCpIndex
58         values and CPs is an implementation dependent matter.
59
60         The contents of this table SHALL be maintained across a restart
61         of the system, except as noted in the description of
62         ieee8021CnCpIfIndex.
63     "
64     REFERENCE
65     "12.21.4"
66     ::= { ieee8021CnMIBObjects 5 }
67
68 ieee8021CnCpEntry
69 OBJECT-TYPE
70     SYNTAX      Ieee8021CnCpEntry
71     MAX-ACCESS  not-accessible
72     STATUS      current

```

```

1  DESCRIPTION
2      "An entry in the Congestion Point table controls a single
3      Congestion Point on a port in a Bridge component or end station.
4      "
5  REFERENCE
6      "12.21.4"
7  INDEX { ieee8021CnCpComponentId,
8          ieee8021CnCpIfIndex,
9          ieee8021CnCpIndex }
10 ::= { ieee8021CnCpTable 1 }
11
12 Ieee8021CnCpEntry
13 ::= SEQUENCE {
14     ieee8021CnCpComponentId      IEEE8021PbbComponentIdentifier,
15     ieee8021CnCpIfIndex          InterfaceIndex,
16     ieee8021CnCpIndex            Unsigned32,
17     ieee8021CnCpPriority          IEEE8021PriorityValue,
18     ieee8021CnCpMacAddress       MacAddress,
19     ieee8021CnCpIdentifier       OCTET STRING,
20     ieee8021CnCpQueueSizeSetPoint Unsigned32,
21     ieee8021CnCpFeedbackWeight   Integer32,
22     ieee8021CnCpMinSampleBase    Unsigned32,
23     ieee8021CnCpDiscardedFrames  Counter64,
24     ieee8021CnCpTransmittedFrames Counter64,
25     ieee8021CnCpTransmittedCnms  Counter64,
26     ieee8021CnCpMinHeaderOctets  Unsigned32
27 }
28
29 ieee8021CnCpComponentId
30 OBJECT-TYPE
31     SYNTAX      IEEE8021PbbComponentIdentifier
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35         "The Bridge component within the system to which the information
36         in this ieee8021CnCpEntry applies. If the system is
37         not a Bridge, or if only one component is present in the
38         Bridge, then this variable (index) MUST be equal to 1.
39         "
40     REFERENCE
41         "12.21.4"
42     ::= { ieee8021CnCpEntry 1 }
43
44 ieee8021CnCpIfIndex
45 OBJECT-TYPE
46     SYNTAX      InterfaceIndex
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "This object represents the port or aggregated port
51         to which the entry applies.
52
53         Upon a restart of the system, the system SHALL, if necessary,
54         change the value of this object, and rearrange the order of the
55         ieee8021CnCpTable, so that the value in ieee8021CnCpIfIndex
56         references the row in the ifXTable with the same value for
57         ifAlias that it referenced before the system restart. If no
58         such entry exists in the ifXTable, then the system SHALL delete
59         the row in the ieee8021CnCpTable.
60         "
61     REFERENCE
62         "12.21.4"
63     ::= { ieee8021CnCpEntry 2 }
64
65 ieee8021CnCpIndex
66 OBJECT-TYPE
67     SYNTAX      Unsigned32 (1..4096)
68     MAX-ACCESS  not-accessible
69     STATUS      current
70     DESCRIPTION
71         "This object is an arbitrary integer indexing the entries in
72         this table among the entries for the same component and

```

```
1      interface. In a system that supports no more than one
2      Congestion Point per priority per interface, ieee8021CnCpIndex
3      SHALL be equal to the lowest numerical
4      Congestion Notification Priority Value served by this
5      Congestion Point. Otherwise, it SHOULD be a small integer
6      value.
7      "
8  REFERENCE
9      "12.21.4"
10     ::= { ieee8021CnCpEntry 3 }
11
12 ieee8021CnCpPriority
13 OBJECT-TYPE
14     SYNTAX      IEEE8021PriorityValue
15     MAX-ACCESS  read-only
16     STATUS      current
17     DESCRIPTION
18         "This object indicates the lowest numerical
19         Congestion Notification Priority Value that this entry's
20         Congestion Point serves.
21         "
22     REFERENCE
23         "12.21.4"
24     ::= { ieee8021CnCpEntry 4 }
25
26 ieee8021CnCpMacAddress
27 OBJECT-TYPE
28     SYNTAX      MacAddress
29     MAX-ACCESS  read-only
30     STATUS      current
31     DESCRIPTION
32         "This object indicates the MAC address used as the source
33         address in Congestion Notification Message transmitted
34         by this Congestion Point.
35         "
36     REFERENCE
37         "32.8.1"
38     ::= { ieee8021CnCpEntry 5 }
39
40 ieee8021CnCpIdentifier
41 OBJECT-TYPE
42     SYNTAX      OCTET STRING (SIZE(8))
43     MAX-ACCESS  read-only
44     STATUS      current
45     DESCRIPTION
46         "This object indicates the Congestion Point Identifier (CPID)
47         transmitted in Congestion Notification Message by this
48         Congestion Point.
49
50         It is not specified whether the CPID reported in a CNM by a CP
51         that serves multiple CNPVs does or does not have the same value
52         for its different CNPVs.
53         "
54     REFERENCE
55         "32.8.2"
56     ::= { ieee8021CnCpEntry 6 }
57
58 ieee8021CnCpQueueSizeSetPoint
59 OBJECT-TYPE
60     SYNTAX      Unsigned32 (100..4294967295)
61     UNITS       "octets"
62     MAX-ACCESS  read-write
63     STATUS      current
64     DESCRIPTION
65         "This object is the set point for the queue managed by this
66         Congestion Point (CP). Congestion Notification Messages are
67         transmitted to the sources of frames queued in this CP's
68         queue in order to keep the total number of octets stored in
69         the queue at this set point.
70         "
71     REFERENCE
72         "30.2, 32.8.3"
```

```
1  DEFVAL { 26000 }
2  ::= { ieee8021CnCpEntry 7 }
3
4  ieee8021CnCpFeedbackWeight
5  OBJECT-TYPE
6      SYNTAX      Integer32 (-10..10)
7      MAX-ACCESS  read-write
8      STATUS      current
9      DESCRIPTION
10         "This object controls the weight (cpW) change in queue length
11         in the calculation of cpFb when the Congestion Point is
12         generating a Congestion Notification Message.
13
14         The weight cpW is equal to two to the power of this object.
15         Thus, if this object contains a -1, cpW = 1/2.
16
17         "
17     REFERENCE
18         "32.8.6"
19     DEFVAL { 1 }
20     ::= { ieee8021CnCpEntry 8 }
21
22  ieee8021CnCpMinSampleBase
23  OBJECT-TYPE
24      SYNTAX      Unsigned32 (10000..4294967295)
25      UNITS       "octets"
26      MAX-ACCESS  read-write
27      STATUS      current
28      DESCRIPTION
29         "This object determines the minimum number of octets to
30         enqueue in the Congestion Point's queue between transmissions
31         of Congestion Notification Messages.
32
33         "
33     REFERENCE
34         "32.8.11"
35     DEFVAL { 150000 }
36     ::= { ieee8021CnCpEntry 9 }
37
38  ieee8021CnCpDiscardedFrames
39  OBJECT-TYPE
40      SYNTAX      Counter64
41      UNITS       "frames"
42      MAX-ACCESS  read-only
43      STATUS      current
44      DESCRIPTION
45         "The number of data frames discarded by the queue controlled
46         by this Congestion Point due to queue congestion.
47
48         Discontinuities in the value of this counter can occur
49         at re-initialization of the management system, and at
50         other times as indicated by the value of
51         ifCounterDiscontinuityTime object of the associated
52         interface (if any).
53
54         "
54     REFERENCE
55         "32.8.12"
56     ::= { ieee8021CnCpEntry 10 }
57
58  ieee8021CnCpTransmittedFrames
59  OBJECT-TYPE
60      SYNTAX      Counter64
61      UNITS       "frames"
62      MAX-ACCESS  read-only
63      STATUS      current
64      DESCRIPTION
65         "The number of data frames passed on to the queue controlled by
66         this Congestion Point that were not discarded due to queue
67         congestion.
68
69         Discontinuities in the value of this counter can occur
70         at re-initialization of the management system, and at
71         other times as indicated by the value of
72         ifCounterDiscontinuityTime object of the associated
```

```

1      interface (if any).
2      "
3      REFERENCE
4      "32.8.13"
5      ::= { ieee8021CnCpEntry 11 }
6
7  ieee8021CnCpTransmittedCnms
8  OBJECT-TYPE
9      SYNTAX      Counter64
10     UNITS       "frames"
11     MAX-ACCESS  read-only
12     STATUS      current
13     DESCRIPTION
14         "The number of Congestion Notification Message transmitted
15         by this Congestion Point.
16
17         Discontinuities in the value of this counter can occur
18         at re-initialization of the management system, and at
19         other times as indicated by the value of
20         ifCounterDiscontinuityTime object of the associated
21         interface (if any).
22     "
23     REFERENCE
24     "32.8.14"
25     ::= { ieee8021CnCpEntry 12 }
26
27  ieee8021CnCpMinHeaderOctets
28  OBJECT-TYPE
29     SYNTAX      Unsigned32 (0..64)
30     UNITS       "octets"
31     MAX-ACCESS  read-write
32     STATUS      current
33     DESCRIPTION
34         "Specifies the minimum number of octets to be returned in a
35         Congestion Notification Message from the mac_service_data_unit
36         of the data frame that triggered transmission of the CNM. If
37         the mac_service_data unit has fewer octets than the value of
38         this object, then all of the mac_service_data_unit is returned
39         in the CNM.
40     "
41     REFERENCE
42     "32.8.15, 32.9.4 k)"
43     DEFVAL { 0 }
44     ::= { ieee8021CnCpEntry 13 }
45
46
47
48 -- *****
49 -- The CPID to {Interface, CP index} table. One table per system.
50 -- One entry per CPID.
51 -- *****
52
53  ieee8021CnCpidToInterfaceTable
54  OBJECT-TYPE
55     SYNTAX      SEQUENCE OF Ieee8021CnCpidToInterfaceEntry
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59         "This table allows the network manager to obtain the
60         interface index and CP index needed to access an entry in
61         the ieee8021CnCpTable, given a Congestion Point Identifier
62         (CPID) received a Congestion Notification Messages (CNMs).
63
64         Upon a restart of the system, the system SHALL, if necessary,
65         update this table to be consistent with the ieee8021CnCpTable.
66     "
67     REFERENCE
68     "17.2.13, 12.21.4, 32.8.2"
69     ::= { ieee8021CnMIBObjects 6 }
70
71  ieee8021CnCpidToInterfaceEntry
72  OBJECT-TYPE

```

```

1  SYNTAX      Ieee8021CnCpidToInterfaceEntry
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "An entry in the ieee8021CnCpidToInterfaceTable. Translates
6      a Congestion Point Identifier to a component identifier,
7      interface index, and CP index
8      "
9  REFERENCE
10     "17.2.13"
11  INDEX { ieee8021CnCpidToIfCpid }
12  ::= { ieee8021CnCpidToInterfaceTable 1 }
13
14 Ieee8021CnCpidToInterfaceEntry
15 ::= SEQUENCE {
16     ieee8021CnCpidToIfCpid      OCTET STRING,
17     ieee8021CnCpidToIfComponentId IEEE8021PbbComponentIdentifier,
18     ieee8021CnCpidToIfIfIndex   InterfaceIndex,
19     ieee8021CnCpidToIfCpIndex   Unsigned32
20 }
21
22 ieee8021CnCpidToIfCpid
23 OBJECT-TYPE
24     SYNTAX      OCTET STRING (SIZE(8))
25     MAX-ACCESS  not-accessible
26     STATUS      current
27     DESCRIPTION
28         "This object is the Congestion Point Identifier (CPID)
29         transmitted in Congestion Notification Message by a
30         Congestion Point residing in this Bridge component or
31         end station.
32         "
33     REFERENCE
34         "17.2.13, 32.8.2"
35     ::= { ieee8021CnCpidToInterfaceEntry 1 }
36
37 ieee8021CnCpidToIfComponentId
38 OBJECT-TYPE
39     SYNTAX      IEEE8021PbbComponentIdentifier
40     MAX-ACCESS  read-only
41     STATUS      current
42     DESCRIPTION
43         "The Bridge component within the system to which the information
44         in this ieee8021CnCpidToInterfaceEntry applies. If the system
45         is not a Bridge, or if only one component is present in the
46         Bridge, then this variable (index) MUST be equal to 1.
47         "
48     REFERENCE
49         "17.2.13"
50     ::= { ieee8021CnCpidToInterfaceEntry 2 }
51
52 ieee8021CnCpidToIfIfIndex
53 OBJECT-TYPE
54     SYNTAX      InterfaceIndex
55     MAX-ACCESS  read-only
56     STATUS      current
57     DESCRIPTION
58         "This object indicates the interface on which the selected
59         Congestion Point resides. This value can be used, along
60         with ieee8021CnCpidToIfCpIndex, to find the Congestion Point
61         in the ieee8021CnCpTable.
62         "
63     REFERENCE
64         "17.2.13"
65     ::= { ieee8021CnCpidToInterfaceEntry 3 }
66
67 ieee8021CnCpidToIfCpIndex
68 OBJECT-TYPE
69     SYNTAX      Unsigned32 (1..4096)
70     MAX-ACCESS  read-only
71     STATUS      current
72     DESCRIPTION

```

```

1      "This object indicates the Congestion Point's index on the
2      interface on which the selected Congestion Point resides.
3      This value can be used, along with ieee8021CnCpidToIfIfIndex,
4      to find the Congestion Point in the ieee8021CnCpTable.
5      "
6  REFERENCE
7      "17.2.13"
8  ::= { ieee8021CnCpidToInterfaceEntry 4 }
9
10
11
12 -- *****
13 -- The Reaction Point Port table. One table per end station. One
14 -- entry per Port per CNPV.
15 -- *****
16
17 ieee8021CnRpPortPriTable
18 OBJECT-TYPE
19     SYNTAX      SEQUENCE OF Ieee8021CnRpPortPriEntry
20     MAX-ACCESS  not-accessible
21     STATUS      current
22     DESCRIPTION
23         "Each row in this table supplies values for all of the
24         Reaction Points (RPs) on one Port and one priority of an end
25         station or Bridge component. This table is indexed by
26         component, port (interface), and priority.
27
28         Creating an entry in ieee8021CnCompntPriTable can create an
29         entry in this table, with the default values, on all ports
30         in the end station.
31
32         Deleting the an entry in ieee8021CnCompntPriTable for a
33         CNPV passing through the RP controlled by this entry deletes
34         entries on some or all of the ports in the end station.
35
36         The contents of this table SHALL be maintained across a restart
37         of the system, except as noted in the description of
38         ieee8021CnRpPortPriIfIndex.
39     "
40     REFERENCE
41         "12.21.5"
42     ::= { ieee8021CnMIBObjects 7 }
43
44 ieee8021CnRpPortPriEntry
45 OBJECT-TYPE
46     SYNTAX      Ieee8021CnRpPortPriEntry
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "An entry in the Reaction Point table controls all of the
51         Reaction Points on a port in an end station that share the same
52         priority value.
53     "
54     REFERENCE
55         "12.21.5"
56     INDEX { ieee8021CnRpPortPriComponentId, ieee8021CnRpPortPriPriority,
57             ieee8021CnRpPortPriIfIndex
58         }
59     ::= { ieee8021CnRpPortPriTable 1 }
60
61 Ieee8021CnRpPortPriEntry
62 ::= SEQUENCE {
63     ieee8021CnRpPortPriComponentId  IEEE8021PbbComponentIdentifier,
64     ieee8021CnRpPortPriPriority      IEEE8021PriorityValue,
65     ieee8021CnRpPortPriIfIndex       InterfaceIndex,
66     ieee8021CnRpPortPriMaxRps        Unsigned32,
67     ieee8021CnRpPortPriCreatedRps    Counter32,
68     ieee8021CnRpPortPriCentiseconds Counter64
69 }
70
71 ieee8021CnRpPortPriComponentId
72 OBJECT-TYPE

```

```

1  SYNTAX      IEEE8021PbbComponentIdentifier
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "The Bridge component within the system to which the information
6      in this ieee8021CnRpGroupEntry applies. If the system is
7      not a Bridge, or if only one component is present in the
8      Bridge, then this variable (index) MUST be equal to 1.
9      "
10 ::= { ieee8021CnRpPortPriEntry 1 }
11
12 ieee8021CnRpPortPriPriority
13 OBJECT-TYPE
14     SYNTAX      IEEE8021PriorityValue
15     MAX-ACCESS  not-accessible
16     STATUS      current
17     DESCRIPTION
18         "This object indicates the lowest numerical
19         Congestion Notification Priority Value that this entry's
20         Reaction Point serves.
21         "
22     REFERENCE
23         "12.21.5"
24     ::= { ieee8021CnRpPortPriEntry 2 }
25
26 ieee8021CnRpPortPriIfIndex
27 OBJECT-TYPE
28     SYNTAX      InterfaceIndex
29     MAX-ACCESS  not-accessible
30     STATUS      current
31     DESCRIPTION
32         "This object indicates the interface on which the selected
33         Reaction Points reside.
34
35         Upon a restart of the system, the system SHALL, if necessary,
36         change the value of this object, and rearrange the order of the
37         ieee8021CnRpPortPriTable, so that the value in
38         ieee8021CnRpPortPriIfIndex references the row in the ifXTable
39         with the same value for ifAlias that it referenced before the
40         system restart. If no such entry exists in the ifXTable, then
41         the system SHALL delete the row in the
42         ieee8021CnRpPortPriTable.
43         "
44     REFERENCE
45         "12.21.5"
46     ::= { ieee8021CnRpPortPriEntry 3 }
47
48 ieee8021CnRpPortPriMaxRps
49 OBJECT-TYPE
50     SYNTAX      Unsigned32 (1..100)
51     MAX-ACCESS  read-write
52     STATUS      current
53     DESCRIPTION
54         "An integer controlling the maximum number of Reaction Points
55         allowed for this CNPV on this Port. An end station SHALL
56         not create more than this many Reaction Point on this Port,
57         but it MAY create fewer.
58         "
59     REFERENCE
60         "32.10.1"
61     DEFVAL { 1 }
62     ::= { ieee8021CnRpPortPriEntry 4 }
63
64 ieee8021CnRpPortPriCreatedRps
65 OBJECT-TYPE
66     SYNTAX      Counter32
67     MAX-ACCESS  read-only
68     STATUS      current
69     DESCRIPTION
70         "This object returns the number of times any of the
71         Reaction Points (RPs) controlled by this entry has had
72         its rpEnabled variable set TRUE by the reception of a

```



```
1      Congestion Notification Message.
2
3      Dividing the change in ieee8021CnRpPortPriCentiseconds by the
4      change in this object over a time interval yields the average
5      lifetime of an active RP during that interval.
6  "
7  REFERENCE
8  "32.10.2, 32.10.3, 32.13.1"
9  ::= { ieee8021CnRpPortPriEntry 5 }
10
11 ieee8021CnRpPortPriCentiseconds
12 OBJECT-TYPE
13     SYNTAX      Counter64
14     UNITS       "centiseconds"
15     MAX-ACCESS  read-only
16     STATUS      current
17     DESCRIPTION
18         "This object returns the total number of centi-seconds that
19         any of the Reaction Points (RPs) controlled by this entry
20         has had its rpEnabled variable in the TRUE state. That is,
21         once each centi-second, this counter is incremented by the
22         number of RPs this entry controls that are actively rate
23         limiting output frames.
24
25         Dividing the change in this object over a time interval by the
26         length of the interval yields the average number of RPs active
27         over that interval. Dividing the change in this object by the
28         change in ieee8021CnRpPortPriCreatedRps over that same time
29         interval yields the average lifetime of an active RP during that
30         interval.
31     "
32     REFERENCE
33     "32.10.3, 32.13.1"
34     ::= { ieee8021CnRpPortPriEntry 6 }
35
36
37
38 -- *****
39 -- The Reaction Point Group table. One table per end station.
40 -- One entry per Reaction Point.
41 -- *****
42
43 ieee8021CnRpGroupTable
44 OBJECT-TYPE
45     SYNTAX      SEQUENCE OF Ieee8021CnRpGroupEntry
46     MAX-ACCESS  not-accessible
47     STATUS      current
48     DESCRIPTION
49         "Each row in this table supplies values for one or more
50         Reaction Points (RPs). This table is indexed by component,
51         port (interface), and an arbitrary RP index.
52
53         Creating an entry in ieee8021CnCompntPriTable can create an
54         entry in this table, with the default values, on all ports
55         in the end station. An end station can have more than one
56         RP for the same Congestion Notification Priority Value
57         (CNPV), so creating an entry in ieee8021CnCompntPriTable can
58         create multiple entries in this table.
59
60         Because each port in a Bridge component or end station can have
61         a different relationship between CNPVs and RPs, the entries
62         created or deleted on each port can be different.
63
64         Deleting the an entry in ieee8021CnCompntPriTable for a
65         CNPV passing through the RP controlled by this entry deletes
66         entries on some or all of the ports in the end station.
67
68         Because each port in an end station can have a
69         different relationship between CNPVs and RPs, the entries
70         created or deleted on each port can be different.
71
72     The relationship between ieee8021CnRpgIdentifier
```

```

1      values and RPs is an implementation dependent matter.
2
3      The contents of this table SHALL be maintained across a restart
4      of the system, except as noted in the description of
5      ieee8021CnRpgIfIndex.
6      "
7      REFERENCE
8      "12.21.6"
9      ::= { ieee8021CnMIBObjects 8 }
10
11 ieee8021CnRpGroupEntry
12 OBJECT-TYPE
13     SYNTAX      Ieee8021CnRpGroupEntry
14     MAX-ACCESS  not-accessible
15     STATUS      current
16     DESCRIPTION
17         "An entry in the Reaction Point table controls a group of
18         Reaction Points, on a port in an end station. All of the
19         Reaction Point controlled by this entry serve the same
20         Congestion Notification Priority Value.
21         "
22     INDEX { ieee8021CnRpgComponentId, ieee8021CnRpgPriority,
23             ieee8021CnRpgIfIndex, ieee8021CnRpgIdentifier }
24     ::= { ieee8021CnRpGroupTable 1 }
25
26 Ieee8021CnRpGroupEntry
27 ::= SEQUENCE {
28     ieee8021CnRpgComponentId  IEEE8021PbbComponentIdentifier,
29     ieee8021CnRpgPriority      IEEE8021PriorityValue,
30     ieee8021CnRpgIfIndex      InterfaceIndex,
31     ieee8021CnRpgIdentifier    Unsigned32,
32     ieee8021CnRpgEnable       TruthValue,
33     ieee8021CnRpgTimeReset     TimeInterval,
34     ieee8021CnRpgByteReset     Unsigned32,
35     ieee8021CnRpgThreshold     Unsigned32,
36     ieee8021CnRpgMaxRate       Unsigned32,
37     ieee8021CnRpgAiRate        Unsigned32,
38     ieee8021CnRpgHaiRate       Unsigned32,
39     ieee8021CnRpgGd            Integer32,
40     ieee8021CnRpgMinDecFac     Unsigned32,
41     ieee8021CnRpgMinRate       Unsigned32
42 }
43
44 ieee8021CnRpgComponentId
45 OBJECT-TYPE
46     SYNTAX      IEEE8021PbbComponentIdentifier
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "The Bridge component within the system to which the information
51         in this ieee8021CnRpGroupEntry applies. If the system is
52         not a Bridge, or if only one component is present in the
53         Bridge, then this variable (index) MUST be equal to 1.
54         "
55     ::= { ieee8021CnRpGroupEntry 1 }
56
57 ieee8021CnRpgPriority
58 OBJECT-TYPE
59     SYNTAX      IEEE8021PriorityValue
60     MAX-ACCESS  not-accessible
61     STATUS      current
62     DESCRIPTION
63         "This object indicates the lowest numerical
64         Congestion Notification Priority Value that this entry's
65         Reaction Point serves.
66         "
67     REFERENCE
68     "12.21.5"
69     ::= { ieee8021CnRpGroupEntry 2 }
70
71 ieee8021CnRpgIfIndex
72 OBJECT-TYPE

```

```

1  SYNTAX      InterfaceIndex
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION
5      "This object indicates the interface on which the group of
6      Reaction Points reside.
7
8      Upon a restart of the system, the system SHALL, if necessary,
9      change the value of this object, and rearrange the order of the
10     ieee8021CnRpGroupTable, so that the value in
11     ieee8021CnRpIfIndex references the row in the ifXTable with
12     the same value for ifAlias that it referenced before the system
13     restart. If no such entry exists in the ifXTable, then the
14     system SHALL delete the row in the ieee8021CnRpGroupTable.
15     "
16  REFERENCE
17      "12.21.5"
18      ::= { ieee8021CnRpGroupEntry 3 }
19
20 ieee8021CnRpgIdentifier
21 OBJECT-TYPE
22     SYNTAX      Unsigned32 (1..4096)
23     MAX-ACCESS  not-accessible
24     STATUS      current
25     DESCRIPTION
26         "This object is an arbitrary integer indexing the entries in
27         this table among the entries for the same interface. This
28         index SHOULD, where possible, be equal to the
29         Congestion Notification Priority Value served by this
30         Reaction Point.
31         "
32     REFERENCE
33         "12.21.6"
34     ::= { ieee8021CnRpGroupEntry 4 }
35
36 ieee8021CnRpgEnable
37 OBJECT-TYPE
38     SYNTAX      TruthValue
39     MAX-ACCESS  read-write
40     STATUS      current
41     DESCRIPTION
42         "Controls the rpEnabled variable of the Reaction Point state
43         machines of the Reaction Points (RPs) controlled by this
44         entry as follows:
45             true(1)    The rpEnabled variable for the RPs controlled by
46                       this object are not held in the FALSE state,
47                       thus enabling them to pay attention to received
48                       CNMs.
49             false(2)   The rpEnabled variable for the RPs controlled by
50                       this object are held in the FALSE state, thus
51                       disabling them from paying attention to received
52                       CNMs.
53         "
54     REFERENCE
55         "32.11.1, 32.13.1"
56     DEFVAL { true }
57     ::= { ieee8021CnRpGroupEntry 5 }
58
59 ieee8021CnRpgTimeReset
60 OBJECT-TYPE
61     SYNTAX      TimeInterval
62     UNITS        "milliseconds"
63     MAX-ACCESS  read-write
64     STATUS      current
65     DESCRIPTION
66         "This object controls the value for all of the state machine
67         variables, rpgTimeReset, used to reset the timers RpWhile.
68         "
69     REFERENCE
70         "32.11.2"
71     DEFVAL { 15 }
72     ::= { ieee8021CnRpGroupEntry 6 }

```

```

1
2 ieee8021CnRpgByteReset
3 OBJECT-TYPE
4     SYNTAX      Unsigned32
5     UNITS       "octets"
6     MAX-ACCESS  read-write
7     STATUS      current
8     DESCRIPTION
9         "This object controls the value for all of the state machine
10        variables, rpgByteReset, used to reset the counters
11        rpByteCount.
12        "
13     REFERENCE
14         "32.11.3"
15     DEFVAL { 150000 }
16     ::= { ieee8021CnRpGroupEntry 7 }
17
18 ieee8021CnRpgThreshold
19 OBJECT-TYPE
20     SYNTAX      Unsigned32
21     MAX-ACCESS  read-write
22     STATUS      current
23     DESCRIPTION
24         "This object controls the number of times rpByteStage or
25        rpTimeStage can count before the
26        RP rate control state machine advances states.
27        "
28     REFERENCE
29         "32.11.4"
30     DEFVAL { 5 }
31     ::= { ieee8021CnRpGroupEntry 8 }
32
33 ieee8021CnRpgMaxRate
34 OBJECT-TYPE
35     SYNTAX      Unsigned32
36     UNITS       "Mb/s"
37     MAX-ACCESS  read-write
38     STATUS      current
39     DESCRIPTION
40         "This object controls the maximum rate, in multiples of 1 Mb/s,
41        at which an RP can transmit. Default value is the speed of the
42        port. A system SHALL support a minimum value for this object
43        that is no larger than 5 Mbits/s (object value 5). This rate
44        includes all bits consequent to transmitting the frame on the
45        LAN, including preamble, inter-frame gap, etc.
46        "
47     REFERENCE
48         "32.11.5"
49     ::= { ieee8021CnRpGroupEntry 9 }
50
51 ieee8021CnRpgAiRate
52 OBJECT-TYPE
53     SYNTAX      Unsigned32
54     UNITS       "Mb/s"
55     MAX-ACCESS  read-write
56     STATUS      current
57     DESCRIPTION
58         "This object controls the transmission rate increment in the
59        RPR_ACTIVE_INCREASE state (rpgAiRate) in multiples of 1 Mb/s.
60        This rate includes all bits consequent to transmitting the
61        frame on the LAN, including preamble, inter-frame gap, etc.
62        "
63     REFERENCE
64         "32.11.6"
65     DEFVAL { 5 }
66     ::= { ieee8021CnRpGroupEntry 10 }
67
68 ieee8021CnRpgHaiRate
69 OBJECT-TYPE
70     SYNTAX      Unsigned32
71     UNITS       "Mb/s"
72     MAX-ACCESS  read-write

```

```

1  STATUS      current
2  DESCRIPTION
3      "This object controls the transmission rate increment in the
4      RPR_HYPER_INCREASE state (rpgHaiRate) in multiples of 1 Mb/s.
5      This rate includes all bits consequent to transmitting the
6      frame on the LAN, including preamble, inter-frame gap, etc.
7      "
8  REFERENCE
9      "32.11.7"
10 DEFVAL { 50 }
11 ::= { ieee8021CnRpGroupEntry 11 }
12
13 ieee8021CnRpgGd
14 OBJECT-TYPE
15     SYNTAX      Integer32
16     MAX-ACCESS  read-write
17     STATUS      current
18     DESCRIPTION
19         "This object controls the number of bits that the value of the
20         Quantized Feedback field received in a CNM PDU is shifted to
21         the right to decrease rpTargetRate. rpgGd is thus 2 to the
22         negative power of this object, e.g., 7 means rpgGd = 1/128.
23         "
24     REFERENCE
25         "32.11.8"
26     DEFVAL { 7 }
27     ::= { ieee8021CnRpGroupEntry 12 }
28
29 ieee8021CnRpgMinDecFac
30 OBJECT-TYPE
31     SYNTAX      Unsigned32 (1..100)
32     UNITS       "percent"
33     MAX-ACCESS  read-write
34     STATUS      current
35     DESCRIPTION
36         "This object controls the minimum factor by which the current
37         RP transmit rate rpCurrentRate can be changed by reception
38         of a Congestion Notification Message. Its integer value
39         represents a percentage, from 1% to 100%.
40         "
41     REFERENCE
42         "32.11.9"
43     DEFVAL { 50 }
44     ::= { ieee8021CnRpGroupEntry 13 }
45
46 ieee8021CnRpgMinRate
47 OBJECT-TYPE
48     SYNTAX      Unsigned32
49     UNITS       "Mb/s"
50     MAX-ACCESS  read-write
51     STATUS      current
52     DESCRIPTION
53         "This object controls the minimum transmission rate (rpgMinRate)
54         in multiples of 1 Mb/s. A system SHALL support a value for
55         this object that is no larger than 5 Mb/s per second.
56         This rate includes all bits consequent to transmitting the
57         frame on the LAN, including preamble, inter-frame gap, etc.
58         "
59     REFERENCE
60         "32.11.10"
61     DEFVAL { 5 }
62     ::= { ieee8021CnRpGroupEntry 14 }
63
64
65
66 -- *****
67 -- IEEE 802.1Qau MIB Module - Conformance Information
68 -- *****
69
70 ieee8021CnCompliances
71 OBJECT IDENTIFIER ::= { ieee8021CnConformance 1 }
72 ieee8021CnGroups

```

```

1      OBJECT IDENTIFIER ::= { ieee8021CnConformance 2 }
2
3 -- *****
4 -- Units of conformance
5 -- *****
6
7 ieee8021CnGlobalReqdGroup
8 OBJECT-GROUP
9   OBJECTS {
10      ieee8021CnGlobalMasterEnable,
11      ieee8021CnComPriLldpInstanceChoice,
12      ieee8021CnComPriLldpInstanceSelector
13   }
14   STATUS      current
15   DESCRIPTION
16      "Objects in the global required group."
17   ::= { ieee8021CnGroups 1 }
18
19 ieee8021CnCpGlobalGroup
20 OBJECT-GROUP
21   OBJECTS {
22      ieee8021CnGlobalCnmTransmitPriority,
23      ieee8021CnGlobalDiscardedFrames
24   }
25   STATUS      current
26   DESCRIPTION
27      "Objects in the Congestion Point global group."
28   ::= { ieee8021CnGroups 2 }
29
30 ieee8021CnCpidTranslateGroup
31 OBJECT-GROUP
32   OBJECTS {
33      ieee8021CnCpidToIfComponentId,
34      ieee8021CnCpidToIfIfIndex,
35      ieee8021CnCpidToIfCpIndex
36   }
37   STATUS      current
38   DESCRIPTION
39      "Objects in the CPID translate group."
40   ::= { ieee8021CnGroups 3 }
41
42 ieee8021CnEplGroup
43 OBJECT-GROUP
44   OBJECTS {
45      ieee8021CnEpIfIndex
46   }
47   STATUS      current
48   DESCRIPTION
49      "Objects for the Errored Ports Table group."
50   ::= { ieee8021CnGroups 4 }
51
52 ieee8021CnComPriGroup
53 OBJECT-GROUP
54   OBJECTS {
55      ieee8021CnComPriDefModeChoice,
56      ieee8021CnComPriAdminDefenseMode,
57      ieee8021CnComPriCreation,
58      ieee8021CnComPriRowStatus
59   }
60   STATUS      current
61   DESCRIPTION
62      "Objects for the global per-priority group."
63   ::= { ieee8021CnGroups 5 }
64
65 ieee8021CnCpPriGroup
66 OBJECT-GROUP
67   OBJECTS {
68      ieee8021CnComPriAlternatePriority,
69      ieee8021CnComPriAutoAltPri
70   }
71   STATUS      current
72   DESCRIPTION

```

```
1      "Objects for the Congestion Point per-priority group."
2      ::= { ieee8021CnGroups 6 }
3
4 ieee8021CnGlobalPriPortGroup
5 OBJECT-GROUP
6     OBJECTS {
7         ieee8021CnPortPriDefModeChoice,
8         ieee8021CnPortPriAdminDefenseMode,
9         ieee8021CnPortPriAutoDefenseMode,
10        ieee8021CnPortPriLldpInstanceChoice,
11        ieee8021CnPortPriLldpInstanceSelector
12    }
13     STATUS      current
14     DESCRIPTION
15         "Objects for the global per-priority per-port group."
16     ::= { ieee8021CnGroups 7 }
17
18 ieee8021CnCpPriPortGroup
19 OBJECT-GROUP
20     OBJECTS {
21         ieee8021CnPortPriAlternatePriority
22     }
23     STATUS      current
24     DESCRIPTION
25         "Objects for the Congestion Point per-priority per-port
26         group."
27         "
28     ::= { ieee8021CnGroups 8 }
29
30 ieee8021CnCpGroup
31 OBJECT-GROUP
32     OBJECTS {
33         ieee8021CnCpPriority,
34         ieee8021CnCpMacAddress,
35         ieee8021CnCpIdentifier,
36         ieee8021CnCpQueueSizeSetPoint,
37         ieee8021CnCpFeedbackWeight,
38         ieee8021CnCpMinSampleBase,
39         ieee8021CnCpDiscardedFrames,
40         ieee8021CnCpTransmittedFrames,
41         ieee8021CnCpTransmittedCnms,
42         ieee8021CnCpMinHeaderOctets
43     }
44     STATUS      current
45     DESCRIPTION
46         "Objects for the Congestion Point group."
47     ::= { ieee8021CnGroups 9 }
48
49 ieee8021CnRpppGroup
50 OBJECT-GROUP
51     OBJECTS {
52         ieee8021CnRpPortPriMaxRps,
53         ieee8021CnRpPortPriCreatedRps,
54         ieee8021CnRpPortPriCentiseconds
55     }
56     STATUS      current
57     DESCRIPTION
58         "Objects for the Reaction Point per-Port per-priority group."
59     ::= { ieee8021CnGroups 10 }
60
61 ieee8021CnRpGroup
62 OBJECT-GROUP
63     OBJECTS {
64         ieee8021CnRpgEnable,
65         ieee8021CnRpgTimeReset,
66         ieee8021CnRpgByteReset,
67         ieee8021CnRpgThreshold,
68         ieee8021CnRpgMaxRate,
69         ieee8021CnRpgAiRate,
70         ieee8021CnRpgHaiRate,
71         ieee8021CnRpgGd,
72         ieee8021CnRpgMinDecFac,
```

```

1      ieee8021CnRpgMinRate
2      }
3      STATUS      current
4      DESCRIPTION
5          "Objects for the Reaction Point group."
6      ::= { ieee8021CnGroups 11 }
7
8
9  -- *****
10 -- MIB Module Compliance statements
11 -- *****
12
13 ieee8021CnBridgeCompliance
14 MODULE-COMPLIANCE
15     STATUS      current
16     DESCRIPTION
17         "The compliance statement for support by a Bridge of
18         the IEEE8021-MIRP-MIB module."
19
20     MODULE SNMPv2-MIB -- The SNMPv2-MIB, RFC 3418
21         MANDATORY-GROUPS {
22             systemGroup
23         }
24
25     MODULE IF-MIB -- The interfaces MIB, RFC 2863
26         MANDATORY-GROUPS {
27             ifGeneralInformationGroup
28         }
29
30     MODULE
31         MANDATORY-GROUPS {
32             ieee8021CnGlobalReqdGroup,
33             ieee8021CnCpGlobalGroup,
34             ieee8021CnCpidTranslateGroup,
35             ieee8021CnEplGroup,
36             ieee8021CnComPriGroup,
37             ieee8021CnCpPriGroup,
38             ieee8021CnGlobalPriPortGroup,
39             ieee8021CnCpPriPortGroup,
40             ieee8021CnCpGroup
41         }
42
43     OBJECT ieee8021CnComPriRowStatus
44         SYNTAX      RowStatus { active(1), notInService(2),
45                             notReady(3) }
46         WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
47                             destroy(6) }
48         DESCRIPTION "Support for createAndWait is not required."
49
50     ::= { ieee8021CnCompliances 1 }
51
52 ieee8021CnStationCompliance
53 MODULE-COMPLIANCE
54     STATUS      current
55     DESCRIPTION
56         "The compliance statement for support by an end station of
57         the IEEE8021-MIRP-MIB module."
58
59     MODULE SNMPv2-MIB -- The SNMPv2-MIB, RFC 3418
60         MANDATORY-GROUPS {
61             systemGroup
62         }
63
64     MODULE IF-MIB -- The interfaces MIB, RFC 2863
65         MANDATORY-GROUPS {
66             ifGeneralInformationGroup
67         }
68
69     MODULE
70         MANDATORY-GROUPS {
71             ieee8021CnGlobalReqdGroup,
72             ieee8021CnComPriGroup,

```



```
1         ieee8021CnGlobalPriPortGroup,
2         ieee8021CnRpppGroup,
3         ieee8021CnRpGroup
4     }
5
6     GROUP ieee8021CnCpGlobalGroup
7         DESCRIPTION
8             "This group is optional and supports end stations that
9             choose to implement Congestion Points."
10
11     GROUP ieee8021CnCpidTranslateGroup
12         DESCRIPTION
13             "This group is optional and supports end stations that
14             choose to implement Congestion Points."
15
16     GROUP ieee8021CnEplGroup
17         DESCRIPTION
18             "This group is optional and supports end stations that
19             choose to implement Congestion Points."
20
21     GROUP ieee8021CnCpPriGroup
22         DESCRIPTION
23             "This group is optional and supports end stations that
24             choose to implement Congestion Points."
25
26     GROUP ieee8021CnCpPriPortGroup
27         DESCRIPTION
28             "This group is optional and supports end stations that
29             choose to implement Congestion Points."
30
31     GROUP ieee8021CnCpGroup
32         DESCRIPTION
33             "This group is optional and supports end stations that
34             choose to implement Congestion Points."
35
36 ::= { ieee8021CnCompliances 2 }
37 END
```

1 17.7.14 Definitions for the IEEE8021-SRP-MIB module

```
2 IEEE8021-SRP-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for support of IEEE 802.1Qat Stream Reservation Protocol
6 -- (SRP) in IEEE 802.1Q Bridges.
7 -- =====
8
9 IMPORTS
10     MODULE-IDENTITY,
11     OBJECT-TYPE,
12     Counter64,
13     Unsigned32
14     FROM SNMPv2-SMI
15     MacAddress,
16     TEXTUAL-CONVENTION,
17     TruthValue
18     FROM SNMPv2-TC
19     MODULE-COMPLIANCE,
20     OBJECT-GROUP
21     FROM SNMPv2-CONF
22     ieee802dot1mibs,
23     IEEE8021VlanIndex
24     FROM IEEE8021-TC-MIB
25     IEEE8021FqtssTrafficClassValue
26     FROM IEEE8021-FQTSS-MIB
27     ieee8021BridgeBaseComponentId,
28     ieee8021BridgeBaseEntry,
29     ieee8021BridgeBasePort,
30     ieee8021BridgeBasePortEntry
31     FROM IEEE8021-BRIDGE-MIB
32
33 ;
34
35 ieee8021SrpMib MODULE-IDENTITY
36     LAST-UPDATED "202309260000Z" -- September 26, 2023
37     ORGANIZATION "IEEE 802.1 Working Group"
38     CONTACT-INFO
39         " WG-URL: http://www.ieee802.org/1/
40         WG-EMail: stds-802-1-1@ieee.org
41         Contact: IEEE 802.1 Working Group Chair
42         Postal: C/O IEEE 802.1 Working Group
43               IEEE Standards Association
44               445 Hoes Lane
45               Piscataway, NJ 08854
46               USA
47         E-mail: stds-802-1-chairs@ieee.org"
48     DESCRIPTION
49         "The Bridge MIB module for managing devices that support
50         the IEEE Std 802.1Q Stream Reservation Protocol.
51
52         Unless otherwise indicated, the references in this MIB
53         module are to IEEE Std 802.1Q-2022.
54
55         Copyright (C) IEEE (2023).
56         This version of this MIB module is part of IEEE Std 802.1Q;
57         see that standard for full legal notices."
58
59     REVISION "202309260000Z" -- September 26, 2023
60     DESCRIPTION
61         "Published as part of IEEE 802.1Qcw.
62         Changed the applicable SYNTAX of object definitions
63         ieee8021SrpStreamDataFramePriority, and
64         ieee8021SrpStreamPreloadDataFramePriority
65         from IEEE8021PriorityCodePoint to INTEGER
66         to correct an error identifying PCP encoding SYNTAX
67         instead of PCP values as intended."
68
69     REVISION "202211080000Z" -- November 8, 2022
70     DESCRIPTION
71         "Published as part of IEEE Std 802.1Q-2022.
```

1 Cross references and contact information updated."
2
3 REVISION "201810040000Z" -- October 4, 2018
4 DESCRIPTION
5 "Published as part of IEEE 802.1Qcc-2018.
6 Added managed objects for Stream Reservation
7 Protocol (SRP) Enhancements and Performance
8 Improvements"
9
10 REVISION "201806280000Z" -- June 28, 2018
11 DESCRIPTION
12 "Published as part of IEEE Std 802.1Q 2018.
13 Cross references updated. "
14
15 REVISION "201512020000Z" -- December 2, 2015
16 DESCRIPTION
17 "Published as part of IEEE Std 802.1Q-2014 Cor-1.
18 ieee8021SrpReservationFailureBridgeId changed to
19 ieee8021SrpReservationFailureSystemId."
20
21 REVISION "201412150000Z" -- December 15, 2014
22 DESCRIPTION
23 "Published as part of IEEE Std 802.1Q 2014 revision.
24 Cross references updated and corrected."
25
26 REVISION "201102270000Z" -- February 27, 2011
27 DESCRIPTION
28 "Minor edits to contact information etc. as part of
29 2011 revision of Std 802.1Q."
30
31 REVISION "201004190000Z" -- April 19, 2010
32 DESCRIPTION
33 "Initial revision, included in IEEE 802.1Qat"
34 ::= { ieee802dot1mibs 19 }
35
36 -- =====
37 -- Textual Conventions
38 -- =====
39
40 IEEE8021SrpStreamRankValue ::= TEXTUAL-CONVENTION
41 STATUS current
42 DESCRIPTION
43 "An 802.1 SRP Stream Rank value. This is an integer,
44 with the following interpretation placed on the value:
45
46 0: Emergency, high-rank stream,
47 1: Non-emergency stream."
48 REFERENCE "35.2.2.8.5b"
49 SYNTAX INTEGER {
50 emergency(0),
51 nonEmergency(1)
52 }
53
54
55 IEEE8021SrpStreamIdValue ::= TEXTUAL-CONVENTION
56 DISPLAY-HINT "1x:1x:1x:1x:1x:1x:1x:1x"
57 STATUS current
58 DESCRIPTION
59 "Represents an SRP Stream ID, which is often defined
60 as a MAC Address followed by a unique 16-bit ID."
61 SYNTAX OCTET STRING (SIZE (8))
62
63 IEEE8021SrpReservationDirectionValue ::= TEXTUAL-CONVENTION
64 STATUS current
65 DESCRIPTION
66 "An 802.1 SRP Stream Reservation Direction value. This is
67 an integer, with the following interpretation placed on
68 the value:
69
70 0: Talker registrations,
71 1: Listener registrations."
72 REFERENCE "35.2.1.2"

```

1  SYNTAX      INTEGER {
2              talkerRegistrations(0),
3              listenerRegistrations(1)
4          }
5
6
7  IEEE8021SrpReservationDeclarationTypeValue ::= TEXTUAL-CONVENTION
8      STATUS      current
9      DESCRIPTION
10         "An 802.1 SRP Stream Reservation Declaration Type value.
11         This is an integer, with the following interpretation
12         placed on the value:
13
14         0: Talker Advertise,
15         1: Talker Failed,
16         2: Listener Asking Failed,
17         3: Listener Ready,
18         4: Listener Ready Failed."
19  REFERENCE    "35.2.1.3"
20  SYNTAX      INTEGER {
21              talkerAdvertise(0),
22              talkerFailed(1),
23              listenerAskingFailed(2),
24              listenerReady(3),
25              listenerReadyFailed(4)
26          }
27
28
29  IEEE8021SrpReservationFailureCodeValue ::= TEXTUAL-CONVENTION
30      STATUS      current
31      DESCRIPTION
32         "An 802.1 SRP Stream Reservation Failure Code value.
33         This is an integer, with the following interpretation
34         placed on the value:
35
36         0: No failure,
37         1: Insufficient bandwidth,
38         2: Insufficient Bridge resources,
39         3: Insufficient bandwidth for Traffic Class,
40         4: StreamID in use by another Talker,
41         5: Stream destination address already in use,
42         6: Stream pre-empted by higher rank,
43         7: Reported latency has changed,
44         8: Egress port is not AVBCapable,
45         9: Use a different destination_address,
46         10: Out of MSRP resources,
47         11: Out of MMRP resources,
48         12: Cannot store destination_address,
49         13: Requested priority is not an SR Class priority,
50         14: MaxFrameSize is too large for media,
51         15: maxFanInPorts limit has been reached,
52         16: Changes in FirstValue for a registered StreamID,
53         17: VLAN is blocked on this egress port (Registration Forbidden),
54         18: VLAN tagging is disabled on this egress port (untagged set),
55         19: SR class priority mismatch."
56  REFERENCE    "35.2.2.8.7"
57  SYNTAX      INTEGER {
58              noFailure(0),
59              insufficientBandwidth(1),
60              insufficientResources(2),
61              insufficientTrafficClassBandwidth(3),
62              streamIDInUse(4),
63              streamDestinationAddressInUse(5),
64              streamPreemptedByHigherRank(6),
65              latencyHasChanged(7),
66              egressPortNotAVBCapable(8),
67              useDifferentDestinationAddress(9),
68              outOfMSRPResources(10),
69              outOfMMRPResources(11),
70              cannotStoreDestinationAddress(12),
71              priorityIsNoAnSRClass(13),
72              maxFrameSizeTooLarge(14),

```

```

1          maxFanInPortsLimitReached(15),
2          firstValueChangedForStreamID(16),
3          vlanBlockedOnEgress(17),
4          vlanTaggingDisabledOnEgress(18),
5          srClassPriorityMismatch(19)
6      }
7
8
9  -- =====
10 -- subtrees in the SRP MIB
11 -- =====
12
13 ieee8021SrpNotifications
14     OBJECT IDENTIFIER ::= { ieee8021SrpMib 0 }
15
16 ieee8021SrpObjects
17     OBJECT IDENTIFIER ::= { ieee8021SrpMib 1 }
18
19 ieee8021SrpConformance
20     OBJECT IDENTIFIER ::= { ieee8021SrpMib 2 }
21
22 ieee8021SrpConfiguration
23     OBJECT IDENTIFIER ::= { ieee8021SrpObjects 1 }
24
25 ieee8021SrpLatency
26     OBJECT IDENTIFIER ::= { ieee8021SrpObjects 2 }
27
28 ieee8021SrpStreams
29     OBJECT IDENTIFIER ::= { ieee8021SrpObjects 3 }
30
31 ieee8021SrpReservations
32     OBJECT IDENTIFIER ::= { ieee8021SrpObjects 4 }
33
34
35 -- =====
36 -- The ieee8021SrpConfiguration subtree
37 -- This subtree defines the objects necessary for the
38 -- operational management of SRP.
39 -- =====
40
41 ieee8021SrpBridgeBaseTable OBJECT-TYPE
42     SYNTAX      SEQUENCE OF Ieee8021SrpBridgeBaseEntry
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46         "A table for SRP main control and status information.
47         All writable objects in this table must be persistent
48         over power up restart/reboot. These objects augment
49         the ieee8021BridgeBasePortTable."
50     ::= { ieee8021SrpConfiguration 1 }
51
52 ieee8021SrpBridgeBaseEntry OBJECT-TYPE
53     SYNTAX      Ieee8021SrpBridgeBaseEntry
54     MAX-ACCESS  not-accessible
55     STATUS      current
56     DESCRIPTION
57         "SRP control and status information for a Bridge."
58     AUGMENTS { ieee8021BridgeBaseEntry }
59     ::= { ieee8021SrpBridgeBaseTable 1 }
60
61 Ieee8021SrpBridgeBaseEntry ::=
62     SEQUENCE {
63         ieee8021SrpBridgeBaseMsrpEnabledStatus
64             TruthValue,
65         ieee8021SrpBridgeBaseMsrpTalkerPruning
66             TruthValue,
67         ieee8021SrpBridgeBaseMsrpMaxFanInPorts
68             Unsigned32,
69         ieee8021SrpBridgeBaseMsrpLatencyMaxFrameSize
70             Unsigned32,
71         ieee8021SrpBridgeBaseMsrpTalkerVlanPruning
72             TruthValue,

```

```

1         ieee8021SrpBridgeBaseMsrpMaxSRClasses
2         Unsigned32
3     }
4
5 ieee8021SrpBridgeBaseMsrpEnabledStatus OBJECT-TYPE
6     SYNTAX      TruthValue
7     MAX-ACCESS  read-create
8     STATUS      current
9     DESCRIPTION
10        "The administrative status requested by management for
11        MSRP. The value true(1) indicates that MSRP should
12        be enabled on this device, in all VLANs, on all ports
13        for which it has not been specifically disabled. When
14        false(2), MSRP is disabled, in all VLANs and on all
15        ports, and all MSRP frames will be forwarded
16        transparently. This objects affects both Applicant and
17        Registrar state machines. A transition from false(2)
18        to true(1) will cause a reset of all MSRP state
19        machines on all ports.
20
21        This object may be modified while the corresponding
22        instance of ieee8021BridgeBaseRowStatus is active(1).
23
24        The value of this object MUST be retained across
25        reinitializations of the management system."
26     REFERENCE   "35.2.1.4d"
27     DEFVAL      { true }
28     ::= { ieee8021SrpBridgeBaseEntry 1 }
29
30 ieee8021SrpBridgeBaseMsrpTalkerPruning OBJECT-TYPE
31     SYNTAX      TruthValue
32     MAX-ACCESS  read-create
33     STATUS      current
34     DESCRIPTION
35        "The value of the talkerPruning parameter, which
36        controls the propagation of Talker declarations.
37        The value true(1) indicates that Talker attributes
38        are only declared on ports that have the Stream
39        destination_address registered in the MMRP MAC
40        Address Registration Entries. When false(2),
41        Talker attribute are declared on all egress ports
42        in the active topology.
43
44        The value of this object MUST be retained across
45        reinitializations of the management system."
46     REFERENCE   "12.22.1, 35.2.1.4b, 35.2.4.3.1"
47     DEFVAL      { false }
48     ::= { ieee8021SrpBridgeBaseEntry 2 }
49
50 ieee8021SrpBridgeBaseMsrpMaxFanInPorts OBJECT-TYPE
51     SYNTAX      Unsigned32
52     MAX-ACCESS  read-create
53     STATUS      current
54     DESCRIPTION
55        "The value of the msrpMaxFanInPorts parameter, which
56        limits the total number of ports on a Bridge that
57        are allowed to establish reservations for inbound
58        Streams. A value of zero (0) indicates no fan-in
59        limit is being specified and calculations involving
60        fan-in will only be limited by the number of MSRP
61        enabled ports.
62
63        The value of this object MUST be retained across
64        reinitializations of the management system."
65     REFERENCE   "12.22.1, 35.2.1.4f"
66     DEFVAL      { 0 }
67     ::= { ieee8021SrpBridgeBaseEntry 3 }
68
69 ieee8021SrpBridgeBaseMsrpLatencyMaxFrameSize OBJECT-TYPE
70     SYNTAX      Unsigned32
71     MAX-ACCESS  read-create
72     STATUS      current

```

```

1  DESCRIPTION
2      "The value of msrpLatencyMaxFrameSize parameter
3      which is used in the calculation of the maximum
4      latency through a Bridge. The maximum size is
5      defined to be 2000 octets by default, but may be
6      set to a smaller or larger value dependent on the
7      particular Bridge configuration. This parameter
8      does not imply any type of policing of frame size,
9      it is only used in the latency calculations.
10
11      The value of this object MUST be retained across
12      reinitializations of the management system."
13  REFERENCE   "12.22.1, 35.2.1.4g"
14  DEFVAL      { 2000 }
15  ::= { ieee8021SrpBridgeBaseEntry 4 }
16
17 ieee8021SrpBridgeBaseMsrpTalkerVlanPruning OBJECT-TYPE
18     SYNTAX      TruthValue
19     MAX-ACCESS   read-create
20     STATUS      current
21     DESCRIPTION
22         "This parameter allows to limit the Talker declaration
23         to ports, that have the Stream's VLAN identifier
24         registered as a member in the VLAN Registration
25         Entries. The value true(1) indicates that Talker
26         declarations are only sent out on ports, that have the
27         Stream's VLAN identifier registered as a member in the
28         VLAN Registration Entries. When false(2), Talker
29         declarations are propagated according to the VLAN
30         spanning tree."
31     REFERENCE   "12.22.1, 35.2.1.4l"
32     DEFVAL      { false }
33     ::= { ieee8021SrpBridgeBaseEntry 5 }
34
35 ieee8021SrpBridgeBaseMsrpMaxSRClasses OBJECT-TYPE
36     SYNTAX      Unsigned32
37     MAX-ACCESS   read-only
38     STATUS      current
39     DESCRIPTION
40         "This attribute provides the maximum number of SR classes
41         supported by the Bridge."
42     REFERENCE   "12.22.1, 35.2.1.4m"
43     ::= { ieee8021SrpBridgeBaseEntry 6 }
44
45
46 ieee8021SrpBridgePortTable OBJECT-TYPE
47     SYNTAX      SEQUENCE OF Ieee8021SrpBridgePortEntry
48     MAX-ACCESS   not-accessible
49     STATUS      current
50     DESCRIPTION
51         "A table for SRP control and status information about
52         every Bridge Port. Augments the ieee8021BridgeBasePortTable."
53     ::= { ieee8021SrpConfiguration 2 }
54
55 ieee8021SrpBridgePortEntry OBJECT-TYPE
56     SYNTAX      Ieee8021SrpBridgePortEntry
57     MAX-ACCESS   not-accessible
58     STATUS      current
59     DESCRIPTION
60         "SRP control and status information for a Bridge Port."
61     AUGMENTS { ieee8021BridgeBasePortEntry }
62     ::= { ieee8021SrpBridgePortTable 1 }
63
64 Ieee8021SrpBridgePortEntry ::=
65     SEQUENCE {
66         ieee8021SrpBridgePortMsrpEnabledStatus
67             TruthValue,
68         ieee8021SrpBridgePortMsrpFailedRegistrations
69             Counter64,
70         ieee8021SrpBridgePortMsrpLastPduOrigin
71             MacAddress,
72         ieee8021SrpBridgePortSrPvid

```

```

1         IEEE8021VlanIndex,
2         ieee8021SrpBridgePortMsrpTalkerPrunningPerPort
3         TruthValue
4     }
5
6 ieee8021SrpBridgePortMsrpEnabledStatus OBJECT-TYPE
7     SYNTAX      TruthValue
8     MAX-ACCESS  read-create
9     STATUS      current
10    DESCRIPTION
11        "The administrative state of MSRP operation on this port. The
12        value true(1) indicates that MSRP is enabled on this port
13        in all VLANs as long as ieee8021BridgeMsrpEnabledStatus is
14        also true(1). A value of false(2) indicates that MSRP is
15        disabled on this port in all VLANs: any MSRP frames received
16        will be silently discarded, and no MSRP registrations will be
17        propagated from other ports. Setting this to a value of
18        true(1) will be stored by the agent but will only take
19        effect on the MSRP protocol operation if
20        ieee8021BridgeMsrpEnabledStatus
21        also indicates the value true(1). This object affects
22        all MSRP Applicant and Registrar state machines on this
23        port. A transition from false(2) to true(1) will
24        cause a reset of all MSRP state machines on this port.
25
26        The value of this object MUST be retained across
27        reinitializations of the management system."
28    REFERENCE   "35.2.1.4e"
29    DEFVAL      { true }
30    ::= { ieee8021SrpBridgePortEntry 1 }
31
32 ieee8021SrpBridgePortMsrpFailedRegistrations OBJECT-TYPE
33     SYNTAX      Counter64
34     UNITS       "failed MSRP registrations"
35     MAX-ACCESS  read-only
36     STATUS      current
37     DESCRIPTION
38         "The total number of failed MSRP registrations, for any
39         reason, in all VLANs, on this port.
40
41         Discontinuities in the value of the counter can occur at
42         re-initialization of the management system, and at other
43         times as indicated by the value of ifCounterDiscontinuityTime
44         object of the associated interface (if any)."

```



```

1 ieee8021SrpBridgePortMsrpTalkerPruningPerPort OBJECT-TYPE
2     SYNTAX      TruthValue
3     MAX-ACCESS  read-create
4     STATUS      current
5     DESCRIPTION
6         "This parameter controls the forwarding behavior for
7         Talker declarations on the port when the TalkerPruning
8         parameter is disabled for the bridge. The value true(1)
9         indicates, that Talker declarations are only forwarded
10        on that port, if the destination_address of the Stream
11        is found in the MAC Address Registration Entries for the
12        port. When false(2), Talker declarations are forwarded
13        on that port regardless of the destination address."
14    REFERENCE    "12.22.2, 35.2.1.4k"
15    DEFVAL       { false }
16    ::= { ieee8021SrpBridgePortEntry 5 }
17
18 -- =====
19 -- The ieee8021SrpLatency subtree
20 -- This subtree defines the objects necessary for retrieving
21 -- the latency of the various traffic classes on a port.
22 -- =====
23
24 -- =====
25 -- the ieee8021SrpLatencyTable
26 -- =====
27 ieee8021SrpLatencyTable OBJECT-TYPE
28     SYNTAX      SEQUENCE OF Ieee8021SrpLatencyEntry
29     MAX-ACCESS  not-accessible
30     STATUS      current
31     DESCRIPTION
32         "A table containing a set of latency measurement
33         parameters for each traffic class."
34     REFERENCE    "35.2.2.8.6"
35     ::= { ieee8021SrpLatency 1 }
36
37 ieee8021SrpLatencyEntry OBJECT-TYPE
38     SYNTAX      Ieee8021SrpLatencyEntry
39     MAX-ACCESS  not-accessible
40     STATUS      current
41     DESCRIPTION
42         "A list of objects containing latency information
43         for each traffic class. Rows in the table are
44         automatically created for ports that are not an
45         SRP domain boundary port (i.e., SRPdomainBoundaryPort
46         is FALSE). See 35.1.4, 8.8.2, 12.22.3."
47     INDEX { ieee8021BridgeBaseComponentId,
48             ieee8021BridgeBasePort,
49             ieee8021SrpTrafficClass }
50     ::= { ieee8021SrpLatencyTable 1 }
51
52 Ieee8021SrpLatencyEntry ::=
53     SEQUENCE {
54         ieee8021SrpTrafficClass
55         IEEE8021FqtssTrafficClassValue,
56         ieee8021SrpPortTcLatency
57         Unsigned32
58     }
59
60 ieee8021SrpTrafficClass OBJECT-TYPE
61     SYNTAX      IEEE8021FqtssTrafficClassValue
62     MAX-ACCESS  not-accessible
63     STATUS      current
64     DESCRIPTION
65         "The traffic class number associated with the
66         row of the table.
67
68         Rows in the table are automatically created for
69         ports that are not an SRP domain boundary port
70         (i.e., SRPdomainBoundaryPort is FALSE)."
71     REFERENCE    "35.1.4, 8.8.2, 12.22.3"
72     ::= { ieee8021SrpLatencyEntry 1 }

```

```

1
2
3 ieee8021SrpPortTcLatency OBJECT-TYPE
4     SYNTAX      Unsigned32
5     UNITS       "nano-seconds"
6     MAX-ACCESS  read-only
7     STATUS      current
8     DESCRIPTION
9         "The value of the portTcMaxLatency parameter for the
10        traffic class. This value is expressed in
11        nano-seconds."
12     REFERENCE   "35.2.1.4, 35.2.2.8.6"
13     ::= { ieee8021SrpLatencyEntry 2 }
14
15
16 -- =====
17 -- The ieee8021SrpStreams subtree
18 -- This subtree defines the objects necessary for retrieving
19 -- the characteristics of the various Streams currently registered.
20 -- =====
21
22 -- =====
23 -- the ieee8021SrpStreamTable
24 -- =====
25 ieee8021SrpStreamTable OBJECT-TYPE
26     SYNTAX      SEQUENCE OF Ieee8021SrpStreamEntry
27     MAX-ACCESS  not-accessible
28     STATUS      current
29     DESCRIPTION
30         "A table containing a set of characteristics
31         for each registered Stream."
32     REFERENCE   "35.2.2.8"
33     ::= { ieee8021SrpStreams 1 }
34
35 ieee8021SrpStreamEntry OBJECT-TYPE
36     SYNTAX      Ieee8021SrpStreamEntry
37     MAX-ACCESS  not-accessible
38     STATUS      current
39     DESCRIPTION
40         "A list of objects containing characteristics
41         for each registered Stream. Rows in the table are
42         automatically created for Streams registered on any
43         port of a Bridge."
44     INDEX { ieee8021SrpStreamId }
45     ::= { ieee8021SrpStreamTable 1 }
46
47 Ieee8021SrpStreamEntry ::=
48     SEQUENCE {
49         ieee8021SrpStreamId
50         IEEE8021SrpStreamIdValue,
51         ieee8021SrpStreamDestinationAddress
52         MacAddress,
53         ieee8021SrpStreamVlanId
54         IEEE8021VlanIndex,
55         ieee8021SrpStreamTspecMaxFrameSize
56         Unsigned32,
57         ieee8021SrpStreamTspecMaxIntervalFrames
58         Unsigned32,
59         ieee8021SrpStreamDataFramePriority
60         INTEGER,
61         ieee8021SrpStreamRank
62         IEEE8021SrpStreamRankValue
63     }
64
65 ieee8021SrpStreamId OBJECT-TYPE
66     SYNTAX      IEEE8021SrpStreamIdValue
67     MAX-ACCESS  not-accessible
68     STATUS      current
69     DESCRIPTION
70         "The Stream ID associated with the row of the table.
71
72         Rows in the table are automatically created when

```

```
1      Streams are registered via MSRP."
2      REFERENCE    "35.2.2.8.2"
3      ::= { ieee8021SrpStreamEntry 1 }
4
5 ieee8021SrpStreamDestinationAddress OBJECT-TYPE
6     SYNTAX      MacAddress
7     MAX-ACCESS  read-only
8     STATUS      current
9     DESCRIPTION
10      "The MAC destination address for the Stream described
11      by this reservation."
12     REFERENCE   "35.2.2.8.3a"
13     ::= { ieee8021SrpStreamEntry 2}
14
15 ieee8021SrpStreamVlanId OBJECT-TYPE
16     SYNTAX      IEEE8021VlanIndex
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20      "The VLAN ID associated with the MSRP registration
21      for this Stream."
22     REFERENCE   "35.2.2.8.3b"
23     ::= { ieee8021SrpStreamEntry 3}
24
25 ieee8021SrpStreamTspecMaxFrameSize OBJECT-TYPE
26     SYNTAX      Unsigned32 (0..65535)
27     MAX-ACCESS  read-only
28     STATUS      current
29     DESCRIPTION
30      "The maximum size frame that will be sent by
31      a Talker for this Stream. This value is part
32      of the Traffic Specification for the Stream."
33     REFERENCE   "35.2.2.8.4a"
34     ::= { ieee8021SrpStreamEntry 4}
35
36 ieee8021SrpStreamTspecMaxIntervalFrames OBJECT-TYPE
37     SYNTAX      Unsigned32 (0..65535)
38     MAX-ACCESS  read-only
39     STATUS      current
40     DESCRIPTION
41      "The maximum number of frame that will be sent
42      during a class measurement interval (L.2). This
43      value is part of the Traffic Specification for
44      the Stream."
45     REFERENCE   "35.2.2.8.4b, L.2"
46     ::= { ieee8021SrpStreamEntry 5}
47
48 ieee8021SrpStreamDataFramePriority OBJECT-TYPE
49     SYNTAX      INTEGER (0..7)
50     MAX-ACCESS  read-only
51     STATUS      current
52     DESCRIPTION
53      "The Priority Code Point (PCP) value that the
54      referenced Stream will be tagged with. This value
55      is used to distinguish Class A and Class B traffic."
56     REFERENCE   "35.2.2.8.5a"
57     ::= { ieee8021SrpStreamEntry 6}
58
59 ieee8021SrpStreamRank OBJECT-TYPE
60     SYNTAX      IEEE8021SrpStreamRankValue
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64      "SRP supports emergency and non-emergency.
65      Emergency traffic will interrupt non-emergency
66      traffic if there is insufficient bandwidth or
67      resources available for the emergency traffic."
68     REFERENCE   "35.2.2.8.5b"
69     ::= { ieee8021SrpStreamEntry 7}
70
71
72 -- =====
```

```

1  -- the ieee8021SrpStreamPreloadTable
2  -- =====
3  ieee8021SrpStreamPreloadTable OBJECT-TYPE
4      SYNTAX      SEQUENCE OF Ieee8021SrpStreamPreloadEntry
5      MAX-ACCESS  not-accessible
6      STATUS      current
7      DESCRIPTION
8          "A table containing a set of parameters for each StreamID
9          that is preloaded on the Bridge as it initializes."
10     REFERENCE   "12.22.6"
11     ::= { ieee8021SrpStreams 2 }
12
13  ieee8021SrpStreamPreloadEntry OBJECT-TYPE
14      SYNTAX      Ieee8021SrpStreamPreloadEntry
15      MAX-ACCESS  not-accessible
16      STATUS      current
17      DESCRIPTION
18          "A list of objects containing characteristics
19          for each registered Stream. Rows in the table are
20          automatically created for Streams registered on any
21          port of a Bridge."
22      INDEX      { ieee8021SrpStreamPreloadId }
23      ::= { ieee8021SrpStreamPreloadTable 1 }
24
25  Ieee8021SrpStreamPreloadEntry ::=
26      SEQUENCE {
27          ieee8021SrpStreamPreloadId
28              IEEE8021SrpStreamIdValue,
29          ieee8021SrpStreamPreloadDestinationAddress
30              MacAddress,
31          ieee8021SrpStreamPreloadVlanId
32              IEEE8021VlanIndex,
33          ieee8021SrpStreamPreloadTspecMaxFrameSize
34              Unsigned32,
35          ieee8021SrpStreamPreloadTspecMaxIntervalFrames
36              Unsigned32,
37          ieee8021SrpStreamPreloadDataFramePriority
38              INTEGER,
39          ieee8021SrpStreamPreloadRank
40              IEEE8021SrpStreamRankValue
41      }
42
43  ieee8021SrpStreamPreloadId OBJECT-TYPE
44      SYNTAX      IEEE8021SrpStreamIdValue
45      MAX-ACCESS  not-accessible
46      STATUS      current
47      DESCRIPTION
48          "The 64-bit StreamID is used to match Talker
49          registrations with their corresponding Listener
50          registrations(35.2.4)."
```

REFERENCE "12.22.6, 35.2.2.8.2"

```

51  ::= { ieee8021SrpStreamPreloadEntry 1 }
52
53  ieee8021SrpStreamPreloadDestinationAddress OBJECT-TYPE
54      SYNTAX      MacAddress
55      MAX-ACCESS  read-write
56      STATUS      current
57      DESCRIPTION
58          "The MAC destination address for the Stream described
59          by this reservation."
60      REFERENCE   "12.22.6, 35.2.2.8.3a"
61      ::= { ieee8021SrpStreamPreloadEntry 2 }
62
63  ieee8021SrpStreamPreloadVlanId OBJECT-TYPE
64      SYNTAX      IEEE8021VlanIndex
65      MAX-ACCESS  read-write
66      STATUS      current
67      DESCRIPTION
68          "The VLAN ID associated with the MSRP registration
69          for this Stream."
70      REFERENCE   "12.22.6, 35.2.2.8.3b"
71      ::= { ieee8021SrpStreamPreloadEntry 3 }
72

```

```
1
2 ieee8021SrpStreamPreloadTspecMaxFrameSize OBJECT-TYPE
3     SYNTAX      Unsigned32 (0..65535)
4     MAX-ACCESS  read-write
5     STATUS      current
6     DESCRIPTION
7         "The maximum size frame that will be sent by
8         a Talker for this Stream. This value is part
9         of the Traffic Specification for the Stream."
10    REFERENCE   "12.22.6, 35.2.2.8.4a"
11    ::= { ieee8021SrpStreamPreloadEntry 4}
12
13 ieee8021SrpStreamPreloadTspecMaxIntervalFrames OBJECT-TYPE
14     SYNTAX      Unsigned32 (0..65535)
15     MAX-ACCESS  read-write
16     STATUS      current
17     DESCRIPTION
18         "The maximum number of frames that the Talker may
19         transmit in one classMeasurementInterval (34.3).
20         This value is part of the Traffic Specification
21         for the Stream."
22    REFERENCE   "12.22.6, 35.2.2.8.4b"
23    ::= { ieee8021SrpStreamPreloadEntry 5}
24
25 ieee8021SrpStreamPreloadDataFramePriority OBJECT-TYPE
26     SYNTAX      INTEGER (0..7)
27     MAX-ACCESS  read-write
28     STATUS      current
29     DESCRIPTION
30         "The Priority Code Point (PCP) value that the
31         referenced Stream will be tagged with. This value
32         is used to distinguish Class A and Class B traffic."
33    REFERENCE   "12.22.6, 35.2.2.8.5a"
34    ::= { ieee8021SrpStreamPreloadEntry 6}
35
36 ieee8021SrpStreamPreloadRank OBJECT-TYPE
37     SYNTAX      IEEE8021SrpStreamRankValue
38     MAX-ACCESS  read-write
39     STATUS      current
40     DESCRIPTION
41         "SRP supports emergency and non-emergency.
42         Emergency traffic will interrupt non-emergency
43         traffic if there is insufficient bandwidth or
44         resources available for the emergency traffic."
45    REFERENCE   "12.22.6, 35.2.2.8.5b"
46    ::= { ieee8021SrpStreamPreloadEntry 7}
47
48
49 -- =====
50 -- The ieee8021SrpReservations subtree
51 -- This subtree defines the objects necessary for retrieving
52 -- the Stream attribute registrations on each port of a Bridge.
53 -- =====
54
55 -- =====
56 -- the ieee8021SrpReservationsTable
57 -- =====
58 ieee8021SrpReservationsTable OBJECT-TYPE
59     SYNTAX      SEQUENCE OF Ieee8021SrpReservationsEntry
60     MAX-ACCESS  not-accessible
61     STATUS      current
62     DESCRIPTION
63         "A table containing Stream attribute
64         registrations per port."
65    REFERENCE   "35.2.4"
66    ::= { ieee8021SrpReservations 1 }
67
68 ieee8021SrpReservationsEntry OBJECT-TYPE
69     SYNTAX      Ieee8021SrpReservationsEntry
70     MAX-ACCESS  not-accessible
71     STATUS      current
72     DESCRIPTION
```

```

1      "A list of objects containing Stream attribute
2      registrations per port. Rows in the table are
3      automatically created for Streams registered on any
4      port of a Bridge."
5  INDEX { ieee8021SrpReservationStreamId,
6          ieee8021SrpReservationDirection,
7          ieee8021BridgeBaseComponentId,
8          ieee8021BridgeBasePort }
9  ::= { ieee8021SrpReservationsTable 1 }
10
11 Ieee8021SrpReservationsEntry ::=
12     SEQUENCE {
13         ieee8021SrpReservationStreamId
14         IEEE8021SrpStreamIdValue,
15         ieee8021SrpReservationDirection
16         IEEE8021SrpReservationDirectionValue,
17         ieee8021SrpReservationDeclarationType
18         IEEE8021SrpReservationDeclarationTypeValue,
19         ieee8021SrpReservationAccumulatedLatency
20         Unsigned32,
21         ieee8021SrpReservationFailureSystemId
22         OCTET STRING,
23         ieee8021SrpReservationFailureCode
24         IEEE8021SrpReservationFailureCodeValue,
25         ieee8021SrpReservationDroppedStreamFrames
26         Counter64,
27         ieee8021SrpReservationStreamAge
28         Unsigned32
29     }
30
31 ieee8021SrpReservationStreamId OBJECT-TYPE
32     SYNTAX      IEEE8021SrpStreamIdValue
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "The Stream ID associated with the row of the table.
37
38         Rows in the table are automatically created when
39         Streams are registered via MSRP."
40     REFERENCE   "35.2.2.8.2"
41     ::= { ieee8021SrpReservationsEntry 1 }
42
43 ieee8021SrpReservationDirection OBJECT-TYPE
44     SYNTAX      IEEE8021SrpReservationDirectionValue
45     MAX-ACCESS  not-accessible
46     STATUS      current
47     DESCRIPTION
48         "The source of this Stream registration, either
49         Talker or Listener."
50     REFERENCE   "35.2.1.2"
51     ::= { ieee8021SrpReservationsEntry 2 }
52
53 ieee8021SrpReservationDeclarationType OBJECT-TYPE
54     SYNTAX      IEEE8021SrpReservationDeclarationTypeValue
55     MAX-ACCESS  read-only
56     STATUS      current
57     DESCRIPTION
58         "The type of Talker or Listener registration."
59     REFERENCE   "35.2.1.3"
60     ::= { ieee8021SrpReservationsEntry 3 }
61
62 ieee8021SrpReservationAccumulatedLatency OBJECT-TYPE
63     SYNTAX      Unsigned32
64     UNITS        "nano-seconds"
65     MAX-ACCESS  read-only
66     STATUS      current
67     DESCRIPTION
68         "The Accumulated Latency associated with the current
69         registration.
70
71         For Talker registrations this represents the accumulated
72         latency from the Talker to the ingress port of this

```

```

1      Bridge.
2
3      For Listener registrations this represents the accumulated
4      latency to the ingress port of the neighbor Bridge or
5      end stations. This include the latency of the media
6      attached to this egress port."
7      REFERENCE      "35.2.2.8.6"
8      ::= { ieee8021SrpReservationsEntry 4 }
9
10 ieee8021SrpReservationFailureSystemId OBJECT-TYPE
11     SYNTAX          OCTET STRING(SIZE(8))
12     MAX-ACCESS      read-only
13     STATUS          current
14     DESCRIPTION
15         "The first system that changes a Talker Advertise to a
16         Talker Failed registration will report its System
17         Identification in this field. That single System
18         Identification is then propagated from system to system."
19     REFERENCE      "35.2.2.8.7a"
20     ::= { ieee8021SrpReservationsEntry 5 }
21
22 ieee8021SrpReservationFailureCode OBJECT-TYPE
23     SYNTAX          IEEE8021SrpReservationFailureCodeValue
24     MAX-ACCESS      read-only
25     STATUS          current
26     DESCRIPTION
27         "The first Bridge that changes a Talker Advertise to a
28         Talker Failed registration will report the Failure Code
29         in this field. That single Failure Code is then propagated
30         from Bridge to Bridge."
31     REFERENCE      "35.2.2.8.7b"
32     ::= { ieee8021SrpReservationsEntry 6 }
33
34 ieee8021SrpReservationDroppedStreamFrames OBJECT-TYPE
35     SYNTAX          Counter64
36     UNITS           "frames"
37     MAX-ACCESS      read-only
38     STATUS          current
39     DESCRIPTION
40         "A count of the number of data stream frames that have
41         been dropped for whatever reason. These are not MSRP
42         frames, but the stream data frames that are carried by
43         the MSRP Reservation.
44
45         Discontinuities in the value of the counter can occur at
46         re-initialization of the management system, and at other
47         times as indicated by the value of ifCounterDiscontinuityTime
48         object of the associated interface (if any)."

```

```

1      registrations per port."
2      REFERENCE    "12.22.7"
3      ::= { ieee8021SrpReservations 2 }
4
5 ieee8021SrpReservationsPreloadEntry OBJECT-TYPE
6     SYNTAX      Ieee8021SrpReservationsPreloadEntry
7     MAX-ACCESS  not-accessible
8     STATUS      current
9     DESCRIPTION
10      "A list of objects containing Stream attribute
11      registrations per port. Rows in the table are
12      automatically created for Streams registered on any
13      port of a Bridge."
14     INDEX { ieee8021SrpReservationsPreloadStreamId,
15             ieee8021SrpReservationPreloadDirection,
16             ieee8021BridgeBaseComponentId,
17             ieee8021BridgeBasePort }
18     ::= { ieee8021SrpReservationsPreloadTable 1 }
19
20 Ieee8021SrpReservationsPreloadEntry ::=
21     SEQUENCE {
22         ieee8021SrpReservationsPreloadStreamId
23         IEEE8021SrpStreamIdValue,
24         ieee8021SrpReservationPreloadDirection
25         IEEE8021SrpReservationDirectionValue,
26         ieee8021SrpReservationPreloadAccumulatedLatency
27         Unsigned32
28     }
29
30 ieee8021SrpReservationsPreloadStreamId OBJECT-TYPE
31     SYNTAX      IEEE8021SrpStreamIdValue
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35      "The 64-bit StreamID is used to match Talker
36      registrations with their corresponding Listener
37      registrations(35.2.4)."
38     REFERENCE   "12.22.7, 35.2.2.8.2"
39     ::= { ieee8021SrpReservationsPreloadEntry 1 }
40
41 ieee8021SrpReservationPreloadDirection OBJECT-TYPE
42     SYNTAX      IEEE8021SrpReservationDirectionValue
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46      "The source of this Stream registration, either
47      Talker or Listener"
48     REFERENCE   "12.22.7, 35.2.1.1"
49     ::= { ieee8021SrpReservationsPreloadEntry 2 }
50
51 ieee8021SrpReservationPreloadAccumulatedLatency OBJECT-TYPE
52     SYNTAX      Unsigned32
53     UNITS       "nano-seconds"
54     MAX-ACCESS  read-write
55     STATUS      current
56     DESCRIPTION
57      "The Accumulated Latency associated with the current
58      registration.
59
60      For Talker registrations this represents the accumulated
61      latency from the Talker to the ingress port of this
62      Bridge.
63
64      For Listener registrations this represents the accumulated
65      latency to the ingress port of the neighbor Bridge or
66      end stations. This include the latency of the media
67      attached to this egress port."
68     REFERENCE   "12.22.7, 35.2.2.8.6"
69     ::= { ieee8021SrpReservationsPreloadEntry 3 }
70
71
72 -- =====

```



```

1 -- IEEE8021 SRP MIB - Conformance Information
2 -- =====
3
4 ieee8021SrpCompliances
5   OBJECT IDENTIFIER ::= { ieee8021SrpConformance 1 }
6 ieee8021SrpGroups
7   OBJECT IDENTIFIER ::= { ieee8021SrpConformance 2 }
8
9 -- =====
10 -- units of conformance
11 -- =====
12
13 -- =====
14 -- the ieee8021SrpConfiguration group
15 -- =====
16
17 ieee8021SrpConfigurationGroup OBJECT-GROUP
18   OBJECTS {
19     ieee8021SrpBridgeBaseMsrpEnabledStatus,
20     ieee8021SrpBridgeBaseMsrpTalkerPruning,
21     ieee8021SrpBridgeBaseMsrpMaxFanInPorts,
22     ieee8021SrpBridgeBaseMsrpLatencyMaxFrameSize,
23     ieee8021SrpBridgeBaseMsrpTalkerVlanPruning,
24     ieee8021SrpBridgeBaseMsrpMaxSRClasses,
25     ieee8021SrpBridgePortMsrpEnabledStatus,
26     ieee8021SrpBridgePortMsrpFailedRegistrations,
27     ieee8021SrpBridgePortMsrpLastPduOrigin,
28     ieee8021SrpBridgePortSrPvid,
29     ieee8021SrpBridgePortMsrpTalkerPrunningPerPort
30   }
31   STATUS      current
32   DESCRIPTION
33     "Objects that define configuration of SRP."
34   ::= { ieee8021SrpGroups 1 }
35
36 -- =====
37 -- the ieee8021SrpLatency group
38 -- =====
39
40 ieee8021SrpLatencyGroup OBJECT-GROUP
41   OBJECTS {
42     ieee8021SrpPortTcLatency
43   }
44   STATUS      current
45   DESCRIPTION
46     "Objects that define latency for SRP."
47   ::= { ieee8021SrpGroups 2 }
48
49 -- =====
50 -- the ieee8021SrpStreams group
51 -- =====
52
53 ieee8021SrpStreamsGroup OBJECT-GROUP
54   OBJECTS {
55     -- ieee8021SrpStreamId,
56     ieee8021SrpStreamDestinationAddress,
57     ieee8021SrpStreamVlanId,
58     ieee8021SrpStreamTspecMaxFrameSize,
59     ieee8021SrpStreamTspecMaxIntervalFrames,
60     ieee8021SrpStreamDataFramePriority,
61     ieee8021SrpStreamRank
62   }
63   STATUS      current
64   DESCRIPTION
65     "Objects that define Streams for SRP."
66   ::= { ieee8021SrpGroups 3 }
67
68 -- =====
69 -- the ieee8021SrpReservations group
70 -- =====
71
72 ieee8021SrpReservationsGroup OBJECT-GROUP

```

```

1  OBJECTS {
2      -- ieee8021SrpReservationStreamId,
3      -- ieee8021SrpReservationDirection,
4      ieee8021SrpReservationDeclarationType,
5      ieee8021SrpReservationAccumulatedLatency,
6      ieee8021SrpReservationFailureSystemId,
7      ieee8021SrpReservationFailureCode,
8      ieee8021SrpReservationDroppedStreamFrames,
9      ieee8021SrpReservationStreamAge
10 }
11 STATUS      current
12 DESCRIPTION
13     "Objects that define Stream Reservations for SRP."
14 ::= { ieee8021SrpGroups 4 }
15
16 -- =====
17 -- the ieee8021SrpConfigurationPruning group
18 -- =====
19
20 ieee8021SrpConfigurationPruningGroup OBJECT-GROUP
21     OBJECTS {
22         ieee8021SrpBridgeBaseMsrpTalkerVlanPruning,
23         ieee8021SrpBridgePortMsrpTalkerPruningPerPort
24     }
25     STATUS      current
26     DESCRIPTION
27         "Objects that allow configuration of pruning behavior
28         for SRP."
29     ::= { ieee8021SrpGroups 5 }
30
31 -- =====
32 -- the ieee8021SrpMonitoringSRclasses group
33 -- =====
34
35 ieee8021SrpMonitoringSRclassesGroup OBJECT-GROUP
36     OBJECTS {
37         ieee8021SrpBridgeBaseMsrpMaxSRClasses
38     }
39     STATUS      current
40     DESCRIPTION
41         "Objects that provides information on the maximum number
42         of SR classes supported on the Bridge."
43     ::= { ieee8021SrpGroups 6 }
44
45 -- =====
46 -- the ieee8021SrpStreamsPreload group
47 -- =====
48
49 ieee8021SrpStreamsPreloadGroup OBJECT-GROUP
50     OBJECTS {
51         ieee8021SrpStreamPreloadDestinationAddress,
52         ieee8021SrpStreamPreloadVlanId,
53         ieee8021SrpStreamPreloadTspecMaxFrameSize,
54         ieee8021SrpStreamPreloadTspecMaxIntervalFrames,
55         ieee8021SrpStreamPreloadDataFramePriority,
56         ieee8021SrpStreamPreloadRank
57     }
58     STATUS      current
59     DESCRIPTION
60         "Objects that allow to preload parameters for each
61         StreamId on Bridge Ports as the Bridge initializes."
62     ::= { ieee8021SrpGroups 7 }
63
64 -- =====
65 -- the ieee8021SrpReservationsPreload group
66 -- =====
67
68 ieee8021SrpReservationsPreloadGroup OBJECT-GROUP
69     OBJECTS {
70         ieee8021SrpReservationPreloadAccumulatedLatency
71     }
72     STATUS      current

```

```
1  DESCRIPTION
2      "Objects that allow to initialize Streams within each
3      Bridge as it powers up, to preload the Stream
4      registrations that will later be provided by operation
5      of SRP."
6  ::= { ieee8021SrpGroups 8 }
7
8  -- =====
9  -- compliance statements
10 -- =====
11
12 ieee8021SrpCompliance MODULE-COMPLIANCE
13     STATUS      current
14     DESCRIPTION
15         "The compliance statement for devices supporting
16         Stream Reservation Protocol.
17
18         Support of the objects defined in the IEEE8021-SRP MIB
19         also requires support of the IEEE8021-BRIDGE-MIB; the
20         provisions of 17.3.2 apply to implementations claiming
21         support of the IEEE8021-SRP MIB."
22
23     MODULE -- this module
24         MANDATORY-GROUPS {
25             ieee8021SrpConfigurationGroup,
26             ieee8021SrpLatencyGroup,
27             ieee8021SrpStreamsGroup,
28             ieee8021SrpReservationsGroup
29         }
30
31     GROUP      ieee8021SrpConfigurationPruningGroup
32     DESCRIPTION
33         "Implementation of this group is optional. Implementation
34         will allow configuration of pruning behavior for SRP."
35
36     GROUP      ieee8021SrpMonitoringSRclassesGroup
37     DESCRIPTION
38         "Implementation of this group is optional. Implementation
39         will allow configuration of pruning behavior for SRP."
40
41     GROUP      ieee8021SrpStreamsPreloadGroup
42     DESCRIPTION
43         "Implementation of this group is optional. Implementation
44         will allow to preload parameters for each StreamId on
45         Bridge Ports as the Bridge initializes."
46
47     GROUP      ieee8021SrpReservationsPreloadGroup
48     DESCRIPTION
49         "Implementation of this group is optional. Implementation
50         will allow to initialize Streams within each Bridge as it
51         powers up, to preload the Stream registrations that will
52         later be provided by operation of SRP."
53
54     ::= { ieee8021SrpCompliances 1 }
55
56 END
```

1 17.7.15 Definitions for the IEEE8021-MVRPX-MIB module

```
2 IEEE8021-MVRPX-MIB
3 DEFINITIONS ::= BEGIN
4
5 -- *****
6 -- IEEE P802.1Q(TM) Multiple VLAN Registration Protocol Extension MIB
7 -- *****
8
9 IMPORTS
10     MODULE-IDENTITY,
11     OBJECT-TYPE
12         FROM SNMPv2-SMI      -- [RFC2578]
13     TruthValue
14         FROM SNMPv2-TC      -- [RFC2579]
15     MODULE-COMPLIANCE,
16     OBJECT-GROUP            FROM SNMPv2-CONF  -- [RFC2580]
17     systemGroup
18         FROM SNMPv2-MIB     -- [RFC3418]
19     ieee8021BridgeBasePortEntry
20
21         FROM IEEE8021-BRIDGE-MIB -- IEEE Std 802.1Q
22 ;
23
24 ieee8021MvrpxMib
25 MODULE-IDENTITY
26     LAST-UPDATED "202103050000Z" -- March 5, 2021
27     ORGANIZATION "IEEE 802.1 Working Group"
28     CONTACT-INFO
29         " WG-URL: http://www.ieee802.org/1/
30         WG-EMail: stds-802-1-1@ieee.org
31         Contact: IEEE 802.1 Working Group Chair
32         Postal: C/O IEEE 802.1 Working Group
33               IEEE Standards Association
34               445 Hoes Lane
35               Piscataway, NJ 08854
36               USA
37         E-mail: stds-802-1-chairs@ieee.org"
38 DESCRIPTION
39     "Multiple VLAN Registration Protocol extension module for
40     managing MVRP extensions defined in IEEE Std 802.1Q
41
42     Unless otherwise indicated, the references in this MIB
43     module are to IEEE Std 802.1Q-2022.
44
45     Copyright (C) IEEE (2022).
46     This version of this MIB module is part of IEEE Std 802.1Q;
47     see that standard for full legal notices."
48
49 REVISION "202103050000Z" -- March 5, 2021
50 DESCRIPTION
51     "Published as part of IEEE Std 802.1Q-2022.
52     Cross references and contact information updated."
53
54 REVISION "201807010000Z" -- July 1, 2018
55 DESCRIPTION
56     "Published as part of IEEE Std 802.1Q 2018 revision.
57     Cross references updated and corrected."
58
59 REVISION "201412150000Z" -- December 15, 2014
60 DESCRIPTION
61     "Published as part of IEEE Std 802.1Q 2014 revision.
62     Cross references updated and corrected."
63
64 REVISION      "201104050000Z"      -- (YYYYMMDDHHMM Zulu=GMT)
65 DESCRIPTION
66     "Included in IEEE Std 802.1Qbe-2011
67
68     Copyright (C) IEEE (2011)."
```

```
 ::= { iso(1) org(3) ieee(111)
        standards-association-numbers-series-standards (2)
        lan-man-stds (802) ieee802dot1 (1) ieee802dot1mibs (1) 22 }
```

```

1
2 ieee8021MvrpxMIBObjects
3   OBJECT IDENTIFIER ::= { ieee8021MvrpxMib 1 }
4 ieee8021MvrpxConformance
5   OBJECT IDENTIFIER ::= { ieee8021MvrpxMib 2 }
6
7
8
9 -- =====
10 -- MVRP extension augmentation of the Generic Bridge Port Table
11 -- =====
12
13 ieee8021MvrpxPortTable
14 OBJECT-TYPE
15   SYNTAX SEQUENCE OF Ieee8021MvrpxPortEntry
16   MAX-ACCESS not-accessible
17   STATUS current
18   DESCRIPTION
19     "A table that contains controls for the Multiple VLAN
20      Registration Protocol (MVRP) state machines for all of the Ports
21      of a Bridge."
22   REFERENCE "12.9.2"
23   ::= { ieee8021MvrpxMIBObjects 1 }
24
25 ieee8021MvrpxPortEntry
26 OBJECT-TYPE
27   SYNTAX Ieee8021MvrpxPortEntry
28   MAX-ACCESS not-accessible
29   STATUS current
30   DESCRIPTION
31     "Each entry contains the MVRP Registrar controls for one Port."
32 AUGMENTS { ieee8021BridgeBasePortEntry }
33   ::= { ieee8021MvrpxPortTable 1 }
34
35 Ieee8021MvrpxPortEntry
36 ::= SEQUENCE {
37     ieee8021MvrpxPortNewOnly          TruthValue,
38     ieee8021MvrpxPortMvrpNewPropagated TruthValue,
39     ieee8021MvrpxPortXmitZero         TruthValue
40 }
41
42 ieee8021MvrpxPortNewOnly
43 OBJECT-TYPE
44   SYNTAX TruthValue
45   MAX-ACCESS read-create
46   STATUS current
47   DESCRIPTION
48     "The mode of operation of the MVRP state machines on
49     this port, if enabled. The value of this object and the value
50     of the individual Port+Attribute type enable object
51     ieee8021QBridgePortMvrpEnabledStatus combine to control the
52     state machines as follows:
53
54     ieee8021MvrpxPortNewOnly          ieee8021QBridgePortMvrpEnabledStatus
55                                     MVRP state machines
56
57     not implemented      true(1)      Full participant
58     false(2)             true(1)      Full participant
59     true(1)              true(1)      New-only participant
60     not implemented      false(2)     MVRP disabled
61     false(2)             false(2)     MVRP disabled
62     true(1)              false(2)     MVRP disabled
63
64     This object affects all MVRP Applicant and Registrar state
65     machines on this port. A change to the value of this object
66     will cause a reset of all MVRP state machines for this attribute
67     type on this port.
68
69     The value of this object MUST be retained across
70     reinitializations of the management system."
71   REFERENCE "12.9.2.1.3, 12.9.2.2.2"
72

```

```

1  DEFVAL      { false }
2  ::= { ieee8021MvrpxPortEntry 1 }
3
4  ieee8021MvrpxPortMvrpNewPropagated
5  OBJECT-TYPE
6      SYNTAX      TruthValue
7      MAX-ACCESS  read-create
8      STATUS      current
9      DESCRIPTION
10         "The mode of operation of the MVRP on this port, if enabled.
11         If this object contains the value true(1), then all Static VLAN
12         Registration Entries that are Registration Fixed are treated as
13         Registration Fixed (New propagated), and if false(2), as
14         Registration Fixed (New ignored)
15
16         This object affects only the MVRP Applicant and Registrar state
17         machines on this port. A change to the value of this object
18         will cause a reset of all MVRP state machines on this port.
19
20         The value of this object MUST be retained across
21         reinitializations of the management system."
22  REFERENCE   "12.7.7.1.2:d, 12.7.7.3.3:d"
23  DEFVAL      { false }
24  ::= { ieee8021MvrpxPortEntry 2 }
25
26  ieee8021MvrpxPortXmitZero
27  OBJECT-TYPE
28      SYNTAX      TruthValue
29      MAX-ACCESS  read-create
30      STATUS      current
31      DESCRIPTION
32         "Selects whether MVRP is enabled to transmit 0 as the attribute
33         value for the one VLAN ID for which this Port is in the untagged
34         set, true(1) to enable transmit 0, and false(2) to transmit the
35         VLAN ID. The value 0 is not transmitted unless
36         ieee8021MvrpxPortNewOnly is true(1).
37
38         This feature is optional. If not supported, the system SHALL
39         NOT allow this object to be set to the value true(1).
40
41         The value of this object MUST be retained across
42         reinitializations of the management system."
43  REFERENCE   "12.9.2.1.3:c, 12.9.2.2.2:e, 11.2.3.1.7"
44  DEFVAL      { false }
45  ::= { ieee8021MvrpxPortEntry 3 }
46
47
48
49 -- *****
50 -- IEEE 802.1Qbe MVRP extension Module - Conformance Information
51 -- *****
52
53  ieee8021MvrpxCompliances
54      OBJECT IDENTIFIER
55      ::= { ieee8021MvrpxConformance 1 }
56  ieee8021MvrpxGroups
57      OBJECT IDENTIFIER
58      ::= { ieee8021MvrpxConformance 2 }
59
60
61
62 -- *****
63 -- Units of conformance
64 -- *****
65
66  ieee8021MvrpxReqdGroup
67  OBJECT-GROUP
68      OBJECTS {
69          ieee8021MvrpxPortNewOnly,
70          ieee8021MvrpxPortMvrpNewPropagated,
71          ieee8021MvrpxPortXmitZero
72      }

```

```
1  STATUS current
2  DESCRIPTION
3  "Objects in the MVRP extension augmentation table required
4  group."
5  ::= { ieee8021MvrpxGroups 1 }
6
7
8
9 -- *****
10 -- MIB Module Compliance statements
11 -- *****
12
13 ieee8021MvrpxCompliance
14 MODULE-COMPLIANCE
15     STATUS current
16     DESCRIPTION
17         "The compliance statement for support by a Bridge of
18         the IEEE8021-MVRPX-MIB module."
19
20     MODULE SNMPv2-MIB -- The SNMPv2-MIB, RFC 3418
21         MANDATORY-GROUPS {
22             systemGroup
23         }
24
25     MODULE
26         MANDATORY-GROUPS {
27             ieee8021MvrpxReqdGroup
28         }
29
30     ::= { ieee8021MvrpxCompliances 1 }
31
32 END
```

1 17.7.16 Definitions for the IEEE8021-MIRP-MIB module

```
2 IEEE8021-MIRP-MIB
3 DEFINITIONS ::= BEGIN
4
5 -- *****
6 -- IEEE P802.1Q(TM) Multiple I-SID Registration Protocol MIB
7 -- *****
8
9 IMPORTS
10     MODULE-IDENTITY,
11     OBJECT-TYPE
12         FROM SNMPv2-SMI      -- [RFC2578]
13     TruthValue
14         FROM SNMPv2-TC      -- [RFC2579]
15     MODULE-COMPLIANCE,
16     OBJECT-GROUP            FROM SNMPv2-CONF  -- [RFC2580]
17     systemGroup
18         FROM SNMPv2-MIB      -- [RFC3418]
19     VlanIdOrNone
20         FROM Q-BRIDGE-MIB    -- [RFC4363]
21     IEEE8021BridgePortNumberOrZero
22
23         FROM IEEE8021-TC-MIB  -- IEEE Std 802.1ap
24     ieee8021PbbBackboneEdgeBridgeObjects
25
26         FROM IEEE8021-PBB-MIB -- IEEE Std 802.1ap
27     ieee8021BridgeBasePortEntry
28
29         FROM IEEE8021-BRIDGE-MIB -- IEEE Std 802.1ap
30 ;
31
32 ieee8021MirpMib
33 MODULE-IDENTITY
34     LAST-UPDATED "202211080000Z" -- November 8, 2022
35     ORGANIZATION "IEEE 802.1 Working Group"
36     CONTACT-INFO
37         " WG-URL: http://www.ieee802.org/1/
38         WG-EMail: stds-802-1-1@ieee.org
39         Contact: IEEE 802.1 Working Group Chair
40         Postal: C/O IEEE 802.1 Working Group
41               IEEE Standards Association
42               445 Hoes Lane
43               Piscataway, NJ 08854
44               USA
45         E-mail: stds-802-1-chairs@ieee.org"
46 DESCRIPTION
47     "The Bridge MIB module for managing devices that support the
48     Multiple I-SID Registration Protocol module for IEEE 802.1Q Bridges
49
50     Unless otherwise indicated, the references in this MIB
51     module are to IEEE Std 802.1Q-2022.
52
53     Copyright (C) IEEE (2022).
54     This version of this MIB module is part of IEEE Std 802.1Q;
55     see that standard for full legal notices."
56
57 REVISION "202211080000Z" -- November 8, 2022
58 DESCRIPTION
59     "Published as part of IEEE Std 802.1Q-2022.
60     Description, cross references, and contact information updated."
61
62 REVISION "201807010000Z" -- July 1, 2018
63 DESCRIPTION
64     "Published as part of IEEE Std 802.1Q 2018 revision.
65     Cross references updated and corrected."
66
67 REVISION "201412150000Z" -- December 15, 2014
68 DESCRIPTION
69     "Published as part of IEEE Std 802.1Q 2014 revision.
70     Cross references updated and corrected.
71     Objects renamed and attached to the proper MIB arc.
```



```
1      NOTE THAT the original MIB version incorrectly
2      attached new objects to the CN MIB arc. The old objects
3      and object names SHOULD NOT be used. The old names/objects
4      concerned are:
5      ieee8021MirpMIBObjects
6      ieee8021MirpConformance
7      ieee8021MirpPortTable
8      ieee8021MirpCompliances
9      ieee8021MirpGroups"
10
11     REVISION      "201104050000Z"      -- (YYYYMMDDHHMM Zulu=GMT)
12     DESCRIPTION
13         "Included in IEEE Std 802.1Qbe-2011
14
15         Copyright (C) IEEE Std 802.1."
16     ::= { iso(1) org(3) ieee(111)
17         standards-association-numbers-series-standards (2)
18         lan-man-stds (802) ieee802dot1 (1) ieee802dot1mibs (1) 23 }
19
20 ieee8021MirpV2MIBObjects
21     OBJECT IDENTIFIER ::= { ieee8021MirpMib 1 }
22 ieee8021MirpV2Conformance
23     OBJECT IDENTIFIER ::= { ieee8021MirpMib 2 }
24
25
26 -- =====
27 -- MIRP augmentation of the Generic Bridge Port Table
28 -- =====
29
30 ieee8021MirpV2PortTable
31     OBJECT-TYPE
32         SYNTAX SEQUENCE OF Ieee8021MirpV2PortEntry
33         MAX-ACCESS not-accessible
34         STATUS current
35         DESCRIPTION
36             "A table that contains controls for the Multiple I-SID
37             Registration Protocol (MIRP) state machines for all of the Ports
38             of a Bridge."
39         REFERENCE "12.9.2"
40         ::= { ieee8021MirpV2MIBObjects 1 }
41
42 ieee8021MirpV2PortEntry
43     OBJECT-TYPE
44         SYNTAX Ieee8021MirpV2PortEntry
45         MAX-ACCESS not-accessible
46         STATUS current
47         DESCRIPTION
48             "Each entry contains the MIRP Participant controls for one Port."
49     AUGMENTS { ieee8021BridgeBasePortEntry }
50     ::= { ieee8021MirpV2PortTable 1 }
51
52 Ieee8021MirpV2PortEntry
53     ::= SEQUENCE {
54         ieee8021MirpV2PortEnabledStatus      TruthValue
55     }
56
57 ieee8021MirpV2PortEnabledStatus
58     OBJECT-TYPE
59         SYNTAX TruthValue
60         MAX-ACCESS read-create
61         STATUS current
62         DESCRIPTION
63             "The state of MIRP operation on this port. The value
64             true(1) indicates that MIRP is enabled on this port,
65             as long as ieee8021PbbMirpEnableStatus is also enabled
66             for this component. When false(2) but
67             ieee8021PbbMirpEnableStatus is still
68             enabled for the device, MIRP is disabled on this port.
69
70             If MIRP is enabled on a VIP, then the MIRP Participant is
71             enabled on that VIP's PIP. If MIRP is enabled on none of the
72             VIPs on a PIP, then the MIRP Participant on the PIP is
```

```

1      disabled; any MIRP packets received will be silently discarded,
2      and no MIRP registrations will be propagated from the PIP. A
3      transition from all VIPs on a PIP false(2) to at least one VIP
4      on the PIP true(1) will cause a reset of all MIRP state
5      machines on this PIP.
6
7      If MIRP is enabled on any port not a VIP, then the MIRP
8      Participant is enabled on that port. If MIRP is
9      disabled on a non-VIP port, then MIRP packets received will be
10     silently discarded, and no MIRP registrations will be
11     propagated from the port. A transition from false(2) to
12     true(1) will cause a reset of all MIRP state machines on a
13     non-VIP port.
14
15     The value of this object MUST be retained across
16     reinitializations of the management system."
17     REFERENCE    "12.7.7.1, 12.7.7.2, 39.2.1.11"
18     DEFVAL       { true }
19     ::= { ieee8021MirpV2PortEntry 1 }
20
21
22
23 -- =====
24 -- MIRP augmentation of BEB subtree
25 -- =====
26
27
28 ieee8021PbbMirpEnableStatus
29 OBJECT-TYPE
30     SYNTAX      TruthValue
31     MAX-ACCESS  read-write
32     STATUS      current
33     DESCRIPTION
34         "The administrative status requested by management for
35         MIRP. The value true(1) indicates that MIRP should be enabled
36         on this component, on all ports for which it has not been
37         specifically disabled. When false(2), MIRP is disabled on all
38         ports. This object affects all MIRP Applicant and Registrar
39         state machines. A transition from false(2) to true(1) will
40         cause a reset of all MIRP state machines on all ports.
41
42         The value of this object MUST be retained across
43         reinitializations of the management system."
44     REFERENCE    "12.16.1.1.3:i, 12.16.1.2.2:b"
45     DEFVAL       { false }
46     ::= { ieee8021PbbBackboneEdgeBridgeObjects 7 }
47
48 ieee8021PbbMirpBvid
49 OBJECT-TYPE
50     SYNTAX      VlanIdOrNone
51     MAX-ACCESS  read-write
52     STATUS      current
53     DESCRIPTION
54         "The B-VID to which received MIRPDUs are to be assigned, or 0,
55         if they are to be sent on the CBP PVID."
56     REFERENCE    "12.14.7.7.2 j), 12.14.7.7.2 c)"
57     DEFVAL       { 0 }
58     ::= { ieee8021PbbBackboneEdgeBridgeObjects 8 }
59
60 ieee8021PbbMirpDestSelector
61 OBJECT-TYPE
62     SYNTAX      INTEGER {
63         cbpMirpGroup (1),
64         cbpMirpVlan (2),
65         cbpMirpTable (3)
66     }
67     MAX-ACCESS  read-write
68     STATUS      current
69     DESCRIPTION
70         "An enumerated value specifying what destination_address and
71         vlan_identifier are to be used when the MIRP Participant
72         transmits an MIRPDU towards the MAC relay entity:

```

```

1
2      cbpMirpGroup (1) Use the Nearest Customer Bridge group
3                      address from Table 8-1 with the MIRP B-VID.
4      cbpMirpVlan  (2) Use the Nearest Customer Bridge group
5                      address from Table 8-1 with the Backbone
6                      VLAN Identifier field from the Backbone
7                      Service Instance table.
8      cbpMirpTable (3) Use the Default Backbone Destination and
9                      Backbone VLAN Identifier fields from the
10                     Backbone Service Instance table.
11
12      The value of this object MUST be retained across
13      reinitializations of the management system."
14      REFERENCE      "Table 8-1, 12.14.7.7.2 k), 12.14.7.7.2 d)"
15      DEFVAL         { cbpMirpGroup }
16      ::= { ieee8021PbbBackboneEdgeBridgeObjects 9 }
17
18 ieee8021PbbMirpPnpEnable
19 OBJECT-TYPE
20     SYNTAX      TruthValue
21     MAX-ACCESS  read-write
22     STATUS      current
23     DESCRIPTION
24         "A Boolean value specifying the administrative status requested
25         by management for attaching a MIRP Participant to a PNP if
26         and only if this system is a Backbone Edge Bridge (BEB):
27
28         true(1) The BEB is to attach a MIRP Participant to exactly one
29                 Port, either a management Port with no LAN connection
30                 external to the BEB, or a PNP.
31         false(2) No MIRP Participant is to be present on any PNP (or
32                 on the MAC Relay-facing side of a CBP).
33
34         The value of this object MUST be retained across
35         reinitializations of the management system.
36         "
37     REFERENCE      "12.14.7.7.2 j), 12.14.7.7.2 c)"
38     DEFVAL         { true }
39     ::= { ieee8021PbbBackboneEdgeBridgeObjects 10 }
40
41 ieee8021PbbMirpPnpPortNumber
42 OBJECT-TYPE
43     SYNTAX      IEEE8021BridgePortNumberOrZero
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "The Bridge Port Number of the Provider Network Port (PNP) that
48         has an associated MIRP Participant, or 0, if no Bridge Port has
49         an associated MIRP Participant. This object indexes an entry in
50         the Bridge Port Table. The system SHALL ensure that either
51         ieee8021PbbMirpPnpPortNumber contains 0, or that the indexed
52         ieee8021BridgeBasePortType object contains the value
53         providerNetworkPort(3)."
54     REFERENCE      "12.14.7.7.2 j), 12.14.7.7.2 c)"
55     DEFVAL         { 0 }
56     ::= { ieee8021PbbBackboneEdgeBridgeObjects 11 }
57
58
59
60
61 -- *****
62 -- IEEE 802.1Qbe MIB Module - Conformance Information
63 -- *****
64
65 ieee8021MirpV2Compliances
66     OBJECT IDENTIFIER
67     ::= { ieee8021MirpV2Conformance 1 }
68 ieee8021MirpV2Groups
69     OBJECT IDENTIFIER
70     ::= { ieee8021MirpV2Conformance 2 }
71
72 -- *****

```

```
1 -- Units of conformance
2 -- *****
3
4 ieee8021MirpV2ReqdGroup
5 OBJECT-GROUP
6   OBJECTS {
7     ieee8021MirpV2PortEnabledStatus,
8     ieee8021PbbMirpEnableStatus,
9     ieee8021PbbMirpBvid,
10    ieee8021PbbMirpDestSelector,
11    ieee8021PbbMirpPnpEnable,
12    ieee8021PbbMirpPnpPortNumber
13  }
14  STATUS current
15  DESCRIPTION
16    "Objects in the MIRP augmentation required group."
17  ::= { ieee8021MirpV2Groups 1 }
18
19 -- *****
20 -- MIB Module Compliance statements
21 -- *****
22
23 ieee8021MirpV2BridgeCompliance
24 MODULE-COMPLIANCE
25   STATUS current
26   DESCRIPTION
27     "The compliance statement for support by a Bridge of
28     the IEEE8021-MIRP-MIB module."
29
30   MODULE SNMPv2-MIB -- The SNMPv2-MIB, RFC 3418
31     MANDATORY-GROUPS {
32       systemGroup
33     }
34
35   MODULE
36     MANDATORY-GROUPS {
37       ieee8021MirpV2ReqdGroup
38     }
39
40   ::= { ieee8021MirpV2Compliances 1 }
41
42 END
```

1 17.7.17 Definitions for the IEEE8021-PFC-MIB module

```
2 IEEE8021-PFC-MIB DEFINITIONS ::= BEGIN
3
4 -- *****
5 -- IEEE P802.1Q(TM) Priority-based Flow Control MIB
6 -- *****
7
8 IMPORTS
9     MODULE-IDENTITY,
10     OBJECT-TYPE,
11     Counter32,
12     Unsigned32          FROM SNMPv2-SMI    -- [RFC2578]
13     ieee802dot1mibs     FROM IEEE8021-TC-MIB
14     MODULE-COMPLIANCE,
15     OBJECT-GROUP        FROM SNMPv2-CONF   -- [RFC2580]
16     ifEntry,
17     ifGeneralInformationGroup
18     FROM IF-MIB          -- [RFC2863]
19     systemGroup         FROM SNMPv2-MIB     -- [RFC3418]
20 ;
21
22 ieee8021PFCMib MODULE-IDENTITY
23     LAST-UPDATED "202103050000Z" -- March 5, 2021
24     ORGANIZATION "IEEE 802.1 Working Group"
25     CONTACT-INFO
26         " WG-URL: http://www.ieee802.org/1/
27         WG-EMail: stds-802-1-1@ieee.org
28         Contact: IEEE 802.1 Working Group Chair
29         Postal: C/O IEEE 802.1 Working Group
30                 IEEE Standards Association
31                 445 Hoes Lane
32                 Piscataway, NJ 08854
33                 USA
34         E-mail: stds-802-1-chairs@ieee.org"
35     DESCRIPTION
36         "Priority-based Flow Control module for IEEE Std 802.1Q.
37
38         Unless otherwise indicated, the references in this MIB
39         module are to IEEE Std 802.1Q-2022.
40
41         Copyright (C) IEEE (2022).
42         This version of this MIB module is part of IEEE Std 802.1Q;
43         see that standard for full legal notices."
44
45     REVISION "202103050000Z" -- March 5, 2021
46     DESCRIPTION
47         "Published as part of IEEE Std 802.1Q-2022.
48         Cross references and contact information updated."
49
50     REVISION "201807010000Z" -- July 1, 2018
51     DESCRIPTION
52         "Published as part of IEEE Std 802.1Q 2018 revision.
53         Cross references updated and corrected."
54
55     REVISION "201412150000Z" -- December 15, 2014
56     DESCRIPTION
57         "Published as part of IEEE Std 802.1Q 2014 revision.
58         Cross references updated and corrected."
59
60     REVISION "201002080000Z" -- 02/08/2010 00:00GMT
61     DESCRIPTION
62         "Included in IEEE P802.1Qbb
63
64         Copyright (C) IEEE."
65     ::= { ieee802dot1mibs 21 }
66
67
68 ieee8021PfcMIBObjects OBJECT IDENTIFIER ::= { ieee8021PFCMib 1 }
69 ieee8021PfcConformance OBJECT IDENTIFIER ::= { ieee8021PFCMib 2 }
70
71
```

```

1 ieee8021PfcIfTable OBJECT-TYPE
2     SYNTAX      SEQUENCE OF Ieee8021PfcIfEntry
3     MAX-ACCESS  not-accessible
4     STATUS      current
5     DESCRIPTION
6         "A table of PFC information for all interfaces of a system."
7     REFERENCE
8         "12.22.6"
9     ::= { ieee8021PfcMIBObjects 1 }
10
11 ieee8021PfcIfEntry OBJECT-TYPE
12     SYNTAX      Ieee8021PfcIfEntry
13     MAX-ACCESS  not-accessible
14     STATUS      current
15     DESCRIPTION
16         "Each entry contains information about
17         the PFC function on a single interface."
18     REFERENCE
19         "12.22.6"
20     AUGMENTS { ifEntry }
21     ::= { ieee8021PfcIfTable 1 }
22
23 Ieee8021PfcIfEntry ::= SEQUENCE {
24     ieee8021PfcLinkDelayAllowance      Unsigned32,
25     ieee8021PfcRequests                 Counter32,
26     ieee8021PfcIndications              Counter32
27 }
28
29 ieee8021PfcLinkDelayAllowance OBJECT-TYPE
30     SYNTAX      Unsigned32
31     MAX-ACCESS  read-write
32     STATUS      current
33     DESCRIPTION
34         "The allowance made for round-trip propagation delay
35         of the link in bits.
36
37         The value of this object MUST be retained across
38         reinitializations of the management system."
39     ::= { ieee8021PfcIfEntry 1 }
40
41 ieee8021PfcRequests OBJECT-TYPE
42     SYNTAX      Counter32
43     UNITS        "Requests"
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "A count of the invoked PFC M_CONTROL.request primitives.
48
49         Discontinuities in the value of this counter can occur at
50         re-initialization of the management system, and at other
51         times as indicated by the value of
52         ifCounterDiscontinuityTime."
53     ::= { ieee8021PfcIfEntry 2 }
54
55 ieee8021PfcIndications OBJECT-TYPE
56     SYNTAX      Counter32
57     UNITS        "Indications"
58     MAX-ACCESS  read-only
59     STATUS      current
60     DESCRIPTION
61         "A count of the received PFC M_CONTROL.indication primitives.
62
63         Discontinuities in the value of this counter can occur at
64         re-initialization of the management system, and at other
65         times as indicated by the value of
66         ifCounterDiscontinuityTime."
67     ::= { ieee8021PfcIfEntry 3 }
68
69
70 -- *****
71 -- IEEE 802.1Qbb MIB Module - Conformance Information
72 -- *****

```

```
1
2  ieee8021PfcCompliances
3      OBJECT IDENTIFIER ::= { ieee8021PfcConformance 1 }
4  ieee8021PfcGroups
5      OBJECT IDENTIFIER ::= { ieee8021PfcConformance 2 }
6
7
8 -- *****
9 -- Units of conformance
10 -- *****
11
12 ieee8021PfcGlobalReqdGroup OBJECT-GROUP
13     OBJECTS {
14         ieee8021PfcLinkDelayAllowance,
15         ieee8021PfcRequests,
16         ieee8021PfcIndications
17     }
18     STATUS      current
19     DESCRIPTION
20         "Objects in the global required group."
21     ::= { ieee8021PfcGroups 1 }
22
23
24 -- *****
25 -- MIB Module Compliance statements
26 -- *****
27
28 ieee8021PfcCompliance MODULE-COMPLIANCE
29     STATUS      current
30     DESCRIPTION
31         "The compliance statement for support by a system of
32         the IEEE8021-PFC-MIB module."
33
34     MODULE SNMPv2-MIB -- The SNMPv2-MIB, RFC 3418
35         MANDATORY-GROUPS {
36             systemGroup
37         }
38
39     MODULE IF-MIB -- The interfaces MIB, RFC 2863
40         MANDATORY-GROUPS {
41             ifGeneralInformationGroup
42         }
43
44     MODULE
45         MANDATORY-GROUPS {
46             ieee8021PfcGlobalReqdGroup
47         }
48     ::= { ieee8021PfcCompliances 1 }
49
50
51 END
```

1 17.7.18 Definitions for the IEEE8021-TEIPS-V2-MIB module

```
2 IEEE8021-TEIPS-V2-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for IEEE 802.1Q TEIPS Devices
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY,
10    NOTIFICATION-TYPE,
11    OBJECT-TYPE,
12    Unsigned32
13        FROM SNMPv2-SMI
14    RowStatus,
15    StorageType,
16    TruthValue
17        FROM SNMPv2-TC
18    ieee802dot1mibs,
19    IEEE8021BridgePortNumber,
20    IEEE8021TeipsIpgConfigActiveRequests,
21    IEEE8021TeipsIpgid,
22    IEEE8021TeipsIpgConfigAdmin,
23    IEEE8021PbbTeTSidId
24        FROM IEEE8021-TC-MIB
25    ieee8021BridgeBaseComponentId
26        FROM IEEE8021-BRIDGE-MIB
27    MODULE-COMPLIANCE,
28    NOTIFICATION-GROUP,
29    OBJECT-GROUP
30        FROM SNMPv2-CONF;
31
32 ieee8021TeipsV2Mib MODULE-IDENTITY
33     LAST-UPDATED "202211080000Z" -- November 8, 2022
34     ORGANIZATION "IEEE 802.1 Working Group"
35     CONTACT-INFO
36         " WG-URL: http://www.ieee802.org/1/
37         WG-EMail: stds-802-1-1@ieee.org
38         Contact: IEEE 802.1 Working Group Chair
39         Postal: C/O IEEE 802.1 Working Group
40             IEEE Standards Association
41             445 Hoes Lane
42             Piscataway, NJ 08854
43             USA
44         E-mail: stds-802-1-chairs@ieee.org"
45
46     DESCRIPTION
47         "MIB Module for managing systems that provide Provider
48         Backbone Bridge Traffic Engineering (PBB-TE) Infrastructure
49         Segment Protection.
50
51         Unless otherwise indicated, the references in this MIB
52         module are to IEEE Std 802.1Q-2022.
53
54         Copyright (C) IEEE (2022).
55         This version of this MIB module is part of IEEE Std 802.1Q;
56         see that standard for full legal notices."
57
58     REVISION "202211080000Z" -- November 8, 2022
59     DESCRIPTION
60         "Published as part of IEEE Std 802.1Q-2022.
61         Cross references and contact information updated."
62
63     REVISION "201807010000Z" -- July 1, 2018
64     DESCRIPTION
65         "Published as part of IEEE Std 802.1Q 2018 revision.
66         Cross references updated and corrected."
67
68     REVISION "201412150000Z" -- December 15, 2014
69     DESCRIPTION
70         "Published as part of IEEE Std 802.1Q 2014 revision.
71         Cross references updated and corrected."
```



```
1      Module and object names changed to V2 and base arc
2      changed from 24 to 27 to remove conflicting OID
3      allocation (see comments below). The old version of
4      this MIB, and its object names, SHOULD NOT BE USED."
5
6      REVISION "201108170000Z" -- (YYYYMMDDHHMM Zulu=GMT)
7
8      DESCRIPTION
9      "Version 1 of the TEIPS MIB module based upon IEEE 802.1Qbf"
10     ::= { iso(1) org(3) ieee(111)
11           standards-association-numbers-series-standards (2)
12           lan-man-stds (802) ieee802dot1 (1) ieee802dot1mibs (1) 27 }
13
14 --
15 -- An earlier version of this MIB was
16 -- inadvertently published under the wrong root arc:
17 -- { iso(1) org(3) ieee(111)
18 --   standards-association-numbers-series-standards (2)
19 --   lan-man-stds (802) ieee802dot1 (1) ieee802dot1mibs (1) 24 }
20 -- That version of the MIB, and its object names, SHOULD NOT BE USED.
21 --
22
23 ieee8021TeipsV2Notifications OBJECT IDENTIFIER ::= { ieee8021TeipsV2Mib 0 }
24 ieee8021TeipsV2Objects      OBJECT IDENTIFIER ::= { ieee8021TeipsV2Mib 1 }
25 ieee8021TeipsV2Conformance  OBJECT IDENTIFIER ::= { ieee8021TeipsV2Mib 2 }
26
27 --
28 --TEIPS MIB Objects
29 --
30
31 -- =====
32 -- the ieee8021TeipsV2IpgTable
33 -- =====
34 ieee8021TeipsV2IpgTable OBJECT-TYPE
35     SYNTAX      SEQUENCE OF Ieee8021TeipsV2IpgEntry
36     MAX-ACCESS  not-accessible
37     STATUS      current
38     DESCRIPTION
39         "The IPG table. Each entry in this table corresponds to an
40         Infrastructure Protection Group (IPG) associated with a PBB
41         supporting Infrastructure Protection Switching (IPS)."
```

```

1  REFERENCE
2  "12.24.1.1.3 a"
3  ::= { ieee8021TeipsV2IpgEntry 1 }
4
5
6  ieee8021TeipsV2IpgWorkingMA OBJECT-TYPE
7      SYNTAX      Unsigned32
8      MAX-ACCESS  read-create
9      STATUS      current
10     DESCRIPTION
11         "Identifies the Segment MA that corresponds to
12         the IPG's working entity. The MA index in
13         this column must hold a value that is the
14         value of dotlagCfmStackMaIndex column for
15         some entry in the dotlagCfmStackTable before
16         the RowStatus for this row can be set to
17         Active. Furthermore, this column may not be
18         modified when the RowStatus for this row is
19         Active."
20     REFERENCE
21         "12.24.1.1.3 b)"
22 ::= { ieee8021TeipsV2IpgEntry 2 }
23
24 ieee8021TeipsV2IpgProtectionMA OBJECT-TYPE
25     SYNTAX      Unsigned32
26     MAX-ACCESS  read-create
27     STATUS      current
28     DESCRIPTION
29         "Identifies the Segment MA that corresponds to the
30         IPG's protection entity. The MA index in this
31         column must hold a value that is the value of
32         dotlagCfmStackMaIndex column for some entry in
33         the dotlagCfmStackTable before the RowStatus
34         for this row can be set to Active. Furthermore,
35         this column may not be modified when the
36         RowStatus for this row is Active."
37     REFERENCE
38         "12.24.1.1.3 c)"
39 ::= { ieee8021TeipsV2IpgEntry 3 }
40
41 ieee8021TeipsV2IpgWorkingPortNumber OBJECT-TYPE
42     SYNTAX      IEEE8021BridgePortNumber
43     MAX-ACCESS  read-only
44     STATUS      current
45     DESCRIPTION
46         "Identifies the local Port associated with the
47         IPG Working Segment."
48     REFERENCE
49         "12.24.1.1.3 b)"
50 ::= { ieee8021TeipsV2IpgEntry 4 }
51
52 ieee8021TeipsV2IpgProtectionPortNumber OBJECT-TYPE
53     SYNTAX      IEEE8021BridgePortNumber
54     MAX-ACCESS  read-only
55     STATUS      current
56     DESCRIPTION
57         "Identifies the local Port associated with the
58         IPG Protection Segment."
59     REFERENCE
60         "12.24.1.1.3 c)"
61 ::= { ieee8021TeipsV2IpgEntry 5 }
62
63 ieee8021TeipsV2IpgStorageType OBJECT-TYPE
64     SYNTAX      StorageType
65     MAX-ACCESS  read-create
66     STATUS      current
67     DESCRIPTION
68         "This object indicates the persistence of this
69         entry. All read-create columns must be
70         writable if this column is set to permanent."
71     DEFVAL { nonVolatile }
72 ::= { ieee8021TeipsV2IpgEntry 6 }

```

```

1
2 ieee8021TeipsV2IpgRowStatus OBJECT-TYPE
3     SYNTAX      RowStatus
4     MAX-ACCESS  read-create
5     STATUS      current
6     DESCRIPTION
7         "The status of this row.
8         The writable columns in a row cannot be
9         changed if the row is active. The
10        TeipsIpgWorkingMA and TeipsIpgProtectionMA
11        columns must be specified before the row
12        can be activated."
13     REFERENCE
14         "12.24.1.2"
15     ::= { ieee8021TeipsV2IpgEntry 7 }
16
17 -- =====
18 -- the ieee8021TeipsV2TesiTable
19 -- =====
20 ieee8021TeipsV2TesiTable OBJECT-TYPE
21     SYNTAX      SEQUENCE OF Ieee8021TeipsV2TesiEntry
22     MAX-ACCESS  not-accessible
23     STATUS      current
24     DESCRIPTION
25         "The IPG TE-SID table contains identifies
26         the TE service instances associated with
27         an IPG."
28     REFERENCE
29         "12.24.2.1.3 e)"
30     ::= { ieee8021TeipsV2Objects 2 }
31
32 ieee8021TeipsV2TesiEntry OBJECT-TYPE
33     SYNTAX      Ieee8021TeipsV2TesiEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "The TE-IPS TESI entry. Each entry identifies
38         a TESI associated with an IPG."
39     INDEX { ieee8021TeipsV2Ipgid,
40             ieee8021TeipsV2TesiIndex }
41     ::= { ieee8021TeipsV2TesiTable 1 }
42
43 Ieee8021TeipsV2TesiEntry ::=
44     SEQUENCE {
45         ieee8021TeipsV2TesiIndex      Unsigned32,
46         ieee8021TeipsV2TesiId         IEEE8021PbbTeTSidId,
47         ieee8021TeipsV2TesiStorageType StorageType,
48         ieee8021TeipsV2TesiRowStatus RowStatus
49     }
50
51 ieee8021TeipsV2TesiIndex OBJECT-TYPE
52     SYNTAX      Unsigned32 (1..4294967295)
53     MAX-ACCESS  not-accessible
54     STATUS      current
55     DESCRIPTION
56         "This is an identifier, of local significance to a
57         particular PBB-TE TE-SID associated with an IPG."
58     REFERENCE
59         "12.24.2.1.3 e)"
60     ::= { ieee8021TeipsV2TesiEntry 1 }
61
62 ieee8021TeipsV2TesiId OBJECT-TYPE
63     SYNTAX      IEEE8021PbbTeTSidId
64     MAX-ACCESS  read-create
65     STATUS      current
66     DESCRIPTION
67         "This column holds the TESI identifier corresponding
68         to a TE service instance associated with an IPG."
69     REFERENCE
70         "12.24.2.1.3 e)"
71     ::= { ieee8021TeipsV2TesiEntry 2 }
72

```

```

1 ieee8021TeipsV2TesiStorageType OBJECT-TYPE
2     SYNTAX      StorageType
3     MAX-ACCESS  read-create
4     STATUS      current
5     DESCRIPTION
6         "This object indicates the persistence of this
7         entry. All read-create columns must be
8         writable for permanent rows."
9     DEFVAL { nonVolatile }
10    ::= { ieee8021TeipsV2TesiEntry 3 }
11
12 ieee8021TeipsV2TesiRowStatus OBJECT-TYPE
13     SYNTAX      RowStatus
14     MAX-ACCESS  read-create
15     STATUS      current
16     DESCRIPTION
17         "This column holds the status for this row.
18         When the status is active, no columns of
19         this table may be modified. All columns
20         must have a valid value before the row
21         can be activated."
22    ::= { ieee8021TeipsV2TesiEntry 4 }
23
24 -- =====
25 -- the ieee8021TeipsV2CandidatePsTable
26 -- =====
27 ieee8021TeipsV2CandidatePsTable OBJECT-TYPE
28     SYNTAX      SEQUENCE OF Ieee8021TeipsV2CandidatePsEntry
29     MAX-ACCESS  not-accessible
30     STATUS      current
31     DESCRIPTION
32         "The Candidate PS table lists, in priority order,
33         from highest priority to lowest priority, the
34         Maintenance Associations corresponding to
35         candidate Protection Segments associated with
36         an IPG."
37     REFERENCE
38         "12.24.2.1.3 d)"
39    ::= { ieee8021TeipsV2Objects 3 }
40
41 ieee8021TeipsV2CandidatePsEntry OBJECT-TYPE
42     SYNTAX      Ieee8021TeipsV2CandidatePsEntry
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46         "A Candidate PS entry. Each entry identifies a
47         candidate Protection Segment associated with an IPG."
48     INDEX { ieee8021TeipsV2Ipgid,
49             ieee8021TeipsV2CandidatePsIndex }
50    ::= { ieee8021TeipsV2CandidatePsTable 1 }
51
52 Ieee8021TeipsV2CandidatePsEntry ::=
53     SEQUENCE {
54         ieee8021TeipsV2CandidatePsIndex Unsigned32,
55         ieee8021TeipsV2CandidatePsMA    Unsigned32,
56         ieee8021TeipsV2CandidatePsPort  IEEE8021BridgePortNumber,
57         ieee8021TeipsV2CandidatePsOper  TruthValue,
58         ieee8021TeipsV2CandidatePsStorageType StorageType,
59         ieee8021TeipsV2CandidatePsRowStatus RowStatus
60     }
61
62 ieee8021TeipsV2CandidatePsIndex OBJECT-TYPE
63     SYNTAX      Unsigned32 (1..4294967295)
64     MAX-ACCESS  not-accessible
65     STATUS      current
66     DESCRIPTION
67         "This is an identifier, of local significance
68         to a particular candidate Protection Segment
69         associated with an IPG."
70     REFERENCE
71         "12.24.2.1.3 d)"
72    ::= { ieee8021TeipsV2CandidatePsEntry 1 }

```

```

1
2 ieee8021TeipsV2CandidatePsMA OBJECT-TYPE
3     SYNTAX      Unsigned32
4     MAX-ACCESS  read-create
5     STATUS      current
6     DESCRIPTION
7         "This column holds the candidate Protection
8         Segment MA corresponding to a candidate
9         Protection Segment associated with an IPG."
10    REFERENCE
11        "12.24.2.1.3 d)"
12    ::= { ieee8021TeipsV2CandidatePsEntry 2 }
13
14 ieee8021TeipsV2CandidatePsPort OBJECT-TYPE
15     SYNTAX      IEEE8021BridgePortNumber
16     MAX-ACCESS  read-only
17     STATUS      current
18     DESCRIPTION
19         "This column holds the Port Number
20         corresponding to the candidate Protection
21         Segment associated with an IPG."
22    REFERENCE
23        "12.24.2.1.3 d)"
24    ::= { ieee8021TeipsV2CandidatePsEntry 3 }
25
26 ieee8021TeipsV2CandidatePsOper OBJECT-TYPE
27     SYNTAX      TruthValue
28     MAX-ACCESS  read-only
29     STATUS      current
30     DESCRIPTION
31         "This column indicates whether or not
32         the candidate Protection Segment is
33         operational."
34    REFERENCE
35        "12.24.2.1.3 d)"
36    ::= { ieee8021TeipsV2CandidatePsEntry 4 }
37
38 ieee8021TeipsV2CandidatePsStorageType OBJECT-TYPE
39     SYNTAX      StorageType
40     MAX-ACCESS  read-create
41     STATUS      current
42     DESCRIPTION
43         "This object indicates the persistence
44         of this entry. All read-create
45         columns must be writable for permanent rows."
46     DEFVAL { nonVolatile }
47    ::= { ieee8021TeipsV2CandidatePsEntry 5 }
48
49 ieee8021TeipsV2CandidatePsRowStatus OBJECT-TYPE
50     SYNTAX      RowStatus
51     MAX-ACCESS  read-create
52     STATUS      current
53     DESCRIPTION
54         "This column holds the status for this row.
55         When the status is active, no columns
56         of this table may be modified. All
57         columns must have a valid value before the row
58         can be activated."
59    ::= { ieee8021TeipsV2CandidatePsEntry 6 }
60
61 -- =====
62 -- the ieee8021TeipsV2IpgConfigTable
63 -- =====
64 ieee8021TeipsV2IpgConfigTable OBJECT-TYPE
65     SYNTAX      SEQUENCE OF Ieee8021TeipsV2IpgConfigEntry
66     MAX-ACCESS  not-accessible
67     STATUS      current
68     DESCRIPTION
69         "The PBB-TE IPS IPG config table contains
70         configuration and status information for
71         each IPG configured in the system.
72         Entries in this table are created implicitly

```

```

1         by the creation of entries in the
2         ieee8021TeipsV2IpgTable."
3     REFERENCE
4         "12.24.2.1.3 f,g,h,i,j,k)"
5     ::= { ieee8021TeipsV2Objects 4 }
6
7 ieee8021TeipsV2IpgConfigEntry OBJECT-TYPE
8     SYNTAX      Ieee8021TeipsV2IpgConfigEntry
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "The IPG configuration table entry. Rows are
13         created in this table implicitly when a row
14         is added to the ieee8021TeipsV2IpgTable."
15    INDEX { ieee8021BridgeBaseComponentId,
16            ieee8021TeipsV2Ipgid }
17    ::= { ieee8021TeipsV2IpgConfigTable 1 }
18
19 Ieee8021TeipsV2IpgConfigEntry ::=
20     SEQUENCE {
21         ieee8021TeipsV2IpgConfigState INTEGER,
22         ieee8021TeipsV2IpgConfigCommandStatus
23             IEEE8021TeipsIpgConfigAdmin,
24         ieee8021TeipsV2IpgConfigCommandLast
25             IEEE8021TeipsIpgConfigAdmin,
26         ieee8021TeipsV2IpgConfigCommandAdmin
27             IEEE8021TeipsIpgConfigAdmin,
28         ieee8021TeipsV2IpgConfigActiveRequests
29             IEEE8021TeipsIpgConfigActiveRequests,
30         ieee8021TeipsV2IpgConfigWTR             Unsigned32,
31         ieee8021TeipsV2IpgConfigHoldOff         Unsigned32,
32         ieee8021TeipsV2IpgM1ConfigState         INTEGER,
33         ieee8021TeipsV2IpgConfigMWTR           Unsigned32,
34         ieee8021TeipsV2IpgConfigNotifyEnable   TruthValue,
35         ieee8021TeipsV2IpgConfigStorageType     StorageType
36     }
37 ieee8021TeipsV2IpgConfigState OBJECT-TYPE
38     SYNTAX      INTEGER {
39         workingSegment(1),
40         protectionSegment(2),
41         waitToRestore(3),
42         protAdmin(4)
43     }
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "This column indicates the current state of the
48         protection switching state machine for an IPG.
49         The value can be one of the following:
50
51         workingSegment(1)   The protection switching state machine
52                             is in the WORKING_PATH state.
53         protectionSegment(2) The protection switching state machine
54                             is in the PROTECTION_PATH state.
55         waitToRestore(3)    The protection switching state machine
56                             is in the WTR state.
57         protAdmin(4)        The protection switching state machine
58                             is in the PROT_ADMIN state."
59
60     REFERENCE "12.24.2.1.3 f)"
61     ::= { ieee8021TeipsV2IpgConfigEntry 1 }
62
63 ieee8021TeipsV2IpgConfigCommandStatus OBJECT-TYPE
64     SYNTAX      IEEE8021TeipsIpgConfigAdmin
65     MAX-ACCESS  read-only
66     STATUS      current
67     DESCRIPTION
68         "This column indicates the status of
69         administrative commands within the
70         protection group. It reflects the current
71         operational administrative command being
72         acted upon by the IPG."

```

```

1  REFERENCE "12.24.2.1.3 f)"
2  ::= { ieee8021TeipsV2IpgConfigEntry 2 }
3
4  ieee8021TeipsV2IpgConfigCommandLast OBJECT-TYPE
5  SYNTAX      IEEE8021TeipsIpgConfigAdmin
6  MAX-ACCESS  read-only
7  STATUS      current
8  DESCRIPTION
9      "This column indicates the last attempted administrative
10     command applied to the IPG. It is changed
11     whenever a write is made to the CommandAdmin column of
12     this table and is essentially record of the last attempted
13     administrative operation."
14  REFERENCE "12.24.2.1.3 f)"
15  ::= { ieee8021TeipsV2IpgConfigEntry 3 }
16
17  ieee8021TeipsV2IpgConfigCommandAdmin OBJECT-TYPE
18  SYNTAX      IEEE8021TeipsIpgConfigAdmin
19  MAX-ACCESS  read-create
20  STATUS      current
21  DESCRIPTION
22      "This column is used by the operator to request
23      that the IPG state machine perform some
24      administrative operation. The operator requests
25      a command by writing the command value to this
26      column. The state machine indicates the command
27      that it is performing by setting the value of the
28      CommandStatus column of this table. This column
29      always reads back as clear(1)."
```

REFERENCE "12.24.2.1.3 f)"

```

31  DEFVAL { clear }
32  ::= { ieee8021TeipsV2IpgConfigEntry 4 }
33
34  ieee8021TeipsV2IpgConfigActiveRequests OBJECT-TYPE
35  SYNTAX      IEEE8021TeipsIpgConfigActiveRequests
36  MAX-ACCESS  read-only
37  STATUS      current
38  DESCRIPTION
39      "This column shows the status of active requests
40      associated with the IPG."
41  REFERENCE "12.24.2.1.3 f)"
42  ::= { ieee8021TeipsV2IpgConfigEntry 5 }
43
44  ieee8021TeipsV2IpgConfigWTR OBJECT-TYPE
45  SYNTAX      Unsigned32 ( 0 | 5..12 )
46  UNITS       "minutes"
47  MAX-ACCESS  read-create
48  STATUS      current
49  DESCRIPTION
50      "This column is used to configure the
51      wait-to-restore timer for the IPG operation.
52      The timer may be configured in steps of 1 minute
53      between 5 and 12 minutes, the default being 5.
54      Additionally, the value 0 is used to indicate
55      that the IPG is to operate non-revertively. The
56      value 0 is not permitted if the IPG is configured
57      for M:1 IPS operation."
58  REFERENCE "12.24.2.1.3 h)"
59  DEFVAL { 5 }
60  ::= { ieee8021TeipsV2IpgConfigEntry 6 }
61
62  ieee8021TeipsV2IpgConfigHoldOff OBJECT-TYPE
63  SYNTAX      Unsigned32( 0..100 )
64  UNITS       "deciseconds"
65  MAX-ACCESS  read-create
66  STATUS      current
67  DESCRIPTION
68      "This column is used to configure the hold off
69      timer. The purpose is to allow IPS to fix the problem
70      before a higher-layer mechanism, such as PBB-TE TESI
71      protection, is invoked or to allow an inner IPG to fix
72      the problem before IPS is invoked by the outer IPG when
```

```

1      IPGs are nested. The hold off timer has a period of
2      from 0 to 10 seconds, the default being 0, with a 100ms
3      granularity."
4      REFERENCE "12.24.2.1.3 i)"
5      DEFVAL { 0 }
6      ::= { ieee8021TeipsV2IpgConfigEntry 7 }
7
8 ieee8021TeipsV2IpgM1ConfigState OBJECT-TYPE
9     SYNTAX      INTEGER {
10         psAssigned(1),
11         segmentOk(2),
12         segmentFailed(3),
13         assignNewPs(4),
14         revertToBetterPs(5)
15     }
16     MAX-ACCESS read-only
17     STATUS      current
18     DESCRIPTION
19         "This column indicates the current state of the M:1 protection
20         switching state machine for an IPG if M:1 IPS is supported.
21         The value can be one of the following:
22
23         psAssigned(1)      The protection switching state machine
24                             is in the PS_ASSIGNED state.
25         segmentOk(2)       The protection switching state machine
26                             is in the SEGMENT_OK state.
27         segmentFailed(3)   The protection switching state machine
28                             is in the SEGMENT_FAILED state.
29         assignNewPs(4)     The protection switching state machine
30                             is in the ASSIGN_NEW_PS state.
31         revertToBetterPs(5) The protection switching state machine
32                             is in the REVERT_TO_BETTER_PS state."
33
34     REFERENCE "12.24.2.1.3 j)"
35     ::= { ieee8021TeipsV2IpgConfigEntry 8 }
36
37 ieee8021TeipsV2IpgConfigMWTR OBJECT-TYPE
38     SYNTAX      Unsigned32 ( 0 | 5..12 )
39     UNITS       "minutes"
40     MAX-ACCESS read-create
41     STATUS      current
42     DESCRIPTION
43         "This column is used to configure the M:1 wait-to-restore
44         timer for the IPG operation if M:1 protection is
45         supported. The timer may be configured in steps of
46         1 minute between 5 and 12 minutes, the default being 5.
47         Additionally, the value 0 is used to indicate that the
48         IPG is to operate non-revertively."
49     REFERENCE "12.24.2.1.3 k)"
50     DEFVAL { 5 }
51     ::= { ieee8021TeipsV2IpgConfigEntry 9 }
52
53
54 ieee8021TeipsV2IpgConfigNotifyEnable OBJECT-TYPE
55     SYNTAX      TruthValue
56     MAX-ACCESS read-create
57     STATUS      current
58     DESCRIPTION
59         "This column is used to enable or disable transmission
60         of ieee8021TeipsV2IpgAdminFailure notifications.
61         These notifications are generated whenever an
62         administrative command cannot be performed by the IPG."
63     DEFVAL { false }
64     ::= { ieee8021TeipsV2IpgConfigEntry 10 }
65
66 ieee8021TeipsV2IpgConfigStorageType OBJECT-TYPE
67     SYNTAX      StorageType
68     MAX-ACCESS read-create
69     STATUS      current
70     DESCRIPTION
71         "This object indicates the persistence of this entry. For
72         permanent objects the ieee8021TeipsV2IpgConfigCommandAdmin

```



```

1      column must be writable."
2
3      DEFVAL { nonVolatile }
4      ::= { ieee8021TeipsV2IpgConfigEntry 11 }
5
6
7  -- *****
8  -- NOTIFICATIONS (TRAPS)
9  -- These notifications will be sent to the management entity
10 -- whenever an IPG admin command cannot be performed
11 -- *****
12
13 ieee8021TeipsV2IpgAdminFailure NOTIFICATION-TYPE
14     OBJECTS {
15         ieee8021TeipsV2IpgConfigState,
16         ieee8021TeipsV2IpgConfigCommandStatus,
17         ieee8021TeipsV2IpgConfigCommandLast
18     }
19     STATUS current
20     DESCRIPTION
21         "An IPG generates this notification whenever
22         an administrative command cannot be
23         executed by the IPS state machine. For
24         example, when a requested manual switch
25         cannot be performed because of a signal
26         failure condition.
27
28         The management entity receiving the
29         notification can identify
30         the system from the network source
31         address of the notification and can
32         identify the IPG by the indices of
33         the OID of the ieee8021TeipsV2IpgConfigState
34         variable in the notification:
35
36         ieee8021BridgeBaseComponentId - Identifies
37         the component on the Bridge where the
38         protection group is configured.
39
40         ieee8021TeipsV2Ipgid - The ID of the protection group.
41         "
42     ::= { ieee8021TeipsV2Notifications 1 }
43
44 --
45 -- MIB Module Compliance Statements
46 --
47
48 ieee8021TeipsV2Compliances OBJECT IDENTIFIER ::= { ieee8021TeipsV2Conformance 1 }
49 ieee8021TeipsV2Groups      OBJECT IDENTIFIER ::= { ieee8021TeipsV2Conformance 2 }
50
51 --
52 -- Units of Conformance
53
54 ieee8021TeipsV2IpgGroup OBJECT-GROUP
55     OBJECTS {
56         ieee8021TeipsV2IpgWorkingMA,
57         ieee8021TeipsV2IpgProtectionMA,
58         ieee8021TeipsV2IpgWorkingPortNumber,
59         ieee8021TeipsV2IpgProtectionPortNumber,
60         ieee8021TeipsV2IpgStorageType,
61         ieee8021TeipsV2IpgRowStatus
62     }
63     STATUS current
64     DESCRIPTION
65         "Objects for the IPG group."
66     ::= { ieee8021TeipsV2Groups 1 }
67
68 ieee8021TeipsV2CandidatePsGroup OBJECT-GROUP
69     OBJECTS {
70         ieee8021TeipsV2CandidatePsMA,
71         ieee8021TeipsV2CandidatePsPort,
72         ieee8021TeipsV2CandidatePsOper,

```

```

1         ieee8021TeipsV2CandidatePsStorageType,
2         ieee8021TeipsV2CandidatePsRowStatus
3     }
4     STATUS current
5     DESCRIPTION
6         "Objects for the Candidate PS group."
7     ::= { ieee8021TeipsV2Groups 2 }
8
9 ieee8021TeipsV2IpgTesiGroup OBJECT-GROUP
10    OBJECTS {
11        ieee8021TeipsV2TesiId,
12        ieee8021TeipsV2TesiStorageType,
13        ieee8021TeipsV2TesiRowStatus
14    }
15    STATUS current
16    DESCRIPTION
17        "Objects for the IPG Tuple group."
18    ::= { ieee8021TeipsV2Groups 3 }
19
20 ieee8021TeipsV2IpgConfigManGroup OBJECT-GROUP
21    OBJECTS {
22        ieee8021TeipsV2IpgConfigState,
23        ieee8021TeipsV2IpgConfigCommandStatus,
24        ieee8021TeipsV2IpgConfigCommandLast,
25        ieee8021TeipsV2IpgConfigCommandAdmin,
26        ieee8021TeipsV2IpgConfigActiveRequests,
27        ieee8021TeipsV2IpgConfigNotifyEnable,
28        ieee8021TeipsV2IpgConfigStorageType
29    }
30    STATUS current
31    DESCRIPTION
32        "Mandatory objects for the TeipsConfiguration group."
33    ::= { ieee8021TeipsV2Groups 4 }
34
35 ieee8021TeipsV2IpgConfigOptGroup OBJECT-GROUP
36    OBJECTS {
37        ieee8021TeipsV2IpgConfigWTR,
38        ieee8021TeipsV2IpgConfigMWTR,
39        ieee8021TeipsV2IpgM1ConfigState,
40        ieee8021TeipsV2IpgConfigHoldOff
41    }
42    STATUS current
43    DESCRIPTION
44        "Optional Objects for the TeipsConfiguration group."
45    ::= { ieee8021TeipsV2Groups 5 }
46
47 ieee8021TeipsV2NotificationsGroup NOTIFICATION-GROUP
48    NOTIFICATIONS {
49        ieee8021TeipsV2IpgAdminFailure
50    }
51    STATUS current
52    DESCRIPTION
53        "Objects for the notifications group."
54    ::= { ieee8021TeipsV2Groups 6 }
55
56 ieee8021TeipsV2Compliance MODULE-COMPLIANCE
57    STATUS current
58    DESCRIPTION
59        "The compliance statement for support
60         of the TEIPS MIB module."
61    MODULE
62        MANDATORY-GROUPS {
63            ieee8021TeipsV2IpgGroup,
64            ieee8021TeipsV2IpgTesiGroup,
65            ieee8021TeipsV2IpgConfigManGroup,
66            ieee8021TeipsV2NotificationsGroup
67        }
68    GROUP ieee8021TeipsV2IpgConfigOptGroup
69    DESCRIPTION
70        "This group allows implementation to
71         optionally change the WaitToRestore,
72         M:1 WaitToRestore, and HoldOff timers

```

```
1         for IPGs."
2
3     GROUP ieee8021TeipsV2CandidatePsGroup
4     DESCRIPTION
5         "This group allows implementation to
6         optionally list candidate Protection
7         Segments when M:1 IPS is deployed."
8
9
10    OBJECT ieee8021TeipsV2IpgConfigWTR
11    MIN-ACCESS not-accessible
12    DESCRIPTION "This object is optional."
13
14    OBJECT ieee8021TeipsV2IpgConfigHoldOff
15    MIN-ACCESS not-accessible
16    DESCRIPTION "This object is optional."
17
18    OBJECT ieee8021TeipsV2IpgConfigMWTR
19    MIN-ACCESS not-accessible
20    DESCRIPTION "This object is optional."
21
22    OBJECT ieee8021TeipsV2IpgRowStatus
23    SYNTAX      RowStatus { active(1), notInService(2) }
24    WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
25                          destroy(6) }
26    DESCRIPTION "Support for createAndWait is not required."
27
28    OBJECT ieee8021TeipsV2TesiRowStatus
29    SYNTAX      RowStatus { active(1), notInService(2) }
30    WRITE-SYNTAX RowStatus { notInService(2), createAndGo(4),
31                          destroy(6) }
32    DESCRIPTION "Support for createAndWait is not required."
33
34    ::= { ieee8021TeipsV2Compliances 1 }
35
36 END
```

1 17.7.19 Definitions for the IEEE8021-SPB-MIB module

```
2 IEEE8021-SPB-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- IEEE 802.1 Shortest Path Bridging (SPB) MIB --
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE, Integer32, Unsigned32
10    FROM SNMPv2-SMI
11    RowStatus, MacAddress, TruthValue, TEXTUAL-CONVENTION
12    FROM SNMPv2-TC
13    ieee802dot1mibs, IEEE8021PbbIngressEgress,
14    IEEE8021BridgePortNumber, IEEE8021PbbServiceIdentifier,
15    IEEE8021PbbTeEsp
16    FROM IEEE8021-TC-MIB
17    ieee8021BridgeBasePort
18    FROM IEEE8021-BRIDGE-MIB
19    VlanId, VlanIdOrNone, VlanIdOrAny
20    FROM Q-BRIDGE-MIB
21    dotlagCfmMepEntry, dotlagCfmMdIndex, dotlagCfmMaIndex
22    FROM IEEE8021-CFM-MIB
23    InterfaceIndexOrZero
24    FROM IF-MIB
25    SnmpAdminString
26    FROM SNMP-FRAMEWORK-MIB
27    MODULE-COMPLIANCE, OBJECT-GROUP
28    FROM SNMPv2-CONF;
29
30 ieee8021SpbMib MODULE-IDENTITY
31     LAST-UPDATED "202211080000Z" -- November 8, 2022
32     ORGANIZATION "IEEE 802.1 Working Group"
33     CONTACT-INFO
34         " WG-URL: http://www.ieee802.org/1/
35         WG-EMail: stds-802-1-1@ieee.org
36         Contact: IEEE 802.1 Working Group Chair
37         Postal: C/O IEEE 802.1 Working Group
38             IEEE Standards Association
39             445 Hoes Lane
40             Piscataway, NJ 08854
41             USA
42         E-mail: stds-802-1-chairs@ieee.org"
43
44     DESCRIPTION
45         "The MIB module for managing the support of
46         Shortest Path Bridging (SPB) in IEEE 802.1Q Bridges.
47
48         Unless otherwise indicated, the references in this MIB
49         module are to IEEE Std 802.1Q-2022.
50
51         Copyright (C) IEEE (2022).
52         This version of this MIB module is part of IEEE Std 802.1Q;
53         see that standard for full legal notices."
54
55     REVISION "202211080000Z" -- November 8, 2022
56     DESCRIPTION
57         "Published as part of IEEE Std 802.1Q-2022.
58         Cross references and contact information updated."
59
60     REVISION "201807010000Z" -- July 1, 2018
61     DESCRIPTION
62         "Published as part of IEEE Std 802.1Q 2018 revision.
63         Cross references updated and corrected.
64         Contact details corrected. Syntax error in
65         ieee8021SpbEctDynamicEntryIngressCheckDiscards
66         corrected."
67
68     REVISION "201506230000Z" -- June 23, 2015
69     DESCRIPTION "802.1Qca additions"
70
71     REVISION "201305130000Z" -- May 13, 2013
```

```

1      DESCRIPTION  "802.1Qbp additions and corrections"
2
3      REVISION  "201202030000Z"  -- February 3, 2012
4      DESCRIPTION  "802.1 Shortest Path Bridging MIB Initial Version"
5
6      ::= { ieee802dot1mibs 26 }
7
8  -- =====
9  --  TYPE DEFINITIONS                                     --
10 -- =====
11
12 IEEE8021SpbAreaAddress ::= TEXTUAL-CONVENTION
13     DISPLAY-HINT "1x:"
14     STATUS current
15     DESCRIPTION
16         "This identifier is the 3 Byte IS-IS Area Address.
17         Domain Specific part(DSP)."

```

```

1 IEEE8021SpbmSPsourceId ::= TEXTUAL-CONVENTION
2     DISPLAY-HINT "1x:"
3     STATUS current
4     DESCRIPTION
5         "It is the high order 3 bytes for Group Address DA from this
6         bridge.
7         Note that only the 20 bits not including the top 4 bits are
8         the SPSourceID."
9     REFERENCE "27.15"
10    SYNTAX OCTET STRING (SIZE(3))
11
12 IEEE8021SpbDigest ::= TEXTUAL-CONVENTION
13     DISPLAY-HINT "1x"
14     STATUS current
15     DESCRIPTION
16         "The Topology Agreement digest hex string."
17     REFERENCE "28.4"
18     SYNTAX OCTET STRING (SIZE(32))
19
20 IEEE8021SpbMCID ::= TEXTUAL-CONVENTION
21     DISPLAY-HINT "1x"
22     STATUS current
23     DESCRIPTION
24         "MST Configuration Identifier digest hex string."
25     REFERENCE "13.8"
26     SYNTAX OCTET STRING (SIZE(51))
27
28 IEEE8021SpbBridgePriority ::= TEXTUAL-CONVENTION
29     DISPLAY-HINT "1x"
30     STATUS current
31     DESCRIPTION
32         "The Bridge priority is the top 2 bytes of the Bridge Identifier.
33         Lower values represent a better priority."
34     REFERENCE "13.26.3"
35     SYNTAX OCTET STRING (SIZE(2))
36
37 IEEE8021SpbMTID ::= TEXTUAL-CONVENTION
38     DISPLAY-HINT "d"
39     STATUS current
40     DESCRIPTION
41         "The IS-IS Multi Topology Identifier."
42     REFERENCE "3.167, 3.168"
43     SYNTAX Unsigned32
44
45 IEEE8021SpbServiceIdentifierOrAny ::= TEXTUAL-CONVENTION
46     DISPLAY-HINT "d"
47     STATUS current
48     DESCRIPTION
49         "The service instance identifier is used at the Customer Backbone
50         port in SPBM to distinguish a service instance.
51         The special value of 0xFFFFF is used for wildcard.
52         This range also includes the default I-SID. "
53     REFERENCE "3.167, 3.168"
54     SYNTAX Unsigned32 (255..16777215)
55
56 -- =====
57 -- OBJECT DEFINITIONS --
58 -- =====
59
60 -- =====
61 -- ieee8021SpbObjects:
62 -- =====
63 ieee8021SpbObjects OBJECT IDENTIFIER
64     ::= { ieee8021SpbMib 1 }
65
66 -- =====
67 -- ieee8021PcrObjects:
68 -- =====
69 ieee8021PcrObjects OBJECT IDENTIFIER ::= { ieee8021SpbMib 3 }
70
71 -- =====
72 -- ieee8021SpbSys:

```

```

1 -- =====
2 ieee8021SpbSys OBJECT IDENTIFIER
3     ::= { ieee8021SpbObjects 1 }
4
5 ieee8021SpbSysAreaAddress OBJECT-TYPE
6     SYNTAX IEEE8021SpbAreaAddress
7     MAX-ACCESS read-write
8     STATUS current
9     DESCRIPTION
10        "The three byte IS-IS Area Address to join. Normally
11         SPB will use area 00:00:00 however if SPB is being
12         used in conjunction with IPV4/V6 it can operate
13         using the IS-IS area address already in use.
14         This object is persistent."
15     REFERENCE "12.25.1.3.2, 12.25.1.3.3"
16     ::= { ieee8021SpbSys 1 }
17
18 ieee8021SpbSysId OBJECT-TYPE
19     SYNTAX MacAddress
20     MAX-ACCESS read-write
21     STATUS current
22     DESCRIPTION
23        "SYS ID used for all SPB instances on this bridge.
24         A six byte network wide unique identifier. This is
25         defaulted to the Bridge Address initially but can
26         be overridden.
27         This object is persistent."
28     REFERENCE "12.25.1.3.3, 3.253"
29     ::= { ieee8021SpbSys 2 }
30
31 ieee8021SpbSysControlAddr OBJECT-TYPE
32     SYNTAX MacAddress
33     MAX-ACCESS read-write
34     STATUS current
35     DESCRIPTION
36        "Group MAC that the ISIS control plane will use. SPB can
37         use a number of different addresses for SPB Hello and
38         LSP exchange. Section 27.2, 8.13.5.1 and Table 8-16 cover
39         the different choices. The choices are as follows:
40         01:80:C2:00:00:14 = All Level 1 Intermediate Systems
41         01:80:C2:00:00:15 = All Level 2 Intermediate Systems
42         09:00:2B:00:00:05 = All Intermediate Systems.
43         01:80:C2:00:00:2E = All Provider Bridge Intermediate Systems.
44         01:80:C2:00:00:2F = All Customer Bridge Intermediate Systems.
45         This object is persistent."
46     REFERENCE "12.25.1.1.2, 8.13.5.1"
47     ::= { ieee8021SpbSys 3 }
48
49 ieee8021SpbSysName OBJECT-TYPE
50     SYNTAX SnmpAdminString (SIZE(0..32))
51     MAX-ACCESS read-only
52     STATUS current
53     DESCRIPTION
54        "Name to be used to refer to this SPB bridge. This is advertised
55         in IS-IS and used for management."
56     REFERENCE "12.25.1.3.3"
57     ::= { ieee8021SpbSys 4 }
58
59 ieee8021SpbSysBridgePriority OBJECT-TYPE
60     SYNTAX IEEE8021SpbBridgePriority
61     MAX-ACCESS read-only
62     STATUS current
63     DESCRIPTION
64        "This is a 16-bit quantity that ranks this SPB Bridge
65         relative to others when breaking ties. This priority
66         is the high 16 bits of the Bridge Identifier. Its impact
67         depends on the tie breaking algorithm. Recommend
68         values 0..15 be assigned to core switches to ensure
69         diversity of the ECT Algorithms."
70     REFERENCE "12.25.1.3.3, 13.26.3"
71     ::= { ieee8021SpbSys 5 }
72

```

```
1 ieee8021SpbmSysSPSourceId OBJECT-TYPE
2     SYNTAX IEEE8021SpbmSPsourceId
3     MAX-ACCESS read-write
4     STATUS current
5     DESCRIPTION
6         "The Shortest Path Source Identifier.
7         It is the high order 3 bytes for Group Address DA from this
8         bridge.
9         Note that only the 20 bits not including the top 4 bits are
10        the SPSourceID.
11        This object is persistent."
12    REFERENCE "12.25.1.3.3, 3.259, 27.15"
13    ::= { ieee8021SpbSys 6 }
14
15 ieee8021SpbvSysMode OBJECT-TYPE
16     SYNTAX IEEE8021SpbMode
17     MAX-ACCESS read-write
18     STATUS current
19     DESCRIPTION
20         "Indication of supporting SPBV mode
21         auto(=1)/manual(=2)
22         auto => auto allocate SPVIDs.
23         manual => manually assign SPVIDs.
24         This object is persistent."
25    REFERENCE "12.25.1.3.3, 3.264"
26    DEFVAL {auto}
27    ::= { ieee8021SpbSys 7 }
28
29 ieee8021SpbmSysMode OBJECT-TYPE
30     SYNTAX IEEE8021SpbMode
31     MAX-ACCESS read-write
32     STATUS current
33     DESCRIPTION
34         "Indication of supporting SPBM mode
35         auto(=1)/manual(=2)
36         auto => enable SPBM mode and auto allocate SPsourceID.
37         manual => enable SPBM mode and manually assign SPsourceID.
38         This object is persistent."
39    REFERENCE "12.25.1.3.3, 3.254"
40    DEFVAL {auto}
41    ::= { ieee8021SpbSys 8 }
42
43 ieee8021SpbSysDigestConvention OBJECT-TYPE
44     SYNTAX IEEE8021SpbDigestConvention
45     MAX-ACCESS read-write
46     STATUS current
47     DESCRIPTION
48         "The Agreement Digest convention setting
49         off(=1)/loopFreeBoth(=2)/loopFreeMcastOnly(=3)
50         off => disable agreement digest checking in hellos
51         loopFreeBoth => block unsafe group and individual
52         traffic when digests disagree.
53         loopFreeMcastOnly => block unsafe group traffic when digests
54         disagree.
55         This object is persistent."
56    REFERENCE "12.25.1.3.3, 28.4.3"
57    DEFVAL {loopFreeBoth}
58    ::= { ieee8021SpbSys 9 }
59
60 -- =====
61 -- ieee8021SpbMtidStaticTable:
62 -- =====
63 ieee8021SpbMtidStaticTable OBJECT-TYPE
64     SYNTAX SEQUENCE OF Ieee8021SpbMtidStaticTableEntry
65     MAX-ACCESS not-accessible
66     STATUS current
67     DESCRIPTION
68         "A Table of multiple logical topologies - MT."
69    REFERENCE "12.25.2"
70    ::= { ieee8021SpbObjects 2 }
71
72 ieee8021SpbMtidStaticTableEntry OBJECT-TYPE
```



```

1      SYNTAX Ieee8021SpbMtidStaticTableEntry
2      MAX-ACCESS not-accessible
3      STATUS current
4      DESCRIPTION
5          "This table configures a MTID to a topology index. A
6          topology index uniquely identifies a specific ISIS
7          topology enabling multiple instances or multiple
8          MTIDs within an instance. "
9      REFERENCE "12.25.2"
10     INDEX {
11         ieee8021SpbMtidStaticEntryMtid,
12         ieee8021SpbTopIx
13     }
14     ::= { ieee8021SpbMtidStaticTable 1 }
15
16 Ieee8021SpbMtidStaticTableEntry ::=
17     SEQUENCE {
18         ieee8021SpbMtidStaticEntryMtid IEEE8021SpbMTID,
19         ieee8021SpbMTidStaticEntryMtidOverload TruthValue,
20         ieee8021SpbMtidStaticEntryRowStatus RowStatus,
21         ieee8021SpbTopIx IEEE8021SpbMTID
22     }
23
24 ieee8021SpbMtidStaticEntryMtid OBJECT-TYPE
25     SYNTAX IEEE8021SpbMTID
26     MAX-ACCESS not-accessible
27     STATUS current
28     DESCRIPTION
29         "ISIS Multi Topology Identifier MTID
30         Each MTID defines logical topology and is used
31         to enable multiple SPB instances within one ISIS instance."
32     REFERENCE "12.25.1.3.2, 12.25.2.3.3, 28.12"
33     ::= { ieee8021SpbMtidStaticTableEntry 1 }
34
35 ieee8021SpbMTidStaticEntryMtidOverload OBJECT-TYPE
36     SYNTAX TruthValue
37     MAX-ACCESS read-create
38     STATUS current
39     DESCRIPTION
40         "When set for this logical topology this bridge can only
41         originate or terminate traffic. It cannot transit SPB
42         encapsulated traffic. This is the IS-IS overload feature
43         specific to an SPB IS-IS MTID logical topology.
44         This object is persistent."
45     REFERENCE "12.25.2.3.3, 27.8.1"
46     DEFVAL {false}
47     ::= { ieee8021SpbMtidStaticTableEntry 2 }
48
49 ieee8021SpbMtidStaticEntryRowStatus OBJECT-TYPE
50     SYNTAX RowStatus
51     MAX-ACCESS read-create
52     STATUS current
53     DESCRIPTION
54         "The object indicates the status of an entry, and is used
55         to create/delete entries. This object is persistent.
56         This object is persistent."
57     REFERENCE "12.25.2.3.3"
58     ::= { ieee8021SpbMtidStaticTableEntry 3 }
59
60 ieee8021SpbTopIx OBJECT-TYPE
61     SYNTAX IEEE8021SpbMTID
62     MAX-ACCESS not-accessible
63     STATUS current
64     DESCRIPTION
65         "Unique identifier of this SPB topology
66         This is index is allocated for this ISIS/MT instance.
67         It is used as an index to most other SPB tables below and to
68         select the exact ISIS instance and the MT instance together."
69     REFERENCE "12.25.2.3.3"
70     ::= { ieee8021SpbMtidStaticTableEntry 4 }
71
72 -- =====

```

```

1 -- ieee8021SpbTopIxDynamicTable:
2 -- =====
3 ieee8021SpbTopIxDynamicTable OBJECT-TYPE
4     SYNTAX SEQUENCE OF Ieee8021SpbTopIxDynamicTableEntry
5     MAX-ACCESS not-accessible
6     STATUS current
7     DESCRIPTION
8         "This table is for SPB dynamic information. The dynamic
9         information that is sent in this bridges Hellos."
10    REFERENCE "12.25.3"
11    ::= { ieee8021SpbObjects 3 }
12
13 ieee8021SpbTopIxDynamicTableEntry OBJECT-TYPE
14     SYNTAX Ieee8021SpbTopIxDynamicTableEntry
15     MAX-ACCESS not-accessible
16     STATUS current
17     DESCRIPTION
18         "This table displays the digest information computed for this bridge.
19         A bridge configures this information in MTID 0 only. "
20    REFERENCE "12.25.3"
21    INDEX {
22        ieee8021SpbTopIxDynamicEntryTopIx
23    }
24    ::= { ieee8021SpbTopIxDynamicTable 1 }
25
26 Ieee8021SpbTopIxDynamicTableEntry ::=
27     SEQUENCE {
28         ieee8021SpbTopIxDynamicEntryTopIx IEEE8021SpbMTID,
29         ieee8021SpbTopIxDynamicEntryAgreeDigest IEEE8021SpbDigest,
30         ieee8021SpbTopIxDynamicEntryMCID IEEE8021SpbMCID,
31         ieee8021SpbTopIxDynamicEntryAuxMCID IEEE8021SpbMCID
32     }
33
34 ieee8021SpbTopIxDynamicEntryTopIx OBJECT-TYPE
35     SYNTAX IEEE8021SpbMTID
36     MAX-ACCESS not-accessible
37     STATUS current
38     DESCRIPTION
39         "ISIS-SPB Topology Index identifier
40         Each Topology Index defines logical topology and is used
41         to enable multiple SPB instances within several ISIS instances."
42    REFERENCE "12.25.3.1.2, 28.12"
43    ::= { ieee8021SpbTopIxDynamicTableEntry 1 }
44
45 ieee8021SpbTopIxDynamicEntryAgreeDigest OBJECT-TYPE
46     SYNTAX IEEE8021SpbDigest
47     MAX-ACCESS read-only
48     STATUS current
49     DESCRIPTION
50         "The topology agreement digest value. Digest of all
51         topology information, as in clause 28.4."
52    REFERENCE "12.25.3.1.3, 28.4"
53    ::= { ieee8021SpbTopIxDynamicTableEntry 2 }
54
55 ieee8021SpbTopIxDynamicEntryMCID OBJECT-TYPE
56     SYNTAX IEEE8021SpbMCID
57     MAX-ACCESS read-only
58     STATUS current
59     DESCRIPTION
60         "The MST Identifier MCID. The MCID is a digest of the
61         VID to MSTID configuration table, which determines the Base VIDs
62         enabled for SPBV and SPBM."
63    REFERENCE "12.25.3.1.3, 13.8"
64    ::= { ieee8021SpbTopIxDynamicTableEntry 3 }
65
66 ieee8021SpbTopIxDynamicEntryAuxMCID OBJECT-TYPE
67     SYNTAX IEEE8021SpbMCID
68     MAX-ACCESS read-only
69     STATUS current
70     DESCRIPTION
71         "The aux MST Identifier for migration.
72         Either MCID or AuxMCID has to match for adjacency to form."

```

```

1      REFERENCE "12.25.3.1.3, 28.9"
2      ::= { ieee8021SpbTopIxDynamicTableEntry 4 }
3
4 -- =====
5 -- ieee8021SpbEctStaticTable:
6 -- =====
7 ieee8021SpbEctStaticTable OBJECT-TYPE
8     SYNTAX SEQUENCE OF Ieee8021SpbEctStaticTableEntry
9     MAX-ACCESS not-accessible
10    STATUS current
11    DESCRIPTION
12        "The Equal Cost Tree (ECT) static configuration table."
13    REFERENCE "12.25.4"
14    ::= { ieee8021SpbObjects 4 }
15
16 ieee8021SpbEctStaticTableEntry OBJECT-TYPE
17     SYNTAX Ieee8021SpbEctStaticTableEntry
18     MAX-ACCESS not-accessible
19     STATUS current
20     DESCRIPTION
21         "The Equal Cost Tree static configuration Table defines the
22         ECT-ALGORITHM for the Base VID and if SPBV is used for the SPVID. "
23     REFERENCE "12.25.4"
24     INDEX {
25         ieee8021SpbEctStaticEntryTopIx,
26         ieee8021SpbEctStaticEntryBaseVid
27     }
28     ::= { ieee8021SpbEctStaticTable 1 }
29
30 Ieee8021SpbEctStaticTableEntry ::=
31     SEQUENCE {
32         ieee8021SpbEctStaticEntryTopIx IEEE8021SpbMTID,
33         ieee8021SpbEctStaticEntryBaseVid VlanIdOrAny,
34         ieee8021SpbEctStaticEntryEctAlgorithm IEEE8021SpbEctAlgorithm,
35         ieee8021SpbvEctStaticEntrySpvid VlanIdOrNone,
36         ieee8021SpbEctStaticEntryRowStatus RowStatus
37     }
38
39 ieee8021SpbEctStaticEntryTopIx OBJECT-TYPE
40     SYNTAX IEEE8021SpbMTID
41     MAX-ACCESS not-accessible
42     STATUS current
43     DESCRIPTION
44         "The ISIS Topology Index identifier to which this
45         instance belongs. Each Topology Index defines logical topology
46         and is used to enable multiple SPB instances within several
47         ISIS instances."
48     REFERENCE "12.25.4.2.2, 12.25.4.2.3, 28.12"
49     ::= { ieee8021SpbEctStaticTableEntry 1 }
50
51 ieee8021SpbEctStaticEntryBaseVid OBJECT-TYPE
52     SYNTAX VlanIdOrAny
53     MAX-ACCESS not-accessible
54     STATUS current
55     DESCRIPTION
56         "Base VID to use for this ECT-ALGORITHM.
57         Traffic B-VID (SPBM) or Management VID (SPBV).
58         A Base VID value of 4095 is a wildcard for any Base VID
59         assigned to SPB operation."
60     REFERENCE "12.25.4.2.3, 3.25"
61     ::= { ieee8021SpbEctStaticTableEntry 2 }
62
63 ieee8021SpbEctStaticEntryEctAlgorithm OBJECT-TYPE
64     SYNTAX IEEE8021SpbEctAlgorithm
65     MAX-ACCESS read-create
66     STATUS current
67     DESCRIPTION
68         "This identifies the method and the algorithm used
69         to determine the active topology. The standard ECT
70         Algorithm values are specified by Table 28-1,
71         Table 44-1, and Table 45-1.
72         Table 28-1 values identify the tie-breaking algorithms

```

```

1          used in Shortest Path Tree computation;
2          values range from 00-80-c2-01 to 00-80-c2-10.
3          Table 44-1 values (00-80-c2-11 and 00-80-c2-12)
4          identify ECMP operations.
5          Table 45-1 values identify explicit path control;
6          values are 00-80-c2-17, 00-80-c2-18, 00-80-c2-19,
7          and the range from 00-80-c2-21 to 00-80-c2-40.
8          The default is 00-80-c2-01, which is the LowPATHID
9          from Table 28-1.
10         This object is persistent."
11     REFERENCE "12.25.4.1, 12.25.4.2.3, 3.167"
12     DEFVAL {"00-80-c2-01"}
13     ::= { ieee8021SpbEctStaticTableEntry 3 }
14
15 ieee8021SpbvEctStaticEntrySpvid OBJECT-TYPE
16     SYNTAX VlanIdOrNone
17     MAX-ACCESS read-create
18     STATUS current
19     DESCRIPTION
20         "If SPBV mode this is the VID originating from this bridge.
21         This input is ignored if ieee8021SpbvSysMode is auto(1),
22         but the output always returns the SPVID in use.
23         Otherwise in SPBM this is empty, should be set = 0.
24         This object is persistent."
25     REFERENCE "12.25.4.2.3, 3.261"
26     ::= { ieee8021SpbEctStaticTableEntry 4 }
27
28 ieee8021SpbEctStaticEntryRowStatus OBJECT-TYPE
29     SYNTAX RowStatus
30     MAX-ACCESS read-create
31     STATUS current
32     DESCRIPTION
33         "The object indicates the status of an entry, and is used
34         to create/delete entries.
35         This object is persistent."
36     REFERENCE "12.25.4.2.3"
37     ::= { ieee8021SpbEctStaticTableEntry 5 }
38
39 -- =====
40 -- ieee8021SpbEctDynamicTable:
41 -- =====
42 ieee8021SpbEctDynamicTable OBJECT-TYPE
43     SYNTAX SEQUENCE OF Ieee8021SpbEctDynamicTableEntry
44     MAX-ACCESS not-accessible
45     STATUS current
46     DESCRIPTION
47         "A table containing Data about the ECT behavior on this bridge"
48     REFERENCE "12.25.5"
49     ::= { ieee8021SpbObjects 5 }
50
51 ieee8021SpbEctDynamicTableEntry OBJECT-TYPE
52     SYNTAX Ieee8021SpbEctDynamicTableEntry
53     MAX-ACCESS not-accessible
54     STATUS current
55     DESCRIPTION
56         "This table can be used to check that neighbor bridges are
57         using the same ECT Algorithm. "
58     REFERENCE "12.25.5"
59     INDEX {
60         ieee8021SpbEctDynamicEntryTopIx,
61         ieee8021SpbEctDynamicEntryBaseVid
62     }
63     ::= { ieee8021SpbEctDynamicTable 1 }
64
65 Ieee8021SpbEctDynamicTableEntry ::=
66     SEQUENCE {
67         ieee8021SpbEctDynamicEntryTopIx IEEE8021SpbMTID,
68         ieee8021SpbEctDynamicEntryBaseVid VlanId,
69         ieee8021SpbEctDynamicEntryMode IEEE8021SpbEctMode,
70         ieee8021SpbEctDynamicEntryLocalUse TruthValue,
71         ieee8021SpbEctDynamicEntryRemoteUse TruthValue,
72         ieee8021SpbEctDynamicEntryIngressCheckDiscards Unsigned32

```

```

1      }
2
3 ieee8021SpbEctDynamicEntryTopIx OBJECT-TYPE
4     SYNTAX IEEE8021SpbMTID
5     MAX-ACCESS not-accessible
6     STATUS current
7     DESCRIPTION
8         "The ISIS Topology Index identifier to which this
9         instance belongs. Each Topology Index defines logical topology
10        and is used to enable multiple SPB instances within several
11        ISIS instances."
12    REFERENCE "12.25.5.1.2, 12.25.5.1.3, 28.12"
13    ::= { ieee8021SpbEctDynamicTableEntry 1 }
14
15 ieee8021SpbEctDynamicEntryBaseVid OBJECT-TYPE
16     SYNTAX VlanId
17     MAX-ACCESS not-accessible
18     STATUS current
19     DESCRIPTION
20         "The Base VID being queried. Base VID
21         define the mode in the VID to MSTID table. "
22    REFERENCE "12.25.5.1.2, 12.25.5.1.3, 3.25"
23    ::= { ieee8021SpbEctDynamicTableEntry 2 }
24
25 ieee8021SpbEctDynamicEntryMode OBJECT-TYPE
26     SYNTAX IEEE8021SpbEctMode
27     MAX-ACCESS read-only
28     STATUS current
29     DESCRIPTION
30         "The Operating mode of this Base VID.
31         SPBM (=2), SPBV (=3), or disabled or none (1)."

```

```

1 -- ieee8021SpbAdjStaticTable:
2 -- =====
3 ieee8021SpbAdjStaticTable OBJECT-TYPE
4     SYNTAX SEQUENCE OF Ieee8021SpbAdjStaticTableEntry
5     MAX-ACCESS not-accessible
6     STATUS current
7     DESCRIPTION
8         "A table containing the SPB configuration data for a neighbor"
9     REFERENCE "12.25.6"
10    ::= { ieee8021SpbObjects 6 }
11
12 ieee8021SpbAdjStaticTableEntry OBJECT-TYPE
13     SYNTAX Ieee8021SpbAdjStaticTableEntry
14     MAX-ACCESS not-accessible
15     STATUS current
16     DESCRIPTION
17         "This table can be used to display the interfaces and metrics
18         of a neighbor bridge. "
19     REFERENCE "12.25.6"
20     INDEX {
21         ieee8021SpbAdjStaticEntryTopIx,
22         ieee8021SpbAdjStaticEntryIfIndex
23     }
24     ::= { ieee8021SpbAdjStaticTable 1 }
25
26 Ieee8021SpbAdjStaticTableEntry ::=
27     SEQUENCE {
28         ieee8021SpbAdjStaticEntryTopIx IEEE8021SpbMTID,
29         ieee8021SpbAdjStaticEntryIfIndex InterfaceIndexOrZero,
30         ieee8021SpbAdjStaticEntryMetric IEEE8021SpbLinkMetric,
31         ieee8021SpbAdjStaticEntryIfAdminState IEEE8021SpbAdjState,
32         ieee8021SpbAdjStaticEntryRowStatus RowStatus
33     }
34
35 ieee8021SpbAdjStaticEntryTopIx OBJECT-TYPE
36     SYNTAX IEEE8021SpbMTID
37     MAX-ACCESS not-accessible
38     STATUS current
39     DESCRIPTION
40         "The ISIS Topology Index identifier to which this
41         instance belongs. Each Topology Index defines logical topology
42         and is used to enable multiple SPB instances within several
43         ISIS instances."
44     REFERENCE "12.25.6.1.2, 12.25.6.1.3, 28.12"
45     ::= { ieee8021SpbAdjStaticTableEntry 1 }
46
47 ieee8021SpbAdjStaticEntryIfIndex OBJECT-TYPE
48     SYNTAX InterfaceIndexOrZero
49     MAX-ACCESS not-accessible
50     STATUS current
51     DESCRIPTION
52         "The System interface/index that defines this
53         adjacency. A value of 0 is a wildcard for any
54         interface on which SPB Operation is supported."
55     REFERENCE "12.25.6.1.2, 12.25.6.1.3"
56     ::= { ieee8021SpbAdjStaticTableEntry 2 }
57
58 ieee8021SpbAdjStaticEntryMetric OBJECT-TYPE
59     SYNTAX IEEE8021SpbLinkMetric
60     MAX-ACCESS read-create
61     STATUS current
62     DESCRIPTION
63         "The ieee8021Spb metric (incremental cost) to this peer.
64         The contribution of this link to total path cost.
65         Recommended values are inversely proportional to link speed.
66         Range is (1..16777215) where 16777215 (0xFFFFF) is
67         infinity; infinity signifies that the adjacency is
68         UP, but is not to be used for traffic.
69         This object is persistent."
70     REFERENCE "12.25.6.1.2, 12.25.6.1.3, 28.12.7"
71     ::= { ieee8021SpbAdjStaticTableEntry 3 }
72

```

```

1 ieee8021SpbAdjStaticEntryIfAdminState OBJECT-TYPE
2     SYNTAX IEEE8021SpbAdjState
3     MAX-ACCESS read-create
4     STATUS current
5     DESCRIPTION
6         "The administrative state of this interface/port.
7         Up is the default.
8         This object is persistent."
9     REFERENCE "12.25.6.1.2, 12.25.6.1.3"
10    ::= { ieee8021SpbAdjStaticTableEntry 4 }
11
12 ieee8021SpbAdjStaticEntryRowStatus OBJECT-TYPE
13     SYNTAX RowStatus
14     MAX-ACCESS read-create
15     STATUS current
16     DESCRIPTION
17         "The object indicates the status of an entry, and is used
18         to create/delete entries.
19         This object is persistent."
20     REFERENCE "12.25.6.1.3"
21    ::= { ieee8021SpbAdjStaticTableEntry 5 }
22
23 -- =====
24 -- ieee8021SpbAdjDynamicTable:
25 -- =====
26 ieee8021SpbAdjDynamicTable OBJECT-TYPE
27     SYNTAX SEQUENCE OF Ieee8021SpbAdjDynamicTableEntry
28     MAX-ACCESS not-accessible
29     STATUS current
30     DESCRIPTION
31         "The SPB neighbor dynamic information table."
32     REFERENCE "12.25.7"
33    ::= { ieee8021SpbObjects 7 }
34
35 ieee8021SpbAdjDynamicTableEntry OBJECT-TYPE
36     SYNTAX Ieee8021SpbAdjDynamicTableEntry
37     MAX-ACCESS not-accessible
38     STATUS current
39     DESCRIPTION
40         "This table is used to determine operational values of digests
41         and interfaces of neighbor bridges."
42     REFERENCE "12.25.7"
43     INDEX {
44         ieee8021SpbAdjDynamicEntryTopIx,
45         ieee8021SpbAdjDynamicEntryIfIndex,
46         ieee8021SpbAdjDynamicEntryPeerSysId
47     }
48    ::= { ieee8021SpbAdjDynamicTable 1 }
49
50 Ieee8021SpbAdjDynamicTableEntry ::=
51     SEQUENCE {
52         ieee8021SpbAdjDynamicEntryTopIx IEEE8021SpbMTID,
53         ieee8021SpbAdjDynamicEntryIfIndex InterfaceIndexOrZero,
54         ieee8021SpbAdjDynamicEntryPeerSysId MacAddress,
55         ieee8021SpbAdjDynamicEntryPort IEEE8021BridgePortNumber,
56         ieee8021SpbAdjDynamicEntryIfOperState IEEE8021SpbAdjState,
57         ieee8021SpbAdjDynamicEntryPeerSysName SnmpAdminString,
58         ieee8021SpbAdjDynamicEntryPeerAgreeDigest IEEE8021SpbDigest,
59         ieee8021SpbAdjDynamicEntryPeerMCID IEEE8021SpbMCID,
60         ieee8021SpbAdjDynamicEntryPeerAuxMCID IEEE8021SpbMCID,
61         ieee8021SpbAdjDynamicEntryLocalCircuitID Unsigned32,
62         ieee8021SpbAdjDynamicEntryPeerLocalCircuitID Unsigned32,
63         ieee8021SpbAdjDynamicEntryPortIdentifier Unsigned32,
64         ieee8021SpbAdjDynamicEntryPeerPortIdentifier Unsigned32,
65         ieee8021SpbAdjDynamicEntryIsisCircIndex Unsigned32
66     }
67
68 ieee8021SpbAdjDynamicEntryTopIx OBJECT-TYPE
69     SYNTAX IEEE8021SpbMTID
70     MAX-ACCESS not-accessible
71     STATUS current
72     DESCRIPTION

```

```

1         "The ISIS Topology Index identifier to which this
2         instance belongs. Each Topology Index defines logical topology
3         and is used to enable multiple SPB instances within several
4         ISIS instances."
5     REFERENCE "12.25.7.1.2, 12.25.7.1.3, 28.12"
6     ::= { ieee8021SpbAdjDynamicTableEntry 1 }
7
8 ieee8021SpbAdjDynamicEntryIfIndex OBJECT-TYPE
9     SYNTAX InterfaceIndexOrZero
10    MAX-ACCESS not-accessible
11    STATUS current
12    DESCRIPTION
13        "System interface/index that defines this adjacency
14        A value of 0 is a wildcard for any interface
15        on which SPB Operation is enabled."
16    REFERENCE "12.25.7.1.2, 12.25.7.1.3"
17    ::= { ieee8021SpbAdjDynamicTableEntry 2 }
18
19 ieee8021SpbAdjDynamicEntryPeerSysId OBJECT-TYPE
20     SYNTAX MacAddress
21     MAX-ACCESS not-accessible
22     STATUS current
23     DESCRIPTION
24         "The SPB System Identifier of this peer. This is used to
25         identify a neighbor uniquely."
26     REFERENCE "12.25.7.1.3, 3.253"
27     ::= { ieee8021SpbAdjDynamicTableEntry 3 }
28
29 ieee8021SpbAdjDynamicEntryPort OBJECT-TYPE
30     SYNTAX IEEE8021BridgePortNumber
31     MAX-ACCESS read-only
32     STATUS current
33     DESCRIPTION
34         "The port number to reach this adjacency."
35     REFERENCE "12.25.7.1.3"
36     ::= { ieee8021SpbAdjDynamicTableEntry 4 }
37
38 ieee8021SpbAdjDynamicEntryIfOperState OBJECT-TYPE
39     SYNTAX IEEE8021SpbAdjState
40     MAX-ACCESS read-only
41     STATUS current
42     DESCRIPTION
43         "The operational state of this port.
44         up, down, or testing (in test)."
45     REFERENCE "12.25.7.1.3"
46     ::= { ieee8021SpbAdjDynamicTableEntry 5 }
47
48 ieee8021SpbAdjDynamicEntryPeerSysName OBJECT-TYPE
49     SYNTAX SnmpAdminString (SIZE(0..32))
50     MAX-ACCESS read-only
51     STATUS current
52     DESCRIPTION
53         "IS-IS system name of peer.
54         This is the ASCII name assigned to the bridge to aid
55         management. It is the same as the ieee8021SpbSysName. "
56     REFERENCE "12.25.7.1.3"
57     ::= { ieee8021SpbAdjDynamicTableEntry 6 }
58
59 ieee8021SpbAdjDynamicEntryPeerAgreeDigest OBJECT-TYPE
60     SYNTAX IEEE8021SpbDigest
61     MAX-ACCESS read-only
62     STATUS current
63     DESCRIPTION
64         "The peer topology agreement digest value
65         (all of the elements defined in 28.4).
66         If it does not match this bridge's digest it indicates loss of
67         synchronization."
68     REFERENCE "12.25.7.1.3, 28.4"
69     ::= { ieee8021SpbAdjDynamicTableEntry 7 }
70
71 ieee8021SpbAdjDynamicEntryPeerMCID OBJECT-TYPE
72     SYNTAX IEEE8021SpbMCID

```



```
1      MAX-ACCESS read-only
2      STATUS current
3      DESCRIPTION
4          "The peer MST Identifier MCID. The MCID is a digest of the
5          VID to MSTID configuration table, which determines the Base VIDs
6          enabled for SPBV and SPBM."
7      REFERENCE "12.25.7.1.3, 13.8"
8      ::= { ieee8021SpbAdjDynamicTableEntry 8 }
9
10 ieee8021SpbAdjDynamicEntryPeerAuxMCID OBJECT-TYPE
11     SYNTAX IEEE8021SpbMCID
12     MAX-ACCESS read-only
13     STATUS current
14     DESCRIPTION
15         "The peer auxiliary MST Identifier. This MCID is
16         used for migration."
17     REFERENCE "12.25.7.1.3, 27.4.1, 28.12.2"
18     ::= { ieee8021SpbAdjDynamicTableEntry 9 }
19
20 ieee8021SpbAdjDynamicEntryLocalCircuitID OBJECT-TYPE
21     SYNTAX Unsigned32
22     MAX-ACCESS read-only
23     STATUS current
24     DESCRIPTION
25         "The value used by IS-IS to identify this adjacency locally."
26     REFERENCE "12.25.7.1.3, 28.11"
27     ::= { ieee8021SpbAdjDynamicTableEntry 10 }
28
29 ieee8021SpbAdjDynamicEntryPeerLocalCircuitID OBJECT-TYPE
30     SYNTAX Unsigned32
31     MAX-ACCESS read-only
32     STATUS current
33     DESCRIPTION
34         "The value used by peer IS-IS to identify this adjacency remotely."
35     REFERENCE "12.25.7.1.3, 28.11"
36     ::= { ieee8021SpbAdjDynamicTableEntry 11 }
37
38 ieee8021SpbAdjDynamicEntryPortIdentifier OBJECT-TYPE
39     SYNTAX Unsigned32
40     MAX-ACCESS read-only
41     STATUS current
42     DESCRIPTION
43         "The value for this bridge that has been selected by
44         IS-IS to form this adjacency if there is more than 1 candidate link."
45     REFERENCE "12.25.7.1.3, 28.11"
46     ::= { ieee8021SpbAdjDynamicTableEntry 12 }
47
48 ieee8021SpbAdjDynamicEntryPeerPortIdentifier OBJECT-TYPE
49     SYNTAX Unsigned32
50     MAX-ACCESS read-only
51     STATUS current
52     DESCRIPTION
53         "The value for peer port Identifier selected by IS-IS
54         to form this adjacency if there is more than 1 candidate link."
55     REFERENCE "12.25.7.1.3, 28.11"
56     ::= { ieee8021SpbAdjDynamicTableEntry 13 }
57
58 ieee8021SpbAdjDynamicEntryIscircIndex OBJECT-TYPE
59     SYNTAX Unsigned32
60     MAX-ACCESS read-only
61     STATUS current
62     DESCRIPTION
63         "The isiscircTable reference. This allows cross referencing
64         to an IS-IS MIB."
65     REFERENCE "12.25.7.1.3"
66     ::= { ieee8021SpbAdjDynamicTableEntry 14 }
67
68 -- =====
69 -- ieee8021SpbTopNodeTable:
70 -- =====
71 ieee8021SpbTopNodeTable OBJECT-TYPE
72     SYNTAX SEQUENCE OF Ieee8021SpbTopNodeTableEntry
```

```

1      MAX-ACCESS not-accessible
2      STATUS current
3      DESCRIPTION
4          "Table of network specific bridge information."
5      REFERENCE "12.25.8"
6      ::= { ieee8021SpbObjects 8 }
7
8  ieee8021SpbTopNodeTableEntry OBJECT-TYPE
9      SYNTAX Ieee8021SpbTopNodeTableEntry
10     MAX-ACCESS not-accessible
11     STATUS current
12     DESCRIPTION
13         "This table is used to display system level information about
14         bridges in the network. "
15     REFERENCE "12.25.8"
16     INDEX {
17         ieee8021SpbTopNodeEntryTopIx,
18         ieee8021SpbTopNodeEntrySysId
19     }
20     ::= { ieee8021SpbTopNodeTable 1 }
21
22 Ieee8021SpbTopNodeTableEntry ::=
23     SEQUENCE {
24         ieee8021SpbTopNodeEntryTopIx IEEE8021SpbMTID,
25         ieee8021SpbTopNodeEntrySysId MacAddress,
26         ieee8021SpbTopNodeEntryBridgePriority IEEE8021SpbBridgePriority,
27         ieee8021SpbmTopNodeEntrySPsourceID IEEE8021SpbmSPsourceId,
28         ieee8021SpbTopNodeEntrySysName SnmpAdminString
29     }
30
31 ieee8021SpbTopNodeEntryTopIx OBJECT-TYPE
32     SYNTAX IEEE8021SpbMTID
33     MAX-ACCESS not-accessible
34     STATUS current
35     DESCRIPTION
36         "The ISIS Topology Index identifier to which this
37         instance belongs. Each Topology Index defines logical topology
38         and is used to enable multiple SPB instances within several
39         ISIS instances."
40     REFERENCE "12.25.8.1.2, 12.25.8.1.3, 28.12"
41     ::= { ieee8021SpbTopNodeTableEntry 1 }
42
43 ieee8021SpbTopNodeEntrySysId OBJECT-TYPE
44     SYNTAX MacAddress
45     MAX-ACCESS not-accessible
46     STATUS current
47     DESCRIPTION
48         "The IS-IS System ID of a bridge in the SPB
49         LSP database and hence the network.
50         A value of 0 is a wildcard for all System identifiers."
51     REFERENCE "12.25.8.1.2, 12.25.8.1.3, 3.253"
52     ::= { ieee8021SpbTopNodeTableEntry 2 }
53
54 ieee8021SpbTopNodeEntryBridgePriority OBJECT-TYPE
55     SYNTAX IEEE8021SpbBridgePriority
56     MAX-ACCESS read-only
57     STATUS current
58     DESCRIPTION
59         "The Bridge Priority of the bridge in the LSP database.
60         This is a 16-bit quantity that ranks this SPB Bridge
61         relative to others when breaking ties. This priority
62         is the high 16 bits of the Bridge Identifier. Its impact
63         depends on the tie breaking algorithm. Recommend
64         values 0..15 be assigned to core switches to ensure
65         diversity of the ECT Algorithms."
66     REFERENCE "12.25.8.1.3, 13.26.3"
67     ::= { ieee8021SpbTopNodeTableEntry 3 }
68
69 ieee8021SpbmTopNodeEntrySPsourceID OBJECT-TYPE
70     SYNTAX IEEE8021SpbmSPsourceId
71     MAX-ACCESS read-only
72     STATUS current

```

```

1      DESCRIPTION
2          "The Shortest Path Source Identifier.
3          It is the high order 3 bytes for Group Address DA from this
4          bridge. Note that only the 20 bits not including the
5          top 4 bits are the SPSourceID."
6      REFERENCE "12.25.8.1.3, 3.262"
7      ::= { ieee8021SpbTopNodeTableEntry 4 }
8
9  ieee8021SpbTopNodeEntrySysName OBJECT-TYPE
10     SYNTAX SnmpAdminString (SIZE(0..32))
11     MAX-ACCESS read-only
12     STATUS current
13     DESCRIPTION
14         "The System Name. A Human readable name of this bridge
15         This is used to aid in management and is used in
16         place of the System identifier in many commands and displays."
17     REFERENCE "12.25.8.1.3"
18     ::= { ieee8021SpbTopNodeTableEntry 5 }
19
20 -- =====
21 -- ieee8021SpbTopEctTable:
22 -- =====
23 ieee8021SpbTopEctTable OBJECT-TYPE
24     SYNTAX SEQUENCE OF Ieee8021SpbTopEctTableEntry
25     MAX-ACCESS not-accessible
26     STATUS current
27     DESCRIPTION
28         "Table of all ECT use in the network"
29     REFERENCE "12.25.9"
30     ::= { ieee8021SpbObjects 9 }
31
32 ieee8021SpbTopEctTableEntry OBJECT-TYPE
33     SYNTAX Ieee8021SpbTopEctTableEntry
34     MAX-ACCESS not-accessible
35     STATUS current
36     DESCRIPTION
37         "This table lists bridges and the ECT Algorithms configured and in use. "
38     REFERENCE "12.25.9"
39     INDEX {
40         ieee8021SpbTopEctEntryTopIx,
41         ieee8021SpbTopEctEntrySysId,
42         ieee8021SpbTopEctEntryBaseVid
43     }
44     ::= { ieee8021SpbTopEctTable 1 }
45
46 Ieee8021SpbTopEctTableEntry ::=
47     SEQUENCE {
48         ieee8021SpbTopEctEntryTopIx IEEE8021SpbMTID,
49         ieee8021SpbTopEctEntrySysId MacAddress,
50         ieee8021SpbTopEctEntryBaseVid VlanIdOrAny,
51         ieee8021SpbTopEctEntryEctAlgorithm IEEE8021SpbEctAlgorithm,
52         ieee8021SpbTopEctEntryMode IEEE8021SpbEctMode,
53         ieee8021SpbvTopEctSysMode IEEE8021SpbMode,
54         ieee8021SpbvTopEctEntrySpvid VlanIdOrNone,
55         ieee8021SpbTopEctEntryLocalUse TruthValue
56     }
57
58 ieee8021SpbTopEctEntryTopIx OBJECT-TYPE
59     SYNTAX IEEE8021SpbMTID
60     MAX-ACCESS not-accessible
61     STATUS current
62     DESCRIPTION
63         "The ISIS Topology Index identifier to which this
64         instance belongs. Each Topology Index defines logical topology
65         and is used to enable multiple SPB instances within several
66         ISIS instances."
67     REFERENCE "12.25.9.1.2, 12.25.9.1.3"
68     ::= { ieee8021SpbTopEctTableEntry 1 }
69
70 ieee8021SpbTopEctEntrySysId OBJECT-TYPE
71     SYNTAX MacAddress
72     MAX-ACCESS not-accessible

```

```
1      STATUS current
2      DESCRIPTION
3          "The system ID that is using a particular ECT.
4          A value of 0 is a wildcard for all System identifiers."
5      REFERENCE "12.25.9.1.2, 12.25.9.1.3, 3.253"
6      ::= { ieee8021SpbTopEctTableEntry 2 }
7
8  ieee8021SpbTopEctEntryBaseVid OBJECT-TYPE
9      SYNTAX VlanIdOrAny
10     MAX-ACCESS not-accessible
11     STATUS current
12     DESCRIPTION
13         "Base VID related to this algorithm.
14         In the case of SPBM this is the B-VID that carries
15         traffic for this ECT-ALGORITHM. In the case of SPBV
16         this is the Base-VID used for management.
17         A Base VID value of 4095 is a wildcard for any Base VID
18         assigned to SPB operation."
19     REFERENCE "12.25.9.1.2, 12.25.9.1.3, 3.25"
20     ::= { ieee8021SpbTopEctTableEntry 3 }
21
22  ieee8021SpbTopEctEntryEctAlgorithm OBJECT-TYPE
23     SYNTAX IEEE8021SpbEctAlgorithm
24     MAX-ACCESS read-only
25     STATUS current
26     DESCRIPTION
27         "The ECT-ALGORITHM in use.
28         A 32 bit number. The ISIS Topology Index identifier to which this
29         instance belongs. Each Topology Index defines logical topology
30         and is used to enable multiple SPB instances within several
31         ISIS instances.; the upper 24 bits are an OUI
32         and the lower 8 bits are an index. This creates a
33         world-wide unique identity for the computation that
34         will be using the VID thus ensuring consistency."
35     REFERENCE "12.25.9.1.3, 3.167"
36     ::= { ieee8021SpbTopEctTableEntry 4 }
37
38  ieee8021SpbTopEctEntryMode OBJECT-TYPE
39     SYNTAX IEEE8021SpbEctMode
40     MAX-ACCESS read-only
41     STATUS current
42     DESCRIPTION
43         "Operating mode : SPBM (=2) or SPBV (=3)"
44     REFERENCE "12.25.9.1.3"
45     ::= { ieee8021SpbTopEctTableEntry 5 }
46
47  ieee8021SpbvTopEctSysMode OBJECT-TYPE
48     SYNTAX IEEE8021SpbMode
49     MAX-ACCESS read-only
50     STATUS current
51     DESCRIPTION
52         "Indication of supporting SPBV mode
53         auto(=1)/manual(=2)
54         auto => SPBV mode and auto allocate SPVIDs.
55         manual => SPBV mode and manually assign SPVIDs."
56     REFERENCE "12.25.9.1.3, 3.263"
57     ::= { ieee8021SpbTopEctTableEntry 6 }
58
59  ieee8021SpbvTopEctEntrySpvid OBJECT-TYPE
60     SYNTAX VlanIdOrNone
61     MAX-ACCESS read-only
62     STATUS current
63     DESCRIPTION
64         "SPVID in V mode
65         The VID this bridge will use to originate traffic
66         using this ECT-ALGORITHM when running in SPBV mode."
67     REFERENCE "12.25.9.1.3, 3.259"
68     ::= { ieee8021SpbTopEctTableEntry 7 }
69
70  ieee8021SpbTopEctEntryLocalUse OBJECT-TYPE
71     SYNTAX TruthValue
72     MAX-ACCESS read-only
```

```

1      STATUS current
2      DESCRIPTION
3          "Is this ECT-ALGORITHM in use locally by advertising
4              bridge :- TRUE or FALSE. This is used to help with
5              disruption-free migration between ECT-ALGORITHMS.
6              Changes are only allowed if this flag is FALSE."
7      REFERENCE "12.25.9.1.3, 28.12.5"
8      ::= { ieee8021SpbTopEctTableEntry 8 }
9
10 -- =====
11 -- ieee8021SpbTopEdgeTable:
12 -- =====
13 ieee8021SpbTopEdgeTable OBJECT-TYPE
14     SYNTAX SEQUENCE OF Ieee8021SpbTopEdgeTableEntry
15     MAX-ACCESS not-accessible
16     STATUS current
17     DESCRIPTION
18         "A Table of edges in network (not duplicated),
19         but each link will appear as two entries, one
20         ordered {near-far}, the other {far-near}."
21     REFERENCE "12.25.10"
22     ::= { ieee8021SpbObjects 10 }
23
24 ieee8021SpbTopEdgeTableEntry OBJECT-TYPE
25     SYNTAX Ieee8021SpbTopEdgeTableEntry
26     MAX-ACCESS not-accessible
27     STATUS current
28     DESCRIPTION
29         "The table lists information about bridge edges (links)."
30     REFERENCE "12.25.10"
31     INDEX {
32         ieee8021SpbTopEdgeEntryTopIx,
33         ieee8021SpbTopEdgeEntrySysIdNear,
34         ieee8021SpbTopEdgeEntrySysIdFar
35     }
36     ::= { ieee8021SpbTopEdgeTable 1 }
37
38 Ieee8021SpbTopEdgeTableEntry ::=
39     SEQUENCE {
40         ieee8021SpbTopEdgeEntryTopIx IEEE8021SpbMTID,
41         ieee8021SpbTopEdgeEntrySysIdNear MacAddress,
42         ieee8021SpbTopEdgeEntrySysIdFar MacAddress,
43         ieee8021SpbTopEdgeEntryMetricNear2Far IEEE8021SpbLinkMetric,
44         ieee8021SpbTopEdgeEntryMetricFar2Near IEEE8021SpbLinkMetric
45     }
46
47 ieee8021SpbTopEdgeEntryTopIx OBJECT-TYPE
48     SYNTAX IEEE8021SpbMTID
49     MAX-ACCESS not-accessible
50     STATUS current
51     DESCRIPTION
52         "The ISIS Topology Index identifier to which this
53         instance belongs. Each Topology Index defines logical topology
54         and is used to enable multiple SPB instances within several
55         ISIS instances."
56     REFERENCE "12.25.10.1.2, 12.25.10.1.3, 28.12"
57     ::= { ieee8021SpbTopEdgeTableEntry 1 }
58
59 ieee8021SpbTopEdgeEntrySysIdNear OBJECT-TYPE
60     SYNTAX MacAddress
61     MAX-ACCESS not-accessible
62     STATUS current
63     DESCRIPTION
64         "The System ID of near bridge (the bridge
65         reporting the adjacency).
66         A value of 0 is a wildcard for all System identifiers."
67     REFERENCE "12.25.10.1.2, 12.25.10.1.3, 3.253"
68     ::= { ieee8021SpbTopEdgeTableEntry 2 }
69
70 ieee8021SpbTopEdgeEntrySysIdFar OBJECT-TYPE
71     SYNTAX MacAddress
72     MAX-ACCESS not-accessible

```

```
1      STATUS current
2      DESCRIPTION
3          "The System ID of far bridge (the neighbor
4              of the bridge reporting).
5              A value of 0 is a wildcard for all System identifiers."
6      REFERENCE "12.25.10.1.2, 12.25.10.1.3, 3.253"
7      ::= { ieee8021SpbTopEdgeTableEntry 3 }
8
9  ieee8021SpbTopEdgeEntryMetricNear2Far OBJECT-TYPE
10     SYNTAX IEEE8021SpbLinkMetric
11     MAX-ACCESS read-only
12     STATUS current
13     DESCRIPTION
14         "The metric used on this edge advertised by near end
15             This is the raw value. If it is less than the
16             MetricFar2Near (below), the MetricFar2Near is
17             used as the SPF metric in both directions."
18     REFERENCE "12.25.10.1.3, 28.12.7"
19     ::= { ieee8021SpbTopEdgeTableEntry 4 }
20
21 ieee8021SpbTopEdgeEntryMetricFar2Near OBJECT-TYPE
22     SYNTAX IEEE8021SpbLinkMetric
23     MAX-ACCESS read-only
24     STATUS current
25     DESCRIPTION
26         "The metric used on this edge advertised by far end
27             This is the raw value. If it is less than the
28             MetricNear2Far (above), the MetricNear2Far is
29             used as the SPF metric in both directions."
30     REFERENCE "12.25.10.1.3, 28.12.7"
31     ::= { ieee8021SpbTopEdgeTableEntry 5 }
32
33 -- =====
34 -- ieee8021SpbmTopSrvTable:
35 -- =====
36 ieee8021SpbmTopSrvTable OBJECT-TYPE
37     SYNTAX SEQUENCE OF Ieee8021SpbmTopSrvTableEntry
38     MAX-ACCESS not-accessible
39     STATUS current
40     DESCRIPTION
41         "All SPBM PBB encapsulated services in this network."
42     REFERENCE "12.25.11"
43     ::= { ieee8021SpbObjects 11 }
44
45 ieee8021SpbmTopSrvTableEntry OBJECT-TYPE
46     SYNTAX Ieee8021SpbmTopSrvTableEntry
47     MAX-ACCESS not-accessible
48     STATUS current
49     DESCRIPTION
50         "This table displays information about PBB services received
51             in the LSP data base. The Service Identifier is associated with
52             the MAC address and Base VID of the bridge that originates or
53             terminates the service. "
54     REFERENCE "12.25.11"
55     INDEX {
56         ieee8021SpbmTopSrvEntryTopIx,
57         ieee8021SpbmTopSrvEntrySysId,
58         ieee8021SpbmTopSrvEntryIsid,
59         ieee8021SpbmTopSrvEntryBaseVid,
60         ieee8021SpbmTopSrvEntryMac
61     }
62     ::= { ieee8021SpbmTopSrvTable 1 }
63
64 Ieee8021SpbmTopSrvTableEntry ::=
65     SEQUENCE {
66         ieee8021SpbmTopSrvEntryTopIx IEEE8021SpbMTID,
67         ieee8021SpbmTopSrvEntrySysId MacAddress,
68         ieee8021SpbmTopSrvEntryIsid IEEE8021SpbServiceIdentifierOrAny,
69         ieee8021SpbmTopSrvEntryBaseVid VlanIdOrAny,
70         ieee8021SpbmTopSrvEntryMac MacAddress,
71         ieee8021SpbmTopSrvEntryIsidFlags IEEE8021PbbIngressEgress
72 }
```

```

1      }
2
3 ieee8021SpbmTopSrvEntryTopIx OBJECT-TYPE
4     SYNTAX IEEE8021SpbMTID
5     MAX-ACCESS not-accessible
6     STATUS current
7     DESCRIPTION
8         "Entry of one The ISIS Topology Index identifier to which this
9         instance belongs. Each Topology Index defines logical topology
10        and is used to enable multiple SPB instances within several
11        ISIS instances."
12    REFERENCE "12.25.11.1.2, 12.25.11.1.3, 28.12"
13    ::= { ieee8021SpbmTopSrvTableEntry 1 }
14
15 ieee8021SpbmTopSrvEntrySysId OBJECT-TYPE
16     SYNTAX MacAddress
17     MAX-ACCESS not-accessible
18     STATUS current
19     DESCRIPTION
20         "The System identifier this service originates/terminates on.
21         A value of 0 is a wildcard for all System identifiers."
22    REFERENCE "12.25.11.1.2, 12.25.11.1.3, 3.253"
23    ::= { ieee8021SpbmTopSrvTableEntry 2 }
24
25 ieee8021SpbmTopSrvEntryIsid OBJECT-TYPE
26     SYNTAX IEEE8021SpbServiceIdentifierOrAny
27     MAX-ACCESS not-accessible
28     STATUS current
29     DESCRIPTION
30         "An ISID (service) originating/terminating on this bridge.
31         A value of 0 is a wildcard for any ISID."
32    REFERENCE "12.25.11.1.2, 12.25.11.1.3, 28.12.10"
33    ::= { ieee8021SpbmTopSrvTableEntry 3 }
34
35 ieee8021SpbmTopSrvEntryBaseVid OBJECT-TYPE
36     SYNTAX VlanIdOrAny
37     MAX-ACCESS not-accessible
38     STATUS current
39     DESCRIPTION
40         "The Base VID associated with this service. The Base VID determines
41         the ECT Algorithm that is associated with this service.
42         A Base VID value of 4095 is a wildcard for any Base VID
43         assigned to SPB operation."
44    REFERENCE "12.25.11.1.2, 12.25.11.1.3, 28.12.10"
45    ::= { ieee8021SpbmTopSrvTableEntry 4 }
46
47 ieee8021SpbmTopSrvEntryMac OBJECT-TYPE
48     SYNTAX MacAddress
49     MAX-ACCESS not-accessible
50     STATUS current
51     DESCRIPTION
52         "The MAC address associated with a service.
53         An additional nodal MAC address by which an I-SID
54         can be reached can be advertised, in which case
55         traffic to this MAC follows a forwarding path identical
56         to that taken to reach the corresponding SYSID (nodal) MAC.
57         If no additional MAC is advertised, this will be the SYSID MAC.
58         A value of 0 is a wildcard for the MAC address."
59    REFERENCE "12.25.11.1.3, 28.12.10"
60    ::= { ieee8021SpbmTopSrvTableEntry 5 }
61
62 ieee8021SpbmTopSrvEntryIsidFlags OBJECT-TYPE
63     SYNTAX IEEE8021PbbIngressEgress
64     MAX-ACCESS read-only
65     STATUS current
66     DESCRIPTION
67         "A pair of flags defining the attributes of this
68         service. These specify independently whether
69         ingress frames to the SPBM region should be
70         transmitted within it, and whether frames
71         received from the SPBM region are required
72         egress it."

```

```

1      REFERENCE "12.25.11.1.2, 12.25.11.1.3, 28.12.10"
2      ::= { ieee8021SpbmTopSrvTableEntry 6 }
3
4  -- =====
5  -- ieee8021SpbvTopSrvTable:
6  -- =====
7  ieee8021SpbvTopSrvTable OBJECT-TYPE
8      SYNTAX SEQUENCE OF Ieee8021SpbvTopSrvTableEntry
9      MAX-ACCESS not-accessible
10     STATUS current
11     DESCRIPTION
12         "The SPBV group services in this network"
13     REFERENCE "12.25.12"
14     ::= { ieee8021SpbObjects 12 }
15
16  ieee8021SpbvTopSrvTableEntry OBJECT-TYPE
17      SYNTAX Ieee8021SpbvTopSrvTableEntry
18      MAX-ACCESS not-accessible
19      STATUS current
20      DESCRIPTION
21          "This table displays information about SPBV group address.
22           The group address is a associated with MAC address and Base
23           VID of the bridge that originates or terminates the service."
24      REFERENCE "12.25.12"
25      INDEX {
26          ieee8021SpbvTopSrvEntryTopIx,
27          ieee8021SpbvTopSrvEntrySysId,
28          ieee8021SpbvTopSrvEntryMMac
29      }
30      ::= { ieee8021SpbvTopSrvTable 1 }
31
32  Ieee8021SpbvTopSrvTableEntry ::=
33      SEQUENCE {
34          ieee8021SpbvTopSrvEntryTopIx IEEE8021SpbMTID,
35          ieee8021SpbvTopSrvEntrySysId MacAddress,
36          ieee8021SpbvTopSrvEntryMMac MacAddress,
37          ieee8021SpbvTopSrvEntryBaseVid VlanId,
38          ieee8021SpbvTopSrvEntryMMacFlags IEEE8021PbbIngressEgress
39      }
40
41  ieee8021SpbvTopSrvEntryTopIx OBJECT-TYPE
42      SYNTAX IEEE8021SpbMTID
43      MAX-ACCESS not-accessible
44      STATUS current
45      DESCRIPTION
46          "The ISIS Topology Index identifier to which this
47           instance belongs. Each Topology Index defines logical topology
48           and is used to enable multiple SPB instances within several
49           ISIS instances."
50      REFERENCE "12.25.12.1.2, 12.25.12.1.3, 28.12"
51      ::= { ieee8021SpbvTopSrvTableEntry 1 }
52
53  ieee8021SpbvTopSrvEntrySysId OBJECT-TYPE
54      SYNTAX MacAddress
55      MAX-ACCESS not-accessible
56      STATUS current
57      DESCRIPTION
58          "The System identifier advertising this group address.
59           A value of 0 is a wildcard for all System identifiers."
60      REFERENCE "12.25.12.1.2, 12.25.12.1.3, 3.253"
61      ::= { ieee8021SpbvTopSrvTableEntry 2 }
62
63  ieee8021SpbvTopSrvEntryMMac OBJECT-TYPE
64      SYNTAX MacAddress
65      MAX-ACCESS not-accessible
66      STATUS current
67      DESCRIPTION
68          "This Group MAC address entry.
69           A value of 0 is a wildcard for any Group MAC address. "
70      REFERENCE "12.25.12.1.2, 12.25.12.1.3, 28.12.9"
71      ::= { ieee8021SpbvTopSrvTableEntry 3 }
72

```



```
1
2 ieee8021SpbvTopSrvEntryBaseVid OBJECT-TYPE
3     SYNTAX VlanId
4     MAX-ACCESS read-only
5     STATUS current
6     DESCRIPTION
7         "The Base VID associated with this service. The Base VID determines
8         the ECT Algorithm that is associated with this service."
9     REFERENCE "12.25.12.1.3, 3.25"
10    ::= { ieee8021SpbvTopSrvTableEntry 4 }
11
12 ieee8021SpbvTopSrvEntryMMacFlags OBJECT-TYPE
13     SYNTAX IEEE8021PbbIngressEgress
14     MAX-ACCESS read-only
15     STATUS current
16     DESCRIPTION
17         "A pair of {ingress, egress} flags for this
18         Group Address, defining transmit/receive or both. This enables
19         filtering of Group addresses to interwork with MMRP."
20     REFERENCE "12.25.12.1.3, 28.12.9"
21    ::= { ieee8021SpbvTopSrvTableEntry 5 }
22
23 -- =====
24 -- ieee8021SpbmBsiStaticTable:
25 -- =====
26 ieee8021SpbmBsiStaticTable OBJECT-TYPE
27     SYNTAX SEQUENCE OF Ieee8021SpbmBsiStaticEntry
28     MAX-ACCESS not-accessible
29     STATUS current
30     DESCRIPTION
31         "Table of BSIs configured on this system and assigned to
32         an SPBM VID.
33         The table is indexed by
34         - ieee8021SpbTopIx from ieee8021SpbMtidStaticTable
35           indicating the ISIS-SPB topology instance into
36           which the BSI will be advertised,
37         - ieee8021BridgeBasePort from ieee8021PbbCbpTable
38           identifying the CPB on which the BSI is configured,
39         - an I-SID value identifying the BSI, and
40         - a VID value identifying a B-VID for which forwarding
41           state is to be installed for the BSI"
42     REFERENCE "12.25.8"
43    ::= { ieee8021SpbObjects 13 }
44
45 ieee8021SpbmBsiStaticEntry OBJECT-TYPE
46     SYNTAX Ieee8021SpbmBsiStaticEntry
47     MAX-ACCESS not-accessible
48     STATUS current
49     DESCRIPTION
50         "This table contains information about backbone services
51         configured on this system to be advertised by ISIS-SPB."
52     REFERENCE "12.25.8"
53     INDEX {
54         ieee8021SpbTopIx,
55         ieee8021BridgeBasePort,
56         ieee8021SpbmBsiStaticEntryIsid,
57         ieee8021SpbmBsiStaticEntryBaseVid
58     }
59    ::= { ieee8021SpbmBsiStaticTable 1 }
60
61 Ieee8021SpbmBsiStaticEntry ::=
62     SEQUENCE {
63         ieee8021SpbmBsiStaticEntryIsid
64             IEEE8021PbbServiceIdentifier,
65         ieee8021SpbmBsiStaticEntryBaseVid    VlanId,
66         ieee8021SpbmBsiStaticEntryTBit       TruthValue,
67         ieee8021SpbmBsiStaticEntryRBit       TruthValue,
68         ieee8021SpbmBsiStaticEntryTsBit      TruthValue,
69         ieee8021SpbmBsiStaticEntryTieBreakMask Integer32,
70         ieee8021SpbmBsiStaticEntryRowStatus  RowStatus
71     }
72
```

```

1 ieee8021SpbmBsiStaticEntryIsid OBJECT-TYPE
2     SYNTAX      IEEE8021PbbServiceIdentifier
3     MAX-ACCESS  not-accessible
4     STATUS      current
5     DESCRIPTION
6         "An I-SID registered on the CBP identified
7         by ieee8021BridgeBasePort."
8     ::= { ieee8021SpbmBsiStaticEntry 1 }
9
10 ieee8021SpbmBsiStaticEntryBaseVid OBJECT-TYPE
11     SYNTAX      VlanId
12     MAX-ACCESS  not-accessible
13     STATUS      current
14     DESCRIPTION
15         "An B-VID registered on the CBP identified
16         by ieee8021BridgeBasePort."
17     ::= { ieee8021SpbmBsiStaticEntry 2 }
18
19 ieee8021SpbmBsiStaticEntryTBit OBJECT-TYPE
20     SYNTAX      TruthValue
21     MAX-ACCESS  read-create
22     STATUS      current
23     DESCRIPTION
24         "If true(1), indicates the BSI transmits multicast
25         frames from this CBP.
26         This object is persistent."
27     ::= { ieee8021SpbmBsiStaticEntry 3 }
28
29 ieee8021SpbmBsiStaticEntryRBit OBJECT-TYPE
30     SYNTAX      TruthValue
31     MAX-ACCESS  read-create
32     STATUS      current
33     DESCRIPTION
34         "If true(1), indicates the BSI wishes to receive
35         multicast frames at this CBP.
36         This object is persistent."
37     ::= { ieee8021SpbmBsiStaticEntry 4 }
38
39 ieee8021SpbmBsiStaticEntryTsBit OBJECT-TYPE
40     SYNTAX      TruthValue
41     MAX-ACCESS  read-create
42     STATUS      current
43     DESCRIPTION
44         "If true(1), indicates the BSI transmits multicast
45         frames on a shared tree from this CBP.
46         This object is persistent."
47     ::= { ieee8021SpbmBsiStaticEntry 5 }
48
49 ieee8021SpbmBsiStaticEntryTieBreakMask OBJECT-TYPE
50     SYNTAX      Integer32 (0..15)
51     MAX-ACCESS  read-create
52     STATUS      current
53     DESCRIPTION
54         "The value used to create the Tie-Break Mask
55         for calculating multicast trees.
56         This object is persistent."
57     ::= { ieee8021SpbmBsiStaticEntry 6 }
58
59 ieee8021SpbmBsiStaticEntryRowStatus OBJECT-TYPE
60     SYNTAX      RowStatus
61     MAX-ACCESS  read-create
62     STATUS      current
63     DESCRIPTION
64         "This column holds the status for this row.
65         When the status is active, no columns of this table can be
66         modified.
67         This object is persistent."
68     ::= { ieee8021SpbmBsiStaticEntry 7 }
69
70 -- =====
71 -- SPBM MEP configurable objects
72 -- =====

```

```

1 dotlagCfmMepSpbmTable OBJECT-TYPE
2   SYNTAX SEQUENCE OF DotlagCfmMepSpbmEntry
3   MAX-ACCESS not-accessible
4   STATUS current
5   DESCRIPTION
6       "The additional objects configurable in SPBM MEPs"
7   REFERENCE "27.18"
8   ::= { ieee8021SpbObjects 14 }
9
10 dotlagCfmMepSpbmEntry OBJECT-TYPE
11   SYNTAX DotlagCfmMepSpbmEntry
12   MAX-ACCESS not-accessible
13   STATUS current
14   DESCRIPTION
15       "The SPBM MEP table additions."
16
17   AUGMENTS { dotlagCfmMepEntry }
18   ::= { dotlagCfmMepSpbmTable 1 }
19
20 DotlagCfmMepSpbmEntry ::=
21   SEQUENCE {
22       dotlagCfmMepTransmitLbmSpbmDA  MacAddress,
23       dotlagCfmMepTransmitLtmSpbmDA  MacAddress
24   }
25
26 dotlagCfmMepTransmitLbmSpbmDA OBJECT-TYPE
27   SYNTAX MacAddress
28   MAX-ACCESS read-create
29   STATUS current
30   DESCRIPTION
31       "The MAC Address to be used as the LBM destination address
32       in an SPBM MA: A unicast or multicast address."
33   REFERENCE
34       "12.14.7.3.2:g"
35   ::= { dotlagCfmMepSpbmEntry 1 }
36
37 dotlagCfmMepTransmitLtmSpbmDA OBJECT-TYPE
38   SYNTAX MacAddress
39   MAX-ACCESS read-create
40   STATUS current
41   DESCRIPTION
42       "The MAC Address to be used as the LTM destination address
43       in an SPBM MA: A unicast or multicast address."
44   REFERENCE
45       "12.14.7.4.2:f"
46   ::= { dotlagCfmMepSpbmEntry 2 }
47
48 -- =====
49 -- SPBM path MA and ECMP path MA TE-SIDs
50 -- =====
51 dotlagCfmMepSpbmEspTable OBJECT-TYPE
52   SYNTAX SEQUENCE OF DotlagCfmMepSpbmEspEntry
53   MAX-ACCESS not-accessible
54   STATUS current
55   DESCRIPTION
56       "The SPBM ESP table contains path-tesid information for each
57       SPBM path MA known to a system."
58
59       This table uses three indices. The first two indices are the
60       indices of the Maintenance Domain and MA tables, the reason
61       being that a path-tesid is always related to an MA and
62       Maintenance Domain."
63   REFERENCE
64       "12.14.5.3.2:c, 27.18.1"
65   ::= { ieee8021SpbObjects 15 }
66
67 dotlagCfmMepSpbmEspEntry OBJECT-TYPE
68   SYNTAX DotlagCfmMepSpbmEspEntry
69   MAX-ACCESS not-accessible
70   STATUS current
71   DESCRIPTION
72       "The SPBM path MA ESP entry. Each entry refers to an

```

```

1      ESP by identifier and contains information about
2      one of the ESPs that comprise an SPBM path MA.
3      The "
4      INDEX { dotlagCfmMdIndex,
5               dotlagCfmMaIndex,
6               dotlagCfmMepSpbmEspIndex
7             }
8      ::= { dotlagCfmMepSpbmEspTable 1 }
9
10     dotlagCfmMepSpbmEspEntry ::=
11         SEQUENCE {
12             dotlagCfmMepSpbmEspIndex      Unsigned32,
13             dotlagCfmMepSpbmEspEsp        IEEE8021PbbTEsp,
14             dotlagCfmMepSpbmEspRowStatus  RowStatus
15         }
16
17     dotlagCfmMepSpbmEspIndex OBJECT-TYPE
18         SYNTAX      Unsigned32 (1..4294967295)
19         MAX-ACCESS  not-accessible
20         STATUS      current
21         DESCRIPTION
22             "This is an identifier, of local significance to a particular
23             SPBM path MA that is used to index the ESPs associated
24             with that MA."
25         ::= { dotlagCfmMepSpbmEspEntry 1 }
26
27     dotlagCfmMepSpbmEspEsp OBJECT-TYPE
28         SYNTAX      IEEE8021PbbTEsp
29         MAX-ACCESS  read-create
30         STATUS      current
31         DESCRIPTION
32             "This column holds the ESP identifier for one of the Ethernet
33             Switched Paths that define the SPBM path MA.
34             This object is persistent."
35         REFERENCE
36             "12.14.5.3.2:c, 27.18.1"
37         ::= { dotlagCfmMepSpbmEspEntry 2 }
38
39     dotlagCfmMepSpbmEspRowStatus OBJECT-TYPE
40         SYNTAX      RowStatus
41         MAX-ACCESS  read-create
42         STATUS      current
43         DESCRIPTION
44             "This column holds the status for this row.
45             When the status is active, no columns of this table can be
46             modified.
47             This object is persistent."
48         ::= { dotlagCfmMepSpbmEspEntry 3 }
49
50 -- =====
51 -- PCR objects:
52 -- =====
53
54 -- =====
55 -- ieee8021PcrEctStaticTable:
56 -- =====
57     ieee8021PcrEctStaticTable OBJECT-TYPE
58         SYNTAX SEQUENCE OF Ieee8021PcrEctStaticTableEntry
59         MAX-ACCESS not-accessible
60         STATUS current
61         DESCRIPTION
62             "The Path Control and Reservation (PCR)
63             static configuration table."
64         REFERENCE "12.28.1"
65         ::= { ieee8021PcrObjects 1 }
66
67     ieee8021PcrEctStaticTableEntry OBJECT-TYPE
68         SYNTAX Ieee8021PcrEctStaticTableEntry
69         MAX-ACCESS not-accessible
70         STATUS current
71         DESCRIPTION
72             "The PCR static configuration Table defines the

```

```
1           MRT VIDs for the Base VID if MRT is used."
2 REFERENCE "12.28.1"
3 INDEX {
4     ieee8021PcrEctStaticEntryTopIx,
5     ieee8021PcrEctStaticEntryBaseVid
6 }
7 ::= { ieee8021PcrEctStaticTable 1 }
8
9 Ieee8021PcrEctStaticTableEntry ::=
10 SEQUENCE {
11     ieee8021PcrEctStaticEntryTopIx IEEE8021SpbMTID,
12     ieee8021PcrEctStaticEntryBaseVid VlanIdOrAny,
13     ieee8021PcrEctStaticEntryMrtBlueVid VlanIdOrNone,
14     ieee8021PcrEctStaticEntryMrtRedVid VlanIdOrNone,
15     ieee8021PcrEctStaticEntryRowStatus RowStatus
16 }
17
18 ieee8021PcrEctStaticEntryTopIx OBJECT-TYPE
19 SYNTAX IEEE8021SpbMTID
20 MAX-ACCESS not-accessible
21 STATUS current
22 DESCRIPTION
23     "The IS-IS Topology Index identifier to which this
24     instance belongs."
25 REFERENCE "12.28.1.1.2"
26 ::= { ieee8021PcrEctStaticTableEntry 1 }
27
28 ieee8021PcrEctStaticEntryBaseVid OBJECT-TYPE
29 SYNTAX VlanIdOrAny
30 MAX-ACCESS not-accessible
31 STATUS current
32 DESCRIPTION
33     "Base VID to use for the MRT ECT-Algorithm or for the
34     MRTG ECT Algorithm.
35     In the case of a non-learning VLAN, this is the VID
36     that carries traffic. In the case of a learning VLAN,
37     this is the Base-VID used for management.
38     A Base VID value of 4095 is a wildcard for any Base VID
39     assigned to MRT operation."
40 REFERENCE "12.28.1.1.2"
41 ::= { ieee8021PcrEctStaticTableEntry 2 }
42
43 ieee8021PcrEctStaticEntryMrtBlueVid OBJECT-TYPE
44 SYNTAX VlanIdOrNone
45 MAX-ACCESS read-write
46 STATUS current
47 DESCRIPTION
48     "MRT-Blue VID.
49     The VID this bridge will use to originate traffic
50     on MRT-Blue for the VLAN if the VLAN is associated
51     with MRT operation.
52     This object is persistent."
53 REFERENCE "12.28.1.1.2, 45.3.3, 45.3.4"
54 ::= { ieee8021PcrEctStaticTableEntry 3 }
55
56 ieee8021PcrEctStaticEntryMrtRedVid OBJECT-TYPE
57 SYNTAX VlanIdOrNone
58 MAX-ACCESS read-write
59 STATUS current
60 DESCRIPTION
61     "MRT-Red VID.
62     The VID this bridge will use to originate traffic
63     on MRT-Red for the VLAN if the VLAN is associated
64     with MRT operation.
65     This object is persistent."
66 REFERENCE "12.28.1.1.2, 45.3.3, 45.3.4"
67 ::= { ieee8021PcrEctStaticTableEntry 4 }
68
69 ieee8021PcrEctStaticEntryRowStatus OBJECT-TYPE
70 SYNTAX RowStatus
71 MAX-ACCESS read-create
72 STATUS current
```

```

1      DESCRIPTION
2          "The object indicates the status of an entry and is used
3          to create/delete entries.
4          This object is persistent."
5      REFERENCE "12.28.1.2.3"
6      ::= { ieee8021PcrEctStaticTableEntry 5 }
7
8  -- =====
9  -- ieee8021PcrTopEctTable:
10 -- =====
11 ieee8021PcrTopEctTable OBJECT-TYPE
12     SYNTAX SEQUENCE OF Ieee8021PcrTopEctTableEntry
13     MAX-ACCESS not-accessible
14     STATUS current
15     DESCRIPTION
16         "Table of MRT use in the network."
17     REFERENCE "12.28.2"
18     ::= { ieee8021PcrObjects 2 }
19
20 ieee8021PcrTopEctTableEntry OBJECT-TYPE
21     SYNTAX Ieee8021PcrTopEctTableEntry
22     MAX-ACCESS not-accessible
23     STATUS current
24     DESCRIPTION
25         "This table lists bridges configured to use MRT."
26     REFERENCE "12.25.9"
27     INDEX {
28         ieee8021PcrTopEctEntryTopIx,
29         ieee8021PcrTopEctEntrySysId,
30         ieee8021PcrTopEctEntryBaseVid
31     }
32     ::= { ieee8021PcrTopEctTable 1 }
33
34 Ieee8021PcrTopEctTableEntry ::=
35     SEQUENCE {
36         ieee8021PcrTopEctEntryTopIx IEEE8021SpbMTID,
37         ieee8021PcrTopEctEntrySysId MacAddress,
38         ieee8021PcrTopEctEntryBaseVid VlanIdOrAny,
39         ieee8021PcrTopEctEntryMode IEEE8021SpbEctMode,
40         ieee8021PcrTopEctEntryMrtBlueVid VlanIdOrNone,
41         ieee8021PcrTopEctEntryMrtRedVid VlanIdOrNone
42     }
43
44 ieee8021PcrTopEctEntryTopIx OBJECT-TYPE
45     SYNTAX IEEE8021SpbMTID
46     MAX-ACCESS not-accessible
47     STATUS current
48     DESCRIPTION
49         "The IS-IS Topology Index identifier to which this
50         instance belongs."
51     REFERENCE "12.28.2.1.2, 12.28.2.1.3"
52     ::= { ieee8021PcrTopEctTableEntry 1 }
53
54 ieee8021PcrTopEctEntrySysId OBJECT-TYPE
55     SYNTAX MacAddress
56     MAX-ACCESS not-accessible
57     STATUS current
58     DESCRIPTION
59         "The System ID that is using MRT.
60         A value of 0 is a wildcard for all System identifiers."
61     REFERENCE "12.28.2.1.2, 12.28.2.1.3"
62     ::= { ieee8021PcrTopEctTableEntry 2 }
63
64 ieee8021PcrTopEctEntryBaseVid OBJECT-TYPE
65     SYNTAX VlanIdOrAny
66     MAX-ACCESS not-accessible
67     STATUS current
68     DESCRIPTION
69         "Base VID related to this algorithm.
70         In the case of a non-learning VLAN, this is the VID
71         that carries traffic. In the case of a learning VLAN,
72         this is the Base-VID used for management."

```

```
1           A Base VID value of 4095 is a wildcard for any Base VID
2           assigned to MRT operation."
3   REFERENCE "12.28.2.1.2, 12.28.2.1.3"
4   ::= { ieee8021PcrTopEctTableEntry 3 }
5
6   ieee8021PcrTopEctEntryMode OBJECT-TYPE
7       SYNTAX IEEE8021SpbEctMode
8       MAX-ACCESS read-only
9       STATUS current
10      DESCRIPTION
11          "Operating mode : non-learning (=2) or learning (=3)"
12      REFERENCE "12.28.2.1.3"
13      ::= { ieee8021PcrTopEctTableEntry 4 }
14
15  ieee8021PcrTopEctEntryMrtBlueVid OBJECT-TYPE
16      SYNTAX VlanIdOrNone
17      MAX-ACCESS read-only
18      STATUS current
19      DESCRIPTION
20          "MRT-Blue VID.
21           The VID this bridge will use to originate traffic
22           on MRT-Blue for the VLAN if the VLAN is associated
23           with MRT operation."
24      REFERENCE "12.28.2.1.3, 45.3.3, 45.3.4"
25      ::= { ieee8021PcrTopEctTableEntry 5 }
26
27  ieee8021PcrTopEctEntryMrtRedVid OBJECT-TYPE
28      SYNTAX VlanIdOrNone
29      MAX-ACCESS read-only
30      STATUS current
31      DESCRIPTION
32          "MRT-Red VID.
33           The VID this bridge will use to originate traffic
34           on MRT-Red for the VLAN if the VLAN is associated
35           with MRT operation."
36      REFERENCE "12.28.2.1.3, 45.3.3, 45.3.4"
37      ::= { ieee8021PcrTopEctTableEntry 6 }
38
39  -- =====
40  -- Conformance Information
41  -- =====
42
43  ieee8021SpbConformance OBJECT IDENTIFIER ::= { ieee8021SpbMib 2}
44  ieee8021SpbGroups      OBJECT IDENTIFIER ::= { ieee8021SpbConformance 1}
45  ieee8021SpbCompliances OBJECT IDENTIFIER ::= { ieee8021SpbConformance 2}
46
47  ieee8021PcrConformance OBJECT IDENTIFIER ::= { ieee8021SpbMib 4 }
48  ieee8021PcrGroups      OBJECT IDENTIFIER ::= { ieee8021PcrConformance 1}
49  ieee8021PcrCompliances OBJECT IDENTIFIER ::= { ieee8021PcrConformance 2}
50
51  -- =====
52  -- SPBM Units of conformance
53  -- =====
54
55  ieee8021SpbSysGroupSPBM OBJECT-GROUP
56      OBJECTS {
57          ieee8021SpbSysAreaAddress,
58          ieee8021SpbSysId,
59          ieee8021SpbSysControlAddr,
60          ieee8021SpbSysName,
61          ieee8021SpbSysBridgePriority,
62          ieee8021SpbmSysSPSourceId,
63          ieee8021SpbmSysMode,
64          ieee8021SpbSysDigestConvention
65      }
66
67      STATUS current
68      DESCRIPTION
69          "The collection of objects used to represent ieee8021SpbSys"
70      ::= { ieee8021SpbGroups 1 }
71
72  ieee8021SpbMtidStaticTableGroupSPBM OBJECT-GROUP
```

```
1      OBJECTS {
2          ieee8021SpbMTidStaticEntryMtidOverload,
3          ieee8021SpbMTidStaticEntryRowStatus
4      }
5
6      STATUS current
7      DESCRIPTION
8      "The collection of objects used to represent ieee8021SpbMtidStaticTable"
9      ::= { ieee8021SpbGroups 2 }
10
11 ieee8021SpbTopIxDynamicTableGroupSPBM OBJECT-GROUP
12     OBJECTS {
13         ieee8021SpbTopIxDynamicEntryAgreeDigest,
14         ieee8021SpbTopIxDynamicEntryMCID,
15         ieee8021SpbTopIxDynamicEntryAuxMCID
16     }
17
18     STATUS current
19     DESCRIPTION
20     "The collection of objects used to represent ieee8021SpbTopIxDynamicTable"
21     ::= { ieee8021SpbGroups 3 }
22
23 ieee8021SpbEctStaticTableGroupSPBM OBJECT-GROUP
24     OBJECTS {
25         ieee8021SpbEctStaticEntryEctAlgorithm,
26         ieee8021SpbEctStaticEntryRowStatus
27     }
28
29     STATUS current
30     DESCRIPTION
31     "The collection of objects used to represent ieee8021SpbEctStaticTable"
32     ::= { ieee8021SpbGroups 4 }
33
34 ieee8021SpbEctDynamicTableGroupSPBM OBJECT-GROUP
35     OBJECTS {
36         ieee8021SpbEctDynamicEntryMode,
37         ieee8021SpbEctDynamicEntryLocalUse,
38         ieee8021SpbEctDynamicEntryRemoteUse,
39         ieee8021SpbEctDynamicEntryIngressCheckDiscards
40     }
41
42     STATUS current
43     DESCRIPTION
44     "The collection of objects used to represent ieee8021SpbEctDynamicTable"
45     ::= { ieee8021SpbGroups 5 }
46
47 ieee8021SpbAdjStaticTableGroupSPBM OBJECT-GROUP
48     OBJECTS {
49         ieee8021SpbAdjStaticEntryMetric,
50         ieee8021SpbAdjStaticEntryIfAdminState,
51         ieee8021SpbAdjStaticEntryRowStatus
52     }
53
54     STATUS current
55     DESCRIPTION
56     "The collection of objects used to represent ieee8021SpbAdjStaticTable"
57     ::= { ieee8021SpbGroups 6 }
58
59 ieee8021SpbAdjDynamicTableGroupSPBM OBJECT-GROUP
60     OBJECTS {
61         ieee8021SpbAdjDynamicEntryPort,
62         ieee8021SpbAdjDynamicEntryIfOperState,
63         ieee8021SpbAdjDynamicEntryPeerSysName,
64         ieee8021SpbAdjDynamicEntryPeerAgreeDigest,
65         ieee8021SpbAdjDynamicEntryPeerMCID,
66         ieee8021SpbAdjDynamicEntryPeerAuxMCID,
67         ieee8021SpbAdjDynamicEntryLocalCircuitID,
68         ieee8021SpbAdjDynamicEntryPeerLocalCircuitID,
69         ieee8021SpbAdjDynamicEntryPortIdentifier,
70         ieee8021SpbAdjDynamicEntryPeerPortIdentifier,
71         ieee8021SpbAdjDynamicEntryIisisCircIndex
72     }
```



```

1
2     STATUS current
3     DESCRIPTION
4     "The collection of objects used to represent ieee8021SpbAdjDynamicTable"
5     ::= { ieee8021SpbGroups 7 }
6
7 ieee8021SpbTopNodeTableGroupSPBM OBJECT-GROUP
8     OBJECTS {
9         ieee8021SpbTopNodeEntryBridgePriority,
10        ieee8021SpbmTopNodeEntrySPsourceID,
11        ieee8021SpbTopNodeEntrySysName
12    }
13
14     STATUS current
15     DESCRIPTION
16     "The collection of objects used to represent ieee8021SpbTopNodeTable"
17     ::= { ieee8021SpbGroups 8 }
18
19 ieee8021SpbTopEctTableGroupSPBM OBJECT-GROUP
20     OBJECTS {
21        ieee8021SpbTopEctEntryEctAlgorithm,
22        ieee8021SpbTopEctEntryMode,
23        ieee8021SpbTopEctEntryLocalUse
24    }
25
26     STATUS current
27     DESCRIPTION
28     "The collection of objects used to represent ieee8021SpbTopEctTable"
29     ::= { ieee8021SpbGroups 9 }
30
31 ieee8021SpbTopEdgeTableGroupSPBM OBJECT-GROUP
32     OBJECTS {
33        ieee8021SpbTopEdgeEntryMetricNear2Far,
34        ieee8021SpbTopEdgeEntryMetricFar2Near
35    }
36
37     STATUS current
38     DESCRIPTION
39     "The collection of objects used to represent ieee8021SpbTopEdgeTable"
40     ::= { ieee8021SpbGroups 10 }
41
42 ieee8021SpbmTopSrvTableGroupSPBM OBJECT-GROUP
43     OBJECTS {
44        ieee8021SpbmTopSrvEntryIsidFlags
45    }
46
47     STATUS current
48     DESCRIPTION
49     "The collection of objects used to represent ieee8021SpbmTopSrvTable"
50     ::= { ieee8021SpbGroups 11 }
51
52 -- See below for additional SPBM Units of conformance (after SPBV section)
53
54 -- =====
55 -- SPBV Units of conformance
56 -- =====
57
58 ieee8021SpbSysGroupSPBV OBJECT-GROUP
59     OBJECTS {
60        ieee8021SpbSysAreaAddress,
61        ieee8021SpbSysId,
62        ieee8021SpbSysControlAddr,
63        ieee8021SpbSysName,
64        ieee8021SpbSysBridgePriority,
65        ieee8021SpbvSysMode,
66        ieee8021SpbSysDigestConvention
67    }
68
69     STATUS current
70     DESCRIPTION
71     "The collection of objects used to represent ieee8021SpbSys"
72     ::= { ieee8021SpbGroups 12 }

```

```
1
2 ieee8021SpbMtidStaticTableGroupSPBV OBJECT-GROUP
3     OBJECTS {
4         ieee8021SpbMtidStaticEntryMtidOverload,
5         ieee8021SpbMtidStaticEntryRowStatus
6     }
7
8     STATUS current
9     DESCRIPTION
10    "The collection of objects used to represent ieee8021SpbMtidStaticTable"
11    ::= { ieee8021SpbGroups 13 }
12
13 ieee8021SpbTopIxDynamicTableGroupSPBV OBJECT-GROUP
14     OBJECTS {
15         ieee8021SpbTopIxDynamicEntryAgreeDigest,
16         ieee8021SpbTopIxDynamicEntryMCID,
17         ieee8021SpbTopIxDynamicEntryAuxMCID
18     }
19
20     STATUS current
21     DESCRIPTION
22    "The collection of objects used to represent ieee8021SpbTopIxDynamicTable"
23    ::= { ieee8021SpbGroups 14 }
24
25 ieee8021SpbEctStaticTableGroupSPBV OBJECT-GROUP
26     OBJECTS {
27         ieee8021SpbEctStaticEntryEctAlgorithm,
28         ieee8021SpbvEctStaticEntrySpvid,
29         ieee8021SpbEctStaticEntryRowStatus
30     }
31
32     STATUS current
33     DESCRIPTION
34    "The collection of objects used to represent ieee8021SpbEctStaticTable"
35    ::= { ieee8021SpbGroups 15 }
36
37 ieee8021SpbEctDynamicTableGroupSPBV OBJECT-GROUP
38     OBJECTS {
39         ieee8021SpbEctDynamicEntryMode,
40         ieee8021SpbEctDynamicEntryLocalUse,
41         ieee8021SpbEctDynamicEntryRemoteUse,
42         ieee8021SpbEctDynamicEntryIngressCheckDiscards
43     }
44
45     STATUS current
46     DESCRIPTION
47    "The collection of objects used to represent ieee8021SpbEctDynamicTable"
48    ::= { ieee8021SpbGroups 16 }
49
50 ieee8021SpbAdjStaticTableGroupSPBV OBJECT-GROUP
51     OBJECTS {
52         ieee8021SpbAdjStaticEntryMetric,
53         ieee8021SpbAdjStaticEntryIfAdminState,
54         ieee8021SpbAdjStaticEntryRowStatus
55     }
56
57     STATUS current
58     DESCRIPTION
59    "The collection of objects used to represent ieee8021SpbAdjStaticTable"
60    ::= { ieee8021SpbGroups 17 }
61
62 ieee8021SpbAdjDynamicTableGroupSPBV OBJECT-GROUP
63     OBJECTS {
64         ieee8021SpbAdjDynamicEntryPort,
65         ieee8021SpbAdjDynamicEntryIfOperState,
66         ieee8021SpbAdjDynamicEntryPeerSysName,
67         ieee8021SpbAdjDynamicEntryPeerAgreeDigest,
68         ieee8021SpbAdjDynamicEntryPeerMCID,
69         ieee8021SpbAdjDynamicEntryPeerAuxMCID,
70         ieee8021SpbAdjDynamicEntryLocalCircuitID,
71         ieee8021SpbAdjDynamicEntryPeerLocalCircuitID,
72         ieee8021SpbAdjDynamicEntryPortIdentifier,
```

```

1         ieee8021SpbAdjDynamicEntryPeerPortIdentifier,
2         ieee8021SpbAdjDynamicEntryIsisCircIndex
3     }
4
5     STATUS current
6     DESCRIPTION
7     "The collection of objects used to represent ieee8021SpbAdjDynamicTable"
8     ::= { ieee8021SpbGroups 18 }
9
10 ieee8021SpbTopNodeTableGroupSPBV OBJECT-GROUP
11     OBJECTS {
12         ieee8021SpbTopNodeEntryBridgePriority,
13         ieee8021SpbTopNodeEntrySysName
14     }
15
16     STATUS current
17     DESCRIPTION
18     "The collection of objects used to represent ieee8021SpbTopNodeTable"
19     ::= { ieee8021SpbGroups 19 }
20
21 ieee8021SpbTopEctTableGroupSPBV OBJECT-GROUP
22     OBJECTS {
23         ieee8021SpbTopEctEntryEctAlgorithm,
24         ieee8021SpbTopEctEntryMode,
25         ieee8021SpbvTopEctSysMode,
26         ieee8021SpbvTopEctEntrySpvid,
27         ieee8021SpbTopEctEntryLocalUse
28     }
29
30     STATUS current
31     DESCRIPTION
32     "The collection of objects used to represent ieee8021SpbTopEctTable"
33     ::= { ieee8021SpbGroups 20 }
34
35 ieee8021SpbTopEdgeTableGroupSPBV OBJECT-GROUP
36     OBJECTS {
37         ieee8021SpbTopEdgeEntryMetricNear2Far,
38         ieee8021SpbTopEdgeEntryMetricFar2Near
39     }
40
41     STATUS current
42     DESCRIPTION
43     "The collection of objects used to represent ieee8021SpbTopEdgeTable"
44     ::= { ieee8021SpbGroups 21 }
45
46 ieee8021SpbvTopSrvTableGroupSPBV OBJECT-GROUP
47     OBJECTS {
48         ieee8021SpbvTopSrvEntryBaseVid,
49         ieee8021SpbvTopSrvEntryMMacFlags
50     }
51
52     STATUS current
53     DESCRIPTION
54     "The collection of objects used to represent ieee8021SpbvTopSrvTable"
55     ::= { ieee8021SpbGroups 22 }
56
57 -- =====
58 -- Additional SPBM Units of conformance
59 -- =====
60
61 ieee8021SpbmBsiStaticTableGroupSPBM OBJECT-GROUP
62     OBJECTS {
63         ieee8021SpbmBsiStaticEntryTBit,
64         ieee8021SpbmBsiStaticEntryRBit,
65         ieee8021SpbmBsiStaticEntryTsBit,
66         ieee8021SpbmBsiStaticEntryTieBreakMask,
67         ieee8021SpbmBsiStaticEntryRowStatus
68     }
69
70     STATUS current
71     DESCRIPTION
72     "The collection of objects used to represent ieee8021SpbmBsiStaticTable"
73     ::= { ieee8021SpbGroups 23 }

```

```

1
2 dotlagCfmMepSpbmTableGroupSPBM OBJECT-GROUP
3     OBJECTS {
4         dotlagCfmMepTransmitLbmSpbmDA,
5         dotlagCfmMepTransmitLtmSpbmDA
6     }
7     STATUS current
8     DESCRIPTION
9     "The collection of objects used to represent dotlagCfmMepSpbmTable"
10    ::= { ieee8021SpbGroups 24 }
11
12 dotlagCfmMepSpbmEspTableGroupSPBM OBJECT-GROUP
13     OBJECTS {
14         dotlagCfmMepSpbmEspEsp,
15         dotlagCfmMepSpbmEspRowStatus
16     }
17     STATUS current
18     DESCRIPTION
19     "The collection of objects used to represent dotlagCfmMepSpbmEspTable"
20    ::= { ieee8021SpbGroups 25 }
21
22 -- =====
23 -- PCR Units of conformance
24 -- =====
25
26 ieee8021PcrSysGroup OBJECT-GROUP
27     OBJECTS {
28         ieee8021SpbSysAreaAddress,
29         ieee8021SpbSysId,
30         ieee8021SpbSysControlAddr,
31         ieee8021SpbSysName,
32         ieee8021SpbSysBridgePriority,
33         ieee8021SpbmSysSPSourceId,
34         ieee8021SpbmSysMode,
35         ieee8021SpbvSysMode,
36         ieee8021SpbSysDigestConvention
37     }
38     STATUS current
39     DESCRIPTION
40     "The collection of objects used to represent ieee8021SpbSys for PCR."
41    ::= { ieee8021PcrGroups 1 }
42
43 ieee8021PcrMtidStaticTableGroup OBJECT-GROUP
44     OBJECTS {
45         ieee8021SpbMTidStaticEntryMtidOverload,
46         ieee8021SpbMtidStaticEntryRowStatus
47     }
48     STATUS current
49     DESCRIPTION
50     "The collection of objects used to represent ieee8021SpbMtidStaticTable for PCR."
51    ::= { ieee8021PcrGroups 2 }
52
53 ieee8021PcrTopIxDynamicTableGroup OBJECT-GROUP
54     OBJECTS {
55         ieee8021SpbTopIxDynamicEntryAgreeDigest,
56         ieee8021SpbTopIxDynamicEntryMCID,
57         ieee8021SpbTopIxDynamicEntryAuxMCID
58     }
59     STATUS current
60     DESCRIPTION
61     "The collection of objects used to represent ieee8021SpbTopIxDynamicTable for PCR."
62    ::= { ieee8021PcrGroups 3 }
63
64 ieee8021PcrEctStaticTableGroupMAC OBJECT-GROUP
65     OBJECTS {
66         ieee8021SpbEctStaticEntryEctAlgorithm,
67         ieee8021SpbEctStaticEntryRowStatus
68     }
69     STATUS current
70     DESCRIPTION
71     "The collection of objects used to represent ieee8021SpbEctStaticTable for PCR,
72     for non-learning VLAN, i.e., MAC-based."

```

```
1      ::= { ieee8021PcrGroups 4 }
2
3 ieee8021PcrEctStaticTableGroupVID OBJECT-GROUP
4     OBJECTS {
5         ieee8021SpbEctStaticEntryEctAlgorithm,
6         ieee8021SpbvEctStaticEntrySpvid,
7         ieee8021SpbEctStaticEntryRowStatus
8     }
9     STATUS current
10    DESCRIPTION
11    "The collection of objects used to represent ieee8021SpbEctStaticTable for PCR,
12    for learning VLAN, i.e., VID-based."
13    ::= { ieee8021PcrGroups 5 }
14
15 ieee8021PcrEctStaticTableGroupMrt OBJECT-GROUP
16     OBJECTS {
17         ieee8021PcrEctStaticEntryMrtBlueVid,
18         ieee8021PcrEctStaticEntryMrtRedVid,
19         ieee8021PcrEctStaticEntryRowStatus
20     }
21     STATUS current
22     DESCRIPTION
23     "The collection of objects used to represent ieee8021PcrEctStaticTable,
24     for MRT operation."
25     ::= { ieee8021PcrGroups 6 }
26
27 ieee8021PcrEctDynamicTableGroup OBJECT-GROUP
28     OBJECTS {
29         ieee8021SpbEctDynamicEntryMode,
30         ieee8021SpbEctDynamicEntryLocalUse,
31         ieee8021SpbEctDynamicEntryRemoteUse,
32         ieee8021SpbEctDynamicEntryIngressCheckDiscards
33     }
34     STATUS current
35     DESCRIPTION
36     "The collection of objects used to represent ieee8021SpbEctDynamicTable for PCR."
37     ::= { ieee8021PcrGroups 7 }
38
39 ieee8021PcrAdjStaticTableGroup OBJECT-GROUP
40     OBJECTS {
41         ieee8021SpbAdjStaticEntryMetric,
42         ieee8021SpbAdjStaticEntryIfAdminState,
43         ieee8021SpbAdjStaticEntryRowStatus
44     }
45     STATUS current
46     DESCRIPTION
47     "The collection of objects used to represent ieee8021SpbAdjStaticTable for PCR."
48     ::= { ieee8021PcrGroups 8 }
49
50 ieee8021PcrAdjDynamicTableGroup OBJECT-GROUP
51     OBJECTS {
52         ieee8021SpbAdjDynamicEntryPort,
53         ieee8021SpbAdjDynamicEntryIfOperState,
54         ieee8021SpbAdjDynamicEntryPeerSysName,
55         ieee8021SpbAdjDynamicEntryPeerAgreeDigest,
56         ieee8021SpbAdjDynamicEntryPeerMCID,
57         ieee8021SpbAdjDynamicEntryPeerAuxMCID,
58         ieee8021SpbAdjDynamicEntryLocalCircuitID,
59         ieee8021SpbAdjDynamicEntryPeerLocalCircuitID,
60         ieee8021SpbAdjDynamicEntryPortIdentifier,
61         ieee8021SpbAdjDynamicEntryPeerPortIdentifier,
62         ieee8021SpbAdjDynamicEntryIsisCircIndex
63     }
64     STATUS current
65     DESCRIPTION
66     "The collection of objects used to represent ieee8021SpbAdjDynamicTable for PCR."
67     ::= { ieee8021PcrGroups 9 }
68
69 ieee8021PcrTopNodeTableGroup OBJECT-GROUP
70     OBJECTS {
71         ieee8021SpbTopNodeEntryBridgePriority,
72         ieee8021SpbmTopNodeEntrySPsourceID,
```

```

1         ieee8021SpbTopNodeEntrySysName
2     }
3     STATUS current
4     DESCRIPTION
5     "The collection of objects used to represent ieee8021SpbTopNodeTable for PCR."
6     ::= { ieee8021PcrGroups 10 }
7
8 ieee8021PcrTopEctTableGroup OBJECT-GROUP
9     OBJECTS {
10         ieee8021PcrTopEctEntryMode,
11         ieee8021PcrTopEctEntryMrtBlueVid,
12         ieee8021PcrTopEctEntryMrtRedVid
13     }
14     STATUS current
15     DESCRIPTION
16     "The collection of objects used to represent ieee8021PcrTopEctTable."
17     ::= { ieee8021PcrGroups 11 }
18
19 ieee8021PcrTopEdgeTableGroup OBJECT-GROUP
20     OBJECTS {
21         ieee8021SpbTopEdgeEntryMetricNear2Far,
22         ieee8021SpbTopEdgeEntryMetricFar2Near
23     }
24     STATUS current
25     DESCRIPTION
26     "The collection of objects used to represent ieee8021SpbTopEdgeTable for PCR."
27     ::= { ieee8021PcrGroups 12 }
28
29 ieee8021PcrTopSrvTableGroupVid OBJECT-GROUP
30     OBJECTS {
31         ieee8021SpbvTopSrvEntryBaseVid,
32         ieee8021SpbvTopSrvEntryMMacFlags
33     }
34     STATUS current
35     DESCRIPTION
36     "The collection of objects used to represent ieee8021SpbvTopSrvTable for PCR,
37     i.e., when the service is provided by a VID."
38     ::= { ieee8021PcrGroups 13 }
39
40 -- =====
41 -- Compliance statements SPBM
42 -- =====
43
44 ieee8021SpbComplianceSPBM MODULE-COMPLIANCE
45     STATUS current
46     DESCRIPTION
47     "Compliance to IEEE 802.1 SPBM mode"
48     MODULE
49         MANDATORY-GROUPS {
50             ieee8021SpbSysGroupSPBM ,
51             ieee8021SpbMtidStaticTableGroupSPBM ,
52             ieee8021SpbTopIxDynamicTableGroupSPBM ,
53             ieee8021SpbEctStaticTableGroupSPBM ,
54             ieee8021SpbEctDynamicTableGroupSPBM ,
55             ieee8021SpbAdjStaticTableGroupSPBM ,
56             ieee8021SpbAdjDynamicTableGroupSPBM ,
57             ieee8021SpbTopNodeTableGroupSPBM ,
58             ieee8021SpbTopEctTableGroupSPBM ,
59             ieee8021SpbTopEdgeTableGroupSPBM ,
60             ieee8021SpbmTopSrvTableGroupSPBM ,
61             ieee8021SpbmBsiStaticTableGroupSPBM
62         }
63
64         GROUP dotlagCfmMepSpbmTableGroupSPBM
65         DESCRIPTION
66         "This group is mandatory ONLY for devices supporting
67         SPBM VID MAS."
68
69         GROUP dotlagCfmMepSpbmEspTableGroupSPBM
70         DESCRIPTION
71         "This group is mandatory ONLY for devices supporting

```

```

1           SPBM path MAs or ECMP path MAs."
2
3       ::= { ieee8021SpbCompliances 1 }
4
5 -- =====
6 -- Compliance statements SPBV
7 -- =====
8
9 ieee8021SpbComplianceSPBV MODULE-COMPLIANCE
10     STATUS current
11     DESCRIPTION
12         "Compliance to IEEE 802.1 SPBV mode"
13     MODULE
14         MANDATORY-GROUPS {
15             ieee8021SpbSysGroupSPBV ,
16             ieee8021SpbMtidStaticTableGroupSPBV ,
17             ieee8021SpbTopIxDynamicTableGroupSPBV ,
18             ieee8021SpbEctStaticTableGroupSPBV ,
19             ieee8021SpbEctDynamicTableGroupSPBV ,
20             ieee8021SpbAdjStaticTableGroupSPBV ,
21             ieee8021SpbAdjDynamicTableGroupSPBV ,
22             ieee8021SpbTopNodeTableGroupSPBV ,
23             ieee8021SpbTopEctTableGroupSPBV ,
24             ieee8021SpbTopEdgeTableGroupSPBV ,
25             ieee8021SpbvTopSrvTableGroupSPBV
26         }
27     ::= { ieee8021SpbCompliances 2 }
28
29 -- =====
30 -- Compliance statements PCR
31 -- =====
32
33 ieee8021PcrCompliance MODULE-COMPLIANCE
34     STATUS current
35     DESCRIPTION
36         "Compliance to IEEE 802.1 PCR"
37     MODULE
38         MANDATORY-GROUPS {
39             ieee8021PcrSysGroup ,
40             ieee8021PcrMtidStaticTableGroup ,
41             ieee8021PcrTopIxDynamicTableGroup ,
42             ieee8021PcrEctStaticTableGroupMAC ,
43             ieee8021PcrEctStaticTableGroupVID ,
44             ieee8021PcrEctStaticTableGroupMrt,
45             ieee8021PcrEctDynamicTableGroup ,
46             ieee8021PcrAdjStaticTableGroup ,
47             ieee8021PcrAdjDynamicTableGroup ,
48             ieee8021PcrTopNodeTableGroup ,
49             ieee8021PcrTopEctTableGroup ,
50             ieee8021PcrTopEdgeTableGroup ,
51             ieee8021PcrTopSrvTableGroupVid
52         }
53     ::= { ieee8021PcrCompliances 1 }
54
55 END

```

1 17.7.20 Definitions for the IEEE8021-EVB-MIB module

```
2 IEEE8021-EVB-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- IEEE 802.1Q MIB for EVB Bridges and EVB Stations
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE,
10     Integer32, Counter32, Unsigned32, TimeTicks
11     FROM SNMPv2-SMI
12     MacAddress, TruthValue, RowStatus, StorageType
13     FROM SNMPv2-TC
14
15     ieee802dot1mibs, IEEE8021PbbComponentIdentifier,
16     IEEE8021BridgePortNumber
17     FROM IEEE8021-TC-MIB
18     VlanIndex
19     FROM Q-BRIDGE-MIB
20     InterfaceIndexOrZero
21     FROM IF-MIB
22     ieee8021BridgePhyPort
23     FROM IEEE8021-BRIDGE-MIB
24     MODULE-COMPLIANCE, OBJECT-GROUP
25     FROM SNMPv2-CONF;
26
27 ieee8021BridgeEvbMib MODULE-IDENTITY
28     LAST-UPDATED "202211080000Z" -- November 8, 2022
29     ORGANIZATION "IEEE 802.1 Working Group"
30     CONTACT-INFO
31         " WG-URL: http://www.ieee802.org/1/
32         WG-EMail: stds-802-1-1@ieee.org
33         Contact: IEEE 802.1 Working Group Chair
34         Postal: C/O IEEE 802.1 Working Group
35             IEEE Standards Association
36             445 Hoes Lane
37             Piscataway, NJ 08854
38             USA
39         E-mail: stds-802-1-chairs@ieee.org"
40     DESCRIPTION
41         "The EVB MIB module for managing devices that support
42         Edge Virtual Bridging.
43
44         Unless otherwise indicated, the references in this MIB
45         module are to IEEE Std 802.1Q-2022.
46
47         Copyright (C) IEEE (2022).
48         This version of this MIB module is part of IEEE Std 802.1Q;
49         see that standard for full legal notices."
50
51     REVISION "202211080000Z" -- November 8, 2022
52     DESCRIPTION
53         "Published as part of IEEE Std 802.1Q-2022.
54         Cross references and contact information updated."
55
56     REVISION "201807010000Z" -- July 1, 2018
57     DESCRIPTION
58         "Published as part of IEEE Std 802.1Q 2018 revision.
59         Cross references updated and corrected."
60
61     REVISION "201412150000Z" -- December 15, 2014
62     DESCRIPTION
63         "Published as part of IEEE Std 802.1Q 2014 revision.
64         Cross references updated and corrected.
65         ieee8021BridgeEvbVSIMvFormat added.
66         ieee8021BridgeEvbVsiMgrID16 added and
67         ieee8021BridgeEvbVsiMgrID deprecated.
68         ieee8021BridgeEvbVDPCounterDiscontinuity description
69         clarified. Conformance and groups fixed.
70         Fixed maintenance item to IEEE Std 802.1Qbg-2012."
71
```



```

1  REVISION      "201202150000Z" -- February 15, 2012
2  DESCRIPTION
3      "Initial version published in IEEE Std 802.1Qbg."
4      ::= { ieee802dot1mibs 24 }
5
6  -- =====
7  -- subtrees in the EVB MIB
8  -- =====
9
10 ieee8021BridgeEvbNotifications
11     OBJECT IDENTIFIER ::= { ieee8021BridgeEvbMib 0 }
12
13 ieee8021BridgeEvbObjects
14     OBJECT IDENTIFIER ::= { ieee8021BridgeEvbMib 1 }
15
16 ieee8021BridgeEvbConformance
17     OBJECT IDENTIFIER ::= { ieee8021BridgeEvbMib 2 }
18
19 -- =====
20 -- EVB Bridge managed object
21 -- =====
22
23 ieee8021BridgeEvbSys OBJECT IDENTIFIER ::= { ieee8021BridgeEvbObjects 1 }
24
25 ieee8021BridgeEvbSysType OBJECT-TYPE
26     SYNTAX      INTEGER {
27         evbBridge (1),
28         evbStation (2)
29     }
30     MAX-ACCESS   read-only
31     STATUS       current
32     DESCRIPTION  "The evbSysType determines if this is an EVB Bridge
33                 or EVB station."
34     REFERENCE    "5.23, 5.24"
35
36     ::= { ieee8021BridgeEvbSys 1 }
37
38 ieee8021BridgeEvbSysNumExternalPorts OBJECT-TYPE
39     SYNTAX      Unsigned32 (1..4095)
40     MAX-ACCESS   read-only
41     STATUS       current
42     DESCRIPTION  "The evbSysNumExternalPorts parameter indicates how
43                 many externally accessible port are available."
44     REFERENCE    "12.4.2, 12.5.1"
45     ::= { ieee8021BridgeEvbSys 2 }
46
47 ieee8021BridgeEvbSysEvbLldpTxEnable OBJECT-TYPE
48     SYNTAX      TruthValue
49     MAX-ACCESS   read-write
50     STATUS       current
51     DESCRIPTION  "This object is used to initialize the LLDP EVB
52                 objects for new SBPs and URPS.
53                 When set to 'true' a new SBP or URP will place the local
54                 EVB objects in the LLDP nearest Customer database;
55                 when set to 'false' a new SBP or URP will not place
56                 the local EVB objects in the LLDP database."
57     REFERENCE    "D.2.12"
58     DEFVAL      { true }
59     ::= { ieee8021BridgeEvbSys 3 }
60
61
62 ieee8021BridgeEvbSysEvbLldpManual OBJECT-TYPE
63     SYNTAX      TruthValue
64     MAX-ACCESS   read-write
65     STATUS       current
66     DESCRIPTION  "This object is used to initialize the LLDP EVB
67                 objects for new SBPs and URPS.
68                 When set to 'false' the operating configuration
69                 will be determined by the comparison between
70                 the local and remote LLDP EVB objects
71                 (automatic), regardless of the setting of
72                 ieee8021BridgeEvbSysLldpTxEnable.

```

```

1           When ieee8021BridgeEvbSysLldpManual is 'true' the
2           configuration will be determined by the setting
3           of the local EVB objects only (manual)."
```

4 REFERENCE "D.2.12"

5 DEFVAL { false }

6 ::= {ieee8021BridgeEvbSys 4}

7

8 ieee8021BridgeEvbSysEvlldpGidCapable OBJECT-TYPE

9 SYNTAX TruthValue

10 MAX-ACCESS read-write

11 STATUS current

12 DESCRIPTION "The value of this object is used as the default

13 value of the BGID or SGID bit of the EVB LLDP TLV string."

14 REFERENCE "D.2.12"

15 ::= {ieee8021BridgeEvbSys 5}

16

17 ieee8021BridgeEvbSysEcpAckTimer OBJECT-TYPE

18 SYNTAX Integer32

19 MAX-ACCESS read-write

20 STATUS deprecated

21 DESCRIPTION

22 "A value indicating the Bridge Proposed ECP ackTimer."

23 REFERENCE

24 "D.2.12.6, 43.3.6.1"

25 ::= { ieee8021BridgeEvbSys 6 }

26

27 ieee8021BridgeEvbSysEcpDfltAckTimerExp OBJECT-TYPE

28 SYNTAX Integer32 (0..31)

29 MAX-ACCESS read-write

30 STATUS current

31 DESCRIPTION

32 "The exponent of 2 indicating the Bridge Proposed ECP ackTimer

33 in tens of microseconds."

34 REFERENCE

35 "D.2.12.6, 43.3.6.1"

36 ::= { ieee8021BridgeEvbSys 11 }

37

38 ieee8021BridgeEvbSysEcpMaxRetries OBJECT-TYPE

39 SYNTAX Integer32 (0..7)

40 MAX-ACCESS read-write

41 STATUS current

42 DESCRIPTION

43 "A value indicating the Bridge ECP maxRetries."

44 REFERENCE

45 "D.2.12.5, 43.3.7.4"

46 DEFVAL { 3 }

47 ::= { ieee8021BridgeEvbSys 7 }

48

49

50 ieee8021BridgeEvbSysVdpDfltRsrcWaitDelay OBJECT-TYPE

51 SYNTAX Integer32

52 MAX-ACCESS read-write

53 STATUS deprecated

54 DESCRIPTION

55 "A value indicating the Bridge Resource VDP Timeout."

56 REFERENCE "D.2.12, 41.5.5.7"

57 ::= { ieee8021BridgeEvbSys 8 }

58

59 ieee8021BridgeEvbSysVdpDfltRsrcWaitDelayExp OBJECT-TYPE

60 SYNTAX Integer32 (0..31)

61 MAX-ACCESS read-write

62 STATUS current

63 DESCRIPTION

64 "The exponent of 2 indicating the Bridge Resource VDP

65 Timeout in tens of microseconds."

66 REFERENCE "D.2.12, 41.5.5.7"

67 ::= { ieee8021BridgeEvbSys 12 }

68

69 ieee8021BridgeEvbSysVdpDfltReinitKeepAlive OBJECT-TYPE

70 SYNTAX Integer32

71 MAX-ACCESS read-write

72 STATUS deprecated

```

1  DESCRIPTION
2  "A value indicating the Bridge Proposed VDP Keep Alive Timeout."
3  REFERENCE "D.2.12, 41.5.5.5"
4  ::= { ieee8021BridgeEvbSys 9 }
5
6  ieee8021BridgeEvbSysVdpDfltReinitKeepAliveExp OBJECT-TYPE
7      SYNTAX      Integer32 (0..31)
8      MAX-ACCESS   read-write
9      STATUS       current
10     DESCRIPTION
11         "The exponent of 2 indicating the Bridge Proposed VDP Keep
12         Alive Timeout in tens of microseconds."
13     REFERENCE "D.2.12, 41.5.5.5"
14     ::= { ieee8021BridgeEvbSys 13 }
15
16 -- =====
17 -- Station facing Bridge Port table
18 -- =====
19
20  ieee8021BridgeEvbSbpTable OBJECT-TYPE
21      SYNTAX      SEQUENCE OF Ieee8021BridgeEvbSbpEntry
22      MAX-ACCESS   not-accessible
23      STATUS       current
24      DESCRIPTION
25          "A table that contains Station-facing Bridge Port (SBP)
26          details."
27      REFERENCE   "12.26.2"
28      ::= { ieee8021BridgeEvbSys 10}
29
30  ieee8021BridgeEvbSbpEntry OBJECT-TYPE
31      SYNTAX      Ieee8021BridgeEvbSbpEntry
32      MAX-ACCESS   not-accessible
33      STATUS       current
34      DESCRIPTION
35          "A list of objects describing SBP."
36      INDEX { ieee8021BridgeEvbSbpComponentID,
37              ieee8021BridgeEvbSbpPortNumber
38            }
39      ::= { ieee8021BridgeEvbSbpTable 1 }
40
41  Ieee8021BridgeEvbSbpEntry ::=
42      SEQUENCE {
43          ieee8021BridgeEvbSbpComponentID
44              IEEE8021PbbComponentIdentifier,
45          ieee8021BridgeEvbSbpPortNumber
46              IEEE8021BridgePortNumber,
47          ieee8021BridgeEvbSbpLldpManual          TruthValue,
48          ieee8021BridgeEvbSbpVdpOperRsrcWaitDelay Unsigned32,
49          ieee8021BridgeEvbSbpVdpOperReinitKeepAlive Unsigned32,
50          ieee8021BridgeEvbSbpVdpOperToutKeepAlive Unsigned32,
51          ieee8021BridgeEvbSbpVdpOperRsrcWaitDelayExp Unsigned32,
52          ieee8021BridgeEvbSbpVdpOperReinitKeepAliveExp Unsigned32
53      }
54
55  ieee8021BridgeEvbSbpComponentID OBJECT-TYPE
56      SYNTAX      IEEE8021PbbComponentIdentifier
57      MAX-ACCESS   not-accessible
58      STATUS       current
59      DESCRIPTION
60          "The SBP component ID"
61      REFERENCE   "12.4.1.5"
62      ::= { ieee8021BridgeEvbSbpEntry 1 }
63
64  ieee8021BridgeEvbSbpPortNumber OBJECT-TYPE
65      SYNTAX      IEEE8021BridgePortNumber
66      MAX-ACCESS   not-accessible
67      STATUS       current
68      DESCRIPTION   "The SBP port number."
69      REFERENCE     "12.4.2"
70      ::= { ieee8021BridgeEvbSbpEntry 2 }
71
72  ieee8021BridgeEvbSbpLldpManual OBJECT-TYPE

```

```

1  SYNTAX          TruthValue
2  MAX-ACCESS      read-write
3  STATUS          current
4  DESCRIPTION
5  "The evbSbpLldpManual parameter switches EVB TLVs to manual mode.
6   In manual mode the running parameters are determined solely from
7   the local LLDP database values."
8  ::= { ieee8021BridgeEvbSbpEntry 3 }
9
10 ieee8021BridgeEvbSbpVdpOperRsrcWaitDelay OBJECT-TYPE
11 SYNTAX          Unsigned32
12 UNITS           "micro-seconds"
13 MAX-ACCESS      read-only
14 STATUS          deprecated
15 DESCRIPTION     "The value used to initialize the waitWhile timer
16                  (41.5.5.7) by the station VDP state machine when
17                  the state machine is waiting for a response."
18 REFERENCE       "D.2.12, 41.5.5.7"
19 ::= { ieee8021BridgeEvbSbpEntry 4 }
20
21 ieee8021BridgeEvbSbpVdpOperRsrcWaitDelayExp OBJECT-TYPE
22 SYNTAX          Unsigned32 (0..31)
23 MAX-ACCESS      read-only
24 STATUS          current
25 DESCRIPTION     "The exponent of 2 used to calculate the value to initialize
26                  the waitWhile timer
27                  (41.5.5.7) by the station VDP state machine when
28                  the state machine is waiting for a response."
29 REFERENCE       "D.2.12, 41.5.5.7"
30 ::= { ieee8021BridgeEvbSbpEntry 7 }
31
32
33 ieee8021BridgeEvbSbpVdpOperReinitKeepAlive OBJECT-TYPE
34 SYNTAX          Unsigned32
35 UNITS           "micro-seconds"
36 MAX-ACCESS      read-only
37 STATUS          deprecated
38 DESCRIPTION     "The value used to initialize the waitWhile timer
39                  (41.5.5.5) by the station VDP state machine in
40                  order to determine when to transmit a keep alive
41                  message."
42 REFERENCE       "D.2.12, 41.5.5.5"
43 ::= { ieee8021BridgeEvbSbpEntry 5 }
44
45 ieee8021BridgeEvbSbpVdpOperReinitKeepAliveExp OBJECT-TYPE
46 SYNTAX          Unsigned32 (0..31)
47 MAX-ACCESS      read-only
48 STATUS          current
49 DESCRIPTION     "The exponent of 2 used to calculate the value to initialize
50                  the waitWhile timer
51                  (41.5.5.5) by the station VDP state machine in
52                  order to determine when to transmit a keep alive
53                  message."
54 REFERENCE       "D.2.12, 41.5.5.5"
55 ::= { ieee8021BridgeEvbSbpEntry 8 }
56
57 ieee8021BridgeEvbSbpVdpOperToutKeepAlive OBJECT-TYPE
58 SYNTAX          Unsigned32
59 UNITS           "micro-seconds"
60 MAX-ACCESS      read-only
61 STATUS          current
62 DESCRIPTION     "The value used to initialize the waitWhile timer
63                  (41.5.5.13) by the EVBCB VDP state machine in order to
64                  determine when to transmit a keep alive message."
65 REFERENCE       "D.2.12, 41.5.5.13"
66 ::= { ieee8021BridgeEvbSbpEntry 6 }
67
68 -- =====
69 -- VSI Database
70 -- =====
71
72 ieee8021BridgeEvbVSIDBObjects OBJECT IDENTIFIER ::= { ieee8021BridgeEvbObjects 2 }

```

```

1
2 ieee8021BridgeEvbVSIDBTable OBJECT-TYPE
3   SYNTAX      SEQUENCE OF Ieee8021BridgeEvbVSIDBEntry
4   MAX-ACCESS  not-accessible
5   STATUS      current
6   DESCRIPTION
7       "A table that contains database of the active Virtual Station
8       Interfaces."
9   REFERENCE   "12.26.3"
10  ::= { ieee8021BridgeEvbVSIDBObjects 1 }
11
12 ieee8021BridgeEvbVSIDBEntry OBJECT-TYPE
13   SYNTAX      Ieee8021BridgeEvbVSIDBEntry
14   MAX-ACCESS  not-accessible
15   STATUS      current
16   DESCRIPTION
17       "A list of objects containing database of the active Virtual Station
18       Interfaces."
19   INDEX { ieee8021BridgeEvbVSIComponentID,
20           ieee8021BridgeEvbVSIPortNumber,
21           ieee8021BridgeEvbVSIIDType,
22           ieee8021BridgeEvbVSIID
23         }
24  ::= { ieee8021BridgeEvbVSIDBTable 1 }
25
26 Ieee8021BridgeEvbVSIDBEntry ::=
27   SEQUENCE {
28       ieee8021BridgeEvbVSIComponentID
29       IEEE8021PbbComponentIdentifier,
30       ieee8021BridgeEvbVSIPortNumber
31       IEEE8021BridgePortNumber,
32       ieee8021BridgeEvbVSIIDType      INTEGER,
33       ieee8021BridgeEvbVSIID          OCTET STRING,
34       ieee8021BridgeEvbVSITimeSinceCreate Unsigned32,
35       ieee8021BridgeEvbVsiVdpOperCmd  INTEGER,
36       ieee8021BridgeEvbVsiOperRevert  TruthValue,
37       ieee8021BridgeEvbVsiOperHard    TruthValue,
38       ieee8021BridgeEvbVsiOperReason  BITS,
39       ieee8021BridgeEvbVSIIMgrID      OCTET STRING,
40       ieee8021BridgeEvbVSIType        Integer32,
41       ieee8021BridgeEvbVSITypeVersion OCTET STRING,
42       ieee8021BridgeEvbVSIIMvFormat   INTEGER,
43       ieee8021BridgeEvbVSIINumMACs    Integer32,
44       ieee8021BridgeEvbVDPMachineState INTEGER,
45       ieee8021BridgeEvbVDPCCommandsSucceeded Counter32,
46       ieee8021BridgeEvbVDPCCommandsFailed Counter32,
47       ieee8021BridgeEvbVDPCCommandReverts Counter32,
48       ieee8021BridgeEvbVDPCounterDiscontinuity TimeTicks,
49       ieee8021BridgeEvbVSIIMgrID16    OCTET STRING,
50       ieee8021BridgeEvbVSIFilterFormat INTEGER
51   }
52
53 ieee8021BridgeEvbVSIComponentID OBJECT-TYPE
54   SYNTAX      IEEE8021PbbComponentIdentifier
55   MAX-ACCESS  not-accessible
56   STATUS      current
57   DESCRIPTION "The evbVSIComponentID is the ComponentID for the
58               C-VLAN component of the EVB Bridge or for the edge
59               relay of the EVB station."
60   REFERENCE   "12.4.1.5"
61  ::= { ieee8021BridgeEvbVSIDBEntry 1 }
62
63 ieee8021BridgeEvbVSIPortNumber OBJECT-TYPE
64   SYNTAX      IEEE8021BridgePortNumber
65   MAX-ACCESS  not-accessible
66   STATUS      current
67   DESCRIPTION "The evbVSIPortNumber is the Port Number for the SBP
68               or URP where the VSI is accessed."
69   REFERENCE   "12.4.2"
70  ::= { ieee8021BridgeEvbVSIDBEntry 2 }
71
72 ieee8021BridgeEvbVSIIDType OBJECT-TYPE

```

```

1  SYNTAX          INTEGER{
2                      vsiidIpv4 (1),
3                      vsiidIpv6 (2),
4                      vsiidMAC (3),
5                      vsiidLocal (4),
6                      vsiidUUID (5)
7                      }
8  MAX-ACCESS      not-accessible
9  STATUS          current
10 DESCRIPTION
11 "This object specifies the VSIID Type for the VSIID in the DCN "
12 REFERENCE       "41.2.6"
13 ::= { ieee8021BridgeEvbVSIDBEntry 3 }
14
15
16 ieee8021BridgeEvbVSIID OBJECT-TYPE
17 SYNTAX          OCTET STRING (SIZE (16))
18 MAX-ACCESS      not-accessible
19 STATUS          current
20 DESCRIPTION
21 "This object specifies the VSIID that uniquely identifies the VSI
22 in the DCN "
23 REFERENCE       "41.2.7"
24 ::= { ieee8021BridgeEvbVSIDBEntry 4 }
25
26
27 ieee8021BridgeEvbVSITimeSinceCreate OBJECT-TYPE
28 SYNTAX          Unsigned32
29 UNITS           "centi-seconds"
30 MAX-ACCESS      read-only
31 STATUS          current
32 DESCRIPTION
33 "This object specifies the time since creation "
34 REFERENCE       "41"
35 ::= { ieee8021BridgeEvbVSIDBEntry 5 }
36
37
38 ieee8021BridgeEvbVsiVdpOperCmd OBJECT-TYPE
39 SYNTAX          INTEGER
40                {
41                  preAssociate (1),
42                  preAssociateWithRsrcReservation (2),
43                  associate (3),
44                  deAssociate (4)
45                }
46 MAX-ACCESS      read-only
47 STATUS          current
48 DESCRIPTION
49 "This object identifies the type of TLV."
50 REFERENCE       "41.2.1"
51 ::= { ieee8021BridgeEvbVSIDBEntry 6 }
52
53 ieee8021BridgeEvbVsiOperRevert OBJECT-TYPE
54 SYNTAX          TruthValue
55 MAX-ACCESS      read-only
56 STATUS          current
57 DESCRIPTION     "The evbOperRevert status indicator shows the most
58                  recent value of the KEEP indicator from the VDP
59                  protocol exchange."
60 REFERENCE       "41.2.3"
61 ::= { ieee8021BridgeEvbVSIDBEntry 7 }
62
63 ieee8021BridgeEvbVsiOperHard OBJECT-TYPE
64 SYNTAX          TruthValue
65 MAX-ACCESS      read-only
66 STATUS          current
67 DESCRIPTION     "The evbVsiHard status indicator shows the most
68                  recent value of the HARD indicator from the VDP
69                  protocol exchange."
70 REFERENCE       "41.2.3"
71
72 ::= { ieee8021BridgeEvbVSIDBEntry 8 }

```

```

1
2 ieee8021BridgeEvbVsiOperReason  OBJECT-TYPE
3     SYNTAX          BITS
4         {
5             success (0),
6             invalidFormat (1),
7             insufficientResources (2),
8             otherfailure(3)
9         }
10    MAX-ACCESS      read-only
11    STATUS          current
12    DESCRIPTION
13    "This object indicates the outcome of a request."
14    REFERENCE       "41.2.3"
15
16
17    ::= { ieee8021BridgeEvbVSIDBEntry 9 }
18
19
20 ieee8021BridgeEvbVSiMgrID        OBJECT-TYPE
21     SYNTAX          OCTET STRING (SIZE (1))
22     MAX-ACCESS      read-only
23     STATUS          deprecated
24     DESCRIPTION
25     "This object identifies the VSI Manager with a database that holds
26     the detailed VSI type and or instance definitions."
27
28     REFERENCE       "41.1.3"
29     ::= { ieee8021BridgeEvbVSIDBEntry 10 }
30
31
32 ieee8021BridgeEvbVSIType OBJECT-TYPE
33     SYNTAX          Integer32
34     MAX-ACCESS      read-only
35     STATUS          current
36     DESCRIPTION " The VTID is an integer value used to identify
37                 a pre-configured set of controls and attributes
38                 that are associated with a set of VSIs."
39     REFERENCE       "41.2.4"
40     ::= { ieee8021BridgeEvbVSIDBEntry 11 }
41
42
43 ieee8021BridgeEvbVSITypeVersion  OBJECT-TYPE
44     SYNTAX          OCTET STRING (SIZE (1))
45     MAX-ACCESS      read-only
46     STATUS          current
47     DESCRIPTION
48     "The VSI Type Version is an integer identifier designating the
49     expected/desired VTID version. The VTID version allows a VSI
50     Manager Database to contain multiple versions of a given VSI
51     Type, allowing smooth migration to newer VSI types."
52
53     REFERENCE       "41.2.5"
54     ::= { ieee8021BridgeEvbVSIDBEntry 12 }
55
56
57 ieee8021BridgeEvbVSiMvFormat      OBJECT-TYPE
58     SYNTAX          INTEGER
59         {
60             basic (1),
61             partial (2),
62             vlanOnly (3)
63         }
64     MAX-ACCESS      read-only
65     STATUS          deprecated
66     DESCRIPTION
67     "This object specifies the MAC/VLAN format.
68     basic - Basic MAC/VLAN format
69     partial - Partial MAC/VLAN format
70     vlanOnly - Vlan-only MAC/VLAN format
71     "
72     REFERENCE       "41.2.8"

```

```

1  ::= { ieee8021BridgeEvbVSIDBEntry 13 }
2
3 ieee8021BridgeEvbVSINumMACs    OBJECT-TYPE
4      SYNTAX          Integer32
5      MAX-ACCESS      read-only
6      STATUS          current
7      DESCRIPTION
8      "This object specifies the the number of MAC address/VLAN ID pairs
9      contained in the repeated portion of the MAC/VLANs field in the
10     VDP TLV."
11     REFERENCE        "41.2.9"
12     ::= { ieee8021BridgeEvbVSIDBEntry 14 }
13
14
15 ieee8021BridgeEvbVDPMachineState  OBJECT-TYPE
16     SYNTAX          INTEGER
17     {
18         preAssociate (1),
19         preAssociateWithRsrcReservation (2),
20         associate (3),
21         deAssociate (4)
22     }
23     MAX-ACCESS      read-only
24     STATUS          current
25     DESCRIPTION
26     "This object specifies the VDP state machine. "
27     REFERENCE        "41.5.5.14"
28     ::= { ieee8021BridgeEvbVSIDBEntry 15 }
29
30
31 ieee8021BridgeEvbVDPCommandsSucceeded  OBJECT-TYPE
32     SYNTAX          Counter32
33     MAX-ACCESS      read-only
34     STATUS          current
35     DESCRIPTION
36     "This object specifies the VDP number of successful commands since
37     creation."
38     REFERENCE        "41.5"
39     ::= { ieee8021BridgeEvbVSIDBEntry 16 }
40
41 ieee8021BridgeEvbVDPCommandsFailed  OBJECT-TYPE
42     SYNTAX          Counter32
43     MAX-ACCESS      read-only
44     STATUS          current
45     DESCRIPTION
46     "This object specifies the VDP number of failed commands since
47     creation "
48     REFERENCE        "41.5"
49     ::= { ieee8021BridgeEvbVSIDBEntry 17 }
50
51 ieee8021BridgeEvbVDPCommandReverts  OBJECT-TYPE
52     SYNTAX          Counter32
53     MAX-ACCESS      read-only
54     STATUS          current
55     DESCRIPTION
56     "This object specifies the VDP command reverts since creation  "
57     REFERENCE        "41.5"
58     ::= { ieee8021BridgeEvbVSIDBEntry 18 }
59
60
61 ieee8021BridgeEvbVDPCounterDiscontinuity OBJECT-TYPE
62     SYNTAX          TimeTicks
63     UNITS            "hundredths of a second"
64     MAX-ACCESS      read-only
65     STATUS          current
66     DESCRIPTION
67     "The time (in hundredths of a second) since the
68     last counter discontinuity for any of the counters
69     in the row."
70     ::= { ieee8021BridgeEvbVSIDBEntry 19}
71
72

```



```

1 ieee8021BridgeEvbVSIMgrID16      OBJECT-TYPE
2   SYNTAX      OCTET STRING (SIZE (16))
3   MAX-ACCESS   read-only
4   STATUS      current
5   DESCRIPTION
6   "This object identifies the VSI Manager with a database that holds
7   the detailed VSI type and or instance definitions."
8
9   REFERENCE    "41.1.3"
10  ::= { ieee8021BridgeEvbVSIDBEntry 20 }
11
12
13 ieee8021BridgeEvbVSIFilterFormat  OBJECT-TYPE
14   SYNTAX      INTEGER
15   {
16       vid (1),
17       macVid (2),
18       groupidVid (3),
19       groupidMacVid (4)
20   }
21   MAX-ACCESS   read-only
22   STATUS      current
23   DESCRIPTION
24   "This object specifies the MAC/VLAN format:
25   vid (see 41.2.9.1)
26   macVid (see 41.2.9.2)
27   groupidVid (see 41.2.9.3)
28   groupidMacVid (see 41.2.9.4)
29   "
30   REFERENCE    "41.2.8, 41.2.9"
31  ::= { ieee8021BridgeEvbVSIDBEntry 21 }
32
33
34 -- =====
35 -- List of MAC/VLANs
36 -- =====
37
38 ieee8021BridgeEvbVSIDBMacTable OBJECT-TYPE
39   SYNTAX      SEQUENCE OF Ieee8021BridgeEvbVSIDBMacEntry
40   MAX-ACCESS   not-accessible
41   STATUS      current
42   DESCRIPTION
43   "A table that contains database of the active Virtual Station
44   Interfaces."
45   REFERENCE    "12.26.3"
46  ::= { ieee8021BridgeEvbVSIDBObjects 2 }
47
48 ieee8021BridgeEvbVSIDBMacEntry OBJECT-TYPE
49   SYNTAX      Ieee8021BridgeEvbVSIDBMacEntry
50   MAX-ACCESS   not-accessible
51   STATUS      current
52   DESCRIPTION
53   "A list of objects containing database of the MAC/VLANs
54   associated with Virtual Station Interfaces."
55
56   INDEX { ieee8021BridgeEvbVSIComponentID,
57           ieee8021BridgeEvbVSIPortNumber,
58           ieee8021BridgeEvbVSIIDType,
59           ieee8021BridgeEvbVSIID,
60           ieee8021BridgeEvbGroupID,
61           ieee8021BridgeEvbVSIMac,
62           ieee8021BridgeEvbVSIvlanId
63
64   }
65  ::= { ieee8021BridgeEvbVSIDBMacTable 1 }
66
67 Ieee8021BridgeEvbVSIDBMacEntry ::=
68   SEQUENCE {
69       ieee8021BridgeEvbGroupID      Unsigned32,
70       ieee8021BridgeEvbVSIMac       MacAddress,
71       ieee8021BridgeEvbVSIvlanId    VlanIndex
72   }

```

```

1
2 ieee8021BridgeEvbGroupID OBJECT-TYPE
3     SYNTAX      Unsigned32
4     MAX-ACCESS   not-accessible
5     STATUS      current
6     DESCRIPTION  "Group ID"
7     REFERENCE    "41.2.9"
8     ::= { ieee8021BridgeEvbVSIDBMacEntry 1}
9
10 ieee8021BridgeEvbVSIMac OBJECT-TYPE
11     SYNTAX      MacAddress
12     MAX-ACCESS   not-accessible
13     STATUS      current
14     DESCRIPTION  "The mac-address part of the MAC/VLANs for a VSI."
15     REFERENCE    "41.2.9"
16     ::= { ieee8021BridgeEvbVSIDBMacEntry 2}
17
18
19
20 ieee8021BridgeEvbVSIVlanId OBJECT-TYPE
21     SYNTAX      VlanIndex
22     MAX-ACCESS   read-only
23     STATUS      current
24     DESCRIPTION  "The Vlan ID part of the MAC/VLANs for a VSI."
25     REFERENCE    "41.2.9"
26     ::= { ieee8021BridgeEvbVSIDBMacEntry 3}
27
28
29
30 -- =====
31 -- Uplink Access Port table entry managed object
32 -- =====
33
34 ieee8021BridgeEvbSChannelObjects OBJECT IDENTIFIER ::=
35     { ieee8021BridgeEvbObjects 3 }
36
37
38 ieee8021BridgeEvbUAPConfigTable OBJECT-TYPE
39     SYNTAX      SEQUENCE OF Ieee8021BridgeEvbUAPConfigEntry
40     MAX-ACCESS   not-accessible
41     STATUS      current
42     DESCRIPTION  "A table that contains configuration parameters for UAP."
43     REFERENCE    "12.26.4.1"
44     ::= { ieee8021BridgeEvbSChannelObjects 1 }
45
46
47 ieee8021BridgeEvbUAPConfigEntry OBJECT-TYPE
48     SYNTAX      Ieee8021BridgeEvbUAPConfigEntry
49     MAX-ACCESS   not-accessible
50     STATUS      current
51     DESCRIPTION  "A list of objects containing information to configure the
52     attributes for UAP."
53     INDEX {
54         ieee8021BridgePhyPort
55     }
56     ::= { ieee8021BridgeEvbUAPConfigTable 1 }
57
58
59 Ieee8021BridgeEvbUAPConfigEntry ::=
60     SEQUENCE {
61         ieee8021BridgeEvbUAPComponentId
62             IEEE8021PbbComponentIdentifier,
63         ieee8021BridgeEvbUAPPort
64             IEEE8021BridgePortNumber,
65         ieee8021BridgeEvbUapConfigIfIndex
66             InterfaceIndexOrZero,
67         ieee8021BridgeEvbUAPSchCdcAdminEnable    INTEGER,
68         ieee8021BridgeEvbUAPSchAdminCDCPRole     INTEGER,
69         ieee8021BridgeEvbUAPSchAdminCDCPChanCap  Integer32,
70         ieee8021BridgeEvbUAPSchOperCDCPChanCap   Integer32,
71         ieee8021BridgeEvbUAPSchAdminCDCPSVIDPoolLow  VlanIndex,
72         ieee8021BridgeEvbUAPSchAdminCDCPSVIDPoolHigh VlanIndex,

```

```

1      ieee8021BridgeEvbUAPSchOperState          INTEGER,
2      ieee8021BridgeEvbSchCdcRemoteEnabled      INTEGER,
3      ieee8021BridgeEvbSchCdcRemoteRole         INTEGER,
4      ieee8021BridgeEvbUAPConfigStorageType     StorageType,
5      ieee8021BridgeEvbUAPConfigRowStatus       RowStatus
6      }
7
8 ieee8021BridgeEvbUAPComponentId OBJECT-TYPE
9     SYNTAX      IEEE8021PbbComponentIdentifier
10    MAX-ACCESS  read-only
11    STATUS      current
12    DESCRIPTION
13     "The ComponentID of the port for the UAP."
14     ::= { ieee8021BridgeEvbUAPConfigEntry 1 }
15
16
17 ieee8021BridgeEvbUAPPort OBJECT-TYPE
18     SYNTAX      IEEE8021BridgePortNumber
19     MAX-ACCESS  read-only
20     STATUS      current
21     DESCRIPTION
22     "The port number of the port for the UAP."
23     ::= { ieee8021BridgeEvbUAPConfigEntry 2 }
24
25 ieee8021BridgeEvbUapConfigIfIndex OBJECT-TYPE
26     SYNTAX      InterfaceIndexOrZero
27     MAX-ACCESS  read-only
28     STATUS      current
29     DESCRIPTION
30     "The value of the instance of the IfIndex object,
31     defined in the IF-MIB, for the interface corresponding
32     to this port, or the value 0 if the port has not been
33     bound to an underlying frame source and sink."
34     ::= { ieee8021BridgeEvbUAPConfigEntry 3 }
35
36
37 ieee8021BridgeEvbUAPSchCdcAdminEnable OBJECT-TYPE
38     SYNTAX      INTEGER
39     {
40         enable (1),
41         disable (2)
42     }
43     MAX-ACCESS  read-create
44     STATUS      current
45     DESCRIPTION "Administrative status of CDCP."
46     REFERENCE  "42.4.2"
47
48     ::= { ieee8021BridgeEvbUAPConfigEntry 4 }
49
50
51 ieee8021BridgeEvbUAPSchAdminCDCPRole OBJECT-TYPE
52     SYNTAX      INTEGER
53     {
54         cdcRoleB (1),
55         cdcRoleS (2)
56     }
57     MAX-ACCESS  read-create
58     STATUS      current
59     DESCRIPTION "The administratively configured value for the local
60     port's role parameter. The value of AdminRole is not reflected in
61     the S-channel TLV. The AdminRole may take the value S or B.
62     S indicates the sender is unwilling to accept S-channels
63     configuration (mode, # channels supported, channel index) from
64     its neighbor and that the sender is willing to accept SVID
65     assignments from the neighbor. Stations usually take the S role.
66     B indicates the sender is willing to accept S-channels
67     configuration (mode, # channels supported, channel index)
68     from its neighbor and that the sender is willing do the best
69     it can to fill the SVID assignments
70     from the neighbor. Bridges usually take the B role."
71
72     REFERENCE  "42.4.2"

```

```

1  DEFVAL { 1 }
2
3 ::= { ieee8021BridgeEvbUAPConfigEntry 5 }
4
5
6 ieee8021BridgeEvbUAPSchAdminCDCPChanCap OBJECT-TYPE
7     SYNTAX      Integer32 (1 .. 167)
8     MAX-ACCESS   read-create
9     STATUS       current
10    DESCRIPTION  "The administratively configured value for the
11                  Number of Channels supported parameter. This
12                  value is included as the ChanCap parameter in
13                  the S-channel TLV."
14    REFERENCE    "42.4.1"
15
16 ::= { ieee8021BridgeEvbUAPConfigEntry 6 }
17
18
19 ieee8021BridgeEvbUAPSchOperCDCPChanCap OBJECT-TYPE
20     SYNTAX      Integer32 (1 .. 167)
21     MAX-ACCESS   read-only
22     STATUS       current
23     DESCRIPTION  "The operational value for the Number of Channels
24                  supported parameter. This value is included
25                  as the ChnCap parameter in the S-channel TLV."
26     REFERENCE    "42.4.8"
27 ::= { ieee8021BridgeEvbUAPConfigEntry 7 }
28
29 ieee8021BridgeEvbUAPSchAdminCDCPSVIDPoolLow OBJECT-TYPE
30     SYNTAX      VlanIndex
31     MAX-ACCESS   read-create
32     STATUS       current
33     DESCRIPTION  "Determines the lowest S-VIDs available for
34                  assignment by CDCP."
35
36     REFERENCE    "42.4.7"
37 ::= { ieee8021BridgeEvbUAPConfigEntry 8 }
38
39
40 ieee8021BridgeEvbUAPSchAdminCDCPSVIDPoolHigh OBJECT-TYPE
41     SYNTAX      VlanIndex
42     MAX-ACCESS   read-create
43     STATUS       current
44     DESCRIPTION  "Determines the highest S-VIDs available for
45                  assignment by CDCP."
46     REFERENCE    "42.4.7"
47
48 ::= { ieee8021BridgeEvbUAPConfigEntry 9 }
49
50
51 ieee8021BridgeEvbUAPSchOperState OBJECT-TYPE
52     SYNTAX      INTEGER
53     {
54         running (1),
55         notRunning (2)
56     }
57     MAX-ACCESS   read-only
58     STATUS       current
59     DESCRIPTION  "The current running state of CDCP."
60     REFERENCE    "42.4.14"
61
62 ::= { ieee8021BridgeEvbUAPConfigEntry 10 }
63
64
65
66 ieee8021BridgeEvbSchCdcPRemoteEnabled OBJECT-TYPE
67     SYNTAX      INTEGER
68     {
69         enable (1),
70         disable (2)
71     }
72     MAX-ACCESS   read-only

```

```

1  STATUS          current
2  DESCRIPTION     "CDCP state for the remote S-channel."
3  REFERENCE       "42.4.14"
4  ::= { ieee8021BridgeEvbUAPConfigEntry 11 }
5
6
7 ieee8021BridgeEvbSchCdcRemoteRole OBJECT-TYPE
8   SYNTAX          INTEGER
9   {
10      cdcRoleB (1),
11      cdcRoleS (2)
12   }
13   MAX-ACCESS     read-only
14   STATUS          current
15   DESCRIPTION     "The value for the remote port's role parameter."
16   REFERENCE       "42.4.12"
17   ::= { ieee8021BridgeEvbUAPConfigEntry 12 }
18
19 ieee8021BridgeEvbUAPConfigStorageType OBJECT-TYPE
20   SYNTAX          StorageType
21   MAX-ACCESS     read-create
22   STATUS          current
23   DESCRIPTION
24      "The storage type for this row. Rows in this table that
25      were created through an external process may have a storage
26      type of readOnly or permanent.
27      For a storage type of permanent, none of the columns have
28      to be writable."
29   DEFVAL { nonVolatile }
30   ::= { ieee8021BridgeEvbUAPConfigEntry 13 }
31
32
33 ieee8021BridgeEvbUAPConfigRowStatus OBJECT-TYPE
34   SYNTAX          RowStatus
35   MAX-ACCESS     read-create
36   STATUS          current
37   DESCRIPTION     "RowStatus for creating a UAP table entry."
38   ::= { ieee8021BridgeEvbUAPConfigEntry 14 }
39
40
41
42 -- =====
43 -- S-Channel Interface Table
44 -- =====
45
46 ieee8021BridgeEvbCAPConfigTable OBJECT-TYPE
47   SYNTAX          SEQUENCE OF Ieee8021BridgeEvbCAPConfigEntry
48   MAX-ACCESS     not-accessible
49   STATUS          current
50   DESCRIPTION
51      "A table that contains configuration information for
52      the S-Channel Access Ports (CAP)."
```

REFERENCE "12.26.4.2"

```

53   ::= { ieee8021BridgeEvbSChannelObjects 2 }
54
55
56 ieee8021BridgeEvbCAPConfigEntry OBJECT-TYPE
57   SYNTAX          Ieee8021BridgeEvbCAPConfigEntry
58   MAX-ACCESS     not-accessible
59   STATUS          current
60   DESCRIPTION
61      "A list of objects containing information for the S-Channel
62      Access Ports (CAP)"
63   INDEX { ieee8021BridgePhyPort,
64          ieee8021BridgeEvbSchID
65   }
66   ::= { ieee8021BridgeEvbCAPConfigTable 1 }
67
68 Ieee8021BridgeEvbCAPConfigEntry ::=
69   SEQUENCE {
70     ieee8021BridgeEvbSchID
71     Unsigned32,
72     ieee8021BridgeEvbCAPComponentId

```

```

1         IEEE8021PbbComponentIdentifier,
2     ieee8021BridgeEvbCapConfigIfIndex
3         InterfaceIndexOrZero,
4     ieee8021BridgeEvbCAPPort
5         IEEE8021BridgePortNumber,
6     ieee8021BridgeEvbCAPSChannelID
7         Unsigned32,
8     ieee8021BridgeEvbCAPAssociateSBPOrURPCompID
9         IEEE8021PbbComponentIdentifier,
10    ieee8021BridgeEvbCAPAssociateSBPOrURPPort
11        IEEE8021BridgePortNumber,
12    ieee8021BridgeEvbCAPRowStatus
13        RowStatus
14    }
15
16 ieee8021BridgeEvbSchID OBJECT-TYPE
17     SYNTAX      Unsigned32 (1..4094)
18     MAX-ACCESS  not-accessible
19     STATUS      current
20     DESCRIPTION
21     "This object represents the SVID for a ieee8021BridgeEvbSysType
22     of evbBridge and a SCID(S-Channel ID) for a
23     ieee8021BridgeEvbSysType of evbStation."
24     REFERENCE   "42.4.3"
25     ::= { ieee8021BridgeEvbCAPConfigEntry 1 }
26
27
28 ieee8021BridgeEvbCAPComponentId OBJECT-TYPE
29     SYNTAX      IEEE8021PbbComponentIdentifier
30     MAX-ACCESS  read-only
31     STATUS      current
32     DESCRIPTION "Component ID for S-channel Access Port."
33     ::= { ieee8021BridgeEvbCAPConfigEntry 2 }
34
35 ieee8021BridgeEvbCapConfigIfIndex OBJECT-TYPE
36     SYNTAX      InterfaceIndexOrZero
37     MAX-ACCESS  read-only
38     STATUS      current
39     DESCRIPTION "The value of the instance of the IfIndex object,
40     defined in the IF-MIB, for the interface corresponding
41     to this port, or the value 0 if the port has not been
42     bound to an underlying frame source and sink.
43     The underlying IfEntry indexed by this column MUST be persistent
44     across reinitializations of the management system."
45     ::= { ieee8021BridgeEvbCAPConfigEntry 3 }
46
47
48 ieee8021BridgeEvbCAPPort OBJECT-TYPE
49     SYNTAX      IEEE8021BridgePortNumber
50     MAX-ACCESS  read-only
51     STATUS      current
52     DESCRIPTION "Port number for the S-Channel Access Port."
53     ::= { ieee8021BridgeEvbCAPConfigEntry 4 }
54
55
56 ieee8021BridgeEvbCAPSChannelID OBJECT-TYPE
57     SYNTAX      Unsigned32
58     MAX-ACCESS  read-only
59     STATUS      current
60     DESCRIPTION "S-Channel ID (SCID) for this CAP."
61     REFERENCE   "42.4.2"
62     ::= { ieee8021BridgeEvbCAPConfigEntry 5 }
63
64 ieee8021BridgeEvbCAPAssociateSBPOrURPCompID OBJECT-TYPE
65     SYNTAX      IEEE8021PbbComponentIdentifier
66     MAX-ACCESS  read-write
67     STATUS      current
68     DESCRIPTION "Component ID of the Server Edge Port to be
69     associated with the CAP."
70     REFERENCE   "12.4.1.5"
71     ::= { ieee8021BridgeEvbCAPConfigEntry 6 }
72

```

```

1
2
3 ieee8021BridgeEvbCAPAssociateSBPOrURPPort      OBJECT-TYPE
4     SYNTAX          IEEE8021BridgePortNumber
5     MAX-ACCESS      read-write
6     STATUS          current
7     DESCRIPTION     "Port number of the Server Edge Port to be
8     associated with the CAP."
9     REFERENCE       "12.4.2"
10 ::= { ieee8021BridgeEvbCAPConfigEntry 7 }
11
12
13 ieee8021BridgeEvbCAPRowStatus                  OBJECT-TYPE
14     SYNTAX          RowStatus
15     MAX-ACCESS      read-create
16     STATUS          current
17     DESCRIPTION     "RowStatus to create/destroy this table."
18
19     ::= { ieee8021BridgeEvbCAPConfigEntry 8 }
20
21 -- =====
22 -- Uplink Relay Port table entry
23 -- =====
24
25 ieee8021BridgeEvbURPTable OBJECT-TYPE
26     SYNTAX          SEQUENCE OF Ieee8021BridgeEvbURPEntry
27     MAX-ACCESS      not-accessible
28     STATUS          current
29     DESCRIPTION     "A table that contains configuration information for
30     the Uplink Relay Ports (URP)."
31     REFERENCE       "12.26.5.1"
32     ::= { ieee8021BridgeEvbSChannelObjects 3 }
33
34
35 ieee8021BridgeEvbURPEntry OBJECT-TYPE
36     SYNTAX          Ieee8021BridgeEvbURPEntry
37     MAX-ACCESS      not-accessible
38     STATUS          current
39     DESCRIPTION     "A list of objects containing information for the Uplink
40     Relay Ports (URP)."
41     INDEX { ieee8021BridgeEvbURPComponentId,
42             ieee8021BridgeEvbURPPort
43     }
44     ::= { ieee8021BridgeEvbURPTable 1 }
45
46
47 Ieee8021BridgeEvbURPEntry ::=
48     SEQUENCE {
49         ieee8021BridgeEvbURPComponentId
50             IEEE8021PbbComponentIdentifier,
51         ieee8021BridgeEvbURPPort
52             IEEE8021BridgePortNumber,
53         ieee8021BridgeEvbURPIfIndex
54             InterfaceIndexOrZero,
55         ieee8021BridgeEvbURPBindToISSPort
56             IEEE8021BridgePortNumber,
57         ieee8021BridgeEvbURPLdpManual
58             TruthValue,
59         ieee8021BridgeEvbURPVdpOperRsrcWaitDelay
60             Unsigned32,
61         ieee8021BridgeEvbURPVdpOperRespWaitDelay
62             Unsigned32,
63         ieee8021BridgeEvbURPVdpOperReinitKeepAlive
64             Unsigned32,
65         ieee8021BridgeEvbURPVdpOperRsrcWaitDelayExp
66             Unsigned32,
67         ieee8021BridgeEvbURPVdpOperReinitKeepAliveExp
68             Unsigned32
69     }
70
71
72 ieee8021BridgeEvbURPComponentId                  OBJECT-TYPE

```

```

1  SYNTAX      IEEE8021PbbComponentIdentifier
2  MAX-ACCESS  not-accessible
3  STATUS      current
4  DESCRIPTION "Component ID that the URP belongs to."
5  ::= { ieee8021BridgeEvbURPEntry 1 }
6
7  ieee8021BridgeEvbURPPort          OBJECT-TYPE
8  SYNTAX      IEEE8021BridgePortNumber
9  MAX-ACCESS  not-accessible
10 STATUS      current
11 DESCRIPTION "port number of the urp."
12 ::= { ieee8021BridgeEvbURPEntry 2 }
13
14
15 ieee8021BridgeEvbURPIfIndex        OBJECT-TYPE
16 SYNTAX      InterfaceIndexOrZero
17 MAX-ACCESS  read-write
18 STATUS      current
19 DESCRIPTION "The value of the instance of the IfIndex object,
20              defined in the IF-MIB, for the interface corresponding
21              to this port, or the value 0 if the port has not been
22              bound to an underlying frame source and sink.
23
24              It is an implementation specific decision as to whether
25              this object may be modified if it has been created or
26              if 0 is a legal value.
27
28              The underlying IfEntry indexed by this column MUST be
29              persistent across reinitializations of the management
30              system. "
31 ::= { ieee8021BridgeEvbURPEntry 3 }
32
33 ieee8021BridgeEvbURPBindToISSPort  OBJECT-TYPE
34 SYNTAX      IEEE8021BridgePortNumber
35 MAX-ACCESS  read-write
36 STATUS      current
37 DESCRIPTION "The evbURPBindToISSPort is the ISS Port Number where
38              the URP is attached.
39              This binding is optional and only required in some
40              systems."
41 ::= { ieee8021BridgeEvbURPEntry 4 }
42
43
44 ieee8021BridgeEvbURPLldpManual      OBJECT-TYPE
45 SYNTAX      TruthValue
46 MAX-ACCESS  read-write
47 STATUS      current
48 DESCRIPTION "The evbUrpLldpManual parameter control how the EVB
49              TLV determines the operating values for parameters.
50              When set TRUE only the local EVB TLV will be used to
51              determine the parameters."
52 ::= { ieee8021BridgeEvbURPEntry 6 }
53
54
55 ieee8021BridgeEvbURPVdpOperRsrcWaitDelay  OBJECT-TYPE
56 SYNTAX      Unsigned32
57 UNITS       "micro-seconds"
58 MAX-ACCESS  read-only
59 STATUS      deprecated
60 DESCRIPTION "The parameter evbURPVdpOperRsrcWaitDelay is the
61              exponent of 2 used to set the VDP resourceWaitDelay
62              timer at the EVB Bridge."
63 ::= { ieee8021BridgeEvbURPEntry 9 }
64
65 ieee8021BridgeEvbURPVdpOperRsrcWaitDelayExp  OBJECT-TYPE
66 SYNTAX      Unsigned32 (0..31)
67 MAX-ACCESS  read-only
68 STATUS      current
69 DESCRIPTION "The parameter evbURPVdpOperRsrcWaitDelay is the
70              exponent of 2 used to set the VDP resourceWaitDelay
71              timer at the EVB Bridge."
72 ::= { ieee8021BridgeEvbURPEntry 12 }

```



```

1
2 ieee8021BridgeEvpURPVdpOperRespWaitDelay OBJECT-TYPE
3     SYNTAX      Unsigned32
4     UNITS       "micro-seconds"
5     MAX-ACCESS  read-write
6     STATUS      current
7     DESCRIPTION "The evbUrpVdpOperRespWaitDelay is how long a
8                 EVb station VDP will wait for a response from
9                 the EVB Bridge VDP."
10    ::= { ieee8021BridgeEvpURPEntry 10 }
11
12
13 ieee8021BridgeEvpURPVdpOperReinitKeepAlive OBJECT-TYPE
14     SYNTAX      Unsigned32
15     UNITS       "micro-seconds"
16     MAX-ACCESS  read-write
17     STATUS      deprecated
18     DESCRIPTION "The evbURPVdpOperReinitKeepAlive is the exponent
19                 of 2 used to determine the time interval of Keep
20                 Alive transmitted by the EVB station."
21    ::= { ieee8021BridgeEvpURPEntry 11 }
22
23 ieee8021BridgeEvpURPVdpOperReinitKeepAliveExp OBJECT-TYPE
24     SYNTAX      Unsigned32 (0..31)
25     MAX-ACCESS  read-write
26     STATUS      current
27     DESCRIPTION "The evbURPVdpOperReinitKeepAlive is the exponent
28                 of 2 used to determine the time interval of Keep
29                 Alive transmitted by the EVB station."
30    ::= { ieee8021BridgeEvpURPEntry 13 }
31
32 -- =====
33 -- Edge Control Protocol Table
34 -- =====
35
36 ieee8021BridgeEvpEcpTable OBJECT-TYPE
37     SYNTAX      SEQUENCE OF Ieee8021BridgeEvpEcpEntry
38     MAX-ACCESS  not-accessible
39     STATUS      current
40     DESCRIPTION
41     "A table that contains configuration information for
42     the Edge Control Protocol (ECP)."
```

REFERENCE "12.26.4.2"

```

44    ::= { ieee8021BridgeEvpSChannelObjects 4 }
45
46 ieee8021BridgeEvpEcpEntry OBJECT-TYPE
47     SYNTAX      Ieee8021BridgeEvpEcpEntry
48     MAX-ACCESS  not-accessible
49     STATUS      current
50     DESCRIPTION
51     "A list of objects containing information for the Edge Control
52     Protocol (ECP)."
```

INDEX { ieee8021BridgeEvpEcpComponentId,
ieee8021BridgeEvpEcpPort

```

55 }
56 ::= { ieee8021BridgeEvpEcpTable 1 }
57
58 Ieee8021BridgeEvpEcpEntry ::=
59     SEQUENCE {
60         ieee8021BridgeEvpEcpComponentId
61         IEEE8021PbbComponentIdentifier,
62         ieee8021BridgeEvpEcpPort
63         IEEE8021BridgePortNumber,
64         ieee8021BridgeEvpEcpOperAckTimerInit Unsigned32,
65         ieee8021BridgeEvpEcpOperMaxRetries Unsigned32,
66         ieee8021BridgeEvpEcpTxFrameCount Counter32,
67         ieee8021BridgeEvpEcpTxRetryCount Counter32,
68         ieee8021BridgeEvpEcpTxFailures Counter32,
69         ieee8021BridgeEvpEcpRxFrameCount Counter32,
70         ieee8021BridgeEvpEcpOperAckTimerInitExp Unsigned32
71     }
72

```

```

1
2  ieee8021BridgeEvpEcpComponentId          OBJECT-TYPE
3      SYNTAX      IEEE8021PbbComponentIdentifier
4      MAX-ACCESS  not-accessible
5      STATUS      current
6      DESCRIPTION "Component ID ."
7      ::= { ieee8021BridgeEvpEcpEntry 1 }
8
9  ieee8021BridgeEvpEcpPort                  OBJECT-TYPE
10     SYNTAX      IEEE8021BridgePortNumber
11     MAX-ACCESS  not-accessible
12     STATUS      current
13     DESCRIPTION "Port number."
14     ::= { ieee8021BridgeEvpEcpEntry 2 }
15
16
17  ieee8021BridgeEvpEcpOperAckTimerInit      OBJECT-TYPE
18     SYNTAX      Unsigned32
19     UNITS       "micro-seconds"
20     MAX-ACCESS  read-only
21     STATUS      deprecated
22     DESCRIPTION "The initial value used to initialize ackTimer
23                  (43.3.6.1)."
24     ::= { ieee8021BridgeEvpEcpEntry 3 }
25
26  ieee8021BridgeEvpEcpOperAckTimerInitExp    OBJECT-TYPE
27     SYNTAX      Unsigned32 (0..31)
28     MAX-ACCESS  read-only
29     STATUS      current
30     DESCRIPTION "The initial value used to initialize ackTimer.
31                  Expressed as exp where 10*2exp microseconds is the
32                  duration of the ack timer(43.3.6.1)."
33     ::= { ieee8021BridgeEvpEcpEntry 9 }
34
35  ieee8021BridgeEvpEcpOperMaxRetries         OBJECT-TYPE
36     SYNTAX      Unsigned32
37     MAX-ACCESS  read-only
38     STATUS      current
39     DESCRIPTION "This integer variable defines the maximum number
40                  of times that the ECP transmit state machine will
41                  retry a transmission if no ACK is received."
42     ::= { ieee8021BridgeEvpEcpEntry 4 }
43
44  ieee8021BridgeEvpEcpTxFrameCount          OBJECT-TYPE
45     SYNTAX      Counter32
46     MAX-ACCESS  read-only
47     STATUS      current
48     DESCRIPTION "The evbECPTxFrameCount is the number of ECP frame
49                  transmitted since ECP was instanciated."
50     ::= { ieee8021BridgeEvpEcpEntry 5 }
51
52
53  ieee8021BridgeEvpEcpTxRetryCount          OBJECT-TYPE
54     SYNTAX      Counter32
55     MAX-ACCESS  read-only
56     STATUS      current
57     DESCRIPTION "The evbECPTxRetryCount is the number of times
58                  ECP re-tried transmission since ECP was
59                  instanciated."
60     ::= { ieee8021BridgeEvpEcpEntry 6 }
61
62  ieee8021BridgeEvpEcpTxFailures            OBJECT-TYPE
63     SYNTAX      Counter32
64     MAX-ACCESS  read-only
65     STATUS      current
66     DESCRIPTION "The evbECPTxFailures is the number of times ECP
67                  failed to successfully deliver a frame since ECP
68                  was instanciated."
69     ::= { ieee8021BridgeEvpEcpEntry 7 }
70
71
72  ieee8021BridgeEvpEcpRxFrameCount          OBJECT-TYPE

```

```

1      SYNTAX      Counter32
2      MAX-ACCESS  read-only
3      STATUS      current
4      DESCRIPTION "The evbECPRxFrameCount is the number
5                  of frames received since ECP was instantiated."
6      ::= { ieee8021BridgeEvpEcpEntry 8 }
7
8
9
10 -- =====
11 -- Conformance Information
12 -- =====
13
14 ieee8021BridgeEvpGroups
15     OBJECT IDENTIFIER ::= { ieee8021BridgeEvpConformance 1 }
16
17 ieee8021BridgeEvpCompliances
18     OBJECT IDENTIFIER ::= { ieee8021BridgeEvpConformance 2 }
19
20 -- =====
21 -- Units of conformance
22 -- =====
23
24 ieee8021BridgeEvpSysGroup OBJECT-GROUP
25     OBJECTS {
26         ieee8021BridgeEvpSysType,
27         ieee8021BridgeEvpSysNumExternalPorts,
28         ieee8021BridgeEvpSysEvpLldpTxEnable,
29         ieee8021BridgeEvpSysEvpLldpGidCapable,
30         ieee8021BridgeEvpSysEvpLldpManual,
31         ieee8021BridgeEvpSysEcpAckTimer,
32         ieee8021BridgeEvpSysEcpMaxRetries,
33         ieee8021BridgeEvpSysVdpDfltRsrcWaitDelay,
34         ieee8021BridgeEvpSysVdpDfltReinitKeepAlive
35     }
36
37     STATUS      deprecated
38     DESCRIPTION
39         "The collection of objects used to represent a EVB
40         management objects."
41     ::= { ieee8021BridgeEvpGroups 1 }
42
43
44
45 ieee8021BridgeEvpSbpGroup OBJECT-GROUP
46     OBJECTS {
47         ieee8021BridgeEvpSbpLldpManual,
48         ieee8021BridgeEvpSbpVdpOperRsrcWaitDelay ,
49         ieee8021BridgeEvpSbpVdpOperReinitKeepAlive,
50         ieee8021BridgeEvpSbpVdpOperToutKeepAlive
51     }
52     STATUS deprecated
53     DESCRIPTION
54         "The collection of objects used to represent a SBP
55         management objects."
56     ::= { ieee8021BridgeEvpGroups 3 }
57
58
59 ieee8021BridgeEvpVSIDBGroup OBJECT-GROUP
60     OBJECTS {
61         ieee8021BridgeEvpVSItimeSinceCreate ,
62         ieee8021BridgeEvpVsiVdpOperCmd,
63         ieee8021BridgeEvpVsiOperRevert,
64         ieee8021BridgeEvpVsiOperHard,
65         ieee8021BridgeEvpVsiOperReason,
66         ieee8021BridgeEvpVSIIMgrID,
67         ieee8021BridgeEvpVSIType,
68         ieee8021BridgeEvpVSITypeVersion ,
69         ieee8021BridgeEvpVSIIMvFormat,
70         ieee8021BridgeEvpVSIINumMACs ,
71         ieee8021BridgeEvpVDPMachineState ,
72         ieee8021BridgeEvpVDPCommandsSucceeded ,

```

```

1         ieee8021BridgeEvbVDPCCommandsFailed ,
2         ieee8021BridgeEvbVDPCCommandReverts ,
3         ieee8021BridgeEvbVDPCounterDiscontinuity,
4         ieee8021BridgeEvbVSIVlanId
5     }
6     STATUS      deprecated
7     DESCRIPTION
8         "The collection of objects used to represent a EVB VSI
9         DB table."
10    ::= { ieee8021BridgeEvbGroups 4 }
11
12
13    ieee8021BridgeEvbUAPGroup OBJECT-GROUP
14    OBJECTS {
15        ieee8021BridgeEvbUAPComponentId,
16        ieee8021BridgeEvbUAPPort,
17        ieee8021BridgeEvbUapConfigIfIndex,
18        ieee8021BridgeEvbUAPSchCdcAdminEnable,
19        ieee8021BridgeEvbUAPSchAdminCDCPRole,
20        ieee8021BridgeEvbUAPSchAdminCDCPChanCap,
21        ieee8021BridgeEvbUAPSchOperCDCPChanCap,
22        ieee8021BridgeEvbUAPSchAdminCDCPSVIDPoolLow,
23        ieee8021BridgeEvbUAPSchAdminCDCPSVIDPoolHigh,
24        ieee8021BridgeEvbUAPSchOperState,
25        ieee8021BridgeEvbSchCdcRemoteEnabled,
26        ieee8021BridgeEvbSchCdcRemoteRole,
27        ieee8021BridgeEvbUAPConfigStorageType ,
28        ieee8021BridgeEvbUAPConfigRowStatus
29    }
30    STATUS      current
31    DESCRIPTION
32        "The collection of objects used to represent a EVB UAP
33        table."
34    ::= { ieee8021BridgeEvbGroups 5 }
35
36
37    ieee8021BridgeEvbCAPConfigGroup OBJECT-GROUP
38    OBJECTS {
39        ieee8021BridgeEvbCAPComponentId,
40        ieee8021BridgeEvbCapConfigIfIndex,
41        ieee8021BridgeEvbCAPPort,
42        ieee8021BridgeEvbCAPSChannelID,
43        ieee8021BridgeEvbCAPAssociateSBPForURPCompID,
44        ieee8021BridgeEvbCAPAssociateSBPForURPPort,
45        ieee8021BridgeEvbCAPRowStatus
46    }
47    STATUS      current
48    DESCRIPTION
49        "The collection of objects used to represent a EVB
50        CAP management objects."
51    ::= { ieee8021BridgeEvbGroups 6 }
52
53
54
55    ieee8021BridgeEvbsURPGroup OBJECT-GROUP
56    OBJECTS {
57        ieee8021BridgeEvbURPIfIndex,
58        ieee8021BridgeEvbURPBindToISSPort ,
59        ieee8021BridgeEvbURPLldpManual,
60        ieee8021BridgeEvbURPVdpOperRsrcWaitDelay,
61        ieee8021BridgeEvbURPVdpOperRespWaitDelay ,
62        ieee8021BridgeEvbURPVdpOperReinitKeepAlive
63    }
64    STATUS      deprecated
65    DESCRIPTION
66        "The collection of objects used to represent a EVBS URP
67        management objects."
68    ::= { ieee8021BridgeEvbGroups 7 }
69
70
71    ieee8021BridgeEvbEcpGroup OBJECT-GROUP
72    OBJECTS {

```

```

1      ieee8021BridgeEvpEcpOperAckTimerInit,
2      ieee8021BridgeEvpEcpOperMaxRetries ,
3      ieee8021BridgeEvpEcpTxFrameCount,
4      ieee8021BridgeEvpEcpTxRetryCount,
5      ieee8021BridgeEvpEcpTxFailures ,
6      ieee8021BridgeEvpEcpRxFrameCount
7      }
8      STATUS      deprecated
9      DESCRIPTION
10     "The collection of objects used to represent a EVB CAP
11     management objects."
12     ::= { ieee8021BridgeEvpGroups 8 }
13
14
15     ieee8021BridgeEvpSysV2Group OBJECT-GROUP
16     OBJECTS {
17         ieee8021BridgeEvpSysType,
18         ieee8021BridgeEvpSysNumExternalPorts,
19         ieee8021BridgeEvpSysEvpLldpTxEnable,
20         ieee8021BridgeEvpSysEvpLldpGidCapable,
21         ieee8021BridgeEvpSysEvpLldpManual,
22         ieee8021BridgeEvpSysEcpDfltAckTimerExp,
23         ieee8021BridgeEvpSysEcpMaxRetries,
24         ieee8021BridgeEvpSysVdpDfltRsrcWaitDelayExp,
25         ieee8021BridgeEvpSysVdpDfltReinitKeepAliveExp
26     }
27     STATUS      current
28     DESCRIPTION
29     "The collection of objects used to represent a EVB
30     management objects."
31     ::= { ieee8021BridgeEvpGroups 9 }
32
33     ieee8021BridgeEvpSbpV2Group OBJECT-GROUP
34     OBJECTS {
35         ieee8021BridgeEvpSbpLldpManual,
36         ieee8021BridgeEvpSbpVdpOperRsrcWaitDelayExp,
37         ieee8021BridgeEvpSbpVdpOperReinitKeepAliveExp,
38         ieee8021BridgeEvpSbpVdpOperToutKeepAlive
39     }
40     STATUS      current
41     DESCRIPTION
42     "The collection of objects used to represent a SBP
43     management objects."
44     ::= { ieee8021BridgeEvpGroups 10 }
45
46     ieee8021BridgeEvpVSIDBV2Group OBJECT-GROUP
47     OBJECTS {
48         ieee8021BridgeEvpVSItimeSinceCreate ,
49         ieee8021BridgeEvpVsiVdpOperCmd,
50         ieee8021BridgeEvpVsiOperRevert,
51         ieee8021BridgeEvpVsiOperHard,
52         ieee8021BridgeEvpVsiOperReason,
53         ieee8021BridgeEvpVSIIMgrID16,
54         ieee8021BridgeEvpVSIType,
55         ieee8021BridgeEvpVSITypeVersion ,
56         ieee8021BridgeEvpVSIFilterFormat,
57         ieee8021BridgeEvpVSINumMACs ,
58         ieee8021BridgeEvpVDPMachineState ,
59         ieee8021BridgeEvpVDPCCommandsSucceeded ,
60         ieee8021BridgeEvpVDPCCommandsFailed ,
61         ieee8021BridgeEvpVDPCCommandReverts ,
62         ieee8021BridgeEvpVDPCounterDiscontinuity,
63         ieee8021BridgeEvpVSIvlanId
64     }
65     STATUS      current
66     DESCRIPTION
67     "The collection of objects used to represent a EVB VSI
68     DB table."
69     ::= { ieee8021BridgeEvpGroups 11 }
70
71
72     ieee8021BridgeEvbsURPV2Group OBJECT-GROUP

```

```
1      OBJECTS {
2          ieee8021BridgeEvbURPIfIndex,
3          ieee8021BridgeEvbURPBindToISSPort ,
4          ieee8021BridgeEvbURPLldpManual,
5          ieee8021BridgeEvbURPVdpOperRsrcWaitDelayExp,
6          ieee8021BridgeEvbURPVdpOperRespWaitDelay ,
7          ieee8021BridgeEvbURPVdpOperReinitKeepAliveExp
8      }
9      STATUS current
10     DESCRIPTION
11         "The collection of objects used to represent a EVBS URP
12         management objects."
13     ::= { ieee8021BridgeEvbGroups 12 }
14
15 ieee8021BridgeEvbEcpV2Group OBJECT-GROUP
16     OBJECTS {
17         ieee8021BridgeEvbEcpOperAckTimerInitExp,
18         ieee8021BridgeEvbEcpOperMaxRetries ,
19         ieee8021BridgeEvbEcpTxFrameCount,
20         ieee8021BridgeEvbEcpTxRetryCount,
21         ieee8021BridgeEvbEcpTxFailures ,
22         ieee8021BridgeEvbEcpRxFrameCount
23     }
24     STATUS current
25     DESCRIPTION
26         "The collection of objects used to represent a EVB CAP
27         management objects."
28     ::= { ieee8021BridgeEvbGroups 13 }
29
30 -- =====
31 -- compliance statements
32 -- =====
33
34 ieee8021BridgeEvbbCompliance MODULE-COMPLIANCE
35     STATUS deprecated
36     DESCRIPTION
37         "The compliance statement for devices supporting EVB
38         as defined in IEEE 802.1Q."
39     MODULE
40         MANDATORY-GROUPS {
41             ieee8021BridgeEvbSysGroup,
42             ieee8021BridgeEvbVSIDBGroup,
43             ieee8021BridgeEvbSbpGroup,
44             ieee8021BridgeEvbEcpGroup
45         }
46
47         GROUP ieee8021BridgeEvbUAPGroup
48         DESCRIPTION "This group is mandatory when S-Channels
49         are present."
50
51         GROUP ieee8021BridgeEvbCAPConfigGroup
52         DESCRIPTION "This group is mandatory when S-Channels
53         are present."
54
55     ::= { ieee8021BridgeEvbCompliances 1 }
56
57 ieee8021BridgeEvbsCompliance MODULE-COMPLIANCE
58     STATUS deprecated
59     DESCRIPTION
60         "The compliance statement for devices supporting EVBS
61         as defined in IEEE 802.1Q."
62     MODULE
63         MANDATORY-GROUPS {
64             ieee8021BridgeEvbSysGroup,
65             ieee8021BridgeEvbVSIDBGroup,
66             ieee8021BridgeEvbsURPGroup,
67             ieee8021BridgeEvbEcpGroup
68         }
69
70         GROUP ieee8021BridgeEvbUAPGroup
71         DESCRIPTION "This group is mandatory when S-Channels
72         are present."
```

```
1
2     GROUP ieee8021BridgeEvbCAPConfigGroup
3     DESCRIPTION "This group is mandatory when S-Channels
4                 are present."
5
6     ::= { ieee8021BridgeEvbCompliances 2 }
7
8 ieee8021BridgeEvbbComplianceV2 MODULE-COMPLIANCE
9     STATUS         current
10    DESCRIPTION
11        "The compliance statement for devices supporting EVB
12        as defined in IEEE 802.1Q."
13    MODULE
14        MANDATORY-GROUPS {
15            ieee8021BridgeEvbSysV2Group,
16            ieee8021BridgeEvbVSIDBV2Group,
17            ieee8021BridgeEvbSbpV2Group,
18            ieee8021BridgeEvbEcpV2Group
19        }
20
21    GROUP ieee8021BridgeEvbUAPGroup
22    DESCRIPTION "This group is mandatory when S-Channels
23                are present."
24
25    GROUP ieee8021BridgeEvbCAPConfigGroup
26    DESCRIPTION "This group is mandatory when S-Channels
27                are present."
28
29    ::= { ieee8021BridgeEvbCompliances 3 }
30
31
32 ieee8021BridgeEvbsComplianceV2 MODULE-COMPLIANCE
33     STATUS         current
34     DESCRIPTION
35         "The compliance statement for devices supporting EVBS
36         as defined in IEEE 802.1Q."
37     MODULE
38         MANDATORY-GROUPS {
39             ieee8021BridgeEvbSysV2Group,
40             ieee8021BridgeEvbVSIDBV2Group,
41             ieee8021BridgeEvbsURPV2Group,
42             ieee8021BridgeEvbEcpV2Group
43         }
44
45     GROUP ieee8021BridgeEvbUAPGroup
46     DESCRIPTION "This group is mandatory when S-Channels
47                 are present."
48
49     GROUP ieee8021BridgeEvbCAPConfigGroup
50     DESCRIPTION "This group is mandatory when S-Channels
51                 are present."
52
53     ::= { ieee8021BridgeEvbCompliances 4 }
54
55 END
```

1 17.7.21 Definitions for the IEEE8021-ECMP-MIB module

```
2 IEEE8021-ECMP-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- IEEE 802.1 Equal Cost Multiple Paths (ECMP) MIB
6 -- =====
7
8 IMPORTS
9     MODULE-IDENTITY, OBJECT-TYPE, Integer32, Counter64
10    FROM SNMPv2-SMI
11    RowStatus, TruthValue
12    FROM SNMPv2-TC
13    ieee802dot1mibs
14    FROM IEEE8021-TC-MIB
15    PortList, VlanId
16    FROM Q-BRIDGE-MIB
17    ieee8021BridgeBasePortComponentId, ieee8021BridgeBasePort
18    FROM IEEE8021-BRIDGE-MIB
19    ieee8021QBridgeTpFdbEntry, ieee8021QBridgePortVlanStatisticsEntry
20    FROM IEEE8021-Q-BRIDGE-MIB
21    IEEE8021SpbBridgePriority, ieee8021SpbmTopSrvTableEntry,
22    ieee8021SpbTopIx
23    FROM IEEE8021-SPB-MIB
24    MODULE-COMPLIANCE, OBJECT-GROUP
25    FROM SNMPv2-CONF;
26
27 ieee8021EcmpMib MODULE-IDENTITY
28     LAST-UPDATED "202211080000Z" -- November 8, 2022
29     ORGANIZATION "IEEE 802.1 Working Group"
30     CONTACT-INFO
31         " WG-URL: http://www.ieee802.org/1/
32         WG-EMail: stds-802-1-l@ieee.org
33         Contact: IEEE 802.1 Working Group Chair
34         Postal: C/O IEEE 802.1 Working Group
35               IEEE Standards Association
36               445 Hoes Lane
37               Piscataway, NJ 08854
38               USA
39         E-mail: stds-802-1-chairs@ieee.org"
40     DESCRIPTION
41         "MIB Module for managing systems that provide
42         802.1Q Equal Cost Multiple Paths.
43
44         Unless otherwise indicated, the references in this MIB
45         module are to IEEE Std 802.1Q-2022.
46
47         Copyright (C) IEEE (2022).
48         This version of this MIB module is part of IEEE Std 802.1Q;
49         see that standard for full legal notices."
50
51     REVISION "202211080000Z" -- November 8, 2022
52     DESCRIPTION
53         "Published as part of IEEE Std 802.1Q-2022.
54         Cross references and contact information updated."
55
56     REVISION "201807010000Z" -- July 1, 2018
57     DESCRIPTION
58         "Published as part of IEEE Std 802.1Q 2018 revision.
59         Cross references updated and corrected.
60         Description replaced."
61
62     REVISION "201412150000Z" -- December 15, 2014
63     DESCRIPTION "Incorporated into IEEE Std 802.1Q 2014 Revision.
64         Cross-references and front matter updated."
65
66     REVISION "201305130000Z" -- May 13, 2013
67     DESCRIPTION "802.1 Equal Cost Multiple Paths MIB Initial Version"
68     ::= { ieee802dot1mibs 28 }
69
70 ieee8021EcmpNotifications OBJECT IDENTIFIER ::= { ieee8021EcmpMib 0 }
71 ieee8021EcmpObjects       OBJECT IDENTIFIER ::= { ieee8021EcmpMib 1 }
```



```

1 ieee8021EcmpConformance OBJECT IDENTIFIER ::= { ieee8021EcmpMib 2 }
2
3 -- =====
4 -- OBJECT DEFINITIONS
5 -- =====
6
7 -- =====
8 -- ECMP FDB object for Individual Addresses
9 -- =====
10
11 ieee8021QBridgeEcmpFdbTable OBJECT-TYPE
12     SYNTAX      SEQUENCE OF Ieee8021QBridgeEcmpFdbEntry
13     MAX-ACCESS  not-accessible
14     STATUS      current
15     DESCRIPTION
16         "A table that contains information about unicast entries
17         for which the device has forwarding and/or filtering
18         information. This information is used by the
19         ECMP next hop selection function in determining how to
20         propagate a received frame."
21     REFERENCE   "12.7.7.3, 8.8.3:c"
22     ::= { ieee8021EcmpObjects 1 }
23
24 ieee8021QBridgeEcmpFdbEntry OBJECT-TYPE
25     SYNTAX      Ieee8021QBridgeEcmpFdbEntry
26     MAX-ACCESS  not-accessible
27     STATUS      current
28     DESCRIPTION
29         "Information about a specific unicast MAC address for
30         which the device has some forwarding and/or filtering
31         information."
32     AUGMENTS { ieee8021QBridgeTpFdbEntry }
33     ::= { ieee8021QBridgeEcmpFdbTable 1 }
34
35 Ieee8021QBridgeEcmpFdbEntry ::=
36     SEQUENCE {
37         ieee8021QBridgeEcmpFdbPortList  PortList
38     }
39
40 ieee8021QBridgeEcmpFdbPortList OBJECT-TYPE
41     SYNTAX      PortList
42     MAX-ACCESS  read-only
43     STATUS      current
44     DESCRIPTION
45         "The complete set of ports, in this FID, to which
46         frames destined for this individual MAC address may be
47         forwarded."
48     ::= { ieee8021QBridgeEcmpFdbEntry 1 }
49
50 -- =====
51 -- Flow Filtering Control Table
52 -- =====
53
54 ieee8021EcmpFlowFilterCtlTable OBJECT-TYPE
55     SYNTAX      SEQUENCE OF Ieee8021EcmpFlowFilterCtlEntry
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59         "A table flow filtering control information for ports
60         in a Bridge supporting F-Tag processing."
61     REFERENCE   "12.16.5.4, 12.16.5.5"
62     ::= { ieee8021EcmpObjects 2 }
63
64 ieee8021EcmpFlowFilterCtlEntry OBJECT-TYPE
65     SYNTAX      Ieee8021EcmpFlowFilterCtlEntry
66     MAX-ACCESS  not-accessible
67     STATUS      current
68     DESCRIPTION
69         "An entry in the Flow Filtering Control Table for a
70         port (CPB or PNP)."
71     INDEX       { ieee8021BridgeBasePortComponentId,
72                 ieee8021BridgeBasePort,

```

```

1         ieee8021EcmpFlowFilterCtlVid }
2     ::= { ieee8021EcmpFlowFilterCtlTable 1 }
3
4 Ieee8021EcmpFlowFilterCtlEntry ::=
5     SEQUENCE {
6         ieee8021EcmpFlowFilterCtlVid          VlanId,
7         ieee8021EcmpFlowFilterCtlEnabled      TruthValue,
8         ieee8021EcmpFlowFilterCtlHashGen      TruthValue,
9         ieee8021EcmpFlowFilterCtlTtl          Integer32
10    }
11
12 ieee8021EcmpFlowFilterCtlVid OBJECT-TYPE
13     SYNTAX      VlanId
14     MAX-ACCESS  not-accessible
15     STATUS      current
16     DESCRIPTION
17         "A B-vID registered on the port."
18     ::= { ieee8021EcmpFlowFilterCtlEntry 1 }
19
20 ieee8021EcmpFlowFilterCtlEnabled OBJECT-TYPE
21     SYNTAX      TruthValue
22     MAX-ACCESS  read-only
23     STATUS      current
24     DESCRIPTION
25         "Indicates whether or not flow filtering behavior
26         is enabled on the port for the VID
27         (true(1) is enabled, false(2) is disabled)."

```

```

1  SYNTAX Ieee8021EcmpEctStaticEntry
2  MAX-ACCESS not-accessible
3  STATUS current
4  DESCRIPTION
5      "This entry contains information about backbone services
6      configured on this system to be advertised by ISIS-SPB."
7  REFERENCE "12.25.8"
8  INDEX {
9      ieee8021SpbTopIx,
10     ieee8021EcmpEctStaticEntryTieBreakMask
11 }
12 ::= { ieee8021EcmpEctStaticTable 1 }
13
14 Ieee8021EcmpEctStaticEntry ::=
15     SEQUENCE {
16         ieee8021EcmpEctStaticEntryTieBreakMask    Integer32,
17         ieee8021EcmpEctStaticEntryBridgePriority   IEEE8021SpbBridgePriority,
18         ieee8021EcmpEctStaticEntryRowStatus       RowStatus
19     }
20
21 ieee8021EcmpEctStaticEntryTieBreakMask OBJECT-TYPE
22     SYNTAX      Integer32 (0..15)
23     MAX-ACCESS  not-accessible
24     STATUS      current
25     DESCRIPTION
26         "The value used to create the Tie-Break Mask
27         for selecting a shared tree root bridge."
28     ::= { ieee8021EcmpEctStaticEntry 1 }
29
30 ieee8021EcmpEctStaticEntryBridgePriority OBJECT-TYPE
31     SYNTAX      IEEE8021SpbBridgePriority
32     MAX-ACCESS  read-create
33     STATUS      current
34     DESCRIPTION
35         "A Bridge Priority value to be used
36         for selecting a shared tree root bridge,
37         i.e., the most significant 4 bits of the
38         Bridge Identifier.
39         This object is persistent."
40     ::= { ieee8021EcmpEctStaticEntry 2 }
41
42 ieee8021EcmpEctStaticEntryRowStatus OBJECT-TYPE
43     SYNTAX      RowStatus
44     MAX-ACCESS  read-create
45     STATUS      current
46     DESCRIPTION
47         "This column holds the status for this row.
48
49         When the status is active, no columns of this table may be
50         modified. All columns must have a valid value before the row
51         can be activated.
52         This object is persistent."
53     ::= { ieee8021EcmpEctStaticEntry 3 }
54
55 -- =====
56 -- ECMP extensions to ieee8021SpbmTopSrvTable
57 -- =====
58
59 ieee8021EcmpTopSrvTable OBJECT-TYPE
60     SYNTAX      SEQUENCE OF Ieee8021EcmpTopSrvEntry
61     MAX-ACCESS  not-accessible
62     STATUS      current
63     DESCRIPTION
64         "Added info for SPBM PBB encapsulated services in this network."
65     REFERENCE "12.25.8"
66     ::= { ieee8021EcmpObjects 4 }
67
68 ieee8021EcmpTopSrvEntry OBJECT-TYPE
69     SYNTAX      Ieee8021EcmpTopSrvEntry
70     MAX-ACCESS  not-accessible
71     STATUS      current
72     DESCRIPTION

```

```

1      "This table contains additional information about backbone
2      services configured on this system to be advertised by
3      ISIS-SPB."
4  REFERENCE "12.25.8"
5  AUGMENTS { ieee8021SpbmTopSrvTableEntry }
6  ::= { ieee8021EcmpTopSrvTable 1 }
7
8  Ieee8021EcmpTopSrvEntry ::=
9  SEQUENCE {
10     ieee8021EcmpTopSrvEntryTsBit      TruthValue,
11     ieee8021EcmpTopSrvEntryTieBreakMask Integer32
12  }
13
14  ieee8021EcmpTopSrvEntryTsBit OBJECT-TYPE
15  SYNTAX      TruthValue
16  MAX-ACCESS  read-only
17  STATUS      current
18  DESCRIPTION
19      "If true(1), indicates the BSI transmits multicast
20      frames on a shared tree from this CBP."
21  ::= { ieee8021EcmpTopSrvEntry 1 }
22
23  ieee8021EcmpTopSrvEntryTieBreakMask OBJECT-TYPE
24  SYNTAX      Integer32 (0..15)
25  MAX-ACCESS  read-only
26  STATUS      current
27  DESCRIPTION
28      "The value used to create the Tie-Break Mask
29      for calculating multicast trees."
30  ::= { ieee8021EcmpTopSrvEntry 2 }
31
32  -- =====
33  -- Per port VLAN TTL Statistics Table
34  -- =====
35
36  ieee8021QBridgePortVlanTtlStatisticsTable OBJECT-TYPE
37  SYNTAX      SEQUENCE OF Ieee8021QBridgePortVlanTtlStatisticsEntry
38  MAX-ACCESS  not-accessible
39  STATUS      current
40  DESCRIPTION
41      "A table containing per-port, per-VID TTL discard statistics."
42  ::= { ieee8021EcmpObjects 5 }
43
44  ieee8021QBridgePortVlanTtlStatisticsEntry OBJECT-TYPE
45  SYNTAX      Ieee8021QBridgePortVlanTtlStatisticsEntry
46  MAX-ACCESS  not-accessible
47  STATUS      current
48  DESCRIPTION
49      "TTL discard statistics for a VID on an interface."
50  AUGMENTS { ieee8021QBridgePortVlanStatisticsEntry }
51  ::= { ieee8021QBridgePortVlanTtlStatisticsTable 1 }
52
53  Ieee8021QBridgePortVlanTtlStatisticsEntry ::=
54  SEQUENCE {
55     ieee8021QBridgeTpVlanPortTtlDiscards Counter64
56  }
57
58  ieee8021QBridgeTpVlanPortTtlDiscards OBJECT-TYPE
59  SYNTAX      Counter64
60  UNITS       "frames"
61  MAX-ACCESS  read-only
62  STATUS      current
63  DESCRIPTION
64      "The number of valid frames received by this port from
65      its segment that were classified as belonging to this
66      VLAN and that were discarded due to TTL expiry.
67      Discontinuities in the value of the counter can occur
68      at re-initialization of the management system, and at
69      other times as indicated by the value of
70      ifCounterDiscontinuityTime object of the associated
71      interface (if any)."
72  REFERENCE "12.6.1.1.3"

```

```

1      ::= { ieee8021QBridgePortVlanTtlStatisticsEntry 1 }
2
3 -- =====
4 -- Conformance Information
5 -- =====
6
7 ieee8021EcmpGroups      OBJECT IDENTIFIER ::= { ieee8021EcmpConformance 1 }
8 ieee8021EcmpCompliances OBJECT IDENTIFIER ::= { ieee8021EcmpConformance 2 }
9
10 -- =====
11 -- Units of conformance
12 -- =====
13
14 ieee8021QBridgeEcmpFdbGroup OBJECT-GROUP
15     OBJECTS {
16         ieee8021QBridgeEcmpFdbPortList
17     }
18     STATUS current
19     DESCRIPTION
20     "FDB Port Map for ECMP Individual address"
21     ::= { ieee8021EcmpGroups 1 }
22
23 ieee8021EcmpFlowFilterCtlGroup OBJECT-GROUP
24     OBJECTS {
25         ieee8021EcmpFlowFilterCtlEnabled,
26         ieee8021EcmpFlowFilterCtlHashGen,
27         ieee8021EcmpFlowFilterCtlTtl
28     }
29     STATUS current
30     DESCRIPTION
31     "Flow filtering control parameters on a CBP or PNP"
32     ::= { ieee8021EcmpGroups 2 }
33
34 ieee8021EcmpEctStaticGroup OBJECT-GROUP
35     OBJECTS {
36         ieee8021EcmpEctStaticEntryBridgePriority,
37         ieee8021EcmpEctStaticEntryRowStatus
38     }
39     STATUS current
40     DESCRIPTION
41     "Optional Bridge Priority for selecting shared tree root"
42     ::= { ieee8021EcmpGroups 3 }
43
44 ieee8021EcmpTopSrvGroup OBJECT-GROUP
45     OBJECTS {
46         ieee8021EcmpTopSrvEntryTsBit,
47         ieee8021EcmpTopSrvEntryTieBreakMask
48     }
49     STATUS current
50     DESCRIPTION
51     "Advertised I-SID parameters controlling multicast routing"
52     ::= { ieee8021EcmpGroups 4 }
53
54 ieee8021QBridgePortVlanTtlStatisticsGroup OBJECT-GROUP
55     OBJECTS {
56         ieee8021QBridgeTpVlanPortTtlDiscards
57     }
58     STATUS current
59     DESCRIPTION
60     "TTL discard statistics"
61     ::= { ieee8021EcmpGroups 5 }
62
63 -- =====
64 -- Compliance statements
65 -- =====
66
67 ieee8021EcmpCompliance MODULE-COMPLIANCE
68     STATUS current
69     DESCRIPTION
70     "Compliance to IEEE 802.1 SPBM ECMP"
71     MODULE
72     MANDATORY-GROUPS {

```

```
1         ieee8021QBridgeEcmpFdbGroup,  
2         ieee8021EcmpEctStaticGroup,  
3         ieee8021EcmpTopSrvGroup  
4     }  
5     ::= { ieee8021EcmpCompliances 1 }  
6  
7 ieee8021EcmpFlowFilteringCompliance MODULE-COMPLIANCE  
8     STATUS current  
9     DESCRIPTION  
10        "Compliance to IEEE 802.1 SPBM ECMP with flow filtering"  
11     MODULE  
12        MANDATORY-GROUPS {  
13            ieee8021EcmpFlowFilterCtlGroup,  
14            ieee8021QBridgePortVlanTtlStatisticsGroup  
15        }  
16        ::= { ieee8021EcmpCompliances 2 }  
17  
18 END
```

1 17.7.22 Definitions for the IEEE8021-ST-MIB module

```
2 IEEE8021-ST-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for support of the Scheduled Traffic Enhancements
6 -- for IEEE 802.1Q Bridges.
7 -- =====
8
9
10 IMPORTS
11     MODULE-IDENTITY,
12     OBJECT-TYPE,
13     Unsigned32,
14     Counter64
15     FROM SNMPv2-SMI
16     TEXTUAL-CONVENTION,
17     TruthValue
18     FROM SNMPv2-TC
19     MODULE-COMPLIANCE,
20     OBJECT-GROUP
21     FROM SNMPv2-CONF
22     ieee802dot1mibs
23     FROM IEEE8021-TC-MIB
24     ieee8021BridgeBaseComponentId,
25     ieee8021BridgeBasePort
26     FROM IEEE8021-BRIDGE-MIB
27 ;
28
29 ieee8021STMib MODULE-IDENTITY
30     LAST-UPDATED "202309260000Z" -- September 26, 2023
31     ORGANIZATION "IEEE 802.1 Working Group"
32     CONTACT-INFO
33         " WG-URL: http://www.ieee802.org/1/
34         WG-EMail: stds-802-1-1@ieee.org
35         Contact: IEEE 802.1 Working Group Chair
36         Postal: C/O IEEE 802.1 Working Group
37             IEEE Standards Association
38             445 Hoes Lane
39             Piscataway, NJ 08854
40             USA
41         E-mail: stds-802-1-chairs@ieee.org"
42     DESCRIPTION
43         "The Bridge MIB module for managing devices that support
44         the Scheduled Traffic Enhancements
45         for IEEE 802.1Q Bridges.
46
47         Unless otherwise indicated, the references in this MIB
48         module are to IEEE Std 802.1Q-2022 as amended by
49         IEEE Std 802.1Qcz and IEEE Std 802.1Qcw.
50
51         Copyright (C) IEEE (2023).
52         This version of this MIB module is part of IEEE Std 802.1Q;
53         see that standard for full legal notices."
54
55     REVISION "202309260000Z" -- September 26, 2023
56     DESCRIPTION
57         "Update to include SupportedCycleMax and
58         SupportedIntervalMax parameters.
59         Published as part of IEEE Std 802.1Qcw-2023."
60
61     REVISION "202211080000Z" -- November 8, 2022
62     DESCRIPTION
63         "Published as part of IEEE Std 802.1Q-2022.
64         Cross references and contact information updated."
65
66     REVISION "201807010000Z" -- July 1, 2018
67     DESCRIPTION
68         "Published as part of IEEE Std 802.1Q 2018 revision.
69         Cross references updated and corrected."
70
71     REVISION "201608150000Z" -- August 15, 2016
```

```
1      DESCRIPTION
2      "Revised to include Set-And-Hold-MAC and
3      Set-And-Release-MAC in the description of
4      ieee8021STAdminControlList and
5      ieee8021STOperControlList.
6      Published as part of IEEE Std 802.1Qbu."
7
8 REVISION "201602190000Z" -- February 19, 2016
9      DESCRIPTION
10     "Initial version published as part of IEEE Std 802.1Qbv."
11
12     ::= { ieee802dot1mibs 30 }
13
14 -- =====
15 -- Textual Conventions
16 -- =====
17
18 IEEE8021STTrafficClassValue ::= TEXTUAL-CONVENTION
19     DISPLAY-HINT "d"
20     STATUS      current
21     DESCRIPTION
22         "A traffic class value.
23         This is the numerical value associated with a traffic
24         class in a Bridge. Larger values are associated with
25         higher priority traffic classes."
26     REFERENCE   "12.29.1"
27     SYNTAX      Unsigned32 (0..7)
28
29 IEEE8021STPTPtimeValue ::= TEXTUAL-CONVENTION
30     STATUS      current
31     DESCRIPTION
32         "A PTPtime value, represented as a 48-bit unsigned integer
33         number of seconds and a 32-bit unsigned integer number of
34         nanoseconds.
35         The first 6 octets represent the number of seconds: the
36         first octet is the most significant
37         octet of the 48-bit seconds value and the sixth octet
38         is the least significant octet of the seconds value.
39         The remaining octets, 7 through 10, represent the
40         number of nanoseconds: the seventh octet
41         is the most significant octet of the 32-bit nanoseconds
42         value and the tenth octet is the
43         least significant octet of the nanoseconds value."
44     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
45     SYNTAX      OCTET STRING (SIZE(10))
46
47
48 -- =====
49 -- subtrees in the ST MIB
50 -- =====
51
52 ieee8021STNotifications
53     OBJECT IDENTIFIER ::= { ieee8021STMib 0 }
54
55 ieee8021STObjects
56     OBJECT IDENTIFIER ::= { ieee8021STMib 1 }
57
58 ieee8021STConformance
59     OBJECT IDENTIFIER ::= { ieee8021STMib 2 }
60
61 ieee8021STMaxSDUSubtree
62     OBJECT IDENTIFIER ::= { ieee8021STObjects 1 }
63
64 ieee8021STParameters
65     OBJECT IDENTIFIER ::= { ieee8021STObjects 2 }
66
67 -- =====
68 -- The ieee8021STMaxSDUSubtree subtree
69 -- This subtree defines the objects necessary for the management
70 -- of the max SDU size parameters for each traffic class on a Port.
71 -- =====
72
```



```

1 -- =====
2 -- the ieee8021STMaxSDUTable
3 -- =====
4
5 ieee8021STMaxSDUTable OBJECT-TYPE
6     SYNTAX      SEQUENCE OF Ieee8021STMaxSDUEntry
7     MAX-ACCESS  not-accessible
8     STATUS      current
9     DESCRIPTION
10        "A table containing a set of max SDU
11         parameters, one for each traffic class.
12         All writable objects in this table must be
13         persistent over power up restart/reboot."
14     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
15     ::= { ieee8021STMaxSDUSubtree 1 }
16
17 ieee8021STMaxSDUEntry OBJECT-TYPE
18     SYNTAX      Ieee8021STMaxSDUEntry
19     MAX-ACCESS  not-accessible
20     STATUS      current
21     DESCRIPTION
22        "A list of objects containing Max SDU size
23         for each traffic class supported by the Port."
24     INDEX { ieee8021BridgeBaseComponentId,
25             ieee8021BridgeBasePort,
26             ieee8021STTrafficClass }
27     ::= { ieee8021STMaxSDUTable 1 }
28
29 Ieee8021STMaxSDUEntry ::=
30     SEQUENCE {
31         ieee8021STTrafficClass
32         IEEE8021STTrafficClassValue,
33         ieee8021STMaxSDU
34         Unsigned32,
35         ieee8021TransmissionOverrun
36         Counter64
37     }
38
39 ieee8021STTrafficClass OBJECT-TYPE
40     SYNTAX      IEEE8021STTrafficClassValue
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44        "The traffic class number associated with the row of
45         the table.
46
47         A row in this table is created for each traffic class
48         that is supported by the Port"
49     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
50     ::= { ieee8021STMaxSDUEntry 1 }
51
52
53 ieee8021STMaxSDU OBJECT-TYPE
54     SYNTAX      Unsigned32
55     UNITS       "octets"
56     MAX-ACCESS  read-write
57     STATUS      current
58     DESCRIPTION
59        "The value of the MaxSDU parameter for the traffic class.
60         This value is represented as an unsigned integer. A value
61         of 0 is interpreted as the max SDU size supported by
62         the underlying MAC.
63
64         The default value of the MaxSDU parameter is 0.
65
66         The value of this object MUST be retained across
67         reinitializations of the management system."
68     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
69     DEFVAL { 0 }
70     ::= { ieee8021STMaxSDUEntry 2 }
71
72

```

```

1 ieee8021TransmissionOverrun OBJECT-TYPE
2     SYNTAX      Counter64
3     MAX-ACCESS  read-only
4     STATUS      current
5     DESCRIPTION
6         "A counter of transmission overrun events, where
7         a PDU is still being transmitted by a MAC at the
8         time when the transmission gate for the queue closed."
9     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1, 12.29.1.1.2"
10    DEFVAL { 0 }
11    ::= { ieee8021STMaxSDUEntry 3}
12
13
14 -- =====
15 -- The ieee8021STParameters subtree
16 -- This subtree defines the objects necessary for the management
17 -- of the traffic scheduling mechanism for IEEE Std 802.1Q.
18 -- =====
19
20 -- =====
21 -- the ieee8021STParametersTable
22 -- =====
23
24 ieee8021STParametersTable OBJECT-TYPE
25     SYNTAX      SEQUENCE OF Ieee8021STParametersEntry
26     MAX-ACCESS  not-accessible
27     STATUS      current
28     DESCRIPTION
29         "A table that contains the per-port manageable parameters for
30         traffic scheduling.
31
32         For a given Port, a row in the table exists.
33
34         All writable objects in this table must be
35         persistent over power up restart/reboot."
36     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
37     ::= { ieee8021STParameters 1 }
38
39 ieee8021STParametersEntry OBJECT-TYPE
40     SYNTAX      Ieee8021STParametersEntry
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44         "A list of objects that contains the manageable parameters for
45         traffic scheduling for a port."
46     INDEX { ieee8021BridgeBaseComponentId,
47             ieee8021BridgeBasePort
48           }
49     ::= { ieee8021STParametersTable 1 }
50
51 Ieee8021STParametersEntry ::=
52     SEQUENCE {
53         ieee8021STGateEnabled
54             TruthValue,
55         ieee8021STAdminGateStates
56             OCTET STRING,
57         ieee8021STOperGateStates
58             OCTET STRING,
59         ieee8021STAdminControlListLength
60             Unsigned32,
61         ieee8021STOperControlListLength
62             Unsigned32,
63         ieee8021STAdminControlList
64             OCTET STRING,
65         ieee8021STOperControlList
66             OCTET STRING,
67         ieee8021STAdminCycleTimeNumerator
68             Unsigned32,
69         ieee8021STAdminCycleTimeDenominator
70             Unsigned32,
71         ieee8021STOperCycleTimeNumerator
72             Unsigned32,

```

```

1      ieee8021STOperCycleTimeDenominator
2          Unsigned32,
3      ieee8021STAdminCycleTimeExtension
4          Unsigned32,
5      ieee8021STOperCycleTimeExtension
6          Unsigned32,
7      ieee8021STAdminBaseTime
8          IEEE8021STPTPtimeValue,
9      ieee8021STOperBaseTime
10         IEEE8021STPTPtimeValue,
11      ieee8021STConfigChange
12         TruthValue,
13      ieee8021STConfigChangeTime
14         IEEE8021STPTPtimeValue,
15      ieee8021STTickGranularity
16         Unsigned32,
17      ieee8021STCurrentTime
18         IEEE8021STPTPtimeValue,
19      ieee8021STConfigPending
20         TruthValue,
21      ieee8021STConfigChangeError
22         Counter64,
23      ieee8021STSupportedListMax
24         Unsigned32,
25      ieee8021STSupportedCycleMaxNumerator
26         Unsigned32,
27      ieee8021STSupportedCycleMaxDenominator
28         Unsigned32,
29      ieee8021STSupportedIntervalMax
30         Unsigned32
31      }
32
33 ieee8021STGateEnabled OBJECT-TYPE
34     SYNTAX      TruthValue
35     MAX-ACCESS  read-write
36     STATUS      current
37     DESCRIPTION
38         "The GateEnabled parameter determines whether traffic scheduling
39         is active (true) or inactive (false).
40
41         The value of this object MUST be retained across
42         reinitializations of the management system."
43     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
44     DEFVAL { false }
45     ::= { ieee8021STParametersEntry 1 }
46
47 ieee8021STAdminGateStates OBJECT-TYPE
48     SYNTAX      OCTET STRING (SIZE(1))
49     MAX-ACCESS  read-write
50     STATUS      current
51     DESCRIPTION
52         "The administrative value of the GateStates parameter for the Port.
53         The bits of the octet represent the gate states for the
54         corresponding traffic classes; the MS bit corresponds to traffic class 7,
55         the LS bit to traffic class 0. A bit value of 0 indicates closed; a
56         bit value of 1 indicates open.
57
58         The value of this object MUST be retained across
59         reinitializations of the management system."
60     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
61     ::= { ieee8021STParametersEntry 2 }
62
63 ieee8021STOperGateStates OBJECT-TYPE
64     SYNTAX      OCTET STRING (SIZE(1))
65     MAX-ACCESS  read-only
66     STATUS      current
67     DESCRIPTION
68         "The operational value of the GateStates parameter for the Port.
69         The bits of the octet represent the gate states for the
70         corresponding traffic classes; the MS bit corresponds to traffic class 7,
71         the LS bit to traffic class 0. A bit value of 0 indicates closed; a
72         bit value of 1 indicates open."

```

```

1  REFERENCE    "8.6.8.4, 8.6.9.4, 12.29.1"
2  ::= { ieee8021STParametersEntry 3 }
3
4  ieee8021STAdminControlListLength OBJECT-TYPE
5  SYNTAX      Unsigned32
6  MAX-ACCESS  read-write
7  STATUS      current
8  DESCRIPTION
9      "The administrative value of the ListMax parameter for the Port.
10     The integer value indicates the number of entries (TLVs) in the
11     AdminControlList.
12
13     The value of this object MUST be retained across
14     reinitializations of the management system."
15  REFERENCE    "8.6.8.4, 8.6.9.4, 12.29.1"
16  ::= { ieee8021STParametersEntry 4 }
17
18  ieee8021STOperControlListLength OBJECT-TYPE
19  SYNTAX      Unsigned32
20  MAX-ACCESS  read-only
21  STATUS      current
22  DESCRIPTION
23      "The operational value of the ListMax parameter for the Port.
24      The integer value indicates the number of entries (TLVs) in the
25      OperControlList."
26  REFERENCE    "8.6.8.4, 8.6.9.4, 12.29.1"
27  ::= { ieee8021STParametersEntry 5 }
28
29  ieee8021STAdminControlList OBJECT-TYPE
30  SYNTAX      OCTET STRING
31  MAX-ACCESS  read-write
32  STATUS      current
33  DESCRIPTION
34      "The administrative value of the ControlList parameter for the Port.
35      The octet string value represents the contents of the control list as
36      an ordered list of entries, each encoded as a TLV, as follows.
37      The first octet of each TLV is interpreted as an
38      unsigned integer representing a gate operation name:
39          0: SetGateStates
40          1: Set-And-Hold-MAC
41          2: Set-And-Release-MAC
42          3-255: Reserved for future gate operations
43
44      The second octet of the TLV is the length field,
45      interpreted as an unsigned integer, indicating the number of
46      octets of the value that follows the length. A length of
47      zero indicates that there is no value
48      (i.e., the gate operation has no parameters).
49
50      The third through (3 + length -1)th octets encode the
51      parameters of the gate operation, in the order that they
52      appear in the definition of the operation
53      in Table 8-6. Two parameter types are currently defined:
54
55      - GateState:
56          A GateState parameter is encoded in a single octet.
57          The bits of the octet represent the gate states for the
58          corresponding traffic classes; the MS bit corresponds
59          to traffic class 7,
60          the LS bit to traffic class 0. A bit value of 0 indicates
61          closed; a bit value of 1 indicates open.
62
63      - TimeInterval:
64          A TimeInterval is encoded in 4 octets as a 32-bit
65          unsigned integer, representing a number of nanoseconds.
66          The first octet encodes the most significant 8 bits of the
67          integer, and the fourth octet encodes the least
68          significant 8 bits.
69
70      The value of this object MUST be retained across
71      reinitializations of the management system."
72  REFERENCE    "8.6.8.4, 8.6.9.4, 12.29.1"

```

```

1 ::= { ieee8021STParametersEntry 6 }
2
3 ieee8021STOperControlList OBJECT-TYPE
4     SYNTAX      OCTET STRING
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "The operational value of the ListMax parameter for the Port.
9         The octet string value represents the contents of the control list as
10        an ordered list of TLVs, as follows.
11        The first octet of each TLV is interpreted as a gate operation name:
12            0: SetGateStates
13            1: Set-And-Hold-MAC
14            2: Set-And-Release-MAC
15            3-255: Reserved for future gate operations
16
17        The second octet of the TLV is the length field,
18        interpreted as an unsigned integer,
19        indicating the number of octets of the value that follows
20        the length. A length of zero indicates that there is no value
21        (i.e., the gate operation has no parameters).
22
23        The third through (3 + length -1)th octets encode the
24        parameters of the gate operation, in the order that they
25        appear in the definition of the operation
26        in Table 8-6. Two parameter types are currently defined:
27
28        - GateState:
29            A GateState parameter is encoded in a single octet.
30            The bits of the octet represent the gate states for the
31            corresponding traffic classes; the MS bit corresponds to
32            traffic class 7, the LS bit to traffic class 0.
33            A bit value of 0 indicates closed; a
34            bit value of 1 indicates open.
35
36        - TimeInterval:
37            A TimeInterval is encoded in 4 octets as a 32-bit
38            unsigned integer, representing
39            a number of nanoseconds. The first octet encodes the
40            most significant 8 bits of the integer, and the fourth
41            octet encodes the least significant 8 bits."
42     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
43 ::= { ieee8021STParametersEntry 7 }
44
45 ieee8021STAdminCycleTimeNumerator OBJECT-TYPE
46     SYNTAX      Unsigned32
47     MAX-ACCESS  read-write
48     STATUS      current
49     DESCRIPTION
50         "The administrative value of the numerator of the CycleTime
51         parameter for the Port.
52         The numerator and denominator together represent the cycle time as
53         a rational number of seconds.
54
55         The value of this object MUST be retained across
56         reinitializations of the management system."
57     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
58 ::= { ieee8021STParametersEntry 8 }
59
60 ieee8021STAdminCycleTimeDenominator OBJECT-TYPE
61     SYNTAX      Unsigned32
62     MAX-ACCESS  read-write
63     STATUS      current
64     DESCRIPTION
65         "The administrative value of the denominator of the
66         CycleTime parameter for the Port.
67         The numerator and denominator together represent the cycle time as
68         a rational number of seconds.
69
70         The value of this object MUST be retained across
71         reinitializations of the management system."
72     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"

```

```

1 ::= { ieee8021STParametersEntry 9 }
2
3 ieee8021STOperCycleTimeNumerator OBJECT-TYPE
4     SYNTAX      Unsigned32
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "The operational value of the numerator of the
9         CycleTime parameter for the Port.
10        The numerator and denominator together represent the cycle
11        time as a rational number of seconds."
12     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
13 ::= { ieee8021STParametersEntry 10 }
14
15 ieee8021STOperCycleTimeDenominator OBJECT-TYPE
16     SYNTAX      Unsigned32
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20         "The operational value of the denominator of the
21         CycleTime parameter for the Port.
22        The numerator and denominator together represent the
23        cycle time as a rational number of seconds."
24     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
25 ::= { ieee8021STParametersEntry 11 }
26
27 ieee8021STAdminCycleTimeExtension OBJECT-TYPE
28     SYNTAX      Unsigned32
29     UNITS       "nanoseconds"
30     MAX-ACCESS  read-write
31     STATUS      current
32     DESCRIPTION
33         "The administrative value of the CycleTimeExtension
34         parameter for the Port.
35        The value is an unsigned integer number of nanoseconds.
36
37        The value of this object MUST be retained across
38        reinitializations of the management system."
39     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
40 ::= { ieee8021STParametersEntry 12 }
41
42 ieee8021STOperCycleTimeExtension OBJECT-TYPE
43     SYNTAX      Unsigned32
44     UNITS       "nanoseconds"
45     MAX-ACCESS  read-only
46     STATUS      current
47     DESCRIPTION
48         "The operational value of the CycleTimeExtension parameter for the Port.
49        The value is an unsigned integer number of nanoseconds."
50     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
51 ::= { ieee8021STParametersEntry 13 }
52
53 ieee8021STAdminBaseTime OBJECT-TYPE
54     SYNTAX      IEEE8021STPTPtimeValue
55     UNITS       "PTP time"
56     MAX-ACCESS  read-write
57     STATUS      current
58     DESCRIPTION
59         "The administrative value of the BaseTime parameter for the Port.
60        The value is a representation of a PTPtime value,
61        consisting of a 48-bit integer
62        number of seconds and a 32-bit integer number of nanoseconds.
63
64        The value of this object MUST be retained across
65        reinitializations of the management system."
66     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
67 ::= { ieee8021STParametersEntry 14 }
68
69 ieee8021STOperBaseTime OBJECT-TYPE
70     SYNTAX      IEEE8021STPTPtimeValue
71     UNITS       "PTP time"
72     MAX-ACCESS  read-only

```

```

1  STATUS      current
2  DESCRIPTION
3      "The operational value of the BaseTime parameter for the Port.
4      The value is a representation of a PTPtime value,
5      consisting of a 48-bit integer
6      number of seconds and a 32-bit integer number of nanoseconds."
7  REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
8  ::= { ieee8021STParametersEntry 15 }
9
10 ieee8021STConfigChange OBJECT-TYPE
11     SYNTAX      TruthValue
12     MAX-ACCESS   read-write
13     STATUS      current
14     DESCRIPTION
15         "The ConfigChange parameter signals the start of a
16         configuration change
17         when it is set to TRUE. This should only be done
18         when the various administrative parameters
19         are all set to appropriate values."
20     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
21     ::= { ieee8021STParametersEntry 16 }
22
23 ieee8021STConfigChangeTime OBJECT-TYPE
24     SYNTAX      IEEE8021STPTPtimeValue
25     UNITS       "PTP time"
26     MAX-ACCESS   read-only
27     STATUS      current
28     DESCRIPTION
29         "The PTPtime at which the next config change is scheduled to occur.
30         The value is a representation of a PTPtime value,
31         consisting of a 48-bit integer
32         number of seconds and a 32-bit integer number of nanoseconds.
33
34         The value of this object MUST be retained across
35         reinitializations of the management system."
36     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
37     ::= { ieee8021STParametersEntry 17 }
38
39     ieee8021STTickGranularity OBJECT-TYPE
40     SYNTAX      Unsigned32
41     MAX-ACCESS   read-only
42     STATUS      current
43     DESCRIPTION
44         "The granularity of the cycle time clock, represented as an
45         unsigned number of tenths of nanoseconds.
46
47         The value of this object MUST be retained across
48         reinitializations of the management system."
49     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
50     ::= { ieee8021STParametersEntry 18 }
51
52 ieee8021STCurrentTime OBJECT-TYPE
53     SYNTAX      IEEE8021STPTPtimeValue
54     MAX-ACCESS   read-only
55     STATUS      current
56     DESCRIPTION
57         "The current time, in PTPtime, as maintained by the local system.
58         The value is a representation of a PTPtime value,
59         consisting of a 48-bit integer
60         number of seconds and a 32-bit integer number of nanoseconds."
61     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"
62     ::= { ieee8021STParametersEntry 19 }
63
64 ieee8021STConfigPending OBJECT-TYPE
65     SYNTAX      TruthValue
66     MAX-ACCESS   read-only
67     STATUS      current
68     DESCRIPTION
69         "The value of the ConfigPending state machine variable.
70         The value is TRUE if a configuration change is in progress
71         but has not yet completed."
72     REFERENCE   "8.6.8.4, 8.6.9.4, 12.29.1"

```

```

1      ::= { ieee8021STParametersEntry 20 }
2
3  ieee8021STConfigChangeError OBJECT-TYPE
4      SYNTAX      Counter64
5      MAX-ACCESS  read-only
6      STATUS      current
7      DESCRIPTION
8          "A counter of the number of times that a re-configuration
9          of the traffic schedule has been requested with the old
10         schedule still running and the requested base time was
11         in the past."
12     REFERENCE   "8.6.8.4, 8.6.9.3, 12.29.1"
13     ::= { ieee8021STParametersEntry 21 }
14
15 ieee8021STSupportedListMax OBJECT-TYPE
16     SYNTAX      Unsigned32
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20         "The maximum value supported by this Port of the
21         AdminControlListLength and OperControlListLength
22         parameters."
23     REFERENCE   "12.29.1.5"
24     ::= { ieee8021STParametersEntry 22 }
25
26 ieee8021STSupportedCycleMaxNumerator OBJECT-TYPE
27     SYNTAX      Unsigned32
28     MAX-ACCESS  read-only
29     STATUS      current
30     DESCRIPTION
31         "The numerator of the SupportedCycleMax parameter for
32         the Port. The numerator and denominator together
33         represent the maximum value supported by this Port of
34         the AdminCycleTime and OperCycleTime parameters"
35     REFERENCE   "12.29.1.6"
36     ::= { ieee8021STParametersEntry 23 }
37
38 ieee8021STSupportedCycleMaxDenominator OBJECT-TYPE
39     SYNTAX      Unsigned32
40     MAX-ACCESS  read-only
41     STATUS      current
42     DESCRIPTION
43         "The denominator of the SupportedCycleMax parameter for
44         the Port. The numerator and denominator together
45         represent the maximum value supported by this Port of
46         the AdminCycleTime and OperCycleTime parameters"
47     REFERENCE   "12.29.1.6"
48     ::= { ieee8021STParametersEntry 24 }
49
50 ieee8021STSupportedIntervalMax OBJECT-TYPE
51     SYNTAX      Unsigned32
52     MAX-ACCESS  read-only
53     STATUS      current
54     DESCRIPTION
55         "The maximum value supported by this Port of the
56         TimeIntervalValue parameter."
57     REFERENCE   "12.29.1.7"
58     ::= { ieee8021STParametersEntry 25 }
59
60 -- =====
61 -- IEEE8021 ST MIB - Conformance Information
62 -- =====
63
64 ieee8021STCompliances
65     OBJECT IDENTIFIER ::= { ieee8021STConformance 1 }
66 ieee8021STGroups
67     OBJECT IDENTIFIER ::= { ieee8021STConformance 2 }
68
69 -- =====
70 -- units of conformance
71 -- =====
72

```



```
1 -- =====
2 -- the ieee8021STObjectsGroup group
3 -- =====
4
5 ieee8021STObjectsGroup OBJECT-GROUP
6     OBJECTS {
7         ieee8021STMaxSDU,
8         ieee8021STTransmissionOverrun,
9         ieee8021STGateEnabled,
10        ieee8021STAdminGateStates,
11        ieee8021STOperGateStates,
12        ieee8021STAdminControlListLength,
13        ieee8021STOperControlListLength,
14        ieee8021STAdminControlList,
15        ieee8021STOperControlList,
16        ieee8021STAdminCycleTimeNumerator,
17        ieee8021STAdminCycleTimeDenominator,
18        ieee8021STOperCycleTimeNumerator,
19        ieee8021STOperCycleTimeDenominator,
20        ieee8021STAdminCycleTimeExtension,
21        ieee8021STOperCycleTimeExtension,
22        ieee8021STAdminBaseTime,
23        ieee8021STOperBaseTime,
24        ieee8021STConfigChange,
25        ieee8021STConfigChangeTime,
26        ieee8021STTickGranularity,
27        ieee8021STCurrentTime,
28        ieee8021STConfigPending,
29        ieee8021STConfigChangeError,
30        ieee8021STSupportedListMax,
31        ieee8021STSupportedCycleMaxNumerator,
32        ieee8021STSupportedCycleMaxDenominator,
33        ieee8021STSupportedIntervalMax
34    }
35     STATUS      current
36     DESCRIPTION
37         "Objects that allow management of scheduled traffic."
38     ::= { ieee8021STGroups 1 }
39
40
41 -- =====
42 -- compliance statements
43 -- =====
44
45 ieee8021STCompliance MODULE-COMPLIANCE
46     STATUS      current
47     DESCRIPTION
48         "The compliance statement for devices supporting
49         scheduled traffic.
50
51         Support of the objects defined in this MIB module
52         also requires support of the IEEE8021-BRIDGE-MIB; the
53         provisions of 17.3.2 apply to implementations claiming
54         support of this MIB. "
55
56     MODULE -- this module
57         MANDATORY-GROUPS {
58             ieee8021STObjectsGroup
59         }
60
61     ::= { ieee8021STCompliances 1 }
62
63 END
```

1 17.7.23 Definitions for the IEEE8021-Preemption-MIB module

```
2 IEEE8021-Preemption-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for support of the frame preemption enhancements
6 -- for IEEE 802.1Q Bridges.
7 -- =====
8
9 IMPORTS
10     MODULE-IDENTITY,
11     OBJECT-TYPE,
12     Unsigned32
13         FROM SNMPv2-SMI
14     MODULE-COMPLIANCE,
15     OBJECT-GROUP
16         FROM SNMPv2-CONF
17     ieee802dot1mibs,
18     IEEE8021PriorityValue
19         FROM IEEE8021-TC-MIB
20     ieee8021BridgeBaseComponentId,
21     ieee8021BridgeBasePort
22         FROM IEEE8021-BRIDGE-MIB
23 ;
24
25 ieee8021PreemptionMib MODULE-IDENTITY
26     LAST-UPDATED "202211080000Z" -- November 8, 2022
27     ORGANIZATION "IEEE 802.1 Working Group"
28     CONTACT-INFO
29         " WG-URL: http://www.ieee802.org/1/
30         WG-EMail: stds-802-1-1@ieee.org
31         Contact: IEEE 802.1 Working Group Chair
32         Postal: C/O IEEE 802.1 Working Group
33               IEEE Standards Association
34               445 Hoes Lane
35               Piscataway, NJ 08854
36               USA
37         E-mail: stds-802-1-chairs@ieee.org"
38     DESCRIPTION
39         "The Bridge MIB module for managing devices that support
40         the frame preemption enhancements
41         for IEEE 802.1Q Bridges.
42
43         Unless otherwise indicated, the references in this MIB
44         module are to IEEE Std 802.1Q-2022.
45
46         Copyright (C) IEEE (2022).
47         This version of this MIB module is part of IEEE Std 802.1Q;
48         see that standard for full legal notices."
49
50     REVISION "202211080000Z" -- November 8, 2022
51     DESCRIPTION
52         "Published as part of IEEE Std 802.1Q-2022.
53         Cross references and contact information updated."
54
55     REVISION "201807010000Z" -- July 1, 2018
56     DESCRIPTION
57         "Published as part of IEEE Std 802.1Q 2018 revision.
58         Cross references updated and corrected."
59
60     REVISION "201608150000Z" -- August 15, 2016
61     DESCRIPTION
62         "Initial version published as part of IEEE Std 802.1Qbu."
63
64     ::= { ieee802dot1mibs 29 }
65
66
67 -- =====
68 -- subtrees in the Preemption MIB
69 -- =====
70
71 ieee8021PreemptionNotifications
```

```

1  OBJECT IDENTIFIER ::= { ieee8021PreemptionMib 0 }
2
3  ieee8021PreemptionObjects
4  OBJECT IDENTIFIER ::= { ieee8021PreemptionMib 1 }
5
6  ieee8021PreemptionConformance
7  OBJECT IDENTIFIER ::= { ieee8021PreemptionMib 2 }
8
9  ieee8021PreemptionParameters
10 OBJECT IDENTIFIER ::= { ieee8021PreemptionObjects 1 }
11
12 -- =====
13 -- The ieee8021PreemptionNotifications subtree
14 -- This subtree defines any notifications necessary for the management
15 -- of frame preemption on a Port. This subtree is currently unused,
16 -- but is retained as a place-holder for future standardization.
17 -- =====
18
19
20 -- =====
21 -- The ieee8021PreemptionParameters subtree
22 -- This subtree defines the objects necessary for the management
23 -- of the frame preemption parameters for each priority value
24 -- on a Port.
25 -- =====
26
27 -- =====
28 -- the ieee8021PreemptionParameterTable
29 -- =====
30
31 ieee8021PreemptionParameterTable OBJECT-TYPE
32 SYNTAX      SEQUENCE OF Ieee8021PreemptionParameterEntry
33 MAX-ACCESS  not-accessible
34 STATUS      current
35 DESCRIPTION
36     "A table containing a set of frame preemption
37     parameters, one for each priority value.
38     All writable objects in this table must be
39     persistent over power up restart/reboot."
40 REFERENCE   "6.7.2, 12.30.1"
41 ::= { ieee8021PreemptionParameters 1 }
42
43 ieee8021PreemptionParameterEntry OBJECT-TYPE
44 SYNTAX      Ieee8021PreemptionParameterEntry
45 MAX-ACCESS  not-accessible
46 STATUS      current
47 DESCRIPTION
48     "A list of objects containing preemption parameters
49     for each priority value."
50 INDEX { ieee8021BridgeBaseComponentId,
51         ieee8021BridgeBasePort,
52         ieee8021PreemptionPriority }
53 ::= { ieee8021PreemptionParameterTable 1 }
54
55 Ieee8021PreemptionParameterEntry ::=
56 SEQUENCE {
57     ieee8021PreemptionPriority
58     IEEE8021PriorityValue,
59     ieee8021FramePreemptionAdminStatus
60     INTEGER
61 }
62
63 ieee8021PreemptionPriority OBJECT-TYPE
64 SYNTAX      IEEE8021PriorityValue
65 MAX-ACCESS  not-accessible
66 STATUS      current
67 DESCRIPTION
68     "The priority number associated with the row of
69     the table.
70
71     A row in this table is created for each priority value."
72 REFERENCE   "6.7.2, 12.30.1"

```

```

1 ::= { ieee8021PreemptionParameterEntry 1 }
2
3
4 ieee8021FramePreemptionAdminStatus OBJECT-TYPE
5     SYNTAX      INTEGER {express (1), preemptable (2)}
6     MAX-ACCESS  read-write
7     STATUS      current
8     DESCRIPTION
9         "The value of the framePreemptionAdminStatus parameter
10        for the traffic class.
11
12        The default value of the framePreemptionAdminStatus parameter
13        is express (1).
14
15        The value of this object MUST be retained across
16        reinitializations of the management system."
17     REFERENCE   "6.7.2, 12.30.1"
18 ::= { ieee8021PreemptionParameterEntry 2 }
19
20
21 -- =====
22 -- the ieee8021PreemptionConfigTable
23 -- =====
24
25 ieee8021PreemptionConfigTable OBJECT-TYPE
26     SYNTAX      SEQUENCE OF Ieee8021PreemptionConfigEntry
27     MAX-ACCESS  not-accessible
28     STATUS      current
29     DESCRIPTION
30         "A table containing a set of frame preemption
31         parameters, one for each Port.
32         All writable objects in this table must be
33         persistent over power up restart/reboot."
34     REFERENCE   "6.7.2, 12.30.1"
35 ::= { ieee8021PreemptionParameters 2 }
36
37 ieee8021PreemptionConfigEntry OBJECT-TYPE
38     SYNTAX      Ieee8021PreemptionConfigEntry
39     MAX-ACCESS  not-accessible
40     STATUS      current
41     DESCRIPTION
42         "A list of objects containing preemption parameters
43         for each Port."
44     INDEX { ieee8021BridgeBaseComponentId,
45            ieee8021BridgeBasePort }
46 ::= { ieee8021PreemptionConfigTable 1 }
47
48 Ieee8021PreemptionConfigEntry ::=
49     SEQUENCE {
50         ieee8021FramePreemptionHoldAdvance
51             Unsigned32,
52         ieee8021FramePreemptionReleaseAdvance
53             Unsigned32,
54         ieee8021FramePreemptionActive
55             INTEGER,
56         ieee8021FramePreemptionHoldRequest
57             INTEGER
58     }
59
60 ieee8021FramePreemptionHoldAdvance OBJECT-TYPE
61     SYNTAX      Unsigned32
62     MAX-ACCESS  read-only
63     STATUS      current
64     DESCRIPTION
65         "The value of the holdAdvance parameter
66         for the Port in nanoseconds.
67
68         There is no default value; the holdAdvance is
69         a property of the underlying MAC."
70     REFERENCE   "6.7.2, 12.30.1.2"
71 ::= { ieee8021PreemptionConfigEntry 1 }
72

```

```

1
2 ieee8021FramePreemptionReleaseAdvance OBJECT-TYPE
3     SYNTAX      Unsigned32
4     MAX-ACCESS  read-only
5     STATUS      current
6     DESCRIPTION
7         "The value of the releaseAdvance parameter
8         for the Port in nanoseconds.
9
10        There is no default value; the releaseAdvance is
11        a property of the underlying MAC."
12     REFERENCE   "6.7.2, 12.30.1.3"
13     ::= { ieee8021PreemptionConfigEntry 2 }
14
15 ieee8021FramePreemptionActive OBJECT-TYPE
16     SYNTAX      INTEGER {idle (1), active (2)}
17     MAX-ACCESS  read-only
18     STATUS      current
19     DESCRIPTION
20         "The value is active (2) when preemption is operationally
21         active for the Port, and idle (1) otherwise."
22     REFERENCE   "6.7.2, 12.30.1.4"
23     ::= { ieee8021PreemptionConfigEntry 3 }
24
25 ieee8021FramePreemptionHoldRequest OBJECT-TYPE
26     SYNTAX      INTEGER {hold (1), release (2)}
27     MAX-ACCESS  read-only
28     STATUS      current
29     DESCRIPTION
30         "The value is hold (1) when the sequence of gate operations
31         for the Port has executed a Set-And-Hold-MAC operation,
32         and release (2) when the sequence of gate operations has
33         executed a Set-And-Release-MAC operation. The
34         value of this object is release (2) on system
35         initialization."
36     REFERENCE   "6.7.2, Table 8-7, 12.30.1.5"
37     ::= { ieee8021PreemptionConfigEntry 4 }
38
39
40 -- =====
41 -- IEEE8021 Preemption MIB - Conformance Information
42 -- =====
43
44 ieee8021PreemptionCompliances
45     OBJECT IDENTIFIER ::= { ieee8021PreemptionConformance 1 }
46 ieee8021PreemptionGroups
47     OBJECT IDENTIFIER ::= { ieee8021PreemptionConformance 2 }
48
49 -- =====
50 -- units of conformance
51 -- =====
52
53 -- =====
54 -- the ieee8021PreemptionGroup group
55 -- =====
56
57 ieee8021PreemptionGroup OBJECT-GROUP
58     OBJECTS {
59         ieee8021FramePreemptionAdminStatus,
60         ieee8021FramePreemptionHoldAdvance,
61         ieee8021FramePreemptionReleaseAdvance,
62         ieee8021FramePreemptionActive,
63         ieee8021FramePreemptionHoldRequest
64     }
65     STATUS      current
66     DESCRIPTION
67         "Objects that allow management of frame preemption."
68     ::= { ieee8021PreemptionGroups 1 }
69
70
71 -- =====
72 -- compliance statements

```

```
1 -- =====
2
3 ieee8021PreemptionCompliance MODULE-COMPLIANCE
4     STATUS          current
5     DESCRIPTION
6         "The compliance statement for devices supporting
7         frame preemption.
8
9         Support of the objects defined in this MIB module
10        also requires support of the IEEE8021-BRIDGE-MIB; the
11        provisions of 17.3.2 apply to implementations claiming
12        support of this MIB. "
13
14    MODULE -- this module
15        MANDATORY-GROUPS {
16            ieee8021PreemptionGroup
17        }
18
19    ::= { ieee8021PreemptionCompliances 1 }
20
21 END
```

1 17.7.24 Definitions for the IEEE8021-PSFP-MIB module

```
2 IEEE8021-PSFP-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for support of the Per-Stream Filtering and Policing
6 -- Enhancements for IEEE 802.1Q Bridges.
7 -- =====
8
9 IMPORTS
10     MODULE-IDENTITY,
11     OBJECT-TYPE,
12     Unsigned32,
13     Integer32,
14     Counter64
15     FROM SNMPv2-SMI
16     TruthValue, RowStatus
17     FROM SNMPv2-TC
18     MODULE-COMPLIANCE,
19     OBJECT-GROUP
20     FROM SNMPv2-CONF
21     ieee802dot1mibs
22     FROM IEEE8021-TC-MIB
23     ieee8021BridgeBaseComponentId
24     FROM IEEE8021-BRIDGE-MIB
25     IEEE8021STPTptimeValue
26     FROM IEEE8021-ST-MIB
27     ;
28
29 ieee8021PSFPMib MODULE-IDENTITY
30     LAST-UPDATED "202309260000Z" -- September 26, 2023
31     ORGANIZATION "IEEE 802.1 Working Group"
32     CONTACT-INFO
33         " WG-URL: http://www.ieee802.org/1/
34         WG-EMail: stds-802-1-1@ieee.org
35         Contact: IEEE 802.1 Working Group Chair
36         Postal: C/O IEEE 802.1 Working Group
37             IEEE Standards Association
38             445 Hoes Lane
39             Piscataway, NJ 08854
40             USA
41         E-mail: stds-802-1-chairs@ieee.org"
42     DESCRIPTION
43         "The Bridge MIB module for managing devices that support
44         the Per-Stream Filtering and Policing enhancements
45         for IEEE 802.1Q Bridges.
46
47         Unless otherwise indicated, the references in this MIB
48         module are to IEEE Std 802.1Q-2022.
49
50         Copyright (C) IEEE (2023).
51         This version of this MIB module is part of IEEE Std 802.1Q;
52         see that standard for full legal notices."
53
54     REVISION "202309260000Z" -- September 26, 2023
55     DESCRIPTION
56         "Update to include SupportedCycleMax and
57         SupportedIntervalMax parameters.
58         Published as part of IEEE Std 802.1Qcw-2023."
59
60     REVISION "202201130000Z" -- January 13, 2022
61     DESCRIPTION
62         "Published as part of IEEE Std 802.1Q-2022.
63         Cross references and contact information updated."
64
65     REVISION "202011060000Z" -- November 6, 2020
66     DESCRIPTION
67         "Published as part of IEEE Std 802.1Qcr-2020.
68         Cross references and contact information updated."
69
70     REVISION "201807010000Z" -- July 1, 2018
71     DESCRIPTION
```

```

1      "Published as part of IEEE Std 802.1Q 2018 revision.
2      Cross references updated and corrected."
3
4      REVISION "201709080000Z" -- September 29, 2017
5      DESCRIPTION
6          "Initial version published as part of IEEE Std 802.1Qci."
7
8          ::= { ieee802dot1mibs 31 }
9
10
11 -- =====
12 -- subtrees in the PSFP MIB
13 -- =====
14
15 ieee8021PSFPNotifications
16     OBJECT IDENTIFIER ::= { ieee8021PSFPMib 0 }
17
18 ieee8021PSFPObjects
19     OBJECT IDENTIFIER ::= { ieee8021PSFPMib 1 }
20
21 ieee8021PSFPConformance
22     OBJECT IDENTIFIER ::= { ieee8021PSFPMib 2 }
23
24 ieee8021PSFPStreamFilterParameters
25     OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 1 }
26
27 ieee8021PSFPStreamGateParameters
28     OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 2 }
29
30 ieee8021PSFPFlowMeterParameters
31     OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 3 }
32
33 ieee8021PSFPStreamParameters
34     OBJECT IDENTIFIER ::= { ieee8021PSFPObjects 4 }
35
36
37 -- =====
38 -- The ieee8021PSFPStreamFilterParameters subtree
39 -- This subtree defines the objects necessary for the management
40 -- of the stream filters for IEEE Std 802.1Q.
41 -- =====
42
43 -- =====
44 -- the ieee8021PSFPStreamFilterTable
45 -- =====
46
47 ieee8021PSFPStreamFilterTable OBJECT-TYPE
48     SYNTAX      SEQUENCE OF Ieee8021PSFPStreamFilterEntry
49     MAX-ACCESS   not-accessible
50     STATUS      current
51     DESCRIPTION
52         "A table that contains the per-filter instance
53         manageable parameters for stream filters.
54
55         A row in the table exists for each stream filter instance.
56         associated with a Bridge component.
57
58         All writable objects in this table must be
59         persistent over power up restart/reboot."
60     REFERENCE    "8.6.5.2.1, 8.6.5.3, 12.31.2"
61     ::= { ieee8021PSFPStreamFilterParameters 1 }
62
63 ieee8021PSFPStreamFilterEntry OBJECT-TYPE
64     SYNTAX      Ieee8021PSFPStreamFilterEntry
65     MAX-ACCESS   not-accessible
66     STATUS      current
67     DESCRIPTION
68         "A list of objects that contains the manageable parameters for
69         stream filters for a Bridge component."
70     INDEX       { ieee8021BridgeBaseComponentId,
71                 ieee8021PSFPStreamFilterInstance
72                 }

```



```

1 ::= { ieee8021PSFPStreamFilterTable 1 }
2
3 Ieee8021PSFPStreamFilterEntry ::=
4     SEQUENCE {
5         ieee8021PSFPStreamFilterInstance
6         Unsigned32,
7         ieee8021PSFPStreamHandleSpec
8         Integer32,
9         ieee8021PSFPPrioritySpec
10        Integer32,
11        ieee8021PSFPStreamGateInstanceID
12        Unsigned32,
13        ieee8021PSFPFilterSpecificationList
14        OCTET STRING,
15        ieee8021PSFPMatchingFramesCount
16        Counter64,
17        ieee8021PSFPPassingFramesCount
18        Counter64,
19        ieee8021PSFPNotPassingFramesCount
20        Counter64,
21        ieee8021PSFPPassingSDUCount
22        Counter64,
23        ieee8021PSFPNotPassingSDUCount
24        Counter64,
25        ieee8021PSFPPREDFramesCount
26        Counter64,
27        ieee8021PSFPStreamBlockedDueToOversizeFrameEnable
28        TruthValue,
29        ieee8021PSFPStreamBlockedDueToOversizeFrame
30        TruthValue,
31        ieee8021PSFPStreamFilterEntryRowStatus
32        RowStatus
33    }
34
35 ieee8021PSFPStreamFilterInstance OBJECT-TYPE
36     SYNTAX      Unsigned32
37     MAX-ACCESS  not-accessible
38     STATUS      current
39     DESCRIPTION
40         "The StreamFilterInstance parameter is an index into the
41         StreamFilterTable.
42
43         The value of this object MUST be retained across
44         reinitializations of the management system."
45     REFERENCE   "8.6.5.2.1, 8.6.5.3, 12.31.2"
46     ::= { ieee8021PSFPStreamFilterEntry 1 }
47
48 ieee8021PSFPStreamHandleSpec OBJECT-TYPE
49     SYNTAX      Integer32 (-1..2147483647)
50     MAX-ACCESS  read-create
51     STATUS      current
52     DESCRIPTION
53         "The StreamHandleSpec parameter contains a stream identifier
54         specification value. A value of -1 denotes the wild card value;
55         all positive values denote stream identifier values.
56
57         The value of this object MUST be retained across
58         reinitializations of the management system."
59     REFERENCE   "8.6.5.2.1, 8.6.5.3, 12.31.2"
60     ::= { ieee8021PSFPStreamFilterEntry 2 }
61
62 ieee8021PSFPPrioritySpec OBJECT-TYPE
63     SYNTAX      Integer32 (-1..2147483647)
64     MAX-ACCESS  read-create
65     STATUS      current
66     DESCRIPTION
67         "The PrioritySpec parameter contains a priority
68         specification value. A value of -1 denotes the wild card value;
69         zero or positive values denote priority values.
70
71         The value of this object MUST be retained across
72         reinitializations of the management system."

```

```

1  REFERENCE    "8.6.5.2.1, 8.6.5.3, 12.31.2"
2  ::= { ieee8021PSFPStreamFilterEntry 3}
3
4  ieee8021PSFPStreamGateInstanceID OBJECT-TYPE
5      SYNTAX      Unsigned32
6      MAX-ACCESS   read-create
7      STATUS       current
8      DESCRIPTION
9          "The StreamGateInstance parameter contains the index of an
10         entry in the Stream Gate Table.
11
12         The value of this object MUST be retained across
13         reinitializations of the management system."
14  REFERENCE    "8.6.5.2.1, 8.6.5.3, 12.31.2"
15  ::= { ieee8021PSFPStreamFilterEntry 4}
16
17  ieee8021PSFPFilterSpecificationList OBJECT-TYPE
18      SYNTAX      OCTET STRING
19      MAX-ACCESS   read-create
20      STATUS       current
21      DESCRIPTION
22          "The FilterSpecificationList parameter contains a list of
23         filter specifications associated with this stream filter.
24
25         The octet string value represents the contents of the list as
26         an ordered list of entries, each encoded as a TLV, as follows.
27
28         The first octet of each TLV is interpreted as an
29         unsigned integer representing a filter specification type:
30             0: Maximum SDU Size.
31             1: Flow meter instance identifier.
32             2-255: Reserved for future filter specification types
33
34         The second and third octets of the TLV are the length field,
35         interpreted as an unsigned integer, indicating the number of
36         octets of the value that follows the length. A length of
37         zero indicates that there is no value
38         (i.e., the filter specification has no parameters).
39
40         The fourth through (4 + length -1)th octets encode the
41         parameters of the filter specification, as defined for each
42         filter specification type.
43
44         - Maximum SDU Size:
45             A single SDU size parameter is encoded in four octets, and
46             is interpreted as an unsigned integer value.
47
48         - Flow meter instance identifier:
49             A single flow meter instance identifier is encoded in
50             four octets, and is interpreted as an unsigned integer value.
51
52         The value of this object MUST be retained across
53         reinitializations of the management system."
54  REFERENCE    "8.6.5.2.1, 8.6.5.3, 12.31.2"
55  ::= { ieee8021PSFPStreamFilterEntry 5}
56
57  ieee8021PSFPMatchingFramesCount OBJECT-TYPE
58      SYNTAX      Counter64
59      MAX-ACCESS   read-only
60      STATUS       current
61      DESCRIPTION
62          "The MatchingFramesCount counter counts received frames that
63         match this stream filter.
64         "
65  REFERENCE    "8.6.5.2.1, 8.6.5.3, 12.31.2"
66  ::= { ieee8021PSFPStreamFilterEntry 6}
67
68  ieee8021PSFPPassingFramesCount OBJECT-TYPE
69      SYNTAX      Counter64
70      MAX-ACCESS   read-only
71      STATUS       current
72      DESCRIPTION

```

```

1      "The PassingFramesCount counter counts received frames that
2      pass the gate associated with this stream filter.
3      "
4      REFERENCE      "8.6.5.2.1, 8.6.5.3, 12.31.2"
5      ::= { ieee8021PSFPStreamFilterEntry 7}
6
7 ieee8021PSFPNotPassingFramesCount OBJECT-TYPE
8     SYNTAX          Counter64
9     MAX-ACCESS      read-only
10    STATUS           current
11    DESCRIPTION
12        "The NotPassingFramesCount counter counts received frames that
13        do not pass the gate associated
14        with this stream filter.
15        "
16    REFERENCE      "8.6.5.2.1, 8.6.5.3, 12.31.2"
17    ::= { ieee8021PSFPStreamFilterEntry 8}
18
19 ieee8021PSFPPassingSDUCount OBJECT-TYPE
20    SYNTAX          Counter64
21    MAX-ACCESS      read-only
22    STATUS           current
23    DESCRIPTION
24        "The PassingSDUCount counter counts received frames that
25        pass the maximum SDU size filter specification associated
26        with this stream filter.
27        "
28    REFERENCE      "8.6.5.2.1, 8.6.5.3, 12.31.2"
29    ::= { ieee8021PSFPStreamFilterEntry 9}
30
31 ieee8021PSFPNotPassingSDUCount OBJECT-TYPE
32    SYNTAX          Counter64
33    MAX-ACCESS      read-only
34    STATUS           current
35    DESCRIPTION
36        "The NotPassingSDUCount counter counts received frames that
37        do not pass the maximum SDU size filter specification associated
38        with this stream filter.
39        "
40    REFERENCE      "8.6.5.2.1, 8.6.5.3, 12.31.2"
41    ::= { ieee8021PSFPStreamFilterEntry 10}
42
43 ieee8021PSFPPREDFramesCount OBJECT-TYPE
44    SYNTAX          Counter64
45    MAX-ACCESS      read-only
46    STATUS           current
47    DESCRIPTION
48        "The REDFramesCount counter counts received
49        frames that were discarded as a result of the
50        operation of the flow meter.
51        "
52    REFERENCE      "8.6.5.2.1, 8.6.5.3, 12.31.2"
53    ::= { ieee8021PSFPStreamFilterEntry 11}
54
55 ieee8021PSFPStreamBlockedDueToOversizeFrameEnable OBJECT-TYPE
56    SYNTAX          TruthValue
57    MAX-ACCESS      read-create
58    STATUS           current
59    DESCRIPTION
60        "The ieee8021PSFPStreamBlockedDueToOversizeFrameEnable object
61        contains a Boolean value that indicates whether the
62        StreamBlockedDueToOversizeFrame function is
63        enabled (TRUE) or disabled (FALSE).
64
65        The value of this object MUST be retained across
66        reinitializations of the management system."
67    REFERENCE      "8.6.5.2.1, 8.6.5.3, 12.31.2"
68    DEFVAL { false }
69    ::= { ieee8021PSFPStreamFilterEntry 12 }
70
71 ieee8021PSFPStreamBlockedDueToOversizeFrame OBJECT-TYPE
72    SYNTAX          TruthValue

```

```

1  MAX-ACCESS  read-create
2  STATUS      current
3  DESCRIPTION
4      "The ieee8021PSFPStreamBlockedDueToOversizeFrame object
5      contains a Boolean value that indicates whether, if the
6      StreamBlockedDueToOversizeFrame function is
7      enabled, all frames are to be discarded (TRUE)
8      or not (FALSE).
9
10     The value of this object MUST be retained across
11     reinitializations of the management system."
12  REFERENCE   "8.6.5.2.1, 8.6.5.3, 12.31.2"
13  DEFVAL { false }
14  ::= { ieee8021PSFPStreamFilterEntry 13 }
15
16 ieee8021PSFPStreamFilterEntryRowStatus OBJECT-TYPE
17     SYNTAX      RowStatus
18     MAX-ACCESS  read-create
19     STATUS      current
20     DESCRIPTION
21         "The status of the row.
22
23         The writable columns in a row cannot be changed if the row
24         is active. All columns MUST have a valid value before a row
25         can be activated.
26         "
27     ::= { ieee8021PSFPStreamFilterEntry 14 }
28
29
30 -- =====
31 -- The ieee8021PSFPStreamGateParameters subtree
32 -- This subtree defines the objects necessary for the management
33 -- of the stream gate scheduling mechanism for IEEE Std 802.1Q.
34 -- =====
35
36 -- =====
37 -- the ieee8021PSFPStreamGateTable
38 -- =====
39
40 ieee8021PSFPStreamGateTable OBJECT-TYPE
41     SYNTAX      SEQUENCE OF Ieee8021PSFPStreamGateEntry
42     MAX-ACCESS  not-accessible
43     STATUS      current
44     DESCRIPTION
45         "A table that contains the per-gate instance
46         manageable parameters for stream gate scheduling.
47
48         For a given Bridge component, a row in the table exists for
49         each stream gate instance.
50
51         All writable objects in this table must be
52         persistent over power up restart/reboot."
53     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
54     ::= { ieee8021PSFPStreamGateParameters 1 }
55
56 ieee8021PSFPStreamGateEntry OBJECT-TYPE
57     SYNTAX      Ieee8021PSFPStreamGateEntry
58     MAX-ACCESS  not-accessible
59     STATUS      current
60     DESCRIPTION
61         "A list of objects that contains the manageable parameters for
62         stream gate scheduling for a Bridge component."
63     INDEX { ieee8021BridgeBaseComponentId,
64            ieee8021PSFPStreamGateInstance
65            }
66     ::= { ieee8021PSFPStreamGateTable 1 }
67
68 Ieee8021PSFPStreamGateEntry ::=
69     SEQUENCE {
70         ieee8021PSFPStreamGateInstance
71         Unsigned32,
72         ieee8021PSFPGateEnabled

```

```

1      TruthValue,
2      ieee8021PSFPAdminGateStates
3      INTEGER,
4      ieee8021PSFPOperGateStates
5      INTEGER,
6      ieee8021PSFPAdminControlListLength
7      Unsigned32,
8      ieee8021PSFPOperControlListLength
9      Unsigned32,
10     ieee8021PSFPAdminControlList
11     OCTET STRING,
12     ieee8021PSFPOperControlList
13     OCTET STRING,
14     ieee8021PSFPAdminCycleTimeNumerator
15     Unsigned32,
16     ieee8021PSFPAdminCycleTimeDenominator
17     Unsigned32,
18     ieee8021PSFPOperCycleTimeNumerator
19     Unsigned32,
20     ieee8021PSFPOperCycleTimeDenominator
21     Unsigned32,
22     ieee8021PSFPAdminCycleTimeExtension
23     Unsigned32,
24     ieee8021PSFPOperCycleTimeExtension
25     Unsigned32,
26     ieee8021PSFPAdminBaseTime
27     IEEE8021STPTPtimeValue,
28     ieee8021PSFPOperBaseTime
29     IEEE8021STPTPtimeValue,
30     ieee8021PSFPConfigChange
31     TruthValue,
32     ieee8021PSFPConfigChangeTime
33     IEEE8021STPTPtimeValue,
34     ieee8021PSFPTickGranularity
35     Unsigned32,
36     ieee8021PSFPCurrentTime
37     IEEE8021STPTPtimeValue,
38     ieee8021PSFPConfigPending
39     TruthValue,
40     ieee8021PSFPConfigChangeError
41     Counter64,
42     ieee8021PSFPAdminIPV
43     Integer32,
44     ieee8021PSFPOperIPV
45     Integer32,
46     ieee8021PSFPGateClosedDueToInvalidRxEnable
47     TruthValue,
48     ieee8021PSFPGateClosedDueToInvalidRx
49     TruthValue,
50     ieee8021PSFPGateClosedDueToOctetsExceededEnable
51     TruthValue,
52     ieee8021PSFPGateClosedDueToOctetsExceeded
53     TruthValue,
54     ieee8021PSFPStreamGateEntryRowStatus
55     RowStatus
56     }
57
58 ieee8021PSFPStreamGateInstance OBJECT-TYPE
59     SYNTAX      Unsigned32
60     MAX-ACCESS  not-accessible
61     STATUS      current
62     DESCRIPTION
63         "The StreamGateInstance parameter is an index into the
64         StreamGateTable.
65
66         The value of this object MUST be retained across
67         reinitializations of the management system."
68     REFERENCE   "8.6.5.2.1, 8.6.5.4, 12.31.3"
69     ::= { ieee8021PSFPStreamGateEntry 1}
70
71 ieee8021PSFPGateEnabled OBJECT-TYPE
72     SYNTAX      TruthValue

```

```

1  MAX-ACCESS    read-create
2  STATUS        current
3  DESCRIPTION
4      "The GateEnabled parameter determines whether the stream gate
5      is active (true) or inactive (false).
6
7      The value of this object MUST be retained across
8      reinitializations of the management system."
9  REFERENCE     "8.6.8.4, 8.6.9.4, 12.31.3"
10 DEFVAL { false }
11 ::= { ieee8021PSFPStreamGateEntry 2 }
12
13 ieee8021PSFPAdminGateStates OBJECT-TYPE
14     SYNTAX      INTEGER { open(1), closed(2) }
15     MAX-ACCESS  read-create
16     STATUS      current
17     DESCRIPTION
18         "The administrative value of the GateStates parameter for the
19         stream gate.
20         The open value indicates that the gate is open,
21         the closed value indicates that the gate is closed.
22
23         The value of this object MUST be retained across
24         reinitializations of the management system."
25     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
26     ::= { ieee8021PSFPStreamGateEntry 3 }
27
28 ieee8021PSFPOperGateStates OBJECT-TYPE
29     SYNTAX      INTEGER { open(1), closed(2) }
30     MAX-ACCESS  read-only
31     STATUS      current
32     DESCRIPTION
33         "The operational value of the GateStates parameter for the
34         stream gate.
35         The open value indicates that the gate is open,
36         the closed value indicates that the gate is closed.
37 "
38     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
39     ::= { ieee8021PSFPStreamGateEntry 4 }
40
41 ieee8021PSFPAdminControlListLength OBJECT-TYPE
42     SYNTAX      Unsigned32
43     MAX-ACCESS  read-create
44     STATUS      current
45     DESCRIPTION
46         "The administrative value of the ListMax parameter for the gate.
47         The integer value indicates the number of entries (TLVs) in the
48         AdminControlList.
49
50         The value of this object MUST be retained across
51         reinitializations of the management system."
52     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
53     ::= { ieee8021PSFPStreamGateEntry 5 }
54
55 ieee8021PSFPOperControlListLength OBJECT-TYPE
56     SYNTAX      Unsigned32
57     MAX-ACCESS  read-only
58     STATUS      current
59     DESCRIPTION
60         "The operational value of the ListMax parameter for the gate.
61         The integer value indicates the number of entries (TLVs) in the
62         OperControlList."
63     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
64     ::= { ieee8021PSFPStreamGateEntry 6 }
65
66 ieee8021PSFPAdminControlList OBJECT-TYPE
67     SYNTAX      OCTET STRING
68     MAX-ACCESS  read-create
69     STATUS      current
70     DESCRIPTION
71         "The administrative value of the ControlList parameter for the gate.
72         The octet string value represents the contents of the control list as

```

1 an ordered list of entries, each encoded as a TLV, as follows.
2 The first octet of each TLV is interpreted as an
3 unsigned integer representing a gate operation name:
4 0: SetGateAndIPV
5 1-255: Reserved for future gate operations
6
7 The second octet of the TLV is the length field,
8 interpreted as an unsigned integer, indicating the number of
9 octets of the value that follows the length. A length of
10 zero indicates that there is no value
11 (i.e., the gate operation has no parameters).
12
13 The third through (3 + length -1)th octets encode the
14 parameters of the gate operation, in the order that they
15 appear in the definition of the operation in Table 8-4.
16 The first parameter is a StreamGateState value; the second
17 parameter is an IPV value; the third parameter is a
18 TimeInterval value; and the fourth parameter is an
19 IntervalOctetMax value. IntervalOctetMax is optional.
20
21 - StreamGateState:
22 A GateState parameter is encoded in a single octet, and
23 is interpreted as an integer value.
24 The value 1 indicates open; the value 2 indicates closed.
25
26 - IPV:
27 An IPV is encoded in four octets as a 32-bit
28 signed integer. A negative denotes the null value;
29 zero or positive values denote internal priority values.
30
31 - TimeInterval:
32 A TimeInterval is encoded in 4 octets as a 32-bit
33 unsigned integer, representing a number of nanoseconds.
34 The first octet encodes the most significant 8 bits of the
35 integer, and the fourth octet encodes the least
36 significant 8 bits.
37 - IntervalOctetMax:
38 An integer representing the maximum number of MSDU octets
39 that are permitted to pass the gate during the specified
40 TimeInterval. If this parameter is omitted, there is
41 no maximum.
42
43 The value of this object MUST be retained across
44 reinitializations of the management system."
45 REFERENCE "8.6.8.4, 8.6.9.4, 12.31.3"
46 ::= { ieee8021PSFPStreamGateEntry 7 }
47
48 ieee8021PSFPOperControlList OBJECT-TYPE
49 SYNTAX OCTET STRING
50 MAX-ACCESS read-only
51 STATUS current
52 DESCRIPTION
53 "The operational value of the ControlList parameter for the gate.
54 The octet string value represents the contents of the control list as
55 an ordered list of entries, each encoded as a TLV, as follows.
56 The first octet of each TLV is interpreted as an
57 unsigned integer representing a gate operation name:
58 0: SetGateAndIPV
59 1-255: Reserved for future gate operations
60
61 The second octet of the TLV is the length field,
62 interpreted as an unsigned integer, indicating the number of
63 octets of the value that follows the length. A length of
64 zero indicates that there is no value
65 (i.e., the gate operation has no parameters).
66
67 The third through (3 + length -1)th octets encode the
68 parameters of the gate operation, in the order that they
69 appear in the definition of the operation in Table 8-4.
70 The first parameter is a StreamGateState value; the second
71 parameter is an IPV value; the third parameter is a
72 TimeInterval value; and the fourth parameter is an

```
1      IntervalOctetMax value. IntervalOctetMax is optional.
2
3      - StreamGateState:
4          A GateState parameter is encoded in a single octet, and
5          is interpreted as an integer value.
6          The value 1 indicates open; the value 2 indicates closed.
7
8      - IPV:
9          An IPV is encoded in four octets as a 32-bit
10         signed integer. A negative value denotes the null value;
11         zero and positive values denote internal priority values.
12
13     - TimeInterval:
14         A TimeInterval is encoded in 4 octets as a 32-bit
15         unsigned integer, representing
16         a number of nanoseconds. The first octet encodes the
17         most significant 8 bits of the integer, and the fourth
18         octet encodes the least significant 8 bits.
19     - IntervalOctetMax:
20         An integer representing the maximum number of MSDU octets
21         that are permitted to pass the gate during the specified
22         TimeInterval. If this parameter is omitted, there is
23         no maximum.
24     "
25     REFERENCE      "8.6.8.4, 8.6.9.4, 12.31.3"
26     ::= { ieee8021PSFPStreamGateEntry 8 }
27
28 ieee8021PSFPAdminCycleTimeNumerator OBJECT-TYPE
29     SYNTAX          Unsigned32
30     MAX-ACCESS      read-create
31     STATUS          current
32     DESCRIPTION
33         "The administrative value of the numerator of the CycleTime
34         parameter for the gate.
35         The numerator and denominator together represent the cycle time as
36         a rational number of seconds.
37
38         The value of this object MUST be retained across
39         reinitializations of the management system."
40     REFERENCE      "8.6.8.4, 8.6.9.4, 12.31.3"
41     ::= { ieee8021PSFPStreamGateEntry 9 }
42
43 ieee8021PSFPAdminCycleTimeDenominator OBJECT-TYPE
44     SYNTAX          Unsigned32
45     MAX-ACCESS      read-create
46     STATUS          current
47     DESCRIPTION
48         "The administrative value of the denominator of the
49         CycleTime parameter for the gate.
50         The numerator and denominator together represent the cycle time as
51         a rational number of seconds.
52
53         The value of this object MUST be retained across
54         reinitializations of the management system."
55     REFERENCE      "8.6.8.4, 8.6.9.4, 12.31.3"
56     ::= { ieee8021PSFPStreamGateEntry 10 }
57
58 ieee8021PSFPOperCycleTimeNumerator OBJECT-TYPE
59     SYNTAX          Unsigned32
60     MAX-ACCESS      read-only
61     STATUS          current
62     DESCRIPTION
63         "The operational value of the numerator of the
64         CycleTime parameter for the gate.
65         The numerator and denominator together represent the cycle
66         time as a rational number of seconds."
67     REFERENCE      "8.6.8.4, 8.6.9.4, 12.31.3"
68     ::= { ieee8021PSFPStreamGateEntry 11 }
69
70 ieee8021PSFPOperCycleTimeDenominator OBJECT-TYPE
71     SYNTAX          Unsigned32
72     MAX-ACCESS      read-only
```



```

1  STATUS      current
2  DESCRIPTION
3      "The operational value of the denominator of the
4      CycleTime parameter for the gate.
5      The numerator and denominator together represent the
6      cycle time as a rational number of seconds."
7  REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
8  ::= { ieee8021PSFPStreamGateEntry 12 }
9
10 ieee8021PSFPAdminCycleTimeExtension OBJECT-TYPE
11     SYNTAX      Unsigned32
12     UNITS        "nanoseconds"
13     MAX-ACCESS   read-create
14     STATUS      current
15     DESCRIPTION
16         "The administrative value of the CycleTimeExtension
17         parameter for the gate.
18         The value is an unsigned integer number of nanoseconds.
19
20         The value of this object MUST be retained across
21         reinitializations of the management system."
22     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
23     ::= { ieee8021PSFPStreamGateEntry 13 }
24
25 ieee8021PSFPOperCycleTimeExtension OBJECT-TYPE
26     SYNTAX      Unsigned32
27     UNITS        "nanoseconds"
28     MAX-ACCESS   read-only
29     STATUS      current
30     DESCRIPTION
31         "The operational value of the CycleTimeExtension
32         parameter for the gate.
33         The value is an unsigned integer number of nanoseconds."
34     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
35     ::= { ieee8021PSFPStreamGateEntry 14 }
36
37 ieee8021PSFPAdminBaseTime OBJECT-TYPE
38     SYNTAX      IEEE8021STPTPtimeValue
39     UNITS        "PTP time"
40     MAX-ACCESS   read-create
41     STATUS      current
42     DESCRIPTION
43         "The administrative value of the BaseTime parameter for the gate.
44         The value is a representation of a PTPtime value,
45         consisting of a 48-bit integer
46         number of seconds and a 32-bit integer number of nanoseconds.
47
48         The value of this object MUST be retained across
49         reinitializations of the management system."
50     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
51     ::= { ieee8021PSFPStreamGateEntry 15 }
52
53 ieee8021PSFPOperBaseTime OBJECT-TYPE
54     SYNTAX      IEEE8021STPTPtimeValue
55     UNITS        "PTP time"
56     MAX-ACCESS   read-only
57     STATUS      current
58     DESCRIPTION
59         "The operational value of the BaseTime parameter for the gate.
60         The value is a representation of a PTPtime value,
61         consisting of a 48-bit integer
62         number of seconds and a 32-bit integer number of nanoseconds."
63     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
64     ::= { ieee8021PSFPStreamGateEntry 16 }
65
66 ieee8021PSFPConfigChange OBJECT-TYPE
67     SYNTAX      TruthValue
68     MAX-ACCESS   read-create
69     STATUS      current
70     DESCRIPTION
71         "The ConfigChange parameter signals the start of a
72         configuration change for the gate

```

```
1      when it is set to TRUE. This should only be done
2      when the various administrative parameters
3      are all set to appropriate values."
4  REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
5  ::= { ieee8021PSFPStreamGateEntry 17 }
6
7  ieee8021PSFPConfigChangeTime OBJECT-TYPE
8      SYNTAX      IEEE8021STPTPtimeValue
9      UNITS        "PTP time"
10     MAX-ACCESS   read-only
11     STATUS        current
12     DESCRIPTION
13         "The PTPtime at which the next config change is scheduled to occur.
14         The value is a representation of a PTPtime value,
15         consisting of a 48-bit integer
16         number of seconds and a 32-bit integer number of nanoseconds.
17
18         The value of this object MUST be retained across
19         reinitializations of the management system."
20     REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
21     ::= { ieee8021PSFPStreamGateEntry 18 }
22
23     ieee8021PSFPTickGranularity OBJECT-TYPE
24         SYNTAX      Unsigned32
25         MAX-ACCESS   read-only
26         STATUS        current
27         DESCRIPTION
28             "The granularity of the cycle time clock, represented as an
29             unsigned number of tenths of nanoseconds.
30
31             The value of this object MUST be retained across
32             reinitializations of the management system."
33         REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
34         ::= { ieee8021PSFPStreamGateEntry 19 }
35
36     ieee8021PSFPCurrentTime OBJECT-TYPE
37         SYNTAX      IEEE8021STPTPtimeValue
38         MAX-ACCESS   read-only
39         STATUS        current
40         DESCRIPTION
41             "The current time, in PTPtime, as maintained by the local system.
42             The value is a representation of a PTPtime value,
43             consisting of a 48-bit integer
44             number of seconds and a 32-bit integer number of nanoseconds."
45         REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
46         ::= { ieee8021PSFPStreamGateEntry 20 }
47
48     ieee8021PSFPConfigPending OBJECT-TYPE
49         SYNTAX      TruthValue
50         MAX-ACCESS   read-only
51         STATUS        current
52         DESCRIPTION
53             "The value of the ConfigPending state machine variable.
54             The value is TRUE if a configuration change is in progress
55             but has not yet completed."
56         REFERENCE   "8.6.8.4, 8.6.9.4, 12.31.3"
57         ::= { ieee8021PSFPStreamGateEntry 21 }
58
59     ieee8021PSFPConfigChangeError OBJECT-TYPE
60         SYNTAX      Counter64
61         MAX-ACCESS   read-only
62         STATUS        current
63         DESCRIPTION
64             "A counter of the number of times that a re-configuration
65             of the traffic schedule has been requested with the old
66             schedule still running and the requested base time was
67             in the past."
68         REFERENCE   "8.6.8.4, 8.6.9.3, 12.31.3"
69         ::= { ieee8021PSFPStreamGateEntry 23 }
70
71     ieee8021PSFPAdminIPV OBJECT-TYPE
72         SYNTAX      Integer32 (-1..2147483647)
```

```

1  MAX-ACCESS  read-create
2  STATUS      current
3  DESCRIPTION
4      "The administrative value of the IPV parameter for the gate.
5      A value of -1 denotes the null value.
6      "
7  REFERENCE   "8.6.5.4, 8.6.10, 12.31.3"
8  ::= { ieee8021PSFPStreamGateEntry 24 }
9
10 ieee8021PSFPOperIPV OBJECT-TYPE
11 SYNTAX      Integer32 (-1..2147483647)
12 MAX-ACCESS  read-create
13 STATUS      current
14 DESCRIPTION
15     "The operational value of the IPV parameter for the gate.
16     A value of -1 denotes the null value.
17     "
18 REFERENCE   "8.6.5.4, 8.6.10, 12.31.3"
19 ::= { ieee8021PSFPStreamGateEntry 25 }
20
21 ieee8021PSFPGateClosedDueToInvalidRxEnable OBJECT-TYPE
22 SYNTAX      TruthValue
23 MAX-ACCESS  read-create
24 STATUS      current
25 DESCRIPTION
26     "The PSFPGateClosedDueToInvalidRxEnable object contains
27     a Boolean value that indicates whether the
28     GateClosedDueToInvalidRx function is enabled (TRUE) or
29     disabled (FALSE).
30
31     The value of this object MUST be retained across
32     reinitializations of the management system."
33 REFERENCE   "8.6.5.4, 12.31.3"
34 DEFVAL { false }
35 ::= { ieee8021PSFPStreamGateEntry 26 }
36
37 ieee8021PSFPGateClosedDueToInvalidRx OBJECT-TYPE
38 SYNTAX      TruthValue
39 MAX-ACCESS  read-create
40 STATUS      current
41 DESCRIPTION
42     "The PSFPGateClosedDueToInvalidRx object contains
43     a Boolean value that indicates whether, if the
44     GateClosedDueToInvalidRx function is enabled,
45     all frames are to be discarded (TRUE) or not (FALSE).
46
47     The value of this object MUST be retained across
48     reinitializations of the management system."
49 REFERENCE   "8.6.5.4, 12.31.3"
50 DEFVAL { false }
51 ::= { ieee8021PSFPStreamGateEntry 27 }
52
53 ieee8021PSFPGateClosedDueToOctetsExceededEnable OBJECT-TYPE
54 SYNTAX      TruthValue
55 MAX-ACCESS  read-create
56 STATUS      current
57 DESCRIPTION
58     "The PSFPGateClosedDueToOctetsExceededEnable object contains
59     a Boolean value that indicates whether the
60     GateClosedDueToOctetsExceeded function is enabled (TRUE)
61     or disabled (FALSE).
62
63     The value of this object MUST be retained across
64     reinitializations of the management system."
65 REFERENCE   "8.6.5.4, 12.31.3"
66 DEFVAL { false }
67 ::= { ieee8021PSFPStreamGateEntry 28 }
68
69 ieee8021PSFPGateClosedDueToOctetsExceeded OBJECT-TYPE
70 SYNTAX      TruthValue
71 MAX-ACCESS  read-create
72 STATUS      current

```

```

1  DESCRIPTION
2      "The PSFPGateClosedDueToOctetsExceeded parameter contains
3      a Boolean value that indicates whether, if the
4      GateClosedDueToOctetsExceeded function is enabled, all
5      frames are to be discarded (TRUE) or not (FALSE).
6
7      The value of this object MUST be retained across
8      reinitializations of the management system."
9  REFERENCE   "8.6.5.4, 12.31.3"
10 DEFVAL { false }
11 ::= { ieee8021PSFPStreamGateEntry 29}
12
13
14 ieee8021PSFPStreamGateEntryRowStatus OBJECT-TYPE
15     SYNTAX      RowStatus
16     MAX-ACCESS   read-create
17     STATUS      current
18     DESCRIPTION
19         "The status of the row.
20
21         The writable columns in a row cannot be changed if the row
22         is active. All columns MUST have a valid value before a row
23         can be activated.
24         "
25     ::= { ieee8021PSFPStreamGateEntry 30 }
26
27
28
29 -- =====
30 -- The ieee8021PSFPFlowMeterParameters subtree
31 -- This subtree defines the objects necessary for the management
32 -- of the flow meters for IEEE Std 802.1Q.
33 -- =====
34
35 -- =====
36 -- the ieee8021PSFPFlowMeterTable
37 -- =====
38
39 ieee8021PSFPFlowMeterTable OBJECT-TYPE
40     SYNTAX      SEQUENCE OF Ieee8021PSFPFlowMeterEntry
41     MAX-ACCESS   not-accessible
42     STATUS      current
43     DESCRIPTION
44         "A table that contains the per-meter instance
45         manageable parameters for flow meters.
46
47         For a given Bridge component, a row in the table exists for
48         each flow meter instance.
49
50         All writable objects in this table must be
51         persistent over power up restart/reboot."
52     REFERENCE   "8.6.5.5, 12.31.4"
53     ::= { ieee8021PSFPFlowMeterParameters 1 }
54
55 ieee8021PSFPFlowMeterEntry OBJECT-TYPE
56     SYNTAX      Ieee8021PSFPFlowMeterEntry
57     MAX-ACCESS   not-accessible
58     STATUS      current
59     DESCRIPTION
60         "A list of objects that contains the manageable parameters for
61         flow meters for a Bridge component."
62     INDEX { ieee8021BridgeBaseComponentId,
63            ieee8021PSFPFlowMeterInstance
64            }
65     ::= { ieee8021PSFPFlowMeterTable 1 }
66
67 Ieee8021PSFPFlowMeterEntry ::=
68     SEQUENCE {
69         ieee8021PSFPFlowMeterInstance
70         Unsigned32,
71         ieee8021PSFPFlowMeterCIR
72         Unsigned32,

```

```
1      ieee8021PSFPFlowMeterCBS
2          Unsigned32,
3      ieee8021PSFPFlowMeterEIR
4          Unsigned32,
5      ieee8021PSFPFlowMeterEBS
6          Unsigned32,
7      ieee8021PSFPFlowMeterCF
8          Integer32,
9      ieee8021PSFPFlowMeterCM
10         INTEGER,
11      ieee8021PSFPFlowMeterDropOnYellow
12         TruthValue,
13      ieee8021PSFPFlowMeterMarkAllFramesRedEnable
14         TruthValue,
15      ieee8021PSFPFlowMeterMarkAllFramesRed
16         TruthValue,
17      ieee8021PSFPFlowMeterEntryRowStatus
18         RowStatus
19      }
20
21 ieee8021PSFPFlowMeterInstance OBJECT-TYPE
22     SYNTAX      Unsigned32
23     MAX-ACCESS  not-accessible
24     STATUS      current
25     DESCRIPTION
26         "The FlowMeterInstance parameter is an index into the
27         FlowMeterTable.
28
29         The value of this object MUST be retained across
30         reinitializations of the management system."
31     REFERENCE   "8.6.5.5, 12.31.4"
32     ::= { ieee8021PSFPFlowMeterEntry 1}
33
34 ieee8021PSFPFlowMeterCIR OBJECT-TYPE
35     SYNTAX      Unsigned32
36     MAX-ACCESS  read-create
37     STATUS      current
38     DESCRIPTION
39         "The FlowMeterCIR parameter contains an integer value that
40         represents the CIR value for the flow meter, in bit/second.
41
42         The value of this object MUST be retained across
43         reinitializations of the management system."
44     REFERENCE   "8.6.5.5, 12.31.4"
45     ::= { ieee8021PSFPFlowMeterEntry 2}
46
47 ieee8021PSFPFlowMeterCBS OBJECT-TYPE
48     SYNTAX      Unsigned32
49     MAX-ACCESS  read-create
50     STATUS      current
51     DESCRIPTION
52         "The FlowMeterCBS parameter contains an integer value that
53         represents the CBS value for the flow meter, in octets.
54
55         The value of this object MUST be retained across
56         reinitializations of the management system."
57     REFERENCE   "8.6.5.5, 12.31.4"
58     ::= { ieee8021PSFPFlowMeterEntry 3}
59
60 ieee8021PSFPFlowMeterEIR OBJECT-TYPE
61     SYNTAX      Unsigned32
62     MAX-ACCESS  read-create
63     STATUS      current
64     DESCRIPTION
65         "The FlowMeterEIR parameter contains an integer value that
66         represents the EIR value for the flow meter, in bit/second.
67
68         The value of this object MUST be retained across
69         reinitializations of the management system."
70     REFERENCE   "8.6.5.5, 12.31.4"
71     ::= { ieee8021PSFPFlowMeterEntry 4}
72
```

```

1 ieee8021PSFPFlowMeterEBS OBJECT-TYPE
2     SYNTAX      Unsigned32
3     MAX-ACCESS  read-create
4     STATUS      current
5     DESCRIPTION
6         "The FlowMeterEBS parameter contains an integer value that
7         represents the EBS value for the flow meter, in octets.
8
9         The value of this object MUST be retained across
10        reinitializations of the management system."
11     REFERENCE   "8.6.5.5, 12.31.4"
12     ::= { ieee8021PSFPFlowMeterEntry 5}
13
14 ieee8021PSFPFlowMeterCF OBJECT-TYPE
15     SYNTAX      Integer32 (0..1)
16     MAX-ACCESS  read-create
17     STATUS      current
18     DESCRIPTION
19         "The FlowMeterCF parameter contains an integer value that
20         represents the CF value for the flow meter, as an integer
21         value 0 or 1.
22
23         The value of this object MUST be retained across
24         reinitializations of the management system."
25     REFERENCE   "8.6.5.5, 12.31.4"
26     ::= { ieee8021PSFPFlowMeterEntry 6}
27
28 ieee8021PSFPFlowMeterCM OBJECT-TYPE
29     SYNTAX      INTEGER {colorBlind(1), colorAware(2)}
30     MAX-ACCESS  read-create
31     STATUS      current
32     DESCRIPTION
33         "The FlowMeterCM parameter contains an integer value that
34         represents the CM value for the flow meter, as an enumerated
35         value indicating colorBlind(1) or colorAware(2).
36
37         The value of this object MUST be retained across
38         reinitializations of the management system."
39     REFERENCE   "8.6.5.5, 12.31.4"
40     ::= { ieee8021PSFPFlowMeterEntry 7}
41
42 ieee8021PSFPFlowMeterDropOnYellow OBJECT-TYPE
43     SYNTAX      TruthValue
44     MAX-ACCESS  read-create
45     STATUS      current
46     DESCRIPTION
47         "The FlowMeterDropOnYellow parameter contains a Boolean value that
48         indicates whether yellow frames are dropped (TRUE) or
49         have drop_eligible set to TRUE (FALSE).
50
51         The value of this object MUST be retained across
52         reinitializations of the management system."
53     REFERENCE   "8.6.5.5, 12.31.4"
54     ::= { ieee8021PSFPFlowMeterEntry 8}
55
56 ieee8021PSFPFlowMeterMarkAllFramesRedEnable OBJECT-TYPE
57     SYNTAX      TruthValue
58     MAX-ACCESS  read-create
59     STATUS      current
60     DESCRIPTION
61         "The FlowMeterMarkAllFramesRedEnable parameter contains
62         a Boolean value that indicates whether the MarkAllFramesRed
63         function is enabled (TRUE) or disabled (FALSE).
64
65         The value of this object MUST be retained across
66         reinitializations of the management system."
67     REFERENCE   "8.6.5.5, 12.31.4"
68     DEFVAL { false }
69     ::= { ieee8021PSFPFlowMeterEntry 9}
70
71 ieee8021PSFPFlowMeterMarkAllFramesRed OBJECT-TYPE
72     SYNTAX      TruthValue

```

```

1  MAX-ACCESS  read-create
2  STATUS      current
3  DESCRIPTION
4      "The FlowMeterMarkAllFramesRed parameter contains
5      a Boolean value that indicates whether, if the
6      MarkAllFramesRed function is enabled, all frames are to
7      be discarded (TRUE) or not (FALSE).
8
9      The value of this object MUST be retained across
10     reinitializations of the management system."
11  REFERENCE   "8.6.5.5, 12.31.4"
12  DEFVAL { false }
13  ::= { ieee8021PSFPFlowMeterEntry 10}
14
15 ieee8021PSFPFlowMeterEntryRowStatus OBJECT-TYPE
16     SYNTAX      RowStatus
17     MAX-ACCESS  read-create
18     STATUS      current
19     DESCRIPTION
20         "The status of the row.
21
22         The writable columns in a row cannot be changed if the row
23         is active. All columns MUST have a valid value before a row
24         can be activated.
25         "
26     ::= { ieee8021PSFPFlowMeterEntry 11 }
27
28
29
30
31 -- =====
32 -- The ieee8021PSFPStreamParameters subtree
33 -- This subtree defines the objects necessary for the management
34 -- of the flow meters for IEEE Std 802.1Q.
35 -- =====
36
37 -- =====
38 -- the ieee8021PSFPStreamParameterTable
39 -- =====
40
41 ieee8021PSFPStreamParameterTable OBJECT-TYPE
42     SYNTAX      SEQUENCE OF Ieee8021PSFPStreamParameterEntry
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46         "A table that contains per-Bridge component
47         manageable parameters for PSFP.
48
49         A row in the table exists for each Bridge component.
50
51         All writable objects in this table must be
52         persistent over power up restart/reboot."
53     REFERENCE   "8.6.5.2, 12.31.1"
54     ::= { ieee8021PSFPStreamParameters 1 }
55
56 ieee8021PSFPStreamParameterEntry OBJECT-TYPE
57     SYNTAX      Ieee8021PSFPStreamParameterEntry
58     MAX-ACCESS  not-accessible
59     STATUS      current
60     DESCRIPTION
61         "A list of objects that contains the manageable parameters for
62         flow meters for a Bridge component."
63     INDEX { ieee8021BridgeBaseComponentId
64           }
65     ::= { ieee8021PSFPStreamParameterTable 1 }
66
67 Ieee8021PSFPStreamParameterEntry ::=
68     SEQUENCE {
69         ieee8021PSFPMaxStreamFilterInstances
70             Unsigned32,
71         ieee8021PSFPMaxStreamGateInstances
72             Unsigned32,

```

```

1      ieee8021PSFPMaxFlowMeterInstances
2      Unsigned32,
3      ieee8021PSFPSupportedListMax
4      Unsigned32,
5      ieee8021PSFPSupportedCycleMaxNumerator
6      Unsigned32,
7      ieee8021PSFPSupportedCycleMaxDenominator
8      Unsigned32,
9      ieee8021PSFPSupportedIntervalMax
10     Unsigned32
11     }
12
13 ieee8021PSFPMaxStreamFilterInstances OBJECT-TYPE
14     SYNTAX      Unsigned32
15     MAX-ACCESS  read-only
16     STATUS      current
17     DESCRIPTION
18         "The MaxStreamFilterInstances parameter defines the
19         maximum number of stream filter instances that are
20         supported by this Bridge component."
21     REFERENCE   "8.6.5.3, 12.31.2"
22     ::= { ieee8021PSFPStreamParameterEntry 1}
23
24 ieee8021PSFPMaxStreamGateInstances OBJECT-TYPE
25     SYNTAX      Unsigned32
26     MAX-ACCESS  read-only
27     STATUS      current
28     DESCRIPTION
29         "The MaxStreamGateInstances parameter defines the
30         maximum number of stream gate instances that are
31         supported by this Bridge component."
32     REFERENCE   "8.6.5.4, 12.31.3"
33     ::= { ieee8021PSFPStreamParameterEntry 2}
34
35 ieee8021PSFPMaxFlowMeterInstances OBJECT-TYPE
36     SYNTAX      Unsigned32
37     MAX-ACCESS  read-only
38     STATUS      current
39     DESCRIPTION
40         "The MaxFlowMeterInstances parameter defines the
41         maximum number of flow meter instances that are
42         supported by this Bridge component."
43     REFERENCE   "8.6.5.5, 12.31.4"
44     ::= { ieee8021PSFPStreamParameterEntry 3}
45
46 ieee8021PSFPSupportedListMax OBJECT-TYPE
47     SYNTAX      Unsigned32
48     MAX-ACCESS  read-only
49     STATUS      current
50     DESCRIPTION
51         "The SupportedListMax parameter defines the
52         The maximum value supported by this Bridge component of
53         the AdminControlListLength and
54         OperControlListLength parameters."
55     REFERENCE   "8.6.5.4, 12.31.3"
56     ::= { ieee8021PSFPStreamParameterEntry 4}
57
58 ieee8021PSFPSupportedCycleMaxNumerator OBJECT-TYPE
59     SYNTAX      Unsigned32
60     MAX-ACCESS  read-only
61     STATUS      current
62     DESCRIPTION
63         "The numerator of the SupportedCycleMax parameter for
64         the Bridge component. The numerator and denominator
65         together represent the maximum value supported of
66         the AdminCycleTime and OperCycleTime parameters"
67     REFERENCE   "12.31.1.7"
68     ::= { ieee8021PSFPStreamParameterEntry 5}
69
70 ieee8021PSFPSupportedCycleMaxDenominator OBJECT-TYPE
71     SYNTAX      Unsigned32
72     MAX-ACCESS  read-only

```



```

1  STATUS      current
2  DESCRIPTION
3      "The denominator of the SupportedCycleMax parameter for
4      the Bridge component. The numerator and denominator
5      together represent the maximum value supported of
6      the AdminCycleTime and OperCycleTime parameters"
7  REFERENCE   "12.31.1.7"
8  ::= { ieee8021PSFPStreamParameterEntry 6}
9
10 ieee8021PSFPSupportedIntervalMax OBJECT-TYPE
11     SYNTAX      Unsigned32
12     MAX-ACCESS  read-only
13     STATUS      current
14     DESCRIPTION
15         "The maximum value supported by this Bridge component
16         of the TimeIntervalValue parameter."
17     REFERENCE   "12.31.1.8"
18     ::= { ieee8021PSFPStreamParameterEntry 7}
19
20
21 -- =====
22 -- IEEE8021 PSFP MIB - Conformance Information
23 -- =====
24
25 ieee8021PSFPCompliances
26     OBJECT IDENTIFIER ::= { ieee8021PSFPConformance 1 }
27 ieee8021PSFPGroups
28     OBJECT IDENTIFIER ::= { ieee8021PSFPConformance 2 }
29
30 -- =====
31 -- units of conformance
32 -- =====
33
34 -- =====
35 -- the ieee8021PSFPObjectsGroup group
36 -- =====
37
38 ieee8021PSFPObjectsGroup OBJECT-GROUP
39     OBJECTS {
40         ieee8021PSFPStreamHandleSpec,
41         ieee8021PSFPPrioritySpec,
42         ieee8021PSFPStreamGateInstanceID,
43         ieee8021PSFPFilterSpecificationList,
44         ieee8021PSFPMatchingFramesCount,
45         ieee8021PSFPPassingFramesCount,
46         ieee8021PSFPNotPassingFramesCount,
47         ieee8021PSFPPassingSDUCount,
48         ieee8021PSFPNotPassingSDUCount,
49         ieee8021PSFPPREDFramesCount,
50         ieee8021PSFPStreamBlockedDueToOversizeFrameEnable,
51         ieee8021PSFPStreamBlockedDueToOversizeFrame,
52         ieee8021PSFPStreamFilterEntryRowStatus,
53         ieee8021PSFPGateEnabled,
54         ieee8021PSFPAdminGateStates,
55         ieee8021PSFPOperGateStates,
56         ieee8021PSFPAdminControlListLength,
57         ieee8021PSFPOperControlListLength,
58         ieee8021PSFPAdminControlList,
59         ieee8021PSFPOperControlList,
60         ieee8021PSFPAdminCycleTimeNumerator,
61         ieee8021PSFPAdminCycleTimeDenominator,
62         ieee8021PSFPOperCycleTimeNumerator,
63         ieee8021PSFPOperCycleTimeDenominator,
64         ieee8021PSFPAdminCycleTimeExtension,
65         ieee8021PSFPOperCycleTimeExtension,
66         ieee8021PSFPAdminBaseTime,
67         ieee8021PSFPOperBaseTime,
68         ieee8021PSFPConfigChange,
69         ieee8021PSFPConfigChangeTime,
70         ieee8021PSFPTickGranularity,
71         ieee8021PSFPCurrentTime,
72         ieee8021PSFPConfigPending,

```

```

1      ieee8021PSFPConfigChangeError,
2      ieee8021PSFPAdminIPV,
3      ieee8021PSFPOperIPV,
4      ieee8021PSFPGateClosedDueToInvalidRxEnable,
5      ieee8021PSFPGateClosedDueToInvalidRx,
6      ieee8021PSFPGateClosedDueToOctetsExceededEnable,
7      ieee8021PSFPGateClosedDueToOctetsExceeded,
8      ieee8021PSFPStreamGateEntryRowStatus,
9      ieee8021PSFPFlowMeterCIR,
10     ieee8021PSFPFlowMeterCBS,
11     ieee8021PSFPFlowMeterEIR,
12     ieee8021PSFPFlowMeterEBS,
13     ieee8021PSFPFlowMeterCF,
14     ieee8021PSFPFlowMeterCM,
15     ieee8021PSFPFlowMeterDropOnYellow,
16     ieee8021PSFPFlowMeterMarkAllFramesRedEnable,
17     ieee8021PSFPFlowMeterMarkAllFramesRed,
18     ieee8021PSFPFlowMeterEntryRowStatus,
19     ieee8021PSFPMaxStreamFilterInstances,
20     ieee8021PSFPMaxStreamGateInstances,
21     ieee8021PSFPMaxFlowMeterInstances,
22     ieee8021PSFPSupportedListMax,
23     ieee8021PSFPSupportedCycleMaxNumerator,
24     ieee8021PSFPSupportedCycleMaxDenominator,
25     ieee8021PSFPSupportedIntervalMax
26     }
27
28     STATUS          current
29     DESCRIPTION
30         "Objects that allow management of PSFP."
31     ::= { ieee8021PSFPGroups 1 }
32
33
34 -- =====
35 -- compliance statements
36 -- =====
37
38 ieee8021PSFPCompliance MODULE-COMPLIANCE
39     STATUS          current
40     DESCRIPTION
41         "The compliance statement for devices supporting
42         PSFP.
43
44         Support of the objects defined in this MIB module
45         also requires support of the IEEE8021-BRIDGE-MIB; the
46         provisions of 17.3.2 apply to implementations claiming
47         support of this MIB. "
48
49     MODULE -- this module
50         MANDATORY-GROUPS {
51             ieee8021PSFPObjectsGroup
52         }
53
54     ::= { ieee8021PSFPCompliances 1 }
55
56 END

```

1 17.7.25 Definitions for the IEEE8021-TSN-REMOTE-MANAGEMENT-MIB module

```
2 IEEE8021-TSN-REMOTE-MANAGEMENT-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB module supporting TSN stream reservations
6 -- in IEEE 802.1Q Bridges.
7 -- =====
8
9 IMPORTS
10     OBJECT-GROUP,
11     MODULE-COMPLIANCE
12         FROM SNMPv2-CONF
13     MODULE-IDENTITY,
14     OBJECT-TYPE,
15     Counter64,
16     Unsigned32
17         FROM SNMPv2-SMI
18     TEXTUAL-CONVENTION,
19     TruthValue,
20     RowStatus
21         FROM SNMPv2-TC
22     IEEE8021BridgePortNumber,
23     ieee802dot1mibs
24         FROM IEEE8021-TC-MIB
25     ieee8021BridgeBasePort,
26     ieee8021BridgeBaseComponentId,
27     ieee8021BridgeTrafficClass
28         FROM IEEE8021-BRIDGE-MIB
29     ieee8021QBridgeVlanIndex
30         FROM IEEE8021-Q-BRIDGE-MIB
31 ;
32
33 ieee8021TsnRemoteMgmtMib MODULE-IDENTITY
34     LAST-UPDATED "202211080000Z" -- November 8, 2022
35     ORGANIZATION "IEEE 802.1 Working Group"
36     CONTACT-INFO
37         " WG-URL: http://www.ieee802.org/1/
38         WG-EMail: stds-802-1-1@ieee.org
39         Contact: IEEE 802.1 Working Group Chair
40         Postal: C/O IEEE 802.1 Working Group
41                 IEEE Standards Association
42                 445 Hoes Lane
43                 Piscataway, NJ 08854
44                 USA
45         E-mail: stds-802-1-chairs@ieee.org"
46     DESCRIPTION
47         "IEEE 802.1Q Bridge MIB module supporting
48         TSN stream reservations.
49
50         Unless otherwise indicated, the references in this MIB
51         module are to IEEE Std 802.1Q-2022.
52
53         Copyright (C) IEEE (2022).
54         This version of this MIB module is part of IEEE Std 802.1Q;
55         see that standard for full legal notices."
56
57     REVISION "202211080000Z" -- November 8, 2022
58     DESCRIPTION
59         "Published as part of IEEE Std 802.1Q-2022.
60         Cross references and contact information updated."
61
62     REVISION "201810040000Z" -- October 4, 2018
63     DESCRIPTION
64         "Initial revision, included in IEEE 802.1Qcc-2018"
65     ::= { ieee802dot1mibs 32 }
66
67 -- =====
68 -- subtrees in the TSN Remote Management MIB
69 -- =====
70
71 ieee8021TsnRemoteMgmtNotifications
```

```

1   OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtMib 0 }
2
3   ieee8021TsnRemoteMgmtObjects
4   OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtMib 1 }
5
6   ieee8021TsnRemoteMgmtConformance
7   OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtMib 2 }
8
9   ieee8021TsnRemoteMgmtBridgeDelay
10  OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtObjects 1 }
11
12  ieee8021TsnRemoteMgmtPropagationDelay
13  OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtObjects 2 }
14
15  ieee8021TsnRemoteMgmtStaticTrees
16  OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtObjects 3 }
17
18  ieee8021TsnRemoteMgmtMrpExternalControl
19  OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtObjects 4 }
20
21  -- =====
22  -- the ieee8021TsnRemoteBridgeDelayTable
23  -- =====
24
25  ieee8021TsnRemoteMgmtBridgeDelayTable OBJECT-TYPE
26      SYNTAX      SEQUENCE OF Ieee8021TsnRemoteMgmtBridgeDelayEntry
27      MAX-ACCESS  not-accessible
28      STATUS      current
29      DESCRIPTION
30          "A table containing a set of parameters necessary to
31          determine the delay of frames as they pass through the
32          Bridge's relay.
33          There is one Bridge Delay managed object per Port pair of
34          a Bridge component. The Port pair consists of three indices,
35          an ingress Port followed by an egress Port and a traffic
36          class associated with the Port pair."
37      REFERENCE   "12.32.1"
38      ::= { ieee8021TsnRemoteMgmtBridgeDelay 1 }
39
40  ieee8021TsnRemoteMgmtBridgeDelayEntry OBJECT-TYPE
41      SYNTAX      Ieee8021TsnRemoteMgmtBridgeDelayEntry
42      MAX-ACCESS  not-accessible
43      STATUS      current
44      DESCRIPTION
45          "A list of objects containing information necessary to
46          determine the delay of frames as they pass through the
47          Bridge's relay."
48      INDEX { ieee8021BridgeBaseComponentId,
49              ieee8021BridgeTrafficClass,
50              ieee8021TsnRemoteMgmtBridgeIngressPort,
51              ieee8021TsnRemoteMgmtBridgeEgressPort
52          }
53      ::= { ieee8021TsnRemoteMgmtBridgeDelayTable 1 }
54
55  Ieee8021TsnRemoteMgmtBridgeDelayEntry ::=
56      SEQUENCE {
57          ieee8021TsnRemoteMgmtBridgeIngressPort
58              IEEE8021BridgePortNumber,
59          ieee8021TsnRemoteMgmtBridgeEgressPort
60              IEEE8021BridgePortNumber,
61          ieee8021TsnRemoteMgmtIndependentDelayMin
62              Unsigned32,
63          ieee8021TsnRemoteMgmtIndependentDelayMax
64              Unsigned32,
65          ieee8021TsnRemoteMgmtDependentDelayMin
66              Unsigned32,
67          ieee8021TsnRemoteMgmtDependentDelayMax
68              Unsigned32
69      }
70
71  ieee8021TsnRemoteMgmtBridgeIngressPort OBJECT-TYPE
72      SYNTAX      IEEE8021BridgePortNumber

```

```

1  MAX-ACCESS    not-accessible
2  STATUS        current
3  DESCRIPTION
4      "The ingress port of the port pair for which the bridge delay is being provided."
5  REFERENCE     "12.32.1"
6  ::= { ieee8021TsnRemoteMgmtBridgeDelayEntry 1 }
7
8  ieee8021TsnRemoteMgmtBridgeEgressPort OBJECT-TYPE
9      SYNTAX      IEEE8021BridgePortNumber
10     MAX-ACCESS   not-accessible
11     STATUS        current
12     DESCRIPTION
13         "The egress port of the port pair for which the bridge delay is being provided."
14     REFERENCE     "12.32.1"
15     ::= { ieee8021TsnRemoteMgmtBridgeDelayEntry 2 }
16
17  ieee8021TsnRemoteMgmtIndependentDelayMin OBJECT-TYPE
18     SYNTAX      Unsigned32
19     MAX-ACCESS   read-only
20     STATUS        current
21     DESCRIPTION
22         "This attribute provides the minimum delay independent
23         from frame length for a frame to forward from ingress
24         port to egress port.
25
26         The delay begins when the message timestamp point of the
27         ingress frame passes the reference plane marking the
28         boundary between the network media and PHY. The delay
29         ends when the message timestamp point of the egress
30         frame passes the reference plane marking the boundary
31         between the network media and PHY. The message timestamp
32         point is specified by IEEE Std 802.1AS for various media,
33         near the start of the frame.
34
35         Note: This delay includes all aspects of length-independent
36         delay for a frame that is forwarded, including handling of
37         error conditions."
38     REFERENCE     "12.32.1.1"
39     ::= { ieee8021TsnRemoteMgmtBridgeDelayEntry 3 }
40
41  ieee8021TsnRemoteMgmtIndependentDelayMax OBJECT-TYPE
42     SYNTAX      Unsigned32
43     MAX-ACCESS   read-only
44     STATUS        current
45     DESCRIPTION
46         "This attribute provides the maximum delay independent
47         from frame length for a frame to forward from ingress
48         port to egress port.
49
50         The delay begins when the message timestamp point of
51         the ingress frame passes the reference plane marking
52         the boundary between the network media and PHY. The delay
53         ends when the message timestamp point of the egress frame
54         passes the reference plane marking the boundary between
55         the network media and PHY. The message timestamp point is
56         specified by IEEE Std 802.1AS for various media, near the
57         start of the frame."
58     REFERENCE     "12.32.1.1"
59     ::= { ieee8021TsnRemoteMgmtBridgeDelayEntry 4 }
60
61  ieee8021TsnRemoteMgmtDependentDelayMin OBJECT-TYPE
62     SYNTAX      Unsigned32
63     MAX-ACCESS   read-only
64     STATUS        current
65     DESCRIPTION
66         "This attribute provides the minimum length-dependent
67         delay from ingress port to egress port.
68
69         It provides the portion of delay that is dependent on
70         frame length, where frame length is the number of octets
71         that transfer across the MAC Service interfaces. Each
72         length-dependent delay attribute specifies the time for

```

```

1         a single octet of the frame to transfer from ingress
2         to egress."
3     REFERENCE    "12.32.1.2"
4     ::= { ieee8021TsnRemoteMgmtBridgeDelayEntry 5 }
5
6 ieee8021TsnRemoteMgmtDependentDelayMax OBJECT-TYPE
7     SYNTAX      Unsigned32
8     MAX-ACCESS  read-only
9     STATUS      current
10    DESCRIPTION
11        "This attribute provides the maximum length-dependent
12        delay from ingress port to egress port.
13
14        It provides the portion of delay that is dependent on
15        frame length, where frame length is the number of octets
16        that transfer across the MAC Service interfaces. Each
17        length-dependent delay attribute specifies the time for
18        a single octet of the frame to transfer from ingress
19        to egress."
20    REFERENCE    "12.32.1.2"
21    ::= { ieee8021TsnRemoteMgmtBridgeDelayEntry 6 }
22
23 -- =====
24 -- the ieee8021TsnRemoteMgmtPropagationDelayTable
25 -- =====
26
27 ieee8021TsnRemoteMgmtPropagationDelayTable OBJECT-TYPE
28     SYNTAX      SEQUENCE OF Ieee8021TsnRemoteMgmtPropagationDelayEntry
29     MAX-ACCESS  not-accessible
30     STATUS      current
31     DESCRIPTION
32        "A table containing a set of parameters necessary to
33        determine the delay along the network media (e.g. cable)
34        for a frame transmitted from the specified Port of this
35        Bridge to the neighboring Port on a different Bridge.
36        There is one Propagation Delay managed object per
37        egress Port of a Bridge."
38    REFERENCE    "12.32.2"
39    ::= { ieee8021TsnRemoteMgmtPropagationDelay 1 }
40
41 ieee8021TsnRemoteMgmtPropagationDelayEntry OBJECT-TYPE
42     SYNTAX      Ieee8021TsnRemoteMgmtPropagationDelayEntry
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46        "A list of objects containing information necessary to
47        determine the delay along the network media (e.g. cable)
48        for a frame transmitted from the specified Port of this
49        Bridge to the neighboring Port on a different Bridge."
50     INDEX      { ieee8021BridgeBasePort }
51     ::= { ieee8021TsnRemoteMgmtPropagationDelayTable 1 }
52
53 Ieee8021TsnRemoteMgmtPropagationDelayEntry ::=
54     SEQUENCE {
55         ieee8021TsnRemoteMgmtTxPropagationDelay
56         Unsigned32
57     }
58
59 ieee8021TsnRemoteMgmtTxPropagationDelay OBJECT-TYPE
60     SYNTAX      Unsigned32
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64        "This attribute provides the transmission propagation delay.
65
66        The propagation delay begins when the message timestamp
67        point of an egress frame passes the reference plane marking
68        the boundary between the network media and PHY. It ends
69        when the message timestamp point of an ingress frame on the
70        neighboring Bridge's Port passes the reference plane marking
71        the boundary between the network media and PHY. The message
72        timestamp point is specified by IEEE Std 802.1AS for

```

```

1      various media."
2      REFERENCE    "12.32.2.1"
3      ::= { ieee8021TsnRemoteMgmtPropagationDelayEntry 1 }
4
5
6  -- =====
7  -- The Static Tree subtree
8  -- This subtree defines the objects necessary to determine if
9  -- the static trees feature is supported by the Bridge.
10 -- =====
11
12 ieee8021TsnRemoteMgmtStaticTreesSupported OBJECT-TYPE
13     SYNTAX      TruthValue
14     MAX-ACCESS  read-only
15     STATUS      current
16     DESCRIPTION
17         "This attribute is used by the TSN CNC to determine that
18         TE-MSTID is supported by the Bridge."
19     REFERENCE   "12.32.3.1"
20     ::= { ieee8021TsnRemoteMgmtStaticTrees 1 }
21
22 -- =====
23 -- the ieee8021TsnRemoteMgmtMsrpMrpExternalControlTable
24 -- =====
25
26 ieee8021TsnRemoteMgmtMsrpMrpExternalControlTable OBJECT-TYPE
27     SYNTAX      SEQUENCE OF Ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry
28     MAX-ACCESS  not-accessible
29     STATUS      current
30     DESCRIPTION
31         "A table containing a set of parameters necessary for
32         a network manager to 1) disable MRP attribute propagation (MAP)
33         for the MRP Participant of a bridge port, 2) read MRP attribute
34         registrations that the MRP Participant receives, and 3) write
35         MRP attribute values for the MRP Participant to declare."
36     REFERENCE   "12.32.4"
37     ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControl 1 }
38
39 ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry OBJECT-TYPE
40     SYNTAX      Ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44         "A list of objects necessary for a network manager to
45         1) disable MRP attribute propagation (MAP) for the
46         MRP Participant of a bridge port, 2) read MRP attribute
47         registrations that the MRP Participant receives, and 3) write
48         MRP attribute values for the MRP Participant to declare."
49     INDEX      { ieee8021BridgeBaseComponentId,
50                  ieee8021BridgeBasePort,
51                  ieee8021QBridgeVlanIndex }
52     ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlTable 1 }
53
54 Ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry ::=
55     SEQUENCE {
56         ieee8021TsnRemoteMgmtMsrpMrpExternalControl
57             TruthValue,
58         ieee8021TsnRemoteMgmtMrpIndicationList
59             OCTET STRING,
60         ieee8021TsnRemoteMgmtMrpIndicationListLength
61             Unsigned32,
62         ieee8021TsnRemoteMgmtMrpIndicationChangeCounter
63             Counter64,
64         ieee8021TsnRemoteMgmtMrpAdminRequestList
65             OCTET STRING,
66         ieee8021TsnRemoteMgmtMrpAdminRequestListLength
67             Unsigned32,
68         ieee8021TsnRemoteMgmtMrpOperRequestList
69             OCTET STRING,
70         ieee8021TsnRemoteMgmtMrpOperRequestListLength
71             Unsigned32
72     }

```

```

1
2 ieee8021TsnRemoteMgmtMsrpMrpExternalControl OBJECT-TYPE
3     SYNTAX      TruthValue
4     MAX-ACCESS  read-write
5     STATUS      current
6     DESCRIPTION
7         "This attribute is used to indicate, whether MRP
8         attributes are propagated on the MRP Participant,
9         according to the specifications for MRP Attribute
10        Propagation (MAP) and specifications of the
11        MRP Application. When true(1), the MRP Participant is
12        removed from the MRP Application's MAP Context. The
13        MRP Participant performs all other aspects of MRP,
14        including MRP operation, MRP specifications, and
15        MRPDU encodings. The application component stores MAD
16        indications for registration received on the Port,
17        and invokes MAD requests for declarations on the Port.
18        When false(2), MRP attributes propagate on the
19        MRP Participant according to the specifications for
20        MRP Attribute Propagation (MAP) and specifications of
21        the MRP Application. Ports with the externalControl
22        attribute false(2) are considered as candidates for
23        the MRP Application's MAP Context. The remaining
24        attributes of this subtree are ignored by Ports with
25        the externalControl attribute false(2).
26        This managed object applies to the MSRP application.
27        A table is provided for each MAP Context (VLAN ID)."
28     REFERENCE   "12.32.4.1"
29     DEFVAL { false }
30     ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 1 }
31
32 ieee8021TsnRemoteMgmtMrpIndicationList OBJECT-TYPE
33     SYNTAX      OCTET STRING
34     MAX-ACCESS  read-only
35     STATUS      current
36     DESCRIPTION
37         "This attribute is used to store the list of all joined
38         MRP attributes for the MRP Participant when the
39         ieee8021TsnRemoteMgmtMrpExternalControl attribute is
40         true(1). When the ieee8021TsnRemoteMgmtMrpExternalControl
41         attribute is false(2), this attribute is ignored by the
42         MRP Participant, and returns the empty octet string."
43     REFERENCE   "12.32.4.2"
44     ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 2 }
45
46 ieee8021TsnRemoteMgmtMrpIndicationListLength OBJECT-TYPE
47     SYNTAX      Unsigned32
48     MAX-ACCESS  read-only
49     STATUS      current
50     DESCRIPTION
51         "This attribute is used to provide the number of octets
52         in the ieee8021TsnRemoteMgmtMrpIndicationListLength
53         attribute. When the ieee8021TsnRemoteMgmtMrpExternalControl
54         attribute is false(2), this attribute returns zero."
55     REFERENCE   "12.32.4.3"
56     ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 3 }
57
58 ieee8021TsnRemoteMgmtMrpIndicationChangeCounter OBJECT-TYPE
59     SYNTAX      Counter64
60     MAX-ACCESS  read-only
61     STATUS      current
62     DESCRIPTION
63         "This attribute is used to provide the number of changes
64         done to the ieee8021TsnRemoteMgmtMrpIndicationList. When
65         the ieee8021TsnRemoteMgmtMrpExternalControl attribute is
66         false(2), this attribute returns zero."
67     REFERENCE   "12.32.4.4"
68     ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 4 }
69
70 ieee8021TsnRemoteMgmtMrpAdminRequestList OBJECT-TYPE
71     SYNTAX      OCTET STRING
72     MAX-ACCESS  read-only

```



```

1  STATUS      current
2  DESCRIPTION
3      "This attribute is used to provide the administrative
4      value of the current list of MAD requests for the
5      MRP Participant (operRequestList). Each entry in this
6      attribute is encoded as the attribute_type parameter as
7      a single octet, followed by the length of the
8      attribute_value parameter as a single octet, followed
9      by a sequence of octets for the attribute_value parameter.
10     When the ieee8021TsnRemoteMgmtMrpExternalControl attribute
11     is true(1), this attribute is copied to the
12     ieee8021TsnRemoteMgmtMrpOperRequestList attribute as soon
13     as possible according to the implementation. When the
14     ieee8021TsnRemoteMgmtMrpExternalControl attribute is
15     false(2), this attribute is ignored by the MRP Participant,
16     but it retains its value."
17  REFERENCE   "12.32.4.5"
18  DEFVAL { "" }
19  ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 5 }
20
21 ieee8021TsnRemoteMgmtMrpAdminRequestListLength OBJECT-TYPE
22  SYNTAX      Unsigned32
23  MAX-ACCESS  read-only
24  STATUS      current
25  DESCRIPTION
26      "This attribute is used to provide the administrative
27      value for the number of octets in the
28      ieee8021TsnRemoteMgmtMrpAdminRequestList attribute.
29      When the ieee8021TsnRemoteMgmtMrpExternalControl attribute
30      is true(1), this attribute is copied to the
31      ieee8021TsnRemoteMgmtMrpOperRequestListLength attribute at
32      the same time that the ieee8021TsnRemoteMgmtMrpAdminRequestList
33      attribute is copied to the ieee8021TsnRemoteMgmtMrpOperRequestList
34      attribute. When the ieee8021TsnRemoteMgmtMrpExternalControl
35      attribute is false(2), this attribute is ignored by the
36      MRP Participant, but it retains its value."
37  REFERENCE   "12.32.4.6"
38  DEFVAL { 0 }
39  ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 6 }
40
41 ieee8021TsnRemoteMgmtMrpOperRequestList OBJECT-TYPE
42  SYNTAX      OCTET STRING
43  MAX-ACCESS  read-only
44  STATUS      current
45  DESCRIPTION
46      "This attribute is used to provide the operational value
47      of the current list of MAD requests for the MRP Participant."
48  REFERENCE   "12.32.4.7"
49  ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 7 }
50
51 ieee8021TsnRemoteMgmtMrpOperRequestListLength OBJECT-TYPE
52  SYNTAX      Unsigned32
53  MAX-ACCESS  read-only
54  STATUS      current
55  DESCRIPTION
56      "This attribute is used to provide the operational value
57      of the ieee8021TsnRemoteMgmtMrpAdminRequestListLength
58      attribute, and it is copied at the same time that
59      ieee8021TsnRemoteMgmtMrpAdminRequestList attribute is
60      copied to ieee8021TsnRemoteMgmtMrpOperRequestList."
61  REFERENCE   "12.32.4.8"
62  ::= { ieee8021TsnRemoteMgmtMsrpMrpExternalControlEntry 8 }
63
64
65 -- =====
66 -- IEEE802 TSN REMOTE MANAGEMENT MIB - Conformance Information
67 -- =====
68
69 ieee8021TsnRemoteMgmtCompliances
70  OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtConformance 1 }
71 ieee8021TsnRemoteMgmtGroups
72  OBJECT IDENTIFIER ::= { ieee8021TsnRemoteMgmtConformance 2 }

```

```

1
2 -- =====
3 -- units of conformance
4 -- =====
5
6 -- =====
7 -- the ieee8021TsnRemoteMgmtBridgeDelay group
8 -- =====
9
10 ieee8021TsnRemoteMgmtBridgeDelayGroup OBJECT-GROUP
11     OBJECTS {
12         ieee8021TsnRemoteMgmtIndependentDelayMin,
13         ieee8021TsnRemoteMgmtIndependentDelayMax,
14         ieee8021TsnRemoteMgmtDependentDelayMin,
15         ieee8021TsnRemoteMgmtDependentDelayMax
16     }
17     STATUS      current
18     DESCRIPTION
19         "Objects that define the delay of frames as they pass
20         through the Bridge's relay."
21     ::= { ieee8021TsnRemoteMgmtGroups 1 }
22
23
24 -- =====
25 -- the ieee8021TsnRemoteMgmtPropagationDelay group
26 -- =====
27
28 ieee8021TsnRemoteMgmtPropagationDelayGroup OBJECT-GROUP
29     OBJECTS {
30         ieee8021TsnRemoteMgmtTxPropagationDelay
31     }
32     STATUS      current
33     DESCRIPTION
34         "Objects that define delay of frames along the network
35         media (e.g. cable)."
36     ::= { ieee8021TsnRemoteMgmtGroups 2 }
37
38
39 -- =====
40 -- the ieee8021TsnRemoteMgmtStaticTrees group
41 -- =====
42
43 ieee8021TsnRemoteMgmtStaticTreesGroup OBJECT-GROUP
44     OBJECTS {
45         ieee8021TsnRemoteMgmtStaticTreesSupported
46     }
47     STATUS      current
48     DESCRIPTION
49         "Objects that define static tree support."
50     ::= { ieee8021TsnRemoteMgmtGroups 3 }
51
52
53 -- =====
54 -- the ieee8021TsnRemoteMgmtMrpExternalControl group
55 -- =====
56
57 ieee8021TsnRemoteMgmtMrpExternalControlGroup OBJECT-GROUP
58     OBJECTS {
59         ieee8021TsnRemoteMgmtMsrpMrpExternalControl,
60         ieee8021TsnRemoteMgmtMrpIndicationList,
61         ieee8021TsnRemoteMgmtMrpIndicationListLength,
62         ieee8021TsnRemoteMgmtMrpIndicationChangeCounter,
63         ieee8021TsnRemoteMgmtMrpAdminRequestList,
64         ieee8021TsnRemoteMgmtMrpAdminRequestListLength,
65         ieee8021TsnRemoteMgmtMrpOperRequestList,
66         ieee8021TsnRemoteMgmtMrpOperRequestListLength
67     }
68     STATUS      current
69     DESCRIPTION
70         "Objects that define configuration of MRP External control."
71     ::= { ieee8021TsnRemoteMgmtGroups 4 }
72

```

```
1
2 -- =====
3 -- compliance statements
4 -- =====
5
6 ieee8021TsnRemoteMgmtCompliance MODULE-COMPLIANCE
7     STATUS          current
8     DESCRIPTION
9         "The compliance statement for devices supporting
10         TSN Remote management.
11
12         Support of the objects defined in the IEEE8021-TSN REMOTE
13         MANAGEMENT MIB also requires support of the IEEE8021-BRIDGE-MIB;
14         the provisions of 17.3.2 apply to implementations claiming
15         support of the IEEE8021-TSN REMOTE MANAGEMENT MIB."
16
17 MODULE -- this module
18     MANDATORY-GROUPS {
19         ieee8021TsnRemoteMgmtBridgeDelayGroup,
20         ieee8021TsnRemoteMgmtPropagationDelayGroup,
21         ieee8021TsnRemoteMgmtStaticTreesGroup,
22         ieee8021TsnRemoteMgmtMrpExternalControlGroup
23     }
24
25 ::= { ieee8021TsnRemoteMgmtCompliances 1 }
26
27 END
```

1 17.7.26 Definitions for the IEEE8021-PBBN-AA-MIB module

```
2 IEEE8021-PBBN-AA-MIB DEFINITIONS ::= BEGIN
3
4 -- =====
5 -- MIB for support of PBBN Auto Attach to Provider Backbone
6 -- Bridging (PBB) services.
7 -- =====
8
9 IMPORTS
10     MODULE-IDENTITY, OBJECT-TYPE, Integer32, Gauge32, Unsigned32, Counter32,
11     NOTIFICATION-TYPE
12     FROM SNMPv2-SMI
13     SnmpAdminString
14     FROM SNMP-FRAMEWORK-MIB
15     RowStatus, TruthValue, MacAddress
16     FROM SNMPv2-TC
17     MODULE-COMPLIANCE, OBJECT-GROUP, NOTIFICATION-GROUP
18     FROM SNMPv2-CONF;
19
20 ieee8021PbbnAutoAttachMib MODULE-IDENTITY
21     LAST-UPDATED "202309210000Z" -- September 21, 2023
22     ORGANIZATION "IEEE 802.1 Working Group"
23     CONTACT-INFO
24         " WG-URL: http://www.ieee802.org/1/
25         WG-EMail: stds-802-1-1@ieee.org
26         Contact: IEEE 802.1 Working Group Chair
27         Postal: C/O IEEE 802.1 Working Group
28                 IEEE Standards Association
29                 445 Hoes Lane
30                 Piscataway, NJ 08854
31                 USA
32         E-mail: stds-802-1-chairs@ieee.org"
33     DESCRIPTION
34         "The PBBN Auto Attach MIB module for configuration and statistics support.
35
36         Unless otherwise indicated, the references in this MIB module are to
37         IEEE Std 802.1Q Bridges and Bridged Networks:
38         IEEE Std 802.1Q-2022 as amended by IEEE Std 802.1Qcz, IEEE Std 802.1Qcw,
39         and IEEE Std 802.1Qcj.
40
41         Copyright (C) IEEE (2023).
42         This version of this MIB module is part of IEEE Std 802.1Q;
43         see that standard for full legal notices."
44
45     REVISION "202309210000Z" -- September 21, 2023
46     DESCRIPTION
47         "Initial version published as part of IEEE Std 802.1Qcj."
48
49     ::= { iso (1) iso-identified-organization (3) ieee (111)
50         standards-association-numbered-series-standard1s (2)
51         lan-man-stds (802) ieee802dot1 (1)
52         ieee802dot1mibs (1) ieee8021PbbnAutoAttachMib (37) }
53
54 ieee8021AaNotifications OBJECT IDENTIFIER ::= { ieee8021PbbnAutoAttachMib 0 }
55 ieee8021AaObjects OBJECT IDENTIFIER ::= { ieee8021PbbnAutoAttachMib 1 }
56 ieee8021AaNotifyObjects OBJECT IDENTIFIER ::= { ieee8021PbbnAutoAttachMib 2 }
57
58 ieee8021AaConfig OBJECT IDENTIFIER ::= { ieee8021AaObjects 1 }
59 ieee8021AaStats OBJECT IDENTIFIER ::= { ieee8021AaObjects 2 }
60
61 -- =====
62 -- Auto Attach scalar attributes
63 -- =====
64
65 ieee8021AaSystemEnable OBJECT-TYPE
66     SYNTAX INTEGER {
67         enabled(1),
68         disabled(2)
69     }
70     MAX-ACCESS read-write
71     STATUS current
```

```

1  DESCRIPTION
2  "Exports the status of the Auto Attach service on this system."
3  ::= { ieee8021AaConfig 1 }
4
5  ieee8021AaSystemType OBJECT-TYPE
6      SYNTAX      INTEGER {
7          aaBeb(1),
8          aaDeviceVlanAware(2),
9          aaDeviceVlanUnaware(3)
10         }
11      MAX-ACCESS   read-only
12      STATUS       current
13      DESCRIPTION
14          "Defines the AAS type indicating the services supported by the system.
15           May be read-only on platforms that only support a single system type."
16      ::= { ieee8021AaConfig 2 }
17
18  ieee8021AaSystemMAC OBJECT-TYPE
19      SYNTAX      MacAddress
20      MAX-ACCESS   read-only
21      STATUS       current
22      DESCRIPTION
23          "Defines the Auto Attach System MAC Address used to uniquely identify
24           this system."
25      ::= { ieee8021AaConfig 3 }
26
27  ieee8021AaSystemResetTime OBJECT-TYPE
28      SYNTAX      Integer32 (1..1200)
29      MAX-ACCESS   read-write
30      STATUS       current
31      DESCRIPTION
32          "AAP reset time delay used to re-synchronize the associated AASs"
33      ::= { ieee8021AaConfig 4 }
34
35  -- =====
36  -- Auto Attach VID/I-SID assignment table support
37  -- =====
38
39  ieee8021AaVidIsidAsgnsTable OBJECT-TYPE
40      SYNTAX      SEQUENCE OF Ieee8021AaVidIsidAsgnsEntry
41      MAX-ACCESS   not-accessible
42      STATUS       current
43      DESCRIPTION
44          "This table contains the VID <-> I-SID assignments for the
45           Auto Attach application."
46      ::= { ieee8021AaConfig 5 }
47
48  ieee8021AaVidIsidAsgnsEntry OBJECT-TYPE
49      SYNTAX      Ieee8021AaVidIsidAsgnsEntry
50      MAX-ACCESS   not-accessible
51      STATUS       current
52      DESCRIPTION
53          "Information about the I-SID <-> VID assignments for the
54           Auto Attach application."
55      INDEX { ieee8021AaVidIsidAsgnsIfIndex,
56              ieee8021AaVidIsidAsgnsIsid,
57              ieee8021AaVidIsidAsgnsVid }
58      ::= { ieee8021AaVidIsidAsgnsTable 1 }
59
60  Ieee8021AaVidIsidAsgnsEntry ::= SEQUENCE {
61      ieee8021AaVidIsidAsgnsIfIndex      Integer32,
62      ieee8021AaVidIsidAsgnsIsid         Integer32,
63      ieee8021AaVidIsidAsgnsVid          Integer32,
64      ieee8021AaVidIsidAsgnsStatus       INTEGER,
65      ieee8021AaVidIsidAsgnsRowStatus    RowStatus }
66
67  ieee8021AaVidIsidAsgnsIfIndex OBJECT-TYPE
68      SYNTAX      Integer32 (0..65535)
69      MAX-ACCESS   not-accessible
70      STATUS       current
71      DESCRIPTION
72          "Interface identifier component of the I-SID <-> VID assignment."

```

```

1 ::= { ieee8021AaVidIsidAsgnsEntry 1 }
2
3 ieee8021AaVidIsidAsgnsIsid OBJECT-TYPE
4     SYNTAX      Integer32 (1256..16777214)
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "I-SID component of the I-SID <-> VID assignment."
9     ::= { ieee8021AaVidIsidAsgnsEntry 2 }
10
11 ieee8021AaVidIsidAsgnsVid OBJECT-TYPE
12     SYNTAX      Integer32 (1..4094)
13     MAX-ACCESS  not-accessible
14     STATUS      current
15     DESCRIPTION
16         "VLAN ID component of the I-SID <-> VID assignment."
17     ::= { ieee8021AaVidIsidAsgnsEntry 3 }
18
19 ieee8021AaVidIsidAsgnsStatus OBJECT-TYPE
20     SYNTAX      INTEGER {
21         pending(1),
22         accepted(2),
23         rejected(3),
24         rejectedAutoAttachResourcesUnavailable(4),
25         rejectedInvalidVLANID(5),
26         rejectedVLANResourcesUnavailable(6),
27         rejectedInvalidISID(7),
28         rejectedISIDResourcesUnavailable(8),
29         rejectedApplicationInteractionIssue(9),
30         rejectedAssignmentNotAllowed(10)
31     }
32     MAX-ACCESS  read-only
33     STATUS      current
34     DESCRIPTION
35         "The current state of the Auto Attach I-SID <-> VID assignment.
36         Rejections can be from multiple causes including resources
37         unavailable, duplicate, application interaction, etc."
38     ::= { ieee8021AaVidIsidAsgnsEntry 4 }
39
40 ieee8021AaVidIsidAsgnsRowStatus OBJECT-TYPE
41     SYNTAX      RowStatus
42     MAX-ACCESS  read-write
43     STATUS      current
44     DESCRIPTION
45         "This object allows entries to be created and deleted in the
46         ieee8021AaVidIsidAsgnsTable."
47     ::= { ieee8021AaVidIsidAsgnsEntry 5 }
48
49 -- =====
50 -- Auto Attach port table support
51 -- =====
52
53 ieee8021AaPortTable OBJECT-TYPE
54     SYNTAX      SEQUENCE OF Ieee8021AaPortEntry
55     MAX-ACCESS  not-accessible
56     STATUS      current
57     DESCRIPTION
58         "This table contains port-specific data for the
59         Auto Attach application."
60     ::= { ieee8021AaConfig 6 }
61
62 ieee8021AaPortEntry OBJECT-TYPE
63     SYNTAX      Ieee8021AaPortEntry
64     MAX-ACCESS  not-accessible
65     STATUS      current
66     DESCRIPTION
67         "Port-specific data related to the Auto Attach application."
68     INDEX { ieee8021AaPortIfIndex,
69             ieee8021AaPortNetId }
70     ::= { ieee8021AaPortTable 1 }
71
72 Ieee8021AaPortEntry ::= SEQUENCE {

```

```

1  ieee8021AaPortIfIndex      Integer32,
2  ieee8021AaPortNetId        OCTET STRING,
3  ieee8021AaPortEnable       INTEGER,
4  ieee8021AaPortRowStatus    RowStatus }
5
6  ieee8021AaPortIfIndex OBJECT-TYPE
7      SYNTAX      Integer32 (0..65535)
8      MAX-ACCESS   not-accessible
9      STATUS      current
10     DESCRIPTION
11         "IfIndex on the interface (port) identifier attribute."
12     ::= { ieee8021AaPortEntry 1 }
13
14  ieee8021AaPortNetId OBJECT-TYPE
15      SYNTAX      OCTET STRING (SIZE (0..12))
16      MAX-ACCESS   read-only
17      STATUS      current
18      DESCRIPTION
19         "Identifies an AAS port uniquely within the administrative domain.
20         The object is the concatenation of the ieee8021AaSystemMAC and
21         ieee8021AaPortIfIndex. The identified port can be simple, a LAG,
22         or a DRNI."
23     ::= { ieee8021AaPortEntry 2 }
24
25  ieee8021AaPortEnable OBJECT-TYPE
26      SYNTAX      INTEGER {
27          enabled(1),
28          disabled(2)
29      }
30      MAX-ACCESS   read-create
31      STATUS      current
32      DESCRIPTION
33         "The current port Auto Attach Protocol enable. Indicates whether
34         Auto Attach TLVs will be included in LLDPDUs generated on
35         the port (enabled) or not (disabled)."
36     ::= { ieee8021AaPortEntry 3 }
37
38  ieee8021AaPortRowStatus OBJECT-TYPE
39      SYNTAX      RowStatus
40      MAX-ACCESS   read-create
41      STATUS      current
42      DESCRIPTION
43         "This object allows entries to be created, deleted, and modified in the
44         ieee8021AaPortTable, if the operation is supported by the agent."
45     ::= { ieee8021AaPortEntry 4 }
46
47 -- =====
48 -- Discovered Auto Attach Systems table support
49 -- =====
50
51  ieee8021AaDiscSystemsTable OBJECT-TYPE
52      SYNTAX      SEQUENCE OF Ieee8021AaDiscSystemsEntry
53      MAX-ACCESS   not-accessible
54      STATUS      current
55      DESCRIPTION
56         "This table contains details about locally connected AASs that have
57         been discovered."
58     ::= { ieee8021AaConfig 7 }
59
60  ieee8021AaDiscSystemsEntry OBJECT-TYPE
61      SYNTAX      Ieee8021AaDiscSystemsEntry
62      MAX-ACCESS   not-accessible
63      STATUS      current
64      DESCRIPTION
65         "Contains details about a locally connected AAS."
66      INDEX { ieee8021AaDiscLocPortIfIndex,
67              ieee8021AaDiscRemPortNetId }
68     ::= { ieee8021AaDiscSystemsTable 1 }
69
70  Ieee8021AaDiscSystemsEntry ::= SEQUENCE {
71      ieee8021AaDiscLocPortIfIndex      Integer32,
72      ieee8021AaDiscRemSystemType        INTEGER,

```

```

1  ieee8021AaDiscRemPortNetId      OCTET STRING,
2  ieee8021AaDiscRemPortTagging     BITS,
3  ieee8021AaDiscRemPortAssocState  INTEGER,
4  ieee8021AaDiscLocPortAssocState  INTEGER }
5
6  ieee8021AaDiscLocPortIfIndex OBJECT-TYPE
7      SYNTAX          Integer32 (0..65535)
8      MAX-ACCESS      not-accessible
9      STATUS          current
10     DESCRIPTION
11         "Identifies the interface through which the AAS
12         represented by this entry was discovered."
13     ::= { ieee8021AaDiscSystemsEntry 1 }
14
15  ieee8021AaDiscRemSystemType OBJECT-TYPE
16      SYNTAX          INTEGER {
17          aaBeb(1),
18          aaDeviceCVlanAware(2),
19          aaDeviceVlanUnaware(3),
20          aaDeviceSVlanAware(4)
21      }
22      MAX-ACCESS      read-only
23      STATUS          current
24      DESCRIPTION
25         "Identifies the AAS type as advertised through LLDP."
26     ::= { ieee8021AaDiscSystemsEntry 2 }
27
28  ieee8021AaDiscRemPortNetId OBJECT-TYPE
29      SYNTAX          OCTET STRING (SIZE (0..12))
30      MAX-ACCESS      read-only
31      STATUS          current
32      DESCRIPTION
33         "Identifies an AAS port uniquely within the administrative domain.
34         The object is the concatenation of the ieee8021AaSystemMAC and
35         port IfIndex. The identified port can be simple, a LAG, or a DRNI."
36     ::= { ieee8021AaDiscSystemsEntry 3 }
37
38  ieee8021AaDiscRemPortTagging OBJECT-TYPE
39      SYNTAX          BITS {
40          trafficTagged(0),
41          trafficTaggedAndUntagged(1),
42          trafficUntaggedOnly(2)
43      }
44      MAX-ACCESS      read-only
45      STATUS          current
46      DESCRIPTION
47         "Exports the tagging data associated with the discovered
48         Auto Attach System as advertised through LLDP. When bit 0 is set
49         to 0, all traffic is tagged on the link; when bit 0 is set to 1,
50         the link supports both tagged and untagged traffic. When bit 2 is
51         set to 1, the link supports only untagged traffic."
52     ::= { ieee8021AaDiscSystemsEntry 4 }
53
54  ieee8021AaDiscRemPortAssocState OBJECT-TYPE
55      SYNTAX          INTEGER {
56          notReady(0), -- initializing, not ready
57          readyToAssoc(1), -- ready to associate
58          readyToAttach(2), -- viable partner
59          assocAttached(3), -- port is active
60          assocFailTypes(18), -- System type mismatched
61          assocStandby(19), -- at AAD, active elsewhere
62          assocFailTags(34), -- Tagging mismatched
63          assocInvalid(35), -- at AAB, AAD active elsewhere
64          assocFailTopo(50), -- Multipoint link detected
65          assocFailOther(66) -- Parsing or resourcing error
66      }
67      MAX-ACCESS      read-only
68      STATUS          current
69      DESCRIPTION
70         "The current remote operational status for the associated interface."
71     ::= { ieee8021AaDiscSystemsEntry 5 }
72

```



```

1 ieee8021AaDiscLocPortAssocState OBJECT-TYPE
2   SYNTAX      INTEGER {
3       notReady(0), -- initializing, not ready
4       readyToAssoc(1), -- advertising ready to associate
5       readyToAttach(2), -- viable partner
6       assocAttached(3), -- port is active
7       assocFailTypes(18), -- System type mismatched
8       assocStandby(19), -- at AAD, active elsewhere
9       assocFailTags(34), -- Tagging mismatched
10      assocInvalid(35), -- at AAB, AAD active elsewhere
11      assocFailTopo(50), -- Multipoint link detected
12      assocFailOther(66) -- Parsing or resourcing error
13      }
14   MAX-ACCESS   read-only
15   STATUS       current
16   DESCRIPTION
17       "The current local operational status for the associated interface."
18   ::= { ieee8021AaDiscSystemsEntry 6 }
19
20 -- =====
21 -- AAS statistics and error counters table
22 -- =====
23
24 ieee8021AaStatsTable OBJECT-TYPE
25   SYNTAX      SEQUENCE OF Ieee8021AaStatsEntry
26   MAX-ACCESS   not-accessible
27   STATUS       current
28   DESCRIPTION
29       "This table contains Auto Attach port statistics data."
30   ::= { ieee8021AaStats 1 }
31
32 ieee8021AaStatsEntry OBJECT-TYPE
33   SYNTAX      Ieee8021AaStatsEntry
34   MAX-ACCESS   not-accessible
35   STATUS       current
36   DESCRIPTION
37       "Each entry contains Auto Attach operational statistics for a
38       specific Auto Attach System port. Ports are identified by their MIB II
39       ifIndex value."
40   INDEX       { ieee8021AaStatsPortIfIndex }
41   ::= { ieee8021AaStatsTable 1 }
42
43 Ieee8021AaStatsEntry ::= SEQUENCE {
44     ieee8021AaStatsPortIfIndex      Integer32,
45     ieee8021AaStatsAssocAttached    Counter32,
46     ieee8021AaStatsAsgnsRequested   Counter32,
47     ieee8021AaStatsAsgnsAccepted    Counter32,
48     ieee8021AaStatsAsgnRejected     Counter32,
49     ieee8021AaStatsAssocFailed      Counter32,
50     ieee8021AaStatsAsgnsWithdrawn   Counter32,
51     ieee8021AaStatsAssocReset       Counter32,
52     ieee8021AaStatsAssocStandby     Counter32 }
53
54 ieee8021AaStatsPortIfIndex OBJECT-TYPE
55   SYNTAX      Integer32 (0..65535)
56   MAX-ACCESS   not-accessible
57   STATUS       current
58   DESCRIPTION
59       "An index value that uniquely identifies a port. This value
60       corresponds to a MIB II ifIndex."
61   ::= { ieee8021AaStatsEntry 1 }
62
63 ieee8021AaStatsAssocAttached OBJECT-TYPE
64   SYNTAX      Counter32
65   MAX-ACCESS   read-only
66   STATUS       current
67   DESCRIPTION
68       "Indicates the number of PBBN Auto Attach System TLVs received on the
69       identified port."
70   ::= { ieee8021AaStatsEntry 2 }
71

```

```

1 ieee8021AaStatsAsgnsRequested OBJECT-TYPE
2     SYNTAX          Counter32
3     MAX-ACCESS      read-only
4     STATUS          current
5     DESCRIPTION
6         "Indicates the number of I-SID/VID bindings received in
7         PBBN Auto Attach Assignment TLVs on the identified port."
8     ::= { ieee8021AaStatsEntry 3 }
9
10 ieee8021AaStatsAsgnsAccepted OBJECT-TYPE
11     SYNTAX          Counter32
12     MAX-ACCESS      read-only
13     STATUS          current
14     DESCRIPTION
15         "Indicates the number of I-SID/VID bindings from received
16         PBBN Auto Attach Assignment TLVs that are accepted (activated)
17         on the identified port. Counter is incremented when the binding
18         transitions from a non-accepted state (e.g., 'pending', 'rejected')
19         to the accepted state. The AAD counts the number received from the
20         AAB while the AAB counts the number transmitted to the AAD."
21     ::= { ieee8021AaStatsEntry 4 }
22
23 ieee8021AaStatsAsgnRejected OBJECT-TYPE
24     SYNTAX          Counter32
25     MAX-ACCESS      read-only
26     STATUS          current
27     DESCRIPTION
28         "Indicates the number of I-SID/VID bindings from received Auto Attach
29         Assignment TLVs that are rejected on the identified port. Counter is
30         incremented when the binding transitions from a non-rejected state
31         (e.g., 'pending', 'accepted') to the rejected state. The AAD counts
32         the number received from the AAB while the AAB counts the number
33         transmitted to the AAD."
34     ::= { ieee8021AaStatsEntry 5 }
35
36 ieee8021AaStatsAssocFailed OBJECT-TYPE
37     SYNTAX          Counter32
38     MAX-ACCESS      read-only
39     STATUS          current
40     DESCRIPTION
41         "Indicates the number of discovered AAS from received AA System TLVs
42         that have failed association or attachment due to errors such as
43         system mismatches, multi-point link, incompatible tagging, etc."
44     ::= { ieee8021AaStatsEntry 6 }
45
46 ieee8021AaStatsAsgnsWithdrawn OBJECT-TYPE
47     SYNTAX          Counter32
48     MAX-ACCESS      read-only
49     STATUS          current
50     DESCRIPTION
51         "Indicates the number of I-SID/VID bindings from received PBBN
52         Auto Attach Assignment TLVs that have been deleted on the identified
53         port. Counter is only incremented when bindings are deleted for
54         reasons other than expiration."
55     ::= { ieee8021AaStatsEntry 7 }
56
57 ieee8021AaStatsAssocReset OBJECT-TYPE
58     SYNTAX          Counter32
59     MAX-ACCESS      read-only
60     STATUS          current
61     DESCRIPTION
62         "Indicates the number of discovered AAS from received AA System TLVs
63         that have reset an association or attachment due deleted
64         discovered system TLVs, etc. This condition is most commonly a result
65         of an AAS being reset or powered down."
66     ::= { ieee8021AaStatsEntry 8 }
67

```

```

1 ieee8021AaStatsAssocStandby OBJECT-TYPE
2     SYNTAX      Counter32
3     MAX-ACCESS   read-only
4     STATUS      current
5     DESCRIPTION
6         "Indicates the number of discovered AAS from received AA System TLVs
7         that have had an association in standby."
8     ::= { ieee8021AaStatsEntry 9 }
9
10 -- =====
11 -- Auto Attach notification objects
12 -- =====
13
14 ieee8021AaDiscSystemsDescr OBJECT-TYPE
15     SYNTAX      SnmpAdminString (SIZE(0..255))
16     MAX-ACCESS   accessible-for-notify
17     STATUS      current
18     DESCRIPTION
19         "The string value used to identify the LLDP system description of the
20         remote system distributed through LLDPDUs."
21     REFERENCE
22         "IEEE 802.1AB-2016 8.5.7.2"
23     ::= { ieee8021AaNotifyObjects 1 }
24
25 ieee8021AaDiscSystemsMgmtOid OBJECT-TYPE
26     SYNTAX      OBJECT IDENTIFIER
27     MAX-ACCESS   accessible-for-notify
28     STATUS      current
29     DESCRIPTION
30         "The OID value used to identify the type of hardware component or
31         protocol entity associated with the management address advertised by
32         the remote system agent distributed through LLDP."
33     REFERENCE
34         "IEEE 802.1AB-2016 8.5.9.8"
35     ::= { ieee8021AaNotifyObjects 2 }
36
37 -- =====
38 -- Auto Attach notifications
39 -- =====
40
41 ieee8021AaDiscoveredSystem NOTIFICATION-TYPE
42     OBJECTS      {
43         ieee8021AaDiscRemSystemType,
44         ieee8021AaDiscRemPortNetId,
45         ieee8021AaDiscSystemsDescr,
46         ieee8021AaDiscSystemsMgmtOid
47     }
48     STATUS      current
49     DESCRIPTION
50         "This notification is generated when an AAS is discovered."
51     ::= { ieee8021AaNotifications 1 }
52
53
54 -- =====
55 -- IEEE 802.1 - Auto Attach conformance information
56 -- =====
57
58 ieee8021AaConformance OBJECT IDENTIFIER ::= { ieee8021PbbnAutoAttachMib 4 }
59
60 ieee8021AaCompliances
61     OBJECT IDENTIFIER ::= { ieee8021AaConformance 1 }
62 ieee8021AaGroups
63     OBJECT IDENTIFIER ::= { ieee8021AaConformance 2 }
64
65

```

```
1 --
2 -- Auto Attach - compliance statements
3 --
4
5 ieee8021AaCompliance MODULE-COMPLIANCE
6     STATUS          current
7     DESCRIPTION
8         "A compliance statement for SNMP entities that implement
9         the IEEE 802.1Qcj Auto Attach MIB.
10
11         This group is mandatory for agents that implement the
12         Auto Attach aaSet TLV set."
13     MODULE          -- this module
14     MANDATORY-GROUPS { ieee8021AaGroup }
15     ::= { ieee8021AaCompliances 1 }
16
17
18 --
19 -- Auto Attach - MIB groupings
20 --
21
22 ieee8021AaGroup  OBJECT-GROUP
23     OBJECTS {
24         ieee8021AaSystemEnable,
25         ieee8021AaSystemType,
26         ieee8021AaSystemMAC,
27         ieee8021AaVidIsidAsgnsStatus,
28         ieee8021AaVidIsidAsgnsRowStatus,
29         ieee8021AaPortNetId,
30         ieee8021AaPortEnable,
31         ieee8021AaDiscRemPortNetId,
32         ieee8021AaStatsAssocAttached,
33         ieee8021AaStatsAsgnsRequested,
34         ieee8021AaStatsAsgnsAccepted,
35         ieee8021AaStatsAsgnRejected,
36         ieee8021AaDiscSystemsDescr,
37         ieee8021AaDiscSystemsMgmtOid
38     }
39     STATUS          current
40     DESCRIPTION
41         "The collection of objects that support configuration of the PBBN
42         Auto Attach service"
43     ::= { ieee8021AaGroups 1 }
44
45 END
46
```

1 18. Principles of Connectivity Fault Management operation

2 Connectivity Fault Management (CFM) comprises capabilities for detecting, verifying, and isolating
3 connectivity failures in Virtual Bridged Networks. These capabilities can be used in networks operated by
4 multiple independent organizations, each with restricted management access to each other's equipment.

5 CFM is designed to be transparent to the customer data transported by a network and to be capable of
6 providing maximum fault coverage. Accordingly, CFM Entities (Clause 19) are specified as shims that make
7 use of and provide the ISS or EISS (IEEE Std 802.1AC, 6.8, 6.17) at SAPs within the network. They can, in
8 principle, be added between any of the other media-independent protocol entities that compose a Bridge
9 Port, without requiring changes to those entities. Customer data is forwarded transparently by CFM Entities
10 while CFM PDUs are generated and processed as specified in Clause 20. The formats of the various CFM
11 PDUs are described in Clause 21, and the creation and placement of CFM Entities in Bridges are specified in
12 Clause 22.

13 This clause provides the context necessary to understand each of the CFM protocols, and how CFM Entities
14 in Bridges are selected and configured as Maintenance Points (MPs, 19.1) to participate in and operate those
15 protocols.

16 CFM introduces the following concepts to support multiple independent operators, each supporting service
17 instances for multiple independent customers:

- 18 a) A Maintenance Domain (18.1) is a part of a network that is controlled by a single operator and used
19 to support connectivity between the Domain Service Access Points (DoSAPs, 18.1) that bound the
20 Maintenance Domain.
- 21 b) A Maintenance Domain Level (MD Level, 18.3), carried in CFM PDUs, allows each of an
22 operator's customers also to use CFM and to function as an operator if desired, multiplexing
23 provided service instances over its own connectivity.
- 24 c) A Maintenance Association (MA, 18.2) is created by configuring CFM Entities that support an
25 individual service instance's DoSAPs as MA Endpoints (MEPs), and is used to monitor connectivity
26 provided by that instance through the Maintenance Domain.

27 MP (19.1) configuration of CFM Entities supports the hierarchical nesting of Maintenance Domains. The
28 DoSAPs for a given service instance are Intermediate Service Access Points (ISAPs) for MAs monitoring
29 connectivity through an enclosing Maintenance Domain. When a MEP is configured, a MA Intermediate
30 Point (MIP) can also be configured at the appropriate level for its immediately enclosing Maintenance
31 Domain. Below the innermost Maintenance Domain, every physical LAN can serve as an implied
32 Maintenance Domain, and every Bridge Port as an implied MEP. In the innermost Maintenance Domain,
33 every Bridge Port is an ISAP and can be configured with a MIP.

34 Maintenance Domains are nested, not overlapped. That is, if a given DoSAP for service instance 1
35 in Maintenance Domain *x* is an ISAP for Maintenance Domain *y*, then all DoSAPs for service instance 1 in
36 Maintenance Domain *x* are either ISAPs or DoSAPs for Maintenance Domain *y*, and no other Maintenance
37 Domain. Conformance to this rule is not a condition for the correct operation of CFM. Rather, if this rule is
38 violated, then CFM will detect the violation as an error condition, whatever the intentions of the
39 administrators.

40 The information about individual service instances that is configured and recorded within a network to
41 support CFM is entirely associated with MEPs, and thus scales linearly with the number of SAPs provided
42 to customers. The transmission of CFM PDUs is stimulated by state machines associated with MEPs as are
43 actions taken that require recording information from received PDUs. MIPs can add, check, and respond to
44 information in received PDUs, thus supporting discovery of paths among MEPs and location of faults along
45 those paths. MIPs can be configured per Maintenance Domain, thus allowing the amount of MIP

1 configuration information to scale linearly with the size of the network, and is thus a constant function of the
2 ability of the network to support those MIPs, rather than being a product of the number of service instances
3 supported. In contrast each MEP is associated with a SAP that provides access to a single service instance.

4 CFM functions are partitioned as follows:

- 5 — Path discovery
- 6 — Fault detection
- 7 — Fault verification and isolation
- 8 — Fault notification
- 9 — Fault recovery

10 Path discovery uses the Linktrace protocol (20.3) to determine the path taken to a target MAC address, MIP
11 by MIP, from one MEP to another MP across an MA. This target MAC address can be that of a MIP, a MEP,
12 or any other individual MAC address. A Linktrace Message (LTM, 20.3.1) is transmitted from a MEP to its
13 neighboring MIPs, and from MIP to MIP, to the MP terminating the path. Each MIP along the path and the
14 terminating MP return unicast Linktrace Replies (LTR, 20.3.2) to the originating MEP. LTMs are triggered
15 by administrative action. The replies are cataloged by the originating MEP for examination by the
16 administrator.

17 Fault detection uses the Continuity Check protocol (20.1) to detect both connectivity failures and unintended
18 connectivity between service instances. Each MEP can periodically transmit a Continuity Check Message
19 (CCM) announcing the identity of the MEP and its MA, and tracks the CCMs received from the other MEPs.
20 All connectivity faults that can misdirect a CCM show up as differences between the CCMs received and the
21 MEP's configured expectations. The state of the tracked CCMs is available to the administrator for
22 examination.

23 Fault verification and fault isolation are administrative actions, typically performed after fault detection.
24 Fault verification can also confirm successful initiation or restoration of connectivity. The administrator uses
25 the Loopback protocol (20.2) to perform fault verification. A MEP can be ordered to transmit a Loopback
26 Message (LBM, 20.2.1) to a MEP or MIP in the MA, whose MAC address can be discovered from CCMs
27 (MEPs only) or LTMs/LTRs (MEPs or MIPs). The receiving MP responds by transforming the LBM into a
28 unicast Loopback Reply (LBR, 20.2.2) sent back to the originating MEP. That MEP records the responses
29 for examination by the administrator.

30 Fault notification is provided by a MEP that has detected a connectivity fault in its MA, because expected
31 CCMs were not received, because unexpected or invalid CCMs were received, or because a CCM carried a
32 notification of the failure of its associated Bridge Port. A single fault may be detected by multiple MEPs and
33 may result in the generation of multiple Fault Alarms. In the absence of alarm suppression mechanisms in
34 CFM, network management systems receiving such Fault Alarms can deploy other means to identify if a
35 Fault Alarm identifies the primary fault for which a fault corrective action can be initiated, or represents a
36 secondary fault for which no corrective action can be taken.

37 For VLANs, fault recovery is provided by the active topology protocols of Clause 13 while fault recovery
38 for TESIs is provided by the Protection Switching mechanisms of 26.10. Fault recovery actions performed
39 by the network administrator, such as the correction of configuration errors or replacement of failed
40 components, are outside the scope of this standard.

1 18.1 Maintenance Domains and DoSAPs

2 A “domain” may be defined as a network or part of a network that is capable of offering connectivity to
3 systems outside this region. It comprises a set of DoSAPs at which the domain is capable of or intended to
4 offer connectivity to systems outside the domain. Each DoSAP is an instance either of the EISS or of the
5 ISS. A Maintenance Domain is then a domain or part of a domain for which faults in connectivity are to be
6 managed.

7 A Maintenance Domain is, or is intended to be, fully connected internally. That is, a DoSAP associated with
8 a Maintenance Domain has connectivity to every other DoSAP in the Maintenance Domain, in the absence
9 of faults. It is, in principle at least, capable of connecting (i.e., supporting service between) any of its
10 DoSAPs. The factors (apart from faults) that can prevent desired connectivity are second-order, e.g.,
11 exhaustion of resources that would support a proposed connection, an administrative decision to limit
12 multipoint services, misconfiguration of security parameters, unwillingness of customers or attached
13 Maintenance Domains to use a particular connection configuration. The purpose of CFM is to monitor and
14 diagnose the ability of a Maintenance Domain to meet its operators’ intentions for connectivity, as expressed
15 in the configuration of CFM.

16 Figure 18-1 illustrates a single Maintenance Domain, consisting of five Bridges, from the point of view of
17 an operator. The operator has six DoSAPs to offer to customers.

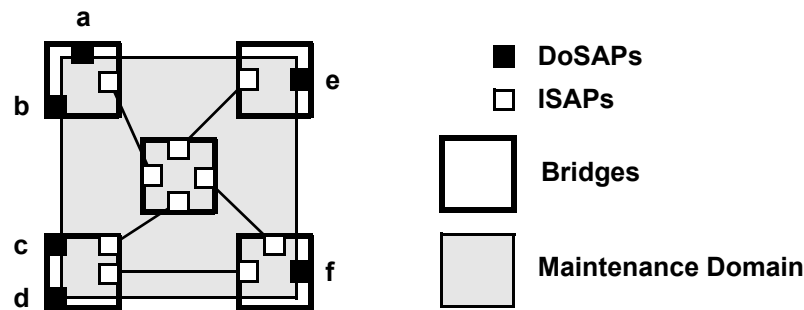


Figure 18-1—One Maintenance Domain: operator’s view

18 Each Maintenance Domain can be separately administered. Each Maintenance Domain is assigned a
19 Maintenance Domain Name. This should be chosen to be unique among all those used or available to an
20 operator, and to facilitate easy identification of administrative responsibility for the Maintenance Domain.
21 CFM PDU formats support the creation of Maintenance Domain Names that are unique over the domain for
22 which CFM is to protect against accidental concatenation of service instances. Ideally, these names are
23 globally unique.

24 There are a number of methods to fulfill the requirements to both maintain connectivity within, and control
25 access to, a Maintenance Domain. A Maintenance Domain can be, for example,

- 26 a) An MST Region, identified and kept separate from other Maintenance Domains, by its MST
27 Configuration Name (8.9.2, 13.8).
- 28 b) A number of such MST Regions (13.5.3), interconnected via the CIST (13.5.2), and kept separate
29 from other Maintenance Domains by configuring the managed objects controlling the CIST
30 topology.
- 31 c) An entire Virtual Bridged Network, and kept separate from other Maintenance Domains by means
32 not specified in this standard.
- 33 d) A subset of the DoSAPs of a VLAN Bridged Network, assigned to the use of a particular higher-
34 level provider among many.

- e) A subset of the DoSAPs of the concatenated network in item c).
- f) A single physical link between two providers or between a provider and a customer.
- g) A single VLAN on a single physical link between two providers or between a provider and a customer.

18.2 Service instances and MAs

When a network administrator configures a service instance for a customer, the network administrator configures some number of the DoSAPs to be accessible by that customer, assigning whatever identifiers the systems require to segregate that service instance from others supported by that Maintenance Domain (e.g., VLANs), as well as configuring other service properties (such as bandwidth profiles) for the DoSAPs. Creation of a service instance establishes a connectionless connectivity (IEEE Std 802.1AC) among the selected DoSAPs.

Configuring a MEP at each of the DoSAPs of a service instance creates an MA to monitor that connectionless connectivity. The MA is identified by a Short MA Name that is unique within the Maintenance Domain. Together, the Maintenance Domain Name and the Short MA Name form the Maintenance Association Identifier (MAID) that is carried in CFM PDUs to identify inadvertent connections among MEPs. A small integer, the MA Endpoint Identifier (MEPID) uniquely identifies each MEP among those configured on a single MA.

NOTE 1—In order to accommodate the requirements of service providers as expressed in ITU-T G.8013/Y.1731, the Maintenance Domain Name can be configured to be null, in which case the Short MA Name needs to be globally unique, in order to provide complete protection against the accidental concatenation of service instances. Suitable choices for globally unique Short MA Names are defined in ITU-T G.8013/Y.1731.

Figure 18-2 illustrates a single service instance created from the Maintenance Domain of Figure 18-1, again from the point of view of an operator. It offers four DoSAPs to a customer (C1). Each DoSAP is marked

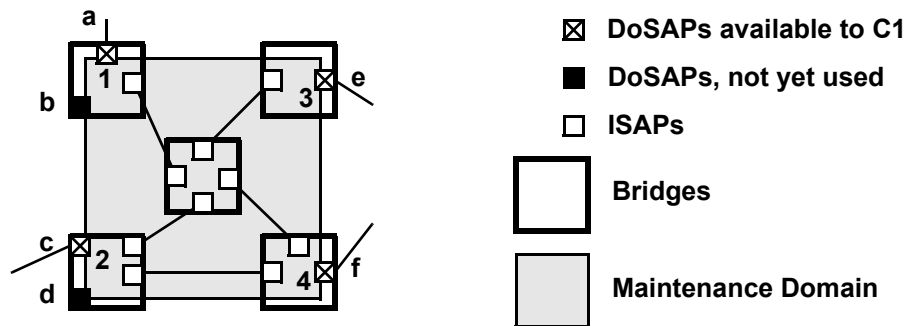


Figure 18-2—One service instance: operator's view

Figure 18-3 illustrates that same service instance from the point of view of customer C1. The customer has four items of customer equipment attached to the four DoSAPs 1–4. The means by which the operator connects the four DoSAPs to create the service instance are invisible to the customer.

NOTE 2—The term “service instance” has been used to mean an “end-to-end service supplied to a customer.” One cannot predict whether an apparently end-to-end service instance will be used by a customer to create another service instance at a still higher level to yet another customer. Therefore, no specific term is provided by this standard to label an “end-to-end” service instance.

The CFM protocol state machines and variables (see Clause 20) are used by MEPs to conduct protocol exchanges within a Maintenance Domain. Each MIP functions symmetrically with respect to the SAPs of its containing MA and service instance, and is modeled as comprising two MIP Half Functions (MHFs). Figure 18-4 specifies the symbols used for MEPs, MHFs, and MIPs in this standard, e.g., in Figure 18-7.

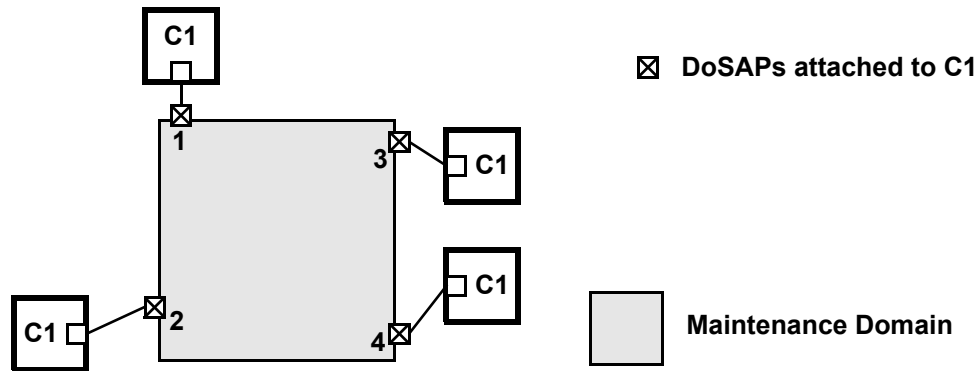


Figure 18-3—One service instance: customer's view

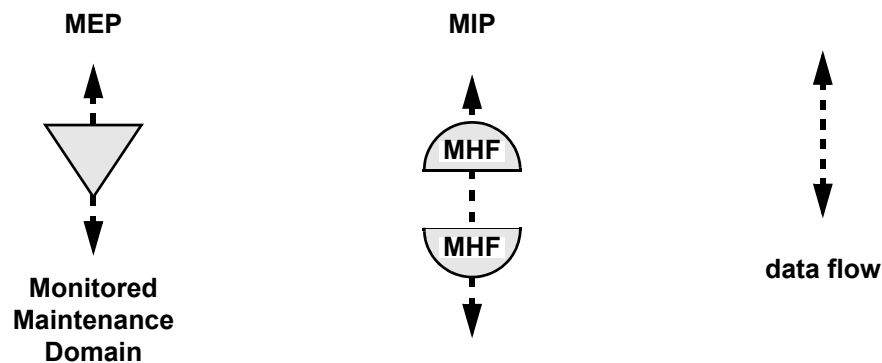


Figure 18-4—MEP and MIP Symbols

18.3 Maintenance Domain Levels

A Maintenance Domain can provide service instances to an enclosing Maintenance Domain or utilize service instances provided by an enclosed Maintenance Domain. An explicit Maintenance Domain Level (MD Level) is included in CFM PDUs to indicate the nesting relationships among Maintenance Domains, because systems can be organized into Maintenance Domains as administrative needs dictate, without necessarily adding headers to data frames to enforce a layered organization.

As part of the nested decomposition facility provided by Maintenance Domains, the CFM protocols hide information. The components of a given Maintenance Domain interior to the DoSAPs presented to the enclosing Maintenance Domain are invisible, through the CFM protocols and managed objects, to the Maintenance Domains enclosing that given Maintenance Domain, except for configured points of visibility. No part of a Maintenance Domain lower than (nested within) a given Maintenance Domain is unintentionally visible to any higher Maintenance Domain. No part of a Maintenance Domain is ever visible to any Maintenance Domain except its immediately higher (enclosing) Maintenance Domain. This enables the separation of responsibility for network administration. The administrator of an end-to-end service on the largest scale can be insulated from the administration of the networks comprising that service, and so on, in principle, down to the administration of individual LANs. This prevents, for example, a customer from learning the internal topology of an operator's network.

1 In order to facilitate the diagnosis of connectivity failures, an administrator can make a DoSAP visible as an
2 ISAP to the immediately enclosing Maintenance Domain by configuring it as a MIP. The diagnostic
3 functions provided by CFM detect connectivity failures between any pair of MEPs in an MA. The MIPs
4 allow these failures to be isolated to smaller segments of the network; loss of connectivity between a pair of
5 MIPs corresponds to a MEP connectivity failure at a lower MD Level. In the lowest Maintenance Domains,
6 the MIPs can be configured on individual Bridge Ports.

7 Figure 18-5 illustrates the nesting of Maintenance Domains, service instances, and MAs. It illustrates nested
8 service instances, each configured with one MA in a Maintenance Domain. There are seven Maintenance
9 Domains, and hence seven service instances illustrated. At the (white) “provider” MD Level, the service
10 instance is offered to a single “customer” C1. That customer creates a (gray) service instance of his own,
11 configured with an MA that verifies the integrity of the service instance offered by the provider. Several
12 levels of nested Maintenance Domains are shown:

- 13 a) Covering the widest physical extent, a Maintenance Domain (and MA and service instance) at the
14 customer MD Level has a MEP in each of four devices labeled C1 in Figure 18-5. The four DoSAPs
15 a, b, g, and h can be configured as MIPs for this MA.
- 16 b) Covering the physical range corresponding to the limits of the end-to-end provider’s network are the
17 MEPs a, b, g, and h belonging to the provider MD Level. The four DoSAPs c, d, e, and f can be
18 configured as MIPs for this MA.
- 19 c) Each of the two operator networks has a Maintenance Domain at the operator MD Level. These
20 DoSAPs are marked a through h. The unlabeled SAPs in both operators’ Maintenance Domains can
21 be configured as MIPs for their respective MAs.
- 22 d) Three “physical” layer Maintenance Domains are shown as well. (Two are labeled “PMDL,” for
23 “physical MD Level.”) Note that not all physical links are given Maintenance Domains.
 - 24 1) A network of devices is being used, in some unspecified fashion, to offer the operators’ Bridges
25 an emulated shared-medium LAN with four DoSAPs. A Maintenance Domain has been created
26 to verify the connectivity of that LAN.

27 NOTE—This central emulated LAN could equally be a network of Bridges that are separated via nonstandard
28 configuration from the operator clouds on either side, a network of PBBs, a network of Ethernet over SONET circuits
29 with a central interconnect device, or an ATM Emulated LAN.³³

- 30 2) Physical-level Maintenance Domains have been created at the edge of the network (on the left
31 side) to monitor two LANs, perhaps because they include Repeaters or are otherwise
32 unreliable.

33 Figure 18-6 expands the upper right corner of Figure 18-5, including Bridges 4 and 5 and device 6.
34 Examining device 6 (C1), readers can see that the highest MA, at the “customer” MD Level, requires a MEP
35 in its Port q. The side of the DoSAP of this service instance that offers access to the service instance faces
36 the center of device 6, just like the service offered by a LAN. On the other hand, in Port g, there are two
37 service instances, each with a MEP: the white service instance at the provider MD Level, and the gray
38 service instance at the operator MD Level. Neither of these service instances include the wire attached to
39 Port g. Thus, ensuring the proper operation of the LAN connecting Port q to Port g is the responsibility of
40 the MA at the customer MD Level, and not the other MAs. Port e has two MEPs, one facing in each
41 direction.

42 Port g in Figure 18-6 also is an example of the fact that only the MD Level distinguishes the provider and
43 operator Level Maintenance Domains at Port g. The single service instance in the example network shown
44 in Figure 18-5 and Figure 18-6 is carried on a single VLAN with a single VID through the right-hand
45 operator’s Maintenance Domain. Within that Maintenance Domain, it cannot be distinguished whether a

³³ See ATM Forum af-lane-0021.000; “LAN Emulation over ATM 1.0”;
<https://www.broadband-forum.org/ftp/pub/approved-specs/af-lane-0021.000.pdf>.

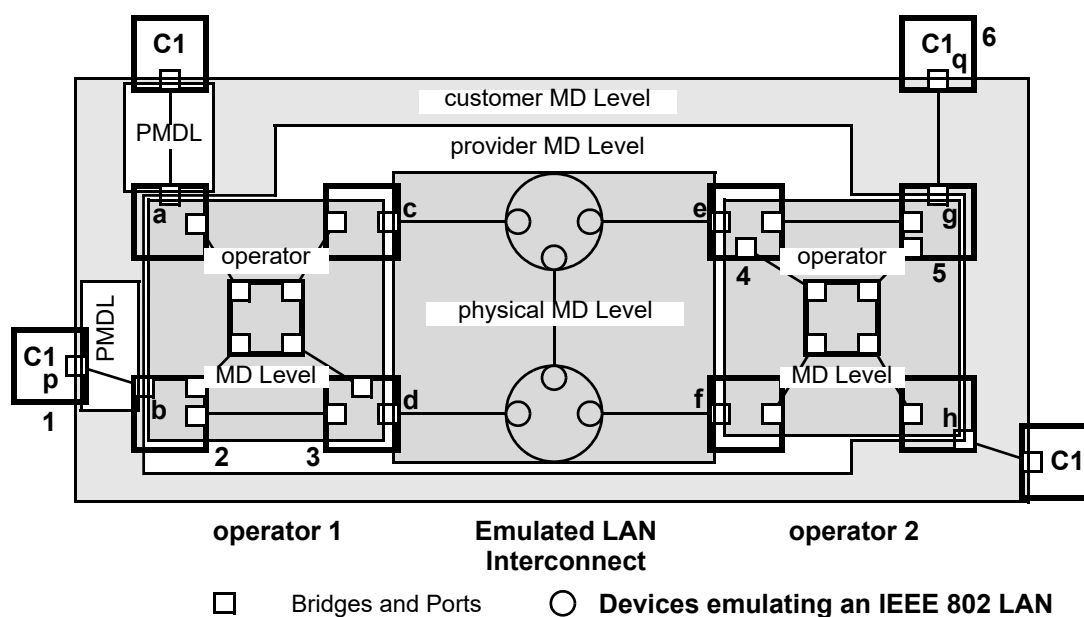


Figure 18-5—MAs: one service instance in a provider network

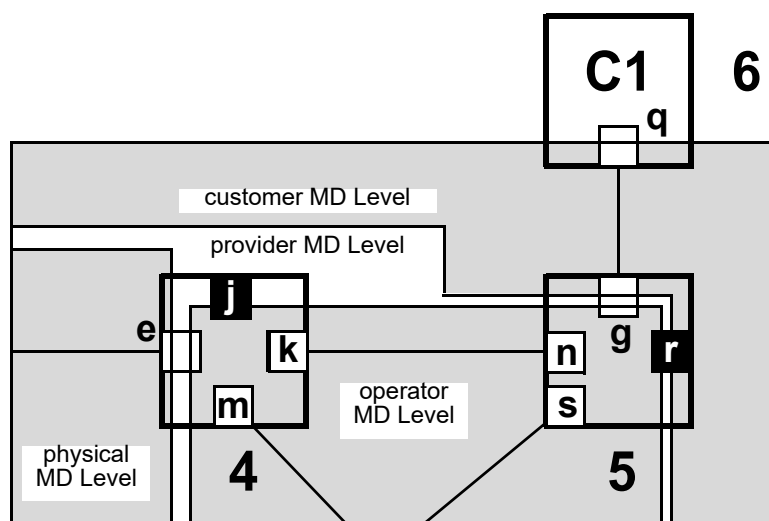


Figure 18-6—MAs: Expansion of Figure 18-5

1 data frame on that VID belongs to the operator, provider, or customer Maintenance Domain; it belongs to all
2 of them. However, the MD Level permits CFM PDUs to be associated with different Maintenance Domains
3 even when they share the same VID.

4 Port j and Port r have been added to Figure 18-6; they are not present in Figure 18-5. Both represent
5 DoSAPs that have not been configured to support either a service instance or an MA. For the integrity of the
6 network to be ensured by CFM, these DoSAPs should be configured to be Disabled.

1 Ports k, m, n, and s can be configured as ISAPs for the gray service instance, providing a MIP at the operator
2 MD Level. Port e can be configured with a MIP for the provider MD Level, and Port g with a MIP for the
3 customer MD Level.

4 Figure 18-7 illustrates the use of MD Levels to allow the use of CFM by a user of connectivity provided by
5 two bridged Maintenance Domains and by the operators of each of those Maintenance Domains. It
6 illustrates a vertical slice through Figure 18-5, including two of the customer’s devices, and the path through
7 the network connecting them. One can see from Figure 18-7 why MD Levels are required to identify CFM
8 PDUs; there would otherwise be no way for Bridge Port b, for example, to tell whether a given CFM PDU is
9 to be sunk (MD Levels 2 or 3) or passed (MD Level 5).

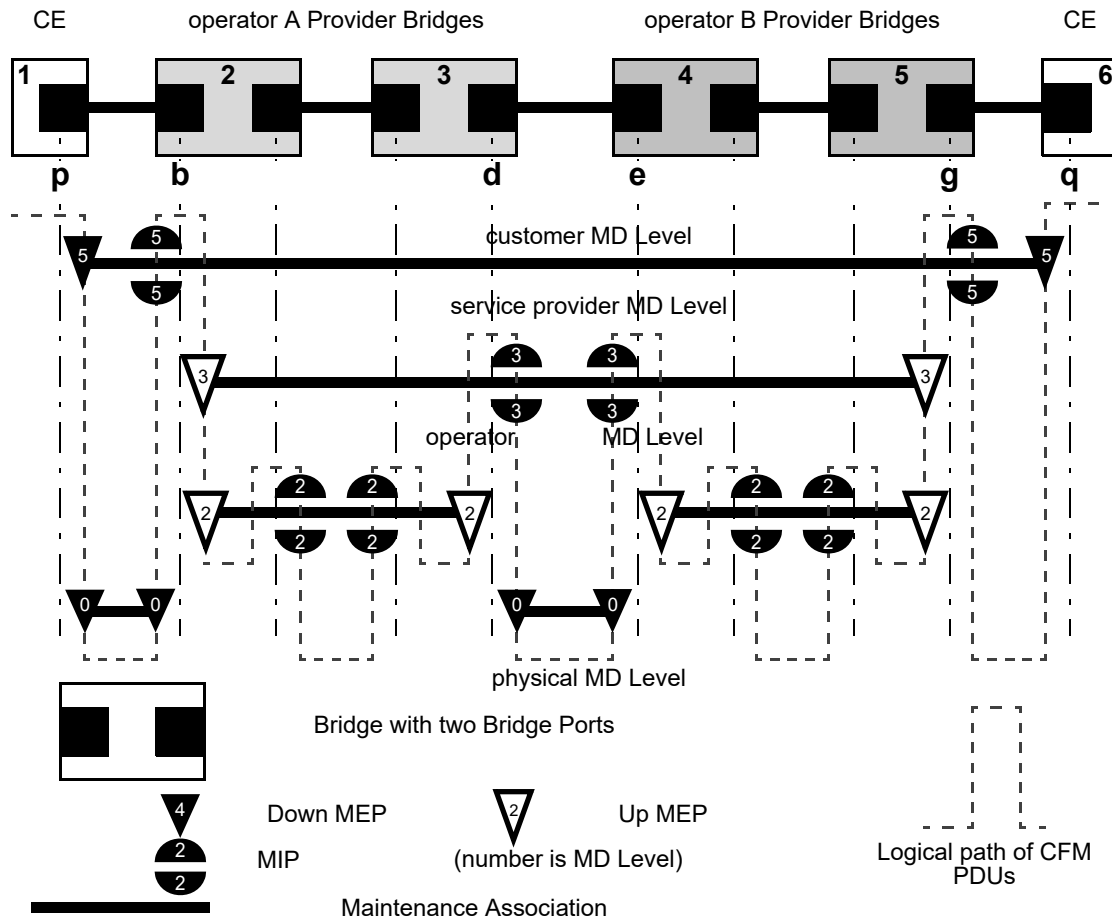


Figure 18-7—MEPs, MIPs, and MD Levels

10 Assignment of numerical MD Levels to the “customer” role, the “service provider” role, or the “operator”
11 MD Levels is somewhat arbitrary, since those terms imply business relationships that cannot be
12 standardized. See J.3 for a more detailed discussion of the assignment of MD Levels.

19. CFM entity operation

This clause specifies:

- a) How the MPs that instantiate a CFM MA attach to ISS-SAPs or EISS-SAPs (19.1).
- b) How MPs are addressed in networks (19.4).
- c) The architecture of MEPs (19.2), MIPs (19.3), MHFs (19.3), Linktrace Output Multiplexers (LOMs, 19.5), and Linktrace Responders (19.6), including their identification, addressing, and decomposition into components that provide the CFM functions.

NOTE—Clause 18 introduces the principles of CFM operation and the network architectural concepts that support it. Clause 20 specifies the protocols operated by the components of each MP, and Clause 21 specifies the PDU formats used by those protocols. The use of CFM within systems and networks is further described in Clause 22.

The models of operation in this clause provide a basis for specifying the externally observable behavior of MEPs and MIPs, and are not intended to place additional constraints on implementations; these can adopt any internal model of operation compatible with the externally observable behavior specified. Conformance of equipment to this standard is purely in respect of observable protocol.

19.1 Maintenance Points (MPs)

The primary CFM protocol shims are called Maintenance Points (MPs). (See 19.6 for the other CFM protocol shim.) An MP can be either a MEP or an MHF. (See also 19.5 for a third type of CFM shim.) A MEP or an MHF each have two principle SAPs, an Active SAP, through which CFM PDUs produced and consumed by the MP are exchanged, and a Passive SAP, through which data frames and unprocessed CFM PDUs pass. Figure 19-1 illustrates the principle SAPs, and the symbols used for MEPs, MIPs, and MHFs in other figures, e.g., Figure 22-1. Note that an MP can be a Down MP or an Up MP, depending on whether the Passive SAP is above or below the Active SAP, respectively, in a shim diagram.

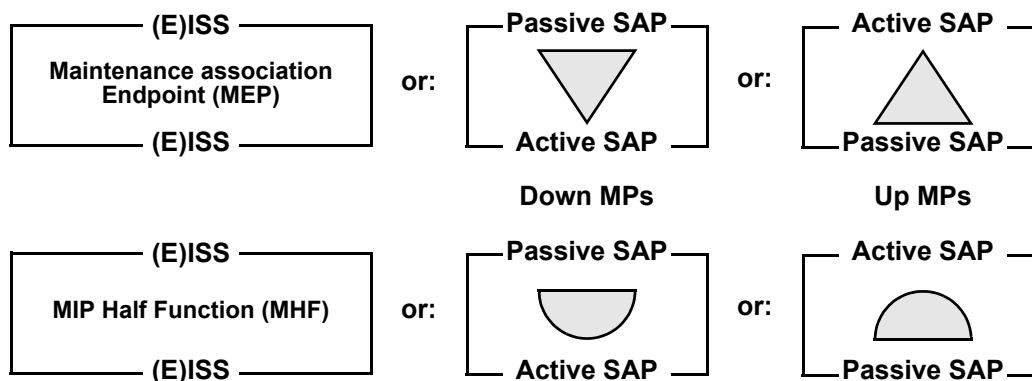


Figure 19-1—CFM Protocol shims

NOTE—The difference between Up MPs and Down MPs is explained more fully in 19.2.3 and 22.1.1.

19.2 MA Endpoints (MEPs)

A MEP can be created in either a Bridge Port (22.2) or an end station attachment to a LAN.

19.2.1 MEP identification

A MEP is associated with exactly one MA. It is identified for management purposes by two parameters, although other, additional parameters are used to manage the MEP:

- a) The MAID, identifying the MA
- b) A MEPID specific to this MEP

The MEP is identified for data forwarding purposes by:

- c) A set of VIDs, including a Primary VID, inherited from the MA; or
- d) An I-SID in the case of a Backbone Service Instance MA; or
- e) A set of 3-tuples <ESP-DA, ESP-SA, ESP-VID> inherited from the MA when the MA is associated with a TESI; or
- f) A SEG-ID inherited from the MA when the MA is associated with an Infrastructure Segment.

From the MA, the MEP inherits a Maintenance Domain, and from this it inherits:

- g) A Maintenance Domain Level (MD Level), inherited from its Maintenance Domain and
- h) One of the following inherited from its MA:
 - 1) A Primary VID or
 - 2) an I-SID in the case of a backbone service instance or
 - 3) a TE-SID in the case of a TESI or
 - 4) a SEG-ID in the case of an Infrastructure Segment

A MEP is associated with two Bridge Ports. For a Down MEP, both associations are with the same Bridge Port. For an Up MEP, the two associations can be with either the same, or with different, Bridge Ports (see 19.4, J.6). The two Bridge Ports are:

- i) The Bridge Port on which the MEP is configured.
- j) The Bridge Port on which the MEP is implemented, and from which it inherits its individual MAC address.

The set of MEPs configured with identical values for MAID define an MA. Thus, one can say that the MA, and not the MEPs, possesses this parameter. However, an MA is a diffuse entity that can be spread among a number of geographically separated Bridges. Each Bridge has its own Maintenance Association managed object for an MA, from which its MEP's MAID, MD Level, and Primary VID, I-SID, series of <ESP-DA, ESP-SA, ESP-VID> 3-tuples, or SEG-ID are derived. In the case of a VLAN-associated MA, the selection of which of the MA's VIDs is the Primary VID can be overridden for a specific MEP. Although this information can be incorrectly configured (i.e., configured differently than some other Bridge), the Maintenance Association managed object ensures the uniform configuration of MEPs in a single Bridge.

Each individual MEP is configured with a MEPID that is unique within that MA. Again, the Maintenance Association managed object helps to maintain uniqueness, as it ties together the configuration parameters of the MEPs in a single Bridge and the same MA and ensures the uniqueness of the MEPIDs, at least in that Bridge.

The MAID can take a number of forms (see 21.6.5.1), including a Domain Name-based string. To prevent incorrect operation in the face of unanticipated connections among systems, the MAID is unique over the domain for which CFM is to provide such protection. If the MAID is globally unique, then that domain is global. CFM can detect connectivity errors only over a set of MEPs in which MAIDs are unique.

The MEPID is an integer in the range 1–8191. It provides each MEP with the unique identity required for the MEPs to verify the correct connectivity of the MA.

NOTE—The range of the MEPID has been chosen to be large enough to accommodate most network needs, but small enough to be used as an index into a table of remote MEPs, and not require a table lookup, in a MEP Continuity Check Receiver (19.2.8) implemented in hardware.

19.2.2 MEP functions

The MEP:

- a) Periodically transmits CCMs according to the managed objects defined in item e) in 12.14.6.1.3.
 - b) Validates received CFM PDUs as specified in 20.1, 20.2, 20.3, and 20.51.
 - c) Discards those CFM PDUs at a lower MD Level to that configured in the MEP (20.7.1).
 - d) Transmits LBMs and receives LBRs as defined in 20.2, according to the managed objects defined in 12.14.7.3.
 - e) Transmits LTMs, and transmits and receives LTRs as defined in 20.3, according to the managed objects defined in 12.14.7.4.
 - f) Responds to LBMs with LBRs as defined in 20.2.2.
 - g) Responds to LTMs with LTRs as defined in 20.3.2.
 - h) Maintains the MEP CCM Database as defined in 20.1.3.
 - i) May maintain the MIP CCM Database as defined in 20.1.3.
 - j) Maintains the variables in 20.9.
- a) May decapsulate and transmit the payload of received Send Frame Messages (SFMs) (29.2.6).
 - b) May pass the received Reflected Frame Messages (RFMs) to RFM-Receiver (29.2.4).

Although defined as having either ISS or EISS interfaces, a MEP is highly constrained in its use of the EISS. Specifically,

- c) Every frame passing through the MEP, in either direction, is emitted with its `vlan_identifier` and `drop_eligible` parameters unchanged from the value received.
- d) All PDUs generated by the MEP, namely CCMs, LBMs, LBRs, LTMs, and LTRs, use the MEP's configured Primary VID as the `vlan_identifier` parameter.

19.2.3 MEP architecture

Figure 19-2 illustrates the component entities of a MEP. Entities that simply pass, redirect, or filter frames without altering them, and that have no internal state, are shown with dashed outlines; those that generate or transform frames, or that have an internal state, are shown with continuous outlines. State machines describing the latter entities are specified in Clause 20. Figure 19-2 shows that the two enclosing SAPs are not equivalent. The upper SAP in the figure is the Passive SAP. It lies closer to the DoSAP of the monitored service instance. The lower SAP in the figure is the Active SAP, the one that faces the monitored service instance. Thus, Figure 19-2 illustrates one of the Down MEPs shown in Figure 22-1. To illustrate one of the Up MEPs of Figure 22-1, Figure 19-2 would have to be inverted. Note also in the diagram that there are two paths, one of which is present only in the Up MEP, and one of which is present only in the Down MEP.

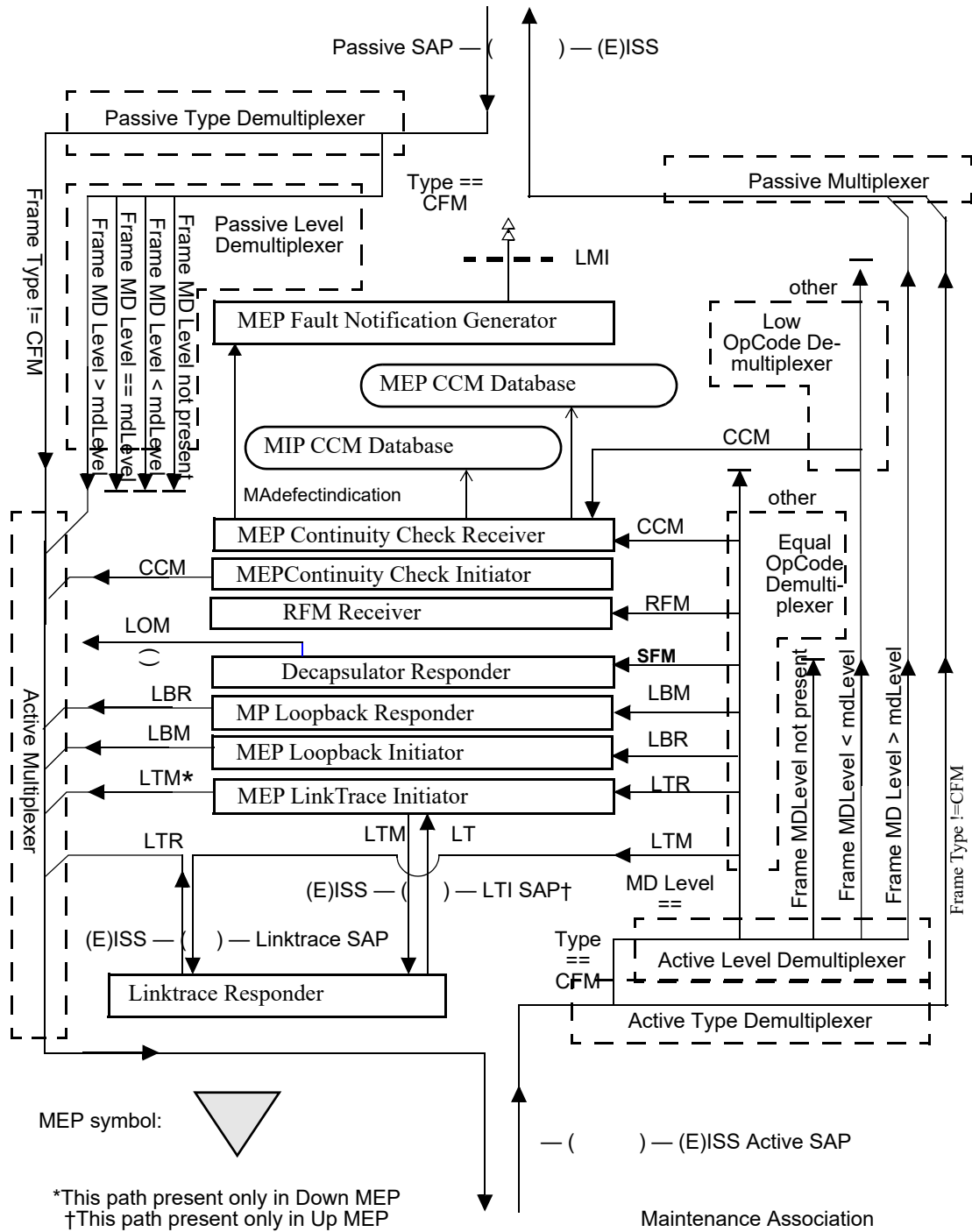


Figure 19-2—MA Endpoint (MEP)

1 19.2.4 MP Type Demultiplexer

2 There are two MP Type Demultiplexers in a MEP, the Active and the Passive Type Demultiplexers, and one
3 in an MHF. All have the same purpose: to separate the data frames from those carrying CFM PDUs using the
4 CFM encapsulation (21.2), output non-CFM frames to one output, and output the CFM frames to a second
5 output. Frames containing CFM PDUs are sent to the Level Demultiplexer, and non-CFM frames are passed
6 through the MEP intact.

7 NOTE—The model used for functional entities in this standard assumes that the operations of the components shown in
8 Figure 19-2 and Figure 19-3 are atomic with respect to each other, so that the demultiplexing and multiplexing of
9 frames, e.g., of CFM and non-CFM frames, cannot change the order of delivery of these frames. Although such
10 reordering has no adverse consequences on this standard, and although it would not normally violate the frame ordering
11 requirements of 6.5.3 (because CFM and non-CFM frames would seldom have the same source and destination MAC
12 address pairs), this reordering could cause the frame loss measurement operations of ITU-T G.8013/Y.1731 to fail to
13 operate correctly. IEEE Std 802.1AX also can disrupt the operation of ITU-T G.8013/Y.1731 frame loss measurement
14 operations.

15 19.2.5 MP Multiplexer

16 There are two MP Multiplexers in a MEP or in an MHF, the Active and the Passive Multiplexers. An MP
17 Multiplexer merges frames from multiple inputs and outputs them to a single output. No frame queuing is
18 specified or implied by this standard.

19 19.2.6 MP Level Demultiplexer

20 There are two MP Level Demultiplexers in a MEP, the Active and Passive Level Demultiplexers. Both
21 separate CFM PDUs according to the MD Level contained within the PDU, and the MD Level configured in
22 the MEP, and output them to different outputs. Specifically,

- 23 a) Any CFM PDU received that is too short to contain an MD Level header field shall be discarded.
- 24 b) Otherwise, the MD Level header field of the CFM PDU is compared to the MD Level configured for
25 the MP [item b) in 12.14.5.1.3, item b) in 12.14.3.2.2, 20.7.1], and the CFM PDU is directed to one
26 of the three outputs of the MP Level Demultiplexer, according to whether the CFM PDU's MD
27 Level is less than, equal to, or greater than the MP's MD Level.

28 19.2.7 MP OpCode Demultiplexer

29 There are two MP OpCode Demultiplexers in a MEP, the Equal and the Low OpCode Demultiplexers, and
30 one in an MHF. Each MP OpCode Demultiplexer separates CFM PDUs that carry different values in their
31 OpCode fields (see 21.4.3), and either discards them or passes them to the appropriate state machine. Those
32 PDUs that do not match any of the OpCodes recognized by an MP OpCode Demultiplexer, or that are too
33 short to contain an OpCode, are discarded by either of a MEP's MP OpCode Demultiplexers, and passed to
34 the Passive MHF Multiplexer by an MHF's MP OpCode Demultiplexer. Those PDUs that do match a
35 recognized OpCode are stored in a variable and cause another Boolean variable to be set, according to
36 Table 19-1. These variables, in turn, trigger state machine actions. The three MP OpCode Demultiplexers
37 separate different sets of OpCodes; see Figure 19-2 (in 19.2.3) and Figure 19-3 (in 19.3.3).

38 19.2.8 MEP Continuity Check Receiver

39 The MEP Continuity Check Receiver receives CCMs whose MD Levels are equal to or less than the MD
40 Level of the MEP. The MEP Continuity Check Receiver processes those at or lower than its own MD Level,
41 to update the MEP CCM Database. The MEP Continuity Check Receiver maintains one instance of the
42 Remote MEP state machine (20.20) for each active remote MEP configured for this MA. In addition, the
43 MEP Continuity Check Receiver maintains a single instance each of the Remote MEP Error state machine
44 (20.22) and the MEP Cross Connect state machine (20.24).

Table 19-1—Actions taken by MP OpCode Demultiplexers

OpCode	MEP Equal OpCode Demultiplexer	MEP Low OpCode Demultiplexer	MHF OpCode Demultiplexer
CCM	Sets CCMreceivedEqual and CCMequalPDU for 20.18	Sets CCMreceivedLow and CCMlowPDU for 20.18	Sets MHFrecvdCCM and MHFCCMPDU for 20.15 and transfers CCM to Passive MHF Multiplexer
LBM	Sets LBMreceived and LBMPDU for 20.29	Discards PDU	Sets LBMreceived and LBMPDU for 20.29 and in non-PBB-TE MHFs transfers LBM to Passive MHF Multiplexer
LBR	Sets LBRreceived and LBRPDU for 20.34	Discards PDU	Transfers LBR to Passive MHF Multiplexer
LTM	Transfers LTM to MEP Linktrace SAP	Discards PDU	Transfers LTM to MHF Linktrace SAP
LTR	Sets LTRreceived and LTRPDU for 20.45	Discards PDU	Transfers LTR to Passive MHF Multiplexer
Other	Discards PDU	Discards PDU	Transfers PDU to Passive MHF Multiplexer
SFM	Sets SFMreceived (29.3.8.5) and SFMPDU (29.3.8.3)	Discards PDU	Sets SFMreceived (29.3.8.5) and SFMPDU (29.3.8.3). Transfers the SFM to Passive MHF Multiplexer
RFM	Sets RFMreceived (29.3.6.1) and RFMPDU (29.3.6.2)	Discards PDU	Sets RFMreceived (29.3.6.1) and RFMPDU (29.3.6.2). Transfers the RFM to Passive MHF Multiplexer

1 If a MEP's Continuity Check Receiver state machine's allRMEPsDead variable (20.9.4) is set, and if the
2 MEP is not associated with any VID or I-SID and is a Down MEP (i.e., in position 5 in Figure 22-9), then
3 the MAC_Operational parameter presented to the Passive SAP by the MEP is false, whatever the state of the
4 MAC_Operational parameter received from the Active SAP. Thus, a completely failed MA causes the
5 attached Bridge Port to appear as a failed link.

6 Optionally, the MEP Continuity Check Receiver may maintain the MIP CCM Database as described in
7 19.3.10.

8 Various error conditions, including the receipt of CCMs at a lower MD Level than that of the MEP, cause the
9 Remote MEP state machine to detect a defect that can in turn generate a Fault Alarm.

10 **19.2.9 MEP Continuity Check Initiator**

11 The Continuity Check Initiator transmits CCMs at the MD Level of the MEP only. The state machine for the
12 Continuity Check Initiator is described in 20.12.

13 **19.2.10 MP Loopback Responder**

14 The Loopback Responder receives LBMs at the MD Level of the MP only, validates and filters them
15 according to destination_address (19.4) and the PBB-TE MIP TLV (21.7.5) in the case of a PBB-TE MA
16 and transmits LBRs in response. The state machine for the MP Loopback Responder is described in 20.29.

1 **19.2.11 MEP Loopback Initiator**

2 The Loopback Initiator transmits LBMs at the MD Level of the MEP only, and validates, filters according to
3 destination_address (19.4), and counts the LBRs returned by other MPs' Loopback Responders. The state
4 machine for the Loopback Initiator is described in 20.32.

5 **19.2.12 MEP Linktrace Initiator**

6 The Linktrace Initiator transmits LTMs at the MD Level of the MEP only, and validates, filters according to
7 destination_address (19.4), and catalogs the LTRs returned by other Bridges' Linktrace Responders. The
8 state machine for the Linktrace Initiator is described in 20.45. In an Up MEP the Linktrace Initiator
9 transmits its LTMs to the Bridge's Linktrace Responder. In a Down MEP, the Linktrace Initiator transmits its
10 LTMs through the MEP's own Active Multiplexer toward the MEP's Active SAP.

11 **19.2.13 MEP LTI SAP**

12 An Up MEP, but not a Down MEP, has a MEP LTI SAP, an instance of the ISS or EISS. This SAP is used to:

- 13 a) Transmit LTMs originated by the MEP's Linktrace Initiator.
- 14 b) Receive LTRs from the Linktrace Responder.

15 **19.2.14 MEP Linktrace SAP**

16 Every MEP has a MEP Linktrace SAP, an instance of the ISS or EISS. This SAP is used to relay received
17 LTMs to the Linktrace Responder.

18 NOTE—Both the MEP LTI SAP and MEP Linktrace SAP are required so that the MEP knows, when an LTR is returned
19 to it by the Linktrace Responder, whether the LTR is a response to an LTM generated by the MEP and thus to direct it to
20 the MEP Linktrace Initiator, or whether the LTR is a response to an LTM previously received by the MEP and thus to
21 forward it out the Active SAP.

22 **19.2.15 MEP CCM Database**

23 The MEP CCM Database is maintained by the Remote MEP state machines (20.20), one entry for each of
24 the other MEPs configured in the MA. In addition to the Remote MEP variables (20.19) used by the Remote
25 MEP state machine, each entry holds information obtained from the remote MEP's CCMs as listed in
26 12.14.7.6.3. See also 20.1.3.

27 **19.2.16 MEP Fault Notification Generator**

28 The Fault Notification Generator uses the MAdefectIndication variable (see 20.9.3) in order to generate a
29 Fault Alarm to the administrator of the Maintenance Domain to which the MA is attached. This can be
30 accomplished by sending an SNMP Notification. The state machine for the Fault Notification Generator is
31 described in 20.37.

32 **19.2.17 MEP Decapsulator Responder (DR)**

33 The MEP DR is an optional component entity within an MEP shim required only by DDCFM. Its position
34 relative to other MP component entities is shown in Figure 19-2. When the destination_address of the SFM
35 matches with the MEP's MAC address, the MEP DR (29.2.6) will decapsulate the SFM, change the
36 source_address field of the decapsulated data frame if it is required, and send the frame to the LOM SAP of
37 selected port(s) (29.3.9.1).

19.2.18 MEP RFM Receiver

The MEP RFM Receiver is an optional component entity within a MEP shim required only by DDCFM. It is placed next to the OpCode Demultiplexer functions receiving RFM frames. When the destination_address of the RFM matches with the MP's MAC address, the MEP RFM Receiver will pass the received RFM to the corresponding, nonstandardized RFM Analyzer.

19.3 MIP Half Function

A MIP consists of two MIP Half Functions (MHFs) on a single Bridge Port, an Up MHF and a Down MHF. (See J.6 for a more detailed explanation.) An MHF may maintain a MIP CCM Database, separate from the MEP CCM Databases.

19.3.1 MHF identification

Every MHF is characterized by one configuration parameter for management purpose, and all of the MHFs of a MIP have identical values for that configuration parameter:

a) The Maintenance Domain to which the MHF is assigned.

For data flow purposes, the MHF is characterized by:

b) A Maintenance Domain Level (MD Level), inherited from its Maintenance Domain or from the Default MD Level managed object.

c) A set of Primary VIDs, one from each MA associated with the MHF's Maintenance Domain; or a set of I-SIDs, one for each MA associated with the MHF's Maintenance Domain.

NOTE—When multiple VIDs are used for a single service instance, an MHF is aware of each MA's VID set and Primary VID. Unlike the MEP, however, the MHF has no occasion to use a MAID, so is not identified by a MAID.

An MHF is associated with two Bridge Ports. For a Down MHF, both associations are with the same Bridge Port; for an Up MHF, the two associations can be with either the same, or with different, Bridge Ports (see 19.4, J.6). The two Bridge Ports are:

d) The Bridge Port on which the MHF is configured.

e) The Bridge Port on which the MHF is implemented and from which it inherits its individual MAC address.

19.3.2 MHF functions

Each MHF:

a) Validates received LTMs as specified in 20.1, 20.2, 20.3, and 20.51.

b) Optionally, validates received CCMs and builds the MIP CCM Database.

c) Responds to LBMs with LBRs as defined in 20.2.2.

d) Responds to LTMs by forwarding them and responding with LTRs as defined in 20.3.2.

e) May decapsulate and transmit the payload of the received SFM (29.4.3).

f) May pass the received RFM (29.4.2).

Although defined as having either ISS or EISS interfaces, an MHF is highly constrained in its use of the EISS. Specifically,

g) Every frame passing through the MHF, in either direction, is emitted with its vlan_identifier and drop_eligible parameters unchanged from the value received.

h) All LBMs forwarded by the MHF carry the same vlan_identifier and drop_eligible parameters as the received LBM.

1 **19.3.5 MHF Type Demultiplexer**

2 The MHF Type Demultiplexer is identical to the MP Type Demultiplexer described in 19.2.4.

3 **19.3.6 MHF OpCode Demultiplexer**

4 An example of the MP OpCode Demultiplexer described in 19.2.7. The MHF OpCode Demultiplexer is
5 fully described in that subclause.

6 NOTE—As indicated by the split arrows labeled “CCM” and “LBM” in Figure 19-3, some of the frames output from the
7 MHF OpCode Demultiplexer are supplied to two other entities, and not just one.

8 **19.3.7 MHF Multiplexer**

9 The MHF Multiplexer is identical to the MP Multiplexer described in 19.2.5. There are two MHF
10 Multiplexers in an MHF, the Active MHF Multiplexer and the Passive MHF Multiplexer.

11 **19.3.8 MHF Loopback Responder**

12 The MHF Loopback Responder is identical to the MP Loopback Responder described in 19.2.10.

13 **19.3.9 MHF Continuity Check Receiver**

14 The optional MHF Continuity Check Receiver receives CCMs, validates them, filters them by
15 destination_address parameter (19.4), and contributes to the MIP CCM Database. From Figure 19-3, readers
16 can see that all CCMs at the same MD Level as the MHF are presented to the MHF Continuity Check
17 Receiver. No other information in the received CCMs is examined. In particular, the MAID in the CCM is
18 not compared to the MHF’s MAID.

19 The MHF Continuity Check Receiver, and the MIP CCM Database it maintains, are not required for
20 conformance to this standard.

21 **19.3.10 MIP CCM Database**

22 The MIP CCM Database is optionally maintained by an MHF or MEP. It is a list of every distinct {FID,
23 source_address, Port number} triple from all of the CCMs presented to all of the Bridge’s MPs’ Continuity
24 Check Receivers since the Bridge was last reset. Entries in the MIP CCM Database are timed out very
25 slowly, on the order of one day, so that their information is available for fault isolation long after the
26 information is no longer used for frame forwarding.

27 **19.3.11 MHF Linktrace SAP**

28 Every MHF has an MHF Linktrace SAP, an instance of the ISS or EISS. This SAP is used to relay received
29 LTMs to the Linktrace Responder, and to pass LTRs from Linktrace Responder to the MHF’s Active SAP.

30 **19.3.12 MHF DR**

31 The MHF DR is identical to the MEP DR described in 19.2.17.

32 **19.3.13 MHF RFM Receiver**

33 The MHF RFM Receiver is identical to the MEP RFM Receiver described in 19.2.18.

19.4 MP addressing

The CFM entities within an MP use the Group addresses for CCM and LTM PDUs listed in Table 8-18 and Table 8-19 and in the case of PBB-TE, SPBM, and ECMP path MAs, the individual MAC addresses or the group MAC addresses that are associated with the monitored services (20.1, 20.2, 20.3). In addition, they recognize and in the case of PBB-TE, SPBM, and ECMP path MEPs use the individual MAC address of the port on which the MP is operating. This is the port on which the MP is configured, if the Bridge created the MP using the Individual MP address model (J.6.1), and is a CFM Port, if the Bridge created the MP using the Shared MP address model (J.6.2). CFM entities associated with a TESI may also use the ESP-SA parameter in the tuple identifying the corresponding ESP.

A Down MP shall not be assigned an individual MAC address [item i) in 12.14.7.1.3] that is the same as the individual MAC address configured for an MP on any other Bridge Port, because if two Bridge Ports are connected to the same LAN, but have the same MAC address, then both would respond to an LBM sent to that MAC address. The CFM protocol is specifically designed to allow an Up MP to be assigned an individual MAC address that is the same as the individual MAC address configured for an Up MP on some other Bridge Port, because it is impossible to tell, from outside the Bridge, from which Up MP a given CFM PDU was actually transmitted, or to which physical entity a given CFM PDU was delivered. See J.6 for a discussion of the assignment of individual MAC addresses to MPs.

All the CFM entities within one MP use the same CCM and LTM addresses, and individual MAC address. The various multiplexing and demultiplexing entities pay no attention to the destination_address parameter of a received CFM PDU. Only the entities responsible for processing received CFM PDUs check a received frame's destination_address parameter. Each of these entities recognizes frames whose destination_address parameters match the individual MAC address of the MP or are a group address. Each of these entities (except the MEP Continuity Check Receiver) filters and does not process frames whose destination_address parameters are not a group address and do not match the individual MAC address of the MP.

NOTE—Even though in the original version of IEEE Std 802.1ag-2007, the MEP Continuity Check Initiator could only transmit multicast CCMs, the MEP Continuity Check Receiver recognized unicast CCMs. In general the definitions of the various CFM receiving entities were more liberal in recognizing certain Individual or group MAC addresses than the corresponding definitions of the CFM transmitting entities a property that allows for the PBB-TE enhancements of CFM, and for improved compatibility with ITU-T G.8013/Y.1731.

19.5 Linktrace Output Multiplexer (LOM)

The third type of CFM protocol shim is the LOM. An LOM has two principle SAPs, an Active SAP, through which CFM PDUs are transmitted, and a Passive SAP, through which pass data frames and unprocessed CFM PDUs. Figure 19-4 illustrates these SAPs and the symbol used for the LOM in other figures, e.g., Figure 22-1. An LOM can be only a Down LOM; its Passive SAP is always above the Active SAP in a shim diagram.

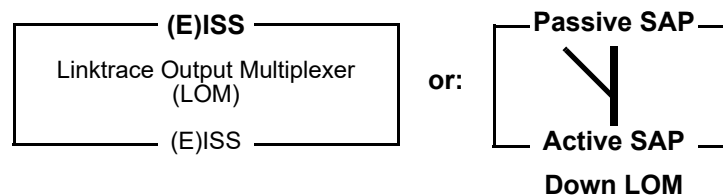


Figure 19-4—LOM shim

1 As illustrated in Figure 19-5, the LOM has only a single active entity, the LOM Multiplexer. It injects LTMs
2 from the Linktrace Responder among the frames passing from the LOM's Passive SAP to its Active SAP.
3 Frames received from the LOM's Linktrace SAP are transmitted through only the Active SAP, and not
4 through the Passive SAP. All frames received on either the Active or Passive SAP are passed through the
5 LOM unchanged, and only to the Passive or Active SAP, respectively.

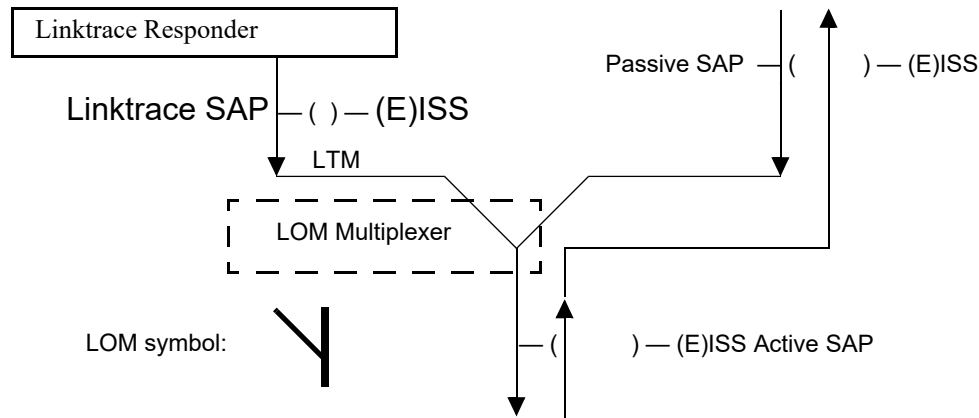


Figure 19-5—LOM architecture

6 NOTE 1—An LOM is not an MP, but an entity that assists the Linktrace Responder (19.6), also not an MP.

7 NOTE 2—The LOM exists on every Bridge Port in order to allow the Linktrace Responder to transmit an LTM on a
8 particular Bridge Port without the possibility of that LTM being reflected back toward the Bridge's Forwarding process.
9 However, the LOM has no constituent entity that differentiates CFM frames from any other frames.

10 19.6 Linktrace Responder

11 A single Linktrace Responder serves all of the MHFs and MEPs in a Bridge. It is connected to every MP and
12 LOM in the Bridge, via the Linktrace SAPs and LTI SAPs in those MPs and LOMs.

13 NOTE—A Linktrace Responder is not an MP, but an entity that assists the MPs (MEP and MHF, 19.2 and 19.3) in
14 performing their functions.

15 In order to minimize the possibility of generating an unbounded number of responses to an LTM, an LTM is
16 always transmitted from a Bridge on a single Bridge Port (20.3). If the LLC entity of a Bridge Port (see
17 Figure 22-1) were used to transmit an LTM on a particular Bridge Port, the Bridge Port connectivity entity
18 (8.5.1) would direct the LTM to the LAN, but it would also pass that frame up toward the Frame filtering
19 entity (8.6.3), as it does for BPDUs. BPDUs are never transmitted to other Bridge Ports, however, because
20 the Frame filtering entity filters them, based on their destination MAC addresses. The placement of MHFs at
21 different MD Levels on different Bridge Ports, all for the same VID, precludes filtering LTMs in this
22 manner. Therefore, the Linktrace Responder filters incoming LTMs by the destination_address parameter
23 (19.4), and a path is required that will direct an LTM only down the Bridge Port, toward the LAN, and not
24 toward the Frame filtering entity.

25 This path is provided by the Linktrace SAPs in the MEP (Figure 19-2), the MHF (Figure 19-3), and the
26 LOM (Figure 19-5). The Linktrace SAPs are also used to direct a received LTM to the Linktrace Responder
27 when the LTM first encounters an MP at the appropriate MD Level.

28 Referring to Figure 22-1, readers can see that an LTM originated by a Down MEP Linktrace Initiator is
29 directed out to the LAN and is thus output on a single Bridge Port, as desired. The situation is different for
30 an LTM originated by an Up MEP. The Bridge in which an Up MEP resides is, in effect, the first hop of the
31 Linktrace protocol. The MEP LTI SAP is required in order to provide a path through which the MEP can
32 direct its originated LTM to the Linktrace Responder. This path can also be used by the Linktrace Responder

1 to return the resultant LTR to the MEP Linktrace Initiator. See Figure 20-16 for a diagrammatic explanation
2 of these paths.
3 Table 19-2 summarizes the SAP connections between the MPs and LOM on the one hand, and the Linktrace
4 Responder, on the other.

Table 19-2—SAP use for LTMs and LTRs

Entity taking action		Sends originated LTM to	Relays received LTM to	Sends to Active SAP an LTR received on	Sends to Active SAP an LTM received on
Up	MEP	MEP LTI SAP	MEP Linktrace SAP	MEP Linktrace SAP	Not applicable ^a
	MHF	Not applicable ^b	MHF Linktrace SAP	MHF Linktrace SAP	Not applicable ^a
Down	MEP	Not applicable ^c	MEP Linktrace SAP	MEP Linktrace SAP	Not applicable ^a
	MHF	Not applicable ^b	MHF Linktrace SAP	MHF Linktrace SAP	Not applicable
LOM		Not applicable ^b	Not applicable ^d	Not applicable ^a	LOM Linktrace SAP

^a The Linktrace Responder only forwards LTMs through an LOM.

^b The MHF and LOM have no MEP Linktrace Initiator, so cannot originate LTMs.

^c The Down MEP transmits the LTMs it originates through its Active SAP.

^d The LOM is not an MP; it cannot detect received LTMs.

5 A single instance of the LTR Transmitter state machine is instantiated by the Linktrace Responder. It is
6 described in 20.50.

20. CFM protocols

MEPs and MIPs can participate in the following CFM protocols:

- a) Continuity Check protocol (20.1)
- b) Loopback protocol (20.2)
- c) Linktrace protocol (20.3)
- d) DDCFM protocol (29.3)

The detailed operation of these protocols by the MEP (19.2) and MHF (19.3) components introduced in Clause 19 is specified in terms of a number of state machines, and the variables and procedures associated with each machine. The state machines defined adhere to the conventions in Annex E and Figure 13-14.

This clause specifies the following:

- e) Relationships among the various CFM state machines and their variables (20.4).
 - f) Requirements for decrementing the timer counters used by a number of CFM state machines (20.5).
 - g) Maintenance Domain variables (20.7) that control all MAs belonging to the Maintenance Domain.
 - h) MA variables (20.8) that control all MEPs and MHFs belonging to the MA.
 - i) MEP variables (20.9) that control the MEP's overall operation, and either apply to the majority of state machines or facilitate communication of information between the individual MEP state machines.
 - j) MEP Continuity Check Initiator (19.2.9) variables (20.10), procedures (20.11), and state machine (20.12).
 - k) (optional) MHF Continuity Check Receiver (19.3.9) variables (20.13), procedures (20.14), and state machine (20.15).
 - l) MEP Continuity Check Receiver (19.2.8) variables (20.16), procedures (20.17), and state machine (20.18).
 - m) Remote MEP variables (20.19) and state machines (20.20) used by the MEP Continuity Check Receiver to track each active Remote MEP [item ae) in 12.14.7.1.3].
 - n) Remote MEP Error variables (20.21) and the Remote MEP Error state machine (20.22) used by MEP Continuity Check Receiver to track received invalid CCMs.
 - o) MEP Cross Connect variables (20.23) and state machine (20.24) used by the MEP Continuity Check Receiver to track received CCMs that could indicate a cross connect defect.
 - p) MP Loopback Responder (19.2.10) variables, (20.27), procedures (20.28), and state machine (20.29).
 - q) MEP Loopback Initiator (19.2.11) variables (20.30), transmit procedures (20.31) and state machine (20.32), and receive procedures (20.33) and state machine (20.34).
 - r) MEP Fault Notification Generator (19.2.16) variables (20.35), procedures (20.36), and state machine (20.37).
 - s) MEP Linktrace Initiator (19.2.12) variables (20.41) and transmit procedures (20.42), and receive variables (20.43), procedures (20.44), and state machine (20.45).
 - t) Linktrace Responder variables (20.46), procedures (20.47), and state machine (20.48), used by the Linktrace Responder (19.6).
 - u) LTR Transmitter state machine (20.50) used only by the Linktrace Responder.
- To facilitate extension of this standard while maximizing interoperability, this clause further specifies:
- v) Rules for CFM PDU validation and versioning (20.51) that support specific goals for extensibility and interoperability (20.51.1).
 - w) Rules for the identification of incoming CFM PDUs (20.52).
 - x) Rules for the use of transaction identifier and sequence number fields (20.53).

1 The models of operation in this clause provide a basis for specifying the externally observable behavior of
2 MHFs and MIPs, and are not intended to place additional constraints on implementations; these can adopt
3 any internal model of operation compatible with the externally observable behavior specified. Conformance
4 of equipment to this standard is purely in respect of observable protocol.

5 NOTE—Clause 18 introduces the principles of CFM operation and the network architectural concepts that support it.
6 Clause 19 breaks down the CFM protocol entities into their components. Clause 21 specifies the PDU formats used by
7 the CFM protocols. The use of CFM within systems and networks is further described in Clause 22.

8 **20.1 Continuity Check protocol**

9 The Continuity Check protocol is performed by the MEP Continuity Check Initiator (19.2.9), the MEP
10 Continuity Check Receiver (19.2.8) of the MEP, and optionally, in the MHF Continuity Check Receiver
11 (19.3.9) of the MHF.

12 For the purposes of Fault detection and the Continuity Check protocol, readers can define an MA as “a set of
13 MEPs, all of which are configured with the same MAID and MD Level, each of which is configured with a
14 MEPID unique within that MAID and MD Level, and all of which are configured with the complete list of
15 MEPIDs.” The CCM (21.6) provides a means to detect connectivity failures in an MA. Each MEP can be
16 configured to periodically transmit a CCM. A connectivity failure is then defined as one of the following:

- 17 a) Inability of a MEP to receive three consecutive CCMs from any one of the other MEPs in its MA,
18 indicating either a MEP failure or a network failure.
- 19 b) Reception by a MEP of a CCM with an incorrect transmission interval, indicating a configuration
20 error.
- 21 c) Reception by a MEP of a CCM with an incorrect MEPID or MAID, indicating a configuration error
22 or a cross connect error.
- 23 d) Reception by a MEP of a CCM with an MD Level lower than that of the MEP, indicating a
24 configuration error or a cross connect error.
- 25 e) Reception by a MEP of a CCM containing a Port Status TLV or Interface Status TLV indicating a
26 failed Bridge Port or aggregated port.
- 27 f) Reception by a PBB-TE MEP implementing the Traffic field, over a period of time that is 4.75 times
28 the CCM interval or equal to 50 ms, whichever is larger, of CCMs with Traffic fields not matching
29 the local Traffic field declaration, indicating a mismatch on the presence or absence of Backbone
30 Service Instance entries on the CBPs that are associated with the monitored TESI.

31 In order to take the fullest advantage of CFM, the operator defines “correct connectivity” as being the
32 configuration of the MEPs. If an MA is defined for every service instance, then the Continuity Check
33 protocol can detect 100% of all connectivity faults, cross connection errors, or misconfigurations that occur
34 within the boundaries of the configured MAs.

35 The transmission rate for CCMs is configurable, to accommodate different needs for error reporting, and
36 different capabilities of network devices. The loss of three CCMs can be detected in as little as 10.8 ms (see
37 Table 21-15). The reception of a cross connected CCM is an instantaneous occurrence.

38 A MEP uses its own configured timer to detect the loss of CCMs, rather than using the transmission rate of
39 the sender (signaled in the CCM), in order to minimize the complexity of a MEP tracking hundreds of other
40 MEPs. The timer is reset when a valid CCM is received, and a defect declared if the timer expires. The timer
41 value is chosen such that the loss of three consecutive CCMs from a given remote MEP triggers a defect for
42 that remote MEP, and the timer accuracy chosen to ensure against a premature defect indication. In addition,
43 state machines in the MEP track the receipt of CCMs from remote MEPs with unexpected information,
44 CCMs from remote MEPs not configured in the receiving MEP, CFM PDUs from a lower MD Level, or
45 CCMs from MEPs in another MA.

1 Depending on the associated MA, a CCM is carried:

- 2 g) In a multicast frame, with a destination_address parameter chosen according to the transmitting
3 MEP's MD Level, according to Table 8-18; or
- 4 h) In the case of a PBB-TE, SPBM, or ECMP associated MA, in a unicast or multicast frame with a
5 destination_address identified by the monitored MA.

6 The CCM does not generate a response. Therefore, in an MA with n MEPs, n CCMs need to be transmitted
7 periodically, one from each MEP. Compared to a full mesh of point-to-point messages in terms of frames
8 carried on wires, multicast CCMs require less bandwidth than the point-to-point messages, except of course
9 for MAs with only two MEPs. And for each MEP, only one transmission and $n - 1$ receptions are required,
10 instead of $n - 1$ of each. In addition, even in MAs with only two MEPs, using multicast instead of unicasts
11 for CCMs obviates the need for configuration or discovery of those MEPs' MAC addresses.

12 Of more importance than frame count is the fact that making the CCM a multicast frame enables the
13 detection of a cross connect error when MEPs from two different service instances are accidentally
14 connected. This type of error can result from a wiring error or a misconfiguration of VID mapping
15 parameters at the boundary between two Maintenance Domains. Were CCMs unicast to specific MAC
16 addresses, then a merging of two service instances, e.g., the accidental concatenation of two point-to-point
17 services into a 4-SAP LAN service, would go undetected; unicast CCMs would still be delivered properly.
18 This scenario also points out why globally unique MAIDs are important.

19 NOTE—The TESI Multiplex Entity (6.19) helps PBB-TE associated MEPs to avoid this type of cross connect problem
20 for unicast CCMs by checking the source_address to determine CCMs destined to PBB-TE MAs. Correspondingly the
21 assignment of globally unique MAIDs is not as important for PBB-TE MAs as it is for other types of MAs but in general
22 a globally unique MAID will provide guaranties for full coverage on all possible misconfiguration errors.

23 Cataloging the receipt of CCMs ensures that a MEP, say, MEP 1, knows that it is receiving them from
24 exactly the correct set of remote MEPs; it does not ensure that the CCMs transmitted by that same MEP are
25 being received by the remote MEPs. For example, suppose that in a 10-MEP MA, all of MEP 1's CCMs are
26 being lost. All of the other MEPs in the MA would know about MEP 1's problem, because they would not
27 be receiving 1's CCMs. But, MEP 1 itself would be unaware of the problem. In order for the operator to
28 decide whether an MA is working properly, every MEP in the MA would have to be inspected.

29 For this reason, the Remote Defect Indication (RDI), a single bit, is carried by the CCM. The absence of RDI
30 in a CCM indicates that the transmitting MEP is receiving CCMs from all configured MEPs. In the case of
31 MEP 1's unidirectional connectivity, MEP 1 would receive the RDI in the CCM from each remote MEP that
32 is not receiving MEP 1's (or any other MEP's) CCMs. Thus, every MEP, including the receive-only MEP 1,
33 is aware that a problem exists. The presence of RDI enables a system administrator to inspect any single
34 MEP belonging to an MA, and discover whether there is any MEP in the MA that is detecting a defect, as
35 well as which particular MEPs have defects.

36 A sequence number should be transmitted in every CCM. The receiver may use this sequence number to
37 detect and count CCM losses. Detection of occasional CCM losses can enable a network administrator to
38 notice suboptimal network conditions such as overloaded links, and correct them before connectivity defects
39 and their consequent Fault Alarms occur.

40 An ECMP path MA is used to test multiple paths between a pair of endpoints. This type of MA has only two
41 MEPs, at two SPT Region boundary ports. An ECMP path MEP cycles through a set of flow hash values
42 intended to direct the CCM frames along (and thus test) different paths to the other MEP in the MA. The
43 path taken by CCM frames with a given flow hash depends on the current state of the network (i.e., paths are
44 subject to rerouting triggered by topology changes). Each flow hash is used in at least three consecutive
45 CCM frames so that a fault on a particular path will be detected and reported back via an RDI signal. To
46 avoid false positive or false negative detections of a loss of connectivity, the rate of CCM transmission
47 should be greater than the maximum difference in latency between any pair of paths.

1 20.1.1 MAC status reporting in the CCM

2 The CCM can carry information about the status of the port on which the transmitting MEP is configured in
3 the Port Status TLV (21.5.4) and Interface Status TLV (21.5.5). This information is not strictly within the
4 scope of the MA bounded by the MEPs transmitting and receiving the CCM. But, by providing port status
5 information in an Up MEP's CCMs, this TLV allows the detection of an error immediately outboard of the
6 MA. This in turn supports fault isolation where a problem has arisen at the point of connection to a service
7 instance, rather than within the equipment supporting the service instance. The Port Status TLV and
8 Interface Status TLV are particularly useful where the boundary between the equipment used by two
9 operators to support a service instance is a link or connection that could fail. Use of the Port Status TLV and
10 Interface Status TLV allows monitoring to be uninterrupted from end to end, without requiring management
11 access by both operators to a single item of interface equipment, or relying on monitoring by the user of the
12 concatenated service. The Port Status TLV and Interface Status TLV allow errors that are immediately
13 outboard of the MA to be distinguished easily from those properly attributed to the MA itself.

14 Figure 18-7 provides an example. Consider a failure of the access link that connects the customer equipment
15 (1) directly to Port b of Provider Bridge 2. Without the Interface Status TLV, none of the MEPs at the
16 operator or service provider MD Level would be notified of the error. The Interface Status TLV allows the
17 failure to be signaled in CCMs transmitted by the Port b service provider and operator MD Level MEPs.

18 20.1.2 Defects and Fault Alarms

19 Defects are separated from Fault Alarms (19.2.16), as is standard practice for service providers.³⁴ The loss
20 of CCMs or the reception of a cross connected CCM is a defect. A Fault Alarm is a Management operation
21 (12.14.7.7), an unsolicited announcement by a Bridge. If the CFM MIB module (17.5) is implemented and is
22 operating, an SNMP Notification is sent. If the CFM YANG module (Clause 48) is implemented and is
23 operating, a YANG Notification is sent. Notifications are sent when the MEP Fault Notification Generator
24 state machine (20.37), or the MEP Mismatch Fault Notification Generator state machine (20.40), detects that
25 a configured time period (default, 2.5 s) has passed with one or more defects indicated, and Fault Alarms are
26 enabled. Each of these state machine can transmit no further Fault Alarms until it is reset by the passage of a
27 configured time period (default, 10 s) during which no defect indication is present. The normal operator
28 procedure upon receiving a Fault Alarm is to inspect the reporting MEP's managed objects, diagnose the
29 fault, correct the fault, examine the MEP's managed objects to see whether the state machine has reset, and
30 repeat those steps until the Fault Alarm has cleared.

31 A number of separate defects are maintained by a MEP, as shown in Table 20-1. The defects are ranked by
32 priority. If a higher priority defect occurs after a lower priority defect has triggered a Fault Alarm, but before
33 the Fault Alarm has reset, then the MEP will immediately issue another Fault Alarm. This enables the
34 operator to reliably prioritize Fault Alarms. For example, cross connect errors are typically of greater
35 concern in a service provider environment than loss of connectivity errors.

36 Only the highest priority defect is reported in the Fault Alarm. Table 20-1 shows the relationships between
37 the variables that indicate defects, the priorities of these defects, and the enumerated value reported in the
38 Fault Alarm for each defect. The mmdefectIndication (20.38.2) is independent from the other CFM defects
39 and is reported independently by the operation of the MEP Mismatch Fault Notification Generator state
40 machine (20.40).

³⁴ See, for example, ITU-T G.806 (2006) [B49].

Table 20-1—Fault Alarm defects and priorities

Defect		Priority	
Variable	highestDefect (20.35.9)	highestDefectPri (20.35.8)	Importance
xconCCMdefect (20.23.3)	DefXconCCM	5	Most
errorCCMdefect (20.21.3)	DefErrorCCM	4	
someRMEPCCMdefect (20.35.5)	DefRemoteCCM	3	
someMACstatusDefect (20.35.6)	DefMACstatus	2	
someRDIddefect (20.35.7)	DefRDICCM	1	Least

20.1.3 CCM reception

Every active MEP receives and catalogs CCMs in its MEP CCM Database. Every CCM shall be examined to be sure that its MAID matches that configured in the receiving MP. This check is optional for CCMs associated with PBB-TE MAs. A receiving MEP checks to ensure that its own MEPID does *not* match that in the received CCM. (This could indicate either a duplicate MEPID or a network forwarding loop.) The information in the CCM is cataloged in the MEP CCM Database, indexed by the received MEPID. The information kept in the MEP CCM Database is listed in 12.14.7.6.3.

The MIP CCM Database is optionally maintained by a MEP Continuity Check Receiver or an MHF Continuity Check Receiver. It is a list of every distinct {FID, source_address, Port number} triple from all of the CCMs presented to all of the Bridge's Down MHFs' MHF Continuity Check Receivers since the Bridge was last reset. (See 8.8.8 for a discussion of VIDs and FIDs.) An entry in the CCM Database is retained for at least 24 h after the last CCM with its triple is received and is removed from the MIP CCM Database after, at most, 48 h. If the resources allocated for the MIP CCM Database are not sufficient to maintain all of the entries for the required time, then the least-recently updated entries should preferentially be removed to make room for new entries. See 20.3.2 for a description of the use of the MIP CCM Database.

20.2 Loopback protocol

The Loopback protocol is used for Fault verification and isolation. To verify the connectivity between MEPs and MIPs, a MEP can be instructed by a system administrator to issue one or more LBMs. The LBM is initiated by a MEP with specified destination address, priority, and drop_eligible parameters, the destination_address being the individual MAC address of another MP within the same MA as the transmitting MEP. In the case of PBB-TE and SPBM VID MAs, the destination_address parameter for an LBM is always a MAC address associated with the terminating CBP(s). In order to identify intermediate MHFs in PBB-TE and SPBM VID MAs, an additional TLV, the PBB-TE MIP TLV (21.7.5), is carried as the first TLV in an LBM, with the receiving MHF's MAC address in its MIP MAC address field (21.7.5.1). The receiving MP responds to the LBM with an LBR.

The Loopback protocol is performed by the MEP Loopback Initiator (19.2.11) and the MP Loopback Responder (19.2.10), the latter residing in either a MEP or an MHF.

1 20.2.1 LBM transmission

2 LBMs are transmitted by operator command as specified in 12.14.7.3. Each LBM transmitted contains a
3 Loopback Transaction Identifier field (21.7.3) that is incremented with each transmission. That enables the
4 transmitting MEP to correlate the returned LBRs with the transmitted LBMs. In a PBB-TE or SPBM VID
5 MA, a PBB-TE MIP TLV is carried by the transmitted LBM as the first TLV if the targeted MP is a MIP.
6 The MIP MAC address field (21.7.5.1) in the TLV is set to the `destination_address` parameter of the
7 appropriate managed object [item b) in 12.14.7.3.2]. The PBB-TE MIP TLV provides also the Reverse VID
8 field [21.7.5.2, item f) in 12.14.7.3.2] to be used in the VID field by a MIP transmitted LBR, and, when the
9 LBM is to be sent on a point-to-multipoint TESI, the Reverse MAC field (21.7.5.3) associated with MAC
10 addresses to be used by the LBR. The LBM can carry an arbitrary amount of data to help diagnose faults that
11 are sensitive to the amount or pattern of data in a frame [item d) in 12.14.7.3.2]. It can be sent with any
12 priority or drop_eligible parameters [item e) in 12.14.7.3.2].

13 No means for specifying the rate at which the LBMs are to be sent is provided. A Bridge shall not transmit
14 LBMs at a rate that would cause the queues serving that Bridge Port to overflow and drop LBMs, were there
15 no other traffic being inserted into those queues.

16 20.2.2 LBM reception and LBR transmission

17 When an LBM is received by an MP Loopback Responder (19.2.10), it may be examined for validity and
18 discarded if invalid. Regardless of whether the LBM is examined for validity, it shall be discarded if the
19 `source_address` is a Group and not an individual MAC address. Also, if the `destination_address` does not
20 match the MAC address of the receiving MP, and the MP Loopback Responder does not reside in a PBB-TE
21 or SPBM VID MHF, the MP shall discard the LBM. If the receiving MP Loopback Responder resides in a
22 non-PBB-TE and non-SPBM VID associated MHF and the `destination_address` is a group MAC address, the
23 MHF shall discard the LBM. If the MP Loopback Responder state machine resides in a PBB-TE MEP and
24 the received LBM carries a PBB-TE MIP TLV, the MEP shall discard the LBM. If the MP Loopback
25 Responder state machine resides in an SPBM VID MEP and the received LBM carries a PBB-TE MIP TLV,
26 the MEP may process the LBM if the MIP MAC address field matches the MEP's address and shall
27 otherwise discard the LBM. PBB-TE and SPBM VID MHFs forward all received LBMs and only stop and
28 process those that carry a PBB-TE MIP TLV containing in their MIP MAC address field the MAC address
29 of the receiving MIP [item f) in 20.8.1]. If the frame passes these tests, the receiving MP generates an LBR
30 and transmits it to the originating MEP. Every `M_UNITDATA.indication` (or `EM_UNITDATA.indication`)
31 parameter and octet in the `mac_service_data_unit` in the LBM is copied to the LBR's
32 `M_UNITDATA.request` (or `EM_UNITDATA.request`) with the following exceptions:

- 33 a) The `source_address` parameter of the received LBM is used as the `destination_address` parameter for
34 the transmitted LBR. In the case of a PBB-TE MHF, this `destination_address` value will be
35 overwritten by the value of the group MAC address in the Reverse MAC field of the PBB-TE MIP
36 TLV, if this is present. In the case of a PBB-TE MEP, the `destination_address` parameter for the
37 transmitted LBR is the value of the ESP-DA of the MA's ESP that has the replying MEP's MAC
38 address [item i) in 12.14.7.1.3] in its ESP-SA field.
- 39 b) The `source_address` parameter for the LBR is the MAC address of the replying MP. If the MP is
40 associated with a PBB-TE MHF, the `source_address` is set to the `destination_address` of the received
41 LBM if this is an individual MAC address or otherwise to the value carried in the Reverse MAC
42 field, contained in the PBB-TE MIP TLV.
- 43 c) If the LBM was received at a PBB-TE MHF and contains a PBB-TE MIP TLV, then the
44 `vlan_identifier` in the LBR is set to the value in the Reverse VID field; otherwise, the `vlan_identifier`
45 in the LBR is set to the value of the Primary VID associated with the replying MP [item d) in
46 12.14.7.1.3, item a) in 12.14.3.2.2, item c) in 12.14.5.3.2].
- 47 d) The `OpCode` field is changed from LBM to LBR.

- 1 e) (Only in PBB-TE MAs) The `vlan_identifier` for the LBR is the Primary VID [item d) in 12.14.7.1.3]
2 associated with the replying PBB-TE MEP; otherwise, if the received LBM contained a PBB-TE
3 MIP TLV, the value in the Reverse VID field is used as the `vlan_identifier` parameter.

4 The receiver of an LBM shall not interpret any of the other fields or TLVs in the LBM. The contents of any
5 TLVs that do not violate the validation criteria of 20.51, but are not specified in this standard, and any valid
6 Organization-Specific TLV not known to the receiving MP, shall be ignored, and not interpreted, by the
7 receiver, except that their contents shall be copied to the LBR.

8 NOTE—This standard provides no means for transmitting an LBM with a group MAC address. The MP Loopback
9 Responder responds to LBMs sent to the CCM Group address because ITU-T G.8013/Y.1731 provides for transmitting
10 such messages. See J.4.

11 20.2.3 LBR reception

12 When an LBR is received by a MEP Loopback Initiator (19.2.11), it is checked to see whether the
13 `destination_address` parameter matches the MAC address of the receiving MEP, and discarded if not. Next it
14 is checked to see whether its Loopback Transaction Identifier field matches that of a recently transmitted
15 LBM, and the appropriate counter incremented, either the in-sequence counter [item y) in 12.14.7.1.3] or the
16 out-of-sequence counter [item z) in 12.14.7.1.3].

17 If the Shared MP address model for implementing CFM is used by a particular Bridge, then as mentioned in
18 20.52, there is an ambiguity in the identification of the particular Up MP to which an LBR is destined.
19 Therefore, a Bridge using this model uses a single variable for managing the Loopback Transaction
20 Identifier fields for all MPs sharing the same MAC address, MAID, and MD Level.

21 The receiver of an LBR may compare its contents bit-for-bit with the remembered contents of the
22 corresponding LBM, and increment a variable [item aa) in 12.14.7.1.3] if they do not match.

23 20.3 Linktrace protocol

24 The Linktrace protocol is performed by the MEP Linktrace Initiator (19.2.12), associated with a single MEP,
25 and the Linktrace Responder (19.6) associated with a Bridge.

26 A Linktrace Message (LTM) is transmitted by a MEP Linktrace Initiator in order to perform path discovery
27 and fault isolation. The LTM carries a target MAC address. It is carried in a frame, with a
28 `destination_address` taken from Table 8-19 according to the MD Level of the transmitting MEP, or the
29 ESP-DA field of the appropriate ESP in the case of a PBB-TE MA (20.31.1), or an individual or group
30 address in the case of an SPBM VID MA, and is relayed as such through the bridged network until it reaches
31 an MP at the appropriate MD Level. That MP intercepts the LTM and deflects it to a Bridge's Linktrace
32 Responder. The Linktrace Responder determines whether its Bridge's MAC Relay Entity would forward an
33 ordinary data frame with the specified target MAC address and `vlan_identifier` to an egress Bridge Port, or
34 would filter or flood it. If the single egress port is found, or if the receiving MP is the terminating MP, or if
35 the MP is associated with a PBB-TE MA or an SPBM VID MA, the Linktrace Responder sends a unicast
36 Linktrace Reply (LTR) to the originator of the LTM, whose MAC address was also carried as payload in the
37 LTM. The LTR is sent after a random delay in the range $0 < \text{delay} \leq 1 \text{ s}$, to mitigate the load on the
38 originating MEP. In addition, if the MP through which the LTM was received was an MHF, the Linktrace
39 Responder forwards an altered version of the LTM in the direction of the target MAC address. The MEP
40 Linktrace Initiator that originated the initial LTM collects the LTRs. These provide sufficient information to
41 construct the sequence of MPs that would be traversed by a data frame sent to the target MAC address.

1 Path discovery in a connectionless environment is more challenging than a connection-oriented
2 environment. In Bridged Networks, it can be especially challenging, since the MAC address of the target of
3 an LTM can be deleted from the FDB by a TCN immediately after a network topology change, and ages out
4 and is deleted a few minutes (Max Age) after last being learned when a fault results in isolating the target
5 from the MEP originating the LTM.

6 CFM offers three ways to address the problem of ageing out MAC addresses:

- 7 a) Launching the Linktrace protocol automatically, immediately following the detection of a fault, such
8 that it gets exercised within the Max Age window.
- 9 b) Maintaining information about the destination MP at the intermediate points along the path by using
10 the optional MIP CCM Database, 19.3.9 and 20.1.3.
- 11 c) Maintaining normal path information by issuing periodic LTMs during normal operations.

12 NOTE—Periodic LTMs allow a system administrator to determine the MIPs along the paths among MEPs during
13 normal operations, thus establishing a baseline for subsequent fault isolation. However, triggering LTMs at rates
14 approaching the CCM transmission interval could overwhelm the Bridges' ability to process CFM PDUs. See also 22.5.

15 20.3.1 LTM origination

16 LTMs are transmitted by a MEP Linktrace Initiator in response to an operator command (12.14.7.4). The
17 nextLTMtransID [item b) in 12.14.7.4.3, 20.41.1] of each LTM transmitted is retained for at least 5 s after
18 the LTM is transmitted. The LTM Transaction Identifier values transmitted by a given MEP are unique over
19 this time period in order to match LTRs returned by slow MPs to the LTM that triggered the slow LTR.
20 Initializing the nextLTMtransID to a random value³⁵ and incrementing it once per LTM transmitted satisfies
21 this criterion. For each LTM transmitted, the MEP Linktrace Initiator creates an entry in a Linktrace
22 database (20.41.2). This entry is indexed by nextLTMtransID for retrieval when a corresponding LTR is
23 received.

24 The transmitting Bridge can maintain a separate sequence of LTM Transaction Identifiers per MEP if
25 multiple MEPs in the same service instance and at the same MD Level have different MAC addresses, but
26 uses a common sequence of LTM Transaction Identifiers for MEPs that share both of these characteristics.

27 In the case of a VID-based MA, the destination_address of an LTM is the group MAC address reserved for
28 LTMs and appropriate to the MD Level of the originating MEP according to Table 8-19. In the case of a
29 PBB-TE MEP, the destination_address of an LTM is the entry in the ESP-DA field of the 3-tuple of the MA's
30 ESP that has the MEP's MAC address in its ESP-SA field. In the case of an SPBM VID MEP, the
31 destination_address of an LTM is the individual B-MAC address of a remote system or the SPBM group
32 MAC address (27.15) associated with an I-SID associated with the source MEP.

33 The Target MAC Address field [item c) in 12.14.7.4.2, 21.8.6] in the LTM can be any individual MAC
34 address. However, since MEPs are required to periodically transmit CCMs, a MEP's MAC address is likely
35 to be known to the MIPs along the path. Conversely, a MIP's MAC address, because the MIP does not
36 initiate any CFM PDUs, can be unknown to intermediate Bridges unless it has recently replied to an LBM or
37 LTM.

38 In the case of a PBB-TE MA, a PBB-TE MIP TLV is carried by the LTM, containing the Reverse VID
39 [21.7.5.2, item e) in 12.14.7.4.2], to be used in the VID field by a MIP transmitted LTR, and, when the LTM
40 is to be sent on a point-to-multipoint TESI, the Reverse MAC field (21.7.5.3) associated with MAC
41 addresses to be used by the corresponding LTRs. The MIP MAC address field (21.7.5.1) in a PBB-TE MIP
42 TLV associated with an LTM is null.

³⁵ For this purpose, a pseudo-random value is sufficient, provided that the same value is not produced each time a system is reinitialized.

1 The MEP Linktrace Initiator in a Down MEP transmits the LTM through its Active SAP, toward the LAN
2 attached to its Bridge Port. The MEP Linktrace Initiator in an Up MEP transmits the LTM through its MEP
3 LTI SAP to its own Bridge's Linktrace Responder. That Linktrace Responder handles the LTM as described
4 in 20.3.2. The Linktrace Responder can forward the LTM through the LOM Linktrace SAP and can respond
5 through the MEP LTI SAP with an LTR. Thus, when an LTM is initiated in an Up MEP, the originating
6 Bridge is the first hop.

7 **20.3.2 LTM reception, forwarding, and replying**

8 An LTM at a particular MD Level is forwarded as an ordinary data frame until it encounters a MEP at an
9 equal or higher MD Level, or an MHF at an equal MD Level. A MEP at a higher MD Level discards the
10 LTM without further processing. An MP at an equal MD Level directs the LTM to its Bridge's Linktrace
11 Responder. Since, for a given MD Level, different Bridge Ports on a given Bridge can be configured with
12 MHFs, MEPs, both, or neither, an LTM can be both forwarded as ordinary data and processed by the
13 Bridge's Linktrace Responder. An LTM that is originated by an Up MEP is forwarded directly to its Bridge's
14 Linktrace Responder and not through the MEP's Active SAP toward the Frame filtering process.

15 The Linktrace Responder is responsible for processing and forwarding LTMs, and for replying to them with
16 LTRs. All LTMs are subject to the validation criteria of 20.51.4; invalid or incorrectly addressed LTMs are
17 discarded without further processing. After processing the LTM, the Linktrace Responder can return an LTR
18 through the SAP on which the LTM was received and can forward at most one altered copy of the received
19 LTM through an LOM on a different Bridge Port. It never forwards an LTM past a MEP at the LTM's own
20 MD Level or higher, i.e., a MEP that bounds the LTM's MA. An LTM carries an LTM TTL field (21.8.4)
21 that is decremented each time the LTM passes through a Linktrace Responder. The LTM is not forwarded if
22 this field is 0 or 1 when the LTM is received.

23 The Linktrace Responder replies to the LTM if:

- 24 a) An ordinary data frame, with the same `vlan_identifier` as the LTM, a `destination_address` equal to the
25 Target MAC Address field (21.8.6) of the LTM and received on the same Bridge Port as the LTM,
26 would be forwarded through exactly one other Bridge Port (or to just the Management Port); or
- 27 b) The `UseFDBOnly` bit of the Flags field of the LTM is 0, the Target MAC Address field (21.8.6) of
28 the LTM is found in the Bridge's MIP CCM Database (19.3.10), and that entry in the MIP CCM
29 Database identifies a Bridge Port (or the Management Port), other than the one on which the LTM
30 was received; or
- 31 c) The MAC address of the MP that received the LTM equals the Target MAC Address field of the
32 LTM;

33 and

- 34 d) LTM TTL field is nonzero when the LTM is received.

35 The single output Port identified in the preceding item a) or item b) is called the Egress Port. The Linktrace
36 Responder forwards a copy of the LTM through the LOM on the Egress Port, with a new `source_address`
37 parameter and the LTM TTL field decremented by 1, only if all of the following conditions are met:

- 38 e) An LTR was generated for the reasons specified in the preceding item a) or item b), but not item c);
- 39 f) The LTM was received via an MHF Linktrace SAP or a MEP LTI SAP, and not a MEP Linktrace
40 SAP;
- 41 g) The Egress Port does not have a MEP at or above the LTM's MD Level; and
- 42 h) The received LTM TTL field was greater than 1.

1 The source_address parameter in the forwarded LTM does not change if the MA is associated to a TESI.
2 NOTE 1—When tracing the path to a known MAC address, all of the Bridges on a shared medium will receive the LTM.
3 Making transmission of the LTR and forwarding of the LTM conditional on the FDB query ensures that at most one of
4 those Bridges will reply. Similarly an LTM can be flooded, through a region of the network that has not yet learned the
5 target MAC address, without provoking a storm of LTRs.

6 LTMs are forwarded immediately in order to minimize the time required to complete a Linktrace operation.
7 LTRs are enqueued for delivery at a later time. If the LTM was received by a Down MEP or a Down MHF,
8 the LTR includes a Reply Ingress TLV (21.9.8) describing that MP. If not received by a Down MEP, and if
9 the Egress Port has an Up MEP or Up MHF configured for the LTM's MA, the LTR includes a Reply Egress
10 TLV (21.9.9). These TLVs report, to the originating MEP, the MIPs and/or the MEP along the path to the
11 targeted MAC address.

12 If an LTM is forwarded to a shared medium LAN, and if one or more of the other Bridges attached to that
13 LAN have only two Bridge Ports forwarding data for the LTM's vlan_identifier (including that shared
14 medium), and if the MAC address in the LTM's Target MAC Address field (21.8.6) is not present in those
15 Bridge's FDBs (or if the targeted MAC address is present in more than one Bridge's MIP CCM Database),
16 then more than one of those receiving Bridges can respond to the LTM with an LTR. Also, an anomaly in the
17 operation of Bridge or LAN, e.g., an FDB is recording its information incorrectly, can cause LTR responses
18 along multiple paths.

19 The Reply TTL field in the LTR is sufficient for the originating MEP Linktrace Initiator to order the LTRs
20 returned along a single path. The LTMs (21.8) also include the LTM Egress Identifier TLV (21.8.8). The
21 LTRs (21.9) also include the LTR Egress Identifier TLV (21.9.7). The information in these TLVs allows the
22 originating MEP to unambiguously construct the tree of paths taken by the LTM, if LTRs are returned along
23 multiple paths.

24 NOTE 2—If an MA cannot be configured to transmit CCMs more quickly than Max Age causes MAC addresses to be
25 removed from the Bridges' FDBs, then the best practice for the system administrator is to trigger an LTM transmission
26 only after triggering a few LBMs to the target MAC address. This helps to ensure that the target MAC address is known
27 to the intermediate Bridges, so that the LBM will return a result, and so that LTRs will be returned only along a single
28 path.

29 NOTE 3—If unusual conditions prompt responses along multiple paths, the delay in transmitting the LTR reduces the
30 chance that the originating MEP's receiving capabilities will be overwhelmed by the returned LTRs.

31 NOTE 4—The forwarded LTM uses, as its source address, the address of the MP through which the LTM is forwarded,
32 not the source address of the original LTM. This means that a Bridge's Linktrace Responder need not be fully
33 synchronized with its Relay Entity, as the latter can change the active topology rapidly without the risk of an adjacent
34 Bridge learning the wrong direction for the source MAC address from an LTM. However, in the case of TESIs, the
35 forwarded LTM simply uses the source address of the original LTM since the previous considerations do not apply in a
36 PBB-TE Region. The LTM TTL field ensures that an LTM that is being processed while the active topology is changing
37 will not be forwarded in a persistent loop.

38 The MEP Linktrace Initiator ignores any TLVs received in an LTR that do not violate the validation criteria
39 of 20.51, but are not specified in this standard, and it ignores any valid Organization-Specific TLV not
40 known to it. Except as otherwise specified in 20.47.3 and 20.47.4, all TLVs, whether known or ignored by
41 the Linktrace Responder, are copied from the received LTM to the forwarded LTM, and from the received
42 LTM to the LTR.

43 NOTE 5—Forwarding all unknown TLVs in both the LTM and the LTR enables future revisions of this standard, and
44 designers of Organization-Specific TLVs, to add new capabilities.

45 20.3.3 LTR reception

46 LTRs are transmitted by a Linktrace Responder in response to a received LTM and returned in a unicast
47 frame to the MEP Linktrace Initiator that originated the LTM. The MEP Linktrace Initiator validates the
48 received LTR, discarding it if invalid, if the destination_address parameter does not match the MAC address
49 of the Bridge Port on which the MEP resides, or if the LTR Transaction Identifier field does not match any
50 of those stored in the MEP's Linktrace database (20.41.2), created when the LTM was originated.

Each validated LTR triggers the creation of a new entry in the Linktrace database, accessible as a managed object through the Read Linktrace Reply command (12.14.7.5). If no entry exists in the Linktrace database for this MEP, corresponding to the LTR Transaction Identifier field, the MEP Linktrace Initiator creates a new entry, with an index value [item c) in 12.14.7.5.2] of 1. If an entry does exist matching that field, it creates a new entry, with an index value one greater than the last matching LTR. In either case, the MEP Linktrace Initiator stores the contents of the LTR in this new entry.

See J.5 for a description of a method to reconstruct the path(s) taken by an LTM, based on the information in the Linktrace database.

20.4 CFM state machines

The relationships among a MEP's state machines are illustrated in Figure 20-1. That figure uses conventions similar to those of 13.24 and of Figure 13-13: open arrows denote variables that are set by one state machine and both read and set by another; closed arrows denote variables that are set by the owning state machine, and only read by other state machines. Not included in Figure 20-1 are the MEP Continuity Check Initiator state machine, the MP Loopback Responder state machine used in the MEP, and all of the MHF and Linktrace Responder state machines, because they operate independently.

20.5 CFM state machine timers

The timer variables declared in this subclause are part of the specification of the operation of CFM. The accompanying descriptions of their meaning and use are provided to aid in the comprehension of the protocol only and are not part of the specification.

One instance of the following shall be implemented:

- a) LTFwhile (20.5.1)

One instance of the following shall be implemented per MEP:

- b) CCIwhile (20.5.2)
- c) errorCCMwhile (20.5.3)
- d) xconCCMwhile (20.5.4)
- e) LBIwhile (20.5.5)
- f) FNGwhile (20.5.6)

One instance of the following shall be implemented per PBB-TE MEP implementing the Traffic field (21.6.1.4):

- g) mmCCMwhile (20.5.7)
- h) mmLocwhile (20.5.8)
- i) mmFNGwhile (20.5.9)

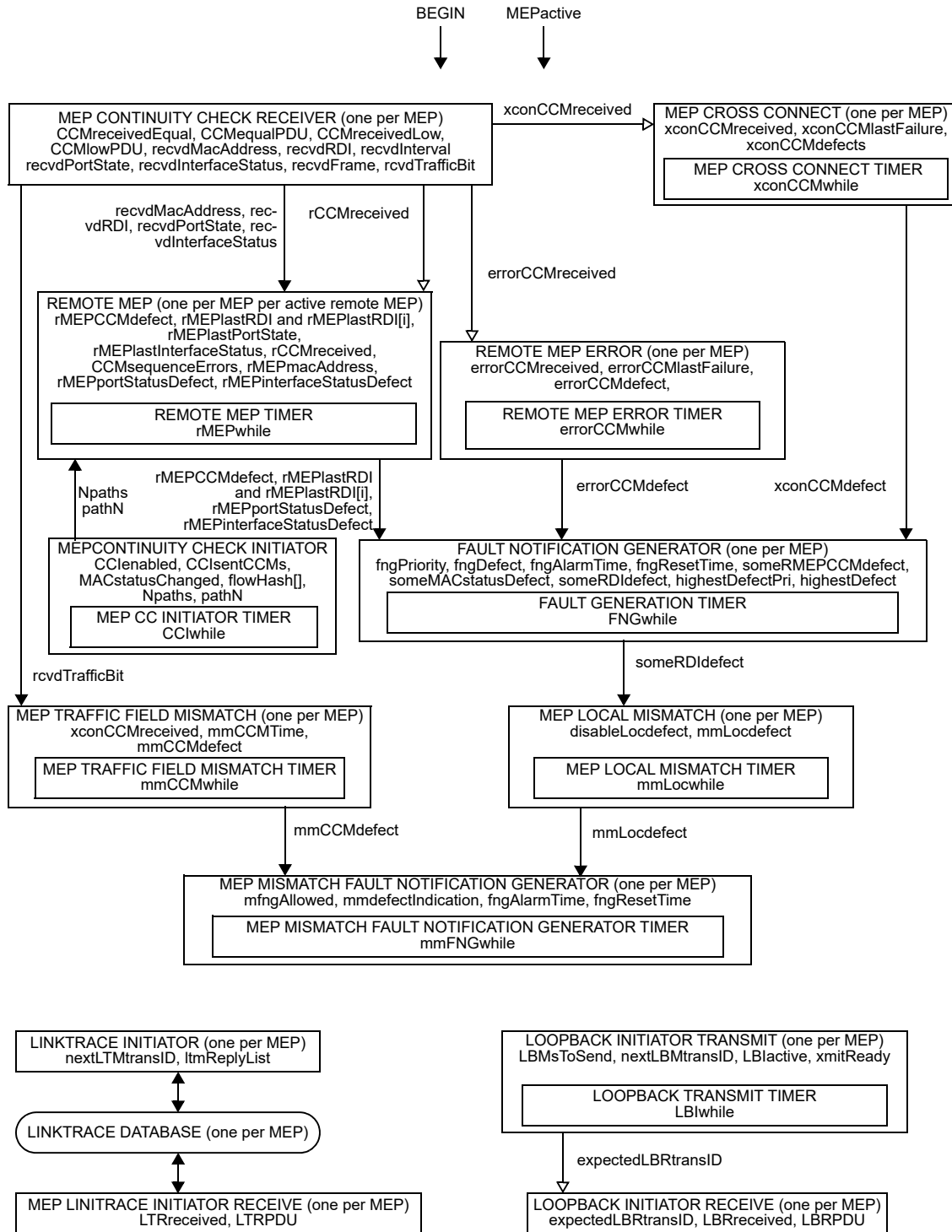
One instance per MEP of the following shall be implemented for each remote MEP in the MEP's MA (those MEPs in the MA other than the MEP, itself):

- j) rMEPwhile (20.5.10)

The "granularity" of a timer variable is the periodicity with which it is decremented.

20.5.1 LTFwhile

A timer variable used by the LTR Transmitter state machine to time out the expected transmission of LTRs.



1 **20.5.2 CCIwhile**

2 Timer counter for transmitting CCMs. CCIwhile has a granularity finer than or equal to 1/4 of the time
3 represented by the CCMinterval variable.

4 **20.5.3 errorCCMwhile**

5 Timer counter for timing out invalid CCMs. errorCCMwhile has a granularity finer than or equal to 1 ms.

6 **20.5.4 xconCCMwhile**

7 Timer counter for timing out cross connect CCMs. xconCCMwhile has a granularity finer than or equal to
8 1 ms.

9 **20.5.5 LBIwhile**

10 A timer variable used by the MEP Loopback Initiator transmit state machine to time out the expected
11 reception of LBRs.

12 **20.5.6 FNGwhile**

13 A timer variable used by the MEP Fault Notification Generator state machine to wait for defects to stabilize
14 and disappear.

15 **20.5.7 mmCCMwhile**

16 Timer counter for declaring mismatch defects based on the received Traffic field. mmCCMwhile has a
17 granularity finer than or equal to 1 ms.

18 **20.5.8 mmLocwhile**

19 Timer counter for declaring mismatch defects based on inconsistencies in locally declared values (20.9.9).
20 mmLocwhile has a granularity finer than or equal to 1 ms.

21 **20.5.9 mmFNGwhile**

22 A timer variable used by the MEP Mismatch Fault Notification Generator state machine (20.40) to wait for
23 defects to stabilize and disappear.

24 **20.5.10 rMEPwhile**

25 Timer counter for timing out CCMs. rMEPwhile has a granularity finer than or equal to 1/4 of the time
26 represented by the CCMinterval variable. A Bridge shall not set rMEPCCMdefect within
27 $(3.25 \times \text{CCMtime}(\text{CCMinterval}))$ seconds of the receipt of a CCM, and shall set rMEPCCMdefect
28 within $(3.5 \times \text{CCMtime}(\text{CCMinterval}))$ seconds after the receipt of the last CCM. This variable is readable
29 as a managed object [item c) in 12.14.7.6.3].

30 NOTE—In order to generate a Defect upon the loss of three CCMs, but not generate a Defect when only two CCMs
31 have been lost, a minimum resolution is required for rMEPwhile. The granularity specified for rMEPwhile, when used
32 with the state machine in Figure 20-5, generates a Defect in a maximum of $(3.5 \times \text{CCMtime}(\text{CCMinterval}))$ seconds, and
33 a minimum of $(3.25 \times \text{CCMtime}(\text{CCMinterval}))$ seconds.

1 20.6 CFM procedures

2 The following procedure is global to the system:

- 3 a) CCMtime() (20.6.1)

4 20.6.1 CCMtime()

5 CCMtime() takes, as a parameter, a value of the form used in the CCMinterval variable (20.8.1) and the
6 CCM Interval field in a CCM PDU (21.6.1.3). It returns, as a value, the corresponding time interval, in the
7 form used by the CCIwhile or rMEPwhile timer, as appropriate (20.5.2, 20.5.10).

8 20.7 Maintenance Domain variable

9 The following variable is local to a single Maintenance Domain and is accessible by more than one state
10 machine:

- 11 a) mdLevel (20.7.1)

12 20.7.1 mdLevel

13 The integer MD Level of the Maintenance Domain. This variable is available as a managed object [item b]
14 in 12.14.5.1.3].

15 20.8 MA variables

16 The following variable is local to a single MA and is accessible by more than one state machine:

- 17 a) CCMinterval (20.8.1)

18 20.8.1 CCMinterval

19 The configured time between CCM transmissions. The value configured for this variable shall be one of the
20 nonzero values in Table 21-15. This variable is available as a managed object [item e) in 12.14.6.1.3].

21 20.9 MEP variables

22 The following variables are local to a single MEP and are accessible by more than one state machine:

- 23 a) MEPactive (20.9.1)
24 b) enableRmepDefect (20.9.2)
25 c) MAdefectIndication (20.9.3)
26 d) allRMEPsDead (20.9.4)
27 e) lowestAlarmPri (20.9.5)
28 f) presentRDI (20.9.6)
29 g) MEPprimaryVID (20.9.7)

30 The following variables may be present only in PBB-TE MAs, are local to a single PBB-TE MEP and are
31 accessible by more than one state machine:

- 32 h) presentTraffic (20.9.8)
33 i) presentmmLoc (20.9.9)

The following variables are present only in Infrastructure Segment MAs, are local to a single Infrastructure Segment MEP, and are accessible by more than one state machine:

- j) ISpresentTraffic (20.9.10)
- k) ISpresentmmLoc (20.9.11)

The following variable is present only in ECMP path MAs, is local to a single ECMP path MEP, and is accessible by more than one state machine:

- l) EpMEP (20.9.12)

20.9.1 MEPactive

A Boolean indicating the administrative state of the MEP. True indicates that the MEP is to function normally, and false that it is to cease functioning. This variable is available as a managed object [item e) in 12.14.7.1.3].

NOTE—MEPactive controls the BEGIN input to the MEP state machines. Therefore, for any MEP that has been completely configured, it is as much an indication of the operative state of the MEP as a control over that state.

20.9.2 enableRmepDefect

A Boolean indicating whether frames on the service instance monitored by this MEP's MA are enabled to pass through this Bridge Port by the spanning tree protocol (Clause 13) and VLAN topology management (Clause 11). Set true either if they are enabled, or if the MEP is not associated with a single value of the Port State and VID member set, else false. In a Bridge, the value of enableRmepDefect is determined from the Port State (8.4) of the spanning tree instance that controls the MEP's Primary VID, and that VID's member set (8.8.10), as shown in Table 20-2. This variable also controls the value output in the Port Status TLV (21.5.4).

Table 20-2—Deriving enableRmepDefect and Port Status TLV in a Bridge

Port State (8.4)	Bridge Port in Primary VID's member set	Port Status TLV (21.5.4)	enableRmepDefect
Disabled, blocked, listening, broken, discarding ^a or learning		psBlocked	False
Forwarding	No	psBlocked	False
	Yes	psUp	True

^a “Discarding” is used in place of the values disabled, blocked, listening, or broken defined in IETF RFC 4318.

A MEP that is:

- a) Configured in an MA that is not associated with a VID
- b) On a Bridge that is running multiple instances of the spanning tree
- c) On a Bridge Port that is not excluded by Static VLAN Registration Entries from membership in VIDs belonging to more than one spanning tree instance

can be unable to generate an unambiguous value for the Port Status TLV, because different MSTIs can be in different Port States. For a MEP in that position, enableRmepDefect is always true.

1 **20.9.3 MAdefectIndication**

2 A Boolean indicating the operational state of the MEP's MA. True indicates that at least one of the remote
3 MEPs configured on this MEP's MA has failed, and false indicates either that all are functioning, or that the
4 MEP has been active for less than the time-out period. MAdefectIndication is true whenever an enabled
5 defect is indicated. That is, MAdefectIndication is true if and only if, for one or more of the variables
6 someRDId defect, someRMEPCCMdefect, someMACstatusDefect, errorCCMdefect, or xconCCMdefect,
7 that variable is true and the corresponding priority of that variable from Table 20-1 is greater than or equal to
8 the value of the variable lowestAlarmPri.

9 **20.9.4 allRMEPsDead**

10 A Boolean indicating that this MEP is receiving none of the remote MEPs' CCMs. allRMEPsDead is the
11 logical AND of all of the rMEPCCMdefect variables.

12 **20.9.5 lowestAlarmPri**

13 An integer value indicating the lowest defect priority (see Table 20-1) that can trigger the generation of a
14 Fault Alarm. This variable is a managed object [item k) in 12.14.7.1.3].

15 **20.9.6 presentRDI**

16 A Boolean value indicating the state of the RDI bit in CCMs transmitted by this MEP. presentRDI is true if
17 and only if one or more of the variables someRMEPCCMdefect, someMACstatusDefect, errorCCMdefect,
18 or xconCCMdefect is true, and if the corresponding priority of that variable, from Table 20-1, is greater than
19 or equal to the value of the variable lowestAlarmPri.

20 **20.9.7 MEPprimaryVID**

21 An integer specifying a VID, one of the VIDs assigned to the MEP's MA, which is to be used as the Primary
22 VID for this MEP. This variable is related to the managed object item d) in 12.14.7.1.3. It is not necessarily
23 numerically equal to that object, however. MEPprimaryVID always contains the numerical value of the
24 Primary VID. The managed object may contain 0, to indicate that the Primary VID is that of the MEP's MA,
25 or contain the Primary VID itself, if the MA's VID is overridden for this particular MEP. In the case of a
26 MEP associated with a PBB-TE MA, this value is not directly configurable but it takes the value of the
27 ESP-VID field in the 3-tuple of the PBB-TE MA's ESP that has the MEP's MAC address in its ESP-SA
28 field.

29 **20.9.8 presentTraffic**

30 A Boolean value indicating if at least one backbone service instance is configured to use the TESI's ESP
31 upon which this PBB-TE MEP is transmitting CCMs. presentTraffic is TRUE if and only if the Backbone
32 Service Instance table, of the CBP associated with this MEP, contains an entry that has in its B-VID and
33 Default Backbone Destination fields the values of ESP-VID and ESP-DA of the monitored TESI's ESP that
34 originates at the MEP.

35 **20.9.9 presentmmLoc**

36 presentmmLoc is the logical AND of presentRDI (20.9.6) and presentTraffic (20.9.8).

1 20.9.10 ISpresentTraffic

2 A Boolean value indicating if at least one TESI protected by the IPG is configured to use the segment
3 monitored by the Infrastructure Segment MEP. ISpresentTraffic is TRUE if and only if the port upon which
4 this MEP is configured is the outbound port of the entry in the FDB corresponding to the TESI protected by
5 the IPG.

6 20.9.11 ISpresentmmLoc

7 ISpresentmmLoc is the logical AND of presentRDI (20.9.6) and ISpresentTraffic (20.9.10).

8 20.9.12 EpMEP

9 Boolean. This variable is used to condition actions related to ECMP path MEPs. It is true for ECMP path
10 MEPs and false otherwise.

11 20.10 MEP Continuity Check Initiator variables

12 The following variables are local to the MEP Continuity Check Initiator state machine:

- 13 a) CClenabled (20.10.1)
- 14 b) CCIsentCCMs (20.10.2)
- 15 c) MACstatusChanged (20.10.3)
- 16 d) Npaths (20.10.4)
- 17 e) flowHash[] (20.10.5)
- 18 f) pathN (20.10.6)
- 19 g) CCMcnt (20.10.7)

20 20.10.1 CClenabled

21 Controls the transmission of CCMs. When set to true, CCMs are transmitted. When set to false, CCM
22 transmission ceases. This variable is available as a managed object [item g) in 12.14.7.1.3].

23 20.10.2 CCIsentCCMs

24 Integer value. CCIsentCCMs is initialized to 1 when the MEP is created and may be used thereafter by
25 xmitCCM() to fill the Sequence Number field of a transmitted CCM. This variable is available as a managed
26 object [item w) in 12.14.7.1.3].

27 20.10.3 MACstatusChanged

28 Boolean. MACstatusChanged triggers the transmission of one extra CCM. If the Port Status TLV (21.5.4) or
29 Interface Status TLV (21.5.5) is being transmitted by the MEP, and if CCMinterval indicates a transmission
30 interval of 10 s or slower, MACstatusChanged is set to true whenever the value reported in either of those
31 two TLVs changes. MACstatusChanged is reset to false by the MEP Continuity Check Initiator state
32 machine.

33 20.10.4 Npaths

34 Integer value. The number of paths being tested by this MEP. The value is 1 (default) except in the case of
35 ECMP path MEPs that may test a set of paths by using one destination address and different flow hash
36 values.

1 20.10.5 flowHash[]

2 This variable is applicable only for ECMP path MEPs.

3 Array of 16-bit values. A set of flow hash values used to test a set of paths to a given destination address.

4 The default is a single NULL value.

5 20.10.6 pathN

6 This variable is applicable only for ECMP path MEPs.

7 Integer value in the range 0..(Npaths – 1) and all arithmetic using this variable is modulo Npaths. The index
8 of the current path (flow hash) being used in CCMs sent from the MEP. This value is shared with the Remote
9 MEP state machine to enable correlation of received RDIs with the cycle of paths being tested.

10 20.10.7 CCMcnt

11 This variable is applicable only for ECMP path MEPs.

12 Integer value in the range 0..3 and all arithmetic using this variable is modulo 4. This variable is a counter
13 used to send each flowHash[] value in four successive CCMs.

14 20.11 MEP Continuity Check Initiator procedures

15 The following procedure is local to the MEP Continuity Check Initiator state machine:

16 a) xmitCCM() (20.11.1)

17 20.11.1 xmitCCM()

18 xmitCCM() constructs and transmits a CCM on the Active SAP using an M_UNITDATA.request as follows:

- 19 a) Sets the destination_address parameter to the value from Table 8-18 corresponding to the MEP's
20 MD Level. In the case of a PBB-TE associated MEP, SPBM path MEP, or ECMP path MEP, the
21 destination_address parameter is set to the MAC address indicated by the ESP-DA field of the ESP
22 having in its ESP-SA field the MAC address of the MEP. In the case of an SPBM VID MEP
23 (27.18.1), the destination_address parameter is set to the SPBM group MAC address (27.15)
24 associated with the source MEP and the SPBM default I-SID. In the case of an SPBM group MEP,
25 the destination_address parameter is set to the SPBM group MAC address (27.15) assigned to the
26 source MEP's CBP for the SPBM group MA's I-SID.
- 27 b) Sets the source_address parameter to the MAC address of the MEP [item i) in 12.14.7.1.3].
- 28 c) Sets the priority parameter according to the MEP's managed objects [item h) in 12.14.7.1.3].
- 29 d) Sets the drop_eligible parameter to false.
- 30 e) In the case of ECMP with flow filtering, the time_to_live and flow_hash parameters must be set.
31 The time_to_live parameter is set to the TTL Value in the Flow Filtering Control Table entry on the
32 CBP for the source MEP's VID [item d) in 12.16.5.5.2]. In the case of an SPBM VID MEP or
33 SPBM group MEP, the flow_hash parameter is set to zero. In the case of an ECMP path MEP, the
34 flow_hash parameter is set to the value of flowHash[pathN].
- 35 f) Places the MEP's MD Level (20.7.1) in the MD Level field (21.4.1).
- 36 g) Fills the CCM Interval field (21.6.1.3) with the CCM transmission interval [20.8.1, item e) in
37 12.14.6.1.3].
- 38 h) Fills the RDI field (21.6.1.1) with the presentRDI variable (20.9.6).
- 39 i) Should copy CCI sent CCMs (20.10.2) to the Sequence Number field of the CCM (21.6.3), else
40 copies 0 into that field.

- 1 j) Places the MEP's MAID into the appropriate fields of the CCM [item a) in 12.14.1.2.2, item b) in
- 2 12.14.5.3.2, and 21.6.5.1 through 21.6.5.6].
- 3 k) Places the MEP's MEPID [item g) in 12.14.6.1.3] into the Maintenance association Endpoint
- 4 Identifier field (21.6.4).
- 5 l) As controlled by the managed objects item e) in 12.14.3.1.3, item d) in 12.14.5.1.3, and item d) in
- 6 12.14.6.1.3, places a Sender ID TLV (21.5.3) in the CCM, identifying the transmitting system.
- 7 m) Optionally, places a Port Status TLV (21.5.4) in the CCM, reporting the status of the Bridge Port.
- 8 n) Optionally, places an Interface Status TLV (21.5.5) in the CCM, reporting the status of the Bridge
- 9 Port.
- 10 o) Increments CCI_{sentCCMs} by 1, wrapping around from 232 – 1 to 0.
- 11 p) Only in the case of a PBB-TE MEP, optionally fills the Traffic field (21.6.1.4) with the
- 12 presentTraffic variable (20.9.8).
- 13 q) Only in the case of an Infrastructure Segment MEP, optionally fills the Traffic field (21.6.1.4) with
- 14 the value IS_{presentTraffic} (20.9.10).

15 20.12 MEP Continuity Check Initiator state machine

16 Each MEP Continuity Check Initiator (19.2.9) instantiates one MEP Continuity Check Initiator state
17 machine. The MEP Continuity Check Initiator state machine implements the function specified by the state
18 diagram in Figure 20-2, the variable declarations in 20.10 and the procedures in 20.11.

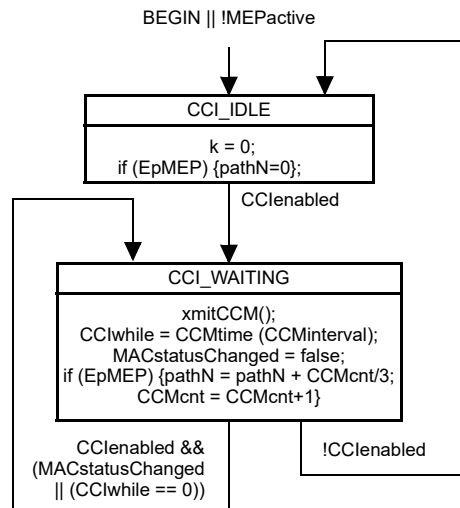


Figure 20-2—MEP Continuity Check Initiator state machine

19 NOTE—Figure 20-2 indicates that the setting of MACstatusChanged causes CCIwhile to be reset and thus changes the
20 phase of the CCM transmission cycle. An alternative formulation of state machine could transmit an extra CCM, but not
21 reset CCIwhile. Whether setting MACstatusChanged in a MEP causes CCIwhile to be reset is not specified by this
22 standard.

23 20.13 MHF Continuity Check Receiver variables

24 The following variables are defined for the MHF Continuity Check Receiver:

- 25 a) MHFrecvdCCM (20.13.1)
- 26 b) MHFCCMPDU (20.13.2)

20.13.1 MHFrecvdCCM

A Boolean value set to true by the MHF OpCode Demultiplexer (19.3.6) when a CCM at the MHF's MD Level is received. Cleared by the MHF Continuity Check Receiver state machine.

20.13.2 MHFCCMPDU

Structure. Loaded with the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the CCM PDU received by the MHF OpCode Demultiplexer (19.3.6) when a CCM at the MHF's MD Level is received.

20.14 MHF Continuity Check Receiver procedures

The following procedure is defined for the MHF Continuity Check Receiver:

- a) MHFprocessCCM() (20.14.1)

20.14.1 MHFprocessCCM()

MHFprocessCCM() shall validate the received CCM according to 20.51.4 and discard the received CCM if invalid. Otherwise, MHFprocessCCM() records the FID, source_address, and Port number of the received CCM in the MHF's MIP CCM Database.

20.15 MHF Continuity Check Receiver state machine

The MHF Continuity Check Receiver state machine implements the functions specified by the state diagram in Figure 20-3, using the variables in 20.13 and procedures in 20.14.

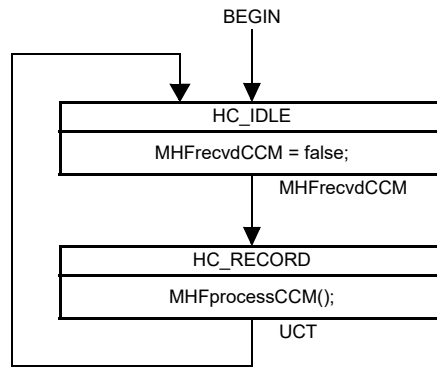


Figure 20-3—MHF Continuity Check Receiver state machine

20.16 MEP Continuity Check Receiver variables

The following variables are defined for the MEP Continuity Check Receiver:

- a) CCMreceivedEqual (20.16.1)
- b) CCMequalPDU (20.16.2)
- c) CCMreceivedLow (20.16.3)
- d) CCMlowPDU (20.16.4)

- 1 e) recvdMacAddress (20.16.5)
- 2 f) recvdRDI (20.16.6)
- 3 g) recvdInterval (20.16.7)
- 4 h) recvdPortState (20.16.8)
- 5 i) recvdInterfaceStatus (20.16.9)
- 6 j) recvdSenderId (20.16.10)
- 7 k) recvdFrame (20.16.11)
- 8 l) CCMsequenceErrors (20.16.12)

9 The following variable is present only in PBB-TE MEP or Infrastructure Segment MEP that supports the
10 Traffic field (21.6.1.4), is local to a single PBB-TE MEP or Infrastructure Segment MEP, and is accessible
11 by the MEP Mismatch state machines (20.26):

- 12 m) recvdTrafficBit (20.16.13)

13 **20.16.1 CCMreceivedEqual**

14 Boolean flag. Set by the MEP Equal OpCode Demultiplexer (19.2.7) when a CCM at the MEP's MD Level
15 is received. Cleared by the MEP Continuity Check Receiver state machine.

16 **20.16.2 CCMequalPDU**

17 Structure. Loaded with the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the
18 CCM PDU received by the MEP Equal OpCode Demultiplexer (19.2.7) when a CCM at the MEP's MD
19 Level is received.

20 **20.16.3 CCMreceivedLow**

21 Boolean flag. Set by the MEP Low OpCode Demultiplexer (19.2.7) when a CCM below the MEP's MD
22 Level is received. Cleared by the MEP Continuity Check Receiver state machine.

23 **20.16.4 CCMlowPDU**

24 Structure. Loaded with the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the
25 CCM PDU received by the MEP Low OpCode Demultiplexer (19.2.7) when a CCM below the MEP's MD
26 Level is received.

27 **20.16.5 recvdMacAddress**

28 Set by the MEPprocessEqualCCM() procedure with the source_address from the last-received CCM from
29 the remote MEP served by this copy of the Remote MEP state machine.

30 **20.16.6 recvdRDI**

31 Boolean flag. Set by MEPprocessEqualCCM() according to the RDI field of the last-received valid CCM.

32 **20.16.7 recvdInterval**

33 Timer counter indicating timer counter equivalent of the value encoded in the CCM Interval field of a
34 received CCM. Set by MEPprocessEqualCCM().

1 **20.16.8 recvdPortState**

2 (optional) Enumerated variable indicating the contents of the Port Status TLV (21.5.4) of the last-received
3 valid CCM or psNoPortStateTLV: Indicates either that no CCM has been received or that no Port Status TLV
4 was present in the last CCM received. Set by MEPprocessEqualCCM().

5 **20.16.9 recvdInterfaceStatus**

6 (optional) Enumerated variable indicating the contents of the Interface Status TLV (21.5.5) of the
7 last-received valid CCM or isNoInterfaceStatusTLV: Indicates either that no CCM has been received or that
8 no Interface Status TLV was present in the last CCM received. Set by MEPprocessEqualCCM().

9 **20.16.10 recvdSenderId**

10 (optional) Enumerated variable indicating the contents of the Sender ID TLV (21.5.3) of the last-received
11 valid CCM or isNoSenderIdTLV: Indicates either that no CCM has been received or that no Sender ID TLV
12 was present in the last CCM received. Set by MEPprocessEqualCCM().

13 **20.16.11 recvdFrame**

14 Octet string holding a frame received by MEPprocessEqualCCM(). recvdFrame can be reconstructed either
15 from the M_UNITDATA.indication information presented to the MEP through the Active SAP, or from the
16 EM_UNITDATA.indication presented to the EISS Multiplex Entity. The size of recvdFrame indicates the
17 size of the reconstructed frame. When set to the value NULL, the size of recvdFrame is 0.

18 **20.16.12 CCMsequenceErrors**

19 The total number of out-of-sequence CCMs received from all remote MEPs. This variable is readable as a
20 managed object [item v) in 12.14.7.1.3].

21 **20.16.13 rcvdTrafficBit**

22 Boolean flag that is applicable only for PBB-TE MEPs and Infrastructure Segment MEPs supporting the
23 Traffic field (21.6.1.4). Set by MEPprocessEqualCCM() according to the Traffic field of the last-received
24 valid CCM.

25 **20.17 MEP Continuity Check Receiver procedures**

26 The following procedures are defined for the Continuity Check Receiver:

- 27 a) MEPprocessEqualCCM() (20.17.1)
- 28 b) MEPprocessLowCCM() (20.17.2)

29 **20.17.1 MEPprocessEqualCCM()**

30 Called by the MEP Continuity Check Receiver state machine whenever a CCM is received at the MD Level
31 of the MEP. MEPprocessEqualCCM() processes the CCM contained in CCMequalPDU as follows:

- 32 a) MEPprocessEqualCCM() shall process the CCM according to 20.51.4.2, and may validate the CCM
33 according to 20.51.4.3, and discard any frames that fail the validation.
- 34 b) Otherwise, if the MAID of the received CCM does not exactly match the MAID configured in the
35 receiving MEP [item a) in 12.14.1.2.2, item b) in 12.14.5.3.2], then MEPprocessEqualCCM() sets
36 xconCCMreceived (20.23.1) true (this procedural step being optional in the case of a PBB-TE MEP
37 or Infrastructure Segment MEP, SPBM path MEP, or ECMP path MEP), reconstructs the frame

- 1 containing the CCM into recvdFrame, and places a timer counter value into recvdInterval
2 corresponding to the value of the CCM Interval field in the received CCM.
- 3 c) Otherwise, if:
- 4 1) MEPID in the received CCM is not configured in the receiving MEP [item g) in 12.14.6.1.3],
5 or
6 2) MEPID in the received CCM matches the MEPID of the receiving MEP [item b) in
7 12.14.6.3.2], or
8 3) CCM Interval field in the received CCM does not match that configured for the receiving MEP
9 [item e) in 12.14.6.1.3],
10 then MEPprocessEqualCCM() sets errorCCMreceived (20.21.1) true, reconstructs the frame
11 containing the CCM into recvdFrame, and places a timer counter value into recvdInterval
12 corresponding to the value of the CCM Interval field in the received CCM.
- 13 d) Otherwise, MEPprocessEqualCCM()
- 14 1) Copies the source_address parameter into recvdMacAddress.
15 2) Copies the RDI field to recvdRDI.
16 3) Optionally copies the Port Status TLV to recvdPortState.
17 4) Optionally copies the Interface Status TLV to recvdInterfaceStatus.
18 5) Optionally copies the Sender ID TLV to recvdSenderId.
19 6) Optionally updates the Bridge's MIP CCM Database.
20 7) Sets the rCCMreceived variable (20.19.6) for the particular instance of the Remote MEP state
21 machine corresponding to the Maintenance association Endpoint Identifier field in the CCM.
22 8) Only in the case of a PBB-TE MEP or Infrastructure Segment MEP, optionally copies the
23 Traffic field (21.6.1.4) to recvdTrafficBit (20.16.13).

24 MEPprocessEqualCCM() should also:

- 25 e) Compare the Sequence Number field in the received CCM to the value saved in the MEP CCM
26 Database.
27 f) If both values are not 0, and if the new value is not 1 greater than the last, increment
28 CCMsequenceErrors (20.16.12).
29 g) Store the received Sequence Number in the MEP CCM Database.

30 **20.17.2 MEPprocessLowCCM()**

31 Called by the MEP Continuity Check Receiver state machine whenever a CCM is received with an MD
32 Level that is less than the MEP's MD Level (mdLevel, 20.7.1). MEPprocessLowCCM() processes the CCM
33 contained in CCMlowPDU as follows:

- 34 a) MEPprocessLowCCM() shall process the CCM according to 20.51.4.2, and may validate the CCM
35 according to 20.51.4.3, and discard any frames that fail the validation.
36 b) Otherwise, MEPprocessLowCCM() sets xconCCMreceived (20.23.1) true, reconstructs the frame
37 containing the CCM into recvdFrame, and places a timer counter value into recvdInterval
38 corresponding to the value of the CCM Interval field in the received CCM.

39 **20.18 MEP Continuity Check Receiver state machine**

40 The MEP Continuity Check Receiver state machine implements the functions specified by the state diagram
41 in Figure 20-4, using the variables in 20.16 and procedures in 20.17.

20.19 Remote MEP variables

The following variables are local to the Remote MEP state machine:

- a) rMEPCCMdefect (20.19.1)
- b) rMEPlastRDI and rMEPlastRDI[i] (20.19.2)
- c) rMEPlastPortState (20.19.3)
- d) rMEPlastInterfaceStatus (20.19.4)
- e) rMEPlastSenderId (20.19.5)
- f) rCCMreceived (20.19.6)
- g) rMEPmacAddress (20.19.7)
- h) rMEPportStatusDefect (20.19.8)
- i) rMEPinterfaceStatusDefect (20.19.9)
- j) lastPathN (20.19.10)

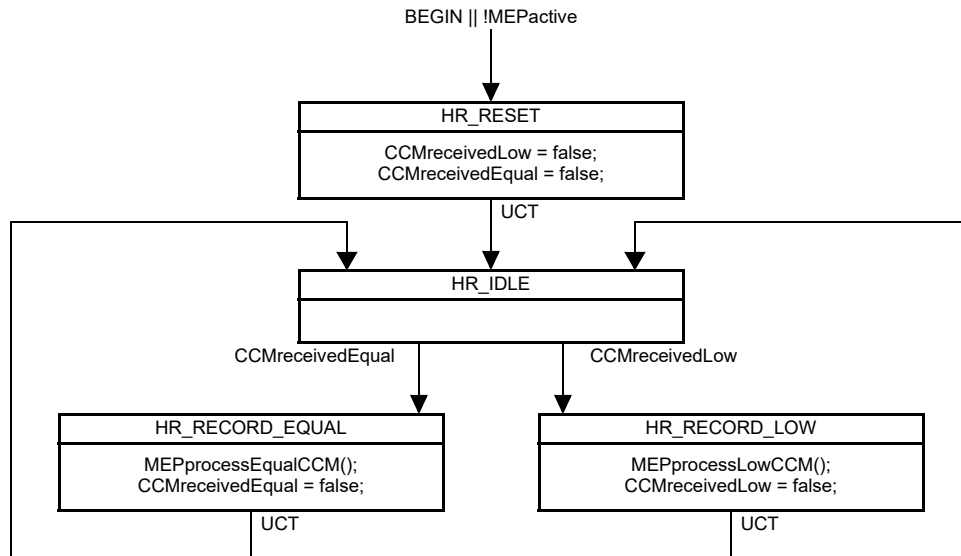


Figure 20-4—MEP Continuity Check Receiver state machine

20.19.1 rMEPCCMdefect

Reports the state of the remote MEP. When true, no CCM has been received from the remote MEP for at least $(3.25 \times \text{CCMtime}(\text{CCMinterval}))$ seconds.

20.19.2 rMEPlastRDI and rMEPlastRDI[i]

Boolean flag. Contains the RDI flag from the last-received CCM. This variable is readable as a managed object [item e) in 12.14.7.6.3]. In the case of an ECMP path MEP, if $N_{\text{paths}} > 1$, there is a set of variables $\text{rMEPlastRDI}[i]$ of size N_{paths} (20.10.4). In this case the variable rMEPlastRDI is the logical OR of the members of the set $\text{rMEPlastRDI}[i]$. Each member of the set $\text{rMEPlastRDI}[i]$ is set if an RDI is received while $\text{pathN} = i$ and is reset if no RDI is received while $\text{pathN} = i$ in the last CCM cycle of the MEP Continuity Check Initiator state machine.

1 **20.19.3 rMEPlastPortState**

2 Enumerated value. Contains the value obtained from the Port Status TLV (21.5.4) of the last-received CCM
3 or a value indicating that the last-received CCM contained no Port Status TLV. This variable is readable as a
4 managed object [item f) in 12.14.7.6.3].

5 **20.19.4 rMEPlastInterfaceStatus**

6 Enumerated value. Contains the value obtained from the Interface Status TLV (21.5.5) of the last-received
7 CCM or a value indicating that the last-received CCM contained no Interface Status TLV. This variable is
8 readable as a managed object [item g) in 12.14.7.6.3].

9 **20.19.5 rMEPlastSenderId**

10 Enumerated value. Contains the value obtained from the Sender ID TLV (21.5.3) of the last-received CCM
11 or a value indicating that the last-received CCM contained no Sender ID TLV. This variable is readable as a
12 managed object [item h) in 12.14.7.6.3].

13 **20.19.6 rCCMreceived**

14 Boolean flag set true by MEPprocessEqualCCM() to indicate that a CCM has been received that matches the
15 configured expectations of a particular Remote MEP state machine. Cleared to false by the Remote MEP
16 state machine.

17 **20.19.7 rMEPmacAddress**

18 One field in each entry in the MEP CCM Database for a remote MEP, containing source_address of the
19 last-received CCM from that remote MEP. This variable is readable as a managed object [item d) in
20 12.14.7.6.3].

21 **20.19.8 rMEPportStatusDefect**

22 A Boolean indicating that the remote MEP is reporting a failure in its Port Status TLV (21.5.4). It is true only
23 if the last received CCM contained a Port Status TLV that contained some value other than psUp, i.e., the
24 remote MEP's Bridge Port is not forwarding data. It is equal to (rMEPlastPortState!= psUp &&
25 rMEPlastPortState!= psNoPortStateTLV: Indicates either that no CCM has been received or that no Port
26 Status TLV was present in the last CCM received).

27 **20.19.9 rMEPinterfaceStatusDefect**

28 A Boolean indicating that the remote MEP is reporting a failure in its Interface Status TLV (21.5.5). It is true
29 only if the last received CCM contained an Interface Status TLV that contained some value other than isUp,
30 i.e., the remote MEP's Bridge Port is not available for forwarding data. It is equal to
31 (rMEPlastInterfaceStatus!= isUp && rMEPlastInterfaceStatus!= isNoInterfaceStatusTLV: Indicates either
32 that no CCM has been received or that no Interface Status TLV was present in the last CCM received).

33 **20.19.10 lastPathN**

34 This variable is applicable only for ECMP path MEPs.

35 Integer value. The variable lastPathN holds the previous value of the pathN variable.

20.20 Remote MEP state machine

The Remote MEP state machine implements the function specified by the state diagram in Figure 20-5 and the variable declarations in 20.19, and utilizes the procedures in 20.17.

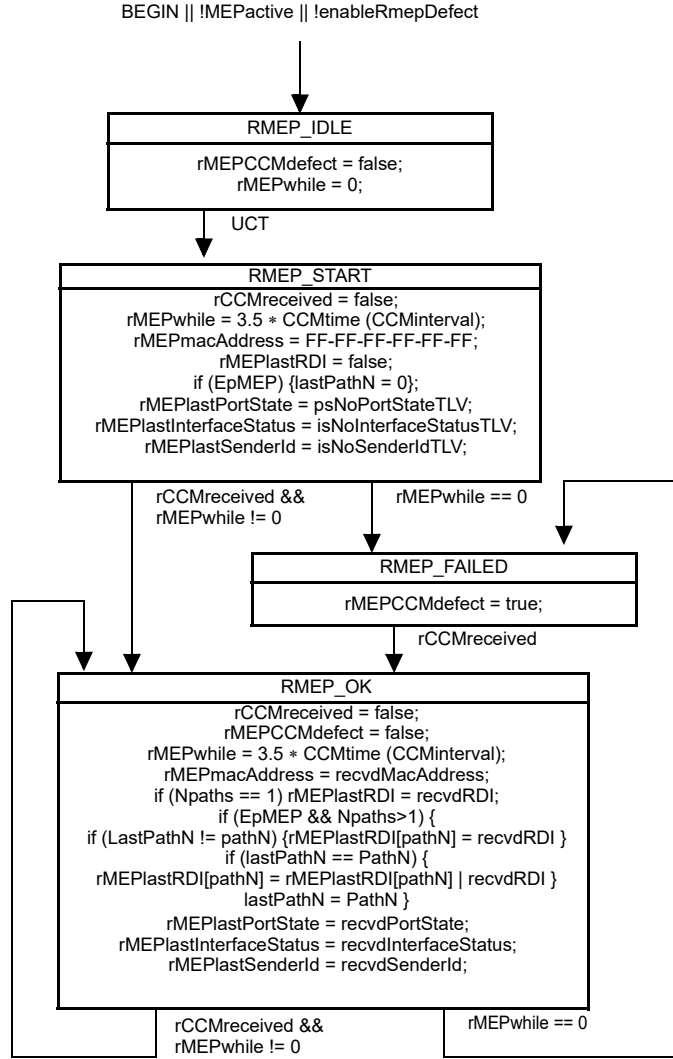


Figure 20-5—Remote MEP state machine

20.21 Remote MEP Error variables

The following variables are local to the Remote MEP Error state machine:

- a) errorCCMreceived (20.21.1)
- b) errorCCMlastFailure (20.21.2)
- c) errorCCMdefect (20.21.3)

20.21.1 errorCCMreceived

Boolean flag set true by MEPprocessEqualCCM() if an invalid CCM is received. Cleared to false by the Remote MEP Error state machine.

20.21.2 errorCCMlastFailure

Octet string with the same characteristics as recvdFrame. Value controlled by the Remote MEP Error state machine. Readable as a managed object [item t) in 12.14.7.1.3]. Size is sufficient to retain a frame containing a maximum-length CCM in errorCCMlastFailure.

20.21.3 errorCCMdefect

A Boolean flag set and cleared by the Remote MEP Error state machine to indicate that one or more invalid CCMs has been received, and that 3.5 times that CCM's transmission interval has not yet expired. This variable is readable as a managed object [item r) in 12.14.7.1.3].

20.22 Remote MEP Error state machine

The Remote MEP Error state machine implements the function specified by the state diagram in Figure 20-6 and the variable declarations in 20.21.

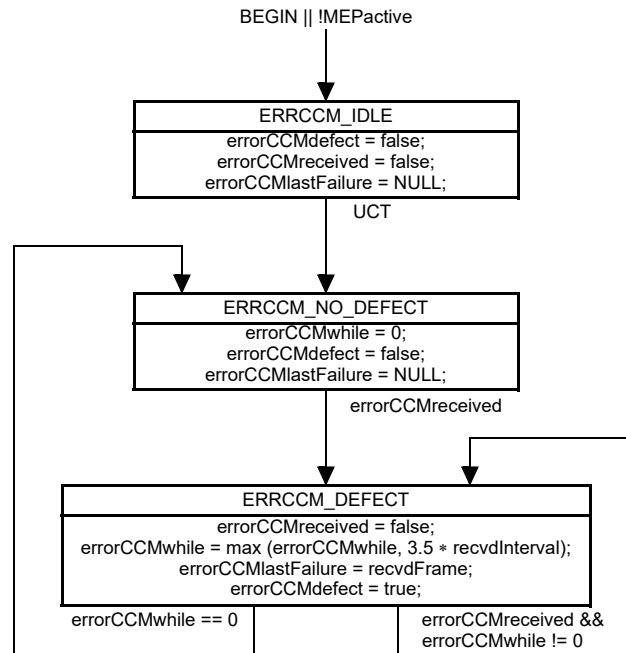


Figure 20-6—Remote MEP Error state machine

20.23 MEP Cross Connect variables

The following variables are local to the MEP Cross Connect state machine:

- xconCCMreceived (20.23.1)
- xconCCMlastFailure (20.23.2)
- xconCCMdefect (20.23.3)

20.23.1 xconCCMreceived

Boolean flag set true by MEPprocessEqualCCM() when a cross connect CCM is received. Cleared to false by the MEP Cross Connect state machine.

20.23.2 xconCCMlastFailure

Octet string with the same characteristics as recvdFrame. Value controlled by the MEP Cross Connect state machine. Readable as a managed object [item u) in 12.14.7.1.3]. Size is sufficient to retain a frame containing a maximum-length CCM.

20.23.3 xconCCMdefect

A Boolean flag set and cleared by the MEP Cross Connect state machine to indicate that one or more cross connect CCMs has been received, and that 3.5 times of at least one of those CCMs' transmission interval has not yet expired. This variable is readable as a managed object [item s) in 12.14.7.1.3].

20.24 MEP Cross Connect state machine

The MEP Cross Connect state machine implements the function specified by the state diagram in Figure 20-7 and the variable declarations in 20.23.

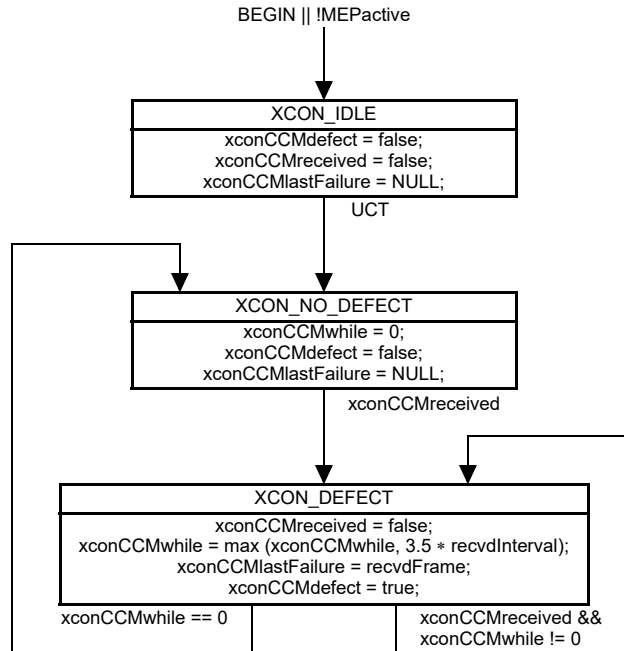


Figure 20-7—MEP Cross Connect state machine

20.25 MEP Mismatch variables

The following variables are local to the MEP Mismatch state machines for a PBB-TE MEP or Infrastructure Segment MEP implementing the Traffic field (21.6.1.4):

- a) mmCCMreceived (20.25.1)
- b) mmCCMdefect (20.25.2)
- c) mmCCMTime (20.25.3)

- 1 d) disableLocdefect (20.25.4)
- 2 e) mmLocdefect (20.25.5)

3 **20.25.1 mmCCMreceived**

4 Boolean flag set to TRUE when rcvdTrafficBit (20.16.13) does not match the presentTraffic (20.9.8) in the
5 case of a PBB-TE MEP or rcvdTrafficBit (20.16.13) does not match the ISpresentTraffic (20.9.10) in the
6 case of an Infrastructure segment MEP.

7 **20.25.2 mmCCMdefect**

8 A Boolean flag set and cleared by the MEP Mismatch state machines to indicate that one or more CCMs
9 with Traffic fields not matching the presentTraffic (20.9.8) have been received in the case of a state machine
10 associated with a PBB-TE MEP or that one or more CCMs with ISpresentTraffic (20.9.10) has been
11 received in the case of a state machine associated with an Infrastructure Segment MEP, over a period that is
12 greater than 3.5 times the configured CCM transmission rate and given by the mmCCMTime (20.25.3). This
13 variable is readable as a managed object [item ah] in 12.14.7.1.3].

14 **20.25.3 mmCCMTime**

15 The time used to initiate the mmCCMwhile timer. This is the time taken for a remote IB-BEB to send a
16 CCM with the correct information in the Traffic field and is tied to the target protection switching time, i.e.,
17 50 ms, as a mismatch would be normal prior to the completion of a protection switch. In the most generic
18 case it is equal to the Detection Time + Restoration Procedure Time + Restoration Transfer time. This gives
19 a time of $3.5 \times \text{CCM interval} + \text{Restoration Procedure Time} + 1.25 \times \text{CCM interval}$. As a result its value is
20 set to max (50 ms, $4.75 \times \text{CCMtime} (\text{CCMinterval}) + 10 \text{ ms}$).

21 **20.25.4 disableLocdefect**

22 A Boolean that disables the indication of the mmLocdefect. The variable is always set to TRUE for PBB-TE
23 MAs that are not part of a TE protection group. On a PBB-TE MEP associated with the working entity in
24 one or more TE protection groups to which its monitoring TESI MA is assigned, disableLocdefect is TRUE
25 if and only if either p.SF (26.10.3.3.2) or LoP (26.10.3.3.4) is TRUE on any of these TE protection groups.
26 On a PBB-TE MEP associated with only one protection entity in all the TE protection groups to which it is
27 assigned, disableLocdefect is TRUE if and only if FS (26.10.3.3.5) is TRUE on this TE protection group.

28 **20.25.5 mmLocdefect**

29 A Boolean flag set and cleared by the MEP Local Mismatch state machine to indicate that presentmmLoc
30 (20.9.9) or ISpresentmmLoc (20.9.11) is set to TRUE, over a period of 50 ms [item ai] in 12.14.7.1.3].

31 **20.26 MEP Mismatch state machines**

32 The MEP Mismatch state machines implement the functions specified by the state diagrams in Figure 20-8,
33 Figure 20-9, and the variable declarations in 20.25. There is one MEP Traffic Field Mismatch state machine
34 and one MEP Local Mismatch state machine per PBB-TE MEP or Infrastructure Segment MEP
35 implementing the Traffic field (21.6.1.4).

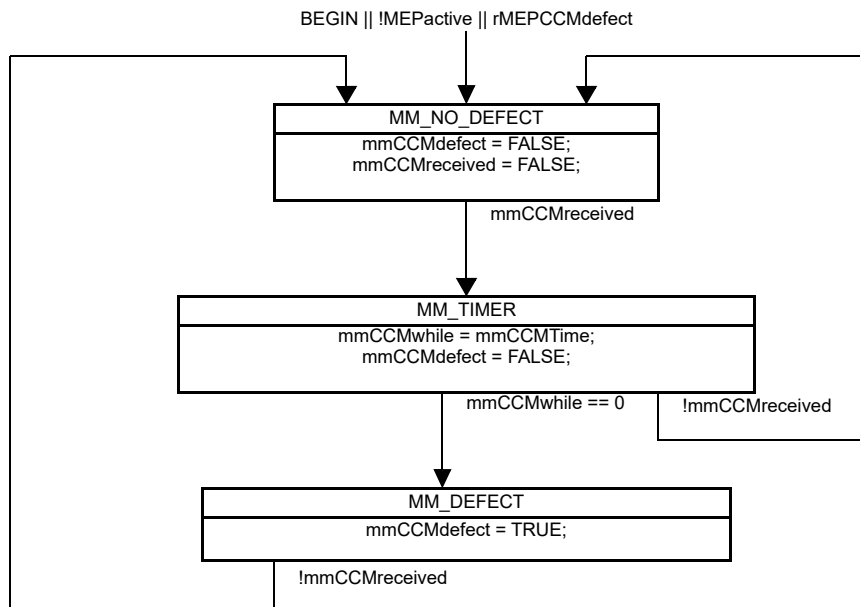


Figure 20-8—MEP Traffic Field Mismatch state machine

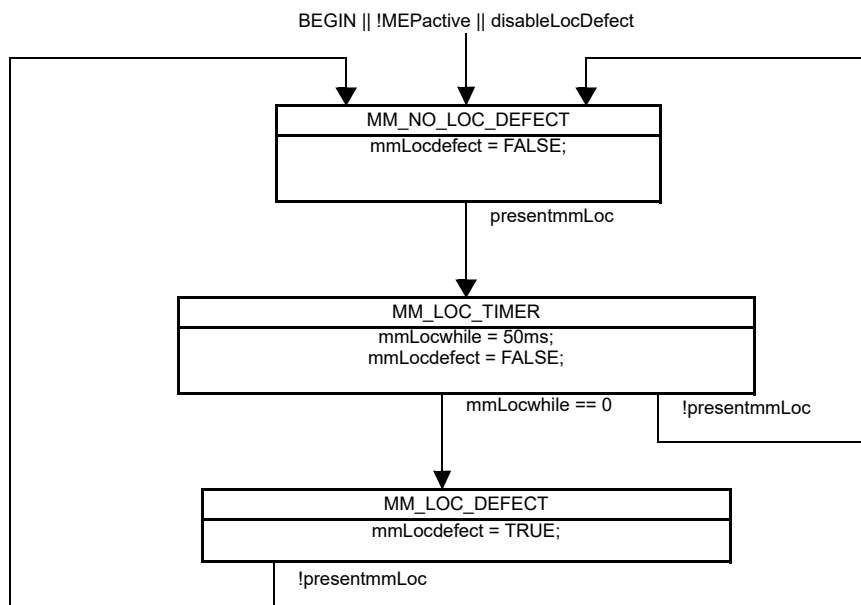


Figure 20-9—MEP Local Mismatch state machine

1 20.27 MP Loopback Responder variables

2 The following variables are local to the MP Loopback Responder state machine:

- 3 a) LBMreceived (20.27.1)
- 4 b) LBMPDU (20.27.2)

5 20.27.1 LBMreceived

6 Boolean variable, set to true by an MP OpCode Demultiplexer when an LBM is received, and cleared by an
7 MP Loopback Responder state machine.

8 20.27.2 LBMPDU

9 Structure. Loaded with the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the
10 LBM PDU received by the an MP OpCode Demultiplexer (19.2.7) when an LBM is received.

11 20.28 MP Loopback Responder procedures

12 The following procedures are defined for the Loopback Responder:

- 13 a) ProcessLBM() (20.28.1)
- 14 b) xmitLBR() (20.28.2)

15 20.28.1 ProcessLBM()

16 Called by the MP Loopback Responder state machine whenever an LBM is received. ProcessLBM()
17 processes the LBM in LBMPDU as follows:

- 18 a) If :
 - 19 1) the destination_address parameter contains an individual MAC address that does not match the
 - 20 MAC address of the receiving MP and
 - 21 2) the MP Loopback Responder state machine does not reside in a PBB-TE or SPBM VID MHF,
 - 22 ProcessLBM() discards the LBM and performs no further processing.
- 23 b) If the destination_address parameter contains a Group address and the MP Loopback Responder
- 24 state machine resides in an MHF associated with neither PBB-TE nor SPBM (rather than in a MEP),
- 25 ProcessLBM() discards the LBM and performs no further processing.
- 26 c) If the source_address parameter is a Group, and not an individual MAC address, ProcessLBM()
- 27 discards the frame and performs no further processing.
- 28 d) ProcessLBM() shall process the LBM according to 20.51.4.2, and may validate the LBM according
- 29 to 20.51.4.3, and discard any frames that fail the validation.
- 30 e) If the MP Loopback Responder state machine resides in a PBB-TE MEP and the LBM carries a
- 31 PBB-TE MIP TLV, ProcessLBM() discards the LBM and performs no further processing.
- 32 f) If the MP Loopback Responder state machine resides in a PBB-TE or SPBM VID MHF and:
 - 33 1) There is no PBB-TE MIP TLV or
 - 34 2) There is a PBB-TE MIP TLV but the address carried in the MIP MAC address field does not
 - 35 match the MAC address of the receiving MIP,
 - 36 ProcessLBM() forwards the LBM to the Passive MHF Multiplexer and performs no further
 - 37 processing.
- 38 g) Otherwise, ProcessLBM() calls xmitLBR() to generate and transmit an LBR.

1 20.28.2 xmitLBR()

2 Called by ProcessLBM() to transmit an LBR. xmitLBR() constructs an LBR from the LBM contained in
3 LBMPDU, and transmits it to the Active SAP using an M_UNITDATA.request as follows:

- 4 a) Sets the destination_address parameter to the source_address of the received LBM if the MP is not
5 associated with a PBB-TE MA; otherwise to:
 - 6 1) The value of the ESP-DA of the MA's ESP that has the replying MEP's MAC address [item i)
7 in 12.14.7.1.3] in its ESP-SA field, if the MEP is a PBB-TE MEP; or otherwise to
 - 8 2) The value carried in the Reverse MAC field contained in the PBB-TE MIP TLV of the received
9 LBM, if this is a group MAC address; otherwise to
 - 10 3) The source_address of the received LBM.
- 11 b) Sets the source_address parameter to the MAC address of the replying MP if the MP is not a
12 PBB-TE MHF; or otherwise to:
 - 13 1) The destination_address of the received LBM if this is an individual MAC address; or
14 otherwise to
 - 15 2) The value carried in the Reverse MAC field contained in the PBB-TE MIP TLV of the received
16 LBM.
- 17 c) If the LBM was received at a PBB-TE MHF and contains a PBB-TE MIP TLV, then the
18 vlan_identifier in the LBR is set to the value in the Reverse VID field; otherwise, the vlan_identifier
19 in the LBR is set to the value of the Primary VID associated with the replying MP [item d) in
20 12.14.7.1.3, item a) in 12.14.3.2.2, item c) in 12.14.5.3.2].
- 21 d) In the case of ECMP with flow filtering, the flow_hash parameter is set to zero and the time_to_live
22 parameter is set to 63.
- 23 e) The priority parameter and the drop-eligible parameter are copied from the LBM.
- 24 f) Changes the OpCode field (21.4.3) from LBM to LBR.
- 25 g) Copies the remainder of the LBM's mac_service_data_unit verbatim to the LBR.
- 26 h) If the replying MP is a MEP, increments the LBR transmission counter by 1 [item ad) in
27 12.14.7.1.3].

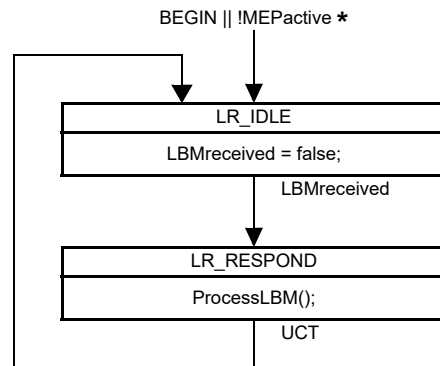
28 20.29 MP Loopback Responder state machine

29 The MP Loopback Responder state machine implements the functions specified by the state diagram in
30 Figure 20-10, using the variables in 20.27 and the procedures in 20.28. Although the definition of the MP
31 Loopback Responders in 19.2.10 and 19.3.8 requires the instantiation of an MP Loopback Responder state
32 machine in every MP, in practice, the number of MP Loopback Responder state machines implemented in a
33 system cannot be determined from external observation of the system.

34 20.30 MEP Loopback Initiator variables

35 The following variables are local to the MEP Loopback Initiator transmit state machine and the MEP
36 Loopback Initiator receive state machine:

- 37 a) LBMsToSend (20.30.1)
- 38 b) nextLBMtransID (20.30.2)
- 39 c) expectedLBRtransID (20.30.3)
- 40 d) LBIactive (20.30.4)
- 41 e) xmitReady (20.30.5)
- 42 f) LBRreceived (20.30.6)
- 43 g) LBRPDU (20.30.7)



* MEPactive in a MEP; not present (always true) in an MHF

Figure 20-10—MP Loopback Responder state machine

1 20.30.1 LBMstoSend

2 The integer number of LBMs that the MEP Loopback Initiator transmit state machine is to transmit. Set by a
3 management operation [item c) in 12.14.7.3.2]. Setting this variable to a nonzero value starts transmission of
4 LBMs. LBMsToSend is decremented by 1 by the MEP Loopback Initiator transmit state machine with each
5 transmission.

6 20.30.2 nextLBMtransID

7 The value to place in the Loopback Transaction Identifier field of the next LBM transmitted by xmitLBM().
8 nextLBMtransID is incremented by 1 by the MEP Loopback Initiator transmit state machine with each
9 transmission. This variable is available as a managed object [item x) in 12.14.7.1.3].

10 20.30.3 expectedLBRtransID

11 The value expected to be found in the Loopback Transaction Identifier field of the next LBR received by
12 ProcessLBR(). Altered by ProcessLBR() (see 20.33.1).

13 20.30.4 LBIactive

14 A Boolean flag indicating whether the MEP Loopback Initiator transmit state machine is (true) or is not
15 (false) actively engaged in a requested operation. Set to true by the MEP Loopback Initiator transmit state
16 machine after LBMsToSend is set to a nonzero value by a management operation. Reset to false by the MEP
17 Loopback Initiator transmit state machine 5 s after the last LBM is transmitted or the last LBR is received,
18 whichever comes later.

19 20.30.5 xmitReady

20 A Boolean flag set to true by the Bridge Port to indicate that another LBM can be transmitted. Reset to false
21 by the MEP Loopback Initiator transmit state machine.

22 20.30.6 LBRreceived

23 Boolean flag. Set by the MEP Equal OpCode Demultiplexer (19.2.7) when an LBR at the MEP's MD Level
24 is received. Cleared by the MEP Loopback Initiator receive state machine.

1 20.30.7 LBRPDU

2 Structure. Loaded with the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the LBR
3 PDU received by the MEP Equal OpCode Demultiplexer (19.2.7) when an LBR at the MEP's MD Level is
4 received.

5 20.31 MEP Loopback Initiator transmit procedures

6 The following procedure is local to the MEP Loopback Initiator transmit state machine:

- 7 a) xmitLBM() (20.31.1)

8 20.31.1 xmitLBM()

9 xmitLBM() is called by the MEP Loopback Initiator transmit state machine. It constructs and transmits an
10 LBM on the Active SAP using an M_UNITDATA.request as follows:

- 11 a) Sets the destination_address parameter from the appropriate managed object [item b) in 12.14.7.3.2]
12 or in the case of an SPBM VID MEP [item g) in 12.14.7.3.2]. If the MEP is configured on a PBB-TE
13 MA, the destination_address parameter is set to the MAC address indicated by the value of the
14 ESP-DA field of the MA's ESP that has the MEP's MAC address indicated in its ESP-SA field.
- 15 b) Sets the source_address parameter to the MAC address of the MEP [item i) in 12.14.7.1.3].
- 16 c) Sets the priority and drop_eligible parameters from the appropriate managed object [item e) in
17 12.14.7.3.2].
- 18 d) In the case of ECMP with flow filtering, the flow_hash parameter is set from the appropriate
19 managed object [item h) in 12.14.7.3.2] and the time_to_live parameter is set to 63.
- 20 e) Copies nextLBMtransID (20.30.2) to the Loopback Transaction Identifier field (21.7.3) of the LBM.
- 21 f) If the MEP is configured on a PBB-TE MA, constructs a PBB-TE MIP TLV using in the MIP MAC
22 address field the target address parameter from the appropriate managed object [item b) in
23 12.14.7.3.2] and in the Reverse VID field the parameter [item f) in 12.14.7.3.2]; if the MEP is a root
24 in a point-to-multipoint TESI, the Reverse MAC field is also included in the PBB-TE MIP TLV
25 carrying the ESP-SA value of any of the MA's point-to-point ESPs; if the MEP is a leaf in a
26 point-to-multipoint TESI, the Reverse MAC field carries the ESP-DA of the point-to-multipoint
27 ESP; the PBB-TE MIP TLV is not used if the LBM target address in 12.14.7.3.2 is associated with
28 any of the values in the ESP-DA field of the monitored MA's ESPs.
- 29 g) If the MEP is configured on an SPBM VID MA, constructs a PBB-TE MIP TLV using in the MIP
30 MAC address field the target address parameter from the appropriate managed object [item b) in
31 12.14.7.3.2].
- 32 h) Constructs a Data TLV from the appropriate managed object [item d) in 12.14.7.3.2] if and only if
33 that managed object has a nonzero length.
- 34 i) As controlled by the managed objects item e) in 12.14.3.1.3, item d) in 12.14.5.1.3, and item d) in
35 12.14.6.1.3, places a Sender ID TLV (21.5.3), identifying the transmitting system, in the LBM.
- 36 j) Increments nextLBMtransID (20.30.2) by 1, wrapping around from $2^{32} - 1$ to 0.

37 20.32 MEP Loopback Initiator transmit state machine

38 A MEP creates a single instance of the MEP Loopback Initiator transmit state machine. The MEP Loopback
39 Initiator transmit state machine implements the function specified by the state diagram in Figure 20-11 and
40 the procedures in 20.31. It shares with the MEP Loopback Initiator receive state machine the variable
41 declarations in 20.30.

20.33 MEP Loopback Initiator receive procedures

The following procedure is local to the MEP Loopback Initiator receive state machine:

a) ProcessLBR() (20.33.1)

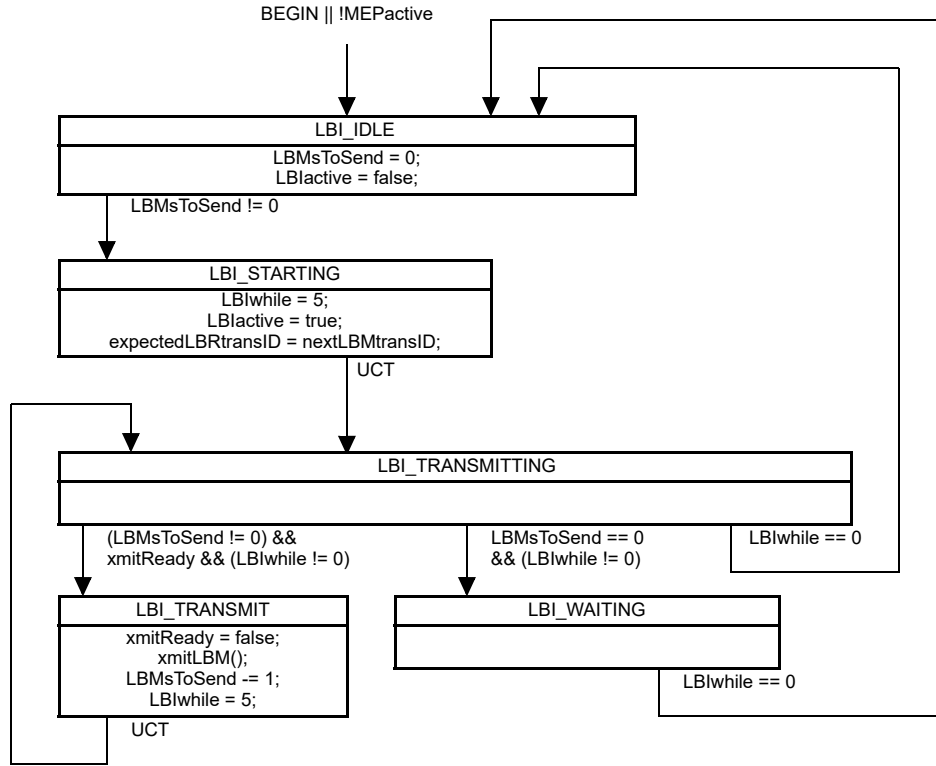


Figure 20-11—MEP Loopback Initiator transmit state machine

20.33.1 ProcessLBR()

Called by the MEP Loopback Initiator receive state machine whenever an LBR is received. ProcessLBR() processes the LBR in LBRPDU as follows:

- a) If the I/G bit of the source_address indicates a Group address, or if the destination_address does not match the MAC address of the receiving MP, ProcessLBR() discards the received LBR.
- b) ProcessLBR() shall process the LBM according to 20.51.4.2, and may validate the received LBR according to 20.51.4.3, and discard it if invalid.
- c) If the LBR is not discarded, and if and only if LBIactive is true, the Loopback Transaction Identifier field of the LBR is compared to expectedLBRtransID:
 - 1) If the two values are equal, then the number of correct LBRs received [item y) in 12.14.7.1.3] is incremented by 1.
 - 2) If the two values are unequal, then the value from the received Loopback Transaction Identifier field is copied into expectedLBRtransID, and the number of incorrect LBRs received [item z) in 12.14.7.1.3] is incremented by 1.
 - 3) Regardless of whether the two values were equal, expectedLBRtransID is then incremented by 1.

- 1 d) ProcessLBR() may perform a bit-by-bit comparison of the received LBR against the LBM with the
- 2 matching Loopback Transaction Identifier, except for the OpCode field, and increment a managed
- 3 object [item aa) in 12.14.7.1.3] if they do not match.

4 20.34 MEP Loopback Initiator receive state machine

5 A MEP creates a single instance of the MEP Loopback Initiator receive state machine. The MEP Loopback
6 Initiator receive state machine implements the function specified by the state diagram in Figure 20-12 and
7 the procedures in 20.33. It shares with the MEP Loopback Initiator transmit state machine the variable
8 declarations in 20.30.

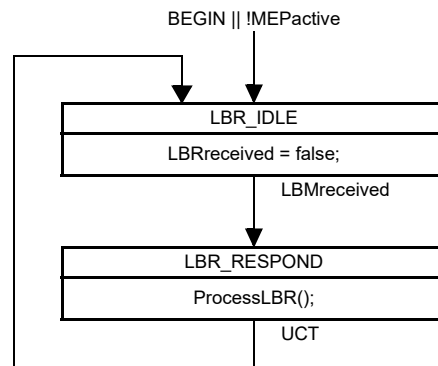


Figure 20-12—MEP Loopback Initiator receive state machine

9 20.35 MEP Fault Notification Generator variables

10 The following variables are local to the MEP Fault Notification Generator state machine:

- 11 a) fngPriority (20.35.1)
- 12 b) fngDefect (20.35.2)
- 13 c) fngAlarmTime (20.35.3)
- 14 d) fngResetTime (20.35.4)
- 15 e) someRMEPCCMdefect (20.35.5)
- 16 f) someMACstatusDefect (20.35.6)
- 17 g) someRDIdefect (20.35.7)
- 18 h) highestDefectPri (20.35.8)
- 19 i) highestDefect (20.35.9)

20 20.35.1 fngPriority

21 An integer specifying the priority of the last defect reported in a Fault Alarm. fngPriority takes the same
22 values as highestDefectPri (20.35.8).

23 20.35.2 fngDefect

24 An enumerated value specifying the last defect reported in a Fault Alarm. fngDefect takes the same values
25 as highestDefect (20.35.9).

1 **20.35.3 fngAlarmTime**

2 The time that one or more defects must be present before a Fault Alarm is issued. Default value 2.5 s. Also a
3 managed object [item l) in 12.14.7.1.3].

4 **20.35.4 fngResetTime**

5 The time, after a Fault Alarm, that no defects must be present before another Fault Alarm is enabled. Default
6 value 10 s. Also a managed object [item m) in 12.14.7.1.3].

7 **20.35.5 someRMEPCCMdefect**

8 A Boolean indicating the aggregate state of the Remote MEP state machines. True indicates that at least one
9 of the Remote MEP state machines is not receiving valid CCMs from its remote MEP, and false that all
10 Remote MEP state machines are receiving valid CCMs. someRMEPCCMdefect is the logical OR of all of
11 the rMEPCCMdefect variables for all of the Remote MEP state machines on this MEP. This variable is
12 readable as a managed object [item q) in 12.14.7.1.3].

13 **20.35.6 someMACstatusDefect**

14 A Boolean indicating that one or more of the remote MEPs is reporting a failure in its Port Status TLV
15 (21.5.4) or Interface Status TLV (21.5.5). It is true either if some remote MEP is reporting that its interface is
16 not isUp (i.e., at least one remote MEP's interface is unavailable), or if all remote MEPs are reporting a Port
17 Status TLV that contains some value other than psUp (i.e., all remote MEPs' Bridge Ports are not forwarding
18 data). It is thus the logical OR of the following two terms:

- 19 a) The logical AND, across all remote MEPs, of the rMEPportStatusDefect variable OR
- 20 b) The logical OR, across all remote MEPs, of the rMEPinterfaceStatusDefect variable.

21 This variable is readable as a managed object [item p) in 12.14.7.1.3].

22 **20.35.7 someRDId defect**

23 A Boolean indicating the aggregate health of the remote MEPs. True indicates that at least one of the
24 Remote MEP state machines is receiving valid CCMs from its remote MEP that has the RDI bit set, and
25 false that no Remote MEP state machines are receiving valid CCMs with the RDI bit set. someRDId defect is
26 the logical OR of all of the rMEPlastRDI and rMEPlastRDI[i] variables for all of the Remote MEP state
27 machines on this MEP. This variable is readable as a managed object [item o) in 12.14.7.1.3].

28 **20.35.8 highestDefectPri**

29 An integer value indicating the priority of the defect named in the variable highestDefect. See also Table
30 20-1.

31 **20.35.9 highestDefect**

32 An enumerated value indicating the highest priority defect among the variables xconCCMdefect (20.23.3),
33 errorCCMdefect (20.21.3), someRMEPCCMdefect (20.35.5), someMACstatusDefect (20.35.6), and
34 someRDId defect (20.35.7), as limited by lowestAlarmPri (20.9.5). This variable is readable as a managed
35 object [item c) in 12.14.7.7.2]. The variables, their priorities, and the enumerated values of highestDefect are
36 shown in Table 20-1.

20.36 MEP Fault Notification Generator procedures

The following procedure is local to the MEP Fault Notification Generator state machine:

a) `xmitFaultAlarm()` (20.36.1).

20.36.1 `xmitFaultAlarm()`

Transmits a Fault Alarm (12.14.7.7). The identity of the MEP and the variable `fngDefect` (20.35.2), specifying the cause of the Fault Alarm, are transmitted in the Fault Alarm PDU. The format and method of transmission of the Fault Alarm is not specified in this standard.

20.37 MEP Fault Notification Generator state machine

A MEP creates a single instance of the MEP Fault Notification Generator state machine. The MEP Fault Notification Generator state machine implements the function specified by the state diagram in Figure 20-13, the variables in 20.35, and the procedure in 20.36. The current state of the MEP Fault Notification Generator state machine is available in a managed object [item f) in 12.14.7.1.3].

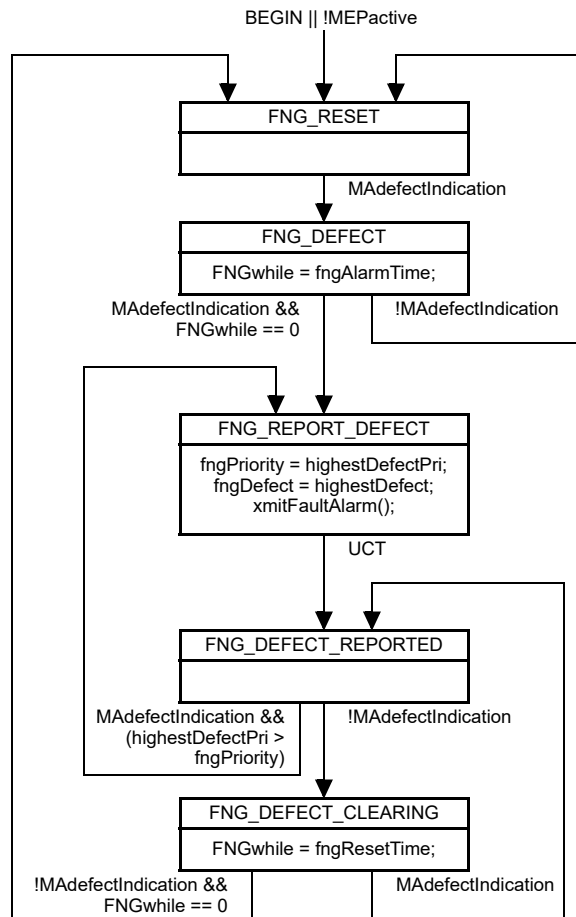


Figure 20-13—MEP Fault Notification Generator state machine

1 **20.38 MEP Mismatch Fault Notification Generator variables**

2 The following variables are local to the MEP Mismatch Fault Notification Generator state machine for a
3 PBB-TE MEP or Infrastructure Segment MEP implementing the Traffic field (21.6.1.4):

- 4 a) mfngAllowed (20.38.1)
- 5 b) mmdefectIndication (20.38.2)
- 6 c) mfngAlarmTime (20.38.3)
- 7 d) mfngResetTime (20.38.4)

8 **20.38.1 mfngAllowed**

9 A Boolean indicating the mismatch defect is allowed to generate a Fault Alarm. Its value is configurable as
10 a managed object [item ag] in 12.14.7.1.3].

11 **20.38.2 mmdefectIndication**

12 A Boolean indicating the presence of the mismatch defect that is allowed to generate a Fault Alarm. It
13 corresponds to the logical AND of the variables mfngAllowed (20.38.1), mmCCMdefect (20.25.2), and
14 mmLocdefect (20.25.5).

15 **20.38.3 mfngAlarmTime**

16 The time that one or more defects must be present before a mismatch Fault Alarm is issued. Default value
17 2.5 s. Also a managed object [item l] in 12.14.7.1.3].

18 **20.38.4 mfngResetTime**

19 The time, after a mismatch Fault Alarm, that no defects must be present before another mismatch Fault
20 Alarm is enabled. Default value 10 s. Also a managed object [item m] in 12.14.7.1.3].

21 **20.39 MEP Mismatch Fault Notification Generator procedures**

22 The following procedure is local to the MEP Mismatch Fault Notification Generator state machine for a
23 PBB-TE MEP or Infrastructure Segment MEP implementing the Traffic field (21.6.1.4):

- 24 a) xmitFaultAlarm() (20.36.1)

25 **20.39.1 xmitFaultAlarm()**

26 Transmits a Fault Alarm (12.14.7.7). The identity of the MEP and a value indicating that the cause of the
27 Fault Alarm is a mismatch defect, are transmitted in the Fault Alarm PDU. The format and method of
28 transmission of the Fault Alarm is not specified in this standard.

29 **20.40 MEP Mismatch Fault Notification Generator state machine**

30 A PBB-TE MEP or Infrastructure Segment MEP implementing the Traffic field (21.6.1.4) creates a single
31 instance of the MEP Mismatch Fault Notification Generator state machine. The MEP Mismatch Fault
32 Notification Generator state machine implements the function specified by the state diagram in
33 Figure 20-14, the variables in 20.38, and the procedure in 20.39. The current state of the MEP Mismatch
34 Fault Notification Generator state machine is available in a managed object [item ak] in 12.14.7.1.3].

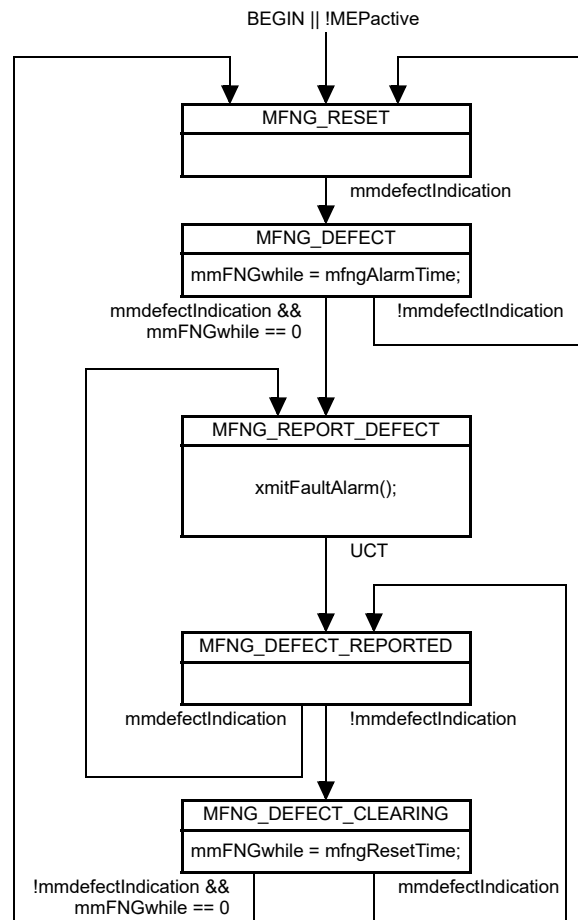


Figure 20-14—MEP Mismatch Fault Notification Generator state machine

20.41 MEP Linktrace Initiator variables

The following variables are local to the MEP Linktrace Initiator variables:

- a) nextLTMtransID (20.41.1)
- b) ltmReplyList (20.41.2)

20.41.1 nextLTMtransID

The value to place in the LTM Transaction Identifier of the next LTM transmitted by xmitLTM(). nextLTMtransID is incremented by 1 by the MEP Linktrace Initiator variables with each transmission. This variable is also a managed object [item ab) in 12.14.7.1.3].

1 **20.41.2 ltmReplyList**

2 The list of recently-issued LTM entries and their corresponding LTRs. An LTM entry, with no attached LTR
3 entries, is added to this list by xmitLTM() each time an LTM is transmitted, and an LTR entry is attached to
4 an LTM entry in this list, by ProcessLTR(), each time an LTR corresponding to an LTM in the list is
5 received. The MEP does not remove any entry from this list before 5 s have elapsed since the transmission
6 of the corresponding LTM, unless the addition of another LTM or LTR entry would exceed the maximum
7 resources allocated for this list. This variable constitutes the Linktrace database for a given MEP and is also
8 a managed object (12.14.7.5.3).

9 Each LTR entry contains the following variables:

- 10 a) ltrFlags (20.41.2.1)
- 11 b) ltrReplyTTL (20.41.2.2)
- 12 c) ltrLastEgressId (20.41.2.3)
- 13 d) ltrNextEgressId (20.41.2.4)
- 14 e) ltrRelayAction (20.41.2.5)
- 15 f) ltrIngressAction (20.41.2.6)
- 16 g) ltrIngressAddress (20.41.2.7)
- 17 h) ltrIngressPortIdSubtype (20.41.2.8)
- 18 i) ltrIngressPortId (20.41.2.9)
- 19 j) ltrEgressAction (20.41.2.10)
- 20 k) ltrEgressAddress (20.41.2.11)
- 21 l) ltrEgressPortIdSubtype (20.41.2.12)
- 22 m) ltrEgressPortId (20.41.2.13)
- 23 n) ltrSenderIdTlv (20.41.2.14)
- 24 o) ltrOrgSpecTlv (20.41.2.15)

25 **20.41.2.1 ltrFlags**

26 The bit string returned in the Flags field (21.9.1) of the LTR, including the FwdYes and TerminalMEP bits.

27 **20.41.2.2 ltrReplyTTL**

28 The integer value returned in the Reply TTL field (21.9.4) of the LTR.

29 **20.41.2.3 ltrLastEgressId**

30 The octet string returned in the Last Egress Identifier field (21.9.7.1) of the LTR Egress Identifier TLV.

31 **20.41.2.4 ltrNextEgressId**

32 The integer value returned in the Next Egress Identifier field (21.9.7.2) of the LTR Egress Identifier TLV.

33 **20.41.2.5 ltrRelayAction**

34 The enumerated value returned in the Relay Action field (21.9.5) of the LTR. The enumerated values are
35 listed in Table 21-27.

36 **20.41.2.6 ltrIngressAction**

37 The enumerated value returned in the Ingress Action field (21.9.8.1) of the Reply Ingress TLV (21.9.8) of
38 the LTR. The enumerated values are listed in Table 21-30. An enumerated value of 0 indicates that no Reply
39 Ingress TLV was returned in the LTR.

1 **20.41.2.7 ltrIngressAddress**

2 The MAC address of the MP on the Ingress Port. Contents are undefined if no Reply Ingress TLV was
3 returned in the LTR.

4 **20.41.2.8 ltrIngressPortIdSubtype**

5 An enumerated value as specified by IEEE Std 802.1AB, indicating the format of ltrIngressPortId. Contents
6 are undefined if no Reply Ingress TLV was returned in the LTR.

7 **20.41.2.9 ltrIngressPortId**

8 An octet string as specified by IEEE Std 802.1AB, identifying the Ingress Port, in the format identified in
9 ltrIngressPortIdSubtype.

10 **20.41.2.10 ltrEgressAction**

11 The enumerated value returned in the Egress Action field (21.9.9.1) of the Reply Egress TLV (21.9.9) of the
12 LTR. The enumerated values are listed in Table 21-32. A value of 0 indicates that no Reply Egress TLV was
13 returned in the LTR.

14 **20.41.2.11 ltrEgressAddress**

15 The MAC address of the MP on the Egress Port. Contents are undefined if no Reply Egress TLV was
16 returned in the LTR.

17 **20.41.2.12 ltrEgressPortIdSubtype**

18 An enumerated value as specified by IEEE Std 802.1AB, indicating the format of ltrEgressPortId. Contents
19 are undefined if no Reply Ingress TLV was returned in the LTR.

20 **20.41.2.13 ltrEgressPortId**

21 An octet string as specified by IEEE Std 802.1AB, identifying the Egress Port, in the format identified in
22 ltrEgressPortIdSubtype.

23 **20.41.2.14 ltrSenderIdTlv**

24 An octet string identifying the transmitting system, as received in the Sender ID TLV (21.5.3), if one was
25 present in the LTR.

26 **20.41.2.15 ltrOrgSpecTlv**

27 An octet string containing the Organization-Specific TLVs (21.5.2), if any were present in the LTR.

28 **20.42 MEP Linktrace Initiator procedures**

29 The following procedure is local to the MEP Linktrace Initiator:

30 a) xmitLTM() (20.42.1)

31 **20.42.1 xmitLTM()**

32 xmitLTM() is called whenever the Transmit Linktrace Message management operation (12.14.7.4) is
33 invoked. It constructs and transmits an LTM on the Active SAP, or on the MEP LTI SAP if the operation is
34 invoked on an Up MEP, using an M_UNITDATA.request as follows:

35 a) Sets the destination_address parameter to the value from Table 8-19 corresponding to the MEP's
36 MD Level. If the MEP is configured on a PBB-TE MA, the destination_address parameter is the

- 1 MAC address indicated by the value of the ESP-DA field of the MA's ESP that has the MEP's MAC
2 address indicated in its ESP-SA field. If the MEP is configured on an SPBM VID MA, the
3 destination_address parameter is set to the value configured in the LTM management request [item
4 f) in 12.14.7.4.2].
- 5 b) Sets the source_address parameter and Original MAC Address field (21.8.5) to the MAC address of
6 the MEP [item i) in 12.14.7.1.3].
- 7 c) If the MEP is configured on a PBB-TE MA, constructs a PBB-TE MIP TLV using in the MIP MAC
8 address field a null field and in the Reverse VID field the parameter [item e) in 12.14.7.4.2]; if the
9 MEP is a root in a point-to-multipoint TESI, the Reverse MAC field is also included in the PBB-TE
10 MIP TLV carrying the ESP-SA value of any of the MA's point-to-point ESPs; if the MEP is a leaf in
11 a point-to-multipoint TESI, the Reverse MAC field carries the ESP-DA of the point-to-multipoint
12 ESP.
- 13 d) In the case of ECMP with flow filtering, the flow_hash parameter is set to the value specified in the
14 appropriate managed object [item g) in 12.14.7.4.2] and the time_to_live parameter is set to 63.
- 15 e) Sets the priority parameter to the same value as for CCMs [item h) in 12.14.7.1.3].
- 16 f) Copies nextLTMtransID [item ab) in 12.14.7.1.3] to the LTM Transaction Identifier field (21.8.3) of
17 the LTM.
- 18 g) Sets the LTM Egress Identifier TLV (21.8.8) to a value that is unique among all Bridges in the
19 Management Domain.
- 20 h) Sets the Target MAC Address field (21.8.6) from the appropriate managed object [item c) in
21 12.14.7.4.2].
- 22 i) Sets the LTM TTL field (21.8.4) from the appropriate managed object [item d) in 12.14.7.4.2].
- 23 j) Sets the UseFDBOnly bit of the Flags field (21.8.1) from the appropriate managed object [item b) in
24 12.14.7.4.2], and sets all other bits of the Flags field to 0.
- 25 k) As controlled by the managed objects item e) in 12.14.3.1.3, item d) in 12.14.5.1.3, and item d) in
26 12.14.6.1.3, places a Sender ID TLV (21.5.3), identifying the transmitting system, in the LTM.
- 27 l) Creates a new entry in the ltmReplyList variable (20.41.2) for this LTM, identified by the LTM
28 Transaction Identifier in nextLTMtransID (20.41.1).
- 29 m) Increments nextLTMtransID (20.41.1) by 1, wrapping around from $2^{32} - 1$ to 0.

30 If the addition of this LTM entry would exceed the resources allocated to ltmReplyList, then the oldest LTM
31 entries in ltmReplyList are deleted until sufficient resources are available to hold the new LTM entry.

32 **20.43 MEP Linktrace Initiator receive variables**

33 The following variables are local to the MEP Linktrace Initiator receive state machine:

- 34 a) LTRreceived (20.43.1)
35 b) LTRPDU (20.43.2)

36 **20.43.1 LTRreceived**

37 Boolean flag. Set by the MEP Equal OpCode Demultiplexer (19.2.7) when an LTR at the MEP's MD Level
38 is received. Cleared by the MEP Linktrace Initiator receive state machine.

39 **20.43.2 LTRPDU**

40 Structure. Loaded with the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the LTR
41 PDU received by the MEP Equal OpCode Demultiplexer (19.2.7) when an LTR at the MEP's MD Level is
42 received.

20.44 MEP Linktrace Initiator receive procedures

The following procedure is local to the MEP Linktrace Initiator receive state machine:

- a) ProcessLTR() (20.44.1)

20.44.1 ProcessLTR()

The ProcessLTR() procedure is called by the MEP Linktrace Initiator receive state machine whenever an LTR is received, and processes the LTR contained in the LTRPDU variable as follows:

- a) If the destination_address of the LTR does not match the MAC address of the MEP, or if the LTR fails the validation criteria of 20.51.4, ProcessLTR() shall discard the LTR without counting it, and no further processing takes place.
- b) Otherwise, if the LTR Transaction Identifier field matches an LTM entry in the ltmReplyList variable, then a new LTR entry is attached to that LTM entry, containing the information returned in the LTR.
- c) Otherwise, the number of unexpected LTRs received [item ac) in 12.14.7.1.3] is incremented by 1.

If the addition of this LTR entry would exceed the resources allocated to ltmReplyList, then the oldest LTM entries in ltmReplyList (the oldest LTR entries, if only one LTM entry is present) are deleted until sufficient resources are available to hold the new LTR entry.

20.45 MEP Linktrace Initiator receive state machine

One instance of the MEP Linktrace Initiator receive state machine is instantiated by each MEP Linktrace Initiator. The MEP Linktrace Initiator receive state machine implements the function specified by the state diagram in Figure 20-15, the variables in 20.43, and the procedures in 20.44.

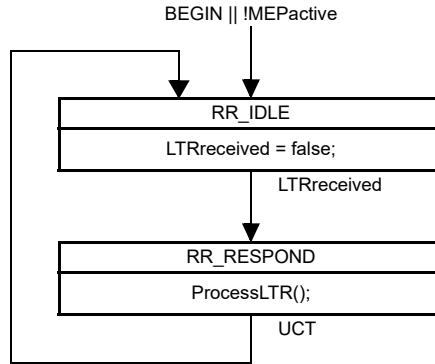


Figure 20-15—MEP Linktrace Initiator receive state machine

20.46 Linktrace Responder variables

The following variables are local to the Linktrace Responder:

- a) nPendingLTRs (20.46.1)
- b) LTMreceived (20.46.2)
- c) LTMPDU (20.46.3)

20.46.1 nPendingLTRs

An integer value used to track the number of LTRs that have been enqueued for transmission by enqueueLTR() and not yet transmitted by xmitOldestLTR(). Can be reset to 0 by clearPendingLTRs().

1 **20.46.2 LTMreceived**

2 A Boolean value set when a valid LTM is received and cleared by the LTM Receiver state machine.

3 **20.46.3 LTMPDU**

4 Structure. Loaded with the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the
5 LTM PDU received by the MEP Equal OpCode Demultiplexer (19.2.7) when an LTM at the MEP's MD
6 Level is received.

7 **20.47 LTM Receiver procedures**

8 The following procedures are local to the LTM Receiver state machine:

- 9 a) ProcessLTM() (20.47.1)
- 10 b) clearPendingLTRs() (20.47.2)
- 11 c) ForwardLTM() (20.47.3)
- 12 d) enqueueLTR() (20.47.4)

13 **20.47.1 ProcessLTM()**

14 Called by the LTM Receiver state machine when an LTM is received and processes the LTM contained in
15 the LTMPDU, making the decision whether to call ForwardLTM() to forward the LTM, and whether to call
16 enqueueLTR() to enqueue an LTR for transmission, according to 20.3.2. The received LTM is first validated.

- 17 a) ProcessLTM() validates the received LTM according to 20.51.3. If the LTM is invalid, no further
18 validation steps are performed.
- 19 b) Otherwise, the LTM TTL field (21.8.4) of the received LTM is examined. If its value is 0, the LTM
20 is invalid, and no further validation steps are performed.
- 21 c) Otherwise, if the vlan_identifier identifies an ESP-VID or SPBM VID the LTM is valid, and no
22 further validation steps are performed.
- 23 d) Otherwise, if the destination_address parameter of the LTM is a Group address, the LTM is valid,
24 and no further validation steps are performed.
- 25 e) Otherwise, if both:
 - 26 1) The LTM was received from a Linktrace SAP, and not from the originating Up MEP through its
27 MEP LTI SAP and
 - 28 2) The destination_address parameter of the LTM is the individual MAC address of the receiving
29 MP [item i) in 12.14.7.1.3 for a MEP, the MAC address of the Bridge Port for an MHF],
30 then the LTM is valid.
- 31 f) Otherwise, the LTM is invalid.

32 **20.47.1.1 LTM paths through a Bridge**

33 If the LTM is valid, processing proceeds according to the following subclauses. Otherwise, ProcessLTM()
34 discards the LTM, and no further processing takes place.

35 Figure 20-16 illustrates a number of (but not all) possible paths taken by an LTM through a Bridge. The gray
36 circles indicate the processing of an LTM and the possible forwarding of a modified copy of the LTM. The
37 LTR path is not shown.

1

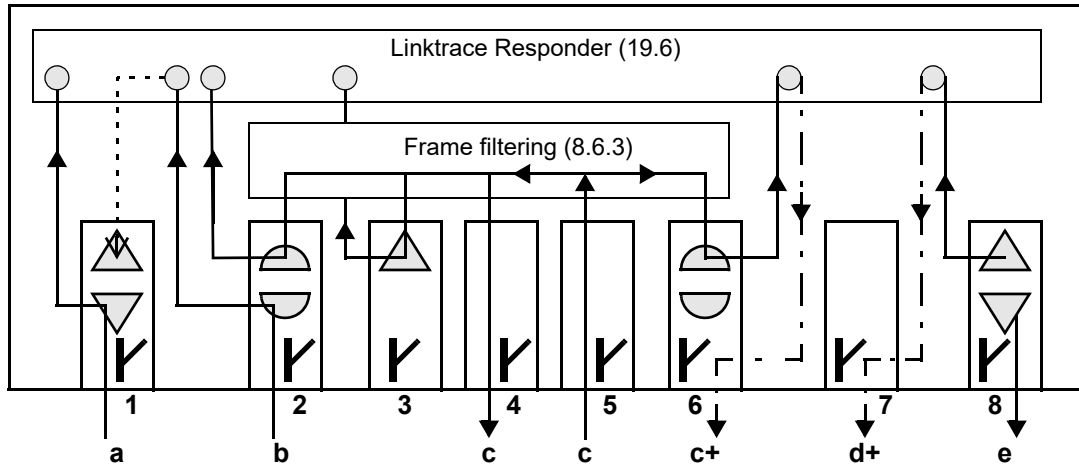


Figure 20-16—Linktrace Responder, MEPs, MHFs, and LOMs

2 NOTE—All of the MPs shown in Figure 20-16 are at the same MD Level.

- 3 a) An LTM is detected by a Down MEP as it enters Port 1. It is deflected to the Linktrace Responder through the MEP Linktrace SAP.
- 4
- 5 b) An LTM is detected by an MHF as it enters Port 2. It is deflected to the Linktrace Responder through the MHF Linktrace SAP, which determines that the Target MAC Address would be forwarded out Port 1, on which there is a MEP. The LTM is not actually forwarded to the MEP.
- 6
- 7
- 8 c) An LTM enters on Port 5, which has no MHF, and if it is a multicast, the LTM is distributed to Ports 2, 3, 4, and 6 by the Frame filtering function (8.6.3). The original, unaltered LTM passes out Port 4. The Up MEP in Port 3 and the MHFs in Ports 2 and 6 deliver the LTM to the Linktrace Responder. The Linktrace Responder discards the LTMs received on Port 2 and 3, but forwards an updated version of the LTM through the LOM on Port 6.
- 9
- 10
- 11
- 12
- 13 d) The Up MEP on Port 8 generates an LTM, which it forwards to the Linktrace Responder. The Linktrace Responder decides to forward an updated version of the LTM through the LOM on Port 7. (The Linktrace Responder might equally well have forwarded it through some other Port's MHF.)
- 14
- 15
- 16 e) The Down MEP on Port 8 generates an LTM and transmits it through its Active SAP to the LAN attached to Port 8.
- 17

18 If an LTM is forwarded, the forwarding takes place immediately. If an LTR is generated in response to the
19 LTM, it is enqueued for later transmission by the LTR Transmitter state machine (20.50).

20 20.47.1.2 Ingress Port, vlan_identifier, and Egress Port determination

21 In 20.3.2 and Figure 20-16, item a) through item e) all require ProcessLTM() to determine the Ingress and
22 Egress Ports for this received LTM. The Ingress Port is the Bridge Port on which the LTM entered the
23 Bridge, whether through a MEP, an MHF, or an LOM. In item d)), where an LTM is generated by an Up
24 MEP and passed through a MEP LTI SAP to the Linktrace Responder, the Ingress Port is the Bridge Port of
25 the originating MEP.

26 If received from an EISS SAP, the vlan_identifier of the received LTM is included in the
27 EM_UNITDATA.indication. If received from an ISS SAP, the vlan_identifier of the received LTM is that
28 configured in the MEP, MHF, or LOM that received the LTM, or if none, is the PVID (IEEE Std 802.1AC)
29 of the Ingress Port.

1 The Egress Port is the Bridge Port on which a data frame whose destination_address is equal to the Target
2 MAC Address carried in the LTM, or in the case of an MP associated with a PBB-TE MA, the
3 destination_address of the LTM, and whose vlan_identifier matched that of the LTM, would be forwarded.
4 This determination is made in two steps:

5 a) ProcessLTM() first queries the FDB (8.8). The set of potential transmission ports, normally created
6 by Active topology enforcement (8.6.1), is the set of all Bridge Ports that are both in the active set of
7 the vlan_identifier of the LTM and that are in the Forwarding state for that vlan_identifier, except
8 that the Ingress Port is excluded from the set. The query uses the Target MAC Address field of the
9 LTM as the destination_address of the lookup, the Original MAC Address field of the LTM as the
10 source_address, and the vlan_identifier of the LTM. In the case of an MP associated with a PBB-TE
11 or SPBM VID MA, the query uses the destination_address and the vlan_identifier of the LTM as the
12 corresponding parameters for the lookup. If the MP is associated with ECMP with flow filtering, the
13 query uses the flow_hash parameter in the lookup process, when required. The output from this
14 query is a (perhaps reduced) set of potential transmission ports. If the resultant set contains one and
15 only one Bridge Port, that Bridge Port is the Egress Port, and item b)) is not performed. If the
16 resultant set contains more than one Bridge Port, and the MP is associated with a PBB-TE MA,
17 those Bridge Ports are all Egress Ports, and step b is not performed.

18 NOTE 1—If there are only two Bridge Ports that are members of the VID's member set, the Ingress Port was
19 one of those two Bridge Ports, and the Ingress Port is not connected to a shared medium, then this step can
20 identify the Egress Port even if the FDB is not used for this vlan_identifier.

21 NOTE 2—This query cannot produce an Egress Port that is the same as the Ingress Port, since the Ingress Port
22 is not included in the set of potential transmission ports.

23 b) If the FDB could not produce a unique Egress Port, and the MPs serving the vlan_identifier of the
24 LTM are maintaining a MIP CCM Database, and the UseFDBonly bit of the Flags field of the LTM
25 is 0, then ProcessLTM() queries the MIP CCM Database to see whether the target MAC address and
26 vlan_identifier have been retained in that database. If so, and if the Port number in the MIP CCM
27 Database is not the same as the Ingress Port, that Port number identifies the Egress Port.

28 It is possible that neither of these two steps are able to determine the Egress Port; when a unique Egress Port
29 cannot be determined, an LTM is never forwarded by MHFs other than PBB-TE MHFs or SPBM VID
30 MHFs. PBB-TE MHFs associated with point-to-multipoint TESIs forward LTMs even if multiple Egress
31 Ports are identified. MHFs associated with SPBM VID MAs forward group addressed LTMs even if
32 multiple Egress Ports are identified.

33 20.47.1.3 LTM is received by a Down MEP

34 In case a) in 20.47.1.1, illustrated in Figure 20-16, the LTM is received by a Down MEP, and delivered to the
35 Linktrace Responder through its MEP Linktrace SAP (19.2.14). In this case, ProcessLTM() performs the
36 following steps:

- 37 a) If the Target MAC Address carried in the LTM is the MAC address of the receiving MEP, then the
38 LTM has reached its target. ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the
39 MEP Linktrace SAP of the receiving Down MEP.
- 40 b) Otherwise, if the spanning tree state of the Bridge Port and vlan_identifier of the LTM is not
41 Forwarding, the LTM is discarded and no further processing takes place.
- 42 c) Otherwise, the Ingress and Egress Ports are determined according to 20.47.1.2. If a unique Egress
43 Port cannot be determined, then the LTM is discarded, and no further processing takes place.
- 44 d) Otherwise, i.e., a unique Egress Port was found, ProcessLTM() calls enqueueLTR() (20.47.4) to
45 enqueue an LTR for the Linktrace SAP through which the LTM was received. The LTM is
46 discarded, and no further processing takes place.

1 **20.47.1.4 LTM is received by a Down MHF or originated by an Up MEP**

2 In case b) in 20.47.1.1, illustrated in Figure 20-16, the LTM is received by a Down MHF. In case d) in
3 20.47.1.1, illustrated in Figure 20-16, the LTM is originated by an Up MEP. In either case, ProcessLTM()
4 performs the following steps:

- 5 a) If the LTM was generated by an Up MEP, item b)) and item c)) are skipped, and processing
6 continues with item d)), as follows.
- 7 b) If the Target MAC Address carried in the LTM is the MAC address of the receiving MHF, then the
8 LTM has reached its target. ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the
9 Linktrace SAP through which the LTM was received. The LTM is discarded, and no further
10 processing takes place.
- 11 c) Otherwise, if the spanning tree state of the Bridge Port and vlan_identifier of the LTM is not
12 Forwarding, the LTM is discarded and no further processing takes place.

13 NOTE—This test prevents spurious LTRs from being generated on Backup Ports on a shared medium.

- 14 d) Otherwise, the Ingress and Egress Ports are determined according to 20.47.1.2. If an Egress Port
15 cannot be determined, then the LTM is discarded, and no further processing takes place.
- 16 e) There are three cases for further processing of the LTM, depending on whether, as it passes through
17 the Egress Port, a frame with the LTM's vlan_identifier would first encounter an Up MEP, an Up
18 MHF, or neither. These described in item f)), item g)), item h)), and item i)), as follows.
- 19 f) If an Up MHF would be encountered on the Egress Port, then:
 - 20 1) ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the MHF Linktrace SAP of
21 the Egress Port's Up MHF.
 - 22 2) If the target MAC address carried in the LTM is the Egress Port Up MHF's MAC address, then
23 the LTM is discarded, and no further processing takes place.
 - 24 3) Otherwise, if the LTM TTL field equals 0 or 1, the LTM is discarded and no further processing
25 takes place.
 - 26 4) Otherwise, ProcessLTM() calls ForwardLTM() (20.47.3) to forward an altered copy of the LTM
27 through the LOM Linktrace SAP of the LOM on the identified Egress Port.
- 28 g) If an Up MEP at the MD Level of the LTM would be encountered on the Egress Port, then:
 - 29 1) ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the MHF Linktrace SAP of
30 the Ingress Port's Down MHF, or the MEP LTI SAP of the Up MEP, that delivered the LTM to
31 the Linktrace Responder.
 - 32 2) The LTM is discarded, and no further processing takes place.
- 33 h) Otherwise, if an Up MEP higher than the MD Level of the LTM would be encountered on the Egress
34 Port, then the LTM is discarded, and no further processing takes place.
- 35 i) If neither an Up MHF, nor an Up MEP at an MD Level higher than or equal to the LTM, would be
36 encountered on the Egress Port, then:
 - 37 1) ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the MHF Linktrace SAP of
38 the Ingress Port's Down MHF.
 - 39 2) If the LTM TTL field equals 0 or 1, the LTM is discarded and no further processing takes place.
 - 40 3) Otherwise, ProcessLTM() calls ForwardLTM() (20.47.3) to forward an altered copy of the LTM
41 through the LOM Linktrace SAP of the LOM that a frame with the LTM's vlan_identifier
42 would first encounter when passing out through the identified Egress Port.

1 20.47.1.5 LTM is received by an Up MEP

2 In case c) in 20.47.1.1, illustrated in Figure 20-16, an LTM enters a Bridge on a Bridge Port that has no MEP
3 or MHF and therefore only an LOM. The LTM can therefore be received by an Up MEP such as the one on
4 Port 3 of Figure 20-16. Processing of an LTM received on an Up MEP proceeds as follows:

- 5 a) If the Target MAC Address carried in the LTM is the MAC address of the receiving MEP, then the
6 LTM has reached its target. ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the
7 MEP Linktrace SAP of the receiving Up MEP. The LTM is discarded, and no further processing
8 takes place.
- 9 b) Otherwise, the Ingress and Egress Ports are determined according to 20.47.1.2. If the Egress Port
10 cannot be determined, or if the Egress Port is not the Bridge Port on which the receiving MEP is
11 configured, then the LTM is discarded and no further processing takes place.
- 12 c) Otherwise, (i.e., the Egress Port is that of the receiving Up MEP) ProcessLTM() calls enqueueLTR()
13 (20.47.4) to enqueue an LTR for the MEP Linktrace SAP of the receiving Up MEP. The LTM is then
14 discarded.

15 20.47.1.6 LTM is received by an Up MHF

16 In case c) in 20.47.1.1, illustrated in Figure 20-16, an LTM enters a Bridge on a Bridge Port that has no MEP
17 or MHF and therefore only an LOM. The LTM can therefore be received by an Up MHF such as the ones on
18 Port 2 and Port 6 of Figure 20-16. Processing of an LTM received on an Up MHF proceeds as follows:

- 19 a) If the Target MAC Address carried in the LTM is the MAC address of the receiving Up MHF (i.e.,
20 that of the Bridge Port on which that MHF resides), then the LTM has reached its target.
21 ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the MHF Linktrace SAP of the
22 receiving Up MHF. The LTM is discarded, and no further processing takes place.
- 23 b) Otherwise, the Ingress and Egress Ports are determined according to 20.47.1.2. If the Egress Port
24 cannot be determined, or if the Egress Port is not the Bridge Port on which the receiving MHF is
25 configured, then the LTM is discarded, and no further processing takes place.
- 26 c) Otherwise,
 - 27 1) ProcessLTM() calls enqueueLTR() (20.47.4) to enqueue an LTR for the MHF Linktrace SAP of
28 the receiving Up MHF.
 - 29 2) If the LTM TTL field equals 0 or 1, the LTM is discarded and no further processing takes place.
 - 30 3) Otherwise, ProcessLTM() calls ForwardLTM() (20.47.3) to forward an altered copy of the LTM
31 through the MHF Linktrace SAP of the LOM on the same Bridge Port as the Up MHF that
32 received the LTM.

33 20.47.2 clearPendingLTRs()

34 Clears the queue of pending LTRs for this MP. Resets nPendingLTRs to 0.

35 20.47.3 ForwardLTM()

36 Constructs and transmits a single LTM. Using the original input LTM and the Linktrace Responder SAP
37 through which the LTM (LTR) is to be output, ForwardLTM() performs the following steps:

- 38 a) Uses the MAC address of the LOM owning the SAP specified for output as the source_address
39 parameter of the LTM.
- 40 b) Uses the destination_address and priority parameters of the input LTM as the destination_address
41 and priority of the forwarded LTM.
- 42 c) Sets the drop_eligible parameter of the forwarded LTM to false.
- 43 d) Uses the same value for the vlan_identifier parameter for the forwarded LTM that was presented
44 with the received LTM.

- 1 e) In the case of ECMP with flow filtering, the `flow_hash` parameter and the `time_to_live` parameter of
2 the forwarded LTM are set to values presented with the received LTM.
- 3 f) In PBB-TE related MAs, uses the `destination_address`, `source_address`, and `vlan_identifier` of the
4 input LTM and sets the `drop_eligible` parameter of the forwarded LTM to `FALSE`.
- 5 g) Copies, verbatim, all fields and TLVs, regardless of whether known to the Linktrace Responder,
6 from the input LTM to the forwarded LTM, except that `ForwardLTM()`:
 - 7 1) Places in the forwarded LTM TTL field the value from the input LTM TTL field decremented
8 by 1.
 - 9 2) Changes the value in the LTM Egress Identifier TLV to identify the forwarding Linktrace
10 Responder.
 - 11 3) Deletes the Sender ID TLV (21.5.3), if present in the LTM.
 - 12 4) Optionally, places a Sender ID TLV (21.5.3), identifying the forwarding system, in the LTM.
- 13 h) Transmits the forwarded LTM on the specified SAP.

14 **20.47.4 enqueueLTR()**

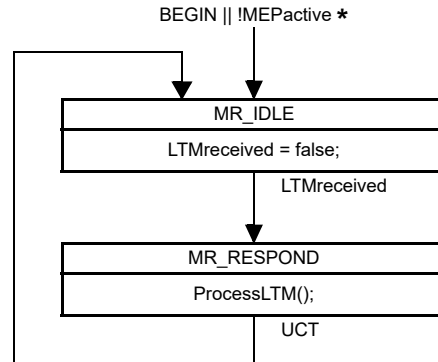
15 Constructs and enqueues a single LTR for later transmission by `xmitOldestLTR()` as follows. Using the input
16 LTM and the Linktrace Responder SAP through which the LTR is to be output, `enqueueLTR()` performs the
17 following steps:

- 18 a) Uses the MAC address contained in the LTM's Original MAC Address field as the
19 `destination_address` of the LTR. If the MP is associated with a PBB-TE MA, the LTR uses as
20 `destination_address`,
 - 21 1) The value carried in the Reverse MAC field, contained in the PBB-TE MIP TLV of the
22 received LTM, if this is a group MAC address; otherwise,
 - 23 2) The `source_address` of the received LTM.
- 24 b) Uses the MAC address of the MP owning the SAP specified for output as the `source_address` of the
25 LTR, if the MP is not a PBB-TE MHF; otherwise, the LTR uses as the `source_address`:
 - 26 1) The value of the `destination_address` of the received LTM if this is an individual MAC address;
27 otherwise,
 - 28 2) The value carried in the Reverse MAC field, contained in the PBB-TE MIP TLV of the
29 received LTM,
- 30 c) If the LTR includes a Reply Egress TLV [see the following item r)], uses the Primary VID of the
31 MP on that Egress Port as the `vlan_identifier` of the LTR, else it uses the Primary VID of the MP on
32 the Ingress Port as the `vlan_identifier` of the LTR. In PBB-TE related MA's, sets the `vlan_identifier`
33 parameter as the value carried in the Reverse VID field contained in the PBB-TE MIP TLV of the
34 received LTM.
- 35 d) Sets the priority parameter to the same value as for CCMs [item h) in 12.14.7.1.3].
- 36 e) In the case of ECMP with flow filtering, the `flow_hash` parameter is set to zero and the `time_to_live`
37 parameter is set to 63.
- 38 f) Sets the `drop_eligible` parameter to `FALSE`.
- 39 g) Places its own version in the Version field.
- 40 h) If its own version is lower than that of the received LTM, sets all bits and fields in the transmitted
41 PDU that are reserved in its own version, including all bits between the portion of the last header
42 field defined for its own version and the first TLV, to 0.
- 43 i) Sets the FwdYes bit of the Flags field to 1 if the LTM was forwarded by the Linktrace Responder, or
44 0 if not.
- 45 j) Sets the TerminalMEP bit of the Flags field to 1 if the MP reported in either the Reply Ingress TLV
46 or the Reply Egress TLV is a MEP, or 0 if not.
- 47 k) Copies the Flags field, excepting the FwdYes bit and the TerminalMEP bit, from the LTM to the
48 LTR.

- 1 l) Copies the LTM Transaction Identifier field from the LTM to the LTR Transaction Identifier field of
2 the LTR.
- 3 m) Copies the LTM Egress Identifier TLV (21.8.8) value from the LTM to the Last Egress Identifier
4 field (21.9.7.1) of the LTR Egress Identifier TLV, or places 0 in that field, if there is no LTM Egress
5 Identifier TLV in the received LTM (see J.4).
- 6 n) Sets the Next Egress Identifier field (21.9.7.2) of the LTR Egress Identifier TLV to a value that
7 identifies the forwarding Linktrace Responder (see 21.8.8).
- 8 o) Places one less than the value in the LTM TTL field of the LTM in the Reply TTL field of the LTR;
9 p) Sets the Relay Action field according to Table 21-27.
- 10 q) If the LTM was not received by a Down MEP or Down MHF, does not place a Reply Ingress TLV in
11 the LTR; otherwise,
12 1) Fills the Ingress Action field of a Reply Ingress TLV (21.9.8) with the appropriate value
13 according to Table 21-30;
14 2) Places the receiving MP's MAC address in the Ingress MAC Address field of the Reply Ingress
15 TLV; and
16 3) Optionally, fills the remainder of the Reply Ingress TLV with the receiving MP's Port ID
17 information.
- 18 r) If the LTM was received by a Down MEP, or if no Egress Port was identified, or if no Up MEP nor
19 Up MHF belonging to the LTM's MA is configured on the Egress Port, does not place a Reply
20 Egress TLV in the LTR; otherwise,
21 1) Fills the Egress Action field of a Reply Egress TLV (21.9.9) with the appropriate value
22 according to Table 21-32;
23 2) Places the Egress Port's Up MP's MAC address in the Egress MAC Address field of a Reply
24 Egress TLV in the LTM; and
25 3) Optionally, fills the remainder of the Reply Egress TLV with Egress Port's Port ID information.
- 26 s) As controlled by the managed objects item e) in 12.14.3.1.3, item d) in 12.14.5.1.3, and item d) in
27 12.14.6.1.3, places a Sender ID TLV (21.5.3), identifying the replying Bridge, in the LTM.
- 28 t) Copies, verbatim, all other TLVs in the input LTM to the LTR, except for the Sender ID TLV
29 (21.5.3), which is not copied.
- 30 NOTE 1—Forwarding all unknown TLVs in both the LTM and the LTR enables future revisions of this standard, and
31 designers of Organization-Specific TLVs, to add new capabilities.
- 32 NOTE 2—The resultant LTR can be larger than the maximum frame size for the media-dependent sublayer on which the
33 LTR is to be transmitted, causing the LTR to be discarded.
- 34 u) Increments nPendingLTRs by 1.

20.48 LTM Receiver state machine

One instance of the LTM Receiver state machine is instantiated by Bridge's Linktrace Responder. The LTM Receiver state machine implements the function specified by the state diagram in Figure 20-17, the variables in 20.46, and the procedures in 20.47.



* MEPActive in a MEP; not present (always true) in an MHF

Figure 20-17—LTM Receiver state machine

20.49 LTR Transmitter procedure

The following procedure is local to the LTR Transmitter state machine:

- a) xmitOldestLTR() (20.49.1)

20.49.1 xmitOldestLTR()

If and only if nPendingLTRs is nonzero, dequeues a single LTR and transmits it, and decrements nPendingLTRs by 1.

20.50 LTR Transmitter state machine

One instance of the LTR Transmitter state machine is instantiated by Bridge's Linktrace Responder. The LTR Transmitter state machine implements the function specified by the state diagram in Figure 20-18, the variables in 20.46, and the procedures in 20.47.

This state machine implements the requirement of (20.3.2) to wait a random time interval after receiving an LTM before transmitting an LTR by means of a free running 1 s timer, the expiry of which triggers all LTR transmissions. This method avoids the need to create and manage a timer for every LTM received. Any other implementation that meets the requirements of 20.3.2 can be used in place of this state machine.

20.51 CFM PDU validation and versioning

The purpose of this subclause is to state specific goals to be achieved by CFM with respect to the relationship of this and future versions of this standard, and to define the procedures necessary to meet those goals.

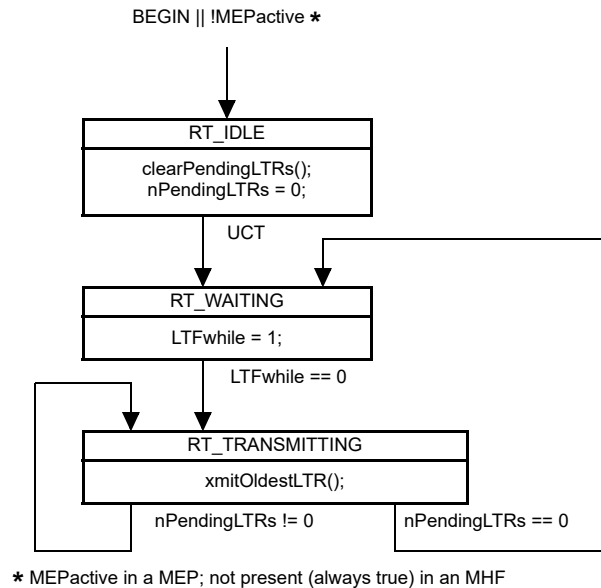


Figure 20-18—LTR Transmitter state machine

20.51.1 Goals of CFM PDU versioning

The goals of CFM with respect to the relationship of this and future versions of this standard are to ensure that:

- a) Implementations of this standard will interoperate with implementations of future versions of this standard.
- b) Implementers will be able to offer proprietary, nonstandard extensions to this standard with enhanced functionality.
- c) Conformant but extended implementations of this standard will not restrict the ability of future versions of this standard to extend the standard functionality.

20.51.2 PDU transmission

In order to ensure that future versions of CFM will be compatible with implementations of this standard, certain requirements are placed on transmitted CFM PDUs:

- a) The fixed header fields shall be transmitted exactly as specified in this standard.
- b) All bits defined as “reserved” in this standard, e.g., unused bits in the Flags field, shall be transmitted as 0.
- c) Additional fields shall not be added to the fixed header specified in this standard.
- d) Code points reserved in this standard, or in ITU-T G.8013/Y.1731, e.g., additional values for the OpCode field in the fixed header (Table 21-3), the Type field in a TLV (Table 21-5), or the reserved values in Table 21-18 for the Maintenance Domain Name Format, shall not be transmitted in any CFM PDU.
- e) Additional fields shall not be added to any TLV specified in this standard.

Organization-Specific TLVs can be defined by any organization possessing an OUI or a CID, and may be transmitted by an implementation conformant to this standard.

NOTE—The preceding is not an exhaustive list of the restrictions on the transmission of CFM PDUs; it is only a list of the specifications that are most important to future compatibility.

1 20.51.3 PDU validation

2 CFM PDUs that fail certain specified tests are discarded by MPs. This requirement is described here as a
3 two-pass processing algorithm, first a Validation Pass, and then an Execution Pass. The Validation Pass is
4 performed first. If and only if it succeeds, the Execution Pass is performed. This description does not
5 preclude other algorithms. For example, an MP could process a CFM PDU in one pass, building a tentative
6 list of changes to the affected state machines and databases, and commit the changes only after the
7 processing has succeeded. However, the behavior of such an implementation shall be indistinguishable, in
8 terms of externally observable behavior (i.e., Management Objects and protocol frames) from the two-pass
9 algorithm described in this subclause.

10 20.51.4 Validation pass

11 During the Validation pass, no changes are made to any CFM database other than the updating of certain
12 management counter variables. No protocol state machines are affected, and no changes are made to any
13 CFM PDUs that can be transmitted by the receiving MP. The description of the Validation pass is split into
14 three parts:

- 15 a) Operations required of an MP Level Demultiplexer (20.51.4.1)
- 16 b) Operations required of all MP components when performing the Validation pass (20.51.4.2)
- 17 c) Operations that are required of some MP components, and optional for others (20.51.4.3)

18 20.51.4.1 Validation pass operations required of an MP Level Demultiplexer

19 If the following test fails, the receiving MP Level Demultiplexer (19.2.6) shall consider the CFM PDU
20 invalid and discard it:

- 21 a) The length of the `mac_service_data_unit` is long enough to contain a complete MD Level field.

22 20.51.4.2 Validation pass operations required of MP components

23 All bits declared “reserved” in this standard, e.g., unused bits in the Flags field, shall be ignored by the
24 receiving MP.

25 The receiving MP processes the CFM PDU in accordance with the numerically lower of 1) the Version field
26 in the CFM PDU and 2) the highest Version number known to the receiving implementation. That is, a
27 Version 1 implementation receiving a Version 0 CFM PDU processes it according to Version 0 of this
28 standard, and it processes a Version 2 CFM PDU according to Version 1. The IEEE 802.1 Working Group
29 places the imposition on future versions of this standard that all earlier version implementations can process
30 their CFM PDU properly.

31 As an example of the use of this rule, if a new Flags field bit is defined in Versions 2 and 3, then a Version 3
32 implementation receiving a Version 1 CFM PDU will ignore that bit, even though the bit is defined in both
33 Version 2 and Version 3, and even though the bit is actually set.

34 NOTE 1—One effect of this rule is that an implementation conformant to this Version of this standard, Version 0,
35 ignores the Version field in a received CFM PDU.

36 The following criteria shall not be used by a CFM entity to validate a received CFM PDU:

- 37 a) The fixed header can be longer than the length specified by the version of this standard indicated by
38 the CFM PDU’s Version field.
- 39 b) Bits can be set in reserved bits of the Flags field.
- 40 c) A TLV can have a Type field not specified by this standard.

1 d) A TLV's Length field can be larger than the value (if any) specified in this standard. (This allows
2 TLVs to be extended in future Versions.)

3 e) Either the First TLV Offset field, or the Length field of the last TLV, in the CFM PDU, can indicate
4 a position for the first (next) TLV that coincides with the end of the `mac_service_data_unit`
5 containing the CFM PDU. That is, the End TLV can be missing from the CFM PDU if the frame is
6 longer than or equal to the minimum frame size for the media-dependent MAC sublayer.

7 NOTE 2—These latter criteria cannot be used by a CFM entity to validate a received CFM PDU. This helps to ensure
8 that the CFM PDUs transmitted by future versions of this standard will be considered valid by implementations of the
9 current version (0) of this standard. These criteria can, however, be used by test equipment in order to test the
10 conformance of a system to this standard, e.g., to the rules in 20.51.2.

11 **20.51.4.3 Validation pass operations required of some receiving MP components**

12 The following Validation pass tests are required of some receiving MP components, and are optional for
13 other MP components, as specified in other subclauses. If performed, and if any test fails, the receiving
14 System shall consider the CFM PDU invalid and discard it.

- 15 a) The fixed header length, as determined by the First TLV Offset field (21.4.5), is not shorter than the
16 length specified by the version selected according to 20.51.4.2.
- 17 b) The fixed-length header does not run over the end of the `mac_service_data_unit`.
- 18 c) A TLV Length field does not run over the end of the `mac_service_data_unit`.
- 19 d) A TLV Length field does not indicate a length that is shorter than the minimum length for that TLV
20 as specified by the Version field of the CFM PDU.
- 21 e) Every header field and TLV in the CFM PDU meets the validity criteria specified in the header field
22 and TLV definitions in Clause 21 (see the Validation Test headings in that clause).

23 **20.51.5 Execution pass**

24 The CFM PDU is processed in the Execution Pass. Changes to the CFM databases can be made, the state
25 machines can change states, and the contents of future CFM PDUs can be affected by these changes. The
26 receiving MP:

- 27 a) Processes the CFM PDU in accordance with the numerically lower of 1) the Version field in the
28 CFM PDU; and 2) the highest Version number known to the receiving implementation.
- 29 b) Processes only those fields in the fixed header portion of the CFM PDU that are defined in the
30 selected Version of the standard, and ignores any extra octets in the fixed header, if the fixed header
31 is longer than the length specified by the selected Version.
- 32 c) Ignores any TLV with a Type field not specified by the selected Version.
- 33 d) Does not process any part of the CFM PDU following the End TLV (the lack of an End TLV is not
34 an error).
- 35 e) Ignores any octets following those specified by the selected Version, if any TLV's Length field is
36 larger than the value (if any) specified the selected Version.
- 37 f) Ignores all bits undefined in this standard, e.g., unused bits in the Flags field.

38 Unlike the loss of ST BPDUs, the loss of CFM PDUs cannot completely disrupt the function of the network.
39 Also, unlike other control protocols such as the IEEE 802.3 Slow Protocols, CFM PDUs can be presented at
40 an arbitrarily high rate. Therefore, a Bridge shall ensure that receiving CFM PDUs on all Bridge Ports at the
41 maximum possible rate supported by the MAC Services underlying those ports shall not significantly
42 increase the probability of the failure of the Bridge to maintain loop-free forwarding paths in the bridged
43 network. Rather, the Bridge's CPU can be protected against excess CFM PDUs in a similar manner as for
44 other potential Denial of Service attacks.

1 20.51.6 Future extensions

2 It is expected that future versions of this protocol will utilize the above versioning rules in order to extend
3 the CFM PDU format in any number of ways, including perhaps:

- 4 a) Adding new fields to the header by increasing the minimum size of the First TLV Offset field, and
5 placing the new header fields after the First TLV Offset field in the Version 0 header.
- 6 b) Adding new TLVs and extending the lengths of existing TLVs with new fields.
- 7 c) Adding a new TLV, that is the first TLV to be processed, by renaming the First TLV Offset field to
8 the “First Version 0 TLV Position,” adding a new “Version X Header Length” field following the
9 fields in the Version 0 header, and inserting one or more Version X TLVs just ahead of the first
10 required Version 0 TLV.
- 11 d) Adding new information, perhaps fixed-length information, that is required to follow the End TLV
12 specified in Version 0.

13 Other methods for extending the CFM PDU format are certainly possible. It is therefore critical that
14 Version 0 implementations adhere to the versioning requirements of this clause.

15 NOTE—The extension options discussed in this clause are reserved for use by future versions of this standard. Each one
16 is a violation of 20.51.2, 20.51.4, and/or 20.51.5. The Organization-Specific TLV is the only available compliant
17 extension to this standard.

18 20.52 PDU identification

19 A received CFM PDU is associated with a particular MP. Every data frame, including one carrying a CFM
20 PDU, has some means, either explicit or implicit, by which that frame can be associated by the receiver with
21 a particular service instance. In a Bridge that is not VLAN aware, there is only one service instance. In a
22 VLAN Bridge, the `vlan_identifier` (6.8) identifies the service instance. The means by which the appropriate
23 receiving MP for processing a received CFM PDU is selected are as follows:

- 24 a) The direction (PHY side or MAC Relay Entity side) from which the CFM PDU was received.
- 25 b) The implicit or explicit association of the frame with a particular service instance (the
26 `vlan_identifier`).
- 27 c) The MD Level field in the CFM Header.

28 Once the frame containing the CFM PDU has been delivered to the appropriate entity within an MP, the
29 `destination_address` can be used to decide whether the CFM PDU is to be processed or discarded, as
30 specified in the description of each entity.

31 The MAID and/or Maintenance association Endpoint Identifier field is not used to identify the receiving MP.
32 Thus, if the Individual MP address model (see J.6) is used by a particular Bridge, and each MP uses its
33 Bridge Port’s individual MAC address, then the VID, MD Level, and flow direction uniquely identify the
34 receiving MEP. If the Shared MP address model is used by a Bridge, so that Up MPs in the Bridge can share
35 a MAC address, then the specific Bridge Port that is the target of an LBM can be genuinely ambiguous. This
36 ambiguity allows the designer to trade the value of detailed information against cost of obtaining that
37 information.

38 20.53 Use of transaction IDs and sequence numbers

39 Each CCM, LBM, and LTM has a field (21.6.3, 21.7.3, and 21.8.3) that is incremented for each PDU
40 transmitted of each type, so that consecutively transmitted PDUs are in numerical order. The variables that
41 control these fields are `CCIsentCCMs` (20.10.2, for CCM), `nextLBMtransID` (20.30.2, for LBM), and
42 `nextLTMtransID` (20.41.1, for LTM).

1 In theory, every MEP in a Bridge has a CCI_{sent}CCMs variable, a nextLBMtransID variable, and a
2 nextLTMtransID variable, independently from every other MEP. Therefore, any number of MEPs in a single
3 Bridge could be transmitting independent streams of LBMs and LTMs as long as they have different MAC
4 addresses. (If they had the same MAC address, they could not tell each others' LBRs and LTRs apart.)

5 In practice, there can be fewer instances of these variables and their corresponding state machines than one
6 each per MEP. The managed objects in Clause 12, particularly the definitions of the MEP's LBR counters
7 [item y) in 12.14.7.1.3 and item z) in 12.14.7.1.3] and the responses permitted to the Transmit Loopback
8 Messages command (12.14.7.3.3), are specified so that any number of MEP Loopback Initiator transmit
9 state machines, from one to the number of MEPs, can be implemented. Each state machine can increment its
10 own nextLBMtransID variable to determine the next transmitted Loopback Transaction Identifier field.
11 Similarly, the allowed responses to the Transmit Linktrace Message command (12.14.7.4.3) provide the
12 same latitude for the number of nextLTMtransID variables implemented. The resources for transmitting
13 LTMs or streams of LBMs can then be used serially, rather than simultaneously, by different MEPs in the
14 same Bridge.

15 On the other hand, every MEP that is incrementing its CCI_{sent}CCMs variable has its own instance of that
16 variable, and thus its own MEP Continuity Check Initiator state machine, because the Remote MEP state
17 machines receiving those CCMs are tracking each MEP's sequence of CCMs independently. A MEP can
18 also transmit 0 in every CCM's Sequence Number field.

21. Encoding of CFM PDUs

This clause specifies the method of encoding CFM PDUs. The specifications include the following:

- a) Format used to encapsulate or decapsulate a CFM PDU in a frame (21.2).
- b) Format of the Common CFM Header, used in all CFM PDUs (21.4).
- c) Format used for all Type, Length, Value (TLV) information elements that can be included in CFM PDUs (21.5).
- d) Formats of the Continuity Check Message (CCM, 21.6), the Loopback Message and Loopback Reply (LBM and LBR, 21.7), the Linktrace Message (LTM, 21.8), Linktrace Reply (LTR, 21.9), Send Frame Message (SFM, 29.4.3), and the Reflected Frame Message (RFM, 29.4.2).

NOTE—Clause 18 introduces the principles of CFM operation and the network architectural concepts that support it. Clause 19 breaks down the CFM protocol entities into their components. Clause 20 specifies the protocols operated by the components of each MP. The use of CFM within systems and networks is further described in Clause 22.

21.1 Structure, representation, and encoding

All CFM PDUs shall contain an integral number of octets.

The octets in a CFM PDU are numbered starting from 1 and increasing in the order they are put into the MSDU that accompanies a request to or indication from the instance of the MAC Internal Sublayer Service (ISS or EISS) used by a CFM entity.

The bits in an octet are numbered from 1 to 8 in order of increasing bit significance, where 1 is the LSB in the octet.

Where octets and bits within a CFM PDU are represented using a diagram, octets shown higher on the page than subsequent octets and octets shown to the left of subsequent octets at the same height on the page are lower numbered; bits shown to the left of other bits within the same octet are higher numbered.

Where two or more consecutive octets are represented as hexadecimal values, lower numbered octet(s) are shown to the left and each octet following the first is preceded by a hyphen, e.g., 01-80-C2-00-00-00.

When consecutive octets are used to encode a binary number, the lower octet number has the more significant value. When consecutive bits within an octet are used to encode a binary number, the higher bit number has the most significant value. When bits within consecutive octets are used to encode a binary number, the lower octet number composes the more significant bits of the number. A flag is encoded as a single bit, and is set (True) if the bit takes the value 1, and clear (False) otherwise. The remaining bits within the octet can be used to encode other protocol fields.

21.2 CFM encapsulation

The means for identifying CFM PDUs consists of two octets containing the EtherType value shown (in hexadecimal notation) in Table 21-1.

Table 21-1—CFM PDU Encapsulation EtherType

Name	Value
IEEE 802.1Q CFM PDU Encapsulation EtherType	89-02

1 21.3 CFM request and indication parameters

2 21.3.1 destination_address parameter

3 There are three classes of CFM PDUs in terms of the CFM entities to which they are addressed:

- 4 a) Those addressed to all MEPs in a service instance (CCM).
- 5 b) Those addressed to the set of MPs immediately adjacent to the transmitting MP, at a certain MD
6 Level, in a service instance (LTM).
- 7 c) Those addressed to a single, specific MP (LBM, LBR, LTR).

8 Frames carrying CFM PDUs belonging to two different MAs, but at the same MD Level, are distinguished
9 from each other in the same manner as data frames are distinguished from each other in the corresponding
10 service instances monitored by those MAs.

11 In a VLAN Bridge, it is the VID that distinguishes service instances. CCMs monitoring a service instance
12 distinguished by its VID use the group MAC addresses listed in Table 8-18 as the destination_address.
13 LTMs monitoring a VID-distinguished service instance use those shown in Table 8-19. All but the last three
14 bits of the destination_address are fixed by the OpCode, either CCM or LTM. As shown in these two tables,
15 the last three bits of the destination_address match the MD Level field of the CFM PDU Header. CFM PDUs
16 belonging to MAs monitoring service instances distinguished by their I-SID use the same set of destination
17 addresses as those belonging to MAs monitoring VLAN service instances. MAs monitoring TESIs are
18 distinguished from MAs monitoring VLAN services by the vlan_identifier parameters and from each other
19 by the 3-tuples <ESP-DA, ESP-SA, ESP-VID> of their ESPs. CFM PDUs belonging to PBB-TE MAs use
20 the MAC address identified in the ESP-DA fields of their ESP identifiers.

21 21.3.2 source_address parameter

22 The individual MAC address of the MP transmitting the PDU. This is the MAC address of the transmitting
23 MP. An MP's MAC address is not necessarily unique; MPs configured on the same Bridge Port can share the
24 same MAC address. The principles used to allocate Organization Unique Identifiers (OUIs) required that
25 universally unique MAC addresses (those with the U/L bit = 0, see 8.2 of IEEE Std 802-2014 [B7]) not be
26 used to create MP MAC addresses separately from a physical instance of an IEEE 802 MAC. See J.6 for a
27 discussion of possible assignments of MAC addresses to MPs.

28 **Validation Test:** The source_address parameter contains an Individual, and not a Group, MAC address.

29 21.4 Common CFM Header

30 The format of a CFM PDU is shown in Table 21-2. A CFM PDU is identified by the encapsulation described
31 in 21.2. Octets 1 through 4, including the MD Level, Version, OpCode, Flags, and First TLV Offset fields,
32 constitute the Common CFM Header.

33 21.4.1 MD Level

34 (most significant 3 bits) Integer identifying the Maintenance Domain Level (MD Level) of the frame.
35 Higher numbers correspond to higher MAs, those with the greatest physical reach, with the highest values
36 for customers' CFM frames. Lower numbers correspond to lower MAs, those with more limited physical
37 reach, with the lowest values for single Bridges or physical links.

38 21.4.2 Version

39 (least significant 5 bits) The protocol version number, always 0. Ignored on receipt.

Table 21-2—Common CFM Header format

Field	Octet
MD Level	1 (high-order 3 bits)
Version	1 (low-order 5 bits)
OpCode	2
Flags	3
First TLV Offset	4
Varies with value of OpCode	5
End TLV (0)	First TLV Offset + 5

1 21.4.3 OpCode

2 (1 octet) The OpCode field specifies the format and meaning of the remainder of the CFM PDU. Certain
3 values of the OpCode field are reserved for allocation by IEEE 802.1 and/or other organizations. The values
4 for the various CFM PDU types are shown in Table 21-3. When assigning new OpCode values, pairs of
5 CFM PDUs that operate in stimulus-response fashion, such as the LBM/LBR or the LTM/LTR pairs, will
6 use even/odd pairs of values such that the odd (numerically larger) of the two values is the stimulus, and the
7 even (numerically smaller) the response.

Table 21-3—OpCode Field range assignments

CFM PDU or organization	OpCode range
Reserved for IEEE Std 802.1	0
Continuity Check Message (CCM)	1
Loopback Reply (LBR)	2
Loopback Message (LBM)	3
Linktrace Reply (LTR)	4
Linktrace Message (LTM)	5
Reflected Frame Message (RFM)	6
Send Frame Message (SFM)	7
Reserved for IEEE Std 802.1Q	8–31
Defined by ITU-T G.8013/Y.1731	32–63
Reserved for IETF (RFC 7319 [B44])	64–95
Reserved for IEEE Std 802.1	96–255

8 21.4.4 Flags

9 (1 octet) The use of the Flags field is defined separately for each OpCode.

21.4.5 First TLV Offset

(1 octet) The offset, starting from the first octet following the First TLV Offset field, up to the first TLV in the CFM PDU. The value of the First TLV Offset field shall be transmitted as indicated in the specification for each of the OpCode field values, as follows (21.6.2, 21.7.2, 21.8.2, and 21.9.2).

21.5 TLV format

TLV stands for *Type*, *Length*, *Value* and denotes a method of encoding variable-length and/or optional information in a PDU. TLVs are not aligned to any particular word or octet boundary. TLVs follow each other with no padding between TLVs.

21.5.1 General format for CFM TLVs

The TLV format is shown in Table 21-4.

Table 21-4—TLV format

Field	Octet
Type	1
Length	2–3
(Value)	4

21.5.1.1 Type

(1 octet) Required. If 0, no Length or Value fields follow. If not 0, at least the Length field follows the Type field. The Type field is encoded as shown in Table 21-5.

21.5.1.2 Length

(2 octets) Required if the Type field is not 0. Not present if the Type field is 0. The 16 bits of the Length field indicate the size, in octets, of the Value field. 0 in the Length field indicates that there is no Value field.

21.5.1.3 Value

(Length specified by the Length field) Optional. Not present if the Type field is 0 or if Length field is 0.

21.5.2 Organization-Specific TLV

Type = 31. Any organization can define TLVs for use in CFM. The format for such TLVs is shown in Table 21-6. The “OUI/CID” is an organizationally unique identifier or a Company ID, obtainable from IEEE.³⁶ The Subtype is required, so that an additional OUI/CID will not be required if more Organization-Specific TLV are required by an owner of an OUI/CID.

³⁶ Interested applicants should contact the IEEE Standards Department of The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/regauth/index.html>).

Table 21-5—Type Field values

TLV or organization	Type field
End TLV	0
Sender ID TLV	1
Port Status TLV	2
Data TLV	3
Interface Status TLV	4
Reply Ingress TLV	5
Reply Egress TLV	6
LTM Egress Identifier TLV	7
LTR Egress Identifier TLV	8
PBB-TE MIP TLV	9
Data Part 1 TLV (29.3.3.2)	10
Data Part 2 TLV (29.3.3.2)	11
Truncated Data TLV (29.3.3.2)	12
Reserved for IEEE Std 802.1Q	13–30
Organization-Specific TLV	31
Defined by ITU-T G.8013/Y.1731	32–63
Reserved for IETF (RFC 7319 [B44])	64–95
Reserved for IEEE Std 802.1	96–255

Table 21-6—Organization-Specific TLV format

Field	Octet
Type = 31	1
Length	2–3
OUI/CID	4–6
Subtype	7
Value (optional)	8 – (Length + 3)

¹ This TLV category is provided to allow different organizations, such as IEEE 802.1, IEEE 802.3, ITU-T, as
² well as individual software and equipment vendors, to define TLVs that advertise information to remote
³ entities attached to the same media, subject to the following restrictions:

- ⁴ a) Information transmitted in an Organization-Specific TLV shall not require the recipient to violate
⁵ any requirement in this standard.
- ⁶ b) Information transmitted in one Organization-Specific TLV shall not be used to provide a means for
⁷ sending messages that are larger than would fit within a single CFM PDU.
- ⁸ c) Organization-Specific TLVs shall conform to 20.51.

21.5.3 Sender ID TLV

Type = 1. Identifies the Bridge on which the transmitting MP is configured and may also include a management address for that Bridge. The allowed values are those contained in the Chassis ID TLV and Management Address TLV in IEEE Std 802.1AB, modified to fit into a CFM TLV, as shown in Table 21-7. Whether the Sender ID TLV is transmitted, and what information is included in the TLV, are controlled by managed objects [item e) in 12.14.3.1.3, item d) in 12.14.5.1.3, item d) in 12.14.6.1.3].

Table 21-7—Sender ID TLV format

Field	Octet
Type = 1	1
Length	2–3
Chassis ID Length	4
Chassis ID Subtype	5
Chassis ID	6 – (Chassis ID Length + 5)
Management Address Domain Length	(Chassis ID Length + 6)
Management Address Domain	(Chassis ID Length + 7) – (Chassis ID Length + Management Address Domain Length + 6)
Management Address Length	(Chassis ID Length + Management Address Domain Length + 7)
Management Address	(Chassis ID Length + Management Address Domain Length + 8) – (Length + 3)

Validation Test: The Length field is large enough to contain all of the fields indicated as being present by the Chassis ID Length, Management Address Domain Length, and/or Management Address Length fields.

21.5.3.1 Chassis ID Length

(1 octet) The length, in octets, of the Chassis ID field. 0 if no Chassis ID is specified. Always present.

Validation Test: The Chassis ID Length field either is 0, or is less than (TLV Length field value – 1).

21.5.3.2 Chassis ID Subtype

(1 octet) The **chassis ID subtype**, as specified by IEEE Std 802.1AB, identifying the type of the component referenced by the Chassis ID field. Not present if the Chassis ID Length field contains 0.

21.5.3.3 Chassis ID

(Length specified by the Chassis ID Length field) Identifies the chassis. Specified by IEEE Std 802.1AB. Not present if the Chassis ID Length field contains 0.

21.5.3.4 Management Address Domain Length

(1 octet) The Management Address Domain Length field contains the length, in octets, of the Management Address Domain field. If 0, or if the TLV's Length field indicates that the Management Address Domain Length field is not present, then the Management Address Domain, Management Address Length, and Management Address fields are not present.

21.5.3.5 Management Address Domain

(Length specified by the Management Address Domain Length field) The Management Address Domain field specifies the format and type of the contents of the Management Address field, as well as a management mechanism. The format of the Management Address Domain field shall be that of an Object Identifier (OID), as specified by section 8.19 in ITU-T X.690 (2002). The OID references a TDomain (IETF RFC 2579). Standard values that are useful for the Management Address Domain field include, but are not limited to,

- transportDomainUdpIpv4, indicating Simple Network Management Protocol (SNMP) over User Datagram Protocol (UDP) over IPv4 (IETF RFC 3419).
- transportDomainUdpIpv6, indicating SNMP over UDP over IPv6 (IETF RFC 3419).
- snmpIeee802Domain, indicating SNMP over the MAC Service (IETF RFC 4789).

This field is not present if the Management Address Domain Length field is not present or contains a 0.

21.5.3.6 Management Address Length

(1 octet) The length, in octets, of the Management Address field. This field is not present if the Management Address Domain Length field is not present or contains a 0.

21.5.3.7 Management Address

(Length specified by the Management Address Length field) Identifies a TransportDomain (IETF RFC 3419, IETF RFC 4789) through which the Bridge in which the transmitting MP is configured can be managed. The format of the Management Address field is specified by the MIB module defining the TransportDomain. This field is not present if the Management Address Domain Length field is not present or contains a 0, or if the Management Address Length field is not present or contains a 0.

NOTE—The contents of the Management Address Domain and Management Address are defined in terms of MIB modules. Therefore, a published MIB module is required in order to define a specific value that can be used in the Management Address Domain field. However, the use of Management Address Domain field by a system does not require that the system be manageable via SNMP; the published MIB module could define, for example, a TransportDomain for a proprietary Command Line Interpreter over ITU-T X.25 [B52].

21.5.4 Port Status TLV

Type = 2. The Port Status TLV indicates the ability of the Bridge Port on which the transmitting MEP resides to pass ordinary data, regardless of the status of the MAC. The value of this TLV is driven by the MEP variable enableRmepDefect (20.9.2), as shown in Table 21-9. The format of this TLV is shown in Table 21-8. Any change in the Port Status TLV's value triggers one extra transmission of that Bridge Port's MEPs' CCMs.

Table 21-8—Port Status TLV format

Field	Octet
Type = 2	1
Length	2–3
See Table 21-9	4

Validation Test: The Port Status TLV contains one of the values listed in Table 21-9.

Table 21-9—Port Status TLV values

mnemonic	Ordinary data passing freely through the Port	value
psBlocked	No: enableRmepDefect = false	1
psUp	Yes: enableRmepDefect = true	2

¹ A MEP that is configured in a Bridge in a position that is not associated with a single value for the Port State
² and VID member set, e.g., a MEP in position 5 in Figure 22-9, on a Bridge running multiple spanning tree
³ instances via MSTP, shall not transmit the Port Status TLV.

⁴ 21.5.5 Interface Status TLV

⁵ Type = 4. The Interface Status TLV indicates the status of the interface on which the MEP transmitting the
⁶ CCM is configured (which is not necessarily the interface on which it resides, see J.6), or the next lower
⁷ interface in the IETF RFC 2863 IF-MIB. The format of this TLV is shown in Table 21-10. The enumerated
⁸ values are shown in Table 21-11. These values correspond to the values for ifOperStatus in IETF RFC 2863.

Table 21-10—Interface Status TLV format

Field	Octet
Type = 4	1
Length	2–3
See Table 21-11	4

⁹ **Validation Test:** The Interface Status TLV field contains one of the values listed in Table 21-11.

Table 21-11— Interface Status TLV values

mnemonic	Interface Status (IETF RFC 2863 ifOperStatus)	value
isUp	up	1
isDown	down	2
isTesting	testing	3
isUnknown	unknown	4
isDormant	dormant	5
isNotPresent	notPresent	6
isLowerLayerDown	lowerLayerDown	7

21.5.6 Data TLV

Type = 3. Zero or more octets of arbitrary data. Serves several purposes, including reflected data frame in RFM, to be decapsulated data in SFM, the transmission of different frame sizes to test Maximum Service Data Unit Size capabilities and the testing for data-specific error dependencies. The Data TLV may be included in the LBM and the LBR, the RFM, and the SFM, but not in any other CFM PDU. The contents of the Data TLV shall not be examined or interpreted by the receiver of any CFM PDU except an LBR. The format of the Data TLV is shown in Table 21-12.

Table 21-12—Data TLV format

Field	Octet
Type = 3	1
Length	2–3
Data	4 – (Length + 3)

21.5.7 End TLV

Required. Type = 0. Length and Value fields are not present. The End TLV is the last TLV in the CFM PDU. (Lack of the End TLV does not invalidate a received CFM PDU, but certain cases where frames receive extra headers, e.g., MACsec or additional VLAN tags, can cause errors if it is not present.) The format of the End TLV is shown in Table 21-13.

Table 21-13—End TLV format

Field	Octet
Type = 0	1

21.6 CCM format

The format of the CCM is shown in Table 21-14. In order to limit the resources required to generate and receive CCMs, the following restrictions apply to the format of the CCM:

- a) The Maintenance Domain Name and Short MA Name are encoded in a manner such that an MP can treat all of the fields from the Maintenance Domain Name Format field through the Short MA Name field as a unit when deciding whether a received CCM does or does not belong to the receiving MP's MA. That is, the MP can perform a binary comparison of these fields in a received CCM to the fields it would transmit in its own CCMs. This means, for example, that two Maintenance Domain Names differing, e.g., only by trailing spaces or by upper/lower case substitutions, can be interpreted to denote different MAs.
- b) The portion of the PDU allocated for the MAID is limited to 48 octets, in order to minimize the memory requirements of a hardware implementation of a MEP.

An MP shall be able to receive and process any valid CCM PDU that is 128 octets in length or less, starting with the MD Level/Version octet and including the End TLV. An MP shall not transmit a CCM PDU exceeding this length. An MP may discard as invalid any received CCM PDU that exceeds this length.

Table 21-14—CCM format

Field	Octet
Common CFM Header	1–4
Sequence Number	5–8
Maintenance association Endpoint Identifier	9–10
Maintenance Association Identifier	11–58
Defined by ITU-T G.8013/Y.1731	59–74
Reserved for definition in future versions of the protocol ^a	
Optional CCM TLVs	First TLV Offset + 5 ^b
End TLV (0)	First TLV Offset + 5, if no Optional CCM TLVs are present

^a This field has 0 length in this version 0 of CFM. It is shown in order to stress that additional information can be present in future versions of CFM, and that a version 0 receiver ignores its contents, if present.

^b Octet 75 for transmitted CCMs.

21.6.1 Flags

(1 octet) The Flags field of the Common CFM Header is split into four parts for the CCM as follows:

- a) RDI field (21.6.1.1)
- b) Reserved field (21.6.1.2)
- c) CCM Interval field (21.6.1.3)
- d) Traffic field (21.6.1.4)

21.6.1.1 RDI

The MSB of the Flags field is the RDI bit. This bit is set to 1 if the transmitting MEP's presentRDI variable (20.9.6) is set, and 0 if not.

21.6.1.2 Reserved

The bits of the Flags field not including the RDI field, the Traffic field, and the CCM Interval field are set to 0 by the transmitting MP, and are not to be examined by the receiving MP [item b) in 20.51.2].

21.6.1.3 CCM Interval

The least significant 3 bits of the Flags field constitute the CCM Interval field. The CCM Interval field is encoded as specified in Table 21-15.

NOTE—The maximum CCM Lifetime in Table 21-15 is equal to 3.5 times the CCM Interval. The minimum CCM Lifetime is the maximum CCM Lifetime minus 1/4 of the CCM Interval, the minimum required granularity of the timer variables CCIwhile, rMEPwhile, errorCCMwhile, and xconCCMwhile.

Validation Test: The CCM Interval field does not contain the value 0.

21.6.1.4 Traffic field

In the case of a PBB-TE MA or Infrastructure Segment MA, the second most significant bit of the Flags field is the Traffic bit. This bit if supported, is used to indicate the presence of backbone service instances in the monitored TESI or the assignment of TESIs in the monitored Infrastructure Segments. This bit is set to 1

Table 21-15—CCM Interval field encoding

Transmission Interval	max. CCM Lifetime	min. CCM Lifetime	CCM Interval field
invalid			0
3 1/3 ms (300 Hz)	11 2/3 ms	10 5/6 ms	1
10 ms	35 ms	32.5 ms	2
100 ms	350 ms	325 ms	3
1 s	3.5 s	3.25 s	4
10 s	35 s	32.5 s	5
1 min	3.5 min	3.25 min	6
10 min	35 min	32.5 min	7

1 if the transmitting MEP's presentTraffic variable (20.9.8) is set in the case of a PBB-TE MEP or the
2 transmitting MEP's ISpresentTraffic variable (20.9.10) is set in the case of a Infrastructure Segment MEP,
3 and 0 if not. This field is not examined if the receiving MPs are not PBB-TE or Infrastructure Segment
4 MEPs supporting the traffic field.

5 21.6.2 First TLV Offset

6 (1 octet) The First TLV Offset field of the Common CFM Header in a CCM is transmitted as 70.

7 **Validation Test:** The First TLV Offset field of the Common CFM Header in a CCM contains a value greater
8 than or equal to 70.

9 21.6.3 Sequence Number

10 (4 octets) A MEP either transmits a 0 in this field or copies to it the contents of the CCIsentCCMs variable
11 (20.10.2).

12 21.6.4 Maintenance association Endpoint Identifier

13 (2 octets) Contains an integer value. This TLV specifies from which MEP the CCM was transmitted.

14 **Validation Test:** MEPID is in the range 1–8191.

15 21.6.5 Maintenance Association Identifier

16 (48 octets) This field contains the MAID of the transmitting MEP. It is divided into six subfields, plus, if
17 necessary, a pad of octets contains 0 to fill out its fixed length. The subfields are as follows:

- 18 a) Maintenance Domain Name Format (21.6.5.1)
- 19 b) Maintenance Domain Name Length (21.6.5.2)
- 20 c) Maintenance Domain Name (21.6.5.3)
- 21 d) Short MA Name Format (21.6.5.4)
- 22 e) Short MA Name Length (21.6.5.5)
- 23 f) Short MA Name (21.6.5.5)

¹ The total length of these fields, including padding, if present, shall be exactly 48 octets. There are two
² possible formats of the Maintenance Association Identifier field, as shown in Table 21-16 and Table 21-17,
³ depending on whether the Maintenance Domain Name is present.

**Table 21-16—CCM Maintenance Association Identifier field format:
Maintenance Domain present**

Field	Octet
Maintenance Domain Name Format (not 1)	1
Maintenance Domain Name Length	2
Maintenance Domain Name	$3 - (\text{mdnl}^a + 2)$
Short MA Name Format	$(\text{mdnl} + 3)$
Short MA Name Length	$(\text{mdnl} + 4)$
Short MA Name	$(\text{mdnl} + 5) - (\text{mdnl} + \text{smanl}^b + 4)$
0 pad, if necessary	$(\text{mdnl} + \text{smanl} + 5) - 48$

^a Maintenance Domain Name Length.

^b Short MA Name Length.

**Table 21-17—CCM Maintenance Association Identifier field format:
Maintenance Domain not present**

Field	Octet
Maintenance Domain Name Format (1)	1
Short MA Name Format	2
Short MA Name Length	3
Short MA Name	$4 - (\text{smanl}^a + 3)$
0 pad, if necessary	$(\text{smanl} + 4) - 48$

^a Short MA Name Length.

⁴ **21.6.5.1 Maintenance Domain Name Format**

⁵ (1 octet) Specifies the format of the Maintenance Domain Name field. The Maintenance Domain Name
⁶ Format is encoded according to Table 21-18.

⁷ **21.6.5.2 Maintenance Domain Name Length**

⁸ (1 octet) Specifies the length in octets of the Maintenance Domain Name field. Not present if the
⁹ Maintenance Domain Name Format field specifies “No Maintenance Domain Name present” (1).

¹⁰ **Validation Test:** The Maintenance Domain Name Length in a CCM, if present, is greater than or equal to 1,
¹¹ and less than or equal to 43.

Table 21-18—Maintenance Domain Name Format

Maintenance Domain Name Format field	Value
Reserved for IEEE Std 802.1	0
No Maintenance Domain Name present	1
Domain Name-based string ^a	2
MAC address + 2-octet integer	3
Character string ^b	4
Reserved for IEEE Std 802.1	5–31
Defined by ITU-T G.8013/Y.1731	32–63
Reserved for IEEE Std 802.1	64–255

^a This is a string the terminal substring of which is an IETF RFC 1035 DNS Name, and the remainder of which is a text string, following the syntax of a DNS Name, denoting the identity of a particular Maintenance Domain.

^b This is an IETF RFC 2579 DisplayString, with the exception that character codes 0–31 (decimal) are not used.

21.6.5.3 Maintenance Domain Name

(Length specified by the Maintenance Domain Name Length field) Contains the Maintenance Domain Name, in the format specified by the Maintenance Domain Name Format field. Not present if the Maintenance Domain Name Format field specifies “No Maintenance Domain Name present” (1).

21.6.5.4 Short MA Name Format

(1 octet) Specifies the format of the Short MA Name field. The Short MA Name Format is encoded according to Table 21-19.

Table 21-19—Short MA Name Format

Short MA Name Format field	Value
Reserved for IEEE Std 802.1	0
Primary VID	1
Character string ^a	2
2-octet integer	3
IETF RFC 2685 VPN ID	4
Reserved for IEEE Std 802.1	5–31
ICC-based format as specified in ITU-T G.8013/Y.1731	32
Defined by ITU-T G.8013/Y.1731	33–63
Reserved for IEEE Std 802.1	64–255

^a This is an IETF RFC 2579 DisplayString, with the exception that character codes 0–31 (decimal) are not used.

21.6.5.5 Short MA Name Length

(1 octet) Specifies the length in octets of the Short MA Name field.

Validation Test: The Short MA Name Length in a CCM contains a value greater than or equal to 1.

Validation Test: The Short MA Name Length in a CCM does not indicate that the Short MA Name runs over the 48-octet limit for the MAID.

21.6.5.6 Short MA Name

(Length specified by the Short MA Name Length field) Contains the Short MA Name, in the format specified by the Short MA Name Format field.

21.6.6 Defined by ITU-T G.8013/Y.1731

(16 octets) The use of this field is defined by ITU-T G.8013/Y.1731. A value of 0 should be transmitted in this field.

21.6.7 Optional CCM TLVs

The following TLVs should be included, as allowed by their specifications, in every CCM transmitted:

- a) Sender ID TLV (21.5.3)
- b) Port Status TLV (21.5.4)
- c) Interface Status TLV (21.5.5)

The following TLV may be included in any CCM transmitted:

- d) Organization-Specific TLV (21.5.2)

The Optional CCM TLVs in this subclause may be placed in any order by the MEP transmitting the CCM. The receiving MP shall not depend upon any particular ordering of the Optional CCM TLVs in a CCM for its proper operation.

21.7 LBM and LBR formats

The format of the LBM and LBR is shown in Table 21-20.

Table 21-20—LBM and LBR formats

Field	Octet
Common CFM Header	1–4
Loopback Transaction Identifier	5–8
Reserved for definition in future versions of the protocol ^a	
Additional LBM/LBR TLVs	First TLV Offset + 5 ^b
End TLV (0)	First TLV Offset + 5, if no Additional LBM/LBR TLVs are present

^a This field has 0 length in this version 0 of CFM. It is shown in order to stress that additional information can be present in future versions of CFM, and that a version 0 receiver ignores its contents, if present.

^b Octet 9 for transmitted LBMs.

21.7.1 Flags

(1 octet) In an LBM, the Flags field of the Common CFM Header is set to 0 by the transmitting MP, and is not examined by the receiving MP.

21.7.2 First TLV Offset

(1 octet) The First TLV Offset field of the Common CFM Header in an LBM or LBR is transmitted as 4.

Validation Test: The First TLV Offset field of the Common CFM Header in an LBM or LBR contains a value greater than or equal to 4.

21.7.3 Loopback Transaction Identifier

(4 octets) A MEP copies the contents of the nextLBMtransID variable (20.30.2) to this field.

21.7.4 Additional LBM/LBR TLVs

The following TLV shall be included in every LBM transmitted by a MEP to a MIP configured on a PBB-TE MA as specified in 21.7.5. The TLV will not be present in any other type of MA.

a) PBB-TE MIP TLV (21.7.5)

The following TLV should be included in every LBM transmitted:

b) Sender ID TLV (21.5.3)

The following TLV can be included by management action in any LBM transmitted:

c) Data TLV (21.5.6)

The following TLV may be included in any LBM transmitted:

d) Organization-Specific TLV (21.5.2)

The PBB-TE MIP TLV, if present, shall be placed first. The other Additional LBM/LBR TLVs may be placed in any order by the MEP transmitting the LBM.

21.7.5 PBB-TE MIP TLV

Type = 9. Used only in MAs monitoring PBB-TE services. It enables MIPs to selectively intercept LBMs that are targeting them. The PBB-TE MIP TLV is not used if the target address of the LBM or LTM is associated with any of the values in the ESP-DA field of the monitored MA's ESPs; otherwise, it shall be present; The format of the PBB-TE MIP TLV is illustrated in Table 21-21.

Table 21-21—PBB-TE MIP TLV format

Field	Octet
Type = 9	1
Length	2–3
MIP MAC address	4–9
Reverse VID	10–11
Reverse MAC	12–17

1 **Validation Test:** The MIP MAC address field contains an individual MAC address.

2 **Validation Test:** The Reverse MAC field contains an Individual, or a group MAC address as specified in
3 21.7.5.3.

4 **Validation Test:** The Length field either is 8, indicating that no Reverse MAC field is present, or is 14, and
5 thus contains the Reverse MAC field.

6 **21.7.5.1 MIP MAC address**

7 The MIP MAC address contains the destination MAC address for LBMs transmitted by the MEP [item b) in
8 12.14.7.3.2]. In the case of LTMs, this field is null.

9 **21.7.5.2 Reverse VID**

10 The Reverse VID field is used by the MIPs in order to set the `vlan_identifier` parameters on LBRs and LTRs.
11 Its value is provided by the operator on the initiation of the LBM [item f) in 12.14.7.3.2] or the LTM [item e)
12 in 12.14.7.4.2]. In the case of point-to-point TESIs, its value is the ESP-VID of the MA's ESP that has the
13 MEP's MAC address in its ESP-DA field.

14 **21.7.5.3 Reverse MAC**

15 This Reverse MAC field refers to the ESP-SA of any of the point-to-point ESPs of the MA, if the LBM/LTM
16 is initiated on a root PBB-TE MEP or to the ESP-DA of the point-to-multipoint ESP of the MA, if the
17 LBM/LTM is initiated on any of the leaf MEPs.

18 **21.8 LTM format**

19 The format of the LTM is shown in Table 21-22.

Table 21-22—LTM format

Field	Octet
Common CFM Header	1–4
LTM Transaction Identifier	5–8
LTM TTL	9
Original MAC Address	10–15
Target MAC Address	16–21
Reserved for definition in future versions of the protocol ^a	
Additional LTM TLVs	First TLV Offset + 5 ^b
End TLV (0)	First TLV Offset + 5, if no Additional LTM TLVs are present

^a This field has 0 length in this version 0 of CFM. It is shown in order to stress that additional information can be present in future versions of CFM and that a version 0 receiver ignores its contents, if present.

^b Octet 22 for transmitted LTMs.

21.8.1 Flags

(1 octet) In the LTM, the Flags field of the Common CFM Header specifies certain options as shown in Table 21-23.

Table 21-23—LTM Flags field

Mnemonic	Meaning	Bit
UseFDBonly	If set, indicates that only MAC addresses learned in a Bridge's FDB, and not information saved in the MIP CCM Database, is to be used to determine the Egress Port.	8 (MSB)
Reserved	Transmitted as 0, and ignored by the receiver.	7–1

21.8.2 First TLV Offset

(1 octet) The First TLV Offset field of the Common CFM Header in an LTM is transmitted as 17.

Validation Test: The First TLV Offset field of the Common CFM Header in an LTM contains a value greater than or equal to 17.

21.8.3 LTM Transaction Identifier

(4 octets) A MEP copies the contents of the nextLTMtransID variable (20.41.1) to this field.

21.8.4 LTM TTL

(1 octet) The number of hops remaining to this LTM. Decrement by 1 by each Linktrace Responder that handles the LTM. One less than this value is returned in the LTR. If 0 or 1 on input, the LTM is not transmitted to the next hop. If 0 on input, no LTR is returned. The value of the LTM TTL field in the LTM transmitted by the originating MEP is controlled by a managed object [item d) in 12.14.7.4.2]; the default value if none is specified is 64.

21.8.5 Original MAC Address

(6 octets) The MAC address of the MEP that originated the LTM. This can be different from the source MAC address of an LTM because each MIP along the path puts its own MAC address in the source MAC address field, while retaining the Original MAC Address Field.

Validation Test: The Original MAC Address field contains an Individual, and not a Group, MAC address.

21.8.6 Target MAC Address

(6 octets) Specifies an individual MAC address, the path to which the LTM is intended to trace. Any individual MAC address can be specified in this field. See also 20.3.1.

Validation Test: The Target MAC Address field contains an Individual, and not a Group, MAC address.

21.8.7 Additional LTM TLVs

The following TLV shall be included in every LTM by each MP originating or forwarding an LTM:

- a) LTM Egress Identifier TLV (21.8.8)

The following TLVs may be included in an LTM by each MP originating or forwarding an LTM:

- b) Sender ID TLV (21.5.3)
- c) Organization-Specific TLV(s) (21.5.2)

The following TLV shall be included in every LTM transmitted by a MEP configured on a PBB-TE MA.

The TLV will not be present in any other type of MA.

- d) PBB-TE MIP TLV (21.7.5)

The Additional LTM TLVs may be placed in any order by the MP transmitting the LTM.

21.8.8 LTM Egress Identifier TLV

Type = 7. Identifies the MEP Linktrace Initiator that is originating, or the Linktrace Responder that is forwarding, this LTM. The low-order (highest numbered) six octets contain an EUI-48 IEEE MAC address unique to the system in which the MEP Linktrace Initiator or Linktrace Responder resides. The high-order (lowest numbered) two octets contain an integer value sufficient to uniquely identify the transmitted LTM; a constant value (e.g., 0) is sufficient for this purpose, since a Bridge initiates or forwards only a single copy of an LTM. The format of the LTM Egress Identifier TLV is illustrated in Table 21-24.

Table 21-24—LTM Egress Identifier TLV format

Field	Octet
Type = 7	1
Length	2–3
Egress Identifier	4–11

NOTE—For most Bridges, the address of any MAC attached to the Bridge will suffice for the low-order six octets, and 0 for the high-order octets. In some situations, e.g., if multiple virtual Bridges utilizing emulated LANs are implemented in a single physical system, the high-order two octets can be used to differentiate among the transmitting entities.

21.9 LTR format

The format of the LTR is shown in Table 21-25.

Validation Test: The Reply Ingress TLV (21.9.8), the Reply Egress TLV (21.9.9), or both, are present.

21.9.1 Flags

(1 octet) In the LTR, the Flags field of the Common CFM Header specifies certain options as shown in Table 21-26.

Table 21-25—LTR format

Field	Octet
Common CFM Header	1–4
LTR Transaction Identifier	5–8
Reply TTL	9
Relay Action	10
Reserved for definition in future versions of the protocol ^a	
Additional LTR TLVs	First TLV Offset + 5 ^b
End TLV (0)	First TLV Offset + 5, if no Additional LTR TLVs are present

^a This field has 0 length in this version 0 of CFM. It is shown in order to stress that additional information can be present in future versions of CFM, and that a version 0 receiver ignores its contents, if present.

^b Octet 11 for transmitted LTRs.

Table 21-26—LTR Flags field

Mnemonic	Meaning	Bit
UseFDBonly	Copied from LTM.	8 (MSB)
FwdYes	The LTM was (1) or was not (0) forwarded.	7
TerminalMEP	The MP reported in the Reply Egress TLV (Reply Ingress TLV, if the Reply Egress TLV is not present) is a MEP.	6
Reserved	Copied from LTM.	5–1

1 21.9.2 First TLV Offset

2 (1 octet) The First TLV Offset field of the Common CFM Header in an LTR is transmitted as 6.

3 **Validation Test:** The First TLV Offset field of the Common CFM Header in an LTR contains a value greater
4 than or equal to 6.

5 21.9.3 LTR Transaction Identifier

6 (4 octets) The value from the LTM Transaction Identifier field in the LTM that triggered the transmission of
7 this LTR.

8 21.9.4 Reply TTL

9 (1 octet) One less than the value from the LTM TTL field in the LTM that triggered the transmission of this
10 LTR. If the LTM TTL field contained a 0, no LTR is transmitted.

21.9.5 Relay Action

(1 octet) Reports how the data frame targeted by the LTM would be passed through the MAC Relay Entity to the Egress Bridge Port, as shown in Table 21-27.

Validation Test: The value in the Relay Action field is one of the values enumerated in 21.9.5.

Table 21-27—Relay Action field values

Mnemonic	Relay action	Value
RlyHit	The LTM reached an MP whose MAC address matches the target MAC address.	1
RlyFDB	The Egress Port was determined by consulting the FDB [item a) in 20.47.1.2].	2
RlyMPDB	The Egress Port was determined by consulting the MIP CCM Database [item b) in 20.47.1.2].	3

21.9.6 Additional LTR TLVs

The following TLV shall be included in an LTR according to 20.3.2:

a) LTR Egress Identifier TLV (21.9.7)

One or both of the following TLVs shall be included in an LTR according to 20.3.2:

b) Reply Ingress TLV (21.9.8)

c) Reply Egress TLV (21.9.9)

The following TLVs may be included in an LTR:

d) Sender ID TLV (21.5.3)

e) Organization-Specific TLV (21.5.2)

The Additional LTR TLVs may be placed in any order by the MP transmitting the LTR. The receiving MEP shall not depend upon any particular ordering of the Additional LTR TLVs in an LTR for its proper operation.

21.9.7 LTR Egress Identifier TLV

Type = 8. Identifies the source and destination of the LTM that triggered transmission of this LTR. The format of the LTR Egress Identifier TLV is illustrated in Table 21-28.

Table 21-28—LTR Egress Identifier TLV format

Field	Octet
Type = 8	1
Length	2–3
Last Egress Identifier	4–11
Next Egress Identifier	12–19

21.9.7.1 Last Egress Identifier

(8 octets) Identifies the MEP Linktrace Initiator that originated, or the Linktrace Responder that forwarded, the LTM to which this LTR is the response. This field takes the same value as the LTM Egress Identifier TLV of that LTM, or the value 0, if no LTM Egress Identifier TLV was present in that LTM.

21.9.7.2 Next Egress Identifier

(8 octets) Identifies the Linktrace Responder that transmitted this LTR, and can forward the LTM to the next hop. This field takes the same value as the LTM Egress Identifier TLV of the forwarded LTM, if any. If the FwdYes bit of the Flags field is false, the contents of this field are undefined, i.e., any value can be transmitted, and the field is ignored by the receiver.

21.9.8 Reply Ingress TLV

Type = 5. The format of the Reply Ingress TLV is shown in Table 21-29. The Reply Ingress TLV is transmitted if and only if the Bridge Port on which the LTM was received has an MP in the MA specified by the LTM.

Table 21-29—Reply Ingress TLV format

Field	Octet
Type = 5	1
Length	2–3
Ingress Action	4
Ingress MAC Address	5–10
Ingress Port ID Length	11
Ingress Port ID Subtype	12
Ingress Port ID	13 – (Length + 3)

Validation Test: The value in the Ingress Action field is one of the values enumerated in 21.9.8.1.

Validation Test: The Ingress MAC Address field contains an Individual, and not a Group, MAC address.

Validation Test: The Length field either is 7, indicating that no Ingress Port ID field is present, or is (9 + the Ingress Port ID Length field) or larger, and thus contains the Ingress Port ID field.

21.9.8.1 Ingress Action

(1 octet) Reports how the data frame targeted by the LTM would be received on the receiving MP, as shown in Table 21-30.

21.9.8.2 Ingress MAC Address

(6 octets) The MAC address that can be used as the Destination MAC address of an LBM intended for the ingress MP.

Table 21-30—Ingress Action field values

Mnemonic	Ingress action	Value
IngOK	The target data frame would be passed through to the MAC Relay Entity.	1
IngDown	The Bridge Port's MAC_Operational parameter is false. ^a	2
IngBlocked	The target data frame would not be forwarded if received on this Port due to active topology enforcement (8.6.1). (See Figure 22-1 to see how this is possible.)	3
IngVID	The ingress port is not in the member set of the LTM's VID, and ingress filtering is enabled, so the target data frame would be filtered by ingress filtering (8.6.2).	4

^a This value could be returned, for example, by an operational Down MEP that has another Down MEP at a higher MD Level on the same Bridge Port that is causing the Bridge Port's MAC_Operational parameter to be false.

21.9.8.3 Ingress Port ID Length

(1 octet) The length, in octets, of the Ingress Port ID field. Cannot be 0. If the Length field of the Reply Ingress TLV is less than 8, then the Ingress Port ID Length field, Ingress Port ID Subtype field, and Ingress Port ID field are not present.

21.9.8.4 Ingress Port ID Subtype

(1 octet) The **port ID subtype**, as specified by IEEE Std 802.1AB, identifying the format of the Ingress Port ID field. If the Length field of the Reply Ingress TLV is less than 8, then the Ingress Port ID Subtype field is not present.

21.9.8.5 Ingress Port ID

(Length specified by the Ingress Port ID Length field) The Ingress Port, identified by the **port ID** specified by IEEE Std 802.1AB, in the chassis identified in the Sender ID TLV. If the Length field of the Reply Ingress TLV is less than 8, then the Ingress Port ID field is not present.

21.9.9 Reply Egress TLV

Type = 6. The format of the Reply Egress TLV is shown in Table 21-31. The Reply Egress TLV is included in the LTR if and only if the Bridge Port on which the LTM would be relayed could be identified, and is an MP in the MA specified by the LTM.

Validation Test: The value in the Egress Action field is one of the values enumerated in 21.9.9.1.

Validation Test: The Egress MAC Address field contains an Individual, and not a Group, MAC address.

Validation Test: The Length field either is 7, indicating that no Egress Port ID field is present, or is (9 + the Egress Port ID Length field) or larger, and thus contains the Egress Port ID field.

21.9.9.1 Egress Action

(1 octet) Reports how the data frame targeted by the LTM would be passed through Egress Bridge Port, as shown in Table 21-32.

Table 21-31—Reply Egress TLV format

Field	Octet
Type = 6	1
Length	2–3
Egress Action	4
Egress MAC Address	5–10
Egress Port ID Length	11
Egress Port ID Subtype	12
Egress Port ID	13 – (Length + 3)

Table 21-32—Egress Action field values

Mnemonic	Egress action	Value
EgrOK	The targeted data frame would be forwarded.	1
EgrDown	The Egress Port can be identified, but that Bridge Port’s MAC_Operational parameter is false.	2
EgrBlocked	The Egress Port can be identified, but the data frame would not pass through the Egress Port due to active topology management (8.6.1), i.e., the Bridge Port is not in the Forwarding state.	3
EgrVID	The Egress Port can be identified, but the Bridge Port is not in the LTM’s VID’s member set, so would be filtered by egress filtering (8.6.4).	4

1 21.9.9.2 Egress MAC Address

2 (6 octets) The MAC address that can be used as the Destination MAC address of an LBM intended for the
3 egress MP.

4 21.9.9.3 Egress Port ID Length

5 (1 octet) The length, in octets, of the Egress Port ID field. Cannot be 0. If the Length field of the Reply
6 Egress TLV is less than 8, then the Egress Port ID Length field, Egress Port ID Subtype field, and Egress
7 Port ID field are not present.

8 21.9.9.4 Egress Port ID Subtype

9 (1 octet) The **port ID subtype**, as specified by IEEE Std 802.1AB, identifying the format of the Egress Port
10 ID field. If the Length field of the Reply Egress TLV is less than 8, then the Egress Port ID Subtype field is
11 not present.

12 21.9.9.5 Egress Port ID

13 (Length specified by the Egress Port ID Length field) The Egress Port, identified by the **port ID** specified by
14 IEEE Std 802.1AB, in the chassis identified in the Sender ID TLV. If the Length field of the Reply Egress
15 TLV is less than 8, then the Egress Port ID field is not present.

22. CFM in systems

The relationships among MPs, and between the MPs and the other entities in a Bridge, are configurable. This configuration determines how CFM is used in a network and determines what parts of the Bridges and LANs comprising a network are protected by an MA. This clause provides both specifications and explanatory text to assist the implementor and system administrator to make efficient use of CFM. This clause:

- a) Specifies how the EISS Multiplex Entity (6.17) is used to support MPs for multiple MAs associated with service instances supported by EISS-SAPs (22.1).
- b) Specifies the creation of MPs in a Bridge (22.2).
- c) Suggests methods for the efficient assignment of MD Levels to Maintenance Domains (22.3).
- d) Specifies the use of MPs in stations (22.4).
- e) Clarifies issues related to the scaling of resources required to run CFM (22.5).
- f) Shows how MPs can be placed in Provider Bridges (22.6).
- g) Suggests methods for using CFM in the enterprise environment, as opposed to the provider environment (22.7).
- h) Provides guidance for the implementation of CFM in Bridges designed before the development of this standard (22.8).

NOTE—Clause 18 introduces the principles of CFM operation and the network architectural concepts that support it. Clause 19 breaks down the CFM protocol entities into their components. Clause 20 specifies the protocols operated by the components of each MP, and Clause 21 specifies the PDU formats used by those protocols.

22.1 CFM shims in Bridges

CFM is instantiated by configuring an MA, which maintains a single service instance. If multiple MAs, monitoring VLAN service instances, are configured on a single Bridge Port, the `vlan_identifier` parameter of the EISS and the MD Level in the CFM PDUs are used to multiplex CFM PDUs for the various MAs. Non-CFM data frames for these service instances are distinguished only by `vlan_identifier`: the particular service instance to which a non-CFM data frame belongs, if multiple service instances share the same VID, distinguished only by MD Level, is undetermined. MAs that monitor backbone or TESIs use I-SIDs or a combination of destination addresses, source addressees, and ESP-VIDs, respectively as service distinguishing parameters, instead of VIDs.

22.1.1 Preliminary positioning of MPs

Figure 22-1 illustrates how MPs are associated with particular VIDs or TESIs. The EISS Multiplex Entity (6.17) distributes frames across an array of ISS- or EISS-SAPs, each of which can have a “column” of MPs at different MD Levels and orientations (Up or Down). Several MEPs and MHFs are shown, distinguished by VID and MD Level in this manner.

NOTE—For MPs monitoring backbone service instances, it is the Backbone Service Instance Multiplex Entity (6.18) that is used to distinguish MPs by I-SIDs and associate MPs with particular backbone service instances as illustrated in Figure 26-2.

An MP that has its Passive SAP closer to the MAC Relay Entity than its Active SAP is called a “Down MEP” or “Down MHF,” because it points down, away from the MAC Relay Entity, in Figure 22-1. An MP that has its Passive SAP further away from the MAC Relay Entity than its Active SAP is called an “Up MEP” or “Up MHF.” The Bridge Port in Figure 22-1 is configured with five Up MEPs, three Up MHFs, three Down MHFs, and three Down MEPs. A single LOM is positioned below the per-VID Down MEPs.

Figure 22-1 illustrates a Down MEP, at MD Level 1, configured on the ISS below the Bridge Port connectivity entity. The service instance its MA protects encompasses all data frames passing through the Bridge Port, no matter what VID they are associated with.

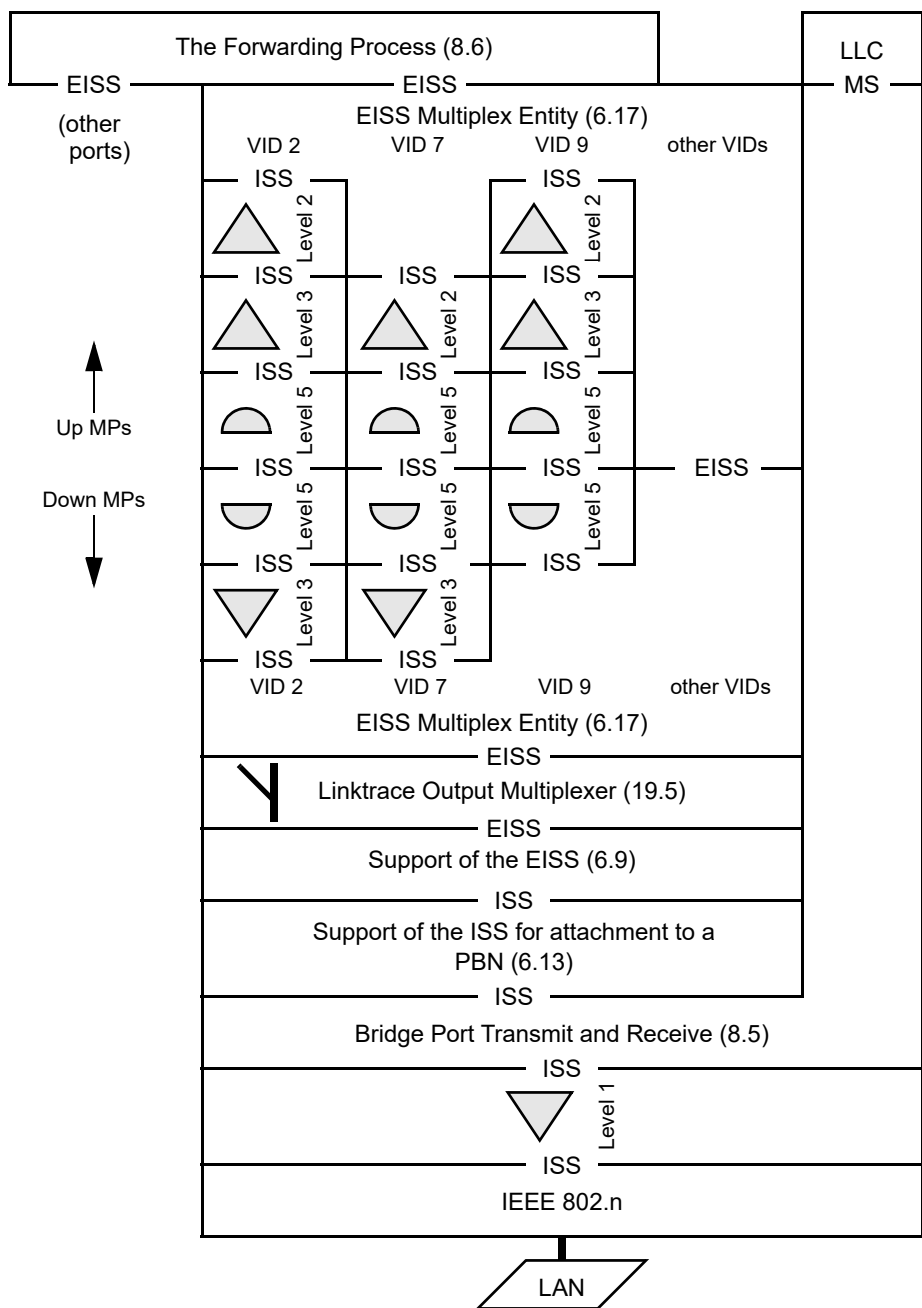


Figure 22-1—MEPs and MIPs distinguished by VID (incomplete picture)

22.1.2 CFM and the Forwarding Process

Figure 22-1 is incomplete, because it does not illustrate the relationship of CFM to the various components of the Forwarding Process (8.6). These components are illustrated in Figure 8-12. Readers can notice two facts about Figure 8-12:

- a) The functions in Figure 8-12 can be divided into two groups. The first group, described in 8.6.1 through 8.6.5, consists of filtering functions that determine a set of potential transmission ports for

- 1 each frame received at a reception port. The second group, described in 8.6.6 through 8.6.8, consists
2 of queuing functions that control the transmission of the frame at each port in the resulting set of
3 potential transmission ports.
4 b) The parameters passed from component to component in Figure 8-12 are exactly those of the EISS;
5 therefore, each of the components in Figure 8-12 could be described as a shim.

6 From these facts, readers can reconstruct Figure 8-12 in a manner that is functionally identical, but is better
7 suited to illustrate the relationship between CFM and the Forwarding Process. This reconstruction is shown
8 in Figure 22-2.

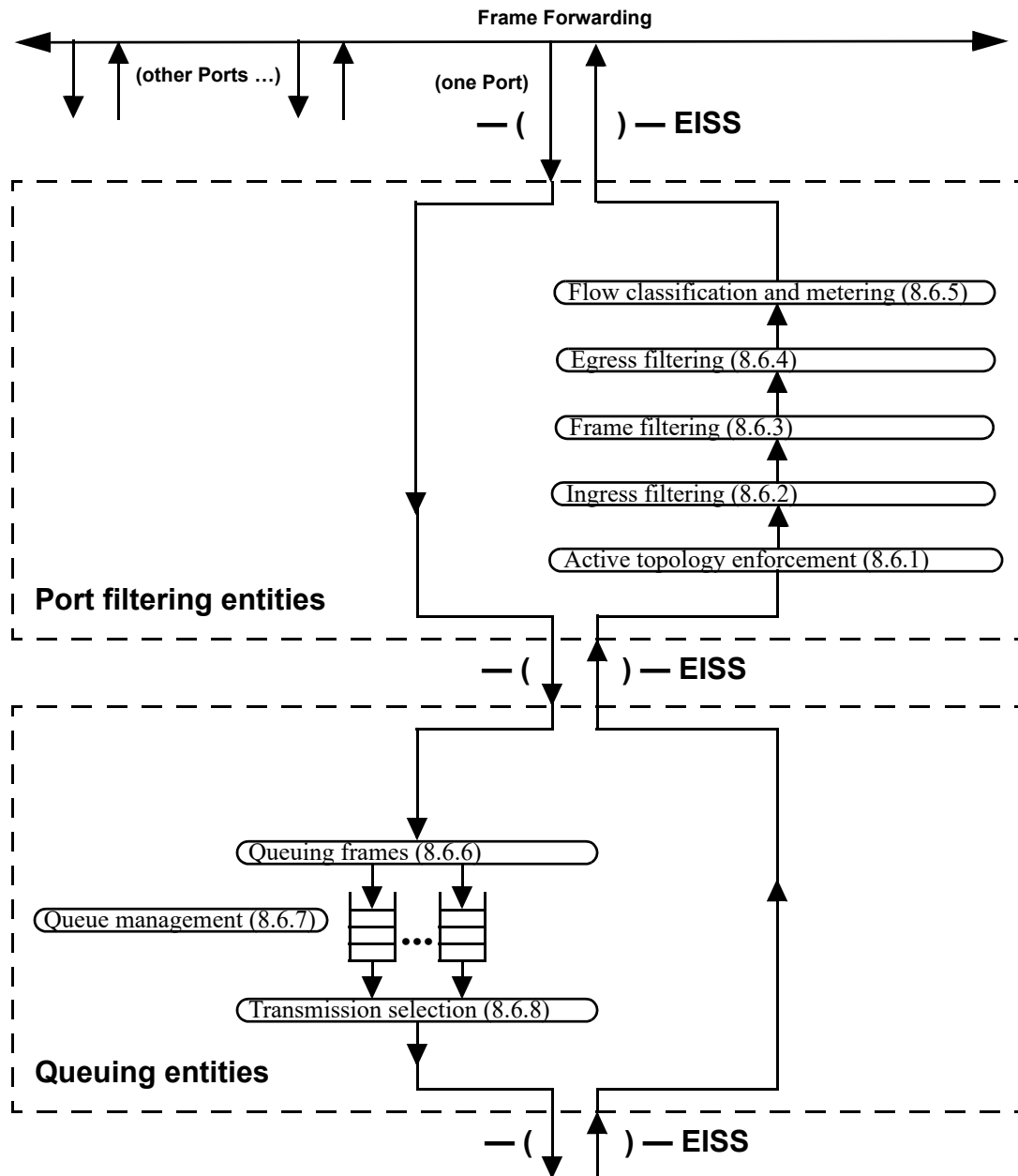


Figure 22-2—Alternate view of Forwarding process

22.1.3 Up/Down separation of MPs

An additional transformation to Figure 22-1 is required before the relationship between the Forwarding Process and CFM can be shown clearly. By adding a further pair of EISS Multiplex Entities between the Up MPs and the Down MPs in Figure 22-1, the entire set of Up MPs can be shown as a single shim, as can the entire set of Down MPs. This transformation is illustrated in Figure 22-3.

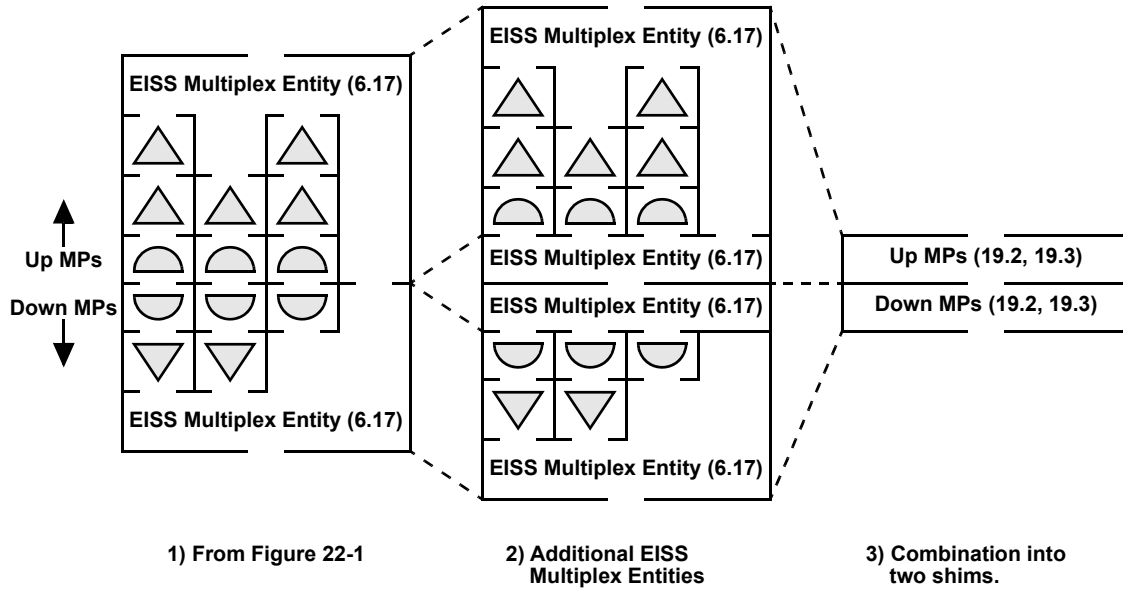


Figure 22-3—Combining per-VLAN MPs into two shims

Figure 22-4 combines Figure 22-1 with Figure 22-2, and shows the relationship of CFM to the various components of the Forwarding Process. MEPs and MHFs source and sink CFM PDUs. CFM PDUs are treated by the Port filtering entities in the same manner as ordinary data. Thus, the five Up MEPs of Figure 22-4 can transmit and receive CFM PDUs through the Bridge even when the Bridge Port is blocked. Similarly, the two Down MEPs in Figure 22-4 operate normally, transmitting CFM PDUs to and receiving them from the LAN, even if the Bridge Port is blocked. However, no frames, neither CFM PDUs nor ordinary data, can pass through the Port filtering entities when the VID to which those frames or CFM PDUs belong is filtered.

Also shown in Figure 22-4 are the positions of the DoSAPs and ISAPs introduced in Clause 18. The DoSAP provides access to a service instance (an instance of the MAC Service) and exists independently of any MA that might be created to protect it. The DoSAPs and ISAPs therefore can be associated with the specific inter-shim SAPs at the points indicated in Figure 22-4. The DoSAPs and ISAPs are attached to the Port filtering entities, because that is where the decisions are made on whether frames belonging to the service instance are or are not allowed to pass through the Bridge Port.³⁷ It does not matter to which side of the EISS Multiplex Entity one considers the DoSAPs and ISAPs to be attached. Except for the topmost Down MEP in each stack of Down MPs, the DoSAP is not coincident with the Passive SAP of the MEP protecting the DoSAP's service instance; the DoSAP is separated from its MA by the MHFs and/or MEPs at higher MD Levels. Even for that topmost Down MEP, an MHF for a higher MD Level can separate that MEP from its DoSAP. This gap is illustrated in Figure 22-5 and Figure 22-6, where the service instance, DoSAP to DoSAP, is shown in gray, but the MEPs protecting that service are separated from the DoSAPs by MHFs belonging to a service instance at a higher MD Level.

³⁷ The DoSAP is not attached to an MP's Passive SAP because the DoSAP and its associated service instance exist regardless of whether the service instance is protected by an MA.

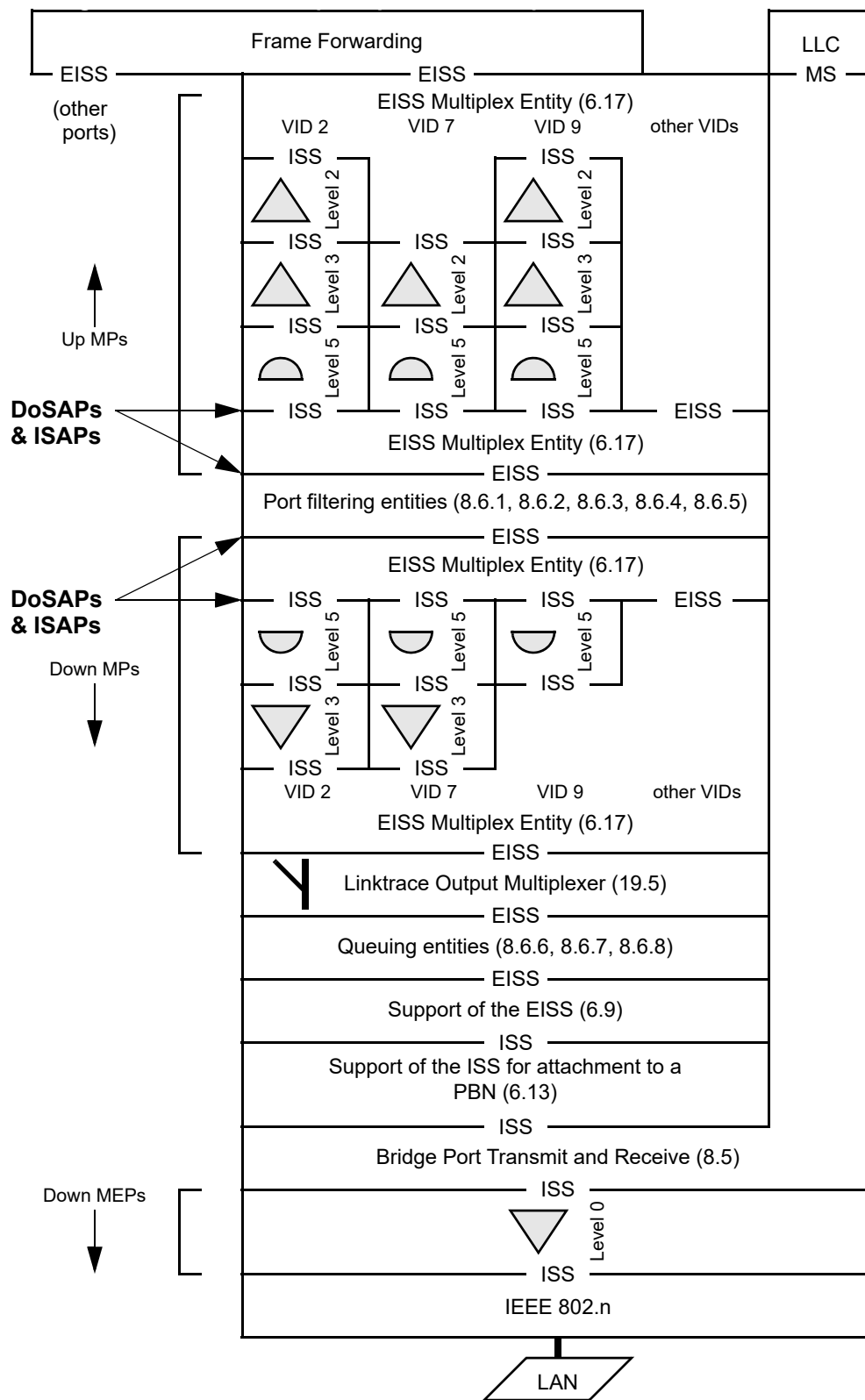


Figure 22-4—More complete picture of MP placement in a Bridge Port

1 In the “stack” of MPs on a single Bridge Port, the following rules apply:

- 2 a) All Up MEPs belonging to MAs that are attached to specific VIDs are placed above the Port filtering
3 entities (8.6.1, 8.6.2, and 8.6.4). Separately for each VID, there can be from zero to eight Up MEPs,
4 ordered by increasing MD Level, from the upper EISS toward Port filtering. In the case of PBB-TE
5 MAs, more than one Up MEP can share the VID and MD Level.
- 6 b) For each MA there can be at most one MIP, consisting of an Up MHF just above, and a Down MHF
7 just below, the Port filtering entities. Both MHFs are at the same MD Level, higher than the highest
8 MEP on the same VID, if any. Backbone Service Instance MHFs use the Backbone Service Instance
9 Multiplex Entity (6.18) and are placed as depicted in Figure 26-2.
- 10 c) Up to eight Down MEPs belonging to MAs that are attached to the same list of VIDs are placed
11 between the Down MHF, if any, and the Queuing entities (8.6.6, 8.6.7, and 8.6.8). They are placed in
12 order of decreasing MD Level, from the Port filtering entities to the Queuing entities, including any
13 Down MHF.
- 14 d) No Down MEPs exist for PBB-TE MAs.
- 15 e) An LOM is positioned below the Down MPs belonging to MAs that are attached to specific VIDs to
16 handle forwarded LTMs.
- 17 f) Up to eight Down MEPs, ordered from highest MD Level to lowest, belonging to MAs not attached
18 to any specific VID, can be placed below Bridge Port connectivity (8.5.1). They are ordered by
19 decreasing MD Level going away from the MAC Relay Entity (see also 22.1.8).
- 20 g) Up to 8 Up MEPs, ordered from highest MD Level to lowest, belonging to Backbone Service
21 Instance MAs that are identified by the same I-SID, can be placed below the Bridge Port
22 connectivity (8.5.1) using the Backbone Service Instance Multiplex Entity (6.18) as depicted in
23 Figure 26-2. They are ordered by decreasing MD Level going away from the multiplexed SAP.
- 24 h) Up to 8 Down MEPs, ordered from highest MD Level to lowest, belonging to Backbone Service
25 Instance MAs that are identified by the same I-SID, can be placed below the Bridge Port
26 connectivity (8.5.1) using the Backbone Service Instance Multiplex Entity (6.18) as depicted in
27 Figure 26-2. They are ordered by decreasing MD Level going away from the MAC Relay Entity.
- 28 i) All of the Down MEPs below Bridge Port connectivity have lower MD Levels than any Down MP
29 attached to a VID that is emitted untagged by the Support of the EISS.
- 30 j) Any shim that introduces an EtherType tag insulates the MPs above and below it, removing any MD
31 Level restriction between the separated groups of MPs.

32 A single Port on a Bridge could, in theory, have MHFs for multiple service instances at different MD Levels.
33 However, that would require a vector of 4094 MD Level registers instead of one register to identify the CFM
34 PDUs to be deflected to the Bridge’s Higher Layer Entities. This capability is optional.

35 22.1.4 Service instances over multiple Bridges

36 Figure 22-4 completes the sequence of expansions starting with Figure 18-5 and continuing with
37 Figure 18-6 and Figure 18-7. In this sequence, Figure 22-4 illustrates Bridge Port b of Bridge 2 in Figure 18-
38 5 through Figure 18-7. The service instances illustrated in those three figures all use VID 9 in Figure 22-4.
39 The Up MEPs for VID 9 at MD Levels 3 and 2 are the Up MEPs protecting the provider MD Level and the
40 operator MD Level service instances, respectively, and the pair of MHFs at MD Level 5 on VID 9 provide a
41 customer MD Level MIP. Figure 22-4 also illustrates two other sets of service instances sharing the same
42 Bridge Port using VIDs 2 and 7.

43 Additional views of service instances spanning two Bridges are shown in Figure 22-5 and Figure 22-6.
44 Figure 22-5 shows a service instance protected by Up MPs, and Figure 22-6 shows a service instance
45 protected by Down MPs.

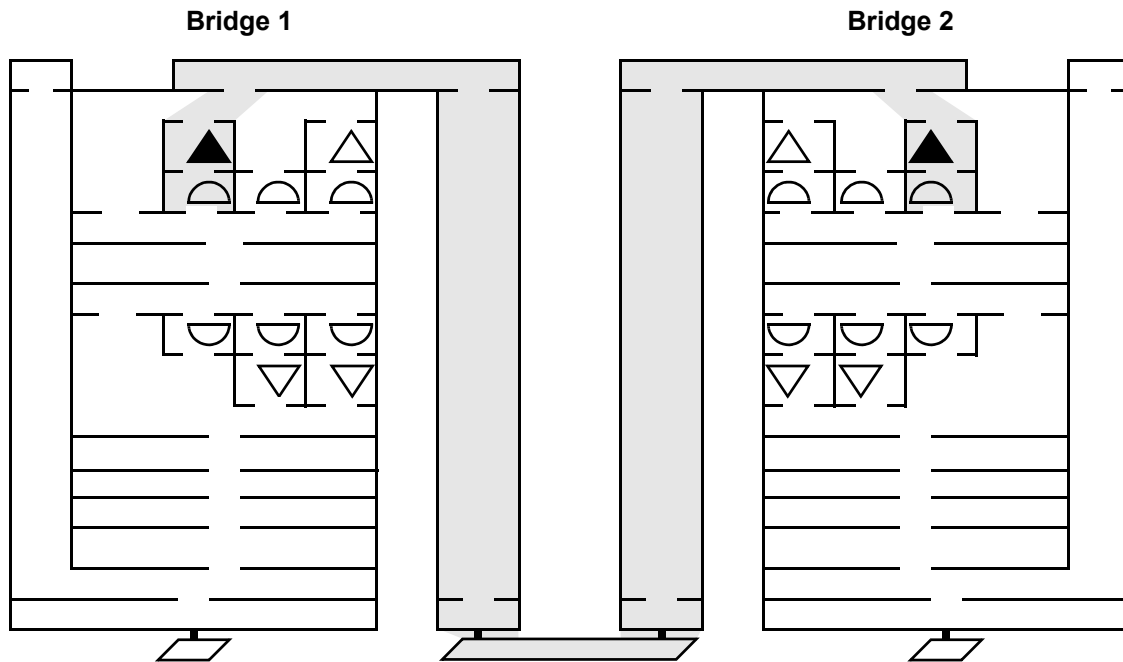


Figure 22-5—Service instance spanning two Bridges protected by Up MPs

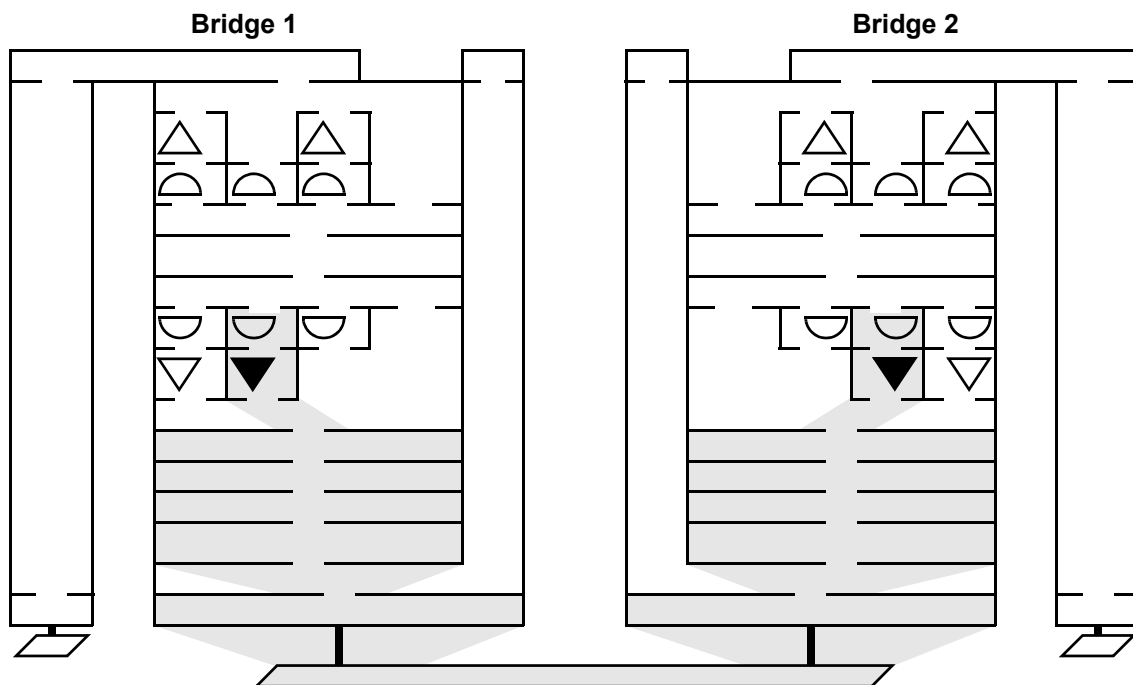


Figure 22-6—Service instance spanning two Bridges protected by Down MPs

1 22.1.5 Multiple VID service instances

2 A service instance can be implemented using multiple VIDs. It is always the case that the VLAN tag
3 uniquely determines the VLAN service instance for a given frame. A VID-based MA or a PBB-TE MA with
4 multiple VIDs does not require any more MIPs or MEPs than the corresponding MA with a single VID. All
5 of the VIDs belonging to a given VID or TESI share the same MA. That is, one MAID is used for all CFM
6 PDUs on the various VIDs in the same VID or TESI and Maintenance Domain. The choice of VID used to
7 tag a given CFM PDU depends on the OpCode and the ESP, as specified in Clause 20.

8 22.1.6 Untagged CFM PDUs

9 A MEP placed anywhere below the Support of the EISS entity (6.9) or the Bridge Port connectivity entity
10 (8.5.1) differs in three important respects from a MEP placed above the Support of the EISS entity:

- 11 a) The lower position would not have the Queuing entities available to it. For example, the reflection of
12 a high-speed stream of LBMs (as LBRs) could seriously disrupt the flow of data frames, and thus
13 bring into question the utility of the Queuing entities, CFM, or both.
- 14 b) A MEP in the lower position is necessarily VLAN untagged; the VLAN tagging functions are
15 performed in the Support of the EISS entity.
- 16 c) A MEP in the lower position protects all data passing through the Bridge Port or if it is associated
17 with a backbone service instance only data with this instance as specified in (26.8); it does not
18 separately protect the different VLANs.

19 Point c) above does not mean that the lower position is the only one that can protect all of the traffic on a
20 single link. MEPs placed above the Queuing entities can be associated with a VLAN whose VID is the
21 PVID, for which this Bridge Port belongs to the untagged set. That VID can be statically configured in the
22 member set to not be allowed to pass through the Frame filtering entity (8.6.3). MEPs so configured can then
23 transmit and receive only untagged frames, and thus protect all of the data on the link. Their untagged CFM
24 PDUs can be visible to complex “links” such as PBNs, even though the C-VLAN-tagged CFM PDUs are not
25 visible inside the PBN.

26 The option to create MEPs below the Bridge Port connectivity entity for VID identified service instances
27 cannot be omitted, however, because of considerations given in 22.1.8. This subclause also explains why
28 only MEPs, and not MHFs, can be placed below the Bridge Port connectivity entity.

29 22.1.7 MPs and non-VLAN-aware Bridges

30 The positioning of MPs in Bridges that are not VLAN aware (i.e., MAC Bridges) is illustrated in Figure 22-
31 7. Because there are no VLANs, no EISS Multiplex Entity is needed. However, the Up MPs are still above
32 the Port filtering entities, so the transformation introduced in 22.1.2 is still retained.

33 Down MPs can be placed either above or below the Queuing entities and Bridge Port connectivity in
34 Figure 22-7; the difference between the two locations is that:

- 35 a) In the upper location, frame queues are available to CFM, so that CFM frames and data frames can
36 be scheduled appropriately for output, and
- 37 b) Placing a Down MEP in the lower location protects the LLC path used by the Higher Layer Entities,
38 e.g., BPDUs, as well as the normal data forwarding path.

39

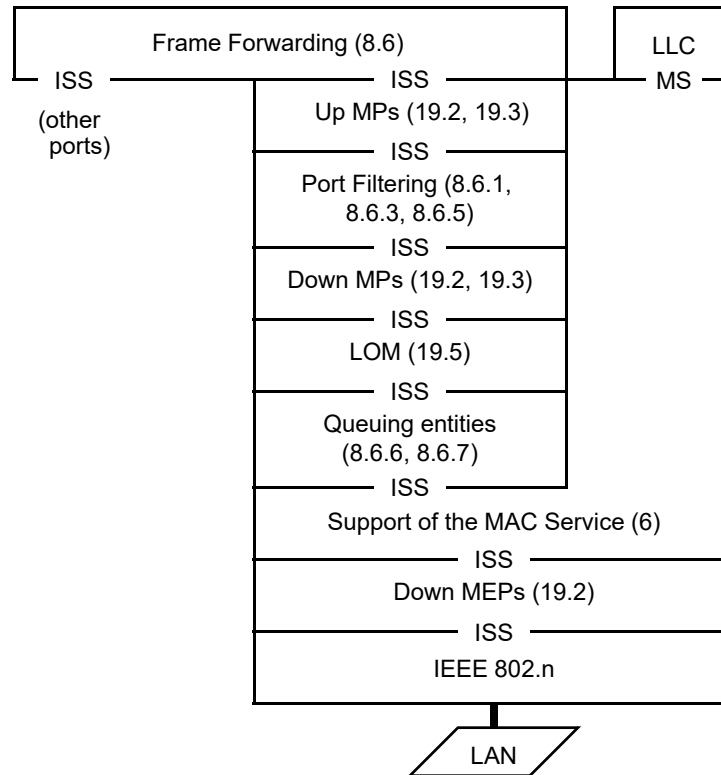


Figure 22-7—MP placement in a non-VLAN-aware Bridge Port

22.1.8 MPs and other standards

The relationship of MPs and the shims defined by two other standards, IEEE Std 802.1AX (Link Aggregation) and IEEE Std 802.1AE (MAC Security), is illustrated in Figure 22-8. This figure shows Down MEPs positioned both above and below Link Aggregation and illustrates a number of points.

MPs could, in theory, be placed either above or below the MAC Security shim. No loss of data integrity beyond the protected link would be incurred by placing CFM below MAC Security; the only attacks that could be generated would be variants of a denial of service attack. However, there are several reasons for restricting MPs to positions above the MAC Security shim:

- a) The denial of service attacks available by the introduction of bogus CFM PDUs have more potential for disrupting management operations, in more complex ways, than do attacks based on other unprotected protocols, namely those defined by IEEE Std 802.3. For example, bogus cross connect CCMs could be introduced, leading to undesirable interactions between different providers' management systems.
- b) MAC Security ensures that all data frames are legitimate, in the sense that they were last transmitted by a trusted device. Allowing the introduction of unprotected CFM PDUs violates the principle that data frames and CFM PDUs follow the same paths, distinguished only by the CFM protocol encapsulation.
- c) Allowing the CFM and MAC Security shims to be interchanged by management operations could significantly complicate the implementation of a Bridge, with no commensurate benefit to the user.

As described in 22.1.6, placing an MP below the Queuing entities, whether above or below the Link Aggregation shim, is not desirable, as that MP is not able to use the Queuing entities. For this reason, it is preferable to use a Down MEP that is attached to a specific VID, and is thus above the Queuing entities, to protect the link.

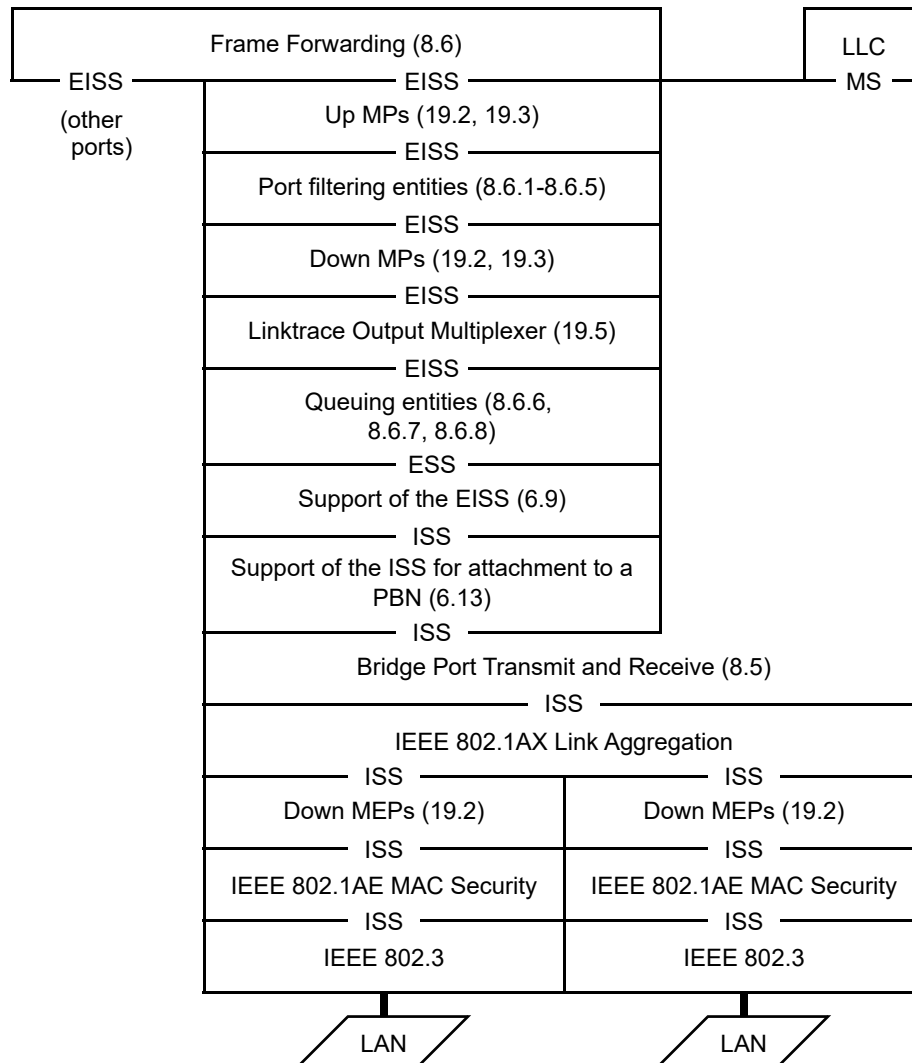


Figure 22-8—MP placement relative to other standards

1 When Link Aggregation is employed, Down MEPs attached to no VID can be useful for protecting the
2 individual IEEE 802.3 LANs. The method used in Link Aggregation to select the outgoing IEEE 802.3 link
3 is undefined. It is not necessarily possible for an MP above the Queuing entities to protect all of a Bridge
4 Port's data; CCM PDUs could take one IEEE 802.3 link, and data frames another. Therefore, provision is
5 made in this standard for configuring Down MEPs in an MA attached to no VID below the Bridge Port
6 connectivity entity, and below Link Aggregation, when employed. The following considerations are
7 important when configuring Down MEPs:

- 8 d) It is best to disable the Bridge Port before issuing a high-speed stream of LBRs (relative to the
9 physical link speed), either from or toward a Down MEP attached to no VID; otherwise, the
10 Queuing entities might be unable to achieve any guarantees for throughput or latency.
- 11 e) The rate of CCMs generated by Down MEPs attached to VIDs should be taken into account when
12 configuring the Queuing entities to achieve particular guarantees for throughput or latency.
- 13 f) If Link Aggregation is present, Down MEPs attached to no VID should be configured below Link
14 Aggregation.

- 1 g) The creation of MHFs below Link Aggregation presents two problems that preclude it from this
2 standard:
- 3 1) IEEE Std 802.3 provides no means for Link Aggregation to use a destination MAC address to
4 direct a CFM PDU down the correct link to reach a particular MP at one or the other end of
5 that link.
- 6 2) The utility of isolating the short segment between the Port filtering entities and the links being
7 aggregated, which is completely internal to a Bridge, is very limited, especially considering the
8 wide range of methods employed for implementing Link Aggregation.

9 22.1.9 CFM and IEEE 802.3 OAM

10 The following differences between CFM and the IEEE Std 802.3 specification of Operations,
11 Administration, and Maintenance (OAM) are relevant to making a decision whether to employ one, the
12 other, or both of these protocols.

- 13 a) IEEE 802.3 OAM has provision for a loopback mode in which all frames are returned to the sender.
14 CFM does not.
- 15 NOTE—IEEE 802.3 OAM confines loopbacked frames to the individual IEEE 802.3 LAN, effectively
16 disabling any attached Bridge Port. Use of simple frame loopback through a Bridged Network could be
17 expected to seriously impact the operation of the network.
- 18 b) IEEE 802.3 OAM has provision for acquiring statistics from a remote interface. CFM does not.
- 19 c) IEEE 802.3 OAM has provision for generating error conditions when configured thresholds are
20 exceeded for parameters such as the number of frames received with Cyclic Redundancy Check
21 (CRC) errors. CFM does not.
- 22 d) IEEE 802.3 OAM has provision for organizations other than IEEE 802 to define both new TLVs and
23 new operation codes using OUIs/CIDs. CFM allows only TLVs to be defined using OUIs/CIDs.
- 24 e) IEEE 802.3 OAM performs a continuity check once per second. CFM can be programmed to
25 perform the continuity check over a wide range of intervals.
- 26 f) IEEE 802.3 OAM is defined only for IEEE 802.3 media. CFM can be used over any medium, or
27 bridged network, that can carry IEEE 802 frames.
- 28 g) IEEE 802.3 OAM protects a single link. CFM can protect a single link or a network of bridged
29 networks, over multiple nested ranges.
- 30 h) CFM can protect the connectivity of shared media. IEEE 802.3 OAM has no such capability.
- 31 i) CFM can perform in-band loopback tests at any rate up to the physical line rate, during normal
32 operations. IEEE 802.3 OAM has no such capability.
- 33 j) CFM can perform Linktrace functions through a Bridged Network. IEEE 802.3 OAM has no such
34 capability.
- 35 k) CFM can detect unintentional connectivity, as well as lack of connectivity. IEEE 802.3 OAM
36 cannot.

37 22.2 Maintenance Entity creation

38 MPs are created using the managed objects described in 12.14. Before a MEP or MHF can be created, either
39 an entry in the Default MD Level managed object, or the Maintenance Domain and MA to which the MEP
40 or MHF belongs, are created. In this context, a Default MD Level managed object, Maintenance Domain, or
41 an MA are strictly managed objects in a single Bridge. In order to realize the network-wide definitions of
42 Maintenance Domains and MAs given in 18.1 and 18.2, the Bridges in a Maintenance Domain are
43 configured with identical information in these managed objects. During transitional periods when this
44 information is changing, Fault Alarms will likely be generated.

45 NOTE—The correct operation of CFM does not depend on the configuration restrictions in this subclause. Rather,
46 violating these restrictions is likely to generate Fault Alarms that might or might not be erroneous. In other words, if no
47 Fault Alarms are generated, then all monitored MAs have connectivity. If any Fault Alarm is generated, then either

1 connectivity has been lost or the network is improperly configured. Determining which of these two is the cause of a
2 Fault Alarm depends on the intentions of the operators and is therefore beyond the scope of this standard.

3 **22.2.1 Creating Maintenance Domains and MAs**

4 The operator creates either a Default MD Level managed object or a Maintenance Domain managed object
5 in a Bridge for every Maintenance Domain whose CFM PDUs can pass through that Bridge and are to be
6 processed. The operator can also create a MA managed object in a Bridge for some or all of the MAs whose
7 CFM PDUs can pass through that Bridge and are to be processed. Whether a service instance can pass
8 through a Bridge depends on the following:

- 9 a) Enable Ingress Filtering per-Bridge Port parameter (8.6.2).
- 10 b) Static Filtering Entries (8.8.1).
- 11 c) Static VLAN Registration Entries (8.8.2).
- 12 d) PIP configuration parameters (6.10).
- 13 e) CBP configuration parameters (6.11).
- 14 f) Dynamic VLAN Registration Entries (8.8.5).
- 15 g) Operation of MSTP (Clause 13), which in turn, can be affected by the state of the MAC_Operational
16 parameters (IEEE Std 802.1AC) of the ISSs in the Bridge Port.

17 The first five of these methods are altered only by modifying managed objects and can therefore be
18 considered static for the purposes of configuring CFM. The last two methods are dynamic, in that they
19 depend not only on configuration, but on protocols that respond to network events, such as the addition or
20 loss of a link or Bridge. For example, a VLAN can pass through a Bridge unless there is a Static VLAN
21 Registration Entry prohibiting its VID from passing through every Bridge Port, and Ingress Filtering is
22 enabled on every Bridge Port. If a Bridge's Maintenance Domain requires only insignificant resources to
23 support MHF configuration, then the most reliable network is one that assumes that all service instances will
24 require MHFs at some MD Level higher than that of that Maintenance Domain.

25 **22.2.2 Creating MEPs**

26 There are four kinds of Maintenance Association managed objects: those that are attached to one or more
27 specific VIDs in the Bridge, those that are attached to no VID, those that are attached to an I-SID, and those
28 that are attached to a TESI or ECMP path.

29 Once the appropriate Maintenance Domain and MA have been created, a MEP can be created, modified, or
30 deleted using the MA managed object defined in 12.14.6. MEPs are always created explicitly. MEP creation
31 is controlled by creating a Maintenance association Endpoint managed object using the following
32 parameters:

- 33 a) Whether the MA to which the MEP belongs is associated with specific VID(s), I-SID, TESI, or
34 ECMP path, and if so, which parameters [item b) in 12.14.6.1.3], including a specification of the
35 MA's Primary VID [item d) in 12.14.7.1.3].
- 36 b) Whether the MEP is a Down MEP (pointing away from the MAC Relay Entity), or an Up MEP,
37 (pointing toward the MAC Relay Entity) [item c) in 12.14.6.3.2].

38 Given those two values in the MEP managed object, Table 22-1 shows where, in Figure 22-9, the MEP is
39 created. In Figure 22-9, which shows a PNP or a CNP, the VID selects which of the MEP positions is
40 selected for Table 22-1 column 3 values 1 and 4. Only one row of MEPs is shown in positions 1, 4, and 5,
41 even though up to eight rows can exist at each of these points. Because the MEPs are always placed in order
42 by MD Level, with the highest MD Level nearest the MHFs, at the center of each stack, there is thus no
43 ambiguity. The third column, starting from the left, in the EISS Multiplex Entity depicted in Figure 22-9

1 corresponds to an ESP-VID or SPBM VID. Only MIPs can be placed in this column for a PNP. MEPs can be
2 created on a CBP and PIP as shown in Figure 26-2.

Table 22-1—MEP creation

MA has VID [item b) in 12.14.6.1.3]	MEP direction [item c) in 12.14.7.1.3]	Position of MEP in Figure 22-9
Yes	Up	1
	Down	4
No	Up	disallowed
	Down	5

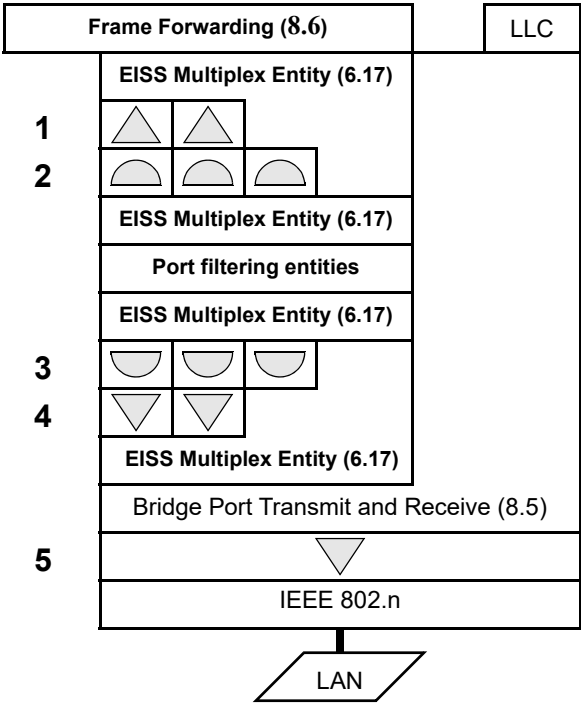


Figure 22-9—Creating MEPs and MIPs

3 If any MEPs are configured on an MA, then in any given Bridge, either all of those MEPs are Up MEPs or
4 all of them are Down MEPs; a Bridge refuses to create an Up (Down) MEPs if a Down (Up) MEP already is
5 configured in that MA in that Bridge. An MA can be configured with Up MEPs in one Bridge and Down
6 MEPs in another Bridge or Station. Any given VID can be configured on any number of MAs (including
7 zero) containing either Down MEPs or no MEPs at all, plus zero or one MA containing Up MEPs. No VID
8 can be configured on more than one MA on which are configured Up MEPs.

9 These restrictions, along with the configuration errors in 22.2.4, enable MEPs to be configured so that they
10 bound an MA, no matter what action is taken by the spanning tree protocols within a Maintenance Domain.

22.2.3 Creating MIPs

Managed objects control the creation of MIPs, but indirectly, rather than explicitly, as for MEPs. Every MA defined in a Bridge can cause the management entity to create MIPs on every Bridge Port. On each Bridge Port, the following conditions trigger the management entity to reevaluate the creation of MIPs for a given VID (or list of VIDs) or I-SID x on that port:

- a) The Bridge is initialized.
- b) A MEP is created or deleted on VID(s) or I-SID x on that Bridge Port.
- c) An Up MEP is created or deleted on VID(s) x , which is (are) not allocated to a TESI, on any Bridge Port or an Up MEP is created or deleted on I-SID x .
- d) A change occurs in whether VID(s) or I-SID x can or cannot pass through the Bridge Port (22.2.1).
- e) A change occurs in the Default MD Level managed object (12.14.3).
- f) A change occurs in a Maintenance Association managed object associated with VID(s) or I-SID x .

A MIP for VID-based service instances and TESIs is created by creating a pair of MHFs in positions 2 and 3 in Figure 22-9. MIPs associated with I-SIDs are created on the two sides of the multiplexed SAPs in the Backbone Service Instance Multiplex Entity as depicted in Figure 26-2. MAs that are associated with no VID are not considered in this subclause. For a given VID on a given Bridge Port, and in the absence of any of the configuration errors in 22.2.4, there are at most two MAs at each MD Level that can affect MIP creation, plus at most one entry in the Default MD Level managed object. If one or more of the configuration errors in 22.2.4 are present, this standard does not specify what MIPs the Bridge creates.

When required, the Maintenance Entity performs the MIP evaluation on each Bridge Port for each VID. For each Bridge Port p and service identifier x , which can be a list of VIDs, or an I-SID, the Maintenance Entity creates a list of active MD Levels. This list contains the following:

- g) MD Level of each of the MAs (if any) that monitors service x and has a MEP configured on Bridge Port p .
- h) MD Level of each of the MAs (if any) that monitors a service x , and has at least one Up MEP configured on some Bridge Port in this Bridge other than Bridge Port p .
- i) MD Level of each of the MAs (if any) that monitors service x and has no MEPs configured on any Bridge Port in this Bridge.
- j) MD Level of the entry in the Default MD Level managed object (12.14.3) for service x .

Exactly one MD Level d in this active list is eligible for MIP creation. It is the lowest MD Level in the list of active MD Levels such that there is no MEP configured on Bridge Port p and service x at MD Level d or at any higher MD Level. This is the lowest MD Level in the list, if there is no such MEP configured. The Maintenance Entity then uses Table 22-2 to determine whether the MIP is to be created.

Table 22-2—MIP creation

Enumeration [item d) in 12.14.3.1.3, item c) in 12.14.6.1.3, or item c) in 12.14.6.1.3]	A MEP is configured on Bridge Port p	MIP created
defMHFnone:	—	No
defMHFexplicit	False	No
	True	Yes
defMHFdefault	—	Yes

1 The enumeration controlling whether the MIP is created is taken from:

- 2 k) The Maintenance Association managed object [item c) in 12.14.6.1.3] for the MA that monitors
3 service x , and has at least one Up MEP configured on some Bridge Port in the Bridge [see item h)];
4 or
- 5 l) That MA's Maintenance Domain managed object [item c) in 12.14.5.1.3], if that MA exists, but its
6 Maintenance Association managed object contains the value `defMHFdefer`; or
- 7 m) The Default MD Level managed object [item d) in 12.14.3.1.3] if no such MA exists [see item j)].

8 The enumerated values used in the managed objects item d) in 12.14.3.1.3, item c) in 12.14.5.1.3, and
9 item c) in 12.14.6.1.3 to control the creation of a MIP for a service instance identified by the parameter x on
10 a Bridge Port are as follows:

- 11 n) **defMHFnone**—No MHFs can be created for this VID(s) or I-SID (the default value).
- 13 o) **defMHFdefault**—MHFs can be created for this VID(s) or I-SID on any Bridge Port through which
15 the VID(s) or I-SID can pass where:
 - 16 1) There are no lower active MD levels, or
 - 17 2) There is a MEP at the next lower active MD level on the port.
- 18 p) **defMHFexplicit**—MHFs can be created for this VID(s) or I-SID only on Bridge Ports through
20 which this VID(s) or I-SID can pass, and only if there is a MEP at the next lower active MD level on
21 the port.
- 22 q) **defMHFdefer**—In the Maintenance Association managed object [item c) in 12.14.6.1.3] only, the
24 control of MHF creation is deferred to the corresponding variable in the enclosing Maintenance
25 Domain [item c) in 12.14.5.1.3].

26 22.2.4 CFM configuration errors

27 While making the MIP creation evaluation described in 22.2.3, or whenever a MEP is created or deleted, the
28 management entity can encounter errors in the configuration. Because of this, the Management entity
29 maintains a Configuration Error List managed object (12.14.4) that lists, for every VID or I-SID, the Bridge
30 Ports that have MIP configuration errors. Every time the MIP creation evaluation is performed or a MEP is
31 created or deleted, this list is updated to reflect which Bridge Ports and VIDs are in error and which are not.
32 A Bridge Port is placed in the list if and only if one of the following errors occurs:

- 33 a) **CFMleak**—MA x is associated with a specific VID list or I-SID, one or more of the VIDs or I-SID
35 in MA x can pass through the Bridge Port, no Up MEP is configured for MA x on the Bridge Port,
36 no Down MEP is configured on any Bridge Port for MA x , and some other MA y , at a higher MD
37 Level than MA x , and associated with at least one of the VID(s) or the I-SID also in MA x , does
38 have a MEP configured on the Bridge Port.
- 39 b) **ConflictingVIDs**—MA x is associated with a specific VID list or I-SID, an Up MEP is configured
41 on MA x on the Bridge Port, and some other MA y , associated with at least one of the VID(s) or I-
42 SID also in MA x , and at the same MD Level as MA x , also has an Up MEP configured on some
43 Bridge Port.
- 44 c) **ExcessiveLevels**—The number of different MD Levels at which MIPs are to be created on this port
46 exceeds the Bridge's capabilities (see 22.3).
- 47 d) **OverlappedLevels**—A MEP is created for one VID or I-SID at one MD Level, but a MEP is
49 configured on another VID or I-SID at that MD Level or higher, exceeding the Bridge's capabilities
50 (see 22.3).

51 A Bridge uses detection of the ExcessiveLevels error to refuse an operation on a managed object and thus
52 avoids adding Bridge Ports to the error lists. This is because this incremental operation is known to be the
53 one that exceeds the capabilities of the Bridge. However, a Bridge uses only the detection of the
54 ExcessiveLevels error to refuse an operation on a managed object. That is because the other errors can arise
55 as a result of a management operation, e.g., the deletion of a Static VLAN Registration Entry, that has no

1 knowledge of CFM and cannot report the error. Furthermore, the root cause of either error might not be the
2 last MA to be configured and could be under the control of another administration. Therefore, management
3 operations that trigger errors other than ExcessiveLevels are performed, not rejected. The lists of Bridge
4 Ports in error are maintained (12.14.4), so that the system administrator(s) can agree on a course of action,
5 outside this standard, to correct the problem.

6 **22.3 MPs, Ports, and MD Level assignment**

7 It is possible to configure MPs at different MD Levels for different VIDs on a single Bridge Port. However,
8 this imposes upon an implementation the burden to be able to configure up to 8189 MD Levels for the VIDs
9 on a Port, 4094 for MHFs and 4095 for MEPs, each requiring 3 bits of storage. An implementation may not
10 support different MD Levels for MIPs on different VIDs on a single Port, hence the ExcessiveLevels and
11 OverlappedLevels error responses (22.2.4). At least one MD Level shall be supported for each port in a
12 Bridge, such that MIPs can operate at that MD Level, and all CFM PDUs below that MD Level are filtered
13 or processed, according to the configuration of MEPs on the port.

14 In Figure 18-7, the left-most Maintenance Entity at the lowest MD Level (0) has one MEP in the customer's
15 equipment 1 and one in Provider Bridge 2. When Down MEPs are used in this manner to create a
16 Maintenance Entity that interconnects two administrations, those administrations can agree upon a particular
17 MD Level (and MAID and MEPIDs) to use. To enhance the chances for interoperability, the MD Level for
18 Down MEPs should be the lowest not already in use by an existing MA, typically 0.

19 Assigning MD Levels with gaps between the levels used, as in Figure 18-7, can be wasteful of resources,
20 especially MAC Address Registration Entries in the FDB. Minimizing the number of MD Levels, and
21 concentrating the MD Levels used toward MD Level 0, minimizes the required resources, since CFM PDUs
22 carrying higher MD Levels than those used in a Bridge are treated as ordinary multicasts by that Bridge.
23 However, if one creates a Maintenance Domain at MD Level 0, and subsequently needs a level beneath that
24 one, for example if a PBN is substituted for a physical connection, then the MD Level of the original
25 Maintenance Domain might have to be changed. If there is another Maintenance Domain above the one at
26 MD Level 0, that one would have to be changed, first, and so on. Similarly, if an existing Maintenance
27 Domain at MD Level 3 is partitioned into several Domains, a new Maintenance Domain at MD Level 4
28 would be needed to ensure end-to-end connectivity across the extent of the old Maintenance Domain.

29 Such difficulties can be avoided in several ways:

- 30 a) The MD Levels of different Maintenance Domains can be insulated from each other by VLAN tags.
31 Appropriate placement of the MEPs in Bridge Ports can ensure that the tags insulate the MD Levels,
32 and that no reconfiguration will be necessary.
- 33 b) If it is expected that an additional MD Level will be needed because a physical link will likely
34 become virtualized, MD Level 0 can be left vacant.
- 35 c) If it is expected that an additional MD Level will be needed because a Maintenance Domain will
36 likely be partitioned, a gap can be left between that Maintenance Domain's MD Level and the lowest
37 MD Level offered to a customer of that Maintenance Domain.

38 See J.2 for a description of an MD Level assignment plan that can be used in the most general case.

22.4 Stations and CFM

Stations, as well as Bridges, can implement CFM. For the purposes of CFM, and for no other purposes, a Station is modeled as a Bridge, with the following exceptions:

- a) This standard defines no entities above the layer of Down MPs in Figure 22-8. In particular, a Station has no Port filtering entities, Up MPs, or Forwarding Process. A CFM-capable Station is therefore a set of one or more disconnected Bridge Ports.
- b) A Station shall not support the creation of MIPs or Up MEPs. Only Down MEPs can be implemented in a Station.
- c) CFM PDUs not processed and not discarded by a MEP shall be ignored, and not processed, by the Station.
- d) A Station shall create no entries in the Configuration Error List managed object (12.14.4) except for CFMleak errors.

22.5 Scalability of CFM

CFM can create a considerable load on the resources of a Bridge. Ideally, a Bridge has the capability to reflect LBMs as LBRs at line rate, so that the capacity and reliability of a network can be tested. The total load imposed by LBMs, LBRs, LTMs, and LTRs cannot be predicted, as these are all under the control of the system administrator, and any number of streams of these PDUs can be generated.

The load imposed by CCMs is more regular, so reasonable predictions of the load they impose can be made. Based on a figure of 109 octets as the minimum frame size, from the start of the destination MAC address of one frame to the start of the destination MAC address of the next frame,³⁸ Table 22-3 shows the load that is imposed by CCM transmission and reception upon a Bridge Port with a single configured MEP, as a function of the CCM transmission interval, the number of MEPs in the MA, and the speed of the link. One row in the table shows the number of CCMs per second transmitted or received. The number in each of the other rows is the fraction of the total bandwidth required by those CCMs. Table 22-4 extends this calculation by multiplying all numbers by 1000, for the case where 1000 VIDs, each with a MEP, are configured on the Bridge Port, again with the specified transmission interval and average number of MEPs per MA.

Counting 4094 allowed VIDs in a VLAN tag, three kinds of MEPs, Up VID MEPs, Down VID MEPs, and Down no-VID MEPs, and eight MD Levels, a single Bridge Port with no Link Aggregation can, in theory, support 65 504 MEPs and 8188 MHFs.

Table 22-3—Bandwidth required for CCMs for 1 MA

Transmit interval	3.3 ms		10 ms		1 s		60 s	
MEPs in each MA	10	2	10	2	10	2	10	2
CCM / s	3000	600	1000	200	10	2	0.167	0.033
10 Mb/s	26.160%	5.232%	8.720%	1.744%	0.087%	0.017%	0.001%	0.000%
100 Mb/s	2.616%	0.523%	0.872%	0.174%	0.009%	0.002%	0.000%	0.000%
1 Gb/s	0.262%	0.052%	0.087%	0.017%	0.001%	0.000%	0.000%	0.000%
10 Gb/s	0.026%	0.005%	0.009%	0.002%	0.000%	0.000%	0.000%	0.000%
100 Gb/s	0.003%	0.001%	0.001%	0.000%	0.000%	0.000%	0.000%	0.000%

³⁸ First TLV Offset = 70, 12 for MAC addresses, 6 for Type and fixed header, 1 for End TLV, 20 for inter-frame gap and preamble.

Table 22-4—Bandwidth required for CCMs for 1000 MAs

Transmit interval	3.3 ms		10 ms		1 s		60 s	
MEPs in each MA	10	2	10	2	10	2	10	2
CCM / s	3 000 000	600 000	1 000 000	200 000	10 000	2000	167	33
10 Mb/s	> 100%	> 100%	> 100%	> 100%	87.200%	17.440%	1.453%	0.291%
100 Mb/s	> 100%	> 100%	> 100%	> 100%	8.720%	1.744%	0.145%	0.029%
1 Gb/s	> 100%	52.320%	87.200%	17.440%	0.872%	0.174%	0.015%	0.003%
10 Gb/s	26.160%	5.232%	8.720%	1.744%	0.087%	0.017%	0.001%	0.000%
100 Gb/s	2.616%	0.523%	0.872%	0.174%	0.009%	0.002%	0.000%	0.000%

22.6 CFM in Provider Bridges

The S-VLAN component of a Provider Bridge is, for the purposes of this standard, an example of a VLAN Bridge. Fortunately, the many combinations possible when configuring CFM on Customer Bridges and Provider Bridges, connected using Port-based, C-tagged, or Remote Customer Service Interfaces (RCSIs), can be simplified.

22.6.1 MPs and C-VLAN components

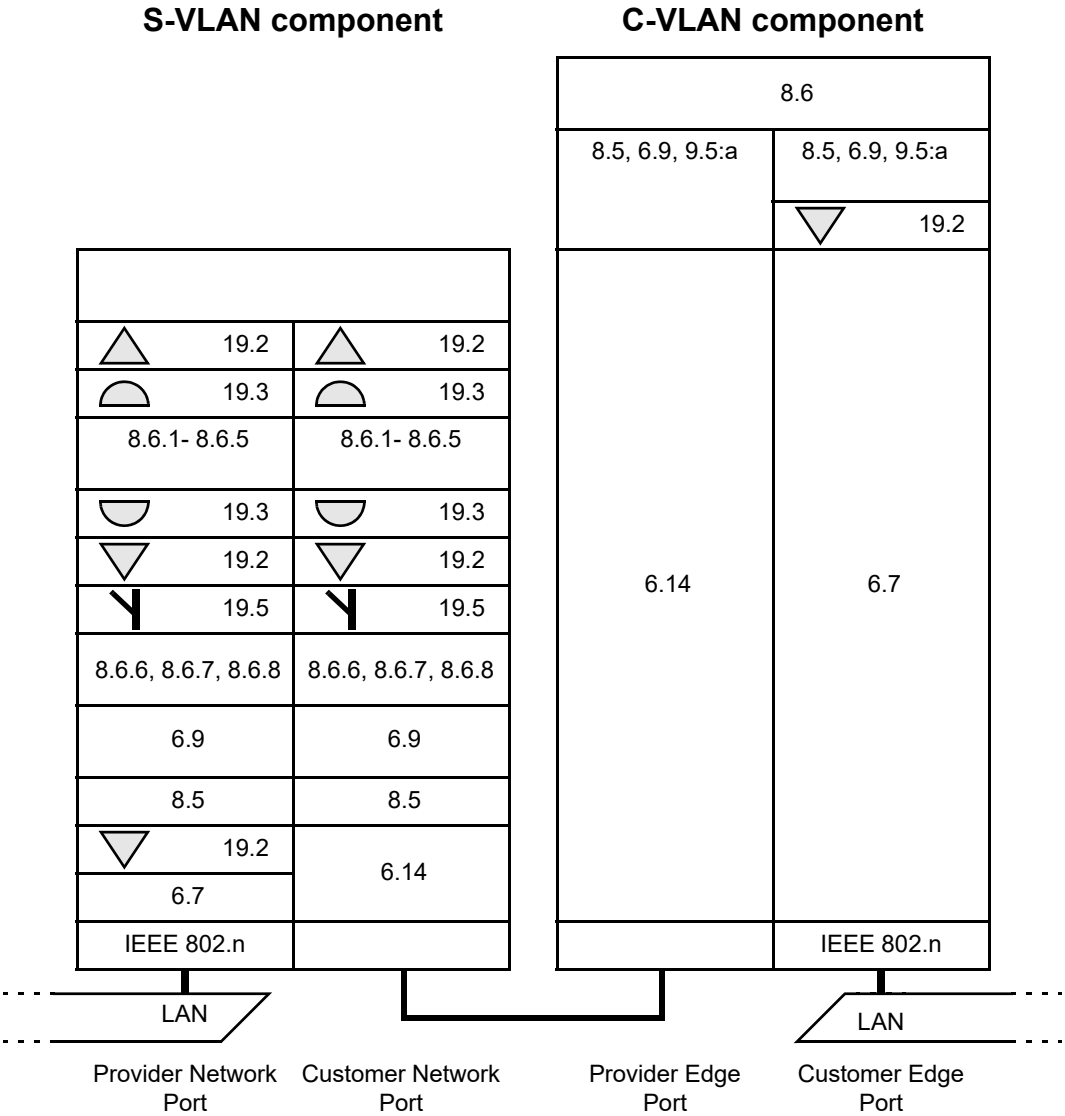
When a C-VLAN component is integrated into a Provider Edge Bridge to form a C-tagged service interface, there is little justification for making all possible CFM entities manageable. Subclause 15.4 describes a C-tagged service interface, illustrated in Figure 15-4. Figure 22-10 recasts Figure 15-4, illustrating the minimum set of CFM functions that a C-tagged service interface is required to support, if its Provider Bridge supports CFM. Almost all of the CFM components have been removed from the C-VLAN component. As a result, the transformations made in 22.1.2 and 22.1.3 are unnecessary, and the subclause references collapse to match those in Figure 15-4. The S-VLAN component contains almost the full suite of CFM entities. One set of shims has been moved from the CNP to the CEP, however.

The managed objects in 12.14 support the creation of up to eight MEPs below the Bridge Port Transmit and Receive process (8.5). These MEPs belong to MAs associated with no VID. A MEP in this position would be of little use in a CNP, since there is no need to pair it with a MEP in the PEP to protect an Internal LAN. However, a MEP attached to no VID in the CEP could be used to protect the LAN to the customer equipment.

In order to support protection of the CEP's LAN, while maximizing interoperability, whenever (Down) MEPs are created (12.14.6.3) in an MA associated with no VID, a Provider Edge Bridge shall allow those MEPs to be created using the interface number [item d) in 12.14.6.3.2] for a CEP.

22.6.2 Maintenance C-VLAN on a Port-based service interface

Assume that a customer's C-VLAN Bridge is attached to an S-VLAN Bridge as in Figure 22-4, and that the customer wants to pair a Down MEP in the C-VLAN Bridge with a similar Down MEP in the C-VLAN Bridge on the other side of the PBN, thus creating an MA to provide end-to-end protection of the service instance offered by the Provider. In order to obtain the advantage of inserting the CFM PDUs above the Queuing entities, the customer would not use MEPs below the Bridge Port Transmit and Receive process (8.5). The customer could create one MA for each C-VID. However, this would have drawbacks in that:



1 **22.6.3 Maintenance C-VLAN on a C-tagged service interface**

2 Assume that a customer's C-VLAN Bridge, illustrated by Figure 22-4 (less, of course, the box for "Support
3 of the ISS for attachment to a PBN (6.13)"), is attached to a Provider Edge Bridge as in Figure 22-10.
4 Assume that the customer has purchased several service instances, which have C-tagged service interfaces to
5 other of the customer's C-VLAN Bridges at several different locations. Let us further suppose that the
6 customer wants to pair Down MEPs in the C-VLAN Bridge to create MAs to provide end-to-end protection
7 of the service instances. In order to obtain the advantage of inserting the CFM PDUs above the Queuing
8 entities, the customer would not use MEPs below the Bridge Port Transmit and Receive process (8.5). The
9 customer could create one MA for each C-VID.

10 There would be more justification in creating an MA per C-VLAN than for the Port-based service instance
11 described in 22.6.2. However, if the customer is reasonably confident that the provider will carry each
12 C-VLAN in the correct S-VLAN, then the customer can create multiple Maintenance C-VLANs, one per
13 provider service instance (S-VLAN), and thus one per PEP in the C-VLAN component. By configuring each
14 of these PEPs to pass that one C-VLAN untagged to the corresponding CNP in the S-VLAN component, the
15 CFM PDUs in that one customer MA are visible to the S-VLAN component. The provider can configure a
16 MIP for that S-VID that is visible to the customer's MA. The customer can have one MA protecting each
17 service instance.

18 **22.6.4 MPs and Port-mapping S-VLAN components**

19 When a Port-mapping S-VLAN component is used to provide RCSI, there is little justification for making
20 all possible CFM entities manageable. The other Provider Bridge components can provide the necessary
21 MPs for service OAM. There is one MP that can only be provided by the Port-mapping S-VLAN component
22 and that is the point below the Bridge Port Transmit and Receive process (8.5) on the RCAP. This MP can be
23 used to protect the LAN to the neighboring PBN.

24 Figure 22-11 extends Figure 22-10, showing the minimum set of CFM entities required for a C-tagged
25 RCSI, if the Provider Bridge supports CFM. The S-VLAN component and C-VLAN component support the
26 same set of CFM entities as in Figure 22-10. The Port-mapping S-VLAN component that provides the
27 RCSI supports at least one more CFM entity, as shown, that can protect the LAN to the neighboring PBN.

28 Similarly, in the case of a Port-based RCSI or an internal PNP the required service CFM entities can be
29 provided by the S-VLAN component. Thus the only CFM entity required from the Port-mapping S-VLAN
30 component is the one shown in Figure 22-11 which is independent of the number and type of internal Ports
31 provided.

1

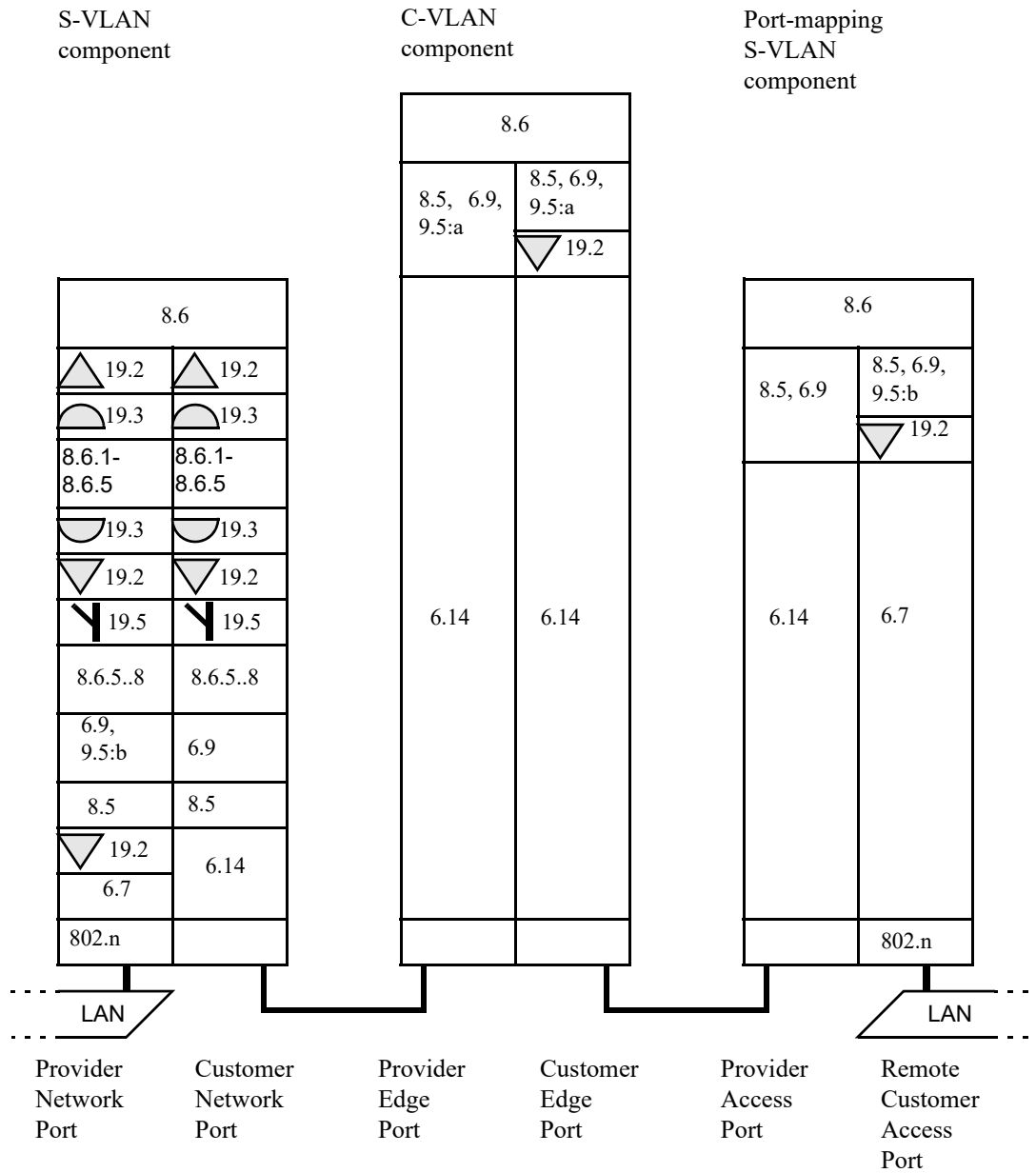


Figure 22-11—CFM in a Provider Edge Bridge C-tagged RCSI

22.7 Management Port MEPs and CFM in the enterprise environment

Figure 22-12 illustrates the placement of Management Port MEPs. It is similar to the CFM Port illustrated in Figure J-1. However, the Management Port MEPs are configured explicitly for the Management Port. They do not represent information for any other Bridge Port in their CFM PDUs. Therefore, there is only one MEP for each MA. Furthermore, there can be no MHFs configured on a Management Port, because there can be no MEP further along the path to terminate the MA.

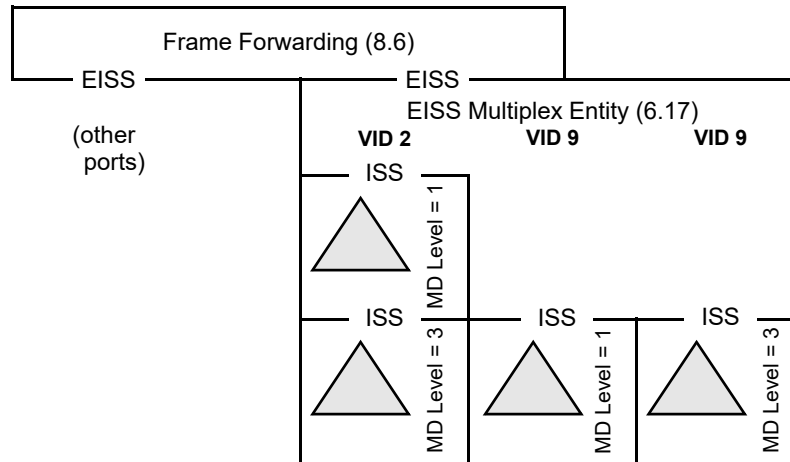


Figure 22-12—Up MEPs in a Management Port

NOTE—The fact that a CFM Port could be configured on a Management Port, and thus instantiate MHFs on the Management Port, does not contradict the preceding paragraph. Those MHFs are configured on a Bridge Port, and instantiated on the Management Port (CFM Port), not configured on the Management Port.

Although this standard uses the terminology of Provider Bridges and describes the functions of CFM in terms of Provider Bridges, it can be applied to any Bridge, whether a Provider Bridge, C-VLAN Bridge, or MAC Bridge. The normative definitions and functions specified by this standard do not depend on, for example, whether the VLAN tag used to associate frames with VLANs use the customer format or the provider format. Certainly, no business relationships are required to implement this standard.

Given that both the users and the providers of an enterprise network are, by definition, all members of the same company, the utility of CFM in the enterprise environment is less than in the provider/customer environment. The network administrator of an enterprise network can find a number of uses for CFM, however.

The default values for MEP and MIP configuration make CFM easy to configure for the typical enterprise network. Figure 22-13 illustrates two Bridge Ports, the Management Port, and one Bridge Port connected to a LAN, using the default configuration as follows:

- A single Maintenance Domain can be configured at MD Level 0.
- An MA for each VID is configured in each Bridge.
- A single MEP for each VID is configured on a Management Port in each Bridge. Typically, this is a port with no external interface, often the one used to manage the Bridge.
- Through the Maintenance Domain managed object previously created, it is easy to define a MIP on every Bridge Port on every VID.

1 This simple configuration enables the administrator to use LBMs and LTMs to diagnose network faults
2 among the Bridges of the network. In addition, specific high-priority VLANs, e.g., a VLAN interconnecting
3 key routers and file servers for a large enterprise, could be explicitly monitored via CFM.

4 In a large enterprise, separate Maintenance Domains can be established, along with an accompanying
5 enterprise-wide Maintenance Domain at a higher MD Level, in order to define and enforce separate spheres
6 of responsibility for separate network administrations, e.g., in different departments in a university. The
7 boundaries between Maintenance Domains might well coincide with administratively imposed boundaries
8 between MSTP Regions.

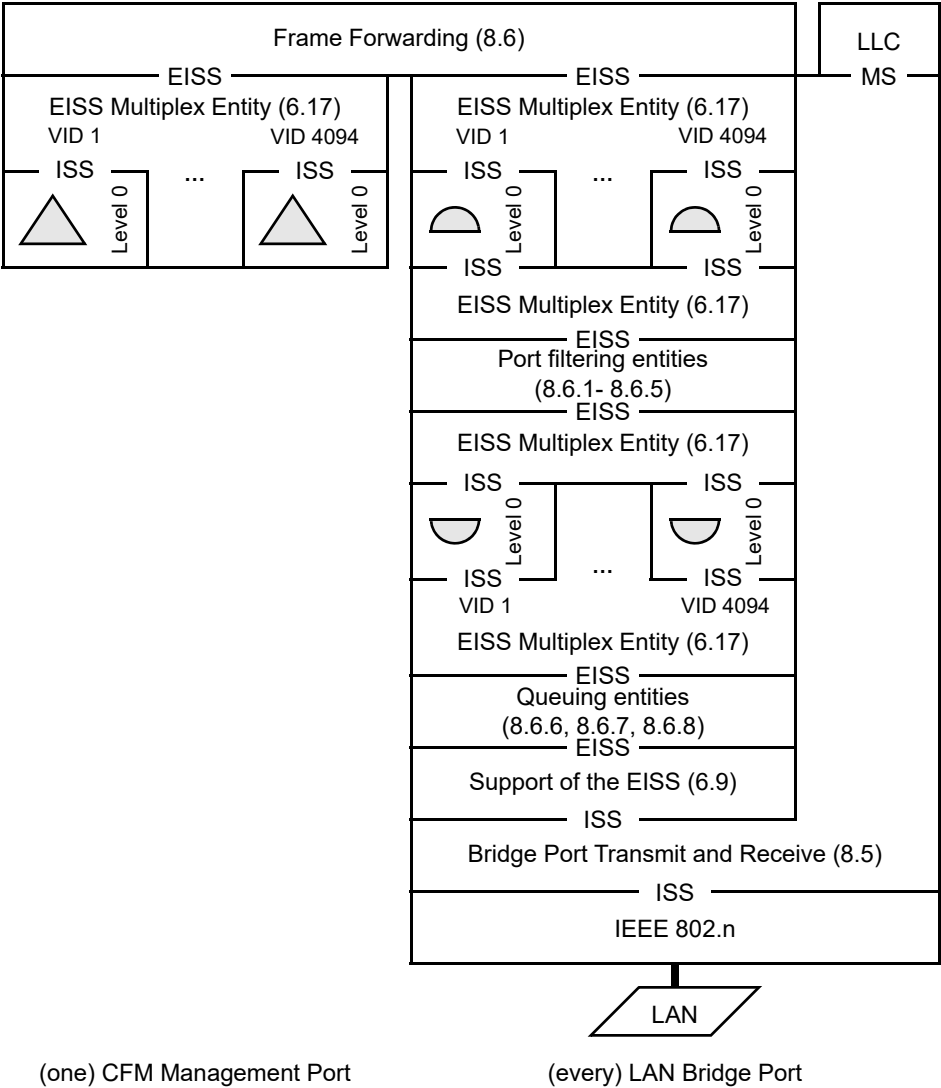


Figure 22-13—CFM in the enterprise environment

22.8 Implementing CFM on Bridges that implement earlier revisions of IEEE Std 802.1Q

It might or might not be easy to implement CFM in a VLAN Bridge that implement earlier revisions of IEEE Std 802.1Q. The problems are best illustrated by a three-port Bridge as illustrated in Figure 22-14. Each of the three Bridge Ports A, B, and C is configured with MEPs and MIPs at the MD Levels shown in the figure.

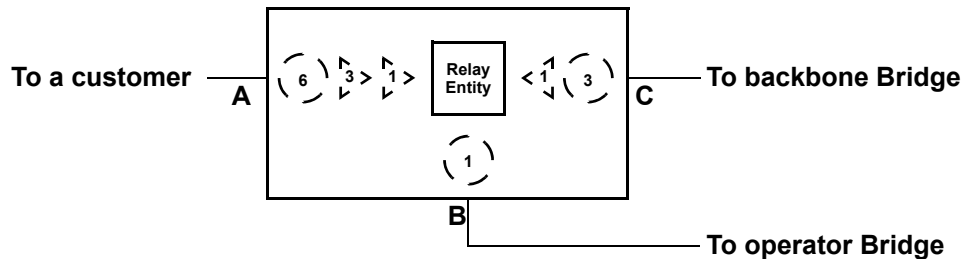


Figure 22-14—CFM in the enterprise environment

If a CCM at MD Level 3 enters the Bridge from Bridge Port C, that CCM will be passed to both of the other Bridge Ports A and B. It will not exit Bridge Port A, however, because there is a MEP at MD Level 3 on that Bridge Port. It will, however, pass through Bridge Port B to the other operator Bridge. One can easily imagine configuring MAC Address Registration Entry in the Bridge for the MD Level 3 group address shown in Table 8-18 that permits that group address to pass through the Bridge Ports B and C, but not through Bridge Port A, and also sends the CCM to a CFM module in the Bridge's Higher Layer Entities, perhaps residing on the Management Port of 22.7, for processing. It is precisely so that MAC Address Registration Entries can be used in this manner that CCMs and LTMs are required to use the group addresses in Table 8-19 and Table 8-18.

A potential problem for some implementations can be seen if readers ask what happens if that same CCM at MD Level 3 enters from Bridge Port A. In that case, the CCM does not exit either of Bridge Ports B or C, because Bridge Port A has a MEP at MD Level 3 to stop it; MAC Address Registration Entries prevent frames from leaving a Bridge Port, not from entering one. Many Bridges have the ability to filter incoming frames based on the destination MAC address and/or the contents of the frame; either facility will enable this CCM to be stopped. In order to support both a Down MEP attached to no VID, and a Down MEP attached to a tagged VID at a lower MD Level than the no-VID MEP, the Bridge filters frames based on both the MAC address and whether the frame is tagged.

23. MAC status propagation

Individual LANs, each operating its own MAC and media access method-specific procedures, can be connected by one or more TPMRs to form a composite LAN that is transparent to other Bridges and stations and their configuration protocols. The MAC Status Protocol (MSP) ensures that changes in the connectivity provided by the composite LAN results in changes in the MAC_Operational status parameter (IEEE Std 802.1AC) at each of the attached Bridges and stations, just as if they were connected to an individual LAN.

NOTE 1—MSP is designed only for use with point-to-point LANs.

MAC_Operational provides rapid notification of connectivity failures and prompts the necessary initial protocol behavior to ensure that new connectivity has not caused an instantaneous data loop. If this MAC status parameter were not propagated by a TPMR, Bridge protocols would have to rely on periodic transmissions to detect connectivity changes. On their own these periodic transmissions take longer to detect failures, and cannot detect the possibility of data loops until they have been created, as illustrated by the following examples.

Figure 23-1 shows a TPMR connecting two Bridge Ports.

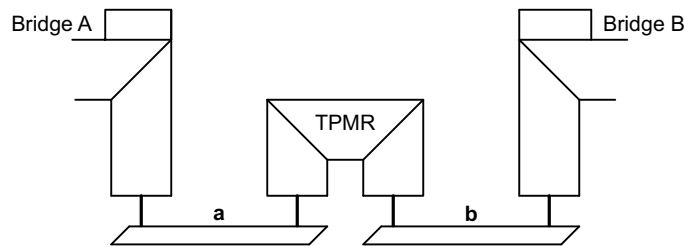


Figure 23-1—TPMR connecting two Bridge Ports

Assume that Bridge A is the spanning tree Designated Bridge for the LAN that comprises the individual LANs **a** and **b** and the TPMR, and that Bridge B's spanning tree Root Port is shown. If **a** fails and the TPMR does not propagate MAC_Operational, Bridge B will not reselect its Root Port until it has timed out the last BPDU from A.

Figure 23-2 shows a link that uses two TPMRs, perhaps because LAN **c** uses a non-IEEE-802 technology together with an appropriate convergence function.

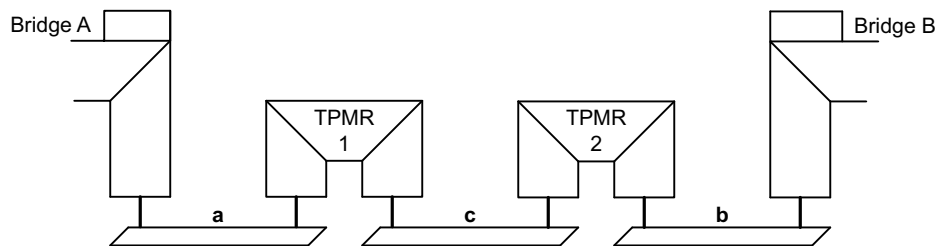


Figure 23-2—TPMR chain connecting Bridge Ports

Without MAC_Operational propagation, failure of **c** will not be immediately visible to either Bridge. Worse, if **c** fails and is restored after a while, both A and B will believe themselves to be Designated Bridge for the composite LAN, and will forward frames until one receives a BPDU from the other, even if a data loop has been created.

1 When working correctly the MAC Service provides bidirectional connectivity or no connectivity at all.
2 MAC_Operational is TRUE for each of two peers connected to the same LAN if they can communicate, and
3 FALSE for either or both if they cannot. Protocols, such as the spanning tree protocols, that make use of
4 MAC_Operational to detect new connectivity and initialize state machines rely on the last peer that sees
5 MAC_Operational transition TRUE to enforce any necessary behavior after a connectivity change. For
6 example, it is the last of two connected Bridge Ports to be powered on that enforces the necessary delay prior
7 to setting operEdge TRUE (13.27).

8 NOTE 2—MAC_Operational being TRUE within a single station does not guarantee connectivity to any peer. Even if
9 connected by a point-to-point LAN, the peer could have just reinitialized. It is not possible, given only the use of the
10 LAN medium for communication, to arrange for two peers to transition MAC_Operational at exactly the same time.

11 This clause defines media-independent propagation of MAC status between TPMRs and to attached
12 stations, as follows. It:

- 13 a) Models MAC status propagation within the Bridge architecture used to describe a TPMR (23.1).
- 14 b) Provides an overview of MSP (23.2).
- 15 c) Specifies state machines for MSP operation (23.3–23.9).
- 16 d) Specifies the addressing, protocol identification, format, encoding, and validation of MAC Status
17 Protocol Data Units (MSPDUs, 23.13–23.16).

18 The term *MAC status propagation* (3.163) describes the overall process of communicating a
19 MAC_Operational parameter value through one or more TPMRs. MSP can use *link status notification*³⁹
20 (3.132) between some Bridges and end stations and *MAC status notification* (3.162) between others. These
21 two notification methods differ, as follows:

- 22 e) Link status notification uses frames to convey information about MAC_Operational. It does not
23 interrupt or prevent other communication between adjacent Bridges. However, it requires both the
24 source and destination of the notification to implement additional protocol. Since it does not prevent
25 communication it cannot, by itself, prevent loops caused by new connectivity.
- 26 f) MAC status notification uses a layer management interaction with the local MAC Entity to change
27 MAC_Operational for a peer user of the MAC Service provided by an individual LAN. It is
28 equivalent to bringing a link down, and is generally effective for MACs with specific point-to-point
29 support. It also interrupts all other communication, including the use of in-band management to
30 rectify an underlying problem.

31 Where management of a TPMR is permitted through one of its ports, the failure of an individual LAN not in
32 the communication path does not cause MAC status propagation to prevent management connectivity. In
33 Figure 23-2, for example, the failure of LAN **b** does not prevent connectivity to TPMR 2 from Bridge A.
34 When **b** recovers, MAC_Operational for Bridge A's Port is "blipped," i.e., taken FALSE for a brief period
35 and allowed to return TRUE, thus meeting the requirement for transition when connectivity changes. While
36 this slows the recognition of newly available connectivity, that is rarely an issue since several seconds
37 hysteresis should be applied to any MAC_Operational status transition to prevent higher layer protocols
38 "flapping."

³⁹ The use of the term link status notification in this standard is not to be confused with the term used in SNMP to refer to a type of trap notification.

23.1 Model of operation

The model of operation presented in this clause allows the description of MSP functionality to be consistent with that of the general Bridge architecture and operation (8.2, 8.3). Implementations of a TPMR may adopt any internal model of operation compatible with the externally visible behavior specified.

The additional entities that model MSP operation, and their relationship to the other processes and entities that model the operation of a TPMR are illustrated in Figure 23-3. They comprise the following:

- a) A MAC Status Shim (MSS, 23.1.1) for each Port, which allows MSP to control the value of MAC_Operational presented to the TPMR Port connectivity function (8.5.2) and hence to the frame transmission and reception functions of the MAC Relay Entity, Higher Layer Entities, and MAC Status Propagation Entity (MSPE).
- b) The MSPE, which implements MSP, controlling each MSS to ensure that frames are not forwarded by the MAC Relay Entity until status propagation is complete, transmitting and receiving frames to support link status notification, and controlling the individual LAN's MAC to provide MAC status notification.

NOTE—In a TPMR the priority signaling functions shown in Figure 23-3 are provided by 6.20 Support of the ISS with signaled priority.

MSPDUs are sent to a standard group address. They are received by the MSPE, but are forwarded by the MAC Relay Entity (like any other frame) in order to communicate without a delay in each TPMR. The MSPE attaches to each Port by a TPMR Port connectivity function (8.5.2) that allows it to transmit to and receive from the attached individual LAN only, as shown in Figure 23-3.

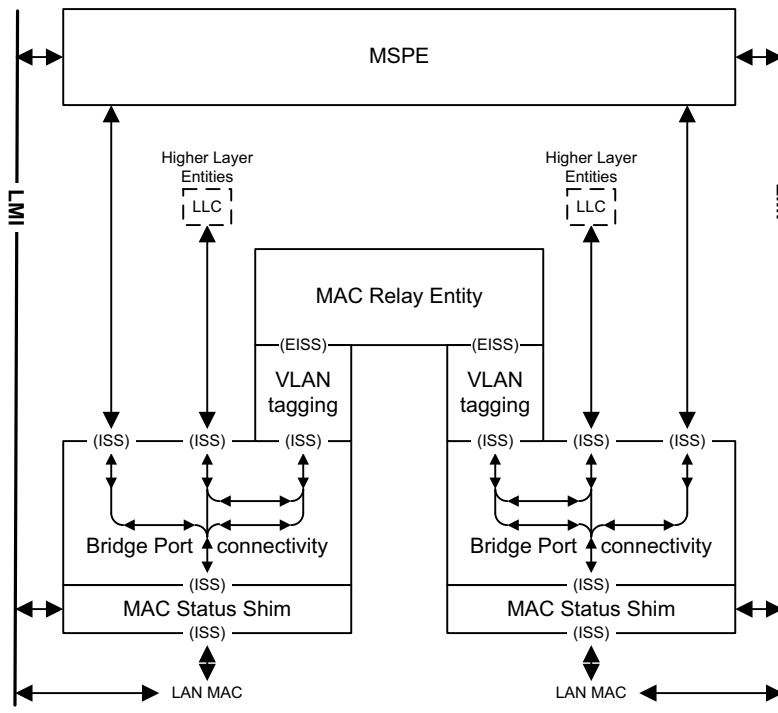


Figure 23-3—MSSs and the MSPE

1 23.1.1 MAC Status Shim (MSS)

2 The MSS allows MSP to control the transmission and reception of frames through a Port by all the entities
3 (including the MAC Relay Entity) supported by the TPMR Port Connectivity function (8.5.2). Each transmit
4 request from the latter at the MSS's upper ISS-SAP results in a corresponding transmit request at the lower
5 ISS-SAP supported by the LAN MAC and each receive indication from the lower ISS-SAP results in a
6 corresponding receive indication at the upper ISS-SAP, without omission or duplication, and with identical
7 parameters, if and only if MAC_Operational for the upper ISS-SAP is TRUE.

8 The values of the MAC status parameters presented at the upper ISS-SAP are determined as follows:

- 9 — **MAC_Enabled:** The value of this parameter is FALSE if the value of STM's disableMSS variable is
10 TRUE. Otherwise, the value of this parameter is TRUE. STM communicates the value of its
11 disableMSS variable using a Layer Management Interface (LMI).
- 12 — **MAC_Operational:** The value of this parameter is TRUE if and only if the MSS's MAC_Enabled
13 parameter is TRUE and the value of the MAC_Operational parameter provided by the LAN MAC at
14 the lower ISS-SAP is TRUE.

15 The MSPE uses the LMI to control and receive status from each MSS and individual LAN MAC, allowing
16 generic management requests and indications to be tailored to the requirements of different MACs, as well
17 as providing a way for one MSS to propagate status to another (via the MSPE), even though
18 MAC_Operational for the MSS's upper ISS-SAP is FALSE.

19 23.1.2 Relationship of CFM to the MSS

20 MEPs may be placed above or below the MSS. A MEP placed in the lower position and partnered with
21 another MEP so that the pair spans a chain of TPMRs would obscure the LAN MAC's MAC_Operational
22 parameter and interfere with the operation of MSP. For this reason, MEPs that protect individual LANs
23 should be placed below MSS, and MEPs that protect chains of TPMRs should be placed above MSS.

24 23.2 MAC Status Protocol (MSP) overview

25 This clause provides examples of MSP operation as time sequence diagrams that show the following:

- 26 a) Exchange of MSPDUs transmitted and received to support link status notification.
- 27 b) The values of MAC_Operational and MAC_Enabled provided by the following:
 - 28 1) Each MSS, at its upper ISS-SAP, to the TPMR Port connectivity function.
 - 29 2) Each LAN MAC to the MSS, at the latter's lower ISS-SAP.

30 The value of MAC_Operational provided by the MSS is TRUE if, and only if the MSS's MAC_Enabled is
31 TRUE and the value of MAC_Operational provided to the MSS by the LAN MAC is TRUE. The latter can
32 be TRUE only if MAC_Enabled is TRUE for both the LAN MAC and its peer in the other station connected
33 to the LAN. The MSPE can use the LMI to disable the MSS in order to prevent communication until MAC
34 status propagation has occurred, and to disable the LAN MAC in order to provide MAC status notification
35 to a peer service user. This clause uses the symbols defined in Table 23-1 to show combinations of the MAC
36 status parameter values for each Port.

Table 23-1—Time sequence diagram symbols

						
MSS	MAC_Operational	T	F	F	F	F
	MAC_Enabled	T	T	F	F	—
LAN MAC	MAC_Operational	T	F	F	T	F
	MAC_Enabled	T	T	T	T	F

T = TRUE, F = FALSE, — = either TRUE or FALSE

1 Throughout this protocol description, the term *LAN* is used to refer to individual LANs connecting adjacent
2 TPMRs or a TPMR and an adjacent Bridge. A *chain* is a series or part of a series of TPMRs connected by
3 LANs, and a *link* is the connectivity provided by a chain between non-TPMR devices that communicate at a
4 higher (sub) layer and perceive the entire link as a single transparent LAN.

5 The number of TPMRs in the following examples is deliberately high, four where one or two might be more
6 common, in order to show all the necessary aspects of protocol behavior.

7 Signaling the addition of new or restored connectivity is considered first, as it is more difficult than signaling
8 failure. Figure 23-4 shows the response to a new or recovered LAN connection in a link between two
9 Bridges that do not implement MSP—and thus require MAC status notification.

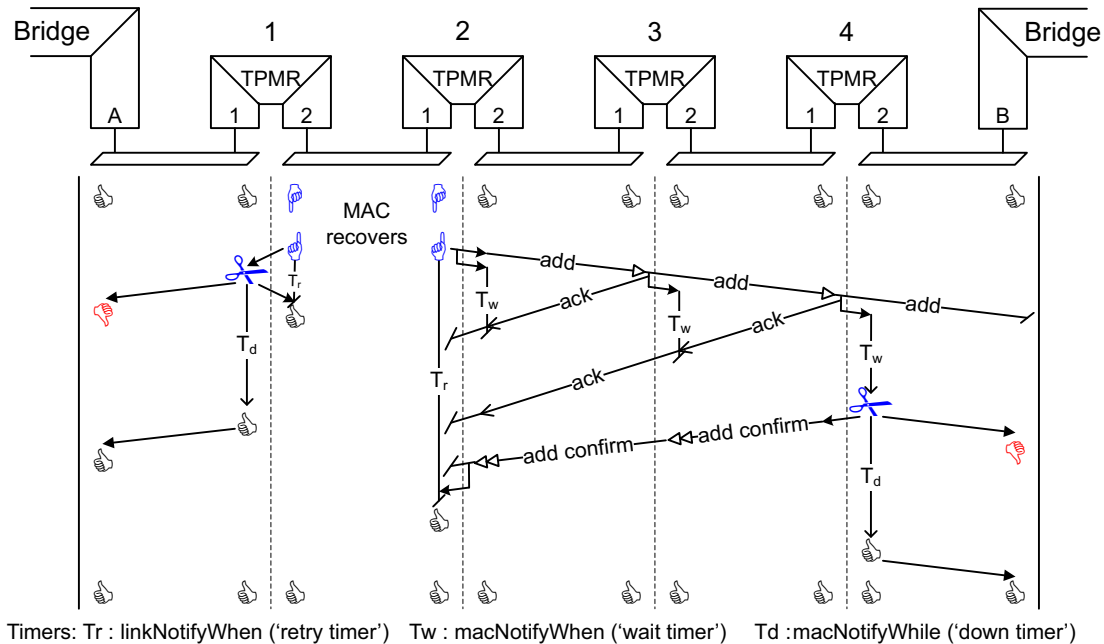


Figure 23-4—Adding connectivity

10 Before the LAN that connects TPMR 1 to TPMR 2 is MAC_Operational, each MSPE disables its MSS ()
11 This allows the MSPE to intercept the new connectivity as the LAN MAC asserts MAC_Operational ()
12 TPMR 2 propagates the new status to its other Port, transmitting an *add* MSPDU. TPMR 2 does not

1 necessarily know that TPMR 3 implements MSP (or indeed is a TPMR) but uses the default MSP
2 configuration—waiting before resorting to MAC status notification and starting a *linkNotifyWhile* timer (as
3 do TPMRs 3 and 4) on receipt of the *add* (which is forwarded by the Relay Entity). Each TPMR responds to
4 the *add* with an *ack* that clears the timer, but the Bridge at the end of the chain does not (as it does not
5 implement MSP). When TPMR 4's timer expires, its MSPE disables the LAN MAC (✂), ensuring that
6 *MAC_Operational* becomes FALSE (👎) for the Bridge's Port, and starts a *macNotifyWhile* timer. When
7 that timer expires, the MSPE reenables the LAN MAC so that the Bridge Port's *MAC_Operational* can
8 become TRUE (👍) once more. As well as disabling the LAN MAC for its Port 2, TPMR 4 also transmitted
9 an *add confirm* through Port 1, and receipt of this *add confirm* by TPMR 2 causes the latter to cancel its retry
10 timer (*linkNotifyWhen*). The initial value of *macNotifyWhile* (*MacNotifyTime*) is sufficient for the *add*
11 *confirm* to reach the TPMR 2, and for that TPMR's MSPE to enable the MSS, thus ensuring that there is
12 connectivity between the Bridges connected by the link when they have both reported *MAC_Operational*
13 TRUE to their local protocol clients.

14 In Figure 23-4 the MSP configuration of TPMR 2's Port 2 differs from that for Port 1 of TPMR 1. The latter
15 is explicitly configured to use MAC status notification immediately without waiting to see if its peer
16 implements MSP.

17 Figure 23-5 illustrates the operation of MSP when the connectivity provided by an individual LAN is lost.
18 The MSPE for each of the TPMR ports directly attached to the failed LAN begins by disabling the MSS for
19 that Port (👎) to ensure that connectivity does not 'flap'. Otherwise, the protocol proceeds as for connectivity
20 addition, except that *loss* and *loss confirm* are used instead of *add* and *add confirm*, and the MSSs attached
21 to the failed LAN are left disabled. The final state of the link components is the initial state assumed in
22 Figure 23-4—both attached Bridges have seen *MAC_Operational* transition to indicate that the connectivity
23 has changed, and the TPMRs in each of the two chains can be reached and managed (if their individual
24 configuration permits) through the attached Bridge Ports.

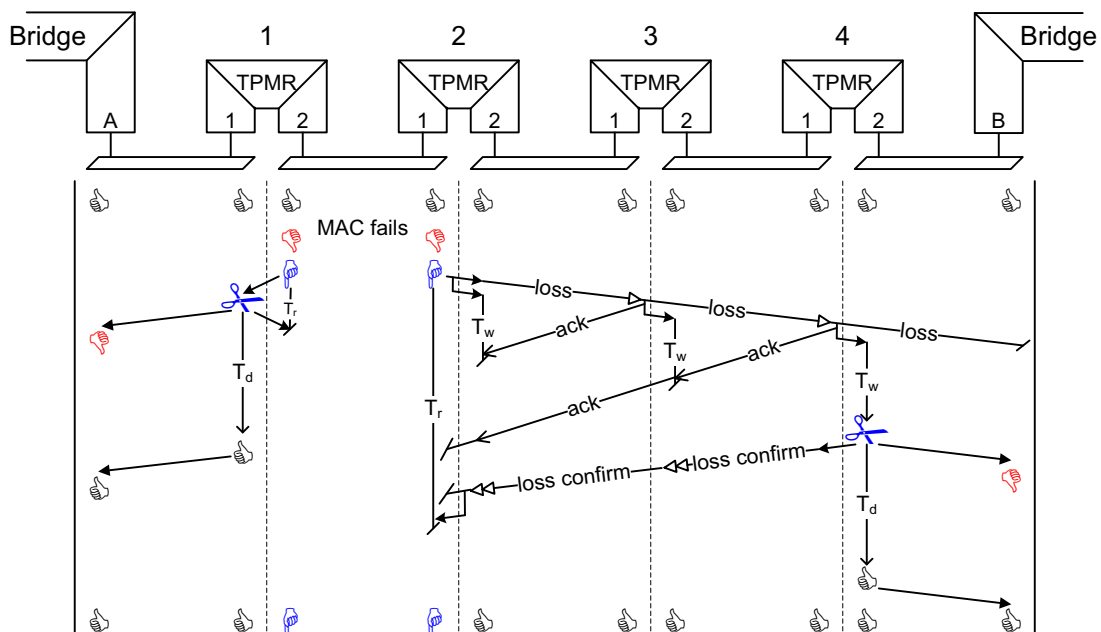


Figure 23-5—Losing connectivity

1 In the examples above, if an *add*, or *loss* is lost then the TPMR Port that transmitted or relayed that MSPDU
2 will revert to using MAC status notification, as will a TPMR that fails to receive the *ack* from the next (or a
3 subsequent) TPMR in the chain. If the MSPDU is lost on a LAN removed from the TPMR that initiated the
4 link status notification, then that notification will be retried on expiry of the *linkNotifyWhen* timer, which
5 also serves to protect against the loss of an *add confirm* or *loss confirm*.

6 Loss of a frame due to physical corruption is rare in LAN technologies, and frame losses due to buffer
7 overrun are not expected when connectivity is being added (as the link is not usable prior to the addition) nor
8 when a loss of connectivity is being signaled (as that connectivity loss will have prevented other frames
9 from being added to the link). The loss of an MSPDU is most likely to be due to failure of one of the LANs,
10 or to interruption of a TPMR's relay functionality by another MSS whose MSPE is also waiting for
11 confirmation that a connectivity change that it has detected has been propagated to the end of the link. MSP
12 does not, and cannot, ensure that information about each and every connectivity change reaches both ends of
13 the link. Unless link status notification or MAC status notification is disabled or the individual LANs fail to
14 report MAC_Operational correctly, MSP does ensure that any change in connectivity is accompanied by one
15 or more notifications at each end of the link, and that a continuous period during which MAC_Operational is
16 reported TRUE at both ends of the link provides unchanged connectivity from 1 second after the start of the
17 period to 1 second before the end. This guarantee meets the requirements implicit in the initial transmission
18 and retransmission strategies of well designed protocols. Protocol clients of the MAC Service should not use
19 the difference between *loss* and *add* MSPDUs to take different actions on receipt, though the distinction can
20 be useful to a network administrator when investigating connectivity changes.

21 Figure 23-6 provides a common example of simultaneous change. TPMR 2 is powered on and initializes
22 both its ports. When the LAN MAC becomes operational, each Port attempts to send a notification through
23 the other, but cannot as they are both waiting for the connectivity addition to be confirmed. However, in this
24 eventuality, each Port can provide the other with the confirmation required, because the peer system for each
25 of the individual LANs will also have seen the initial MAC_Operational transition and will have initiated a
26 notification toward its end of the link.

27 NOTE—The SNM state machine transition direct from SNM:LINK_NOTIFICATION to SNM:MAC_NOTIFYING
28 supports this behavior (see Figure 23-11).

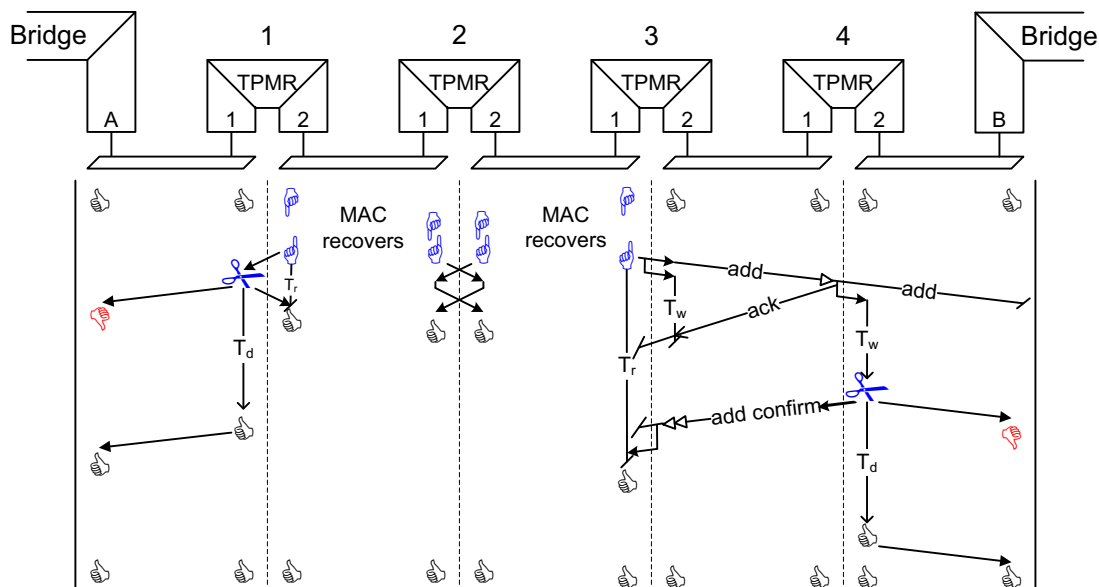


Figure 23-6—TPMR recovery

1 If the individual LAN that recovers (or loses) connectivity is at the end of the link, MSP ensures that the
2 other end of the link is also aware of the status change as illustrated in Figure 23-7. This figure also shows
3 the effect on MSP of including two nonstandard relays in the chain. Provided they (and their attached LANs)
4 never fail, MSP can continue to operate as intended.

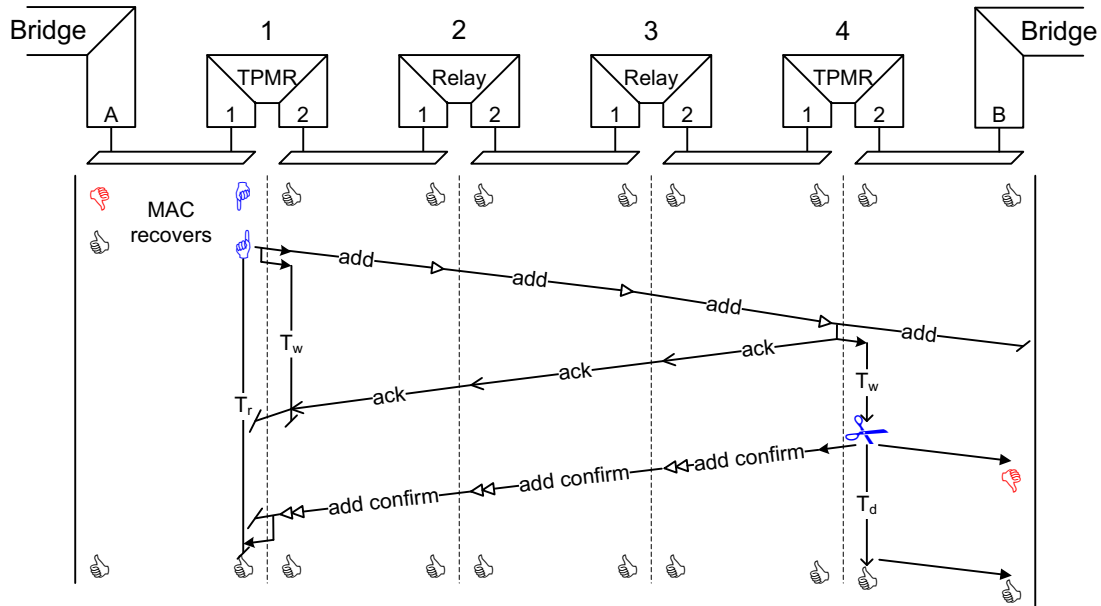


Figure 23-7—Notification from one end of the link to the other

5 If TPMR 4 is configured to use MAC status notification immediately, it does not return an *ack* as the *add*
6 *confirm* can be sent almost immediately (see Figure 23-8).

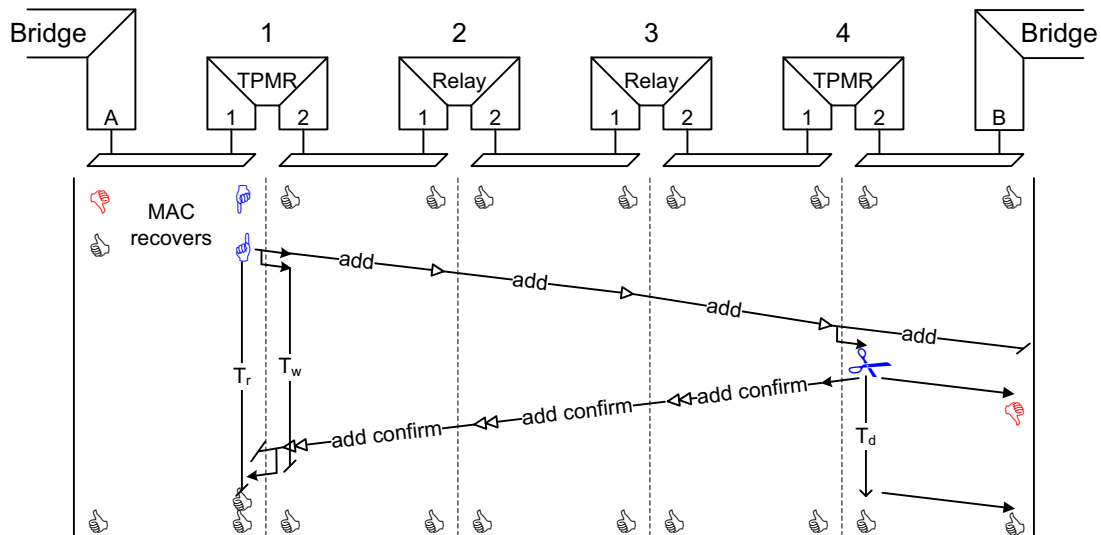


Figure 23-8—Immediate MAC status notification at the end of a link

23.3 MSP state machines

The operation of the MSPE is represented by an instance of each of the following for each Port of the TPMR:

- A Status Transition state machine (STM, 23.8)
- A Status Notification state machine (SNM, 23.9)
- A Receive Process (23.10)
- A Transmit Process (23.11)

Figure 23-9 shows the state machine variables that are used to communicate between these machines and processes, and that support management control over their operation. Variables prefixed with ‘r.’ are those of the corresponding state machines of the other Port of the TPMR.

The notational conventions used in Figure 23-9 and in the specification of the state machine are identical to those used in the specification of MSTP (Clause 13) and RSTP and are defined in Annex E.

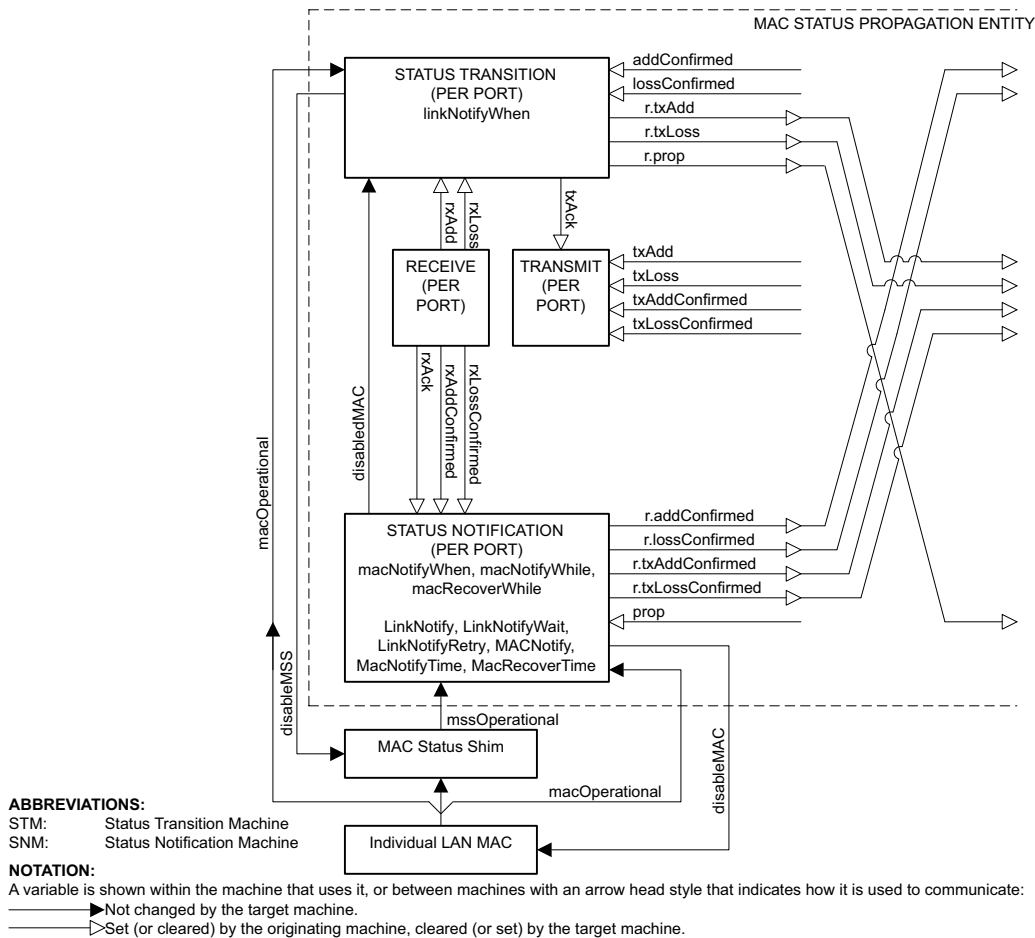


Figure 23-9—MSPE state machine overview

Port 1’s STM monitors MAC_Operational transitions for its own LAN, and tells Port 2’s Transmit process to send *add* or *loss* link notifications and Port 2’s SNM to monitor the progress of status notification, using MAC status notification if necessary. Port 2’s SNM receives the *ack* that indicates that it should wait for link status notification to complete, and the *add confirm* or *loss confirm* that indicates that completion. Whether

1 link status or MAC status notification is used, Port 2's SNM tells Port 1's STM when the notification has
2 been confirmed, so the latter does not have to retry the notification. Port 1's SNM provides the same service
3 to Port 2's STM.

4 A *loss* or *add* notification that is received from Port 1's own LAN is handled by its STM, which propagates
5 the notification to Port 2's SNM (in the same way as a local MAC_Operational transition) while transmitting
6 an *ack* on Port 1's LAN. Similarly Port 2's STM transmits an *ack* for a link status notification received on its
7 own LAN, and propagates the notification to Port 1's SNM.

8 **23.4 State machine timers**

9 Timers are implemented by variables that are decremented on each timer tick, with timer expiry occurring
10 when they reach zero.

11 **23.4.1 linkNotifyWhen**

12 Causes a link status notification to be sent on each expiry until the original status transition is confirmed.

13 **23.4.2 linkNotifyWhile**

14 Started when a change is first propagated through the Port, on expiry allows MAC status notification.

15 **23.4.3 macNotifyWhile**

16 Sets the time for which the MAC is disabled for MAC status propagation.

17 **23.4.4 macRecoverWhile**

18 Sets the time for which the MAC is permitted to be nonoperational, after being disabled, before the link is
19 reported as lost.

20 **23.5 MSP performance parameters**

21 These parameters are not modified by the operation of MSP but are treated as constants by the state
22 machines. They can be managed independently for each TPMR Port—default values and permissible ranges
23 are specified in Table 23-2.

Table 23-2—MSP performance parameters

Parameter	Recommended or default value	Permitted range
LinkNotify	TRUE	TRUE or FALSE
LinkNotifyWait	0.4 s	0.2–1.0 s
LinkNotifyRetry	1.0 s	0.1–1.0 s
MACNotify	TRUE	TRUE or FALSE
MACNotifyTime	0.2 s	0.01–0.5 s
MACRecoverTime	0.1 s	0.02–0.5 s

1 **23.5.1 LinkNotify**

2 TRUE if the Port uses link status notification to propagate MAC status, and will wait to allow link status
3 notification to succeed before using MAC status notification.

4 NOTE—If LinkNotify is FALSE, the TPMR still forwards *loss* and *add* notifications transmitted by other TPMRs prior
5 to using MAC status notification, but the TPMR will not originate link status notifications.

6 **23.5.2 LinkNotifyWait**

7 The initial value of the linkNotifyWhile timer.

8 **23.5.3 LinkNotifyRetry**

9 The initial value of the linkNotifyWhen timer of the other TPMR Port.

10 **23.5.4 MACNotify**

11 TRUE if the Port uses MAC status notification.

12 **23.5.5 MACNotifyTime**

13 The initial value of the macNotifyWhile timer.

14 **23.5.6 MACRecoverTime**

15 The initial value of the macRecoverWhile timer.

16 **23.6 State machine variables**

17 **23.6.1 BEGIN**

18 This is a Boolean variable controlled by the system initialization process. A value of TRUE causes all
19 TPMR state machines to continuously execute their initial state. A value of FALSE allows all state machines
20 to perform transitions out of their initial state, in accordance with the relevant state machine definitions.

21 **23.6.2 addConfirmed**

22 Set by the other Port's SNM to tell STM that the addition has been confirmed. Cleared by STM.

23 **23.6.3 disableMAC**

24 Set by SNM to instruct the individual LAN MAC (via an LMI) to disable itself in a way that will cause
25 MAC_Operational to be FALSE for the peer user of the MAC Service provided by that LAN.

26 NOTE—The state machines do not assume that a client of the MAC can tell whether the MAC is not operational because
27 that client has disabled it, or whether some other client has disabled it.

28 **23.6.4 disabledMAC**

29 Set by the SNM when it has set disableMAC and for MACRecoverTime after disableMAC has been reset,
30 so that the STM does not conclude that a loss notification should be sent through the other TPMR Port.

31 **23.6.5 disableMSS**

32 Set (or reset) by the STM to instruct the MSS (via an LMI) to set its MAC_Enabled status parameter to
33 FALSE (correspondingly, TRUE).

34 **23.6.6 lossConfirmed**

35 Similar to addConfirmed but confirms a loss.

1 **23.6.7 macOperational**

2 The value of MAC_Operational for the individual LAN MAC.

3 **23.6.8 mssOperational**

4 The value of MAC_Operational provided by the MSS to the TPMR Port's Bridge transmit and receive
5 function.

6 **23.6.9 prop**

7 Set by the other Port's STM to Add or Loss to notify SNM that a change is being propagated through the
8 Port. Reset by SNM to None.

9 **23.6.10 rxAck**

10 Set by the Receive process to tell SNM that an acknowledgment has been received. Cleared by SNM.

11 **23.6.11 rxAdd**

12 Set by the Receive process to tell STM that an *add* notification is being propagated through the TPMR.
13 Cleared by STM.

14 **23.6.12 rxAddConfirm**

15 Set by the Receive process to tell SNM that an *add confirm* has been received. Cleared by SNM.

16 **23.6.13 rxLoss**

17 Similar to rxAdd, but for a *loss*.

18 **23.6.14 rxLossConfirm**

19 Similar to rxAddConfirm, but for a *loss confirm*.

20 **23.6.15 txAck**

21 Set by the STM to instruct the Transmit process to send an acknowledgment. Cleared by the Transmit
22 process.

23 **23.6.16 txAdd**

24 Set by the other Port's STM to cause transmission of an *add* notification. Cleared by the Transmit process.

25 **23.6.17 txAddConfirm**

26 Set by the other Port's SNM to cause transmission of an *add confirm* (through this Port) confirming that a
27 received *add* message has been acted upon. Cleared by the Transmit process.

28 **23.6.18 txLoss**

29 Similar to a txAdd but causes a *loss* message to be transmitted.

30 **23.6.19 txLossConfirm**

31 Similar to a txAddConfirm but causes a *loss confirm* message to be transmitted.

23.7 State machine procedures

No procedures are defined beyond those represented in the state machines.

23.8 Status Transition state machine (STM)

The STM shall implement the function specified by the state diagram in Figure 23-10 and the attendant definitions contained in 23.4 through 23.7.

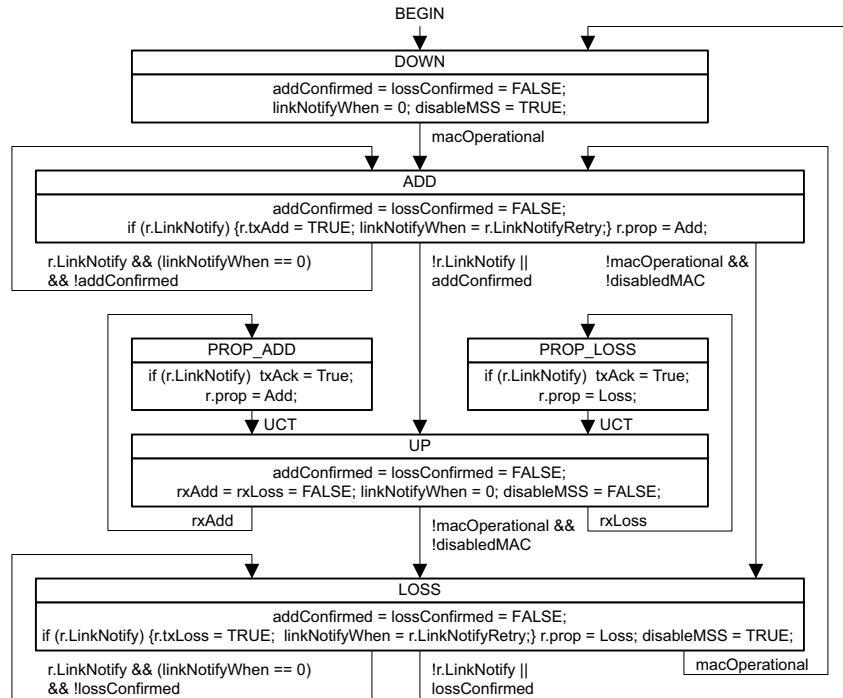


Figure 23-10—Status Transition state machine (STM)

23.9 Status Notification state machine (SNM)

The SNM shall implement the function specified by the state diagram in Figure 23-11 and the attendant definitions contained in 23.4 through 23.7.

23.10 Receive Process

The Receive Process shall receive and validate MSPDUs as specified in 23.16.

23.11 Transmit Process

The Transmit Process shall transmit and encode MSPDUs as specified in 23.13 through 23.15. If the Transmit Process is instructed to transmit an MSPDU before it has had the opportunity to transmit a prior MSPDU, that prior MSPDU shall be discarded and not transmitted. If the MAC_Operational status provided by the MSS for the Port is FALSE, the MSPDU shall be discarded and not transmitted.

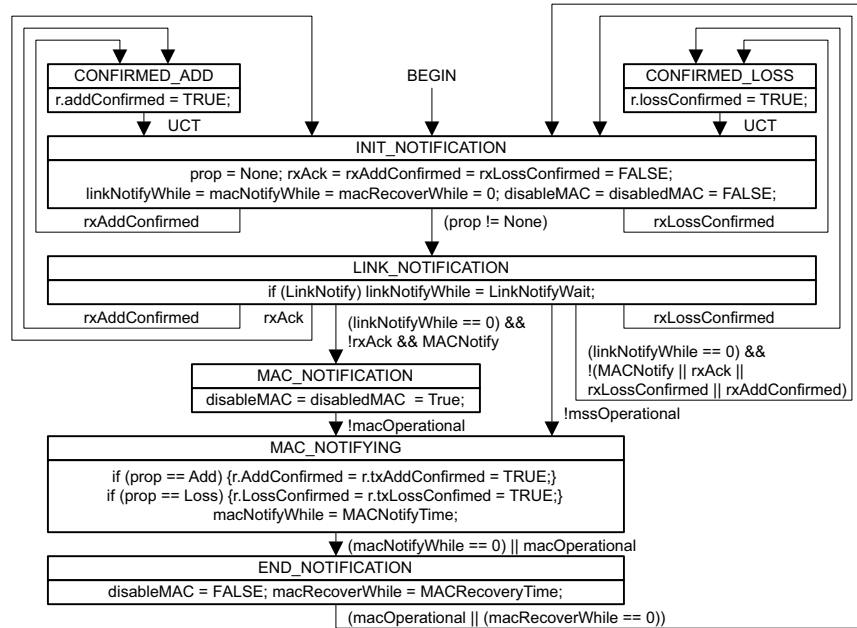


Figure 23-11—Status Notification state machine (SNM)

23.12 Management of MSP

An implementation of MSP in a TPMR:

- a) May allow the performance parameters (23.5) to be read by management.
- b) May allow the performance parameters (23.5) to be modified by management.
- c) May maintain each of the following counts for one or both ports of the TPMR:
 - 1) acksTransmitted: The number of *acks* transmitted (23.6.15) by the Port's Transmit Process as a consequence of *txAck* being set.
 - 2) addNotificationsTransmitted: The number of *adds* transmitted (23.6.16) by the Port's Transmit Process as a consequence of *txAdd* being set.
 - 3) addConfirmationsTransmitted: The number of *add confirms* transmitted (23.6.17) by the Port's Transmit Process as a consequence of *txAddConfirm* being set.
 - 4) lossNotificationsTransmitted: The number of *losses* transmitted (23.6.18) by the Port's Transmit Process as a consequence of *txLoss* being set.
 - 5) lossConfirmationsTransmitted: The number of *loss confirms* transmitted (23.6.19) by the Port's Transmit Process as a consequence of *txLossConfirm* being set.
 - 6) acksReceived: The number of *acks* received (23.6.10) by the Port's Transmit Process.
 - 7) addNotificationsReceived: The number of *adds* received (23.6.11) by the Port's Receive Process.
 - 8) addConfirmationsReceived: The number of *add confirms* received (23.6.12) by the Port's Receive Process.
 - 9) lossNotificationsReceived: The number of *losses* received (23.6.13) by the Port's Receive Process.
 - 10) lossConfirmationsReceived: The number of *loss confirms* received (23.6.14) by the Port's Receive Process.
 - 11) addEvents: The number of transitions to STM:ADD directly from STM:DOWN or STM:LOSS (23.8).

- 1 12) lossEvents: The number of transitions to STM:LOSS directly from STM:UP or STM:ADD
2 (23.8).
3 13) macStatusNotifications: The number of transitions to SNM:MAC_NOTIFICATION (23.9).

4 **23.13 MSPDU transmission, addressing, and protocol identification**

5 MSPDUs are transmitted and received using the service provided by an LLC entity that uses, in turn, a
6 single instance of the MAC Service provided at an MSAP. In a TPMR the MSPE transmits and receives the
7 MSPDUs, and the MSAP is provided by the TPMR Port connectivity function as illustrated in Figure 23-3.
8 Each MSPDU is transmitted as a single MAC Service request, and received as a single MAC Service
9 indication, with the following parameters:

- 10 a) destination address (23.13.1)
11 b) source address (23.13.2)
12 c) MSDU
13 d) priority (23.13.3)

14 The MSDU of each request and indication comprises an number of octets that provide EtherType protocol
15 identification (23.13.4) followed by the MSPDU proper (23.15).

16 NOTE 1—For the purposes of this standard, the term “LLC entity” includes entities that support protocol discrimination
17 using the EtherType field as specified in IEEE Std 802.

18 NOTE 2—The complete format of an MSP frame “on the wire” or “through the air” depends not only on the MSPDU
19 format, as specified in this clause, but also on the media access method-dependent procedures used to support the MAC
20 Service.

21 **23.13.1 Destination MAC Address**

22 The destination address for each MAC Service request used to transmit an MSPDU shall be the group
23 address identified in Table 8-1 and Table 8-2 as “Nearest non-TPMR Bridge group address”⁴⁰.

24 **23.13.2 Source MAC Address**

25 The source address for each MAC Service request used to transmit an MSPDU shall be an individual
26 address associated with the MSAP at which the request is made.

27 **23.13.3 Priority**

28 The priority associated with each MAC Service request should be the default associated with the MSAP.
29 Transmitted MSPDUs are not VLAN tagged or priority tagged.

30 **23.13.4 EtherType use and encoding**

31 All MSPDUs are identified by the EtherType specified in Table 23-3.

Table 23-3—MSP EtherType assignment

Assignment	Value
MAC Status Protocol EtherType	22-E2

⁴⁰ This address was originally assigned by IEEE Std 802.1X-2001 and was identified as an S-VLAN component Reserved Address (Table 8-2) by the IEEE Std 802.1ad amendment to IEEE Std 802.1Q-2005. This standard identifies it as the “nearest non-TPMR Bridge group address,” to be filtered by all relay functions above the sublayer specified for TPMRs (see Table 8-1 and Table 8-2).

23.14 Representation and encoding of octets

All MSPDUs consist of an integral number of octets, numbered starting from 1 and increasing in the order that they are put into a MAC frame. The bits in each octet are numbered from 1 to 8, where 1 is the low-order bit. When consecutive octets are used to encode a binary number, the lower numbered octet contains the more significant bits of the binary number.

When the encoding of (an element of) an MSPDU is represented using a diagram in this clause, the following representations are used:

- a) Octet 1 is shown toward the top of the page, higher numbered octets being toward the bottom.
- b) Where more than one octet appears on a given line, octets are shown with the lowest numbered octet to the left, higher numbered octets being to the right.
- c) Within an octet, bits are shown with bit 8 to the left and bit 1 to the right.

23.15 MSPDU structure

The MSPDU comprises the octets following the MSP EtherType. All MSPDUs comprise a Protocol Version (23.15.1) and a Packet Type (23.15.2). See Figure 23-12.

	Octet
Protocol Version (23.15.1)	1
Packet Type (23.15.2)	2

Figure 23-13—MSPDU structure

23.15.1 Protocol Version

The MSP Protocol Version is encoded in all MSPDUs as a single octet, representing an unsigned binary number. Its value identifies the version of MSP supported by originator of the MSPDU. An implementation conforming to this specification shall encode the value 0000 0000 in this field. All other values are reserved.

NOTE—TPMRs that relay an MSPDU do not change its Protocol Version.

23.15.2 Packet Type

The MSP Packet Type is encoded as a single octet, representing an unsigned binary number. Table 23-4 lists the Packet Types specified by this standard, and the state machine variables set to indicate reception and transmission of MSPDUs of that type. All other possible values of the Packet Type field are reserved and shall not be used.

Table 23-4—MSP Packet Types

Packet Type	Value	Transmission	Reception
MSP-Add	0	txAdd (23.6.16)	rxAdd (23.6.11)
MSP-Loss	1	txLoss (23.6.18)	rxLoss (23.6.13)
MSP-Add Confirmed	2	txAddConfirm (23.6.17)	rxAddConfirm (23.6.12)
MSP-Loss Confirmed	3	txLossConfirm (23.6.19)	rxLossConfirm (23.6.14)
MSP-Ack	4	txAck (23.6.15)	rxAck (23.6.10)

1 23.16 Validation of received MSPDUs

2 To ensure that backward compatibility is maintained for future versions of this protocol, the validation and
3 protocol version handling for all MSPDUs, follows general rules developed for this and other protocols. A
4 received MSPDU shall be processed as specified by Table 23-4 if and only if:

- 5 a) The destination MAC address is the group address specified (23.13.1); and
- 6 b) The MSPDU is identified by the MSP EtherType encoded as specified in 23.13.4; and
- 7 c) The received MSPDU contains at least two octets, i.e., at least the Protocol Version and Packet
8 Type; and
- 9 d) The Packet Type is one of the values specified in Table 23-4.

10 Otherwise, the received MSPDU shall be discarded. No other checks shall be applied to received MSPDUs,
11 in particular the value of the Protocol Version is not checked and MSPDUs of length greater than the
12 minimum of two octets are accepted as valid.

13 23.17 Other MSP participants

14 An end station or non-TPMR Bridge attached to the end of a TPMR link can participate in link status
15 notification, avoiding the need for the last TPMR in the chain to use MAC status notification and thus
16 speeding the transition of the link to an operational state. Such a participant receives MSPDUs, but acts only
17 on a received *loss* or *add*, notifying its protocol clients of the change in connectivity, and responding
18 immediately by transmitting an *add confirm* or *loss confirm* as appropriate. There is no need for such a
19 participant to transmit an *add*, *loss*, or *ack*, or act upon a received *ack*, *add confirm* or *loss confirm*, or to
20 initiate MAC status notification.

1 24. Bridge performance

2 This clause specifies a set of parameters that represent the performance of a Bridge. These parameters have
3 been selected to allow a basic level of confidence to be established in a Bridge, for use in an initial
4 determination of its suitability for a given application. They cannot be considered to provide an exhaustive
5 description of the performance of a Bridge. It is recommended that further performance information be
6 provided and sought concerning the applicability of a Bridge implementation.

7 The following set of performance parameters is defined:

- 8 a) Guaranteed Port Filtering Rate, and a related time interval T_F , that together characterize the traffic
9 for which filtering is guaranteed.
- 10 b) Guaranteed Bridge Relaying Rate, and a related time interval T_R .

11 24.1 Guaranteed Port Filtering Rate

12 For a specific Bridge Port, a valid Guaranteed Port Filtering Rate, in frames per second, is a value that, given
13 any set of frames from the specific Bridge Port to be filtered during any T_F interval, the Forwarding Process
14 shall filter all of the set as long as all of the following are true:

- 15 a) The number of frames in the set does not exceed the specific Bridge Port's Guaranteed Port Filtering
16 Rate multiplied by T_F .
- 17 b) The Guaranteed Port Filtering Rate of each of the other Bridge Port(s) is not exceeded.
- 18 c) The Guaranteed Bridge Relaying Rate is not exceeded.
- 19 d) Relayed frames are not discarded due to output congestion (8.6.7).
- 20 e) The information upon which the filtering decisions are based has been configured in the FDB prior
21 to the start of time interval T_F .

22 24.2 Guaranteed Bridge Relaying Rate

23 For a Bridge, a valid Guaranteed Bridge Relaying Rate, in frames per second, is a value that given any set of
24 frames from the specific Bridge Port to be relayed during any T_R interval, the Forwarding Process shall relay
25 all of the set as long as all of the following are true:

- 26 a) The number of frames in the set does not exceed the Bridge's Guaranteed Bridge Relaying Rate
27 multiplied by T_R .
- 28 b) The Guaranteed Port Filtering Rate of each Bridge Port is not exceeded.
- 29 c) Relayed frames are not discarded due to output congestion (8.6.7).
- 30 d) The information upon which the forwarding decisions are based has been configured in the FDB
31 prior to the start of time interval T_R .

24.3 RSTP performance requirements

This subclause places requirements on the performance of the Spanning Tree Protocol Entities of Bridges in a Bridged Network to ensure that RSTP operates correctly.

The delay between the occurrence of an external event and the action or actions mandated by the RSTP specification as a consequence of the event shall not exceed the Maximum RSTP processing delay specified in Table 24-1.

Table 24-1—Transmission and reception delays

Parameter	Absolute maximum value (s)
Maximum RSTP processing delay	1.0
Maximum BPDU transmission delay	0.2

External events subject to this provision of this specification shall include the following:

- a) Transmission of a BPDU by another Bridge on a LAN to which the Bridge is attached.

Specified actions subject to this provision shall include the following:

- b) Transmissions of BPDUs on all Ports mandated by the RSTP specification, with the exception of transmissions delayed by the Port Transmit state machine (13.34) to enforce transmit rate limits.
- c) Ceasing to learn or forward frames.

The delay between internal timer related events and the transmission of all BPDUs on Ports mandated by the RSTP specification as a consequence shall not exceed the Maximum BPDU transmission delay specified in Table 24-1. Timer events subject to this provision of this specification shall include the following:

- d) Decrementing of txCount by the Port Transmit machine, thus allowing a BPDU transmission if transmit rate limits were being enforced.
- e) Expiry of the helloWhen timer (13.25.3).

25. Support of the MAC Service by PBBNs

A PBBN comprises a set of BEBs interconnected by some or all of the S-VLANs supported by a PBN (see Clause 16). Each BEB provides interfaces that encapsulate (or verify the encapsulation of) customer frames, thus allowing Customer MAC (C-MAC) addresses and VLAN IDs to be independent of the Backbone MAC (B-MAC) addresses and VLANs administered by the PBBN operator and used to relay those frames across the backbone. The S-VLANs used to encapsulate customer frames are known as Backbone VLANs (B-VLANs), and the resources that support those VLANs are usually considered to be part of the PBBN.

NOTE 1—The term Customer used in the context of PBBs may refer to a first Provider who is purchasing service from a second Provider. The first Provider deploys a PBN or a PBBN within its domain. The second Provider deploys a PBBN within its domain. In this case, the first Provider is identified as a Customer with respect to the second Provider. The term Customer used here may alternatively refer to each of a pair of Providers communicating as peers.

Figure 25-1 shows peer BEBs within the MAC sublayer and their relationship to customer and backbone bridging functions. The right-most BEB shown is modeled as comprising two types of VLAN Bridge components, an I-component and a B-component, connected together. The two BEBs on the left of the figure each comprise a single VLAN Bridge component of different types. The left-most BEB has a single I-component, while the BEB just to the right of the left-most BEB has a B-component. A T-component may also be used in place of an I-component.

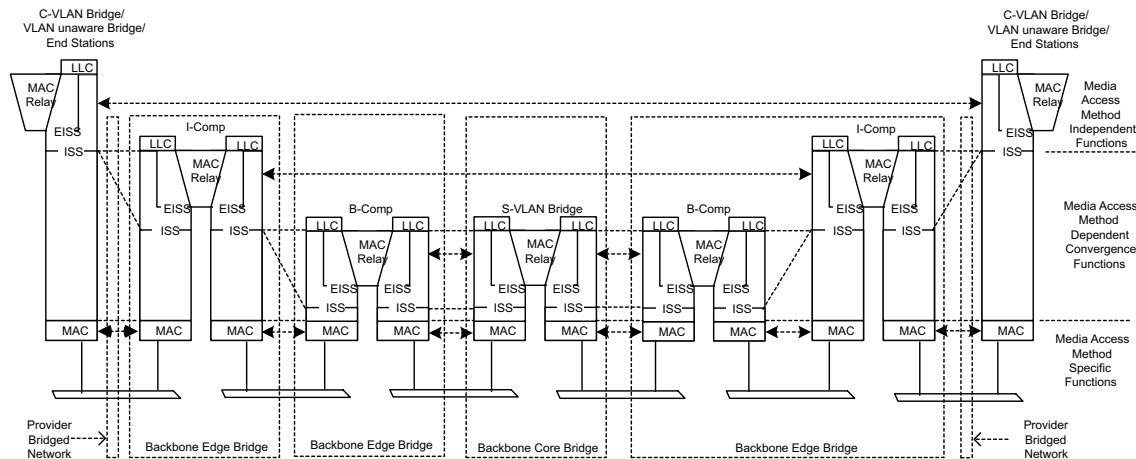


Figure 25-1—Internal organization of the MAC sublayer in a PBBN

Each I-component or T-component is responsible for encapsulating frames received from customers and assigning each frame to a backbone service instance. The backbone service instance consists of a set of BEBs that support a given customer's S-VLANs, and is uniquely identified within the PBBN by an I-SID. The customer frame is encapsulated by an I-TAG, which includes the I-SID, and a set of source and destination B-MAC addresses. The B-MAC addresses identify the BEBs of the backbone service instance where the customer frame will enter and exit the PBBN. If the I-component or T-component does not know which of the other BEBs provides connectivity to a given customer address, it uses a default encapsulating B-MAC address that reaches all the other BEBs in the backbone service instance. Each I-component learns the association between customer source addresses received (encapsulated) from the backbone and the backbone source address, so subsequent frames to that address can be transmitted to the correct BEB.

A PBBN or a series of PBBNs providing the MAC Service to attached end stations is typically modeled as a symmetric sequence of relay functions, as illustrated in Figure 25-1. The outermost peer relay functions are identified as I-components. The next peer relay functions in the sequence are identified as B-components.

1 Between the peer B-components are one or more S-VLAN relay functions. A B-component relay forms the
2 service layer to an I-component relay. A B-component relay forwards frames taking into account the identity
3 of a B-VLAN (B-VID), while an I-component relay forwards frames taking into account the identity of an
4 S-VLAN (S-VID).

5 A single B-component is responsible for relaying encapsulated customer frames to and from I-components
6 and T-components, either within the same BEB or externally connected, checking that ingress/egress is
7 permitted for frames with that I-SID, translating the I-SID (if necessary) and using it to assign the supporting
8 connection parameters (backbone addresses if necessary and VIDs) for the PBBN, and relaying the frame to
9 and from the PNP(s) that provide connectivity to the other Bridges within and attached to the backbone. A
10 B-component performs the same functions when relaying frames to and from another B-component when
11 two PBBNs interconnect (26.6.2).

12 The number of I-components (zero or more), T-components (zero or more), and B-components (zero or
13 one) within a given BEB is a direct consequence of the type and number of external interfaces supported by
14 that BEB. The I-components, T-components, and B-component may be in the same BEB or may be in
15 different BEBs. The types of BEBs may be classified as I-BEB, B-BEB, T-BEB, and ITB-BEB as depicted
16 in Figure 25-2, where ITB-BEB covers all valid combinations of multi-component BEBs, e.g., IB-BEB.

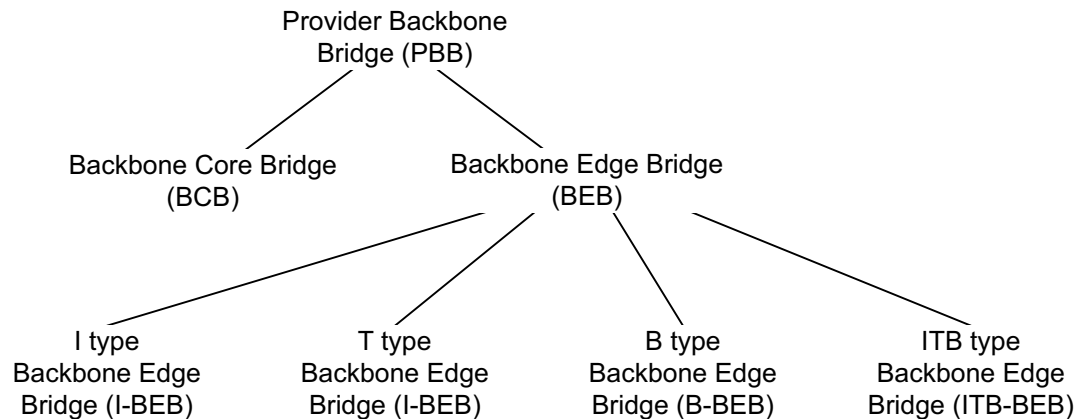


Figure 25-2—PBB terminology

17 This clause discusses the following aspects of PBBN service:

- 18 a) Service transparency (25.1)
- 19 b) Customer service interfaces (25.2, 25.3, 25.4, 25.5)
- 20 c) Service instance segregation (25.6)
- 21 d) Service instance and backbone service instance selection and identification (25.7)
- 22 e) Service priority and drop eligibility selection (25.8)
- 23 f) Service access protection (25.9)

24 NOTE 2—This standard makes use of the term *service* as defined by the OSI Reference Model (ISO/IEC 7498-1). In this
25 sense, a service comprises a set of primitives and associated parameters, provided by one protocol layer in the
26 architectural model to the protocol layer above, and the causal relationships between the primitives invoked by an upper
27 layer protocol entity in one system with those resulting indications to a peer entity in another system. The term service
28 used by backbone providers, while including layering concepts, goes far beyond this formal definition, and commonly
29 specifies some or all of the following: interfacing considerations across multiple protocol layers (including physical
30 connectors, for example); selection of interface points; interfacing equipment; QoS guarantees and measurement
31 methods; charging methods and responsibilities; connectivity verification and other management tools; and regulatory
32 issues.

25.1 Service transparency

The operation of PBBNs is, by design, largely transparent to Provider Bridges and PBNs as illustrated by Figure 25-1. The service provided by BEBs is transparent to the use of the MAC Service by end stations attached to the CBNs through PBNs and transparent to the operation of media access method-independent functions by Customer Bridges.

The service is not transparent to the operation of media access method-dependent convergence functions or to the operation of the media access method-specific functions specified by standards for each media access method. Media access method-dependent and -specific functions operate between Bridges—whether Customer Bridges, Provider Bridges, or PBBs—attached to the same LAN.

25.2 Customer service interface

A backbone provider can offer to customers one or more types of service interfaces, each providing different capabilities for service selection, priority selection, and service access protection (25.7, 25.8, 25.9). In some cases it is assumed the customer provides an S-VLAN component of a Provider Bridge while in other cases, more generic customer systems are also allowed. There are four basic types of customer service interfaces—Port-based, S-tagged, I-tagged, and transparent service interface. The customer service interface types are summarized by Figure 25-3.

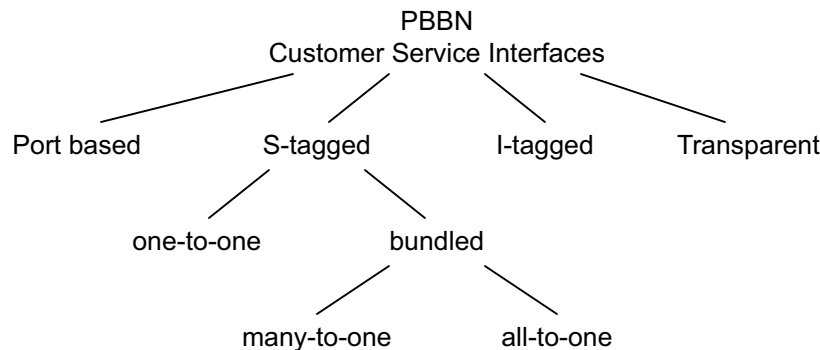


Figure 25-3—Customer service interface types

In all cases, segregation of different service instances is achieved at an interface wholly under the control of the backbone provider, and by verification of customer provided parameters that provide service instance selection. Stronger authentication and authorization of the attached customer systems can be achieved by use of IEEE Std 802.1X.

NOTE—The term service access protection describes provision of service access over multiple access LANs and (or) nodes with redundancy and rapid failover in case of failure of an access LAN or attached equipment.

Access to a given backbone service instance can be provided through different types of customer interfaces.

25.3 Port-based service interface

A PBBN may provide a Port-based service interface for customer attachment. The PBBN Port-based interface provides the same type of service to a customer as the PBN Port-based interface described in 15.3. A Port-based service interface is delivered on a CNP provided by a BEB as illustrated in Figure 25-4 and Figure 25-5. A Port-based service interface may attach to a C-VLAN Bridge (5.9), MAC Bridge, router, or end-station. The service provided by this interface forwards all frames without an S-TAG over the backbone on a single backbone service instance. All frames with an S-TAG that has a non-null VID are discarded by a Port-based service interface.

The Port-based service interface requires specific constraints on the configuration of the I-component. Each I-component CNP providing a Port-based interface is associated with one and only one VIP with exactly the same parameters for their tagging functions support. More specifically both CNP and VIP

- Have the Acceptable Frame Types parameter configured to Admit Only Untagged and Priority-tagged frames.
- Have equal PVID values.
- Are exclusive members of the same PVID member set.
- Are members of the PVID untagged egress set.

Figure 25-4 illustrates a customer system attached to a PBBN over a Port-based service interface.

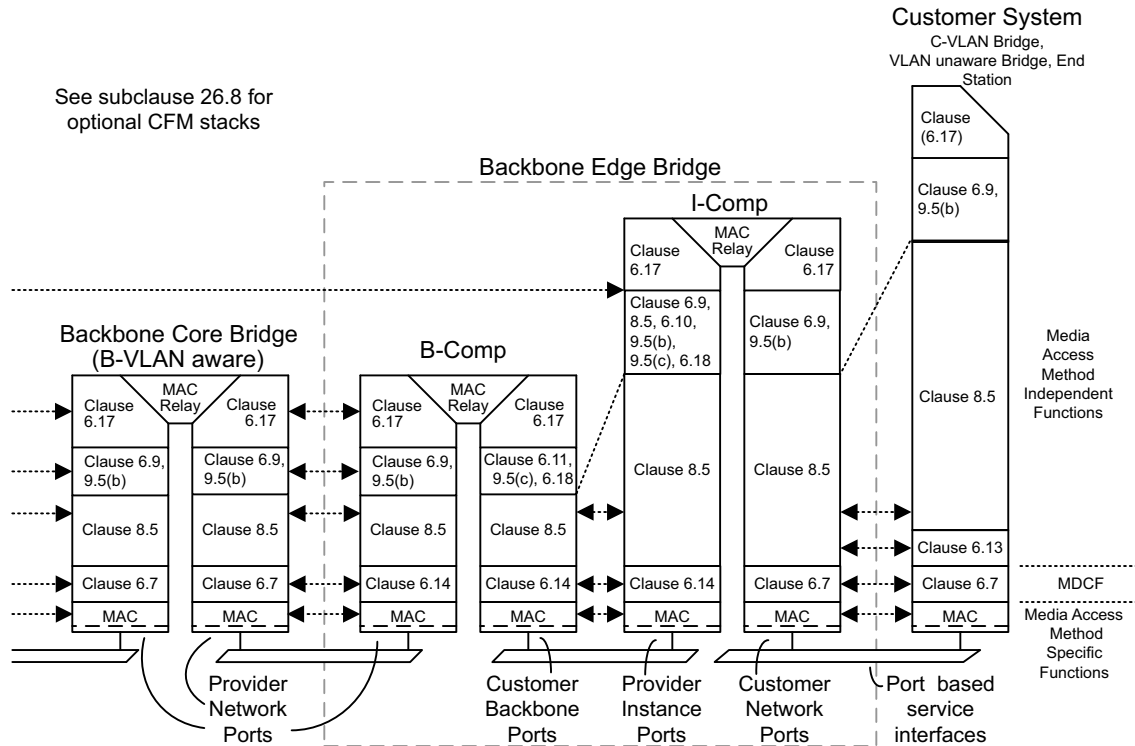


Figure 25-4—Port-based service interface

Figure 25-5 shows an example of equipment used to implement an unprotected Port-based service interface. For details on redundant connections and equipment, see 25.9. An I-component may support one or more Port-based service interfaces. Each CNP may be associated with one Port-based service interface. A CNP is connected to a customer system (or network) using a LAN.

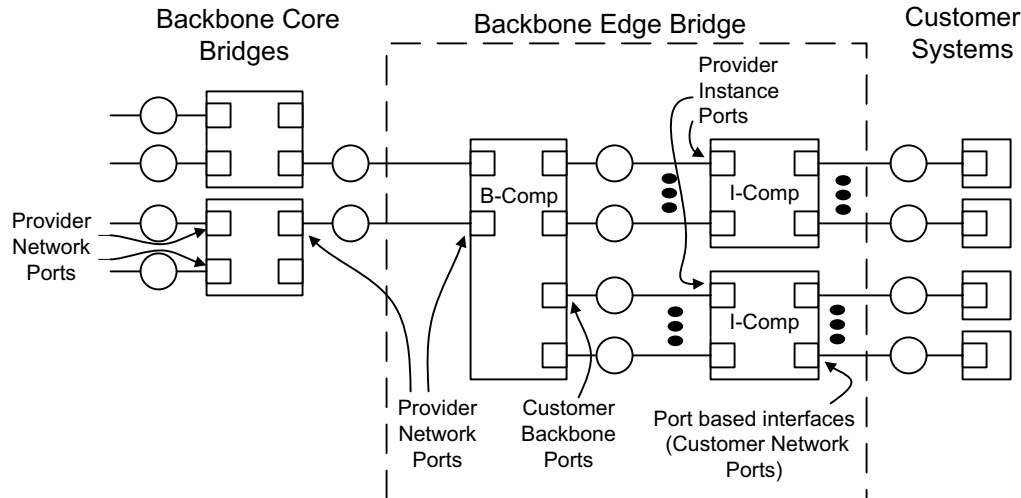


Figure 25-5—Port-based interface equipment

25.4 S-tagged service interface

The S-tagged service interface maps a service instance from a PBN, identified by an S-VID, to a backbone service instance on the PBBN, identified by an I-SID. There are two types of S-tagged service interfaces—one performing a one-to-one mapping of S-VIDs to I-SIDs and the other bundling S-VIDs to I-SIDs. Frames that are mapped to the I-SID are carried over the PBBN while frames that are not mapped to an I-SID are not carried over the PBBN.

NOTE 1—The restriction that each PBN S-VLAN map to a single backbone service instance on the PBBN allows the PBN equipment receiving frames to correctly identify the service instance used to deliver that frame and prevents the configuration of the I-component to create a multipoint service from point-to-point service instances, which could result in accidental creation of data loops. The backbone provider can offer a multipoint service through appropriate configuration of the B-VLAN component.

A PBBN may provide an S-tagged service interface for attachment to customer PBNs (15.5). An S-tagged service interface is provided by a BEB over a CNP as illustrated by Figure 25-7 and Figure 25-8. The attached Provider Bridges can in turn provide Port-based, C-tagged, or S-tagged service interfaces to their customers as described in 15.2, 15.3, 15.4, and 15.5.

The S-tagged service interface has the variations shown in Figure 25-3 under the S-tagged branch. The first variation, called a one-to-one S-tagged interface, uses a one-to-one mapping between S-VIDs and I-SIDs. This interface variation maps each S-VID to a single I-SID for use over the PBBN. The one-to-one mapped interface does not carry the S-TAG over the PBBN. The DEI and PCP bits may be regenerated on ingress and are then carried in the I-DEI and I-PCP bits in the I-TAG across the PBBN. On egress from the one-to-one S-tagged interface, the S-TAG can be deduced from the I-TAG received from the PBBN (the I-SID is mapped to an S-VID, the I-DEI and I-PCP bits may be regenerated and are then carried in the DEI and PCP bits).

The second S-tagged service interface variation is the bundling S-tagged service interface. This interface variation maps multiple S-VIDs to a single I-SID for delivery over the PBBN. To allow the remote end to reconstruct the S-VID, this interface variation will carry an S-TAG over the PBBN. On a bundled S-tagged interface, the DEI and PCP bits of the S-TAG may be regenerated and are then carried in both the DEI and PCP bits of the S-TAG and the I-DEI and I-PCP bits of the I-TAG over the PBBN.

A special case of the bundling S-tagged service interface is where all S-VIDs are mapped to a single I-SID. This special case is called an all-to-one bundling S-tagged service interface. The I-component used for an all-to-one bundled S-tagged service interface is constrained to supporting a single S-tagged service interface.

1 Figure 25-6 illustrates the information passed over each of the ISS interfaces of a BEB.

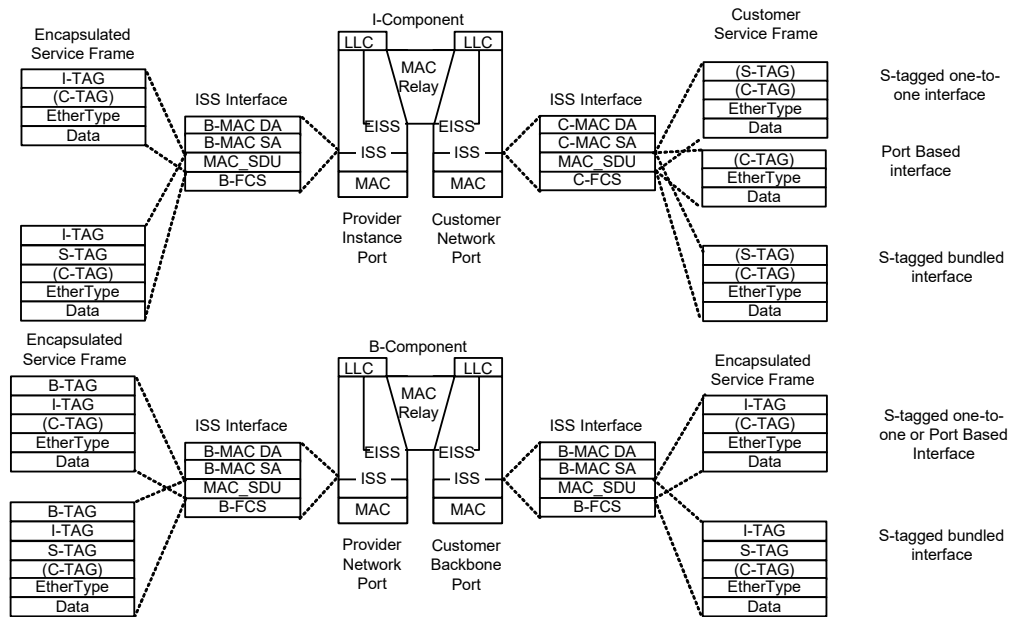


Figure 25-6—Encapsulated service frames at ISS

2 Figure 25-7 illustrates a customer network attached to a PBBN using an S-tagged service interface. The
3 customer network uses Provider Bridges with S-VLAN components for connecting to the PBBN. The
4 PBBN in turn is composed of BEBs interfacing to the customer Provider Bridges and BCBs used to forward
5 frames between the BEBs.

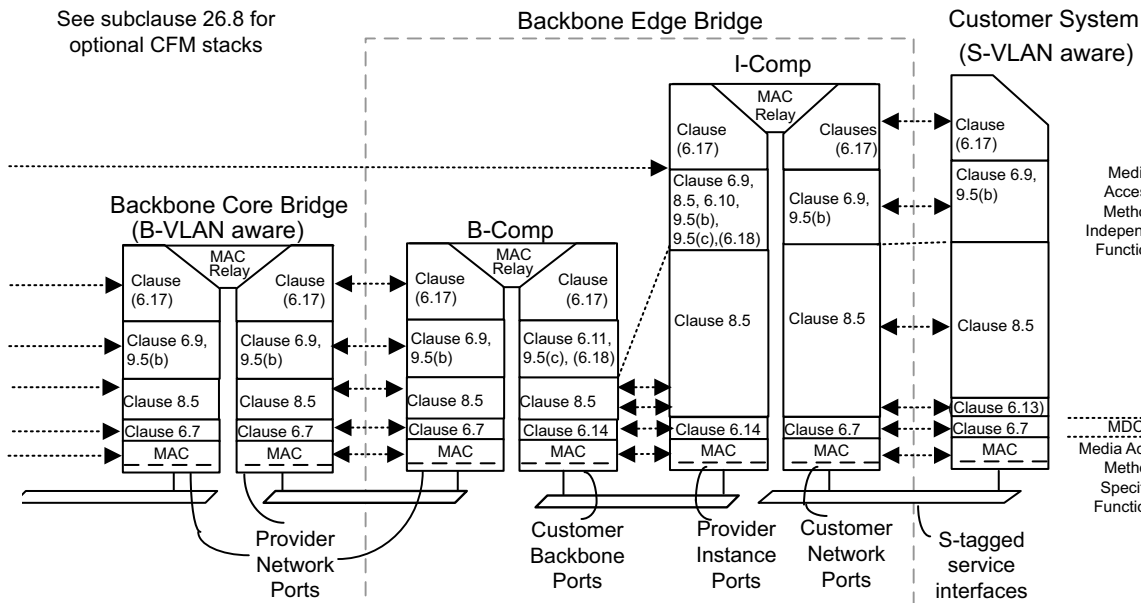


Figure 25-7—S-tagged service interface

1 Figure 25-8 shows an example of equipment used to implement an unprotected S-tagged service interface.
2 For details on redundant connections and equipment, see 25.9. In this diagram, a BEB is formed by
3 connecting two I-components and a B-component. These connections may be over a backplane or over a
4 LAN. An I-component may support one or more S-tagged service interfaces. When an I-component supports
5 an all-to-one bundled S-tagged service interface, the entire I-component must be dedicated to a single S-
6 tagged service interface. Each CNP may be associated with one S-tagged service interface. A CNP is
7 connected to a customer system (or network) using a LAN.

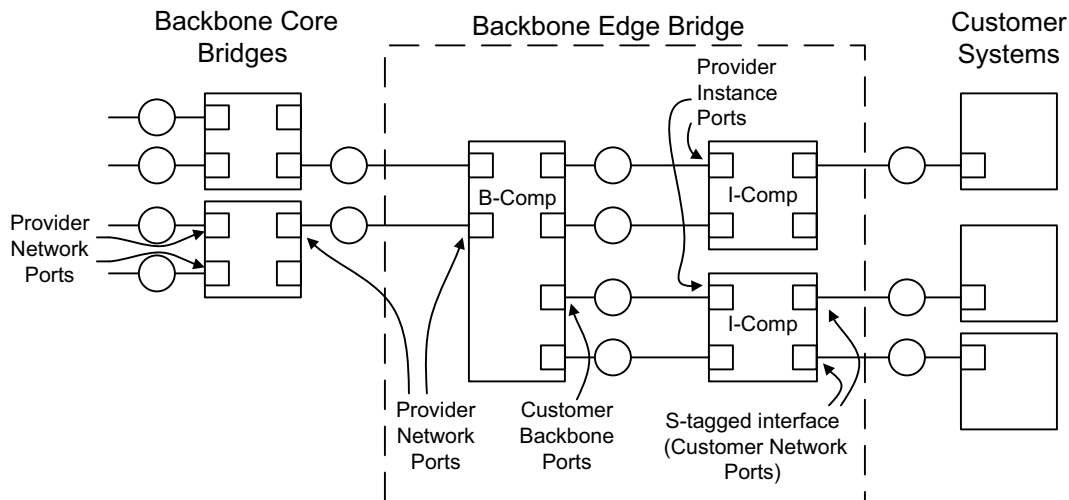


Figure 25-8—S-tagged service interface equipment

8 NOTE 2—It is always possible to build equipment that includes both BEB and Provider Edge Bridge components. In
9 particular, it is possible for a BEB to support a C-tagged service interface (15.4) for attachment to a CBN by including a
10 C-VLAN component with PEPs that connect to the CNPs of an I-component. A BEB can also support an RCSI (15.6) by
11 including a C-VLAN component and a Port-mapping S-VLAN component.

12 25.5 I-tagged service interface

13 A PBBN may provide a native I-tagged service interface for attachment to another PBBN or for attachment
14 to a customer's PIP. An I-tagged service interface can provide access to all the backbone service instances
15 within a PBBN. Access to backbone service instances is controlled by the configuration of the CBP service
16 instance tables (6.11). Each I-SID delivered over the I-tagged service interface by a customer identifies
17 a service instance that will be carried over the PBBN. Service instances are carried over the PBBN inside a
18 B-VLAN selected by the CBP. The customer must provide the B-DA for frames delivered to an I-tagged
19 service interface.

20 Figure 25-9 illustrates a customer network attached to a PBBN using an I-tagged service interface. The
21 customer network uses a BEB with an I-component (26.6.1) or a B-component (26.6.2) connecting to the
22 PBBN. The PBBN in turn is composed of BEBs interfacing to the customer and BCBs in the core of the
23 PBBN used to forward frames between the BEBs.

24 An I-tagged service interface may be provided at a CBP of a BEB as illustrated by Figure 25-9 and
25 Figure 25-10. In this service interface, the I-SID provided over the service interface within the I-TAG is
26 mapped 1-1 to an I-SID within the PBBN. As illustrated, a customer (or peer) may attach to the I-tagged
27 service interface through a CBP. In this configuration, each PBBN is attached at a CBP.

28 Figure 25-10 shows an example of equipment used to implement an unprotected I-tagged service interface.
29 For details on redundant connections and equipment, see 25.9. A B-component may support one or more
30 I-tagged service interfaces. Each CBP may be associated with one I-tagged service interface. These CBP are
31 connected to a customer system (or network) using an Ethernet LAN.

The diagram illustrates the mapping of MAC Relay and MAC Access Method Independent Functions (MAC MF) across different network components. It shows the flow of traffic from the Backbone Core Bridge (B-VLAN aware) through the Backbone Edge Bridge (B-Comp) to the Provider Network Ports, and then to the Customer Backbone Ports, I-tagged service interfaces, and Provider Instance Ports.

Backbone Core Bridge (B-VLAN aware):

- MAC Relay: Clause (6.17)
- MAC Access Method Independent Functions (MAC MF): Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

Backbone Edge Bridge (B-Comp):

- MAC Relay: Clause (6.17)
- MAC Access Method Independent Functions (MAC MF): Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

Provider Network Ports:

- MAC Relay: Clause (6.17)
- MAC Access Method Independent Functions (MAC MF): Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

Customer Backbone Ports:

- MAC Relay: Clause (6.17)
- MAC Access Method Independent Functions (MAC MF): Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

I-tagged service interfaces:

- MAC Relay: Clause (6.17)
- MAC Access Method Independent Functions (MAC MF): Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

Provider Instance Ports:

- MAC Relay: Clause (6.17)
- MAC Access Method Independent Functions (MAC MF): Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

Media Access Method Independent Functions (MAC MF):

- MAC MF: Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

Media Access Method Specific Functions (MAC MSF):

- MAC MSF: Clause 6.9, 9.5(b), Clause 8.5, Clause 6.7, MAC

The diagram illustrates a network architecture. On the left, 'Provider Network Ports' are represented by four circles. These connect to 'Backbone Core Bridges', shown as two vertical rectangles with four ports each. The output of the core bridges connects to a 'Backbone Edge Bridge', a rectangle with a 'B-Comp' block and two ports. This bridge has 'Provider Network Ports' on its left and an 'I-tagged service interface (Customer Backbone Port)' on its right. This interface connects to a 'Customer Controlled BEB', a rectangle with two ports, which then connects to two final output circles on the right.

Copyright © 2025 IEEE. All rights reserved.
This is an unapproved IEEE Standards draft, subject to change.

1 25.6 Service instance segregation

2 Segregation of data frames associated with different instances of MAC Service is achieved by supporting
3 each service instance with a backbone service instance identified by an I-SID and ensuring that:

- 4 a) No service frames are transmitted through a CBP without an I-TAG.
- 5 b) No frames are accepted, i.e., received and relayed, from any customer system without first being
6 subject to service instance selection.
- 7 c) No frames are delivered to any customer system without explicit service instance identification.
- 8 d) Prior to transmission through a PNP of a BEB, service frames are received either through an
9 I-component's CNP or through a CBP. When the customer is attached to a CNP, the port is under the
10 control of the backbone provider and is exclusively accessed by a single customer. In this case, the
11 I-component used to support the CNP is under the control of the Backbone provider and is attached
12 to a CBP through a PIP that is also under the control of the backbone provider and that may be
13 shared with multiple customers. When the customer is attached to a CBP, the port is under the
14 control of the provider and is exclusively accessed by a single customer. All frames received through
15 the CNP or CBP (depending on how the customer is attached) must correspond to a service instance
16 or instances that the customer is permitted to access. (The method used to ensure the port is attached
17 to the specified customer and therefore secure is beyond the scope of this standard. See
18 IEEE Std 802.1X.)
- 19 e) The BCBs and the B-component of each BEB within the PBBN can only be controlled by the
20 provider.

21 A single backbone service instance may support many customer service instances. Each customer service
22 instance is supported by a single backbone service instance.

23 25.7 Service instance selection and identification

24 Service instance selection is provided to the attached customer system by the Port-based, S-tagged, or
25 I-tagged service interfaces (25.3, 25.4, 25.5). In the Port-based interface, only one backbone service instance
26 is offered to the attached customer and the CNP of the I-component is configured to accept only untagged or
27 priority tagged frames. In all S-tagged service interfaces, the CNP of the I-component is configured with
28 Enable Ingress Filtering (8.6.2) and the Port is only included in the Member Set for S-VLANs corresponding
29 to service instances that the customer is permitted to use. The service instance for each frame received by the
30 attached customer system is identified in the same way as frames transmitted using the same interface, but
31 not necessarily in the same way that the service instance is selected or identified at other interfaces. In the I-
32 tagged service interface, the CBP of the B-component is configured (12.16.5, 6.11) with the acceptable
33 backbone service instances and I-SID values for use by the customer. A single backbone service instance
34 can support any combination of Port-based, S-tagged, or I-tagged service interfaces.

35 NOTE—The means used by a backbone provider and a customer to determine the S-VIDs and I-SIDs used by the
36 customer to select and identify a given service instance are outside the scope of this standard.

37 25.8 Service priority and drop eligibility selection

38 For all service interface types, the service priority is selected using the received priority for each frame. For
39 S-tagged and Port-based service interfaces, the priority and drop eligibility may be regenerated using the
40 Priority Regeneration Table (6.9.4). For I-tagged service interfaces, the priority and drop eligibility may be
41 regenerated using procedures of 6.11.

42 The mechanism for determining the received priority varies with the type of service interface. Service
43 priority selection is provided for Port-based service interfaces using the received priority signaled from the
44 media access method of the port. If the media access method used to attach to the interface does not directly

1 support priority, this will result in the selection of a single value for all frames. An attached system may also
2 signal priority to a Port-based service interface on a per-frame basis by priority-tagging frames with an
3 S-TAG that has a null VID (6.13).

4 Service priority selection is provided by S-tagged service interfaces using the PCP and DEI parameters
5 conveyed in the S-TAG of each frame. An S-tagged service interface can provide a single backbone service
6 instance for all S-VIDs received, and in this way functions like a Port-based service interface with the
7 addition of the capability of delivering S-tagged service frames (whose S-TAG has a non-null VID).

8 Service priority selection is provided by I-tagged interfaces using the received priority decoded from the
9 I-PCP and I-DEI fields in the I-TAG.

10 25.9 Service access protection

11 A customer system (or systems) that is part of a single PBN or that contains a customer operated I-
12 component or B-component can attach to a PBBN using a protected service interface providing multiple
13 LANs and (or) nodes, thus providing fault tolerance through redundancy of the service interface
14 components. Three classes of service access protection may be supported (see Figure 25-11 and
15 Figure 25-12), which provide for unprotected service interface, for link-level redundancy at the access link
16 or link and node redundancy at the access point. An access service interface may be a Class I, II, III access
17 service interface.

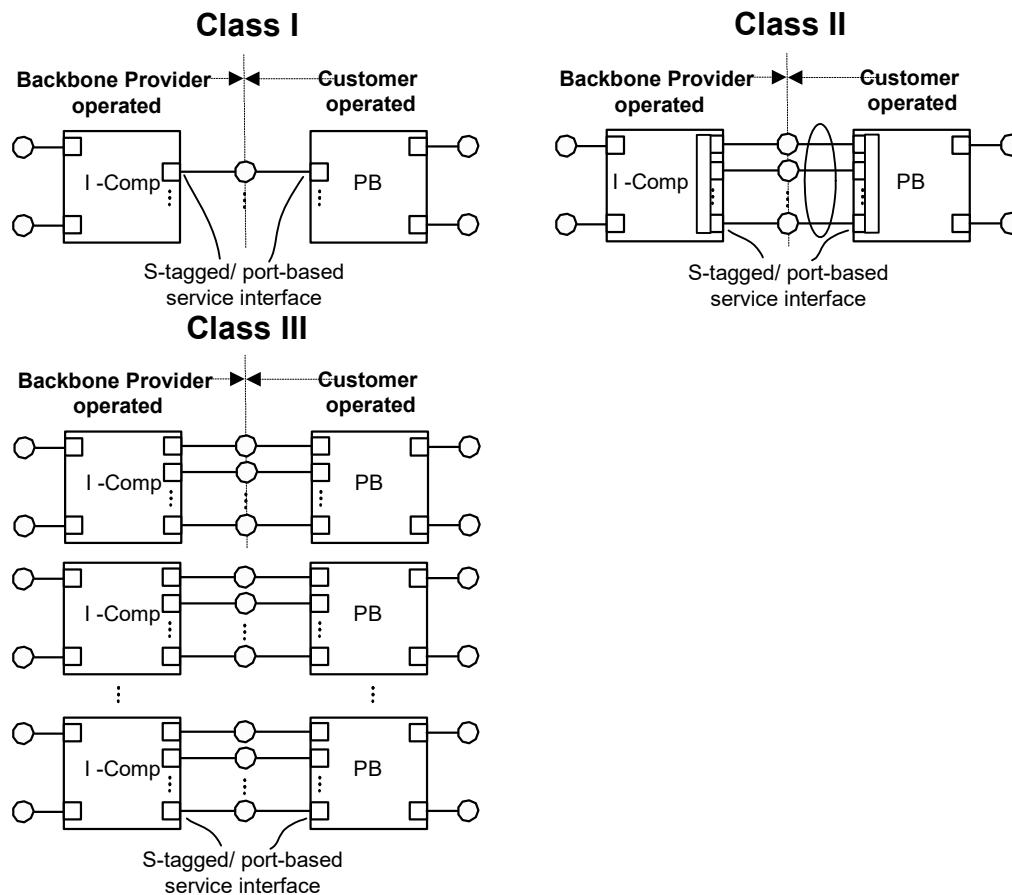


Figure 25-11—S-tagged and Port-based service interface access classifications

1 Protected Port-based, S-tagged, and I-tagged service interfaces are depicted in Figure 25-11 and
2 Figure 25-12, respectively.

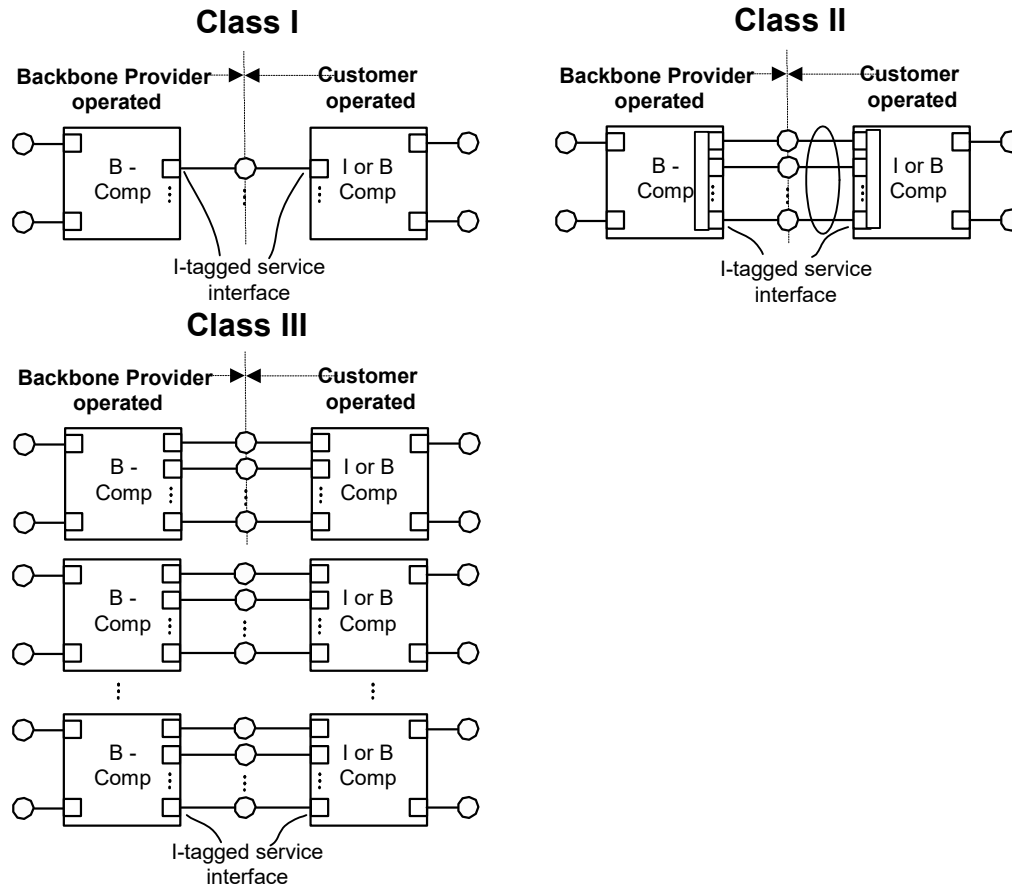


Figure 25-12—I-tagged service interface access protection classifications

3 Class I service interfaces depicted in these figures are unprotected service interfaces providing a single LAN
4 and node. Class I service interface equipment is depicted in Figure 25-5, Figure 25-8, and Figure 25-10.
5 These provide no redundancy and therefore will fail if any LAN or component fails.

6 Class II service interfaces are link protected service interfaces that provide multiple LANs; however, they do
7 not provide any node redundancy. Class II service interfaces are tolerant of single or multiple LAN failures.
8 These provide no node redundancy and therefore will fail if any node fails.

9 Class III service interfaces are link and node protected service interfaces providing multiple LANs and
10 multiple nodes. Class III service interfaces are tolerant of single or multiple LAN failures and/or single or
11 multiple node failures. The operation of a Class III service interface switches between access nodes within
12 both the customer and provider networks; therefore, protection switching over a Class III service interface
13 always results in state changes spreading over the customer and provider networks.

14 To attach a PBN to a PBBN, the following rules must be maintained to ensure loop-free operation over the
15 extended network:

- 16 a) Each PBN and PBBN prevents forwarding loops by running an independent spanning tree.
- 17 b) Each PBN connects to other PBNs only through a PBBN.

- 1 c) Each PBN ensures that no data frame passes through more than one BEB attachment point into or
2 out of the PBBN. An attachment point is a single LAN connection to a PBBN. This connection may
3 be part of a protected or unprotected Port-based (see Figure 25-4), S-tagged (see Figure 25-7), or I-
4 tagged service interface (see Figure 25-9). A PBN may use L2GP (13.40) to ensure only a single
5 link is active when using a backup link to the PBBN.

- 6 d) Each PBN ensures that it attaches any given S-VLAN to no more than one PBBN.

7 NOTE—When an attached PBN uses the L2GP (13.40) to enforce criterion c), then the PBBN may implement a Class I,
8 II, or III protection interface. If the PBBN is configured for Class I or II protection, then it will provide multipoint
9 services with the services delivered to two or more Class I or II interfaces for attachment to the PBN. In this case, the
10 PBN is responsible for selecting one of the two attachments using the L2GP to ensure criterion c). If the PBBN is
11 configured for Class III protection, then the PBN must support MVRP over the access links to inform the PBBN which
12 S-VIDs are active on which LAN connections.

13 25.9.1 Class II redundant LANs access protection

14 A Class II redundant interface is implemented using IEEE Std 802.1AX (Link Aggregation). Using Link
15 Aggregation, a single CNP and associated PNP of a Provider Bridge, or CBP and associated PIP, may be
16 implemented using multiple LANs. A failure on any of the LANs implementing the link will cause all traffic
17 to migrate to the remaining LAN. CFM may run both on the aggregate and on each individual link as shown
18 in Figure 22-8.

19 It is desirable for the distribution algorithm used on a Class II interface to divide traffic between the multiple
20 links based on the service instance. One distribution algorithm that always retains a service on a single LAN
21 is to place all the traffic on one of N links. In the event of a failure, all traffic is shifted to the highest priority
22 remaining link. This distribution algorithm has the advantage that it provides for full reserved bandwidth on
23 the backup link.

24 Another distribution algorithm that divides traffic by service and also allows traffic on all redundant links is
25 to divide the traffic by either the S-VID or the I-SID. Distributing traffic based on the S-VID would be used
26 for traffic over multiple LANs comprising an S-tagged interface, while distributing traffic based on the
27 I-SID would be used for traffic over multiple LANs comprising an I-tagged interface. With either of these
28 distribution schemes, services are allocated to one of the LANs used to implement the provider CNP or CBP
29 and the customer PNP, PIP, or CBP. In the event of a failure, the services are moved to the remaining LANs.

30 25.9.2 Class III simple redundant LANs and nodes access protection

31 A Class III interface uses redundant LANs to connect a primary and one or more secondary customer nodes
32 to a primary and one or more secondary BEBs. The operation protects the interface against failures on any of
33 the interconnecting LANs or nodes used on each side of the interface. It is possible to use Class II protected
34 interfaces for each link of the Class III interface. In this case, the Class II protection procedures must happen
35 first and complete before a Class III protection switch begins.

36 For a Class III interface, each customer node is connected to a BEB using a single LAN. The interface is
37 provisioned with a priority for each LAN that is part of the interface. Each of the LANs is constantly
38 monitored using either PHY management, CFM, or both. In operation, a Class III interface uses only the
39 highest priority operating LAN for traffic. The other LANs are protection paths. In a Class III interface, any
40 switch in the LAN used for traffic will also switch both the customer node and the PBBN BEB since the
41 connections of a Class III interface are sets composed of a customer node, a LAN, and a BEB.

42 When providing a Class III protected S-tagged interface, multiple CNPs, each on a different BEB, are used
43 to create a single S-tagged interface. Each LAN of the S-tagged interface executes CFM at the port level to
44 detect failures over the interface LANs. Information from CFM is used by the customer to determine which
45 LANs of the S-tagged interface are operating and which are inactive. Each customer network runs MSTP to
46 determine which S-VLANs are active on which LANs of the Class III interface. To avoid loops through the

1 Class III interface, either the I-components participate in the customer network spanning tree (13.42), or the
2 ports on the customer equipment that connects to the I-components are configured as Layer 2 Gateway Ports
3 (13.40). Upon an active link failure between customer node and its corresponding BEB or upon the customer
4 node failure, the customer network switches to one (or more) of the backup links and informs the PBBN
5 which S-VLANs are active on which links using MVRP (11.2) on the access LANs. The BEB(s), upon
6 receiving the MVRP message(s) from the customer network, generate the corresponding MRP-based
7 message(s) for the affected I-SIDs and send these messages over the associated B-VLANs. The far-end
8 BEBs, upon receiving these messages, will flush their C-MAC entries in their FDB corresponding to the
9 affected I-SIDs.

10 When providing a Class III protected I-tagged interface, multiple CBPs, each on a different B-BEB, are used
11 to create the single I-tagged interface. In this scenario, each B-BEB is connected via a single link to the
12 customer I-BEB where the link can be Class II protected. The customer I-BEB is in turn connected via
13 multiple CNPs to the customer network. Each customer network runs MSTP to determine which S-VLANs
14 are active on which LANs of the Class III interface. To avoid loops through the Class III interface, either the
15 I-components participate in the customer network spanning tree (13.42), or the ports on the customer
16 equipment that connects to the I-components are configured as Layer 2 Gateway Ports (13.40).
17 PHY management or port-level CFM is used to detect failures over the interface LANs between the
18 customer I-BEBs and the customer network. When a failure is detected, the customer network informs the
19 customer I-BEBs which S-VLANs are active on which links using MVRP (11.2). The customer I-BEBs in
20 turn notify the provider B-BEBs using MRP-based messages corresponding to the effected I-SIDs. The
21 provider B-BEBs relay this message to the other BEBs in the PBBN over the associated B-VLANs. The far-
22 end BEBs, upon receiving these messages, will flush their C-MAC entries in their FDB corresponding to the
23 effected I-SIDs. Each LAN of the I-tagged interface also executes PHY management or CFM at the port
24 level to detect failures over the I-tagged interface. In case of a complete I-tagged interface failure between
25 the customer I-BEB and the provider B-BEB, the I-BEB uses this info to deactivate all its CNP ports toward
26 the customer network.

27 **25.10 Support of the MAC Service by a PBB-TE Region**

28 A PBB-TE Region comprises a continuous set of IB-BEBs and BCBs, capable of providing TESIs, that have
29 allocated a common subset of B-VIDs to the control of an external agent (8.9) responsible for setting up
30 Ethernet Switched Paths (ESPs). Each IB-BEB that belongs to a PBB-TE Region can provide interfaces that
31 encapsulate customer frames in ESPs, thus allowing the PBBN operator to offer TESIs.

32 A PBB-TE Region providing the MAC Service to attached stations is modeled as a sequence of relay
33 functions, as illustrated in Figure 25-13. The MST Configuration Table associated with each BCB or B-
34 component of an IB-BEB specifies a set of VIDs, each of which may be exclusively used throughout the
35 PBB-TE Region to form part of an ESP identifier. Such a VID is called an ESP-VID and is associated with a
36 special value of the MSTID in the MST Configuration Table, the TE-MSTID (8.9), indicating that the VID
37 is not under the control of a spanning tree protocol.

38 The I-components of IB-BEBs supporting PBB-TE are responsible for encapsulating frames received from
39 customers and allocating each frame to a backbone service instance and for decapsulating frames received
40 from the network via backbone service instances. The allocation of customer frames to backbone service
41 instances that will be transported by TESIs is done at the PIP as described in 6.10. At the PIP the customer
42 frame is encapsulated by an I-TAG, which includes the I-SID, and a set of source and destination B-MAC
43 addresses. The source B-MAC address is the PIP MAC address that is configured to take the same value as
44 the CBP MAC address of the internally connected CBP on the B-component. As a result, the B-MAC
45 addresses of frames associated with TESIs, the ESP MAC addresses, are always equal to the CBP MAC
46 addresses.

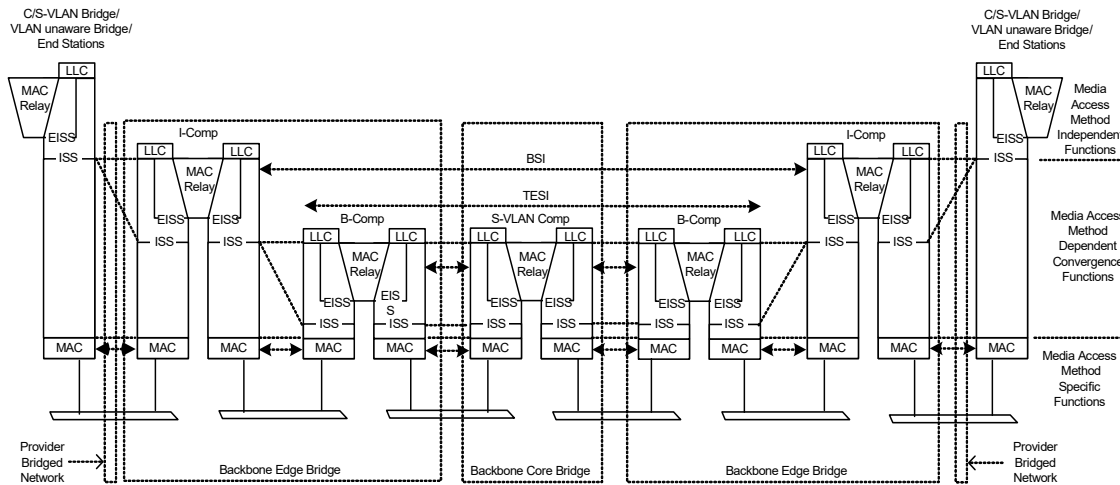


Figure 25-13—Internal organization of the MAC sublayer in a PBB-TE Region

1 CBPs on B-components in IB-BEBs supporting PBB-TE use the I-SID to map the encapsulated customer
2 frames to TESIs or B-VLANs (6.11). Figure 25-13 depicts a TESI providing the MAC Service at two CBPs.
3 For an I-SID mapped to a TESI, the Backbone Service Instance table contains the terminating CBP's MAC
4 address, the ESP-DA, and the ESP-VID of the associated ESP. I-SIDs mapped to a B-VLAN must not use a
5 B-VID from the ESP-VID space. The B-Component MAC relay forwards frames belonging to an ESP
6 between the CBP and PNPs or between PNPs using the ESP-DA and ESP-VID as described in 8.6.

7 25.10.1 Provisioning TESIs

8 ESPs are provisioned in a PBB-TE Region by an external agent. This replaces the spanning tree protocols
9 and populates the filtering tables of the corresponding B-components and BCBs by creating static filtering
10 table entries for the ESP MAC addresses and ESP-VIDs (Figure 25-14).

11 The ability of a PBB-TE Region to utilize an external management or control plane is facilitated by PBB
12 because the PIP MACs and correspondingly the ESP MAC addresses can be managed by the Provider and
13 therefore can all be identified in the Provider's topology by the external management or control plane.

14 The external PBB-TE management/control plane is responsible for maintaining and controlling all the active
15 topology information to support point-to-point or point-to-multipoint ESPs over the PBB-TE Region. The
16 PBB-TE active topology can co-exist with active topologies associated with spanning tree protocols and the
17 allocation to a specific active topology is governed by the B-VID value. The PBB-TE external agent takes
18 control of the ESP-VID range on the Bridges in a PBB-TE Region.

19 Similar to the forwarding connectivity created by the operation of spanning tree and learning, the PBB-TE
20 management/control plane can form an active topology of CBP rooted trees, from each CBP belonging to a
21 PBB-TE Region to a specific subset of any other CBPs in this region. These traffic engineered trees define
22 the path(s) taken by the frames that belong to ESPs within the PBB-TE Region. Furthermore each such tree
23 is further qualified by the PBB-TE reserved ESP-VID, thus enabling the construction of independent trees
24 per ESP-VID. As a result each such CBP rooted tree can be identified by a CBP MAC address associated
25 with the root of the tree, the CBP MACs associated with the leaves of the tree and the ESP-VID. The PBB-
26 TE management/control plane routes the ESPs along these trees by explicitly populating the FDBs in the
27 Bridges along each tree with the Static Filtering Entries containing the CBP MAC DA and ESP-VIDs. The
28 mechanism by which the PBB-TE management/control plane selects the path for a routing tree is not within

1 the scope of this standard. The PBB-TE management/control plane also manages the bandwidth of all ESPs
2 along each routing tree. For each destination CBP, which is part of a routing tree maintained by the PBB-TE
3 management/control plane, PBB-TE will maintain tree(s) which provide symmetric paths from the CBP
4 MAC DA to the CBP MAC SA. The ESP-VID(s) used in this reverse ESP(s) need not have the same ESP-
5 VID used for the forward ESP. Accordingly each of the ESPs can be identified by a 3-tuple:

6 $\langle \text{ESP-DA}, \text{ESP-SA}, \text{ESP-VID} \rangle$,

7 where the three component fields are described as follows:

- 8 a) The ESP-SA is the address of the PIP encapsulating the customer service instance that by
9 configuration is the same as the address of the internally connected source CBP.
- 10 b) The ESP-DA identifies the CBP destination address(es).
- 11 c) The ESP-VID is a VID value distinguishing among ESPs having the same $\langle \text{destination}, \text{origin} \rangle$ pair.
12 It can only take values that are allocated to the PBB-TE Region identified by the TE-MSTID (8.9).

13 An ESP is point-to-point or point-to-multipoint:

- 14 — A point-to-point ESP is an ESP where the ESP-DA and the ESP-SA in its 3-tuple identifier are
15 individual MAC addresses.
- 16 — A point-to-multipoint ESP is an ESP between one root CBP to n leaf CBPs, identified by a 3-tuple
17 where the ESP-DA is a group MAC address identifying the n leaves CBPs, and the ESP-SA is the
18 individual MAC address of the root.

19 A TESI, is an instance of the MAC Service supported by a set of ESPs, collectively identified by a single
20 TE-SID, forming a bidirectional service. The following two types of TESIs are described:

- 21 — A point-to-point TESI that is supported by two point-to-point ESPs where the ESPs' endpoints have
22 the same CBP MAC addresses.
- 23 — A point-to-multipoint TESI that is supported by a set of ESPs that constitute one point-to-multipoint
24 ESP from a root to each of n leaves plus a point-to-point ESP from each of the n leaves to the root.

25 NOTE—For reasons relating to TESI monitoring diagnostics, operational simplicity, etc., this standard assumes the
26 point-to-point ESPs associated with a point-to-point TESI are symmetric and that the point-to-point ESPs associated
27 with a point-to-multipoint TESI are routed from each of the n leaves to the root along the branches of the point-to-
28 multipoint ESP. Support for a TESI that comprises ESPs not routed in this way is problematic, and is not defined in this
29 standard.

30 Although an ESP is identified by the 3-tuple $\langle \text{ESP-DA}, \text{ESP-SA}, \text{ESP-VID} \rangle$ it is only the ESP-DA, ESP-
31 VID pair that are used for forwarding decisions. The combination (ESP-DA, ESP-VID) is treated as though
32 it was a single 60-bit address, where 12 bits are the ESP-VID and 48 bits are the ESP-DA. This ESP-VID
33 field is used as a path selector to the destination CBP rather than a B-VID, allowing up to 2^{12} unique routing
34 trees to any single CBP. In Figure 25-14, two paths are configured to reach S2. These two paths are
35 separated by using a different ESP-VID in combination with the ESP-DA for a second path.

36 If the ESP-VID range delegated is the 4094 possible values, then each CBP termination can sink $(2^{12} - 2)$
37 routing trees, and the theoretical network maximum is about 2^{60} ESPs. The number of ESPs supported by
38 PBB-TE is constrained by issues related to coordinated management, independent management of groups of
39 nodes or independent routing choices through parts of the path, limit through intermediate nodes, etc. In the
40 cases where B-VLAN services are also offered by the same network, it is expected that most B-VIDs will be
41 allocated to those B-VLANs since the use of the MAC addresses as part of the ESP identifiers allows at least
42 2^{48} ESPs and the ESP-VIDs can be used mainly to provide path diversity.

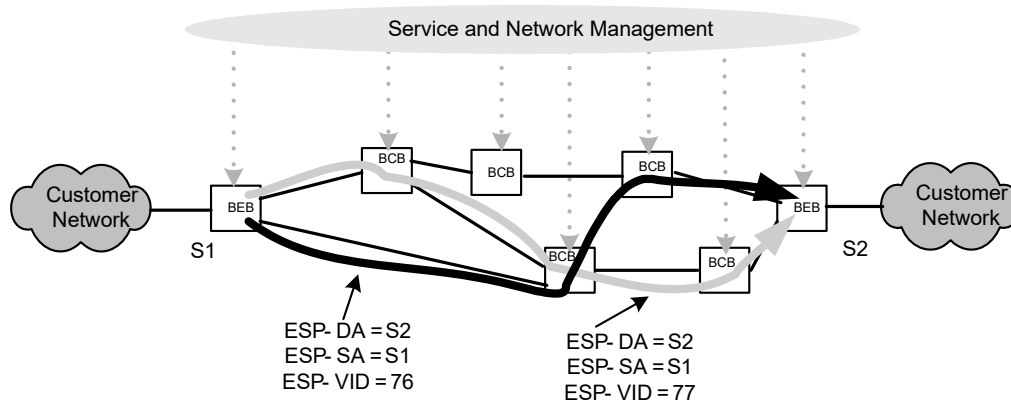


Figure 25-14—PBB-TE Region

25.10.2 ESP forwarding behavior

Forwarding of frames belonging to an ESP-VID differs from forwarding of frames belonging to B-VLANs in the following ways:

- a) Frames associated with the ESP-VID are discarded when a Static Filtering Entry corresponding to the (ESP-DA, ESP-VID) is not found in the FDB [item b) in 5.4.5]. Proper operation of PBB-TE requires the provisioning of an FDB entry in each Bridge through which an ESP passes in order to specify the forwarding of traffic associated with that ESP. Such an entry may be absent due to a provisioning error. In the event of such an error, it is important that frames associated with the improperly configured ESP not be flooded, as PBB-TE would be violated if traffic intended for a particular ESP is sent on one or more other ESPs. Further, flooding based on the ESP-VID may result in unbounded looping and replication as the ESP-VID is not associated with a spanning tree. For this reason, frames associated with the ESP-VID are discarded when a Static Filtering Entry corresponding to the (ESP-DA, ESP-VID) is not found in the FDB. Such an ESP MAC address may represent an individual address, for which no more specific Static Filtering Entry exists or a group address, for which no more specific Static Filtering Entry exists (8.8.1).
- b) MAC learning is not performed on receipt of a frame carrying an ESP-VID [item a) in 5.4.5]. Where ESPs have been correctly provisioned, learning would not interfere with the proper operation of Bridges, but in the presence of provisioning errors, learning may result in undesired behavior.
- c) The active topology for the ESP-VIDs allocated to PBB-TE is no longer controlled by spanning tree protocols but by an external agent that is responsible for setting up the ESPs [item b) in 5.4.5].

NOTE 1—In a network supporting both VLAN (i.e., PBB) and ESP (PBB-TE) traffic, the establishment of an ESP and the policing of traffic at the ingress of the ESP are necessary but not sufficient conditions to ensure that traffic associated with an ESP receives the QoS intended by the network operator. The operator should ensure that the QoS requirements of an ESP are satisfied and that sufficient network capacity is available to accommodate broadcast-unknown traffic, spanning tree control traffic, and other traffic associated with VLAN usage.

Key properties of an ESP are as follows:

- The ESP is identified at all points along its path by a single identifier <ESP-DA, ESP-SA, ESP-VID>.
- The information referenced for forwarding <ESP-DA, ESP-VID> is contained within the ESP Identifier.
- The information referenced for forwarding <ESP-DA, ESP-VID> does not change along the length of the ESP.

NOTE 2—The static and provisioned nature of an ESP provides support for traffic engineering. These properties of the ESP, together with policing and queuing functions supported by a Bridge (8.6) can provide per ESP QoS.

25.11 Transparent service interface

A PBBN may provide a transparent service interface for customer attachment. A transparent service interface is delivered on a CNP provided by a BEB as illustrated in Figure 25-15 and Figure 25-16. A transparent service interface may attach to a Provider Bridge, C-VLAN Bridge (5.9), MAC Bridge, router, or end-station. The service provided by this interface forwards all frames over the backbone on a single backbone service instance.

A transparent service interface does not recognize, add, remove, or modify either S-tags or C-tags on customer frames. It does not learn customer addresses, nor does it filter frames based on any MAC addresses except the TPMR component Reserved addresses listed in Table 8-3 and the management address of the TPMR component.

Figure 25-15 illustrates a customer network attached to a PBBN using a transparent service interface. The customer network uses End Stations, Customer Bridges, or Provider Bridges for connecting to the PBBN.

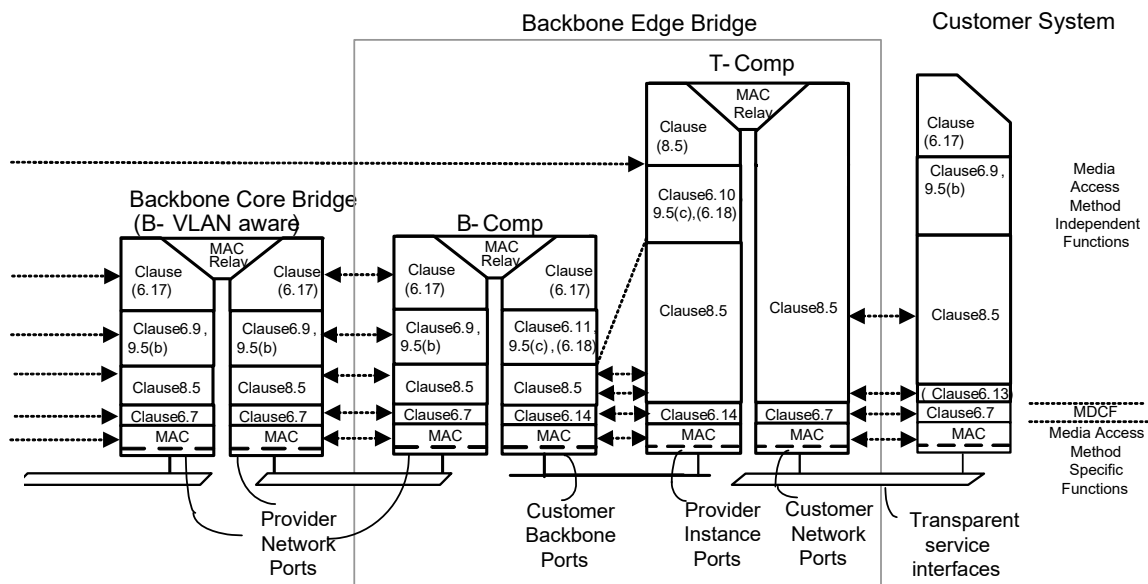


Figure 25-15—Transparent service interface

Figure 25-16 shows an example of equipment used to implement a transparent service interface. In this diagram a BEB is formed by connecting two T-components and a B-component. These connections may be over a backplane or over a LAN. A T-component supports a single transparent service interface associated with a CNP. A CNP is connected to a customer system (or network) using a LAN.

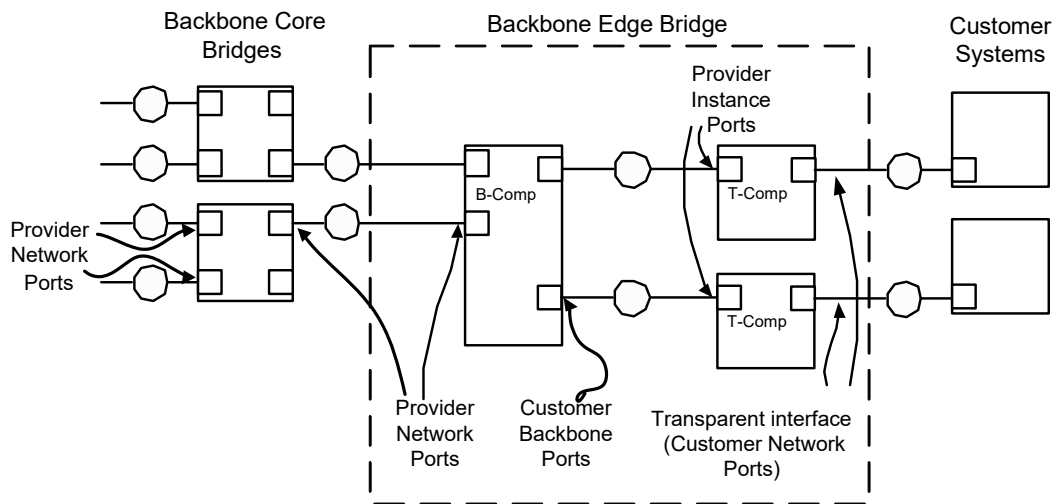


Figure 25-16—Transparent service interface equipment

26. Principles of Provider Backbone Bridged Network (PBBN) operation

This clause establishes the principles and a model of PBBN operation. It provides the context necessary to understand how the:

- a) Operation of individual Backbone Edge Bridges (BEBs) and Backbone Core Bridges (BCBs) (Clause 8, Clause 25);
- b) Configuration and management of individual BEBs and BCBs (Clause 12); and
- c) Management of spanning tree and VLAN topologies within a PBBN (Clause 7, Clause 11, and Clause 13)

support, preserve, and maintain the quality of each of the instances of the MAC Service offered to the customers of the PBBN (Clause 6, Clause 25) including the following:

- d) Independence of each service instance supported by a PBBN from other service instances (25.6).
- e) Identification of service instances and backbone service instance within the PBBN (25.7, 8.8).
- f) Maintenance of service availability in the event of the failure, restoration, removal, or insertion of LAN components connecting a customer network to a PBBN (Clause 6, Clause 11, Clause 13, 25.9).

A PBBN is a Virtual Bridged Network that comprises BEBs at the edge of the PBBN and BCBs at the core of the PBBN and attached LANs, under the administrative control of a single backbone provider. The principal elements of PBBN operation are those specified in Clause 16, as amended by this clause.

NOTE 1—The term PBBN is used exclusively to refer to networks configured and managed as specified by this clause and comprising only (a) BCBs and BEBs and (b) communications media and equipment providing the ISS (IEEE Std 802.1AC). While the requirements of Clause 25 are generally applicable to similar services, a generalized framework for all the network designs that could support these requirements, while useful in the context of other equipment and services, is outside the scope of this standard. This clause describes specific best practice for PBBNs to ensure that the requirements for Bridge functionality are clear. Conformance of a BEB implementation to this standard does not require that the implementation be used as specified in this clause; merely, that it is capable of being so used.

NOTE 2—Within a PBBN, an instance or instances of the MAC Service are reserved for the backbone provider's own use to configure and manage the network. All frames associated with such service instances, and that are not confined to an individual LAN, are subject to service instance selection, segregation, and identification as specified in 26.1.

26.1 PBBN overview

The principal elements of PBBN operation comprise the following:

- a) Encapsulation/de-encapsulation of service frames in B-MAC Frames (6.10).
- b) Service instance segregation within a PBBN for service frames (25.6).
- c) Service instance selection on ingress, and service instance identification on egress, for each customer frame (25.7).
- d) Service access protection (25.9).
- e) Resource allocation and configuration to provide service instance connectivity (26.3).

The following may also be included:

- f) Management of customer end station address learning (26.4)
- g) Prevention of connectivity loops formed through attached networks (26.5)
- h) Scaling of PBBNs (26.6)
- i) Provisioning of PBBN services (26.7)
- j) Management of PBBN service connectivity (26.8)

26.2 PBBN example

An example PBBN is illustrated by the Figure 26-1. The example network is divided into three major areas: the customer equipment area, which is owned by the customer and is interfaced to the PBBN; the PBBN area, which is owned by the backbone provider and interconnects to the customer equipment and to other PBBNs; and the other PBBN area, which is owned by the same or other backbone providers and interconnects to multiple PBBNs allowing the network to scale indefinitely.

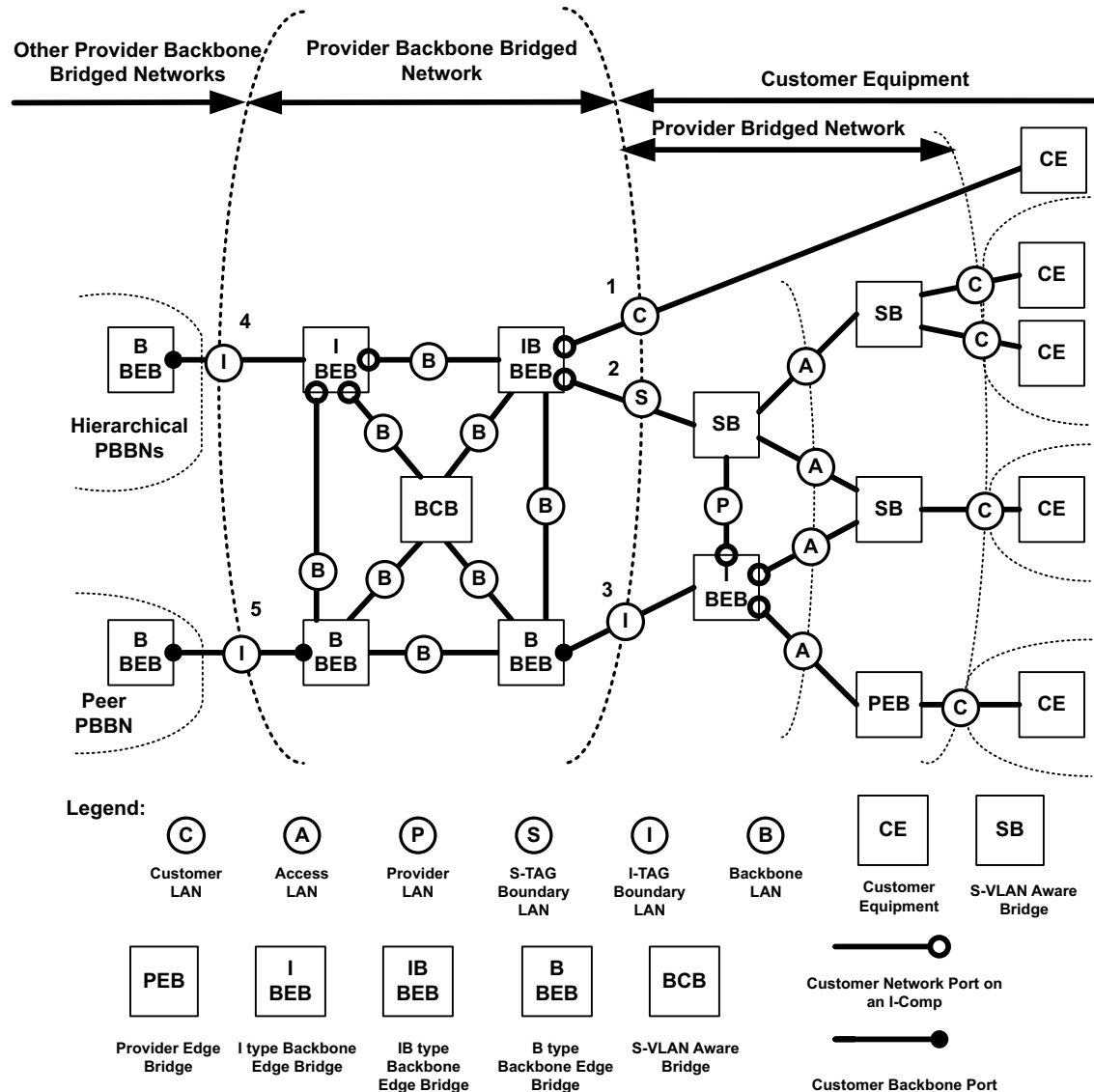


Figure 26-1—PBBN example

The interface between a BEB and a Provider Bridge is a CNP of an I-Component as specified in 6.10. The interfaces between BEBs and BCBs and between BCBs and BCBs are PNPs. In Figure 26-1, these interfaces are over LANs labeled S and B. The interfaces between B-BEB and I-BEBs, or between B-BEBs in different PBBNs, are over LANs labeled I. All exterior interfaces from a PBBN are over S, I, or C LANs; whereas, all interior interfaces within a PBBN are over B LANs. The S, I, and C LANs (exterior interfaces) are at the boundary between the PBBN and the attached customer networks.

1 The B LANs (interior interface) are backbone LANs, which interconnect all the PBBs within a PBBN.

2 Within the network example, BEBs, BCBs, and B LANs are secured so that only the backbone provider
3 operating the PBBN can manage the reception, transmission, and relay of frames between BEBs and BCBs.
4 The arbitrary physical network topology of the PBBN and the connectivity that it provides to support
5 segregated instances of the MAC Service is designed and managed by the backbone provider to meet
6 bandwidth and service availability requirements at the CBPs and PIPs. Application of the service instance
7 ingress and egress rules at these Ports in support of service instance selection and identification ensures that
8 frames cannot be transmitted or received on any backbone service instance by any customer's equipment
9 without prior agreement with the provider.

10 The active MSTP topology of the PBBN area is separated from the active topology of the customer
11 equipment area. This is accomplished by isolating the MST BPDUs for each customer network from the
12 PBBN at the BEBs that surround the perimeter of the PBBN. The BEBs also stop the propagation of MST
13 BPDUs, used to support the active topology of the PBBN, into the Provider Bridged area. In Figure 26-1,
14 Provider Bridge MST BPDUs are propagated over the A, P, C, and S LANs but not over the I or B LANs,
15 while PBB MST BPDUs are propagated over the B LANs but not over the I, C, A, P, or S LANs.

16 For Port-based service interfaces, the entire C LAN is treated as a single instance of the MAC Service. The
17 BEBs extend the C LAN over the backbone by mapping all service frames on a single backbone service
18 instance identified by an I-SID. The I-SID used is provisioned by the provider operating the PBBN. The
19 provider sets the I-SID on each BEB port attached to a C LAN that is part of the port based service. The
20 BEBs encapsulate the service frames with an I-TAG, B-TAG, B-SA, and B-DA. During the encapsulation,
21 any C-TAG delivered to the CNP is retained in the encapsulated frame as part of the SDU. The BEBs then
22 map the frames onto a B-VLAN with its B-VID contained in the B-TAG and which interconnects the BEBs
23 provisioned for the service. These new frames are transmitted over the B LANs by the BEBs and by the
24 BCBs that carry the B-VLAN. Since the initial octets of the data conveyed in each backbone frame comprise
25 an S-TAG, the frames may be forwarded by BCBs of the PBBN until they reach the next BEB where they
26 might be de-encapsulated. During de-encapsulation, the B-DA, B-SA, B-TAG, and I-TAG are stripped. The
27 C-DA and C-SA from the I-TAG will become the DA and SA transmitted to the receiving C LAN. If the
28 encapsulated SDU contains a C-TAG, it becomes the outside tag as the frame is transmitted to the receiving
29 C LAN.

30 For S-tagged service interfaces, each instance of MAC Service is carried over the S LANs on one or more
31 S-VLANs. The BEBs preserve the S-VLAN over the backbone by mapping them onto an I-SID and in the
32 case of S-VLAN bundling carrying the S-TAG. This operation is performed by the backbone provider
33 operating the PBBN by configuring the VIP-ISID on each BEB attached to an S LAN. The BEB maps the
34 S-VID to I-SID and encapsulates the original service frame with a new I-TAG, B-SA, and B-DA. The BEB
35 then maps the frame onto a B-VLAN, which interconnects BEBs. This new frame is transmitted over the B
36 LANs by the BEBs and by the BCBs. Since the initial octets of the data conveyed in each backbone frame
37 comprise an S-TAG, the frames may be forwarded by BCBs of the PBBN until they reach the next BEB
38 where they might be de-encapsulated.

39 For I-tagged service interfaces, each instance of MAC Service is identified by an I-SID value carried over
40 the I LAN in an I-TAG. The BEB maps the frame onto a B-VLAN, which interconnects BEBs. This new
41 frame is transmitted over the B LANs by the BEBs and by the BCBs. Since the initial octets of the data
42 conveyed in each backbone frame comprise an S-TAG, the frames may be forwarded by BCBs of the PBBN
43 until they reach the next BEB where they might be de-encapsulated.

44 B-MAC addresses are used to identify the destination BEB's PIP. These B-MAC addresses are learned by
45 each PBB as frames are exchanged over B-VLANs. To perform the encapsulation and de-encapsulation of
46 service frames, BEBs use the connection identifier stored in the FDB (see 8.8) to correlate C-MAC
47 addresses to B-MAC addresses. At startup, the BEBs have not learned the B-MAC or C-MAC addresses yet.
48 When the B-MAC address is unknown, the BEB encapsulates the service frames using the Default

1 Backbone Destination address. The same holds true for the service frames having a group or broadcast
2 destination address. An individual B-DA is used for forwarding unknown or group frames when the service
3 is point-to-point and where the far end individual address is known. Both the B-MAC and C-MAC addresses
4 are learned by the provider BEB (see Clause 8).

5 Frames containing encapsulated service data are carried within the B-VLANs created over the PBBN. It is
6 also possible to carry unencapsulated S-VLAN traffic in the PBBN core by allocating some of the PBBN
7 VID space to S-VLANs. The B-VLAN determines the route that the frames will take and limits broadcasting
8 within the backbone. The B-TAG is added to the frame at the CBP. The selection of B-VLAN used to form
9 the B-TAG is determined by the configuration of the CBP Backbone Service Instance table. This table maps
10 I-SIDs to B-VIDs and is created as part of service provisioning.

11 If a BEB's PIP receives a B-MAC Frame whose B-SA is the same as the B-MAC address of the PIP, then a
12 loop exists through the BEBs. Optionally, the BEB may use this condition for loop detection.

13 **26.3 B-VLAN connectivity**

14 The B-VLAN topology of each B-VLAN is established by the mechanisms introduced in 7.1 and
15 Figure 7-1. The backbone provider can use and configure MSTP to provide a number of independent
16 spanning tree active topologies and can assign each B-VLAN to one of these active topologies to best use
17 the resources in the network. MVRP running in the context of each spanning tree active topology configures
18 the extent of each B-VLAN to the subset of that active topology necessary to support connectivity between
19 the customer points of attachment to the instance of MAC Service provided, and can reconfigure that
20 connectivity as required if the spanning tree active topology changes.

21 The operation of MSTP within a Backbone Provider's network is independent of the operation of any
22 spanning tree protocol within attached provider or customer networks. This is achieved by removing all
23 MST BPDUs received or to be transmitted at the service access interfaces.

24 The operation of MVRP within a PBBN is independent of the operation of any configuration protocol within
25 attached customer networks. The Provider Bridge MVRP address (Table 8-1) is used as the destination
26 address of all MRPDUs transmitted in support of the MVRP Application. Frames received by CBPs and
27 addressed to the Customer Bridge MVRP address (Table 10-1) are subject to service instance selection and
28 relay in the same way as service frames. The MVRP Administrative Control for each B-VLAN is either
29 Registration Fixed (New Ignored) or Registration Forbidden on all CBPs, so no information is received from
30 any BEB MVRPDU that has been erroneously transmitted by a customer system.

31 **26.4 Backbone addressing**

32 When customer frames enter a PBBN they are encapsulated by the addition of B-MAC addresses and the
33 creation of an I-TAG. The encapsulation is the result of the actions of a PIP as specified in 6.10. The PIPs
34 represent the endpoints of a backbone service instance, and the B-MAC addresses identify these endpoints.
35 (More specifically the VIPs of the PIP represent the endpoints of a backbone service instance, and the
36 combination of the B-MAC addresses and the I-SID identify these endpoints. Since the I-SID identifies
37 which VIP of a PIP is the Backbone Service Instance endpoint, the B-MAC address need only identify
38 the PIP. Therefore, all VIPs of a PIP can share the same MAC address.) This encapsulation ensures that only
39 I-components learn C-MAC addresses, while B-components and Bridges in the PBBN core learn only
40 B-MAC addresses.

41 The B-SA of an encapsulated frame identifies the PIP that performed the encapsulation and is referred to as
42 the PIP MAC address. For a PIP that connects to an external LAN, this is the address of the PIP itself. For a
43 PIP that has an internal connection to a CBP in the same BEB, this can be any address that results in

1 delivering received I-tagged frames to the I-component, such as the Bridge address of that I-component.
2 When the PIP MAC address is also used by CFM MPs instantiated on the PIP, there are additional
3 constraints on the uniqueness of that address (26.4.3).

4 The B-DA of an encapsulated frame identifies the PIP(s) to which the frame should be delivered. The
5 default value for the B-DA is the Backbone Service Instance Group address constructed by concatenating
6 the three octet OUI shown in Table 26-1 with the three octet I-SID, and asserting the I/G bit in the first octet
7 of the resultant value to signify a group MAC address. In a Backbone Service Instance Group address
8 formed in this way, the OUI value is encoded in the first three octets of the address, with the most significant
9 octet of the OUI encoded in the first octet, with the I/G bit (the LSB) set, and the least significant octet of the
10 OUI encoded in the third octet; the most significant octet of the I-SID is encoded in the fourth octet of the
11 address and the least significant octet of the I-SID in the sixth octet.

Table 26-1—Backbone Service Instance Group address OUI

Name	Value
IEEE 802.1Q Backbone Service Instance Group address OUI	00-1E-83

12 NOTE—Table 26-1 uses the Hexadecimal Representation specified in IEEE Std 802.

13 When the B-DA of a frame is the Backbone Service Instance Group address, the nominal behavior of the
14 PBBN is to deliver the frame to all CBPs reachable within the B-VLAN to which the backbone service
15 instance is mapped. Filtering based on I-SID by the egress CBP ensures that frames are not transmitted by
16 CBPs that are not part of the backbone service instance. Flooding frames throughout the B-VLAN and
17 filtering at egress provides the correct connectivity for the backbone service instance, but may result in
18 inefficient use of PBBN resources. More efficient frame delivery can be achieved by configuring Static
19 Filtering Entries in the backbone Bridges, or using MMRP to create MAC Address Registration Entries, that
20 restrict the delivery to just the endpoints of the backbone service instance. More efficient delivery can also
21 be achieved by learning individual B-MAC addresses at a PIP, and in some cases by configuring the CBPs to
22 translate the Backbone Service Instance Group address.

23 **26.4.1 Learning individual backbone addresses at a PIP**

24 Individual backbone addresses are learned by means of the connection_identifier parameter generated by the
25 PIP and stored in the FDB. For each received frame, the PIP includes in the EM_UNITDATA.indication a
26 connection_identifier parameter that identifies the backbone source address of the frame. When the
27 I-component Learning Process creates or updates a Dynamic Filtering Entry for a Customer Source MAC
28 address (C-SA) of a frame received through a PIP, it stores the connection_identifier parameter of the
29 EM_UNITDATA.indication for that frame. The connection_identifier parameter is included in the
30 EM_UNITDATA.request for subsequent frames destined to that customer address. Within the PIP, the
31 connection_identifier determines the individual backbone address associated with the customer address.
32 This individual address is then used as the B-DA in place of the Backbone Service Instance Group address.

33 When a backbone service instance includes only two VIPs, it is possible to use a learned individual
34 backbone address as the default B-DA instead of the Backbone Service Instance Group address. This is
35 accomplished by configuring the AdminPointToPointMAC parameter for the VIP to ForceTrue, which
36 results in the operPointToPointMAC parameter having a value of TRUE. In this case, the PIP copies the
37 B-SA of frames received for that VIP (i.e., the individual MAC address of the peer PIP) to the Default
38 Backbone Destination parameter of the VIP. Subsequent frames transmitted by the PIP on that backbone
39 service instance will then use the individual MAC address of the peer PIP as the B-DA even if the C-DA is
40 unknown or broadcast. Enabling the enhanced filtering criteria (8.7.2) on a VIP will prevent the

1 I-component from creating Dynamic Filtering Entries for C-MAC addresses unnecessarily when the
2 backbone service instance is point-to-point and the S-VLAN that maps to the backbone service instance has
3 only the VIP and one CNP in its member set.

4 When the enableConnectionIdentifier of a VIP is set to FALSE, the connection_identifier is kept to a NULL
5 value and correspondingly the B-DA used by frames associated with this VIP is always the Backbone
6 Service Instance Group address.

7 **26.4.2 Translating backbone destination addresses at a CBP**

8 In some cases, when the B-DA is the Backbone Service Instance Group address it may be advantageous that
9 the CBP translate this to a different address. Examples where such translation may be useful include the
10 following:

- 11 a) When a backbone service instance has only two endpoints within a PBBN, the Backbone Service
12 Instance Group address may be translated at the ingress CBP to the individual address of the egress
13 CBP.
- 14 b) When multiple backbone service instances connect the same set of CBPs, the ingress CBP may
15 translate the Backbone Service Instance Group address to a single group address chosen for the set
16 of backbone service instances.

17 These translations can be accomplished by configuring the Default Backbone Destination parameter in the
18 service instance table of the CBP (6.11). Whenever an ingress CBP is configured to translate the B-DA, the
19 egress CBP must be configured to translate the B-DA back to the Backbone Service Instance Group address.
20 This allows multiple PBBNs connected through Peer E-NNIs to use addresses local to each PBBN.

21 The CBP is also capable of translating the I-SID on frames entering and exiting the PBBN. This allows the
22 values of the I-SID to be local to a PBBN. If the B-DA of the frame is the Backbone Service Instance Group
23 address and the CBP translates the I-SID, the CBP will also reconstruct the Backbone Service Instance
24 Group address using the new value of the I-SID.

25 **26.4.3 Backbone addressing considerations for CFM MPs**

26 Possible placement of CFM shims in a PIP and a CBP are shown in Figure 26-2. In the PIP, these include
27 MPs to monitor backbone service instances (identified by I-SIDs) in the PBBN, as well as MPs to monitor
28 service instances (identified by VIDs of S-VLANs) in the attached PBNs. In the CBP, these include MPs to
29 monitor backbone service instance (identified by I-SIDs) in the PBBN, as well as MPs to monitor B-VLANs
30 in the PBBN. All of these MPs are assigned individual addresses subject to the constraints specified in 19.4;
31 however, there are additional considerations for the addresses used by MPs in PIPs and CBPs.

32 In a PIP, the individual address assigned to the backbone service instance-level MPs must be unique within
33 the PBBN. The individual address assigned to the S-VLAN-level MPs must be unique within the set of
34 PBNs interconnected by any backbone service instance supported by the PIP. Therefore, these two groups of
35 MPs may use a common address that is globally unique; however, they may need distinct addresses if local
36 addressing is used in the PBBN.

37 In the CBP, both the backbone service instance-level MPs and the B-VLAN-level MPs require an individual
38 address that is unique within the PBBN. Therefore, these two groups may use a common address whether it
39 is a global or local address.

1 26.5 Detection of connectivity loops through attached networks

2 The transmission and reception of MST BPDUs through CNPs will detect accidental direct connection of
3 those ports. A backbone provider cannot rely on any customer network relaying such frames, and should
4 develop a policy and mechanisms to deal with potential data loops that can arise if the attached customer
5 systems do not operate their own instance or instances of a spanning tree protocol.

6 A bridged network can be redundantly attached to a PBBN by the means of several Layer Two Gateway
7 Ports (13.40). By way of communication within their region and without any influence from the PBBN,
8 those ports will provide connectivity to the PBBN through a single Layer Two Gateway Port, thus avoiding
9 any bridging loop between the PBBN and the network.

10 NOTE 1—Use of the restrictedRole parameter at ingress ports ensures that receipt of BPDUs addressed to the Provider
11 Bridge Group address cannot disrupt internal connectivity within the PBBN.

12 NOTE 2—Specification of PBBN policies, mechanisms, and heuristics used to detect or minimize the impact of data
13 loops created by customer systems is not addressed by this standard. They can include, but are not limited to, bandwidth
14 limitation, charging policies, detection of the repetitive movement of the apparent location of customer stations, and
15 customer agreement to allow the use of PBBN loop detection protocols by not filtering the associated frames.

16 NOTE 3—A data loop is not the only possible cause of excess bandwidth consumption by a given customer of a PBBN,
17 and the PBBN is usually required to meet service guarantees to other customers irrespective of the cause of the excess
18 bandwidth demand. Data loops are not a unique threat to satisfactory overall network performance. Their distinct
19 characteristic is consumption of discretionary bandwidth without benefiting any customer. The customer that creates the
20 loop can suffer particularly serious network degradation or excess cost as the PBBN limits the total bandwidth consumed
21 by that customer. It is therefore in the interests of each individual customer and the PBBN to raise service satisfaction by
22 preventing and detecting loops.

23 26.6 Scaling of PBBNs

24 For PBBN deployments, it is important to be able to interconnect PBBNs operated by different
25 organizations. The interfaces between PBBNs are examples of the MEF Forum External Network to
26 Network Interface (E-NNI) referred to in the MEF reference model (section 7.2 and Figure 3
27 of MEF 4 [B53]). E-NNIs for interconnection of PBBNs can be further classified into two types:
28 hierarchal interconnect and peer interconnect. Examples of hierarchal and peer interconnects are depicted in
29 Figure 26-1 at the interfaces marked 4 and 5.

30 26.6.1 Hierarchal PBBNs

31 The hierarchal interconnect connects the PBBNs in a leveled hierarchy. In this interconnect model one
32 PBBN acts as a second-level ($n + 1$) PBBN to a first-level (n) PBBN. This model allows scaling the number
33 of services by additional hierarchal level. Each level of a hierarchal interconnect adds an additional
34 encapsulation layer.

35 In the hierarchical model, the ($n + 1$)-level PBBN carries the n -level PBBN B-VLANs as services in the
36 same manner that the first-level PBBN carries the customer S-VLANs as services. In this model an S-VLAN
37 is presented to a level n PBBN for transport. The PBBN transports the S-VLAN as an instance of MAC
38 Service within a B-VLAN generated by the n -level PBBN. The n -level PBBN is connected to an
39 ($n + 1$)-level PBBN over a hierarchal E-NNI interface. The ($n + 1$)-level PBBN carries B-VLANs from the
40 attached n -level PBBN as service instances in ($n + 1$)-level B-VLANs through the ($n + 1$)-level PBBN.

41 Since each hierarchal level requires an additional encapsulation, the nesting depth is limited by the
42 maximum defined frame length for the MAC type used.

1 26.6.2 Peer PBBNs

2 The peer interconnect between PBBNs results in a flat service space that does not increase the encapsulation
3 nesting depth. The network scales the number of services over the combined PBBNs by service localization.
4 Further localization and scaling of services is supported by the I-SID filtering functions in addition to the
5 optional I-SID translation functions of the CBP interfaces on each side of the peer E-NNI (6.11). Services
6 that are limited in extent to a single PBBN do not consume any resources within the other connected
7 PBBNs.

8 The peer interconnect will extend backbone service instances across the peer connected PBBNs. In the peer
9 interconnect a service instance in PBBN A is connected to a service instance in PBBN B by the peer E-NNI
10 between the PBBNs. The peer E-NNI is formed by an I-tagged Service Interface between two CBPs (6.11),
11 one in each of the peer PBBNs. The I-SID used over the E-NNI must be agreed to mutually by both
12 providers. At each end of the E-NNI the CBP may optionally translate the I-SID (and the B-DA if it is the
13 default multicast address generated from the I-SID) used over the E-NNI to a new I-SID for use within the
14 PBBN. The CBPs also map the I-SID to a B-VLAN within the PBBN, as no B-TAGs are transmitted across
15 the peer interconnect.

16 26.7 Network management

17 Management of a PBB is directly under the control of the backbone provider. PBB customers shall not have
18 access to managed objects related to elements of PBBs within the PBBN.

19 In SNMP management systems, this can be realized by implementing one management entity acting as an
20 agent in the Provider Bridge and separating the management views that include customer networks
21 information from the management views that include provider networks information using the mechanisms
22 defined by the SNMP management framework in IETF RFC 3411.

23 26.8 CFM in PBBs

24 CFM (Clause 18 through Clause 22) may operate over LANs, C-VLANs, S-VLANs, B-VLANs, and
25 backbone service instances identified by I-SIDs. CFM Maintenance domain Endpoints (MEPs) and
26 Maintenance domain Intermediate Points (MIPs) may be inserted at any of the EISS service interfaces
27 described by 6.9, 6.10, and 6.11, at the ISS service interface described in IEEE Std 802.1AC, and within the
28 EISS and Backbone Service Instance Multiplex Entity described in 6.17 and 6.18.

29 Within a PBBN, the encapsulation performed by PIPs also encapsulates CFM frames sourced by customers
30 attached to CNPs. Encapsulation of S-VLAN and C-VLAN CFM frames hides them from the PBBN
31 enabling a new set of independent MD levels to be defined in the PBBN for CFM, which may be used for
32 managing B-VLANs spanning between the B-components and BCBs of the PBBN, and over LANs that
33 span between PBB ports. In addition, the backbone may support MAs per backbone service instance that are
34 identified by I-SIDs and span between the PIPs and CBPs of I-components and B-components of the PBBN.
35 A full set of eight maintenance levels exists within each PBBN for use by B-VLAN CFM frames. The
36 Backbone Service Instance CFM frames may extend over a E-NNI, therefore providing eight levels of
37 Backbone Service Instance CFM for the extended backbone, which includes all peer connected PBBNs. All
38 eight levels of CFM frames generated in customer networks are carried over the backbone as encapsulated
39 data and may be used by customer networks. Another eight maintenance levels exist for the LANs for which
40 no frames are untagged. This may be the case for the PNP-PNP LAN links. The CFM stacks for the PIP and
41 CBP are depicted in Figure 26-2. The stacks for the CNP and PNP are as depicted in 22.1.1.

1

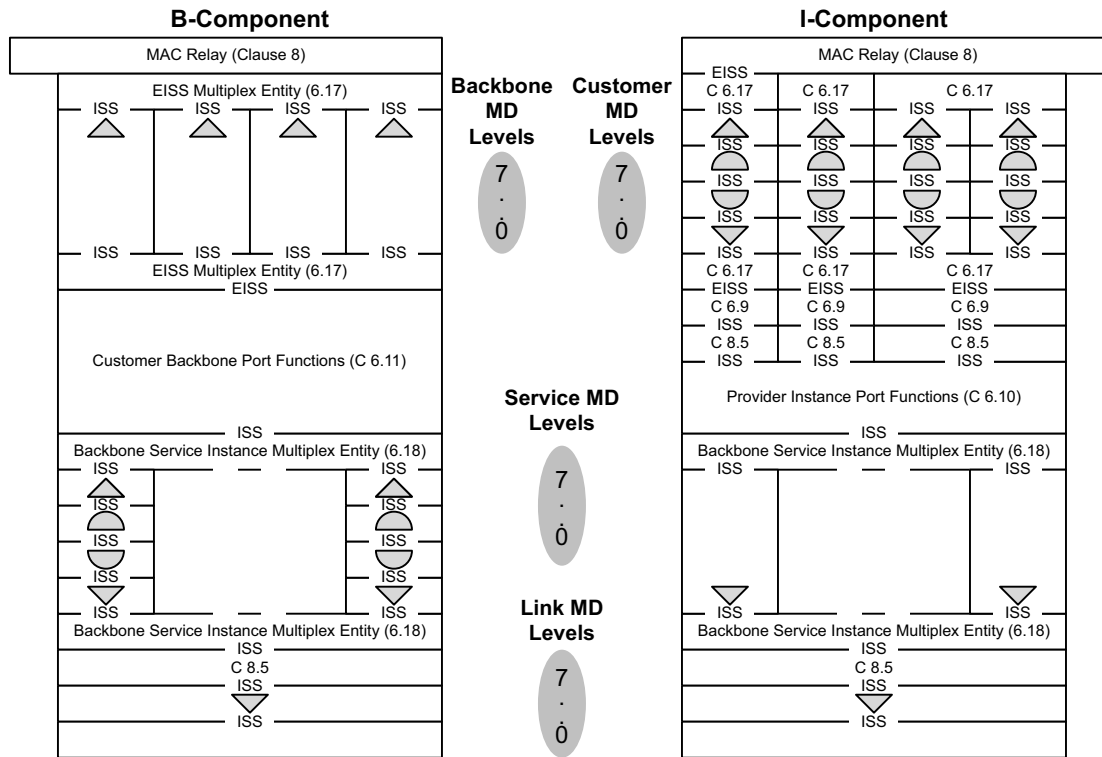


Figure 26-2—CFM shim model

2 Figure 26-3 and Figure 26-4 show possible MAs at the three types of PBBN service interfaces described in
3 25.3, 25.4, and 25.5, respectively. These three interfaces are depicted in Figure 26-1, where they are
4 labeled 1, 2, and 3. Figure 26-5 and Figure 26-6 show possible MAs at the two types of PBBN E-NNI
5 interfaces, which are described in 26.6.1 and 26.6.2, respectively. These two types of E-NNIs are depicted in
6 Figure 26-1, where they are labeled 4 and 5. The MAs indicated in Figure 26-3 through Figure 26-6 are
7 example MAs for each PBBN service interface type and PBBN E-NNI type. Typically not all these MAs
8 would be supported within any given PBBN since some MAs illustrated provide redundant capabilities and
9 since the selection of MAs will depend on the service-level agreements used for a particular service, on the
10 operational environment of a particular PBBN and on the equipment used in the particular PBBN. In these
11 figures, four distinct categories of MAs are depicted. These are the S-VLAN and C-VLAN MAs, the
12 Backbone Service Instance MAs, the B-VLAN MAs, and the LAN MAs. Each of these four classes of MAs
13 is associated with independent sets of MDs introducing four independent sets of eight maintenance levels.
14 The selection of which levels for each of these MDs to use in a particular network depends on the specific
15 network implementation. The S-VLAN MAs are only visible within the I-component where customer
16 frames are un-encapsulated. Customer CFM frames are encapsulated along with the other customer frames
17 at the VIPs within the I-component. Once the S-VLAN CFM frames are encapsulated, they appear just like
18 any data frame within the PBBN; therefore, they do not activate any CFM functions within the PBBN past
19 the VIPs. The Backbone Service Instance CFM frames may optionally be generated at the Backbone Service
20 Instance Multiplex Entity within the PIP or CBP. These CFM frames are only visible within the PBBN
21 where the I-TAG is being processed, that is at the CBP and at the PIP. Each Backbone Service Instance MD
22 may be assigned to one of eight maintenance levels. Since the Backbone Service Instance CFM maintenance
23 domain may extend over E-NNI boundaries, these eight levels extend over all interconnected PBBNs until
24 they are terminated at a VIP. B-VLAN MAs manage the B-VLANs within a single PBBN. The B-VLAN
25 MDs have a new set of eight maintenance levels, which are separate for each PBBN and never extend out of

the PBBN. Finally, the LAN links within a PBBN (PIP to CBP, PNP to PNP), between two PBBNs (CBP to CBP, PIP to CBP) and entering/egressing the PBBN (to/from CNP) may optionally be monitored by LAN link MAs, generated/terminated within the PIP, CBP, PNP, and CNP. LAN link MAs are typically confined to the LAN link.

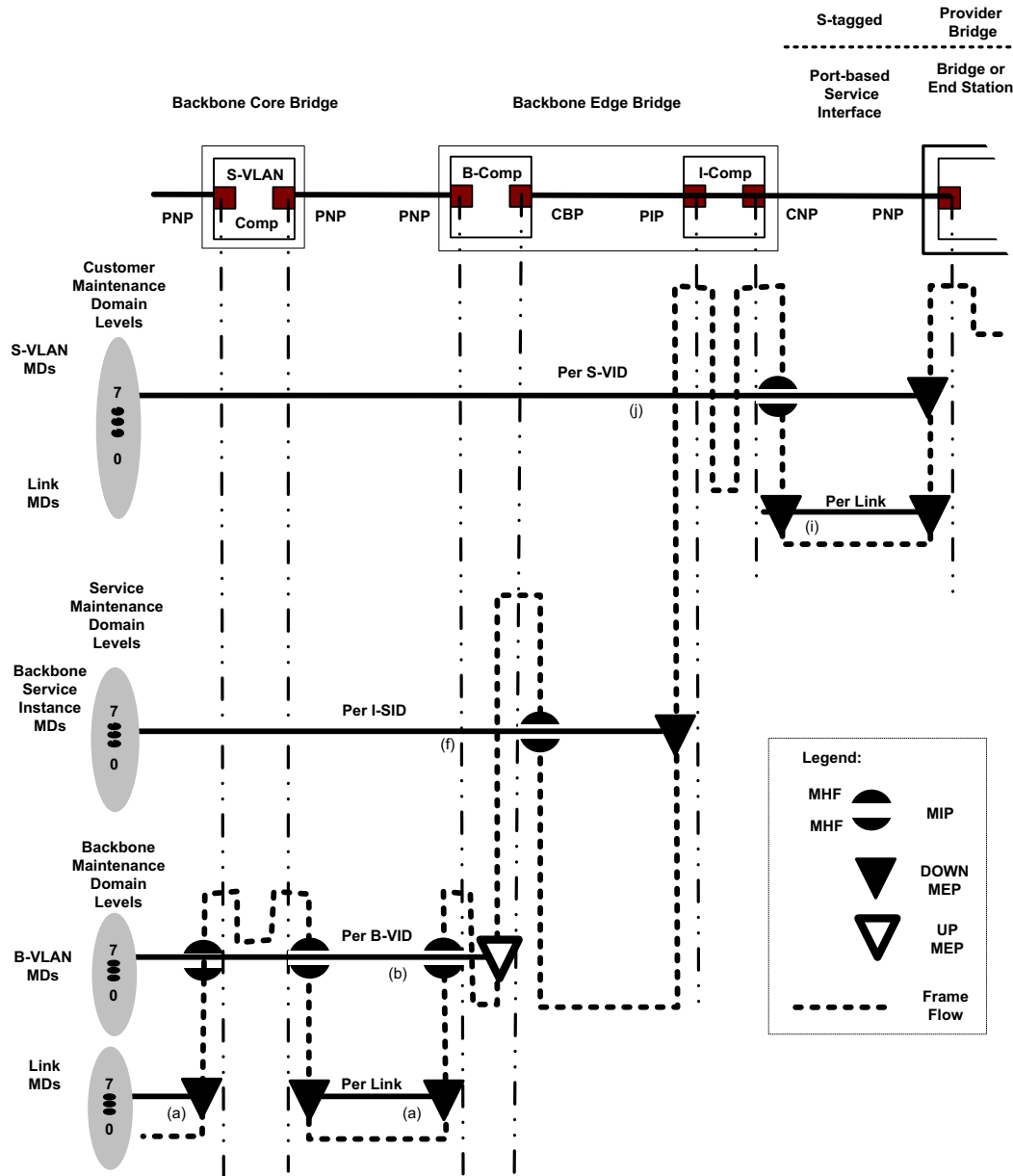


Figure 26-3—CFM example applied to a Port-based and S-tagged service interface

1

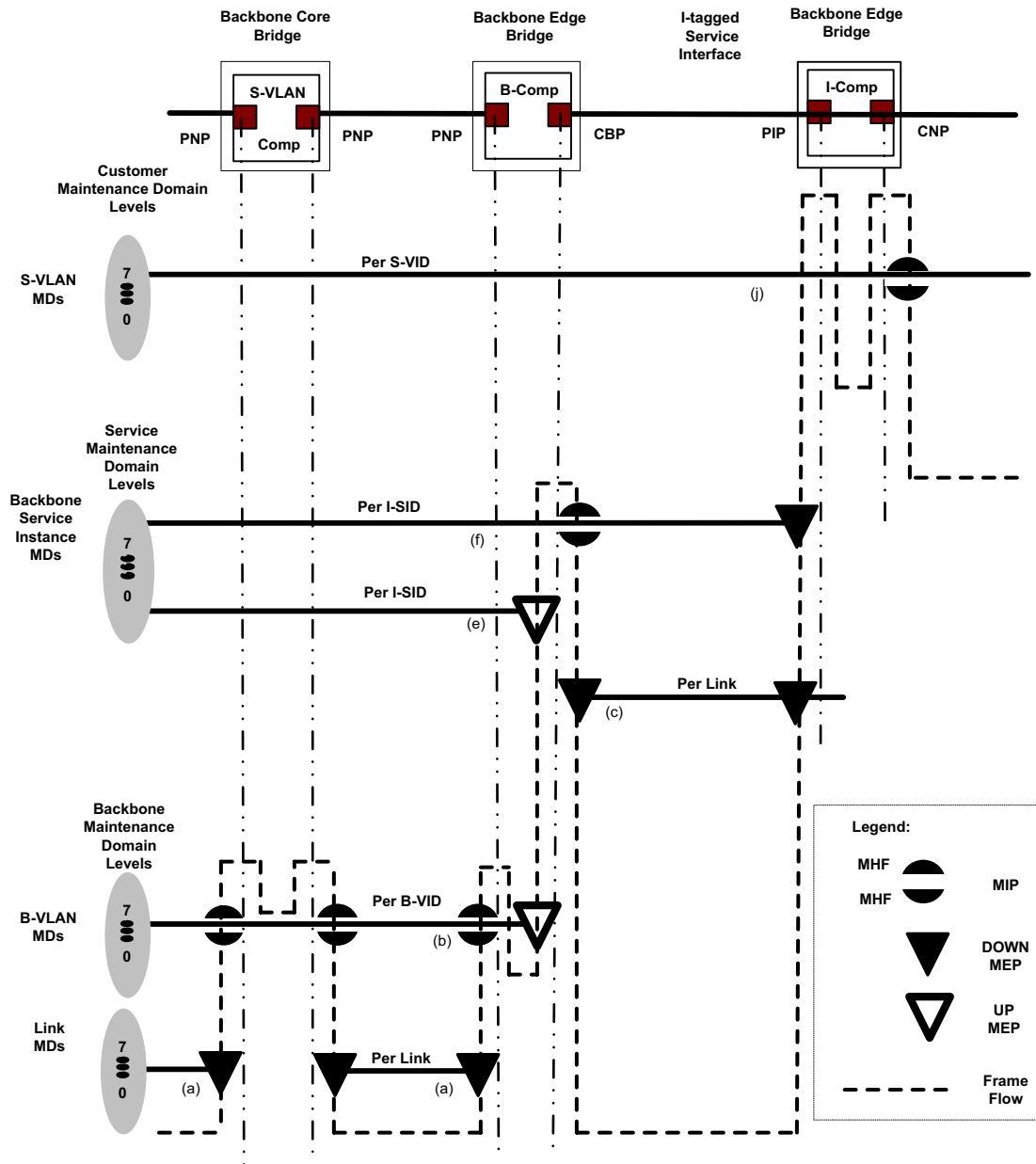
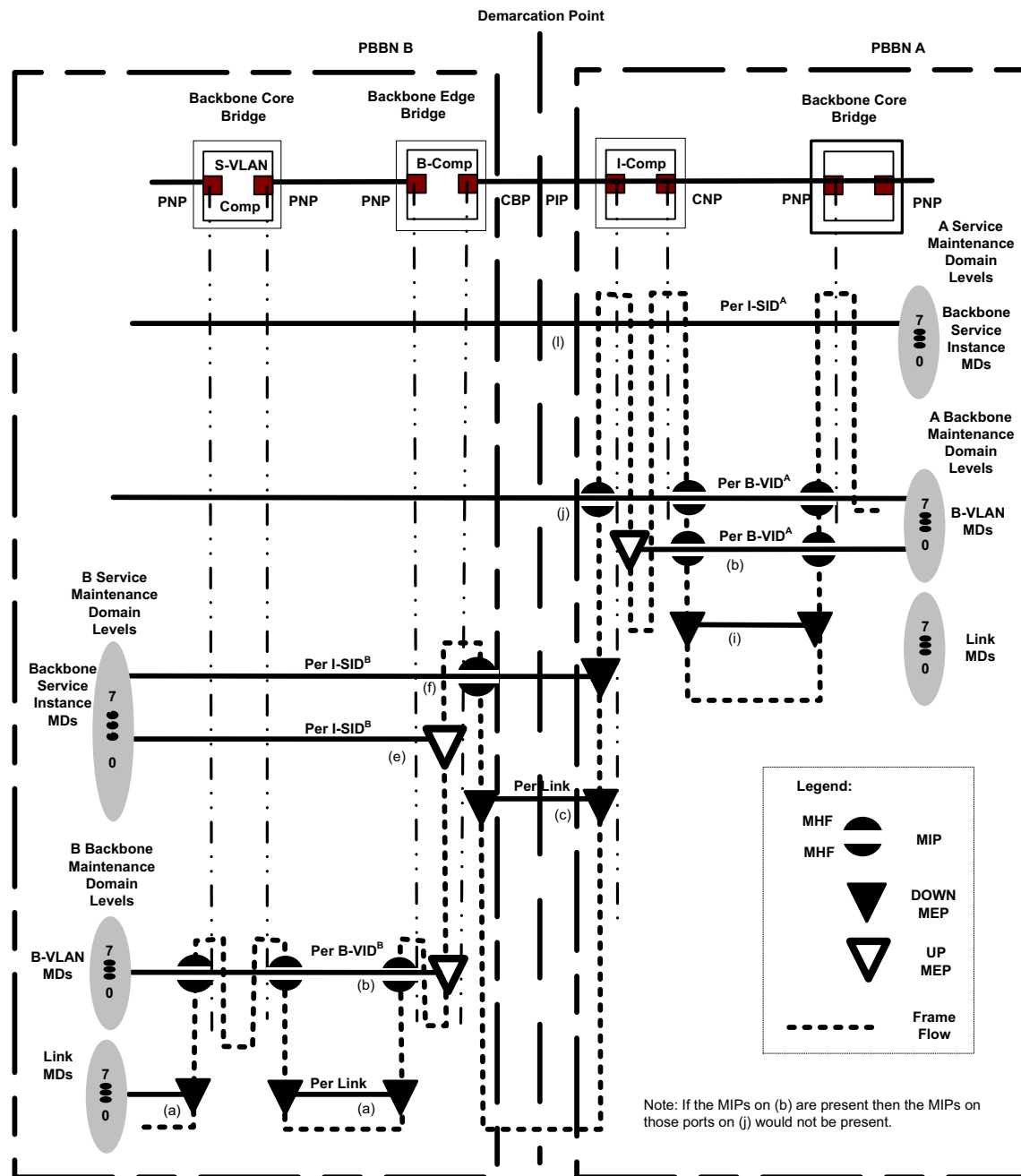


Figure 26-4—CFM example applied to an I-tagged Service Interface

1



1

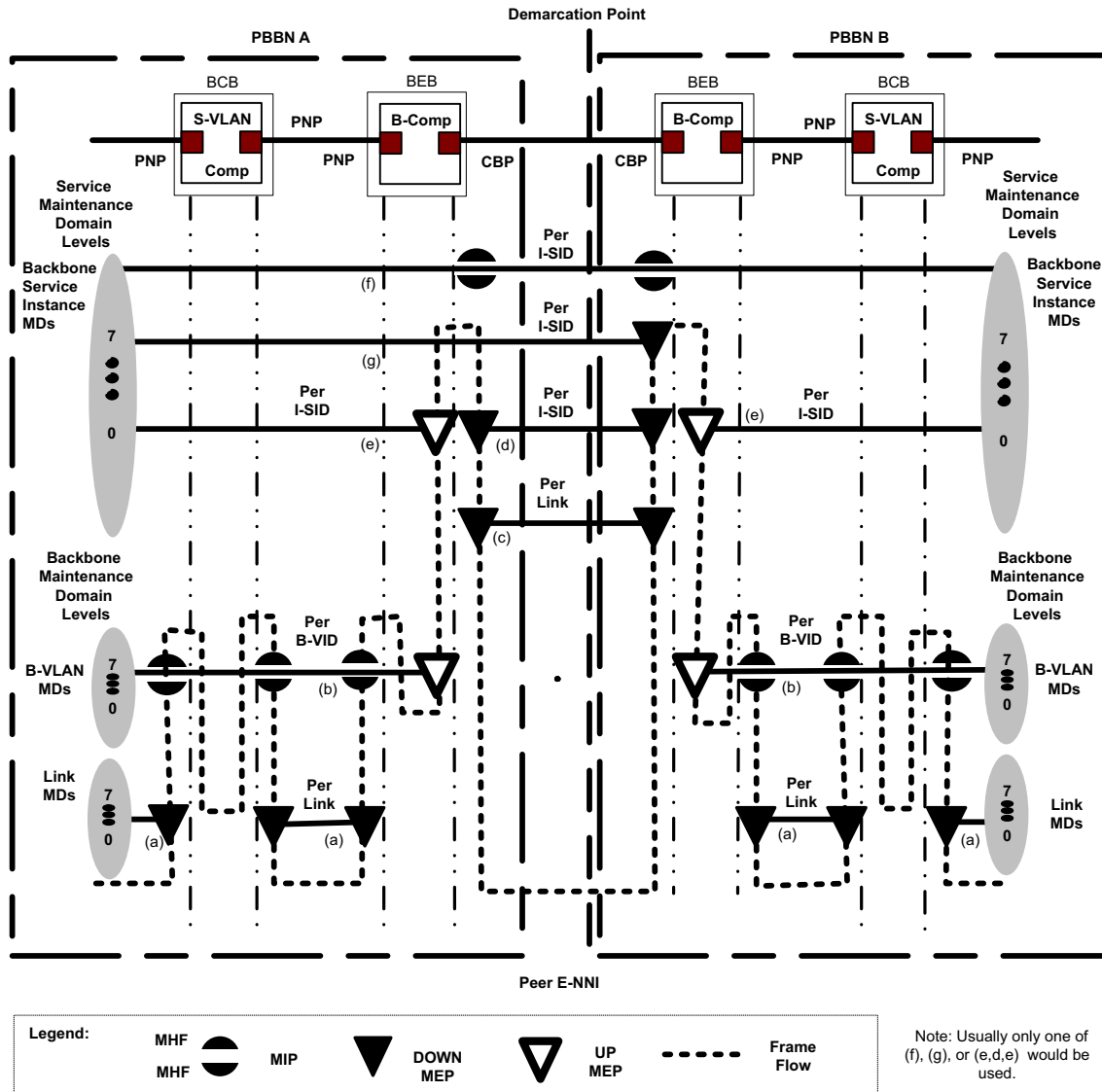


Figure 26-6—CFM example applied to a peer E-NNI, CBP-PIP

26.8.1 CFM over Port-based and S-tagged service interfaces

Figure 26-3 shows example MAs for the Port-based or S-tagged service interface described in 25.3 and 25.4 and depicted in Figure 26-1 labeled 1 and 2.

MAs (a) and (b) illustrated in Figure 26-3 through Figure 26-6 are used to monitor the PBBN core. Each PBBN has an independent backbone MD-level space. MAs (a) and (b) are assigned Backbone Maintenance Levels from the backbone MD-level space of their PBBN in the same manner used to assign S-VLAN MAs described in Clause 22. The MAs (a) are used to monitor each individual LAN that interconnects PNPs. The MAs (b) monitor specific B-VLANs. The B-VLAN monitoring, illustrated in the example, begins at the up MEP located at the top of the CBP. A given B-VLAN may exit a B-component on multiple PNPs, each of which may have a MIP on MA (b) if desired.

1 MAs (i) and (j) illustrated in Figure 26-3 through Figure 26-6 are two possible customer MAs. Attached
2 customers have an independent customer MD-level space. MAs (i) and (j) are assigned Customer
3 Maintenance Domain Levels from their customer MD-level space in the manner used for S-VLAN MAs
4 described in Clause 22. The MAs (i) monitor the links between the customer equipment and the CNP or the
5 PBBN. At the Port-based or the S-tagged service interface, MAs (i) may be terminated by the provider by an
6 MEP at the CNP and used to monitor the customer's access link. The MAs (j) monitor the service instance
7 from the customer over the PBBN. MAs (j) extend between all the customer attachments of the extended
8 PBBN, passing through any PBBN E-NNI interfaces. At the VIPs located at the edge of the extended PBBN,
9 the CFM frames for MA(j) will be mapped to/from an I-SID and carried over the extended PBBN as
10 encapsulated frames. These CFM frames are encapsulated by the PBBN in the same manner as data frames.
11 The encapsulation makes them invisible to any MPs within the extended PBBN.

12 In the case of a Port-based service interface, the MA(j) CFM frames are untagged or C-tagged over the
13 interface formed between the CNP and the customer equipment. In this case, the MA(j) CFM frames have an
14 S-VID assigned from the PVID at the CNP, which will be carried to the VIP within the I-component. In the
15 case of an S-tagged service interface, the MA(j) CFM frames are untagged, C-tagged, or S-tagged over the
16 interface formed between the CNP and the customer equipment. In this case, the MA(j) CFM frames have
17 the S-VID carried over the service interface or, if the CFM frame is not S-tagged, it will have an S-VID
18 assigned from the PVID of the CNP.

19 MA (f) depicted in Figure 26-3 through Figure 26-6 is used for backbone service instance management.
20 MAs (f) are within the Backbone Service Instance MD-level space, which is shared over the entire extended
21 PBBN. MAs (f) are assigned Service Maintenance Domain Levels from this Backbone Service Instance
22 MD-level space. The MEPs for MA(f) are formed at the PIP within the Backbone Service Instance
23 Multiplex Entity (6.18) as seen in Figure 26-2. MA(f) may have MIPs located in the CBP also formed by a
24 Backbone Service Instance Multiplex Entity as depicted in Figure 26-2.

25 26.8.2 CFM over I-tagged Service Interfaces

26 Figure 26-4 shows possible MAs for the I-tagged service interface described in 25.5 and depicted in
27 Figure 26-1 labeled 3. The MAs (a) and (b) in Figure 26-4 are used for PBBN core monitoring as described
28 in 26.8.1.

29 MAs (c) in Figure 26-4, Figure 26-5, and Figure 26-6 are used to monitor the link between the PIP and CBP.
30 The MAs (c) CFM frames are unique among the other CFM frames because they are transmitted without an
31 I-TAG over the CBP to PIP LAN.

32 MAs (f) depicted in Figure 26-4 are used for service management as described in 26.8.1. In an I-tagged
33 service interface, MAs (f) have an MEP in the customer network located at the Backbone Service Instance
34 Multiplex Entities (6.18) of the customer PIP and may have MIP within the extended provider network
35 located at the Backbone Service Instance Multiplex Entities of the CBPs. The provider may use MAs (e) to
36 monitor each service instance by placing a MEP at the Backbone Service Instance Multiplex Entities of the
37 CBP. The backbone service instance-level space is shared between the customer and provider who must
38 agree on the levels used for provider and customer Backbone Service Instance CFM messages.

39 Customer S-VLAN MAs (j) are encapsulated by the customer equipment and are indistinguishable from
40 data within the PBBN.

41 26.8.3 CFM over hierarchal E-NNI

42 Figure 26-5 shows possible MAs for the hierarchal E-NNI described in 26.6.1. In Figure 26-1 the hierarchal
43 E-NNI labeled 4 depicts an E-NNI with the demarcation point between the CBP and PIP. Example MAs for
44 a hierarchal E-NNI with demarcation between CBP and PIP are illustrated in Figure 26-5. The MAs (a) and
45 (b) in Figure 26-5 are used for PBBN core monitoring as described in 26.8.1.

1 The operation of the backbone service instance and LAN MAs (f), (e), and (c) are as described in 26.8.2.
2 Since each PBBN-A B-VLAN MA (j) is mapped into a PBBN-B backbone service instance monitored by
3 MAs (f) and/or (e) the MAs (f) and (e) may be used to monitor the extended B-VLAN from PBBN-A.

4 MAs (j) are not visible to PBBN-B as described in 26.8.2. The per PBBN-A Backbone Service Instance
5 MAs (l) illustrated in Figure 26-5 are never visible to PBBN-B since they are encapsulated along with the
6 B-VLAN.

7 **26.8.4 CFM over peer E-NNI**

8 Figure 26-6 shows possible MAs for the peer E-NNI described in 26.6.2 and depicted in Figure 26-1 labeled
9 5. The MAs (a) and (b) in Figure 26-6 are used for PBBN core monitoring as described in 26.8.1. The LAN
10 MAs (c) and Backbone Service Instance MAs (e) and (f) in Figure 26-6 are as described in 26.8.2.

11 The MAs (f), (g), (d), and (c) extend over the demarcation point between PBBN-A and PBBN-B, while the
12 MAs (e) are within PBBN-A and PBBN-B and do not extend over the demarcation point. The MA (f) is a
13 Backbone Service Instance MA that extends through both PBBN-A and PBBN-B providing monitoring for
14 the service from VIP to VIP. MAs (f) may have a MIP on each side of the demarcation point to allow
15 isolation of a fault to an individual PBBN. The MAs (g) also are Backbone Service Instance MAs that
16 extend over the demarcation point. MAs (g) are launched by PBBN-B at the CBP facing the E-NNI rather
17 than at the service originating VIP and extend to the far edge of PBBN-A. MAs (g) may be used by PBBN-B
18 to monitor the backbone service instance over PBBN-A, but only provides information to PBBN-A on the
19 operation of the backbone service instance over the demarcation point to PBBN-B. The MA (d) is a
20 Backbone Service Instance MA that monitors the service over the demarcation point, but does not provide
21 any monitoring deeper in PBBN-A or PBBN-B. MAs (e) provide internal monitoring of the backbone
22 service instance within PBBN-A and PBBN-B, respectively.

23 **26.9 CFM in a PBB-TE Region**

24 CFM as specified in Clause 18 through Clause 22, provides capabilities useful in detecting, isolating, and
25 reporting connectivity faults in VID-based service instances, backbone service instances, TESIS, and
26 Infrastructure Segments. The list of features that are specifically related to TESIS and Infrastructure
27 Segments is summarized here.

28 NOTE—MIP function is not supported for an Infrastructure Segment MA. Loopback is supported on Infrastructure
29 Segment MEPs. It is not useful to apply the Linktrace operation to an Infrastructure Segment MA. No extension is made
30 to the Loopback or Linktrace operations supporting their use for an Infrastructure Segment MA. This does not preclude
31 the use of MIPs associated with a PBB-TE MA.

32 In particular, this subclause summarizes PBB-TE considerations related to the following:

- 33 a) Addressing PBB-TE MEPs (26.9.1)
- 34 b) TESI identification (26.9.2)
- 35 c) PBB-TE MEP placement in a Bridge Port (26.9.3)
- 36 d) PBB-TE MIP placement in a Bridge Port (26.9.4)
- 37 e) TESI Maintenance Domains (26.9.5)
- 38 f) PBB-TE enhancements of the CFM protocols (26.9.6)
- 39 g) Addressing Infrastructure Segment MEPs (26.9.7)
- 40 h) Infrastructure Segment identification (26.9.8)
- 41 i) Infrastructure Segment MEP placement in a Bridge Port (26.9.9)
- 42 j) Infrastructure Segment Maintenance Domains (26.9.10)
- 43 k) IPS extensions to Continuity Check operation (26.9.11)

26.9.1 Addressing PBB-TE MEPs

The configuration of an MA requires a parameter (or a list of parameters) that associates it with the monitored service. A TESI is identified by the list of ESPs supporting it. Correspondingly, the configuration of a PBB-TE MA, requires the list of the monitored TESI's ESPs, each identified by the 3-tuple <ESP-DA, ESP-SA, ESP-VID> or collectively by the TE-SID (12.14.5.3.2).

The MEPs related to a TESI require the same set of parameters as a VID-based MEP requires, with the following changes:

- a) The Primary VID is not writable but is always set to the value of the ESP-VID parameter that corresponds to the MA's ESP that has the MEP's MAC address in its ESP-SA field [item d) in 12.14.7.1.3, 20.9.7].
- b) The MAC address of the MEP is the MAC address of the CBP upon which the MEP is operating [item i) in 12.14.7.1.3].

26.9.2 TESI identification

In contrast to VID-based services where the `vlan_identifier` is used to identify the service, independent ESPs could have the same VID value in their ESP-VID field if some other parameters in their identifying 3-tuples are different. Figure 26-7 depicts such a case where the two ESPs are distinguished by their source addresses. The TESI Multiplex Entity (6.19) allows shims defined for PBB-TE to be instantiated per TESI at a SAP that supports multiple TESIs.

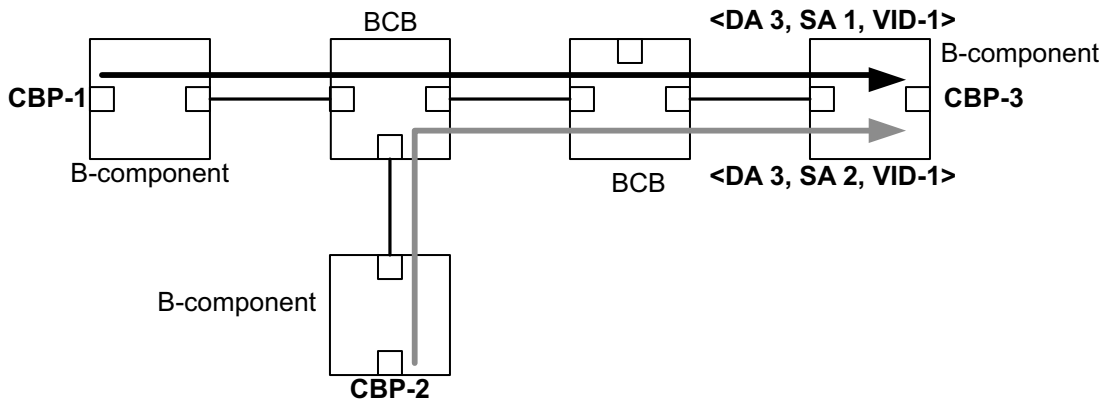


Figure 26-7—Independent ESPs using the same ESP-DAs and ESP-VIDs

26.9.3 PBB-TE MEP placement in a Bridge Port

PBB-TE MEPs are always Up MEPs and can only be placed on CBPs as these correspond to the demarcation points of the ESPs supporting the PBB-TE service. The PBB-TE MEPs are placed between the Frame filtering entity (8.6.3) and the Port filtering entities (8.6.1, 8.6.2, and 8.6.4) in the ESP-VID space identified by the EISS Multiplex Entity (6.17). Since the ESP MAC addresses can be reused by different ESPs, the PBB-TE MEPs need to be further differentiated by the TESI Multiplex Entity (6.19). In principle, separately for each TESI, there can be from zero to eight Up MEPs, ordered by increasing MD Level, from Frame filtering toward Port filtering, even though not more than one MD Level is expected for PBB-TE MAs. Figure 26-8 depicts an example of PBB-TE MEPs on a CBP.

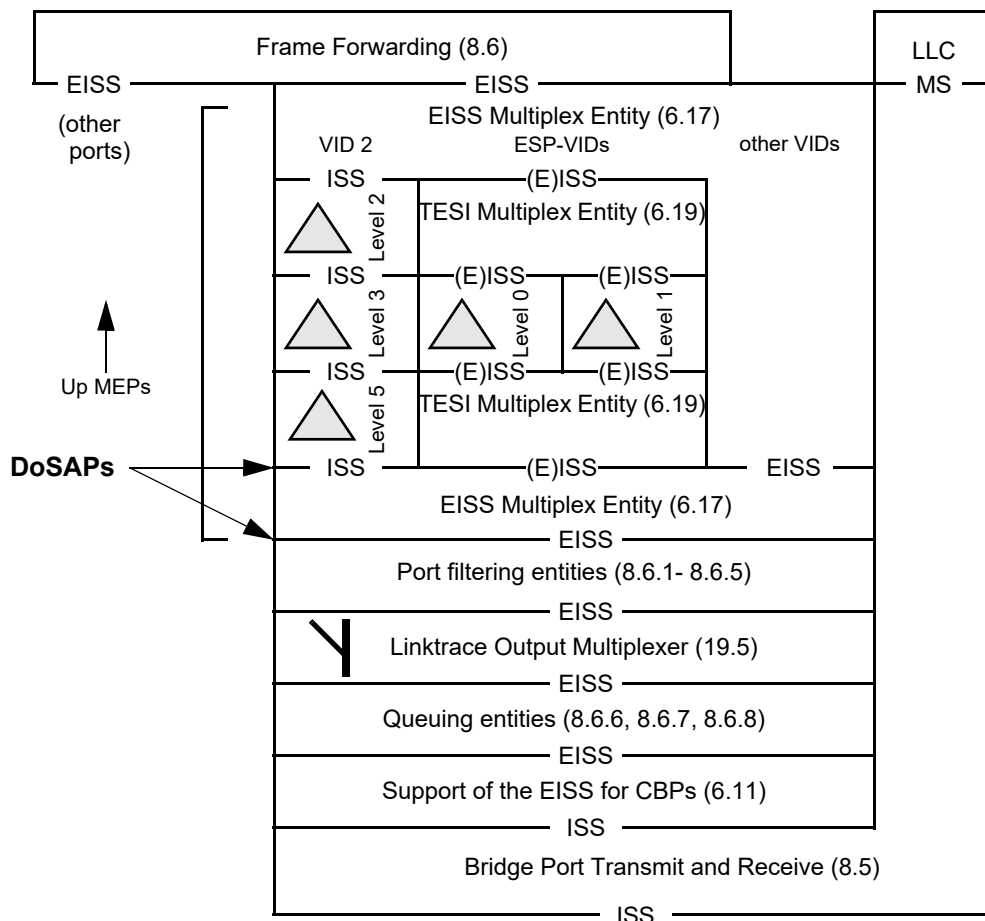


Figure 26-8—PBB-TE MEP placement in a CBP

1 26.9.4 PBB-TE MIP placement in a Bridge Port

2 PBB-TE MIPs creation is based on the algorithm described in 22.2.3. PBB-TE MIPs are created per
3 identified ESP-VIDs on PNPs. Figure 22-9 depicts an example of a pair of PBB-TE MHFs on a PNP.

4 26.9.5 TESI Maintenance Domains

5 A PBB that is capable of providing TESIS supports a new set of TESI MDs in addition to those described in
6 26.8. In particular the CFM stacks in Figure 26-2 will need to have TESI MD levels in parallel to the
7 depicted Backbone MD levels for CBPs initiating TESIS while TESI MDs will have to be present in parallel
8 to the Backbone MD levels depicted in Figure 26-3.

1 26.9.6 PBB-TE enhancements of the CFM protocols

2 26.9.6.1 Continuity Check protocol in a PBB-TE MA

3 The Continuity Check protocol is described in 20.1 and the corresponding state machines in Clause 20. The
4 only enhancement required by the PBB-TE MA is in the procedure that is responsible for constructing and
5 transmitting a CCM (20.11.1):

- 6 a) The `destination_address` parameter is set to the MAC address indicated by the value of the ESP-DA
7 field of the ESP having as ESP-SA the MAC address of the MEP emitting the CCM.

8 PBB-TE MAs may also implement the following enhancements to the Continuity Check protocol:

- 9 b) The procedure `MEPprocessEqualCCM()` on PBB-TE MEPs does not include the check of the MAID
10 on received CCMs [item b) in 20.17.1].
- 11 c) PBB-TE MEP's may:
- 12 1) Set the Traffic field (21.6.1.4) bit on transmitted CCMs based on information from the
13 Backbone Service Instance table associated with the CBP upon which the MEP is configured.
14 In particular, the procedure `xmitCCM()` (20.11.1) is performing the additional action to fill the
15 Traffic field with the `presentTraffic` variable (20.9.8); and
- 16 2) Check the Traffic field bit on received CCMs (20.16.13); and
- 17 3) Implement the MEP Mismatch state machines (20.26).

18 All other Continuity Check processes are the same as those for a VID-based MA.

19 26.9.6.2 Loopback protocol in a PBB-TE MA

20 The Loopback protocol is described in 20.2 and the corresponding state machines in Clause 20. The
21 enhancements required by the PBB-TE MA are summarized below:

- 22 a) The LBMs transmitted by a MEP associated with a PBB-TE MA use as the `destination_address`
23 parameter, the MAC address indicated by the value of the ESP-DA field of the MA's ESP that has
24 the MEP's MAC address indicated in its ESP-SA field [item a) in 20.31.1]. To enable MIPs to
25 selectively intercept LBMs that are targeting them, the PBB-TE MIP TLV (21.7.5) is inserted as the
26 first TLV [item f) in 20.31.1] in an LBM. The PBB-TE MIP TLV is not included if the LBM
27 destination address in 12.14.7.3.2 is associated with any of the values in the ESP-DA field of the
28 monitored MA's ESPs. The format of the PBB-TE MIP TLV is described in 21.7.5 and is
29 constructed as follows:
- 30 1) The MIP MAC address field contains the MAC address of the MIP to which the LBM is
31 targeted [item b) in 12.14.7.3.2].
- 32 2) The Reverse VID field contains the parameter provided in item f) in 12.14.7.3.2, which it uses
33 as the `vlan_identifier` in the replied LBR.
- 34 3) The Reverse MAC field is only included if the MEP is associated with a point-to-multipoint
35 TESI. If the MEP is the root MEP, the Reverse MAC field contains the ESP-SA value of any of
36 the point-to-point ESPs in the TESI. If the MEP is a leaf MEP, the Reverse MAC field contains
37 the ESP-DA of the point-to-multipoint ESP in the TESI.
- 38 b) LBMs received by PBB-TE MEPs are not processed and are discarded if the received LBM carries a
39 PBB-TE MIP TLV. A PBB-TE MHF forwards all received LBMs except those carrying a PBB-TE
40 MIP TLV containing in their MIP MAC address field the MAC address of the MIP associated with
41 that PBB-TE MHF [item f) in 20.28.1].
- 42 c) The `destination_address` parameter used by LBRs transmitted by PBB-TE MEPs is the value of the
43 ESP-DA of the MA's ESP that has the MEP's own address in its ESP-SA field. In the case of a
44 PBB-TE MHF, the `destination_address` parameter for the transmitted LBR is:

- 1) The value carried in the Reverse MAC field contained in the PBB-TE MIP TLV of the received LBM, if this is a group MAC address; otherwise,
- 2) The value of the source_address of the received LBM.
- d) The source_address of LBRs transmitted by PBB-TE MHFs is set to:
 - 1) The destination_address of the received LBM if this destination address is an individual MAC address; otherwise,
 - 2) The value carried in the Reverse MAC field contained in the PBB-TE MIP TLV if the received LBM's destination address is a group MAC address.
- e) The vlan_identifier used by LBRs transmitted by PBB-TE MEPs is set to the value of its Primary VID [item d) in 12.14.7.1.3]. The vlan_identifier used by LBRs transmitted by PBB-TE MHFs is set to the value carried in the Reverse VID field contained in the PBB-TE MIP TLV of the received LBM.

All other Loopback processes are the same as those for a VID-based MA.

26.9.6.3 Linktrace protocol in a PBB-TE MA

The Linktrace protocol is described in 20.3 and the corresponding state machines in Clause 20. The enhancements required by the PBB-TE MA are summarized as follows:

- a) The LTM's transmitted by a MEP associated with a PBB-TE MA use as the destination_address parameter, the MAC address indicated by the value of the ESP-DA field of the MA's ESP that has the MEP's MAC address indicated in its ESP-SA field [item a) in 20.42.1]. LTM's carry the PBB-TE MIP TLV constructed as follows:
 - 1) The MIP MAC address field is null.
 - 2) The Reverse VID field contains the parameter provided in item e) in 12.14.7.4.2, which is used as the vlan_identifier of the replied LTR.
 - 3) The Reverse MAC field is only included if the MEP is associated with a point-to-multipoint TESI. If the MEP is the root MEP, the Reverse MAC field contains the ESP-SA value of any of the point-to-point ESPs in the TESI. If the MEP is a leaf MEP, the Reverse MAC field contains the ESP-DA of the point-to-multipoint ESP in TESI.
- b) No special destination_address validation tests are performed by the ProcessLTM() procedure in the case of PBB-TE MAs [item c) in 20.47.1].
- c) The process to identify a possible egress port by an intermediate device that implements a MIP associated with a PBB-TE MA, queries the FDB of the corresponding Bridge, using the destination_address, and the vlan_identifier of the LTM as the parameters for the lookup (20.47.1.2). The LTM's can be further forwarded by PBB-TE MIPs even if the lookup identifies more than one Egress port.
- d) Finally an LTR issued by a PBB-TE MP:
 - 1) Uses as destination_address: the value carried in the Reverse MAC field contained in the PBB-TE MIP TLV of the received LTM, if this is a group MAC address; otherwise, the source_address of the received LTM [item a) in 20.47.4].
 - 2) Uses as source_address: its MAC address if the LTR is issued by a PBB-TE MEP; otherwise, the destination_address of the received LTM if this is destination address is an individual MAC address; otherwise, the value carried in the Reverse MAC field contained in the PBB-TE MIP TLV of the received LTM [item b) in 20.47.4].
 - 3) Sets the vlan_identifier parameter to the value carried in the Reverse VID field contained in the PBB-TE MIP TLV of the received LTM [item c) in 20.47.4].

All other Linktrace processes are the same as those for a VID-based MA.

26.9.7 Addressing Infrastructure Segment MEPs

The configuration of an Infrastructure Segment MA requires a parameter identifying the associated Infrastructure Segment. An Infrastructure Segment is composed of a pair of counterdirectional and co-routed Segment Monitoring Paths (SMPs) (26.11.1). Each SMP is identified by an SMP Identifier (SMP-ID). Thus, an Infrastructure Segment is identified by a pair of SMP-IDs that is known as an Infrastructure Segment Identifier (SEG-ID). The SEG-ID is specified on configuration of the associated Infrastructure Segment MA (12.14.5.3.2).

A MEP associated with an Infrastructure Segment requires the same set of parameters required by a VID-based MEP, with the following changes:

- a) The Primary VID is not writable but is always set to the value of the SMP-VID parameter that corresponds to the MA's SMP that has the MEP's MAC address in its SMP-SA field [item d) in 12.14.7.1.3].
- b) The MAC address of the MEP is the MAC address of the PNP upon which the MEP is operating [item i) in 12.14.2.1.2].

26.9.8 Infrastructure Segment identification

Two SMPs are distinct if their corresponding SMP-ID values (3-tuples) differ in at least one parameter. Thus, independent Infrastructure Segments can be associated with SMPs having identical values of SMP-DA and SMP-VID. Figure 26-9 depicts a case in which two SMPs, having identical values of SMP-DA and SMP-VID, are distinguished by their SMP-SA values. The Infrastructure Segment Multiplex Entity (6.21) allows shims defined for PBB-TE IPS to be instantiated per Infrastructure Segment at a SAP that supports multiple Infrastructure Segments.

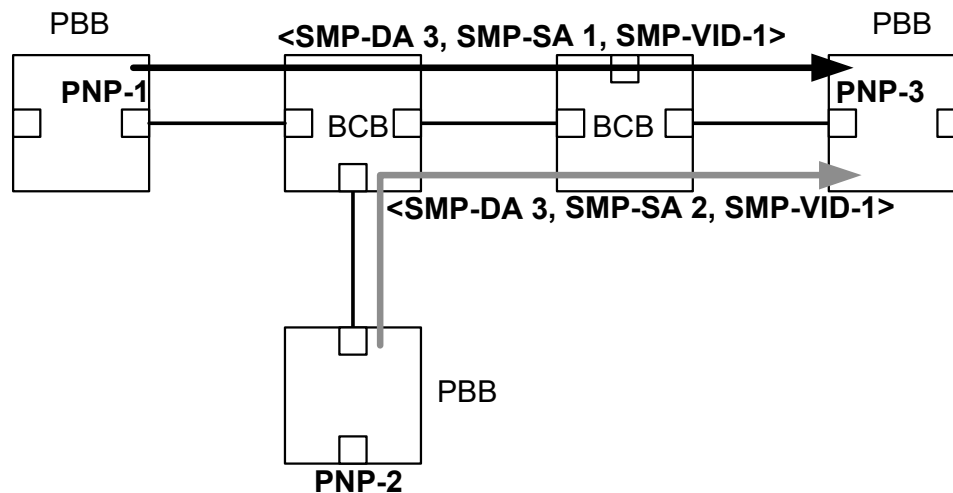


Figure 26-9—Independent Infrastructure Segments distinguished by SMP-SA

26.9.9 Infrastructure Segment MEP placement in a Bridge Port

Infrastructure Segment MEPs are always Down MEPs and are placed only on PNPs as these correspond to the demarcation points of the Infrastructure Segment. The Infrastructure Segment MEPs are placed between the Port filtering entities (8.6.1, 8.6.2, and 8.6.4) and the Queuing entities (8.6.5, 8.6.6, 8.6.7, and 8.6.8) in the SMP-VID space identified by the EISS Multiplex Entity (6.17). Since the Infrastructure Segment MAC addresses can be reused by different Infrastructure Segments, the Infrastructure Segment MEPs need to be further differentiated by the Infrastructure Segment Multiplex Entity (6.21). In principle, separately for each

- 1 Infrastructure Segment, there can be from zero to eight Down MEPs, ordered by increasing MD Level, from
- 2 Frame filtering towards Port filtering, even though not more than one MD Level is expected for
- 3 Infrastructure Segment MAs. Figure 26-10 depicts an example of Infrastructure Segment MEPs on a PNP.

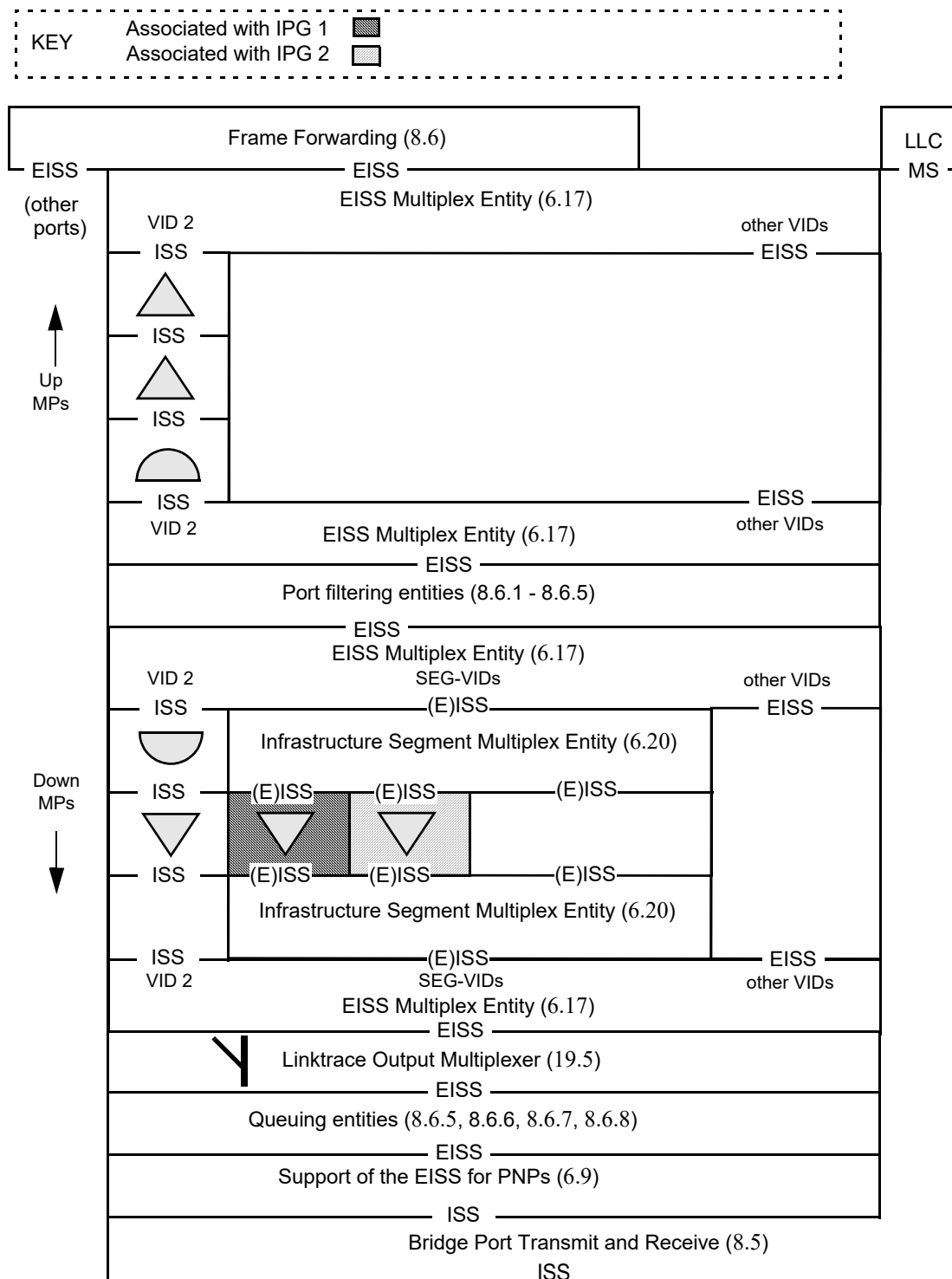


Figure 26-10—Infrastructure Segment MEP placement in a PNP

26.9.10 Infrastructure Segment Maintenance Domains

A PBB deploying IPS specifies a set of Infrastructure Segment MDs independent of those described in 26.8 and 26.9.5. In particular the CFM stacks in Figure 26-2 have Infrastructure Segment MD levels in parallel to the depicted Backbone MD levels for Ports instantiating Infrastructure Segments while Infrastructure Segment MDs will be present in parallel to the Backbone MD levels depicted in Figure 26-3.

26.9.11 IPS extensions to Continuity Check operation

The Continuity Check protocol is described in 20.1 and the corresponding state machines in Clause 20. The only enhancement required by the Infrastructure Segment MA is in the procedure that is responsible for constructing and transmitting a CCM (20.11.1):

- a) The destination_address parameter is set to the MAC address indicated by the value of the SMP-DA field of the SMP-ID having as SMP-SA the MAC address of the MEP emitting the CCM.

All other Continuity Check processes are the same as those for a VID-based MA.

26.10 Protection switching for point-to-point TESI

26.10.1 Introduction

In contrast to PBB, spanning tree protocols and broadcasting/flooding are not used in PBB-TE. FDBs are populated using a network management system or a control plane, allowing ESPs to be engineered and provisioned across the network. PBB-TE provides end-to-end linear protection for point-to-point TESI, where a dedicated protection point-to-point TESI is established for one particular working point-to-point TESI, and the traffic is automatically switched from the working TESI to the protection TESI when a failure occurs on the working entity. The protection entity is preestablished, ensuring the availability of those resources when a defect is detected on the working entity, and allowing for a corresponding sub-50 ms switchover.

Figure 26-11 depicts the essential elements of the PBB-TE protection scheme and its arrangement. In this, 1:1 (i.e., 1 for 1) arrangement, traffic is sent on only one of the paths at any point in time based on information from OAM processes or network operators.

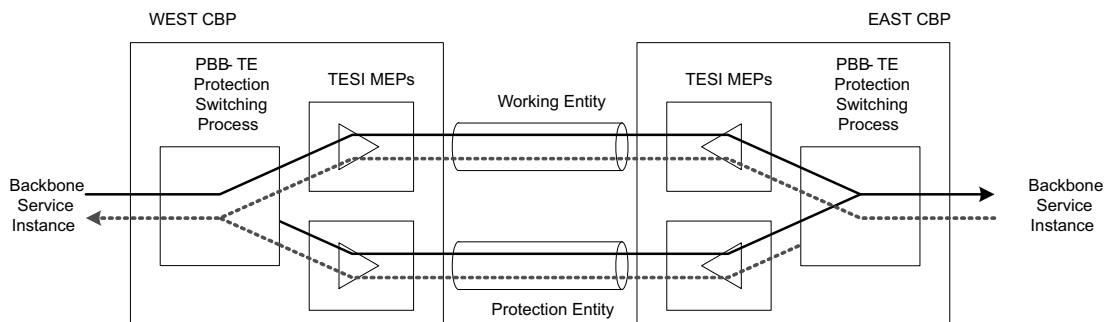


Figure 26-11—Protection switching architecture

Protection switching may be triggered by manual operation or by CFM information arising from, periodic monitoring of the working and protection paths, or from physical layer monitoring, such as loss of signal or other defects detected through CFM.

1 The PBB-TE protection switching mechanism aims to offer the capability to switch completely (both ends)
2 in less than 50 ms following the detection of the fault.

3 NOTE 1—The capability of supporting the above switchover requirement on TESI, is provided by configuring the
4 interval of CCMs on the MAs associated these TESI to be equal to or lower than 10 ms enabling the detection of the
5 fault and the signaling of the coordination information end to end in less than 50 ms.

6 The PBB-TE protection scheme may be configurable to be “revertive” or “nonrevertive,” where traffic
7 transmission reverts, or not, to the working path automatically once CFM indicates the fault or defect has
8 been cleared. The protection switching mechanism may also incorporate hold-off and wait to restore timers,
9 the first to allow the fault to be protected by a lower layer protection switching mechanism for instance,
10 while the latter ensures the performance of the working path is fully restored before switching back to it. The
11 hold-off timer obviously slows the overall recovery time for a fault within the protected domain.

12 NOTE 2—A complete discussion on the various protection switching mechanisms and terminology is provided by
13 ITU-T G.8031 [B50]. As a caution it should be noted that ITU-T uses the term “Bridge” for the logical entity that selects
14 either or both of the transmit paths at the sending end of a protected domain. This is not to be confused with the term
15 Bridge used in this standard.

16 The description herein, defines and provides a scalable end-to-end resiliency mechanism that offers
17 end-to-end 1:1 bidirectional linear protection switching capable of load sharing for point-to-point TESI in a
18 PBB-TE Region.

19 NOTE 3—Some scenarios exist where multiple input events may cause the protection switching state machines in a
20 protection group to select diverse TESI on which to send traffic. The Mismatch defect informs the operator of such an
21 occurrence thereby allowing the appropriate corrective action to be taken.

22 Protection for point-to-multipoint TESI may be provided by external mechanisms or agents, which are out
23 of scope for this specification.

24 **26.10.2 1:1 point-to-point TESI protection switching**

25 In a PBB-TE Region, IB-BEBs constitute the demarcation between the PBBN of interest and the networks
26 attached to it. The protected domain is defined to be the area between the CBPs on the different
27 B-Components of the involved IB-BEBs.

28 The starting point in providing protection switching is the creation and configuration of a TE protection
29 group. A TE protection group is a group of two point-to-point TESI between a pair of CBPs, which carries
30 an assigned set of backbone service instances, and continues to carry these backbone service instances if any
31 one of the TESI in the group is failed or disabled. The creation of the TE protection group requires the
32 assignment of one TESI as a working entity and a second TESI with the same terminating points as a
33 protection entity providing a different bidirectional connectivity path (12.14.1.2). The configuration of the
34 TE protection group is completed with the assignment of the list of the backbone service instances that are to
35 be protected by the TE protection group [item b) in 12.18.2.1.3].

36 The protection switching mechanism is capable of load sharing as the TESI that are assigned to a TE
37 protection group can be reused in a number of TE protection groups enabling a list of I-SIDs to be
38 distributed among a set of interdependent TE protection groups [item c) in 12.18.2.1.3]. A set of
39 interdependent TE protection groups forms a coordinated protection group. Protection switching requests to
40 interdependent TE protection groups must be coordinated for an operator to manage the TESI in a coherent
41 manner and to avoid potentially competing requests for each TESI. For example, to unconditionally remove
42 traffic from a TESI, one request to Lock Out the TESI results in a request to each TE protection group to
43 which the TESI belongs, either LoP or FS according the role of the TESI in each group. Similarly, to
44 conditionally remove traffic from a TESI, one Manual Switch request for the TESI results in the appropriate
45 Manual Switch request to each TE protection group to which the TESI belongs. Allowing a single operator
46 request per coordinated protection group and coordinating the resulting requests to each TE protection group

1 reduces the operational complexity of the set of interrelated TE protection groups. Using a coordinated
2 protection group, the service traffic (backbone service instance) load can be shared across a set of TESIs and
3 at the same time be protected against single faults by 1:1 protection.

4 NOTE 1—If the operator initiates switching of an individual load shared TE protection group the mmCCMdefect may
5 not detect a switch mismatch.

6 NOTE 2—The load sharing mode of operation corresponds to 1:1 ETH Sub Network Connection Group protection with
7 Inherent monitoring (ETH SNCG/I) in the ITU-T protection switching terminology.

8 The following description of the PBB-TE protection switching mechanism is an example provided for
9 illustrative purposes. Figure 26-12 depicts a network where two point-to-point TESIs have been
10 provisioned. The TE-1 service instance on the top of the figure consists of two ESPs each identified by a
11 3-tuple: <CBP-B, CBP-A, VID-1> is depicted in black and <CBP-A, CBP-B, VID-2> in gray. The TE-2
12 service instance below consists of two other ESPs identified by the 3-tuples <CBP-B, CBP-A, VID-3> and
13 <CBP-A, CBP-B, VID-3> which are depicted in white. The use of the same ESP-VID on the TE-2 ESPs is
14 only for illustrative purposes. There are no explicit requirements on using the same or different ESP-VIDs
15 on the ESPs comprising the TESIs that are assigned to a TE protection group. One set of backbone service
16 instances, shown by the black arrows just outside the two CBPs, has TE-1 assigned as its working TESI and
17 TE-2 assigned as its protecting TESI.

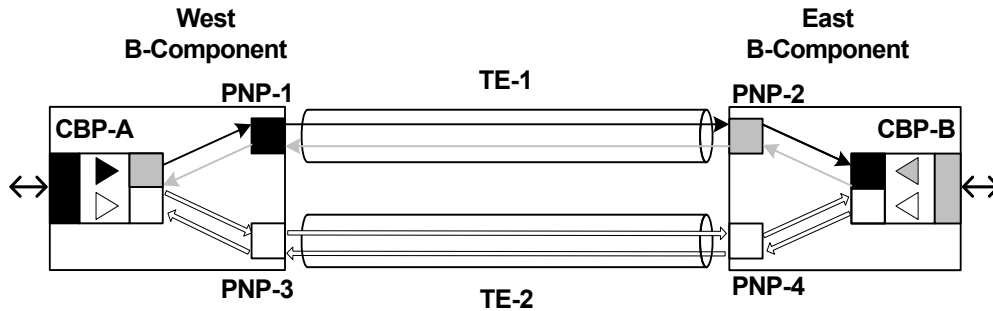


Figure 26-12—PBB-TE point-to-point protection switching

18 The ESPs are provided by configuring entries in the FDBs on all the Bridges that these ESPs need to pass
19 through. On the terminating IB-BEBs, the VLAN membership of each participating port has to be
20 configured for the B-components. In Figure 26-12, this is depicted by the color of the boxes that are placed
21 inside each of the B-Component boxes. For example, the upper right PNP-1 port on the West B-component
22 is a member of the VID-1 (black), while the CBP on the same component is a member of VID-2 (gray) and
23 VID-3 (white). Only frames tagged with the appropriate VID parameter can egress these ports.

24 NOTE 3—The PNPs are shown for illustrative purposes only and that there is no explicit requirement to diversify PNPs
25 on the terminating IB-BEBs.

26 Each of the TESIs is monitored by an independent MA. One MA is set to monitor the top (TE-1) TESI and a
27 second to monitor the bottom (TE-2) TESI. Each of these two MAs is associated with a pair of ESPs and
28 identified by their corresponding 3-tuples. Two Up MEPs, associated with the MA monitoring the TE-1
29 service instance, are configured on the CBPs that terminate the TE-1 service instance. Each of these MEPs
30 has its own primary VID, VID-1 (black) for the MEP on the West B-component associated with the TE-1
31 service instance, and VID-2 (gray) for the MEP on the East B-component. In this configuration each MEP
32 receives frames that are tagged with the ESP-VID corresponding to the primary VID of the associated
33 remote MEP and can only send frames that are tagged with its own primary VID. In particular, in the
34 depicted example the MEP for the top entity on the West B-component can send only CCMs tagged with
35 VID-1 (black) and receive CCMs tagged with VID-2 (gray), while the corresponding MEP on the East
36 component can only send CCMs tagged with VID-2 (gray) and receive CCMs tagged with VID-1 (black).

1 The primary VID of each MEP is depicted by the color of the MEP. It should be also noted that all the
2 depicted ports in Figure 26-12 have their Enable Ingress Filtering parameter (8.6.2) reset to its default value,
3 i.e., Disable Ingress Filtering.

4 Data traffic is mapped to a TESI by configuring the CBP parameters (6.11). In particular, the CBP I-SID is
5 used to allow specific service instances to be carried by the TESI while the entries of the B-VID column in
6 the Backbone Service Instance table are indicating how to map the identified service instances to a specific
7 ESP. The CBP's B-VID value is depicted in Figure 26-12 by the color of the vertical bars (black for CBP-A,
8 gray for CBP-B) placed at the tips of the arrows representing the set of backbone service instances.

9 As a result customer frames that need to be transported by a TESI and are identified by a specific I-SID and
10 reach the CBP on the West B-component from the left can be mapped to the black ESP or the white ESP
11 while frames on the same service that reach the CBP on the East B-component from the right will be mapped
12 to the gray ESP or the white ESP based on the CBP's configured B-VID value. If all the services are
13 normally mapped to the black ESP and gray ESP, then TE-1 would correspond to the working entity and
14 TE-2 to a stand-by protection entity. Irrespective of how the data traffic is mapped to the TESIs, CCM
15 frames are exchanged on both TESIs in order to regularly check the provided connectivity.

16 If a fault occurs on any of the ESPs, the MEP on the receiving end will be notified. In particular if a fault on
17 the black ESP occurs, as shown in Figure 26-13, the MEP on the East B-component will declare a remote
18 MEP defect by setting the rMEPCCMdefect parameter (20.19.1). The timer counter for timing out CCMs
19 has a granularity finer than or equal to $1/4$ of the time represented by the CCMinterval variable (20.8.1). A
20 Bridge does not set rMEPCCMdefect within $(3.25 \times \text{CCMinterval})$ seconds of the receipt of a CCM, and
21 sets rMEPCCMdefect within $(3.5 \times \text{CCMinterval})$ seconds after the receipt of the last CCM. The setting of
22 the rMEPCCMdefect parameter will result in a change of the appropriate B-VID entries in the Backbone
23 Service Instance table, of the east CBP from the gray VID-2 to the white VID-3, which is the ESP-VID of
24 the associated ESP on the protection TESI.

25 NOTE 4—The B-VID parameter will also change when the xConCCMdefect (20.23.3) or the errorCCMdefect (20.21.3)
26 parameters are set as these indicate a very serious misconfiguration problem.

27 All subsequent CCMs sent by the east gray MEP on TE-1 will have the RDI field set for as long as proper
28 CCMs are not received by the MEP. A reception of a CCM frame with the RDI field set (or an event that
29 causes setting of the someRMEPCCMdefect, xConCCMdefect, or errorCCMdefect) by the MEP associated
30 with TE-1 on CBP-A will cause a change of the appropriate B-VID entries in the Backbone Service Instance
31 table of the CBP-A on the West B-component, to the pre-configured value of the protection ESP. The
32 result will be to move the affected service instances to the protection TE-2 service instance as depicted
33 Figure 26-13. Consequently, the approximate maximum time before a bidirectional switch occurs following

1 a unidirectional failure would be approximately equal to $(3.5 \times \text{CCMinterval})$ + the time required for a CCM
2 PDU to travel between the MEPs monitoring the TESI + the time to update the associated Backbone Service
3 Instance table entries.

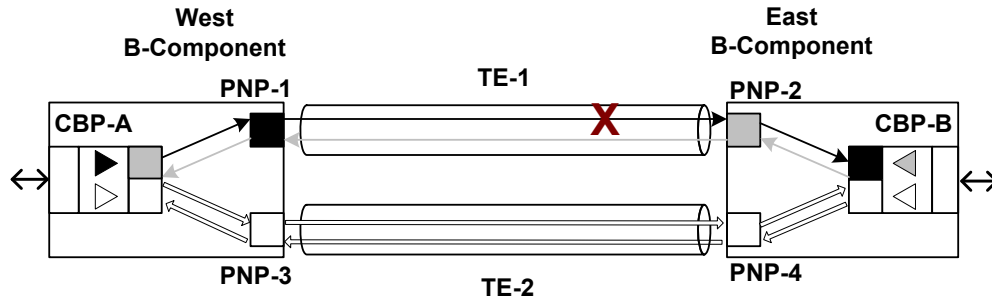


Figure 26-13—Mapping data traffic to the protection entity

26.10.3 Protection Switching state machines

5 The local protection commands and protection behavior specified here follow the architectural model used
6 in ITU-T G.8031 [B50]. The relationships among the protection switching state machines are illustrated in
7 Figure 26-14. That figure uses conventions similar to those of 13.24, and of Figure 13-13: open arrows
8 denote variables that are set by one state machine and both read and set by another; closed arrows denote
9 variables that are set by the owning state machine, and only read by other state machines.

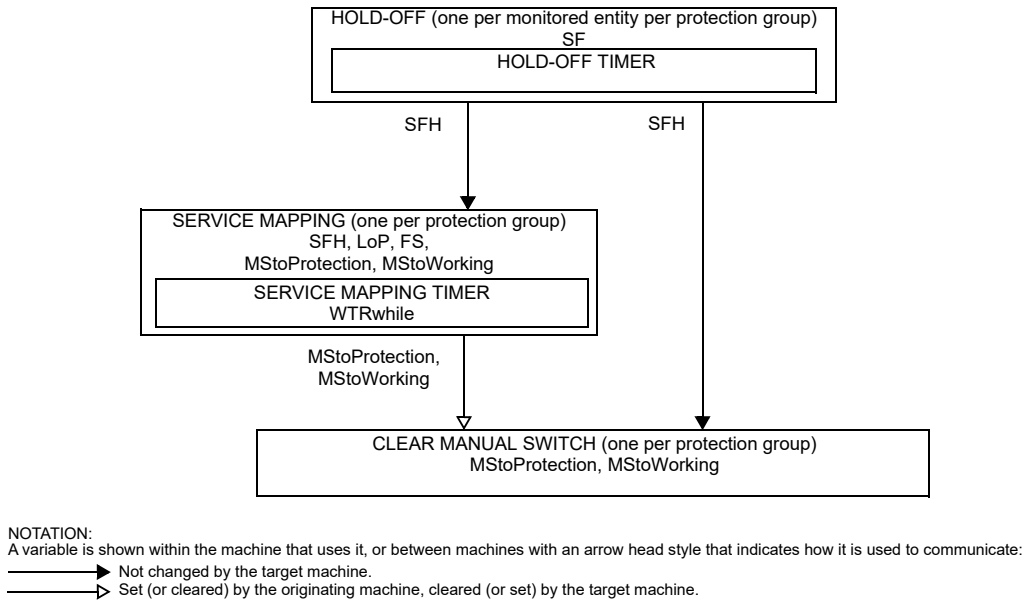


Figure 26-14—Relationships of the Protection switching state machines—overview

10 There is one set of Protection Switching state machines per protection group on each Bridge that terminates
11 the protection group. The protection group consists of the working and the protection entities. The
12 Protection Switching state machine reuses the defect variables that are presented in Table 20-1. Variables

1 and procedures that are preceded with “w.” refer to the working entity while those that are preceded with
2 “p.” refer to the protection entity. These internal prefix/entity associations remain the same following a
3 protection switch to the protection entity.

4 **26.10.3.1 Notational conventions used in state diagrams**

5 The Protection Switching state machines are specified using the notational conventions defined in Annex E.

6 **26.10.3.2 State machine timers**

7 The timer variables declared in this subclause are part of the specification of the operation of Protection
8 Switching. The accompanying descriptions of their meaning and use are provided to aid in the
9 comprehension of the protocol only, and are not part of the specification.

10 One instance of the following may be implemented per Service Mapping state machine:

11 a) WTRwhile (26.10.3.2.1)

12 One instance of the following may be implemented per Hold-off state machine:

13 b) HoldOffWhile (26.10.3.2.2)

14 **26.10.3.2.1 WTRwhile**

15 A timer to be used to prevent frequent operation of the protection switch due to an intermittent defect. This
16 timer allows for a fixed period of time to elapse before data traffic is mapped from the protection entity to
17 the working entity when in revertive mode.

18 **26.10.3.2.2 HoldOffWhile**

19 In order to coordinate timing of protection switches at multiple layers or across cascaded protected domains,
20 a hold-off timer may be required. The purpose is either to allow a server layer protection switch to have a
21 chance to fix the problem before switching at a client layer, to allow an upstream protected domain to switch
22 before a downstream domain, or to allow the inner protected domain to switch before the outer protected
23 domain in the case of IPS nested protection domains.

24 **26.10.3.3 Protection Switching variables**

25 The following variables are defined for the Protection Switching state machines:

- 26 a) BEGIN (26.10.3.3.1)
- 27 b) SF (26.10.3.3.2)
- 28 c) SFH (26.10.3.3.3)
- 29 d) LoP (26.10.3.3.4)
- 30 e) FS (26.10.3.3.5)
- 31 f) MStoProtection (26.10.3.3.6)
- 32 g) MStoWorking (26.10.3.3.7)
- 33 h) WTRTime (26.10.3.3.8)
- 34 i) HoldOffTime (26.10.3.3.9)

35 Table 26-2 illustrates the inherent priority of each protection request.

36 NOTE—The priorities associated with the various requests in this table are in general alignment with the corresponding
37 priorities in ITU-T G.8031 [B50]. The only difference is that the precedence of p.SF and FS are inverted since ITU-T
38 G.8031 relies on an APS protocol to be running on the protection path.

Table 26-2—Protection Requests Hierarchy

Priority	Request
highest	LoP
	FS
	p.SF
	w.SF
	MStoProtection, MStoWorking
	WTR
lowest	NoRequest

1 **26.10.3.3.1 BEGIN**

2 This is a Boolean variable controlled by the system initialization process. A value of TRUE causes all
3 Protection Switching state machines per protection group to continuously execute their initial state. A value
4 of FALSE allows these state machines to perform transitions out of their initial state, in accordance with the
5 relevant state machine definitions.

6 **26.10.3.3.2 SF**

7 SF is the logical OR of someRMEPCCMdefect (20.35.5), someRDId defect (20.35.7), xConCCMdefect
8 (20.23.3), and errorCCMdefect (20.21.3).

9 NOTE—Locally detected MAC/PHY faults may also be components of SF (e.g., to enable potentially faster fault
10 detection), but are outside the scope of this standard.

11 **26.10.3.3.3 SFH**

12 A Boolean flag set and cleared by the Hold-off state machine to indicate that SF is set for a period that is
13 equal to, or larger than, the HoldOffTime. The variable will be set equal to SF if the Hold-Off timer is not
14 supported.

15 **26.10.3.3.4 LoP**

16 A Boolean flag indicating the administrative state of the protection entity. If set, it prohibits the use of the
17 protection entity. Its value can only be controlled by an administrator action [item b2) in 12.18.2.3.2].

18 **26.10.3.3.5 FS**

19 A Boolean flag indicating the presence of an administrative command to force switch the data traffic to the
20 protection entity. Its value is set to 1 by an administrator action [item b3) in 12.18.2.3.2 or 12.24.2.3.2]. It
21 can be reset by an administrator action that corresponds to a request of the same or higher priority according
22 to Table 26-2.

1 26.10.3.3.6 MStoProtection

2 A Boolean flag indicating the presence of an administrative command to manually switch the data traffic to
3 the protection entity, in the absence of a failure of the working or the protection entity. Its value is set to 1 by
4 an administrator action [item b4) in 12.18.2.3.2 or 12.24.2.3.2]. It can be reset by an administrator action
5 that corresponds to a request of the same or higher priority according to Table 26-2 and by the operation of
6 the Clear Manual Switch state machine.

7 26.10.3.3.7 MStoWorking

8 A Boolean flag indicating the presence of an administrative command to manually switch the data traffic to
9 the working entity in the absence of a failure of the working or the protection entity. Its value is set to 1 by an
10 administrator action [item b5) in 12.18.2.3.2 or 12.24.2.3.2]. It can be reset by an administrator action that
11 corresponds to a request of the same or higher priority according to Table 26-2 and by the operation of the
12 Clear Manual Switch state machine.

13 26.10.3.3.8 WTRTime

14 The wait-to-restore (WTR) period, as provided by the corresponding managed object item e) in 12.18.2.1.3
15 or item h) in 12.24.2.1.3]. May be configured by the operator in 1 min steps between 5 and 12 min; the
16 default value is 5 min. A value of 0 indicates nonrevertive mode. The overall accuracy of the WTR timer
17 (e.g., ± 25 ms) should be sufficient to allow both ports at the protection domain termination to revert to the
18 working entity in less than 50 ms.

19 26.10.3.3.9 HoldOffTime

20 The Hold-Off period as provided by the corresponding managed object [item f) in 12.14.6.1.3 or item i) in
21 12.24.2.1.3]. The suggested range of the hold-off timer is 0 to 10 s in steps of 100 ms (accuracy of ± 5 ms);
22 the default value is 0.

23 26.10.3.4 Protection Switching procedures for PBB-TE TESI Protection

24 The following procedures are defined for the Service Mapping state machine:

- 25 a) mapDataToWorking() (26.10.3.4.1)
- 26 b) mapDataToProtection() (26.10.3.4.2)

27 NOTE 1—Since a merging selector is used, there is the potential for misordered frames at egress due to differential
28 delay between the working and protection TESI following a protection switch. If this is a concern, then briefly
29 discarding frames prior to executing either mapDataToX() processes described in 26.10.3.4.1 or 26.10.3.4.2 will avoid
30 the issue (e.g., 1 ms is equivalent to a 20% differential delay across 1000 km of fiber).

31 NOTE 2—The procedure descriptions in this subclause are specific to PBB-TE TESI Protection. A description of the
32 corresponding procedures for IPS can be found in 26.11.4.1.1 and 26.11.4.1.2.

33 26.10.3.4.1 mapDataToWorking()

34 Maps the customer service(s) that are to be transported by a TE protection group [item b) in 12.18.2.1.3], to
35 the working TESI identified by the 3-tuples <remote CBP MAC, local CBP MAC, ESP-VID(w)>, <local
36 CBP MAC, remote CBP MAC, ESP-VID(w')>, by setting the VID values of the corresponding I-SID
37 entry(-ies) in the Backbone Service Instance table to the ESP-VID(w).

38 26.10.3.4.2 mapDataToProtection()

39 Maps the customer service(s) that are to be transported by a TE protection group [item b) in 12.18.2.1.3],
40 to the protection TESI identified by the 3-tuples <remote CBP MAC, local CBP MAC, ESP-VID(p)>,
41 <local CBP MAC, remote CBP MAC, ESP-VID(p')>, by setting the VID values of the corresponding I-SID
42 entry(ies) in the Backbone Service Instance table to the ESP-VID(p).

26.10.3.5 Protection Switching state machine diagram

- The Protection Switching state machines implement the function specified by the state diagrams in Figure 26-15, Figure 26-16, and Figure 26-17, the variables in 26.10.3.3, and the procedures in 26.10.3.4.
- The Hold-off state machine shall be present only when the Hold-off timer is supported.
- The current state of the Service Mapping state machine is available as a managed object [item c) in 12.18.2.1.3].

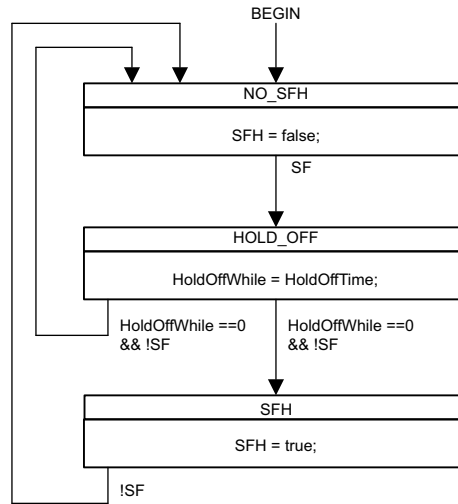


Figure 26-15—Hold-off state machine

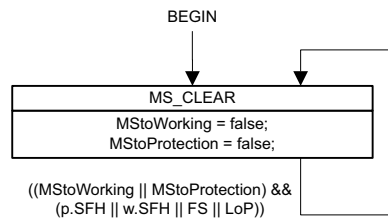


Figure 26-16—Clear Manual Switch state machine

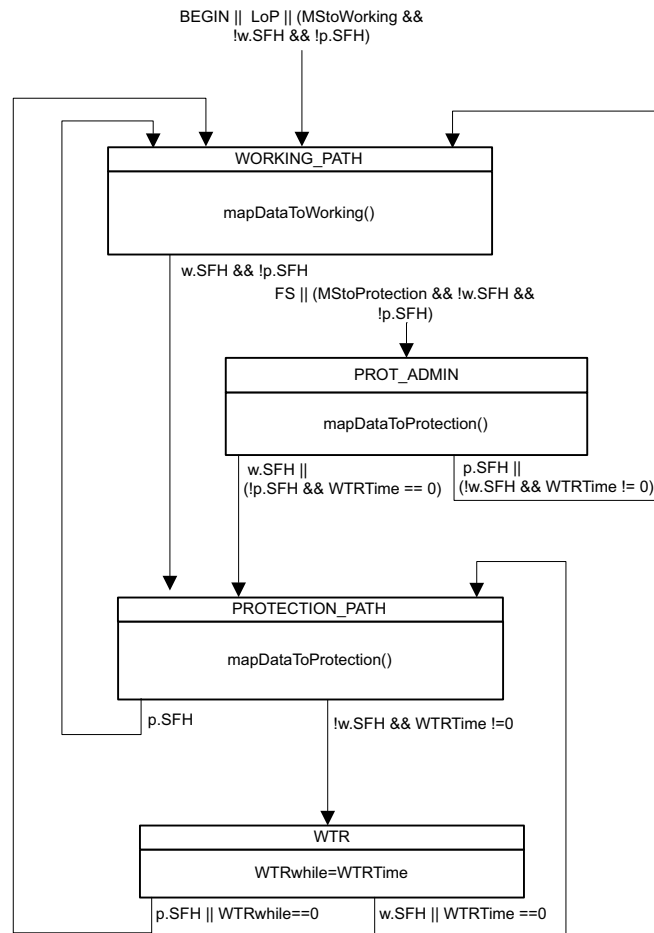


Figure 26-17—Service Mapping state machine

26.11 IPS in PBB-TE Region

In addition to supporting end-to-end linear protection for TESIs (26.10), PBB-TE supports localized protection of selected TESIs traversing a common sequence of PNPs. Such a sequence of PNPs, together with the intervening MAC relay entities and LANs, is called an Infrastructure Segment. The group of TESIs associated with the Infrastructure Segment is protected from the failure of one or more components (i.e., port, LAN, or MAC Relay Entity) of the Infrastructure Segment. The method of providing such protection is called Infrastructure Protection Switching (IPS). 1:1 IPS and M:1 IPS are described by this standard. Support for IPS is optional. If IPS is supported, 1:1 IPS is required, and M:1 IPS is optional. IPS may be triggered automatically by a change in the operational state of an Infrastructure Segment or manually by administrative command.

NOTE—A complete discussion on the various protection switching mechanisms and terminology is provided by ITU-T G.8031 [B50]. The ITU-T uses the term “bridge” for the logical entity that selects either or both of the transmit paths at the sending end of a protected domain. This is not to be confused with the term “Bridge” used in this standard.

As illustrated in Figure 26-18, a Bridge supporting IPS is a PBB. Definitions of entities associated with an Infrastructure Segment and properties of such entities are as follows:

- a) A PBB terminating the Infrastructure Segment is a Segment Endpoint Bridge (SEB).
- b) A PBB in the interior of the Infrastructure Segment is a Segment Intermediate Bridge (SIB).
- c) A PNP terminating the Infrastructure Segment is called a Segment Endpoint Port (SEP).
- d) A PNP in the interior of the Infrastructure Segment is called a Segment Intermediate Port (SIP).
- e) A SEB contains one SEP and does not contain a SIP.
- f) A SIB contains exactly two SIPs and, by definition, cannot contain a SEP.
- g) A SEP is connected to a single SIP or SEP in a neighboring PBB via a LAN.
- h) A SIP is connected to one other SIP within the SIB via the MAC Relay Entity and is connected to a single SEP or SIP in a neighboring PBB via a LAN.

The role of SEB or SIB is assigned to a PBB only with respect to a particular Infrastructure Segment. Similarly, the role of SEP or SIP is assigned to a Port only with respect to a particular Infrastructure Segment.

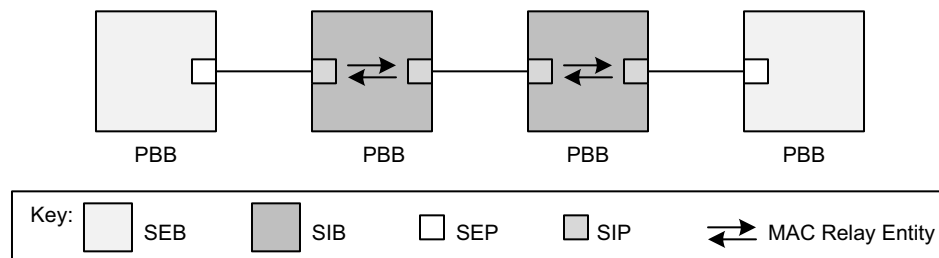


Figure 26-18—Segment terminology and properties

26.11.1 Infrastructure Segment monitoring

As illustrated in Figure 26-19, each Infrastructure Segment is associated with an MA for purposes of determining the connectivity state of that Infrastructure Segment. The Infrastructure Segment MA is associated with a pair of co-routed counterdirectional Infrastructure Segment Monitoring Paths (SMPs). Each SMP is identified by an <SMP-DA, SMP-SA, SMP-VID> 3-tuple that is known as an SMP-ID. The SMP-SA takes the value of the MAC address of the SEP from which the SMP originates. The SMP-DA takes the value of the MAC address of the SEP in which the SMP terminates. It follows from the co-routed and counterdirectional properties of the pair of SMPs that the SMP-DA value of one SMP in the pair of SMPs is equal to the SMP-SA value of the other SMP of the pair. The SMP-VID is a VID associated with a special value of the Multiple Spanning Tree Instance Identifier (MSTID) in the MST Configuration Table, the TE-MSTID, indicating that the VID is under the control of an external agent (8.9). The pair of SMP-IDs associated with an Infrastructure Segment is known as an Infrastructure Segment Identifier (SEG-ID). The MA managed object (12.14.6) associated with an Infrastructure Segment specifies the SEG-ID identifying that Infrastructure Segment. The value of the SEG-ID is inherited by the MEP associated with the Infrastructure Segment MA.

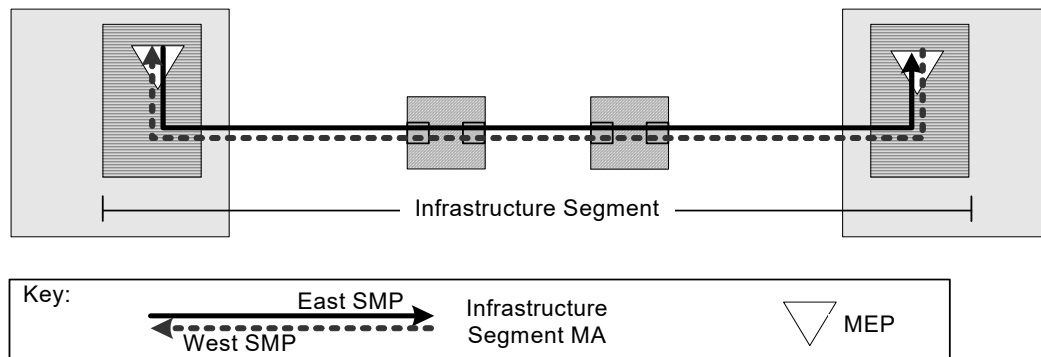


Figure 26-19—Infrastructure Segment monitoring

The operational state of the Infrastructure Segment is indicated by the Signal Fail (SF) variable (26.10.3.3.2) of the Protection Switching state machines (26.10.3). A value of TRUE indicates that the associated Infrastructure Segment is not operational. A value of FALSE indicates that the associated Infrastructure Segment is operational.

NOTE—For proper network operation, each SMP, of the pair of SMPs identifying an Infrastructure Segment MA, is represented by a Static Filtering Entry configured via the Create Filtering Entry operation (12.7.7.1) in each SIB along the path of the Infrastructure Segment. The SMP-ID <SMP-DA, SMP-SA, SMP-VID> would be represented by the Static Filtering Entry <SMP-DA, SMP-VID, outbound port value>. The outbound port value is the port number of the port by which the SMP exits the SIB. Proper operation further includes the creation, at each SEP, of a Maintenance Domain managed object (12.14.5), a Maintenance Association managed object (12.14.6), and a Maintenance association Endpoint managed object (12.14.7) associated with the Infrastructure Segment MA.

26.11.2 1:1 IPS

Figure 26-20 depicts two Infrastructure Segments with a TESI that has been provisioned to fully traverse the upper Infrastructure Segment. The Infrastructure Segment traversed by the provisioned TESI is known as the Working Segment with respect to that TESI. The lower Infrastructure Segment terminates in the same pair of SEBs that terminates the Working Segment but is otherwise diverse from the Working Segment. The lower Infrastructure Segment is known as the Protection Segment with respect to the TESI that traverses the Working Segment.

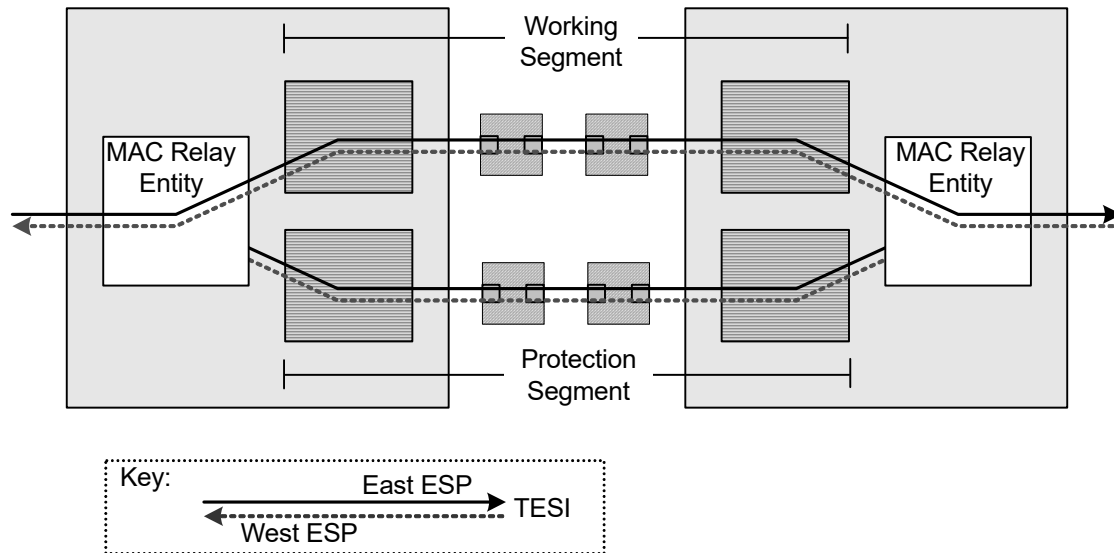


Figure 26-20—Working Segment and Protection Segment

26.11.2.1 Infrastructure Protection Group (IPG)

The associated Working and Protection Segments depicted in Figure 26-20 are said to form an IPG. The Working and Protection Segments are fully disjoint except for the common SEBs in which they terminate; that is, they share no common SEP, SIB, SIP, LAN, or MAC Relay Entity. One or more TESI, each identified by a TE-SID are associated with an IPG at the SEB by which the TESI enters that IPG. A TESI is associated with the IPG via the Write IPG managed object (12.24.2.2.) Multiple TESI may be associated with an IPG. TESI associated with an IPG may have diverse end-to-end paths, but they follow a common path traversing the IPG. Two or more TESI may be associated with ESPs having the same <ESP-DA, ESP-VID> 2-tuple. In particular this can occur if the TESI are distinguished only by different values of ESP-SA. In such cases where TESI share a common value of <ESP-DA, ESP-VID>, the association of one such TESI with an IPG at a SEB implies that all such TESI are associated with that IPG. In other words, all such TESI are protected by the IPG, or no such TESI are protected by the IPG.

NOTE—An Infrastructure Segment can be associated with more than one IPG. For example, the operator can designate Infrastructure Segment 1 as the Working Segment of IPG1 and Infrastructure Segment 2 as the Protection Segment of IPG1. The operator can concurrently designate Infrastructure Segment 2 as the Working Segment of IPG2 and Infrastructure Segment 1 as an Protection Segment of IPG2. This property can be utilized as a method of load sharing across multiple Infrastructure Segments during normal operation.

26.11.2.1.1 Nested IPGs

IPGs are said to be nested with respect to a TESI if:

- a) The TESI is associated with both IPGs, and
- b) The set of SEPs and SIPs comprising the Working Segment of one IPG, the inner IPG, is identical to a contiguous sequence of SIPs contained within the Working Segment of the other IPG, the outer IPG.

In Figure 26-21, IPG2 (dashed) is nested within IPG1 (solid). IPG1 is the outer IPG, and IPG2 is the inner IPG. The Working Segment of the inner IPG is fully contained within the Working Segment of the outer IPG. The SEBs associated with the outer IPG are different from those associated with the inner IPG. Each nested IPG deploys an independent set of MAs for the purpose of monitoring the operational state of Infrastructure Segments associated with that IPG.

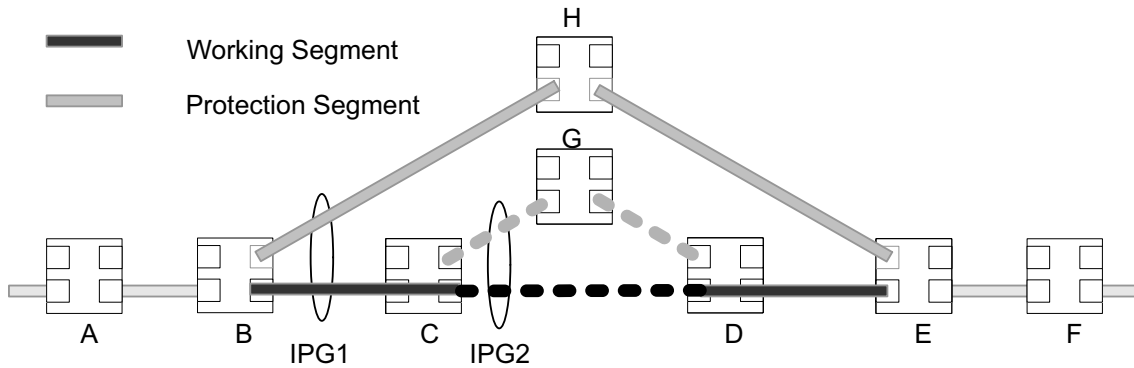


Figure 26-21—Nested IPGs

NOTE—Two IPGs are said to overlap with respect to a TESI if (a) the TESI is associated with both IPGs and (b) the IPGs share a sequence of PNPs. Designing networks having IPGs that overlap, but are not nested, requires careful study of issues not specifically identified in this standard.

26.11.2.2 Protection Switching method

When a TESI is assigned to an IPG by provisioning, traffic associated with that TESI transits the IPG via the Working Segment or the Protection Segment as directed by the Protection Switching state machines (26.10.3). In the absence of faults or outstanding administrative commands, TESI traffic transits the IPG via the Working Segment. The Protection Switching state machines can cause the TESI traffic to transit the IPG via the Protection Segment by invoking the Protection Switching procedure `mapDataToProtection()` (26.11.4.1.2). This procedure causes the value of the outbound Port field of the FDB entry corresponding to the $\langle \text{ESP}_{\text{in}}\text{-DA}, \text{ESP}_{\text{in}}\text{-VID} \rangle$ 2-tuple of each TESI associated with the IPG to be assigned the Port Number associated with the Protection Segment. The notation “ ESP_{in} ” denotes the ESP, of the pair of ESPs associated with the TESI, that enters the IPG via the SEB. The Protection Switching state machines can cause the traffic to transit the IPG via the Working Segment by invoking the Protection Switching procedure `mapDataToWorking()` (26.11.4.1.1) (26.11.4.1.2). This procedure causes the value of the outbound Port field of the FDB entry corresponding to each $\langle \text{ESP}_{\text{in}}\text{-DA}, \text{ESP}_{\text{in}}\text{-VID} \rangle$ 2-tuple of each TESI associated with the IPG to be assigned the Port Number associated with Working Segment. The two SEBs associated with the IPG perform protection activities independently. At any point in time, traffic associated with an IPG is carried on either the Working Segment or the Protection Segment. The Infrastructure Segment carrying the TESI traffic is known as the active Infrastructure Segment.

NOTE—Proper operation of the network requires that the FDB entry provisioned in a SIB to forward traffic along an Infrastructure Segment associated with the IPG identify only a single outbound Port. In other words, the portion of a TESI transiting an IPG is provisioned as point-to-point, without regard to whether the TESI is point-to-point or point-to-multipoint from an end-to-end perspective.

1 26.11.2.3 Hold-off timer

2 An IPG is provisioned with a “hold-off timer” value. The hold-off timer delays protection action associated
3 with the failure of an Infrastructure Segment. The delay provides a lower layer with the opportunity to take
4 corrective action before IPS is performed. Use of the hold-off timer can increase the total time required to
5 recover from a failure.

6 In the case of nested IPGs, proper operation of the network requires that the outer IPG be provisioned with a
7 “hold-off timer” value greater than that of the inner IPG. The outer IPG Working Segment is provisioned to
8 coincide with the path of the inner IPG Working Segment. CCM traffic monitoring the outer IPG Working
9 Segment traverses the Active Segment of the Inner IPG. The TESI list associated with the Inner IPG
10 Endpoint Bridge is provisioned to include the SEG-ID associated with the CCM on the outer IPG Working
11 Segment MA that ingresses the Inner IPG SEB. Thus, on a failure of the inner IPG Working Segment, traffic
12 associated with the outer IPG Working Segment, including the CCM traffic associated with the outer IPG
13 Working Segment MA, is protected. The outer IPG will not be aware of the protection switch of the inner
14 IPG and will not perform protection switching activities.

15 26.11.2.4 Reversion

16 An IPG is provisioned to operate in revertive mode or nonrevertive mode. In revertive mode, traffic is
17 switched from the Protection Segment to the Working Segment when the Working Segment becomes
18 operational. In nonrevertive mode, traffic remains associated with the Protection Segment when the Working
19 Segment becomes operational unless:

- 20 a) The Protection Segment fails, or
- 21 b) Traffic is switched to the Working Segment by administrative command.

22 An IPG operates in revertive mode if that IPG is provisioned to support M:1 IPS.

23 26.11.2.4.1 Wait-to-restore timer

24 An IPG operating in revertive mode is provisioned with a “wait-to-restore (WTR) timer” value. The WTR
25 timer delays reversion from the Protection Segment to the Working Segment on recovery of the Working
26 Segment. Use of the WTR timer reduces the severity of “flapping,” or rapid oscillation, that may occur if the
27 Working Segment is experiencing intermittent connectivity failure. Setting the WTR time to zero indicates
28 that the IPG is operating in nonrevertive mode.

29 26.11.2.5 Administrative commands

30 IPS supports the following administrative commands:

- 31 — **Lockout of Protection (LoP)** allows the operator to suppress automatic switching from Working
32 Segment to Protection Segment.
- 33 — **Forced Switch (FS)** allows the operator to force traffic to the Protection Segment as if a
34 connectivity failure exists on the Working Segment. The behavior persists until the command is
35 withdrawn.
- 36 — **Manual Switch to Protection (MStoProtection)** switches traffic to the Protection Segment, but
37 does not force traffic to remain associated with the Protection Segment.
- 38 — **Manual Switch to Working (MStoWorking)** switches traffic to the Working Segment, but does not
39 force traffic to remain associated with the Working Segment.

1 26.11.3 IPS Control entity

2 The IPS Control entity contains the 1:1 IPS state machines (26.11.4) and, if M:1 IPS (26.11.5) is supported,
3 the M:1 IPS state machines (26.11.5.1). There is an instance of the IPS Control entity for each SEB.

4 As shown in Figure 26-22, the operation of the IPS state machines depends upon input from Bridge
5 Management and from each Infrastructure Segment MA associated with the IPG. The input from Bridge
6 Management includes the list of TESIIs [item e) in 12.14.2.1.2] associated with the IPG and Administrative
7 commands (26.11.2.5) associated with the IPG. The outbound Port field of a Static Filtering Entry associated
8 with an IPG can be modified by IPS Control. When a Static Filtering Entry is no longer associated with an
9 IPG, the Outbound Port value of that Static Filtering Entry is restored by IPS control to the Port associated
10 with the Working Segment of the IPG and IPS Control cannot subsequently modify the Outbound Port
11 value. While the Outbound Port value can be modified by IPS Control, addition and deletion of FDB entries
12 remains under the control of Bridge Management.

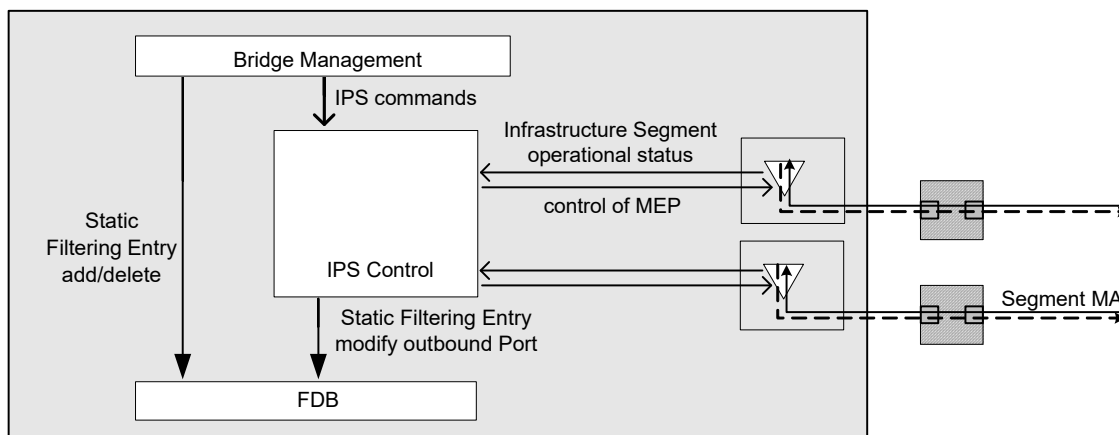


Figure 26-22—IPS Control entity

13 IPS Control maintains the following entities associated with the IPG:

- 14 a) The SEG-ID identifying the Working Segment MA
- 15 b) The SEG-ID identifying the Protection Segment MA
- 16 c) The 1:1 IPS state machines
- 17 d) The list of the TESIIs describing traffic protected by the IPG

18 When an IPG is associated with optional M:1 IPS, the following additional entities are maintained:

- 19 e) The list of SEG-IDs identifying the MAs of candidate Protection Segments
- 20 f) The M:1 IPS state machines

1 **26.11.4 1:1 IPS state machines**

2 Protection switching for point-to-point TESI (26.10) and 1:1 IPS (26.11.2) both utilize Protection Switching
3 state machines (26.10.3). In the case of 1:1 IPS, the “Protection Group” is an “IPG,” the “working entity” is
4 the “Working Segment,” and the “protection entity” is the “Protection Segment.” One set of Protection
5 Switching state machines is instantiated per IPG per SEB.

6 **26.11.4.1 Procedures referenced by the 1:1 IPS state machines**

7 The procedures `mapDataToWorking()` and `mapDataToProtection()` referenced by the 1:1 IPS state machines
8 and described in the subclauses that follow, differ from the procedures of the same name (26.10.3.4)
9 referenced by the TESI Protection state machines.

10 **26.11.4.1.1 `mapDataToWorking()`**

11 This procedure changes the value of the outbound Port field of each Static Filtering Entry identified by the
12 $\langle \text{ESP}_{\text{in}}\text{-DA}, \text{ESP}_{\text{in}}\text{-VID} \rangle$ 2-tuple associated with each TESI on the list of TESIs associated with an IPG to
13 the port Number associated with the Working Segment of the IPG.

14 **26.11.4.1.2 `mapDataToProtection()`**

15 This procedure changes the value of the outbound Port field of each Static Filtering Entry identified by the
16 $\langle \text{ESP}_{\text{in}}\text{-DA}, \text{ESP}_{\text{in}}\text{-VID} \rangle$ 2-tuple associated with each TESI on the list of TESIs associated with an IPG to
17 the port Number associated with the Protection Segment of the IPG.

1 26.11.5 M:1 IPS

2 This subclause describes extensions to 1:1 IPS (26.11.2) supporting M:1 IPS when this optional feature is
3 implemented. An IPG is constrained to operate in 1:1 revertive mode when that IPG supports M:1 IPS.

4 As illustrated in Figure 26-23, a list of candidate Protection Segments, ordered by priority value, is
5 provisioned for each IPG. deploying M:1 IPS. The list contains at least one candidate Protection Segment.
6 Candidate Protection Segments are mutually disjoint and disjoint from the Working Segment. The candidate
7 Protection Segment having the highest (lowest numeric) priority among the operational candidate Protection
8 Segments is the Protection Segment referenced by the 1:1 IPS state machine. This selected candidate
9 Protection Segment is called the “current” Protection Segment. In the absence of any operational candidate
10 Protection Segments, the current Protection Segment retains the value it had prior to the failure of the last
11 candidate Protection Segment. In this case, the 1:1 IPS Protection Segment is declared to have failed. If the
12 1:1 IPS Protection Segment is not operational and a candidate Protection Segment becomes operational, that
13 candidate Protection Segment assumes the role of 1:1 IPS Protection Segment, and the 1:1 IPS Protection
14 Segment becomes operational. If a candidate Protection Segment of higher priority than the current
15 Protection Segment becomes operational, then that candidate Protection Segment assumes the role of the 1:1
16 IPS Protection Segment following expiration of the M:1 wait-to-restore timer. If the current Protection
17 Segment has changed and the 1:1 IPS Protection Segment is the active Infrastructure Segment, then Static
18 Filtering Entries associated with TESIs protected by the IPG are updated with the Port number of the current
19 Protection Segment as described in 26.11.2.2. Thus, M:1 IPS is deployed using an extension of 1:1 IPS that
20 allows the current Protection Segment to be replaced by a candidate Protection Segment of higher priority,
21 provided that such a candidate exists and is operational.

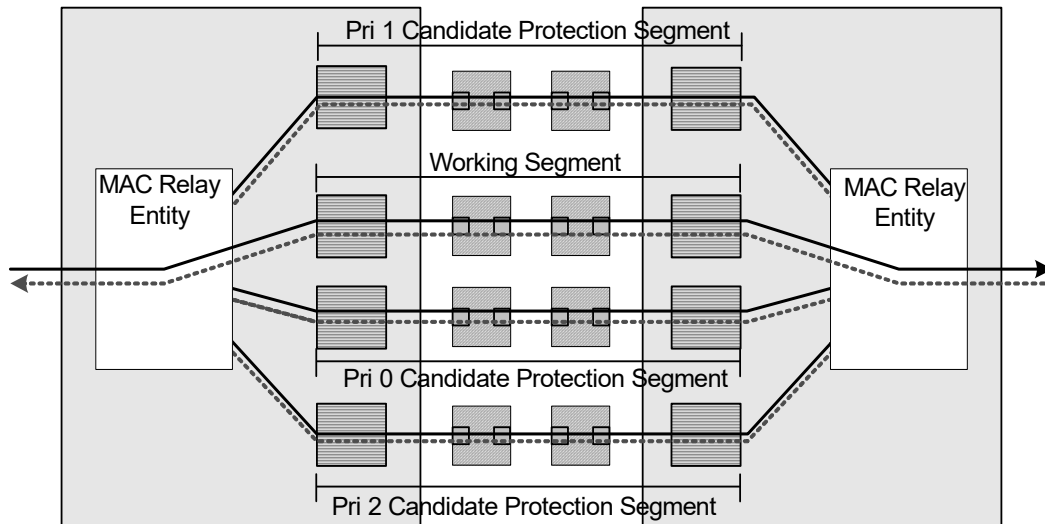


Figure 26-23—M:1 IPS

26.11.5.1 M:1 IPS state machines

There is one set of M:1 IPS state machines per IPG.

The M:1 IPS state machines comprise one M:1 Hold-off state machine (26.11.5.6) per candidate Protection Segment and one Protection Segment Selection state machine (26.11.5.7) per IPG, in addition to the 1:1 IPS state machines (26.11.4). The relationship between the M:1 IPS state machines and the 1:1 IPS state machines is illustrated in Figure 26-24.

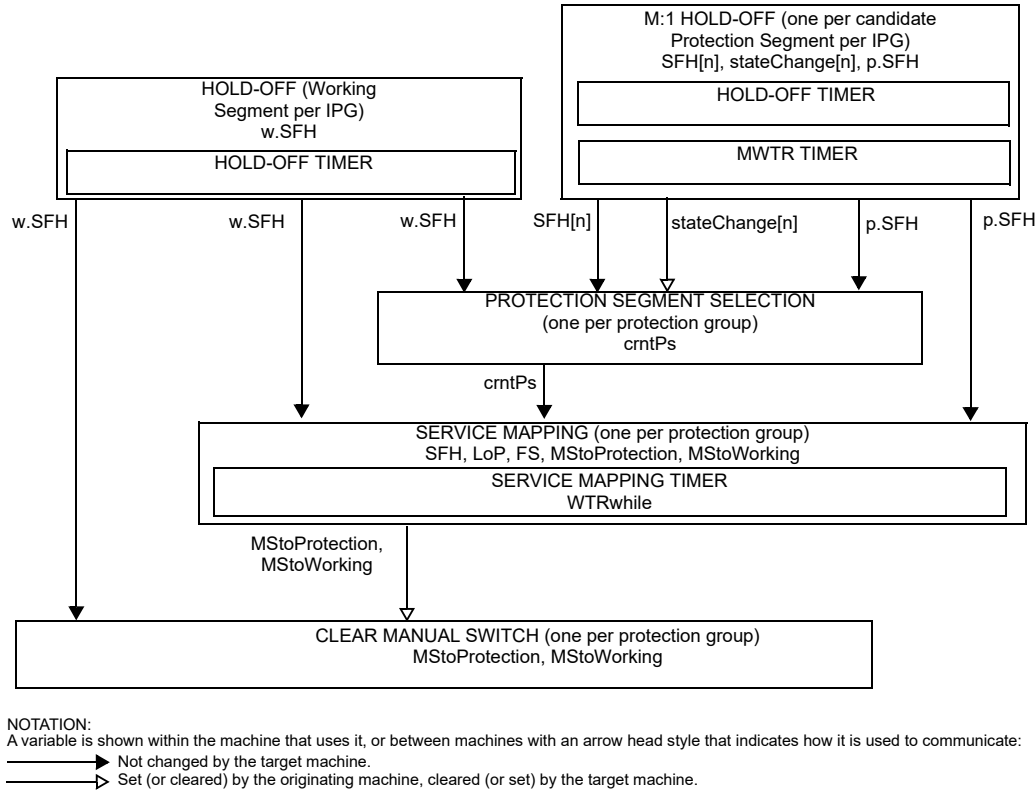


Figure 26-24—M:1 IPS state machines

26.11.5.2 Notational conventions used in state diagrams

The Protection Switching state machines are specified using the notational conventions defined in Annex E. Additionally, in cases of multiple instances of a state variable, each instance is represented by adding the suffix “[n]” to indicate the nth instance. If instances of a state variable may be associated with either the Working Segment or the Protection Segment, the references can be distinguished by prefixing “w.” or “p.” to distinguish the two cases.

26.11.5.3 State machine timers

The timer variables declared in this subclause are part of the specification of the operation of Protection Switching. The accompanying descriptions of their meaning and use are provided to aid in the comprehension of the protocol only and are not part of the specification. Timer variables deployed by the M:1 IPS state machines are as follows:

- a) HoldOffWhile (26.10.3.2.2): one instance per M:1 Hold-off state machine
- b) MWTRwhile (26.11.5.3.1): one instance per M:1 Hold-off state machine

1 **26.11.5.3.1 MWTRwhile**

2 A timer to be used to prevent frequent operation of the protection switch among candidate Protection
3 Segments due to an intermittent defect. This timer allows for a fixed period of time to elapse before data
4 traffic is mapped from a lower priority Protection Segment to a higher priority Protection Segment when in
5 M:1 revertive mode.

6 **26.11.5.4 Variables referenced by M:1 IPS state machines**

7 The following variables are referenced by the M:1 IPS state machines:

- 8 a) BEGIN (26.11.5.4.1)
- 9 b) SF (26.11.5.4.2)
- 10 c) SFH (26.11.5.4.3)
- 11 d) stateChange (26.11.5.4.4)
- 12 e) pri (26.11.5.4.5)
- 13 f) allPsSFH (26.11.5.4.8)
- 14 g) crntPs (26.11.5.4.6)
- 15 h) WTRTime (26.10.3.3.8)
- 16 i) MWTRTime (26.11.5.4.7)
- 17 j) HoldOffTime (26.10.3.3.9)

18 **26.11.5.4.1 BEGIN**

19 This is a Boolean variable controlled by the system initialization process. A value of TRUE causes all M:1
20 IPS state machines per IPG to continuously execute their initial state. A value of FALSE allows these state
21 machines to perform transitions out of their initial state, in accordance with the relevant state machine
22 definitions.

23 **26.11.5.4.2 SF**

24 SF is the logical OR of someRMEPCCMdefect (20.35.5), someRDId defect (20.35.7), xConCCMdefect
25 (20.23.3), and errorCCMdefect (20.21.3). An instance of this variable, identified by the notation SF[n], is
26 specified per candidate Protection Segment.

27 **26.11.5.4.3 SFH**

28 A Boolean flag set and cleared by the M:1 Hold-off state machine to indicate that SF is set for a period that
29 is equal to, or larger than, the HoldOffTime. The variable is set equal to SF if the hold-off timer is not
30 supported. An instance of this variable, identified by the notation SFH[n], is specified per candidate
31 Protection Segment.

32 **26.11.5.4.4 stateChange**

33 A Boolean flag set by the M:1 Hold-off state machine and cleared by the Protection Segment Selection state
34 machine indicating the value of SFH associated with a candidate Protection Segment has changed. An
35 instance of the variable, identified by the notation stateChange[n], is specified per candidate Protection
36 Segment.

1 **26.11.5.4.5 pri**

2 An integer value associated with each candidate Protection Segment and unique within the IPG indicating
3 the selection priority of a candidate Protection Segment. Lower numeric values are associated with higher
4 priority. An instance of the variable, identified by the notation pri[n], is specified per candidate Protection
5 Segment.

6 **26.11.5.4.6 crntPs**

7 An integer value identifying the operational candidate Protection Segment having the highest (lowest
8 numeric) selection priority. In the event that no candidate Protection Segment is available, the value is that
9 of the last candidate Protection Segment to be available.

10 **26.11.5.4.7 MWTRTime**

11 The M:1 wait-to-restore (MWTR) period associated with reverting from a lower priority to a higher priority
12 candidate Protection Segment, as provided by the corresponding managed object [item k] in 12.24.2.1.3].
13 May be configured by the operator in 1 min steps between 5 and 12 min; the default value is 5 min. A value
14 of 0 indicates M:1 nonrevertive mode. The overall accuracy of the MWTR timer (e.g., ± 25 ms) should be
15 sufficient to allow both SEBs at the termination of the IPG to revert to the same value of the Protection
16 Segment in less than 50 ms.

17 **26.11.5.4.8 allPsSFH**

18 The logical AND of the value of SFH[n] over the M candidate Protection Segments.

19 **26.11.5.5 Procedures referenced by M:1 IPS state machines**

20 The following procedures are referenced by the M:1 IPS state machines:

- 21 a) highestPriOperPs() (26.11.5.5.1)
- 22 b) setPs(n) (26.11.5.5.2) (26.11.5.5.2)
- 23 c) mapDataToProtection() (26.10.3.4.2)

24 **26.11.5.5.1 highestPriOperPs()**

25 Returns the identity of the highest priority candidate Protection Segment for which SFH[n] == FALSE or, in
26 the event that SFH[n] == TRUE for all candidate Protection Segments, the value NULL.

27 **26.11.5.5.2 setPs(n)**

28 Establishes the n^{th} candidate Protection Segment as the Protection Segment referenced by the Service
29 Mapping state machine.

30 **26.11.5.6 M:1 Hold-off state machine**

31 The M:1 Hold-off state machine is specified by the state diagram in Figure 26-25 and the variables in
32 26.11.5.4.

1

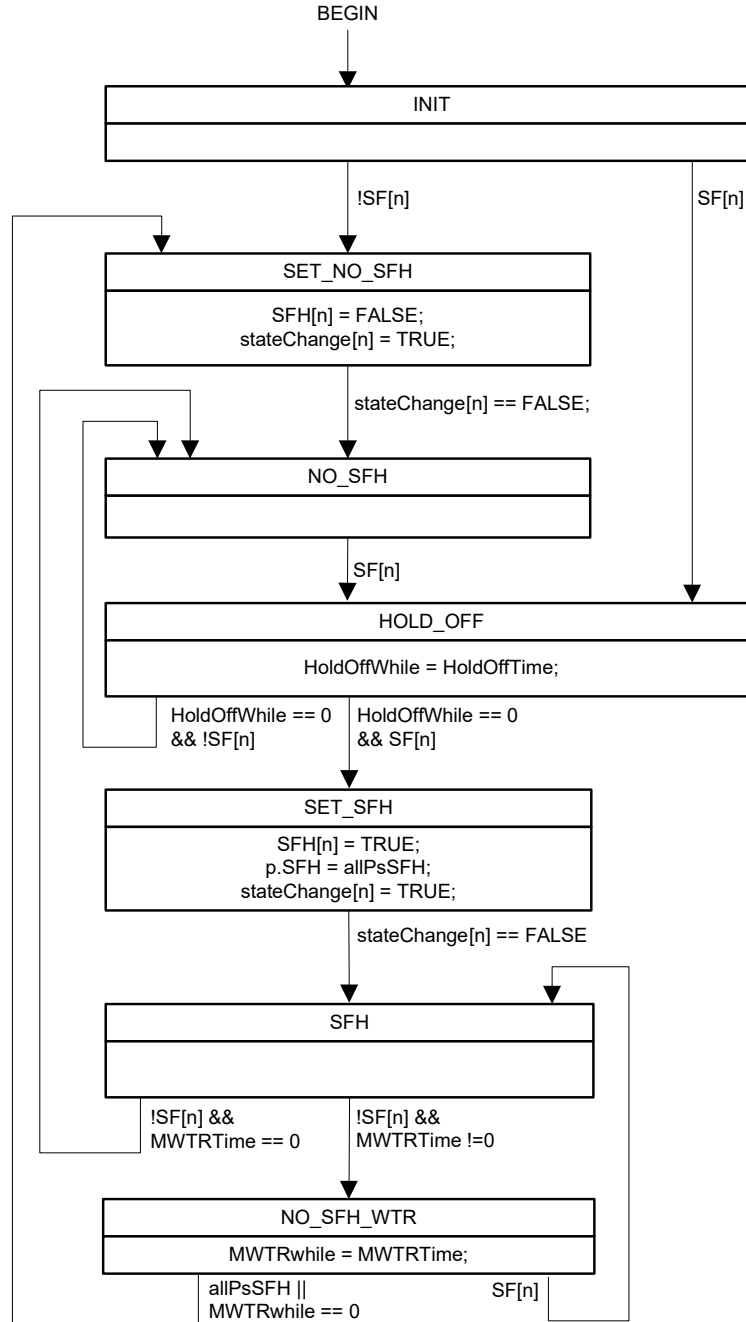


Figure 26-25—M:1 Hold-off state machine

2 NOTE—The state NO_SFH_WTR is entered only in the case that M:1 reversion has been provisioned (i.e., MWTRTime
3 != 0) and the candidate Protection Segment “n” is operational (i.e., !SF[n]). The state allows the machine to wait a period
4 of time (i.e., MWTRTime) before permitting the candidate Protection Segment “n” to become available for selection as
5 the Protection Segment, thus increasing stability. If, during this time, it is determined that no candidate Protection
6 Segments are available for selection (i.e., allPsSFH), but candidate Protection Segment “n” is operational, then it is
7 prudent not to wait MWTRTime but instead to proceed directly to the selection of candidate Protection Segment “n” as
8 the current Protection Segment. In other words, when no candidate Protection Segment has met the criterion for
9 availability (i.e., allPsSFH), but there is a candidate Protection Segment that is operational (i.e., !SF[n]), then it is not
10 necessary to wait MWTRTime before proceeding with selection.

26.11.5.7 Protection Segment Selection state machine

The Protection Segment Selection state machine is specified by the state diagram in Figure 26-26 and the variables in 26.11.5.4.

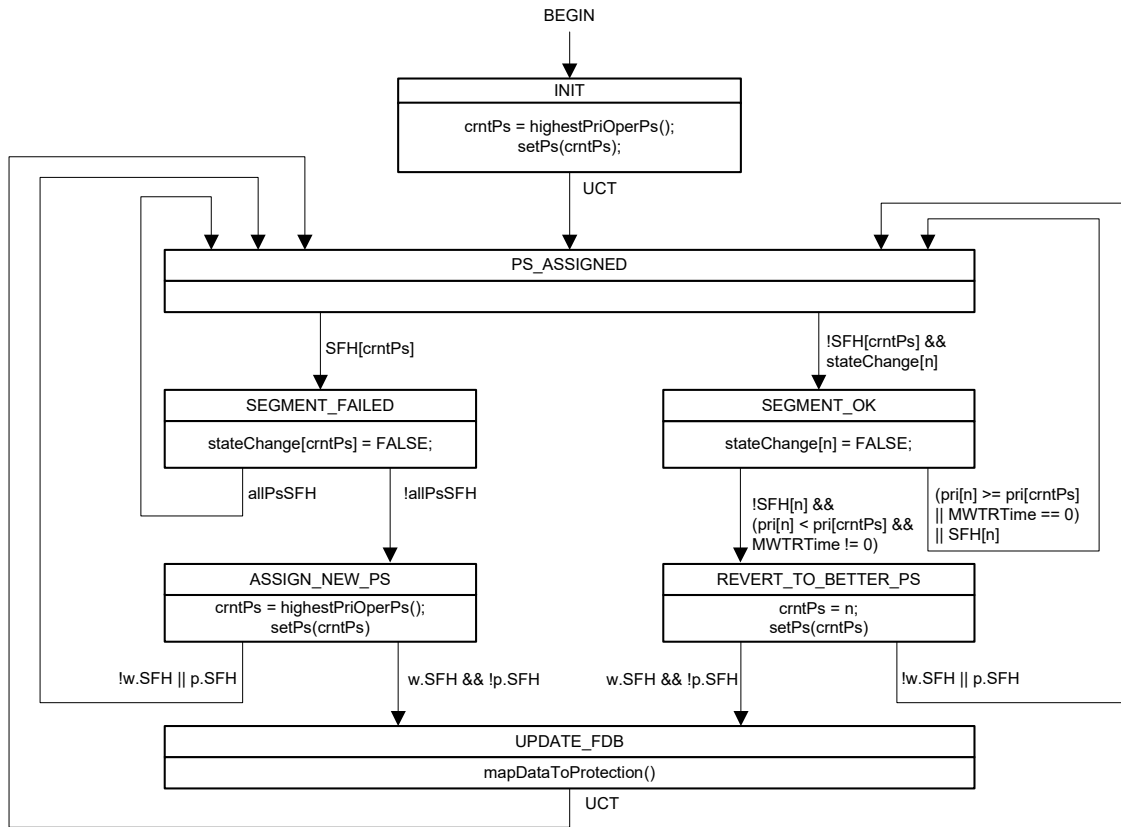


Figure 26-26—Protection Segment Selection state machine

26.12 Mismatch defect

Under equipment malfunction conditions and/or incorrect configuration, a mismatch between the mappings of the backbone service instances to the appropriate TESI at the terminating CBPs can happen. Similarly, a mismatch can occur in the mapping of a TESI to the appropriate Infrastructure Segment at the terminating PNP. To maintain the proper operation of the network, such a mismatch should be detected and reported. There can be two types of mismatch. They are as follows:

- Protection switching incomplete
- Working/protection configuration mismatch

An example of protection switching incomplete mismatch is when, due to equipment malfunction, the near end (East B-Component in Figure 26-13) fails to switch over but it sends a CCM with the RDI field set to the far end (West B-Component in Figure 26-13). The far end transmits to the protection TESI while the near end is still transmitting in the working TESI. Similarly a mismatch can also happen when the near end transmits to the protection TESI but the far end fails to start transmitting to the protection TESI when it receives a CCM with the RDI field set.

1 The mismatch can also happen because of incorrect configuration. For example, one end is configured to
2 send traffic on working TESI while the other end is configured to send traffic on protection TESI. Or one
3 end is configured as revertive mode while the other end is configured as nonrevertive mode. The mismatch
4 occurs when a failure is cleared.

5 NOTE—Since a merging selector is used, in the case of TESI protection, there is not necessarily a traffic loss in all
6 mismatch cases.

7 PBB-TE MEPs and Infrastructure Segment MEPs that support the Traffic field (21.6.1.4) can detect such a
8 mismatch. The Mismatch state machine in 20.26 provides this capability.

9 **26.13 Signaling VLAN registrations among I-components**

10 A VIP is a Bridge Port. MVRP can therefore be enabled in an I-component and used to convey dynamic
11 VLAN Registrations from I-component to I-component across a PBBN.

12 In the particular case of VIPs that do not emit an S-TAG, and thus use only the I-TAG to differentiate among
13 service instances, the Multiple I-SID Registration Protocol (MIRP, Clause 39) can be enabled instead of
14 MVRP (11.2) to signal the need to flush learned C-MAC address information from one Icomponent to
15 another. The network administrator can select between using MVRP or MIRP to perform this signaling
16 according to the need to minimize the number of control frames sent (MIRP) or the number of remote
17 I-components affected by any given control frame (MVRP).

1 27. Shortest Path Bridging (SPB)

2 SPB provides shortest path communication for user data frames in SPT regions (Clause 3). ISIS-SPB
3 interoperates with MSTP and RSTP (Clause 13) and STP (Clause 8 of IEEE Std 802.1D, 1998 Edition
4 [B13]), exchanging BPDUs to determine the extent of each SPT Region and to configure full, simple, and
5 symmetric connectivity throughout a network comprising arbitrarily interconnected Bridges and individual
6 point-to-point LANs (13.6, 13.7). IS-IS supports point-to-point links and shared LANs; however, only point-
7 to-point is considered for ISIS-SPB in this specification. A future revision may add shared media. The
8 Clause 13 SPB description allows for both point-to-point links and shared media.

9 Within an SPT Region (7.4, 13.6), the active topology that supports the frames for any given Base VID can
10 be chosen (27.4) by the network administrator to be one of the following:

- 11 — The Internal Spanning Tree (IST).
- 12 — A Multiple Spanning Tree Instance (MSTI).
- 13 — A set of Traffic Engineering service instances (TESIs).
- 14 — A set of shortest path trees (SPT Set), one SPT rooted at each SPT Bridge and supporting frames
15 transmitted into the region through that Bridge.
- 16 — Multiple equal cost unicast paths and multicast trees that spread traffic rooted at each SPT Bridge
17 over multiple SPTs. This ECMP connectivity is only supported by SPT Bridges using SPBM mode.
- 18 — A set of Explicit Trees (ETs).

19 Clause 28 specifies the use of ISIS-SPB to calculate the IST and multiple SPT Sets (supporting different
20 ECTs) within an SPT Region. Clause 13 specifies the use of MSTP to calculate MSTIs, and state machines
21 that use the results provided by ISIS-SPB for each SPT to prevent loops within and between regions while
22 maintaining or recovering connectivity as the network reconfigures (13.12, 13.14–13.40, 27.5). The
23 umbrella term ISIS-SPB refers to the coordinated use of the protocols and algorithms specified in Clause 28
24 and Clause 13 to support SPB.

25 Clause 45 specifies the use of IS-IS to establish explicit trees within an SPT Region. These trees can differ
26 from the SPTs.

27 To allow SPB to support plug-and-play operation for some VLANs, while providing the administrative
28 controls and scalability required for large scale operations, this standard specifies two complementary
29 modes of SPT Bridges (SPBV and SPBM) for assigning each shortest path bridged frame to the correct SPT.
30 All the Bridges in a given SPT Region agree on the method to be used to support any given VLAN.
31 Subclause 27.19 describes network scenarios that use one or both of these methods.

32 If a VLAN is supported by an SPT Bridge using SPBV mode, Shortest Path VIDs (SPVIDs) are
33 automatically allocated by ISIS-SPB and used in VLAN tags within an SPT Region. Each SPVID identifies
34 both the VLAN and the SPT supporting transmission of a tagged frame, and Dynamic VLAN Registration
35 Entries are created to provide connectivity while preventing loops (13.38.2). The SPVIDs for a given VLAN
36 use a single FID that is associated with the Base VID for that VLAN (Shared VLAN Learning (SVL), 7.5,
37 8.8.8). VID translation (6.9) at the region boundary (7.4, 13.5, 13.12) maps SPVIDs to and from the
38 associated Base VID.

39 If a VLAN is supported by an SPT Bridge in SPBM mode without ECMP, the Base VID is used in each
40 frame's VLAN tag within the SPT Region and identifies the VLAN and the SPT Set. Each Bridge mitigates
41 and prevents loops (6.5.4, 13.38.2) by using the source and destination MAC address of each frame with
42 Dynamic Filtering Entries for the Base VID, MAC address tuples. The source MAC address identifies the
43 particular SPT from the SPT Set, and supports active topology enforcement (13.38.2, 8.6.1). When
44 forwarding unicast frames it is not necessary to consider the particular SPT since all the SPTs in a given set
45 transiting a given Bridge en route to a destination take the same path from that given Bridge to the

1 destination (in other words, they are all reverse path congruent with the SPT rooted at that destination).
2 Therefore, for unicast frames the destination MAC address is used for egress filtering as normal. SPT
3 Bridging using SPBM mode requires the use of SPT-specific destination group MAC addresses, so loop-free
4 multicast forwarding is also supported without requiring the source MAC address or source Bridge Port to
5 be part of the FDB query. ISIS-SPB automatically allocates SPSourceIDs (27.10) that are used as part of
6 each group address. Since an SPT Bridge using SPBM mode inspects source MAC addresses as a loop
7 mitigation mechanism, those addresses have to be known before forwarding can take place and are
8 advertised using ISIS-SPB. Learning of B-MAC addresses is not enabled when SPBM mode is used, and
9 frames destined for unknown addresses are discarded and not flooded. The individual MAC addresses used
10 are those of the SPT Bridges themselves, or are associated with functions provided by Provider Bridges or
11 BEB functions that include an SPT Bridge component. SPBM group multicast addresses are local to the SPT
12 Bridge domain [using the Locally administered (U/L) address MAC address flag (IEEE Std 802)], so the use
13 of SPT-specific group MAC addresses does not intrude on the well known global multicast space.

14 With ECMP (Clause 44), ISIS-SPB can configure multiple potential forwarding Ports for each individual
15 MAC address, and forwarding port selection is performed dynamically. Forwarding Port selection is
16 controlled by a deterministic hash function whose inputs are the SPB System Identifier and the Flow Hash
17 carried in an F-TAG. This tends to spread traffic flows over the set of equal cost paths between the source
18 Bridge and destination Bridge in the SPT Region while preserving frame ordering of individual flows. As a
19 substitute for the loop prevention and mitigation techniques mentioned above, an alternative loop quenching
20 mechanism for ECMP is realized with a TTL field included in the F-TAG. With ECMP, bidirectional and
21 unicast-multicast congruence is no longer guaranteed (since there is no longer a single path in use between a
22 source and destination). This enables each Backbone Service Instance group address to be independently
23 routed and multicast traffic to be spread over a set of equal cost SPTs.

24 ISIS-SPB uses the same mechanism to allocate SPVIDs (for SPBV) and SPSourceIDs (for SPBM) (27.10).
25 The MST Configuration Identifier (MCID) (8.9, 13.8) is used to ensure that all the Bridges in an SPT
26 Region agree on the method used to support each shortest path VLAN and on the pool of SPVIDs, as well as
27 on the other uses of VIDs (27.4). An SPSourceID is allocated to each Bridge in an SPT Region,
28 independently of SPBM mode VLANs, while an SPVID for a given SPBV VLAN is only allocated to the
29 SPT rooted at a given Bridge if VLAN Registration Entries and controls allow that Bridge to transmit frames
30 assigned to that VLAN into the region (8.8, 13.6).

31 An SPT Bridge using SPBV mode is typically used to support a C-VLAN or S-VLAN for a single customer.
32 An SPT Bridge using SPBM mode is typically used to support B-VLANs in PBBNs (Clause 25). Each
33 B-VLAN can support many backbone service instances, each requiring connectivity between a subset of
34 the BEBs participating in the B-VLAN, and each identified by an I-SID that is not processed by BCBs. An
35 SPT Bridge using SPBM mode does not use source address learning, so unicast B-MAC frames
36 conveying customer data are never flooded throughout the B-VLAN. Multicast frames and flooding of
37 unknown unicast frames for a given backbone service instance are confined to paths that reach the BEBs
38 supporting that service instance by including the I-SID in the SPT-specific destination group MAC addresses
39 (Figure 27-3). The destination address filtering in each BCB prunes transmission of the encapsulated
40 customer multicast frames as well as preventing loops.

41 MVRP (11.2) is not used within SPT Regions, as ISIS-SPB communicates VLAN registration information
42 for VLANs supported by SPB from the SPT Bridges that provide ingress to and egress from the region to the
43 other Bridges, thus allowing SPT Bridges using SPBV and SPBM to configure Dynamic VLAN
44 Registration Entries for Bridge Ports within the region to provide correct VLAN connectivity when SPTs are
45 recalculated—instead of waiting for registration information to propagate on those SPTs. SPT Bridges can
46 use the registration information communicated within the SPT Region to participate in MVRP with Bridges
47 outside the region (27.13) by supporting MVRP interfaces on the Region boundaries.

1 This clause specifies the following:

- 2 a) Protocol design and support requirements (27.1, 27.2) and design goals (27.3).

3 This clause further specifies how:

- 4 b) Each Bridge's configuration mechanisms specify the supporting method (SPBV, SPBM, MSTI,
5 CIST, or PBB-TE) and SPT Set (for SPBV and SPBM) for each VLAN (27.4).
6 c) The boundaries of each SPT Region and ISIS-SPB adjacencies are determined dynamically by ISIS-
7 SPB (Clause 28).
8 d) The information exchanged by ISIS-SPB is used (27.5), including CIST calculation (27.6, 27.7),
9 SPT calculations (27.8), allocating SPVIDs and SPSourceIDs (27.10), and controlling the
10 connectivity supporting each VLAN (27.13) and backbone service instance (27.15, 27.16).
11 e) The results of ISIS-SPB's CIST and SPT calculations are used by the Clause 13 state machines to
12 determine when to discard frames to ensure the calculated trees are loop-free (27.9).
13 f) SPVIDs are allocated to FIDs (27.11) and translated to and from Base VIDs at the boundary of an
14 SPT Region (6.9, 13.6).
15 g) VLAN Registration Entries are used to forward or discard frames for each VLAN (27.13).
16 h) FDB entries for individual addresses are used to prevent or mitigate loops for VLANs supported by
17 SPBM (27.14).
18 i) SPBM Group Addresses are assigned, and FDB entries for those addresses used to forward or
19 discard frames for each backbone service instance for B-VLANs supported by SPBM (27.15,
20 27.16).
21 j) The SPT calculations choose between Equal Cost Trees (ECTs), making different choices for
22 different SPT Sets, each associated with a different VLAN, enabling load spreading (27.17, 28.9.2).

23 This clause also:

- 24 k) Provides examples of when and how SPT Bridges using SPBV and SPBM modes can be used
25 (27.19).

26 **27.1 Protocol design requirements**

27 ISIS-SPB configures entries in each SPT Bridge's FDB (8.8) to meet the following requirements:

- 28 a) The active topology will fully connect all physically connected LANs and Bridges, and stabilize
29 (with high probability) within a short, known bounded interval after any change in the physical
30 topology, maximising service availability (8.4).
31 b) Except in the case of ECMP, the active topology for any given frame remains simply connected at all
32 times (6.5.3, 6.5.4).
33 c) The same symmetric active topology is used, in a stable network, for all frames for the same VLAN,
34 i.e., between any two individual LANs those frames are forwarded through the same Bridge Ports.
35 d) The active topology supporting a given VLAN within an SPT Region can be chosen by the network
36 administrator to be shortest path, the IST, an MSTI, or an ET.
37 e) Each active topology is predictable and reproducible, and may be influenced by management, thus
38 allowing the application of Configuration Management following traffic analysis, to meet the goals
39 of Performance Management (6.5 and 6.5.10).
40 f) The configured network can support VLAN-unaware end stations, such that they are unaware of
41 their attachment to a single LAN or a bridged network, or their use of a VLAN (6.3).
42 g) The SPB mode, SPBV or SPBM, can be chosen by the network administrator independently for
43 each VLAN using shortest paths.

1 SPBV mode meets the following requirements:

- 2 h) No additional constraint is placed on the values of the Individual or group MAC addresses used, and
3 those addresses do not have to be known before being used in data frames.
- 4 i) SPVIDs are allocated to every SPT Bridge that is a potential source of frames on VLANs supported
5 by SPBV.
- 6 j) FDB entries for a given SPVID will only be populated if they are on the shortest path to a
7 participating boundary node.

8 SPBM mode meets the following requirements:

- 9 k) Multicast flooding for each backbone service instance supported by a PBBN is restricted to paths
10 necessary to reach BEBs supporting that service instance.
 - 11 l) ECMP operation enables multiple equal cost paths in the SPT Region to carry a share of unicast and
12 multicast traffic.
- 13 NOTE—The ECMP ECT Algorithm supports multicast load spreading over up to 16 source trees per node and
14 up to 16 shared trees.

15 Additionally, SPB algorithms and protocols meet the following goals, which limit the complexity of Bridges
16 and their configuration:

- 17 m) The memory requirements associated with each Bridge Port are either a constant or a linear function
18 of the number of Bridges and LANs in the network.
- 19 n) The communications bandwidth consumed by ISIS-SPB on any particular LAN is always a small
20 fraction of the total available bandwidth (6.5.10).
- 21 o) Bridges do not have to be individually configured before being added to a network, other than
22 having their MAC addresses assigned through normal procedures.

23 27.2 Protocol support

24 ISIS-SPB is configured to use one of the Group addresses from Table 8-16 to establish adjacencies and
25 exchange PDUs. SPB running on C-VLAN components use the Customer Bridge address. SPB running on
26 S-VLAN components use the Provider Bridge address. SPB running on B-VLAN components may use the
27 Provider Bridge Address or one of the existing IS-IS addresses.

28 The use of the ISIS-SPB adjacency between Bridges is contingent on the Bridges inclusion within the same
29 SPT Region, and thus requires they have an MCID and Auxiliary MCID (13.8, 28.12.2) where at least one
30 matches on every adjacency in the Region (8.9.4, 13.8). MCID and Auxiliary MCID enable migration of the
31 MCID definition allowing these Bridges to operate as one SPT Region under certain small changes. The
32 operational set of Base VLANs and SPVIDs must be consistent during migration. This standard specifies
33 default configurations resulting in a MCID that enables plug-and-play support of both the CIST and SPB
34 using SPT Bridges using SPBV mode. No default recommendations are made for SPT Bridges using SPBM
35 mode, since it is only expected to be used in managed environments.

36 This standard recommends defaults for Bridge and Bridge Port priorities and path costs to support the
37 calculation of active topologies without further configuration. To allow management of those active
38 topologies, a means of assigning values to the following are required:

- 39 a) The relative priority of each Bridge in the network (13.26.3)
- 40 b) The relative priority of each Port of a Bridge (13.27.47)
- 41 c) A Port Path Cost for each Port (13.27.25, 13.27.33)

1 Management of SPT topologies and of the use of SPBV or SPBM mode to support particular VLANs, also
2 requires the following:

- 3 d) Management of the SPB objects (12.25)
- 4 e) Administrative agreement on the MCID Configuration Name and Revision Level (Clause 28)

5 **27.3 Protocol design goals**

6 It is highly desirable that the operation of ISIS-SPB allows the following:

- 7 a) Addition of Bridges to an existing SPT Region without disrupting existing communication
- 8 b) SPB support for additional VLANs without disrupting existing communication
- 9 c) Enabling or disabling SPB support for individual VLANs with minimum frame loss

10 **27.4 ISIS-SPB VLAN configuration**

11 The configuration mechanisms used by MST Bridges to allocate each VLAN to a specified MSTI or the IST
12 within an MST Region and used to allocate VIDs for PBB-TE (25.10) can be summarized as follows:

- 13 a) The VID to FID allocation table (8.8.8, 12.12.2) is used to allocate each VID to a FID.
- 14 b) The FID to MSTID Allocation Table is used to associate an MSTID with a FID (8.9.3, 12.12.2):
 - 15 1) The IST is identified by the reserved MSTID value 0.
 - 16 2) The use of PBB-TE is identified by the reserved MSTID value TE-MSTID (0xFFE)
 - 17 3) Each MSTID in the MSTI List identifies an MSTI.
 - 18 The reserved MSTID values 0 and TE-MSTID, SPBV-MSTID, SPBM-MSTID, and SPVID-
19 Pool-MSTID are never used in the MSTI List.

20 SPT Bridges extend these configuration mechanisms as follows:

- 21 c) The VID to FID allocation table is used to allocate Base VIDs to FIDs, and to allocate VIDs for use
22 as SPVIDs. SPVIDs are allocated to the reserved FID value 0xFFFF. SPVIDs in use are reported by
23 management as allocated to a FID supported by the implementation (27.11).
- 24 d) The reserved FID value SPVID-Pool-MSTID is allocated to the reserved MSTID value 0xFFFF.
- 25 e) The following MSTID values identify the SPT Bridge mode used by SPB:
 - 26 1) 0xFFC—SPBM
 - 27 2) 0xFFD—SPBV

28 This standard recommends default VID to FID and FID to MSTID allocations (27.2, 8.9.3) for SPT Bridges
29 that enable plug-and-play support of both the CIST and SPB using SPBV mode.

30 Table 27-1 shows the VID to FID and the FID to MSTID allocation tables in a single table for the
31 configuration example shown in Figure 27-1 for an SPT Region.

32 Base VID 1 is allocated to the IST and VIDs from 0x100 to 0x1FF form the SPVID Pool. Base VIDs 0x002,
33 0x003, and 0x004 are configured to operate in SPBM mode. Base VIDs 0x005, 0x006, and 0x007 operate in
34 SPBV mode in the example.

Table 27-1—Allocation of VIDs to FIDs and FIDs to MSTIDs in an SPT Region (example)

VID	FID	MSTID
0x001	0x001	0x000
0x002	0x002	0xFFC
0x003	0x003	0xFFC
0x004		
0x005	0x005	0xFFD
0x006	0x006	0xFFD
0x007		
0x100–0x1FF	0xFFF	0xFFF

1

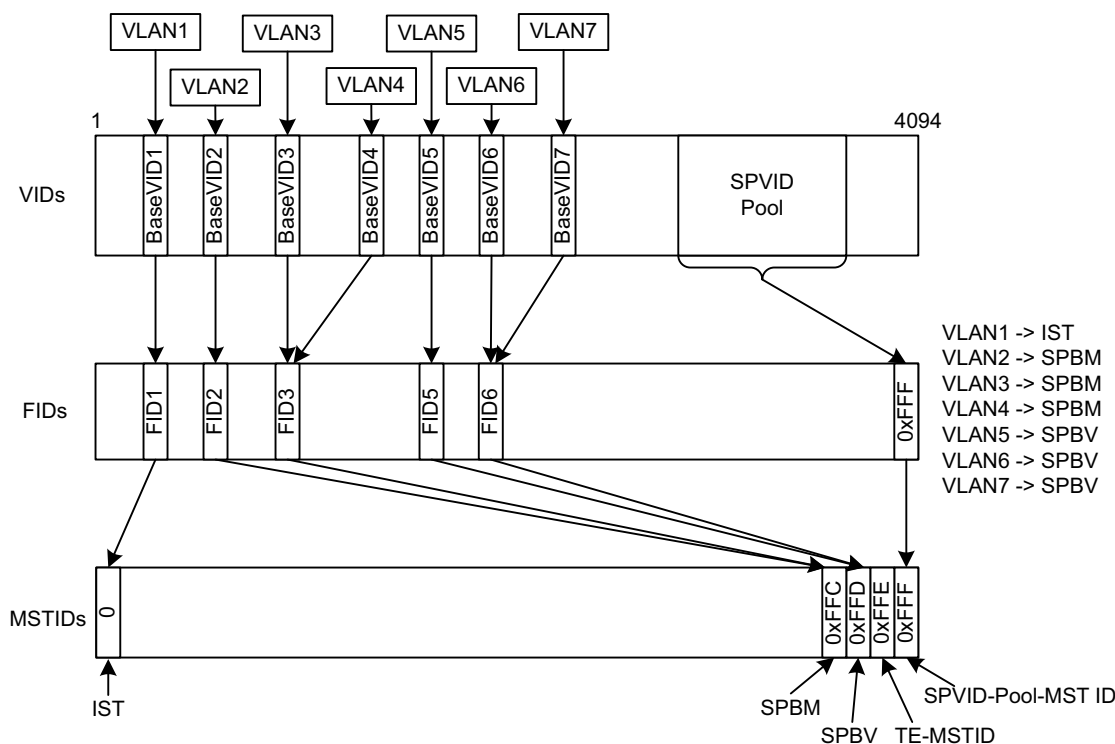


Figure 27-1—Configuring VLAN support in an SPT Region (example)

27.4.1 SPT Region and ISIS-SPB adjacency determination

It is essential that all Bridges within an SPT Region agree on the allocation of VIDs to spanning trees prior to transmission and reception of frames tagged with those VIDs. This configuration requirement is common to both SPB (13.6) and MSTP operation (13.5), and MSTP can coexist with ISIS-SPB within an SPT Region, supporting different VLANs with different VIDs. ISIS-SPB uses and extends MSTP's configuration mechanisms and use of the MCID (8.9, 13.5, 13.8). Note the MCID (13.8) includes a digest derived from the mapping of VIDs to MSTIDs. To allow dynamic manipulation of SPVIDs without impacting the MCIDs, the mapping of SPVIDs to the FID associated with the supported Base VID is not captured in this table.

1 SPB configuration is highly automated. An SPT Region is created by configuring at least two adjacent
2 Bridges with compatible information. The region grows dynamically as more SPT Bridges with the same
3 region configuration are introduced. Once a region has been created, other VLANs can also be created. ISIS-
4 SPB will populate forwarding only between Bridges that support the same services. The SPT region is an
5 interconnected set of Bridges supporting the Region configuration [MCID (13.8), ISIS-SPB Configuration
6 (27.4)] but the forwarding for any one VLAN is simply the Bridges configured to support services on that
7 VLAN or the Bridges that lie on the shortest path between those Bridges.

8 The same is true of SPT Bridges operating in SPBM mode that can be added to support new VLANs on a
9 Bridge by Bridge basis. One common operation for SPT Bridges operating in SPBM mode is to create a new
10 ECT for a new VLAN. To support a graceful migration of services to a new Base VID, SPB allows the
11 configuration of the Base VID before configuring services on those Base VIDs. The new Base VIDs in SPT
12 Bridges need to be configured on all Bridges where the services reside. The U-bit (28.12.5) and Use-flag
13 (28.12.4) are used to identify configured Base VIDs that do not have active services associated with them.

14 In the cases where the allocation of Base VIDs is insufficient a graceful way to support an increase in the
15 number of Base VIDs is supported by a second (auxiliary) MCID (13.26.8, 28.12.2). Two such Bridges
16 consider themselves to be in the same MST Region provided that at least one of their MCIDs matches with
17 one of their SPB neighbor's MCIDs. Non critical changes in the MCID, can be made while maintaining
18 forwarding on the previous configuration. This is done by configuring a Bridge with a revised VID
19 allocation that is compatible with the current allocation in use for SPB forwarding. The Bridge advertises
20 this new MCID and continues to advertise the previous MCID as the Auxiliary MCID, maintaining its
21 adjacencies.

22 When an adjacent Bridge is similarly configured, it also advertises the new MCID as its MCID to its
23 adjacencies. The Auxiliary MCID is always the outgoing MCID. Bridges that have not been updated will
24 continue to match their updated neighbors' Auxiliary MCID until they have been updated, whereupon their
25 MCID values will match again. When a Bridge sees that all its adjacencies are advertising the value of its
26 own new MCID as their MCID, then the Bridge shall start advertising the value of its new MCID as its
27 Auxiliary MCID also. When all Bridges have been verified to have installed the new configuration, the
28 transfer of services to use the new VLAN allocations may be administratively initiated.

29 NOTE 1—The MCID verifies that each Bridge in the region agrees how each and every Base VID is to be assigned to
30 the IST, to an MSTI, for use by PBB-TE, for support by SPBV mode, for use as an SPVID, or for support by SPBM
31 mode. The MCID does not specify the SPT Set for VLANs supported by SPB, or preallocate SPVIDs to SPT Sets as
32 disagreement on the former or temporary differences in the latter will not cause permanent loss of connectivity, loops, or
33 protracted flooding. ISIS-SPB verifies that SPT Bridges agree on the SPT Set to be used to support any given SPB
34 supported VLAN, and the Agreement Digest ensures that new connectivity is not created until that agreement has been
35 reached.

36 NOTE 2—The MCID and the Agreement Digest (13.8, 28.4) can both influence the connectivity provided, and their
37 respective roles need to be clear. MCID differences identify configuration mismatches that could cause network
38 malfunction (e.g., data loops) if connectivity were to be created, and that are not amenable to timely resolution by
39 normal protocol operation. MCID differences act as a fail-safe mechanism, forcing lowest common denominator
40 connectivity (to a single spanning tree, the CST) so at least some connectivity is provided, albeit with lower throughput.
41 The fallback connectivity can be used to investigate or correct configuration issues. The Agreement Digest and related
42 mechanisms simply verify that delays in propagating and using protocol information do not cause loops in permitted
43 connectivity.

44 SPT Regions are similar to MST Regions, and can also be modeled as single Bridges (13.5.3). They do not
45 overlap, superset, or subset other regions but participate in the same active topology solely through the CST.
46 Each region can be identified by its Master Port, i.e., the Bridge Port that provides connectivity from the
47 region to a CIST Root that lies outside the region, or, in the case where the CIST Root lies within the region,
48 by the CIST Root Bridge itself. The Master Port is uniquely and globally identified by its Bridge Identifier
49 and Port Identifier on an SPT Bridge, and is communicated by the CIST Regional Root Identifier and its
50 associated CIST Port Identifier fields in SPT BPDUs.

1 IS-IS adjacencies are formed according to IS-IS protocol rules independent of ISIS-SPB. ISIS-SPB
2 forwarding is only allowed on a more strict basis when the MCID, Base VID, and ECT Algorithm
3 information lines up in the SPB Hello PDUs.

4 When two SPT Bridge Ports attached to the same LAN have no match between their primary or Auxiliary
5 MCIDs in transmitted SPT BPDUs, each is a Boundary Port (13.12) for its SPT Region, and forwards user
6 data frames only if it is a CST Root Port or Designated Port. Such a network configuration is usually
7 temporary and infrequent, a result of as yet incomplete administrative action, or simply a configuration error
8 (13.6). To minimize network disruption during such administrative episodes, ISIS-SPB communicates
9 information throughout an SPT Domain comprising a transitive closure of SPT Bridges attached to
10 individual LANs.

11 Each SPT Bridge Port uses BPDUs and/or SPB Hello PDUs to transmit and receive MCIDs and
12 Agreements. BPDUs also carry currently calculated CIST Information, so that all the Bridges (not just SPT
13 Bridges) attached to each individual LAN can be detected and can participate in the information exchange
14 required at SPT Region boundaries. SPT BPDUs or SPB Hello PDUs also convey the CIST Agreement and
15 Agreement Digest information that SPB uses to prevent temporary loops (28.4). This Agreement
16 information can only be sent after the Bridge's ISIS-SPB calculations have been used to modify its FDB
17 entries, and is thus not naturally tied to the transmission of the ISIS-SPB Link State PDUs (LSPs) that
18 convey the adjacency information that makes those calculations possible (LSPs can be disseminated
19 throughout the SPT Domain without waiting for any single Bridge to complete its calculations—each
20 proceeding in parallel). Including the Agreement information in the SPT BPDUs helps determine the SPT
21 Region's extent, ensuring backwards compatibility to non SPT Bridges. Exchanging SPB Hello PDUs (to
22 establish adjacencies) and transmitting ISIS-SPB LSPs (to propagate information for SPT Identifier
23 allocation and SPT calculation throughout the SPT Domain) as soon each Bridge Port is MAC_Operational
24 is supported in parallel with BPDUs. Bridges that are not SPB-capable will not support SPB Hello PDUs
25 and it is up to the SPT BPDU logic to determine what type of spanning tree protocol the non-SPT Bridges
26 support. In most networks the SPT Region's boundary will coincide with that of the SPT Domain, in other
27 cases LSPs for those Bridge Ports that bound a region but lie within the domain might have to be
28 transmitted.

29 NOTE 3—Clause 13 specifies the state machines and procedures that determine each SPT's region boundaries,
30 including interoperability with Bridges implementing STP (Clause 8 of IEEE Std 802.1D, 1998 Edition [B13]), RSTP, or
31 MSTP as well as the use of SPT BPDUs to address other error conditions that can occur (e.g., disputes with other
32 Bridges due to one-way connectivity).

33 27.5 ISIS-SPB information

34 ISIS-SPB can be viewed primarily as a means of sharing information between nodes in a network, providing
35 each node with a copy of a common database of information contributed by each and every node, and
36 ensuring (with high probability) that the copies are kept up to date though not necessarily in lock step
37 synchronization. Each node can then execute its own independent copy of one or more previously agreed
38 algorithms that use that node's copy of the database to compute results that are common to all the nodes, or
39 that are part of a common or complementary set. The result is a set of forwarding tables for each node. ISIS-
40 SPB shares information throughout an SPT Domain so that any given SPT Bridge in the domain can
41 calculate or identify the following:

- 42 a) The CIST connectivity provided to each SPT Bridge in the domain (27.6, 27.7). This calculation
43 identifies each of the SPT Regions within the domain (if more than one, 8.9, 13.6), the Bridges in
44 each region, and the CST connectivity between regions.
- 45 b) The SPB connectivity provided by each SPT Set between Bridges in the given Bridge's region
46 (27.8).
- 47 c) The SPVIDs (for SPBV mode) and SPSourceIDs (for SPBM mode) used to support SPB
48 connectivity (27.10).
- 49 d) The VLAN connectivity provided for each Base VID over the calculated tree(s) (27.13).
- 50 e) The connectivity provided for each I-SID by the SPT Bridge over the calculated tree(s) (27.16).

1 27.6 Calculating CIST connectivity

2 ISIS-SPB calculates CIST connectivity (13.4.1, 13.5.3) using the following information from each Bridge in
3 the SPT Domain:

- 4 a) The Bridge Identifier and, if the Bridge has currently selected its CST Root Port using information
5 received from a Bridge that is not in the SPT Domain:
 - 6 1) The CIST Root Identifier and CIST External Root Path Cost.
- 7 b) The Port Identifier and the following information for each of the Bridge's Ports:
 - 8 1) Its IS-IS adjacencies, i.e., a list of the Bridge Identifier, Port Identifier tuples that identify each
9 of the other SPT Bridges attached to the same LAN.
 - 10 2) Whether the Bridge Port is or is not a Boundary Port (13.12).
- 11 c) For each of the Bridge's Ports that is not a Boundary Port:
 - 12 1) The Internal Port Path Cost.
- 13 d) For each of the Bridge's Ports that is a Boundary Port:
 - 14 1) The External Port Path Cost.

15 If the `restrictedDomainRole` (13.27.63) parameter is set for an SPT Bridge Port, then that port will not be
16 selected as a CIST Root Port if the Designated Bridge for the attached LAN is not in the SPT Domain. Used
17 arbitrarily `restrictedDomainRole` and `restrictedRole` (13.27.64) can cause lack of connectivity. If used to
18 support the network design these parameters can speed network convergence and protect against disruption
19 if inappropriately configured Bridges are attached. If configured for all SPT Bridge Ports,
20 `restrictedDomainRole` ensures that the CIST Root and all connectivity to and from that CIST Root will be
21 provided solely by Bridges within the SPT Domain, thus simplifying CIST calculation. An SPT Bridge
22 implementation may choose to always set `restrictedDomainRole`, managing it as a read only parameter. This
23 choice restricts network design, but can include all scenarios intended for the implementation (27.19).

24 The CIST calculation uses the Dijkstra algorithm to compute the same active topology that would be
25 produced by the distributed distance vector calculation specified in 13.10. Computation begins with the best
26 priority CIST Bridge priority vector from those that represent each of the Bridges within the domain as a
27 potential root and from those reported as received from a Bridge outside the domain [item a1) above]. If the
28 External Path Cost accumulated during the computation would exceed that for a Bridge selecting its CST
29 Root Port using information from a Bridge outside the domain, then that Bridge is also marked as having
30 been reached in spreading computation and can be the best path to further Bridges, indicating that the CST
31 path between SPT Regions lies outside the domain (not always a temporary or undesirable result, 27.19).

32 NOTE 1—The purpose of defining *priority vectors* (13.9) is to allow all the components that contribute to the choice of
33 one path or another to reach a given Bridge or LAN from the CIST Root to form part of a single metric.

34 Each Bridge in the SPT Domain performs the CIST calculation independently, computing and retaining the
35 following information for use by the state machines specified in Clause 13 (13.10, 13.17, 13.36, 13.29.34):

- 36 e) The CIST root path priority vector for every Bridge Port on the Bridge and the CIST root priority
37 vector for the Bridge.
- 38 f) The IST Port Role for each of the Bridge's Bridge Ports that is not a Boundary Port.
- 39 g) The CIST designated priority vector for each of the Bridge's Bridge Ports.

40 Each Bridge's CIST calculation also identifies the following:

- 41 h) The SPT Region for itself, and for each of the other Bridges in the domain.
- 42 i) The CST Port Role for each Bridge Port that is a Boundary Port for a region.

43 NOTE 2—Each SPT Region can be identified by the region's Master Port. Connectivity from all Bridges in a given
44 region, R_1 say, passes through the same Boundary Port BP_{12} , to reach any Bridge in a region R_2 . Recording the region
45 and boundary port information during CIST calculation facilitates identification of the boundary Bridges.

1 27.7 Connectivity between regions in the same domain

2 If the SPT Domain has been found to comprise more than one region, then, prior to computing shortest
3 paths, each Bridge uses the SPT Region and CST Port Role information [item h) and item i) in 27.6]
4 computed for each Boundary Port together with the adjacency information for those ports to determine the
5 following:

- 6 a) For each of the regions ($R_2, R_3, \dots$ say) other than the computing Bridge's own region (R_1 , say), the
7 Boundary Bridge ($B_{12}, B_{13}, \dots$) within R_1 that provides CST connectivity to that region.

8 If the connectivity between regions lies outside the domain and either has (or has IST connectivity to a
9 region that has) more than one Bridge that provides connectivity to a Bridge outside the domain, then ISIS-
10 SPB does not have the information necessary to select a particular Boundary Bridge. Address learning (for
11 SPBV) or registration protocols can provide the necessary information. If more than one possibility remains,
12 an SPT Bridge using SPBM mode cannot provide connectivity between the regions.

13 NOTE—SPBM mode will often be configured to permit connectivity only if the entire path lies within the domain.

14 27.8 Calculating SPT connectivity

15 Each SPT Bridge uses ISIS-SPB to calculate the shortest paths from each of the Bridges [item h) in 27.6] in
16 its own SPT Region including itself, to each of the other Bridges in that region. The information used by
17 ISIS-SPB to perform the necessary Dijkstra calculations is a subset of that used for the CIST calculation
18 [item a), item b1), item c1) in 27.6]. The following information is computed for each SPT and retained for
19 use by the state machines specified in Clause 13 and 27.9, 13.17, 13.36, and 13.29.34:

- 20 a) The root priority vector for the Bridge.

21 and for each of the Bridge's ports that is not a Boundary Port:

- 22 b) The Port Role.

- 23 c) The designated priority vector.

24 and for each of the Bridge's ports that is not a Boundary Port and has a Port Role of Designated Port:

- 25 d) The best root path priority vector out of those that would be calculated, using the same ISIS-SPB
26 information, by the other SPT Bridges.

27 There can be several equal cost shortest paths between any given pair of Bridges, but unless ECMP is in use,
28 only one path between the pair can be used in a given SPT Set. Ties are broken using an algorithm (27.17,
29 28.5) that is independent of the order of SPT computation, that ensures that the SPTs in each SPT Set are
30 symmetric, that allows path selection to be managed using the same variables that are used to manage active
31 topologies in the absence of multiple paths, and that is likely to select different equal cost paths for different
32 SPT Sets if configuration defaults are used.

33 When a VLAN is supported by an SPT Bridge using SPBM mode and loop mitigation (6.5.4.2, 27.14) is
34 being used, each of the FDB entries that control forwarding for unicast frames can be created or modified as
35 soon as the port providing the shortest path to the Bridge or end station using that address is known, without
36 the need to use additional protocol or to complete the SPT computation. In that case, restoration of unicast
37 connectivity after a physical topology failure is most rapidly achieved by beginning the SPT
38 (re)computation with the SPT rooted at the calculating Bridge.

1 27.8.1 ISIS-SPB overload

2 IS-IS has an overload capability that applies to a node that does not allow transit traffic through this node for
3 various operational reasons. This capability has also been added to ISIS-SPB [item c in 28.12.1] TLVs. Just
4 as in normal IS-IS Overload modifies the shortest path results by not allowing Bridges with the overload bit
5 set to be eligible to forward transit traffic. For the purpose of computing paths, any paths through Bridges
6 with overload set are not considered shortest paths and the computation uses the remaining topology and the
7 next shortest paths. Use of the overload condition is typically reserved for edge devices that would not
8 benefit the network if they were to become transit switches.

9 In ISIS-SPB if overload were set in the only Bridge connecting an SPT Region this would result in a region
10 split where the overload Bridge was connected to both halves of the region but traffic could not transit
11 through the overload Bridge. Each section of the SPT Region that is split would behave normally but there
12 would be incomplete connectivity between the split sections. ISIS-SPB would have a complete view of the
13 region and this condition is relatively easy to detect.

14 27.9 Loop prevention

15 The state machines in Clause 13 use the calculated priority vectors and Port Roles for each tree (27.6, 27.8)
16 to determine the tree's Port State (i.e., the setting of the per port learn (13.27.34) and forward (13.27.29)
17 variables), and exchange SPB BPDUs or SPB Hello PDUs to convey Agreements (13.17) that allow these
18 variables to become set for Root Ports, Designated Ports, and Master Ports without the risk of temporary
19 loops.

20 The Port State Transition state machine (13.38) signals changes to the learn and forward variables for a port
21 for the CIST through calling disableLearning(), disableForwarding(), enableLearning(), and
22 enableForwarding() procedures. These implementation-dependent procedures prompt changes to the
23 mechanisms that perform forwarding and/or learning for frames assigned to the CIST.

24 Similarly changes to forward for a port for an SPT prompt changes in the FDB entries that allow (or prevent)
25 the ingress or egress of frames assigned to that SPT (27.13, 27.14, 27.16). In turn these prompt changes to
26 the implementation-dependent mechanisms that use the entries to control forwarding.

27 These mechanisms can take time to respond to changes in learn or forward, and report their current state to
28 the state machines (by setting the learning (13.27.35) and forwarding (13.27.30) variables) so that new
29 connectivity on some ports can be made contingent on other ports not forwarding frames. The forwarding
30 variable (for any given tree) is only cleared if frames assigned to that tree neither ingress nor egress the port.

31 27.10 SPVID and SPSourceID allocation

32 Each SPT Bridge uses ISIS-SPB to support the distributed allocation of SPVIDs (used by SPBV mode) and
33 the 20-bit SPSourceIDs (used by SPBM mode), ensuring that all Bridges in the SPT Domain agree on the
34 allocated values. The allocation protocol for these two types of identifier differs slightly, as a consequence of
35 the differences in the number of values available and the number required for each Bridge.

36 Candidate values for these identifiers may be pre-configured, in some or all of the Bridges in the domain, but
37 by default they are allocated dynamically. Each Bridge attempts to allocate the following values:

- 38 a) An SPSourceID for the Bridge.
- 39 b) An SPVID for each VLAN that is supported by an SPT Bridge using SPBV mode and for which:
 - 40 1) The Bridge is a potential source of frames, either from a protocol entity within the Bridge or by
41 forwarding from a port on the Bridge that is a Boundary Port; or
 - 42 2) The Bridge has a pre-configured allocation, even if conflict resolution means that this allocated
43 value is not used.

1 NOTE 1—Each SPT Bridge allocates an SPSourceID even if no VLANs are to be supported by SPBM mode, or the
2 Bridge is configured not to transmit frames for those VLANs. The number of available SPSourceIDs is much greater
3 than the number to be allocated; therefore, there is no need to complicate allocation scenarios by attempting to optimize
4 some cases, however frequent.

5 NOTE 2—If the SPT Bridge does not have a pre-configured SPVID for a VLAN supported by SPBV mode, and has no
6 need to transmit frames for that VLAN (possibly because there is no other Bridge in the domain that wishes to receive
7 frames for the VLAN or because a VLAN Registration Entry does not allow the Bridge to transmit frames for that
8 VLAN), then an SPVID is not allocated. These conditions can arise even if the Base VID to FID to MSTID
9 configuration for the Bridge, and the resulting MCID, identifies the VLAN as supported by SPBV mode.

10 The allocation protocol resolves conflicting attempts by different Bridges. Pre-configured or dynamic
11 allocations are communicated by the setting of the Configuration Flags in the ISIS-SPB TLV used for
12 allocation. Conflicts for allocations of a given type are resolved by granting the allocation to the Bridge with
13 the best priority (numerically least) Bridge Identifier (14.2.5). When the conflict occurs on an auto allocated
14 value, the other Bridges attempt to satisfy their need for a value by changing their allocation to a new
15 dynamically allocated value. In the case that two configured values conflict, these must be resolved by
16 operator action. All Bridges are aware of these conflict resolution rules, so there is no need for additional
17 protocol to confirm or announce the allocation.

18 Candidate values for dynamic allocation are chosen using a pseudo-random function of each Bridge's
19 Bridge Identifier (thus reducing the chance of only one identifier being allocated each time all the Bridges
20 transmit fresh allocation information). Values that the allocating Bridge knows to be already in use are
21 excluded (thus ensuring that an allocation that has been in existence long enough to be included in the IS-IS
22 database of all Bridges in the domain will not be disrupted). Assuming that the unallocated values are
23 arranged as an ordered list, the Bridge performs a modulo operation on its Bridge Identifier by the number of
24 free values, and selects the indicated value from the list. This operation is repeated to select the next
25 candidate value.

26 NOTE 3—When a Bridge changes its allocations, i.e., allocates new values to itself as required for (a) or (b) above or
27 releases existing values, ISIS-SPB will communicate the updated information to the other Bridges in the domain.
28 However, the detailed transmission timing and the other contents of IS-IS PDUs is under the control of ISIS-SPB itself,
29 and is not directly controlled by the allocation protocol. A single PDU will usually convey a number of allocation
30 changes.

31 NOTE 4—SPT Bridges can use different pseudo-random candidate selection functions, without imperiling the eventual
32 success of the protocol. Nonetheless, SPVID allocation conflicts are likely to be fewer, and more quickly resolved, if the
33 Bridges in a domain use the same algorithm.

34 On first discovering an ISIS-SPB adjacency (27.2, Clause 3) an SPT Bridge does not perform any allocation
35 before acquiring the IS-IS database. The SPT Bridge becomes aware of existing allocation through normal
36 IS-IS operation. While acquiring the full IS-IS Database, the Bridge advertises SPSourceID = 0 in its LSPs,
37 to indicate to other Bridges that it shall not be included in Shortest Path First (SPF) calculations. As soon as
38 the Bridge has successfully allocated itself an SPSourceID, this allocated value shall be advertised in ISIS-
39 SPB. If the Bridge is aware of dynamic allocations that it has been granted in the past, it may begin by
40 retrying those allocations rather than performing a pseudo-random selection for each that is free.

41 Each SPT Bridge ages out the allocations, both configured and dynamic, made by other Bridges exactly as
42 the allocation information ages out or is replaced in the IS-IS database.

43 Each SPT Bridge's SPSourceID can be any one of the possible 2^{20} values, of which a small number (less
44 than 2^{10}) are expected to be used, and no exceptional provision is made for the case of a Bridge not being
45 able to allocate an SPSourceID. If an SPSourceID is not available, frames for VLANs supported by SPBM
46 mode are discarded.

The pool of SPVIDs are those available for allocation and reserved in the MST Configuration Table. The MCID (13.8) communicated in SPT BPDUs ensures that each Bridge in an SPT Region has agreed to the same pool of SPVIDs. A quantity of 400 SPVIDs are pre-configured for auto allocation (13.8) for small SPT networks running SPBV. Additional SPVID allocations can be made out of the remaining free VID. An MST Configuration Table change that extends the available SPVID pool does not disturb existing allocations, but will affect the MCID—so the connectivity between adjacent Bridges will revert to the CST representing a region boundary, unless the procedures for hitless migration of MCIDs using Auxiliary MCID (27.4.1) are followed. If a Bridge is added to a region or a new SPBV VLAN configured and there are not enough SPVIDs to satisfy the demand, any SPBV VLAN that does not have a full allocation of SPVIDs falls back to CST operation using its Base VID. When enough SPVIDs become available, either by being freed from other SPBV VLANs or by increasing the size of the SPVID pool, and an SPBV VLAN obtains a full allocation of SPVIDs it advances to SPBV operation using the SPVID set. During periods when there are insufficient SPVIDs to satisfy all SPBV VLANs, Bridges that have successfully obtained allocations of SPVIDs to SPBV VLANs must maintain them to ensure stable behavior.

27.11 Allocation of VIDs to FIDs

The allocation of SPVIDs to FIDs (8.8.8) for SPBV is a local decision, subject to the following constraints:

- a) All SPVIDs (if any) supporting a given VLAN (identified by Base VID V_I , say) shall be allocated to the same FID (F_{SVI} , say).
- b) SPVIDs shall be allocated to FIDs ($F_{SV1}, F_{SV2}, \dots$) that are different from those allocated to any VLAN's Base VID ($F_{V1}, F_{VN} \dots$).
- c) VLANs supported by different SPT Sets always use different FIDs (both for SPBV and SPBM modes).
- d) If the VID to FID allocation table (8.8.8) has been configured to allocate two different Base VIDs to different FIDs (F_{V1}, F_{V2}), then the SPVIDs used to support each of the VLANs identified by those Base VIDs shall be allocated to different FIDs ($F_{SV1} \neq F_{SV2}$).
- e) If the VID to FID allocation table (8.8.8) configuration allocates two different Base VIDs to the same FID ($F_{V3} = F_{V4}$), then the SPVIDs shall also be allocated to the same FID ($F_{SV3} = F_{SV4}$). This is shared VLANs where two or more VLANs can share the same topology.
- f) SPVIDs allocated to FID in this manner do not show up in the MCID as changed.

NOTE 1—Frames for VLANs supported by SPBM mode are always assigned to an SPT Set within an SPT Region, never to the IST, and are always identified by each VLAN's Base VID.

NOTE 2—Constraint (b) allows the active topology for any SPBV supported VLAN to be changed from the IST to an SPT Set without requiring topology change signaling.

These constraints, and the fact that the MST Configuration Table's entries for VLANs supported by SPB are not configured directly but are a consequence of configuring the relationship between Base VIDs and FIDs (8.8.8) and between FIDs and MSTIDs (8.9.3), also ensure the following:

- g) SPT Bridges using SPBV and SPBM modes always use different FIDs, so SPBV learning cannot disrupt SPBM operation.
- h) VLANs with shared learning constraints (expressed by allocation of a common FID) are assigned to the same SPT Set.

If all the MAC addresses for a VLAN supported by SPBM mode are those of BCBs and BEBs, then the use of different FIDs for B-VLANs supported by the same SPT Set can seem unnecessary. However, the constraints could support additional capabilities and shall be followed.

1 27.12 SPBV SPVID translation

2 An SPT Bridge Port that is a Boundary Port (13.12) translates, on both ingress and egress, the VID assigned
3 to a VLAN that is supported by SPBV. On reception, the Boundary Port uses the VID Translation Table (6.9)
4 to translate the Base VID received in the tag to the SPVID for the SPT rooted at that Bridge. On
5 transmission, the Boundary Port uses the Egress VID Translation Table (6.9) to map all of the SPVIDs for
6 that VLAN to one Base VID. The Boundary Port's VID Translation Table is also configured to prevent any
7 local VID from being translated to a relay VID within the SPVID range.

8 NOTE 1—One way to prevent local VIDs within the SPVID range from being translated to relay VIDs in the SPVID
9 range is to change the default relay VID value for these entries to the reserved value 0xFFF, which would cause these
10 frames to be discarded.

11 If it has not been possible to allocate SPVIDs to support a given VLAN, no translation is performed and the
12 frame is carried on the IST within the region using the unmodified Base VID.

13 If a Boundary Port's untagged or priority tagged frames belong to a VLAN supported by SPBV, the PVID
14 for that port is set to the SPVID for the SPT rooted at that Bridge, and the port should belong to the untagged
15 set for each of the SPVIDs for that VLAN.

16 Frames received on ports internal to the region are not translated, whether they carry SPVIDs or Base VIDs.

17 NOTE 2—A frame within any Bridge within an SPT Region is considered as being forwarded within that region.
18 Consequently a frame that is forwarded between two Designated Ports at the boundary of a region can have its VID
19 translated on reception, from the VLAN's Base VID to the SPVID for the VLAN and SPT rooted at the Bridge, and on
20 transmission, back to the Base VID. Translation is not dependent on forwarding, so the frame could have been equally
21 forwarded through a port within the region.

22 27.13 VLAN topology management

23 An SPT Bridge using SPBV mode uses Dynamic VLAN Registration entries, for each VLAN it supports, to
24 create a VLAN topology that is the pruned subset of the active topology that connects participants in that
25 VLAN. The VLAN registration entries thus serve to prevent both loops and unnecessary flooding of frames
26 through Bridge Ports and LANs that cannot provide connectivity to their destinations.

27 Each SPT Bridge uses ISIS-SPB with the information calculated for the CIST and each SPT (27.6, 27.7,
28 27.8), and the following information from each Bridge in the SPT Domain with one or more Boundary
29 Ports:

30 a) VLAN membership registration for each VLAN registered on a Boundary Port, subject to the
31 registration controls defined for MVRP (11.2.3.2.3).

32 For each of its Bridge Ports that is not a Boundary Port, and for each VLAN supported by SPBV mode, the
33 SPT Bridge computes the following potential values of control in VLAN registration entries:

- 34 b) Forwarded or Filtered, for the IST, for each port that is a Root Port or a Designated Port
- 35 c) Filtered, for the IST, for each port that is an Alternate Port or Backup Port
- 36 d) Forwarded or Filtered, for each SPT, for each port that is a Designated Port
- 37 e) Forwarded, for each SPT, for each port that is a Root Port
- 38 f) Filtered, for each SPT, for each port that is an Alternate Port or Backup Port

39 The potential control element for the CIST specifies that frames for the VLAN are to be Forwarded, if the
40 port provides CIST connectivity to a Boundary Port that has registered the VLAN. Similarly the potential
41 control element for an SPT Designated Port specifies Forwarded if that port provides the shortest path to a
42 Bridge with a Boundary Port that has registered the VLAN, either directly (if that Bridge is in the same
43 region) or via a Boundary Port that provides CST connectivity to the Bridge in another region in the domain.

1 VLAN Registration Entries support ingress filtering, so the potential control element always specifies
2 Forwarded for each SPT's Root Port, thus supporting Open Host Groups (10.10.1). However, an SPVID is
3 only allocated (27.10), to support a given VLAN over a given SPT, if frames for that VLAN can enter the
4 SPT Region at the root of that SPT, so the potential control element might not be used.

5 Each SPT Bridge creates Dynamic VLAN Registration Entries using the potential control elements and the
6 Port State variable forward (27.9) for each tree for each VLAN supported by SPBV as follows:

- 7 g) An entry for the Base VID specifies Registered for a port if:
 - 8 1) forward is True for the IST for the port and
 - 9 2) the potential control element for the port for the IST and the VLAN specifies Forwarded
 - 10 and specifies Not Registered for the port otherwise.
- 11 h) An entry for each SPVID specifies Registered for a port if:
 - 12 1) forward is True for the port for the SPT identified by the SPVID and
 - 13 2) the potential control element for the port for the SPT and the VLAN specifies Forwarded
 - 14 and specifies Not Registered for the port otherwise.

15 These Dynamic VLAN Registration control elements for each of the SPT Bridge's ports are subject to
16 restrictions imposed by Static VLAN Registration Entries. Static VLAN Registration Entries shall not be
17 administratively created for any SPVID, so the controls applied to shortest path forwarding are solely those
18 determined at the boundaries of each SPT Region.

19 VLAN topology management for an SPT Bridge using SPBV, as specified in this subclause (27.13), makes
20 use of Static and Dynamic VLAN Registration Entries that can be used with MVRP. However, their use in
21 this clause is independent of whether MVRP is implemented or used, and does not require the Bridges in an
22 SPT Domain to agree on its use. If MVRP is implemented and used on Boundary Ports, then MVRPDUs are
23 transmitted and received on those ports (as permitted by the controls specified for MVRP) to exchange
24 registration information with Bridges or end stations outside the SPT Domain.

25 Since SPBM mode only forwards frames whose destination and source addresses are known to ISIS-SPB,
26 the filtering entries installed for these addresses can be used to control all forwarding for each SPBM
27 supported VLAN. SPBM mode creates a Dynamic VLAN Registration Entry for the VLAN used by the SPT
28 Bridge. Operators should not create Static VLAN Registration Entries for the VLAN used by SPBM mode.

29 **27.14 Individual addresses and SPBM**

30 Each SPT Bridge that supports SPBM mode may be configured to use loop prevention (6.5.4.1) and/or loop
31 mitigation (6.5.4.2) for unicast frames assigned to VLANs supported by the SPT Bridge. Loop prevention is
32 always used for multicast frames. Both loop prevention and loop mitigation use Dynamic Filtering Entries
33 for individual MAC addresses for each VLAN supported by SPBM mode. These filtering entries serve both
34 to implement ingress checking, effectively assigning each received frame to an SPT and ensuring that the
35 frame is only forwarded if received on the Root Port of that tree, and to implement egress checking, ensuring
36 that the frame is only forwarded through the port providing the shortest path to its destination.

37 SPBM mode uses VLAN membership configuration information distributed by ISIS-SPB to reduce the
38 number of individual address filtering entries, i.e., {VID, Individual MAC Address} tuples that need to be
39 made in each SPT Bridge. Entries for a given VID are made only for ports that are in the pruned subset of
40 the active topology that connects participants in the VLAN with that Base VID.

41 NOTE—The reduction in individual address entries is most effective in very large SPBM-capable networks, using
42 hundreds (or thousands) of Bridges, and hundreds of B-VLANs each covering a small subset of the Bridges, each subset
43 supporting a number of backbone service instances. Using a large number of B-VLANs in such a network could
44 significantly reduce the scope of the control flows used to create and maintain backbone service instances.

1 27.14.1 Loop mitigation

2 If loop mitigation is selected, each SPT Bridge can create or modify an individual address filtering entry (for
3 a given VLAN) once the calculation of the SPT rooted at that Bridge (and in the SPT Set for that VLAN) has
4 associated the address with one of the Bridge's ports, subject to satisfying constraints that prevent multicast
5 loops. These constraints are those that govern the Port State transition of a Root Port to Forwarding (i.e., on
6 setting the Port State variable forward true for SPT's Root Port, as specified in Clause 13). This subclause
7 (27.14.1) specifies a sufficient, though not necessary way to meet those constraints. If any discrepancy exists
8 between this summary and the Clause 13 specification, the latter takes precedence.

9 The Port Role Transition machine ensures that there are no disputes with the adjacent Bridge(s) with respect
10 to the active topology, such as might arise from partial or one-way connectivity (13.21). Otherwise, the
11 Dynamic Filtering Entry can be created or modified provided:

- 12 a) The entry was previously unused, or the operation of changing the entry ensures that no two ports
13 can be receiving frames from or sending frames to the same {VID, Individual Address} tuple at the
14 same time; and
- 15 b) The Static Filtering Entries that limit the propagation of the SPBM SPT-specific group addresses for
16 that SPT have been deleted or specify Filter on all ports (27.15).

17 NOTE—Where a large number of SPT-specific group addresses are used, the effectiveness of loop mitigation as a
18 strategy (for more rapidly restoring unicast connectivity after a component failure than might be achieved by using loop
19 prevention) depends on the ability of the implementation to rapidly suspend multicast connectivity, as required by (b)
20 above, possibly by distinguishing unicast and multicast forwarding.

21 27.14.2 Loop prevention

22 If loop prevention is selected, Dynamic Filtering Entries for individual MAC addresses for VLANs
23 supported by SPBM mode reflect SPT Port States for both Root Ports and Designated Ports, for different
24 SPTs in the SPT Set for each VLAN. The state machines specified in Clause 13 attempt to minimize the
25 changes to filtering entries necessary following a network component failure by identifying conditions when
26 the change in active topology constitutes a loop-free alternative. Otherwise, Dynamic Filtering Entries are
27 removed as necessary as part of making the forwarding variable False, so that an Agreement Digest can be
28 transmitted to adjacent Bridges and full connectivity restored.

29 NOTE—The Clause 13 state machines specify necessary conditions for creating new connectivity. These conditions can
30 be satisfied in a number of ways, by processes that can execute in parallel in adjacent Bridges and in the same Bridge.
31 The most effective way (time taken, resource expended) is implementation dependent, and is not specified as a set of
32 fixed steps. For example, Static VLAN Registration Entries (or an implementation-dependent equivalent) could be used
33 to disable forwarding to allow a fresh Agreement Digest to be sent at the earliest opportunity. Alternatively, where
34 SPBM mode uses a large number of SPT-specific group MAC addresses to support many backbone service instances
35 between rather fewer SPT Bridges, removing (or disabling) Dynamic Filtering Entries for individual addresses entries
36 being used as an ingress check may be more expedient.

37 27.15 SPBM group addressing

38 When a B-VLAN is supported by an SPT Bridge using SPBM, the individual destination and source MAC
39 addresses of each frame are those used by entities within SPT Bridges or within systems that include an SPT
40 Bridge component, such as BEBs (Clause 3, Clause 25). SPBM mode assigns group addresses of local
41 significance to the SPT Domain for use by those entities. For symmetric ECT Algorithms (28.8), group
42 addresses are assigned to source rooted trees and include an SPSourceID (27.10). Taken together each
43 frame's B-VID and SPSourceID identify the source rooted SPT used to forward the frame. For the ECMP
44 ECT Algorithm (44.1.2), group addresses may be assigned to one of 16 source rooted trees per bridge or to
45 one of 16 shared trees. Group addresses assigned to source rooted trees include an SPSourceID and I-SID,
46 and the I-SID may be associated with a Tie-Break Mask. Together the B-VID, SPSourceID, and Tie-Break
47 Mask identify the source rooted SPT used to forward the frame. Group addresses assigned to a shared tree
48 include the Tie-Break Mask used in root node selection, and the I-SID. The B-VID and Tie-Break Mask

1 together identify a particular shared tree used to forward the frame. In all cases ISIS-SPB's knowledge of the
2 addressed group and its assigned tree allows it to create a MAC Address Registration Entry (8.8.4) for the
3 address, restricting propagation to the pruned subset of the tree that reaches the group's members.

4 In particular, a distinct group MAC address is assigned to each VIP (Clause 3) for use as the B-DA of the
5 multicast frames that it transmits into an SPT Domain. Each VIP supports a single backbone service
6 instance, so the B-DA identifies the I-SID for each customer frame that is multicast and its transmission is
7 confined to LANs providing connectivity between BEBs supporting that service instance.

8 Figure 27-2 illustrates the general format of SPBM group MAC addresses. The octet and bit ordering and
9 identification conventions used in this figure follow those of Figure 10 of IEEE Std 802-2014 [B7]. The
10 conventions used to encode parameters with numeric values, such as the SPSourceID or I-SID, are those
11 used elsewhere in this standard (14.1.1): the number is encoded as an unsigned binary numeral with bit
12 positions in lower octet numbers having more significance. The least significant and the next to least
13 significant bits of the first octet of the address, the Individual/Group and Universally/Locally administered
14 bits, are both set denoting a locally administered group address. The two next most significant bits compose
15 an SPBM address type, permitting alternative uses of the remaining bits of the address.

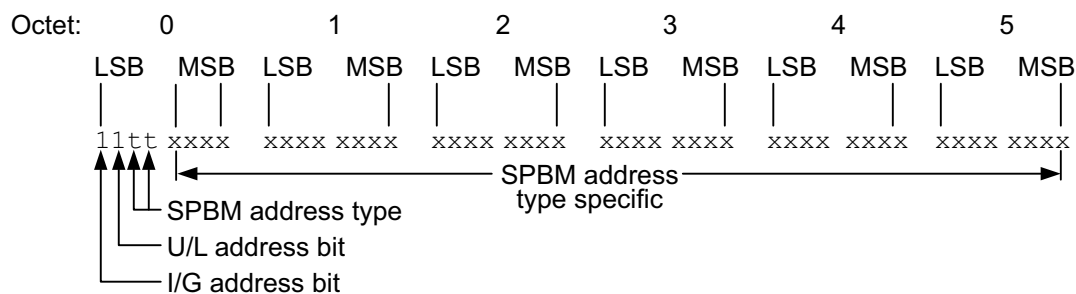


Figure 27-2—SPBM group MAC address—general format

16 This standard encodes the I-SID in the last three octets, and the value 0 or 1 in the address type field. Other
17 values (2 and 3) are reserved for future standardization. The address format for group addresses assigned to
18 a source rooted SPT is shown in Figure 27-3. In this format the SPSourceID is encoded in the first three
19 octets.

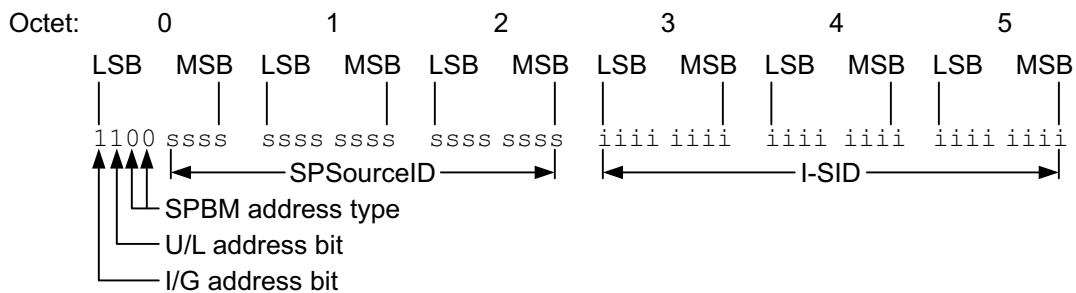


Figure 27-3—SPBM group MAC addresses—source rooted SPT

20 Figure 27-4 shows the format for group addresses assigned to a shared tree. Four bits of octet 2 are used to
21 encode the Tie-Break Mask identifying the shared tree. If the same group address were to be used in two
22 shared trees, forwarding loops could be created. The Tie-Break Mask is included in the group address so that
23 the addresses used in different shared trees are distinct and forwarding loops are not created. The remaining
24 bits are set to zero, leaving a substantial portion of the SPBM type 1 address space available for future use.

The use of locally assigned group MAC addresses by SPBM mode does not affect the use of the same or different group MAC addresses for other VLANs. In addition to creating and managing the required FDB entries for each specific group address (GA_{SPBM}) for any given Base VID (V_{SPBM} , say) that identifies an SPBM mode supported VLAN, an SPT Bridge using SPBM mode also manages the following entries:

- If the FDB contains a static entry specifying Forward (for any Bridge Port) for the wildcard VID and any specific group MAC address not required by this subclause (27.15) then an entry specifying Filter (for all Bridge Ports) is also created for that group address and each Base VID supported by SPBM mode.
- A Static Filtering Entry for All Group Addresses (for which no more specific Static Filtering Entry exists, (8.8.1) is created for each Base VID supported by SPBM mode, specifying that those addresses are to be filtered.

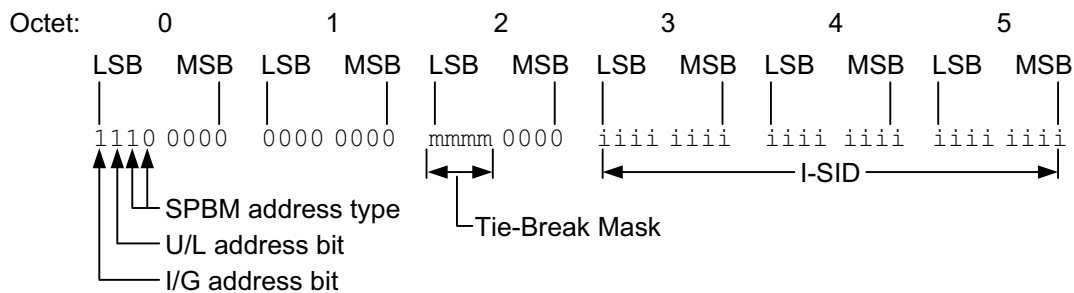


Figure 27-4—SPBM group MAC addresses—shared tree

27.16 Backbone service instance topology management

An SPT Bridge using SPBM mode uses Dynamic Filtering Entries to explicitly permit unicast communication between stations, thus restricting that communication to the shortest path between those stations. SPBM mode also uses Static Filtering Entries, for each VLAN it supports, to create a multicast topology that is the pruned subset of the active topology that connects BEBs supporting the same backbone service instance using that VLAN.

Each SPT Bridge uses ISIS-SPB with the information calculated for the CIST and each SPT (27.6, 27.7, 27.8), and the following information from each BEB in the SPT Domain:

- a) I-SID membership registration for each VLAN supported by SPBM mode

For each of its Bridge Ports, for each I-SID and each SPT supporting the VLAN supporting that I-SID, the SPT Bridge computes the potential filtering control element values, Forward or Filter, for each port that is a Designated Port or Root Port. The potential control element specifies Forwarded if that port provides the shortest path to a BEB that has registered the VLAN, either directly (if that Bridge is in the same region) or via a Boundary Port (if the Bridge is in another region in the domain). A value of Filter is associated with each port that is an Alternate Port or Backup Port for the SPT.

The SPT Bridge creates a Static Filtering Entry for the VLAN's B-VID and SPBM Group Address (27.15) for the I-SID and SPT, with a control element that specifies Forward if both the potential control element specifies Forward and the Port State variable forward (27.9) is True for the SPT for the port.

1 27.17 Equal cost shortest paths, ECTs, and load spreading

2 There can be several equal cost shortest paths between any given pair of Bridges, and the paths used by the
3 SPTs in a given SPT Set are required to be symmetric (13.1), so the local tie-breaker [the lowest ISIS Local
4 Circuit ID on the Bridge with the best (numerically least) SPB System Identifier (Clause 3)] is only used to
5 select between links providing alternative connectivity between immediately adjacent Bridges. Equal cost
6 tie breaking between paths that traverse different sets of Bridges is performed by calculating a unique
7 identifier for each path that is independent of order used to compute trees. See 28.5 for a description of the
8 Algorithm and the tie breakers.

9 The capability to support multiple VLANs using different ECTs allows a network administrator to choose
10 different equal cost paths for different services by assigning services to selected VLANs. ECMP (Clause 44)
11 provides the capability to spread load across equal cost shortest paths using a single SPBM VID.

12 A set of ECT Algorithms (28.8) is specified, each using a different tie breaker. The tie-breaking rules have
13 been selected to provide a reasonable probability that different ECT Algorithms will choose different paths
14 from a set of equal cost paths. ECT path selection can be controlled by selection of ECT Algorithm and by
15 adjustment of Bridge Priorities.

16 SPB makes use of all nodes in the SPT region for forwarding.

17 27.18 Connectivity Fault Management for SPBM

18 Connectivity Fault Management (CFM) as specified in Clause 18 through Clause 22 provides capabilities
19 useful in detecting, isolating, and reporting connectivity faults in VID-based service instances, backbone
20 service instances, and TESIs. As with PBB-TE, SPBM disables flooding of frames with unknown
21 destination address. This and other changes in behavior with SPBM require some modification to VLAN
22 CFM functions. The modifications related to SPBM are summarized here.

23 In particular, this subclause summarizes SPBM considerations related to the following:

- 24 a) SPBM MA types (27.18.1)
- 25 b) SPBM MEP placement in a Bridge Port (27.18.2)
- 26 c) SPBM MIP placement in a Bridge Port (27.18.3)
- 27 d) SPBM modifications of the CFM protocols (27.18.4)

28 27.18.1 SPBM MA types

29 SPBM CFM uses three types of MA: SPBM VID MAs, SPBM path MAs, and SPBM group MAs.

30 An SPBM VID MA operates like a VLAN MA. For example, SPBM MEPs multicast CCM frames to all
31 other MEPs and expect to receive CCM frames from all other MEPs in an SPBM VID MA. There are minor
32 differences from VLAN MAs, primarily in the destination addresses used for CFM frames. SPBM VID
33 MAs can be used to monitor and diagnose SPBM connectivity.

34 SPBM specifies two MA types for testing forwarding state installed by ISIS-SPB for specific addresses. In
35 SPBM path MA CCMs are used to test the path for an individual destination MAC address between a pair of
36 endpoints. This type of MA has only two MEPs, at two SPT Region boundary ports. An SPBM path MA is
37 identified by a TE-SID comprising an ESP in each direction between the two endpoints. In an SPBM group
38 MA CCMs are used to test a set of multicast trees used by a backbone service instance. This type of MA has
39 multiple MEPs, each located at a CBP supporting the backbone service instance whose multicast trees are to
40 be tested. An SPBM group MA is identified by the I-SID whose multicast trees it tests. The path taken by
41 CCM frames depends on the current state of the network (i.e., paths are subject to rerouting triggered by

1 topology changes). A CCM defect detected by an SPBM path MA or SPBM group MA may indicate a
2 problem with the forwarding state for the MAC addresses being tested. This standard does not specify the
3 mechanisms that would be necessary to support Loopback and Linktrace on an SPBM path MA or SPBM
4 group MA since full Loopback and Linktrace coverage can be provided by an SPBM VID MA. Therefore,
5 there are no MHFs associated with an SPBM path MA or SPBM group MA.

6 27.18.2 SPBM MEP placement in a Bridge Port

7 SPBM MEPs are always Up MEPs and can only be placed on CBPs as these correspond to the demarcation
8 points of the SPT Region. The SPBM MEPs are placed between the Frame filtering entity (8.6.3) and the
9 Port filtering entities (8.6.1, 8.6.2, and 8.6.4) in the VID space identified by the EISS Multiplex Entity
10 (6.17). An SPBM VID MEP is placed in the same location as a VLAN MEP as shown in Figure 27-5.

11 Since SPBM path MEPs and SPBM group MEPs must only see traffic using specific addresses, these MEPs
12 need to be further differentiated by the TESI Multiplex Entity (6.19). This locates an SPBM path MEP or
13 SPBM group MEP so that it receives CCMs only from the other MEPs in its MA. Figure 27-5 also shows an
14 example of an SPBM path MEP and SPBM group MEP on a CBP.

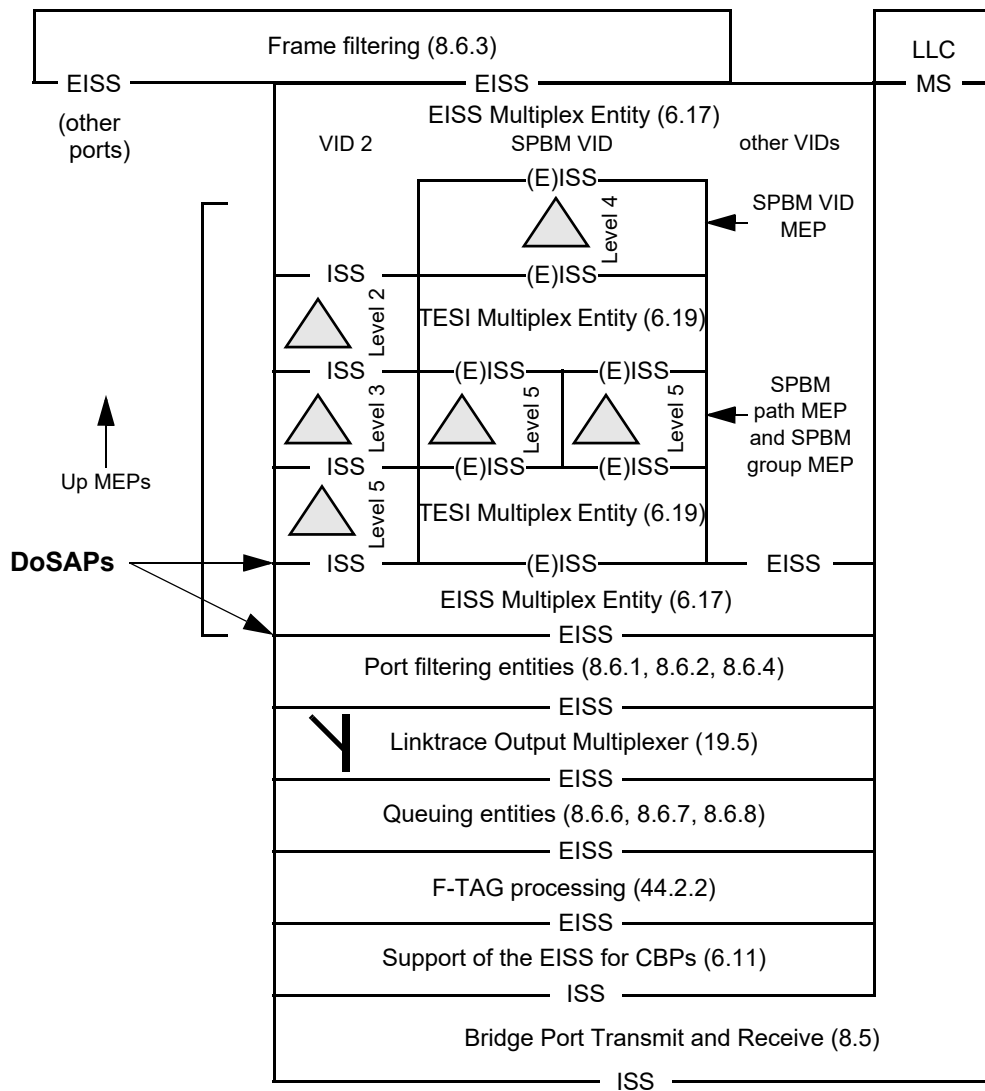


Figure 27-5—SPBM MEP placement in a CBP

1 27.18.3 SPBM MIP placement in a Bridge Port

2 SPBM MIP creation is based on the algorithm described in 22.2.3. SPBM MIPs are created per identified
3 VID on PNPs. Figure 22-9 depicts an example of a pair of SPBM MHFs on a PNP.

4 27.18.4 SPBM modifications of the CFM protocols

5 27.18.4.1 Continuity Check protocol in an SPBM MA

6 The Continuity Check protocol is described in 20.1, and the corresponding state machines are presented in
7 Clause 20. The modifications required to realize SPBM MAs include the following:

- 8 a) In the case of an SPBM VID MEP, the `destination_address` parameter is set to the group MAC
9 address for the SPBM default I-SID (Table 9-3) for the Bridge on which the MEP is configured. In
10 the case of an SPBM path MEP, the `destination_address` parameter is set to the MAC address
11 indicated by the ESP-DA field of the ESP having in its ESP-SA field the MAC address of the MEP.
12 In the case of an SPBM group MEP the `destination_address` parameter is set to the SPBM group
13 MAC address assigned to the source MEP's CBP for the SPBM group MA's I-SID [item a) in
14 20.11.1].
- 15 b) In the case of ECMP with flow filtering, the `flow_hash` and `time_to_live` parameters are set [item e)
16 in 20.11.1].

17 All other Continuity Check processes are the same as those for a VID-based MA.

18 27.18.4.2 Loopback protocol in an SPBM MA

19 The Loopback protocol is described in 20.2 and the corresponding state machines in Clause 20. The
20 modifications required to realize the SPBM MA are summarized below:

- 21 a) To enable MIPs to selectively intercept LBMs that are targeting them, the PBB-TE MIP TLV
22 (21.7.5) is inserted as the first TLV [item g) in 20.31.1] in an LBM. The format of the PBB-TE MIP
23 TLV is described in 21.7.5 and is constructed as follows:
 - 24 1) The MIP MAC address field contains the MAC address of the MIP to which the LBM is
25 targeted [item b) in 12.14.7.3.2].
 - 26 2) The Reverse VID field and Reverse MAC field are not used in SPBM.
- 27 b) An SPBM MHF forwards all received LBMs except those carrying a PBB-TE MIP TLV
28 containing in their MIP MAC address field the MAC address of the MIP associated with that SPBM
29 MHF [item f) in 20.28.1].
- 30 c) In the case of ECMP with flow filtering, the `flow_hash` and `time_to_live` parameters are set [item d)
31 in 20.31.1].

32 NOTE—The MIP MAC address for an LBM does not have to be advertised by ISIS-SPB since it is only used locally to
33 match the value provided in a PBB-TE MIP TLV.

34 All other Loopback processes are the same as those for a VID-based MA.

35 27.18.4.3 Linktrace protocol in an SPBM MA

36 The Linktrace protocol is described in 20.3 and the corresponding state machines in Clause 20. The
37 modifications required to realize the SPBM MA are summarized as follows:

- 38 a) The LTMs transmitted by a MEP associated with an SPBM MA use as the `destination_address`
39 parameter the MAC address indicated in the LTM management request [item c) in 12.14.7.4.2].
- 40 b) No special `destination_address` validation tests are performed by the `ProcessLTM()` procedure in the
41 case of SPBM MAs [item c) in 20.47.1].

- 1 c) To identify a possible egress port, an SPBM MIP queries the FDB of its associated bridge using the
2 destination_address and the vlan_identifier of the LTM as the parameters for the lookup (20.47.1.2).
3 In the case of ECMP with flow filtering, the flow_hash value of the LTM is also used for the lookup,
4 if required. For group addresses, LTMs can be further forwarded by SPBM MIPs even if the lookup
5 identifies more than one Egress port.
- 6 d) In the case of ECMP with flow filtering, the flow_hash and time_to_live parameters are set [item d]
7 in 20.42.1].

8 All other Linktrace processes are the same as those for a VID-based MA.

9 **27.19 Using SPBV and SPBM modes**

10 This subclause provides one example topology of SPT Bridges using SPBV mode and one example
11 topology of SPT Bridges using SPBM mode.

12 **27.19.1 Shortest Path Bridging—VID**

13 The primary advantage SPT Bridges operating in SPBV mode bring over spanning tree protocols is the more
14 flexible shortest path forwarding. SPBV accomplishes this by using SPVIDs and VID translation (6.9) at
15 SPT Region boundaries. All traffic within an SPT Region always takes a shortest path. Loops are prevented
16 by never allowing traffic to be accepted on a non symmetric shortest path and enforced with loop prevention
17 (13.17, 6.5.4.1).

18 An example campus network using SPBV is illustrated by Figure 27-6. The SPT Region contains eight SPT
19 Bridges with two RST Bridges connected to the SPT Region. All VLANs within the SPT Region are
20 supported by the SPT Bridges. The dashed lines represent the SPT for the SPVID of a Base VID (VID10)
21 for which Access Bridge 1 is source (root of the SPT). The solid lines to the RST Bridges indicates spanning
22 tree forwarding. VID translation from and to the respective SPVIDs occurs on the region boundary ports
23 SPT Bridge D port 5 and SPT Bridge E port 3. SPT forwarding makes better use of a mesh network;
24 however, it only will use a single path per Base VID. For example the Bridge D port 2 to Bridge B port 4
25 link is not used by access Bridge 1 because the ECT Algorithm has chosen paths through Bridge C. If
26 another VLAN is supported in this network the recommendation is to use a different ECT Algorithm that
27 could choose D-B over C-B by using an appropriate tie breaker (28.8).

28 ISIS-SPB uses SPB Hellos to establish adjacencies within the SPT region and BPDUs to establish peering
29 with RST Bridges. BPDUs from legacy Bridges L1 and L2 will indicate that they are able to use RSTP.
30 Therefore, RSTP would be enabled and SPB Hello would be transmitted periodically but dropped. The ports
31 connected to the legacy Bridges are enabled and block according to RSTP logic.

32 SPT Bridges using SPBV mode use the same principles as MSTP to form a logical Bridge with a CIST root
33 port on Bridge A, which has the best (numerically least) Bridge Identifier in the SPT Region. The CIST in
34 Figure 27-6 is a spanning tree rooted on Bridge A (not illustrated).

- 35 a) The eight Bridges in the SPT Region have the same MCID (13.8) or Auxiliary MCID (28.9).
36 b) The CIST is computed in the SPT Region and maintains the spanning tree root costs (13.17).
37 c) SPT BPDUs are exchanged by the SPT Bridges within the SPT Region (13.17).
38 d) VLANs identified by other than the Base VIDs are supported by the CIST (13.17).

39 Legacy Bridges L1 and L2 can be aware of the SPB operation of Bridges but they only support RSTP and
40 interface to the aggregation Bridges D and E using RST BPDUs. L1 and L2 learn from the received BPDUs
41 that aggregation Bridge A is the Designated Bridge for VLAN 10 while a Bridge somewhere above the SPT
42 region is the CIST Root for the network. Aggregation Bridge D receives traffic from L1 but uses an SPVID
43 and the corresponding SPT from D to reach all nodes within the region and beyond. Normal learning of
44 MAC addresses occurs.

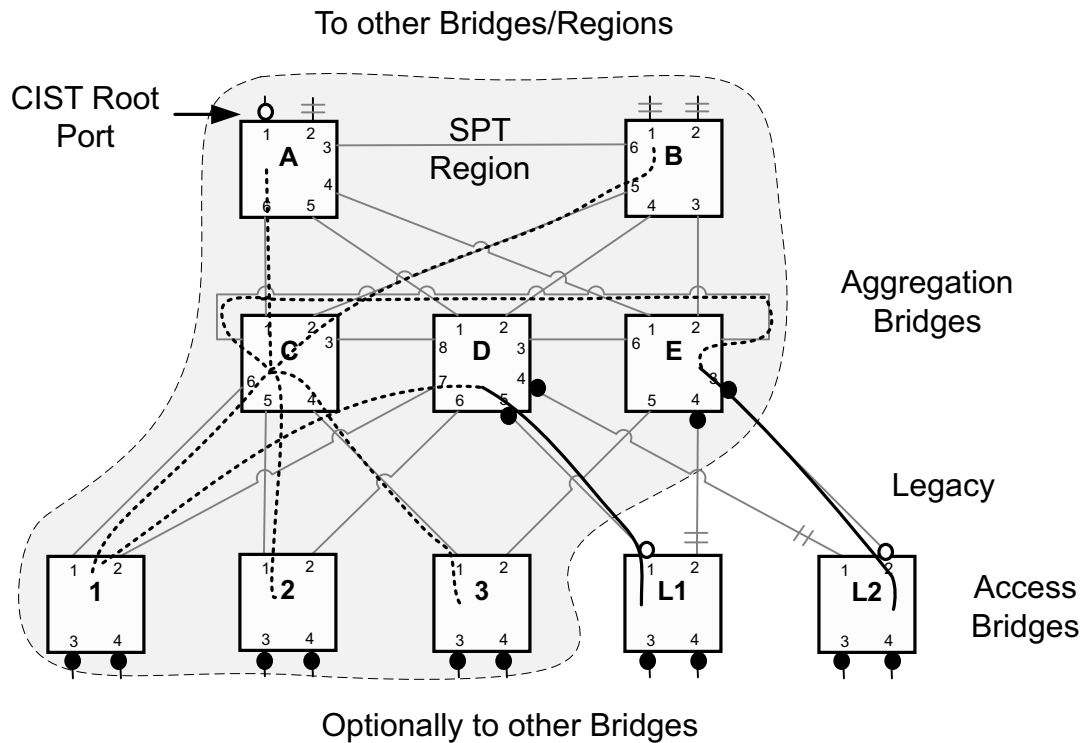


Figure 27-6—SPBV campus network example

1 Dynamic VLAN Registration capabilities are supported natively for Base VLANs by utilizing the link state to
2 auto enable forwarding on intermediate Bridges. Referring to Figure 27-6, if access Bridge 1 and
3 aggregation Bridge A were the only Bridges with VID 10 registered on their boundary ports forwarding
4 would automatically be setup on aggregation Bridge C to complete the data path between Bridge 1 and
5 Bridge A. Only Bridges configured with the Base VID, Bridge 1 and Bridge A require SPVIDs. MVRP for
6 other VLANs is supported on the CIST.

7 MAC Address Registration (8.8.4) is supported for the Base VLANs as outlined in 28.10. Group addresses for
8 Base VLANs are distributed in ISIS-SPB. At the region boundaries MMRP is required to propagate
9 registration to and from the SPT Region.

10 Since the SPT Region follows very similar design rules as the MST Region, RSTP and STP interwork with
11 both. SPB interworks with MSTP and other SPT Regions with MSTP.

12 27.19.2 Shortest Path Bridging—MAC

13 An example SPT Bridge network using SPBM mode is illustrated by Figure 27-7. The SPBM mode example
14 network is divided into two major areas: the customer equipment area, which is owned by the “customer”
15 and is connected to the SPT BEBs using C-tagged or S-tagged interfaces; and the SPT Region, which is
16 owned by the backbone provider and interconnects to the customer equipment and possibly to other SPT
17 Bridges. There is no visibility of either backbone topology or B-MAC addresses outside the SPT Region,
18 and consequently connections to other SPB or PBB areas will be identical to those to customer equipment.
19 While this is primarily leveraging the capabilities of the PBBN the emphasis is that the separation of the
20 address spaces proved by the PBB encapsulation provided useful capabilities of scaling even larger
21 enterprise or data center networks (DCNs) where bridging and delivery of optimized unicast and multicast
22 frames on an S-VLAN basis is desired.

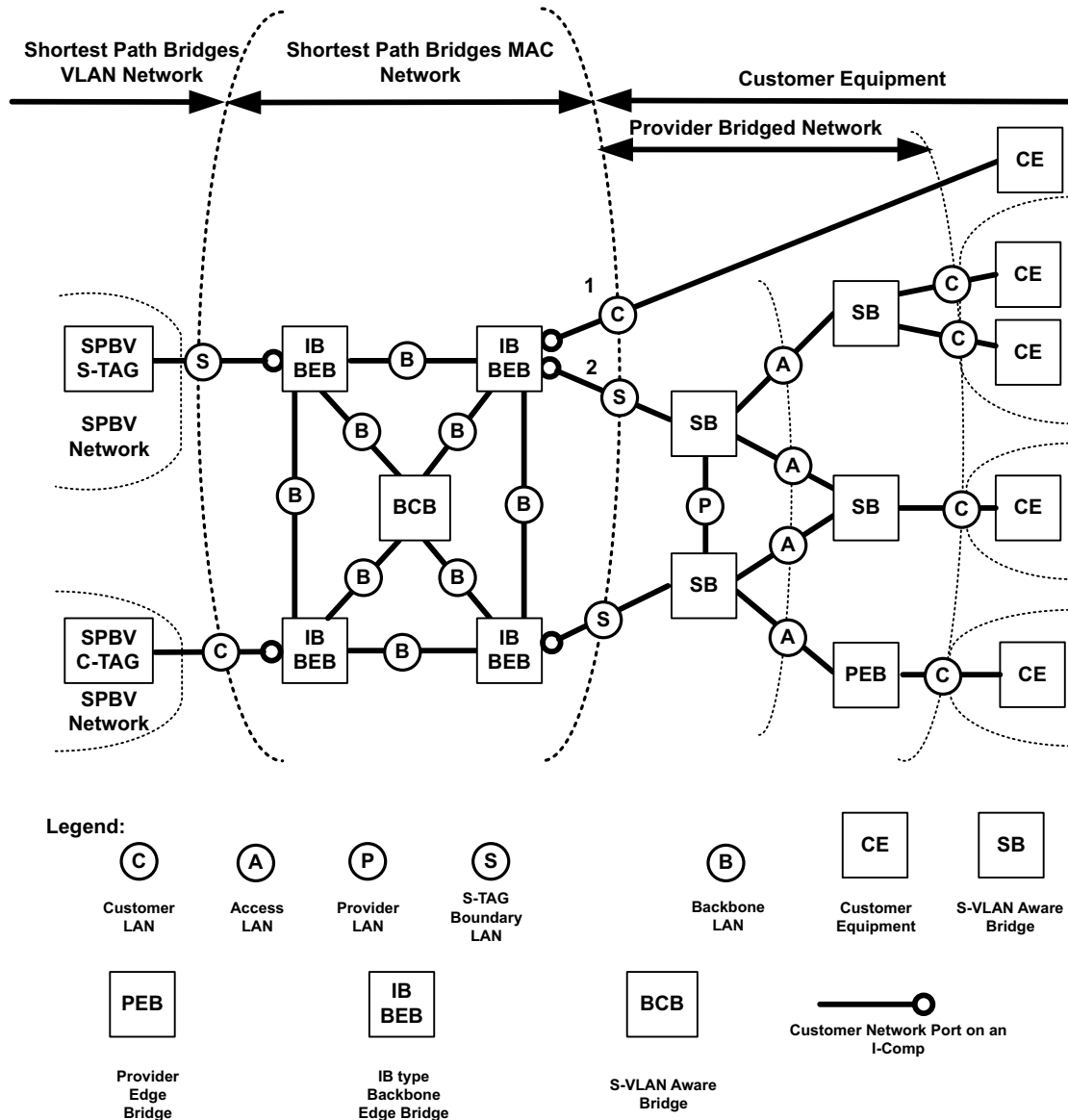


Figure 27-7—SPT Bridge Network using SPBM example

1 The interface between an IB BEB and a customer Bridge is a CNP of an I-Component as specified in 6.10.
2 The interfaces between BEBs and BCBs and between BCBs and other BCBs are PNPs. In Figure 27-7, these
3 interfaces are over LANs labeled S and B. All exterior interfaces from a SPT Bridge network using SPBM
4 mode are over S, or C LANs; whereas all interior interfaces within a SPT Bridge network support
5 appropriate segments of SPB SPTs, each set of which is identified by one Base VID per forwarding
6 topology. Each individual SPT within a set is identified by the B-MAC address of its source Bridge. The S
7 and C VLANs (exterior interfaces) are at the boundary between the SPT Bridge network and the attached
8 customer networks.

9 Within the network example, BEBs, BCBs, and the SPTs for each Base VID are secured so that only the
10 network administrator operating the SPT Bridge network can manage the reception, transmission, and relay
11 of frames between BEBs and BCBs. The arbitrary physical network topology of the SPT Bridge network
12 and the connectivity that it provides to support segregated instances of the MAC Service is designed and
13 managed by the network administrator to meet bandwidth and service availability requirements at the CBPs

1 and PIPs. Application of the service instance ingress and egress rules at these Ports in support of service
2 instance selection and identification ensures that frames cannot be transmitted or received on any backbone
3 service instance by any customer's equipment without prior agreement with the administrator of the SPT
4 Bridge network.

5 The active topology of the SPT Bridge network is separated from the active topology of the customer
6 equipment area. This is accomplished by isolating the BPDUs for each customer network from the SPT
7 Bridge network at the BEBs that surround the perimeter of the SPT Region. The ISIS-SPB LSPs and SPT
8 BPDUs, that support the active topology of the SPT Bridge network, are only exchanged on PNP and hence
9 are not leaked into the customer bridged area.

10 B-MAC addresses are used to identify the destination BEB's PIP. These B-MAC addresses are advertised by
11 each BEB to all other BEBs and BCBs in the SPT Bridge area in LSPs as part of the IS-IS routing
12 exchanges. I-SID instances associated with each PIP are also advertised using the same mechanism. To
13 perform the encapsulation and de-encapsulation of service frames, BEBs use the connection identifier stored
14 in the FDB (8.8) to correlate C-MAC addresses to B-MAC addresses. At startup, after convergence of ISIS-
15 SPB, the BEBs are aware of the location of B-MAC addresses in the topology, but have not learned the C-
16 MAC addresses yet. When the C-MAC address is unknown, the BEB encapsulates the service frames using
17 an SPBM Group address, which is analogous to the Backbone Service Instance Group address (26.4). This
18 address is then multicast to the set of registered receivers that are those BEBs where that I-SID has been
19 configured. The same holds true for the service frames having a group or broadcast destination address. An
20 individual B-DA is used for forwarding unknown or group frames when the service is point-to-point and
21 where the far end individual address is known. C-MAC addresses are learned by the provider BEB against
22 the B-MAC address associated with the service interface on the BEB that sourced them (see Clause 25).

23 **27.20 Security considerations**

24 This clause contains no exceptional provisions for securing the operation of ISIS-SPB. ISIS has a number of
25 optional capabilities that may be utilized. For example, ISIS-SPB does not prevent the use of the IS-IS
26 Authentication TLV. If stronger security is required, IEEE 802.1AE™ MAC Security should be deployed on
27 all SPT Bridge Ports, and the configuration of the controls provided by this standard to
28 regulate the acceptance of spanning tree protocol information [e.g., restrictedRole and restrictedTen
29 (13.27.64, 13.27.65)] should be set to reflect the degree of trust (i.e., authorization) accorded to the mutually
30 authenticated securely communicating peer Bridges. SPT BPDUs include a flag that indicates that the
31 attached LAN is not considered to be a part of the transmitting Bridge's SPT Region: this avoids one way
32 participation in a region if one Bridge considers the authorization of another inadequate. Use of MAC
33 Security in this way can ensure that only authenticated and authorized (i.e., trusted) Bridges participate in a
34 region, as well as preventing unauthorized transmission of ISIS-SPB information or user data on LANs
35 within the region.

1 28. ISIS-SPB Link State Protocol

2 Intermediate System to Intermediate System (IS-IS) is the link state routing protocol used for SPB
3 (Clause 27). IS-IS is easily extended to carry the required Ethernet addresses, VLAN and Service
4 membership information introduced by SPB. IS-IS is defined in ISO/IEC 10589:2002. When used for SPB,
5 IS-IS has no IP dependencies. IS-IS also supports Multi-Topology (MT) and logical instances for easy
6 virtualization of the link state protocol. This subclause specifies extensions to IS-IS for SPB. IS-IS extended
7 for SPB will be referred to as *ISIS-SPB*. The SPB extensions are additive to IS-IS and do not preclude the
8 IS-IS protocol being used for other purposes at the same time.

9 In VLANs that are configured to use SPB, ISIS-SPB controls forwarding (Clause 8) of individual and group
10 addressed MAC frames. The Agreement Protocol (13.17), utilizes the ISIS-SPB link state database along
11 with a synchronization mechanism carried in SPT BPDUs or SPB Hello PDUs. Locally on each Bridge, the
12 Agreement Protocol logic is used to control the ISIS-SPB FDB updates and implement loop prevention
13 logic. ISIS-SPB uses normal IS-IS procedures to update the link state database.

14 IS-IS uses SPB Hello PDUs and Link State PDUs (LSPs) to communicate between SPT Bridges. This clause
15 (Clause 28) describes the relevant TLVs in the ISIS-SPB PDUs. Note that ISIS-SPB does not add any new
16 PDU's to IS-IS it extends the currently defined PDUs.

17 Figure 28-2 through Figure 28-12 (in 28.12) illustrate the new TLVs and sub-TLVs for SPBV and SPBM.
18 Note that SPBM, since it derives from PBB has the capability to indicate Service instance membership using
19 the 24-bit I-SIDs. A default I-SID (I-SID 0x0000ff) maps to all SPT Bridges operating in SPBM mode
20 allowing a default broadcast and group address capability for ISIS-SPB. The default I-SID is assigned to the
21 default ECT Algorithm (LowPATHID, 28.7). I-SID group registrations via ISIS-SPB are similar to the
22 MMRP capability but because they are delivered by IS-IS they involve no additional messaging. Group
23 addresses are constructed by SPBM for each service instance/source and therefore do not require explicit
24 advertisement.

25 The I-SID value of 0 is reserved by SPBM and is used as a deleted or free I-SID value in the TLVs. This
26 reserved value may be used to avoid reorganizing LSPs when only one or a few I-SIDs are changed. This
27 provides an optimization by allowing actual deletion (shuffling of the LSP contents) to be decoupled from
28 the ISIS-SPB updates triggered by topology changes. LSPs may be reorganized during periodic updates
29 thereby minimizing impacts.

30 SPBV and SPBM will operate either within an IS-IS level 1, or an ISIS level 2. As a result the TLVs
31 specified here may propagate either in level 1 OR level 2 LSPs. IS-IS SPBM implementations shall support
32 level 1 and may support level 2 operations. Multi-level ISIS-SPB is, however, for further study.

33 NOTE—IS-IS supports point-to-point links and shared LANs; however, only point-to-point is considered for ISIS-SPB
34 in this specification. A future revision may add shared media. Clause 13 SPB description allow both point-to-point links
35 and shared media.

36 28.1 ISIS-SPB control plane MAC

37 ISIS-SPB makes no changes to the IS-IS control plane addressing principles. However, ISIS-SPB defines
38 two further addresses for the propagation of LSPs besides the three addresses already reserved by ISO (see
39 Table 8-16). ISIS-SPB differentiates itself from IS-IS for Ipv4 etc. by the contents of the SPB Hello PDUs.
40 The most important differentiation is the NLPID value for SPB (0xC1) [Annex X] which is carried in the
41 SPB Hello protocols supported TLV and used to decide if a given adjacency should support SPB. ISIS-SPB
42 carries this value in the Protocols Supported TLV (129) of IS-IS.

1 SPB operates either stand alone within an IS-IS instance, or shares an IS-IS instance and its adjacencies with
2 other protocols as distinguished by the appropriate Network Layer Protocol Identifiers (NLPIDs). In order to
3 accomplish this, ISIS-SPB may as one option use the same control plane MAC addressing as the other
4 NLPIDs.

5 SPB does not advertise its NLPID on shared media interfaces.

6 **28.2 Formation and maintenance of ISIS-SPB adjacencies**

7 ISIS-SPB uses the IS-IS three-way handshake for IS-IS point-to-point adjacencies described in IETF
8 RFC 5303 to form adjacencies when they can be formed. This subclause specifies the conditions under
9 which they may be formed, and the conditions under which the link may be used for forwarding after an
10 adjacency has formed.

11 a) The initial exchange of SPB Hello PDUs between a pair of IS-IS instances advertises the NLPID
12 value(s) supported by each Bridge. If any NLPID values match, it is anticipated that an adjacency
13 will be established to support the matched Network Layer Protocol(s).

14 This standard specifies subsequent behavior only for adjacencies that support SPB (NLPID value 0xC1).
15 ISIS-SPB supports MTs, and if more than one topology is instantiated, the procedures below are applied
16 independently to each topology instance.

17 b) The link supporting the adjacency is initialized to “SPB Operationally down.” This is signaled by
18 setting the SPB Link Metric in the Link Metric Sub TLV (28.12.7) to the value $(2^{24} - 1)$. All SPB
19 Link Metric values $< (2^{24} - 1)$ signify “SPB Operationally up,” and the value advertised is that to be
20 used in path computation.

21 c) The SPB Hello PDUs exchange MCID and Auxiliary MCID (28.4) values to determine whether
22 VLANs are consistently allocated to an operational mode. If a Bridge detects that neither its MCID
23 nor its Auxiliary MCID match the adjacency’s MCID or Auxiliary MCID, then the adjacency cannot
24 be put into service, and no further tests are performed until an acceptable match has been achieved.

25 d) After an acceptable MCID match is achieved, configuration consistency for each Base VID in the
26 topology is verified. The SPB Base VLAN-Identifiers sub-TLV (28.12.4) carried in SPB Hello
27 PDUs carries, for each Base VID supported,

- 28 1) The Base VID value.
- 29 2) The ECT-ALGORITHM identifier (OUI or CID/Index) for that Base VID.
- 30 3) The SPB Hello Use-Flag (1-bit), set if this Bridge or any Bridge in the SPT Region is using
31 this Base VID for traffic.
- 32 4) The M-Bit (1 bit), which indicates SPBM mode or SPBV mode.

33 Consistency is determined as follows:

- 34 — If either the local end or the far end SPB Hello Use-Flag of the adjacency are set, then only if
35 the respective Base VID, ECT-ALGORITHM, and Mode bits match exactly is “matched” set
36 for that Base VID.
- 37 — If both local end and far end SPB Hello Use-Flags are clear, then “matched” is set (to allow
38 maintenance of the adjacency during ECT Algorithm and mode reassignment).

39 Only if “matched” is true for all Base VIDs in the topology is the link supporting the adjacency set to “SPB
40 Operationally up,” and usable for forwarding.

41 e) Once the link supporting the adjacency has advanced to “SPB Operationally up” as above, a Bridge
42 shall only modify the ECT-ALGORITHM or Mode bit associated with any Base VID when both the
43 local end and far end SPB Hello Use-Flags are clear. A Bridge shall not modify the
44 ECT-ALGORITHM or Mode bit associated with any Base VID when either local or far end SPB
45 Hello Use-Flag is set. To do so is indicative of a serious error condition.

- 1 f) If a Bridge detects that an adjacent Bridge has altered its advertisement of the ECT-ALGORITHM
2 or Mode bit associated with any Base VID while either the local or far end SPB Hello Use-Flags is
3 set, it shall declare the link supporting the adjacency “SPB Operationally down.”
4 NOTE—If the error condition arises because of aberrant behavior by a single Bridge, the condition described
5 will be detected by all its adjacencies, and the behavior specified will cause the aberrant Bridge to be isolated
6 from the SPT Region. This is the least damaging mitigation of such a condition.

7 **28.3 Loop prevention**

8 The Group Address FDB entries for SPBV and SPBM are under the control of the ISIS-SPB database
9 exchange. Group address entries can be ensured to be loop-free when two neighboring Bridges agree never
10 to install potentially loop-forming state derived from an unsynchronized topology in the FDB (13.17,
11 6.5.4.1). This occurs by allowing ISIS-SPB to exchange link state messages and remove any older FDB
12 entries that are no longer valid to any destination. Native IS-IS does not have a method for synchronizing the
13 removal of state from the forwarding but it does have a method for synchronizing databases. ISIS-SPB
14 augments this by using a digest of the database and adding the FDB condition. ISIS-SPB needs two
15 modifications from regular IS-IS.

- 16 a) ISIS-SPB needs to remove any potential loop causing forwarding entries if its link state database is
17 not synchronized with its neighbor Bridges.
18 b) ISIS-SPB needs to synchronize with its neighbors Bridges before installing Group Address FDB
19 entries to prevent any potential loops.

20 The Agreement Protocol (13.17) specifies how this applies to ISIS-SPB.

21 **28.4 The Agreement Digest**

22 The Agreement Digest specified in this clause includes a Computed Topology Digest used to determine
23 whether a neighboring Bridge is operating with identical network topology information and thereby
24 determine whether frames may be safely forwarded to the neighbor.

25 The Agreement Digest field (Figure 28-1) comprises six elements:

- 26 — The Agreement Digest Format Identifier
27 — The Agreement Digest Format Capabilities
28 — The Agreement Digest Convention Identifier
29 — The Agreement Digest Convention Capabilities
30 — The Agreement Digest Edge Count
31 — The Computed Topology Digest

32 The Agreement Digest is carried as a structured 32-byte field in the ISIS-SPB Digest sub-TLV. Byte 1 is the
33 most significant, Byte 32 is the least significant byte. These and all parameters carried in ISIS-SPB TLVs and
34 sub-TLVs are encoded using the “big-endian” convention, in which lower numbered bits within each byte
35 (those to the left) have higher significance.

36 **28.4.1 Agreement Digest Format Identifier**

37 The Agreement Digest Format Identifier (4 bits) is carried in the most significant 4 bits of Byte 1 of the
38 Agreement Digest field. Its purpose is to allow alternative digests to be defined at some time in the future.
39 This release of the standard defines Digest Format 0. All future upgrades to the standard protocol will
40 require recipients to be able to receive and process digests of this format.

	Octet	Length
Agreement Digest Format Identifier	1	4 bits
Agreement Digest Format Capabilities	1	4 bits
Agreement Digest Convention Identifier	2	4 bits
Agreement Digest Convention Capabilities	2	4 bits
Agreement Digest Edge Count	3–4	2
Reserved (set to 0)	5–12	8
Computed Topology Digest	13–32	20

Figure 28-1—Agreement Digest field format

1 The Agreement Protocol (13.17) will not declare a topology match if the recipient does not support the
2 Format value of this field, but otherwise it will operate as normal, so that the transmitter will receive an
3 indication of the Digest Format Identifier and Capabilities of the receiver as a result of normal protocol
4 operation.

5 **28.4.2 Agreement Digest Format Capabilities**

6 The Agreement Digest Format Capabilities (4 bits) is carried in the least significant 4 bits of Byte 1 of the
7 Agreement Digest field. Its purpose is to allow alternative digest Formats to be defined and advertised at
8 some time in the future. In this version of the standard it is transmitted as value 0 and ignored on receipt.

9 **28.4.3 Agreement Digest Convention Identifier**

10 The Agreement Digest Convention Identifier (4 bits) is carried in the most significant 4 bits of Byte 2 of the
11 Agreement Digest field. Its purpose is to advertise the set of loop-free forwarding rules, for which the link
12 state database information identified by the Digest acts as input, which are currently in use by the transmitter
13 (13.17). The following rules are defined in this version of the standard:

- 14 — 1. indicates no digest match. Digests are exchanged but match is not required. Configured as off
15 (Clause 17).
- 16 — 2. means the transmitter will continue loop-free forwarding of both multicast and unicast traffic
17 either only after any change which occurs has been reflected in matched digests (strict agreement);
18 or up to the limits of change specified by the rules in this version of the standard. This is the default.
19 Configured as loopFreeBoth (Clause 17).
- 20 — 3. means the transmitter will continue loop-free forwarding of multicast traffic up to the limits of
21 change specified by the rules in this version of the standard, and will continue forwarding of unicast
22 traffic unconditionally. Configured as loopFreeMCastOnly (Clause 17).

23 NOTE—While it is recommended that all Bridges use the same convention, Bridges operate independently and the
24 conventions can be different on individual Bridges. For loopFreeBoth and loopFreeMCastOnly as indicated, a Bridge
25 may block all the respective traffic on change or allow “safe” traffic during change. Individual Bridges can have
26 different definitions of “safety” applied to the forwarding allowed during digest mismatch.

27 **28.4.4 Agreement Digest Convention Capabilities**

28 The Agreement Digest Convention Capabilities (4 bits) is carried in the least significant 4 bits of Byte 2 of
29 the Agreement Digest field. Its purpose is to advertise the set of loop-free forwarding rule conventions that
30 are understood by the transmitter. The value 0 declares that the transmitter understands conventions 1 to 3
31 (above). Bridges conforming to this version of the standard transmit the value 0 and ignore this field on
32 receipt.

1 28.4.5 Agreement Digest Edge Count

2 The Agreement Digest Edge Count (16 bits) is an unsigned integer carried in Byte 3 and Byte 4 of the
3 Agreement Digest field. Its purpose is to provide one component of the Agreement Digest, which is simple
4 to compute and powerful in detecting many simple topology mismatches.

5 This value is the sum modulo 2^{16} of all Edges in the SPB topology. Each point-to-point physical link is
6 counted as two Edges, corresponding to its advertisement by ISIS-SPB in an LSP flooded from either end of
7 the link.

8 If more than one ISIS-SPB topology is configured, the Agreement Digest Edge Count accumulates all Edges
9 in all configured topologies. If Edges are absent in a particular topology, they have the value 0 for this
10 summation.

11 NOTE—MT SPB allows different Link Metrics to be used in different topologies. The Agreement Digest format
12 specified here can only achieve a match between topologies when exactly the same set of Bridges are present in all
13 topologies but the requirement that MCIDs match for all Bridges in an SPT Region means that this is not an additional
14 constraint.

15 28.4.6 The Computed Topology Digest

16 The overall procedure for constructing the Computed Topology Digest is to:

- 17 — Form a signature of each Edge in the topology by computing the MD5 hash (IETF RFC 1321 [B20])
18 of the significant parameters of the Edge, as defined below.
- 19 — Compute the Digest as the arithmetic sum of all Edges in the topology.

20 If more than one topology is declared in ISIS-SPB, the Computed Topology Digest covers all topologies
21 declared, and this single value is advertised in the ISIS-SPB Digest sub-TLV. The SPB Digest sub-TLV is
22 carried within the MT-Port-Cap TLV [IETF RFC 6165] with the MTID value of 0, which in turn is carried in
23 an SPB Hello PDU.

24 NOTE 1—MD5 is widely reported to be cryptographically weak. This is not relevant in this application. What is
25 required is a function exhibiting good avalanche properties such that signatures with potentially very similar input
26 parameters have an infinitesimal probability of collision.

27 NOTE 2—This strategy allows the Digest to be incrementally computed when the topology changes, by subtracting the
28 signatures of vanished Edges from the Digest and adding the signatures of new Edges. In general, the signature of an
29 Edge therefore needs only to be computed once, when it is first advertised.

30 The input message to the MD5 hash for each Edge is constructed by concatenating the following fields in
31 order, with the first field being the beginning of the message.

- 32 a) The Bridge Identifier (13.26.2) of the Bridge advertising the Edge with the greater Bridge Identifier
33 value (8 bytes).
- 34 b) The Bridge Identifier of the Bridge advertising the Edge with the lesser Bridge Identifier value
35 (8 bytes).
- 36 c) A variable number of 8-byte 3-tuples, one 3-tuple for each MTID (Clause 3) declared in ISIS-SPB.
37 The 3-tuples are declared in descending order of MTID value, with the largest MTID declared first.

38 Each 3-tuple is constructed by concatenating the following fields in the order below:

- 39 d) A 2-byte field containing the 12-bit MTID value [IETF RFC 5120] in the least significant bits, with
40 the most significant 4 bits of the field set to zero.
- 41 e) The SPB Link Metric [item c) in 12.25.6.1.2] for the Edge in this topology that has been advertised
42 by the Bridge with the greater Bridge Identifier value (3 bytes).
- 43 f) The SPB Link Metric for the Edge in this topology that has been advertised by the Bridge with the
44 lesser Bridge Identifier value (3 bytes).

1 If an Edge is not present in a topology, its SPB Link Metric is set to zero in that topology.

2 The value of the Computed Topology Digest is the arithmetic sum of all of the signatures returned by
3 presenting every Edge message to MD5, treating the signature as an unsigned 16-byte integer and
4 accumulating into a 20-byte integer. Every physical link is seen as two Edges, one advertised by each Bridge
5 comprising the adjacency, and formally the Computed Topology Digest includes both.

6 NOTE 3—Although the Computed Topology Digest contains signatures for both Edges associated with each link, the
7 construction defined above means that these are always identical. An implementation may choose to compute a single
8 signature per link, and then double it before accumulating it into the Computed Topology Digest.

9 The value of the Computed Topology Digest is placed in Bytes 13 to 32 of the Agreement Digest field for
10 transmission in the ISIS-SPB Digest sub-TLV.

11 Bytes 5 to 12 of the Agreement Digest field are unused in this version of this standard. They are set to zero
12 on transmission and ignored on reception.

13 **28.5 Symmetric shortest path tie breaking**

14 When performing shortest paths computation there may be several equal cost shortest paths between the
15 same Bridge pairs (27.17). In order to meet the congruency requirements, SPB requires that the computation
16 produces a deterministic unique shortest path per SPT set independent of order or direction of computation.

17 The SPB symmetric (Clause 3) shortest path tie-breaking algorithm needs to be accurately specified or
18 different versions of SPB would compute different SPTs. SPB requires symmetry and congruence in its
19 unicast and multicast routing, so the operation of the algorithms defined for the base specification and
20 constraints on their future extension are specified unambiguously.

21 NOTE—Exploration of path computation that produces various types of ECTs is beyond the scope of this standard.

22 Two mechanisms are used to provide for symmetric and congruent SPT calculation. The first mechanism
23 addresses the possibility that different ends (nodes) of an adjacency advertise different values for the SPB
24 Link Metric (28.12.7). To maintain determinism, the SPB shortest path calculation always uses the
25 maximum of the two advertised SPB Link Metric values when accumulating path costs. The second
26 mechanism addresses the situation in which two equal cost paths are found. To maintain determinism in this
27 case, a tie-breaking algorithm is required that achieves the same result regardless of order of computation.

28 SPB is designed to support many tie-breaking algorithms for ECTs, and SPBM can assign traffic by I-SID to
29 different B-VIDs assigned to different ECTs. SPB defines one required tie-breaking algorithm and an open
30 framework for the creation of a very large number of additional algorithms. As part of the base standard, a
31 number pre-defined tie-breaking algorithms are described using this framework with extensions for
32 expansion in the future.

33 SPB accomplishes basic equal cost tie breaking by logically assigning a computation-order-independent
34 identifier to each possible equal cost shortest path that it computes. This tie breaking is symmetric because it
35 is independent of which direction the selection is done and it preserves reverse path congruency. This is
36 referred to as the Path Identifier (PATHID). A PATHID is defined as the sorted list (ascending lexicographic
37 order) of Bridge IDs that the path traverses including the endpoints.

38 When there are different equal cost shortest paths output by the SPF computation, each of them will have a
39 unique PATHID. The PATHID for each distinct path will be the same whichever Bridge computes that
40 PATHID. The PATHID will be the same for both the forward and reverse of a path (because they are sorted
41 lists of Bridge-IDs). PATHIDs may be ranked and compared by successively comparing the sorted list of
42 Bridge-IDs they contain. PATHID A is ranked “less than” PATHID B if it contains fewer Bridge-IDs:

43 e.g.: {9,15,22} < {7,8,9,10,22}

1 PATHID A is ranked “less than” PATHID B (of equal length) if its i ’th Bridge-ID is less than B’s i ’th
2 Bridge-ID and all other Bridge-IDs up to the i ’th are equal (lexicographic order).

3 e.g.: $\{9,15,22,99\} < \{9,15,22,100\}$

4 Both SPBV’s and SPBM’s default tie-breaking algorithm picks the equal cost path with the lowest PATHID
5 as defined in the above ranking.

6 The above algorithm can be easily implemented efficiently by simply back tracking the two or more
7 competing equal cost paths and picking the path that has the minimum PATHID between the fork and join
8 points of the competing paths. A path selected in this manner will also be part of any longer shortest paths
9 that transit the fork and join points, and so intermediate resolution of shortest path permutations can be
10 performed and the state for discarded permutations need not be carried forward.

11 The Bridge Priority by default is the middle value equal to 8 of the 4-bit priority in the most significant part
12 of the 2-byte field [$8 \times 4096 = 32\,768$ middle of the priority range] on all Bridges, and so does not affect the
13 tie-breaking results. Bridge Priority can be tuned by the operator and will therefore affect the pre-defined
14 tie-breaking algorithm(s) in a deterministic manner.

15 **28.6 Symmetric ECT framework**

16 Each tie-breaking method is uniquely identified by an ECT Algorithm. The ECT Algorithm is a 32-bit
17 number that contains an OUI or CID and an index. This document specifies an initial set of SPB ECT
18 Algorithms together with a framework for a large number of other algorithms. The OUI or CID allows
19 organizations to specify and manage their own algorithms and behaviors and to document them
20 independently either through the IEEE, or through other SDOs, or to keep them proprietary/experimental
21 should they desire. The different SPB ECT Algorithms defined in this document use the IEEE 802.1 OUI
22 (00-80-C2). Index values 1 through 16 are associated with symmetric ECT Algorithms while one additional
23 non-ECT spanning tree algorithm is defined with Index value 0.

24 The ECT Algorithm is unique for an SPT Set. One or more Base VIDs may be associated with an ECT
25 Algorithm; and this association is advertised both in LSPs to all other Bridges in the SPT region and also is
26 exchanged in the SPB Hellos. A Base VID should not be used within the SPT region until all Bridges agree
27 on the relationship. Temporary disagreement between the Base VIDs assigned is expected and permitted
28 between Bridges as Base VIDs with new ECT Algorithms are configured. Disagreements are allowed when
29 the Base VIDs is not in use. Base VIDs may share the same ECT Algorithm regardless of mode.

30 In SPBM, an I-SID is associated with a Base VID and thus by inference with an ECT Algorithm as well.
31 This permits an operator to assign traffic to a certain ECT-ALGORITHM and to move it as required to
32 achieve a degree of traffic balancing. In SPBM mode this Base VID is the B-VID directly.

33 In SPBV mode the Base VID shall be used to lookup and find the proper SPVID for the source under
34 consideration during the setting of VLAN Registration Entries. A VLAN supported by SPBV can be forced
35 to use the spanning tree of the IST in cases where the Bridges do not agree on the configuration or are short
36 on SPVID resources.

37 Bridges must consider all SPT sets and populate forwarding even if they do not define a VID. An ECT
38 Algorithm can be defined to operate using only data already advertised by ISIS-SPB (for example the Low
39 and High PathID algorithms), or, it may require additional tuning parameters that cannot be foreseen. To
40 accommodate a future ECT Algorithm need for tuning parameters SPB supports the concept of Opaque ECT
41 data.

1 To allow for nodal- and link-based ECT behavior in future requirement the Opaque ECT data may be
2 advertised under both SPB nodal and/or link data. The Opaque ECT data begins with the 32-bit
3 ECT-ALGORITHM identifier but the remainder of the data format is specific to future ECT Algorithms and
4 is therefore opaque to this document.

5 An example use of this framework could be as follows. Consider the tie breaking that can be done by using
6 a minimum sum over a secondary link metric and a hash-based but deterministic assignment of such
7 secondary metrics to every link in the network. Further consider these secondary link metrics as tunable.
8 Initially a new value for this ECT Algorithm is assigned and registering it with the appropriate numbering
9 authority for the OUI or CID. The hash functions that create the secondary link metrics based for example
10 on the two Bridge-IDs at each end of the link, could be defined. Then a format for a 32-bit opaque secondary
11 metric could be defined which would be advertised as opaque data under the link data and prefixed with the
12 ECT-ALGORITHM index for the new tie breaker. Migration to this new tie breaker would proceed in a
13 manner consistent with any ECT Algorithm (28.9).

14 **28.7 Symmetric ECT**

15 In the case of multiple equal cost paths, multiple ECTs may be computed, in fact most efficient shortest path
16 computations like Dijkstra actually produce all shortest paths from a root, i.e., an SPF tree, and different
17 unique trees emerge depending on the tie breaking employed at the branch points. These different trees are
18 distinguished in the forwarding plane by using SPVIDs (SPBV) or B-VIDs and B-MAC SAs (SPBM) for
19 each “active topology”. In order to be repeatable a unique tie breaker (ECT Algorithm) is chosen for each
20 SPT set as described in 28.5. Typically a small number of B-VIDs would satisfy most requirements for equal
21 cost routes in the case of SPBM.

22 SPBV and SPBM support a set of symmetric (Clause 3) equal cost paths between any pair of Bridges for a
23 given SPB instance/MTID. The symmetric shortest path algorithms are identified by the ECT Algorithm
24 using the IEEE 802.1 OUI=00-80-C2 and with Index values 0..16. Index value 0 is somewhat special in that
25 it relates the VID used for the CIST and is not a shortest path algorithm but is instead the spanning tree
26 algorithm. The remaining algorithms are shortest path algorithms: The LowPATHID algorithm (index = 1) is
27 the default SPT path computation tie breaker. SPB uses LowPATHID as the default SPT tie-breaking
28 algorithm. SPB can use any alternate tie-breaking algorithm for another ECT when it is configured. The
29 other defined algorithms use a computed shuffle of the LowPATHID algorithm. For example the
30 HighPATHID ECT-ALGORITHM=<OUI=00-80-C2:index = 2> is just a rank inversion, which
31 ones-complements the Bridge-IDs prior to doing the same comparisons as the LowPATHID algorithm. The
32 remaining 14 pre-defined algorithms have indexes 3..16 and are defined in terms of a bit mask that they
33 XOR the Bridge-IDs with prior to finding the minimum PATHID. Since they XOR over all 8 bytes, which
34 include the Bridge-Priority and the SPB System Identifier, these algorithms can be tuned in deterministic
35 ways by adjusting the Bridge-Priority. SPBM can advertise a Base VID for each of these unique symmetric
36 shortest paths through the ECT SPB Instance sub-TLV (28.12.5).

37 SPB assigns a unique ECT Algorithm to a Base VID in the Base VID sub-TLV (28.12.4) and the SPB
38 Instance sub-TLV (28.12.5). To allow migration between different ECT Algorithms, further ECT Algorithm
39 and Base VID bindings may be advertised, unused, without interfering with the operation of currently used
40 ECT Algorithms and Base VIDs. Once Bridges are configured to support the new ECT Algorithm and share
41 a consistent Base VID for this algorithm, the algorithm may then be used by mapping I-SIDs to the Base
42 VID for the desired ECT Algorithm.

43 SPBV also assigns a unique Base VID to each ECT Algorithm; however, the actual data path VID used
44 depends on the source. Therefore, an SPVID shall be specified in the SPVID field in conjunction with the
45 Base VID and ECT Algorithm in the SPB Instance sub-TLV. A similar procedure can be used with SPBV to
46 introduce a Base VID that would not carry traffic until fully configured. Then services could be switched to
47 the new VLAN. This is no different than introducing a new VLAN on RSTP for example.

28.8 Symmetric ECT Algorithm details

The exact method applied for computing the active topology of a VLAN assigned to ISIS-SPB control is determined by the ECT Algorithm configured for the given VLAN. A set of symmetric ECT Algorithms is defined to calculate SPT Sets for SPBV or SPBM VLANs. Two ECMP ECT Algorithms (44.1.2) are defined to calculate SPTs and shared trees for ECMP (Clause 44). Five explicit ECT Algorithms (45.1) are defined for VLANs associated with Explicit Trees (Clause 45).

Each of the standard symmetric ECT-ALGORITHM values is formed using the OUI=00-80-C2 and the Index=1..16. An Algorithm Mask is specified for each Index, and is exclusive-ORed with each octet of each of the Bridge Identifiers (2 octet Bridge Priority concatenated with the 6 octet SPB System Identifier) to generate Masked Bridge Identifiers for each of the candidate paths. Given any two paths, the one chosen for a given ECT Algorithm has the lowest PATHID formed from the ranked masked Bridge Identifiers, which will necessarily include a lowest masked Bridge Priority, excluding those common to both paths. This comparison gives the greatest weight to the (masked) Bridge Priority of each Bridge Identifier, allowing the path selection to be managed/tuned without creating additional identifiers or identifier components for each Bridge. The ECT-ALGORITHM values and corresponding Masks and their effects, in terms of prioritizing path selection by Bridge Priority, are given in Table 28-1.

Table 28-1—Bridge Priority Masking

ECT-ALGORITHM	Algorithm MASK	Effect of using MASK on priority nibble
00-80-C2-01	0x00	The selected path includes Bridge with best (numerically lowest) Bridge Identifier after masking. In this case the masking 0x00 no effect and original administered Bridge priority values apply. When Bridge priority value is equal for two Bridge identifiers the lower system identifier determines the priority. (0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15).
00-80-C2-02	0xFF	The selected path includes Bridge with best priority (numerically lowest) Bridge Identifier after masking. In this case the masking 0xFF reverses the original administered Bridge priority values. When Bridge priority value is equal for two Bridge identifiers the lower system identifier determines the priority. (15,14,13,12,11,10,9,8,7,6,5,4,3,2,1,0). Read vector as administered priority 0 equals 15, administered priority 15 equals 0 and so on.
00-80-C2-03	0x88	Bridges with Bridge Priority 8 will be treated as best priority for path selection, followed by Bridge Priorities of 9, 10, 11, and so on. (8,9,10,11,12,13,14,15,0,1,2,3,4,5,6,7). Read vector as administered priority 0 equals 8, administered priority 8 equals 0.
00-80-C2-04	0x77	Bridges with Bridge Priority 7 will be treated as best priority for path selection, followed by Bridge Priorities of 6, 5, 4, and so on. (7,6,5,4,3,2,1,0,15,14,13,12,11,10,9,8). Read vector as administered priority 0 equals 7, administered priority 7 equals 0.
00-80-C2-05	0x44	(4,5,6,7,0,1,2,3,12,13,14,15,8,9,10,11)
00-80-C2-06	0x33	(3,2,1,0,7,6,5,4,11,10,9,8,15,14,13,12)
00-80-C2-07	0xCC	(12,13,14,15,8,9,10,11,4,5,6,7,0,1,2,3)
00-80-C2-08	0xBB	(11,10,9,8,15,14,13,12,3,2,1,0,7,6,5,4)
00-80-C2-09	0x22	(2,3,0,1,6,7,4,5,10,11,8,9,14,15,12,13)
00-80-C2-0A	0x11	(1,0,3,2,5,4,7,6,9,8,11,10,13,12,15,14)
00-80-C2-0B	0x66	(6,7,4,5,2,3,0,1,14,15,12,13,10,11,8,9)

Table 28-1—Bridge Priority Masking (continued)

ECT-ALGORITHM	Algorithm MASK	Effect of using MASK on priority nibble
00-80-C2-0C	0x55	(5,4,7,6,1,0,3,2,13,12,15,14,9,8,11,10)
00-80-C2-0D	0xAA	(10,11,8,9,14,15,12,13,2,3,0,1,6,7,4,5)
00-80-C2-0E	0x99	(9,8,11,10,13,12,15,14,1,0,3,2,5,4,7,6)
00-80-C2-0F	0xDD	(13,12,15,14,9,8,11,10,5,4,7,6,1,0,3,2)
00-80-C2-10	0xEE	(14,15,12,13,10,11,8,9,6,7,4,5,2,3,0,1)

1 NOTE 1—In these algorithms, the Algorithm Mask is applied to all octets of each Bridge Identifier. Thus if two
2 alternative paths include only Bridges with one or a few Bridge Priorities, the Algorithm Mask will still reorder their
3 priorities for inclusion on the path as it acts on the rest of the SPB System Identifier component of the Bridge Identifier
4 field.

5 NOTE 2—If the two priority bytes in the 8-byte Bridge Identifier are all equal, then all Bridges have equal priority and
6 ECT-ALGORITHM values 00-80-C2-01 and 00-80-C2-02 are exactly Low and High PATHID as described in 28.5.

7 28.9 ECT Migration

8 For orderly migration, each Base VID to ECT Algorithm assignment also has associated flags indicating
9 active usage of the Base VID by services. The Use-Flag in the SPB Base VLAN-Identifiers sub-TLV
10 (28.12.4) indicates a Bridge's awareness of use of the Base VID somewhere in the SPT Region, and the
11 U-bit in the SPB Instance sub-TLV (28.12.5) indicates local use of this Base VID. The Use-Flag in the SPB
12 Base VLAN-Identifiers sub-TLV is carried in the SPB Hello PDU, and is set once the Bridge advertising the
13 algorithm has started either using or seeing the use of that ECT-ALGORITHM. ISIS-SPB enforces the
14 following rules with respect to ECT Algorithms and their usage in order to protect the SPB network from
15 outages created by incorrect configuration. If neighbors disagree on a Base VID to ECT-ALGORITHM
16 assignment, then use of the adjacency for SPB traffic is permitted (i.e., the SPB Link Metric may be set less
17 than $2^{24} - 1$) only if both neighbors' Use-Flags associated with the Base VID are cleared (28.2).

18 If a new ECT Algorithm is implemented in all Bridges of an SPT Domain and intended to be used by a Base
19 VID, then the assignment of the Base VID to the new ECT-ALGORITHM with Use-Flag cleared is first
20 configured in the SPT Domain on a Bridge by Bridge basis. As soon as this configuration is complete,
21 services (I-SIDs in SPBM or SPVIDs in SPBV) may be assigned to the given Base VID, which causes the
22 local U-bit to be set, which will in turn cause the Use-Flag to be set for the given Base VID.

23 SPBV and SPBM have different migration capabilities after the introduction of a new ECT Algorithm. The
24 following subclauses describe how SPBV and SPBM use a newly introduced ECT Algorithm.

25 28.9.1 Use of a new ECT Algorithm in SPBV

26 A new ECT Algorithm, which is implemented by each Bridge of an SPT Domain, gets to be used if it is
27 configured to support a VLAN. In SPBV mode, the application of a new ECT Algorithm typically coincides
28 with the introduction of a new VLAN supported by SPBV. That is, the Base VID of the new VLAN is
29 assigned to the new ECT-ALGORITHM in the SPB Hello PDU Base VID and SPB Instance sub-TLVs
30 (28.9). ISIS-SPB then allocates (27.10) the SPVIDs necessary for the support of the new VLAN. To support
31 in-service upgrade from one set of VID / ECT Algorithm allocations to another, SPT Bridges allow graceful
32 configuration of those services. Provided there is an operating SPBV or SPBM VLAN in a region, SPT
33 Bridges can be configured within the Region for a new VLAN on a Bridge by Bridge basis. As long as the
34 SPT Region is connected the Bridges supporting the new VLAN do not need to be directly connected.
35 Intermediate Bridges are capable of populating forwarding without terminating services allowing them to
36 participate in the VLAN.

1 The migration of a VLAN from an old ECT Algorithm to a new ECT Algorithm can be performed by the
2 reassignment of the Base VID of the VLAN. That is, the Base VID is assigned to the new
3 ECT-ALGORITHM instead of the old ECT-ALGORITHM on a Bridge by Bridge basis. In SPBV the
4 forcing of one of the Bridges Base-VIDs to use the spanning tree algorithm as opposed to another algorithm
5 forces all of the Bridges in the SPT region using that Base VID to spanning tree. Migration can then proceed
6 to the new Base VID assignment until the last Bridge is configured to use the new ECT-ALGORITHM. The
7 new assignment is then propagated in SPB Hello PDU Base VID and SPB Instance sub-TLVs (28.9) by
8 ISIS-SPB within the SPT Domain. As soon as each SPT Bridge is configured with the Base VID assignment
9 to the new ECT-ALGORITHM, the spanning tree assignment on the last Bridge is cleared.

10 NOTE 1—ISIS-SPB automatically performs the allocation and release of SPVIDs during the migration. The SPVIDs
11 allocated for the Base VID old ECT-ALGORITHM assignment are released as soon as the assignment has been broken.
12 These SPVIDs might be then allocated for the Base VID new ECT-ALGORITHM assignment. That is, there is no need
13 for additional SPVID allocations if a Base VID supported by SPBV is migrated from an old ECT-ALGORITHM to a
14 new one.

15 NOTE 2—SPBV does not support hitless migration. That is, the migration of a VLAN from an old ECT-ALGORITHM
16 to a new ECT-ALGORITHM may cause traffic outage.

17 **28.9.2 Use of a new ECT Algorithm in SPBM**

18 SPBM offers the ability to map I-SIDs to Base VIDs as a load balancing option. In topologies with multiple
19 equal cost paths the ability to spread traffic on a per I-SID basis comes from using the Base VIDs with
20 different algorithms as specified in this clause.

21 When an I-SID is configured to be associated with a Base VID, that Base VID shall be consistently
22 associated with the same ECT-ALGORITHM throughout the SPT Domain. Once at least one I-SID has been
23 assigned to a Base VID the U-bit is set and advertised in subsequent LSPs, which also causes the Use-Flag
24 to be set in SPB Hello PDUs.

25 SPBM supports in-service migration of an I-SID from an old ECT Algorithm to a new ECT Algorithm
26 supported by each Bridge of an SPT Domain. In order to accomplish this, two complete Base VIDs must be
27 configured in the SPT Domain. Then an I-SID under migration can be assigned to two Base VIDs at the
28 same time: the old Base VID (old ECT-ALGORITHM), and the new Base VID (new ECT-ALGORITHM).
29 The U-bit (in the SPB Instance sub-TLV, 28.12.5) is set by the advertising Bridge when there is at least one
30 service using this VID. This propagates throughout the SPT domain, and any Bridge seeing at least one
31 U-bit set in received LSPs (signifying that the indicated Base VID is being used by a service somewhere)
32 sets the Use-Flag in its SPB Base VLAN-Identifiers sub-TLV (28.12.4).

33 I-SIDs will have originally been configured to be associated with the old Base VID (28.12.10). No filtering
34 entry is set for the new Base VID in the FDBs as there is no Transmitter set on the new Base VID yet. I-SIDs
35 can be configured to be associated with the new Base VID. Care must be taken to ensure the new Base VID
36 is configured in all the required places. When a particular I-SID has been added to a new Base VID,
37 ISIS-SPB detects that the I-SID is now associated with two Base VIDs and at each location that I-SID may
38 receive traffic on either Base-VID. During this period individual addresses are available under both Base
39 VIDs. The setting of the R and T bits still determines whether the group addresses are populated for each
40 Bridge supporting that I-SID in either Base VID. In other words the same rules for a single Base VID are
41 followed but two base VIDs are temporarily in use for received traffic. Traffic is “rolled” from the old Base
42 VID to the new one by administrative reassignment of the I-SID at each service end-point from the old to the
43 new B-VID. This can be an atomic action, and so at any instant only one Base VID will be used by any
44 service originating frames. The association of the I-SID to the old Base VID is removed once all I-SIDs are
45 configured to one Base VID. Furthermore, if this is the last I-SID removed from a Base VID the U-bit is
46 cleared (in the SPB Instance sub-TLV) for the old Base VID to ECT-ALGORITHM assignment. Eventually
47 the Use-Flag carried in the SPB Hello PDU will be cleared when all the Bridges have been configured.

1 28.10 MAC address registration

2 ISIS-SPB creates MAC Address Registration Entries (8.8.4) for group addresses in order to reach the
3 members of a group. The group addresses are locally administered in case of SPBM (27.16), thus, they are
4 only significant inside the SPT Region.

5 ISIS-SPB also installs MAC Address Registration Entries for universally administered group addresses as
6 they might be used by SPBV. Furthermore, ISIS-SPB provides interworking with MMRP at the Boundary
7 Ports of an SPT Region. That is, ISIS-SPB interprets MMRPDUs received at a Boundary Port and assembles
8 the proper ISIS-SPB TLVs in order to advertise the corresponding MAC address registration inside the SPT
9 Region. Furthermore, ISIS-SPB assembles and issues MMRPDUs on Boundary Ports if MAC address
10 registration needs to be propagated outside of the SPT Region.

11 When an MMRPDU is received on a Boundary Port, ISIS-SPB disseminates the registration by sending
12 LSPs with an SPBV MAC Address sub-TLV (28.12.9) filled according to the registration received from the
13 MMRPDU. The MAC Address and Base VID values shall be copied from the MMRPDU, with the R bit set
14 and T bit cleared. The SR bits (28.12.9) are set according to the Group service requirement information
15 carried in the MMRPDU. When processing received SPBV MAC Address sub-TLVs that have the R bit set,
16 SPT Bridges at the boundary of an SPT Region shall check whether they should filter or forward the frames
17 received on their Boundary Port in the VLAN for which the registration applies. If the option is to forward
18 frames, then they also send the SPBV MAC Address sub-TLV into the SPT Region for the MAC address
19 under registration; but set the T bit, clear the R bit and for the VID value use the SPVID allocated to the
20 given VLAN. ISIS-SPB then creates MAC Address Registration Entries inside the SPT Region based on the
21 SPBV MAC Address sub-TLVs. Furthermore, if the Boundary Port is in the given VLAN's member set,
22 then ISIS-SPB assembles the proper MMRPDU and issues it on the Boundary Port.

23 Besides interworking with MMRP at Boundary Ports, ISIS-SPB also takes into account the Registrar
24 Administrative Control parameters (10.7.2).

25 NOTE—Creation of unnecessary MAC Address Registration Entries can be prevented by using the “Registration
26 Forbidden” parameter for the proper Ports and VLANs.

27 28.11 Circuit IDs and Port Identifiers

28 IS-IS allows the formation of multiple adjacencies between two routers. ISIS-SPB supports only a single
29 logical adjacency per MTID. Link Aggregation allows a single ISIS-SPB adjacency to cover multiple links,
30 and this is the only method defined in this version of this standard by which multiple links between
31 neighbors are supported. In cases where multiple potential adjacencies exist, each pair of peers using
32 ISIS-SPB should choose the link with the lowest Local Circuit ID on the Bridge with the better (numerically
33 lower) SPB System Identifier as the link over which to send and receive frames.

34 Procedures for using multiple adjacencies are not specified by this standard.

35 28.12 ISIS-SPB TLVs

36 All SPB TLVs and sub-TLVs are described. SPBV and SPBM share most of the same TLVs.
37 Multi-Topology (MT) [Annex X] is introduced as well, but multiple SPT instances, each supporting a
38 different VLAN, and each using either SPBV or SPBM can be described in a single topology instance. MT
39 instances allow different topologies to be advertised, for example if there is a desire to use different metrics
40 since there is only a single metric scheme within a topology.

1 SPB adds one new TLV and several new sub-TLVs that can be added to IS-IS PDUs, as described below.
2 The new ISIS-SPB data have been organized to support multiple topologies within a single IS-IS instance.
3 Each of the new sub-TLVs can be included in a TLV containing a topology instance identifier (MTID) value
4 (as per [MT]) so that the computations can operate only on sub-TLVs specific to their topology instance. The
5 intent of MT is to support the use of multiple metric sets when required. The construction of the MCID and
6 the Agreement Digest (13.17, 28.4) requires that all SPT Bridges in an SPT region appear in all MT
7 instances.

8 All parameters carried in the ISIS-SPB (sub-)TLVs defined below are encoded using the “big-endian”
9 convention, in which higher significance bytes and bits within each byte appear to the left of and above
10 bytes and bits of lower significance.

11 28.12.1 MT-Capability TLV

12 The MT-Capability TLV (Figure 28-2) is a top-level TLV for LSPs that provides Multi-Topology context. It
13 identifies the MTID for sub-TLVs in LSPs.

	Octet	Length
Type (144)	1	1
Length (2)	2	1
Overload Bit	3	1 bit
reserved	3	3 bits
MTID	3–4	12 bits

Figure 28-2—MT-Capability TLV

- 14 a) Type (8 bits) Value 144
- 15 b) Length (8 bits)
- 16 Total number of bytes contained in the value field.
- 17 c) Overload Bit (1 bit)
- 18 The overload bit is an ISIS-SPB MT instance-specific overload bit.
- 19 d) MTID (12 bits)
- 20 This is used to provide the ISIS-SPB Multi-Topology context for SPB sub-TLVs. This should be 0
- 21 when there is only one topology instance.

22 28.12.2 SPB MCID sub-TLV

23 This sub-TLV (Figure 28-3) shall be included in an MT-Port-Cap TLV [IETF RFC 6165] with MTID equal
24 to 0 in SPB Hello PDUs to announce the Bridge’s MCID (13.8). This information should be the same on all
25 Bridges in the topology being controlled by this ISIS-SPB instance. The MCID is controlled solely by
26 configuration and is a digest of the protocol assignments for all possible VIDs. Two MCIDs are carried to
27 allow transitions between configurations when the changes are non-critical. If neither MCID advertised by a
28 neighbor matches either MCID on this Bridge the adjacency is a Region boundary.

	Octet	Length
Type (4)	1	1
Length (102)	2	1
MCID	3–53	51
Aux MCID	54–104	51

Figure 28-3—SPB MCID sub-TLV

- 1 a) Type (8 bits) Value 4
- 2 b) Length (8 bits)
- 3 Total number of bytes contained in the value field.
- 4 c) MCID (51 bytes)
- 5 The complete MCID defined in (13.8) that identifies an SPT Region.
- 6 d) AUX-MCID (51 bytes)
- 7 In the case of migration and where the MCID will change but in a non-critical way, this MCID
- 8 contains the previous advertised MCID. The neighbor will not drop an adjacency if at least one of
- 9 these MCIDs matches either of its MCIDs.

10 28.12.3 SPB Digest sub-TLV

11 This sub-TLV (Figure 28-4) shall be included in an MT-Port-Cap TLV with MTID 0 in SPB Hello PDUs.
12 The ISIS-SPB Agreement Digest (13.26.1, 28.4) is computed based on current topology and it changes when
13 significant topology changes. The Agreement Digest is a key input into the Agreement Protocol (13.17),
14 which provides multicast loop prevention. During the propagation of LSPs the Agreement Digest will vary
15 between neighbors until the LSPs are common. The digest is a summarized means of determining agreement
16 between nodes on the distance to all multicast roots, hence is essential for loop prevention. For each SPT
17 where it has been determined the distance to the root has changed, unsafe multicast forwarding is blocked
18 until the exchanged digests match.

	Octet	Length
Type (5)	1	1
Length (33)	2	1
reserved	3	3 bits
V	3	1 bit
A	3	2 bits
D	3	2 bits
Agreement Digest	4–35	32

Figure 28-4—SPB Digest sub-TLV

- 19 a) Type (8 bits) Value 5
- 20 b) Length (8 bits)
- 21 Total number of bytes contained in the value field.
- 22 c) V (1 bit)
- 23 The Agreed Digest Valid Bit (13.27.7, 13.27.9). This bit is populated and read from the Agreement
- 24 Logic.
- 25 d) A (2 bits)
- 26 The agreement number 0–3, which aligns with BPDUs agreement number concept (13.27.11). When
- 27 the Agreement Digest for this node changes this number is updated and sent in the SPB Hello.
- 28 e) D (2 bits)
- 29 The discarded agreement number 0–3, which aligns with BPDUs agreement number concept
- 30 (13.27.12). When the Agreement Digest for this node changes this number is updated. Once an
- 31 Agreement has been sent it is considered outstanding until a matching or more recent Discarded
- 32 Agreement Number is received.
- 33 f) Agreement Digest (32 bytes)
- 34 An Agreement Digest as described in 28.4.

28.12.4 SPB Base VLAN-Identifiers sub-TLV

This sub-TLV (Figure 28-5) is included in an MT-Port-Cap TLV in an SPB Hello PDU to indicate the ECT Algorithms for the Base VIDs (and by implication the VID(s) used on the forwarding path for each SPT Set for a VLAN identified by a Base VID) that are in use. This information should be the same on all Bridges in the topology identified by the MT-Port-Cap TLV in which it is carried. Discrepancies between neighbors with respect to this sub-TLV are temporarily allowed during reconfiguration but at all times the active Base-VIDs (as determined by the state of the Use-Flag carried in this sub-TLV) shall agree and use the same ECT-ALGORITHM.

In the case of SPBM, the Base VID is the B-VID used to forward frames. In the case of SPBV, each source uses a different SPVID and a Base VID is used for frames transmitted on the IST. One or more Base VIDs are associated with an ECT Algorithm. This structure supports multiple SPT Sets within an IS-IS topology instance for both SPBV and SPBM. It also allows ECMP (Clause 44) for SPBM and the use of Explicit Trees (Clause 45) both for SPBV and SPBM within the same IS-IS topology instance supporting the SPT Sets.

		Octet	Length
ECT-VID Tuple 1	Type (6)	1	1
	Length (6n)	2	1
	ECT Algorithm	3–6	4
	Base VID	7–8	12 bits
	U	8	1 bit
	M	8	1 bit
	reserved	8	2 bits
...			
ECT-VID Tuple n	ECT Algorithm	(6n–3)–6n	4
	Base VID	(6n+1)–(6n+2)	12 bits
	U	6n+2	1 bit
	M	6n+2	1 bit
	reserved	6n+2	2 bits

Figure 28-5—SPB Base VLAN-Identifiers sub-TLV

- a) Type (8 bits) Value 6
- b) Length (8 bits)
The size of the value is ECT-VID Tuples*6 bytes. Each 6-byte part of the ECT-VID tuple is formatted as described below.
- c) ECT-ALGORITHM (4 bytes)
The ECT-ALGORITHM is advertised when the Bridge supports a given ECT-ALGORITHM (by OUI or CID/Index) on a given Base VID.
- d) Base VID (12 bits)
The Base VID that is associate with the SPT Set.
- e) Use-Flag (1 bit)
The Use-Flag is set if this Bridge, or any Bridge in the SPT Region is currently using this ECT-ALGORITHM and Base VID. The Use-Flag is calculated as the logical OR of the U-bit values of all Bridges in the region, as found in their SPB Instance sub-TLV (28.12.5) in the link state database. This definition includes the U-bit value of this Bridge.
- f) M-Bit (1 bit)
The M-bit indicates if this is SPBM when set to 1 or SPBV mode when set to 0.
- g) 2 Reserved Bits set to 0.

28.12.5 SPB Instance sub-TLV

This sub-TLV (Figure 28-6) is carried within an MT-Capability TLV in the fragment ZERO LSP. It identifies the Bridge uniquely and identifies the ECT-ALGORITHM values supported by the Bridge and the Base VIDs and SPVIDs assigned to those algorithms. For SPBM, only the Base VID is valid and the SPVID is set to zero. In the case of SPBV, the Base VID is associated with the SPVID used for forwarding by the Bridge originating the TLV. There can be multiple ECT-ALGORITHM values specifying a number of ECTs. Alternatively, the ECT-ALGORITHM value can indicate that the VLAN is assigned to ECMP operation (Table 44-1) or to explicit path control (Table 45-1).

		Octet	Length
VID Tuple 1	Type (1)	1	1
	Length	2	1
	CIST Root Identifier	3–10	8
	CIST External Root Path Cost	11–14	4
	Bridge Priority	15–16	2
	reserved	17–18	11 bits
	V	18	1 bit
	SPSourceID	18–20	20 bits
	Number of Trees	21	1
	U	22	1 bit
	M	22	1 bit
	A	22	1 bit
	reserved	22	5 bits
	ECT Algorithm	23–26	4
VID Tuple n	Base VID	27–28	12 bits
	SPVID	28–29	12 bits
	...		
	U	8n+14	1 bit
	M	8n+14	1 bit
	A	8n+14	1 bit
	reserved	8n+14	5 bits
	ECT Algorithm	(8n+15)–(8n+18)	4
	Base VID	(8n+19)–(8n+20)	12 bits
	SPVID	(8n+20)–(8n+21)	12 bits

Figure 28-6—SPB Instance sub-TLV

- a) Type (8 bits) Value 1
- b) Length (8 bits)
Total number of bytes contained in the value field.
- c) CIST Root Identifier (64 bits)
The CIST Root Identifier is for SPB interworking with RSTP and MSTP at SPT Region Boundaries. This is an imported value from a spanning tree.
- d) CIST External Root Path Cost (32 bits)
The CIST External Root Path Cost is the cost from the spanning tree algorithm to the Root.
- e) Bridge Priority (16 bits)
Bridge priority is the 16 bits that together with the low 6 bytes of the SPB System Identifier form the Bridge Identifier. The Bridge Identifier is the spanning tree compatible Bridge identifier. This is configured exactly as specified in IEEE Std 802.1D-2004 [B14]. This allows SPB to build a

compatible spanning tree using link state by combining the Bridge Priority and the SPB System Identifier to form the 8-byte Bridge Identifier. The 8-byte Bridge Identifier is also the input to the 16 pre-defined ECT tie-breaker algorithms.

f) V bit (1 bit)

The V bit (SPBM) indicates this SPSrcID is auto allocated (27.10). If the V bit is clear the SPSrcID has been configured and shall be unique. When a Bridge joining an SPT Region has formed at least one SPB adjacency, it can discover the previously allocated SPSrcIDs and it will allocate a SPSrcID according to the allocation logic (27.10).

g) SPSrcID (20 bits)

The SPSrcID (SPBM) is a 20-bit value used to construct group MAC address (27.15) for multicast frames originating from the origin (SPT Bridge) of the LSP that contains this sub-TLV. SPSrcID may be 0 if it has not been allocated (27.10) or if there is no SPBM service configured. When SPSrcID is 0 all information pertaining to this MTID SPB instance will be distributed but not acted upon until a valid SPSrcID is populated. If the SPSrcID conflicts with another SPSrcID from another Bridge the tie-breaker rules determine the winning Bridge (27.10).

h) Number of Trees (8 bits)

The Number of Trees is set to the number of VID tuples that follow (minimum one). The following seven fields make up a VID tuple. A sequence of VID tuples can occur in any order.

1) U-Bit (1 bit)

The U-bit is set if this Bridge is currently using this ECT-ALGORITHM for I-SIDs it sources or sinks or the SPVIDs it sources. This is importantly different from the Use-Flag found in the SPB Hello, which is set if a Bridge sees other nodal U-bits are set OR it sources or sinks itself.

2) M-Bit (1 bit)

The M-bit indicates if this is SPBM when set to 1 or SPBV mode when set to 0.

3) A bit (1 bit)

The A bit (SPB) when set declares this is an SPVID with auto allocation (27.10). If the SPVID value is zero, VID will be allocated once the Bridge has synchronized the IS-IS LSPs. Neighbor Bridges can distribute the LSPs but shall not populate FDBs (forwarding) for traffic from a Bridge that has an SPVID of 0. When the Bridge allocating is synchronized with the IS-IS adjacency, it will allocate one or more SPVIDs according to the allocation logic (27.10).

4) Reserved (5 bits)

Five bits reserved for future use shall be set to 0.

5) ECT-ALGORITHM (4 bytes)

ECT-ALGORITHM is advertised when the Bridge supports a given ECT-ALGORITHM (by OUI or CID/Index) on a given VID. This declaration shall match the declaration in the SPB Hello PDU originating from the same Bridge. When migrating from one algorithm to another discrepancies between algorithm and Base VID on different Bridges are allowed if the Use-Flag is clear.

6) Base VID (12 bits)

The Base VID that associated the SPT Set via the ECT-ALGORITHM.

7) SPVID (12 bits)

The SPVID is the Shortest Path VID when using SPBV mode. In SPBM mode it is not used and should be set to 0.

28.12.6 SPB Instance Opaque ECT Algorithm sub-TLV

There are multiple ECT Algorithms defined for SPB, and in the future additional algorithms may be defined. These algorithms may use this optional sub-TLV (Figure 28-7) to carry new algorithm parameters, for example, tie-breaking data. There are two broad classes of ECT Algorithm: one which uses nodal data to

- 1 break ties and one which uses link data to break ties. This sub-TLV can associate opaque data with a node.
2 This sub-TLV is carried in an MT-Capability TLV in an LSP (along with a valid SPB Instance sub-TLV).
3 Multiple copies of this sub-TLV may be carried for different ECT-ALGORITHM values related to the node.

	Octet	Length
Type (2)	1	1
Length	2	1
ECT-ALGORITHM	3–6	4
ECT Information	7–(Length+2)	variable

Figure 28-7—SPB Instance Opaque ECT-ALGORITHM sub-TLV

- 4 a) Type (8 bits) Value 2
5 b) Length (8 bits)
6 Total number of bytes contained in the value field.
7 c) ECT-ALGORITHM (4 bytes)
8 ECT-ALGORITHM is advertised when the Bridge supports a given ECT-ALGORITHM (by OUI or
9 CID/Index) on a given VID.
10 d) ECT Information (variable)
11 ECT-ALGORITHM Information of variable length.

12 28.12.6.1 ECMP ECT Algorithm sub-TLV

13 This TLV (Figure 28-8) is not required for SPBM symmetric ECT operation and is optional for SPBM
14 ECMP. This sub-TLV can be used to advertise a Bridge Priority value (most significant 2 bytes of the Bridge
15 Identifier) to be used in shared tree root Bridge selection (44.1.2) when using the normal Bridge Priority
16 (13.26.2) will not produce the desired result. Since Bridge Priority influences both source rooted tree
17 structure and shared tree root selection, this optional sub-TLV is provided for situations in which the desired
18 outcome for both of these selections cannot be achieved using a single Bridge Priority. This sub-TLV
19 associates an alternate Bridge Priority [item c) in 12.25.14.1.2] with a node for use with a particular
20 Tie-Break Mask. The alternate Bridge Priority is used instead of the normal Bridge Priority when the
21 associated Tie-Break Mask is used to select a shared tree root Bridge. This enables greater control over root
22 Bridge selection for shared trees, when needed. This sub-TLV is carried in an MT-Capability TLV in an LSP
23 (along with a valid SPB Instance sub-TLV).

		Octet	Length
Tuple 1	Type (2)	1	1
	Length	2	1
	ECT-ALGORITHM (00-80-C2-11)	3–6	4
	Tie-Break Mask	7	4 bits
	reserved	7	4 bits
...	Bridge Priority	8–9	2
Tuple n	Tie-Break Mask	(3n+4)	4 bits
	reserved	(3n+4)	4 bits
	Bridge Priority	(3n+5)–(3n+6)	2

Figure 28-8—ECMP ECT-ALGORITHM sub-TLV

- 24 a) Type (8 bits) Value 2
25 b) Length (8 bits)
26 Total number of bytes contained in the value field.
27 c) ECT-ALGORITHM (4 bytes)

- 1 ECT-ALGORITHM is 00-80-C2-11 for ECMP.
- 2 d) Tie-Break Mask (4 bits)
- 3 A 4-bit mask that is repeated to form a 64-bit mask used in selecting the low Bridge Identifier from
- 4 a set of Bridge Identifiers. This selection process is used to select a shared tree root Bridge.
- 5 e) Bridge Priority (16 bits)
- 6 Bridge priority is the 16 bits that together with the low 6 bytes of the SPB System Identifier form the
- 7 Bridge Identifier. This Bridge Priority is used to form the Bridge Identifier used to select a shared
- 8 tree root bridge using the mask identified by the accompanying Tie-Break Mask field.

9 28.12.7 SPB Link Metric sub-TLV

10 This sub-TLV (Figure 28-9) is included in an Extended IS Reachability TLV (type 22) [IETF RFC 5305] or

11 MT Intermediate Systems TLV (type 222) [IETF RFC 5120]. If this sub-TLV is not present for an IS-IS

12 adjacency then that adjacency must not carry SPB traffic for the given topology instance. The maximum

13 metric value ($2^{24} - 1$) signifies that the link shall not be used for SPB traffic (it is treated as “SPB

14 operationally down”).

15 NOTE—In ECMP the elimination of symmetry requirements could allow the use of a different metric for each link

16 direction; however, to maintain consistency ECMP shortest path algorithms continue to use the maximum metric value

17 in cases where the metrics advertised by adjacent nodes for a given link are different.

	Octet	Length
Type (29)	1	1
Length (6)	2	1
SPB Link Metric	3–5	3
Number PORTs	6	1
Port Identifier	7–8	2

Figure 28-9—SPB Link Metric sub-TLV

- 18 a) Type (8 bits) Value 29
- 19 b) Length (8 bits)
- 20 Total number of bytes contained in the value field.
- 21 c) SPB Link Metric (24 bits)
- 22 This indicates SPB Link Metric the administrative cost or weight of using this link as a 24-bit
- 23 unsigned number. Smaller numbers indicate lower weights and are more likely to carry SPB traffic.
- 24 Only one metric is allowed per SPB instance per link. If multiple metrics are required multiple SPB
- 25 instances are required, either within IS-IS or within several independent IS-IS instances. If there are
- 26 multiple links the metric for all links are the same.
- 27 d) Number PORTs (8 bits)
- 28 Number PORTs is the number of Port identifiers associated with this link.
- 29 e) Port Identifier (16 bits)
- 30 Port Identifier is the unique Port identifier comprising two parts, the Port Number and the Port
- 31 Priority field (13.27.47).

32 28.12.8 SPB Adjacency Opaque ECT Algorithm sub-TLV

33 There are multiple ECT Algorithms defined for SPB, and in the future additional algorithms may be defined.

34 The SPB Adjacency Opaque ECT Algorithm sub-TLV (Figure 28-10) may be included in an Extended IS

35 Reachability TLV (type 22) or MT Intermediate System TLV (type 222) to carry new algorithm parameters

36 associated with an adjacency. Multiple copies of this sub-TLV may be carried for different ECT Algorithms

37 related to the adjacency.

1

	Octet	Length
Type (30)	1	1
Length	2	1
ECT-ALGORITHM	3–6	4
ECT Information	7–(Length+2)	variable

Figure 28-10—SPB Adjacency Opaque ECT-ALGORITHM sub-TLV

- a) Type (8 bits) Value 30
- b) Length (8 bits)
Total number of bytes contained in the value field.
- c) ECT-ALGORITHM (4 bytes)
ECT-ALGORITHM is advertised when the Bridge supports a given ECT-ALGORITHM (by OUI or CID/Index) on a given VID.
- d) ECT Information (variable)
ECT-ALGORITHM Information of variable length.

28.12.9 SPBV MAC address sub-TLV

The SPBV MAC Address sub-TLV (Figure 28-11) is carried in an MT-Capability TLV in an LSP. It should be used for advertisement of group MAC addresses in SPBV mode. Individual MAC addresses will normally be distributed by reverse path learning, but carrying them in this sub-TLV is not precluded. It has the following format:

		Octet	Length
MAC Address Tuple 1	Type (4)	1	1
	Length	2	1
	reserved	3	2 bits
	S-R	3	2 bits
	SPVID	3–4	12 bits
	T	5	1 bit
	R	5	1 bit
	reserved	5	6 bits
	MAC Address	6–11	6
MAC Address Tuple n	...		
	T	(7n–2)	1 bit
	R	(7n–2)	1 bit
	reserved	(7n–2)	6 bits
	MAC Address	(7n–1)–(7n+4)	6

Figure 28-11—SPBV MAC Address sub-TLV

- a) Type (8 bits) Value 4
- b) Length (8 bits)
Total number of bytes contained in the value field.
- c) SR bits (2 bits)
The SR bits are the service requirement parameter from MMRP. The service requirement parameters have the value 0 (Forward all Groups) (10.12.1.7) and 1 (Forward All Unregistered Groups) defined. However, this attribute may also be missing. So the SR bits are defined as 0 not declared, 1 Forward all Groups and 2 Forward All Unregistered Groups.

- 1 d) SPVID (12 bits)
2 This is the SPVID and by association Base VID and the ECT-ALGORITHM and SPT Set that the
3 MAC addresses defined below will use. If the SPVID is not allocated the SPVID Value is 0.
4 NOTE—If the ECT algorithm in use is a spanning tree algorithm, this value should be populated with the Base
5 VID.
- 6 e) T Bit (1 bit)
7 This is the Transmit allowed Bit for the following group MAC address. This is an indication that
8 SPBV group MAC address with SPVID of source should be populated (for the Bridge advertising
9 this Group MAC), and installed in the FDB of transit Bridges, when the Bridge computing the trees
10 is on the corresponding ECT-ALGORITHM shortest path between the Bridge advertising this group
11 MAC address with the T bit set, and any receiver of this group MAC address. A Bridge that does not
12 advertise this bit set for a group MAC address should have no forwarding state installed for traffic
13 originating from that Bridge on other transit Bridges in the network.
- 14 f) R Bit (1 bit)
15 This is the Receive allowed Bit for the following group MAC address. This is an indication that
16 SPBV group MAC addresses as receiver should be populated (for Bridges advertising this group
17 MAC address with the T bit set) and installed when the Bridge computing the trees lies on the
18 corresponding shortest path for this ECT-ALGORITHM between this receiver and any transmitter
19 on this group MAC address. An entry that does not have this bit set for an group MAC address is
20 prevented from receiving on this group MAC address because transit Bridges will not install
21 multicast forwarding state towards it in their FDBs or the traffic is explicitly filtered.
- 22 g) MAC Address (48 bits)
23 The MAC Address is either a group address or an individual address. When the MAC address is a
24 group address it declares this Bridge as part of the multicast interest for this destination MAC
25 address. Multicast trees can be efficiently constructed for destination by populating Group Address
26 FDB entries for the subset of the SPT that connects the Bridges supporting the multicast address.
27 This replaces the function of MMRP for SPTs. The T and R bits above have meaning if this is a
28 group address. Individual addresses are populated only as if the R bit was set. (Note that individual
29 addresses could also be learned normally.)

30 **28.12.10 SPBM Service Identifier and Unicast Address (ISID-ADDR) sub-TLV**

31 This sub-TLV (Figure 28-12) declares an individual B-MAC address and maps I-SIDs in the context of a
32 B-VID to that B-MAC, allowing automatic creation of efficient group trees that are subsets of the SPT
33 rooted at the node identified by that individual B-MAC address. In a symmetric ECT environment, the
34 I-SIDs are mapped to a B-VID that is associated with a symmetric ECT Algorithm specifying the SPT Set.
35 In ECMP the I-SIDs are mapped to a B-VID that is associated with the ECMP ECT Algorithm, which may
36 specify a source rooted SPT or a shared tree for group addressed frames. Multicast trees can be selected per
37 I-SID for maximum diversity. In the case of explicit trees, the I-SID is mapped to a B-VID allocated to one
38 of the explicit ECT Algorithms (Table 45-1), and the B-MAC can be either an Individual MAC address or
39 null. The null value indicates that the Backbone Service Instance Group address (26.4) is used by all sources
40 of multicast traffic onto the given I-SID in (*,G) mode on one simple tree as specified in 45.1.5. This
41 sub-TLV is carried in an MT-Capability TLV in an LSP.

- 42 a) Type (8 bits) Value 3
43 b) Length (8 bits)
44 Total number of bytes contained in the value field.

		Octet	Length
	Type (3)	1	1
	Length	2	1
	B-MAC Address	3–8	6
	reserved	9	4 bits
	Base VID	9–10	12 bits
I-SID Tuple 1	T	11	1 bit
	R	11	1 bit
	Ts	11	1 bit
	reserved	11	1 bit
	Tie-Break Mask	11	4 bits
	I-SID	12–14	3
	...		
I-SID Tuple n	T	(4n+7)	1 bit
	R	(4n+7)	1 bit
	Ts	(4n+7)	1 bit
	reserved	(4n+7)	1 bit
	Tie-Break Mask	(4n+7)	4 bits
	I-SID	(4n+8)–(4n+10)	3

Figure 28-12—SPBM Service Identifier and Unicast Address sub-TLV

- 1 c) B-MAC Address (48 bits)
- 2 B-MAC Address is an individual MAC address on this Bridge. It may be either the single nodal
- 3 address, or may address a port or any other level of granularity relative to the Bridge. In the case
- 4 where the Bridge only has one B-MAC address this may be the same as the MAC address portion
- 5 SPB System Identifier of the Bridge. To add multiple B-MAC addresses, this sub-TLV shall be
- 6 repeated for each additional B-MAC address.
- 7 Each individual MAC address has one or more I-SIDs that are associated with this address. In PBB
- 8 terms this corresponds to a PIP B-MAC address. All B-MAC addresses that are used for individual
- 9 MAC addresses and for the Sources of Group MACs shall be associated with at least one I-SID. This
- 10 sub-TLV is initiated by IB-BEBs.
- 11 d) Base VID (12 bits)
- 12 The Base VID identifies the B-VID and by association the ECT-ALGORITHM and SPT Set that the
- 13 I-SIDs defined below will use.
- 14 e) T Bit (1 bit)
- 15 This is the Transmit allowed Bit for the following I-SID. This is an indication that a group MAC
- 16 Address Registration Entry (8.8.4) for this I-SID as source should be constructed and installed in the
- 17 FDB of a Bridge, if the Bridge is on the corresponding ECT-ALGORITHM path between the Bridge
- 18 advertising this I-SID with the T bit set and any receiver of this I-SID. A Bridge that does not
- 19 advertise this bit or the Ts bit set for an I-SID should have no forwarding state installed for traffic
- 20 originating from that Bridge on other Bridges in the network.
- 21 f) R Bit (1 bit)
- 22 This is the Receive allowed Bit for the following I-SID. This is an indication that a group MAC
- 23 Address Registration Entry (8.8.4) for this I-SID as receiver should be constructed and installed (for
- 24 Bridges advertising this I-SID with the T bit or Ts bit set) if a Bridge lies on the corresponding path
- 25 for this ECT-ALGORITHM between this receiver and any transmitter on this I-SID. An entry that
- 26 does not have this bit set for an I-SID is prevented from receiving on this group MAC addresses for
- 27 this I-SID because Bridges will not install multicast forwarding state towards it in their FDBs.

g) Ts Bit (1 bit)

The Ts bit indicates that this I-SID source uses a shared tree for multicast frames. The group MAC address for this I-SID source is constructed using the Tie-Break Mask and the I-SID (27.15, Figure 27-6). A group MAC Address Registration Entry (8.8.4) is installed in any bridge on the shared tree, selected by the Tie-Break Mask, between the advertising bridge and bridges that have the R bit set for the same I-SID. If both the T bit and Ts bit are set, the Ts bit takes precedence. This bit is ignored by ECT-ALGORITHM values in Table 28-1.

h) Tie-Break Mask (4 bits)

A 4-bit mask that is repeated to form a 64-bit mask used in selecting the low Bridge Identifier from a set of Bridge Identifiers. This selection process is used in multicast tree calculations to select a shared tree root Bridge and parent nodes in SPTs where equal cost paths are available. This field is ignored by ECT-ALGORITHMS in Table 28-1.

i) I-SID (24 bits)

I-SID is the 24-bit group service membership identifier. If two Bridges have an I-SID in common, intermediate Bridges on the unique shortest path between them will create forwarding state for the related B-MAC addresses. They will also construct multicast forwarding state using the I-SID and the Bridge's SPSrcID to construct a multicast DA. Each I-SID has a Transmit (T) and Receive (R) bit that indicates if the membership is as a Transmitter/Receiver or both (with both bits set). In the case where the Transmit (T) and Receive (R) bits are both zero, the I-SID is ignored (this forces a point-to-point capability that does not require Group address installation, but the advertised individual B-MAC address shall be processed and installed if required on the SPT). If more I-SIDs are associated with a particular B-MAC address than can fit in a single sub-TLV, this sub-TLV can be repeated with the same B-MAC address and MTID but with additional I-SID values. I-SID values can be in any order. Duplicate I-SIDs for a B-MAC address are allowed (within a sub-TLV, across two or more sub-TLVs, across two or more LSP fragments, or even across two or more LSPs). If the duplicates are in different B-VIDs, they are processed; otherwise, the T and R bits processed are the logical OR of the T and R bits of the two entries and it is processed as a single entry. (I-SID=0 is an exception that is always ignored, and allows a fast method to remove an I-SID without restructuring all the LSPs). Repacking an LSP (when I-SID=0 entries should be collapsed) can be done by moving entries from the high end into the empty I-SID=0 entries. To avoid outages temporary duplicates are allowed during this repacking and the entry further up in the LSP may later be removed without impact. The I-SID value 0x0000ff is reserved for SPB broadcast to all Bridges.

Item d), item h), and item i) are set to zero on transmit and ignored on receipt for SPBM symmetric ECT operation.

29. DDCFM operations and protocols

29.1 Principles of DDCFM operation

Data-driven and data-dependent connectivity fault management (DDCFM) comprises tools to facilitate the diagnosis and isolation of data-driven and data-dependent faults (DDFs) in Virtual Bridged Networks. This clause describes the functions of DDCFM and how they can be operated and managed. DDCFM is an extension to CFM specified by Clause 18 through Clause 22. As in the case of CFM, DDCFM capabilities can be used in networks operated by multiple independent organizations, each with restricted management access to others' equipment.

29.1.1 Data-driven and data-dependent faults (DDFs)

There are two broad types of faults in Virtual Bridged Networks that affect only certain frames or sequences of frames. Simple data-dependent faults are those that result in the repetitive loss or misdirection of each of those frames, independently of any other frames, and are usually the result of simple misconfiguration or of a failure to appreciate the consequences of a configuration option (installing protocol-specific filters, for example). Data-driven faults are more complex: the presence (or absence) of some data frames causes or contributes to the loss of other frames. While the services supported by Virtual Bridged Networks are conceptually data independent, the use of data-driven techniques enables enhanced service delivery. To give three examples: multicast frame filtering and consequent bandwidth saving is facilitated by Internet Group Membership Protocol (IGMP) snooping; stateful firewalls are used to protect users connected to managed services; and efficient allocation of frames to the individual links of an aggregation (IEEE 802.1AX Link Aggregation) is often based on spotting conversations by looking at frame data.

29.1.2 Basic principle to diagnose and isolate DDFs

The basic principle to diagnose DDFs is to isolate them to a small enough segment of the network, such as a Bridge Port, a member port of Link Aggregation Group (LAG) within a Bridge Port, or a LAN. Once DDFs are isolated, understanding the cause of the fault at this location becomes much easier.

The basic procedure to achieve isolation of a DDF is to divide the network into segments and determine whether certain data frames can traverse through each segment as expected. When a network segment is identified as faulty, the segment is further divided into smaller segments until a Bridge, a Bridge Port, or a member port of LAG within a Bridge Port is identified as not passing data frames as expected.

A DDF may not be apparent in the absence of live traffic (that is, when test data are used). DDCFM is designed to carry out the diagnosis while live traffic is being exchanged.

DDCFM facilitates diagnosis and consequent isolation of DDFs with the following two techniques: forward path test (FPT) and return path test (RPT).

29.1.2.1 Forward path test (FPT)

The goal of FPT, as depicted in Figure 29-1, is to determine whether specified data frames (frames associated with a service instance, or data frames with certain destination addresses or VID, etc.) can reach a particular location (which could be a Bridge Port, or a member port of LAG within a Bridge Port) without error. FPT's Reflection Responder (RR) (29.2.2) reflects the identified data frames (e.g., a service instance or selected data frames) to a specific target location, which could be an end station, a Bridge, or a test device. The data frames to be reflected can also be continued to their original destination(s). This option, called the Continue option throughout this standard, is to support testing of a data stream in a manner transparent to the application sourcing or sinking that data stream.

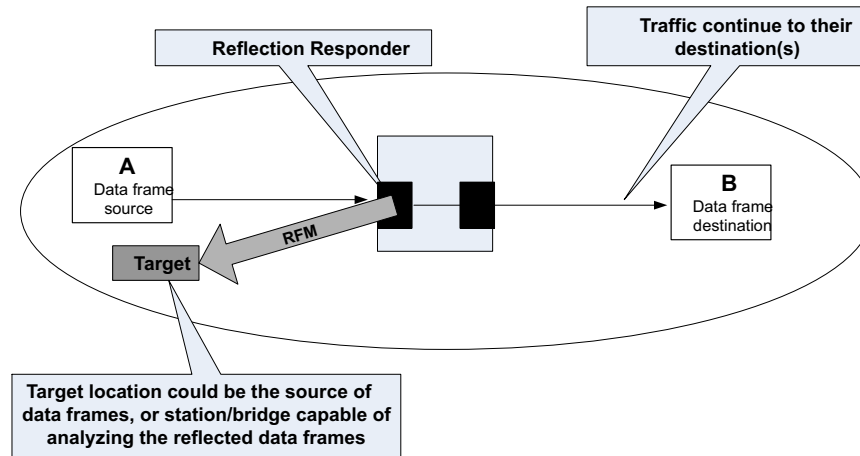


Figure 29-1—Forward path test (FPT)

1 FPT can be configured on ports where no MPs exist. However, when they are implemented within MPs or
2 configured on ports with MPs, better fault isolation can be achieved. For example, if the target location does
3 not receive the expected reflected data frames in the specified time span, it can send a LBM (21.7) to verify
4 the connectivity between the Target location and the MP on which the RR (29.2.2) is configured. For a
5 network that supports MEPs but not MIPs, FPT can be used to diagnose segment connectivity by
6 configuring an RR on an intermediate node to encapsulate the received CCM in RFM format (29.4.2) and
7 forward them to a target location.

8 The receiver at the target location records the reflected frames for examination by the operator and may also
9 transfer the reflected frames to an analysis function. Some examples of target location analyzers are as
10 follows: comparing the reflected frames with the original ones to verify if there are any errors, running a
11 proxy application at the target location to simulate the handshakes as if those frames actually reach their
12 original destinations, etc.

13 As in CFM (Clause 18 through Clause 22), a network operator can only set or activate FPT's RR within its
14 own maintenance domain. At the customer's request, a network provider may use DDCFM to verify
15 whether DDFs occur within its domain. FPT's RR (29.2.2) can only be created or activated by a network
16 provider's own Network Management System via secure interface. Therefore, customers or other operators
17 cannot set an FPT RR (29.2.2) in a maintenance domain that does not belong to them.

18 In order for intermediate Bridges and/or a target location to distinguish the reflected data frames from other
19 traffic, each reflected frame is encapsulated with an RFM header. The RFM header added to the reflected
20 data frame carries the same Maintenance Domain level associated with the RR, so that the reflected data
21 frames, each encapsulated with a proper RFM header, are contained within the same Maintenance Domain.
22 The target location has to be in the same maintenance domain to receive the RFMs.

23 Depending on the conditions specified for selecting data frames to be reflected, some conditions can cause
24 large amount of data frames to be reflected. If there are multiple locations within a network reflecting a large
25 number of data frames simultaneously, excessive traffic could be added to some segments of the network,
26 which may impact network performance. To avoid excessive traffic being added to the network by an RR,
27 this standard recommends only one RR be activated at a time within one maintenance domain.

28 FPT consists of RR configuration, the action to reflect identified data frames, and the analysis of the
29 reflected data frames. The last step, i.e., analysis of the reflected data frames, is beyond the scope of this
30 standard.

29.1.2.2 Return path test (RPT)

The RPT is to determine whether a flow of data frames can be sent without error from a specific location within a network to a station or stations specified by the destination_address associated with each of the frames under test. The RPT is achieved by an originating station encapsulating each frame of the flow under test with an SFM header, a DR (29.2.6) removing the SFM header, optionally placing its own MAC in the source_address field of the decapsulated data frames, and sending the data frames to a station or stations as specified by the destination_address field. By default, RPT's DR places its own MAC address in the source_address field of the decapsulated data frames before sending them out. However, a network operator can force a DR not to change the source_address field of the decapsulated data frame before sending it out to conduct special purpose testing.

The RPT allows a network operator to inject specific data frames into the network for testing purposes. To prevent one network operator from injecting traffic to other networks, it is necessary for the associated managed object to be created under secure mode and for SFM originator to be in the same Maintenance Domain as the MP on which DR is defined.

The Figure 29-2 illustrates a simple case of injected traffic from Node B to reach Node A. Multiple data frames can be injected and injected data frames could have different addresses or multicast addresses in their corresponding destination_address field.

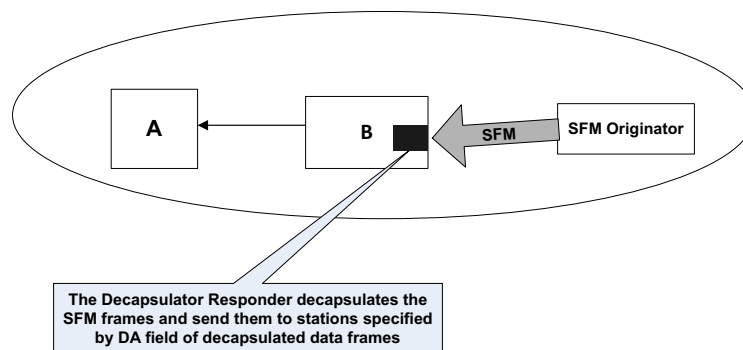


Figure 29-2—Return path test (RPT)

The RPT consists of the DR configuration, the sending of SFMs, the action to decapsulate and forward, and the analysis of the received data frames at their destination(s). The last step, i.e., the analysis of the received data frames at their destination(s), is beyond the scope of this standard.

29.1.2.3 Derived testing scenarios

FPT and RPT can be used together in various ways to achieve more sophisticated testing to diagnose and isolate DDFs. It is beyond the scope of this standard to elaborate different ways of combining FPT and RPT. This subclause only illustrates one example of using both FPT and RPT.

Figure 29-3 shows how to achieve segment fault isolation in the middle of a network using FPT and RPT together. By creating a DR and RR on two Bridge Ports in the middle of a network, a network operator can test whether the specified data frames can traverse through the segment. This type of testing is especially useful to diagnose DDFs out of firewall. This type of testing can also be used to diagnose a connectivity fault between two intermediate Bridge Ports (or two MIPs) by setting the RR filter (29.2.2.1) to the specified CCMs.

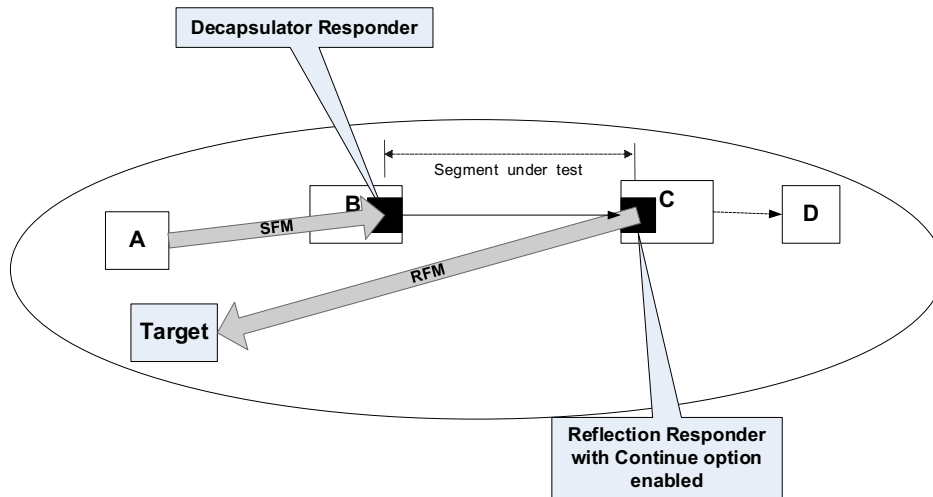


Figure 29-3—Combination of FPT and RPT

29.2 DDCFM Entity operation

This subclause specifies the following:

- a) FPT RR
- b) FPT RFM Receiver
- c) RPT DR
- d) RPT frames' SFM Originator

29.2.1 DDCFM implementation

The DDCFM protocol is performed by the RR, the DR, the RFM Receiver, and the SFM Originator. All of these entities are created and activated by operator commands. The operation of RR, DR, and SFM Originator are timer limited.

Both RFMs and SFMs are CFM PDUs and their format is described in 21.4 and 29.4. They are forwarded in the same way as regular CFM messages. An MEP at higher MD level shall drop the received RFMs and SFMs. An MEP at the same level shall process, but not forward, RFMs and SFMs received from its Active SAP, and shall drop RFMs/SFMs received from its Passive SAP. An MHF shall always forward RFMs/SFMs received from its SAPs and shall also process those at the same MD level received from its Active SAP.

The RR, RFM Receiver, DR, and SFM originator are CFM entities that are associated with a specific MD, enabling access only to the administrators of this MD. They can be placed at any point in the network that is bounded by any DoSAP of their corresponding MDs. Their relationship to MPs is guided by the MP component entities that the DDCFM entities also require. In particular, the RR does not require any of the MP's sub-functions (19.2 and 19.3) and correspondingly it is defined as an independent CFM shim. It can be placed at any Bridge Port bounded by the DoSAP of the RR's associated MD. It does not even require the EISS Multiplex Entity (6.17) or Backbone Service Instance Multiplex Entity (6.18). The same holds true for an SFM Originator. Note that both of them transmit CFM frames that are associated with a specific MA but the creation of these DDCFM entities themselves is not associated with an MA. This in practice means that one RR or SFM Originator can send RFMs or SFMs (respectively), which are associated with different MAs than the encapsulated frames. The other two DDCFM entities have a number of common subentities to the MPs. In particular, on top of their unique (MP) RFM Receiver and (MP) Decapsulator Responder state

machines, they require the functions provided by the Type Demultiplexer, Level Demultiplexer, and the OpCode Demultiplexer. In addition if the service is VLAN based, they also require the EISS Multiplex Entity. All of these make the implementation of the RFM receiver or the DR very simple in the case of an MP where they can reuse the MP component entities and provide the additional functions required by the MP RFM Receiver or MP Decapsulator Responder entities. Implementing these DDCFM entities on non-MPs would require all the previously mentioned MP subentities. Figure 29-5 shows an MP-independent RFM Receiver and Figure 29-6 shows an MP-independent DR. On the other hand, implementing these DDCFM entities on MPs would require only the addition of the MP Decapsulator Responder and MP RFM Receiver component entities in the MP architecture with the appropriate changes in the Opcode Demultiplexer as specified in 19.2 and 19.3.

29.2.2 FPT RR

The RR is a CFM protocol shim for filtering frames to be reflected, copying the filtered frames to the Passive SAP if the Continue option is set, and encapsulating each filtered frame with the RFM header (29.4.2). For an interface supporting DDCFM, an RR shim shall be allowed at EISS/ISS SAPs where MPs are allowed on the protocol stack shown in Figure 22-4, Figure 22-8, Figure 26-2, and Figure 26-8. By placing an RR shim on an aggregated port within a Bridge Port, the RR can reflect data frames from an member link of a LAG to test if specific data frames can go through the link.

The Figure 29-4 illustrates the component entities of an RR.

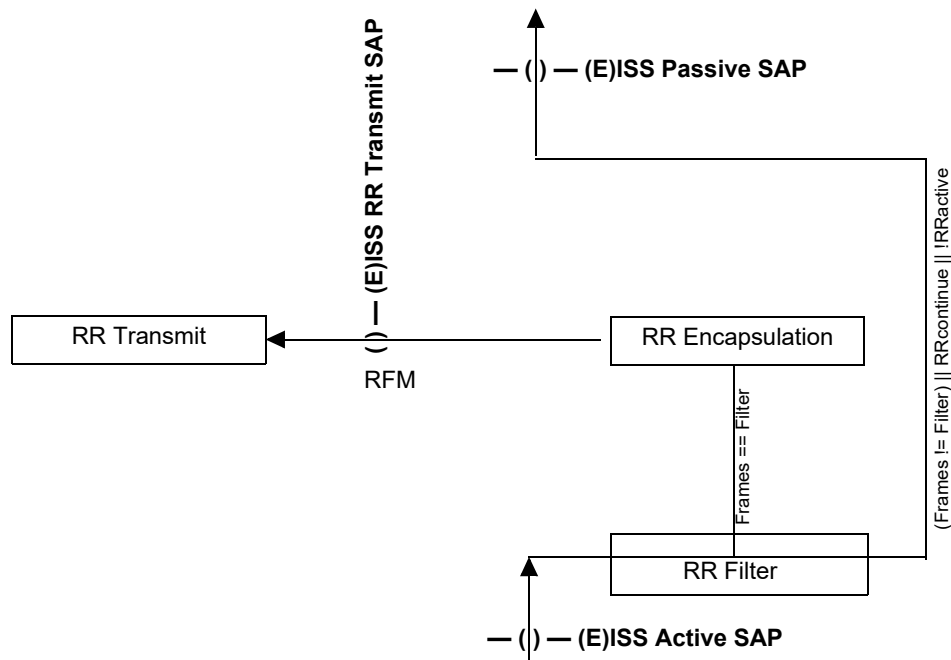


Figure 29-4—Detailed functions of RR

29.2.2.1 RR Filter

The RR Filter function within the RR filters frames from Active SAP that match the Reflection Filtering definition(s) [item b2) in 29.2.3]. When the RR is active and the sampling interval has expired, a frame that matches the Reflection Filter definition(s) is sent to RR Encapsulation to be reflected. A frame is passed to the Passive SAP when RR is not active, the frame does not match the Reflection Filter definition(s), or the Continue option is true.

1 To prevent data frames from being reflected multiple times within a network, RFM frames cannot be
2 reflected. An RR Filter definition implicitly includes the condition of the OpCode in common CFM header
3 not equal to RFM, in addition to all other conditions specified.

4 **29.2.2.2 RR Encapsulation**

5 The RR Encapsulation function within the RR is for encapsulating the filtered frame with the appropriate
6 RFM header. If a sampling interval is specified, once a frame meeting the Reflection Filtering definitions is
7 identified within one sampling interval, no more frames shall be encapsulated until the next sampling
8 interval starts.

9 **29.2.2.3 RR Transmit**

10 RR Transmit identifies the egress port by querying FDB and forwards the RFM frames to the LOM entity of
11 the identified egress port. The query uses the Reflection Target address [item b3) in 12.17.2.1.2] and the
12 vlan_identifier associated with the RFM [item c) in 29.3.3.3]. If there is no vlan_identifier associated with
13 the RFM, the PVID value of the port where RR is configured is used. If an egress port cannot be identified
14 and the FloodingEnabled flag of the RR is true, then the RFM frame is sent to the LOM entity on every port
15 associated with the VID set of the RFM frame. If the FloodingEnabled flag of the RR is false and an egress
16 port cannot be identified, the RFM frame is dropped.

17 **29.2.3 RR-related parameters**

18 **29.2.3.1 RR identification**

19 An RR is identified by the interface upon which the RR is created, a Maintenance Domain that identifies the
20 administrative boundaries, and a value indicating the direction (Up or Down) in which the RR is facing. The
21 interface upon which the RR is created can be a Bridge Port or a member port of a LAG of a Bridge Port.

22 **29.2.3.2 MA for RR**

23 The primary VID associated with RR's MA is in the RFM emitted by the RR. When MA for an RR is set to
24 0, then the VID of the filtered frame are used in the corresponding RFM.

25 **29.2.3.3 RR Filter definition**

26 The RR Filter is for selecting data frames to be reflected. Multiple filters can be combined together by "&&
27 (and)", "|| (or)", or "! (negation)" operations.

28 All Bridges supporting DDCFM shall support the following reflection filters (and may support additional
29 Reflection Filters):

- 30 a) Reflect All—all frames are selected.
- 31 b) VLAN-based selection—all data frames associated with the specified VID are selected.
- 32 c) Destination address-based selection—all data frames with the specified destination address are
33 selected.
- 34 d) Source address-based selection—all data frames with the specified source address are selected.
- 35 e) EtherType.

1 **29.2.3.4 Sampling interval**

2 The sampling interval controls the maximum rate at which frames can be reflected. If no sampling interval is
3 set, then all frames that arrive while the RR is active and match the filter criteria are reflected. If a sampling
4 interval is specified then the first matching frame that arrives during the interval is reflected, other matching
5 frames arriving in that interval are not reflected.

6 **29.2.3.5 Continue option**

7 The Continue option [item b5) in 12.17.2.2.2] allows the selected data frames to be continued to their
8 original destinations by making a copy of the selected frames to be reflected.

9 The Continue option is set to be true by default, which means that the selected data frames are branched to
10 two directions—one direction to the RR Encapsulation function, and the other direction to the passive SAP.
11 When the Continue option is set to be false, the selected frames shall only be forwarded to the RR
12 Encapsulation function.

13 **29.2.3.6 Duration for RR to remain active**

14 Reflecting data frames back into network can create extra traffic in the network. To prevent this action
15 running forever, a duration [item b7) in 12.17.2.2.2] is used to limit the maximum amount of time for the RR
16 to be running. Operator can also specify the maximum number of frames to be reflected. Once activated, RR
17 shall remain active until either its specified duration expires or the maximum number of frames reflected is
18 reached, whichever comes first. An active RR can also be deactivated by operator command before its
19 duration expires.

20 **29.2.3.7 Truncation flag**

21 If the received data frame causes the RFM encapsulated frame to exceed the Maximum Service Data Unit
22 Size (6.5.8), either the data frame is truncated, or the data frame is split into two smaller frames and those
23 two smaller frames are encapsulated by two separate RFM frames. If the Truncation flag is set to be true, the
24 data frame shall be truncated to be encapsulated in one RFM.

25 **29.2.4 Reflection Target and RFM Receiver**

26 The Reflection Target is where the RFM encapsulated reflected frames are forwarded to. The RFM Receiver
27 is a function running on a Reflection Target to pass the received RFMs to their corresponding analyzers.

28 The Reflection Target has to be within the same Maintenance Domain as the RR in order to receive the RFM
29 encapsulated data frames.

30 When an RFM Receiver is defined on a Bridge interface that does not support MP, the RFM Receiver is
31 implemented as a separate CFM entity, as depicted in Figure 29-5 for receiving RFMs from Active SAP. It
32 requires a number of the component entities that are required by an MP. If the RFMs are associated with a
33 VID-based MA, the RFM receiver needs to be placed in a demultiplexer SAP provided by an EISS
34 multiplex entity (6.17).

35 **29.2.5 RPT-related parameters**

36 RPT configuration has two parts: one part is to configure SFM Originator and the other part is to configure
37 DR.

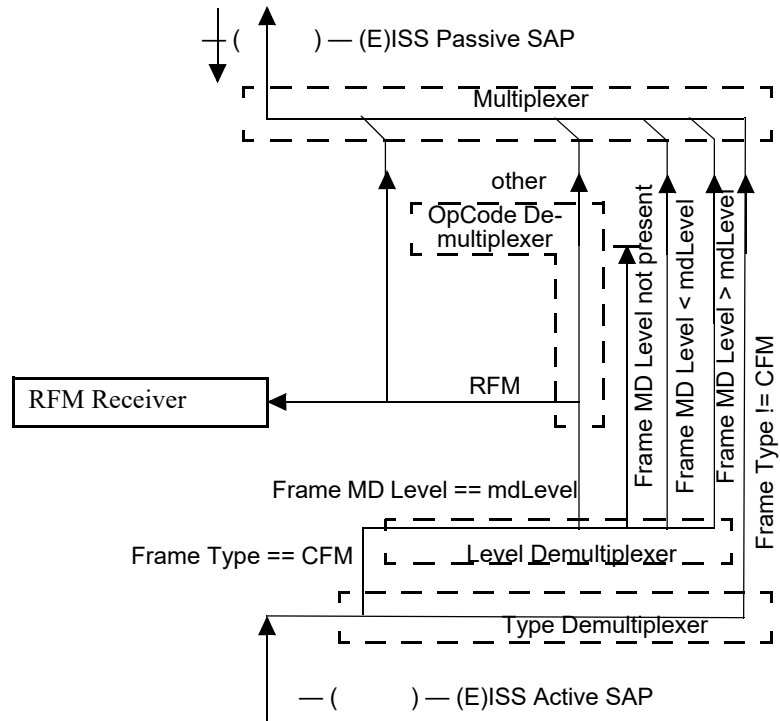


Figure 29-5—RFM Receiver on a non-MP

29.2.6 Decapsulator Responder (DR)

The DR decapsulates SFMs and sends the decapsulated frames toward destinations specified by the destination_address field of the decapsulated data frames. Under normal circumstance, the source_address field of the decapsulated frame is replaced by the DR's own MAC address to avoid confusing other Bridges' learning in the network. However, a network operator can enforce the DR not to modify the source_address field of the decapsulated frame for special purpose fault diagnosis. The DR configuration is specified in 12.17.4.1.

DR shim shall be allowed at EISS/ISS SAPs where MPs are allowed on the protocol stack shown in Figure 22-4, Figure 22-8, Figure 26-2, and Figure 26-8.

Multiple DRs can be created on one Bridge interface to receive SFMs at different MD levels or SFMs originated from different SFM Originators.

When an SFM is received by an active DR, it is examined for validity and discarded if invalid. If valid, the received SFM shall be decapsulated. The source_address field of the decapsulated frame is replaced by DR's own MAC address unless SourceAddressStayFlag is set. The modified frame shall be used to identify the egress port by querying the FDB. The query uses the destination_address and vlan_identifier values of the decapsulated data frame. If the decapsulated frame is untagged, the PVID value of the port where DR is configured is used. The Boolean FloodingEnabled flag of the DR indicates if the frame is to be flooded or discarded if an egress port cannot be identified by FDB. Figure 29-6 illustrates the detailed functions of DR. DR shim can be placed at any places where MPs are allowed on the protocol stack as shown in Figure 22-8.

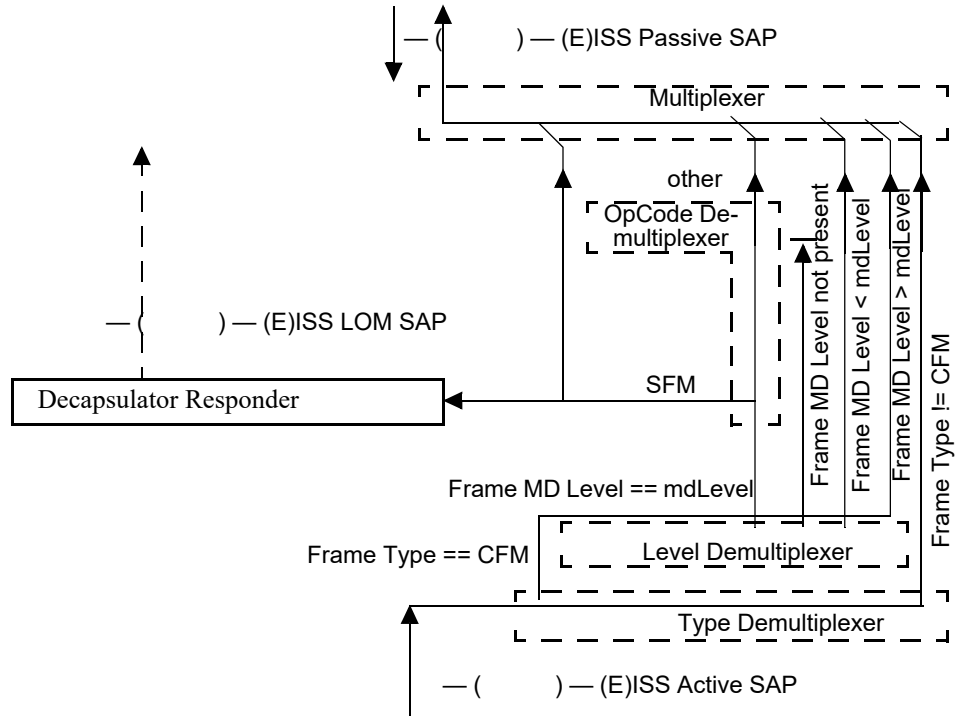


Figure 29-6—Return Path DR

29.2.7 SFM Originator

SFM Originator is for generating SFMs and sending the SFMs to the corresponding DR. Each SFM encapsulates one data frame. The SFM Originator configuration is specified in 12.17.5.1.

The Bridge interface on which SFM Originator is created does not have to support any MP. However, the SFM Originator has to be configured with the same Maintenance Domain level as the DR, so that the SFM can be forwarded to the DR. It is beyond the scope of this standard to address how SFMs are generated.

29.3 DDCFM protocols

This subclause specifies the DDCFM protocols in terms of a number of state machines, the variables and procedures associated with each machine.

29.3.1 RR variables

There is one instance of the following parameters per RR, except the nPendingRFMs variable, that is common to all RRs on the Bridge components (i.e., per Bridge component, but not per RR).

29.3.1.1 RRcontinue

A Boolean variable for the RR's Continue option [item b5) in 12.17.2.2.2] attribute.

29.3.1.2 nPendingRFMs

An integer value used to track the number of RFMs to be transmitted. nPendingRFMs is a variable that can be incremented by every RR's Encapsulation state machine and can be decremented by RR Transmit state machine.

nPendingRFMs is set to 0 when all of the RRs on the Bridge are deactivated.

1 **29.3.1.3 dataReceived**

2 dataReceived is a Boolean variable, which is set to true when there is a data frame received from (E)ISS, and
3 set to false by the RR Filter state machine.

4 **29.3.1.4 dataFrame**

5 This variable is the data frame received from (E)ISS.

6 **29.3.1.5 nFilteredFrameList**

7 This variable keeps track of the number of filtered data frames to be reflected. This variable is incremented
8 by RR Filter and decremented by RR's Encapsulation state machine.

9 **29.3.1.6 filteredFrameList**

10 This is a list of received data frames from RR Filter that have not been processed yet.

11 **29.3.1.7 nextFilteredFrame**

12 This variable copies the first one on the filteredFrameList to be reflected as RFM. Once a frame is copied to
13 this variable, the frame is removed from the filteredFrameList.

14 **29.3.1.8 RRtime**

15 RRtime is the value of RR Duration (29.2.3.6).

16 **29.3.1.9 RRwhile**

17 RRwhile is a timer variable, which is initialized to RR Duration [item b7) in 12.17.2.2.2] when RR is
18 activated and used by the RR Filter state machine to time out the active RR. RRwhile has granularity in
19 seconds.

20 **29.3.1.10 RRsampInt**

21 RRsampInt is the sampling interval (29.2.3.4) during which the RR only reflects a maximum of one frame
22 that meets the filter condition. When the sampling interval is not configured, RRsampInt is set to 0.

23 **29.3.1.11 RRsampWhile**

24 A timer variable used by RR Filter state machine to time out sampling interval.

25 **29.3.1.12 RRmaxFrames**

26 The maximum number of frames for RR to reflect after RR is activated [item b8) in 12.17.2.2.2]. Once
27 activated, RR will stay active until either RRmaxFrames is reached or RRwhile (29.3.1.9) expires,
28 whichever comes first. If RRmaxFrames is set to 0, the RR stays active until RRwhile (29.3.1.9) expires.

29 **29.3.1.13 RRframeCount**

30 A counter used by the RR's Encapsulation state machine to keep track of the number of data frames to be
31 reflected. Once RRframeCount is equal to or greater than RRmaxFrames, RR is deactivated.

32 **29.3.1.14 filterMatched**

33 A Boolean variable that is set to true if the data frame received matches with the RR's filter conditions and
34 false if the data frame received does not match the RR's filter conditions.

1 **29.3.1.15 RRactive**

2 RRactive is true when the corresponding RR is active and false otherwise.

3 **29.3.1.16 nextRFMtransID**

4 The value to place in the RFM Transaction Identifier field of the next RFM to be transmitted.
5 nextRFMtransID is incremented by 1 by the RR's processRRencap() procedure with each transmission.

6 This variable is readable as a managed object [item b14) in 12.17.2.3.3].

7 **29.3.1.17 maxDataTLVvalueSize**

8 This is the maximum number of bytes allowed to be copied into Reflected Data TLV's value field in order to
9 keep the RFM's SDU size not exceeding its Maximum Service Data Unit Size (MSDUS) (6.5.8).

10 The length of an RFM PDU is 12 octets (4 bytes for CFM header + 1 byte for End TLV(0) + 4 bytes
11 Transaction Identifier + 3 bytes Reflected Data TLV Length/Type field) + the actual number of bytes of the
12 reflected data frame. If there is no other optional TLV for the RFM data frame⁴¹, then:

13 a) $\text{maxDataTLVvalueSize} = \text{MSDUS} - (5 \text{ (CFM common header length)} + 4 \text{ (Transaction Identifier)} +$
14 $3 \text{ (Reflected Data Frame TLV Length/Type field)})$.

15 **29.3.1.18 reflectedDataLength**

16 This variable is to hold the value for the Length field of Reflected Data TLV (29.4.2.4).

17 **29.3.2 RR Filter procedures**

18 The following procedures are defined for the RR Filter state machine:

- 19 a) filterFrame()
- 20 b) forwardFrame()
- 21 c) loadFilteredFrameList()

22 **29.3.2.1 forwardFrame()**

23 This procedure passes the frame received to (E)ISS Passive SAP.

24 **29.3.2.2 filterFrame()**

25 filterFrame() is called when RRactive is true and there is a data frame received from EISS. This procedure
26 performs the following steps:

- 27 a) If the received data frame from (E)ISS SAP meets the Reflection Filter definition(s), then returns
28 true.
- 29 b) Otherwise, returns false.

30 **29.3.2.3 loadFilteredFrameList()**

31 This procedure adds the received frame to the filteredFrameList.

⁴¹ Even though not required, implementers may choose to include other optional TLVs in RFM. If extra TLVs are included in a RFM, the maxDataTLVvalueSize has to be decremented by the number of bytes taken by the extra TLV.

1 29.3.3 RR Encapsulation procedures

2 The following procedures are defined for the RR Encapsulation state machine:

- 3 a) processRRencap()
- 4 b) splitFilteredFrame()
- 5 c) constructRFM()

6 29.3.3.1 processRRencap()

7 processRRencap() is called when RR is activated and filteredFrameList is not empty. There are two local
8 variables in this procedure: DataFrame_1 and DataFrame_2, which are used when the filtered frame has to
9 be truncated or split into two frames to be reflected.

10 processRRencap() processes the incoming data frame from the RR Filter as follows:

- 11 a) Set the reflectedDataLength variable to the length of the nextFilteredFrame.
- 12 b) If the value of reflectedDataLength is larger than the Bridge's maxDataTLVvalueSize (29.3.1.17),
13 then call splitFilteredFrame (nextFilteredFrame, DataFrame_1, DataFrame_2) and then perform the
14 following two steps:
 - 15 1) If Truncation flag is set, call constructRFM(TLV_type, maxDataTLVvalueSize,
16 DataFrame_1), where the TLV_type is set to the "Truncated Data TLV" type value defined by
17 Table 21-5.
 - 18 2) If the Truncation flag is not set, perform the following two steps:
 - 19 i) Call constructRFM(TLV_type, maxDataTLVvalueSize, DataFrame_1), where the TLV_
20 type is set to the "Data Part 1 TLV" type value defined by Table 21-5.
 - 21 ii) Call constructRFM(TLV_type, reflectedDataLength – maxDataTLVvalueSize, Data-
22 Frame_2), where the TLV_type is set to the "Data Part 2 TLV" type value defined by
23 Table 21-5.
- 24 c) Otherwise, call constructRFM(TLV_type, reflectedDataLength, nextFilteredFrame), where
25 TLV_type is set to the "Data TLV" type value defined by Table 21-5.

26 29.3.3.2 splitFilteredFrame()

27 This procedure is called when the nextFilteredFrame causes the length of RFM's SDU size to be larger than
28 the Maximum Service Data Unit Size (6.5.8). There are three parameters for this procedure:

- 29 a) nextFilteredFrame, which is an input to the procedure.
- 30 b) DataFrame_1, which contains the first maxDataTLVvalueSize number of bytes from the
31 nextFilteredFrame.
- 32 c) DataFrame_2, which contains the remaining bytes from the nextFilteredFrame.

33 splitFilteredFrame() performs the following steps:

- 34 — Copy the maxDataTLVvalueSize number of bytes from the nextFilteredFrame to DataFrame_1.
- 35 — Copy the remaining bytes from the nextFilteredFrame to DataFrame_2.

36 29.3.3.3 constructRFM()

37 This procedure is to construct a RFM frame and increment the nPendingRFMs by 1. There are three
38 parameters passed into this procedure:

- 39 a) TLV_Type, which could be 3 (i.e., Data TLV) (21.5.6), 9 (Truncated Data TLV) (21.5.1.1), 10 (Data
40 Part 1 TLV), or 11 (Data Part 2 TLV).
- 41 b) TLV_Length, which is the length of the reflected data frame.
- 42 c) Data frame to be included in the TLV value field.

1 This procedure constructs the RFM (E)M_UNITDATA.request by the following steps:

- 2 d) Set the source_address parameter to the MP's or Bridge interface's own MAC address.
- 3 e) If the Reflection Target Address [item b4) in 12.17.2.2.2] is not specified, then source_address of the
4 selected data frame is copied to the RFM's destination_address parameter; otherwise, Reflection
5 Target Address is copied to the RFM's destination_address parameter.
- 6 f) Set the priority and drop_eligible parameters to the priority and drop_eligible attributes of the
7 Reflection Responder managed object (12.17.2.2.2).
- 8 g) In a VLAN-aware Bridge, the primary VID associated with the RR's MA [item b1) in 12.17.2.1.2] is
9 used as the vlan_identifier. If the RR's MA is set to 0, the vlan_identifier is determined by the
10 following:
 - 11 1) If the RR is defined on an S-VLAN component, use the S-VID if the S-VID is present in the
12 selected data frame; otherwise, use the PVID of the port where RR is configured.
 - 13 2) If the RR is defined on a C-VLAN component, use the C-VID if the C-VID is present in the
14 selected data frame; otherwise, use the PVID of the port where the RR is configured.
- 15 h) Copy RFM OpCode to the RFM's OpCode Field.
- 16 i) Copy the RR's Maintenance Domain Level [item a) in 12.17.2.1.2] to the RFM's MDLevel field.
- 17 j) Copy the first parameter (TLV_Type) to the Reflected Data TLV Type field.
- 18 k) Copy the second parameter (TLV_Length) to the Reflected Data TLV Length field.
- 19 l) Copy the third parameter (Data frame) to the Reflected Data TLV value field.
- 20 m) Copy the nextRFMtransID to the RFM Transaction Identifier field. Increment nextRFMtransID by
21 1, wrapping around from $2^{32} - 1$ to 0.
- 22 n) Increment nPendingRFMs by 1.
- 23 o) Set the frame_check_sequence parameter to the unspecified value; sets the connection_identifier to
24 null.

25 **29.3.4 RR Transmit procedure**

26 The following procedure is defined for the RR Transmit state machine:

- 27 a) xmitRFM()

28 **29.3.4.1 xmitRFM()**

29 xmitRFM() is called by the RR Transmit state machine. It queries the FDB in order to identify an egress
30 port. The set of potential transmission ports, normally created by the Active Topology enforcement (8.6.1),
31 is the set of all Bridge interfaces that are both in the active set of the vlan_identifier associated with the
32 RFM, and that are in the Forwarding state for that vlan_identifier.

- 33 a) Query FDB using the RFM's destination_address field and vlan_identifier parameter values.
- 34 b) If an egress port is identified and the identified Egress port meets the following conditions:
 - 35 1) The Egress port does not have a Down MEP with higher MD level than the RFM's MD,
 - 36 2) The Egress port does not have an Up MEP with same or higher MD level as the received
37 RFM's MD,
 - 38 1) then send an RFM EM_UNITDATA.request to the egress port's LOM SAP. Otherwise, drop
39 the received RFM frame.
- 40 c) If an egress port cannot be identified:
 - 41 2) When the FloodingEnabled flag of the RR [item b7) in 12.17.2.1.2] is not set, drop the frame.

42 When the FloodingEnabled flag of the RR is set, send M_UNITDATA.request to the LOM's SAP of every
43 forwarding port that meets the conditions 1) and 2) of item b) above.

1 Figure 29-8 describes the state machine for RR Encapsulation.

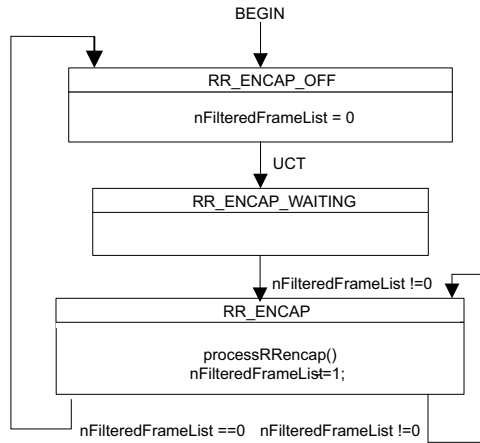


Figure 29-8—RR Encapsulation state machine

2 29.3.5.2 RR Transmit state machine

3 RR Transmit state machine (Figure 29-9) can be shared by multiple RRs activated on the Bridge.

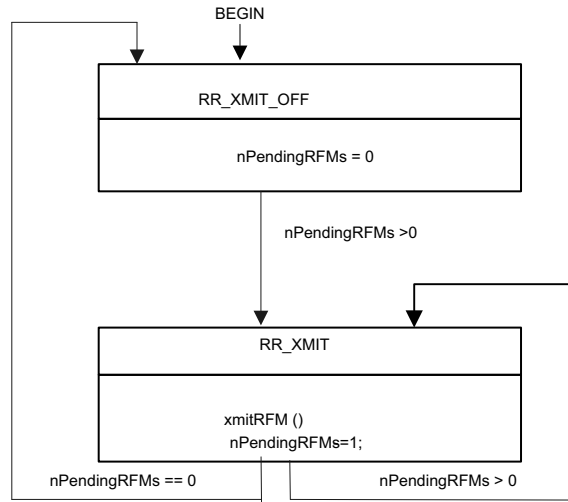


Figure 29-9—RR Transmit state machine

4 29.3.6 RFM Receiver variables

5 The following variables are local to the RFM Receiver state machine. There is one instance of these
6 variables per RFM Receiver.

7 29.3.6.1 RFMreceived

8 RFMreceived is a Boolean variable, which is set to true by the MP OpCode Demultiplexer when an RFM is
9 received, and cleared by the RFM Receiver state machine.

29.3.6.2 RFMPDU

RFMPDU is a record of the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the last RFM received by the MP OpCode Demultiplexer (19.2.7).

29.3.7 RFM Receiver procedure

The procedure listed in 29.3.7.1 and 29.3.7.2 is defined for the RFM receiver.

29.3.7.1 ReceiveRFM()

ReceiveRFM() procedure performs the following steps:

- a) If the destination_address field of the received RFM does not match the RFM Receiver's MAC address, drops the frame.
- b) Otherwise, pass the received RFMPDU to RFM Analyzer.⁴²

Even though RFM Analyzer is out of the scope of this standard, RFM Analyzer has to be able to recognize the value in Reflected Data TLV (29.3.3.2), which could be:

- A whole data frame being reflected; or
- A truncated data frame being reflected; or
- The first portion of the data frame being reflected; or
- The second portion of the data frame being reflected.

If an RFM Receiver is configured on an MP and the RFM Receiver does not receive any RFMs within the expected time frame, it can send a LBM (21.7) to the MP on which the RR is defined to verify the connectivity to the MP. If there is no LBR received, ReceiveRFM() can report an alarm to the network administrator.

29.3.7.2 RFM Receiver state machine

Figure 29-10 describes the state machine for RFM Receivers.

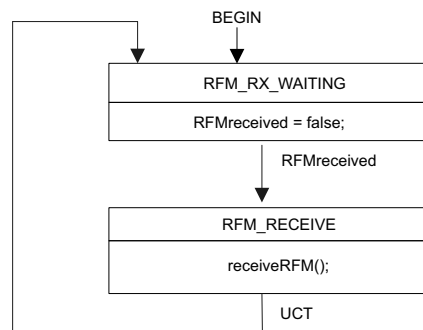


Figure 29-10—RFM Receiver state machine

⁴² RFM Analyzer is out of the scope of this standard.

1 **29.3.8 DR variables**

2 There is one instance of the parameters listed in 29.3.8.1 through 29.3.8.7 per DR.

3 **29.3.8.1 DRwhile**

4 A timer variable used by the Decapsulator Responder state machine to time out the active DR. DRwhile has
5 a granularity finer than or equal to 1 s.

6 **29.3.8.2 DRtime**

7 The time that the DR can remain active after its activation.

8 **29.3.8.3 SFMPDU**

9 SFMPDU is a record of the M_UNITDATA.indication or EM_UNITDATA.indication parameters of the
10 SFM received by the MP OpCode Demultiplexer (19.2.7) when an SFM is received.

11 **29.3.8.4 DRactive**

12 DRactive variable represents the activation status of the corresponding DR. DRactive is set to true when the
13 corresponding DR is activated and set to false when the corresponding DR is deactivated or its duration
14 expires. When DRactive is false, all SFMs received by the Bridge interface are dropped.

15 **29.3.8.5 SFMreceived**

16 SFMreceived is set to true by the OpCodeDemultiplexer (19.2.7) when an SFM is received.

17 **29.3.8.6 previousSFMtransId**

18 previousSFMtransId variable keeps the value of SFM Transaction Identifier of the previously received SFM
19 frame. The previousSFMtransId variable is initialized to zero when the DR is activated.

20 **29.3.8.7 SFMsequenceErrors**

21 SFMsequenceErrors variable keeps the total number of out-of-sequence SFMs received.
22 SFMsequenceErrors is initialized to zero when the DR is activated.

23 **29.3.9 DR procedures**

24 The DR has the following procedures:

- 25 a) ProcessSFM()
- 26 b) DropSFM()

27 **29.3.9.1 processSFM()**

28 Called by the Decapsulator Responder state machine when an SFM is received and the DR is active.
29 processSFM() processes the incoming SFM in SFMPDU as follows:

- 30 a) The SFM is examined for validity using the methods used for CFM as described in (20.46) and
31 discarded if invalid.
- 32 b) If the destination_address of the SFM does not match the DR's own MAC address or the
33 source_address of the SFM does not match with DR's SFM Originator [item b2) in 12.17.4.3.2], the
34 frame is discarded.

- 1 c) If the SFM frame passes these tests, the frame carried in its Original Data TLV (29.4.3.4) is
2 decapsulated to create an (E)M_UNITDATA.request as follows:
- 3 1) The destination_address parameter is set to the destination_address of the frame in the Original
4 Data TLV.
- 5 2) If the SourceAddressStayFlag [item b1) in 12.17.4.3.2] is false, the source_address parameter
6 is set to the DR's MAC address; otherwise, the source_address parameter is set to the
7 source_address of the frame in the Original Data TLV.
- 8 3) The vlan_identifier, priority, drop_eligible, and mac_service_data_unit parameters are set as
9 follows:
- 10 i) If the DR is configured on a port of an S-VLAN component, and the TPID of the frame in
11 the Original Data TLV indicates the S-TAG, the vlan_identifier, priority, and drop_eligible
12 parameters are taken from the corresponding values in the Tag Control Information (TCI)
13 and the mac_service_data_unit is taken from the remainder of the frame following the TCI
14 and excluding the last four octets (FCS).
- 15 ii) If the DR is configured on a port of a C-VLAN component, and the TPID in the Original
16 Data TLV indicates a C-TAG, the vlan_identifier, priority, and drop_eligible parameters
17 are taken from the corresponding values in the TCI, and the mac_service_data_unit is
18 taken from the remainder of the frame in the Original Data TLV following the TCI and
19 excluding the last four octets (FCS).
- 20 iii) Otherwise, the vlan_identifier is set to the PVID for the port on which the DR is config-
21 ured, the priority is set to the priority of the SFM, the drop_eligible parameter is set to the
22 value of drop_eligible for the SFM, and the mac_service_data_unit is taken from the
23 frame in the Original Data TLV immediately following the source_address and excluding
24 the last four octets (FCS).
- 25 4) Sets the frame_check_sequence parameter to the unspecified value (IEEE Std 802.1AC).
- 26 5) Sets the connection_identifier parameter to null.
- 27 d) ProcessSFM() queries the FDB using the destination_address and vlan_identifier parameters of the
28 (E)M_UNITDATA.request. The set of potential transmission ports, normally created by the Active
29 Topology enforcement (8.6.1), is the set of all Bridge Ports that are both in the member set of the
30 vlan_identifier and that are in the Forwarding state for that vlan_identifier.
- 31 1) If an Egress Port is identified, the decapsulated frame is carried to the related LOM SAP for
32 transmission.
- 33 2) If an Egress Port is not identified and the FloodingEnabled flag of the DR is set, the
34 decapsulated frame is flooded to the LOM SAP of all forwarding ports in the VID set;
35 otherwise, the frame is discarded.
- 36 e) If the received SFM is not the first frame received by the DR and the value of SFM Transaction
37 Identifier field in the received SFM is not 1 greater than the value saved in the previousSFMtransId,
38 increment SFMsequenceErrors by 1.
- 39 f) Set the value of previousSFMtransId to the value of the SFM Transaction Identifier field of the
40 received SFM.

41 29.3.9.2 DropSFM()

42 Called by Decapsulator Responder state machine whenever the DR is not active in order to silently discard
43 the received SFM frames.

29.3.10 Decapsulator Responder state machine

Once the DR is configured and activated by network administrator, it remains active until the configured timer expires or being deactivated by the network administrator (see Figure 29-11).

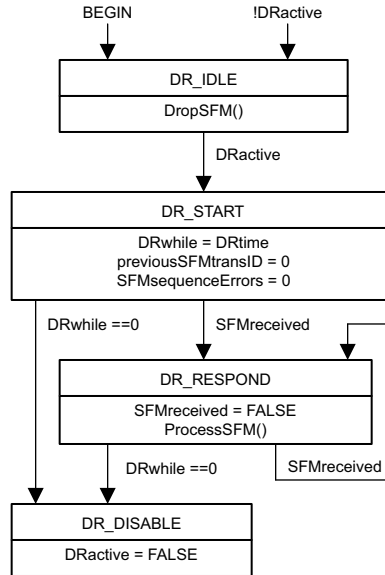


Figure 29-11—Decapsulator Responder state machine

29.4 Encoding of DDCFM PDUs

This subclause specifies the method of encoding DDCFM PDUs. The specifications include the following:

- a) The format of RFM
- b) The format of SFM

29.4.1 RFM and SFM Header

The RFM Header and SFM Header are same as the Common CFM Header defined by Table 21-2, with extra OpCodes defined for RFM and SFM (Table 21-3).

29.4.2 RFM format

The RFM format is shown in Table 29-1.

29.4.2.1 Flags

(1 octet) in RFM. The Flags field of the Common CFM Header for RFM is reserved for future use.

29.4.2.2 First TLV Offset

(1 octet) The First TLV Offset field of the Common CFM Header in a RFM is transmitted as 4.

Validation Test: The First TLV Offset field of the Common CFM Header in a RFM contains a value greater than or equal to 4.

Table 29-1—RFM format

Field	Octet
Common CFM Header	1–4
RFM Transaction Identifier	5–8
Reserved for definition in future versions of the protocol ^a	
Reflected Data TLV	First TLV Offset + 5
Additional TLV	First TLV Offset + 5 + Length of Reflected Data TLV
End TLV (0)	

^a This field has 0 length in this version 0 of DDCFM. It is shown in order to stress that additional information can be present in future versions of DDCFM, and that a version 0 receiver ignores its contents, if present.

1 29.4.2.3 RFM Transaction Identifier

2 (4 octets) An RR copies the content of nextRFMtransID variable (29.3.1.16) to this field.

3 29.4.2.4 Reflected Data TLV

4 Reflected Data TLV in RFM is a Data TLV (21.5.6) to contain the reflected data frame. The Length field is
5 the total octets of the reflected data frame.

6 The Type field of the Reflected Data TLV could be one of the following four values (Table 21-5):

- 7 a) Data TLV Type value, which is 3
- 8 b) Truncated Data TLV Type value, which is 9
- 9 c) Data-Part-1 TLV Type value, which is 10
- 10 d) Data-Part-2 TLV Type value, which is 11

11 29.4.2.5 Additional RFM TLVs

12 The following TLVs may be included in a RFM by each MP originating RFM (RR):

- 13 a) Send ID TLV (21.5.3)
- 14 b) Organization-Specific TLVs (21.5.2)

15 The Additional RFM TLVs may be placed in any order.

16 29.4.3 SFM format

17 The SFM format is shown in Table 29-2.

18 29.4.3.1 Flags

19 (1 octet) in SFM. The Flags field of the Common CFM Header for SFM is reserved for future use.

20 29.4.3.2 First TLV Offset

21 (1 octet) The First TLV Offset field of the Common CFM Header in an SFM is transmitted as 4.

22 **Validation Test:** The First TLV Offset field of the Common CFM Header in an SFM contains a value
23 greater than or equal to 4.

Table 29-2—SFM format

Field	Octet
Common CFM Header	1–4
SFM Transaction Identifier	5–8
Reserved for definition in future versions of the protocol ^a	
Original Data TLV	First TLV Offset + 5
Additional TLVs	First TLV Offset + 5 + Length of Original Data TLV
End TLV (0)	

^a This field has 0 length in this version 0 of DDCFM.

1 29.4.3.3 SFM Transaction Identifier

2 (4 octets) SFM originator may use this field to distinguish the order of SFMs sent. DR may use this identifier
3 to check the order of the received SFMs.

4 29.4.3.4 SFM Original Data TLV

5 SFM Original Data TLV is a Data TLV (21.5.6) to contain the original data frame. The Length field is the
6 total octets of the original data frame, including its destination_address, source_address, TPID, TCI,
7 Length/Type, client data, pad, and FCS fields.

8 29.4.3.5 Additional SFM TLVs

9 The following TLVs may be included in an SFM by each SFM Originator:

- 10 a) Send ID TLV (21.5.3)
- 11 b) Organization-Specific TLVs (21.5.2)

12 The Additional SFM TLVs may be placed in any order.

1 30. Principles of congestion notification

2 Congestion notification comprises capabilities for detecting and mitigating queue congestion for selected
3 classes of traffic in Virtual Bridged Networks. These capabilities can be used in networks with a bandwidth-
4 delay product on the order of 5 Mbits or less in order to decrease the likelihood of frame loss for Congestion
5 Controlled Flows (CCFs). As the geographical size, per-hop delay, and/or maximum hop count of a network
6 grow, causing the bandwidth-delay product of the network to increase beyond this value, oscillations in
7 buffer occupancy begin, and their amplitude increases gracefully with bandwidth-delay product. If the
8 parameters controlling the algorithm are not adjusted accordingly and the sizes of the Bridges' buffers are
9 not increased, frames are then likely to be lost due to congestion.

10 Congestion notification depends on the formation of a cooperating set of systems including VLAN Bridges
11 and end stations to achieve the reduction in frame loss. By partitioning the Bridges' and end stations'
12 resources, frames sourced or sunk by noncooperating end stations or Bridges can be carried with minimal
13 contention with CCFs for those resources. Accordingly, congestion notification entities (Clause 31) are
14 specified as shims that make use of and provide the ISS or EISS (IEEE Std 802.1AC, 6.8) at SAPs within the
15 network. The operation of the congestion notification protocol is described in Clause 32, and the formats of
16 the CNM and Congestion Notification TLV are described in Clause 33.

17 This clause provides the context necessary to understand the congestion notification protocol: how
18 congestion controlled regions of networks are identified; how CCFs are identified and separated from other
19 frames; and how congestion notification entities in Bridges and end stations operate.

20 This clause introduces the concepts essential to congestion notification as follows:

- 21 a) The problem and requirements on the solution are presented (30.1).
- 22 b) The basic principles of the Quantized Congestion Notification protocol (QCN) are presented,
23 including descriptions of a CP that sends a CNM to a RP (30.2).
- 24 c) A CCF consists of frames, all with the same priority value, and all assigned to a single Flow queue
25 and RP in the originating end station (30.3).
- 26 d) A Congestion Notification Priority Value (CNPV) is one of the eight priority values that has been
27 configured exclusively for the use of CCFs in the congestion-aware systems of a Virtual Bridged
28 Network (30.4).
- 29 e) A CN-TAG can be transmitted with every data frame in a CCF, and is returned in every CNM (30.5).
- 30 f) A CND is constructed as a subset of Bridges and end stations in a Virtual Bridged Network, all of
31 which are configured to handle a particular CNPV, the resources of which subset are defended by the
32 Bridges whenever the offered load exceeds the network capacity (30.6).
- 33 g) The relationship of Multicast data to congestion notification is explained (30.7).
- 34 h) The peering relationship of RPs and CPs in PBNs and PBBNs is explained (30.8).

35 30.1 Congestion notification design requirements

36 The congestion notification protocol, congestion notification algorithm (QCN), and supporting protocols
37 specified in this standard meet requirements for the following:

- 38 — CCF and network performance.
- 39 — Limiting the implementation complexity, of both RPs and CPs.
- 40 — Not over-constraining the design of Bridged Networks using CCF.
- 41 — Facilitating interoperability, coexistence, and deployment into existing networks, and limiting the
42 administrative effort associated with using CCF in a network.

43 NOTE 1—QCN is designed to operate over a wide range of conditions. This subclause (30.2) necessarily uses
44 qualitative terms such as “low probability,” “small,” “stable,” and “modest” (for brevity). The bibliography (Annex X)
45 provides references (Alizadeh et al. [B1]) to studies of QCN performance for various network and traffic configurations,
46 using both simulation and a fluid model (see 30.3 for the necessary equations, and for network sizing recommendations).

1 The following performance requirements are met for s CCFs (30.3):

- 2 a) There is a very low probability of frame discard due to buffer overflow at a CP.
- 3 b) The average buffer occupancy at each CP that is a Current Congestion Point (CCP) for one or more
4 CCFs is a small fraction of the bandwidth delay product, and largely independent of the number of
5 CCFs, thus ensuring that loss avoidance does not result in large and fluctuating transit delays.
- 6 c) The buffer occupancy at a CCP has a low probability of underflow, i.e., of becoming empty and thus
7 failing to use available bandwidth when an RP is rate limiting a CCF as a result of CNMs received
8 from that CCP.
- 9 d) When multiple CCFs with a single CCP share that CCP, they obtain very nearly the same share of
10 the bandwidth. When multiple CCFs have one or more CCPs in common they each obtain a share of
11 bandwidth resources proportional to their share of contested resources.
- 12 e) The low probability of buffer overflow and underflow at a CCP is maintained for both small and
13 large numbers of CCFs, and when CCFs or the load offered by a given CCF changes.
- 14 f) The bandwidth provided to each CP by its attached LAN will be used, at all times, i.e., if the CP's
15 buffer is not empty it will transmit, given the available bandwidth. The fact that the bandwidth
16 available to one CCF passing through a given CP can be constrained by another CCP will not reduce
17 the bandwidth through that CP for other CCFs not sharing the CCP.

18 NOTE 2—The probability of frame loss in the absence of CCF naturally depends on the response of higher layer
19 protocols to increasing transit delay, and the importance of avoiding loss on their response. Highly loss sensitive
20 protocols typically lack loss avoidance mechanisms, and no such mechanisms are assumed in the analysis of QCN.

21 NOTE 3—Requirements a) through d) can be summarized by characterizing the performance of QCN as stable,
22 responsive, and (proportionally) fair with respect to buffer occupancy processes. Requirements e) and f) demand that
23 available resources are used, and that stability not depend on having a large number of slowly changing CCFs.

24 The following requirements limit the complexity and hence the cost of Bridge Port (CP) support for CCF:

- 25 g) The CCF state retained by each Bridge Port is fixed, and determined solely by the number of
26 congestion management transmission queues (i.e., by the number of CPs, one per CNPV, up to a
27 maximum of 7) and the per CP requirements of the congestion notification algorithm (QCN).
- 28 h) A CP does not retain any per CCF or per RP state, nor is it required to allocate any data frame
29 passing through the CP to a CCF or an RP. All the information necessary for a CP to address a CNM
30 to an end station, and for that end station to identify the CCF and RP from the CNM, is contained in
31 each frame of a CCF.
- 32 i) The buffering required, at each CP, to avoid frame loss should be a small fraction of the network's
33 bandwidth delay product.

34 The following requirements limit the complexity and hence the cost of end station (RP) support for CCF:

- 35 j) The CCF state retained by each end station for each congestion managed transmission queue, i.e.,
36 for each RP, is fixed.
- 37 k) The end station state requirements of QCN are determined solely by the number of RPs, and are
38 independent of the number of CCFs and the number of CPs in the network.
- 39 l) The number of CCFs, the identification of each CCF, the allocation of transmitted frames to CCFs,
40 and the allocation of CCFs to RPs is determined by each end station independently.
- 41 m) The end station is not required to transmit any additional frames, as a consequence either of
42 transmitting or of receiving a frame for a CCF.
- 43 n) The CN-TAG for each CCF is independent of changes in RP state information.
- 44 o) The CCF protocol and QCN do not use any higher-layer protocol information carried in frames.
- 45 p) QCN calculations are simple and can be performed rapidly: i.e., the number of operations required
46 to process each event is fixed, and all multiplications and divisions can be reduced to the use of
47 factors of 2 or the use of a fixed and small lookup table. An RP does not attempt to calculate control
48 loop gains and delays and other, potentially rapidly changing CCF-dependent variables.

1 NOTE 4—A CP or an RP can retain additional state for the purposes of management, but this state is not required by the
2 CCF protocol or QCN.

3 NOTE 5—An end station can use higher layer protocol information to decide whether a frame is to be allocated to a
4 CCF, and to select a CCF.

5 The following requirements ensure that CCF does not over-constrain the design of the network, or limit the
6 use of protocols that determine the active topology of the network:

- 7 q) RPs and CPs are unaware of the path taken by frames through the network, require no signaling
8 from the network of changes in path, and take no exceptional action when paths change.
- 9 r) An RP does not assume that all frames that it transmits, or any identifiable subset of those frames
10 such as those identified by the end station as belonging to a single CCF or transmitted to the same
11 destination, are constrained to follow the same path through the network. An RP can transmit frames
12 to many different destinations, and transmission to a given destination can be multi-path.

13 The following requirements support interoperability, facilitate deployment, and limit the administrative
14 effort required to use CCF:

- 15 s) The boundaries of each CND are determined automatically.
- 16 t) The destination of a CCF can lie outside the transmitting end station's CND, and the destination end
17 station can be unaware of the use of CCF within the CND. Incremental deployment of CCF is
18 therefore possible.
- 19 u) The feedback provided in each CNM provides the receiving RP with the information needed to act
20 on that CNM, and does not require the RP to know or interpret information about the sending CP.
- 21 v) PBBNs can be interposed into a CND, in order to reduce the number of MAC addresses learned by
22 Bridges in environments with large numbers of (perhaps virtual) end stations.

23 These requirements have a number of additional consequences for the design of QCN, including the
24 following:

- 25 w) Additional mechanisms to determine round trip time are not required, as the algorithm is robust over
26 its targeted range of round trip time.
- 27 x) QCN cannot regulate frame transmission by acknowledgments, as does Transmission Control
28 Protocol/Internet Protocol (TCP/IP), see IETF RFC 791 and IETF RFC 793 [B19].

29 30.2 Quantized Congestion Notification protocol (QCN)

30 This subclause provides an overview of the baseline simulation of the QCN algorithm (Alizadeh et al. [B1])
31 used to develop this standard. This introduction provides no normative text. It will focus on the key features
32 of the QCN algorithm, omitting the normative details given in the rest of this standard.

33 The QCN algorithm is composed of the following two parts:

- 34 a) **CP algorithm:** the mechanism by which a congested Bridge or end station buffer samples outgoing
35 frames and generates a feedback message (CNM, 33.3, in this standard) addressed to the source of
36 the sampled frame. The feedback message contains information about the extent of congestion at the
37 CP.
- 38 b) **RP algorithm:** the mechanism by which a Rate Limiter (RL) associated with a source decreases its
39 sending rate based on feedback received from the CP, and increases its rate *unilaterally* (without
40 further feedback) to recover lost bandwidth and probe for extra available bandwidth.

30.2.1 The CP algorithm

A Bridge containing a CP is modeled as an ideal output-queued Bridge. The CP buffer is shown in Figure 30-1. The goal of the CP is to maintain the buffer occupancy at a desired operating point, Q_{eq} .⁴³ The CP computes a congestion measure F_b (defined below) and, with a probability depending on the severity of congestion, selects a frame from the incoming stream and sends the value of F_b in a feedback message to the source of the selected frame.

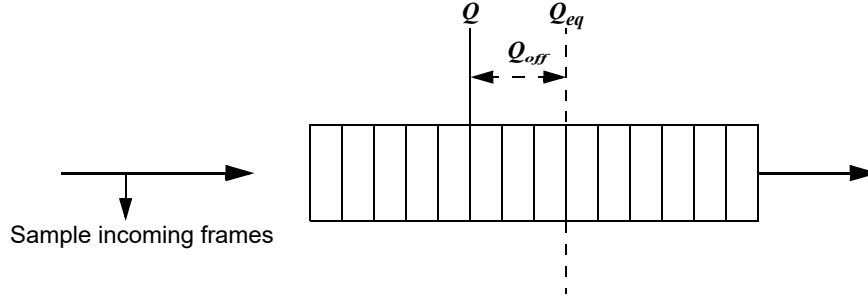


Figure 30-1—Congestion detection in QCN CP

Let Q denote the instantaneous queue size and Q_{old} denote the queue size when the last feedback message was generated. Let $Q_{off} = Q - Q_{eq}$ and $Q_{\delta} = Q - Q_{old}$.

Then F_b is given by Equation (30-1).

$$F_b = -(Q_{off} + wQ_{\delta}) \quad (30-1)$$

where w is a non-negative constant, taken to be 2 for the baseline simulation. The value of F_b is quantized to 6 bits before transmission, based on Q_{eq} and w .

The interpretation is that F_b captures a combination of queue size excess (Q_{off}) and rate excess (Q_{δ}). Indeed, $Q_{\delta} = Q - Q_{old}$ is the *derivative* of the queue size and equals input rate less output rate. Thus, when F_b is negative, the buffer is oversubscribed. When $F_b < 0$, Figure 30-2 shows the probability with which a congestion message is reflected back to the source as a function of $|F_b|$. The feedback message contains the value of F_b , quantized to 6 bits. When $F_b \geq 0$, there is no congestion and no feedback messages are sent.

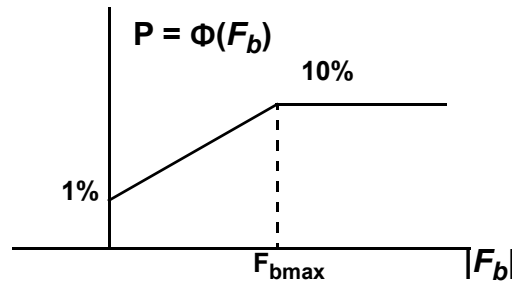


Figure 30-2—Sampling (reflection) probability in QCN CP as a function of $|F_b|$

⁴³ In simulation, setting Q_{eq} to 20% of the physical buffer size (150 000 octets) produced good results.

30.2.2 Basic RP algorithm

The RP is not given positive rate-increase signals by the network. Also, there are no acks at Layer 2 to trigger rate increases. Therefore, the RP requires a local mechanism to determine when, and by how much, the send rate is increased. Before proceeding to explain the RP algorithm, readers need the following terminology:

- **Current Rate (CR):** The transmission rate of the Rate Limiter (RL) at any time.
- **Target Rate (TR):** The sending rate of the RL *just before* the arrival of the last feedback message, and the new goal for the Current Rate.
- **Byte Counter:** A counter at the RP for counting the number of bytes transmitted by the RL. It triggers rate increases by the RL. See 30.2.2.2.
- **Timer:** A clock at the RP that is also used for triggering rate increases at the RL. The main purpose of the timer is to allow the RL to rapidly increase the rate when its sending rate is very low and bandwidth becomes available. See 30.2.3.

This clause now explains the RP algorithm assuming that only the byte counter is available. Later, how the timer is integrated into the RP algorithm is briefly explained. Figure 30-3 shows the basic RP behavior.

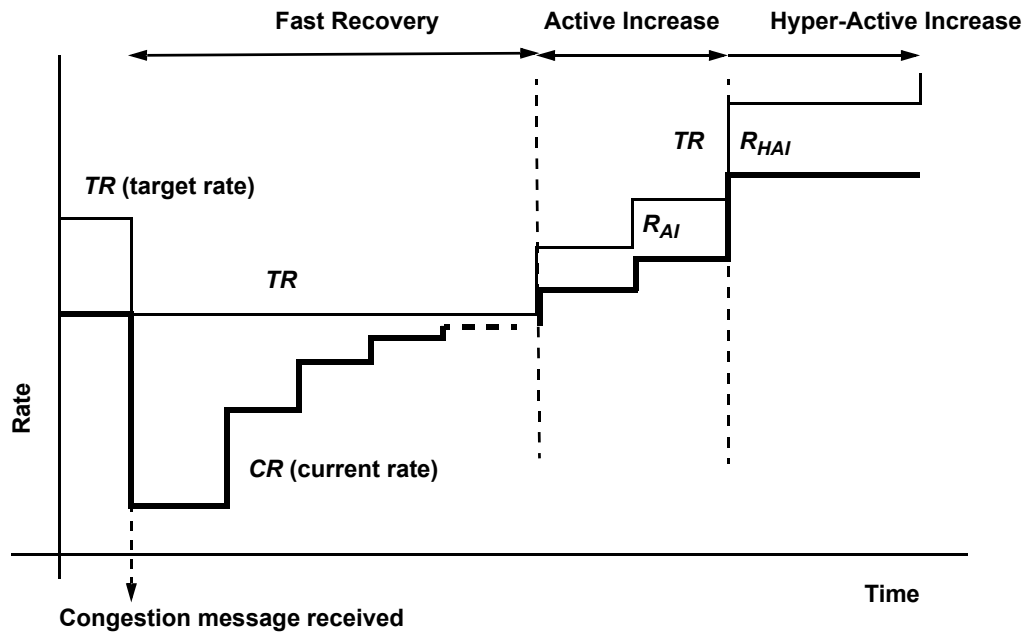


Figure 30-3—QCN RP operation

30.2.2.1 Rate decreases

Rate decreases occur only when a feedback message is received, in which case CR and TR are updated as follows in Equation (30-2) and Equation (30-3):

$$TR \leftarrow CR \quad (30-2)$$

$$CR \leftarrow CR(1 - G_d|F_b|) \quad (30-3)$$

where the constant G_d is chosen so that $G_d|F_{bmax}| = 1/2$, i.e., the sending rate can decrease by at most 50%.

30.2.2.2 Rate increases

Rate increases occur in two phases: Fast Recovery and Active Increase.

Fast Recovery (FR). The byte counter is reset every time a rate decrease is applied and the RP enters the FR state. FR consists of 5 cycles, each cycle equal to 150 kB of data transmission by the RL. The RP counts the cycles of the byte counter. At the end of each cycle, TR remains unchanged while CR is updated as follows in Equation (30-4):

$$CR \leftarrow \frac{1}{2}(CR + TR) \quad (30-4)$$

The cycle duration of 150 kB is chosen to correspond to the transmission of 100 frames, each 1500 bytes long. The idea is that when the RL has transmitted 100 frames and, given that the minimum sampling probability at the CP is 1%, if it has not received a feedback message then it may infer that the CP is uncongested. Therefore, it increases its rate as above, recovering some of the bandwidth it lost at the previous rate decrease episode. Thus, the goal of the RP in FR is to *rapidly recover* the rate it lost at the last rate decrease episode.⁴⁴

Active Increase (AI). After 5 cycles of FR have completed, the RP enters the Active Increase (AI) phase where it probes for extra bandwidth on the path. During AI, the byte counter counts out cycles of 50 frames (this can be set to 100 frames for a less frequent probing). At the end of each cycle, the RL updates TR and CR as follows in Equation (30-5) and Equation (30-6):

$$TR \leftarrow TR + R_{AI} \quad (30-5)$$

$$CR \leftarrow \frac{1}{2}(CR + TR) \quad (30-6)$$

where R_{AI} is a constant chosen to be 5 Mb/s in the baseline simulation.

30.2.3 RP algorithm with timer

Since rate increases using the Byte Counter occur at a frequency proportional to the current sending rate of the RL, when CR is very small, the duration of Byte Counter cycles when measured in seconds can become unacceptably large. Since the speed of bandwidth recovery (or responsiveness) is a key performance metric, a Timer was included in QCN.

The Timer functions similarly as the Byte Counter: it is reset when a feedback message arrives, enters FR and counts out 5 cycles of T ms duration (T is 10 ms long in the baseline), and then enter AI where each cycle is $T/2$ ms long.

The Byte Counter and Timer jointly determine rate increases at the RL as shown in Figure 30-4. After a feedback message is received, they each operate independently and execute their respective cycles of FR and AI. Together, they determine the state of the RL and its rate changes as follows:

- a) The RL is in FR if *both* the Byte Counter and the Timer are in FR. In this case, when either the Byte Counter or the Timer completes a cycle, CR is updated according to Equation (30-4).
- b) The RL is in AI if *only one of* the Byte Counter and the Timer is in AI. In this case, when either completes a cycle, TR and CR are updated according Equation (30-5) and Equation (30-6).
- c) The RL is in HAI (for Hyper-Active Increase) if *both* the Byte Counter and the timer are in AI. In this case, the i th time that a cycle for either the Byte Counter or the Timer is completed, TR and CR are updated as shown in Equation (30-7) and Equation (30-8):

$$TR \leftarrow TR + iR_{HAI} \quad (30-7)$$

$$CR \leftarrow \frac{1}{2}(CR + TR) \quad (30-8)$$

where R_{HAI} is constant set to 50 Mb/s in the baseline simulation (Figure 30-3). So the increments to TR in HAI occur in multiples of 50 Mb/s.

⁴⁴ The BIC-TCP algorithm [B68] has a similar Fast Recovery mechanism.

- 1 Thus, the Byte Counter and Timer should be viewed as providing the RL opportunities to increase the rate.
- 2 Their state determines the state of, and hence the amount of rate increase at, the RL.

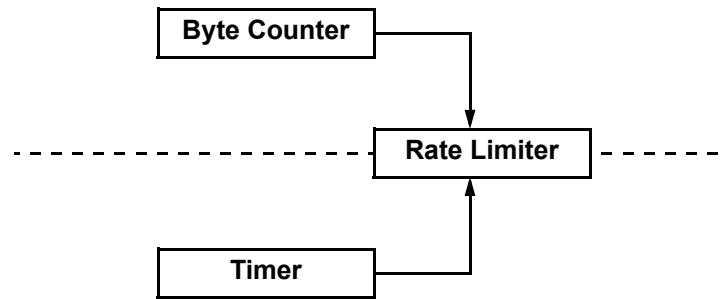


Figure 30-4—Byte Counter and Timer interaction with Rate Limiter

3 It is very important to note that the RL goes to HAI only after at least 500 frames have been sent and
 4 50 ms have passed since the last congestion feedback message was received. This doubly ensures that
 5 aggressive rate increases occur only after the RL provides the network adequate opportunity (in frames sent
 6 for possible sampling) for sending rate decrease signals should there be congestion. This is vital to ensure
 7 the stability of the algorithm, and while optimizations can be performed to improve its responsiveness, in the
 8 interests of stability and simplicity, the baseline simulation (Alizadeh et al. [B1]) took no further
 9 optimization steps.

10 **30.3 Congestion Controlled Flow (CCF)**

11 A CCF consists of all of the frames passing through a single Flow queue (31.2.2.1) in an originating end
 12 station. Examples of CCFs include, but are not limited to,

- 13 a) The frames carrying data for a single User Datagram Protocol (UDP, IETF RFC 768 STD0006)
 14 connection.
- 15 b) All of the frames generated by a particular process running in an end station.
- 16 c) All of the frames being transmitted by an end station that produce the same value when a hash
 17 computation is performed on the source and destination IP addresses, UDP vs. Transmission Control
 18 Protocol (TCP) selection, and source and destination UDP/TCP port numbers of the IP packets
 19 carried by those frames.
- 20 d) All of the frames passing through an RP that have the same destination_address parameter.
- 21 e) All of the frames being transmitted by an end station.
- 22 f) All of the frames passing through a Bridge Port sharing identical source_address,
 23 destination_address, and priority parameters.
- 24 g) All of the frames leaving a router that contain IP packets, and have identical values for their source
 25 and destination IP addresses, UDP vs. TCP selection, and source and destination UDP/TCP port
 26 numbers.
- 27 h) All of the frames with a certain value for the priority parameter.

28 All of the frames in a given CCF are sent with the same priority value, which is a CNPV (30.4).

29 Not all frames transmitted by an end station with an RP need be CCFs. Frames not belonging to CCFs
 30 bypass the Flow queues and RPs.

1 30.4 Congestion Notification Priority Value (CNPV)

2 A CNPV consists of one value of the priority parameter such that all of the Bridges' and end stations' ports
3 in a Congestion Notification Domain (CND) are configured to assign frames at that value to the same CP
4 and/or an RP. Different CNPVs correspond to classes of applications, or even single applications, such as
5 interprocess communications or disk storage data, that have different requirements for such network
6 resources as latency or bandwidth. Since there are eight values for the priority parameter, a single port in a
7 Virtual Bridged Network can support as many as seven CNPV values; at least one value is required for best-
8 effort traffic.

9 30.5 Congestion Notification tag (CN-TAG)

10 An end station may add a CN-TAG to every frame it transmits from a CCF. The CN-TAG contains a Flow
11 Identifier field. The destination_address, Flow ID, and a portion of the frame that triggered the transmission
12 of the CNM are the means provided by this standard by which a station can determine to which RP a CNM
13 applies. How the station makes that determination is beyond the scope of this standard. The reasons for
14 adding a CN-TAG include the following:

- 15 a) An end station can identify a particular flow within a CCF that triggered the CNM.
- 16 b) It can be simpler for the end station to use a Flow ID, rather than to parse the returned
17 mac_service_data_unit of the original triggering frame, in order to identify either the RP or the
18 particular flow.
- 19 c) Depending on the application, e.g., if fragmented IP datagrams are transmitted, it can be impossible
20 to determine either the RP or the particular flow based solely on a returned mac_service_data_unit.

21 An end station that has only a single RP per CNPV, or that is able to identify the RP on the basis of the
22 returned portion of the triggering frame, can omit transmitting the CN-TAG. The CP always returns a
23 CN-TAG in a CNM with the triggering frame's Flow ID, or with a 0 Flow ID, if the triggering frame has no
24 CN-TAG. The fixed format for the CNM minimizes the effort required of the CP to generate the CNM.

25 30.6 Congestion Notification Domain (CND)

26 In order for congestion notification to successfully control congestion in a Virtual Bridged Network, the
27 managed objects controlling the congestion notification state machines in the Bridges and end stations in
28 that network have to be configured with values that are appropriate to the characteristics of the CCFs
29 generated by the applications that expect congestion controlled services. For example,

- 30 a) If frames that were not originated from an RP can enter a CP experiencing congestion, then the
31 CNMs generated by that CP upon receipt of those frames cannot correct the problem, and
32 congestion notification cannot operate correctly.
- 33 b) Congestion notification cannot operate correctly if a CP's configuration is inappropriate for the
34 CCFs passing through it, or if priority values are regenerated in a manner that moves frames in and
35 out of CNPVs.
- 36 c) Frames transmitted from an end station with a CN-TAG cannot be understood by an end station that
37 is not congestion aware.

38 Congestion-aware Bridges therefore construct a CND, within which a particular CNPV is supported. A
39 CND is a connected subset of the Bridges and end stations in a Virtual Bridged Network that are
40 appropriately configured to serve a particular CNPV. This standard does not assume that every Bridge in a
41 Virtual Bridged Network will be capable of congestion control, does not assume that every capable Bridge
42 will be so configured, and does not assume that the subset of Bridges in a Virtual Bridged Network forming
43 a CND that serves one CNPV will be the same subset that form a CND serving another CNPV.

1 CNDs can be created by configuring the Bridges and end stations in a network, or they can be created
2 automatically, using an additional Type Length Value element, the Congestion Notification TLV (D.2.7),
3 that this standard adds to the IEEE 802.1AB Link Layer Discovery Protocol (LLDP). Either way, every
4 Bridge Port knows, for each CNPV, whether its neighbor(s) are or are not all configured with a matching
5 CNPV.

6 NOTE—If CND boundaries are configured individually, rather than through LLDP, and there is an error in the
7 configuration, there is a risk that frames not originating from RPs will pass through a CP, or that RP-originated frames
8 will pass through a congested Port that has no CP. Either case can result in congestion that cannot be alleviated, with
9 consequent excessive discarding of RP-originated data frames. In addition, such misconfiguration can result in a
10 congestion-aware end station being unable to communicate with a congestion unaware end station, because the latter is
11 receiving CN-TAGs that it does not understand.

12 Making each CND/CNPV independent, rather than making a single determination of neighbor capabilities
13 based on whether all CNPVs are configured the same for all Ports attached to the LAN, allows a CND to be
14 added to or removed from a network already using another CND, without disrupting the operation of the
15 first CND.

16 30.7 Multicast data

17 Although control of multicast streams is not an object of this standard, and no simulations or experiments
18 involving multicast data were utilized in the writing of this standard, the transmission of multicast data
19 through a CCF is not prohibited. Some applications primarily send unicast data, but send occasional
20 multicast frames for purposes such as discovery. Transmission of multicast data has been allowed in order to
21 allow well-designed applications using a very small percentage of multicast traffic to operate on a CNPV.

22 Multicast streams are often sent to multiple destinations along multiple paths through the network, and a
23 unicast stream normally travels along a single path. Therefore, a multicast stream is more likely to encounter
24 any congested ports that exist in a network than is a unicast stream. The consequences of sending multicast
25 streams on a CCF include the following:

- 26 a) The RP could limit the CCF to the rate of the slowest (most congested) path taken by the multicast.
- 27 b) The chances of unintended fate sharing are higher in a CCF that included multicast streams. That is,
28 a multicast stream is more likely to unnecessarily slow down a unicast stream sharing the same CCF
29 than one unicast stream is likely to slow down another.

30 30.8 Congestion notification and additional tags

31 The transmission of a CNM from a CP to an RP makes the CP and RP peer entities in the sense discussed in
32 IEEE Std 802.1AC. An example of this peering relationship for a VLAN Bridged Network is illustrated in
33 Figure 30-5, with the CP-RP relationship shown by a heavy dashed line.

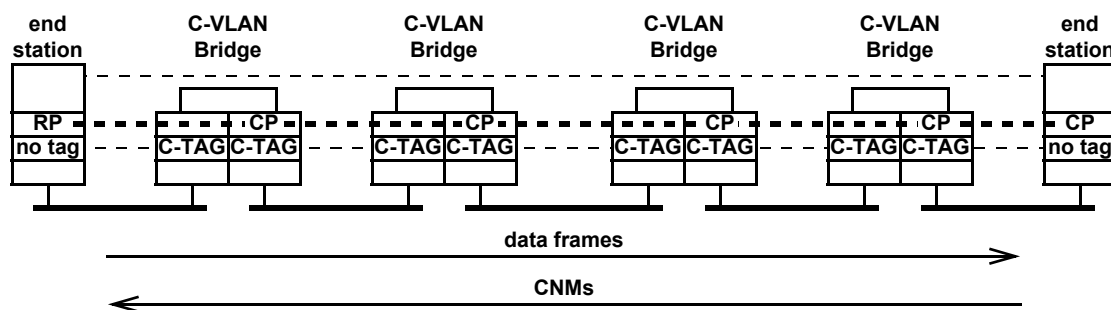


Figure 30-5—CP–RP peering in VLAN Bridged Network

PBBs (Clause 25) could be used in a data center environment, in order to reduce the number of MAC addresses that have to be learned in the core of the network. The I-component (5.7) of a PBB adds and removes I-TAGs (9.7) on data frames. However, as illustrated in Figure 30-6,⁴⁵ adding an I-TAG to a data frame by an I-component impairs the ability of a CP in the core of that network to return a CNM that will be meaningful to the originating end station. The following two difficulties follow from the fact that a CP in a BCB is not a peer of the RP in an end station:

- a) The BCB CP cannot parse the I-TAG and C-TAG, which is necessary in order to find the RP's MAC address and the data frame's CN-TAG, in order to build an I-TAG-encapsulated, CNM addressed to the RP.
- b) There is no MAC address available to the BCB CP that would be a valid source_address in the CNM when received by the RP.

In order to provide the benefits of PBBNs or other hierarchical bridging technologies to the data center environment, an interworking function (32.16) is defined to allow core CPs to peer with end stations.

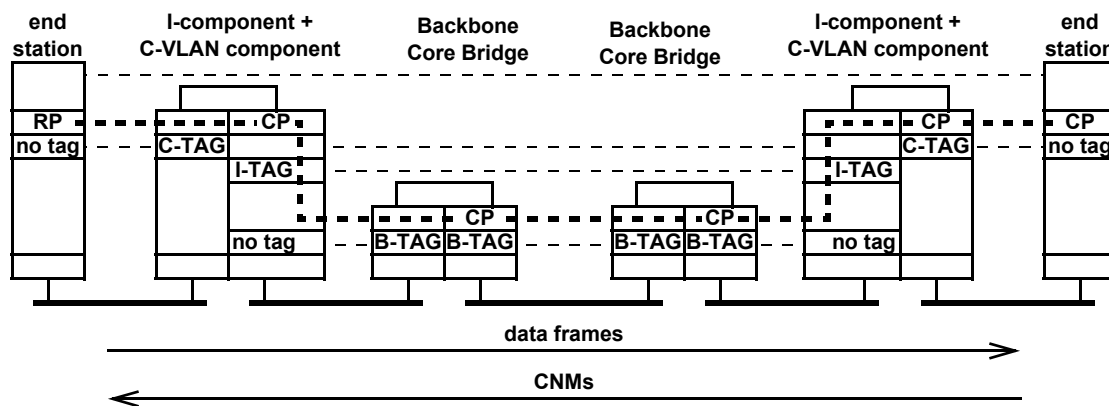


Figure 30-6—CP–RP peering in PBBN

⁴⁵ In the configuration chosen for illustration for the data center environment in Figure 30-6, each S-VLAN component of an I-components is, in effect, a Provider Bridge (5.10). That is, each CNP in an I-component is virtualized and multiplied, and connected to a corresponding virtual PEP of a C-VLAN component offering a single CEP facing the end station. This offers a C-TAG interface to the end station, without incurring the burden of an additional layer of S-TAGs.

31. Congestion notification entity operation

This clause specifies the following:

- a) The architecture of the CP in the Forwarding Process in a congestion-aware Bridge component or end station (31.1).
- b) The architecture of a CP and an RP in a congestion-aware end station (31.2).

NOTE—Clause 30 introduces the principles of congestion notification operation and the network architectural concepts that support it. Clause 32 specifies the protocols operated by entities defined in this clause. Clause 33 specifies the encoding of the PDUs used by congestion notification.

The models of operation in this clause provide a basis for specifying the externally observable behavior of congestion notification, and are not intended to place additional constraints on implementations; these can adopt any internal model of operation compatible with the externally observable behavior specified. Conformance of equipment to this standard is purely in respect of observable protocol.

31.1 Congestion-aware Bridge Forwarding Process

Figure 8-12 illustrates the details of the Bridge Forwarding Process. Figure 22-1 shows a different view of those details, rearranged so that those entities of the Forwarding Process that are concerned only with a single Bridge Port can be put in the proper relationship to entities such as CFM shims. Figure 31-1 shows only the queuing functions of Figure 22-1, as modified for a congestion-aware Bridge. Two new entities are added for the congestion-aware Bridge: the CP (31.1.1) and the CP ingress multiplexer (31.1.2).

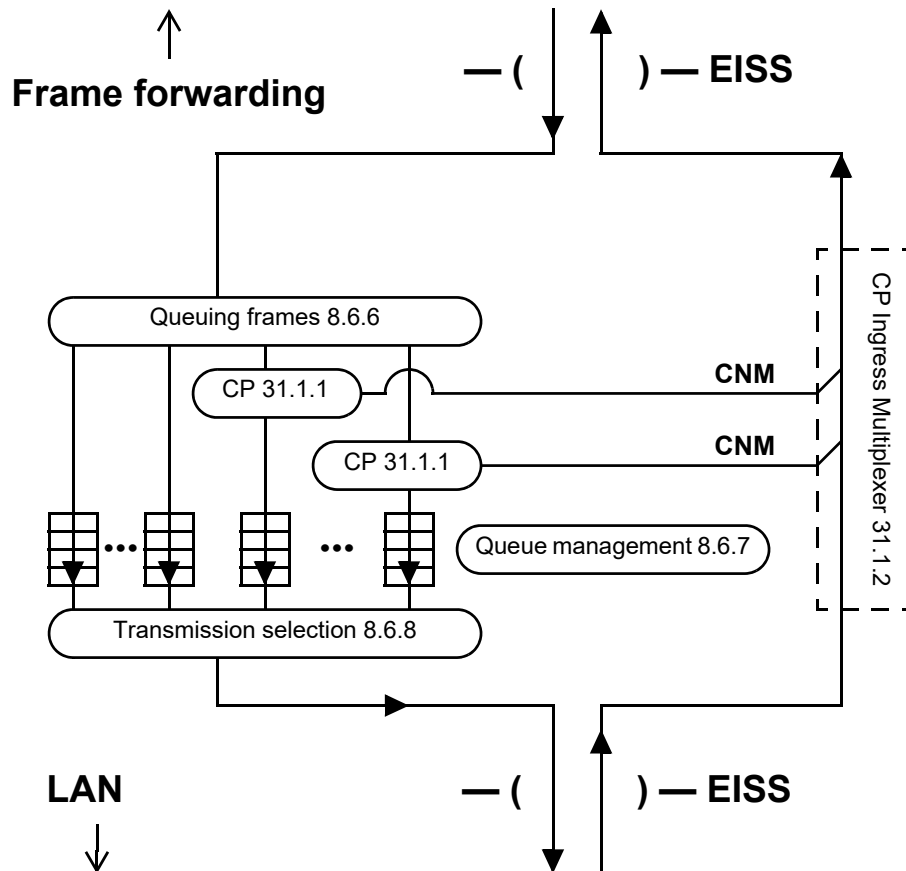


Figure 31-1—CPs and congestion-aware queues in a Bridge

1 31.1.1 Congestion Point (CP)

2 As described in 32.8 and 32.9, based on the state of its internal variables, and a frame given the CP by the
3 Queuing frames entity (8.6.6) in an EM_UNITDATA.request (parameters) (32.9.3), a CP either inserts the
4 frame into its attached queue or discards the frame, and can generate a CNM, based on the frame. Discarded
5 frames are counted in the variable cpDiscardedFrames (32.8.12). Frames successfully queued are counted in
6 the variable cpTransmittedFrames (32.8.13). Any CNM generated is passed to the CP ingress multiplexer
7 (31.1.2, Figure 31-1) or Flow multiplexer (31.2.4, Figure 31-2) for transmission back to the RP that sourced
8 the frame.

9 Congestion notification does not operate correctly if frames not belonging to a CCF are allowed to pass
10 through a CP. The CND operations (32.1) help to ensure that the only frames passing through a CP belong to
11 CCFs.

12 The CP shall remove the CN-TAG from every frame passing through it with a priority value (a CNPV) for
13 which the port on which the CP resides is acting in cptEdge or cptInterior CND defense mode.

14 The functions of Figure 22-1 could be implemented in different ways than that illustrated, without altering
15 the externally observed behavior of the conformant equipment. For example, the CP could pass a copy of the
16 output frame triggering a CNM to a function in the higher layer interfaces of a Bridge, which in turn, could
17 generate the CNM. However, in order to provide a management interface that is useful over a wide range of
18 implementations, the following provisions are made:

- 19 a) CNMs generated on a Bridge Port are not counted in the Frames Received [item a) in 12.6.1.1.3] or
20 Octets Received [item b) in 12.6.1.1.3] counters.
- 21 b) CNMs generated on one Bridge Port are counted, as are ordinary data frames, on the Bridge Ports (if
22 any) on which they are output.

23 If a Bridge supports the dot1agCfmVlanTable or the ieee8021CfmVlanTable, its CPs transmit each CNM to
24 the Primary VID associated with the vlan_identifier of the frame triggering the CNM's transmission.

25 31.1.2 CP ingress multiplexer

26 Inserts the CNMs generated by the CPs (31.1.1) among the frames received from the LAN.

27 31.2 Congestion-aware end station functions

28 Figure 31-2 illustrates the architecture of the queue functions of a congestion-aware end station. These
29 functions offer a service to higher layers through a single instance of the ISS or EISS, and utilize an instance
30 of the ISS or EISS to connect to the lower layers. The Output flow segregation function is presumed to
31 exercise control over the higher layers' ability to present frames for transmission through the ISS or the
32 EISS, and the Reception selection function to control the flow of frames to the higher layers, by means of
33 additional parameters local to this instance of the ISS or the EISS and/or locally significant LMI parameters
34 (IEEE Std 802.1AC), neither of which are specified by this standard.

35 In Figure 31-2, the outbound queue block, in the lower left of Figure 31-2, is exactly the same as the output
36 queues in a Bridge Port that is not congestion aware, as illustrated in Figure 22-1.

37 The remaining entities illustrated in Figure 31-2 are those that are peculiar to a congestion-aware end
38 station, and are further described in the following subclauses.

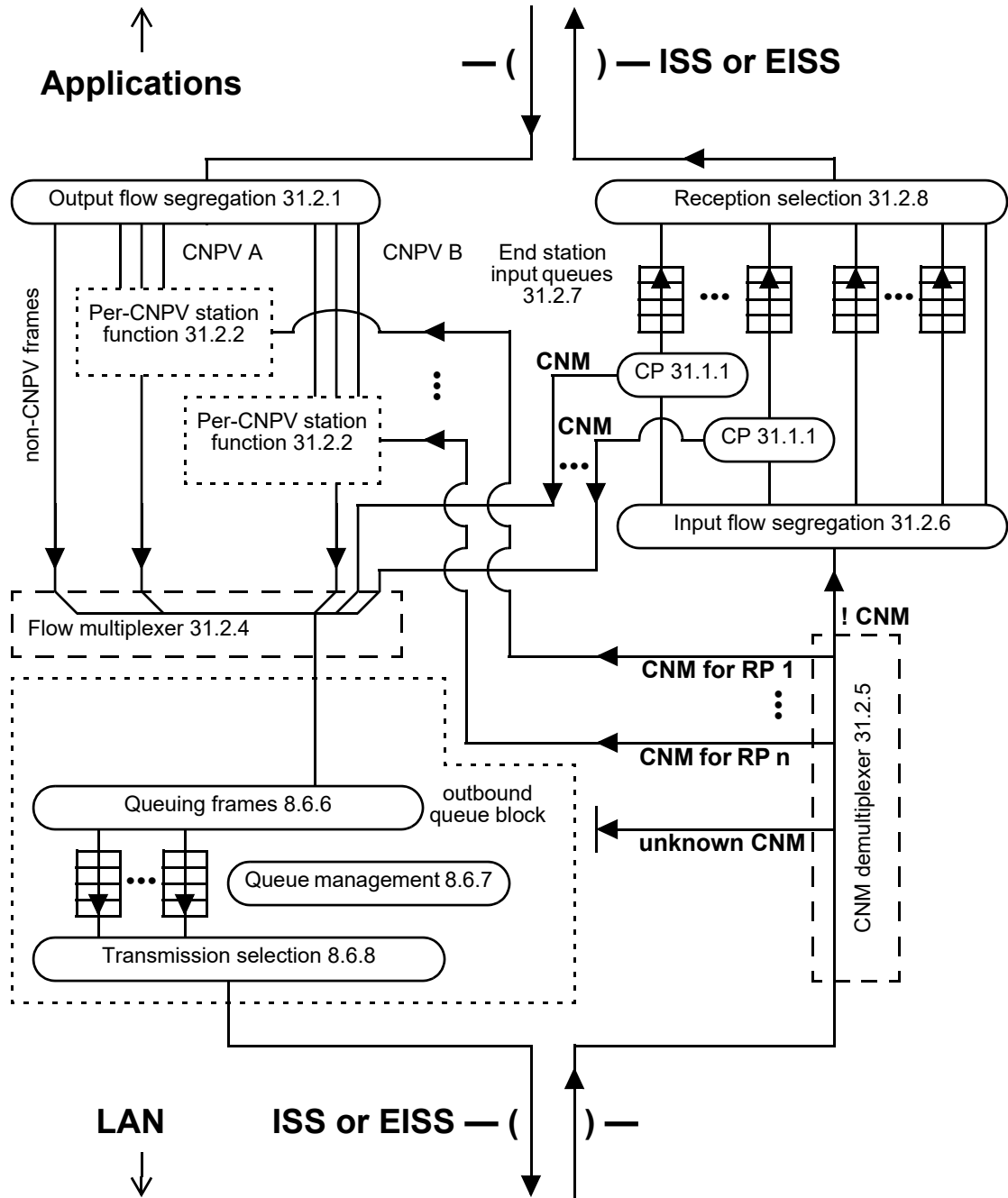


Figure 31-2—Congestion-aware queue functions in an end station

- 1
- 31.2.1 Output flow segregation
- 2
- For each frame received from the upper layers for transmission, the Output flow segregation entity:
- 3
- a)
- 4
- Assigns the frame a priority parameter based on the priority presented from the upper layer and/or upon other criteria, unspecified by this standard.
- 5
- b)
- 6
- Determines, using priority value and the Flow Select Database (31.2.3), to which Flow queue, if any, the frame is to be assigned.

- 1 c) If the frame is not assigned to any Flow queue (the frame's priority is not a CNPV), passes the frame
2 directly to the Flow multiplexer entity (31.2.4).
- 3 d) If the frame is assigned to a particular Flow queue (the frame's priority is a CNPV), enqueues the
4 frame in the selected Flow queue.
- 5 e) For frames that are assigned to a Flow queue, depending on the state of the Flow queue to which the
6 frame is assigned, and through means unspecified by this standard, can influence the higher layers'
7 presentation of further frames to the Output flow segregation entity.

8 The Output flow segregation entity never discards a frame. Every frame presented to it is either passed to the
9 Flow multiplexer entity (31.2.4) or stored in a Flow queue, as described previously. This standard assumes
10 that the Output flow segregation entity can apply means, unspecified by this standard, to prevent the upper
11 layers from offering a frame that should be stored in a Flow queue, but for which the Flow queue has
12 insufficient resources.

13 The default configuration for an end station shall have a single Flow queue per CNPV. An end station may
14 support additional Flow queues, or support more than one CNPV with a single Flow queue.

15 NOTE—Requirements for a default configurations do not imply that explicit action via SNMP is required to change
16 the default. For example, managed objects in an end station can be adjusted by an application program.

17 If the Output flow segregation function in an end station can change its method for assigning frames to Flow
18 queues, it shall not do so in a manner that impairs the ability of any compliant CP implementation to throttle
19 the data from the sourcing end station, and shall not materially increase the likelihood of misordered frame
20 transmissions in a manner that adversely impacts the higher layers' functions.

21 **31.2.2 Per-CNPV station function**

22 There is one Per-CNPV station function on each Port of a congestion-aware end station for each priority
23 value that is a CNPV. All of the frames passing through a Per-CNPV station function have the same CNPV
24 priority parameter value.

25 As illustrated in Figure 31-3, each Per-CNPV station function is composed of the following entities:

- 26 a) One or more Flow queues (31.2.2.1).
- 27 b) One or more RPs (31.2.2.2), each associated with exactly one Flow queue, each comprising:
 - 28 1) A RP state machine (31.2.2.3) for each RP.
 - 29 2) A Rate Limiter (31.2.2.4) for each RP.
- 30 c) One Flow Selection function (31.2.2.5).

31 **31.2.2.1 Flow queue**

32 A Flow queue is a first-in-first-out queue of frames, all for the same CNPV. An Output flow segregation
33 entity selects the Flow queue into which a given frame is inserted, and an RP determines when a frame is
34 removed from a Flow queue.

35 NOTE—The Flow queue is a notional convention used to unambiguously describe the behavior of the RP. An end
36 station can implement the Flow queue without explicitly storing frames in a queue, based on its knowledge of the state of
37 the higher-layer applications supplying the frames.

38 **31.2.2.2 Reaction Point (RP)**

39 A single RP is associated with each Flow queue. An RP is composed of a RP state machine (31.2.2.3) and a
40 Rate Limiter (31.2.2.4).

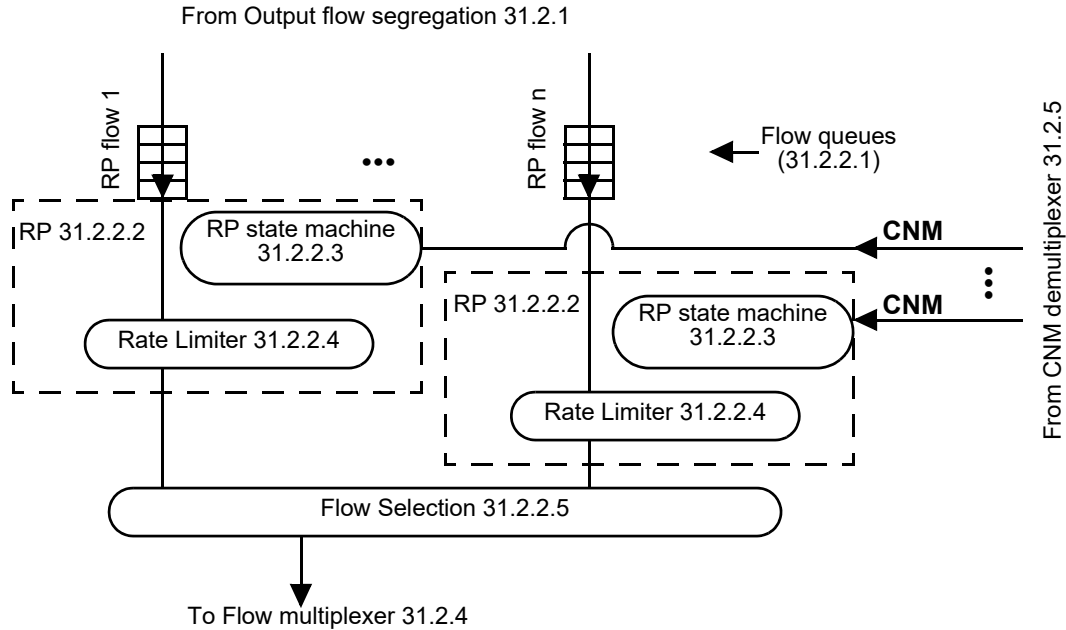


Figure 31-3—Per-CNPV station function

31.2.2.3 RP state machine

A single RP state machine is associated with each Flow queue. As described in detail in Clause 32, the RP state machine consists of a timer (32.12), variables (32.10, 32.11, and 32.13), procedures (32.14), and a state machine (32.15) that determine the value of the `rpLimiterRate` variable (32.13.8), which, via the Rate Limiter (31.2.2.4), controls the transmission of frames from the Flow queue. The state machine and variables, in turn, are controlled by the frames input, managed objects (12.20.4), the Output flow segregation entity (31.2.1), and the CNMs received from the CNM demultiplexer (31.2.5).

31.2.2.4 Rate Limiter

A single Rate Limiter is associated with each Flow queue. The Rate Limiter enforces the output rate, `rpLimiterRate` (32.13.8), determined by the RP state machine (31.2.2.3). The octets in the frames passed from the Rate Limiter to the Flow Selection function (31.2.2.5) are counted in `rpByteCount` (32.13.2). The details of the operation of the Rate Limiter are not specified by this standard, but the measurable output rate is specified.

For a given measurement period T , R_T is defined as the integral of `rpLimiterRate` over T , and L_T as the number of octets actually output to the LAN, including preambles, inter-frame gaps, etc., stemming from frames that pass through that RP's Rate Limiter. An end station shall ensure that:

$$L_T \leq (1.05 \times R_T) + (16 \text{ maximum-sized frames}) \quad (31-1)$$

for the two values of T = one second and T = 1 million bits divided by the physical line rate (in bits per second), where any number of simultaneous measurements can start at any time.

NOTE 1—The rate computations and variables [Equation (31-1), 32.13.6, 32.11.6, etc.] include all of the octets associated with a frame on the LAN, including preambles, inter-frame gaps, etc., and thus reflect actual physical data rates. However, the resource actually protected by congestion notification is octets in buffers, not octet times on wires. Per-frame overhead on the LAN is added to serve as a reasonable approximation, made at the RP, to the actual size of the

1 frame in the Bridges' or end stations' buffers. The extra 5% margin in Equation (31-1) is provided because this
2 approximation is inexact.

3 NOTE 2—Although the simulations have been studied with a burst of one maximum-sized frame, this standard allows
4 the end stations to send data in bursts of 16 maximum-sized frames, primarily to ease implementations. Larger bursts of
5 data have a negative impact on the overall performance of the QCN algorithm, and can lead to lower link utilization.

6 **31.2.2.5 Flow Selection**

7 The Flow Selection function decides from which Flow queue in its Per-CNPV station function, if any, a
8 frame will be offered to the Flow multiplexer. Whenever the output queue that receives frames from the
9 Flow Selection function (via the Flow multiplexer) is full, the Flow Selection function shall not present any
10 frame to the Flow multiplexer; this ensures that no frames in CNPVs will be discarded in the Port due to a
11 lack of room in the output queue.

12 By means unspecified in this standard, the Flow Selection function selects frames to pass to the Flow
13 multiplexer from the uncontrolled Flow queues (31.2.2.1) and Rate Limiters (31.2.2.4). Whenever a frame is
14 passed from a Rate Limiter through the Flow Selection function to the Flow multiplexer, the Flow Selection
15 function calls TransmitDataFrame (32.14.3) to update the RP variables (32.13). If the port's neighbor is
16 ready to receive frames with CN-TAGs for the frame's CND (defense mode is `cptInteriorReady`), the Flow
17 Selection function may insert a CN-TAG (30.5) at the beginning of the frame's `mac_service_data_unit`, and
18 if not (any CND defense mode other than `cptInteriorReady`), it shall not insert a CN-TAG. (See 33.2 for a
19 further explanation of CN-TAG insertion.)

20 **31.2.3 Flow Select Database**

21 The Flow Select Database is used by the Output flow segregation (31.2.1) entity to determine to which Flow
22 queue (31.2.2.1) a given frame is assigned.

23 The means by which the Flow Select Database selects a Flow queue is unspecified by this standard.

24 **31.2.4 Flow multiplexer**

25 The Flow multiplexer merges the frames from the Per-CNPV station functions' Flow Selection functions
26 (31.2.2.5), the frames bypassing the Per-CNPV station functions, and the CNM frames from the CPs
27 (31.1.1), and delivers the frames to the Queuing frames (8.6.6) entity in the end station's output path.

28 **31.2.5 CNM demultiplexer**

29 The CNM demultiplexer identifies CNMs received from the LAN, uses the Flow ID (33.2.1) and/or
30 Encapsulated priority (33.4.7) to determine to which RP (31.2.2.2) each CNM is intended, and directs them
31 to the correct RPs. This standard does not specify whether or how the end station can use the Flow ID to
32 further identify one or more individual flows within a Flow queue and its associated RP. If a CNM is
33 received that has no CN-TAG, or a CN-TAG whose Flow ID is unknown to the RP, the CNM is discarded.
34 All non-CNM frames are passed to the Input flow segregation entity (31.2.6).

35 **31.2.6 Input flow segregation**

36 For each frame received from the lower layers for reception, the Input flow segregation entity:

- 37 a) If the frame's priority is not a CNPV, decides, in an unspecified manner, whether to pass the frame to
38 an unspecified queue, pass it to the higher layers, or discard it.
- 39 b) If the frame's priority is a CNPV, passes the frame to a CP (31.1.1), selected by unspecified means,
40 for processing.

41 NOTE 1—For example, an end station might support a single CP, or support one CP per CNPV, per VID, per
42 application, or per data flow. Neither these nor any choices for CP selection are required by this standard.

1 The Input flow segregation entity never discards a CNPV frame. Every CNPV frame presented to it is
2 passed to a CP, as described previously.

3 NOTE 2—This does not mean that a station cannot drop CNPV frames. See 31.2.7.

4 **31.2.7 End station input queue**

5 The number and operation of end station input queues and the method used to drain these queues (i.e., to
6 select and transfer frames to the receiving higher layer applications) are not specified by this standard.
7 However, the procedures performed by a CP (32.9) require that its associated end station input queue exist,
8 and possess certain operational parameters (32.8).

9 In the unfortunate event that a misconfiguration or an unfortunate sequence of frame transmissions results in
10 more CNM frames arriving at a station than it can process, an end station input queue can fill up, and discard
11 a frame presented to it by its CP or by the Input flow segregation (31.2.6) entity.

12 **31.2.8 Reception selection**

13 The Reception selection entity selects frames for delivery to the upper layers. It takes the place of the
14 Transmission selection entity (8.6.8) of a congestion-aware Bridge. The selection algorithm depends on
15 interactions between the applications receiving the data, the operating system supporting those applications,
16 and the received frames that are not specified by this standard. The Reception selection entity is shown in
17 Figure 31-2 for completeness, to indicate that frames enqueued by the Queuing frames entity (8.6.6) are, in
18 fact, removed from the queues.

19 If the first octets of the `mac_service_data_unit` of a frame selected by the Reception selection entity are a
20 CN-TAG, Reception selection removes the CN-TAG from the `mac_service_data_unit` before passing the
21 frame to the higher layers. (See 33.2 for a further explanation of CN-TAG removal.)

1 32. Congestion notification protocol

2 Congestion-aware systems can participate in the congestion notification protocols specified in this clause:

- 3 a) CND operations (32.1).
- 4 b) The variables controlling congestion notification at the end station or Bridge component (32.2) and
5 Port and CNPV levels (32.3).
- 6 c) The variables (32.4), procedures (32.5), and state machine (32.6) that control the defense of a CND.
- 7 d) The congestion notification protocol (32.7).
- 8 e) The variables (32.8) and procedures (32.9) that define a CP.
- 9 f) The variables associated with all (32.10), or a subset of (32.11), the RPs on a given Port and CNPV.
- 10 g) The timer (32.12), variables (32.13), procedures (32.14), and state machine (32.15) that define the
11 operation of a single a RP.
- 12 h) The processing of CNMs by a congestion notification and encapsulation interworking function
13 (32.16).

14 NOTE—Clause 30 introduces the principles of congestion notification operation and the network architectural concepts
15 that support it. Clause 31 breaks down the congestion notification entities into their components. Clause 33 specifies the
16 encoding of the PDUs used by congestion notification.

17 32.1 CND operations

18 As described in 30.6, a CND is a connected subset of a Virtual Bridged Network configured to support a
19 particular CNPV. This standard provides a means whereby a VLAN Bridge or end station can recognize the
20 like-configured ports of a CND (32.1.1), and defend its CND against incoming frames from outside the
21 CND.

22 32.1.1 CND defense

23 Unless every Bridge along a path between two congestion-aware end stations using a particular CNPV is
24 configured for congestion notification (belongs to a CND), and unless the Bridges ensure that frames not in
25 CNPVs use different queues than the queues used by those two end stations, those end stations will not
26 accrue the advantages that congestion notification offers. Therefore, steps must be taken at the boundaries of
27 a CND to protect both the CND and the network outside the CND.

28 Every frame received on a Port passes through the ISS Support by specific MAC procedures
29 (IEEE Std 802.1AC), which can change the priority parameter of the received frame, using the Port's
30 Priority regeneration table, Table 6-4. This table can be controlled by a managed object (12.6). In addition,
31 in order to prevent frames not in CNPVs from entering CP-controlled queues, each Port's Priority
32 regeneration table can be modified.

33 If a Port's neighbor is known to also be configured for a particular CNPV, the entry in the Port's Priority
34 regeneration table for that CNPV is ignored, and the priority is never changed on input. The Bridge
35 component or end station connected to that Port can be trusted to use that CNPV. If the Port's neighbor is
36 known to not be configured for a particular CNPV, the entry in the Port's Priority regeneration table for that
37 CNPV shall be overridden to translate the CNPV to an alternate non-CNPV value. The neighboring system
38 is not trusted, so the priority values of any frames carrying the CNPV's value are changed. Altering the
39 priority values of received frames defends the CND from frames not originating from an RP. Similarly, if a
40 CNPV is configured on any Port, then on that Port, the Port's Priority regeneration table shall be overridden
41 to prevent any other priority value from being remapped into that CNPV.

42 If a port has multiple neighbors, or if the neighbor on a Port is not congestion aware, then the CN-TAGs
43 shall be removed from any frames transmitted on a CNPV. This ensures that the benefits of congestion
44 notification can be provided as far as possible along a path from a congestion-aware source to a

1 noncongestion-aware destination. Furthermore, CN-TAGs shall be removed from any frames transmitted
2 from a CNPV toward a system that is congestion aware, but is still remapping the CNPV to a non-CNPV
3 priority.

4 Thus, for each priority value, the CND defense mode can take one of the following four values, whether
5 derived from LLDP or set by managed variables:

6 — **cptDisabled:** The congestion notification capability is administratively disabled for this priority
7 value and port. This priority is not a CNPV. The priority regeneration table (6.9.4) controls the
8 remapping of input frames on this port to or from this priority. CN-TAGs are neither added by an
9 end station nor stripped by a Bridge.

10 — **cptEdge:** On this port and for this CNPV, the priority parameters of input frames are remapped to an
11 alternate (non-CNPV) value, and no priority value is remapped to this CNPV, regardless of the
12 priority regeneration table. CN-TAGs are not added by an end station and are removed from frames
13 before being output by a Bridge.

14 NOTE—The cptEdge CND defense mode is optional for an end station. See 32.4.9.

15 — **cptInterior:** On this port and for this CNPV, the priority parameters of input frames are not
16 remapped to another value, and no priority value is remapped to this CNPV, regardless of the
17 priority regeneration table. CN-TAGs are not added by an end station, and are removed from frames
18 before being output by a Bridge.

19 — **cptInteriorReady:** On this port and for this CNPV, the priority parameters of input frames are not
20 remapped to another value, and no priority value is remapped to this CNPV, regardless of the
21 priority regeneration table. CN-TAGs can be added by an end station and are not removed from
22 frames by a Bridge.

23 The determination of the operational state of a CNPV on a given port is determined by variables and by
24 LLDP (32.1.1, 32.2.1, 32.3.7, 32.3.4, 32.4.1, 32.4.3).

25 Taken together, these capabilities ensure that as long as the default values of the managed variables
26 controlling CND defense are not overridden with incorrect values:

- 27 a) A data frame received by an end station on a CNPV has been governed by congestion notification
28 along its entire path from the transmitting end station.
- 29 b) Uncontrolled data streams sourced by an end station that is congestion unaware cannot pass through
30 a CP, with a resultant excessive loss of congestion-aware frames.
- 31 c) A data frame carrying a CN-TAG cannot be delivered to a system that is congestion unaware, and
32 thus be unreadable by that system.

33 It is therefore recommended that the network administrator allow CND recognition to operate automatically
34 (32.1.2).

35 CND defense operates on each LAN separately; it does not ensure the existence of an end-to-end
36 congestion-aware path from end station to end station through a VLAN Bridged Network. Misconfiguration,
37 the presence along the path of a congestion unaware Bridge, or the activation of a new Bridge or LAN
38 (temporarily) can cause frames to exit a CND, and hence the following to occur:

- 39 d) Data frames can be transmitted by a congestion-aware end station on a CNPV, then be uncontrolled
40 for congestion along some part of their path, and be delivered to an end station on a non-CNPV
41 priority value with no CN-TAG.
- 42 e) If a congestion-aware end station transmits data frames with CN-TAGs, and is connected to a
43 congestion unaware end station along a path consisting entirely of congestion unaware or
44 misconfigured Bridges, the receiving end station will be unable to understand the CN-TAG, and thus
45 unable to process the data frames.

- 1 f) If a system generates CNMs on a CNPV (not recommended, see 32.2.2), they can be delivered on a
2 non-CNPV priority value with no CN-TAG.

3 32.1.2 Automatic CND recognition

4 Congestion notification can use the Link Layer Discovery Protocol (LLDP), defined in IEEE Std 802.1AB,
5 to advertise the CNPVs supported on a Bridge's or end station's port, to determine whether the other
6 Bridges' or end stations' ports connected to the same LAN are configured for those same CNPVs, and to
7 control the operation of the CND defenses (32.1.1).

8 Multiple instances of LLDP can be run on a single port. Each instance's LLDPDUs use a different
9 destination_address and therefore can reach different distances through a Virtual Bridged Network,
10 depending on the types of the devices in that network and their connectivity. Since any Bridge that is not
11 congestion aware can spoil the ability of the Virtual Bridged Network to minimize frame loss, the selection
12 of which LLDP instance is used for CND defense defaults to the Nearest Bridge Address (01-80-C2-00-00-
13 0E). This instance minimizes the reach of LLDP, and hence the risk that a noncongestion-aware Bridge is
14 present.

15 As described in D.2.7, a single Congestion Notification TLV can be inserted into the LLDPDUs transmitted
16 from any Bridge or end station port that is configured to support a CNPV. The Congestion Notification TLV
17 carries two fields that together provide for automatic control the of the CND defenses:

- 18 a) Eight per-priority CNPV indicators (D.2.7.3), one per priority level, indicating whether each priority
19 is or is not a CNPV on this port; and
20 b) Eight per-priority Ready indicators (D.2.7.4), one per priority level, indicating whether the priority
21 remap defenses are disabled on this port.

22 For every CNPV configured on a port, if LLDP finds exactly one remote neighbor, and if that neighbor's
23 database in the port's LLDP table carries a Congestion Notification TLV (D.2.7) with that CNPV's bit set in
24 the per-priority CNPV indicators, then that CNPV is known to be configured on the remote system's port.
25 This condition is summarized by the cnpdRcvdCnpv variable (32.4.11). Similarly, the cnpdRcvdReady
26 variable (32.4.11) indicates whether the remote neighbor has or has not turned off its priority remapping
27 defense. The read-only managed object cnpdAutoDefenseMode (32.4.3) indicates the results of the LLDP.

28 32.1.3 Variables controlling CND defense

29 In order to control CND defense, CN component managed objects (12.21.1), CN component priority
30 managed objects (12.21.2), CN Port priority managed objects (12.21.3), and Congestion Point managed
31 objects (12.21.4) override and are overridden by each other as follows:

- 32 a) Altering the CN component managed object (12.21.1) or CN component priority managed object
33 (12.21.2) does not alter the values of any of the other managed objects (12.21.3, 12.21.4, 12.21.6).
34 b) If cngMasterEnable (32.2.1) is FALSE, all congestion notification activity is suppressed; the Priority
35 Regeneration Table (6.9.4) operates normally, and CNMs, CN-TAGs, and LLDP Congestion
36 Notification TLVs are never generated and are always ignored on receipt. If TRUE, the other
37 managed objects control the operation of congestion notification.
38 c) The Port/priority and component objects override each other as shown in Table 32-1 and Table 32-2.
39 In each case, if the Port/priority object contains the specified value, the component or component/
40 priority object controls the specified function.

Table 32-1—LLDP instance selection managed object overrides

(Per-port per-priority) cnpdLdpInstanceChoice 32.4.4	(Component per-priority) cncpLdpInstanceChoice 32.3.6	Congestion Notification TLV is sent/received	LLDP instance is selected by
cnlNone	any	No	None selected
cnlAdmin	any	Yes	(Per-port per-priority) cnpdLdpInstanceSelector 32.4.5
cnlComponent	cnlNone	No	None selected
cnlComponent	cnlAdmin	Yes	(Component per-priority) cncpLdpInstanceSelector 32.3.7

Table 32-2—CND defense mode selection managed object overrides

(Per-port per-priority) cnpdDefModeChoice 32.4.1	(Component per-priority) cncpDefModeChoice 32.3.1	CND defense mode is selected by	Alternate priority is selected by
cpcAdmin	any	(Per-port per-priority) cnpdAdminDefenseMode 32.4.2	(Per-port per-priority) cnpdAlternatePriority 32.4.6
cpcAuto	any	(Per-port per-priority) cnpdAutoDefenseMode 32.4.3	(Component per-priority) cncpAutoAltPri 32.3.3
cpcComp	cpcAdmin	(Component per-priority) cncpAdminDefenseMode 32.3.4	(Component per-priority) cncpAlternatePriority 32.3.2
cpcComp	cpcAuto	(Per-port per-priority) cnpdAutoDefenseMode 32.4.3	(Component per-priority) cncpAutoAltPri 32.3.3

1 32.2 CN component variables

2 Every congestion-aware end station or Bridge component has a set of CN component variables to control the
3 overall operation of congestion notification. All of these variables are included in the CN component
4 managed object (12.21.1). Variables marked, “CP only” are not required for an end station that does not
5 support CPs. The CN component variables include the following:

- 6 a) cngMasterEnable (32.2.1)
- 7 b) cngCnmTransmitPriority (32.2.2)
- 8 c) cngDiscardedFrames (32.2.3)
- 9 d) cngErroredPortList (32.2.4)

1 32.2.1 cngMasterEnable

2 A Boolean value specifying whether congestion notification is enabled in this Bridge component or end
3 station.

4 32.2.2 cngCnmTransmitPriority

5 (CP only) The priority value to be used when transmitting CNMs from this Bridge component or end station
6 (default 6) [item h) in 32.9.4].

7 NOTE—If cngCnmTransmitPriority is itself a CNPV, then it is possible that another CP could generate a CNM in
8 response to a CNM sent by this Port. Although this is not desirable, the fact that CNMs are sent in response to only a
9 sampling of frames means that the network will not fail due to an excessive number of CNMs.

10 32.2.3 cngDiscardedFrames

11 (CP only) The total number of frames discarded from full CP queues. This is the total of the values of all
12 cpDiscardedFrames (32.8.12) variables for this end station or Bridge component.

13 32.2.4 cngErroredPortList

14 (CP only) A list of Ports whose alternate priority values (cnpdAlternatePriority, 32.4.6) specify a CNPV,
15 where 0 indicates that the Bridge's alternate priority value (cncpAlternatePriority, 32.3.2) specifies a CNPV.

16 NOTE—The alternate priority (cncpAlternatePriority, 32.3.2, or cnpdAlternatePriority, 32.4.6) for a given CNPV cannot
17 itself be a CNPV. But, because alternate priority values can be set using different managed objects for different CNPVs,
18 invalid configurations can be specified. cngErroredPortList provides the network administrator with a list of Ports that
19 could have invalid configurations.

20 32.3 Congestion notification per-CNPV variables

21 An instance of the congestion notification per-CNPV variables exists for each CNPV in a congestion-aware
22 end station or Bridge component to control CN for all Ports in the end station or Bridge component.
23 Variables marked “CP only” are not required for an end station that does not support CPs. The congestion
24 notification per-CNPV variables include the following:

- 25 a) cncpDefModeChoice (32.3.1)
- 26 b) cncpAlternatePriority (32.3.2)
- 27 c) cncpAutoAltPri (32.3.3)
- 28 d) cncpAdminDefenseMode (32.3.4)
- 29 e) cncpCreation (32.3.5)
- 30 f) cncpLldpInstanceChoice (32.3.6)
- 31 g) cncpLldpInstanceSelector (32.3.7)

32 32.3.1 cncpDefModeChoice

33 An enumerated value specifying how the CND defense mode and alternate priority for all Ports for this
34 CNPV are to be selected, unless overridden (32.1.3) by the variables in the CND defense per-Port per-CNPV
35 variables (32.4), either one of the following:

- 36 1) **cpcAdmin:** The Port's default CND defense mode for this priority is controlled by
37 cncpAdminDefenseMode (32.3.4) and the alternate priority by cncpAlternatePriority (32.3.2);
38 or
- 39 2) **cpcAuto:** The Port's default CND defense mode for this priority is controlled by LLDP and
40 the Congestion Notification TLV (cnpdAutoDefenseMode, 32.4.3) and the alternate priority by
41 cncpAutoAltPri (32.3.3).

1 32.3.2 cncpAlternatePriority

2 (CP only) An alternate priority value to which this priority value is to be remapped when the Port's CND
3 defense mode is cptEdge for this priority (default value 0). This can be overridden by cnpdAlternatePriority
4 (32.1.3, 32.4.6).

5 32.3.3 cncpAutoAltPri

6 An integer indicating the next lower priority value than this CNPV that is not a CNPV in an end station or
7 Bridge component, or the next higher non-CNPV, if all lower values are CNPVs (32.1.1).

8 32.3.4 cncpAdminDefenseMode

9 An enumerated value controlling the CND defense mode (30.6) for all Ports for this CNPV in this Bridge
10 component or end station: cptDisabled (1), cptInterior (2, the default value), cptInteriorReady (3), or
11 cptEdge (4) (32.1.1). This can be overridden by cnpdAdminDefenseMode (32.1.3, 32.4.2).

12 32.3.5 cncpCreation

13 An enumerated value controlling the applicability of this CN component priority managed object to the
14 Ports in that Bridge component or end station, either one of the following:

- 15 1) **cncpAutoEnable:** The cnpdDefModeChoice variable (32.4.1) in any CN Port priority
16 managed object created as a result of the creation of this CN component priority managed
17 object or the creation of a Port is given the value cpcComp (3); or
- 18 2) **cncpAutoDisable:** The cnpdDefModeChoice variable (32.4.1) in any CN Port priority
19 managed object created as a result of the creation of this CN component priority managed
20 object or the creation of a Port is given the value cpcAdmin (1).

21 32.3.6 cncpLldpInstanceChoice

22 Determines the method by which the component LLDP instance selector, specifying on which LLDP
23 instance the Congestion Notification TLV is to carry information for this priority, is determined (32.1.1),
24 either one of the following:

- 25 1) **cnlNone:** if LLDP is not to carry Congestion Notification TLVs on any Port or priority; or
- 26 2) **cnlAdmin:** If cncpLldpInstanceSelector (32.3.7) governs LLDP instance selection for all Ports
27 and priorities (this is the default value).

28 This choice can be overridden (32.1.3) on a per-Port per-priority basis by cnpdLldpInstanceChoice (32.4.4).

29 32.3.7 cncpLldpInstanceSelector

30 A reference to the LLDP destination address table entry whose LLDPDUs are to carry Congestion
31 Notification TLVs for this Bridge component or end station (32.1.1). The use of this variable is controlled by
32 cncpLldpInstanceChoice (32.3.6).

1 32.4 CND defense per-Port per-CNPV variables

2 For each Port in a congestion-aware end station or Bridge component, and for each priority value for which
3 a set of congestion notification per-CNPV variables exists, there is an instance of the CND defense per-Port
4 per-CNPV variables. These variables control congestion notification on the port for a CNPV, including CND
5 defense. Variables marked, “CP only” are not required for an end station that does not support CPs. The
6 CND defense per-Port per-CNPV variables include the following:

- 7 a) `cnpdDefModeChoice` (32.4.1)
- 8 b) `cnpdAdminDefenseMode` (32.4.2)
- 9 c) `cnpdAdminDefenseMode` (32.4.2)
- 10 d) `cnpdAutoDefenseMode` (32.4.3)
- 11 e) `cnpdLldpInstanceChoice` (32.4.4)
- 12 f) `cnpdLldpInstanceSelector` (32.4.5)
- 13 g) `cnpdAlternatePriority` (32.4.6)
- 14 h) `cnpdXmitCnpvCapable` (32.4.7)
- 15 i) `cnpdXmitReady` (32.4.8)
- 16 j) `cncpDoesEdge` (32.4.9)
- 17 k) `cnpdAcceptsCnTag` (32.4.10)
- 18 l) `cnpdRcvdCnpv` (32.4.11)
- 19 m) `cnpdRcvdReady` (32.4.12)
- 20 n) `cnpdIsAdminDefMode` (32.4.13)
- 21 o) `cnpdDefenseMode` (32.4.14)

22 32.4.1 `cnpdDefModeChoice`

23 An enumerated value specifying how the CND defense mode and the alternate priority of the Port for this
24 priority is to be determined, (30.6, 32.1.1, 32.1.1):

- 25 1) **cpcAdmin:** The Port’s CND defense mode for this priority is controlled by
26 `cnpdAdminDefenseMode` (32.4.2), and the alternate priority by `cnpdAlternatePriority`
27 (32.4.6);
- 28 2) **cpcAuto:** The Port’s CND defense mode for this priority is controlled by LLDP and the
29 Congestion Notification TLV (`cnpdAutoDefenseMode`, 32.4.3) and the alternate priority by
30 `cncpAutoAltPri` (32.3.3); or
- 31 3) **cpcComp:** The Port’s CND defense mode for this priority is controlled by
32 `cncpDefModeChoice` (32.3.1) and the alternate priority by `cncpAutoAltPri` (32.3.3) (this is the
33 default value).

34 32.4.2 `cnpdAdminDefenseMode`

35 An enumerated value specifying the CND defense mode of the Port for this priority, if and only if
36 `cnpdDefModeChoice` (32.4.1) has the value `cpcAdmin` (1) (30.6, 32.1.1, 32.1.1):

- 37 1) **cptDisabled:** The CP is disabled on this port for this CNPV. Priority regeneration on input is
38 controlled by the priority regeneration table. This is the same behavior as when
39 `cngMasterEnable` (32.2.1) has the value FALSE. (This is the default value.);
- 40 2) **cptInterior:** The priority parameter of frames input are not remapped to or from this priority,
41 and CN-TAGs are not output;
- 42 3) **cptInteriorReady:** The priority parameter of frames input are not remapped to or from this
43 priority, and CN-TAGs can be output; or
- 44 4) **cptEdge:** The priority parameter of frames input at this priority are remapped to an alternate
45 value, frames at other priorities are not remapped to this priority, and CN-TAGs are not output.

1 32.4.3 cnpdAutoDefenseMode

2 An enumerated value indicating the operating mode: cptInterior (2), cptInteriorReady (3), or cptEdge (4), in
3 which the Port would operate for this CNPV if cnpdDefModeChoice (32.4.1) had the value cpcAuto (2).

4 NOTE—cncpAdminDefenseMode (32.3.4) is controlled by the network administrator. The object
5 cnpdAutoDefenseMode (32.4.3) indicates what the CND defense mode would be, if it were controlled by the LLDP
6 Congestion Notification TLV.

7 32.4.4 cnpdLldpInstanceChoice

8 Port and priority LLDP instance selector, specifying on which LLDP instance the Congestion Notification
9 TLV is to carry information for this priority on this Port (32.1.1):

- 10 1) **cnlNone:** No LLDP Congestion Notification TLV is to carry Per-priority CNPV indicators or
11 Per-priority Ready indicators on this Port for this priority;
- 12 2) **cnlAdmin:** cnpdLldpInstanceSelector (32.4.5) governs which LLDP instance is to carry Per-
13 priority CNPV indicators and Per-priority Ready indicators for this priority in its Congestion
14 Notification TLV on this Port; or
- 15 3) **cnlComponent:** cncpLldpInstanceSelector (32.3.7) governs LLDP instance selection for this
16 Port and priority (this is the default value).

17 32.4.5 cnpdLldpInstanceSelector

18 A reference to the LLDP destination address table entry for an LLDP instance selector, specifying on which
19 LLDP instance the Congestion Notification TLV is to carry information for this priority on this Port, if and
20 only if cnpdLldpInstanceChoice (32.4.4) is set to cnlAdmin (2). Default value is 1. (32.1.1);

21 32.4.6 cnpdAlternatePriority

22 (CP only) An alternate priority value to which this priority value is to be remapped when the Port's CND
23 defense mode is cptEdge for this priority, if and only if cnpdDefModeChoice (32.4.1) is set to cpcAdmin (1)
24 (default value 0).

25 32.4.7 cnpdXmitCnpvCapable

26 Per-port per-priority Boolean variable indicating whether a given priority on this Port is (TRUE) or is not
27 (FALSE) currently operating as a CNPV. cnpdXmitCnpvCapable is transmitted in the per-priority CNPV
28 indicators in the Congestion Notification TLV (D.2.7.3). The variable is TRUE if and only if all of the
29 following conditions are met:

- 30 a) cngMasterEnable (32.2.1) is TRUE; and
- 31 b) The CND defense mode, as selected by cncpDefModeChoice, cncpAdminDefenseMode,
32 cnpdDefModeChoice, and cnpdAdminDefenseMode, according to Table 32-2, is not cptDisabled.

33 32.4.8 cnpdXmitReady

34 Per-port per-priority Boolean variable indicating whether the priority remap defenses for this port and
35 CNPV have been disabled. cnpdXmitReady is transmitted in the per-priority Ready indicators in the
36 Congestion Notification TLV (D.2.7.4). This variable is set and reset by the CND defense state machine
37 (32.6).

1 **32.4.9 cncpDoesEdge**

2 Boolean variable indicating whether the cptEdge CND defense mode (32.1.1) is (TRUE) or is not (FALSE)
3 implemented on this port for this priority. This variable is TRUE in a Bridge. In an end station, it may be
4 either TRUE or FALSE. This variable is not a managed object.

5 NOTE—This variable, when FALSE, causes the state machine in Figure 32-1 to pass through the
6 CNDD_EDGE state directly to the CNDD_INTERIOR state. Since the state machine takes, in theory, no
7 time to execute, it is impossible to tell from any measurement taken outside the system whether the actions
8 specified in the CNDD_EDGE state are actually executed.

9 **32.4.10 cnpdAcceptsCnTag**

10 Boolean variable indicating whether the CN-TAG is (TRUE) or is not (FALSE) acceptable on data frames
11 received on this port for this CNPV. This variable is TRUE in a Bridge. In an end station, it may be either
12 TRUE or FALSE. This variable is not a managed object.

13 NOTE—This variable can be used by an end station that finds it convenient for the adjacent Bridge to remove CN-TAGs
14 from frames on a CNPV. By preventing this Port and CNPV's cnpdXmitReady variable from being set TRUE, (see
15 Figure 32-1) cnpdAcceptsCnTag prevents the adjacent device's CND defense state machine (32.6) from advancing from
16 the CNDD_INTERIOR state to the CNDD_INTERIOR_READY state, and thus prevents it from transmitting a
17 CN-TAG on this CNPV.

18 **32.4.11 cnpdRcvdCnpv**

19 Per-port per-priority Boolean variable indicating the presence of an LLDP neighbor's Congestion
20 Notification TLV (D.2.7). cnpdRcvdCnpv is TRUE only if all of the following are true:

- 21 a) An LLDP instance is selected by cncpLldpInstanceSelector (32.3.7) and cnpdLldpInstanceSelector
22 (32.4.5);
- 23 b) The selected LLDP instance indicates that the Port has a single neighbor;
- 24 c) The selected LLDP instance's neighbor information includes a Congestion Notification TLV; and
- 25 d) That Congestion Notification TLV's Per-priority CNPV indicators field has a value of 1 in the
26 position corresponding to this CNPV.

27 **32.4.12 cnpdRcvdReady**

28 Per-port per-priority Boolean variable indicating that the neighboring system has turned off its CND
29 defenses. cnpdRcvdReady is TRUE only if all of the following are true:

- 30 a) cnpdRcvdCnpv is TRUE; and
- 31 b) The bit in the same Congestion Notification TLV's (D.2.7) Per-priority Ready indicators field
32 (D.2.7.4) has a value of 1 in the position corresponding to this CNPV.

33 **32.4.13 cnpdIsAdminDefMode**

34 Boolean variable derived from the managed objects, indicating to the CND defense state machine (32.6)
35 whether the CND defense mode is being forced to a particular state by the managed objects, as shown in
36 Table 32-3.

37 NOTE—The purpose of this variable is to simplify the Boolean expressions in Figure 32-1.

38 **32.4.14 cnpdDefenseMode**

39 Enumerated value, indicating the management choice for the CND defense mode, if any, as shown in
40 Table 32-3. The value of this variable is not defined if the managed variables indicate no choice for the CND
41 defense mode.

Table 32-3—Determining cnpdIsAdminDefMode and cnpdDefenseMode

(Per-port per-priority) cnpdDefModeChoice (32.4.1)	(Component per-priority) cncpDefModeChoice (Clause 32)	cnpdIsAdminDefMode (32.4.13) is	cnpdDefenseMode (32.4.14) is obtained from
cpcAdmin	Any	TRUE	(Per-port per-priority) cnpdAdminDefenseMode 32.4.2
cpcAuto	Any	FALSE	Not defined
cpcComp	cpcAdmin	TRUE	(Component per-priority) cncpAdminDefenseMode 32.3.4
cpcComp	cpcAuto	FALSE	Not defined

1 NOTE—The purpose of this variable is to simplify the exit condition expressions in Figure 32-1.

2 32.5 CND defense procedures

3 There are three procedures associated with the CND defense state machine. They are as follows:

- 4 a) DisableCnpvRemapping() (32.5.1)
- 5 b) TurnOnCnDefenses() (32.5.2)
- 6 c) TurnOffCnDefenses() (32.5.3)

7 32.5.1 DisableCnpvRemapping()

8 Removes any overrides by congestion notification of the Priority Regeneration Table (6.9.4).

9 32.5.2 TurnOnCnDefenses()

10 Overrides the Priority Regeneration Table (6.9.4) to use the alternate priority specified by
11 cncpAlternatePriority (32.3.2), cncpAutoAltPri (32.3.3), and cnpdAlternatePriority (32.4.6) for frames with
12 this CNPV, instead of the value in the Priority Regeneration Table.

13 32.5.3 TurnOffCnDefenses()

14 Overrides the Priority Regeneration Table (6.9.4) to not alter the priority of incoming frames with this
15 CNPV, instead of using the value in the Priority Regeneration Table.

16 32.6 CND defense state machine

17 The CND defense state machine is illustrated in Figure 32-1. A congestion-aware Bridge component or end
18 station shall implement one CND defense state machine per port per priority level.

19 NOTE—In Figure 32-1, global transitions are associated with each value of cnpdDefenseMode when the CND defense
20 mode is chosen by the managed variables; the nonglobal transitions from state to state can take place only when the
21 CND defense mode is chosen by LLDP.

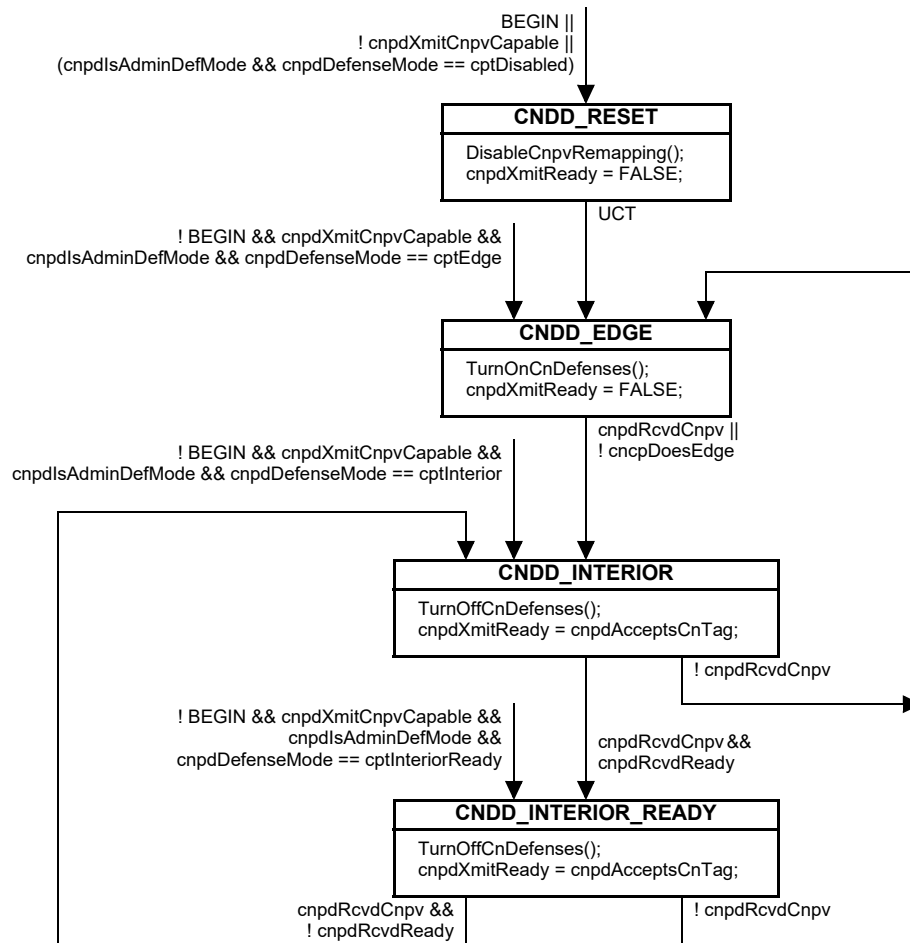


Figure 32-1—CND defense state machine

32.7 Congestion notification protocol

Clause 32 is a detailed specification of the state machines required to implement the QCN algorithm introduced in 30.2. Table 32-4 shows the correlation between the description of QCN in 30.2, the state machine variables in this clause, and the CNM PDU fields in 33.4.

There are two stateful participants in the congestion notification protocol:

- a) The CP (31.1.1) shall behave in a manner that is indistinguishable, from an observer external to a system, from a strict implementation of the variables in 32.8 and the procedures in 32.9.
- b) The RP (31.2.2.2) shall behave in a manner that is indistinguishable, from an observer external to a system, from a strict implementation of the timer in 32.12, variables in 32.13, the procedures in 32.14, and the state machine in 32.15.

Table 32-4—Correspondence of QCN and CCF message fields

Quantized Congestion Notification protocol, 30.2	Congestion notification protocol, 32.7	Congestion Notification Message, 33.3
Quantized Feedback calculation (32.8.9, item e) in 32.9.4)		
F_b (30.2.1)	—	cpFb (33.4.3)
Q_{off} (30.2.1)	– cpQOffset (32.8.7)	cnmQOffset (33.4.5)
Q (30.2.1)	cpQLen (32.8.4)	—
Q_{eq} (30.2.1)	cpQSp (32.8.3)	—
Q_{δ} (30.2.1)	cpQDelta (32.8.8)	cnmQDelta (33.4.6)
Q_{old} (30.2.1)	cpQLenOld (32.8.5)	—
w (30.2.1)	cpW (32.8.6)	—
RP rate calculations (32.14.5, 32.14.6, Figure 32-2)		
CR (30.2.2)	rpCurrentRate (32.13.6)	—
TR (30.2.2)	rpTargetRate (32.13.5)	—
Byte Counter (30.2.3)	rpByteCount (32.13.2)	—
Timer (30.2.3)	RpWhile (32.12.1)	—
G_d (30.2.2.1)	rpgGd (32.11.8)	—
R_{AI} (30.2.2.1)	rpgAiRate (32.11.6)	—
R_{HAI} (30.2.3)	rpgHaiRate (32.11.7)	—
i (30.2.3)	min(rpByteStage, rpTimeStage) (RPR_HYPER_INCREASE)	—

1 32.8 CP variables

2 The following variables control the operation of a CP:

- 3 a) cpMacAddress (32.8.1)
- 4 b) cpId (32.8.2)
- 5 c) cpQSp (32.8.3)
- 6 d) cpQLen (32.8.4)
- 7 e) cpQLenOld (32.8.5)
- 8 f) cpW (32.8.6)
- 9 g) cpQOffset (32.8.7)
- 10 h) cpQDelta (32.8.8)
- 11 i) cpFb (32.8.9)
- 12 j) cpEnqueued (32.8.10)
- 13 k) cpSampleBase (32.8.11)
- 14 l) cpDiscardedFrames (32.8.12)
- 15 m) cpTransmittedFrames (32.8.13)
- 16 n) cpTransmittedCnms (32.8.14)
- 17 o) cpMinHeaderOctets (32.8.15)

1 32.8.1 cpMacAddress

2 The MAC address, belonging to the system transmitting the CNM PDU, used as the source_address of
3 CNMs sent from this CP [item b) in 32.9.4].

4 32.8.2 cpId

5 Unsigned integer. A number that uniquely identifies a CP in a Virtual Bridged Network [item p) in 32.9.4].

6 NOTE—This standard does not specify whether the Congestion Point Identifier (CPID) reported in a CNM by a CP that
7 serves multiple CNPVs does or does not have the same value for its different CNPVs.

8 32.8.3 cpQSp

9 Unsigned integer. The set-point for the queue. This is the target number of octets in the CP's queue [item e)
10 in 32.9.4]. Default value 26000.

11 32.8.4 cpQLen

12 Unsigned integer. The current number of octets in the CP's queue [item c) in 32.9.4].

13 32.8.5 cpQLenOld

14 Unsigned integer. The previous value of cpQLen. cpQLenOld is updated from cpQLen each time
15 GenerateCnmPdu() is called [item c) in 32.9.4].

16 32.8.6 cpW

17 Real number. cpW is the weight to be given to the change in queue length in the calculation of cpFb
18 (32.8.9). Default value 2. Although cpW is specified as a real number, it is constrained to be a power of 2,
19 e.g., 1/2, 1, 2, 4. In practice, therefore, it can be represented as a shift distance (plus an addition) in item e) in
20 32.9.4.

21 32.8.7 cpQOffset

22 The signed integer value of the transmitting CP's cpQSp (32.8.3) – cpQLen (32.8.4) used to calculate cpFb
23 (32.8.9), and thus the Quantized Feedback field (33.4.3).

24 32.8.8 cpQDelta

25 The signed integer value of the transmitting CP's cpQLen (32.8.4) – cpQLenOld (32.8.5) used to calculate
26 cpFb (32.8.9), and thus the Quantized Feedback field (33.4.3).

27 32.8.9 cpFb

28 Signed integer. Calculated just before the CP attempts to enqueue a frame:

29
$$\text{cpFb} = \text{cpQOffset} - \text{cpW} \times \text{cpQDelta},$$

30 where

31
$$\text{cpQDelta} = \text{cpQLen} - \text{cpQLenOld} \quad \text{cpQOffset} = \text{cpQSp} - \text{cpQLen}$$

32 cpFb has two terms. The first term is the difference between the current and the desired queue lengths
33 (cpQOffset, 32.8.7). The second is a weight factor cpW times the difference cpQDelta (32.8.8) between the
34 current and the previous queue lengths. Thus, a multiple of the first derivative of the queue size is subtracted
35 from the current nonoptimality of the queue, so that if the queue length is moving toward the set point
36 cpQSp, cpFb will be closer to 0 than if the queue length is moving away from cpQSp.

1 **32.8.10 cpEnqueued**

2 Signed integer. The number of octets remaining to be enqueued by the CP before a CNM PDU is to be
3 generated (32.9.3).

4 **32.8.11 cpSampleBase**

5 Unsigned integer. The minimum number of octets to enqueue in the CP's queue between CNM PDU
6 transmissions (32.9.3). Default value 150 000.

7 **32.8.12 cpDiscardedFrames**

8 The number of frames offered to this CP that were discarded because of a full output queue (31.1.1).

9 **32.8.13 cpTransmittedFrames**

10 The number of data frames enqueued for transmission on this CP's output queue (31.1.1).

11 **32.8.14 cpTransmittedCnms**

12 The number of CNMs transmitted by this CP [item s] in 32.9.4].

13 **32.8.15 cpMinHeaderOctets**

14 The minimum number of octets that the CP is to return in the Encapsulated MSDU field (33.4.10) of each
15 CNM it generates [item k] in 32.9.4]. Default value 0.

16 **32.9 CP procedures**

17 The procedures used in a CP include the following:

- 18 a) Random() (32.9.1)
- 19 b) NewCpSampleBase() (32.9.2)
- 20 c) EM_UNITDATA.request (parameters) (32.9.3)
- 21 d) GenerateCnmPdu() (32.9.4)

22 A CP needs no state machine; the procedure EM_UNITDATA.request (parameters), called each time a
23 frame is presented for queuing, triggers all of the CP's functions.

24 **32.9.1 Random**

25 The Random function takes two parameters, min and max, and returns a pseudo-random number in the range
26 $\text{min} \leq \text{number} < \text{max}$. This function shall be initialized so that it generates a different value each time the
27 system is reset.

28 **32.9.2 NewCpSampleBase()**

29 Called by EM_UNITDATA.request (parameters) (32.9.3) when generating a new value for cpSampleBase
30 (32.8.11) from the CNM PDU Quantized Feedback field (33.4.3) last generated by GenerateCnmPdu().
31 NewCpSampleBase() returns a real number according to Table 32-5.

32 NOTE—The values in Table 32-5 are chosen so that the value returned by NewCpSampleBase() can be an integer to be
33 used in a division, rather than a real number to be used in a multiplication.

Table 32-5—NewCpSampleBase() return value as a function of cpFb

Quantized Feedback / 8	Value returned by NewCpSampleBase()
0	1
1	1/2
2	1/3
3	1/4
4	1/5
5	1/6
6	1/7
7	1/8

32.9.3 EM_UNITDATA.request (parameters)

A CP offers an instance of the EISS (6.8) to the Queuing frames function (8.6.6). When called upon to enqueue a frame, the CP:

- a) Determines the number of octets of buffer space required to enqueue the frame, and subtracts that number from cpEnqueued (32.8.10).
- b) If cpEnqueued ≤ 0 , then:
 - 1) Calculates a new value for cpFb according to 32.8.9.
 - 2) Calls GenerateCnmPdu() to conditionally transmit a CNM PDU.
 - 3) Replaces cpEnqueued with cpSampleBase $\times$ NewCpSampleBase() $\times$ Random(0.85, 1.15), and if the result is less than zero, resets cpEnqueued to 0.

The accuracy of the sample jitter introduced by the function Random(0.85, 1.15) is not critical to the success of congestion notification. The amount of jitter introduced by this function should be no more than $0.75 \leq \text{jitter} < 1.25$ and no less than $0.875 \leq \text{jitter} < 1.125$.

32.9.4 GenerateCnmPdu()

Called by the CP to conditionally generate a CN-TAGged CNM PDU for output. GenerateCnmPdu()

- a) Uses the source_address of the frame triggering the CNM PDU as the destination_address of the generated CNM PDU.
- b) Uses cpMacAddress (32.8.1) as the source_address of the CNM PDU.
- c) Replaces the value of cpQLenOld (32.8.5) with cpQLen (32.8.4).
- d) If cpFb is greater than or equal to 0, or if the destination_address parameter of the CNM PDU is a Group, and not an Individual, address, does nothing; no further processing takes place, and no CNM PDU is transmitted.
- e) Fills the Quantized Feedback field (33.4.3) of the CNM PDU as follows:

if (cpFb $< -\text{cpQSp} \times (2 \times \text{cpW} + 1)$)

Quantized Feedback = 63

else

Quantized Feedback = $-\text{cpFb} \times 63 / (\text{cpQSp} \times (2 \times \text{cpW} + 1))$
- f) If the Quantized Feedback field equal to 0 does nothing; no further processing takes place, and no CNM PDU is transmitted.

- 1 g) If the first octets of the `mac_service_data_unit` of the frame triggering the CNM PDU are a
2 CN-TAG, copies the Flow Identifier field (33.2.1) from that CN-TAG to the CN-TAG of the CNM,
3 else fills the Flow Identifier field of the CNM's CN-TAG with 0.

- 4 h) Sets the priority parameter of the CNM PDU from `engCnmTransmitPriority` (32.2.2).

5 NOTE 1—The default value for the priority of a CNM PDU is 6. CNM PDUs are not expected to be sent in high
6 volumes, but undelivered CNM PDUs reduce the ability of congestion notification to prevent lost data frames. Priority 7
7 is typically reserved for control traffic that, if not delivered, can result in the loss of connectivity in the Bridged Network.

- 8 i) Sets the `vlan_identifier` of the CNM PDU from the `vlan_identifier` of the frame triggering the CNM
9 PDU, if the `dot1agCfmVlanTable` is not implemented, else the Primary VID of that `vlan_identifier`.

- 10 j) Fills the Encapsulated destination MAC address field (33.4.8) of the CNM PDU with the
11 `destination_address` parameter of the frame triggering the CNM PDU.

- 12 k) Fills the Encapsulated MSDU field (33.4.10) of the CNM PDU with the first octets of the remainder
13 of the `mac_service_data_unit` following the CN-TAG, if present, of the frame triggering the CNM
14 PDU. The minimum number of octets is the value of `cpMinHeaderOctets` (32.8.15), or the whole of
15 the `mac_service_data_unit` following the CN-TAG, if it is shorter, and the maximum is 64.

- 16 l) Fills the Encapsulated MSDU length field (33.4.9) of the CNM PDU with the number of octets
17 inserted into the Encapsulated MSDU field (33.4.10).

- 18 m) Inserts the encapsulation appropriate to the CP's port, as specified in 33.3, at the beginning of the
19 `mac_service_data_unit`.

20 NOTE 2—This encapsulation is changed by the Bridge, if required, before the CNM PDU is output on a port that uses a
21 different encapsulation from the CP's port.

- 22 n) Fills the Version field (33.4.1) of the CNM PDU with 0.

- 23 o) Fills the ReservedV field (33.4.2) of the CNM PDU with 0.

- 24 p) Fills the Congestion Point Identifier field (33.4.4) of the CNM PDU from the variable `cpId` (32.8.2).

- 25 q) Computes and fills the `cnmQOffset` and `cnmQDelta` fields (33.4.5, 33.4.6) from the variables
26 `cpQOffset` and `cpQDelta`.

- 27 r) Fills the Encapsulated priority field (33.4.7) from the priority parameter of the frame triggering the
28 CNM PDU.

- 29 s) Increments `cpTransmittedCnms` (32.8.14).

- 30 t) Passes the CN-TAG and CNM PDU as an `EM_UNITDATA.indication` to the higher layers
31 (Figure 31-1), if the CP is in a Bridge, or to the Flow multiplexer (31.2.4), if the CP is in an end
32 station (Figure 31-2).

33 32.10 RP per-Port per-CNPV variables

34 There is one set of these variables per Port per CNPV in a congestion-aware end station. A set of RP per-Port
35 per-CNPV variables is created or deleted whenever a CN Port priority managed object is created or deleted.
36 All of the RPs (if more than one) associated with a given Port and CNPV share a single set of the RP per-
37 Port per-CNPV variables.

38 32.10.1 rpppMaxRps

39 An unsigned integer controlling the maximum number of RPs allowed for this CNPV on this Port
40 (default 1). An end station shall not create more than `rpppMaxRps` on a given Port, but it can create fewer.

41 32.10.2 rpppCreatedRps

42 The number of times an RP's `rpEnabled` variable has been set TRUE at this priority level on this Port
43 [item e) in 32.4.14].

1 32.10.3 rpppRpCentiseconds

2 An integer containing the number of RP-centiseconds accumulated by RPs at this priority level on this Port.
3 This variable is incremented once per centisecond (10 ms) for each RP on this Port whose rpEnabled
4 variable (32.13.1) is TRUE.

5 NOTE—Reading rpppRpCentiseconds and SysUpTime⁴⁶ at two different times allows a management entity to compute
6 the average number of RPs present for a priority and Port by dividing the change in rpppRpCentiseconds by the change
7 in SysUpTime (which is measured in centiseconds).

8 32.11 RP group variables

9 There is one set of RP group variables for each group of RPs belonging to a particular CNPV on a Port in a
10 congestion-aware end station. It is an implementation choice whether each RP rate control state machine
11 (32.15) has its own set of RP group variables or shares them with another RP rate control state machine on
12 the same Port and the same CNPV. This standard does not specify how RPs are grouped together for
13 management purposes, except that all of the RPs controlled by a single set of RP group variables shall serve
14 a single CNPV. At the extremes, a single set of RP group variables could control all of the RPs serving the
15 same CNPV, or each RP could be controlled by its own set of RP group variables. The RP group variables
16 include the following:

- 17 a) rpgEnable (32.11.1)
- 18 b) rpgTimeReset (32.11.2)
- 19 c) rpgByteReset (32.11.3)
- 20 d) rpgThreshold (32.11.4)
- 21 e) rpgMaxRate (32.11.5)
- 22 f) rpgAiRate (32.11.6)
- 23 g) rpgHaiRate (32.11.7)
- 24 h) rpgGd (32.11.8)
- 25 i) rpgMinDecFac (32.11.9)
- 26 j) rpgMinRate (32.11.10)

27 32.11.1 rpgEnable

28 A Boolean value controlling the rpEnabled variables for all of the RP rate control state machines controlled
29 by this Reaction Point group managed object. If rpgEnable is TRUE, the controlled rpEnabled variables are
30 not held in the FALSE state, thus enabling the RPs to pay attention to received CNMs. If rpgEnable is
31 FALSE, the controlled rpEnabled variables are held in the FALSE state, thus disabling the RPs.

32 32.11.2 rpgTimeReset

33 Unsigned integer. RpWhile (32.12.1) is reset to either rpgTimeReset or rpgTimeReset / 2 (with random
34 jitter—see Figure 32-2 in 32.15, RPR_ACTIVE_TIME) each time it reaches 0, according to the state of the
35 RP rate control state machine. Default value 15 ms.

36 32.11.3 rpgByteReset

37 Unsigned integer. rpByteCount (32.13.2) is reset to either rpgByteReset or rpgByteReset / 2 (with random
38 jitter—see Figure 32-2 in 32.15, RPR_ACTIVE_BYTE) each time it reaches 0, according to the state of the
39 RP rate control state machine. Default value 150 000 octets.

⁴⁶ From IETF RFC 3418.

1 **32.11.4 rpgThreshold**

2 Unsigned integer. Specifies the number of times rpByteStage or rpTimeStage can count before the RP rate
3 control state machine advances states. Default value is 5.

4 **32.11.5 rpgMaxRate**

5 Unsigned integer. The maximum rate, in bits per second, at which an RP can transmit. Default value is the
6 speed of the Port. A system shall support a nonzero minimum value for rpgMaxRate no larger than 5 Mb/s.
7 This rate includes all bits consequent to transmitting the frame on the LAN, including preamble, inter-frame
8 gap, etc. rpgMaxRate is configurable in multiples of 1 Mb/s.

9 **32.11.6 rpgAiRate**

10 Unsigned integer. The rate, in bits per second, used to increase rpTargetRate in the
11 RPR_ACTIVE_INCREASE state in Figure 32-2 in 32.15. Default value 5 Mb/s. This rate includes all bits
12 consequent to transmitting the frame on the LAN, including preamble, inter-frame gap, etc. rpgAiRate is
13 configurable in multiples of 1 Mb/s.

14 **32.11.7 rpgHaiRate**

15 Unsigned integer. The rate, in bits per second, used to increase rpTargetRate in the
16 RPR_HYPER_INCREASE state in Figure 32-2 in 32.15. Default value 50 Mb/s. This rate includes all bits
17 consequent to transmitting the frame on the LAN, including preamble, inter-frame gap, etc. rpgHaiRate is
18 configurable in multiples of 1 Mb/s.

19 **32.11.8 rpgGd**

20 Real number. Multiplied times the Quantized Feedback field (33.4.3) received in a CNM PDU to decrease
21 rpTargetRate. Default value 1/128. rpgGd is configurable as a power of 2, e.g., 1/256, 1/128, 1/64.

22 **32.11.9 rpgMinDecFac**

23 Real number. The minimum factor by which the current RP transmit rate rpCurrentRate can be changed by
24 reception of a CNM. Default value is 0.5.

25 **32.11.10 rpgMinRate**

26 Unsigned integer. The minimum value, in bits per second, for rpCurrentRate (32.13.6) Default value is
27 10 Mb/s. A system shall support a minimum value for rpgMinRate no larger than 10 Mb/s. This rate includes
28 all bits consequent to transmitting the frame on the LAN, including preamble, inter-frame gap, etc.
29 rpgMinDecFac is configurable in multiples of 1 Mb/s.

30 **32.12 RP timer**

31 An RP implements one timer as follows:

- 32 a) RpWhile (32.12.1).

33 **32.12.1 RpWhile**

34 Timer counter for controlling the RP rate control state machine (32.15). RpWhile operates normally when
35 rpFreeze (32.13.7) is FALSE, and shall not be decremented when rpFreeze is TRUE.

1 32.13 RP variables

2 Each RP rate control state machine (32.15) has its own set of RP variables. The RP variables include the
3 following:

- 4 a) rpEnabled (32.13.1)
- 5 b) rpByteCount (32.13.2)
- 6 c) rpByteStage (32.13.3)
- 7 d) rpTimeStage (32.13.4)
- 8 e) rpTargetRate (32.13.5)
- 9 f) rpCurrentRate (32.13.6)
- 10 g) rpFreeze (32.13.7)
- 11 h) rpLimiterRate (32.13.8)
- 12 i) rpFb (32.13.9)

13 32.13.1 rpEnabled

14 Boolean. Controls RP rate control state machine. If TRUE, state machine is running. If FALSE, state
15 machine is held in the RPR_RESET state. Initialized to FALSE when an RP rate control state machine is
16 created. Held in the FALSE state if rpgEnable (32.11.1) has the value FALSE.

17 32.13.2 rpByteCount

18 Unsigned integer. Counts octets passed from the Rate Limiter (31.2.2.4) to the Flow Selection function
19 (31.2.2.5). The counting down of rpByteCount to or past 0 triggers a reevaluation of rpCurrentRate
20 (32.13.6).

21 32.13.3 rpByteStage

22 Unsigned integer. Counts the number of times rpByteCount has counted down to or past 0 since the last
23 CNM was received.

24 32.13.4 rpTimeStage

25 Unsigned integer. Counts the number of times RpWhile has counted down to 0 since the last CNM was
26 received.

27 32.13.5 rpTargetRate

28 Unsigned integer. The target rate, in bits per second, toward which rpCurrentRate is heading. This rate
29 includes all bits consequent to transmitting the frame on the LAN, including preamble, inter-frame gap, etc.

30 32.13.6 rpCurrentRate

31 Unsigned integer. The rate, in bits per second, at which the Rate Limiter (31.2.2.4) transmits data from its
32 Flow queue when rpFreeze (32.13.7) is FALSE. (The actual output of the Rate Limiter is controlled by
33 rpLimiterRate, 32.13.8.) This rate includes all bits consequent to transmitting the frame on the LAN,
34 including preamble, inter-frame gap, etc.

35 32.13.7 rpFreeze

36 Boolean. When FALSE, the RP and Rate Limiter operate normally. When TRUE, rpLimiterRate is held to
37 the value 0, to suppress the transmission of frames. rpFreeze is FALSE when there is room for frames from
38 this RP in the output queue, and TRUE when not.

1 **32.13.8 rpLimiterRate**

2 Unsigned integer. The rate, in bits per second, at which the Rate Limiter (31.2.2.4) transmits data from its
3 Flow queue. rpLimiterRate is equal to rpCurrentRate (32.13.6) as long as rpFreeze (32.13.7) is FALSE, and
4 0, when rpFreeze is TRUE. This rate includes all bits consequent to transmitting the frame on the LAN,
5 including preamble, inter-frame gap, etc.

6 **32.13.9 rpFb**

7 Unsigned integer. The value last received in the Quantized Feedback field (33.4.3) of a CNM PDU or 0.

8 **32.14 RP procedures**

9 The following procedures are used by the RP rate control state machine and the Output flow segregation
10 function (31.2.1):

- 11 a) ResetCnm (32.14.1)
- 12 b) TestRpTerminate (32.14.2)
- 13 c) TransmitDataFrame (32.14.3)
- 14 d) ReceiveCnm (32.14.4)
- 15 e) ProcessCnm (32.14.5)
- 16 f) AdjustRates (32.14.6)

17 **32.14.1 ResetCnm**

18 Called whenever the RP rate control state machine enters the RPR_RESET state (Figure 32-2 in 32.15) to
19 set the following variables:

- 20 — rpCurrentRate = rpgMaxRate
- 21 — rpTargetRate = rpgMaxRate
- 22 — rpByteCount = rpgByteReset
- 23 — rpFb = 0

24 **32.14.2 TestRpTerminate**

25 Called by the Flow Selection function (31.2.2.5) each time a frame is passed from the Per-CNPV station
26 function's Rate Limiter (31.2.2.4) to the Flow multiplexer (31.2.4). If:

- 27 a) rpCurrentRate equals rpgMaxRate
- 28 b) The Flow queue is empty
- 29 c) rpEnabled is TRUE

30 Then TestRpTerminate sets rpEnabled (32.13.1) to FALSE, which disables the RP until another CNM is
31 received.

32 NOTE—Condition b) can be met when all frames limited by a given Rate Limiter have been processed. See 31.2.2.1.

33 **32.14.3 TransmitDataFrame**

34 TransmitDataFrame() is called each time a frame is passed from a Rate Limiter (31.2.2.4) to the Port's Flow
35 multiplexer (31.2.4). TransmitDataFrame() subtracts the length of the transmitted frame from rpByteCount
36 (32.13.2) and may insert a CN-TAG containing the Flow ID of the Flow queue at the beginning of the
37 frame's mac_service_data_unit.

1 If an end station is VLAN-aware or priority-aware, and if a CN-TAG is added to a frame, the CN-TAG shall
2 follow the VLAN or priority tag in the frame.

3 **32.14.4 ReceiveCnm**

4 Called whenever a CNM is received. ReceiveCnm() performs the following actions:

- 5 a) The CNM is validated according to 33.4.11 and is discarded if invalid.
- 6 b) If the CNM frame carries a CN-TAG (33.2), the Flow ID from that CN-TAG (33.2.1) is used to
7 identify the particular RP rate control state machine to which this CNM applies.
- 8 c) If the receiving end station is unable to identify the particular RP rate control state machine to which
9 this CNM applies, it discards the CNM, and no further processing takes place.
- 10 d) Sets the rpFb variable from the CNM's Quantized Feedback field (33.4.3).
- 11 e) If the selected RP rate control state machine's rpEnabled variable is FALSE and is not held FALSE
12 because rpgEnable (32.11.1) has the value FALSE, and the CNM's cnmQOffset field (33.4.5) is
13 positive, then rpEnabled is reset to TRUE, and the variable rpppCreatedRps (32.10.2) is
14 incremented.

15 These actions activate the selected RP rate control state machine, if it was not active (rpEnabled == FALSE),
16 and also cause the RP rate control state machine to process the received CNM.

17 **32.14.5 ProcessCnm**

18 Called whenever the RP rate control state machine enters the RPR_CNM_RECEIVED state (Figure 32-2 in
19 32.15) to perform the following actions:

```
20     if (rpByteStage != 0) {
21         rpTargetRate = rpCurrentRate;
22         rpByteCount = rpgByteReset;
23     }
24     rpByteStage = 0;
25     rpTimeStage = 0;
26     dec_factor = (1 - (rpgGd * rpFb));
27     if (dec_factor < rpgMinDecFac)
28         dec_factor = rpgMinDecFac;
29     rpCurrentRate = rpCurrentRate * dec_factor;
30     if (rpCurrentRate < rpgMinRate)
31         rpCurrentRate = rpgMinRate;
32     RpWhile = rpgTimeReset;
```

33 **32.14.6 AdjustRates**

34 AdjustRates is called whenever the RPR_ADJUST_RATES, RPR_ACTIVE_INCREASE, or
35 RPR_HYPER_INCREASE states are entered. It takes one parameter, *increase*, that specifies how much to
36 increase rpTargetRate, and performs the following actions:

```
37     if ((rpByteStage == 1) || (rpTimeStage == 1)) && (rpTargetRate > 10 * rpCurrentRate))
38         rpTargetRate = rpTargetRate / 8;
39     else
40         rpTargetRate = rpTargetRate + increase;
41     rpCurrentRate = (rpTargetRate + rpCurrentRate)/2;
42     if (rpCurrentRate > rpgMaxRate)
43         rpCurrentRate = rpgMaxRate;
```

1 **32.15 RP rate control state machine**

2 The RP rate control state machine is illustrated in Figure 32-2. In this figure, the RPR_RESET state has no
3 exit. The state machine is held in the RPR_RESET state as long as either the general system initialization
4 variable “BEGIN” is TRUE or the rpEnabled variable, local to the particular instance of the RP rate control
5 state machine, is FALSE. When “BEGIN” becomes FALSE and rpEnabled TRUE, the state machine
6 remains in the RPR_RESET state until a CNM is received by ReceiveCnm (32.14.4) with a Quantized
7 Feedback field (33.4.3) that is less than 0. ReceiveCnm places that field into rpFb (32.13.9), which condition
8 forces the state machine into the RPR_CNM_RECEIVED state. When that state resets rpFb to 0, progress
9 through the state machine continues either until another CNM is received and rpFb is set to a nonzero value,
10 or until the state machine is deactivated by TestRpTerminate (32.14.2) setting rpEnabled to FALSE.

1

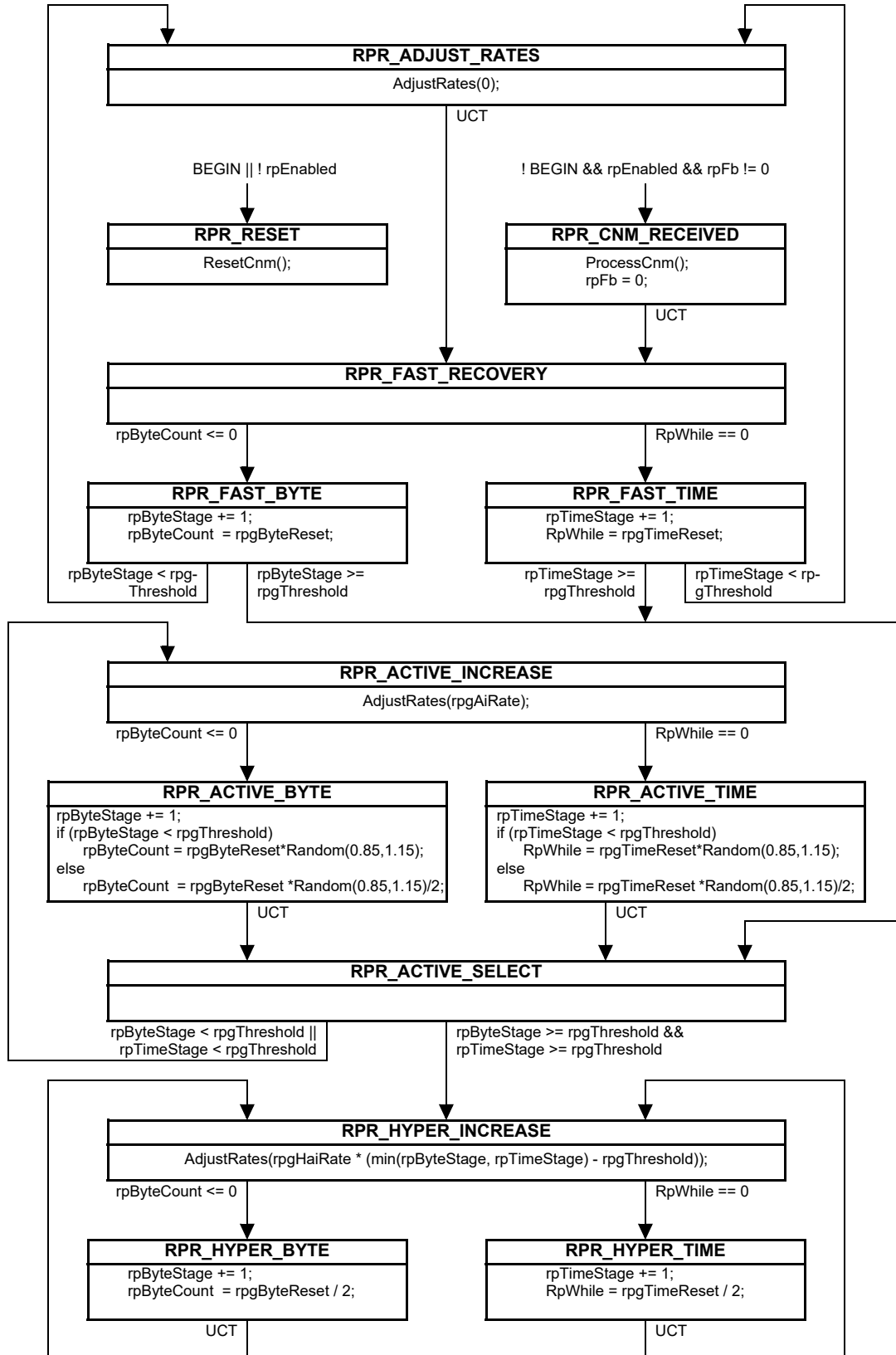


Figure 32-2—RP rate control state machine

32.16 Congestion notification and encapsulation interworking function

As mentioned in 30.8, a hierarchical Bridged Network, e.g., a PBBN, can require a congestion notification and encapsulation interworking function for a CP in the core of the network to generate a CNM that can reach the RP, which is presumably outside the core. This general case is illustrated in Figure 32-3. In the network illustrated, encapsulation and decapsulation functions peer with each other to encapsulate the data frames inside some kind of wrapper, e.g., an S-TAG and/or an I-TAG.

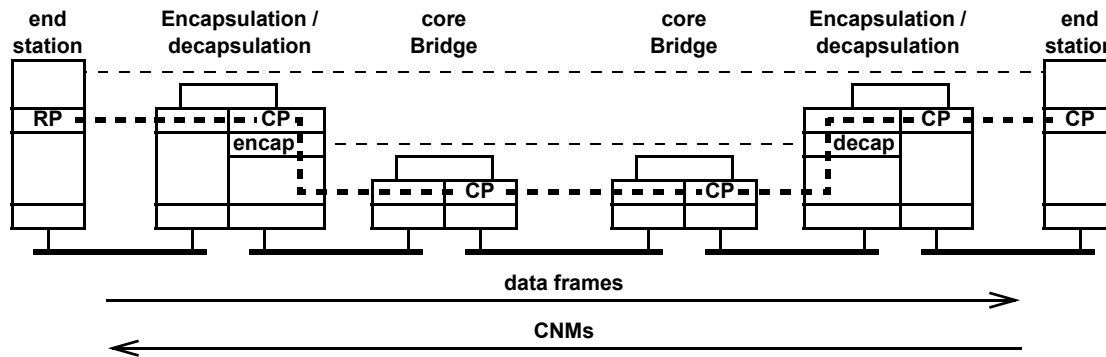


Figure 32-3—CP–RP peering in any hierarchical Bridged Network

Not all encapsulation schemes are equivalent. It is useful to divide hierarchical Bridged Networks into two categories for the purpose of defining congestion notification and encapsulation interworking functions, based on the `mac_service_data_unit` of a data frame carrying (optionally) a CN-TAG as seen by a CP residing in a core Bridge, with one of these cases further subdivided as follows:

- a) In a network where the original end station's `mac_service_data_unit`, including the optional CN-TAG, comprises the first octets of the `mac_service_data_unit` at the core Bridge's CP, there is no need for a congestion notification and encapsulation interworking function. For example, a PBN in which the C-TAGs are not carried across the core needs no interworking function.
- b) In a network where an encapsulation comprises the first octets of the `mac_service_data_unit` at the core Bridge's CP, a congestion notification and encapsulation interworking function is required. It is useful to make a further distinction, in this case:
 - 1) The MAC addresses of Ports in the Bridges where the encapsulation and decapsulation functions are performed are the destination and source addresses of frames crossing the core Bridges. For example, in a PBBN, the I-TAG, and perhaps an S-TAG and/or a C-TAG as well, can be present as the first octets of the `mac_service_data_unit`, and the `source_address` and `destination_address` parameters visible to the core Bridge's CP are the addresses of PIPs in BEBs.
 - 2) In other networks, although an encapsulation is in the first octets of the `mac_service_data_unit` at the core Bridge's CP, the source and destination MAC addresses are not altered by the encapsulation and decapsulation operations. For example, in a PBN in which both a C-TAG and an S-TAG are carried across the core, the C-TAG occupies the first octets of the `mac_service_data_unit` as seen by a core Bridge's CP.

In either type of network covered by case b), the core Bridge's CP is configured by the network administrator to return, in the Encapsulated MSDU field of the CNM PDU (33.4.10), the leading octets of the `mac_service_data_unit` of the data frame triggering transmission of the CNM. Since the CN-TAG of the data frame, if present, is buried behind encapsulations unknown to the CP, when the CN-TAG and CNM are built according to 32.9.4, the Flow ID returned in the CN-TAG of the CNM is filled with 0. The network administrator configures the CP to return at least enough octets of the data frame's `mac_service_data_unit` to include the CN-TAG of the original data frame transmitted by the RP.

1 The difference between case 1) and case 2), preceding, lies in the means used by the Port containing the
2 congestion notification and encapsulation interworking function to recognize the CNM in order to translate
3 it. In case 1), the CNM is addressed to that Port, and has a CN-TAG and the CNM encapsulation, instead of
4 a data frame tag and encapsulation. In case 2), the only reliable indicator of a CNM that needs to be
5 transformed, as opposed for example, to a CNM returned by the destination end station's CP, can be the
6 presence of an encapsulation known to the interworking function in the Encapsulated MSDU field (33.4.10)
7 of the CNM PDU.

8 When the CNM generated by the core Bridge's CP reaches a congestion notification and encapsulation
9 interworking function, that interworking function uses the information in the CNM PDU, including the
10 encapsulated data frame information, to construct a CN-TAG and CNM PDU to send to the RP. The steps
11 taken are as follows:

- 12 c) In case 1), where the CNM is returned to the MAC address of the interworking function's Port, the
13 destination_address parameter is obtained from the appropriate source address found in the
14 Encapsulated MSDU field, and the source_address parameter is a MAC address belonging to the
15 Bridge containing the interworking function, and valid in the context of the newly generated CNM,
16 for example, a management port address.
- 17 d) In case 2), where the CNM is recognized by the presence of the encapsulation in the Encapsulated
18 MSDU field, the source_address, and destination_address parameters are unchanged.
- 19 e) The vlan_identifier is obtained from the Encapsulated MSDU field.
- 20 f) The drop_eligible parameter is False.
- 21 g) The priority parameter used is the value configured for a CP on a Port in the Bridge in which the
22 interworking function resides, and through which the CNM passes.
- 23 h) The Encapsulated priority field (33.4.7) is obtained from the Encapsulated MSDU field, or if not
24 present there, the Encapsulated priority is left unchanged.

25 NOTE—In spite of the provisions for CND defense in this clause, it is possible for a network administrator to configure
26 a network so that the priority of a CCF data frame is not preserved. For example, in a PBN, two CNPVs could be
27 mapped to the same S-TAG priority, and the C-TAG not configured to be carried across the network. In that case, an end
28 station with an RP on each of those CNPVs, and that does not distinguish between those RPs using CN-TAGs, would be
29 unable to assign returned CNMs to the right RP. At least some of its CCFs would not be constrained, and uncontrolled
30 congestion would likely result. It is therefore recommended that the CN-TAG be included across a PBN, or that each
31 CNPV be mapped to a separate priority in the backbone.

- 32 i) If a CN-TAG is found in the Encapsulated MSDU, its Flow ID is copied to the CN-TAG of the
33 newly generated CNM, else 0 is used for that CN-TAG's Flow ID.
- 34 j) The encapsulation in the Encapsulated MSDU field of the CNM PDU, up to and including the
35 CN-TAG (which is not necessarily present), is elided, the remainder (if any) of the Encapsulated
36 MSDU field moved up to the first octets of the field, and the Encapsulated MSDU length field
37 reduced accordingly.

33. Encoding of congestion notification PDUs

This clause specifies the method of encoding congestion notification PDUs. The specifications include the following:

- a) The general structure and encoding of congestion notification PDUs (33.1)
- b) The format used for the CN-TAG (33.2)
- c) The encapsulation used for a CNM (33.3)
- d) The format used for the CNM (33.4)

The format of the IEEE 802.1AB Type Length Value (TLV) used to signal that a VLAN Bridge or end station is congestion aware, and the state of its CN defense mode, is specified in D.2.7.

NOTE—Clause 30 introduces the principles of congestion notification operation and the network architectural concepts that support it. Clause 31 breaks down the congestion notification entities into their components. Clause 32 specifies the protocols operated by these components.

33.1 Structure, representation, and encoding

All congestion notification PDUs contain an integral number of octets.

The octets in a congestion notification PDU are numbered starting from 1 and increasing in the order they are put into the MSDU that accompanies a request to or indication from the instance of the MAC Internal Sublayer Service (ISS or EISS) used by a congestion notification entity.

The bits in an octet are numbered from 1 to 8 in order of increasing bit significance, where 1 is the LSB in the octet.

Where octets and bits within a congestion notification PDU are represented using a diagram, octets shown higher on the page than subsequent octets and octets shown to the left of subsequent octets at the same height on the page are lower numbered; bits shown to the left of other bits within the same octet are higher numbered.

Where two or more consecutive octets are represented as hexadecimal values, lower numbered octet(s) are shown to the left and each octet following the first is preceded by a hyphen, e.g., 01-80-C2-00-00-00.

When consecutive octets are used to encode a binary number, the lower octet number has the more significant value. When consecutive bits within an octet are used to encode a binary number, the higher bit number has the most significant value. When bits within consecutive octets are used to encode a binary number, the lower octet number composes the more significant bits of the number. A flag is encoded as a single bit, and is set (TRUE) if the bit takes the value 1, and clear (FALSE) otherwise. The remaining bits within the octet can be used to encode other protocol fields.

33.2 CN-TAG format

The means for identifying CN-TAG PDUs consist of two octets containing the EtherType value shown (in hexadecimal notation) in Table 33-1.

In a VLAN Bridge, the shim that supports the EISS (IEEE Std 802.1AC, 6.8, Figure 22-4) is below the Queuing shim (8.6.6, 8.6.7, 8.6.8, 31.1.1). The CN-TAG is examined by the CP, which is part of the Queuing shim. Therefore, the CN-TAG will be further from the source_address and destination_address fields, and closer to the end of the frame, than a VLAN tag added by 6.8, if any. In order for a Bridge to be able to recognize the CN-TAG in a data frame, an end station shall insert the CN-TAG and VLAN tag in the relative order required by a Bridge: addresses, VLAN tag, CN-TAG, data.⁴⁷

Table 33-1—CN-TAG Encapsulation

	Octet	Length
Tag EtherType (22-E9)	1	2
Flow Identifier	3	2

33.2.1 Flow Identifier

The meaning and encoding of the Flow Identifier field are not specified by this standard. A CP shall not interpret the value in a Flow Identifier field. The Flow ID returned in the CN-TAG of a CNM is used by an end station's CNM demultiplexer (31.2.5) to identify the RP, the Flow queue, or a group of one or more individual flows, from which was transmitted the data frame that triggered the CNM.

33.3 Congestion Notification Message (CNM)

The means for identifying CNM PDUs consist of two octets containing the EtherType value shown (in hexadecimal notation) in Table 33-2.

Table 33-2—CNM Encapsulation

	Octet	Length
PDU EtherType (22-E7)	1	2
CNM PDU	3	24 to 88

33.4 Congestion Notification Message PDU format

The format of a Congestion Notification Message (CNM) PDU is illustrated in Table 33-3. If the frame that triggered the transmission of the CNM carried a CN-TAG, then that CN-TAG is retained in the frame carrying the CNM PDU; otherwise, a CN-TAG with a FlowID of 0 is used in the frame carrying the CNM PDU.

33.4.1 Version

This field, 4 bits in length, shall be transmitted with the value 0, and shall be ignored on receipt. The Version field occupies the most significant bits of the first octet of the CNM PDU.

33.4.2 ReservedV

This field, 6 bits in length, shall be transmitted with the value 0, and shall be ignored on receipt. The ReservedV field occupies the least significant 4 bits of the first octet, and the most significant 2 bits of the second octet, of the CNM PDU.

⁴⁷ There are other tags, e.g., the IEEE 802.1AE MAC Security tag, not included in this abbreviated list.

Table 33-3—Congestion Notification Message PDU

	Octet	Length
Version	1	4 bits
ReservedV	1, 2	6 bits
Quantized Feedback	2	6 bits
Congestion Point Identifier	3	8
cnmQOffset	11	2
cnmQDelta	13	2
Encapsulated priority	15	2
Encapsulated destination MAC address	17	6
Encapsulated MSDU length	23	2
Encapsulated MSDU	25	0–64

33.4.3 Quantized Feedback

This field, 6 bits in length, contains the Quantized Feedback value calculated by the CP's EM_UNITDATA.request (parameters) procedure (32.9.3). The Quantized Feedback field occupies the least significant bits of the first two octets of the CNM PDU.

33.4.4 Congestion Point Identifier

This field, 8 octets in length, uniquely identifies the CP that triggered the transmission of this Congestion Notification Message PDU format within the CND. Because the transmitting system can format the Congestion Point Identifier (CPID) in any manner, a receiving RP shall not assign any meaning to subfields within a CPID.

NOTE—Making the CPID 8 octets long removes the connection between CP identity and the source MAC address of the CNM PDU. One way to achieve the requirements for uniqueness of the CPID would be to use a Port's MAC address as the high-order 6 octets of the CPID, and use the low-order two octets of the CPID to hold the priority of the CP's queue and/or an indication of the physical port.

33.4.5 cnmQOffset

The two's-complement signed integer value of $(-1 \times \text{cpQOffset})$ (32.8.7) of the transmitting CP in units of 64 octets. This standard does not specify whether cpQOffset is rounded or truncated to obtain cnmQOffset. If the value of cpQOffset is less than $-32\,768$, a value of $-32\,768$ is used in cnmQOffset, and if greater than $32\,767$, then $32\,767$ is used.

33.4.6 cnmQDelta

The two's-complement signed integer value of the transmitting CP's cpQDelta (32.8.8) in units of 64 octets. This standard does not specify whether cpQDelta is rounded or truncated to obtain cnmQDelta. If the value of cpQDelta is less than $-32\,768$, a value of $-32\,768$ is used in cnmQDelta, and if greater than $32\,767$, then $32\,767$ is used.

NOTE—Although cnmQDelta is not used by the RP, it is included in the CNM PDU to aid the network administrator when adjusting configuration parameters.

1 **33.4.7 Encapsulated priority**

2 This field, 2 octets in length, contains the priority parameter of the frame that triggered the transmission of
3 this CNM in the most significant 3 bits of the field. The remaining bits of the field are transmitted as 0, and
4 ignored on receipt.

5 **33.4.8 Encapsulated destination MAC address**

6 This field, 6 octets in length, contains the destination_mac_address parameter of the frame that triggered the
7 transmission of this CNM.

8 **33.4.9 Encapsulated MSDU length**

9 This field, 2 octets in length, contains the number of octets returned in the Encapsulated MSDU field
10 (33.4.10).

11 **33.4.10 Encapsulated MSDU**

12 This field, a maximum of 64 octets in length, contains the initial octets of the mac_service_data_unit of the
13 frame that triggered the transmission of this CNM.

14 **33.4.11 CNM Validation**

15 A CNM PDU received by a CNM demultiplexer (31.2.5) shall be considered invalid and be discarded if the
16 following condition is TRUE:

- 17 a) There are fewer than 24 octets in the mac_service_data_unit.

18 A CNM PDU received by a CNM demultiplexer (31.2.5) may be considered invalid and be discarded if the
19 following condition is TRUE:

- 20 b) The CNM PDU has no CN-TAG.

21 NOTE 1—The variable cnpdAcceptsCnTag (32.4.10), when set FALSE in an end station, indicates to the neighboring
22 Bridge that the CN-TAG is to be removed from all frames, including CNMs, transmitted to the end station. It is not
23 recommended that an end station set cnpdAcceptsCnTag to FALSE if this would result in the end station discarding
24 CNMs for lack of a CN-TAG.

25 The following condition shall not cause a received CNM PDU to be considered invalid:

- 26 c) There are nonzero bits in the Version (33.4.1) or ReservedV (33.4.2).

27 NOTE 2—These rules permit information to be added to the CNM PDU following the Encapsulated MSDU field
28 (33.4.10) in later revisions of this standard.

1 34. Forwarding and Queuing Enhancements for time-sensitive streams 2 (FQTSS)

3 34.1 Overview

4 This clause describes a set of tools that can be used to support the forwarding and queuing requirements of
5 time-sensitive streams. In this context, a “time-sensitive stream” is taken to be a stream (flow) of traffic,
6 transmitted from a single source station (Talker), destined for one or more destination stations (Listeners),
7 where the traffic is sensitive to timely delivery, and in particular, requires transmission latency to be
8 bounded. Such streams include video or audio data streams, where there is a desire to limit the amount of
9 buffering required in the receiving station (Listener). The description in this clause (Clause 34) applies when
10 the transmission gates for scheduled traffic (8.6.8.4) are continuously open, as closing those gates affects the
11 availability of bandwidth.

12 NOTE 1—An example of the need to support time-sensitive stream requirements would be a live performance where a
13 video image of the performance is simultaneously projected on a screen in the auditorium, and it is desirable for the
14 sound and image to be “in sync” with the performance.

15 The term “stream reservation class” (SR class) refers to a traffic class whose bandwidth can be reserved for
16 time-sensitive streams using the Stream Reservation Protocol (SRP). FQTSS specifies the relationship
17 between an SR class and the underlying tools of this standard, such as traffic class, priority, and transmission
18 selection.

19 NOTE 2—The FQTSS functions described in this clause are intended for use with SRP, with the exception of
20 `adminIdleSlope` [item c) in 34.3]. The `adminIdleSlope` parameter is intended for TSN configuration models that do not
21 use SRP. In the fully centralized model of TSN configuration (46.1.3.3), the CNC entity can use `adminIdleSlope` to
22 reserve bandwidth for a traffic class.

23 In order to address the needs of such traffic in Bridges, the following are provided:

- 24 a) A means of detecting the boundary between a set of Bridges that support SRP (an SRP domain) and
25 surrounding Bridges that do not support SRP. This mechanism is described in 34.2.
- 26 b) A set of bandwidth availability parameters for each port that are used to record the maximum
27 bandwidth available to a given outbound queue, and the actual bandwidth reserved, for that queue.
28 These parameters are described in 34.3.
- 29 c) A discussion of the relationship between the size of the layer 2 “payload” (the MSDU) carried in a
30 frame and how that relates to the actual bandwidth that will be consumed when that MSDU is
31 transmitted on a particular Port (34.4).
- 32 d) Specifications for the default configuration of SR classes, including the mapping of the priorities
33 associated with received frames onto the traffic classes available on the transmission Ports of a
34 Bridge (34.5).
- 35 e) Specifications for use of transmission selection algorithms for SR classes (34.6).

36 34.2 Detection of SRP domains

37 The concepts of time-sensitive streams, the Stream Reservation Protocol (SRP), and the traffic forwarding
38 and shaping functions that support stream transmission (e.g., 8.6.8.1), rely on the ability of each Bridge to
39 detect whether each of its Ports is at the edge of a region of connected Bridges that support SRP on a
40 particular priority, so that the Priority Code Point values associated with traffic entering an *SRP domain*
41 (3.269) can be properly regenerated at the boundary of the domain, as described in 6.9.4.

42 Bridges detect the edge of an SRP domain by observing SRP behavior. If a Bridge receives SRP
43 registrations using a particular priority, then it is reasonable to believe that they are being received from an
44 SRP-capable device; the SRP engine can therefore signal which Ports of a Bridge are at the boundary of an
45 SRP domain. The per-Port, per-SR class, *SRPdomainBoundaryPort* parameter determines whether a Port is

1 considered to be at the edge of an SRP domain or within the core of the domain, as defined in 35.1.4. This
2 parameter is controlled by the operation of SRP.

3 NOTE—SRP domain detection is based on the assumption that a device connected to a Port either is SRP capable for a
4 given SR class, or is not SRP capable for that SR class. SRP provides a boundary detection mechanism through the
5 exchange of MSRPDU; the boundary of a domain therefore expands to include Ports as SRP attributes are declared.
6 The position of the domain boundary has no effect on the transmission of SRP frames; rather, it reflects where SRP
7 activity is occurring. Ports are removed from the SRP domain when they are removed from the active topology.

8 34.3 The bandwidth availability parameters

9 The following bandwidth availability parameters exist for each Port, and for each traffic class, N , that is
10 configured as an SR class (34.5):

- 11 a) ***portTransmitRate*** as defined in 8.6.8.2;
- 12 b) ***deltaBandwidth(N)***: The bandwidth, represented as a percentage of *portTransmitRate*, that can be
13 reserved by SRP for use by the queue associated with traffic class N . The interpretation of
14 *deltaBandwidth(N)* is determined by the value of the associated *lockClassBandwidth(N)* as specified
15 in 34.3.1 and 34.3.3.
- 16 c) ***adminIdleSlope(N)***: The bandwidth, in bits per second, that has been requested by management to
17 be reserved for use by the queue associated with traffic class N . If SRP is in operation for traffic
18 class N , this parameter has no effect. If SRP is not in operation for traffic class N , then the value of
19 *operIdleSlope(N)* is always equal to the value of *adminIdleSlope(N)*.
- 20 d) ***operIdleSlope(N)***: The actual bandwidth, in bits per second, that is currently reserved for use by the
21 queue associated with traffic class N (see 34.6.1 and 34.6.2).
- 22 e) ***classMeasurementInterval(N)***: The interval of time, in seconds, over which SRP's TSpec (34.4,
23 35.2.2.8.4) is measured for traffic class N at the Talker.
- 24 f) ***lockClassBandwidth(N)***: The Boolean parameter that determines the interpretation of
25 *deltaBandwidth(N)*. For the value false (default), *deltaBandwidth(N)* is specified in 34.3.1. For the
26 value true, *deltaBandwidth(N)* is specified in 34.3.3.

27 In all cases, bandwidth is defined in terms of the actual bandwidth consumed when frames are transmitted
28 through the Port, not the size of the MSDU "payload" carried within those frames. Subclause 34.4 discusses
29 the relationship between MSDU size and actual bandwidth consumed. The bandwidth availability
30 parameters and their use as described in this clause (Clause 34) do not take into account any interruption to
31 bandwidth availability resulting from the operation of stream gates (8.6.5.2).

32 34.3.1 *deltaBandwidth* when *lockClassBandwidth* is false

33 When *lockClassBandwidth(N)* is false, *deltaBandwidth(N)* is the additional bandwidth, represented as a
34 percentage of *portTransmitRate*, that can be reserved by SRP for use by the queue associated with traffic
35 class N , in addition to the *deltaBandwidth(N)* values associated with higher priority queues with
36 *lockClassBandwidth(N)* false. For a given traffic class N , the total bandwidth that can be reserved is the sum
37 of the *deltaBandwidth* values for traffic class N and all higher traffic classes with *lockClassBandwidth(N)*
38 false, minus any bandwidth reserved by higher traffic classes (i.e., configured as SR classes).

39 The default value of *lockClassBandwidth(N)* is false. This default can be changed using the managed
40 objects of the Bandwidth Availability Parameter Table (12.20.1).

41 The default value of *deltaBandwidth(N)* for the highest numbered traffic class supported is 75%, and for any
42 lower numbered traffic classes, the default value is 0%. The *deltaBandwidth(N)* for a given N , plus the
43 *deltaBandwidth(N)* values for any higher priority queues (larger values of N) defines the total percentage of
44 the Port's bandwidth that can be reserved for that queue and all higher priority queues. For the highest
45 priority queue, this means that the maximum value of *operIdleSlope(N)* is *deltaBandwidth(N)*% of

portTransmitRate. However, if *operIdleSlope(N)* is actually less than this maximum value, any lower priority queue with *lockClassBandwidth(N)* false can make use of the reservable bandwidth that is unused by the higher priority queue. So, for queue $N - 1$, the maximum value of $(\text{operIdleSlope}(N) + \text{operIdleSlope}(N - 1))$ is $(\text{deltaBandwidth}(N) + \text{deltaBandwidth}(N - 1))\%$ of *portTransmitRate*.

NOTE 1—For example, suppose two queues, 3 and 2, are configured as SR classes A and B, respectively. Suppose *deltaBandwidth(3)* for SR class A is currently 20%, and *deltaBandwidth(2)* for SR class B is currently 30%, and *lockClassBandwidth* is false for both. If *operIdleSlope(3)* is currently 10% of *portTransmitRate*, then half of queue 3's maximum allocation is unused, and the maximum value of *operIdleSlope(2)* is therefore currently 40% of *portTransmitRate*. However, if *operIdleSlope(3)* increases to the full 20% that it is entitled to use, the maximum value of *operIdleSlope(2)* reduces to 30% of *portTransmitRate*.

NOTE 2—The sum of the *deltaBandwidth(N)* values for all values of N should be chosen such that there is sufficient bandwidth available for any nonreserved (best-effort, strict-priority) traffic; the default values are chosen such that the sum of the *deltaBandwidth(N)* values is 75%, so no more than 75% of the Port's available bandwidth is permitted to be reserved. This ensures that when using default settings, there is at least 25% of the Port's bandwidth available for nonreserved traffic. However, as these default settings may be inappropriate for some situations (e.g., links that offer very high bandwidth, or networks with very low levels of nonreserved traffic), they can be modified by management.

34.3.2 *deltaBandwidth* when *lockClassBandwidth* is true

When *lockClassBandwidth(N)* is true, *deltaBandwidth(N)* is the maximum bandwidth, represented as a percentage of *portTransmitRate*, that can be reserved by SRP for use by the queue associated with traffic class N . The bandwidth limit of *deltaBandwidth(N)* is not related to higher or lower priority traffic classes (i.e., it is not truly a delta).

NOTE—For example, suppose two queues, 3 and 2, are configured as SR classes A and B, respectively. Suppose *deltaBandwidth(3)* for SR class A is currently 20%, *deltaBandwidth(2)* for SR class B is currently 30%, and *lockClassBandwidth* is true for both. If *operIdleSlope(3)* is currently 10% of *portTransmitRate*, then half of queue 3's maximum allocation is unused. Nevertheless, the maximum value of *operIdleSlope(2)* remains 30% of *portTransmitRate*. The value of *operIdleSlope(2)* cannot increase to 40%, because 20% is locked to use by *operIdleSlope(3)*.

34.3.3 Bandwidth availability parameter management

The values of *deltaBandwidth(N)* and *adminIdleSlope(N)* can be changed by management, using the management operations defined in 12.20. If the stream reservation mechanisms defined in SRP are supported, then the values of *operIdleSlope(N)* are determined solely by the operation of SRP. If the stream reservation mechanisms defined in SRP are not supported, then the values of *operIdleSlope(N)* are equal to the values requested by management in the corresponding *adminIdleSlope(N)* parameters.

It is possible for the value of *portTransmitRate* for a Port to change as a result of the normal operation of the underlying MAC Service (e.g., as a result of the operation of IEEE 802.1AX Link Aggregation, or as a result of dynamic changes in bandwidth of the physical layer technology itself, as can occur in wireless LAN technologies); it is also possible for management action to change the values of *deltaBandwidth(N)* for a Port. In either case, the consequence could be one of the following:

- a) The sum of the *operIdleSlope(N)* values for the Port could now exceed the total reservable bandwidth allowed for the Port, or the *operIdleSlope(N)* value for a given queue could now exceed the reservable bandwidth allowed for the queue, as defined in 34.3.1 and 34.3.2. Consequently, there could be streams currently active on the Port that can no longer be supported.
- b) The bandwidth now available to a given queue could mean that there are streams that are currently inactive that could be supported on the Port.
- c) Active streams that continue to be supported after the change could see their latency guarantee change.

In either case, corrective action, either by management or by the stream reservation mechanisms defined in SRP, is required in order to restore the parameters to a consistent set of values. In order to indicate that there have been changes in the bandwidth availability database, a signal, *bandwidthAvailabilityChanged*, is

generated for each Port whenever the bandwidth availability parameters *portTransmitRate* or *deltaBandwidth(N)* are changed; this signal is used by SRP to trigger recalculation of the set of streams that can be reserved on the Port.

NOTE—During recalculation of stream reservations, the Bridge might be temporarily unable to honor bandwidth reservation commitments or forward best-effort traffic.

34.4 Deriving actual bandwidth requirements from the size of the MSDU

The forwarding and queuing mechanisms defined in this clause use bandwidth parameters that are defined in terms of the actual bandwidth used when frames are transmitted on the medium that supports the MAC Service available through the Port. In contrast, the SRP makes use of a traffic specification (TSpec) for each stream that defines the maximum number of bits per frame (*MaxFrameSize*), of the *mac_service_data_unit* parameter that is relayed by the relay function of the Bridge, and a maximum frame rate (*MaxIntervalFrames*), in frames per class measurement interval, for that stream; i.e., the TSpec takes no account of the per-frame overhead associated with transmitting the MSDU over a given medium. However, when SRP determines the value to be used for the *operIdleSlope(N)* parameter associated with a given queue, it is necessary for this value to include the per-frame overhead that will be incurred when frames are transmitted on that Port.

NOTE 1—The frame rate in a TSpec is measured over the *classMeasurementInterval* (34.3) that depends upon the SR class associated with the stream. Default values for *classMeasurementInterval* are specified in 34.5.

For the purposes of calculating the bandwidth consumption of a stream, it is assumed that the stream data is essentially of constant size and transmission rate, so these maxima can be used to directly define an assumed maximum payload size and the maximum frame rate in frames per second; i.e.,

$$\text{assumedPayloadSize} = \text{MaxFrameSize} \quad (34-1)$$

$$\text{maxFrameRate} = \text{MaxIntervalFrames} \times (1/\text{classMeasurementInterval}) \quad (34-2)$$

where *classMeasurementInterval* is measured in seconds.

NOTE 2—As stated, the calculation of bandwidth from TSpec parameters assumes that the stream data is essentially of constant frame size, and hence, the approximations shown in this section are valid. If the data varies significantly in frame size, then the calculation of per-frame overhead using these assumptions could be significantly in error.

From this, and also from local knowledge of the protocol stack that supports the Bridge Port, it is possible to determine the overhead that is added to the per-frame MSDU payload when a frame is transmitted. There are at least the following sources of per-frame overhead:

- a) Any VLAN tags and security tags (see IEEE Std 802.1AE) that are added to the layer 2 payload as it passes through the various service interfaces in the Port's protocol stack.
- b) The MAC framing (header and trailer octets, plus any padding octets that are required to meet minimum frame size limitations) that is added by the underlying MAC Service.
- c) Any physical layer overhead, such as preamble characters and inter-frame gaps.

The precise per-frame overhead will therefore depend upon the protocol stack and the underlying MAC technology.

The actual bandwidth needed to support a given stream is therefore defined as follows [using *assumedPayloadSize* from Equation (34-1)]:

$$\text{actualBandwidth} = (\text{perFrameOverhead} + \text{assumedPayloadSize}) \times \text{maxFrameRate} \quad (34-3)$$

34.5 Default SR class configuration

This subclause (34.5) makes traffic class, priority, transmission selection algorithm, and classMeasurementInterval recommendations for use with FQTSS.

The recommended mappings of priorities to traffic classes meet the following constraints:

- a) Priority values that correspond to SR classes are mapped onto traffic classes that support the credit-based shaper algorithm as the transmission selection algorithm.
- b) Traffic classes that support the credit-based shaper algorithm have a higher priority than traffic classes that support the strict priority (or any other) transmission selection algorithm.
- c) At least one traffic class supports the credit-based shaper algorithm, and at least one traffic class supports the strict priority transmission selection algorithm.

NOTE 1—The constraint that there is at least one traffic class that supports the strict priority transmission selection ensures that there is at least one traffic class that can support traffic that is not subject to bandwidth reservation, such as “best effort” traffic.

The recommended priority to traffic class mappings for a network that supports SR class A (using priority 3) and SR class B (using priority 2) are shown in Table 34-1. The recommended priority to traffic class mappings for a system that supports only SR class B (using priority 2) are shown in Table 34-2. These mappings can be changed using the managed objects of the Traffic Class Table for each Port (12.6.3).

The corresponding configuration for the Transmission Selection Algorithm Table (see 8.6.8) is that the traffic classes that are shaded in the tables are configured to use the credit-based shaper algorithm, and the remaining traffic classes are configured to use the strict priority algorithm. Traffic classes that are shaded correspond to default SR classes. These defaults can be changed using the managed objects of the Transmission Selection Algorithm Table (12.20.2).

Table 34-1—Default priority to traffic class mappings for SR classes A and B

		Number of available traffic classes						
		2	3	4	5	6	7	8
Priority	0 (Default)	0	0	0	0	0	0	1
	1	0	0	0	0	0	0	0
	2	1	1	2	3	4	5	6
	3	1	2	3	4	5	6	7
	4	0	0	1	1	1	1	2
	5	0	0	1	1	1	2	3
	6	0	0	1	2	2	3	4
	7	0	0	1	2	3	4	5

NOTE 2—The mapping shown in Table 34-1 for the case of only two available traffic classes maps two SR classes to a single traffic class. While this is a permissible configuration, in general it is desirable, and in some applications, can be a requirement, to assign SR classes to distinct traffic classes, as is done in this table for the cases where three or more traffic classes are available. The mappings shown deal only with one or two supported SR classes; a similar mapping strategy can be adopted if more than two SR classes are supported.

Table 34-2—Default priority to traffic class mappings for SR class B only

		Number of available traffic classes						
		2	3	4	5	6	7	8
Priority	0 (Default)	0	0	0	0	0	1	1
	1	0	0	0	0	0	0	0
	2	1	2	3	4	5	6	7
	3	0	0	0	1	1	2	2
	4	0	1	1	2	2	3	3
	5	0	1	1	2	2	3	4
	6	0	1	2	3	3	4	5
	7	0	1	2	3	4	5	6

1 Table 34-1 and Table 34-2 specify a default configuration of SR class A as priority 3, and SR class B as
2 priority 2. These defaults can be changed using the managed objects of the SR Class to Priority Mapping
3 Table (12.20.4).

4 The default classMeasurementInterval (34.3) of SR class A is 125 μ s. The default classMeasurementInterval
5 of SR class B is 250 μ s. These defaults can be changed using the managed objects of the Bandwidth
6 Availability Parameter Table (12.20.1).

7 34.6 Transmission selection

8 Transmission selection (8.6.8) is used to shape the transmission of a Stream's frames in accordance with the
9 bandwidth that has been reserved on a given outbound queue for a traffic class.

10 The following transmission selection algorithm shall be supported for FQTSS, configured as the default for
11 SR class A and/or B:

- 12 a) Credit-based shaper algorithm, specified in 8.6.8.2

13 The following transmission selection algorithm may be supported for FQTSS, configured using the
14 managed objects of the Transmission Selection Algorithm Table (12.20.2):

- 15 a) Strict priority algorithm, specified in 8.6.8.1

16 The following transmission technique may be supported for FQTSS, configured using the managed objects
17 for scheduled traffic (12.29):

- 18 a) Scheduled traffic, specified in 8.6.8.4

34.6.1 Credit-based shaper

The *operIdleSlope(N)* parameter (34.3) is used by the credit-based shaper algorithm (8.6.8.2) as its idleSlope for the corresponding queue, to shape outbound traffic.

The *operIdleSlope(N)* parameter is also used to reserve space in the queue of the traffic class that is assigned to the SR class.

In order for an end station to successfully participate in the transmission and reception of time-sensitive streams, it is necessary for their behavior to be compatible with the operation of the forwarding and queuing mechanisms employed in Bridges. The requirements for end stations that participate as “Talkers”—i.e., sources of time-sensitive streams—are different from the requirements that apply to “Listeners”—i.e., the destination station(s) for the streams.

34.6.1.1 Talker behavior

In order for Talker-originated data streams to make use of the credit-based shaper behavior in Bridges, it is a requirement for a Talker to use the priorities that the Bridges in the network recognize as being associated with SR classes exclusively for transmitting stream data. It is also necessary for the Talker, and the Bridges in the path to the Listener(s), to have a common view of the bandwidth required in order to transmit the Talker’s streams, and for that bandwidth to be reserved along the path to the Listener(s). This latter requirement can be met by means of stream reservation mechanisms, such as defined in SRP, or by other management means.

End stations that are Talkers shall exhibit transmission behavior for frames that are part of time-sensitive streams that is consistent with the operation of the credit-based shaper algorithm, both in terms of the way they transmit frames that are part of an individual data stream, and in terms of the way they transmit stream data frames from a Port. In effect, the queuing model for a Talker Port, and for a given priority, can be considered to look like Figure 34-1.

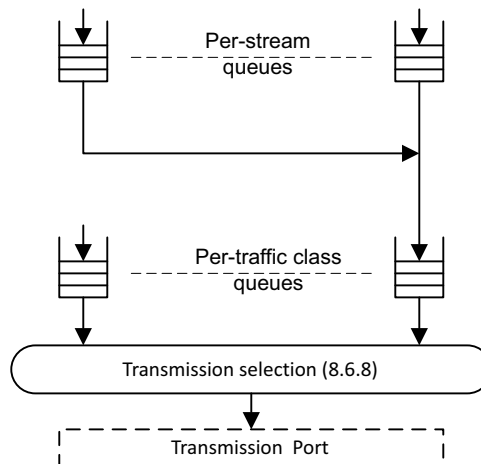


Figure 34-1—Queuing model for a Talker station

The Talker places frames into the queue associated with an individual stream based on the Tspec for that stream; i.e., during each *classMeasurementInterval*, it can place up to *MaxIntervalFrames* data frames, each no longer than *MaxFrameSize* into that stream’s queue.

1 The queue associated with each individual stream uses the credit-based shaper algorithm, with the *idleSlope*
2 set to the bandwidth requirement of the stream on that transmission Port, as the means of determining the
3 rate at which data frames for that stream are placed in the outbound queue for the priority that the stream is
4 using. The outbound queue for that priority, in turn, makes use of the credit-based shaper algorithm, with the
5 *idleSlope* set to the sum of the *idleSlope* values for all streams using that priority on that transmission Port,
6 as the means of determining the rate at which data frames for that stream are selected for transmission.

7 Transmission selection in a Talker operates in the same way as in a Bridge; from this point of view, a Talker
8 can be thought of as if it is a single-port Bridge.

9 For streams that make use of SR class A or SR class B, it is a requirement that the rate at which frames for
10 any given stream are selected for placement in its per-stream queue does not exceed the bandwidth reserved
11 for the stream, measured over the *classMeasurementInterval* (34.3) for the SR class. For some combinations
12 of stream bandwidth requirement and transmission Port data rate, this can place a limit on the frame size that
13 can be used when transmitting stream data.

14 NOTE—The sole implication of the *classMeasurementInterval* is its effect on frame size, because the shaper behavior
15 itself is independent of the *classMeasurementInterval*. The intent in limiting the *classMeasurementInterval* is to limit
16 frame size, as this is a major contribution to the latency experienced by a frame in transit through a time-sensitive
17 network (see discussion in Annex L).

18 **34.6.1.2 Listener behavior**

19 The primary requirement for a Listener end station using the credit-based shaper is that it is capable of
20 buffering the amount of data that could be transmitted for a stream during a time period equivalent to the
21 accumulated maximum jitter that could be experienced by stream data frames in transmission between
22 Talker and Listener. From the point of view of the specification of the forwarding and queuing requirements
23 for time-sensitive streams, it is assumed that the Listener will assess the buffering required for a stream as
24 part of the stream bandwidth reservation mechanisms employed by the implementation.

25 **34.6.2 Strict priority**

26 The *operIdleSlope(N)* parameter (34.3) represents the total outbound bandwidth for the queue of traffic
27 class N, and *operIdleSlope(N)* shall be used to reserve space in the queue of the associated traffic class N.

28 NOTE 1—The *operIdleSlope(N)* parameter is not used by the algorithm for transmission selection (8.6.8.1).

29 When SRP is used for reservation of a stream, the egress Port that uses strict priority will place frames into
30 its outbound queue based on the Tspec for that Stream. The TSpecs of all Streams that egress the Port are
31 used to compute the *operIdleSlope(N)*.

32 When SRP is not used, the worst-case bandwidth for traffic class N is computed by a management client
33 (e.g., CNC of 46.1.3.3, using TSpecs from the CUC) for the queue of the associated traffic class N. The
34 management client uses *adminIdleSlope(N)* to configure this bandwidth in the station (12.20.1). Since
35 *operIdleSlope(N)* is equal to *adminIdleSlope(N)*, *operIdleSlope(N)* is used to reserve space in the queue of
36 the associated traffic class N. Since the priority associated with traffic class N is not using SRP, that priority
37 does not need to be configured as an SR class (i.e., assigned an SRclassID using 12.20.4).

38 NOTE 2—As an example of a traffic class configured using *adminIdleSlope(N)*, consider a traffic class that is protected
39 using scheduled traffic (8.6.8.4). This protected traffic class can use the strict priority algorithm. Traffic in the protected
40 window requires queue reservation in order to ensure that adequate storage exists in its queue. The CNC entity of
41 46.1.3.3 can use *adminIdleSlope(N)* to ensure that the queue for the protected traffic class is of sufficient size.

42 NOTE 3—Although *adminIdleSlope(N)* is specified as bits per second, with regard to reserving queue space for a shorter
43 interval of time, it is best to assume that *adminIdleSlope(N)* is uniform over the entire second. For example, for a traffic
44 class that is protected using scheduled traffic (8.6.8.4), if AdminCycleTime is 1 ms, *adminIdleSlope(N)* is divided by
45 1000 to determine the worst-case traffic for that AdminCycleTime.

1 **34.6.3 Scheduled traffic**

2 When SRP uses Centralized Network Configuration (CNC), scheduled traffic (8.6.8.4) can be used in a
3 Talker as well as the Bridges to each Listener. For information, refer to the specifications for externalControl
4 TRUE in 35.2.2.10.

5 When SRP is fully distributed (externalControl FALSE in 35.2.2.10), use of scheduled traffic in Bridges is
6 not specified for SRP's Streams. If scheduled traffic is used in a Talker, that Talker appends the
7 TSpecTimeAware TLV to its Talker Advertise. If the nearest Bridge supports the optional capability of
8 translating scheduled traffic in the Talker to the transmission selection used in Bridges (i.e., 34.6.1 or
9 34.6.2), the nearest Bridge propagates Talker Advertise. As specified in 35.2.2.10.6, if the nearest Bridge
10 does not support the TSpecTimeAware TLV, the nearest Bridge fails the Stream (propagates Talker Failed).

11 NOTE—For an example using scheduled traffic in a Talker, refer to U.1.1.

1 35. Stream Reservation Protocol (SRP)

2 SRP utilizes three signaling protocols [MMRP (10.9), MVRP (Clause 11), and MSRP (35.1)] to establish
3 stream reservations across a bridged network.

4 Within SRP the Multiple MAC Registration Protocol (MMRP) may be used to control the propagation of
5 Talker registrations throughout the bridged network (35.2.4.3.2).

6 The Multiple VLAN Registration Protocol (MVRP) is used by end stations and Bridges to declare
7 membership in a VLAN where a Stream is being sourced. This allows the Data Frame Priority
8 [35.2.2.8.5(a)] to be propagated along the path from Talker to Listener(s) in tagged frames. MSRP will not
9 allow Streams to be established across Bridge Ports that are members of the untagged set (8.8.10) for the
10 related VID.

11 The Multiple Stream Registration Protocol (MSRP) is a signaling protocol that provides end stations with
12 the ability to reserve network resources that will guarantee the transmission and reception of data streams
13 across a network with the requested QoS. These end stations are referred to as Talkers (devices that produce
14 data streams) and Listeners (devices that consume data streams).

15 Talkers declare attributes that define the stream characteristics so Bridges have the necessary information
16 available when they need to allocate resources for a Stream. Listeners declare attributes that request
17 reception of those same streams. Bridges along the path from Talker to Listener process, possibly adjust, and
18 then forward these MSRP attribute declarations. Bridges associate Talker and Listener attributes via the
19 StreamID present in each of those attributes, which result in changes to the extended filtering services and
20 allocation of internal resources when streams are “brought up.”

21 In order to establish the SRP domain boundaries, Bridges exchange SR class characteristics with each other
22 and with end stations via MSRP. Neighboring devices that have identical SR class characteristics are
23 considered to be in the same SRP domain and streams may be established between those devices.

24 MSRP provides a limited error reporting capability that is utilized when a Listener’s request to receive a
25 stream cannot be honored because of some resource constraint within the network.

26 MSRP also supports the concept of data stream importance. For example, an emergency announcement
27 would be flagged with a more important “rank” than a stream providing background music. This ranking
28 ability allows the Bridges to replace less important streams with more important streams without requiring
29 intervention from the end stations.

30 There is a considerable body of experience in supplying data streams with guarantees for QoS parameters
31 such as latency, latency variation, or bandwidth. In particular, routers and hosts use the Internet Protocol (IP)
32 and the Resource Reservation Protocol (RSVP, IETF RFC 2205, and IETF RFC 2750) to achieve such
33 guarantees. Supplying guarantees to a data stream requires the following two components:

- 34 a) A definition of the resources to be allocated and configured, by end stations and network nodes, for
35 the support of a data stream; and
- 36 b) A protocol for end stations to signal to the network nodes their data streams’ requirements, for
37 network nodes to distribute those requirements among each other, and for the network nodes to
38 signal the success or failure of the attempt to reserve resources to support the guarantees.

39 RSVP supplies the signaling protocol for routers to support data streams in routed networks. This and the
40 following clauses define a protocol to support data streams in bridged networks.

41 The preceding description of SRP as a signaling protocol corresponds to the Fully distributed model
42 (46.1.3.1) for TSN configuration. SRP also supports the Centralized network/distributed user model
43 (46.1.3.2) for TSN configuration, which can be used to enable enhanced features (see 35.2.2.10). In order to

1 use the Centralized network/distributed user model, the MRP External Control (12.32.4) feature is enabled
2 in the Nearest Bridge (8.6.3) to each Talker and Listener. Instead of propagating (signaling) through Bridges,
3 SRP messages are exchanged with a Centralized Network Configuration (CNC) entity. The CNC uses
4 remote management protocols (e.g., SNMP, NETCONF) in order to configure Bridges to meet the Stream
5 requirements of each Talker and Listener.

6 **35.1 Multiple Stream Registration Protocol (MSRP)**

7 MSRP supports the reservation of resources for streams, each destined for one or more Listeners, and each
8 from a single source, across a bridged network. Transmitted data that conforms to a successful stream
9 reservation will not be discarded by any Bridge due to congestion on a LAN. In order to propagate requests
10 for reservations, MSRP defines an *MRP application* that provides the Stream resource registration service
11 defined in 35.2.3. MSRP makes use of the MRP Attribute Declaration (MAD) function, which provides the
12 common state machine descriptions defined for use in MRP-based applications. The MRP architecture, and
13 MAD are defined in Clause 10. MSRP defines a new MRP Attribute Propagation (MAP) function, to
14 provide an attribute propagation mechanism.

15 MSRP propagates registrations for stream reservations in a manner similar to the operation of MMRP (10.9)
16 and MVRP (11.2), which are used for registering Group membership and individual MAC address
17 information, and VLAN membership, respectively. Unlike MMRP and MVRP, however, the registered
18 attributes can be combined, discarded, or otherwise altered, as they are propagated by the participating
19 Bridges.

20 In order to make and keep QoS guarantees all devices in a bridged network must participate in the signaling
21 and queuing operations required of Bridges. For example, this would include IEEE 802.11 wireless media
22 access points and stations. Thus, MSRP provides a means for Bridges or end stations running MSRP to
23 cooperate both with higher network layers, such as routers or hosts running RSVP, and with lower network
24 layers, such as wireless media.

25 MSRP is also responsible for establishing the SRP domain boundary for a particular SR class. All systems
26 that support a particular SR class are in the same SRP domain if they use the same priority. An SRP domain
27 boundary exists for an SR class when neighboring devices use different priorities for the SR class.

28 Figure 35-1 illustrates the architecture of MSRP in the case of a two-Port Bridge and an end station.

29 The version of MSRP is distinguished by the ProtocolVersion of MRP. ProtocolVersion 0x00 (MSRPv0) is
30 the original version. ProtocolVersion 0x01 (MSRPv1) or higher specifies new AttributeTypes (35.2.2.4) that
31 provide enhanced capabilities. This standard specifies MSRPv1 (35.2.2.1). In order to meet interoperability
32 goals, implementations of this version shall support the original AttributeTypes specified in MSRPv0 as well
33 as the enhanced AttributeTypes. The interoperability goals require a network of mixed MSRPv0 and
34 MSRPv1 end stations and Bridges to provide a quality of service that is as good as or better than a complete
35 MSRPv0 network.

36 For both MSRPv0 and MSRPv1, each MSRP Participant declares the Domain attribute prior to other MSRP
37 attributes in order to determine SR class boundaries. The MSRPv1 end station or Bridge also uses this
38 Domain exchange to determine the ProtocolVersion of its neighboring MSRP Participant. If the neighbor is
39 MSRPv0, then only the original AttributeTypes are exchanged with that neighbor, although the
40 ProtocolVersion will be set to the value defined in 35.2.2.1 [see item b) in 10.8.3.5]. If the neighbor is
41 MSRPv1, then both original and enhanced AttributeTypes can be exchanged with that neighbor. For a
42 Bridge, this means that it is possible to register an enhanced AttributeType from one Port and propagate and
43 declare that same attribute to another Port as an original AttributeType. To enable this mixture of
44 AttributeTypes, the MSRP Attribute Propagation (35.2.4) specifies the translation from each enhanced
45 AttributeType to its corresponding original AttributeType.

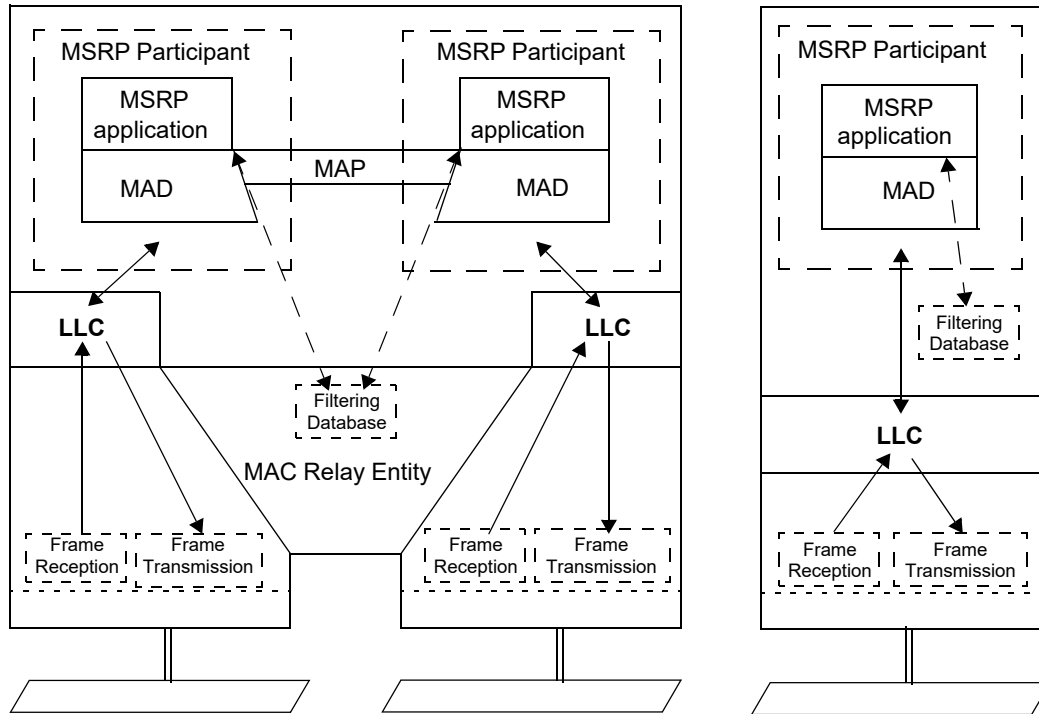


Figure 35-1—Operation of MSRP

1 For propagation of an MSRP attribute from one Port using MSRPv1 to another Port using MSRPv1, the
2 AttributeType is not changed. This allows original AttributeTypes to propagate through a Bridged Network
3 as is, without translation to enhanced AttributeTypes.

4 NOTE—MRP (Clause 10) has limitations on the total amount of attribute data that can be exchanged between
5 participants. Therefore, the larger each MSRP attribute's value, the fewer total Streams can be reserved in the network.
6 Due to the enhanced features, the enhanced AttributeType is larger than the original AttributeType. If the Talker and
7 Listeners of a Stream do not need the enhanced features, use of the original AttributeTypes can allow for a larger number
8 of total Streams. If all Talkers and Listeners in the network use original AttributeTypes, the maximum number of
9 Streams in an all-MSRPv1 network is the same as an all-MSRPv0 network.

10 35.1.1 MSRP and Shared Media

11 Classic shared media, such as IEEE 802.3 half-duplex Carrier Sense Multiple Access with Collision Detect
12 (CSMA/CD), cannot provide latency or bandwidth guarantees, because their operation depends on random
13 timers. Such media are therefore not supported by MSRP.

14 There are other shared media where one node on the medium exercises control over access to the medium by
15 the other nodes. For example, an IEEE 802.11 infrastructure wireless medium has a single Access Point
16 (AP) that controls access by the AP and the stations attached to the wireless medium so that some guarantee
17 of latency and bandwidth can be made, subject to frame loss caused by data corruption errors. Similarly, an
18 IEEE 802.3 Ethernet Passive Optical Network has a single Optical Line Terminal (OLT) that controls access
19 to the optical medium by itself and some number of Optical Network Units (ONUs).

20 Different kinds of shared media use different techniques to allocate opportunities to transmit, and these
21 techniques can have various dependencies on frame sizes, station-to-station vs. station-to-head data paths, or
22 other factors. Rather than introducing the complexities of every such medium into MSRP, this standard takes
23 advantage of the presence of a controlling entity to map medium-specific characteristics to the capabilities
24 of MSRP.

1 MSRP defines and requires the existence of a Designated MSRP Node (DMN) on any shared medium. This
2 DMN provides the MSRP services for the shared medium and determines each station's ability to receive the
3 MSRPDU's transmitted by other stations on the medium. A Non-DMN Port shall be configured to only
4 process the MSRPDU's transmitted by DMN Ports, and ignore the MSRPDU's transmitted by other Non-
5 DMN Ports. Furthermore, the DMN has the absolute control of the resource allocation on the shared
6 medium. Given these two facts, a DMN can effectively control which reservations are and are not successful
7 on the medium it controls.

8 Annex C provides examples on various shared media.

9 **35.1.2 Behavior of end stations**

10 **35.1.2.1 Talkers**

11 To announce the Streams that can be supplied and their characteristics, Talkers use the MAD_Join.request
12 primitive (10.2) to make the Talker Declarations (35.2.1.3). To indicate the Streams that are no longer
13 supplied, Talkers use the MAD_Leave.request primitive (10.2) to withdraw their Talker Declarations.

14 The Talker shall issue an MVRP VLAN membership declaration prior to issuing the Talker Declaration. The
15 MVRP VLAN membership shall contain the vlan_identifier of the DataFrameParameters [item b) in
16 35.2.2.8.3], so that the neighboring Bridge adds the associated Bridge Port to the member set for the VLAN.

17 NOTE—VLAN membership is needed for the Talker when the neighboring Bridge has the Enable Ingress
18 Filtering parameter set (8.6.2).

19 Talker Declarations are propagated by MSRP such that the Listeners and Bridges are aware of the presence
20 of Talkers and the Streams that are offered. Talker Declarations are also used to gather QoS information
21 along their paths. Based on the gathered QoS information, Talker Declarations are classified as follows:

- 22 a) **Talker Advertise:** An advertisement for a Stream that has not encountered any bandwidth or other
23 network constraints along the network path from the Talker. Listeners that request attachment to this
24 Stream are likely to create a reservation with the described QoS. A Talker Advertise will continue to
25 be declared as long as the resources continue to be available.
- 26 b) **Talker Failed:** An advertisement for a Stream that is not available to the Listener because of
27 bandwidth constraints or other limitations somewhere along the path from the Talker.

28 A Talker Advertise shall not be declared if there are not sufficient bandwidth and resources available. If a
29 Talker Advertise is being declared and the required bandwidth or resources become unavailable the Talker
30 Advertise shall be withdrawn and a Talker Failed may be declared. A Talker is allowed to transition directly
31 from a Talker Advertise to a Talker Failed without waiting for the Talker Advertise to be deregistered from
32 the network (see 35.2.6).

33 Talkers respond to the registration and deregistration events of Listener Declarations (35.2.1.3), signaled by
34 MAD as follows:

35 On receipt of a MAD_Join.indication for a Listener Declaration, the Talker first merges (35.2.4.4.3) the
36 Listener Declarations that it has registered for the same Stream. Then the Talker examines the StreamID
37 (35.2.2.8.2) and Declaration Type (35.2.1.3) of the merged Listener Declaration. If the merged Listener
38 Declaration is associated with a Stream that the Talker can supply, and the DeclarationType is either Ready
39 or Ready Failed (i.e., one or more Listeners can receive the Stream), the Talker can start the transmission for
40 this Stream immediately. If the merged Listener Declaration is an Asking Failed, the Talker shall stop the
41 transmission for the Stream, if it is transmitting.

1 On receipt of a MAD_Leave.indication for a Listener Declaration, if the StreamID of the Declaration
2 matches a Stream that the Talker is transmitting, then the Talker shall stop the transmission for this Stream, if
3 it is transmitting.

4 **35.1.2.2 Listeners**

5 To indicate what Streams they want to receive, Listeners use the MAD_Join.request primitive (10.2) to make
6 the Listener Declarations (35.2.1.3). To indicate the Streams that are no longer wanted, Listeners use the
7 MAD_Leave.request primitive (10.2) to withdraw their Listener Declarations.

8 The Listener Declaration also conveys the results of the bandwidth and resource allocation along its path
9 back to the Talker. Based on those results, Listener declarations are classified as follows:

- 10 a) **Listener Ready:** One or more Listeners are requesting attachment to the Stream. There is sufficient
11 bandwidth and resources available along the path(s) back to the Talker for all Listeners to receive the
12 Stream.
- 13 b) **Listener Ready Failed:** Two or more Listeners are requesting attachment to the Stream. At least
14 one of those Listeners has sufficient bandwidth and resources along the path to receive the Stream,
15 but one or more other Listeners are unable to receive the stream because of network bandwidth or
16 resource allocation problems.
- 17 c) **Listener Asking Failed:** One or more Listeners are requesting attachment to the Stream. None of
18 those Listeners are able to receive the Stream because of network bandwidth or resource allocation
19 problems.

20 NOTE 1—The reader will notice that the Talker response to Ready and Ready Failed declarations is the same: the Talker
21 can begin transmitting the Stream. Talkers might choose to pass the Ready Failed response to a higher layer protocol that
22 could notify the user that the Stream is flowing, but not all Listeners are receiving it. It would be the responsibility of that
23 higher layer to respond to this information as appropriate.

24 If the Listener receives a Talker Advertise declaration, and the Listener is ready to receive the Stream, the
25 Listener shall issue an MSRP Listener Ready declaration for the Stream.

26 The Listener shall issue an MVRP VLAN membership declaration prior to issuing the MSRP Listener
27 Ready declaration. The MVRP VLAN membership shall contain the vlan_identifier of the Talker Advertise
28 DataFrameParameters [item b) in 35.2.2.8.3], so that the neighboring Bridge adds the associated Bridge Port
29 to the member set for the VLAN.

30 If the *talkerVlanPruning* parameter (35.2.4.3.4) is not enabled in the Bridges in the path from Talker to the
31 Listener, the Listener receives Talker Advertise prior to declaring MVRP VLAN membership. The Listener
32 can then use the registered Talker Advertise to obtain the vlan_identifier for declaration of MVRP VLAN
33 membership. This is the default behavior for the *talkerVlanPruning* parameter.

34 If one or more Bridges in the path from Talker to the Listener have enabled the *talkerVlanPruning*
35 parameter, the Listener declares VLAN membership prior to receiving the Talker Advertise declaration. The
36 *talkerVlanPruning* parameter instructs the Bridge to filter Talker declarations for the VLAN (in addition to
37 the VLAN-tagged data frames of the stream). If the *talkerVlanPruning* parameter is enabled and the VLAN
38 is not Registration Fixed (8.8.10), the Listener obtains knowledge of the Talker's vlan_identifier from a
39 higher layer protocol above IEEE Std 802.1Q and uses that knowledge to declare the membership for the
40 vlan_identifier with MVRP.

41 If *talkerPruning* or *talkerPruningPerPort* is enabled, the Listener must issue an MMRP registration for the
42 Stream's destination MAC address as described in 35.2.4.2 before the Talker Advertise will be received.

1 If the Listener receives a Talker Failed declaration, and the Listener is ready to receive the Stream, the
2 Listener shall issue either an MSRP Listener Asking Failed declaration or an MSRP Listener Ready
3 declaration for the Stream.

4 There is no requirement for the order in which the Talker and Listener declarations are communicated. The
5 Listener Declaration can be made before the Listener receives an associated Talker Declaration, in which
6 case the Listener shall issue a Listener Asking Failed declaration.

7 NOTE 2—One reason for the Listener to issue Listener Ready in response to Talker Failed is that the Listener wants to
8 inform the network that it is ready to proceed in case the Talker's state changes from failed to ready in the future.

9 **35.1.3 Behavior of Bridges**

10 MSRP-aware Bridges register and deregister Talker and Listener declarations on the Bridge Ports according
11 to the procedures defined in MRP (Clause 10), and automatically generate deregistration of stale
12 registrations. Any changes in the state of registration are processed by the MSRP Attribute Propagation
13 (35.2.4) function, and disseminated in the network by making or withdrawing Talker and Listener
14 declarations as defined in the Talker attribute propagation (35.2.4.3) and Listener attribute propagation
15 (35.2.4.4).

16 In general, Talker declarations are propagated to all other Bridge Ports. There are *talkerPruning* [item b) in
17 35.2.1.4], *talkerPruningPerPort* [item k) in 35.2.1.4], and *talkerVlanPruning* [item l) in 35.2.1.4]
18 parameters that limit the scope of Talker declaration propagation. Listener declarations are only propagated
19 to the Bridge Port with the associated Talker declaration (i.e., matching StreamID). If there is no associated
20 Talker declaration registered on any Bridge Port then Listener declaration will not be propagated.

21 **35.1.3.1 Blocked Declarations**

22 For the purposes of MSRP Attribute Propagation (35.2.4), a Declaration is said to be “blocked” on a Bridge
23 Port if the state of the spanning tree instance identified by the *vlan_identifier* in the *DataFrameParameters*
24 (35.2.2.8.3) of the Declaration, on that Bridge Port, has any value other than Forwarding. In an end station's
25 MSRP Participant, no Declaration is ever blocked.

26 **35.1.4 SRP domains and status parameters**

27 An SRP domain is a set of stations (end stations and/or Bridges), their Ports, and the attached individual
28 LANs, that satisfy all of the following conditions for a given SR class:

- 29 a) Those stations that transmit streams all support the credit-based shaper algorithm, defined in 8.6.8,
30 as the transmission selection method for the SR class.
- 31 b) The stations all support SRP as the means of creating bandwidth reservations for the SR class.
- 32 c) Those stations that transmit streams all associate the same priority value with the SR class.
- 33 d) Each Port in the set is either an SRP domain core port or an SRP domain boundary port.
- 34 e) Each SRP domain core Port in the set is connected, via an individual LAN that is part of the active
35 topology, to an SRP domain core Port of another station in the set.

36 In Bridges that support the SRP, and for each SR class supported by the Bridge, an
37 **SRPdomainBoundaryPort** parameter is associated with each Port of the Bridge.

38 Each **SRPdomainBoundaryPort** parameter has a Boolean value. The value of the **SRPdomainBoundaryPort**
39 parameter for a given SR class is TRUE if the operation of SRP has determined that the Port is an SRP
40 domain boundary port (3.269) for that SR class; otherwise, the Port either is an SRP domain core port
41 (3.270) for that SR class or is not part of that SRP domain, and the **SRPdomainBoundaryPort** parameter
42 value is FALSE.

1 35.2 Definition of the MSRP application

2 MSRP maintains two categories of variables. The first category is used internally by the application state
3 machines. These are defined in detail in the subclauses that follow.

4 MSRP also defines another category of variables identified as MRP elements that are communicated in
5 MSRPDUs between stations on a network. These protocol elements include the MRP frame addressing and
6 other fields defined in the MRPDUs. The MSRP FirstValue fields, which are used to exchange the MSRP
7 attributes, are also defined here.

8 35.2.1 Definition of internal state variables

9 The following variables and parameters are utilized by various state machines within MSRP:

- 10 a) Port Media Type (35.2.1.1);
- 11 b) Direction (35.2.1.2);
- 12 c) Declaration Type (35.2.1.3);
- 13 d) SRP parameters (35.2.1.4);

14 35.2.1.1 Port Media Type

15 MSRPDU processing on a port is handled differently depending on the type of medium the port is attached
16 to. For example, the DMN on a shared medium port that receives MSRPDUs from one station shall update
17 and retransmit those attributes so that all stations on that medium are updated appropriately. The possible
18 values are as follows:

- 19 a) **Access Control Port:** Transmitter controls access to the medium on which it is sending, so either it
20 is the DMN for a shared medium, or it is a port on a full-duplex point-to-point medium.
- 21 b) **Non-DMN shared medium Port:** Transmitter is attached to a shared medium, but does not control
22 access to the medium.

23 35.2.1.2 Direction

24 The Direction field is derived from the MSRP AttributeType definitions (35.2.2.4). The Direction indicates
25 whether this is a Talker: MSRP AttributeType definitions of type Talker Advertise Vector Attribute Type
26 [item a) in 35.2.2.4], Talker Failed Vector Attribute Type [item b) in 35.2.2.4], or Talker Enhanced Vector
27 Attribute Type [item e) in 35.2.2.4]. Set Direction to zero for Talker attributes. or a Listener MSRP
28 Declaration, and takes one of the following two values:

- 29 a) **Talker:** MSRP AttributeType definitions of type Talker Advertise Vector Attribute Type [item a) in
30 35.2.2.4], Talker Failed Vector Attribute Type [item b) in 35.2.2.4], or Talker Enhanced Vector
31 Attribute Type [item e) in 35.2.2.4]. Set Direction to zero for Talker attributes.
- 32 b) **Listener:** MSRP AttributeType definitions of type Listener Vector Attribute Type [item c) in
33 35.2.2.4] or Listener Enhanced Vector Attribute Type [item f) in 35.2.2.4]. Set Direction to one for
34 Listener attributes.

35 35.2.1.3 Declaration Type

36 The Declaration Type field is derived from the MSRP AttributeType definitions (35.2.2.4) and the MSRP
37 FourPackedEvents (35.2.2.7.2). The Declaration Type indicates the specific type of the Talker or Listener
38 MSRP Declaration.

1 For a Talker, the value of the Declaration Type component is either:

- 2 a) **Advertise:** MSRP AttributeType definitions of Talker Advertise Vector Attribute Type [item a) in
3 35.2.2.4] or Talker Enhanced Vector Attribute Type [item e) in 35.2.2.4] with
4 Status.StatusInfo.TalkerStatus equal to Ready (35.2.2.10.9). Set Declaration Type to zero for Talker
5 Advertise.
- 6 b) **Failed:** MSRP AttributeType definitions of Talker Failed Vector Attribute Type [item b) in 35.2.2.4]
7 or Talker Enhanced Vector Attribute Type [item e) in 35.2.2.4] with Status.StatusInfo.TalkerStatus
8 equal to Failed (35.2.2.10.9). Set Declaration Type to one for Talker Failed.

9 For a Listener, the value of the Declaration Type component is one of the following:

- 10 c) **Asking Failed:** MSRP AttributeType definitions of Listener Vector Attribute Type [item c) in
11 35.2.2.4] with MSRP FourPackedType equal to Asking Failed [item b) in 35.2.2.7.2] or Listener
12 Enhanced Vector Attribute Type [item f) in 35.2.2.4] with Status.StatusInfo.ListenerStatus equal to
13 Failed (35.2.2.10.9). Set Declaration Type to two for Listener Asking Failed.
- 14 d) **Ready:** MSRP AttributeType definitions of Listener Vector Attribute Type with MSRP
15 FourPackedType equal to Ready [item c) in (35.2.2.7.2) or Listener Enhanced Vector Attribute Type
16 [item f) in 35.2.2.4] with Status.StatusInfo.ListenerStatus equal to Ready (35.2.2.10.9). Set
17 Declaration Type to three for Listener Ready.
- 18 e) **Ready Failed:** MSRP AttributeType definitions of Listener Vector Attribute Type with MSRP
19 FourPackedType equal to Ready Failed [item d) in 35.2.2.7.2] or Listener Enhanced Vector
20 Attribute Type [item f) in 35.2.2.4] with Status.StatusInfo.ListenerStatus equal to PartialFailed
21 (35.2.2.10.9). Set Declaration Type to four for Listener Ready Failed.

22 35.2.1.4 SRP parameters

23 The following parameters are used by SRP:

- 24 a) **portTcMaxLatency:** The maximum per-port per-traffic class latency, expressed in nanoseconds, a
25 frame may experience through the underlying MAC Service. There may be different latency
26 numbers for different traffic classes on the same port.
- 27 b) **talkerPruning:** Enabling this parameter on the Bridge will limit the Talker declarations to Ports that
28 have the Streams destination_address [item a) in 35.2.2.8.3] in the MAC Address Registration
29 Entries (8.8.4), such as by using MMRP.
- 30 c) **streamAge:** A per-port per-stream 32-bit unsigned value used to represent the time, in seconds, since
31 the control element for the associated port most recently became forwarding in the Dynamic
32 Reservations Entries (8.8.7) corresponding to the stream's destination_address. This value is used
33 when determining which streams have been configured the longest. Streams with a numerically
34 larger streamAge are considered to be configured earlier than other streams and therefore carry a
35 higher implicit importance.

36 NOTE 1—A 32-bit unsigned value allows for expressing a streamAge of up to 136 years.

- 37 d) **msrpEnabledStatus:** MSRP shall have the ability to be enabled (true) or disabled (false) on a
38 device. When MSRP is enabled on a device it shall cause a reset of all MSRP state machines on all
39 ports. This affects the Applicant and Registrar state machines. The state of this parameter shall be
40 persistent over power-up restart/reboot.
- 41 e) **msrpPortEnabledStatus:** MSRP shall have the ability to be enabled (true) or disabled (false) on the
42 ports of a device. When MSRP is enabled or disabled on a port of a device it shall cause MAP to be
43 rerun on all MSRP enabled ports so existing attributes can be propagated to the port just enabled.
44 This affects the Applicant and Registrar state machines. The state of this parameter shall be
45 persistent over power-up restart/reboot.

- 1 f) **msrpMaxFanInPorts:** The total number of ports on a Bridge that are allowed to establish
2 reservations for inbound Streams. This number may be less than the total number of ports with
3 msrpPortEnabledStatus set TRUE, which will result in lower maximum latency because of limits on
4 the amount of possible interfering traffic. A value of zero (0) indicates no fan-in limit is being
5 specified and calculations involving fan-in will only be limited by the number of MSRP enabled
6 ports. Example calculations of delay associated with fan-in can be found in the paper “Calculating
7 the Delay Added by Qav Stream Queue” [B3].
- 8 g) **msrpLatencyMaxFrameSize:** Calculation of the maximum latency through a Bridge is in-part
9 related to the maximum size of an interfering frame. The maximum size is defined to be 2000 octets
10 by default. This parameter allows a smaller or larger value to be used in the latency calculations for
11 the particular Bridge implementation. msrpLatencyMaxFrameSize does not imply any type of
12 policing of frame size, it is only used in the latency calculations.
- 13 h) **SRPdomainBoundaryPort (35.1.4):** A per-port, per-SR class, Boolean parameter that contains the
14 value TRUE if the port is an SRP Domain Boundary Port; otherwise, it contains the value FALSE.
15 The parameter for a given SR class and Port shall be set to TRUE if any of the following conditions
16 are met:
 - 17 1) The port is declaring an MSRP Domain attribute for that SR class, and the port has no MSRP
18 Domain attribute registrations for that SR class, or
 - 19 2) The port is declaring an MSRP Domain attribute for that SR class, and the port has at least one
20 MSRP Domain attribute registration for that SR class with a different priority, or
 - 21 3) One or more ports which support that SR class are declaring MSRP Domain attributes for that
22 SR class, and this port does not support that SR class.
- 23 In all other cases the parameter shall be set to FALSE.
- 24 i) **SR_PVID:** The Stream Reservation Port VLAN Identifier is a per-port parameter that contains the
25 default VID for Stream-related traffic. It shall contain a valid VID value (Table 9-2) and may be
26 configured by management. If the value has not been explicitly configured, the SR_PVID shall
27 assume the default SR_PVID defined in Table 9-2. This value is passed to the Talker via the
28 SRclassVID (35.2.2.9.4) contained in the MSRP Domain attribute.
- 29 j) **neighborProtocolVersion:** Each Port provides a neighborProtocolVersion parameter that specifies
30 the MSRP ProtocolVersion of the neighboring MSRP Participant on that Port. The initial value is
31 0x00. When an MSRP Domain attribute is registered (received) on the Port, the value of the
32 attribute’s MRP ProtocolVersion (35.2.2.1) is copied to this parameter.
- 33 k) **talkerPruningPerPort:** When the talkerPruning parameter is disabled (false), each Port provides a
34 talkerPruningPerPort parameter to control MAC address pruning for that Port. For more
35 information, refer to 35.2.4.3.3.
- 36 l) **talkerVlanPruning:** Enabling this parameter on the Bridge will limit the Talker declarations to Ports
37 that have the Stream’s vlan_identifier [item b) in 35.2.2.8.3] registered as a member in the VLAN
38 Registration Entries (8.8). For more information, refer to 35.2.4.3.4.
- 39 m) **maxSRclasses:** This parameter provides the maximum number of SR classes supported by the
40 Bridge.

41 35.2.2 Definition of MRP elements

42 35.2.2.1 MSRP application address

43 The group MAC address used as the destination address for MRPDUs destined for MSRP Participants shall
44 be the group MAC address for “Individual LAN Scope group address, Nearest Bridge group address” as
45 specified in Table 8-1, Table 8-2, and Table 8-3 (C-VLAN, S-VLAN, and TPMR component Reserved
46 addresses, respectively).

47 NOTE—Using this address will guarantee that MSRPDUs are never forwarded by an 802.1 Bridge, although MSRP-
48 aware Bridges do propagate the MSRP attributes.

1 35.2.2.2 MSRP application EtherType

2 The EtherType used for MRPDUs destined for MSRP Participants shall be the MSRP EtherType identified
3 in Table 10-2.

4 35.2.2.3 MSRP ProtocolVersion

5 The ProtocolVersion for the version of MSRP defined in this standard takes the hexadecimal value 0x01.

6 NOTE—Although an MSRP Participant for this standard transmits MRPDUs using ProtocolVersion 0x01, the MSRP
7 Participant must process received MRPDUs using ProtocolVersion 0x01 or 0x00, as specified in 10.8.3.5.

8 35.2.2.4 MSRP AttributeType definitions

9 MSRP defines six AttributeTypes (10.8.2.2) that are carried in MRP exchanges. The numeric values for the
10 AttributeType are shown in Table 35-1 and their use is defined by the following list:

- 11 a) **Talker Advertise Vector Attribute Type:** Attributes identified by the Talker Advertise Vector
12 Attribute Type are instances of VectorAttributes (10.8.1), used to identify a sequence of values of
13 Talker advertisements for related Streams that have not been constrained by insufficient bandwidth
14 or resources. This Attribute Type applies to all versions of MSRP.
- 15 b) **Talker Failed Vector Attribute Type:** Attributes identified by the Talker Failed Vector Attribute
16 Type are instances of VectorAttributes, used to identify a sequence of values of Talker
17 advertisements for related Streams that have been constrained by insufficient bandwidth or
18 resources. This Attribute Type applies to all versions of MSRP.
- 19 c) **Listener Vector Attribute Type:** Attributes identified by the Listener Vector Attribute Type are
20 instances of VectorAttributes, used to identify a sequence of values of Listener requests for related
21 Streams regardless of bandwidth constraints. Listener Vector Attribute Types are subdivided into
22 individual Declaration Types via the MSRP FourPackedEvents (35.2.2.7.2). This Attribute Type
23 applies to all versions of MSRP.
- 24 d) **Domain Vector Attribute Type:** Attributes identified by the Domain Vector Attribute Type are
25 instances of VectorAttributes, used to identify a sequence of values that describe the characteristics
26 of an SR class. This Attribute Type applies to all versions of MSRP.
- 27 e) **Talker Enhanced Vector Attribute Type:** Attributes identified by the Talker Enhanced Vector
28 Attribute Type are instances of VectorAttributes (10.8.1), used to identify a sequence of values of
29 Talker advertisements for related Streams. Talker Enhanced Vector Attribute Types are subdivided
30 into individual Declaration Types (35.2.1.3) via the Status.StatusInfo.TalkerStatus (35.2.2.10.9).
31 This Attribute Type applies to ProtocolVersion 0x01 or higher.
- 32 f) **Listener Enhanced Vector Attribute Type:** Attributes identified by the Listener Enhanced Vector
33 Attribute Type are instances of VectorAttributes, used to identify a sequence of values of Listener
34 requests for related Streams. Listener Enhanced Vector Attribute Types are subdivided into
35 individual Declaration Types (35.2.1.3) via the Status.StatusInfo.ListenerStatus (35.2.2.10.9). This
36 Attribute Type applies to ProtocolVersion 0x01 or higher.

37 35.2.2.5 MSRP AttributeLength definitions

38 The AttributeLength field (10.12.1.8) in instances of the Talker Advertise Vector Attribute Type shall be
39 encoded in MRPDUs (10.8) as an unsigned binary number, equal to the value shown in Table 35-2.

40 The AttributeLength field in instances of the Talker Failed Vector Attribute Type shall be encoded in
41 MRPDUs as an unsigned binary number, equal to the value shown in Table 35-2.

42 The AttributeLength field in instances of the Listener Vector Attribute Type shall be encoded in MRPDUs as
43 an unsigned binary number, equal to the value shown in Table 35-2.

Table 35-1—AttributeType Values

AttributeType	Value	Used when neighborProtocolVersion is 0x00	Used when neighborProtocolVersion is 0x01 or higher
Talker Advertise Vector	1	yes	yes
Talker Failed Vector	2	yes	yes
Listener Vector	3	yes	yes
Domain Vector	4	yes	yes
Talker Enhanced Vector	5	no	yes
Listener Enhanced Vector	6	no	yes

Table 35-2—AttributeLength Values

AttributeType	Value
Talker Advertise Vector	25 (0x19)
Talker Failed Vector	34 (0x22)
Listener Vector	8
Domain Vector	4
Talker Enhanced Vector	variable
Listener Enhanced Vector	variable

1 The AttributeLength field in instances of the Domain Vector Attribute Type shall be encoded in MRPDUs as
2 an unsigned binary number, equal to the value shown in Table 35-2.

3 The AttributeLength field in instances of the Talker Enhanced Vector Attribute Type and Listener Enhanced
4 Vector Attribute Type shall be encoded in MRPDUs as an unsigned binary number. The value of
5 AttributeLength varies according to the TLVs contained in the attribute's value, but shall not change during
6 the duration of the attribute's declaration.

7 **35.2.2.6 MSRP AttributeListLength definitions**

8 The AttributeListLength field (10.12.1.9) shall be encoded in MRPDUs (10.8) as an unsigned binary
9 number, equal to the number of octets contained within the AttributeList. This field can be used when
10 calculating the number of octets to skip to proceed to the next Message (or Message EndMark) in the
11 MRPDUs.

12 **35.2.2.7 MSRP Vector definitions**

13 **35.2.2.7.1 MSRP ThreePackedEvents**

14 The ThreePackedEvent vectors are encoded as defined in 10.8.2.10.1.

35.2.2.7.2 MSRP FourPackedEvents

MSRP FourPackedEvents are only used by the Listener Vector Attribute Type [item c) in 35.2.2.4]. Within the FourPackedEvent, there are four possible values for the FourPackedType (10.8.2.10.2) as explained in the following list. The numeric values for the MSRP FourPackedEvents are shown in Table 35-3.

- a) **Ignore:** The StreamID referenced by FirstValue+n is not defined in this MSRPDU. When using this FourPackedType, the AttributeEvent (10.8.2.10.1) value, encoded in the ThreePackedEvent, shall be set to zero on transmit and ignored on receive.
- b) **Asking Failed:** The StreamID referenced by FirstValue+n has a declaration type of Listener Asking Failed.
- c) **Ready:** The StreamID referenced by FirstValue+n has a declaration type of Listener Ready.
- d) **Ready Failed:** The StreamID referenced by FirstValue+n has a declaration of type Listener Ready Failed.

Table 35-3—FourPackedEvent Values

FourPackedType	Value
Ignore	0
Asking Failed	1
Ready	2
Ready Failed	3

NOTE 1—In terms of efficient use of octets within an MSRP packet, placing nineteen (19) Ignores between two Listener declarations uses less octets than using two VectorAttributes to declare the Listener attributes separately. A single Listener VectorAttribute takes 12 octets, two attributes would take 24 octets. Declaring 21 attributes (two valid attributes with 19 Ignores in between) takes 12 octets, plus 6 additional ThreePackedEvents, plus 5 additional FourPackedEvents for a total of 23 octets.

NOTE 2—MSRP FourPackedEvents are not used for the Listener Enhanced Vector Attribute Type [item f) in 35.2.2.4], which uses the Status.StatusInfo.ListenerStatus (35.2.2.10.9) to distinguish Declaration Types (35.2.1.3).

35.2.2.8 MSRP FirstValue definitions (Stream reservations, original)

There are four Attribute Declarations defined for the original ProtocolVersion 0x00 of MSRP (35.2.2.4), three of which are related to stream reservations: Talker Advertise, Talker Failed, and Listener. The fourth attribute type, Domain, is used to discover the SRP domain, as well as the ProtocolVersion of each Port's neighbor, and is described in 35.2.2.9.

NOTE—This version of MSRP requires implementation of the original attributes of 35.2.2.8 and the enhanced attributes of 35.2.2.10. The specifications for enhanced attributes are based on the specifications for original attributes.

The Talker Advertise attribute contains all the characteristics that a Bridge needs in order to understand the resource requirements and importance of the referenced stream.

The Talker Failed attribute contains all the fields carried in the Talker Advertise Attribute, plus additional information regarding resource or bandwidth availability failures.

Listener attributes carry three subtypes: Ready, Ready Failed, and Asking Failed. These Listener subtypes are encoded in the FourPackedEvent (35.2.2.7.2).

1 FirstValue shall be incremented one or more times when NumberOfValues is greater than one. Incrementing
2 FirstValue within MSRP is defined as follows:

- 3 a) Add 1 to Unique ID [item b) in 35.2.2.8.2], and
- 4 b) Add 1 to Stream destination_address [item a) in 35.2.2.8.3].

5 The example shown in Table 35-4 illustrates the use of FirstValue and NumberOfValues within MSRP. This
6 example shows four Streams (a, b, c, and d) to be declared. In order to use the efficient packing techniques
7 of MRP it would be preferable to assign these Streams sequential StreamIDs and destination_addresses as
8 shown. Notice that StreamID yy-yy-yy-yy-yy-yy:00-04 is missing from this table (between Stream “c” and
9 “d”). MSRP allows declaration of all four StreamIDs in a single VectorAttribute by setting
10 NumberOfValues=5, with the StreamID = yy-yy-yy-yy-yy-yy:00-01 and a destination_address of xx-xx-xx-
11 xx-xx-25. Setting the fourth FourPackedEvent to Ignore [item a) in 35.2.2.7.2] notifies MSRP that the
12 StreamID between “c” and “d” is not in use.

Table 35-4—MSRP FirstValue NumberOfValues example

Stream	StreamID	destination_address
a	yy-yy-yy-yy-yy-yy:00-01	xx-xx-xx-xx-xx-25
b	yy-yy-yy-yy-yy-yy:00-02	xx-xx-xx-xx-xx-26
c	yy-yy-yy-yy-yy-yy:00-03	xx-xx-xx-xx-xx-27
d	yy-yy-yy-yy-yy-yy:00-05	xx-xx-xx-xx-xx-29

13 The FirstValue field within MSRP comprises several components: StreamID (35.2.2.8.2),
14 DataFrameParameters (35.2.2.8.3), TSpec (35.2.2.8.4), PriorityAndRank (35.2.2.8.5), Accumulated Latency
15 (35.2.2.8.6), and FailureInformation (35.2.2.8.7). MSRP does not support changes in any of the FirstValue
16 fields for an existing StreamID. If a Talker wishes to tear an old Stream down and bring a new Stream
17 up, with a different FirstValue, utilizing the same StreamID, there must be at least two LeaveAllTime
18 (Table 10-7) time periods between when the Talker removes the existing Stream registration and declares the
19 new Stream. This guarantees that MRP has enough time to remove the current attribute from all devices in
20 the network. If the new declaration were to occur too quickly the associated Streaming data could be
21 corrupted because the FDB may allow the new Stream data to start flowing while the old Stream bandwidth
22 constraints are still configured. When the Bridge detects this occurring it will fail the Talker Advertise with
23 the appropriate FailureInformation (35.2.2.8.7). Talkers may tear a Stream down and bring the same Stream
24 back up immediately, as long as the FirstValue has not changed.

25 35.2.2.8.1 Structure definition

26 The FirstValue for Talker Advertise, Talker Failed, and Listener attributes in MSRPDU's exchanged
27 according to the protocol specified in this subclause shall have the following structure:

- 28 a) The first eight octets contain the *StreamID* (35.2.2.8.2).
- 29 This is the end of the Listener attribute. If this is a Talker Advertise or Talker Failed attribute
30 continue as follows:
- 31 b) Following the StreamID are eight octets containing the *DataFrameParameters* (35.2.2.8.3).
- 32 c) Following the DataFrameParameters are four octets containing the *TSpec* (35.2.2.8.4).
- 33 d) Following the TSpec is one octet containing the *PriorityAndRank* (35.2.2.8.5).
- 34 e) Following the PriorityAndRank are four octets containing the *AccumulatedLatency* (35.2.2.8.6).
- 35 This is the end of the Talker Advertise attribute. If this is a Talker Failed attribute continue as
36 follows:

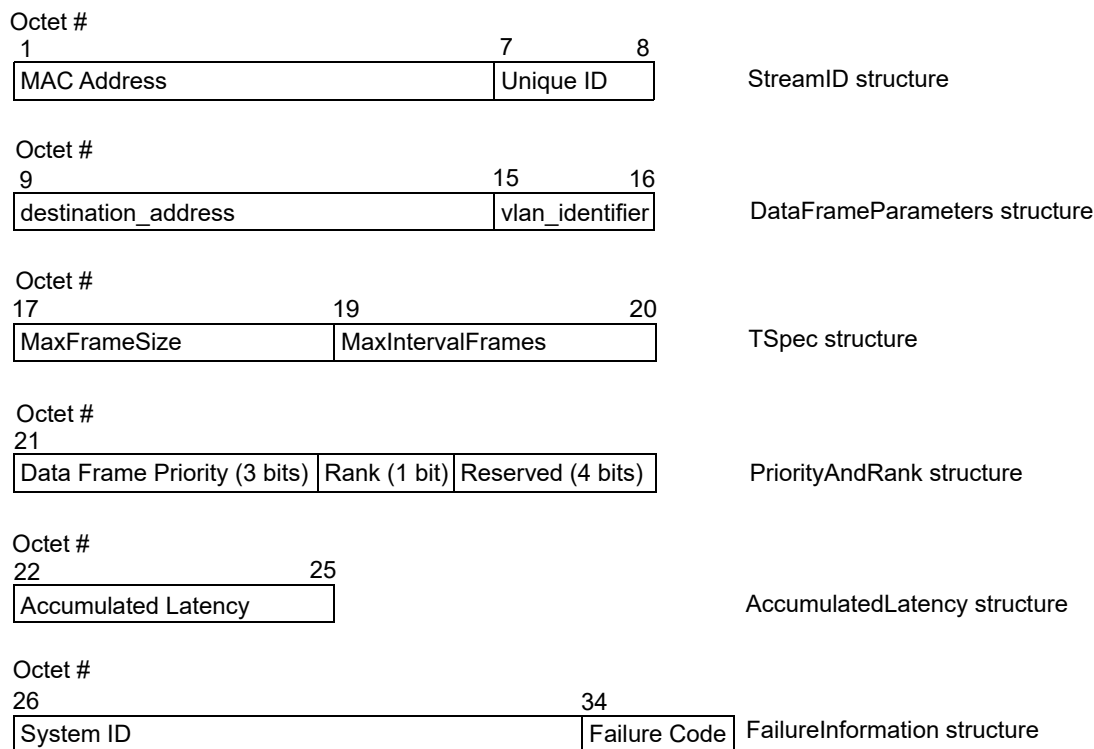


Figure 35-2—Format of the components of the reservation FirstValue fields

- 1 f) Following the AccumulatedLatency are nine octets containing the *FailureInformation* (35.2.2.8.7).
- 2 The following partial BNF production gives the formal description of the MSRPDU FirstValue structure for
- 3 the Talker Advertise attribute:
4 FirstValue ::=StreamID, DataFrameParameters, TSpec, PriorityAndRank, AccumulatedLatency
- 5 The following partial BNF production gives the formal description of the MSRPDU FirstValue structure for
- 6 the Talker Failed attribute:
7 FirstValue ::=StreamID, DataFrameParameters, TSpec, PriorityAndRank, AccumulatedLatency,
- 8 FailureInformation
- 9 The following partial BNF production gives the formal description of the MSRPDU FirstValue structure for
- 10 the Listener attribute:
11 FirstValue ::=StreamID
- 12 Figure 35-2 illustrates the structure of the MSRPDU FirstValue components. Each MSRP Attribute shall
- 13 only use those structures as defined by the partial BNF productions described previously. The octet numbers
- 14 shown represent the octet location within the FirstValue field.

1 35.2.2.8.2 StreamID

2 The 64-bit StreamID is used to match Talker registrations with their corresponding Listener registrations
3 (35.2.4). The StreamID comprises the following two subcomponents:

- 4 a) A 48-bit MAC Address associated with the Talker sourcing the stream to the bridged network. The
5 entire range of MAC addresses are acceptable.
- 6 b) A 16-bit unsigned integer value, Unique ID, used to distinguish among multiple streams sourced by
7 the same Talker.

8 StreamIDs are unique across the entire bridged network and are generated by the system offering the stream,
9 or possibly a device controlling that system. A system reserving resources for more than one stream in the
10 same bridged network shall use a StreamID that is unique among all StreamIDs in that bridged network. The
11 combination of these two subcomponents ensure that such an assignment is possible.

12 NOTE 1—The spanning tree protocol ensures that there can be at most one path from a Talker to its Listener(s). Multiple
13 Declarations for the same StreamID can therefore occur briefly, during changes in the active topology of the bridged
14 network.

15 NOTE 2—The MAC address component of the StreamID can, but does not necessarily, have the same value as the
16 source_address parameter of any frame in the actual data stream. For example, the StreamID can be assigned by a TSN
17 CUC (see 46.1.3.3), using a pool of MAC addresses that the TSN CUC maintains.

18 NOTE 3—If the MAC addresses used to construct StreamIDs are not unique within the network, duplicate StreamIDs
19 can be generated, with unpredictable results for SRP.

20 35.2.2.8.3 DataFrameParameters

21 The DataFrameParameters component of the MSRP Attribute specifies the EISS parameters that are
22 common to all frames belonging to the data stream for which this MAD is reserving resources. This
23 information is used by Bridges to create Dynamic Reservation Entries (8.8.7). The parameters are as
24 follows:

- 25 a) The destination_address
- 26 b) The vlan_identifier

27 The destination_address specifies the destination MAC address of the streaming data frames. Only one
28 Stream is allowed per destination_address. MSRP does not describe the actual streaming data, only the
29 bandwidth associated with that stream.

30 The use of destination_address for both a Stream and “best effort” traffic (34.5) is outside the scope of SRP.
31 SRP only supports destination_addresses that are multicast or locally administered addresses.

32 NOTE—MSRP enforces reserved bandwidth guarantees by filtering Stream destination addresses (35.2.4.4.2) for
33 Streams that do not have a reservation. This blocks all traffic to that destination address. If that destination address was
34 also being used for “best effort” traffic that device would no longer be reachable.

35 Systems that are not VLAN aware shall use the value SRclassVID (35.2.2.9.4) for the vlan_identifier in the
36 DataFrameParameters. VLAN-aware systems may use any valid VID (1 through 4094).

37 35.2.2.8.4 TSpec

38 The 32-bit TSpec component is the TSpec associated with a Stream. It consists of the following two
39 elements (which are encoded as described in 10.8.1.1):

- 40 a) **MaxFrameSize:** The 16-bit unsigned MaxFrameSize component is used to allocate resources and
41 adjust queue selection parameters in order to supply the QoS requested by an MSRP Talker
42 Declaration. It represents the maximum frame size that the Talker will produce, excluding any
43 overhead for media-specific framing (e.g., preamble, IEEE 802.3 header, Priority/VID tag, CRC,
44 interframe gap). As the Talker or Bridge determines the amount of bandwidth to reserve on the

egress port it will calculate the media-specific framing overhead on that port and add it to the number specified in the MaxFrameSize field.

b) **MaxIntervalFrames:** The 16-bit unsigned MaxIntervalFrames component is used to allocate resources and adjust queue selection parameters in order to supply the QoS requested by an MSRP Talker Declaration. It represents the maximum number of frames that the Talker can initiate in one classMeasurementInterval (34.3).

NOTE—Consider the example of a Class A 48kHz stereo audio stream encapsulated in an Ethernet frame (see IEEE Std 1722™ [B18]). The audio data within the frame would contain two sets of six 32-bit samples, plus a 32-octet header, for a total of 80 octets per frame sent once every class measurement interval (34.4). Therefore, MaxFrameSize=80, and MaxIntervalFrames=1. An IEEE 802.3 port on a Bridge would also add 42 octets of media-specific framing overhead (8-octet preamble, 14-octet IEEE 802.3 header, 4-octet IEEE 802.1Q priority/VID Tag, 4-octet CRC, 12-octet IFG). When the Bridge calculates the amount of bandwidth to reserve it would combine 42 octets of media-specific framing overhead with the MaxFrameSize of 80 octets, to arrive at a total frame size of 122 octets per class measurement interval. This represents a total bandwidth of approximately 7.7 Mb/s (122 octets × 8 b/octet × 8000 frames/s).

Table 35-5 contains some examples of various forms of audio and video Streams with their associated TSpec components.

Table 35-5—TSpec components examples

Source	Raw bit rate	Media-specific framing overhead	TSpec MaxFrameSize	TSpec MaxIntervalFrames
48kHz stereo audio stream (32-bit samples) Class A (IEEE Std 1722 [B18])	~3 Mb/s	~4.7 Mb/s	80	1 (8 000 frames/s)
96kHz stereo audio stream (32-bit samples) Class A (IEEE Std 1722 [B18])	~6 Mb/s	~4.7 Mb/s	128	1 (8 000 frames/s)
MPEG2-TS video Class B (IEEE Std 1722 [B18])	~24 Mb/s	~2.5 Mb/s	786	1 (4 000 frames/s) ^a
SD SDI (Level C) uncompressed Class A (SMPTE 259M-2008 [B63])	270 Mb/s	~15 Mb/s	1442	3 (24 000 frames/s)
SD SDI (Level D) uncompressed Class A (SMPTE 259M-2008 [B63])	360 Mb/s	~20 Mb/s	1442	4 (32 000 frames/s)
HD SDI 1080i uncompressed Class A (SMPTE 292M-2008 [B64])	1.485 Gb/s	~80 Mb/s	1486	16 (128 000 frames/s)
HD SDI 1080p uncompressed Class A (SMPTE 424M-2008 [B65])	2.97 Gb/s	~160 Mb/s	1486	32 (256 000 frames/s)

^a The MPEG-2 TS entry in this table (third row) is shown as Class B traffic, which runs at a default frame rate of 250 µs/frame or 4000 frames/s. Class A traffic runs at a default rate of 8000 frames/s.

35.2.2.8.5 PriorityAndRank

a) **Data Frame Priority:** The 3-bit Data Frame Priority component specifies the priority value used to generate the Priority Code Point the referenced data streams will be tagged with. It indicates the priority EISS or ISS parameter that will be used in all frames belonging to the data stream for which this MAD is reserving resources. This parameter determines which queue the frame is placed into on

1 an output Bridge Port. In accordance with 10.8.1.1 these three bits are in the most significant
2 positions (8, 7, and 6). The priority specified here is associated with the SR Classes as described in
3 34.5.

4 b) **Rank:** The single-bit Rank component is used by systems to decide which streams can and cannot
5 be served, when the MSRP registrations exceed the capacity of a Port to carry the corresponding
6 data streams. If a Bridge becomes oversubscribed (e.g., network reconfiguration, IEEE 802.11
7 bandwidth reduction), the Rank will also be used to help determine which Stream or Streams can be
8 dropped. A lower numeric value is more important than a higher numeric value. In accordance with
9 10.8.1.1 this bit is in position 5.

10 For streams that carry emergency data such as North America 911 emergency services telephone
11 calls, or fire safety announcements, the Rank shall be 0. Nonemergency traffic shall set this bit
12 to a 1.

13 NOTE—It is expected that higher layer applications and protocols can use the Rank to indicate the relative
14 importance of streams based on user preferences expressed by means beyond the scope of this standard. The
15 values and defaults provided by this Standard are sufficient to order streams on a first-come-first-served basis,
16 with special priority provided for emergency services.

17 c) **Reserved:** This 4-bit field shall be zero filled on transmit and ignored on receive. In accordance
18 with 10.8.1.1 these four bits are in the least significant positions (4, 3, 2, and 1).

19 35.2.2.8.6 Accumulated Latency

20 The 32-bit unsigned Accumulated Latency component is used to determine the worst-case latency that a
21 Stream can encounter in its path from the Talker to a given Listener. The latency reported here is not
22 intended to increase during the life of the reservation. If some event occurs that would increase the latency
23 beyond the original guarantee, MSRP will change the Talker Advertise to a Talker Failed and report Failure
24 Code=7 (Table 35-6).

25 NOTE 1—An example of how latency could increase is if the speed of the underlying media were to decrease, such as
26 one might see on a wireless link.

27 The initial value sent by the Talker is set to *portTcMaxLatency* plus any amounts specified in the
28 REGISTER_STREAM.request, and its value is increased by each Bridge as the Talker Declaration
29 propagates through the network.

30 The *portTcMaxLatency* per hop is equal to the sum of the following:

- 31 a) (equal or higher priority traffic) the time required to empty the queue in which frames of that priority
32 are placed, if that queue and all higher priority queues are full.
- 33 b) (lower priority traffic) the time required to transfer one lower priority frame of maximum size that
34 could have just started transmitting as the current priority frame was queued up.
- 35 c) (internal processing) the worst-case time required by the Bridge to transfer a received frame from
36 the input port to the output queue.
- 37 d) (wire propagation time) the time required for the first bit of the frame to propagate from the output
38 port to the receiving device.
- 39 e) (media access delay) the time required to wait for the media to become available for transmission.

40 For item a) the total number of ports with *msrpPortEnabledStatus* set TRUE, and the *msrpMaxFanInPorts*
41 will effect these calculations.⁴⁸

42 For item a) and item b) the maximum size of the interfering traffic is limited to *msrpLatencyMaxFrameSize*
43 octets.

⁴⁸ Example calculations for latency are contained in the paper “Calculating the Delay Added by Qav Stream Queue” [B3].

1 For item d), the propagation time, if the managed object for Propagation Delay (12.32.2) is supported,
2 txPropagationDelay shall be used. Otherwise, in the absence of better information, a value of 500 ns shall be
3 used.

4 NOTE 2—This implies that no type of frame flow control can be used on the associated data stream frames.

5 The Listener can use this information to decide if the Latency is too large for acceptable presentation of the
6 stream. The Accumulated Latency component is in units of nanoseconds.

7 **35.2.2.8.7 FailureInformation**

8 At the point when a Talker Advertise Declaration is transformed into a Talker Failed Declaration, the system
9 making the transformation adds information that indicates, to the Listeners registering the Talker Failed
10 Declaration, the cause of the failure and the identity of the system at which the failure occurred. The
11 subcomponents of the FailureInformation include:

- 12 a) The system identifier, which is the Bridge Identifier (13.26.2) of the Bridge or the 48-bit MAC
13 Address of the end station's port extended to 64-bits by prepending 16 bits of zero, that changed the
14 Declaration Type from Advertise to Failed.

15 NOTE 1—Bridge Identifiers are normally constructed from MAC Addresses that are unique in the bridged LAN but are
16 not required to be constructed in that manner; therefore, there is a possibility of an end station MAC Address colliding
17 with the Bridge ID.

- 18 b) The Failure Code, which is represented by a single octet containing the value shown in Table 46-16.

19 NOTE 2—Although the table of failure codes (Table 46-16) is located in a subclause associated with MSRPv1, the
20 codes are compatible with MSRPv0.

21 **35.2.2.9 MSRP FirstValue definitions (Domain discovery)**

22 The Domain attribute contains all the information that a Bridge Port needs in order to determine the location
23 of the SRP domain boundary [item h) in 35.2.1.4]. The Domain attribute is also used to detect the
24 ProtocolVersion of the neighboring MSRP Participant on each Bridge Port [item j) in 35.2.1.4].

25 The MSRP Participant shall declare (transmit) its Domain attribute after MAC_Operational (IEEE Std
26 802.1AC) transitions to TRUE and prior to declaring a Talker Advertise, Talker Failed, or Listener attribute.

27 FirstValue shall be incremented one or more times when NumberOfValues is greater than one. Incrementing
28 FirstValue for the MSRP Domain attribute is defined as follows:

- 29 a) Add 1 to SRclassID (35.2.2.9.2), and
30 b) Add 1 to SRclassPriority (35.2.2.9.3).

31 The choice of encoding and incrementing with this rule means that if one class (e.g., class B) is supported,
32 with the default values, then the FirstValue will be {5,2,VID} (class B, priority 2, and a VID) and the
33 NumberOfValues field will be set to 1. If class A and class B are supported, with the default values, the
34 FirstValue will again be {5,2,VID}, but the NumberOfValues fields will be set to 2. Applying the above
35 incrementing rule to {5,2,VID} generates the value {6,3,VID}, i.e., class A, priority 3, and a VID, which is
36 needed for the default case.

37 NOTE—If the SR Class to Priority Mapping Table (12.20.4) is configured through management such that SR class ID
38 values do not increment contiguously with priority values, the declared domains need to be encoded in multiple Domain
39 Vector Attributes. For example, if the SR class ID and priority rows are {2,4}, {3,0}, {4,1}, {5,2}, and {6,3}, two
40 Domain attributes are needed: one Domain with FirstValue {2,4,VID} and NumberOfValues one, and a second Domain
41 with FirstValue {3,0,VID} and NumberOfValues four.

1 **35.2.2.9.1 Structure definition**

- 2 The FirstValue for the Domain attribute in MSRPDU exchanged according to the protocol specified in this
- 3 subclause shall have the following structure:
- 4 a) The first octet contains the *SRclassID* (35.2.2.9.2).
- 5 b) Following the *SRclassID* is an octet containing the *SRclassPriority* (35.2.2.9.3).
- 6 c) Following the *SRclassPriority* are two octets containing the *SRclassVID* (35.2.2.9.4).
- 7 The following BNF production gives the formal description of the MSRPDU FirstValue structure for the
- 8 Domain attribute:
- 9 FirstValue ::= SRclassID, SRclassPriority, SRclassVID

10 Figure 35-3 illustrates the structure of the MSRPDU FirstValue components for the Domain attribute.

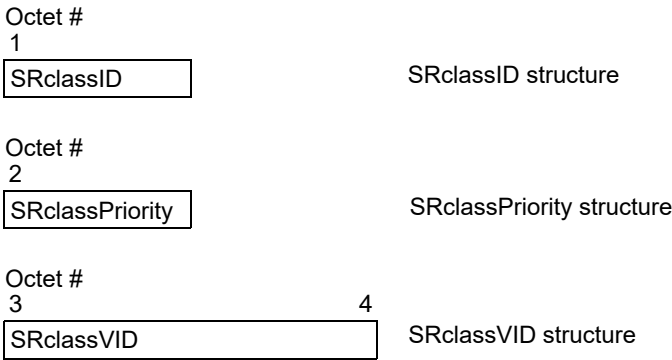


Figure 35-3—Format of the components of the Domain FirstValue

11 **35.2.2.9.2 SRclassID**

12 SRclassID is a numeric representation of the SR classes that are supported by a particular Bridge Port. The

13 mapping of SR class letter to SR class ID is shown in Table 35-6.

Table 35-6—SR class ID

SR class	SR class ID
A	6
B	5
C	4
D	3
E	2
F	1
G	0

14 NOTE—When SRP is supported, only one SR class is required. Only SR classes A and B can be enabled by default.

15 Additional SR classes must be configured through use of management (12.20.4, 34.5).

1 35.2.2.9.3 SRclassPriority

2 This field holds the Data Frame Priority [item a) in 35.2.2.8.5] value that will be used for streams that
3 belong to the associated SR class. Whenever an end station's MAC_Operational (IEEE Std 802.1AC)
4 transitions to TRUE, its SRclassPriority shall be set to the default SR class priority (Table 6-5) until an
5 SRclassPriority declaration is received from a neighboring device, at which time it shall be set to the
6 declared value. An end station cannot change its default mapping of SR class to priority (see 12.20.4).

7 NOTE—This allows two end stations that are connected back-to-back to share streams, while also providing an end
8 station the flexibility to change to the SRclassPriority of a attached network (i.e., neighboring Bridge).

9 35.2.2.9.4 SRclassVID

10 This field contains the SR_PVID [item i) in 35.2.1.4] that the associated streams will be tagged with by the
11 Talker. Whenever an end station's MAC_Operational (IEEE Std 802.1AC) transitions to TRUE, its
12 SRclassVID shall be set to the default SR_PVID (Table 9-2) until an SRclassVID declaration is received
13 from a neighboring device, at which time it shall be set to the declared value.

14 NOTE—This allows two end stations that are connected back-to-back to share streams, while also providing an end
15 station the flexibility to change to the SRclassVID of a attached network.

16 35.2.2.10 MSRP FirstValue definitions (Stream reservations, enhanced)

17 This subclause specifies two Attribute Types for enhanced stream reservations: Talker Enhanced and
18 Listener Enhanced.

19 The specifications of the enhanced Attribute Types (35.2.2.10) are based on the specifications of the original
20 Attribute Types (35.2.2.8), with differences stated explicitly.

21 As compared to the original Attribute Types, the enhanced Attribute Types use a Type-Length-Value (TLV)
22 encoding for MSRP FirstValue. MSRP FirstValue contains a list of TLVs. Each TLV consists of a Type field
23 that specifies what the Value field contains, a Length field that specifies the number of octets in the Value
24 field, and the Value field. The Value contains one or more elements encoded in binary. The TLV encoding
25 enables flexibility in the structure of MSRP FirstValue, such as support for optional features.

26 This subclause specifies the structure of MSRP FirstValue as well as the Type and Length for each TLV. This
27 subclause also specifies the binary encoding of the elements within the Value of each TLV. The semantic
28 specification of each element in the TLV's Value is provided as a reference to the corresponding group of
29 elements in TSN User/network configuration information (46.2).

30 The transmission of octets in MSRP FirstValue is specified in 10.8.1.1. The encoding of the Value of each
31 TLV is specified in a figure, and the representation in the figure is specified in 10.8.1.1. Elements marked as
32 “reserved” shall be transmitted as zero and ignored on reception. When an element is not a multiple of 8 bits
33 in length, the element is organized from most significant bit of the octet to least significant bit. Boolean
34 elements are a single bit. When an element's Length in the figure is a decimal number only, Length specifies
35 the number of octets for the element. When an element's Length in the figure is a decimal number followed
36 by “bits,” Length specifies the number of bits for the element.

37 FirstValue shall be incremented one or more times when NumberOfValues is greater than one, as follows:

- 38 a) Add 1 to Unique ID of the StreamID TLV (46.2.3.1), and
- 39 b) Add 1 to each element of the DataFrameSpecification TLV (46.2.3.4) that identifies the Stream:
 - 40 1) If the IEEE802-MacAddresses TLV is used, add 1 to DestinationMacAddress.
 - 41 2) If the IPv4-tuple TLV is used, add 1 to DestinationIpAddress.
 - 42 3) If the IPv6-tuple TLV is used, add 1 to DestinationIpAddress.

1 This rule for incrementing FirstValue is the same as the original Attribute Types (35.2.2.8), with the addition
2 of optional support for IP destination address as the unique identification of the Stream.

3 NOTE 1—For an example of incrementing FirstValue for DestinationMacAddress, refer to 35.2.2.8.

4 NOTE 2—As compared to the fixed attribute values of the original AttributeTypes, the use of TLVs in the enhanced
5 AttributeTypes allows each attribute value to vary based on the use of sub-TLVs. This imposes an additional constraint
6 when NumberOfValues is greater than one such that all FirstValues must use the same sub-TLVs.

7 NOTE 3—As specified in 35.2.3.1, when a Talker or Listener has a *neighborProtocolVersion* of 0x00 (i.e., Nearest
8 Bridge is MSRPv0), declarations must use only the original Attribute Types (35.2.2.8). If the application of the Talker/
9 Listener is using the enhanced features of this subclause (35.2.2.10), an internal translation is needed in order to
10 determine if the original Attribute Type can be declared. This internal translation is similar to the translation specified for
11 MAP in a Bridge (Table 35-11 and Table 35-16), but it is outside the scope of this standard. If the enhanced features do
12 not translate successfully to an original Attribute Type, the Talker or Listener can choose to either a) declare the failure
13 to the network using an original Attribute Type (e.g., Talker Failed Vector of Table 35-1) or b) return an error up to the
14 application of the Talker/Listener.

15 Some of the enhanced features of this subclause require the assistance of a Centralized Network
16 Configuration (CNC) entity. As described in the introduction to Clause 35, in order to use a CNC with SRP,
17 the MRP External Control (12.32.4) feature is enabled in the Nearest Bridge (8.6.3) to each Talker and
18 Listener. Therefore, when a Talker or Listener uses an enhanced feature, the success or failure of the Stream
19 reservation depends on the externalControl (12.32.4.1) parameter on the Port connected to the Talker/
20 Listener, as follows:

21 **a) externalControl TRUE**

22 SRP messages are exchanged with the CNC, and the CNC uses those SRP messages to learn the
23 requirements for each Stream. The CNC uses remote management protocols (e.g., SNMP,
24 NETCONF) in order to learn the capabilities of the Bridges between each Talker and its Listeners.
25 The CNC is not required to support all Bridge features, but for the Bridge features that it supports, it
26 compares the Bridge's capabilities to the configuration needed in order to meet each Stream's
27 requirements. If the CNC successfully configures the Bridges for the Stream, the CNC shall use
28 MRP External Control to declare a Talker Advertise attribute to each Listener and a Listener Ready
29 attribute to the Talker. If the CNC fails to configure the Bridges for the Stream, the CNC shall use
30 MRP External Control to declare a Talker Failed attribute to each Listener and a Listener Ready
31 Failed or Listener Asking Failed attribute to the Talker.

32 **b) externalControl FALSE**

33 When MRP External Control is not used, SRP messages propagate through Bridges, and the Bridges
34 configure themselves for the Stream reservation. Some enhanced features can be supported with this
35 model, and other enhanced features cannot be supported. In the specifications of this subclause, the
36 phrase “When the externalControl attribute is FALSE” is used to specify failure conditions when
37 MRP External Control is not used for the Talker/Listener. If this phrase is not used, the feature is
38 fully supported in the absence of MRP External Control (i.e., CNC).

35.2.2.10.1 Structure definition

- The FirstValue of each attribute shall consist of a list of one or more TLVs. Each TLV shall contain a single octet TLV type, followed by a single octet TLV length, followed by multiple octets for the TLV value.
- The TLV type shall use a value specified in Table 35-7.

Table 35-7—TLV types

TLV	TLV type	TLV length
Talker	1	variable
StreamID	2	8
StreamRank	3	1
EndStationInterfaces	4	variable
InterfaceID	5	variable
DataFrameSpecification	6	18
IEEE802-MacAddresses	7	12
IEEE802-VlanTag	8	2
IPv4-tuple	9	15
IPv6-tuple	10	39
TrafficSpecification	11	9
TSpecTimeAware	12	12
UserToNetworkRequirements	13	5
InterfaceCapabilities	14	variable
Listener	15	variable
Status	16	variable
StatusInfo	17	3
AccumulatedLatency	18	4
InterfaceConfiguration	19	variable
TimeAwareOffset	20	4
FailedInterfaces	21	variable

- The TLV length shall contain the number of octets in the TLV value that follows. The TLV value is specified in subsequent subclauses for each TLV.

1 The following Backus-Naur Form (BNF) production gives the formal description of the MSRPDU
2 FirstValue structure for the Talker Enhanced attribute:

```

3     TalkerEnhanced_FirstValue ::= Talker, Status
4     Talker ::= Type, Length,
5         StreamID,
6         StreamRank,
7         [ EndStationInterfaces, ]
8         [ DataFrameSpecification, ]
9         TrafficSpecification,
10        [ TSpecTimeAware, ]
11        [ UserToNetworkRequirements, ]
12        [ InterfaceCapabilities, ]
13    Status ::= Type, Length,
14        StatusInfo,
15        AccumulatedLatency
16        [ InterfaceConfiguration ]
17        [ FailedInterfaces ]
18    StreamID ::= Type, Length, value specified in 35.2.2.10.2
19    StreamRank ::= Type, Length, value specified in 35.2.2.10.3
20    EndStationInterfaces ::= Type, Length, value specified in 35.2.2.10.4, 35.2.2.10.4
21    DataFrameSpecification ::= Type, Length, value specified in 35.2.2.10.5, 35.2.2.10.5
22    TrafficSpecification ::= Type, Length, value specified in 35.2.2.10.6
23    TSpecTimeAware ::= Type, Length, value specified in 35.2.2.10.6
24    UserToNetworkRequirements ::= Type, Length, value specified in 35.2.2.10.7
25    InterfaceCapabilities ::= Type, Length, value specified in 35.2.2.10.8, 35.2.2.10.8
26    StatusInfo ::= Type, Length, value specified in 35.2.2.10.9
27    AccumulatedLatency ::= Type, Length, value specified in 35.2.2.10.10
28    InterfaceConfiguration ::= Type, Length, value specified in 35.2.2.10.11
29    FailedInterfaces ::= Type, Length, value specified in 35.2.2.10.12
30    Type BYTE ::= corresponding TLV Type from Table 35-7
31    Length BYTE ::= corresponding TLV Length (number of octets in TLV Value)

```

32 The following Backus-Naur Form (BNF) production gives the formal description of the MSRPDU
33 FirstValue structure for the Listener Enhanced attribute:

```

34    ListenerEnhanced_FirstValue ::= Listener, Status
35    Listener ::= Type, Length,
36        StreamID,
37        [ EndStationInterfaces, ]
38        [ UserToNetworkRequirements, ]
39        [ InterfaceCapabilities ]
40    StatusGroup ::= Type, Length,
41        StatusInfo,
42        [ InterfaceConfiguration ]
43        [ FailedInterfaces ]
44    StreamID ::= Type, Length, value specified in 35.2.2.10.2
45    EndStationInterfaces ::= Type, Length, value specified in 35.2.2.10.4, 35.2.2.10.4
46    UserToNetworkRequirements ::= Type, Length, value specified in 35.2.2.10.7
47    InterfaceCapabilities ::= Type, Length, value specified in 35.2.2.10.8, 35.2.2.10.8
48    StatusInfo ::= Type, Length, value specified in 35.2.2.10.9
49    InterfaceConfiguration ::= Type, Length, value specified in 35.2.2.10.11
50    FailedInterfaces ::= Type, Length, value specified in 35.2.2.10.12
51    Type BYTE ::= corresponding TLV Type from Table 35-7
52    Length BYTE ::= corresponding TLV Length (number of octets in TLV value)

```

35.2.2.10.2 StreamID

The StreamID group is specified in 46.2.3.1.

Figure 35-4 specifies the encoding of the value for the StreamID TLV.

	Octet	Length
MacAddress	1	6
UniqueID	7	2

Figure 35-4—Value of StreamID TLV

The MacAddress is encoded in a manner consistent with the MAC address of a frame header.

The semantics of the MacAddress and UniqueID elements of the enhanced StreamID TLV are the same as the semantics of the Mac Address and Unique ID of the original StreamID structure (35.2.2.8.2).

Requirements for incrementing UniqueID are specified in 35.2.2.10.

35.2.2.10.3 StreamRank

The StreamRank group is specified in 46.2.3.2.

Figure 35-5 specifies the encoding of the value for the StreamRank TLV.

	Octet	Length
reserved	1	7 bits
Rank	1	1 bit

Figure 35-5—Value of StreamRank TLV

The semantics of the Rank element of the enhanced StreamRank TLV are the same as the semantics of the Rank of the original PriorityAndRank structure (35.2.2.8.5).

35.2.2.10.4 EndStationInterfaces

The EndStationInterfaces group is specified in 46.2.3.3.

The Value of the EndStationInterfaces TLV shall consist of one or more InterfaceID TLVs. An InterfaceID TLV is provided for each interface used by the Stream's Talker/Listener.

Figure 35-6 specifies the encoding of the value for the InterfaceID TLV.

Group length: 6 + NameLength

	Octet	Length
MacAddress	1	6
InterfaceName	7	NameLength

Figure 35-6—Value of InterfaceID TLV

The MacAddress is encoded in a manner consistent with the MAC address of a frame header.

If the InterfaceName element is not used, its NameLength is zero, and InterfaceName does not exist in the InterfaceID's Value.

EndStationInterfaces does not exist in the original Attribute Types (MSRPv0). EndStationInterfaces enhances MSRP for optional support of CNC.

1 EndStationInterfaces specifies the identification of physical interfaces (distinct points of attachment) in the
2 end station acting as a Talker/Listener. EndStationInterfaces is used by the CNC to locate the Talker/Listener
3 in the topology. EndStationInterfaces can also be used by end stations with two or more interfaces.

4 When the externalControl attribute is TRUE (12.32.4.1), EndStationInterfaces should be included in the
5 attribute declared by the Talker or Listener. If the CNC does not have the information needed to identify the
6 Talker/Listener Ports in the topology, the CNC is likely to fail the Stream reservation (35.2.2.10).

7 When the externalControl attribute is FALSE (12.32.4.1), if EndStationInterfaces contains more than one
8 interface, the Nearest Bridge shall fail the Stream reservation. The fully distributed model does not specify
9 support for multiple interfaces. To fail the Stream reservation, the Nearest Bridge shall change Talker
10 Advertise to Talker Failed and report Failure Code=25 (Table 46-16) and also change Listener Ready or
11 Listener Ready Failed to Listener Asking Failed and report Failure Code=25.

12 Since the EndStationInterfaces is only used by MRP External Control (12.32.4) for communication with a
13 CNC, the EndStationInterfaces TLV shall be removed from the MSRP FirstValue prior to propagation with
14 MAP (35.2.4).

15 35.2.2.10.5 DataFrameSpecification

16 The DataFrameSpecification group is specified in 46.2.3.4.

17 The TLV for DataFrameSpecification shall consist of Type and Length, followed by a Value that consists of
18 a list of one or more TLVs for each field in the user's frame. The list of TLVs within the
19 DataFrameSpecification value shall be ordered from start of frame to end of header.

20 Figure 35-7 specifies the encoding of the value for the IEEE802-MacAddresses TLV. The MAC addresses
21 are encoded in a manner consistent with the MAC address of a frame header.

	Octet	Length
DestinationMacAddress	1	6
SourceMacAddress	7	6

Figure 35-7—Value of IEEE802-MacAddresses TLV

22 Figure 35-8 specifies the encoding of the value for the IEEE802-VlanTag TLV.

	Octet	Length
Priority Code Point	1	3 bits
Reserved	1	1 bit
VLAN ID	1	12 bits

Figure 35-8—Value of IEEE802-VlanTag TLV

Figure 35-9 specifies the encoding of the value for the IPv4-tuple TLV. The IP addresses are encoded in a manner consistent with the IP address of a packet header.

	Octet	Length
SourceIpAddress	1	4
DestinationIpAddress	5	4
Dscp	9	1
Protocol	10	2
SourcePort	12	2
DestinationPort	14	2

Figure 35-9—Value of IPv4-tuple TLV

Figure 35-10 specifies the encoding of the value for the IPv6-tuple TLV. The IP addresses are encoded in a manner consistent with the IP address of a packet header.

	Octet	Length
SourceIpAddress	1	16
DestinationIpAddress	17	16
Dscp	33	1
Protocol	34	2
SourcePort	36	2
DestinationPort	38	2

Figure 35-10—Value of IPv6-tuple TLV

The semantics of the DestinationMacAddress of the enhanced DataFrameSpecification TLV (IEEE802-MacAddresses sub-TLV) are the same as the semantics of the destination_address of the original DataFrameParameters structure (35.2.2.8.3). The semantics of the VlanId of the enhanced DataFrameSpecification TLV (IEEE802-VlanTag sub-TLV) are the same as the semantics of the vlan_identifier of the original DataFrameParameters structure (35.2.2.8.3). The semantics of the PriorityCodePoint element of the enhanced DataFrameSpecification TLV are the same as the semantics of the Data Frame Priority of the original PriorityAndRank structure (35.2.2.8.5).

Requirements for incrementing destination addresses of DataFrameSpecification are specified in 35.2.2.10.

In order to apply TSN behavior to Streams (e.g., reserved bandwidth guarantees), MSRP must be able to distinguish one Stream from another Stream and to distinguish Streams from non-TSN traffic (e.g., best-effort). This requires unique identification of each Stream within the user (i.e., Talker and Listener) as well as the network (i.e., Bridges).

As compared to the original Attribute Types (MSRPv0), the enhancements of the DataFrameSpecification TLV support Stream transformation (46.1.4).

Stream transformation is an optional feature in MSRP Participants. Since Stream Transformation can be supported in either end station or Bridge, one of the following methodologies may be chosen:

a) No Stream Transformation

When Stream transformation (46.1.4) is not supported, the user's identification of the Stream must be the same as the network's identification (i.e., destination MAC address and VLAN ID). This is the methodology used for MSRPv0. This methodology requires the following in each end station (Talker/Listener):

- 1) The destination MAC address of the user's data frame shall be either multicast (group) or a locally administered unicast (individual). A multicast MAC address is allocated by the user (e.g., a higher layer protocol like IEEE Std 1722 [B18]). Only one Stream is allowed per destination MAC address. The destination MAC address shall be provided as the DestinationMacAddress element of the IEEE802-MacAddresses sub-TLV of the DataFrameSpecification TLV.

NOTE—When Stream transformation is not supported, MSRP prohibits a DestinationMacAddress element that is universal unicast (i.e., individual MAC address of the Listener). Since the Listener's universal unicast MAC address is used for non-TSN traffic, it does not uniquely identify the Stream. The VLAN ID alone does not distinguish the Stream's traffic because Bridges can be configured to egress non-TSN traffic using the same VLAN ID that the Stream is using.

- 2) The Talker uses its individual MAC address as the source MAC address of the data frame. The source MAC address shall be provided as the SourceMacAddress element of the IEEE802-MacAddresses sub-TLV of the DataFrameSpecification TLV.
- 3) The data frame shall use a VLAN tag containing a Priority Code Point (PCP) and a VLAN ID, both provided in the IEEE802-VlanTag sub-TLV of the DataFrameSpecification TLV. The Talker uses the value SRclassPriority of the Domain attribute (35.2.2.9.3) for the PriorityCodePoint. If the Talker has knowledge of a specific VLAN ID to use, any valid VlanId can be specified (1 to 4094). If the Talker does not have knowledge of a specific VLAN ID to use, the VlanId uses the value from the SRclassVID of the Domain attribute (35.2.2.9.4).

b) Stream Transformation in end station

When Stream transformation (46.1.4) is performed in the end station (Talker/Listener), the end station relies on the network to provide the network's identification of the Stream (i.e., destination MAC address and VLAN ID). The network identification is provided by the CNC through the use of MRP External Control (12.32.4) in the Nearest Bridge (8.6.3). This methodology requires the following in each end station (Talker/Listener):

- 1) The Talker's DataFrameSpecification TLV shall not be included in the Talker attribute. This indicates to the CNC that Stream transformation is performed in the end station. Since the user's data frame identification is not used in the network, the DataFrameSpecification is not relevant to Bridges.
- 2) The end station's InterfaceCapabilities TLV (46.2.3.7) shall set VlanTagCapable=true and CB-StreamIdenTypeList containing the Active Destination MAC and VLAN Stream identification type (46.2.3). If the user's data frame uses either IPv4-tuple or IPv6-tuple, the CB-StreamIdenTypeList shall contain the IP Stream identification type. These InterfaceCapabilities specify that the end station supports the IEEE 802.1CB stream identification functions that transform between the user's identification and the corresponding network identification.
- 3) When the externalControl attribute is FALSE (12.32.4.1), if the Talker's DataFrameSpecification TLV is missing from Talker Advertise, the Nearest Bridge shall fail the Stream reservation. The fully distributed model does not specify support for stream transformation in end station. To fail the Stream reservation, the Nearest Bridge shall change Talker Advertise to Talker Failed and report Failure Code=22 (Table 46-16).
- 4) The CNC shall allocate a multicast MAC address for the network identification and use MRP External Control to declare an MSRP attribute containing a InterfaceConfiguration sub-TLV (35.2.2.10.11) with configuration values for IEEE802-MacAddresses and IEEE802-VlanTag. These configuration values provide the network identification for the end station's transformation. For an end station acting as Talker, the Nearest Bridge includes InterfaceConfiguration in the declared Listener Ready or Listener Ready Failed. For an end station acting as Listener, the Nearest Bridge includes InterfaceConfiguration in the declared Talker Advertise.
- 5) When the Stream is withdrawn (i.e., MRP Leave of Talker Advertise), the CNC can free the multicast MAC address for use by subsequent Streams.

c) Stream Transformation in Bridge

When Stream transformation (46.1.4) is performed in the Bridge, the end station relies on the CNC to configure the Bridge to perform all transformation. The user identification (i.e., `DataFrameSpecification`) is provided to the CNC through the use of MRP External Control (12.32.4) in the Nearest Bridge (8.6.3). The CNC uses management to configure IEEE 802.1CB features in the Nearest Bridge to transform the Stream between user and network identification. This methodology requires the following:

- 1) The Talker's `DataFrameSpecification` TLV provides unique user identification of the Stream, but that user identification does not meet the requirements for no stream transformation [item a) in 35.2.2.10.5]. If the destination address alone does not distinguish the Stream from other Streams and non-TSN traffic, other elements of the `DataFrameSpecification` shall be unique. For example, if the data frame is untagged with unicast `DestinationMacAddress` and unicast `DestinationIpAddress`, a unique `DestinationPort` can distinguish the Stream.
- 2) When the `externalControl` attribute is `FALSE` (12.32.4.1), since the Talker's `DataFrameSpecification` TLV does not meet the requirements for no stream transformation [item a) in 35.2.2.10.5], the Nearest Bridge shall fail the Stream reservation. The fully distributed model does not specify support for stream transformation in Bridge. To fail the Stream reservation, the Nearest Bridge shall change Talker Advertise to Talker Failed and report Failure Code=22 (Table 46-16).
- 3) If the CNC determines that the Nearest Bridge does not support IEEE Std 802.1CB, it shall use MRP External Control to declare an MSRP attribute containing a `StatusInfo` sub-TLV (35.2.2.10.9) with Failure Code=23 (Table 46-16). For an end station acting as Talker, the Nearest Bridge declares Listener Asking Failed. For an end station acting as Listener, the Nearest Bridge declares Talker Failed.
- 4) If the CNC determines that the Nearest Bridge does not support the IEEE 802.1CB stream identification types for the `DataFrameSpecification`, it shall use MRP External Control to declare an MSRP attribute containing a `StatusInfo` sub-TLV (35.2.2.10.9) with Failure Code=24 (Table 46-16). For an end station acting as Talker, the Nearest Bridge declares Listener Asking Failed. For an end station acting as Listener, the Nearest Bridge declares Talker Failed.
- 5) If the CNC successfully configures the IEEE 802.1CB stream identification types for the `DataFrameSpecification`, it shall use MRP External Control to declare a successful MSRP attribute (assuming no other failures). For an end station acting as Talker, the Nearest Bridge declares Listener Ready or Listener Ready Failed. For an end station acting as Listener, the Nearest Bridge declares Talker Advertise.
- 6) When the Stream is withdrawn (i.e., MRP Leave of Talker Advertise), the CNC can free the multicast MAC address for use by subsequent Streams. The CNC can also remove IEEE 802.1CB configurations for the withdrawn Stream.

35.2.2.10.6 TrafficSpecification and TSpecTimeAware

The `TrafficSpecification` group and optional `TSpecTimeAware` group are specified in 46.2.3.5.

The `TrafficSpecification` shall consist of a required `TrafficSpecification` TLV, followed by an optional `TSpecTimeAware` TLV.

1 Figure 35-11 specifies the encoding of the value for the TrafficSpecification TLV.

	Octet	Length
Interval	1	4
MaxFramesPerInterval	5	2
MaxFrameSize	7	2
TransmissionSelection	9	1

Figure 35-11—Value of TrafficSpecification TLV

2 Figure 35-12 specifies the encoding of the value for the TSpecTimeAware TLV. The presence of the optional
3 TSpecTimeAware TLV is handled as specified in 46.2.3.5 for the presence of the TSpecTimeAware group.

	Octet	Length
EarliestTransmitOffset	1	4
LatestTransmitOffset	5	4
Jitter	9	4

Figure 35-12—Value of TSpecTimeAware TLV

4 The semantics of the MaxFramesPerInterval element of the enhanced TrafficSpecification TLV are the same
5 as the semantics of the MaxIntervalFrames of the original TSpec structure (35.2.2.8.4). The semantics of the
6 MaxFrameSize element of the enhanced TrafficSpecification TLV are the same as the semantics of the
7 MaxFrameSize of the original TSpec structure (35.2.2.8.4).

8 A Talker that assumes usage of the fully distributed model should use TrafficSpecification as follows:

- 9 a) Set the Interval element of the TrafficSpecification TLV to the classMeasurementInterval (34.3) of
10 the stream's SR class (35.2.2.9.2).
- 11 b) Set the TransmissionSelection element to the value 1 for use of the credit-based shaper (8.6.8.2).
- 12 c) Do not use the TSpecTimeAware TLV.

13 When the externalControl attribute is FALSE (12.32.4.1), if the Talker's TrafficSpecification TLV does not
14 meet the preceding recommendations a) through c), a Bridge may fail the Stream reservation. Bridges using
15 the fully distributed model are not required to support Talker-specific intervals or shaping beyond those of
16 MSRPv0. To fail the Stream reservation, the Bridge shall change Talker Advertise to Talker Failed and
17 report Failure Code=25 (Table 46-16).

18 NOTE—In IEEE Std 802.1BA [B11], the AccumulatedLatency computation assumes consistent use of the credit-based
19 shaper at the classMeasurementInterval for the Talker and all Bridges using the SR class. This assumption aligns with
20 the preceding recommendations. The Talker's TrafficSpecification specifies the interval, shaping, and/or scheduling of
21 the Talker only, and TrafficSpecification does not change as it propagates through Bridges. Therefore, when MRP
22 External Control is not in use, in order to support TrafficSpecifications beyond those of IEEE Std 802.1BA, each Bridge
23 must be able to 1) distinguish whether the Port registering Talker Advertise is connected directly to the Talker (i.e., end
24 station) or another Bridge, to determine where TrafficSpecification impacts the AccumulatedLatency, and 2) understand
25 the shaping and scheduling used in each Bridge in the path from Talker to the current Bridge since that also impacts
26 AccumulatedLatency. Specifications for these tasks are not provided in this standard.

1 35.2.2.10.7 UserToNetworkRequirements

2 The UserToNetworkRequirements group is specified in 46.2.3.6.

3 Figure 35-13 specifies the encoding of the value for the UserToNetworkRequirements TLV.

	Octet	Length
NumSeamlessTrees	1	1
MaxLatency	2	4

Figure 35-13—Value of UserToNetworkRequirements TLV

4 UserToNetworkRequirements does not exist in the original Attribute Types (MSRPv0).
5 UserToNetworkRequirements provides an optional mechanism for the Stream's Talker and/or Listeners to
6 specify requirements to the network.

7 As specified in the MSRP structure definition (35.2.2.10.1), the UserToNetworkRequirements TLV may be
8 used for either Talker or Listener attributes.

9 If UserToNetworkRequirements.MaxLatency is greater than zero, when an MSRP Participant detects that
10 the AccumulatedLatency (35.2.2.10.10) of the Stream is greater than
11 UserToNetworkRequirements.MaxLatency, the Stream's status is reported as failed (35.2.4.6).

12 If UserToNetworkRequirements.MaxLatency is zero (or UserToNetworkRequirements is not provided),
13 when an MSRP Participant detects that the AccumulatedLatency of the Stream is greater than the initial
14 AccumulatedLatency when the Stream was successfully configured, the Stream's status is reported as failed.
15 This technique is compatible with MSRPv0 (35.2.2.8.6).

16 As specified in 46.2.3.6 and 35.2.4.6, UserToNetworkRequirements.MaxLatency specified by a Talker
17 applies to all Listeners of the stream. UserToNetworkRequirements.MaxLatency specified by a Listener
18 applies to that Listener alone.

19 Additional specifications for UserToNetworkRequirements.MaxLatency are provided in 35.2.6.

20 A Talker that assumes usage of the fully distributed model should use UserToNetworkRequirements as
21 follows:

22 a) Set the NumSeamlessTrees element to one.

23 When the externalControl attribute is FALSE (12.32.4.1), if the Talker's UserToNetworkRequirements TLV
24 does not meet the preceding recommendation a), a Bridge may fail the Stream reservation. Bridges using the
25 fully distributed model are not required to support configuration of seamless redundancy. To fail the
26 Stream reservation, the Bridge shall change Talker Advertise to Talker Failed and report Failure Code=25
27 (Table 46-16).

28 NOTE—NumSeamlessTrees greater than one is intended for use with MRP External Control (i.e., externalControl
29 TRUE), which passes SRP attributes to a CNC, and the CNC configures Bridges for seamless redundancy (e.g.,
30 IEEE Std 802.1CB or IEC 62439-3:2016 [B6]). When MRP External Control is not in use, in order to support
31 NumSeamlessTrees greater than one, each Bridge must be able to either 1) determine that seamless redundancy has
32 already been configured for all Bridges between the Talker and its Listeners or 2) solicit such configuration from an
33 external entity. Specifications for these tasks are not provided in this standard.

35.2.2.10.8 InterfaceCapabilities

The InterfaceCapabilities group is specified in 46.2.3.7.

Figure 35-14 specifies the encoding of the value for the InterfaceCapabilities TLV. NumITL represents the number of elements in the CB-StreamIdenTypeList. NumSTL represents the number of elements in the CB-SequenceTypeList.

TLV Length: $3 + (4 \times \text{NumITL}) + (4 \times \text{NumSTL})$	Octet	Length
reserved	1	7 bits
VlanTagCapable	1	1 bit
NumITL	2	1
NumSTL	3	1
CB-StreamIdenTypeList	4	$4 \times \text{NumITL}$
CB-SequenceTypeList	4	$4 \times \text{NumITL}$

Figure 35-14—Value of InterfaceCapabilities TLV

InterfaceCapabilities does not exist in the original Attribute Types (MSRPv0). InterfaceCapabilities provides an optional mechanism for stream transformation as specified in 35.2.2.10.5.

The InterfaceCapabilities TLV is used only for negotiation between an end station and the neighboring MRP Participant on the link. The InterfaceCapabilities TLV shall be removed from the MSRP FirstValue prior to propagation with MAP (35.2.4).

35.2.2.10.9 StatusInfo

The StatusInfo group is specified in 46.2.5.1.

Figure 35-15 specifies the encoding of the value for the StatusInfo TLV.

	Octet	Length
TalkerStatus	1	1
ListenerStatus	2	1
FailureCode	3	1

Figure 35-15—Value of StatusInfo TLV

The semantics of the FailureCode element of the enhanced StatusInfo TLV are the same as the semantics of the Failure Code of the original FailureInformation structure (35.2.2.8.7), and both use the same Table 46-16 for values.

The original Attribute Types (MSRPv0) did not provide a FailureCode from a Listener. The enhanced StatusInfo provides a mechanism for a Listener to provide a FailureCode as status to the Talker.

1 The StatusInfo in the Talker attribute is used to communicate status information to Listeners. The StatusInfo
2 in the Listener attribute is used to communicate status information to the Talker. In SRP, status flows in one
3 direction. Unless stated otherwise in these specifications, elements of StatusInfo are not copied from a
4 Talker attribute to a Listener attribute, or vice versa.

5 A StatusInfo TLV for Declaration Type (35.2.1.3) of Talker Advertise shall use a TalkerStatus of Ready.

6 A StatusInfo TLV for Declaration Type of Talker Failed shall use a TalkerStatus of Failed.

7 A StatusInfo TLV for Declaration Type of Listener Asking Failed shall use a ListenerStatus of Failed.

8 A StatusInfo TLV for Declaration Type of Listener Ready shall use a ListenerStatus of Ready.

9 A StatusInfo TLV for Declaration Type of Listener Ready Failed shall use a ListenerStatus of PartialFailed.

10 For Talker and Listener attributes, the value for the FailureCode element is specified in Table 46-16.

11 Refer to 35.2.4.4.6 for information on merging nonzero FailureCode values from multiple Listeners.

12 NOTE—MSRPv1 adds new values for FailureCode (Table 46-16) that did not exist in MSRPv0 (i.e., 20 and higher).

13 MSRPv0 participants that register Talker Failed with these values are expected to handle the failure without complete
14 understanding of the cause.

15 **35.2.2.10.10 AccumulatedLatency**

16 The AccumulatedLatency group is specified in 46.2.5.2.

17 Figure 35-16 specifies the encoding of the value for the AccumulatedLatency TLV.

	Octet	Length
AccumulatedLatency	1	4

Figure 35-16—Value of AccumulatedLatency TLV

18 The semantics of the enhanced AccumulatedLatency TLV are the same as the semantics of the original
19 Accumulated Latency structure (35.2.2.8.6).

20 The original Attribute Types (MSRPv0) failed the Stream when Accumulated Latency increased beyond its
21 initial value. In addition to supporting that mechanism by default, the enhanced AccumulatedLatency
22 (MSRPv1) provides an optional mechanism to fail the Stream when AccumulatedLatency exceeds specified
23 UserToNetworkRequirements for MaxLatency (35.2.2.10.7).

24 **35.2.2.10.11 InterfaceConfiguration**

25 The InterfaceConfiguration group is specified in 46.2.5.3.

26 The encoding of the value for the InterfaceConfiguration TLV consists of one or more interface
27 configurations. Each interface configuration consists of a single InterfaceID TLV (35.2.2.10.4), followed by
28 a list of configuration values for that interface.

29 The values in each InterfaceID shall match one of the InterfaceID entries in the Talker/Listener
30 EndStationInterfaces group.

31 The list of configuration values uses zero or more of the following TLVs:

- 32 — IEEE802-MacAddresses TLV (35.2.2.10.5)
- 33 — IEEE802-VlanTag TLV (35.2.2.10.5)
- 34 — IPv4-tuple TLV (35.2.2.10.5)
- 35 — IPv6-tuple TLV (35.2.2.10.5)
- 36 — TimeAwareOffset TLV with a value as specified in Figure 35-17

Group length: 4		Octet	Length
	TimeAwareOffset	1	4

Figure 35-17—Value of TimeAwareOffset TLV

InterfaceConfiguration does not exist in the original Attribute Types (MSRPv0). InterfaceConfiguration provides an optional mechanism for stream transformation as specified in 35.2.2.10.5.

The InterfaceConfiguration TLV is used only for negotiation between an end station and the neighboring MRP Participant on the link. The InterfaceConfiguration TLV shall be removed from the MSRP FirstValue prior to propagation with MAP (35.2.4).

35.2.2.10.12 FailedInterfaces

The FailedInterfaces group is specified in 46.2.5.4.

The FailedInterfaces TLV may be contained within the Talker or Listener attribute. When a failure occurs in network configuration (i.e., nonzero FailureCode in StatusInfo TLV), FailedInterfaces provides a list of one or more physical interfaces (distinct points of attachment) in the failed Bridge or end station. Each entry in the list is sufficient to locate the interface in the physical topology.

The Value of the FailedInterfaces TLV consists of a sequence of zero or more TLV entries, and each TLV entry uses the InterfaceID TLV specified in 35.2.2.10.4.

The semantics of the enhanced FailedInterfaces TLV is similar to the system identifier of the original FailureInformation structure (35.2.2.8.7), but with enhanced ability to identify more than one failed interface.

The FailedInterfaces TLV is associated with the single FailureCode in the StatusInfo TLV, and it is not used to communicate multiple failures.

Elements of FailedInterfaces are not copied from a Talker attribute to a Listener attribute, or vice versa.

35.2.3 Provision and support of Stream registration service

35.2.3.1 Initiating MSRP registration and deregistration

MSRP utilizes the following five declaration types (35.2.1.3) to communicate: Talker Advertise, Talker Failed, Listener Ready, Listener Ready Failed, and Listener Asking Failed.

When a Port's *neighborProtocolVersion* is 0x00, declarations shall use only the original Attribute Types (35.2.2.8). When a Port's *neighborProtocolVersion* is 0x01 or higher, declarations shall use either the original Attributes Types or the enhanced Attributes Types (35.2.2.10).

An end station behaving as a Talker will send a Talker Advertise declaration to inform the network about the characteristics of the Stream it can provide. Bridges register this declaration, update some of the information contained within the Talker declaration, and forward it out the nonblocked (35.1.3.1) Ports on the Bridge. If *talkerPruning* is enabled and the Bridge has the Stream's destination_address registered on one or more Ports (e.g., using MMRP) it shall only forward the declarations out those Ports. Eventually the Talker declaration will be registered by other end stations.

1 If *talkerPruning* is enabled and the Stream's *destination_address* is not registered on any Ports the Talker
2 declaration shall not be forwarded (35.2.4.3.2). When the *talkerPruning* parameter is disabled (false) for the
3 Bridge, each Port provides a *talkerPruningPerPort* parameter to control MAC address pruning for that Port
4 (35.2.4.3.3).

5 If *talkerVlanPruning* is enabled (35.2.4.3.4), the Bridge will limit the Talker declarations to Ports that have
6 the Stream's VLAN identifier [item b) in 35.2.2.8.3] registered as a member in the VLAN Registration
7 Entries (8.8).

8 An end station behaving as a Listener will receive the Talker Advertise declaration and register it. If the
9 Listener is interested in receiving that Stream it will send a Listener Ready declaration back toward the
10 Talker. The Bridge's MSRP MAP function will use the StreamID (35.2.2.8.2, 35.2.2.10.2) to associate the
11 Listener Ready with the Talker Advertise and forward the Listener Ready declaration only on the Port that
12 registered the Talker Advertise. This is referred to as Listener Pruning since the declarations are not
13 forwarded out any other Ports. The Bridge will also reserve the required bandwidth and configure its queues
14 and the FDB.

15 If any Bridge along the path from Talker to Listener does not have sufficient bandwidth or resources
16 available its MSRP MAP function will change the Talker Advertise declaration to a Talker Failed
17 declaration before forwarding it. Similarly a Listener Ready declaration will be changed to a Listener
18 Asking Failed declaration if there is not sufficient bandwidth or resources available. This way both the
19 Talker and Listener will know whether the reservation was successful.

20 In the case where there is a Talker attribute and Listener attribute(s) registered within a Bridge for a
21 StreamID and a MAD_Leave.request is received for the Talker attribute, the Bridge shall act as a proxy for
22 the Listener(s) and automatically generate a MAD_Leave.request back toward the Talker for those Listener
23 attributes. This is a special case of the behavior described in 35.2.4.4.1.

24 Finally, there is the Listener Ready Failed declaration. This is used when there are two or more Listeners for
25 a Stream. To simplify the explanation assume there are only two Listeners and one has sufficient bandwidth
26 back to the Talker (signified by a Listener Ready), and the other does not (signified by a Listener Asking
27 Failed). At some point in the network topology there will be a single Bridge that will receive the Listener
28 Ready on one port from the first Listener and the Listener Asking Failed on another port from the second
29 Listener. That Bridge's MSRP MAP function will merge (35.2.4.4.3) those two declarations into a single
30 Listener Ready Failed declaration that will be forwarded to the Talker. When the Talker receives the Listener
31 Ready Failed it will know there are one or more Listeners that want the Stream and can receive it, and there
32 are one or more Listeners that want the Stream but have insufficient bandwidth or resources somewhere
33 along the path to receive it. The Talker may still send the Stream, but it will realize that not all Listeners are
34 going to receive it.

35 MSRP provides a set of Service Primitives that control the declarations of the attributes defined previously.
36 The primitives associated with the Talker application entities are summarized in Table 35-8. Listener
37 application entities have a corresponding set of primitives summarized in Table 35-9.

Table 35-8—Summary of Talker primitives

Name	Request	Indication
REGISTER_STREAM	35.2.3.1.1	—
DEREGISTER_STREAM	35.2.3.1.3	—
REGISTER_ATTACH	—	35.2.3.1.6
DEREGISTER_ATTACH	—	35.2.3.1.8

Table 35-9—Summary of Listener primitives

Name	Request	Indication
REGISTER_STREAM	—	35.2.3.1.2
DEREGISTER_STREAM	—	35.2.3.1.4
REGISTER_ATTACH	35.2.3.1.5	—
DEREGISTER_ATTACH	35.2.3.1.7	—

1 **35.2.3.1.1 REGISTER_STREAM.request**

2 A Talker application entity shall issue a REGISTER_STREAM.request to the MSRP Participant to initiate
3 the advertisement of an available Stream.

4 A Talker may choose to enter a nonzero value in the Accumulated Latency that indicates the amount of
5 latency in nanoseconds that a Stream will encounter before being passed to the MAC Service interface.

6 On receipt of a REGISTER_STREAM.request the MSRP Participant shall issue a MAD_Join.request
7 service primitive (10.2, 10.3). The attribute_type (10.2) parameter of the request shall carry the appropriate
8 Talker Attribute Type (35.2.2.4), depending on the Declaration Type and *neighborProtocolVersion*. The
9 attribute_value (10.2) parameter shall carry the values from the REGISTER_STREAM.request primitive.

```

10 REGISTER_STREAM.request (
11     StreamID,
12     Declaration Type,
13     DataFrameParameters,
14     TSpec,
15     Data Frame Priority,
16     Rank,
17     Accumulated Latency,
18     FailureInformation
19 )

```

1 **35.2.3.1.2 REGISTER_STREAM.indication**

2 A REGISTER_STREAM.indication notifies the Listener application entity that the referenced Stream is
3 being advertised by a Talker somewhere on the attached network.

4 On receipt of a MAD_Join.indication service primitive (10.2, 10.3) with an attribute_type of Talker
5 Advertise, Talker Failed, or Talker Enhanced (35.2.2.4), the MSRP application shall issue a
6 REGISTER_STREAM.indication to the Listener application entity. The REGISTER_STREAM.indication
7 shall carry the values from the attribute_value parameter.

```
8     REGISTER_STREAM.indication (
9         StreamID,
10        Declaration Type,
11        DataFrameParameters,
12        TSpec,
13        Data Frame Priority,
14        Rank,
15        Accumulated Latency,
16        FailureInformation
17    )
```

18 **35.2.3.1.3 DEREGISTER_STREAM.request**

19 A Talker application entity shall issue a DEREGISTER_STREAM.request to the MSRP Participant to
20 remove the Talker's advertisement declaration, and thus remove the advertisement of a Stream, from the
21 network.

22 On receipt of a DEREGISTER_STREAM.request the MSRP Participant shall issue a MAD_Leave.request
23 service primitive (10.2, 10.3) with the attribute_type set to the Declaration Type currently associated with
24 the StreamID. The attribute_value parameter shall carry the StreamID and other values that were in the
25 associated REGISTER_STREAM.request primitive.

```
26     DEREGISTER_STREAM.request (
27         StreamID
28     )
```

29 **35.2.3.1.4 DEREGISTER_STREAM.indication**

30 A DEREGISTER_STREAM.indication notifies the Listener application entity that the referenced Stream is
31 no longer being advertised by a Talker.

32 On receipt of a MAD_Leave.indication service primitive (10.2, 10.3) with an attribute_type of Talker
33 Advertise, Talker Failed, or Talker Enhanced (35.2.2.4), the MSRP application shall issue a
34 DEREGISTER_STREAM.indication to the Listener application entity.

```
35     DEREGISTER_STREAM.indication (
36         StreamID
37     )
```

38 **35.2.3.1.5 REGISTER_ATTACH.request**

39 A Listener application entity shall issue a REGISTER_ATTACH.request to the MSRP Participant to request
40 attachment to the referenced Stream.

1 On receipt of a REGISTER_ATTACH.request the MSRP Participant shall issue a MAD_Join.request
2 service primitive (10.2, 10.3). The attribute_type parameter of the request shall carry the appropriate
3 Listener Attribute Type (35.2.2.4), depending on *neighborProtocolVersion*. The attribute_value shall contain
4 the StreamID and the Declaration Type.

```
5 REGISTER_ATTACH.request (
6     StreamID,
7     Declaration Type
8 )
```

9 **35.2.3.1.6 REGISTER_ATTACH.indication**

10 A REGISTER_ATTACH.indication notifies the Talker application entity that the referenced Stream is being
11 requested by one or more Listeners.

12 On receipt of a MAD_Join.indication service primitive (10.2, 10.3) with an attribute_type of Listener
13 (35.2.2.4), the MSRP application shall issue a REGISTER_ATTACH.indication to the Talker application
14 entity. The REGISTER_ATTACH.indication shall carry the values from the attribute_value parameter.

```
15 REGISTER_ATTACH.indication (
16     StreamID,
17     Declaration Type
18 )
```

19 **35.2.3.1.7 DEREGISTER_ATTACH.request**

20 A Listener application entity shall issue a DEREGISTER_ATTACH.request to the MSRP Participant to
21 remove the request to attach to the referenced Stream.

22 On receipt of a DEREGISTER_ATTACH.request the MSRP Participant shall issue a MAD_Leave.request
23 service primitive (10.2, 10.3) with the attribute_type set to the appropriate Listener Attribute Type
24 (35.2.2.4). The attribute_value parameter shall carry the StreamID and the Declaration Type currently
25 associated with the StreamID.

```
26 DEREGISTER_ATTACH.request (
27     StreamID
28 )
```

29 **35.2.3.1.8 DEREGISTER_ATTACH.indication**

30 A DEREGISTER_ATTACH.indication notifies the Talker application entity that the referenced Stream is no
31 longer being requested by any Listeners.

32 On receipt of a MAD_Leave.indication service primitive (10.2, 10.3) with an attribute_type of Listener
33 (35.2.2.4), the MSRP application shall issue a DEREGISTER_ATTACH.indication to the Talker application
34 entity. The DEREGISTER_ATTACH.indication shall contain the StreamID.

```
35 DEREGISTER_ATTACH.indication (
36     StreamID
37 )
```

1 35.2.4 MSRP Attribute Propagation

2 There is no MSRP MAP function for Domain attributes. MSRP simply declares the characteristics of the SR
3 classes that are supported on the Bridge Port regardless of what has been learned from Domain registrations
4 on other Bridge Ports. Registration of the neighbor's Domain attribute determines the value of the
5 *SRPDomainBoundaryPort* and *neighborProtocolVersion* variables (35.2.1.4).

6 This clause describes the following:

- 7 a) Rules for combining and propagating Listener attributes toward the associated Talker
- 8 b) How MSRP adjusts the Talker and Listener attributes before propagating them
- 9 c) How MSRP translates Attribute Types when propagating from a ProtocolVersion (35.2.2.1) on a
10 registering Port to a different ProtocolVersion on declaring Ports.

11 Unless stated otherwise, Talker and Listener attributes are propagated as described in 10.3.

12 In principle, the MAP performs MSRP Attribute Propagation when any of the following conditions occur:

- 13 d) A *MAD_Join.indication* adds a new attribute to MAD (with the *new* parameter, 10.2, set to TRUE).
- 14 e) A *MAD_Leave.indication* is issued by the MAD.
- 15 f) An internal application declaration or withdrawal is made in a station.
- 16 g) When the bandwidth of the underlying media changes (see *bandwidthAvailabilityChanged*
17 notification in 34.3.3).
- 18 h) A Port becomes an SRP domain core port (3.270).
- 19 i) If *talkerPruning* (35.2.4.3.2) or *talkerPruningPerPort* (35.2.4.3.3) is enabled and there is a change in
20 the MAC Address Registration Entries (8.8.4) of a Port.
- 21 j) If *talkerVlanPruning* (35.2.4.3.4) is enabled and there is a change in the VLAN Registration Entries
22 on a Port (8.8).
- 23 k) A Port changes its *neighborProtocolVersion* parameter (35.2.1.4(j)).

24 The MSRP MAP function is responsible for adjusting and propagating Talker and Listener attributes
25 throughout the bridged network. It also updates the Dynamic Reservation Entries (8.8.7) to specify which
26 Streams shall be filtered and which shall be forwarded, along with updating the associated *streamAge*
27 [item c) in 35.2.1.4]. The bandwidth associated with those Streams is reported to the queuing algorithms via
28 the *operIdleSlope(N)* parameter [item d) in 34.3] on a per-port per-traffic class basis.

29 Streams of higher importance are given available bandwidth before streams of lower importance.

30 If insufficient bandwidth or resources are available the streams *destination_address* will be filtered and the
31 failure will be noted in the Talker and Listener attributes declared from that Bridge.

32 A port shall only forward MSRP declarations for SR classes it supports. This will eliminate unnecessary
33 priority remapping for traffic related to unsupported SR classes.

34 The following subclauses describe what the MSRP MAP function shall accomplish.

35 35.2.4.1 Stream importance

36 MSRP utilizes the stream Rank [item b) in 35.2.2.8.5, 35.2.2.10.3] to decide which stream is more important
37 than another.

1 In the case where two streams have the same Rank, the *streamAge* will be compared. If the Ranks are
2 identical and the *streamAges* are identical the StreamIDs (35.2.2.8.2, 35.2.2.10.2) will be compared and the
3 numerically lower StreamID is considered to be more important.

4 **35.2.4.2 Stream bandwidth calculations**

5 As referenced in Table 35-5, the bandwidth requirements of a Stream include more than just the amount
6 specified in the REGISTER_STREAM.request (35.2.3.1.1). SRP shall add the *perFrameOverhead* (34.4)
7 associated with the media attached to the port.

8 If this port is on a Shared Media (35.1.1) the bandwidth requirements may need to be further increased. For
9 example, a Stream transmitted from one station to another may have to be sent across the media twice. One
10 time from the Talker station to the DMN, and a second time from the DMN to the Listener station. In this
11 case the bandwidth requirements would need to be doubled.

12 The *totalFrameSize* for a stream on an outbound Port is therefore the sum of the following three amounts
13 (doubled if each frame is transmitted on the media twice):

- 14 a) *MaxFrameSize* [item a) in 35.2.2.8.4].
- 15 b) *perFrameOverhead* (34.4) associated with the media attached to the port.
- 16 c) One (1) additional octet to account for slight differences (up to 200 ppm) in the class measurement
17 interval between neighboring devices.

18 Multiply *totalFrameSize* by *MaxIntervalFrames* [item b) in 35.2.2.8.4] to arrive at the associated bandwidth
19 (in bits per second), which is then used to update *operIdleSlope(N)* as shown in Table 35-14.

20 Streams that are in the Listener Ready or Listener Ready Failed state reduce the amount of bandwidth
21 available to other Streams. Streams that have no Listeners, or the Listeners are in the Asking Failed state, do
22 not reduce available bandwidth for other Streams since the Stream data will not be flowing through this
23 outbound Port. These details are considered when calculating how many Streams can flow through a
24 particular Port. The total amount of bandwidth available to a particular traffic class on a Port is represented
25 by *deltaBandwidth(N)* [item b) in 34.3]. Subclauses 34.3.1 and 34.3.2 describe the relationship of available
26 bandwidth between traffic classes.

27 **35.2.4.3 Talker attribute propagation**

28 Table 35-10 describes the propagation of Talker attributes for a StreamID from one port of a Bridge to
29 another. If no Talker attributes are registered for a StreamID then no Talker attributes for that StreamID will
30 be declared on any other port of the Bridge. If a Talker Failed is registered then it will be propagated as a
31 Talker Failed out all other nonblocked (35.1.3.1) ports on the Bridge.

32 Talker Advertise registrations require further processing by the MAP function. MAP will analyze available
33 bandwidth and other factors to determine if the outbound port has enough resources available to support the
34 Stream. MAP will also verify *msrpMaxFanInPorts*, if nonzero, will not be exceeded. If there are sufficient
35 resources and bandwidth available MAP will declare a Talker Advertise. Otherwise, MAP will declare a
36 Talker Failed on the outbound port and add appropriate FailureInformation.

37 If the outbound Port is an SRP domain boundary port for a given SR class, then for that SR class, the MAP
38 function shall operate such that any Talker Advertise declaration for that SR class that would otherwise
39 propagate through that Port is converted to a Talker Failed declaration with a failure code of 8 (Table 46-16),
40 meaning “Egress Port is not AVB-capable.” The consequence of this behavior is that stream reservations can
41 be established only in circumstances where the Talker and the Listener(s) for the stream are located within
42 the same SRP domain.

Table 35-10—Talker attribute propagation per port

		Talker		
		(none)	Advertise	Failed
Listener	(none)	(none)	Talker Advertise or Talker Failed	Talker Failed
	Ready	(none)	Talker Advertise or Talker Failed	Talker Failed
	Ready Failed	(none)	Talker Advertise or Talker Failed	Talker Failed
	Asking Failed	(none)	Talker Advertise or Talker Failed	Talker Failed

1 35.2.4.3.1 ProtocolVersion Translation

2 Ports with *neighborProtocolVersion* [item j) in 35.2.1.4] of 0x00 shall use only the original Attribute Types
3 (35.2.2.4) of Talker Advertise and Talker Failed. Ports with *neighborProtocolVersion* of 0x01 or higher shall
4 use either the original Attribute Types or the enhanced Attribute Type of Talker Enhanced.

5 In order to propagate an enhanced Attribute Type (e.g., registered from a Port with *neighborProtocolVersion*
6 of 0x01) to an original Attribute Type (e.g., declared to a Port with *neighborProtocolVersion* of 0x00), fields
7 of the attribute's FirstValue shall be translated as specified in Table 35-11.

8 NOTE—Original Attribute Types always propagate without translation, even if *neighborProtocolVersion* is 0x01 or
9 higher.

Table 35-11—Translation of Talker attributes

Enhanced field	Original field	Enhanced to original translation
StreamID, MacAddress	StreamID, MAC Address	No change: Copy the field's contents from enhanced to original without change.
StreamID, UniqueID	StreamID, Unique ID	No change: Copy the field's contents from enhanced to original without change.
StreamRank, Rank	PriorityAndRank, Rank	No change: Copy the field's contents from enhanced to original without change.
EndStationInterfaces	—	Not propagated: This TLV is not propagated through MAP, so translation is not applicable.
DataFrameSpecification, IEEE802-MacAddresses, DestinationMacAddress	DataFrameParameters, destination_address	No change: Copy the field's contents from enhanced to original without change.
DataFrameSpecification, IEEE802-MacAddresses, SourceMacAddress	—	Lost: The declaration type does not change (i.e., fail), because MSRPv0 does not use the source MAC address for Stream identification.
DataFrameSpecification, IEEE802-VlanTag, PriorityCodePoint	PriorityAndRank, Data Frame Priority	No change: Copy the field's contents from enhanced to original without change.

Table 35-11—Translation of Talker attributes (continued)

Enhanced field	Original field	Enhanced to original translation
DataFrameSpecification, IEEE802-VlanTag, VlanId	DataFrameParameters, vlan_identifier	No change: Copy the field's contents from enhanced to original without change.
TrafficSpecification, Interval	—	Lost: If this field does not match the default classMeasurementInterval (34.3) of the stream's SR class, a Talker Advertise declaration is converted to a Talker Failed declaration with a failure code of 20.
TrafficSpecification, MaxFramesPerInterval	TSpec, MaxIntervalFrames	No change: Copy the field's contents from enhanced to original without change.
TrafficSpecification, MaxFrameSize	TSpec, MaxFrameSize	No change: Copy the field's contents from enhanced to original without change.
TrafficSpecification, TransmissionSelection	—	Lost: If this field is not one, a Talker Advertise declaration is converted to a Talker Failed declaration with a failure code of 20.
TrafficSpecification, TSpecTimeAware TLV	—	Lost: If this TLV is present, a Talker Advertise declaration is converted to a Talker Failed declaration with a failure code of 20.
UserToNetworkRequirements, NumSeamlessTrees	—	Lost: If this field is not one, a Talker Advertise declaration is converted to a Talker Failed declaration with a failure code of 20.
UserToNetworkRequirements, MaxLatency	—	Lost: If this field is not zero, a Talker Advertise declaration is converted to a Talker Failed declaration with a failure code of 20.
InterfaceCapabilities	—	Not propagated: This TLV is not propagated through MAP, so translation is not applicable.
StatusInfo, TalkerStatus	—	No change: This field determines the original Attribute Type to declare (Talker Advertise or Talker Failed).
StatusInfo, ListenerStatus	—	No change: This field is set according to the ListenerStatus that the Bridge last propagated in the Listener attribute of the Stream. If no Listener attribute has been propagated, this field is set to zero (None).
StatusInfo, FailureCode	FailureInformation, Failure Code	For Talker Failed, this field has no change (i.e., copy from enhanced to original). For Talker Advertise, this field is lost, but the declaration type does not change (i.e., fail).
AccumulatedLatency	AccumulatedLatency, AL	No change: Copy the field's contents from enhanced to original without change.
InterfaceConfiguration	—	Not propagated: This TLV is not propagated through MAP, so translation is not applicable.

Table 35-11—Translation of Talker attributes (continued)

Enhanced field	Original field	Enhanced to original translation
FailedInterfaces, MacAddress	FailureInformation, system identifier	No change: This field applies to Talker Failed only. The most significant 16 bits of system identifier are set to zero, and the MacAddress is copied to the least significant 48 bits of system identifier.
FailedInterfaces, InterfaceName	FailureInformation	Lost: This field applies to Talker Failed only.

35.2.4.3.2 Talker Pruning

The *talkerPruning* parameter (35.2.1.4) is disabled by default, and Talker declarations are sent out all nonblocked Ports. If *talkerPruning* is enabled and the destination_address [item a) in 35.2.2.8.3] of the Stream is found in the MAC Address Registration Entries (8.8.4) for the Port, the declaration shall be forwarded. If the destination_address is not found the declaration shall be blocked and no Talker declaration of any type shall be forwarded.

35.2.4.3.3 Talker Pruning Per Port

When the *talkerPruning* parameter is disabled (false) for the Bridge, each Port may provide a *talkerPruningPerPort* parameter (35.2.1.4) to control MAC address pruning for that Port.

When the *talkerPruningPerPort* feature is provided, all *talkerPruningPerPort* parameters are disabled by default.

When *talkerPruning* is enabled (true), the *talkerPruningPerPort* parameters are ignored.

Each *talkerPruningPerPort* parameter controls MAC address pruning for that Port when *talkerPruning* is disabled (false). If *talkerPruningPerPort* is disabled for the Port, Talker declarations are forwarded on that Port regardless of the destination_address [item a) in 35.2.2.8.3] of the Stream. If *talkerPruningPerPort* is enabled for the Port and the destination_address of the Stream is found in the MAC Address Registration Entries (8.8.4) for the Port, the declaration shall be forwarded. If *talkerPruningPerPort* is enabled for the Port and the destination_address of the Stream is not found in the MAC Address Registration Entries for the Port, the declaration shall be blocked, and no Talker declaration of any type shall be forwarded.

35.2.4.3.4 Talker VLAN Pruning

The *talkerVlanPruning* parameter (35.2.1.4) may be provided on the Bridge in order to limit the Talker declarations to Ports that have the Stream's vlan_identifier [item b) in 35.2.2.8.3] registered as a member in the VLAN Registration Entries (8.8).

When the *talkerVlanPruning* feature is provided, it is disabled (false) by default.

When *talkerVlanPruning* is disabled (false), the MAP context for MSRP propagates Talker declarations according to the VLAN spanning tree (35.1.3.1), regardless of the registration of the Stream's vlan_identifier in the Filtering Database (8.8.10). This allows potential Listeners to receive Talker declarations prior to registering the VLAN with MVRP.

1 When *talkerVlanPruning* is enabled (true), in addition to propagating along the VLAN spanning tree, the
2 MAP context for MSRP filters propagation according to the registration of the Stream's *vlan_identifier* in
3 the Filtering Database (8.8.10). If *talkerVlanPruning* is enabled and registration of the Stream's
4 *vlan_identifier* on a Port is not member (Registration Forbidden, Not Registered, or No Dynamic
5 Registration Entry Present), MSRP shall not propagate the declaration on that Port. This propagation is
6 effectively the same as propagation for the Stream's data frames. Potential Listeners can join the VLAN
7 dynamically with MVRP in order to receive Talker declarations for that *vlan_identifier* (35.1.2.1).

8 The *talkerVlanPruning* parameter filters by VLAN, and it operates independently from the parameters that
9 filter by MAC address (*talkerPruning* and *talkerPruningPerPort*).

10 35.2.4.4 Listener attribute propagation

11 Listener Attributes, unlike Talker Attributes, can be merged (35.2.4.4.3) from several Listeners on different
12 ports into a single Listener declaration. There are two steps involved:

- 13 a) Processing of incoming Listener attribute registration on a port based upon the status of the
14 associated Talker attribute registration,
- 15 b) Merging all the individual ports Listener attributes gathered in step 1, above, into a single Listener
16 attribute to be declared on the nonblocked (35.1.3.1) outbound port which has the associated Talker
17 registration. If no Talker attribute is registered within the Bridge for the StreamID associated with
18 the Listener, the Listener attribute will not be propagated.

19 35.2.4.4.1 Incoming Listener attribute processing

20 Table 35-12 describes how Listener attributes are propagated from the incoming ports.

Table 35-12—Incoming Listener attribute propagation per port

		Talker		
		(none)	Advertise	Failed
Listener	(none)	(none)	(none)	(none)
	Ready	(none)	Listener Ready	Listener Asking Failed
	Ready Failed	(none)	Listener Ready Failed	Listener Asking Failed
	Asking Failed	(none)	Listener Asking Failed	Listener Asking Failed

21 If no Listener attributes are registered on a particular port then no Listener attribute will be propagated to the
22 Listener attribute merging (35.2.4.4.3) from that port.

23 If a Listener Asking Failed is registered on a port then it will be propagated as a Listener Asking Failed and
24 merged with other Listener Attributes from other ports.

25 If no Talker attributes are associated with the Listener attribute, the Listener attribute will not be propagated.

26 If a Talker Failed is registered on another port for the associated Listener Ready or Listener Ready Failed
27 then a Listener Asking Failed will be propagated and merged with other Listener attributes from other ports.

28 If a Talker Advertise is registered on another port for the StreamID associated with a Listener Ready or
29 Listener Ready Failed then that Listener attribute will be forwarded as-is and merged with Listener
30 Attributes from other ports.

35.2.4.4.2 Updating Queuing and Forwarding information

- When Incoming Listener attribute processing (35.2.4.4.1) has been completed for a port the Dynamic Reservation Entries (8.8.7) shall be updated as shown in Table 35-13.

Table 35-13—Updating Dynamic Reservation Entries

		Talker		
		(none)	Advertise	Failed
Listener	(none)	(no entry)	Filtering	Filtering
	Ready	Filtering	Forwarding	Filtering
	Ready Failed	Filtering	Forwarding	Filtering
	Asking Failed	Filtering	Filtering	Filtering

- The *operIdleSlope(N)* [item d) in 34.3] shall be updated as shown in Table 35-14, dependent on the change to the Dynamic Reservation Entries. MSRP MAP processing can occur at any time (35.2.4), resulting in a change to which Streams are filtered and to which streams are forwarded. These changes in bandwidth requirements shall be reflected in the *operIdleSlope(N)* variable. Streams that have had their bandwidth removed shall decrease *operIdleSlope(N)*. Streams that have just been allocated bandwidth shall increase *operIdleSlope(N)*.

Table 35-14—Updating *operIdleSlope(N)*

		Dynamic Reservation Entries prior to MSRP MAP running		
		(none)	Filtering	Forwarding
Dynamic Reservation Entries after MSRP MAP running	(none)	(no change)	(no change)	Decrease <i>operIdleSlope(N)</i>
	Filtering	(no change)	(no change)	Decrease <i>operIdleSlope(N)</i>
	Forwarding	Increase <i>operIdleSlope(N)</i>	Increase <i>operIdleSlope(N)</i>	(no change)

- NOTE—The order of operations is important when updating the Dynamic Reservation Entries and the *operIdleSlope(N)* in order not to allow any associated streaming frames to be dropped. If the bandwidth utilization of a port is going to be increased (i.e., a Stream is going to be forwarded) the *operIdleSlope(N)* is updated before the Dynamic Reservation Entries. If the bandwidth utilization of a port is going to be decreased (i.e., a Stream is going to be filtered) the Dynamic Reservation Entries are updated before the *operIdleSlope(N)*.

35.2.4.4.3 Merge Listener Declarations

Listener Registrations with the same StreamID shall be merged into a single Listener Declaration that will be declared on the port with the associated Talker registration. If no such Talker registration exists the Listener attribute is not declared. When this Declaration is sent,

- a) The Direction (35.2.1.2) is Listener.
- b) The Declaration Type (35.2.1.3) is determined according to Table 35-15.
- c) The StreamID (35.2.2.8.2) is that of the Listener Registrations.

Table 35-15—Listener Declaration Type Summation

First Declaration Type	Second Declaration Type	Resultant Declaration Type
Ready	none or Ready	Ready
	Ready Failed or Asking Failed	Ready Failed
Ready Failed	any	Ready Failed
Asking Failed	Ready or Ready Failed	Ready Failed
	none or Asking Failed	Asking Failed

35.2.4.4.4 ProtocolVersion Translation

Ports with *neighborProtocolVersion* [item j) in 35.2.1.4] of 0x00 shall use only the original Attribute Type (35.2.2.4) of Listener. Ports with *neighborProtocolVersion* of 0x01 or higher shall use either the original Attribute Type or the enhanced Attribute Type of Listener Enhanced.

In order to propagate an enhanced Attribute Type (e.g., registered from a Port with *neighborProtocolVersion* of 0x01) to an original Attribute Type (e.g., declared to a Port with *neighborProtocolVersion* of 0x00), fields of the attribute's FirstValue shall be translated as specified in Table 35-16.

NOTE—Original Attribute Types always propagate without translation, even if *neighborProtocolVersion* is 0x01 or higher.

35.2.4.4.5 Merge MaxLatency

When more than one Listener provides the UserToNetworkRequirements TLV, multiple values of MaxLatency shall be merged by using the largest MaxLatency value in that set of multiple values. This ensures that Bridges closer to the Talker support the most flexible requirement for any Listener. The requirements of an individual Listener continue to be met in Bridges closer to that Listener.

35.2.4.4.6 Merge Listener Failures

As specified in 35.2.4.4.3, it is possible for a Bridge to merge failed Declaration Types for Listeners (e.g., Ready Failed with Asking Failed). Each failed Declaration Type uses a nonzero FailureCode (35.2.2.10.9) and a FailedInterfaces TLV associated with that FailureCode.

When merging failures, the FailedInterfaces TLV shall always be transferred with its corresponding StatusInfo TLV since the FailureCode is associated with the FailedInterfaces.

Table 35-16—Translation of Listener attributes

Enhanced field	Original field	Enhanced to original translation
StreamID, MacAddress	StreamID, MAC Address	No change: This field determines the original Attribute Type to declare (Talker Advertise or Talker Failed).
StreamID, UniqueID	StreamID, Unique ID	No change: This field determines the original Attribute Type to declare (Talker Advertise or Talker Failed).
EndStationInterfaces	—	Not propagated: This TLV is not propagated through MAP, so translation is not applicable.
UserToNetworkRequirements, NumSeamlessTrees	—	Lost: If this field is not one, a Listener Ready or Listener Ready Failed declaration is converted to a Listener Asking Failed declaration.
UserToNetworkRequirements, MaxLatency	—	Lost: If this field is not zero, a Listener Ready or Listener Ready Failed declaration is converted to a Listener Asking Failed declaration.
InterfaceCapabilities	—	Not propagated: This TLV is not propagated through MAP, so translation is not applicable.
StatusInfo, TalkerStatus	—	No change: This field is set according to the TalkerStatus that the Bridge last propagated in the Talker attribute of the Stream. If no Talker attribute has been propagated, this field is set to zero (None).
StatusInfo, ListenerStatus	—	No change: This field determines the original FourPackedType (35.2.2.10) to declare.
StatusInfo, FailureCode	—	Lost: The declaration type does not change (i.e., fail).
InterfaceConfiguration	—	Not propagated: This TLV is not propagated through MAP, so translation is not applicable.

1 When merging a Ready Failed declaration with an Asking Failed declaration, the StatusInfo and
2 FailedInterfaces shall be transferred from the Asking Failed declaration. When merging failed declarations
3 with the same Declaration Type, the StatusInfo and FailedInterfaces shall be transferred from the declaration
4 with the Failure Code of the lowest numeric value. When merging failed declarations with the same
5 Declaration Type and Failure Code, the decision of which failure to transfer is left to the Bridge's
6 implementation.

7 **35.2.4.5 MAP Context for MSRP**

8 MSRPDU's can carry information about Streams in multiple VLANs, which in an MST environment, can be
9 in different spanning tree instances. Queue resources, however, are allocated and used according to priority
10 parameters, not according to VID. Furthermore, on a shared medium, Streams can use the shared medium
11 even on VLANs that are blocked on the Bridge's Port to that shared medium (e.g., consider an IEEE 802.11
12 AP that transmits frames between two stations that are on a VLAN that is blocked on the AP). Therefore,
13 there is a single context for MSRP Attribute Propagation that includes all Bridge Ports. The Declarations are
14 filtered according to the requirements of 35.2.4 and its subclauses and according to the state of the spanning
15 tree per 35.1.3.1.

16 All MSRPDU's are transmitted as untagged frames.

1 **35.2.4.6 MaxLatency Comparison**

2 The comparison for `UserToNetworkRequirements.MaxLatency` greater than zero determines whether to fail
3 the Stream due to `AccumulatedLatency` (35.2.2.8.6, 35.2.2.10.10) that exceeds the requirement specified by
4 Talkers and/or Listeners. The comparison is performed separately for the Talker and each Listener.

5 For the Talker, the `UserToNetworkRequirements.MaxLatency` from the Talker is compared to
6 `AccumulatedLatency` for all Listeners of the Stream. For example, if the Bridge forwards the Stream to three
7 Ports (three Listeners registered), the Talker's `UserToNetworkRequirements.MaxLatency` is compared to
8 three values of `AccumulatedLatency`. For any of the comparisons, if any Listener's `AccumulatedLatency` is
9 greater than the Talker's `UserToNetworkRequirements.MaxLatency`, the Talker fails the `MaxLatency`
10 comparison.

11 If the Talker fails the `MaxLatency` comparison and the Declaration Type of the Talker is `Advertise`, the
12 Bridge shall change Talker `Advertise` to `Talker Failed` and report Failure Code=21 (Table 46-16).

13 For the Listener, the `UserToNetworkRequirements.MaxLatency` from the Listener is compared to
14 `AccumulatedLatency` for the corresponding Listener. If the Listener's `AccumulatedLatency` is greater than
15 the Listener's `UserToNetworkRequirements.MaxLatency`, the Listener fails the `MaxLatency` comparison.

16 If the Listener fails the `MaxLatency` comparison, the Bridge shall change Listener `Ready` to `Listener Asking`
17 `Failed` for the failed Listener and report Failure Code=21. The Listener `Asking Failed` is merged with
18 declarations of other Listeners to determine the resultant Declaration Type to the Talker (35.2.4.4.3).

19 In the fully distributed model (46.1.3.1), the comparison for the Talker's
20 `UserToNetworkRequirements.MaxLatency` is performed in each Bridge in the path from the Talker to each
21 Listener, in the same direction as `AccumulatedLatency`. In the fully distributed model, the comparison for
22 the Listener's `UserToNetworkRequirements.MaxLatency` is performed in each Bridge in the path from the
23 Listener to each Talker, using the `AccumulatedLatency` from the Talker's attribute.

24 In the centralized network/distributed user model (46.1.3.2), the comparison for the Talker's and Listener's
25 `UserToNetworkRequirements.MaxLatency` is performed in the CNC, using the capabilities of each Bridge
26 as learned through remote management (e.g., Bridge Delay of 12.32.1). When the CNC detects failure of the
27 `MaxLatency` comparison, Failure Code 21 is declared to the Talker and Listener as specified in item a) in
28 35.2.2.10.

29 **35.2.5 Operational reporting and statistics**

30 **35.2.5.1 Dropped Stream Frame Counter**

31 An implementation may support the ability to maintain a per-port per-traffic class count of data stream
32 frames that are dropped for any reason. These are not MRP frames, but the data stream frames that flow
33 between Talker and Listener(s) through the reservations established by MSRP.

34 **35.2.6 Encoding**

35 If an MSRP message is received from a Port with an event value specifying the `JoinIn` or `JoinMt` message,
36 and if the `StreamID` (35.2.2.8.2, 35.2.2.10.2), and `Direction` (35.2.1.2) all match those of an attribute already
37 registered on that Port, and the `Attribute Type` (35.2.2.4) or `FourPackedEvent` (35.2.2.7.2) has changed, then
38 the Bridge should behave as though an `rLv!` event (with immediate leavetimer expiration in the Registrar
39 state table) was generated for the MAD in the Received MSRP Attribute Declarations before the `rJoinIn!` or
40 `rJoinMt!` event for the attribute in the received message is processed. This allows an Applicant to indicate a

1 change in a stream reservation, e.g., a change from a Talker Failed to a Talker Advertise registration, without
2 having to issue both a withdrawal of the old attribute, and a declaration of the new. A Listener attribute is
3 also updated this way, for example, when changing from a Listener Ready to a Listener Ready Failed.

4 NOTE—This rule ensures that there is at most one Listener Declaration or one Talker Declaration for any given value of
5 StreamID on any given port. In the unlikely situation where a Talker Advertise and a Talker Failed are received for the
6 same Stream on the same port, the Talker Failed declaration takes precedence.

7 The following examples will help clarify the intent of this subclause. The first example describes the
8 behavior when one attribute (Talker Advertise) is replaced by another (Talker Failed). The second example
9 describes the behavior when the MSRP FourPackedEvents changes with a single Listener attribute.

10 This example illustrates processing of Talker Declaration changes. Assume a Bridge is receiving Talker
11 Advertise declarations on an inbound Port. An emergency situation occurs and a 911 call is being placed via
12 an entirely different Stream. The bandwidth that was available for the first Stream is now no longer available
13 so the Bridge begins receiving Talker Failed declarations for that original stream. The MAP function realizes
14 the Talker declaration has changed for the Stream and generates an internal leave event for the Talker
15 Advertise and a join event for the Talker Failed. This behavior guarantees there will not be a declaration for
16 a Talker Advertise and a Talker Failed for a single Stream existing within the Bridge at the same time.

17 As another example assume the same situation has occurred as described in the example above (a 911 call).
18 For this scenario consider the Listener declarations flowing in the opposite direction. When there was
19 bandwidth available the Bridge was declaring a Listener Ready, which was then changed to a Listener
20 Asking Failed as soon as the 911 call came through. The other Bridge receiving these Listener declarations
21 realized that the Listener attribute MSRP FourPackedEvents had changed and acted as if the Listener
22 declaration had been withdrawn and replaced by the updated Listener declaration.

23 **35.2.7 Attribute value support requirements**

24 Implementations of MSRP shall maintain state information for all attribute values that support the Stream
25 registrations (35.2.2.8, 35.2.2.10).

26 Implementations of MSRP shall be capable of supporting any attribute value in the range of possible values
27 that can be registered using Stream registrations; however, the maximum number of attribute values for
28 which the implementation is able to maintain current state information is an implementation decision, and
29 may be different for Talker attributes and Listener attributes. The number of values that the implementation
30 can support shall be stated in the PICS.

36. Priority-based Flow Control (PFC)

This clause specifies the operation of PFC (see 36.1) and the architecture of Priority-based Flow Control in a PFC-aware system (see 36.2).

The models of operation in this clause provide a basis for specifying the externally observable behavior of PFC and are not intended to place additional constraints on implementations; these can adopt any internal model of operation compatible with the externally observable behavior specified.

36.1 PFC operation

36.1.1 Overview

Operation of PFC is limited to a data center environment. PFC enables to not discard frames due to congestion for protocols that require this property. However, PFC can cause congestion spreading behavior; therefore, it is intended for use on networks of limited extent. When PFC is used, deployment of congestion notification (see Clause 30) can reduce the frequency with which PFC is invoked.

PFC is a function defined only for a pair of full duplex MACs (e.g., IEEE 802.3 MACs operating in point-to-point full-duplex mode) connected by one point-to-point link. Use of PFC on shared media such as EPON is out of the scope of this standard. Figure 36-1 shows an example of PFC peering when IEEE 802.3 point-to-point full-duplex MACs are used.

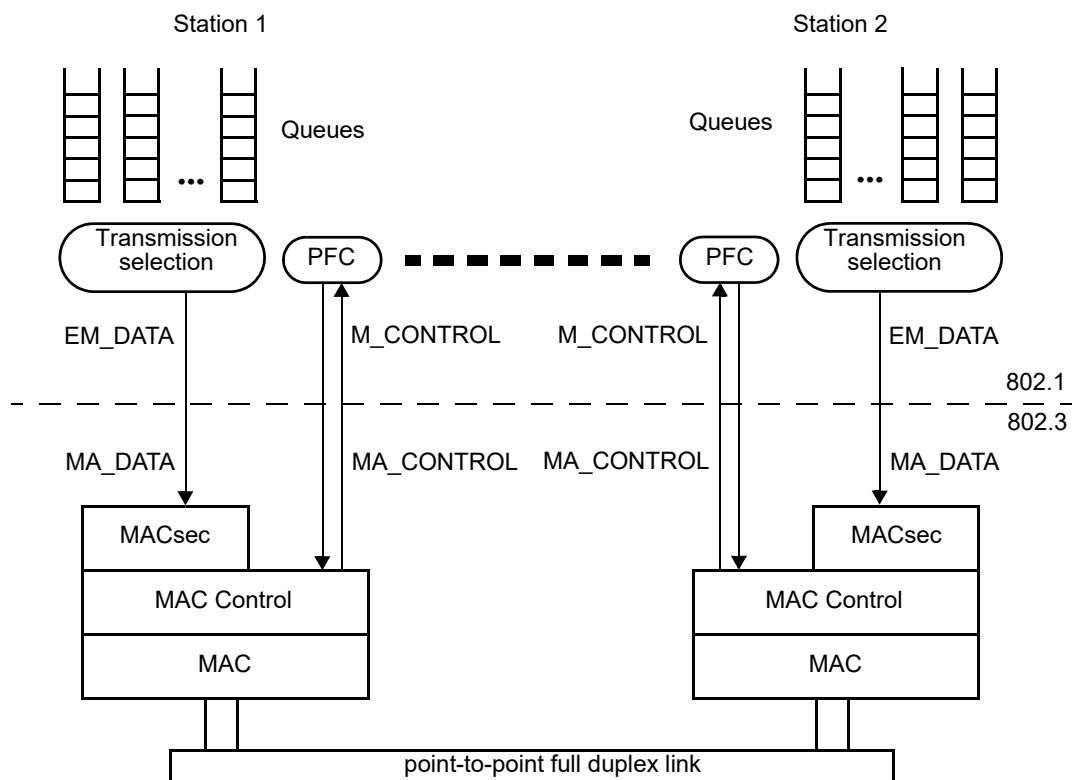


Figure 36-1—PFC peering

PFC allows link flow control to be performed on a per-priority basis. In particular, PFC is used to inhibit transmission of data frames associated with one or more priorities for a specified period of time. PFC can be enabled for some priorities on the link and disabled for others.

1 Given that BPDUs, for example, are sent untagged and can bypass the output queues, it is strongly
2 recommended for the default priority of a port to not have PFC enabled.

3 NOTE—The LLC-SAP of a Bridge Port can host a management protocol stack that uses PFC-enabled priorities, and
4 these management frames can bypass the output queues. In this situation PFC can fail to provide insurance against these
5 frames overflowing the buffer in the remote station of the link.

6 36.1.2 PFC primitives

7 PFC is invoked through the M_CONTROL PFC primitives (11.4 of IEEE Std 802.1AC-2016 [B10]). A
8 system client wishing to inhibit transmission of data frames on certain priorities from the remote system on
9 the link generates an M_CONTROL.request primitive specifying the following:

- 10 a) The globally assigned 48-bit multicast address 01-80-C2-00-00-01.
- 11 b) The PFC opcode (i.e., 01-01).
- 12 c) A request_operand_list with two operands indicating, respectively, the set of priorities addressed
13 and the lengths of time for which it wishes to inhibit data frame transmission of the corresponding
14 priorities.

15 NOTE—By definition, a point-to-point full-duplex link comprises exactly two stations, thus there is no ambiguity
16 regarding the destination station's identity. The use of a well-known multicast address does not require a station to know,
17 and maintain knowledge of, the individual 48-bit address of the other station.

18 As specified in IEEE Std 802.3, when PFC is enabled on a port for at least one priority over an IEEE 802.3
19 link layer, the IEEE Std 802.3 PAUSE mechanism is not used for that port.

20 As a result of the processing of the PFC M_CONTROL.request, the peering PFC station receives a PFC
21 M_CONTROL.indication primitive.

22 The parameters of the PFC M_CONTROL.indication are as follows:

- 23 d) The PFC opcode (i.e., 01-01).
- 24 e) A indication_operand_list with two operands indicating, respectively, the set of priorities addressed
25 and the lengths of time for which data frame transmission of the corresponding priorities has to be
26 inhibited.

27 The request_operand_list of a PFC M_CONTROL.request and the indication_operand_list of a PFC
28 M_CONTROL.indication are composed of the following operands:

- 29 f) priority_enable_vector: a 2-octet field, with the most significant octet being reserved (i.e., set to zero
30 on transmission and ignored on receipt). Each bit of the least significant octet indicates if the
31 corresponding field in the time_vector parameter is valid. The bits of the least significant octet are
32 named e[0] (the LSB) to e[7] (the MSB). Bit e[n] refers to priority n. For each e[n] bit set to one, the
33 corresponding time[n] value is valid. For each e[n] bit set to zero, the corresponding time[n] value is
34 invalid.
- 35 g) time_vector: a list of eight 2-octet fields, named time[0] to time[7]. The eight time[n] values are
36 always present regardless of the value of the corresponding e[n] bit. Each time[n] field is a 2-octet,
37 unsigned integer containing the length of time for which the receiving station is requested to inhibit
38 transmission of data frames associated with priority n. The field is transmitted most significant octet
39 first, and least significant octet second. The time[n] fields are transmitted sequentially, with time[0]
40 transmitted first and time[7] transmitted last. Each time[n] value is measured in units of
41 pause_quanta, equal to the time required to transmit 512 bits of a frame at the data rate of the MAC.
42 Each time[n] field can assume a value in the range of 0 to 65 535 pause_quanta.

36.1.3 Detailed specification of PFC operation

36.1.3.1 Processing PFC M_CONTROL.requests

Invoking the PFC M_CONTROL.request results in the invocation of the appropriate link layer service request. For IEEE 802.3 link layers the PFC M_CONTROL.request is mapped to a PFC MA_CONTROL.request (IEEE Std 802.1AC). If PFC is not enabled for priority n , then PFC requests with $e[n]$ set to one and $time[n]$ different than zero (see 36.1.2) should not be generated.

NOTE—In the IEEE 802.1Q architecture frames coming from the LLC, including BPDUs, bypass the priority queues and therefore are not subject to PFC. However, in some implementations frames coming from the LLC can pass through the priority queues. In this case, it is not recommended to enable PFC for the priority to which BPDUs are assigned (usually priority 7).

36.1.3.2 Processing PFC M_CONTROL.indications

The PFC Receiver entity (see 36.2.2) maintains and makes available to Transmission Selection the vector of the Priority_Paused[n] variables, indicating the state of each of the eight priorities. Each Priority_Paused[n] variable is a boolean. When Priority_Paused[n] is FALSE, priority n is not in paused state. When Priority_Paused[n] is TRUE, priority n is in paused state.

Figure 36-2 shows the PFC state diagram for priority n . If PFC is not enabled for priority n , then the PFC state diagram does not apply to priority n and Priority_Paused[n] is FALSE.

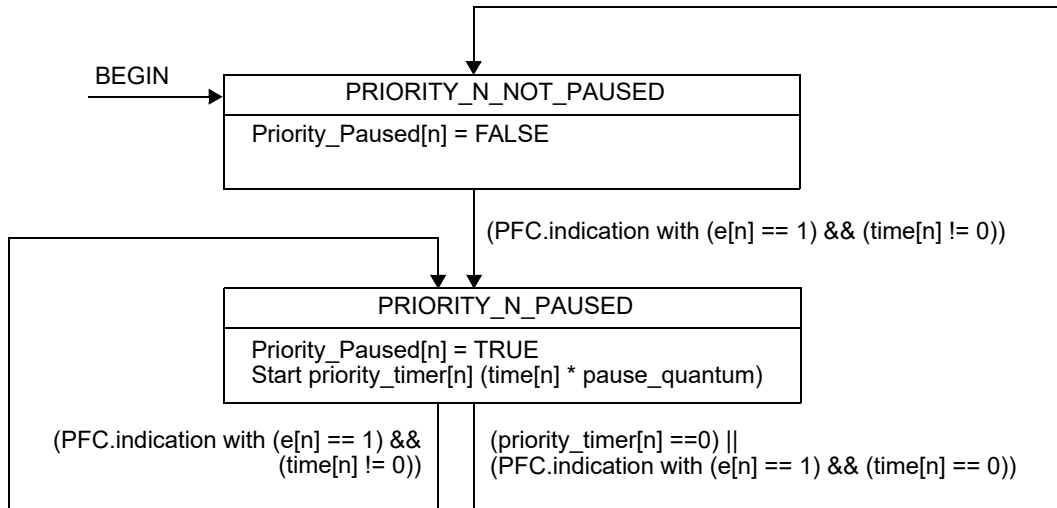


Figure 36-2—PFC Receiver state diagram for priority n

Upon receipt of a PFC M_CONTROL.indication, the PFC Receiver programs up to eight separate timers, each associated with a different priority, depending on the priority_enable_vector. For each bit in the priority_enable_vector that is set to one, the corresponding timer value is set to the corresponding time value in the time_vector parameter. Priority_Paused[n] is set to TRUE when the corresponding timer value (i.e., priority_timer[n]) is nonzero. Priority_Paused[n] is set to FALSE when the corresponding timer value (i.e., priority_timer[n]) counts down to zero. A time value of zero in the time_vector parameter has the same effect as the timer having counted down to zero. If PFC is not enabled for priority n and a PFC indication is received with $e[n]$ set to one, then the $time[n]$ parameter is ignored (i.e., the primitive is processed as if $e[n]$ was set to zero).

NOTE—A priority_enable_vector with all bits set to zero is legal and equivalent to a no-op.

1 36.1.3.3 Timing considerations

2 For effective flow control on a point-to-point full-duplex link, it is necessary to place an upper bound on the
3 length of time that a device can transmit data frames after receiving a PFC M_CONTROL.indication with
4 e[n] set to one in the priority_enable_vector and a nonzero time[n] in the time_vector operands.

5 If MACsec is not supported, a queue shall go into paused state in no more than 614.4 ns since the reception
6 of a PFC M_CONTROL.indication that paused that priority. This delay is equivalent to 12 pause quanta
7 (i.e., 6144 bit times) at the speed of 10 Gb/s, 48 pause quanta (i.e., 24 576 bit times) at the speed of 40 Gb/s,
8 and 120 pause quanta (i.e., 61 440 bit times) at the speed of 100 Gb/s.

9 If MACsec is used, a queue shall go into paused state in no more than 614.4 ns + ‘SecY transmit delay’ (see
10 IEEE Std 802.1AE) since the reception of a PFC M_CONTROL.indication that paused that priority. The
11 ‘SecY transmit delay’ is defined as the wire transmit time for a maximum sized MPDU + 4 times the wire
12 transmit time for 64 octet MPDUs. For a 2000-byte frame the ‘SecY transmit delay’ is $8 \times (2000 + 20) + 8 \times$
13 $4 \times (64 + 12 + 4 + 20) = 19\,360$ bit times.

14 NOTE 1—19 360 bit times is an appropriate value for ‘SecY transmit delay’ for speeds up to 10 Gb/s. Support for the
15 speeds of 40 Gb/s and 100 Gb/s can require a higher value.

16 If MACsec is supported but not used, the delay computation has to take into account the MACsec Bypass
17 Capability (MBC) bit in the PFC configuration TLV of DCBX (D.2.10), that indicates if the link peer needs
18 the extra time for MACsec. If the MBC bit is set to zero, the maximum PFC delay is 614.4 ns. If the MBC
19 bit is set to one, the maximum PFC delay is 614.4 ns + ‘SecY transmit delay’.

20 NOTE 2—In addition to the above delays, system designers should take into account the delay of the PHY and of the
21 link segment when designing devices that implement the PFC operation to ensure frames are not lost due to congestion
22 (see Annex O for additional discussion on this topic).

23 36.2 PFC-aware system queue functions

24 Figure 36-3 illustrates the architecture of the queue functions of a PFC-aware system when Link
25 Aggregation is not used. These functions offer a service to higher layers that utilizes a single instance of the
26 ISS or EISS to connect to the lower layers. In Figure 36-3, two major blocks are outlined with dotted
27 boundaries:

- 28 a) The PFC Initiator block, in the right of Figure 36-3 (see 36.2.1)
- 29 b) The outbound queue block, in the left of Figure 36-3 (see Figure 22-2)

30 The remaining entities illustrated in Figure 36-3, other than the PFC Receiver entity, are part of the
31 IEEE 802.1 architecture and are not further discussed here.

32 36.2.1 PFC Initiator

33 The PFC Initiator entity generates M_CONTROL PFC requests using the M_CONTROL.request primitive
34 (see 36.1.3.1) when appropriate (e.g., when an input buffer reaches a certain threshold).

35 36.2.2 PFC Receiver

36 The PFC Receiver entity processes the M_CONTROL.indication primitives as specified in 36.1.3.2. In
37 addition, the PFC Receiver maintains and makes available to Transmission Selection the vector of the
38 Priority_Paused[n] variables, indicating the state of each of the eight priorities.

39 The PFC Receiver entity acts per physical port. When Transmission Selection is running above Link
40 Aggregation, each PFC Receiver entity processes the M_CONTROL.indication primitives as specified in
41 36.1.3.2, and maintains and makes available to Transmission Selection the vector of the Priority_Paused[n]

1 variables, indicating the state of each of the eight priorities of that physical link, as shown in Figure 36-
2 4.

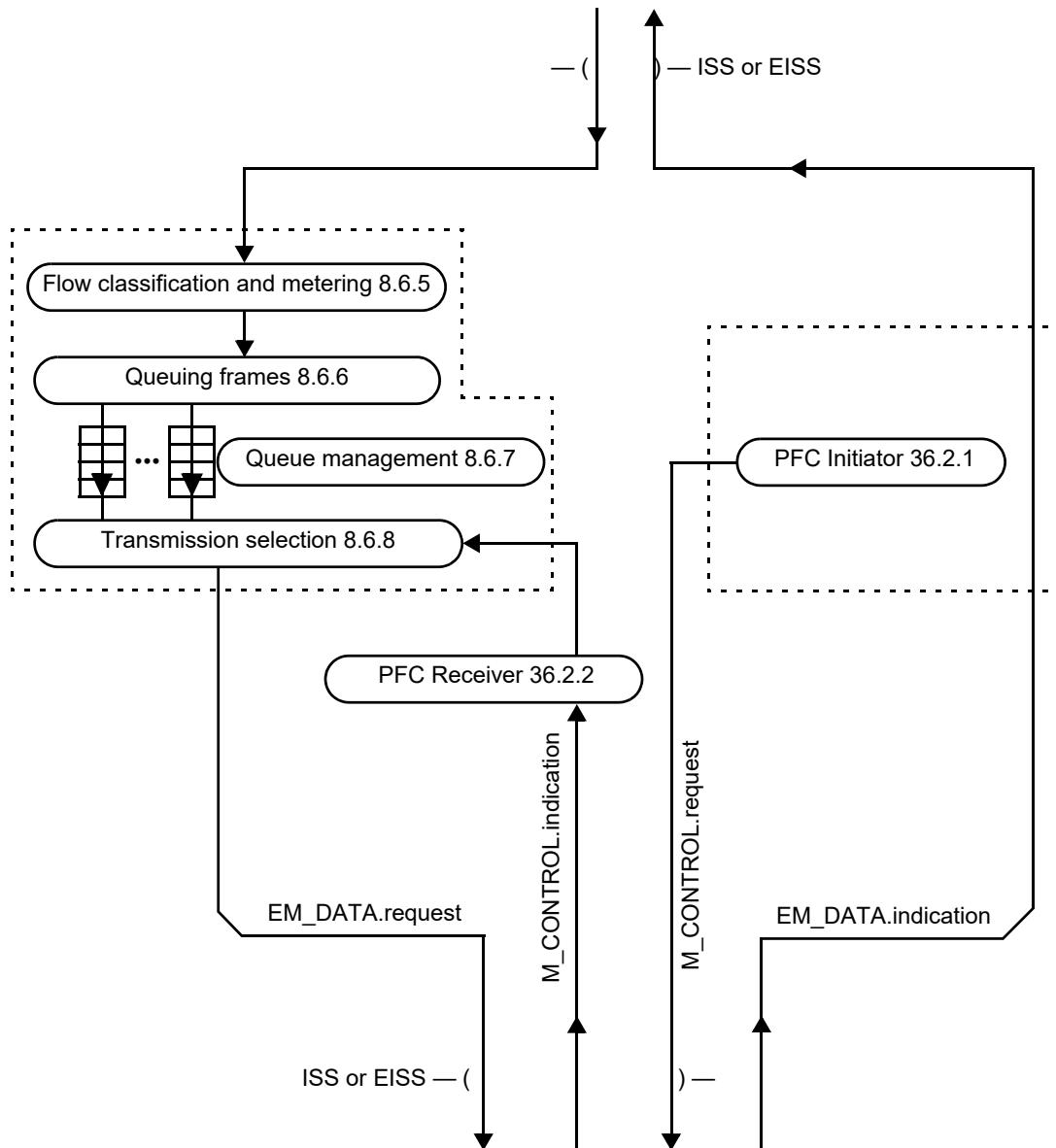


Figure 36-3—PFC-aware system queue functions

1

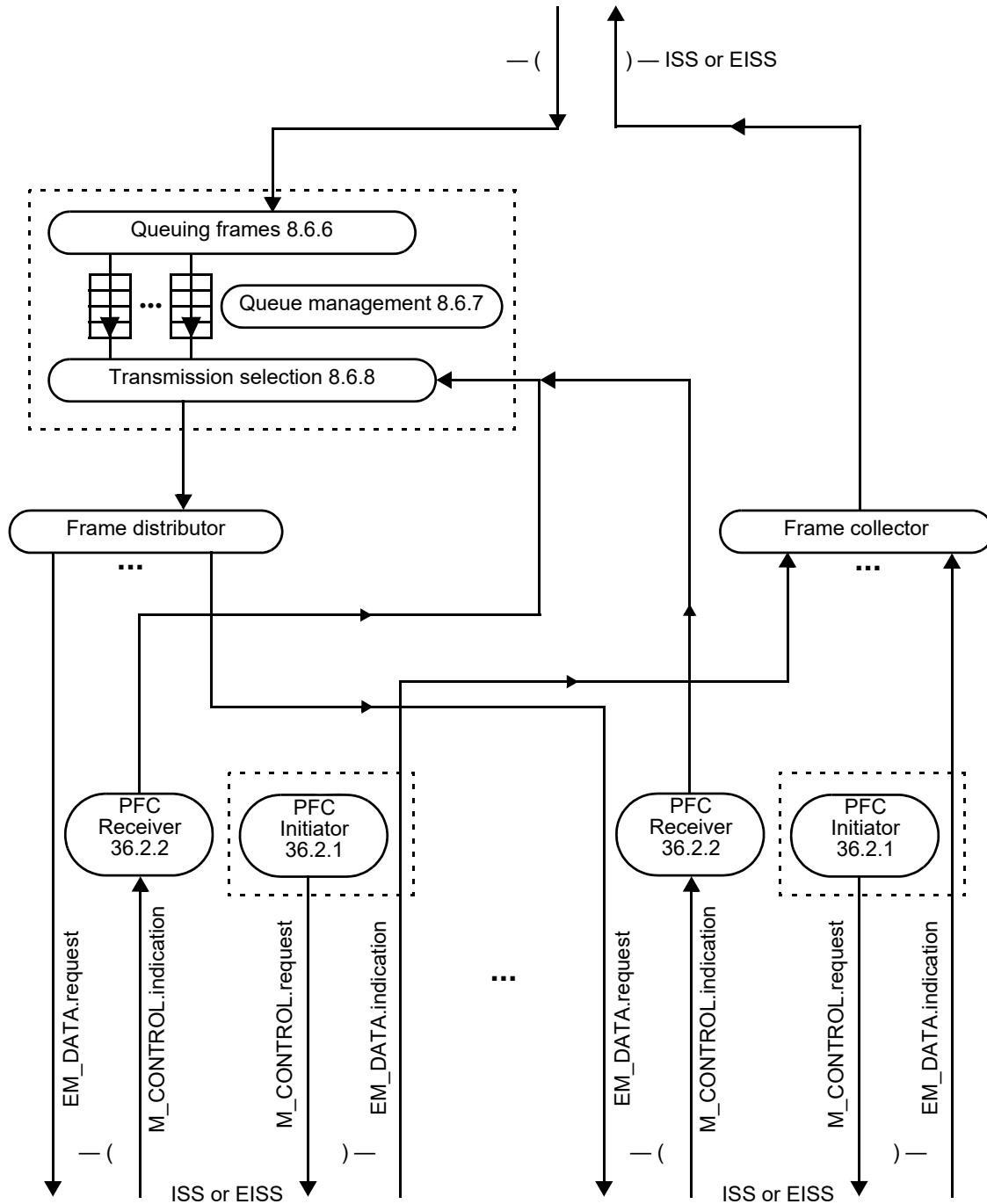


Figure 36-4—PFC-aware system queue functions with Link Aggregation

1 37. Enhanced Transmission Selection (ETS)

2 37.1 Overview

3 This clause provides definitions and an operational model for priority processing and bandwidth allocation
4 in end stations and Bridges in a Data Center Bridging (DCB) environment. Using Priority-based processing
5 and bandwidth allocations, different traffic classes with different types of traffic such as LAN, Storage
6 Networking, Clustering, and management can be configured to provide bandwidth allocation, or best effort
7 transmit characteristics.

8 In order to accomplish this, the following are provided:

- 9 a) A set of bandwidth configuration parameters that are used to configure the percentage of bandwidth
10 assigned to each traffic class.
- 11 b) A set of characteristics that an ETS enabled transmitter is required to follow.
- 12 c) A means of communicating priority-to-traffic class allocations to neighboring systems (DCBX; see
13 Clause 38).

14 37.1.1 Relationship to other transmission selection algorithms

15 If ETS is used in conjunction with the strict priority and credit-based shaper algorithms, then attention
16 should be paid to assignment of traffic classes as specified in 34.5.

17 37.2 ETS configuration parameters

18 The following ETS configuration parameters exist for each ETS enabled port:

- 19 a) *numTrafficClassesSupported*: Indicates the number of traffic classes supported by a port. This value
20 has a minimum of 3 (see 37.3) and a maximum of 8.
- 21 b) *TCPriorityAssignment(P)*: For each supported priority, P, the traffic class to which that priority P is
22 assigned (see 37.1.1). A traffic class only operates with the ETS algorithm if it is assigned to do so
23 by the Transmission Selection Algorithm Table (see 8.6.8); i.e., the entry in the table for that traffic
24 class is assigned the value 2 (see Table 8-6).
- 25 c) *TCBandwidth(N)*: For each traffic class N, the percentage of the available bandwidth assigned to
26 that traffic class. The total for all traffic classes is equal to 100% (see 37.3).

27 Configuration of ETS parameters shall be done via DCBX (see Clause 38).

28 37.3 ETS algorithm

29 The ETS algorithm provides allocation of bandwidth to traffic classes. The algorithm allows bandwidth
30 intensive and loss sensitive traffic to share the network while allowing coexistence with low latency traffic
31 using the strict priority and credit-based shaper algorithms. Since there are a number of variants of
32 bandwidth sharing algorithms (such as weighted round robin) that provide appropriate bandwidth sharing, a
33 detailed algorithm is not specified. Any algorithm is allowed as long as it meets the specified performance
34 requirements.

35 NOTE 1—The ETS algorithm is used to determine when an ETS traffic class has a frame available to transmit. The
36 determination of which frame in the traffic class to send can be done in an implementation specific manner, subject to
37 the ordering requirements in 8.6.6.

38 For the purposes of the ETS algorithm, Available Bandwidth is defined as the link bandwidth remaining
39 after traffic classes not assigned to Transmission selection algorithm 2, and Vendor Specific (see Table 8-6)

are serviced. The effect of Vendor Specific algorithms to Available Bandwidth is outside the scope of this standard.

The ETS algorithm shall provide the ability to configure traffic classes to share bandwidth according to the following:

- a) Allow one or more priorities to be assigned to a traffic class.

NOTE 2—All priorities within a traffic class typically share similar traffic handling requirements (e.g., loss and bandwidth).

- b) Allow bandwidth to be configured for each traffic class with a granularity of 1%. The configured bandwidth indicates the percentage of Available Bandwidth that can be used by the traffic class when all other traffic classes are consuming their configured bandwidth.

- c) For traffic classes for which the transmission selection algorithm is ETS (i.e., Transmission selection algorithm 2 in Table 8-6), select frames for transmission from the traffic class queues such that the bandwidth consumed by the traffic class approaches its percentage of Available Bandwidth. The traffic class consumes less than its percentage of Available Bandwidth when the offered load for that traffic class is less than its percentage of Available Bandwidth, in that case the remainder of its percentage of Available Bandwidth can be used by other traffic classes. A traffic class can consume more than its percentage of Available Bandwidth when the offered load of other traffic classes results in those traffic classes consuming less than their percentage of Available Bandwidth. Allocation of any excess Available Bandwidth is implementation specific.

- d) When all traffic classes are offered enough load to consume their share of available bandwidth, and there is no offered load in the traffic classes using transmission selection algorithms other than ETS, the bandwidth received from an ETS traffic class shall deviate from its allocation by no more than 10% of the available bandwidth when measured over a period of 10 000 000 bit times using maximum sized frames during a time period when no PFC frames are received (i.e., no congestion).

NOTE 3—The measurement window is derived from an approximate number of bits in 500 frames (i.e., 500 frames × 2000 bytes per frame × 8 bits per byte rounded up to 10 000 000).

NOTE 4—Allowed deviation of ± 10% allows for variations of traffic (e.g., burstiness) over the specified measurement window and provides balance between precision and implementation flexibility.

- e) At least 3 traffic classes shall be supported per port. A minimum of 3 traffic classes allows a minimum configuration such that one ETS traffic class contains priorities that have PFC enabled, one ETS traffic class contains priorities that have PFC disabled, and one traffic class using strict priority.

37.4 Legacy configuration

Subclause 8.6.8 specifies strict priority scheduling as default behavior. This can be achieved by configuring the Transmission selection algorithm value for a traffic class to 0 (see Table 8-6).

Subclause 8.6.6 specifies the default priority to traffic class mapping. There is no change to such mapping when ETS is used.

1 38. Data Center Bridging eXchange protocol (DCBX)

2 38.1 Overview

3 This clause details the DCBX, which is used by DCB devices to exchange configuration information with
4 directly connected peers. The protocol may also be used for misconfiguration detection and for
5 configuration of the peer.

6 This standard describes the base protocol, which comprises state machines and TLVs for capability
7 exchange. For each feature that is supported by DCBX, the attributes that are to be exchanged specify the
8 following:

- 9 a) The attributes to be exchanged
- 10 b) How the attributes are used for detecting misconfiguration
- 11 c) What action needs to be taken when a misconfiguration is detected

12 The information listed above is specified for the following:

- 13 d) ETS
- 14 e) PFC
- 15 f) Application Priority TLV
- 16 g) Application VLAN TLV

17 38.2 Goals

18 The goals of DCBX are as follows:

- 19 a) Discovery of DCB capability in a peer port; for example, it can be used to determine if two link peer
20 ports support PFC.
- 21 b) DCB feature misconfiguration detection: DCBX can be used to detect misconfiguration of a feature
22 between the peers on a link. Misconfiguration detection is feature-specific because some features
23 allow asymmetric configuration.
- 24 c) Peer configuration of DCB features: DCBX can be used by a device to perform configuration of
25 DCB features in its peer port if the peer port is willing to accept configuration.

26 38.3 Types of DCBX attributes

27 Three types of DCBX attributes are exchanged:

- 28 a) Informational attributes (see 38.3.1)
- 29 b) Asymmetric attributes (see 38.4.1)
- 30 c) Symmetric attributes (see 38.4.2)

31 38.3.1 Informational attributes

32 Informational attributes are exchanged via LLDP without any participation in a DCBX state machine.

33 38.4 DCBX and LLDP

34 DCBX uses Link Layer Discovery Protocol (LLDP) (see IEEE Std 802.1AB) to exchange attributes between
35 two link peers. LLDP is a unidirectional protocol. It advertises connectivity and management information
36 about the local station to adjacent stations on the same IEEE 802 LAN.

1 DCB exchanged attributes are packaged into Organizationally Specific TLVs. The OUI used for the DCBX
2 TLV is the IEEE 802.1 OUI (i.e., 00-80-C2).

3 DCBX state machine transitions are based on the DCBX objects in the LLDP MIB module. Operation of the
4 DCBX state machine may affect the values of the DCBX objects in the LLDP MIB module.

5 A port capable of any DCB feature shall have the capability for DCBX to be administratively disabled. The
6 default state for DCBX is enabled.

7 DCBX is expected to operate over a point-to-point link. If multiple LLDP peer ports running DCBX are
8 detected, then DCBX should behave as if the peer port's DCBX TLVs are not present until the multiple
9 LLDP peer port condition is no longer present. However, a transition in LLDP peer port may occur in some
10 circumstances (e.g., such as a transition from system boot to system operation). Therefore, when it is
11 detected that the number of peer ports running DCBX exceeds 1 for a period longer than the longest TTL of
12 any of the peers, a multi-peer condition is detected. During the time when the multi-peer condition has not
13 been detected the DCBX data from the most recent DCBX peer shall be used. An LLDP peer port is
14 identified by a concatenation of the chassis ID and port ID values transmitted in the LLDPDU. A DCBX
15 peer port is a LLDP peer port that is sending DCBX TLVs.

16 DCBX defines two different types of attribute passing mechanisms:

- 17 a) Asymmetric: the passing of a attribute from one port to its peer port. In this case, the desired
18 configuration for the peer may not match the local configuration.
- 19 b) Symmetric: the passing of a attribute from one port to its peer port with objective of both ports
20 utilizing the same attribute value.

21 **38.4.1 Asymmetric attribute passing**

22 **38.4.1.1 Overview**

23 Two types of TLV are passed for Asymmetric attribute Passing:

- 24 a) Configuration TLV: Provides current operational state and Willing bit. The received Willing bit is
25 not used by the state machine, but instead provided to higher layers to provide an indication of the
26 expected behavior of the remote port. The presence of the willing bit set to 1 indicates to the remote
27 port that the local port is willing to accept a configuration for the specific attribute for which the
28 willing bit is set.
- 29 b) Recommendation TLV: Provides recommendation for the operational state of remote port.
30 Transmitted only if the local port is configured to make recommendations in which case it is
31 transmitted in all LLDPDUs. Transmitted regardless of the “willingness” of the remote port.

32 NOTE—An implementation is allowed to send a recommendation TLV or a configuration TLV or both.

33 **38.4.1.2 Asymmetric state variables**

34 **LocalWilling:** Indicates that the local port has been administratively configured to accept recommendations.
35 This value is included in the Willing bit of Configuration TLVs transmitted by this port. A local port may be
36 configured as not Willing.

37 **OperParam:** The current operational value of the attribute on the local port. This value is included as the
38 attribute in the Configuration TLVs.

39 **LocalAdminParam:** The administratively configured value for the attribute. This becomes the operational
40 value of the attribute by default, and the operational value may be overridden if the local port accepts a
41 recommendation from the remote port.

1 **RemoteParam:** The attribute received in the last Recommendation TLV. This variable is set to NULL if the
2 remote LLDP database contains no Recommendation TLV.

3 **DCBXMapping:** A function that maps the received parameters to the capabilities of the receiving port.

4 38.4.1.3 Asymmetric state machine

5 The Asymmetric state machine is depicted in Figure 38-1.

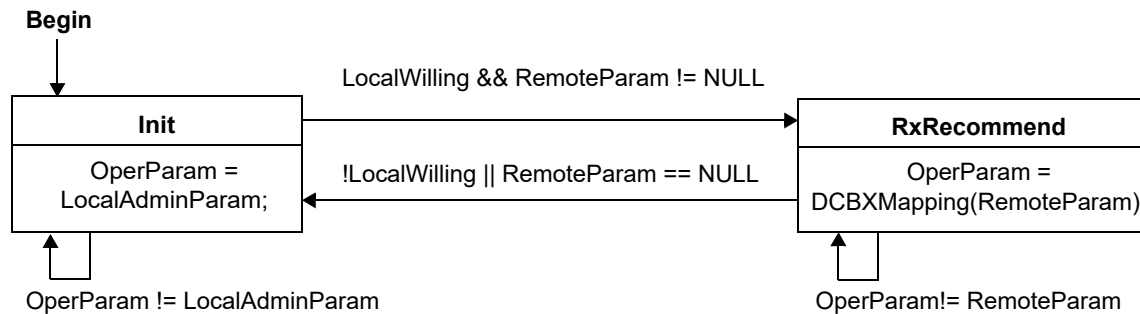


Figure 38-1—DCBX Asymmetric state machine

6 38.4.2 Symmetric attribute passing

7 38.4.2.1 Overview

8 For Symmetric attribute passing one type of TLV is used:

- 9 a) Configuration TLV. This TLV always carries the current local operational state and a Willing (W)
10 bit.

11 A port that sets the W bit is considered Willing. A Willing port shall set its operational attribute to that
12 indicated in the received TLV if the received TLV has the W bit set to zero. If both the local port and remote
13 port are willing, then the attribute values of the port with the lower numerical MAC address shall take
14 precedence.

15 38.4.2.2 Symmetric state variables

16 **LocalWilling:** Indicates that the local port has been administratively configured to accept the attribute from
17 the remote port. This value is included in the Willing bit of DCBX TLVs transmitted by this port.

18 **RemoteWilling:** rwTrue indicates that the Willing bit was set in the last TLV received. rwFalse indicates
19 that the Willing bit was not set in the last TLV received. This variable is set to rwNull if the remote LLDP
20 database contains no Willing bit.

21 **LocalMAC:** The MAC address of the local port.

22 **RemoteMAC:** The MAC address of the remote port.

23 **OperParam:** The current operational value of the attribute on the local port. This value is included as the
24 attribute in the DCBX TLV.

25 **RemoteParam:** Contains the value of the last attribute (i.e., the operational value of the remote port)
26 received in the TLV. This variable is set to NULL if the remote LLDP database contains no TLV.

1 **LocalAdminParam:** The administratively configured value for the attribute. This becomes the operational
2 value of the attribute by default, and may be overridden if the local port accepts the attribute from the remote
3 port.

4 **DCBXMapping:** A function that maps the received parameters to the capabilities of the receiving port.

5 **38.4.2.3 Symmetric state machine**

6 The Symmetric state machine is depicted in Figure 38-2.

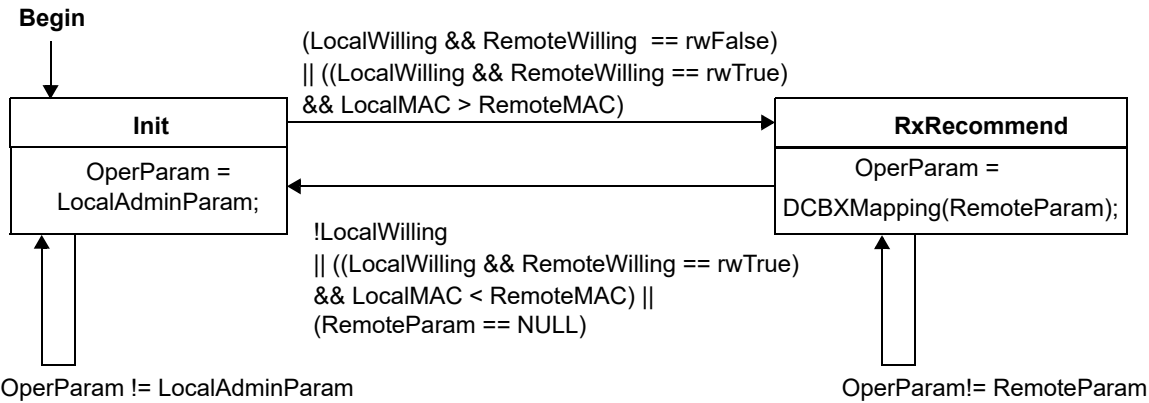


Figure 38-2—Symmetric state machine

7 NOTE—Through observation of the states and state variables it is possible to determine that the state machine is in the
8 process of passing attributes. This knowledge may be useful for clients of DCBX. For example, a client may wish to
9 delay use of the link while the DCBX state machine is in the process of passing and possibly setting attributes. A
10 “Pending” indication indicating that the state machine is in this process may be created by the following equations:

11 Pending = RemoteParam == NULL
12 || !LocalWilling && RemoteWilling == rwTrue && OperParam != RemoteParam;

1 39. Multiple I-SID Registration Protocol (MIRP)

2 MIRP defines an MRP application (10.1) that provides the ability to flush learned MAC Address Entries
3 held in the FDB of an I-component on a per-I-SID basis. MIRP makes use of MRP Attribute Declaration
4 (MAD) and MRP Attribute Propagation (MAP), which provide the common state machine descriptions and
5 the common attribute propagation mechanisms defined for use in MRP-based applications. The MRP
6 architecture, MAD, and MAP are defined in Clause 10.

7 In a PBBN, the assignment of S-VIDs and I-SIDs to VIPs is often fixed, and the overhead of signaling the
8 creation of Dynamic VLAN Registration Entries among I-components is undesirable. It can nevertheless be
9 necessary to signal the need to flush the I-components' FDBs of learned MAC address information. For
10 example, one could configure a point-to-point service that is dual-homed (has two I-components) at each
11 end of the service, only two of which (one at each end) are active at any one time. A failure that causes one
12 I-component to take over from the other requires that the active I-component at the other end of the service
13 forget its associations of C-MAC addresses to the failed I-component's B-MAC address. Therefore, the
14 MIRP application supports the Registration Fixed (New propagated), instead of the Registration Fixed (New
15 ignored), value in the Registrar Administrative Controls.

16 The function of MIRP can be performed by MVRP. However, in an I-component with one S-VID per VIP
17 requires 4094 MVRPDUs, one for each VIP, to signal a simultaneous topology change on all 4094 S-VIDs,
18 whereas it can signal those same topology changes with a single Multiple I-SID Registration Protocol Data
19 Unit (MIRPDU) through the PIP.

20 39.1 MIRP overview

21 MIRP is an extension of MVRP (11.2) for use in I-components and B-components. The extensions in both
22 I-components and B-components are as follows:

- 23 a) The attribute values carried by the MIRPDUs are 24-bit I-SID values, rather than 12-bit VIDs in an
24 MVRPDU.
- 25 b) A MIRP Participant can only be a New-Only Participant (10.6), so the Registrar Administrative
26 Controls can take only the value, Registration Fixed (New propagated).
- 27 c) MIRP uses the Nearest Customer Bridge group address (Table 8-1) or the Default Backbone
28 Destination from the CBP's Backbone Service Instance table (6.11) instead of an MRP Application
29 address from Table 10-1, so a Provider Bridge forwards frames containing MIRPDUs normally,
30 while a Customer Bridge filters them.

31 In an I-component only, the extensions are as follows:

- 32 d) A VIP in an I-component can be attached to an MVRP Participant or to a MIRP Participant, but not
33 to both.
- 34 e) An MVRP to MIRP attributes translation function (39.2.1.1) enables interoperability between the
35 MVRP and MIRP Participants in the I-component.
- 36 f) There is only one MIRP Participant and one instance of the MIRP application per PIP.

37 In a B-component only, the extensions are as follows:

- 38 g) MIRP Participants are placed on zero or more CBPs, and on at most one non-CBP Port in the
39 B-component.
- 40 h) MAP (10.3) operates among MIRP MAD components only, using 24-bit MIRP I-SID values.

41 Figure 39-1 illustrates the architecture of MVRP and MIRP in an I-component. The arrows showing the
42 relationship between the MVRP or MIRP applications and the FDB in the MAC Relay Entity have been
43 omitted from Figure 39-1, but these relationships are the same as those shown in Figure 11-1.

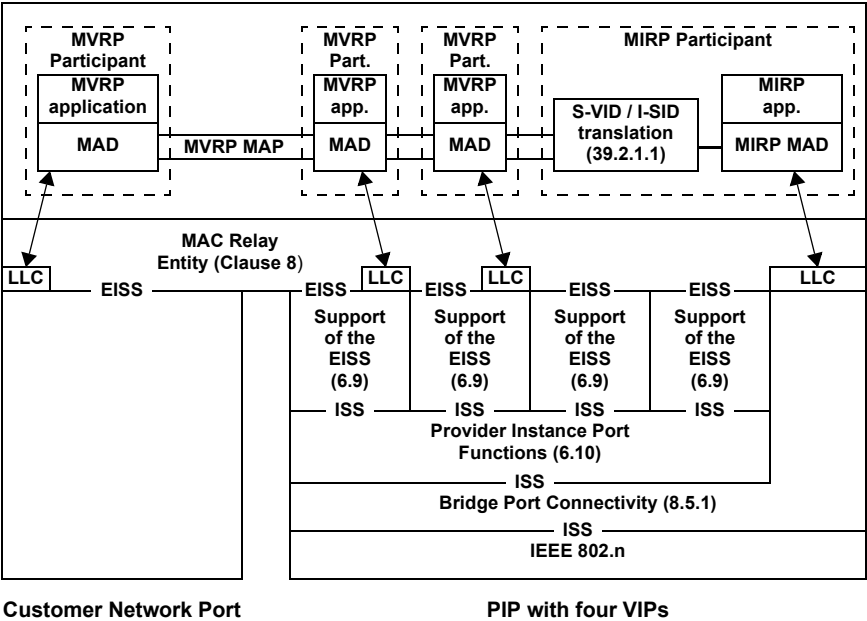


Figure 39-1—Operation of MIRP in an I-component

1 The architecture of MIRP in a B-component is illustrated in Figure 39-2. One management Port and one
2 CBP are shown, although any number of CBPs can be supported. (See 39.2.2 for an alternate model.)

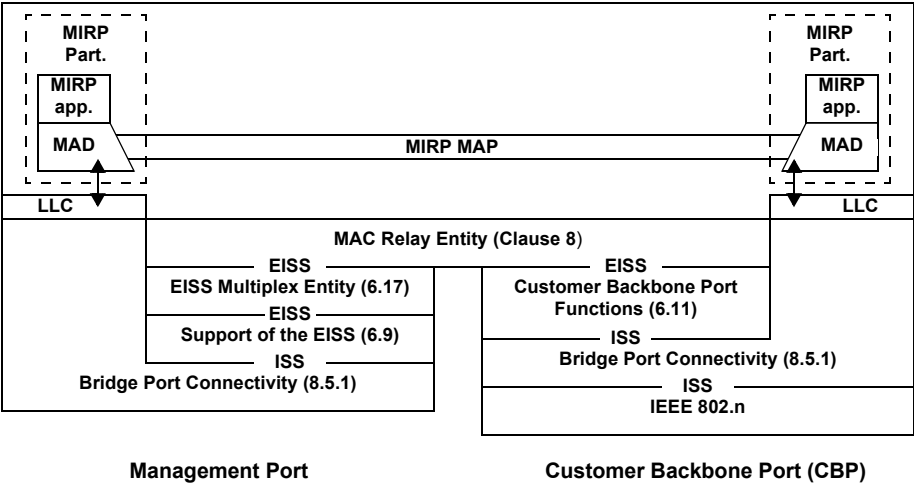


Figure 39-2—Operation of MIRP in a B-component

1 39.1.1 Behavior of I-components

2 Since PBBs are insulated from C-MAC addresses by the I-TAG, MVRP Participants in I-components are
3 peers of MVRP Participants in other I-components, and not with MVRP Participants attached to PNP in
4 B-components or BCBs. In the case that an I-component is configured for some number of VIPs, each
5 serving some number of VLANs, a single MIRPDU transmitted or received over the PIP takes the place of
6 the individual MVRPDUs that would otherwise be transmitted or received over those VIPs.

7 An I-component supporting MIRP shall:

- 8 a) Implement the MIRP application as indicated in 39.2;
- 9 b) Implement an MVRP/MIRP attributes translation function as indicated in 39.2.1.1; and
- 10 c) Support operation only as an MRP New-Only Participant (10.6).

11 39.1.2 Behavior of B-components

12 As illustrated in Figure 39-2, there are any number of MIRP Participants in a B-component, each on a CBP,
13 and each containing an instance of the MIRP application and a MIRP MAD, with the MADs interacting via
14 MAP. In addition, at most one MIRP Participant is attached to a Port that is not a CBP. This port can be a
15 PNP, or a Management Port with no attachment to an external LAN. The latter case is shown in Figure 39-2.

16 A B-component supporting MIRP shall:

- 17 a) Implement the MIRP application as indicated in 39.2; and
- 18 b) Support operation as an MRP New-Only Participant (10.6).

19 39.2 Definition of the MIRP application

20 39.2.1 Definition of MRP elements

21 39.2.1.1 S-VID and I-SID mapping in an I-Component

22 MVRP MAP (11.2.1, 10.3) propagates information among the MVRP MADs and the MIRP MAD. But,
23 since MVRP MADs support 12-bit VID values, and the MIRP MAD supports 24-bit I-SID values, a
24 translation function is required to map the VIP-ISID to or from S-VIDs statically configured on the VIP.

- 25 c) When a “new” MIRP declaration is received on a given VIP as a result of receiving an MIRPDU
26 from the attached LAN (MAD_Join.indication), the MAD_Join.request with the *new* parameter set
27 that is generated by the MIRP MAD is translated to a MAD_Join.request for each S-VLAN with a
28 Static VLAN Registration Entry of Registration Fixed (New propagated) on that VIP. When any
29 MIRP declaration with the *new* parameter is received on a given VIP as a result of a request from
30 MAP (MAD_Join.request) for any S-VLAN with a Static VLAN Registration Entry of Registration
31 Fixed (New propagated) on that VIP, it is translated to a MAD_Join.request for that VIP’s I-SID
32 with the *new* parameter set. All other requests are discarded.

33 NOTE—If many S-VLANs are configured per VIP, then using an MVRP New-Only Participant (11.2.6) offers more
34 detailed control of MAC address flushing than does MIRP, at some cost in bandwidth and processing effort.

35 39.2.1.2 I-SID translation in a B-component

36 If a Backbone Service Instance table (6.11) on a CBP implements the Local Service Instance Identifier
37 (Local-SID) field, the attribute values encoded in MIRPDUs by the MIRP Participants in CBPs shall be
38 Local-SID values. The attribute values transferred by MAP in the B-component and encoded in MIRPDUs
39 by the MIRP Participant on a non-CBP Port shall be Backbone-SID values.

1 39.2.1.3 MAP Context for MIRP

2 The MIRP MAP Context identifies the set of Ports that form the applicable topology for the propagation of
3 the MIRPDUs among the MIRP Participants. Within a single I-component, the MIRP Participants on PIPs
4 behave as independent end stations from an I-SID registration perspective and correspondingly there is no
5 further propagation of the MIRPDUs. As described in 39.2.1.1, the S-VID/I-SID translation function allows
6 the MIRP MADs to communicate with MVRP MADs. Thus, New messages can propagate throughout a
7 concatenated network of PBNs (via MVRP) and PBBNs (via MIRP).

8 Within a single B-component, there may be multiple MIRP MAP Contexts. The reason for having separate
9 MIRP MAP Contexts is to collect and distribute MIRP information bound for or received from MIRPDUs
10 that use different combinations of B-VIDs and MAC addresses, as chosen by the network administrator via
11 managed objects (12.16.1). A Port in a B-component supporting a MIRP Participant, either an
12 administratively chosen PNP or Management Port, or a CBP, can participate in more than one MIRP
13 Context. The Ports of a B-component comprising a given MIRP Context is a PNP or Management Port, plus
14 the set of CBPs on the B-component for which the following are true:

- 15 a) The CBP is in the Member set (8.8.9) for the B-VID.
- 16 b) If the MIRPDU destination_address for the CBP is the Default Backbone Destination address, the
17 FDB does not filter that B-VID and MAC address on the Port.

18 In a B-component, the parameters controlling MIRPDU addressing (39.2.1.6) control the number of MIRP
19 MAP Contexts as follows:

- 20 c) If a single destination_address and B-VID are used for all MIRPDUs [item a) in 39.2.1.6] then there
21 is a single MIRP MAP Context.
- 22 d) If the Nearest Customer Bridge group address (Table 8-1) used with a B-VID from a Backbone
23 Service Instance table [item b) in 39.2.1.6], then there is a separate MIRP MAP Context for each
24 B-VID that appears in the CBP Backbone Service Instance table (6.11) of any CBP with a
25 configured MIRP Participant.
- 26 e) If the Default Backbone Destination option for the MIRPDU destination_address is used [item c) in
27 39.2.1.6], then there is a separate MIRP MAP Context for each combination of Default Backbone
28 Destination and B-VID in the CBP Backbone Service Instance table (6.11) of any CBP with a
29 configured MIRP Participant.

30 The MIRP Participant on a CBP propagates information for a particular I-SID between the CBP MIRP
31 MAD and the appropriate MIRP MAP Context, as determined by the CBP's Backbone Service Instance
32 table and the addressing selection parameters (39.2.1.6). Thus, a CBP MIRP MAD may exchange requests
33 and indications with multiple MIRP MAP Contexts.

34 NOTE 1—MIRPDUs cannot pass through the CBP functions (6.11) because they have no I-TAG. Readers can see from
35 Figure 39-2 and Figure 39-3 that this isolates the MIRPDUs exchanged among PNPs, Management Ports (and CBPs in
36 Figure 39-3) over the backbone from the MIRPDUs exchanged over the various I-component-to-CBP links.

37 NOTE 2—See 39.2.2 for an alternate model for MIRP operation in a B-component and the MIRP MAP Context.

38 39.2.1.4 MAP Context identification for MIRP

39 In an I-component, there is only one MAP Context.

40 In a B-component, the vlan_identifier and destination_address of an MIRPDU received on a PNP or
41 Management Port serve to identify the MIRP MAP Context as defined in 39.2.1.3. In a CBP, the MIRP MAP
42 Context used to propagate information is determined by CBP Backbone Service Instance table (6.11) entry
43 for each I-SID value propagated. An MIRPDU that corresponds to no MAP Context configured on the
44 receiving B-component is discarded.

1 39.2.1.5 MIRP application addressing in an I-component

2 The source MAC address for an MIRPDU is the MAC address of the PIP.

3 The destination MAC address for an MIRPDU transmitted from a PIP shall be the Nearest Customer Bridge
4 group address, (01-80-C2-00-00-00, see Table 8-1).

5 MIRPDUs transmitted and received on a PIP carry no VLAN tag and have no `vlan_identifier`.

6 NOTE—An MIRPDU transmitted by a PIP needs no B-TAG because it is connected to a CBP (see Figure 25-4) that
7 either has an MIRP Participant to receive the untagged MIRPDU, or if not, discards the MIRPDU (39.2.1.3).

8 39.2.1.6 MIRP application addressing in a B-component

9 The source MAC address for an MIRPDU is the MAC address of the Port from which it is transmitted.

10 For a MIRP Participant attached to a CBP, the destination MAC address for an MIRPDU transmitted shall
11 be the Nearest Customer Bridge group address, (01-80-C2-00-00-00, see Table 8-1). MIRPDUs shall be
12 transmitted from a CBP without a VLAN tag.

13 For the MIRP Participant attached to a PNP or Management Port, there are three choices, selectable via
14 managed variables [see item k) in 12.16.1.1.3], for the destination MAC address and B-VLAN on
15 transmitted MIRPDUs:

- 16 a) The `destination_address` is the Nearest Customer Bridge group address (Table 8-1), and the
17 `vlan_identifier` is the MIRP B-VID [item c) in 12.16.1.2.2].
- 18 b) The `destination_address` is the Nearest Customer Bridge group address (Table 8-1), and the
19 `vlan_identifier` is a B-VID from the CBPs' Backbone Service Instance tables (6.11).
- 20 c) The `destination_address` is a Default Backbone Destination from the CBPs' Backbone Service
21 Instance tables (6.11), and the `vlan_identifier` is a B-VID from those tables.

22 In case b) or case c), it is possible that different CBPs in the serving I-SIDs in the same MIRP MAP Context
23 (39.2.1.3) can be configured differently. For each I-SID value's message in the MIRPDU, the MIRP
24 Participant on the PNP or Management Port uses the Backbone Service Instance table from the CBP with the
25 lowest numerical Port Number to determine the {`destination_address`, `vlan_identifier`} pair to use for that
26 I-SID value.

27 A separate MIRPDU is transmitted for each separate MIRP MAP Context. If multiple {`destination_address`,
28 `vlan_identifier`} pairs are generated by different I-SIDs within a single MIRP MAP Context, then the MIRP
29 Participant shall transmit at least one MIRPDU for each distinct pair.

30 NOTE 1—The network administrator can use the `destination_address` and `vlan_identifier` configuration choice to
31 optimize either for the fewest transmitted MIRPDUs by using case a), for the fewest unnecessarily addressed
32 I-components by using case c), or for a balance of these two conflicting optimizations by using b).

33 NOTE 2—If different B-components in a single PBBN are configured differently for MIRP application addressing, then
34 MIRPDUs can be lost or misinterpreted, because the different B-components can classify MIRPDUs in different MAP
35 Contexts.

36 NOTE 3—See 39.2.2 for an alternate model for MIRP operation in a B-component and MIRPDU addressing.

37 39.2.1.7 MIRP application EtherType

38 The EtherType used for MIRPDUs shall be the MIRP EtherType identified in Table 10-2.

39 39.2.1.8 MIRP ProtocolVersion

40 The ProtocolVersion for the version of MIRP defined in this standard takes the hexadecimal value 0x00.

1 39.2.1.9 MIRP AttributeType definitions

2 MIRP defines a single Attribute Type (10.8.2.2) that is carried in MRP exchanges, as follows:

3 Attributes identified by the I-SID Vector Attribute Type are instances of VectorAttributes (10.8.1), used to
4 identify a sequence of values of I-SIDs. The value of AttributeType used to identify the I-SID Vector
5 Attribute Type in MRPDUs (10.8.2.2) shall be 1.

6 39.2.1.10 MIRP FirstValue definitions

7 The FirstValue field in instances of the I-SID Vector Attribute Type shall be encoded in MRPDUs (10.4) as
8 three octets, taken to represent an unsigned binary number, and equal to the value of the VIP-ISID parameter
9 for the VIP.

10 The range of permitted I-SID values that can be encoded in the FirstValue fields in MIRP is restricted to
11 those not listed in Table 9-3.

12 39.2.1.11 Administrative controls

13 MIRP supports only the Registration Fixed (New propagated) value of the Registrar Administrative
14 Controls (10.7.2) and this value cannot be manipulated by management. Therefore, the MRP Registrar
15 Administrative Control (12.7.7.1, 12.7.7.2) can only take the value New propagated.

16 The provision of static control over the ability of Applicant state machines to participate in protocol
17 exchanges is achieved by means of the Applicant Administrative Control parameters associated with the
18 operation of MRP (10.7.3). Where management capability is implemented, the Applicant Administrative
19 Control parameters can be applied and modified by means of the management functionality defined in 12.9.

20 The management controls in Clause 12 operate per Bridge Port, but a MIRP Participant is attached to a PIP,
21 which is not a Bridge Port. Therefore, if MIRP is enabled on any VIP on a PIP, then the MIRP Participant is
22 enabled on that PIP. If MIRP is disabled on all of the VIPs on a PIP, then the MIRP Participant on the PIP is
23 disabled.

24 The choice of destination_address and vlan_identifier used in an MIRPDU from a B-component are made
25 by an enumerated configuration parameter [item d) in 12.16.1.2.2] and a B-VLAN configuration parameter
26 [item c) in 12.16.1.2.2].

27 39.2.2 Alternate MIRP model for B-components

28 An alternate model for the operation of MIRP in a B-component is illustrated in Figure 39-3. This model is
29 perfectly interoperable with that presented in 39.2.1 and Figure 39-2, is indistinguishable in terms of
30 required protocol behavior from that model, and thus can be used to implement MIRP. In this alternate
31 model, there is a pair of communicating MIRP Participants in each CBP, separate from a similar pair in
32 every other CBP, with no MIRP Participant on any PNP or management Port. The LAN-facing MIRP
33 Participant (the right-hand MIRP Participant in Figure 39-3) corresponds to the CBP MIPR Participant
34 described in 39.2.1. The Relay-facing MIRP Participant (the left-hand MIRP Participant in Figure 39-3) is
35 attached to the EISS Connectivity (39.2.2.1). Its functions correspond to those of the MIRP Participant on a
36 PNP or Management Port in 39.2.1 and Figure 39-2.

37 NOTE 1—Because the CBP functions (6.11) do not pass frames without I-TAGs, the Relay-facing MIRP Participant can
38 exchange MRPDUs only through the MAC Relay Entity, and the LAN-facing MIRP Participant only through the LAN.

39 NOTE 2—The only difference between the models in Figure 39-2 and Figure 39-3 that is visible to an observer external to
40 the B-component is that the alternate model can issue MRPDUs with different source_address parameters, while a the
41 PNP model issues all MRPDUs with the same source_address. Since MRP and MIRP make no use of that parameter,
42 this difference does not affect the operation of the protocol.

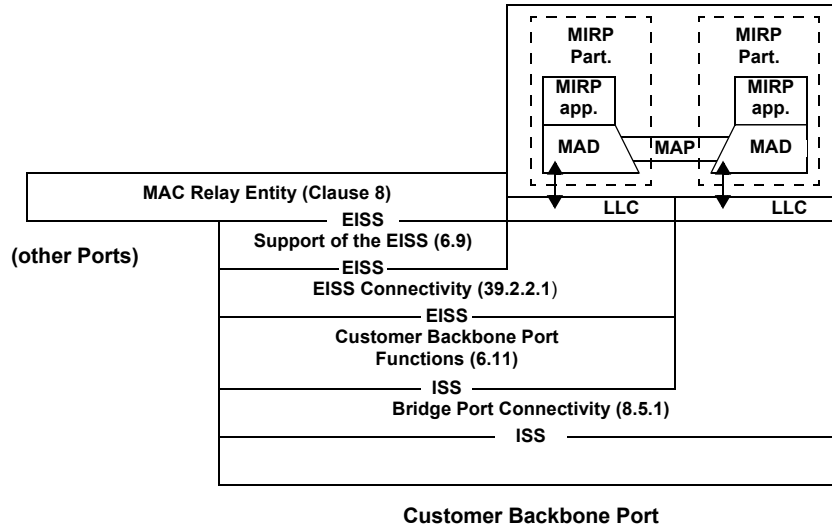


Figure 39-3—Alternate model for MIRP in a B-component

1 39.2.2.1 EISS Connectivity

2 The EISS Connectivity function is identical to the Bridge Port Connectivity of 8.5.1, except that it supports
3 the EISS (6.8) instead of the ISS (IEEE Std 802.1AC).

4 Each EM_UNITDATA.indication provided by the lower EISS access point (Figure 39-3) shall result in a
5 corresponding EM_UNITDATA.indication with identical parameters at each of the access points supporting
6 the MAC Relay and Higher Layer Entities. (Only one such Higher Layer Entity is shown in Figure 39-3.)
7 Each EM_UNITDATA.request from the EISS access point supporting the MAC Relay Entity shall result in
8 a corresponding EM_UNITDATA.indication with identical parameters at each of the access points for the
9 Higher Layer Entities, and a corresponding EM_UNITDATA.request with identical parameters at the lower
10 access point. Each EM_UNITDATA.request from an ISS access point supporting a Higher Layer Entity
11 shall result in a corresponding EM_UNITDATA.indication with identical parameters at the access points for
12 the MAC Relay Entity, and at other access points for Higher Layer Entities, and a corresponding
13 EM_UNITDATA.request with identical parameters at the access point for the LAN.

14 The MAC_Enabled, MAC_Operational, and operPointToPointMAC status parameters for the EISS access
15 point for the MAC Relay Entity and Higher Layer Entities shall take the same value as that for the lower
16 access point if that is present, and shall be True otherwise (i.e., if the Port is a Management Port).

17 39.2.2.2 Alternate MIRP MAP Context

18 There is a single MIRP MAP Context per CBP, that includes both of the MIRP Participants on that CBP but
19 no MIRP Participant on any other CBP.

20 39.2.2.3 Alternate MIRPDU addressing

21 The addressing requirements for the two MIRP Participants on a CBP are different. The LAN-facing MIRP
22 Participant behaves as described in 39.2.1.6. The Relay-facing MIRP Participant follows the rules described
23 for the PNP or Management Port in 39.2.1.6. Since there is only one Backbone Service Instance table (6.11)
24 per CBP, there is no ambiguity with regard to MIRPDU addressing. The MIRPDU addressing control
25 parameters described in 39.2.1.6 apply to every Relay-facing MIRP Participant.

26 NOTE—The model in Figure 39-3 results in the CBPs communicating MIRP information via MIRPDUs, instead of via
27 the MIRP MAP, as in Figure 39-2, and all of those MIRPDUs would be output to the rest of the PBBN, as well.

1 39.2.3 Use of “new” declaration capability

2 MIRP supports the Registration Fixed (New propagated) value for the Registrar Administrative Controls for
3 a Port in a Static VLAN Registration Entry. MIRP will accept and propagate a New Message received on
4 that Port for that VLAN and propagate it to all other Ports.

5 When any MIRP declaration marked as “new” is received on a given VIP, either as a result of receiving an
6 MIRPDU from the attached LAN (MAD_Join.indication), or as a result of receiving a request from the
7 MAP (via the S-VID/I-SID translation function [39.2.1.1]) for the MIRP Application (MAD_Join.request),
8 any entries in the I-component’s FDB for that Port and for the VLANs corresponding to the attribute value in
9 the MAD_Join primitive are removed.

10 39.2.4 Attribute value support requirements

11 Implementations of MIRP shall be capable of supporting all attribute values in the range of possible values
12 that can be registered using MIRP. An I-component shall be capable of maintaining current state information
13 for at least 4094 attributes in the range of possible values. A CBP implementation shall support the full
14 range of attribute values that are supported, in turn, by its Backbone Service Instance table.

15 39.2.5 MRP Message filtering

16 A MIRP Participant operates in Registration Fixed (New propagated) mode only. Therefore, any MIRP
17 Message for an attribute that is not a VIP-ISID (for an I-component) or a Local-SID or Backbone-SID (for a
18 CBP) is ignored by a MIRP Participant.

40. Edge Virtual Bridging (EVB)

Figure 40-1 provides an overview of the EVB architecture. An end station that supports the attachment of one or more virtual stations is said to be an EVB station. Each virtual station has at least one Virtual Station Interface (VSI). Each virtual station communicates with other virtual stations or other stations on the Bridged Network via the edge relay (ER) to which it is attached (see 3.85).

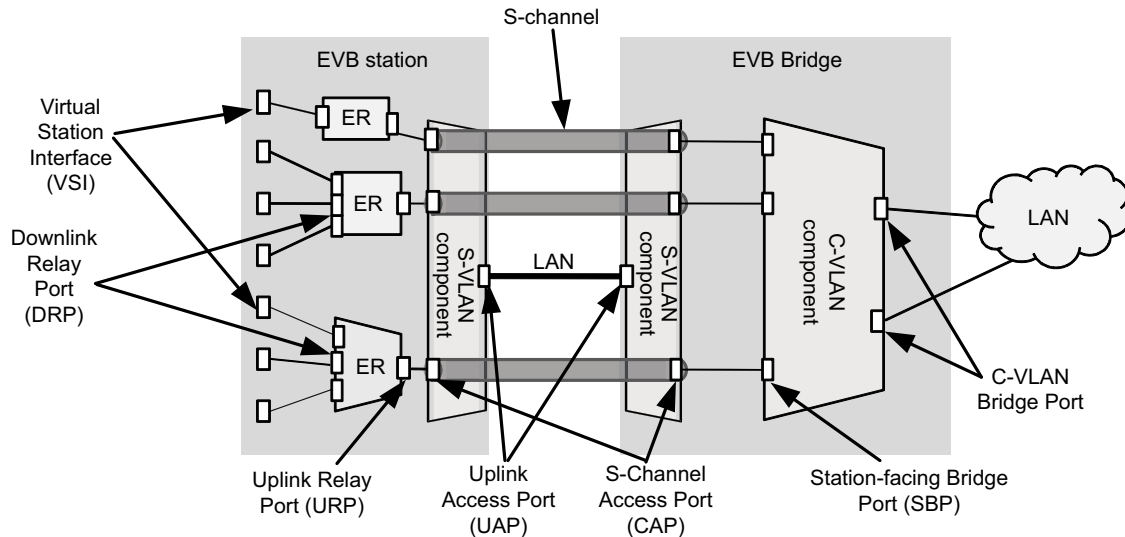


Figure 40-1—EVB architecture overview

An ER supports local relay among virtual stations and/or relay between a virtual station and other stations on the Bridged Network through an EVB Bridge. When forwarding of frames from one DRP to one or more other DRPs associated with the same ER (i.e., local relay) is not supported, then forwarding of traffic from one DRP to one or more other DRPs is performed by the EVB Bridge, utilizing reflective relay (8.6.1).

Connection between a DRP and a virtual station is achieved via a VSI. Traffic from a VSI traverses an internal LAN connecting the DRP to the virtual station. The operation of an ER does not result in any modifications to relayed frames over and above the normal tagging and untagging functions of a VLAN Bridge. ERs do not participate in, or affect, spanning tree operation; it is therefore necessary that the logical connectivity maintained within the station is always loop-free (5.24.1).

Figure 40-1 shows a 2-port ER within the EVB station; this illustrates the fact that even where a single VSI is supported by an S-channel, an ER is present in order to provide C-tagging, reserved address filtering (per Table 8-1) and support for the VSI Discovery and Configuration Protocol (VDP, Clause 41), the Edge Control Protocol (ECP), the EVB TLV (D.2.12), and LLDP (IEEE Std 802.1AB).

Each VSI instance is assigned a VSI manager ID, VSI Type Identifier (VTID), and VSI Instance Identifier (VSIID). VDP associates a VSI instance and its related VID(s), MAC address(es), GroupID(s), VSI manager ID, VTID,⁴⁹ and VSIID with an SBP. Similarly, VDP de-associates a VSI instance from an SBP.

VDP can also be used to associate a single VTID with, or de-associate a single VTID from the SBP. In this case, the VSI instance does not contain any MAC addresses, VIDs, or Group IDs, and uses the wildcard VID format (41.2.9.1). Only the most recent associate command is used to configure the VTID for the SBP.

⁴⁹ The meaning of the VTID is decided by local system and network management.

1 An ER supports relaying of frames associated with one or more VSIs. In order to achieve this, an ER can
2 support two types of operation. In the first type, referred to as virtual edge bridge (VEB), traffic transferred
3 from one DRP to another DRP of the same ER is forwarded directly by that ER. In the second type, referred
4 to as VEPA, traffic transferred from one DRP to another DRP of the same ER is forwarded onto a single
5 URP beyond the ER to the EVB Bridge. In this case, the EVB Bridge's SBP is enabled with reflective relay
6 (40.4, 8.6.1); this allows the frame to be reflected back to the same ER from which it was received by the
7 EVB Bridge. The ER can then forward the frame to the destination. Thus, in the second mode, all traffic
8 transits the EVB Bridge's SBP and is subject to, for example, filtering or policing behavior associated with
9 the EVB Bridge.

10 NOTE—Connection between an EVB Bridge and an ordinary end station takes place via a C-VLAN Bridge Port, not an
11 SBP.

12 An S-channel is a point-to-point S-VLAN that spans a pair of Port-mapping S-VLAN components (22.6.4)
13 and can be used to interconnect an ER and the C-VLAN component of an EVB Bridge. Multiple S-channels
14 can share the use of a LAN. The use of multiple S-channels allows the EVB station to support multiple ERs.
15 The endpoint of an S-channel is known as an S-channel Access Port (CAP); frames are S-tagged on entry to,
16 and are untagged on exit from, the S-VLAN component through a CAP.

17 EVB TLVs (D.2.12) exchanged via LLDP allow an EVB station and an EVB Bridge to exchange
18 information related to the use of reflective relay and other operational parameters. Each ER has an LLDP
19 database at its URP. Each ER can also have an LLDP database at each DRP.

20 Each URP and each SBP has an instance of ECP (Clause 43) used to support the VDP. These instances of
21 ECP use the Nearest Customer Bridge address as the destination for frames exchanged between the URP and
22 SBP. VDP TLVs are packed into PDUs that are handed to ECP for delivery. ECP provides reliable delivery
23 of VDP PDUs.

24 **40.1 EVB architecture without S-channels**

25 Figure 40-2 illustrates the relationship of the EVB entities to the Bridge architecture when no S-channels are
26 supported and no Port-mapping S-VLAN components are implemented. In this configuration, the EVB
27 station and EVB Bridge may exchange an S-channel Discovery and Configuration Protocol (CDCP) TLV
28 over the Nearest non-TMPR Bridge address indicating that the S-VLAN component is not present. If the
29 CDCP TLV managed object does not exist in the LLDP database, then the transmitting station or Bridge is
30 assumed to not support S-channels. A URP or an SBP can send a CDCP TLV. If the EVB station supports
31 the CDCP TLV then the nearest non-TMPR LLDP database is located at the URP. If the EVB Bridge
32 supports the CDCP TLV then the nearest non-TMPR LLDP database is located at the SBP.

1

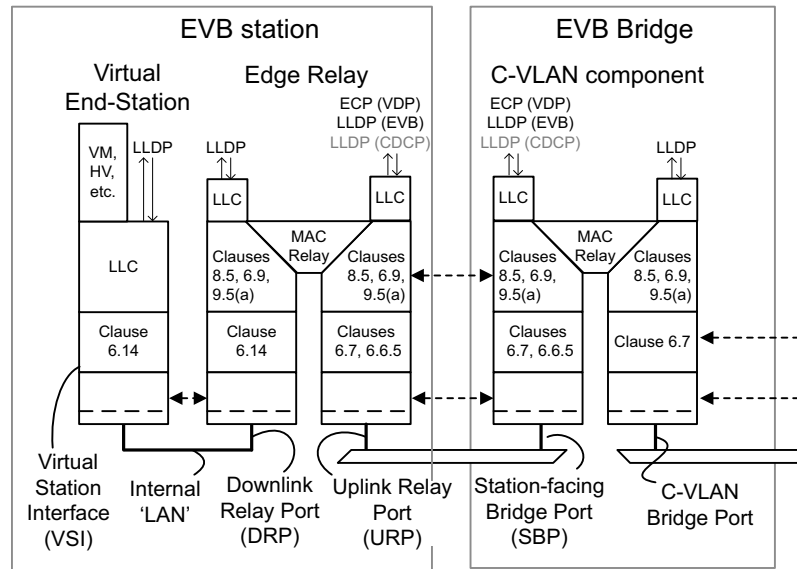


Figure 40-2—EVB architecture without S-channels

2 40.2 EVB architecture with S-channels

3 Figure 40-3 shows the relationship of the EVB entities to the Bridge architecture when S-channels are
4 supported. In this configuration, the EVB station and Bridge build nearest non-TPMR LLDP databases at
5 their Uplink Access Ports (UAPs) and use them to exchange CDCP TLVs. Both the EVB station and EVB
6 Bridge set the SComp parameter in the CDCP TLV to TRUE indicating they have an S-VLAN component.
7 The CDCP operating on the CDCP TLVs exchanged by LLDP is used to configure the S-channels.

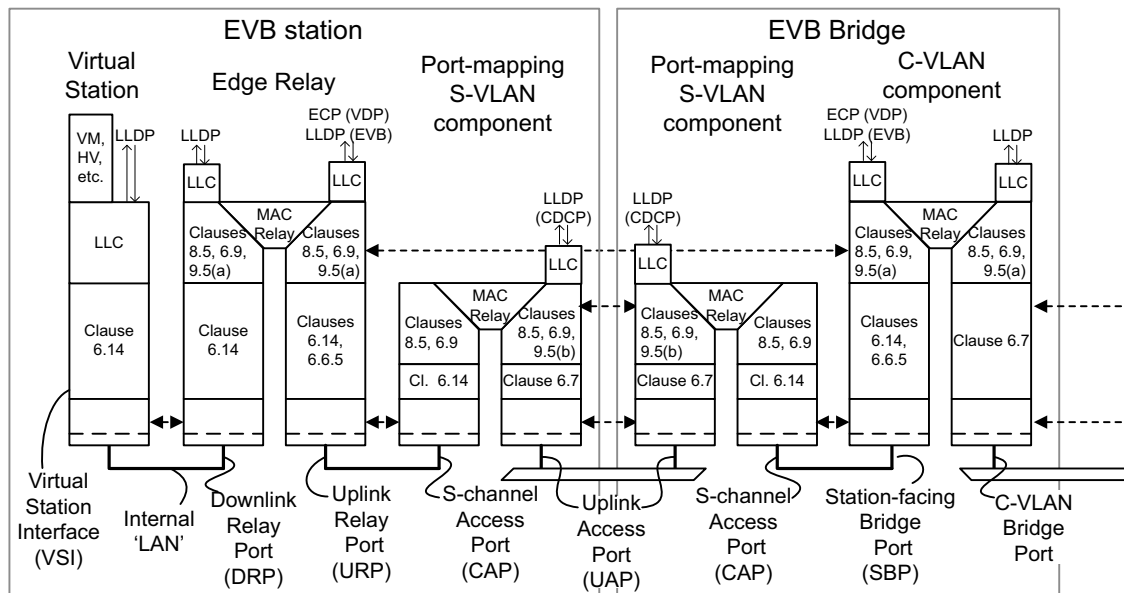


Figure 40-3—EVB architecture with S-channel

1 EVB stations and EVB Bridges use Port-mapping S-VLAN components (5.6, 22.6.4) to instantiate S-
2 channels. Each S-channel connects a CAP on the EVB station to a CAP on the EVB Bridge. The CAP on an
3 EVB station connects to a single URP on an ER via an internal LAN. The CAP on an EVB Bridge connects
4 to a single SBP on the C-VLAN component via an internal LAN. There is a 1:1 relationship between a CAP
5 and an SBP of the EVB Bridge, and a 1:1 relationship between a CAP and a URP of the EVB station. S-
6 channel support allows the EVB station and EVB Bridge to support multiple ERs on a LAN. Each S-channel
7 is associated with the URP of a distinct ER.

8 Figure 40-4 shows the relationship between S-channels and CAPs and the positioning of a station's internal
9 and external LANs. When S-channels are supported, each physical LAN can be used to support multiple S-
10 channels identified on the LAN by S-tagging. The S-channels are supported by one Port-mapping S-VLAN
11 component for each UAP. Each Port-mapping S-VLAN component within an EVB station can be identified
12 by its single UAP. A CAP is uniquely identified by the combination of the UAP of the S-VLAN component
13 and the S-VID of the S-channel. Each CAP attaches by internal C-tagged LANs to a single URP or SBP. The
14 C-VLANs carried over each S-channel are determined by configuration of the EVB station and of the C-
15 VLAN component within the EVB Bridge.

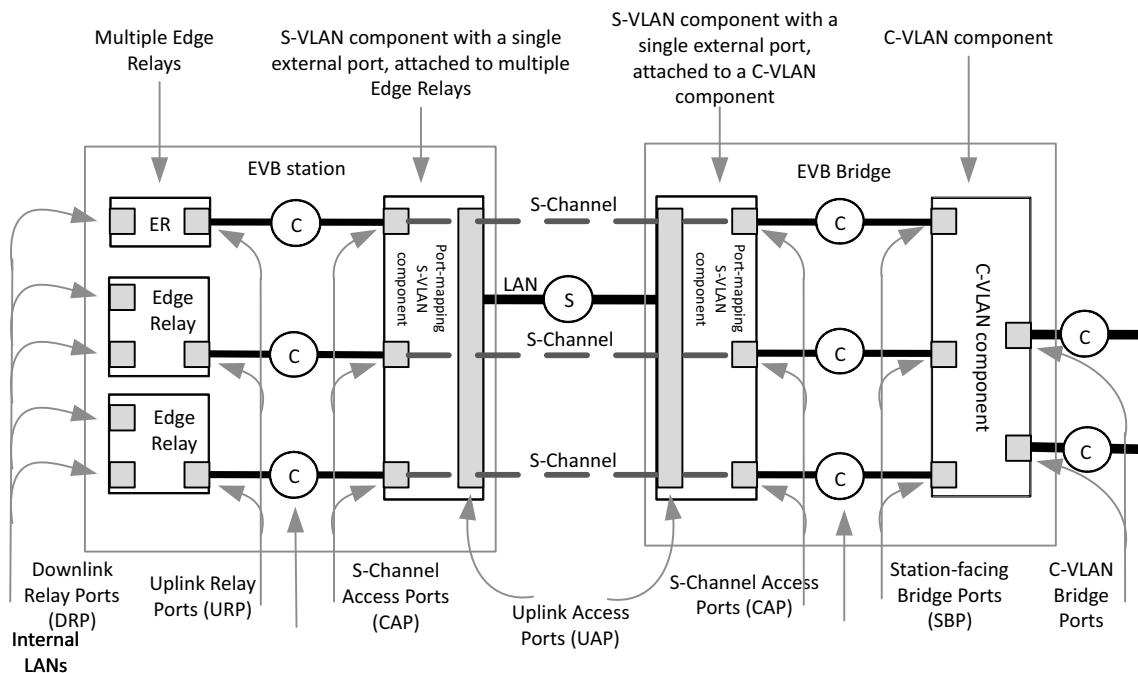


Figure 40-4—EVB components and internal LANs with S-channels

16 NOTE—As a result of normal Bridge behavior described in 6.9, the priority carried in the C-TAG is regenerated at the
17 CAP to form the S-TAG priority.

18 The C-VLAN component of the EVB Bridge is a standard C-VLAN component (5.5) that additionally
19 supports reflective relay (40.4, 8.6.1), the EVB LLDP TLV, and VDP.

1 When a UAP table entry (12.26.4.1) is created, a Port-mapping S-VLAN component is instantiated and the
2 following actions are taken automatically:

- 3 a) The UAP is configured to perform as follows:
 - 4 1) Admit all frames (6.9).
 - 5 2) Have a PVID parameter equal to the default S-channel S-VID (6.9, 40.3).
 - 6 3) Be included in the member set for the default S-channel S-VID (8.8.10).
 - 7 4) Be a member of the untagged set for the default S-channel S-VID (8.8.2).
 - 8 5) Be included in the member set for all S-VIDs of active S-channels.
- 9 b) An S-channel Interface table entry is created if one does not already exist for the default S-channel.
10 This table provides the equivalent functionality of the following:
 - 11 1) Creating a CAP for the default S-channel.
 - 12 2) Configuring the CAP to accept only un-S-tagged frames (6.9).
 - 13 3) Setting the member set for the default S-channel's S-VID to include the CAP.
 - 14 4) Setting the CAP's PVID to the default S-channel's S-VID.
 - 15 5) Adding the CAP to the default S-channel S-VID's untagged set.
 - 16 6) Setting filters on the CAP for the Nearest Bridge and Nearest non-TPMR Bridge group MAC
17 addresses.
 - 18 7) In the case of an EVB Bridge allocating (or creating) an SBP on the C-VLAN component
19 attached to the CAP by an internal LAN.
- 20 c) An instance of LLDP is started on the UAP transmitting a local database on the Nearest Non-TPMR
21 Bridge Address and including the CDCP TLV.
- 22 d) CDCP is started on the UAP and configured with the parameters specified when the UAP was
23 created.
 - 24 1) If the CDCP role is "B," then CDCP will wait for new S-channel creation requests. As new
25 requests are found, CDCP creates new S-channel interface table entries for each new S-channel
26 and deletes entries when S-channels are removed.
 - 27 2) If the CDCP role is "S," then CDCP uses the S-channel interface table to create the list of S-
28 channel Identifiers (SCIDs) for the S-channels it is requesting from the "B" side.

29 **40.3 Asymmetric EVB architecture without S-channels**

30 Figure 40-5 and Figure 40-6 illustrate the relationship of the EVB entities to the Bridge architecture when
31 S-channels are supported by only one side at a time; either the EVB Bridge or EVB station, but not both
32 simultaneously. In these configurations, the EVB entity with S-channel support will advertise it has an
33 S-VLAN component by building a nearest non-TPMR LLDP databases at its UAP and including the CDCP
34 TLV with the parameter SComp set to TRUE. The EVB entity without S-channel support may advertise a
35 Nearest non-TPMR Bridge LLDP database with the CDCP TLV indicating an SComp parameter set to
36 FALSE. CDCP is assumed not to be supported by the peer EVB entity until a CDCP TLV has been received.

37 Each Port-mapping S-VLAN component within an EVB entity supports an internal default S-channel
38 identified by S-VID 1 and uses it to pass untagged frames to its UAP. This default S-channel is always
39 present in the entity supporting an S-VLAN component. In the asymmetric configurations, frames from the
40 system without S-channel support are carried over the default S-channel within the system that has
41 S-channel support.

1

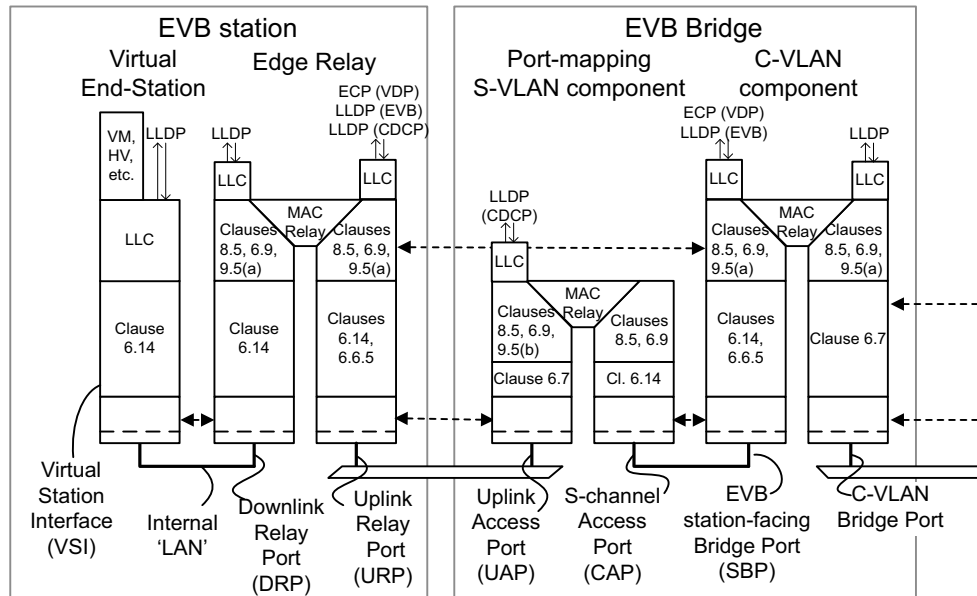


Figure 40-5—EVB architecture without S-channels, with EVB Bridge S-VLAN component

2

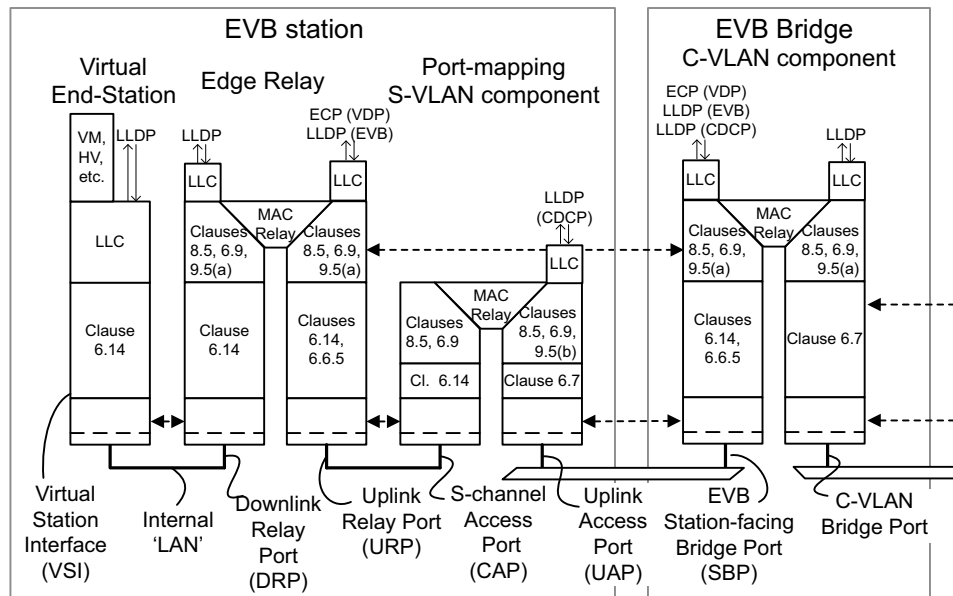


Figure 40-6—EVB architecture without S-channels, with EVB station S-VLAN component

1 40.4 EVB status parameters

2 In EVB Bridges and EVB Stations, an **EVBMode** parameter is associated with each port that provides EVB
3 functionality. The parameter represents the EVB status of the port.

4 The **EVBMode** parameter determines whether EVB functionality is supported, and in what mode. The
5 parameter can take one of the following values:

- 6 a) **EVB Bridge.** The port supports the functionality of an EVB Bridge.
- 7 b) **EVB station.** The port supports the functionality of an EVB station.
- 8 c) **Not Supported.** The port does not support EVB functionality. This value is assumed if the EVB
9 status parameters are not implemented.
- 10 d) **NVO3 Mode.** The port supports the revised EVB functionality to be used in an NVO3 Split-NVE
11 scenario.

12 40.4.1 EVBMode = Not supported

13 If the value of the **EVBMode** parameter is **Not Supported**, then no further status parameters are available,
14 EVB functionality is not supported, and the operation of the service associated with the port follows the
15 normal forwarding rules for a Bridge.

16 40.4.2 EVBMode = EVB Bridge

17 If the value of the **EVBMode** parameter is **EVB Bridge**, then further parameters are available, as follows:

- 18 a) **reflectiveRelayCapable.** If this parameter is TRUE, then the active topology enforcement function
19 is capable of performing reflective relay, as specified in 8.6.1; if FALSE, the active topology
20 enforcement function is not capable of performing reflective relay.
21 NOTE 1—The value of the reflectiveRelayCapable parameter is an inherent property of the implementation
22 and is not subject to administrative control.
- 23 b) **operReflectiveRelayControl.** If this parameter is TRUE, then reflective relay is enabled; if FALSE,
24 reflective relay is disabled.
25 NOTE 2—Reflective relay is enabled if a remote EVB station has requested that it be provided (as determined
26 by protocol exchanges between the EVB station and EVB Bridge) and the EVB Bridge is capable of providing
27 it, or disabled if the EVB station has not requested that it be provided or the EVB Bridge is not capable of
28 providing it.

29 40.4.3 EVBMode = EVB station

30 If the value of the **EVBMode** parameter is **EVB station**, then further parameters are available, as follows:

- 31 a) **adminReflectiveRelayRequest.** This parameter can take one of two values:
 - 32 1) **TRUE.** The attached EVB Bridge is requested to enable Reflective relay.
 - 33 2) **FALSE.** The attached EVB Bridge is requested to disable Reflective relay.
34 NOTE 1—The value of adminReflectiveRelayRequest is used in the EVB TLV exchanges described in D.2.12
35 to indicate to an attached EVB Bridge that the EVB station needs reflective relay to be provided. A given EVB
36 station is not required to support both possible values of adminReflectiveRelayRequest.
- 37 b) **operReflectiveRelayStatus.** This parameter can take one of three values:
 - 38 1) **TRUE.** The EVB Bridge has enabled reflective relay.
 - 39 2) **FALSE.** The EVB Bridge has disabled reflective relay.
 - 40 3) **Unknown.** It is not known whether the EVB Bridge has enabled reflective relay.
41 NOTE 2—The value of operReflectiveRelayStatus indicates whether the EVB Bridge has enabled reflective
42 relay, or whether the EVB Bridge status is not currently known, as determined by protocol exchanges between
43 the EVB station and EVB Bridge. The EVB Bridge status can be unknown during initialization or until the
44 protocol exchanges have completed.

1 40.4.4 EVBMode = NVO3 Mode

2 If the value of the **EVBMode** parameter is **NVO3 Mode**, then further parameters are available in 40.5.

3 40.5 EVB Status Parameter for NVO3 Mode Support

4 In an NVO3 Split-NVE scenario, tNVE is located in a station that may not be an EVB Station and nNVE is
5 located in a bridge or router that may not be an EVB Bridge. Hence tNVE and nNVE will be used to refer to
6 EVB Station and EVB Bridge revised to support NVO3.

7 The parameters of the EVB TLV provide the required parameter list. EVB TLV (Annex D) and status
8 parameters are amended to support NVO3.

9 When **EVBMode** is set to **NVO3 Mode**, an **NVERole** parameter is to be further inspected. The parameter
10 represents the NVE role of the port.

11 40.5.1 NVERole = nNVE

12 If the value of the **NVERole** parameter is **nNVE**, then further parameters are available, as follows:

13 a) **reflectiveRelayCapable**. This parameter is FALSE in NVO3 as tNVE has no requirement to
14 support reflective relay.

15 NOTE 1—The value of the reflectiveRelayCapable parameter is an inherent property of the implementation
16 and is not subject to administrative control.

17 b) **operReflectiveRelayControl**. If this parameter is TRUE, then reflective relay is enabled; if FALSE,
18 reflective relay is disabled. In an NVO3 scenario, it is always FALSE since tNVE will not request
19 reflective relay to be enabled.

20 NOTE 2—Reflective relay is enabled if a remote EVB station has requested that it be provided (as determined
21 by protocol exchanges between the EVB station and EVB Bridge) and the EVB Bridge is capable of providing
22 it, or disabled if the EVB station has not requested that it be provided or the EVB Bridge is not capable of
23 providing it.

24 c) **BGID**. This parameter is set to TRUE, which indicates that the nNVE wishes to control VID
25 assignments and use the GroupID in VDP exchanges.

26 NOTE 3—GroupID in VDP is equivalent to VNI ID in NVO3, which is a mandatory parameter to be supported.
27 Both bridge and station sides support it.

28 40.5.2 NVERole = tNVE

29 If the value of the **EVBMode** parameter is **tNVE**, then further parameters are available, as follows:

30 a) **adminReflectiveRelayRequest**. This parameter is set to FALSE, which indicates the attached
31 nNVE is requested to disable reflective relay.

32 b) **operReflectiveRelayStatus**. This parameter can be FALSE or Unknown.

33 NOTE 1—The value of operReflectiveRelayStatus indicates whether the nNVE has enabled reflective relay, or
34 whether the nNVE status is not currently known, as determined by protocol exchanges between the tNVE and
35 nNVE. The nNVE status can be unknown during initialization or until the protocol exchanges have completed.
36 However The nNVE status cannot be reflective relay enabled.

37 c) **SGID**. This parameter is set to TRUE, which indicates that the tNVE can support the use of the
38 GroupID.

39 NOTE 2—GroupID in VDP is equivalent to VNI ID in NVO3, which is a mandatory parameter to be supported.
40 Both bridge and station sides should support it.

41. VSI Discovery and Configuration Protocol (VDP)

VDP associates (registers) a VSI instance with an Station-facing Bridge Port (SBP) of an EVB Bridge. VDP simplifies and automates virtual station configuration by enabling the movement of a VSI instance (and its related VSI Type information) from one virtual station to another or from one EVB Bridge to another. VDP supports VSI discovery and configuration across a channel interconnecting an EVB station and an EVB Bridge. VDP TLVs are exchanged between the station and the Bridge in support of this protocol.

This clause defines the VDP TLV structure and state machines.

When VDP is used between EVB Station and EVB Bridge, VDP uses ECP (Clause 43) as a transport protocol for VDP TLV exchanges. When ECP is used as a transport protocol for VDP, ECP uses the Nearest Customer Bridge group MAC address (Table 8-1) as the destination address for ECPDUs. Three VDP TLVs are defined as follows:

- a) The VSI manager ID TLV (41.1). There is a single instance of this TLV in any ECPDU that carries VDP, and it appears as the first TLV in the ECPDU.
- b) The VDP association TLV (41.2). One or more of these TLVs can appear in any ECPDU, following the VSI manager ID TLV.
- c) The organizationally defined TLV (41.3).

NOTE 1—If there are multiple VSI managers, then their TLVs are transmitted in separate ECPDUs.

NOTE 2—Beyond the requirement stated, that the VSI manager ID TLV appears as the first TLV in ECPDUs carrying VDP, there are no further constraints placed upon how an implementation chooses to pack VDP TLVs into an ECPDU.

NOTE 3—VDP TLVs are not LLDP TLVs, and the TLV type values used in VDP TLVs are assigned from a distinct number space from those used in LLDP TLVs.

When VDP is used between tNVE and nNVE in an NVO3 Split-NVE scenario, VDP may use ECP (Clause 43) as a transport protocol for VDP TLV exchanges. When ECP is used as a transport protocol for VDP in NVO3, ECP uses a specific unicast MAC address or the Nearest Customer Bridge group MAC address (Table 8-1) as the destination address for ECPDUs. The VDP TLVs used between a tNVE and nNVE are as follows:

- d) The VDP association TLV (41.2). One or more of these TLVs can appear in any ECPDU or other transport protocol.
- e) The organizationally defined TLV (41.3).

41.1 VSI manager ID TLV definition

Figure 41-1 illustrates the format of the VSI manager ID TLV.

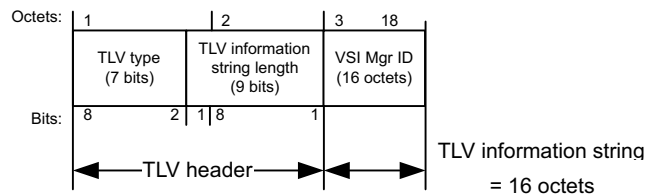


Figure 41-1—VSI manager ID TLV

The VSI manager ID TLV field definitions are contained in 41.1.1 through 41.1.3.

41.1.1 TLV type

The TLV type field takes the value shown in Table 41-1 for VSI manager ID.

Table 41-1—VDP TLV types

TLV type	Value
Pre-Associate	0x01
Pre-Associate with resource reservation	0x02
Associate	0x03
De-associate	0x04
VSI manager ID	0x05
Organizationally defined TLV	0x7F
Reserved for future standardization	0x00, 0x06–0x7E

41.1.2 TLV information string length

This field contains the length of the TLV information string, which is 16 octets.

41.1.3 VSI Manager ID

Identifies the database that should be accessed to get the VSI Type. The value 0 means that the station does not know what VSI Manager ID to use, indicating that the Bridge should select a default value. Any other value is interpreted as an IPv6 address, as defined in IETF RFC 4291.

41.2 VDP association TLV definitions

Figure 41-2 illustrates the format of the VDP association TLV.

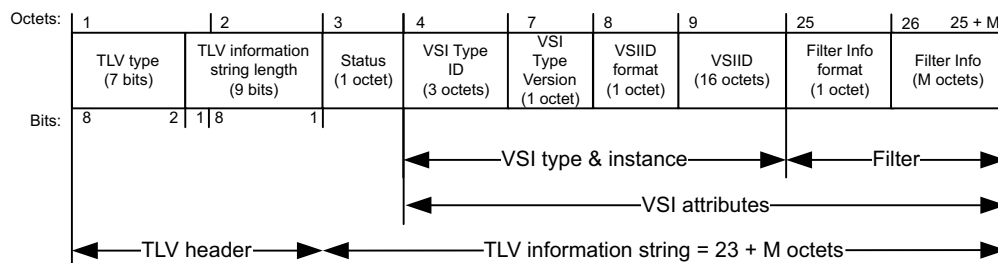


Figure 41-2—VDP association TLV

The VDP association TLV field definitions are contained in 41.2.1 through 41.2.9. The semantics of the VDP association TLV types are defined in 41.2.10.

When the VDP association TLV is sent as a response, the Status field indicates the outcome of the requested operation, and the remaining fields are populated using the information provided in the request or information provided by the EVB Bridge.

41.2.1 TLV type

The TLV type field identifies the type of the VDP association TLV, and can take any of the values shown in Table 41-1 for pre-associate, pre-associate with resource reservation, associate, or de-associate.

41.2.2 TLV information string length

This field contains the length of the TLV information string, calculated as $23 + M$ octets, where M is the number of octets in the filter info field (41.2.9).

41.2.3 Status

The Status field contains a 4-bit error type, encoded in bits 1–4, and four individual Boolean flags, encoded in bits 5–8.

For all requests, the error type field is reserved for future standardization; it is transmitted as 0x0 and is ignored on receipt.

For all requests, the Boolean flags are interpreted as shown in Table 41-2.

Table 41-2—Flag values in VDP requests

Name	Bit position	Interpretation															
M-bit	Bit 5	Indicates that the user of the VSI (e.g., the virtual station) is migrating (M-bit = 1) or provides no guidance on the migration of the user of the VSI (M-bit = 0). The M-bit is used as an indicator relative to the VSI to which the user is migrating.															
S-bit	Bit 6	Indicates that the VSI user (e.g., the virtual station) is suspended (S-bit = 1) or provides no guidance about whether the user of the VSI is suspended (S-bit = 0). A keep-alive Associate request with S-bit = 1 can be sent when the VSI user is suspended. The S-bit is used as an indicator relative to the VSI that the user is migrating from.															
Req/Ack	Bit 7	Set to 0 to indicate that the TLV contains a request.															
N-bit	Bit 8	Indicates that the user of the VSI is NOT migrating (N-bit = 1) or provides no guidance on the migration of the user of the VSI (N-bit = 0).															
NOTE—The M-bit is restored to 0 when migration has stopped, either because the migration has succeeded, or it has failed. The S-bit is restored to 0 when the VSI user is no longer suspended. NOTE 2—Interpretation of M and N bits is described in the following table: <table> <tr> <th>M bit</th><th>N bit</th><th>Interpretation</th></tr> <tr> <td>0</td><td>0</td><td>No guidance on the migration of the user of the VSI</td></tr> <tr> <td>0</td><td>1</td><td>User of the VSI (e.g., the virtual station) is NOT migrating</td></tr> <tr> <td>1</td><td>0</td><td>User of the VSI (e.g., the virtual station) is migrating</td></tr> <tr> <td>1</td><td>1</td><td>Reserved</td></tr> </table>			M bit	N bit	Interpretation	0	0	No guidance on the migration of the user of the VSI	0	1	User of the VSI (e.g., the virtual station) is NOT migrating	1	0	User of the VSI (e.g., the virtual station) is migrating	1	1	Reserved
M bit	N bit	Interpretation															
0	0	No guidance on the migration of the user of the VSI															
0	1	User of the VSI (e.g., the virtual station) is NOT migrating															
1	0	User of the VSI (e.g., the virtual station) is migrating															
1	1	Reserved															

For all responses, the value of the error type indicates the outcome of the request, as shown in Table 41-3, and the Boolean flags are interpreted as shown in Table 41-4.

Table 41-3—Error types in VDP responses

Name	Value	Interpretation
Success	0x0	The VDP Request was successfully completed by the Bridge.
Invalid Format	0x1	The VDP TLV format is invalid.
Insufficient Resources	0x2	The Bridge does not have enough resources to complete the VDP operation successfully.
Unable to contact VSI manager	0x3	The Bridge was unable to contact the VSI manager.
Other failure	0x4	The operation failed for some other reason.
Invalid VID, GroupID, or MAC address	0x5	The operation failed because the VID, GroupID, or MAC address was invalid.
Reserved	0x6–0xF	Reserved for future standardization.
NOTE—“Success” is only interpreted as success by the state machines if all of the flag bits (Table 41-4) are zero.		

Table 41-4—Flag values in VDP responses

Name	Bit position	Interpretation
Hard error	Bit 5	Set to 1 to indicate that the operation failed, and if the same operation is re-tried, it is likely to fail in the same way.
Keep	Bit 6	Set to 1 to indicate that the command was rejected and the state prior to the requested command has been kept.
Req/Ack	Bit 7	Set to 1 to indicate that the TLV contains a response.
Reserved	Bit 8	Reserved for future standardization.

1 41.2.4 VSI Type ID (VTID)

2 The VTID is an integer value used to identify a VSI Type.

3 NOTE—One VTID could describe the VSI Type configuration of multiple VSIs. A VTID is only unique per VSI
4 manager ID.

5 41.2.5 VSI Type Version

6 The VSI Type Version is an integer identifier that allows a VSI Manager Database to contain multiple
7 versions of a given VSI Type.

8 41.2.6 VSIID format

9 The VSIID format field defines the format of the VSIID field that follows it (41.2.7). The possible values of
10 VSIID format are as shown in Table 41-5.

Table 41-5—VSIID format values

Name	Description	Value
IPv4	An IPv4 address, encoded as specified in IETF RFC 4291.	0x01
IPv6	An IPv6 address, encoded as specified in IETF RFC 4291.	0x02
MAC	An IEEE 802 MAC address (6 octets), with 10 leading octets containing all zeros.	0x03
Local	The interpretation of the VSIID is locally defined.	0x04
UUID	A Universally Unique Identifier (UUID) as specified in IETF RFC 4122.	0x05
Reserved	Reserved for future standardization.	0x00, 0x06 through 0xFF

1 41.2.7 VSIID

2 The VSIID is an identifier for the VSI instance. A VSIID is generated when a VSI instance is created. The
3 VSIID remains constant during virtual station migration. The format of the VSIID is determined by the
4 VSIID format field (41.2.6). In cases where the format uses an identifier value that has fewer than 16 octets,
5 the VSIID field is packed out to 16 octets with leading octets containing zeros.

6 41.2.8 Filter Info format

7 The Filter Info format field determines the format of the Filter Info field (41.2.9). The Filter Info formats
8 defined by this standard are shown in Table 41-6.

Table 41-6—Filter Info format values

Format	Value
VID (41.2.9.1)	0x01
MAC/VID (41.2.9.2)	0x02
GroupID/VID (41.2.9.3)	0x03
GroupID/MAC/VID (41.2.9.4)	0x04
GroupID/VID/IPv4 (41.2.9.5)	0x05
GroupID/MAC/VID/IPv4 (41.2.9.6)	0x06
GroupID/VID/IPv6 (41.2.9.7)	0x07
GroupID/MAC/VID/IPv6 (41.2.9.8)	0x08
Reserved for future standardization	0x00, 0x09 through 0xFF

1 41.2.9 Filter Info field

2 The Filter Info field contains information from which a filter can be constructed. The filter is a set of VID
3 values or a set of MAC/VID values. The MAC address in a MAC/VID value is an individual MAC address.
4 The filter is applied to traffic transiting ports that do not have direct knowledge of the associated VSI, such
5 as an EVB SBP, in order to identify the traffic associated with a particular VSI. This allows such ports to
6 apply a VSI Type to the traffic of an individual VSI. Other devices that have direct knowledge of the traffic
7 associated with a VSI, for example devices that form a 1:1 relationship between a port and VSI, simply
8 provide this information via management interfaces.

9 The Filter Info field can also contain information that is not part of the filter. In particular, the Filter Info
10 field can contain GroupID values. Like the VID, the GroupID identifies a VLAN. When the number of
11 VLANs in the network is less than 4095, each VLAN can be assigned a VID value that is global within the
12 network. When the number of VLANs in the network exceeds 4094, a globally-scoped VID can no longer be
13 used to uniquely identify each VLAN. Instead, overlapping VIDs may be used in different regions of the
14 network, and a per-region mapping between the global VLAN and the region-specific VID is maintained. In
15 this case, the VLAN is uniquely and globally identified by a GroupID.

16 When VLANs are identified by GroupID, the station has knowledge of the GroupID but it does not, in
17 general, know the corresponding VID to be used by traffic associated with the VLAN. The Bridge is aware
18 of, or can obtain knowledge of, the VID associated with the specified GroupID. Thus, the station can send
19 GroupID values to the Bridge via the Filter Info field of the VDP Request. The Bridge can map GroupID
20 values to local VID values. The VID is included in the filter constructed by the Bridge and is returned with
21 its corresponding GroupID to the station via the VDP Response.

22 In the NVO3 Split-NVE scenario, the VNI ID is carried in the lower 3 octets of the GroupID. The upper octet of the
23 GroupID should be all zeros.

24 NOTE 1—The mechanism by which the EVB Bridge determines the GroupID to local VID associations is outside the
25 scope of this standard.

26 Additionally, the Filter Info field of a VDP TLV in a VDP Response can specify a Priority Code Point (PCP)
27 value associated with any, or all, of the VID values carried by that VDP Response. The PCP value, if
28 specified, is used by the EVB station as the default PCP value associated with the VSI and VID. The Filter
29 Info field contains a PCP Significant (PS) bit associated with each PCP field, indicating whether the PCP
30 field carries a PCP value (binary 1) or does not carry a PCP value (binary 0). If the PCP field carries a PCP
31 value, then the EVB station can adopt that value as the default PCP value associated with the VSI and VID.
32 When sending data frames associated with a given VSI and VID, the EVB station can determine the PCP
33 value associated with each frame by using an algorithm local to the EVB station. For example, the PCP
34 value can be based on the identity of an application associated with the frame as determined by examining
35 higher layer information. For any given frame, it is possible that the algorithm does not provide a specific
36 value of PCP. In such cases, the PCP field is assigned the value of the default PCP associated with the VSI
37 and VID.

38 NOTE 2—Specification of a PCP value in the VDP Response does not imply that all frames sent by the EVB station,
39 associated with the VSI and VID, carry the specified PCP. It implies only that, if the EVB station has no other
40 information regarding the PCP value that should appear in that particular frame, then the specified default PCP value is
41 used.

1 41.2.9.1 VID Filter Info format

2 The VID Filter Info format specifies that the Format Info field contains a set of VID values to be associated
3 with the VSI instance (41.2.7). Figure 41-3 illustrates the VID Filter Info format.

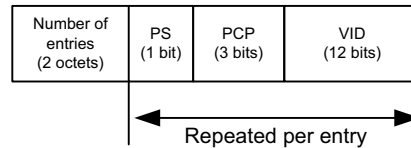


Figure 41-3—VID Filter Info format

4 The number of VID values in the sequence is specified by the Number of entries field.

5 The VID field can specify the null VID (see Table 9-2). When the null VID is specified, it is the only VID
6 specified in the Filter Info field (i.e., the Number of entries field is assigned the value 0x0001). Use of the
7 null VID indicates that the set of VID values associated with the VSI is supplied by the Bridge. The Bridge
8 can obtain VID values from the VSI Type whose identity is specified by the VSI Type information in the
9 VDP Request. The set of VID values is returned to the station via the VDP Response.

10 NOTE—In the case that more than one VID is assigned, the policy that determines how the VIDs are used is outside the
11 scope of this standard.

12 The Filter Info field can specify the wildcard VID (see Table 9-2). When the wildcard VID is specified, it is
13 the only VID specified in the Filter Info field (i.e., the Number of entries field is assigned the value 0x0001).
14 Use of the wildcard VID value indicates that the VSI Type specified by the VDP Request is designated as
15 the channel VSI Type applied to the EVB SBP associated with the S-channel.

16 41.2.9.2 MAC/VID Filter Info format

17 The MAC/VID Filter Info format indicates that the Format Info field specifies a sequence of MAC/VID
18 value pairs to be associated with the VSI instance (41.2.7). Figure 41-4 illustrates the MAC/VID Filter Info
19 format of the Filter Info field.

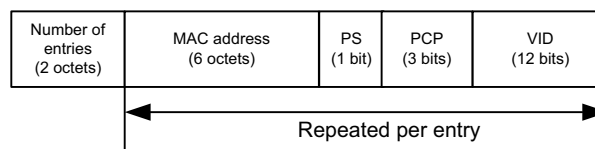


Figure 41-4—MAC/VID filter format

20 The number of MAC/VID pair values is specified by the field Number of Filter Info entries. Each MAC/
21 VID pair value carries a 6-octet individual MAC address and a 2-octet VID value.

22 The Filter Info field can specify the null VID for any entry. Use of the null VID indicates that the VID value
23 is supplied by the Bridge.

24 41.2.9.3 GroupID/VID Filter Info format

25 The GroupID/VID Filter Info format indicates that the Format Info field specifies a sequence of GroupID/
26 VID pairs to be associated with the VSI instance (41.2.7).

27 Figure 41-5 illustrates the GroupID/VID Filter Info format of the Filter Info field.

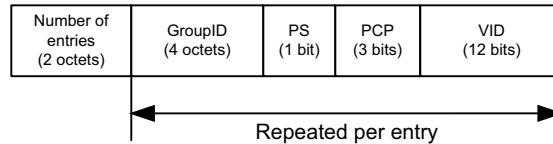


Figure 41-5—GroupID/VID filter format

- 1 The number of GroupID/VID pairs is specified by the Number of entries field.
- 2 The null VID (see Table 9-2) can be used in a GroupID/VID pair when the GroupID/VID filter format is
- 3 specified in the VDP Request. In this case, the Bridge is expected to supply the corresponding local VID
- 4 value in the VDP Response. For this purpose, the Bridge maintains, or has access to, the mapping between
- 5 GroupID and local VID.

6 41.2.9.4 GroupID/MAC/VID Filter Info format

- 7 The GroupID/MAC/VID Filter Info format indicates that the Filter Info field specifies a sequence of
- 8 GroupID/MAC/VID triples to be associated with the VSI instance (41.2.7). Figure 41-6 illustrates the
- 9 GroupID/MAC/VID Filter Info format of the Filter Info field.

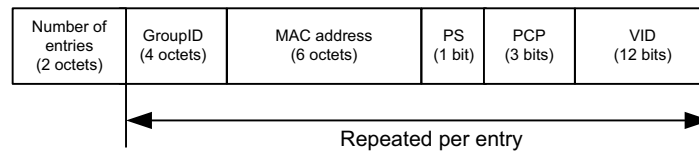


Figure 41-6—GroupID/MAC/VID filter format

- 10 The number of GroupID/MAC/VID triples is specified by the value of the Number of entries field. The null
- 11 VID (see Table 9-2) can be used in a GroupID/MAC/VID triple when the GroupID/MAC/VID filter format
- 12 is specified in the VDP Request. In this case, the Bridge is expected to supply the corresponding local VID
- 13 value in the VDP Response. For this purpose, the Bridge maintains, or has access to, the mapping between
- 14 GroupID and local VID.

15 41.2.9.5 GroupID/VID/IPv4 Filter Info format

- 16 The GroupID/VID/IPv4 Filter Info format indicates that the Format Info field specifies a sequence of
- 17 GroupID/VID/IPv4 triples to be associated with the VSI instance (41.2.7).

- 18 Figure 41-7 illustrates the GroupID/VID/IPv4 Filter Info format of the Filter Info field.

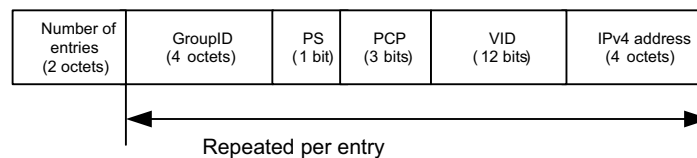


Figure 41-7—GroupID/VID/IPv4 filter format

1 The number of GroupID/VID/IPv4 triples is specified by the Number of entries field.

2 The null VID (see Table 9-2) can be used in a GroupID/VID/IPv4 triple when the GroupID/VID/IPv4 filter
3 format is specified in the VDP Request. In this case, the Bridge is expected to supply the corresponding local
4 VID value in the VDP Response. For this purpose, the Bridge maintains, or has access to, the mapping
5 between GroupID and local VID.

6 41.2.9.6 GroupID/MAC/VID/IPv4 Filter Info format

7 The GroupID/MAC/VID/IPv4 Filter Info format indicates that the Format Info field specifies a sequence of
8 GroupID/MAC/VID/IPv4 values to be associated with the VSI instance (41.2.7).

9 Figure 41-8 illustrates the GroupID/MAC/VID/IPv4 Filter Info format of the Filter Info field.

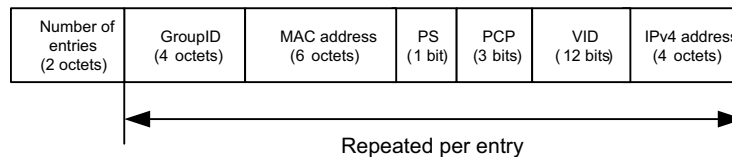


Figure 41-8—GroupID/MAC/VID/IPv4 filter format

10 The number of GroupID/MAC/VID/IPv4 values is specified by the Number of entries field. The null VID
11 (see Table 9-2) can be used in a GroupID/MAC/VID/IPv4 value when the GroupID/MAC/VID/IPv4 filter
12 format is specified in the VDP Request. In this case, the Bridge is expected to supply the corresponding local
13 VID value in the VDP Response. For this purpose, the Bridge maintains, or has access to, the mapping
14 between GroupID and local VID.

15 41.2.9.7 GroupID/VID/IPv6 Filter Info format

16 The GroupID/VID/IPv6 Filter Info format indicates that the Format Info field specifies a sequence of
17 GroupID/VID/IPv6 triples to be associated with the VSI instance (41.2.7).

18 Figure 41-9 illustrates the GroupID/VID/IPv6 Filter Info format of the Filter Info field.

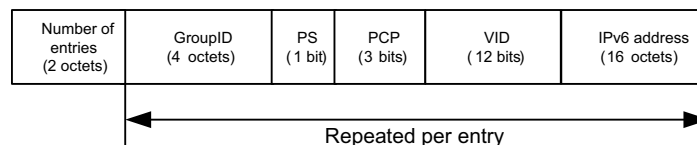


Figure 41-9—GroupID/VID/IPv6 filter format

19 The number of GroupID/VID/IPv6 triples is specified by the Number of entries field.

20 The null VID (see Table 9-2) can be used in a GroupID/VID/IPv6 triple when the GroupID/VID/IPv6 filter
21 format is specified in the VDP Request. In this case, the Bridge is expected to supply the corresponding local
22 VID value in the VDP Response. For this purpose, the Bridge maintains, or has access to, the mapping
23 between GroupID and local VID.

1 41.2.9.8 GroupID/MAC/VID/IPv6 Filter Info format

2 The GroupID/MAC/VID/IPv6 Filter Info format indicates that the Format Info field specifies a sequence of
3 GroupID/MAC/VID/IPv6 values to be associated with the VSI instance (41.2.7).

4 Figure 41-10 illustrates the GroupID/MAC/VID/IPv6 Filter Info format of the Filter Info field.

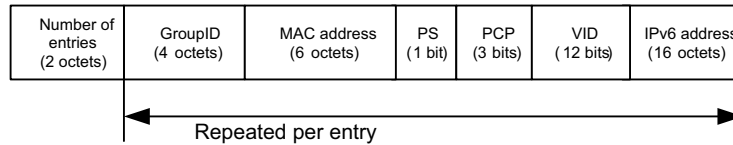


Figure 41-10—GroupID/MAC/VID/IPv6 filter format

5 The number of GroupID/MAC/VID/IPv6 values is specified by the Number of entries field. The null VID
6 (see Table 9-2) can be used in a GroupID/MAC/VID/IPv6 value when the GroupID/MAC/VID/IPv6 filter
7 format is specified in the VDP Request. In this case, the Bridge is expected to supply the corresponding local
8 VID value in the VDP Response. For this purpose, the Bridge maintains, or has access to, the mapping
9 between GroupID and local VID.

10 41.2.10 VDP TLV type and status semantics

11 The following subclauses define the semantics associated with each VDP TLV type.

12 41.2.10.1 Pre-Associate

13 The Pre-Associate TLV type is used to pre-associate a VSI instance with a Bridge Port. The Bridge validates
14 the request (see below) and returns a failure Status in case of errors. Successful pre-association does not
15 imply that the VSI Type will be applied to any traffic flowing through the VSI. The pre-associate enables
16 faster response to an associate by allowing the Bridge to obtain the VSI Type prior to an association.

17 The station can send a Pre-Associate TLV to roll back the station and bridge from associated to pre-associate
18 status.

19 NOTE—If the VSI Type changes without a corresponding change to its version, then inconsistent behavior can result.

20 41.2.10.2 Pre-Associate with Resource Reservation

21 Pre-Associate with Resource Reservation involves the same steps as Pre-Associate (41.2.10.1), but on
22 successful pre-association also reserves resources in the Bridge to prepare for a subsequent Associate
23 request.

24 The station can send Pre-Associate with a Resource Reservation TLV to roll back the station and bridge
25 from associated to pre-associate with Resource Reservation status.

26 41.2.10.3 Associate

27 The Associate TLV Type creates and activates an association between a VSI instance and a Bridge Port. The
28 Bridge allocates any required Bridge resources for the referenced VSI. The Bridge activates the
29 configuration for the VTID. This association is then applied to the traffic flow to/from the VSI instance.

30 NOTE—The mechanism used by a Bridge to determine the required resources associated with a VTID is outside the
31 scope of this standard.

1 For a given VSIID, a station may issue an Associate without having previously issued a Pre-Associate or
2 Pre-Associate with Resource Reservation. During normal operations a VSI instance is associated on only
3 one port. During network transitions (e.g., virtual station migration) a VSI instance might be associated with
4 more than one port.

5 If a Pre-Associate or a Pre-Associate with Resource Reservation had previously been received for a given
6 VSI instance, the Bridge establishes the association and allocates resources based only on the information
7 contained in the Associate TLV. Any resources that had been reserved in order to satisfy a previous Pre-
8 Associate with Resource Reservation, and that are not required in order to establish the association as
9 specified in the Associate, are released.

10 **41.2.10.4 De-Associate**

11 The de-associate TLV Type is used to remove an association between a VSI instance and a Bridge Port. Pre-
12 Associated and Associated VSIs can be de-associated. De-associate releases any resources that were
13 reserved as a result of prior Associate or Pre-Associate operations for that VSI instance.

14 A de-associate can be initiated either by the station or the Bridge. In the latter case, the Bridge sends a de-
15 associate TLV as if it was a response to a request from the station.

16 NOTE 1—A Bridge could, for example, issue a de-associate as a consequence of changes in the Bridge's status or
17 configuration.

18 NOTE 2—The result of the above semantics is that a de-associate can be initiated at any time and by either party.

19 **41.3 Organizationally defined TLV definitions**

20 Figure 41-11 illustrates the format of the organizationally defined TLV.

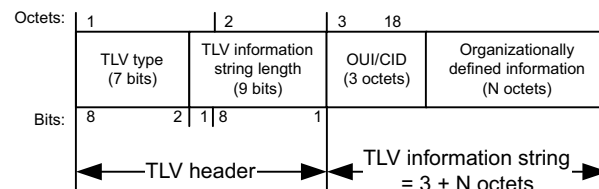


Figure 41-11—Organizationally defined TLV

21 The organizationally defined TLV field definitions are contained in 41.3.3 and 41.3.4.

22 **41.3.1 TLV type**

23 The TLV type field takes the value shown in Table 41-1 for the organizationally defined TLV.

24 **41.3.2 TLV information string length**

25 This field contains the length of the TLV information string, which is 3 + N octets, where N is the number of
26 octets in the organizationally defined information field (41.3.4).

27 **41.3.3 Organizationally unique identifier (OUI) or Company ID (CID)**

28 Identifies the organization that is responsible for defining the content of the organizationally defined
29 information field (41.3.4). The value of the OUI/CID field is an OUI (see IEEE Std 802) or a CID assigned
30 to that organization by the IEEE registration authority.

1 **41.3.4 Organizationally defined information**

2 The content and interpretation of this field is specified by the organization that owns the OUI or CID value
3 contained in the OUI/CID field (41.3.3).

4 **41.4 Validation rules for VDP TLVs**

5 The following rules apply to the validation of received ECPDUs that carry VDP TLVs:

- 6 a) If the first TLV in the ECPDU is not a VSI manager ID TLV (41.1), then the entire ECPDU is
7 discarded without further processing.
- 8 b) If the ECPDU contains a TLV of a type that is not recognized by the implementation, then that TLV
9 is discarded and is ignored by the VDP state machines.
- 10 c) If a TLV extends past the physical end of the ECPDU, then that TLV is discarded.

11 **41.5 VDP state machines**

12 The station VDP state machine is defined in 41.5.3. A station that supports VDP shall support one instance
13 of the station VDP state machine for each active VSI.

14 The Bridge VDP state machine is defined in 41.5.2. A Bridge that supports VDP shall support one instance
15 of the Bridge VDP state machine for each active VSI.

16 **41.5.1 State machine conventions**

17 The notational conventions used in the specification of VDP are as stated in Annex E.

1 41.5.2 Bridge VDP state machine

2 The Bridge VDP state machine shall implement the function defined in Figure 41-12 and the attendant
3 definitions in 41.5.4 through 41.5.7.

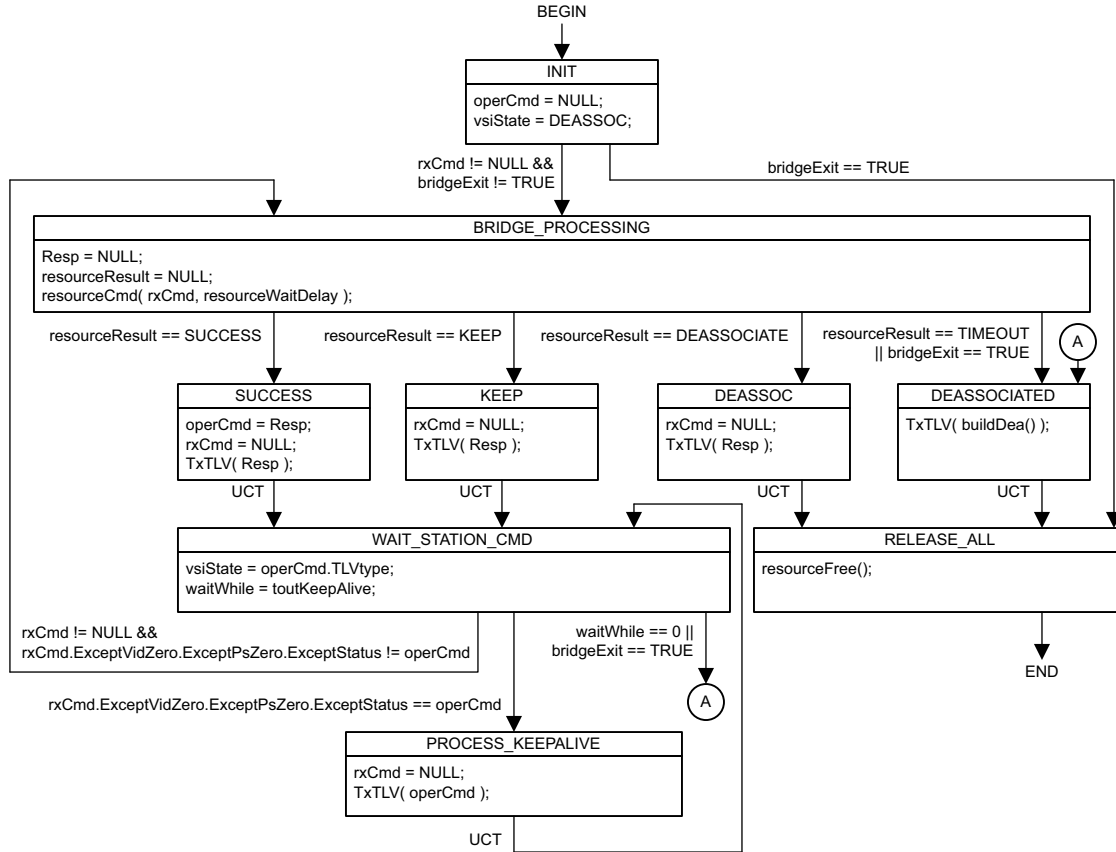


Figure 41-12—Bridge VDP state machine

2 The station VDP state machine shall implement the function defined in Figure 41-13 and the attendant
3 definitions in 41.5.4 through 41.5.7.

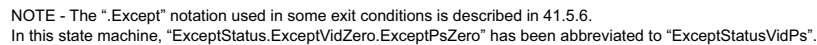


Figure 41-13—Station VDP state machine

1 **41.5.4 VDP state machine timers**

2 A set of timers is used by the VDP state machines; these operate as countdown timers (i.e., they expire when
3 their value reaches zero). These timers are 32-bit countdown timers. They:

- 4 a) Have a resolution of ten microseconds, with a tolerance of $\pm 20\%$.
- 5 b) Are started by loading an initial integer value, n , where $0 < n \leq 2^{31}$.
- 6 c) Are decremented by one per timer tick, as long as $n > 0$; the interval between timer ticks is the same
7 as the timer resolution.
- 8 d) Represent the remaining time in the period.

9 NOTE—Where timers are used in the VDP state machines to initiate keep-alive messages, it is recommended that a
10 small random component is added to the timer interval in order to avoid the possibility that timers associated with
11 different VSIs become synchronized.

12 **41.5.4.1 waitWhile**

13 An instance of the waitWhile timer exists for each instance of the station VDP state machine (41.5.3) and for
14 each instance of the Bridge VDP state machine (41.5.2).

15 **41.5.5 VDP state machine variables and parameters**

16 **41.5.5.1 bridgeExit**

17 A Boolean signal from the Bridge. When TRUE, this variable indicates that the Bridge VDP state machine
18 should exit.

19 **41.5.5.2 newCmd**

20 This variable is set to TRUE by the system to indicate to the state machine that there is a command ready to
21 be transmitted. The state machine sets newCmd FALSE when that command has been processed and is
22 ready to process a further command.

23 **41.5.5.3 NULL**

24 A null value. If NULL is assigned to a TLV variable, it indicates that the variable contains no TLV.

25 **41.5.5.4 operCmd**

26 The command TLV (the TLV that carried the current operating command—associate, pre-associate, or de-
27 associate), at the station or Bridge, or NULL if there is no current operating command.

28 **41.5.5.5 reinitKeepAlive**

29 The value used to initialize the waitWhile timer (41.5.4.1) by the station VDP state machine in order to
30 determine when to transmit a keep alive message. This value is derived from the value of the management
31 variable `urpVdpOperReinitKeepAlive` (12.26.5), which is type `timer_exp`, by using
32 `urpVdpOperReinitKeepAlive` as an exponent of 2. The variable `urpVdpOperReinitKeepAlive` is the larger of
33 the values proposed by the station and Bridge.

34 The default value used by the station for `urpVdpOperReinitKeepAlive` is an exponent value of 20,
35 representing a timer interval of about 10.5 s.

1 41.5.5.6 resourceCmdResult

2 This variable is used to record the result of a resourceCmd() procedure call (41.5.7.2). The possible result
3 values are as follows:

- 4 a) Success
- 5 b) timeOut
- 6 c) Fail (insufficient resources)
- 7 d) Fail (invalid format)
- 8 e) Fail (other)

9 41.5.5.7 resourceWaitDelay

10 The value used to initialize the waitWhile timer (41.5.4.1) by the Bridge VDP state machine when the state
11 machine is waiting for a response. This value is derived from the values of the management variable
12 sbpVdpOperRsrcWaitDelay (12.26.2), which is type timer exp, by using sbpVdpOperRsrcWaitDelay as an
13 exponent of 2. The variable sbpVdpOperRsrcWaitDelay is the larger of the values proposed by the EVB
14 station and EVB Bridge.

15 The default value used by the station and Bridge for sbpVdpOperRsrcWaitDelay is an exponent value of 20,
16 representing a timer interval of about 10.5 s.

17 41.5.5.8 Resp

18 A response TLV returned from the procedure resourceCmd() (41.5.7.2). Resp can be set to NULL prior to
19 issuing resourceCmd(). The variable is always non-NULL when resourceCmd() completes.

20 41.5.5.9 respWaitDelay

21 The value used to initialize the waitWhile timer (41.5.4.1) by the station VDP state machine when the state
22 machine is waiting for a response. This value is derived from the values of the management variables
23 urpVdpOperRsrcWaitDelay (12.26.5), ecpOperAckTimerInit (12.27) and ecpOperMaxRetries (12.27). The
24 value is expressed in units of VDP timer tics of 10 usec by the following equation:

$$25 \quad \text{respWaitDelay} = 1.5 \times (2^{\text{urpVdpOperRsrcWaitDelay}} + (2 + 2 \times \text{maxRetries} + 1) \times \text{ackTimerInit})$$

26 NOTE 1—The factor of 1.5 allows for a 20% tolerance in the timer values.

27 where maxRetries is the ECP state machine variable that is ecpVdpOperMaxRetries and ackTimerInit is the
28 ECP state machine variable that is 2ecpOperAckTimerInit. The values of urpVdpOperRscWaitDelay,
29 ecpOperAckTimerInit, and ecpOperMaxRetries that are used are the larger of the values proposed by the
30 station and Bridge.

31 The default value used by the station and Bridge is about 17.4 s. The default value for
32 urpVdpOperRscWaitDelay is an exponent of 20 representing a timer interval of about 10.5 s and for
33 ecpOperAckTimerInit is an exponent of 14 representing a timer interval of about 164 ms. The default value
34 of the ECP state machine variable ackTimerInit is 16,384 ECP/VDP timer tics. The default value for
35 ecpOperMaxRetries and the ECP state machine variable maxRetries is 3.

36 NOTE 2—The factor of 1.5 allows for a 20% tolerance in the timer values.

37 41.5.5.10 rxCmd

38 The last received command TLV, or NULL if no command TLV has been received. The rxCmd variable is
39 updated only if it is NULL.

1 41.5.5.11 rxResp

2 The last received response TLV. The rxResp variable contains the last received Resp TLV at the station or
3 NULL. RxResp is NULL if no TLVs have been received or if the variable has been cleared by the state
4 machine.

5 NOTE—It is possible to have a race condition when clearing rxResp since it can be updated asynchronously if the
6 Bridge issues an unsolicited DEASSOC command. If the race condition occurs, the de-associate will occur as a result of
7 the station timer expiring.

8 41.5.5.12 sysCmd

9 A command TLV from the system or hypervisor, or NULL if there is no pending command. The VDP state
10 machine is ready to accept a new command when the value of sysCmd is NULL.

11 41.5.5.13 toutKeepAlive

12 The value used to initialize the waitWhile timer (41.5.4.1) by the Bridge VDP state machine in order to
13 determine when to expect to receive a keep alive message. This variable is derived from the values of the
14 management variables sbpVdpOperReinitKeepAlive (12.26.2), ecpOperAckTimerInit (12.27), and
15 ecpOperMaxRetries (12.27). The value is expressed in units of VDP timer tics of 10 usec by the following
16 equation:

$$17 \quad \text{toutKeepAlive} = 1.5 \times (2^{\text{sbpVdpOperReinitKeepAlive}} + (2 + 2 \times \text{maxRetries} + 1) \times \text{ackTimerInit})$$

18 NOTE—The factor of 1.5 allows for a 20% tolerance in the timer values.

19 Where maxRetries is the ECP state machine variable that is ecpVdpOperMaxRetries and ackTimerInit is the
20 ECP state machine variable that is $2^{\text{ecpOperAckTimerInit}}$. The values of sbpVdpOperReinitKeepAlive,
21 ecpOperAckTimerInit, and ecpOperMaxRetries that are used are the larger of the values proposed by the
22 EVB station and EVB Bridge.

23 The default value used by the station and Bridge is about 17.4 s. The default for
24 sbpVdpOperReinitKeepAlive is an exponent of 20 representing a timer interval of about 10.5 s and for
25 ecpOperAckTimerInit is an exponent of 14 representing a timer interval of about 164 ms. The default value
26 of the ECP state machine variable ackTimerInit is 16,384 ECP/VDP timer tics. The default value for
27 ecpOperMaxRetries and the ECP state machine variable maxRetries is 3.

28 41.5.5.14 vsiState

29 The current association state of the VDP state machine. This variable may take the values DEASSOC (de-
30 associated), PREASSOC (pre-associated), PREASSOCR (pre-associated with resource reservation), or
31 ASSOC (associated).

32 41.5.6 Command-Response TLV field references in state machines

33 The state machines can make use of the value of individual fields within the value of a TLV by using the
34 following notation:

35 Tlv-variable-name.Field-name

36 In practice, only the following two Field-names are used in the state machines:

- 37 a) TLVtype, which references the TLV type field of the TLV (41.2.1)
- 38 b) Keep, which references the Keep bit of the Status field in the TLV (41.2.3)

1 So, for example, a reference in the state diagram to sysCmd.Keep is a reference to the Keep bit of the Status
2 field of the TLV value contained in the sysCmd variable.

3 The state machines also make use of the ability to compare TLV values for equality or inequality while
4 ignoring a specific field, using the following notation:

5 Tlv-variable-name.ExceptField-name

6 This is interpreted as meaning “The value of the TLV contained in the Tlv-variable-name variable, ignoring
7 the value of Field-Name in the comparison.”

8 The reserved Field-name “VidZero” is used to specify that if any VID in the Filter Info field (41.2.9.1)
9 contains zero, then that VID is ignored in the comparison. The reserved Field-name “PsZero” is used to
10 specify that if PS in the Filter Info field contains zero, then the PS and PCP are both ignored in the
11 comparison.

12 Multiple fields can be excepted by concatenating ExceptField-name items with a separating period.

13 The value of the Flag bits (Table 41-2 and Table 41-4) are always ignored in field comparison operations.

14 **41.5.7 VDP state machine procedures**

15 **41.5.7.1 buildDea()**

16 The buildDea() procedure builds a DEASSOCIATE TLV for the VSI as the return parameter.

17 **41.5.7.2 resourceCmd(rxCmd, delay)**

18 This procedure makes a resource request from the Bridge, waits for a response, builds a response TLV and
19 places it in the variable Resp. The response values reflect the requested resource action (PREASSOC,
20 ASSOC, or DEASSOC), conditioned by return variable resourceResult, which is set to NULL before calling
21 the resourceCmd procedure and set to one of the following values by the procedure:

- 22 a) SUCCESS
- 23 b) KEEP
- 24 c) DEASSOCIATE
- 25 d) TIMEOUT

26 The response constructed by the procedure in the Resp variable can be PREASSOC, ASSOC, or DEASSOC
27 with a Status, keep indicator and hard error indicator. For a successful completion the procedure will copy
28 the rxCmd parameter into the Resp variable. If the Bridge is selecting VIDs based on GroupIDs, then the
29 procedure also replaces zero VIDs with valid VIDs.

30 The delay parameter specifies how long the procedure should wait for a response. If the delay is exceeded,
31 no response is received, and the VSI is not associated, then the procedure returns a value of TIMEOUT. If
32 the delay is exceeded, no response is received and the VSI is associated, then the procedure returns a value
33 of KEEP along with a Resp equal to the rxCmd parameter. If the delay is not exceeded, then the procedure
34 returns SUCCESS, KEEP, or DEASSOCIATE depending on the response received along with the rxCmd in
35 the Resp and the Status set as follows:

36 e) **DEASSOC:**

37 The procedure returns DEASSOCIATE along with Resp.Status set to Success.

- 1 f) **PREASSOC:**
- 2 1) If the request can be satisfied, the procedure returns SUCCESS along with Resp.Status set to
- 3 Success.
- 4 2) If the request cannot be satisfied, the procedure returns DEASSOCIATE along with
- 5 Resp.Status set to a code other than Success and the Resp.hard set to TRUE if a retry will not
- 6 change the situation or FALSE if a retry might change the situation.
- 7 g) **PREASSOCR:**
- 8 1) If the request can be satisfied, and the resources requested are available and reserved for this
- 9 VSI, the procedure returns SUCCESS along with the Resp.Status set to Success.
- 10 2) If the request cannot be satisfied, or the resources are unavailable or not reserved for this VSI,
- 11 the procedure returns DEASSOCIATE along with the Resp.Status code other than Success and
- 12 the Resp.hard set to TRUE if a retry will not change the situation or FALSE if a retry might
- 13 change the situation.
- 14 h) **ASSOC:**
- 15 1) If the request can be satisfied, and the resources requested are available and enabled for this
- 16 VSI, the procedure returns SUCCESS along with the Resp.Status set to Success.
- 17 2) If the request cannot be satisfied, or the resources are unavailable or not reserved for this VSI,
- 18 and the VSI is not currently Associated, the procedure returns DEASSOCIATE along with the
- 19 Resp.Status code other than Success and the Resp.hard set to TRUE if a retry will not change
- 20 the situation or FALSE if a retry might change the situation.
- 21 3) If the request cannot be satisfied, or the resources are unavailable or not reserved for this VSI,
- 22 and the VSI is currently Associated, the procedure returns KEEP along with the Resp.Status
- 23 code other than Success and the Resp.hard set to TRUE if a retry will not change the situation
- 24 or FALSE if a retry might change the situation.

25 NOTE—A de-associate can happen at any time, initiated by either party to an association.

26 **41.5.7.3 resourceFree()**

27 The resourceFree() procedure frees all resources associated with this VSI.

28 **41.5.7.4 TxTlv(tlv)**

29 The TxTlv() procedure causes the TLV passed in the tlv parameter to be transmitted.

1 42. S-Channel Discovery and Configuration Protocol (CDCP)

2 This clause provides an overview, detailed semantics, and state machines for CDCP.

3 42.1 CDCP discovery and configuration

4 CDCP is used to configure S-channels (see 40.2). S-channels are implemented in stations and Bridges using
5 a Port-mapping S-VLAN component (22.6.4). Figure 40-1 illustrates the use of S-channels.

6 When the Port-mapping S-VLAN components used to create S-channels exist, they can exchange
7 un-S-tagged frames that are assigned to S-VID 1 and are considered to be assigned to the default S-channel,
8 which has an S-channel Identifier (SCID) of 1. The default S-channel is always un-S-tagged even when
9 S-channels are enabled.

10 NOTE 1—SCIDs are locally assigned identifiers.

11 The S-channel configuration is determined by the Bridge's capabilities and by requests made using CDCP
12 described in this clause. The station requests S-channels using CDCP. CDCP in turn uses an LLDP TLV
13 exchange to coordinate the creation and deletion of S-channels. The LLDP database used by CDCP is
14 addressed using the Nearest non-TPMR Bridge address. The Port-mapping S-VLAN component filters both
15 the Nearest Bridge and the Nearest non-TPMR Bridge addresses on all ports and passes the Nearest
16 Customer Bridge address.

17 The SCID value 1 and S-VID value 1 are always reserved for the exclusive use as the un-S-tagged default
18 S-channel. CDCP reports the default S-channel in the CDCP TLV as the first SCID, S-VID pair (i.e., <1,1>).
19 The Bridge shall not assign this S-VID except to the default S-channel.

20 NOTE 2—For the default S-channel, any QoS information that is necessary is extracted from the PCP bits of the C-tag.

21 42.2 CDCP state machine overview

22 CDCP requires each side of the configuration be assigned a role as a Bridge or a station. This is done by
23 setting the AdminRole variable. In most cases the station or Bridge role will not be settable, though the
24 protocol allows for systems that can take either role. For CDCP to configure an S-channel, one side takes the
25 station role and the other side takes the Bridge role. If both sides of the LAN have equipment configured as
26 stations or as Bridges the protocol will not configure S-channels.

27 NOTE—The role adopted by a given system can be fixed if the system is only capable of operating in a given role.

28 The CDCP state machine (Figure 42-1) operates on data contained in LLDP MIBs that is updated as a result
29 of the reception of CDCP TLVs, exchanged using LLDP operating on the Nearest non-TPMR Bridge
30 address. The structure of CDCP TLVs is defined in D.2.13.

31 The configuration proceeds by the Bridge providing the best match it can to the station's requested channels
32 and configuration. The station makes the resource request, the Bridge responds with its best matching
33 resources, the station then goes operational and reports its running configuration to the Bridge, and finally
34 the Bridge goes operational with the running configuration of the station.

35 In the event the station wishes to change its configuration it alters the request in its CDCP TLV and then
36 follows the same process as previously stated. If the 'B' loses its ability to support the current configuration
37 it can alter the current configuration in its CDCP TLV at which time the station drops down to the resources
38 supplied by the Bridge.

42.3 CDCP configuration state machine

The notational conventions used in the specification of CDCP are as stated in Annex E.

In an implementation that supports the station role, the CDCP configuration state machine shall implement the function specified by the state diagram in Figure 42-1 and the attendant definitions in 42.4 and 42.5.

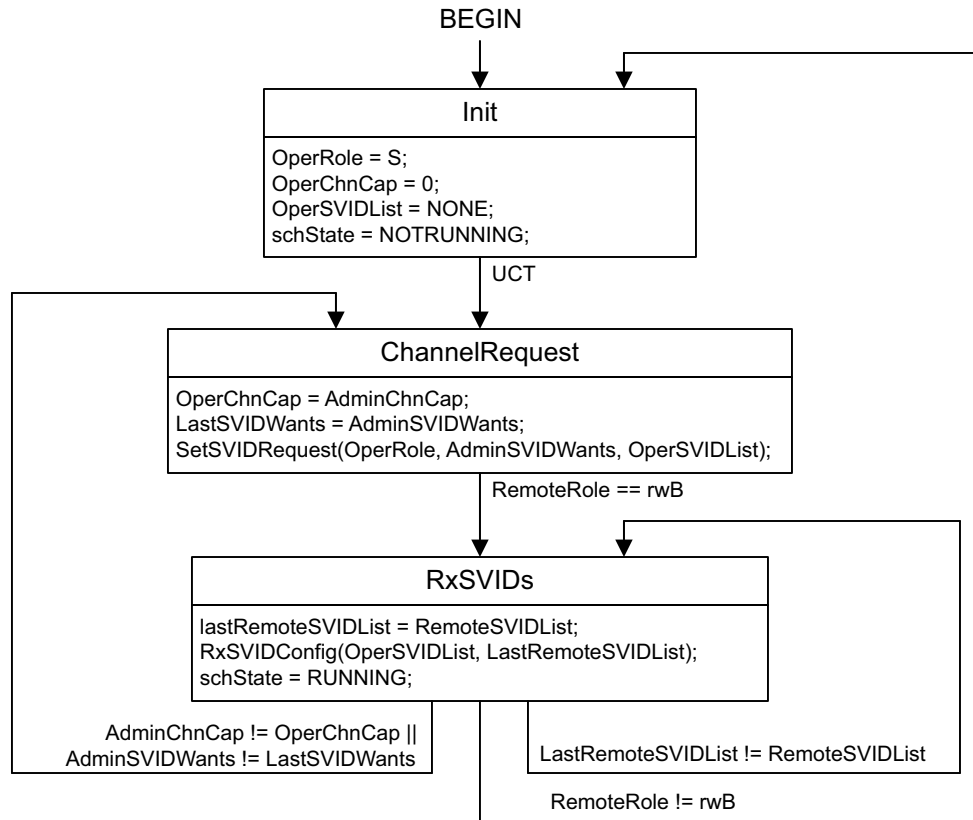


Figure 42-1—CDCP state machine—Station role

1 In an implementation that supports the Bridge role, the CDCP configuration state machine shall implement
2 the function specified by the state diagram in Figure 42-2 and the attendant definitions in 42.4 and 42.5.

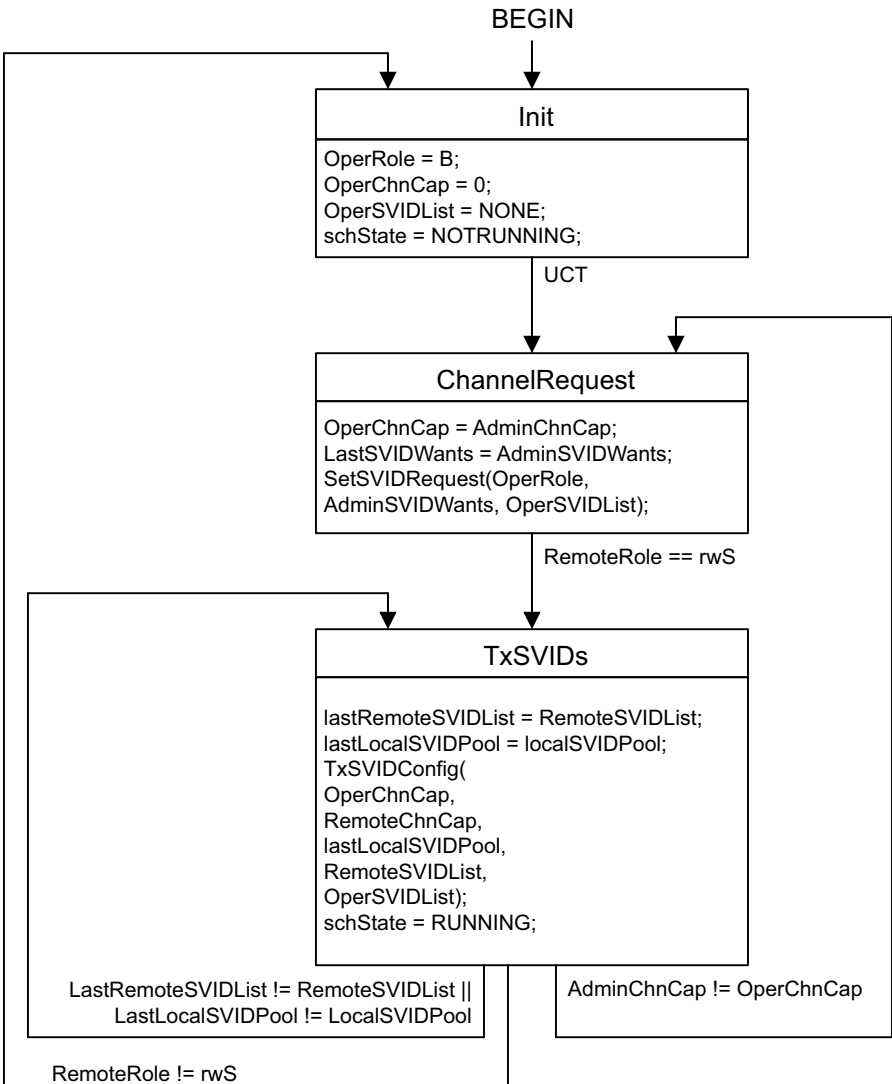


Figure 42-2—CDCP state machine—Bridge role

3 42.4 CDCP configuration variables

4 42.4.1 AdminChnCap

5 The administratively configured value for the Number of Channels supported parameter. This value is
6 included as the ChnCap parameter in the CDCP TLV.

1 42.4.2 AdminRole

2 The administratively configured value for the local port's role parameter. The value of AdminRole is not
3 reflected in the CDCP TLV. The AdminRole can take the value S or B; the value determines which of the
4 CDCP state machines is instantiated. If AdminRole is S, the Station role state machine (Figure 42-1) is
5 instantiated. If AdminRole is B, the Bridge role state machine (Figure 42-2) is instantiated. S indicates the
6 sender is unwilling to accept S-channels configuration (mode, number of channels supported, channel index)
7 from its neighbor and that the sender is willing to accept S-VID assignments from the neighbor. Stations
8 usually take the S role. B indicates the sender is willing to accept S-channels configuration (mode, number
9 of channels supported, channel index) from its neighbor and that the sender is willing do the best it can to fill
10 the S-VID requests from the neighbor. Bridges usually take the B role.

11 42.4.3 AdminSVIDWants

12 The administratively configured value for (SCID, S-VID) pairs wanted by a station; it is not used by a
13 Bridge. The first value is always the pair (1, 1) for the default S-channel assignment. The S-channel numbers
14 may be any valid number in the range 0–167. A 0 S-channel number indicates reserved space in the TLV. If
15 the S-VID value is 0 it means the station is requesting any available S-VID. S-VID value 1 is reserved for
16 exclusive use for the default S-channel S-VID. No other values of the SVID are permitted. The
17 AdminSVIDWants parameter is used to form the (SCID, S-VID) pairs in the CDCP TLV. This list is formed
18 from the EVB station's S-channel interface table (12.26.4) and is used to build the EVB Bridge's S-channel
19 interface table.

20 42.4.4 LastLocalSVIDPool

21 A temporary copy of the LocalSVIDPool.

22 42.4.5 LastRemoteSVIDList

23 Temporary local copy of the RemoteSVIDList. This variable is not included in the CDCP TLV. The
24 LastRemoteSVIDList has the same syntax as RemoteSVIDList.

25 42.4.6 LastSVIDWants

26 A local temporary copy of the AdminSVIDWants.

27 42.4.7 LocalSVIDPool

28 The set of S-VIDs and Bridge Ports available for S-channel assignment. These are determined by both
29 administrative resource assignments and by resource availability. The OperSVIDList for a B role is drawn
30 from the LocalSVIDPool.

31 42.4.8 OperChnCap

32 The current value for the ChnCap parameter. This value is included as the ChnCap parameter in the local
33 CDCP TLV. The range for this variable is 1–167.

34 42.4.9 OperRole

35 The current operational value of the Role parameter in the local port. This value is included as the Role
36 parameter in the CDCP TLV and may take values S or B as described for AdminRole.

1 **42.4.10 OperSVIDList**

2 The current value for (SCID, S-VID) assignments. This is the list of (SCID, S-VID) pairs included in the
3 local CDCP TLV. The total size of the list cannot exceed 167 pairs. The list always includes the default
4 S-channel pair (1,1). The valid range for each S-channel of this list is from 1–167. The valid range for each
5 S-VID in the list is from 0 to 0xffe. For the S role a S-VID of 0 indicates a request for a channel. For the B
6 role an S-VID of 0 indicates a non-configured channel.

7 **42.4.11 RemoteChnCap**

8 The current value for the ChnCap parameter. This value is included as the ChnCap parameter in the remote
9 CDCP TLV. NULL means no remote CDCP TLV exists in the local LLDP database. The range for this
10 variable is 1–167.

11 **42.4.12 RemoteRole**

12 Indicates the value in the remote CDCP TLV role field. rrNull indicates either the TLV was not present in the
13 last LLDPDU or that no LLDPDUs have been received. rwS and rwB indicate that the Role field was set in
14 the CDCP TLV received and that it had a value of S or B, respectively, as described for the AdminRole
15 variable.

16 **42.4.13 RemoteSVIDList**

17 The current value for (SCID, S-VID) assignments. This is the list of (SCID, S-VID) pairs included in the
18 remote CDCP TLV. NULL means no remote CDCP TLV exists in the local LLDP database. If the list is
19 empty but the CDCP TLV is present, its value is NONE. The total size of the list cannot exceed 167 pairs.
20 The valid range for each S-channel of this list is from 1–167. The valid range for each S-VID in the list is
21 from 0 to 0xffe. When the S-VID is value is 0 the S-VID is not configured. For the S role, a S-VID of 0
22 indicates a request for a channel. For the B role, an S-VID of 0 indicates a non-configured channel. The
23 RemoteSVIDList is reflected within the EVB Bridge in the S-channel interface table (12.26.4).

24 **42.4.14 schState**

25 The current running state of the S-channel. The values for this variable are NOTRUNNING or RUNNING.
26 This variable can be read using the management functionality defined in Clause 12.

27 **42.5 CDCP configuration procedures**

28 **42.5.1 SetSVIDRequest (OperRole, AdminSVIDWants, OperSVIDList)**

29 This function creates the OperSVIDList placed in the Local LLDP database, as follows:

- 30 a) If the OperRole for the equipment is B, then the OperSVIDList remains unchanged.
- 31 b) If the OperRole for the equipment is S, the function compares the AdminSVIDWants with the
32 OperSVIDList and amends the OperSVIDList, as follows:
 - 33 1) All active S-channels in the OperSVIDList that are in the AdminSVIDWants are kept active,
34 and in addition, any channels not currently in the OperSVIDList are requested by including
35 them in the OperSVIDList along with a 0 S-VID number. The OperSVIDList S-channel order
36 is changed to match the AdminSVIDWants.
 - 37 2) Any S-channels in the OperSVIDList that are not in AdminSVIDWants are made inactive and
38 are removed from the OperSVIDList.

1 42.5.2 RxSVIDConfig (OperSVIDList, LastRemoteSVIDList)

2 This function creates the OperSVIDList placed in the Local LLDP database for an S role port.

3 The function compares the AdminSVIDWants with the LastRemoteSVIDList. For each AdminSVIDWants
4 S-channel with an S-VID assignment in the LastRemoteSVIDList, a (SCID, S-VID) pair is generated in the
5 OperSVIDList. For each AdminSVIDWants S-channel without an S-VID assignment in the
6 LastRemoteSVIDList, a (SCID,0) pair is generated in the OperSVIDList. The OperSVIDList S-channel
7 order is set to match the AdminSVIDWants.

8 42.5.3 TxSVIDConfig (OperChnCap, RemoteChnCap, LastLocalSVIDPool, 9 RemoteSVIDList, OperSVIDList)

10 This function creates the OperSVIDList placed in the Local LLDP database for a B role port.

11 First the function takes the smaller of the OperChnCap and RemoteChnCap and truncates the
12 RemoteSVIDList to the smaller of the two.

13 A new OperSVIDList is created as follows:

- 14 a) For each S-channel in the RemoteSVIDList with a (SCID, S-VID) pair in the OperSVIDList, the
15 (SCID, S-VID) remains unchanged unless the S-VID is no longer part of the LastLocalSVIDPool. If
16 the S-VID is no longer in the pool, a new one is selected if available. If no S-VID is available, the
17 (SCID, S-VID) pair will be deleted from the OperSVIDList.
- 18 b) For each S-channel in the OperSVIDList without a (SCID, SVID) pair in the RemoteSVIDList, the
19 (SCID, SVID) pair will be deleted from the OperSVIDList.
- 20 c) For a (SCID, SVID) pair in the remote list, where the S-VID is zero, an S-VID is assigned if it is
21 available and the pair is inserted in the OperSVIDList. If an S-VID is not available, the pair is not
22 inserted in the OperSVIDList.
- 23 d) For a (SCID, SVID) pair in the remote list, where the S-VID is nonzero, there is no effect on the
24 OperSVIDList.

43. Edge Control Protocol (ECP)

This clause provides an overview, detailed semantics, and state machines for ECP.

43.1 ECP operation

Figure 43-1 depicts, at a high level, ECP operation. In step 1, the upper layer protocol (ULP) passes an outgoing ULP Data Unit (ULPDU) to ECP by invoking a transmit request procedure. In step 2, the ULPDU, which for some ULPs (e.g., VDP) may contain a set of ULP TLVs, is transmitted and an ECP low-level acknowledgment (ACK) timer is set. The frame is not yet deleted from the transmit buffer until an ACK (L-ACK in the diagram) is received for that ECPDU. In step 3, the arriving ECP frame is received into a receive buffer, where it is held until it is removed by an ECP indication procedure that passes the ULP Data Unit to the associated upper-level protocol. In step 4, when the receive buffer is emptied, a L-ACK is sent to the sender. In step 5, if the L-ACK is received before the L-ACK timer expires, then the transmit buffer is cleared and ECP can process another ULPDU through the ECP procedure. However, if the L-ACK timer expires before the L-ACK is received, then the frame in the transmit buffer is re-sent and the L-ACK timer is re-initialized. This timeout and re-sending can occur up to a maximum number of retries determined by the value of the *maxRetries* parameter of the transmit state machine. If this number of retries is reached and there is still no response, then the transmit buffer is cleared, a failure counter is incremented, and the transmit state machine is then ready to process another ULPDU. There is no indication to the ULP that a transmission failure has occurred; it is the ULP's responsibility both to detect the failure condition and to recover from it in an appropriate way.

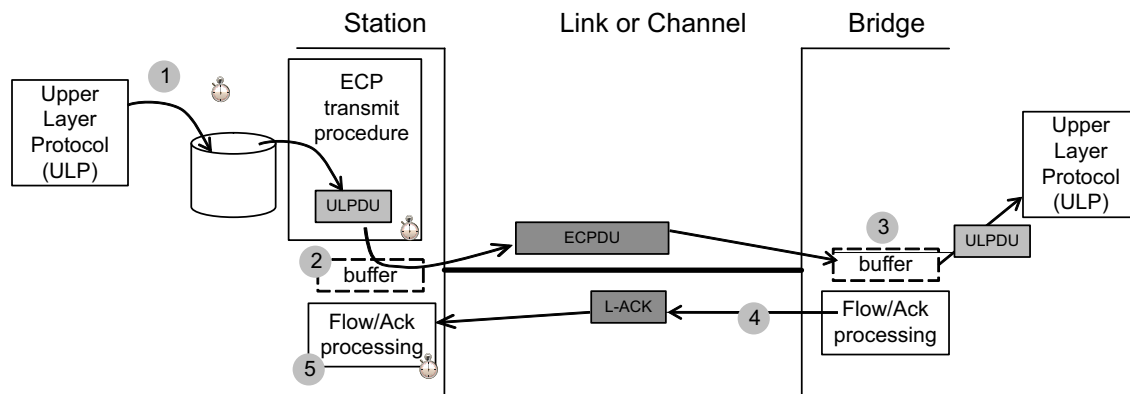


Figure 43-1—Example ECP exchange

ECP is intended to operate between two peers over an IEEE 802 LAN. ECP delivers the following service characteristics:

- Reliable delivery of ULPDUs, resilient against frame loss. The value of the *maxRetries* parameter determines the number of sequential lost frames that the protocol can sustain.
- Delivery of ULPDUs to the recipient ULP in the order that they were transmitted by the sending ULP.
- Delivery of a single copy of each ULPDU to the recipient.
- Flow control that provides protection against buffer overrun on the receive side.

1 43.2 Edge Control Sublayer Service (ECSS)

2 Two service primitives model the hand-off of data units between the ULP and ECP:
3 ECP_UNITDATA.request and ECP_UNITDATA.indication.

4 ECP_UNITDATA.request (ulptype, ulpdu)

5 The ECP_UNITDATA.request primitive is invoked by the ULP to notify ECP that a ULPDU is ready to be
6 transmitted. The **ulpdu** parameter is the ULPDU that the ULP wishes to transmit. The **ulptype** parameter
7 identifies the type of the ULP (see 43.3.3).

8 NOTE—For example, for VDP the ULPDU consists of a set of VDP TLVs passed to ECP for transmission. The
9 maximum size of the ULPDU, and therefore the set of TLVs that it can contain, is determined by the maximum SDU
10 size supported by the underlying MAC (see 6.5.8).

11 ECP_UNITDATA.indication (ulptype, ulpdu)

12 The ECP_UNITDATA.indication is invoked by ECP to indicate a ULPDU has been received and is
13 available for ULP processing. The **ulpdu** parameter is the ULPDU that has been received. The **ulptype**
14 parameter identifies the type of the ULP, as indicated in the received ECPDU (see 43.3.3).

15 43.3 ECP state machines

16 43.3.1 State machine conventions

17 The notational conventions used in the specification of ECP are as stated in Annex E.

18 43.3.2 Overview

19 There are two state machines used by each ECP instance: the ECP transmit state machine (43.3.4) and the
20 ECP receive state machine (43.3.5). A Bridge Port that supports ECP shall support one instance of the ECP
21 transmit state machine and one instance of the ECP receive state machine.

22 Initialization of the transmit and receive state machines occurs when portEnabled (43.3.7.5) is FALSE, or
23 when a BEGIN global event occurs. The transmit state machine transmits an ECPDU in response to an
24 ECP_UNITDATA.request from the ULP that indicates there is a PDU ready to be transmitted. The PDU is
25 transmitted with a sequence number that is used by the (remote) receive state machine in a responding ACK
26 ECPDU. If no ACK with the correct sequence number is received within a defined time period, and if the
27 maximum number of retries has not been reached, the transmit state machine retransmits the ECPDU. If the
28 maximum number of retries is exceeded, or if an ACK is received that matches the last sequence number
29 sent, then the transmit state machine increments the sequence number and waits for the next
30 ECP_UNITDATA.request.

31 NOTE—The sequence number for the first ECPDU transmitted after a state machine initialization (which occurs when
32 BEGIN is TRUE or portEnabled [43.3.7.5] is FALSE) is an implementation choice; for example, it could be a pre-
33 determined number, a random number, or it could continue the sequence from the last sequence number used.

34 When the first ECPDU is received following initialization, the receive state machine initializes its local
35 record of the last sequence number received to be one less than the sequence number in the received
36 ECPDU. This record of the last sequence number received allows the state machine to detect whether the
37 received ECPDU has been received already (current and last sequence numbers match) or this is a new
38 ECPDU (current and last sequence numbers differ). In both cases, the receive state machine sends an ACK
39 ECPDU, using the current sequence number. In the case that the received ECPDU is new, the last received
40 sequence number is updated to reflect the sequence number of the received ECPDU, and an
41 ECP_UNITDATA.indication is sent to the ULP to pass the contents of the ECPDU to the service user.

43.3.3 Edge Control Protocol Data Unit (ECPDU)

This subclause specifies the format of a ECPDU, along with the header that is added to and removed from ECP frames by the ECP function. The ECP header allows each ECPDU from the sender to be identified through a sequence number, which the receiver acknowledges by sending a ECP Acknowledgment frame. The format of the ECPDU is illustrated in Figure 43-2.

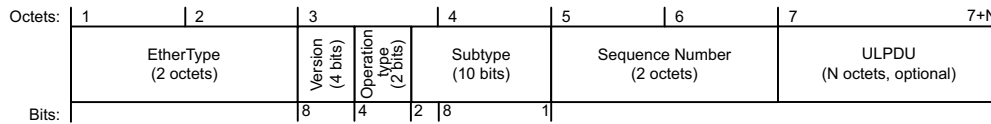


Figure 43-2—ECPDU structure

The destination address of the Ethernet frame that contains a ECPDU is specified by the ULP. The address used is either an individual MAC address or one of the reserved addresses specified in Table 8-1.

The source address shall be the individual MAC address of the sending station or port.

The fields of the ECPDU are defined in the following subclauses.

43.3.3.1 EtherType

A 16-bit field that contains the EtherType assigned for use by ECP (89-40).

43.3.3.2 Version

A 4-bit field that identifies the protocol version. The version shall be 0x01.

43.3.3.3 Operation type

A 2-bit field that identifies the operation type as follows:

- a) ECP request (0x0)
- b) ECP acknowledgment (0x1)

43.3.3.4 Subtype

A 10-bit field that defines the ULP type included in the PDU. For ACKs the subtype is ignored at the station.

The subtype used by VDP is as shown in Table 43-1.

Table 43-1—ECP subtypes

Use	Reference	Subtype
VDP	Clause 41	0x0001
Port Extender Control and StatusProtocol (PE CSP)	IEEE Std 802.1BR	0x0002
Link-local Registration Protocol (LRP)	IEEE Std 802.1CS	0x0003
Reserved for future standardization	—	All other values

43.3.3.5 Sequence Number

A 2-octet field that identifies the sequential order of the PDU, with respect to other ECPDUs. The starting sequence number can start anywhere for the first ECPDU, but the sequence number for each subsequent new request ECPDU is incremented by 1 modulo 65536.

NOTE—The sequence numbers used by each instance of the ECP transmit state machine are independent of each other.

43.3.3.6 ULPGU

This field contains an upper layer protocol data unit (ULPGU) if the operation type in the Mode field is ECP request; the field is absent if the operation type is ECP acknowledgment.

43.3.4 ECP transmit state machine

The ECP transmit state machine shall implement the function specified by the state diagram in Figure 43-3 and the attendant definitions in 43.3.6 through 43.3.8.

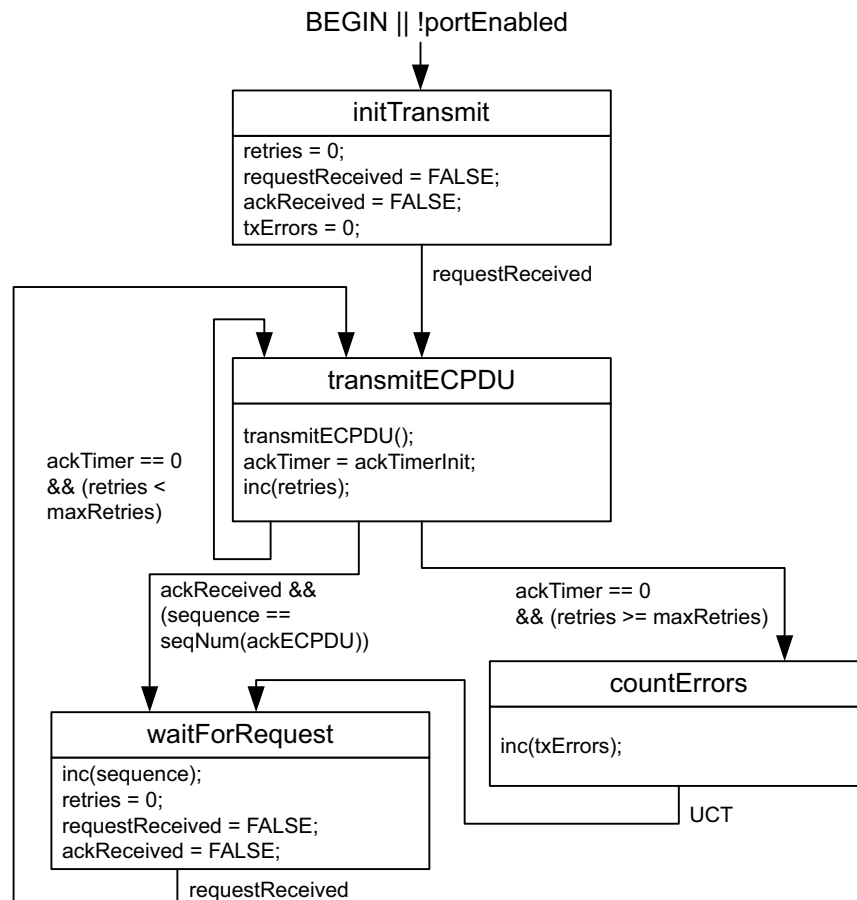


Figure 43-3—ECP transmit state machine

43.3.5 ECP receive state machine

The ECP receive state machine shall implement the function specified by the state diagram in Figure 43-4 and the attendant definitions in 43.3.6 through 43.3.8.

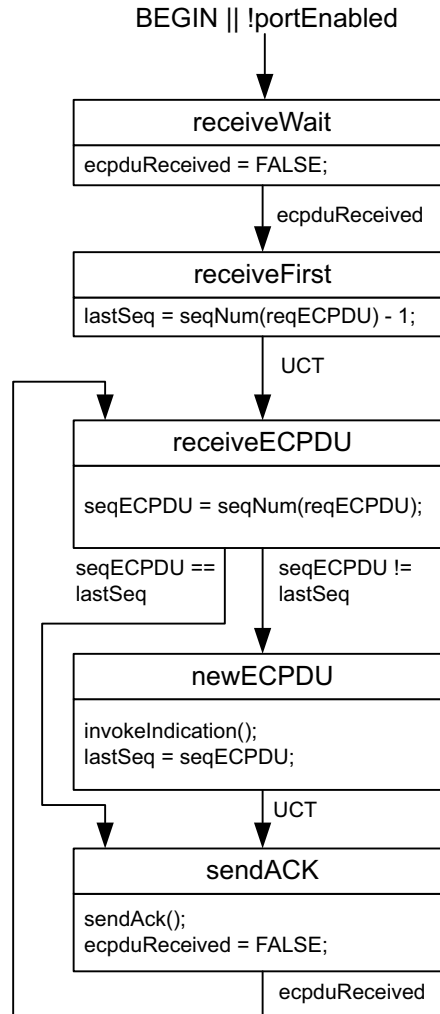


Figure 43-4—ECP receive state machine

43.3.6 ECP state machine timers

A set of timers is used by the ECP state machines; these operate as countdown timers (i.e., they expire when their value reaches zero). These timers are based on the Timer Exp data type (12.3). They:

- Have a resolution of ten microseconds, with a tolerance of $\pm 20\%$.
- Are started by loading an initial integer value, n , where $0 < n \leq 2^{31}$.
- Are decremented by one per timer tick, as long as $n > 0$; the interval between timer ticks is the same as the timer resolution.
- Represent the remaining time in the period.

1 **43.3.6.1 ackTimer**

2 The ackTimer is used to determine how long the transmit state machine will wait for an acknowledgment
3 PDU to be received before it either retries a transmission or aborts a transmission due to too many retries.
4 This timer is initialized using the value of ackTimerInit determined as stated in D.2.12.6.

5 **43.3.7 ECP state machine variables and parameters**

6 **43.3.7.1 ackReceived**

7 This Boolean variable is set to TRUE when an ECPDU is received with a MODE field indicating that the
8 PDU is an ACK. The variable is set FALSE by the ECP transmit state machine once the ACK has been
9 processed.

10 **43.3.7.2 ec pduReceived**

11 This Boolean variable is set to TRUE when an ECPDU is received with a MODE field indicating that the
12 PDU is a request. The variable is set FALSE by the ECP receive state machine once the request has been
13 processed and the ACK has been sent.

14 **43.3.7.3 lastSeq**

15 This integer variable is used to record the previous received sequence number.

16 **43.3.7.4 maxRetries**

17 This integer variable defines the maximum number of times that the ECP transmit state machine will retry a
18 transmission if no ACK is received. The default value of maxRetries is 3; this variable can be changed by
19 management as documented in 12.26.2. The value is derived from ecOperMaxRetries (Table 12-31).

20 **43.3.7.5 portEnabled**

21 This Boolean variable is set to the value of the MAC_Operational parameter (IEEE Std 802.1AC) for the
22 Port.

23 **43.3.7.6 requestReceived**

24 This Boolean variable is set to TRUE when a ULP issues an ECP_UNITDATA.request primitive. The
25 variable is set FALSE by the state machine once the request has been processed.

26 **43.3.7.7 retries**

27 This integer variable counts the number of transmission retries that have been made for the current ECPDU.

28 **43.3.7.8 seqECPDU**

29 This integer variable is used to record the sequence number contained in the most recent received request
30 ECPDU.

31 **43.3.7.9 sequence**

32 This integer variable is used to record the current sequence number that is used in transmitted request
33 ECPDUs.

1 **43.3.7.10 txErrors**

2 This integer variable is used to count the number of times that the ECP transmit state machine has
3 retransmitted an ECPDU.

4 **43.3.8 ECP state machine procedures**

5 **43.3.8.1 inc(counter)**

6 This procedure increments the counter variable by 1 modulo 65 536.

7 **43.3.8.2 transmitECPDU()**

8 This procedure causes an ECPDU to be transmitted, using the PDU structure defined in 43.3.3. The
9 sequence number field is set to the least significant 16 bits of the current sequence number contained in the
10 sequence variable (43.3.7.9). The mode field is set to ECP request. The ULPDU field is set to the value of
11 the ulpdu parameter of the request primitive. The subtype field is set to the value of the ulptype parameter of
12 the request primitive.

13 **43.3.8.3 invokeIndication()**

14 This procedure causes an ECP_UNITDATA.indication primitive to be invoked in order to pass the contents
15 of an incoming ECPDU to the ECP service user. The ulptype parameter carries the value of the ULP type
16 carried in the ECPDU. The ulpdu parameter carries the value of the ULPDU field of the ECPDU.

17 **43.3.8.4 sendAck()**

18 This procedure causes an ECPDU to be transmitted, using the PDU structure defined in 43.3.3. The
19 sequence number field is set to the least significant 16 bits of the sequence number contained in the
20 seqECPDU variable (43.3.7.8). The mode field is set to ECP acknowledgment. The ULPDU field is absent.
21 The subtype field is set to the value of the ulptype parameter of the received request ECPDU.

22 **43.3.8.5 seqNum(pdu-type)**

23 This procedure returns an integer value equal to the value of the most recently received request ECPDU
24 (pdu-type = reqECPDU) or acknowledgment ECPDU (pdu-type = ackECPDU).

25

1 44. Equal Cost Multiple Paths (ECMP)

2 SPB supports two approaches to spreading traffic load. The first approach, ECT VLANs, spreads traffic by
3 assigning each service instance to a VLAN configured to use one of the supported SPT Sets. This approach
4 is applicable to both SPBV and SPBM. Each Base VID is assigned to an ECT-ALGORITHM (tie-breaker)
5 that defines the SPT Set for that VLAN, and different ECT Algorithms may thus be used to select different
6 shortest paths in the SPT Region. In case of a PBBN, each I-SID is assigned to a Base VID. This approach
7 can spread traffic load while maintaining traffic path congruence in each VLAN.

8 The second approach, ECMP, spreads traffic by employing multiple equal cost paths with a single VID. This
9 approach is applicable only to SPBM. Unicast frame filtering dynamically selects a next hop from the set of
10 equal cost next hops for the frame's destination MAC address. Multicast frame filtering is controlled by
11 selecting a multicast routing mode and a tie-break mask for each backbone service instance and advertising
12 that selection in ISIS-SPB. Multicast filtering entries for the backbone service instance are generated by
13 calculations that select a multicast tree from the set of Equal Cost Trees (ECTs) in the SPT Region based on
14 the advertised multicast mode and tie-break mask.

15 44.1 SPBM ECMP

16 44.1.1 ECMP Operation

17 SPBM ECMP provides a mechanism to spread traffic load across a set of equal cost shortest paths in an SPT
18 Region using a single VID. Rather than using one equal cost tie-breaking function to create a single SPT for
19 all individual addresses, ECMP uses a hash function to spread individual address forwarding over a set of
20 equal cost forwarding ports.

21 If flow filtering (44.2) is not supported, the ECMP ECT Algorithm selects one port from the set of equal cost
22 forwarding ports for each individual address (44.1.2) and creates a Dynamic Filtering Entry for the address
23 indicating the selected port.

24 If flow filtering (44.2) is supported, a set of equal cost forwarding ports may be recorded in the FDB in a
25 Dynamic Filtering Entry (8.8.3). During unicast frame filtering the MAC Relay Entity selects one
26 forwarding port from the applicable set using a dynamic port selection process (44.2.3).

27 The individual address forwarding port selection processes, both with and without flow filtering, are
28 designed to:

- 29 a) Spread traffic (i.e., make varied choices) for different flows traversing the SPT Region, and
- 30 b) Maintain frame ordering for each flow by making the same choice for all frames belonging to any
31 single flow.

32 ECMP is supported by Dynamic Filtering Entries in the FDB and, optionally, the flow filtering tag (F-TAG,
33 44.2.1) that carries Flow Hash and TTL information. The flow hash serves both to distinguish frames
34 belonging to different flows and to identify frames that belong to (or may belong to) the same flow. The flow
35 hash is used as input to the dynamic port selection process to support the design goals noted above. The TTL
36 is used to mitigate traffic loops, if they happen to occur (44.2.4).

37 While unicast frames are spread across equal cost paths, multicast frames cannot be treated this way. Instead
38 multicast traffic load is spread by assigning a multicast mode and tie-breaker to each group address. This
39 selection enables the use of varied SPTs for each group address or set of group addresses. The default ECMP
40 ECT Algorithm selects a source rooted SPT for each node that may transmit multicast traffic. This default
41 source rooted SPT is used for all group addresses assigned to that node. Finer grained control of multicast
42 load spreading is done by selecting a multicast mode and tie-break mask for a particular I-SID. This

1 selection governs the multicast routing used for that I-SID's assigned group address. The selectable
2 multicast modes include the following:

- 3 c) Source rooted SPT
- 4 d) Shared tree
- 5 e) Head-end replication

6 Up to 16 source rooted SPTs can be supported per node by using one of 16 different tie-break masks in
7 calculating the SPT. During SPT calculation the tie-break mask is used to select one from the set of equal
8 cost parents (neighbor nodes closer to the root) for each node (28.9). The default multicast treatment is
9 equivalent to using a tie-break mask of zero. An I-SID on a CBP can be assigned to a specific SPT by
10 configuring the Tie-Break Mask for that I-SID [item h) in 12.25.6.1.2 and item h) in 28.12.10].

11 To reduce the amount of multicast forwarding state required, shared (spanning) trees are supported. Up to 16
12 shared trees can be configured using the 16 tie-break masks. For shared trees the tie-break mask is used both
13 to select a shared tree root Bridge and to select from the set of equal cost parents for each node in the tree.
14 The shared tree root for a given tie-break mask is determined by selecting the best (least) masked Bridge ID
15 (i.e., the Bridge ID value masked using the tie-break mask). The SPT calculation from the shared tree root
16 also uses the tie-break mask to select between equal cost parents (again selecting the least masked Bridge
17 ID). Each I-SID on a CBP can be assigned to a shared tree by setting a shared tree transmit indicator bit and
18 configuring the tie-break mask to identify the shared tree (28.12.10). All CBPs for a given I-SID that
19 transmit on a shared tree use the same group address (Figure 27-6), and share the forwarding state for this
20 address in the FDB.

21 It is possible to avoid using group addresses altogether and handle multicast frames by head-end replication.
22 An I-SID on a CBP using head-end replication sends a unicast frame to each destination instead of sending a
23 group addressed frame. If an I-SID is using head-end replication it is configured to indicate no multicast
24 transmission from that CBP. In this case no group address forwarding state will be installed for that I-SID's
25 assigned address at that CBP (28.9). An I-SID using head-end replication at a CBP should nevertheless
26 indicate that it wishes to receive multicast frames since other CBPs supporting the same I-SID may still use
27 multicast.

28 NOTE—ISIS-SPB does not check for the use of the same tree type for all endpoints of an I-SID, but use of different tree
29 types within a single I-SID is not recommended.

30 44.1.2 ECMP ECT Algorithm

31 The content of FDB entries for individual addresses in ECMP depends on whether flow filtering (44.2) is
32 supported by a Bridge.

33 If flow filtering is not supported or the ECT-ALGORITHM value indicates no flow filtering (Table 44-1),
34 then for each individual address the ECMP ECT Algorithm selects one port from the set of equal cost
35 forwarding ports using a form of the FNV non-cryptographic hash algorithm and creates a Dynamic
36 Filtering Entry (8.8.3) indicating that port as forwarding. The forwarding port is selected by the following
37 process:

38 For a set of k equal cost forwarding Ports, let P be an ordered list of the forwarding Ports sorted from
39 greatest to least neighbor SPB System Identifier and P_i be the i th Port in P . Thus P_0 is the Port whose
40 neighbor has the greatest SPB System Identifier and P_{k-1} is the Port whose neighbor has the least SPB
41 System Identifier. The following algorithm is used to select a value for i for the individual MAC address,
42 thus selecting Port P_i to be the forwarding port:

- 43 a) Set a 32-bit variable *bestHash32* to 0xFFFFFFFF.
- 44 b) Set i to 0 (zero).

- 1 c) For each n in $0..k-1$, (in order) perform the following steps:
 - 2 1) Set a 32-bit variable $hash32$ to $0x811C9DC5$.
 - 3 2) For each octet O_j (extended to 32 bits with leading 0's) in the Bridge's SPB System Identifier (8.13.8), starting with the least significant octet (octet 5) and progressing to the most significant octet (octet 0), set $hash32$ to $((hash32 \text{ XOR } O_j) \times 0x01000193)$.
 - 4 3) For each octet O_j (extended to 32 bits with leading 0's) in the neighbor SPB System Identifier for Port P_n , starting with the least significant octet (octet 5) and progressing to the most significant octet (octet 0), set $hash32$ to $((hash32 \text{ XOR } O_j) \times 0x01000193)$.
 - 5 4) For each octet O_j (extended to 32 bits with leading 0's) in the individual MAC address, starting with the least significant octet (octet 5) and progressing to the most significant octet (octet 0), set $hash32$ to $((hash32 \text{ XOR } O_j) \times 0x01000193)$.
 - 6 5) If $hash32 < bestHash32$, set i to n and set $bestHash32$ to $hash32$.

13 If flow filtering (44.2) is supported and the ECT-ALGORITHM value indicates flow filtering (Table 44-1),
14 ISIS-SPB configures the Flow Filtering Control Table (44.2.2) for each port on which the ECMP SPBM
15 VID is registered (27.13). For CBPs the Base VID is configured with both Flow Filtering and Flow Hash
16 Generation enabled. For PNPs the Base VID is configured with Flow Filtering enabled and Flow Hash
17 Generation disabled. For each individual address the ECMP ECT Algorithm calculates a Port Map
18 specifying forwarding for a set of equal cost hops to that address and creates a corresponding Dynamic
19 Filtering Entry (8.8.3). These Dynamic Filtering Entries are used in the dynamic selection of forwarding
20 ports by the flow filtering process (44.2.3).

21 For each I-SID on a CBP transmitting multicast frames, the ECMP ECT Algorithm may calculate a source
22 rooted SPT. If equal cost paths are found to any node on the tree, the path whose parent node (neighbor node
23 toward the root) has the least Bridge Identifier is selected. A group MAC Address Registration Entry (8.8.4)
24 is created for the I-SID's source rooted tree SPBM group MAC address (Figure 27-3) if the node performing
25 the calculation is located between the source node and any multicast receiver(s) for the I-SID. The group
26 MAC Registration Entry indicates forwarding for each port that leads to at least one multicast receiver for
27 the I-SID on the calculated SPT (except the port from the source).

28 The ECMP ECT Algorithm supports two additional parameters that control multicast forwarding state
29 calculation. First, a Tie-Break Mask value may be specified for an I-SID on a CBP [item i) in 28.12.10 and
30 item h) in 12.25.6.1.2]. When a nonzero Tie-Break Mask is advertised, a 64-bit mask (formed by repeating
31 the 4-bit Tie-Break Mask value in each nibble) is applied to (XORed with) each Bridge Identifier before
32 selecting the lowest value as described in the SPT calculation in the paragraph above. Thus the default
33 behavior described above is equivalent to using a Tie-Break Mask of zero. This parameter enables up to 16
34 different SPTs to be calculated from a given source node. Each I-SID on a CBP may be assigned
35 independently to one of these 16 source trees.

36 The second parameter controlling multicast forwarding state calculation is the shared tree indicator. An
37 I-SID on a CBP may be configured to use a shared tree instead of a source-specific SPT [item h) in 28.12.10
38 and item g) in 12.25.6.1.2]. There may be up to 16 shared trees, corresponding to the 16 Tie-Break Mask
39 values. Each shared tree is calculated by selecting a root Bridge using the least masked Bridge Identifier. If a
40 node advertises a Bridge Priority for the given Tie-Break Mask (28.12.6.1) the advertised value is used
41 instead of the normal Bridge Priority in forming its Bridge Identifier (before masking). In calculating the
42 SPT from the selected root, the same Tie-Break Mask is used to form masked Bridge Identifiers (using the
43 normal Bridge Priority), and the least value of masked Bridge Identifier is used to choose between equal cost
44 parents for any node. If an I-SID on a CBP is configured to use a shared tree, the group address used in the
45 group MAC Address Registration Entry is constructed using the Tie-Break Mask and the I-SID (27.15,
46 Figure 27-6). In this case the ECMP ECT Algorithm creates a group MAC Address Registration Entry if the
47 node performing the calculation is between the CBP and any multicast receivers on the SPT calculated from
48 the selected root node. The Port Map indicates forwarding for all Ports that lead from a multicast transmitter
49 to at least one multicast receiver for the I-SID.

1 If an I-SID on a CBP does not indicate transmission of multicast frames (e.g., head-end replication is to be
2 used instead), then no group MAC Address Registration Entry is configured for multicast from that CBP for
3 that I-SID.

4 Table 44-1 shows the ECT-ALGORITHM values for the ECMP ECT Algorithm.

Table 44-1—ECMP ECT-ALGORITHM values

ECT-ALGORITHM	ECMP behavior
00-80-C2-11	ECMP without flow filtering
00-80-C2-12	ECMP with flow filtering

5 **44.1.3 Loop prevention for ECMP**

6 Loop prevention for unicast ECMP operates as specified in Clause 13 with the extension of allowing
7 multiple Root Ports in case of multiple equal cost paths in order to support the per frame tie-breaking of
8 ECMP. The ECMP connectivity to a destination bridge comprises the shortest paths destined to that bridge
9 when the topology is Agreed in the meaning of the Agreement Protocol. If a bridge is not the destination
10 bridge, then the one or more ports that ISIS-SPB has calculated as respectively providing a shortest path to
11 the destination is a Root Port in the ECMP connectivity. Each port, other than a Root Port, is a Designated
12 Port if ISIS-SPB has calculated that the port provides a shortest path for frames forwarded from the attached
13 bridge to the destination in the ECMP connectivity. With these extensions to the definitions of Port Roles,
14 Clause 13 specifications apply to ECMP as well.

15 NOTE 1—Despite allowing more than one Root Port for an SPT Bridge, a single Root Port is enforced on a per-frame
16 basis as ECMP operation ensures that a frame is forwarded through only one of the Root Ports towards the destination.

17 Loop prevention for the shared trees used for ECMP multicast operates as specified by Clause 13 for shared
18 spanning trees except for that instead of a Spanning Tree Protocol Entity (STPE), it is ISIS-SPB that
19 configures the per-tree Port States and Roles based on its calculations and the operation of the Agreement
20 Protocol; furthermore, the Learning Process (8.7) neither creates nor deletes Dynamic Filtering Entries.

21 NOTE 2—The per-tree Port States and Roles do not belong to an MSTI having a distinct MSTID because an ECMP B-
22 VID is assigned to the SPBM MSTI. The Port States and Roles are associated with a single SPT used as a shared
23 spanning tree supporting bidirectional forwarding (not as a unidirectional tree as for SPB).

24 **44.2 Support for Flow Filtering**

25 Flow filtering support enables Bridges to distinguish frames belonging to different client flows and to use
26 this information in the forwarding process. Information related to client flows may be used at the boundary
27 of an SPT Domain to generate a flow hash value. The flow hash, carried in an F-TAG, serves to distinguish
28 frames belonging to different flows and can be used in the forwarding process to distribute frames over
29 equal cost paths. This provides for finer granularity load spreading while maintaining frame order for each
30 client flow.

31 Flow filtering behavior is controlled by ISIS-SPB. To allow Bridges that support flow filtering to be used
32 alongside Bridges that do not support flow filtering, ISIS-SPB requires that all Bridges at the SPT Domain
33 boundary that advertise I-SIDs for an ECMP Base VID using flow filtering support F-TAG processing.
34 Bridges that do not advertise I-SIDs for that Base VID are not required to support flow filtering. Therefore,

1 network operators are able to upgrade selected bridges to support flow filtering and use flow filtering for
2 selected VIDs without requiring the simultaneous upgrade of all bridges in the SPT Domain.

3 NOTE—Support for F-TAG processing is indicated in I-SID advertisements and so is only available via ISIS-SPB once
4 I-SIDs are assigned to the Base-VID. A “dummy” ISID configuration can be used to discover via ISIS-SPB whether
5 F-TAG processing is supported on a Bridge.

6 This clause specifies the following:

- 7 a) A flow filtering tag (F-TAG) that carries a flow hash value (44.2.1).
- 8 b) F-TAG processing on Bridge Ports (44.2.2).
- 9 c) A forwarding process using the flow hash to select one forwarding port from a set of equal cost
10 options (44.2.3).
- 11 d) TTL loop mitigation (44.2.4).
- 12 e) CFM for ECMP with flow filtering (44.2.5).
- 13 f) Operation of bridged LANs with selective support for flow filtering (44.2.6).

14 44.2.1 Flow filtering tag (F-TAG)

15 An F-TAG is specified to carry additional information in support of flow filtering functions. The F-TAG
16 comprises a Tag Protocol Identifier (TPID) (EtherType) and Tag Control Information (TCI).

Table 44-2—F-TAG EtherType

Tag Type	Name	Value
Flow Filtering Tag	IEEE 802.1Q Flow Filtering Tag EtherType	89-4B

17 The F-TAG TCI field (Figure 44-1) is 4 octets in length and encodes priority, drop_eligible, TTL, and flow
18 hash information for a service primitive.

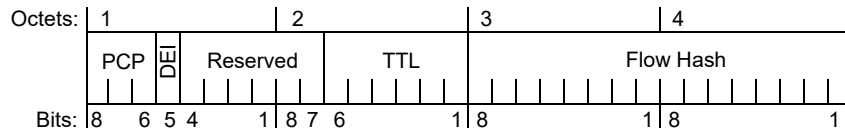


Figure 44-1—Flow Filtering TCI format

19 The F-TAG TCI contains the following fields:

- 20 a) *PCP (Priority Code Point)*—This 3-bit field encodes the priority and drop_eligible parameters of
21 the service request primitive associated with this frame using the same encoding as specified for
22 VLAN tags in 6.9.3.
- 23 b) *DEI (Drop Eligible Indicator)*—This 1-bit field carries the drop_eligible parameter of the service
24 request primitive associated with this frame.
- 25 c) *Reserved*—This 6-bit field may be used for future format variations. The Reserved field contains a
26 value of zero when the tag is encoded and is ignored when the tag is decoded.
- 27 d) *TTL (Time To Live)*—This 6-bit field holds a counter value that is decremented each time the F-TAG
28 is read (44.2.2.1) and provides a mechanism to limit the number of network hops a frame may take.
- 29 e) *Flow Hash*—This 16-bit field carries a flow hash value used in the dynamic port selection process
30 for ECMP (8.8.3).

1 44.2.2 F-TAG processing

2 This standard specifies flow filtering behavior for SPBM ECMP (44.2). To support this behavior, a flow
3 hash may be conveyed in a `mac_service_data_unit` using an F-TAG (44.2.1). This clause specifies a Support
4 for Flow Filtering shim that produces and processes F-TAG information. The shim has two EISS SAPs—a
5 lower EISS toward a LAN and an upper EISS toward the MAC Relay Entity.

6 At the boundary of a flow filtering SPT Domain, for example, at CBPs, the flow hash is generated for
7 `EM_UNITDATA.indication` primitives. At intermediate ports, for example, PNPs, the F-TAG is generated
8 using the flow hash in `EM_UNITDATA.request` primitives or processed to retrieve the flow hash value from
9 `EM_UNITDATA.indication` primitives. The F-TAG also includes a TTL field, used for loop mitigation, and
10 PCP and DEI fields that are processed by this shim.

11 Flow filtering support is governed by a Flow Filtering Control Table for each port containing an entry for
12 each VID with the following fields:

- 13 a) Flow Filtering (enabled/disabled). This field indicates whether flow filtering behavior is enabled on
14 the port for the VID. The default value is disabled.
- 15 b) Flow Hash Generation (enabled/disabled). This field indicates whether flow hash generation is
16 enabled on the port for the VID. The default value is disabled.
- 17 c) TTL Value (1..63). This field is the initial TTL value for frames entering the flow filtering SPT
18 Domain [item d) in 12.16.5.5.2]. The default value is 8.

19 The Flow Filtering Control Table is provisioned by ISIS-SPB (44.1.2). For flow filtering, a CBP will have
20 both the Flow Filtering field and the Flow Hash Generation field set to enabled and the TTL Value field set
21 to provide the initial `time_to_live` value for indication primitives. For flow filtering, a PNP will have the
22 Flow Filtering field set to enabled and the Flow Hash Generation field set to disabled. The TTL Value field
23 (initial TTL value) is not used on a PNP.

24 NOTE—Based on the description above, for flow filtering, the condition “Flow Hash Generation field indicates
25 enabled” is equivalent to indicating a CBP, and the condition “Flow Hash Generation field indicates disabled” is
26 equivalent to indicating a PNP.

27 44.2.2.1 Data indications

28 On receipt of an `EM_UNITDATA.indication` primitive from the lower EISS on a PNP, the received frame
29 shall be discarded if:

- 30 a) For the `vlan_identifier` in the Flow Filtering Control Table, the Flow Filtering field indicates
31 enabled, and the initial octets of the `mac_service_data_unit` do not contain a valid F-TAG; or
- 32 b) The value in the TTL field in the F-TAG is 0.

33 NOTE 1—A frame received with a TTL value of 0 will be discarded, i.e., not relayed, even if it is addressed to another
34 port on the receiving Bridge.

35 Otherwise, an `EM_UNITDATA.indication` primitive is invoked at the upper EISS with parameter values
36 determined from the received primitive based on the Flow Filtering Control Table for the received
37 `vlan_identifier` as follows:

- 38 c) If the Flow Filtering field indicates disabled, the invoked primitive is identical to the received
39 primitive.
- 40 d) If the Flow Filtering field indicates enabled, the **destination_address**, **source_address**,
41 **vlan_identifier**, and **connection_identifier** in the invoked primitive are identical to those
42 parameters in the received primitive.
- 43 e) If the Flow Filtering field indicates enabled and the Flow Hash Generation field indicates disabled,
44 the **mac_service_data_unit** in the invoked primitive is created by removing the F-TAG from the

1 received **mac_service_data_unit**; otherwise, the **mac_service_data_unit** in the invoked primitive
2 is identical to the received primitive.

3 The value of the **flow_hash** and **time_to_live** parameters are determined as follows:

4 f) If the Flow Hash Generation field indicates enabled, the value of **flow_hash** is generated from the
5 received primitive (using a process selected by the implementation). **time_to_live** is set to the TTL
6 Value from the Flow Filtering Control Table for the received **vlan_identifier**.

7 NOTE 2—Procedures for calculating the flow hash are not covered by this standard. The procedure is typically a
8 protocol-dependent hash of the client protocol header being carried within the **mac_service_data_unit**.

9 NOTE 3—There is often a compromise between the speed of a hash calculation and the quality of value spreading. This
10 standard makes no assurances regarding the quality of the distribution of **flow_hash** values.

11 g) If the Flow Hash Generation field indicates disabled, the value of **flow_hash** is taken from the Flow
12 Hash field in the received F-TAG, and **time_to_live** is set to the TTL value in the received F-TAG
13 decremented by 1.

14 The value of the **drop_eligible** and **priority** parameters are determined as follows:

15 h) If the **mac_service_data_unit** was tagged with an F-TAG, the value of the **drop_eligible** parameter
16 and the received priority value are decoded as described in 6.9.3, using the PCP and DEI values
17 from the F-TAG. Otherwise,

18 i) The received priority value and the **drop_eligible** parameter value are the values in the
19 **EM_UNITDATA.indication**.

20 j) The value of the priority parameter is then regenerated from the received priority, as specified in
21 6.9.4.

22 44.2.2.2 Data requests

23 On invocation of an **EM_UNITDATA.request** primitive at the upper EISS, an **EM_UNITDATA.request**
24 primitive is invoked at the lower EISS, with parameter values determined based on the Flow Filtering
25 Control Table for the received **vlan_identifier**.

26 If the Flow Filtering field indicates disabled or the Flow Hash Generation field indicates enabled, the
27 invoked primitive is identical to the received primitive.

28 If the Flow Filtering field indicates enabled and the Flow Hash Generation field indicates disabled, all
29 parameters in the invoked primitive match those in the received primitive except the **mac_service_data_unit**.
30 The **mac_service_data_unit** in the invoked primitive is formed by inserting an F-TAG as the initial octets to
31 the received **mac_service_data_unit**. The values of the **flow_hash**, **time_to_live**, **priority**, and
32 **drop_eligible** parameters are used to determine the contents of the F-TAG, in accordance with 44.2.1.

33 44.2.3 Forwarding process extension for flow filtering

34 Frame filtering (8.6.3) reduces the set of potential transmission ports in accordance with applicable FDB
35 entries. When flow filtering is enabled, the ISIS-SPB control plane creates Dynamic Filtering Entries with
36 multiple forwarding Ports in the Port Map when there are multiple equal cost hops to a given individual
37 address. Frame filtering must select only one of these forwarding ports on which to transmit each frame with
38 a matching **destination_address** and VID.

39 Selection of one Port from a set indicated by the Port Map shall be performed using a form of the FNV non-
40 cryptographic hash algorithm as follows:

41 For a Port Map specifying k potential forwarding Ports, let **P** be an ordered list of potential forwarding Ports
42 sorted from greatest to least neighbor SPB System Identifier and P_i be the i th Port in **P**. Thus P_0 is the Port

1 whose neighbor has the greatest SPB System Identifier, and P_{k-1} is the Port whose neighbor has the least
2 SPB System Identifier. The following dynamic port selection process is used to select a value for i , thus
3 selecting Port P_i to be the transmission port:

- 4 a) Set a 32-bit variable *hash32* to 0x811C9DC.
- 5 b) For each octet O_j (extended to 32 bits with leading 0's) in the Bridge's SPB System Identifier
6 (8.13.8), starting with the least significant octet (octet 5) and progressing to the most significant
7 octet (octet 0), set *hash32* to $((hash32 \text{ XOR } O_j) \times 0x01000193)$.
- 8 c) For each octet O_j (extended to 32 bits with leading 0's) in the flow_hash parameter, starting with the
9 least significant octet (octet 4 of Figure 44-1) and progressing to the most significant octet (octet 3
10 of Figure 44-1), set *hash32* to $((hash32 \text{ XOR } O_j) \times 0x01000193)$.
- 11 d) Set a 16-bit variable *hash16* to $((hash32 \text{ and } 0x0000FFFF) \text{ XOR } (hash32 / 0x00010000))$.
- 12 e) Set i to $(hash16 \bmod k)$, selecting Port P_i .

13 44.2.4 TTL Loop mitigation

14 The F-TAG contains a TTL field whose value is decremented each time a frame is forwarded by an SPT
15 Bridge supporting flow filtering [item g) in 44.2.2]. A frame is filtered (discarded) if it is received with a
16 TTL value of 0. This mechanism mitigates the impact of looping frames by limiting the number of hops
17 these frames may travel in the network. This method of loop mitigation protects network resources by
18 ensuring frames are forwarded at most a fixed number of times.

19 NOTE—The TTL carried in the F-TAG is decremented for all frames; however, the Flow Hash is applicable only to
20 unicast frames. The Flow Hash has no effect on the filtering of group addressed frames.

21 44.2.5 CFM for ECMP with flow filtering

22 CFM as specified in Clause 18 through Clause 22 provides capabilities useful in detecting, isolating, and
23 reporting connectivity faults. SPBM MAs (27.18) can be used with an ECMP VID and provide most of the
24 necessary CFM functionality. One additional MA type is specified for ECMP, the ECMP path MA, because
25 an SPBM path MA can test only one path between a pair of endpoints.

26 An ECMP path MA is used to continuously test multiple paths between two endpoints. This type of MA has
27 only two MEPs at two SPT Region boundary ports. The MA is identified by a TE-SID comprising an ESP in
28 each direction between the two endpoints. An ECMP path MEP cycles through a set of flow hash values
29 intended to direct the CCM frames along (and thus test) different paths to the other MEP in the MA. Each
30 flow hash is used in multiple consecutive CCM frames so that a fault on a particular path will be detected
31 and reported back via an RDI signal. This standard does not specify the mechanisms that would be necessary
32 to support Loopback and Linktrace on an ECMP path MA since full Loopback and Linktrace coverage can
33 be provided by an SPBM VID MA. Therefore, there are no MHFs associated with an ECMP path MA.

44.2.5.1 ECMP path MEP placement in a Bridge Port

Since ECMP path MEPs must only see traffic from one other endpoint, these MEPs need to be further differentiated by the TESI Multiplex Entity (6.19). This locates an ECMP path MEP so that it receives CCMs only from the other MEP in its MA. Figure 44-2 shows an example of ECMP path MEPs on a CBP.

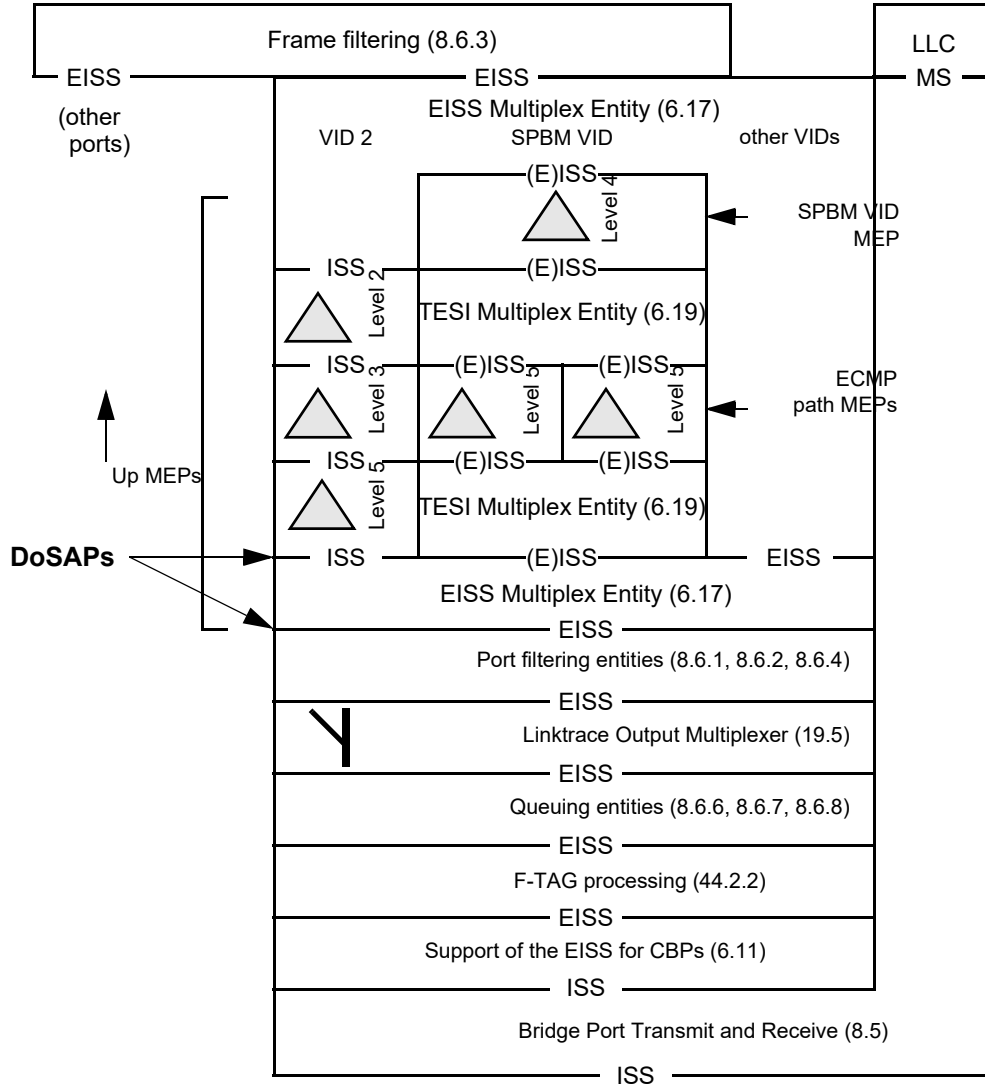


Figure 44-2—SPBM VID MEP and ECMP path MEP placement in a CBP

44.2.5.2 Continuity Check protocol in an ECMP path MA

The Continuity Check protocol is described in 20.1 and the corresponding state machines in Clause 20. The modifications required to realize ECMP path MAs include the following:

- The MEP Continuity Check Initiator state machine adds three variables for controlling ECMP path testing (20.10.4, 20.10.5, 20.10.6) and shares two of these with the Remote MEP state machine (Figure 20-1).
- The destination_address parameter is set to the individual MAC address of the other MEP in the MA [item a) in 20.11.1].

- 1 c) The flow_hash parameter is set to one of the values to be tested, and the time_to_live parameter is
2 set as specified in item e) in 20.11.1.
- 3 d) The MEP Continuity Check initiator state machine is modified to support a cycle of path tests for an
4 ECMP path MEP (20.12).
- 5 e) The Remote MEP state machine is modified to track a cycle of path test RDI values and report
6 rMEPlastRDI as set if any RDI has been seen in the last path test cycle for an ECMP path MEP
7 (20.19.2, 20.20).

8 NOTE—In an ECMP path MEP testing a set of paths, the RDI signal reflects back the state of far end CCM reception.
9 Using the current CCM sending cycle position (pathN), the CCM sending rate, and the path round trip delay, one can
10 calculate roughly the position in the sending cycle corresponding to a received RDI. This approach makes minimal
11 changes to the CCM state machines at the CCM sending MEP and no changes at the CCM receiving MEP. It is subject to
12 some ambiguity based on differences in path delay, the sending rate, and cycle length. In the case of ambiguous results,
13 further testing would be required to identify a faulty path.

14 SPBM MAs may also implement the following enhancements to the Continuity Check protocol:

- 15 f) The procedure MEPprocessEqualCCM() on ECMP path MEPs does not include the check of the
16 MAID on received CCMs [item b) in 20.17.1].

17 All other Continuity Check processes are the same as those for a VID-based MA.

18 44.2.6 Operation with selective support for flow filtering

19 When adding new capabilities to a network, it is desirable to make changes incrementally rather than
20 perform a wholesale upgrade (e.g., incur a “flag day” event). Flow filtering is optional functionality that can
21 be used with selected Base-VIDs while not adversely affecting other services in the network. However, flow
22 filtering changes frame encoding (i.e., adds new header information in an F-TAG) and therefore requires that
23 a Base-VID using ECMP with flow filtering have all its boundary points capable of processing the F-TAG.
24 To enable incremental incorporation of flow filtering, ISIS-SPB requires F-TAG support for any Bridges
25 with a CBP associated with a Base VID using ECMP with flow filtering while allowing other Bridges in the
26 SPT region to operate without flow filtering support.

27 When the ECMP ECT-ALGORITHM value indicates support for flow filtering (Table 28-1), an I-SID
28 cannot be configured to use a Base VID using ECMP with flow filtering unless the Bridge supports flow
29 filtering (12.16.5.2). Requiring all Bridges advertising service instances using ECMP with flow filtering to
30 support F-TAG processing provides for proper handling of F-tagged frames at the boundary of the SPT
31 Domain.

32 Within the SPT Domain, it is possible to use Bridges that support the ECMP ECT Algorithm but do not
33 support flow filtering (i.e., do not recognize F-TAGs). In this case, the ECMP ECT Algorithm (28.9) selects
34 one port from the set of equal cost forwarding ports for each individual address and creates a Dynamic
35 Filtering Entry for the address indicating the selected port.

36 NOTE—Bridges that do not support flow filtering should not change PCP and DEI values for frames carrying F-TAGs
37 as this may not provide the desired behavior. If the PCP and DEI values are changed in the VLAN tag only, the PCP and
38 DEI values carried in the F-TAG will be reasserted when the F-TAG is next processed.

39 These ECMP behaviors enable incremental upgrade of edge bridges to support flow filtering and selective
40 use of this capability with I-SIDs registered only at these upgraded edge bridges. ECMP operation with
41 selective support for flow filtering enables incremental introduction of flow filtering capabilities into an
42 existing network and earlier introduction of flow filtering support for selected services.

1 45. Path Control and Reservation (PCR)

2 This clause specifies IS-IS extensions to provide the following:

- 3 a) Establishment of explicit trees for frame forwarding in an SPT Region (45.1),
- 4 b) Use of IS-IS to communicate bandwidth assignments made by the Path Computation Element (PCE)
- 5 (45.2), and
- 6 c) Redundancy with the establishment of the corresponding trees (45.3).

7 The Path Control and Reservation (PCR) IS-IS extensions specified in this clause are compatible with ISIS-
8 SPB specified in Clause 27 and Clause 28. Furthermore, IS-IS with PCR extensions (ISIS-PCR) relies on
9 the SPB architecture and terminology; and ISIS-PCR also leverages some of the ISIS-SPB sub-TLVs (see,
10 e.g., 5.5) as specified by this clause. This specification considers only point-to-point links for PCR although
11 IS-IS also supports shared media LANs.

12 NOTE 1—ISIS-PCR does not require the implementation of the full ISIS-SPB protocol; but in addition to IS-IS, ISIS-
13 PCR requires the support of the ISIS-SPB sub-TLVs listed in 5.5 (also listed as PCR-2–PCR-5 in A.43), whose use is
14 specified by this clause. Nonetheless, if an SPT Bridge supports both ISIS-SPB and ISIS-PCR, then both of them are
15 implemented by the same IS-IS Higher Layer Entity.

16 A VID can be associated with an explicit tree, i.e., with an explicit active topology within an SPT Region.
17 The Base VID of the VLAN is then associated with explicit path control mode of IS-IS operation, i.e., ISIS-
18 PCR. A VID can be associated with multiple explicit trees if the considerations explained in 45.1.4 are taken
19 into account. VIDs controlled by ISIS-PCR do not participate in ECMP operation (Clause 44).

20 NOTE 2—A VLAN can provide a point-to-point, point-to-multipoint, or multipoint-to-point service using the
21 multipoint-to-multipoint connectivity provided by an active topology. Similarly the Transmit and Receive IS-IS sub-
22 TLV flags (28.12.10) of Edge Bridges allow an I-SID to use a multipoint-to-multipoint VLAN to provide point-to-point,
23 point-to-multipoint, or multipoint-to-point service.

24 Path Computation Element (PCE) entities are external to the IS-IS protocol and fully or partially determine
25 and describe each explicit tree to be established by ISIS-PCR throughout an SPT Region. There can be
26 multiple PCEs in a region; nevertheless, any given explicit tree is under the control of one PCE. Explicit
27 trees can be used, for example, for placing selected traffic on a precisely defined route, usually off the
28 shortest path tree.

29 ISIS-PCR is able to record and communicate bandwidth assignments if instructed to do so by a PCE. ISIS-
30 PCR can be used for bandwidth assignments only if MSRP is not used in the SPT Region. If ISIS-PCR
31 communicates bandwidth assignments, then the tree descriptor assembled by the PCE also includes the
32 details of the bandwidth assignment to be recorded by the Bridges. This mode of operation is expected to be
33 used to divert traffic aggregates off their shortest path route, in order to avoid potential congestion on the
34 default shortest path.

35 The redundancy provided by a physical topology can be leveraged by various resiliency schemes. A PCE
36 can be used for the computation, and ISIS-PCR can be used for the establishment of the trees required for a
37 given resiliency solution.

38 45.1 Explicit trees

39 This subclause specifies IS-IS extensions that provide explicit forwarding trees for data frames. An explicit
40 tree is determined by a Path Computation Element (PCE) and is not required to follow the shortest path.
41 PCE is defined by IETF RFC 4655 [B34]. A PCE is an entity that is capable of computing a topology for
42 forwarding based on a network topology, its corresponding attributes, and potential constraints. A PCE
43 explicitly describes a forwarding tree as specified in 45.1.9. Either a single PCE or multiple PCEs determine
44 explicit trees for a region. Even if there are multiple PCEs in a region, each explicit tree is determined by
45 only one PCE, which is referred to as the owner PCE of the tree. PCEs and ISIS-PCR can be used in
46 combination with ISIS-SPB shortest path routing.

1 A PCE is a higher layer entity in an SPT Bridge or an end station. The PCE interacts with the active
2 topology control protocol, i.e., with ISIS-PCR. The collaboration with ISIS-PCR can be provided by a Path
3 Control Agent (PCA) on behalf of a PCE. Either the PCE or the corresponding PCA is part of the IS-IS
4 Domain. If the PCE is not part of the IS-IS Domain, then the PCE has to be associated with a PCA that
5 resides either in an SPT Bridge or in an end station directly connected to at least one Bridge of the SPT
6 Region. The PCE or its PCA establishes IS-IS adjacency (45.1.7) in order to receive all the LSPs transmitted
7 by the Bridges in the region. The PCE, either on its own or via its PCA, can control the establishment of
8 explicit trees in that region by injecting an LSP conveying an explicit tree and thus instruct ISIS-PCR to set
9 up the explicit tree determined by the PCE. Each PCE, whether located in a Bridge or end station, has access
10 to the link state topology and resource information common throughout the region. If instructed to do so by
11 a PCE, ISIS-PCR can also record and communicate bandwidth assignments, which can be applied only if
12 MSRP is not used in the region. Different PCE and PCA locations are illustrated in Figure 45-1 and
13 Figure 45-2.

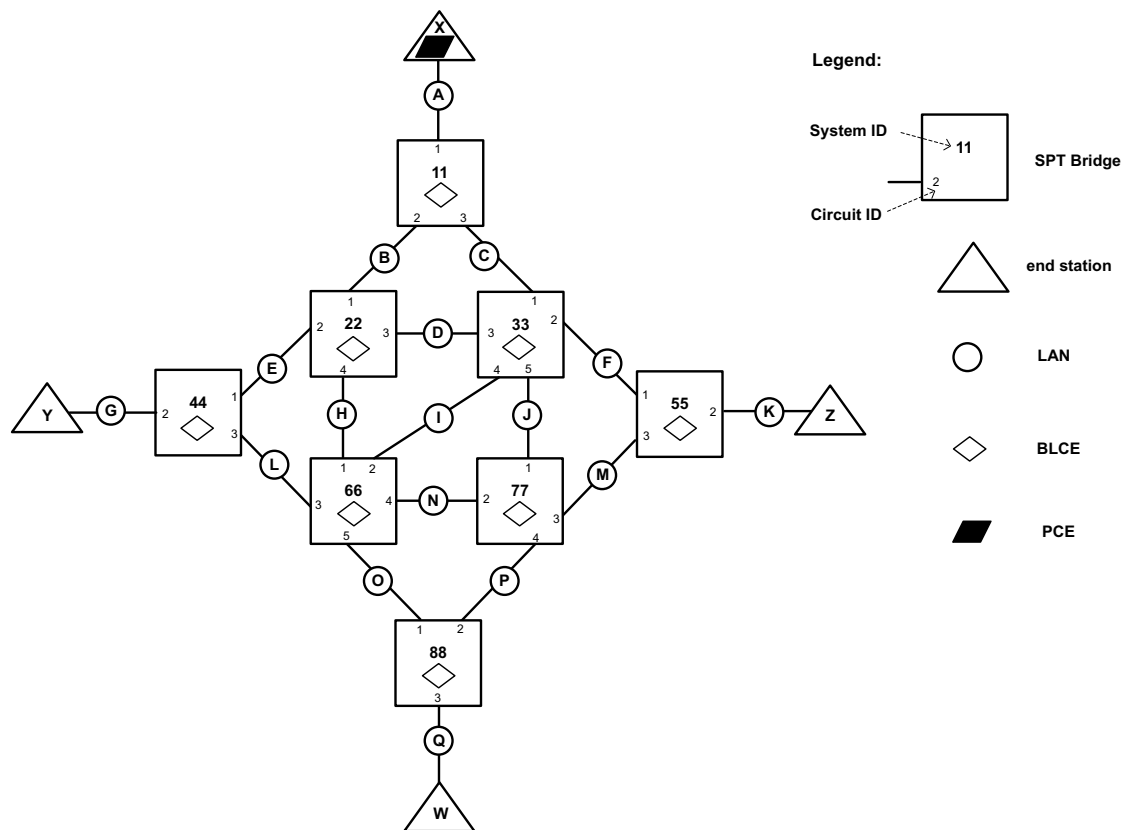


Figure 45-1—An SPT Region controlled by a single PCE

14 NOTE 1—The operation details of the PCE are not specified by this standard. If the PCE is part of the IS-IS Domain,
15 then the PCE uses IS-IS PDUs to communicate with the IS-IS Domain, and the PCE has a live IS-IS LSDB (i.e., the PCE
16 implements the PCA functions, too). A PCE can instead communicate with the IS-IS Domain via a PCA, e.g., to retrieve
17 the Link State Database (LSDB) or instruct the creation of an explicit tree. However, the means of communication
18 between the PCE and the PCA is not specified by this standard. A PCE could operate on a network topology retrieved by
19 other means, e.g., configuration, instead of retrieving it from a live IS-IS LSDB; which operation mode is not specified
20 by this standard. Having no live LSDB, the PCE instructs its PCA to flood the LSP conveying the appropriate Topology
21 sub-TLV.

22 ISIS-PCR implements Software Defined Networking (SDN) through hosting the PCE in an external agent,
23 e.g., an SDN Controller. In this case IS-IS is the protocol used to control the Bridges by the SDN Controller;
24 the explicit trees are programmed in the Bridges via IS-IS.

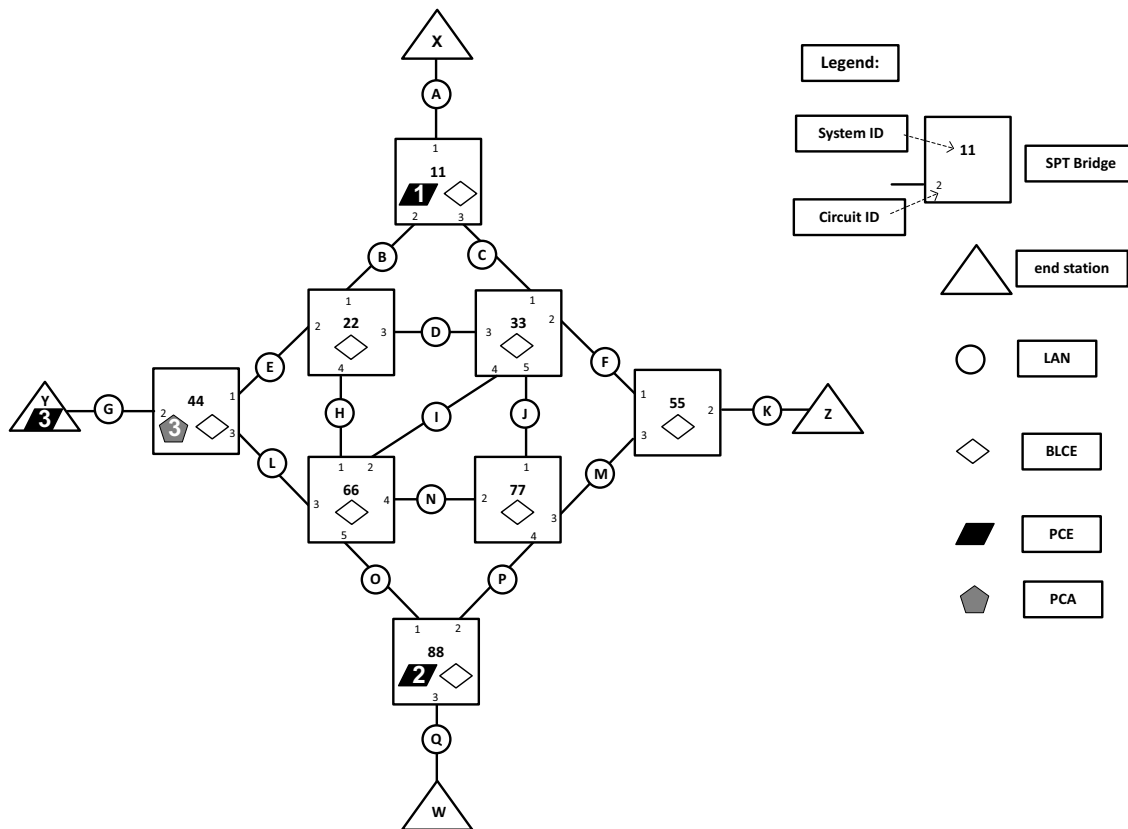


Figure 45-2—An SPT Region controlled by multiple PCEs

1 Figure 45-1 shows a region controlled by a single PCE residing in end station X connected to SPT Bridge
2 11. The PCE has IS-IS adjacency established with SPT Bridge 11, i.e., the PCE is part of the IS-IS Domain.

3 The IS-IS Domain coincides with the SPT Region in Figure 45-1 and Figure 45-2; they comprise SPT
4 Bridges 11, 44, 55, and 88 (Edge Bridges) and SPT Bridges 22, 33, 66, and 77 (Core Bridges). As shown in
5 the figures, each SPT Bridge implements a Bridge Local Computation Engine (BLCE).

6 Figure 45-2 shows a region controlled by multiple PCEs. Some of the SPT Bridges implement a PCE in
7 addition to their BLCE; PCE 1 and PCE 2 reside in SPT Bridge 11 and SPT Bridge 88, respectively. PCE 3
8 resides in End Station Y. PCE 3 is not part of the IS-IS Domain; PCE 3 uses PCA 3 to instruct ISIS-PCR for
9 the establishment of explicit trees and to retrieve link state data from the domain.

10 An Explicit Tree (ET) is an undirected loop-free topology, whose use is under the control of the owner PCE
11 by means of associating VIDs and MAC addresses with it. As it is undirected, the ET contains no
12 assumptions about the direction of any flows that use it; it can be used in either direction as specified by the
13 VIDs and MAC addresses associated with it. It is the responsibility of the PCE to ensure reverse path
14 congruency (3.222) and unicast multicast congruency (3.295) if that is required, e.g., making an explicit tree
15 symmetric (3.273) by setting it up so that it is used in both directions between a pair of Bridges.

16 An explicit tree is either strict or loose. A strict explicit tree specifies all Bridges and paths it comprises. A
17 loose tree specifies only the Bridges that have a special role in the tree, e.g., an Edge Bridge, and no path or
18 path segment is specified between the Bridges, which are therefore loose hops even if Edge Bridges are
19 adjacent neighbors. The special role of a hop can be an Edge Bridge, a root, a leaf, a Bridge to be avoided, or

1 a transit hop in the case of a tree with a single leaf. The path for a loose hop is determined by the BLCE of
2 the SPT Bridges. The shortest path is used for a loose hop unless specified otherwise by the descriptor
3 (45.1.9) of the tree or by the corresponding ECT Algorithm (45.1.2).

4 NOTE 2—PCE and BLCE are architecturally different entities. A BLCE is implemented in each SPT Bridge, where the
5 BLCE implements the SPF algorithm at the minimum and can also implement more complex path computation
6 algorithms, e.g., for determining constrained or redundant trees. For instance, the BLCE of SPT Bridges supporting
7 ISIS-SPB at least implements an all pairs shortest path algorithm in addition to SPF. Forwarding trees are computed by
8 the BLCEs unless they are explicitly given. Conversely, a PCE can implement more complex computation algorithms,
9 and its main task is to determine explicit trees even if the PCE resides in a Bridge.

10 NOTE 3—If shortest paths are used for loose hops, then there is no need for any computation in addition to SPB as ISIS-
11 SPB performs all pairs shortest path computation.

12 A loose explicit tree is constrained if the tree descriptor includes one or more constraints, e.g., the
13 administrative group to which the links of the tree have to belong. The BLCE of the SPT Bridges then
14 performs constrained routing (CR) to determine a loose hop instead of shortest path routing. CR relies on
15 distributed link state operation similarly to shortest path routing. CR essentially performs shortest path
16 routing on the topology that contains only the links meeting the constraint(s); therefore, the applied path
17 computation algorithm is also referred to as Constrained Shortest Path First (CSPF).

18 NOTE 4—If a tree is a strict explicit tree, then it is fully specified; therefore, no constraint is included in the tree
19 descriptor (45.1.9) as no further computation is needed.

20 An explicit tree is specified by a Topology sub-TLV (45.1.9). The Topology sub-TLV associates one or more
21 VIDs with an explicit tree. The Topology sub-TLV includes two or more Hop sub-TLVs (45.1.10), and a hop
22 is specified by an IS-IS System ID. A strict tree is decomposed to branches, and each branch is specified by
23 an ordered list of Hop sub-TLVs. A loose tree is not fully specified, but the Topology sub-TLV conveys
24 some of its hops. The Hop sub-TLV of a loose hop can include a delay constraint [item h) in 45.1.10]. A
25 Topology sub-TLV can also include further sub-TLVs to constrain (45.1.11, 45.1.12) loose hops. The
26 Bridges involved in an explicit tree store the corresponding Topology sub-TLVs in their Explicit Tree
27 Database (ETDB).

28 Explicit trees are propagated and set up by ISIS-PCR in an SPT Region. The PCE or its PCA assembles the
29 Topology sub-TLVs (45.1.9) and then instructs ISIS-PCR to establish the tree. If the PCE resides in an SPT
30 Bridge, then the PCE entity passes the Topology sub-TLV (45.1.9) to the ISIS-PCR entity, which shall then
31 flood an LSP including the Topology sub-TLV throughout the SPT Domain. If the PCE resides in an end
32 station, then either the PCE or its PCA adds the Topology sub-TLV (45.1.9) into an LSP, which is flooded
33 throughout the SPT Domain. The Topology sub-TLV is flooded by the same techniques used for the SPB
34 LSPs. The SPT Bridges then shall process the Topology sub-TLV (45.1.9) upon reception. If the Topology
35 sub-TLV specifies one or more loose trees, then the path for the loose hops is determined by the BLCE of the
36 SPT Bridges. The SPT Bridges then install the appropriate FDB entries (45.1.6) for frame forwarding along
37 the tree described by the Topology sub-TLV (45.1.9) or the trees computed based on the Topology sub-TLV.
38 Dynamic Filtering Entries are maintained by ISIS-PCR for the VID, MAC address tuples associated with
39 an ET.

40 NOTE 5—Due to the LSP aging of IS-IS, the Topology sub-TLVs (45.1.9) have to be refreshed similar to other IS-IS
41 TLVs in order to keep the integrity of the LSDB. The corresponding Dynamic Filtering Entries are also refreshed when a
42 Topology sub-TLV is refreshed. Refreshing Topology sub-TLVs is the task of the entity being part of the IS-IS Domain,
43 i.e., either the PCE or the PCA.

44 The owner PCE can withdraw an explicit tree by sending an updated LSP that does not include the Topology
45 sub-TLV. If a Topology sub-TLV is removed from an LSP (or has been changed), so that (previous)
46 Topology sub-TLV is no longer present (or has been changed) in the LSDB, then that (previous) Topology
47 sub-TLV is implicitly withdrawn. ISIS-PCR then removes (or updates) the explicit tree.

48 ISIS-PCR provides precedence order among Topology sub-TLVs as specified in 45.2.3 if it is needed in a
49 region with multiple PCEs.

1 45.1.1 Tree structures

2 Two types of tree structures can be used for explicit trees, as follows:

- 3 — Ad-hoc trees
- 4 — Template trees

5 Ad-hoc explicit trees do not follow any template; they can comprise any path. A particular explicit tree is
6 defined, and a VID is assigned to the explicit tree. The administratively straightforward way to do this is to
7 use a bidirectional VID for the explicit tree. This approach constructs a shared bidirectional (*,G) tree
8 joining all members of a group.

9 NOTE 1—A single common (*,G) spanning tree can be constructed this way.

10 Alternatively, explicit trees can follow templates. It can be helpful to define a tree rooted on every Bridge,
11 where each tree spans every Bridge of a domain and the complete set of trees, one rooted on each Bridge of
12 a domain, are constructed consistently, so that collectively they form a single forwarding plane offering any-
13 to-any connectivity within that domain. Such trees are used as a source rooted tree for multicast traffic
14 emitted from a Bridge, and they are also used as a destination rooted tree for unicast traffic being sent to
15 their Bridge. There is a fundamental constraint that has to be obeyed for all such structures individually in
16 that plane. They have to be simply connected trees, with only one path from any point in the network domain
17 to the root of the tree. If paths to the root cross at any point, or merge and subsequently diverge, a unicast
18 forwarding inconsistency or an unwanted multicast frame replication point is created.

19 This connectivity style (i.e., a tree per Bridge per plane) is, for example, enforced by construction by
20 Shortest Path Bridging. In SPB, a Base VID is associated with one of the standard symmetric ECT
21 Algorithms (28.8), which identifies a plane. In SPBM, the individual trees within each Base VID are defined
22 by the MAC address of the root Bridge; in SPBV each Bridge is allocated an individual SPVID to define its
23 tree.

24 Any preferred algorithm can be used when performing explicit routing, and the required constraint can be
25 replicated by maintaining the simply connected tree construct. Thus the template-tree-based approach
26 constructs one or more explicitly routed forwarding planes with tree sets, one tree rooted on each Bridge
27 within each plane.

28 In the case of a learning VLAN, a unidirectional VID can be assigned to each tree (similarly to the
29 unidirectional SPVID model), so that each tree is independent and identifies its root Bridge. Shared VLAN
30 Learning (3.247) takes place among the VIDs associated with the same explicit tree set, where reverse
31 congruence (3.222) has to be enforced by the function that computes the explicit trees on a particular
32 forwarding plane (Base VID).

33 In the case of a non-learning VLAN, the MAC of the root Bridge identifies the tree, and a single Base VID
34 identifies the entire forwarding plane.

35 Care has to be taken when constructing multicast trees with multiple sources if the template tree approach is
36 used. Source-specific multicast (S,G) has to be applied if multiple sources participate in a single multicast
37 group. A separate source rooted tree is constructed for every source of the group, and the complete multicast
38 structure is the superposition of all these separate source rooted trees. The multicast traffic of the different
39 sources has to be distinguished either by a VID (as in SPBV) or by a source-specific Group MAC address
40 (as in SPBM, see 27.15).

41 NOTE 2—There is no requirement here that the initial template construction actually generates spanning trees. If
42 Bridges only initially require connectivity to a subset of the SPT Bridges of a region, only the relevant part of the
43 template need initially be constructed. This leaves freedom to add hitherto unconnected Bridges by extension of trees
44 using a preferred route at a later stage.

45.1.2 Explicit ECT Algorithms

Five explicit path control modes are specified by this standard; each of them is identified by a distinct ECT-ALGORITHM. Table 45-1 summarizes the ECT Algorithms that can be used for explicit trees. The ECT-ALGORITHMS specified by this standard for explicit trees are formed using OUI=00-80-C2 and Index values 0x17, 0x18, 0x19, and the range from 0x21 to 0x40. SPT Bridges that support PCR shall support the Strict Tree ECT Algorithm (00-80-C2-17) and may support the other ECT Algorithms of Table 45-1. VLANs under explicit path control are associated with one of the ECT Algorithms as specified by 45.1.3.

NOTE 1—The ECT-ALGORITHM is in fact the identifier of the method and the algorithm used to determine the active topology. Although the active topology specified by this clause is not an Equal Cost Tree, the terminology and the ECT Algorithm framework of Clause 27 and Clause 28 is kept. The Opaque ECT concept (28.6) is also supported for explicit path control, which can be used for further algorithms.

Table 45-1—ECT-ALGORITHM values for explicit trees

ECT-ALGORITHM	Algorithm Name	Behavior
00-80-C2-17	Strict Tree (ST ECT Algorithm)	A single strict explicit tree. No restoration or update is performed by IS-IS on its own; only the PCE can initiate the update of a strict ET.
00-80-C2-21 ... 00-80-C2-30	Loose Tree (LT ECT Algorithm)	A single loose explicit tree. The loose hops are computed by the BLCE of SPT Bridges applying constrained or shortest path routing. The loose hops are restored by IS-IS upon a topology change if loop-free paths are available. The Bridge Priority Mask to be used in the case of constrained or shortest paths is specified by Table 45-2.
00-80-C2-31 ... 00-80-C2-40	Loose Tree Set (LTS ECT Algorithm)	A set of loose explicit trees. The set comprises an individual tree for each Edge Bridge included in the descriptor of the explicit tree, i.e., they are template trees. Each tree is computed by the BLCE of SPT Bridges applying constrained routing. These trees are restored by IS-IS upon a topology change. The Bridge Priority Mask to be used in the case of constrained or shortest paths is specified by Table 45-2.
00-80-C2-18	Maximally Redundant Trees (MRT ECT Algorithm)	Maximally Redundant Trees (MRTs) are loose trees for each MRT Root, which are computed together with the corresponding GADAG by the BLCE of SPT Bridges and cautiously restored by ISIS-PCR. If the topology view is identical throughout the SPT Domain, then IS-IS restores these trees one by one, only one of each redundant set at a time (45.3.3).
00-80-C2-19	Maximally Redundant Trees with GADAG (MRTG ECT Algorithm)	Maximally Redundant Trees (MRTs) are loose trees for each MRT Root, which are computed by the BLCE of SPT Bridges based on the GADAG received from the single GADAG Computer. MRTs are cautiously restored by ISIS-PCR upon reception of a new GADAG from the GADAG Computer. IS-IS restores these trees one by one, only one of each redundant set at a time (45.3.4).

The Strict Tree (ST) ECT Algorithm is used for a strict explicit tree. A strict ET is static as no other entity can update it but the tree owner PCE. In the case of a topology change, it is the task of the owner PCE to detect the topology change, e.g., based on the changes in the LSDB, and to update the strict trees if needed. In other words, the owner PCE computes the new tree, assembles its descriptor, and then instructs ISIS-PCR to install it. The use of VIDs for strict trees is described in 45.1.4, and redundant strict trees are explained in 45.3.2.

The Loose Tree (LT) ECT Algorithm is used for a single loose explicit tree. The path for loose hops is determined by the BLCE of the SPT Bridges; therefore, the Topology sub-TLV (45.1.9) specifying the tree

1 has to indicate which hop is the root of the tree. The loose hops are maintained by IS-IS, i.e., restored upon a
2 topology change if a loop-free path is available. If the tree computed by the BLCE visits the same Bridge
3 twice (implying that a loop or hairpin has been created), then that loop or hairpin has to be pruned from the
4 tree even if it contains a hop specified by the Topology sub-TLV. Constrained routing can be applied for the
5 loose hops based on the attributes listed in 45.1.8. If a Bridge is not to be included is also a constraint, which
6 can be specified by the Exclude flag [item c6) in 45.1.10] of a Hop sub-TLV (45.1.10) conveyed by the
7 Topology sub-TLV specifying the tree.

8 The Loose Tree Set (LTS) ECT Algorithm is used if connectivity among the Edge Bridges specified by the
9 Topology sub-TLV (45.1.9) is to be provided by a set of loose trees so that one tree is rooted at each Edge
10 Bridge, i.e., the loose trees are template trees (45.1.1). The BLCEs of the SPT Bridges compute the loose
11 trees, which are maintained by IS-IS, i.e., restored upon a topology change. Avoiding some bridges in these
12 trees is a constraint, which can be specified by the Exclude flag [item c6) in 45.1.10]. Further constraints can
13 be specified by the Topology sub-TLV based on the attributes listed in 45.1.8.

14 NOTE 2—A Loose Tree Set is similar to a Shortest Path Tree Set with the following differences: The trees of an LTS
15 span only the Edge Bridges specified by the Topology sub-TLV; they do not span the entire SPT Region. Furthermore,
16 constrained routing is applied instead of shortest path routing.

Table 45-2—Bridge Priority Masking for the LT and LTS ECT Algorithms

ECT-ALGORITHM (LT ECT Algorithm)	ECT-ALGORITHM (LTS ECT Algorithm)	Algorithm MASK
00-80-C2-21	00-80-C2-31	0x00
00-80-C2-22	00-80-C2-32	0xFF
00-80-C2-23	00-80-C2-33	0x88
00-80-C2-24	00-80-C2-34	0x77
00-80-C2-25	00-80-C2-35	0x44
00-80-C2-26	00-80-C2-36	0x33
00-80-C2-27	00-80-C2-37	0xCC
00-80-C2-28	00-80-C2-38	0xBB
00-80-C2-29	00-80-C2-39	0x22
00-80-C2-2A	00-80-C2-3A	0x11
00-80-C2-2B	00-80-C2-3B	0x66
00-80-C2-2C	00-80-C2-3C	0x55
00-80-C2-2D	00-80-C2-3D	0xAA
00-80-C2-2E	00-80-C2-3E	0x99
00-80-C2-2F	00-80-C2-3F	0xDD
00-80-C2-30	00-80-C2-40	0xEE

17 In the case of the LT and LTS ECT Algorithms, the symmetric shortest path tie-breaking specified in 28.5,
18 28.6, 28.7, and 28.8 is used during shortest path computation to manipulate the lexicographic ordering of
19 hops on a path after pruning the topology according to the constraints. The Bridge Priority Masks for the LT
20 and LTS ECT Algorithms are shown in Table 45-2. The mask identification method specified for the

1 Symmetric ECT Algorithms is used. The IEEE 802.1 OUI (00-80-C2) is used, Index values 0x21 through
2 0x30 are used for the LT ECT Algorithm, and Index values 0x31 through 0x40 are used for the LTS ECT
3 Algorithm as shown in Table 45-2.

4 The Maximally Redundant Trees (MRT) ECT Algorithm or the Maximally Redundant Trees with
5 Generalized Almost Directed Acyclic Graph (GADAG) (MRTG) ECT Algorithm is used if maximally
6 redundant loose explicit trees have to be maintained together. The use of the MRT ECT Algorithm and the
7 MRTG ECT Algorithm is described in 45.3.3 and 45.3.4, respectively.

8 ISIS-PCR uses the link metrics specified by the SPB Link Metric sub-TLVs (28.12.7) if the LT, the LTS, the
9 MRT, or the MRTG ECT Algorithm is used. The SPB Link Metric sub-TLV is used as specified in 28.12.7;
10 therefore, the maximum metric value is used in cases where the metrics advertised by adjacent Bridges for a
11 given link are different.

12 A topology change can cause the need for recomputing and updating multiple loose trees. If constrained
13 routing based on available bandwidth (45.1.12) is used for multiple loose trees, then there can be a race
14 hazard for the same resources if they are updated at the same time. Furthermore, the computation order can
15 influence the trees. Therefore, ISIS-PCR applies the tie-breaking method specified in 45.2.3 in order to
16 determine the computation order among the loose trees based on their descriptor Topology sub-TLV. If a
17 Topology sub-TLV specifies multiple loose trees, e.g., LTS or MRTs, then the computation order of these
18 trees follows the ascending order of the IS-IS System ID of the Bridges rooting the trees.

19 **45.1.3 ISIS-PCR VLAN configuration**

20 A VLAN provided by an explicit tree controlled by IS-IS is associated with IS-IS by means of allocating the
21 VLAN's Base VID to the appropriate MSTID, i.e., either to the SPBM-MSTID or to the SPBV-MSTID. The
22 explicit path control mode is then selected by associating the Base VID with the corresponding explicit ECT
23 Algorithm.

24 The Base VID of VLANs controlled by IS-IS is allocated either to the SPBM-MSTID (0xFFC) or to the
25 SPBV-MSTID (0xFFD) as specified in 27.4, which also applies to VLANs under explicit path control via
26 IS-IS. If multiple VIDs belong to a VLAN under explicit path control, then each VID has to be allocated to
27 the same MSTID; otherwise, the explicit trees do not get installed for the VLAN.

28 The exact active topology enforcement method applied for the VLAN is determined by the ECT Algorithm
29 (28.8) with which the VLAN is associated. Explicit path control mode is configured by means of associating
30 the Base VID with one of the ECT Algorithms specified in 45.1.2. The association of the VLAN with the
31 ECT Algorithm shall be provided by the SPB Base VLAN-Identifiers sub-TLV (28.12.4).

32 Each VID belongs only to one PCE, which is the owner PCE that has full control on the use of the VID. The
33 VIDs of a PCE are to be configured at the PCE. If multiple PCEs try to use the same VID, then ISIS-PCR
34 provides the precedence as specified in 45.2.3.

35 An IS-IS controlled VLAN is a non-learning VLAN if the VLAN's Base VID is allocated to the SPBM-
36 MSTID (0xFFC). MAC addresses are distributed explicitly for non-learning VIDs by IS-IS. The M flag is
37 set in the SPB Instance sub-TLV (28.12.5) for non-learning VLANs. If a non-learning VLAN is associated
38 with a symmetric ECT Algorithm (28.8), then the VLAN is under SPBM control as specified by Clause 27
39 and Clause 28 even if the VLAN is not a B-VLAN. If a non-learning VLAN is associated with one of the
40 explicit ECT Algorithms (45.1.2), then it is under explicit path control as specified by this clause.

41 An IS-IS controlled VLAN is a learning VLAN if the VLAN's Base VID is allocated to the SPBV-MSTID
42 (0xFFD). If a learning VLAN is supported by multiple VIDs, then Shared VLAN Learning (3.247) takes
43 place among the VLAN's VIDs associated with the same shortest path or explicit tree set. The M flag is
44 cleared in the SPB Instance sub-TLV (28.12.5) for learning VLANs. If a learning VLAN is associated with a

1 symmetric ECT Algorithm (28.8), then the VLAN is under SPBV control as specified by Clause 27 and
2 Clause 28; correspondingly, SPVIDs can be auto-allocated to SPTs from the pool of SPVIDs, i.e., from the
3 VIDs allocated to the SPVID-Pool-MSTID (0xFFFF) as specified by 27.4. If a learning VLAN is associated
4 with one of the explicit ECT Algorithms (45.1.2), then it is under explicit path control as specified by this
5 clause; and each VID of the VLAN is allocated to the SPBV-MSTID (0xFFD), i.e., none of the VLAN's
6 VIDs is allocated automatically. When a VLAN is under explicit path control, it is the responsibility of the
7 PCE to provide reverse congruent paths (3.222) so that Shared VLAN Learning (3.247) operates correctly.

8 NOTE 1—The learning VIDs used for explicit trees are not allocated to the SPVID-Pool-MSTID. Therefore, they are
9 not taken from the SPVID pool; they are not Shortest Path VIDs even if conveyed by an SPVID field of an IS-IS sub-
10 TLV. To emphasize, a learning VID is an SPVID only if it is taken from the SPVID pool.

11 The Topology sub-TLV (45.1.9) conveys only the VLAN's Base-VID for each ECT Algorithm. Further
12 VIDs, if any, are associated with the VLAN and its Base VID by ISIS-SPB sub-TLVs as specified below.

13 If the ST ECT Algorithm (Table 45-1) is used for either a learning or a non-learning VLAN, then the VLAN
14 is supported only by its Base VID. The VLAN's Base VID shall be associated with the ST ECT Algorithm
15 in the SPB Base VLAN-Identifiers sub-TLV (28.12.4) and in the SPB Instance sub-TLV (28.12.5).

16 If the LT ECT Algorithm (Table 45-1) is used for either a learning or a non-learning VLAN, then the VLAN
17 is supported only by its Base VID, which is associated with the LT ECT Algorithm in the SPB Base VLAN-
18 Identifiers sub-TLV (28.12.4) and in the SPB Instance sub-TLV (28.12.5).

19 If the LTS ECT Algorithm (Table 45-1) is used for a non-learning VLAN, then the VLAN is supported only
20 by its Base VID, which is associated with the LTS ECT Algorithm in the SPB Base VLAN-Identifiers sub-
21 TLV (28.12.4) and in the SPB Instance sub-TLV (28.12.5).

22 NOTE 2—The use of the LTS ECT Algorithm for a non-learning VLAN is similar to SPBM (Clause 27 and Clause 28)
23 operations, but constrained trees are used instead of SPTs. One VID is sufficient for the support of a set of constrained
24 trees, which is specified as a set of loose explicit trees.

25 However, if the LTS ECT Algorithm (Table 45-1) is used for a learning VLAN, then the VLAN is supported
26 by multiple VIDs because each tree of an LTS is required to have its own VID. The VLAN's Base-VID is
27 associated with the LTS ECT Algorithm in the SPB Base VLAN-Identifiers sub-TLV (28.12.4) and in the
28 SPB Instance sub-TLV (28.12.5). In addition to the Base VID, as many VIDs are needed to support the
29 VLAN as the number of transmitter Edge Bridges specified by the Topology sub-TLV (45.1.9). The
30 individual VID of a transmitter Edge Bridge is configured to support the VLAN at the given SPT Bridge.
31 The VID to be used for the loose tree of a transmitter Edge Bridge is conveyed by the SPVID field of the
32 VLAN ID Tuple of the corresponding Base VID in the SPB Instance sub-TLV (28.12.5) of the given Edge
33 Bridge. The A flag of the given VLAN ID Tuple is cleared to indicate that auto-allocation is not used for the
34 SPVID parameter. If no VID is configured for the loose tree in the SPB Instance sub-TLV of a transmitter
35 Edge Bridge, then ISIS-PCR does not install the loose tree for that Edge Bridge. If the same local VID value
36 is configured at multiple Bridges, then ISIS-PCR does not install the loose tree for either Bridge, and the
37 conflict has to be resolved by operator action. Although the field is called SPVID, the VID it conveys is not
38 a Shortest Path VID but a loose tree VID as indicated by the ECT Algorithm field of the given VLAN ID
39 Tuple and by the VID to MSTID allocation.

40 NOTE 3—The use of the LTS ECT Algorithm for a learning VLAN is similar to SPBV (Clause 27 and Clause 28)
41 operations, but with loose trees, i.e., constrained trees are used instead of SPTs. Therefore, each Bridge rooting a loose
42 tree has to have its own VID for the support of a learning VLAN on that tree. The SPB Instance sub-TLV is used to bind
43 the VIDs of the individual root Bridges to the Base VID; the SPVID parameter conveys the VID to be used for the loose
44 tree rooted at the given Bridge. The VID is not a Shortest Path VID because it is used for a loose explicit tree as
45 indicated by the ECT Algorithm parameter and by the VID to MSTID allocation.

46 If the MRT ECT Algorithm (Table 45-1, 45.3.3) or the MRTG ECT Algorithm (Table 45-1, 45.3.4) is used
47 for a VLAN, then a distinct VID is required for the two MRTs: MRT-Blue and MRT-Red. In the case of a
48 non-learning VLAN, a single VID is used for MRT-Blue of all MRT Roots, and another VID is used for all
49 MRT-Reds. Two VIDs support a non-learning VLAN if the MRTs protect each other, whereas the VID of the

1 SPT Set is also needed as the third VID if the MRTs protect SPTs. The MRT VIDs are also allocated to the
2 SPBM-MSTID (0xFFC) in the case of a non-learning VLAN. However, each MRT Root has to be
3 configured with its own unique VID-pair for its MRT-Blue and MRT-Red in the case of a learning VLAN,
4 and all the MRT VIDs are allocated to the SPBV-MSTID (0xFFD). Independent VLAN Learning (3.104) is
5 applied among the VIDs associated with MRTs. Twice as many MRT VIDs support a learning VLAN as
6 MRT Roots, which are all the VIDs that are needed if the MRTs protect each other. The Shortest Path VIDs
7 (real SPVIDs) of the SPTs are also needed in addition to the MRT VIDs for the support of a learning VLAN
8 if the MRTs protect SPTs. As the two MRT algorithms differ only in GADAG handling, the VLAN
9 configuration is the same for these two ECT Algorithms once the appropriate ECT-ALGORITHM value is
10 used in the ECT Algorithm field of the IS-IS sub-TLVs. The VLAN's Base VID is associated with the MRT
11 ECT Algorithm or with the MRTG ECT Algorithm in the SPB Base VLAN identifier sub-TLV (28.12.4).

12 The SPB Instance sub-TLV (28.12.5) provides the VIDs for MRT-Blue and MRT-Red and also associates
13 them with the VLAN. The same VID values are used in the corresponding SPVID fields in the SPB Instance
14 sub-TLV of all MRT Roots in the case of a non-learning VLAN; whereas the VID values in the SPVID
15 fields are unique for each MRT Root in the case of a learning VLAN. The SPB Instance sub-TLV also
16 specifies whether the MRTs are used to protect each other or whether they protect an SPT.

17 The SPB Instance sub-TLV (28.12.5) conveys three VLAN ID Tuples for the VLAN. The ECT Algorithm
18 parameter of the second and the third VLAN ID Tuple is either the MRT ECT Algorithm or the MRTG ECT
19 Algorithm, accordingly. If the MRTs are used to protect SPTs, then the ECT Algorithm parameter of the first
20 VLAN ID Tuple specifies the ECT Algorithm to be used for shortest path computation, e.g., one of the
21 standard symmetric ECT Algorithms (Table 45-1). The first VLAN ID Tuple also specifies the VID to be
22 used for the shortest paths, which is the Base VID for a non-learning VLAN or the Shortest Path VID
23 (allocated to the SPVID-Pool-MSTID) conveyed by the SPVID parameter for a learning VLAN. If the
24 MRTs are used to protect each other, then the ECT Algorithm parameter of the first VLAN ID Tuple is the
25 same as that of the second and the third VLAN ID Tuple. The second and the third VLAN ID Tuples
26 associate the MRT VIDs with the Base VID. The SPVID parameter of the second VLAN ID Tuple provides
27 the VID for MRT-Blue. The SPVID parameter of the third VLAN ID Tuple provides the VID for MRT-Red.
28 In other words, the SPVID parameters of the second and third VLAN ID tuples convey the MRT VIDs, not
29 Shortest Path VIDs. If ISIS-PCR detects that the same local MRT VID value is configured at multiple
30 Bridges, then ISIS-PCR does not install the MRT for either Bridge, and the conflict has to be resolved by
31 operator action.

32 If the MRTs are used to protect SPTs, then the MRT VIDs differ from the Base VID and also differ from
33 Shortest Path VID conveyed by the SPVID parameter of a learning VLAN's first VLAN ID Tuple. In the
34 case of non-learning VLANs, if the MRTs are used to protect each other, then the VLAN's Base VID is used
35 as the VID for MRT-Blue; consequently, the SPVID parameter of the VLAN's second VLAN ID Tuple
36 conveys the VLAN's Base VID. If the MRTs of a learning VLAN are used to protect each other, then no
37 Shortest Path VID is required for the VLAN; therefore, the SPVID parameter of the VLAN's first VLAN ID
38 Tuple is not used but ignored. The use of the SPB instance sub-TLV for the MRT and MRTG ECT
39 Algorithms is illustrated in Figure 45-3.

40 NOTE 4—The SPVID parameter conveys only a Shortest Path VID if the ECT Algorithm parameter of the given VLAN
41 ID Tuple identifies shortest path operations and the VID is allocated to the SPVID-Pool-MSTID. The SPVID parameter
42 is not a Shortest Path VID if the ECT Algorithm parameter identifies explicit path control, i.e., taken from Table 45-1,
43 which is also indicated by the fact that the VID is not allocated to the SPVID-Pool-MSTID.

44 An I-SID of a PBBN shall be associated with an explicit tree by the SPBM Service Identifier and Unicast
45 Address sub-TLV (28.12.10) by means of associating the I-SID with a Base VID that is allocated to one of
46 the explicit ECT Algorithms of Table 45-1.

	Parameter	Value			
		<i>non-learning VLAN</i>		<i>learning VLAN</i>	
		<i>MRTs protect SPT</i>	<i>MRTs protect each other</i>	<i>MRTs protect SPT</i>	<i>MRTs protect each other</i>
	Type	1			
	Length	43			
	CIST Root Identifier	CIST Root Identifier (imported from RSTP or MSTP)			
	CIST External Root Path Cost	CIST External Root Path Cost (imported from RSTP or MSTP)			
	Bridge Priority	Bridge Priority			
	reserved	0			
	V	1 or 0			
	SPSourceID	SPSourceID or 0			
	Number of Trees	3			
	U	1			
	M	1	1	0	0
VLAN ID Tuple 1	A	0	0	1 or 0	0
	reserved	0			
	ECT Algorithm	Symmetric ECT Alg.	MRT or MRTG	Symmetric ECT Alg.	MRT or MRTG
	Base VID	Base VID			
	SPVID	–	–	SPVID	–
	U	1			
	M	1	1	0	0
VLAN ID Tuple 2	A	0			
	reserved	0			
	ECT Algorithm	MRT or MRTG			
	Base VID	Base VID			
	SPVID	MRT-Blue domain VID	MRT-Blue domain VID = Base VID	MRT-Blue local VID	MRT-Blue local VID
	U	1			
	M	1	1	0	0
VLAN ID Tuple 3	A	0			
	reserved	0			
	ECT Algorithm	MRT or MRTG			
	Base VID	Base VID			
	SPVID	MRT-Red domain VID	MRT-Red domain VID	MRT-Red local VID	MRT-Red local VID
	U	1			
	M	1	1	0	0

Figure 45-3—The use of the SPB Instance sub-TLV for MRT

1 45.1.4 Use of VIDs for strict explicit trees

2 The use of a distinct VID for each explicit tree does not scale in some cases. A more flexible and scalable
3 method of VID assignment is available for explicit trees. This subclause explains the rules to be observed
4 and suggests two schemes for the use of VIDs for strict explicit trees, i.e., for VIDs associated with the ST
5 ECT Algorithm (Table 45-1). Nonetheless, ensuring that VIDs associated with strict explicit trees actually
6 follow these rules in order to provide unambiguous and loop-free frame forwarding is the responsibility of
7 the network administrator and the owner PCE controlling the VID; this standard provides no method for
8 policing this. As each VID belongs to a single owner PCE, a VID can be used only for multiple explicit trees
9 that are controlled by the same PCE.

10 NOTE 1—Use of VIDs associated with MRTs is explained in 45.3.3.

11 Ensuring unambiguous filtering entries and providing forwarding to the appropriate destination are the
12 fundamental requirements to be met when assigning VIDs with explicit trees.

13 For unicast frames, there has to be a single egress port for each Individual MAC, VID tuple in each SPT
14 Bridge. In other words, different explicit trees associated with the same bidirectional VID within an SPT
15 Domain are not allowed to have any SPT Bridges in common when MAC learning is being used.
16 Unidirectional VIDs or bidirectional non-learning VIDs associated with different explicit trees to a
17 particular destination can have common SPT Bridges along the merged segments of those explicit trees as
18 long as every such tree uses a single egress port for each unicast destination in every Bridge.

19 NOTE 2—A VID can be unidirectional by means of asymmetric use, e.g., as explained in F.1.3 or like an SPVID
20 (3.257). Also, a bidirectional non-learning VID is in fact used in unidirectional fashion with respect to any given
21 destination MAC address.

22 For multicast frames, it has to be ensured that each member of the multicast group receives only a single
23 copy of a particular frame, even if there are multiple potential sources within a group.

24 Congruency (3.222), if required, has to be enforced by the PCE. In the case of unidirectional VIDs, the same
25 path has to be used for both directions between a source and destination pair in order to provide reverse path
26 congruency, essential when MAC learning is employed. Unicast and multicast traffic have to be placed on
27 the same tree in order to provide unicast multicast congruency.

28 The tree structures explained in 45.1.1 can be applied for strict explicit trees, and VIDs can be used on top of
29 these strict trees as follows.

30 An ad-hoc explicit tree (45.1.1) with a bidirectional VID provides a shared bidirectional (*,G) tree joining
31 all members of a group. This structure minimizes VID consumption (as a local instance-by-instance
32 optimization) compared to cases when each member has its own VID. On the other hand, there is the
33 constraint that trees using the same VID are not allowed to touch or cross; hence the number of explicit trees
34 is limited by the available VID space.

35 The template tree (45.1.1) approach constructs one or more explicitly routed forwarding planes with tree
36 sets, one tree rooted on each Bridge within each plane. In the case of learning VIDs, each tree is associated
37 with a unidirectional VID, and Shared VLAN Learning (3.247) takes place among the VIDs of a tree set. In
38 the case of non-learning VIDs, the MAC of the root Bridge identifies the tree; therefore, one VID (the Base
39 VID) is enough for an entire plane. Furthermore, source-specific multicast (S,G) has to be applied if
40 multiple sources participate in a single multicast group as explained in 45.1.1. The desired connectivity can
41 be then laid for VIDs associated with strict template trees with the complete assurance that the required
42 simply connected tree constraint is obeyed.

43 The efficiency of the template trees approach can be improved substantially in terms of the usage of local
44 VIDs if Edge Bridges “inherit” the tree and VID rooted at their directly connected Core Bridge. This is
45 possible because a loop or forwarding ambiguity cannot be created in a single Ethernet hop; to guarantee
46 this, multi-homed Edge Bridges have to be always configured as non-transit Bridges.

1 45.1.5 MAC addresses and ISIS-PCR

2 Propagation of MAC address information by IS-IS is required for explicit trees in order to create Dynamic
3 Filtering Entries for VID, MAC tuples, except for learning VIDs. The SPBV MAC Address sub-TLV
4 (28.12.9) shall be used for the advertisement of both Individual and Group MAC Addresses for S-VLANs
5 and C-VLANs associated with explicit trees. The VLAN's VID is conveyed by Octets 3 and 4 of the SPBV
6 MAC Address sub-TLV. This local VID (if required) and the VLAN's Base VID are allocated either to the
7 SPBV-MSTID (for learning VIDs) or to the SPBM-MSTID (for non-learning VIDs); furthermore, the Base
8 VID is allocated to one of the explicit ECT Algorithms (Table 45-1) as explained in 45.1.2 in more detail.

9 NOTE 1—Multiple SPBV MAC Address sub-TLVs are used if different MAC addresses are mapped to different VIDs
10 at an SPT Bridge, i.e., MAC addresses associated with an SPT Bridge are not bound together.

11 The SPBM Service Identifier and Unicast Address sub-TLV (28.12.10) shall be used to associate Individual
12 addresses with an I-SID; Octets 3 through 8 of the sub-TLV convey an Individual B-MAC address. Thus, the
13 SPT Bridges populate their FDB with the Individual MAC addresses according to their T/R flags for the B-
14 VLANs allocated to the SPBM-MSTID and associated with explicit path control.

15 Either a Group MAC address associated with an I-SID is source specific, or it is the Backbone Service
16 Instance Group address (26.4). If Octets 3 through 8 of any SPBM Service Identifier and Unicast Address
17 sub-TLV convey the null value throughout the SPT Domain, then the Backbone Service Instance Group
18 address corresponding to the given I-SID is to be used for all multicast sources in (*,G) mode on one simple
19 explicit tree. If no SPBM Service Identifier and Unicast Address sub-TLV with null value in its B-MAC
20 address field is present in the SPT Domain, then the source-specific group addressing specified in 27.15 has
21 to be applied, exactly as used for SPBM. In the source-specific case, the SPB Instance sub-TLV (28.12.5)
22 shall be used to propagate the SPSourceID (27.10) of the Bridges that are Edge Bridges of the explicit tree
23 and can be a source of the multicast traffic. In order to avoid forwarding anomalies, source-specific group
24 addressing has to be used if there are multiple multicast sources for a given I-SID using the same B-VID and
25 the template tree model (45.1.4) is followed, i.e., multiple trees are used.

26 NOTE 2—Multiple SPBM Service Identifier and Unicast Address sub-TLVs are used if different MAC addresses are
27 mapped to different I-SIDs at an SPT Bridge, i.e., MAC addresses associated with an SPT Bridge are not bound together.

28 45.1.6 Filtering Database entries for explicit trees

29 The Topology sub-TLV (45.1.9) provides the information needed for ISIS-PCR to create the appropriate
30 Dynamic VLAN Registration Entries, i.e., it provides the explicit tree, the VIDs associated with the tree, and
31 the direction of the directed VIDs.

32 The per Bridge association of a MAC address with a VID is provided by the SPBV MAC Address sub-TLV
33 (28.12.9) as described in 45.1.5. Based on the Topology sub-TLVs and SPBV MAC Address sub-TLVs,
34 ISIS-PCR can create the VID, MAC tuple Dynamic Filtering Entries for VLANs associated with explicit
35 trees.

36 The association of a B-VID with an explicit tree is provided by the Topology sub-TLV (45.1.9), and the per
37 Bridge association of an I-SID to a B-VID is provided by the SPBM Service Identifier and Unicast Address
38 sub-TLV (28.12.10), which also provides the association of a B-MAC to an I-SID as described in 45.1.5.
39 The SPBM Service Identifier and Unicast Address sub-TLVs also make it clear whether source-specific
40 group addressing is to be used. If that is the case, then the SPB Instance sub-TLVs (28.12.5) provide the
41 SPSourceIDs (27.10) that are needed for the creation of the automatically generated Group MAC addresses
42 of the I-SID as described in 45.1.5. Based on these sub-TLVs, ISIS-PCR can create the B-VID, B-MAC
43 tuple Dynamic Filtering Entries for I-SIDs associated with explicit trees.

1 45.1.7 ISIS-PCR support

2 ISIS-PCR and PCEs are configured to use one of the Group addresses from Table 8-17 to establish
3 adjacencies and exchange PDUs. ISIS-PCR running on C-VLAN components use the Customer Bridge
4 address. ISIS-PCR running on S-VLAN components use the Provider Bridge address. ISIS-PCR running on
5 B-VLAN components can use the Provider Bridge Address or one of the existing IS-IS addresses.

6 The use of the ISIS-PCR adjacency between Bridges is contingent on the Bridges inclusion within the same
7 SPT Region and thus requires that they have an MCID and Auxiliary MCID (13.8, 28.12.2) where at least
8 one matches on every adjacency in the Region (8.9.4, 13.8). MCID and Auxiliary MCID enable migration of
9 the MCID definition allowing these Bridges to operate as one SPT Region under certain small changes. The
10 operational set of Base VLANs has to be consistent during migration. The MCID and the Auxiliary MCID
11 are exchanged in IS-IS Hello PDUs, which also include the SPB Base VLAN-Identifiers sub-TLV (28.12.4).
12 ISIS-PCR adjacencies are formed and maintained as specified by 28.2, i.e., in exactly the same way as ISIS-
13 SPB adjacencies. Either a PCE hosted by an end station has a PCA that is part of the IS-IS Domain, or the
14 PCE forms and maintains an ISIS-PCR adjacency between the PCE and the SPT Bridge to which the end
15 station is attached.

16 Management of explicit trees also requires the following:

- 17 a) Management of the PCR objects (12.26)
- 18 b) Administrative agreement on the MCID Configuration Name and Revision Level (Clause 28)

19 NOTE—The addresses used by IS-IS PDUs and the operations with respect to IS-IS adjacencies are the same for ISIS-
20 PCR and ISIS-SPB. This subclause (45.1.7) simply reiterates provisions of Clause 27 and Clause 28 (see 27.2, 28.2).

21 45.1.8 Attributes for path computation

22 More attributes than the link metric are needed for PCEs or BLCEs in order to be able to determine paths
23 that meet the requirements. Extended link attributes are specified by the Traffic Engineering (TE) extensions
24 for IS-IS in IETF RFC 5305. The TE information is flooded in LSPs by IS-IS and it is stored in the Traffic
25 Engineering Database (TED). The TED contains the topology and the resource information of the IS-IS
26 Domain.

27 SPT Bridges may support the Extended IS Reachability TLV (type 22) specified in IETF RFC 5305, which
28 provides the following link attribute IS-IS sub-TLVs:

- 29 a) Administrative Group (color, resource class) (sub-TLV type 3)
- 30 b) Maximum Link Bandwidth (sub-TLV type 9)
- 31 c) Maximum Reservable Link bandwidth (sub-TLV type 10)
- 32 d) Unreserved Bandwidth (sub-TLV type 11)
- 33 e) Traffic Engineering Default Metric (sub-TLV type 18)

34 SPT Bridges may support the following IS-IS TE Metric Extension link attribute sub-TLVs specified in
35 IETF RFC 7810:

- 36 f) Unidirectional Link Delay (sub-TLV type 33)
- 37 g) Min/Max Unidirectional Link Delay (sub-TLV type 34)
- 38 h) Unidirectional Delay Variation (sub-TLV type 35)
- 39 i) Unidirectional Link Loss (sub-TLV type 36)
- 40 j) Unidirectional Residual Bandwidth (sub-TLV type 37)
- 41 k) Unidirectional Available Bandwidth (sub-TLV type 38)
- 42 l) Unidirectional Utilized Bandwidth (sub-TLV type 39)

1 The IS-IS TE Metric Extensions sub-TLVs are conveyed by IS-IS Extended IS Reachability TLV (type 22).
2 The bandwidth already allocated determines the Residual Bandwidth and the Available Bandwidth. When
3 they are used in a Bridged Network, reservations performed by MSRP or bandwidth assignments recorded
4 by ISIS-PCR are taken into account to determine these sub-TLVs.

5 NOTE—The Residual Bandwidth and the Unreserved Bandwidth sub-TLVs can be used concurrently; see
6 IETF RFC 7810.

7 If Shared Risk Link Group (SRLG) information is to be applied, then the SRLG TLV (type 138) of IETF
8 RFC 5307 may be used as specified here. Figure 45-4 shows the format of the SRLG TLV.

	Octet	Length
Type (138)	1	1
Length	2	1
System ID	3–8	6
Pseudonode num	9	1
Flag	10	1 bit
reserved	10	7 bits
Link Local Identifier	11–14	4
Link Remote Identifier	15–18	4
SRLG <i>l</i>	Shared Risk Link Group Value	19–22 4
	...	
SRLG <i>n</i>	Shared Risk Link Group Value	(4 <i>n</i> +15)–(4 <i>n</i> +18) 4

Figure 45-4—Shared Risk Link Group (SRLG) TLV

9 The SRLG parameters are encoded as follows:

- 10 m) Type (8 bits). Octet 1 conveys the type of the sub-TLV, and its value is 138.
- 11 n) Length (8 bits). Octet 2 conveys the total number of bytes contained in the Value field, which is the
12 number of SRLG Values multiplied by 4 plus 16 bytes.
- 13 o) System ID (48 bits). Octets 3 through 8 convey the 6-byte IS-IS System Identifier of the adjacent
14 neighbor.
- 15 p) Pseudonode num (8 bits). Octet 9 conveys the pseudonode number if the neighbor is connected
16 through a shared media.
- 17 q) Flag (1 bit). Octet 10 conveys a flag, which is encoded as a bit in a single octet. A flag is set if the bit
18 takes the value 1; otherwise, it is reset. The Flag of Octet 10 indicates whether an IP address is used
19 to identify the link. It is not an IP address in the case of ISIS-PCR, i.e., the link is unnumbered.
20 Therefore, the flag is reset in an SPT Domain, i.e., each bit of Octet 10 takes value 0.
- 21 r) reserved (7 bits). Seven bits of Octet 10 are reserved for future use, transmitted as 0, and ignored on
22 receipt.
- 23 s) Link Local Identifier (32 bits). Octets 11 through 14 convey the local identifier of the interface,
24 which is an Extended Local Circuit ID as specified by IETF RFC 5303.
- 25 t) Link Remote Identifier (32 bits). Octets 15 through 18 convey the identifier of the remote interface,
26 which is a Neighbor Extended Local Circuit ID as specified by IETF RFC 5303.
- 27 u) Shared Risk Link Group Value (32 bits). Octets 4*n*+15 through 4*n*+18 convey the value assigned to
28 SRLG *n* to which the link belongs. The same SRLG value is used for all links that share a resource
29 whose failure affects each link of the group. A link can belong to multiple SRLGs; therefore,
30 multiple Shared Risk Link Group Values can be present in the same SRLG TLV.

45.1.9 Topology sub-TLV

The variable-length Topology sub-TLV shall be used to describe an explicit tree. The Topology sub-TLV may be also used for describing a Generalized Almost Directed Acyclic Graph (GADAG) as explained in 45.3.4 in detail. The Topology sub-TLV is carried in an MT-Capability TLV (type 144) in a Link State PDU. A Topology sub-TLV specifying an explicit tree conveys one or more Base VIDs and two or more Hop sub-TLVs (45.1.10) and can convey further sub-TLVs (45.1.11, 45.1.12) to constrain loose hops. The Topology sub-TLV also specifies the algorithm to be used for constrained routing. Figure 45-5 shows the format of the Topology sub-TLV.

		Octet	Length	
	Type (21)	1	1	
	Length	2	1	
	Number of Base VIDs	3	1	
Base VID Tuple <i>l</i>	reserved	4	4 bits	0 or
	Base VID	4–5	12 bits	16 bits
	...			
Base VID Tuple <i>n</i>	reserved	(2 <i>n</i> +2)	4 bits	0 or
	Base VID	(2 <i>n</i> +2)– (2 <i>n</i> +3)	12 bits	16 bits
sub-TLV <i>l</i>	sub-TLV			
	...			
sub-TLV <i>m</i>	sub-TLV			

Figure 45-5—Topology sub-TLV

The parameters of explicit trees are encoded by the Topology sub-TLV as follows:

- a) Type (8 bits). Octet 1 conveys the type of the sub-TLV, and its value is 21.
- b) Length (8 bits). Octet 2 conveys the total number of bytes contained in the Value field.
- c) Number of Base VIDs (8 bits). Octet 3 conveys the number of Base VIDs carried in the Topology sub-TLV. Its minimum value is 1 if the Topology sub-TLV specifies an active topology. Its value can be 0 if the Topology sub-TLV specifies only a GADAG.
- d) reserved (4 bits). Four bits of Octet 2*n*+2 are reserved for future use, transmitted as 0, and ignored on receipt.
- e) Base VID (12 bits). Octet 2*n*+2 and Octet 2*n*+3 convey the *n*th Base VID parameter. The Base VID parameter provides the Base VID of the VLAN that is associated with the explicit tree. Multiple Base VIDs can be associated with the same explicit tree.

Some of the explicit ECT Algorithms (45.1.2) require further VIDs, in addition to the Base VID, which are associated with the VLAN by using the SPB Instance sub-TLV as specified in 45.1.3.

By default, each Edge Bridge is both transmitter and receiver for all VIDs associated with an explicit tree. Different behavior can be specified in the Hop sub-TLV (45.1.10) of the Edge Bridge.

If VIDs of a learning VLAN associated with the LTS ECT Algorithm (Table 45-1) are conveyed by the Topology sub-TLV, then the default behavior for each Edge Bridge is that it is the only transmitter on its own VID and a receiver on the VIDs of all the other Edge Bridges. For such a learning VLAN, a per Bridge local VID is announced by a transmitter Edge Bridge via the SPB Instance sub-TLV (28.12.5) as specified in 45.1.3. If no VID is allocated to the Edge Bridge, then it is not a transmitter but only a receiver.

A Topology sub-TLV specifying a GADAG can have zero Base VID parameters. In this case, the given GADAG has to be applied for each VLAN associated with the MRTG ECT Algorithm.

f) sub TLVs. The rest of the octets convey further sub-TLVs that specify the hops of the topology and can also specify constraints, bandwidth assignment, and timestamp as specified in the following subclauses.

A topology is specified by a list of Hop sub-TLVs (45.1.10), and a hop is specified by an IS-IS System ID. An ill-formed Topology sub-TLV, e.g., specifying an invalid strict tree or having a conflict in its specification, is ignored; no tree is installed but a management report is generated.

The Topology sub-TLV specifies a strict tree by decomposing the tree to branches. Each branch is a point-to-point path specified by an ordered list of hops where the end of each branch is a leaf. Each element of a branch is the direct link between adjacent neighbor Bridges whose Hop sub-TLV is next to each other in the Topology sub-TLV. The first hop of the Topology sub-TLV is the root; hence, the first branch originates from the root. The rest of the branches fork from another branch. The first hop of a branch is a Bridge that is already part of a former branch, and the last hop is a leaf Bridge. Therefore, the hop after a leaf hop is the beginning of a new branch, if any. A hop of a branch is created if and only if the Bridge specified for that hop is directly connected to the preceding Bridge of the same branch. The order of the branches does not matter within the Topology sub-TLV, but the first branch begins with the root. Figure 45-6 shows an example for the specification of a strict tree.

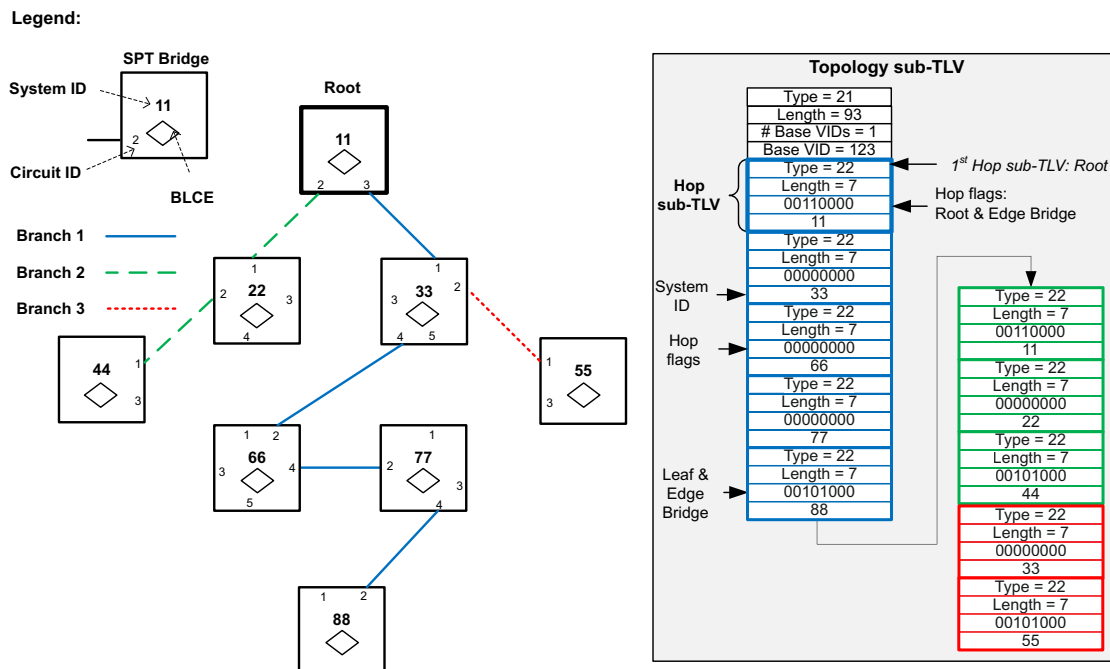


Figure 45-6—A strict tree and its descriptor Topology sub-TLV

The Topology sub-TLV of a loose tree does not provide any path or path segment, but the hops that are to participate. The root is the first hop. The leaves of a single loose tree are also specified. Hop sub-TLVs can be included in a Topology sub-TLV to specify Bridges that have to be avoided. If the Topology sub-TLV specifies only a single leaf, then one or more transit hops can be specified by the Topology sub-TLV to direct the path along a sequence of Bridges, specified by the order of hops. If Bridges whose respective Hop sub-TLVs are adjacent to each other in the Topology sub-TLV but are not topology neighbors, then it is a loose hop. If a Topology sub-TLV conveys one or more loose hops, then that sub-TLV defines a loose explicit tree, and each hop is considered as a loose hop. If a loose tree specified by a Topology sub-TLV would create a loop or hairpin, then the loop is pruned, and the remaining hops are installed. An example for the specification of a single loose tree is shown in Figure 45-7, where Bridge 77 is the root; Bridges 11, 44, 55,

1 and 88 are leaves; and Bridge 22 is excluded from the tree. Bridges 11, 44, 55, and 88 are the Edge Bridges.
2 Bridges 11, 44, 55, 77, and 88 are part of the tree because each of them has its special role as specified by the
3 Topology sub-TLV. Conversely, the role of Bridges 33 and 66 is determined by the BLCE of the SPT
4 Bridges only when the tree is computed after the reception of the Topology sub-TLV.

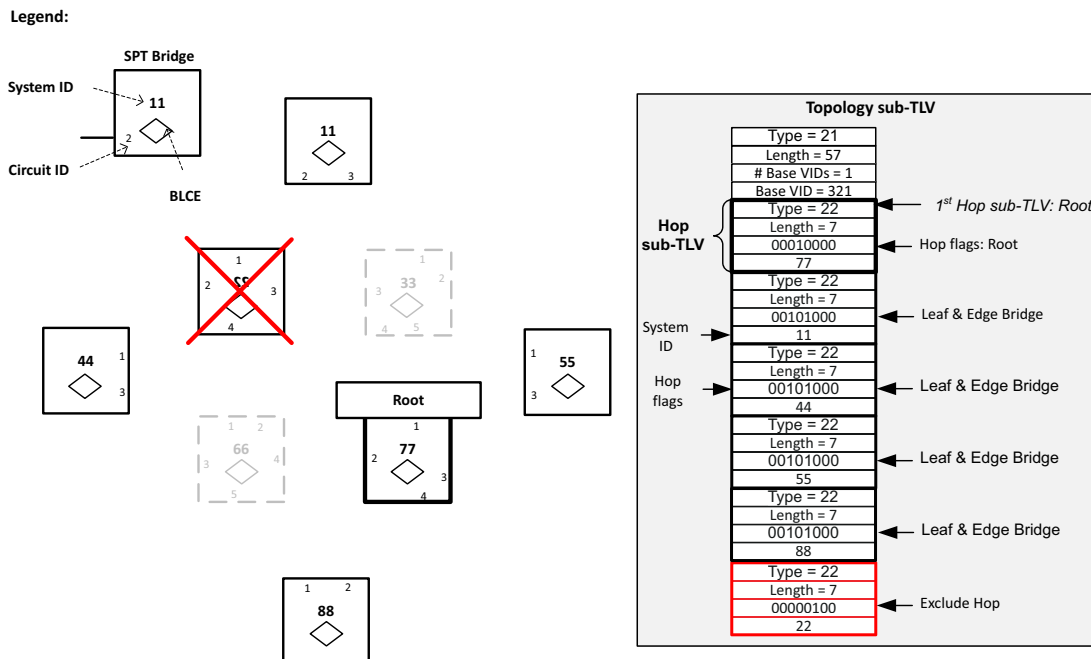


Figure 45-7—Topology sub-TLV of a loose tree

5 If the Base VIDs of the Topology sub-TLV are associated with the LTS ECT Algorithm or the MRT ECT
6 Algorithm, then the Hop sub-TLVs conveyed by the Topology sub-TLV belong to Edge Bridges or Bridges
7 to be excluded [item c6) in 45.1.10]. The BLCEs compute the loose trees, e.g., MRTs, so that they span the
8 Edge Bridges and are rooted at an Edge Bridge.

9 The Topology sub-TLV specifies a GADAG if the Base VIDs conveyed by the Topology sub-TLV are
10 associated with the MRTG ECT Algorithm (Table 45-1). The Topology sub-TLV specifies a GADAG by
11 directed ear decomposition. A directed ear is a directed point-to-point path whose end points can coincide,
12 but no other element of the path is repeated in the ear. Each ear is specified by an ordered list of hops, and
13 that order of hops is according to the direction of the arcs in the GADAG. There are no leaves in a GADAG;
14 hence, the Leaf flag is used to ease the parsing of the Topology sub-TLV and to mark the end of a topology
15 block (3.281). The sequence of ears in the Topology sub-TLV is such that the end points of an ear belong to
16 former ears. The GADAG Root is not marked by any flag, but the GADAG Root is the first hop in the
17 Topology sub-TLV; correspondingly the first ear starts and ends with the GADAG Root. MRT Roots are
18 marked by the Root flag, and all other Edge Bridges are leaves of the MRTs. If no MRT Root is specified,
19 then each SPT Root is also an MRT Root. An example GADAG is shown in Figure 45-13 (in 45.3.4).

20 Each Edge Bridge of an explicit tree is always specified in the Topology sub-TLV by the inclusion of the
21 Hop sub-TLVs corresponding to the Edge Bridges. The Edge Bridges of a tree are identified by setting the
22 Edge Bridge flag [item c3) in 45.1.10] in the appropriate Hop sub-TLVs.

23 If the explicit tree is loose, then Administrative Group (45.1.11) and Bandwidth Constraint (45.1.12) sub-
24 TLVs can be also conveyed by the Topology sub-TLV. In that case, each path of the tree has to meet the
25 constraints specified by these sub-TLVs.

1 If ISIS-PCR is used for recording bandwidth assignment, then the Topology sub-TLV conveys the
2 Bandwidth Assignment sub-TLV (45.2.1), and it can also convey a Timestamp sub-TLV (45.2.2). If the
3 assignment of the bandwidth indicated by the Bandwidth Assignment sub-TLV of the Topology sub-TLV
4 would result in overbooking any link of the explicit tree, then bandwidth assignment is not performed, and a
5 management report is generated. If such a Topology sub-TLV describes a new valid explicit tree, then the
6 paths of the tree are installed, but no bandwidth assignment is performed by ISIS-PCR.

7 NOTE 1—The LSP ID can be also considered as an identifier of an explicit tree if an LSP describes only a single tree,
8 i.e., includes a single Topology sub-TLV.

9 NOTE 2—Allocation of VIDs to individual Aggregation Links between adjacent IS-IS Systems has to be done by
10 the explicit configuration of conversations in that LAG at the two Systems in accordance with IEEE Std 802.1AX.

11 **45.1.10 Hop sub-TLV**

12 The Hop sub-TLV shall be used to specify a hop of a topology. Each Hop sub-TLV conveys an IS-IS System
13 ID, which specifies a hop. A Hop sub-TLV is conveyed by a Topology sub-TLV (45.1.9). A strict explicit
14 tree is decomposed to branches where each branch is a point-to-point path specified by an ordered list of
15 Hop sub-TLVs as specified in 45.1.9 and illustrated in Figure 45-6. A hop of a branch is created if and only
16 if the Bridge specified for that hop is directly connected to the preceding Bridge in the path. In other words,
17 a point-to-point LAN is identified by the two Bridges it interconnects; and the LAN is part of the strict tree
18 if and only if the Hop sub-TLVs of the two Bridges are next to each other in the Topology sub-TLV. A Hop
19 sub-TLV can convey a Circuit ID in order to distinguish multiple links between adjacent neighbor Bridges.
20 A Hop sub-TLV also specifies the role of a Bridge, e.g., if it is the root or an Edge Bridge. The Topology
21 sub-TLV of a loose tree comprises only the Hop sub-TLVs of the Bridges that have special roles in the tree.
22 The Hop sub-TLV may also specify a delay budget for a loose hop.

23 By default, the Edge Bridges both transmit and receive with respect to each VID associated with an explicit
24 tree, except for an LTS associated with a learning VLAN [45.1.3, item e) in 45.1.9], which uses a
25 unidirectional VID per Bridge. The Hop sub-TLV allows different configuration by means of the VID's
26 Transmit (T) and Receive (R) flags conveyed in the sub-TLV. The VID and its T/R flags are present in the
27 Hop sub-TLV only if the behavior of the Edge Bridges differs from the default [defined in item e) in 45.1.9].
28 The T/R flags of an I-SID associated with an explicit tree in a PBBN are set in the SPBM Service Identifier
29 and Unicast Address sub-TLV (28.12.10), which also provides the association of the I-SID with the B-VID.

30 Figure 45-8 shows the format of the variable-length Hop sub-TLV, which shall be conveyed by a Topology
31 sub-TLV (45.1.9).

		Octet	Length	
	Type (22)	1	1	
	Length	2	1	
Hop Flags	Circuit	3	1 bit	8 bits
	VID	3	1 bit	
	Edge Bridge	3	1 bit	
	Root	3	1 bit	
	Leaf	3	1 bit	
	Exclude	3	1 bit	
	reserved	3	2 bits	
	System ID	4–9	6	
	Extended Local Circuit ID	10–13	0 or 4	
	Number of VIDs	10 or 14	0 or 1	
VID Tuple <i>l</i>	T	11 or 15	1 bit	0 or 16 bits
	R	11 or 15	1 bit	
	reserved	11 or 15	2 bits	
	VID	11–12 or 15–16	12 bits	
	...			
VID Tuple <i>n</i>	T	$(2n+9)$ or $(2n+13)$	1 bit	0 or 16 bits
	R	$(2n+9)$ or $(2n+13)$	1 bit	
	reserved	$(2n+9)$ or $(2n+13)$	2 bits	
	VID	$(2n+9)–(2n+10)$ or $(2n+13)–(2n+14)$	12 bits	
	Delay Constraint	10–15 or 14–19 or $(2n+13)–(2n+16)$ or $(2n+17)–(2n+20)$	0 or 6	

Figure 45-8—Hop sub-TLV

1 The parameters of a hop are encoded as follows:

- 2 a) Type (8 bits). Octet 1 conveys the type of the sub-TLV, and its value is 22.
- 3 b) Length (8 bits). Octet 2 conveys the total number of bytes contained in the Value field.
- 4 c) Hop Flags (8 bits). Octet 3 conveys the Hop flags. The Circuit and the VID flags influence the
5 length of the Hop sub-TLV. Table 45-3 summarizes the use of the flags in the case of the different
6 explicit ECT Algorithms.
 - 7 1) Circuit flag (1 bit). The Circuit flag indicates whether the Extended Local Circuit ID parameter
8 is present. If the flag is set, then an Extended Local Circuit ID is also included in the Hop sub-
9 TLV.
 - 10 2) VID flag (1 bit). The VID flag indicates whether one or more VIDs are conveyed by the Hop
11 sub-TLV. If the flag is set, then the Number of VIDs parameter is present and indicates how
12 many VIDs are conveyed by the Hop sub-TLV. If the VID flag is reset, then neither the Number
13 of VIDs parameter nor VIDs are present in the Hop sub-TLV.
 - 14 3) Edge Bridge flag (1 bit). The Edge Bridge flag indicates whether the given System is an Edge
15 Bridge, i.e., transmitter and/or receiver. If the System is an Edge Bridge, then the Edge Bridge
16 flag is set. The Edge Bridge flag indicates that FDB entries have to be installed for the given
17 hop as specified by its SPBV MAC address sub-TLV or SPBM Service Identifier and Unicast
18 Address sub-TLV.

Table 45-3—Hop sub-TLV flags

		Flag			
		Edge Bridge	Root	Leaf	Exclude
ECT Algorithm	Strict Tree	The Bridge is an Edge Bridge.	The Bridge is the root of the tree. (first hop)	The Bridge is a leaf of the tree. (always an Edge Bridge)	—
	Loose Tree	The Bridge is an Edge Bridge. (set for every hop of multi-point trees except for exclude hops)	The Bridge is the root of the tree. (first hop)	The Bridge is a leaf of the tree. (always an Edge Bridge)	The Bridge has to be excluded from the tree.
	Loose Tree Set	The Bridge is an Edge Bridge. (set for every hop of multi-point trees except for exclude hops)	The Bridge roots a tree. (every Bridge that is Edge Bridge)	—	The Bridge has to be excluded from each tree.
	MRT	The Bridge is an Edge Bridge.	The Bridge is an MRT Root. (always an Edge Bridge)	—	The Bridge has to be excluded from the MRTs.
	MRT with GADAG	The Bridge is an Edge Bridge.	The Bridge is an MRT Root. (always an Edge Bridge)	The Bridge is the end of a topology block.	—

- 4) Root flag (1 bit). The Root flag indicates whether the given System is a root of the explicit tree specified by the Topology sub-TLV. If the System is a root of a tree, then the Root flag is set.

If the Topology sub-TLV specifies a single tree, i.e., the Base VIDs conveyed by the Topology sub-TLV are associated with either the ST ECT Algorithm or the LT ECT Algorithm, then the Root flag is set only for one of the Systems conveyed by the Topology sub-TLV. Furthermore, the first Hop sub-TLV of the Topology sub-TLV conveys the System that is the root of the tree. If the Topology sub-TLV specifies a Loose Tree Set, i.e., the Base VIDs conveyed by the Topology sub-TLV are associated with the LTS ECT Algorithm (Table 45-1), then the Root flag is set for each Edge Bridge as each of them roots a tree. The first Hop sub-TLV of the Topology sub-TLV conveys a root.

If the Topology sub-TLV is used for MRT operations, i.e., the Base VIDs conveyed by the Topology sub-TLV are associated with either the MRT ECT Algorithm or the MRTG ECT Algorithm (Table 45-1), then the Root flag is set for each MRT Root. If no MRT Root is specified by a Topology sub-TLV specifying a GADAG, then each SPT Root is an MRT Root as well.

If the Base VIDs conveyed by the Topology sub-TLV are associated with the MRTG ECT Algorithm (Table 45-1), then the Topology sub-TLV specifies a GADAG, and the very first Hop sub-TLV specifies the GADAG Root. There is no flag for indicating the GADAG Root.

- 5) Leaf flag (1 bit). The Leaf flag indicates whether the given System is a leaf of the explicit tree specified by the Topology sub-TLV. If the System is a leaf, then the Leaf flag is set.

The Leaf flag is used to mark a leaf of a tree only if the Topology sub-TLV specifies a single tree, i.e., the Base VIDs are associated with either the ST ECT Algorithm or the LT ECT Algorithm (Table 45-1).

The Leaf flag is not used for marking leaves if the ECT Algorithm produces multiple trees because a Bridge that is leaf in one tree can be the root of another tree.

- 1 The Leaf flag is used to indicate the end of a topology block (3.281) if the Topology sub-TLV
2 specifies a GADAG; see item f) in 45.1.9 and see 45.3.4.
- 3 6) Exclude flag (1 bit). The Exclude flag indicates whether the given System has to be excluded
4 from the topology. The Exclude flag and the Root flag cannot be set for a given hop at the same
5 time.
- 6 7) reserved (2 bits). Two bits of Octet 3 are reserved for future use, transmitted as 0, and ignored
7 on receipt.
- 8 d) System ID (48 bits). Octets 4 through 9 convey the 6-byte IS-IS System Identifier of the Bridge to
9 which the Hop sub-TLV refers.
- 10 e) Extended Local Circuit ID (32 bits). Octets 10 through 13 convey the Extended Local Circuit ID as
11 specified by IETF RFC 5303 if Extended Local Circuit ID is present in the Hop sub-TLV, as
12 indicated by the Circuit flag.
- 13 Parallel links corresponding to different IS-IS adjacencies between a pair of neighbor SPT Bridges
14 can be distinguished by means of the Extended Local Circuit ID. The Extended Local Circuit ID is
15 conveyed by the Hop sub-TLV specifying the Bridge nearer to the root of the tree and identifies a
16 circuit that attaches the given Bridge to its neighbor cited by the next Hop sub-TLV of the Topology
17 sub-TLV. The Extended Local Circuit ID can be used only in strict trees.
- 18 f) Number of VIDs (8 bits). A single octet conveys the Number of VIDs parameter, which is Octet 10
19 if the Extended Local Circuit ID parameter is absent whereas it is Octet 14 if the Extended Local
20 Circuit ID parameter is present. If the Hop sub-TLV does not convey VIDs, then the Number of
21 VIDs parameter is not present, as indicated by the VID flag.
- 22 g) VID and its T/R flags (14 bits). Two octets convey the VID and its T/R flags parameters. The first
23 one is Octet $2n+9$ and the second one is Octet $2n+10$ if the Extended Local Circuit ID parameter is
24 absent whereas the first one is Octet $2n+13$ and the second one is Octet $2n+14$ if the Extended Local
25 Circuit ID parameter is present.
- 26 The VID and its T/R flags are present in the Hop sub-TLV only if the given Bridge is an Edge
27 Bridge and it behaves differently from the default with respect to that particular VID. (The default
28 behavior is defined in item e of the Topology sub-TLV (45.1.9).)
- 29 1) T flag (1 bit). This bit is the Transmit allowed flag for the VID following the flag.
- 30 2) R flag (1 bit). This bit is the Receive allowed flag, i.e., R flag for the VID following the flag.
- 31 3) reserved (2 bits). Two bits of the first Octet are reserved for future use, transmitted as 0, and
32 ignored on receipt.
- 33 4) VID (12 bits). The VID can be a Base VID, an MRT VID, or a VID in support of a learning
34 VLAN associated with the LTS ECT Algorithm.
- 35 h) Delay Constraint (48 bits). The last six octets specify a delay constraint if they convey a
36 Unidirectional Link Delay sub-TLV [item f) in 45.1.8]. The delay constraint can be used in a
37 Topology sub-TLV that specifies a single loose tree, i.e., the Base VIDs are associated with the LT
38 ECT Algorithm (Table 45-1). If delay constraint is applied, then the loose hop has to fit in the delay
39 budget specified by the Delay parameter of the Unidirectional Link Delay sub-TLV [item f) in
40 45.1.8] conveyed by the Hop sub-TLV. If the Topology sub-TLV specifies a single leaf, then the path
41 between the preceding Hop sub-TLV and the current Hop sub-TLV has to meet the delay budget. If
42 the Topology sub-TLV specifies multiple leaves, then the path between the root and the current Hop
43 sub-TLV has to meet the delay budget. If the tree is used as a reverse congruent tree, then the delay
44 constraint applies in both directions. If the tree is used as a directed tree, then the delay constraint
45 applies in the direction of the tree.

46 NOTE 1—The T and R flags can be used to establish asymmetric VLANs, e.g., for rooted-multipoint connectivity.

47 NOTE 2—The Root and Leaf flags specify the role of the Bridges only in the explicit tree, i.e., the active topology.
48 Roots and leaves with respect to data traffic are specified by the T/R flags of VIDs and MAC addresses associated
49 with the Edge Bridges. Roots and leaves of the active topology and data flows do not necessarily coincide, e.g., see
50 Figure 45-7.

1 45.1.11 Administrative Group sub-TLV

2 The Administrative Group sub-TLV may be included in a Topology sub-TLV (45.1.9) in order to put a
3 constraint on loose hops. In this case, each LAN included in a loose hop has to be a member of the
4 Administrative Group(s) indicated by the Administrative Group sub-TLV. The format of the Administrative
5 Group sub-TLV is specified by IETF RFC 5305 as shown in Figure 45-9. In other words, the same format is
6 used for the constraint as for the TE attribute; therefore, their comparison is easy.

	Octet	Length
Type (3)	1	1
Length (4)	2	1
Administrative Group	3–6	4

Figure 45-9—Administrative Group sub-TLV

7 The administrative groups are encoded as follows:

- 8 a) Type (8 bits). Octet 1 conveys the type of the sub-TLV, and its value is 3.
- 9 b) Length (8 bits). Octet 2 conveys the total number of bytes contained in the Value field. The value of
10 the Length field is 4.
- 11 c) Administrative Group (32 bits). Octets 3 through 6 convey the Administrative Group parameter. It
12 comprises 32 flags, and each flag belongs to a distinct group. Group membership is indicated by the
13 flags set.

14 45.1.12 Bandwidth Constraint sub-TLV

15 The Bandwidth Constraint sub-TLV may be included in a Topology sub-TLV (45.1.9) in order to specify
16 how much available bandwidth is to be provided by the tree. Each loose hop has to meet the bandwidth
17 constraint. The bandwidth value of the constraint is a total value, or it refers only to a single PCP as specified
18 by the sub-TLV. Figure 45-10 shows the format of the Bandwidth Constraint sub-TLV.

	Octet	Length
Type (23)	1	1
Length (5)	2	1
PCP	3	3 bits
DEI	3	1 bit
PCP flag	3	1 bit
reserved	3	3 bits
Available Bandwidth	4–7	4

Figure 45-10—Bandwidth Constraint sub-TLV

19 The parameters of the bandwidth constraint are encoded as follows:

- 20 a) Type (8 bits). Octet 1 conveys the type of the sub-TLV, and its value is 23.
- 21 b) Length (8 bits). Octet 2 conveys the total number of bytes contained in the Value field, The value of
22 the Length field is 5 bytes.
- 23 c) PCP (3 bits). Octet 3 conveys the PCP parameter, which determines the traffic class to which the
24 Available Bandwidth parameter refers.
- 25 d) DEI (1 bit). Octet 3 also conveys the DEI parameter.
- 26 If the DEI parameter is clear, then the bandwidth constraint refers to committed information rate.
- 27 If the DEI parameter is set, then the bandwidth constraint refers to peak information rate.

- 1 e) PCP flag (1 bit). Octet 3 also conveys the PCP flag. If it is set, then the PCP parameter is taken into
2 account.
- 3 f) reserved (3 bits). Three bits of Octet 3 are reserved for future use, transmitted as 0, and ignored on
4 receipt.
- 5 g) Available Bandwidth (32 bits). Octets 4 through 7 convey the Available Bandwidth. The Available
6 Bandwidth is specific to the traffic class identified by the PCP parameter if the PCP flag is set;
7 otherwise, it is total bandwidth. In line with the bandwidth parameters specified in IETF RFC 5305,
8 the Available Bandwidth is encoded as a 32-bit IEEE floating point number, and the units are bytes
9 (not bits!) per second. Thus, the Available Bandwidth constraint applied for a traffic class is easily
10 comparable with the Unreserved Bandwidth stored in the TED for the given traffic class (see sub-
11 TLV 11 specified by IETF RFC 5305).
- 12 The bandwidth constraint applies for both directions in the case of symmetric (3.273) explicit trees.
13 Nevertheless, a VID associated with an explicit tree can be made unidirectional by means of the T/R
14 flags belonging to the VID in the Hop sub-TLV (45.1.10) of the Edge Bridges. If all the VIDs of the
15 Topology sub-TLV (45.1.9) are unidirectional and all belong to the traffic class identified by the
16 PCP parameter of the Bandwidth Constraint sub-TLV, then it is enough to meet the bandwidth
17 constraint in the direction applied for those VIDs.
- 18 NOTE—The peak information rate is the committed information rate plus the excess information rate.

19 45.2 Reservation

20 ISIS-PCR may be used for recording bandwidth assignment for explicitly placed data traffic in an SPT
21 Region if MSRP is not used within the region. If MSRP is used in an SPT Region, then only MSRP
22 performs reservations and IS-IS does not.

23 45.2.1 Bandwidth Assignment sub-TLV

24 ISIS-PCR can record bandwidth assignment for explicitly placed data traffic. ISIS-PCR can be used for
25 recording bandwidth assignment only within an SPT Region where MSRP is not used.

26 If precedence order has to be determined among bandwidth assignments in an SPT Region with multiple
27 PCEs, then ISIS-PCR provides it as specified in 45.2.3.

28 The Bandwidth Assignment sub-TLV can be used to define the amount of bandwidth whose assignment is to
29 be recorded by ISIS-PCR at each hop of the explicit tree described by the corresponding Topology sub-TLV
30 (45.1.9). The Bandwidth Assignment sub-TLV is conveyed by a Topology sub-TLV (45.1.9).

31 The Bandwidth Assignment sub-TLV is used by ISIS-PCR for the recording of bandwidth assignment for a
32 traffic class. Figure 45-11 shows the format of the Bandwidth Assignment sub-TLV.

	Octet	Length
Type (24)	1	1
Length (5)	2	1
PCP	3	3 bits
DEI	3	1 bit
Importance	3	3 bits
reserved	3	1 bit
Bandwidth	4–7	4

Figure 45-11—Bandwidth Assignment sub-TLV

1 The parameters for bandwidth assignment recorded by ISIS-PCR are encoded as follows:

- 2 a) Type (8 bits). Octet 1 conveys the type of the sub-TLV, and its value is 24.
- 3 b) Length (8 bits). Octet 2 conveys the total number of bytes contained in the Value field. The value of
4 the Length field is 5.
- 5 c) PCP (3 bits). Octet 3 conveys the PCP parameter, which specifies the traffic class for which the
6 bandwidth is to be assigned.
- 7 d) DEI (1 bit). Octet 3 also conveys the DEI parameter.
8 If the DEI parameter is clear, then the bandwidth assignment is performed to provide committed
9 information rate.
10 If the DEI parameter is set, then the bandwidth assignment is performed to provide peak information
11 rate.
- 12 e) Importance (3 bits). Octet 3 also conveys the Importance parameter for determining precedence
13 order among bandwidth assignments within a PCP as specified in 45.2.3. A lower numerical value
14 indicates a more important bandwidth assignment within a PCP. The default value of the Importance
15 parameter is 7.
- 16 f) reserved (1 bit). One bit of Octet 3 is reserved for future use, transmitted as 0, and ignored on
17 receipt.
- 18 g) Bandwidth (32 bits). Octets 4 through 7 convey the amount of bandwidth to be assigned for the
19 traffic class identified by the PCP parameter. In line with the bandwidth values specified in IETF
20 RFC 5305, the Bandwidth parameter is encoded as a 32-bit IEEE floating point number, and the
21 units are bytes (not bits!) per second.
22 The bandwidth assignment applies for both directions in the case of symmetric (3.273) explicit trees.
23 Nevertheless, a VID associated with an explicit tree can be made unidirectional by means of the T/R
24 flags belonging to the VID in the Hop sub-TLV (45.1.10) of the Edge Bridges. If all the VIDs having
25 the same PCP in the Topology sub-TLV (45.1.9) are unidirectional, then the bandwidth is assigned
26 only in the direction applied for those VIDs.

27 If bandwidth is to be assigned both for committed information rate and for peak information rate, then the
28 Topology sub-TLV conveys two Bandwidth Assignment sub-TLVs: one with DEI cleared and another with
29 DEI set.

30 45.2.2 Timestamp sub-TLV

31 The Timestamp sub-TLV may be included in a Topology sub-TLV (45.1.9) in order to provide precedence
32 order among equally important bandwidth assignments within a PCP as specified in 45.2.3. Figure 45-12
33 shows the format of the Timestamp sub-TLV.

	Octet	Length
Type (25)	1	1
Length (4)	2	1
Time	3–6	4

Figure 45-12—Timestamp sub-TLV

34 The timestamp represents a positive time with respect to the Precision Time Protocol (PTP) epoch, and it is
35 encoded as follows:

- 36 a) Type (8 bits). Octet 1 conveys the type of the sub-TLV, and its value is 25.
- 37 b) Length (8 bits). Octet 2 conveys the total number of bytes contained in the Value field. The value of
38 the Length field is 4.

- 1 c) Time (32 bits). Octets 3 through 6 convey the time in units of seconds with respect to the PTP epoch.
2 NOTE—The Timestamp sub-TLV carries the seconds portion of PTP as specified by IEEE Std 1588 [B17]. The
3 epoch is 1970-01-01 00:00:00 TAI (i.e., the PTP time does not include leap seconds).

4 **45.2.3 Precedence ordering**

5 The PCEs are collectively responsible for making a consistent set of bandwidth assignments when ISIS-PCR
6 is used for recording bandwidth allocations. Despite that responsibility, if precedence ordering is required
7 among bandwidth assignments, then ordering based on the following parameters has to be applied:

- 8 1) PCP parameter of Bandwidth Assignment Sub-TLV
9 2) Importance parameter of Bandwidth Assignment Sub-TLV
10 3) Timestamp sub-TLV (if present in the Topology sub-TLV)

11 A bandwidth assignment takes precedence if it has higher PCP, or higher Importance within a PCP, or earlier
12 timestamp in the case of equal Importance within a PCP. A bandwidth assignment associated with a
13 timestamp takes precedence over a bandwidth assignment without timestamp when PCP and Importance of
14 different bandwidth assignments are both equal.

15 If resolution is not possible based on the above parameters or they are not available, e.g., each bandwidth
16 assignment lacks timestamp or the precedence order has to be determined for the use of a VID, then the item
17 is granted to the PCE whose LSP has the numerically least LSP ID.

18 **45.3 Redundancy**

19 This subclause specifies IS-IS extensions in support of redundancy schemes, i.e., protection and restoration
20 for data flows.

21 Local protection for unicast data flows based on loop-free alternates is described in 45.3.1.

22 Protection schemes are often based on two or more redundant trees, or if that is not possible, then on
23 maximally redundant trees. Maximally Redundant Trees (MRTs) can be used either in a static fashion or
24 with cautious restoration for ISIS-PCR. This subclause specifies how MRTs are installed and maintained
25 leveraging the tools provided by 45.1. The use of static MRTs is described in 45.3.2. Setting up and
26 maintaining MRTs with cautious restoration is described in 45.3.3 and 45.3.4.

27 **45.3.1 Loop-free alternates for unicast data flows**

28 Downstream Loop-Free Alternates (LFAs) provide a simple redundancy scheme, which may be supported
29 by ISIS-SPB for unicast flows of an SPBM VLAN and for SPBV VLANs. A path toward an individual
30 destination MAC address is downstream if it is ensured that the distance to the destination decreases at each
31 hop along the path. Downstream paths are always loop-free alternates to each other with respect to an
32 individual destination MAC address. Therefore, a downstream alternate path can be safely used for local
33 protection as a loop-free backup path.

34 NOTE—The Alternate Ports of RSTP and MSTP leverage the loop-free alternate paths provided by the physical
35 topology to the Root Bridge. In the case of spanning trees, ‘up’ or ‘upward’ to the Root is used to refer to the same
36 direction as ‘downstream’ for destination rooted trees.

37 LFA can be provided for SPBV by SPTs (13.17, Clause 27, Clause 28). The LFA mechanism is the same for
38 an SPT as for an MSTI where Alternate Ports leverage LFA paths.

39 Loop mitigation (6.5.4.2) may be relaxed in order to allow LFAs for unicast SPBM traffic. Ingress checking
40 is modified and relaxed in order to allow the reception of a unicast frame from any SPT Bridge that is
41 upstream with respect to the given individual destination MAC address. Thus, relaxed ingress checking

1 implements a split horizon for loop mitigation by ensuring that a frame is admitted only if it is received from
2 an SPT Bridge that is farther from the destination than the receiving SPT Bridge, which determines the valid
3 set of ingress ports.

4 In the case of a failure at an SPT Bridge's Port or LAN on the shortest path toward individual MAC
5 addresses, the SPT Bridge acts as a Point of Local Repair (PLR) and forwards frames through ports
6 providing downstream LFA paths toward the given individual MAC addresses, if any.

7 An SPT Bridge updates its relaxed ingress checking after detecting a mismatch in the Agreement Digest. An
8 ingress port is removed from the valid set of ingress ports with respect to a particular individual destination
9 MAC address if a formerly upstream SPT Bridge is no longer farther from the destination than the given
10 SPT Bridge. An LFA path is removed if the corresponding formerly downstream neighbor is not closer any
11 more to the destination than the given SPT Bridge.

12 An SPT Bridge updates its LFA paths and relaxed ingress checking after finishing the updates of multicast
13 states. If LFA paths with respect to an individual MAC address have been changed, then the new LFA paths
14 are added. The valid set of ingress ports is extended with the new upstream Bridges with respect to an
15 individual MAC address, if any.

16 45.3.2 Static redundant trees

17 The trees associated with the ST ECT Algorithm (00-80-C2-17, Table 45-1) are static in the sense that no
18 entity other than the owner PCE can update them. In other words, ISIS-PCR only installs the strict trees and
19 does not take any further action on them; i.e., ISIS-PCR neither restores them nor maintains them. In the
20 case of a topology change, it is the task of the owner PCE to detect the topology change, e.g., based on the
21 changes in its LSDB, and also update the strict trees if needed. This requires the owner PCE to compute the
22 new tree and to assemble the corresponding new Topology sub-TLV (45.1.9). The PCE then sends the
23 updated LSP conveying the new Topology sub-TLV, and ISIS-PCR updates the strict explicit tree. The
24 owner PCE can withdraw an explicit tree by sending an updated LSP that does not include the Topology
25 sub-TLV. If a Topology sub-TLV is removed from an LSP (or has been changed), so that (previous)
26 Topology sub-TLV is no longer present (or has been changed) in the LSDB, then that (previous) Topology
27 sub-TLV is implicitly withdrawn. ISIS-PCR then removes (or updates) the explicit tree.

28 NOTE 1—The PCE preferably does not initiate the update of explicit trees if there is any reason to suspect that the SPT
29 Bridges of the SPT Region do not share a common view on network topology.

30 NOTE 2—The PCE can have a hot standby backup PCE in order to avoid a single point of failure.

31 Strict explicit trees can be used for Maximally Redundant Trees (MRTs), especially if more than two
32 redundant trees are needed. Each tree is described by a distinct Topology sub-TLV (45.1.9). The VLAN
33 configuration of strict trees is specified in 45.1.3.

34 The PCE or its PCA assembles and floods the corresponding Topology sub-TLVs (45.1.9); thus ISIS-PCR
35 installs the explicit trees throughout the SPT Region as specified in 45.1. All of the maximally redundant
36 trees are installed independently of each other, i.e., ISIS-PCR is not aware of any relationship between them.
37 Maintaining the association among the redundant strict trees is not the responsibility of ISIS-PCR, but the
38 entity that requests and uses them, e.g., the owner PCE and the management entity responsible for
39 configuring the protection scheme in use. Therefore, no parameters for association of static explicit trees are
40 present in the ISIS-PCR sub-TLVs.

41 NOTE 3—For instance, two static redundant trees can be used for CFM-based protection switching as specified in
42 26.10, where the association and the use of the paths are provided. Static redundant trees can be also used for 1+1 types
43 of protection schemes, where a copy of each data frame is forwarded on each of the redundant paths.

44 The algorithm used by the owner PCE for the computation of static redundant trees is not specified by this
45 standard. Number K , the number of trees to be computed, is an input for the owner PCE. The Topology sub-
46 TLVs (45.1.9) are the output of the owner PCE. Other outputs can include the number of common links and
47 their identity.

1 45.3.3 Maximally Redundant Trees (MRTs)

2 SPT Bridges may support MRTs. The BLCE of SPT Bridges is also involved in the computation of MRTs.
3 The MRTs can be, for example, used for point-to-multipoint protection if the MRTs are source rooted or for
4 multipoint-to-point protection if the MRTs are destination rooted. The MRTs can be used to protect an SPT
5 rooted at the MRT Root. The MRTs can be also used so that they protect each other, e.g., in a 1+1 type of
6 protection scheme where data frames are sent on both MRTs. An MRT provides an undirected active
7 topology just like an SPT or a spanning tree instance (which is also a shortest path tree rooted at the
8 spanning tree root). Correspondingly, MRTs can be used for bidirectional or unidirectional traffic, as
9 determined by the T/R parameters of the data traffic carried on the tree.

10 The MRT ECT Algorithm (00-80-C2-18, Table 45-1) is used for the establishment and maintenance of
11 MRTs in a distributed fashion as specified by this subclause. The MRT Lowpoint Algorithm specified by
12 IETF RFC 7811 has to be used for the computation of MRTs that span more than two Edge Bridges. If only
13 two Edge Bridges are to be connected, then the algorithm specified in this subclause has to be used for path
14 computation. The MRT Lowpoint Algorithm produces two MRTs for an MRT Root: MRT-Blue and MRT-
15 Red. If the MRTs are used in a primary and backup fashion, then MRT-Blue is the primary and MRT-Red is
16 the backup. The MRTs are cautiously restored after a topology change as described in this subclause.

17 NOTE 1—If more than two maximally redundant trees are required, then static redundant trees (45.3.2) have to be used.

18 When the MRT Lowpoint Algorithm is used by ISIS-PCR, then the input parameters of the
19 Interface_Compare function are as follows: The Interface_compare_metric is the SPB Link Metric
20 (28.12.7), and the maximum metric value is used in cases where the metrics advertised by adjacent Bridges
21 for a given link are different. The Interface_compare_unique_node_identifier is the 8-byte Bridge Identifier,
22 which comprises a 6-byte IS-IS System Identifier and a 2-byte Bridge Priority.

23 The MRT Lowpoint Algorithm first computes the GADAG, and the computation requires the selection of the
24 GADAG Root. The Bridge with the best Bridge Identifier is selected as the GADAG Root, where the
25 numerically lower value indicates the better identifier. The manageable Bridge Priority component of the
26 Bridge Identifier allows the configuration of the GADAG Root. The Bridge Priority is conveyed by Octets
27 15 and 16 of the SPB Instance sub-TLV (28.12.5). Based on its LSDB, each SPT Bridge can locally
28 determine which Bridge is the GADAG Root and then compute the GADAG. The MRTs are then computed
29 based on the GADAG. Examples for a GADAG and two corresponding MRTs are shown in Figure 45-13
30 and Figure 45-14, respectively. If tie-breaking is required among equal cost paths while the MRTs are
31 computed, then the tie-breaking specified by the LowPATHID algorithm (28.7) has to be used. If MRTs
32 support a VLAN having multiple multicast sources, then the BLCEs of the Bridges have to compute the
33 MRTs of other Bridges in addition to their own MRTs due to the need to apply source-specific (S,G)
34 multicast. This is similar to the application of All Pairs Shortest Path computation in ISIS-SPB. (See 45.1.1
35 for further details on source-specific (S,G) multicast.)

36 The MRT Lowpoint Algorithm is specified in terms of determining next hop FDB entries toward a particular
37 unicast destination. This generates a tree directed to the root, which is the destination. By inverting the
38 direction of this destination rooted tree, a source rooted tree can be retrieved.

39 If MRTs are required for a VLAN so that each Bridge performs all the MRT computation steps on its own
40 including GADAG computation, i.e., the MRT computation is fully distributed, then the VLAN has to be
41 associated with the MRT ECT Algorithm (Table 45-1) as specified in 45.1.3.

42 Just using the MRT ECT Algorithm creates MRT-Blue and MRT-Red for each SPT Root, i.e., each SPT
43 Root is also an MRT Root, and IS-IS maintains all the MRTs. Furthermore, the MRTs are spanning trees in
44 this case, i.e., each Bridge of the SPT Region is included in both MRT-Blue and MRT-Red. This operation
45 mode is selected by providing the VLAN configuration in the SPB Base VLAN Identifiers sub-TLV
46 (28.12.4) and in the SPB Instance sub-TLV (28.12.5) as specified in 45.1.3. The Topology sub-TLV (45.1.9)
47 is not used in this case, i.e., only ISIS-SPB sub-TLVs are used.

1 If the level of redundancy provided by each SPT Root also being an MRT Root is not required, then the
2 MRT Roots can be specified by a Topology sub-TLV (45.1.9); this approach can be also applied when MRTs
3 protect each other and SPTs are not used. The Topology sub-TLV describes loose explicit trees in this case,
4 i.e., it specifies only the Edge Bridges that the MRTs are to span, and each Edge Bridge is a loose hop. An
5 MRT Root is specified by setting the Root flag [item c4] in 45.1.10] in the corresponding Hop sub-TLV
6 (45.1.10). An MRT Root is typically an Edge Bridge too. MRTs provide a path from each Edge Bridge to the
7 MRT Root, i.e., each Edge Bridge is a leaf except for the MRT Root. If a Topology sub-TLV specifies
8 multiple MRT Roots, then an Edge Bridge has different roles in the MRTs rooted at different Bridges.
9 Therefore, the Leaf flag [item c5] in 45.1.10] is not used to mark leaves of MRTs. The Topology sub-TLV
10 can include a hop with the Exclude flag [item c6] in 45.1.10] set for a Bridge that is to be avoided by the
11 MRTs, in which case the Bridge is excluded from the GADAG, too. The Topology sub-TLV can convey one
12 or more Base VIDs that are associated with the MRT ECT Algorithm. Whether the MRTs are unidirectional
13 or bidirectional is determined by the T/R flags corresponding to the VIDs, I-SIDs, and MAC addresses
14 associated with the given MRTs.

15 NOTE 2—For instance, if a VLAN is supported by ISIS-SPB in SPBM mode (Clause 27, Clause 28) and MRTs are used
16 for the protection of SPTs, then only three VIDs are required to support the VLAN. The Base VID is used for all the
17 SPTs. Another VID is used for all MRT-Blues, and a third VID is used for all MRT-Reds. The MRTs are then used as
18 destination rooted trees for unicast traffic and as source rooted trees for multicast traffic just like the SPTs. The T/R
19 parameters for the MRT VIDs are exactly the same as for the Base VID; therefore, IS-IS populates the FDBs for the
20 MRTs the same way as for the SPTs, i.e., just uses the corresponding MRT VID instead of the Base VID.

21 If the Topology sub-TLV specifies only two Edge Bridges, then the following algorithm is used for the
22 computation of the two maximally redundant paths instead of the MRT Lowpoint Algorithm. As they
23 interconnect only two Edge Bridges, the MRTs are reverse path congruent maximally redundant paths in this
24 case. According to the algorithm below, one of the two maximally redundant paths is the shortest path. The
25 two paths are used to protect each other. A VLAN associated with maximally redundant paths has to be
26 configured as specified in 45.1.3 for the MRT ECT Algorithm. The VID of MRT-Blue is used for the
27 shortest path, and the VID of MRT-Red is used for the other path. The establishment of the redundant paths
28 is initiated by the flooding of a Topology sub-TLV (45.1.9). The paths are then determined by the BLCE of
29 the SPT Bridges and then installed by ISIS-PCR. The Topology sub-TLV can convey constraint sub-TLVs,
30 which have to be also taken into account when determining the paths. The BLCEs perform the following
31 path computation algorithm:

- 32 1) Prune the topology according to the constraints of the Topology sub-TLV (45.1.9), if any. The
33 rest of the computation steps are performed on the pruned topology. If pruning splits the SPT
34 Region, then the outcome can be that no path is available.
- 35 2) Run Dijkstra to get the (constrained) shortest path, which is the primary path.
- 36 3) Install the FDB entries corresponding to the VID of MRT-Blue along the primary path.
- 37 4) Update the network graph by the appropriate method as follows:
38 a) Prune the links of the primary path if that does not partition the SPT Region.
39 b) Otherwise, multiply by 1000 the link metric of the links taking part in the primary path.
- 40 5) Run Dijkstra to get the backup path.
- 41 6) Install the FDB entries corresponding to the VID of MRT-Red along the backup path.

42 It is recommended to use distinct VIDs for MRT-Blue and MRT-Red, which are loose trees and therefore
43 computed by multiple entities. As a strict tree is controlled by a single PCE, it is easier to ensure
44 unambiguous forwarding and allow the use of the same VID for different strict trees as described in 45.1.4
45 than to do so for loose trees.

46 MRTs have to be restored cautiously after a topology change. The restoration should not be initiated if there
47 is any reason to suspect that the SPT Bridges of the SPT Region do not share a common view on the network
48 topology. BLCEs then perform path computation, and MRTs are updated as follows.

1 If MRTs are used for SPT protection, then MRTs are restored only after the restoration of the SPTs. Upon a
2 failure event, Point of Local Repair (PLR) Bridges redirect data traffic to the MRT that is not affected by the
3 failure, if any. As the topology change is announced by the corresponding LSPs, the SPT Bridges compute
4 and install the new SPTs. If each SPT Bridge has installed the new SPT, then data traffic is directed back to
5 the SPTs. After that, the new MRTs are computed and installed by the Bridges.

6 If MRTs support a non-learning VLAN, then the PLRs can redirect traffic by replacing the FDB entry
7 belonging to the tree of normal operation (i.e., the SPT or an MRT) with the FDB entry belonging the
8 protection tree (i.e., with MRT-Blue or MRT-Red) upon a failure event. If MRTs support a learning VLAN,
9 then the PLRs can redirect traffic by updating VID translation at the Ingress Port from the VID of normal
10 operation to the VID of the protection tree (i.e., to MRT-Blue or MRT-Red) upon a failure event.

11 If MRTs protect each other, then only one of the MRTs is restored at a time. Upon a failure event, PLR
12 Bridges direct all data traffic to the MRT (e.g., MRT-Red) that is not affected by the failure. If a 1+1 type of
13 protection scheme is used, then there is no need to react to a failure event as data frames are forwarded on
14 both MRTs, i.e., there are no PLRs. When each SPT Bridge has the same view on the network topology
15 again, then the SPT Bridges compute the new MRTs. The broken MRT (e.g., MRT-Blue) then is restored
16 while the other MRT (e.g., MRT-Red) is not touched. If each SPT Bridge has installed the new MRT (e.g.,
17 the new MRT-Blue), then data traffic is directed to this newly restored MRT. The SPT Bridges then install
18 the other MRT (e.g., the new MRT-Red). The same restoration procedure is followed if both MRTs are
19 affected by the failure so that MRT-Blue is restored first.

20 Loop prevention (6.5.4.1, 27.9) has to be applied for MRTs during the period when they are being restored
21 by ISIS-PCR.

22 There is a non-zero probability of out-of-order frame delivery while ISIS-PCR restores loose explicit trees,
23 e.g., MRTs. This probability is similar to the probability of out-of-order frame delivery while ISIS-SPB
24 restores SPTs.

25 If there are two maximally redundant paths between a pair of Edge Bridges and only the backup path went
26 down, then just Steps 4–6 of the above algorithm are performed again. If only the primary path went down,
27 then data traffic takes the backup path, i.e., only the backup path is in use by the applied protection
28 mechanism. The primary path is then updated according to Steps 1–3. No further steps are taken until the
29 primary path is up again. The protection scheme can then revert to the primary path, and the backup path can
30 be updated according to Steps 4–6. If both paths went down, then Steps 1–6 are performed again.

31 NOTE 3—The method to be used for verifying whether the SPT Bridges have a common view on the topology and
32 whether SPTs and MRTs have been restored is left to the implementer and the network operator. A common method is to
33 wait some time after the reception of the last LSP indicating a topology change. Alternatively, OAM can be used for
34 instance.

35 **45.3.4 MRTs with centralized GADAG computation**

36 The GADAG is identical for all the MRTs within a network domain, as a consequence of the use of the MRT
37 Lowpoint Algorithm (IETF RFC 7811). The GADAG is computed by a single entity for VLANs associated
38 with the MRTG ECT Algorithm (Table 45-1). This entity is referred to as the GADAG Computer and is
39 either a PCE or the BLCE of the GADAG Root.

40 SPT Bridges may support the MRTG ECT Algorithm (00-80-C2-19, Table 45-1). MRTs are then computed
41 by the SPT Bridges based on the GADAG provided by the GADAG Computer in a Topology sub-TLV
42 (45.1.9). The GADAG and the MRTs are computed as specified by the MRT Lowpoint Algorithm
43 (IETF RFC 7811). The MRTs produced by the MRTG ECT Algorithm are cautiously restored after a failure
44 event as described in this subclause. A VLAN is associated with the MRTG ECT Algorithm as specified in
45 45.1.3 if MRTs with central GADAG computation are to be used for that VLAN.

1 The GADAG Root is determined based on the Bridge Identifier as described in 45.3.4, which also
2 determines the sole SPT Bridge computing the GADAG if the GADAG Root is the GADAG Computer.
3 Otherwise, the GADAG Computer PCE determines the GADAG Root, which also allows direct
4 configuration of the GADAG Root.

5 The GADAG Computer performs the GADAG computation as specified by the MRT Lowpoint Algorithm
6 (IETF RFC 7811). The GADAG Computer then encodes the GADAG in a Topology sub-TLV (45.1.9),
7 which is then flooded throughout the domain. A GADAG is encoded in a Topology sub-TLV by means of
8 directed ear decomposition as follows: A directed ear is a directed point-to-point path whose end points can
9 coincide, but no other element of the path is repeated in the ear. Each ear is specified by an ordered list of
10 hops, and that order of hops is according to the direction of the arcs in the GADAG. There are no leaves in a
11 GADAG; hence, the Leaf flag [item c5] in 45.1.10] is used to ease the parsing of the Topology sub-TLV and
12 to mark the end of a topology block (3.281). The sequence of ears in the Topology sub-TLV is such that the
13 end points of an ear belong to former ears. The GADAG Root is not marked by any flag, but the GADAG
14 Root is the first hop in the Topology sub-TLV; correspondingly the first ear starts and ends with the GADAG
15 Root. MRT Roots are marked by the Root flag [item c4] in 45.1.10], and all other Edge Bridges are leaves of
16 the given MRTs. If no MRT Root is specified, then each SPT Root is also an MRT Root. If a Bridge has to be
17 excluded from the MRTs, then it is not part of the GADAG.

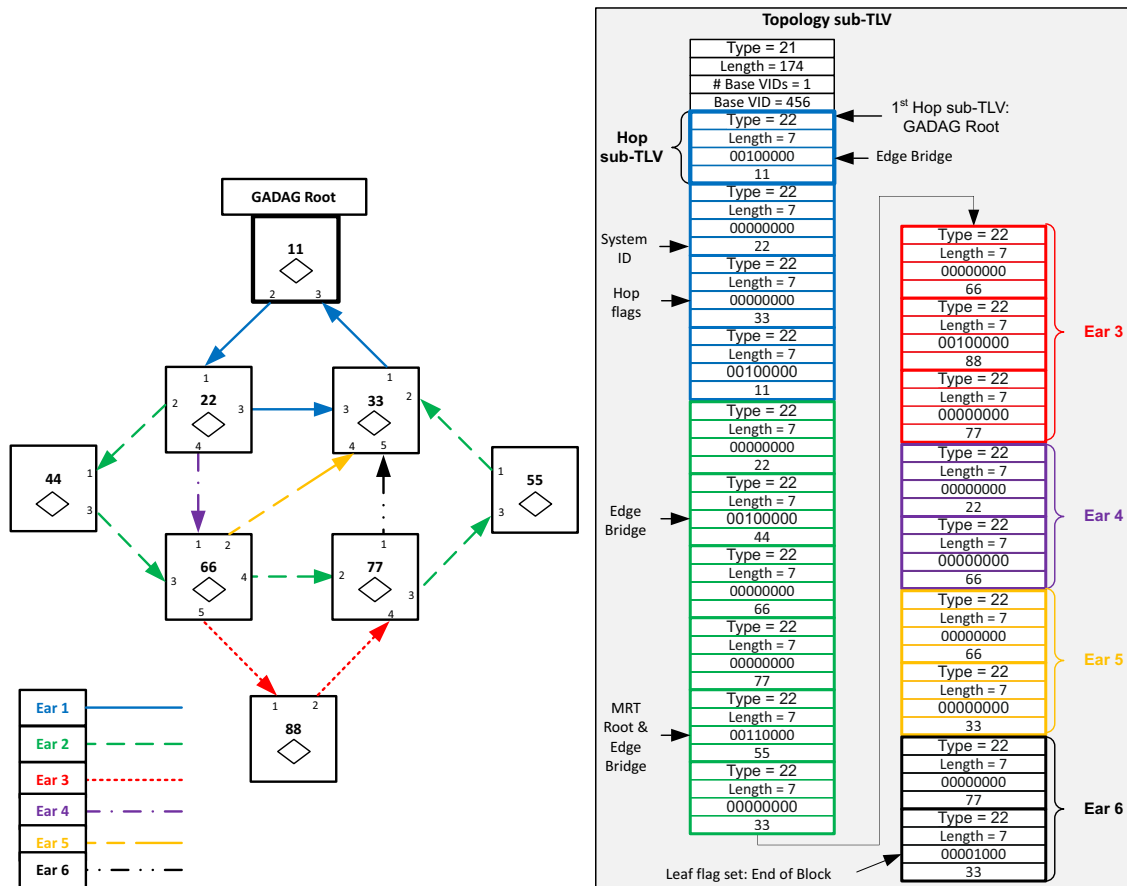


Figure 45-13—A GADAG and its descriptor Topology sub-TLV

18 Figure 45-13 shows the GADAG for the network topology depicted in Figure 45-1 and Figure 45-2 when
19 SPT Bridge 11 is the GADAG Root. The graph shown in Figure 45-13 is in fact an ADAG, and removing the
20 arc going from Bridge 33 to Bridge 11 (the GADAG Root) makes it a DAG. Figure 45-13 also shows the

1 Topology sub-TLV that specifies the GADAG. The first directed ear starts and ends at the GADAG Root,
2 i.e., Bridge 11. The second ear (Ear 2) starts at Bridge 22 and ends at Bridge 33. The third ear (Ear 3)
3 comprises Bridges 66, 88, and 77. Each of the remaining arcs is an individual ear added by a pair of Hop
4 sub-TLVs according to the direction of the arc. Bridges 11, 44, 55, and 88 are Edge Bridges. Bridge 55 is the
5 only MRT Root.

6 The MRTs are then computed by the SPT Bridges as specified by the MRT Lowpoint Algorithm
7 (IETF RFC 7811) based on the GADAG descriptor received from the GADAG Computer. Whether the
8 MRTs are unidirectional or bidirectional is determined by the T/R flags corresponding to the VIDs, I-SIDs,
9 and MAC addresses associated with the given MRTs. Figure 45-14 shows MRT-Blue and MRT-Red if the
10 GADAG descriptor Topology sub-TLV of Figure 45-13 is used for MRT computation, i.e., SPT Bridge 55 is
11 the only MRT Root. Figure 45-14 also shows that an Edge Bridge is not necessarily a leaf; see Bridge 11.

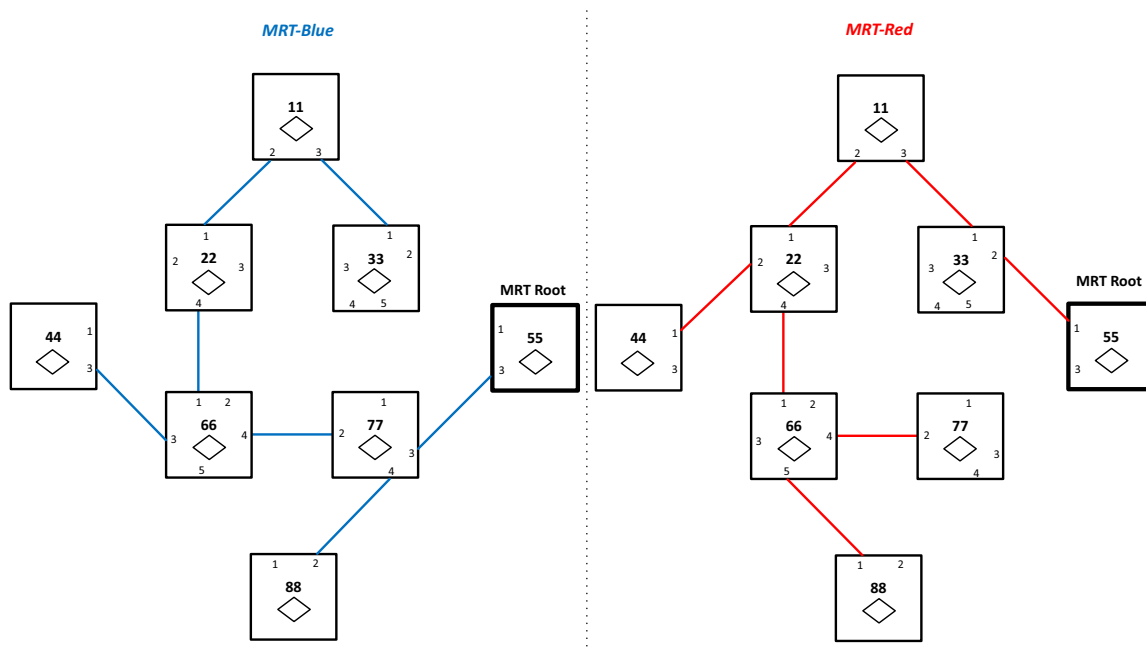


Figure 45-14—MRT-Blue and MRT-Red for MRT Root 55

12 The Topology sub-TLV specifying the GADAG can convey Base VIDs associated with the MRTG ECT
13 Algorithm, which specifies the VLANs for which the GADAG is relevant. If there is no Base VID parameter
14 in the Topology sub-TLV specifying a GADAG, then the given GADAG has to be used for all VLANs
15 whose Base VID is associated with the MRTG ECT Algorithm.

16 A network topology can comprise multiple topology blocks (3.281), like the one illustrated in Figure 45-15,
17 which comprises four blocks because each cut-link with the two Bridges at either end is a block. A GADAG
18 is also shown in Figure 45-15. Note that two arcs with opposite direction represent a cut-link in a GADAG,
19 e.g., see the cut-link between Bridges 75 and 77. The encoding starts with the block (ADAG) involving the
20 GADAG Root as illustrated in Figure 45-15. The first hop in the Topology sub-TLV is the GADAG Root
21 (Bridge 11 in this example). The ADAG of the first block is then described using the ear decomposition, as
22 described above. In this example, the first block has been completely traversed at the second occurrence of
23 Bridge 11 in the GADAG descriptor. The end of a block is indicated by setting the Leaf flag for the last hop
24 of the block, e.g., for the second occurrence of Bridge 11 in the example GADAG descriptor. The next
25 Bridge that appears in the GADAG descriptor (Bridge 44 in this case) is the localroot for the Bridges in the

1 next block. Continuing this process, the Leaf flag is set for the third occurrence of Bridge 44, the third
2 occurrence of Bridge 75, and the third occurrence of Bridge 77, each indicating the end of a block.

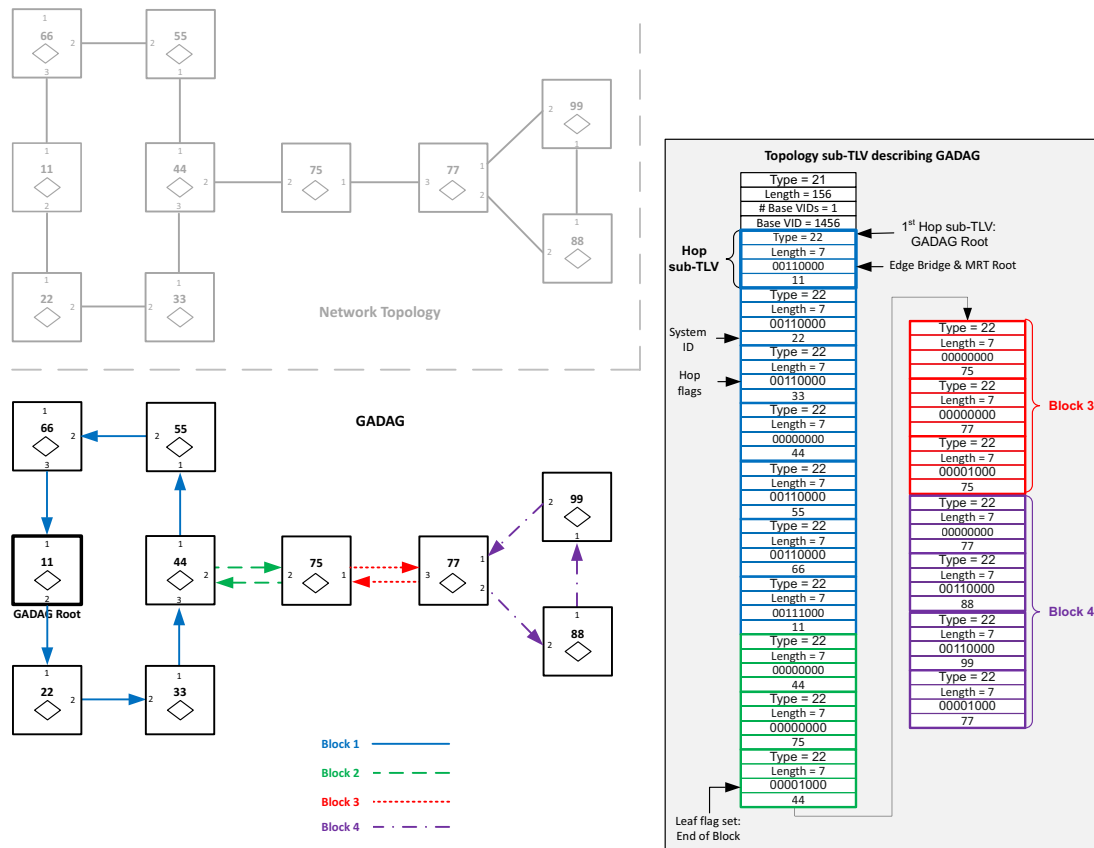


Figure 45-15—A GADAG for a topology with multiple blocks

3 The first hop of the first block is the GADAG Root; the first hop in the rest of the blocks is the localroot. The
4 position of the set Leaf flags helps to determine the localroot, which is the next hop. In the example GADAG
5 descriptor, one can determine that Bridge 11 is the localroot for Bridges 22, 33, 44, 55, and 66 (and Bridge
6 11 is the GADAG Root). Bridge 44 is the localroot for Bridge 75. Bridge 75 is the localroot for Bridge 77.
7 Bridge 77 is the localroot for Bridges 88 and 99. The GADAG Root is assigned a localroot of None.

8 Block IDs are reconstructed while parsing a Topology sub-TLV specifying a GADAG. The current Block ID
9 starts at 0 and is assigned to the GADAG Root. A Bridge appearing in the GADAG descriptor without a
10 previously assigned Block ID value is assigned the current Block ID. The current Block ID is incremented
11 by 1 after processing the localroot of a block. Note that the localroot of a block will keep the Block ID of
12 the firsts block in which it is assigned a Block ID. In the example shown in Figure 45-15, Bridge 11
13 has Block ID=0. Bridges 22, 33, 44, 55, and 66 have Block ID=1. Bridge 75 has Block ID=2. Bridge 77 has
14 Block ID=3. Bridges 88 and 99 have Block ID=4.

15 MRTs have to be restored cautiously after a topology change. The restoration should not be initiated if there
16 is any reason to suspect that the SPT Bridges of the SPT Region do not share a common view on the network
17 topology.

1 If MRTs are used for SPT protection, then MRTs are restored only after the restoration of the SPTs. Upon a
2 failure event, PLR Bridges redirect data traffic to the MRT that is not affected by the failure, if any. As the
3 topology change is announced by the corresponding LSPs, SPT Bridges and the GADAG Computer become
4 aware of the change. The SPT Bridges then compute and install the new SPTs. When each SPT Bridge has
5 installed the new SPT, then data traffic is directed back to the SPTs. The GADAG Computer then computes
6 the new GADAG and floods its descriptor. The SPT Bridges then compute and install the new MRTs if each
7 SPT Bridge has already directed traffic back to the SPTs.

8 If MRTs protect each other, then only one of the MRTs is restored at a time. Upon a failure event, PLR
9 Bridges direct all data traffic to the MRT that is not affected by the failure (e.g., MRT-Red). If a 1+1 type of
10 protection scheme is used, then there is no need to react to a failure event as data frames are forwarded on
11 both MRTs, i.e., there are no PLRs. When the GADAG Computer and the SPT Bridges have the same view
12 on the network topology again, then the GADAG Computer computes the new GADAG and floods its
13 descriptor. Upon reception of the Topology sub-TLV describing the new GADAG, the SPT Bridges compute
14 the new MRTs. The broken MRT (e.g., MRT-Blue) then is restored while the other MRT (e.g., MRT-Red) is
15 not touched. If each SPT Bridge has installed the new MRT (e.g., the new MRT-Blue), then data traffic is
16 directed to this newly restored MRT. The SPT Bridges then install the other MRT (e.g., the new MRT-Red).
17 The same restoration procedure is followed if both MRTs are affected by the failure so that MRT-Blue is
18 restored first.

19 NOTE—The method to be used for verifying whether the SPT Bridges have a common view on the topology and
20 whether SPTs and MRTs have been restored is left to the implementer and the network operator. A common method is to
21 wait some time after the reception of the last LSP indicating a topology change. Alternatively, OAM can be used, for
22 instance.

23

1 46. Time-Sensitive Networking (TSN) configuration

2 Time-Sensitive Networking (TSN) is a collection of features in IEEE 802.1 standards that provide the
3 following:

- 4 — Time synchronization among Bridges and end stations
- 5 — Significant reduction in frame loss due to faults in network equipment
- 6 — Significant reduction in, or the elimination of, frame loss due to egress Port congestion
- 7 — Bounded latency

8 This clause provides specifications for the configuration of TSN features in a network. The configuration
9 process begins when Talkers and Listeners pass their requirements to the network and proceeds with the
10 configuration of TSN features in Bridges along a tree from each Talker to its Listener(s).

11 46.1 Overview of TSN configuration

12 46.1.1 Streams, Talkers, and Listeners

13 TSN configuration uses the concept of a Stream that is transmitted by a Talker to one or more Listeners. The
14 Talkers and Listeners are located within end stations.

15 This clause specifies configuration information for Streams requested by Talkers and Listeners, or by
16 network management entities acting on their behalf. Stream requirements are initially specified by, or for,
17 Talker and Listener without detailed knowledge of the network. Network entities or administrators obtain
18 these requirements, analyze the topology and capabilities of the Bridges, and configure Bridges to meet the
19 requirements. The network returns the success or failure of each Stream's configuration to its Talker(s) and
20 Listener(s).

21 46.1.2 TSN configuration protocols

22 A variety of protocols can be used for the exchange of configuration information (e.g., signaling protocols,
23 remote network management protocols).

24 Specific TSN-capable products list the protocol that is supported as part of their conformance [e.g., 5.18.3,
25 item c) in 5.29]. Each user/network protocol will specify a specific schema and/or encoding for the
26 configuration information in 46.2. Examples of these protocols are described for each of the TSN
27 configuration models in 46.1.3.

28 46.1.3 TSN configuration models

29 Three models for TSN user/network configuration are described. These models provide an architectural
30 context for subsequent specifications. Each model specification shows the logical flow of configuration
31 information between various entities in the network.

32 46.1.3.1 Fully distributed model

33 In the fully distributed model, the end stations that contain Talkers and Listeners communicate the user
34 requirements directly to the neighboring Bridge. The network is configured in a fully distributed manner,
35 without a centralized network configuration entity. The distributed network configuration is performed using
36 a protocol that propagates configuration information along the active topology for the Stream (i.e., Bridges
37 in the tree from Talker to Listeners).

1 As Stream requirements propagate through each Bridge, management of the Bridge's resources is effectively
2 performed locally. This local management is limited to the information that the Bridge has knowledge of and
3 does not necessarily include knowledge of the entire network.

4 Figure 46-1 provides a graphical representation of the fully distributed model.

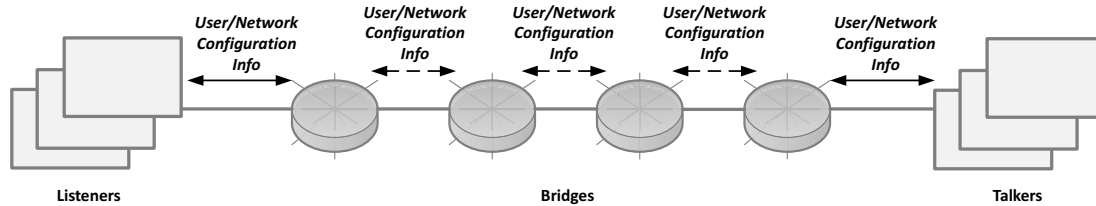


Figure 46-1—Fully distributed model

5 In the figure, the solid arrows represent the protocol that is used to exchange information between Talkers/
6 Listeners (users) and Bridges (network). This information is specified in 46.2.

7 In the figure, the dashed arrows represent the protocol that propagates information within the network. This
8 protocol carries Stream configuration information (46.2) as well as additional information specific to
9 network configuration.

10 The following TSN features can be configured by Bridges using this model:

- 11 a) Credit-based shaper algorithm (8.6.8.2) and its configuration (Clause 34)

12 The Stream Reservation Protocol (SRP) of Clause 35 can be used by Talkers and Listeners, and to propagate
13 configuration information throughout the network of Bridges.

14 **46.1.3.2 Centralized network/distributed user model**

15 Some TSN use cases are computationally complex. For example, for scheduled traffic (8.6.8.4),
16 computation of the gate control list of each Port can take significant time. For such use cases, it is helpful to
17 centralize the computation in a single entity (Bridge or end station), rather than perform the computation in
18 all Bridges.

19 Some TSN use cases can benefit from a complete knowledge of all Streams in the network. For example, if
20 the bandwidth for multiple Streams is greater than the available bandwidth along the shortest path between
21 Talkers and Listeners, it is helpful to forward a subset of those Streams along a path other than the shortest.
22 For these use cases, a centralized entity can gather information for the entire network in order to find the best
23 configuration.

24 In the centralized network/distributed user model, end stations communicate their Talker/Listener
25 requirements directly to the neighboring Bridge, just as in the fully distributed model. However, in the
26 centralized network/distributed user model, a Centralized Network Configuration (CNC, 46.1.6) entity
27 configures Bridges to pass Talker/ Listener Stream requirements directly to the CNC, rather than
28 propagating that information along the path to be taken by Stream data. All configuration of Bridges for
29 Streams is performed by this CNC using a remote network management protocol.

30 The CNC has a complete view of the physical topology of the network as well as the capabilities of each
31 Bridge. This enables the CNC to centralize complex computations. The CNC can exist in either an end
32 station or a Bridge.

Figure 46-2 provides a graphical representation of the centralized network/distributed user model.

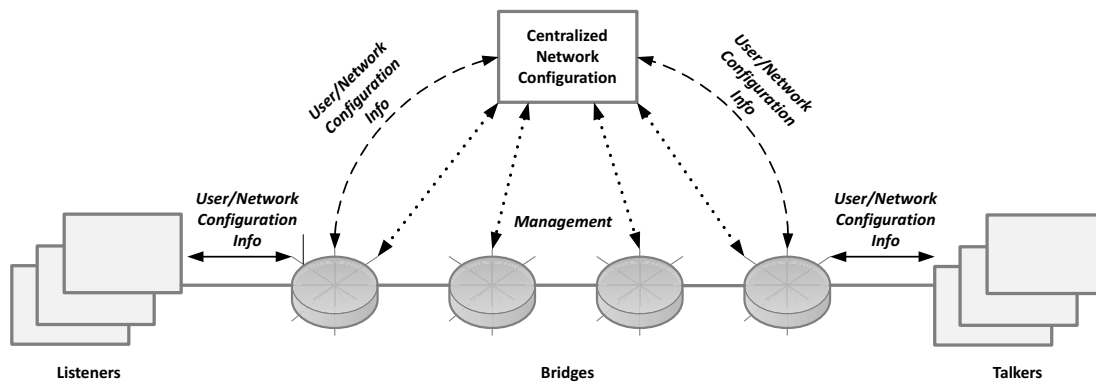


Figure 46-2—Centralized network/distributed user model

In the figure, the solid arrows represent the protocol that is used to exchange Stream information between Talkers/Listeners (users) and Bridges (network). This information is specified in 46.2.

In the figure, the dashed arrows represent the protocol that transfers Stream configuration information between Bridges and the CNC. This information is specified in 46.2.

In the figure, dotted arrows represent the remote network management protocol. The CNC acts as the management client, and each Bridge acts as the management server. The CNC uses remote management to discover physical topology, retrieve Bridge capabilities, and configure TSN features in each Bridge. Talkers and Listeners are not required to participate in this remote network management protocol. The information carried by the remote network management protocol is specified in Clause 12.

NOTE 1—If the Talker/Listener protocol of the fully distributed model is selected to be the same as the Talker/Listener protocol of the centralized network/distributed user model, end stations can support both models without explicit knowledge of how the network is configured.

The following TSN features can be configured by the CNC using this model:

- a) Credit-based shaper algorithm (8.6.8.2) and its configuration (Clause 34)
- b) Frame preemption (6.7.2)
- c) Scheduled traffic (8.6.8.4, 8.6.9)
- d) Frame Replication and Elimination for Reliability (IEEE Std 802.1CB)
- e) Per-Stream Filtering and Policing (8.6.5.1)
- f) Cyclic queuing and forwarding (Annex T)

SRP (Clause 35) can be used as the Talker/Listener protocol (solid arrows of Figure 46-2). SRP's MRP External Control (12.32.4) feature can be used to exchange configuration information with the CNC component (dashed arrows of Figure 46-2). SRP exchanges configuration information using the TLV technique to reference elements in 46.2 (see 46.1.3.1). Examples of a remote network management protocol (dotted arrows of Figure 46-2) include Simple Network Management Protocol (SNMP), NETCONF (IETF RFC 6241 [B39]), and RESTCONF (IETF RFC 8040 [B47]).

NOTE 2—NETCONF and RESTCONF specify a startup datastore: nonvolatile configuration that is applied when the Bridge powers on. The startup datastore feature enables a CNC to configure Bridges and then remove itself from the network. SNMP does not specify a startup datastore feature. If SNMP is used by a CNC, this can be mitigated by a) using a proprietary (Bridge-specific) startup datastore feature or b) ensuring that the CNC is always active in the network in order to reconfigure Bridges that cycle power.

46.1.3.3 Fully centralized model

Many TSN use cases require significant user configuration in the end stations that act as Talkers and Listeners. For example, in many automotive and industrial control applications, the timing of physical inputs and outputs (I/Os) is determined by the physical environment under control, and the timing requirements for Streams are derived from that I/O timing. In some use cases, these I/O timing requirements can be computationally complex and involve detailed knowledge of the application software/hardware within each end station.

In order to accommodate this sort of TSN use case, the fully centralized model enables a Centralized User Configuration (CUC, 46.1.5) entity to discover end stations, retrieve end station capabilities and user requirements, and configure TSN features in end stations. The protocols that the CUC uses for this purpose are not specified in this standard.

From a network perspective, the primary difference between the fully centralized model and the centralized network/distributed user model is that all user requirements are exchanged between the CNC and CUC.

Figure 46-3 provides a graphical representation of the fully centralized model with multiple CUCs.

15

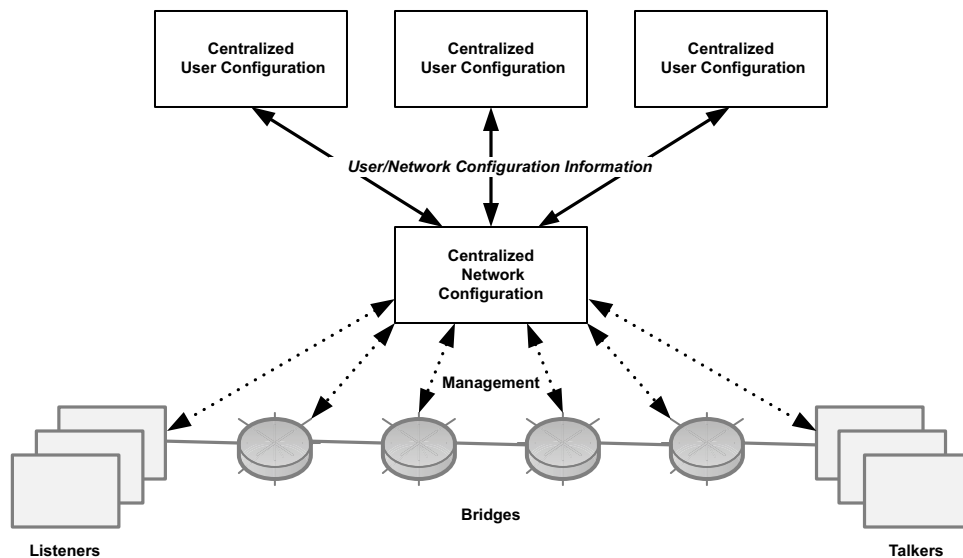


Figure 46-3—Fully centralized model

In the figure, the solid arrows represent the protocol that is used to exchange Stream information between the CUC and the CNC. This information is specified in 46.2.

In the figure, the dotted arrows represent the remote network management protocol. The CNC acts as the management client, and each Bridge acts as the management server. The CNC uses remote management to discover physical topology, retrieve Bridge capabilities, and configure TSN features in each Bridge. Talkers and Listeners are not required to participate in this remote network management protocol. The information carried by the remote network management protocol is specified in Clause 12.

In this fully centralized model, a protocol is used between the CUC and end stations (Talkers and Listeners) to retrieve end station capabilities and requirements and to configure the end stations.

1 The following TSN features can be configured by the CNC using this model:

- 2 a) Credit-based shaper algorithm (8.6.8.2) and its configuration (Clause 34)
- 3 b) Frame preemption (6.7.2)
- 4 c) Scheduled traffic (8.6.8.4, 8.6.9)
- 5 d) Frame Replication and Elimination for Reliability (IEEE Std 802.1CB)
- 6 e) Per-Stream Filtering and Policing (8.6.5.1)
- 7 f) Cyclic queuing and forwarding (Annex T)

8 YANG (IETF RFC 7950) is a data modeling language used to model configuration data and state data for
9 remote network management protocols. The remote network management protocol uses a specific encoding
10 such as XML or JSON. For a particular feature, a YANG module specifies the organization and rules for the
11 feature's management data, and a mapping from YANG to the specific encoding enables the data to be
12 understood correctly by both client (e.g., network manager) and server (e.g., Bridge).

13 In order to support the use of YANG-based protocols for the fully centralized model, 46.3 specifies a YANG
14 module. The YANG module specifies a YANG typedef/grouping for each group of information in 46.2.

15 For an informative example workflow using the fully centralized model, refer to U.2.

16 **46.1.4 Stream identification and transformation**

17 TSN configuration uses the concept of a Stream of data that is transmitted by a Talker to one or more
18 Listeners.

19 In order to apply TSN behavior to a Stream (e.g., to reserve bandwidth), the network distinguishes Streams
20 from each other and from non-TSN traffic (e.g., best-effort). Each frame of a Stream contains fields that
21 uniquely identify the Stream.

22 The goal of TSN configuration is to allow Talkers and Listeners to use their existing transport layer and
23 application layer protocols for data, rather than requiring a TSN-specific frame format. TSN achieves this
24 goal by identifying each Stream using fields from well-established frame formats such as Transmission
25 Control Protocol (TCP), User Datagram Protocol (UDP), and IEEE 802.1 (i.e., MAC addresses and VLAN
26 identifier).

27 As the frames of each Stream cross the user/network boundary, the identification of the Stream in its frames
28 can be different between the network and the user. For example, the user can use UDP without an awareness
29 of VLAN IDs, but the network can require a specific VLAN ID in order to apply TSN features. In order to
30 support this sort of difference in frame format, the TSN user/network configuration information (46.2)
31 provides features to enable transformation of the Stream's identification at the user/network boundary. The
32 user identification translates to/from the network identification at the boundary, either within the end station
33 or a nearest Bridge. This transformation has the benefit of allowing the user's identification to match its
34 higher layer application protocol and the network's identification to match the bridging technology.

Stream transformation can be accomplished using the functions specified in IEEE Std 802.1CB. The functions of IEEE Std 802.1CB can be implemented in the end station (Talker/Listener) or within the nearest Bridge. The descriptions in this clause focus on Stream transformation in the end station and use features of the TSN user/network configuration information (46.2).

NOTE 1—In this clause, Stream transformation refers to changes to the fields of a frame that identify the Stream. IEEE Std 802.1CB specifies Stream transformation for identification as well as for frame replication and elimination (redundancy).

NOTE 2—Stream transformation is an optional capability of end stations and Bridges. If stream transformation is not supported, the user's identification of the Stream must be the same as the network's identification, and the user must use an identification that is consistent with bridging as specified for TSN features in this standard (e.g., VLAN tag and group destination MAC address).

Figure 46-4 provides an example of Stream transformation in the Talker end station. Stream transformation in the Listener end station is similar. The example of Figure 46-4 assumes use of the centralized network/distributed user model (46.1.3.2). Use of the fully centralized model (46.1.3.3) is similar.

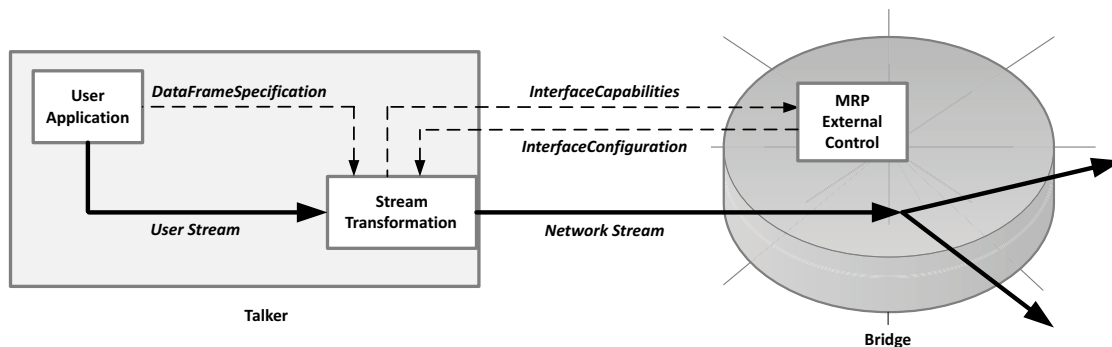


Figure 46-4—Example of Stream transformation in Talker end station

For Stream transformation in the end station, the end station's interface to the network can provide the transformation capability, acting as a network entity within the user's end station.

The identification of the Stream in the user originates from the User Application block, specified by the Talker as the DataFrameSpecification (46.2.3.4). The end station knows this user identification, but not the identification that the network requires. To negotiate the network identification, the Talker uses SRP to transmit InterfaceCapabilities (46.2.3.7) that describe its Stream transformation capabilities to the nearest Bridge. The Bridge uses MRP External Control (12.32.4) to send the InterfaceCapabilities to a CNC. The CNC consults its configuration of network identification and uses MRP External Control to send InterfaceConfiguration (46.2.5.3) along with successful Status (46.2.5) back to the Bridge. When the Bridge receives this information, it propagates back to the Talker using SRP. The InterfaceConfiguration provides the network identification, which the end station uses to perform Stream transformation for data frames.

NOTE 3—The network identification typically entails allocation of a group MAC address for the Stream. If a CNC is used, the CNC can allocate a group MAC address from a pool that it maintains.

Figure 46-5 provides an example of IEEE 802.1CB functions within the Stream Transformation block in the Talker end station. The example assumes that the user identification uses an Internet Protocol (IP) packet for identification and that the frame conveying the IP packet does not use the appropriate MAC address and VLAN tag for TSN features in Bridges (e.g., IP packet unicast destination MAC address, untagged). The IEEE 802.1CB function for IP Stream identification uses fields of the IP packet to identify the packet as a specific Stream. That stream identification is then applied to the IEEE 802.1CB function for Active Destination MAC and VLAN Stream identification to replace the destination MAC address and add a

1 VLAN tag for TSN Bridge features. The IP fields of the packet are not changed. The IEEE 802.1CB
2 functions can be implemented in software (e.g., operating system driver) or in hardware.

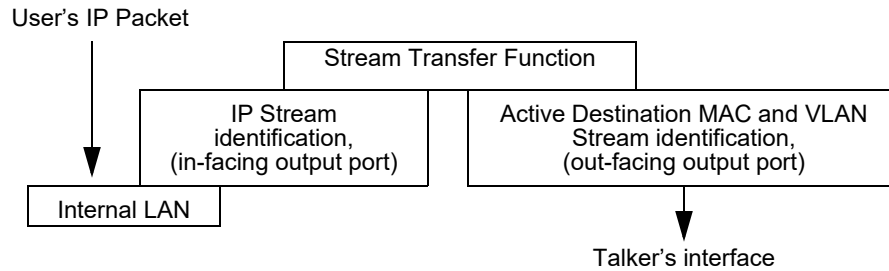


Figure 46-5—Example of IEEE 802.1CB functions in Talker end station

3 Figure 46-6 provides a corresponding example of IEEE 802.1CB functions within the Stream
4 Transformation block in the Listener end station. In this direction, the IEEE 802.1CB function for Active
5 Destination MAC and VLAN Stream identification provides both functions. The IEEE 802.1CB function
6 uses the group destination MAC address and VLAN tag of the received frame to identify a specific Stream.
7 The IEEE 802.1CB function then transforms the destination MAC address and VLAN tag to restore the
8 Stream's frame to its original format (as transmitted by the Talker).

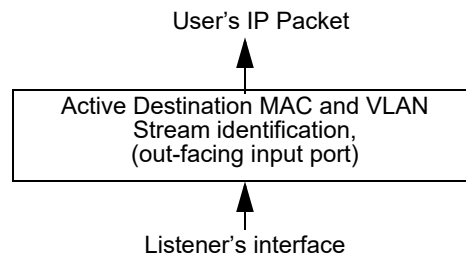


Figure 46-6—Example of IEEE 802.1CB functions in Listener end station

9 46.1.5 Centralized User Configuration

10 A Centralized User Configuration (CUC) delivers user requirements to the CNC. The CUC delivers
11 information for configuring TSN features to end stations. It is a logical entity that can be located in any
12 network system.

13 The CUC is responsible for:

- 14 a) Reconciling the requirements from Talkers and Listeners to Stream requirements, if necessary.
- 15 b) Sending the Stream requirements to the CNC.
- 16 c) Receiving the end station communication-configuration from the CNC.
- 17 d) Distributing the end station communication-configuration to Talkers and Listeners.

18 NOTE—The CNC is responsible for the assignment of a unique StreamID to each Stream. For this a remote procedure
19 call (RPC) RequestFreeStreamId (46.2.7.5) is available so the CUC can request a free (i.e., available) StreamID from the
20 CNC.

21 Stream requirements, in the context of the CUC, result from combining the Stream requirements of one
22 Talker with the Stream requirements of one or multiple Listeners that, together, apply to form a Stream.
23 Reconciling the requirements for the Stream does not change the parameters in the Stream request
24 originating from the Talker or the Listener(s).

1 The end station communication-configuration that is received by the CUC from the CNC and then
2 distributed to the Talkers and Listeners does not directly configure features on the end stations. It consists of
3 configuration information that a CUC can provide for a Talker and Listeners to configure the Stream. An end
4 station could, for example, make use of the information it receives in the communication-configuration from
5 the CUC to configure an application in a way that ensures different Streams are sent by the application in a
6 specific order that correlates with the expected Stream's transmission on the network.

7 A CUC affects only one Configuration Domain. Talkers and Listeners can only make use of the CUC to
8 reconcile their Stream requirements into a Stream request, if they are part of the same Configuration
9 Domain. If a Talker wants to communicate with one or more Listeners in a different Configuration Domain,
10 this needs to be done through dedicated inter-domain communication mechanisms. Such inter-domain
11 communication mechanisms are not specified by this standard.

12 The protocols that the CUC uses for communication with end stations are not specified by this standard. A
13 CUC exchanges information with a CNC in order to configure TSN features on behalf of its end stations
14 (46.2). The CUC can request computation of paths and configurations for Streams in the following ways:

- 15 e) Request computation of the paths and configurations for a set of Streams, using the protocol
16 operation described in 46.2.7.1. The computation is performed by the CNC on the complete set of
17 Streams of this request. This allows for optimized scheduling of Streams in the network.
- 18 f) Request computation of the paths and configurations for new or modified Streams, using the
19 protocol operation described in 46.2.7.2. The computation is performed by the CNC on all Streams
20 in a Configuration Domain that have a StreamStatus (46.2.3.8) of either planned or modified.
- 21 g) Request computation of the paths and configurations for all Streams of a CUC, using the protocol
22 operation described in 46.2.7.3. The computation is performed by the CNC on all Streams in a
23 Configuration Domain that belong to the CUC specified in the request.
- 24 h) Request the joining of a set of Listeners to an already existing Stream. The paths are extended to
25 allow forwarding of the Stream to the new Listeners. Computation for the changes has to be
26 triggered via RPC.
- 27 i) Request the removal of an existing Stream, using the protocol operation described in 46.2.8.1.
- 28 j) Request the removal of one or more Listeners from an existing Stream. Computation for the changes
29 has to be triggered via RPC.

30 A CUC can be present for initial configuration, to manage changes to a running network, or both. Multiple
31 CUCs can co-exist and operate in parallel in the same Configuration Domain as shown in Figure 46-3.

32 **46.1.6 Centralized Network Configuration**

33 The Centralized Network Configuration (CNC) is a logical entity that configures network resources on
34 behalf of applications (users) and can be located in any station of a network.

35 The CNC is responsible for:

- 36 a) Receiving the Stream requirements for one or more Streams from the corresponding CUC.
- 37 b) Providing a way for a CUC to request a free StreamID.
- 38 c) Assigning a unique destination MAC address in the Configuration Domain it is responsible for to
39 each of the requested Streams.
- 40 d) Computing paths for requested Streams.
- 41 e) Performing computation of scheduling and/or shaping configuration for the requested Streams.
- 42 f) Configuring the network devices to provide the required resources for the Streams (e.g. Filtering
43 Database (8.8) entries, configuration of transmission gates, etc.), using remote management.

- 1 g) Providing the end station communication-configuration for the Streams to the corresponding CUC.
- 2 If the paths for the Streams impact existing Streams the CNC is also responsible for providing that
- 3 information to the CUCs that originally requested the impacted Streams.
- 4 h) Removing of Streams as requested by a CUC.
- 5 i) Discovering physical topology, using remote management.
- 6 j) Retrieving station capabilities, using remote management.

7 The CNC communicates with a CUC through the user/network configuration information specified in 46.2.
8 It communicates with the stations using the managed objects defined in this and other IEEE 802.1 standards.
9 There can only be one active CNC per Configuration Domain.

10 **46.1.7 Configuration Domain**

11 A Configuration Domain is a set of stations that are under a common configuration and management
12 scheme, and a single administration. The Configuration Domain provides boundary information for the
13 common management scheme and to support the responsibilities of the CUC and CNC regarding Streams.
14 Whether a CNC and one or more CUCs are present in a Configuration Domain depends on the TSN
15 configuration model (46.1.3) that is used in the domain (e.g., whether the fully centralized model or a
16 different configuration model is used). The CNC and the CUCs required for the configuration of a
17 Configuration Domain affect only one Configuration Domain.

18 **46.2 User/network configuration information**

19 User/network configuration information for the three TSN configuration models (46.1.3) is specified as
20 follows. For the two distributed models, SRP specifies TLVs in 35.2.2.10 that reference information in 46.2.
21 For the fully centralized model, the YANG module in 46.3 references information in 46.2.

22 Within this subclause, the word element refers to a single item of information used for TSN configuration.
23 The word group refers to a collection of related elements. Groups are organized hierarchically, such that a
24 group can be contained within another group. A single low-level group can be contained within multiple
25 higher level groups. The dot-separated notation is used to refer to a specific element in text. For example,
26 “Talker.StreamID.UniqueId” refers to the UniqueId element of the StreamID that is contained within the
27 Talker group.

28 This subclause specifies each group in a table. Each row of the table specifies an element of the group. Each
29 element’s row specifies its name, data type, and a reference to normative text that specifies its semantics
30 (i.e., meaning). The data type of each element uses one of the values from 46.2.1. Other specifications such
31 as conformance (i.e., required or optional), direction of transfer (i.e., user to network or network to user),
32 default value, and range limitations are specified with normative text instead of with the table.

33 Each element name uses a camel-case naming convention (e.g., “MacAddress”) to align with naming
34 conventions used in other clauses of IEEE Std 802.1Q. A specific protocol can use a different naming
35 convention (e.g., 46.3) as long as the protocol’s name for the element can be associated with the element’s
36 specification in 46.2.

37 **46.2.1 Data types**

38 Data types in the tables include the following:

- 39 a) Boolean
- 40 b) int8, for a signed 8-bit integer
- 41 c) int16, for a signed 16-bit integer
- 42 d) int32, for a signed 32-bit integer

- 1 e) uint8, for an unsigned 8-bit integer
- 2 f) uint16, for an unsigned 16-bit integer
- 3 g) uint32, for an unsigned 32-bit integer
- 4 h) string
- 5 i) enumeration, for a collection of named values
- 6 j) rational, for a rational number consisting of a uint32 numerator and uint32 denominator
- 7 k) mac-address-type, for an IEEE 802 MAC address
- 8 l) ipv4-address-type, for an IPv4 address (IETF RFC 791)
- 9 m) ipv6-address-type, for an IPv6 address (IETF RFC 8200)
- 10 n) sequence of <X>, for a list of zero or more instances of data type <X> (e.g., sequence of uint32)

11 If the data type shown in the table is not from the preceding list, then the data type is specified in the
12 normative text for the element.

13 46.2.2 Protocol integration

14 The TSN user/network configuration information consists of three high-level groups:

- 15 — **Talker:** Elements from user to network that specify the Talker for a single Stream.
- 16 — **Listener:** Elements from user to network that specify the Listener for a single Stream.
- 17 — **Status:** Elements from network to user that specify the status of the Stream's network configuration
18 for a Talker or Listener. This group notifies the user when the Stream is ready for use (or a failure
19 occurred).

20 A protocol that supports TSN user/network configuration information shall integrate the three groups using
21 the specifications in this clause. The integration is provided in each protocol's specifications (not in this
22 clause). For examples of such protocol integration, see Clause 35 and 46.3.

23 The StreamID of 46.2.3.1 uniquely identifies the Stream. The StreamID identifies configuration, not data, so
24 it has no formal relation to the data frame encoding for the Stream.

25 TSN configuration can be viewed conceptually as a request/response exchange:

- 26 — Request: End station or CUC transmits a protocol message that contains a Talker or Listener group.
- 27 — Response: Bridge or CNC transmits a protocol message that contains a Status group.
- 28 NOTE—The Response can be unsolicited in order to update configuration, e.g., to address a change in the
29 network.

30 The StreamID of each group correlates each request to its corresponding response, such that configuration of
31 multiple Streams can overlap in time. For example, a possible sequence of messages is Talker request for
32 StreamID 1, Talker request for StreamID 2, Listener request for StreamID 2, Status to Talker for StreamID 2,
33 Status to Listener for StreamID 2, Listener request for StreamID 1, and so on.

34 For the fully distributed model (46.1.3.1), the Status response to a Talker is communicated in a message
35 along with the Listener request(s). When there is more than one Listener per Stream, the Listener and Status
36 groups must be merged as they propagate through Bridges to the Talker. The methodology for this merger is
37 specified in the TSN configuration protocol (not in this clause).

38 For the fully centralized model (46.1.3.3), the CUC combines multiple Talker groups and Listener groups
39 into a single transfer from CUC to CNC for computation together in the CNC. After the CNC computation is
40 complete, a single transfer from CNC to CUC combines multiple Status groups. Each Status group can
41 provide sub-groups for each Talker/Listener and merged groups for the entire Stream. The methodology for
42 these combinations and mergers is specified in the TSN configuration protocol (not in this clause).

1 In addition to the request/response exchange, end stations can transmit Talker/Listener messages as
2 advertisement (e.g., for a Talker to notify potential Listeners of its presence). The methodology for this
3 advertisement is specified in the TSN configuration protocol (not in this clause).

4 For each Talker group and Listener group, there are two operations for each Stream:

- 5 — **Join:** Talker/Listener request to join the Stream and to configure network resources for this Stream's
6 data transfer.
- 7 — **Leave:** Talker/Listener request to leave the Stream and to release the network resources for this
8 Stream.

9 The groups in this clause specify requirements for the join and leave operation, but the encoding of the
10 operation is not specified in this clause. Each TSN configuration protocol shall specify the encoding of the
11 join/leave operation as part of its integration. For example, SRP specifications of this standard (Clause 35)
12 encode the join/leave operation within the Multiple Registration Protocol (MRP) (Clause 10) that serves as
13 its foundation.

14 The protocol message(s) that invoke the join or leave operation are not required to coincide with the protocol
15 message(s) that contain the associated groups (Talker, Listener, or Status). Nevertheless, the groups specify
16 elements that are required for a subsequent join or leave operation to be valid. For example, for the fully
17 centralized model (46.1.3.3), the CUC can transfer a list of Talker/Listener groups to the CNC, followed by
18 a separate protocol message with a join request that applies to the entire list. For the join request to succeed,
19 each of the Talker/Listener groups contains the required elements. At a later time, the CUC can read the
20 resulting list of Status groups from the CNC, which provides the response to the join.

21 **46.2.2.1 DomainID**

22 DomainID uniquely identifies the Configuration Domain of a CUC, and the Streams associated with that
23 CUC. DomainID is only used if the centralized network/distributed user model (46.1.3.2) or the fully
24 centralized model (46.1.3.3) is used.

25 **46.2.2.2 CucID**

26 CucID uniquely identifies a CUC within a Configuration Domain and is used in configuration models that
27 include a CNC. It is used along with the DomainID to associate Streams with a CUC.

28 **46.2.2.3 CncEnabled**

29 CncEnabled is used to enable or disable the CNC functionality of a station capable of acting as a CNC. If
30 CncEnabled is set to TRUE the CNC functionality is enabled. If it is set to FALSE the CNC functionality is
31 disabled. The default value for CncEnabled is FALSE.

32 **46.2.3 Talker**

33 The Talker group specifies the following:

- 34 — Talker's behavior for Stream (how/when transmitted)
- 35 — Talker's requirements from the network
- 36 — TSN capabilities of the Talker's interface(s)

37 In the fully distributed model and the centralized network/distributed user model, this group originates from
38 the Talker's end station. In the fully centralized model, this group originates from the CUC.

39 The Talker group contains the following groups:

- 1 — StreamID (46.2.3.1)
- 2 — StreamRank (46.2.3.2)
- 3 — EndStationInterfaces (46.2.3.3)
- 4 — DataFrameSpecification (46.2.3.4)
- 5 — TrafficSpecification (46.2.3.5)
- 6 — UserToNetworkRequirements (46.2.3.6)
- 7 — InterfaceCapabilities (46.2.3.7)
- 8 — StreamStatus (46.2.3.8)

9 For the join operation, the StreamRank and TrafficSpecification groups shall be included within the Talker
10 group.

11 For the join operation, the UserToNetworkRequirements and InterfaceCapabilities groups may be included
12 within the Talker group. If a group is omitted, the network uses the default values specified for elements of
13 that group.

14 For the join operation, the DataFrameSpecification shall be included within the Talker group unless all of the
15 following conditions apply:

- 16 a) The InterfaceCapabilities group is included.
- 17 b) InterfaceCapabilities.CB-StreamIdenTypeList includes the type for Active Destination MAC and
18 VLAN Stream identification.

19 The preceding conditions enable the Talker and Listener to perform Stream transformation (46.1.4), and
20 therefore the network does not need the user's data frame identification in DataFrameSpecification. The
21 absence of the DataFrameSpecification group notifies the network that Stream transformation is performed
22 in the end stations for this Stream.

23 For the join and leave operation, the StreamID group shall be included in the TSN user/network protocol in
24 a location associated with the Stream (e.g., Talker group).

25 For the join and leave operation, EndStationInterfaces shall be included within the Talker group for the fully
26 centralized model (46.1.3.3). For the join and leave operation, EndStationInterfaces should be included
27 within the Talker group for the fully distributed model (46.1.3.1) or centralized network/distributed user
28 model (46.1.3.2).

29 For the join and leave operation, StreamStatus shall be included.

30 **46.2.3.1 StreamID**

31 The StreamID group provides a unique identifier of the Stream's configuration and is used by protocols in
32 the network to associate the user's Stream with TSN resources. The StreamID group is used by Talker,
33 Listener, and Status groups.

34 The elements of the StreamID group are listed in Table 46-1.

Table 46-1—StreamID elements

Name	Data type	Reference
MacAddress	mac-address-type	46.2.3.1.1
UniqueID	uint16	46.2.3.1.2

1 46.2.3.1.1 MacAddress

2 MacAddress is a 48-bit IEEE 802 MAC address associated with the Talker sourcing the Stream to the
3 bridged network. The entire range of MAC addresses are acceptable.

4 NOTE 1—The MAC address component of the StreamID can, but does not necessarily, have the same value as the
5 source_address parameter of any frame in the actual data Stream. For example, the StreamID can be assigned by a TSN
6 CUC (see 46.1.3.3), using a pool of MAC addresses that the TSN CUC maintains.

7 NOTE 2—If the MAC addresses used to construct StreamIDs are not unique within the network, duplicate StreamIDs
8 can be generated, with unpredictable results.

9 46.2.3.1.2 UniqueID

10 UniqueID is used to distinguish between multiple Streams within the end station identified by MacAddress.

11 46.2.3.2 StreamRank

12 The StreamRank group provides the rank of this Stream relative to other Streams in the network. This rank is
13 used to determine success/failure of Stream resource configuration and is unrelated to the Stream's data.

14 The elements of the StreamRank group are listed in Table 46-2.

Table 46-2—StreamRank elements

Name	Data type	Reference
Rank	uint8	46.2.3.2.1

15 46.2.3.2.1 Rank

16 The Rank is used by the network to decide which Streams can and cannot exist when TSN resources reach
17 their limit. If a Bridge's Port becomes oversubscribed (e.g., network reconfiguration, IEEE 802.11
18 bandwidth reduction), the Rank is used to help determine which Streams can be dropped (i.e., removed from
19 Bridge configuration).

20 The only valid values for Rank shall be zero and one. The configuration of a Stream with Rank zero is more
21 important than the configuration of a Stream with Rank one. The Rank value of zero is intended for
22 emergency traffic, and the Rank value of one is intended for non-emergency traffic.

23 NOTE—It is expected that higher layer applications and protocols can use the Rank to indicate the relative importance
24 of Streams based on user preferences. Those user preferences are expressed by means beyond the scope of this standard.
25 When multiple applications exist in a network (e.g., audio/video along with industrial control), it can be challenging for
26 the varied applications and vendors to agree on multiple Rank values. To mitigate such challenges, this Rank uses a
27 simple concept of emergency (zero) and non-emergency (one) that can be applied over all applications. For example, in
28 a network that carries audio Streams for fire safety announcements, all applications are likely to agree that those Streams
29 use Rank of zero.

30 46.2.3.3 EndStationInterfaces

31 The EndStationInterfaces group provides a list of one or more InterfaceID groups. Each InterfaceID group
32 identifies a physical interface (distinct point of attachment) in the end station acting as a Talker/Listener.

33 Although many end stations contain a single interface, this list allows for multiple interfaces. Some TSN
34 features allow a single Stream to span multiple interfaces (e.g., seamless redundancy).

1 In centralized network models, EndStationInterfaces is used by the CNC to locate the interfaces of the
2 Talker/Listener in the topology.

3 The InterfaceID group is used to identify a distinct point of attachment in EndStationInterfaces,
4 InterfaceConfiguration (46.2.5.3), and FailedInterfaces (46.2.5.4). In the context of EndStationInterfaces
5 and InterfaceConfiguration, each interface is located in an end station only. In the context of
6 FailedInterfaces, each interface can be located within an end station or a Bridge.

7 The elements of the InterfaceID group are listed in Table 46-3.

Table 46-3—InterfaceID elements

Name	Data type	Reference
MacAddress	mac-address-type	46.2.3.3.1
InterfaceName	string	46.2.3.3.2

8 **46.2.3.3.1 MacAddress**

9 MacAddress is the MAC address (IEEE Std 802) of the interface in the station (end station or Bridge). This
10 MAC address uniquely identifies the station within the local network.

11 MacAddress shall be included in the InterfaceID group.

12 NOTE—This MAC address can be discovered in the physical topology using protocols such as IEEE Std 802.1AB
13 (LLDP). LLDP supports MAC address as a subtype for the station's Chassis ID and Port ID. If the station does not use
14 MAC address for its LLDP IDs, remote management can be used to associate this MacAddress to the values provided in
15 the LLDP IDs.

16 **46.2.3.3.2 InterfaceName**

17 InterfaceName is the name of the interface that is assigned locally by the station (end station or Bridge).

18 InterfaceName may be included in the InterfaceID group.

19 IEEE Std 802 recommends that each distinct point of attachment to an IEEE 802 network have its own
20 universal MAC address. If the identified station follows this IEEE 802 recommendation, the MacAddress
21 element uniquely identifies the interface as well as the station, and InterfaceName is not needed.

22 If the MacAddress applies to more than one interface (distinct point of attachment) within the station,
23 InterfaceName provides a locally assigned name that can help to identify the interface.

24 When YANG is used for management of the station, InterfaceName is the interface name that serves as the
25 key for the station's interface list (IETF RFC 7223 [B43]).

26 NOTE 1—The TSN CNC is typically located in a different physical product than the station identified by this
27 InterfaceID. Since the InterfaceName is assigned locally by the identified station, it is possible that the station's product
28 will change InterfaceName in a manner that the TSN CNC cannot detect. For example, IETF RFC 7223 [B43] mentions
29 that the YANG interface name can change when a physical attachment point is inserted or removed.

30 NOTE 2—This interface name can be discovered in the physical topology using protocols such as IEEE Std 802.1AB
31 (LLDP). LLDP supports interface name as a subtype for its Port ID. If the station does not use interface name for its
32 LLDP Port ID, remote management can be used to associate this InterfaceName to the values provided in the LLDP
33 Port ID.

1 46.2.3.4 DataFrameSpecification

2 The DataFrameSpecification group specifies the frame that carries the Talker's Stream data. The network
3 uses the specification to identify this Stream's frames as TSN in order to apply the required TSN
4 configuration.

5 The DataFrameSpecification is based on the user's knowledge of the frame, without any network specifics.
6 In other words, this specifies the frame that the Talker would use in the absence of TSN.

7 The DataFrameSpecification is provided as a list of fields that the user knows. Each field is provided as one
8 of the following groups:

- 9 — IEEE802-MacAddresses (46.2.3.4.1)
- 10 — IEEE802-VlanTag (46.2.3.4.2)
- 11 — IPv4-tuple (46.2.3.4.3)
- 12 — IPv6-tuple (46.2.3.4.4)

13 The list of fields is ordered from start of frame to end of header. For example, if the Talker uses a VLAN-
14 tagged Ethernet frame (not IP), the list consists of IEEE802-MacAddresses followed by IEEE802-VlanTag.
15 For example, if the Talker uses a UDP/IPv4 packet without knowledge of the Ethernet header, the list
16 consists of IPv4-tuple only.

17 This group is optional, and its absence indicates that Stream transformation is performed in the Talker and
18 Listeners of this Stream (46.2.3).

19 46.2.3.4.1 IEEE802-MacAddresses

20 The IEEE802-MacAddresses group specifies a pair of IEEE 802 MAC addresses for Stream identification.

21 The use of these fields for Stream identification corresponds to the managed objects for Stream
22 identification in IEEE Std 802.1CB. If inconsistency arises between this specification and
23 IEEE Std 802.1CB, IEEE Std 802.1CB takes precedence.

24 The elements of the IEEE802-MacAddresses group are listed in Table 46-4.

Table 46-4—IEEE802-MacAddresses elements

Name	Data type	Reference
DestinationMacAddress	mac-address-type	46.2.3.4.1 item a)
SourceMacAddress	mac-address-type	46.2.3.4.1 item b)

25 The elements of the IEEE802-MacAddresses group:

- 26 a) **DestinationMacAddress**—This field specifies a destination MAC address. An address of all 1's
28 specifies that the destination MAC address is ignored for purposes of Stream identification.
- 29 b) **SourceMacAddress**—This field specifies a source MAC address. An address of all 1's specifies
31 that the source MAC address is ignored for purposes of Stream identification.

32 46.2.3.4.2 IEEE802-VlanTag

33 The IEEE802-VlanTag group specifies a customer VLAN Tag (C-TAG of Clause 9) for Stream
34 identification.

- 1 The Drop Eligible Indicator (DEI) field is not relevant from the perspective of a TSN Talker/Listener.
- 2 The use of these fields for Stream identification corresponds to the managed objects for Stream
3 identification in IEEE Std 802.1CB. If inconsistency arises between this specification and
4 IEEE Std 802.1CB, IEEE Std 802.1CB takes precedence.
- 5 The elements of the IEEE802-VlanTag group are listed in Table 46-5.

Table 46-5—IEEE802-VlanTag elements

Name	Data type	Reference
PriorityCodePoint	uint8	46.2.3.4.2 item a)
VlanId	uint16	46.2.3.4.2 item b)

- 6 The elements of the IEEE802-VlanTag group:
- 7 a) **PriorityCodePoint**—This field specifies the Priority Code Point (PCP) field of the VLAN tag. The
9 value range is 0 to 7 inclusive. The PriorityCodePoint is not used to identify the Stream, but it does
10 identify a traffic class (queue) in Bridges.
- 11 b) **VlanId**—This field specifies the VLAN ID (VID) field of the VLAN tag. The value range is 0 to
13 4095 inclusive. If only the PriorityCodePoint is known, the VlanId is specified as 0.

14 46.2.3.4.3 IPv4-tuple

- 15 The IPv4-tuple group specifies fields to identify an IPv4 (RFC791) Stream.
- 16 The use of these fields for Stream identification corresponds to the managed objects for Stream
17 identification in IEEE Std 802.1CB. If inconsistency arises between this specification and
18 IEEE Std 802.1CB, IEEE Std 802.1CB takes precedence.
- 19 The elements of the IPv4-tuple group are listed in Table 46-6.

Table 46-6—IPv4-tuple elements

Name	Data type	Reference
SourceIpAddress	ipv4-address-type	46.2.3.4.3 item a)
DestinationIpAddress	ipv4-address-type	46.2.3.4.3 item b)
Dscp	uint8	46.2.3.4.3 item c)
Protocol	uint16	46.2.3.4.3 item d)
SourcePort	uint16	46.2.3.4.3 item e)
DestinationPort	uint16	46.2.3.4.3 item f)

- 20 The elements of the IPv4-tuple group:
- 21 a) **SourceIpAddress**—This field specifies the source IPv4 address. An address of all 0's specifies that
23 the IP source address is ignored for purposes of Stream identification.

- 1 b) **DestinationIpAddress**—This field specifies the destination IPv4 address.
- 3 c) **Dscp**—This field specifies the Differentiated Services Code Point (DSCP) (IETF RFC 2474). A
5 value of 64 decimal specifies that the DSCP is ignored for purposes of Stream identification.
- 6 d) **Protocol**—This field specifies the IPv4 Protocol (e.g., UDP). The special value of all 1's (FFFF
8 hex) represents 'None', i.e., Protocol, SourcePort, and DestinationPort are ignored for purposes of
9 Stream identification. For any value other than all 1's, the lower octet is used to match IPv4
10 Protocol.
- 11 e) **SourcePort**—This field specifies the source port of the Protocol.
- 13 f) **DestinationPort**—This field specifies the destination port of the Protocol.

15 46.2.3.4.4 IPv6-tuple

16 The IPv6-tuple group specifies fields to identify an IPv6 (RFC 8200) Stream.

17 The use of these fields for Stream identification corresponds to the managed objects for Stream
18 identification in IEEE Std 802.1CB. If inconsistency arises between this specification and
19 IEEE Std 802.1CB, IEEE Std 802.1CB takes precedence.

20 The elements of the IPv6-tuple group are listed in Table 46-7.

Table 46-7—IPv6-tuple elements

Name	Data type	Reference
SourceIpAddress	ipv6-address-type	46.2.3.4.3 item a)
DestinationIpAddress	ipv6-address-type	46.2.3.4.3 item b)
Dscp	uint8	46.2.3.4.3 item c)
Protocol	uint16	46.2.3.4.3 item d)
SourcePort	uint16	46.2.3.4.3 item e)
DestinationPort	uint16	46.2.3.4.3 item f)

21 The elements of the IPv6-tuple group:

- 22 a) **SourceIpAddress**—This field specifies the source IPv6 address. An address of all 0's specifies that
24 the IP source address is ignored for purposes of Stream identification.
- 25 b) **DestinationIpAddress**—This field specifies the destination IPv6 address.
- 27 c) **Dscp**—This field specifies the DSCP (IETF RFC 2474). A value of 64 decimal specifies that the
29 DSCP is ignored for purposes of Stream identification.
- 30 d) **Protocol**—This field specifies the IPv6 Next Header (e.g., UDP). The special value of all 1's (FFFF
32 hex) represents 'None', i.e., Protocol, SourcePort, and DestinationPort are ignored for purposes of
33 Stream identification. For any value other than all 1's, the lower octet is used to match IPv6 Next
34 Header.
- 35 e) **SourcePort**—This field specifies the source port of the Protocol.
- 37 f) **DestinationPort**—This field specifies the destination port of the Protocol.

46.2.3.5 TrafficSpecification

The TrafficSpecification group specifies how the Talker transmits frames for the Stream. This is effectively the Talker's promise to the network. The network uses this traffic specification to allocate resources and adjust queue parameters in Bridges.

The TrafficSpecification group consists of the following:

- a) Its required elements, listed in Table 46-8.

Table 46-8—TrafficSpecification elements

Name	Data type	Reference
Interval	rational	46.2.3.5.1
MaxFramesPerInterval	uint16	46.2.3.5.2
MaxFrameSize	uint16	46.2.3.5.3
TransmissionSelection	uint8	46.2.3.5.4

- b) An optional TSpecTimeAware group, whose elements are listed in Table 46-9.

Table 46-9—TSpecTimeAware elements

Name	Data type	Reference
EarliestTransmitOffset	uint32	46.2.3.5.5
LatestTransmitOffset	uint32	46.2.3.5.6
Jitter	uint32	46.2.3.5.7

The presence of TSpecTimeAware specifies that the Talker's traffic is synchronized to a known time on the network (e.g., using IEEE Std 802.1AS). The TSpecTimeAware group provides elements to specify the Talker's time-aware transmit to the network. The Talker and Listeners of a Stream are assumed to coordinate using user (application) mechanisms, such that each Listener is aware that its Talker transmits in a time-aware manner. If MaxFramesPerInterval is greater than one, the time-aware Talker shall transmit multiple frames as a burst within the Interval, with the minimum inter-frame gap allowed by the media.

NOTE—Although scheduled traffic (8.6.8.4) specifies a valid implementation of a time-aware Talker, the TSpecTimeAware group is intended to support alternate implementations of scheduling.

For informative examples using the TSpecTimeAware group of TrafficSpecification, refer to U.1.

46.2.3.5.1 Interval

Interval specifies the period of time in which the traffic specification cannot be exceeded. The traffic specification is specified by MaxFramesPerInterval and MaxFrameSize.

The Interval is a rational number of seconds, defined by an unsigned 32-bit integer numerator and an unsigned 32-bit integer denominator.

1 If the TSpecTimeAware group is not present, the Interval specifies a sliding window of time. The Talker's
2 transmission is not synchronized to a time on the network, and therefore the traffic specification cannot be
3 exceeded during any Interval in time.

4 If the TSpecTimeAware group is present, the Interval specifies a window of time that is aligned with the
5 time epoch that is synchronized on the network. For example, if IEEE Std 802.1AS is used with the PTP
6 timescale, the first Interval begins at 1 January 00:00:00 TAI. If CurrentTime represents the current time,
7 then the start of the next Interval (StartOfNextInterval) is:

8
$$\text{StartOfNextInterval} = N \times \text{Interval}$$

9 where N is the smallest integer for which the relation:

10
$$\text{StartOfNextInterval} \geq \text{CurrentTime}$$

11 would be TRUE.

12 **46.2.3.5.2 MaxFramesPerInterval**

13 MaxFramesPerInterval specifies the maximum number of frame transmissions that the Talker can initiate in
14 one Interval.

15 **46.2.3.5.3 MaxFrameSize**

16 MaxFrameSize specifies maximum frame size that the Talker will transmit, excluding any overhead for
17 media-specific framing (e.g., preamble, IEEE 802.3 header, Priority/VID tag, CRC, interframe gap). As the
18 Talker or Bridge determines the amount of bandwidth to reserve on the egress Port (interface), it will
19 calculate the media-specific framing overhead on that Port and add it to the number specified in the
20 MaxFrameSize element.

21 **46.2.3.5.4 TransmissionSelection**

22 TransmissionSelection specifies the algorithm that the Talker uses to transmit this Stream's traffic class. This
23 algorithm is often referred to as the shaper for the traffic class.

24 The value for this element uses Table 8-6 (in 8.6.8). If no algorithm is known, the value zero (strict priority)
25 can be used.

26 The Talker's shaping and scheduling of the Stream is considered to be on the user side of the user/network
27 boundary, and this specifies the Talker's behavior to the network. In the fully distributed model, this value
28 can be ignored by a Bridge, and it is not changed during propagation through the Bridge.

29 **46.2.3.5.5 EarliestTransmitOffset**

30 EarliestTransmitOffset specifies the earliest offset within each Interval at which the Talker is capable of
31 starting transmit of its frames. As part of the Status group, the network will return a specific
32 TimeAwareOffset to the Talker (within the earliest/latest range), which the Talker uses to schedule its
33 transmit.

34 EarliestTransmitOffset is specified as an integer number of nanoseconds.

35 The Talker's transmit offsets include EarliestTransmitOffset, LatestTransmitOffset, and the
36 TimeAwareOffset returned to the Talker in the Status group. Each of the Talker's offsets is specified at the
37 point when the message timestamp point of the first frame of the Stream in each Interval passes the reference
38 plane marking the boundary between the network media and PHY. The message timestamp point is specified
39 by IEEE Std 802.1AS for various media.

1 46.2.3.5.6 LatestTransmitOffset

2 LatestTransmitOffset specifies the latest offset within the Interval at which the Talker is capable of starting
3 transmit of its frames. As part of the Status group, the network will return a specific TimeAwareOffset to the
4 Talker (within the earliest/latest range), which the Talker uses to schedule its transmit.

5 LatestTransmitOffset is specified as an integer number of nanoseconds.

6 46.2.3.5.7 Jitter

7 Jitter specifies the maximum difference in time between the Talker's transmit offsets and the ideal
8 synchronized network time (e.g., IEEE 802.1AS time). Jitter is specified as an unsigned integer number of
9 nanoseconds.

10 The maximum difference means sooner or later than the ideal (e.g., transmit ± 500 ns relative to
11 IEEE 802.1AS time results in Jitter of 500).

12 The ideal synchronized network time refers to time at the source (e.g., IEEE 802.1AS grandmaster). The
13 Jitter does not include inaccuracies as time is propagated from the time source to the Talker because those
14 inaccuracies are assumed to be known by the network and time synchronization is a network technology.
15 The Jitter element is intended to specify inaccuracies in the Talker's implementation. For example, if the
16 Talker's IEEE 802.1AS time is ± 812 ns relative to the grandmaster and the Talker schedules using a 100 μ s
17 timer tick driven by IEEE 802.1AS time, Jitter is 50 000 (not 50 812).

18 The Talker's transmit offsets include EarliestTransmitOffset, LatestTransmitOffset, and the
19 TimeAwareOffset returned to the Talker in the Status group.

20 46.2.3.6 UserToNetworkRequirements

21 The UserToNetworkRequirements group specifies user requirements for the Stream, such as latency and
22 redundancy.

23 The network (e.g., CNC) will merge all UserToNetworkRequirements for a Stream to ensure that all
24 requirements are met.

25 The elements of the UserToNetworkRequirements group are listed in Table 46-10.

Table 46-10—UserToNetworkRequirements elements

Name	Data type	Reference
NumSeamlessTrees	uint8	46.2.3.6.1
MaxLatency	uint32	46.2.3.6.2

26 46.2.3.6.1 NumSeamlessTrees

27 NumSeamlessTrees specifies the number of trees that the network will configure to deliver seamless
28 redundancy for the Stream.

29 The value zero is interpreted as one (i.e., no seamless redundancy).

30 This requirement is provided from the Talker only. Listeners shall set this element to one.

1 From each Talker to a single Listener, the network configures a path that relays Stream data through Bridges.
2 If the Talker has more than one Listener, the network configures a tree of multiple paths.

3 NumSeamlessTrees specifies the number of maximally disjoint trees that the network shall configure from
4 the Talker to all Listeners. Each tree is disjoint from other trees, in that the network evaluates the physical
5 topology to avoid sharing the same Bridge and links in each tree's paths. This computation of disjoint trees
6 is maximal, in that shared Bridges and links are avoided to the maximum extent allowed by the physical
7 topology. For example, if a single link exists from a Bridge to a Listener and NumSeamlessTrees is three,
8 then all three trees will share that link to the Listener.

9 When NumSeamlessTrees is greater than one, the transfer of the Stream's data frames shall use a seamless
10 redundancy standard, such as IEEE Std 802.1CB. When a link shared by multiple trees diverges to multiple
11 disjoint links, the seamless redundancy standard replicates (i.e., forwards a distinct copy of each data frame
12 to the disjoint trees). When disjoint trees converge to a single link, the seamless redundancy standard
13 eliminates the duplicate copies of each data frame. Assuming that other sources of frame loss are mitigated
14 (e.g., congestion), failure of a link or Bridge in one disjoint tree does not result in frame loss as long as at
15 least one remaining disjoint tree is operational.

16 If the Talker sets this element to one, the network may make use of redundancy standards that are not
17 seamless (i.e., failure of link results in lost frames), such as MSTP and IS-IS.

18 If the Talker sets this element to greater than one and seamless redundancy is not possible in the current
19 network (no disjoint paths or no seamless redundancy standard in Bridges), the Status group returns a
20 nonzero FailureCode (46.2.5.1).

21 If the UserToNetworkRequirements group is not provided within the Talker or Listener group, the network
22 shall use the default value of one for this element.

23 **46.2.3.6.2 MaxLatency**

24 MaxLatency specifies the maximum latency from Talker to Listener(s) for a single frame of the Stream.

25 MaxLatency is specified as an integer number of nanoseconds.

26 Latency shall use the definition of 3.128, with additional context as follows: The "known reference point in
27 the frame" is the message timestamp point specified in IEEE Std 802.1AS for various media (i.e., start of the
28 frame). The "first point" is in the Talker at the reference plane marking the boundary between the network
29 media and PHY (see IEEE Std 802.1AS). The "second point" is in the Listener at the reference plane
30 marking the boundary between the network media and PHY.

31 When this requirement is specified by the Talker, it must be satisfied for all Listeners.

32 When this requirement is specified by the Listener, it must be satisfied for this Listener only.

33 If the UserToNetworkRequirements group is not provided within the Talker or Listener group, the network
34 shall use the default value zero for this element.

35 The special value of zero represents usage of the initial value of Status.AccumulatedLatency as the
36 maximum latency requirement. This effectively locks down the initial latency that the network calculates
37 after successful configuration of the Stream, such that any subsequent increase in latency beyond that value
38 causes the Stream to fail.

1 The assumption for when the “first point” occurs in the Talker depends on the presence of the
2 TSpecTimeAware group in the Talker’s TrafficSpecification.

3 a) When TSpecTimeAware is not present:

4 1) The Talker is assumed to transmit at an arbitrary time (not scheduled).

5 b) When TSpecTimeAware is present:

6 1) The ‘first point’ is assumed to occur at the start of the Interval as if the Talker’s offsets
7 (EarliestTransmitOffset and LatestTransmitOffset of 46.2.3.5) are both zero. The Talker’s
8 offsets are not typically zero, but use of the start of Interval for purposes of MaxLatency allows
9 the Listener(s) to schedule their application independently from the Talker’s offset
10 configuration.

11 2) The Listener determines MaxLatency based on its scheduling of a read function in the
12 application. Nevertheless, the time from frame reception (i.e., “second point”) to execution of
13 the read function is in the user’s scope and therefore not included in MaxLatency.

14 3) MaxLatency can be set to a value greater than the Talker’s Interval in order to specify a longer
15 latency requirement. For example, if the Talker’s Interval is 500 μ s and MaxLatency is 700 μ s,
16 the Listener receives the frame no later than 200 μ s into the interval that follows the Talker’s
17 Interval.

18 46.2.3.7 InterfaceCapabilities

19 The InterfaceCapabilities group specifies the network capabilities of all interfaces (Ports) contained in the
20 EndStationInterfaces group.

21 The network may provide configuration of these capabilities in the InterfaceConfiguration group of the
22 Status group.

23 NOTE—If an end station contains multiple interfaces with different network capabilities, each interface should be
24 specified in a distinct Talker or Listener group (i.e., one interface in EndStationInterfaces). Use of multiple interface in
25 EndStationInterfaces is intended for network capabilities that span multiple interfaces (e.g., seamless redundancy).

26 The elements of the InterfaceCapabilities group are listed in Table 46-11.

Table 46-11—InterfaceCapabilities elements

Name	Data type	Reference
VlanTagCapable	Boolean	46.2.3.7.1
CB-StreamIdenTypeList	sequence of uint32	46.2.3.7.2
CB-SequenceTypeList	sequence of uint32	46.2.3.7.3

27 46.2.3.7.1 VlanTagCapable

28 When VlanTagCapable is true, the interface supports the ability to tag/untag frames using a Customer
29 VLAN Tag (C-TAG of Clause 9) provided by the network.

30 For a Talker, the network’s tag replaces the tag specified by the DataFrameSpecification. If the
31 DataFrameSpecification is untagged (no IEEE802-VlanTag group), the network’s tag is inserted in the frame
32 as it passes through the interface.

1 For a Listener, the user's tag from the DataFrameSpecification replaces the network's tag as the frame
2 passes through the interface. If the DataFrameSpecification is untagged (no IEEE802-VlanTag group), the
3 network's tag is removed from the frame as it passes through the interface.

4 If the end station supports more than one interface (i.e., more than one entry in EndStationInterfaces),
5 VlanTagCapable of true means that a distinct VLAN tag can be applied to each interface. The list of VLAN
6 tags (one for each interface) can be provided by the network in InterfaceConfiguration.InterfaceList
7 (IEEE802-VlanTag choice).

8 When VlanTagCapable is false, the interface does not support the capability to tag/untag frames using a
9 Customer VLAN Tag (C-TAG of Clause 9) provided by the network.

10 If the InterfaceCapabilities group is not provided within the Talker or Listener group, the network shall use
11 the default value of false for this element.

12 **46.2.3.7.2 CB-StreamIdenTypeList**

13 CB-StreamIdenTypeList provides a list of the supported Stream Identification types as specified in IEEE Std
14 802.1CB.

15 Each Stream Identification type is provided as a 32-bit unsigned integer. The upper 3 octets contain the OUI/
16 CID, and the lowest octet contains the type number.

17 NOTE—If the Talker/Listener end system supports IEEE Std 802.1CB, Null Stream identification is required, and that
18 Stream Identification type is included in this list. If the Talker/Listener end system does not support IEEE Std 802.1CB,
19 this list is empty.

20 If the end station supports more than one interface (i.e., more than one InterfaceID in EndStationInterfaces),
21 an empty CB-StreamIdenTypeList means that the end station is capable of transferring the Stream on any
22 one of its interfaces (not all). When this is specified, the network shall decide which interface is best
23 used for TSN purposes and communicate that decision by returning a single interface in
24 InterfaceConfiguration.InterfaceList. The Talker/Listener uses this interface alone for the Stream.

25 If the InterfaceCapabilities group is not provided within the Talker or Listener group, the network shall use
26 an empty list as the default value for this element.

27 **46.2.3.7.3 CB-SequenceTypeList**

28 CB-SequenceTypeList provides a list of the supported Sequence Encode/Decode types as specified in
29 IEEE Std 802.1CB.

30 Each sequence type is provided as a 32-bit unsigned integer. The upper 3 octets contain the OUI/CID, and
31 the lowest octet contains the type number.

32 If the InterfaceCapabilities group is not provided within the Talker or Listener group, the network shall use
33 an empty list as the default value for this element.

1 46.2.3.8 StreamStatus

2 StreamStatus is an enumeration specified in Table 46-12 that indicates the status of a Stream. The status is
3 maintained by the CNC and is used to determine which Streams are computed by calling the RPC
4 ComputePlannedAndModifiedStreams (46.2.7.2).

Table 46-12—StreamStatus enumeration

Name	Value	Description
Planned	0	Stream has been requested but has not yet been configured.
Configured	1	Stream has been computed and configured.
Modified	2	Stream has been configured but Stream parameters have been modified after configuration..

5 46.2.4 Listener

6 The Listener group specifies the following:

- 7 — Listener's requirements from the network
- 8 — TSN capabilities of the Listener's interface(s)

9 In the fully distributed model and the centralized network/distributed user model, this group originates from
10 the Listener's end station. In the fully centralized model, this group originates from the CUC.

11 The Listener group contains the following groups:

- 12 — StreamID (46.2.3.1)
- 13 — EndStationInterfaces (46.2.3.3)
- 14 — UserToNetworkRequirements (46.2.3.6)
- 15 — InterfaceCapabilities (46.2.3.7)

16 For the join operation, the UserToNetworkRequirements and InterfaceCapabilities groups may be included
17 within the Listener group. If a group is omitted, the network uses the default values specified for elements of
18 that group.

19 For the join and leave operation, the StreamID group shall be included in the TSN user/network protocol in
20 a location associated with the Stream (e.g., Listener group).

21 For the join and leave operation, EndStationInterfaces shall be included within the Listener group for the
22 fully centralized model (46.1.3.3). For the join and leave operation, EndStationInterfaces should be included
23 within the Listener group for the fully distributed model (46.1.3.1) or centralized network/distributed user
24 model (46.1.3.2).

25 46.2.5 Status

26 The Status group provides the status of a Stream's configuration from the network to each user (Talker or
27 Listener).

28 In the fully distributed model and the centralized network/distributed user model, this group is delivered to
29 each Talker end station and Listener end station of the Stream. In the fully centralized mode, this group is
30 delivered to the CUC.

1 The Status group contains the following groups:

- 2 — StreamID (46.2.3.1)
- 3 — StatusInfo (46.2.5.1)
- 4 — AccumulatedLatency (46.2.5.2)
- 5 — InterfaceConfiguration (46.2.5.3)
- 6 — FailedInterfaces (46.2.5.4)

7 The AccumulatedLatency and InterfaceConfiguration groups are distinct for each Talker or Listener, and
8 therefore Status is distinct.

9 The protocol for TSN configuration shall specify that one Status group is received in response to the
10 transmit of a Talker group or Listener group. This provides at least one response to each join or leave
11 request.

12 StreamID (46.2.3.1) identifies the Stream for which status is provided.

13 StatusInfo (46.2.5.1) provides the status of the Stream's configuration in the network.

14 For the join and leave operation, the StreamID and StatusInfo groups shall be included in the TSN user/
15 network protocol in a location associated with the Stream (e.g., Status group).

16 For the join operation, the AccumulatedLatency group shall be included within the Status group that is sent
17 to a Listener. For the join operation, the AccumulatedLatency group may be included within the Status
18 group that is sent to a Talker.

19 For the join operation, the InterfaceConfiguration group may be included within the Status group.
20 InterfaceConfiguration provides configuration of the interfaces in a specific end station.

21 If a group is omitted, the network uses the default values specified for elements of that group.

22 When a failure occurs in network configuration, FailedInterfaces identifies one or more interfaces (Ports) on
23 which the failure occurred. If the FailureCode of the StatusInfo group (46.2.5.1) is zero, the FailedInterfaces
24 group shall not be included within the Status group. If the FailureCode is nonzero, the FailedInterfaces
25 group may be included within the Status group.

26 When a Talker and at least one Listener have joined the stream and network resources are configured for the
27 Stream (success or failure), the network shall transmit Status to the Talker and Listener(s). After the initial
28 Status, upon any change in the status of the network configuration (e.g., new Listeners, change in
29 FailureCode), the network shall transmit Status to the Talker and Listener(s). When the Stream does not have
30 a Talker or a Listener, the network configuration is removed, and the network shall transmit a final Status to
31 the remaining Talker and/or Listener(s). Beyond these requirements, additional transmit of Status is
32 specified by the protocol that carries the Status group.

33 **46.2.5.1 StatusInfo**

34 The StatusInfo group provides information regarding the status of a Stream's configuration in the network.

¹ The elements of the StatusInfo group are listed in Table 46-13.

Table 46-13—StatusInfo elements

Name	Data type	Reference
TalkerStatus	enumeration	46.2.5.1.1
ListenerStatus	enumeration	46.2.5.1.2
FailureCode	uint8	46.2.5.1.3

² **46.2.5.1.1 TalkerStatus**

³ TalkerStatus provides the status of the network configuration for the Stream’s Talker.

⁴ TalkerStatus uses the enumeration specified in Table 46-14.

Table 46-14—TalkerStatus enumeration

Name	Value	Description
None	0	No Talker detected.
Ready	1	Talker ready (configured).
Failed	2	Talker failed.

⁵ **46.2.5.1.2 ListenerStatus**

⁶ ListenerStatus provides the status of the network configuration for the Stream’s Listener(s). If there is more
⁷ than one Listener for the Stream, ListenerStatus provides the status of all Listeners.

⁸ ListenerStatus uses the enumeration specified in Table 46-15.

Table 46-15—ListenerStatus enumeration

Name	Value	Description
None	0	No Listener detected.
Ready	1	All Listeners ready (configured).
PartialFailed	2	One or more Listeners ready, and one or more Listeners failed. If Talker is ready, Stream can be used.
Failed	3	All Listeners failed.

1 **46.2.5.1.3 FailureCode**

2 If the Stream encounters a failure (TalkerStatus is Failed, or ListenerStatus is Failed, or ListenerStatus is
3 PartialFailed), FailureCode provides a nonzero code that specifies the problem. If the Stream is configured
4 without a failure, FailureCode is zero.

5 Table 46-16 specifies the nonzero values for FailureCode.

6 **46.2.5.2 AccumulatedLatency**

7 The AccumulatedLatency group provides the worst-case latency that a single frame of the Stream can encounter along
8 its current path(s).

9 The elements of the AccumulatedLatency group are listed in Table 46-17.

Table 46-16—TSN Failure Codes

Failure Code	Description of cause
1	Insufficient bandwidth
2	Insufficient Bridge resources
3	Insufficient bandwidth for traffic class
4	StreamID in use by another Talker
5	Stream destination_address already in use

Table 46-16—TSN Failure Codes (*continued*)

Failure Code	Description of cause
6	Stream preempted by higher rank
7	Reported latency has changed
8	Egress Port is not AVB capable ^a
9	Use a different destination_address (i.e., MAC DA hash table full)
10	Out of MSRP resources
11	Out of MMRP resources
12	Cannot store destination_address (i.e., Bridge is out of MAC DA resources)
13	Requested priority is not an SR Class (3.271) priority
14	MaxFrameSize [item a) in 35.2.2.8.4] is too large for media
15	msrpMaxFanInPorts [item f) in 35.2.1.4] limit has been reached
16	Changes in FirstValue, other than AccumulatedLatency, for a registered StreamID
17	VLAN is blocked or filtered on this egress Port ^b
18	VLAN tagging is disabled on this egress Port (untagged set)
19	SR class priority mismatch
20	Enhanced feature cannot be propagated to original Port
21	MaxLatency exceeded
22	Nearest Bridge cannot provide network identification for stream transformation
23	Stream transformation not supported
24	Stream identification type not supported for stream transformation
25	Enhanced feature cannot be supported without a CNC

^a A device can use the IEEE Std 802.1AS asCapable variable to help determine if its neighboring device is AVB capable. If the asCapable variable is FALSE for a particular Port, then the neighboring device is not a time-aware system and therefore not AVB capable.

^b This Failure Code is never declared in a Talker Failed message since Talker attributes are not propagated on egress Ports that have the associated VLAN blocked in the VLAN spanning tree (7.3) or filtered in VLAN Registration Entries (8.8). The Bridge can still be queried by other means to learn why the Talker attribute was not declared.

1 46.2.5.2.1 AccumulatedLatency

Table 46-17—AccumulatedLatency elements

Name	Data type	Reference
AccumulatedLatency	uint32	46.2.5.2.1

2 The AccumulatedLatency element provides the worst-case maximum latency that a single frame of the
3 Stream can encounter along its current path(s).

4 AccumulatedLatency is distinct for each Talker or Listener of the Stream.

5 When provided to a Listener, AccumulatedLatency is the worst-case maximum latency for that Listener
6 only.

7 When provided to a Talker, AccumulatedLatency is the worst-case maximum latency for all Listeners (worst
8 path).

9 AccumulatedLatency is specified as an integer number of nanoseconds.

10 AccumulatedLatency uses the same definition for latency as UserToNetworkRequirements.MaxLatency
11 (46.2.3.6).

12 For a successful StatusInfo (46.2.5.1), the network returns a value less than or equal to
13 UserToNetworkRequirements.MaxLatency.

14 If the TSpecTimeAware group is present in the TrafficSpecification group (46.2.3.5) of the Talker, the value
15 is expressed as nanoseconds after the start of the Talker's Interval.

16 If the TSpecTimeAware group is not present in the TrafficSpecification group of the Talker, the value is
17 expressed as nanoseconds after the Talker's transmit of any frame in the Stream, at any arbitrary time.

18 If UserToNetworkRequirements.NumSeamlessTrees is one, AccumulatedLatency shall provide the worst-
19 case maximum latency for the current path from Talker to each Listener. If the path is changed (e.g., by a
20 spanning tree protocol), AccumulatedLatency changes accordingly.

21 If UserToNetworkRequirements.NumSeamlessTrees is greater than one, AccumulatedLatency shall provide
22 the worst-case maximum latency for all paths in use from the Talker to each Listener.

23 46.2.5.3 InterfaceConfiguration

24 The InterfaceConfiguration group provides configuration of interfaces in the Talker/Listener. This
25 configuration assists the network in meeting the Stream's requirements. The InterfaceConfiguration meets
26 the capabilities of the interface as provided in the InterfaceCapabilities group.

27 The InterfaceConfiguration group is distinct for each Talker or Listener of the Stream.

28 A distinct configuration is provided for each interface in the Talker/Listener (even if multiple interfaces use
29 the same configuration). Each interface configuration consists of a single InterfaceID (46.2.3.3) group,
30 followed by a list of configuration values for that interface.

1 The values in InterfaceID shall match one of the InterfaceID entries in the Talker/Listener
2 EndStationInterfaces group.

3 The list of configuration values uses zero or more of the following groups:

- 4 — IEEE802-MacAddresses (46.2.5.3.1)
- 5 — IEEE802-VlanTag (46.2.5.3.2)
- 6 — IPv4-tuple (46.2.5.3.3)
- 7 — IPv6-tuple (46.2.5.3.4)
- 8 — TimeAwareOffset (46.2.5.3.5)

9 If the InterfaceConfiguration group is not provided within the Status group, the network shall assume zero
10 configuration values as the default (no interface configuration).

11 **46.2.5.3.1 IEEE802-MacAddresses**

12 The IEEE802-MacAddresses group provides the source and destination MAC addresses that apply to the
13 network side of the user/network boundary.

14 NOTE 1—On the user side, the MAC addresses are in DataFrameSpecification.IEEE802-MacAddresses.

15 NOTE 2—The source MAC address of the network is typically the same as the user. The destination MAC address can
16 be different. For example, the user can use an individual address, but the network can use a group (multicast) address.

17 This group uses the specifications from DataFrameSpecification.IEEE802-MacAddresses (46.2.3.4.1).

18 This configuration value is not provided unless IEEE Std 802.1CB is supported and a value for Active
19 Destination MAC and VLAN Stream identification is provided in CB-StreamIdentTypeList of
20 InterfaceCapabilities.

21 **46.2.5.3.2 IEEE802-VlanTag**

22 The IEEE802-VlanTag group provides the Customer VLAN Tag (C-TAG of Clause 9) that applies to the
23 network side of the user/network boundary.

24 NOTE—On the user side, the VLAN tag is in DataFrameSpecification.IEEE802-VlanTag (including untagged if that
25 field is not provided).

26 This group uses the specifications from DataFrameSpecification.IEEE802-VlanTag (46.2.3.4.2).

27 If the user provides a VLAN ID in the IEEE802-VlanTag of DataFrameSpecification, the Stream's data
28 frames are assumed to be limited to the active topology for that VLAN ID. Therefore, if the network uses a
29 different VLAN ID in this configuration value, the network shall ensure that the replacement VLAN ID is
30 limited to the equivalent active topology.

31 This configuration value is not provided unless VlanTagCapable of InterfaceCapabilities is true.

32 **46.2.5.3.3 IPv4-tuple**

33 The IPv4-tuple group provides the IPv4 identification that applies to the network side of the user/network
34 boundary.

35 This group uses the specifications from DataFrameSpecification.IPv4-tuple (46.2.3.4.3).

36 This configuration value is not provided unless IEEE Std 802.1CB is supported and a value for IP Stream
37 identification is provided in CB-StreamIdentTypeList of InterfaceCapabilities.

1 46.2.5.3.4 IPv6-tuple

2 The IPv6-tuple group provides the IPv6 identification that applies to the network side of the user/network
3 boundary.

4 This group uses the specifications from DataFrameSpecification.IPv6-tuple (46.2.3.4.4).

5 This configuration value is not provided unless IEEE Std 802.1CB is supported and a value for IP Stream
6 identification is provided in CB-StreamIdenTypeList of InterfaceCapabilities.

7 46.2.5.3.5 TimeAwareOffset

8 If the TSpecTimeAware group is present in the TrafficSpecification group (46.2.3.5) of the Talker, this
9 configuration value shall be provided by the network to the Talker.

10 If the TSpecTimeAware group is not present in the TrafficSpecification group (46.2.3.5) of the Talker, this
11 configuration value shall not be provided by the network.

12 This configuration value shall not be provided to Listeners as it is not applicable.

13 TimeAwareOffset specifies the offset that the Talker shall use for transmit. The network returns a value
14 between EarliestTransmitOffset and LatestTransmitOffset of the Talker's TrafficSpecification. The value is
15 expressed as nanoseconds after the start of the Talker's Interval. The data type is uint32.

16 46.2.5.4 FailedInterfaces

17 When a failure occurs in network configuration (i.e., nonzero FailureCode in StatusInfo group),
18 FailedInterfaces provides a list of one or more physical interfaces (distinct points of attachment) in the failed
19 end station or Bridge. Each identifier is sufficient to locate the interfaces in the physical topology.

20 The FailedInterfaces group is optional.

21 FailedInterfaces consists of a sequence of zero or more entries, each entry using the InterfaceID group
22 specified in 46.2.3.3.

23 46.2.6 Protocol operations

24 The TSN user/network configuration makes use of protocol operations to request specific actions and to
25 receive notifications. The following operations are supported:

- 26 — **Remote Procedure Calls (RPC):** this protocol operation allows requesting an action for the
27 complete YANG data model.
- 28 — **Actions:** this protocol operation allows requesting an action on a specific part of the YANG data
29 model.
- 30 — **Notifications:** this protocol operation provides information, e.g., it allows the CNC to inform the
31 CUC that computing the configuration has finished.

32 46.2.7 Remote Procedure Calls

33 The TSN user/network configuration provides the following RPCs:

- 34 — ComputeStreams (46.2.7.1)
- 35 — ComputePlannedAndModifiedStreams (46.2.7.2)
- 36 — ComputeAllStreams (46.2.7.3)
- 37 — RequestDomainId (46.2.7.4)
- 38 — RequestFreeStreamId (46.2.7.5)

1 **46.2.7.1 ComputeStreams**

2 This RPC starts the computation of path and resource allocation for one or more Streams. The Streams that
3 are to be included in the computation are specified by providing their associated DomainID (46.2.2.1),
4 CucID (46.2.2.2), and StreamID (46.2.3.1). This RPC can be applied to compute new Streams as well as
5 recompute already configured Streams.

6 The RPC returns information that indicates only if the Stream computation has been started successfully or
7 not. It does not return information on whether the Stream configuration itself has been successful or not,
8 because computation and configuration can take an arbitrary amount of time. The notifications
9 ComputeStreamsCompleted (46.2.9.1) and ConfigureStreamsCompleted (46.2.9.2) are available to the CNC
10 to return information on success or failure of the Stream computation and configuration, after the actions
11 have finished.

12 **46.2.7.2 ComputePlannedAndModifiedStreams**

13 This RPC starts the computation of path and resource allocation for Streams that have not been configured
14 or that have been configured and have been modified since configuration. The Streams that are to be
15 included in the computation are specified by providing their associated DomainID (46.2.2.1) and CucID
16 (46.2.2.2). The object StreamStatus (46.2.3.8) is used to determine if a Stream is included in the computation
17 initiated by this RPC.

18 The RPC returns information that indicates only if the Stream computation has been started successfully or
19 not. It does not return information on whether the Stream configuration itself has been successful or not,
20 because computation and configuration can take an arbitrary amount of time. The notifications
21 ComputeStreamsCompleted (46.2.9.1) and ConfigureStreamsCompleted (46.2.9.2) are available to the CNC
22 to return information on success or failure of the Stream computation and configuration, after the actions
23 have finished.

24 **46.2.7.3 ComputeAllStreams**

25 This RPC starts the computation of path and resource allocation for all Streams in a Configuration Domain
26 and that are belonging to a specified CUC. The Streams that are to be included in the computation are
27 specified by providing their associated DomainID (46.2.2.1) and CucID (46.2.2.2).

28 The RPC returns information that indicates only if the Stream computation has been started successfully or
29 not. It does not return information on whether the Stream configuration itself has been successful or not,
30 because computation and configuration can take an arbitrary amount of time. The notifications
31 ComputeStreamsCompleted (46.2.9.1) and ConfigureStreamsCompleted (46.2.9.2) are available to the CNC
32 to return information on success or failure of the Stream computation and configuration, after the actions
33 have finished.

34 **46.2.7.4 RequestDomainId**

35 This RPC allows a CUC to request the DomainID (46.2.2.1) of the Configuration Domain that the CUC
36 belongs to from the CNC. If a CUC already knows the Configuration Domain it belongs to, this RPC can be
37 used to verify that the information the CUC has is correct.

38 **46.2.7.5 RequestFreeStreamId**

39 This RPC allows a CUC to request a free StreamID (46.2.3.1) from a CNC. Requesting a free StreamID
40 allows a CUC to provide an unused, i.e., unique, StreamID for a Stream when requesting that Stream from
41 the CNC.

1 **46.2.8 Actions**

2 The TSN user/network configuration provides the following actions:

- 3 — RemoveStreams (46.2.8.1)

4 **46.2.8.1 RemoveStreams**

5 This action starts the removal of one or more Streams. Each Stream that is to be removed is specified by
6 providing its associated StreamID (46.2.3.1). This action returns information that indicates only if the
7 Stream removal has been started successfully or not. It does not return information on whether the Stream
8 removal itself has been successful or not, because execution can take an arbitrary amount of time. When a
9 Stream is successfully removed, the StreamID associated with that Stream can be used as a free StreamID by
10 the RPC RequestFreeStreamId (46.2.7.5) again.

11 The notification RemoveStreamsCompleted (46.2.9.3) is available to the CNC to return information on
12 success or failure of the Stream removal.

13 **46.2.9 Notifications**

14 The TSN user/network configuration provides the following notifications:

- 15 — ComputeStreamsCompleted (46.2.9.1)
16 — ConfigureStreamsCompleted (46.2.9.2)
17 — RemoveStreamsCompleted (46.2.9.3)

18 **46.2.9.1 ComputeStreamsCompleted**

19 This notification is used by the CNC to inform a CUC that has requested the computation of one or more
20 Streams, that the computation for these Streams has finished. If the computation of these Streams impacts
21 other Streams that are already configured in the network, it can also be used to notify the CUCs that
22 originally requested the impacted Streams about the modification.

23 NOTE—ComputeStreamsCompleted returns only information on the computation of Streams. This does not provide any
24 information on whether the configuration of these Streams has been performed successfully or not.

25 It returns a list of Domains, identified by their DomainIDs (46.2.2.1), CUCs in that domain, identified by
26 their CucIDs (46.2.2.2) and Streams associated with a CUC, identified by their StreamIDs (46.2.3.1). For
27 each Stream it also returns either 0, if the Stream computation was successful, or a FailureCode (46.2.5.1.3),
28 if it was not.

29 **46.2.9.2 ConfigureStreamsCompleted**

30 This notification is used by the CNC to inform a CUC that has requested the computation of one or more
31 Streams, that the computation and configuration for these Streams has finished. If the computation or
32 configuration of these Streams impacts other Streams that are already configured in the network, it can also
33 be used to notify the CUCs that originally requested the impacted Streams about the modification.

34 It returns a list of Domains, identified by their DomainIDs (46.2.2.1), CUCs in that domain, identified by
35 their CucIDs (46.2.2.2) and Streams associated with a CUC, identified by their StreamIDs (46.2.3.1). For
36 each Stream it also returns either 0, if the Stream computation and configuration was successful, or a
37 FailureCode (46.2.5.1.3), if it was not.

1 **46.2.9.3 RemoveStreamsCompleted**

2 This notification is used by the CNC to inform a CUC that has requested the removal of one or more
3 Streams, that the removal of these Streams has finished. It returns a list of Domains, identified by their
4 DomainIDs (46.2.2.1), CUCs in that domain, identified by their CucIDs (46.2.2.2) and Streams associated
5 with a CUC, identified by their StreamIDs (46.2.3.1). For each Stream it also returns either 0, if the Stream
6 removal was successful, or 1, if it was not.

7 **46.3 YANG for TSN user/network configuration**

8 In order to support the use of YANG-based protocols for the fully centralized model (46.1.3.3), 48.6.3, and
9 48.6.23 specify YANG modules.

10 The YANG module of 48.6.3 provides YANG text for each group of elements in 46.2. Each element is
11 specified using a YANG `leaf`. Each group is specified as a YANG `typedef` or `grouping`. The YANG
12 module for user/network configuration (48.6.23) imports the YANG module of 48.6.3 and uses the
13 `typedef` and `grouping` nodes in order to specify the schema tree used for communication between CUC
14 and CNC.

15 YANG identifiers use a naming convention of hyphens between lowercase names (e.g., “mac-address”).
16 Identifiers for elements and groups in 46.2 use a naming convention of camel case (e.g., “MacAddress”).
17 The specifications for an identifier in 48.6.3 and 48.6.23 apply to the corresponding identifier in 46.2
18 regardless of differences in naming convention (e.g., requirements for “MacAddress” in 46.2 apply to “mac-
19 address” in 48.6.3).

20 In the YANG module definitions of 48.6.3 and 48.6.23, if any discrepancy between the “description” text
21 and the corresponding specifications in 46.2 occurs, the specifications in 46.2 take precedence.

22

1 47. Asynchronous Traffic Shaping (ATS) in end stations

2 47.1 Talker transmission behavior

3 The operation of ATS traffic classes in Bridges is specified as a combination of per-stream classification and
4 metering for ATS (8.6.5.2.2) and the ATS transmission selection algorithm (8.6.8.5). In order for end station
5 originated data streams to make use of ATS traffic classes in Bridges, it is required that Talkers emit data
6 streams in a compliant manner, and that the parameters of ATS scheduler state machines and ATS scheduler
7 groups in Bridges are consistently set.

8 47.1.1 ATS traffic class model in Talkers

9 End stations that are Talkers have to emit streams consistent with the operation of the ATS scheduler state
10 machines in Bridges (8.6.11). This subclause specifies a model of ATS traffic classes that satisfies this
11 purpose. End stations shall exhibit the transmission behavior of this model at the egress ports.

12 In this model, Talker ports are composed by ATS scheduler instances, which assign eligibility times to the
13 frames of all emitted streams at the egress port, followed by a queue per traffic class, for which the ATS
14 transmission selection algorithm is executed. The relationship between ATS scheduler instances and streams
15 is one-to-one, and each scheduler state machine is associated with a dedicated scheduler group. This
16 effectively eliminates the association of multiple streams to an ATS scheduler instance, as well as the
17 coupling of multiple ATS scheduler instances by ATS scheduler groups.

18 47.1.2 Simplified ProcessFrame(frame) procedure

19 Due to the one-to-one relationship between streams and ATS scheduler instances, as well as ATS scheduler
20 instances and scheduler groups, the ProcessFrame(frame) procedure of ATS scheduler state machines, as
21 specified in 8.6.11.3, can be simplified. The simplified procedure is described by the following pseudo-code.

```
22 ProcessFrame(frame) {
    lengthRecoveryDuration    = length(frame) /
        CommittedInformationRate;
    emptyToFullDuration       = CommittedBurstSize /
        CommittedInformationRate;
    schedulerEligibilityTime  = BucketEmptyTime +
        lengthRecoveryDuration;
    bucketFullTime           = BucketEmptyTime +
        emptyToFullDuration;
    eligibilityTime           = max(arrivalTime(frame),
        schedulerEligibilityTime);

    BucketEmptyTime          = (eligibilityTime < bucketFullTime) ?
        schedulerEligibilityTime :
        schedulerEligibilityTime + eligibilityTime - bucketFullTime;
    AssignAndProceed(frame, eligibilityTime);
}
```

39 47.1.3 System clock functions and processing delays

40 This model has a single implementation specific local system clock function, which is used as:

- 41 a) ATS scheduler clock (8.6.11.1) for all ATS scheduler instances and
- 42 b) Transmission selection clock (8.6.8.5) of all ATS traffic classes.

43 The processing delay from the ATS scheduler instances to the transmission selection is zero in this model
44 (8.6.11.3.2):

$$0 = \text{ProcessingDelayMin} = \text{ProcessingDelayMax}$$

47.2 Scheduler parameter consistency

The ATS scheduler parameters CommittedBurstSize (8.6.11.3.5) and CommittedInformationRate (8.6.11.3.6) of Talker streams have to be consistently set to the respective parameters in the Bridge component connected to the Talker. In the Bridge component, a single ATS scheduler instance can be associated with multiple streams of this Talker.

For n streams associated with a single ATS scheduler instance in Bridge component, Equation (47-1) and Equation (47-2) can be used for scheduler parameter consistency.

$$CBS_{Bridge} \geq \sum_{i=1}^n CBS_i \quad (47-1)$$

$$CIR_{Bridge} \geq \left(\frac{1 + 10^{-6} ClkDev_{Talker}}{1 - 10^{-6} ClkDev_{Bridge}} \right) \sum_{i=1}^n CIR_i \quad (47-2)$$

where

- CIR_{Bridge} is the CommittedInformationRate parameter of the scheduler instance in the Bridge
- CIR_i is the CommittedInformationRate parameter of the i th stream in the Talker
- CBS_{Bridge} is the CommittedBurstSize parameter of the scheduler instance in the Bridge
- CBS_i is the CommittedBurstSize parameter of the i th stream in the Talker
- $ClkDev_{Bridge}$ is the maximum deviation of the associated ATS scheduler clock (8.6.11.1) in the Bridge from its nominal frequency, in ppm
- $ClkDev_{Talker}$ is the maximum deviation of the transmission selection clock (8.6.8.5 and 47.1.3) in the Talker from its nominal frequency, in ppm

1 48. YANG Data Models

2 This clause specifies YANG data models that provide control and status monitoring of systems and system
3 components that implement functionality specified in this standard. These data models are based on the set
4 of managed objects and their functionality specified in Clause 12.

5 This clause:

- 6 a) Introduces the YANG framework that governs the naming and hierarchy of configuration and
7 operational data structures in the data models, and the modeling of network interfaces (48.1)
- 8 b) Describes each of the data models and its relationship to the operational processes and managed
9 objects specified in the other clauses of this standard, and provides a UML representation of each
10 data model (48.2)
- 11 c) Describes the structure of the data models, each of which comprises or makes use of one or more
12 YANG modules (48.3)
- 13 d) Reviews security considerations applicable to each of the data models, with specific reference to
14 data nodes in the YANG modules that compose the model (48.4)
- 15 e) Includes a schema tree for each of the YANG modules (48.5)
- 16 f) Includes each of the YANG modules (48.6)

17 NOTE 1—The MIB modules specified in Clause 17 were also derived from Clause 12. Consequently, the capabilities
18 and structure of the YANG data models align closely with the MIBs. However the YANG data model has not been
19 derived from the MIB, and there has been no attempt to include data or modeling constructs that might appear in the
20 MIB but not in the information model.

21 NOTE 2—OMG UML 2.5 [B60] conventions together with C++ language constructs are used in this clause as a
22 representation to convey model structure and relationships.

23 The YANG data models specified in this clause include the following:

- 24 — A VLAN Bridge components data model (48.2.1) that allows control and status monitoring of one or
25 more C-VLAN or S-VLAN Bridge components (8.2) that compose all or part of a system's
26 functionality, and the Bridge Port interfaces that support those components.
- 27 — A Two-Port MAC Relay data model (48.2.2) that both subsets and augments the VLAN Bridge
28 components model to model a VLAN-unaware TPMR (3.292)
- 29 — A Customer VLAN Bridge model (48.2.3) that comprises a single VLAN Bridge component from
30 the VLAN Bridge components model.
- 31 — A Provider Bridges model that uses one or multiple components from the VLAN Bridge
32 components model to compose an S-VLAN component Provider Bridge or a Provider Edge Bridge.
- 33 — Connectivity Fault Management (CFM) models (48.2.3) for use with the VLAN Bridge components
34 and related models in systems that provide CFM functionality.
- 35 — A Stream filters and stream gates model (48.2.6) that augments the VLAN Bridge components
36 model.
- 37 — An Asynchronous Traffic Shaping (ATS) model that augments the VLAN Bridge components
38 model and the Stream filters and stream gates model.

39 48.1 YANG Framework

40 This clause has been developed according to the YANG guidelines published in IETF RFC 6087 [B38] as
41 applicable to IEEE standards.

42 The YANG framework applies hierarchy in the following areas:

- 43 1) The uniform resource name (URN), as specified in IEEE Std 802d-2017. The structure of the
44 URN is such that ieee is the root (i.e., name-space identifier), followed by the standard, then the
45 working group developing the standard.

- 1 2) The YANG objects form a hierarchy of configuration and operational data structures that define
- 2 the YANG model. These hierarchical relationships are described in 48.3.
- 3 The general YANG framework hierarchy is illustrated in Figure 48-1.

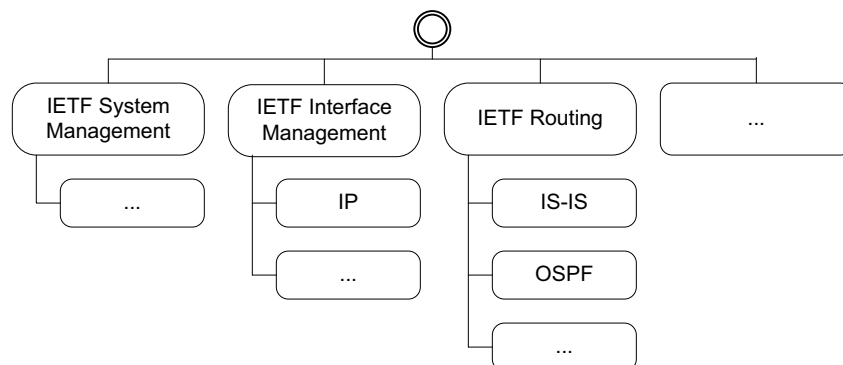


Figure 48-1—General YANG hierarchy

- 4 The YANG hierarchical structure that incorporates the IEEE 802.1Q YANG models supported by this
- 5 standard is represented in Figure 48-2.

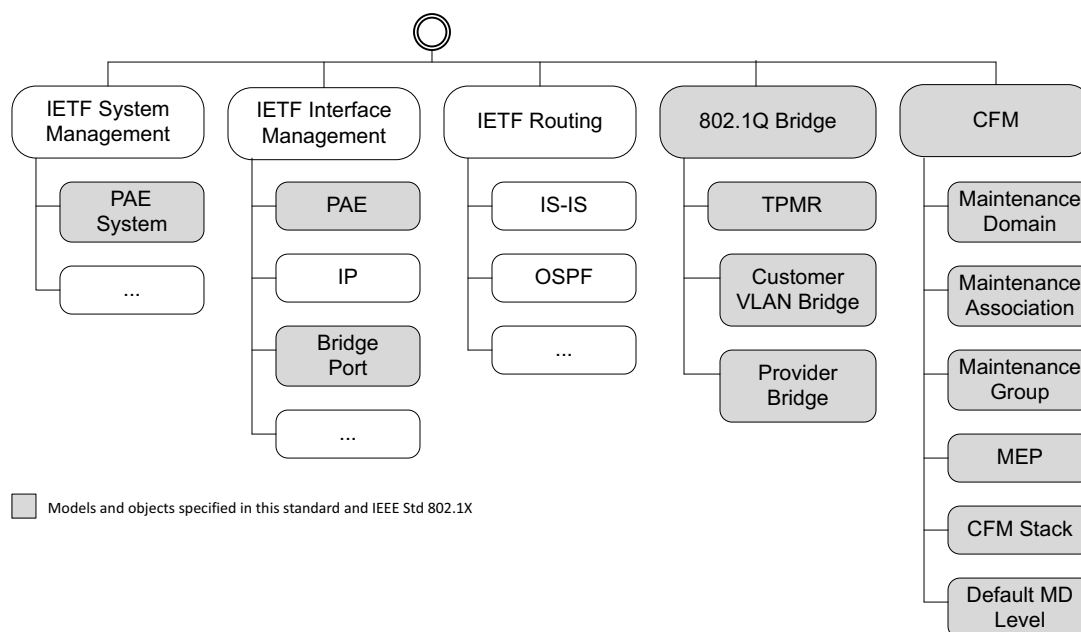


Figure 48-2—YANG root hierarchy with IEEE 802.1Q YANG modules

48.1.1 Interface Management (IETF RFC 8343) Model

Network interfaces are central to the management of protocols. Thus, it is important to establish a common data model for how interfaces are identified, configured, and monitored. The IETF Interface Management model (IETF RFC 8343) defines a generic YANG data model for the management of network interfaces. The Interface Management YANG model is augmented by the IEEE 802.1Q YANG models, and specifically the Bridge Port. A UML representation of the Interface Management model is illustrated in Figure 48-3.⁵⁰

1

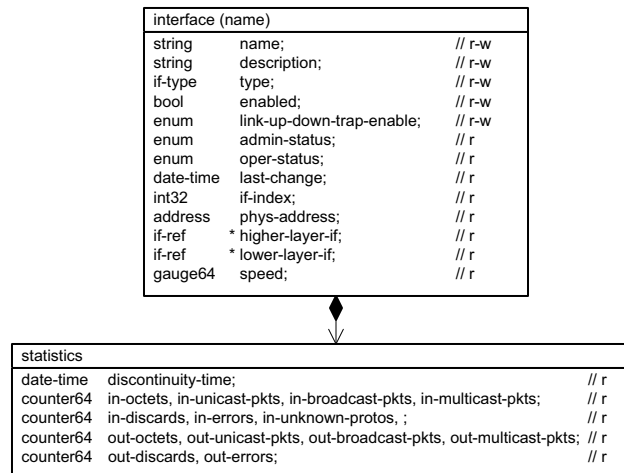


Figure 48-3— Interface YANG model

⁵⁰ The representation of the Interface YANG model is derived from IETF RFC 8343.

1 48.2 IEEE 802.1Q YANG models

2 The YANG data models are based on the management model outline in Clause 12. UML representations of
3 the models are provided in the following subclauses.

4 48.2.1 VLAN Bridge components model

5 A network system can include one or more VLAN Bridge components.

6 Each VLAN Bridge component (5.4–5.6) comprises a single VLAN-aware MAC Relay Entity and its
7 associated functionality (8.2, Figure 48-4), and is supported by a number of Bridge Ports (3.30,
8 Figure 48-5). Each Bridge Port is associated with a single MAC Relay Entity, presents a single interface (a
9 service access point) to that MAC Relay Entity, and comprises the interface stack that supports that service
10 access point (6.1, 7.4 of IEEE Std 802.1AC-2016). In the 802.1Q YANG models, that interface is
11 represented by the Interface Management (IETF RFC 8343) model augmented as shown in Figure 48-5.

12 Each MAC Relay Entity has a *bridge-type* attribute (e.g., *customer-vlan-bridge*), and the manageable
13 attributes for the component can reflect that type: TPMR components (48.2.2), for example, are VLAN
14 unaware.

15 A Bridge Port and the attributes associated with a media access method specific MAC entity (e.g., as
16 specified by IEEE Std 802.3) supporting that Bridge Port can both be represented by augmentations of the
17 same IETF_RFC_8343 Interface and identified by the same Interface index (*if-index*). In other cases
18 *higher-layer-if* and *lower-layer-if* attributes define the interface stacking relationships within the Bridge
19 Port. For example, an Aggregator (as defined by IEEE Std 802.1AX) Interface could be a Bridge Port. The
20 *lower-layer-if* attributes associated with the Aggregator would point to the Aggregation Ports.

21 Each Bridge Port is uniquely identified, amongst those Bridge Ports supporting a given VLAN Bridge
22 component, by a Port Number (*port-number*). A port number has no mandatory relationship to an Interface
23 index (*if-index*).

1

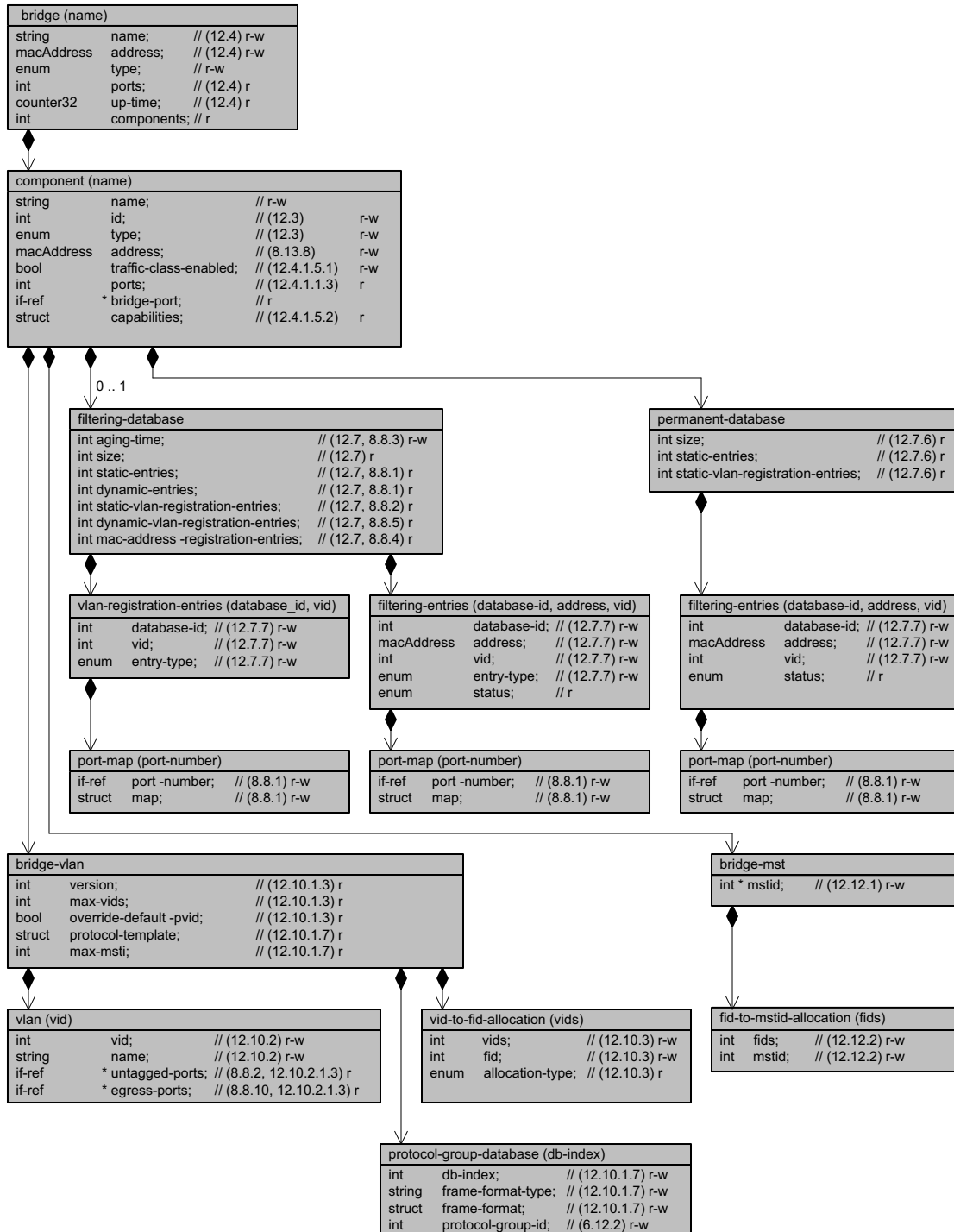


Figure 48-4—VLAN Bridge components model (MAC Relay Entities)

1

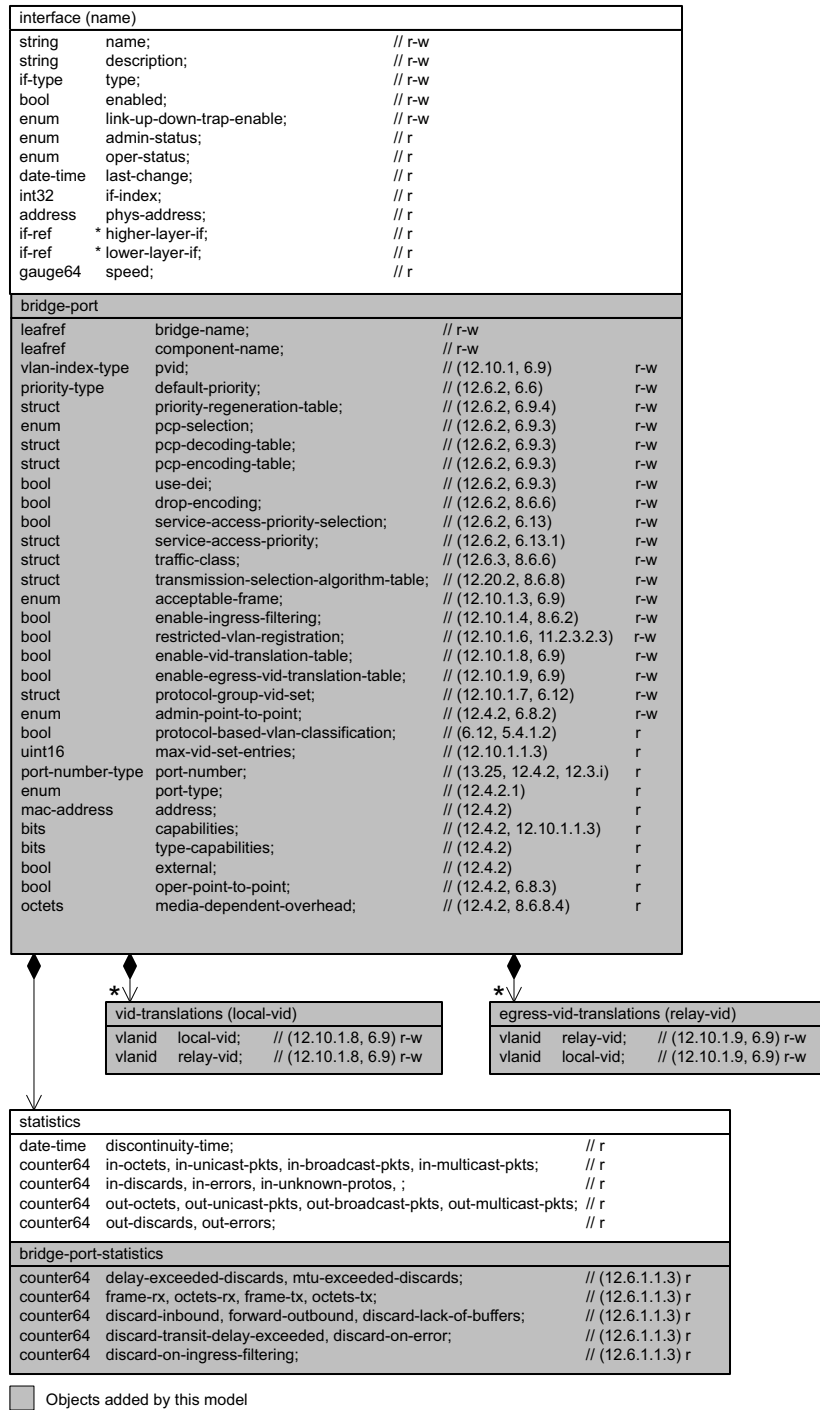


Figure 48-5—Bridge Port model

1 48.2.2 Two-Port MAC Relay (TPMR) model

2 A TPMR (5.15, 5.16) comprises a single VLAN-unaware MAC Bridge component supported by only two
3 TPMR Ports (Clause 8, 8.5.2). The TPMR’s VLAN-unaware MAC Relay Entity is modeled (Figure 48-6) as
4 a subset of a VLAN Bridge component’s MAC Relay Entity (Figure 48-4). This subset does not include the
5 bridge-vlan, bridge-mst, and filtering-database nodes.

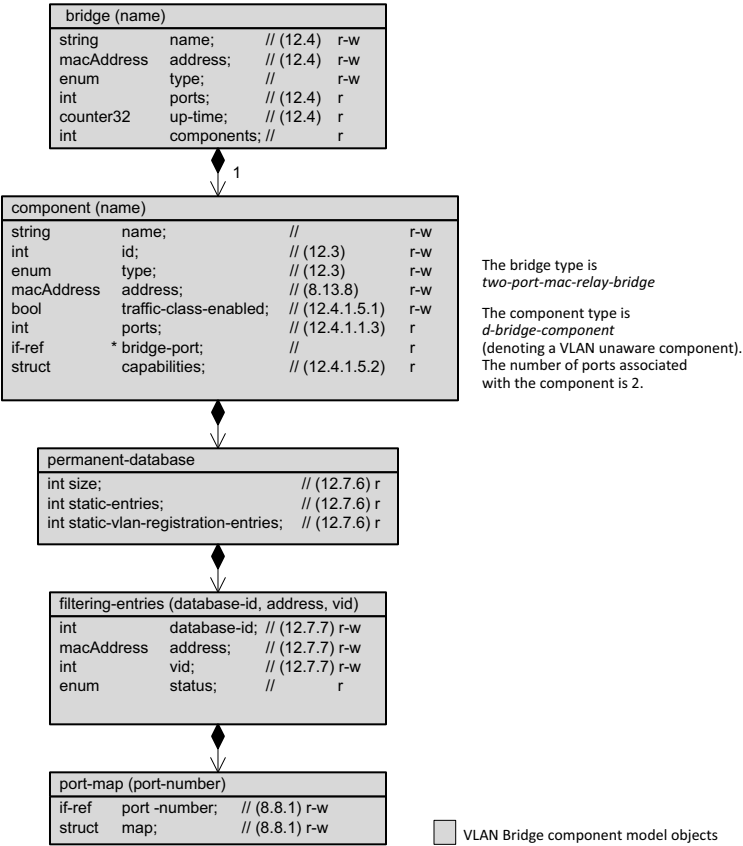


Figure 48-6—TPMR model (MAC Relay Entity)

¹ The TPMR model extends the VLAN Bridge components' Bridge Port model as illustrated in Figure 48-7.

²

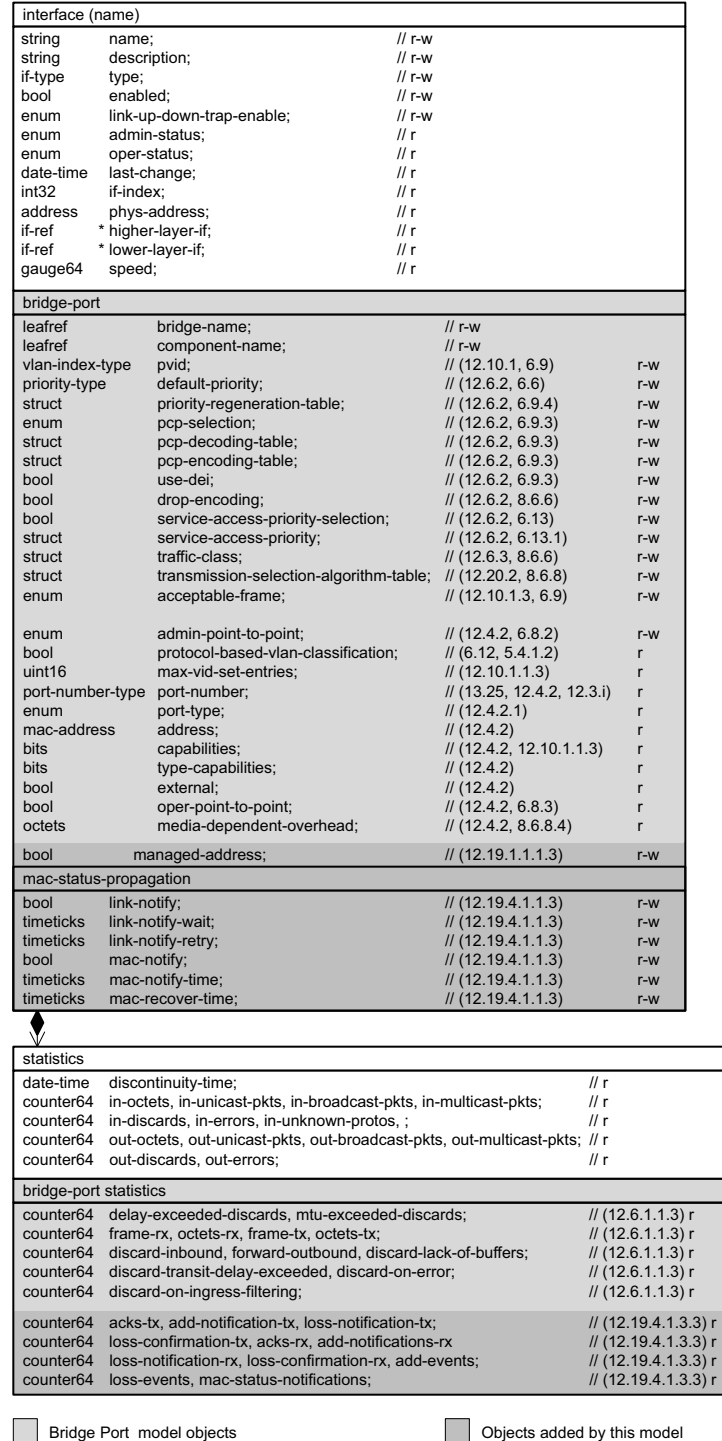


Figure 48-7—TPMR Port model

1 48.2.3 Customer VLAN Bridge model

2 A Customer VLAN Bridge (5.5) comprises a single C-VLAN Bridge component (Clause 8). The C-VLAN
3 Bridge component's MAC Relay Entity and its Bridge Ports are modeled by the VLAN Bridge components
4 model (48.2.1), with a single bridge component node with a type-of-bridge of customer-vlan-bridge (48.6.4)
5 [a ComponentType of cVlanComponent in 12.3].

1 48.2.4 Provider Bridge model

2 A Provider Bridge (5.10, Clause 16) can be an S-VLAN Bridge with a single S-VLAN component or a
3 Provider Edge Bridge with an S-VLAN component and multiple C-VLAN components. A Provider Bridge's
4 MAC Relay Entities and its Bridge Ports (S-VLAN and C-VLAN Interfaces) are modeled (Figure 48-8,
5 Figure 48-9, Figure 48-10) by extending the VLAN Bridge components model (48.2.1).

6

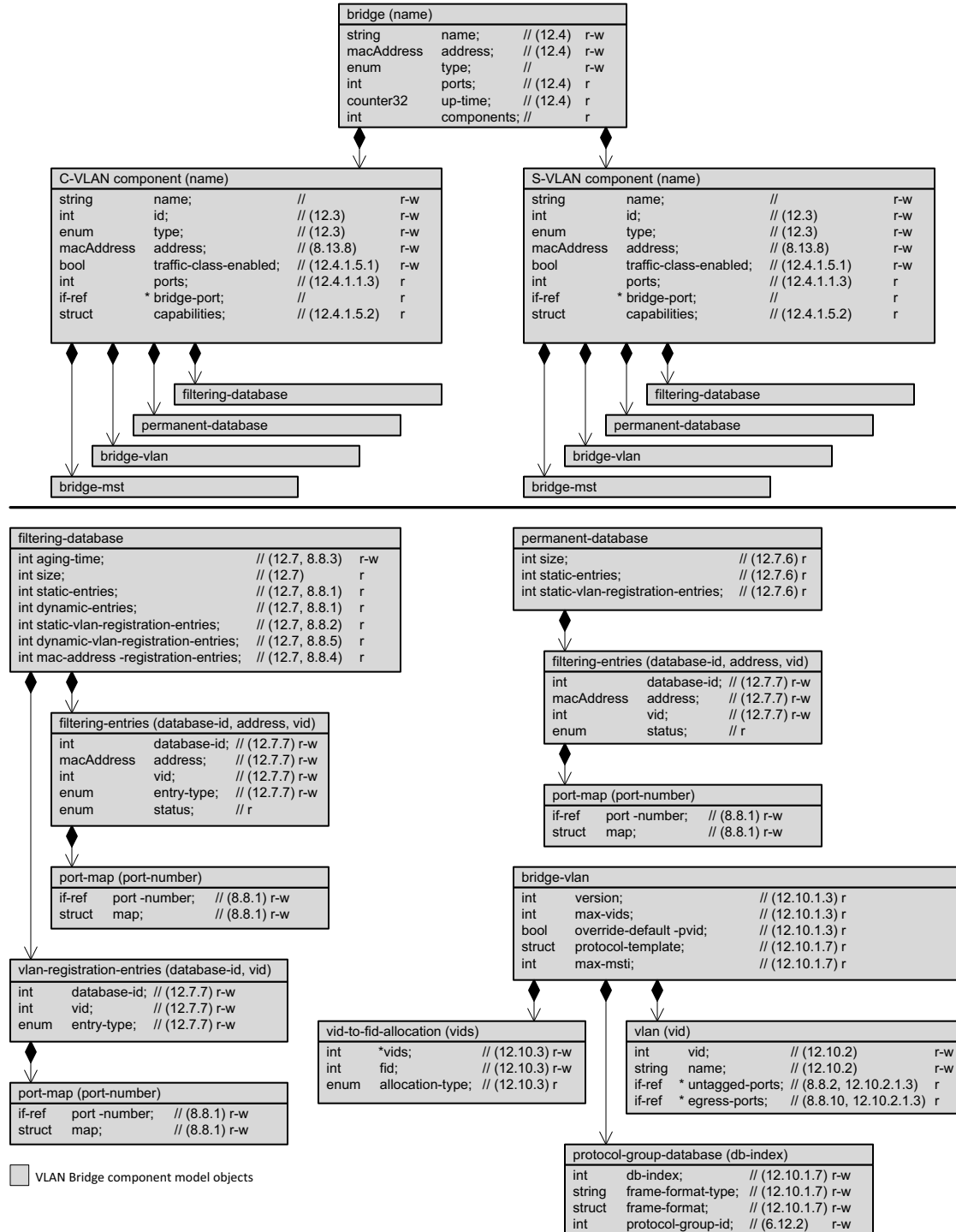


Figure 48-8—Provider Bridge model (MAC Relay Entities)

1

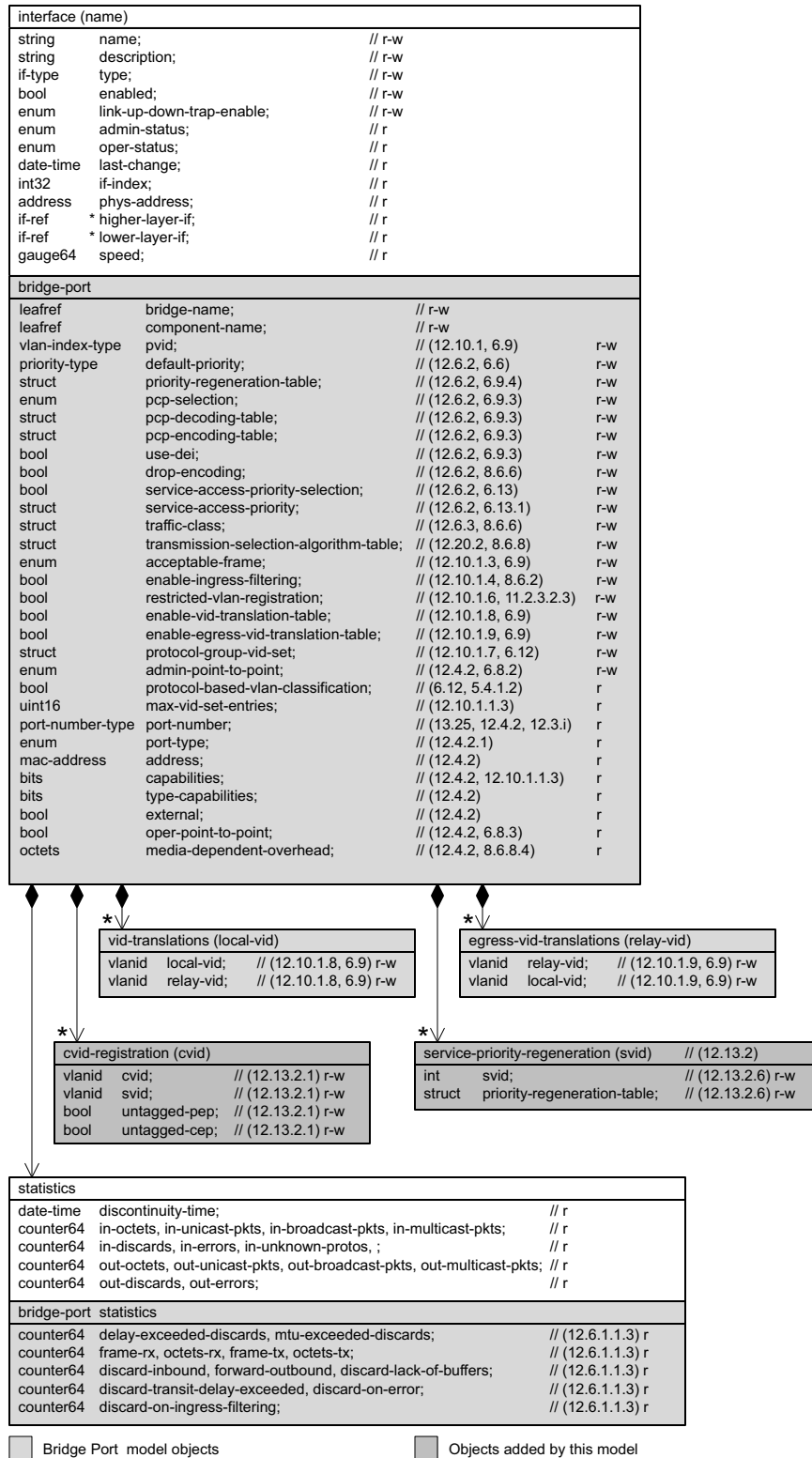


Figure 48-9—Provider Edge Bridge C-VLAN Interface model

1

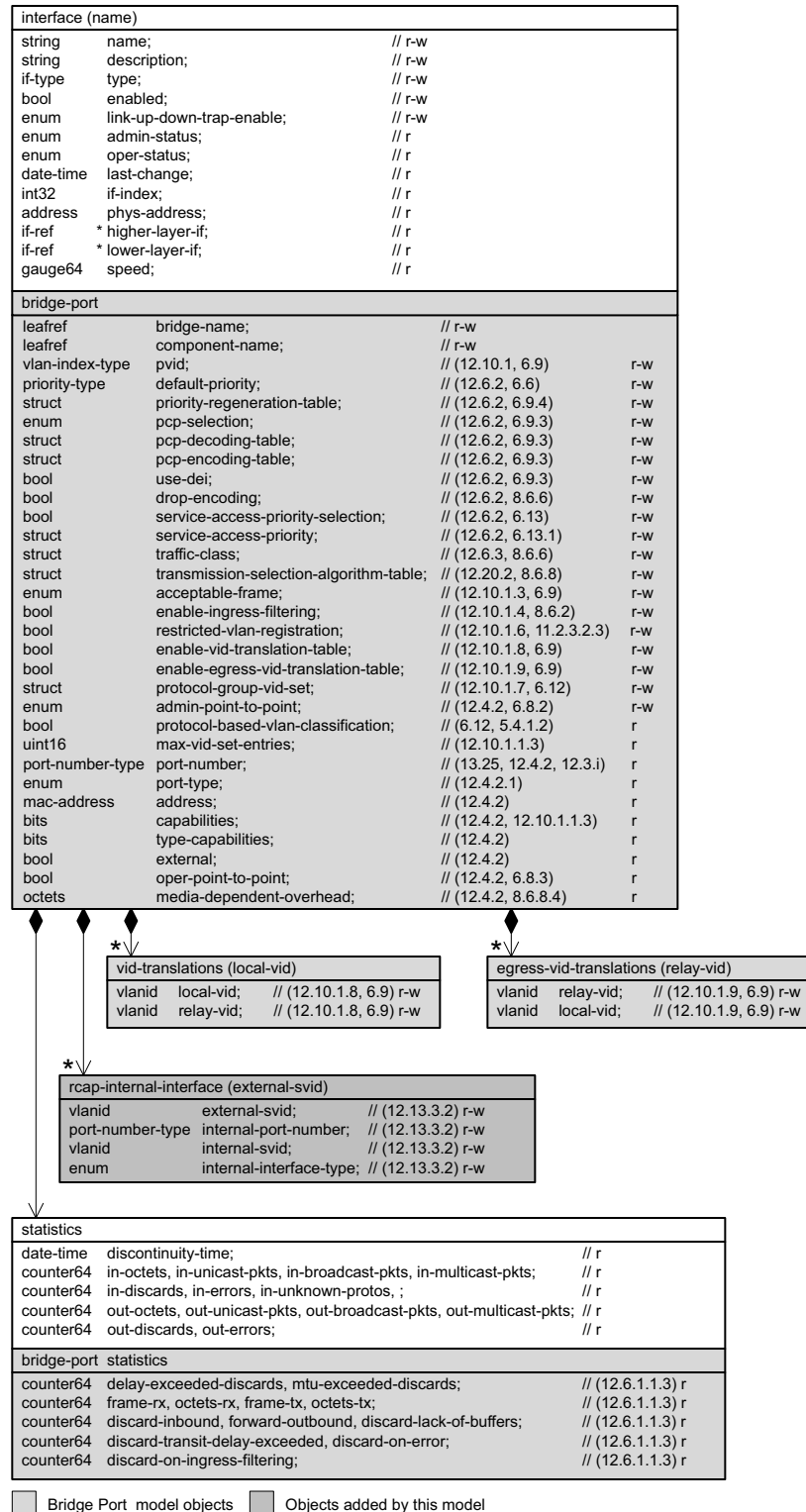


Figure 48-10—Provider Edge Bridge S-VLAN Interface model

1 48.2.5 CFM Model

2 The CFM UML data model is composed of the following set of major objects:

- 3 — The **Maintenance Domain** object is associated with a device (e.g., Bridge). It contains information
4 used to identify the maintenance domain used to bind/contain Maintenance Association objects.
- 5 — The **Maintenance Association** objects are contained within a Maintenance Domain object. They
6 contain maintenance point information related to the service instance being monitored.
- 7 — The **Maintenance Group** objects provide a convenient binding of a Maintenance Association object
8 with a Maintenance Domain object.
- 9 — The **Maintenance association Endpoint** (MEP) object is associated with an Interface (e.g., Bridge
10 Port) and a Maintenance Association Group object. It contains all information to manage a CFM
11 entity, associated with a specific DoSAP of a service instance, which can generate and receive CFM
12 PDUs and track any responses.
- 13 — The **Maintenance domain Intermediate Point** (MIP) object is associated with an Interface (e.g.,
14 Bridge Port) and a service (e.g., VID) to allow the explicit creation of an MHF.
- 15 — The **CFM Stack** object is associated with a device (e.g., Bridge). It contains information about
16 maintenance points on a particular device Interface (e.g., Bridge Port or Aggregation Port).
- 17 — The **Default MD Level** object is associated with a device (e.g., Bridge) component. It contains
18 MHF information needed for the creation of these functions on the services (e.g., VIDs) that are not
19 attached to any Maintenance Association.

20 The **Configuration Error List** is associated with a device (e.g., Bridge). It contains configuration error
21 information that is identifiable on a per-Interface (e.g., Bridge Port) and per-service (e.g., VID) basis.

22 The UML model representing the relationship between relevant Bridge objects to CFM objects is illustrated
23 in Figure 48-11.

24 The CFM objects (e.g., cfm-stack, configuration-error-list, default-md-level, maintenance-domain)
25 reference the device (i.e., Bridge and/or Bridge Component) to which they belong. This reference allows
26 CFM objects to be associated with other types of devices (e.g., non-Bridges, end stations) where
27 appropriate.

28 The Maintenance Group is modeled as a list that contains a convenient reference to a Maintenance
29 Association object and its corresponding Maintenance Domain object. The identification of this collection
30 of objects allows other CFM objects to more easily attach to a maintenance group and, in turn, simplifies
31 configuration and operation retrieval procedures supported by the data model.

1

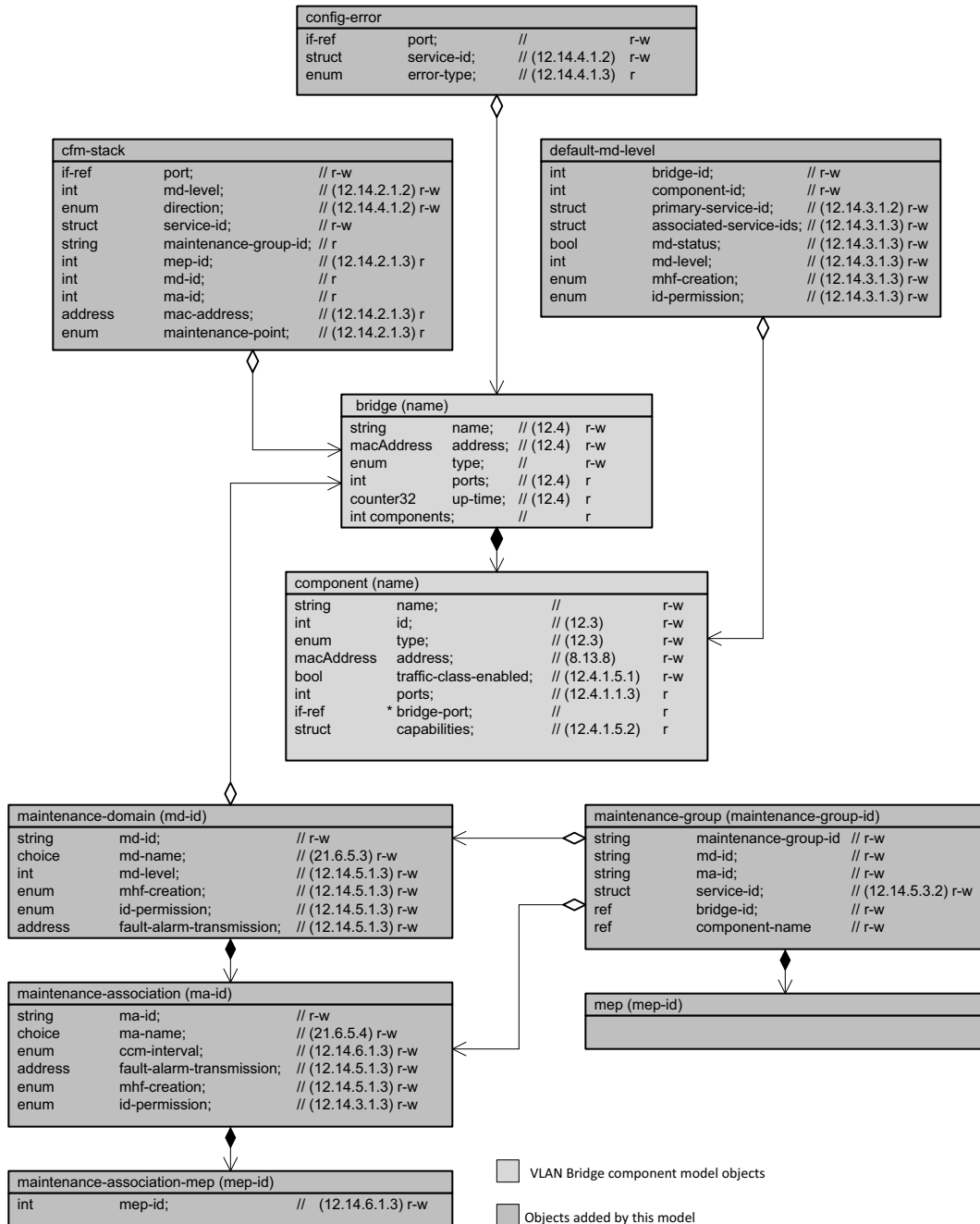


Figure 48-11—Bridge to CFM YANG model

As illustrated in Figure 48-12, MEP objects reference an Interface and are associated with a Maintenance Group. The interface relationship allows a MEP to be associated with a Bridge Port, as specified by this standard. This model also allows MEPs to be associated with other Interface types (for other devices).

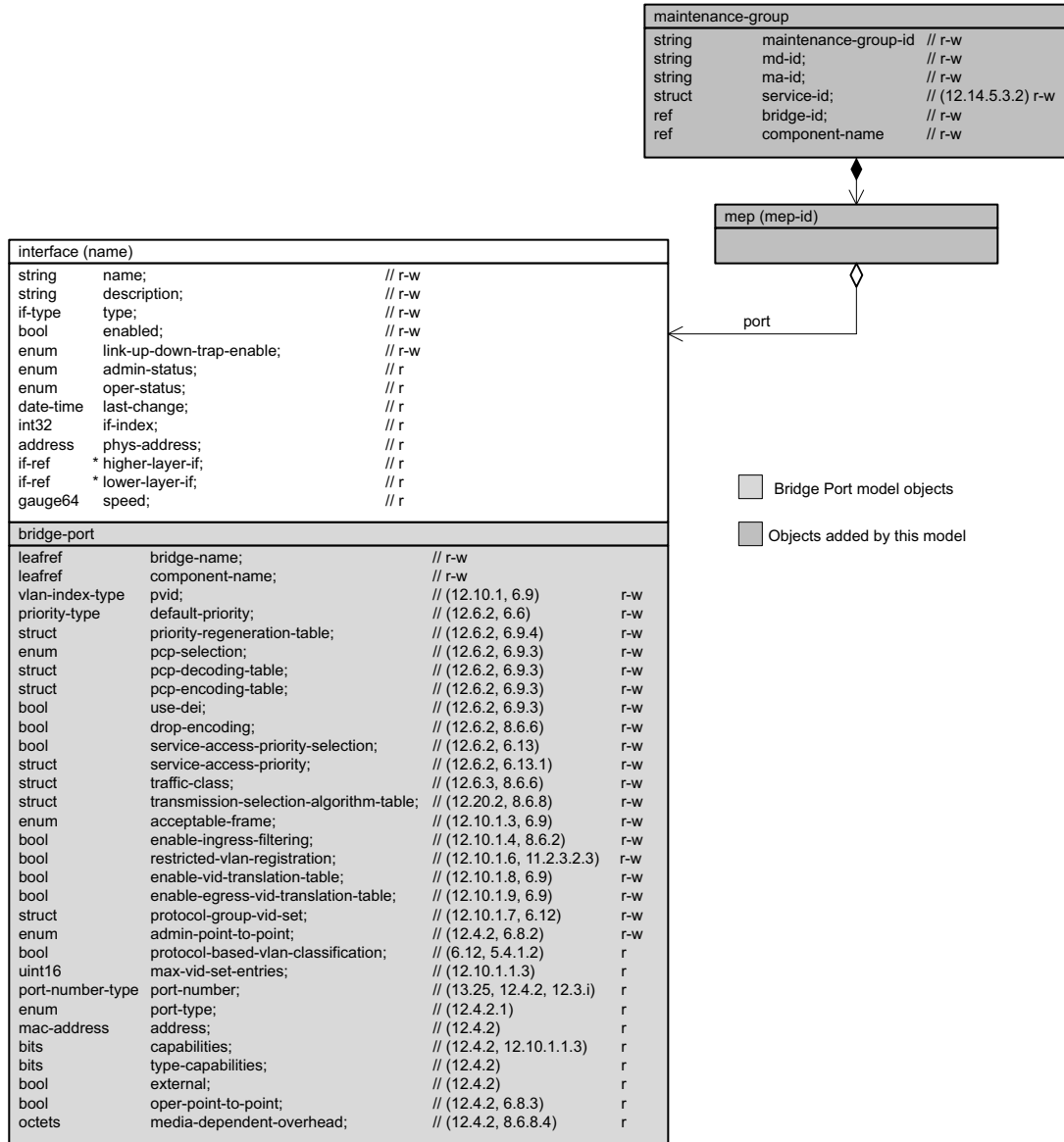


Figure 48-12—CFM MEP model relationships

The MEP object has relationships with other MEP-related objects. As illustrated in Figure 48-13, the CFM continuity check protocol suite-specific data needed to initiate protocol-specific sessions or maintain protocol-specific state information are grouped together. In addition, the MEP object will also reference the MEP CCM database and Stats objects.

48.2.5.1 CFM model support of CFM Operations

The CFM model also supports the initiation of Continuity Check, Loopback, and Linktrace protocol sessions. The general structure of the CFM session action data model is illustrated in Figure 48-14.

4

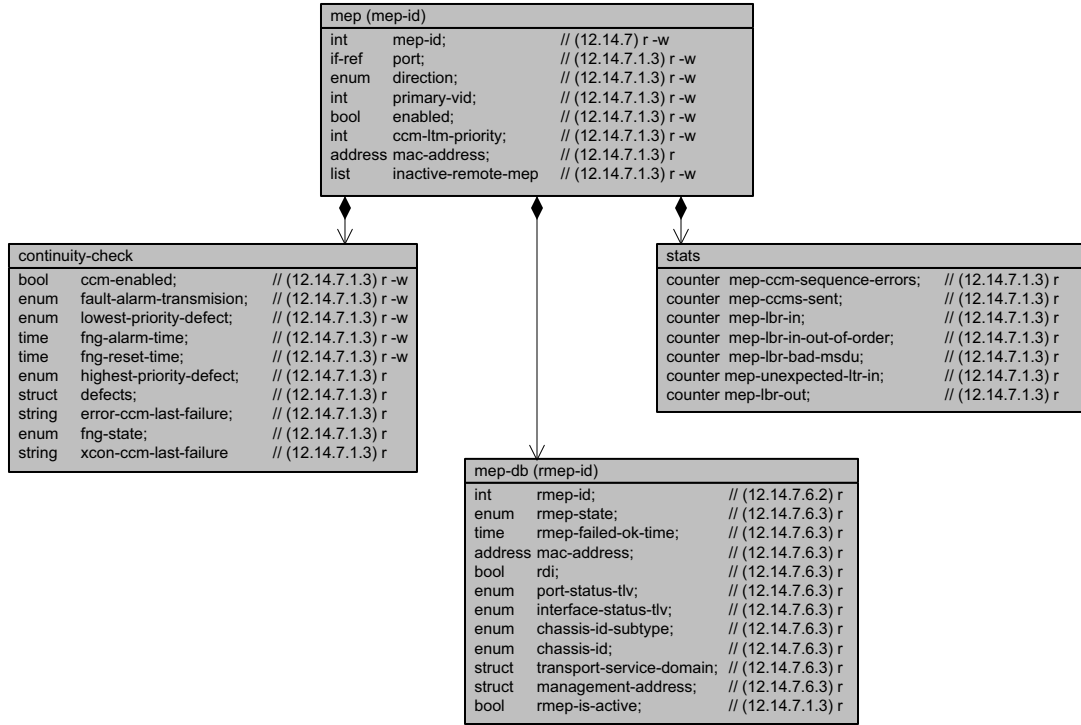


Figure 48-13—CFM MEP model

5

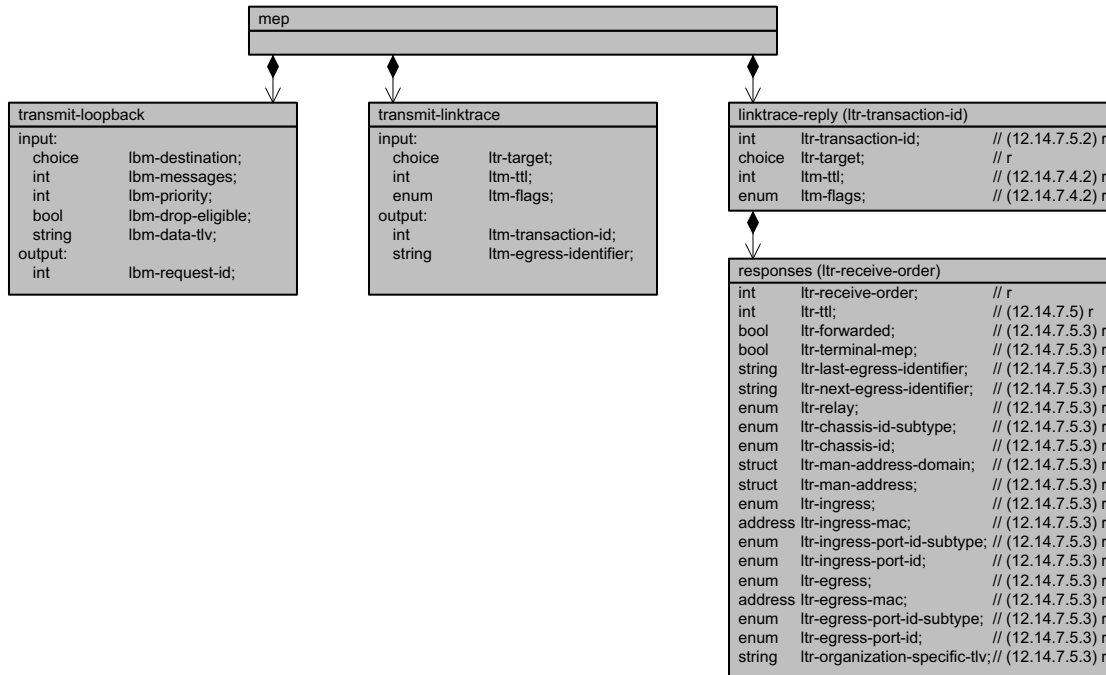


Figure 48-14—CFM operations structure

1 48.2.6 Stream filters and stream gates model

2 The stream filters and stream gates model augments the Bridge component model (48.3.1) by nodes that
3 represent the following managed objects:

- 4 a) Stream Filter Instance Table (12.31.2)
- 5 b) Stream Gate Instance Table (12.31.3)

6 The UML representation of the stream filters and stream gates model is illustrated in Figure 48-15.

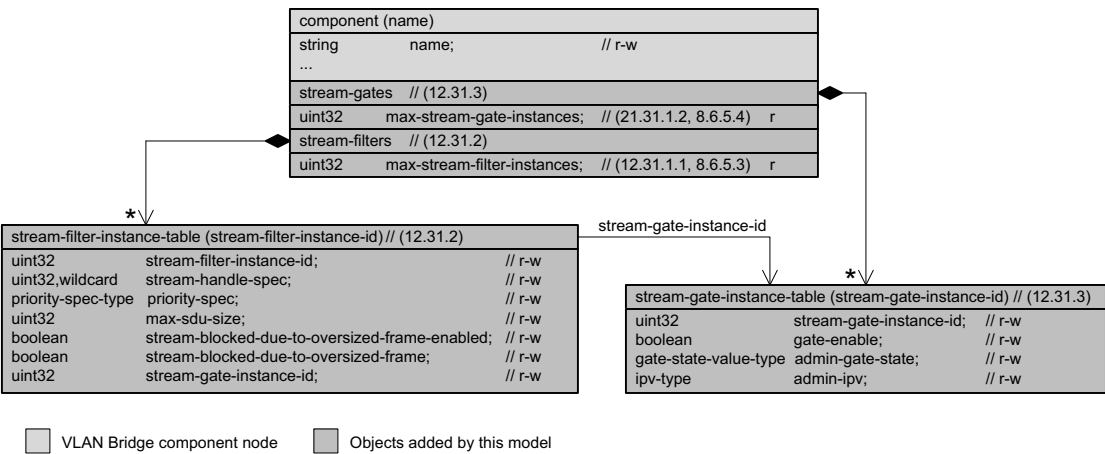


Figure 48-15—Stream filters and stream gates model

48.2.7 Asynchronous Traffic Shaping (ATS) model

The ATS model augments the Bridge component model (48.3.1) and the stream filters and stream gates model (48.3.6) by nodes that represent to following managed objects:

- The Scheduler Instance Table (12.31.5)
- The Scheduler Group Instance Table (12.31.6)
- The Scheduler Port Parameter Table (12.31.7)
- The Scheduler Timing Characteristics Table (12.31.8)
- A Stream Filter specification type representing an ATS scheduler instance identifier (12.31.2.5)

The UML representation of the ATS model is illustrated in Figure 48-16.

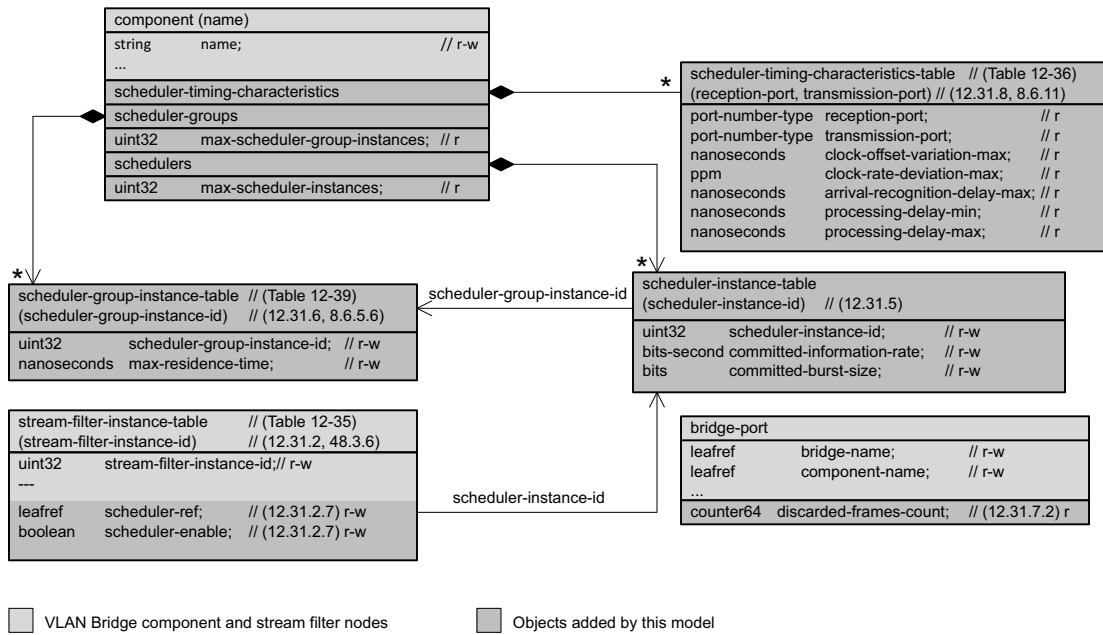


Figure 48-16—Asynchronous Traffic Shaping model

48.2.8 Congestion Isolation (CI) model

The CI model augments the Bridge components model (48.2.1) and the Interface Management model for Bridge ports (48.3.1) by nodes that represent the following managed objects:

- CI entity managed object (12.33.1)
- CI Peer Table (12.33.2)
- CI Stream Table (12.33.3)
- CIP entity managed object (12.33.4)

The UML representation of the CI model is illustrated in Figure 48-17.

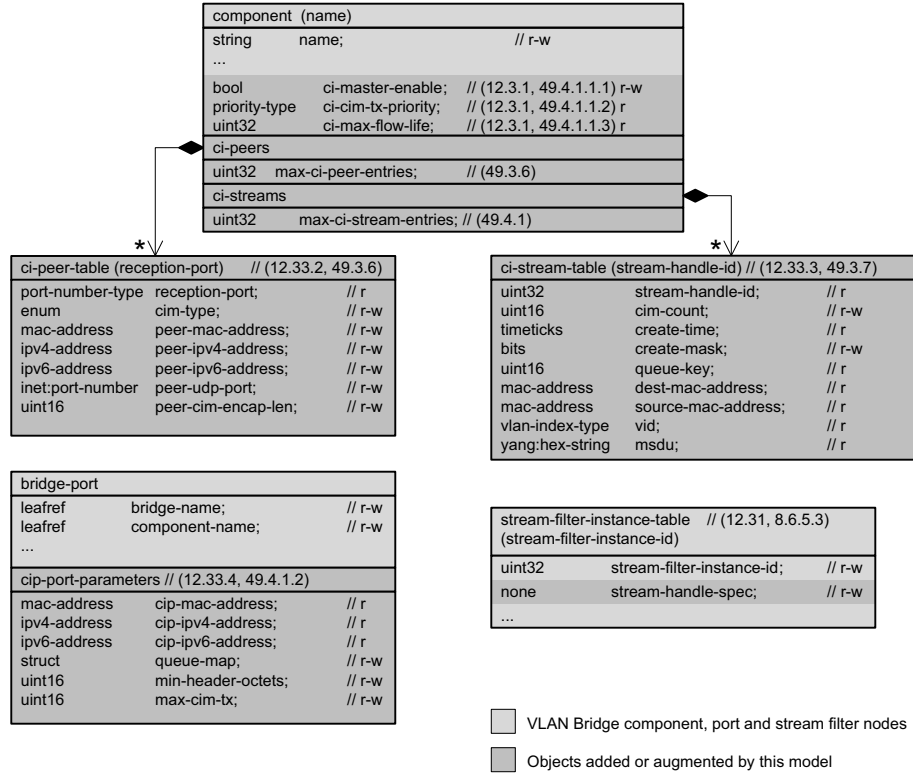


Figure 48-17—Congestion Isolation model

1 48.2.9 Scheduled Traffic model

2 The Scheduled Traffic model augments the Bridge Port model (48.2.1) with nodes that represent the
3 managed objects contained in the following table:

4 a) Gate Parameter Table (12.29.1)

5 The Scheduled Traffic model is illustrated in Figure 48-18.

6 There is one gate-parameter-table per Port of a Bridge component. Each object of the table represents
7 parameters that support the enhancements for scheduled traffic (8.6.8.4) as detailed in Table 12-32. The
8 admin-control-list and gate-control-list are sequences of gate-control-entry as specified in 12.29.1.2. Each
9 gate-control-entry consists of an operation name followed by a set of parameters associated with that
10 operation as defined in Table 8-7. The basic operation for Scheduled Traffic is SetGateStates. Additionally,
11 if Frame Preemption is supported, two additional operations can be performed Set-And-Hold-MAC and
12 Set-And-Release-MAC.

13

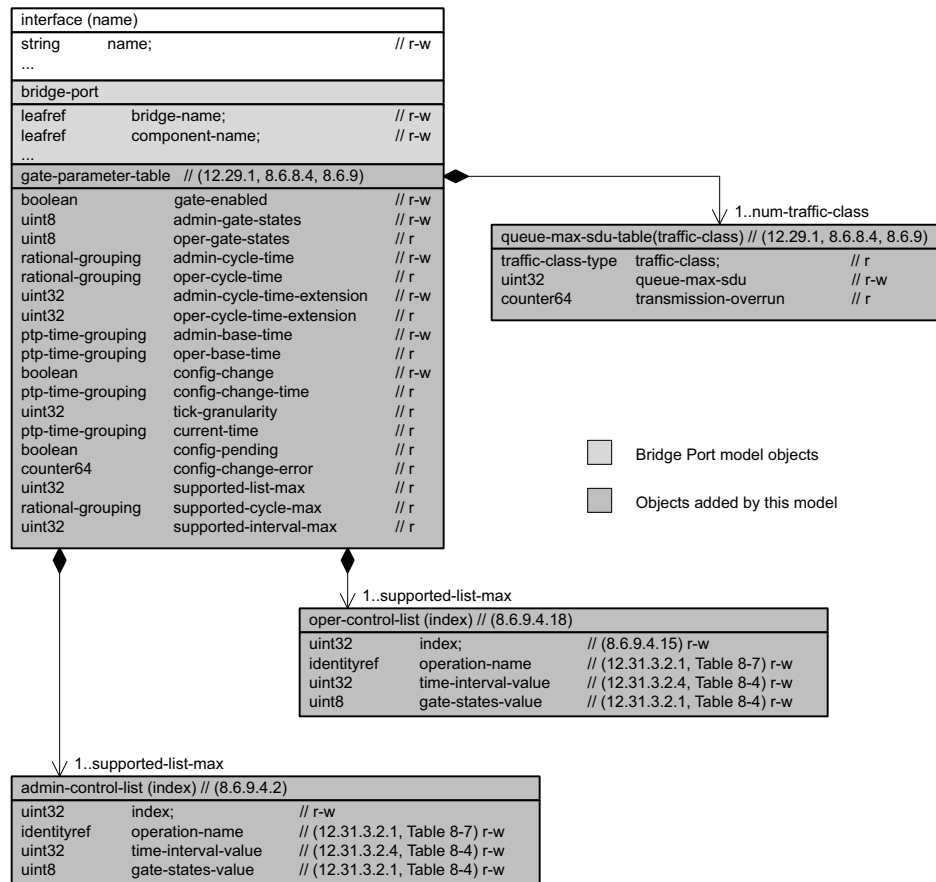


Figure 48-18—Scheduled Traffic model

48.2.10 Frame Preemption model

The Frame Preemption model augments the Bridge Port model (48.2.1) with nodes that represent the managed objects contained in the following table:

a) Frame Preemption Parameter Table (12.30.1)

The Frame Preemption model is illustrated in Figure 48-19.

There is one Frame Preemption Parameter Table per Port of a Bridge component or end station. Each object of the table represents parameters that support frame preemption (6.7.2) as detailed in Table 12-33.

8

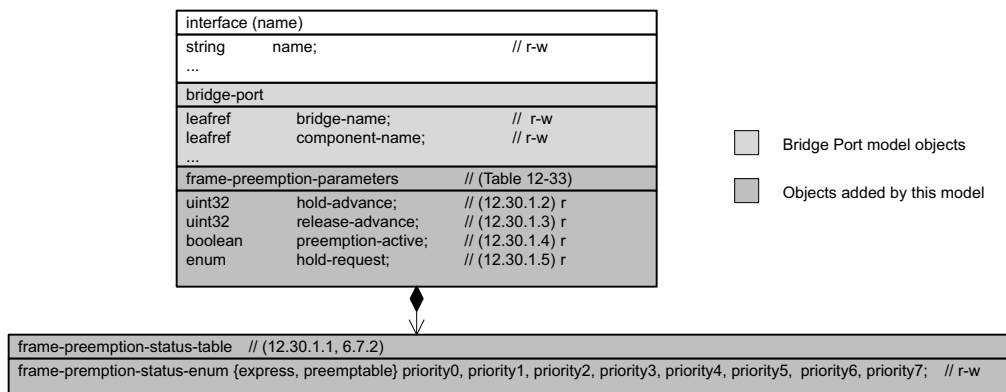


Figure 48-19—Frame Preemption model

48.2.11 Per-Stream Filtering and Policing model

The Per-Stream Filtering and Policing (PSFP) model augments the Bridge components model (48.2.1) and the Stream filters and stream gates model (48.2.6) with nodes that represent the managed objects contained in the following tables:

- The Stream Parameter Table (12.31.1)
- The Stream Filter Instance Table (12.31.2)
- The Stream Gate Instance Table (12.31.3)
- The Flow Meter Instance Table (12.31.4)

The PSFP model is illustrated in Figure 48-20.

There is one stream-filter-instance-table per Bridge component. Each object of the table contains a set of parameters that defines a single Stream Filter (8.6.5.3), as detailed in Table 12-35. The table rows form an ordered list of filter instances, the order being determined by the StreamFilterInstance parameter.

There is one stream-gate-instance-table per Bridge component. Each object of the table contains a set of parameters that defines a single Stream Gate Instance (8.6.5.4), as detailed in Table 12-36.

There is one flow-meter-instance-table per Bridge component. Each object of the table contains a set of parameters that defines a single Flow Meter Instance (8.6.5.5), as detailed in Table 12-37.

1

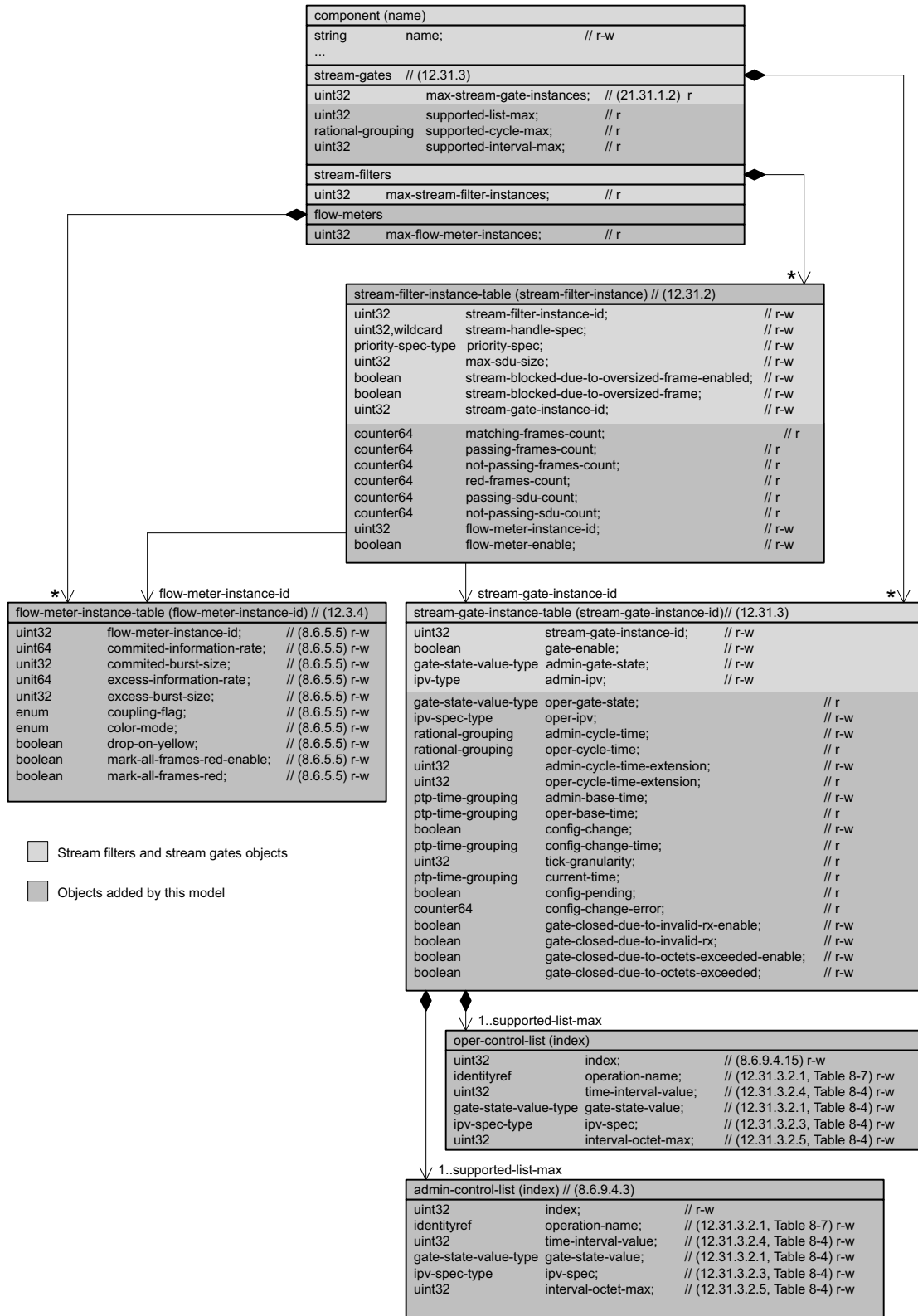


Figure 48-20—PSFP model

48.2.12 CNC-configuration model

- The CNC-configuration model allows communication between a CUC and a CNC and can be implemented in an end station or Bridge.
- The CNC-configuration model consists of three high-level groups, Talker (46.2.3), Listener (46.2.4), and Status (46.2.5) and is modeled as illustrated in Figure 48-21 and Figure 48-22.

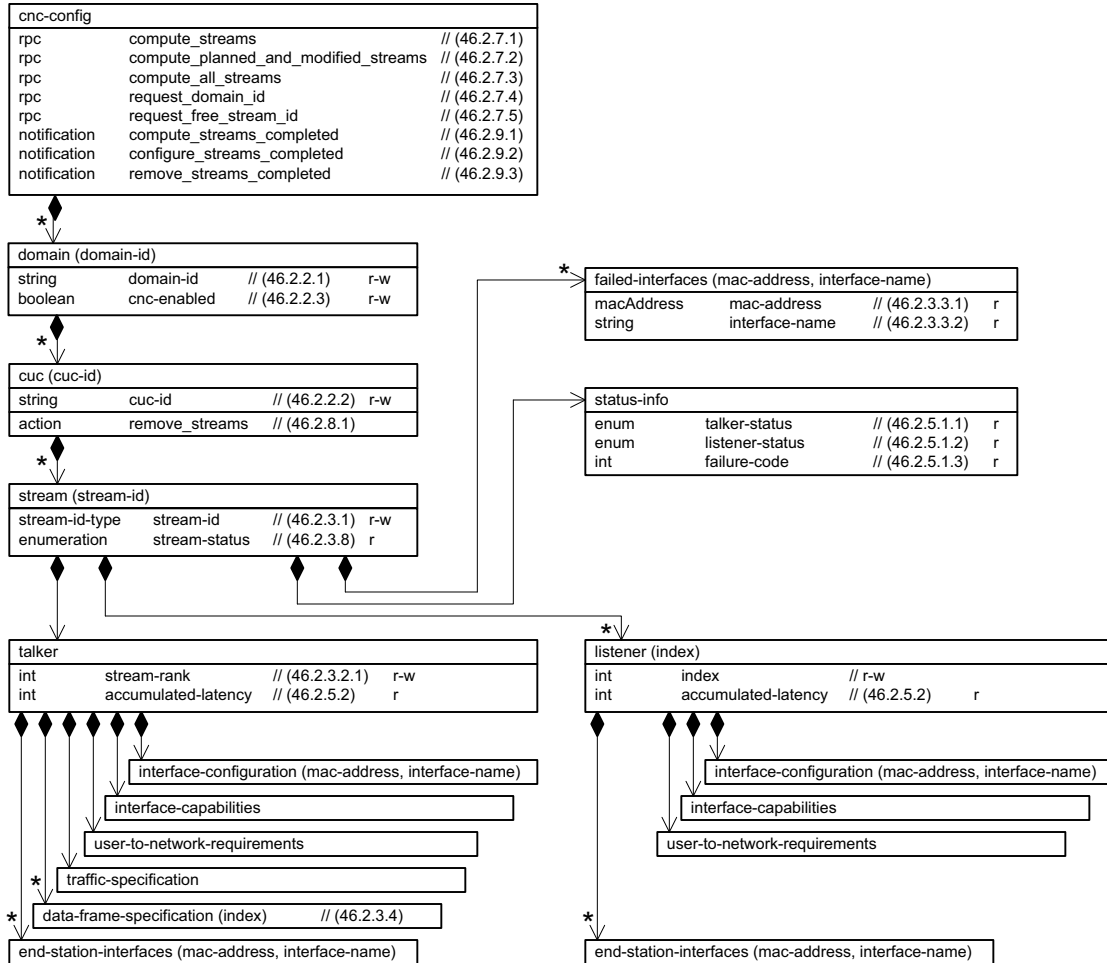


Figure 48-21—CNC-configuration model A

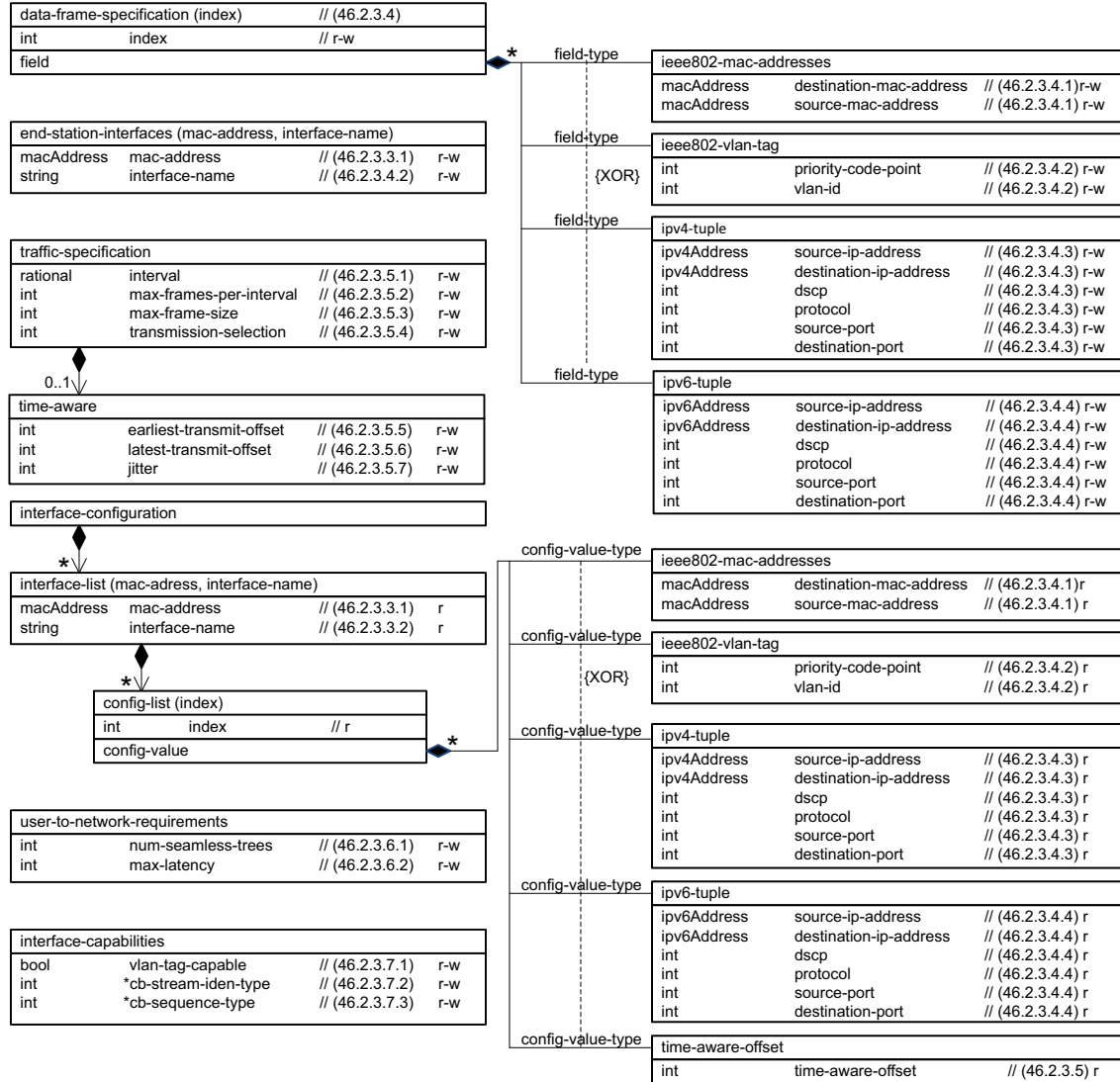


Figure 48-22—CNC-configuration model B

1 **48.2.13 Credit-based shaper algorithm model**

- 2 The credit-based shaper algorithm model augments the Bridge Port model (48.2.1) with nodes that represent
- 3 the managed objects of the credit-based shaper algorithm in Table 12-4.
- 4 The credit-based shaper algorithm model is illustrated in Figure 48-23.

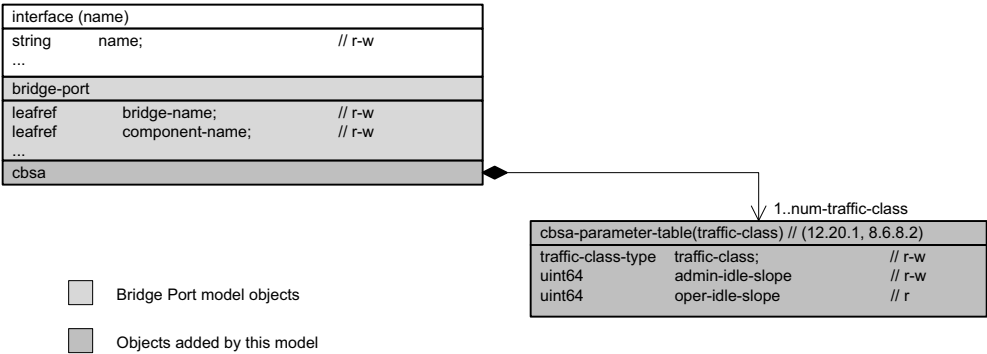


Figure 48-23—Credit-based shaper algorithm model

1 48.2.14 Rapid Spanning Tree Protocol (RSTP) model

2 The RSTP model augments the VLAN Bridge component model (48.2.1, Figure 48-4) and the Interface
3 management model for Bridge Ports (48.2.1, Figure 48-5) with nodes common to both RSTP and MSTP.
4 These nodes control the configuration of the CST (and the CIST, when augmented by the Multiple Spanning
5 Trees model, 48.2.15) and report on protocol operation. The RSTP model is illustrated in Figure 48-24.

component (name)			
string	name;		// r-w
...			
rstp // (13.4)			
enum	force-protocol-version;	// (13.7.2)	r-w
bridge-id	bridge-id;		
{ uint64	bridge-id;	// (13.26.2)	r
id-priority	bridge-priority;	// (14.2)	r-w
uint16	system-id-extension;	// (13.26.2)	r
ieee:macaddress	bridge-address; }	// (8.13.8)	r
bridge-id	root-id		
{ uint64	bridge-id;	// (13.4)	r
id-priority	bridge-priority;	// (14.2)	r
uint16	system-id-extension;	// (13.26)	r
ieee:macaddress	bridge-address; }	// (8.13.8)	r
uint32	root-path-cost;	// (13.4)	r
union	root-port;	// (13.4)	r
uint8	max-age;	// (13.15)	r
uint8	hello-time;	// (13.25)	r
uint8	forward-delay;	// (13.25)	r
uint8	bridge-max-age;	// (13.25)	rw
uint8	bridge-forward-delay;	// (13.25)	rw
uint8	tx-hold-count;	// (13.25)	rw
yang:date-and-time	last-topology-change;	// (13.25)	r

bridge-port			
leafref	bridge-name;		// r-w
leafref	component-name;		// r-w
...			
rstp // (13.4)			
bool	admin-bridge-port-enabled;	// (8.4)	rw
port-state	port-state;	// (8.4)	r
port-role	port-role;	// (13.4)	r
bool	restricted-role;	// (13.20)	rw
bool	restricted-tcn;	// (13.20)	rw
port-id	port-id		
{ uint16	port-id;	// (13.26)	r
id-priority	port-priority;	// (14.2)	rw
id-port-number	port-number; }	// (12.3)	r
uint32	fix-port-path-cost;	// (13.10)	r-w
uint32	port-path-cost;	// (13.4)	r-w
uint8	designated-protocol-version;	// (14.2)	r
bridge-id	root-id		
{ uint64	bridge-id;	// (13.4)	r
id-priority	bridge-priority;	// (14.2)	r
uint16	system-id-extension;	// (13.26)	r
ieee:macaddress	bridge-address; }	// (8.13.8)	r
uint32	root-path-cost;	// (13.4)	r
bridge-id	designated-bridge-id		
{ uint64	bridge-id;	// (13.4)	r
id-priority	bridge-priority;	// (14.2)	r-w
uint16	system-id-extension;	// (13.26)	r
ieee:macaddress	bridge-address; }	// (8.13.8)	r
port-id	designated-port-id		
{ uint16	port-id;	// (13.26)	r
id-priority	port-priority;	// (14.2)	r
id-port-number	port-number; }	// (12.3)	r
bool	admin-edge-port;	// (13.33)	r-w
bool	oper-edge-port;	// (13.33)	r
bool	auto-edge-port;	// (13.33)	r-w
bool	disputed-port;	// (13.21)	r
bool	isolate-port;	// (13.23)	r
action	port-protocol-migration-check;	// (13.32)	

- ☐ VLAN Bridge component and port nodes
☐ Objects added or augmented by this model

Figure 48-24—RSTP model

48.2.15 Multiple Spanning Tree Protocol (MSTP) model

The Multiple Spanning Tree Protocol model augments the bridge-mst container of the VLAN Bridge components model (48.2.1) and the RSTP model (48.2.14) with nodes for MSTP. These nodes control the configuration of the CIST and MSTIs and the assignment of VLANs and VIDs to MSTIs (8.9) within MST Regions. They also report on MSTP protocol operation. The Multiple Spanning Tree Protocol model is illustrated in Figure 48-25.

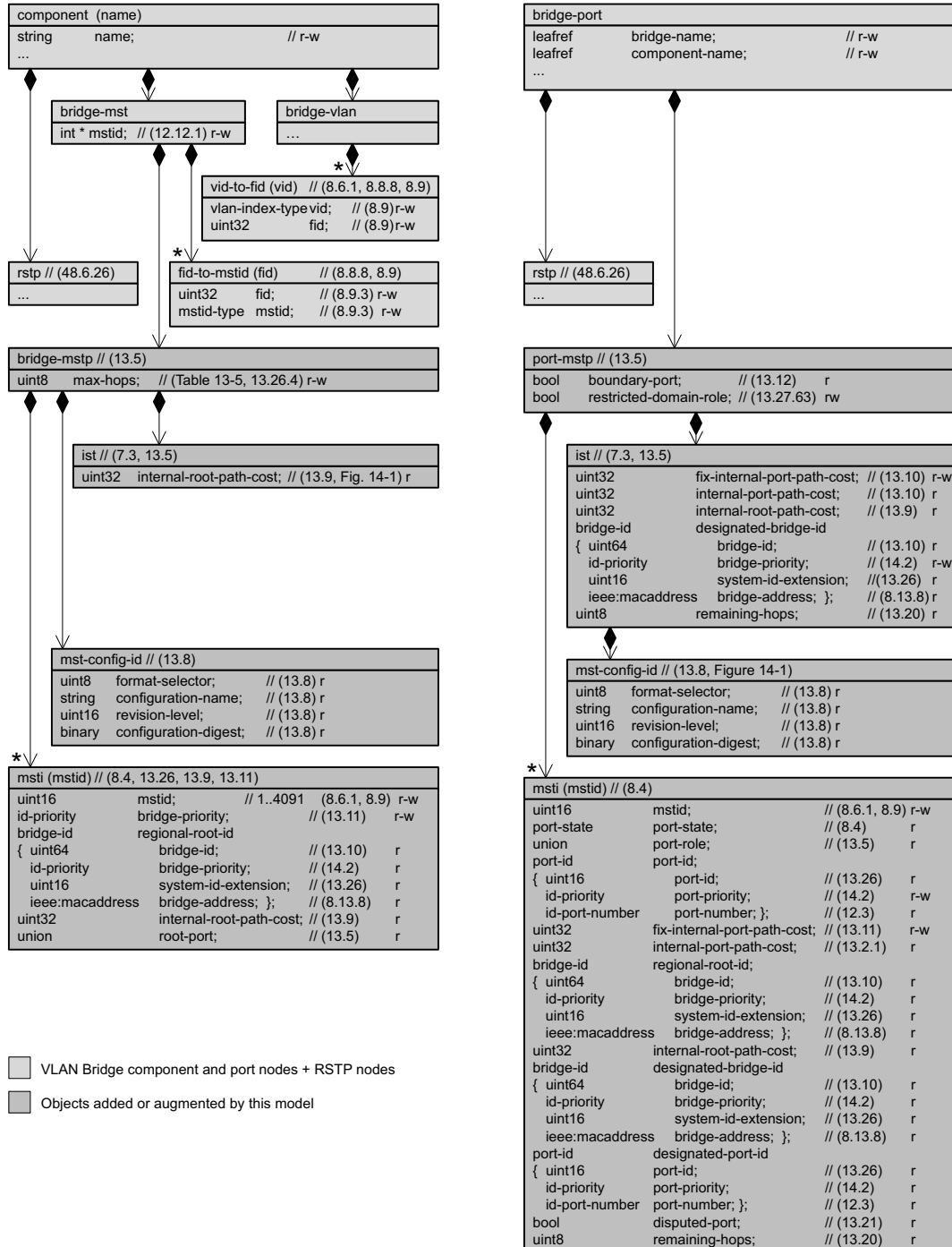


Figure 48-25—MSTP model

1 48.3 Structure of the YANG models

2 The YANG models specified by this standard use the YANG modules summarized in Table 48-1.⁵¹

3 In the YANG module definitions, if any discrepancy between the DESCRIPTION text and the
4 corresponding definition in any other part of this standard occur, the definitions outside this clause
5 (Clause 48) take precedence.

6

Table 48-1—Summary of the YANG modules

Module	References	Managed functionality	Initial YANG specification Notes
ieee802-types	48.6.1	Type definitions	IEEE Std 802.1Qcp General type definitions for IEEE 802 standards.
ieee802-dot1q-types	48.6.2	Type definitions	IEEE Std 802.1Qcp General type definitions used by IEEE Std 802.1Q.
ieee-802-dot1q-tsn-types	48.6.3	Type definitions	IEEE Std 802.1Qcp Type definitions and groupings for TSN user/network components.
ieee802-dot1q-bridge	48.5.4, 48.6.4	Clause 8	IEEE Std 802.1Qcp VLAN bridge component capabilities.
ieee802-dot1q-tpmr	48.5.5, 48.6.5	5.13, 5.15	IEEE Std 802.1Qcp TPMR augmentation of ieee802-dot1q-bridge.
ieee802-dot1q-pb	48.5.6, 48.6.6	Clause 20	IEEE Std 802.1Qcp Provider Bridge augmentation of ieee802-dot1q-bridge.
ieee802-dot1q-cfm-types	48.5.7, 48.6.7	Clause 20	IEEE Std 802.1Qcx General type definitions for CFM modules.
ieee802-dot1q-cfm	48.5.8, 48.6.8	Clause 20	IEEE Std 802.1Qcx Base CFM module defining Maintenance Domains, Maintenance Associations, Maintenance Groups, MEPs, and CFM session actions.
ieee802-dot1q-cfm-bridge	48.5.9, 48.6.9	Clause 20	IEEE Std 802.1Qcx Bridge and Bridge Port extension/ augmentation of ieee802-dot1q-cfm including CFM Stack, Default MD Levels, and Configuration Error Lists.
ieee802-dot1q-cfm-alarm	48.5.10, 48.6.10	Clause 20	IEEE Std 802.1Qcx Common CFM alarms, also supports alarms based on different frameworks (e.g., ietf-alarms).
ieee802-dot1q-stream-filters-gates	48.5.11, 48.6.11	8.6, 8.6.5.3, 8.6.5.4	IEEE Std 802.1Qcr Basic stream filtering and stream gating capabilities.
ieee802-dot1q-stream-filters-gates-bridge	48.5.12, 48.6.12	8.6, 8.6.5.3, 8.6.5.4	IEEE Std 802.1Qcz Bridge component augmentation with stream filtering and stream gating capabilities.

⁵¹ An amendment's designation is often used to refer to functionality in an IEEE standard after the amendment has been incorporated in a revision of the standard, even if the functionality has been revised. The amendment that added each YANG module is identified to help locate the relevant provisions of this standard.

Table 48-1—Summary of the YANG modules (*continued*)

Module	References	Managed functionality	Initial YANG specification Notes
ieee802-dot1q-ats	48.5.13, 48.6.13	8.6	IEEE Std 802.1Qcr ATS extensions to ieee802-dot1q-stream-filters-gates and ieee802-dot1q-bridge modules.
ieee802-dot1q-ats-bridge	48.5.14, 48.6.14	8.6	IEEE Std 802.1Qcz Bridge component augmentation with ATS module.
ieee802-dot1q-congestion-isolation	48.5.15, 48.6.15	8.6.5, Clause 49	IEEE Std 802.1Qcz Congestion isolation specific parameters and extensions to the ieee802-dot1q-stream-filters-gates.
ieee802-dot1q-congestion-isolation-bridge	48.5.16, 48.6.16	8.6.5, Clause 49	IEEE Std 802.1Qcz Bridge and Bridge port extension/augmentation with congestion isolation module.
ieee802-dot1q-sched	48.5.17, 48.6.16	8.6.8.4	IEEE Std 802.1Qcw Scheduled traffic module.
ieee802-dot1q-sched-bridge	48.5.18, 48.6.16	8.6.8.4	IEEE Std 802.1Qcw Augments Bridge Port with Scheduled Traffic module.
ieee802-dot1q-preemption	48.5.19, 48.6.19	6.7.2	IEEE Std 802.1Qcw Frame preemption module.
ieee802-dot1q-preemption-bridge	48.5.20, 48.6.20	6.7.2	IEEE Std 802.1Qcw Augments Bridge Port with Frame Preemption.
ieee802-dot1q-psfp	48.5.21, 48.6.21	8.6.5.3, 8.6.5.4, 8.6.5.5	IEEE Std 802.1Qcw PSFP module.
ieee802-dot1q-psfp-bridge	48.5.22, 48.6.22	8.6.5.3, 8.6.5.4, 8.6.5.5	IEEE Std 802.1Qcw Augments Bridge Port with PSFP module.
ieee802-dot1q-cnc-config	48.5.23, 48.6.23	46.1.5, 46.1.6, 46.2	IEEE Std 802.1Qdj CNC configuration
ieee802-dot1q-cbsa	48.5.24, 48.6.24	8.6.8.2	IEEE Std 802.1Qdx Credit-based shaper algorithm module
ieee802-dot1q-cbsa-bridge	48.5.25, 48.6.25	8.6.8.2	IEEE Std 802.1Qdx Augments Bridge Port with credit-based shaper algorithm module.
ieee802-dot1q-rstp	48.5.26, 48.6.26	8.4, 8.7, 8.8, 8.10, Clause 13	IEEE Std 802.1Qdy Bridge and Bridge Port RSTP parameters.
ieee802-dot1q-rstp-bridge	48.5.27, 48.6.27		IEEE Std 802.1Qdy Augments Bridge components and Bridge Ports with RSTP parameters.
ieee802-dot1q-mstp	48.5.28, 48.6.28	8.4, 8.7, 8.8, 8.8, 8.9, 8.10, Clause 13	IEEE Std 802.1Qdy Bridge and Bridge Port MSTP parameters.
ieee802-dot1q-mstp-bridge	48.5.29, 48.6.29		IEEE Std 802.1Qdy Augments RSTP-capable Bridge components and Bridge Ports with MSTP parameters.

The relationship between the models listed in 48.2 and the YANG modules listed in Table 48-1 is described in 48.3.1 through 48.3.15.

48.3.1 VLAN Bridge components model

This model provides basic bridging capabilities and allows for augmentation by specific YANG models (e.g., Two-Port MAC Relay model, Customer VLAN Bridge model, Provider Bridge model).

A system implementing the VLAN Bridge components model implements the YANG modules listed in Table 48-2.

48.3.2 Two-Port MAC Relay model

A system implementing the TPMR YANG model (48.3.2) implements the YANG modules listed in Table 48-3.

48.3.3 Customer VLAN Bridge model

A system implementing the Customer VLAN Bridge model (48.3.3) implements the YANG modules listed in Table 48-4.

48.3.4 Provider Bridge model

A system implementing the Provider Bridge model (48.3.4) implements the YANG modules listed in Table 48-5.

Table 48-2—VLAN Bridge component model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge

Table 48-3—Two-Port MAC Relay (TPMR) model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-tpmr

17

Table 48-4—Customer VLAN Bridge model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge

Table 48-5—Provider Bridge model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-pb

48.3.5 CFM model

A system implementing the CFM models (48.3.5) implements the YANG modules listed in Table 48-6.

Table 48-6—CFM model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-cfm-types
ieee802-dot1q-bridge
ieee802-dot1q-cfm
ieee802-dot1q-cfm-bridge
ieee802-dot1q-cfm-alarm

48.3.6 Stream filters and stream gates model

The Stream filters and stream gates model (48.3.6) provides basic stream filter (8.6.5.3) and stream gate (8.6.5.4) capabilities and allows for augmentation by specific YANG models (e.g., the ATS model). A system implementing the stream filters and stream gates model implements the YANG modules as described in Table 48-7.

Table 48-7—Stream filters and stream gates model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-stream-filters-gates
ieee802-dot1q-stream-filters-gates-bridge

1 48.3.7 Asynchronous Traffic Shaping (ATS) model

2 A system implementing the ATS model (48.3.7) implements the YANG modules as described in
3 Table 48-8.

Table 48-8—ATS model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-stream-filters-gates
ieee802-dot1q-ats
ieee802-dot1q-ats-bridge

4 48.3.8 Congestion Isolation (CI) model

5 A system implementing the CI model (48.3.8) implements the YANG modules in Table 48-6.

Table 48-9—YANG module dependencies for the CI model

YANG Module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-stream-filter-gates
ieee802-dot1q-congestion-isolation
ieee802-dot1q-congestion-isolation-bridge

6 48.3.9 Scheduled Traffic model

7 A Bridge implementing the Scheduled Traffic model (48.2.9) implements the YANG modules in
8 Table 48-10.

Table 48-10—Scheduled Traffic model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-sched
ieee802-dot1q-sched-bridge

1 48.3.10 Frame Preemption model

2 A Bridge implementing the Frame Preemption model (48.2.10) implements the YANG modules in
3 Table 48-11.

Table 48-11—Frame Preemption model YANG modules

YANG module
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-preemption
ieee802-dot1q-preemption-bridge

4 48.3.11 Per-Stream Filtering and Policing model

5 A Bridge implementing the Per-Stream Filtering and Policing model (48.2.11) implements the YANG
6 modules in Table 48-12.

Table 48-12—Per-Stream Filtering and Policing model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-stream-filters-gates
ieee802-dot1q-stream-filters-gates-bridge
ieee802-dot1q-psfp
ieee802-dot1q-psfp-bridge

1 48.3.12 User/Network Interface model

2 A station implementing the CNC-configuration model (48.2.12) implements the YANG modules in
3 Table 48-13.

Table 48-13—CNC-configuration model YANG modules

YANG module
ieee802-dot1q-tsn-types
ieee802-dot1q-cnc-config

4 48.3.13 Credit-based shaper algorithm model

5 A bridge implementing the credit-based shaper algorithm model (48.2.13) implements the YANG modules
6 in Table 48-14.

Table 48-14—Credit-based shaper algorithm model YANG modules

YANG Module
ieee802-types
ieee802-dot1q-types
ieee-802-dot1q-tsn-types
ieee802-dot1q-bridge
ieee802-dot1q-cbsa
ieee802-dot1q-cbsa-bridge

7 48.3.14 RSTP model

8 A bridge implementing the RSTP model (48.2.14) implements the YANG modules in Table 48-15.

Table 48-15—RSTP model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-rstp
ieee802-dot1q-rstp-bridge

1 **48.3.15 MSTP model**

2 A bridge implementing the MSTP model (48.2.15) implements the YANG modules in Table 48-16.

Table 48-16—MSTP model YANG modules

YANG module
ieee802-types
ieee802-dot1q-types
ieee802-dot1q-bridge
ieee802-dot1q-rstp
ieee802-dot1q-rstp-bridge
ieee802-dot1q-mstp
ieee802-dot1q-mstp-bridge

3

1 48.4 Security considerations

2 The YANG modules defined in this clause are designed to be accessed via a network configuration protocol,
3 e.g., NETCONF protocol (IETF RFC 6241 [B39]). In the case of NETCONF, the lowest NETCONF layer is
4 the secure transport layer and the mandatory to implement secure transport is SSH (IETF RFC 6242 [B40]).
5 The NETCONF access control model (IETF RFC 6536 [B41]) provides the means to restrict access for
6 particular NETCONF users to a pre-configured subset of all available NETCONF protocol operations and
7 content.

8 It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the
9 system that support NETCONF, and any other remote configuration protocols that make use of these YANG
10 modules, are properly configured to allow access only to those principals (users) that have legitimate rights
11 to read or write data nodes. This standard does not specify how the credentials of those users are to be stored
12 or validated.

13 There are a number of management objects defined in the YANG modules specified in this standard that are
14 configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects may be considered sensitive
15 or vulnerable in some network environments. A network configuration protocol, such as NETCONF, can
16 support protocol operations that can edit or delete YANG module configuration data (e.g., edit-config,
17 delete-config, copy-config). If this is done in a non-secure environment without proper protection, then
18 negative effects on the network operation are possible.

19 48.4.1 Security considerations of the VLAN Bridge components model

20 The following objects in the ieee802-dot1q-bridge YANG module can be manipulated to interfere with the
21 operation of VLANs and priority classes. This could, for example, be used to force a reinitialization of state
22 machines, thus causing network instability, or to change the forwarding and filtering policies. Another
23 possibility would be for an attacker to override established policy on Port priorities, thus giving a user (or an
24 attacker) unauthorized preferential treatment.

25 interfaces/interface/bridge-port
26 interfaces/interface/bridge-port/priority-regeneration
27 interfaces/interface/bridge-port/transmission-selection-table
28 bridges/bridge/component/traffic-class-enabled
29 bridges/bridge/component/bridge-vlan/protocol-group-database
30 interfaces/interface/bridge-port/traffic-class
31 interfaces/interface/bridge-port/protocol-group-vid-set

- 32 a) The configurable object bridges/bridge/component/filtering-database/aging-time controls how fast
33 dynamically learned forwarding information is aged out. Setting this object to a large value may
34 simplify FDB overflow attacks. Setting this object to too small a value may compromise the
35 throughput of the network by causing excessive flooding.
- 36 b) The configurable object bridges/bridge/component/filtering-database/filtering-entries/entry-type
37 provides a filtering mechanism controlling which Ports frames originating from a specific source
38 may be forwarded to. Write access to this table can be used to turn provisioned filtering off or to add
39 filters to prevent rightful use of the network.

40 Some of the readable data in this YANG module may be considered sensitive or vulnerable in some network
41 environments. It is thus important to control all types of access (e.g., including NETCONF get, get-config
42 operations) to these objects and possibly to even encrypt the values of these objects when sending them over
43 the network. These tables and objects and their sensitivity/vulnerability are described as follows:

- 44 — The object bridges/bridge/component/capabilities could be used by an attacker to determine which
45 attacks might be useful to attempt against a given device.

1 — The readable objects defined in the `ieee802-dot1q-bridge` module provide information about the
2 topology of a bridged network and the attached active stations. The addresses listed in the
3 `bridges/bridge/component/filtering-database/filtering-entries` usually reveal information about the
4 manufacturer of the MAC hardware, which can be useful information for mounting other specific
5 attacks. In some networks, information about attached active stations can be considered personal
6 identifying information about the user of the station. Unauthorized use of these objects can be
7 considered a privacy threat.

8 **48.4.2 Security considerations of the Two-Port MAC Relay model**

9 The following objects in the `ieee802-dot1q-tpmr` YANG module could be manipulated to interfere with the
10 operation of MAC status propagation on a TPMR port and, for example, be used to cause network
11 instability:

12 `interfaces/interface/bridge-port/mac-status-propagation/link-notify`
13 `interfaces/interface/bridge-port/mac-status-propagation/link-notify-wait`
14 `interfaces/interface/bridge-port/mac-status-propagation/link-notify-retry`
15 `interfaces/interface/bridge-port/mac-status-propagation/mac-notify`
16 `interfaces/interface/bridge-port/mac-status-propagation/mac-notify-time`
17 `interfaces/interface/bridge-port/mac-status-propagation/mac-recover-time`

18 **48.4.3 Security considerations of the Customer VLAN Bridge model**

19 The Customer VLAN Bridge YANG model is based on, and has the same security considerations as, the
20 VLAN Bridge components model. See 48.4.1.

21 **48.4.4 Security considerations of the Provider Bridge model**

22 The following objects in the `ieee802-dot1q-pb` YANG module can be manipulated to interfere with the
23 operation of VLANs. This could, for example, be used to force a reinitialization of state machines, thus
24 causing network instability, or to change the forwarding and filtering policies.

25 `interfaces/interface/bridge-port`
26 `interfaces/interface/bridge-port/cvid-registration`
27 `interfaces/interface/bridge-port/service-priority-regeneration`

28 **48.4.5 Security considerations of the CFM model**

29 The following objects in the `ieee802-dot1q-cfm` YANG module could be manipulated to interfere with the
30 operation of a CFM port. This could, for example, be used to force a reinitialization of CFM state machines,
31 thus causing network instability, or to change the management policies.

32 `cfm/maintenance-domain`
33 `cfm/maintenance-group`

34 See 48.4.1 for related `ieee802-dot1q-bridge` YANG model security considerations.

35 **48.4.6 Security considerations of the Stream filters and stream gates model**

36 The following objects in the `ieee802-dot1q-stream-filters-gates` YANG module could be manipulated to
37 interfere with the operation of stream filtering and gating. This could, for example, be used to force a
38 reinitialization of PSFP or ATS state machines, thus causing network instability.

39 `bridges/bridge/component/stream-filters`

1 bridges/bridge/component/stream-gates

2 See 48.4.1 for related ieee802-dot1q-bridge YANG model security considerations.

3 **48.4.7 Security considerations of the Asynchronous Traffic Shaping model**

4 The following objects in the ieee802-dot1q-ats YANG module could be manipulated to interfere with the
5 operation of stream filtering. This could, for example, be used to force a reinitialization of ATS state
6 machines, thus causing network instability.

7 bridges/bridge/component/stream-filters

8 interfaces/interface/bridge-port/ats-port-parameters

9 bridges/bridge/component/schedulers

10 bridges/bridge/component/scheduler-groups

11 See 48.4.1 and 48.4.6 for related ieee802-dot1q-bridge and ieee802-dot1q-stream-filters-gates YANG model
12 security considerations.

13 **48.4.8 Security considerations of the Congestion Isolation model**

14 The following objects in the ieee802-dot1q-congestion-isolation YANG module could be manipulated to
15 interfere with the operation of MAC status propagation on a TPMR port and, for example, be used to cause
16 network instability:

17 bridges/bridge/component/ci-master-enable

18 bridges/bridge/component/ci-peers/ci-peer-table

19 bridges/bridge/component/ci-streams

20 bridges/bridge/component/stream-gates/stream-gate-instance-table

21 bridges/bridge/component/stream-filters/stream-filter-instance-table

22 bridges/bridge/component/queue-map

23 bridges/bridge/component/min-header-octets

24 bridges/bridge/component/min-cim-tx

25 **48.4.9 Security considerations of the Scheduled Traffic model**

26 The following objects in the ieee802-dot1q-sched YANG module could be manipulated to interfere with the
27 operation of egress gating. This could, for example, be used to force a reinitialization of Scheduled Traffic
28 state machines, thus causing network instability.

29 interfaces/interface/bridge-port/gate-parameter-table

30 See 48.4.1 for related ieee802-dot1q-bridge YANG model security considerations.

31 **48.4.10 Security considerations of the Frame Preemption model**

32 The following objects in the ieee802-dot1q-preemption YANG module could be manipulated to interfere
33 with the operation of preemption. This could, for example, be used to force a reinitialization of the Frame
34 Preemption status table, thus causing network instability.

35 interfaces/interface/bridge-port/frame-preemption-parameters

36 See 48.4.1 for related ieee802-dot1q-bridge YANG model security considerations.

1 **48.4.11 Security considerations of the Per-Stream Filtering and Policing model**

2 The following objects in the ieee802-dot1q-psfp YANG module could be manipulated to interfere with the
3 operation of stream filtering and gating. This could, for example, be used to force a reinitialization of PSFP
4 state machines, thus causing network instability.

5 bridges/bridge/component/stream-filters

6 bridges/bridge/component/stream-gates

7 bridges/bridge/component/flow-meters

8 See 48.4.1 for related ieee802-dot1q-bridge YANG model security considerations.

9 **48.4.12 Security considerations of the CNC-configuration model**

10 The ieee802-dot1q-cnc-config YANG module is structured in a way that allows access to specific parts of
11 the YANG tree to be restricted, e.g., by using standard mechanisms as described in IETF RFC 8341 [B48].

12 The following objects in the ieee802-dot1q-cnc-config YANG module could be manipulated to interfere
13 with the operation of streams in a configuration domain and, for example, be used to cause network
14 instability:

15 cnc-config/domain/cuc/stream

16 cnc-config/domain/cuc/remove_streams

17 **48.4.13 Security considerations of the credit-based shaper algorithm model**

18 The following objects in the ieee802-dot1q-cbsa YANG module could be manipulated to interfere with the
19 operation of the credit-based shaper algorithm. This could, for example, be used to force a reinitialization of
20 credit-based shaper algorithm state, thus causing network instability.

21 interfaces/interface/bridge-port/cbsa/cbsa-parameter-table

22 **48.4.14 Security considerations of the RSTP model**

23 All writeable nodes in the ieee802-dot1q-rstp YANG module could be manipulated to interfere with basic
24 networking connectivity.

25 See 48.4.1 for related ieee802-dot1q-bridge YANG model security considerations.

26 **48.4.15 Security considerations of the Multiple Spanning Trees model**

27 All writeable nodes in the ieee802-dot1q-mstp YANG module could be manipulated to interfere with basic
28 networking connectivity.

29 See 48.4.1 for related ieee802-dot1q-bridge YANG model security considerations.

30

1 48.5 YANG schema tree definitions

2 A simplified graphical representation of the data model is used in this document. The meaning of the
3 symbols in these diagrams is as follows:

- 4 — Brackets “[” and “]” enclose list keys.
- 5 — Abbreviations before data node names: “rw” means configuration (read-write), and “ro” means state
6 data (read-only).
- 7 — Symbols after data node names: “?” means an optional node, “!” means a presence container, and
8 “*” denotes a list and leaf-list.
- 9 — Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (“:”).
- 10 — Ellipsis (“...”) stands for contents of subtrees that are not shown.

11 48.5.1 Schema for the ieee802-types YANG module

12 This YANG module does not have a YANG schema tree.

13 48.5.2 Schema for the ieee802-dot1q-types YANG module

14 This YANG module does not have a YANG schema tree.

15 48.5.3 Schema for the ieee802-dot1q-tsn-types YANG module

16 This YANG module does not have a YANG schema tree.

17 48.5.4 Schema for the ieee802-dot1q-bridge YANG module

```

18 module: ieee802-dot1q-bridge
19   +--rw bridges
20     +--rw bridge* [name]
21       +--rw name          dot1qtypes:name-type
22       +--rw address        ieee:mac-address
23       +--rw bridge-type    identityref
24       +--ro ports?         uint16
25       +--ro up-time?       yang:zero-based-counter32
26       +--ro components?    uint32
27       +--rw component* [name]
28         +--rw name          string
29         +--rw id?            uint32
30         +--rw type           identityref
31         +--rw address?       ieee:mac-address
32         +--rw traffic-class-enabled? boolean
33         +--ro ports?         uint16
34         +--ro bridge-port*    if:interface-ref
35         +--ro capabilities
36           | +--ro extended-filtering?    boolean
37           | +--ro traffic-classes?        boolean
38           | +--ro static-entry-individual-port? boolean
39           | +--ro ivl-capable?            boolean
40           | +--ro svl-capable?            boolean
41           | +--ro hybrid-capable?         boolean
42           | +--ro configurable-pvid-tagging? boolean
43           | +--ro local-vlan-capable?     boolean
44         +--rw filtering-database
45           | +--rw aging-time?            uint32
46           | +--ro size?                  yang:gauge32
47           | +--ro static-entries?        yang:gauge32
48           | +--ro dynamic-entries?       yang:gauge32
49           | +--ro static-vlan-registration-entries? yang:gauge32
50           | +--ro dynamic-vlan-registration-entries? yang:gauge32
51           | +--ro mac-address-registration-entries? yang:gauge32
52 {extended-filtering-services}?
53   | +--rw filtering-entry* [database-id vids address]
54   | | +--rw database-id    uint32

```

```

1      | | +--rw address          ieee:mac-address
2      | | +--rw vids          dot1qtypes:vid-range-type
3      | | +--rw entry-type?   enumeration
4      | | +--rw port-map* [port-ref]
5      | | | +--rw port-ref          port-number-type
6      | | | +--rw (map-type)?
7      | | | | +--:(static-filtering-entries)
8      | | | | | +--rw static-filtering-entries
9      | | | | | +--rw control-element?   enumeration
10     | | | | | +--rw connection-identifier? port-number-type
11     | | | | +--:(static-vlan-registration-entries)
12     | | | | | +--rw static-vlan-registration-entries
13     | | | | | +--rw registrar-admin-control?   enumeration
14     | | | | | +--rw vlan-transmitted?   enumeration
15     | | | | +--:(mac-address-registration-entries)
16     | | | | | +--rw mac-address-registration-entries
17     | | | | | +--rw control-element?   enumeration
18     | | | | +--:(dynamic-vlan-registration-entries)
19     | | | | | +--rw dynamic-vlan-registration-entries
20     | | | | | +--rw control-element?   enumeration
21     | | | | +--:(dynamic-reservation-entries)
22     | | | | | +--rw dynamic-reservation-entries
23     | | | | | +--rw control-element?   enumeration
24     | | | | +--:(dynamic-filtering-entries)
25     | | | | | +--rw dynamic-filtering-entries
26     | | | | | +--rw control-element?   enumeration
27     | | +--ro status?   enumeration
28     | +--rw vlan-registration-entry* [database-id vids]
29     | | +--rw database-id   uint32
30     | | +--rw vids          dot1qtypes:vid-range-type
31     | | +--rw entry-type?   enumeration
32     | | +--rw port-map* [port-ref]
33     | | | +--rw port-ref          port-number-type
34     | | | +--rw (map-type)?
35     | | | | +--:(static-filtering-entries)
36     | | | | | +--rw static-filtering-entries
37     | | | | | +--rw control-element?   enumeration
38     | | | | | +--rw connection-identifier? port-number-type
39     | | | | +--:(static-vlan-registration-entries)
40     | | | | | +--rw static-vlan-registration-entries
41     | | | | | +--rw registrar-admin-control?   enumeration
42     | | | | | +--rw vlan-transmitted?   enumeration
43     | | | | +--:(mac-address-registration-entries)
44     | | | | | +--rw mac-address-registration-entries
45     | | | | | +--rw control-element?   enumeration
46     | | | | +--:(dynamic-vlan-registration-entries)
47     | | | | | +--rw dynamic-vlan-registration-entries
48     | | | | | +--rw control-element?   enumeration
49     | | | | +--:(dynamic-reservation-entries)
50     | | | | | +--rw dynamic-reservation-entries
51     | | | | | +--rw control-element?   enumeration
52     | | | | +--:(dynamic-filtering-entries)
53     | | | | | +--rw dynamic-filtering-entries
54     | | | | | +--rw control-element?   enumeration
55     +--rw permanent-database
56     | +--ro size?   yang:gauge32
57     | +--ro static-entries?   yang:gauge32
58     | +--ro static-vlan-registration-entries?   yang:gauge32
59     | +--rw filtering-entry* [database-id vids address]
60     | | +--rw database-id   uint32
61     | | +--rw address          ieee:mac-address
62     | | +--rw vids          dot1qtypes:vid-range-type
63     | | +--ro status?   enumeration
64     | | +--rw port-map* [port-ref]
65     | | | +--rw port-ref          port-number-type
66     | | | +--rw (map-type)?
67     | | | | +--:(static-filtering-entries)
68     | | | | | +--rw static-filtering-entries
69     | | | | | +--rw control-element?   enumeration
70     | | | | | +--rw connection-identifier? port-number-type
71     | | | | +--:(static-vlan-registration-entries)
72     | | | | | +--rw static-vlan-registration-entries

```

```

1      |      |      +---rw registrar-admin-control?  enumeration
2      |      |      +---rw vlan-transmitted?  enumeration
3      |      |      +---:(mac-address-registration-entries)
4      |      |      |      +---rw mac-address-registration-entries
5      |      |      |      +---rw control-element?  enumeration
6      |      |      +---:(dynamic-vlan-registration-entries)
7      |      |      |      +---rw dynamic-vlan-registration-entries
8      |      |      |      +---rw control-element?  enumeration
9      |      |      +---:(dynamic-reservation-entries)
10     |      |      |      +---rw dynamic-reservation-entries
11     |      |      |      +---rw control-element?  enumeration
12     |      |      +---:(dynamic-filtering-entries)
13     |      |      +---rw dynamic-filtering-entries
14     |      |      +---rw control-element?  enumeration
15     +---rw bridge-vlan
16     |      +---ro version?                uint16
17     |      +---ro max-vids?                uint16
18     |      +---ro override-default-pvid?   boolean
19     |      |      +---ro protocol-template? dot1qtypes:protocol-frame-format-type
20 {port-and-protocol-based-vlan}?
21     |      +---ro max-msti?                uint16
22     |      +---rw vlan* [vid]
23     |      |      +---rw vid                dot1qtypes:vlan-index-type
24     |      |      +---rw name?              dot1qtypes:name-type
25     |      |      +---ro untagged-ports*    if:interface-ref
26     |      |      +---ro egress-ports*     if:interface-ref
27     |      |      +---rw protocol-group-database* [db-index] {port-and-protocol-based-vlan}?
28     |      |      |      +---rw db-index          uint16
29     |      |      |      +---rw frame-format-type? dot1qtypes:protocol-frame-format-type
30     |      |      |      +---rw (frame-format)?
31     |      |      |      |      +---:(ethernet-rfc1042-snap8021H)
32     |      |      |      |      |      +---rw ethertype? dot1qtypes:ethertype-type
33     |      |      |      |      +---:(snap-other)
34     |      |      |      |      |      +---rw protocol-id? string
35     |      |      |      |      +---:(llc-other)
36     |      |      |      |      +---rw dsap-ssap-pairs
37     |      |      |      |      +---rw llc-address? string
38     |      |      |      +---rw group-id?      uint32
39     |      |      +---rw vid-to-fid-allocation* [vids]
40     |      |      |      +---rw vids            dot1qtypes:vid-range-type
41     |      |      |      +---ro fid?            uint32
42     |      |      |      +---ro allocation-type? enumeration
43     |      |      +---rw fid-to-vid-allocation* [fid]
44     |      |      |      +---rw fid            uint32
45     |      |      |      +---ro allocation-type? enumeration
46     |      |      |      +---ro vid*           dot1qtypes:vlan-index-type
47     |      |      +---rw vid-to-fid* [vid]
48     |      |      |      +---rw vid            dot1qtypes:vlan-index-type
49     |      |      |      +---rw fid?          uint32
50     +---rw bridge-mst
51     |      +---rw mstid*                    dot1qtypes:mstid-type
52     |      +---rw fid-to-mstid* [fid]
53     |      |      +---rw fid            uint32
54     |      |      +---rw mstid?         dot1qtypes:mstid-type
55     |      +---rw fid-to-mstid-allocation* [fids]
56     |      |      +---rw fids            dot1qtypes:vid-range-type
57     |      |      +---rw mstid?         dot1qtypes:mstid-type
58
59     augment /if:interfaces/if:interface:
60     +---rw bridge-port
61     |      +---rw bridge-name?              -> /bridges/bridge/name
62     |      +---rw component-name?            ->
63 /bridges/bridge[dot1q:name=current()/../bridge-name]/component/name
64     +---rw port-type?                        identityref
65     +---rw pvid?                            dot1qtypes:vlan-index-type
66     +---rw default-priority?                 dot1qtypes:priority-type
67     +---rw priority-regeneration
68     |      +---rw priority0?    priority-type
69     |      +---rw priority1?    priority-type
70     |      +---rw priority2?    priority-type
71     |      +---rw priority3?    priority-type
72     |      +---rw priority4?    priority-type

```

```

1      |  +--rw priority5?   priority-type
2      |  +--rw priority6? priority-type
3      |  +--rw priority7? priority-type
4      +--rw pcp-selection?                               dot1qtypes:pcp-selection-type
5      +--rw pcp-decoding-table
6      |  +--rw pcp-decoding-map* [pcp]
7      |  |  +--rw pcp                               pcp-selection-type
8      |  |  +--rw priority-map* [priority-code-point]
9      |  |  |  +--rw priority-code-point           priority-type
10     |  |  |  +--rw priority?                     priority-type
11     |  |  |  +--rw drop-eligible?                 boolean
12     +--rw pcp-encoding-table
13     |  +--rw pcp-encoding-map* [pcp]
14     |  |  +--rw pcp                               pcp-selection-type
15     |  |  +--rw priority-map* [priority dei]
16     |  |  |  +--rw priority                       priority-type
17     |  |  |  +--rw dei                           boolean
18     |  |  |  +--rw priority-code-point?           priority-type
19     +--rw use-dei?                                   boolean
20     +--rw drop-encoding?                             boolean
21     +--rw service-access-priority-selection?         boolean
22     +--rw service-access-priority
23     |  +--rw priority0?   priority-type
24     |  +--rw priority1?   priority-type
25     |  +--rw priority2?   priority-type
26     |  +--rw priority3?   priority-type
27     |  +--rw priority4?   priority-type
28     |  +--rw priority5?   priority-type
29     |  +--rw priority6?   priority-type
30     |  +--rw priority7?   priority-type
31     +--rw traffic-class
32     |  x--rw traffic-class-map* [priority]
33     |  |  +--rw priority           priority-type
34     |  |  +--rw available-traffic-class* [num-traffic-class]
35     |  |  |  +--rw num-traffic-class uint8
36     |  |  |  +--rw traffic-class?   traffic-class-type
37     |  +--rw traffic-class-table
38     |  |  +--rw number-of-traffic-classes? uint8
39     |  |  +--rw priority0?                 traffic-class-type
40     |  |  +--rw priority1?                 traffic-class-type
41     |  |  +--rw priority2?                 traffic-class-type
42     |  |  +--rw priority3?                 traffic-class-type
43     |  |  +--rw priority4?                 traffic-class-type
44     |  |  +--rw priority5?                 traffic-class-type
45     |  |  +--rw priority6?                 traffic-class-type
46     |  |  +--rw priority7?                 traffic-class-type
47     +--rw transmission-selection-algorithm-table
48     |  +--rw transmission-selection-algorithm-map* [traffic-class]
49     |  |  +--rw traffic-class           traffic-class-type
50     |  |  +--rw transmission-selection-algorithm? identityref
51     +--rw acceptable-frame?             enumeration
52     +--rw enable-ingress-filtering?      boolean
53     +--rw enable-restricted-vlan-registration? boolean
54     +--rw enable-vid-translation-table?  boolean
55     +--rw enable-egress-vid-translation-table? boolean
56     +--rw protocol-group-vid-set* [group-id] {port-and-protocol-based-vlan}?
57     |  +--rw group-id   uint32
58     |  +--rw vid*       dot1qtypes:vlanid
59     +--rw admin-point-to-point?          enumeration
60     |  +--ro protocol-based-vlan-classification? boolean
61 {port-and-protocol-based-vlan}?
62     +--ro max-vid-set-entries?            uint16 {port-and-protocol-based-vlan}?
63     +--ro port-number?                   dot1qtypes:port-number-type
64     +--ro address?                       ieee:mac-address
65     +--ro capabilities?                  bits
66     +--ro type-capabilities?             bits
67     +--ro external?                      boolean
68     +--ro oper-point-to-point?           boolean
69     +--ro media-dependent-overhead?      uint8
70     +--ro statistics
71     |  +--ro delay-exceeded-discards?    yang:counter64
72     |  +--ro mtu-exceeded-discards?      yang:counter64

```

```

1      | +--ro frame-rx?                yang:counter64
2      | +--ro octets-rx?            yang:counter64
3      | +--ro frame-tx?            yang:counter64
4      | +--ro octets-tx?            yang:counter64
5      | +--ro discard-inbound?     yang:counter64
6      | +--ro forward-outbound?    yang:counter64
7      | +--ro discard-lack-of-buffers? yang:counter64
8      | +--ro discard-transit-delay-exceeded? yang:counter64
9      | +--ro discard-on-error?     yang:counter64
10     | +--ro discard-on-ingress-filtering? yang:counter64 {ingress-filtering}?
11     +--rw vid-translations* [local-vid]
12     | +--rw local-vid      dot1qtypes:vlanid
13     | +--rw relay-vid?     dot1qtypes:vlanid
14     +--rw egress-vid-translations* [relay-vid]
15     +--rw relay-vid      dot1qtypes:vlanid
16     +--rw local-vid?     dot1qtypes:vlanid
17

```

48.5.5 Schema for the ieee802-dot1q-tpmr YANG module

```

19 module: ieee802-dot1q-tpmr
20
21   augment /if:interfaces/if:interface/dot1q:bridge-port:
22     +--rw managed-address?      boolean
23     +--rw mac-status-propagation
24       +--rw link-notify?        boolean
25       +--rw link-notify-wait?   yang:timeticks
26       +--rw link-notify-retry?  yang:timeticks
27       +--rw mac-notify?         boolean
28       +--rw mac-notify-time?    yang:timeticks
29       +--rw mac-recover-time?   yang:timeticks
30   augment /if:interfaces/if:interface/dot1q:bridge-port/dot1q:statistics:
31     +--ro acks-tx?              yang:counter64
32     +--ro add-notifications-tx? yang:counter64
33     +--ro add-confirmations-tx? yang:counter64
34     +--ro loss-notification-tx? yang:counter64
35     +--ro loss-confirmation-tx? yang:counter64
36     +--ro acks-rx?              yang:counter64
37     +--ro add-notifications-rx? yang:counter64
38     +--ro add-confirmations-rx? yang:counter64
39     +--ro loss-notification-rx? yang:counter64
40     +--ro loss-confirmation-rx? yang:counter64
41     +--ro add-events?           yang:counter64
42     +--ro loss-events?          yang:counter64
43     +--ro mac-status-notifications? yang:counter64
44

```

48.5.6 Schema for the ieee802-dot1q-pb YANG module

```

46 module: ieee802-dot1q-pb
47
48   augment /if:interfaces/if:interface/dot1q:bridge-port:
49     +--rw svid?                  dot1qtypes:vlanid
50     +--rw cvid-registration* [cvid]
51     | +--rw cvid                dot1qtypes:vlanid
52     | +--rw svid?               dot1qtypes:vlanid
53     | +--rw untagged-pep?        boolean
54     | +--rw untagged-cep?        boolean
55     +--rw service-priority-regeneration* [svid]
56     | +--rw svid                dot1qtypes:vlanid
57     | +--rw priority-regeneration
58     |   +--rw priority0?        priority-type
59     |   +--rw priority1?        priority-type
60     |   +--rw priority2?        priority-type
61     |   +--rw priority3?        priority-type
62     |   +--rw priority4?        priority-type
63     |   +--rw priority5?        priority-type
64     |   +--rw priority6?        priority-type
65     |   +--rw priority7?        priority-type
66     +--rw rcap-internal-interface* [external-svid]
67     +--rw external-svid          dot1qtypes:vlanid
68     +--rw internal-port-number?  dot1qtypes:port-number-type

```

```

1      +--rw internal-svid?          dot1qtypes:vlanid
2      +--rw internal-interface-type? enumeration
3

```

4 48.5.7 Schema for the ieee802-dot1q-cfm-types YANG module

5 This YANG module does not have a YANG schema tree.

6 48.5.8 Schema for the ieee802-dot1q-cfm YANG module

```

7 module: ieee802-dot1q-cfm
8   +--rw cfm
9     +--rw maintenance-domain* [md-id]
10    |   +--rw md-id          cfm-types:name-key-type
11    |   +--rw (md-name)?
12    |   |   +--:(none)
13    |   |   |   +--rw none?          empty
14    |   |   +--:(dns-like-name)
15    |   |   |   +--rw dns-like-name? string
16    |   |   +--:(mac-address-and-uint)
17    |   |   |   +--rw mac-address-and-uint-type
18    |   |   |   |   +--rw address      ieee:mac-address
19    |   |   |   |   +--rw int          uint16
20    |   |   +--:(char-string)
21    |   |   +--rw char-string?        string
22    |   +--rw md-level?                cfm-types:md-level-type
23    |   +--rw mhfc-creation?            cfm-types:mhf-creation-type
24    |   +--rw id-permission?            cfm-types:sender-id-permission-type
25    |   +--rw fault-alarm-transmission? cfm-types:fault-alarm-type
26    +--rw maintenance-association* [ma-id]
27    |   +--rw ma-id          cfm-types:name-key-type
28    |   +--rw (ma-name)
29    |   |   +--:(primary-vid)
30    |   |   |   +--rw primary-vid?    dot1q-types:vlanid
31    |   |   +--:(char-string)
32    |   |   |   +--rw char-string?     string
33    |   |   +--:(unsigned-int16)
34    |   |   |   +--rw unsigned-int16?  uint16
35    |   |   +--:(rfc2865-vpn-id)
36    |   |   |   +--rw vpn-id
37    |   |   |   |   +--rw vpn-oui      uint32
38    |   |   |   |   +--rw vpn-index   uint32
39    |   +--rw ccm-interval?            cfm-types:ccm-interval-type
40    |   +--rw fault-alarm-transmission? cfm-types:fault-alarm-type
41    |   +--rw mhfc-creation?            cfm-types:mhf-creation-type
42    |   +--rw id-permission?            cfm-types:sender-id-permission-type
43    |   +--rw maintenance-association-mep* [mep-id]
44    |   |   +--rw mep-id          cfm-types:mep-id-type
45    +--rw maintenance-group* [maintenance-group-id]
46    |   +--rw maintenance-group-id cfm-types:name-key-type
47    |   +--rw md-id                -> /cfm/maintenance-domain/md-id
48    |   +--rw ma-id                -> /cfm/maintenance-domain[md-id] =
49    current() / ../md-id / maintenance-association / ma-id
50    |   +--rw mep* [mep-id]
51    |   |   +--rw mep-id          -> /cfm/maintenance-domain[md-id] =
52    current() / ../md-id / maintenance-association [ma-id] =
53    current() / ../ma-id / maintenance-association-mep / mep-id
54    |   +--rw direction            cfm-types:mp-direction-type
55    |   +--rw enabled?              boolean
56    |   +--rw ccm-ltm-priority?     dot1q-types:priority-type
57    |   +--ro mac-address            ieee:mac-address
58    |   +--rw inactive-remote-mep* [inactive-rmep-id]
59    |   |   +--rw inactive-rmep-id -> /cfm/maintenance-domain[md-id] =
60    current() / ../ma-id / maintenance-association [ma-id] =
61    current() / ../ma-id / maintenance-association-mep / mep-id
62    |   +--ro mep-db* [rmep-id]
63    |   |   +--ro rmep-id          cfm-types:mep-id-type
64    |   |   +--ro rmep-state        cfm-types:remote-mep-state-type
65    |   |   +--ro rmep-failed-ok-time yang:timeticks
66    |   |   +--ro mac-address      ieee:mac-address

```



```

1      | +--ro rdi                                boolean
2      | +--ro port-status-tlv?                cfm-types:port-status-tlv-value-type
3      | +--ro interface-status-tlv?           cfm-types:interface-status-tlv-value-type
4      | +--ro chassis-id-subtype?             ieee:chassis-id-subtype-type
5      | +--ro chassis-id?                     ieee:chassis-id-type
6      | +--ro transport-service-domain
7      | | +--ro domain?                       yang:object-identifier-128
8      | | +--ro (management-address)
9      | | | +--:(ip)
10     | | | | +--ro ip-address                 inet:ip-address
11     | | | | +--ro ip-port                     inet:port-number
12     | | | +--:(local)
13     | | | | +--ro local-address               string
14     | | | +--:(dns)
15     | | | | +--ro dns-address                 string
16     | | | +--:(other)
17     | | | +--ro unknown-address?             binary
18     | +--ro rmep-is-active?                  boolean
19 +--rw continuity-check
20     | +--rw ccm-enabled?                      boolean
21     | +--ro fng-state?                        cfm-types:fng-state-type
22     | +--rw fault-alarm-transmission?        cfm-types:fault-alarm-type
23     | +--rw lowest-priority-defect?          cfm-types:lowest-alarm-priority-type
24     | +--rw fng-alarm-time?                  uint16
25     | +--rw fng-reset-time?                  uint16
26     | +--ro highest-priority-defect          cfm-types:highest-defect-priority-type
27     | +--ro defects                          cfm-types:mep-defects-type
28     | +--ro error-ccm-last-failure?          binary
29     | +--ro xcon-ccm-last-failure?           binary
30 +--ro stats
31     | +--ro mep-ccm-sequence-errors           yang:counter64
32     | +--ro mep-ccms-sent                     yang:counter64
33     | +--ro mep-lbr-in                       yang:counter64
34     | +--ro mep-lbr-in-out-of-order          yang:counter64
35     | +--ro mep-lbr-bad-msdu                 yang:counter64
36     | +--ro mep-unexpected-ltr-in            yang:counter64
37     | +--ro mep-lbr-out                      yang:counter64
38 +--ro linktrace-reply* [ltr-transaction-id]
39     | +--ro ltr-transaction-id                cfm-types:seq-number-type
40     | +--ro linktrace-input
41     | | +--ro (ltr-target)
42     | | | +--:(target-ucast-mac-address)
43     | | | | +--ro ltm-target-mac-address?    cfm-types:unicast-mac-address-type
44     | | | | +--:(target-mep-id)
45     | | | | +--ro ltm-target-mep-id?         cfm-types:mep-id-type
46     | | +--ro ltm-ttl?                        uint8
47     | | +--ro ltm-flags?                      cfm-types:mep-tx-ltm-flags-type
48     | +--ro responses* [ltr-receive-order]
49     | +--ro ltr-receive-order                  uint32
50     | +--ro ltr-ttl                            uint8
51     | +--ro ltr-forwarded                      boolean
52     | +--ro ltr-terminal-mep                  boolean
53     | +--ro ltr-last-egress-identifier
54     | | +--ro int                             uint16
55     | | +--ro address                         ieee:mac-address
56     | +--ro ltr-next-egress-identifier
57     | | +--ro int                             uint16
58     | | +--ro address                         ieee:mac-address
59     | | | | |
60 cfm-types:relay-action-field-value-type
61     | +--ro ltr-chassis-id-subtype?           ieee:chassis-id-subtype-type
62     | +--ro ltr-chassis-id                     ieee:chassis-id-type
63     | +--ro ltr-transport-service-domain
64     | | +--ro domain?                         yang:object-identifier-128
65     | | +--ro (management-address)
66     | | | +--:(ip)
67     | | | | +--ro ip-address                 inet:ip-address
68     | | | | +--ro ip-port                     inet:port-number
69     | | | +--:(local)
70     | | | | +--ro local-address               string
71     | | | +--:(dns)
72     | | | | +--ro dns-address                 string

```

```

1      |      |      +---:(other)
2      |      |      +---ro unknown-address?      binary
3      |      |      |      +---ro      ltr-ingress?
4 cfm-types:ingress-action-field-value-type
5      |      +---ro ltr-ingress-mac      ieee:mac-address
6      |      +---ro ltr-ingress-port-id-subtype?      ieee:port-id-subtype-type
7      |      +---ro ltr-ingress-port-id      ieee:port-id-type
8      |      |      |      +---ro      ltr-egress?
9 cfm-types:egress-action-field-value-type
10     |      +---ro ltr-egress-mac      ieee:mac-address
11     |      +---ro ltr-egress-port-id-subtype?      ieee:port-id-subtype-type
12     |      +---ro ltr-egress-port-id      ieee:port-id-type
13     |      +---ro ltr-organization-specific-tlv?      binary
14     +---x transmit-loopback
15     | +---w input
16     | | +---w (lbm-destination)
17     | | | +---:(dest-ucast-mac-address)
18     | | | |      |      |      |      +---w      lbm-dest-ucast-mac-address?
19 cfm-types:unicast-mac-address-type
20     | | | +---:(dest-mcast-class1-mac-address)
21     | | | |      |      |      |      +---w      lbm-dest-mcast-class1-mac-address?
22 cfm-types:multicast-class1-mac-address-type
23     | | | +---:(dest-mep-id)
24     | | | +---w lbm-dest-mep-id?      cfm-types:mep-id-type
25     | | +---w lbm-messages?      uint16
26     | | +---w lbm-priority?      dot1q-types:priority-type
27     | | +---w lbm-drop-eligible?      boolean
28     | | +---w lbm-data-tlv?      cfm-types:lbm-data-tlv-type
29     | +---ro output
30     | +---ro lbm-request-id      cfm-types:seq-number-type
31     +---x transmit-linktrace
32     +---w input
33     | +---w (ltr-target)
34     | | +---:(target-ucast-mac-address)
35     | | | +---w ltm-target-mac-address?      cfm-types:unicast-mac-address-type
36     | | | +---:(target-mep-id)
37     | | | +---w ltm-target-mep-id?      cfm-types:mep-id-type
38     | | +---w ltm-ttl?      uint8
39     | | +---w ltm-flags?      cfm-types:mep-tx-ltm-flags-type
40     +---ro output
41     +---ro ltm-transaction-id      cfm-types:seq-number-type
42     +---ro ltm-egress-identifier
43     +---ro int      uint16
44     +---ro address      ieee:mac-address
45

```

48.5.9 Schema for the ieee802-dot1q-cfm-bridge YANG module

```

47 module: ieee802-dot1q-cfm-bridge
48
49   augment /dot1q-cfm:cfm:
50     +---ro cfm-stack* [port service-selector service-id md-level direction]
51     | +---ro port      port-ref
52     | +---ro md-level      cfm-types:md-level-type
53     | +---ro direction      cfm-types:mp-direction-type
54     | +---ro service-selector      cfm-types:service-selector-type
55     | +---ro service-id      cfm-types:service-selector-value-type
56     | |      +---ro      maintenance-group-id      ->
57 /dot1q-cfm:cfm/maintenance-group/maintenance-group-id
58     | +---ro      mep-id      ->
59 /dot1q-cfm:cfm/maintenance-group[dot1q-cfm:maintenance-group-id]
60 current() /../maintenance-group-id] /mep/mep-id      =
61     | +---ro md-id?      -> /dot1q-cfm:cfm/maintenance-domain/md-id
62     | +---ro ma-id?      -> /dot1q-cfm:cfm/maintenance-domain[dot1q-cfm:md-id =
63 current() /../md-id] /maintenance-association/ma-id
64     | +---ro mac-address      ieee:mac-address
65     | +---ro maintenance-point      cfm-types:mp-type
66     +---rw default-md-level* [bridge-id component-id service-selector primary-service-id]
67     | +---rw bridge-id      bridge-ref
68     | +---rw component-id      -> /dot1q:bridges/bridge[dot1q:name =
69 current() /../bridge-id] /component/name
70     | +---rw service-selector      cfm-types:service-selector-type

```

```

1 | +--rw primary-service-id          cfm-types:service-selector-value-type
2 | +--rw associated-service-ids
3 | | +--rw (service-id)?
4 | |   +---:(vids)
5 | | | +--rw vid* [vlan-id]
6 | | | +--rw vlan-id          dot1q-types:vlanid
7 | | |   +---:(isid)
8 | | | | +--rw isid?          uint32
9 | | |   +---:(tesid)
10 | | | | +--rw tesid?         uint32
11 | | |   +---:(segid)
12 | | | | +--rw segid?        uint32
13 | | |   +---:(path-tesid)
14 | | | | +--rw path-tesid?   uint32
15 | | |   +---:(group-isid)
16 | | | | +--rw group-isid?   uint32
17 | +--ro md-status              boolean
18 | +--rw md-level               cfm-types:md-level-type
19 | +--rw mhf-creation?          cfm-types:mhf-creation-type
20 | +--rw id-permission?          cfm-types:sender-id-permission-type
21 +--ro config-error* [port service-selector service-id]
22   +--ro port                  port-ref
23   +--ro service-selector      cfm-types:service-selector-type
24   +--ro service-id            cfm-types:service-selector-value-type
25   +--ro error-type            cfm-types:config-error-type
26 augment /dot1q-cfm:cfm/dot1q-cfm:maintenance-group:
27   +--rw bridge-id?            bridge-ref
28   +--rw component-name        -> /dot1q:bridges/bridge[dot1q:name =
29 current()/../cfm-bridge:bridge-id]/dot1q:component/name
30   +--rw service-id
31   +--rw (service-id)?
32   +---:(vids)
33   | +--rw vid* [vlan-id]
34   | +--rw vlan-id          dot1q-types:vlanid
35   +---:(isid)
36   | +--rw isid?            uint32
37   +---:(tesid)
38   | +--rw tesid?           uint32
39   +---:(segid)
40   | +--rw segid?           uint32
41   +---:(path-tesid)
42   | +--rw path-tesid?     uint32
43   +---:(group-isid)
44   +--rw group-isid?        uint32
45 augment /dot1q-cfm:cfm/dot1q-cfm:maintenance-group/dot1q-cfm:mep:
46   +--rw port                  port-ref
47   +--rw primary-vid?         -> ../../service-id/vid/vlan-id
48

```

48.5.10 Schema for the ieee802-dot1q-cfm-alarm YANG module

```

50 module: ieee802-dot1q-cfm-alarm
51
52 augment /dot1q-cfm:cfm/dot1q-cfm:maintenance-group/dot1q-cfm:mep:
53   +---n mep-fault-alarm
54   +---ro mep-priority-defect ->
55   ../../dot1q-cfm:continuity-check/highest-priority-defect
56

```

48.5.11 Schema for the ieee802-dot1q-stream-filters-gates YANG module

This YANG module does not have a YANG schema tree

48.5.12 Schema for the ieee802-dot1q-stream-filters-gates-bridge YANG module

```

60 module: ieee802-dot1q-stream-filters-gates-bridge
61
62 augment /dot1q:bridges/dot1q:bridge/dot1q:component:
63   +--rw stream-gates
64   | +--rw stream-gate-instance-table* [stream-gate-instance-id]

```

```

1  | | +--rw stream-gate-instance-id      uint32
2  | | +--rw gate-enable?                boolean
3  | | +--rw admin-gate-states?          gate-state-value-type
4  | | +--rw admin-ipv?                  ipv-spec-type
5  | +--ro max-stream-gate-instances?    uint32
6  +--rw stream-filters
7  +--rw stream-filter-instance-table* [stream-filter-instance-id]
8  | +--rw stream-filter-instance-id      uint32
9  | +--rw (stream-handle-spec)?
10 | | +--:(wildcard)
11 | | | +--rw wildcard?                  empty
12 | | +--:(stream-handle)
13 | | +--rw stream-handle                uint32
14 | +--rw priority-spec
15 | | priority-spec-type
16 | +--rw max-sdu-size                    uint32
17 | +--rw stream-blocked-due-to-oversize-frame-enabled? boolean
18 | +--rw stream-blocked-due-to-oversize-frame? boolean
19 | +--rw stream-gate-ref                 leafref
20 +--ro max-stream-filter-instances?     uint32
21

```

48.5.13 Schema for the ieee802-dot1q-ats YANG module

```

23 module: ieee802-dot1q-ats
24 +--rw stream-gates
25 | +--rw stream-gate-instance-table* [stream-gate-instance-id]
26 | | +--rw stream-gate-instance-id      uint32
27 | | +--rw gate-enable?                boolean
28 | | +--rw admin-gate-states?          gate-state-value-type
29 | | +--rw admin-ipv?                  ipv-spec-type
30 | +--ro max-stream-gate-instances?    uint32
31 +--rw stream-filters
32 +--rw stream-filter-instance-table* [stream-filter-instance-id]
33 | +--rw stream-filter-instance-id      uint32
34 | +--rw (stream-handle-spec)?
35 | | +--:(wildcard)
36 | | | +--rw wildcard?                  empty
37 | | +--:(stream-handle)
38 | | +--rw stream-handle                uint32
39 | +--rw priority-spec
40 | | priority-spec-type
41 | +--rw max-sdu-size                    uint32
42 | +--rw stream-blocked-due-to-oversize-frame-enabled? boolean
43 | +--rw stream-blocked-due-to-oversize-frame? boolean
44 | +--rw stream-gate-ref                 leafref
45 +--rw schedulers
46 | +--rw scheduler-instance-table* [scheduler-instance-id]
47 | | +--rw scheduler-instance-id        uint32
48 | | +--rw committed-information-rate    uint64
49 | | +--rw committed-burst-size         uint32
50 | | +--rw scheduler-group-ref           leafref
51 | +--ro max-scheduler-instances?       uint32
52 +--rw scheduler-groups
53 | +--rw scheduler-group-instance-table*
54 | | | [scheduler-group-instance-id]
55 | | | +--rw scheduler-group-instance-id uint32
56 | | | +--rw max-residence-time          uint32
57 | | +--ro max-scheduler-group-instances? uint32
58 | | +--rw scheduler-timing-characteristics
59 | | +--ro scheduler-timing-characteristics-table*
60 | | | [reception-port transmission-port]
61 | | | +--ro reception-port
62 | | | | dot1qtypes:port-number-type
63 | | | +--ro transmission-port
64 | | | | dot1qtypes:port-number-type
65 | | +--ro clock-offset-variation-max    uint32
66 | | +--ro clock-rate-deviation-max      uint32
67 | | +--ro arrival-recognition-delay-max uint32
68 | | +--ro processing-delay-min          uint32
69 | | +--ro processing-delay-max          uint32
70 +--rw scheduler

```

```
1      |      +--rw scheduler-ref?      leafref
2      |      +--rw scheduler-enable?  boolean
3      +--ro max-stream-filter-instances? uint32
4
```

1 48.5.14 Schema for the ieee802-dot1q-ats-bridge YANG module

```

2 module: ieee802-dot1q-ats-bridge
3
4   augment /if:interfaces/if:interface/dot1q:bridge-port:
5     +---ro discarded-frames-count?   yang:counter64
6   augment /dot1q:bridges/dot1q:bridge/dot1q:component:
7     +---rw stream-gates
8       | +---rw stream-gate-instance-table* [stream-gate-instance-id]
9       | | +---rw stream-gate-instance-id   uint32
10      | | +---rw gate-enable?               boolean
11      | | +---rw admin-gate-states?         gate-state-value-type
12      | | +---rw admin-ipv?                 ipv-spec-type
13      | +---ro max-stream-gate-instances?   uint32
14    +---rw stream-filters
15      +---rw stream-filter-instance-table* [stream-filter-instance-id]
16      | +---rw stream-filter-instance-id   uint32
17      | +---rw (stream-handle-spec)?
18      | | +---:(wildcard)
19      | | | +---rw wildcard?               empty
20      | | +---:(stream-handle)
21      | | | +---rw stream-handle           uint32
22      | +---rw priority-spec
23      | | priority-spec-type
24      | +---rw max-sdu-size                 uint32
25      | +---rw stream-blocked-due-to-oversize-frame-enabled? boolean
26      | +---rw stream-blocked-due-to-oversize-frame?   boolean
27      | +---rw stream-gate-ref              leafref
28      +---rw schedulers
29        | +---rw scheduler-instance-table* [scheduler-instance-id]
30        | | +---rw scheduler-instance-id   uint32
31        | | +---rw committed-information-rate   uint64
32        | | +---rw committed-burst-size       uint32
33        | | +---rw scheduler-group-ref        leafref
34        | | +---ro max-scheduler-instances?   uint32
35        +---rw scheduler-groups
36          | +---rw scheduler-group-instance-table*
37          | | [scheduler-group-instance-id]
38          | | +---rw scheduler-group-instance-id   uint32
39          | | +---rw max-residence-time           uint32
40          | | +---ro max-scheduler-group-instances?   uint32
41          | +---rw scheduler-timing-characteristics
42          | | +---ro scheduler-timing-characteristics-table*
43          | | | [reception-port transmission-port]
44          | | | +---ro reception-port
45          | | | | dot1qtypes:port-number-type
46          | | | +---ro transmission-port
47          | | | | dot1qtypes:port-number-type
48          | | | +---ro clock-offset-variation-max   uint32
49          | | | +---ro clock-rate-deviation-max     uint32
50          | | | +---ro arrival-recognition-delay-max   uint32
51          | | | +---ro processing-delay-min          uint32
52          | | | +---ro processing-delay-max          uint32
53          +---rw scheduler
54            +---rw scheduler-ref?   leafref
55            +---rw scheduler-enable? boolean
56          +---ro max-stream-filter-instances?   uint32
57

```

1 48.5.15 Schema for the ieee802-dot1q-congestion-isolation YANG module

```
2 module: ieee802-dot1q-congestion-isolation
3   +--rw stream-gates
4     | +--rw stream-gate-instance-table* [stream-gate-instance-id]
5     | | +--rw stream-gate-instance-id      uint32
6     | | +--rw gate-enable?                 boolean
7     | | +--rw admin-gate-states?           gate-state-value-type
8     | | +--rw admin-ipv?                   ipv-spec-type
9     | +--ro max-stream-gate-instances?     uint32
10  +--rw stream-filters
11    +--rw stream-filter-instance-table* [stream-filter-instance-id]
12    | +--rw stream-filter-instance-id      uint32
13    | +--rw (stream-handle-spec)?
14    | | +--:(wildcard)
15    | | | +--rw wildcard?                  empty
16    | | +--:(stream-handle)
17    | | | +--rw stream-handle              uint32
18    | | +--:(null-handle) {congestion-isolation}?
19    | | +--rw null-handle?                 empty
20    | +--rw priority-spec
21    | | priority-spec-type
22    | +--rw max-sdu-size                    uint32
23    | +--rw stream-blocked-due-to-oversize-frame-enabled? boolean
24    | +--rw stream-blocked-due-to-oversize-frame? boolean
25    | +--rw stream-gate-ref                 leafref
26    +--ro max-stream-filter-instances?     uint32
27
28
```

1 48.5.16 Schema for the ieee802-dot1q-congestion-isolation-bridge YANG module

```

2 module: ieee802-dot1q-congestion-isolation-bridge
3
4   augment /dot1q:bridges/dot1q:bridge/dot1q:component:
5     +--rw ci-master-enable?          boolean {congestion-isolation-bridge}?
6     +--ro ci-cim-tx-priority?        dot1q-types:priority-type
7     | {congestion-isolation-bridge}?
8     +--ro ci-max-flow-life?          uint32 {congestion-isolation-bridge}?
9     +--rw ci-peers {congestion-isolation-bridge}?
10    | +--rw ci-peer-table* [reception-port]
11    | | +--rw reception-port          dot1q-types:port-number-type
12    | | +--rw cim-type?                enumeration
13    | | +--rw peer-mac-address?        ieee:mac-address
14    | | +--rw peer-ipv4-address?       inet:ipv4-address
15    | | +--rw peer-ipv6-address?       inet:ipv6-address
16    | | +--rw peer-udp-port?           inet:port-number
17    | | +--rw peer-cim-encap-len?      uint16
18    | +--ro max-ci-peer-entries?       uint32
19    +--rw ci-streams {congestion-isolation-bridge}?
20    | +--ro ci-stream-table* [stream-handle-id]
21    | | +--ro stream-handle-id         uint32
22    | | +--ro cim-count?               uint16
23    | | +--ro create-time?             yang:timeticks
24    | | +--ro create-mask?             bits
25    | +--ro queue-key?                 uint16
26    | +--ro dest-mac-address?          ieee:mac-address
27    | +--ro source-mac-address?        ieee:mac-address
28    | +--ro vid?                      dot1q-types:vlan-index-type
29    | +--ro msdu?                     yang:hex-string
30    +--ro max-ci-stream-entries?       uint32 {congestion-isolation-bridge}?
31    +--rw stream-gates {congestion-isolation-bridge}?
32    | +--rw stream-gate-instance-table* [stream-gate-instance-id]
33    | | +--rw stream-gate-instance-id  uint32
34    | | +--rw gate-enable?             boolean
35    | | +--rw admin-gate-states?       gate-state-value-type
36    | | +--rw admin-ipv?               ipv-spec-type
37    | +--ro max-stream-gate-instances? uint32
38    +--rw stream-filters {congestion-isolation-bridge}?
39    | +--rw stream-filter-instance-table* [stream-filter-instance-id]
40    | | +--rw stream-filter-instance-id uint32
41    | | +--rw (stream-handle-spec)?
42    | | | +--:(wildcard)
43    | | | | +--rw wildcard?            empty
44    | | | +--:(stream-handle)
45    | | | | +--rw stream-handle        uint32
46    | | | +--:(null-handle) {congestion-isolation-bridge}?
47    | | | | +--rw null-handle?         empty
48    | | +--rw priority-spec
49    | | | priority-spec-type
50    | +--rw max-sdu-size                uint32
51    | +--rw stream-blocked-due-to-oversize-frame-enabled? boolean
52    | +--rw stream-blocked-due-to-oversize-frame?      boolean
53    | +--rw stream-gate-ref             leafref
54    +--ro max-stream-filter-instances?  uint32
55   augment /if:interfaces/if:interface/dot1q:bridge-port:
56     +--ro cip-mac-address?            ieee:mac-address
57     | {congestion-isolation-bridge}?
58     +--ro cip-ipv4-address?           inet:ipv4-address
59     | {congestion-isolation-bridge}?
60     +--ro cip-ipv6-address?           inet:ipv6-address
61     | {congestion-isolation-bridge}?
62     +--ro cip-cim-port?               inet:port-number
63     | {congestion-isolation-bridge}?
64     +--rw queue-map* [priority] {congestion-isolation-bridge}?
65     | +--rw priority                  dot1q-types:priority-type
66     | +--rw abs-traffic-class-plus-one?
67     | | abs-traffic-class-plus-one-type
68     +--rw min-header-octets?          uint16 {congestion-isolation-bridge}?
69     +--rw max-cim-tx?                 uint16 {congestion-isolation-bridge}?
70

```


1 48.5.17 Schema for the ieee802-dot1q-sched YANG module

2 This YANG module does not have a YANG schema tree.

3 48.5.18 Schema for the ieee802-dot1q-sched-bridge YANG module

```

4 module: ieee802-dot1q-sched-bridge
5
6   augment /if:interfaces/if:interface/dot1q:bridge-port:
7     +--rw gate-parameter-table
8       +--rw queue-max-sdu-table* [traffic-class]
9         | +--rw traffic-class          dot1q-types:traffic-class-type
10        | +--rw queue-max-sdu?         uint32
11        | +--ro transmission-overflow? yang:counter64
12      +--rw gate-enabled?              boolean
13      +--rw admin-gate-states?         uint8
14      +--ro oper-gate-states?          uint8
15      +--rw admin-control-list
16        | +--rw gate-control-entry* [index]
17        |   +--rw index                uint32
18        |   +--rw operation-name       identityref
19        |   +--rw time-interval-value? uint32
20        |   +--rw gate-states-value    uint8
21      +--ro oper-control-list
22        | +--ro gate-control-entry* [index]
23        |   +--ro index                uint32
24        |   +--ro operation-name       identityref
25        |   +--ro time-interval-value? uint32
26        |   +--ro gate-states-value    uint8
27      +--rw admin-cycle-time
28        | +--rw numerator?             uint32
29        | +--rw denominator?          uint32
30      +--ro oper-cycle-time
31        | +--ro numerator?             uint32
32        | +--ro denominator?          uint32
33      +--rw admin-cycle-time-extension? uint32
34      +--ro oper-cycle-time-extension?  uint32
35      +--rw admin-base-time
36        | +--rw seconds?               uint64
37        | +--rw nanoseconds?          uint32
38      +--ro oper-base-time
39        | +--ro seconds?               uint64
40        | +--ro nanoseconds?          uint32
41      +--rw config-change?              boolean
42      +--ro config-change-time
43        | +--ro seconds?               uint64
44        | +--ro nanoseconds?          uint32
45      +--ro tick-granularity?           uint32
46      +--ro current-time
47        | +--ro seconds?               uint64
48        | +--ro nanoseconds?          uint32
49      +--ro config-pending?             boolean
50      +--ro config-change-error?        yang:counter64
51      +--rw supported-list-max?         uint32
52      +--rw supported-cycle-max
53        | +--rw numerator?             uint32
54        | +--rw denominator?          uint32
55      +--rw supported-interval-max?     uint32
56

```

1 48.5.19 Schema for the ieee802-dot1q-preemption YANG module

2 This YANG module does not have a YANG schema tree.

3 48.5.20 Schema for the ieee802-dot1q-preemption-bridge YANG module

```
4 module: ieee802-dot1q-preemption-bridge
5
6   augment /if:interfaces/if:interface/dot1q:bridge-port:
7     +--rw frame-preemption-parameters {frame-preemption}?
8       +--rw frame-preemption-status-table
9         | +--rw priority0?   frame-preemption-status-enum
10        | +--rw priority1?   frame-preemption-status-enum
11        | +--rw priority2?   frame-preemption-status-enum
12        | +--rw priority3?   frame-preemption-status-enum
13        | +--rw priority4?   frame-preemption-status-enum
14        | +--rw priority5?   frame-preemption-status-enum
15        | +--rw priority6?   frame-preemption-status-enum
16        | +--rw priority7?   frame-preemption-status-enum
17        +--ro hold-advance?   uint32
18        +--ro release-advance? uint32
19        +--ro preemption-active? boolean
20        +--ro hold-request?   enumeration
21
```

22 48.5.21 Schema for the ieee802-dot1q-psfp YANG module

23 This YANG module does not have a YANG schema tree.

24 48.5.22 Schema for the ieee802-dot1q-psfp-bridge YANG module

```
25 module: ieee802-dot1q-psfp-bridge
26
27   augment /dot1q:bridges/dot1q:bridge/dot1q:component:
28     +--rw flow-meters
29       | +--rw flow-meter-instance-table* [flow-meter-instance-id]
30       | | +--rw flow-meter-instance-id   uint32
31       | | +--rw committed-information-rate   uint64
32       | | +--rw committed-burst-size   uint32
33       | | +--rw excess-information-rate   uint64
34       | | +--rw excess-burst-size   uint32
35       | | +--rw coupling-flag   enumeration
36       | | +--rw color-mode   enumeration
37       | | +--rw drop-on-yellow   boolean
38       | | +--rw mark-all-frames-red-enable?   boolean
39       | | +--rw mark-all-frames-red?   boolean
40       | +--rw max-flow-meter-instances?   uint32
41     +--rw stream-gates
42       | +--rw stream-gate-instance-table* [stream-gate-instance-id]
43       | | +--rw stream-gate-instance-id   uint32
44       | | +--rw gate-enable?   boolean
45       | | +--rw admin-gate-states?   gate-state-value-type
46       | | +--rw admin-ipv?   ipv-spec-type
47       | | +--ro oper-gate-state?   sfsg:gate-state-value-type
48       | | +--rw oper-ipv?   sfsg:ipv-spec-type
49       | | +--rw admin-control-list
50       | | | +--rw gate-control-entry* [index]
51       | | | | +--rw index   uint32
52       | | | | +--rw operation-name   identityref
53       | | | | +--rw time-interval-value?   uint32
54       | | | | +--rw gate-state-value   sfsg:gate-state-value-type
55       | | | | +--rw ipv-spec   sfsg:ipv-spec-type
56       | | | | +--rw interval-octet-max?   uint32
57       | | +--ro oper-control-list
58       | | | +--ro gate-control-entry* [index]
59       | | | | +--ro index   uint32
60       | | | | +--ro operation-name   identityref
61       | | | | +--ro time-interval-value?   uint32
62       | | | | +--ro gate-state-value   sfsg:gate-state-value-type
```

```

1      | | |      +--ro ipv-spec          sfsg:ipv-spec-type
2      | | |      +--ro interval-octet-max?  uint32
3      | | |      +--rw admin-cycle-time
4      | | |      +--rw numerator?          uint32
5      | | |      +--rw denominator?        uint32
6      | | |      +--ro oper-cycle-time
7      | | |      | +--ro numerator?          uint32
8      | | |      | +--ro denominator?        uint32
9      | | |      +--rw admin-cycle-time-extension?  uint32
10     | | |      +--ro oper-cycle-time-extension?  uint32
11     | | |      +--rw admin-base-time
12     | | |      | +--rw seconds?            uint64
13     | | |      | +--rw nanoseconds?        uint32
14     | | |      +--ro oper-base-time
15     | | |      | +--ro seconds?            uint64
16     | | |      | +--ro nanoseconds?        uint32
17     | | |      +--rw config-change?        boolean
18     | | |      +--ro config-change-time
19     | | |      | +--ro seconds?            uint64
20     | | |      | +--ro nanoseconds?        uint32
21     | | |      +--ro tick-granularity?      uint32
22     | | |      +--ro current-time
23     | | |      | +--ro seconds?            uint64
24     | | |      | +--ro nanoseconds?        uint32
25     | | |      +--ro config-pending?        boolean
26     | | |      +--ro config-change-error?    yang:counter64
27     | | |      +--rw gate-closed-due-to-invalid-rx-enable?  boolean
28     | | |      +--rw gate-closed-due-to-invalid-rx?  boolean
29     | | |      +--rw gate-closed-due-octets-exceeded-enable?  boolean
30     | | |      +--rw gate-closed-due-octets-exceeded?  boolean
31     | | |      +--ro max-stream-gate-instances?  uint32
32     | | |      +--rw supported-list-max?        uint32
33     | | |      +--rw supported-cycle-max
34     | | |      | +--rw numerator?          uint32
35     | | |      | +--rw denominator?        uint32
36     | | |      +--rw supported-interval-max?    uint32
37     +--rw stream-filters
38         +--rw stream-filter-instance-table* [stream-filter-instance-id]
39             | +--rw stream-filter-instance-id  uint32
40             | +--rw (stream-handle-spec)?
41             | | +--:(wildcard)
42             | | | +--rw wildcard?              empty
43             | | +--:(stream-handle)
44             | | | +--rw stream-handle          uint32
45             | +--rw priority-spec              priority-spec-type
46             | +--rw max-sdu-size              uint32
47             | +--rw stream-blocked-due-to-oversize-frame-enabled?  boolean
48             | +--rw stream-blocked-due-to-oversize-frame?  boolean
49             | +--rw stream-gate-ref            ->
50     ../.../stream-gates/stream-gate-instance-table/stream-gate-instance-id
51         | +--ro matching-frames-count?        yang:counter64
52         | +--ro passing-frames-count?          yang:counter64
53         | +--ro not-passing-frames-count?      yang:counter64
54         | +--ro red-frames-count?              yang:counter64
55         | +--ro passing-sdu-count?              yang:counter64
56         | +--ro not-passing-sdu-count?          yang:counter64
57         | +--rw flow-meter-ref?                ->
58     ../.../flow-meters/flow-meter-instance-table/flow-meter-instance-id
59         | +--rw flow-meter-enable?              boolean
60         +--ro max-stream-filter-instances?    uint32
61

```

48.5.23 Schema for the ieee802-dot1q-cnc-config YANG module

```

63 module: ieee802-dot1q-cnc-config
64     +--rw cnc-config
65         +--rw domain* [domain-id]
66             +--rw domain-id      string
67             +--rw cnc-enabled?    boolean
68             +--rw cuc* [cuc-id]
69                 +--rw cuc-id      string
70             +--rw stream* [stream-id]

```

```

1      | +--rw stream-id          tsn:stream-id-type
2      | +--ro stream-status? enumeration
3      | +--rw talker
4      | | +--rw stream-rank
5      | | | +--rw rank?   uint8
6      | | +--rw end-station-interfaces* [mac-address interface-name]
7      | | | +--rw mac-address      string
8      | | | +--rw interface-name   string
9      | | +--rw data-frame-specification* [index]
10     | | | +--rw index              uint8
11     | | | +--rw (field)?
12     | | |   +--:(ieee802-mac-addresses)
13     | | |   | +--rw ieee802-mac-addresses
14     | | |   | | +--rw destination-mac-address?  string
15     | | |   | | +--rw source-mac-address?      string
16     | | |   +--:(ieee802-vlan-tag)
17     | | |   | +--rw ieee802-vlan-tag
18     | | |   | | +--rw priority-code-point?    uint8
19     | | |   | | +--rw vlan-id?                uint16
20     | | |   +--:(ipv4-tuple)
21     | | |   | +--rw ipv4-tuple
22     | | |   | | +--rw source-ip-address?      inet:ipv4-address
23     | | |   | | +--rw destination-ip-address? inet:ipv4-address
24     | | |   | | +--rw dscp?                  uint8
25     | | |   | | +--rw protocol?              uint16
26     | | |   | | +--rw source-port?           uint16
27     | | |   | | +--rw destination-port?      uint16
28     | | |   +--:(ipv6-tuple)
29     | | |   | +--rw ipv6-tuple
30     | | |   | | +--rw source-ip-address?      inet:ipv6-address
31     | | |   | | +--rw destination-ip-address? inet:ipv6-address
32     | | |   | | +--rw dscp?                  uint8
33     | | |   | | +--rw protocol?              uint16
34     | | |   | | +--rw source-port?           uint16
35     | | |   | | +--rw destination-port?      uint16
36     | | +--rw traffic-specification
37     | | | +--rw interval
38     | | | | +--rw numerator?    uint32
39     | | | | +--rw denominator?  uint32
40     | | | +--rw max-frames-per-interval?  uint16
41     | | | +--rw max-frame-size?          uint16
42     | | | +--rw transmission-selection?  uint8
43     | | | +--rw time-aware!
44     | | | | +--rw earliest-transmit-offset?  uint32
45     | | | | +--rw latest-transmit-offset?    uint32
46     | | | | +--rw jitter?                  uint32
47     | | +--rw user-to-network-requirements
48     | | | +--rw num-seamless-trees?    uint8
49     | | | +--rw max-latency?           uint32
50     | | +--rw interface-capabilities
51     | | | +--rw vlan-tag-capable?      boolean
52     | | | +--rw cb-stream-iden-type-list*  uint32
53     | | | +--rw cb-sequence-type-list*    uint32
54     | | +--ro accumulated-latency?      uint32
55     | | +--ro interface-configuration
56     | | | +--ro interface-list* [mac-address interface-name]
57     | | | +--ro mac-address      string
58     | | | +--ro interface-name   string
59     | | | +--ro config-list* [index]
60     | | | | +--ro index              uint8
61     | | | | +--ro (config-value)?
62     | | | |   +--:(ieee802-mac-addresses)
63     | | | |   | +--ro ieee802-mac-addresses
64     | | | |   | | +--ro destination-mac-address?  string
65     | | | |   | | +--ro source-mac-address?      string
66     | | | |   +--:(ieee802-vlan-tag)
67     | | | |   | +--ro ieee802-vlan-tag
68     | | | |   | | +--ro priority-code-point?    uint8
69     | | | |   | | +--ro vlan-id?                uint16
70     | | | |   +--:(ipv4-tuple)
71     | | | |   | +--ro ipv4-tuple
72     | | | |   | | +--ro source-ip-address?

```

```

1 | | | | | inet:ipv4-address
2 | | | | | +---ro destination-ip-address?
3 | | | | | | inet:ipv4-address
4 | | | | | +---ro dscp? uint8
5 | | | | | +---ro protocol? uint16
6 | | | | | +---ro source-port? uint16
7 | | | | | +---ro destination-port? uint16
8 | | | | | +---:(ipv6-tuple)
9 | | | | | | +---ro ipv6-tuple
10 | | | | | | +---ro source-ip-address?
11 | | | | | | | inet:ipv6-address
12 | | | | | | +---ro destination-ip-address?
13 | | | | | | | inet:ipv6-address
14 | | | | | | +---ro dscp? uint8
15 | | | | | | +---ro protocol? uint16
16 | | | | | | +---ro source-port? uint16
17 | | | | | | +---ro destination-port? uint16
18 | | | | | +---:(time-aware-offset)
19 | | | | | +---ro time-aware-offset? uint32
20 | | | +---rw listener* [index]
21 | | | | +---rw index uint32
22 | | | | +---rw end-station-interfaces* [mac-address interface-name]
23 | | | | | +---rw mac-address string
24 | | | | | | +---rw interface-name string
25 | | | | | +---rw user-to-network-requirements
26 | | | | | | +---rw num-seamless-trees? uint8
27 | | | | | | +---rw max-latency? uint32
28 | | | | | +---rw interface-capabilities
29 | | | | | | +---rw vlan-tag-capable? boolean
30 | | | | | | | +---rw cb-stream-iden-type-list* uint32
31 | | | | | | | +---rw cb-sequence-type-list* uint32
32 | | | | | +---ro accumulated-latency? uint32
33 | | | | | +---ro interface-configuration
34 | | | | | | +---ro interface-list* [mac-address interface-name]
35 | | | | | | +---ro mac-address string
36 | | | | | | +---ro interface-name string
37 | | | | | | +---ro config-list* [index]
38 | | | | | | | +---ro index uint8
39 | | | | | | | +---ro (config-value)?
40 | | | | | | | +---:(ieee802-mac-addresses)
41 | | | | | | | | +---ro ieee802-mac-addresses
42 | | | | | | | | | +---ro destination-mac-address? string
43 | | | | | | | | | +---ro source-mac-address? string
44 | | | | | | | +---:(ieee802-vlan-tag)
45 | | | | | | | | +---ro ieee802-vlan-tag
46 | | | | | | | | | +---ro priority-code-point? uint8
47 | | | | | | | | | +---ro vlan-id? uint16
48 | | | | | | | +---:(ipv4-tuple)
49 | | | | | | | | +---ro ipv4-tuple
50 | | | | | | | | | +---ro source-ip-address?
51 | | | | | | | | | | inet:ipv4-address
52 | | | | | | | | | +---ro destination-ip-address?
53 | | | | | | | | | | inet:ipv4-address
54 | | | | | | | | | +---ro dscp? uint8
55 | | | | | | | | | +---ro protocol? uint16
56 | | | | | | | | | +---ro source-port? uint16
57 | | | | | | | | | +---ro destination-port? uint16
58 | | | | | | | +---:(ipv6-tuple)
59 | | | | | | | | +---ro ipv6-tuple
60 | | | | | | | | | +---ro source-ip-address?
61 | | | | | | | | | | inet:ipv6-address
62 | | | | | | | | | +---ro destination-ip-address?
63 | | | | | | | | | | inet:ipv6-address
64 | | | | | | | | | +---ro dscp? uint8
65 | | | | | | | | | +---ro protocol? uint16
66 | | | | | | | | | +---ro source-port? uint16
67 | | | | | | | | | +---ro destination-port? uint16
68 | | | | | | | +---:(time-aware-offset)
69 | | | | | | | +---ro time-aware-offset? uint32
70 | | | +---ro status-info
71 | | | | +---ro talker-status? enumeration
72 | | | | +---ro listener-status? enumeration

```

```

1      | | +--ro failure-code?      uint8
2      | | +--ro failed-interfaces* [mac-address interface-name]
3      | | +--ro mac-address        string
4      | | +--ro interface-name     string
5      +---x remove_streams
6          +---w input
7              | +---w stream-list* [stream-id]
8              | | +---w stream-id    tsn:stream-id-type
9              +--ro output
10             +--ro result?   string
11
12  rpcs:
13      +---x compute_streams
14          | +---w input
15              | | +---w domain* [domain-id]
16              | | +---w domain-id  -> /cnc-config/domain/domain-id
17              | | +---w cuc* [cuc-id]
18              | | +---w cuc-id      -> /cnc-config/domain/cuc/cuc-id
19              | | +---w stream-list* [stream-id]
20              | | +---w stream-id
21              | | -> /cnc-config/domain/cuc/stream/stream-id
22              +--ro output
23              +--ro result?   string
24      +---x compute_planned_and_modified_streams
25          | +---w input
26              | | +---w domain* [domain-id]
27              | | +---w domain-id  string
28              | | +---w cuc* [cuc-id]
29              | | +---w cuc-id      string
30              +--ro output
31              +--ro result?   string
32      +---x compute_all_streams
33          | +---w input
34              | | +---w domain* [domain-id]
35              | | +---w domain-id  string
36              | | +---w cuc* [cuc-id]
37              | | +---w cuc-id      string
38              +--ro output
39              +--ro result?   string
40      +---x request_domain_id
41          | +---w input
42              | | +---w cuc-id?   string
43              +--ro output
44              +--ro result?   string
45      +---x request_free_stream_id
46          +---w input
47              | +---w domain-id?   string
48              | +---w cuc-id?      string
49              +--ro output
50              +--ro result?   string
51
52  notifications:
53      +---n compute_streams_completed
54          | +--ro domain* [domain-id]
55          | +--ro domain-id  string
56          | +--ro cuc* [cuc-id]
57          | +--ro cuc-id      string
58          | +--ro stream* [stream-id]
59          | +--ro stream-id    tsn:stream-id-type
60          | +--ro failure-code? uint8
61      +---n configure_streams_completed
62          | +--ro domain* [domain-id]
63          | +--ro domain-id  string
64          | +--ro cuc* [cuc-id]
65          | +--ro cuc-id      string
66          | +--ro stream* [stream-id]
67          | +--ro stream-id    tsn:stream-id-type
68          | +--ro failure-code? uint8
69      +---n remove_streams_completed
70          +--ro domain* [domain-id]
71          +--ro domain-id  string
72          +--ro cuc* [cuc-id]

```

```

1      +--ro cuc-id      string
2      +--ro stream* [stream-id]
3          +--ro stream-id      tsn:stream-id-type
4
5      +--ro failure-code?  uint8

```

48.5.24 Schema for the ieee802-dot1q-cbsa YANG module

This YANG module does not have a YANG schema tree.

48.5.25 Schema for the ieee802-dot1q-cbsa-bridge YANG module

```

8 module: ieee802-dot1q-cbsa-bridge
9
10  augment /if:interfaces/if:interface/dot1q:bridge-port:
11      +--rw cbsa {credit-based-shaper-algorithm}?
12          +--rw cbsa-parameter-table* [traffic-class]
13              +--rw traffic-class      dot1q-types:traffic-class-type
14              +--rw admin-idle-slope?  uint64
15              +--ro oper-idle-slope?   uint64
16

```

48.5.26 Schema for the ieee802-dot1q-rstp YANG module

This YANG module does not have a YANG schema tree.

48.5.27 Schema for the ieee802-dot1q-rstp-bridge YANG module

```

20 module: ieee802-dot1q-rstp-bridge
21
22  augment /dot1q:bridges/dot1q:bridge/dot1q:component:
23      +--rw rstp!
24          +--rw force-protocol-version?  enumeration
25          +--rw bridge-id
26              | +--ro bridge-id?          uint64
27              | +--rw bridge-priority?    id-priority
28              | +--ro system-id-extension? uint16
29              | +--ro bridge-address?     ieee:mac-address
30          +--ro root-id
31              | +--ro bridge-id?          uint64
32              | +--ro bridge-priority?    id-priority
33              | +--ro system-id-extension? uint16
34              | +--ro bridge-address?     ieee:mac-address
35          +--ro root-path-cost?           uint32
36          +--ro root-port?                union
37          +--ro max-age?                  uint8
38          +--ro hello-time?               uint8
39          +--ro forward-delay?            uint8
40          +--rw bridge-max-age?           uint8
41          +--rw bridge-forward-delay?     uint8
42          +--rw tx-hold-count?            uint8
43          +--ro last-topology-change?     yang:date-and-time
44  augment /if:interfaces/if:interface/dot1q:bridge-port:
45      +--rw rstp!
46          +--rw admin-bridge-port-enabled?  boolean
47          +--ro port-state?                 port-state
48          +--ro port-role?                  port-role
49          +--rw restricted-role?             boolean
50          +--rw restricted-tcn?              boolean
51          +--rw port-id
52              | +--ro port-id?              uint16
53              | +--rw port-priority?        id-priority
54              | +--ro port-number?         id-port-number
55          +--rw fix-port-path-cost?         uint32
56          +--ro port-path-cost?             uint32
57          +--ro designated-protocol-version? uint8
58          +--x port-protocol-migration-check
59          +--ro root-id
60          | +--ro bridge-id?                uint64

```

```

1      | +--ro bridge-priority?      id-priority
2      | +--ro system-id-extension? uint16
3      | +--ro bridge-address?     ieee:mac-address
4      +--ro root-path-cost?       uint32
5      +--ro designated-bridge-id
6      | +--ro bridge-id?          uint64
7      | +--ro bridge-priority?    id-priority
8      | +--ro system-id-extension? uint16
9      | +--ro bridge-address?     ieee:mac-address
10     +--ro designated-port-id
11     | +--ro port-id?            uint16
12     | +--ro port-priority?      id-priority
13     | +--ro port-number?        id-port-number
14     +--rw admin-edge-port?      boolean
15     +--ro oper-edge-port?       boolean
16     +--rw auto-edge-port?       boolean
17     +--ro disputed-port?        boolean
18     +--ro isolate-port?         boolean
19

```

48.5.28 Schema for the ieee802-dot1q-mstp YANG module

This YANG module does not have a YANG schema tree.

48.5.29 Schema for the ieee802-dot1q-mstp-bridge YANG module

```

23 module: ieee802-dot1q-mstp-bridge
24
25   augment /dot1q:bridges/dot1q:bridge/dot1q:component/dot1q:bridge-mst:
26     +--rw bridge-mstp!
27       +--rw mst-config-id
28         | +--ro format-selector?      uint8
29         | +--rw configuration-name?    string
30         | +--ro revision-level?       uint16
31         | +--ro configuration-digest?  binary
32       +--rw max-hops?                 uint8
33       +--rw ist
34         | +--ro internal-root-path-cost? uint32
35       +--rw msti* [mstid]
36         +--rw mstid                  uint16
37         +--rw bridge-priority?        rstp:id-priority
38         +--ro regional-root-id
39         | +--ro bridge-id?            uint64
40         | +--ro bridge-priority?      id-priority
41         | +--ro system-id-extension?  uint16
42         | +--ro bridge-address?       ieee:mac-address
43         +--ro internal-root-path-cost? uint32
44         +--ro root-port?              union
45   augment /if:interfaces/if:interface/dot1q:bridge-port:
46     +--rw port-mstp!
47       +--ro boundary-port?            boolean
48       +--rw restricted-domain-role?   boolean
49       +--rw ist
50         | +--ro mst-config-id
51         | | +--ro format-selector?      uint8
52         | | +--ro configuration-name?    string
53         | | +--ro revision-level?       uint16
54         | | +--ro configuration-digest?  binary
55         | +--rw fix-internal-port-path-cost? uint32
56         | +--ro internal-port-path-cost? uint32
57         | +--ro internal-root-path-cost? uint32
58         | +--ro designated-bridge
59         | | +--ro bridge-id?            uint64
60         | | +--ro bridge-priority?      id-priority
61         | | +--ro system-id-extension?  uint16
62         | | +--ro bridge-address?       ieee:mac-address
63         | +--ro remaining-hops?        uint8
64       +--rw msti* [mstid]
65         +--rw mstid                  uint16
66         +--ro port-state?            rstp:port-state

```



```
1      +---ro port-role?                                union
2      +---rw port-id
3      |  +---ro port-id?                                uint16
4      |  +---rw port-priority?  id-priority
5      |  +---ro port-number?    id-port-number
6      +---rw fix-internal-port-path-cost?  uint32
7      +---ro internal-port-path-cost?      uint32
8      +---ro regional-root-id
9      |  +---ro bridge-id?                uint64
10     |  +---ro bridge-priority?          id-priority
11     |  +---ro system-id-extension?      uint16
12     |  +---ro bridge-address?          ieee:mac-address
13     +---ro internal-root-path-cost?      uint32
14     +---ro designated-bridge-id
15     |  +---ro bridge-id?                uint64
16     |  +---ro bridge-priority?          id-priority
17     |  +---ro system-id-extension?      uint16
18     |  +---ro bridge-address?          ieee:mac-address
19     +---ro designated-port-id
20     |  +---ro port-id?                  uint16
21     |  +---ro port-priority?            id-priority
22     |  +---ro port-number?              id-port-number
23     +---ro disputed-port?                boolean
24     +---ro remaining-hops?                uint8
25
```

1 48.6 YANG modules^{52 53 54}

2 48.6.1 The ieee802-types YANG module

```
3 module ieee802-types {
4   yang-version "1.1";
5   namespace urn:ieee:std:802.1Q:yang:ieee802-types;
6   prefix ieee;
7   organization
8     "IEEE 802.1 Working Group";
9   contact
10    "WG-URL: http://ieee802.org/1/
11    WG-EMail: stds-802-1-1@ieee.org
12
13    Contact: IEEE 802.1 Working Group Chair
14    Postal: C/O IEEE 802.1 Working Group
15           IEEE Standards Association
16           445 Hoes Lane
17           Piscataway, NJ 08854
18           USA
19
20    E-mail: stds-802-1-chairs@ieee.org";
21  description
22    "This module contains a collection of generally useful derived data
23    types for IEEE YANG models.
24
25    Copyright (C) IEEE (2023). All rights reserved.
26
27    This version of this YANG module is part of IEEE Std 802.1Q; see the
28    standard itself for full legal notices.";
29  revision 2023-10-22 {
30    description
31      "Published as part of IEEE Std 802.1Qcw-2023.
32
33      The following reference statement identifies each referenced IEEE
34      Standard as updated by applicable amendments.";
35    reference
36      "IEEE Std 802.1Q Bridges and Bridged Networks:
37      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.
38      IEEE Std 802 Overview and Architecture:
39      IEEE Std 802-2014.
40      IEEE 802.1AS Timing and Synchronization for Time-Sensitive Applications:
41      IEEE Std 802.1AS-2020.";
42  }
43  revision 2022-10-29 {
44    description
45      "Published as part of IEEE Std 802.1Q-2022.";
46    reference
47      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
48  }
49  revision 2022-03-16 {
50    description
51      "Published as part of IEEE Std 802.1ABcu.";
52    reference
53      "IEEE Std 802.1AB-2016";
54  }
55  revision 2020-06-04 {
56    description
57      "Published as part of IEEE Std 802.1Qcx-2020. Second version.";
58    reference
59      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
60      Model for Connectivity Fault Management.";
61  }
```

⁵² Copyright release for YANG: Users of this standard may freely reproduce the YANG modules contained in this standard so that they can be used for their intended purpose.

⁵³ An ASCII version of each YANG module is attached to the PDF of this standard and can also be obtained from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

⁵⁴ References in this standard's YANG module definitions are not clickable, as each module has been incorporated unchanged after development and verification using YANG tools.

```

1  revision 2018-03-07 {
2      description
3          "Published as part of IEEE Std 802.1Q-2018. Initial version.";
4      reference
5          "IEEE Std 802.1Q-2018, Bridges and Bridged Networks.";
6  }
7  typedef mac-address {
8      type string {
9          pattern "[0-9a-fA-F]{2}(-[0-9a-fA-F]{2}){5}";
10     }
11     description
12         "The mac-address type represents a MAC address in the canonical format
13         and hexadecimal format specified by IEEE Std 802. The hexadecimal
14         representation uses uppercase characters.";
15     reference
16         "3.1, 8.1 of IEEE Std 802";
17 }
18 typedef chassis-id-subtype-type {
19     type enumeration {
20         enum chassis-component {
21             value 1;
22             description
23                 "Represents a chassis identifier based on the value of the
24                 entPhysicalAlias object (defined in IETF RFC 2737) for a chassis
25                 component (i.e., an entPhysicalClass value of chassis(3)).";
26         }
27         enum interface-alias {
28             value 2;
29             description
30                 "Represents a chassis identifier based on the value of the ifAlias
31                 object (defined in IETF RFC 2863) for an interface on the
32                 containing chassis.";
33         }
34         enum port-component {
35             value 3;
36             description
37                 "Represents a chassis identifier based on the value of the
38                 entPhysicalAlias object (defined in IETF RFC 2737) for a port or
39                 backplane component (i.e., entPhysicalClass value of port(10) or
40                 backplane(4)), within the containing chassis.";
41         }
42         enum mac-address {
43             value 4;
44             description
45                 "Represents a chassis identifier based on the value of a unicast
46                 source address (encoded in network byte order and IEEE 802.3
47                 bit significance), of a port on the containing chassis as
48                 defined in IEEE Std 802.";
49         }
50         enum network-address {
51             value 5;
52             description
53                 "Represents a chassis identifier based on a network address,
54                 associated with a particular chassis. The encoded address is
55                 actually composed of two fields. The first field is a single
56                 octet, representing the IANA AddressFamilyNumbers value for the
57                 specific address type, and the second field is the network address
58                 value.";
59         }
60         enum interface-name {
61             value 6;
62             description
63                 "Represents a chassis identifier based on the value of the ifName
64                 object (defined in IETF RFC 2863) for an interface on the
65                 containing chassis.";
66         }
67         enum local {
68             value 7;
69             description
70                 "Represents a chassis identifier based on a locally defined value.";
71         }
72     }

```

```

1  description
2  "The source of a chassis identifier.";
3  reference
4  "IEEE Std 802
5  IETF RFC 2737
6  IETF RFC 2863";
7  }
8  typedef chassis-id-type {
9  type string {
10     length "1..255";
11  }
12  description
13     "The format of a chassis identifier string. Objects of this type are
14     always used with an associated chassis-id-subtype object, which
15     identifies the format of the particular chassis-id object
16     instance.
17
18     If the associated chassis-id-subtype object has a value of
19     chassis-component, then the octet string identifies a particular
20     instance of the entPhysicalAlias object (defined in IETF RFC 2737) for
21     a chassis component (i.e., an entPhysicalClass value of chassis(3)).
22
23     If the associated chassis-id-subtype object has a value of
24     interface-alias, then the octet string identifies a particular
25     instance of the ifAlias object (defined in IETF RFC 2863) for an
26     interface on the containing chassis. If the particular ifAlias object
27     does not contain any values, another chassis identifier type should be
28     used.
29
30     If the associated chassis-id-subtype object has a value of
31     port-component, then the octet string identifies a particular instance
32     of the entPhysicalAlias object (defined in IETF RFC 2737) for a port
33     or backplane component within the containing chassis.
34
35     If the associated chassis-id-subtype object has a value of
36     mac-address, then this string identifies a particular unicast source
37     address (encoded in network byte order and IEEE 802.3 bit
38     significance), of a port on the containing chassis as defined in
39     IEEE Std 802.
40
41     If the associated chassis-id-subtype object has a value of
42     network-address, then this string identifies a particular network
43     address, encoded in network byte order, associated with one or more
44     ports on the containing chassis. The first octet contains the IANA
45     Address Family Numbers enumeration value for the specific address
46     type, and octets 2 through N contain the network address value in
47     network byte order.
48
49     If the associated chassis-id-subtype object has a value of
50     interface-name, then the octet string identifies a particular instance
51     of the ifName object (defined in IETF RFC 2863) for an interface on
52     the containing chassis. If the particular ifName object does not
53     contain any values, another chassis identifier type should be used.
54
55     If the associated chassis-id-subtype object has a value of local,
56     then this string identifies a locally assigned Chassis ID.";
57  reference
58  "IEEE Std 802
59  IETF RFC 2737
60  IETF RFC 2863";
61  }
62  typedef port-id-subtype-type {
63  type enumeration {
64  enum interface-alias {
65  value 1;
66  description
67  "Represents a port identifier based on the ifAlias MIB object,
68  defined in IETF RFC 2863.";
69  }
70  enum port-component {
71  value 2;
72  description

```

```

1      "Represents a port identifier based on the value of
2      entPhysicalAlias (defined in IETF RFC 2737) for a port component
3      (i.e., entPhysicalClass value of port(10)), within the containing
4      chassis.";
5  }
6  enum mac-address {
7      value 3;
8      description
9          "Represents a port identifier based on a unicast source address
10         (encoded in network byte order and IEEE 802.3 bit significance),
11         which has been detected by the agent and associated with a
12         particular port (IEEE Std 802).";
13  }
14  enum network-address {
15      value 4;
16      description
17          "Represents a port identifier based on a network address, detected
18          by the agent and associated with a particular port.";
19  }
20  enum interface-name {
21      value 5;
22      description
23          "Represents a port identifier based on the ifName MIB object,
24          defined in IETF RFC 2863.";
25  }
26  enum agent-circuit-id {
27      value 6;
28      description
29          "Represents a port identifier based on the agent-local identifier
30          of the circuit (defined in RFC 3046), detected by the agent and
31          associated with a particular port.";
32  }
33  enum local {
34      value 7;
35      description
36          "Represents a port identifier based on a value locally assigned.";
37  }
38  }
39  description
40      "The source of a particular type of port identifier.";
41  reference
42      "IEEE Std 802
43      IETF RFC 2737
44      IETF RFC 2863
45      IETF RFC 3046";
46  }
47  typedef port-id-type {
48      type string {
49          length "1..255";
50      }
51  }
52  description
53      "The format of a port identifier string. Objects of this type are
54      always used with an associated port-id-subtype object, which
55      identifies the format of the particular port-id object instance.
56
57      If the associated port-id-subtype object has a value of
58      interface-alias, then the octet string identifies a particular
59      instance of the ifAlias object (defined in IETF RFC 2863). If the
60      particular ifAlias object does not contain any values, another port
61      identifier type should be used.
62
63      If the associated port-id-subtype object has a value of
64      port-component, then the octet string identifies a particular instance
65      of the entPhysicalAlias object (defined in IETF RFC 2737) for a port
66      or backplane component.
67
68      If the associated port-id-subtype object has a value of
69      mac-address, then this string identifies a particular unicast source
70      address (encoded in network byte order and IEEE 802.3 bit
71      significance) associated with the port (IEEE Std 802).
72
73      If the associated port-id-subtype object has a value of

```

```

1  network-address, then this string identifies a network address
2  associated with the port. The first octet contains the IANA
3  AddressFamilyNumbers enumeration value for the specific address type,
4  and octets 2 through N contain the networkAddress address value in
5  network byte order.
6
7  If the associated port-id-subtype object has a value of
8  interface-name, then the octet string identifies a particular instance
9  of the ifName object (defined in IETF RFC 2863). If the particular
10 ifName object does not contain any values, another port identifier
11 type should be used.
12
13 If the associated port-id-subtype object has a value of
14 agent-circuit-id, then this string identifies a agent-local identifier
15 of the circuit (defined in RFC 3046).
16
17 If the associated port-id-subtype object has a value of local,
18 then this string identifies a locally assigned port ID.";
19 reference
20 "IEEE Std 802
21 IETF RFC 2737
22 IETF RFC 2863
23 IETF RFC 3046";
24 }
25 grouping rational-grouping {
26   description
27     "Definition of a non-negative rational number.";
28   leaf numerator {
29     type uint32;
30     description
31       "Numerator of the rational number.";
32   }
33   leaf denominator {
34     type uint32 {
35       range "1..4294967295";
36     }
37     description
38       "Denominator of the rational number.";
39   }
40 }
41 grouping ptp-time-grouping {
42   description
43     "This grouping specifies a PTP timestamp, represented as a 48-bit
44     unsigned integer number of seconds and a 32-bit unsigned integer
45     number of nanoseconds.";
46   reference
47     "6.4.3.4 of IEEE Std 802.1AS";
48   leaf seconds {
49     type uint64;
50     description
51       "This is the integer portion of the timestamp in units of seconds.
52       The upper 16 bits are always zero.";
53   }
54   leaf nanoseconds {
55     type uint32;
56     description
57       "This is the fractional portion of the timestamp in units of
58       nanoseconds. This value is always less than 10^9.";
59   }
60 }
61 }
62
63

```

1 48.6.2 The ieee802-dot1q-types YANG module

```
2 module ieee802-dot1q-types {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-types;
5   prefix dot1q-types;
6   import ietf-yang-types {
7     prefix yang;
8   }
9   organization
10    "IEEE 802.1 Working Group";
11   contact
12    "WG-URL: http://ieee802.org/1/
13    WG-EMail: stds-802-1-1@ieee.org
14
15    Contact: IEEE 802.1 Working Group Chair
16    Postal: C/O IEEE 802.1 Working Group
17    IEEE Standards Association
18    445 Hoes Lane
19    Piscataway, NJ 08854
20    USA
21
22    E-mail: stds-802-1-chairs@ieee.org";
23   description
24    "Common types used within dot1Q-bridge modules.
25
26    Copyright (C) IEEE (2023).
27
28    This version of this YANG module is part of IEEE Std 802.1Q; see the
29    standard itself for full legal notices.";
30   revision 2023-10-26 {
31     description
32      "Published as part of IEEE Std 802.1Qcj-2023. This revision adds
33      the isid-type.
34
35      The following reference statement identifies each referenced IEEE
36      Standard as updated by applicable amendments.";
37     reference
38      "IEEE Std 802.1Q Bridges and Bridged Networks:
39      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023,
40      IEEE Std 802.1Qcj-2023.
41      IEEE Std 802 Overview and Architecture: IEEE Std 802-2014.";
42   }
43   revision 2023-10-22 {
44     description
45      "Published as part of IEEE Std 802.1Qcw-2023.
46
47      The following reference statement identifies each referenced IEEE
48      Standard as updated by applicable amendments.";
49     reference
50      "IEEE Std 802.1Q Bridges and Bridged Networks:
51      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.
52      IEEE Std 802 Overview and Architecture:
53      IEEE Std 802-2014.";
54   }
55   revision 2023-07-03 {
56     description
57      "Published as part of IEEE Std 802.1Qcz-2023.";
58     reference
59      "IEEE Std 802.1Qcz-2023, Bridges and Bridged Networks - Congestion
60      Isolation.";
61   }
62   revision 2022-10-29 {
63     description
64      "Published as part of IEEE Std 802.1Q-2022.";
65     reference
66      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
67   }
68   revision 2020-06-04 {
69     description
70      "Published as part of IEEE Std 802.1Qcx-2020.";
71     reference
```

```

1      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
2      Model for Connectivity Fault Management.";
3  }
4  revision 2018-03-07 {
5      description
6          "Published as part of IEEE Std 802.1Q-2018.";
7      reference
8          "IEEE Std 802.1Q-2018, Bridges and Bridged Networks.";
9  }
10 identity dot1q-vlan-type {
11     description
12         "Base identity from which all 802.1Q VLAN tag types are derived.";
13 }
14 identity c-vlan {
15     base dot1q-vlan-type;
16     description
17         "An 802.1Q Customer VLAN, using the 81-00 EtherType.";
18     reference
19         "5.5 of IEEE Std 802.1Q";
20 }
21 identity s-vlan {
22     base dot1q-vlan-type;
23     description
24         "An 802.1Q Service VLAN, using the 88-A8 EtherType originally
25         introduced in 802.1ad, and incorporated into 802.1Q (2011)";
26     reference
27         "5.6 of IEEE Std 802.1Q";
28 }
29 identity transmission-selection-algorithm {
30     description
31         "Specify the transmission selection algorithms of IEEE Std 802.1Q
32         Table 8-6";
33 }
34 identity strict-priority {
35     base transmission-selection-algorithm;
36     description
37         "Indicates the strict priority transmission selection algorithm.";
38     reference
39         "Table 8-6 of IEEE Std 802.1Q";
40 }
41 identity credit-based-shaper {
42     base transmission-selection-algorithm;
43     description
44         "Indicates the credit-based shaper transmission selection algorithm.";
45     reference
46         "Table 8-6 of IEEE Std 802.1Q";
47 }
48 identity enhanced-transmission-selection {
49     base transmission-selection-algorithm;
50     description
51         "Indicates the enhanced transmission selection algorithm.";
52     reference
53         "Table 8-6 of IEEE Std 802.1Q";
54 }
55 identity asynchronous-traffic-shaping {
56     base transmission-selection-algorithm;
57     description
58         "Indicates the asynchronous transmission selection algorithm.";
59     reference
60         "Table 8-6 of IEEE Std 802.1Q";
61 }
62 identity vendor-specific {
63     base transmission-selection-algorithm;
64     description
65         "Indicates a vendor specific transmission selection algorithm.";
66     reference
67         "Table 8-6 of IEEE Std 802.1Q";
68 }
69 identity type-of-operation {
70     description
71         "Represents the operation type (name).";
72 }

```



```

1  typedef name-type {
2      type string {
3          length "0..32";
4      }
5      description
6          "A text string of up to 32 characters, of locally determined
7          significance.";
8  }
9  typedef port-number-type {
10     type uint32 {
11         range "1..4095";
12     }
13     description
14         "The port number of the Bridge port for which this entry contains
15         Bridge management information.";
16 }
17 typedef priority-type {
18     type uint8 {
19         range "0..7";
20     }
21     description
22         "A range of priorities from 0 to 7 (inclusive). The Priority Code
23         Point (PCP) is a 3-bit field that refers to the class of service
24         associated with an 802.1Q VLAN tagged frame. The field specifies a
25         priority value between 0 and 7, these values can be used by quality of
26         service (QoS) to prioritize different classes of traffic.";
27 }
28 typedef num-traffic-class-type {
29     type uint8 {
30         range "1..8";
31     }
32     description
33         "The number of traffic classes supported or participating in a
34         particular feature. There are between 1 and 8 supported traffic
35         classes defined by IEEE Std 802.1Q.";
36 }
37 typedef vid-range-type {
38     type string {
39         pattern
40             "([1-9]" +
41             "[0-9]{0,3}" +
42             "(-[1-9][0-9]{0,3})?" +
43             "(, [1-9][0-9]{0,3}(-[1-9][0-9]{0,3})?)*)";
44     }
45     description
46         "A list of VLAN IDs, or non overlapping VLAN ranges, in ascending
47         order, between 1 and 4094.
48
49         This type is used to match an ordered list of VLAN IDs, or contiguous
50         ranges of VLAN IDs. Valid VLAN IDs must be in the range 1 to 4094, and
51         included in the list in non overlapping ascending order.
52
53         For example: 1,10-100,250,500-1000";
54 }
55 typedef vlanid {
56     type uint16 {
57         range "1..4094";
58     }
59     description
60         "The vlanid type uniquely identifies a VLAN. This is the 12-bit
61         VLAN-ID used in the VLAN Tag header. The range is defined by the
62         referenced specification. This type is in the value set and its
63         semantics equivalent to the VlanId textual convention of the SMIV2.";
64 }
65 typedef vlan-index-type {
66     type uint32 {
67         range "1..4094 | 4096..4294967295";
68     }
69     description
70         "A value used to index per-VLAN tables. Values of 0 and 4095 are not
71         permitted. The range of valid VLAN indices. If the value is greater
72         than 4095, then it represents a VLAN with scope local to the

```

```

1      particular agent, i.e., one without a global VLAN-ID assigned to it.
2      Such VLANs are outside the scope of IEEE 802.1Q, but it is convenient
3      to be able to manage them in the same way using this YANG module.";
4      reference
5          "9.6 of IEEE Std 802.1Q";
6  }
7  typedef mstid-type {
8      type uint32 {
9          range "1..4094";
10     }
11     description
12         "In an MSTP Bridge, an MSTID, i.e., a value used to identify a
13         spanning tree (or MST) instance";
14     reference
15         "13.8 of IEEE Std 802.1Q";
16 }
17 typedef pcp-selection-type {
18     type enumeration {
19         enum 8P0D {
20             description
21                 "8 priorities, 0 drop eligible";
22         }
23         enum 7P1D {
24             description
25                 "7 priorities, 1 drop eligible";
26         }
27         enum 6P2D {
28             description
29                 "6 priorities, 2 drop eligible";
30         }
31         enum 5P3D {
32             description
33                 "5 priorities, 3 drop eligible";
34         }
35     }
36     description
37         "Priority Code Point selection types.";
38     reference
39         "12.6.2.5.3, 6.9.3 of IEEE Std 802.1Q";
40 }
41 typedef protocol-frame-format-type {
42     type enumeration {
43         enum Ethernet {
44             description
45                 "Ethernet frame format";
46         }
47         enum rfc1042 {
48             description
49                 "RFC 1042 frame format";
50         }
51         enum snap8021H {
52             description
53                 "SNAP 802.1H frame format";
54         }
55         enum snapOther {
56             description
57                 "Other SNAP frame format";
58         }
59         enum llcOther {
60             description
61                 "Other LLC frame format";
62         }
63     }
64     description
65         "A value representing the frame format to be matched.";
66     reference
67         "12.10.1.7.1 of IEEE Std 802.1Q";
68 }
69 typedef ethertype-type {
70     type string {
71         pattern "[0-9a-fA-F]{2}-[0-9a-fA-F]{2}";
72     }

```

```

1     description
2         "The EtherType value represented in the canonical order defined by
3         IEEE 802. The canonical representation uses uppercase characters.";
4     reference
5         "9.2 of IEEE Std 802";
6 }
7 typedef dot1q-tag-type {
8     type identityref {
9         base dot1q-vlan-type;
10    }
11    description
12        "Identifies a specific 802.1Q tag type";
13    reference
14        "9.5 IEEE Std 802.1Q";
15 }
16 typedef traffic-class-type {
17     type uint8 {
18         range "0..7";
19     }
20    description
21        "This is the numerical value associated with a traffic class in a
22        Bridge. Larger values are associated with higher priority traffic
23        classes.";
24    reference
25        "3.273 of IEEE Std 802.1Q";
26 }
27 typedef isid-type {
28     type uint32 {
29         range "1|255..16777214";
30     }
31    description
32        "The i-sid type represents a Backbone Service Instance Identifier
33        (I-SID). This is the 24-bit I-SID field used in the I-TAG TCI of a
34        Provider Backbone Bridging frame. The values 0, 2-254, and 16777215
35        are reserved for future standardization. The value 255 is dedicated
36        to use by the SPB protocol.";
37    reference
38        "3.108, 9.7 of IEEE Std 802.1Q-2022";
39 }
40 grouping dot1q-tag-classifier-grouping {
41    description
42        "A grouping which represents an 802.1Q VLAN, matching both the
43        EtherType and a single VLAN ID.";
44    leaf tag-type {
45        type dot1q-tag-type;
46        mandatory true;
47        description
48            "VLAN type";
49    }
50    leaf vlan-id {
51        type vlanid;
52        mandatory true;
53        description
54            "VLAN ID";
55    }
56 }
57 grouping dot1q-tag-or-any-classifier-grouping {
58    description
59        "A grouping which represents an 802.1Q VLAN, matching both the
60        EtherType and a single VLAN ID or 'any' to match on any VLAN ID.";
61    leaf tag-type {
62        type dot1q-tag-type;
63        mandatory true;
64        description
65            "VLAN type";
66    }
67    leaf vlan-id {
68        type union {
69            type vlanid;
70            type enumeration {
71                enum any {
72                    value 4095;

```

```

1      description
2      "Matches 'any' VLAN in the range 1 to 4094 that is not matched
3      by a more specific VLAN ID match";
4    }
5  }
6  }
7  mandatory true;
8  description
9    "VLAN ID or any";
10 }
11 }
12 grouping dot1q-tag-ranges-classifier-grouping {
13   description
14     "A grouping which represents an 802.1Q VLAN that matches a range of
15     VLAN IDs.";
16   leaf tag-type {
17     type dot1q-tag-type;
18     mandatory true;
19     description
20       "VLAN type";
21   }
22   leaf vlan-ids {
23     type vid-range-type;
24     mandatory true;
25     description
26       "VLAN IDs";
27   }
28 }
29 grouping dot1q-tag-ranges-or-any-classifier-grouping {
30   description
31     "A grouping which represents an 802.1Q VLAN, matching both the
32     EtherType and a single VLAN ID, ordered list of ranges, or 'any' to
33     match on any VLAN ID.";
34   leaf tag-type {
35     type dot1q-tag-type;
36     mandatory true;
37     description
38       "VLAN type";
39   }
40   leaf vlan-id {
41     type union {
42       type vid-range-type;
43       type enumeration {
44         enum any {
45           value 4095;
46           description
47             "Matches 'any' VLAN in the range 1 to 4094.";
48         }
49       }
50     }
51     mandatory true;
52     description
53       "VLAN IDs or any";
54   }
55 }
56 grouping priority-regeneration-table-grouping {
57   description
58     "The priority regeneration table provides the ability to map incoming
59     priority values on a per-Port basis, under management control.";
60   reference
61     "6.9.4 of IEEE Std 802.1Q";
62   leaf priority0 {
63     type priority-type;
64     default "0";
65     description
66       "Priority 0";
67     reference
68       "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
69   }
70   leaf priority1 {
71     type priority-type;
72     default "1";

```

```

1      description
2      "Priority 1";
3      reference
4      "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
5  }
6  leaf priority2 {
7      type priority-type;
8      default "2";
9      description
10     "Priority 2";
11     reference
12     "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
13 }
14 leaf priority3 {
15     type priority-type;
16     default "3";
17     description
18     "Priority 3";
19     reference
20     "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
21 }
22 leaf priority4 {
23     type priority-type;
24     default "4";
25     description
26     "Priority 4";
27     reference
28     "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
29 }
30 leaf priority5 {
31     type priority-type;
32     default "5";
33     description
34     "Priority 5";
35     reference
36     "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
37 }
38 leaf priority6 {
39     type priority-type;
40     default "6";
41     description
42     "Priority 6";
43     reference
44     "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
45 }
46 leaf priority7 {
47     type priority-type;
48     default "7";
49     description
50     "Priority 7";
51     reference
52     "12.6.2.3, 6.9.4 of IEEE Std 802.1Q";
53 }
54 }
55 grouping pcg-decoding-table-grouping {
56     description
57     "The Priority Code Point decoding table enables the decoding of the
58     priority and drop-eligible parameters from the PCP.";
59     reference
60     "6.9.3 of IEEE Std 802.1Q";
61     list pcg-decoding-map {
62         key "pcp";
63         description
64         "This map associates the priority code point field found in the VLAN
65         with a priority and drop eligible value based upon the priority code
66         point selection type.";
67         leaf pcg {
68             type pcg-selection-type;
69             description
70             "The priority code point selection type.";
71             reference
72             "12.6.2.7, 6.9.3 of IEEE Std 802.1Q";

```

```

1      }
2      list priority-map {
3          key "priority-code-point";
4          description
5              "This map associates a priority code point value with priority
6              and drop eligible parameters.";
7          leaf priority-code-point {
8              type priority-type;
9              description
10                 "Priority associated with the pcpc.";
11              reference
12                 "12.6.2.7, 6.9.3 of IEEE Std 802.1Q";
13          }
14          leaf priority {
15              type priority-type;
16              description
17                 "Priority associated with the pcpc.";
18              reference
19                 "12.6.2.7, 6.9.3 of IEEE Std 802.1Q";
20          }
21          leaf drop-eligible {
22              type boolean;
23              description
24                 "Drop eligible value for pcpc";
25              reference
26                 "12.6.2.7, 6.9.3 of IEEE Std 802.1Q";
27          }
28      }
29  }
30  }
31  grouping pcpc-encoding-table-grouping {
32      description
33          "The Priority Code Point encoding table encodes the priority and
34          drop-eligible parameters in the PCP field of the VLAN tag.";
35      reference
36          "12.6.2.9, 6.9.3 of IEEE Std 802.1Q";
37      list pcpc-encoding-map {
38          key "pcpc";
39          description
40              "This map associates the priority and drop-eligible parameters
41              with the priority used to encode the PCP of the VLAN based upon
42              the priority code point selection type.";
43          leaf pcpc {
44              type pcpc-selection-type;
45              description
46                 "The priority code point selection type.";
47              reference
48                 "12.6.2.7, 6.9.3 of IEEE Std 802.1Q";
49          }
50          list priority-map {
51              key "priority dei";
52              description
53                  "This map associates the priority and drop-eligible parameters
54                  with the priority code point field of the VLAN tag.";
55              leaf priority {
56                  type priority-type;
57                  description
58                     "Priority associated with the pcpc.";
59                  reference
60                     "12.6.2.7, 6.9.3 of IEEE Std 802.1Q";
61              }
62              leaf dei {
63                  type boolean;
64                  description
65                     "The drop eligible value.";
66                  reference
67                     "12.6.2, 8.6.6 of IEEE Std 802.1Q";
68              }
69              leaf priority-code-point {
70                  type priority-type;
71                  description
72                     "PCP value for priority when DEI value";

```

```
1         reference
2         "12.6.2.9, 6.9.3 of IEEE Std 802.1Q";
3     }
4 }
5 }
6 }
7 grouping service-access-priority-table-grouping {
8     description
9         "The Service Access Priority Table associates a received priority with
10        a service access priority.";
11    reference
12        "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
13    leaf priority0 {
14        type priority-type;
15        default "0";
16        description
17            "Service access priority value for priority 0";
18        reference
19            "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
20    }
21    leaf priority1 {
22        type priority-type;
23        default "1";
24        description
25            "Service access priority value for priority 1";
26        reference
27            "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
28    }
29    leaf priority2 {
30        type priority-type;
31        default "2";
32        description
33            "Service access priority value for priority 2";
34        reference
35            "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
36    }
37    leaf priority3 {
38        type priority-type;
39        default "3";
40        description
41            "Service access priority value for priority 3";
42        reference
43            "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
44    }
45    leaf priority4 {
46        type priority-type;
47        default "4";
48        description
49            "Service access priority value for priority 4";
50        reference
51            "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
52    }
53    leaf priority5 {
54        type priority-type;
55        default "5";
56        description
57            "Service access priority value for priority 5";
58        reference
59            "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
60    }
61    leaf priority6 {
62        type priority-type;
63        default "6";
64        description
65            "Service access priority value for priority 6";
66        reference
67            "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
68    }
69    leaf priority7 {
70        type priority-type;
71        default "7";
72        description
```

```

1      "Service access priority value for priority 7";
2      reference
3      "12.6.2.17, 6.13.1 of IEEE Std 802.1Q";
4  }
5  }
6  grouping traffic-class-table-grouping {
7      description
8      "The Traffic Class Table models the operations that can be
9      performed on, or can inquire about, the current contents of the
10     Traffic Class Table (8.6.6) for a given Port.";
11     reference
12     "12.6.3, 8.6.6 of IEEE Std 802.1Q";
13     list traffic-class-map {
14         key "priority";
15         status "deprecated";
16         description
17         "The priority index into the traffic class table. This list modeled
18         the Traffic Class Table incorrectly. available-traffic-class should
19         be a single value per port and not a list of all possible
20         available-traffic-class. For more information see maintenance issue
21         0230. The status of this object is deprecated. It is replaced by
22         sibling container traffic-class-table.";
23         leaf priority {
24             type priority-type;
25             description
26             "The priority of the traffic class entry.";
27             reference
28             "8.6.6 of IEEE Std 802.1Q";
29         }
30         list available-traffic-class {
31             key "num-traffic-class";
32             description
33             "The traffic class index associated with a given priority within
34             the traffic class table.";
35             reference
36             "8.6.6 of IEEE Std 802.1Q";
37             leaf num-traffic-class {
38                 type uint8 {
39                     range "1..8";
40                 }
41                 description
42                 "The available number of traffic classes.";
43                 reference
44                 "8.6.6 of IEEE Std 802.1Q";
45             }
46             leaf traffic-class {
47                 type traffic-class-type;
48                 description
49                 "The traffic class index associated with a given traffic class
50                 entry.";
51                 reference
52                 "8.6.6 of IEEE Std 802.1Q";
53             }
54         }
55     }
56     container traffic-class-table {
57         description
58         "The priority to traffic class mapping.";
59         reference
60         "8.6.6 of IEEE Std 802.1Q";
61         leaf number-of-traffic-classes {
62             type uint8 {
63                 range "1..8";
64             }
65             description
66             "The number of egress traffic classes supported on this port. This
67             object may be optionally read-only.";
68             reference
69             "12.6.3.1 of IEEE Std 802.1Q";
70         }
71         leaf priority0 {
72             type traffic-class-type;

```



```

1      must
2      "current() < ../number-of-traffic-classes";
3      description
4      "The traffic class index associated with priority 0";
5      reference
6      "8.6.6 of IEEE Std 802.1Q";
7  }
8  leaf priority1 {
9      type traffic-class-type;
10     must
11     "current() < ../number-of-traffic-classes";
12     description
13     "The traffic class index associated with priority 1";
14     reference
15     "8.6.6 of IEEE Std 802.1Q";
16 }
17 leaf priority2 {
18     type traffic-class-type;
19     must
20     "current() < ../number-of-traffic-classes";
21     description
22     "The traffic class index associated with priority 2";
23     reference
24     "8.6.6 of IEEE Std 802.1Q";
25 }
26 leaf priority3 {
27     type traffic-class-type;
28     must
29     "current() < ../number-of-traffic-classes";
30     description
31     "The traffic class index associated with priority 3";
32     reference
33     "8.6.6 of IEEE Std 802.1Q";
34 }
35 leaf priority4 {
36     type traffic-class-type;
37     must
38     "current() < ../number-of-traffic-classes";
39     description
40     "The traffic class index associated with priority 4";
41     reference
42     "8.6.6 of IEEE Std 802.1Q";
43 }
44 leaf priority5 {
45     type traffic-class-type;
46     must
47     "current() < ../number-of-traffic-classes";
48     description
49     "The traffic class index associated with priority 5";
50     reference
51     "8.6.6 of IEEE Std 802.1Q";
52 }
53 leaf priority6 {
54     type traffic-class-type;
55     must
56     "current() < ../number-of-traffic-classes";
57     description
58     "The traffic class index associated with priority 6";
59     reference
60     "8.6.6 of IEEE Std 802.1Q";
61 }
62 leaf priority7 {
63     type traffic-class-type;
64     must
65     "current() < ../number-of-traffic-classes";
66     description
67     "The traffic class index associated with priority 7";
68     reference
69     "8.6.6 of IEEE Std 802.1Q";
70 }
71 }
72 }

```

```

1  grouping transmission-selection-table-grouping {
2      description
3          "The Transmission Selection Algorithm Table models the operations that
4          can be performed on, or can inquire about, the current contents of the
5          Transmission Selection Algorithm Table (12.20.2) for a given Port.";
6      reference
7          "12.20.2, 8.6.8 of IEEE Std 802.1Q";
8      list transmission-selection-algorithm-map {
9          key "traffic-class";
10         description
11             "The traffic class to index into the transmission selection table.";
12         leaf traffic-class {
13             type traffic-class-type;
14             description
15                 "The traffic class of the entry.";
16             reference
17                 "8.6.6 of IEEE Std 802.1Q";
18         }
19         leaf transmission-selection-algorithm {
20             type identityref {
21                 base dot1q-types:transmission-selection-algorithm;
22             }
23             description
24                 "Transmission selection algorithm";
25             reference
26                 "8.6.8, Table 8-6 of IEEE Std 802.1Q";
27         }
28     }
29 }
30 grouping port-map-grouping {
31     description
32         "A set of control indicators, one for each Port. A Port Map,
33         containing a control element for each outbound Port";
34     reference
35         "8.8.1, 8.8.2 of IEEE Std 802.1Q";
36     list port-map {
37         key "port-ref";
38         description
39             "The list of entries composing the port map.";
40         leaf port-ref {
41             type port-number-type;
42             description
43                 "The interface port reference associated with this map.";
44             reference
45                 "8.8.1 of IEEE Std 802.1Q";
46         }
47         choice map-type {
48             description
49                 "Type of port map";
50             container static-filtering-entries {
51                 description
52                     "Static filtering entries attributes.";
53                 leaf control-element {
54                     type enumeration {
55                         enum forward {
56                             description
57                                 "Forwarded, independently of any dynamic filtering
58                                 information held by the FDB.";
59                         }
60                         enum filter {
61                             description
62                                 "Filtered, independently of any dynamic filtering
63                                 information.";
64                         }
65                         enum forward-filter {
66                             description
67                                 "Forwarded or filtered on the basis of dynamic filtering
68                                 information, or on the basis of the default Group
69                                 filtering behavior for the outbound Port (8.8.6) if no
70                                 dynamic filtering information is present specifically for
71                                 the MAC address.";
72                         }

```

```

1      }
2      description
3      "A control element for each outbound Port, specifying that
4      a frame with a destination MAC address, and in the case of
5      VLAN Bridge components, VID that meets this specification.";
6      reference
7      "8.8.1 of IEEE Std 802.1Q";
8  }
9  leaf connection-identifier {
10     type port-number-type;
11     description
12     "A Port MAP may contain a connection identifier (8.8.12) for
13     each outbound port. The connection identifier may be
14     associated with the Bridge Port value maintained in a Dynamic
15     Filtering Entry of the FDB for Bridge Ports.";
16     reference
17     "8.8.1, 8.8.12 of IEEE Std 802.1Q";
18 }
19 }
20 container static-vlan-registration-entries {
21     description
22     "Static VLAN registration entries.";
23     leaf registrar-admin-control {
24         type enumeration {
25             enum fixed-new-ignored {
26                 description
27                 "Registration Fixed (New ignored).";
28             }
29             enum fixed-new-propagated {
30                 description
31                 "Registration Fixed (New propagated).";
32             }
33             enum forbidden {
34                 description
35                 "Registration Forbidden.";
36             }
37             enum normal {
38                 description
39                 "Normal Registration.";
40             }
41         }
42         description
43         "The Registrar Administrative Control values for MVRP and MIRP
44         for the VID.";
45         reference
46         "8.8.2 of IEEE Std 802.1Q";
47     }
48     leaf vlan-transmitted {
49         type enumeration {
50             enum tagged {
51                 description
52                 "VLAN-tagged";
53             }
54             enum untagged {
55                 description
56                 "VLAN-untagged";
57             }
58         }
59         description
60         "Whether frames are to be VLAN-tagged or untagged when
61         transmitted.";
62         reference
63         "8.8.2 of IEEE Std 802.1Q";
64     }
65 }
66 container mac-address-registration-entries {
67     description
68     "MAC address registration entries attributes.";
69     leaf control-element {
70         type enumeration {
71             enum registered {
72                 description

```

```

1         "Forwarded, independently of any dynamic filtering
2         information held by the FDB.";
3     }
4     enum not-registered {
5         description
6         "Filtered, independently of any dynamic filtering
7         information.";
8     }
9 }
10 description
11 "A control element for each outbound Port, specifying that
12 a frame with a destination MAC address, and in the case of
13 VLAN Bridge components, VID that meets this specification.";
14 reference
15 "8.8.4 of IEEE Std 802.1Q";
16 }
17 }
18 container dynamic-vlan-registration-entries {
19     description
20     "Dynamic VLAN registration entries attributes.";
21     leaf control-element {
22         type enumeration {
23             enum registered {
24                 description
25                 "Forwarded, independently of any dynamic filtering
26                 information held by the FDB.";
27             }
28         }
29         description
30         "A control element for each outbound Port, specifying that
31         a frame with a destination MAC address, and in the case of
32         VLAN Bridge components, VID that meets this specification.";
33         reference
34         "8.8.5 of IEEE Std 802.1Q";
35     }
36 }
37 container dynamic-reservation-entries {
38     description
39     "Dynamic reservation entries attributes.";
40     leaf control-element {
41         type enumeration {
42             enum forward {
43                 description
44                 "Forwarded, independently of any dynamic filtering
45                 information held by the FDB.";
46             }
47             enum filter {
48                 description
49                 "Filtered, independently of any dynamic filtering
50                 information.";
51             }
52         }
53         description
54         "A control element for each outbound Port, specifying that
55         a frame with a destination MAC address, and in the case of
56         VLAN Bridge components, VID that meets this specification.";
57         reference
58         "8.8.7 of IEEE Std 802.1Q";
59     }
60 }
61 container dynamic-filtering-entries {
62     description
63     "Dynamic filtering entries attributes.";
64     leaf control-element {
65         type enumeration {
66             enum forward {
67                 description
68                 "Forwarded, independently of any dynamic filtering
69                 information held by the FDB.";
70             }
71         }
72         description

```

```

1         "A control element for each outbound Port, specifying that
2         a frame with a destination MAC address, and in the case of
3         VLAN Bridge components, VID that meets this specification.";
4     reference
5         "8.8.3 of IEEE Std 802.1Q";
6     }
7 }
8 }
9 }
10 }
11 grouping bridge-port-statistics-grouping {
12     description
13         "Grouping of bridge port statistics.";
14     reference
15         "12.6.1.1.3 of IEEE Std 802.1Q";
16     leaf delay-exceeded-discards {
17         type yang:counter64;
18         description
19             "The number of frames discarded by this port due to excessive
20             transit delay through the Bridge. It is incremented by both
21             transparent and source route Bridges.";
22         reference
23             "12.6.1.1.3, 8.6.6 of IEEE Std 802.1Q";
24     }
25     leaf mtu-exceeded-discards {
26         type yang:counter64;
27         description
28             "The number of frames discarded by this port due to an excessive
29             size. It is incremented by both transparent and source route
30             Bridges.";
31         reference
32             "Item g) in 12.6.1.1.3 of IEEE Std 802.1Q";
33     }
34     leaf frame-rx {
35         type yang:counter64;
36         description
37             "The number of frames that have been received by this port from its
38             segment. Note that a frame received on the interface corresponding
39             to this port is only counted by this object if and only if it is for
40             a protocol being processed by the local bridging function, including
41             Bridge management frames.";
42         reference
43             "12.6.1.1.3 of IEEE Std 802.1Q";
44     }
45     leaf octets-rx {
46         type yang:counter64;
47         description
48             "The total number of octets in all valid frames received (including
49             BPDUs, frames addressed to the Bridge as an end station, and frames
50             that were submitted to the Forwarding Process).";
51         reference
52             "12.6.1.1.3 of IEEE Std 802.1Q";
53     }
54     leaf frame-tx {
55         type yang:counter64;
56         description
57             "The number of frames that have been transmitted by this port to its
58             segment. Note that a frame transmitted on the interface
59             corresponding to this port is only counted by this object if and
60             only if it is for a protocol being processed by the local bridging
61             function, including Bridge management frames.";
62     }
63     leaf octets-tx {
64         type yang:counter64;
65         description
66             "The total number of octets that have been transmitted by this port
67             to its segment.";
68     }
69     leaf discard-inbound {
70         type yang:counter64;
71         description
72             "Count of received valid frames that were discarded (i.e., filtered)

```

```

1      by the Forwarding Process.";
2      reference
3      "12.6.1.1.3 of IEEE Std 802.1Q";
4  }
5  leaf forward-outbound {
6      type yang:counter64;
7      description
8      "The number of frames forwarded to the associated MAC Entity (8.5).";
9      reference
10     "12.6.1.1.3 of IEEE Std 802.1Q";
11 }
12 leaf discard-lack-of-buffers {
13     type yang:counter64;
14     description
15     "The count of frames that were to be transmitted through the
16     associated Port but were discarded due to lack of buffers.";
17     reference
18     "12.6.1.1.3 of IEEE Std 802.1Q";
19 }
20 leaf discard-transit-delay-exceeded {
21     type yang:counter64;
22     description
23     "The number of frames discarded by this port due to excessive
24     transit delay through the Bridge. It is incremented by both
25     transparent and source route Bridges.";
26     reference
27     "12.6.1.1.3 of IEEE Std 802.1Q";
28 }
29 leaf discard-on-error {
30     type yang:counter64;
31     description
32     "The number of frames that were to be forwarded on the associated
33     MAC but could not be transmitted (e.g., frame would be too large,
34     6.5.8).";
35     reference
36     "12.6.1.1.3 of IEEE Std 802.1Q";
37 }
38 }
39 grouping base-gate-control-entries {
40     description
41     "Grouping of gate control entries.";
42     list gate-control-entry {
43         key "index";
44         description
45         "List of gate control entries.";
46         leaf index {
47             type uint32;
48             description
49             "The index for this entry.";
50         }
51         leaf operation-name {
52             type identityref {
53                 base type-of-operation;
54             }
55             mandatory true;
56             description
57             "The name (type) of the operation for this entry.";
58         }
59         leaf time-interval-value {
60             type uint32;
61             description
62             "timeIntervalValue is a 32-bit unsigned integer, representing a
63             number of nanoseconds. After timeIntervalValue nanoseconds have
64             elapsed since the completion of the previous entry in the gate
65             control list, control passes to the next entry.";
66             reference
67             "12.29.1.2.3, 8.6.8.4 of IEEE Std 802.1Q";
68         }
69     }
70 }
71 }
72

```

1 48.6.3 The ieee802-dot1q-tsn-types YANG module

```
2 module ieee802-dot1q-tsn-types {
3   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-tsn-types;
4   prefix dot1q-tsn-types;
5   import ietf-inet-types {
6     prefix inet;
7   }
8   organization
9     "Institute of Electrical and Electronics Engineers";
10  contact
11    "WG-URL: http://ieee802.org/1/
12    WG-EMail: stds-802-1@ieee.org
13
14    Contact: IEEE 802.1 Working Group Chair
15    Postal: C/O IEEE 802.1 Working Group
16            IEEE Standards Association
17            445 Hoes Lane
18            Piscataway
19            NJ 08854
20            USA
21
22    E-mail: stds-802-1@ieee.org";
23  description
24    "Common typedefs and groupings for TSN user/network configuration in
25    IEEE Std 802.1Q.
26
27    Copyright (C) IEEE (2022).
28
29    This version of this YANG module is part of IEEE Std 802.1Q; see the
30    standard itself for full legal notices.";
31  revision 2024-03-21 {
32    description
33      "TO BE Published as part of IEEE Std 802.1Q-2022-Revision.
34      Deprecated stream-id-type, added stream-id-type-upper.";
35    reference
36      "IEEE Std 802.1Q-2022-Revision, Bridges and Bridged Networks.
37      Maintenance Task Group database item 0312.";
38  }
39  revision 2022-10-29 {
40    description
41      "Published as part of IEEE Std 802.1Q-2022. Second version";
42    reference
43      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
44  }
45  revision 2018-02-15 {
46    description
47      "Amendment: Stream Reservation Protocol (SRP) Enhancements and
48      Performance Improvements. Initial version";
49    reference
50      "46.3 of IEEE Std 802.1Qcc-2018";
51  }
52  typedef stream-id-type {
53    type string {
54      pattern
55        "[0-9a-fA-F]{2}" +
56        "(-[0-9a-fA-F]{2}){5}" +
57        ":" +
58        "[0-9a-fA-F]{2}" +
59        "-" +
60        "[0-9a-fA-F]{2}";
61    }
62    status "deprecated";
63    description
64      "This typedef is deprecated. Use stream-id-type-upper instead.
65
66      This typedef specifies a Stream ID, a unique identifier of the
67      Stream's configuration, used by protocols in the network to
68      associate the user's Stream with TSN resources.
69
70      The Stream ID is a string that represents two fields:
71
```

1 MAC Address:

2

3 A 48-bit IEEE 802 MAC address associated with the Talker sourcing

4 the Stream to the bridged network. The entire range of MAC

5 addresses are acceptable.

6

7 NOTE 1: The MAC address component of the StreamID can, but does not

8 necessarily, have the same value as the source_address parameter

9 of any frame in the actual data Stream. For example, the Stream ID

10 can be assigned by a TSN CUC (see 46.1.3.3 of IEEE Std

11 802.1Q-2022), using a pool of MAC addresses that the TSN CUC

12 maintains.

13

14 NOTE 2: If the MAC addresses used to construct Stream IDs are not

15 unique within the network, duplicate Stream IDs can be generated,

16 with unpredictable results.

17

18 Unique ID:

19

20 A 16-bit unique ID that is used to distinguish between multiple

21 Streams within the station identified by MAC Address.

22

23 The string specifies eight octets, with each octet represented as

24 two hexadecimal characters. The first six octets specify the MAC

25 Address, using the canonical format of IEEE Std 802, with a dash

26 separating each octet. The last two octets specify the Unique ID,

27 with the high-order octet, a dash, and then the low-order octet.

28 The MAC Address and Unique ID are separated by colon.

29

30 stream-id-type is intended for use by other modules as the type

31 for a key to a list of Stream configurations (using group-talker

32 and group-listener) and a list of Stream status (using

33 group-status-stream and group-status-talker-listener).";

34 reference

35 "46.2.3.1 of IEEE Std 802.1Q-2022";

36 }

37 typedef stream-id-type-upper {

38 type string {

39 pattern

40 "[0-9A-F]{2}" +

41 "(-[0-9A-F]{2}){5}" +

42 ": " +

43 "[0-9A-F]{2}" +

44 "- " +

45 "[0-9A-F]{2}";

46 }

47 description

48 "This typedef specifies a Stream ID, a unique identifier of the

49 Stream's configuration, used by protocols in the network to

50 associate the user's Stream with TSN resources.

51

52 The Stream ID is a string that represents two fields:

53

54 MAC Address:

55

56 A 48-bit IEEE 802 MAC address associated with the Talker sourcing

57 the Stream to the bridged network. The entire range of MAC

58 addresses are acceptable.

59

60 NOTE 1: The MAC address component of the StreamID can, but does not

61 necessarily, have the same value as the source_address parameter

62 of any frame in the actual data Stream. For example, the Stream ID

63 can be assigned by a TSN CUC (see 46.1.3.3 of IEEE Std

64 802.1Q-2022), using a pool of MAC addresses that the TSN CUC

65 maintains.

66

67 NOTE 2: If the MAC addresses used to construct Stream IDs are not

68 unique within the network, duplicate Stream IDs can be generated,

69 with unpredictable results.

70

71 Unique ID:

72

1 A 16-bit unique ID that is used to distinguish between multiple
2 Streams within the station identified by MAC Address.
3
4 The string specifies eight octets, with each octet represented as
5 two hexadecimal characters. The first six octets specify the MAC
6 Address, using the canonical format of IEEE Std 802, with a dash
7 separating each octet. The last two octets specify the Unique ID,
8 with the high-order octet, a dash, and then the low-order octet.
9 The MAC Address and Unique ID are separated by colon. Only upper
10 case characters are allowed to be used for the hexadecimal
11 characters.
12
13 stream-id-type-upper is intended for use by other modules as the type
14 for a key to a list of Stream configurations (using group-talker
15 and group-listener) and a list of Stream status (using
16 group-status-stream and group-status-talker-listener).";
17 reference
18 "46.2.3.1 of IEEE Std 802.1Q-2022";
19 }
20 grouping group-interface-id {
21 description
22 "This YANG grouping specifies the identification of a distinct
23 point of attachment (interface) in a station (end station or
24 Bridge).";
25 reference
26 "46.2.3.3 of IEEE Std 802.1Q-2022";
27 leaf mac-address {
28 type string {
29 pattern "[0-9a-fA-F]{2}(-[0-9a-fA-F]{2}){5}";
30 }
31 description
32 "mac-address is the unique individual MAC address (IEEE Std 802) of
33 the interface in the station (end station or Bridge). This MAC
34 address uniquely identifies the station within the local
35 network.
36
37 mac-address shall be included in an instance of a container
38 using group-interface-id.
39
40 NOTE: This MAC address can be discovered in the physical topology
41 using protocols such as IEEE Std 802.1AB (LLDP). LLDP supports
42 MAC address as a subtype for the stations Chassis ID and Port
43 ID. If the station does not use MAC address for its LLDP IDs,
44 remote management can be used to associate this mac-address to
45 the values provided in the LLDP IDs.
46
47 The string uses the hexadecimal representation specified in IEEE
48 Std 802 (i.e. canonical format).";
49 }
50 leaf interface-name {
51 type string;
52 description
53 "interface-name is the name of the interface that is assigned
54 locally by the station (end station or Bridge).
55
56 interface-name may be included in an instance of a container
57 using group-interface-id.
58
59 IEEE Std 802 recommends that each distinct point of attachment
60 to an IEEE 802 network have its own EUI MAC address. If the
61 identified station follows this IEEE 802 recommendation, the
62 mac-address leaf uniquely identifies the interface as well as
63 the station, and interface-name is not needed.
64
65 If the mac-address applies to more than one interface (distinct
66 point of attachment) within the station, interface-name provides
67 a locally assigned name that can help to identify the interface.
68
69 When YANG is used for management of the station, interface-name
70 is the interface name that serves as the key for the stations
71 interface list (RFC7223).
72

```

1     NOTE 1: The TSN CNC is typically located in a different physical
2     product than the station identified by this group-interface-id.
3     Since the interface-name is assigned locally by the identified
4     station, it is possible that the stations product will change
5     interface-name in a manner that the TSN CNC cannot detect. For
6     example, RFC7223 mentions that the YANG interface name can
7     change when a physical attachment point is inserted or removed.
8
9     NOTE 2: This interface name can be discovered in the physical
10    topology using protocols such as IEEE Std 802.1AB (LLDP). LLDP
11    supports interface name as a subtype for its Port ID. If the
12    station does not use interface name for its LLDP Port ID, remote
13    management can be used to associate this interface-name to the
14    values provided in the LLDP Port ID.";
15    }
16  }
17  grouping group-ieee802-mac-addresses {
18    description
19      "This YANG grouping specifies the pair of IEEE 802 MAC addresses
20      for Stream identification.
21
22      The use of these fields for Stream identification corresponds to
23      the managed objects for Stream identification in IEEE Std 802.1CB.
24      If inconsistency arises between this specification and IEEE Std
25      802.1CB, IEEE Std 802.1CB takes precedence.";
26    reference
27      "46.2.3.4.1 of IEEE Std 802.1Q-2022";
28    leaf destination-mac-address {
29      type string {
30        pattern "[0-9a-fA-F]{2}(-[0-9a-fA-F]{2}){5}";
31      }
32      description
33        "Destination MAC address.
34
35        An address of all 1's specifies that the destination MAC address
36        is ignored for purposes of Stream identification.
37
38        The string uses the hexadecimal representation specified in IEEE
39        Std 802 (i.e. canonical format).";
40    }
41    leaf source-mac-address {
42      type string {
43        pattern "[0-9a-fA-F]{2}(-[0-9a-fA-F]{2}){5}";
44      }
45      description
46        "Source MAC address.
47
48        An address of all 1's specifies that the source MAC address is
49        ignored for purposes of Stream identification.
50
51        The string uses the hexadecimal representation specified in IEEE
52        Std 802 (i.e. canonical format).";
53    }
54  }
55  grouping group-ieee802-vlan-tag {
56    description
57      "This YANG grouping specifies a customer VLAN Tag (C-TAG of clause
58      9) for Stream identification.
59
60      The Drop Eligible Indicator (DEI) field is not relevant from the
61      perspective of a TSN Talker/Listener.
62
63      The use of these fields for Stream identification corresponds to
64      the managed objects for Stream identification in IEEE Std 802.1CB.
65      If inconsistency arises between this specification and IEEE Std
66      802.1CB, IEEE Std 802.1CB takes precedence.";
67    reference
68      "46.2.3.4.2 of IEEE Std 802.1Q-2022";
69    leaf priority-code-point {
70      type uint8 {
71        range "0 .. 7";
72      }

```

```

1      description
2          "Priority Code Point (PCP) field.
3
4          The priority-code-point is not used to identify the Stream, but
5          it does identify a traffic class (queue) in Bridges.";
6      }
7      leaf vlan-id {
8          type uint16 {
9              range "0 .. 4095";
10         }
11         description
12             "VLAN ID (VID) field.
13
14             If only the priority-code-point is known, the vlan-id is
15             specified as 0.";
16     }
17 }
18 grouping group-ipv4-tuple {
19     description
20         "This YANG grouping specifies parameters to identify an IPv4
21         (RFC791) Stream.
22
23         The use of these fields for Stream identification corresponds to
24         the managed objects for Stream identification in IEEE Std 802.1CB.
25         If inconsistency arises between this specification and IEEE Std
26         802.1CB, IEEE Std 802.1CB takes precedence.";
27     reference
28         "46.2.3.4.3 of IEEE Std 802.1Q-2022";
29     leaf source-ip-address {
30         type inet:ipv4-address;
31         description
32             "Source IPv4 address.
33
34             An address of all 0's specifies that the IP source address is
35             ignored for purposes of Stream identification.";
36     }
37     leaf destination-ip-address {
38         type inet:ipv4-address;
39         description
40             "Destination IPv4 address.";
41     }
42     leaf dscp {
43         type uint8;
44         description
45             "Differentiated services code point, DSCP (RFC2474).
46
47             A value of 64 decimal specifies that the DSCP is ignored for
48             purposes of Stream identification.";
49     }
50     leaf protocol {
51         type uint16;
52         description
53             "IPv4 Protocol (e.g. UDP).
54
55             The special value of all 1s (FFFF hex) represents None, meaning
56             that protocol, source-port, and destination-port are ignored for
57             purposes of Stream identification.
58
59             For any value other than all 1s, the lower octet is used to
60             match IPv4 Protocol.";
61     }
62     leaf source-port {
63         type uint16;
64         description
65             "This matches the source port of the protocol.";
66     }
67     leaf destination-port {
68         type uint16;
69         description
70             "This matches the destination port of the protocol.";
71     }
72 }

```

```

1  grouping group-ipv6-tuple {
2      description
3          "This YANG grouping specifies parameters to identify an IPv6
4          (RFC8200) Stream.
5
6          The use of these fields for Stream identification corresponds to
7          the managed objects for Stream identification in IEEE Std 802.1CB.
8          If inconsistency arises between this specification and IEEE Std
9          802.1CB, IEEE Std 802.1CB takes precedence.";
10     reference
11         "46.2.3.4.4 of IEEE Std 802.1Q-2022";
12     leaf source-ip-address {
13         type inet:ipv6-address;
14         description
15             "Source IPv6 address.
16
17             An address of all 0's specifies that the IP source address is
18             ignored for purposes of Stream identification.";
19     }
20     leaf destination-ip-address {
21         type inet:ipv6-address;
22         description
23             "Destination IPv6 address.";
24     }
25     leaf dscp {
26         type uint8;
27         description
28             "Differentiated services code point, DSCP (RFC2474).
29
30             A value of 64 decimal specifies that the DSCP is ignored for
31             purposes of Stream identification.";
32     }
33     leaf protocol {
34         type uint16;
35         description
36             "IPv6 Next Header (e.g. UDP).
37
38             The special value of all 1s (FFFF hex) represents None, meaning
39             that protocol, source-port, and destination-port are ignored for
40             purposes of Stream identification.
41
42             For any value other than all 1s, the lower octet is used to
43             match IPv6 Next Header.";
44     }
45     leaf source-port {
46         type uint16;
47         description
48             "This matches the source port of the protocol.";
49     }
50     leaf destination-port {
51         type uint16;
52         description
53             "This matches the destination port of the protocol.";
54     }
55 }
56 grouping group-user-to-network-requirements {
57     description
58         "This YANG grouping specifies user requirements for the
59         Stream, such as latency and redundancy.
60
61         The network (e.g. CNC) will merge all user-to-network-requirements
62         for a Stream to ensure that all requirements are met.";
63     reference
64         "46.2.3.6 of IEEE Std 802.1Q-2022";
65     leaf num-seamless-trees {
66         type uint8;
67         default "1";
68         description
69             "num-seamless-trees specifies the number of trees that the
70             network will configure to deliver seamless redundancy for the
71             Stream.
72 
```

1 The value zero is interpreted as one (i.e. no seamless
2 redundancy).
3
4 This requirement is provided from the Talker only. Listeners
5 shall set this leaf to one.
6
7 From each Talker to a single Listener, the network configures a
8 path that relays Stream data through Bridges. If the Talker has
9 more than one Listener, the network configures a tree of
10 multiple paths.
11
12 num-seamless-trees specifies the number of maximally disjoint
13 trees that the network shall configure from the Talker to all
14 Listeners. Each tree is disjoint from other trees, in that the
15 network evaluates the physical topology to avoid sharing the
16 same Bridge and links in each trees paths. This computation of
17 disjoint trees is maximal, in that shared Bridges and links are
18 avoided to the maximum extent allowed by the physical topology.
19 For example, if a single link exists from a Bridge to a
20 Listener, and num-seamless-trees is 3, then all 3 trees will
21 share that link to the Listener.
22
23 When num-seamless-trees is greater than one, the transfer of the
24 Streams data frames shall use a seamless redundancy standard,
25 such as IEEE Std 802.1CB. When a link shared by multiple trees
26 diverges to multiple disjoint links, the seamless redundancy
27 standard replicates (i.e. forwards a distinct copy of each data
28 frame to the disjoint trees). When disjoint trees converge to a
29 single link, the seamless redundancy standard eliminates the
30 duplicate copies of each data frame. Assuming that other sources
31 of frame loss are mitigated (e.g. congestion), failure of a link
32 or Bridge in one disjoint tree does not result in frame loss as
33 long as at least one remaining disjoint tree is operational.
34
35 If the Talker sets this leaf to one, the network may make use of
36 redundancy standards that are not seamless (i.e. failure of link
37 results in lost frames), such as MSTP and IS-IS.
38
39 If the Talker sets this leaf to greater than one, and seamless
40 redundancy is not possible in the current network (no disjoint
41 paths, or no seamless redundancy standard in Bridges),
42 group-status-stream.status-info.failure-code is non-zero
43 (46.2.4.1 of IEEE Std 802.1Q-2022).
44
45 If group-user-to-network-requirements is not provided by the
46 Talker or Listener, the network shall use the default value of
47 one for this leaf."
48 reference
49 "46.2.3.6.1 of IEEE Std 802.1Q-2022";
50 }
51 leaf max-latency {
52 type uint32;
53 default "0";
54 description
55 "Maximum latency from Talker to Listener(s) for a single frame
56 of the Stream.
57
58 max-latency is specified as an integer number of nanoseconds.
59
60 Latency shall use the definition of 3.102, with additional
61 context as follows: The Known reference point in the frame is
62 the message timestamp point specified in IEEE Std 802.1AS for
63 various media (i.e. start of the frame). The first point is in
64 the Talker, at the reference plane marking the boundary between
65 the network media and PHY (see IEEE Std 802.1AS). The second
66 point is in the Listener, at the reference plane marking the
67 boundary between the network media and PHY.
68
69 When this requirement is specified by the Talker, it must be
70 satisfied for all Listeners.
71
72 When this requirement is specified by the Listener, it must be

1 satisfied for this Listener only.

2

3 If group-user-to-network-requirements is not provided by the

4 Talker or Listener, the network shall use the default value of

5 zero for this leaf.

6

7 The special value of zero represents usage of the initial value

8 of group-status-talker-listener.accumulated-latency as the

9 maximum latency requirement. This effectively locks-down the

10 initial latency that the network calculates after successful

11 configuration of the Stream, such that any subsequent increase

12 in latency beyond that value causes the Stream to fail.

13

14 The assumption for when the first point occurs in the Talker

15 depends on the presence of the time-aware container in the

16 Talkers traffic-specification.

17

18 When time-aware is not present:

19

20 The Talker is assumed to transmit at an arbitrary time (not

21 scheduled).

22

23 When time-aware is present:

24

25 The first point is assumed to occur at the start of each

26 traffic-specification.interval, as if the Talkers offsets

27 (earliest-transmit-offset and latest-transmit-offset) are both

28 zero. The Talkers offsets are not typically zero, but use of the

29 start of interval for purposes of max-latency allows the

30 Listener(s) to schedule their application independently from the

31 Talkers offset configuration.

32

33 The Listener determines max-latency based on its scheduling of a

34 read function in the application. Nevertheless, the time from

35 frame reception (i.e. second point) to execution of the read

36 function is in the user scope, and therefore not included in

37 max-latency.

38

39 max-latency can be set to a value greater than the Talkers

40 interval, in order to specify a longer latency requirement. For

41 example, if the Talkers interval is 500 microsec, and

42 max-latency is 700 microsec, the Listener receives the frame no

43 later than 200 microsec into the interval that follows the

44 Talkers interval.";

45 reference

46 "46.2.3.6.2 of IEEE Std 802.1Q-2022";

47 }

48 }

49 grouping group-interface-capabilities {

50 description

51 "This YANG grouping specifies the network capabilities of all

52 interfaces (Ports) contained in end-station-interfaces.

53

54 The network may provide configuration of these capabilities in

55 group-status-talker-listener.interface-configuration.

56

57 NOTE: If an end station contains multiple interfaces with different

58 network capabilities, each interface should be specified as a

59 distinct Talker or Listener (i.e. one entry in

60 end-station-interfaces). Use of multiple entries in

61 end-station-interfaces is intended for network capabilities that

62 span multiple interfaces (e.g. seamless redundancy).";

63 reference

64 "46.2.3.7 of IEEE Std 802.1Q-2022";

65 leaf vlan-tag-capable {

66 type boolean;

67 default "false";

68 description

69 "When vlan-tag-capable is true, the interface supports the

70 ability to tag/untag frames using a Customer VLAN Tag (C-TAG of

71 clause 9) provided by the network.

72

1 For a Talker, the networks tag replaces the tag specified by the
2 data-frame-specification. If the data-frame-specification is
3 untagged (no group-ieee802-vlan-tag), the networks tag is
4 inserted in the frame as it passes through the interface.
5
6 For a Listener, the users tag from the data-frame-specification
7 replaces the networks tag as the frame passes through the
8 interface. If the data-frame-specification is untagged (no
9 group-ieee802-vlan-tag), the networks tag is removed from the
10 frame as it passes through the interface.
11
12 If the end station supports more than one interface (i.e. more
13 than one entry in end-station-interfaces), vlan-tag-capable of
14 true means that a distinct VLAN tag can be applied to each
15 interface. The list of VLAN tag (one for each interface) can be
16 provided by the network in
17 interface-configuration.interface-list (ieee802-vlan-tag
18 choice).
19
20 When vlan-tag-capable is false, the interface does not support
21 the capability to tag/untag frames using a Customer VLAN Tag
22 (C-TAG of clause 9) provided by the network.
23
24 If interface-capabilities is not provided by the Talker or
25 Listener, the network shall use the default value of false for
26 this leaf.";
27 reference
28 "46.2.3.7.1 of IEEE Std 802.1Q-2022";
29 }
30 leaf-list cb-stream-iden-type-list {
31 type uint32;
32 description
33 "cb-stream-iden-type-list provides a list of the supported
34 Stream Identification types as specified in IEEE Std 802.1CB.
35
36 Each Stream Identification type is provided as a 32-bit unsigned
37 integer. The upper three octets contain the OUI/CID, and the
38 lowest octet contains the type number.
39
40 NOTE: If the Talker/Listener end system supports IEEE Std 802.1CB,
41 Null Stream identification is required, and that Stream
42 Identification type is included in this list. If the
43 Talker/Listener end system does not support IEEE Std 802.1CB,
44 this list is empty.
45
46 If the end station supports more than one interface (i.e. more
47 than one interface-id in end-station-interfaces, an empty
48 cb-stream-iden-type-list means that the end station is capable
49 of transferring the Stream on any one of its interfaces (not
50 all). When this is specified, the network shall decide which
51 interface is best used for TSN purposes, and communicate that
52 decision by returning a single interface in
53 interface-configuration.interface-list. The Talker/Listener uses
54 this interface alone for the Stream.
55
56 If interface-capabilities is not provided within group-talker or
57 group-listener, the network shall use an empty list as the
58 default value for this element.";
59 reference
60 "46.2.3.7.2 of IEEE Std 802.1Q-2022";
61 }
62 leaf-list cb-sequence-type-list {
63 type uint32;
64 description
65 "cb-sequence-type-list provides a list of the supported Sequence
66 Encode/Decode types as specified in IEEE Std 802.1CB.
67
68 Each sequence type is provided as a 32-bit unsigned integer. The
69 upper three octets contain the OUI/CID, and the lowest octet
70 contains the type number.
71
72 If interface-capabilities is not provided within group-talker or

```

1      group-listener, the network shall use an empty list as the
2      default value for this element.";
3      reference
4      "46.2.3.7.3 of IEEE Std 802.1Q-2022";
5  }
6  }
7  grouping group-interface-configuration {
8      description
9      "This YANG grouping provides configuration of interfaces in the
10      Talker/Listener. This configuration assists the network in meeting
11      the Streams requirements. The interface-configuration meets the
12      capabilities of the interface as provided in
13      interface-capabilities.";
14      reference
15      "46.2.5.3 of IEEE Std 802.1Q-2022";
16      list interface-list {
17          key "mac-address interface-name";
18          description
19          "A distinct configuration is provided for each interface in the
20          Talker/Listener (even if multiple interfaces use the same
21          configuration). Each entry in this interface-list consists of an
22          interface identification (group-interface-id), followed by a
23          list of configuration values for that interface (config-list).
24
25          If interface-configuration is not provided within
26          group-status-talker-listener, the network shall assume zero
27          entries as the default (no interface configuration).
28
29          Since the interface-name leaf is optional, empty string can be
30          used for its key value.";
31          uses group-interface-id;
32          list config-list {
33              key "index";
34              description
35              "List of configuration values for the interface.";
36              leaf index {
37                  type uint8;
38                  description
39                  "This index is provided in order to provide a unique key per
40                  list entry. The value of index for each entry shall be
41                  unique (but not necessarily contiguous).";
42              }
43              choice config-value {
44                  description
45                  "One of the following choices is provided for each
46                  configuration value. Each container name acts as the case
47                  name for the choice.";
48                  container ieee802-mac-addresses {
49                      description
50                      "Source and destination MAC addresses that apply to the
51                      network side of the user/network boundary.
52
53                      NOTE 1: On the user side, the MAC addresses correspond to
54                      the ieee802-mac-addresses of data-frame-specification.
55
56                      NOTE 2: The source MAC address of the network is typically
57                      the same as the user. The destination MAC address can be
58                      different. For example, the user can use an individual
59                      address, but the network can use a group (multicast)
60                      address.
61
62                      This configuration value is not provided unless IEEE Std
63                      802.1CB is supported, and a value for Active Destination
64                      MAC and VLAN Stream identification is provided in
65                      cb-stream-iden-type-list of interface-capabilities.";
66                      reference
67                      "46.2.5.3.1 of IEEE Std 802.1Q-2022";
68                      uses group-ieee802-mac-addresses;
69                  }
70                  container ieee802-vlan-tag {
71                      description
72                      "Customer VLAN Tag (C-TAG of clause 9) that applies to the

```



```
1      network side of the user/network boundary.
2
3      NOTE: On the user side, the VLAN tag corresponds to the
4      ieee802-vlan-tag of data-frame-specification (including
5      untagged if this field is not provided).
6
7      If the user provides a VLAN ID in the ieee802-vlan-tag of
8      data-frame-specification, the Streams data frames are
9      assumed to be limited to the active topology for that VLAN
10     ID. Therefore, if the network uses a different VLAN ID in
11     this config-value, the network shall ensure that the
12     replacement VLAN ID is limited to the equivalent active
13     topology.
14
15     This configuration value is not provided unless
16     vlan-tag-capable of interface-capabilities is true.";
17     reference
18     "46.2.5.3.2 of IEEE Std 802.1Q-2022";
19     uses group-ieee802-vlan-tag;
20 }
21 container ipv4-tuple {
22     description
23     "IPv4 identification that applies to the network side of
24     the user/network boundary.
25
26     This configuration value is not provided unless IEEE Std
27     802.1CB is supported, and a value for IP Stream
28     identification is provided in cb-stream-iden-type-list of
29     interface-capabilities.";
30     reference
31     "46.2.5.3.3 of IEEE Std 802.1Q-2022";
32     uses group-ipv4-tuple;
33 }
34 container ipv6-tuple {
35     description
36     "IPv6 identification that applies to the network side of
37     the user/network boundary.
38
39     This configuration value is not provided unless IEEE Std
40     802.1CB is supported, and a value for IP Stream
41     identification is provided in cb-stream-iden-type-list of
42     interface-capabilities.";
43     reference
44     "46.2.5.3.4 of IEEE Std 802.1Q-2022";
45     uses group-ipv6-tuple;
46 }
47 leaf time-aware-offset {
48     type uint32;
49     description
50     "If the time-aware container is present in the
51     traffic-specification of the Talker, this config-value
52     shall be provided by the network to the Talker.
53
54     If the time-aware container is not present in the
55     traffic-specification of the Talker, this config-value
56     shall not be provided by the network.
57
58     This config-value shall not be provided to Listeners, as
59     it is not applicable.
60
61     time-aware-offset specifies the offset that the Talker
62     shall use for transmit. The network returns a value
63     between earliest-transmit-offset and
64     latest-transmit-offset of the Talkers
65     traffic-specification. The value is expressed as
66     nanoseconds after the start of the Talkers interval.";
67     reference
68     "46.2.5.3.5 of IEEE Std 802.1Q-2022";
69 }
70 }
71 }
72 }
```

```

1  }
2  grouping group-talker {
3    description
4      "This YANG grouping specifies: - Talkers behavior for Stream
5      (how/when transmitted) - Talkers requirements from the network -
6      TSN capabilities of the Talkers interface(s)
7
8      In the fully centralized model of TSN configuration, this grouping
9      originates from the CUC, and is delivered to the CNC.";
10   reference
11     "46.2.3 of IEEE Std 802.1Q-2022";
12   container stream-rank {
13     description
14       "Rank of this Stream's configuration relative to other Streams
15       in the network. This rank is used to determine success/failure
16       of Stream resource configuration, and it is unrelated to the
17       Streams data.";
18     reference
19       "46.2.3.2 of IEEE Std 802.1Q-2022";
20     leaf rank {
21       type uint8;
22       description
23         "The Rank is used by the network to decide which Streams can
24         and cannot exist when TSN resources reach their limit. If a
25         Bridges Port becomes oversubscribed (e.g. network
26         reconfiguration, IEEE 802.11 bandwidth reduction), the Rank is
27         used to help determine which Streams can be dropped (i.e.
28         removed from Bridge configuration).
29
30         The only valid values for Rank shall be zero and one. The
31         configuration of a Stream with Rank zero is more important
32         than the configuration of a Stream with Rank one. The Rank
33         value of zero is intended for emergency traffic, and the Rank
34         value of one is intended for non-emergency traffic.
35
36         NOTE: It is expected that higher layer applications and
37         protocols can use the Rank to indicate the relative importance
38         of Streams based on user preferences. Those user preferences
39         are expressed by means beyond the scope of this standard. When
40         multiple applications exist in a network (e.g. audio/video
41         along with industrial control), it can be challenging for the
42         varied applications and vendors to agree on multiple Rank
43         values. To mitigate such challenges, this Rank uses a simple
44         concept of emergency (zero) and non-emergency (one) that can
45         be applied over all applications. For example, in a network
46         that carries audio Streams for fire safety announcements, all
47         applications are likely to agree that those Streams use Rank
48         of zero.";
49     reference
50       "46.2.3.2.1 of IEEE Std 802.1Q-2022";
51   }
52 }
53 list end-station-interfaces {
54   key "mac-address interface-name";
55   min-elements 1;
56   description
57     "List of identifiers, one for each physical interface (distinct
58     point of attachment) in the end station acting as a Talker.
59
60     Although many end stations contain a single interface, this list
61     allows for multiple interfaces. Some TSN features allow a single
62     Stream to span multiple interfaces (e.g. seamless redundancy).
63
64     Each entry of end-station-interfaces is used by the CNC to
65     locate the Talker in the topology.
66
67     Since the interface-name leaf is optional, empty string can be
68     used for its key value.";
69   reference
70     "46.2.3.3 of IEEE Std 802.1Q-2022";
71   uses group-interface-id;
72 }

```

```
1 list data-frame-specification {
2     key "index";
3     min-elements 1;
4     description
5         "data-frame-specification specifies the frame that carries the
6         Talkers Stream data. The network uses the specification to
7         identify this Streams frames as TSN, in order to apply the
8         required TSN configuration.
9
10        The specification is based on the users knowledge of the frame,
11        without any network specifics. In other words, this specifies
12        the frame that the Talker would use in the absence of TSN.
13
14        The specification is provided as a list of fields that the user
15        knows. The list is ordered from start of frame to end of header.
16        For example, if the Talker uses a VLAN-tagged Ethernet frame
17        (not IP), the list consists of ieee802-mac-addresses followed by
18        ieee802-vlan-tag. For example, if the Talker uses a UDP/IPv4
19        packet without knowledge of the Ethernet header, the list
20        consists of ipv4-tuple.
21
22        This list is optional, and its absence indicates that Stream
23        transformation is performed in the Talker and Listeners of this
24        Stream (46.2.2 of IEEE Std 802.1Q-2022).";
25     reference
26         "46.2.3.4 of IEEE Std 802.1Q-2022";
27     leaf index {
28         type uint8;
29         description
30             "This index is provided in order to provide a unique key per
31             list entry. The value of index for each entry shall be unique
32             (but not necessarily contiguous).";
33     }
34     choice field {
35         description
36             "One of the following choices is provided for each field that
37             the user knows. Each container name acts as the case name for
38             the choice.";
39         container ieee802-mac-addresses {
40             description
41                 "IEEE 802 MAC addresses.";
42             uses group-ieee802-mac-addresses;
43         }
44         container ieee802-vlan-tag {
45             description
46                 "IEEE 802.1 CTAG";
47             uses group-ieee802-vlan-tag;
48         }
49         container ipv4-tuple {
50             description
51                 "IPv4 packet identification";
52             uses group-ipv4-tuple;
53         }
54         container ipv6-tuple {
55             description
56                 "IPv6 packet identification";
57             uses group-ipv6-tuple;
58         }
59     }
60 }
61 container traffic-specification {
62     description
63         "This traffic-specification specifies how the Talker transmits
64         frames for the Stream. This is effectively the Talkers promise
65         to the network. The network uses this traffic spec to allocate
66         resources and adjust queue parameters in Bridges.";
67     reference
68         "46.2.3.5 of IEEE Std 802.1Q-2022";
69     container interval {
70         description
71             "This interval specifies the period of time in which the
72             traffic specification cannot be exceeded. The traffic
```

specification is specified by max-frames-per-interval and max-frame-size.

The interval is a rational number of seconds, defined by an integer numerator and an integer denominator.

If the time-aware container is not present, the interval specifies a sliding window of time. The Talkers transmission is not synchronized to a time on the network, and therefore the traffic specification cannot be exceeded during any interval in time.

If the time-aware container is present, the interval specifies a window of time that is aligned with the time epoch that is synchronized on the network. For example, if IEEE Std 802.1AS-2011 is used with the PTP timescale, the first interval begins at 1 January 00:00:00 TAI. If CurrentTime represents the current time, then the start of the next interval (StartOfNextInterval) is: StartOfNextInterval = N * interval where N is the smallest integer for which the relation StartOfNextInterval >= CurrentTime would be TRUE.";

```

reference
  "46.2.3.5.1 of IEEE Std 802.1Q-2022";
leaf numerator {
  type uint32;
  description
    "intervals numerator.";
}
leaf denominator {
  type uint32;
  description
    "intervals denominator.";
}
}
leaf max-frames-per-interval {
  type uint16;
  description
    "max-frames-per-interval specifies the maximum number of
    frames that the Talker can transmit in one interval.";
  reference
    "46.2.3.5.2 of IEEE Std 802.1Q-2022";
}
leaf max-frame-size {
  type uint16;
  description
    "max-frame-size specifies maximum frame size that the Talker
    will transmit, excluding any overhead for media-specific
    framing (e.g., preamble, IEEE 802.3 header, Priority/VID tag,
    CRC, interframe gap). As the Talker or Bridge determines the
    amount of bandwidth to reserve on the egress Port (interface),
    it will calculate the media-specific framing overhead on that
    Port and add it to the number specified in the max-frame-size
    leaf.";
  reference
    "46.2.3.5.3 of IEEE Std 802.1Q-2022";
}
leaf transmission-selection {
  type uint8;
  description
    "transmission-selection specifies the algorithm that the
    Talker uses to transmit this Streams traffic class. This
    algorithm is often referred to as the shaper for the traffic
    class.

    The value for this leaf uses Table 8-5 (Transmission selection
    algorithm identifiers) of 8.6.8 of IEEE Std 802.1Q-2022. If no
    algorithm is known, the value zero (strict priority) can be
    used.

    The Talkers shaping and scheduling of the Stream is considered
    to be on the user side of the user/network boundary, and this
    leaf specifies the Talkers behavior to the network.";
```

```
1     reference
2     "46.2.3.5.4 of IEEE Std 802.1Q-2022";
3 }
4 container time-aware {
5     presence
6     "Specifies that the Talkers traffic is synchronized to"+
7     "a known time on the network (e.g. using IEEE Std 802.1AS)";
8     description
9     "The time-aware container provides leafs to specify the
10     Talkers time-aware transmit to the network.
11
12     The Talker and Listeners of a Stream are assumed to coordinate
13     using user (application) mechanisms, such that each Listener
14     is aware that its Talker transmits in a time-aware manner.
15
16     If max-frames-per-interval is greater than one, the Talker
17     shall transmit multiple frames as a burst within the interval,
18     with the minimum inter-frame gap allowed by the media.
19
20     NOTE: Although scheduled traffic (8.6.8.4 of IEEE Std
21     802.1Q-2022) specifies a valid implementation of a time-aware
22     Talker, the time-aware container is intended to support
23     alternate implementations of scheduling.";
24     reference
25     "46.2.3.5 of IEEE Std 802.1Q-2022";
26     leaf earliest-transmit-offset {
27         type uint32;
28         description
29         "earliest-transmit-offset specifies the earliest offset
30         within each interval at which the Talker is capable of
31         starting transmit of its frames. As part of
32         group-status-talker-listener.interface-configuration, the
33         network will return a specific time-aware-offset to the
34         Talker (within the earliest/latest range), which the Talker
35         uses to schedule its transmit.
36
37         earliest-transmit-offset is specified as an integer number
38         of nanoseconds.
39
40         The Talkers transmit offsets include
41         earliest-transmit-offset, latest-transmit-offset, and the
42         time-aware-offset returned to the Talker. Each of the
43         Talkers offsets is specified at the point when the message
44         timestamp point of the first frame of the Stream passes the
45         reference plane marking the boundary between the network
46         media and PHY. The message timestamp point is specified by
47         IEEE Std 802.1AS for various media.";
48         reference
49         "46.2.3.5.5 of IEEE Std 802.1Q-2022";
50     }
51     leaf latest-transmit-offset {
52         type uint32;
53         description
54         "latest-transmit-offset specifies the latest offset within
55         the interval at which the Talker is capable of starting
56         transmit of its frames. As part of
57         group-status-talker-listener.interface-configuration, the
58         network will return a specific time-aware-offset to the
59         Talker within the earliest/latest range), which the Talker
60         uses to schedule its transmit.
61
62         latest-transmit-offset is specified as an integer number of
63         nanoseconds.";
64         reference
65         "46.2.3.5.6 of IEEE Std 802.1Q-2022";
66     }
67     leaf jitter {
68         type uint32;
69         description
70         "The jitter leaf specifies the maximum difference in time
71         between the Talkers transmit offsets, and the ideal
72         synchronized network time (e.g. IEEE 802.1AS time). Jitter
```

```

1      is specified as an unsigned integer number of nanoseconds.
2
3      The maximum difference means sooner or later than the ideal
4      (e.g. transmit +/- 500 nanoseconds relative to IEEE 802.1AS
5      time results in jitter of 500).
6
7      The ideal synchronized network time refers to time at the
8      source (e.g. IEEE 802.1AS grandmaster). The jitter does not
9      include inaccuracies as time is propagated from the time
10     source to the Talker, because those inaccuracies are assumed
11     to be known by the network, and time synchronization is a
12     network technology. The jitter leaf is intended to specify
13     inaccuracies in the Talkers implementation. For example, if
14     the Talkers IEEE 802.1AS time is +/- 812 nanoseconds
15     relative to the grandmaster, and the Talker schedules using
16     a 100 microsecond timer tick driven by IEEE 802.1AS time,
17     Jitter is 50000 (not 50812).
18
19     The Talkers transmit offsets include
20     earliest-transmit-offset, latest-transmit-offset, and the
21     time-aware-offset returned to the Talker in
22     group-status-talker-listener.interface-configuration.";
23     reference
24     "46.2.3.5.7 of IEEE Std 802.1Q-2022";
25   }
26 }
27 }
28 container user-to-network-requirements {
29   description
30     "user-to-network-requirements specifies user requirements for
31     the Stream, such as latency and redundancy. The network (CNC)
32     will merge all user-to-network-requirements for a Stream to
33     ensure that all requirements are met.";
34   reference
35     "46.2.3.6 of IEEE Std 802.1Q-2022";
36   uses group-user-to-network-requirements;
37 }
38 container interface-capabilities {
39   description
40     "interface-capabilities specifies the network capabilities of
41     all interfaces (Ports) contained in end-station-interfaces.
42
43     The network may provide configuration of these capabilities in
44     group-status-talker-listener.interface-configuration.
45
46     NOTE: If an end station contains multiple interfaces with
47     different network capabilities, each interface should be
48     specified as a distinct Talker or Listener (i.e. one entry in
49     end-station-interfaces). Use of multiple entries in
50     end-station-interfaces is intended for network capabilities that
51     span multiple interfaces (e.g. seamless redundancy).";
52   reference
53     "46.2.3.7 of IEEE Std 802.1Q-2022";
54   uses group-interface-capabilities;
55 }
56 }
57 grouping group-listener {
58   description
59     "This YANG grouping specifies: - Listeners requirements from the
60     network - TSN capabilities of the Listeners interface(s)
61
62     In the fully centralized model of TSN configuration, this grouping
63     originates from the CUC, and is delivered to the CNC.";
64   reference
65     "46.2.4 of IEEE Std 802.1Q-2022";
66   list end-station-interfaces {
67     key "mac-address interface-name";
68     min-elements 1;
69     description
70       "List of identifiers, one for each physical interface (distinct
71       point of attachment) in the end station acting as a Listener.
72

```

```

1      Although many end stations contain a single interface, this list
2      allows for multiple interfaces. Some TSN features allow a single
3      Stream to span multiple interfaces (e.g. seamless redundancy).
4
5      Each entry of end-station-interfaces is used by the CNC to
6      locate the Listener in the topology.
7
8      Since the interface-name leaf is optional, empty string can be
9      used for its key value.";
10     reference
11     "46.2.3.3 of IEEE Std 802.1Q-2022";
12     uses group-interface-id;
13 }
14 container user-to-network-requirements {
15     description
16     "user-to-network-requirements specifies user requirements for
17     the Stream, such as latency and redundancy. The network (CNC)
18     will merge all user-to-network-requirements for a Stream to
19     ensure that all requirements are met.";
20     reference
21     "46.2.3.6 of IEEE Std 802.1Q-2022";
22     uses group-user-to-network-requirements;
23 }
24 container interface-capabilities {
25     description
26     "interface-capabilities specifies the network capabilities of
27     all interfaces (Ports) contained in end-station-interfaces.
28
29     The network may provide configuration of these capabilities in
30     group-status-talker-listener.interface-configuration.
31
32     NOTE: If an end station contains multiple interfaces with
33     different network capabilities, each interface should be
34     specified as a distinct Talker or Listener (i.e. one entry in
35     end-station-interfaces). Use of multiple entries in
36     end-station-interfaces is intended for network capabilities that
37     span multiple interfaces (e.g. seamless redundancy).";
38     reference
39     "46.2.3.7 of IEEE Std 802.1Q-2022";
40     uses group-interface-capabilities;
41 }
42 }
43 grouping group-status-stream {
44     description
45     "This YANG grouping provides the status of a Streams configuration
46     from the network to each user. The status in this grouping applies
47     to the entire Stream (Talker and all Listeners).
48
49     In the fully centralized model of TSN configuration, this grouping
50     originates from the CNC, and is delivered to the CUC.
51
52     The group-status-stream and group-status-talker-listener groupings
53     are intended to be used by other modules within a list of status
54     (state) for each Stream, with each list entry using: - leaf of
55     type stream-id-type, used as key to the list - container using
56     group-status-stream - container for Talker, using
57     group-status-talker-listener - list for Listeners, using
58     group-status-talker-listener";
59     reference
60     "46.2.5 of IEEE Std 802.1Q-2022";
61     container status-info {
62         description
63         "status-info provides information regarding the status of a
64         Streams configuration in the network.";
65         reference
66         "46.2.5.1 of IEEE Std 802.1Q-2022";
67         leaf talker-status {
68             type enumeration {
69                 enum none {
70                     value 0;
71                     description
72                     "No Talker detected.";

```

```

1      }
2      enum ready {
3          value 1;
4          description
5              "Talker ready (configured).";
6      }
7      enum failed {
8          value 2;
9          description
10             "Talker failed.";
11     }
12 }
13 description
14     "This is an enumeration for the status of the Streams Talker.";
15 reference
16     "46.2.5.1.1 of IEEE Std 802.1Q-2022";
17 }
18 leaf listener-status {
19     type enumeration {
20         enum none {
21             value 0;
22             description
23                 "No Listener detected.";
24         }
25         enum ready {
26             value 1;
27             description
28                 "All Listeners ready (configured).";
29         }
30         enum partial-failed {
31             value 2;
32             description
33                 "One or more Listeners ready, and one or more Listeners
34                 failed. If Talker is ready, Stream can be used.";
35         }
36         enum failed {
37             value 3;
38             description
39                 "All Listeners failed";
40         }
41     }
42     description
43         "This is an enumeration for the status of the Streams
44         Listener(s).";
45     reference
46         "46.2.5.1.2 of IEEE Std 802.1Q-2022";
47 }
48 leaf failure-code {
49     type uint8;
50     description
51         "If the Stream encounters a failure (talker-status is failed,
52         or listener-status is failed, or listener-status is
53         partial-failed), failure-code provides a non-zero code that
54         specifies the problem. Table 46-15 of IEEE Std 802.1Q-2022
55         describes each code.";
56     reference
57         "46.2.5.1.3 of IEEE Std 802.1Q-2022";
58 }
59 }
60 list failed-interfaces {
61     key "mac-address interface-name";
62     description
63         "When a failure occurs in network configuration (i.e. non-zero
64         failure-code in status-info), failed-interfaces provides a list
65         of one or more physical interfaces (distinct points of
66         attachment) in the failed end station or Bridge. Each
67         identifier is sufficient to locate the interface in the physical
68         topology.
69
70         The failed-interfaces list is optional.
71
72         Since the interface-name leaf is optional, empty string can be

```



```

1      used for its key value.";
2      reference
3      "46.2.5.4 of IEEE Std 802.1Q-2022";
4      uses group-interface-id;
5  }
6  }
7  grouping group-status-talker-listener {
8      description
9      "This YANG grouping provides the status for a specific Talker or
10     Listener.
11
12     In the fully centralized model of TSN configuration, this grouping
13     originates from the CNC, and is delivered to the CUC.";
14     reference
15     "46.2.5 of IEEE Std 802.1Q-2022";
16     leaf accumulated-latency {
17         type uint32;
18         description
19         "accumulated-latency provides the worst-case maximum latency
20         that a single frame of the Stream can encounter along its
21         current path(s).
22
23         When provided to a Listener, accumulated-latency is the
24         worst-case maximum latency for that Listener only.
25
26         When provided to a Talker, accumulated-latency is the worst-case
27         maximum latency for all Listeners (worst path).
28
29         accumulated-latency is specified as an integer number of
30         nanoseconds.
31
32         accumulated-latency uses the same definition for latency as
33         user-to-network-requirements.max-latency.
34
35         For successful status-info, the network returns a value less
36         than or equal to user-to-network-requirements.max-latency.
37
38         If the time-aware container is present in the
39         traffic-specification of the Talker, the value is expressed as
40         nanoseconds after the start of the Talkers
41         traffic-specification.interval.
42
43         If the time-aware container is not present in the
44         traffic-specification of the Talker, the value is expressed as
45         nanoseconds after the Talkers transmit of any frame in the
46         Stream, at any arbitrary time.
47
48         If user-to-network-requirements.num-seamless-trees is one,
49         accumulated-latency shall provide the worst-case maximum latency
50         for the current path from Talker to each Listener. If the path
51         is changed (e.g. by a spanning tree protocol),
52         accumulated-latency changes accordingly.
53
54         If user-to-network-requirements.num-seamless-trees is greater
55         than one, accumulated-latency shall provide the worst-case
56         maximum latency for all paths configured from the Talker to each
57         Listener.";
58     reference
59     "46.2.5.2 of IEEE Std 802.1Q-2022";
60 }
61 container interface-configuration {
62     description
63     "interface-configuration provides configuration of interfaces in
64     the Talker/Listener. This configuration assists the network in
65     meeting the Streams requirements. The interface-configuration
66     meets the capabilities of the interface as provided in
67     interface-capabilities.";
68     reference
69     "46.2.5.3 of IEEE Std 802.1Q-2022";
70     uses group-interface-configuration;
71 }
72 }

```

1 }
2

1 48.6.4 The ieee802-dot1q-bridge YANG module

```
2 module ieee802-dot1q-bridge {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge;
5   prefix dot1q;
6   import ieee802-types {
7     prefix ieee;
8   }
9   import ietf-yang-types {
10    prefix yang;
11  }
12  import ietf-interfaces {
13    prefix if;
14  }
15  import iana-if-type {
16    prefix ianaif;
17  }
18  import ieee802-dot1q-types {
19    prefix dot1qtypes;
20  }
21  organization
22    "IEEE 802.1 Working Group";
23  contact
24    "WG-URL: http://ieee802.org/1/
25    WG-EMail: stds-802-1-1@ieee.org
26
27    Contact: IEEE 802.1 Working Group Chair
28    Postal: C/O IEEE 802.1 Working Group
29            IEEE Standards Association
30            445 Hoes Lane
31            Piscataway, NJ 08854
32            USA
33
34    E-mail: stds-802-1-chairs@ieee.org";
35  description
36    "This YANG module describes the Bridge configuration model for the
37    following IEEE 802.1Q Bridges:
38      1) Two Port MAC Relays
39      2) Customer VLAN Bridges
40      3) Provider Bridges.
41
42    Copyright (C) IEEE (2023).
43
44    This version of this YANG module is part of IEEE Std 802.1Q; see the
45    standard itself for full legal notices.";
46  revision 2023-10-26 {
47    description
48      "Published as part of IEEE Std 802.1Qcw-2023.
49
50    The following reference statement identifies each referenced IEEE
51    Standard as updated by applicable amendments.";
52    reference
53      "IEEE Std 802.1Q Bridges and Bridged Networks:
54      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.
55      IEEE Std 802 Overview and Architecture:
56      IEEE Std 802-2014.
57      IEEE Std 802.1AC Media Access Control (MAC) Service Definition:
58      IEEE Std 802.1AC-2016, IEEE Std 802.1AC-2016-Cor1-2018.";
59  }
60  revision 2022-10-29 {
61    description
62      "Published as part of IEEE Std 802.1Q-2022.";
63    reference
64      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
65  }
66  revision 2020-11-06 {
67    description
68      "Published as part of IEEE Std 802.1Qcr-2020. Third version.";
69    reference
70      "IEEE Std 802.1Qcr-2020, Bridges and Bridged Networks - Asynchronous
71      Traffic Shaping.";
```

```
1  }
2  revision 2020-06-04 {
3    description
4      "Published as part of IEEE Std 802.1Qcx-2020. Second version.";
5    reference
6      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
7      Model for Connectivity Fault Management.";
8  }
9  revision 2018-03-07 {
10   description
11     "Published as part of IEEE Std 802.1Q-2018. Initial version.";
12   reference
13     "IEEE Std 802.1Q-2018, Bridges and Bridged Networks.";
14 }
15 feature ingress-filtering {
16   description
17     "Each Port may support an Enable Ingress Filtering parameter. A frame
18     received on a Port that is not in the member set (8.8.10) associated
19     with the frames VID shall be discarded if this parameter is set. The
20     default value for this parameter is reset, i.e., Disable Ingress
21     Filtering, for all Ports. Any Port that supports setting this
22     parameter shall also support resetting it. The parameter may be
23     configured by the management operations defined in Clause 12.";
24   reference
25     "8.6.2 of IEEE Std 802.1Q";
26 }
27 feature extended-filtering-services {
28   description
29     "Extended Filtering Services support the filtering behavior required
30     for regions of a network in which potential recipients of multicast
31     frames exist, and where both the potential recipients of frames and
32     the Bridges are able to support dynamic configuration of filtering
33     information for group MAC addresses. In order to integrate this
34     extended filtering behavior with the needs of regions of the network
35     that support only Basic Filtering Services, Bridges that support
36     Extended Filtering Services can be statically and dynamically
37     configured to modify their filtering behavior on a per-group MAC
38     address basis, and also on the basis of the overall filtering service
39     provided by each outbound Port with regard to multicast frames. The
40     latter capability permits configuration of the Port's default
41     forwarding or filtering behavior with regard to group MAC addresses
42     for which no specific static or dynamic filtering information has been
43     configured.";
44   reference
45     "8.8.4, Clause 10 of IEEE Std 802.1Q";
46 }
47 feature port-and-protocol-based-vlan {
48   description
49     "A VLAN-aware Bridge component implementation in conformance to the
50     provisions of this standard for Port-and-Protocol-based VLAN
51     classification (5.4.1) shall 1) Support one or more of the following
52     Protocol Classifications and Protocol Template formats: Ethernet,
53     RFC_1042, SNAP_8021H, SNAP_Other, or LLC_Other (6.12); and may 2)
54     Support configuration of the contents of the Protocol Group Database.";
55   reference
56     "5.4.1.2 of IEEE Std 802.1Q";
57 }
58 feature flow-filtering {
59   description
60     "Flow filtering support enables Bridges to distinguish frames
61     belonging to different client flows and to use this information in the
62     forwarding process. Information related to client flows may be used at
63     the boundary of an SPT Domain to generate a flow hash value. The flow
64     hash, carried in an F-TAG, serves to distinguish frames belonging to
65     different flows and can be used in the forwarding process to
66     distribute frames over equal cost paths. This provides for finer
67     granularity load spreading while maintaining frame order for each
68     client flow.";
69   reference
70     "44.2 of IEEE Std 802.1Q";
71 }
72 feature simple-bridge-port {
```

```

1  description
2  "A simple bridge port allows underlying (MAC) layers to share the same
3  Interface as the Bridge Port.";
4  }
5  feature flexible-bridge-port {
6  description
7  "A flexible Bridge Port provides an interface that is distinct from
8  the interface provided by the underlying (MAC) layer.";
9  }
10 identity type-of-bridge {
11 description
12 "Represents the configured Bridge type.";
13 }
14 identity customer-vlan-bridge {
15 base type-of-bridge;
16 description
17 "Base identity for a Customer VLAN Bridge.";
18 }
19 identity provider-bridge {
20 base type-of-bridge;
21 description
22 "Base identity for a Provider Bridge (PB).";
23 }
24 identity provider-edge-bridge {
25 base type-of-bridge;
26 description
27 "Base identity for a Provider Edge Bridge (PEB).";
28 }
29 identity two-port-mac-relay-bridge {
30 base type-of-bridge;
31 description
32 "Base identity for a Two Port MAC Relay (TPMR).";
33 }
34 identity type-of-component {
35 description
36 "Represents the type of Component.";
37 }
38 identity c-vlan-component {
39 base type-of-component;
40 description
41 "Base identity for a C-VLAN component.";
42 }
43 identity s-vlan-component {
44 base type-of-component;
45 description
46 "Base identity for a S-VLAN component.";
47 }
48 identity d-bridge-component {
49 base type-of-component;
50 description
51 "Base identity for a VLAN unaware component.";
52 }
53 identity edge-relay-component {
54 base type-of-component;
55 description
56 "Base identity for an EVB station ER component.";
57 }
58 identity type-of-port {
59 description
60 "Represents the type of Bridge port.";
61 }
62 identity c-vlan-bridge-port {
63 base type-of-port;
64 description
65 "Indicates the port can be a C-TAG aware port of an enterprise VLAN
66 aware Bridge.";
67 }
68 identity provider-network-port {
69 base type-of-port;
70 description
71 "Indicates the port can be an S-TAG aware port of a Provider Bridge or
72 Backbone Edge Bridge used for connections within a PBN (Provider

```

```

1     Bridged Network) or PBBN (Provider Backbone Bridged Network).";
2 }
3 identity customer-network-port {
4     base type-of-port;
5     description
6         "Indicates the port can be an S-TAG aware port of a Provider Bridge or
7         Backbone Edge Bridge used for connections to the exterior of a PBN
8         (Provider Bridged Network) or PBBN (Provider Backbone Bridged
9         Network).";
10 }
11 identity customer-edge-port {
12     base type-of-port;
13     description
14         "Indicates the port can be a C-TAG aware port of a Provider Bridge
15         used for connections to the exterior of a PBN (Provider Bridged
16         Network) or PBBN (Provider Backbone Bridged Network).";
17 }
18 identity d-bridge-port {
19     base type-of-port;
20     description
21         "Indicates the port can be a VLAN-unaware member of an 802.1Q Bridge.";
22 }
23 identity remote-customer-access-port {
24     base type-of-port;
25     description
26         "Indicates the port can be an S-TAG aware port of a Provider Bridge
27         capable of providing Remote Customer Service Interfaces.";
28 }
29 identity bridge-interface {
30     description
31         "Generic interface property that represents any interface that can be
32         associated with an IEEE 802.1Q compliant Bridge component. Any new
33         Interface types would derive from this identity to automatically pick
34         up Bridge related configuration or operational data.";
35 }
36 container bridges {
37     description
38         "Contains the Bridge(s) configuration information.";
39     list bridge {
40         key "name";
41         unique "address";
42         description
43             "Provides configuration data in support of the Bridge Configuration
44             resources. There is a single bridge data node per Bridge.";
45         leaf name {
46             type dot1qttype:name-type;
47             description
48                 "A text string associated with the Bridge, of locally determined
49                 significance.";
50             reference
51                 "12.4 of IEEE Std 802.1Q";
52         }
53         leaf address {
54             type ieee:mac-address;
55             mandatory true;
56             description
57                 "The MAC address for the Bridge from which the Bridge Identifiers
58                 used by the STP, RSTP, and MSTP are derived.";
59             reference
60                 "12.4 of IEEE Std 802.1Q";
61         }
62         leaf bridge-type {
63             type identityref {
64                 base type-of-bridge;
65             }
66             mandatory true;
67             description
68                 "The type of Bridge.";
69         }
70         leaf ports {
71             type uint16 {
72                 range "1..4095";

```

```

1      }
2      config false;
3      description
4          "The number of Bridge Ports (MAC Entities)";
5      reference
6          "12.4 of IEEE Std 802.1Q";
7  }
8  leaf up-time {
9      type yang:zero-based-counter32;
10     units "seconds";
11     config false;
12     description
13         "The count in seconds of the time elapsed since the Bridge was
14         last reset or initialized.";
15     reference
16         "12.4 of IEEE Std 802.1Q";
17 }
18 leaf components {
19     type uint32;
20     config false;
21     description
22         "The number of components associated with the Bridge.";
23 }
24 list component {
25     key "name";
26     description
27         "The set of components associated with a given Bridge. For
28         example, - A TPMR is associated with a single VLAN unaware
29         component. - A Customer VLAN Bridge is associated with a single
30         VLAN aware component. - A Provider Bridge is associated with a
31         single S-VLAN component and zero or more C-VLAN components.";
32     reference
33         "12.4.1.5 of IEEE Std 802.1Q";
34     leaf name {
35         type string;
36         description
37             "The name of the Component.";
38     }
39     leaf id {
40         type uint32;
41         description
42             "Unique identifier for a particular Bridge component within the
43             system.";
44         reference
45             "Item l) in 12.3 of IEEE Std 802.1Q";
46     }
47     leaf type {
48         type identityref {
49             base type-of-component;
50         }
51         mandatory true;
52         description
53             "The type of component used to classify a particular Bridge
54             component within a Bridge system comprising multiple components.";
55         reference
56             "Item m) in 12.3 of IEEE Std 802.1Q";
57     }
58     leaf address {
59         type ieee:mac-address;
60         description
61             "Unique EUI-48 Universally Administered MAC address assigned to
62             a Bridge component.";
63         reference
64             "8.13.8 of IEEE Std 802.1Q";
65     }
66     leaf traffic-class-enabled {
67         type boolean;
68         default "true";
69         description
70             "Indication of Traffic Classes enablement associated with the
71             Bridge Component. A value of True indicates that Traffic Classes
72             are enabled on this Bridge Component. A value of False indicates

```

```

1         that the Bridge Component operates with a single priority level
2         for all traffic.";
3     reference
4         "Item b) in 12.4.1.5.2 of IEEE Std 802.1Q";
5 }
6 leaf ports {
7     type uint16 {
8         range "1..4095";
9     }
10    config false;
11    description
12        "The number of Bridge Ports associated with the Bridge
13        Component.";
14    reference
15        "Item c) in 12.4.1.1.3 of IEEE Std 802.1Q";
16 }
17 leaf-list bridge-port {
18     type if:interface-ref;
19     config false;
20     description
21         "List of bridge-port references.";
22 }
23 container capabilities {
24     config false;
25     description
26         "Array of Boolean values of the feature capabilities associated
27         with a given Bridge Component.";
28     reference
29         "Item b) in 12.10.1.1.3, 12.4.1.5.2 of IEEE Std 802.1Q";
30     leaf extended-filtering {
31         type boolean;
32         default "false";
33         description
34             "Can perform filtering on individual multicast addresses
35             controlled by MMRP.";
36         reference
37             "12.4.1.5.2 of IEEE Std 802.1Q";
38     }
39     leaf traffic-classes {
40         type boolean;
41         default "false";
42         description
43             "Can map priority to multiple traffic classes.";
44         reference
45             "12.4.1.5.2 of IEEE Std 802.1Q";
46     }
47     leaf static-entry-individual-port {
48         type boolean;
49         default "false";
50         description
51             "Static entries per port.";
52         reference
53             "12.4.1.5.2 of IEEE Std 802.1Q";
54     }
55     leaf ivl-capable {
56         type boolean;
57         default "true";
58         description
59             "Independent VLAN Learning (IVL).";
60         reference
61             "12.4.1.5.2 of IEEE Std 802.1Q";
62     }
63     leaf svl-capable {
64         type boolean;
65         default "false";
66         description
67             "Shared VLAN Learning (SVL).";
68         reference
69             "12.4.1.5.2 of IEEE Std 802.1Q";
70     }
71     leaf hybrid-capable {
72         type boolean;

```



```

1      default "false";
2      description
3          "Both IVL and SVL simultaneously.";
4      reference
5          "12.4.1.5.2 of IEEE Std 802.1Q";
6  }
7  leaf configurable-pvid-tagging {
8      type boolean;
9      default "false";
10     description
11         "Whether the implementation supports the ability to override
12         the default PVID setting and its egress status (VLAN-tagged or
13         Untagged) on each port.";
14     reference
15         "12.4.1.5.2 of IEEE Std 802.1Q";
16 }
17 leaf local-vlan-capable {
18     type boolean;
19     default "false";
20     description
21         "Can support multiple local Bridges, outside the scope of
22         802.1Q defined VLANs.";
23     reference
24         "12.4.1.5.2 of IEEE Std 802.1Q";
25 }
26 }
27 container filtering-database {
28     when
29         "not(derived-from-or-self ../../bridge-type, "+
30         "'two-port-mac-relay-bridge'))" {
31         description
32             "Applies to non TPMRs.";
33     }
34     description
35         "Contains filtering information used by the Forwarding Process
36         in deciding through which Ports of the Bridge frames should be
37         forwarded.";
38     reference
39         "12.7 of IEEE Std 802.1Q";
40     leaf aging-time {
41         type uint32 {
42             range "10..1000000";
43         }
44         units "seconds";
45         default "300";
46         description
47             "The timeout period in seconds for aging out
48             dynamically-learned forwarding information.";
49         reference
50             "12.7, 8.8.3 of IEEE Std 802.1Q";
51     }
52     leaf size {
53         type yang:gauge32;
54         config false;
55         description
56             "The maximum number of entries that can be held in the FDB.";
57         reference
58             "12.7 of IEEE Std 802.1Q";
59     }
60     leaf static-entries {
61         type yang:gauge32;
62         config false;
63         description
64             "The number of Static Filtering entries currently in the FDB.";
65         reference
66             "12.7, 8.8.1 of IEEE Std 802.1Q";
67     }
68     leaf dynamic-entries {
69         type yang:gauge32;
70         config false;
71         description
72             "The number of Dynamic Filtering entries currently in the FDB.";

```

```

1      reference
2      "12.7, 8.8.3 of IEEE Std 802.1Q";
3  }
4  leaf static-vlan-registration-entries {
5      type yang:gauge32;
6      config false;
7      description
8          "The number of Static VLAN Registration entries currently in
9          the FDB.";
10     reference
11         "12.7, 8.8.2 of IEEE Std 802.1Q";
12 }
13 leaf dynamic-vlan-registration-entries {
14     type yang:gauge32;
15     config false;
16     description
17         "The number of Dynamic VLAN Registration entries currently in
18         the FDB.";
19     reference
20         "12.7, 8.8.5 of IEEE Std 802.1Q";
21 }
22 leaf mac-address-registration-entries {
23     if-feature "extended-filtering-services";
24     type yang:gauge32;
25     config false;
26     description
27         "The number of MAC Address Registration entries currently in
28         the FDB.";
29     reference
30         "12.7, 8.8.4 of IEEE Std 802.1Q";
31 }
32 list filtering-entry {
33     key "database-id vids address";
34     description
35         "Information for the entries associated with the Permanent
36         Database.";
37     leaf database-id {
38         type uint32;
39         description
40             "The identity of this Filtering Database.";
41         reference
42             "12.7.7 of IEEE Std 802.1Q";
43     }
44     leaf address {
45         type ieee:mac-address;
46         description
47             "A MAC address (unicast, multicast, broadcast) for which the
48             device has forwarding and/or filtering information.";
49         reference
50             "12.7.7 of IEEE Std 802.1Q";
51     }
52     leaf vids {
53         type dot1qt-types:vid-range-type;
54         description
55             "The set of VLAN identifiers to which this entry applies.";
56         reference
57             "12.7.7 of IEEE Std 802.1Q";
58     }
59     leaf entry-type {
60         type enumeration {
61             enum static {
62                 description
63                     "Static entry type";
64             }
65             enum dynamic {
66                 description
67                     "Dynamic/learnt entry type";
68             }
69         }
70         description
71             "The type of filtering entry. Whether static or dynamic.
72             Static entries can be created, deleted, and retrieved.

```

```

1         However, dynamic entries can only be deleted or retrieved by
2         the management entity. Consequently, a Bridge is not
3         required to accept a command that can alter the dynamic
4         entries except delete a dynamic entry.";
5     reference
6         "12.7.7 of IEEE Std 802.1Q";
7 }
8 uses dot1qtypes:port-map-grouping;
9 leaf status {
10     type enumeration {
11         enum other {
12             description
13                 "None of the following. This may include the case where
14                 some other object is being used to determine if and how
15                 frames addressed to the value of the corresponding
16                 instance of 'address' are being forwarded.";
17         }
18         enum invalid {
19             description
20                 "This entry is no longer valid (e.g., it was learned but
21                 has since aged out), but has not yet been flushed from
22                 the table.";
23         }
24         enum learned {
25             description
26                 "The value of the corresponding instance of the port
27                 node was learned and is being used.";
28         }
29         enum self {
30             description
31                 "The value of the corresponding instance of the address
32                 node representing one of the devices address.";
33         }
34         enum mgmt {
35             description
36                 "The value of the corresponding instance of address node
37                 that is also the value of an existing instance.";
38         }
39     }
40     config false;
41     description
42         "The status of this entry.";
43 }
44 }
45 list vlan-registration-entry {
46     key "database-id vids";
47     description
48         "The VLAN Registration Entries models the operations that can
49         be performed on a single VLAN Registration Entry in the FDB.
50         The set of VLAN Registration Entries within the FDB changes
51         under management control and also as a result of MVRP
52         exchanges";
53     reference
54         "12.7.5 of IEEE Std 802.1Q";
55     leaf database-id {
56         type uint32;
57         description
58             "The identity of this Filtering Database.";
59         reference
60             "12.7.7 of IEEE Std 802.1Q";
61     }
62     leaf vids {
63         type dot1qtypes:vid-range-type;
64         description
65             "The set of VLAN identifiers to which this entry applies.";
66         reference
67             "12.7.7 of IEEE Std 802.1Q";
68     }
69     leaf entry-type {
70         type enumeration {
71             enum static {
72                 description

```

```

1         "Static entry type";
2     }
3     enum dynamic {
4         description
5             "Dynamic/learnt entry type";
6     }
7 }
8 description
9     "The type of filtering entry. Whether static or dynamic.
10    Static entries can be created, deleted, and retrieved.
11    However, dynamic entries can only be deleted or retrieved by
12    the management entity. Consequently, a Bridge is not
13    required to accept a command that can alter the dynamic
14    entries except delete a dynamic entry.";
15 reference
16     "12.7.7 of IEEE Std 802.1Q";
17 }
18 uses dot1qtypes:port-map-grouping;
19 }
20 }
21 container permanent-database {
22     description
23         "The Permanent Database container models the operations that can
24         be performed on, or affect, the Permanent Database. There is a
25         single Permanent Database per FDB.";
26     leaf size {
27         type yang:gauge32;
28         config false;
29         description
30             "The maximum number of entries that can be held in the FDB.";
31         reference
32             "12.7.6 of IEEE Std 802.1Q";
33     }
34     leaf static-entries {
35         type yang:gauge32;
36         config false;
37         description
38             "The number of Static Filtering entries currently in the FDB.";
39         reference
40             "12.7.6 of IEEE Std 802.1Q";
41     }
42     leaf static-vlan-registration-entries {
43         type yang:gauge32;
44         config false;
45         description
46             "The number of Static VLAN Registration entries currently in
47             the FDB.";
48         reference
49             "12.7.6 of IEEE Std 802.1Q";
50     }
51     list filtering-entry {
52         key "database-id vids address";
53         description
54             "Information for the entries associated with the Permanent
55             Database.";
56         leaf database-id {
57             type uint32;
58             description
59                 "The identity of this Filtering Database.";
60             reference
61                 "12.7.7 of IEEE Std 802.1Q";
62         }
63         leaf address {
64             type ieee:mac-address;
65             description
66                 "A MAC address (unicast, multicast, broadcast) for which the
67                 device has forwarding and/or filtering information.";
68             reference
69                 "12.7.7 of IEEE Std 802.1Q";
70         }
71         leaf vids {
72             type dot1qtypes:vid-range-type;

```

```

1      description
2      "The set of VLAN identifiers to which this entry applies.";
3      reference
4      "12.7.7 of IEEE Std 802.1Q";
5  }
6  leaf status {
7      type enumeration {
8          enum other {
9              description
10             "None of the following. This may include the case where
11             some other object is being used to determine if and how
12             frames addressed to the value of the corresponding
13             instance of 'address' are being forwarded.";
14         }
15         enum invalid {
16             description
17             "This entry is no longer valid (e.g., it was learned but
18             has since aged out), but has not yet been flushed from
19             the table.";
20         }
21         enum learned {
22             description
23             "The value of the corresponding instance of the port
24             node was learned and is being used.";
25         }
26         enum self {
27             description
28             "The value of the corresponding instance of the address
29             node representing one of the devices address.";
30         }
31         enum mgmt {
32             description
33             "The value of the corresponding instance of address node
34             that is also the value of an existing instance.";
35         }
36     }
37     config false;
38     description
39     "The status of this entry.";
40 }
41 uses dot1qtypes:port-map-grouping;
42 }
43 }
44 container bridge-vlan {
45     when
46     "not(derived-from-or-self(../bridge-type, '"+
47     "'two-port-mac-relay-bridge'))" {
48         description
49         "Applies to non TPMRs.";
50     }
51     description
52     "The Bridge VLAN container models configuration information that
53     modifies, or inquires about, the overall configuration of the
54     Bridge's VLAN resources. There is a single Bridge VLAN
55     Configuration managed object per Bridge.";
56     reference
57     "12.10 of IEEE Std 802.1Q";
58     leaf version {
59         type uint16;
60         config false;
61         description
62         "The version number supported.";
63         reference
64         "Item a) of 12.10.1.1.3 of IEEE Std 802.1Q";
65     }
66     leaf max-vids {
67         type uint16;
68         config false;
69         description
70         "The maximum number of VIDs supported.";
71         reference
72         "Item b)1) of 12.10.1.1.3 of IEEE Std 802.1Q";

```

```

1      }
2      leaf override-default-pvid {
3          type boolean;
4          default "false";
5          config false;
6          description
7              "Indicates if the default PVID can be overridden, and its
8              egress status (VLAN-tagged or untagged) on each port.";
9          reference
10             "Item b)2) of 12.10.1.1.3 of IEEE Std 802.1Q";
11     }
12     leaf protocol-template {
13         if-feature "port-and-protocol-based-vlan";
14         type dot1qtypes:protocol-frame-format-type;
15         config false;
16         description
17             "The data-link encapsulation format or the detagged_frame_type
18             in a Protocol Template";
19         reference
20             "12.10.1.7 of IEEE Std 802.1Q";
21     }
22     leaf max-msti {
23         type uint16;
24         config false;
25         description
26             "The maximum number of MSTIs supported within an MST region
27             (i.e., the number of spanning tree instances that can be
28             supported in addition to the CIST), for MST Bridges. For SST
29             Bridges, this parameter may be either omitted or reported as
30             0.";
31         reference
32             "Item b)4) of 12.10.1.1.3 of IEEE Std 802.1Q";
33     }
34     list vlan {
35         key "vid";
36         description
37             "List of VLAN related configuration nodes associated with the
38             Bridge.";
39         reference
40             "12.10.2 of IEEE Std 802.1Q";
41         leaf vid {
42             type dot1qtypes:vlan-index-type;
43             description
44                 "The VLAN identifier to which this entry applies.";
45             reference
46                 "12.10.2 of IEEE Std 802.1Q";
47         }
48         leaf name {
49             type dot1qtypes:name-type;
50             description
51                 "A text string of up to 32 characters of locally determined
52                 significance.";
53             reference
54                 "12.10.2 of IEEE Std 802.1Q";
55         }
56         leaf-list untagged-ports {
57             type if:interface-ref;
58             config false;
59             description
60                 "The set of ports in the untagged set for this VID.";
61             reference
62                 "12.10.2.1.3, 8.8.2 of IEEE Std 802.1Q";
63         }
64         leaf-list egress-ports {
65             type if:interface-ref;
66             config false;
67             description
68                 "The set of egress ports in the member set for this VID.";
69             reference
70                 "12.10.2.1.3, 8.8.10 of IEEE Std 802.1Q";
71         }
72     }

```

```

1 list protocol-group-database {
2     if-feature "port-and-protocol-based-vlan";
3     key "db-index";
4     description
5         "List of the protocol group database entries.";
6     reference
7         "12.10.1.7, 6.12.3 of IEEE Std 802.1Q";
8     leaf db-index {
9         type uint16;
10        description
11            "The protocol group database index.";
12    }
13    leaf frame-format-type {
14        type dot1qtys:protocol-frame-format-type;
15        description
16            "The data-link encapsulation format or the
17            detagged_frame_type in a Protocol Template";
18        reference
19            "12.10.1.7 of IEEE Std 802.1Q";
20    }
21    choice frame-format {
22        description
23            "The identification of the protocol above the data-link
24            layer in a Protocol Template. Depending on the frame type,
25            the octet string will have one of the following values:
26            - For Ethernet, rfc1042, and snap8021H, this is the 16-bit
27            (2-octet) EtherType field.
28            - For snapOther, this is the 40-bit (5-octet) PID.
29            - For llcOther, this is the 2-octet IEEE 802.2 Link Service
30            Access Point (LSAP) pair: first octet for Destination Service
31            Access Point (DSAP) and second octet for Source Service Access
32            Point (SSAP).";
33        reference
34            "12.10.1.7 of IEEE Std 802.1Q";
35        case ethernet-rfc1042-snap8021H {
36            when
37                "frame-format-type = 'Ethernet' or frame-format-type = '+'
38                'rfc1042' or frame-format-type = 'snap8021H'" {
39                description
40                    "Applies to Ethernet, RFC 1042, SNAP 8021H frame
41                    formats.";
42            }
43            description
44                "Identifier used if Ethenet, RFC1042, or SNAP 8021H.";
45            leaf ethertype {
46                type dot1qtys:ethertype-type;
47                description
48                    "Format containing the 16-bit IEEE 802 EtherType field.";
49                reference
50                    "9.2 of IEEE Std 802";
51            }
52        }
53        case snap-other {
54            when
55                "frame-format-type = 'snapOther'" {
56                description
57                    "Applies to Snap Other frame formats.";
58            }
59            description
60                "Identifier used if SNAP other.";
61            leaf protocol-id {
62                type string {
63                    pattern "[0-9a-fA-F]{2}(-[0-9a-fA-F]{2}){4}";
64                }
65                description
66                    "Format containing the 40-bit protocol identifier (PID).
67                    The canonical representation uses uppercase characters.";
68                reference
69                    "12.10.1.7.1 of IEEE Std 802.1Q";
70            }
71        }
72        case llc-other {

```

```

1         when
2             "frame-format-type = 'llcOther'" {
3                 description
4                     "Applies to LLC Other frame formats";
5             }
6         description
7             "Identifier used if LLC other.";
8         container dsap-ssap-pairs {
9             description
10                "A pair of ISO/IEC 8802-2 DSAP and SSAP address field
11                values, for matching frame formats of LLC_Other.";
12            leaf llc-address {
13                type string {
14                    pattern "[0-9a-fA-F]{2}-[0-9a-fA-F]{2}";
15                }
16                description
17                    "A pair of ISO/IEC 8802-2 DSAP and SSAP address field
18                    values, for matching frame formats of LLC_Other. The
19                    canonical representation uses uppercase characters.";
20                reference
21                    "12.10.1.7.1 of IEEE Std 802.1Q";
22            }
23        }
24    }
25 }
26 leaf group-id {
27     type uint32;
28     description
29         "Designates a group of protocols in the Protocol Group
30         Database.";
31     reference
32         "6.12.2 of IEEE Std 802.1Q";
33 }
34 }
35 list vid-to-fid-allocation {
36     key "vids";
37     description
38         "This list allows inquiries about VID to FID allocations.";
39     leaf vids {
40         type dot1qt-types:vid-range-type;
41         description
42             "Range of VLAN identifiers.";
43         reference
44             "12.10.3 of IEEE Std 802.1Q";
45     }
46     leaf fid {
47         type uint32;
48         config false;
49         description
50             "The Filtering Database used by a set of VIDs.";
51         reference
52             "12.10.3 of IEEE Std 802.1Q";
53     }
54     leaf allocation-type {
55         type enumeration {
56             enum undefined {
57                 description
58                     "No allocation defined.";
59             }
60             enum fixed {
61                 description
62                     "A fixed allocation to FID is defined.";
63             }
64             enum dynamic {
65                 description
66                     "A dynamic allocation to FID is defined.";
67             }
68         }
69         config false;
70         description
71             "The type of allocation used";
72         reference

```



```

1         "12.10.3 of IEEE Std 802.1Q";
2     }
3 }
4 list fid-to-vid-allocation {
5     key "fid";
6     description
7         "The FID to VID allocations managed object models operations
8         that inquire about FID to VID allocations.";
9     leaf fid {
10        type uint32;
11        description
12            "The Filtering Database used by a set of VIDs.";
13        reference
14            "12.10.3 of IEEE Std 802.1Q";
15    }
16    leaf allocation-type {
17        type enumeration {
18            enum undefined {
19                description
20                    "No allocation defined.";
21            }
22            enum fixed {
23                description
24                    "A fixed allocation to FID is defined.";
25            }
26            enum dynamic {
27                description
28                    "A dynamic allocation to FID is defined.";
29            }
30        }
31        config false;
32        description
33            "The type of allocation used";
34        reference
35            "12.10.3 of IEEE Std 802.1Q";
36    }
37    leaf-list vid {
38        type dot1qtotypes:vlan-index-type;
39        config false;
40        description
41            "The VLAN identifier to which this entry applies.";
42        reference
43            "12.7.7 of IEEE Std 802.1Q";
44    }
45 }
46 list vid-to-fid {
47     key "vid";
48     description
49         "Fixed allocation of a VID to an FID. The underlying system
50         will ensure that subsequent commands that make changes to the
51         VID to FID mapping can override previous associations.";
52     reference
53         "12.10.3.4, 12.10.3.5 of IEEE Std 802.1Q";
54     leaf vid {
55         type dot1qtotypes:vlan-index-type;
56         description
57             "A list of VIDs associated with a given database identifier
58             (i.e., FID).";
59         reference
60             "12.7.7 of IEEE Std 802.1Q";
61     }
62     leaf fid {
63         type uint32;
64         description
65             "The Filtering Database used by this VLAN";
66         reference
67             "12.10.3 of IEEE Std 802.1Q";
68     }
69 }
70 }
71 container bridge-mst {
72     when

```

```

1      "not(derived-from-or-self(../bridge-type, "+
2      "'two-port-mac-relay-bridge'))" {
3      description
4          "Applies to non TPMRs.";
5      }
6      description
7          "The Bridge MST container models configuration information that
8          modifies, or inquires about, the overall configuration of the
9          Bridges MST resources.";
10     reference
11         "12.12 of IEEE Std 802.1Q";
12     leaf-list mstid {
13         type dot1qtypes:mstid-type;
14         description
15             "The list of MSTID values that are currently supported by the
16             Bridge";
17     }
18     list fid-to-mstid {
19         key "fid";
20         description
21             "The FID to MSTID allocation table.";
22         reference
23             "12.12.2 of IEEE Std 802.1Q";
24         leaf fid {
25             type uint32;
26             description
27                 "The Filtering Database identifier.";
28             reference
29                 "12.12.2 of IEEE Std 802.1Q";
30         }
31         leaf mstid {
32             type dot1qtypes:mstid-type;
33             description
34                 "The MSTID to which the FID is to be allocated.";
35             reference
36                 "12.12.2 of IEEE Std 802.1Q";
37         }
38     }
39     list fid-to-mstid-allocation {
40         key "fids";
41         description
42             "The FID to MSTID allocation table";
43         leaf fids {
44             type dot1qtypes:vid-range-type;
45             description
46                 "Range of FIDs.";
47             reference
48                 "12.12.2 of IEEE Std 802.1Q";
49         }
50         leaf mstid {
51             type dot1qtypes:mstid-type;
52             description
53                 "The MSTID to which the FID is allocated.";
54             reference
55                 "12.12.2 of IEEE Std 802.1Q";
56         }
57     }
58 }
59 }
60 }
61 }
62 augment "/if:interfaces/if:interface" {
63     when
64         "derived-from-or-self(if:type,'ianaif:bridge') or "+
65         "derived-from-or-self(if:type,'ianaif:ethernetCsmacd') or "+
66         "derived-from-or-self(if:type,'ianaif:ieee8023adLag') or "+
67         "derived-from-or-self(if:type,'ianaif:ilann')";
68     description
69         "Applies when a Bridge interface.";
70 }
71 description
72     "Augment the interface model with the Bridge Port";

```

```

1  container bridge-port {
2      description
3          "Bridge Port is an extension of the IETF Interfaces model (RFC7223).";
4      leaf bridge-name {
5          type leafref {
6              path '/dot1q:bridges/dot1q:bridge/dot1q:name';
7          }
8
9          description
10             "Used to reference configured Bridge node.";
11     }
12     leaf component-name {
13         type leafref {
14             path
15                 '/dot1q:bridges'+
16                 '/dot1q:bridge[dot1q:name=current()'+
17                 '/..' +
18                 '/bridge-name]'+
19                 '/dot1q:component'+
20                 '/dot1q:name';
21         }
22
23         description
24             "Used to reference configured Component node.";
25     }
26     leaf port-type {
27         type identityref {
28             base type-of-port;
29         }
30         description
31             "The port type. Indicates the capabilities of this port.";
32         reference
33             "12.4.2.1 of IEEE Std 802.1Q";
34     }
35     leaf pvid {
36         when
37             "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
38             "/../dot1q:bridge-name]/dot1q:component[name=current()"+
39             "/../dot1q:component-name]/dot1q:type !="+
40             "'dot1q:d-bridge-component'" {
41             description
42                 "Applies to non TPMRs";
43         }
44         type dot1qtys:vlan-index-type;
45         default "1";
46         description
47             "The primary (default) VID assigned to a specific Bridge Port.";
48         reference
49             "12.10.1, 5.4, item m) of IEEE Std 802.1Q";
50     }
51     leaf default-priority {
52         type dot1qtys:priority-type;
53         default "0";
54         description
55             "The default priority assigned to a specific Bridge Port.";
56         reference
57             "12.6.2 of IEEE Std 802.1Q";
58     }
59     container priority-regeneration {
60         description
61             "The Priority Regeneration Table parameters associated with a
62             specific Bridge Port. A list of Regenerated User Priorities for
63             each received priority on each port of a Bridge. The regenerated
64             priority value may be used to index the Traffic Class Table for
65             each input port. This only has effect on media that support native
66             priority. The default values for Regenerated User Priorities are
67             the same as the User Priorities";
68         reference
69             "12.6.2, 6.9.4 of IEEE Std 802.1Q";
70         uses dot1qtys:priority-regeneration-table-grouping;
71     }
72     leaf pcsp-selection {

```

```

1      type dot1qtotypes:pcp-selection-type;
2      default "8P0D";
3      description
4          "The Priority Code Point selection assigned to a specific Bridge
5          Port. This object identifies the rows in the PCP encoding and
6          decoding tables that are used to remark frames on this port if
7          this remarking is enabled";
8      reference
9          "12.6.2, 6.9.3 of IEEE Std 802.1Q";
10     }
11     container pcp-decoding-table {
12         description
13             "The Priority Code Point Decoding Table parameters associated with
14             a specific Bridge Port.";
15         uses dot1qtotypes:pcp-decoding-table-grouping;
16     }
17     container pcp-encoding-table {
18         description
19             "The Priority Code Point Encoding Table parameters associated with
20             a specific Bridge Port.";
21         uses dot1qtotypes:pcp-encoding-table-grouping;
22     }
23     leaf use-dei {
24         type boolean;
25         default "false";
26         description
27             "The Drop Eligible Indicator. If it is set to True, then the
28             drop_eligible parameter is encoded in the DEI of transmitted
29             frames, and the drop_eligible parameter shall be true(1) for a
30             received frame if the DEI is set in the VLAN tag or the Priority
31             Code Point Decoding Table indicates drop_eligible True for the
32             received PCP value. If this parameter is False, the DEI shall be
33             transmitted as zero and ignored on receipt.";
34         reference
35             "12.6.2, 6.9.3 of IEEE Std 802.1Q";
36     }
37     leaf drop-encoding {
38         type boolean;
39         default "false";
40         description
41             "The Drop Encoding parameter. If a Bridge supports encoding or
42             decoding of drop_eligible from the PCP field of a VLAN tag (6.9.3)
43             on any of its Ports, then it shall implement a Boolean parameter
44             Require Drop Encoding on each of its Ports with default value
45             False. If Require Drop Encoding is True and the Bridge Port cannot
46             encode particular priorities with drop_eligible, then frames
47             queued with those priorities and drop_eligible True shall be
48             discarded and not transmitted.";
49         reference
50             "12.6.2, 6.9.3 of IEEE Std 802.1Q";
51     }
52     leaf service-access-priority-selection {
53         type boolean;
54         default "false";
55         description
56             "The Service Access Priority selection. Indication of whether the
57             Service Access Priority Selection function is supported on the
58             Customer Bridge Port to request priority handling of the frame
59             from a Port-based service interface.";
60         reference
61             "12.6.2, 6.13 of IEEE Std 802.1Q";
62     }
63     container service-access-priority {
64         description
65             "The Service Access Priority table parameters. A table that
66             contains information about the Service Access Priority Selection
67             function for a Provider Bridge. The use of this table enables a
68             mechanism for a Customer Bridge attached to a Provider Bridged
69             Network to request priority handling of frames.";
70         reference
71             "12.6.2, 6.13.1 of IEEE Std 802.1Q";
72         uses dot1qtotypes:service-access-priority-table-grouping;

```

```

1      }
2      container traffic-class {
3          description
4              "The Traffic Class table parameters. A table mapping evaluated
5              priority to Traffic Class, for forwarding by the Bridge";
6          reference
7              "12.6.3, 8.6.6 of IEEE Std 802.1Q";
8          uses dot1qtypes:traffic-class-table-grouping;
9      }
10     container transmission-selection-algorithm-table {
11         description
12             "The Transmission Selection Algorithm Table for a given Port
13             assigns, for each traffic class that the Port supports, the
14             transmission selection algorithm that is to be used to select
15             frames for transmission from the corresponding queue. Transmission
16             Selection Algorithm Tables may be managed, and allow the
17             identification of vendor-specific transmission selection
18             algorithms. The transmission selection algorithms are identified
19             in the Transmission Selection Algorithm Table by means of integer
20             identifiers.";
21         reference
22             "12.20.2, 8.6.8 of IEEE Std 802.1Q";
23         uses dot1qtypes:transmission-selection-table-grouping;
24     }
25     leaf acceptable-frame {
26         when
27             "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
28             "/../dot1q:bridge-name]/dot1q:component[name=current()"+
29             "/../dot1q:component-name]/dot1q:type !="+
30             "'dot1q:d-bridge-component'" {
31             description
32                 "Applies to non TPMRs";
33         }
34         type enumeration {
35             enum admit-only-VLAN-tagged-frames {
36                 description
37                     "Admit only VLAN-tagged frames.";
38             }
39             enum admit-only-untagged-and-priority-tagged {
40                 description
41                     "Admit only untagged and priority-tagged frames.";
42             }
43             enum admit-all-frames {
44                 description
45                     "Admit all frames.";
46             }
47         }
48         default "admit-all-frames";
49         description
50             "To configure the Acceptable Frame Types parameter associated with
51             one or more Ports";
52         reference
53             "12.10.1.3, 6.9 of IEEE Std 802.1Q";
54     }
55     leaf enable-ingress-filtering {
56         when
57             "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
58             "/../dot1q:bridge-name]/dot1q:component[name=current()"+
59             "/../dot1q:component-name]/dot1q:type !="+
60             "'dot1q:d-bridge-component'" {
61             description
62                 "Applies to non TPMRs";
63         }
64         type boolean;
65         default "false";
66         description
67             "To enable the Ingress Filtering feature associated with one or
68             more Ports.";
69         reference
70             "12.10.1.4, 8.6.2 of IEEE Std 802.1Q";
71     }
72     leaf enable-restricted-vlan-registration {

```

```

1      when
2          "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
3          "/../dot1q:bridge-name]/dot1q:component[name=current()"+
4          "/../dot1q:component-name]/dot1q:type !="+
5          "'dot1q:d-bridge-component'" {
6          description
7              "Applies to non TPMRs";
8          }
9      type boolean;
10     default "false";
11     description
12         "To enable the Restricted VLAN Registration associated with one or
13         more Ports.";
14     reference
15         "11.2.3.2.3, 12.10.1.6 of IEEE Std 802.1Q";
16 }
17 leaf enable-vid-translation-table {
18     when
19         "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
20         "/../dot1q:bridge-name]/dot1q:component[name=current()"+
21         "/../dot1q:component-name]/dot1q:type !="+
22         "'dot1q:d-bridge-component'" {
23         description
24             "Applies to non TPMRs";
25         }
26     type boolean;
27     default "false";
28     description
29         "To enable VID Translation table associated with a Bridge Port.
30         This is not applicable to Bridge Ports that do not support a VID
31         Translation Table.";
32     reference
33         "12.10.1.8, 6.9 of IEEE Std 802.1Q";
34 }
35 leaf enable-egress-vid-translation-table {
36     when
37         "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
38         "/../dot1q:bridge-name]/dot1q:component[name=current()"+
39         "/../dot1q:component-name]/dot1q:type !="+
40         "'dot1q:d-bridge-component'" {
41         description
42             "Applies to non TPMRs";
43         }
44     type boolean;
45     default "false";
46     description
47         "To enable Egress VID Translation table associated with a Bridge
48         Port. This is not applicable to Ports that do not support an
49         Egress VID Translation table.";
50     reference
51         "12.10.1.8, 6.9 of IEEE Std 802.1Q";
52 }
53 list protocol-group-vid-set {
54     when
55         "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
56         "/../dot1q:bridge-name]/dot1q:component[name=current()"+
57         "/../dot1q:component-name]/dot1q:type !="+
58         "'dot1q:d-bridge-component'" {
59         description
60             "Applies to non TPMRs";
61         }
62     if-feature "port-and-protocol-based-vlan";
63     key "group-id";
64     description
65         "The list of VID values associated with the Protocol Group
66         Identifier for this port.";
67     reference
68         "12.10.1.1.3 of IEEE Std 802.1Q";
69     leaf group-id {
70         type uint32;
71         description
72             "The protocol group identifier";

```

```

1      reference
2      "12.10.1.7 of IEEE Std 802.1Q";
3  }
4  leaf-list vid {
5      type dot1qttype:vlanid;
6      description
7          "The VLAN identifier to which this entry applies.";
8      reference
9          "12.10.2 of IEEE Std 802.1Q";
10 }
11 }
12 leaf admin-point-to-point {
13     type enumeration {
14         enum force-true {
15             value 1;
16             description
17                 "Indicates that this port should always be treated as if it is
18                 connected to a point-to-point link.";
19         }
20         enum force-false {
21             value 2;
22             description
23                 "Indicates that this port should be treated as having a shared
24                 media connection.";
25         }
26         enum auto {
27             value 3;
28             description
29                 "Indicates that this port is considered to have a
30                 point-to-point link if it is an Aggregator and all of its
31                 members are aggregatable, or if the MAC entity is configured
32                 for full duplex operation, either through auto-negotiation or
33                 by management means.";
34         }
35     }
36     description
37         "For a port running spanning tree, this object represents the
38         administrative point-to-point status of the LAN segment attached
39         to this port, using the enumeration values of IEEE Std 802.1AC. A
40         value of forceTrue(1) indicates that this port should always be
41         treated as if it is connected to a point-to-point link. A value of
42         forceFalse(2) indicates that this port should be treated as having
43         a shared media connection. A value of auto(3) indicates that this
44         port is considered to have a point-to-point link if it is an
45         Aggregator and all of its members are aggregatable, or if the MAC
46         entity is configured for full duplex operation, either through
47         auto-negotiation or by management means. Manipulating this object
48         changes the underlying adminPointToPointMAC.";
49     reference
50         "12.4.2, 6.8.3 of IEEE Std 802.1Q; 11.3 of IEEE Std 802.1AC";
51 }
52 leaf protocol-based-vlan-classification {
53     when
54         "/dot1q:bridges/dot1q:bridge[dot1q:name=current()]" +
55         "../dot1q:bridge-name]/dot1q:component[name=current()]" +
56         "../dot1q:component-name]/dot1q:type !="+
57         "'dot1q:d-bridge-component'" {
58         description
59             "Applies to non TPMRs";
60     }
61     if-feature "port-and-protocol-based-vlan";
62     type boolean;
63     config false;
64     description
65         "A boolean indication indicating if Port-and-Protocol-based VLAN
66         classification is supported on a given Port.";
67     reference
68         "5.4.1.2 of IEEE Std 802.1Q";
69 }
70 leaf max-vid-set-entries {
71     when
72         "/dot1q:bridges/dot1q:bridge[dot1q:name=current()]" +

```

```

1      "../../../dot1q:bridge-name]/dot1q:component[name=current()"+
2      "../../../dot1q:component-name]/dot1q:type !="+
3      "'dot1q:d-bridge-component'" {
4      description
5          "Applies to non TPMRs";
6      }
7      if-feature "port-and-protocol-based-vlan";
8      type uint16;
9      config false;
10     description
11         "The maximum number of entries supported in the VID set on a given
12         Port.";
13     reference
14         "12.10.1.1.3 of IEEE Std 802.1Q";
15 }
16 leaf port-number {
17     type dot1qt-types:port-number-type;
18     config false;
19     description
20         "An integer that uniquely identifies a Bridge Port.";
21     reference
22         "Item i) in 12.3, 17.3.2.2 of IEEE Std 802.1Q";
23 }
24 leaf address {
25     type ieee:mac-address;
26     config false;
27     description
28         "The specific MAC address of the individual MAC Entity associated
29         with the Port.";
30     reference
31         "12.4.2, Item a) in 12.4.1.1.3 of IEEE Std 802.1Q";
32 }
33 leaf capabilities {
34     type bits {
35         bit tagging {
36             position 0;
37             description
38                 "Supports 802.1Q VLAN tagging of frames and MVRP.";
39         }
40         bit configurable-acceptable-frame-type {
41             position 1;
42             description
43                 "Allows modified values of acceptable frame types";
44         }
45         bit ingress-filtering {
46             position 2;
47             description
48                 "Supports the discarding of any frame received on a Port whose
49                 VLAN classification does not include that Port in its member
50                 set.";
51         }
52     }
53     config false;
54     description
55         "The feature capabilities associated with port. Indicates the
56         parts of IEEE 802.1Q that are optional on a per-port basis, that
57         are implemented by this device, and that are manageable.";
58     reference
59         "Item c) in 12.10.1.1.3, 12.4.2 of IEEE Std 802.1Q";
60 }
61 leaf type-capabilities {
62     type bits {
63         bit customer-vlan-port {
64             position 0;
65             description
66                 "Indicates the port can be a C-TAG aware port of an enterprise
67                 VLAN aware Bridge";
68         }
69         bit provider-network-port {
70             position 1;
71             description
72                 "Indicates the port can be an S-TAG aware port of a Provider

```



```

1         Bridge or Backbone Edge Bridge used for connections within a
2         PBN or PBBN.";
3     }
4     bit customer-network-port {
5         position 2;
6         description
7             "Indicates the port can be an S-TAG aware port of a Provider
8             Bridge or Backbone Edge Bridge used for connections to the
9             exterior of a PBN or PBBN.";
10    }
11    bit customer-edge-port {
12        position 3;
13        description
14            "Indicates the port can be a C-TAG aware port of a Provider
15            Bridge used for connections to the exterior of a PBN or PBBN.";
16    }
17    bit customer-backbone-port {
18        position 4;
19        description
20            "Indicates the port can be a I-TAG aware port of a Backbone
21            Edge Bridge's B-component.";
22    }
23    bit virtual-instance-port {
24        position 5;
25        description
26            "Indicates the port can be a virtual S-TAG aware port within a
27            Backbone Edge Bridge's I-component which is responsible for
28            handling S-tagged traffic for a specific backbone service
29            instance.";
30    }
31    bit d-bridge-port {
32        position 6;
33        description
34            "Indicates the port can be a VLAN-unaware member of an 802.1Q
35            Bridge.";
36    }
37    bit remote-customer-access-port {
38        position 7;
39        description
40            "Indicates the port can be an S-TAG aware port of a Provider
41            Bridge capable of providing Remote Customer Service
42            Interfaces.";
43    }
44    bit station-facing-bridge-port {
45        position 8;
46        description
47            "Indicates the station-facing Bridge Port in a EVB Bridge.";
48    }
49    bit uplink-access-port {
50        position 9;
51        description
52            "Indicates the uplink access port in an EVB Bridge or EVB
53            station.";
54    }
55    bit uplink-relay-port {
56        position 10;
57        description
58            "Indicates the uplink relay port in an EVB station.";
59    }
60    }
61    config false;
62    description
63        "The type of feature capabilities supported with port. Indicates
64        the capabilities of this port.";
65    reference
66        "12.4.2 of IEEE Std 802.1Q";
67    }
68    leaf external {
69        type boolean;
70        config false;
71        description
72            "A boolean indicating whether the port is external. A value of

```

```

1      True means the port is external. A value of False means the port
2      is internal.";
3      reference
4      "12.4.2 of IEEE Std 802.1Q";
5  }
6  leaf oper-point-to-point {
7      type boolean;
8      config false;
9      description
10     "For a port running spanning tree, this object represents the
11     operational point-to-point status of the LAN segment attached to
12     this port. It indicates whether a port is considered to have a
13     point-to-point connection.
14
15     If admin-point-to-point is set to auto(2), then the value of
16     oper-point-to-point is determined in accordance with the specific
17     procedures defined for the MAC entity concerned, as defined in
18     IEEE Std 802.1AC.
19
20     The value is determined dynamically; that is, it is re-evaluated
21     whenever the value of admin-point-to-point changes, and whenever
22     the specific procedures defined for the MAC entity evaluate a
23     change in its point-to-point status.";
24     reference
25     "IEEE Std 802.1AC; 12.4.2 of IEEE Std 802.1Q";
26 }
27 leaf media-dependent-overhead {
28     type uint8;
29     units "octets";
30     config false;
31     description
32     "The portMediaDependentOverhead parameter provides the number of
33     additional octets for media-dependent framing. The overhead
34     includes all octets prior the first octet of the Destination
35     Address field and all octets after the last octet of the frame
36     check sequence.";
37     reference
38     "12.4.2 of IEEE Std 802.1Q";
39 }
40 container statistics {
41     config false;
42     description
43     "Container of operational state node information associated with
44     the bridge port.";
45     uses dot1qtypes:bridge-port-statistics-grouping;
46     leaf discard-on-ingress-filtering {
47         when
48         "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
49         "/../dot1q:bridge-name]/dot1q:component[name=current()"+
50         "/../dot1q:component-name]/dot1q:type !="+
51         "'dot1q:d-bridge-component'" {
52             description
53             "Applies to non TPMRs";
54         }
55         if-feature "ingress-filtering";
56         type yang:counter64;
57         description
58         "The number of frames that were discarded as a result of Ingress
59         Filtering being enabled.
60
61         Discontinuities in the value of this counter can occur at
62         re-initialization of the management system, and at other times
63         as indicated by the value of 'discontinuity-time'.";
64         reference
65         "12.6.1.1.3 of IEEE Std 802.1Q";
66     }
67 }
68 list vid-translations {
69     when
70     "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
71     "/../dot1q:bridge-name]/dot1q:component[name=current()"+
72     "/../dot1q:component-name]/dot1q:type !="+

```

```

1      "'dot1q:d-bridge-component'" {
2      description
3          "Applies to non TPMRs";
4      }
5      key "local-vid";
6      description
7          "To configure the VID Translation Table (6.9) associated with a
8          Port. This object is not applicable to Ports that do not support a
9          VID Translation Table. The default configuration of the table has
10         the value of the Relay VID equal to the value of the Local VID. If
11         no local VID is configured, then it is assumed that the relay VID
12         is the same value as the local VID.
13
14         If the port supports an Egress VID translation table, the VID
15         Translation Configuration object configures the Local VID to Relay
16         VID mapping on ingress only. If an Egress VID translation is not
17         supported, the VID Translation Configuration object defines a
18         single bidirectional mapping. In this case, the Bridge should not
19         allow multiple keys ('local-vid') mapped to the same 'relay-vid'
20         value.";
21     leaf local-vid {
22         type dot1qtypes:vlanid;
23         description
24             "The Local VID after translation received at the ISS or EISS.";
25         reference
26             "12.10.1.8, 6.9 of IEEE Std 802.1Q";
27     }
28     leaf relay-vid {
29         type dot1qtypes:vlanid;
30         description
31             "The Relay VID received before translation received at ISS or
32             EISS.";
33         reference
34             "12.10.1.8, 6.9 of IEEE Std 802.1Q";
35     }
36 }
37 list egress-vid-translations {
38     when
39         "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
40         "/../dot1q:bridge-name]/dot1q:component[name=current()"+
41         "/../dot1q:component-name]/dot1q:type !="+
42         "'dot1q:d-bridge-component'" {
43         description
44             "Applies to non TPMRs";
45         }
46         key "relay-vid";
47         description
48             "To configure the Egress VID Translation Table (6.9) associated
49             with a Port. This object is not applicable to Ports that do not
50             support an Egress VID Translation Table. The default configuration
51             of the table has the value of the Local VID equal to the value of
52             the Relay VID. If no Relay VID is configured, then it is assumed
53             that the local VID is the same value as the relay VID.";
54         leaf relay-vid {
55             type dot1qtypes:vlanid;
56             description
57                 "The Relay VID received before translation received at ISS or
58                 EISS.";
59             reference
60                 "12.10.1.9, 6.9 of IEEE Std 802.1Q";
61         }
62         leaf local-vid {
63             type dot1qtypes:vlanid;
64             description
65                 "The Local VID after translation received at the ISS or EISS.";
66             reference
67                 "12.10.1.9, 6.9 of IEEE Std 802.1Q";
68         }
69     }
70 }
71 }
72 }

```

1
2

1 48.6.5 The ieee802-dot1q-tpmr YANG module

```
2 module ieee802-dot1q-tpmr {
3   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-tpmr;
4   prefix dot1q-tpmr;
5   import ieee802-dot1q-bridge {
6     prefix dot1q;
7   }
8   import ietf-yang-types {
9     prefix yang;
10  }
11  import ietf-interfaces {
12    prefix if;
13  }
14  organization
15    "IEEE 802.1 Working Group";
16  contact
17    "WG-URL: http://ieee802.org/1/
18    WG-EMail: stds-802-1-1@ieee.org
19
20    Contact: IEEE 802.1 Working Group Chair
21    Postal: C/O IEEE 802.1 Working Group
22            IEEE Standards Association
23            445 Hoes Lane
24            Piscataway, NJ 08854
25            USA
26
27    E-mail: stds-802-1-chairs@ieee.org";
28  description
29    "This YANG module describes the bridge configuration model for the
30    Two Port MAC Relays.
31
32    Copyright (C) IEEE (2022).
33
34    This version of this YANG module is part of IEEE Std 802.1Q; see the
35    standard itself for full legal notices.";
36  revision 2022-01-19 {
37    description
38      "Published as part of IEEE Std 802.1Q-2022. Third version.";
39    reference
40      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
41  }
42  revision 2020-06-04 {
43    description
44      "Published as part of IEEE Std 802.1Qcx-2020. Second version.";
45    reference
46      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
47      Model for Connectivity Fault Management.";
48  }
49  revision 2018-09-30 {
50    description
51      "Creation for Working Group review.";
52    reference
53      "IEEE Std 802.1Q-2018, Bridges and Bridged Networks.";
54  }
55  revision 2018-03-07 {
56    description
57      "Published as part of IEEE Std 802.1Q-2018. Initial version.";
58    reference
59      "IEEE Std 802.1Q-2018, Bridges and Bridged Networks.";
60  }
61  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
62    when
63      "dot1q:port-type = 'dot1q:d-bridge-port'" {
64      description
65        "Applies to TPMRs ports";
66    }
67    description
68      "Augment Interface model with TPMR port configuration specific
69      nodes.";
70    leaf managed-address {
71      type boolean;
```

```

1  default "true";
2  description
3    "A Boolean value, which is TRUE if the MAC address is the
4    management address for the TPMR, and is otherwise FALSE.
5
6    The TPMR management entity may make use of one or both Ports of
7    a TPMR to transmit and receive management frames. However, the
8    MAC address used by the TPMR management entity as the source MAC
9    address in transmitted management frames (the management MAC
10   address) is the individual MAC address associated with one of
11   the Ports of the TPMR";
12  reference
13    "12.19.1.1.1.3 of IEEE Std 802.1Q-2022";
14 }
15 container mac-status-propagation {
16   description
17     "MAC status propagation configuration node parameters.";
18   leaf link-notify {
19     type boolean;
20     default "true";
21     description
22       "The current value (Boolean) of LinkNotify (23.5.1) being used
23       by the MSP state machines.";
24     reference
25       "12.19.4.1.1.3, 12.19.4.1.2.2 of IEEE Std 802.1Q-2022";
26   }
27   leaf link-notify-wait {
28     type yang:timeticks {
29       range "20..100";
30     }
31     default "40";
32     description
33       "The current value, in centiseconds, of LinkNotifyWait
34       (23.5.2) being used by the MSP state machines.";
35     reference
36       "12.19.4.1.1.3, 12.19.4.1.2.2 of IEEE Std 802.1Q-2022";
37   }
38   leaf link-notify-retry {
39     type yang:timeticks {
40       range "10..100";
41     }
42     default "100";
43     description
44       "The current value, in centiseconds, of LinkNotifyRetry
45       (23.5.3) being used by the MSP state machines.";
46     reference
47       "12.19.4.1.1.3, 12.19.4.1.2.2 of IEEE Std 802.1Q-2022";
48   }
49   leaf mac-notify {
50     type boolean;
51     default "true";
52     description
53       "The current value (Boolean) of MACNotify (23.5.4) being used
54       by the MSP state machines.";
55     reference
56       "12.19.4.1.1.3, 12.19.4.1.2.2 of IEEE Std 802.1Q-2022";
57   }
58   leaf mac-notify-time {
59     type yang:timeticks {
60       range "1..50";
61     }
62     default "20";
63     description
64       "The current value, in centiseconds, of MACNotifyTime (23.5.5)
65       being used by the MSP state machines.";
66     reference
67       "12.19.4.1.1.3, 12.19.4.1.2.2 of IEEE Std 802.1Q-2022";
68   }
69   leaf mac-recover-time {
70     type yang:timeticks {
71       range "2..50";
72     }

```

```

1      default "10";
2      description
3          "The current value, in centiseconds, of MACRecoverTime
4          (23.5.6) being used by the MSP state machines.";
5      reference
6          "12.19.4.1.1.3, 12.19.4.1.2.2 of IEEE Std 802.1Q-2022";
7  }
8  }
9  }
10 augment
11     "/if:interfaces/if:interface/dot1q:bridge-port/dot1q:statistics" {
12     when
13         "../dot1q:port-type = 'dot1q:d-bridge-port'" {
14         description
15             "Applies to TPMRs ports";
16         }
17     description
18         "Augment Interface model with TPMR port operational state specific
19         nodes.";
20     leaf acks-tx {
21         type yang:counter64;
22         config false;
23         description
24             "The number of acks transmitted (23.6.15) by the Ports Transmit
25             Process as a consequence of txAck being set.
26
27             Discontinuities in the value of this counter can occur at
28             re-initialization of the management system, and at other times
29             as indicated by the value of 'discontinuity-time'.";
30         reference
31             "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
32     }
33     leaf add-notifications-tx {
34         type yang:counter64;
35         config false;
36         description
37             "The number of adds transmitted (23.6.16) by the Ports Transmit
38             Process as a consequence of txAdd being set.
39
40             Discontinuities in the value of this counter can occur at
41             re-initialization of the management system, and at other times
42             as indicated by the value of 'discontinuity-time'.";
43         reference
44             "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
45     }
46     leaf add-confirmations-tx {
47         type yang:counter64;
48         config false;
49         description
50             "The number of add confirms transmitted (23.6.17) by the Ports
51             Transmit Process as a consequence of txAddConfirm being set.
52
53             Discontinuities in the value of this counter can occur at
54             re-initialization of the management system, and at other times
55             as indicated by the value of 'discontinuity-time'.";
56         reference
57             "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
58     }
59     leaf loss-notification-tx {
60         type yang:counter64;
61         config false;
62         description
63             "The number of losses transmitted (23.6.18) by the Ports
64             Transmit Process as a consequence of txLoss being set.
65
66             Discontinuities in the value of this counter can occur at
67             re-initialization of the management system, and at other times
68             as indicated by the value of 'discontinuity-time'.";
69         reference
70             "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
71     }
72     leaf loss-confirmation-tx {

```

```
1     type yang:counter64;
2     config false;
3     description
4         "The number of loss confirms transmitted (23.6.19) by the Ports
5         Transmit Process as a consequence of txLossConfirm being set.
6
7         Discontinuities in the value of this counter can occur at
8         re-initialization of the management system, and at other times
9         as indicated by the value of 'discontinuity-time.'";
10    reference
11        "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
12    }
13    leaf acks-rx {
14        type yang:counter64;
15        config false;
16        description
17            "The number of acks received (23.6.10) by the Ports Transmit
18            Process.
19
20            Discontinuities in the value of this counter can occur at
21            re-initialization of the management system, and at other times
22            as indicated by the value of 'discontinuity-time.'";
23        reference
24            "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
25    }
26    leaf add-notifications-rx {
27        type yang:counter64;
28        config false;
29        description
30            "The number of adds received (23.6.11) by the Ports Receive
31            Process.
32
33            Discontinuities in the value of this counter can occur at
34            re-initialization of the management system, and at other times
35            as indicated by the value of 'discontinuity-time.'";
36        reference
37            "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
38    }
39    leaf add-confirmations-rx {
40        type yang:counter64;
41        config false;
42        description
43            "The number of add confirms received (23.6.12) by the Ports
44            Receive Process.
45
46            Discontinuities in the value of this counter can occur at
47            re-initialization of the management system, and at other times
48            as indicated by the value of 'discontinuity-time.'";
49        reference
50            "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
51    }
52    leaf loss-notification-rx {
53        type yang:counter64;
54        config false;
55        description
56            "The number of losses received (23.6.13) by the Ports Receive
57            Process.
58
59            Discontinuities in the value of this counter can occur at
60            re-initialization of the management system, and at other times
61            as indicated by the value of 'discontinuity-time.'";
62        reference
63            "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
64    }
65    leaf loss-confirmation-rx {
66        type yang:counter64;
67        config false;
68        description
69            "The number of loss confirms received (23.6.14) by the Ports
70            Receive Process.
71
72            Discontinuities in the value of this counter can occur at
```



```
1      re-initialization of the management system, and at other times
2      as indicated by the value of 'discontinuity-time.';
3      reference
4      "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
5  }
6  leaf add-events {
7      type yang:counter64;
8      config false;
9      description
10         "The number of transitions to STM:ADD directly from STM:DOWN or
11         STM:LOSS (23.8).
12
13         Discontinuities in the value of this counter can occur at
14         re-initialization of the management system, and at other times
15         as indicated by the value of 'discontinuity-time.';
16     reference
17         "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
18 }
19 leaf loss-events {
20     type yang:counter64;
21     config false;
22     description
23         "The number of transitions to STM:LOSS directly from STM:UP or
24         STM:ADD (23.8).
25
26         Discontinuities in the value of this counter can occur at
27         re-initialization of the management system, and at other times
28         as indicated by the value of 'discontinuity-time.';
29     reference
30         "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
31 }
32 leaf mac-status-notifications {
33     type yang:counter64;
34     config false;
35     description
36         "The number of transitions to SNM:MAC_NOTIFICATION (23.9).
37
38         Discontinuities in the value of this counter can occur at
39         re-initialization of the management system, and at other times
40         as indicated by the value of 'discontinuity-time.';
41     reference
42         "12.19.4.1.3.3 of IEEE Std 802.1Q-2022";
43 }
44 }
45 }
46
47
```

1 48.6.6 The ieee802-dot1q-pb YANG module

```
2 module ieee802-dot1q-pb {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-pb;
5   prefix dot1q-pb;
6   import ieee802-dot1q-bridge {
7     prefix dot1q;
8   }
9   import ieee802-dot1q-types {
10    prefix dot1qtypes;
11  }
12  import ietf-interfaces {
13    prefix if;
14  }
15  organization
16    "IEEE 802.1 Working Group";
17  contact
18    "WG-URL: http://ieee802.org/1/
19    WG-EMail: stds-802-1-1@ieee.org
20
21    Contact: IEEE 802.1 Working Group Chair
22    Postal: C/O IEEE 802.1 Working Group
23            IEEE Standards Association
24            445 Hoes Lane
25            Piscataway, NJ 08854
26            USA
27
28    E-mail: stds-802-1-chairs@ieee.org";
29  description
30    "This YANG module describes the Bridge configuration model for Provider
31    Bridges.
32
33    Copyright (C) IEEE (2023).
34
35    This version of this YANG module is part of IEEE Std 802.1Q; see the
36    standard itself for full legal notices.";
37  revision 2023-10-22 {
38    description
39      "Published as part of IEEE Std 802.1Qcw-2023.
40
41      The following reference statement identifies each referenced IEEE
42      Standard as updated by applicable amendments.";
43    reference
44      "IEEE Std 802.1Q Bridges and Bridged Networks:
45      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.";
46  }
47  revision 2022-01-19 {
48    description
49      "Published as part of IEEE Std 802.1Q-2022.";
50    reference
51      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
52  }
53  revision 2020-06-04 {
54    description
55      "Published as part of IEEE Std 802.1Qcx-2020. Second version.";
56    reference
57      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
58      Model for Connectivity Fault Management.";
59  }
60  revision 2018-03-07 {
61    description
62      "Published as part of IEEE Std 802.1Q-2018. Initial version.";
63    reference
64      "IEEE Std 802.1Q-2018, Bridges and Bridged Networks.";
65  }
66  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
67    description
68      "Augment the interface model with 802.1Q Bridge Port configuration
69      specific nodes.";
70    leaf svid {
71      type dot1qtypes:vlanid;
```

```

1      description
2      "Service VLAN identifier.";
3      reference
4      "12.13.2.1 of IEEE Std 802.1Q";
5  }
6  list cvid-registration {
7      when
8          "/dot1q:bridges/dot1q:bridge[dot1q:name=current()"+
9          "/../dot1q:bridge-name]/dot1q:component[dot1q:name=current()"+
10         "/../dot1q:component-name]/dot1q:type = 'dot1q:c-vlan-component'"+
11         " and ../dot1q:port-type = 'dot1q:customer-edge-port'" {
12         description
13             "Applies when the component associated with this interface is a
14             C-VLAN component and the port-type is a customer edge port.";
15         }
16         key "cvid";
17         description
18             "The C-VID Registration Table, provides a mapping between a C-VID
19             and the service instance represented by an S-VID selected for that
20             C-VLAN. This table provides the equivalent functionality of
21             1) Configuring the PVID of the internal CNP on the S-VLAN
22             component
23             2) Adding the corresponding PEP on the C-VLAN component to the
24             member set of the C-VLAN
25             3) Adding the PEP and/or CEP to the untagged set of the C-VLAN
26             (if it is desired that frames forwarded to that port are
27             transmitted untagged for this C-VLAN).";
28         leaf cvid {
29             type dot1qt-types:vlanid;
30             description
31                 "Customer VLAN identifiers associated with this bridge port.";
32             reference
33                 "12.13.2.1 of IEEE Std 802.1Q";
34         }
35         leaf svid {
36             type dot1qt-types:vlanid;
37             description
38                 "Service VLAN identifier.";
39             reference
40                 "12.13.2.1 of IEEE Std 802.1Q";
41         }
42         leaf untagged-pep {
43             type boolean;
44             default "true";
45             description
46                 "A boolean indicating frames for this C-VLAN should be forwarded
47                 untagged through the Provider Edge Port.";
48             reference
49                 "12.13.2.1 of IEEE Std 802.1Q";
50         }
51         leaf untagged-cep {
52             type boolean;
53             default "true";
54             description
55                 "A boolean indicating frames for this C-VLAN should be forwarded
56                 untagged through the Customer Edge Port.";
57             reference
58                 "12.13.2.1 of IEEE Std 802.1Q";
59         }
60     }
61     list service-priority-regeneration {
62         when
63             "/dot1q:bridges/dot1q:bridge[dot1q:name=current() "+
64             "/../dot1q:bridge-name]/dot1q:component[dot1q:name=current() "+
65             "/../dot1q:component-name]/dot1q:type = 'dot1q:c-vlan-component' "+
66             "and ../dot1q:port-type = 'dot1q:customer-edge-port'" {
67             description
68                 "Applies when the component associated with this interface is a
69                 C-VLAN component and the port-type is a customer edge port.";
70             }
71             key "svid";
72             description

```

```

1      "The Service Priority Regeneration Table, which provides the
2      Priority Regeneration Table (12.6.2) for each internal CNP connected
3      to the C-VLAN component associated with the CEP.";
4  leaf svid {
5      type dot1qttype:vlanid;
6      description
7          "Service VLAN identifier.";
8      reference
9          "12.13.2.6 of IEEE Std 802.1Q";
10 }
11 container priority-regeneration {
12     description
13         "Contains Service Priority Regeneration table nodal information.";
14     reference
15         "12.13.2.6 of IEEE Std 802.1Q";
16     uses dot1qttype:priority-regeneration-table-grouping;
17 }
18 }
19 list rcap-internal-interface {
20     when
21         "/dot1q:bridges/dot1q:bridge[dot1q:name=current() "+
22         "/../dot1q:bridge-name]/dot1q:component[dot1q:name=current() "+
23         "/../dot1q:component-name]/dot1q:type = 'dot1q:s-vlan-component' "+
24         "and ../dot1q:port-type = 'dot1q:remote-customer-access-port'" {
25         description
26             "Applies when the component associated with this interface is a
27             C-VLAN component and the port-type is a customer edge port.";
28     }
29     key "external-svid";
30     description
31         "Designating an external port as an RCAP automatically creates a
32         Port-mapping S-VLAN component associated with that port. This
33         Port-mapping S-VLAN component includes one internal PNP.";
34     leaf external-svid {
35         type dot1qttype:vlanid;
36         description
37             "External Service VLAN identifier.";
38         reference
39             "12.13.3.2 of IEEE Std 802.1Q";
40     }
41     leaf internal-port-number {
42         type dot1qttype:port-number-type;
43         description
44             "The number of the RCAP.";
45         reference
46             "12.13.3.2 of IEEE Std 802.1Q";
47     }
48     leaf internal-svid {
49         type dot1qttype:vlanid;
50         description
51             "Internal Service VLAN Identifier (not applicable for a C-tagged
52             RCSI).";
53         reference
54             "12.13.3.2 of IEEE Std 802.1Q";
55     }
56     leaf internal-interface-type {
57         type enumeration {
58             enum port-based-rcsi {
59                 description
60                     "Port-based RCSI";
61             }
62             enum c-tagged-rcsi {
63                 description
64                     "C-tagged RCSI";
65             }
66             enum pnp {
67                 description
68                     "Provider Network Port";
69             }
70             enum discard {
71                 description
72                     "Discard (external S-VID is not associated with an internal

```

```
1         port).";
2     }
3 }
4 description
5     "A value indicating the type of internal interface associated with
6     the external S-VID.";
7 reference
8     "12.13.3.2 of IEEE Std 802.1Q";
9 }
10 }
11 }
12 }
13
14
```

1 48.6.7 The ieee802-dot1q-cfm-types YANG module

```
2 module ieee802-dot1q-cfm-types {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-cfm-types;
5   prefix cfm-types;
6   organization
7     "IEEE 802.1 Working Group";
8   contact
9     "WG-URL: http://ieee802.org/1/
10    WG-EMail: stds-802-1-1@ieee.org
11
12    Contact: IEEE 802.1 Working Group Chair
13    Postal: C/O IEEE 802.1 Working Group
14            IEEE Standards Association
15            445 Hoes Lane
16            Piscataway, NJ 08854
17            USA
18
19    E-mail: stds-802-1-chairs@ieee.org";
20  description
21    "Common types used within ieee802-dot1q-cfm modules.
22
23    Copyright (C) IEEE (2022).
24
25    This version of this YANG module is part of IEEE Std 802.1Q; see the
26    standard itself for full legal notices.";
27  revision 2022-10-29 {
28    description
29      "Published as part of IEEE Std 802.1Q-2022.";
30    reference
31      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
32  }
33  revision 2020-06-04 {
34    description
35      "Published as part of IEEE Std 802.1Qcx-2020. Initial version.";
36    reference
37      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
38  }
39
40  /* -----
41   * Type definitions used by 802.1Qcx YANG module.
42   * -----
43   */
44  typedef unicast-mac-address-type {
45    type string {
46      pattern "[0-9a-fA-F][02468aAcCeE](-[0-9a-fA-F]{2}){5}";
47    }
48    description
49      "A unicast destination MAC address. The I/G address bit is used to
50      identify the destination MAC address as an individual MAC address
51      or a group MAC address. If the I/G address bit is 0, it indicates
52      that the MAC address field is an individual MAC address. If this
53      bit is 1, the MAC address is a group MAC address that identifies
54      one or more (or all) stations connected to the IEEE 802 network";
55  }
56  typedef multicast-class1-mac-address-type {
57    type string {
58      pattern "(01-80-C2-00-00-3)[0-7]";
59    }
60    description
61      "The multicast class 1 MAC address must take the form of
62      01-80-C2-00-00-3x, where x represents the MEG level, with x being
63      a value in the range of 0..7.";
64  }
65  typedef mp-type {
66    type enumeration {
67      enum mhf {
68        value 1;
69        description
70          "Indicates a MHF.";
71      }
```

```

1      enum mep {
2          value 2;
3          description
4              "Indicates a MEP.";
5      }
6  }
7  description
8      "Indicates the type of Maintenance Point.";
9  }
10 typedef mhf-creation-type {
11     type enumeration {
12         enum mhf-none {
13             value 1;
14             description
15                 "No MHFs can be created for designated VID(s) or ISID.";
16         }
17         enum mhf-default {
18             value 2;
19             description
20                 "MHFs can be created for designated VID(s) or ISID on any
21                 Bridge Port through which the VID(s) or ISID can pass, where:
22                 i) There are no lower active MD levels; or
23                 ii) There is a MEP at the next lower active MD level on the
24                 port.";
25         }
26         enum mhf-explicit {
27             value 3;
28             description
29                 "MHFs can be created for designated VID(s) or ISID only on
30                 Bridge Ports through which the VID(s) or ISID can pass, and
31                 only if there is a MEP at the next lower active MD level on
32                 the port.";
33         }
34         enum mhf-defer {
35             value 4;
36             description
37                 "In the Maintenance Association only, the control of MHF
38                 creation is deferred to the corresponding variable in the
39                 enclosing Maintenance Domain.";
40         }
41     }
42     description
43         "Indicates if the Management Entity can create MHFs.";
44 }
45 typedef mp-direction-type {
46     type enumeration {
47         enum down {
48             value 1;
49             description
50                 "Down maintenance point, where CFM protocol messages are
51                 dispatched away from the MAC Relay entity.";
52         }
53         enum up {
54             value 2;
55             description
56                 "Up maintenance point, where CFM protocol messages are
57                 dispatched towards the MAC Relay entity.";
58         }
59     }
60     description
61         "Indicates the direction in which the Maintenance Point (MEP or
62         MIP) faces on the Bridge Port.";
63 }
64 typedef mep-id-type {
65     type uint16 {
66         range "1..8191";
67     }
68     description
69         "Maintenance association End Point Identifier, which is unique
70         over a given Maintenance Association.";
71 }
72 typedef md-level-type {

```

```

1  type uint8 {
2      range "0..7";
3  }
4  description
5      "Integer identifying the Maintenance Domain Level (MD Level).
6      Higher numbers correspond to higher Maintenance Domains, those
7      with the greatest physical reach, with the highest values for
8      customers' CFM PDUs. Lower numbers correspond to lower Maintenance
9      Domains, those with more limited physical reach, with the lowest
10     values for CFM PDUs protecting single bridges or physical links.";
11 }
12 typedef port-status-tlv-value-type {
13     type enumeration {
14         enum no-port-state-tlv {
15             value 0;
16             description
17                 "Indicates either that no CCM has been received or that no
18                 port status TLV was present in the last CCM received.";
19         }
20         enum blocked {
21             value 1;
22             description
23                 "Ordinary data cannot pass freely through the port on which
24                 the remote MEP resides.";
25         }
26         enum up {
27             value 2;
28             description
29                 "Ordinary data can pass freely through the port on which the
30                 remote MEP resides.";
31         }
32     }
33     description
34         "An enumerated value from the Port Status TLV from the last CCM
35         received from the last MEP. It indicates the ability of the Bridge
36         Port on which the transmitting MEP resides to pass ordinary data,
37         regardless of the status of the MAC.";
38 }
39 typedef interface-status-tlv-value-type {
40     type enumeration {
41         enum no-interface-status-tlv {
42             value 0;
43             description
44                 "Indicates either that no CCM has been received or that no
45                 interface status TLV was present in the last CCM received.";
46         }
47         enum up {
48             value 1;
49             description
50                 "The interface is ready to pass frames.";
51         }
52         enum down {
53             value 2;
54             description
55                 "The interface can not pass frames.";
56         }
57         enum testing {
58             value 3;
59             description
60                 "The interface is in some test mode.";
61         }
62         enum unknown {
63             value 4;
64             description
65                 "The interface status cannot be determined for some reason.";
66         }
67         enum dormant {
68             value 5;
69             description
70                 "The interface is not in a state to pass frames but is in a
71                 pending state, waiting for some external event.";
72         }
73     }

```



```

1      enum not-present {
2          value 6;
3          description
4              "Some component of the interface is missing.";
5      }
6      enum lower-layer-down {
7          value 7;
8          description
9              "The interface is down due to state of the lower layer
10             interface.";
11      }
12  }
13  description
14      "An enumerated value from the Interface Status TLV from the last
15      CCM received from the last MEP. It indicates the status of the
16      Interface within which the MEP transmitting the CCM is configured,
17      or the next lower Interface in the Interface Stack, if the MEP is
18      not configured within an Interface.";
19  }
20  typedef highest-defect-priority-type {
21      type enumeration {
22          enum none {
23              value 0;
24              description
25                  "No defects since Fault Notification Generator state machine
26                  reset.";
27          }
28          enum def-rdi-ccm {
29              value 1;
30              description
31                  "The last CCM received by this MEP from some remote MEP
32                  contained the RDI bit set.";
33          }
34          enum def-mac-status {
35              value 2;
36              description
37                  "The last CCM received by this MEP from some remote MEP
38                  indicating that the transmitting MEP's associated MAC is
39                  reporting its status via the Port Status TLV or Interface
40                  Status TLV.";
41          }
42          enum def-remote-ccm {
43              value 3;
44              description
45                  "This MEP is not receiving CCMs from some other MEP in its
46                  configured list.";
47          }
48          enum def-error-ccm {
49              value 4;
50              description
51                  "This MEP is receiving invalid CCMs.";
52          }
53          enum def-xcon-ccm {
54              value 5;
55              description
56                  "This MEP is receiving CCMs that could be from some other MA.";
57          }
58      }
59      description
60          "An enumerated value, equal to the contents of the variable
61          highestDefect (20.35.9 and Table 20-1), indicating the
62          highest-priority defect that has been present since the MEP Fault
63          Notification Generator State Machine was last in the FNG_RESET
64          state. The integer value assigned to the enum value determines the
65          priority. The higher value corresponds to the higher priority.";
66  }
67  typedef lowest-alarm-priority-type {
68      type enumeration {
69          enum all-def {
70              value 1;
71              description
72                  "Includes def-rdi-ccm, def-mac-status, def-remote-ccm,

```

```

1      def-error-ccm, and def-xcon-ccm.";
2  }
3  enum mac-remote-error-xcon {
4      value 2;
5      description
6          "Only includes def-mac-status, def-remote-ccm, def-error-ccm,
7          and def-xcon-ccm.";
8  }
9  enum remote-error-xcon {
10     value 3;
11     description
12         "Includes def-remote-ccm, def-error-ccm, and def-xcon-ccm.";
13 }
14 enum error-xcon {
15     value 4;
16     description
17         "Includes def-error-ccm and def-xcon-ccm.";
18 }
19 enum xcon {
20     value 5;
21     description
22         "Only def-xcon-ccm";
23 }
24 enum no-xcon {
25     value 6;
26     description
27         "No defects def-xcon or lower are to be reported.";
28 }
29 }
30 description
31     "Specifies the lowest priority defect that is allowed to generate
32     a Fault Alarm (20.9.5). The to be reported defects are identified
33     per enum value.";
34 }
35 typedef sender-id-permission-type {
36     type enumeration {
37         enum send-id-none {
38             value 1;
39             description
40                 "The Sender ID TLV is not to be sent.";
41         }
42         enum send-id-chassis {
43             value 2;
44             description
45                 "The Chassis ID Length, Chassis ID Subtype, and Chassis ID
46                 fields of the Sender ID TLV are to be sent.";
47         }
48         enum send-id-manage {
49             value 3;
50             description
51                 "The Management Address Length and Management Address of the
52                 Sender ID TLV are to be sent.";
53         }
54         enum send-id-chassis-manage {
55             value 4;
56             description
57                 "The Chassis ID Length, Chassis ID Subtype, Chassis ID,
58                 Management Address Length and Management Address fields are
59                 all to be sent.";
60         }
61         enum send-id-defer {
62             value 5;
63             description
64                 "The content of the Sender ID TLV are determined by the
65                 corresponding Maintenance Domain variable.";
66         }
67     }
68     description
69         "Indicates what, if anything, is to be included in the Sender ID
70         TLV transmitted in CCMs, LBMs, LTMs, and LTRs.";
71 }
72 typedef ccm-interval-type {

```

```

1  type enumeration {
2      enum 300hz {
3          value 1;
4          description
5              "CCM PDUs are sent every 3 1/3 milliseconds (300Hz).";
6      }
7      enum 10ms {
8          value 2;
9          description
10             "CCM PDUs are sent every 10 milliseconds.";
11     }
12     enum 100ms {
13         value 3;
14         description
15             "CCM PDUs are sent every 100 milliseconds.";
16     }
17     enum 1sec {
18         value 4;
19         description
20             "CCM PDUs are sent every second.";
21     }
22     enum 10sec {
23         value 5;
24         description
25             "CCM PDUs are sent every 10 seconds.";
26     }
27     enum 1min {
28         value 6;
29         description
30             "CCM PDUs are sent every minute.";
31     }
32     enum 10min {
33         value 7;
34         description
35             "CCM PDUs are sent every 10 minutes.";
36     }
37 }
38 description
39     "Indicates the interval at which CCM PDUs are sent by a MEP.";
40 }
41 typedef fng-state-type {
42     type enumeration {
43         enum fng-reset {
44             value 1;
45             description
46                 "No defect has been present since the MEP's fng-reset-time
47                 timer expired, or since the state machine was last reset.";
48         }
49         enum fng-defect {
50             value 2;
51             description
52                 "A defect is present, but not for a long enough time to be
53                 reported.";
54         }
55         enum fng-report-defect {
56             value 3;
57             description
58                 "A momentary state during which the defect is reported by
59                 sending a fault-alarm notification, if that action is enabled.";
60         }
61         enum fng-defect-reported {
62             value 4;
63             description
64                 "A defect is present, and some defect has been reported.";
65         }
66         enum fng-defect-clearing {
67             value 5;
68             description
69                 "No defect is present, but the MEP's fng-reset-time timer has
70                 not yet expired.";
71         }
72     }

```

```

1     description
2     "Indicates the different states of the MEP Fault Notification
3     Generator State Machine.";
4 }
5 typedef relay-action-field-value-type {
6     type enumeration {
7         enum relay-hit {
8             value 1;
9             description
10            "The LTM reached a Maintenance Point whose MAC address matches
11            the target address.";
12        }
13        enum relay-fdb {
14            value 2;
15            description
16            "The Egress Port was determined by consulting the Filtering
17            Database.";
18        }
19        enum relay-mpdb {
20            value 3;
21            description
22            "The Egress Port was determined by consulting the MIP CCM
23            Database.";
24        }
25    }
26    description
27    "Possible values the Relay action field can take.";
28 }
29 typedef ingress-action-field-value-type {
30     type enumeration {
31         enum ingress-ok {
32             value 1;
33             description
34             "The target data frame would be passed through to the MAC
35             Relay Entity.";
36         }
37         enum ingress-down {
38             value 2;
39             description
40             "The Bridge Ports MAC_Operational parameter is false.";
41         }
42         enum ingress-blocked {
43             value 3;
44             description
45             "The target data frame would not be forwarded if received on
46             this Port due to active topology enforcement.";
47         }
48         enum ingress-vid {
49             value 4;
50             description
51             "The ingress port is not in the member set of the LTMs VID,
52             and ingress filtering is enabled, so the target data frame
53             would be filtered by ingress filtering.";
54         }
55     }
56    description
57    "Possible values returned in the ingress action field.";
58 }
59 typedef egress-action-field-value-type {
60     type enumeration {
61         enum egress-okay {
62             value 1;
63             description
64             "The targeted data frame would be forwarded.";
65         }
66         enum egress-down {
67             value 2;
68             description
69             "The Egress Port can be identified, but that Bridge Port
70             MAC_Operational parameter is false.";
71         }
72         enum egress-blocked {

```

```

1      value 3;
2      description
3          "The Egress Port can be identified, but the data frame would
4          not pass through the Egress Port due to active topology
5          management (i.e., the Bridge Port is not in the Forwarding
6          state.";
7      }
8      enum egress-vid {
9          value 4;
10         description
11             "The Egress Port can be identified, but the Bridge Port is not
12             in the LTMs VIDs member set, so would be filtered by egress
13             filtering.";
14     }
15 }
16 description
17     "Possible values returned in the egress action field.";
18 }
19 typedef remote-mep-state-type {
20     type enumeration {
21         enum rmep-idle {
22             value 1;
23             description
24                 "Momentary state during reset.";
25         }
26         enum rmep-start {
27             value 2;
28             description
29                 "The timer has not expired since the state machine was reset,
30                 and no valid CCM has yet been received.";
31         }
32         enum rmep-failed {
33             value 3;
34             description
35                 "The timer has expired, both since the state machine was
36                 reset, and since a valid CCM was received.";
37         }
38         enum rmep-ok {
39             value 4;
40             description
41                 "The timer has not expired since a valid CCM was received.";
42         }
43     }
44     description
45         "Operational state of the remote MEP state machine. This state
46         machine monitors the reception of valid CCMs from a remote MEP
47         with a specific MEPID. It uses a timer that expires in 3.5 times
48         the length of time indicated by the MA's ccm-interval object.";
49 }
50 typedef mep-defects-type {
51     type bits {
52         bit def-rdi-ccm {
53             position 0;
54             description
55                 "A remote MEP reported that RDI bit in its last CCM.";
56         }
57         bit def-mac-status {
58             position 1;
59             description
60                 "Either some remote MEP is reporting its Interface Status TLV
61                 as not isUp, or all remote MEPs are reporting a Port Status
62                 TLV that contains some value other than psUp.";
63         }
64         bit def-remote-ccm {
65             position 2;
66             description
67                 "The MEP is not receiving valid CCMs from at least one of the
68                 remote MEPs.";
69         }
70         bit def-error-ccm {
71             position 3;
72             description

```

```

1      "The MEP has received at least one invalid CCM whose CCM
2      Interval has not yet timed out.";
3  }
4  bit def-xcon-ccm {
5      position 4;
6      description
7          "The MEP has received at last one CCM from either another MAID
8          or a lower MD level whose CCM Interval has not yet timed out.";
9  }
10 }
11 description
12     "A MEP can detect and report a number of defects, and multiple
13     defects can be present at the same time.";
14 }
15 typedef config-error-type {
16     type bits {
17         bit cfm-leak {
18             position 0;
19             description
20                 "MA x is associated with a specific VID list, one or more of
21                 the VID(s) in MA x can pass through the Bridge Port, no Down MEP
22                 is configured on any Bridge Port for MA x, and some other MA
23                 y, at a higher MD Level than MA x, and associated with at
24                 least one of the VID(s) also in MA x, does have a MEP
25                 configured on the Bridge Port.";
26         }
27         bit conflicting-vids {
28             position 1;
29             description
30                 "MA x is associated with a specific VID list, an Up MEP is
31                 configured on MA x on the Bridge Port, and some other MA y,
32                 associated with at least one of the VID(s) also in MA x, also
33                 has an Up MEP configured on some Bridge Port.";
34         }
35         bit excessive-levels {
36             position 2;
37             description
38                 "The number of different MD Levels at which MIPs are to be
39                 created on this port exceeds the Bridge's capabilities.";
40         }
41         bit overlapped-levels {
42             position 3;
43             description
44                 "A MEP is created for one VID at one MD Level, but a MEP is
45                 configured on another VID at that MD Level or higher,
46                 exceeding the Bridge's capabilities.";
47         }
48     }
49     description
50         "While making the MIP creation evaluation described in 22.2.3, the
51         management entity can encounter errors in the configuration.";
52 }
53 typedef mep-tx-ltm-flags-type {
54     type bits {
55         bit use-fdb-only {
56             position 1;
57             description
58                 "Use FDB only";
59         }
60     }
61     description
62         "The flags field for LTMs transmitted by the MEP.";
63 }
64 typedef fault-alarm-type {
65     type enumeration {
66         enum address {
67             value 1;
68             description
69                 "Indicates that a Network address to which Fault Alarms are to
70                 be transmitted should be used.";
71         }
72         enum not-transmitted {

```

```

1      value 2;
2      description
3          "Indicates that Fault alarms are not to be transmitted.";
4      }
5  }
6  description
7      "The Fault Alarm indicators.";
8  }
9  typedef lbm-data-tlv-type {
10     type binary {
11         length "1..1480";
12     }
13     description
14         "The loopback message Data TLV type.";
15 }
16 typedef name-key-type {
17     type string {
18         length "1..255";
19         pattern '[0-9a-zA-Z\-\_]*';
20     }
21     description
22         "String type with at least 1 and up to 255 of the specified
23         characters.";
24 }
25 typedef seq-number-type {
26     type uint32 {
27         range "0..4294967295";
28     }
29     description
30         "The transaction identifier or sequence number of the CFM PDU.";
31 }
32 typedef service-selector-type {
33     type enumeration {
34         enum ieee-reserved-0 {
35             value 0;
36             description
37                 "Reserved for definition by IEEE 802.1.";
38         }
39         enum vlan-id {
40             value 1;
41             description
42                 "12-bit identifier found in a VLAN tag.";
43         }
44         enum isid {
45             value 2;
46             description
47                 "24-bit identifier found in an I-TAG.";
48         }
49         enum tesid {
50             value 3;
51             description
52                 "32-bit identifier";
53         }
54         enum segid {
55             value 4;
56             description
57                 "32-bit identifier";
58         }
59         enum path-tesid {
60             value 5;
61             description
62                 "32-bit identifier";
63         }
64         enum group-isid {
65             value 6;
66             description
67                 "24 bit identifier";
68         }
69         enum ieee-reserved {
70             value 7;
71             description
72                 "Reserved for definition by IEEE 802.1";

```

```
1     }
2   }
3   default "vlan-id";
4   description
5     "A value that represents a type (and thereby the format) of a
6     service-selector-value.";
7 }
8 typedef service-selector-value-type {
9   type uint32 {
10     range "1..4294967295";
11   }
12   description
13     "An integer that uniquely identifies a generic MAC Service, or
14     none. Examples of service selectors are a VLAN-ID and an I-SID. A
15     service-selector-value value is always interpreted within the
16     context of a service-selector-type value. Every usage of the
17     service-selector-value textual convention is required to specify
18     the service-selector-type object that provides the context. The
19     value of a service-selector-value object must always be consistent
20     with the value of the associated service-selector-type object.
21     Attempts to set a service-selector-value object to a value
22     inconsistent with the associated service-selector-type must fail
23     with an inconsistent-value error.";
24 }
25 }
26
27
```


1 48.6.8 The ieee802-dot1q-cfm YANG module

```
2 module ieee802-dot1q-cfm {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-cfm;
5   prefix dot1q-cfm;
6   import ieee802-dot1q-cfm-types {
7     prefix cfm-types;
8   }
9   import ieee802-dot1q-types {
10    prefix dot1q-types;
11  }
12  import ieee802-types {
13    prefix ieee;
14  }
15  import ietf-yang-types {
16    prefix yang;
17  }
18  import ietf-inet-types {
19    prefix inet;
20  }
21  organization
22    "IEEE 802.1 Working Group";
23  contact
24    "WG-URL: http://ieee802.org/1/
25    WG-EMail: stds-802-1-1@ieee.org
26
27    Contact: IEEE 802.1 Working Group Chair
28    Postal: C/O IEEE 802.1 Working Group
29            IEEE Standards Association
30            445 Hoes Lane
31            Piscataway, NJ 08854
32            USA
33
34    E-mail: stds-802-1-chairs@ieee.org";
35  description
36    "Connectivity Fault Management (CFM) comprises capabilities for
37    detecting, verifying, and isolating connectivity failures in Virtual
38    Bridged Local Area Networks. These capabilities can be used in
39    networks operated by multiple independent organizations, each with
40    restricted management access to each other's equipment.
41
42    Copyright (C) IEEE (2022).
43
44    This version of this YANG module is part of IEEE Std 802.1Q; see the
45    standard itself for full legal notices.";
46  revision 2022-01-19 {
47    description
48      "Published as part of IEEE Std 802.1Q-2022.";
49    reference
50      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
51  }
52  revision 2020-06-04 {
53    description
54      "Published as part of IEEE Std 802.1Qcx-2020. Initial version.";
55    reference
56      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
57      Model for Connectivity Fault Management.";
58  }
59
60  /* -----
61   * Grouping definitions used by 802.1Qcx YANG module
62   * -----
63   */
64  grouping mac-address-and-uint-type-grouping {
65    description
66      "The MAC address and uint type grouping.";
67    container mac-address-and-uint-type {
68      description
69        "The MAC address and uint type definition.";
70      leaf address {
71        type ieee:mac-address;
```

```

1      mandatory true;
2      description
3          "The MAC address.";
4      }
5      leaf int {
6          type uint16;
7          mandatory true;
8          description
9              "The additional 2-octet (unsigned) integer.";
10     }
11 }
12 }
13 grouping md-name-choice-grouping {
14     description
15         "The Maintenance Domain name and name format choice.";
16     choice md-name {
17         default "char-string";
18         description
19             "The Maintenance Domain name type.";
20         case none {
21             leaf none {
22                 type empty;
23                 description
24                     "No format specified, usually because there is not a
25                     Maintenance Domain Name. The Maintenance Domain name
26                     inserted in CFM protocol messages will be a zero length
27                     OCTET string.";
28             }
29         }
30         case dns-like-name {
31             leaf dns-like-name {
32                 type string {
33                     length "1..43";
34                 }
35                 description
36                     "Domain name like string, globally unique text string
37                     derived from a DNS name.";
38             }
39         }
40         case mac-address-and-uint {
41             description
42                 "MAC address plus 2-octet (unsigned) integer.";
43             uses mac-address-and-uint-type-grouping;
44         }
45         case char-string {
46             leaf char-string {
47                 type string {
48                     length "1..43";
49                     pattern "[ -~]*";
50                 }
51                 default "DEFAULT";
52                 description
53                     "RFC2579 DisplayString, except that the character codes 0-31
54                     (decimal) are not used.";
55             }
56         }
57     }
58 }
59 grouping ma-name-choice-grouping {
60     description
61         "The Maintenance Association name and name format choice.";
62     choice ma-name {
63         mandatory true;
64         description
65             "The Maintenance Association name type.";
66         case primary-vid {
67             leaf primary-vid {
68                 type dot1q-types:vlanid;
69                 description
70                     "Primary VLAN ID. 12 bits represented in a 2-octet integer.";
71             }
72         }

```

```

1     case char-string {
2         leaf char-string {
3             type string {
4                 length "1..45";
5                 pattern "[ -~]*";
6             }
7             description
8                 "RFC2579 DisplayString, except that the character codes 0-31
9                 (decimal) are not used.";
10        }
11    }
12    case unsigned-int16 {
13        leaf unsigned-int16 {
14            type uint16;
15            description
16                "2-octet integer.";
17        }
18    }
19    case rfc2865-vpn-id {
20        container vpn-id {
21            description
22                "RFC2685 VPN ID. 3 octet VPN authority Organizationally
23                Unique Identifier followed by 4 octet VPN index identifying
24                VPN according to the OUI.";
25            leaf vpn-oui {
26                type uint32 {
27                    range "0..16777215";
28                }
29                mandatory true;
30                description
31                    "3 octet VPN authority Organizationally Unique Identifier.";
32            }
33            leaf vpn-index {
34                type uint32;
35                mandatory true;
36                description
37                    "4 octet VPN index identifying VPN according to OUI.";
38            }
39        }
40    }
41 }
42 }
43 grouping management-address-grouping {
44     description
45         "Defines the Management Address.";
46     reference
47         "21.5.3.5 of IEEE Std 802.1Q-2022";
48     leaf domain {
49         type yang:object-identifier-128;
50         description
51             "The domain type.";
52     }
53     choice management-address {
54         when
55             "./domain";
56         mandatory true;
57         description
58             "Selects the management address";
59         case ip {
60             description
61                 "Represents an IP TCP, UDP, or SCTP transport address
62                 consisting of an IPv4/v6 address, and a port number,
63                 associated with the domain type defined by the domain leaf
64                 node.";
65             leaf ip-address {
66                 type inet:ip-address;
67                 mandatory true;
68                 description
69                     "IPv4 or IPv6 address.";
70             }
71             leaf ip-port {
72                 type inet:port-number;

```

```

1      mandatory true;
2      description
3          "IP port.";
4  }
5  }
6  case local {
7      leaf local-address {
8          type string {
9              length "1..255";
10         }
11         mandatory true;
12         description
13             "Represents a POSIX Local IPC transport address, associated
14             with the domain type defined by the domain leaf node.";
15     }
16 }
17 case dns {
18     leaf dns-address {
19         type string {
20             length "1..255";
21         }
22         mandatory true;
23         description
24             "The transport domain using fully qualified domain names,
25             associated with the domain type defined by the domain leaf
26             node. Represents a DNS domain name followed by a colon ':'
27             (ASCII character 0x3A) and a port number in ASCII. The name
28             SHOULD be fully qualified whenever possible.";
29     }
30 }
31 case other {
32     leaf unknown-address {
33         type binary {
34             length "1..255";
35         }
36         description
37             "This represents an undefined address, for the case when the
38             domain type provided is an unrecognizable value.";
39     }
40 }
41 }
42 }
43 grouping loopback-input-grouping {
44     description
45         "Defines the group of loopback input parameters.";
46     choice lbm-destination {
47         mandatory true;
48         description
49             "Selects the destination type (which is either MEP identifier or
50             MAC address) used for the Loopback transmissions.";
51         case dest-ucast-mac-address {
52             description
53                 "The target unicast MAC Address field to be transmitted. A
54                 unicast destination MAC address.";
55             leaf lbm-dest-ucast-mac-address {
56                 type cfm-types:unicast-mac-address-type;
57                 description
58                     "The target MAC Address field to be transmitted. A unicast
59                     destination MAC address.";
60                 reference
61                     "Item b) in 12.14.7.3.2 of IEEE Std 802.1Q-2022";
62             }
63         }
64         case dest-mcast-class1-mac-address {
65             description
66                 "The target multicast Class 1 MAC address field to be
67                 transmitted.";
68             leaf lbm-dest-mcast-class1-mac-address {
69                 type cfm-types:multicast-class1-mac-address-type;
70                 description
71                     "The target multicast Class 1 MAC address field to be
72                     transmitted";

```

```

1      }
2    }
3    case dest-mep-id {
4      description
5        "The identifier of a remote MEP in the same MA to which the
6        LBM is to be sent.";
7      leaf lbm-dest-mep-id {
8        type cfm-types:mep-id-type;
9        description
10         "The identifier of a remote MEP in the same MA to which the
11         LBM is to be sent.";
12        reference
13         "Item b) in 12.14.7.3.2 of IEEE Std 802.1Q-2022";
14      }
15    }
16  }
17  leaf lbm-messages {
18    type uint16 {
19      range "1..1024";
20    }
21    default "1";
22    description
23      "The number of Loopback messages to be transmitted.";
24    reference
25      "Item c) in 12.14.7.3.2 of IEEE Std 802.1Q-2022";
26  }
27  leaf lbm-priority {
28    type dot1q-types:priority-type;
29    default "7";
30    description
31      "Priority. 3 bit value to be used in the VLAN tag, if present in
32      the transmitted frame. The default value should be priority 7,
33      which is the default CCM priority (ccm-ltm-priority).";
34    reference
35      "Item e) in 12.14.7.3.2 of IEEE Std 802.1Q-2022";
36  }
37  leaf lbm-drop-eligible {
38    type boolean;
39    default "false";
40    description
41      "Drop eligible bit value to be used in the VLAN tag, if present
42      in the transmitted frame. A value 'true' means inserting value
43      '1' in the DEI field, while a value 'false' means inserting
44      value '0' in the DEI field.";
45    reference
46      "Item e) in 12.14.7.3.2 of IEEE Std 802.1Q-2022";
47  }
48  leaf lbm-data-tlv {
49    type cfm-types:lbm-data-tlv-type;
50    description
51      "An arbitrary amount of data to be included in the Data TLV, if
52      the Data TLV is selected to be sent.";
53    reference
54      "Item d) in 12.14.7.3.2 of IEEE Std 802.1Q-2022";
55  }
56 }
57 grouping linktrace-input-grouping {
58   description
59     "Defines the group of linktrace input parameters. By default, the
60     priority used is that of the CCMs (ccm-ltm-priority) and the drop
61     eligibility is false.";
62   choice ltr-target {
63     mandatory true;
64     description
65       "Selects the target address type (which is either MEP identifier
66       or MAC address) used for the Linktrace transmissions.";
67     case target-ucast-mac-address {
68       description
69         "The target MAC address field to be transmitted. A unicast MAC
70         address.";
71       leaf ltm-target-mac-address {
72         type cfm-types:unicast-mac-address-type;

```

```

1      description
2      "The target MAC address field to be transmitted. A unicast
3      MAC address.";
4      reference
5      "Item c) in 12.14.7.4.2 of IEEE Std 802.1Q-2022";
6  }
7  }
8  case target-mep-id {
9      description
10     "The MEP identifier of another MEP in the same MA.";
11     leaf ltm-target-mep-id {
12         type cfm-types:mep-id-type;
13         description
14         "The MEP identifier of another MEP in the same MA.";
15         reference
16         "Item c) in 12.14.7.4.2 of IEEE Std 802.1Q-2022";
17     }
18 }
19 }
20 leaf ltm-ttl {
21     type uint8 {
22         range "0..255";
23     }
24     default "64";
25     description
26     "The LTM TTL field. Indicates the number of hops remaining to
27     the LTM. Decrement by 1 by each Linktrace Responder that
28     handles the LTM. The value returned in the LTR is one less than
29     that received in the LTM. If the LTM TTL is 0 or 1, the LTM is
30     not forwarded to the next hop, and if 0, no LTR is generated.";
31     reference
32     "Item d) in 12.14.7.4.2, 21.8.4 of IEEE Std 802.1Q-2022";
33 }
34 leaf ltm-flags {
35     type cfm-types:mep-tx-ltm-flags-type;
36     default "";
37     description
38     "The flags field for the LTMs transmitted by the MEP.";
39     reference
40     "Item b) in 12.14.7.4.2, 20.42.1 of IEEE Std 802.1Q-2022";
41 }
42 }
43 grouping ltm-egress-identifier-grouping {
44     description
45     "The grouping used to identify the MEP Linktrace initiator that is
46     originating the LTM. It defines the MAC address and uint type
47     definition.";
48     leaf int {
49         type uint16;
50         mandatory true;
51         description
52         "A value used to uniquely identify the MEP Linktrace Initiator
53         or Linktrace Responder within that system.";
54     }
55     leaf address {
56         type ieee:mac-address;
57         mandatory true;
58         description
59         "A 48-bit IEEE MAC address unique to the system in which the MEP
60         Linktrace Initiator or Linktrace Responder resides.";
61     }
62 }
63
64 /* -----
65  * Configuration objects used by 802.1Qcx YANG module
66  * -----
67  */
68 container cfm {
69     description
70     "Connectivity Fault Management configuration and operational
71     information.";
72     list maintenance-domain {

```

```

1  key "md-id";
2  description
3    "Contains the Maintenance Domain configuration and operational
4    data. A Maintenance Domain is the network or the part of the
5    network for which faults in connectivity can be managed. The
6    boundary of a Maintenance Domain is defined by a set of Domain
7    Service Access Points (DoSAPs), each of which can become a point
8    of connectivity to a service instance.";
9  leaf md-id {
10     type cfm-types:name-key-type;
11     description
12       "The index to the Maintenance Domain list.";
13   }
14   uses md-name-choice-grouping;
15   leaf md-level {
16     type cfm-types:md-level-type;
17     default "0";
18     description
19       "The Maintenance Domain level.";
20     reference
21       "3.123, Item b) in 12.14.5.1.3 of IEEE Std 802.1Q-2022";
22   }
23   leaf mhf-creation {
24     type cfm-types:mhf-creation-type;
25     must
26       ". != 'mhf-defer'" {
27         description
28           "The value mhf-defer is not allowed";
29       }
30     default "mhf-none";
31     description
32       "Value indicating whether the management entity can create
33       MHFs (MIP Half Function) for this Maintenance Domain. Since
34       there is no encompassing Maintenance Domain, the value
35       mhf-defer is not allowed.";
36     reference
37       "3.123, Item c) in 12.14.5.1.3 of IEEE Std 802.1Q-2022";
38   }
39   leaf id-permission {
40     type cfm-types:sender-id-permission-type;
41     must
42       ". != 'send-id-defer'" {
43         description
44           "The value send-id-defer is not allowed";
45       }
46     default "send-id-none";
47     description
48       "Value indicating what, if anything, is to be included in the
49       Sender ID TLV transmitted by Maintenance Points configured in
50       this Maintenance Domain. Since there is no encompassing
51       Maintenance Domain, the value send-id-defer is not allowed.";
52     reference
53       "3.123, item d) in 12.14.5.1.3 of IEEE Std 802.1Q-2022";
54   }
55   leaf fault-alarm-transmission {
56     type cfm-types:fault-alarm-type;
57     default "not-transmitted";
58     description
59       "A value indicating whether Fault Alarms are to be transmitted
60       or not. The default is not transmit.";
61     reference
62       "3.123, item e) in 12.14.5.1.3 of IEEE Std 802.1Q-2022";
63   }
64   list maintenance-association {
65     key "ma-id";
66     description
67       "Provides configuration and operational data for the
68       Maintenance Associations. A Maintenance Association is a set
69       of MEPs, each configured with the same MAID and MD level,
70       established to verify the integrity of a single service
71       instance. A Maintenance Association can be thought of as a
72       full mesh of Maintenance Entities among a set of MEPs so

```

```

1      configured.";
2  leaf ma-id {
3      type cfm-types:name-key-type;
4      description
5          "Key of the Maintenance Association list of entries.";
6  }
7  uses ma-name-choice-grouping;
8  leaf ccm-interval {
9      type cfm-types:ccm-interval-type;
10     default "1sec";
11     description
12         "The interval between CCM transmissions to be used by all
13         MEPs in the Maintenance Association.";
14     reference
15         "Item e) in 12.14.6.1.3 of IEEE Std 802.1Q-2022";
16 }
17 leaf fault-alarm-transmission {
18     type cfm-types:fault-alarm-type;
19     description
20         "A value indicating whether Fault Alarms are to be
21         transmitted or not. If this leaf node is not present then
22         the disposition of the fault-alarm used by the MD should be
23         used.";
24     reference
25         "3.123, Item e) in 12.14.5.1.3 of IEEE Std 802.1Q-2022";
26 }
27 leaf mhf-creation {
28     type cfm-types:mhf-creation-type;
29     default "mhf-defer";
30     description
31         "Value indicating whether the management entity can create
32         MHFs (MIP Half Function) for this Maintenance Association.";
33     reference
34         "3.123, Item c) in 12.14.5.1.3 of IEEE Std 802.1Q-2022";
35 }
36 leaf id-permission {
37     type cfm-types:sender-id-permission-type;
38     default "send-id-defer";
39     description
40         "Enumerated value indicating what, if anything, is to be
41         included in the Sender ID TLV (21.5.3) transmitted by MPs
42         configured in this MA.";
43     reference
44         "Item d) in 12.14.3.1.3 of IEEE Std 802.1Q-2022";
45 }
46 list maintenance-association-mep {
47     key "mep-id";
48     description
49         "The list of all MEPs that belong to this Maintenance
50         Association.";
51     leaf mep-id {
52         type cfm-types:mep-id-type;
53         description
54             "Integer that is unique among all the MEPs in the same
55             Maintenance Association.";
56         reference
57             "Item g) in 12.14.6.1.3 of IEEE Std 802.1Q-2022";
58     }
59 }
60 }
61 }
62
63 // maintenance-domain
64 list maintenance-group {
65     key "maintenance-group-id";
66     description
67         "The list of maintenance association groups, which are uniquely
68         associated with a maintenance domain, maintenance association,
69         for which the MEPs belong.";
70     leaf maintenance-group-id {
71         type cfm-types:name-key-type;
72         description

```



```

1      "The maintenance group provides a handle for the MD and MA
2      combination.";
3  }
4  leaf md-id {
5      type leafref {
6          path '/cfm/maintenance-domain/md-id';
7      }
8      mandatory true;
9      description
10         "A reference to the maintenance domain that this maintenance
11         group is associated with.";
12 }
13 leaf ma-id {
14     type leafref {
15         path
16             '/cfm'+
17             '/maintenance-domain[md-id = current()'+
18             '/..'+'+
19             '/md-id]'+
20             '/maintenance-association'+
21             '/ma-id';
22     }
23     mandatory true;
24     description
25         "A reference to the maintenance association in the specified
26         maintenance domain, that this maintenance group is associated
27         with.";
28 }
29 list mep {
30     key "mep-id";
31     description
32         "A list of local Maintenance association End Points (MEPs). A
33         MEP is an actively managed CFM entity, associated with a
34         specific DoSAP of a service instance, which can generate and
35         receive CFM PDUs and track any responses. It is an end point
36         of a single Maintenance Association (MA) and is an end point
37         of a separate Maintenance Entity for each of the other MEPs in
38         the same MA.";
39     leaf mep-id {
40         type leafref {
41             path
42                 '/cfm'+
43                 '/maintenance-domain[md-id = current()'+
44                 '/..'+'+
45                 '/..'+'+
46                 '/md-id]'+
47                 '/maintenance-association[ma-id = current()'+
48                 '/..'+'+
49                 '/..'+'+
50                 '/ma-id]'+
51                 '/maintenance-association-mep'+
52                 '/mep-id';
53         }
54         description
55             "Integer that is unique among all the MEPs in the same
56             Maintenance Association.";
57         reference
58             "12.14.7, 19.2 of IEEE Std 802.1Q-2022";
59     }
60     leaf direction {
61         type cfm-types:mp-direction-type;
62         mandatory true;
63         description
64             "The direction in which the MEP faces on the Bridge Port.
65             Example, up or down.";
66         reference
67             "Item c) in 12.14.7.1.3, 19.2 of IEEE Std 802.1Q-2022";
68     }
69     leaf enabled {
70         type boolean;
71         default "false";
72         description

```

```

1      "The administrative state of the MEP. TRUE indicates that
2      the MEP is to functional normally, and FALSE indicates that
3      it is to cease functioning.";
4      reference
5      "Item e) in 12.14.7.1.3, 20.9.1 of IEEE Std 802.1Q-2022";
6  }
7  leaf ccm-ltm-priority {
8      type dot1q-types:priority-type;
9      default "7";
10     description
11         "The priority value for CCMs and LTMs transmitted by the
12         MEP. The default value is the highest priority allowed to
13         pass through the Bridge Port for any of the MEPs VID(s).";
14     reference
15         "Item h) in 12.14.7.1.3 of IEEE Std 802.1Q-2022";
16 }
17 leaf mac-address {
18     type ieee:mac-address;
19     config false;
20     mandatory true;
21     description
22         "The MAC address of the MEP.";
23     reference
24         "Item i) in 12.14.7.1.3, 19.4 of IEEE Std 802.1Q-2022";
25 }
26 list inactive-remote-mep {
27     key "inactive-rmep-id";
28     description
29         "A list indicating which of the remote MEPs in the same MA
30         are inactive. The Remote MEP state machines (20.20) are
31         instantiated only for the remote MEPs which are present in
32         the maintenance-association-mep-list, but not in this list.
33         By default, all configured remote MEPs in the same MA are
34         active";
35     leaf inactive-rmep-id {
36         type leafref {
37             path
38                 '/cfm'+
39                 '/maintenance-domain[md-id = current()'+
40                 '/..'+'+
41                 '/..'+'+
42                 '/..'+'+
43                 '/md-id]'+
44                 '/maintenance-association[ma-id = current()'+
45                 '/..'+'+
46                 '/..'+'+
47                 '/..'+'+
48                 '/ma-id]'+
49                 '/maintenance-association-mep'+
50                 '/mep-id';
51         }
52         description
53             "Maintenance association Endpoint Identifier of a remote
54             MEP for which the Remote MEP state machine should not be
55             instantiated.";
56         reference
57             "Item ae) in 12.14.7.1.3 of IEEE Std 802.1Q-2022";
58     }
59 }
60
61 //inactive-remote-mep
62 list mep-db {
63     key "rmep-id";
64     config false;
65     description
66         "The MEP CCM Database. A database, maintained by every MEP,
67         that maintains received information about other MEPs in the
68         Maintenance Association.";
69     leaf rmep-id {
70         type cfm-types:mep-id-type;
71         description
72             "Maintenance association Endpoint Identifier of a remote

```

```

1         MEP whose information from the MEP Database is to be
2         returned.";
3     reference
4         "Item b) in 12.14.7.6.2 of IEEE Std 802.1Q-2022";
5 }
6 leaf rmep-state {
7     type cfm-types:remote-mep-state-type;
8     mandatory true;
9     description
10        "The operational state of the remote MEP state machine";
11    reference
12        "Item b) in 12.14.7.6.3, 20.20 of IEEE Std 802.1Q-2022";
13 }
14 leaf rmep-failed-ok-time {
15     type yang:timeticks;
16     mandatory true;
17     description
18        "The time (SysUpTime) at which the Remote MEP state
19        machine last entered either the RMEP_FAILED or RMEP_OK
20        state";
21    reference
22        "Item c) in 12.14.7.6.3 of IEEE Std 802.1Q-2022";
23 }
24 leaf mac-address {
25     type ieee:mac-address;
26     mandatory true;
27     description
28        "The MAC address of the remote MEP.";
29    reference
30        "Item d) in 12.14.7.6.3, 20.19.7 of IEEE Std 802.1Q-2022";
31 }
32 leaf rdi {
33     type boolean;
34     mandatory true;
35     description
36        "State of the RDI bit in the last received CCM (true for
37        RDI=1), or false if none has been received.";
38    reference
39        "Item e) in 12.14.7.6.3, 20.19.2 of IEEE Std 802.1Q-2022";
40 }
41 leaf port-status-tlv {
42     type cfm-types:port-status-tlv-value-type;
43     description
44        "An enumerated value of the Port status TLV received in
45        the last CCM from the remote MEP or the default value
46        no-port-state-tlv indicating either no CCM has been
47        received, or that no port status TLV was received in the
48        last CCM.";
49    reference
50        "Item f) in 12.14.7.6.3, 20.19.3 of IEEE Std 802.1Q-2022";
51 }
52 leaf interface-status-tlv {
53     type cfm-types:interface-status-tlv-value-type;
54     description
55        "An enumerated value of the Interface status TLV received
56        in the last CCM from the remote MEP or the default value
57        is-no-interface-status-tlv indicating either no CCM has
58        been received, or that no interface status TLV was
59        received in the last CCM.";
60    reference
61        "Item g) in 12.14.7.6.3, 20.19.4 of IEEE Std 802.1Q-2022";
62 }
63 leaf chassis-id-subtype {
64     type ieee:chassis-id-subtype-type;
65     description
66        "This object specifies the format of the Chassis ID
67        received in the last CCM.";
68    reference
69        "Item h) in 12.14.7.6.3, 21.5.3.2 of IEEE Std 802.1Q-2022";
70 }
71 leaf chassis-id {
72     type ieee:chassis-id-type;

```

```

1      description
2          "The Chassis ID. The format of this object is determined
3          by the value of the ltr-chassis-id-subtype object.";
4      reference
5          "Item h) in 12.14.7.6.3, 21.5.3.3 of IEEE Std 802.1Q-2022";
6  }
7  container transport-service-domain {
8      description
9          "The transport management domain and address.";
10     reference
11         "Item h) in 12.14.7.6.3 of IEEE Std 802.1Q-2022";
12     uses management-address-grouping;
13 }
14 leaf rmep-is-active {
15     type boolean;
16     default "true";
17     description
18         "A Boolean value stating if the remote MEP is active.";
19     reference
20         "Item ae) in 12.14.7.1.3 of IEEE Std 802.1Q-2022";
21 }
22 }
23
24 // mep-db
25 container continuity-check {
26     description
27         "Continuity check protocol";
28     leaf ccm-enabled {
29         type boolean;
30         default "false";
31         description
32             "Indicates whether the MEP can generate CCMs. If TRUE, the
33             MEP will generate CCM PDUs.";
34         reference
35             "Item g) in 12.14.7.1.3, 20.10.1 of IEEE Std 802.1Q-2022";
36     }
37     leaf fng-state {
38         type cfm-types:fng-state-type;
39         default "fng-reset";
40         config false;
41         description
42             "The current state of the MEP Fault Notification Generator
43             state machine.";
44         reference
45             "Item f) in 12.14.7.1.3, 20.35 of IEEE Std 802.1Q-2022";
46     }
47     leaf fault-alarm-transmission {
48         type cfm-types:fault-alarm-type;
49         description
50             "A value indicating whether Fault Alarms are to be
51             transmitted or not. If this leaf is not specified, the
52             disposition of the fault-alarm used by the MD should be
53             used.";
54         reference
55             "3.124, Item j) in 12.14.7.1.3 of IEEE Std 802.1Q-2022";
56     }
57     leaf lowest-priority-defect {
58         type cfm-types:lowest-alarm-priority-type;
59         default "mac-remote-error-xcon";
60         description
61             "The lowest priority defect that is allowed to generate
62             fault alarms.";
63         reference
64             "Item k) in 12.14.7.1.3, 20.9.5 of IEEE Std 802.1Q-2022";
65     }
66     leaf fng-alarm-time {
67         type uint16 {
68             range "2500..10000";
69         }
70         units "milliseconds";
71         default "2500";
72         description

```

```

1         "The time that defect must be present before a Fault Alarm
2         is issued.";
3     reference
4         "Item l) in 12.14.7.1.3, 20.35.3 of IEEE Std 802.1Q-2022";
5 }
6 leaf fng-reset-time {
7     type uint16 {
8         range "2500..10000";
9     }
10    units "milliseconds";
11    default "10000";
12    description
13        "The time that defects must be absent before resetting a
14        Fault Alarm.";
15    reference
16        "Item m) in 12.14.7.1.3, 20.35.4 of IEEE Std 802.1Q-2022";
17 }
18 leaf highest-priority-defect {
19     type cfm-types:highest-defect-priority-type;
20     config false;
21     mandatory true;
22     description
23         "The highest priority defect that has been present since
24         the MEPs Fault Notification Generator state machine was
25         last in the FNG_RESET state.";
26     reference
27         "Item n) in 12.14.7.1.3, 20.35.9 of IEEE Std 802.1Q-2022";
28 }
29 leaf defects {
30     type cfm-types:mep-defects-type;
31     config false;
32     mandatory true;
33     description
34         "Vector of boolean error conditions";
35     reference
36         "12.14.7.1.3 of IEEE Std 802.1Q-2022";
37 }
38 leaf error-ccm-last-failure {
39     type binary {
40         length "1..128";
41     }
42     config false;
43     description
44         "The last received CCM that triggered a def-error-ccm
45         fault.";
46     reference
47         "Item t) in 12.14.7.1.3, 20.21.2 of IEEE Std 802.1Q-2022";
48 }
49 leaf xcon-ccm-last-failure {
50     type binary {
51         length "1..128";
52     }
53     config false;
54     description
55         "The last received CCM that triggered a def-xcon-ccm
56         fault.";
57     reference
58         "Item u) in 12.14.7.1.3, 20.23.2 of IEEE Std 802.1Q-2022";
59 }
60 }
61
62 // continuity-check
63 container stats {
64     config false;
65     description
66         "Contains the counters associated with the MEP.";
67     leaf mep-ccm-sequence-errors {
68         type yang:counter64;
69         mandatory true;
70         description
71             "The total number of out-of-sequence CCMs received from
72             all remote MEPs.";

```

```

1         reference
2         "Item v) in 12.14.7.1.3, 20.16.12 of IEEE Std 802.1Q-2022";
3     }
4     leaf mep-ccms-sent {
5         type yang:counter64;
6         mandatory true;
7         description
8             "Total number of CCMS transmitted";
9         reference
10            "Item w) in 12.14.7.1.3, 20.10.2 of IEEE Std 802.1Q-2022";
11    }
12    leaf mep-lbr-in {
13        type yang:counter64;
14        mandatory true;
15        description
16            "Total number of valid, in-order Loopback Replies
17            received.";
18        reference
19            "Item y) in 12.14.7.1.3, 20.31.1 of IEEE Std 802.1Q-2022";
20    }
21    leaf mep-lbr-in-out-of-order {
22        type yang:counter64;
23        mandatory true;
24        description
25            "The total number of valid, out-of-order Loopback Replies
26            received";
27        reference
28            "Item z) in 12.14.7.1.3, 20.31.1 of IEEE Std 802.1Q-2022";
29    }
30    leaf mep-lbr-bad-msdu {
31        type yang:counter64;
32        mandatory true;
33        description
34            "The total number of LBRs received whose
35            mac_service_data_unit did not match (except for the
36            OpCode) that of the corresponding LBM.";
37        reference
38            "Item aa) in 12.14.7.1.3, 20.2.3 of IEEE Std 802.1Q-2022";
39    }
40    leaf mep-unexpected-ltr-in {
41        type yang:counter64;
42        mandatory true;
43        description
44            "The total number of unexpected LTRs received.";
45        reference
46            "Item ac) in 12.14.7.1.3, 20.44.1 of IEEE Std 802.1Q-2022";
47    }
48    leaf mep-lbr-out {
49        type yang:counter64;
50        mandatory true;
51        description
52            "Total number of Loopback Replies transmitted.";
53        reference
54            "Item ad) in 12.14.7.1.3, 20.28.2 of IEEE Std 802.1Q-2022";
55    }
56 }
57
58 // transmit-linktrace
59 list linktrace-reply {
60     key "ltr-transaction-id";
61     config false;
62     description
63         "This list extends the MEP table and reports on accepted
64         transmit-linktrace actions. In case linktrace replies are
65         received it also reports with data from the received
66         Linktrace reply messages.";
67     leaf ltr-transaction-id {
68         type cfm-types:seq-number-type;
69         description
70             "Transaction identifier returned by a previous transmit
71             linktrace message command, indicating which LTMs response
72             is going to be returned.";

```

```

1      reference
2      "Item b) in 12.14.7.5.2 of IEEE Std 802.1Q-2022";
3  }
4  container linktrace-input {
5      description
6          "The linktrace parameter input. By default, the priority
7          used is that of the CCMs (ccm-ltm-priority) and the drop
8          eligibility is false.";
9      uses linktrace-input-grouping;
10 }
11 list responses {
12     key "ltr-receive-order";
13     description
14         "The responses associated with the request.";
15     leaf ltr-receive-order {
16         type uint32 {
17             range "1..4294967295";
18         }
19         description
20             "An index to distinguish among multiple LTRs with the
21             same LTR Transaction Identifier field value. Assigned
22             sequentially from 1, in the order that the Linktrace
23             Initiator received the LTRs.";
24     }
25     leaf ltr-ttl {
26         type uint8 {
27             range "0..255";
28         }
29         mandatory true;
30         description
31             "TTL field value for a returned LTR.";
32         reference
33             "12.14.7.5, 20.41.2.2 of IEEE Std 802.1Q-2022";
34     }
35     leaf ltr-forwarded {
36         type boolean;
37         mandatory true;
38         description
39             "Indicates if an LTM was forwarded by the responding MP,
40             as returned in the FwdYes flag of the flags field.";
41         reference
42             "Item c) in 12.14.7.5.3, 20.41.2.1 of IEEE Std 802.1Q-2022";
43     }
44     leaf ltr-terminal-mep {
45         type boolean;
46         mandatory true;
47         description
48             "A Boolean value stating whether the forwarded LTM
49             reached a MEP enclosing its MA, as returned in the
50             Terminal MEP flag of the Flags field";
51         reference
52             "Item d) in 12.14.7.5.3, 20.41.2.1 of IEEE Std 802.1Q-2022";
53     }
54     container ltr-last-egress-identifier {
55         description
56             "An octet field holding the Last Egress Identifier
57             returned in the LTR Egress Identifier TLV of the LTR.
58             The Last Egress Identifier identifies the MEP Linktrace
59             Initiator that originated, or the Linktrace Responder
60             that forwarded, the LTM to which this LTR is the
61             response. This is the same value as the Egress
62             Identifier TLV of that LTM.";
63         reference
64             "Item e) in 12.14.7.5.3, 20.41.2.3 of IEEE Std 802.1Q-2022";
65         uses ltm-egress-identifier-grouping;
66     }
67     container ltr-next-egress-identifier {
68         description
69             "An octet field holding the Next Egress Identifier
70             returned in the LTR Egress Identifier TLV of the LTR.
71             The Next Egress Identifier Identifies the Linktrace
72             Responder that transmitted this LTR, and can forward the

```

```

1         LTM to the next hop. This is the same value as the
2         Egress Identifier TLV of the forwarded LTM, if any. If
3         the FwdYes bit of the Flags field is false, the contents
4         of this field are undefined, i.e., any value can be
5         transmitted, and the field is ignored by the receiver.";
6     reference
7         "Item f) in 12.14.7.5.3, 20.41.2.4 of IEEE Std 802.1Q-2022";
8     uses ltm-egress-identifier-grouping;
9 }
10 leaf ltr-relay {
11     type cfm-types:relay-action-field-value-type;
12     mandatory true;
13     description
14         "Value returned in the Relay Action field.";
15     reference
16         "Item g) in 12.14.7.5.3, 20.41.2.5 of IEEE Std 802.1Q-2022";
17 }
18 leaf ltr-chassis-id-subtype {
19     type ieee:chassis-id-subtype-type;
20     description
21         "Specifies the format of the Chassis ID returned in the
22         Sender ID TLV of the LTR, if any. This leaf is not
23         present if the LTR did not contain a Sender ID TLV or if
24         the Sender ID TLV did not contain a Chassis ID.";
25     reference
26         "Item h) in 12.14.7.5.3, 21.5.3.2 of IEEE Std 802.1Q-2022";
27 }
28 leaf ltr-chassis-id {
29     when
30         "../ltr-chassis-id-subtype";
31     type ieee:chassis-id-type;
32     mandatory true;
33     description
34         "The Chassis ID returned in the Sender ID TLV of the
35         LTR, if any. The format of this object is determined by
36         the value of the ltr-chassis-id-subtype object. This
37         leaf is not present if the LTR did not contain a Sender
38         ID TLV or if the Sender ID TLV did not contain a Chassis
39         ID.";
40     reference
41         "Item i) in 12.14.7.5.3, 21.5.3.2 of IEEE Std 802.1Q-2022";
42 }
43 container ltr-transport-service-domain {
44     description
45         "The transport management domain and address. This
46         container is empty if the LTR did not contain a Sender
47         ID TLV or if the Sender ID TLV did not contain a
48         Management Address.";
49     reference
50         "Item j) in 12.14.7.5.3 of IEEE Std 802.1Q-2022";
51     uses management-address-grouping;
52 }
53 leaf ltr-ingress {
54     type cfm-types:ingress-action-field-value-type;
55     description
56         "The value returned in the Ingress Action Field of the
57         LTM. This leaf is not present if no Reply Ingress TLV
58         was returned in the LTM.";
59     reference
60         "Item k) in 12.14.7.5.3, 20.41.2.6 of IEEE Std 802.1Q-2022";
61 }
62 leaf ltr-ingress-mac {
63     when
64         "../ltr-ingress";
65     type ieee:mac-address;
66     mandatory true;
67     description
68         "MAC address returned in the ingress MAC address field.
69         This leaf is not present if the ltr-ingress leaf is not
70         present.";
71     reference
72         "Item l) in 12.14.7.5.3, 20.41.2.7 of IEEE Std 802.1Q-2022";

```



```

1      }
2      leaf ltr-ingress-port-id-subtype {
3          when
4              "../ltr-ingress";
5          type ieee:port-id-subtype-type;
6          description
7              "Format of the ingress Port ID. This leaf is not present
8              if the ltr-ingress leaf is not present, or if the Reply
9              Ingress TLV did not contain a Port ID.";
10         reference
11             "Item m) in 12.14.7.5.3, 20.41.2.8 of IEEE Std 802.1Q-2022";
12     }
13     leaf ltr-ingress-port-id {
14         when
15             "../ltr-ingress and ../ltr-ingress-port-id-subtype";
16         type ieee:port-id-type;
17         mandatory true;
18         description
19             "Ingress Port ID. The format of this object is
20             determined by the value of the
21             ltr-ingress-port-id-subtype object. This leaf is not
22             present if the ltr-ingress leaf or the
23             ltr-ingress-port-id-subtype leaf are not present.";
24         reference
25             "Item n) in 12.14.7.5.3, 20.41.2.9 of IEEE Std 802.1Q-2022";
26     }
27     leaf ltr-egress {
28         type cfm-types:egress-action-field-value-type;
29         description
30             "The value returned in the Egress Action Field of the
31             LTM. The node is not present if no Reply Egress TLV was
32             returned in the LTM.";
33         reference
34             "Item o) in 12.14.7.5.3, 20.41.2.10 of IEEE Std 802.1Q-2022";
35     }
36     leaf ltr-egress-mac {
37         when
38             "../ltr-egress";
39         type ieee:mac-address;
40         mandatory true;
41         description
42             "MAC address returned in the egress MAC address field.
43             This leaf is not present if the ltr-egress leaf is not
44             present.";
45         reference
46             "Item p) in 12.14.7.5.3, 20.41.2.11 of IEEE Std 802.1Q-2022";
47     }
48     leaf ltr-egress-port-id-subtype {
49         when
50             "../ltr-egress";
51         type ieee:port-id-subtype-type;
52         description
53             "Format of the egress Port ID. This leaf is not present
54             if the ltr-egress leaf is not present, or if the Reply
55             Egress TLV did not contain a Port ID.";
56         reference
57             "Item q) in 12.14.7.5.3, 20.41.2.12 of IEEE Std 802.1Q-2022";
58     }
59     leaf ltr-egress-port-id {
60         when
61             "../ltr-egress and ../ltr-egress-port-id-subtype";
62         type ieee:port-id-type;
63         mandatory true;
64         description
65             "Egress Port ID. The format of this object is determined
66             by the value of the ltr-egress-port-id-subtype object.
67             This leaf is not present if the ltr-egress leaf or the
68             ltr-egress-port-id-subtype leaf are not present.";
69         reference
70             "Item r) in 12.14.7.5.3, 20.41.2.13 of IEEE Std 802.1Q-2022";
71     }
72     leaf ltr-organization-specific-tlv {

```

```

1         type binary {
2             length "0 | 4..1500";
3         }
4         description
5             "All Organization specific TLVs returned in the LTR, if
6             any. Includes all octets including and following the TLV
7             length field of each TLV, concatenated together.";
8         reference
9             "Item s) in 12.14.7.5.3, 21.5.2 of IEEE Std 802.1Q-2022";
10    }
11 }
12 }
13
14 // stats
15 action transmit-loopback {
16     description
17         "To signal to the MEP to transmit some number of LBMs.
18         Accepting the action means the device will transmit LBM
19         messages according the input leaf nodes.";
20     input {
21         uses loopback-input-grouping;
22     }
23
24     // input
25     output {
26         leaf lbm-request-id {
27             type cfm-types:seq-number-type;
28             mandatory true;
29             description
30                 "The Loopback transaction identifier of the first LBM
31                 (to
32                 be) sent.";
33             reference
34                 "Item b) in 12.14.7.3.3 of IEEE Std 802.1Q-2022";
35         }
36     }
37 }
38
39 // transmit-loopback
40 action transmit-linktrace {
41     description
42         "To signal to the MEP to transmit an LTM and to create an
43         LTM entry in the MEPs Linktrace Database. Accepting the
44         action means the device will transmit LTM messages according
45         to the input leafs. The outcome of sending these LTM
46         messages will be reported through the list
47         'linktrace-reply'.";
48     input {
49         uses linktrace-input-grouping;
50     }
51
52     // input
53     output {
54         leaf ltm-transaction-id {
55             type cfm-types:seq-number-type;
56             mandatory true;
57             description
58                 "The LTM transaction identifier of the LTM sent. The
59                 value returned is undefined if and RPC error is
60                 returned.";
61             reference
62                 "Item b) in 12.14.7.4.3 of IEEE Std 802.1Q-2022";
63         }
64         container ltm-egress-identifier {
65             description
66                 "Identifies the MEP Linktrace Initiator that is
67                 originating this LTM.
68
69                 The low-order six octets contain a 48-bit IEEE MAC
70                 address unique to the system in which the MEP Linktrace
71                 Initiator or Linktrace Responder resides. The high-order
72                 two octets contain a value sufficient to uniquely

```

```
1         identify the MEP Linktrace Initiator or Linktrace
2         Responder within that system.
3
4         For most Bridges, the address of any MAC attached to the
5         Bridge will suffice for the low-order six octets, and 0
6         for the high-order octets. In some situations, e.g., if
7         multiple virtual Bridges utilizing emulated LANs are
8         implemented in a single physical system, the high-order
9         two octets can be used to differentiate among the
10        transmitting entities.
11
12        The value returned is undefined if the
13        transmit-linktrace input action was not accepted.";
14    reference
15        "Item c) in 12.14.7.4.3 of IEEE Std 802.1Q-2022";
16    uses ltm-egress-identifier-grouping;
17    }
18  }
19 }
20 }
21 }
22 }
23 }
24
25
```

1 48.6.9 The ieee802-dot1q-cfm-bridge YANG module

```
2 module ieee802-dot1q-cfm-bridge {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-cfm-bridge;
5   prefix cfm-bridge;
6   import ieee802-dot1q-bridge {
7     prefix dot1q;
8   }
9   import ieee802-dot1q-cfm {
10    prefix dot1q-cfm;
11  }
12  import ietf-interfaces {
13    prefix if;
14  }
15  import ieee802-types {
16    prefix ieee;
17  }
18  import ieee802-dot1q-cfm-types {
19    prefix cfm-types;
20  }
21  import ieee802-dot1q-types {
22    prefix dot1q-types;
23  }
24  organization
25    "IEEE 802.1 Working Group";
26  contact
27    "WG-URL: http://ieee802.org/1/
28    WG-EMail: stds-802-1-1@ieee.org
29
30    Contact: IEEE 802.1 Working Group Chair
31    Postal: C/O IEEE 802.1 Working Group
32            IEEE Standards Association
33            445 Hoes Lane
34            Piscataway, NJ 08854
35            USA
36
37    E-mail: stds-802-1-chairs@ieee.org";
38  description
39    "Connectivity Fault Management (CFM) comprises capabilities for
40    detecting, verifying, and isolating connectivity failures in Virtual
41    Bridged Local Area Networks. This module binds the CFM modules to an
42    IEEE 802.1Q Bridge.
43
44    Copyright (C) IEEE (2022).
45
46    This version of this YANG module is part of IEEE Std 802.1Q; see the
47    standard itself for full legal notices.";
48  revision 2022-01-19 {
49    description
50      "Published as part of IEEE Std 802.1Q-2022.";
51    reference
52      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
53  }
54  revision 2020-06-04 {
55    description
56      "Published as part of IEEE Std 802.1Qcx-2020. Initial version.";
57    reference
58      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
59      Model for Connectivity Fault Management.";
60  }
61  typedef bridge-ref {
62    type leafref {
63      path '/dot1q:bridges/dot1q:bridge/dot1q:name';
64    }
65    description
66      "This type is used by data models that need to reference a
67      configured Bridge.";
68  }
69  typedef port-ref {
70    type if:interface-ref;
71    description
```

```

1      "This type is used to represent interfaces that can be used by a
2      Bridge device.";
3  }
4
5  /* -----
6   * Grouping definitions used by 802.1Qcx YANG module
7   * -----
8   */
9  grouping service-id-grouping {
10     description
11         "The list of VIDs, the I-SID, the TE-SID, or the SEG-ID monitored
12         by this MA, if present. In the case that a list of VIDs is
13         specified, the first VID in the list is the MA's Primary VID
14         (default none). The specification of I-SID is allowed only in the
15         case of I- or B-components. The TE-SID is allowed only in the case
16         that PBB-TE or SPBM is supported. The SEG-ID is allowed only in
17         the case that IPS is supported.";
18     choice service-id {
19         description
20             "The type of service identifier.";
21         case vids {
22             list vid {
23                 key "vlan-id";
24                 description
25                     "A list of VIDs associated with any MHF on the VID, always
26                     including that VID. The first VID is the MA's primary VID.
27                     List is empty if no primary VID specified";
28                 leaf vlan-id {
29                     type dot1q-types:vlanid;
30                     description
31                         "The 12-bit VLAN identifier.";
32                 }
33             }
34         }
35         case isid {
36             leaf isid {
37                 type uint32 {
38                     range "1..16777215";
39                 }
40                 description
41                     "24-bit I-SID identifier.";
42             }
43         }
44         case tesid {
45             leaf tesid {
46                 type uint32 {
47                     range "1..4294967295";
48                 }
49                 description
50                     "The tesid is used as a service selector for MAs that are
51                     present in Bridges that implement PBB-TE functionality.";
52             }
53         }
54         case segid {
55             leaf segid {
56                 type uint32 {
57                     range "1..4294967295";
58                 }
59                 description
60                     "The segid is used as a service selector for MAs that are
61                     present in Bridges that implement IPS functionality.";
62             }
63         }
64         case path-tesid {
65             leaf path-tesid {
66                 type uint32 {
67                     range "1..4294967295";
68                 }
69                 description
70                     "The path-tesid is used as a service selector for SPBM path
71                     MAs.";
72             }
73         }
74     }
75 }

```

```

1      }
2      case group-isid {
3          leaf group-isid {
4              type uint32 {
5                  range "1..4294967295";
6              }
7              description
8                  "The group-isid is used as a service selector for SPBM group
9                  MAs.";
10         }
11     }
12 }
13 }
14
15 /* -----
16  * Augmentations of objects defined in generic CFM YANG module
17  * (ieee802-dot1q-cfm) and generic MEP YANG module
18  * (ieee802-dot1q-cfm-mep).
19  * -----
20  */
21 augment "/dot1q-cfm:cfm" {
22     description
23         "Augment the base/common CFM model with CFM Bridge specific
24         attributes.";
25     list cfm-stack {
26         key "port service-selector service-id md-level direction";
27         config false;
28         description
29             "The CFM Stack contains information about the Maintenance Points
30             configured on a particular Bridge Port (or Aggregation Port). It
31             contains all CFM Stack specific related configuration and
32             operational data.
33
34             Upon a restart of the system, the system SHALL, if necessary,
35             change the value of this variable, and rearrange the cfm-stack,
36             so that it indexes the entry in the interface table with the
37             same value of interface-ref that it indexed before the system
38             restart. If no such entry exists, then the system SHALL delete
39             all entries in the cfm-stack with the interface index.";
40         leaf port {
41             type port-ref;
42             description
43                 "An interface on which maintenance points might be configured.
44                 This object represents the Bridge Port or Aggregation Port on
45                 which MEPs or MHFs might be configured.";
46             reference
47                 "Item a) in 12.14.2.1.2 of IEEE Std 802.1Q-2022";
48         }
49         leaf md-level {
50             type cfm-types:md-level-type;
51             description
52                 "The MD level of the maintenance point";
53             reference
54                 "Item b) in 12.14.2.1.2 of IEEE Std 802.1Q-2022";
55         }
56         leaf direction {
57             type cfm-types:mp-direction-type;
58             description
59                 "The direction in which the maintenance point faces on the
60                 Bridge Port.";
61             reference
62                 "Item c) in 12.14.2.1.2 of IEEE Std 802.1Q-2022";
63         }
64         leaf service-selector {
65             type cfm-types:service-selector-type;
66             description
67                 "The type of the service selector";
68         }
69         leaf service-id {
70             type cfm-types:service-selector-value-type;
71             description
72                 "A VID, I-SID, Traffic Engineering service instance Identifier

```

```

1      (TE-SID), or Segment Identifier (SEG-ID) associated with an
2      MP, or 0, in the case that the MP is associated with no VID,
3      I-SID, TE-SID, or SEG-ID.";
4      reference
5      "Item d) in 12.14.2.1.2 of IEEE Std 802.1Q-2022";
6  }
7  leaf maintenance-group-id {
8      when
9          "../maintenance-point = \"mep\"" {
10         description
11             "This should only exist if the configured maintenance point
12             is a MEP (and not a MHF).";
13         }
14         type leafref {
15             path
16                 '/dot1q-cfm:cfm'+
17                 '/dot1q-cfm:maintenance-group'+
18                 '/dot1q-cfm:maintenance-group-id';
19         }
20         mandatory true;
21         description
22             "The maintenance group that the MEP is associated with. If the
23             Maintenance Point is a MHF created due to an entry in the
24             default-md-levels list, it has no associated maintenance
25             group, and therefore this leaf is not present.";
26     }
27     leaf mep-id {
28         when
29             "../maintenance-point = \"mep\"" {
30                 description
31                     "This should only exist if the configured maintenance-point
32                     is a MEP.";
33                 }
34                 type leafref {
35                     path
36                         "/dot1q-cfm:cfm/dot1q-cfm:maintenance-group[dot1q-cfm:main"+
37                         "tenance-group-id = current()../maintenance-group-id]/dot"+
38                         "1q-cfm:mep/dot1q-cfm:mep-id";
39                 }
40                 mandatory true;
41                 description
42                     "The MEP identifier if a MEP is configured. Does not exist if
43                     the maintenance point is a MHF.";
44                 reference
45                     "Item d) in 12.14.2.1.3 of IEEE Std 802.1Q-2022";
46             }
47             leaf md-id {
48                 when
49                     "../maintenance-point != \"mep\"" {
50                         description
51                             "This should only exist if the configured maintenance point
52                             is a MHF (and not a MEP).";
53                         }
54                         type leafref {
55                             path
56                                 '/dot1q-cfm:cfm'+
57                                 '/dot1q-cfm:maintenance-domain'+
58                                 '/dot1q-cfm:md-id';
59                             }
60                             description
61                                 "The maintenance domain that the MHF is associated with. If
62                                 the MHF is created due to an entry in the default-md-levels
63                                 list, it has no associated maintenance domain, and therefore
64                                 this leaf is not present.";
65                             }
66                             leaf ma-id {
67                                 when
68                                     "../maintenance-point != \"mep\"" {
69                                         description
70                                             "This should only exist if the configured maintenance point
71                                             is a MHF (and not a MEP), and it has a Maintenance Domain.";
72                                     }

```

```

1      type leafref {
2          path
3              '/dot1q-cfm:cfm'+
4              '/dot1q-cfm:maintenance-domain[dot1q-cfm:md-id = current()'+
5              '/..'+'
6              '/md-id]'+
7              '/dot1q-cfm:maintenance-association'+
8              '/dot1q-cfm:ma-id';
9      }
10     description
11         "The maintenance association that the MHF is associated with.
12         If the MHF is created due to an entry in the default-md-levels
13         list, it has no associated maintenance association, and
14         therefore this leaf is not present.";
15 }
16 leaf mac-address {
17     type ieee:mac-address;
18     mandatory true;
19     description
20         "The MAC address of the maintenance point.";
21     reference
22         "Item e) in 12.14.2.1.3 of IEEE Std 802.1Q-2022";
23 }
24 leaf maintenance-point {
25     type cfm-types:mp-type;
26     mandatory true;
27     description
28         "Indicates the type of maintenance point. That is whether a
29         MEP or MHF.";
30     reference
31         "Item a) in 12.14.2.1.3 of IEEE Std 802.1Q-2022";
32 }
33 }
34
35 // cfm-stack
36 list default-md-level {
37     key "bridge-id component-id service-selector primary-service-id";
38     description
39         "For each bridge component, the Default MD Level Managed Object
40         controls MHF creation for VLANs that are not attached to a
41         specific Maintenance Association Managed Object, and Sender ID
42         TLV transmission by those MHFs.
43
44         For each Bridge Port, and for each VLAN ID whose data can pass
45         through that Bridge Port, an entry in this table is used by the
46         algorithm in subclause 22.2.3 only if there is no entry in the
47         Maintenance Association table defining an MA for the same VLAN
48         ID and MD Level as this table's entry, and on which MA an Up MEP
49         is defined. If there exists such an MA, that MA's objects are
50         used by the algorithm in subclause 22.2.3 in place of this table
51         entry objects.
52
53         The agent maintains the value of md-status to indicate whether
54         this entry is overridden by an MA. When first initialized, the
55         agent creates this table automatically with entries for all VLAN
56         IDs, with the default values specified for each object. After
57         this initialization, the writable objects in this table need to
58         be persistent upon reboot or restart of a device.";
59     leaf bridge-id {
60         type bridge-ref;
61         description
62             "The reference to the IEEE 802.1Q Bridge associated with the
63             CFM objects.";
64     }
65     leaf component-id {
66         type leafref {
67             path
68                 '/dot1q:bridges'+
69                 '/dot1q:bridge[dot1q:name = current()'+
70                 '/..'+'
71                 '/bridge-id]'+
72                 '/dot1q:component'+

```



```

1      '/dot1q:name';
2    }
3    description
4      "The Bridge component within the system to which the
5      information in this entry applies. If the system is not a
6      Bridge, or if only one component is present in the Bridge,
7      then this variable (index) MUST be equal to 1.";
8    reference
9      "12.31 of IEEE Std 802.1Q-2022";
10   }
11   leaf service-selector {
12     type cfm-types:service-selector-type;
13     description
14       "The type of the service selector";
15   }
16   leaf primary-service-id {
17     type cfm-types:service-selector-value-type;
18     description
19       "A vid or isid in an I or B component.";
20     reference
21       "Item a) in 12.14.3.1.2 of IEEE Std 802.1Q-2022";
22   }
23   container associated-service-ids {
24     description
25       "A list of VIDs associated with any MHF on the VID, always
26       including that VID, or the Backbone-SID of the B-component or
27       VIP-SID of the I-component associated with any MHF on the
28       I-SID. The first VID is the MAs Primary VID.
29
30       List is empty if no primary VID specified.";
31     reference
32       "Item a) in 12.14.3.1.3 of IEEE Std 802.1Q-2022";
33     uses service-id-grouping;
34   }
35   leaf md-status {
36     type boolean;
37     config false;
38     mandatory true;
39     description
40       "A Boolean value indicating whether this entry is in effect or
41       has been overridden by the existence of a Maintenance
42       Association managed object associated with the same VID or
43       I-SID of I- or B-components and MD Level, and on which is
44       configured an Up MEP. True if this Maintenance Domain managed
45       object is in effect.";
46     reference
47       "Item b) in 12.14.3.1.3 of IEEE Std 802.1Q-2022";
48   }
49   leaf md-level {
50     type cfm-types:md-level-type;
51     mandatory true;
52     description
53       "A value indicating the MD Level at which MHFs are to be
54       created, and Sender ID TLV transmission by those MHFs is to be
55       controlled, for the VLAN to which this entry's objects apply.";
56     reference
57       "Item c) in 12.14.3.1.3, Item b) in 12.14.3.2.2 of IEEE Std
58       802.1Q-2022";
59   }
60   leaf mhf-creation {
61     type cfm-types:mhf-creation-type;
62     must
63       ". != \"mhf-defer\" {
64         description
65           "The value can not be mhf-defer. Must be one of mhf-none,
66           mhf-default, or mfh-explicit.";
67       }
68     default "mhf-none";
69     description
70       "An enumerated value indicating whether the management entity
71       can create MHFs for this VID(s) or I-SID(s) of I- or
72       B-components.";
```

```

1      reference
2      "Item d) in 12.14.3.1.3 of IEEE Std 802.1Q-2022";
3  }
4  leaf id-permission {
5      type cfm-types:sender-id-permission-type;
6      must
7      ". != \"send-id-defer\"" {
8          description
9          "The value can not be send-id-defer. Must be one of
10             send-id-none, send-id-chassis, send-id-manage,
11             send-id-chassis-manage.";
12      }
13      default "send-id-none";
14      description
15      "An enumerated value indicating what, if anything, is to be
16         included in the Sender ID TLV transmitted by MPs configured in
17         the Default Maintenance Domain.";
18      reference
19      "Item e) in 12.14.3.1.3, Item a) in 12.14.3.2.2 of IEEE Std
20         802.1Q-2022";
21  }
22 }
23
24 // default-md-level
25 list config-error {
26     key "port service-selector service-id";
27     config false;
28     description
29     "The CFM Configuration Error List table provides a list of
30         Interfaces and VIDs that are incorrectly configured.";
31     leaf port {
32         type port-ref;
33         description
34         "The interface index of the interface (i.e., Bridge Port).
35
36         Upon a restart of the system, the system SHALL, if necessary,
37         change the value of this variable so that it indexes the entry
38         in the interface table with the same value of the index that
39         it indexed before the system restart. If no such entry exists,
40         then the system SHALL delete any entries in config-error-list
41         indexed by that interface-ref.";
42         reference
43         "Item b) in 12.14.4.1.2 of IEEE Std 802.1Q-2022";
44     }
45     leaf service-selector {
46         type cfm-types:service-selector-type;
47         description
48         "The type of the service selector";
49     }
50     leaf service-id {
51         type cfm-types:service-selector-value-type;
52         description
53         "A vid or isid in an I or B component.";
54         reference
55         "Item a) in 12.14.4.1.2 of IEEE Std 802.1Q-2022";
56     }
57     leaf error-type {
58         type cfm-types:config-error-type;
59         mandatory true;
60         description
61         "vector of Boolean error conditions from 22.2.4.";
62         reference
63         "Item b) in 12.14.4.1.3 of IEEE Std 802.1Q-2022";
64     }
65 }
66 }
67 augment "/dot1q-cfm:cfm/dot1q-cfm:maintenance-group" {
68     when
69     "dot1q-cfm:maintenance-group-id";
70     description
71     "Augment the maintenance group list object with the maintenance
72     association component identifier, since the maintenance

```

```

1      association component is an Bridge specific attributes.";
2  leaf bridge-id {
3      type bridge-ref;
4      description
5          "The reference to the IEEE 802.1Q Bridge associated with the CFM
6          objects.";
7  }
8  leaf component-name {
9      when
10         "../bridge-id";
11     type leafref {
12         path
13             '/dot1q:bridges'+
14             '/dot1q:bridge[dot1q:name = current()'+
15             '/../'+
16             '/cfm-bridge:bridge-id]'+
17             '/dot1q:component'+
18             '/dot1q:name';
19     }
20     mandatory true;
21     description
22         "A reference to the maintenance association component within the
23         provided maintenance domain and maintenance association values.";
24 }
25 container service-id {
26     description
27         "The service identifier grouping.";
28     uses service-id-grouping;
29 }
30 }
31 augment "/dot1q-cfm:cfm/dot1q-cfm:maintenance-group/dot1q-cfm:mep" {
32     when
33         "dot1q-cfm:mep-id";
34     description
35         "Augment the MEP object defined in the ieee802-dot1q-cfm (CFM MEP)
36         YANG module with Bridge specific attributes.";
37     leaf port {
38         type port-ref;
39         must
40             "not(/dot1q-cfm:cfm/dot1q-cfm:maintenance-group/cfm-bridge:com"+
41             "ponent-name) or /dot1q-cfm:cfm/dot1q-cfm:maintenance-group/cf"+
42             "m-bridge:component-name = /if:interfaces/if:interface[current"+
43             "() = ./if:name]/dot1q:bridge-port/dot1q:component-name" {
44             description
45                 "If there is a Bridge Component, then the component of the
46                 port (Interface) reference must be the same as the bridge and
47                 component references by the maintenance association group.";
48         }
49         mandatory true;
50         description
51             "The interface index of the interface (e.g., Bridge Port) to
52             which the MEP is attached.";
53         reference
54             "Item b) in 12.14.7.1.3 of IEEE Std 802.1Q-2022";
55     }
56     leaf primary-vid {
57         type leafref {
58             path '../../service-id/vid/vlan-id';
59         }
60         description
61             "An integer indicating the Primary VID of the MEP. It is always
62             one of the VIDs assigned to the MEPs MA. If not present, then
63             indicates that either the Primary VID is that of the MEPs MA, or
64             that the MEPs MA is associated with no VID.";
65         reference
66             "Item d) in 12.14.7.1.3 of IEEE Std 802.1Q-2022";
67     }
68 }
69 }
70
71

```

1 48.6.10 The ieee802-dot1q-cfm-alarm YANG module

```
2 module ieee802-dot1q-cfm-alarm {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-cfm-alarm;
5   prefix dot1q-cfm-alarm;
6   import ieee802-dot1q-cfm {
7     prefix dot1q-cfm;
8   }
9   organization
10    "IEEE 802.1 Working Group";
11   contact
12    "WG-URL: http://ieee802.org/1/
13    WG-EMail: stds-802-1-1@ieee.org
14
15    Contact: IEEE 802.1 Working Group Chair
16    Postal: C/O IEEE 802.1 Working Group
17            IEEE Standards Association
18            445 Hoes Lane
19            Piscataway, NJ 08854
20            USA
21
22    E-mail: stds-802-1-chairs@ieee.org";
23   description
24    "Connectivity Fault Management (CFM) comprises capabilities for
25    detecting, verifying, and isolating connectivity failures in Virtual
26    Bridged Local Area Networks. These capabilities can be used in
27    networks operated by multiple independent organizations, each with
28    restricted management access to each other's equipment.
29
30    Copyright (C) IEEE (2022).
31
32    This version of this YANG module is part of IEEE Std 802.1Q; see the
33    standard itself for full legal notices.";
34   revision 2022-01-19 {
35     description
36      "Published as part of IEEE Std 802.1Q-2022.";
37     reference
38      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
39   }
40   revision 2020-06-04 {
41     description
42      "Published as part of IEEE Std 802.1Qcx-2020. Initial version.";
43     reference
44      "IEEE Std 802.1Qcx-2020, Bridges and Bridged Networks - YANG Data
45      Model for Connectivity Fault Management.";
46   }
47
48   /* -----
49   * Augmentations of objects defined in generic CFM YANG module
50   * (ieee802-dot1q-cfm) to define CFM alarms.
51   * -----
52   */
53   augment "/dot1q-cfm:cfm/dot1q-cfm:maintenance-group/dot1q-cfm:mep" {
54     when
55      "dot1q-cfm:mep-id";
56     description
57      "Augment the MEP object defined in the ieee802-dot1q-cfm (CFM MEP)
58      YANG module with CFM alarm specific attributes.";
59     notification mep-fault-alarm {
60       description
61        "To alert the Manager to the existence of a fault in a monitored
62        MA by issuing a Fault Alarm.";
63       reference
64        "Item c) in 12.14.7.7.2 of IEEE Std 802.1Q-2022";
65       leaf mep-priority-defect {
66         type leafref {
67           path
68            '..'+
69            '/../'+
70            '/dot1q-cfm:continuity-check'+
71            '/dot1q-cfm:highest-priority-defect';
```

```
1      }
2      config false;
3      mandatory true;
4      description
5          "The highest priority defect that has been present since the
6          MEPS Fault Notification Generator state machine was last in
7          the FNG_RESET state.";
8      }
9  }
10 }
11 }
12
13
```

1 48.6.11 The ieee802-dot1q-stream-filters-gates YANG module

```
2 module ieee802-dot1q-stream-filters-gates {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-stream-filters-gates;
5   prefix sfsg;
6   organization
7     "IEEE 802.1 Working Group";
8   contact
9     "WG-URL: http://ieee802.org/1/
10    WG-EMail: stds-802-1-1@ieee.org
11
12    Contact: IEEE 802.1 Working Group Chair
13    Postal: C/O IEEE 802.1 Working Group
14            IEEE Standards Association
15            445 Hoes Lane
16            Piscataway, NJ 08854
17            USA
18
19    E-mail: stds-802-1-chairs@ieee.org";
20  description
21    "This module provides management of IEEE 802.1Q Bridge components
22    that support Stream Filters and Stream Gates.
23
24    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
25    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
26
27    Copyright (C) IEEE (2023).
28
29    This version of this YANG module is part of IEEE Std 802.1Q; see the
30    standard itself for full legal notices.";
31  revision 2023-07-03 {
32    description
33      "Published as part of IEEE Std 802.1Qcz-2023.";
34    reference
35      "IEEE Std 802.1Qcz-2023 - Bridges and Bridged Networks - Amendment:
36      Congestion Isolation.";
37  }
38  revision 2022-10-29 {
39    description
40      "Published as part of IEEE Std 802.1Q-2022.";
41    reference
42      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
43  }
44  revision 2020-11-06 {
45    description
46      "Published as part of IEEE Std 802.1Qcr-2020.";
47    reference
48      "IEEE Std 802.1Qcr-2020, Bridges and Bridged Networks -
49      Asynchronous Traffic Shaping.";
50  }
51  feature closed-gate-state {
52    description
53      "The bridge component supports gate state closed.";
54    reference
55      "IEEE Std 802.1Q";
56  }
57
58  /* Types and groupings */
59  typedef priority-spec-type {
60    type enumeration {
61      enum zero {
62        value 0;
63        description
64          "Priority 0";
65      }
66      enum one {
67        value 1;
68        description
69          "Priority 1";
70      }
71      enum two {
```

```

1      value 2;
2      description
3      "Priority 2";
4  }
5  enum three {
6      value 3;
7      description
8      "Priority 3";
9  }
10 enum four {
11     value 4;
12     description
13     "Priority 4";
14 }
15 enum five {
16     value 5;
17     description
18     "Priority 5";
19 }
20 enum six {
21     value 6;
22     description
23     "Priority 6";
24 }
25 enum seven {
26     value 7;
27     description
28     "Priority 7";
29 }
30 enum wildcard {
31     description
32     "wildcard value";
33 }
34 }
35 description
36 "The frame's priority value";
37 reference
38 "8.6.5.2 of IEEE Std 802.1Q";
39 }
40 typedef ipv-spec-type {
41     type enumeration {
42         enum zero {
43             value 0;
44             description
45             "Priority 0";
46         }
47         enum one {
48             value 1;
49             description
50             "Priority 1";
51         }
52         enum two {
53             value 2;
54             description
55             "Priority 2";
56         }
57         enum three {
58             value 3;
59             description
60             "Priority 3";
61         }
62         enum four {
63             value 4;
64             description
65             "Priority 4";
66         }
67         enum five {
68             value 5;
69             description
70             "Priority 5";
71         }
72         enum six {

```

```

1      value 6;
2      description
3          "Priority 6";
4  }
5  enum seven {
6      value 7;
7      description
8          "Priority 7";
9  }
10 enum null {
11     description
12         "null value";
13 }
14 }
15 description
16     "An IPV can be either of the following:
17     1) The null value. For a frame that passes through the gate, the
18     priority value associated with the frame is used to determine
19     the frame's traffic class, using the Traffic Class Table as
20     specified in 8.6.6.
21     2) An internal priority value. For a frame that passes through
22     the gate, the IPV is used, in place of the priority value
23     associated with the frame, to determine the frame's traffic
24     class, using the Traffic Class Table as specified in 8.6.6.";
25 reference
26     "8.6.5.2 of IEEE Std 802.1Q";
27 }
28 typedef gate-state-value-type {
29     type enumeration {
30         enum closed {
31             description
32                 "Gate closed";
33         }
34         enum open {
35             description
36                 "Gate open";
37         }
38     }
39     description
40         "The gate-state-value-type indicates a gate state, open or closed,
41         for the stream gate.";
42     reference
43         "12.31.3.2.1 of IEEE Std 802.1Q";
44 }
45 grouping sfsg-parameters {
46     description
47         "The grouping of all stream filter and stream gate parameters.";
48     container stream-gates {
49         description
50             "This container encapsulates all nodes related to Stream Gates.";
51         list stream-gate-instance-table {
52             key "stream-gate-instance-id";
53             description
54                 "Each list entry contains a set of parameters that defines a
55                 single stream gate (8.6.5.4), as detailed in Table 12-36.
56                 Entries in the table can be created or removed dynamically in
57                 implementations that support dynamic configuration of stream
58                 gates.";
59             reference
60                 "12.31.3 of IEEE Std 802.1Q";
61             leaf stream-gate-instance-id {
62                 type uint32;
63                 description
64                     "An integer table index that allows the stream gate to be
65                     referenced from Stream Filter Instance Table entries.";
66                 reference
67                     "12.31.2.5 of IEEE Std 802.1Q
68                     8.6.5.3 of IEEE Std 802.1Q
69                     8.6.5.4 of IEEE Std 802.1Q";
70             }
71             leaf gate-enable {
72                 type boolean;

```



```

1      default "false";
2      description
3          "A Boolean variable that indicates whether the operation of
4          the state machines is enabled (TRUE) or disabled (FALSE).
5          This variable is set by management. The default value of this
6          variable is FALSE.";
7      reference
8          "8.6.9.4.14 of IEEE Std 802.1Q";
9  }
10 leaf admin-gate-states {
11     type gate-state-value-type;
12     default "open";
13     description
14         "The administratively set gate state of this gate.";
15     reference
16         "12.31.3.2.1 of IEEE Std 802.1Q
17         8.6.10.4 of IEEE Std 802.1Q";
18 }
19 leaf admin-ipv {
20     type ipv-spec-type;
21     default "null";
22     description
23         "The administratively set internal priority value
24         specification.";
25     reference
26         "12.31.3.3 of IEEE Std 802.1Q
27         8.6.10.6 of IEEE Std 802.1Q
28         8.6.5.4 of IEEE Std 802.1Q";
29 }
30 }
31 leaf max-stream-gate-instances {
32     type uint32;
33     config false;
34     description
35         "The maximum number of Stream Gate instances supported by this
36         Bridge component.";
37     reference
38         "12.31.1.2 of IEEE Std 802.1Q";
39 }
40 }
41 container stream-filters {
42     description
43         "This container encapsulates all nodes related to stream filters.";
44     reference
45         "12.31 of IEEE Std 802.1Q";
46     list stream-filter-instance-table {
47         key "stream-filter-instance-id";
48         description
49             "Each list entry contains a set of parameters that defines a
50             single stream filter (8.6.5.1) with associated maximum SDU size
51             filtering (8.6.5.3.1), as detailed in Table 12-35. Entries can
52             be created or removed dynamically in implementations that
53             support dynamic configuration of stream filters. The value of
54             the stream-handle-spec and priority-spec parameters associated
55             with a received frame determine which stream filter is selected
56             by the frame, and therefore what combination of filtering and
57             policing actions is applied to the frame. If the
58             stream-handle-spec and priority-spec parameters associated with
59             a received frame match more than one stream filter, the stream
60             filter that is selected is the one that appears earliest in the
61             ordered list. If a received frame's stream-handle-spec and
62             priority-spec does not match any of the stream filters in the
63             list, the frame is processed as if stream filters and stream
64             gates would not be supported.";
65         reference
66             "12.31.2 of IEEE Std 802.1Q";
67         leaf stream-filter-instance-id {
68             type uint32;
69             mandatory true;
70             description
71                 "An integer index value that determines the place of the
72                 stream filter in the ordered list of stream filter instances.

```

```

1      The values are ordered according to their integer value;
2      smaller values appear earlier in the ordered list.";
3  reference
4      "12.31.2.1 of IEEE Std 802.1Q";
5  }
6  choice stream-handle-spec {
7      description
8          "The stream_handle specification data type allows either of
9          the following to be represented:
10         a) A stream_handle value, represented as an integer.
11         b) The wildcard value, which matches any frame";
12  reference
13      "12.31.2.2 of IEEE Std 802.1Q";
14
15      /* NOTE: The mapping of the wildcard literal is
16      *         other than in the MIB definition, where
17      *         the wildcard value is mapped to -1.
18      */
19  case wildcard {
20      leaf wildcard {
21          type empty;
22          description
23              "The stream handle specification represents a wildcard
24              value.";
25      }
26  }
27  case stream-handle {
28      leaf stream-handle {
29          type uint32;
30          mandatory true;
31          description
32              "The stream handle specification refers to a
33              stream_handle value.";
34      }
35  }
36  }
37  leaf priority-spec {
38      type priority-spec-type;
39      mandatory true;
40      description
41          "The priority specification data type allows either of the
42          following to be represented:
43          a) A priority value, represented as an integer.
44          b) The wildcard value, which matches any priority.";
45  reference
46      "12.31.2.3 of IEEE Std 802.1Q";
47  }
48  leaf max-sdu-size {
49      type uint32;
50      units "octets";
51      mandatory true;
52      description
53          "The allowed maximum SDU size, in octets. If set to 0, any
54          SDU size is accepted.";
55  reference
56      "8.6.5.3.1 of IEEE Std 802.1Q";
57  }
58  leaf stream-blocked-due-to-oversize-frame-enabled {
59      type boolean;
60      default "false";
61      description
62          "A value of true indicates that
63          stream-blocked-due-to-oversize-frame is set to true as soon
64          as a frame exceeds max-sdu-size.";
65  reference
66      "8.6.5.3.1 of IEEE Std 802.1Q";
67  }
68  leaf stream-blocked-due-to-oversize-frame {
69      type boolean;
70      default "false";
71      description
72          "Indicates by value true that frames are permanently

```

```

1      discarded as a result of an initial frame exceeding
2      max-sdu-size. The value of
3      stream-blocked-due-to-oversize-frame can be administratively
4      reset to false.";
5      reference
6      "8.6.5.3.1 of IEEE Std 802.1Q";
7  }
8  leaf stream-gate-ref {
9      type leafref {
10         path
11         '..'+
12         '/..' +
13         '/..' +
14         '/stream-gates'+
15         '/stream-gate-instance-table'+
16         '/stream-gate-instance-id';
17     }
18     mandatory true;
19     description
20     "This node refers to the stream gate (12.31.3) that is
21     associated with the stream filter. The relationship between
22     stream filters and stream gates is many to one; a given
23     stream filter can be associated with only one stream gate,
24     but there can be multiple stream filters associated with a
25     given stream gate.";
26     reference
27     "12.31.3.1 of IEEE Std 802.1Q";
28 }
29 }
30 leaf max-stream-filter-instances {
31     type uint32;
32     config false;
33     description
34     "The maximum number of stream filter instances supported by
35     this Bridge component.";
36     reference
37     "12.31.1.1, 8.6.5.1 of IEEE Std 802.1Q";
38 }
39 }
40 }
41 }
42

```

1 48.6.12 The ieee802-dot1q-stream-filters-gates-bridge YANG module

```
2 module ieee802-dot1q-stream-filters-gates-bridge {
3   yang-version "1.1";
4   namespace
5     urn:ieee:std:802.1Q:yang:ieee802-dot1q-stream-filters-gates-bridge;
6   prefix sfsg-bridge;
7   import ieee802-dot1q-bridge {
8     prefix dot1q;
9   }
10  import ieee802-dot1q-stream-filters-gates {
11    prefix sfsg;
12  }
13  organization
14    "IEEE 802.1 Working Group";
15  contact
16    "WG-URL: https://1.ieee802.org/
17    WG-EMail: stds-802-1@ieee.org
18    Contact: IEEE 802.1 Working Group Chair
19    Postal: C/O IEEE 802.1 Working Group
20            IEEE Standards Association
21            445 Hoes Lane
22            Piscataway, NJ 08855
23            USA
24
25    E-mail: STDS-802-1-CHAIRS@LISTSERV.IEEE.ORG";
26  description
27    "This module provides management of IEEE 802.1Q Bridge components
28    that support Stream Filters and Stream Gates.
29
30    Copyright (C) IEEE (2023).
31
32    This version of this YANG module is part of IEEE Std 802.1Q; see the
33    standard itself for full legal notices.";
34  revision 2023-07-03 {
35    description
36      "Published as part of IEEE Std 802.1Qcz-2023.";
37    reference
38      "IEEE Std 802.1Qcz-2023 - Bridges and Bridged Networks - Amendment:
39      Congestion Isolation.";
40  }
41  augment "/dot1q:bridges/dot1q:bridge/dot1q:component" {
42    description
43      "Augments the Bridge component with stream filters and stream
44      gates.";
45    uses sfsg:sfsg-parameters;
46  }
47 }
48
49
```

1 48.6.13 The ieee802-dot1q-ats YANG module

```
2 module ieee802-dot1q-ats {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-ats;
5   prefix ats;
6   import ietf-yang-types {
7     prefix yang;
8   }
9   import ieee802-dot1q-types {
10    prefix dot1qtypes;
11  }
12  import ieee802-dot1q-stream-filters-gates {
13    prefix sfsg;
14  }
15  organization
16    "IEEE 802.1 Working Group";
17  contact
18    "WG-URL: http://ieee802.org/1/
19    WG-EMail: stds-802-1-1@ieee.org
20
21    Contact: IEEE 802.1 Working Group Chair
22    Postal: C/O IEEE 802.1 Working Group
23            IEEE Standards Association
24            445 Hoes Lane
25            Piscataway, NJ 08854
26            USA
27
28    E-mail: stds-802-1-chairs@ieee.org";
29  description
30    "This module provides management of IEEE 802.1Q Bridge components
31    that support Asynchronous Traffic Shaping (ATS).
32
33    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
34    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
35
36    Copyright (C) IEEE (2023).
37
38    This version of this YANG module is part of IEEE Std 802.1Q; see the
39    standard itself for full legal notices.";
40  revision 2023-07-03 {
41    description
42      "Published as part of IEEE Std 802.1Qcz-2023.";
43    reference
44      "IEEE Std 802.1Qcz-2023, Bridges and Bridged Networks - Congestion
45      Isolation.";
46  }
47  revision 2022-01-19 {
48    description
49      "Published as part of IEEE Std 802.1Q-2022.";
50    reference
51      "IEEE Std 802.1Q-2022, Bridges and Bridged Networks.";
52  }
53  revision 2020-11-06 {
54    description
55      "Published as part of IEEE Std 802.1Qcr-2020.";
56    reference
57      "IEEE Std 802.1Qcr-2020, Bridges and Bridged Networks -
58      Asynchronous Traffic Shaping.";
59  }
60  grouping ats-port-parameters {
61    description
62      "This container comprises all ATS per-Port parameters.";
63    leaf discarded-frames-count {
64      type yang:counter64;
65      config false;
66      description
67        "A counter of frames discarded by ATS scheduler instances
68        associated with the Bridge Port.";
69      reference
70        "12.31.7.2 of IEEE Std 802.1Q";
71    }
```

```

1  }
2  grouping ats-parameters {
3    description
4      "System specific configuration for ATS includes:
5      a) ATS schedulers
6      b) ATS scheduler groups";
7    container schedulers {
8      description
9        "This container comprises all nodes related to ATS schedulers.";
10     list scheduler-instance-table {
11       key "scheduler-instance-id";
12       description
13         "Each list entry comprises a set of parameters that defines a
14         single ATS scheduler instance, as detailed in Table 12-38.";
15       reference
16         "12.31.5 of IEEE Std 802.1Q";
17       leaf scheduler-instance-id {
18         type uint32;
19         mandatory true;
20         description
21           "A unique index identifying this ATS scheduler instance.";
22         reference
23           "12.31.5.1, 8.6.5.6 of IEEE Std 802.1Q";
24       }
25       leaf committed-information-rate {
26         type uint64;
27         units "bits/second";
28         mandatory true;
29         description
30           "The committed information rate parameter of this ATS
31           scheduler instance.";
32         reference
33           "12.31.5.3, 8.6.5.6 of IEEE Std 802.1Q";
34       }
35       leaf committed-burst-size {
36         type uint32;
37         units "bits";
38         mandatory true;
39         description
40           "The committed burst size parameter of this ATS scheduler
41           instance.";
42         reference
43           "12.31.5.2, 8.6.5.6 of IEEE Std 802.1Q";
44       }
45       leaf scheduler-group-ref {
46         type leafref {
47           path
48             '..'+
49             '/..' +
50             '/..' +
51             '/scheduler-groups'+
52             '/scheduler-group-instance-table'+
53             '/scheduler-group-instance-id';
54         }
55         mandatory true;
56         description
57           "A reference to the scheduler group (12.31.5.4) associated
58           with this ATS scheduler instance. Multiple ATS scheduler
59           instances can be associated to one scheduler group, as
60           detailed in 8.6.5.6.";
61         reference
62           "12.31.5.4, 8.6.5.6 of IEEE Std 802.1Q";
63       }
64     }
65     leaf max-scheduler-instances {
66       type uint32;
67       config false;
68       description
69         "The maximum number of ATS scheduler instances supported by
70         this Bridge component.";
71       reference
72         "12.31.1.5 of IEEE Std 802.1Q";

```

```

1     }
2   }
3   container scheduler-groups {
4     description
5       "This container comprises all ATS scheduler group related nodes.";
6     list scheduler-group-instance-table {
7       key "scheduler-group-instance-id";
8       description
9         "Each list entry comprises a set of parameters that defines a
10        single ATS scheduler group instance.";
11      reference
12        "12.31.6, 8.6.5.6 of IEEE Std 802.1Q";
13      leaf scheduler-group-instance-id {
14        type uint32;
15        description
16          "A unique index identifying this ATS scheduler group
17          instance.";
18        reference
19          "12.31.6.1, 8.6.5.6 of IEEE Std 802.1Q";
20      }
21      leaf max-residence-time {
22        type uint32;
23        units "nanoseconds";
24        mandatory true;
25        description
26          "The maximum residence time parameter of the ATS scheduler
27          group.";
28        reference
29          "12.31.6.2, 8.6.5.6 of IEEE Std 802.1Q";
30      }
31    }
32    leaf max-scheduler-group-instances {
33      type uint32;
34      config false;
35      description
36        "The maximum number of ATS scheduler group instances supported
37        by this Bridge component.";
38      reference
39        "12.31.1.6, 8.6.5.6 of IEEE Std 802.1Q";
40    }
41    container scheduler-timing-characteristics {
42      description
43        "This container comprises all ATS scheduler timing
44        characteristics related nodes.";
45      list scheduler-timing-characteristics-table {
46        key "reception-port transmission-port";
47        config false;
48        description
49          "Each list entry comprises the timing characteristics of a
50          reception Port transmission Port pair, as detailed in Table
51          12-41.";
52        reference
53          "12.31.8, 8.6.11 of IEEE Std 802.1Q";
54        leaf reception-port {
55          type dot1qttype:port-number-type;
56          config false;
57          mandatory true;
58          description
59            "A reference to the associated reception Port.";
60          reference
61            "12.31.8.1 of IEEE Std 802.1Q";
62        }
63        leaf transmission-port {
64          type dot1qttype:port-number-type;
65          config false;
66          mandatory true;
67          description
68            "A reference to the associated transmission Port.";
69          reference
70            "12.31.8.2 of IEEE Std 802.1Q";
71        }
72      }
73      leaf clock-offset-variation-max {

```

```

1      type uint32;
2      units "nanoseconds";
3      config false;
4      mandatory true;
5      description
6          "The maximum clock offset variation associated with the
7           reception Port transmission Port pair.";
8      reference
9          "12.31.8.3 of IEEE Std 802.1Q";
10     }
11     leaf clock-rate-deviation-max {
12         type uint32;
13         units "ppm";
14         config false;
15         mandatory true;
16         description
17             "The maximum clock rate deviation associated with the
18              reception Port transmission Port pair.";
19         reference
20             "12.31.8.4 of IEEE Std 802.1Q";
21     }
22     leaf arrival-recognition-delay-max {
23         type uint32;
24         units "nanoseconds";
25         config false;
26         mandatory true;
27         description
28             "The maximum arrival time recognition delay associated with
29              the reception Port transmission Port pair.";
30         reference
31             "12.31.8.5 of IEEE Std 802.1Q";
32     }
33     leaf processing-delay-min {
34         type uint32;
35         units "nanoseconds";
36         config false;
37         mandatory true;
38         description
39             "The minimum processing delay associated with the reception
40              Port transmission Port pair.";
41         reference
42             "12.31.8.6 of IEEE Std 802.1Q";
43     }
44     leaf processing-delay-max {
45         type uint32;
46         units "nanoseconds";
47         config false;
48         mandatory true;
49         description
50             "The maximum processing delay associated with the reception
51              Port transmission Port pair.";
52         reference
53             "12.31.8.7 of IEEE Std 802.1Q";
54     }
55 }
56 }
57 }
58 }
59 uses sfsg:sfsg-parameters {
60     augment "stream-filters/stream-filter-instance-table" {
61         description
62             "Augments the system stream filter for ATS schedulers.";
63         uses ats:ats-parameters;
64         container scheduler {
65             description
66                 "This container encapsulates ATS scheduler nodes.";
67             leaf scheduler-ref {
68                 type leafref {
69                     path
70                         '..'+
71                         '/..' +
72                         '/schedulers'+

```



```
1         '/scheduler-instance-table'+
2         '/scheduler-instance-id';
3     }
4     description
5         "A reference to the ATS scheduler associated with this stream
6         filter.";
7     }
8     leaf scheduler-enable {
9         type boolean;
10        default "false";
11        description
12            "If TRUE, this stream filter has an associated ATS scheduler
13            referenced by scheduler-ref. If FALSE, no ATS scheduler is
14            associated with this stream filter (scheduler-ref is
15            ignored).";
16    }
17 }
18 }
19 }
20 }
21
22
```

1 48.6.14 The ieee802-dot1q-ats-bridge YANG module

```
2 module ieee802-dot1q-ats-bridge {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-ats-bridge;
5   prefix ats-bridge;
6   import ietf-interfaces {
7     prefix if;
8   }
9   import ieee802-dot1q-bridge {
10    prefix dot1q;
11  }
12  import ieee802-dot1q-ats {
13    prefix ats;
14  }
15  import ieee802-dot1q-stream-filters-gates {
16    prefix sfsg;
17  }
18  organization
19    "IEEE 802.1 Working Group";
20  contact
21    "WG-URL: http://ieee802.org/1/
22    WG-EMail: stds-802-1-1@ieee.org
23
24    Contact: IEEE 802.1 Working Group Chair
25    Postal: C/O IEEE 802.1 Working Group
26            IEEE Standards Association
27            445 Hoes Lane
28            Piscataway, NJ 08854
29            USA
30
31    E-mail: stds-802-1-chairs@ieee.org";
32  description
33    "This module provides management of IEEE 802.1Q Bridge components
34    that support Asynchronous Traffic Shaping (ATS).
35
36    Copyright (C) IEEE (2023).
37
38    This version of this YANG module is part of IEEE Std 802.1Q; see the
39    standard itself for full legal notices.";
40  revision 2023-07-03 {
41    description
42      "Published as part of IEEE Std 802.1Qcz-2023.";
43    reference
44      "IEEE Std 802.1Qcz-2023, Bridges and Bridged Networks - Congestion
45      Isolation.";
46  }
47  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
48    description
49      "Augments Bridge Ports by ATS per-Port parameters.";
50    uses ats:ats-port-parameters;
51  }
52  augment "/dot1q:bridges/dot1q:bridge/dot1q:component" {
53    description
54      "Augments the Bridge component with ATS parameters.";
55    uses sfsg:sfsg-parameters {
56      augment "stream-filters/stream-filter-instance-table" {
57        description
58          "Augments the Bridge component stream filter for ATS
59          schedulers.";
60        uses ats:ats-parameters;
61        container scheduler {
62          description
63            "This container encapsulates ATS scheduler nodes.";
64          leaf scheduler-ref {
65            type leafref {
66              path
67                '..'+
68                '/../'+
69                '/schedulers'+
70                '/scheduler-instance-table'+
71                '/scheduler-instance-id';
```

```
1      }
2      description
3          "A reference to the ATS scheduler associated with this
4          stream filter.";
5      }
6      leaf scheduler-enable {
7          type boolean;
8          default "false";
9          description
10             "If TRUE, this stream filter has an associated ATS
11             scheduler referenced by scheduler-ref. If FALSE, no ATS
12             scheduler is associated with this stream filter
13             (scheduler-ref is ignored).";
14     }
15 }
16 }
17 }
18 }
19 }
20 }
```

1 48.6.15 The ieee802-dot1q-congestion-isolation YANG module

```
2 module ieee802-dot1q-congestion-isolation {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-congestion-isolation;
5   prefix dot1q-ci;
6   import ietf-yang-types {
7     prefix yang;
8   }
9   import ieee802-types {
10    prefix ieee;
11  }
12  import ieee802-dot1q-stream-filters-gates {
13    prefix sfsg;
14  }
15  import ieee802-dot1q-types {
16    prefix dot1q-types;
17  }
18  import ietf-inet-types {
19    prefix inet;
20  }
21  organization
22    "Institute of Electrical and Electronics Engineers";
23  contact
24    "WG-URL: http://ieee802.org/1/
25    WG-EMail: stds-802-1-1@ieee.org
26    Contact: IEEE 802.1 Working Group Chair
27    Postal: C/O IEEE 802.1 Working Group
28            IEEE Standards Association
29            445 Hoes Lane
30            Piscataway, NJ 08854
31            USA
32
33    E-mail: stds-802-1-chairs@ieee.org";
34  description
35    "This YANG module augments the configuration and operational state
36    data for interfaces for Congestion Isolation.
37
38    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
39    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
40
41    Copyright (C) IEEE (2023).
42
43    This version of this YANG module is part of IEEE Std 802.1Q; see the
44    standard itself for full legal notices.";
45  revision 2023-07-03 {
46    description
47      "Published as part of IEEE Std 802.1Qcz-2023.";
48    reference
49      "IEEE Std 802.1Qcz-2023";
50  }
51
52  /*-----*/
53  /* Feature          */
54  /*-----*/
55  feature congestion-isolation {
56    description
57      "Feature Congestion Isolation";
58  }
59
60  /*-----*/
61  /* Typedefs         */
62  /*-----*/
63  typedef abs-traffic-class-plus-one-type {
64    type enumeration {
65      enum monitored-queue-tc-1 {
66        value 1;
67        description
68          "Traffic class 0 of monitored queue.";
69      }
70      enum monitored-queue-tc-2 {
71        value 2;
```

```
1      description
2      "Traffic class 1 of monitored queue.";
3  }
4  enum monitored-queue-tc-3 {
5      value 3;
6      description
7      "Traffic class 2 of monitored queue.";
8  }
9  enum monitored-queue-tc-4 {
10     value 4;
11     description
12     "Traffic class 3 of monitored queue.";
13 }
14 enum monitored-queue-tc-5 {
15     value 5;
16     description
17     "Traffic class 4 of monitored queue.";
18 }
19 enum monitored-queue-tc-6 {
20     value 6;
21     description
22     "Traffic class 5 of monitored queue.";
23 }
24 enum monitored-queue-tc-7 {
25     value 7;
26     description
27     "Traffic class 6 of monitored queue.";
28 }
29 enum monitored-queue-tc-8 {
30     value 8;
31     description
32     "Traffic class 7 of monitored queue.";
33 }
34 enum congesting-queue-tc-1 {
35     value -1;
36     description
37     "Traffic class 0 of congesting queue.";
38 }
39 enum congesting-queue-tc-2 {
40     value -2;
41     description
42     "Traffic class 1 of congesting queue.";
43 }
44 enum congesting-queue-tc-3 {
45     value -3;
46     description
47     "Traffic class 2 of congesting queue.";
48 }
49 enum congesting-queue-tc-4 {
50     value -4;
51     description
52     "Traffic class 3 of congesting queue.";
53 }
54 enum congesting-queue-tc-5 {
55     value -5;
56     description
57     "Traffic class 4 of congesting queue.";
58 }
59 enum congesting-queue-tc-6 {
60     value -6;
61     description
62     "Traffic class 5 of congesting queue.";
63 }
64 enum congesting-queue-tc-7 {
65     value -7;
66     description
67     "Traffic class 6 of congesting queue.";
68 }
69 enum congesting-queue-tc-8 {
70     value -8;
71     description
72     "Traffic class 7 of congesting queue.";
```

```

1      }
2      enum not-participating-congestion-isolation {
3          value 0;
4          description
5              "Traffic class not participating in congestion isolation.";
6      }
7  }
8  description
9      "Specifies a value that can be translated to the numeric value of
10     the traffic class to be used as either the congesting or monitored
11     queue. The absolute value of the enumerated value is the value of
12     the traffic class plus 1. A value of 0 indicates the traffic class
13     is not participating in congestion isolation. For example, the
14     enumerated value congesting-queue-tc-5 specifies that traffic class
15     4 is used as the congesting queue.";
16 }
17 grouping cip-port-parameters {
18     description
19         "The bridge port specific configuration for Congestion Isolation.";
20     leaf cip-mac-address {
21         type ieee:mac-address;
22         config false;
23         description
24             "The source MAC address of a CIM, belonging to the system
25             transmitting the CIM.";
26         reference
27             "49.4.1.2.1 of IEEE Std 802.1Q";
28     }
29     leaf cip-ipv4-address {
30         type inet:ipv4-address;
31         config false;
32         description
33             "The source IPv4 address of a CIM, belonging to the system
34             transmitting the IPv4 layer-3 CIM.";
35         reference
36             "49.4.1.2.2 of IEEE Std 802.1Q";
37     }
38     leaf cip-ipv6-address {
39         type inet:ipv6-address;
40         config false;
41         description
42             "The source IPv6 address of a CIM, belonging to the system
43             transmitting the IPv6 layer-3 CIM.";
44         reference
45             "49.4.1.2.3 of IEEE Std 802.1Q";
46     }
47     leaf cip-cim-port {
48         type inet:port-number;
49         config false;
50         description
51             "The UDP port number to be used by a peer transmitting a layer-3
52             CIM. This value will be sent to the peer via LLDP in the CI TLV.";
53         reference
54             "49.4.1.2.4 of IEEE Std 802.1Q";
55     }
56     list queue-map {
57         key "priority";
58         description
59             "An array of integers, one entry for each traffic class, 0
60             through 7, specifying a value that can be translated to the
61             numeric value of the traffic class to be used as either the
62             congesting traffic class or the monitored traffic class for the
63             traffic class specified by the index. The integers range in value
64             from -8 to 8. A value of 0 in the table specifies that the
65             traffic class is not participating in congestion isolation. A
66             positive number specifies a traffic class for a monitored queue
67             that is one less than the value (e.g., a value of 5 represents
68             traffic class 4). A negative number specifies a traffic class for
69             a congesting queue that is one less than the absolute value (e.g.,
70             a value of -4 represents traffic class 3).";
71         reference
72             "49.4.1.2.5 of IEEE Std 802.1Q";

```

```

1     leaf priority {
2         type dot1q-types:priority-type;
3         description
4             "There are eight values of Priority that map to an absolute
5             value that is a traffic class plus one, or the value 0 which
6             indicates the traffic class is not used by congestion
7             isolation.";
8         reference
9             "49.4.1.2.5 of IEEE Std 802.1Q";
10    }
11    leaf abs-traffic-class-plus-one {
12        type abs-traffic-class-plus-one-type;
13        description
14            "A value that can be translated to represent a traffic class or
15            an indication of non-use. A value of 0 specifies that the
16            traffic class is not participating in congestion isolation. A
17            positive number specifies a traffic class for a monitored
18            queue, and a negative number specifies a traffic class for a
19            congesting queue.";
20        reference
21            "49.4.1.2.5 of IEEE Std 802.1Q";
22    }
23 }
24 leaf min-header-octets {
25     type uint16;
26     description
27         "The minimum number of octets to include in the Encapsulated MSDU
28         field of each CIM generated. The default value is 48.";
29     reference
30         "49.4.1.2.6 of IEEE Std 802.1Q";
31 }
32 leaf max-cim-tx {
33     type uint16;
34     description
35         "The maximum number of times a CIM PDU will be sent for a
36         congesting flow. The default value is 3.";
37     reference
38         "49.4.1.2.7 of IEEE Std 802.1Q";
39 }
40 }
41 grouping cip-parameters {
42     description
43         "The system specific configuration for Congestion Isolation.";
44     leaf ci-master-enable {
45         type boolean;
46         description
47             "Specifies whether CI is enabled in this system.";
48         reference
49             "49.4.1.1.1 of IEEE Std 802.1Q";
50     }
51     leaf ci-cim-tx-priority {
52         type dot1q-types:priority-type;
53         config false;
54         description
55             "Specifies the priority value to be used when transmitting CIMs
56             from the system. The default is 6.";
57         reference
58             "49.4.1.1.2 of IEEE Std 802.1Q";
59     }
60     leaf ci-max-flow-life {
61         type uint32;
62         config false;
63         description
64             "Specifies the maximum number of centiseconds that a congesting
65             flow entry, created by the receipt of a CIM, can remain in the CI
66             Stream Table after the congesting queue has transitioned from
67             congested back to non-congested. The default value is 100.";
68         reference
69             "49.4.1.1.3 of IEEE Std 802.1Q";
70     }
71 }
72 container ci-peers {
73     description

```

```

1      "Contains information about an immediate peer obtained from a
2      received LLDP Congestion Isolation TLV.";
3  list ci-peer-table {
4      key "reception-port";
5      description
6          "Contains entries for each participating immediate peer and
7          provides the information needed to generate a CIM for
8          transmission to the peer.";
9      reference
10         "49.3.6 of IEEE Std 802.1Q";
11  leaf reception-port {
12      type dot1q-types:port-number-type;
13      description
14          "The port number where the immediate congestion isolation
15          participating peer is attached.";
16      reference
17         "49.3.6 of IEEE Std 802.1Q";
18  }
19  leaf cim-type {
20      type enumeration {
21          enum l2 {
22              description
23                  "Layer 2 CIM encapsulation.";
24          }
25          enum ipv4 {
26              description
27                  "IPv4 CIM encapsulation.";
28          }
29          enum ipv6 {
30              description
31                  "IPv6 CIM encapsulation.";
32          }
33      }
34      description
35          "The format of the CIM expected by the peer.";
36      reference
37         "49.3.6 of IEEE Std 802.1Q";
38  }
39  leaf peer-mac-address {
40      type ieee:mac-address;
41      description
42          "The destination MAC address to use when generating a CIM for
43          the peer.";
44      reference
45         "49.3.6 of IEEE Std 802.1Q";
46  }
47  leaf peer-ipv4-address {
48      type inet:ipv4-address;
49      description
50          "The destination IPv4 address to use when generating an IPv4
51          layer-3 CIM for the peer.";
52      reference
53         "49.3.6 of IEEE Std 802.1Q";
54  }
55  leaf peer-ipv6-address {
56      type inet:ipv6-address;
57      description
58          "The destination IPv6 address to use when generating an IPv6
59          layer-3 CIM for the peer.";
60      reference
61         "49.3.6 of IEEE Std 802.1Q";
62  }
63  leaf peer-udp-port {
64      type inet:port-number;
65      description
66          "The UDP port number to use when generating a layer-3 CIM for
67          the peer.";
68      reference
69         "49.3.6 of IEEE Std 802.1Q";
70  }
71  leaf peer-cim-encap-len {
72      type uint16 {

```



```

1      range "0..512";
2    }
3    description
4      "The number of octets from the MSDU to include in the
5      Encapsulated MSDU field of the CIM PDU";
6    reference
7      "49.3.6 of IEEE Std 802.1Q";
8  }
9  }
10 leaf max-ci-peer-entries {
11   type uint32;
12   config false;
13   description
14     "Specifies the maximum number of CI peer entries that can be
15     stored.";
16   reference
17     "49.3.6 of IEEE Std 802.1Q";
18 }
19 }
20 container ci-streams {
21   description
22     "Contains control information to manage congestion flows.";
23   list ci-stream-table {
24     key "stream-handle-id";
25     config false;
26     description
27       "Contains entries for each congesting flow and has a 1:1
28       mapping to entries in the IEEE Std 802.1CB Stream Identity
29       Table.";
30     reference
31       "49.3.7 of IEEE Std 802.1Q";
32     leaf stream-handle-id {
33       type uint32;
34       config false;
35       description
36         "There is a unique stream handle ID for each congesting flow
37         stored in the CI Stream Table.";
38       reference
39         "49.4.1.5.1 of IEEE Std 802.1Q";
40     }
41     leaf cim-count {
42       type uint16;
43       config false;
44       description
45         "Contains a count of the number of CIMs sent for a congesting
46         flow.";
47       reference
48         "49.4.1.5.2 of IEEE Std 802.1Q";
49     }
50     leaf create-time {
51       type yang:timeticks;
52       config false;
53       description
54         "The time (SysUpTime, IETF RFC 3418) at which the CI Stream
55         Table entry was created.";
56       reference
57         "49.4.1.5.3 of IEEE Std 802.1Q";
58     }
59     leaf create-mask {
60       type bits {
61         bit local-create {
62           position 0;
63           description
64             "CI Stream entry was created because congestion was
65             detected locally by the AQM.";
66         }
67         bit cim-create {
68           position 1;
69           description
70             "CI Stream entry was created because of the receipt of a
71             CIM.";
72         }
73       }
74     }
75   }
76 }

```

```

1      }
2      config false;
3      description
4          "Indicates the reason for creating or updating the CI Stream
5          Table entry. The LSB indicates that the entry was created or
6          updated because the AQM determined that a forwarded frame is
7          part of a congesting flow. The MSB indicates that the entry
8          was created or updated because of the receipt of a CIM.";
9      reference
10         "49.4.1.5.4 of IEEE Std 802.1Q";
11    }
12 }
13 leaf queue-key {
14     type uint16;
15     config false;
16     description
17         "Represents the unique queue identity of a congesting queue at
18         an egress port that a congesting flow is traversing. The key is
19         calculated by the product of the congesting traffic class
20         number plus one and the port number.";
21     reference
22         "49.4.1.5.5 of IEEE Std 802.1Q";
23 }
24 leaf dest-mac-address {
25     type ieee:mac-address;
26     config false;
27     description
28         "The destination MAC address of a congesting flow.";
29     reference
30         "49.4.1.5.6 of IEEE Std 802.1Q";
31 }
32 leaf source-mac-address {
33     type ieee:mac-address;
34     config false;
35     description
36         "The source MAC address of a congesting flow.";
37     reference
38         "49.4.1.5.7 of IEEE Std 802.1Q";
39 }
40 leaf vid {
41     type dot1q-types:vlan-index-type;
42     config false;
43     description
44         "The VID of a congesting flow.";
45     reference
46         "49.4.1.5.8 of IEEE Std 802.1Q";
47 }
48 leaf msdu {
49     type yang:hex-string;
50     config false;
51     description
52         "The initial octets of the MSDU of a congesting flow. The number
53         of octets to keep in the CI Stream Table entry is specified by
54         the peer CIM encapsulation length of the CI Peer Table.";
55     reference
56         "49.4.1.5.9 of IEEE Std 802.1Q";
57 }
58 }
59 leaf max-ci-stream-entries {
60     type uint32;
61     config false;
62     description
63         "Specifies the maximum number of CI stream entries that can be
64         stored.";
65     reference
66         "49.4.1.5 of IEEE Std 802.1Q";
67 }
68 }
69
70 /*-----*/
71 /* Configuration Data */
72 /*-----*/

```

```
1  uses sfsg:sfsg-parameters {
2    augment
3      "stream-filters/stream-filter-instance-table/stream-handle-spec" {
4        if-feature "congestion-isolation";
5        description
6          "Dot1q Congestion Isolation";
7        case null-handle {
8          description
9            "Congestion isolation specific stream_handle specifications.";
10         leaf null-handle {
11           type empty;
12           description
13             "The stream_handle specification represents the value when no
14             stream_handle is provided.";
15         }
16       }
17     }
18   }
19 }
20
```

1 48.6.16 The ieee802-dot1q-congestion-isolation-bridge YANG module

```
2 module ieee802-dot1q-congestion-isolation-bridge {
3   yang-version "1.1";
4   namespace
5     urn:ieee:std:802.1Q:yang:ieee802-dot1q-congestion-isolation-bridge;
6   prefix ci-bridge;
7   import ietf-interfaces {
8     prefix if;
9   }
10  import ieee802-dot1q-bridge {
11    prefix dot1q;
12  }
13  import ieee802-dot1q-congestion-isolation {
14    prefix ci;
15  }
16  import ieee802-dot1q-stream-filters-gates {
17    prefix sfsg;
18  }
19  organization
20    "IEEE 802.1 Working Group";
21  contact
22    "WG-URL: https://1.ieee802.org/
23    WG-EMail: stds-802-1@ieee.org
24    Contact: IEEE 802.1 Working Group Chair
25    Postal: C/O IEEE 802.1 Working Group
26            IEEE Standards Association
27            445 Hoes Lane
28            Piscataway, NJ 08855
29            USA
30
31    E-mail: stds-802-1-chairs@listserv.ieee.org";
32  description
33    "This module provides management of IEEE 802.1Q Bridge components
34    that support Congestion Isolation.
35
36    Copyright (C) IEEE (2023).
37
38    This version of this YANG module is part of IEEE Std 802.1Q; see the
39    standard itself for full legal notices.";
40  revision 2023-07-03 {
41    description
42      "Published as part of IEEE Std 802.1Qcz-2023.";
43    reference
44      "IEEE Std 802.1Qcz-2023 - Bridges and Bridged Networks - Amendment:
45      Congestion Isolation.";
46  }
47  feature congestion-isolation-bridge {
48    description
49      "Feature Congestion Isolation in Bridges";
50  }
51  augment "/dot1q:bridges/dot1q:bridge/dot1q:component" {
52    if-feature "congestion-isolation-bridge";
53    description
54      "Augments the Bridge component with stream filters and stream
55      gates.";
56    uses ci:cip-parameters;
57    uses sfsg:sfsg-parameters {
58      augment
59        "stream-filters/stream-filter-instance-table/stream-handle-spec" {
60        if-feature "congestion-isolation-bridge";
61        description
62          "Dot1q Congestion Isolation";
63        case null-handle {
64          description
65            "Congestion isolation specific stream_handle specifications.";
66          leaf null-handle {
67            type empty;
68            description
69              "The stream_handle specification represents the value when
70              no stream_handle is provided.";
71          }
64
```

```
1         }
2     }
3 }
4 }
5 augment "/if:interfaces/if:interface/dot1q:bridge-port" {
6     if-feature "congestion-isolation-bridge";
7     description
8         "Augment a bridge port with Congestion Isolation specific
9         configuration.";
10    uses ci:cip-port-parameters;
11 }
12 }
13
```

1 48.6.17 The ieee802-dot1q-sched YANG module

```
2 module ieee802-dot1q-sched {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-sched;
5   prefix sched;
6   import ietf-yang-types {
7     prefix yang;
8   }
9   import ieee802-types {
10    prefix ieee802;
11  }
12  import ieee802-dot1q-types {
13    prefix dot1q-types;
14  }
15  organization
16    "IEEE 802.1 Working Group";
17  contact
18    "WG-URL: http://www.ieee802.org/1/
19    WG-EMail: stds-802-1-1@ieee.org
20
21    Contact: IEEE 802.1 Working Group Chair
22    Postal: C/O IEEE 802.1 Working Group
23    IEEE Standards Association
24    445 Hoes Lane
25    Piscataway, NJ 08854
26    USA
27
28    E-mail: STDS-802-1-CHAIRS@IEEE.ORG";
29  description
30    "This module provides for management of IEEE Std 802.1Q Bridges that
31    support Scheduled Traffic Enhancements.
32
33    Copyright (C) IEEE (2024).
34
35    This version of this YANG module is part of IEEE Std 802.1Q; see the
36    standard itself for full legal notices.";
37  revision 2024-10-22 {
38    description
39      "Published as part of IEEE Std 802.1Q-202X. Revised must statements
40      allow succesful validation of non-scheduled interfaces when the
41      module is loaded.
42
43      The following reference statement identifies each referenced IEEE
44      Standard as updated by applicable amendments.";
45    reference
46      "IEEE Std 802.1Q Bridges and Bridged Networks: IEEE Std 802.1Q-202X.";
47  }
48  revision 2023-10-22 {
49    description
50      "Published as part of IEEE Std 802.1Qcw-2023.
51
52      The following reference statement identifies each referenced IEEE
53      Standard as updated by applicable amendments.";
54    reference
55      "IEEE Std 802.1Q Bridges and Bridged Networks:
56      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.";
57  }
58  feature scheduled-traffic {
59    description
60      "Enhancements for Scheduled Traffic supported.";
61    reference
62      "IEEE Std 802.1Q";
63  }
64  identity set-gate-states {
65    base dot1q-types:type-of-operation;
66    description
67      "Operation to set the gate states.";
68  }
69  identity set-and-hold-mac {
70    base dot1q-types:type-of-operation;
71    description
```

```

1     "Operation to set and hold MAC.";
2 }
3 identity set-and-release-mac {
4     base dot1q-types:type-of-operation;
5     description
6         "Operation to set and release MAC.";
7 }
8 grouping sched-gate-control-entries {
9     description
10        "A GateControlEntry consists of an operation name, followed by up to 2
11        parameters associated with the operation. The first parameter is a
12        gateStatesValue; the second parameter is a timeIntervalValue";
13    uses dot1q-types:base-gate-control-entries {
14        refine "gate-control-entry/operation-name" {
15            must
16                "(. = 'sched:set-gate-states') or (. = '+'
17                'sched:set-and-hold-mac') or (. = 'sched:set-and-release-mac')";
18        }
19        refine "gate-control-entry/time-interval-value" {
20            must
21                "not (. > ../../../../supported-interval-max )";
22        }
23        augment "gate-control-entry" {
24            description
25                "Augment gate-control-entry from base-gate-control-entries with
26                parameter gate-states-value.";
27            leaf gate-states-value {
28                type uint8;
29                mandatory true;
30                description
31                    "gateStatesValue is the gate states for this entry for the Port.
32                    The gates are immediately set to the states in gateStatesValue
33                    when this entry executes. The bits of the octet represent the
34                    gate states for the corresponding traffic classes; the
35                    most-significant bit corresponds to traffic class 7, the
36                    least-significant bit to traffic class 0. A bit value of 0
37                    indicates closed; a bit value of 1 indicates open.";
38                reference
39                    "12.29.1.2.2 and 8.6.8.4 of IEEE Std 802.1Q.";
40            }
41        }
42    }
43 }
44 grouping sched-parameters {
45     description
46         "sched-parameters comprises all of the parameters associated with
47         Scheduled Traffic configuration.";
48     container gate-parameter-table {
49         description
50             "A table that contains the per-port manageable parameters for
51             traffic scheduling. For a given Port, an entry in the table exists.
52             All writable objects in this table must be persistent over power up
53             restart/reboot.";
54         reference
55             "12.29.1, 8.6.8.4 and 8.6.9 of IEEE Std 802.1Q";
56         list queue-max-sdu-table {
57             key "traffic-class";
58             description
59                 "A list containing a set of max SDU parameters, one for each
60                 traffic class. All writable objects in this table must be
61                 persistent over power up restart/reboot.";
62             reference
63                 "12.29.1.1, 8.6.8.4 and 8.6.9 of IEEE Std 802.1Q";
64             leaf traffic-class {
65                 type dot1q-types:traffic-class-type;
66                 description
67                     "Traffic class";
68             }
69             leaf queue-max-sdu {
70                 type uint32;
71                 default "0";
72                 description

```

```

1      "The value of the queueMaxSDU parameter for the traffic class. A
2      value of 0 is interpreted as the max SDU size supported by the
3      underlying MAC. The value must be retained across
4      reinitializations of the management system.";
5      reference
6      "12.29.1.1.1, 8.6.8.4 and 8.6.9 of IEEE Std 802.1Q.";
7  }
8  leaf transmission-overflow {
9      type yang:counter64;
10     default "0";
11     config false;
12     description
13         "A counter of transmission overflow events, where a PDU is still
14         being transmitted by a MAC at the time when the transmission
15         gate for the queue closed.";
16     reference
17         "12.29.1.1.2, 8.6.8.4 and 8.6.9 of IEEE Std 802.1Q.";
18 }
19 }
20 leaf gate-enabled {
21     type boolean;
22     default "false";
23     description
24         "The GateEnabled parameter determines whether traffic scheduling
25         is active (true) or inactive (false). The value must be retained
26         across reinitializations of the management system.";
27     reference
28         "12.29.1, 8.6.8.2 and 8.6.9.4.14 of IEEE Std 802.1Q.";
29 }
30 leaf admin-gate-states {
31     type uint8;
32     default "255";
33     description
34         "AdminGateStates is the administrative value of the initial gate
35         states for the Port. The bits of the octet represent the gate
36         states for the corresponding traffic classes; the most-significant
37         bit corresponds to traffic class 7, the least-significant bit to
38         traffic class 0. A bit value of 0 indicates closed; a bit value of
39         1 indicates open. The value must be retained across
40         reinitializations of the management system.";
41     reference
42         "12.29.1 and 8.6.9.4.5 of IEEE Std 802.1Q.";
43 }
44 leaf oper-gate-states {
45     type uint8;
46     config false;
47     description
48         "OperGateStates is the operational value of the current gate
49         states for the Port. The bits of the octet represent the gate
50         states for the corresponding traffic classes; the most-significant
51         bit corresponds to traffic class 7, the least-significant bit to
52         traffic class 0. A bit value of 0 indicates closed; a bit value of
53         1 indicates open.";
54     reference
55         "12.29.1 and 8.6.9.4.21 of IEEE Std 802.1Q.";
56 }
57 container admin-control-list {
58     must
59         "not(count(../gate-control-entry) > ../supported-list-max)" {
60         error-message
61             "Number of elements in admin-control-list must not be greater "+
62             "than supported-list-max";
63         }
64     description
65         "AdminControlList is the administrative value of the gate control
66         list for the Port. The value must be retained across
67         reinitializations of the management system.";
68     reference
69         "12.29.1.2, 8.6.8.4 and 8.6.9.4.2 of IEEE Std 802.1Q.";
70     uses sched-gate-control-entries;
71 }
72 container oper-control-list {

```



```

1      must
2      "not (count (./gate-control-entry) > ../supported-list-max)" {
3      error-message
4      "Number of elements in oper-control-list must not be greater "+
5      "than supported-list-max";
6      }
7      config false;
8      description
9      "OperControlList is the operational value of the gate control list
10     for the Port.";
11     reference
12     "12.29.1.2, 8.6.8.4 and 8.6.9.4.18 of IEEE Std 802.1Q.";
13     uses sched-gate-control-entries;
14 }
15 container admin-cycle-time {
16     must
17     "not (./numerator div ./denominator > "+
18     "../supported-cycle-max/numerator div "+
19     "../supported-cycle-max/denominator )" {
20     error-message
21     "admin-cycle-time must not be greater than supported-cycle-max";
22     }
23     description
24     "AdminCycleTime specifies the administrative value of the gating
25     cycle time for the Port. AdminCycleTime is a rational number of
26     seconds, defined by an integer numerator and an integer
27     denominator. The value must be retained across reinitializations
28     of the management system.";
29     reference
30     "12.29.1, 8.6.8.4 and 8.6.9.4.3 of IEEE Std 802.1Q.";
31     uses ieee802:rational-grouping;
32 }
33 container oper-cycle-time {
34     must
35     "not (./numerator div ./denominator > "+
36     "../supported-cycle-max/numerator div "+
37     "../supported-cycle-max/denominator )" {
38     error-message
39     "oper-cycle-time must not be greater than supported-cycle-max";
40     }
41     config false;
42     description
43     "OperCycleTime specifies the operational value of the gating cycle
44     time for the Port. OperCycleTime is a rational number of seconds,
45     defined by an integer numerator and an integer denominator.";
46     reference
47     "12.29.1, 8.6.8.4 and 8.6.9.4.19 of IEEE Std 802.1Q.";
48     uses ieee802:rational-grouping;
49 }
50 leaf admin-cycle-time-extension {
51     type uint32;
52     units "nanoseconds";
53     description
54     "An unsigned integer number of nanoseconds, defining the maximum
55     amount of time by which the gating cycle for the Port is permitted
56     to be extended when a new cycle configuration is being installed.
57     This is the administrative value. The value must be retained
58     across reinitializations of the management system.";
59     reference
60     "12.29.1 and 8.6.9.4.4 of IEEE Std 802.1Q.";
61 }
62 leaf oper-cycle-time-extension {
63     type uint32;
64     units "nanoseconds";
65     config false;
66     description
67     "An unsigned integer number of nanoseconds, defining the maximum
68     amount of time by which the gating cycle for the Port is permitted
69     to be extended when a new cycle configuration is being installed.
70     This is the operational value.";
71     reference
72     "12.29.1 and 8.6.9.4.20 of IEEE Std 802.1Q.";

```

```

1      }
2      container admin-base-time {
3          description
4              "The administrative value of the base time at which gating cycles
5              begin, expressed as an IEEE 1588 precision time protocol (PTP)
6              timescale. The value must be retained across reinitializations of
7              the management system.";
8          reference
9              "12.29.1 and 8.6.9.4.1 of IEEE Std 802.1Q.";
10         uses ieee802:ptp-time-grouping;
11     }
12     container oper-base-time {
13         config false;
14         description
15             "The operational value of the base time at which gating cycles
16             begin, expressed as an IEEE 1588 precision time protocol (PTP)
17             timescale.";
18         reference
19             "12.29.1 and 8.6.9.4.17 of IEEE Std 802.1Q.";
20         uses ieee802:ptp-time-grouping;
21     }
22     leaf config-change {
23         type boolean;
24         description
25             "The ConfigChange parameter signals the start of a configuration
26             change when it is set to TRUE, indicating that the administrative
27             parameters for the Port are ready to be copied into their
28             corresponding operational parameters. This should only be done
29             when the various administrative parameters are all set to
30             appropriate values.";
31         reference
32             "12.29.1 and 8.6.9.4.7 of IEEE Std 802.1Q.";
33     }
34     container config-change-time {
35         config false;
36         description
37             "The time at which the next config change is scheduled to occur.";
38         reference
39             "12.29.1 and 8.6.9.4.9 of IEEE Std 802.1Q.";
40         uses ieee802:ptp-time-grouping;
41     }
42     leaf tick-granularity {
43         type uint32;
44         config false;
45         description
46             "The granularity of the cycle time clock, represented as an
47             unsigned number of tenths of nanoseconds. The value must be
48             retained across reinitializations of the management system.";
49         reference
50             "12.29.1 of IEEE Std 802.1Q";
51     }
52     container current-time {
53         config false;
54         description
55             "The current time as maintained by the local system.";
56         reference
57             "12.29.1 and 8.6.9.4.10 of IEEE Std 802.1Q.";
58         uses ieee802:ptp-time-grouping;
59     }
60     leaf config-pending {
61         type boolean;
62         config false;
63         description
64             "The value of the ConfigPending state machine variable. The value
65             is TRUE if a configuration change is in progress but has not yet
66             completed.";
67         reference
68             "12.29.1 and 8.6.9.4.8 of IEEE Std 802.1Q.";
69     }
70     leaf config-change-error {
71         type yang:counter64;
72         config false;

```

```
1      description
2      "A counter of the number of times that a re-configuration of the
3      traffic schedule has been requested with the old schedule still
4      running and the requested base time was in the past.";
5      reference
6      "12.29.1 and 8.6.9.3.1 of IEEE Std 802.1Q.";
7  }
8  leaf supported-list-max {
9      type uint32;
10     description
11     "The maximum value supported by this Port for the
12     AdminControlListLength and OperControlListLength parameters. It is
13     available for use by schedule computation software to determine
14     the port's control list capacity prior to computation. The object
15     may optionally be read-only.";
16     reference
17     "12.29.1.5 of IEEE Std 802.1Q";
18 }
19 container supported-cycle-max {
20     description
21     "The maximum value supported by this Port of the AdminCycleTime
22     and OperCycleTime parameters. The object may optionally be
23     read-only.";
24     reference
25     "12.29.1.6 of IEEE Std 802.1Qcw";
26     uses ieee802:rational-grouping;
27 }
28 leaf supported-interval-max {
29     type uint32;
30     description
31     "The maximum value supported by this Port of the TimeIntervalValue
32     parameter. The object may optionally be read-only.";
33     reference
34     "12.29.1.7 of IEEE Std 802.1Qcw";
35 }
36 }
37 }
38 }
39
40
```

1 48.6.18 The ieee802-dot1q-sched-bridge YANG module

```
2 module ieee802-dot1q-sched-bridge {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-sched-bridge;
5   prefix sched-bridge;
6   import ietf-interfaces {
7     prefix if;
8   }
9   import ieee802-dot1q-bridge {
10    prefix dot1q;
11  }
12  import ieee802-dot1q-sched {
13    prefix sched;
14  }
15  organization
16    "IEEE 802.1 Working Group";
17  contact
18    "WG-URL: http://www.ieee802.org/1/
19    WG-EMail: stds-802-1-1@ieee.org
20
21    Contact: IEEE 802.1 Working Group Chair
22    Postal: C/O IEEE 802.1 Working Group
23    IEEE Standards Association
24    445 Hoes Lane
25    Piscataway, NJ 08854
26    USA
27
28    E-mail: STDS-802-1-CHAIRS@IEEE.ORG";
29  description
30    "This module provides for management of IEEE Std 802.1Q Bridges that
31    support Scheduled Traffic Enhancements.
32
33    Copyright (C) IEEE (2023).
34
35    This version of this YANG module is part of IEEE Std 802.1Q; see the
36    standard itself for full legal notices.";
37  revision 2023-10-26 {
38    description
39      "Published as part of IEEE Std 802.1Qcw-2023.
40
41      The following reference statement identifies each referenced IEEE
42      Standard as updated by applicable amendments.";
43    reference
44      "IEEE Std 802.1Q Bridges and Bridged Networks:
45      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.";
46  }
47  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
48    description
49      "Augment bridge-port with Scheduled Traffic configuration.";
50    uses sched:sched-parameters;
51  }
52 }
53
54
```

1 48.6.19 The ieee802-dot1q-preemption YANG module

```
2 module ieee802-dot1q-preemption {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-preemption;
5   prefix preempt;
6   organization
7     "IEEE 802.1 Working Group";
8   contact
9     "WG-URL: http://www.ieee802.org/1/
10    WG-EMail: stds-802-1-1@ieee.org
11
12    Contact: IEEE 802.1 Working Group Chair
13    Postal: C/O IEEE 802.1 Working Group
14    IEEE Standards Association
15    445 Hoes Lane
16    Piscataway, NJ 08854
17    USA
18
19    E-mail: STDS-802-1-CHAIRS@IEEE.ORG";
20  description
21    "This module provides for management of IEEE Std 802.1Q Bridges that
22    support frame preemption.
23
24    Copyright (C) IEEE (2023).
25
26    This version of this YANG module is part of IEEE Std 802.1Q; see the
27    standard itself for full legal notices.";
28  revision 2023-10-26 {
29    description
30      "Published as part of IEEE Std 802.1Qcw-2023.
31
32      The following reference statement identifies each referenced IEEE
33      Standard as updated by applicable amendments.";
34    reference
35      "IEEE Std 802.1Q Bridges and Bridged Networks:
36      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.";
37  }
38  feature frame-preemption {
39    description
40      "Frame preemption supported.";
41    reference
42      "IEEE Std 802.1Q";
43  }
44  typedef frame-preemption-status-enum {
45    type enumeration {
46      enum express {
47        description
48          "Frames queued for the priority are to be transmitted using the
49          express service for the Port.";
50      }
51      enum preemptable {
52        description
53          "Frames queued for the priority are to be transmitted using the
54          preemptable service for the Port.";
55      }
56    }
57    description
58      "Frame preemption status of each priority.";
59  }
60  grouping preemption-parameters {
61    description
62      "preemption-parameters comprises all of the parameters associated with
63      frame preemption configuration.";
64    container frame-preemption-parameters {
65      description
66        "A table containing a set of frame preemption parameters, one for
67        each Port. All writeable objects in this table must be persistent
68        over power up restart/reboot.";
69      reference
70        "12.30.1 of IEEE Std 802.1Q";
71      container frame-preemption-status-table {
```

```
1      description
2          "The framePreemptionStatusTable consists of 8 frame-preemption
3          status parameter leafs, one per priority";
4      reference
5          "12.30.1.1 of IEEE Std 802.1Q";
6      leaf priority0 {
7          type frame-preemption-status-enum;
8          default "express";
9          description
10             "The value of the frame-preemption status parameter for the
11             priority 0. The default value of the parameter is express (1).
12             The value of this object MUST be retained across
13             reinitializations of the management system.";
14          reference
15             "12.30.1.1.1 of IEEE Std 802.1Q";
16      }
17      leaf priority1 {
18          type frame-preemption-status-enum;
19          default "express";
20          description
21             "The value of the frame-preemption status parameter for the
22             priority 1. The default value of the parameter is express (1).
23             The value of this object MUST be retained across
24             reinitializations of the management system.";
25          reference
26             "12.30.1.1.1 of IEEE Std 802.1Q";
27      }
28      leaf priority2 {
29          type frame-preemption-status-enum;
30          default "express";
31          description
32             "The value of the frame-preemption status parameter for the
33             priority 2. The default value of the parameter is express (1).
34             The value of this object MUST be retained across
35             reinitializations of the management system.";
36          reference
37             "12.30.1.1.1 of IEEE Std 802.1Q";
38      }
39      leaf priority3 {
40          type frame-preemption-status-enum;
41          default "express";
42          description
43             "The value of the frame-preemption status parameter for the
44             priority 3. The default value of the parameter is express (1).
45             The value of this object MUST be retained across
46             reinitializations of the management system.";
47          reference
48             "12.30.1.1.1 of IEEE Std 802.1Q";
49      }
50      leaf priority4 {
51          type frame-preemption-status-enum;
52          default "express";
53          description
54             "The value of the frame-preemption-status parameter for the
55             priority 4. The default value of the parameter is express (1).
56             The value of this object MUST be retained across
57             reinitializations of the management system.";
58          reference
59             "12.30.1.1.1 of IEEE Std 802.1Q";
60      }
61      leaf priority5 {
62          type frame-preemption-status-enum;
63          default "express";
64          description
65             "The value of the frame-preemption status parameter for the
66             priority 5. The default value of the parameter is express (1).
67             The value of this object MUST be retained across
68             reinitializations of the management system.";
69          reference
70             "12.30.1.1.1 of IEEE Std 802.1Q";
71      }
72      leaf priority6 {
```

```

1      type frame-preemption-status-enum;
2      default "express";
3      description
4          "The value of the frame-preemption status parameter for the
5          priority 6. The default value of the parameter is express (1).
6          The value of this object MUST be retained across
7          reinitializations of the management system.";
8      reference
9          "12.30.1.1.1 of IEEE Std 802.1Q";
10     }
11     leaf priority7 {
12         type frame-preemption-status-enum;
13         default "express";
14         description
15             "The value of the frame-preemption status parameter for the
16             priority 7. The default value of the parameter is express (1).
17             The value of this object MUST be retained across
18             reinitializations of the management system.";
19         reference
20             "12.30.1.1.1 of IEEE Std 802.1Q";
21     }
22 }
23 leaf hold-advance {
24     type uint32;
25     units "nanoseconds";
26     config false;
27     description
28         "The value of the holdAdvance parameter for the Port in
29         nanoseconds. There is no default value; the holdAdvance is a
30         property of the underlying MAC.";
31     reference
32         "12.30.1.2 of IEEE Std 802.1Q";
33 }
34 leaf release-advance {
35     type uint32;
36     units "nanoseconds";
37     config false;
38     description
39         "The value of the releaseAdvance parameter for the Port in
40         nanoseconds. There is no default value; the releaseAdvance is a
41         property of the underlying MAC.";
42     reference
43         "12.30.1.3 of IEEE Std 802.1Q";
44 }
45 leaf preemption-active {
46     type boolean;
47     config false;
48     description
49         "TRUE if preemption is both supported by the MAC and currently
50         active.";
51     reference
52         "12.30.1.4 of IEEE Std 802.1Q";
53 }
54 leaf hold-request {
55     type enumeration {
56         enum hold {
57             value 1;
58             description
59                 "A hold request has been issued to the MAC.";
60         }
61         enum release {
62             value 2;
63             description
64                 "A release request has been issued to the MAC.";
65         }
66     }
67     config false;
68     description
69         "The value of the holdRequest parameter for the Port, either hold
70         (1) or release (2). The value of this object is release (2) on
71         system initialization.";
72     reference

```

```
1      "12.30.1.5 of IEEE Std 802.1Q";  
2      }  
3  }  
4  }  
5 }  
6  
7
```


1 48.6.20 The ieee802-dot1q-preemption-bridge YANG module

```
2 module ieee802-dot1q-preemption-bridge {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-preemption-bridge;
5   prefix preempt-bridge;
6   import ietf-interfaces {
7     prefix if;
8   }
9   import ieee802-dot1q-bridge {
10    prefix dot1q;
11  }
12  import ieee802-dot1q-preemption {
13    prefix preempt;
14  }
15  organization
16    "IEEE 802.1 Working Group";
17  contact
18    "WG-URL: http://www.ieee802.org/1/
19    WG-EMail: stds-802-1-1@ieee.org
20
21    Contact: IEEE 802.1 Working Group Chair
22    Postal: C/O IEEE 802.1 Working Group
23    IEEE Standards Association
24    445 Hoes Lane
25    Piscataway, NJ 08854
26    USA
27
28    E-mail: STDS-802-1-CHAIRS@IEEE.ORG";
29  description
30    "This module provides for management of IEEE Std 802.1Q Bridges that
31    support frame preemption.
32
33    Copyright (C) IEEE (2023).
34
35    This version of this YANG module is part of IEEE Std 802.1Q; see the
36    standard itself for full legal notices.";
37  revision 2023-10-26 {
38    description
39      "Published as part of IEEE Std 802.1Qcw-2023.
40
41      The following reference statement identifies each referenced IEEE
42      Standard as updated by applicable amendments.";
43    reference
44      "IEEE Std 802.1Q Bridges and Bridged Networks:
45      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.";
46  }
47  feature frame-preemption {
48    description
49      "Frame preemption supported.";
50    reference
51      "IEEE Std 802.1Q";
52  }
53  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
54    if-feature "frame-preemption";
55    description
56      "Augment bridge-port with frame preemption configuration.";
57    uses preempt:preemption-parameters;
58  }
59 }
60
61
```

1 48.6.21 The ieee802-dot1q-psfp YANG module

```
2 module ieee802-dot1q-psfp {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-psfp;
5   prefix psfp;
6   import ietf-yang-types {
7     prefix yang;
8   }
9   import ieee802-types {
10    prefix ieee802;
11  }
12  import ieee802-dot1q-types {
13    prefix dot1q-types;
14  }
15  import ieee802-dot1q-stream-filters-gates {
16    prefix sfsg;
17  }
18  organization
19    "IEEE 802.1 Working Group";
20  contact
21    "WG-URL: http://www.ieee802.org/1/
22    WG-EMail: stds-802-1-1@ieee.org
23
24    Contact: IEEE 802.1 Working Group Chair
25    Postal: C/O IEEE 802.1 Working Group
26    IEEE Standards Association
27    445 Hoes Lane
28    Piscataway, NJ 08854
29    USA
30
31    E-mail: STDS-802-1-CHAIRS@IEEE.ORG";
32  description
33    "This module provides management of IEEE Std 802.1Q Bridge
34    components that support Per-Stream Filtering and Policing (PSFP).
35
36    Copyright (C) IEEE (2024).
37
38    This version of this YANG module is part of IEEE Std 802.1Q; see the
39    standard itself for full legal notices.";
40  revision 2024-10-22 {
41    description
42      "Published as part of IEEE Std 802.1Q-202X. Revised must statements
43      allow succesful validation of non-scheduled interfaces when the
44      module is loaded.
45
46      The following reference statement identifies each referenced IEEE
47      Standard as updated by applicable amendments.";
48    reference
49      "IEEE Std 802.1Q Bridges and Bridged Networks: IEEE Std 802.1Q-202X.";
50  }
51  revision 2023-10-26 {
52    description
53      "Published as part of IEEE Std 802.1Qcw-2023.
54
55      The following reference statement identifies each referenced IEEE
56      Standard as updated by applicable amendments.";
57    reference
58      "IEEE Std 802.1Q Bridges and Bridged Networks:
59      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.";
60  }
61  feature psfp {
62    description
63      "Per-Stream Filtering and Policing supported.";
64    reference
65      "IEEE Std 802.1Q";
66  }
67  identity set-gate-and-ipv {
68    base dot1q-types:type-of-operation;
69    description
70      "The StreamGateState parameter specifies a desired state, open or
71      closed, for the stream gate, and the IPV parameter specifies a desired
```

```

1     value of the IPV associated with the stream. On execution, the
2     StreamGateState and IPV parameter values are used to set the
3     operational values of the stream gate state and internal priority
4     specification parameters for the stream. After TimeInterval ticks have
5     elapsed since the completion of the previous stream gate control
6     operation in the stream gate control list, control passes to the next
7     stream gate control operation. The optional IntervalOctetMax parameter
8     specifies the maximum number of MSDU octets that are permitted to pass
9     the gate during the specified TimeInterval. If the IntervalOctetMax
10    parameter is omitted, there is no limit on the number of octets that
11    can pass the gate.";
12    reference
13    "8.6.5.4 of IEEE Std 802.1Q";
14  }
15  grouping stream-gate-control-entries {
16    description
17    "A GateControlEntry consists of an operation name, followed by three
18    mandatory parameters and one optional parameter associated with the
19    operation. The first parameter is a gateStatesValue; the second
20    parameter is an IPV value; the third parameter is a timeIntervalValue;
21    and the fourth parameter is an IntervalOctetMaxValue.
22    IntervalOctetMaxValue is optional.";
23    uses dot1q-types:base-gate-control-entries {
24      refine "gate-control-entry/operation-name" {
25        must
26        ". = 'psfp:set-gate-and-ipv'";
27      }
28      refine "gate-control-entry/time-interval-value" {
29        must
30        "not ( . > ../../../../../supported-interval-max )";
31      }
32      augment "gate-control-entry" {
33        description
34        "Augment gate-control-entry from base-gate-control-entries with
35        parameters gate-state-value, ipv-spec and interval-octet-max.";
36        leaf gate-state-value {
37          type sfsg:gate-state-value-type;
38          mandatory true;
39          description
40          "The PSFPGateStatesValue indicates the desired gate state, open
41          or closed, for the stream gate.";
42          reference
43          "12.31.3.2.1 of IEEE Std 802.1Q";
44        }
45        leaf ipv-spec {
46          type sfsg:ipv-spec-type;
47          mandatory true;
48          description
49          "The IPV value indicates the IPV to be associated with frames
50          that pass the gate.";
51          reference
52          "12.31.3.2.3 of IEEE Std 802.1Q";
53        }
54        leaf interval-octet-max {
55          type uint32;
56          description
57          "An unsigned integer, denoting a IntervalOctetMax in MSDU
58          octets. If this parameter is omitted, there is no maximum.";
59          reference
60          "12.31.3.2.5 of IEEE Std 802.1Q";
61        }
62      }
63    }
64  }
65  grouping psfp-parameters {
66    description
67    "psfp-parameters comprises all of the parameters associated with
68    Per-Stream Filtering and Policing configuration.";
69    container flow-meters {
70      description
71      "This container comprises all flow meter related nodes.";
72      must

```

```

1      "not (count(/flow-meter-instance-table) > ./max-flow-meter-instances)" {
2      error-message
3      "Number of elements in flow-meter-instance-table must not be "+
4      "greater than max-flow-meter-instances.";
5  }
6  list flow-meter-instance-table {
7      key "flow-meter-instance-id";
8      description
9          "There is one Flow Meter Instance Table per Bridge component. Each
10         table row contains a set of parameters that defines a single Flow
11         Meter Instance. Tables can be created or removed dynamically in
12         implementations that support dynamic configuration of Bridge
13         components. Rows in the table can be created or removed
14         dynamically in implementations that support dynamic configuration
15         of flow meters.";
16      reference
17          "12.31.4 of IEEE Std 802.1Q";
18      leaf flow-meter-instance-id {
19          type uint32;
20          mandatory true;
21          description
22              "An integer table index that allows the Flow Meter to be
23              referenced from Stream Filter Instance Table entries.";
24          reference
25              "8.6.5.5 of IEEE Std 802.1Q";
26      }
27      leaf committed-information-rate {
28          type uint64;
29          units "bits/second";
30          mandatory true;
31          description
32              "Committed information rate (CIR), in bits per second.";
33          reference
34              "8.6.5.5 of IEEE Std 802.1Q";
35      }
36      leaf committed-burst-size {
37          type uint32;
38          units "octets";
39          mandatory true;
40          description
41              "Committed burst size (CBS), in octets.";
42          reference
43              "8.6.5.5 of IEEE Std 802.1Q";
44      }
45      leaf excess-information-rate {
46          type uint64;
47          units "bits/second";
48          mandatory true;
49          description
50              "Excess Information Rate (EIR), in bits per second.";
51          reference
52              "8.6.5.5 of IEEE Std 802.1Q";
53      }
54      leaf excess-burst-size {
55          type uint32;
56          units "octets";
57          mandatory true;
58          description
59              "Excess burst size (EBS) per bandwidth profile flow, in octets.";
60          reference
61              "8.6.5.5 of IEEE Std 802.1Q";
62      }
63      leaf coupling-flag {
64          type enumeration {
65              enum zero {
66                  value 0;
67                  description
68                      "Uncoupled";
69              }
70              enum one {
71                  value 1;
72                  description

```

```

1         "Coupled";
2     }
3 }
4 mandatory true;
5 description
6     "Coupling flag (CF), which takes the value 0 or 1.";
7 reference
8     "8.6.5.5 of IEEE Std 802.1Q";
9 }
10 leaf color-mode {
11     type enumeration {
12         enum color-blind {
13             description
14                 "Color-blind (i.e., the color of a frame is ignored by the
15                 bandwidth profile algorithm).";
16         }
17         enum color-aware {
18             description
19                 "Color-aware (i.e., the color of a frame is considered by
20                 the bandwidth profile algorithm).";
21         }
22     }
23     mandatory true;
24     description
25         "Color mode (CM), which takes the value color-blind or
26         color-aware, as specified in Bandwidth Profile Parameters and
27         Algorithm in MEF 10.3.";
28     reference
29         "8.6.5.5 of IEEE Std 802.1Q";
30 }
31 leaf drop-on-yellow {
32     type boolean;
33     mandatory true;
34     description
35         "DropOnYellow, which takes the value TRUE or FALSE. A value of
36         TRUE indicates that yellow frames are dropped (i.e., discarded);
37         a value of FALSE indicates that yellow frames will have the
38         drop_eligible parameter set to TRUE.";
39     reference
40         "8.6.5.5 of IEEE Std 802.1Q";
41 }
42 leaf mark-all-frames-red-enable {
43     type boolean;
44     default "false";
45     description
46         "MarkAllFramesRedEnable, which takes the value TRUE or FALSE. A
47         value of TRUE indicates that the MarkAllFramesRed function is
48         enabled; a value of FALSE indicates that the MarkAllFramesRed
49         function is disabled. The default value of MarkAllFramesRedEnable
50         is FALSE.";
51     reference
52         "8.6.5.5 of IEEE Std 802.1Q";
53 }
54 leaf mark-all-frames-red {
55     type boolean;
56     default "false";
57     description
58         "MarkAllFramesRed, which takes the value TRUE or FALSE. If
59         MarkAllFramesRedEnable is TRUE, a value of TRUE in
60         MarkAllFramesRed indicates that all frames are dropped (i.e.,
61         discarded). If MarkAllFramesRed is FALSE, it has no effect. The
62         default value of MarkAllFramesRed is FALSE; if the operation of
63         the flow meter causes any frame to be discarded, then
64         MarkAllFramesRed is set TRUE.";
65     reference
66         "8.6.5.5 of IEEE Std 802.1Q";
67 }
68 }
69 leaf max-flow-meter-instances {
70     type uint32;
71     description
72         "The maximum number of Flow Meter instances supported by this

```

```

1      Bridge component. The object may optionally be read-only.";
2  reference
3      "12.31.1.3 of IEEE Std 802.1Q";
4  }
5  }
6  uses sfsg:sfsg-parameters {
7      augment "stream-filters/stream-filter-instance-table" {
8          description
9              "Augment the Bridge component Stream filter by frame counters.";
10         leaf matching-frames-count {
11             type yang:counter64;
12             config false;
13             description
14                 "A count of frames matching both the stream_handle and priority
15                 specifications.";
16             reference
17                 "8.6.5.3 of IEEE Std 802.1Q";
18         }
19         leaf passing-frames-count {
20             type yang:counter64;
21             config false;
22             description
23                 "A count of frames that passed the stream gate.";
24             reference
25                 "8.6.5.3 of IEEE Std 802.1Q";
26         }
27         leaf not-passing-frames-count {
28             type yang:counter64;
29             config false;
30             description
31                 "A count of frames that did not pass the stream gate.";
32             reference
33                 "8.6.5.3 of IEEE Std 802.1Q";
34         }
35         leaf red-frames-count {
36             type yang:counter64;
37             config false;
38             description
39                 "A count of frames that were discarded as a result of the
40                 operation of the flow meter.";
41             reference
42                 "8.6.5.3 of IEEE Std 802.1Q";
43         }
44         leaf passing-sdu-count {
45             type yang:counter64;
46             config false;
47             description
48                 "A count of frames that passed the Maximum SDU size filter.";
49             reference
50                 "8.6.5.3 of IEEE Std 802.1Q";
51         }
52         leaf not-passing-sdu-count {
53             type yang:counter64;
54             config false;
55             description
56                 "A count of frames that did not pass the Maximum SDU size
57                 filter.";
58             reference
59                 "8.6.5.3 of IEEE Std 802.1Q";
60         }
61         leaf flow-meter-ref {
62             type leafref {
63                 path
64                     '..'+
65                     '/..' +
66                     '/..' +
67                     '/flow-meters'+
68                     '/flow-meter-instance-table'+
69                     '/flow-meter-instance-id';
70             }
71             description
72                 "Flow meter instance that is associated with the stream filter.";

```

```

1      reference
2      "8.6.5.5 and 12.31.2.6 of IEEE Std 802.1Q.";
3  }
4  leaf flow-meter-enable {
5      type boolean;
6      description
7          "If it is set to TRUE, flow-meter-instance-id identifies the
8           flow meter instance associated with the stream filter. If it is
9           set to FALSE, no flow meter instance is associated with the
10          stream filter.";
11      reference
12          "8.6.5.5 and 12.31.2.6 of IEEE Std 802.1Q.";
13  }
14  }
15  augment "stream-gates" {
16      description
17          "Augment the Bridge component Stream Gates by maximum control list
18           limits, as used for PTP-controlled open and close transitions";
19      leaf supported-list-max {
20          type uint32;
21          description
22              "The maximum value supported by this Bridge component of the
23               AdminControlListLength and OperControlListLength parameters. It
24               is available for use by schedule computation software to
25               determine the Bridge component's control list capacity prior to
26               computation. The object may optionally be read-only.";
27          reference
28              "12.31.1.4 of IEEE Std 802.1Q";
29      }
30      container supported-cycle-max {
31          description
32              "The maximum value supported by this Bridge component of the
33               AdminCycleTime and OperCycleTime parameters. The object may
34               optionally be read-only.";
35          reference
36              "12.31.1.7 of IEEE Std 802.1Q";
37          uses ieee802:rational-grouping;
38      }
39      leaf supported-interval-max {
40          type uint32;
41          description
42              "The maximum value supported by this Bridge component of the
43               TimeIntervalValue parameter. The object may optionally be
44               read-only.";
45          reference
46              "12.31.1.8 of IEEE Std 802.1Q";
47      }
48  }
49  augment "stream-gates/stream-gate-instance-table" {
50      description
51          "Augment the Bridge component Stream Gate instances by
52           a) Operational gate states
53           b) Operational IPV values
54           c) PTP controlled open and close transitions
55           d) Management for PTP controlled open and close transitions";
56      leaf oper-gate-state {
57          type sfsg:gate-state-value-type;
58          config false;
59          description
60              "The current operational state of the gate. The operational
61               state of the gate is set by the List Execute state machine, and
62               its initial value is determined by the value of the
63               administrative state of the gate.";
64          reference
65              "12.31.3.2.1 and 8.6.10.5 of IEEE Std 802.1Q";
66      }
67      leaf oper-ipv {
68          type sfsg:ipv-spec-type;
69          description
70              "The operational internal priority value specification.";
71          reference
72              "12.31.3.3, 8.6.10.7, and 8.6.5.4 of IEEE Std 802.1Q";

```

```

1      }
2      container admin-control-list {
3          must
4              "not (count(/gate-control-entry) > ../../supported-list-max)" {
5                  error-message
6                      "Number of elements in admin-control-list must not be "+
7                      "greater than supported-list-max";
8              }
9          description
10             "The AdminControlList and OperControlList are ordered lists
11             containing AdminControlListLength or OperControlListLength
12             entries, respectively. Each entry represents a gate operation.
13             Each entry in the list is structured as a GateControlEntry.";
14         reference
15             "12.31.3.2, 12.31.3.2.2, and 8.6.9.4.2 of IEEE Std 802.1Q";
16         uses stream-gate-control-entries;
17     }
18     container oper-control-list {
19         must
20             "not (count(/gate-control-entry) > ../../supported-list-max)" {
21                 error-message
22                     "Number of elements in oper-control-list must not be greater "+
23                     "than supported-list-max";
24             }
25         config false;
26         description
27             "The AdminControlList and OperControlList are ordered lists
28             containing AdminControlListLength or OperControlListLength
29             entries, respectively. Each entry represents a gate operation.
30             Each entry in the list is structured as a GateControlEntry.";
31         reference
32             "12.31.3.2, 12.31.3.2.2 and 8.6.9.4.18 of IEEE Std 802.1Q";
33         uses stream-gate-control-entries;
34     }
35     container admin-cycle-time {
36         must
37             "not (/numerator div ./denominator > "+
38             "../../supported-cycle-max/numerator div "+
39             "../../supported-cycle-max/denominator)" {
40                 error-message
41                     "admin-cycle-time must not be greater than "+
42                     "supported-cycle-max";
43             }
44         description
45             "The administrative value of the gating cycle for the Port. This
46             value can be changed by management, and is used by the List
47             Config state machine to set the value of OperCycleTime. The
48             AdminCycleTime variable is a rational number of seconds, defined
49             by an integer numerator and an integer denominator.";
50         reference
51             "12.29.1.3 and 8.6.9.4.3 of IEEE Std 802.1Q";
52         uses ieee802:rational-grouping;
53     }
54     container oper-cycle-time {
55         must
56             "not (/numerator div ./denominator > "+
57             "../../supported-cycle-max/numerator div "+
58             "../../supported-cycle-max/denominator)" {
59                 error-message
60                     "oper-cycle-time must not be greater than "+
61                     "supported-cycle-max";
62             }
63         config false;
64         description
65             "The operational value of the gating cycle for the Port. This
66             variable is set dynamically from the AdminCycleTime variable
67             under the control of the List Config state machine OperCycleTime
68             is used by the Cycle Timer state machine to enforce the cycle
69             time for the Port. The OperCycleTime variable is a rational
70             number of seconds, defined by an integer numerator and an
71             integer denominator.";
72         reference

```



```

1      "12.29.1.3 and 8.6.9.4.19 of IEEE Std 802.1Q";
2      uses ieee802:rational-grouping;
3  }
4  leaf admin-cycle-time-extension {
5      type uint32;
6      units "nanoseconds";
7      description
8          "An integer number of nanoseconds, defining the maximum amount
9          of time by which the gating cycle for the Port is permitted to
10         be extended when a new cycle configuration is being installed.
11         This administrative value can be changed by management, and is
12         used by the List Config state machine to set the value of
13         OperCycleTimeExtension.";
14     reference
15         "8.6.9.4.4 of IEEE Std 802.1Q";
16 }
17 leaf oper-cycle-time-extension {
18     type uint32;
19     units "nanoseconds";
20     config false;
21     description
22         "An integer number of nanoseconds, defining the maximum amount
23         of time by which the gating cycle for the Port is permitted to
24         be extended when a new cycle configuration is installed. This
25         operational value is set by the List Config state machine to the
26         value of AdminCycleTimeExtension. The value of
27         OperCycleTimeExtension is used by the SetCycleStartTime()
28         procedure.";
29     reference
30         "8.6.9.4.20 of IEEE Std 802.1Q";
31 }
32 container admin-base-time {
33     description
34         "The administrative value of base time, expressed as an IEEE
35         1588 precision time protocol (PTP) timescale (see IEEE Std
36         802.1AS-2011). This value can be changed by management, and is
37         used by the List Config state machine to set the value of
38         OperBaseTime.";
39     reference
40         "12.29.1.4 and 8.6.9.4.1 of IEEE Std 802.1Q";
41     uses ieee802:ptp-time-grouping;
42 }
43 container oper-base-time {
44     config false;
45     description
46         "The operational value of base time, expressed as a PTP
47         timescale (see IEEE Std 802.1AS-2011). This variable is used by
48         the List Config state machine.";
49     reference
50         "12.29.1.4 and 8.6.9.4.17 of IEEE Std 802.1Q";
51     uses ieee802:ptp-time-grouping;
52 }
53 leaf config-change {
54     type boolean;
55     description
56         "A Boolean variable that acts as a start signal to the List
57         Config state machine that the administrative variable values for
58         the Port are ready to be copied into their corresponding
59         operational variables. This variable is set TRUE by management
60         and is set FALSE by the List Config state machine.";
61     reference
62         "8.6.9.4.7 of IEEE Std 802.1Q";
63 }
64 container config-change-time {
65     config false;
66     description
67         "The time at which the administrative variables that determine
68         the cycle are to be copied across to the corresponding
69         operational variables, expressed as a PTP timescale. The value
70         of this variable is set by the SetConfigChangeTime() procedure
71         in the List Config state machine.";
72     reference

```

```

1      "12.29.1.4 and 8.6.9.4.9 of IEEE Std 802.1Q";
2      uses ieee802:ptp-time-grouping;
3  }
4  leaf tick-granularity {
5      type uint32;
6      config false;
7      description
8          "Characteristics of an implementation's cycle timer clock
9          (TickGranularity).";
10     reference
11         "8.6.9.4.16 of IEEE Std 802.1Q";
12 }
13 container current-time {
14     config false;
15     description
16         "The current time maintained by the local system, expressed as a
17         PTP timescale (see IEEE Std 802.1AS-2011).";
18     reference
19         "12.29.1.4 and 8.6.9.4.10 of IEEE Std 802.1Q";
20     uses ieee802:ptp-time-grouping;
21 }
22 leaf config-pending {
23     type boolean;
24     config false;
25     description
26         "A Boolean variable, set TRUE by the List Config state machine
27         to signal that there is a new cycle configuration awaiting
28         installation. The variable is set FALSE when the List Config
29         state machine has installed the new configuration. The variable
30         is used by the SetCycleStartTime() procedure to control the
31         length of the cycle that immediately precedes the first cycle
32         that uses the new configuration values. This value can be read
33         by management.";
34     reference
35         "8.6.9.3 and 8.6.9.4.8 of IEEE Std 802.1Q";
36 }
37 leaf config-change-error {
38     type yang:counter64;
39     config false;
40     description
41         "An error counter that is incremented if AdminBaseTime specifies
42         a time in the past, and the current schedule is running.";
43     reference
44         "8.6.9.3.1 of IEEE Std 802.1Q";
45 }
46 leaf gate-closed-due-to-invalid-rx-enable {
47     type boolean;
48     default "false";
49     description
50         "A value of TRUE indicates that the GateClosedDueToInvalidRx
51         function is enabled; a value of FALSE indicates that the
52         GateClosedDueToInvalidRx function is disabled. The default value
53         of GateClosedDueToInvalidRxEnable is FALSE.";
54     reference
55         "8.6.5.4 of IEEE Std 802.1Q";
56 }
57 leaf gate-closed-due-to-invalid-rx {
58     type boolean;
59     default "false";
60     description
61         "If GateClosedDueToInvalidRxEnable is TRUE, a value of TRUE in
62         GateClosedDueToInvalidRx indicates that all frames are dropped
63         (i.e., the gate behaves as if the operational stream gate state
64         is Closed). If GateClosedDueToInvalidRx is FALSE, it has no
65         effect. The default value of GateClosedDueToInvalidRx is FALSE;
66         if any frame is discarded because the gate is in the Closed
67         state, then GateClosedDueToInvalidRx is set TRUE.";
68     reference
69         "8.6.5.4 of IEEE Std 802.1Q";
70 }
71 leaf gate-closed-due-octets-exceeded-enable {
72     type boolean;

```

```
1      default "false";
2      description
3          "A value of TRUE indicates that the
4          GateClosedDueToOctetsExceeded function is enabled; a value of
5          FALSE indicates that the GateClosedDueToOctetsExceeded function
6          is disabled. The default value of GateClosedDueToOctetsExceed is
7          FALSE.";
8      reference
9          "8.6.5.4 of IEEE Std 802.1Q";
10     }
11 leaf gate-closed-due-octets-exceeded {
12     type boolean;
13     default "false";
14     description
15         "If GateClosedDueToOctetsExceededEnable is TRUE, a value of TRUE
16         in GateClosedDueToOctetsExceeded indicates that all frames are
17         dropped (i.e., the gate behaves as if the operational stream
18         gate state is Closed). If GateClosedDueToOctetsExceeded is
19         FALSE, it has no effect. The default value of
20         GateClosedDueToOctetsExceeded is FALSE; if any frame is
21         discarded because there are insufficient IntervalOctetsLeft,
22         then GateClosedDueToOctetsExceeded is set TRUE.";
23     reference
24         "8.6.5.4 of IEEE Std 802.1Q";
25 }
26 }
27 }
28 }
29 }
30
31
```

1 48.6.22 The ieee802-dot1q-psfp-bridge YANG module

```
2 module ieee802-dot1q-psfp-bridge {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-psfp-bridge;
5   prefix psfp-bridge;
6   import ieee802-dot1q-bridge {
7     prefix dot1q;
8   }
9   import ieee802-dot1q-psfp {
10    prefix psfp;
11  }
12  organization
13    "IEEE 802.1 Working Group";
14  contact
15    "WG-URL: http://www.ieee802.org/1/
16    WG-EMail: stds-802-1-1@ieee.org
17
18    Contact: IEEE 802.1 Working Group Chair
19    Postal: C/O IEEE 802.1 Working Group
20    IEEE Standards Association
21    445 Hoes Lane
22    Piscataway, NJ 08854
23    USA
24
25    E-mail: STDS-802-1-CHAIRS@IEEE.ORG";
26  description
27    "This module provides management of IEEE Std 802.1Q Bridge
28    components that support Per-Stream Filtering and Policing (PSFP).
29
30    Copyright (C) IEEE (2023).
31
32    This version of this YANG module is part of IEEE Std 802.1Q; see the
33    standard itself for full legal notices.";
34  revision 2023-10-26 {
35    description
36      "Published as part of IEEE Std 802.1Qcw-2023.
37
38      The following reference statement identifies each referenced IEEE
39      Standard as updated by applicable amendments.";
40    reference
41      "IEEE Std 802.1Q Bridges and Bridged Networks:
42      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023.";
43  }
44  feature psfp {
45    description
46      "Per-Stream Filtering and Policing supported.";
47    reference
48      "IEEE Std 802.1Q";
49  }
50  augment "/dot1q:bridges/dot1q:bridge/dot1q:component" {
51    description
52      "Augment bridge with Per-Stream Filtering and Policing configuration";
53    uses psfp:psfp-parameters;
54  }
55 }
56
57
```

1 48.6.23 The ieee802-dot1q-cnc-config YANG module

```
2 module ieee802-dot1q-cnc-config {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-cnc-config;
5   prefix dot1q-cnc-config;
6   import ieee802-dot1q-tsn-types {
7     prefix tsn;
8     reference
9       "48.6.3 of IEEE Std 802.1Q";
10  }
11  organization
12    "IEEE 802.1 Working Group";
13  contact
14    "WG-URL: http://ieee802.org/1/
15    WG-EMail: stds-802-1-1@ieee.org
16
17    Contact: IEEE 802.1 Working Group Chair
18    Postal: C/O IEEE 802.1 Working Group
19            IEEE Standards Association
20            445 Hoes Lane
21            Piscataway, NJ 08854
22            USA
23
24    E-mail: stds-802-1-chairs@ieee.org";
25  description
26    "This module supports management of a Time-Sensitive Networking (TSN)
27    Centralized Network Configuration (CNC) component, its Configuration
28    Domains - each supporting one or more Centralized User Configuration (CUC)
29    components, and the Streams created by those CUCs. RPCs support
30    path computation and resource allocation across all or specified
31    Configuration Domains, CUCs, and Streams.
32
33    Copyright (C) IEEE (2024).
34
35    This version of this YANG module is part of IEEE Std 802.1Q; see
36    the standard itself for full legal notices.";
37  revision 2024-01-31 {
38    description
39      "Published as part of IEEE Std 802.1Qdj-2024.
40
41      The following reference statement identifies each referenced IEEE
42      Standard as updated by applicable amendments.";
43    reference
44      "IEEE Std 802.1Q Bridges and Bridged Networks:
45      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023,
46      IEEE Std 802.1Qcj-2023, IEEE Std 802.1Qdj-2024.";
47  }
48  container cnc-config {
49    description
50      "Top-level container for the CNC module.";
51    list domain {
52      key "domain-id";
53      description
54        "A list of Configuration Domains, each supporting one or more CUCs.
55        Access to Streams and resources associated with particular Configuration
56        Domains can be restricted.";
57      leaf domain-id {
58        type string;
59        description
60          "The DomainID uniquely identifies a Configuration Domain.";
61        reference
62          "46.2.2.1 of IEEE Std 802.1Q";
63      }
64      leaf cnc-enabled {
65        type boolean;
66        default "false";
67        description
68          "cnc-enabled is used to enable or disable CNC functionality. If TRUE,
69          the CNC functionality is enabled. If FALSE, the CNC functionality is
70          disabled.";
71        reference
```

```

1      "46.2.2.3 of IEEE Std 802.1Q";
2  }
3  list cuc {
4      key "cuc-id";
5      description
6          "A list of CUCs in the Configuration Domain, each with its associated
7          Streams. Access to Streams and resources associated with particular
8          CUCs can be restricted.";
9      leaf cuc-id {
10         type string;
11         description
12             "The CucID uniquely identifies the CUC.";
13         reference
14             "46.2.2.2 of IEEE Std 802.1Q";
15     }
16     list stream {
17         key "stream-id";
18         description
19             "A list of Streams created by the CUC, with their status and
20             configuration (talker and listener parameters) in the CNC.";
21         leaf stream-id {
22             type tsn:stream-id-type;
23             description
24                 "The StreamID uniquely identifies a Stream.";
25         }
26         leaf stream-status {
27             type enumeration {
28                 enum planned {
29                     value 0;
30                     description
31                         "The Stream has been requested but has not yet been
32                         configured by the CNC.";
33                 }
34                 enum configured {
35                     value 1;
36                     description
37                         "The Stream has been computed and configured by the
38                         CNC.";
39                 }
40                 enum modified {
41                     value 2;
42                     description
43                         "The Stream has been configured but Stream parameters
44                         have been modified after configuration.";
45                 }
46             }
47             config false;
48             description
49                 "The stream-status is the status of the Stream in the CNC.";
50             reference
51                 "46.2.3.8 of IEEE Std 802.1Q";
52         }
53     }
54     container talker {
55         description
56             "The Talker container for the Stream comprises the following:
57             - The Talker's behavior (how/when frames are transmitted).
58             - The Talker's network requirements.
59             - The Talker's interface(s) TSN capabilities.";
60         uses tsn:group-talker;
61         uses tsn:group-status-talker-listener {
62             refine "accumulated-latency" {
63                 config false;
64             }
65             refine "interface-configuration" {
66                 config false;
67             }
68         }
69     }
70     list listener {
71         key "index";
72         description
73             "A list of parameters for each of the Stream's Listeners,
```

```

1         each list entry comprising:
2         - The Listener's interface(s).
3         - The Listener's network requirements.
4         - The Listener's interface(s) TSN capabilities.";
5     leaf index {
6         type uint32;
7         description
8             "This index is provided in order to provide a unique key
9             per list entry.";
10    }
11    uses tsn:group-listener;
12    uses tsn:group-status-talker-listener {
13        refine "accumulated-latency" {
14            config false;
15        }
16        refine "interface-configuration" {
17            config false;
18        }
19    }
20 }
21 uses tsn:group-status-stream {
22     refine "status-info" {
23         config false;
24     }
25     refine "failed-interfaces" {
26         config false;
27     }
28 }
29 }
30 action remove_streams {
31     description
32         "Removes the selected Streams.";
33     reference
34         "46.2.8.1 of IEEE Std 802.1Q";
35     input {
36         list stream-list {
37             key "stream-id";
38             description
39                 "List of stream-ids for the Streams to be removed.";
40             leaf stream-id {
41                 type tsn:stream-id-type;
42                 description
43                     "The stream-id uniquely identifies a Stream.";
44             }
45         }
46     }
47     output {
48         leaf result {
49             type string;
50             description
51                 "Returns status information indicating if Stream removal
52                 has been successfully started.";
53         }
54     }
55 }
56 }
57 }
58 }
59
60 // RPCs
61 rpc compute_streams {
62     description
63         "Starts computation of path and resource allocation for one or more
64         Streams, each identified by its domain-id, cuc-id, and stream-id.
65         This RPC can compute paths and resource allocations for both new and
66         modified Streams.";
67     input {
68         list domain {
69             key "domain-id";
70             description
71                 "A list of Configuration Domains containing the Streams for which
72                 paths and resource allocations are to be computed.";

```

```

1      reference
2      "46.2.7.1 of IEEE Std 802.1Q";
3      leaf domain-id {
4          type leafref {
5              path '/cnc-config/domain/domain-id';
6          }
7          description
8              "The referenced DomainID uniquely identifies a Configuration
9              Domain.";
10     }
11     list cuc {
12         key "cuc-id";
13         description
14             "A list of CUCs in the Configuration Domain, each with Streams whose
15             paths and resource allocations are to be computed. ";
16         leaf cuc-id {
17             type leafref {
18                 path '/cnc-config/domain/cuc/cuc-id';
19             }
20             description
21                 "The referenced CucID uniquely identifies a CUC.";
22         }
23         list stream-list {
24             key "stream-id";
25             description
26                 "A list of Streams, created by the CUC, whose paths and resource
27                 allocations are to be computed.";
28             leaf stream-id {
29                 type leafref {
30                     path '/cnc-config/domain/cuc/stream/stream-id';
31                 }
32                 description
33                     "The referenced StreamID uniquely identifies a Stream.";
34             }
35         }
36     }
37 }
38 }
39 output {
40     leaf result {
41         type string;
42         description
43             "Only returns status information indicating if the computation
44             has been started. It does not return status information on the
45             success or failure of the actual Stream computation. A
46             notification can be used to inform the caller of this RPC on the
47             results of Stream computation after the computation has
48             finished.";
49     }
50 }
51 }
52 rpc compute_planned_and_modified_streams {
53     description
54         "Starts computation of path and resource allocation for all Streams
55         that are in specified CUCs in specified domains, and that have not been
56         computed (i.e., that have a Stream status of planned or modified).";
57     reference
58         "46.2.7.2 of IEEE Std 802.1Q";
59     input {
60         list domain {
61             key "domain-id";
62             description
63                 "A list of Configuration Domains containing the CUCs with Streams
64                 whose paths and resource allocations are to be computed.";
65             leaf domain-id {
66                 type string;
67                 description
68                     "A unique identifier of a Configuration Domain. It is used to
69                     identify the Configuration Domain a CUC belongs to.";
70             }
71             list cuc {
72                 key "cuc-id";

```



```

1      description
2      "A list of CUCs, in the Configuration Domain, that have Streams
3      whose paths and resource allocations are to be computed.";
4      leaf cuc-id {
5          type string;
6          description
7              "The CucID uniquely identifies the CUC.";
8      }
9  }
10 }
11 }
12 output {
13     leaf result {
14         type string;
15         description
16             "Only returns status information indicating if the computation
17             has been started. It does not return status information on the
18             success or failure of the actual Stream computation. A
19             notification can be used to inform the caller of this RPC of the
20             results of Stream computation after the computation has
21             finished.";
22     }
23 }
24 }
25 rpc compute_all_streams {
26     description
27         "Starts computation of path and resource allocation for all Streams
28         that are in specified CUCs in specified domains.";
29     reference
30         "46.2.7.3 of IEEE Std 802.1Q";
31     input {
32         list domain {
33             key "domain-id";
34             description
35                 "A list of Configuration Domains containing the CUCs with Streams
36                 whose paths and resource allocations are to be computed";
37             leaf domain-id {
38                 type string;
39                 description
40                     "The DomainID uniquely identifies a Configuration Domain.";
41             }
42             list cuc {
43                 key "cuc-id";
44                 description
45                     "A list of CUCs in the Configuration Domain, each with Streams whose
46                     paths and resource allocations are to be computed.";
47                 leaf cuc-id {
48                     type string;
49                     description
50                         "The CucID uniquely identifies the CUC.";
51                 }
52             }
53         }
54     }
55     output {
56         leaf result {
57             type string;
58             description
59                 "Only returns status information indicating if the computation
60                 has been started. It does not return status information on the
61                 success or failure of the actual Stream computation. A
62                 notification can be used to inform the caller of this RPC of the
63                 results of Stream computation after the computation has
64                 finished.";
65         }
66     }
67 }
68 rpc request_domain_id {
69     description
70         "Returns the DomainID of the Configuration Domain to which a specified
71         CUC belongs.";
72     reference

```

```

1      "46.2.7.4 of IEEE Std 802.1Q";
2  input {
3      leaf cuc-id {
4          type string;
5          description
6              "The CucID uniquely identifies the CUC.";
7      }
8  }
9  output {
10     leaf result {
11         type string;
12         description
13             "Returns the DomainID of the Configuration Domain for the
14             specified CUC.";
15     }
16 }
17 }
18 rpc request_free_stream_id {
19     description
20         "Returns a free (i.e., available) StreamID for use by a specified
21         CUC in a specified Configuration Domain.";
22     reference
23         "46.2.7.5 of IEEE Std 802.1Q";
24     input {
25         leaf domain-id {
26             type string;
27             description
28                 "The DomainID uniquely identifies the Configuration Domain.";
29         }
30         leaf cuc-id {
31             type string;
32             description
33                 "The CucID uniquely identifies the CUC.";
34         }
35     }
36     output {
37         leaf result {
38             type string;
39             description
40                 "Returns a free (i.e., available) StreamID for use by the
41                 specified CUC in the specified Configuration Domain.";
42         }
43     }
44 }
45
46 // Notifications
47 notification compute_streams_completed {
48     description
49         "Notifies the caller of an RPC or action that initiated computation for
50         one or more Streams, that the computation is complete. Returns information
51         on the success or failure of computation for each of those Streams.";
52     reference
53         "46.2.9.1 of IEEE Std 802.1Q";
54     list domain {
55         key "domain-id";
56         description
57             "The list of Configuration Domains for which computation was requested,
58             with each list entry specifying the CUCs, and the Streams for each of
59             those CUCs for which computation was requested, with the result of the
60             computation (success or failure) for each Stream.";
61         leaf domain-id {
62             type string;
63             description
64                 "The DomainID uniquely identifies a Configuration Domain.";
65         }
66         list cuc {
67             key "cuc-id";
68             description
69                 "The list of CUCs for the specified Configuration Domain, with each
70                 list entry specifying the Streams for which computation was requested
71                 and the result of the computation for each Stream.";
72             leaf cuc-id {

```

```

1      type string;
2      description
3          "The CucID uniquely identifies the CUC.";
4  }
5  list stream {
6      key "stream-id";
7      description
8          "The list of Streams, for which computation was requested, for the
9          specified Configuration Domain and CUC, with the result of the
10         computation (success or failure) for each Stream.";
11     reference
12         "46.2.3 of IEEE Std 802.1Q";
13     leaf stream-id {
14         type tsn:stream-id-type;
15         description
16             "The StreamID uniquely identifies a Stream.";
17     }
18     leaf failure-code {
19         type uint8;
20         description
21             "A code that indicates if the computation for the Stream
22             was successful (0) or not. In the case of a failure a code
23             is returned to indicate what kind of failure occurred.";
24     }
25 }
26 }
27 }
28 }
29 notification configure_streams_completed {
30     description
31         "Notifies the caller of an RPC or action that initiated computation for
32         one or more Streams, that computation and configuration is complete.
33         Returns information on the success or failure information of computation
34         and configuration for each of those Streams.";
35     reference
36         "46.2.9.2 of IEEE Std 802.1Q";
37     list domain {
38         key "domain-id";
39         description
40             "The list of Configuration Domains for which computation was requested,
41             with each list entry specifying the CUCs, and the Streams for each of
42             those CUCs for which computation was requested, with the result of the
43             computation (success or failure) for each Stream.";
44     leaf domain-id {
45         type string;
46         description
47             "The DomainID uniquely identifies a Configuration Domain.";
48     }
49     list cuc {
50         key "cuc-id";
51         description
52             "The list of CUCs for the specified Configuration Domain, with each
53             list entry specifying the Streams for which computation was requested
54             and the result of the computation for each Stream.";
55     leaf cuc-id {
56         type string;
57         description
58             "The CucID uniquely identifies the CUC.";
59     }
60     list stream {
61         key "stream-id";
62         description
63             "The list of Streams, for which computation was requested, for the
64             specified Configuration Domain and CUC, with the result of the
65             computation (success or failure) for each Stream.";
66     reference
67         "46.2.3 of IEEE Std 802.1Q";
68     leaf stream-id {
69         type tsn:stream-id-type;
70         description
71             "The StreamID uniquely identifies a Stream.";
72     }

```

```

1      leaf failure-code {
2          type uint8;
3          description
4              "A code that indicates if the computation and configuration
5              for the Stream was successful (0) or not. In the case of a
6              failure a code is returned to indicate what kind of failure
7              occurred.";
8      }
9  }
10 }
11 }
12 }
13 notification remove_streams_completed {
14     description
15         "Notifies the caller of an RPC or action that initiated computation for
16         one or more Streams, that the removal is complete. Returns information
17         on the success or failure of removal for each of those Streams.";
18     reference
19         "46.2.9.3 of IEEE Std 802.1Q";
20     list domain {
21         key "domain-id";
22         description
23             "A list of Configuration Domains with each list entry specifying the
24             CUCs, and the Streams for each of those CUCs for which removal was
25             requested, with the result of the removal attempt (success or failure)
26             for each Stream.";
27         leaf domain-id {
28             type string;
29             description
30                 "The DomainID uniquely identifies a Configuration Domain.";
31         }
32         list cuc {
33             key "cuc-id";
34             description
35                 "The list of CUCs for the specified Configuration Domain, with each
36                 list entry specifying the Streams for which removal was requested
37                 and the result of the removal attempt for each Stream.";
38             leaf cuc-id {
39                 type string;
40                 description
41                     "The CucID uniquely identifies the CUC.";
42             }
43             list stream {
44                 key "stream-id";
45                 description
46                     "The list of Streams, for which removal was requested, for the
47                     specified Configuration Domain and CUC, with the result of the
48                     removal (success or failure) for each Stream.";
49                 reference
50                     "46.2.3 of IEEE Std 802.1Q";
51                 leaf stream-id {
52                     type tsn:stream-id-type;
53                     description
54                         "The StreamID uniquely identifies a Stream.";
55                 }
56                 leaf failure-code {
57                     type uint8;
58                     description
59                         "A code that indicates if the removal of the Stream was
60                         successful (0) or unsuccessful (1).";
61                 }
62             }
63         }
64     }
65 }
66 }
67

```

1 48.6.24 The ieee802-dot1q-cbsa YANG module

```
2 module ieee802-dot1q-cbsa {
3   namespace "urn:ieee:std:802.1Q:yang:ieee802-dot1q-cbsa";
4   prefix cbsa;
5
6   import ieee802-dot1q-types {
7     prefix dot1q-types;
8   }
9
10  organization
11    "IEEE 802.1 Working Group";
12  contact
13    "WG-URL: http://www.ieee802.org/1/
14    WG-EMail: stds-802-1-L@ieee.org
15
16    Contact: IEEE 802.1 Working Group Chair
17    Postal: C/O IEEE 802.1 Working Group
18    IEEE Standards Association
19    445 Hoes Lane
20    Piscataway, NJ 08854
21    USA
22
23    E-mail: stds-802-1-chairs@ieee.org";
24  description
25    "This module provides for management of IEEE Std 802.1Q components
26    that support the credit-based shaper algorithm.
27
28    Copyright (C) IEEE (2024)
29
30    This version of this YANG module is part of IEEE Std 802.1Q; see the
31    standard itself for full legal notices.";
32
33  revision 2024-08-07 {
34    description
35      "Published as part of IEEE Std 802.1Qdx-2024.
36
37      The following reference statement identifies each referenced IEEE
38      Standard as updated by applicable amendments.";
39    reference
40      "IEEE Std 802.1Q Bridges and Bridged Networks:
41      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023,
42      IEEE Std 802.1Qcj-2023, IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024.";
43  }
44
45  grouping cbsa-parameters {
46    description
47      "cbsa-parameters comprises all the parameters associated
48      with the credit-based shaper algorithm.";
49    container cbsa {
50      description
51        "This container comprises all credit-based shaper algorithm related
52        nodes.";
53      list cbsa-parameter-table {
54        key "traffic-class";
55        max-elements 8;
56        description
57          "There is one cbsa-parameter-table list per Port. Each list entry
58          corresponds to the set of parameters (12.20.1) for each traffic class
59          configured for use with the credit-based shaper algorithm (8.6.8.2).
60          The cbsa-parameter-table list is valid for a Bridge Port only when
61          used in the absence of the stream reservation protocol.";
62        reference
63          "8.6.8.2, 34.3, 34.6, and 12.20.1 of IEEE Std 802.1Q.";
64        leaf traffic-class {
65          type dot1q-types:traffic-class-type;
66          description
67            "An 802.1 traffic class value. This is the numerical value
68            associated with a traffic class. Larger values are associated with
69            higher priority traffic classes.";
70          reference
71            "8.6.8 of IEEE Std 802.1Q.";
```

```
1      }
2      leaf admin-idle-slope {
3          type uint64;
4          units "bits/second";
5          default "0";
6          description
7              "The bandwidth, in bits per second, that has been requested by
8              management to be reserved for use by the queue associated with
9              this traffic class.
10
11              If the stream reservation protocol is in operation for this traffic
12              class, this parameter has no effect.
13
14              If the stream reservation protocol is not in operation for this
15              traffic class, then the value of oper-idle-slope is equal to the
16              value of admin-idle-slope.";
17          reference
18              "34.3 of IEEE Std 802.1Q.";
19      }
20      leaf oper-idle-slope {
21          type uint64;
22          units "bits/second";
23          default "0";
24          config false;
25          description
26              "The actual bandwidth, in bits per second, that is currently
27              reserved for use by the queue associated with this traffic class
28              (see 34.6.1 and 34.6.2).";
29          reference
30              "34.3, 34.6.1, and 34.6.2 of IEEE Std 802.1Q.";
31      }
32  }
33 }
34 }
35 }
36
37
```

1 48.6.25 The ieee802-dot1q-cbsa-bridge YANG module

```
2 module ieee802-dot1q-cbsa-bridge {
3   namespace "urn:ieee:std:802.1Q:yang:ieee802-dot1q-cbsa-bridge";
4   prefix cbsa-bridge;
5
6   import ietf-interfaces {
7     prefix if;
8   }
9   import ieee802-dot1q-cbsa {
10    prefix cbsa;
11  }
12  import ieee802-dot1q-bridge {
13    prefix dot1q;
14  }
15
16  organization
17    "IEEE 802.1 Working Group";
18  contact
19    "WG-URL: http://www.ieee802.org/1/
20    WG-EMail: stds-802-1-L@ieee.org
21
22    Contact: IEEE 802.1 Working Group Chair
23    Postal: C/O IEEE 802.1 Working Group
24    IEEE Standards Association
25    445 Hoes Lane
26    Piscataway, NJ 08854
27    USA
28
29    E-mail: stds-802-1-chairs@ieee.org";
30  description
31    "This module provides for management of IEEE Std 802.1Q Bridges
32    that support the credit-based shaper algorithm.
33
34    Copyright (C) IEEE (2024).
35
36    This version of this YANG module is part of IEEE Std 802.1Q; see the
37    standard itself for full legal notices.";
38
39  revision 2024-08-07 {
40    description
41      "Published as part of IEEE Std 802.1Qdx-2024.
42
43      The following reference statement identifies each referenced IEEE
44      Standard as updated by applicable amendments.";
45    reference
46      "IEEE Std 802.1Q Bridges and Bridged Networks:
47      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023,
48      IEEE Std 802.1Qcj-2023, IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024.";
49  }
50
51  feature credit-based-shaper-algorithm {
52    description
53      "Credit-based shaper algorithm supported.";
54    reference
55      "8.6.8.2 of IEEE Std 802.1Q.";
56  }
57
58  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
59    if-feature "credit-based-shaper-algorithm";
60    description
61      "Augment dot1q:bridge-port with credit-based shaper algorithm
62      configuration.";
63    uses cbsa:cbsa-parameters;
64  }
65 }
66
67
```

1 48.6.26 The ieee802-dot1q-rstp YANG module

```
2 module ieee802-dot1q-rstp {
3   yang-version 1.1;
4   namespace "urn:ieee:std:802.1Q:yang:ieee802-dot1q-rstp";
5   prefix rstp;
6
7   import ietf-yang-types {
8     prefix yang;
9   }
10  import ietf-interfaces {
11    prefix if;
12  }
13  import ieee802-types {
14    prefix ieee;
15  }
16
17  organization
18    "IEEE 802.1 Working Group";
19  contact
20    "WG-URL: http://www.ieee802.org/1/
21    WG-EMail: stds-802-1-1@ieee.org
22
23    Contact: IEEE 802.1 Working Group Chair
24    Postal: C/O IEEE 802.1 Working Group
25            IEEE Standards Association
26            445 Hoes Lane
27            Piscataway, NJ 08854
28            USA
29
30    E-mail: stds-802-1-chairs@ieee.org";
31  description
32    "This module provides management of IEEE Std 802.1Q Bridge components
33    that support the Rapid Spanning Tree Algorithm and Protocol (RSTP).
34    The managed objects specified also support those aspects of Multiple
35    Spanning Tree Algorithm and Protocol (MSTP) and Shortest Path Bridging
36    (SPB) operation that provide plug-and-play interoperability with RSTP.
37
38    References specify constraints on, and consequences of, settings of
39    leaf values and the creation and deletion of list entries. The values
40    of all configured objects are retained across system reinitialization.
41
42    Copyright (C) IEEE (2025).
43
44    This version of this YANG module is part of IEEE Std 802.1Q; see the
45    standard itself for full legal notices.";
46
47  revision 2025-02-04 {
48    description
49      "Published as part of IEEE Std 802.1Qdy-2025.
50
51      The following reference statement identifies each referenced IEEE
52      Standard as updated by applicable amendments.";
53    reference
54      "IEEE Std 802.1Q Bridges and Bridged Networks:
55      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
56      IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qcj-2023,
57      IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024,
58      IEEE Std 802.1Qdy-2025.";
59  }
60
61  typedef id-priority {
62    type uint8 {
63      range "0..15";
64    }
65    default "8";
66    description
67      "Priority for Bridge and Bridge Port identifiers. Lower numeric
68      values indicate the better identifier.";
69  }
70
71  typedef id-port-number{
```



```

1  type uint16 {
2      range "0..4095";
3  }
4  description
5      "A system assigned Bridge Port number, unique in the context of
6      a Bridge component, used in a Port Identifier.";
7  }
8
9  grouping bridge-id {
10     description
11         "A 64-bit Bridge Identifier and its components: a manageable
12         priority in the four most-significant bits, a system ID extension in
13         the next 12 bits, and the Bridge Address in the remaining 48 bits.";
14     reference
15         "13.26.2, 14.1.2, 14.2.5, and item e) in 13.26 of IEEE Std 802.1Q.";
16     leaf bridge-id {
17         type uint64;
18         config false;
19         description
20             "The Bridge Identifier as used in protocol and tree computation.";
21     }
22     leaf bridge-priority {
23         type id-priority;
24         description
25             "The manageable bridge priority.";
26     }
27     leaf system-id-extension {
28         type uint16 {
29             range "0..4095";
30         }
31         config false;
32         description
33             "The system ID extension.";
34         reference
35             "13.26.2, 13.41.5, 13.42, 14.2.5, and 14.4 of IEEE Std 802.1Q.";
36     }
37     leaf bridge-address {
38         type ieee:mac-address;
39         config false;
40         description
41             "A Bridge Address.";
42     }
43 }
44
45 grouping port-id {
46     description
47         "A 16-bit Port Identifier and its components: a priority in the four
48         most-significant bits, and a twelve-bit Bridge Port number.";
49     reference
50         "13.2, 13.18, Table 13-3, and 14.2.7 of IEEE Std 802.1Q.";
51     leaf port-id {
52         type uint16;
53         config false;
54         description
55             "The Port Identifier as used in protocol and tree computation.";
56     }
57     leaf port-priority {
58         type id-priority;
59         description
60             "The manageable port priority, encoded in the four most-significant
61             bits of octet 26 of Configuration BPDUs for the CIST and in the
62             four most-significant bits of octet 15 of the MSTI Configuration
63             Message for an MSTI.";
64         reference
65             "13.2, 13.18, Table 13-3, 14.2.7, and item e) in 14.4.1
66             of IEEE Std 802.1Q.";
67     }
68     leaf port-number {
69         type id-port-number;
70         config false;
71         description
72             "The system assigned Port Number, encoded in the twelve

```

```

1      least-significant bits of octets 26 and 27 of Configuration BPDUs.";
2      reference
3          "12.3, 12.4.2, 13.2, and 14.2.7 of IEEE Std 802.1Q.";
4      }
5  }
6
7  typedef port-state {
8      type enumeration {
9          enum discarding {
10             value 1;
11             description
12                 "Received frames are not submitted to the Learning Process or
13                 to the Forwarding Process. The Forwarding Process does not relay
14                 frames to the port for transmission. The Discarding state
15                 includes an inability to receive or transmit frames or BPDUs
16                 because MAC_Operational is false (if:oper-status not up).";
17             }
18             enum learning {
19                 value 2;
20                 description
21                     "Received frames are submitted to the Learning Process, but not
22                     to the Forwarding Process. The Forwarding Process does not
23                     relay frames to the port for transmission.";
24             }
25             enum forwarding {
26                 value 3;
27                 description
28                     "Received frames are submitted to the Learning Process and the
29                     Forwarding Process. The Forwarding Process can relay frames to
30                     the port for transmission.";
31             }
32         }
33         description
34             "The Port State for frames assigned to a given spanning tree,
35             governing submission of frames received by the port to the Learning
36             and Forwarding Processes and relay of frames to the port by the
37             Forwarding Process.";
38         reference
39             "8.4, 8.6, 8.7, 13.4, and items aw) and ba) in 13.27
40             of IEEE Std 802.1Q.";
41     }
42
43     typedef port-role {
44         type enumeration {
45             enum disabled-port {
46                 value 1;
47                 description
48                     "The port's MAC_Operational status is false (if:oper-status not
49                     up), or admin-bridge-port-enabled is false.";
50             }
51             enum root-port {
52                 value 2;
53                 description
54                     "The Root Port.";
55             }
56             enum designated-port {
57                 value 3;
58                 description
59                     "A Designated Port.";
60             }
61             enum alternate-port {
62                 value 4;
63                 description
64                     "An Alternate Port.";
65             }
66             enum backup-port {
67                 value 5;
68                 description
69                     "A Backup Port.";
70             }
71         }
72         description

```

```

1      "The Port Role assigned by the Bridge's spanning tree protocol entity
2      for a given spanning tree.";
3  reference
4      "13.4, 13.5, 13.12, 13.24, 13.27.66, and item bn) in 13.27
5      of IEEE Std 802.1Q.";
6  }
7
8  grouping bridge-component-parameters {
9      description
10         "This grouping comprises per-Bridge component parameters.";
11     container rstp {
12         presence "The presence of this container indicates that RSTP is
13         supported.";
14         description
15             "Per-Bridge component parameters common to RSTP, MST, and SPB.";
16         leaf force-protocol-version {
17             type enumeration {
18                 enum emulate-stp {
19                     value 0;
20                     description
21                         "Forces transmission of STP BPDUs. Slows reconfiguration and
22                         ageing of FDB entries.";
23                 }
24                 enum rstp {
25                     value 2;
26                     description
27                         "RST BPDUs are transmitted. MCID, MSTI, and SPT information
28                         is not transmitted and is ignored on receipt, treating a
29                         neighboring Bridge as in a separate MST or SPT region even
30                         if MSTP and/or SPB are implemented.";
31                 }
32                 enum rstp-mstp {
33                     value 3;
34                     description
35                         "If MSTP is implemented, full MSTP behavior is enabled and
36                         transmitted BPDUs include RSTP and MSTP information.";
37                 }
38                 enum rstp-mstp-spb{
39                     value 4;
40                     description
41                         "Shortest Path Bridging (SPB) and MSTP (if implemented) are
42                         both enabled. Transmitted BPDUs can include RSTP, MSTP,
43                         and SPB information.";
44                 }
45             }
46         description
47             "Each Bridge implementing RSTP, MSTP, or SPT is plug-and-play
48             interoperable with neighboring Bridges using a different subset
49             of those protocols. Force Protocol Version applies to all ports,
50             and can restrict protocol use. Each Bridge Port also detects a
51             neighbor(s) using STP (IEEE Std 802.1D-1998, now withdrawn)
52             and uses STP BPDUs and slow reconfiguration to interwork with
53             those neighbors independently of Force Protocol Version.
54             Force Protocol Version value 1 was previously assigned to
55             IEEE Std 802.1G-1996 (now withdrawn) and is not to be used.";
56         reference
57             "5.4, 5.4.1.1, 5.4.5, 13.7.2, 13.26, 13.28.21, 13.29.13, and
58             13.29.28 of IEEE Std 802.1Q.";
59     }
60     container bridge-id {
61         uses bridge-id;
62         description
63             "The Bridge Identifier used by this Bridge for the single spanning
64             tree (the Common Spanning Tree, CST) supported by RSTP, and for
65             that tree and its extension (the CIST) through MST and SPT Regions
66             supported by MSTP and SPB.";
67         reference
68             "13.26.2, 14.1.2, 14.2.5, and item e) in 13.26 of IEEE Std 802.1Q.";
69     }
70     container root-id {
71         uses bridge-id;
72         config false;

```

```

1      description
2          "The Root Bridge of the single spanning tree (the CST) supported
3              by RSTP, and its extension (the CIST) through MST and SPT Regions
4              supported by MSTP and SPB, as currently determined by this node.";
5      reference
6          "13.9, 14.1.2, and item f) in 13.26 of IEEE Std 802.1Q.";
7  }
8  leaf root-path-cost {
9      type uint32;
10     config false;
11     description
12         "The Root Path Cost for the single spanning tree (the CST)
13             supported by RSTP and the External Root Path Cost for its
14             extension (the CIST) through MST and SPT Regions supported by
15             MSTP and SPB. Includes the Port Path Cost of the Bridge's Root
16             Port unless the Bridge is the Root Bridge or its Root Port is not
17             a Boundary Port of its MST or SPT Region.";
18     reference
19         "13.9, 13.10, 13.5.3, and item i) in 13.26 of IEEE Std 802.1Q.";
20 }
21 leaf root-port {
22     type union {
23         type if:interface-ref;
24         type empty;
25     }
26     config false;
27     description
28         "A reference to the Root Port for the single spanning tree
29             (the CST) supported by RSTP or its extension (the CIST) through
30             MST and SPT Regions supported by MSTP and SPB, empty if this
31             Bridge is the Root Bridge.";
32     reference
33         "13.5 and 13.26.9 of IEEE Std 802.1Q.";
34 }
35 leaf max-age {
36     type uint8;
37     units "seconds";
38     config false;
39     description
40         "The maximum age of the spanning tree information (Max Age)
41             currently used by this Bridge for the common spanning tree
42             supported by RSTP, MSTP, and SPB. Set to the bridge-max-age if
43             the Bridge is the Root Bridge, and to the whole seconds part of
44             the value received on the Root Port otherwise.";
45     reference
46         "13.15, 13.20, 13.25, item l) in 13.28, and item g) in 13.26 of
47             IEEE Std 802.1Q.";
48 }
49 leaf hello-time {
50     type uint8;
51     units "seconds";
52     config false;
53     description
54         "The interval (Hello Time) between the periodic transmission
55             of Configuration Bridge PDUs on Designated Ports. Set to
56             2 seconds if the Bridge is currently acting as the Root
57             Bridge, and to the whole seconds part of the value received on
58             the Root Port for the common spanning tree supported by RSTP,
59             MSTP, and SPB otherwise.";
60     reference
61         "Table 13-5, 13.25.3, and 13.28.11 of IEEE Std 802.1Q.";
62 }
63 leaf forward-delay {
64     type uint8;
65     units "seconds";
66     config false;
67     description
68         "If a Port is using STP because STP BPDUs are being received or
69             force-protocol-version is set to emulate-stp (0), Forward Delay
70             determines the time to transition from Discarding to Learning
71             and from Learning to Forwarding, and to age FDB entries after a
72             topology change. Set to bridge-forward-delay if the Bridge is the

```

```

1      CIST Root, and to the whole seconds part of the value received on
2      the CIST Root Port otherwise. Does not affect RSTP, MSTP, and SPB
3      rapid Port State transitions and FDB flushing.";
4      reference
5          "Table 13-5, 13.25.2, 13.28.10, and item a) in 13.26.4 of
6          IEEE Std 802.1Q.";
7      }
8      leaf bridge-max-age {
9          type uint8 {
10             range "6..40";
11         }
12         units "seconds";
13         default "20";
14         description
15             "The value of Max Age when this Bridge is the Root Bridge for the
16             common spanning tree supported by RSTP, MSTP, and SPB.";
17         reference
18             "Table 13-5, 13.26.4, and item g) in 13.26 of IEEE Std 802.1Q.";
19     }
20     leaf bridge-forward-delay {
21         type uint8 {
22             range "4..30";
23         }
24         units "seconds";
25         default "15";
26         description
27             "The value of Forward Delay when this Bridge is the Root Bridge
28             for the common spanning tree supported by RSTP, MSTP, and SPB.";
29         reference
30             "Table 13-5, 13.26.4, and item a) in 13.26.4 of IEEE Std 802.1Q.";
31     }
32     leaf tx-hold-count {
33         type uint8 {
34             range "1..10";
35         }
36         default "6";
37         description
38             "Limits the rate of BPDU transmission. If the variable txCount
39             (which is incremented on each transmission and decremented once a
40             second) reaches TxHoldCount, transmission is delayed.";
41         reference
42             "Table 13-5, 13.27.75, 13.26.12, and item b) in 13.26 of
43             IEEE Std 802.1Q.";
44     }
45     leaf last-topology-change {
46         type yang:date-and-time;
47         config false;
48         description
49             "The time at which the Topology Change timer, tcWhile, for any
50             Bridge Port was last running: when Topology Change Notification
51             (TCN) Messages were last transmitted.";
52         reference
53             "13.25.9 of IEEE Std 802.1Q.";
54     }
55 }
56 }
57
58 grouping bridge-port-parameters {
59     description
60         "This grouping comprises per-interface parameters.";
61     container rstp {
62         presence "The presence of this container indicates that RSTP is
63         supported";
64         description
65             "Per-Bridge Port interface parameters common to RSTP, MSTP, and SPB.";
66         leaf admin-bridge-port-enabled {
67             type boolean;
68             default "true";
69             description
70                 "The Administrative Bridge Port State for the port. If false, the
71                 port is excluded from the active topology for all spanning trees
72                 and Ethernet Switched Paths (ESPs), and BPDUs are not transmitted

```

```

1      or received.";
2      reference
3      "8.4, 8.6.1, 8.13.9, 13.12, and 13.27.45 of IEEE Std 802.1Q.";
4  }
5  leaf port-state {
6      type port-state;
7      config false;
8      description
9          "The Port State for RSTP, and for the CIST for MSTP and SPB.";
10     reference
11         "8.4, 8.6, 8.7, 13.4, item aw) and item ba) in 13.27
12         of IEEE Std 802.1Q.";
13 }
14 leaf port-role {
15     type port-role;
16     config false;
17     description
18         "The Port Role for the single spanning tree (the CST) supported
19         by RSTP, and for its extension (the CIST) through MST and SPT
20         Regions supported by MSTP and SPB.";
21     reference
22         "13.4, 13.5, 13.12, 13.24, 13.27.66, and item bn) in 13.27
23         of IEEE Std 802.1Q.";
24 }
25 leaf restricted-role {
26     type boolean;
27     default "false";
28     description
29         "Set to prevent selection of the port as Root Port for the CIST
30         or any MSTI, even if it has the best spanning tree priority
31         vector, at the risk of causing a lack of spanning tree
32         connectivity, thus preventing Bridges external to a core region
33         of the network influencing the spanning tree active topology.
34         The port will be selected as an Alternate Port rather than a
35         Root Port.";
36     reference
37         "13.20, 13.27.64, 13.29.34, 16.2, 16.3, 26.5, 27.6, 27.20,
38         item s) in 13.27 of IEEE Std 802.1Q.";
39 }
40 leaf restricted-tcn {
41     type boolean;
42     default "false";
43     description
44         "Set to prevent propagation of topology changes and topology change
45         notifications to other ports. Set to prevent a Bridge external to
46         a core region of the network causing address flushing. If set can
47         cause temporary loss of connectivity after a topology change.";
48     reference
49         "13.20, 13.27.65, 13.29.25, 16.2, 16.3, item t) in 13.27
50         of IEEE Std 802.1Q.";
51 }
52 container port-id {
53     uses port-id;
54     description
55         "The Port Identifier used by this Bridge Port by RSTP for the
56         CST and by MSTP and SPB for the CIST.";
57     reference
58         "13.27.46, 14.1.2, items bd) and be) in 13.27
59         of IEEE Std 802.1Q.";
60 }
61 leaf fix-port-path-cost {
62     type uint32 {
63         range "0..2000000000";
64     }
65     default "0";
66     description
67         "Zero if the port-path-cost is to be determined by the Bridge,
68         and the value to be used otherwise.";
69     reference
70         "13.10, 13.18, and Table 13-4 of IEEE Std 802.1Q.";
71 }
72 leaf port-path-cost {

```

```

1      type uint32 {
2          range "1..200000000";
3      }
4      config false;
5      description
6          "The port's contribution to the Bridge's External Root Path Cost,
7           when the port is the CST Root Port (for RSTP and MSTP) or the
8           CIST Master Port (for MSTP and ISIS-SPB).";
9      reference
10         "13.5.3, 13.27.25, Table 13-4, and item g) of 13.27
11         of IEEE Std 802.1Q.";
12    }
13    leaf designated-protocol-version {
14        type uint8;
15        config false;
16        description
17            "The Protocol Version Identifier in BPDUs transmitted by the
18             Designated Bridge for the attached LAN.";
19        reference
20            "14.2.2, 14.3, and Figure 14-1 of IEEE Std 802.1Q.";
21    }
22    action port-protocol-migration-check {
23        description
24            "Executing port-protocol-migration-check (mcheck) forces
25             transmission of RST (or MST or SPT) BPDUs for migrate-time to
26             check that all STP Bridges have been removed from the port's
27             attached LAN, so RST/MST/SPT BPDU transmission can persist.
28             Has no effect if force-protocol-version is emulate-stp(0).";
29        reference
30            "13.27.38, 13.32, and item j) in 13.27 of IEEE Std 802.1Q.";
31    }
32    container root-id {
33        uses bridge-id;
34        config false;
35        description
36            "The CIST Root Identifier in Configuration BPDUs transmitted by
37             the Designated Bridge for the attached LAN.";
38        reference
39            "14.1.2, 13.27.20, 13.10, and item ar) in 13.27
40             of IEEE Std 802.1Q.";
41    }
42    leaf root-path-cost {
43        type uint32;
44        config false;
45        description
46            "The CIST External Path Cost advertised in BPDUS by the
47             Designated Bridge for the CIST for the attached LAN.";
48        reference
49            "14.1.2, 13.27.20, item ar) in 13.27, and 27.6
50             of IEEE Std 802.1Q.";
51    }
52    container designated-bridge-id {
53        uses bridge-id;
54        config false;
55        description
56            "The Bridge Identifier of the Bridge that this port considers to
57             be the Designated Bridge for the attached LAN for RSTP, and the
58             CIST Regional Root for MSTP and SPB.";
59        reference
60            "14.1.2, 13.27.20, 13.10, item ar) in 13.27, and Figure 14-1
61             of IEEE Std 802.1Q.";
62    }
63    container designated-port-id {
64        uses port-id;
65        config false;
66        description
67            "The Port Identifier of the Bridge Port that this port considers
68             to be the Designated Port for the CIST for the attached LAN.";
69        reference
70            "13.27.20, 13.10, item ar) in 13.27, and octets 26 and 27 of
71             Figure 14-1 of IEEE Std 802.1Q.";
72    }

```

```
1 leaf admin-edge-port {
2     type boolean;
3     default "false";
4     description
5         "Set if the port is to be identified as an Edge Port on
6         initialization without a delay to detect other Bridges attached
7         to the LAN. Recommended default false(2). The operational value,
8         oper-edge-port, becomes false(2) if a BPDU has been received.";
9     reference
10        "13.33, and item a) in 13.27 of IEEE Std 802.1Q.";
11 }
12 leaf oper-edge-port {
13     type boolean;
14     config false;
15     description
16         "Set if the port is currently identified as an Edge Port.
17         Initialized to the value of admin-edge-port, and controlled
18         by the Bridge Detection state machine.";
19     reference
20        "13.33, and item l) in 13.27 of IEEE Std 802.1Q.";
21 }
22 leaf auto-edge-port {
23     type boolean;
24     default "true";
25     description
26         "Administratively set true(1) to allow oper-edge-port
27         to be maintained automatically, with continuous monitoring of
28         the presence or absence of other Bridges attached to the LAN.
29         If admin-edge-port and auto-edge-port are both false and the port
30         is attached to a point-to-point LAN, the port is presumed to
31         provide connectivity to another Bridge, and a failure to receive
32         BPDUs when the port is attempting to become Forwarding is taken
33         to be an indication of the failure of that other bridge's Spanning
34         Tree Protocol Entity (or its transmission and/or reception),
35         isolate-port will be set and the port will remain Discarding.";
36     reference
37        "13.33, 13.27.18, and item c) in 13.27 of IEEE Std 802.1Q.";
38 }
39 leaf disputed-port {
40     type boolean;
41     config false;
42     description
43         "Set when more than one Bridge Port apparently believes itself to
44         be the CIST Designated Port for the LAN and has incorrectly
45         transitioned to a Learning or Forwarding port state.";
46     reference
47        "13.21, 13.29.16, Figure 13-20 of IEEE Std 802.1Q.";
48 }
49 leaf isolate-port {
50     type boolean;
51     config false;
52     description
53         "Set by the Bridge Detection state machine when the Spanning Tree
54         Protocol Entity of a neighboring Bridge has apparently failed.";
55     reference
56        "13.23, 13.27.27, 13.33, and item i) in 13.27
57        of IEEE Std 802.1Q.";
58 }
59 }
60 }
61 }
62
63
```


1 48.6.27 The ieee802-dot1q-rstp-bridge YANG module

```
2 module ieee802-dot1q-rstp-bridge {
3   yang-version 1.1;
4   namespace "urn:ieee:std:802.1Q:yang:ieee802-dot1q-rstp-bridge";
5   prefix rstp-bridge;
6
7   import ietf-interfaces {
8     prefix if;
9   }
10  import ieee802-dot1q-bridge {
11    prefix dot1q;
12  }
13  import ieee802-dot1q-rstp {
14    prefix rstp;
15  }
16
17  organization
18    "IEEE 802.1 Working Group";
19  contact
20    "WG-URL: http://www.ieee802.org/1/
21    WG-EMail: stds-802-1-1@ieee.org
22
23    Contact: IEEE 802.1 Working Group Chair
24    Postal: C/O IEEE 802.1 Working Group
25            IEEE Standards Association
26            445 Hoes Lane
27            Piscataway, NJ 08854
28            USA
29
30    E-mail: stds-802-1-chairs@ieee.org";
31  description
32    "This module provides management of IEEE Std 802.1Q Bridge components
33    that support the Rapid Spanning Tree Algorithm and Protocol (RSTP).
34
35    Copyright (C) IEEE (2025).
36
37    This version of this YANG module is part of IEEE Std 802.1Q; see the
38    standard itself for full legal notices.";
39
40  revision 2025-02-04 {
41    description
42      "Published as part of IEEE Std 802.1Qdy-2025.
43
44      The following reference statement identifies each referenced IEEE
45      Standard as updated by applicable amendments.";
46    reference
47      "IEEE Std 802.1Q Bridges and Bridged Networks:
48      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
49      IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qcj-2023,
50      IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024,
51      IEEE Std 802.1Qdy-2025.";
52  }
53
54  augment "/dot1q:bridges/dot1q:bridge/dot1q:component" {
55    description
56      "Augment Bridge component with RSTP parameters.";
57    reference
58      "13.24, 13.25, and 13.26 of IEEE Std 802.1Q.";
59    uses rstp:bridge-component-parameters;
60  }
61
62  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
63    description
64      "Augment Bridge Port with RSTP configuration.";
65    reference
66      "13.24, 13.25, and 13.27 of IEEE Std 802.1Q.";
67    uses rstp:bridge-port-parameters;
68  }
69 }
70
71
```

1 48.6.28 The ieee802-dot1q-mstp YANG module

```
2 module ieee802-dot1q-mstp {
3   yang-version 1.1;
4   namespace "urn:ieee:std:802.1Q:yang:ieee802-dot1q-mstp";
5   prefix mstp;
6
7   import ietf-interfaces {
8     prefix if;
9   }
10  import ieee802-dot1q-rstp {
11    prefix rstp;
12  }
13
14  organization
15    "IEEE 802.1 Working Group";
16  contact
17    "WG-URL: http://www.ieee802.org/1/
18     WG-EMail: stds-802-1-1@ieee.org
19
20     Contact: IEEE 802.1 Working Group Chair
21     Postal: C/O IEEE 802.1 Working Group
22             IEEE Standards Association
23             445 Hoes Lane
24             Piscataway, NJ 08854
25             USA
26
27     E-mail: stds-802-1-chairs@ieee.org";
28  description
29    "This module augments ieee802-dot1q-rstp to provide management of
30     IEEE Std 802.1Q Bridge components that support the
31     Multiple Spanning Tree Algorithm and Protocol (MSTP).
32
33     References specify constraints on, and consequences of, settings of
34     leaf values and the creation and deletion of list entries. The values
35     of all configured objects are retained across system reinitialization.
36
37     Copyright (C) IEEE (2025).
38
39     This version of this YANG module is part of IEEE Std 802.1Q; see the
40     standard itself for full legal notices.";
41
42  revision 2025-02-04 {
43    description
44      "Published as part of IEEE Std 802.1Qdy-2025.
45
46      The following reference statement identifies each referenced IEEE
47      Standard as updated by applicable amendments.";
48    reference
49      "IEEE Std 802.1Q Bridges and Bridged Networks:
50       IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
51       IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qcj-2023,
52       IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024,
53       IEEE Std 802.1Qdy-2025.";
54  }
55
56  grouping mst-config-id {
57    description
58      "A system calculated MST Configuration Identifier (MCID), reflecting
59      the assignment of VIDs to the CIST, MSTIs, SPTs, or ESPs.";
60    reference
61      "13.8, 13.9, and item d) in 13.26 of IEEE Std 802.1Q.";
62    leaf format-selector {
63      type uint8;
64      config false;
65      description
66        "The Configuration Identifier Format Selector. A value of 0
67        indicates the format specified in IEEE Std 802.1Q.";
68      reference
69        "Item a) in 13.8 of IEEE Std 802.1Q.";
70    }
71    leaf configuration-name {
```

```

1     type string {
2         length "1..32";
3     }
4     description
5         "The Configuration Name.";
6     reference
7         "Item b) in 13.8 of IEEE Std 802.1Q.";
8 }
9 leaf revision-level {
10     type uint16;
11     config false;
12     description
13         "The Revision Level.";
14     reference
15         "Item c) in 13.8 of IEEE Std 802.1Q.";
16 }
17 leaf configuration-digest {
18     type binary {
19         length "16";
20     }
21     config false;
22     description
23         "The Configuration Digest (16 octets).";
24     reference
25         "Item d) in 13.8 of IEEE Std 802.1Q.";
26 }
27 }
28
29 grouping bridge-component-parameters {
30     description
31         "This grouping comprises MSTP per-Bridge component parameters.";
32     container bridge-mstp {
33         presence "The presence of this container indicates that MSTP is
34         supported.";
35         description
36             "Per-Bridge configuration and management parameters for the IST
37             and each MSTI in an MST Region.";
38         reference
39             "Clause 13 and 13.26 of IEEE Std 802.1Q.";
40         container mst-config-id {
41             uses mst-config-id;
42             description
43                 "This Bridge's MST Configuration Identifier.";
44         }
45         leaf max-hops {
46             type uint8 {
47                 range "6..100";
48             }
49             default "20";
50             description
51                 "In an MSTP Bridge, the MaxHops parameter for the IST, and each
52                 of the MSTIs.";
53             reference
54                 "Clause 13, Table 13-5, 13.26.4, item g) in 13.26
55                 of IEEE Std 802.1Q.";
56         }
57         container ist {
58             description
59                 "CIST parameter(s) additional to those for RSTP.";
60             leaf internal-root-path-cost {
61                 type uint32;
62                 config false;
63                 description
64                     "The Internal Root Path Cost for the CIST including the
65                     Internal Port Path Cost for the Root Port.";
66                 reference
67                     "13.9, 13.10, 13.5.3, 13.26.10, and octets 90-93 in Figure 14-1
68                     of IEEE Std 802.1Q.";
69             }
70         }
71         list msti {
72             key "mstid";

```

```

1      description
2          "Per-Bridge configuration and management parameters for each MST,
3          with an MSTID in the range allocated for MSTP configuration.";
4      reference
5          "13.26, 13.9, and 13.11 of IEEE Std 802.1Q.";
6      leaf mstid {
7          type uint16 {
8              range "1..4091";
9          }
10         description
11             "MSTIDs allocated for use by MSTP (1..4091) to identify MSTIs.
12             Other MSTIDs in the 12-bit range are reserved [for use in
13             the MST Configuration Table, whose configuration is identified
14             by the Configuration Digest in the MST Configuration Identifier
15             (mst-config-id/configuration-digest)] for the CIST-MSTID,
16             SPBM-MSTID, SPBV-MSTID, TE-MSTID, and SPVID-Poll-MSTID.";
17         reference
18             "8.4, 8.6.1, 8.9.3, 13.2.1 of IEEE Std 802.1Q.";
19     }
20     leaf bridge-priority {
21         type rstp:id-priority;
22         description
23             "The priority component of this Bridge's Bridge Identifier for
24             the MSTI, encoded in the four most-significant bits of octet 14
25             of the MST Configuration Message.";
26         reference
27             "13.26.3, 13.11, 14.4.1, and item f) in 13.26
28             of IEEE Std 802.1Q.";
29     }
30     container regional-root-id {
31         uses rstp:bridge-id;
32         config false;
33         description
34             "The MSTI Regional Root Identifier for this MSTI.";
35         reference
36             "13.27.20, 13.10, and item ar) in 13.27 of IEEE Std 802.1Q.";
37     }
38     leaf internal-root-path-cost {
39         type uint32;
40         config false;
41         description
42             "The Internal Root Path Cost for the MSTI including the
43             Internal Port Path Cost for the Root Port for this MSTI.";
44         reference
45             "13.9, 13.10, 13.5.3, 13.26.10 and item i) in 13.26
46             of IEEE Std 802.1Q.";
47     }
48     leaf root-port {
49         type union {
50             type if:interface-ref;
51             type empty;
52         }
53         config false;
54         description
55             "A reference to the Root Port for the MSTI, empty if this
56             Bridge is the MSTI Regional Root.";
57         reference
58             "13.5 and 13.26.9 of IEEE Std 802.1Q.";
59     }
60 }
61 }
62 }
63
64 grouping bridge-port-parameters {
65     description
66         "This grouping comprises MSTP Bridge Port parameters.";
67     reference
68         "13.24, 13.25, and 13.27 of IEEE Std 802.1Q.";
69     container port-mstp {
70         presence "The presence of this container indicates that MSTP is
71         supported.";
72         description

```

```

1      "Per-Bridge Port interface parameters for MST Configuration,
2      for the IST, and for each MSTI.";
3  leaf boundary-port {
4      type boolean;
5      config false;
6      description
7          "True if the port is a Boundary Port of the Region.";
8      reference
9          "13.12 of IEEE Std 802.1Q.";
10 }
11 leaf restricted-domain-role {
12     type boolean;
13     description
14         "If TRUE, causes a port that is a Boundary Port of an SPT Region
15         not to be selected as Root Port for the CIST, any MSTI, or any
16         SPT, even it has the best spanning tree priority vector.";
17     reference
18         "13.27.63 of IEEE Std 802.1Q.";
19 }
20 container ist {
21     description
22         "CIST parameters additional to those for RSTP.";
23     container mst-config-id {
24         uses mst-config-id;
25         config false;
26         description
27             "The MST Configuration Identifier transmitted by the CIST's
28             Designated Bridge for the attached LAN.";
29         reference
30             "Figure 14-1 of IEEE Std 802.1Q.";
31     }
32     leaf fix-internal-port-path-cost {
33         type uint32 {
34             range "0..200000000";
35         }
36         default "0";
37         description
38             "Zero if the internal-port-path-cost is to be determined by
39             the Bridge, and the value to be used otherwise.";
40         reference
41             "13.10, 13.18, and Table 13-4 of IEEE Std 802.1Q.";
42     }
43     leaf internal-port-path-cost {
44         type uint32 {
45             range "1..200000000";
46         }
47         config false;
48         description
49             "The CIST's Internal Port Path Cost for this Bridge Port.";
50         reference
51             "13.10, 13.18, and Table 13-4 of IEEE Std 802.1Q.";
52     }
53     leaf internal-root-path-cost {
54         type uint32;
55         config false;
56         description
57             "The CIST Internal Root Path Cost advertised in BPDUs by the
58             Designated Bridge for the CIST. Zero if the field was not
59             present in the last BPDU received from that Bridge.";
60         reference
61             "13.5.3, 13.9, 13.27.20, 14.4.1, and 27.6 of IEEE Std 802.1Q.";
62     }
63     container designated-bridge {
64         uses rstp:bridge-id;
65         config false;
66         description
67             "The CIST Bridge Identifier advertised in BPDUs by the CIST's
68             Designated Bridge. Zero if the field was not present in the
69             last BPDU received from that Bridge.";
70         reference
71             "Octets 94-101 of Figure 14-1 of IEEE Std 802.1Q.";
72     }

```

```

1      leaf remaining-hops {
2          type uint8;
3          config false;
4          description
5              "The remainingHops for the CIST, received from or to be
6              transmitted by this port's CIST Designated Bridge.";
7          reference
8              "13.20, 13.26.4, 13.26.11, 13.27.40, 14.2.11, and octet 102 in
9              Figure 14-1 of IEEE Std 802.1Q.";
10     }
11 }
12 list msti {
13     key "mstid";
14     description
15         "Per-Bridge Port parameters for each MSTI with an MSTID in the
16         range allocated for MSTP configuration.";
17     reference
18         "13.27 of IEEE Std 802.1Q.";
19     leaf mstid {
20         type uint16 {
21             range "1..4091";
22         }
23         description
24             "MSTIDs allocated for use by MSTP (1..4091) to identify MSTIs.";
25         reference
26             "8.4, 8.6.1, 8.9.3, 13.2.1 of IEEE Std 802.1Q.";
27     }
28     leaf port-state {
29         type rstp:port-state;
30         config false;
31         description
32             "The Port State for this MSTI.";
33         reference
34             "8.4, 8.6, 8.7, 13.4, item aw) and item ba) in 13.27
35             of IEEE Std 802.1Q.";
36     }
37     leaf port-role {
38         type union {
39             type rstp:port-role;
40             type enumeration {
41                 enum master-port {
42                     value 6;
43                     description "A Master Port.";
44                 }
45             }
46         }
47         config false;
48         description
49             "The Port Role for the MSTI.";
50         reference
51             "13.4, 13.5, 13.12, 13.24, 13.27.66, and item bn) in 13.27
52             of IEEE Std 802.1Q.";
53     }
54     container port-id {
55         uses rstp:port-id;
56         description
57             "This Bridge Port's Port Identifier for the MSTI, including the
58             manageable port-priority encoded in the four most-significant
59             bits of octet 15 of the MSTI Configuration Message.";
60         reference
61             "13.18, 13.27.46, and item e) in 14.4.1, of IEEE Std 802.1Q.";
62     }
63     leaf fix-internal-port-path-cost {
64         type uint32 {
65             range "0..200000000";
66         }
67         default "0";
68         description
69             "Zero if the internal-port-path-cost for this Bridge Port and
70             MSTI is to be determined by the Bridge, and the value to be
71             used otherwise.";
72         reference

```

```

1      "13.11, 13.18, and Table 13-4 of IEEE Std 802.1Q.";
2  }
3  leaf internal-port-path-cost {
4      type uint32 {
5          range "1..200000000";
6      }
7      config false;
8      description
9          "The Internal Port Path Cost for this Bridge Port and MSTI.";
10     reference
11         "Item b) of 13.2.1, and item ay) in 13.27 of IEEE Std 802.1Q.";
12 }
13 container regional-root-id {
14     uses rstp:bridge-id;
15     config false;
16     description
17         "The MSTI Regional Root Identifier in Configuration BPDUs
18         transmitted by the Designated Bridge for the attached LAN
19         and this MSTI.";
20     reference
21         "13.27.20, 13.10, and item ar) in 13.27 of IEEE Std 802.1Q.";
22 }
23 leaf internal-root-path-cost {
24     type uint32;
25     config false;
26     description
27         "The MSTI Internal Path Cost advertised in BPDUS by the
28         Designated Bridge for the attached LAN and this MSTI.";
29     reference
30         "13.5.3, 13.9, 13.27.20, 14.4.1, and 27.6 of IEEE Std 802.1Q.";
31 }
32 container designated-bridge-id {
33     uses rstp:bridge-id;
34     config false;
35     description
36         "The Bridge Identifier of the Bridge believed to be the
37         Designated Bridge for the attached LAN and this MSTI. The
38         bridge-priority component is encoded in the MSTI Configuration
39         Message, the system-id-extension component is the MSTID, and
40         the bridge-address is that of the ist/designated-bridge.";
41     reference
42         "13.27.20, 13.10, and item ar) in 13.27 of IEEE Std 802.1Q.";
43 }
44 container designated-port-id {
45     uses rstp:port-id;
46     config false;
47     description
48         "The Port Identifier of the Designated Port for this MSTI for
49         the attached LAN, transmitting the port-priority in the
50         MSTI Configuration Message and the port-number in the CIST
51         Port Identifier field.";
52     reference
53         "13.27.20, 13.10, item ar) in 13.27, octet 15 in Figure 14-3,
54         and octets 26 and 27 of Figure 14-1 of IEEE Std 802.1Q.";
55 }
56 leaf disputed-port {
57     type boolean;
58     config false;
59     description
60         "Set when more than one Bridge Port apparently believes itself
61         to be the MSTI Designated Port for the LAN and has incorrectly
62         transitioned to a Learning or Forwarding port state.";
63     reference
64         "13.21, 13.29.16, Figure 13-20 of IEEE Std 802.1Q.";
65 }
66 leaf remaining-hops {
67     type uint8;
68     config false;
69     description
70         "The remainingHops for the MSTI, received from or transmitted
71         by the MSTI's Designated Bridge for this port.";
72     reference

```

```
1      "13.20, 13.26.4, 13.26.11, 13.27.40, 14.2.11, and octet 16 in
2      Figure 14-3 of IEEE Std 802.1Q.";
3
4      }
5  }
6  }
7 }
8
9
```


1 48.6.29 The ieee802-dot1q-mstp-bridge YANG module

```
2 module ieee802-dot1q-mstp-bridge {
3   yang-version 1.1;
4   namespace "urn:ieee:std:802.1Q:yang:ieee802-dot1q-mstp-bridge";
5   prefix mstp-bridge;
6
7   import ietf-interfaces {
8     prefix if;
9   }
10  import ieee802-dot1q-bridge {
11    prefix dot1q;
12  }
13  import ieee802-dot1q-rstp-bridge {
14    prefix rstp-bridge;
15  }
16  import ieee802-dot1q-mstp {
17    prefix mstp;
18  }
19
20  organization
21    "IEEE 802.1 Working Group";
22  contact
23    "WG-URL: http://www.ieee802.org/1/
24    WG-EMail: stds-802-1-1@ieee.org
25
26    Contact: IEEE 802.1 Working Group Chair
27    Postal: C/O IEEE 802.1 Working Group
28            IEEE Standards Association
29            445 Hoes Lane
30            Piscataway, NJ 08854
31            USA
32
33    E-mail: stds-802-1-chairs@ieee.org";
34  description
35    "This module provides management of IEEE Std 802.1Q Bridge components
36    that support the Multiple Spanning Tree Algorithm and Protocol (MSTP).
37
38    Copyright (C) IEEE (2025).
39
40    This version of this YANG module is part of IEEE Std 802.1Q; see the
41    standard itself for full legal notices.";
42
43  revision 2025-02-04 {
44    description
45      "Published as part of IEEE Std 802.1Qdy-2025.
46
47      The following reference statement identifies each referenced IEEE
48      Standard as updated by applicable amendments.";
49    reference
50      "IEEE Std 802.1Q Bridges and Bridged Networks:
51      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
52      IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qcj-2023,
53      IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024,
54      IEEE Std 802.1Qdy-2025.";
55  }
56
57  augment "/dot1q:bridges/dot1q:bridge/dot1q:component/dot1q:bridge-mst" {
58    when '../rstp-bridge:rstp';
59    description
60      "Augment RSTP-capable Bridge component with MSTP configuration and
61      management.";
62    reference
63      "13.24, 13.25, and 13.26 of IEEE Std 802.1Q.";
64    uses mstp:bridge-component-parameters;
65  }
66
67  augment "/if:interfaces/if:interface/dot1q:bridge-port" {
68    when 'rstp-bridge:rstp';
69    description
70      "Augment RSTP Bridge Port with MSTP configuration.";
71    reference
```

```
1      "13.24, 13.25, and 13.27 of IEEE Std 802.1Q.";
2      uses mstp:bridge-port-parameters;
3  }
4 }
5
6
```

49. Congestion Isolation

Congestion Isolation (CI) mitigates head-of-line blocking caused by the frequent use of PFC in lossless networks and reduces frame loss in lossy networks that are not using PFC. CI identifies the flows that are most likely causing congestion, isolates them to a separate lower priority traffic class and signals to the upstream peer to do the same. CI effectively moves the congesting flows out of the way, temporarily delaying the delivery of congesting frames, while the higher layer congestion control feedback loop has time to take effect.

This clause introduces the concepts and protocols essential to congestion isolation as follows:

- a) The objectives for congestion isolation (49.1).
- b) Principles of congestion isolation (49.2).
- c) Congestion Isolation Aware Forwarding Process (49.3).
- d) Congestion Isolation Protocol (49.4).
- e) Topology Recognition (49.5).

In current data center networks, traffic can be a mix of various multi-tenant TCP and UDP flows across both the physical underlay and virtual overlay network. Intermittent congestion within the network can occur from an unfortunate mix of flows across the network. For example, a small number of long duration ‘elephant’ flows can align in such a way to create queuing delays for the larger number of short, but critical, ‘mice’ flows. In this case, the ‘elephant’ flows would be considered congesting flows to the ‘mice’ flows. Any flow can become a congesting flow when it contributes to congestion in the network.

Queuing delays deter the end-to-end congestion control loop, and in a lossless environment, cannot prevent Priority-based Flow Control (PFC) from being invoked (see Clause 36). When buffers fill and eventual flow-control kicks in (for lossless networks), non-congesting flows can be blocked by the backlog of frames from congesting flows. If PFC is not being used, frame loss for non-congesting flows can result in long retransmission timeouts, significantly penalizing such flows, which are often used for application control and synchronization.

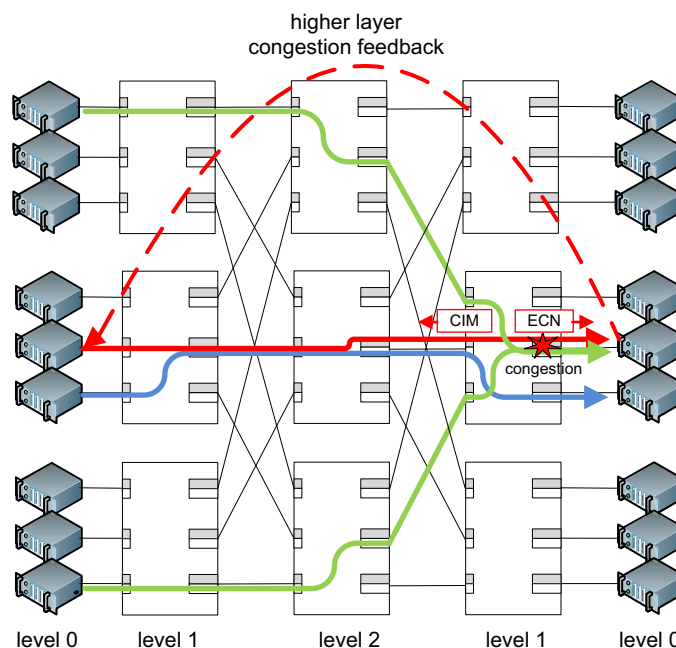


Figure 49-1—Congestion Isolation example operation

1 Figure 49-1 shows an example operation of CI. In the figure, all of the relay systems are layer-3 routers and
2 are congestion isolation aware systems. The example shows a network of layer-3 routers, but congestion
3 isolation also applies to Bridged Networks. Each system is showing two queues on each port; a monitored
4 queue and a congesting queue. Server-to-server traffic is flowing from left to right across a data center
5 network. When congestion is detected CI attempts to identify the congesting flows (49.2.1). For each
6 identified congesting flow (e.g. the red flow in Figure 49-1), the congestion isolation aware system takes
7 local actions to isolate the congesting flow, sends a Congestion Isolation Message (CIM) upstream if the
8 congesting queue is congested, and sends Explicit Congestion Notification (ECN) downstream.

9 The local actions taken by a congestion isolation aware system deprioritize frames from the congesting flow
10 and place them in a dedicated congesting flow queue (49.2.3). An entry is made in the IEEE Std 802.1CB
11 Stream identity table of the local system so that subsequent frames from the congesting flow can be
12 identified and deprioritized (49.2.2).

13 When a flow is isolated the deprioritized frames are queued in the congesting flow queue. When that queue
14 becomes congested (49.2.1) and the upstream peer is congestion isolation aware, the CI functionality signals
15 to the immediate upstream peer using a CIM (49.2.5).

16 The CIM contains flow description information necessary for the upstream peer to identify the same
17 congesting flow. The upstream peer creates a Stream identity table entry using the contents of the CIM
18 (49.2.2). The table entry is used to isolate the same flow by identifying subsequent frames and deprioritizing
19 those frames to the congested flow queue.

20 The IP packets in frames of the congesting flows are marked with higher layer congestion signals such as
21 those defined by the IETF RFC 3168 for Explicit Congestion Notification (ECN). Since CI is performed
22 within the network it is essential that end-to-end higher layer congestion control is combined with CI so the
23 sending sources will reduce their rate of transmission in response to the congestion signals. ECN operates by
24 marking packets where they encounter queuing delay as they travel towards the destination. At the IP
25 destination, feedback is signaled to the source system, so it takes time for the signals to propagate and
26 ultimately impact the transmission rate of the source system.

27 The frames in the congesting flow queue are serviced at a lower priority than other non-congesting queues,
28 so if congestion persists, the congesting flow queue may fill. When the congesting flow queue fills, the
29 ingress port feeding that queue can issue PFC for the lower priority queue to avoid frame loss (49.2.4). In
30 this case, PFC only acts on the lower priority congesting flow queues and other well-behaved flows continue
31 through the network via non-congesting priority queues.

32 **49.1 Congestion isolation objectives**

33 The operation, procedures, and protocols of congestion isolation are designed to meet the following
34 objectives by category:

35 **Functionality**

- 36 a) With high probability, identify the flows that are causing congestion.
- 37 b) Avoid head-of-line blocking of non-congesting flows locally by quickly adjusting the traffic class of
38 congesting flows.
- 39 c) Avoid head-of-line blocking of non-congesting flows upstream by signaling to upstream peers the
40 information needed to isolate the same congesting flows.
- 41 d) Reduce the frequency of invoking PFC in a lossless environment.

42 **Compatibility**

- 43 e) Work in legacy environments by automatically detecting legacy peers and disabling CIM signaling.
- 44 f) Work in existing lossless environments using PFC without requiring additional traffic classes.

- 1 g) Work in conjunction with higher layer end-to-end congestion control protocols (e.g. Explicit
- 2 Congestion Notification (ECN), Remote Direct Memory Access (RDMA) over Converged Ethernet
- 3 (RoCE), Data Center Quantized Congestion Notification (DCQCN [B69])).
- 4 h) Work in both layer-2 and layer-3 networks.
- 5 i) Coexist with existing traffic scheduling paradigms on other traffic classes.

6 Performance

- 7 j) Reduce average flow completion time across the data center network.
- 8 k) Reduce the amount of time frames are paused when PFC is enabled.
- 9 l) Reduce overall frame loss when PFC is not enabled.
- 10 m) Reduce head-of-line blocking of victim flows at upstream peers from PFC.
- 11 n) Reduce overall congestion control signaling.
- 12 o) Increase link utilization.

13 Scale

- 14 p) Work in arbitrary data center network topologies with a mix of link speeds.
- 15 q) Limit messaging overhead by restricting message propagation to hop-by-hop.
- 16 r) Reduce Stream identity table requirements by only requiring the registration of congesting flows and
- 17 providing facilities to rapidly remove flows that are no longer congesting.

18 Implementation complexity

- 19 s) Limit the impact of existing traffic selection algorithms.
- 20 t) Achieve the benefits of congestion isolation without additional buffer requirements.
- 21 u) Support implementations of existing traffic classes.
- 22 v) Leverage existing standard functionality for congesting flow identification and stream identification.

23 Manageability

- 24 w) Only require a small set of configuration parameters that are consistent across multiple Bridge
- 25 deployments.
- 26 x) Limit the ability to configure an inoperable environment.
- 27 y) Provide auto discovery of peer capabilities using existing LLDP messages and without creating
- 28 additional hello and auto-configuration protocols.

29 49.2 Principles of congestion isolation

30 This clause introduces the principles of congestion isolation. Items a) through f) describe the life of a
31 congesting flow from identification through isolation to returning the flow to non-congesting status. Item g)
32 describes a mechanism for determining a device's level in the data center topology that can be used to ensure
33 compatibility and proper operation of existing protocols. Item h) compares and contrasts congestion
34 isolation with Congestion Notification (see Clause 30).

35 The following items describe the principles of congestion isolation:

- 36 a) Congesting flow identification (49.2.1).
- 37 b) IEEE Std 802.1CB stream identification (49.2.2).
- 38 c) Flow priority modification (49.2.3).
- 39 d) Priority-based Flow Control interaction (49.2.4)
- 40 e) Congestion isolation signaling (49.2.5).
- 41 f) Congesting to non-congesting status change (49.2.6).
- 42 g) System topology and port orientation (49.2.7).
- 43 h) Comparison to Congestion Notification (49.2.8).

1 49.2.1 Congesting flow identification

2 An essential step in the process of CI is identifying congesting flows by an Active Queue Management
3 (AQM) scheme that supports Explicit Congestion Notification (ECN) specified in IETF RFC 3168. There
4 are many potential methods of identifying congesting flows and interoperable implementations can exist
5 using different approaches. The CIP Congestion Detection function (49.3.1) of the Congestion Isolation
6 Aware Forwarding Process (49.3) is responsible for implementing the AQM. This standard defines the CP
7 algorithm (30.2.1) for detecting Congestion Controlled Flows (CCFs) in congestion aware Bridges. This
8 approach may be used to detect congesting flows in a CI aware system. A number of other possible
9 approaches, including those that support the end-to-end ECN congestion control, are discussed in IETF
10 RFC 7567 [B45].

11 Many modern data centers utilize encapsulated overlay networks, such as those described in
12 IETF RFC 8014 [B46]. An overlay network can carry multiple encapsulated flows within a single
13 encapsulation flow. The congesting and non-congesting flows identified by CI are the outer encapsulation
14 flow as seen by the underlay network. The inner encapsulated flows might not be visible to the bridges and
15 routers within the data center network, and are therefore not separated into congesting and non-congesting
16 flows.

17 49.2.2 IEEE Std 802.1CB stream identification

18 Subsequent frames of congesting flows are identified by matching in the IEEE Std 802.1CB Stream identity
19 table. Congesting flow entries are added to and removed from the Stream identity table as `tsnStreamIdEntry`
20 objects using the managed object interface specified in Clause 9 of IEEE Std 802.1CB-2017. The
21 `tsnStreamIdIdentificationType` for these objects is IP stream identification. The contents for an IP stream
22 `tsnStreamIdEntry` come from the received frame that has been identified as causing congestion or from the
23 parameters in the CIM PDU (49.4.3.4) from a peer. Flows matching in the Stream identity table will provide
24 a `stream_handle` for congesting flows to the EISS. The `stream_handle` for congesting flows is used as the
25 `tsnStreamIdHandle` entity of the `tsnStreamIdEntry` object.

26 Congestion isolation uses `stream_handles` to identify frames of congesting and non-congesting flows. This
27 use prohibits `stream_handles` from being used by other functionality requiring `stream_handles` on a port
28 where congestion isolation operates.

29 49.2.3 Flow priority modification

30 CI uses the facilities of per-stream classification and metering (8.6.5.2) to modify the priority of congesting
31 flow frames. The `stream_handle`, available as part of the `connection_identifier`, is provided by the Stream
32 identification function and accompanies frames of congesting flows as they pass through a relay. This
33 `stream_handle` is used to select a stream filter instance and stream gate instance that will modify the priority
34 of the frame for congesting flows. A single stream filter instance from the stream filter instance table
35 (8.6.5.2.3), with a wildcard match on the `stream_handle`, is used to select a stream gate instance that contains
36 the new internal priority value (IPV). Configuration of the stream filter instance table and stream gate
37 instance table is described in 8.6.5.

38 The process of modifying the priority of subsequent frames of a congesting flow brings with it the risk of
39 misordering. There may be a period of time where frames for the same flow exist in multiple queues at the
40 same time. Strict priority transmission selection can ensure ordering is maintained for congesting flows
41 because the monitored queue is higher priority than the congesting queue. However, other transmission
42 selection algorithms may be desired. Congestion isolation implementations need to ensure that misordering
43 (6.5.3) remains negligible. Annex W discusses possible approaches for ensuring ordering.

44 49.2.4 Priority-based Flow Control interaction

45 Priority-based Flow Control (PFC) (Clause 36) is used in data centers to create lossless traffic classes in the
46 data center network. PFC allows link flow control to be performed on a per-priority basis. For PFC to
47 operate properly, it is imperative that the downstream system can determine the priority of the frame when it

egresses the upstream peer. The Priority Code Point (PCP—see 6.9.3) field of a VLAN tag encodes a frame's priority. Since an upstream congestion isolation aware system modifies the priority of congesting frames, it must also modify the PCP so the downstream system can assert PFC, if needed, on the correct priority. Therefore all frames must be VLAN tagged when operating CI with PFC in a lossless environment. Additional design considerations for the interaction of PFC and CI are discussed in W.2.

49.2.5 Congestion isolation signaling

Once a flow has been isolated and subsequent frames are forwarded through the congesting queue and the congesting queue becomes congested, it is useful to inform the upstream peer of this event by signaling a Congestion Isolation Message (CIM). Upon receiving a CIM, the upstream peer can isolate the same flow by creating an entry in the IEEE Std 802.1CB Stream identity table. The benefit of isolating a flow upon the directive of a peer is that the congesting flow can be moved out of the way of other non-congesting flows and the potential for head-of-line blocking can be reduced.

The CIM contains parameters and a portion of the received frame so the upstream peer can identify the congesting flow and create an entry in its Stream identity table. The parameters and portion of the received frame included in the CIM must be the same as those of the frame sent by the upstream peer in order for that peer to identify the same flow. Features such as the VID translation table (6.9) can change the frame's parameters from what was sent by the upstream peer and should not be used with CI.

The CIM is addressed to the upstream peer. The system generating the CIM must locate CIM format information and the address parameters of the upstream peer by indexing into the CI Peer Table (49.3.6) using the reception port as the index. The CI Peer Table is built using information provided by LLDP and the Congestion Isolation TLV (D.2.15).

By default, a CIM contains the first 64 bytes of the congesting frame's `mac_service_data_unit`. Systems encapsulating and decapsulating flows for overlay networks may request the peer to include up to 512 octets of the received frame in the CIM. This allows the systems to access and identify encapsulated flows within the congesting flow encapsulation.

49.2.6 Congesting to non-congesting status change

Flows that are no longer causing congestion may be removed from the Stream identity table. Many methods of determining that a flow is no longer causing congestion are possible, but at a minimum, all flows in the Stream identity table for the port can be removed when the congesting queue for that port becomes empty. Waiting until the congesting queue is empty can cause congesting flows to remain in the congesting state longer than necessary, but it allows an implementation to easily abide by the frame misordering requirements of (6.5.3). Other approaches can react quickly and independently of other congesting flows, but may bring the risk of misordering. Implementations need to ensure that misordering is negligible. An example approach for avoiding misordering when returning a flow to non-congesting status can be seen in Annex W.

Frames from congesting flows received downstream from the root of congestion may be safely returned to the non-congesting traffic class. These frames are easily identified downstream because they are received on the congesting traffic class, but do not have a matching entry in the Stream identity table and will therefore not have a `stream_handle` parameter when passed through relay. These frames are isolated to the congesting queue by an upstream peer and the traffic class, as well as the associated Priority Code Point (PCP—see 6.9.3) or other indicators used to select a traffic class, such as the Differentiated Services Code Point (DSCP—see IETF RFC 2474) are modified upstream to indicate they are part of a congesting flow. Downstream from the congestion, however, these frames can be safely forwarded through the monitored queue. A second stream filter instance is used to return these frames to the monitored queue and update the PCP or other indicators used to select a traffic class. This stream filter is defined to match on the absence of

1 a stream_handle and the priority for the congesting queue. The end-to-end ECN congestion markings remain
2 unchanged and the congestion indications in the higher layer protocols will be noted at the source of the flow
3 once the feedback is propagated.

4 NOTE—As noted in 6.8.1, modification to a frame's PCP or other indicators used to select a traffic class, will require the
5 FCS to be regenerated. Options for regenerating the FCS are discussed in Annex O.

6 **49.2.7 System topology and port orientation**

7 Figure 49-1 shows an example of a popular data center topology. Other topologies are possible, but they all
8 can be described as having devices positioned at levels within the topology. The topology levels are
9 distinguished by considering the number of link hops from any interior system to non-relay end stations or
10 servers at the edge of the network. By this definition, end stations and servers are at level 0. The immediate
11 attached device would be at level 1 and further interior only systems could be at level 2 or higher depending
12 on the connectivity of the topology.

13 A port's orientation is distinguished by considering the level in the topology a port is facing. A port facing a
14 lower level has an orientation of a downlink, while a port facing a higher level has an orientation of an
15 uplink. A port facing a system at the same level has an orientation of a crosslink.

16 Congestion isolation benefits from knowing a system's level in the topology. Recognizing the level is
17 automated by the use of the LLDP Topology Recognition TLV (D.2.16) and the procedures defined in
18 49.5.3. During the process of isolating a congesting flow, the frame's priority may be changed from its
19 original value. The corresponding Priority Code Point (PCP—see 6.9.3) or Differentiated Services Code
20 Point (DSCP—see IETF RFC 2474) received at the destination may be different than the value provided by
21 the original sender. To avoid complications with unexpected PCP or DSCP values, the Bridge or router at
22 level 1 in the topology may change the PCP or DSCP back to the original value for the monitored queue.

23 In addition to avoiding complications with changed PCP or DSCP values, knowing a system's level in the
24 topology can assist in identifying the type of congestion experienced and consequentially a strategy for
25 mitigating it. Ports of Bridges or routers at level 1 in the topology are connected to non-relay end stations or
26 servers. Congestion on these ports can often be attributed to incast traffic patterns. Congestion on links of
27 systems that are at level 2 or higher in the topology may be experiencing in-network congestion. It has been
28 shown that load balancing techniques, which may alleviate in-network congestion, are counter productive at
29 addressing incast congestion [B61].

30 Lossless networks enabled by PFC have been shown, in certain circumstances, to have circular buffer
31 dependencies that can cause deadlocks when traffic is re-routed due to link failures [B5]. Again, knowing
32 the position in the topology assists in knowing when traffic has been re-routed and can be used to break
33 circular buffer dependent deadlocks [B4].

34 **49.2.8 Comparison to Congestion Notification**

35 Congestion Isolation and Congestion Notification (Clause 30) share common aspects but are distinctly
36 different. Both generate congestion messages and send them upstream, but CN directs the message across
37 the Bridged Network to the source station, while CI directs the message to the immediate peer. CI is
38 intended to operate in layer 3 and layer 2 networks with higher layer end-to-end congestion control
39 supported by the end stations to regulate the rate of traffic injection into the network. CN is designed to
40 operate across a large layer 2 domain where reaction points in end-station network adapters regulate traffic
41 injection. CI isolates congesting flows by reclassifying their traffic class. CN does not modify the traffic
42 class of frames.

49.3 The Congestion Isolation Aware Forwarding Process

This clause specifies the architecture of the Congestion Isolation Point (CIP) in the Forwarding Process of a congestion isolation aware Bridge. In this architecture, a router is as a higher layer entity that relays frames using layer-3 information but uses the forwarding process of the underlying congestion isolation aware Bridge to deliver frames to peers and end stations.

The models of operation in this clause provide a basis for specifying the externally observable behavior of CI, and are not intended to place additional constraints on implementations; these can adopt any internal model of operation compatible with the externally observable behavior specified. Conformance of equipment to this standard is purely in respect of observable protocol.

Figure 8-12 illustrates the Bridge Forwarding Process at its highest conceptual level. Figure 8-16 shows the specific filtering and assignment functions of the flow classification and metering elements in the Forwarding Process of a congestion isolation aware Bridge. Figure 49-2 focuses on the operation of a single Bridge Port and the relationship of new elements to the queuing and classification functions. Five new elements and two new managed tables are specified for a CI aware Bridge as follows:

- a) CIP Congestion Detection (49.3.1).
- b) CIP transmission gates (49.3.2).
- c) CIM Demultiplexer (49.3.3).
- d) Congesting flow identification (49.3.4).
- e) CIM Multiplexer (49.3.5).
- f) CI Peer Table (49.3.6).
- g) CI Stream Table (49.3.7).

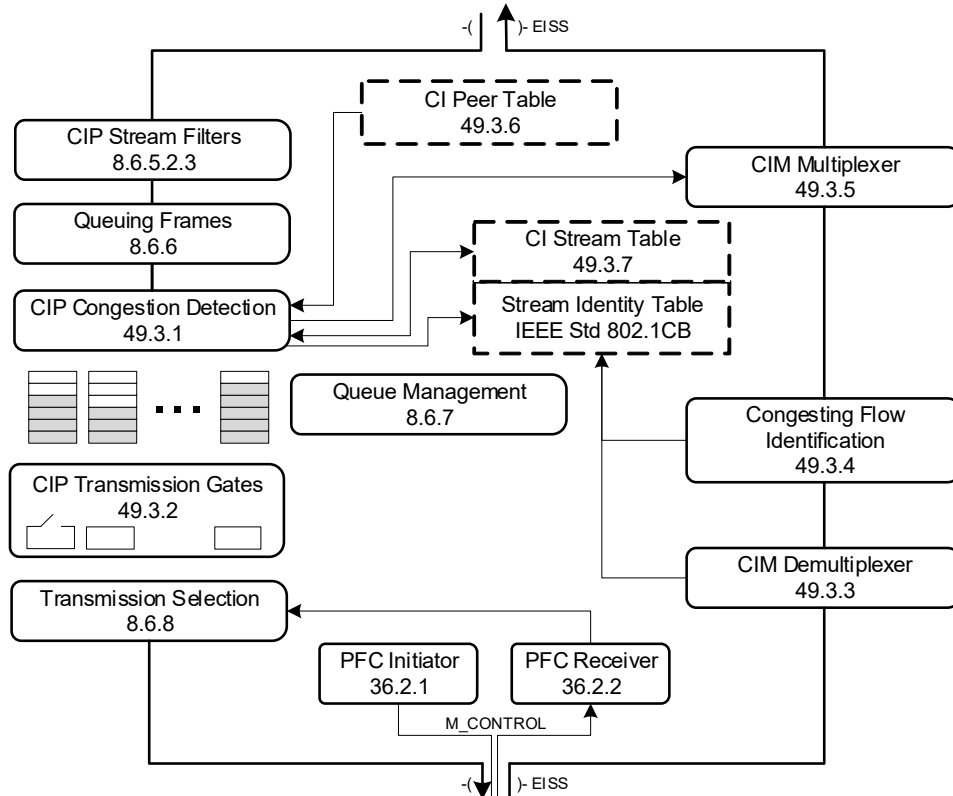


Figure 49-2—Congestion Isolation reference diagram

1 49.3.1 CIP Congestion Detection

2 CIP Congestion Detection identifies frames of a congesting flow, inserts frames passed by the Queuing
3 Frames entity (8.6.6) into the appropriate queue or discards them and can generate a CIM based on the
4 contents of the frame. CIP Congestion Detection is also responsible for creating entries in the Stream
5 identity table and the CI Stream Table (49.3.7) for newly identified congesting flows. As described in 49.2.1,
6 frames of a congesting flow are identified by a suitable AQM implemented by CIP Congestion Detection,
7 such as the algorithm specified by a CP in 30.2.1.

8 Frames given to CIP Congestion Detection by the Queuing Frames entity (8.6.6) in an
9 EM_UNITDATA.request (49.4.2.2) may be identified by the AQM as being part of a congesting flow. CIP
10 Congestion Detection creates a new entry in the Stream identity table and the CI Stream Table for congesting
11 flows received on a monitored queue. A CIM may be generated from the parameters obtained with the
12 received frame and values from the CI Peer Table entry for the source port of the received frame. The CIM is
13 delivered to the upstream peer via the CIM Multiplexer.

14 On each EM_UNITDATA.request, CIP Congestion Detection and its associated AQM indicate to the CIP
15 procedures (49.4) whether a monitored or congesting queue is empty, congested, or not congested. The
16 AQM or other means can be used to provide the queue status.

17 49.3.2 CIP transmission gates

18 CIP transmission gates control whether or not a queued frame can be selected for transmission by the
19 transmission selection algorithm. The purpose of controlling a queue's availability to traffic selection is to
20 allow the CI implementation a means to preserve order of congesting flow frames. The need to preserve
21 order is dependent upon the transmission selection algorithm configured for the congesting and monitored
22 queues. When the transmission selection algorithm is strict priority, the state of the transmission gate is
23 permanently *open*. When the transmission selection algorithm is anything other than strict priority the
24 implementation is responsible for ensuring order. An example approach is discussed in Annex W.

25 49.3.3 CIM Demultiplexer

26 The CIM Demultiplexer identifies CIMs received from the LAN and uses the content of the CIM PDU
27 (49.4.3.4) to create an entry in the IEEE Std 802.1CB Stream identity table. A CIM PDU may be
28 encapsulated by three different CIM encapsulations; layer-2, IPv4, or IPv6. Implementations supporting
29 IPv4 and IPv6 encapsulations must be able to identify and validate IPv4 and/or IPv6 packets in the CIM
30 Demultiplexer. The rules for validating received CIMs are specified in 49.4.3.5.

31 49.3.4 Congesting flow identification

32 Frames received from the LAN belonging to congesting flows are identified by the Stream identification
33 function of IEEE Std 802.1CB. The Stream identification function provides a *stream_handle* subparameter
34 in the *connection_identifier* parameter of an EM_UNITDATA.indication for frames of congesting flows and
35 does not provide a *stream_handle* for frames of non-congesting flows. The presence or non-presence of the
36 *stream_handle* is used to match a stream filter instance in the per-stream classification and metering function
37 to modify the frame's priority (8.6.5.2.3).

38 49.3.5 CIM Multiplexer

39 The CIM Multiplexer inserts CIMs generated by the CIP Congestion Detection element among frames
40 received from the LAN. Layer-2 encapsulated CIMs (49.4.3.1) are delivered to the upstream peer through
41 the Bridge relay. Layer-3 encapsulated CIMs (49.4.3.2 and 49.4.3.3) are routed to the upstream peer by a
42 higher-layer routing function that is beyond the scope of this standard.

1 49.3.6 CI Peer Table

2 The CI Peer Table (12.33.2) contains entries for each participating immediate peer and provides the
3 information needed to generate a CIM for transmission to the peer. The entries are indexed by port number.
4 An entry contains the format type of the CIM needed to reach the peer as well as the addressing information
5 needed to generate the message. This includes the MAC address to use as the destination address and
6 optionally an IPv4 or IPv6 address. The entry provides the UDP port number to use when the CIM
7 encapsulation is a layer-3. An entry also contains the number of bytes of the received congesting flow frame
8 to encapsulate in the CIM PDU itself. The contents of the entry are obtained from the LLDP Congestion
9 Isolation TLV (D.2.15) received from participating peers.

10 49.3.7 CI Stream Table

11 The CI Stream Table (12.33.3) contains entries for each congesting flow and has a 1:1 mapping to entries in
12 the IEEE Std 802.1CB Stream identity table. The entries are indexed by the stream_handle provided as a
13 subparameter of the connection_identifier in EM_UNITDATA.request (49.4.2.2) invocations at the EISS.
14 Each entry contains controlling variables (49.4.1.5) that allow an implementation to manage a congesting
15 flow. This includes a means of identifying the congesting queue used to isolate the congesting flow. It also
16 maintains a count of the CIMs sent on behalf of the congesting flow as well as an indication of why the entry
17 was created or updated.

18 49.4 Congestion Isolation Protocol

19 Congestion isolation aware systems control forwarding elements, participate in congestion isolation
20 protocols, and act upon the LLDP Congestion Isolation TLV as specified in this clause. This includes:

- 21 a) Variables controlling operation (49.4.1).
- 22 b) CIP procedures (49.4.2).
- 23 c) Encoding of the CIM PDU and CIM encapsulations (49.4.3).
- 24 d) LLDP Congestion Isolation TLV (49.4.4).

25 49.4.1 Variables controlling operation

26 The congestion isolation variables control the operation of the CI entity and the CIP entity.

27 49.4.1.1 CI entity variables

28 Every congestion isolation aware system has a set of CI entity variables to control the overall operation of
29 CI. These variables are included in the CI entity managed object (12.33.1). These include the following:

- 30 a) ciMasterEnable (49.4.1.1.1).
- 31 b) ciCIMTransmitPriority (49.4.1.1.2).
- 32 c) ciMaxFlowLife (49.4.1.1.3).

33 49.4.1.1.1 ciMasterEnable

34 A boolean value specifying whether CI is enabled in this system. If ciMasterEnable is FALSE all congestion
35 isolation activity is disabled; CIMs and LLDP Congestion Isolation TLVs are not generated and are ignored
36 on receipt. If ciMasterEnable is TRUE the other managed objects and variables specified in the clause
37 control the operation of CI.

38 49.4.1.1.2 ciCIMTransmitPriority

39 An integer specifying the priority value to be used when transmitting CIMs from the system. The default
40 is 6.

1 **49.4.1.1.3 ciMaxFlowLife**

2 An unsigned integer specifying the maximum number of centiseconds that a congesting flow entry, created
3 by the receipt of a CIM, can remain in the CI Stream Table beyond the time indicated by the per-port
4 per-queue variable cipNonCongestedTime (49.4.1.4.2). The default value is 100.

5 **49.4.1.2 CIP entity variables**

6 The following variables control the operation of a CIP:

- 7 a) cipMacAddress (49.4.1.2.1)
- 8 b) cipIPv4Address (49.4.1.2.2)
- 9 c) cipIPv6Address (49.4.1.2.3)
- 10 d) cipCIMUDPPort (49.4.1.2.4)
- 11 e) cipQueueMap[] (49.4.1.2.5)
- 12 f) cipMinHeaderOctets (49.4.1.2.6)

13 **49.4.1.2.1 cipMacAddress**

14 The MAC address, belonging to the system transmitting the CIM (49.4.3), used as the source_address of
15 CIMs sent from the CIP.

16 **49.4.1.2.2 cipIPv4Address**

17 The IPv4 address, belonging to the system transmitting the IPv4 layer-3 CIM (49.4.3.2), used as the IPv4
18 source address in the IPv4 header (IETF RFC 791) of IPv4 layer-3 CIMs sent from the CIP. The
19 cipIPv4Address is included in the LLDP Congestion Isolation TLV (D.2.15) to allow peers to properly
20 generate IPv4 layer-3 CIMs that can be received by the system.

21 **49.4.1.2.3 cipIPv6Address**

22 The IPv6 address, belonging to the system transmitting the IPv6 layer-3 CIM (49.4.3.3), used as the IPv6
23 source address in the IPv6 header (IETF RFC 8200) of IPv6 layer-3 CIMs sent from the CIP. The
24 cipIPv6Address is included in the LLDP Congestion Isolation TLV (D.2.15) to allow peers to properly
25 generate IPv6 layer-3 CIMs that can be received by the system.

26 **49.4.1.2.4 cipCIMUDPPort**

27 The destination UDP port number in the UDP header (IETF RFC 768) of IPv4 and IPv6 layer-3 CIMs sent
28 by peers. The value is also used as the UDP source port number in layer-3 CIMs sent to peers. The UDP port
29 number must be selected from the range of dynamic port numbers, between 49152 and 65535, as specified in
30 IETF RFC 6335. The port number must be currently available for use by the implementation. For example,
31 an implementation may use UDP port 58622, if it is not currently being used by any other application in the
32 system.

33 **49.4.1.2.5 cipQueueMap[]**

34 An array of integers, one entry for each traffic class, 0 through 7, specifying a value that can be translated to
35 the numeric value of the traffic class to be used as either the congesting traffic class or the monitored traffic
36 class for the traffic class specified by the index. The integers range in value from –8 to 8. A value of 0 in the
37 table specifies that the traffic class is not participating in congestion isolation. A positive number specifies a
38 traffic class for a monitored queue that is one less than the value (e.g., a value of 5 represents traffic class 4).
39 A negative number specifies a traffic class for a congesting queue that is one less than the absolute value
40 (e.g., a value of –4 represents traffic class 3).

41 NOTE—Configuration of the cipQueueMap[] is flexible, allowing the arrangement of multiple monitored and
42 congesting queue sets.

1 Frames from flows that the AQM determines are causing congestion and are currently transferring through
2 the monitored traffic class specified by the index into this table may have their priority remapped to the
3 translated traffic class specified by the entry in the `cipQueueMap[]`. Frames from flows that have an entry in
4 the Stream identity table have a `stream_handle` that specifies a stream filter to remap the priority to the IPV.
5 Frames that are transferring through a congesting traffic class and do not have a `stream_handle` match on the
6 null-handle stream filter entry to have their priority remapped to the monitored traffic class specified in the
7 IPV.

8 The `cipQueueMap[]` is included in the LLDP Congestion Isolation TLV (D.2.15) to inform peers of the
9 system's configuration. Mismatched configurations can be detected by comparing the queue map received in
10 the TLV with configured `cipQueueMap[]`.

11 **49.4.1.2.6 cipMinHeaderOctets**

12 The minimum number of octets that the CIP is to return in the Encapsulated MSDU field of each CIM
13 generated. The default value is 48. The `cipMinHeaderOctets` is included in the LLDP Congestion Isolation
14 TLV (D.2.15) to inform peers of the required minimum. The `cipMinHeaderOctets` value from peers is stored
15 in the CI Peer Table (49.3.6) and used to generate CIMs to be transmitted to the peer.

16 **49.4.1.2.7 cipMaxCIM**

17 The maximum number of times a CIM PDU will be sent for a congesting flow. The default value is 3. A
18 larger value provides more resilience to lost CIMs, but generates more traffic on the network.

19 **49.4.1.3 CIP entity per-port variables**

20 For each port in a congestion isolation aware system there is the following set of variables:

- 21 a) `ciRxPort` (49.4.1.3.1)
- 22 b) `ciAddDel` (49.4.1.3.2)

23 **49.4.1.3.1 ciRxPort**

24 A local variable holding the reception port number for a congesting flow. The CI Stream Table (12.33.3) is
25 indexed by this local variable during the construction of a CIM. The value may be loaded during an
26 `EM_UNITDATA.request` or when entries are being removed from the CI Stream Table and IEEE Std
27 802.1CB Stream identity table.

28 **49.4.1.3.2 ciAddDel**

29 A local variable holding an enumeration `{add, delete}` that provides an indication of the reason to build and
30 send a CIM. The Add/Del field of the CIM PDU informs the peer that a congesting flow is either being
31 added or deleted. The `ciAddDel` variable is used by the common `buildAndSendCim()` procedure (49.4.2.5)
32 to parameterize and properly set the Add/Del field.

33 **49.4.1.4 CIP entity per-port per-traffic class variables**

34 For each port and monitored or congesting queue in a congestion isolation aware system there is the
35 following set of variables:

- 36 a) `cipGateControl` (49.4.1.4.1)
- 37 b) `cipNonCongestedTime` (49.4.1.4.2)
- 38 c) `cipCongesting` (49.4.1.4.3)

1 **49.4.1.4.1 cipGateControl**

2 Determines the state of the transmission gate for a monitored or congesting queue. There is an instance of
3 `cipGateControl` for each queue that is participating in congestion isolation. The transmission gate allows
4 implementations to maintain frame order when isolating a congesting flow. The value of the `cipGateControl`
5 variable is either *open* or *closed*. It is directly tied to the state of the transmission gate of the monitored
6 queues and the congesting queue. When the transmission selection algorithm is strict priority, the
7 `cipGateControl` shall be *open* for all queues. Other transmission selection algorithms may require that order
8 is maintained by controlling the values of `cipGateControl`. An example implementation approach for a
9 traffic selection algorithm other than strict priority is described in Annex W.

10 **49.4.1.4.2 cipNonCongestedTime**

11 The time (`SysUpTime`, IETF RFC 3418) at which the queue last transitioned from congested to
12 non-congested. This timestamp is used by the background `periodicTableCleanup()` procedure (49.4.2.10) to
13 remove stale CI Stream Table entries that were created because of the receipt of a CIM and still remain in the
14 table for a period longer than the `ciMaxFlowLife` (49.4.1.1.3).

15 **49.4.1.4.3 cipCongesting**

16 A boolean value that is set during the processing of a frame by the `EM_UNITDATA.request` (49.4.2.2)
17 procedure. The value is set true when the AQM indicates that the monitored queue is no longer congested or
18 the congesting queue is empty. The variable is initialized to false.

19 **49.4.1.5 CIP entity per-stream variables**

20 For each `tsnStreamIdEntry` object created in the IEEE Std 802.1CB Stream identity table there is a
21 corresponding entry in the CI Stream Table (12.33.3). The entry contains per-stream variables that identify
22 the congesting queue for a congesting flow, count the number of CIMs that have been transmitted for the
23 congesting flow, and register the reasons for creating or updating the `tsnStreamIdEntry` object. Each entry
24 contains the following variables:

- 25 a) `ciStreamIdHandle` (49.4.1.5.1)
- 26 b) `ciCIMCount` (49.4.1.5.2)
- 27 c) `ciCreateTime` (49.4.1.5.3)
- 28 d) `ciStreamCreateMask` (49.4.1.5.4)
- 29 e) `ciQueueKey` (49.4.1.5.5)
- 30 f) `ciDestination_address` (49.4.1.5.6)
- 31 g) `ciSource_address` (49.4.1.5.7)
- 32 h) `ciVlan_identifier` (49.4.1.5.8)
- 33 i) `ciMsdu` (49.4.1.5.9)

34 **49.4.1.5.1 ciStreamIdHandle**

35 The `stream_handle` value used to create the `tsnStreamIdEntry` object in the IEEE Std 802.1CB Stream
36 identity table. This variable provides the 1:1 mapping of the CI Stream Table entry with the
37 `tsnStreamIdEntry` object.

38 **49.4.1.5.2 ciCIMCount**

39 An integer containing a count of the number of CIMs sent for the congesting flow. The count is reset when a
40 frame from a congesting flow is received on a monitored queue.

1 **49.4.1.5.3 ciCreateTime**

2 The time (SysUpTime, IETF RFC 3418) at which the CI Stream Table entry was created.

3 **49.4.1.5.4 ciStreamCreateMask**

4 A 2-bit mask indicating the reason for creating or updating the CI Stream Table entry. The LSB indicates
5 that the entry was created or updated because the AQM determined that a forwarded frame is part of a
6 congesting flow. The MSB indicates that the entry was created or updated because of the receipt of a CIM.
7 The ciStreamCreateMask allows an implementation to take different actions when de-isolating a congesting
8 flow.

9 **49.4.1.5.5 ciQueueKey**

10 An integer that represents the unique queue identity of a congesting queue at an egress port that a congesting
11 flow is traversing. The key is calculated by the product of the congesting traffic class number plus one and
12 the port number. The key is used to help find all congesting flows using the same queue so the
13 flushCongestingFlows() procedure (49.4.2.9) can efficiently remove them.

14 **49.4.1.5.6 ciDestination_address**

15 A MAC address obtained from the destination_address parameter of an EM_UNITDATA.indication for a
16 congesting flow or from the destination_address field of a received CIM PDU.

17 **49.4.1.5.7 ciSource_address**

18 A MAC address obtained from the source_address parameter of an EM_UNITDATA.indication for a
19 congesting flow or from the source_address field of a received CIM PDU.

20 **49.4.1.5.8 ciVlan_identifier**

21 A VLAN Identifier obtained from the vlan_identifier parameter of an EM_UNITDATA.indication for a
22 congesting flow or from the vlan_identifier field of a received CIM PDU.

23 **49.4.1.5.9 ciMsdu**

24 An octet string containing an MSDU obtained from the mac_service_data_unit parameter of an
25 EM_UNITDATA.indication for a congesting flow or from the Encapsulated MSDU field of a received CIM
26 PDU.

27 **49.4.2 CIP procedures**

28 Congestion isolation is implemented through the procedures of a CIP. These include the following:

- 29 a) ciInitialize() (49.4.2.1)
- 30 b) EM_UNITDATA.request (parameters) (49.4.2.2)
- 31 c) condTransmitCimAddPdu() (49.4.2.3)
- 32 d) transmitCimDelPdu() (49.4.2.4)
- 33 e) buildAndSendCim() (49.4.2.5)
- 34 f) processCimPdu() (49.4.2.6)
- 35 g) addCongestingFlow() (49.4.2.7)
- 36 h) delCongestingFlow() (49.4.2.8)
- 37 i) flushCongestingFlows() (49.4.2.9)
- 38 j) periodicTableCleanup() (49.4.2.10)

1 49.4.2.1 ciInitialize()

2 Initializes CI support (8.6.5.2.3) for per-stream classification and metering (8.6.5.2) including the creation of
3 stream filter instances and stream gate instances needed for CI operation. At initialization time the procedure
4 performs the following:

- 5 a) For each entry in the ciQueueMap[] table do the following:
 - 6 1) If the value of the entry in ciQueueMap[] is 0, do nothing and move on to the next entry.
 - 7 2) If the value of the entry in ciQueueMap[] is positive, the index specifies a monitored queue and
8 the value can be translated to a congesting queue. Create a stream gate and stream filter as
9 follows:
 - 10 i) Allocate a new StreamGateInstance.
 - 11 ii) Set the StreamGateEnable attribute to TRUE.
 - 12 iii) Set the StreamGateAdminGateStates to open.
 - 13 iv) Set the StreamGateAdminIPV to the value of the entry minus one.
 - 14 v) Allocate a new StreamFilterInstance.
 - 15 vi) Set the StreamHandleSpec to the wildcard.
 - 16 vii) Set the PrioritySpec to the index of the entry.
 - 17 viii) Set the StreamGateInstanceID to the StreamGateInstance.
 - 18 3) If the value of the entry in ciQueueMap[] is negative, the index specifies a congesting queue
19 and the value can be translated to a monitored queue. Create a stream gate and stream filter as
20 follows:
 - 21 i) Allocate a new StreamGateInstance.
 - 22 ii) Set the StreamGateEnable attribute to TRUE.
 - 23 iii) Set the StreamGateAdminGateStates to open.
 - 24 iv) Set the StreamGateAdminIPV to the absolute value of the entry minus one.
 - 25 v) Allocate a new StreamFilterInstance.
 - 26 vi) Set the StreamHandleSpec to none.
 - 27 vii) Set the PrioritySpec to the index of the entry.
 - 28 viii) Set the StreamGateInstanceID to the StreamGateInstance.
 - 29 4) Set cipCongesting for the queue to false.

30 49.4.2.2 EM_UNITDATA.request (parameters)

31 A CIP offers an instance of the EISS (6.8) to the Queuing frames function (8.6.6). When called upon to
32 enqueue a frame the priority parameter specifies the target queue that represents the received priority of the
33 frame. The CIP determines if the target queue is a congesting queue or a monitored queue by indexing into
34 the cipQueueMap[] table using the priority parameter.

35 If the cipQueueMap[priority] value is 0 then the target queue is not participating in congestion isolation and
36 the frame can be enqueued on the target queue with no further processing.

37 If the cipQueueMap[priority] value is positive then the target queue is a monitored queue. The value from
38 the cipQueueMap[] table determines the congesting queue that will be used for isolating congesting flows
39 from the monitored queue. The congesting queue for this monitored queue is one less than the value of
40 cipQueueMap[priority]

41 If the cipQueueMap[priority] value is negative then the target queue is a congesting queue. The value from
42 the cipQueueMap[] table determines the monitored queue that will be used for de-isolating congesting flows
43 from the congesting queue. The monitored queue for this congesting queue is one less than the absolute
44 value of cipQueueMap[priority].

45 The CIP determines if the frame belongs to a flow that has been causing congestion by first checking for the
46 presence of a stream_handle parameter. Frames with the stream_handle parameter are known to be from
47 congesting flows. If the frame does not have a stream_handle and the target queue is a monitored queue, the

1 CIP interacts with the AQM to determine if the frame is part of a flow that is creating congestion in the
2 monitored queue. Newly identified congesting flows will cause entries to be created in the Stream identity
3 table. If the frame does not have a stream_handle and the target queue is a congesting queue, the CIP remaps
4 the priority to the monitored queue by using the translated value from the cipQueueMap[] table.

5 Frames that have a stream_handle parameter or have been determined by the AQM (see 49.2.1) to be
6 creating congestion may cause a CIM to be transmitted. The conditions by which a CIM is transmitted are
7 described in condTransmitCimAddPdu().

8 The CIP uses the same AQM for both the congesting and monitored queues to ensure consistent end-to-end
9 congestion control for frames from a common flow that traverse both queues.

10 For each frame that is presented for queuing, the CIP performs the following:

- 11 a) Determine if the target queue is a monitored queue, a congesting queue or not participating in CI by
12 indexing into the cipQueueMap[] table using the priority parameter and translating the contents as
13 described above.
- 14 b) If the target queue is a monitored queue and the stream_handle is not present, then determine if the
15 AQM indicates the frame is from a flow causing congestion in the monitored queue.
 - 16 1) If the monitored queue is congested:
 - 17 i) Set the LSB and clear the MSB of the ciStreamCreateMask to indicate an entry is being
18 created because congestion has been detected locally by the AQM.
 - 19 ii) Call addCongestingFlow() to create an entry in the IEEE Std 802.1CB Stream
20 identification function and the CI Stream Table and initialize the ciCIMCount variable to
21 0.
 - 22 iii) Call condTransmitCimAddPdu() to conditionally generate and transmit a CIM to the peer.
 - 23 iv) Modify the frame's PCP to indicate the priority of the congesting queue.
 - 24 v) Enqueue the frame on the congesting queue.
 - 25 vi) Set cipCongesting true.
 - 26 2) If the monitored queue is not congested, enqueue the frame on the monitored queue.
- 27 c) If the target queue is a monitored or a congesting queue and the stream_handle is present, the frame
28 is from a congesting flow. Determine from the AQM if the congesting queue is empty, congested, or
29 not congested.
 - 30 1) If the congesting queue is empty, call flushCongestingFlows() to remove all flows from the
31 IEEE Std 802.1CB Stream identification function and the CI Stream Table for the target queue.
 - 32 2) If the congesting queue is congested:
 - 33 i) Load the ciCIMCount variable with the value from the CI Stream Table.
 - 34 ii) Call condTransmitCimAddPdu() to conditionally generate and transmit a CIM to the peer.
 - 35 3) If the congesting queue is not congested, then optionally:
 - 36 i) Call delCongestingFlow() to remove the flow from the IEEE Std 802.1CB Stream
37 identification function and the CI Stream Table.
 - 38 ii) Ensure the cipGateControl variable for the monitored and congesting queues are properly
39 set to ensure ordering requirements for the transmission selection algorithm in use.
 - 40 iii) Modify the frame's PCP to indicate the priority of the monitored queue.
 - 41 iv) Enqueue the frame on the monitored queue.
 - 42 4) If the target queue is a monitored queue, reset the ciCIMCount variable in the CI Stream Table.
- 43 d) If the target queue is a congesting queue and the stream_handle is not present, the frame is from a
44 flow that may be congesting upstream, but is not congesting locally. The frame priority can be
45 remapped back to the monitored queue.
 - 46 i) The PCP is modified to indicate the priority of the monitored queue.
 - 47 ii) The frame is enqueued on the monitored queue.

- 1 e) Check if the congesting queue is empty and call `flushCongestingFlows()` to remove all flows from
2 the IEEE Std 802.1CB Stream identification function and the CI Stream Table for the congesting
3 queue.

4 **49.4.2.3 condTransmitCimAddPdu()**

5 This is called by the CIP to conditionally generate and transmit a CIM instructing the peer to add a
6 congesting flow. A CIM will be generated and transmitted if fewer than `ciMaxCIM` messages have been
7 sent since the `ciCIMCount` was last reset and the AQM has determined that the congesting queue is
8 congested. The `ciCIMCount` variable is loaded prior to invoking `condTransmitCimAddPdu()`.

9 NOTE—Implementations may have additional conditions to restrict or allow the creation of CIMs for adding congesting
10 flows without risking interoperability.

11 The procedure performs the following:

- 12 a) If the `ciCIMCount` is less than `ciMaxCIM` and the AQM indicates the congesting queue is
13 congested then do the following:
14 1) Set `ciAddDel` to *add*
15 2) Call `buildAndSendCim()`

16 **49.4.2.4 transmitCimDelPdu()**

17 This is called by the CIP to generate and transmit a CIM instructing the peer that a congesting flow is being
18 removed from the IEEE Std 802.1CB Stream identity table and the CI Stream Table.

19 The procedure performs the following:

- 20 a) Set `ciAddDel` to *delete*
21 b) Call `buildAndSendCim()`

22 **49.4.2.5 buildAndSendCim()**

23 This is a common procedure shared by `condTransmitCimAddPdu()` and `transmitCimDelPdu()` to construct
24 and send the CIM. To properly generate the CIM, the CI Peer Table must be searched to retrieve the type of
25 CIM and the address information of the peer that will receive the CIM. There are three formats for a CIM;
26 layer-2, IPv4, and IPv6. All of the formats encapsulate a common CIM PDU. The CIM is constructed in the
27 space provided by the parameters of the `EM_UNITDATA.indication`. The local variables holding the
28 parameters from an `EM_UNITDATA.request` are expected to be loaded before the procedure is invoked,
29 whether those parameters came from an actual `EM_UNITDATA.request` or were loaded from the CI Stream
30 Table. The local `ciAddDel` variable determines the contents of the Add/Del field of the CIM PDU. The local
31 `ciRxPort` determines the source port of the congesting flow that is represented in the CIM.

32 The procedure performs the following:

- 33 a) Search the CI Peer Table (12.33.2) using `ciRxPort` as an index to obtain the `ciCIMType`,
34 `ciPeerMacAddress`, and `ciPeerCIMEncapLen` for the peer. If the `ciCIMType` is `ipv4` then obtain the
35 `ciPeerIPv4Address` and `ciPeerUDPPort` from the entry. If the `ciCIMType` is `ipv6` then obtain the
36 `ciPeerIPv6Address` and `ciPeerUDPPort` from the entry. If the entry is not found the procedure does
37 nothing; no further processing takes place and no CIM will be generated or transmitted.
38 b) Fill the Version (49.4.3.4.1) field of the CIM PDU with 0.
39 c) If `ciAddDel` is *add* then set the Add/Del bit to 1 otherwise set the Add/Del bit to 0.
40 d) Set the `destination_address` (49.4.3.4.4) of the CIM PDU to the value of the `destination_address`.
41 e) Set the `source_address` (49.4.3.4.5) of the CIM PDU to the value of the `source_address`.
42 f) Set the `vlan_identifier` (49.4.3.4.6) of the CIM PDU to the value of the `vlan_identifier`.

- 1 g) Fill the Encapsulated MSDU (49.4.3.4.8) field with the first octets of the `mac_service_data_unit`.
2 The Encapsulated MSDU field should be filled with the lesser of `ciPeerCIMEncapLen` or the length
3 of the `mac_service_data_unit`.
- 4 h) Set the Encapsulated MSDU length (49.4.3.4.7) to the number of octets filled into the Encapsulated
5 MSDU field.
- 6 i) The format of the CIM constructed in the `mac_service_data_unit` of the `EM_UNITDATA.indication`
7 depends upon the `ciCIMType` variable as follows:
- 8 1) If the `ciCIMType` is l2:
- 9 i) Insert the CIM PDU EtherType (Table 49-1) encapsulation at the beginning of the
10 `mac_service_data_unit` of the `EM_UNITDATA.indication`.
- 11 ii) Fill the Version (49.4.3.1.1) field of the layer 2 CIM encapsulation with 0.
- 12 iii) Fill the Subtype (49.4.3.1.2) field of the layer 2 CIM encapsulation with 0.
- 13 2) If the `ciCIMType` is ipv4:
- 14 i) Insert the IPv4 EtherType (Figure 49-4) encapsulation at the beginning of the
15 `mac_service_data_unit` of the `EM_UNITDATA.indication`.
- 16 ii) Use the `ciPeerIPv4Address` as the destination IPv4 address, the `cipIPv4Address` as the
17 source IPv4 address, and the value 17 as the IP protocol field to construct and insert an
18 IETF RFC 791 compliant IPv4 header after the IPv4 EtherType in the
19 `mac_service_data_unit` of the `EM_UNITDATA.indication`.
- 20 iii) Use the `ciPeerUDPPort` as the UDP destination port number and the `cipCIMUDPPort` as
21 the source port number to construct and insert an IETF RFC 768 compliant UDP header
22 after the IPv4 header in the `mac_service_data_unit` of the `EM_UNITDATA.indication`.
- 23 3) If the `ciCIMType` is ipv6:
- 24 i) Insert the IPv6 EtherType (Figure 49-5) encapsulation at the beginning of the
25 `mac_service_data_unit` of the `EM_UNITDATA.indication`.
- 26 ii) Use the `ciPeerIPv6Address` as the destination IPv6 address, the `cipIPv6Address` as the
27 source IPv6 address, and the value 17 as the next header field to construct and insert an
28 IETF RFC 8200 compliant IPv6 header after the IPv6 EtherType in the
29 `mac_service_data_unit` of the `EM_UNITDATA.indication`.
- 30 iii) Use the `ciPeerUDPPort` as the UDP destination port number and the `cipCIMUDPPort` as
31 the source port number to construct and insert an IETF RFC 768 compliant UDP header
32 after the IPv6 header in the `mac_service_data_unit` of the `EM_UNITDATA.indication`.
- 33 j) Use the `ciPeerMacAddress` as the `destination_address` of the `EM_UNITDATA.indication` for the
34 CIM.
- 35 k) Use the `cipMacAddress` (49.4.1.2.1) as the `source_address` of the `EM_UNITDATA.indication` for
36 the CIM.
- 37 l) Set the priority parameter of the `EM_UNITDATA.indication` for the CIM to the value of the
38 `ciCIMTransmitPriority` (49.4.1.1.2) variable.
- 39 m) Pass the CIM as an `EM_UNITDATA.indication` to the CIM Multiplexer (49.3.5) for transmission to
40 the peer.

1 49.4.2.6 processCimPdu()

2 The CIM Demultiplexer (49.3.3) receives CIMs from peers and invokes processCimPdu() to process the
3 CIM and either create entries in the IEEE Std 802.1CB Stream identification function and CI Stream Table
4 or remove entries from the tables. The procedure performs the following actions upon receipt of a CIM:

- 5 a) The CIM is validated according to 49.4.3.5 and is discarded if invalid.
- 6 b) If the Add/Del bit is set do the following:
 - 7 1) Clear the LSB and set the MSB of the ciStreamCreateMask to indicate an entry is being created
8 because of a received CIM.
 - 9 2) Extract the parameters and the Encapsulated MSDU from the CIM PDU and call
10 addCongestingFlow().
- 11 c) If the Add/Del bit is clear then do the following:
 - 12 1) Search the CI Stream Table for an entry that matches the destination_address, source_address,
13 vlan_identifier, and Encapsulated MSDU fields of the CIM.
 - 14 2) If no entry is found, then processing is done.
 - 15 3) If the MSB of the ciStreamCreateMask is set, then clear it. Otherwise processing is done.
 - 16 4) If the LSB of the ciStreamCreateMask is set, then update the CI Stream Table entry with the
17 updated ciStreamCreateMask and processing is done.
 - 18 5) If the LSB of the ciStreamCreateMask is clear then call delCongestingFlow().

19 49.4.2.7 addCongestingFlow()

20 Registers a flow with the IEEE Std 802.1CB Stream identification function as described in 49.2.2. A
21 tsnStreamIdEntry is created by using initialized variables and the contents of the mac_service_data_unit that
22 is obtained from the received frame or the encapsulated header of a received CIM. The set of managed
23 objects that make up the tsnStreamIdEntry are created as follows:

- 24 a) Set the tsnStreamIdHandle object to newly allocated and unique stream_handle.
- 25 b) Set the tsnStreamIdOutFacInputPortList include the list of all ports in the system.
- 26 c) Set the tsnStreamIdOutFacOutputPortList, tsnStreamIdInFacInputPortList, and
27 tsnStreamIdInFacOutputPortList to null.
- 28 d) Set the tsnStreamIdIdentificationType to IP stream identification as specified in IEEE Std 802.1CB.
- 29 e) Set the tsnStreamIdParameters for an IP stream identification type as follows:
 - 30 1) Set the tsnCpeIpIdDestMac to the destination_address of the received frame or parameter from
31 the CIM.
 - 32 2) Set the tsnCpeIpIdTagged to the value **all** as specified in IEEE Std 802.1CB.
 - 33 3) Set the tsnCpeIpIdVlan to the vlan_identifier of the received frame or parameter from the CIM.
 - 34 4) Set the tsnCpeIpIdIpSource to the IPv4 (IETF RFC 791) or IPv6 (IETF RFC 8200) source
35 address parameter that is contained within the mac_service_data_unit.
 - 36 5) Set the tsnCpeIpIdIpDestination to the IPv4 (IETF RFC 791) or IPv6 (IETF RFC 8200)
37 destination address parameter that is contained within the mac_service_data_unit.
 - 38 6) Set the tsnCpeIpIdDscp to a value of 64 decimal to ignore the DSCP (IETF RFC 2474) code
39 point when matching received frames.
 - 40 7) Set the tsnCpeIpIdNextProtocol to UDP (IETF RFC 768) if the mac_service_data_unit
41 contains a UDP packet, TCP (IETF RFC 793) if it contains a TCP packet, or SCTP (IETF RFC
42 4960) if it contains an SCTP packet. If the mac_service_data_unit contains another
43 non-supported IP protocol by the implementation then set the tsnCpeIpIdNextProtocol to none
44 and the tsnCpeIpIdSourcePort and tsnCpeIpIdDestinationPort will be ignored.
 - 45 8) Set the tsnCpeIpIdSourcePort to the UDP, TCP, or SCTP source port number from the
46 mac_service_data_unit. Set the tsnCpeIpIdSourcePort to 0 if the tsnCpeIpIdNextProtocol field
47 is none.

- 1 9) Set the `tsnCpeIpIdDestinationPort` to the UDP, TCP, or SCTP destination port number from the
2 `mac_service_data_unit`. Set the `tsnCpeIpIdDestinationPort` to 0 if the `tsnCpeIpIdNextProtocol`
3 field is none.
- 4 f) Write the `tsnStreamIdEntry` to the IEEE Std 802.1CB Stream identity table using the specified
5 management operations.
- 6 g) Create a new entry in the CI Stream Table:
 - 7 1) Set the entry `ciStreamHandleId` index to the `stream_handle`.
 - 8 2) Set the entry `ciCIMCount` to 0.
 - 9 3) Set the entry `ciStreamCreateMask` to the variable `ciStreamCreateMask`.
 - 10 4) Set the entry `ciQueueKey` to the product of the congesting queue plus one and the port number
11 that the queue is supporting.
 - 12 5) Set the entry `ciDestination_address` to the `destination_address`.
 - 13 6) Set the entry `ciSource_address` to the `source_address`.
 - 14 7) Set the entry `ciVlan_identifier` to the `vlan_identifier`.
 - 15 8) Fill the entry `ciMsdu` with the first octets of the `mac_service_data_unit`. The number of octets
16 to fill the `ciMsdu` attribute with is determined by the lesser of `ciPeerCIMEncapLen` or the
17 length of the `mac_service_data_unit`.
 - 18 9) Set the entry `ciCreateTime` to `sysUpTime` as specified by IETF RFC 3418.

19 **49.4.2.8 delCongestingFlow()**

20 Removes entries from the IEEE Std 802.1CB Stream identification function and the CI Stream Table as
21 discussed in 49.2.6. The procedure performs the following:

- 22 a) Load the `destination_address`, `source_address`, `vlan_identifier`, and `mac_service_data_unit`
23 parameters from the entry `ciDestination_address`, `ciSource_address`, `ciVlan_identifier`, and `ciMsdu`,
24 respectively.
- 25 b) Call `transmitCimDelPdu()` to inform the peer that a congesting flow is no longer congesting.
- 26 c) Set the `tsnStreamIdHandle` object to the `stream_handle` parameter of the frame triggering the
27 deletion.
- 28 d) Remove the `tsnStreamIdEntry` associated with the `tsnStreamIdHandle` in the IEEE Std 802.1CB
29 Stream identification function.
- 30 e) Remove the CI Stream Table entry.
- 31 f) if `cipCongesting` is true, set `cipCongesting` false and set `cipNonCongestedTime` to `sysUpTime`.

32 **49.4.2.9 flushCongestingFlows()**

33 Removes all entries from the IEEE Std 802.1CB Stream identification function that were not created
34 because of the receipt of a CIM and are associated with the current congesting queue at the egress port as
35 discussed in 49.2.6. The procedure performs the following:

- 36 a) Calculate the `ciQueueKey` by taking the product of the congesting traffic class plus one and the
37 egress port number.
- 38 b) For each entry in the CI Stream Table with the same `ciQueueKey` and the MSB of the `ciCreateMask`
39 in the entry is not set:
 - 40 1) Load the `destination_address`, `source_address`, `vlan_identifier`, and `mac_service_data_unit`
41 parameters from the entry `ciDestination_address`, `ciSource_address`, `ciVlan_identifier`, and
42 `ciMsdu`, respectively.
 - 43 2) Call `transmitCimDelPdu()` to inform the peer that a congesting flow is no longer congesting.
 - 44 3) Set the `tsnStreamIdHandle` object to the `ciStreamIdHandle`.
 - 45 4) Remove the `tsnStreamIdEntry` associated with the `tsnStreamIdHandle` in the IEEE Std 802.1CB
46 Stream identification function.
 - 47 5) Remove the CI Stream Table entry.

- 1 c) if `cipCongesting` is true, set `cipCongesting` false and set `cipNonCongestedTime` to `sysUpTime`.

2 49.4.2.10 `periodicTableCleanup()`

3 A background procedure that calculates the number of centiseconds between the creation time of a CI
4 Stream Table entry (49.4.1.5.3) and the `cipNonCongestedTime` (49.4.1.4.2), and removes any entries with
5 only the MSB of the `ciStreamCreateMask` (49.4.1.5.4) set that have a calculated value greater than
6 `ciMaxFlowLife` (49.4.1.1.3). The procedure should be invoked at least once every `ciMaxFlowLife` time
7 intervals. The procedure performs the following:

- 8 a) For each entry in the CI Stream Table with the MSB set and the LSB clear in the
9 `ciStreamCreateMask`:
- 10 1) Calculate the difference between the `cipNonCongestedTime` for the queue specified by the
11 `ciQueueKey` and the entry `ciCreateTime`.
 - 12 2) If the difference is greater than `ciMaxFlowLife`:
 - 13 i) Load the `destination_address`, `source_address`, `vlan_identifier`, and `mac_service_data_unit`
14 parameters from the entry `ciDestination_address`, `ciSource_address`, `ciVlan_identifier`, and
15 `ciMsdu`, respectively.
 - 16 ii) Call `transmitCimDelPdu()` to inform the peer that a congesting flow is no longer
17 congesting.
 - 18 iii) Set the `tsnStreamIdHandle` object to the `ciStreamIdHandle`.
 - 19 iv) Remove the `tsnStreamIdEntry` associated with the `tsnStreamIdHandle` in the IEEE Std
20 802.1CB Stream identification function.
 - 21 v) Remove the CI Stream Table entry.

22 49.4.3 Encoding of the CIM PDU

23 This clause specifies the method of encoding Congestion Isolation Message (CIM) PDUs. There are three
24 ways of encapsulating CIM PDUs; a layer-2 CIM PDU encapsulation that can be delivered to peers via the
25 Bridge relay and IPv4 and IPv6 layer-3 CIM PDU encapsulations that can be delivered to peers via a higher
26 layer entity above Bridge relay such as a router. All CIMs contain an integral number of octets.

27 The octets in a Congestion Isolation Message PDU are numbered starting from 1 and increasing in the order
28 they are put into the MSDU that accompanies a request to or indication from the instance of the MAC
29 Internal Sublayer Service (ISS or EISS) used by a congestion isolation entity. The bits in an octet are
30 numbered from 1 to 8 in order of increasing bit significance, where 1 is the LSB in the octet.

31 Where octets and bits within a Congestion Isolation Message PDU are represented using a diagram, octets
32 shown higher on the page than subsequent octets and octets shown to the left of subsequent octets at the
33 same height on the page are lower numbered; bits shown to the left of other bits within the same octet are
34 higher numbered.

35 Where two or more consecutive octets are represented as hexadecimal values, lower numbered octet(s) are
36 shown to the left and each octet following the first is preceded by a hyphen, for example,
37 01-80-C2-00-00-00. When consecutive octets are used to encode a binary number, the lower octet number
38 has the more significant value. When consecutive bits within an octet are used to encode a binary number,
39 the higher bit number has the most significant value. When bits within consecutive octets are used to encode
40 a binary number, the lower octet number composes the more significant bits of the number. A flag is
41 encoded as a single bit, and is set (TRUE) if the bit takes the value 1, and clear (FALSE) otherwise. The
42 remaining bits within the octet can be used to encode other protocol fields.

49.4.3.1 Layer-2 CIM PDU encapsulation

The means for identifying layer-2 encapsulated CIM PDUs consists of two octets containing the EtherType value shown (in hexadecimal notation) in Table 49-1.

Table 49-1—Congestion Isolation Message EtherType

Name	Value
IEEE 802.1Q Congestion Isolation Message (CIM)	89-A2

The layer-2 CIM PDU encapsulation is shown in Figure 49-3.

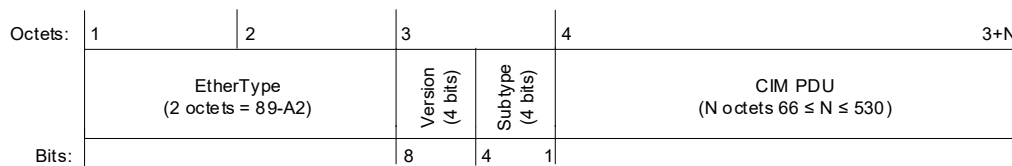


Figure 49-3—Layer-2 CIM Encapsulation

49.4.3.1.1 Version

This field, 4 bits in length, shall be transmitted with the value 0 in this standard. If two Version fields are interpreted as unsigned binary numbers, the greater identifies the more recently defined Version. The Version field occupies the most significant bits of the first octet of the layer-2 CIM encapsulation.

49.4.3.1.2 Subtype

This field, 4 bits in length, shall be transmitted with the value 0 to indicate an encapsulated CIM PDU. The Subtype field occupies the least significant 4 bits of the first octet of the layer-2 CIM encapsulation.

49.4.3.2 IPv4 layer-3 CIM PDU encapsulation

The means of identifying IPv4 layer-3 encapsulated CIM PDUs consist of 2 octets containing the EtherType value for IPv4 packets (08-00) as well as the associated IPv4 header decoding for a UDP datagram carrying the CIM PDU. The encoding of an IPv4 header is defined in IETF RFC 791. IP options are not included in the IPv4 layer-3 encapsulated CIM PDU. The IP protocol field in the IPv4 header consists of 1 octet and identifies the UDP datagram with the value 17. The encoding of a UDP header is defined in IETF RFC 768. The destination port field of the UDP header consists of 2 octets and identifies the encapsulated CIM PDU with the value from the ciPeerUDPPort field of the CI Peer Table (12.33.2) for the row associated with the port of the receiving upstream peer. The source port field of the UDP header is the value of the cipCIMUDPPort variable. The IPv4 encapsulation is shown in Figure 49-4.

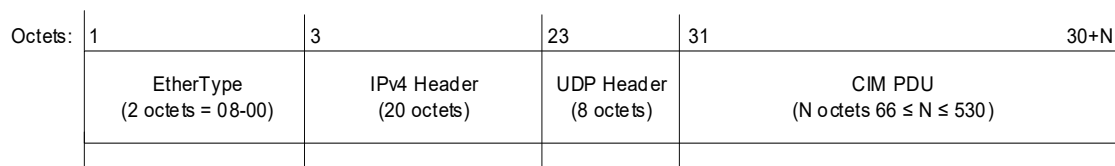


Figure 49-4—IPv4 layer-3 CIM Encapsulation

1 49.4.3.3 IPv6 layer-3 CIM PDU encapsulation

The means of identifying IPv6 layer-3 encapsulated CIM PDUs consist of 2 octets containing the EtherType value for IPv6 packets (86-DD) as well as the associated IPv6 header decoding for a UDP datagram carrying the CIM PDU. The encoding of an IPv6 header is defined in IETF RFC 8200. IPv6 Extension Headers are not used in the IPv6 layer-3 encapsulated CIM PDU. The next header field in the IPv6 headers indicates the upper layer protocol field and consists of 1 octet identifying the UDP datagram with the value 17. The encoding of a UDP header is defined in RFC 768. The destination port field of the UDP header consists of 2 octets and identifies the encapsulated CIM PDU with the value from the ciPeerUDPPort field of the CI Peer Table (12.33.2) for the row associated with the port of the receiving upstream peer. The source port field of the UDP header is the value of the cipCIMUDPPort variable. The IPv6 encapsulation is shown in Figure 49-5.

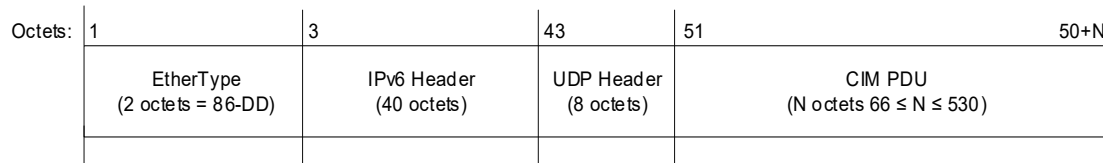


Figure 49-5—IPv6 layer-3 CIM Encapsulation

12 49.4.3.4 Congestion Isolation Message PDU format

¹³ The format of a Congestion Isolation Message (CIM) PDU is illustrated in Figure 49-6.



Figure 49-6—CIM PDU

14 49.4.3.4.1 Version

15 This field, 4 bits in length, shall be transmitted with the value 0 in this standard. If two Version fields are
16 interpreted as unsigned binary numbers, the greater identifies the more recently defined Version. The
17 Version field occupies the most significant bits of the first octet of the CIM PDU.

18 49.4.3.4.2 Reserved

¹⁹ This field, 3 bits in length, shall be transmitted as 0 and ignored on receipt.

20 **49.4.3.4.3 Add/Del**

21 This field, 1 bit in length, instructs the peer to add a congesting flow entry to the IEEE Std 802.1CB Stream
22 identity table and the CI Stream Table, or to remove an entry. When the bit is set, the peer is instructed to
23 add, and when the bit is clear, the peer is instructed to delete. The Add/Del bit occupies the least significant
24 bit of the first octet of the CIM PDU.

25 49.4.3.4.4 destination_address

²⁶ This field, 6 octets in length, contains the destination MAC address of the frame triggering the transmission
²⁷ of this CIM.

1 **49.4.3.4.5 source_address**

2 This field, 6 octets in length, contains the source MAC address of the frame triggering the transmission of
3 this CIM.

4 **49.4.3.4.6 vlan_identifier**

5 This field, 12 bits in length, contains the VLAN Identifier for the frame triggering the transmission of this
6 CIM.

7 **49.4.3.4.7 Encapsulated MSDU length**

8 This field, 2 octets in length, contains the number of octets returned in the Encapsulated MSDU field. The
9 minimum value is 48. The maximum value is 512.

10 **49.4.3.4.8 Encapsulated MSDU**

11 This field, a maximum of 512 octets in length, contains the initial octets of the mac_service_data_unit of the
12 frame that triggered the transmission of this CIM.

13 **49.4.3.5 CIM Validation**

14 A CIM PDU received by a CIM Demultiplexer (49.3.3) shall be considered invalid and be discarded if:

- 15 a) There are fewer than 48 octets in the mac_service_data_unit or greater than 512.
- 16 b) The Encapsulated MSDU length is less than 48 or greater than 512.

17 The following condition shall not cause a received CIM PDU to be considered invalid:

- 18 a) There are nonzero bits in the Version (49.4.3.4.1) field.

19 **49.4.4 LLDP Congestion Isolation TLV**

20 The Congestion Isolation TLV (D.2.15) is used to advertise support for Congestion Isolation to peers on the
21 network. It informs the peers of the configuration of local monitored queues and the congesting queue in
22 order to detect potential misconfiguration. It also advertises the desired length of the encapsulated headers
23 sent in a CIM. The source address of a received LLDPDU is used to populate the CI Peer Table (49.3.6) in
24 order to properly form a CIM.

25 **49.4.4.1 LLDP Congestion Isolation TLV procedures**

26 When the congestion isolation aware LLDP agent detects a change in the set of TLVs advertised by a peer it
27 will invoke the relevant LLDP TLV procedures to add or delete entries in the CI Peer Table (12.33.2). The
28 congestion isolation TLV procedures include:

- 29 a) addCiPeer() (49.4.4.1.1)
- 30 b) delCiPeer() (49.4.4.1.2)

1 **49.4.4.1.1 addCiPeer()**

2 This creates or updates a CI Peer Table entry. The entry is indexed by the port number of the received
3 LLDPDU. The procedure performs the following:

- 4 a) Set the table index to the port of the received LLDPDU.
- 5 b) Set the ciPeerMacAddress to the MAC Address field (D.2.15.5) of the Congestion Isolation TLV.
- 6 c) If the Address Family field (D.2.15.7) of the Congestion Isolation TLV is 1, set the
7 ciPeerIPv4Address to the IP Address field (D.2.15.8) of the Congestion Isolation TLV.
- 8 d) If the Address Family field (D.2.15.7) of the Congestion Isolation TLV is 2, set the
9 ciPeerIPv6Address to the IP Address field (D.2.15.8) of the Congestion Isolation TLV.
- 10 e) Set the ciPeerUDPPort to the UDP Port Number field (D.2.15.6) of the Congestion Isolation TLV.
- 11 f) Set the ciPeerCIMEncapLen to the CIM Encap Length field (D.2.15.4) of the Congestion Isolation
12 TLV.
- 13 g) Write the CI Peer Table entry into the CI Peer Table at the table index.

14 **49.4.4.1.2 delCiPeer()**

15 This deletes a CI Peer Table entry. The entry is indexed by the port number of the received LLDPDU that
16 indicates a congestion isolation peer is no longer present or enabled. The procedure performs the following:

- 17 a) Set the table index to the port where the missing or disabled CI peer is detected.
- 18 b) Delete the CI Peer Table entry at the table index.

19 **49.5 Topology Recognition**

20 Bridges, routers, and end stations in the data center network can determine their level and port orientation
21 within the topology through the use of the LLDP Topology Recognition TLV (D.2.16) and the manipulation
22 of associated managed objects. The level in the topology is relative to non-relay end stations or servers at the
23 edge of the network. Non-relay end stations or servers are at level 0. Bridges or routers attached to non-relay
24 end stations or servers are at level 1 or higher. Virtualization environments, including those that support an
25 edge relay defined by this standard (Clause 40), are end stations that contain a relay and appear at level 1 or
26 higher in the topology. The virtual stations within the virtualization environment appear at level 0.

27 The systems can determine their level in the topology and the receiving port's orientation by examining the
28 current state of the Topology Recognition (TR) variables and manipulating them based on the received
29 values in the LLDP Topology Recognition TLV. This clause specifies how to determine a system's level and
30 a port's orientation within the topology. It includes:

- 31 a) TR theory of operation (49.5.1)
- 32 b) TR variables controlling operation (49.5.2)
- 33 c) TR procedures (49.5.3)

34 **49.5.1 TR theory of operation**

35 The TR functionality is supported by a set of procedures and controlling variables that map to attributes of
36 the LLDP Topology Recognition TLV. Changes to the controlling variables will update the associated
37 objects in the IEEE 802.1/LLDP extension MIB (D.5) and the trSet() procedure defined in this clause will
38 cause the transmission of an LLDPDU to occur by invoking the somethingChangedLocal() procedure
39 defined in IEEE Std 802.1AB. When attributes of a received LLDP Topology Recognition TLV change, the
40 somethingChangedRemote() procedure in IEEE Std 802.1AB is invoked and control is passed to the
41 trUpdate() procedure to process the changes. The TR procedures that act upon the changes ensure that the
42 values of the controlling variables converge to a set of consistent values that represent the topology level and
43 port orientation of all participating devices in the network.

1 The LLDP Topology Recognition TLV contains three attributes that enable the recognition process; the
2 device type, the currently known topology level, and the currently known port orientation. The device type
3 includes non-relay end stations or servers, Bridges, and routers. The topology level attribute contains the
4 level number (0 through 254) or the value of unknown (255). The port orientation indicates whether the port
5 is facing an *uplink*, *downlink*, *crosslink*, or whether the orientation is *unknown*. The trDeviceType variable
6 holds the device type of a system and is a known factory setting of the device. The initial value of the level
7 is held in the trLevel variable and the initial value of the port orientation is held in the trPortOrientation
8 variable. These values are known by non-relay end stations or servers because they are at the edge of the
9 network and are always at level 0 with a port orientation of *uplink*. Bridges and routers initially do not know
10 their level or port orientation, but will recognize them when they receive LLDP Topology Recognition TLVs
11 from peers that have non-relay end stations or servers attached. There is a single trLevel variable for the
12 system, but each port contains a trPortOrientation. As TLVs are exchanged and changes are processed by the
13 procedures, the contents of trLevel and trPortOrientation are updated to converge upon the representation of
14 the stable state of the active topology.

15 49.5.2 TR variables controlling operation

16 The TR variables controlling operation hold the contents of attributes sent in the LLDP Topology
17 Recognition TLV and can be updated by the TR procedures. The TR variables include:

- 18 a) trDeviceType (49.5.2.1)
- 19 b) trLevel (49.5.2.2)
- 20 c) trPortOrientation (49.5.2.3)

21 49.5.2.1 trDeviceType

22 The trDeviceType is a read-only variable that contains the contents of the Device Type field of the LLDP
23 Topology Recognition TLV. The possible values of the trDeviceType are assigned as a factory setting and
24 are specified in Table D-14.

25 49.5.2.2 trLevel

26 A single octet integer holding the currently known level of the system in the topology. Non-relay end
27 stations or servers shall set this variable to 0. Other systems shall initialize this variable to 255 indicating
28 unknown and can set the variable to values between 1 and 254 as determined by the TR procedures (49.5.3).
29 The content of the trLevel variable is used for the Topology Level field of the LLDP Topology Recognition
30 TLV.

31 49.5.2.3 trPortOrientation

32 Each port in a participating system has a trPortOrientation variable that holds the contents of the currently
33 known Port Orientation of the LLDP Topology Recognition TLV. The possible values of the
34 trPortOrientation variable are specified in Table D-15. Non-relay end stations or servers shall set the
35 trPortOrientation variable to *uplink*. Other systems shall initialize this variable to *unknown* and will update
36 the variable to either *uplink*, *downlink*, or *crosslink* as determined by the TR procedures (49.5.3).

37 49.5.3 TR procedures

38 The TR procedures process changes and cause the transmission of LLDP Topology Recognition TLVs. The
39 procedures modify the TR variables and implement the algorithm that allows topology level and port
40 orientation recognition to converge upon a stable set of variables in the topology. The procedures include:

- 41 a) trInit() (49.5.3.1)
- 42 b) trSet() (49.5.3.2)
- 43 c) trUpdate() (49.5.3.3)

1 49.5.3.1 trInit()

2 The trInit() procedure initializes the controlling variables to a known state after system initialization or a
3 restart of the TR functionality. The procedure performs the following:

- 4 a) If trDeviceType is 0, specifying a non-relay end station or server then:
 - 5 1) Set trLevel to 0.
 - 6 2) Set trPortOrientation to *uplink*.
- 7 b) If trDeviceType is not 0:
 - 8 1) Set trLevel to 255, specifying *unknown*.
 - 9 2) Set trPortOrientation to 255, specifying *unknown*.
- 10 c) Call trSet() to cause the transmission of an LLDPDU to peers.

11 49.5.3.2 trSet()

12 This calls the somethingChangedLocal() procedure defined in IEEE Std 802.1AB, which causes the
13 transmission of an LLDPDU. The trLevel and trPortOrientation variables map to objects in the IEEE
14 802.1/LLDP extension MIB (D.5). A change to the single system wide trLevel variable causes the
15 transmission of an LLDPDU on each participating port. A change to the per-port trPortOrientation variable
16 causes the transmission of an LLDPDU on the associated port.

17 49.5.3.3 trUpdate()

18 The trUpdate() procedure is invoked when the somethingChangedRemote() procedure, defined in IEEE Std
19 802.1AB, determines that fields of a received LLDP Topology Recognition TLV have changed. The
20 procedure is responsible for updating the local TR variables and calling trSet() according to the following
21 algorithm:

- 22 a) If trDeviceType is 0, the receiving system is a non-relay end station or server and no action is
23 required. The procedure exits.
- 24 b) If trDeviceType is not 0, the following steps are taken depending upon the current state of trLevel
25 and trPortOrientation for the receiving port:
 - 26 1) If the received Topology Level is *unknown*, the peer is not providing additional information and
27 no further action is required and the procedure exits.
 - 28 2) If the received Topology Level is known with a value of L, where L is the current value of
29 trLevel:
 - 30 i) If trPortOrientation is *crosslink*, no further action is required and the procedure exits.
 - 31 ii) Set trPortOrientation of the receiving port to *crosslink*.
 - 32 3) If the received Topology Level is known with a value of L + 1, where L is the current value of
33 trLevel:
 - 34 i) If trPortOrientation is *uplink*, no further action is required and the procedure exits.
 - 35 ii) Set trPortOrientation of the receiving port to *uplink*.
 - 36 4) If the received Topology Level is known with a value of L – 1, where L is the current value of
37 trLevel:
 - 38 i) If trPortOrientation is *downlink*, no further action is required and the procedure exits.
 - 39 ii) Set trPortOrientation of the receiving port to *downlink*.
 - 40 5) If the received Topology Level is known and is less than trLevel – 1 or trLevel is *unknown*:
 - 41 i) Set trLevel to the received Topology Level plus one.
 - 42 ii) Set trPortOrientation of the receiving port to *downlink*.
 - 43 iii) Set trPortOrientation of all other ports to *unknown*.
 - 44 6) Call trSet().

45

50. PBBN Auto Attach

When a C-VLAN or S-VLAN aware system is connected to a Provider Backbone Bridged Network (PBBN), the Backbone Edge Bridge's (BEB's) mapping of the Backbone Service Instance Identifiers (I-SIDs) to the C-VLAN or S-VLAN Identifiers (C-VIDs or S-VIDs) used by the system needs to be configured. This clause specifies an Auto Attach Protocol (AAP) that runs over the Link Layer Discovery Protocol (LLDP, IEEE Std 802.1AB) and allows Auto Attach capable devices (AADs) and BEBs (AABs) to automate this configuration. Clause 12 (Bridge Management) and Clause 17 (Management Information Base) specify supporting management functionality, and Annex D specifies LLDP TLVs and TLV management for PBBN Auto Attach.

50.1 Overview

In order to automate the attachment of Auto Attach Devices (AADs) to an Auto Attach BEB (AAB), the Auto Attach Protocol (AAP) is used. The AAP, which runs over LLDP, avoids manual configuration by mapping C-VIDs or S-VIDs to I-SIDs to establish a C-VLAN or S-VLAN to BSI mapping at the AAB. The AAP is only supported over a point-to-point single peer-to-peer topology.

Figure 50-1 depicts the Auto Attach architectural framework model that allows an AAD to communicate VLAN to BSI mapping requests for attachment over a PBBN.

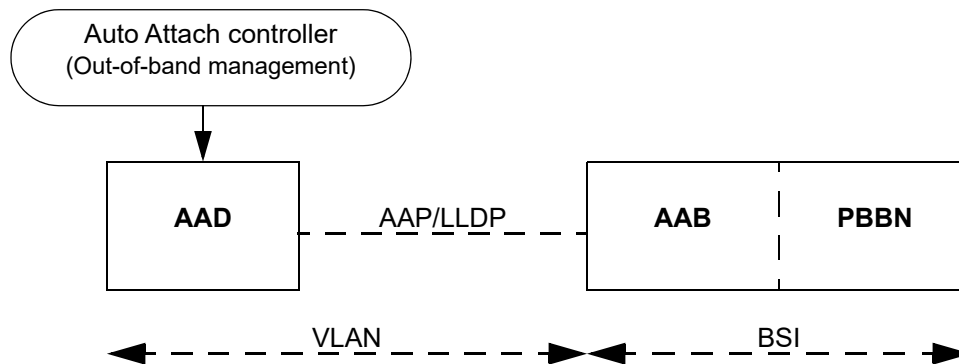


Figure 50-1—PBBN Auto Attach model

The conceptual example in Figure 50-1 depicts the process where an AAD end station or Bridge uses AAP running over LLDP to communicate the need to connect a C-VLAN to the appropriate BSI via the AAB. LLDP agents exchange IEEE 802.1 Organizationally Specific TLVs (D.2.17, D.2.18) to automate VID to I-SID assignment and registration.

An AAD is an end station, C-VLAN Bridge, S-VLAN Bridge, or MAC Bridge that advertises desired C-VID or S-VID to I-SID assignments using the PBBN Auto Attach System TLV (D.2.17) and PBBN Auto Attach Assignment TLV (D.2.18).

An AAB is a BEB along with C-VLAN components that manages and maintains C-VID or S-VID to I-SID assignments used to attach AADs. An AAB uses Auto Attach System and Auto Attach Assignment TLVs to activate C-VID or S-VID to I-SID assignments advertised by the Auto Attach Assignment TLVs.

The AAP operates between an AAB and an AAD port. The AAP controls the exchange of Auto Attach TLVs, and configuration of VLAN and BSI mappings between the participating systems.

1 The AAP invokes LLDP when there is a change to an Auto Attach TLV by invoking the
2 SomethingChangedLocal() procedure, specified by IEEE Std 802.1AB, causing transmission to a peer AAS.
3 Likewise, an AAS is invoked by LLDP when there is an update to an Auto Attach TLV triggering the
4 SomethingChangedRemote() procedure, specified by IEEE Std 802.1AB.

5 Figure 50-2 depicts the Auto Attach functions that invoke the base LLDP transmission and reception of
6 LLDPDUs. The LLDP state machines are defined in Clause 9 of IEEE Std 802.1AB-2016.

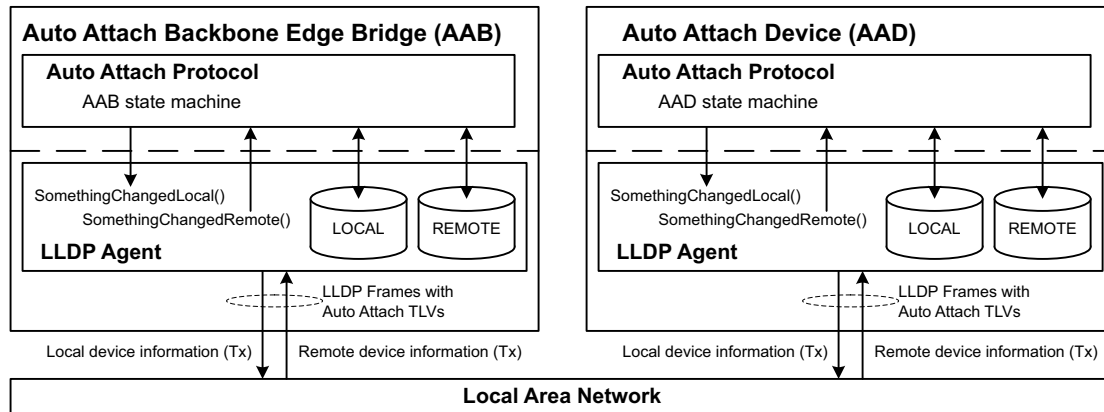


Figure 50-2—PBBN Auto Attach functions diagram

7 50.2 Service interfaces

8 An AAB provides two types of service interface, one for S-tagged service and another for C-tagged service.
9 Which of these interfaces is selected is determined by the AAD system type requesting attachment. If the
10 AAD system type is S-VLAN aware AAD, then the AAB provides an S-tagged service interface and all
11 mappings within the assignment TLV are for S-VID to I-SID. For all other AAD types, a C-tagged service
12 interface is provided and all mappings within the assignment TLV are for C-VID to I-SID.

13 50.2.1 S-tagged service interface

14 The S-tagged service interface maps a service instance from a Provider Bridged Network (PBN), identified
15 by an S-VID, to a backbone service instance on the PBBN, identified by an I-SID. The S-tagged service
16 interface performs a one-to-one mapping of S-VIDs to I-SIDs. Frames that are mapped to the I-SID are
17 carried over the PBBN while frames that are not mapped to an I-SID are not carried over the PBBN.

18 An S-tagged service interface is provided by an AAB over a Customer Network Port provided by the AAB's
19 I-component. The S-tagged interface does not carry the S-TAG over the PBBN. The DEI and PCP bits are
20 regenerated on ingress and are then carried in the I-DEI and I-PCP bits in the I-TAG across the PBBN. On
21 egress to an S-tagged interface, the S-TAG can be deduced from the I-TAG received from the PBBN (the
22 I-SID is mapped to an S-VID, the I-DEI and I-PCP bits are regenerated and then carried in the DEI and PCP
23 bits).

1 50.2.2 C-tagged service interface

2 The C-tagged service interface maps a service instance from a Customer Bridged Network (CBN), identified
3 by a C-VID, to a backbone service instance on the PBBN, identified by an I-SID. The C-tagged service
4 interface performs a one-to-one mapping of C-VIDs to I-SIDs. Frames that are mapped to the I-SID are
5 carried over the PBBN while frames that are not mapped to an I-SID are not carried over the PBBN.

6 A C-tagged service interface is provided by an AAB over a Customer Edge Port (CEP) as illustrated by
7 Figure 50-3. The C-tagged interface does not carry the C-TAG over the PBBN. The DEI and PCP bits are
8 regenerated on ingress and are then carried in the I-DEI and I-PCP bits in the I-TAG across the PBBN. On
9 egress to a C-tagged interface, the C-TAG can be deduced from the I-TAG received from the PBBN (the
10 I-SID is mapped to a C-VID, the I-DEI and I-PCP bits are regenerated and carried in the DEI and PCP bits).

11 Each of the C-VLAN components used to provide CEP service at the edge of an AAB is composed of a
12 single CEP and a single Provider Edge Port (PEP) for each service instance that can be provided through that
13 CEP. Each PEP is connected within the AAB, as specified in 6.14, to a distinct CNP on the I-component.
14 Each C-VLAN component implements RSTP, with the enhancements to support CEPs, as specified in 13.41.

15 NOTE 1—The restriction that each CBN C-VLAN map to a single backbone service instance on the PBBN allows the
16 CBN equipment receiving frames to correctly identify the service instance used to deliver that frame and prevents the
17 configuration of the I-component to create a multi-point service from point-to-point service instances, which could result
18 in the creation of data loops. The backbone provider can offer a multi-point service through appropriate configuration of
19 the B-VLAN component.

20 Figure 50-3 illustrates a CBN attached to a PBBN using a C-tagged service interface. The customer network
21 uses C-VLAN aware end stations or Bridges connecting to the PBBN. The PBBN in turn is composed of
22 AABs or BEBs interfacing to the customer Provider Bridges and BCBs used to forward frames between the
23 AABs or BEBs.

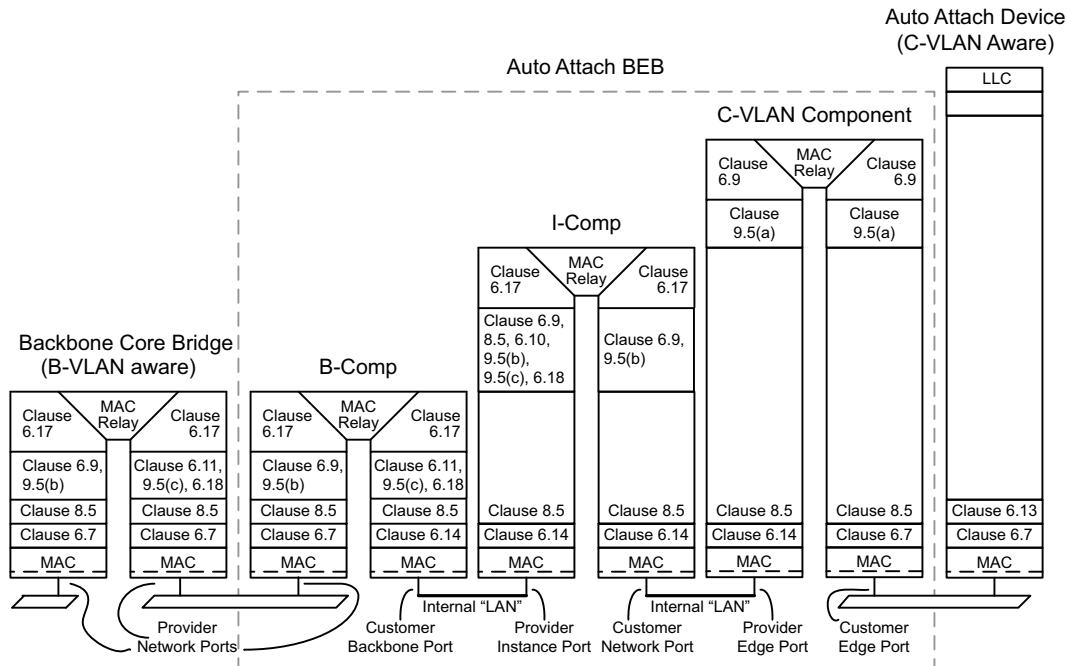


Figure 50-3—PBBN Auto Attach for customer networks

1 Customer VLAN unaware Bridges and MAC relay end station equipment can connect to Customer Network
2 Ports (CNPs) using an unprotected port-based service interface (25.3).

3 Service attachment requests from VLAN unaware (untagged) AADs send an assignment TLV with one
4 VID/I-SID assignment to the AAB. The port-based service interface on the AAB uses the specified VID to
5 update port VLAN membership used by the CNP to map the VID to an I-component.

6 A Port-based service interface allows a single mapping instance of an untagged service to an I-SID.

7 NOTE 2—An AAB includes both BEB and Provider Edge Bridge components. In particular, an AAB supports a
8 C-tagged service interface (15.4) for attachment to a CBN by including a C-VLAN component with PEPs that connect to
9 the CNPs of an I-component.

10 50.3 State machine overview

11 The AAP is specified by the external behavior indicated by the AAD state machine in 50.6 and the AAB
12 state machine in 50.7. These AAP state machines operate over LLDP on a point-to-point single peer-to-peer
13 topology. Each port enabled for PBBN Auto Attach runs an independent copy of the AAP.

14 When Auto Attach is enabled on a port, the AAP begins to write and read information to the local system
15 LLDP database. The initial process for an AAS is to write system information into the local LLDP database.
16 Information written into the LLDP database is transmitted on the port within the PBBN Auto Attach System
17 TLV in the next LLDPDU interval. Once an AAS has learned its neighbors and formed an association with a
18 neighboring AAS, the AAD can update the local LLDP database with assignment mappings that are
19 transmitted on the port within the PBBN Auto Attach Assignment TLV. The AAB receives the assignment
20 mappings advertised by the AAD in the remote LLDP database. When the AAB receives new VID/I-SID
21 assignments from the AAD it determines if it can activate them. The AAB then updates its local database
22 with the VID/I-SID pairs from the AAD and indicates the status as pending, active, or rejected-with reasons.
23 For rejected assignments the AAB supplies some further information on the cause for the rejection.

24 The AAP state machine diagrams (50.6, 50.7) specify state transitions based on changes to the local
25 structures aaRemSysTlv and aaRemAsgnsTlv. These structures are updated whenever the LLDP routine
26 SomethingChangedRemote() calls the Auto Attach LLDP call back procedure (50.5.16). The structure
27 aaRemSysTlv reflects the current PBBN Auto Attach System TLV within the remote LLDP database,
28 however transformed to machine local byte and bit order. The structure aaRemAsgnsTlv reflects the current
29 PBBN Auto Attach Assignment TLV within the remote LLDP database, however transformed to machine
30 local byte and bit order. If no remote database is present, or if either Auto Attach TLV is absent in the remote
31 database, then aaRemSysTlv and(or) aaRemAsgnsTlv is set to NULL.

32 When a system TLV (D.2.17) is received, the AAS enters the VALIDATE_REMOTE_ASSOCIATION state
33 to determine whether an association can be formed with the AAS neighbor.

34 The following actions are performed by the state machine to validate an association:

- 35 — For an AAS verify this port has a point-to-point single peer-to-peer topology on this port by
- 36 verifying there is only a single remote LLDP database
- 37 — For an AAS verify there is only a single system TLV in the remote database
- 38 — For an AAD verify the assoc state of the remote system is READY-TO-ASSOC
- 39 — For an AAD verify the remote system type is AAB
- 40 — For an AAD verify the AAB tagging matches the AAD
- 41 — For an AAB verify the remote system type is AAD
- 42 — For an AAB verify the AAD tagging is supported and indicated in the AAB system TLV

1 If there is an error validating the remote system, such as a system type mismatch,
2 VALIDATE_REMOTE_ASSOCIATION continues to check any received system TLVs until an error free
3 configuration is detected.

4 Once the system TLV validation has completed, the AAD verifies if it already has an active association on a
5 different port. If no other association is active, then the AAD activates the association on this port and begins
6 sending assignment TLVs. If the AAD finds it already has an active association on a different port it holds
7 this association in standby and does not issue an assignment TLV for this port. If the AAD active association
8 is disabled or fails, the AAD state machine resets the port and restarts the association process on that port.
9 On restart the AAD breaks all assignments on the reset port, withdraws all Auto Attach TLVs,
10 resynchronizes with the peer AAB state machine, and starts the association process over. If the AAD has any
11 associations in standby, the first to recognize the active association has broken becomes active and issues an
12 assignment TLV.

13 The AAB can only accept assignment TLVs from an AAD with a valid system TLV. The AAB only allows a
14 single active association for each AAD. If the AAB finds an AAD attempting to activate a port with an
15 invalid system TLV or if the AAD attempts to activate more than one association, the AAB resets its port
16 invalidating the association and forcing a restart of the peer AAD state machine. On restart the AAB breaks
17 all assignments on the reset port, withdraws all Auto Attach TLVs, resynchronizes with the peer AAD state
18 machine, and starts the association process over.

19 An AAD is responsible for initiating one or more VID/I-SID assignment binding requests and sending them
20 to the AAB in the assignment TLV. The AAD uses aaAdminPortVidIsidAsgns[] (50.4.3.2) to determine the
21 requested assignments. Bindings are individually processed by the AAB to determine whether the specified
22 VID can be bound to the I-SID. If the binding is allowed, the AAB creates the VLAN and BSI as indicated
23 by the AAD assignment TLV. Once all assignments are updated, the assignment TLV is updated by the AAB
24 and sent to the AAD with the status of all assignment requests. The AAD updates all VID/I-SID assignments
25 and reflects the current status back to the AAB in its assignment TLV.

26 **50.3.1 System inactivity**

27 If the remote LLDP database for an AAS is deleted, Auto Attach TLVs are deleted from the database, or
28 Auto Attach System TLVs are not received, then all operating VID to I-SID assignments are deactivated and
29 the AAP restarts.

30 **50.3.2 Multiple AAD systems**

31 The Auto Attach function is intended to operate in a point-to-point single peer-to-peer topology. Multiple
32 AADs are not permitted to form an association on a single AAB port. Only one AAD system is permitted to
33 form an association per port interface of an AAB.

34 **50.3.3 AAD assignment requests**

35 Assignment mapping requests are communicated from the AAD to the AAB using the PBBN Auto Attach
36 Assignment TLV (D.2.18). These service assignment requests are a combination of the VID/I-SID mapping
37 and assignment status.

38 Assignment mappings are established when two criteria are met:

- 39 a) An AAB is discovered by an AAD and has a valid remote association determined by the
40 ValidateRemoteAssociation state machine procedures (50.6).
- 41 b) The Auto Attach variable aaAdminPortVidIsidAsgns[] for this port is configured with one or more
42 VID/I-SID assignments using the management interface (12.34.4).

1 50.3.4 AAB assignment request processing

2 Each VLAN that is associated with a VID/I-SID assignment mapping is configured on the AAD by setting
3 the VID/I-SID mapping in the PBBN Auto Attach assignment table (12.34.4). The port associated with the
4 link connecting the AAD to the AAB is a member of the VLAN(s) in the VID/I-SID assignment list received
5 and accepted by the AAB. This allows traffic on these VLANs to pass into the BSI.

6 All initial VID/I-SID assignment mapping requests are sent by the AAD with an assignment status of
7 pending until the AAB completes processing the request. The AAD updates and reflects the assignment
8 status changes for each VID/I-SID according to the last assignment status received from the AAB.

9 Each VID/I-SID assignment request received by the AAB is processed individually and is either accepted or
10 rejected. Assignment requests are processed for interfaces on which the port is in operation, LLDP is
11 enabled, Auto Attach TLV processing is enabled, and AAS valid association has occurred.

12 If an assignment is accepted the following happens:

- 13 — Each VLAN that is associated with an accepted VID/I-SID assignment is instantiated on the AAB if
14 it does not already exist, in order that proper data packet encapsulation and decapsulation is
15 supported.

16 There are two conditions associated with the interface assignment mapping and binding process:

- 17 a) If the I-SID already exists on the AAB, due to a previous configuration from any type of interface
18 binding method, then the AAB configures the associated interface to the pre-existing I-SID.
- 19 b) If the I-SID does not exist on the AAB already, then it is responsible for creating the BSI with the
20 associated I-SID.

21 Port tagging and port VLAN membership updates are performed by the AAD based on assignment
22 acceptance.

23 If an assignment is rejected, a response with a reason is returned to the AAD based on the following:

- 24 — Generic (undefined rejection error)
- 25 — Auto Attach resources unavailable
- 26 — Invalid VLAN ID
- 27 — VLAN resources unavailable
- 28 — Invalid I-SID
- 29 — I-SID resources unavailable
- 30 — Application interaction (VLAN, PBB) issue
- 31 — Assignment not allowed

32 Rejected assignments are returned to the originating AAD with a rejected status and indicate why the
33 rejection occurred using status as defined in D.2.18.6. An AAD continues to request a rejected assignment
34 until the assignment is administratively removed.

35 50.3.5 Assignment updates

36 All active VID/I-SID assignments on the AAB are updated using the current assignment TLV information
37 from the AAD.

38 When a new assignment TLV is received at the AAB containing additional mapping requests, removing
39 existing mapping requests, or changing existing mapping requests, the AAB state machine processes the
40 new requests by adding, deleting, or updating the assignments that were changed in the new assignment
41 TLV.

1 If one or more VID/I-SID assignments are not present in any subsequent assignment TLVs from the AAD,
2 the AAB removes the mapping of those missing assignments from the active list and releases the binding.

3 **50.4 State machine variables**

4 The Auto Attach state machines use both port local (50.4.3) and system wide variables (50.4.2). Updating a
5 system wide variable requires assuring consistency since multiple instances of the state machines (one per
6 port) can access the system wide variables.

7 The Auto Attach state machines use some complex structures (50.4.1). These structures are copied and
8 compared within the state machines in the same manner as strings. Copies result in a new structure of the
9 same length and content duplicated in a new variable. Compares test equal when two structures are exactly
10 the same size and content, otherwise they are not equal.

11 Arrays are indexed with their first element starting at zero.

12 When manipulating structures the state machine diagrams use dot notation to illustrate access to objects
13 within a structure. The syntax for accessing the objects within a structure used in the state machines is:

14 `<structure instance>.<object name>`

15 At times the state machine diagrams need to exclude a field from a structure that is being compared as a
16 string. This is expressed by prefixing the object name with “except”:

17 `<structure instance>.<except><object name>`

18 **50.4.1 Structure type definitions**

19 Type definitions are used to define some of the state machine variables. They are represented by C style
20 typedefs. Though C structures can not be manipulated and compared as though they are strings the state
21 machine both copies and compares these structures to simplify the state machine description.

22 **50.4.1.1 Assignment TLV type definition**

23 The type *ASGNSstring* models the PBBN Auto Attach Assignment TLV (D.2.18). It contains elements for
24 all the fields defined for the assignment TLV, however transformed into machine local byte and bit order.

25 Part of the *ASGNSstring* type is an array of triples, *ASGNStriple*, identifying the assignment
26 status/VID/I-SID. This array is variable in size and so instances of *ASGNSstring* can vary in size.

```
27 typedef struct {
28     Unsigned          status : 4;
29     Unsigned          vid : 12;
30     Unsigned          isid : 24;
31 } ASGNStriple;
32 typedef struct {
33     Unsigned Char      numAsgns;
34     ASGNStriple        asgnsArray[];
35 } ASGNSstring;
```

36 **50.4.1.2 Discovered association table object**

37 This object is the structure used by an AAS state machine for each object of the discovered remote attach
38 port table (50.4.2.2).

```
39 typedef struct {
40     Unsigned          locPortId;
41     PORTNETIDstring   remPortNetId;
42 } DISCport;
```

1 50.4.1.3 System TLV type definition

2 The type *SYSstring* models the PBBN Auto Attach System TLV (D.2.17). It contains elements for all of the
3 fields defined for the system TLV, however transformed into machine local byte and bit order.

```
4 typedef struct {
5     Unsigned          sysMAC : 48;
6     Unsigned16        reserved;
7     Unsigned32        portId;
8 } PORTNETIDstring;
9 typedef struct {
10    Unsigned char      state;
11    Unsigned          sysType : 4;
12    Unsigned          portTagging : 4;
13    Unsigned char      reserved;
14    PORTNETIDstring    portNetId;
15 } SYSstring;
```

16 50.4.1.4 VID to I-SID assignment table object

17 This object is the 2-tuples *VIDISIDpair* containing a single VID to I-SID assignment. This type is used by
18 the state machine for typing arrays of the assignment requests.

```
19 typedef struct {
20     Unsigned          vid : 12;
21     Unsigned          isid : 24;
22 } VIDISIDpair;
```

23 50.4.2 Per AAS variables

24 50.4.2.1 aaAdminSystemEnable

25 When FALSE this Boolean variable causes all AAP state machines on all ports to immediately enter the
26 INIT state. When set to TRUE this variable allows the AAP state machines to progress. The variable
27 aaAdminSystemEnable is set by the management system object aaSystemEnable (12.34.1).

28 50.4.2.2 aaOperDiscAssocTable[]

29 This table lists all of the currently associated ports discovered on remote AASs. It is an array of type
30 DISCport (50.4.1.2). When accessing or updating this table the state machine's implementation takes care of
31 any required interlocking for sharing this table.

32 50.4.2.3 aaOperAadActiveAssocIndex

33 This variable provides an index into the aaOperDiscAssocTable[] for the active association on this AAD. If
34 no association is active, then aaOperAadActiveAssocIndex is set to NO_ACTIVE_ASSOC. When updating
35 or accessing this variable the state machine's implementation takes care of any required interlocking for
36 sharing this variable.

37 50.4.2.4 aaSystemMAC

38 A MAC address selected to uniquely identify this AAS over the administrative domain. This variable is set
39 by the system to an appropriate value (such as the LLDP ChassisID) at system startup. If the system is a
40 DRNI the single aaSystemMAC refers to both devices in the DRNI pair.

41 50.4.2.5 aaSystemType

42 The system type can take the values (1)AAB, (2)CVLAN_AWARE_AAD, (3)VLAN_UNAWARE_AAD,
43 or (4)SVLAN_AWARE_AAD. This variable is set by the system to an appropriate value at system startup.

1 **50.4.2.6 aaSystemResetTime**

2 A variable used to initialize the aaResetTimer during a state machine reset. The time in seconds can range
3 from 1 to 1200 seconds.

4 **50.4.2.7 BEGIN**

5 A Boolean variable controlled by the system initialization process. A value of TRUE causes all state
6 machines to continuously execute their initial state. A value of FALSE allows all state machines to perform
7 transitions out of their initial state.

8 **50.4.3 Per port variables**

9 **50.4.3.1 aaAdminPortEnable**

10 When FALSE this Boolean variable causes the AAP state machine on this port to immediately enter the
11 INIT state. A value of TRUE allows the AAP state machine on the port to progress. The aaAdminPortEnable
12 is set by the management port table object aaPortEnable (12.34.2).

13 **50.4.3.2 aaAdminVidIsidAsgns[]**

14 The administratively assigned VID/I-SID pairs desired by an AAD on this port. The array is loaded by
15 management with the assignments table entries (12.34.4) for this port. This array is only used by the AAD
16 state machine.

17 This variable is an array of 2-tuples each containing a VID to I-SID assignment of type VIDISIDpair
18 (50.4.1.4). The state machine can compare or copy the entire array to an array of the same type with an
19 assignment, compare equal, or compare not equal operator. To be equal the two arrays must be of the same
20 size and contain the same elements in each 2-tuple.

21 The variables aaAdminVidIsidAsgns[n].vid and aaAdminVidIsidAsgns[n].isid of each 2-tuple within
22 aaAdminVidIsidAsgns[] is exposed at the management interface. Each row of the management database
23 assignment table represents an index into aaAdminVidIsidAsgns[]. So, for instance, the first row refers to
24 aaAdminVidIsidAsgns[0] and value of the first VID/ISID pair is set to the objects of the first row of the
25 Auto Attach assignment table entry for this port.

26 **50.4.3.3 aaAsgnsTlvEnable**

27 Enables and disables transmission of the PBBN Auto Attach Assignment TLV on this port. This variable is
28 the LLDP MIB object lldpXdot1AaConfigAsgnsTxEnable (D.5.6) or LLDP YANG object
29 tlvs-tx-org-aa-system-enable.aaAssign (D.6.6.7).

30 PBBN Auto Attach assignment TLV transmission is controlled by the Auto Attach state machines. When
31 aaAsgnsTlvEnable is FALSE, LLDP does not include the Auto Attach Assignment TLV in its transmissions.
32 When the aaAsgnsTlvEnable is TRUE, LLDP includes the Auto Attach Assignment TLV in its
33 transmissions.

34 **50.4.3.4 aaCurVidIsidAsgns[]**

35 An array of 2-tuples each containing a VID to I-SID assignment of type VIDISIDpair (50.4.1.4), containing
36 the VID/I-SID assignments currently advertised by an AAD in LLDP assignment TLVs.

37 **50.4.3.5 aaCurRemAsgnsTlv**

38 A structure of type ASGNSstring (50.4.1.1) containing the state machine's current assignment TLV.

1 **50.4.3.6 aaLocAsgnsTlv**

2 A structure of type ASGNSstring (50.4.1.1) containing the current LLDP local database assignment TLV
3 (D.2.18) in machine local byte and bit format. This structure is transferred to the LLDP local database object
4 lldpV2Xdot1LocAaAsgnsTlvString in network byte and bit order by the procedure aaSetAsgnsTlv()
5 (50.5.14).

6 **50.4.3.7 aaLocSysTlv**

7 A structure of type SYSstring (50.4.1.3) containing the current LLDP local database PBBN Auto Attach
8 System TLV (D.2.17) in machine local byte and bit format. This structure is transferred to the LLDP local
9 database object lldpV2Xdot1LocAaSysTlvString in network byte and bit order by the procedure
10 aaSetSysTlv() (50.5.15).

11 **50.4.3.8 aaOperLocAsgnsTlv**

12 A structure of type ASGNSstring (50.4.1.1) containing the state machine's operational assignment TLV.

13 **50.4.3.9 aaOperLocSysTlv**

14 A structure of type SYSstring (50.4.1.3) containing the state machine's operational PBBN Auto Attach
15 System TLV.

16 **50.4.3.10 aaOperRemAsgnsTlv**

17 A structure of type ASGNSstring (50.4.1.1) containing the state machine's operational assignment TLV.

18 **50.4.3.11 aaOperRemSysTlv**

19 A structure of type SYSstring (50.4.1.3) containing the state machine's operational PBBN Auto Attach
20 System TLV.

21 **50.4.3.12 aaPendingCompleted**

22 An AAB boolean used to indicate when a pending operation has completed. When TRUE a pending
23 operation has completed. The conditions setting this boolean to TRUE are implementation dependent. The
24 AAB state machine uses this boolean to process the completion of operations that are delayed pending
25 completion of requests to management or control functions.

26 **50.4.3.13 aaPortId**

27 Set by the system at startup to the if-ref for this port.

28 **50.4.3.14 aaPortTagging**

29 Set by the system to the tagging options currently configured on the port. This variable can take the values
30 specified for the PBBN Auto Attach System TLV tagging field listed in Table D-18.

31 **50.4.3.15 aaRemAsgnsTlv**

32 A structure of type ASGNSstring (50.4.1.1) containing the current LLDP remote database assignment TLV
33 (D.2.18) derived from lldpV2Xdot1RemAaAsgnsTlvString (D.5.6). If no remote entry exists or if more than
34 a single remote entry exists then aaRemSysTlv == NULL.

35 **50.4.3.16 aaRemSysTlv**

36 A structure of type SYSstring (50.4.1.3) containing the current LLDP remote database PBBN Auto Attach
37 System TLV (D.2.17) derived from lldpV2Xdot1RemAaSysTlvString (D.5.6) for this port. If no remote
38 entry exists or if more than a single remote entry exists then aaRemSysTlv == NULL.

1 **50.4.3.17 aaResetTimer**

2 The Auto Attach state machines use a timer to allow the AAB and AAD to return to synchronization after a
3 reset condition. This timer operates as a countdown timer (i.e., it expires when its value reaches zero). The
4 aaTimer:

- 5 a) Has a resolution of one second.
- 6 b) Is started by loading an initial integer value, n , where $0 < n \leq 1200$.
- 7 c) Is decremented by one per timer tick, as long as $n > 0$; the interval between timer ticks is the same as
8 the timer resolution.
- 9 d) Represents the remaining time in the period.

10 **50.4.3.18 aaStatsAssocAttached**

11 A statistic counter providing the number of times an association attached.

12 **50.4.3.19 aaStatsAssocFailed**

13 A statistic counter providing the number of times an association failed.

14 **50.4.3.20 aaStatsAssocReset**

15 A statistic counter providing the number of times an association was reset.

16 **50.4.3.21 aaStatsAssocStandby**

17 A statistic counter providing the number of times an association was a standby.

18 **50.4.3.22 aaStatsVidIsidAsgnsAccepted**

19 A statistic counter providing the number of VID/ISID assignments accepted by the AAB.

20 **50.4.3.23 aaStatsVidIsidAsgnsRejected**

21 A statistic counter providing the number of VID/ISID assignments rejected by the AAB.

22 **50.4.3.24 aaStatsVidIsidAsgnsRequested**

23 A statistic counter providing the number of VID/ISID assignments requested by the AAD.

24 **50.4.3.25 aaStatsVidIsidAsgnsWithdrawn**

25 A statistic counter providing the number of VID/ISID assignment requests withdrawn by the AAD.

26 **50.4.3.26 aaSysTlvEnable**

27 This variable enables and disables transmission of PBBN Auto Attach System TLVs on this port. This
28 variable is the LLDP MIB object `lldpXdot1AaConfigSysTxEnable` (D.5.6) or LLDP YANG object
29 `tlvs-tx-org-aa-system-enable.aaSystem` (D.6.6.7).

30 PBBN Auto Attach System TLV transmission is controlled by the Auto Attach state machine. When
31 `aaSysTlvEnable` is FALSE, LLDP does not include the Auto Attach System TLV in its transmissions. When
32 the `aaSysTlvEnable` is TRUE, LLDP includes the Auto Attach System TLV in its transmissions.

33 **50.4.3.27 aaTopologyError**

34 A boolean indicating a topology error. If the port is attached to a point-to-point single peer-to-peer network
35 then `aaTopologyError` is set to FALSE, otherwise `aaTopologyError` is set to TRUE. When this variable is set
36 to TRUE, the state machines immediately exit their current state and enter the INIT state.

1 50.5 State machine functions

2 The state machine functions described here are expressed in both words and pseudo-code fragments. Code
3 fragments use C style syntax; however, when implementation specific operations are required that extend
4 outside the scope of this standard, a word description is inserted in the pseudo-code and italicized. These
5 functions are intended to describe the external behavior requirements of the AAP, however are not intended
6 to be an implementation of the protocol.

7 Many of these functions operate on SYSstring and ASGNSstring structures. The functions operating on
8 these structures directly modify the structure instance within the state machine. Some of these functions also
9 provide return codes containing the status of the operations performed.

10 50.5.1 aaBuildSysTlv(SYSstring tlv)

11 This function loads the argument tlv with the current parameters for this port. The following values are set in
12 tlv. The function returns with the modified tlv.

```
13 tlv->state = NOT_READY;  
14 tlv->sysType = aaSystemType;  
15 tlv->portTagging = aaPortTagging;  
16 tlv->reserved = 0;  
17 tlv->portNetId.sysMAC = aaSystemMAC;  
18 tlv->portNetId.reserved = 0;  
19 tlv->portNetId.portId = aaPortId;
```

20 No errors are reported by this routine, however in an implementation there can be internal errors found.

21 50.5.2 aaBuildAsgnsTlv(ASGNSstring tlv, VIDISIDpair vidIsidArray[], ASGNSstring 22 operTlv)

23 This function loads tlv with the current parameters for this port, with the assignment requests contained in
24 vidIsidArray[], and with the current operational status from operTlv. Any new assignment status is set to
25 PENDING. In an AAD the operTlv is an internal copy of the current Auto Attach LLDP remote assignment
26 TLV from the AAB.

27 The function begins by determining the number of elements in the argument vidIsidArray[]. This number is
28 designated here by *n*. The function then initializes the header variables of the tlv argument as follows:

```
29 tlv->numAsngs = n;
```

30 Next the function loads the asgnsArray with the VID/ISID pairs requested by the AAD. For each element of
31 the vidIsidArray[] designated by *i* aaBuildAsgnsTlv() sets:

```
32 if ( n > 0 ) {  
33     for ( i=0; i < n; i++ ) {  
34         tlv->asgnsArray[i].status = PENDING;  
35         tlv->asgnsArray[i].vid = vidIsidArray[i].vid;  
36         tlv->asgnsArray[i].isid = vidIsidArray[i].isid;  
37     };  
38 };
```

39 Next the function loads the asgnsArray with current status for each old VID/ISID pair using the operTlv.

```
40 for ( i=0; i < n; i++ ) {  
41     for ( j=0; j < operTlv->numAsngs; j++ ) {  
42         if (tlv->asgnsArray[i].vid == operTlv->asgnsArray[j].vid &&  
43             tlv->asgnsArray[i].isid == operTlv->asgnsArray[j].isid ) {  
44                 tlv->asgnsArray[i].status = operTlv->asgnsArray[j].status;  
45                 break;  
46             };  
47     };  
48 };
```

49 No errors are reported by this routine, though in an implementation there can be internal system errors.

1 50.5.3 aaConfigAabAsgn(unsigned vid, unsigned isid)

2 This function configures the AAB port as a Customer Edge Port (CEP) or a Customer Network Port (CNP),
3 depending on the type of AAD requesting attachment. If the attachment is to an S-VLAN aware AAD then a
4 CNP is provided and the requested S-VLAN is activated and coupled to the requested backbone service
5 instance (BSI, 50.2). If the attachment is to any other AAD type, a CEP is provided and the requested
6 C-VLAN is activated and coupled to the requested BSI. The function uses the state machine variables
7 aaPortId and aaOperRemSysTlv to identify the AAB port where the VLAN attaches to the AAB and to
8 determine the tagging type to use to interface to the AAD. The settings required to configure the AAB are
9 implementation dependent.

10 To perform the configuration the AAB needs to configure:

- 11 a) The Customer Edge Port, Provider Edge Port, and filtering database are configured in the C-VLAN
12 component providing customer service on this AAB port identified by the aaPortId state machine
13 variable and with tagging parameters specified in the aaOperRemSysTlv for this port (50.2).
- 14 b) The Customer Network Port, Provider Instance Port, and filtering database are configured in the
15 I-component.
- 16 c) The Customer Backbone Port and filtering database are configured in the B-component.
- 17 d) If the desired BSI identified by the I-SID is not already coupled to the AAB, the AAB advertises the
18 I-SID to the network using either Shortest Path Bridging (SPB) or requests to a central management
19 plane (PBB) to form an attachment to the desired BSI.
- 20 e) The function returns ACCEPT if the configuration is complete or PENDING if the configuration is
21 in progress. Once the pending action is complete the system sets the state machine variable
22 aaPendingCompleted = TRUE.

23 50.5.4 aaDisableAabVidIsidAsgns(ASGNSstring tlvNew, ASGNSstring tlvOld)

24 This function determines which VID to I-SID assignments to disable on this port. The function compares the
25 list of VID to I-SID assignments in tlvOld to those in tlvNew and disables all the assignments found in
26 tlvOld with status ACCEPT but either missing in tlvNew or REJECTED in tlvNew. A pseudo-code sketch
27 of aaDisableAabVidIsidAsgns() is as follows:

```
28 if ( tlvOld == NULL || tlvOld->numAsgns == 0 ) return; // no running assignments
29 for ( i=0; i < tlvOld->numAsgns; i++ ) {
30     match = FALSE;
31     if ( tlvNew != NULL && tlvNew->numAsgns != 0 )
32         for ( j=0; j < tlvNew->numAsgns; j++ ) {
33             if ( tlvOld->asgnsArray[i].vid == tlvNew->asgnsArray[j].vid
34                 && tlvOld->asgnsArray[i].isid==tlvNew->asgnsArray[j].isid )
35                 { match = TRUE; k = j; };
36         }; // for each j
37     if ( match == FALSE ) { // tlvOld vid/isid not in tlvNew
38         if ( tlvOld->asgnsArray[i].status == ACCEPT ) {
39             disable tlvOld->asgnsArray[i];
40         };
41         /* the disable operation indicates removing all operating state */
42         /* within the AAB for this VID to I-SID mapping */
43     };
44     if ( match == TRUE // tlvOld vid/isid ACCEPT but tlvNew vid/isid REJECT
45         && tlvOld->asgnsArray[i].status == ACCEPT
46         && tlvOld->asgnsArray[i].status != tlvNew->asgnsArray[k].status )
47         disable tlvOld->asgnsArray[i];
48         /* the disable operation indicates removing all operating */
49         /* state within the AAB for this VID to I-SID mapping */
50 }; // for each i
```

51 In the pseudo-code the italicized “*disable tlvOld->asgnsArray[i]*” indicates execution of an implementation
52 dependent operation to disable the VID to I-SID mapping.

1 50.5.5 aaDisableLldpAaTlvTx()

2 This function disables LLDP transmission of the Auto Attach TLVs. Pseudo-code for
3 aaDisableLldpAaTlvTx() is as follows:

```
4 if ( aaSysTlvEnable == ENABLED ||  
5     aaAsgnsTlvEnable == ENABLED ) {  
6     aaAsgnsTlvEnable = DISABLED;  
7     aaSysTlvEnable = DISABLED;  
8     SomethingChangedLocal();  
9 }
```

10 50.5.6 aaDisableVidIsidAsgns()

11 This function disables all active VID to I-SID assignments on this port. All AAS internal state required to
12 activate the VID to I-SID assignment on this port is deleted.

13 50.5.7 aaDiscAssocAabTblCheck(unsigned port, SYSstring tlv)

14 A function used to search aaOperDiscAssocTable[] for an entry where the Auto Attach System MAC
15 address contained within an aaOperDiscAssocTable.remPortNetId object matches the Auto Attach System
16 MAC address contained within tlv.portNetId or for an existing entry for this AAB port. The function returns
17 ASSOC_ATTACHED if no match was found or if the AAD is already discovered on this AAB port. The
18 function returns ASSOC_INVALID if the AAD is already discovered on a different AAB port or this is a
19 new AAD identified on the AAB port.

20 A pseudo-code sketch of the function is as follows:

```
21 for ( each i an index to aaOperDiscAssocTable[] ) {  
22     if ( aaOperDiscAssocTable[i].remPortNetId == tlv->portNetId &&  
23         aaOperDiscAssocTable[i].locPortId == port ) return ASSOC_ATTACHED;  
24     if ( aaOperDiscAssocTable[i].remPortNetId.sysMAC  
25         == tlv->portNetId.sysMAC ) return ASSOC_INVALID;  
26     if ( aaOperDiscAssocTable[i].locPortId == port ) return ASSOC_INVALID;  
27 } //for each i  
28 return ASSOC_ATTACHED;
```

29 Within the pseudo-code, the italicized text for operator “*each i an index to aaOperDiscAssocTable[]*”
30 replaces the normal C code for loop conditions because the structure of the table aaOperDiscAssocTable is
31 implementation dependent. The for loop within the pseudo-code executes for each member of the
32 aaOperDiscAssocTable, regardless of the table implementation.

1 50.5.8 aaDiscAssocAadTblCheck(unsigned port, SYSstring tlv)

2 An AAD can only have a single active association. It is assumed that the aaOperDiscAssocTable[] has
3 already inserted a discovery entry for this port. The aaOperDiscAssocTable[] entry for the active association
4 is indexed by the variable aaOperAadActiveAssocIndex. The function returns ASSOC_ATTACHED if there
5 is no other active association or if the current active association is this local port with this remote port.
6 Otherwise the function returns ASSOC_STANDBY.

```

7 if ( aaOperAadActiveAssocIndex == NO_ACTIVE_ASSOC ) {
8     for ( each i an index of aaOperDiscAssocTable[] ) {
9         if ( aaOperDiscAssocTable[i].locPortId == port &&
10             aaOperDiscAssocTable[i].remPortNetId == tlv->portNetId ) {
11             aaOperAadActiveAssocIndex = i; return ASSOC_ATTACHED;
12         };
13     };
14 } else {
15     i= aaOperAadActiveAssocIndex;
16     if ( aaOperDiscAssocTable[i].locPortId == port &&
17         aaOperDiscAssocTable[i].remPortNetId == tlv->portNetId )
18         return ASSOC_ATTACHED;
19 }
20 return ASSOC_STANDBY;
```

21 Within the pseudo-code, the italicized text for operator “*each i an index of aaOperDiscAssocTable[]*”
22 replaces the normal C code for loop conditions because the structure of the table aaOperDiscAssocTable is
23 implementation dependent. The for loop within the pseudo-code executes for each member of the
24 aaOperDiscAssocTable, regardless of the table implementation.

25 50.5.9 aaDiscAssocTblDelete(unsigned port)

26 This function deletes an entry in aaOperDiscAssocTable[]. The function deletes the table entry *i* matching
27 the condition aaOperDiscAssocTable[i].locPortId == port.

28 50.5.10 aaDiscAssocTblInsert(unsigned port, SYSstring tlv)

29 This function inserts a new entry in aaOperDiscAssocTable[]. The function builds a DISCport entry and
30 inserts it into a new table entry *i*. It is assumed the table is already checked to assure a duplicate entry is not
31 present. The entry is as follows:

```

32 aaOperDiscAssocTable[i].locPortId = port;
33 aaOperDiscAssocTable[i].remPortNetId = tlv->portNetID;
```

34 50.5.11 aaEnableAabVidIsidAsgns(ASGNSstring tlv)

35 This function determines which VID to I-SID assignments to activate. The function is responsible for
36 determining if each pending VID to I-SID pair contained in the tlv argument can be activated. In addition,
37 this function builds all state required by the AAB to activate a VID to I-SID assignment.

1 The function modifies the status associated with each VID to I-SID pair to reflect the action taken on the
2 pair. The codes that can be returned are listed in Table D-20. A pseudo-code sketch is as follows:

```
3 if ( tlv == NULL || tlv->numAsgns == 0 ) return;
4 for ( i=0; i < tlv->numAsgns; i++ ) {
5     if ( tlv->asgnsArray[i].status == PENDING ) {
6         if ( the VID to I-SID assignment is acceptable to the AAB ) {
7             st=aaConfigAabAsgn(tlv->asgnsArray[i].vid,
8                               tlv->asgnsArray[i].isid);
9             /* The aaConfigAabAsgn() builds all operating state */
10            /* within the AAB for this VID to I-SID mapping */
11            tlv->asgnsArray[i].status = st;
12        } else tlv->asgnsArray[i].status = REJECT; //including reason
13    };
14 }; //for each i
```

15 In this pseudo-code the condition “*the VID to I-SID assignment is acceptable to the AAB*” is a catch-all for
16 any internal checks the AAB performs to allow the assignment before attempting configuration.

17 50.5.12 aaEnableAadVidIsidAsgns(ASGNSstring tlv)

18 This function enables VID to I-SID assignments. It sets all of the AAD internal state required to enable or
19 disable each VID to I-SID pair. Each pair marked with tlv->asgnsArray[i].status == ACCEPT is enabled,
20 otherwise the pair is disabled. The operations required to enable or disable the identified VLAN are system
21 dependent and so not explicitly described in the pseudo-code. Instead the italicized *enable* and *disable*
22 operations are used to indicate the implementation dependent operations.

```
23 if ( tlv == NULL || tlv->numAsgns == 0 ) return; // no assigns received
24 for ( i=0; i < tlv->numAsgns; i++ ) {
25     if ( tlv->asgnsArray[i].status == ACTIVE )
26         enable the VLAN identified by tlv.asgnsArray[i].vid
27     else disable the VLAN identified by tlv.asgnsArray[i].vid;
28 };
```

29 50.5.13 aaPortClearStats()

30 This function clears all the statistic counters for this port.

31 50.5.14 aaSetAsgnsTlv(ASGNSstring tlv)

32 The function is used to place a state machine internal structure of type ASGNSstring into the LLDP
33 database. The ASGNSstring referenced by tlv is transformed from the local machine byte and bit order to
34 the assignment TLV byte and bit order as it is moved into the LLDP database for this port.

35 The function aaSetAsgnsTlv() locks out LLDP from building an LLDPDU during the update. Once the TLV
36 is transfered and LLDP unlocked, the routine sets aaAsgnsTlvEnable = ENABLED then calls
37 SomethingChangedLocal() and returns.

38 50.5.15 aaSetSysTlv(SYSstring tlv)

39 The aaSetSysTlv() function is used to place a state machine internal structure of type SYSstring into the
40 LLDP database. The object referenced by tlv is transformed from the local machine byte and bit order to the
41 PBBN Auto Attach System TLV byte and bit order as it is moved into the LLDP database for this port.

42 The function aaSetSysTlv() locks out LLDP from building an LLDPDU during the update. Once the TLV is
43 transfered and LLDP unlocked, the routine sets aaSysTlvEnable = ENABLED then calls
44 SomethingChangedLocal() and returns.

1 **50.5.16 aaSomethingChangedRemote()**

2 This procedure is called by the LLDP routine SomethingChangedRemote() (IEEE Std 802.1AB-2016
3 clause 9.2.7.9) when something changes in the remote LLDP database. The procedure performs the
4 following operations:

- 5 — The remote LLDP database entries are checked to determine if multiple remote databases are
6 present on this port. If multiple databases are present, then the port is not connected to a
7 point-to-point single peer-to-peer topology and so the procedure sets aaTopologyError = TRUE,
8 aaRemSysTlv = NULL, aaRemAsgnsTlv = NULL. Otherwise the routine sets aaTopologyError =
9 FALSE.
- 10 — If aaTopologyError == FALSE, then the procedure updates the aaRemSysTlv as follows:
11 If a PBBN Auto Attach System TLV is not present in the remote database or if multiple Auto Attach
12 System TLVs are present, then aaRemSysTlv = NULL, otherwise the contents of the Auto Attach
13 System TLV are transformed from the big-endian format used for network transmission into the
14 local machine format and placed in aaRemSysTlv.
- 15 — If aaTopologyError == FALSE, then the procedure updates the aaRemAsgnsTlv as follows:
16 If a PBBN Auto Attach Assignment TLV is not present in the remote database or if multiple copies
17 are present, then aaRemAsgnsTlv = NULL, otherwise the contents of the PBBN Auto Attach
18 Assignment TLV are transferred from the big-endian format used for network transmission into the
19 local machine format and placed in aaRemAsgnsTlv.

20 **50.5.17 aaSomethingChangeRemoteInit()**

21 This procedure used to initialize the LLDP callback routine aaSomethingChangedRemote(). The function
22 registers aaSomethingChangedRemote() with LLDP for call back by the LLDP routine
23 SomethingChangedRemote() and initializes aaTopologyError = FALSE, then calls
24 aaSomethingChangedRemote() to initialize the variables aaTopologyError, aaRemSysTlv, and
25 aaRemAsgnsTlv.

1 50.5.18 aaUpdateReqDelVidIsidStats(ASGNSstring tlvNew, ASGNSstring tlvOld)

2 This routine is used to update the requested and withdrawn VID to I-SID assignment statistics for both the
3 AAB and AAD. The VID to I-SID assignment status indicated in tlvOld is compared with the VID to I-SID
4 status in tlvNew. Only VID to I-SID assignments that are different in tlvOld and tlvNew are considered in
5 the counts. A pseudo-code sketch of aaUpdateReqDelVidIsidStats() is as follows:

```
6 if ( tlvOld == NULL && tlvNew == NULL ) return;
7 if ( tlvOld != NULL && tlvOld->numAsgns != 0 )
8     if ( tlvNew == NULL || tlvNew->numAsgns == 0 ) {
9         aaStatsVidIsidAsgnsWithdrawn += tlvOld->numAsgns;
10        return;
11    };
12 if ( tlvNew != NULL && tlvNew->numAsgns != 0 )
13     if ( tlvOld == NULL || tlvOld->numAsgns == 0 ) {
14         aaStatsVidIsidAsgnsRequested += tlvNew->numAsgns;
15        return;
16    };
17 if ( tlvOld != NULL && tlvOld->numAsgns != 0
18 && tlvNew != NULL && tlvNew->numAsgns != 0 ) {
19     for ( i=0; i < tlvNew->numAsgns; i++ ) {
20         flag = FALSE;
21         for ( j=0; j < tlvOld->numAsgns; j++ ) {
22             if ( tlvNew->asgnsArray[i].vid == tlvOld->asgnsArray[j].vid
23                 && tlvNew->asgnsArray[i].isid == tlvOld->asgnsArray[j].isid )
24                 flag = TRUE;
25         }; // for each j
26         if ( flag == FALSE ) aaStatsVidIsidAsgnsRequested++;
27     }; // for each i
28     for ( i=0; i < tlvOld->numAsgns; i++ ) {
29         flag = FALSE;
30         for ( j=0; j < tlvNew->numAsgns; j++ ) {
31             if ( tlvNew->asgnsArray[i].vid == tlvOld->asgnsArray[j].vid
32                 && tlvNew->asgnsArray[i].isid == tlvOld->asgnsArray[j].isid )
33                 flag = TRUE;
34         }; // for each j
35         if ( flag == FALSE ) aaStatsVidIsidAsgnsWithdrawn++;
36     }; // for each i
37 };
```

1 **50.5.19 aaUpdateAcptRejVidIsidStats(ASGNSstring tlvNew, ASGNSstring tlvOld)**

2 This routine is used to update the accepted and rejected VID to I-SID assignment statistics for both the AAB
3 and AAD. The VID to I-SID assignment status indicated in tlvOld is compared with the VID to I-SID status
4 in tlvNew. Only VID to I-SID assignments that are different in tlvOld and tlvNew are considered in the
5 counts. A pseudo-code sketch of aaUpdateAcptRejVidIsidStats() is as follows:

```

6 if ( tlvNew == NULL || tlvNew->numAsgns == 0 ) return;
7 if ( tlvOld == NULL || tlvOld->numAsgns == 0 ) {
8     for ( i=0; i < tlvNew->numAsgns; i++ ) {
9         if ( tlvNew->asgnsArray[i].status == ACCEPTED )
10             aaStatsVidIsidAsgnsAccepted++
11         else if ( tlvNew->asgnsArray[i].status != PENDING )
12             aaStatsVidIsidAsgnsRejected++;
13     };
14 } else
15     for ( i=0; i < tlvNew->numAsgns; i++ ) {
16         flag = FALSE;
17         for ( j=0; j < tlvOld->numAsgns; j++ ) {
18             if ( tlvNew->asgnsArray[i].vid == tlvOld->asgnsArray[j].vid
19                 && tlvNew->asgnsArray[i].isid == tlvOld->asgnsArray[j].isid
20                 && tlvNew->asgnsArray[i].status ==
21                     tlvOld->asgnsArray[j].status )
22                 flag = TRUE;
23         }; // for each j
24         if ( flag == FALSE ) {
25             if ( tlvNew->asgnsArray[i].status == ACCEPTED )
26                 aaStatsVidIsidAsgnsAccepted++
27             else if ( tlvNew->asgnsArray[i].status != PENDING )
28                 aaStatsVidIsidAsgnsRejected++;
29         };
30 }; //for each i

```

31 **50.5.20 SomethingChangedLocal()**

32 This is an LLDP routine that is called when the Auto Attach state machines make updates to the local LLDP
33 database entries. The routine is defined in the IEEE Std 802.1AB state machines.

50.6 AAD state machine

In an implementation that supports auto attachment, the Auto Attach Device (AAD) state machine implements the functions specified by Figure 50-4. The state machine requires the AAD to form a valid association with one AAB. Therefore, all VID/I-SID requests from an AAD are received on one AAB port supporting an independent VID space.

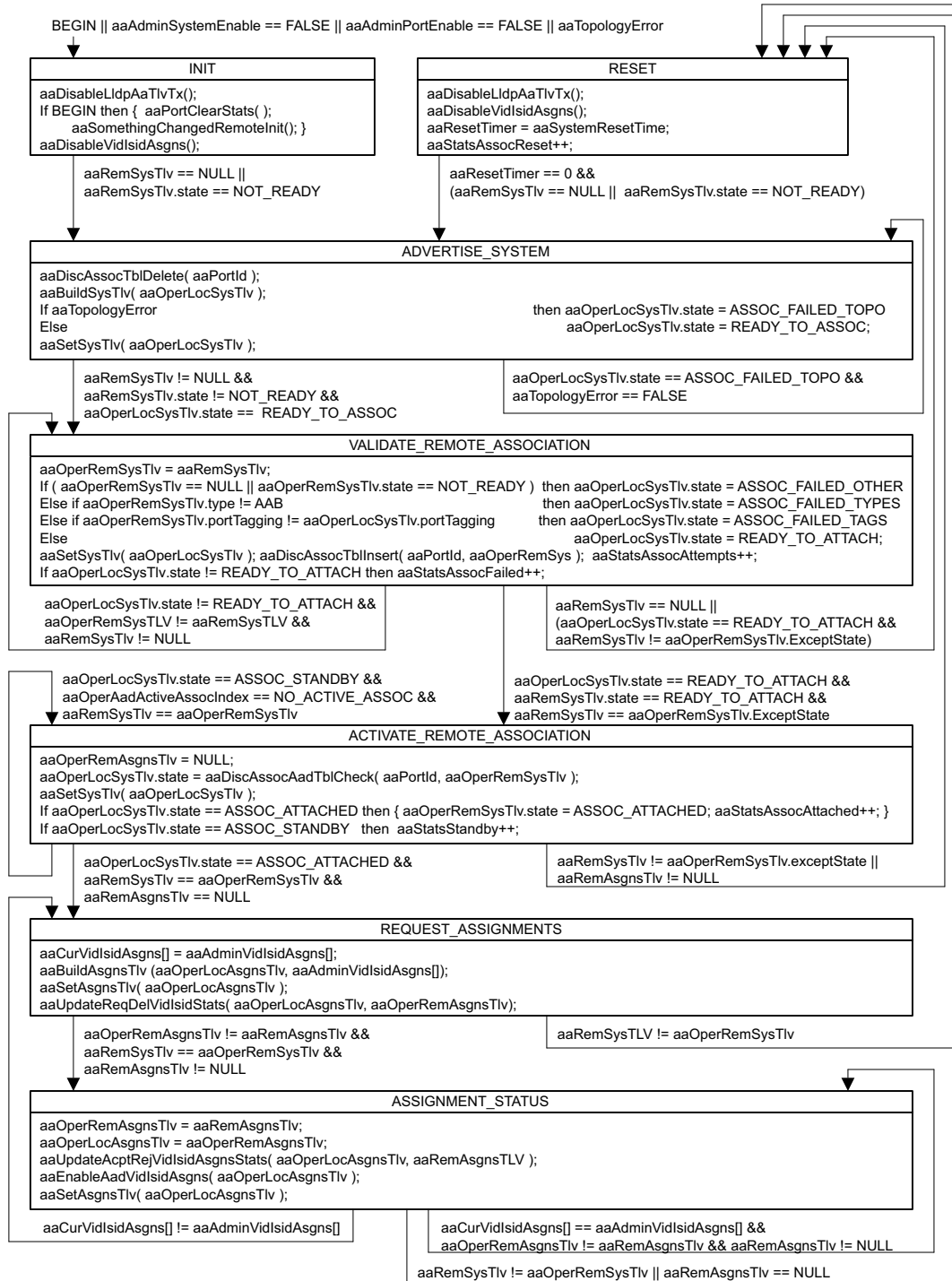


Figure 50-4—PBBN Auto Attach Device (AAD) state machine

50.7 AAB state machine

In an implementation that supports auto attachment, the Auto Attach BEB (AAB) state machine implements the functions specified by Figure 50-5. The state machine requires the AAB to provide a VLAN service specified by the AAD and requires the AAB to have an independent VID space for each port, therefore only one AAD is supported per AAB port.

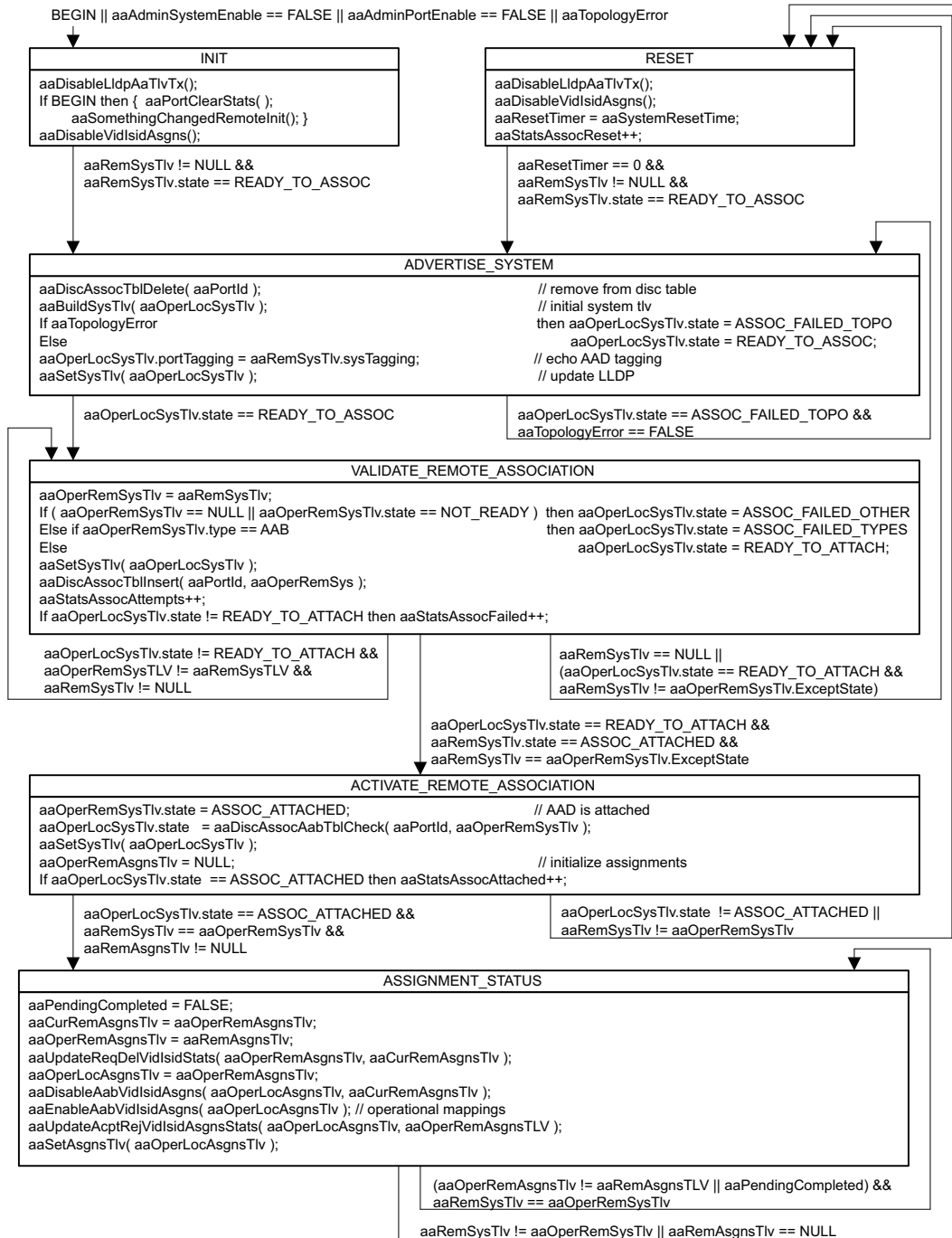


Figure 50-5—PBBN Auto Attach BEB (AAB) state machine

1 **Annex A**

2 (normative)

3 **PICS proforma—Bridge implementations**⁵⁵

4 **A.1 Introduction**

5 The supplier of a protocol implementation that is claimed to conform to this standard shall complete the
6 following Protocol Implementation Conformance Statement (PICS) proforma.

7 A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of
8 which capabilities and options of the protocol have been implemented. The PICS can have a number of uses,
9 including use:

- 10 a) By the protocol implementer, as a checklist to reduce the risk of failure to conform to the standard
11 through oversight.
- 12 b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication
13 of the capabilities of the implementation, stated relative to the common basis for understanding
14 provided by the standard PICS proforma.
- 15 c) By the user—or potential user—of the implementation, as a basis for initially checking the
16 possibility of interworking with another implementation (note that, while interworking can never be
17 guaranteed, failure to interwork can often be predicted from incompatible PICSs).
- 18 d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for
19 conformance of the implementation.

20 **A.2 Abbreviations and special symbols**

21 **A.2.1 Status symbols**

22	M	mandatory
23	O	optional
24	<i>O.n</i>	optional, but support of at least one of the group of options labeled by the same numeral n
25		is required
26	X	prohibited
27	pred:	conditional-item symbol, including predicate identification: see A.3.4
28	¬	logical negation, applied to a conditional item's predicate

29 **A.2.2 General abbreviations**

30	N/A	not applicable
31	PICS	Protocol Implementation Conformance Statement

⁵⁵ *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

1 A.3 Instructions for completing the PICS proforma

2 A.3.1 General structure of the PICS proforma

3 The first part of the PICS proforma, implementation identification and protocol summary, is to be completed
4 as indicated with the information necessary to identify fully both the supplier and the implementation.

5 The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each
6 containing a number of individual items. Answers to the questionnaire items are to be provided in the
7 rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No) or
8 by entering a value or a set or range of values. (Note that there are some items where two or more choices
9 from a set of possible answers can apply; all relevant choices are to be marked.)

10 Each item is identified by an item reference in the first column. The second column contains the question to
11 be answered; the third column records the status of the item—whether support is mandatory, optional, or
12 conditional: see also A.3.4. The fourth column contains the reference or references to the material that
13 specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

14 A supplier may also provide (or be required to provide) further information, categorized as either Additional
15 Information or Exception Information. When present, each kind of further information is to be provided in a
16 further subclause of items labeled A_i or X_i , respectively, for cross-referencing purposes, where i is any
17 unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format
18 and presentation.

19 A completed PICS proforma, including any Additional Information and Exception Information, is the
20 Protocol Implementation Conformation Statement for the implementation in question.

21 NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to
22 describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering
23 some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of
24 the information.

25 A.3.2 Additional information

26 Items of Additional Information allow a supplier to provide further information intended to assist the
27 interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS
28 can be considered complete without any such information. Examples might be an outline of the ways in
29 which a (single) implementation can be set up to operate in a variety of environments and configurations, or
30 information about aspects of the implementation that are outside the scope of this standard but that have a
31 bearing on the answers to some items.

32 References to items of Additional Information may be entered next to any answer in the questionnaire and
33 may be included in items of Exception Information.

34 A.3.3 Exception information

35 It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any
36 conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer
37 will be found in the Support column for this item. Instead, the supplier shall write the missing answer into
38 the Support column, together with an X_i reference to an item of Exception Information, and shall provide the
39 appropriate rationale in the Exception item itself.

40 An implementation for which an Exception item is required in this way does not conform to this standard.

41 NOTE—A possible reason for the situation described previously is that a defect in this standard has been reported, a
42 correction for which is expected to change the requirement not met by the implementation.

1 A.3.4 Conditional status

2 A.3.4.1 Conditional items

3 The PICS proforma contains a number of conditional items. These are items for which both the applicability
4 of the item itself, and its status if it does apply—mandatory or optional—are dependent on whether certain
5 other items are supported.

6 Where a group of items is subject to the same condition for applicability, a separate preliminary question
7 about the condition appears at the head of the group, with an instruction to skip to a later point in the
8 questionnaire if the “Not Applicable” answer is selected. Otherwise, individual conditional items are
9 indicated by a conditional symbol in the Status column.

10 A conditional symbol is of the form “**pred:** S” where **pred** is a predicate as described in A.3.4.2 below, and
11 S is a status symbol, M or O.

12 If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated
13 by the status symbol following the predicate: The answer column is to be marked in the usual way. If the
14 value of the predicate is false, the “Not Applicable” (N/A) answer is to be marked.

15 A.3.4.2 Predicates

16 A predicate is one of the following:

- 17 a) An item-reference for an item in the PICS proforma: The value of the predicate is true if the item is
18 marked as supported and is false otherwise.
- 19 b) A predicate-name, for a predicate defined as a boolean expression constructed by combining item-
20 references using the boolean operator OR: The value of the predicate is true if one or more of the
21 items is marked as supported.
- 22 c) The logical negation symbol “¬” prefixed to an item-reference or predicate-name: The value of the
23 predicate is true if the value of the predicate formed by omitting the “¬” symbol is false, and vice
24 versa.

25 Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for
26 grouped conditional items, is indicated by an asterisk in the Item column.

1 A.4 PICS proforma for IEEE Std 802.1Q—Bridge implementations

A.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification, e.g., name(s) and version(s) of machines and/or operating system names	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification. NOTE 2—The terms “Name” and “Version” should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).	

A.4.2 Protocol summary, IEEE Std 802.1Q

Identification of protocol specification	IEEE Std 802.1Q-2022, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks			
Identification of amendments and corrigenda to the PICS proforma that have been completed as part of the PICS	Amd.	:	Corr.	:
	Amd.	:	Corr.	:
Have any Exception items been required? (See A.3.3: the answer “Yes” means that the implementation does not conform to IEEE Std 802.1Q)	No []		Yes []	
Date of Statement				

1 A.5 Major capabilities

Item	Feature	Status	References	Support
	If the implementation is an end station implementation, mark “N/A” and continue at Annex B.			N/A []
MAC	Do the implementations of MAC technologies and support of the MAC Internal Sublayer Service (ISS) conform to MAC standards as specified in IEEE Std 802.1AC? (If support of a specific MAC technology is claimed, any PICS proforma(s) required by the standard specifying that technology shall also be completed.)	M	A.6, IEEE Std 802.1AC	Yes []
LLC	Is a class of LLC supporting Type 1 operations supported on all Bridge Ports in conformance with ISO/IEC 8802-2? (The PICS proforma required by ISO/IEC 8802-2 shall also be completed.)	M	8.2, 8.3, 8.1.3, ISO/IEC 8802-2	Yes []
RLY	Does the implementation relay and filter frames as specified?	M	8.5, 8.6, 8.7, 6.12, 8.8, A.7	Yes []
BFS	Does the implementation maintain the information required to make frame filtering decisions and support Basic Filtering Services?	M	8.1, 8.5, 8.7, 8.8, A.8	Yes []
ADDR	Does the implementation conform to the provisions for addressing?	M	8.13, A.9	Yes []
MBRIDGE	Can the Bridge be configured to operate as a VLAN-unaware MAC Bridge	O.2	5.14	Yes [] No []
TPMR	Can the Bridge be configured to operate as a Two Port MAC Relay?	O.2	5.16	Yes [] No []
MSP	Is the operation of the MAC Status Propagation Entity (MSPE) supported?	TPMR: M	Clause 23	Yes [] N/A []
RSTP	Is RSTP implemented?	–TPMR:O.1 TPMR:X	Clause 5, Clause 13, Clause 14, A.10	Yes [] No []
BPDU	Are transmitted Bridge Protocol Data Units (BPDUs) encoded and received BPDUs validated as specified?	–TPMR:M TPMR:X	A.11, 13.29.27, 13.29.28, 13.29.29, Clause 14	Yes [] No []
IMP	Are the required implementation parameters included in this completed PICS?	M	8.8, A.12	Yes []
PERF	Are the required performance parameters included in this completed PICS? (Operation of the Bridge within the specified parameters shall not violate any of the other conformance provisions of this standard.)	M	8.5, A.13	Yes []
MGT	Is management of the Bridge supported?	O PBBTE OR TPMR OR SRRM:M	Clause 5, A.14	Yes [] No []

A.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support
RMGT	Is a remote management protocol supported?	MGT:O PBBTE OR TPMR OR SRRM:M	Clause 5, A.15	Yes [] No []
MIB	Does the system implementation support management operations using SMIPv2 MIB modules?	MGT:O	8.12, Clause 17	Yes [] No [] N/A []
TC	Are multiple traffic classes supported for relaying frames?	O	8.6.6, 8.6.7, 8.6.8, A.16	Yes [] No []
EFS	Are Extended Filtering Services supported for relaying and filtering frames?	–TPMR:O TPMR:X	A.17, 6.12	Yes [] No []
MMRP	Is the operation of the Multiple MAC Registration Protocol supported?	EFS:M	5.4.1, A.20	Yes [] N/A []
VLAN	Does the implementation support the ability to insert tag headers into, modify tag headers in, and remove tag headers from relayed frames?	–(MBRIDGE OR TPMR):M (MBRIDGE OR TPMR):X	5.4, 6.3, 6.8, 6.9, 8.6, 6.12, Clause 9	Yes [] No []
VTR	Does the implementation support a VLAN Identifier (VID) translation table?	–(TPMR OR MBRIDGE):O (MBRIDGE OR TPMR):X	6.9	Yes [] No []
MVRP	Is automatic configuration and management of VLAN topology using MVRP supported?	–(TPMR OR MBRIDGE):M TPMR:X	5.4, A.21	Yes [] No []
MRP	Is the Multiple Registration Protocol (MRP) implemented in support of MRP Applications?	MMRP:M MVRP:M	Clause 10, A.20, A.21, A.22	Yes [] N/A []
MRP1	Does the MRP implementation support operation of the Full Participant?	MRP:O.4	Clause 10, A.22	Yes [] No [] N/A []
MRP2	Does the MRP implementation support operation of the Full Participant, point-to-point subset?	MRP:O.4	Clause 10, A.22	Yes [] No [] N/A []
MSTP	Is MSTP implemented?	–TPMR:O.1 TPMR:X	Clause 5, Clause 7, 8.4, 8.6.1, 8.8.8, 8.9, 8.10, 8.13.7, 11.2.3.1.2, Clause 13, Clause 14, A.18	Yes [] No []
VMGT	Does the implementation support VLAN management operations?	–(TPMR OR MBRIDGE) AND MGT:O (MBRIDGE OR TPMR):X	5.4.1, 12.10.2, 12.10.3	Yes [] No [] N/A []
CB	Can the Bridge be configured to operate as a C-VLAN Bridge, recognizing and using C-TAGs?	O.2	5.9	Yes [] No []

A.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support
PB	Can the Bridge be configured to operate as a Provider Bridge, recognizing and using S-TAGs?	O.2	5.10	Yes [] No []
PEB	Can the Bridge be configured to operate as a Provider Edge Bridge with one or more Ports operating as Customer Edge Ports (CEPs)?	PB:O	5.10.2	Yes [] No [] N/A []
RCSI	Can the Bridge be configured to operate as a Provider Edge Bridge with one or more Ports operating as Remote Customer Access Ports (RCAPs)?	PEB:O	5.10.2	Yes [] No [] N/A []
PB-2	State which Ports support the following values for the Provider Bridge Port Type: — PNP — CNP — CEP — RCAP	PB:M	5.10	Ports: _____ Ports: _____ Ports: _____ Ports: _____
CFM	Is Connectivity Fault Management (CFM) implemented?	O PBBTE: M	5.4.1.4, Clause 19, Clause 20, Clause 21, Clause 22	Yes [] No []
CFM-T	Does the implementation support a CFM MD-level zero MIP on each Port?	TPMR:M	5.15, Clause 18–Clause 22	Yes [] N/A [] N/A []
BRG	Is this system a Bridge, and not a Station, for the purposes of CFM?	CFM: O	22.4	Yes [] No []
BEB	Can the Bridge be configured to operate as a Backbone Edge Bridge, recognizing and using I-TAGs?	O.2	5.12	Yes [] No [] N/A []
BEB-B	Can the Bridge be configured to operate as a Backbone Edge Bridge with one or more Ports operating as a Customer Backbone Ports (CBPs)?	BEB: O.3	5.12	Yes [] No [] N/A []
BEB-I	Can the Bridge be configured to operate as a Backbone Edge Bridge with one or more Ports operating as a Provider Instance Port (PIP)?	BEB: O.3	5.12	Yes [] No [] N/A []
BEB-1	State which Ports support the following values for the Backbone Edge Bridge Port Type: — PIP — CNP — PNP — CBP — CEP — RCAP	BEB: M	5.11	PIP: _____ CNP: _____ PNP: _____ CBP: _____ CEP: _____ RCAP: _____
DDCFM	Is management of data-driven and data-dependent connectivity faults implemented?	O	Clause 19, Clause 29	Yes [] No []

A.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support
PBBTE	Can the Bridge be configured by an external agent to provide TESIIs?	O	8.4, 8.9, 25.10	Yes [] No []
EXAG	Is the active topology, learning and forwarding of the TESIIs under the control of an external agent?	PBBTE: O.1	8.4, 8.9	Yes [] No [] N/A []
ESPVID	Are the VIDs associated with ESPs, the ESP-VIDs, allocated to the TE-MSTID?	PBBTE: M	8.9	Yes [] N/A []
ESPFID	Is every ESP-VID allocated to a distinct FID?	PBBTE: M	25.10	Yes [] N/A []
BCBTE	Can the Bridge be configured to operate as a Backbone Core Bridge (BCB) that provides TESIIs?	PB AND PBBTE: O.5	5.10, 5.6.2	Yes [] N/A [] N/A []
BEBTE	Can the Bridge be configured to operate as a Backbone Edge Bridge that provides TESIIs?	BEB AND PBBTE: O.5	5.8.2, 5.12.1	Yes [] No [] N/A []
PS	Is protection switching supported?	BEBTE: M	5.8.2, 26.10.3	Yes [] N/A []
FQTSS	Does the implementation support Forwarding and Queuing Enhancements for time-sensitive streams?	O	5.4.1.5, Clause 34	Yes [] No []
CN	Is congestion notification implemented?	O	5.4.3, Clause 30, Clause 31, Clause 32, Clause 33	Yes [] No []
BRG1	Is this system a Bridge, and not an end station, for the purposes of congestion notification?	CN: O		Yes [] No [] N/A []
SRP	Does the implementation support the Stream Reservation Protocol?	O	Clause 35	Yes [] No []
MIRP	Is the Multiple I-SID Registration Protocol supported?	BEB-I OR BEB-B: O	5.7.1 item e), 5.8.1 item e)	Yes [] No [] N/A []
PFC	Is Priority-based Flow Control implemented?	O	5.11, Clause 36	Yes [] No []
ETS	Does the implementation support bandwidth management using ETS?	O	Clause 37	Yes [] No []
DCBX	Does the implementation support configuration management via DCBX?	O	Clause 38	Yes [] No []
IPS	Is Infrastructure Protection Switching supported?	PBBTE: O	5.6.2, 5.8.2, 26.11	Yes [] No [] N/A []
SPB	Is Shortest Path Bridging supported?	O.1	5.4.5, Clause 27	Yes [] No []
EVB-B	Does the implementation support the functionality of an EVB Bridge?	O	5.23	Yes [] No []
EVB-S	Does the implementation support the functionality of an EVB station?	O	5.24	Yes [] No []
ECMP	Is Equal Cost Multiple Paths supported?	SPBM:O	5.4.5.1, 44.1	Yes [] No [] N/A []

A.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support	
FF	Is flow filtering supported, including inserting, modifying, and removing F-TAGs from relayed frames?	ECMP:O	5.4.5.2, 44.2	Yes []	No []
FF-S	Does the S-VLAN component support F-TAG processing on each PNP?	PB AND FF:M	5.6.4, 44.2.2	Yes []	N/A []
FF-B	Does the B-component support F-TAG processing on each CBP and PNP?	BEB AND FF:M	5.8.4, 44.2.2	Yes []	N/A []
PCR	Is Path Control and Reservation supported?	O	Clause 45	Yes []	No []
SCHED	Does the implementation support scheduled traffic?	O	5.4.1, 5.13.1, 8.6.8, 8.6.9, 12.29, 17.7.22	Yes []	No []
PRE	Does the implementation support frame preemption?	O	5.4.1, 5.13.1, 6.7.2, 8.6.8, 12.30, 17.7.23	Yes []	No []
PSFP	Does the implementation support PSFP?	O	8.6.5.2.1, 8.6.6 items d) and e), 8.6.10, 12.31	Yes []	No []
ATS	Does the implementation support Asynchronous Traffic Shaping?	O	5.4.1.10, 5.13.1.3, 8.6.5.2.2, 8.6.6 items d) and e), 8.6.8, 8.6.8.5, 8.6.11, 12.31	Yes []	No []
CQF	Does the implementation support cyclic queuing and forwarding?	O	5.4.1.9, 5.13.1.2	Yes []	No []
YANG	Does the implementation support management operations using the YANG modules?	MGMT:O	8.12, 48	Yes [] N/A []	No []
SRRM	Does the implementation support Stream reservation remote management?	O	5.4.1.11, 12.32, A.48	Yes []	No []
CNC-S	Does the implementation support the functionality of a Centralized Network Configuration (CNC) station?	O	5.29, 46.2, A.17	Yes []	No []
VDPnNVE	Does the implementation support the VDP NVO3 functionality of an nNVE?	O	5.30.1	Yes []	No []
VDPTNVE	Does the implementation support the VDP NVO3 functionality of a tNVE?	O	5.30.2	Yes []	No []
CI	Does the implementation support Congestion Isolation?	O	5.4.7, Clause 49	Yes []	No []
AAB	Does the implementation support AAB functionality?	(BEB-I AND BEB-1):O	Clause 50, 5.12.2	Yes [] N/A []	No []
AAD	Does the implementation support AAD functionality?	(VLAN OR MBRIDGE):O (BEB OR TMPR):X	Clause 50, 5.9.2, 5.10.3, 5.14.2, 5.33	Yes [] N/A []	No []
CBSA	Does the implementation support the credit-based shaper algorithm?	O	8.6.8.2	Yes []	No []

A.6 Media access control methods

Item	Feature	Status	References	Support
	Which media access control methods are implemented in conformance with the relevant MAC Standards?		5.4, IEEE Std 802.1AC	
MAC-802.3	Ethernet, IEEE Std 802.3	O.2	IEEE Std 802.1AC	Yes [] No []
MAC-802.11-PORT	IEEE 802.11 LAN Portal, IEEE Std 802.11	O.2	G.4.1, IEEE Std 802.1AC	Yes [] No []
MAC-PMPN-N	PMPN multiple port, IEEE Std 802.1AC	O.2	G.4.1, IEEE Std 802.1AC	Yes [] No []
MAC-PMPN-1	PMPN single port, IEEE Std 802.1AC	O.2	G.4.1, IEEE Std 802.1AC	Yes [] No []
MAC-802.20-WB	IEEE 802.20™ Wideband Mode	O.2	IEEE Std 802.1AC	Yes [] No []
MAC-802.20-625	IEEE 802.20 625k-MC Mode	O.2	IEEE Std 802.1AC	Yes [] No []
MAC-1	Has a PICS been completed for each of the media access control methods implemented as required by the relevant MAC standards?	M		Yes []
MAC-2	Do all the media access control methods implemented support the MAC ISS as specified?	M	IEEE Std 802.1AC	Yes []
MAC-3	Are the adminPointToPointMAC and operPointToPointMAC parameters implemented on all Ports?	M	IEEE Std 802.1AC	Yes []
MAC-4	Does the implementation support the use of the adminEdgePort and operEdgePort parameters on any Ports?	O	13.27.1, 13.27.44	Yes [] No []
MAC-4a	State which Bridge Ports support the adminEdgePort and operEdgePort parameters.			Ports _____
MAC-5	Is the priority of received frames set to the Default Priority where specified for the MAC?	M	IEEE Std 802.1AC	Yes []
MAC-6	Can the Default Priority be set for each Port?	O	IEEE Std 802.1AC	Yes [] No []
MAC-7	Can the Default Priority be set to any of 0–7?	MAC-6:M	IEEE Std 802.1AC	Yes [] N/A []
MAC-12	Is the minimum tagged frame length that can be transmitted on IEEE 802.3 Ports less than 68 (but 64 or more) octets?	MAC-802.3:O	IEEE Std 802.1AC	Yes [] No [] N/A []

1

2

3

4

A.7 Relay and filtering of frames

Item	Feature	Status	References	Support
RLY-1	Are received frames with MAC method errors discarded?	M	6.3, 8.5	Yes []
RLY-2	Are user data frames the only type of frame relayed?	M	8.5	Yes []
RLY-3	Is the priority of each frame relayed regenerated as specified?	–TPMR: M TPMR: O	IEEE Std 802.1AC, 6.9.4, 8.5, 6.20	Yes [] No []
RLY-4	Are the default values of the Priority Regeneration Table as specified for each Port?	M	6.9.4, Table 6-4	Yes []
RLY-5	Can the Priority Regeneration Table be modified?	O	6.9.4, Table 6-4	Yes [] No []
RLY-6	Can the entries in the Priority Regeneration Table be set independently for each priority and Port and to any of the full range of values?	RLY-5: M	6.9.4, Table 6-4	Yes [] N/A []
RLY-7	Are frames transmitted by an LLC User attached at a Bridge Port also submitted for relay?	–TPMR: M TPMR: X	8.5	Yes [] No []
RLY-8	Are correctly received user data frames relayed subject to the conditions imposed by the Forwarding Process?	M	8.6, 8.6.2, 8.6.2, 8.6.3, 8.6.5, 8.6.4, 8.8, Table 8-10, Table 8-11, Table 8-12	Yes []
RLY-9	Is the order of relayed frames preserved as required by the forwarding process?	M	8.6.6	Yes []
RLY-10	Is a relayed frame submitted to a MAC Entity for transmission only once?	M	8.6.7	Yes []
RLY-11	Is a maximum Bridge transit delay enforced for relayed frames?	M	8.6.7	Yes []
RLY-12	Are queued frames discarded if a Port leaves the Forwarding State?	M	8.6.7	Yes []
RLY-13	Is the default algorithm for selecting frames for transmission supported?	M	8.6.8	Yes []
RLY-14	Is the access priority of each transmitted frame as specified for each media access method?	M	6.5.9	Yes []
RLY-15	Is the Frame Check Sequence (FCS) of frames relayed between Ports of the same MAC type preserved?	O	6.9	Yes [] No []
RLY-16	Is the undetected frame error rate greater than that achievable by preserving the FCS?	–RLY-15:X	6.5.7, 6.9	No [] N/A []
RLY-17	Are CFM frames discarded if they enter the queue subsequent to the Port leaving the Forwarding state?	CFM: X	8.6.7	No [] N/A []

A.7 Relay and filtering of frames *(continued)*

Item	Feature	Status	References	Support
RLY-18	Are CFM frames discarded when the Port leaves the Forwarding state?	CFM: O	8.6.7	Yes [] No []
RLY-19	Are received frames discarded (or not discarded) in accordance with the specification of a PIP, CBP, or Backbone Service Instance Multiplex Entity?	BEB: M	6.10.1, 6.11.1, 6.18.1	Yes [] N/A []
RLY-20	Is the forwarding process for flow filtering supported?	FF:M	5.4.5.2, 44.2.3	Yes [] N/A []
RLY-21	Does the implementation assign colors to frames before processing by Flow Meters according to 8.6.5.5?	PSFP:M ¬PSFP:O	8.6.5.5	Yes [] No []
RLY-22	Is the frame color assigned according to 8.6.5.5, items k) and l)?	RLY-21:M	8.6.5.5, items k) and l)	Yes [] N/A []
RLY-23	State the algorithm by which the color of frames prior to processing by Flow Meters according to 8.6.5.5 is assigned.	¬RLY-22:M	8.6.5.5	A1:Algorithm

A.8 Basic Filtering Services

Item	Feature	Status	References	Support
BFS-1	Are correctly received user data frames submitted to the Learning Process if the learning variable is set?	¬TPMR: M TPMR: X	8.6, 8.7, 5.15	Yes [] No []
BFS-2	Are correctly received frames of types other than user data frames submitted to the Learning Process?	¬TPMR: O TPMR: X	8.6, 8.7, 5.15	Yes [] No []
BFS-3	Does the Filtering Database (FDB) support creation and update of Dynamic Filtering Entries by the Learning Process?	¬TPMR: M TPMR: X	8.7, 8.8, 8.8.3, 5.15	Yes [] No []
BFS-4	Are Dynamic Filtering Entries created and updated if and only if the Port State permits?	M	8.7, 8.8.3	Yes []
BFS-5	Are Dynamic Filtering Entries created on receipt of frames with a group source address?	X	8.7, 8.8.3	No []
BFS-6	Can a Dynamic Filtering Entry be created that conflicts with an existing Static Filtering Entry?	X	8.7, 8.8, 8.8.2, 8.8.3	No []
BFS-7	Are existing Dynamic Filtering Entries removed to allow creation of a new entry if the FDB is full?	¬TPMR: O TPMR: X	8.7, 8.8.3, 5.15	Yes [] No []
BFS-8	Does the FDB contain Static Filtering Entries?	M	8.8.2	Yes []
BFS-9	Are Static Filtering Entries aged out?	X	8.8	No []

A.8 Basic Filtering Services

Item	Feature	Status	References	Support
BFS-10	Can Static Filtering Entries be created, modified, and deleted by management?	–(TPMR OR PBBTE): O PBBTE: M TPMR: X	8.8, 5.15	Yes [] No []
BFS-11	Can Static Filtering Entries be made for individual and group MAC addresses?	BFS-10:M	8.8.2	Yes [] N/A []
BFS-12	Can a Static Filtering Entry be made for the broadcast MAC address?	BFS-10:M	8.8.2	Yes [] N/A []
BFS-13	Can a Static Filtering Entry specify a forwarding Port Map?	BFS-10:M	8.8.2	Yes [] N/A []
BFS-14	Can a Static Filtering Entry specify a filtering Port Map?	BFS-10:M	8.8.2	Yes [] N/A []
BFS-15	Does the creation of a Static Filtering Entry remove any conflicting information in a Dynamic Filtering Entry for the same address?	M	8.8.2, 8.8.3	Yes []
BFS-16	Can a separate Static Filtering Entry with a Port Map be created for each inbound Port?	–TPMR: O TPMR: X	8.8.2, 5.15	Yes [] No []
BFS-17	Are Dynamic Filtering Entries aged out of the FDB if not updated?	M	8.8.3	Yes []
BFS-18	Can more than one Dynamic Filtering Entry be created for the same MAC address?	X	8.8.3	No []
BFS-19	Can the Bridge be configured to use the recommended default Ageing Time?	–TPMR: O TPMR: X	8.8.3, Table 8-9, 5.15	Yes [] No []
BFS-20	Can the Bridge be configured to use any value in the range specified for Ageing Time?	–TPMR: O TPMR: X	8.8.3, Table 8-9, 5.15	Yes [] No []
BFS-21	Is the FDB initialized with the entries contained in the Permanent Database?	M	8.8.11	Yes []
BFS-22	Are correctly received user data frames submitted to the Learning Process if they are associated with an ESP-VID?	PBBTE: X	8.4, 8.6.1	No [] N/A []
BFS-23	Are correctly received user data frames that are associated with an ESP-VID discarded if their destination MAC address is unknown?	PBBTE: M	8.8.1	Yes [] N/A []

A.9 Addressing

Item	Feature	Status	References	Support
ADDR-1	Does each Port have a separate MAC address?	M	8.13.2	Yes []
ADDR-2	Are frames addressed to a MAC address for a Port and received from or relayed to the attached LAN submitted to LLC Service User for the destination LLC Address?	M	8.5, 8.13.2	Yes []
ADDR-3	Are all BPDUs and MRP PDUs transmitted using the Bridge Spanning Tree Protocol LLC Address?	M	8.13.3, Table 8-15	Yes []
ADDR-4	Are PDUs addressed to the Bridge Spanning Tree Protocol Address with an unknown Protocol Identifier (PID) discarded on receipt?	M	14.5	Yes []
ADDR-5	Are all BPDUs generated by a Spanning Tree Protocol Entity associated with a C-VLAN component transmitted to the Bridge Group Address?	CB OR PEB: M	8.13.3, Table 8-1, Table 8-2	Yes [] N/A []
ADDR-6	Are all BPDUs generated by a Spanning Tree Protocol Entity associated with an S-VLAN component, I-component, or B-component transmitted to the Provider Bridge Group Address?	PB: M BEB:M	Table 8-1	Yes [] N/A []
ADDR-7	Are all MRPDUs transmitted to the group address assigned for the MRP Application?	M	8.13.3, Table 8-1	Yes []
ADDR-8	Is it possible to create entries in the Permanent Database or FDB for unsupported MRP application addresses or delete or modify entries for supported application addresses?	X	8.13.3	No []
ADDR-9	Is the source MAC address of BPDUs and MRPDUs for MRP Applications supported by the Bridge address of the transmitting Port?	M	8.13.3	Yes []
ADDR-10	Is the source MAC address of BPDUs generated by a Spanning Tree Protocol Entity associated with a C-VLAN component of a Provider Edge Bridge the address of the associated CEP?	PEB:M	12.13	Yes [] N/A []
ADDR-11	Is Bridge Management accessible through a Port using the MAC address of the Port?	MGT:O	8.13.7	Yes [] No [] N/A []
ADDR-12	Is an EUI-48 universally administered MAC Address assigned to each Bridge as its Bridge Address?	M	8.13.8	Yes []
ADDR-13	Is the Bridge Address the Address of a Port?	O	8.13.8	Yes [] No []
ADDR-14	Is the Bridge Address the Address of Port 1?	ADDR-13: O	8.13.8	Yes [] No [] N/A []
ADDR-15	Are frames addressed to any of the C-VLAN component Reserved Addresses relayed by a C-VLAN component?	CB or PB:X	8.13.4, Table 8-1	No [] N/A []

A.9 Addressing

Item	Feature	Status	References	Support
ADDR-16	Are frames addressed to any of the S-VLAN component Reserved Addresses relayed by an S-VLAN component, I-component, or B-component?	PB:X BEB:X	Table 8-2	No [] N/A []
ADDR-17	Is it possible to delete or modify entries in the Permanent Database and FDB for the Reserved Addresses?	X	8.13.4	No []
ADDR-18	Are PIPs capable of filtering frames received with a B-SA matching the PIP MAC address?	BEB-I: O	5.7.1	Yes [] No [] N/A []
ADDR-19	Is the address of an internal PIP on Backbone Edge Bridge (BEB) supporting a TESI configured to take the value of the MAC address of the connected CBP?	BEBTE:M	25.10	Yes [] N/A []

A.10 Rapid Spanning Tree Protocol (RSTP)

Item	Feature	Status	References	Support
	If item RSTP is not supported, mark “N/A” and continue at A.11			N/A []
RSTP-1	Does each Bridge have a unique identifier based on the Bridge Address, and a unique identifier for each Port?	RSTP:M	13.2	Yes []
RSTP-2	Can each Port be configured as an edge port by setting the adminEdgePort parameter?	RSTP:O	13.4	Yes [] No []
RSTP-3	Can each Port be configured to automatically determine if it is an edge port by setting the autoEdgePort parameter?	RSTP:O	13.4	Yes [] No []
RSTP-4	Are learned MAC addresses transferred from a retiring Root Port to a new Root Port?	RSTP:O	13.19	Yes [] No []
RSTP-5	Is the Spanning Tree Protocol Entity reinitialized if the Force Protocol Version parameter is modified?	RSTP:M	13.26	Yes []
RSTP-6	Are spanning tree priority vectors and Port Role assignments recomputed if the Bridge Identifier Priority, Port Identifier Priority, or Port Path Costs change?	RSTP:M	13.26	Yes []
RSTP-7	Is the txCount variable for a Port set to zero if the Port’s Transmit Hold Count is modified?	RSTP:M	13.26	Yes []
RSTP-8	Are the recommended default values of Migrate Time, Bridge Hello Time, Bridge Max Age, Bridge Forward Delay, and Transmit Hold Count used?	RSTP:O	13.25	Yes [] No []
RSTP-9	Can the Bridge Max Age, Bridge Forward Delay, and Transmit Hold Count parameters be set?	RSTP:O	13.25, 13.26	Yes [] No []
RSTP-10	Can Bridge Max Age, Bridge Forward Delay, Transmit Hold Count be set to any value in the permitted range?	RSTP-9: M	13.25, 13.30, Table 13-5	Yes [] N/A []
RSTP-11	Are the relationships between Bridge Hello Time, Bridge Max Age, and Bridge Forward Delay enforced?	RSTP-9: M	13.25	Yes [] N/A []
RSTP-12	Are the recommended values of Bridge Identifier Priority, Port Path Costs, and Port Identifier Priorities used?	RSTP:O	13.18	Yes [] No []
RSTP-13	Can the Bridge Identifier Priority, Port Path Costs, and Port Identifier Priorities be set?	RSTP:O	13.1, 13.4.1, 13.18, 13.25, 13.26, 13.27	Yes [] No []
RSTP-14	Can the Bridge Identifier Priority and Port Identifier Priorities be set to any of the values in the ranges specified?	RSTP-13: M	13.18	Yes [] N/A []
RSTP-15	Can the Port Path Cost for each Port be set to any of the values in the specified range?	RSTP-13: M	13.18	Yes [] N/A []
RSTP-16	Are Port Path Costs changed automatically by default if port speeds change?	RSTP:X	13.18	No []

A.10 Rapid Spanning Tree Protocol (RSTP)

Item	Feature	Status	References	Support
RSTP-17	Is one instance of the Port Role Selection state machine implemented for the Bridge; is one instance of each of the Port Timers, Port Receive, Port Protocol Migration, Bridge Detection, Port Transmit, Port Information, Port Role Transition, Port State Transition, and Topology Change state machines implemented per Port; and are the referenced definitions and declarations followed for all machines?	RSTP:M	13.5.2, 13.24, 13.36, 13.30, 13.31, 13.32, 13.33, 13.34, 13.35, 13.37, 13.38, 13.38.2	Yes [] N/A []
RSTP-18	Is it possible to set each Port Protocol Migration state machine's mcheck variable?	RSTP:O	13.27.38	Yes [] No []
RSTP-19	Is a single instance of each of the timer variables implemented per Port?	RSTP:M	13.30	Yes []
RSTP-20	Are the values for maximum RSTP processing delay and maximum BPDU transmission delay ever exceeded for any of the specified external events, actions, internal events, or transmissions?	RSTP:X	24.3, Table 24-1	No []
RSTP-21	Does each C-VLAN component of a Provider Edge Bridge operate an instance of Rapid Spanning Tree (RST) as modified for CEPs?	RSTP AND PEB:M	13.41	Yes [] N/A []
RSTP-22	Does each I-component of a BEB operate an instance of RST as modified for Virtual Instance Ports (VIPs)?	RSTP AND BEB:M	13.42	Yes [] N/A []
RSTP-23	Can each Port be configured to support detection of fragile Bridges by setting the autoIsolatePort parameter?	RSTP:O	13.23, 13.27.19	Yes [] No []
RSTP-24	Can a port be configured as an L2 Gateway Port by setting the isL2gp parameter?	RSTP:O	13.20, 13.27.26	Yes [] No []
RSTP-25	Can the pseudoRootId parameter be set for an L2 Gateway Port?	RSTP-24:M	13.20, 13.27.51	Yes [] N/A []
RSTP-26	Is one instance of the L2 Gateway Port state machine implemented for each L2GP port?	RSTP-24:M	13.40	Yes [] N/A []

A.11 BPDU encoding

Item	Feature	Status	References	Support
BPDU-1	Do all BPDUs contain an integral number of octets?	M	Clause 14	Yes []
BPDU-2	Are all the following BPDU parameter types encoded as specified? Protocol Identifiers Protocol Version Identifiers BPDU Types Flags Bridge Identifiers Root Path Cost Port Identifiers Timer Values	M	Clause 14 Clause 14 Clause 14 Clause 14 Clause 14 Clause 14 Clause 14 Clause 14	Yes []
BPDU-3	Do Configuration BPDUs have the format, parameters, and parameter values specified?	M	Clause 14	Yes []
BPDU-4	Do Topology Change Notification BPDUs have the format, parameters, and parameter values specified?	M	Clause 14	Yes []
BPDU-5	Do RST BPDUs have the format, parameters, and parameter values specified?	M	Clause 14	Yes []
BPDU-6	Are received BPDUs validated as and only as specified?	M	Clause 14	Yes []
BPDU-7	Does the implementation process BPDUs of prior and possible later protocol versions as specified?	M	Clause 14	Yes []
BPDU-8	Do MST BPDUs have the format, parameters, and parameter values specified?	MSTP:M	14.1	Yes [] N/A []

¹A.12 Implementation parameters

Item	Feature	Status	References	Support
IMP-1	State the Filtering Database Size.	M	8.8	____ entries
IMP-2	State the Permanent Database Size.	M	8.8	____ entries
IMP-3	State the maximum number of VIDs supported by the implementation.	M	5.4, 8.8	____ VIDs
IMP-4	State the range of VID values supported by the implementation.	M	5.4, 8.8	0 through ____
IMP-5	State the maximum number of FIDs that can be supported by the implementation.	M	8.8.8	____ FIDs
IMP-6	State the maximum number of VIDs that can be allocated to each FID.	M	8.8.8	____ VIDs
IMP-7	State the number of MSTIs supported.	MSTP:M	5.4.1, 8.8.8, 13.14	____ MSTIs N/A []

2

A.13 Performance

Item	Feature	Status	References	Support
PERF-1	Specify a Guaranteed Port Filtering Rate, and the associated measurement interval T_F , for each Bridge Port in the format specified below.	M	24.1	
PERF-2	Specify a Guaranteed Bridge Relaying Rate, and the associated measurement interval T_R , in the format specified below. Supplementary information shall clearly identify the Ports.	M	24.2	

Guaranteed Bridge Relaying Rate	TR
_____ frames per second	_____ second(s)

1

Port number(s) or other identification	Guaranteed port filtering rate (specify for all ports)	T_F (specify for all ports)
	_____ frames per second	_____ second(s)
	_____ frames per second	_____ second(s)
	_____ frames per second	_____ second(s)
	_____ frames per second	_____ second(s)
	_____ frames per second	_____ second(s)
	_____ frames per second	_____ second(s)
	_____ frames per second	_____ second(s)
	_____ frames per second	_____ second(s)

1 A.14 Bridge management

Item	Feature	Status	References	Support
	If item MGT is not supported, or if item TPMR is supported, mark N/A and continue at A.15.			N/A []
MGT-1	Discover Bridge	MGT:M	12.4.1.1	Yes []
MGT-2	Read Bridge	MGT:M	12.4.1.2	Yes []
MGT-3	Set Bridge Name	MGT:M	12.4.1.3	Yes []
MGT-4	Reset Bridge	MGT:M	12.4.1.4	Yes []
MGT-5	Read Port	MGT:M	12.4.2.1	Yes []
MGT-6	Set Port Name	MGT:M	12.4.2.1	Yes []
MGT-7	Read Forwarding Port Counters	MGT:M	12.6.1.1	Yes []
MGT-8	Read Port Default Priority	MGT:M	12.6.2.1	Yes [] N/A []
MGT-9	Set Port Default Priority	MGT AND MAC-6:M	12.6.2.2	Yes [] N/A []
MGT-10	Read Port Priority Regeneration Table	MGT AND RLY-5:M	12.6.2.3	Yes [] N/A []
MGT-11	Set Port Priority Regeneration Table	MGT AND RLY-5:M	12.6.2.4	Yes [] N/A []
MGT-12	Read Port Traffic Class Table	MGT AND TC:M	12.6.3.1	Yes [] N/A []
MGT-13	Set Port Traffic Class Table	MGT AND TC-3:M	12.6.3.2	Yes [] N/A []
MGT-14	Read Port Priority Code Point Selection	MGT and VLAN-29:M	12.6.2.5	Yes [] N/A []
MGT-15	Set Port Priority Code Point Selection	MGT AND VLAN-29:M	12.6.2.6	Yes [] N/A []
MGT-16	Read Priority Code Point Decoding Table	MGT AND VLAN-29:M	12.6.2.7	Yes [] N/A []
MGT-17	Set Priority Code Point Decoding Table	MGT AND VLAN-29:O	12.6.2.8	Yes [] No [] N/A []
MGT-18	Read Priority Code Point Encoding Table	MGT AND VLAN-29:M	12.6.2.9	Yes [] N/A []
MGT-19	Set Priority Code Point Encoding Table	MGT AND VLAN-29:O	12.6.2.10	Yes [] No [] N/A []
MGT-20	Read Use_DEI parameter	MGT:M	12.6.2.11	Yes [] N/A []
MGT-21	Set Use_DEI parameter	MGT:O	12.6.2.12	Yes [] No [] N/A []
MGT-22	Read Require Drop Encoding parameter	MGT AND VLAN-29:M	12.6.2.13	Yes [] N/A []
MGT-23	Set Require Drop Encoding parameter	MGT AND VLAN-29:M	12.6.2.14	Yes [] N/A []
MGT-24	Read Service Access Priority Selection	MGT AND VLAN-30:M	12.6.2.15	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-25	Set Service Access Priority Selection	MGT AND VLAN-30:M	12.6.2.16	Yes [] N/A []
MGT-26	Read Service Access Priority Table	MGT AND VLAN-30:M	12.6.2.17	Yes [] N/A []
MGT-27	Set Service Access Priority Table	MGT AND VLAN-30:O	12.6.2.18	Yes [] N/A []
MGT-28	Read Filtering Database	MGT:M	12.7.1.1	Yes []
MGT-29	Set Filtering Database Ageing Time	MGT:M	12.7.1.2	Yes []
MGT-30	Read Permanent Database	MGT:M	12.7.6.1	Yes []
MGT-31	Create Filtering Entry	MGT:M	12.7.7.1	Yes []
MGT-32	Delete Filtering Entry	MGT:M	12.7.7.2	Yes []
MGT-33	Read Filtering Entry	MGT:M	12.7.7.3	Yes []
MGT-34	Read Filtering Entry Range	MGT:M	12.7.7.4	Yes []
MGT-35	Read CIST Bridge Protocol Parameters	MGT:M	12.8.1.1	Yes []
MGT-36	Set CIST Bridge Protocol Parameters	MGT:M	12.8.1.3	Yes []
MGT-37	Read CIST Port Parameters	MGT:M	12.8.2.1	Yes []
MGT-38	Set CIST Port Parameters	MGT:M	12.8.2.3	Yes []
MGT-39	Force BPDU Migration Check	MGT:M	12.8.2.5	Yes []
MGT-40	Read MRP Timers	MGT AND MRP:M	12.9.1.1	Yes [] N/A []
MGT-41	Set MRP Timers	MGT AND MRP:M	12.9.1.2	Yes [] N/A []
MGT-42	Read MRP Protocol Controls	MGT AND MRP:M	12.9.2.1	Yes [] N/A []
MGT-43	Set MRP Protocol Controls	MGT AND MRP:M	12.9.2.2	Yes [] N/A []
MGT-44	Read MSTI Bridge Protocol Parameters	MGT AND MSTP:M	12.8.1.2	Yes [] N/A []
MGT-45	Set MSTI Bridge Protocol Parameters	MGT AND MSTP:M	12.8.1.4	Yes [] N/A []
MGT-46	Read MSTI Port Parameters	MGT AND MSTP:M	12.8.2.2	Yes [] N/A []
MGT-47	Set MSTI Port Parameters	MGT AND MSTP:M	12.8.2.4	Yes [] N/A []
MGT-48	Read Bridge VLAN Configuration	MGT:M	12.10.1.1	Yes [] N/A []
MGT-49	Configure PVID values	MGT:M	12.10.1.2	Yes [] N/A []
MGT-50	Configure Acceptable Frame Types parameter	MGT AND VLAN-2:M	12.10.1.3	Yes [] N/A []
MGT-51	Configure Enable Ingress Filtering parameters	MGT AND VLAN-9:M	12.10.1.4	Yes [] N/A []
MGT-52	Reset Bridge	MGT:M	12.10.1.5	Yes [] N/A []
MGT-53	Notify VLAN Registration Failure	MGT:M	12.10.1.6	Yes [] N/A []
MGT-54	Read VLAN Configuration	MGT:M	12.10.2.1	Yes [] N/A []
MGT-55	Create VLAN Configuration	MGT:M	12.10.2.2	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-56	Delete VLAN Configuration	MGT:M	12.10.2.3	Yes [] N/A []
	If item MSTP is not supported, mark N/A and continue at MGT-68			N/A []
MGT-57	Read MSTI List	MGT AND MSTP:M	12.12.1.1	Yes []
MGT-58	Create MSTI	MGT AND MSTP:M	12.12.1.2	Yes []
MGT-59	Delete MSTI	MGT AND MSTP:M	12.12.1.3	Yes []
MGT-60	Read FID to MSTI allocation	MGT AND MSTP:M	12.12.2.1	Yes []
MGT-61	Set FID to MSTI allocation	MGT AND MSTP:M	12.12.2.2	Yes []
MGT-62	Read MST Configuration Table Element	MGT AND MSTP:M	12.12.3.1	Yes []
MGT-63	Read VIDs assigned to MSTID	MGT AND MSTP:M	12.12.3.2	Yes []
MGT-64	Read MSTI Configuration Identifier	MGT AND MSTP:M	12.12.3.3	Yes []
MGT-65	Set MSTI Configuration Identifier	MGT AND MSTP:M	12.12.3.4	Yes []
MGT-66	Does the implementation support configuration of VID to FID allocations via management?	MGT:O	5.4.1, 8.8.8, 12.10.3	Yes [] No [] N/A []
MGT-67	Read VID to FID allocations	MGT-62:M	12.10.3.1	Yes [] N/A []
MGT-68	Read FID allocation for VID	MGT-62:M	12.10.3.2	Yes [] N/A []
MGT-69	Read VIDs allocated to FID	MGT-62:M	12.10.3.3	Yes [] N/A []
MGT-70	Set VID to FID allocation	MGT-62:M	12.10.3.4	Yes [] N/A []
MGT-71	Delete VID to FID allocation	MGT-62:M	12.10.3.5	Yes [] N/A []
MGT-72	Support Bridge management for the Bridge protocol entity in all supported spanning trees	MGT AND MSTP:M	5.4, 12.8	Yes [] N/A []
MGT-73	Support independent management of Bridge and port priority and path cost per spanning tree	MGT AND MSTP:M	5.4, 12.8.1	Yes [] N/A []
MGT-74	Support VLAN management per spanning tree	MGT AND MSTP:M	5.4, 12.10.1	Yes [] N/A []
MGT-75	Support MSTI configuration management	MGT AND MSTP:M	5.4, 12.12	Yes [] N/A []
MGT-76	Read Provider Bridge Port Type	MGT AND PB:M MGT AND BEB:M	12.13.1.1	Yes [] N/A []
MGT-77	Configure Provider Bridge Port Type	MGT AND PB:O MGT AND BEB:O	12.13.1.2	Yes [] No [] N/A []
MGT-78	Read VID Translation Table Entry	MGT AND VLAN-31:M	12.10.1.1	Yes [] No []
MGT-79	Configure VID Translation Table Entry	MGT AND VLAN-31:M	12.10.1.8	Yes [] No []
MGT-80	Read Egress VID Translation Table Entry	MGT AND VLAN-32:M	12.10.1.9	Yes [] No []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-81	Configure Egress VID Translation Table Entry	MGT AND VLAN-32:M	12.10.1.8.2	Yes [] No []
MGT-82	Read C-VID Registration Table Entry	MGT AND PEB:M	12.13.2.1	Yes [] N/A []
MGT-83	Configure C-VID Registration Table Entry	MGT AND PEB:M	12.13.2.2	Yes [] N/A []
MGT-84	Read Provider Edge Port Configuration	MGT AND PEB:M	12.13.2.3	Yes [] N/A []
MGT-85	Set Provider Edge Port Configuration	MGT AND PEB:M	12.13.2.4	Yes [] N/A []
MGT-86	Read Service Priority Regeneration Table	MGT AND PEB:M	12.13.2.5	Yes [] N/A []
MGT-87	Set Service Priority Regeneration Table	MGT AND PEB:O	12.13.2.6	Yes [] No [] N/A []
MGT-87a	Read Internal Interface Table Entry	MGT AND RCSI:M	12.13.3.1	Yes [] N/A []
MGT-87b	Configure Internal Interface Table Entry	MGT AND RCSI:M	12.13.3.2	Yes [] N/A []
MGT-88	Does the Bridge provide control of all of the required CFM managed objects?	BRG AND CFM: M	5.4.1.4 item h), 12.14	Yes [] N/A []
MGT-89	Reserved			
	Questions MGT-90 through MGT-115 refer to operations related to CFM managed objects.			
MGT-90	Read Maintenance Domain list	CFM: M	12.14.1.1	Yes []
MGT-91	Create Maintenance Domain managed object	CFM: M	12.14.1.2	Yes []
MGT-92	Delete Maintenance Domain managed object	CFM: M	12.14.1.3	Yes []
MGT-93	Read CFM Stack managed object	CFM: M	12.14.2.1	Yes []
MGT-94	Read Default MD Level managed object	CFM: M	12.14.3.1	Yes []
MGT-95	Write Default MD Level managed object	CFM: M	12.14.3.2	Yes []
MGT-96	Read Configuration Error List managed object	CFM: M	12.14.4.1	Yes []
MGT-97	Read Maintenance Domain managed object	CFM: M	12.14.5.1	Yes []
MGT-98	Write Maintenance Domain managed object	CFM: M	12.14.5.2	Yes []
MGT-99	Create Maintenance Association managed object	CFM: M	12.14.5.3	Yes []
MGT-100	Delete Maintenance Association managed object	CFM: M	12.14.5.4	Yes []
MGT-101	Read Maintenance Association managed object	CFM: M	12.14.6.1	Yes []
MGT-102	Write Maintenance Association managed object	CFM: M	12.14.6.2	Yes []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-103	Create Maintenance association Endpoint managed object	CFM: M	12.14.6.3	Yes []
MGT-104	Delete Maintenance association Endpoint managed object	CFM: M	12.14.6.4	Yes []
MGT-105	Read Maintenance association Endpoint managed object	CFM: M	12.14.7.1	Yes []
MGT-106	Total number of out-of-sequence CCMs received	CFM: O	12.14.7.1.3 item v), 20.16.12	Yes [] No []
MGT-107	Total number of LBRs received with data match errors	CFM: O	12.14.7.1.3 item aa)	Yes [] No []
MGT-108	Write Maintenance association Endpoint managed object	CFM: M	12.14.7.2	Yes []
MGT-109	Transmit Loopback Messages	CFM: M	12.14.7.3	Yes []
MGT-110	Transmit Linktrace Message	CFM: M	12.14.7.4	Yes []
MGT-111	Read Linktrace Reply	CFM: M	12.14.7.5	Yes []
MGT-112	Read MEP Database	CFM: M	12.14.7.6	Yes []
MGT-113	Read received Port Status TLV	CFM: O	12.14.7.6.3 item f), 20.19.3	Yes [] No []
MGT-114	Read received Interface Status TLV	CFM: O	12.14.7.6.3 item g), 20.19.4	Yes [] No []
MGT-115	Transmit MEP Fault Alarm	CFM: M	12.14.7.7	Yes []
MGT-116	List of active remote MEPs	CFM: O PBBTE: M	12.14.7.1.3 item ae)	No [] N/A [] Yes []
MGT-117	Indication on the capability to report the Traffic field bit	PS-5: M	12.14.7.1.3 item af)	Yes [] N/A []
MGT-118	Configuration to enable generation of a Fault Alarm based on a mismatch defect	PS-5: M	12.14.7.1.3 item ag)	Yes [] N/A []
MGT-119	Indication on the presence of a traffic field mismatch defect	PS-5: M	12.14.7.1.3 item ah), 20.25.2	Yes [] N/A []
MGT-120	Indication on the presence of a local mismatch defect	PS-5: M	12.14.7.1.3 item ai), 20.25.5	Yes [] N/A []
MGT-121	Indication on the presence of the mismatch defect since the MEP Mismatch Fault Notification Generator state machine was last in the MFNG_RESET state	PS-5: M	12.14.7.1.3 item aj)	Yes [] N/A []
MGT-122	Read the current state of the MEP Mismatch Fault Notification Generator state machine	PS-5: M	12.14.7.1.3 item ak), 20.40	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-123	Configuration of the Reverse VID on LBM/LTM transmit on MAs associated with point-to-multipoint TESIs	PBBTE: M	12.14.7.3.2 item f), 12.14.7.4.2 item e), 21.7.5.2	Yes [] N/A []
MGT-124	Read BEB configuration	MGT AND BEB: M	12.16.1.1	Yes [] N/A []
MGT-125	Set BEB configuration	MGT AND BEB: M	12.16.1.2	Yes [] N/A []
MGT-126	Create BEB component	MGT AND BEB: O	12.16.1.3	Yes [] No [] N/A []
MGT-127	Delete BEB component	MGT AND BEB: O	12.16.1.4	Yes [] No [] N/A []
MGT-128	Create BEB Bridge Port	MGT AND BEB: O	12.16.1.5	Yes [] No [] N/A []
MGT-129	Create BEB PIP	MGT AND BEB: O	12.16.1.6	Yes [] No [] N/A []
MGT-130	Delete BEB Bridge Port	MGT AND BEB: O	12.16.1.7	Yes [] No [] N/A []
MGT-131	Delete BEB PIP	MGT AND BEB: O	12.16.1.8	Yes [] No [] N/A []
MGT-132	Read BEB/PB/VLAN Bridge Port configuration	MGT AND BEB: M	12.16.2.1	Yes [] N/A []
MGT-133	Read VIP configuration	MGT AND BEB-I: M	12.16.3.1	Yes [] N/A []
MGT-134	Set VIP configuration	MGT AND BEB-I: M	12.16.3.2	Yes [] N/A []
MGT-135	Read the value of the enableConnectionIdentifier parameter	MGT AND BEB-I: O PBBTE: M	12.14.7.7.2 item f), 6.10	Yes [] N/A [] No []
MGT-136	Configuration of the enableConnectionIdentifier parameter	MGT AND BEB-I: O PBBTE: M	12.14.7.7.2 item d), 6.10	Yes [] N/A [] No []
MGT-137	Read PIP configuration	MGT AND BEB-I: M	12.16.4.1	Yes [] N/A []
MGT-138	Set PIP configuration	MGT AND BEB-I: O	12.16.4.2	Yes [] N/A [] No []
MGT-139	Read VIP to PIP mapping	MGT AND BEB-I: M	12.16.4.3	Yes [] N/A []
MGT-140	Set VIP to PIP mapping	MGT AND BEB-I: O	12.16.4.4	Yes [] N/A [] No []
MGT-141	Read PIP Priority Code Point Selection	MGT AND BEB-I: M	12.16.4.5	Yes [] N/A []
MGT-142	Set PIP Priority Code Point Selection	MGT AND BEB-I: M	12.16.4.6	Yes [] N/A []
MGT-143	Read PIP Priority Code Point Decoding Table	MGT AND BEB-I: M	12.16.4.7	Yes [] N/A []
MGT-144	Set PIP Priority Code Point Decoding Table	MGT AND BEB-I: O	12.16.4.8	Yes [] N/A [] No []
MGT-145	Read PIP Priority Code Point Encoding Table	MGT AND BEB-I: M	12.16.4.9	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-146	Set PIP Priority Code Point Encoding Table	MGT AND BEB-I: O	12.16.4.10	Yes [] N/A [] No []
MGT-147	Read PIP Use_DEI parameter	MGT AND BEB-I: M	12.16.4.11	Yes [] N/A []
MGT-148	Set PIP Use_DEI parameter	MGT AND BEB-I: O	12.16.4.12	YES [] N/A [] No []
MGT-149	Read PIP Require Drop Encoding parameter	MGT AND BEB-I: M	12.16.4.13	Yes [] N/A []
MGT-150	Set PIP Require Drop Encoding parameter	MGT AND BEB-I: O	12.16.4.14	Yes [] N/A [] No []
MGT-151	Read Backbone Service Instance table entry	MGT AND BEB-B: M	12.16.5.1	Yes [] N/A []
MGT-152	Set Backbone Service Instance table entry	MGT AND BEB-B: M	12.16.5.2	Yes [] N/A []
MGT-153	TESI assignment managed object	PBBTE: M	12.16.5.3	Yes [] N/A [] No []
MGT-154	Does the Bridge provide control of all the required DDCFM managed objects?	DDCFM:M	12.17	Yes [] N/A []
	Questions MGT-155 through MGT-174 refer to operations related to DDCFM managed objects.			
MGT-155	Create Reflection Responder managed object	DDCFM:M	12.17.2.1	Yes [] N/A []
MGT-156	Read Reflection Responder managed object	DDCFM:M	12.17.2.3	Yes [] N/A []
MGT-157	Write Reflection Responder managed object's attributes	DDCFM:M	12.17.2.2	Yes [] N/A []
MGT-158	Delete Reflection Responder managed object	DDCFM:M	12.17.2.4	Yes [] N/A []
MGT-159	Activate a Reflection Responder managed object	DDCFM:M	12.17.2.5	Yes [] N/A []
MGT-160	Deactivate a Reflection Responder managed object	DDCFM:M	12.17.2.6	Yes [] N/A []
MGT-161	Create RFM Receiver managed object	DDCFM:M	12.17.3.1	Yes [] N/A []
MGT-162	Delete RFM Receiver managed object	DDCFM:M	12.17.3.2	Yes [] N/A []
MGT-163	Create Decapsulator Responder managed object	DDCFM:M	12.17.4.1	Yes [] N/A []
MGT-164	Read Decapsulator Responder managed object	DDCFM:M	12.17.4.2	Yes [] N/A []
MGT-165	Write Decapsulator Responder managed object's attributes	DDCFM:M	12.17.4.3	Yes [] N/A []
MGT-166	Delete Decapsulator Responder managed object	DDCFM:M	12.17.4.4	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-167	Activate a Decapsulator Responder managed object	DDCFM:M	12.17.4.5	Yes [] N/A []
MGT-168	Deactivate a Decapsulator Responder managed object	DDCFM:M	12.17.4.6	Yes [] N/A []
MGT-169	Create SFM Originator managed object	DDCFM:M	12.17.5.1	Yes [] N/A []
MGT-170	Read SFM Originator managed object	DDCFM:M	12.17.5.2	Yes [] N/A []
MGT-171	Write SFM Originator managed object's attributes	DDCFM:M	12.17.5.4	Yes [] N/A []
MGT-172	Delete SFM Originator managed object	DDCFM:M	12.17.5.3	Yes [] N/A []
MGT-173	Activate SFM Originator managed object	DDCFM:M	12.17.5.5	Yes [] N/A []
MGT-174	Deactivate SFM Originator managed object	DDCFM:M	12.17.5.6	Yes [] N/A []
MGT-175	Read TE protection group list	PBBTE: M	12.18.1.1	Yes [] N/A []
MGT-176	Create TE protection group managed object	PBBTE: M	12.18.1.2	Yes [] N/A []
MGT-177	Delete TE protection group managed object	PBBTE: M	12.18.1.3	Yes [] N/A []
MGT-178	Read TE protection group managed object	PBBTE: M	12.18.2.1	Yes [] N/A []
MGT-179	Write TE protection group managed object	PBBTE: M	12.18.2.2	Yes [] N/A []
MGT-180	TE protection group administrative commands managed object	PBBTE: M	12.18.2.3	Yes [] N/A []
MGT-181	Does the implementation provide control of the TPMR management entity?	MGT AND TPMR: M	12.19.1	Yes [] N/A []
MGT-182	Does the implementation provide control of the individual MAC and PHY entities associated with each TPMR Port?	MGT AND TPMR: M		Yes [] N/A []
MGT-183	Does the implementation provide control of the Forwarding Process of the MAC Relay Entity?	MGT AND TPMR: M	8.6, 12.19.3	Yes [] N/A []
MGT-184	Does the implementation provide control of the MSPE?	MGT AND TPMR: M	23.3, 12.19.4	Yes [] N/A []
MGT-185	Read TPMR	TPMR:M	12.19.1.1.1	Yes [] N/A []
MGT-186	Set TPMR Name	TPMR:M	12.19.1.1.2	Yes [] N/A []
MGT-187	Read Port	TPMR:M	12.19.1.2.1	Yes [] N/A []
MGT-188	Set Port Name	TPMR:M	12.19.1.2.2	Yes [] N/A []
MGT-189	Read Forwarding Port Counters	TPMR:M	12.19.3.1.1	Yes [] N/A []
MGT-190	Read Port Default Priority	TPMR AND TPMR-12:M	12.6.2.1	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-191	Set Port Default Priority	TPMR AND TPMR-12:M	12.6.2.2	Yes [] N/A []
MGT-192	Read Port Priority Regeneration Table	TPMR AND TPMR-12:M	12.6.2.3	Yes [] N/A []
MGT-193	Set Port Priority Regeneration Table	TPMR AND TPMR-12:M	12.6.2.4	Yes [] N/A []
MGT-194	Read Port Traffic Class Table	TPMR AND TPMR-10:M	12.6.3.1	Yes [] N/A []
MGT-195	Set Port Traffic Class Table	TPMR AND TPMR-10:M	12.6.3.2	Yes [] N/A []
MGT-196	Read Port Priority Code Point Selection	TPMR AND TPMR-12:M	12.6.2.5	Yes [] N/A []
MGT-197	Set Port Priority Code Point Selection	TPMR AND TPMR-12:M	12.6.2.6	Yes [] N/A []
MGT-198	Read Priority Code Point Decoding Table	TPMR AND TPMR-12:O	12.6.2.7	Yes [] No [] N/A []
MGT-199	Read Use_DEI parameter	TPMR AND TPMR-12:M	12.6.2.11	Yes [] N/A []
MGT-200	Set Use_DEI parameter	TPMR AND TPMR-12:O	12.6.2.12	Yes [] No [] N/A []
MGT-201	Does the implementation support the management entities defined in 12.20?	FQTSS: O	5.4.1.5 item e), 12.20, 17.7.12, Clause 34	Yes [] N/A []
MGT-202	Does the Bridge support all of the objects in the CN component managed object?	BRG1 AND CN: M	5.4.3 item c), 12.21.1	Yes [] N/A []
MGT-203	Does the Bridge support all of the objects in the CN component priority managed object?	BRG1 AND CN: M	5.4.3 item c), 12.21.2	Yes [] N/A []
MGT-204	Does the Bridge support all of the objects in the CN Port priority managed object?	BRG1 AND CN: M	5.4.3 item c), 12.21.3	Yes [] N/A []
MGT-205	Does the Bridge support all of the objects in the Congestion Point managed object?	BRG1 AND CN: M	5.4.3 item c), 12.21.4	Yes [] N/A []
MGT-206	Does the implementation support the management entities defined in 12.22?	SRP: O	12.22	Yes [] No [] N/A []
MGT-207	Does the BEB support all of the MIRP managed objects?	MIRP AND BEB-I: M	12.16.1	Yes [] N/A []
MGT-208	PFC entities	PFC: O	12.22.6	Yes [] No [] N/A []
MGT-209	ETS Control Entities	ETS:M	12.24	Yes [] N/A []
MGT-210	PFC Control Entities	PFC:M	12.22.6	Yes [] N/A []
MGT-211	Read IPG list	IPS: M	12.24.1.1	Yes [] N/A []
MGT-212	Create IPG managed objects	IPS: M	12.24.1.2	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-213	Delete IPG managed objects	IPS: M	12.24.1.3	Yes [] N/A []
MGT-214	Read IPG managed objects	IPS: M	12.24.2.1	Yes [] N/A []
MGT-215	Write IPG managed objects	IPS: M	12.24.2.2	Yes [] N/A []
MGT-216	Apply administrative command to IPG managed objects	IPS: M	12.24.2.3	Yes [] N/A []
MGT-217	Does the implementation support the SPB managed objects?	MGT AND SPB:M	5.4.5, 12.25	Yes [] N/A []
MGT-218	Discard TTL expired counter	MGT AND FF:M	12.6.1.1.3	Yes [] N/A []
MGT-219	Flow filtering control table	MGT AND FF:M	12.16.5.4, 12.16.5.5	Yes [] N/A []
MGT-220	Does the implementation support the managed objects required for PCR? If item PCR is not supported in A.5, mark N/A and continue at MGT-248	MGT AND PCR:M	5.5, 12.26	Yes [] N/A []
MGT-221	Create SPB System managed object	PCR:M	12.25.1.1	Yes [] N/A []
MGT-222	Write SPB System managed object	PCR:M	12.25.1.2	Yes [] N/A []
MGT-223	Read SPB System managed object	PCR:M	12.25.1.3	Yes [] N/A []
MGT-224	Delete SPB System managed object	PCR:M	12.25.1.4	Yes [] N/A []
MGT-225	Create SPB MTID Static managed object	PCR:M	12.25.2.1	Yes [] N/A []
MGT-226	Write SPB MTID Static managed object	PCR:M	12.25.2.2	Yes [] N/A []
MGT-227	Read SPB MTID Static managed object	PCR:M	12.25.2.3	Yes [] N/A []
MGT-228	Delete SPB MTID Static managed object	PCR:M	12.25.2.4	Yes [] N/A []
MGT-229	Read SPB Topology Instance Dynamic managed object	PCR:M	12.25.3.1	Yes [] N/A []
MGT-230	Create SPB ECT Static Entry managed object	PCR:M	12.25.4.1	Yes [] N/A []
MGT-231	Read SPB ECT Static Entry managed object	PCR:M	12.25.4.2	Yes [] N/A []
MGT-232	Delete SPB ECT Static Entry managed object	PCR:M	12.25.4.3	Yes [] N/A []
MGT-233	Read SPB ECT Dynamic Entry managed object	PCR:M	12.25.5.1	Yes [] N/A []
MGT-234	Write SPB Adjacency Static Entry managed object	PCR:M	12.25.6.1	Yes [] N/A []
MGT-235	Read SPB Adjacency Static Entry managed object	PCR:M	12.25.6.2	Yes [] N/A []
MGT-236	Read SPB Adjacency Dynamic Entry managed object	PCR:M	12.25.7.1	Yes [] N/A []

A.14 Bridge management *(continued)*

Item	Feature	Status	References	Support
MGT-237	Read SPBM BSI Static Entry managed object	BEB AND PCR:M	12.25.8.2	Yes [] N/A []
MGT-238	Read SPBM Topology Service Table managed object	BEB AND PCR:M	12.25.12.1	Yes [] N/A []
MGT-239	Read SPB Topology Node Table managed object	PCR:M	12.25.9.1	Yes [] N/A []
MGT-240	Read SPB Topology ECT Table managed object	PCR:M	12.25.10.1	Yes [] N/A []
MGT-241	Read SPB Topology Edge Table managed object	PCR:M	12.25.11.1	Yes [] N/A []
MGT-242	Read SPBM Topology Service Table managed object	BEB AND PCR:M	12.25.12.1	Yes [] N/A []
MGT-243	Read SPBV Topology Service Table managed object	PCR:M	12.25.13.1	Yes [] N/A []
MGT-244	Create a PCR ECT Static Entry managed object	PCR:O	12.28.1.1	Yes [] No [] N/A []
MGT-245	Read a PCR ECT Static Entry managed object	PCR:O	12.28.1.2	Yes [] No [] N/A []
MGT-246	Delete a PCR ECT Static Entry managed object	PCR:O	12.28.1.4	Yes [] No [] N/A []
MGT-247	Read PCR Topology ECT Table managed object	PCR:O	12.28.2.1	Yes [] No [] N/A []
MGT-248	Does the implementation support the management entities defined in 12.29?	SCHED OR CQF: M	5.4.1 item ad), 5.4.1.9 item c), 5.13.1.2 item c), 12.29	Yes [] N/A []
MGT-249	Does the implementation support the management entities defined in 12.30?	PRE: M	5.4.1 item ae), 12.30	Yes [] N/A []
MGT-250	Does the implementation support the management entities defined in 12.31 for PSFP?	PSFP OR CQF: M	5.4.1.9 item e), 5.13.1.2 item e), 8.6.5.2.1, 8.6.10, 12.31	Yes [] N/A []
MGT-251	Does the implementation support the management entities defined in 12.31 for ATS?	ATS:M	5.4.1.10 item e), 5.13.1.2 item e), 8.6.5.2.1, 8.6.10, 12.31	Yes [] N/A []
MGT-252	Does the implementation support the management entities defined for congestion isolation in 12.33?	CI: M	5.4.7 item e), 12.33, 49.4	Yes [] N/A []
MGT-253	Does the implementation support the PBBN Auto Attach management entities defined in 12.34?	(AAB OR AAD):M	12.34	Yes [] N/A []

A.15 Remote management

Item	Feature	Status	References	Support
	If item RMGT is not supported, mark N/A and continue at A.16.			N/A []
RMGT-1	What Management Protocol standard(s) or specification(s) are supported?	RMGT:M	5.4.1	
RMGT-2	What standard(s) or specifications for Managed Objects and Encodings are supported?	RMGT:M	5.4.1	

1

A.16 Expedited traffic classes

Item	Feature	Status	References	Support
TC-1	Does the implementation provide more than one transmission queue for (a) Bridge Port(s)?	TC: M	8.6.6	Yes [] N/A []
TC-2	Is the recommended mapping of priority to traffic classes supported for each Port?	TC: O	8.6.6	Yes [] No [] N/A []
TC-3	Can the traffic class tables be managed?	TC: O	8.6.6, Table 8-5	Yes [] No [] N/A []
TC-4	Is the default algorithm for selecting frames for transmission supported?	M	8.6.8	Yes []
TC-5	Are additional algorithms for selecting frames for transmission supported?	O	8.6.8	Yes [] No []

2

A.17 Extended Filtering Services

Item	Feature	Status	References	Support
EFS-1	Can MAC Address Registration Entries be created, updated, and removed from the FDB by MMRP?	EFS:M	8.8, 8.8.4, Clause 10	Yes [] N/A []
EFS-2	Can a Static Filtering Entry be created with an address specification that represents a group address, or All Group Addresses, or All Unregistered Group Addresses, and with a control element for each Port that specifies unconditional forwarding, or unconditional filtering, or the use of dynamic or default group filtering information?	EFS:M	8.8.1	Yes [] N/A []
EFS-3	Can a Static Filtering Entry be created with an address specification that represents an individual address and with a control element for each Port that specifies unconditional forwarding, or unconditional filtering?	M	8.8.1	Yes []
EFS-4	Can a Static Filtering Entry be created with an address specification that represents an individual address and with a control element for each Port that specifies unconditional forwarding, or unconditional filtering, or the use of dynamic filtering information?	EFS:O	8.8.1	Yes [] No [] N/A []

3

A.18 Multiple Spanning Tree Protocol (MSTP)

Item	Feature	Status	References	Support
	If item MSTP is not supported, mark N/A and continue at the start of A.19.			N/A []
MSTP-1	Support the CIST plus a stated maximum number of MSTIs, where that number is at least 2 and at most 64	MSTP:M	5.4.1, 8.8.8, 13.14	Yes []
MSTP-2	Support at least as many FIDs as MSTIs	MSTP:M	5.4, 5.4.1, 8.8.8	Yes []
MSTP-3	Associate each FID to a spanning tree	MSTP:M	5.4, 8.9.3	Yes []
MSTP-4	Transmit and receive MST Configuration Identifier (MCID) information	MSTP:M	5.4, 8.9.2	Yes []
MSTP-5	Support a set of port state information per spanning tree per port	MSTP:M	5.4, 8.4, 13.38	Yes []
MSTP-6	Support an instance of spanning tree protocol per spanning tree per port	MSTP:M	5.4, 8.10, Clause 13	Yes []
MSTP-7	Use the Bridge Group Address as specified	MSTP:M	5.4, 8.13.2	Yes []
MSTP-8	Support default Bridge Forward Delay and Bridge Priority parameter values as specified	MSTP:M	5.4, 13.26	Yes []
MSTP-9	Provision of identifiers for Bridge and Ports	MSTP:M	13.2	Yes []
MSTP-10	Not exceed the values in 24.3 for max BPDU transmission delay	MSTP:M	24.3, Table 24-1	Yes []
MSTP-11	Inclusion of active Ports in computation of the active topology for a given spanning tree	MSTP:M	13.9	Yes []
MSTP-12	Processing of BPDUs received on Ports included in the computation of the active topology for a given spanning tree	MSTP:M	13.9	Yes []
MSTP-13	Discarding received frames in the Discarding state for a given spanning tree	MSTP:M	13.9	Yes []
MSTP-14	Incorporating station location information to the FDB in the Learning and Forwarding states for a given spanning tree	MSTP:M	13.9	Yes []
MSTP-15	Not incorporating station location information to the FDB in the Discarding state for a given spanning tree	MSTP:M	13.9	Yes []
MSTP-16	Transfer learned MAC addresses from a retiring Root Port to a new Root Port for a given spanning tree	MSTP:O	13.16	Yes [] No []
MSTP-17	Instances of state machines per Bridge, per Port, and per spanning tree instance, as specified	MSTP:M	13.24	Yes []
MSTP-18	Port Timers state machine support	MSTP:M	13.25, 13.30	Yes []
MSTP-19	Port Receive state machine support	MSTP:M	13.25, 13.31	Yes []
MSTP-20	Port Protocol Migration state machine support	MSTP:M	13.25, 13.32	Yes []
MSTP-21	Port Transmit state machine support	MSTP:M	13.25, 13.34	Yes []

A.18 Multiple Spanning Tree Protocol (MSTP) *(continued)*

Item	Feature	Status	References	Support
MSTP-22	Port Information state machine support	MSTP:M	13.25, 13.35	Yes []
MSTP-23	Port Role Selection state machine support	MSTP:M	13.25, 13.36	Yes []
MSTP-24	Port Role Transitions state machine support	MSTP:M	13.25, 13.37	Yes []
MSTP-25	Port State Transition state machine support	MSTP:M	13.25, 13.38	Yes []
MSTP-26	Topology Change state machine support	MSTP:M	13.25, 13.39	Yes []
MSTP-27	Are the values of Migrate Time, Bridge Hello Time, Bridge Max Age, Bridge Forward Delay, Transmit Hold Count, and Max Hops within the permitted range?	MSTP:M	13.25, Table 13-5	Yes []
MSTP-28	Enforcement of parameter relationships	MSTP:M	13.25, Table 13-5	Yes []
MSTP-29	Range and granularity of priority values	MSTP:M	13.18	Yes []
MSTP-30	Range and granularity of path cost values	MSTP:M	13.18	Yes []
MSTP-31	Are all PNPs capable of supporting the L2GP spanning tree protocol?	PB: O	13.40	Yes [] N/A [] No []
MSTP-32	Are PIPs capable of supporting the encapsulation/decapsulation of BPDUs?	BEB-I: O	5.7.1	Yes [] N/A [] No []
MSTP-33	Can each Port be configured to support detection of fragile Bridges by setting the autoIsolatePort parameter?	MSTP:O	13.23, 13.27.19	Yes [] No []
MSTP-34	Can a port be configured as an L2 Gateway Port by setting the isL2gp parameter?	MSTP:O	13.20, 13.27.26	Yes [] No []
MSTP-35	Can the pseudoRootId parameter be set for an L2 Gateway Port for the CIST and for each MSTI?	MSTP-34:M	13.20, 13.27.51	Yes [] N/A []
MSTP-36	Is one instance of the L2 Gateway Port state machine implemented for each L2GP port?	MSTP-34:M	13.40	Yes [] N/A []

A.19 VLAN support

Item	Feature	Status	References	Support
VLAN-1	Does the implementation support, on each Port, one or more of the permissible combinations of values for the Acceptable Frame Types parameter?	M	5.4, 6.9	Yes []
VLAN-2	Does the implementation support, on each Provider Bridge Port, the <i>Admit All Frames</i> value for the Acceptable Frame Types Parameter?	PB:M	5.6, 6.9	Yes [] N/A []
VLAN-3	State which Ports support the following values for the Acceptable Frame Types parameter: — <i>Admit Only VLAN-tagged frames</i> — <i>Admit Only Untagged and Priority Tagged frames</i> — <i>Admit All frames</i>	M	5.4, 6.9	Ports: _____ Ports: _____ Ports: _____
VLAN-4	On Ports that support both values for the Acceptable Frame Types parameter, is the parameter configurable via management?	M	5.4, 6.9, 12.10	Yes [] N/A []
VLAN-5	Does the implementation support the ability for the FDB to contain static and dynamic configuration information for at least one VID, by means of Static and Dynamic VLAN Registration Entries?	M	5.4, 8.8	Yes []
VLAN-6	Does the implementation support at least one FID?	M	5.4, 8.8.3, 8.8.8, 8.8.9	Yes []
VLAN-7	Can the implementation allocate at least one VID to each FID supported?	M	5.4, 8.8.3, 8.8.8, 8.8.9	Yes []
VLAN-8	Does the implementation take account of the allocation of VID's to FID's when making forwarding decisions relative to group MAC addresses?	O	8.8.9	Yes [] No []
VLAN-9	On Ports that support untagged and priority-tagged frames, does the implementation support:		5.4, 6.9, 8.8.2, 12.10	
VLAN-9.1	— A PVID value?	M		Yes [] N/A []
VLAN-9.2	— The ability to configure one VID whose untagged set includes that Port?	M		Yes [] N/A []
VLAN-9.3	— Configuration of the PVID value via management operations?	M		Yes [] N/A []
VLAN-9.4	— Configuration of Static Filtering Entries via management operations?	M		Yes [] N/A []
VLAN-9.5	— The ability to configure more than one VID whose untagged set includes that Port?	O		Yes [] No [] N/A []
VLAN-10	Does the implementation support the ability to enable and disable Ingress Filtering?	CB:O PB:M BEB:M	5.4.1, 5.6, 8.6.3	Yes [] No [] N/A []
VLAN-11	Can the PVID or the VID in any member of the VID Set for any Port be assigned the value of the null VID?	X	6.9, Table 9-2	No []
VLAN-12	Are frames discarded (or not discarded) in accordance with the settings of the Acceptable Frame Types parameters?	M	6.9	Yes []
VLAN-13	Are all frames received classified as belonging to exactly one VID, as defined in the ingress rules?	M	6.9	Yes []
VLAN-14	Is Ingress Filtering performed in accordance with the value of the Enable Ingress Filtering parameter?	M	8.6.2	Yes []

A.19 VLAN support (continued)

Item	Feature	Status	References	Support
VLAN-15	Are all frames that are not discarded as a result of the application of the ingress rules submitted to the Forwarding Process and to the Learning Process?	M	8.6.2	Yes []
VLAN-16	Does the implementation support Port-and-Protocol-based classification of frames on any or all Ports?	O	6.12	Yes [] No []
VLAN-16.1	State which Ports support Port-and-Protocol- based classification rules.	VLAN-16:M	6.12	Ports: _____
VLAN-16.2	For each Port that supports Port-and-Protocol-based classification rules, is a VID Set supported?	VLAN-16:M	6.12	Port: _____ Yes [] N/A []
VLAN-16.3	For each Port that supports Port-and-Protocol-based classification rules, state how many entries are supported in the VID Set.	VLAN-16:M	6.12	Port: _____ Entries _____
VLAN-16.4	For each Port that supports Port-and-Protocol-based classification rules, is the VID Set configurable via management?	VLAN-16:M	12.10.1.2	Port: _____ Yes [] N/A []
VLAN-17	Does the implementation support a Protocol Group Database?	VLAN-16:M	6.12.3	Yes [] N/A []
VLAN-17.1	State how many entries are supported in the Protocol Group Database.	VLAN-17:M	6.12.3	Entries _____
VLAN-17.2	Is the Protocol Group Database configurable via management?	VLAN-17:O	12.10.2.1	Yes [] No [] N/A []
VLAN-17.3	Does the Protocol Group Database support entries of format Ethernet?	VLAN-17:O	6.12.3	Yes [] No [] N/A []
VLAN-17.4	Does the Protocol Group Database support entries of format RFC_1042?	VLAN-17:O	6.12.3	Yes [] No [] N/A []
VLAN-17.5	Does the Protocol Group Database support entries of format SNAP_8021H?	VLAN-17:O	6.12.3	Yes [] No [] N/A []
VLAN-17.6	Does the Protocol Group Database support entries of format SNAP_Other?	VLAN-17:O	6.12.3	Yes [] No [] N/A []
VLAN-17.7	Does the Protocol Group Database support entries of format LLC_Other?	VLAN-17:O	6.12.3	Yes [] No [] N/A []
VLAN-17.8	Does the Protocol Group Database support entries of at least one of the following formats: Ethernet, RFC_1042, SNAP_8021H, SNAP_Other, LLC_Other?	VLAN-17: M	6.12.3	Yes [] N/A []
VLAN-18	Are frames discarded if the transmission Port is not present in the member set for the frame's VID?	M	6.9.2, 8.8.10	Yes []
VLAN-19	Are frames transmitted as VLAN-tagged frames or as untagged frames in accordance with the value of the untagged set for the frame's VID?	M	8.8.2	Yes []
VLAN-20	Does the implementation support Static VLAN Registration Entries as defined in 8.8.2?	M	8.8.2	Yes []
VLAN-21	Does the implementation support the creation of a separate Static VLAN Registration Entry with a distinct Port Map for each VID from which frames are received by the Forwarding Process?	O	8.8.2	Yes [] No []

A.19 VLAN support (continued)

Item	Feature	Status	References	Support
VLAN-22	Does the implementation support Dynamic VLAN Registration Entries as defined in 8.8.5?	M	8.8.5	Yes []
VLAN-23	Does the implementation support the creation of a separate Dynamic VLAN Registration Entry with a distinct Port Map for each VID from which frames are received by the Forwarding Process?	O	8.8.5	Yes [] No []
VLAN-24	Does the implementation allocate VIDs to FIDs in accordance with the specification in 8.8.8?	M	8.8.8	Yes []
VLAN-25	Does the implementation correctly detect Learning Constraint violations?	M	8.8.8	Yes []
VLAN-26	Is determination of the member set and the untagged set for a given VID achieved as defined in 8.8.10?	M	8.8.10	Yes []
VLAN-27	Do VLAN-tagged frames transmitted by the Bridge conform to the format defined in Clause 9 for the MAC type on which they are transmitted?	M	Clause 9	Yes []
VLAN-28	Are all BPDUs transmitted untagged?	M	8.13.9	Yes []
VLAN-29	Is encoding of the drop_eligible parameter in the PCP field of the VLAN tag supported?	CB:O PB:M BEB:M	5.5.1, 5.6, 6.9.3, 9.6	Yes [] No [] N/A []
VLAN-30	Is Service Access Priority Selection supported?	CB:O PB:X BEB:X	5.5.1, 5.6, 6.13	Yes [] No [] N/A []
VLAN-31	Is the VID translation table supported?	–SPBV:O SPBV:M	5.4.1, 5.4.5, 6.9	Yes [] No []
VLAN-32	Is the Egress VID translation table supported?	–VLAN-31:X –SPBV AND VLAN-31:O SPBV:M	5.4.1, 5.4.5, 6.9	Yes [] No [] N/A []
VLAN-33	Is the VIP-ISID parameter supported?	BEB-I: M	6.10	Yes [] N/A []
VLAN-34	Is the default Backbone Destination supported?	BEB-I: M BEBTE: M	6.10	Yes [] N/A []
VLAN-35	Is the adminPointToPointMAC supported?	BEB-I: O	6.10	Yes [] No [] N/A []
VLAN-36	Is the Backbone-SID field in the Backbone Service Instance table supported?	BEB-B: M	6.11	Yes [] N/A []
VLAN-37	Is the B-VID field in the Backbone Service Instance table supported?	BEB-B: O BEBTE: M	6.11	Yes [] No [] N/A []
VLAN-38	Is the Local-SID field in the Backbone Service Instance table supported?	BEB-B: O	6.11	Yes [] No [] N/A []
VLAN-39	Is the Default Backbone Destination field in the Backbone Service Instance table supported?	BEB-B: O BEBTE: M	6.11	Yes [] No [] N/A []
VLAN-40	Is many-to-one S-VID to I-SID mapping supported?	BEB-I: O	5.7.1	Yes [] No [] N/A []

A.20 Multiple MAC Registration Protocol (MMRP)

Item	Feature	Status	References	Support
	If MMRP is not supported, mark N/A and continue at A.21			N/A []
MMRP1	Does the implementation support the exchange of Multiple MAC Registration Protocol Data Units (MMRPDUs), using the generic MRPDU format defined in 10.8 to exchange MMRP-specific information, as defined in 10.12?	M	5.4.1.3, 10.8, 10.12	Yes []
MMRP2	Is the MMRP Application supported as defined in 10.12?	M	5.4.1.3, 10.12	Yes []
MMRP3	Does the implementation propagate registration information in accordance with the operation of MAP for the Base Spanning Tree Context, as specified in 10.3.1?	M	5.4.1.3, 10.3.1	Yes []
MMRP4	Does the implementation forward, filter, or discard MAC frames carrying any MRP Application address as the destination MAC address in accordance with the requirements of 8.13.6?	M	5.4.1.3, 8.13.6	Yes []
MMRP5	Is the MAP Context Identifier used to identify a VLAN Context equal to the VID used to identify the corresponding VLAN?	M	10.12.1.1	Yes []
MMRP6	Is the set of Ports of a Bridge defined to be part of the active topology for a given VLAN Context as specified in 10.12.1.1?	M	10.12.1.1	Yes []
MMRP7	Is the group MAC address used as the destination address for MRPDUs destined for MMRP Participants the group MAC address identified in Table 10-1 as “Customer and Provider Bridge MMRP address”?	M	10.12.1.3	Yes []
MMRP8	Is the EtherType used for MRPDUs destined for MMRP Participants the MMRP EtherType identified in Table 10-2?	M	10.12.1.4, Table 10-2	Yes []
MMRP9	Does the ProtocolVersion used for the implementation of MMRP take the hexadecimal value 0x00?	M	10.12.1.5	Yes []
MMRP10	Are the Attribute Type values used in the implementation as specified in 10.12.1.6?	M	10.12.1.6	Yes []
MMRP11	Does the implementation encode the values in FirstValue fields in accordance with the definition in 10.12.1.7?	M	10.12.1.7	Yes []
MMRP12	Is management of the Restricted_MAC_Address_Registration control parameter supported?	O	10.12.2	Yes [] No []
MMRP13	If management of the Restricted_MAC_Address_Registration control parameter is not supported, is the value of this parameter FALSE for all Ports?	¬MMRP12:M	10.12.2	Yes [] N/A []

A.20 Multiple MAC Registration Protocol (MMRP) *(continued)*

Item	Feature	Status	References	Support
MMRP14	Does the implementations maintain state information for all attribute values that support the Group service requirement registration?	M	10.10.2	Yes []
MMRP15	Is the implementation capable of supporting any attribute value in the range of possible values that can be registered using Group membership and individual MAC address registration?	M	10.12.4	Yes []
MMRP16	State the number of Group membership and individual MAC address state values that can be supported on each Port.	M	10.12.4	Number_____

1

A.21 Multiple VLAN Registration Protocol (MVRP)

Item	Feature	Status	References	Support
	If MVRP is not supported, mark N/A and continue at A.22			N/A []
MVRP1	Does the implementation support the exchange of MVRPDUs, using the generic MRPDU format defined in 10.8 to exchange MVRP-specific information, as defined in 11.2?	M	5.4.2, 10.8, 11.2	Yes []
MVRP2	Is the MVRP Application supported as defined in 11.2?	M	5.4.2, 11.2	Yes []
MVRP3	Does the implementation propagate registration information in an MST Bridge, in accordance with the operation of MAP for multiple spanning tree contexts, as specified in 11.2.3.1.2?	MSTP:M	5.4.2, 11.2.3.1.2	Yes [] N/A []
MVRP4	Does the implementation propagate registration information in an SST Bridge, in accordance with the operation of MAP for the Base Spanning Tree Context, as specified in 10.3.1?	~MSTP:M	5.4.2, 10.3.1	Yes [] N/A []
MVRP5	Does the implementation forward, filter, or discard MAC frames carrying any MRP Application address as the destination MAC address in accordance with the requirements of 8.13.6?	M	5.4.2, 8.13.6	Yes []
MVRP6	If a VID translation table is in use, does the implementation translate VIDs in MVRPDUs as specified in 11.2.4?	VTR:M	11.2.4	Yes []
MVRP7	Is the group MAC address used as the destination address for MRPDUs destined for MVRP Participants as defined in 11.2.3.1.3?	M	11.2.3.1.3	Yes []
MVRP8	Is the EtherType used for MRPDUs destined for MVRP Participants the MVRP EtherType identified in Table 10-2?	M	11.2.3.1.4, Table 10-2	Yes []
MVRP9	Does the ProtocolVersion used for the implementation of MVRP take the hexadecimal value 0x00?	M	11.2.3.1.5	Yes []
MVRP10	Are the Attribute Type values used in the implementation as specified in 11.2.3.1.6?	M	11.2.3.1.6	Yes []

A.21 Multiple VLAN Registration Protocol (MVRP) *(continued)*

Item	Feature	Status	References	Support
MVRP11	Does the implementation encode the values in FirstValue fields in accordance with the definition in 11.2.3.1.7?	M	11.2.3.1.7	Yes []
MVRP12	Is the implementation of MVRP capable of supporting all attribute values in the range of possible values that can be registered using MVRP?	M	11.2.7	Yes []
MVRP13	Is the implementation capable of maintaining current state information for all attributes in the range of possible values?	M	11.2.7	Yes []
MVRP14	Does the Bridge support an MVRP New-Only Participant?	O	11.2.6	Yes [] No []
MVRP15	Does the MVRP Participant transmit the Attribute value 0 as specified in 11.2.3.1.7?	MVRP14: M	11.2.3.1.7	Yes []
MVRP16	Does the Bridge interpret the Attribute value 0 in an MVRP message as the PVID for the Port?	M	11.2.3.1.7	Yes []
MVRP17	Does MVRP ignore New messages in MVRPDUs for VLANs not Registration Fixed (New propagated) when configured for New propagated?	MVRP14: M	11.2.6	Yes []

1

A.22 Multiple Registration Protocol (MRP)

Item	Feature	Status	References	Support
MRP1	Does the implementation of MRP meet all of the requirements for interoperability stated in 10.5 that apply to Bridge operation?	M	10.5	Yes []
MRP2	Does the implementation support the operation of the complete Applicant state machine?	MRP1:M	10.7, 10.7.7	Yes [] N/A []
MRP3	Does the implementation support the operation of the point-to-point subset of the Applicant state machine?	MRP2:M	10.7, 10.7.7	Yes [] N/A []
MRP4	Does the implementation support the operation of the Registrar state machine?	M	10.7, 10.7.8	Yes []
MRP5	Does the implementation support the operation of the LeaveAll state machine?	M	10.7, 10.7.9	Yes []
MRP6	Does the implementation support the operation of the PeriodicTransmission state machine?	M	10.7, 10.7.10	Yes []

A.23 Connectivity Fault Management (CFM)

Item	Feature	Status	References	Support
CFM-1	Does the Bridge support the creation of Maintenance Domains at eight MD Levels, with multiple Maintenance Domains at each MD Level?	BRG AND CFM: M	5.4.1.4 item a)	Yes [] N/A []
CFM-2	Does the Bridge support the creation of a Maintenance Association (MA) on each VID supported by the Bridge for each MD Level?	BRG AND CFM: M	5.4.1.4 item b)	Yes [] N/A []
CFM-3	Does the Bridge support the creation of a single MIP for each Maintenance Domain on each Port, all MIPs being at the same MD Level?	BRG AND CFM: M	5.4.1.4 item c)	Yes [] N/A []
CFM-4	Does the Bridge support the creation of eight Up MEPs on each VID on each Port, each MEP at a different MD Level?	BRG AND CFM: M	5.4.1.4 item d)	Yes [] N/A []
CFM-5	Does the Bridge support the creation of eight Down MEPs on each VID on each Port, each MEP at a different MD Level?	BRG AND CFM: M	5.4.1.4 item e)	Yes [] N/A []
CFM-6	Does the Bridge support the creation of eight Down MEPs associated with no VID on each Port, each MEP at a different MD Level?	BRG AND CFM: M	5.4.1.4 item f)	Yes [] N/A []
CFM-7	Does the Bridge support the maintenance of a MEP CCM Database?	BRG AND CFM: M	5.4.1.4 item g)	Yes [] N/A []
CFM-8	Does the Bridge conform to the state machines and procedures in Clause 20?	BRG AND CFM: M	5.4.1.4 item i), Clause 20	Yes [] N/A []
CFM-9	Does the Bridge transmit and accept frames in the formats specified in Clause 21?	BRG AND CFM: M	5.4.1.4 item j)	Yes [] N/A []
CFM-10	Does the Bridge support the creation of MIPs at different MD Levels on a single Port?	BRG AND CFM: O	5.4.1.4 item k), 22.3	Yes [] No [] N/A []
CFM-11	Does the Bridge support the creation of MEPs at MD Levels equal to or higher than the MD Levels of MIPs on other VIDs on the same Port?	BRG AND CFM: O	5.4.1.4 item l), 22.3	Yes [] No [] N/A []
CFM-12	Does the Bridge support the maintenance of a MIP CCM Database in MIPs and MEPs?	BRG AND CFM: O	5.4.1.4 item m), 19.3, 19.2.8	Yes [] No [] N/A []
CFM-13	Does the Bridge support the creation of MAs that are associated with more than one VID?	BRG AND CFM: O	5.4.1.4 item o)	Yes [] No [] N/A []
CFM-14	Does the Station support the creation of Maintenance Domains at eight MD Levels, with multiple Maintenance Domains at each MD Level?	–BRG AND CFM: M	5.4.5 item a), 5.19	Yes [] N/A []
CFM-15	Does the Station support the creation of an MA on each VID supported by the Bridge for each MD Level?	–BRG AND CFM: M	5.4.5 item b), 5.19	Yes [] N/A []
CFM-16	Does the Station support the creation of MIPs?	–BRG AND CFM: X	22.4 item b)	No [] N/A []
CFM-17	Does the Station support the creation of Up MEPs?	–BRG AND CFM: X	22.4 item b)	No [] N/A []
CFM-18	Does the Station support the creation of eight Down MEPs on each VID on each Port, each MEP at a different MD Level?	–BRG AND CFM: M	5.4.5 item c), 5.19	Yes [] N/A []

A.23 Connectivity Fault Management (CFM) (continued)

Item	Feature	Status	References	Support
CFM-19	Does the Station support the creation of eight Down MEPs associated with no VID on each Port, each MEP at a different MD Level?	–BRG AND CFM: M	5.4.5 item d), 5.19	Yes [] N/A []
CFM-20	Does the Station support the maintenance of a MEP CCM Database?	–BRG AND CFM: M	5.4.5 item e), 5.19	Yes [] N/A []
CFM-21	Does the Station conform to the state machines and procedures in Clause 20?	–BRG AND CFM: M	5.4.5 item g), 5.19, Clause 20	Yes [] N/A []
CFM-22	Does the Station transmit and accept frames in the formats specified in Clause 21?	–BRG AND CFM: M	5.4.5 item h), 5.19	Yes [] N/A []
CFM-23	Does the Station support the creation of MAs that are associated with more than one VID?	–BRG AND CFM: O	5.4.5 item k), 5.19	Yes [] No [] N/A []
CFM-24	Does the MP Level Demultiplexer discard frames that are too short to contain an MD Level header field?	CFM: M	19.2.6, 20.51.4.1	Yes [] N/A []
CFM-25	Is an LBM always discarded, and not replied to, if its source_address is a Group address?	CFM: M	20.2.2	Yes [] N/A []
CFM-26	Are the contents of an LBM, except for the source_address and OpCode, ignored and not interpreted by a non-PBB-TE receiver?	–CFM-93 AND CFM: M	20.2.2	Yes [] N/A []
CFM-27	Is the LTFwhile timer implemented?	CFM:M	20.5, 20.5.1	Yes [] N/A []
CFM-28	Are the per-MEP timers CCIwhile, errorCCMwhile, xconCCMwhile, LBIwhile, and FNGwhile implemented?	CFM:M	20.5, 20.5.2, 20.5.3, 20.5.4, 20.5.5, 20.5.6	Yes [] N/A []
CFM-29	Is the per-MEP per-remote MEP timer rMEPwhile implemented?	CFM:M	20.5, 20.5.10	Yes [] N/A []
CFM-30	Can a MEP set rMEPCCMdefect within $(3.25 \times \text{CCMtime}(\text{CCMinterval}))$ seconds of the receipt of a CCM?	X	20.5.10	No []
CFM-31	Can a MEP take longer to set rMEPCCMdefect than $(3.5 \times \text{CCMtime}(\text{CCMinterval}))$ seconds of the receipt of the last CCM?	CFM: X	20.5.10	No [] N/A []
CFM-32	Does the system transmit a nonzero value for the CCM Interval in CCMs?	CFM: M	20.8.1, 20.11.1 item g)	Yes [] N/A []
CFM-33	Does the Bridge transmit all CFM PDU fixed header fields in conformance with this standard?	CFM: M	20.51.2	Yes [] N/A []
CFM-34	Does the Bridge transmit all reserved bits and fields in CFM PDUs as 0?	CFM: M	20.51.2	Yes [] N/A []
CFM-35	Does the Bridge transmit additional fixed header fields not defined in this standard in CFM PDUs?	X	20.51.2	No []
CFM-36	Does the Bridge transmit code points in any field that are reserved, either by this standard or by ITU-T G.8013/Y.1731?	X	20.51.2	No []
CFM-37	Does the Bridge transmit additional fields in any CFM PDU in any TLV defined by this standard?	X	20.51.2	No []

A.23 Connectivity Fault Management (CFM) *(continued)*

Item	Feature	Status	References	Support
CFM-38	Does the Bridge determine the validity of those CFM PDUs that are validated, in a manner indistinguishable, by external observation of the Bridge, from the procedures described in this standard?	CFM: M	20.51.3	Yes [] N/A []
CFM-39	Is the version by which each PDU is processed selected correctly, and are the prohibited validation criteria not applied, by the System?	CFM: M	20.17.1, 20.17.2, 20.28.1, 20.33.1, 20.51.4.2	Yes [] N/A []
CFM-40	Does the System determine the validity of a CFM PDU in the manner defined by this standard?	CFM: M	20.17.1, 20.17.2, 20.28.1, 20.33.1, 20.51.4.1, 20.51.4.3	Yes [] N/A []
CFM-41	Are all CFM PDUs transmitted with an integral number of octets?	CFM: M	21.1	Yes [] N/A []
CFM-42	Does the Bridge transmit Organization-Specific TLVs?	CFM: O	20.51.2, 21.5.2	Yes [] No []
CFM-43	Does the Bridge transmit an Organization-Specific TLV that requires it or the receiver to violate any requirement of this standard?	X	21.5.2	No []
CFM-44	Is the information transmitted in an Organization-Specific TLV independent from information in a TLV received from any other port?	CFM-42: M	21.5.2	Yes [] N/A []
CFM-45	Do Organization-Specific TLV(s) transmitted by the Bridge provide a means for sending messages that are larger than would fit within a single CFM PDU?	X	21.5.2	No []
CFM-46	Do the Organization-Specific TLV(s) transmitted by the Bridge conform to the validation and versioning rules of 20.51?	CFM-42: M	21.5.2	Yes [] N/A []
CFM-47	Does the Management Address Domain field in the Sender ID TLV(s) transmitted by the Bridge (if any) conform to section 8.19 of ITU-T X.690 (2002)?	CFM: M	21.5.3.5	Yes [] N/A []
CFM-48	Can a MEP in a multiple spanning tree environment not statically restricted to a single MSTI generate a Port Status TLV?	X	21.5.4	No []
CFM-49	Does the Bridge, in any case except when receiving an LBR, interpret the contents of the Data TLV?	X	21.5.6	No []
CFM-50	Can the Bridge transmit the Data TLV in an LBM?	CFM: O	21.5.6	Yes [] No [] N/A []
CFM-51	Is the Bridge able to receive and process all valid CCM PDUs that are 128 octets or less in length (from MD Level through End TLV)?	CFM: M	21.6	Yes [] N/A []
CFM-52	Does the Bridge discard, as invalid, CCM PDUs that exceed 128 octets in length?	CFM: O	21.6	Yes [] No [] N/A []
CFM-53	Can the Bridge transmit a CCM PDU that is longer than 128 octets in length?	X	21.6	No []

A.23 Connectivity Fault Management (CFM) (continued)

Item	Feature	Status	References	Support
CFM-54	Is the length of the Maintenance Association Identifier (MAID) transmitted in a CCM exactly 48 octets?	CFM: M	21.6.5	Yes [] N/A []
CFM-55	Is the field Defined by ITU-T G.8013/Y.1731 always transmitted as 0?	CFM: M	21.6.6	Yes [] N/A []
CFM-56	Does the First TLV Offset field contain the value as specified for the OpCode for each CFM message transmitted?	CFM: M	21.4.5, 21.6.2, 21.7.2, 21.8.2, 21.9.2	Yes [] N/A []
CFM-57	Does every transmitted LTM contain an LTM Egress Identifier TLV?	CFM: M	21.8.7	Yes [] N/A []
CFM-58	Does every transmitted LTR contain an LTR Egress Identifier TLV?	CFM: M	21.9.6	Yes [] N/A []
CFM-59	Does the receiving Bridge behave differently if the order of TLVs in a CFM PDU, other than the End TLV, or TLVs in an LBR, is altered?	X	21.6.7, 21.9.6	No []
CFM-60	Does the Bridge support the creation of MPs at one or more MD Level on every Port?	CFM: M	22.3	Yes [] N/A []
CFM-61	Is every Down MEP on a Bridge Port assigned a MAC address different than any Down MEP on any other Bridge Port?	CFM: M	19.4	Yes [] N/A []
CFM-62	Does the Provider Edge Bridge support the creation of a Down MEP on the interface corresponding to a CEP?	PEB AND BRG AND CFM: M	22.6.1	Yes [] N/A []
CFM-63	Does the Provider Edge Bridge support the creation of a Down MEP on the interface corresponding to a RCAP?	RCSI AND BRG AND CFM: M	22.6.4	Yes [] No []
CFM-64	Does the Provider Edge Bridge support the creation of a Down MEP on the interface corresponding to a CEP connected to a Provider Access Port (PAP)?	RCSI AND BRG AND CFM: M	22.6.4	Yes [] No []
CFM-65	If the MIP CCM Database has insufficient resources to record a new entry, does it preferentially remove the oldest entry to make room for the new one?	BRG AND CFM-12: O	20.1.3	Yes [] No [] N/A []
CFM-66	Does the Bridge transmit successive integer values in the Sequence Number field of a CCM?	CFM: O	20.1, 20.11.1 item i)	Yes [] No [] N/A []
CFM-67	Does the Bridge transmit neither successive integer values nor 0 in the Sequence Number field of a CCM?	X	20.1, 20.11.1	No []
CFM-68	Does the Bridge keep track of received CCMs' Sequence Number fields in the MEP CCM Database, and count out-of-sequence CCMs in CCMsequenceErrors?	CFM: O	20.1, 20.17.1	Yes [] N/A [] N/A []
CFM-69	Does the Bridge check the validity of every received LTM, and LTR?	CFM: M	20.33.1, 20.44.1, 20.47.1	Yes [] N/A []
CFM-70	Does the Bridge check the validity of every received CCM?	CFM: O	20.17.1, 20.17.2	Yes [] No [] N/A []
CFM-71	Does the Bridge check the validity of every received LBM?	CFM: O	20.2.2, 20.28.1	Yes [] No [] N/A []

A.23 Connectivity Fault Management (CFM) *(continued)*

Item	Feature	Status	References	Support
CFM-72	Does the Bridge check the validity of every received LBR?	CFM: O	20.2.2, 20.33.1	Yes [] No [] N/A []
CFM-73	Does the Bridge compare the received LBR bit-by-bit against the original LBM?	CFM: O	20.2.3, 20.33.1	Yes [] No [] N/A []
CFM-74	Can the Bridge transmit LBMs at a rate fast enough to overflow output queues in the absence of other data traffic?	X	20.2.1	No []
CFM-75	Can a high rate of incoming CFM PDUs increase the probability that the Bridge fails to protect the network against forwarding loops?	BRG: X	20.51.5	No [] N/A []
CFM-76	Are the Optional CCM TLVs included in every CCM?	CFM: O	21.6.7	Yes [] No [] N/A []
CFM-77	Are Organization-Specific TLVs included in CCMs?	CFM: O	21.6.7	Yes [] No [] N/A []
CFM-78	Is the Sender ID TLV included in every LBM?	CFM:O	21.5.3, 21.7.4	Yes [] N/A [] N/A []
CFM-79	Is the Management Address included in the Sender ID TLV?	CFM-78: O	21.5.3	Yes [] No [] N/A []
CFM-80	Are Organization-Specific TLVs included in LBMs?	CFM: O	21.7.4	Yes [] No [] N/A []
CFM-81	Does the Station discard, and not process, all CFM PDUs not discarded or processed by its MEPs?	–BRG AND CFM: M	22.4 item c)	Yes [] N/A []
CFM-82	Does the Station create entries in the Configuration Error List managed object other than CFMleak errors?	X	22.4 item d), 12.14.4	No []
CFM-83	Can CFM monitor backbone service instances using Backbone Service Instance Multiplex Entities placed back-to-back?	BEB: O	5.7.1, 5.8.1	Yes [] No [] N/A []
CFM-84	Does the Bridge support the creation of an MA on each TESI supported by the Bridge for each MD Level?	BEBTE: M	5.8.2, 26.9	Yes [] N/A []
CFM-85	Does the Bridge support the creation of an Up MEP on each TESI on each CBP?	BEBTE: M	5.8.2, 6.19, 26.9.3	Yes [] N/A []
CFM-86	Does the Bridge support the creation of eight Up MEPs on each TESI on each CBP, each MEP at a different MD Level?	BEBTE: O	5.8.2	Yes [] No [] N/A []
CFM-87	Does the Bridge support the creation of MIPs on TESIs?	PBBTE: O	5.6.2, 5.8.2	Yes [] No [] N/A []
CFM-88	Is the PBB-TE MIP TLV included in every LBM that is targeting a PBB-TE MIP?	BEBTE: M	20.2, 21.7.5	Yes [] N/A []
CFM-89	Is the PBB-TE MIP TLV included in every LTM that is associated with a PBB-TE MA?	BEBTE: M	20.3, 21.7.5	Yes [] N/A []
CFM-90	Is the PBB-TE MIP TLV included in LBMs that are targeting a PBB-TE MEP?	BEBTE: X	20.2, 21.7.5	No [] N/A []
CFM-91	Is the PBB-TE MIP TLV the first TLV in an LBM targeting a PBB-TE MIP?	CFM-88: M	20.2, 21.7.5	Yes [] N/A []

A.23 Connectivity Fault Management (CFM) *(continued)*

Item	Feature	Status	References	Support
CFM-92	Is the Reverse MAC field included in the PBB-TE MIP TLVs required by LBMs/LTMs that are associated with Point-to- Multipoint PBB-TE MAs?	CFM-88 OR CFM-89: M	20.2, 20.3, 21.7.5	Yes [] N/A []
CFM-93	Does the PBB-TE MEP check the MAID in received CCMs?	BEBTE: O	20.1.3, 20.17.1	Yes [] No [] N/A []
CFM-94	Does the PBB-TE MEP set the Traffic field bit in the transmitted CCMs and process it in the received CCMs	PS-5: M	20.11.1, 20.17.1, 21.6.1.4	Yes [] N/A []
CFM-95	Does the Bridge keep track of the Traffic field bit in received CCMs?	PS-5: M	20.16.3, 20.17.1, 21.6.1.4	Yes [] N/A []
CFM-96	Are the per-PBB MEP timers mmCCMwhile, mmLocwhile, and mmFNGwhile implemented?	PS-5: M	20.5.7	Yes [] N/A []
CFM-97	Are the per-PBB MEP mismatch defect related, MEP Traffic Field Mismatch state machine, MEP Local Mismatch state machine and MEP Mismatch Fault Notification Generator state machine implemented?	PS-5: M	20.24, 20.40	Yes [] N/A []
CFM-98	Are SPBM VID MAs supported?	SPBM:O	27.18, Clause 20	Yes [] No [] N/A []
CFM-99	Are SPBM group MAs supported?	SPBM:O	27.18, Clause 20	Yes [] No [] N/A []
CFM-100	Are SPBM path MAs supported?	SPBM:O	27.18, Clause 20	Yes [] No [] N/A []
CFM-101	Are ECMP path MAs supported?	FF:O	44.2.5, Clause 20	Yes [] No [] N/A []

1 A.24 Management Information Base (MIB)

Item	Feature	Status	References	Support
	If item MIB is not supported, mark N/A.			N/A []
MIB-1	Is the IEEE8021-TC-MIB module supported?	MIB:M	17.7.1	Yes [] No []
MIB-2	Is the IEEE8021-BRIDGE-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB:M	17.7.2	Yes [] No []
MIB-3	Is the IEEE8021-SPANNING-TREE-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB:O	17.7.3	Yes [] No []
MIB-5	Is the IEEE8021-Q-BRIDGE-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB:M	17.7.4	Yes [] No []
MIB-6	Is the IEEE8021-PB-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND PB:O	17.7.5	Yes [] No []
MIB-7	Is the IEEE8021-MST-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND MSTP:O	17.7.6	Yes [] No [] N/A []
MIB-8	Is the IEEE8021-CFM-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND CFM:O	17.7.7	Yes [] No []
	What method is used by the Bridge to provide control of all of the required CFM managed objects?			
MIB-9	The CFM MIB module defined in 17.7.7?	CFM: O.3	5.4.1.4 item n), 17.3.7, 17.7.7	Yes [] No [] N/A []
MIB-10	Some other method than the MIB module defined in 17.7.7?	CFM: O.3	17.3.7	Yes [] No [] N/A []
MIB-11	If by some other method, what method is used?	MIB-10: M	17.3.7	_____
MIB-12	Does the Station provide control of all of the required CFM managed objects?	–BRG AND CFM: M	5.4.5 item f), 12.14	Yes [] No [] N/A []
	What method is used by the Station to provide control of all of the required CFM managed objects?			
MIB-13	The CFM MIB module defined in 17.7.7?	–BRG AND CFM: O.4	5.4.5 item i), 17.3.7, 17.7.7	Yes [] No [] N/A []
MIB-14	Some other method than the MIB module defined in 17.7.7?	–BRG AND CFM: O.4	17.3.7	Yes [] No [] N/A []
MIB-15	If by some other method, what method is used?	MIB-14: M	17.3.7	_____
MIB-16	Is an entire C-tagged service interface given a single row in the IETF RFC 2863 IF-MIB?	PEB AND CFM: O	17.3.7	Yes [] No [] N/A []
MIB-17	Is every Bridge Port assigned its own conceptual row in the IETF RFC 2863 IF-MIB with its own unique ifIndex?	MIB-13: M	17.3.7	Yes [] N/A []
MIB-18	Does the Bridge support IEEE 802.1AX Link Aggregation?	CFM: O	17.3.7, IEEE Std 802.1AX	Yes [] No [] N/A []
MIB-19	Does every IEEE 802.3 MAC, when aggregated via IEEE 802.1AX Link Aggregation, have its own unique ifIndex, separate from the ifIndex of the Bridge Port as a whole?	MIB-18: M	17.3.7	Yes [] N/A []

A.24 Management Information Base (MIB) (continued)

Item	Feature	Status	References	Support
MIB-20	Is the IEEE8021-PBB-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND BEB:O	17.7.8	Yes [] No [] N/A []
MIB-21	Is the IEEE8021-DDCFM-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND DDCFM:O	17.7.9	Yes [] No [] N/A []
MIB-22	Is the IEEE8021-PBBTE-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND PBBTE: O	17.7.10	Yes [] No [] N/A []
MIB-23	Is the IEEE8021-TPMR-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND TPMR:O	17.7.11	Yes [] No [] N/A []
MIB-24	Is the IEEE8021-FQTSS-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND FQTSS: O	5.4.1.5 item e), 12.20, 17.7.12, Clause 34	Yes [] No [] N/A []
MIB-25	Does the system implement the IEEE8021-CN-MIB?	MIB AND CN: O	5.4.3 item j), 5.21 item n)	Yes [] No [] N/A []
MIB-26	Does the system implement the System Group and Interfaces Group of the SNMPv2-MIB?	MIB-25: M	17.3.7.1	Yes []
MIB-27	Does the system implement the clarifications of the Interfaces group supplied in the descriptions of ieee8021CnEpIfIndex, ieee8021CnPortPriIfIndex, ieee8021CnCpIfIndex, and ieee8021CnCpidToIfIndex?	MIB-25: M	17.3.7.1, 17.7.13	Yes [] N/A []
MIB-28	Does the system assign the same conceptual row in the IF-MIB to both an aggregated port and to one of the ports being aggregated?	MIB-25: M	17.3.7.1	No [] N/A []
MIB-29	Does the system assign a conceptual row in the IF-MIB to the individual aggregated?	MIB-25: O	17.3.7.1	Yes [] No [] N/A []
MIB-30	Does the system allow more than seven rows to be created in the ieee8021CnCompntPriTable?	MIB-25: M	17.7.13	No [] N/A []
MIB-31	Is the minimum value for ieee8021CnRpgMaxRate supported by the system at least equal to the required value?	MIB-25: M	17.7.13, 32.11.5	Yes [] N/A []
MIB-32	Is the minimum value for ieee8021CnRpgMinRate supported by the system at least equal to the required value?	MIB-25: M	17.7.13, 32.11.10	Yes [] N/A []
MIB-33	Is the IEEE8021-SRP-MIB module fully supported (per its MODULE-COMPLIANCE)?	SRP: O	17.2, 35	Yes [] No [] N/A []
MIB-34	Is the IEEE8021-MVRPX-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIRP AND MIB: M	17.7.15	Yes [] N/A []
MIB-35	Is the IEEE8021-MIRP-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIRP AND MIB: M	17.7.16	Yes [] No [] N/A []
MIB-36	Is the IEEE8021-PFC-MIB module fully supported (per its MODULE-COMPLIANCE)?	PFC AND MIB: O	17.7.17	Yes [] No [] N/A []
MIB-37	Is the IEEE8021-TEIPS-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB: O	17.7.10	Yes [] No [] N/A []

A.24 Management Information Base (MIB) (continued)

Item	Feature	Status	References	Support
MIB-38	Is the IEEE8021-SPB-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND SPB:O	5.4.5, 17.7.19	Yes [] No [] N/A []
MIB-39	Are the IEEE8021-ECMP-MIB module objects ieee8021EcmpEctStaticTable and ieee8021EcmpTopSrvTable supported?	MIB AND ECMP:O	5.4.5.1, 17.7.10	Yes [] No [] N/A []
MIB-40	Is the IEEE8021-ECMP-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND FF:O	5.4.5.2, 17.7.21	Yes [] No [] N/A []
MIB-41	Are PCR objects in the IEEE8021-SPB-MIB supported per ieee8021PcrCompliance ?	MIB AND PCR:O	5.5, 17.7.19	Yes [] No [] N/A []
MIB-42	Is the IEEE8021-ST-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND (SCHED OR CQF): O	5.4.1 item ad), 5.4.1.9 item c), 5.13.1.2 item c), 12.29, 17.7.22	Yes [] No [] N/A []
MIB-43	Is the IEEE8021-Preemption-MIB module fully supported (per its MODULE-COMPLIANCE)?	PRE: O	5.4.1 item ae), 12.30, 17.7.23	Yes [] No [] N/A []
MIB-44	Is the IEEE8021-PSFP-MIB module fully supported (per its MODULE-COMPLIANCE)?	PSFP OR CQF: O	5.4.1.9 item e), 5.13.1.2 item e), 8.6.5.2, 8.6.10, 12.31, 17.7.24	Yes [] No [] N/A []
MIB-45	Is the IEEE8021-PBBN-AA-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND (AAB OR AAD):O	17.2.26	Yes [] No [] N/A []
MIB-46	Are the Auto Attach objects in IEEE8021-LLDP-EXT-DOT1-EVB-EXTENSIONS-MIB module supported (per lldpXdot1AaCompliance MODULE-COMPLIANCE)?	MIB AND (AAB OR AAD):O	D.5.6	Yes [] No [] N/A []

¹ A.25 Protection Switching (PS)

Item	Feature	Status	References	Support
	If item PS is not supported, mark N/A and continue at the subsequent subclause.			N/A []
PS-1	Is 1:1 protection switching supported?	PS: M	5.8.2, 26.10.3	Yes []
PS-2	Are the operator commands Forced Switch (FS), Lockout of Protection (LoP), Manual Switch to Working, and Manual Switch to Protection implemented?	PS-1: M	26.10.3.3	Yes []
PS-3	Is the hold-off timer implemented?	PS-1: O	26.10.3.2.2	Yes [] No []
PS-4	Is protection switching with load sharing supported?	PS: O	5.8.2, 12.14.1.2	Yes [] No []
PS-5	Is the detection of mismatch defects supported?	PS: O	26.12	Yes [] No []

1 A.26 Data-driven and data-dependent connectivity fault management (DDCFM)

Item	Feature	Status	References	Support
DDCFM-1	Does the Bridge support the Reflection Responder (RR) function?	DDCFM:M	29.2.2	Yes [] N/A []
DDCFM-2	Does the Bridge support the RFM Receiver function?	DDCFM:M	29.2.4	Yes [] N/A []
DDCFM-3	Does the Bridge support the Decapsulator Responder (DR) function?	DDCFM:M	29.2.6	Yes [] N/A []
DDCFM-4	Does the Bridge support the SFM Originator function?	DDCFM:M	29.2.7	Yes [] N/A []

2 A.27 Two-Port MAC Relay (TPMR)

Item	Feature	Status	References	Support
	If TPMR is not supported, mark N/A and continue at A.28.			N/A []
TPMR-1	Does the TPMR comprise a single TPMR component?	TPMR:M	5.15	Yes []
TPMR-2	Is each TPMR Port capable of attaching directly to an IEEE 802 LAN?	TPMR:M	5.16	Yes []
TPMR-3	Does the implementation support exactly two externally-accessible Bridge Ports?	TPMR:M	5.15	Yes []
TPMR-4	Does the implementation support the operation of the Bridge Port Transmit and Receive process, using the TPMR Port connectivity rules?	TPMR:M	5.15, 8.5, 8.5.2	Yes []
TPMR-5	Does the Forwarding Process support frame filtering, queuing frames, queue management, and transmission selection?	TPMR:M	5.15, 8.6.3, 8.6.6, 8.6.7, 8.6.8	Yes []
TPMR-6	Does the implementation support at least one traffic class on each externally accessible Port?	TPMR:M	5.15	Yes []
TPMR-7	Does the implementation support the management functionality specified in 12.19?	TPMR:M	12.19	Yes []
TPMR-8	Does the implementation support remote management via one of the externally accessible Ports?	TPMR:M	5.15, 8.13.7	Yes []
TPMR-9	Does the implementation support Static Filtering Entries in accordance with the requirements stated in 5.15:f?	TPMR:M	5.15, 8.13.7, Table 8-3	Yes []
TPMR-10	Does the implementation support multiple traffic classes on each externally accessible Port?	TPMR:O	5.15	Yes [] No []
TPMR-11	Does the implementation support remote management via one or both of the externally accessible Ports, using IETF RFC 4789, and the TPMR MIB module?	TPMR:O	5.15, IETF RFC 4789, 8.13.7, 17.7.11	Yes [] No []
TPMR-12	Does the implementation support signaled priority?	TPMR: O	6.20	Yes [] No []

¹ A.28 MAC Status Protocol (MSP)

Item	Feature	Status	References	Support
	If MSP is not supported, mark N/A.			N/A []
MSP-1	Does the implementation support the operation of the Status Transmission state machine (STM)?	MSP:M	23.8	Yes []
MSP-2	Does the implementation support the operation of the Status Notification state machine (SNM)?	MSP:M	23.9	Yes []
MSP-3	Does the Receive Process receive and validate MSPDUs as specified?	MSP:M	23.10, 23.16	Yes []
MSP-4	Does the Transmit Process identify, address, structure, encode, and transmit MSPDUs as specified?	MSP:M	23.11, 23.13, 23.14, 23.15	Yes []
MSP-5	Does the implementation allow performance parameters to be read by management?	MSP:O	23.12	Yes [] No []
MSP-6	Does the implementation allow performance parameters to be modified by management?	MSP:O	23.12	Yes [] No []
MSP-7	Does the implementation maintain the specified counts for at least one Port of the TPMR?	MSP:O	23.12	Yes [] No []
MSP-8	Does the implementation maintain the specified counts for both ports of the TPMR?	MSP:O	23.12	Yes [] No []

1 A.29 Forwarding and Queuing Enhancements for time-sensitive streams (FQTSS)

Item	Feature	Status	References	Support
	If forwarding and queuing for time-sensitive streams (FQTSS in A.6) is not supported, mark N/A and ignore the remainder of this table.		5.4.1.5, Clause 34	N/A []
FQTSSE1	Support a minimum of two traffic classes, of which one supports the strict priority algorithm and the other is an SR class.	FQTSS:M	5.4.1.5, 8.6.8.1, Clause 34	Yes [] N/A []
FQTSSE2	Support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for the SR class.	FQTSS:M	5.4.1.5, 8.6.8.2, Clause 34	Yes [] N/A []
FQTSSE3	Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the default priority regeneration override value defined in Table 6-5, for SR class “B.”	FQTSS:M	5.4.1.5, 6.9.4, Table 6-5, Clause 34	Yes [] N/A []
FQTSSE4	Support the tables and procedures for mapping priorities to traffic classes as defined in 34.5.	FQTSS:M	5.4.1.5, Clause 34, 34.5	Yes [] N/A []
FQTSSE5	Support two or more SR classes (a maximum of seven), and support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for those SR classes. The number of SR classes supported shall be stated in the PICS.	FQTSS:O	5.4.1.5, 8.6.8.2, 34.6	Yes [] No [] Number_____
FQTSSE6	Support SRP domain boundary port priority regeneration override as defined in 6.9.4, and the default priority regeneration override value defined in Table 6-5, for SR class “A.” If more than two SR classes are supported, the default priority regeneration override values used for the additional SR classes shall be stated in the PICS.	FQTSS:O	5.4.1.5, 6.9.4, Table 6-5, 34.6	Yes [] No [] Priority override values_____

2 A.30 Congestion notification

Item	Feature	Status	References	Support
CN-1	Does the system conform to the required provisions of IEEE Std 802.1AB?	CN: M	5.4.3 item e), 5.21 item f)	Yes [] N/A []
CN-2	Does the system support the use of the Congestion Notification TLV in LLDP?	CN: M	5.4.3 item f), 5.21 item g)	Yes [] N/A []
CN-3	Does the system transmit more than one Congestion Notification TLV in a single LLDPDU?	CN: M	D.2.7	No [] N/A []
CN-4	Does the system transmit a Congestion Notification TLV with 0 in all of the Per-priority CNPV indicators?	CN: M	D.2.7	No [] N/A []
CN-5	Does the system implement the Congestion Notification Domain (CND) defense variables, procedures, and state machine?	CN: M	32.4, 32.5, 32.6	Yes [] N/A []
CN-6	Does the Bridge support the creation of at least one CP on at least one Port?	BRG1 AND CN: M	5.4.3 item a)	Yes [] N/A []

A.30 Congestion notification (*continued*)

Item	Feature	Status	References	Support
CN-7	Does the Bridge support the creation of more than one CP on at least one Port?	BRG1 AND CN: O	5.4.3 item i)	Yes [] No [] N/A []
CN-8	Does the Bridge support the creation of more than seven CPs on any Port?	BRG1 AND CN: M	5.4.3 item i)	No [] N/A []
CN-9	Does every CP on the Bridge support all four defense modes separately on each CNPV?	BRG1 AND CN: M	5.4.3 item d), 31.1.1, 32.2.1	Yes [] N/A []
CN-10	Is each CP on the Bridge able to remove CN-TAGs?	BRG1 AND CN: M	5.4.3 item b)	Yes [] N/A []
CN-11	Does the PIP perform CNM translation on the return path?	BEB-I AND CN:M	5.4.3 item g)	Yes [] N/A []
CN-12	Does the Provider Edge Port (PEP) perform CNM translation on the return path?	PEB AND CN:M	5.4.3 item h)	Yes [] N/A []
CN-13	Is each CP on the system able to generate CNMs?	BRG1 AND CN: M	5.4.3 item b)	Yes [] N/A []
CN-14	Does the Bridge override the priority of a frame entering a port on a CNPV when in mode cptEdge?	BRG1 AND CN: M	32.2.1	Yes [] N/A []
CN-15	Does the Bridge allow any other priority to be remapped to a CNPV when in any mode other than cptDisabled?	BRG1 AND CN: M	32.2.1	No [] N/A []
CN-16	Do the system's CPs implement the specified variables and procedures?	BRG1 AND CN: M	32.7 item a), 32.8, 32.9	Yes [] N/A []
CN-17	Is the CP's Random() function initialized to a different value each time the system is reset?	BRG1 AND CN: M	32.9.1	Yes [] N/A []
CN-18	Does a system's CP interpret a CN-TAG to any degree beyond simply copying it to the CNM?	BRG1 AND CN: M	33.2.1	No [] N/A []
CN-19	Does the CP transmit a 0 in the CNM's Version field?	BRG1 AND CN: M	33.4.1	Yes [] N/A []
CN-20	Does the CP transmit a 0 in the CNM's ReservedV field?	BRG1 AND CN: M	33.4.2	Yes [] N/A []

1

A.31 Stream Reservation Protocol (SRP)

Item	Feature	Status	References	Support
	If SRP (in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
SRP-1	Does the implementation support the exchange of Multiple Stream Registration Protocol Data Units (MSRPDU), using the generic Multiple Registration Protocol Data Unit (MRPDU) format defined in 10.8 to exchange MSRP-specific information, as defined in 35.2.2.8.1, 35.2.2.9.1, and 35.2.2.10.1?	M	10.8, 35.2.2.8.1, 35.2.2.9.1, 35.2.2.10.1	Yes []

A.31 Stream Reservation Protocol (SRP) *(continued)*

Item	Feature	Status	References	Support
SRP-2	Is the MSRP Application supported as defined in Clause 35?	M	Clause 35	Yes []
SRP-3	Does the implementation propagate registration information in accordance with the operation of MAP for the Base Spanning Tree Context, as specified in 35.2.4?	M	35.2.4	Yes []
SRP-4	Does the implementation forward, filter, or discard MAC frames carrying any MRP Application address as the destination MAC address in accordance with the requirements of 8.13.6?	M	8.13.6	Yes []
SRP-5	Is the group MAC address used as the destination address for MRPDUs destined for MSRP Participants the group MAC address identified in Table 8-1, Table 8-2, and Table 8-3 as “Individual LAN Scope group address, Nearest Bridge group address”?	M	35.2.2.1, Table 8-1, Table 8-2, Table 8-3	Yes []
SRP-6	Is the EtherType used for MRPDUs destined for MSRP Participants the MSRP EtherType identified in Table 10-2?	M	35.2.2.2, Table 10-2	Yes []
SRP-7	Does the ProtocolVersion used for the implementation of MSRP take the hexadecimal value 0x01?	M	35.2.2.3	Yes []
SRP-8	Are the Attribute Type values used in the implementation as specified in 35.2.2.4 and Table 35-1?	M	35.2.2.4, Table 35-1	Yes []
SRP-9	Are the Attribute Length values used in the implementation as specified in 35.2.2.5 and Table 35-2?	M	35.2.2.5, Table 35-2	Yes []
SRP-10	Are the MSRP Vector FourPackedEvents values used in the implementation as specified in 35.2.2.7.2 and Table 35-3?	M	35.2.2.7.2, Table 35-3	Yes []
SRP-11	Does the implementation encode the values in FirstValue fields in accordance with the definition in 35.2.2.8, 35.2.2.9, and 35.2.2.10?	M	35.2.2.8, 35.2.2.9, 35.2.2.10	Yes []
SRP-12	Does the implementation update Accumulated Latency as the Talker attributes propagate through the Bridge?	M	35.2.2.8.6, 35.2.2.10.10	Yes []
SRP-13	Does the implementation update failure information in the event of insufficient bandwidth or resources through a Bridge?	M	35.2.2.8.7, 35.2.2.10.9, 35.2.2.10.12	Yes []

A.31 Stream Reservation Protocol (SRP) *(continued)*

Item	Feature	Status	References	Support
SRP-14	Does the implementation propagate a Talker Advertise as a Talker Failed in the event of insufficient bandwidth or resources through a Bridge?	M	35.2.4.3, Table 35-9	Yes []
SRP-15	Is talkerPruning and MMRP supported?	O	35.2.4.3.2	Yes [] No []
SRP-16	Are Listener attributes merged and propagated as described in 35.2.4.4?	M	35.2.4.4, Table 35-12, Table 35-15	Yes []
SRP-17	Does the implementation support updates to the Dynamic Reservation Entries as described in 35.2.4.4.2?	M	8.8 item k), 35.2.4.4.2, Table 35-13	Yes []
SRP-18	Does the implementation support updates to operIdleSlope as defined in 35.2.4.4.2?	M	35.2.4.4.2, Table 35-14	Yes []
SRP-19	Does the implementation support the automatic modifications to stream reservations as described in 35.2.6?	M	35.2.6	Yes [] No []
SRP-20	Are MSRPDU s transmitted on all ports?	M	35.2	Yes []
SRP-21	State the number of Talker registrations values that can be supported on each Port.	M	35.2.7	Number _____
SRP-22	State the number of Listener registrations values that can be supported on each Port.	M	35.2.7	Number _____
SRP-23	Can the SR_PVID for any Port be assigned the value of the null VID or VID FFF (hexadecimal)?	X	35.2.1.4 item i), Table 9-2	No []
SRP-24	Does the device support changing the SR_PVID value from the default value shown in Table 9-2?	O	35.2.1.4 item i), Table 9-2	Yes [] No []
SRP-25	Does the implementation support both original and enhanced attribute types?	M	35.1	Yes []
SRP-26	Does the implementation detect domain boundaries?	M	35.2.4.3	Yes []
SRP-27	Does the implementation support Stream transformation in Bridge?	O	35.2.2.10.5	Yes [] No []
SRP-28	Does the implementation support protocol version translation for Talker attributes?	M	35.2.4.3.1	Yes []
SRP-29	Does the implementation support protocol version translation for Listener attributes?	M	35.2.4.4.4	Yes []
SRP-30	Does the implementation declare the Domain attribute prior to Talker/Listener attributes?	M	35.2.2.9	Yes []
SRP-31	Is talker pruning per Port supported?	O	35.2.4.3.3	Yes [] No []

A.31 Stream Reservation Protocol (SRP) *(continued)*

Item	Feature	Status	References	Support
SRP-32	Is talker VLAN pruning supported?	O	35.2.4.3.4	Yes [] No []
SRPMDCSN	Does this device support media-dependent Coordinated Shared Networking (CSN) functionality on one or more ports?	SRPMDMOCA:M OR: SRPMDDOT11:M	Annex C	Yes [] No []
SRPMDMOCA	Does this device support media-dependent Multimedia over Coax Alliance (MoCA) functionality on one or more ports?	O:1	C.2	Yes [] No []
SRPMDDOT11	Does this device support media-dependent IEEE 802.11 Access Point (AP) functionality on one or more ports?	O:1	C.3	Yes [] No []
SRPMDCSN-1	Does this device support a single Designated MSRP Node (DMN)?	SRPMDCSN:M	35.1.1	Yes [] N/A []
SRPMDMOCA-1	Does this device support DMN Device Attribute Information Element to L2ME message?	SRPMDMOCA:M	C.2.1.2	Yes [] N/A []
SRPMDMOCA-2	Does this device support DMN selection?	SRPMDMOCA:M	C.2.1.3	Yes [] N/A []
SRPMDMOCA-3	Does this device support MSRP Attribute Declaration as specified in Table C-1	SRPMDMOCA:M	C.2.2, Table C-1	Yes [] N/A []
SRPMDDOT11-1	Does this device support EDCA-AC?	SRPMDDOT11:M	Table C-5	Yes [] N/A []
SRPMDDOT11-2	Is the DMN and the QAP of the IEEE 802.11 BSS co-located in the same device?	SRPMDDOT11:M	C.3.2	Yes [] N/A []
SRPMDDOT11-3	Does the device support MLME primitives specified in Table C-4?	SRPMDDOT11:M	C.3.3, Table C-4	Yes [] N/A []
SRPMDDOT11-4	Does the device support VLAN tag encapsulation/de-encapsulation on the IEEE 802.11 interface?	SRPMDDOT11:M	C.3.3.1	Yes [] N/A []
SRPMDDOT11-5	Is the reservation process an atomic operation?	SRPMDDOT11:M	C.3.1, Figure C-11, Figure C-12, Figure C-13	Yes [] N/A []

1 A.32 Multiple I-SID Registration Protocol (MIRP)

Item	Feature	Status	References	Support
MIRP-1	Does the Bridge allow both MIRP and MVRP to be enabled on the same VIP?	MIRP AND BEB-I: X	12.9.2.2.2	No [] N/A []
MIRP-2	Does the I-component implement MIRP as specified in Table 39.2?	MIRP AND BEB-I: M	39.1.1 item a)	Yes [] N/A []
MIRP-3	Does the I-component propagate attributes freely between MVRP and MIRP Participants?	MIRP AND BEB-I AND MVRP: M	39.1.1 item b)	Yes [] N/A []
MIRP-4	Does the I-component support operation as an MRP New-Only Participant?	MIRP AND BEB-I: M	39.1.1 item c)	Yes [] N/A []
MIRP-5	Does the B-component implement the MIRP application as specified?	MIRP AND BEB-I: M	39.1.2 item a)	Yes [] N/A []
MIRP-6	Does the B-component support operation as an MRP New-Only Participant?	MIRP AND BEB-I: M	39.1.2 item b)	Yes [] N/A []
MIRP-7	Does the B-component translate I-SIDs in MIRPDUs as required?	MIRP AND BEB-I: M	39.2.1.2	Yes [] N/A []
MIRP-8	Does the I-component transmit every MIRPDU with the Nearest Customer Bridge group address?	MIRP AND BEB-I: M	39.2.1.5	Yes [] N/A []
MIRP-9	Does the I-component or B-component transmit MIRPDUs with the MIRP EtherType from Table 10-2?	MIRP: M	39.2.1.7	Yes [] N/A []
MIRP-10	Does the I-component or B-component use 1 as the attribute type in MIRPDUs?	MIRP: M	39.2.1.9	Yes [] N/A []
MIRP-11	Does the I-component or B-component encode the I-SIDs in the MIRPDU as 3-octet binary values?	MIRP: M	39.2.1.10	Yes [] N/A []
MIRP-12	Does the I-component or B-component support the full range of attribute values in MIRPDUs?	MIRP: M	39.2.4	Yes [] N/A []
MIRP-13	Does the I-component support at least 4094 attribute values?	MIRP AND BEB-I: M	39.2.4	Yes [] N/A []
MIRP-14	Does the B-component support at least as many attribute values as supported by its Backbone Service Instance table?	MIRP AND BEB-B: M	39.2.4	Yes [] N/A []
MIRP-15	Does the B-component transmit at least one MIRPDU for each distinct pair of destination MAC address and VID identified by the managed variables?	MIRP AND BEB-B: M	39.2.1.6	Yes [] N/A []

2

A.33 Priority-based Flow Control (PFC)

Item	Feature	Status	References	Support
PFC-1	Enabling PFC on at least one priority	PFC: M	36.1.2	Yes [] N/A []
PFC-2	Processing PFC Requests	PFC: M	36.1.3.1	Yes [] N/A []
PFC-3	Processing PFC Indications	PFC: M	36.1.3.2	Yes [] N/A []
PFC-4	PFC delay constraints	PFC: M	36.1.3.3	Yes [] N/A []
PFC-5	PFC-aware system queue functions	PFC: M	36.2	Yes [] N/A []
PFC-6	DCBX	PFC: M	5.11	Yes [] N/A []
PFC-7	Enabling PFC on up to eight priorities	PFC: O	36.1.2	Yes [] No [] N/A []
PFC-8	PFC not enabled for traffic classes using the credit-based shaper algorithm	PFC: M	8.6.8.2	Yes [] N/A []

1

A.34 Enhanced Transmission Selection (ETS)

Item	Feature	Status	References	Support
ETS-1	Support at least 3 traffic classes	ETS:M	37.3	Yes [] N/A []
ETS-2	Support bandwidth configuration with a granularity of 1% or finer	ETS:M	37.3	Yes [] N/A []
ETS-3	Support bandwidth allocation with a precision of 10%	ETS:M	37.3	Yes [] N/A []
ETS-4	Support allocation of a portion of available bandwidth to each traffic class	ETS:M	37.3	Yes [] N/A []
ETS-5	Support DCBX.	ETS:M	Clause 38	Yes [] N/A []

1

A.35 Data Center Bridging eXchange protocol (DCBX)

Item	Feature	Status	References	Support
DCBX-1	Support LLDP	DCBX:M	IEEE Std 802.1AB	Yes [] N/A []
DCBX-2	Support the DCBX ETS Configuration TLV	DCBX:M	D.2.8	Yes [] N/A []
DCBX-3	Support the ETS Recommendation TLV	DCBX:M	D.2.9	Yes [] N/A []
DCBX-4	Support the Priority-based Flow Control Configuration TLV	DCBX:M	D.2.10	Yes [] N/A []
DCBX-5	Support the Application Priority TLV	DCBX:M	D.2.11	Yes [] N/A []
DCBX-6	Support the DCBX asymmetric state machine	DCBX:M	38.4.1	Yes [] N/A []
DCBX-7	Support the DCBX symmetric state machine	DCBX:M	38.4.2	Yes [] N/A []
DCBX-8	Support the Application VLAN TLV	DCBX:M	D.2.14	Yes [] N/A []

2

A.36 Infrastructure Protection Switching (IPS)

Item	Feature	Status	References	Support
	If IPS is not supported, mark N/A and continue at A.37.			N/A []
IPS-1	Is 1:1 IPS supported?	IPS: M	26.11.2	Yes []
IPS-2	Is M:1 IPS supported?	IPS: O	26.11.5	Yes [] No [] N/A []
IPS-3	Are the operator commands Forced Switch (FS), Lockout of Protection (LoP), Manual Switch to Working, and Manual Switch to Protection implemented?	IPS: M	26.11.2.5	Yes []
IPS-4	Is the hold-off timer implemented?	IPS: O	26.11.2.3	Yes [] No []
IPS-5	Is the M:1 wait-to-restore (MWTR) timer implemented?	IPS-2: O	26.11.5.4.7	Yes [] No [] N/A []

3

A.37 Shortest Path Bridging (SPB)

Item	Feature	Status	References	Support
SPB-1	Does the Bridge support IS-IS Link State Protocol with procedures to ensure Loop Prevention?	SPB:M	5.4.5, Clause 28	Yes [] N/A []
SPB-2	Encode, decode, and validate IS-IS Hello PDUs or SPT BPDUs for the Agreement Protocol (AP) and support AP logic in IS-IS?	SPB:M	5.4.5, Clause 28	Yes [] N/A []
SPB-3	State the maximum number of FIDs supported. Minimum is three for SPB	SPB:M	5.4.5	_____ FIDs
SPB-4	Support VLAN Registration and filtering of frames with unregistered VIDs	SPB:O	5.4.5	Yes [] No [] N/A []
SPBV	Is Shortest Path Bridging VID mode supported?	SPB: O.6	5.4.5	Yes [] No [] N/A []
SPBM	Is Shortest Path Bridging MAC mode supported?	SPB: O.6	5.4.5	Yes [] No [] N/A []

1

A.38 EVB Bridge

Item	Feature	Status	References	Support
	If EVB Bridge functionality (EVB-B in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
EVB-B-1	Does the implementation comprise a single conformant C-VLAN component?	EVB-B:M	5.5, 5.6, 5.23	Yes []
EVB-B-2	Is each externally accessible port capable of being configured as either a C-VLAN Bridge Port or a Station Facing Bridge Port (SBP)?	EVB-B:M	5.23, Clause 40	Yes []
EVB-B-3	Does the implementation support the functionality of a C-VLAN component?	EVB-B:M	5.5, 5.23	Yes []
EVB-B-4	Does the implementation support at least one SBP on the C-VLAN component?	EVB-B:M	5.23, Clause 40	Yes []
EVB-B-5	Does the implementation support the EVB status parameters for EVBMode = EVB Bridge?	EVB-B:M	5.23, 40.4	Yes []
EVB-B-6	Does the implementation support an LLDP nearest Customer Bridge database including the EVB TLV on each SBP?	EVB-B:M	5.23, D.2.12	Yes []
EVB-B-7	Does the implementation support ECP on each SBP?	EVB-B:M	5.23, Clause 43	Yes []
EVB-B-8	Does the implementation support the Bridge role of VDP on each SBP?	EVB-B:M	5.23, Clause 41	Yes []
EVB-B-9	Does the implementation support at least one Port-mapping S-VLAN component and associated UAP configured as specified in 40.22 (a)–(d)?	EVB-B:O	5.23, 22.6.4, 40.2 items a)–d)	Yes [] No []
EVB-B-10	Is each externally accessible port capable of being configured as an Uplink Access Port (UAP)?	EVB-B:O	5.23, Clause 40	Yes [] No []
EVB-B-11	Does the implementation support CDCP, as specified in Clause 42, operating in Bridge mode?	EVB-B-9: M	Clause 42, 42.3	Yes [] N/A []

A.38 EVB Bridge *(continued)*

Item	Feature	Status	References	Support
EVB-B-12	Does the implementation support the enhanced filtering utility criteria and not support the default filtering utility criteria (8.7)?	EVB-B-9:M	8.7	Yes [] N/A []
EVB-B-13	Does the implementation support configuration of reflective relay on each SBP of the C-VLAN component?	EVB-B:O	5.23, 40.4, 8.6.1	Yes [] No []
EVB-B-14	Does the implementation support management for the EVB components?	EVB-B:O	5.23, 12.4–12.12, 12.26	Yes [] No []
EVB-B-15	Does the implementation support an Simple Network Management Protocol (SNMP) management MIB module?	EVB-B:O	5.23, 17.7.20	Yes [] No []
EVB-B-16	Does the implementation support assignment of VIDs to GroupIDs?	EVB-B:O	5.23, 41.2.9	Yes [] No []
EVB-B-17	Does the implementation support the use of the M and S bits in VDP?	EVB-B:O	5.23, 41.2.3	Yes [] No []
EVB-B-18	Does the Bridge reserve the S-channel Identifier (SCID) value 1 and S-VID value 1 for the exclusive use as the un-S-tagged default S-channel?	EVB-B:M	42.1	Yes []

A.39 EVB station

Item	Feature	Status	References	Support
	If EVB station functionality (EVB-S in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
EVB-S-1	Does the EVB station comprise one or more conformant ER components?	EVB-S:M	5.6, 5.24.1	Yes []
EVB-S-2	Is each externally accessible port capable of being configured as at least one of – A UAP – An uplink relay port (URP)?	EVB-S:M	5.24, Clause 40	Yes []
EVB-S-3	Is each DRP capable of attaching its ER to one or more VSIs?	EVB-S:M	5.24, Clause 40	Yes []
EVB-S-4	Is each URP capable of attaching its ER to a point-to-point LAN connecting the URP to a CAP, or to the LAN connecting to an EVB Bridge in the case where no Port-mapping S-VLAN component is present?	EVB-S:M	5.24, Clause 40	Yes []
EVB-S-5	Does the implementation support at least one ER?	EVB-S:M	5.24, Clause 40	Yes []
EVB-S-6	Does the implementation support at least one accessible URP?	EVB-S:M	5.24, Clause 40	Yes []
EVB-S-7	Does the implementation support the EVB status parameters EVB-S: for EVBMode = EVB station on each URP?	EVB-S:M	5.24, 40.4	Yes []
EVB-S-8	Does the implementation support an LLDP Nearest Customer Bridge database including the EVB TLV on each URP of each ER?	EVB-S:M	5.24, D.2.12	Yes []
EVB-S-9	Does the implementation support ECP on each URP of each ER?	EVB-S:M	5.24, Clause 43	Yes []

A.39 EVB station *(continued)*

Item	Feature	Status	References	Support
EVB-S-10	Does the implementation support the station role of VDP for each URP of each ER?	EVB-S:M	5.24, Clause 41	Yes []
EVB-S-11	Does the implementation support a Port-mapping S-VLAN component on each Port configured as a UAP, configured as specified in 40.2 (a)–(d)?	EVB-S:O	5.24, 22.6.4, 40.2 items a)–d)	Yes [] No []
EVB-S-12	Does the implementation support CDCP, as specified in Clause 42, operating in Station mode?	EVB-S-11: M	Clause 42, 42.3	Yes [] N/A []
EVB-S-13	Does the implementation support the enhanced filtering utility criteria (8.7.2) and not support the default filtering utility criteria (8.7.1)?	EVB-S-11: M	8.7.1, 8.7.2	Yes [] N/A []
EVB-S-14	Does the implementation support multiple ERs?	EVB-S:O	5.24, Clause 40	Yes [] No []
EVB-S-15	Does the implementation support management for the EVB components?	EVB-S:O	5.24, 12.26	Yes [] No []
EVB-S-16	Does the implementation support an EVB station SNMP management MIB module?	EVB-S:O	5.24, 17.7.20	Yes [] No []
EVB-S-17	Does the implementation support assignment of VIDs to GroupIDs?	EVB-S:O	5.24, 41.2.9	Yes [] No []
EVB-S-18	Does the implementation support Support the use of the M and S bits in VDP?	EVB-S:O	5.24, 41.2.3	Yes [] No []

¹ A.40 Edge relay (ER)

Item	Feature	Status	References	Support
	If EVB station functionality (EVB-S in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
ERC-1	Does the ER conform to the relevant standard for the MAC technology implemented at each Port in support of the MAC ISS, as specified in IEEE Std 802.1AC, and 6.14?	EVB-S:M	IEEE Std 802.1AC , 6.14	Yes []
ERC-2	Does the ER support the MAC Enhanced Internal Sublayer Service (EISS) at each Port, as specified in 6.8 and 6.9?	EVB-S:M	6.8, 6.9	Yes []
ERC-3	Does the ER recognize and use C-TAGs?	EVB-S:M	6.9	Yes []
ERC-4	Does the ER relay and filter frames as described in 8.1 and specified in 8.5, 8.6, 8.7, and 8.8?	EVB-S:M	8.5, 8.6, 8.8	Yes []
ERC-5	Does the ER support a PVID value, and configuration of at least one VID whose untagged set includes that Port, on each DRP that supports untagged and priority-tagged frames?	EVB-S:M	6.9, 8.8.2	Yes []
ERC-6	Does the ER support setting the Acceptable Frame Types parameter to <i>Admit Only VLAN-tagged Frames</i> on the URP?	EVB-S:M	5.24.1, 6.9	Yes []

A.40 Edge relay (ER) *(continued)*

Item	Feature	Status	References	Support
ERC-7	Does the ER allow tag headers to be inserted, modified, and removed from relayed frames, as specified in 8.2 and Clause 9, as required by the value(s) of the Acceptable Frame Types parameter supported on each Port, and by the ability of each Port to transmit VLAN-tagged and/or untagged frames?	EVB-S:M	8.1, Clause 9	Yes []
ERC-8	Does the ER support at least one FID?	EVB-S:M	IEEE Std 802.1AC, 8.8.3, 8.8.8, 8.8.9	Yes []
ERC-9	Does the ER allow allocation of at least one VID to each FID that is supported?	EVB-S:M	IEEE Std 802.1AC, 8.8.3, 8.8.8, 8.8.9	Yes []
ERC-10	Does the ER support exactly one URP supporting the parameters of 40.4 for EVBMode = EVB station?	EVB-S:M	5.24.1, 40.4, Clause 40	Yes []
ERC-11	Does the ER support one or more DRPs each supporting access to VSIs?	EVB-S:M	5.24.1, Clause 40	Yes []
ERC-12	Does the ER filter the reserved MAC addresses?	EVB-S:M	5.24.1, Table 8-1	Yes []
ERC-13	Does the ER support more than one DRP?	EVB-S:O	5.24.1	Yes [] No []
ERC-14	Does the ER support setting the Enable Ingress Filtering parameter (8.6.2) on each DRP?	ERC-13:M	5.24.1, 8.6.2	Yes [] N/A []
ERC-15	Does the ER support setting the Enable Ingress Filtering parameter (8.6.2) on each URP?	ERC-13:M	5.24.1, 8.6.2	Yes [] N/A []
ERC-16	Does the ER support the requirements of either a VEB ER or a VEPA ER?	ERC-13:M	5.24.1, 5.24.1.1, 5.24.1.2	Yes []
ERC-17	Does the ER support a PVID value, and configuration of at least one VID whose untagged set includes that Port, if the URP supports untagged and priority-tagged frames?	EVB-S:O	6.9, 8.8.2	Yes []
ERC-18	Does the ER comprise a single conformant C-VLAN component?	EVB-S:O	5.4	Yes [] No []
ERC-19	Does the ER support disabling of learning on each DRP?	EVB-S:O	5.24.1, 8.6.1	Yes [] No []
ERC-20	Does the ER support discarding frames with unregistered source addresses at each DRP?	EVB-S:O	5.24.1, 8.8.1	Yes [] No []
ERC-21	Does the ER support the operation of the learning process?	EVB-S:O	8.7	Yes [] No []

A.41 VEB and VEPA ER components

Item	Feature	Status	References	Support
	If EVB station functionality (EVB-S in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
VERC-1	Does the ER component support VEB functionality?	EVB-S:O.6	5.24.1.1	Yes [] No []
VERC-2	Does the ER component support VEPA functionality?	EVB-S:O.6	5.24.1.2	Yes [] No []
VERC-3	Does the ER component request that reflective relay service not be provided by setting adminReflectiveRelayRequest to FALSE?	VERC-1:M	5.24.1.1	Yes [] N/A []
VERC-4	Does the VEPA ER disable learning on the URP?	VERC-2:M	5.24.1.2, 8.6.1	Yes [] N/A []
VERC-5	Does the VEPA ER filter frames received at each URP that are destined to a DRP that originated the frame?	VERC-2:M	5.24.1.2, 8.6.1	Yes [] N/A []
VERC-6	Does the VEPA ER request reflective relay service by setting adminReflectiveRelayRequest to True?	VERC-2:O	5.24.1.2, 40.4	Yes [] N/A []
VERC-7	Does the ER filter frames received at each DRP that are destined for the URP until reflective relay is enabled?	VERC-2:O	5.24.1.2, 8.6.1.1	Yes [] No [] N/A []
VERC-8	Does the ER forward frames as specified in 8.6.3.1?	VERC-2:M	5.24.1.2, 8.6.3.1	Yes [] N/A []

1

A.42 VDP, CDCP, and ECP

Item	Feature	Status	References	Support
	If neither EVB station functionality (EVB-S in A.5) nor EVB Bridge functionality (EVB-B in A.5) is supported, mark N/A and ignore the remainder of this table.			N/A []
VDP-1	Does the implementation support the Bridge VDP state machine as specified in Clause 41?	EVB-B:M	Clause 41, 41.5.2	Yes []
VDP-2	Does the implementation support the Station VDP state machine as specified in Clause 41?	EVB-S:M	Clause 41, 41.5.3	Yes []
CDCP-1	Does the implementation support the CDCP configuration state machine for the Bridge role, as specified in Clause 42?	EVB-B AND EVB-B-9: M	Clause 42, 42.3	Yes [] N/A []
CDCP-2	Does the implementation support the CDCP configuration state machine for the station role, as specified in Clause 42?	EVB-S AND EVB-S-11: M	Clause 42, 42.3	Yes [] N/A []
ECP-1	Does the implementation support the ECP transmit state machine as specified in Clause 43?	EVB-S AND EVB-B:M	Clause 43, 43.3.4	Yes []
ECP-2	Does the implementation support the ECP receive state machine as specified in Clause 43?	EVB-S AND EVB-B:M	Clause 43, 43.3.5	Yes []

1 A.43 Path Control and Reservation

Item	Feature	Status	References	Support
	If Path Control and Reservation (PCR in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
PCR-1	Does the Bridge support the ISIS-PCR protocol?	PCR:M	Clause 45	Yes []
PCR-2	Does the Bridge support the SPB Link Metric sub-TLV?	PCR:M	28.12.7	Yes []
PCR-3	Does the Bridge support the SPB Base VLAN-Identifiers sub TLV?	PCR:M	28.12.4	Yes []
PCR-4	Does the Bridge support the SPB Instance sub-TLV?	PCR:M	28.12.5	Yes []
PCR-5	Does the Bridge support the SPBV MAC address sub-TLV?	PCR:M	28.12.9	Yes []
PCR-6	Does the Bridge support the SPBM Service Identifier and Unicast Address sub-TLV?	BEB AND PCR:M	28.12.10	Yes [] N/A []
PCR-7	Does the Bridge support the Topology sub-TLV?	PCR:M	45.1.9	Yes []
PCR-8	Does the Bridge support the Hop sub-TLV?	PCR:M	45.1.10	Yes []
PCR-9	Does the Bridge support the ST ECT Algorithm?	PCR:M	45.1.2	Yes []
PCR-10	Does the Bridge support the LT ECT Algorithm?	PCR:O	45.1.2	Yes [] No []
PCR-11	Does the Bridge support the LTS ECT Algorithm?	PCR:O	45.1.2	Yes [] No []
PCR-12	Does the Bridge support the MRT ECT Algorithm?	PCR:O	45.1.2, 45.3.3	Yes [] No []
PCR-13	Does the Bridge support the MRTG ECT Algorithm?	PCR:O	45.1.2, 45.3.4	Yes [] No []
PCR-14	Does the Bridge support the Extended IS Reachability TLV?	PCR:O	45.1.8, IETF RFC 5305	Yes [] No []
PCR-15	Does the Bridge support the Unidirectional Link Delay sub-TLV?	PCR:O	45.1.8, IETF RFC 7810	Yes [] No []
PCR-16	Does the Bridge support the Min/Max Unidirectional Link Delay sub-TLV?	PCR:O	45.1.8, IETF RFC 7810	Yes [] No []
PCR-17	Does the Bridge support the Unidirectional Delay Variation sub-TLV?	PCR:O	45.1.8, IETF RFC 7810	Yes [] No []
PCR-18	Does the Bridge support the Unidirectional Link Loss sub-TLV?	PCR:O	45.1.8, IETF RFC 7810	Yes [] No []
PCR-19	Does the Bridge support the Unidirectional Residual Bandwidth sub-TLV?	PCR:O	45.1.8, IETF RFC 7810	Yes [] No []
PCR-20	Does the Bridge support the Unidirectional Available Bandwidth sub-TLV?	PCR:O	45.1.8, IETF RFC 7810	Yes [] No []
PCR-21	Does the Bridge support the Unidirectional Utilized Bandwidth sub-TLV?	PCR:O	45.1.8, IETF RFC 7810	Yes [] No []

A.43 Path Control and Reservation *(continued)*

Item	Feature	Status	References	Support
PCR-22	Does the Bridge support the Shared Risk Link Group TLV?	PCR:O	45.1.8, IETF RFC 5307	Yes [] No []
PCR-23	Does the Bridge support the Administrative Group sub-TLV?	PCR:O	45.1.11	Yes [] No []
PCR-24	Does the Bridge support the Bandwidth Constraint sub-TLV?	PCR:O	45.1.12	Yes [] No []
PCR-25	Does the Bridge support the Bandwidth Assignment sub-TLV?	PCR:O	45.2.1	Yes [] No []
PCR-26	Does the Bridge support the Timestamp sub-TLV?	PCR:O	45.2.2	Yes [] No []
PCR-27	Does the Bridge support Loop-Free Alternates for unicast data flows supported by SPBM?	SPBM:O PCR:O	45.3.1	Yes [] No [] N/A []
PCR-28	Does the Bridge support relaxed ingress checking?	SPBM:O PCR:O	8.4, 8.6.1, 8.8.3, 45.3.1	Yes [] No [] N/A []

1

A.44 Scheduled traffic

Item	Feature	Status	References	Support
	If neither scheduled traffic (SCHED in A.5) nor cyclic queuing and forwarding (CQF in A.5) are supported, mark N/A and ignore the remainder of this table.		5.4.1, 5.13.1, 8.6.8, 8.6.9, 12.29, 17.7.22	N/A []
SCHED1	Does the implementation support the state machines and associated definitions specified in 8.6.9	SCHED OR CQF:M	5.4.1, 5.13.1, 8.6.8, 8.6.9	Yes [] N/A []
SCHED2	Does the implementation support the management entities defined in 12.29?	SCHED OR CQF:M	5.4.1 item ad), 5.4.1.9 item c), 5.13.1.2 item c), 12.29	Yes [] N/A []
SCHED3	Is the IEEE8021-ST-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND (SCHED OR CQF):O	5.4.1 item ad), 5.4.1.9 item c), 5.13.1.2 item c), 12.29, 17.7.22	Yes [] N/A [] No []

2

A.45 Frame preemption

Item	Feature	Status	References	Support
	If frame preemption (PRE in A.5) is not supported, mark N/A and ignore the remainder of this table.		5.4.1, 5.13.1, 6.7.2, 8.6.8, 12.30, 17.7.23	N/A []
PRE1	Does the implementation support the functionality of frame preemption as specified in 6.7.2 and 8.6.8?	PRE: M	5.4.1, 5.13.1, 6.7.2, 8.6.8	Yes [] N/A []

3

A.46 Per-Stream Filtering and Policing

Item	Feature	Status	References	Support
	If neither Per-Stream Filtering and Policing (PSFP in A.5) nor cyclic queuing and forwarding (CQF in A.5) are supported, mark N/A and ignore the remainder of this table.		5.4.1.9, 5.13.1.2, 8.6.5.2, 8.6.10, 12.31, 17.7.24	N/A []
PSFP1	Does the implementation support the state machines and associated definitions as specified in 8.6.10?	PSFP OR CQF:M	5.4.1.9 item b), 5.13.1.2 item b), 8.6.5, 8.6.10	Yes [] N/A []
PSFP2	Does the implementation support the management entities defined in 12.31?	PSFP OR CQF:M	5.4.1.9 item e), 5.13.1.2 item e), 8.6.5.2, 8.6.10, 12.31	Yes [] N/A []
PSFP3	Is the IEEE8021-PSFP-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND (PSFP OR CQF):O	5.4.1.9 item e), 5.13.1.2 item e), 12.31, 17.7.24	Yes [] N/A [] No []

1 A.47 YANG

Item	Feature	Status	References	Support
	If item YANG is not supported, mark N/A			N/A []
YANG-802-TYPES	Is the <i>ieee802-types</i> module supported?	M	48.6.1	Yes []
YANG-Q-TYPES	Is the <i>ieee802-dot1q-types</i> module supported?	M	48.6.2	Yes []
YANG-TSN-TYPES	Is the <i>ieee802-dot1q-tsn-types</i> module supported?	O	48.6.3	Yes [] No []
YANG-QBRIDGE	Is the <i>ieee802-dot1q-bridge</i> module supported?	M	48.6.4	Yes [] N/A []
YANG-TPMR	Is the <i>ieee802-dot1q-tpmr</i> module supported?	TPMR:O	48.6.5	Yes [] No [] N/A []
YANG-PB	Is the <i>ieee802-dot1q-pb</i> module supported?	PB:O	48.6.6	Yes [] No [] N/A []
YANG-CFM-TYPES	Is the <i>ieee802-dot1q-cfm-types</i> module supported?	CFM:O	48.6.7	Yes [] No [] N/A []
YANG-CFM	Is the <i>ieee802-dot1q-cfm</i> module supported?	CFM:O	48.6.8	Yes [] No [] N/A []
YANG-CFM-BRIDGE	Is the <i>ieee802-dot1q-cfm-bridge</i> module supported?	CFM:O	48.6.9	Yes [] No [] N/A []
YANG-CFM-ALARM	Is the <i>ieee802-dot1q-cfm-alarm</i> module supported?	CFM:O	48.6.10	Yes [] No [] N/A []
YANG-STREAMS	Is the <i>ieee802-dot1q-stream-filters-gates</i> module supported?	ATS:O	48.6.11	Yes [] No [] N/A []
YANG-STREAMS-BRIDGE	Is the <i>ieee802-dot1q-stream-filters-gates-bridge</i> module supported?	ATS:O	48.6.12	Yes [] No [] N/A []
YANG-ATS	Is the <i>ieee802-dot1q-ats</i> module supported?	ATS:O	48.6.13	Yes [] No [] N/A []
YANG-ATS-BRIDGE	Is the <i>ieee802-dot1q-ats-bridge</i> module supported?	ATS:O	48.6.14	Yes [] No [] N/A []
YANG-CI	Is the <i>ieee802-dot1q-congestion-isolation</i> module supported?	CI:O	48.6.15	Yes [] No [] N/A []
YANG-CI-BRIDGE	Is the <i>ieee802-dot1q-congestion-isolation-bridge</i> module supported?	CI:O	48.6.16	Yes [] No [] N/A []
YANG-SCHED	Is the <i>ieee802-dot1q-sched</i> module supported?	SCHED or CQF:O	48.6.17	Yes [] No [] N/A []
YANG-SCHED-BRIDGE	Is the <i>ieee802-dot1q-sched-bridge</i> module supported?	SCHED or CQF:O	48.6.18	Yes [] No [] N/A []
YANG-PREEMP	Is the <i>ieee802-dot1q-preemption</i> module supported?	PRE:O	48.6.19	Yes [] No [] N/A []
YANG-PREEMP-BRIDGE	Is the <i>ieee802-dot1q-preemption-bridge</i> module supported?	PRE:O	48.6.20	Yes [] No [] N/A []

A.47 YANG (*continued*)

YANG-PSFP	Is the <i>ieee802-dot1q-psfp</i> module supported?	PSFP or CQF:O	48.6.21	Yes [] N/A []	No []
YANG-PSFP-BRIDGE	Is the <i>ieee802-dot1q-psfp-bridge</i> module supported?	PSFP or CQF:O	48.6.22	Yes [] N/A []	No []
YANG-LLDP-PBBN-AA	Is the <i>ieee802-dot1q-lldp-pbbn-aa-tlv</i> module supported?	YANG AND (AAB OR AAD): O	D.6.6.7	Yes [] N/A []	No []
YANG-CBSA	Is the <i>ieee802-dot1q-cbsa</i> module supported?	CBSA:O	48.6.24	Yes []	No []
YANG-CBSA-BRIDGE	Is the <i>ieee802-dot1q-cbsa-bridge</i> module supported?	CBSA:O	48.6.25	Yes []	No []
YANG-RSTP	Is the <i>ieee802-dot1q-rstp</i> module supported?	RSTP:O	48.6.26	Yes [] N/A []	No []
YANG-MSTP	Is the <i>ieee802-dot1q-mstp</i> module supported?	MSTP:O	48.6.28	Yes [] N/A []	No []

1

A.48 Stream reservation remote management (SRRM)

Item	Feature	Status	References	Support
	If Stream reservation remote management functionality (SRRM of A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
SRRM-1	What management protocol standard(s) or specification(s) are supported (server side)?	M	5.4.1.11(b)	_____ _____ _____
SRRM-2	Does the implementation report delay through the Bridge?	M	12.32.1	Yes []
SRRM-3	Does the implementation report propagation delay?	M	12.32.2	Yes []
SRRM-4	Does the implementation support Static Trees for static configuration of spanning trees?	M	5.4.1.11 item c), 12.32.3	Yes []
SRRM-5	Does the implementation support MRP External Control for the MSRP application?	O SRP: M	12.32.4	Yes [] No [] N/A []
SRRM-6	Does the implementation support MRP External Control for the MVRP application?	O MVRP: M	12.32.4	Yes [] No [] N/A []
SRRM-7	Does the implementation support MRP External Control for the MMRP application?	O MMRP: M	12.32.4	Yes [] No [] N/A []
SRRM-8	Does the implementation support queue reservation for traffic classes using the strict priority algorithm, through configuration of adminIdleSlope?	M	5.4.1.11 item e), 12.20.1, 34.3	Yes []

2

A.49 TSN Centralized Network Configuration (CNC) station

Item	Feature	Status	References	Support
	If the functionality of a Centralized Network Configuration station (CNC-S of A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
CNC-S-1	What management protocol standard(s) or specification(s) are supported (client side)?	M	5.29 item a)	_____ _____ _____
CNC-S-2	Does the implementation support the managed object definitions and encodings for Stream reservation remote management?	M	5.29 item b), 12.32	Yes []
CNC-S-3	Does the implementation support the managed object definitions and encodings for scheduled traffic?	O	12.29	Yes [] No []
CNC-S-4	Does the implementation support the managed object definitions and encodings for frame preemption?	O	12.30	Yes [] No []
CNC-S-5	Does the implementation support the managed object definitions and encodings for IEEE Std 802.1AS?	O	IEEE Std 802.1AS	Yes [] No []
CNC-S-6	Does the implementation support the managed object definitions and encodings for IEEE Std 802.1CB?	O	IEEE Std 802.1CB	Yes [] No []
CNC-S-7	Does the implementation support MRP External Control for the MSRP application?	O	12.32.4	Yes [] No []
CNC-S-8	Does the implementation support MRP External Control for the MVRP application?	O	12.32.4	Yes [] No []
CNC-S-9	Does the implementation support MRP External Control for the MMRP application?	O	12.32.4	Yes [] No []
CNC-S-10	What user/network configuration protocol standard(s) or specification(s) are supported?	M	5.29 item c), 46.2.2	_____ _____ _____
CNC-S-11	Does the implementation support the YANG modules identified in 46.3?	O	5.29 item e), 46.2, 46.3	Yes []
CNC-S-13	Does the implementation conform to the conditional requirements for use of SRP?	M	5.29 item d), 46.2, 12.32.4	Yes []

A.50 VDP for NVO3 nNVE Devices

Item	Feature	Status	References	Support
	If VDP-NVO3 nNVE functionality (VDP-NVO3-nNVE in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
VDP-nNVE-1	Does the implementation support the Bridge role of VDP on each SBP?	M	5.30.1, Clause 41	Yes []
VDP-nNVE-2	Does the implementation support the Bridge VDP state machine as specified in Clause 41?	M	Clause 41, 41.5.2	Yes []
VDP-nNVE-3	Does the implementation support assignment of VIDs to GroupIDs?	M	5.30.1, 41.2.9	Yes []
VDP-nNVE-4	Does the implementation support at least one SBP on the nNVE?	M	5.30.1, Clause 40	Yes []
VDP-nNVE-5	Does the implementation support an LLDP nearest Customer Bridge database including the EVB TLV on each SBP?	O	5.30.1, D.2.12	Yes [] No []
VDP-nNVE-6	Does the implementation support the EVB status parameters for EVBMode = NVO3 and NVERole = nNVE for the nNVE role?	O	5.30.1, 40.4, 40.5	Yes [] No []
VDP-nNVE-7	Does the implementation support the EVB Bridge status parameters for IPv4 address capability?	VDP-nNVE-6:M	D.2.12.3.5	Yes [] No []
VDP-nNVE-8	Does the implementation support the EVB Bridge status parameters for IPv6 address capability?	VDP-nNVE-6:M	D.2.12.3.4	Yes [] No []
VDP-nNVE-9	Does the implementation support ECP on each SBP?	O	5.30.1, Clause 43	Yes [] No []
VDP-nNVE-10	Does the implementation support the use of M, S, and N bits in VDP?	O	5.30.1, 41.2.3	Yes [] No []
VDP-nNVE-11	Does the implementation support the use of IPv4 addresses in VDP filter info format?	O	5.30.1, 41.2.9	Yes [] No []
VDP-nNVE-12	Does the implementation support the use of IPv6 addresses in VDP filter info format?	O	5.30.1, 41.2.9	Yes [] No []
VDP-nNVE-13	Does the implementation support an LLDP database addressed by a unicast MAC address including the EVB TLV on each SBP?	O	5.30.1, D.2.12	Yes [] No []

A.51 VDP for NVO3 tNVE Devices

Item	Feature	Status	References	Support
	If VDP-NVO3 tNVE functionality (VDP-NVO3-tNVE in A.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
VDP-tNVE-1	Does the implementation support the Station role of VDP on each URP?	M	5.30.1, Clause 41	Yes []
VDP-tNVE-2	Does the implementation support the Station VDP state machine as specified in Clause 41?	M	Clause 41, 41.5.3	Yes []
VDP-tNVE-3	Does the implementation support an LLDP nearest Customer Bridge database including the EVB TLV on each URP?	O	5.30.2, D.2.12	Yes [] No []
VDP-tNVE-4	Does the implementation support the EVB status parameters for EVBMode = NVO3 and NVERole = tNVE for the tNVE role?	O	5.30.2, 40.4, 40.5	Yes [] No []
VDP-tNVE-5	Does the implementation support the EVB Bridge status parameters for IPv4 address capability?	VDP-tNVE-4:M	D.2.12.4.5	Yes [] No []
VDP-tNVE-6	Does the implementation support the EVB Bridge status parameters for IPv6 address capability?	VDP-tNVE-4:M	D.2.12.4.4	Yes [] No []
VDP-tNVE-7	Does the implementation support ECP on each URP?	O	5.30.2, Clause 43	Yes [] No []
VDP-tNVE-8	Does the implementation support the use of M, S, and N bits in VDP?	O	5.30.2, 41.2.3	Yes [] No []
VDP-tNVE-9	Does the implementation support the use of IPv4 addresses in VDP filter info format?	O	5.30.2, 41.2.9	Yes [] No []
VDP-tNVE-10	Does the implementation support the use of IPv6 addresses in VDP filter info format?	O	5.30.2, 41.2.9	Yes [] No []
VDP-tNVE-11	Does the implementation support an LLDP database addressed by a unicast MAC address including the EVB TLV on each URP?	O	5.30.2, D.2.12	Yes [] No []
VDP-tNVE-12	Does the RRREQ in EVB station status parameters set to FALSE to make reflective relay always disabled?	VDP-tNVE-4:M	D.2.12.4	Yes [] No []

A.52 Asynchronous Traffic Shaping

Item	Feature	Status	References	Support
	If Asynchronous Traffic Shaping (ATS in A.5) is not supported, mark N/A and ignore the remainder of this table.		5.4.1.10, 5.13.1.3, 8.6.5.2.2, 8.6.6 items d) and e), 8.6.8.5, 8.6.8, 8.6.8.5, 8.6.11, 12.31	N/A []
ATS-1	Does the implementation support the ATS per-stream classification and metering for ATS as specified in 8.6.5.2.2?	ATS:M	5.4.1.10, 5.13.1.3, 8.6.5.2.2	Yes []
ATS-2	Does the implementation support the ATS transmission selection algorithm as specified in 8.6.8.5?	ATS:M	5.4.1.10, 5.13.1.3, 8.6.8.5	Yes []
ATS-3	Does the implementation support the ATS scheduler state machines as specified in 8.6.11?	ATS:M	5.4.1.10, 5.13.1.3, 8.6.11	Yes []
ATS-4	Does the implementation support the management entities defined in 12.31 for ATS?	ATS:M	5.4.1.10, 5.13.1.3, 12.31	Yes []

1

A.53 Congestion Isolation

Item	Feature	Status	References	Support
	If the functionality of Congestion Isolation (CI of A.5) is not supported, mark N/A and ignore the remainder of this table.		5.4.7, 8.6.5.2.3, 8.6.5.3, 8.6.8.6, 12.33, Clause 49	N/A []
CI-1	Does the Bridge support the creation of at least one CIP on at least one Port?	BRG1 AND CI:M	5.4.75.4.7 item a), 49.3	Yes []
CI-2	Does the implementation support IETF RFC 3168 Explicit Congestion Notification?	CI:M	49.2.1, IETF RFC 3168	Yes []
CI-3	Does the implementation support per-stream classification and metering for CI as specified in 8.6.5.2.3?	CI:M	5.4.75.4.7 item c), 8.6.5.2.3	Yes []
CI-4	Does the implementation support the variables and procedures of the Congestion Isolation Protocol?	CI:M	49.4	Yes []
CI-5	Does the implementation support the ability to configure the variables controlling the operation of Congestion Isolation?	CI:M	12.33	Yes []
CI-6	Does the implementation support LLDP?	CI:M	IEEE Std 802.1AB	Yes []
CI-7	Does the implementation support the Congestion Isolation TLV in LLDP?	CI:M	D.2.15	Yes []
CI-8	Does the implementation support the ability to monitor more than one queue on a Bridge Port?	CI:O	49.4.1.2.5	Yes [] No []
CI-9	Does the implementation support transmission selection algorithms other than strict priority?	CI:O	49.2.3, 49.3.2	Yes [] No []
CI-10	Does the implementation support the ability to return a flow to non-congesting status when the congesting queue is not empty?	CI:O	49.2.6	Yes [] No []
CI-11	Does the implementation support Topology Recognition?	CI:O	49.5	Yes [] No []

2

1

A.54 PBBN auto attach

Item	Feature	Status	References	Support
	If auto attach is not supported, mark N/A	(AAB OR AAD):M	5.9.2, 5.10.3, 5.12.2, 5.14.2, 5.33	N/A []
AAB-1	Does the AAB support one or more I-components?	AAB:M	5.12.2	Yes [] N/A []
AAB-2	Does the AAB support the Link Layer Discovery Protocol (LLDP) transmit and receive mode?	AAB:M	IEEE Std 802.1AB™	Yes [] N/A []
AAB-3	Does the AAB implementation support the PBBN Auto Attach System TLV?	AAB:M	5.12.2, D.2.17	Yes [] N/A []
AAB-4	Does the AAB implementation support the PBBN Auto Attach Assignment TLV?	AAB:M	5.12.2, D.2.18	Yes [] N/A []
AAB-5	Does the AAB support the C-tagging interface?	AAB:M	5.12.2, 50.2, D.2.17.8	Yes [] N/A []
AAB-6	Does the AAB support the AAB State Machine	AAB:M	5.12.2, 50.7	Yes [] N/A []
AAB-7	Does the AAB support the 802.1Q Management VID TLV as specified in D.2.6?	O	D.2.6	Yes [] No [] N/A []
AAD-1	Does the AAD support the Link Layer Discovery Protocol (LLDP) transmit and receive mode?	AAD:M	IEEE Std 802.1AB™	Yes [] N/A []
AAD-2	Does the AAD implementation support the PBBN Auto Attach System TLV?	AAD:M	5.14.2, 5.9.2, D.2.17	Yes [] N/A []
AAD-3	Does the AAD implementation support the PBBN Auto Attach Assignment TLV?	AAD:M	5.14.2, 5.9.2, D.2.18	Yes [] N/A []
AAD-4	Does the AAD support the AAD State Machine	AAD:M	5.14.2, 5.9.2, 50.6	Yes [] N/A []

2

A.55 Credit-based shaper algorithm

Item	Feature	Status	References	Support
	If CBSA is not supported, mark N/A and ignore the remainder of this table.			N/A []
CBSA-1	Does the implementation support the algorithm and associated parameters specified in 8.6.8.2?	CBSA:M	8.6.8.2	Yes [] No []
CBSA-2	Does the implementation support the management parameters adminIdleSlope and operIdleSlope as defined in 12.20.1?	CBSA:M	12.20.1	Yes [] No []

3

1 **Annex B**

2 (normative)

3 **PICS proforma—End station implementations**⁵⁶

4 **B.1 Introduction**

5 The supplier of a protocol implementation that is claimed to conform to this standard shall complete the
6 following Protocol Implementation Conformance Statement (PICS) proforma.

7 A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of
8 which capabilities and options of the protocol have been implemented. The PICS can have a number of uses,
9 including use:

- 10 a) By the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard
11 through oversight.
- 12 b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication
13 of the capabilities of the implementation, stated relative to the common basis for understanding
14 provided by the standard PICS proforma.
- 15 c) By the user—or potential user—of the implementation, as a basis for initially checking the
16 possibility of interworking with another implementation (note that, while interworking can never be
17 guaranteed, failure to interwork can often be predicted from incompatible PICSs).
- 18 d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for
19 conformance of the implementation.

20 **B.2 Abbreviations and special symbols**

21 **B.2.1 Status symbols**

22	M	mandatory
23	O	optional
24	<i>O.n</i>	optional, but support of at least one of the group of options labeled by the same numeral n
25		is required
26	X	prohibited
27	pred:	conditional-item symbol, including predicate identification: see B.3.4
28	¬	logical negation, applied to a conditional item's predicate

29 **B.2.2 General abbreviations**

30	N/A	not applicable
31	PICS	Protocol Implementation Conformance Statement

⁵⁶ *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

1 B.3 Instructions for completing the PICS proforma

2 B.3.1 General structure of the PICS proforma

3 The first part of the PICS proforma, implementation identification and protocol summary, is to be completed
4 as indicated with the information necessary to identify fully both the supplier and the implementation.

5 The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each
6 containing a number of individual items. Answers to the questionnaire items are to be provided in the
7 rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No), or
8 by entering a value or a set or range of values. (Note that there are some items where two or more choices
9 from a set of possible answers can apply; all relevant choices are to be marked.)

10 Each item is identified by an item reference in the first column. The second column contains the question to
11 be answered; the third column records the status of the item—whether support is mandatory, optional, or
12 conditional: see also B.3.4 below. The fourth column contains the reference or references to the material that
13 specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

14 A supplier may also provide (or be required to provide) further information, categorized as either Additional
15 Information or Exception Information. When present, each kind of further information is to be provided in a
16 further subclause of items labeled A_i or X_i , respectively, for cross-referencing purposes, where i is any
17 unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format
18 and presentation.

19 A completed PICS proforma, including any Additional Information and Exception Information, is the
20 Protocol Implementation Conformation Statement for the implementation in question.

21 NOTE—Where an implementation is capable of being configured in more than one way, a single PICS may be able to
22 describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering
23 some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of
24 the information.

25 B.3.2 Additional information

26 Items of Additional Information allow a supplier to provide further information intended to assist the
27 interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS
28 can be considered complete without any such information. Examples might be an outline of the ways in
29 which a (single) implementation can be set up to operate in a variety of environments and configurations, or
30 information about aspects of the implementation that are outside the scope of this standard but that have a
31 bearing upon the answers to some items.

32 References to items of Additional Information may be entered next to any answer in the questionnaire, and
33 may be included in items of Exception Information.

34 B.3.3 Exception information

35 It may occasionally happen that a supplier will wish to answer an item with mandatory status (after any
36 conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer
37 will be found in the Support column for this: instead, the supplier shall write the missing answer into the
38 Support column, together with an X_i reference to an item of Exception Information, and shall provide the
39 appropriate rationale in the Exception item itself.

40 An implementation for which an Exception item is required in this way does not conform to this standard.

41 NOTE—A possible reason for the situation described previously is that a defect in this standard has been reported, a
42 correction for which is expected to change the requirement not met by the implementation.

1 **B.3.4 Conditional status**

2 **B.3.4.1 Conditional items**

3 The PICS proforma contains a number of conditional items. These are items for which both the applicability
4 of the item itself, and its status if it does apply—mandatory or optional—are dependent upon whether
5 certain other items are supported.

6 Where a group of items is subject to the same condition for applicability, a separate preliminary question
7 about the condition appears at the head of the group, with an instruction to skip to a later point in the
8 questionnaire if the “Not Applicable” answer is selected. Otherwise, individual conditional items are
9 indicated by a conditional symbol in the Status column.

10 A conditional symbol is of the form “**pred:** S” where **pred** is a predicate as described in B.3.4.2 below, and
11 S is a status symbol, M or O.

12 If the value of the predicate is true (see B.3.4.2), the conditional item is applicable, and its status is indicated
13 by the status symbol following the predicate: the answer column is to be marked in the usual way. If the
14 value of the predicate is false, the “Not Applicable” (N/A) answer is to be marked.

15 **B.3.4.2 Predicates**

16 A predicate is one of the following:

- 17 a) An item-reference for an item in the PICS proforma: the value of the predicate is true if the item is
18 marked as supported, and is false otherwise.
- 19 b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item-
20 references using the Boolean operator OR: the value of the predicate is true if one or more of the
21 items is marked as supported.
- 22 c) The logical negation symbol “¬” prefixed to an item-reference or predicate-name: the value of the
23 predicate is true if the value of the predicate formed by omitting the “¬” symbol is false, and vice
24 versa.

25 Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for
26 grouped conditional items, is indicated by an asterisk in the Item column.

1 B.4 PICS proforma for IEEE Std 802.1Q—End station implementations

B.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification.	
NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

B.4.2 Protocol summary, IEEE Std 802.1Q

Identification of protocol specification	IEEE Std 802.1Q-2022, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks			
Identification of amendments and corrigenda to the PICS proforma that have been completed as part of the PICS	Amd.	:	Corr.	:
	Amd.	:	Corr.	:
Have any Exception items been required? (See A.3.3: the answer Yes means that the implementation does not conform to IEEE Std 802.1Q)	No []		Yes []	
Date of Statement				

1 B.5 Major capabilities

Item	Feature	Status	References	Support
MRPAP	Does the implementation support any MRP applications? If “No” is marked, continue at FQTSSE.	O	5.18.1	Yes [] No []
MMRP	Is the operation of MMRP supported?	O.1	5.18.1, B.6	Yes [] No []
MVRP	Is automatic configuration and management of VLAN topology using MVRP supported?	O.1	5.18.1, B.7	Yes [] No []
MSRP	Is the operation of MSRP supported?	O.1	5.18.3, B.10	Yes [] No []
MRP	Is the Multiple Registration Protocol implemented in support of MRP Applications?	M	Clause 10, B.6, B.7, B.8	Yes []
SPRU	Does the implementation support Source Pruning?	O	5.18, 10.10.3, 11.2.1.1	Yes [] No []
MRP1	Does the MRP implementation support operation of the Full Participant?	SPRU:O.2 –SPRU:O.3	Clause 10, B.8	Yes [] No []
MRP2	Does the MRP implementation support operation of the Full Participant, point-to-point subset?	SPRU:O.2 –SPRU:O.3	Clause 10, B.8	Yes [] No []
MRP3	Does the MRP implementation support operation of the Applicant-Only Participant?	–SPRU:O.3	Clause 10, B.8	Yes [] No []
MRP4	Does the MRP implementation support operation of the Simple-Applicant Participant, point-to-point subset?	–SPRU:O.3	Clause 10, B.8	Yes [] No []
FQTSSE	Does the implementation support forwarding and queuing for time-sensitive streams?	O	5.20, 34.6, B.9	Yes [] No []
CN	Is congestion notification implemented?	O	5.21, Clauses 30, 31, 32, 33, B.11	Yes [] No []
PFC	Is Priority-based Flow Control implemented?	O	5.11, Clause 36, B.12	Yes [] No []
ETS	Does the implementation support bandwidth management using ETS?	O	Clause 37, B.13	Yes [] No []
DCBX	Does the implementation support configuration management via DCBX?	O	Clause 38, B.14	Yes [] No []
EVB-S	Does the implementation support EVB Station functionality?	O	5.24, A.39	Yes [] No []
SCHED	Does the implementation support scheduled traffic?	O	5.4.1, 5.13.1, 8.6.8, 8.6.9, 12.29, 17.7.22	Yes [] No []
PRE	Does the implementation support frame preemption?	O	5.4.1, 5.13.1, 6.7.2, 8.6.8, 12.30, 17.7.23	Yes [] No []
PSFP	Does the implementation support PSFP?	O	8.6.5.2.1, 8.6.10, 12.31	Yes [] No []
ATS	Does the implementation support Asynchronous Traffic Shaping?	O	8.6.5.2.2, 8.6.8, 8.6.8.5, 8.6.11, 12.31	Yes [] No []
CQF	Does the implementation support cyclic queuing and forwarding?	O	5.25, 5.28	Yes [] No []

B.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support
CNC-S	Does the implementation support Centralized Network Configuration (CNC) station functionality?	O	5.29, 46.2, A.17	Yes [] No []
CI-S	Does the implementation support the functionality of a Congestion Isolation?	O	5.32, Clause 49	Yes [] No []
AAD	Does the implementation support AAD functionality?	O	5.33, Clause 50	Yes [] No []
CBSA	Does the implementation support the credit-based shaper algorithm?	O	8.6.8.2	Yes [] No []

1

B.6 Multiple MAC Registration Protocol (MMRP)

Item	Feature	Status	References	Support
	If MMRP is not supported, mark N/A and continue at B.7.			N/A []
MMRP1	Does the implementation support the exchange of MMRPDUs, using the generic MRPDU format defined in 10.8 to exchange MMRP-specific information, as defined in 10.12?	M	5.4.1.3, 10.8, 10.12	Yes []
MMRP2	Is the MMRP Application supported as defined in 10.12?	M	5.4.1.3, 10.12	Yes []
MMRP5	Is the MAP Context Identifier used to identify a VLAN Context equal to the VID used to identify the corresponding VLAN?	M	10.12.1.1	Yes []
MMRP7	Is the group MAC address used as the destination address for MRPDUs destined for MMRP Participants the group MAC address identified in Table 10-1 as “Customer and Provider Bridge MMRP address”?	M	10.12.1.3	Yes []
MMRP8	Is the EtherType used for MRPDUs destined for MMRP Participants the MMRP EtherType identified in Table 10-2?	M	10.12.1.4, Table 10-2	Yes []
MMRP9	Does the ProtocolVersion used for the implementation of MMRP take the hexadecimal value 0x00?	M	10.12.1.5	Yes []
MMRP10	Are the Attribute Type values used in the implementation as specified in 10.12.1.6?	M	10.12.1.6	Yes []
MMRP11	Does the implementation encode the values in FirstValue fields in accordance with the definition in 10.12.1.7?	M	10.12.1.7	Yes []
MMRP12	Is management of the Restricted_MAC_Address_Registration control parameter supported?	O	10.12.2	Yes [] No []
MMRP13	If management of the Restricted_MAC_Address_Registration control parameter is not supported, is the value of this parameter FALSE for all Ports?	¬MMRP12:M	10.12.2	Yes [] N/A []

B.6 Multiple MAC Registration Protocol (MMRP) *(continued)*

Item	Feature	Status	References	Support
MMRP14	Does the implementations maintain state information for all attribute values that support the Group service requirement registration?	SPRU:M	10.10.2	Yes [] N/A []
MMRP15	Is the implementation capable of supporting any attribute value in the range of possible values that can be registered using Group membership and individual MAC address registration?	M	10.12.4	Yes []
MMRP16	State the number of Group membership and individual MAC address state values that can be supported on each Port.	M	10.12.4	Number _____

1

B.7 Multiple VLAN Registration Protocol (MVRP)

Item	Feature	Status	References	Support
	If MVRP is not supported, mark N/A and continue at B.8.			N/A []
MVRP1	Does the implementation support the exchange of MMRPDUs, using the generic MRPDU format defined in 11.2 to exchange MMRP-specific information, as defined in 10.12?	M	5.4.2, 10.8, 11.2	Yes []
MVRP2	Is the MMRP Application supported as defined in 11.2?	M	5.4.2, 11.2	Yes []
MVRP7	Is the group MAC address used as the destination address for MRPDUs destined for MVRP Participants as defined in 11.2.3.1.3?	M	11.2.3.1.3	Yes []
MVRP8	Is the EtherType used for MRPDUs destined for MVRP Participants the MVRP EtherType identified in Table 10-2?	M	11.2.3.1.4, Table 10-2	Yes []
MVRP9	Does the ProtocolVersion used for the implementation of MVRP take the hexadecimal value 0x00?	M	11.2.3.1.5	Yes []
MVRP10	Are the Attribute Type values used in the implementation as specified in 11.2.3.1.6?	M	11.2.3.1.6	Yes []
MVRP11	Does the implementation encode the values in FirstValue fields in accordance with the definition in 11.2.3.1.7?	M	11.2.3.1.7	Yes []
MVRP12	Is the implementation of MVRP capable of supporting all attribute values in the range of possible values that can be registered using MVRP?	SPRU:M	11.2.7	Yes [] N/A []
MVRP13	Is the implementation capable of maintaining current state information for all attributes in the range of possible values?	SPRU:M	11.2.7	Yes [] N/A []

1 B.8 Multiple Registration Protocol (MRP)

Item	Feature	Status	References	Support
MRP5	Does the implementation of MRP meet all of the requirements for interoperability stated in 10.5 that apply to end station operation?	M	10.5	Yes []
MRP6	Does the implementation support the operation of the complete Applicant state machine?	MRP1:M	10.7, 10.7.7	Yes [] N/A []
MRP7	Does the implementation support the operation of the point-to-point subset of the Applicant state machine?	MRP2:M	10.7, 10.7.7	Yes [] N/A []
MRP8	Does the implementation support the operation of the Applicant-Only subset of the Applicant state machine?	MRP3:M	10.7, 10.7.7	Yes [] N/A []
MRP9	Does the implementation support the operation of the Simple-Applicant subset of the Applicant state machine?	MRP4:M	10.7, 10.7.7	Yes [] N/A []
MRP10	Does the implementation support the operation of the Registrar state machine?	MRP1:M MRP2:M	10.7, 10.7.8	Yes [] N/A []
MRP11	Does the implementation support the operation of the LeaveAll state machine?	MRP1:M MRP2:M	10.7, 10.7.9	Yes [] N/A []
MRP12	Does the implementation support the operation of the PeriodicTransmission state machine?	M	10.7, 10.7.10	Yes []

2 B.9 Forwarding and Queuing Enhancements for time-sensitive streams (FQTSS)

Item	Feature	Status	References	Support
FQTSSE1	Support a minimum of two traffic classes, of which one supports the strict priority algorithm and the other is an SR class?	FQTSSE:M	5.20, 8.6.8.1, 34.6	Yes [] N/A []
FQTSSE2	Support the credit-based shaper algorithm as the transmission selection algorithm for frames transmitted for each stream associated with the SR class.	FQTSSE:M	5.20, 8.6.8.2, 34.6	Yes [] N/A []
FQTSSE3	Support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for the SR class.	FQTSSE:M	5.20, 8.6.8.2, 34.6	Yes [] N/A []
FQTSSE4	Use the default priority associated with SR class “B” as shown in Table 6-5 as the priority value carried in transmitted SR class “B” data frames.	FQTSSE:M	5.20, Table 6-5, 34.6	Yes [] N/A []
FQTSSE5	Support two or more SR classes (a maximum of seven), and support the operation of the credit-based shaper algorithm on all Ports as the transmission selection algorithm used for those SR classes. The number of SR classes supported shall be stated in the PICS.	FQTSSE:O	5.20, 8.6.8.2, 34.6	Yes [] No [] Number_____
FQTSSE6	Use the default priority associated with SR class “A” as shown in Table 6-5 as the priority value carried in transmitted SR class “A” data frames. If more than two SR classes are supported, the priority value carried in transmitted data frames for the additional SR classes shall be stated in the PICS.	FQTSSE:O	5.20, Table 6-5, 34.6	Yes [] No [] Priority override values_____

1 B.10 Stream Reservation Protocol (SRP)

Item	Feature	Status	Reference	Support
	If SRP is not supported, mark N/A and ignore the remainder of this table.			N/A []
SRP-1	Does the implementation support the exchange of Multiple Stream Registration Protocol Data Units (MSRPDU)s, using the generic Multiple Registration Protocol Data Unit (MRPDU) format defined in 10.8 to exchange MSRP-specific information, as defined in 35.2.2.8.1, 35.2.2.9.1, and 35.2.2.10.1?	M	5.4.4, 10.8, 35.2.2.8.1, 35.2.2.9.1, 35.2.2.10.1	Yes []
SRP-2	Is the MSRP Application supported as defined in Clause 35?	M	5.4.4, Clause 35	Yes []
SRP-3	Is the group MAC address used as the destination address for MRPDU's destined for MSRP Participants the group MAC address identified in Table 8-1, Table 8-2, Table 8-3 as "Individual LAN Scope group address, Nearest Bridge group address"?	M	35.2.2.1, Table 8-1, Table 8-2, Table 8-3	Yes []
SRP-4	Is the EtherType used for MRPDU's destined for MSRP Participants the MSRP EtherType identified in Table 10-2?	M	35.2.2.2, Table 10-2	Yes []
SRP-5	Does the ProtocolVersion used for the implementation of MSRP take the hexadecimal value 0x01?	M	35.2.2.3	Yes []
SRP-6	Are the Attribute Type values used in the implementation as specified in 35.2.2.4 and Table 35-1?	M	35.2.2.4, Table 35-1	Yes []
SRP-7	Are the Attribute Length values used in the implementation as specified in 35.2.2.5 and Table 35-2?	M	35.2.2.5, Table 35-2	Yes []
SRP-8	Are the MSRP Vector FourPackedEvents values used in the implementation as specified in 35.2.2.7.2 and Table 35-3?	M	35.2.2.7.2, Table 35-3	Yes []
SRP-9	Does the implementation encode the values in FirstValue fields in accordance with the definition in 35.2.2.8, 35.2.2.9, and 35.2.2.10?	M	35.2.2.8, 35.2.2.9, 35.2.2.10	Yes []
SRP-10	Does the Talker implementation populate the Accumulated Latency with a reasonable, nonzero value?	M	35.2.2.8.6, 35.2.2.10.10	Yes []
SRP-11	Does the implementation update the Failure Information end station MAC address and Code when a Talker Failed is declared?	M	35.2.2.8.7, 35.2.2.10.9, 35.2.2.10.12	Yes []

B.10 Stream Reservation Protocol (SRP) *(continued)*

Item	Feature	Status	Reference	Support	
SRP-12	Does the implementation create a Talker Failed in the event of insufficient bandwidth or resources?	O	35.1.2.1	Yes []	No []
SRP-13	Is talkerPruning and MMRP supported?	O	35.2.4.3.2	Yes []	No []
SRP-14	Are MSRPDU s transmitted on all ports?	M	35.2	Yes []	
SRP-15	State the number of Talker registration values that can be supported on each Port.	M	35.2.7	Number _____	
SRP-16	State the number of Listener registration values that can be supported on each Port.	M	35.2.7	Number _____	
SRP-17	Does the Listener issue an appropriate MVRP VLAN membership request before attaching to a Stream?	M	35.1.2.2	Yes []	
SRP-18	When MAC_Operational transitions to TRUE does the device declare the default SR class priority value as the SRclassPriority prior to receiving an SRclassPriority declaration from its neighbor?	M	IEEE Std 802.1AC, 35.2.2.9.3, Table 6-5	Yes []	
SRP-19	Does the device set SRclassPriority to the value declared by the neighboring device?	M	35.2.2.9.3	Yes []	
SRP-20	When MAC_Operational transitions to TRUE does the device declare the default SR_PVID value as the SRclassVID prior to receiving an SRclassVID declaration from its neighbor?	M	IEEE Std 802.1AC, 35.2.2.9.4, Table 9-2	Yes []	
SRP-21	Does the device set SRclassVID to the value declared by the neighboring device?	M	35.2.2.9.4	Yes []	
SRP-22	Does the implementation support Stream transformation in end station?	O	35.2.2.10.5	Yes []	No []
SRP-23	Does the implementation declare the Domain attribute prior to Talker/Listener attributes?	M	35.2.2.9	Yes []	
SRPMDCSN	Does this device support media-dependent CSN functionality on one or more ports?	SRPMDMOCA:M OR: SRPMDDOT11:M	Annex C	Yes []	No []
SRPMDMOCA	Does this device support media-dependent MoCA functionality on one or more ports?	O:1	C.2	Yes []	No []
SRPMDDOT11	Does this device support media-dependent IEEE 802.11 AP functionality on one or more ports?	O:1	C.3	Yes []	No []

B.10 Stream Reservation Protocol (SRP) *(continued)*

Item	Feature	Status	Reference	Support
SRPMDCSN-1	Does this device support a single Designated MSRP Node (DMN)?	SRPMDCSN:M	35.1.1	Yes []
SRPMDMOCA-1	Does this device support DMN Device Attribute Information Element to L2ME message?	SRPMDMOCA:M	C.2.1.2	Yes []
SRPMDMOCA-2	Does this device support DMN selection?	SRPMDMOCA:M	C.2.1.3	Yes []
SRPMDMOCA-3	Does this device support MSRP Attribute Declaration as specified in Table C-1?	SRPMDMOCA:M	C.2.2 Table C-1	Yes []
SRPMDDOT11-1	Does this device support EDCA-AC?	SRPMDDOT11:M	Table C-5	Yes []
SRPMDDOT11-2	Is the DMN and the QAP of the IEEE 802.11 BSS co-located in the same device?	SRPMDDOT11:M	C.3.2	Yes []
SRPMDDOT11-3	Does the device support MLME primitives specified in Table C-4?	SRPMDDOT11:M	C.3.3, Table C-4	Yes []
SRPMDDOT11-4	Does the device support VLAN tag encapsulation/de-encapsulation on the IEEE 802.11 interface?	SRPMDDOT11:M	C.3.3.1	Yes []
SRPMDDOT11-5	Is the reservation process an atomic operation?	SRPMDDOT11:M	C.3.1, Figure C-11, Figure C-12, Figure C-13	Yes []

B.11 Congestion notification

Item	Feature	Status	References	Support	
CN-1	Does the system conform to the required provisions of IEEE Std 802.1AB?	CN: M	5.21	Yes []	
CN-2	Does the system support the use of the Congestion Notification TLV in LLDP?	CN: M	5.21	Yes []	
CN-3	Does the system transmit more than one Congestion Notification TLV in a single LLDPDU?	CN: M		No []	
CN-4	Does the system transmit a Congestion Notification TLV with 0 in all of the Per-priority CNPV indicators?	CN: M		No []	
CN-5	Does the system implement the Congestion Notification Domain (CND) defense variables, procedures, and state machine?	CN: M	32.4, 32.5, 32.6	Yes []	
CN-6	Does the end station support the creation of at least one RP?	CN: M	5.21	Yes []	
CN-7	Does the end station allow more than rpppMaxRps RPs to be created on one port and priority?	CN: M	32.10.1	No []	
CN-8	Does the end station support at least the cptDisabled and cptInterior defense modes separately on each CNPV?	CN: M	5.21, 32.1.1	Yes []	
CN-9	Does the end station add a tag to frames transmitted from CCFs?	CN: O	30.5, 31.2.2.5	Yes []	No []
CN-10	Does the end station support both the cptInterior, and cptInteriorReady defense modes separately on each CNPV?	CN: O	5.21, 31.2.2.5 , 32.14.3	Yes []	No []
CN-11	Does the end station support the cptEdge defense mode on any CNPV?	CN: O	5.21, 32.4.9	Yes []	No []
CN-12	Does the end station accept CN-TAGs in data frames on any CNPV?	CN: O	5.21, 32.4.10	Yes []	No []
CN-13	Does the end station support the creation of more than one RP?	CN: O	5.21, 31.2.1	Yes []	No []
CN-14	Reserved				
CN-15	Do the end station's RP(s) limit the output frame rate in response to CNMs received?	CN: M	5.21	Yes []	
CN-16	Does the end station distinguish correctly to which RP a given CNM is directed?	CN-13: M	5.21	Yes []	
CN-17	Does the end station support the creation of at least one CP?	CN: O	5.21	Yes []	No []
CN-18	Does the end station's have a single flow queue as its default configuration?	CN: M	31.2.1	Yes []	
CN-19	Does the end station change its method for assigning frames to Flow queues in a manner that impairs the ability of compliant CPs to throttle its flows?	CN: M	31.2.1	No []	

B.11 Congestion notification (*continued*)

Item	Feature	Status	References	Support
CN-20	Does the end station change its method for assigning frames to Flow queues in a manner that significantly increases the likelihood of frame misordering, and thus impacts higher layer functions?	CN: M	31.2.1	No []
CN-21	Does the end station's Rate Limiters always meet the specified formula for L^T ?	CN: M	31.2.2.4, Equation (31-1)	Yes []
CN-22	Can the end station's Flow Selection functions drop frames between the Flow queue and the Flow multiplexer?	CN: M	31.2.2.5	No []
CN-23	Does the end station output frames with CN-TAGs on a priority operating in mode <i>cptEdge</i> or <i>cptInterior</i> ?	CN: M	32.1.1	No []
CN-24	Do the end station's RPs implement the specified timers, variables, procedures, and state machines?	CN: M	32.7 item b), 32.12, 32.13, 32.14, 32.15	Yes []
CN-25	Does the end station insert the CN-TAG and VLAN-TAG in the correct order (addresses, VLAN-TAG, CN-TAG)?	CN: M	33.2	Yes []
CN-26	Does the end station ignore the CNM's Version field?	CN: M	33.4.1	Yes []
CN-27	Does the end station ignore the CNM's ReservedV field?	CN: M	33.4.2	Yes []
CN-28	Does the end station assign meaning to subfields within a received CNM's Congestion Point Identifier field?	CN: M	33.4.4	No []
CN-29	Does the end station ignore received CNMs that are too short?	CN: M	33.4.11 item a)	Yes []
CN-30	Does the end station ignore received CNMs that have no CN-TAG?	CN: O	33.4.11 item b)	Yes [] No []
CN-31	Does the end station ignore received CNMs that have nonzero values in the Version or ReservedV fields?	CN: M	33.4.11 item c)	No []
CN-32	Does the end station support all of the objects required to manage its RP(s)?	CN: M	5.21, 12.21.1, 12.21.6	Yes []
CN-33	Does the end station use the same Reaction Point group managed object to manage RPs serving more than one CNPV?	CN: M	32.11	No []
CN-34	Does the end station support all of the objects in the CN component managed object?	CN-17: M	5.21, 12.21.1	Yes []
CN-35	Does the end station support all of the objects in the CN component priority managed object?	CN-17: M	5.21, 12.21.2	Yes []
CN-36	Does the end station support all of the objects in the CN Port priority managed object?	CN-17: M	5.21, 12.21.3	Yes []
CN-37	Does the end station support all of the objects in the Congestion Point managed object?	CN-17: M	5.21, 12.21.4	Yes []

B.12 Priority-based Flow Control (PFC)

Item	Feature	Status	References	Support
PFC-1	Enabling PFC on at least one priority	PFC: M	36.1.2	Yes []
PFC-2	Processing PFC Requests	PFC: M	36.1.3.1	Yes []
PFC-3	Processing PFC Indications	PFC: M	36.1.3.2	Yes []
PFC-4	PFC delay constraints	PFC: M	36.1.3.3	Yes []
PFC-5	PFC-aware system queue functions	PFC: M	36.2	Yes []
PFC-6	DCBX	PFC: M	5.11	Yes []
PFC-7	Enabling PFC on up to eight priorities	PFC: O	36.1.2	Yes []No []
PFC-8	PFC not enabled for traffic classes using the credit-based shaper algorithm	PFC: M	8.6.8.2	Yes []

¹ B.13 Enhanced Transmission Selection (ETS)

Item	Feature	Status	References	Support
ETS-1	Support at least 3 traffic classes	ETS:M	37.3	Yes []
ETS-2	Support bandwidth configuration with a granularity of 1% or finer	ETS:M	37.3	Yes []
ETS-3	Support bandwidth allocation with a precision of 10%	ETS:M	37.3	Yes []
ETS-4	Support allocation of a portion of available bandwidth to each traffic class	ETS:M	37.3	Yes []
ETS-5	Support DCBX	ETS:M	Clause 38	Yes []

² B.14 Data Center Bridging eXchange protocol (DCBX)

Item	Feature	Status	References	Support
DCBX-1	Support LLDP	DCBX:M	IEEE Std 802.1AB	Yes []
DCBX-2	Support the DCBX ETS Configuration TLV	DCBX:M	D.2.8	Yes []
DCBX-3	Support the ETS Recommendation TLV	DCBX:M	D.2.9	Yes []
DCBX-4	Support the Priority-based Flow Control Configuration TLV	DCBX:M	D.2.10	Yes []
DCBX-5	Support the Application Priority TLV	DCBX:M	D.2.11	Yes []
DCBX-6	Support the DCBX asymmetric state machine	DCBX:M	38.4.1	Yes []
DCBX-7	Support the DCBX symmetric state machine	DCBX:M	38.4.2	Yes []
DCBX-8	Support the Application VLAN TLV	DCBX:M	D.2.14	Yes []

1

B.15 Scheduled traffic

Item	Feature	Status	References	Support
	If neither scheduled traffic (SCHED in B.5) nor cyclic queuing and forwarding (CQF in B.5) are supported, mark N/A and ignore the remainder of this table.		5.25, 5.28, 8.6.8, 8.6.9, 12.29, 17.7.22	N/A []
SCHED1	Does the implementation support the state machines and associated definitions specified in 8.6.9?	SCHED OR CQF:M	5.28 item b), 8.6.8, 8.6.9	Yes [] N/A []
SCHED2	Does the implementation support the management entities defined in 12.29?	SCHED OR CQF:M	5.28 item c), 12.29	Yes [] N/A []
SCHED3	Is the IEEE8021-ST-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND (SCHED OR CQF):O	5.28 item c), 12.29, 17.7.22	Yes [] No []
SCHED4	Is the <i>ieee802-dot1q-sched</i> YANG module supported?	YANG AND (SCHED OR CQF):O	5.28 item c), 12.29, 48.6.17	Yes [] No []

2

B.16 Frame Preemption

Item	Feature	Status	References	Support
	If frame preemption (PRE in B.5) is not supported, mark N/A and ignore the remainder of this table.		5.4.1, 5.13.1, 6.7.2, 8.6.8, 12.30, 17.7.23	N/A []
PRE1	Does the implementation support the functionality of frame preemption as specified in 6.7.2 and 8.6.8?	PRE: M	5.4.1, 5.13.1, 6.7.2, 8.6.8	Yes [] N/A []
PRE2	Does the implementation support the management entities defined in 12.30?	PRE: M	5.4.1 item ae), 12.30	Yes [] N/A []
PRE3	Is the IEEE8021-Preemption-MIB module fully supported (per its MODULE-COMPLIANCE)?	PRE: O	5.4.1 item ae), 12.30, 17.7.23	Yes [] N/A []
PRE4	Is the <i>ieee802-dot1q-preemption</i> YANG module fully supported?	PRE: O	5.4.1 item ae), 12.30, 48.6.19	Yes [] No []

¹B.17 Per-Stream Filtering and Policing

Item	Feature	Status	References	Support	
	If neither Per-Stream Filtering and Policing (PSFP in B.5) nor cyclic queuing and forwarding (CQF in B.5) are supported, mark N/A and ignore the remainder of this table.		5.28 items d) and e), 8.6.5.2.1, 8.6.10, 12.31, 17.7.24	N/A []	
PSFP1	Does the implementation support the state machines and associated definitions as specified in 8.6.10?	PSFP OR CQF:M	5.28 items b) and d), 8.6.5.4, 8.6.10,	Yes []	N/A []
PSFP2	Does the implementation support the management entities defined in 12.31 for PSFP?	PSFP OR CQF:M	5.28 item e), 8.6.5.2, 8.6.10, 12.31	Yes []	N/A []
PSFP3	Is the IEEE8021-PSFP-MIB module fully supported (per its MODULE-COMPLIANCE)?	MIB AND (PSFP OR CQF):O	12.31, 17.7.24	Yes [] No []	N/A []
PSFP4	Is the <i>ieee802-dot1q-psfp</i> module supported?	YANG AND (PSFP OR CQF):O	12.31, 48.6.21	Yes [] No []	N/A []

²

B.18 Asynchronous Traffic Shaping

Item	Feature	Status	References	Support
	If Asynchronous Traffic Shaping (ATS in B.5) is not supported, mark N/A and ignore the remainder of this table.		5.31, 8.6.5.2.2, 8.6.8, 8.6.8.5, 8.6.11, 47.1.2	N/A []
ATS1	Does the implementation support at least one traffic class that supports the strict priority transmission selection algorithm (8.6.8.1) and another traffic class that supports the ATS transmission selection algorithm (8.6.8.5)?	ATS:M	5.31, 8.6.8, 8.6.8.5,	Yes [] N/A []
ATS2	Does the implementation support ATS schedulers as specified in 8.6.5.6?	ATS:M	5.31, 8.6.5.6, 47.1	Yes [] N/A []
ATS3	Does the implementation support the ATS scheduler state machines as specified in 47.1?	ATS:M	5.31, 47.1	Yes [] N/A []

1

B.19 Congestion Isolation

Item	Feature	Status	References	Support
	If Congestion Isolation (CI-S in B.5) is not supported, mark N/A and ignore the remainder of this table.			N/A []
CI-S-1	Does the implementation support IETF RFC 3168 Explicit Congestion Notification?	CI-S:M	49.2.1, IETF RFC 3168	Yes []
CI-S-2	Does the implementation support the variables and procedures of the Congestion Isolation Protocol?	CI-S:M	49.4	Yes []
CI-S-3	Does the implementation support LLDP?	CI-S:M	IEEE Std 802.1AB	Yes []
CI-S-4	Does the implementation support the Congestion Isolation TLV in LLDP?	CI-S:M	D.2.15	Yes []
CI-S-5	Does the implementation support transmission selection algorithms other than strict priority?	CI-S:O	49.2.3, 49.3.2	Yes [] No []
CI-S-6	Does the implementation support the ability to return a flow to non-congesting status when the congesting queue is not empty?	CI-S:O	49.2.6	Yes [] No []
CI-S-7	Is the <i>ieee802-dot1q-congestion-isolation</i> module supported?	CI-S AND YANG:O	48.2.8	Yes [] N/A [] No []
CI-S-8	Does the implementation support Topology Recognition	CI-S:O	49.5	Yes [] No []

1

B.20 PBBN Auto Attach

Item	Feature	Status	References	Support
	If PBBN Auto Attach (AAD in B.5) is not supported, mark N/A and ignore the remainder of this table.	AAD:M		N/A []
AAD-1	Does the AAD support the Link Layer Discovery Protocol (LLDP) transmit and receive mode?	AAD:M	IEEE Std 802.1AB™	Yes [] N/A []
AAD-2	Does the AAD implementation support the PBBN Auto Attach System TLV?	AAD:M	5.14.2, 5.9.2, D.2.17	Yes [] N/A []
AAD-3	Does the AAD implementation support the PBBN Auto Attach Assignment TLV?	AAD:M	5.14.2, 5.9.2, D.2.18	Yes [] N/A []
AAD-4	Does the AAD support the AAD State Machine	AAD:M	5.14.2, 5.9.2, 50.6	Yes [] N/A []
AAD-5	Does the AAD support the Link Layer Discovery Protocol (LLDP) transmit and receive mode?	AAD:M	IEEE Std 802.1AB™	Yes [] N/A []

2

B.21 Credit-based shaper algorithm

Item	Feature	Status	References	Support
	If the credit-based shaper algorithm is not supported, mark N/A and ignore the remainder of this table.			N/A []
CBSA-6	Does the implementation support the credit based shaper algorithm and associated parameters specified in 8.6.8.2?	CBSA:M	8.6.8.2	Yes [] No []
CBSA-7	Does the implementation support the management parameters adminIdleSlope and operIdleSlope as defined in 12.20.1?	CBSA:M	12.20.1	Yes [] No []

3

1 Annex C

2 (normative)

3 Designated MSRP Node (DMN) Implementations

4 This annex describes the DMN implementation on an IEEE 802.11 Network and Coordinated Shared
5 Networks (CSNs).

6 C.1 DMNs on CSNs

7 A CSN is a contention-free, time-division multiplexed-access network, supporting reserved bandwidth
8 based on priority or flow (QoS). One of the nodes of the CSN acts as the Network Coordinator (NC) node,
9 granting transmission opportunities to the other nodes of the network. The NC node also acts as the
10 bandwidth resource manager of the network.

11 C.1.1 CSN characteristics

12 CSNs support two types of transmissions: unicast transmission for node-to-node transmission and multicast/
13 broadcast transmission for one-node-to-other/all-nodes transmission. Each node-to-node link has its own
14 bandwidth characteristics that could change over time due to the periodic ranging of the link. The multicast/
15 broadcast transmission characteristics are the lowest common characteristics of multiple/all the links of the
16 network.

17 A CSN network is physically a shared network, in that a CSN node has a single physical port connected to
18 the half-duplex medium, but is also a logically fully-connected one-hop mesh network, in that every node
19 could transmit to every other node using its own profile over the shared medium.

20 Figure C-1 illustrates a CSN network acting as a backbone interconnecting AV systems.

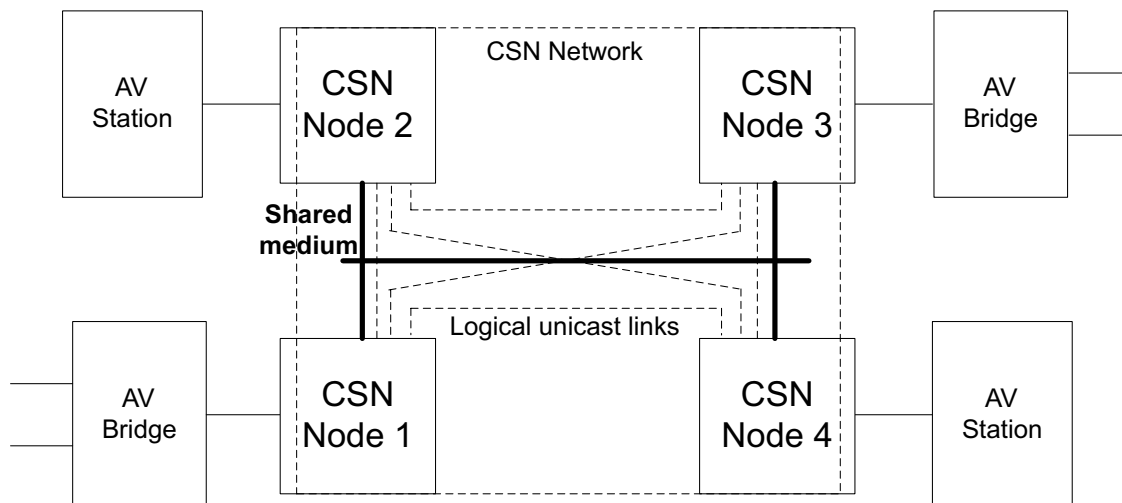


Figure C-1—CSN backbone

21 Depending on the CSN technology, the Network Coordinator node either may be a fixed node or may be
22 dynamically selected during normal operation.

1 C.1.2 DMN handling on CSN

2 From the bandwidth reservation standpoint a CSN network is modeled as a Bridge as illustrated by
3 Figure C-2. Each node-to-node link is equivalent to a Bridge's path from an ingress port to an egress port.

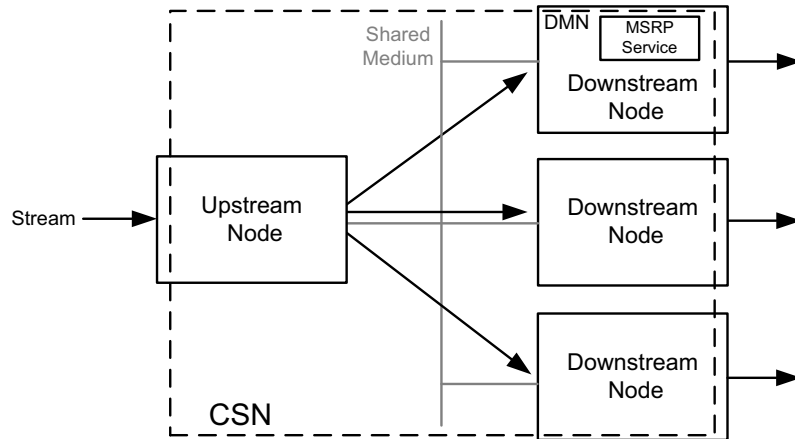


Figure C-2—Bridge's CSN model for bandwidth reservation

4 A CSN shall provide a single entity called the Designated MSRP Node (DMN) (35.1.1), which
5 communicates with the MSRP Service to manage the CSN bandwidth resources for the MSRP streams.

6 NOTE 1—The MSRP Service supplies both MSRP Attribute Propagation (MAP) for the CSN and MSRP Attribute
7 Declaration (MAD) for each MSRP-aware CSN node.

8 NOTE 2—An end station node provides an end station MSRP implementation as outlined in 5.18.3. This MSRP
9 implementation is logically on the upper layer side of this interface. This standard expects that an MSRP-aware CSN
10 node will have a single non-CSN interface. If more interfaces are present, they are expected to be connected through a
11 Bridge function. If this node happens to be the DMN, then this end station MSRP implementation is separate and distinct
12 from the DMN's MSRP Service.

13 C.1.2.1 DMN selection and migration

14 Depending on the CSN technology, the DMN might correspond to a static node or dynamically migrate
15 between nodes during normal operation. The DMN selection is network specific and described in C.2.1 and
16 C.3.2.

17 Over time the DMN constructs its database by handling the MSRP Talker and Listener Declarations
18 generated by the nodes of the CSN. If the DMN migrates, the new DMN broadcasts an MRP LeaveAll
19 message to all the nodes of the CSN, which will force its neighbors to redeclare their attributes. The MSRP
20 Participant nodes answer the MRP LeaveAll message by sending an MRP JoinIn message consumed by the
21 DMN as an MRP Re-Declare! message. These redeclarations permit the new DMN to immediately build its
22 database.

1 C.1.3 MSRPDU handling on a CSN

2 Figure C-3 and Figure C-4 illustrate the flow and handling of MSRPDU messages on a CSN.

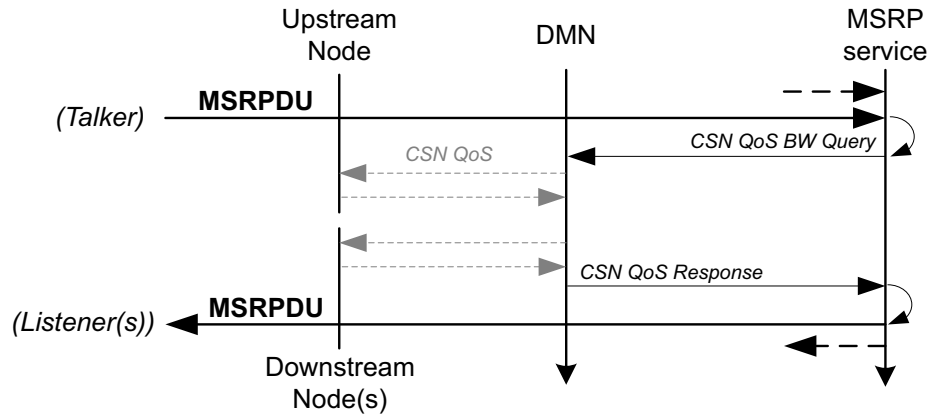


Figure C-3—Talker MSRPDU flow

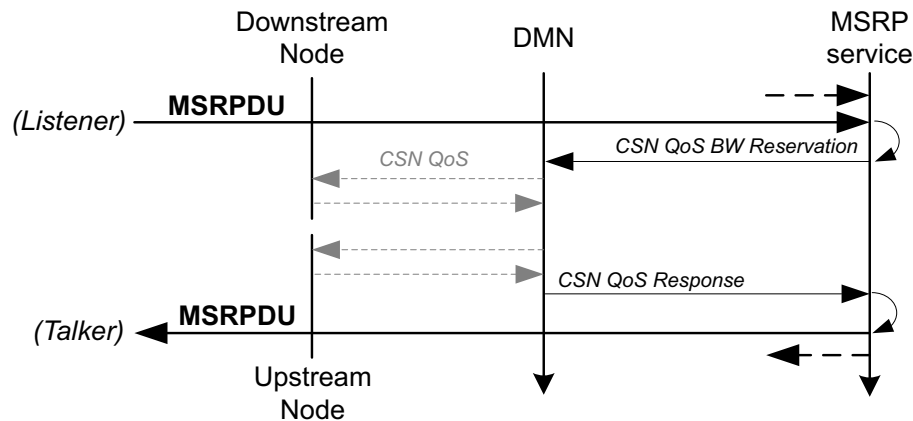


Figure C-4—Listener MSRPDU flow

- 3 a) A MSRP-aware CSN node identifies MSRPDU received on its non-CSN interface (the interface
4 either to another network media or to upper layers of the node) by their destination MAC address
5 (35.2.2.1) and EtherType (35.2.2.2).
- 6 b) Non-DMN nodes send MSRPDU to the DMN over the CSN.
- 7 c) The DMN delivers MSRPDU, along with information about the originating interface, to the MSRP
8 Service.
- 9 d) The DMN translates the MSRP TSpec parameters into CSN QoS parameters and invokes the CSN's
10 Protocol Specific QoS transactions with the CSN Network Coordinator (C.2.2, C.3.3) as follows:
 - 11 1) When the DMN receives a Talker Advertise message originated from an upstream CSN node,
12 the DMN invokes a bandwidth query transaction with the CSN Network Coordinator to check
13 whether the bandwidth advertised in the message's TSpec is available on each upstream to
14 downstream node link of the CSN network. In addition the DMN associates the MSRP
15 attributes with the CSN's native QoS records via the StreamID (which is included in the CSN
16 frames).

- 1 2) When the DMN receives a Listener Ready message originated from a downstream CSN node,
2 the DMN invokes a bandwidth reservation transaction with the CSN QoS manager to reserve
3 the bandwidth associated with the message's StreamID on the downstream to upstream CSN
4 node link.
- 5 e) After the DMN completes the CSN QoS transactions, the DMN behaves as an MSRP application on
6 a Bridge and propagates (MAP) and distributes (MAD) MSRP attributes (35.2).
- 7 NOTE—When the MAD function of the MSRP Service distributes attributes, it will deliver MSRPDU's to the
8 DMN's non-CSN interface and/or send them to the appropriate MSRP-aware CSN nodes.

9 **C.1.4 CSN bandwidth fluctuations**

10 Bandwidth on a CSN may fluctuate depending on interference experienced by the media. The MSRP
11 Attribute Propagation (35.2.4) supports a *bandwidthAvailabilityChanged* notification when bandwidth
12 increases or decreases across the media. When a bandwidth changed is encountered, MAP will reassess
13 reservation requests to maintain appropriate bandwidth utilization.

14 **C.2 DMN on MoCA**

15 NOTE—The discussion that follows is based on terminology found in the MoCA MAC/PHY Specifications v1.1 and
16 v2.0 [B57] and [B58].

17 **C.2.1 DMN Selection on MoCA Network**

18 **C.2.1.1 DMN-capable node discovery**

19 A DMN-capable node shall append the *IEEE DMN Device Attribute Information Element* (C.2.1.2) to the
20 L2ME payload of the Device Discovery Protocol SUBMIT L2ME transaction message specified in the
21 MoCA MAC/PHY Specification v2.0 [B58].

22 Upon completion of the L2ME Device Discovery transaction, all the DMN-capable nodes of the MoCA
23 network share the same information as follows:

- 24 a) Which MoCA nodes are DMN capable
25 b) Which MoCA node is selected as the DMN

26 If no DMN is selected, the DMN selection shall be performed (C.2.1.3).

1 **C.2.1.2 IEEE DMN Device Attribute IE**

2 **C.2.1.2.1 General**

3 The fields of the IEEE DMN Device Attribute IE are specified in Figure C-5 and C.2.1.2.2 through
4 C.2.1.2.7. The general format of the Device Attribute Information Element is described in the MoCA MAC/
5 PHY Specification v2.0 [B58].

Bits								Octets	Offset From Start of IE
8	7	6	5	4	3	2	1		
ATTRIBUTE_ID								1	0
LENGTH								1	1
----- VENDOR_ID -----								2	2
TLV_TYPE								1	4
TLV_LENGTH								1	5
----- TLV_VALUE -----								2	6

Figure C-5—IEEE DMN Device Attribute IE

6 **C.2.1.2.2 ATTRIBUTE_ID (Enumeration8)**

7 The value of the ATTRIBUTE_ID is 0xFF.

8 **C.2.1.2.3 LENGTH (UInteger8)**

9 The value of the LENGTH is 1.

10 NOTE—The actual length of the Attribute IE in bits is (LENGTH + 1) × 32.

11 **C.2.1.2.4 VENDOR_ID (Enumeration16)**

12 The value of VENDOR_ID is 0x0090 (IEEE 802.1 AVB).

13 **C.2.1.2.5 TLV_TYPE (Enumeration16)**

14 The value of the TLV_TYPE is 1 (SRP).

15 **C.2.1.2.6 TLV_LENGTH (UInteger8)**

16 The length of the TLV in octets is 4.

1 **C.2.1.2.7 TLV_VALUE (Octet2)**

2 The meaning of the field of the TLV_VALUE are specified as described in C.2.1.2.7.1 through C.2.1.2.7.3.

3 **C.2.1.2.7.1 DMNcapable (Bit 0 - Boolean)**

4 A value of 1 indicates the node is capable of acting as the DMN of the network. A value of 0 indicates the
5 node is not capable to act as a DMN.

6 **C.2.1.2.7.2 DMNselected (Bit 1 - Boolean)**

7 A value of 1 indicates the node has been selected as the DMN of the network. A value of 0 indicates the
8 node is not the selected DMN.

9 **C.2.1.2.7.3 Reserved (Bit 2..15)**

10 These bits are reserved for future use.

11 **C.2.1.3 DMN selection and confirmation**

12 If either 1) the *NODE_BITMASK* field into MAP frames indicates that the selected DMN has been removed
13 from the network (due to failure, power state/down, etc.) or 2) the DMN node discovery (C.2.1.1) does not
14 indicate a DMN selected node, the DMN-capable node with the lowest node ID will start acting as the DMN
15 and confirm the selection to the other DMN-capable nodes by generating a L2ME DMN Confirmation
16 Transaction (C.2.1.3.1).

17 NOTE—"MAP frame" is defined in the MoCA MAC/PHY Specification v1.1 [B57].

18 **C.2.1.3.1 L2ME DMN Confirmation Transaction**

19 **C.2.1.3.1.1 Overview of DMN Confirmation Transaction**

20 Figure C-6 provides an overview of the signals exchanged among the nodes during an L2ME DMN
21 Confirmation Transaction. As shown in the figure, the NC node starts the transaction either when it receives
22 a Submit L2ME Frame from a node (called the DMN Selected Entry Node for that Transaction) or on its
23 own. The transaction includes two L2ME Waves. The details of each message exchanged during the DMN
24 Confirmation Transaction are provided below.

25 **C.2.1.3.1.2 DMN Confirmation Submit**

26 The DMN Confirmation Transaction starts when the DMN Selected Entry Node sends a Submit L2ME
27 Frame to the NC.

28 The general fields of the Submit L2ME frame are as specified in section 2.2.3.1 of the MoCA MAC/PHY
29 Specification v1.1 [B57]. The parameters in the Submit L2ME Frame are set as follows:

- 30 a) *VENDOR_ID* (see C.2.1.2.4)
- 31 b) *TRANS_TYPE* = 0x1 (SRP)
- 32 c) *TRANS_SUBTYPE* = 0x1 (DMN Confirmation)
- 33 d) *WAVE0_NODEMASK* = (DMN-capable nodes)
- 34 e) *MSG_PRIORITY* = 0xF0
- 35 f) *TXN_LAST_WAVE_NUM* = 0x1
- 36 g) *L2ME_PAYLOAD* = 0 bytes

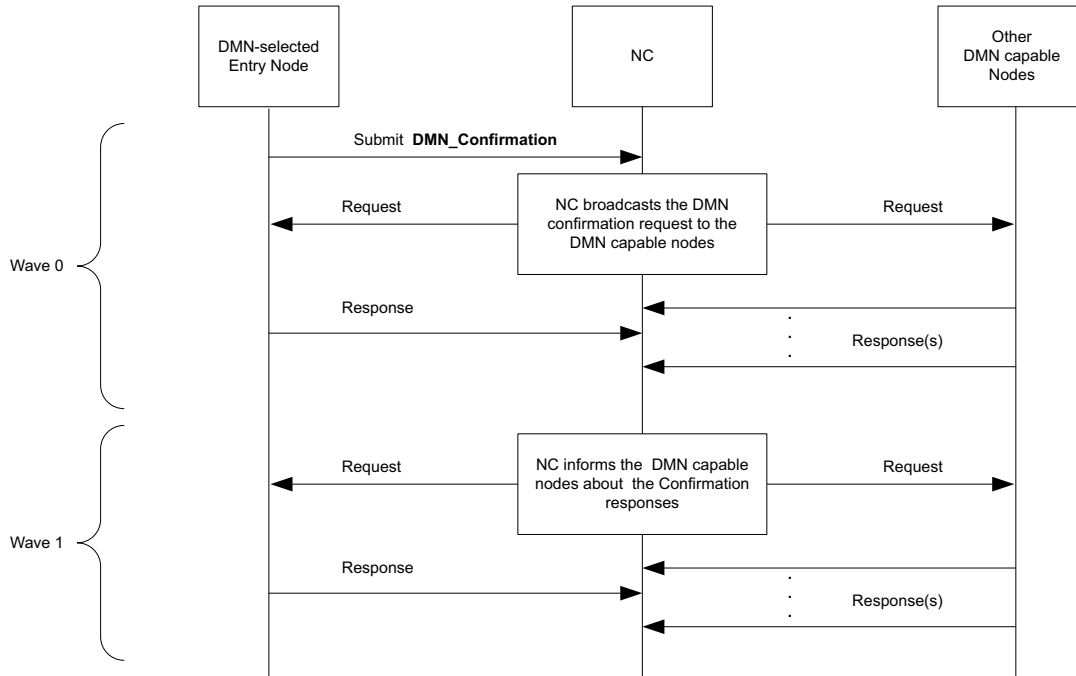


Figure C-6—DMN Confirmation Transaction

1 C.2.1.3.1.3 Request L2ME Frame of Wave 0 of DMN Confirmation Transaction

2 In the Wave 0 that follows the DMN Confirmation Submit message, the NC node initiates a Request L2ME
3 Frame based on the Submit L2ME Frame.

4 The general fields of the Request L2ME frame are as specified in section 2.2.3.2 of the MoCA MAC/PHY
5 Specification v1.1 [B57].

6 C.2.1.3.1.4 Response L2ME Frame of Wave 0 of DMN Confirmation Transaction

7 Each of the requested DMN-capable nodes sends a Response L2ME Frame to acknowledge the confirmation
8 request.

9 The general fields of the Response L2ME frame are as specified in 2.2.3.3 of the MoCA MAC/PHY
10 Specification v1.1 [B57]. The parameters in the Response L2ME Frame are set as follows:

- 11 a) RESP_STATUS <INTERPRETED> = '1'
- 12 b) RESP_STATUS <IN_NEXT_WAVE> = '1'
- 13 c) L2ME_PAYLOAD = 0 bytes

14 C.2.1.3.1.5 Request L2ME Frame of Wave 1 of DMN Confirmation Transaction

15 In Wave 1, the NC node informs the DMN-capable nodes about the acknowledgment results from Wave 0.
16 The NC node initiates Wave 1 using a Request L2ME Frame with the “concatenated” type of
17 L2ME_PAYLOAD.

18 The general fields of the Request L2ME frame are as specified in section 2.2.3.1 of the MoCA MAC/PHY
19 Specification v1.1 [B57]. The “concatenated” L2ME_Payload is as specified in Table 2-4 of the MoCA
20 MAC/PHY Specification v1.1 [B57].

1 C.2.1.3.1.6 Response L2ME Frame of Wave 1 of DMN Confirmation Transaction

2 The DMN Confirmation Transaction is completed when the DMN Selected Entry node and the other DMN-
3 capable nodes send their final Response L2ME Frame to the NC node.

4 The general fields of the Response L2ME frame are as specified in section 2.2.3.3 of the MoCA MAC/PHY
5 Specification v1.1 [B57]. The parameters in the Response L2ME Frame are set as follows:

- 6 a) RESP_STATUS <INTERPRETED> = '1'
- 7 b) RESP_STATUS <IN_NEXT_WAVE> = '0'
- 8 c) L2ME_PAYLOAD = 0 bytes

9 C.2.2 MoCA network bandwidth management

10 The MSRP service within the MoCA network manages the MoCA bandwidth for the MSRP streams by
11 invoking the MoCA native PQoS transactions. The DMN shall map the MSRP Attribute Declaration and the
12 resultant MAD declarations as described in Table C-1.

Table C-1—SRP to MoCA PQoS Transaction mapping

MSRP Attribute	MAD Primitive	MoCA PQoS Transactions	Description
Talker Advertise	MAD_Join.request(new)	Create PQoS Flow	Query bandwidth without reservation (see NOTE)
Listener Ready or Listener Ready Failed	MAD_Join.request(new)	Create PQoS Flow	Reserve bandwidth for a stream
Listener Ready or Listener Ready Failed	MAD_Join.request()	Update PQoS Flow	Renew the bandwidth reservation (leased time) for a stream
Talker or Listener Leave	MAD_Leave.request()	Delete PQoS Flow	Free bandwidth associated with a stream
NOTE—MoCA PQoS APIs do not include a Bandwidth Query-specific API. Therefore, bandwidth is queried by invoking a CreatePQoSFlow reservation for more bandwidth than the network could provide. The request will fail and CreatePQoSFlow will then return a failure status that includes the bandwidth available for reservations.			

13 Table C-2 describes the mapping between SRP TSPEC components and MoCA PQoS TSPEC parameters.

Table C-2—SRP TSPEC to MoCA TSPEC mapping

SRP TSPEC	MoCA PQoS TSPEC
MaxFrameSize	Max Packet Size
MaxFrameSize × MaxIntervalFrames	Peak Data Rate
MaxFrameSize × MaxIntervalFrames × Class B class measurement interval (34.4)	(Max) Burst Size

14 Table C-3 describes the mapping of the SRP StreamID to MoCA PQoS Flow transactions. The MoCA Flow
15 parameters includes a 32-bit field that allows an application, like SRP, to store application-specific
16 information. SRP will use this field (FLOW_TAG) to store the 16-bit unique ID portion of the StreamID.

Table C-3—SRP StreamID to MoCA PQoS Flow transaction mapping

SRP StreamID	MoCA PQoS Flow Transaction		
	Transaction	L2ME Payload	Field
MAC Address	Create PQoS Flow Update PQoS Flow	Submit	PACKET_DA
	Query PQoS Flow	Response	
16-bit Unique ID	Create PQoS Flow Update PQoS Flow	Submit	FLOW_TAG
	Query PQoS Flow	Response	

1 C.3 DMNs on IEEE 802.11 media

2 NOTE—Even though IEEE Std 802.11 is not a CSN, it uses the same DMN concepts described below.

3 From the bandwidth reservation standpoint an IEEE 802.11 BSS network is modeled as a Bridge as
4 illustrated by Figure C-7, Figure C-8, and Figure C-9. Each STA-AP link, STA-AP-STA link and optional
5 STA-STA DLS direct link is equivalent to the path from an input to an output Bridge's port.

6 An IEEE 802.11 BSS provides a single entity called the Designated MSRP Node (DMN) (35.1.1) to manage
7 the BSS bandwidth resources for the MSRP streams.

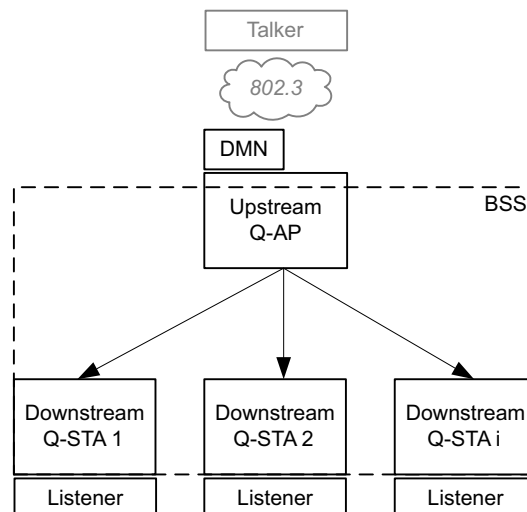


Figure C-7—Bandwidth reservation—bridge model for IEEE 802.11 BSS (STA downstream Port)

1

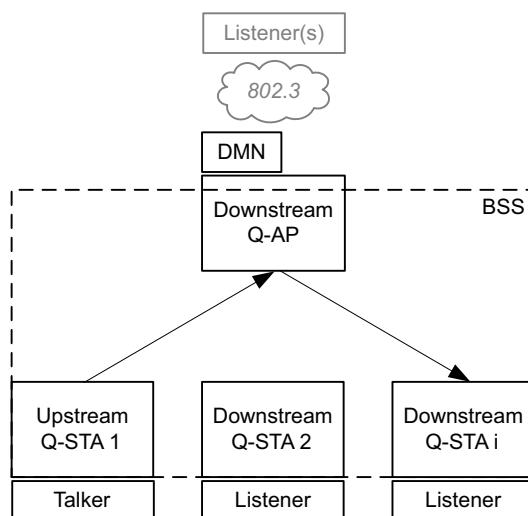


Figure C-8—Bandwidth reservation—bridge model for IEEE 802.11 BSS (STA upstream Port)

2

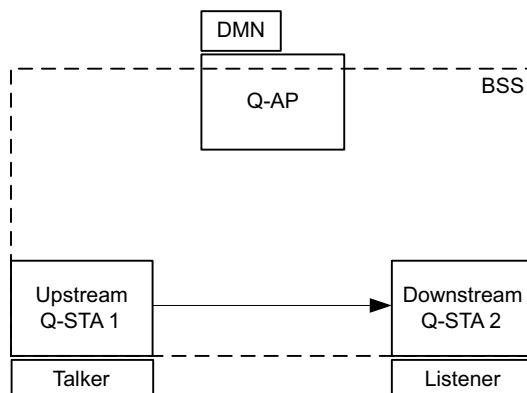


Figure C-9—Bandwidth reservation—bridge model for IEEE 802.11 BSS (direct link setup)

3 C.3.1 MSRP handling

4 MSRPDUs are transparently transported by the IEEE 802.11 BSS network and delivered to the DMN.

5 The DMN maps the MSRP commands into IEEE 802.11 MLME TS commands and interacts with the AP
 6 through the AP's MLME SAP.

7 Figure C-10 through Figure C-13 describe the flow of information between the MSRP and IEEE 802.11
 8 entities, and corresponding over the air IEEE 802.11 frames. Figure C-10 is an example of the IEEE 802.11
 9 bandwidth query process associated with an MSRP Talker Advertise.

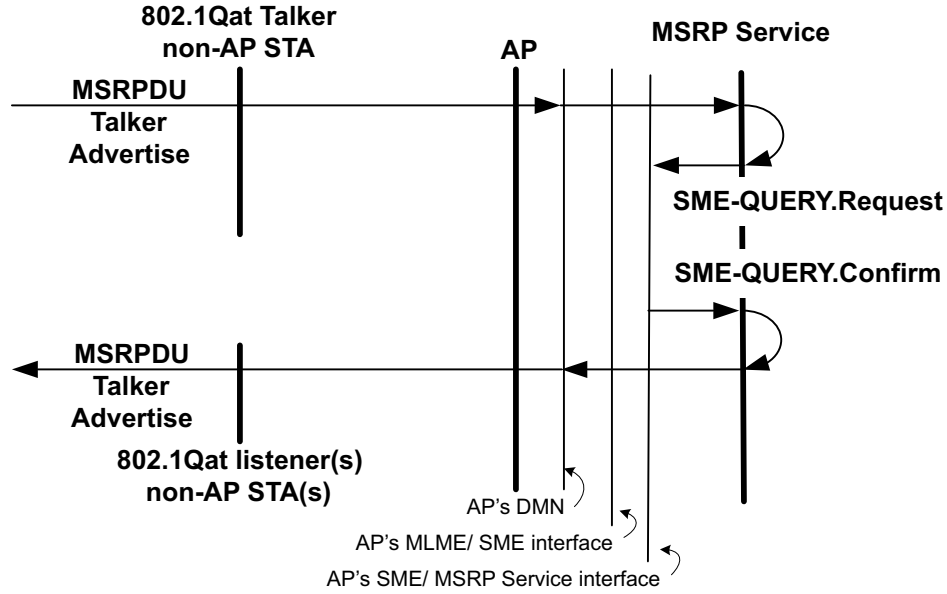


Figure C-10—MSRP/IEEE 802.11 query flows

1 Figure C-10 is an example of the reservation process associated with a Listener non-AP STA requesting a
2 Stream from the Talker non-AP STA. The diagram at the left of the figure shows the Listener non-AP STA
3 sending a Listener Ready (A) through the AP's MSRP Service, which then propagates that Listener Ready
4 (B) to the Talker non-AP STA. The message flow on the right of the figure shows the corresponding
5 IEEE 802.11 message exchanges associated with the two Listener Readys (A & B).

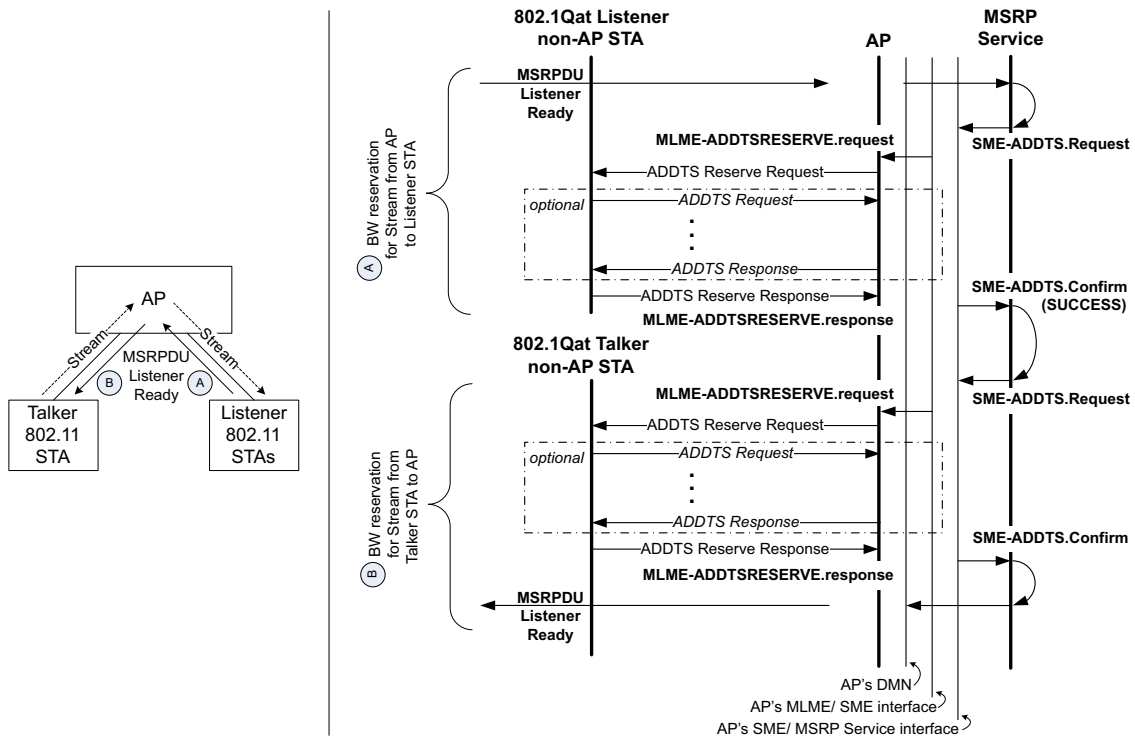


Figure C-11—MSRP/802.11 Talker STA to Listener STA reservation flows

1 There are two reservations (A & B) required to allow the Stream to flow from Talker to Listener. In
2 Figure C-11, Figure C-12, and Figure C-13, the reservation process must be an atomic operation so, if either
3 reservation fails, then both reservations shall be removed and the MSRP attributes updated accordingly.

4 Figure C-12 is an example of the reservation process associated with a “Bridged” Listener requesting a
5 Stream from a Talker non-AP STA. The Listener sends a Listener Ready (A) from the “Bridged” side of the
6 network and bandwidth is reserved as explained in SRP (Clause 35). The IEEE 802.11 message exchanges
7 associated with the Listener Ready (B) propagating to the Talker non-AP STA are also shown.

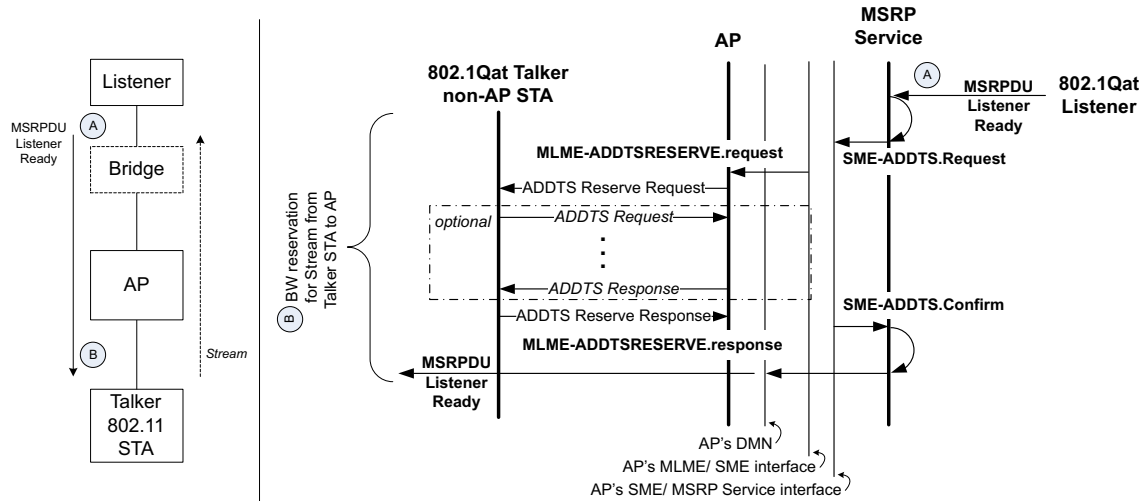


Figure C-12—MSRP/802.11 “Bridged” Listener to Talker STA reservation flows

8 Figure C-13 is an example of the reservation process associated with a Listener non-AP STA requesting a
9 Stream from a “Bridged” Talker. The Listener non-AP STA sends a Listener Ready (A) through the AP’s
10 MSRP Service, resulting in the IEEE 802.11 message exchanges shown. The reservation process associated
11 with the Listener Ready (B) sent to the “Bridged” Talker is explained in SRP (Clause 35).

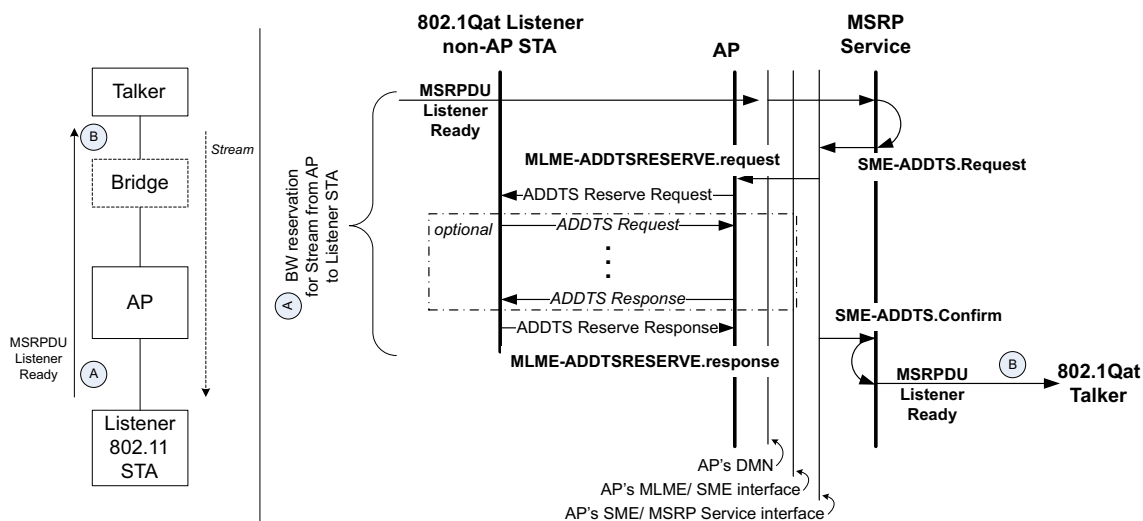


Figure C-13—MSRP/802.11 Listener STA to “Bridged” Talker reservation flows

1 The IEEE 802.11 message exchanges shown in the preceding three figures are explained in more detail as
2 follows:

- 3 a) BBS nodes identify MSRPDUs by their Group Destination Address (35.2.2.1) and EtherType
4 (35.2.2.2) and send these PDUs to the AP.
- 5 b) The AP forwards the MSRPDUs to the MSRP service on the DMN.
- 6 c) The DMN translates the MSRP TSPEC parameters into an equivalent IEEE 802.11 TSPEC and
7 invokes DMN-SME interface primitives with the AP as follows:
 - 8 1) When the DMN receives a Talker Advertise message that originated from an upstream BSS
9 node, the DMN invokes QoS Query transactions (SME-QUERY.Request) with the BSS QoS
10 AP (i.e., HC where dot11RobustAVStreamingImplemented is true) to check whether the
11 bandwidth advertised in the message's TSPEC is available on each upstream to downstream
12 node link of the BSS. In addition the DMN maps the MSRPDU's TSPEC with the message's
13 StreamID.
 - 14 2) When the DMN receives a Listener Ready message originated from a downstream BSS node,
15 the DMN invokes a QoS Reservation transaction (SME-ADDTS.Request) with the BSS QoS
16 manager to reserve the bandwidth associated with the message's StreamID on the downstream
17 to upstream BSS node link.

18 The IEEE 802.11 BSS QoS AP on receipt of a SME-ADDTS.Request from the DMN shall
19 make a determination about whether to accept the request or deny the request. The algorithm to
20 be used by the BSS QoS AP to make this determination is an implementation detail.

21 If the BSS QoS AP decides to accept the request, the AP shall derive a medium time value from
22 the parameters specified in the SME-ADDTS.Request. The BSS QoS AP shall then generate an
23 ADDTS Reserve Request frame in which the medium time value is included and transmit it to
24 the appropriate SRP Talker (BSS upstream) and Listener (BSS downstream) nodes. The
25 appropriate SRP Talker or Listener node replies to this ADDTS Reserve Request frame with an
26 ADDTS Reserve Response frame indicating if the node accepts the reservation request. One or
27 more ADDTS Request and ADDTS Response frames might be exchanged prior to the ADDTS
28 Reserve Response frame.

29 If the BSS QoS AP or Listener/Talker node decides to reject the request, it shall respond to the
30 DMN with SME-ADDTS.confirm with a ResultCode of Rejected. The confirm primitive may
31 also include a TSPEC, which the BSS QoS AP can accept, if specified in a subsequent SME-
32 ADDTS.request.

33 NOTE—The TSPEC included in the SME-ADDTS.confirm is based on the result of the negotiations
34 (labeled optional in Figure C-11 through Figure C-13) with the upstream BSS node. As a result the
35 TSPEC included in the SME-ADDTS.confirm may be different from the one in the SME-ADDTS.request
36 from the DMN.

- 37 d) After the DMN completes the BSS QoS transactions (SME-QUERY.Confirm or SME-
38 ADDTS.Confirm as appropriate), the DMN behaves as an MSRP application on a Bridge and
39 propagates MSRP attributes (35.2).

40 C.3.2 BSS DMN selection

41 The DMN shall be co-located with the device that supports the QoS AP function in the BSS.

1 C.3.3 BSS network bandwidth management

2 The MSRP service within the IEEE 802.11 network manages the BSS bandwidth for the MSRP streams by
3 invoking the MLME QoS services. The DMN shall map the MLME services as described in Table C-4.

Table C-4—SRP to MLME QoS Services mapping

MSRP Attribute	MAD Primitive	SME QoS Services	Description
Talker Advertise	MAD_Join.request(new)	SME-QUERY	Query bandwidth without reservation
Listener Ready or Listener Ready Failed	MAD_Join.request(new)	SME-ADDTS	Reserve bandwidth for a stream
Listener Ready or Listener Ready Failed	MAD_Join.request()	SME-ADDTS ^a	Renew the bandwidth reservation (leased time) for a stream
Talker or Listener Leave	MAD_Leave.request()	SME-DELTS	Free bandwidth associated with a stream

^a Bandwidth renewal is not required as long as the reservation is already established.

4 C.3.3.1 MSRPDU Encapsulation/De-encapsulation

5 In order to preserve the priority of an MSRPDU when traversing through an IEEE 802.11 network not using
6 a Length/Type field in the MSDU, the priority shall be encapsulated while the MSRPDU is in the IEEE
7 802.11 network and shall be de-encapsulated as it exits. See IEEE Std 802.11 for additional information.

8 NOTE—For example if the priority of the MSRPDU is 4, DEI=0 and VID = 1893 the equivalent VLAN tag field
9 (32 bits) is 81-00-87-65. When the frame enters the IEEE 802.11 network, the encapsulated 802.11 LLC header is AA-
10 AA-03-00-00-00-81-00-87-65-08-00, where AA-AA-03-00-00-00-81-00-87-65 is the SNAP encoded VLAN header.
11 When the frame exits the IEEE 802.11 network a de-encapsulation operation is performed and the resulting VLAN tag
12 field is 81-00-87-65.

13 C.3.3.2 QoS Maintenance Report

14 An SRP DMN may obtain QoS Maintenance Report using IEEE 802.11 Transmit Stream/Category
15 Measurement Requests and processing the corresponding Transmit Stream/Category Measurement Reports.
16 The Transmit Stream/Category Measurement Request is sent to both the SRP Talker (BSS upstream) and the
17 SRP Listener (BSS downstream) nodes. Triggers are set on appropriate conditions such that Transmit
18 Stream/Category Measurement Reports are generated only when predefined thresholds are breached. See
19 9.4.2.21.11 of IEEE Std 802.11-2016 [B16] for details.

20 NOTE—This provides an indication to SRP (35.2.4) that bandwidth has changed.

1 C.3.3.3 SRP TSpec to IEEE 802.11 TSPEC mapping

2 SR Class B traffic has three parameters associated with it, namely delay budget per IEEE 802.11 hop
3 (20 ms), **MaxFrameSize** (?1500 bytes) and **MaxIntervalFrames** (units of 4000 frames per second).
4 IEEE 802.11 TSPECs include a Minimum PHY rate that is derived from the SR Class B parameters as
5 described below:

- 6 a) Overhead = 10-byte VLAN tag + 8 byte Protocol definition = 18 bytes.
- 7 b) Mean Data Rate = (SRP TSpec **MaxFrameSize** + overhead) × 4000 × SRP TSpec
8 **MaxIntervalFrames** bytes/second.
- 9 c) The Mean Data Rate is also the Max Data Rate (since it is assumed that MSDU size is fixed).
- 10 d) Assuming 70% × efficiency between the MAC and the PHY this translates into
11 $(10/7) \times (\text{SRP TSpec } \mathbf{MaxFrameSize} + \text{overhead}) \times 4000 \times \text{SRP TSpec } \mathbf{MaxIntervalFrames}$
12 bytes/second, or
13 $(10/7) \times 8 \times (\text{SRP TSpec } \mathbf{MaxFrameSize} + \text{overhead}) \times 4000 \times \text{SRP TSpec } \mathbf{MaxIntervalFrames}$
14 bits/second.
- 15 e) Minimum PHY Rate is therefore
16 $(10/7) \times 8 \times (\text{SRP TSpec } \mathbf{MaxFrameSize} + \text{overhead}) \times 4000 \times \text{SRP TSpec } \mathbf{MaxIntervalFrames}$
17 bits/second.

18 NOTE—For example, with 1500 and 1 for **MaxFrameSize** the above turns into 69.394285 Mb/s. Or, with 64 and 1 for
19 **MaxFrameSize** and **MaxIntervalFrames** the above turns into 3.748571 Mb/s.

20 Table C-5 describes the mapping between SRP TSpec components and IEEE 802.11 QoS TSPEC
21 parameters for the mandatory EDCA-AC mode.

Table C-5—EDCA-AC for AV Streams

TSPEC parameter		SR Class B
TSINFO	TID	5
	Direction	Up, Down
	Access Policy	10 (EDCA)
	ACK Policy	10 (No Ack) / 11 (Block Ack)
	APSD	0
	Aggregation	Yes
	Priority	5
Nominal MSDU Size ^a		SRP TSpec MaxFrameSize+ 18 (also set bit 15 to a 1)
Maximum MSDU Size		SRP TSpec MaxFrameSize + 18
Peak Data Rate		$(\text{SRP TSpec } \mathbf{MaxFrameSize} + 18) \times 4000 \times \text{SRP TSpec } \mathbf{MaxIntervalFrames}$
Minimum PHY Rate ^b		$(10/7) \times 8 \times (\text{SRP TSpec } \mathbf{MaxFrameSize} + 18) \times 4000 \times \text{SRP TSpec } \mathbf{MaxIntervalFrames}$
Delay Bound		20 ms
Surplus Bandwidth Allowance ^c		1.2

^a bit15 set to indicate that the MSDU size is fixed.

^b Assuming 70% efficiency between the MAC and the PHY.

^c 20% surplus allocation.

¹ Table C-6 describes the mapping between SRP TSpec components and IEEE 802.11 QoS TSPEC
² parameters for the optional HCCA mode.

Table C-6—HCCA for AV Streams

TSPEC parameter		SR Class B
TSINFO	TID	5
	Direction	Up, Down
	Access Policy	01 (HCCA)
	ACK Policy	10 (No Ack) / 11 (Block Ack)
	APSD	0
	Aggregation	Yes
	Priority	5
Nominal MSDU Size ^a		0
Maximum MSDU Size		SRP TSpec MaxFrameSize+ 18
Minimum Service Interval		10 ms
Maximum Service Interval		10 ms
Inactivity Interval		0
Minimum Data Rate		0
Mean Data Rate		0
Maximum Burst Size		(SRP TSpec MaxFrameSize + 18) × 4000 × SRP TSpec MaxIntervalFrames × 10−2
Minimum PHY Rate ^b		(10/7) × 8 × (SRP TSpec MaxFrameSize+ 18) × 4000 × SRP TSpec MaxIntervalFrames
Peak Data Rate		(SRP TSpec MaxFrameSize + 18) × 4000 × SRP TSpec MaxIntervalFrames
Delay Bound		20 ms
Surplus Bandwidth Allowance ^c		1.2

^a Bit 15 set to indicate that the MSDU size is fixed.

^b Assuming 70% efficiency between the MAC and the PHY.

^c 20% surplus allocation.

Annex D

(normative)

IEEE 802.1 Organizationally Specific TLVs

D.1 Requirements of the IEEE 802.1 Organizationally Specific TLV sets

The IEEE 802.1 Organizationally Specific TLVs may be supported in conjunction with any of the destination MAC addresses identified by IEEE Std 802.1AB for LLDPDU transmission.

If any IEEE 802.1 Organizationally Specific TLV set is supported, all IEEE 802.1 Organizationally Specific TLVs that are identified as members of that TLV set shall be supported. All IEEE 802.1 Organizationally Specific TLVs shall conform to the LLDPDU bit and octet ordering conventions of IEEE Std 802.1AB.

The IEEE 802.1 Organizationally Specific TLVs specified in this standard are listed in Table D-1. Other standards can also define IEEE 802.1 Organizationally Specific TLVs. The “TLV set name” column identifies the TLV set to which each TLV belongs. Any additions or changes to these TLVs will be included in this annex.

Table D-1—IEEE 802.1 Organizationally Specific TLVs

IEEE 802.1 subtype	TLV name	TLV set name	TLV reference	Feature clause reference
0x01	Port VLAN ID	basicSet	D.2.1	6.9
0x02	Port And Protocol VLAN ID	basicSet	D.2.2	6.12
0x03	VLAN Name	basicSet	D.2.3	12.10.2.1.3
0x04	Protocol Identity	basicSet	D.2.4	D.2.4
0x05	VID Usage Digest	basicSet	D.2.5	D.2.5
0x06	Management VID	basicSet	D.2.6	D.2.6
0x07	Link Aggregation TLV	basicSet	IEEE Std 802.1AX	IEEE Std 802.1AX
0x08	Congestion Notification	cnSet	D.2.7	Clause 33
0x09	ETS Configuration TLV	dcbxSet	D.2.8	Clause 38
0x0A	ETS Recommendation TLV	dcbxSet	D.2.9	Clause 38
0x0B	Priority-based Flow Control Configuration TLV	dcbxSet	D.2.10	Clause 38
0x0C	Application Priority TLV	dcbxSet	D.2.11	Clause 38
0x0D	EVB TLV	evbSet	D.2.12	D.2.12
0x0E	CDCP TLV	evbSet	D.2.13	D.2.13
0x10	Application VLAN TLV	dcbxSet	D.2.14	Clause 38

Table D-1—IEEE 802.1 Organizationally Specific TLVs (continued)

IEEE 802.1 subtype	TLV name	TLV set name	TLV reference	Feature clause reference
0x11	LRP ECP Discovery TLV	lrpSet	IEEE Std 802.1CS	IEEE Std 802.1CS
0x12	LRP TCP Discovery TLV	lrpSet	IEEE Std 802.1CS	IEEE Std 802.1CS
0x13	Congestion Isolation TLV	ciSet	D.2.15	49.4.4
0x14	Topology Recognition TLV	trSet	D.2.16	49.5
0x15	PBBN Auto Attach System TLV	aaSet	D.2.17	Clause 50
0x16	PBBN Auto Attach Assignment TLV	aaSet	D.2.18	Clause 50

1 D.2 Organizationally Specific TLV definitions

2 In the TLV definitions that follow, any fields that are labeled as “Reserved” are transmitted as zero and
3 ignored on receipt.

4 NOTE—In the case of a Port supported by Link Aggregation (IEEE Std 802.1AX), these TLVs apply to the aggregation,
5 not the individual links. Since sending multiple copies is allowed, an implementation can send these TLVs with the same
6 values on each individual link, but a TLV received on any individual link applies to the aggregation.

7 D.2.1 Port VLAN ID TLV

8 The Port VLAN ID TLV is an optional fixed length TLV that allows a VLAN Bridge Port to advertise the
9 port VLAN identifier (PVID) that is associated with untagged or priority tagged frames (see 6.9).

10 Figure D-1 shows the Port VLAN ID TLV format.

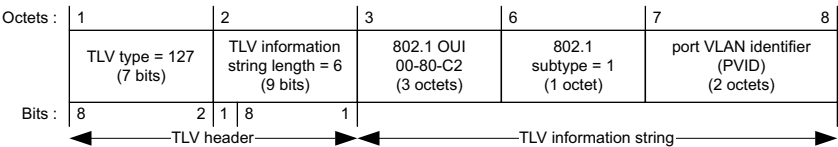


Figure D-1—Port VLAN ID TLV format

11 D.2.1.1 port VLAN identifier (PVID)

12 The PVID field shall contain the VID for the Bridge Port as defined in 6.9. A value of zero shall be used if
13 the system either does not know the PVID or does not support Port-based VLAN operation.

14 D.2.1.2 Port VLAN ID TLV usage rules

15 An LLDPDU should contain no more than one Port VLAN ID TLV.

1 D.2.2 Port And Protocol VLAN ID TLV

2 The Port And Protocol VLAN ID TLV is an optional TLV that allows a Bridge Port to advertise a port and
3 protocol VLAN identifier (PPVID). Figure D-2 shows the Port And Protocol VLAN ID TLV format.

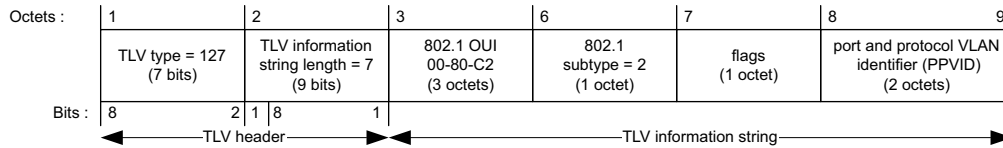


Figure D-2—Port And Protocol VLAN ID TLV format

4 D.2.2.1 flags

5 The flags field shall contain a bit map indicating the port and protocol VLAN capability and status as
6 defined in Table D-2.

Table D-2—Port and protocol capability/status

Bit	Function	Value/meaning
0	Reserved for future standardization	—
1	Port and protocol VLAN supported	1 = supported 0 = not supported
2	Port and protocol VLAN enabled	1 = enabled 0 = not enabled
3–7	Reserved for future standardization	(set to zero)

7 D.2.2.2 port and protocol VLAN identifier (PPVID)

8 The PPVID field shall contain the PPVID number for this IEEE 802 LAN station. If the port is not capable
9 of supporting port and protocol VLANs and/or the port is not enabled with any port and protocol VLAN, the
10 PPVID number should be zero.

11 D.2.2.3 Port And Protocol VLAN ID TLV usage rules

12 Port And Protocol VLAN ID TLVs are subject to the following rules:

- 13 a) If more than one Port And Protocol VLAN ID TLV is defined for a port, the PPVID value shall be
14 different from any other PPVID defined for the port.
- 15 b) If the support bit (bit 1) of the flag field indicates that the port is not capable of supporting port and
16 protocol VLANs but the enabled bit (bit 2) indicates that the port is enabled with one or more port
17 and protocol VLANs, the Port And Protocol VLAN ID TLV is interpreted as containing an error and
18 will be discarded.
- 19 c) If the PPVID reference number is greater than 4094, the Port And Protocol VLAN ID TLV is
20 interpreted as containing an error and is discarded.

1 D.2.3 VLAN Name TLV

2 The VLAN Name TLV is an optional TLV that allows an IEEE 802.1Q-compatible IEEE 802 LAN station
3 to advertise the assigned name of any VLAN with which it is configured.

⁴ Figure D-3 shows the VLAN Name TLV format.

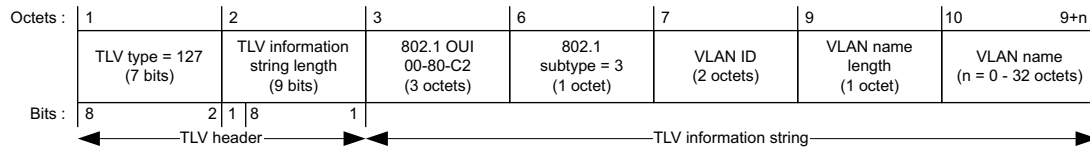


Figure D-3—VLAN Name TLV format

5 D.2.3.1 TLV information string length

⁶ The TLV information string length field shall contain the length, in octets, of the (VLAN name + 7).

7 D.2.3.2 VLAN ID (VID)

⁸ The VLAN ID field shall contain the VID number associated with the VLAN name.

9 D.2.3.3 VLAN name length

10 The VLAN name length field shall contain the length, in octets, of the VLAN name. A VLAN name length
11 of zero indicates that there is no VLAN name associated with this VLAN.

12 D.2.3.4 VLAN name

13 If present, the VLAN name field shall contain the VLAN's name. If implementations support
14 IETF RFC 4363, the dot1QVLANStaticName object should be used for this field.

15 D.2.3.5 VLAN Name TLV usage rules

16 If more than one VLAN Name TLV is defined for a port, the VID and the associated VLAN name
17 combination shall be different from any other VID and VLAN name combination defined for the port.

18 D.2.4 Protocol Identity TLV

19 The Protocol Identity TLV is an optional TLV that allows an IEEE 802 LAN station to advertise particular
20 protocols that are accessible through the port. Figure D-4 shows the protocol identity TLV format.

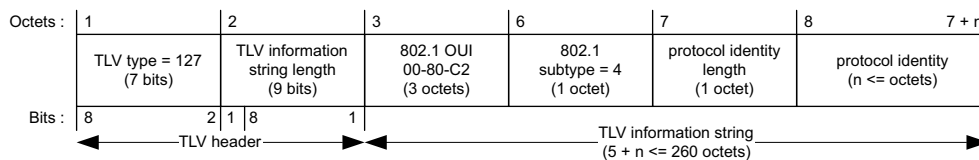


Figure D-4—Protocol Identity TLV format

21 D.2.4.1 TLV information string length

²² The TLV information string length field shall contain the length, in octets, of the (protocol identity + 5).

1 D.2.4.2 protocol identity length

2 The protocol identity length field shall contain the length, in octets, of the protocol identity.

3 D.2.4.3 protocol identity

4 The protocol identity field shall contain the first n octets of the protocol after the layer 2 addresses (i.e., for
5 example, starting with the EtherType field) that the sender would like to advertise. The value of n is
6 determined by the need for the protocol to disambiguate itself. The protocol information string shall include
7 enough octets to allow the receiver to correctly identify the protocol and its version. To advertise spanning
8 tree protocols, for example, the protocol identity field would need to include at least eight octets:
9 IEEE 802.3 length (two octets), LLC addresses (two octets), IEEE 802.3 control (one octet), Protocol ID
10 (two octets), and the protocol version (one octet).

11 D.2.4.4 Protocol Identity TLV usage rules

12 If more than one Protocol Identity TLV is defined for a port, the protocol identity field value shall be
13 different from any other Protocol Identity TLV defined for the port.

14 D.2.5 VID Usage Digest TLV

15 The VID Usage Digest TLV is an optional TLV that allows an IEEE Std 802.1Q-compatible IEEE 802 LAN
16 station to advertise the value of a VID Usage Digest associated with the system. The value of the VID Usage
17 Digest is obtained by applying the CRC32 function (see IEEE Std 802.3) to a VID Usage Table having a
18 fixed length of 512 octets. A bit of the VID Usage Table contains the value PBB-TE-USAGE (binary 1) if
19 the corresponding element of the MST Configuration Table (8.9.1) contains the value PBB-TE MSTID (hex
20 FFE) and otherwise contains the value NON-PBB-TE-USAGE (binary 0).

21 Figure D-5 shows the VID Usage Digest TLV format.

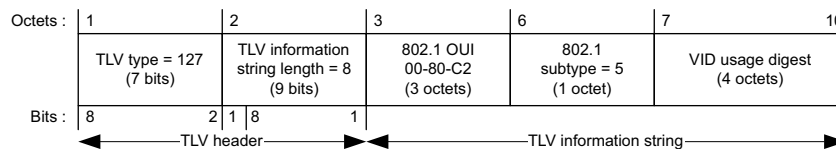


Figure D-5—VID Usage Digest TLV format

22 D.2.5.1 VID Usage Digest

23 The VID Usage Digest field shall contain a VID Usage Digest value obtained by applying the CRC32
24 function to the 512-octet VID Usage Table. A bit of the VID Usage Table contains the value
25 PBB-TE-USAGE (binary 1) if the corresponding element of the MST Configuration Table (8.9.1) contains
26 the value PBB-TE MSTID (hex FFE) and otherwise contains the value NON-PBB-TE-USAGE (binary 0).

27 D.2.6 Management VID TLV

28 The Management VID TLV is an optional TLV that allows an IEEE 802.1Q-compatible IEEE 802 LAN
29 station to advertise the value of a Management VID associated with the system.

1 Figure D-6 shows the Management VID TLV format.

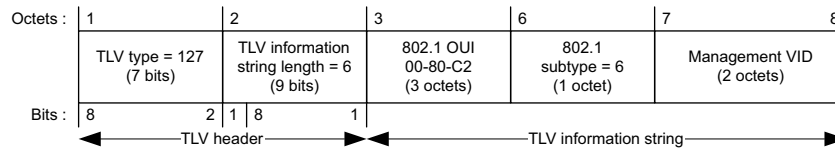


Figure D-6—Management VID TLV format

2 D.2.6.1 Management VID

3 The Management VID field shall contain the value configured for the Management VID, or the value 0 if a
4 Management VID has not been provisioned.

5 D.2.7 Congestion Notification TLV

6 The TLV illustrated in Figure D-7 is encoded into each LLDP message transmitted by a system that is
7 configured to support congestion notification on one or more priority values.

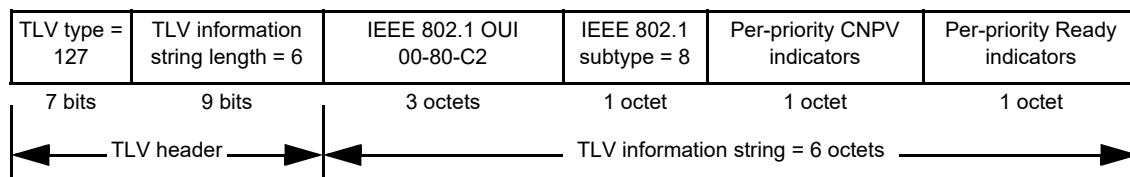


Figure D-7—Congestion Notification TLV format

8 A system shall transmit no more than one Congestion Notification TLV in a single LLDPDU. If a system
9 receives an LLDPDU with more than one Congestion Notification TLV, the behavior of the receiving system
10 is unspecified. A system shall not transmit a Congestion Notification TLV with all of the Per-priority CNPV
11 indicators (D.2.7.3) clear (0).

12 D.2.7.1 TLV type

13 A 7-bit integer value occupying the most significant bits of the first octet of the TLV. Always contains the
14 value 127.

15 D.2.7.2 TLV information string length

16 A 9-bit unsigned integer, occupying the LSB of the first octet of the TLV (the MSB of the TLV information
17 string length) and the entire second octet of the TLV, containing the total number of octets in the TLV
18 information string of the Congestion Notification TLV. This does not count the TLV type and TLV
19 information string length fields. It is equal to 6.

20 D.2.7.3 Per-priority CNPV indicators

21 A bit vector, one per priority value, containing all eight of the `cnpdXmitCnpvCapable` variables (32.4.7) for
22 this port. The LSB of the octet carries priority 0's `cnpdXmitCnpvCapable` variable, and the MSB that of
23 priority 7.

1 D.2.7.4 Per-priority Ready indicators

2 A bit vector, one per priority value, containing all eight of the `cnpdXmitReady` variables (32.4.8) for this
3 port. The LSB of the octet carries priority 0's `cnpdXmitReady` variable, and the MSB that of priority 7.

4 D.2.8 ETS Configuration TLV

5 The TLV illustrated in Figure D-8 is encoded into each LLDP message and may be transmitted by a system
6 in order to indicate the ETS configuration. Shall be sent using Asymmetric attribute passing (38.4.1).

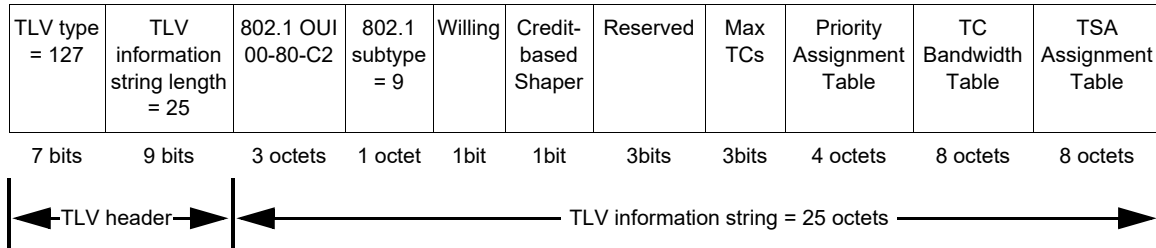


Figure D-8—ETS Configuration TLV format

7 D.2.8.1 TLV type

8 A 7-bit integer value occupying the most significant bits of the first octet of the TLV. Always contains the
9 value 127.

10 D.2.8.2 TLV information string length

11 A 9-bit unsigned integer, occupying the LSB of the first octet of the TLV (the MSB of the TLV information
12 string length) and the entire second octet of the TLV, containing the total number of octets in the TLV
13 information string of the ETS Configuration TLV. This does not count the TLV type and TLV information
14 string length fields. It is equal to 25.

15 D.2.8.3 Willing

16 The Willing bit. A value of one indicates that the station is willing to accept configurations from the remote
17 station.

18 D.2.8.4 Credit-based Shaper

19 The Credit-based Shaper bit. A value of one indicates that the station supports the Credit-based Shaper
20 transmission selection algorithm (see Table 8-6 and Clause 34).

21 D.2.8.5 Max TCs

22 A 3-bit unsigned integer where the value of the integer is the maximum number of traffic classes that the
23 implementation can support with the value 0 used when a device supports 8.

1 D.2.8.6 Priority Assignment Table

2 Table D-3 shows the layout of the priority assignment table.

Table D-3—Priority assignment table

Octets:	1				2				3				4			
	Priority 0		Priority 1		Priority 2		Priority 3		Priority 4		Priority 5		Priority 6		Priority 7	
Bits:	7	4	3	0	7	4	3	0	7	4	3	0	7	4	3	0

3 The table consists of one 4-bit entry per Priority, where:

- 4 a) 0–7 indicate the Priority is mapped to the corresponding traffic class; and
- 5 b) All other values are Reserved.

6 D.2.8.7 TC Bandwidth Table

7 Table D-4 shows the layout of the traffic class bandwidth table.

Table D-4—Traffic class bandwidth assignment table

Octets:	1	2	3	4	5	6	7	8
	TC% 0	TC% 1	TC% 2	TC% 3	TC% 4	TC% 5	TC% 6	TC% 7

8 The table consists of one 8-bit entry per traffic class, and always contains 8 entries. Each entry:

- 9 a) Indicates the current TCBandwidth percentage configured for each traffic class N where N is 0 to 7.
- 10 b) Total shall equal 100 (implies valid range is 0–100 for each entry).

11 NOTE—While it is intended that only TCs configured for ETS will have a bandwidth value associated with them, it is
12 possible, during configuration changes, to have situations where a TC is not configured for ETS but has a nonzero
13 TCBandwidth percentage. In this case, the sum of all the TCBandwidth percentages is still equal to 100, but the TC
14 bandwidth percentages of the non-ETS TCs would be unused bandwidth and reallocated to the ETS TCs.

15 D.2.8.8 TSA Assignment Table

16 Table D-5 shows the layout of the Transmission Selection Algorithm (TSA) Assignment Table.

Table D-5—TSA Assignment Table

Octets:	1	2	3	4	5	6	7	8
	Traffic Class 0	Traffic Class 1	Traffic Class 2	Traffic Class 3	Traffic Class 4	Traffic Class 5	Traffic Class 6	Traffic Class 7

17 The table consists of one 8-bit entry per traffic class. Each entry indicates the Transmission selection
18 algorithm to be used for that traffic class as defined in Table 8-6. A value of 255 indicates a Vendor-specific
19 Transmission selection algorithm.

1 **D.2.9 ETS Recommendation TLV**

2 The TLV illustrated in Figure D-9 is encoded into each LLDP message and may be transmitted by a system
3 in order to indicate a recommendation on how ETS should be configured. A Willing system may receive a
4 recommendation TLV that utilizes more Traffic Classes than the receiving system supports. In this case the
5 receiving system may assign the received recommendations to match its capabilities. Shall be sent using
6 Asymmetric attribute passing (38.4.1).

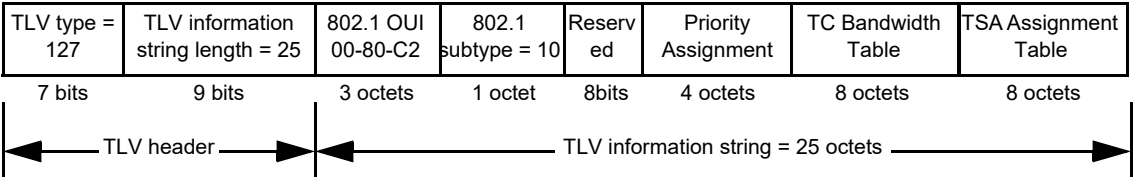


Figure D-9—ETS Recommendation TLV format

7 **D.2.9.1 TLV type**

8 A 7-bit integer value occupying the most significant bits of the first octet of the TLV. Always contains the
9 value 127.

10 **D.2.9.2 TLV information string length**

11 A 9-bit unsigned integer, occupying the LSB of the first octet of the TLV (the MSB of the TLV information
12 string length) and the entire second octet of the TLV, containing the total number of octets in the TLV
13 information string of the ETS Recommendation TLV. This does not count the TLV type and TLV
14 information string length fields. It is equal to 25.

15 **D.2.9.3 Priority Assignment Table**

16 Table D-3 shows the layout of the priority assignment table.

17 See D.2.8.6 for description of table entries.

18 **D.2.9.4 TC Bandwidth Table**

19 Table D-4 shows the layout of the Traffic Class bandwidth assignment table.

20 See D.2.8.7 for description of table entries.

21 **D.2.9.5 TSA Assignment Table**

22 Table D-5 shows the layout of the TSA Assignment Table.

23 See D.2.8.8 for description of table entries.

1 D.2.10 Priority-based Flow Control Configuration TLV

2 The TLV illustrated in Figure D-10 is encoded into each LLDP message and may be transmitted by a system
3 in order to indicate how PFC should be configured. Shall be sent using Symmetric attribute passing.

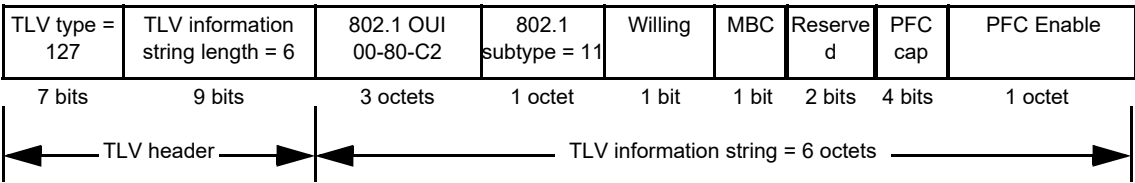


Figure D-10—Priority-based Flow Control Configuration TLV format

4 D.2.10.1 TLV type

5 A 7-bit integer value occupying the most significant bits of the first octet of the TLV. Always contains the
6 value 127.

7 D.2.10.2 TLV information string length

8 A 9-bit unsigned integer, occupying the LSB of the first octet of the TLV (the MSB of the TLV information
9 string length) and the entire second octet of the TLV, containing the total number of octets in the TLV
10 information string of the Priority-based Flow Control Configuration TLV. This does not count the TLV type
11 and TLV information string length fields. It is equal to 6.

12 D.2.10.3 Willing

13 The Willing bit. A value of one indicates that the station is willing to accept configurations from the remote
14 station.

15 D.2.10.4 MBC

16 The MACsec Bypass Capability Bit. If set to zero, the sending station is capable of bypassing MACsec
17 processing when MACsec is disabled. If set to one, the sending station is not capable of bypassing MACsec
18 processing when MACsec is disabled (see Clause 36).

19 D.2.10.5 PFC cap

20 A 4-bit unsigned integer, PFC cap (PFC capability) indicates the device’s limitation of how many traffic
21 classes may simultaneously support PFC.

22 NOTE 1—PFC may be enabled on up to eight traffic classes allowing each priority to be independently paused.

23 NOTE 2—In a DCB environment at least one priority would generally have PFC enabled.

1 D.2.10.6 PFC Enable

2 Table D-6 shows the layout of the PFC Enable bit vector.

Table D-6—PFC Enable bit vector

Octet:	1							
	Priority 7	Priority 6	Priority 5	Priority 4	Priority 3	Priority 2	Priority 1	Priority 0
Bits:	7	6	5	4	3	2	1	0

3 A bit vector of 8 bits, one per priority:

- 4 a) A one indicates PFC is enabled on the priority.
- 5 b) A zero indicates that PFC is disabled on the priority.
- 6 c) Local policy in each end of the link decides whether to use the priority if the configuration does not
- 7 match.

8 D.2.11 Application Priority TLV

9 The TLV illustrated in Figure D-11 is encoded into each LLDP message and may be transmitted by a system
10 in order to indicate the application priority table. This TLV is informational and used to indicate to a peer
11 station the local configuration.

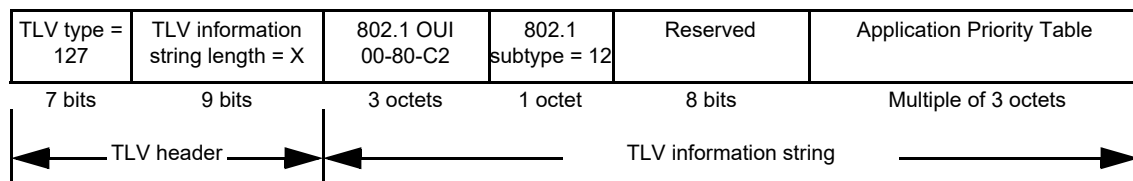


Figure D-11—Application Priority TLV format

12 D.2.11.1 TLV type

13 A 7-bit integer value occupying the most significant bits of the first octet of the TLV. Always contains the
14 value 127.

15 D.2.11.2 TLV information string length

16 A 9-bit unsigned integer, occupying the LSB of the first octet of the TLV (the MSB of the TLV information
17 string length) and the entire second octet of the TLV, containing the total number of octets in the TLV
18 information string of the Application Priority TLV. This does not count the TLV type and TLV information
19 string length fields. The length for the Application Priority TLV is variable depending on the number of
20 Application Priorities specified. The length shall be 5 plus a multiple of 3 octets.

1 D.2.11.3 Application Priority Table

2 Table D-7 shows the layout of a 3-octet entry in the Application Priority Table field.

Table D-7—Application Priority Table Entry format

Octets:	1					2		3	
	Priority		Reserved		Sel	Protocol ID			
Bits:	8	6	5	4	3	1	8	1	8

3 The priority field is a 3-bit unsigned integer indicating the priority for which the Protocol ID is being used.

4 The meaning of the Protocol ID field is determined by the Sel field. Allowed values for the Sel field are
5 shown in Table D-8.

Table D-8—Sel field values

Sel value	Protocol ID value
0	Reserved
1	0: Default application priority. For use when application priority is not otherwise specified. 1–1535: Reserved. 1536–65 535: An EtherType
2	Well Known Port number over TCP, or Stream Control Transmission Protocol (SCTP) [B36].
3	Well Known Port number over UDP, or Datagram Congestion Control Protocol (DCCP).
4	Well Known Port number over TCP, SCTP, UDP, or DCCP.
5	Differentiated Services Code Point (DSCP) value. The 6-bit DSCP value is stored in bits 1–6 of the low order octet of the Protocol ID field (octet 3 of the Application Priority Table Entry). Bits 7 and 8 and the high order octet of the Protocol ID field are set to zero. (See IETF RFC 2474 for the definition of the DSCP value.)
6–7	Reserved

6 NOTE—The port numbers shown are for identification (i.e., as assigned by IANA) instead of the actual port numbers
7 being used in a particular deployment.

8 D.2.12 EVB TLV

9 The EVB TLV is used to:

- 10 a) Advertise a station or Bridge's EVB capabilities.
- 11 b) Negotiate and activate common capabilities.

12 The EVB TLV is exchanged via LLDP and conforms to the LLDP TLV specification. The LLDP database
13 carrying the EVB TLV is addressed using the Nearest Customer Bridge address. One LLDP database is built
14 at the URP of each ER. If the EVB TLV is used for VDP NVO3 support, the LLDP database carrying the
15 EVB TLV can also be accessed using a unicast MAC address.

16 The EVB TLV allows setting the EVB Bridge's C-VLAN component Port to operate in reflective relay.
17 Reflective relay is implemented by changing the active topology enforcement rules described in 8.6.1 to
18 allow forwarding on the reception Bridge Port. When reflective relay is enabled on a given Bridge Port, that
19 port is a potential transmission port for frames received on that port.

1 The EVB TLV structure is illustrated in Figure D-12.

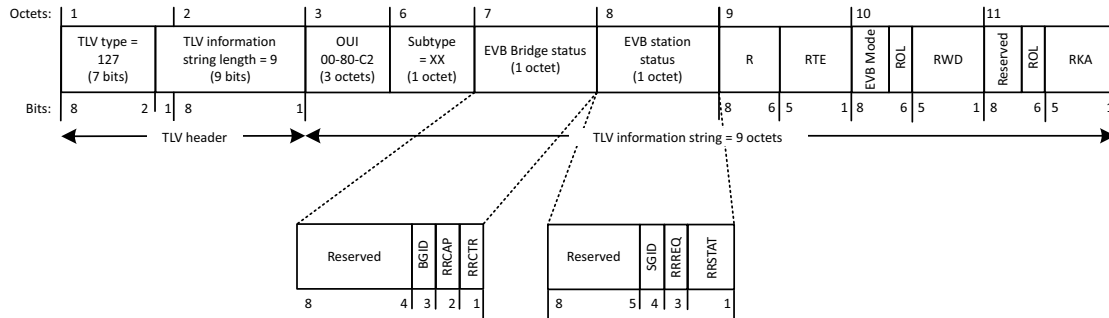


Figure D-12—EVB TLV format

2 The EVB TLV information string fields are as defined in D.2.12.1 through D.2.12.10.

3 D.2.12.1 OUI

4 The OUI used to identify the EVB TLV is the 802.1 OUI 00-80-C2.

5 D.2.12.2 Subtype

6 The subtype used to identify the EVB TLV is as shown in Table D-1.

7 D.2.12.3 EVB Bridge status

8 The EVB Bridge status field describes EVB capabilities that are supported by the EVB Bridge. If the sender
9 of the TLV is an EVB Bridge (EVB Mode = EVB Bridge—see D.2.12.7), then the field reflects its own
10 capabilities; if the sender of the TLV is an EVB station (EVB Mode = EVB station), then the field reflects
11 the capabilities received from an attached EVB Bridge, or a value of zero if no TLV has been received from
12 an attached EVB Bridge.

13 If the sender of the TLV is an nNVE (EVB Mode = NVO3 and NVE Role = nNVE—see D.2.12.7 and
14 D.2.12.10), then the field reflects its own capabilities; if the sender of the TLV is a tNVE (EVB Mode =
15 NVO3 and NVE Role = tNVE), then the field reflects the capabilities received from an attached nNVE, or a
16 value of zero if no TLV has been received from an attached nNVE.

17 Each capability is represented by a single bit flag; a value of TRUE (1) indicates that the capability is
18 supported, a value of FALSE (0) indicates that the capability is not supported. The capabilities are as defined
19 in D.2.12.3.1 through D.2.12.3.5.

20 D.2.12.3.1 BGID

21 A value of TRUE indicates that the EVB Bridge wishes to control VID assignments and use the GroupID in
22 VDP exchanges. A value of FALSE indicates that the EVB Bridge does not wish to make use of the Group
23 ID in VDP exchanges.

24 If the EVB station sets SGID = TRUE, and the EVB Bridge also sets BGID = TRUE, then the EVB Bridge
25 can control VID assignments and use the GroupID in VDP exchanges.

26 If the EVB station does not set SGID = TRUE, or the EVB Bridge does not set BGID = TRUE, then the
27 EVB Bridge cannot control VID assignments or use the GroupID in VDP exchanges.

28 In an NVO3 Split-NVE scenario, BGID should always be set to TRUE by nNVE.

1 **D.2.12.3.2 RRCAP**

2 The RRCAP flag indicates the state of the EVB Bridge's reflectiveRelayCapable parameter (40.4.2).

3 If the EVB Bridge's reflectiveRelayCapable parameter is TRUE, and a TLV has been received by the EVB
4 Bridge from an attached EVB station in which the value of RRREQ (D.2.12.4.2) is also TRUE, then the
5 value of the EVB Bridge's operReflectiveRelayControl parameter (40.4.2) shall be set to TRUE. Otherwise,
6 the value of the EVB Bridge's operReflectiveRelayControl parameter (40.4.2) shall be set to FALSE.

7 In an NVO3 Split-NVE scenario, RRCAP should be FALSE since reflective relay is not a required
8 capability of nNVE.

9 **D.2.12.3.3 RRCTR**

10 The RRCTR flag indicates the state of the EVB Bridge's operReflectiveRelayControl parameter (40.4.2).

11 In an NVO3 Split-NVE scenario, RRCTR should be FALSE since RRREQ (D.2.12.4.2) received by nNVE
12 is always FALSE.

13 **D.2.12.3.4 V6CAP**

14 This capability is valid only when NVE Mode is NVO3. A value of TRUE indicates that the nNVE supports
15 the use of IPv6 addresses in the Filter Info field of the VDP association TLV. A value of FALSE indicates
16 that the nNVE does not support the use of IPv6 addresses in that field. When both nNVE and tNVE set
17 V6CAP = TRUE, then IPv6 addresses can be used in the Filter Info field in VDP exchanges; otherwise, IPv6
18 addresses cannot be used in the Filter Info field in VDP exchanges.

19 **D.2.12.3.5 V4CAP**

20 This capability is valid only when NVE Mode is NVO3. A value of TRUE indicates that the nNVE supports
21 the use of IPv4 addresses in the Filter Info field of the VDP association TLV. A value of FALSE indicates
22 that the nNVE does not support the use of IPv4 addresses in that field. When both nNVE and tNVE set
23 V4CAP = TRUE, then IPv4 addresses can be used in the Filter Info field in VDP exchanges; otherwise, IPv4
24 addresses cannot be used in the Filter Info field in VDP exchanges.

25 **D.2.12.4 EVB station status**

26 The EVB station status field describes EVB capabilities that are supported by the EVB station. If the sender
27 of the TLV is an EVB station (EVB Mode = EVB station—see D.2.12.7), then the field reflects its own
28 capabilities; if the sender of the TLV is an EVB Bridge (EVB Mode = EVB Bridge), then the field reflects
29 the capabilities received from an attached EVB station, or a value of zero if no TLV has been received from
30 an attached EVB station.

31 If the sender of the TLV is a tNVE (EVB Mode = NVO3 and NVE Role = tNVE—see D.2.12.7 and
32 D.2.12.10), then the field reflects its own capabilities; if the sender of the TLV is an nNVE (EVB Mode =
33 NVO3 and NVE Role = nNVE), then the field reflects the capabilities received from an attached tNVE, or a
34 value of zero if no TLV has been received from an attached tNVE.

35 Each capability is represented by a single bit flag; a value of TRUE (1) indicates that the capability is
36 supported, a value of FALSE (0) indicates that the capability is not supported. The capabilities are as defined
37 in D.2.12.4.1 through D.2.12.4.5.

38 **D.2.12.4.1 SGID**

39 A value of TRUE indicates that the EVB station can support the use of the GroupID.

1 If the EVB station sets SGID = TRUE, and the EVB Bridge also sets BGID = TRUE, then the EVB Bridge
2 can control VID assignments and use the GroupID in VDP exchanges.

3 If the EVB station does not set SGID = TRUE, or the EVB Bridge does not set BGID = TRUE, then the
4 EVB Bridge cannot control VID assignments or use the GroupID in VDP exchanges.

5 In an NVO3 Split-NVE scenario, SGID should always be set to TRUE by tNVE.

6 **D.2.12.4.2 RRREQ**

7 The RRREQ flag indicates the state of the EVB station's adminReflectiveRelayRequest parameter (40.4.3).

8 In an NVO3 Split-NVE scenario, RRREQ should always be set to FALSE in order to make sure reflective
9 relay is not enabled.

10 **D.2.12.4.3 RRSTAT**

11 RRSTAT is a composite flag that indicates the state of the EVB station's operReflectiveRelayStatus
12 parameter (40.4.3) as shown in Table D-9.

Table D-9—RRSAT flag values and meanings

Bit 1	Bit 2	Meaning
TRUE	FALSE	operReflectiveRelayStatus is TRUE
FALSE	FALSE	operReflectiveRelayStatus is FALSE
TRUE	TRUE	operReflectiveRelayStatus is Unknown
FALSE	TRUE	operReflectiveRelayStatus is Unknown

13 If a TLV has been received by the EVB station from an attached EVB Bridge in which the value of RRCTR
14 (D.2.12.3.3) is TRUE, then the value of the EVB station's operReflectiveRelayStatus parameter (40.4.3)
15 shall be set to TRUE. If a TLV has been received by the EVB station from an attached EVB Bridge in which
16 the value of RRCTR (D.2.12.3.3) is FALSE, then the value of the EVB station's operReflectiveRelayStatus
17 parameter (40.4.3) shall be set to FALSE. If no TLV has been received by the EVB station from an attached
18 EVB Bridge, then the value of the EVB station's operReflectiveRelayStatus parameter (40.4.3) shall be set
19 to Unknown.

20 In an NVO3 Split-NVE scenario, RRSTAT is always FALSE as RRCTR (D.2.12.3.3) received from nNVE
21 is FALSE.

22 **D.2.12.4.4 V6CAP**

23 This capability is valid only when NVE Mode is NVO3. A value of TRUE indicates that the tNVE supports
24 the use of IPv6 addresses in the Filter Info field of the VDP association TLV. A value of FALSE indicates
25 that the tNVE does not support the use of IPv6 addresses in that field. When both nNVE and tNVE set
26 V6CAP = TRUE, then IPv6 addresses can be used in the Filter Info field in VDP exchanges; otherwise, IPv6
27 addresses cannot be used in the Filter Info field in VDP exchanges.

28 **D.2.12.4.5 V4CAP**

29 This capability is valid only when NVE Mode is NVO3. A value of TRUE indicates that the tNVE supports
30 the use of IPv4 addresses in the Filter Info field of the VDP association TLV. A value of FALSE indicates
31 that the tNVE does not support the use of IPv4 addresses in that field. When both nNVE and tNVE set
32 V4CAP = TRUE, then IPv4 addresses can be used in the Filter Info field in VDP exchanges; otherwise, IPv4
33 addresses cannot be used in the Filter Info field in VDP exchanges.

1 **D.2.12.5 R**

2 This field carries the maxRetries value for the ECP state machine (43.3.7.4). Both sides transmit the local
3 value, and use the largest of the two values of R. If no remote value is available, then the local value is used.

4 **D.2.12.6 RTE (retransmission exponent)**

5 RTE is an EVB link or S-channel attribute used to calculate the minimum ECPDU retransmission time,
6 ackTimerInit. The value of ackTimerInit, in ECP timer tics of 10 usec, is calculated as:

$$7 \quad 2^{\text{RTE}} \text{ ECP timer tics}$$

8 Both sides transmit the local value, and use the largest of the two values of RTE for this calculation. If no
9 remote value is available, then the greater of 2 ms and local value is used.

10 **D.2.12.7 EVB Mode**

11 The EVB Mode field represents the value of the EVBMode parameter (40.4) for the sender of the TLV, as
12 shown in Table D-10.

Table D-10—EVB Mode values

EVBMode (40.4)	Field value
Not Supported	0
EVB Bridge	1
EVB station	2
NVO3	3

13 **D.2.12.8 ROL (remote or local) and RWD (resource wait delay)**

14 The RWD value transmitted by the EVB Bridge indicates the exponent value in use by the EVB Bridge for
15 determining the value of the resourceWaitDelay variable (41.5.5.7). The value of resourceWaitDelay, in
16 VDP timer tics of 10 usec, is calculated as:

$$17 \quad 2^{\text{RWD}} \text{ VDP timer tics}$$

18 Both sides transmit the local value, and use the largest of the local and remote values of RWD in calculation
19 of resourceWaitDelay or respWaitDelay variable (41.5.5.9); if there is no remote value available, the local
20 (proposed) value is used. The Remote or Local (ROL) flag indicates whether the remote RWD value is in
21 use (TRUE) or the local value is in use (FALSE) at the sending node.

22 **D.2.12.9 ROL (remote or local) and RKA (reinit keep alive)**

23 The RKA value transmitted by the EVB station indicates the exponent value in use by the EVB station for
24 determining the value of the reinitKeepAlive variable (41.5.5.5). The value of reinitKeepAlive, in VDP
25 timer tics of 10 usec, is calculated as:

$$26 \quad 2^{\text{RKA}} \text{ VDP timer tics}$$

27 Both sides transmit the local value, and use the largest of the local and remote values of RKA in calculation
28 of reinitKeepAlive or toutKeepAlive variable (41.5.5.13); if there is no remote value available, the local
29 (proposed) value is used. The Remote or Local (ROL) flag indicates whether the remote RKA value is in use
30 (TRUE) or the local value is in use (FALSE) at the sending node.

1 D.2.12.10 NVE Role

2 NVE Role field represents the value of the NVERole parameter (40.5) for the sender of the TLV, as shown in
3 Table D-11. This field is useful only when EVBMode = NVO3.

Table D-11—NVE Role values

NVERole (40.5)	Field values
nNVE	00
tNVE	01
Reserved	10, 11

4 D.2.13 CDCP TLV

5 The EVB station and Bridge both use the same LLDP TLV to configure S-channels (see Figure D-13). This
6 TLV is in LLDP Organizationally Specific TLV format (see IEEE Std 802.1AB). The S-channel's
7 capabilities, requests, and running configuration is encoded in the information string of this TLV as defined
8 in D.2.13.1 through D.2.13.8.

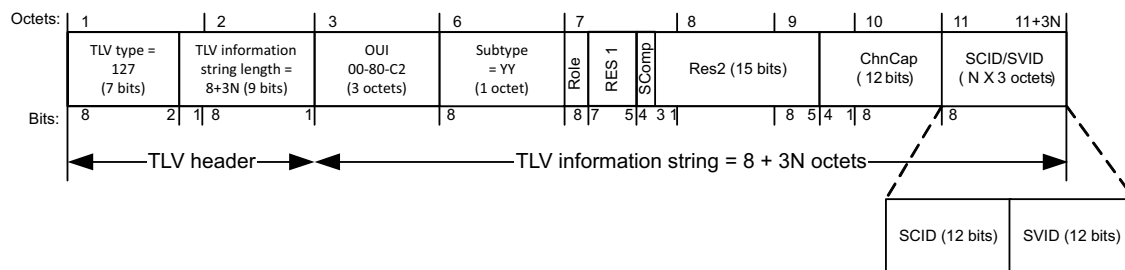


Figure D-13—CDCP TLV structure

9 D.2.13.1 OUI

10 The OUI used to identify the CDCP TLV is the 802.1 OUI 00-80-C2.

11 D.2.13.2 Subtype

12 The subtype used to identify the CDCP TLV is as shown in Table D-1.

13 D.2.13.3 Role

14 Role is a 1-bit field, defined as follows:

- 15 a) S(1)—Indicates the sender is operating in the station role, assigns channels numbers and a default SVID for the default channel 1, and requests SVID assignments from the neighboring 'B'.
- 16
- 17 b) B(0)—Indicates the sender is operating in the Bridge role, accepts S-channel configuration requests from its neighboring 'S' and that the sender will do the best it can to fill the SVID assignment requests from the neighboring 'S'.
- 18
- 19

20 D.2.13.4 RES1

21 RES1 is a 3-bit field, reserved for future standardization. This field is transmitted as zero and ignored on
22 receipt.

1 D.2.13.5 SComp

2 SComp is a 1-bit field that indicates the presence or absence of an S-VLAN component for S-channel
3 support. A value of 1 indicates TRUE, zero indicates FALSE.

4 NOTE—If this bit is zero, then the sender does not have a CDCP state machine, and the other fields in the TLV are not
5 valid.

6 D.2.13.6 Res2

7 RES2 is a 15-bit field, reserved for future standardization. This field is transmitted as zero and ignored on
8 receipt.

9 D.2.13.7 ChnCap

10 Channel capacity. Identifies the total number of S-channels, both assigned and available to be assigned, that
11 the sender has.

12 D.2.13.8 SCID/SVID

13 An SCID/SVID pair exists for each S-channel that is currently supported by the sender. Each SCID/SVID
14 pair consists of two 12-bit values, as follows:

- 15 a) SCID—indicates the index number of the S-channel. The station assigns S-channel numbers in the
16 range 0–167. Zero is reserved. The S-channel index should be between 1 and the maximum number
17 of S-channels supported by the port.
- 18 b) SVID—The VID assigned to the S-channel. The Bridge assigns SVIDs to channels in the range
19 1–0xffe. A station uses the 0 SVID to request an SVID assignment from the Bridge.

20 After the station receives the SVID assignment from the Bridge, it uses the SVID assigned value in all
21 subsequent exchanges for that specific SCID (SCID/SVID pair).

22 NOTE 1—The first entry in the list of SCID/SVID pairs contains the default S-channel. (i.e., the first channel pair is
23 <1,1>).

24 NOTE 2—A maximum of 167 S-channels can be supported. Other formats (assuming sequential SVIDs) could be
25 defined to allow support for 4K+ S-channels.

26 NOTE 3—This listing can be sparse (in order to indicate arrival and removal of S-channels). The S-channel going away
27 is indicated by removing the SCID/SVID pair.

28 NOTE 4—The order of the list determines the priority of SVID assignments. If the Bridge does not have resources for
29 all channels, it assigns the first channels in the list.

30 D.2.14 Application VLAN TLV

31 The TLV illustrated in Figure D-14 is encoded into each LLDP message and may be transmitted by a system
32 in order to indicate the application VLAN table. This TLV is informational and used to indicate to a peer
33 station the local configuration.

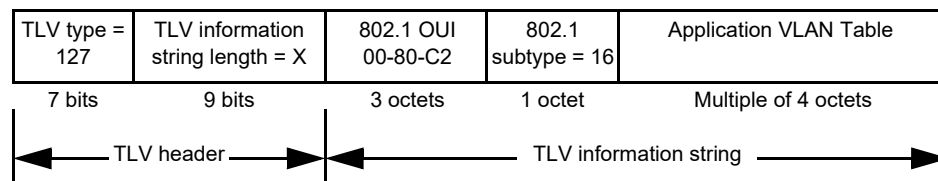


Figure D-14—Application VLAN TLV format

34 D.2.14.1 TLV type

35 A 7-bit integer value occupying the most significant bits of the first octet of the TLV. Always contains the
36 value 127.

1 D.2.14.2 TLV information string length

2 A 9-bit unsigned integer, occupying the LSB of the first octet of the TLV (the MSB of the TLV information
3 string length) and the entire second octet of the TLV, containing the total number of octets in the TLV
4 information string of the Application VLAN TLV. This does not count the TLV type and TLV information
5 string length fields. The length for the Application VLAN TLV is variable depending on the number of
6 Application VLANs specified. The length shall be 4 plus a multiple of 4 octets.

7 D.2.14.3 Application VLAN Table

8 Table D-12 shows the layout of a 4-octet entry in the Application VLAN Table.

Table D-12—Application VLAN Table Entry format

Octets:	1	2	3	4
Field:	VID		Reserved	Sel
Bits:	8	1	8	5
			4	3
			1	8
				1

9 The VID field is a 12-bit unsigned integer indicating the VID of the VLAN for which the Protocol ID is
10 being used.

11 The meaning of the Protocol ID field is determined by the Sel field. Allowed values for the Sel field are
12 shown in Table D-13.

Table D-13—Sel field values

Sel value	Protocol ID value
0	Reserved
1	0: PVID. For use when an application VLAN is not otherwise specified. 1–1535: Reserved 1536–65 535: An EtherType.
2	Well Known Port number over TCP, or SCTP.
3	Well Known Port number over UDP, or DCCP.
4	Well Known Port number over TCP, SCTP, UDP, or DCCP.
5	Differentiated Services Code Point (DSCP) value. The 6-bit DSCP value is stored in bits 1–6 of the low order octet of the Protocol ID field (octet 4 of the Application VLAN Table Entry). Bits 7 and 8 and the high order octet of the Protocol ID field are set to zero. (See IETF RFC 2474 for the definition of the DSCP value.)
6–7	Reserved

13 The Application VLAN TLV may be exchanged between EVB station ERs (5.24.1) and EVB Bridges (5.23)
14 with LLDP using the Nearest Customer Bridge address. This allows ERs sharing a LAN using S-channels
15 (40.2) to use different VLANs for the same application.

16 NOTE 1—The Application VLAN TLV allows multiple VLANs to be associated with an application on a port. Issues
17 involved with using multiple VLANs for an application on a port, such as resolving which IP subnetwork corresponds to
18 which VLAN, are beyond the scope of this standard.

19 NOTE 2—Selection of different IEEE 802.1 paths may use different VLANs, which could be associated with different
20 quality of service levels. This TLV may not apply to deployment cases where the VID is implied by other attributes of
21 the frame, for example, by the IP subnet of the source IP address.

1 D.2.15 Congestion Isolation TLV

2 The Congestion Isolation TLV is an optional TLV that allows an IEEE 802.1Q-compliant Bridge and an
3 IEEE 802.1Q-compatible IEEE 802 LAN station to discover each other and exchange configuration
4 information for congestion isolation.

5 Figure D-15 shows the Congestion Isolation TLV format.

Figure D-15—Congestion Isolation TLV format

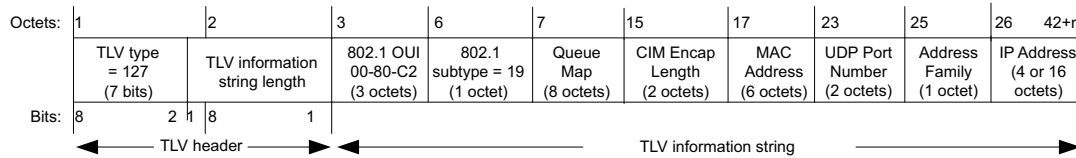


Figure D-15—Congestion Isolation TLV format

6 D.2.15.1 TLV type

7 A 7-bit integer value occupying the most-significant bits of the first octet of the TLV. Always contains the
8 value 127.

9 D.2.15.2 TLV information string length

10 The TLV information string length field of the Congestion Isolation TLV depends on the Address Family
11 specified for the IP Address field and shall contain the value 25 if the Address Family is 1 (IPv4), the value
12 37 if the Address Family is 2 (IPv6), or the value 21 if the Address Family is anything other than 1 or 2.

13 D.2.15.3 Queue Map

14 Eight octets, one for each traffic class supported by the Bridge or end station. The least significant octet
15 represents traffic class 0 and the most significant octet represents traffic class 7. The content of each octet
16 specifies a signed numeric value that can be translated to the value of the traffic class to be used as either the
17 congesting queue or the monitored queue for the represented traffic class. The contents of each octet is a
18 signed numeric value between −8 and 8. A value of 0 specifies that there is no mapping for the traffic class
19 and that it is not participating in congestion isolation. A positive number specifies a traffic class for a
20 monitored queue that is one less than the value (e.g., a value of 5 represents traffic class 4). A negative
21 number specifies a traffic class for a congesting queue that is one less than the absolute value (e.g., a value
22 of −4 represents traffic class 3).

23 D.2.15.4 CIM Encap Length

24 A 2-octet unsigned integer representing the requested length in octets of the data from the frame of a
25 congesting flow to be encapsulated into a CIM by a peer. The default and minimum required value is 48. The
26 maximum value is 512.

27 D.2.15.5 MAC Address

28 The 6-octet MAC address that can be used as the destination MAC address for a CIM to the Bridge or end
29 station sending this TLV.

1 **D.2.15.6 UDP Port Number**

2 An integer value indicating the UDP port number to be used by the peer when creating a layer-3 CIM. The
3 value of the UDP port number is within the dynamic port number range of 49152 to 65535 as specified by
4 IETF RFC 6335 (e.g., 58622).

5 **D.2.15.7 Address Family**

6 An integer value indicating the type of address that is listed in the address field. Enumeration for this field is
7 contained in the ianaAddressFamilyNumbers module of the IETF RFC 3232 on-line database that is
8 accessible through a web page (<http://www.iana.org>). Only two types are recognized with the following
9 definitions; 1=IPv4 and 2=IPv6.

10 **D.2.15.8 IP Address**

11 An octet string with length of 0, 4, or 16 octets dependent upon the Address Family specified. If the Address
12 Family is 1, the address shall be an IPv4 address as specified in IETF RFC 791. If the Address Family is 2,
13 the address shall be an IPv6 address as specified in IETF RFC 8200. No address shall be provided for any
14 other Address Families.
15 NOTE—If the CIM PDU is layer 2 encapsulated, the MAC address field is already provided and the Address Family
16 should be anything other than 1 or 2. For layer 2 encapsulated CIM PDUs, it is recommended to use the Address Family
17 value of 6 for IEEE 802 addresses.

18 **D.2.15.9 Congestion Isolation TLV usage rules**

19 The priority of a congesting queue shall be lower than the priority of all monitored queues.

20 **D.2.16 Topology Recognition TLV**

21 The Topology Recognition TLV is an optional TLV that allows an IEEE 802.1Q-compliant Bridge and an
22 IEEE 802.1Q-compatible IEEE 802 LAN station to discover and determine the position of the system in the
23 data center topology. End-stations are always at level 0 and Bridges or routers are at levels greater than 0
24 depending upon the number of links between the Bridge or router and the end-station.

25 Figure D-16 shows the Topology Recognition TLV format:

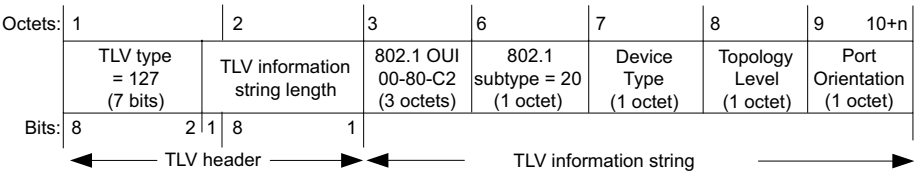


Figure D-16—Topology Recognition TLV Format

26 **D.2.16.1 TLV Type**

27 A 7-bit integer value occupying the most-significant bits of the first octet of the TLV. Always contains the
28 value 127.

1 D.2.16.2 TLV information string length

2 The TLV information string length field of the Congestion Isolation TLV is fixed and shall contain the
3 value 7.

4 D.2.16.3 Device Type

5 The Device Type field indicates what type of device is sending the Topology Recognition TLV. End-stations
6 or servers that are non-relay systems are at the edge of the topology and shall use the value 0. Table D-14
7 specifies valid values for the Device Type field.

Table D-14—Device Type field values

Device Type	Value/meaning
0	end-station/server
1	bridge
2	router
3–254	reserved
255	unknown

8 D.2.16.4 Topology Level

9 An integer indicating the system’s understanding of its current level in the topology. The value of 0 indicates
10 the edge of the topology and the value 255 indicates the level is currently unknown. Other non-zero values
11 indicate the minimum number of links between the system and the edge of the topology. Initially systems
12 may not know their position in the topology and will use the value of unknown by default. As systems
13 discover the device type and topology level of their peers the topology level of the sending system may
14 change. The procedures that implement the recognition of the topology level are described in 49.5.3.

15 D.2.16.5 Port Orientation

16 The Port Orientation field indicates whether the port is facing an *uplink*, *downlink*, *crosslink*, or the
17 orientation is *unknown*. An *uplink* is a port that is facing a system that is deeper in the topology (i.e., has a
18 Topology Level greater than the sending system). A *downlink* is a port that is facing a system closer to the
19 edge of the topology (i.e., has a Topology Level less than the sending system). A *crosslink* is a port that is
20 facing a system at the same level of the topology. Initially systems may not know whether the port
21 orientation is an *uplink*, *downlink*, or *crosslink* and will use the value of *unknown* by default. The procedures
22 that implement the recognition of the Port Orientation are described in 49.5.3. Table D-15 specifies valid
23 values for the Port Orientation field.

Table D-15—Port Orientation field values

Port Orientation	value/meaning
0	uplink
1	downlink
2	crosslink
3–254	reserved
255	unknown

1 D.2.17 PBBN Auto Attach System TLV

2 The PBBN Auto Attach System TLV is an optional TLV that allows Bridges and IEEE 802 LAN stations to
3 discover each other and exchange type and configuration information. Refer to Clause 50 for usage of Auto
4 Attach TLVs.

5 Figure D-17 shows the format of the PBBN Auto Attach System TLV.

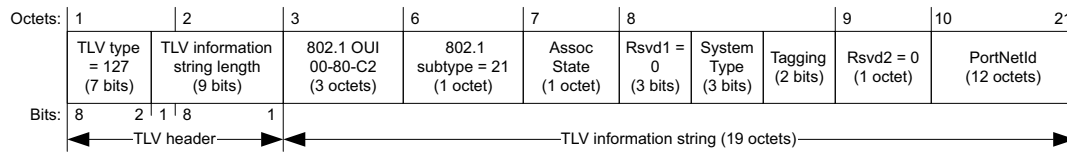


Figure D-17—PBBN Auto Attach System TLV format

6 D.2.17.1 TLV type

7 A 7-bit integer value occupying the most significant bits of the first octet of the PBBN Auto Attach System
8 TLV. Always contains the value 127.

9 D.2.17.2 TLV information string length

10 The TLV information string length field contains the length, in octets, of the fixed length PBBN Auto Attach
11 System TLV fields, which is 19 octets.

12 D.2.17.3 802.1 OUI

13 The 3-octet OUI assigned to IEEE 802.1 (00-80-C2).

14 D.2.17.4 IEEE 802.1 subtype

15 An octet field containing the PBBN Auto Attach System TLV subtype as specified in Table D-1.

16 D.2.17.5 Assoc State

17 An octet field that identifies the current state of the association (Table D-16).

18 D.2.17.6 Rsvd1

19 Three reserved bits, set to zero on transmission. The Rsvd1 field is ignored on receipt.

20 D.2.17.7 System Type

21 The System Type field (Table D-17) is a 3-bit unsigned integer used by all AASs to identify the advertising
22 system's type.

23 An AAB advertises itself as an AAB system so that connected AADs know that it performs the AAB
24 function of receiving and processing VID/I-SID assignment requests and provides responses to such
25 requests.

26 An AAD could be a VLAN-aware or VLAN-unaware system. A VLAN-aware system can be a
27 VLAN-aware Bridge, a VLAN-aware wireless LAN access point, or VLAN-aware end station. A
28 VLAN-unaware system can be a MAC Bridge or an end station.

Table D-16—PBBN Auto Attach association states

Assoc State	Value	Notes
NOT_READY	0x00	Not ready to associate
READY_TO_ASSOC	0x01	Ready to associate
READY_TO_ATTACH	0x02	Associated, viable partner
ASSOC_FAILED_TYPES	0x12	Not AAD to AAB
ASSOC_FAILED_TAGS	0x22	Tagging mismatched
ASSOC_FAILED_TOPO	0x32	Multi-point link detected
ASSOC_FAILED_OTHER	0x42	Parsing or resourcing error
ASSOC_ATTACHED	0x03	Association attached
ASSOC_STANDBY	0x13	Association not attached, AAD already attached
ASSOC_INVALID	0x23	Association rejected at AAB because AAD is already attached elsewhere
Reserved	All other	

Table D-17—PBBN Auto Attach System Type values

System Type	Value
Auto Attach BEB (AAB)	1
C-VLAN aware Auto Attach Device (AAD)	2
VLAN unaware Auto Attach Device (AAD)	3
S-VLAN aware Auto Attach Device (AAD)	4
Reserved for future standardization	0, 5–7

¹ D.2.17.8 Tagging

² The Tagging field (Table D-18) indicates AAD link tagging requirements in AAD-sourced frames and
³ current provisioning mode information. AABs adhere to the Tagging field setting requested by the
⁴ AAD.

Table D-18—PBBN Auto Attach Tagging field values

Bit Numbers	Name	Value
0–1	Link VLAN Tagging requirements	0 - All traffic tagged on link 1 - Untagged and Tagged traffic on Link 2 - Untagged traffic only on Link 3 - Reserved

⁵ D.2.17.9 Rsvd2

⁶ One octet reserved for future alternative formats. The octet is set to zero on transmission. All other values
⁷ are reserved for future standardization. The Rsvd2 field is ignored on receipt.

1 D.2.17.10 PortNetId

2 A 12-octet field (Table D-19) that uniquely identifies an Auto Attach port within the administrative domain
3 used for Auto Attach connection management. This information is particularly important to an AAD to
4 determine to which AAB in a given network it is attached. AADs only form an association to one AAB, and
5 an AAB forms an association to one AAD per AAB port.

6 When the AAB function resides on two physical BEBs comprising a DRNI (IEEE Std 802.1AX), the AAD
7 requires an association with a single logical AAB system. AABs in a DRNI configuration use a single
8 unique PortNetId to ensure the AAD is associated logically with one AAB.

Table D-19—PBBN Auto Attach PortNetId values

Octets	Type	Values
6	System MAC Address	
2	Reserved	0 on transmit, can be anything on receive
4	Integer	Port IfIndex, Aggregator IfIndex

9 D.2.17.11 PBBN Auto Attach System TLV usage rules

10 An LLDPDU shall contain no more than one PBBN Auto Attach System TLV.

11 D.2.18 PBBN Auto Attach Assignment TLV

12 The PBBN Auto Attach Assignment TLV is an optional TLV that allows an AAD to request VID/I-SID
13 assignments that it would like enabled by a directly connected AAB.

14 Figure D-18 depicts a PBBN Auto Attach Assignment TLV containing multiple VID/I-SID assignments.
15 The minimum number of assignments is zero. The maximum number of assignments is determined by the
16 maximum TLV information string size of 511 bytes. This allows 101 VID/I-SID assignments resulting in a
17 TLV size of 510 bytes.

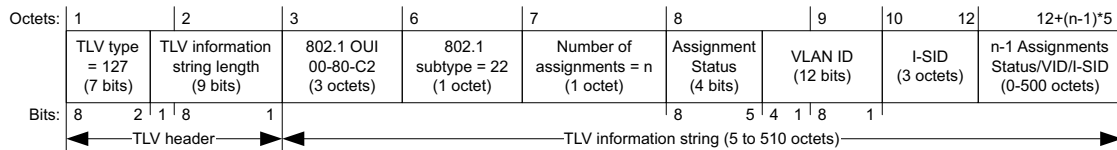


Figure D-18—PBBN Auto Attach Assignment TLV format

18 D.2.18.1 TLV type

19 A 7-bit integer value occupying the most significant bits of the first octet of the TLV. Always contains the
20 value 127.

21 D.2.18.2 TLV information string length

22 The TLV information string length field contains the length, in octets, of the TLV. The length of this TLV is
23 variable between 5 and 510 octets depending on the number of VID/I-SID assignments carried.

24 D.2.18.3 802.1 OUI

25 The 3-octet OUI assigned to IEEE 802.1 (00-80-C2).

1 **D.2.18.4 802.1 TLV subtype**

2 An octet containing the PBBN Auto Attach Assignment TLV subtype specified in Table D-1.

3 **D.2.18.5 Number of assignments**

4 An octet containing the number *n* of Status/VID/I-SID triples in this TLV. The number can range from
5 0 to 101. Zero indicates no Status/VID/I-SID triples are present while 101 is the maximum number allowed
6 in a PBBN Auto Attach Assignment TLV.

7 **D.2.18.6 Assignment Status**

8 The Assignment Status (Table D-20) is returned by the AAB for each VID/I-SID assignment request. If
9 multiple AADs are connected to the AAB, Assignment Status is returned independently on each port. This
10 field is only valid when generated by an AAB.

Table D-20—PBBN Auto Attach Assignment Status values

Assignment Status	Value
Pending	1
Accepted	2
Rejected: Generic	3
Rejected: Auto attach resources unavailable	4
Rejected: Invalid VLAN ID	5
Rejected: VLAN resources unavailable	6
Rejected: Invalid I-SID	7
Rejected: I-SID resources unavailable	8
Rejected: Application interaction issue	9
Rejected: Assignment not allowed	10

11 Pending—AAB is processing assignment; used by the AAD while waiting for a response from the AAB.

12 Accepted—AAB assignment request processing is complete for the VID / I-SID and the VLAN to BSI
13 connection has been established.

14 Rejected: Generic—used when an undefined rejection occurs.

15 Rejected: Auto attach resources unavailable—used when the Auto Attach process is unable to process a
16 request due to system resources being unavailable. For example, memory or CPU processing.

17 Rejected: Invalid VLAN ID—used when the VID value is outside the range of 1 to 4094.

18 Rejected: VLAN resources unavailable—used when VID resource limits have been reached. For example,
19 maximum number of VLANs are already in use for the port or AAB.

20 Rejected: Invalid I-SID—used when the I-SID value is outside the allowed values of 1 or 256 through
21 16 777 214.

22 Rejected: I-SID resources unavailable—used when I-SID resource limits have been reached. For example,
23 the maximum number of I-SIDs are already in use for the port or AAB.

24 Rejected: Application interaction issue—used when Auto Attach assignment processing is not able to
25 complete due to an issue with Auto Attach agent functions on the AAB. For example, a state machine issue.

1 Rejected: Assignment not allowed—used when Auto Attach assignment processing is subject to a policy or
2 rule on the AAB where the assignment requested is not permitted or denied.

3 **D.2.18.7 VLAN ID**

4 The VLAN ID of the VLAN being mapped by this attachment. If this is an attachment with a
5 VLAN-unaware AAD, this field is transmitted as zero and ignored on receive. If this is a C-VLAN aware
6 AAD, then this field is a valid C-VID. If this is an S-VLAN aware AAD, then this field is a valid S-VID.

7 **D.2.18.8 I-SID**

8 The I-SID for the PBBN backbone service instance (BSI).

9 **D.2.18.9 PBBN Auto Attach Assignment TLV usage rules**

10 An LLDPDU shall contain no more than one PBBN Auto Attach Assignment TLV.

11 **D.3 IEEE 802.1 Organizationally Specific TLV management**

12 **D.3.1 IEEE 802.1 Organizationally Specific TLV selection management**

13 TLV selection management consists of providing the network manager with the means to select which
14 specific IEEE 802.1 Organizationally Specific TLVs are enabled for inclusion in an LLDPDU. The
15 following LLDP variables cross reference to LLDP local systems configuration MIB tables indicate which
16 specific TLVs are enabled for the particular port(s) on the system. The specific port(s) through which each
17 TLV is enabled for transmission may be set (or reset) by the network manager:

- 18 a) **mibXdot1PortVlanTxEnable:** This variable lists the VID of the port through which the referenced
19 TLV is enabled for transmission.
- 20 b) **mibXdot1VlanNameConfigTxEnable:** This variable lists the different VLAN name/PPVID TLVs
21 that are defined for the system, each with a bit map indicating the system ports through which the
22 particular VLAN name TLV is enabled for transmission.
- 23 c) **mibXdot1ProtoVlanConfigTxEnable:** This variable lists the Port And Protocol VLAN TLVs that
24 are defined for the system, each with a bit map indicating the system ports through which the
25 particular Port And Protocol VLAN TLV is enabled for transmission.
- 26 d) **mibXdot1ProtocolConfigTxEnable:** This variable lists the protocol identity TLVs that are defined
27 for the system, each with a bit map indicating the system ports through which the particular protocol
28 TLV is enabled for transmission.

29 **D.3.2 IEEE 802.1 managed objects—TLV variables**

30 **D.3.2.1 Port VLAN ID TLV managed objects**

- 31 a) **PVID:** The port VLAN identifier (see D.2.1.1).

32 **D.3.2.2 Port And Protocol VLAN ID TLV managed objects**

- 33 a) **Port and protocol VLAN supported:** A flag indicating whether port and protocol VLANs are
34 supported (see D.2.2.1).
- 35 b) **Port and protocol VLAN enabled:** A flag indicating whether port and protocol VLANs are
36 enabled (see D.2.2.1).
- 37 c) **PPVID:** The advertised port and protocol VLAN identifier (see D.2.2.2).

1 **D.3.2.3 VLAN Name TLV managed objects**

- 2 a) **VID:** The VLAN identifier associated with the VLAN name (see D.2.3.2).
- 3 b) **VLAN name length:** The length of the VLAN name (see D.2.3.3).
- 4 c) **VLAN name:** The VLAN's name (see D.2.3.4).

5 **D.3.2.4 Protocol Identity TLV managed objects**

- 6 a) **protocol identity length:** The length of the protocol identity (see D.2.4.2).
- 7 b) **protocol identity:** The protocol's identity (see D.2.4.3).

8 **D.3.2.5 VID Usage Digest TLV managed objects**

- 9 a) **VID Usage Digest:** The VID Usage Digest value (see D.2.5.1)

10 **D.3.2.6 Management VID TLV managed objects**

- 11 a) **Management VID:** The Management VID value (see D.2.6.1)

12 **D.3.2.7 Link Aggregation TLV managed objects**

- 13 a) **aggregation status:** The capability and current aggregation status of the link (see
- 14 IEEE Std 802.1AX).
- 15 b) **aggregated port ID:** The aggregated port identifier (see IEEE Std 802.1AX).

16 **D.3.2.8 Congestion Notification TLV managed objects**

- 17 a) **CNPV indicators:** A per-priority CNPV indicator vector (see D.2.7.3).
- 18 b) **CN ready indicators:** A per-priority ready indicator vector (see D.2.7.4).

19 **D.3.2.9 EVB TLV managed objects**

- 20 a) **EVB TLV:** see D.2.12.

21 **D.3.2.10 CDCP TLV managed objects**

- 22 a) **CDCP TLV:** see D.2.13.

23 **D.3.2.11 Congestion Isolation TLV managed objects**

- 24 a) **queue map:** see D.2.15.3.
- 25 b) **CIM Encap Length:** see D.2.3.3.
- 26 c) **MAC Address:** see D.2.15.5
- 27 d) **Address Family:** see D.2.15.7
- 28 e) **IP Address:** see D.2.15.8

29 **D.3.2.12 Topology Recognition TLV managed objects**

- 30 a) **Device Type:** see D.2.16.3.
- 31 b) **Topology Level:** see D.2.16.4.
- 32 c) **Port Orientation:** see D.2.16.5

33 **D.3.2.13 PBBN Auto Attach TLV managed objects**

- 34 a) **PBBN Auto Attach System TLV** (D.2.17).
- 35 b) **PBBN Auto Attach Assignment TLV** (D.2.18).

36

1 **D.4 PICS proforma for IEEE 802.1 Organizationally Specific TLV**
2 **extensions**^{57 58}

D.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification.	
NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

D.4.2 Protocol summary, IEEE Std 802.1Q

Identification of protocol specification	IEEE Std 802.1Q-2022, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks			
Identification of amendments and corrigenda to the PICS proforma that have been completed as part of the PICS	Amd.	:	Corr.	:
	Amd.	:	Corr.	:
Have any Exception items been required? (See A.3.3: The answer Yes means that the implementation does not conform to IEEE Std 802.1Q.)	No []		Yes []	

Date of Statement	
-------------------	--

⁵⁷ Instructions for completing the PICS proforma are given in A.3.

⁵⁸ *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

D.4.3 Major capabilities and options

Item	Feature	Status	References	Support
dot1basicSet	Is the IEEE 802.1 Organizationally Specific TLV basicSet implemented?	O.1	D.1, Table D-1	Yes [] No []
dot1basictlv	Is each TLV in the IEEE 802.1 Organizationally Specific TLV basicSet implemented? Port VLAN ID TLV Port And Protocol VLAN ID TLV VLAN Name TLV Protocol Identity TLV VID Usage Digest TLV Management VID TLV Link Aggregation TLV	 dot1basicSet:M dot1basicSet:M dot1basicSet:M dot1basicSet:M dot1basicSet:M dot1basicSet:M dot1basicSet:M	 D.2.1 D.2.2 D.2.3 D.2.4 D.2.5 D.2.6 IEEE Std 802.1AX	 Yes [] Yes [] Yes [] Yes [] Yes [] Yes [] Yes []
dot1cnSet	Is the IEEE 802.1 Organizationally Specific TLV cnSet implemented?	O.1	D.1, Table D-1	Yes [] No []
dot1cntlv	Is each TLV in the IEEE 802.1 Organizationally Specific TLV lagSet implemented? Link Aggregation TLV	 dot1cnSet:M	 IEEE Std 802.1AX	 Yes []
dot1evbSet	Is the IEEE 802.1 Organizationally Specific TLV evbSet implemented?	O.1	D.2.12, D.2.13, Table D-1	Yes [] No []
dot1evbTlv	Is each TLV in the IEEE 802.1 Organizationally Specific TLV evbSet implemented?	dot1evbSet:M	D.2.12, D.2.13, Table D-1	Yes [] N/A []
lldpmib	Which type of MIB is implemented (one is mandatory)? SNMP is supported (if yes, answer item snmpmib and skip equivstor) SNMP is not supported (if yes, answer equivstor and skip snmpmib)	 O.2 O.2	 D.5 D.5	 Yes [] No [] Yes [] No []
snmpmib	If the SNMP MIB is implemented, is the MIB module in conformance with the MIB groups indicated in Table D-21 for the operating mode being implemented?	M	D.5	Yes []
equivstor	If the SNMP is not supported, is the provided storage and retrieval capability functionally equivalent with the indicated specifications of this clause for the operating mode being implemented?	M	D.2.1, D.2.2, D.2.3, D.2.4, and IEEE Std 802.1AX	Yes []
dcbxSet	Is the IEEE 802.1 DCBX TLV set implemented?	O.1	D.4	Yes [] No []

D.4.3 Major capabilities and options *(continued)*

Item	Feature	Status	References	Support
dcbxTlv	Is each TLV in the IEEE 802.1 DCBX TLV set implemented?			
	ETS Configuration TLV	dcbxSet:M	D.2.8	Yes []
	ETS Recommendation TLV	dcbxSet:M	D.2.9	Yes []
	Priority-based Flow Control Configuration TLV	dcbxSet:M	D.2.10	Yes []
	Application Priority TLV	dcbxSet:M	D.2.11	Yes []
	Application VLAN TLV	dcbxSet:M	D.2.14	Yes []
ciSet	Is the IEEE 802.1 Organizationally Specific TLV ciSet implemented?	O.1	D.2.15, Table D-1	Yes [] No []
ciQueuePri	Are the monitored queues higher priority than the congesting queue?	ciSet:M	D.2.15.3	Yes [] N/A []
trSet	Is the IEEE 802.1 Organizationally Specific TLV trSet implemented?	O.1	D.2.16, Table D-1	Yes [] No []
dot1aaSet	Is the IEEE 802.1 Organizationally Specific TLV aaSet implemented?	O.1	D.2.17, D.2.18, Table D-1	Yes [] No []
dot1aaTlv	Is each TLV in the IEEE 802.1 Organizationally Specific TLV aaSet implemented?	dot1aaSet:M	D.2.17, D.2.17, Table D-1	Yes [] No []

1

1 D.5 IEEE 802.1/LLDP extension MIB

2 D.5.1 Internet Standard Management Framework

3 LLDP MIBs are designed to operate in a manner consistent with the principles of the Internet Standard
4 Management Framework, which describes the separation of a data modeling language (for example, SMIV2)
5 from content-specific data models (for example, the LLDP remote systems MIB), and from messages and
6 protocol operations used to manipulate the data (for example, SNMPv3).

7 For a detailed overview of the documents that describe the current Internet Standard Management
8 Framework, refer to section 7 of IETF RFC 3410 (2002).

9 Managed objects are accessed via a virtual information store, termed the *Management Information Base* or
10 *MIB*. MIB objects are generally accessed through the Simple Network Management Protocol (SNMP).
11 Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information
12 (SMI). This clause specifies a MIB module that is compliant to the SMIV2, which is described in
13 IETF STD 58, IETF RFC 2578, IETF RFC 2579, and IETF RFC 2580.

14 D.5.2 Structure of the IEEE 802.1/LLDP extension MIB

15 Table D-21 summarizes the particular object groups that are required for each operating mode. The
16 implemented MIB shall comply with the MIB conformance section for the particular operating mode being
17 supported.

Table D-21—IEEE 802.1 extension MIB object group conformance requirements

MIB group	Rx mode	Tx mode	Tx/Rx mode
lldpV2Xdot1ConfigGroup	M ^a	M	M
lldpV2Xdot1LocSysGroup	M	—	M
lldpV2Xdot1RemSysGroup	—	M	M
ifGeneralInformationGroup	M	M	M
lldpXdot1dcbxETSGroup	DCBX:M	DCBX:M	DCBX:M
lldpXdot1dcbxPFCGroup	DCBX:M	DCBX:M	DCBX:M
lldpXdot1dcbxApplicationPriorityGroup	DCBX:M	DCBX:M	DCBX:M
lldpXdot1dcbxApplicationVlanGroup	DCBX:M	DCBX:M	DCBX:M
lldpXdot1CiGroup	CI:M	CI:M	CI:M
lldpXdot1TrGroup	TR:M	TR:M	TR:M
lldpXdot1AaGroup	—	—	AA:M

^aM=Mandatory.

¹ Table D-22 shows the structure of the MIB and the relationship of the MIB objects to the LLDP operational
² status/control variables, LLDP statistics variables, and TLV variables.

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference

MIB table	MIB object	LLDP reference
<i>Configuration group</i>		
lldpV2Xdot1ConfigPortVlanTable		Augments lldpV2Xdot1ConfigPortVlanTable
	lldpV2Xdot1ConfigPortVlanTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpV2Xdot1ConfigVlanNameTable		Augments lldpV2Xdot1LocVlanNameEntry
	lldpV2Xdot1ConfigVlanNameTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpV2Xdot1ConfigProtoVlanTable		Augments lldpV2Xdot1LocProtoVlanEntry
	lldpV2Xdot1ConfigProtoVlanTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpV2Xdot1ConfigProtocolTable		Augments lldpV2Xdot1LocProtocolEntry
	lldpV2Xdot1ConfigProtocolTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpV2Xdot1ConfigVidUsageDigestTable		Augments lldpV2Xdot1LocVidUsageDigestEntry
	lldpV2Xdot1ConfigVidUsageDigestTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpV2Xdot1ConfigManVidTable		Augments lldpV2Xdot1LocManVidEntry
	lldpV2Xdot1ConfigManVidTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpXdot1CnConfigCnTable		Augments lldpV2Xdot1LocManVidEntry
	lldpXdot1CnConfigCnTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpXdot1EvbConfigEvbTable		Augments lldpV2Xdot1LocManVidEntry
	lldpXdot1EvbConfigEvbTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
lldpXdot1EvbConfigCdcPTable		Augments lldpV2Xdot1LocManVidEntry
	lldpXdot1EvbConfigCdcPTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
lldpXdot1CiConfigCiTable		Augments lldpV2PortConfigEntry
	lldpXdot1CiConfigCiTxEnable	Normal LLDPDUs, 9.1.2.1 of IEEE Std 802.1AB
lldpXdot1AaConfigAaTable		Augments lldpV2Xdot1LocManVidEntry
	lldpXdot1AaConfigSysTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
	lldpXdot1AaConfigAsgnsTxEnable	Normal LLDPDUs, see IEEE Std 802.1AB
<i>Local system information</i>		
lldpV2Xdot1LocTable		D.2.1
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocPortVlanId	PVID, D.2.1.1
lldpV2Xdot1LocProtoVlanTable		D.2.2
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocProtoVlanId	PPVID, D.2.2.2
	lldpV2Xdot1LocProtoVlanSupported	flags, D.2.2.1
	lldpV2Xdot1LocProtoVlanEnabled	flags, D.2.2.1
lldpV2Xdot1LocVlanNameTable		D.2.3
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocVlanId	VID, D.2.3.2 (Table index)
	lldpV2Xdot1LocVlanName	VLAN name, D.2.3.4
lldpV2Xdot1LocProtocolTable		D.2.4
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocProtocolIndex	(Table index)
	lldpV2Xdot1LocProtocolId	protocol identity, D.2.4.3
lldpV2Xdot1LocVidUsageDigestTable		D.2.5
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocVidUsageDigest	VID usage digest, D.2.5.1
lldpV2Xdot1LocManVidTable		D.2.6
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocManVid	Management VID, D.2.6.1
lldpV2Xdot1LocLinkAggTable		IEEE Std 802.1AX
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocLinkAggStatus	aggregation status, IEEE Std 802.1AX
	lldpV2Xdot1LocLinkAggPortId	aggregated port ID, IEEE Std 802.1AX

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
lldpV2Xdot1LocCnTable		D.2.7
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocCNPVIndicators	CPNV indicators, D.2.7.3
	lldpV2Xdot1LocReadyIndicators	Ready indicators, D.2.7.4
lldpV2Xdot1LocEvbTlvTable		D.2.12
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocEvbTlvString	EVb TLV string, D.2.12
lldpV2Xdot1LocCDCPTlvTable		D.2.13
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1LocCDCPTlvString	CDCP TLV string, D.2.13
lldpXdot1LocCiBasicTable		D.2.15
	lldpV2LocPortIfIndex	(Table index)
	lldpXdot1LocCiCIMEncapLen	D.2.15.4
	lldpXdot1LocCiMacAddress	D.2.15.5
	lldpXdot1LocCiNetAddressType	D.2.15.7
	lldpXdot1LocCiNetAddress	D.2.15.8
lldpXdot1CiLocQueueMapTable		D.2.15.3
	lldpV2LocPortIfIndex	(Table index)
	lldpXdot1LocCiQueueId	(Table index)
	lldpXdot1LocCiQueueType	D.2.15.3
	lldpXdot1LocCiMappedQueue	D.2.15.3
lldpV2Xdot1LocAaTlvTable		D.2.17, D.2.18
	lldpV2Xdot1LocPortIfIndex	(Table index)
	lldpV2Xdot1LocAaSysTlvString	PBBN Auto Attach System TLV string, D.2.17
	lldpV2Xdot1LocAaAsgnsTlvString	PBBN Auto Attach Assignment TLV string, D.2.18
<i>Remote system information</i>		
lldpV2Xdot1RemTable		D.2.1
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemPortVlanId	PVID, D.2.1.1
lldpV2Xdot1RemProtoVlanTable		D.2.2

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemProtoVlanId	PPVID, D.2.2.2 (Table index)
	lldpV2Xdot1RemProtoVlanSupported	flags, D.2.2.1
	lldpV2Xdot1RemProtoVlanEnabled	flags, D.2.2.1
lldpV2Xdot1RemVlanNameTable		D.2.3
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemVlanId	VID, D.2.3.2 (Table index)
	lldpV2Xdot1RemVlanName	VLAN name, D.2.3.4
lldpV2Xdot1RemProtocolTable		D.2.4
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemProtocolIndex	(Table index)
	lldpV2Xdot1RemProtocolId	protocol identity, D.2.4.3
lldpV2Xdot1RemVidUsageDigestV2Table		D.2.5
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2Xdot1RemIndex	(Table index)
	lldpV2Xdot1RemVidUsageDigestV2	VID usage digest, D.2.5.1
lldpV2Xdot1RemManVidV2Table		D.2.6
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2Xdot1RemIndex	(Table index)
	lldpV2Xdot1RemManVidV2	Management VID, D.2.6.1
lldpV2Xdot1RemLinkAggTable		IEEE Std 802.1AX

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemLinkAggStatus	aggregation status, IEEE Std 802.1AX
	lldpV2Xdot1RemLinkAggPortId	aggregation port ID, IEEE Std 802.1AX
lldpV2Xdot1RemCnTable		D.2.7
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2LocPortIfIndex	(Table index)
	lldpV2Xdot1RemCNPVIndicators	CPNV indicators, D.2.7.3
	lldpV2Xdot1RemReadyIndicators	Ready indicators, D.2.7.4
lldpV2Xdot1RemEvbTlvTable		D.2.12
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemEvbTlvString	EVb TLV string, D.2.12
lldpV2Xdot1RemCDCPTlvTable		D.2.13
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemCDCPTlvString	CDCP TLV string, D.2.13
lldpXdot1RemCiBasicTable		D.2.3
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpXdot1RemCiCIMEncapLen	D.2.3.3
	lldpXdot1RemCiMacAddress	D.2.15.5
	lldpXdot1RemCiNetAddressType	D.2.15.7
	lldpXdot1RemCiNetAddress	D.2.15.8

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
lldpXdot1CiRemQueueMapTable		D.2.3
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpXdot1RemCiQueueId	(Table index)
	lldpXdot1RemCiQueueType	D.2.15.3
	lldpXdot1RemCiMappedQueue	D.2.15.3
lldpV2Xdot1RemAaTlvTable		D.2.1
	lldpV2RemTimeMark	(Table index)
	lldpV2RemLocalIfIndex	(Table index)
	lldpV2RemLocalDestMACAddress	(Table index)
	lldpV2RemIndex	(Table index)
	lldpV2Xdot1RemAaSysTlvString	PBBN Auto Attach System TLV string, D.2.17
	lldpV2Xdot1RemAaAsgnsTlvString	PBBN Auto Attach Assignment TLV string, D.2.18
<i>lldpXdot1dcbxConfig extension group^a</i>		
lldpXdot1dcbxConfigETSConfigurationEntry		
	lldpXdot1dcbxConfigETSConfigurationTxEnable	D.2.8
lldpXdot1dcbxConfigETSRecommendationTable		
	lldpXdot1dcbxConfigETSRecommendationTxEnable	D.2.9
lldpXdot1dcbxConfigPFCTable		
	lldpXdot1dcbxConfigPFCTxEnable	D.2.10
lldpXdot1dcbxConfigApplicationPriorityTable		
	lldpXdot1dcbxConfigApplicationPriorityTxEnable	D.2.11
lldpXdot1dcbxConfigApplicationVlanTable		
	lldpXdot1dcbxConfigApplicationVlanTxEnable	D.2.14
<i>lldpXdot1dcbxLocalData extension group^a</i>		
lldpXdot1dcbxLocETSConfigurationTable		
	lldpXdot1dcbxLocETSConCreditBasedShaperSupport	D.2.8.4
	lldpXdot1dcbxLocETSConMaxTC	D.2.8.5
	lldpXdot1dcbxLocETSConWilling	D.2.8.3
	lldpXdot1dcbxLocETSConTrafficClassBandwidthTable	D.2.8.7
	lldpXdot1dcbxLocETSConTrafficSelectionAlgorithmTable	D.2.8.8

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
lldpXdot1dcbxLocETSConPriorityAssignmentTable		
	lldpXdot1dcbxLocETSConPriority	D.2.8.6
	lldpXdot1dcbxLocETSConTrafficClass	D.2.8.6
lldpXdot1dcbxLocETSRecommendationTable		
	lldpXdot1dcbxLocETSRecoTrafficClassBandwidthTable	D.2.9.4
lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithmTable		
	lldpXdot1dcbxLocETSRecoTSAPriority	D.2.9.5
	lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithm	D.2.9.5
lldpXdot1dcbxLocPFCBasicTable		
	lldpXdot1dcbxLocPFCWilling	D.2.10.3
	lldpXdot1dcbxLocPFCMBC	D.2.10.4
	lldpXdot1dcbxLocPFCCap	D.2.10.5
lldpXdot1dcbxLocPFCEnableTable		
	lldpXdot1dcbxLocPFCEnablePriority	D.2.10.6
	lldpXdot1dcbxLocPFCEnableEnabled	D.2.10.6
lldpXdot1dcbxLocApplicationPriorityAppTable		
	lldpXdot1dcbxLocApplicationPriorityAESelector	D.2.11.3
	lldpXdot1dcbxLocApplicationPriorityAEProtocol	D.2.11.3
	lldpXdot1dcbxLocApplicationPriorityAEPriority	D.2.11.3
lldpXdot1dcbxLocApplicationVlanAppTable		
	lldpXdot1dcbxLocApplicationVlanAESelector	D.2.14.3
	lldpXdot1dcbxLocApplicationVlanAEProtocol	D.2.14.3
	lldpXdot1dcbxLocApplicationVlanAEVlanId	D.2.14.3
<i>lldpXdot1dcbxRemoteData extension group^a</i>		
lldpXdot1dcbxRemETSConBasicConfigurationTable		
	lldpXdot1dcbxRemETSConCreditBasedShaperSupport	D.2.8.4
	lldpXdot1dcbxRemETSConMaxTC	D.2.8.5
	lldpXdot1dcbxRemETSConWilling	D.2.8.3
	lldpXdot1dcbxRemETSConTrafficClassBandwidthTable	D.2.8.7
	lldpXdot1dcbxRemETSConTrafficSelectionAlgorithmTable	D.2.8.8
lldpXdot1dcbxRemETSConPriorityAssignmentTable		
	lldpXdot1dcbxRemETSConPriority	D.2.8.6
	lldpXdot1dcbxRemETSConTrafficClass	D.2.8.6
lldpXdot1dcbxRemETSRecommendationTable		

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
	lldpXdot1dcbxRemETSRcoTrafficClassBandwidthTable	D.2.9.4
	lldpXdot1dcbxRemETSRcoTrafficSelectionAlgorithmTable	
	lldpXdot1dcbxRemETSRcoTSAPriority	D.2.9.5
	lldpXdot1dcbxRemETSRcoTrafficSelectionAlgorithm	D.2.9.5
	lldpXdot1dcbxRemPFCBasicTable	
	lldpXdot1dcbxRemPFCWilling	D.2.10.3
	lldpXdot1dcbxRemPFCMBC	D.2.10.4
	lldpXdot1dcbxRemPFCCap	D.2.10.5
	lldpXdot1dcbxRemPFCEnableTable	
	lldpXdot1dcbxRemPFCEnablePriority	D.2.10.6
	lldpXdot1dcbxRemPFCEnableEnabled	D.2.10.6
	lldpXdot1dcbxRemApplicationPriorityAppTable	
	lldpXdot1dcbxRemApplicationPriorityAESelector	D.2.11.3
	lldpXdot1dcbxRemApplicationPriorityAEProtocol	D.2.11.3
	lldpXdot1dcbxRemApplicationPriorityAEPriority	D.2.11.3
	lldpXdot1dcbxRemApplicationVlanAppTable	
	lldpXdot1dcbxRemApplicationVlanAESelector	D.2.14.3
	lldpXdot1dcbxRemApplicationVlanAEProtocol	D.2.14.3
	lldpXdot1dcbxRemApplicationVlanAEVlanId	D.2.14.3
<i>lldpXdot1dcbxAdminData extension group^a</i>		
	lldpXdot1dcbxAdminETSBasicConfigurationTable	
	lldpXdot1dcbxAdminETSRcoCreditBasedShaperSupport	D.2.8.4
	lldpXdot1dcbxAdminETSRcoMaxTC	D.2.8.5
	lldpXdot1dcbxAdminETSRcoWilling	D.2.8.3
	lldpXdot1dcbxAdminETSRcoTrafficClassBandwidthTable	D.2.8.7
	lldpXdot1dcbxAdminETSRcoTrafficSelectionAlgorithmTable	D.2.8.8
	lldpXdot1dcbxAdminETSRcoPriorityAssignmentTable	
	lldpXdot1dcbxAdminETSRcoPriority	D.2.8.6
	lldpXdot1dcbxAdminETSRcoTrafficClass	D.2.8.6
	lldpXdot1dcbxAdminETSRRecommendationTable	
	lldpXdot1dcbxAdminETSRcoTrafficClassBandwidthTable	D.2.9.4
	lldpXdot1dcbxAdminETSRcoTrafficSelectionAlgorithmTable	
	lldpXdot1dcbxAdminETSRcoTSAPriority	D.2.9.5
	lldpXdot1dcbxAdminETSRcoTrafficSelectionAlgorithm	D.2.9.5

Table D-22—IEEE 802.1/LLDP extension MIB object cross reference (continued)

MIB table	MIB object	LLDP reference
lldpXdot1dcbxAdminPFCBasicTable		
	lldpXdot1dcbxAdminPFCWilling	D.2.10.3
	lldpXdot1dcbxAdminPFCMBC	D.2.10.4
	lldpXdot1dcbxAdminPFCCap	D.2.10.5
lldpXdot1dcbxAdminPFCEnableTable		
	lldpXdot1dcbxAdminPFCEnablePriority	D.2.10.6
	lldpXdot1dcbxAdminPFCEnableEnabled	D.2.10.6
lldpXdot1dcbxAdminApplicationPriorityAppTable		
	lldpXdot1dcbxAdminApplicationPriorityAESelector	D.2.11.3
	lldpXdot1dcbxAdminApplicationPriorityAEProtocol	D.2.11.3
	lldpXdot1dcbxAdminApplicationPriorityAEPriority	D.2.11.3
lldpXdot1dcbxAdminApplicationVlanAppTable		
	lldpXdot1dcbxAdminApplicationVlanAESelector	D.2.14.3
	lldpXdot1dcbxAdminApplicationVlanAEProtocol	D.2.14.3
	lldpXdot1dcbxAdminApplicationVlanAEVlanId	D.2.14.3

^a The term Extension Group is used here to be consistent with LLDP (see IEEE Std 802.1AB).

1 D.5.3 Relationship to other MIBs

2 Version 2 of the IEEE 802.1 LLDP extension MIB module appears in D.5.5. Version 1 of the MIB module
3 that was published in IEEE Std 802.1AB-2005 [B8] has been superseded by version 2 of the MIB module,
4 and support of the version 2 module is a requirement for conformance to the required or optional capabilities
5 (Clause 5 of IEEE Std 802.1AB-2009 [B9]) in IEEE Std 802.1AB-2009. The version 2 MIB module reflects
6 changes in indexation of the MIB objects that support the use of LLDP with multiple destination MAC
7 addresses, as discussed in Clause 6 of IEEE Std 802.1AB-2009 [B9].

8 The relationship of the IEEE 802.1 LLDP extension MIB module to other MIBs, and coexistence between
9 version 1 and version 2 implementations, is further discussed in 11.3 of IEEE Std 802.1AB-2009 [B9].

10 D.5.4 Security considerations for IEEE 802.1 LLDP extension MIB module

11 There are a number of management objects defined in this MIB module with a MAX-ACCESS clause of
12 read-write.⁵⁹ Such objects may be considered sensitive or vulnerable in some network environments. The
13 support for SET operations in a nonsecure environment without proper protection can have a negative effect
14 on network operations.

⁵⁹ In IETF MIB definitions, the MAX-ACCESS clause defines the type of access that is allowed for particular data elements in the MIB. An explanation of the MAX-ACCESS mappings is given in section 7.3 of IETF RFC 2578 (1999).

1 Setting the following objects to incorrect values can result in improper operation of LLDP when in the
2 transmit mode:

- 3 a) lldpV2Xdot1ConfigPortVlanTxEnableV2
- 4 b) lldpV2Xdot1ConfigVlanNameTxEnable
- 5 c) lldpV2Xdot1ConfigProtoVlanTxEnable
- 6 d) lldpV2Xdot1ConfigProtocolTxEnable
- 7 e) lldpV2Xdot1ConfigVidUsageDigestTxEnable
- 8 f) lldpV2Xdot1ConfigManVidTxEnable

9 The following readable objects in this MIB module may be considered to be sensitive or vulnerable in some
10 network environments:

- 11 g) MIB objects that are related to the transmit mode:
 - 12 1) lldpV2Xdot1LocPortVlanId
 - 13 2) lldpV2Xdot1LocProtoVlanSupported
 - 14 3) lldpV2Xdot1LocProtoVlanEnabled
 - 15 4) lldpV2Xdot1LocVlanName
 - 16 5) lldpV2Xdot1LocProtocolId
 - 17 6) lldpV2Xdot1LocVidUsageDigest
 - 18 7) lldpV2Xdot1LocManVidTxEnable
 - 19 8) lldpV2Xdot1LocLinkAggStatus
 - 20 9) lldpV2Xdot1LocLinkAggPortId
 - 21 10) lldpXdot1dcbxConfigETSConfigurationEntry
 - 22 11) lldpXdot1dcbxConfigPFCtable
 - 23 12) lldpXdot1dcbxLocETSConfigurationTable
 - 24 13) lldpXdot1dcbxLocETSConPriorityAssignmentTable
 - 25 14) lldpXdot1dcbxLocETSConTrafficSelectionAlgorithmTable
 - 26 15) lldpXdot1dcbxLocPFCBasicTable
 - 27 16) lldpXdot1dcbxLocPFCEnableTable
 - 28 17) lldpXdot1dcbxAdminETSConfigurationTable
 - 29 18) lldpXdot1dcbxAdminETSConPriorityAssignmentTable
 - 30 19) lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithmTable
 - 31 20) lldpXdot1dcbxAdminPFCBasicTable
 - 32 21) lldpXdot1dcbxAdminPFCEnableTable
 - 33 22) lldpXdot1dcbxLocApplicationPriorityAppTable
 - 34 23) lldpXdot1dcbxLocApplicationVlanAppTable
 - 35 24) lldpXdot1dcbxAdminApplicationPriorityAppTable
 - 36 25) lldpXdot1dcbxAdminApplicationVlanAppTable
 - 37 26) lldpXdot1LocCiBasicTable
 - 38 27) lldpXdot1CiLocQueueMapTable
 - 39 28) lldpXdot1LocTrTable
 - 40 29) lldpV2Xdot1LocAaTlvTable
- 41 h) MIB objects that are related to the receive mode:
 - 42 1) lldpV2Xdot1RemPortVlanId
 - 43 2) lldpV2Xdot1RemProtoVlanSupported
 - 44 3) lldpV2Xdot1RemProtoVlanEnabled
 - 45 4) lldpV2Xdot1RemVlanName
 - 46 5) lldpV2Xdot1RemProtocolId
 - 47 6) lldpV2Xdot1RemVidUsageDigest
 - 48 7) lldpV2Xdot1RemManVidTxEnable

- 1 8) lldpV2Xdot1RemLinkAggStatus
- 2 9) lldpV2Xdot1RemLinkAggPortId
- 3 10) lldpXdot1dcbxConfigETSConfigurationEntry
- 4 11) lldpXdot1dcbxConfigPFCTable
- 5 12) lldpXdot1dcbxRemETSConfigurationTable
- 6 13) lldpXdot1dcbxRemETSConPriorityAssignmentTable
- 7 14) lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmTable
- 8 15) lldpXdot1dcbxRemPFCBasicTable
- 9 16) lldpXdot1dcbxRemPFCEnableTable
- 10 17) lldpXdot1dcbxAdminETSConfigurationTable
- 11 18) lldpXdot1dcbxAdminETSConPriorityAssignmentTable
- 12 19) lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmTable
- 13 20) lldpXdot1dcbxAdminPFCBasicTable
- 14 21) lldpXdot1dcbxAdminPFCEnableTable
- 15 22) lldpXdot1dcbxRemApplicationPriorityAppTable
- 16 23) lldpXdot1dcbxRemApplicationVlanAppTable
- 17 24) lldpXdot1dcbxAdminApplicationPriorityAppTable
- 18 25) lldpXdot1dcbxAdminApplicationVlanAppTable
- 19 26) lldpXdot1RemCiBasicTable
- 20 27) lldpXdot1CiRemQueueMapTable
- 21 28) lldpXdot1RemTrTable
- 22 29) lldpV2Xdot1RemAaTlvTable

23 This concern applies both to objects that describe the configuration of the local host, as well as for objects
24 that describe information from the remote hosts, acquired via LLDP and displayed by the objects in this
25 MIB module. It is thus important to control even GET and/or NOTIFY access to these objects and possibly
26 to even encrypt the values of these objects when sending them over the network via SNMP.

27 SNMP versions prior to SNMPv3 did not include adequate security. Even if the network itself is secure (for
28 example, by using IPSec), there is no control about who on the secure network is allowed to access and
29 GET/SET (read/change/create/delete) the objects in this MIB module.

30 Implementers should consider the security features as provided by the SNMPv3 framework (see
31 IETF RFC 3410 (2002), section 8), including full support for the SNMPv3 cryptographic mechanisms (for
32 authentication and privacy).

33 Further, implementers should not deploy SNMP versions prior to SNMPv3. Instead, implementers should
34 deploy SNMPv3 to enable cryptographic security. It is then a customer/operator responsibility to ensure that
35 the SNMP entity giving access to an instance of this MIB module is properly configured to give access to the
36 objects only to those principals (users) that have legitimate rights to indeed GET or SET
37 (change/create/delete) them.

1 D.5.5 IEEE 802.1 LLDP extension MIB module—version 2⁶⁰ 61

2 In the following MIB definition, should any discrepancy between the DESCRIPTION text and the
3 corresponding definition in D.2.1 through D.5 occur, the definition in D.2.1 through D.5 shall take
4 precedence.

```

5 LLDP-EXT-DOT1-V2-MIB DEFINITIONS ::= BEGIN
6
7 IMPORTS
8     MODULE-IDENTITY,
9     OBJECT-TYPE,
10    Unsigned32
11        FROM SNMPv2-SMI
12    TruthValue,
13    TEXTUAL-CONVENTION,
14    MacAddress
15        FROM SNMPv2-TC
16    SnmpAdminString
17        FROM SNMP-FRAMEWORK-MIB
18    MODULE-COMPLIANCE,
19    OBJECT-GROUP
20        FROM SNMPv2-CONF
21    ifGeneralInformationGroup
22        FROM IF-MIB
23    lldpV2Extensions,
24    lldpV2LocPortIfIndex,
25    lldpV2RemTimeMark,
26    lldpV2RemLocalIfIndex,
27    lldpV2RemLocalDestMACAddress,
28    lldpV2RemIndex,
29    lldpV2PortConfigEntry
30        FROM LLDP-V2-MIB
31    VlanId
32        FROM Q-BRIDGE-MIB
33    IEEE8021PriorityValue
34        FROM IEEE8021-TC-MIB
35    InetPortNumber
36        FROM INET-ADDRESS-MIB
37    AddressFamilyNumbers
38        FROM IANA-ADDRESS-FAMILY-NUMBERS-MIB;
39
40 lldpV2Xdot1MIB MODULE-IDENTITY
41     LAST-UPDATED "202306050000Z" -- June 5, 2023
42     ORGANIZATION "IEEE 802.1 Working Group"
43     CONTACT-INFO
44         " WG-URL: http://www.ieee802.org/1/
45         WG-EMail: stds-802-1-1@ieee.org
46         Contact: IEEE 802.1 Working Group Chair
47         Postal: C/O IEEE 802.1 Working Group
48                 IEEE Standards Association
49                 445 Hoes Lane
50                 Piscataway, NJ 08854
51                 USA
52         E-mail: stds-802-1-chairs@ieee.org"
53     DESCRIPTION
54         "The LLDP Management Information Base extension module for
55         IEEE 802.1 organizationally defined discovery information.
56
57         In order to ensure the uniqueness of the LLDP-V2-MIB,
58         lldpV2Xdot1MIB is branched from lldpV2Extensions using an
59         Organizationally Unique Identifier (OUI) value as the node.
60         An OUI is a 24 bit globally unique number assigned by the
61         IEEE Registration Authority - see:

```

⁶⁰ Copyright release for MIBs: Users of this standard may freely reproduce the MIB modules in this standard so that they can be used for their intended purpose.

⁶¹ An ASCII version of this MIB module is attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/mib-modules/>.

1
2 <http://standards.ieee.org/develop/regauth/oui/index.html>
3
4 Unless otherwise indicated, the references in this MIB
5 module are to IEEE Std 802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
6
7 Copyright (C) IEEE (2023).
8 This version of this MIB module is part of IEEE Std 802.1Q;
9 see that standard for full legal notices."
10
11 REVISION "202306050000Z" -- June 5, 2023
12 DESCRIPTION
13 "Published as part of IEEE Std 802.1Qcz-2023.
14 Congestion Isolation objects added."
15
16 REVISION "202211080000Z" -- November 8, 2022
17 DESCRIPTION
18 "Published as part of IEEE Std 802.1Q-2022.
19 Cross references and contact information updated."
20
21 REVISION "201807010000Z" -- July 1, 2018
22 DESCRIPTION
23 "Published as part of IEEE Std 802.1Q 2018 revision.
24 Cross references updated and corrected.
25 Changes introduced by IEEE Std 802.1Qcd-2015 and
26 IEEE Std 802.1Q-2014 Cor 1-2015 merged. "
27
28 REVISION "201502160000Z" -- February 16, 2015
29 DESCRIPTION
30 "Published as part of IEEE Std 802.1Q 2014 Cor-1.
31 Updated as a result of maintenance items #0132 and #0152"
32
33 REVISION "201502160000Z" -- February 16, 2015
34 DESCRIPTION
35 "Published as part of IEEE Std 802.1Qcd.
36 Adds Application VLAN TLV objects to the DCBX groups of
37 the MIB module."
38
39 REVISION "201412150000Z" -- December 15, 2014
40 DESCRIPTION
41 "Published as part of IEEE Std 802.1Q 2014 revision.
42 Cross references updated and corrected.
43 New tables lldpV2Xdot1RemVidUsageDigestV2Table
44 and lldpV2Xdot1RemManVidV2Table inserted; old
45 versions deprecated. New versions add an index for
46 lldpV2RemIndex. "
47
48 REVISION "201103250000Z" -- March 25, 2011
49 DESCRIPTION
50 "Published as part of IEEE Std 802.1Qaz-2011. Adds the DCBX
51 objects to the MIB module"
52
53 REVISION "201103230000Z" -- March 23, 2011
54 DESCRIPTION
55 "Published as part of IEEE Std 802.1Q-2011 revision.
56 This revision contains changes associated with
57 relocating the extension MIB from IEEE Std 802.1AB to
58 IEEE Std 802.1Q, minor tweaks to the text of the
59 DESCRIPTION statement above to fix references to
60 IEEE Std 802.1Q, updating of references to refer to
61 Annex D, and addition of object definitions for
62 Congestion Notification TLVs and corresponding
63 compliance statements."
64
65 REVISION "200906080000Z" -- June 08, 2009
66 DESCRIPTION
67 "Published as part of IEEE Std 802.1AB-2009 revision.
68 This revision incorporated changes to the MIB to
69 support the use of LLDP with multiple destination MAC
70 addresses, and to import the Link Aggregation TLV
71 from the IEEE 802.3 extension MIB"
72

```

1 -- OUI for IEEE 802.1 is 32962 (00-80-C2)
2 ::= { lldpV2Extensions 32962 }
3
4 -----
5 -----
6 --
7 -- Organizationally Defined Information Extension - IEEE 802.1
8 -- Definitions to support the basicSet TLV set (Table D-1)
9 --
10 -----
11 -----
12
13 lldpV2Xdot1Objects OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 1 }
14
15 -- LLDP IEEE 802.1 extension MIB groups
16 lldpV2Xdot1Config OBJECT IDENTIFIER ::= { lldpV2Xdot1Objects 1 }
17 lldpV2Xdot1LocalData OBJECT IDENTIFIER ::= { lldpV2Xdot1Objects 2 }
18 lldpV2Xdot1RemoteData OBJECT IDENTIFIER ::= { lldpV2Xdot1Objects 3 }
19
20 -----
21 -- Textual Convention definitions
22 -----
23
24 LldpV2XLinkAggStatusMap ::= TEXTUAL-CONVENTION
25     STATUS      current
26     DESCRIPTION
27         "This TC describes the link aggregation status.
28
29         The bit 'aggCapable(0)' indicates the link is capable of being
30         aggregated if 1, not capable if 0.
31
32         The bit 'aggEnabled(1)' indicates the link is currently in
33         an aggregation if 1, not in an aggregation if 0.
34
35         The bits 'portTypeLS(1)' and portTypeMS(2)' form the LS
36         and MS bits of a Port Type value respectively:
37         00 = no port type specified
38         01 = transmitted from Aggregation Port
39         10 = transmitted from Aggregator
40         11 = transmitted from an Aggregator with a single
41             Aggregation Port.
42
43         The remaining bits are reserved for future standardization."
44     SYNTAX      BITS {
45         aggCapable(0),
46         aggEnabled(1),
47         portTypeLS(2),
48         portTypeMS(3)
49     }
50
51 -----
52 -- IEEE 802.1 - Configuration for the basicSet TLV set
53 -----
54
55 --
56 -- lldpV2Xdot1ConfigPortVlanTable : configure the transmission of the
57 --                               Port VLAN-ID TLVs on set of ports.
58 --
59
60 lldpV2Xdot1ConfigPortVlanTable OBJECT-TYPE
61     SYNTAX      SEQUENCE OF LldpV2Xdot1ConfigPortVlanEntry
62     MAX-ACCESS  not-accessible
63     STATUS      current
64     DESCRIPTION
65         "A table that controls selection of LLDP Port VLAN-ID TLVs
66         to be transmitted on individual ports."
67     ::= { lldpV2Xdot1Config 1 }
68
69 lldpV2Xdot1ConfigPortVlanEntry OBJECT-TYPE
70     SYNTAX      LldpV2Xdot1ConfigPortVlanEntry
71     MAX-ACCESS  not-accessible
72     STATUS      current

```

```

1  DESCRIPTION
2      "LLDP configuration information that controls the
3      transmission of IEEE 802.1 organizationally defined Port
4      VLAN-ID TLV on LLDP transmission-capable ports.
5
6      This configuration object augments the
7      lldpV2PortConfigEntry of the LLDP-MIB, therefore it is only
8      present along with the port configuration defined by the
9      associated lldpV2PortConfigEntry entry.
10
11     Each active lldpConfigEntry is restored from non-volatile
12     storage (along with the corresponding
13     lldpV2PortConfigEntry) after a re-initialization of the
14     management system."
15  AUGMENTS { lldpV2PortConfigEntry }
16  ::= { lldpV2Xdot1ConfigPortVlanTable 1 }
17
18  lldpV2Xdot1ConfigPortVlanEntry ::= SEQUENCE {
19      lldpV2Xdot1ConfigPortVlanTxEnable  TruthValue
20  }
21
22  lldpV2Xdot1ConfigPortVlanTxEnable OBJECT-TYPE
23      SYNTAX      TruthValue
24      MAX-ACCESS  read-write
25      STATUS      current
26      DESCRIPTION
27          "The lldpV2Xdot1ConfigPortVlanTxEnable, which is defined
28          as a truth value and configured by the network management,
29          determines whether the IEEE 802.1 organizationally defined
30          port VLAN TLV transmission is allowed on a given LLDP
31          transmission-capable port.
32
33          The value of this object is restored from non-volatile
34          storage after a re-initialization of the management system."
35      REFERENCE
36          "9.1.2.1 of IEEE Std 802.1AB"
37      DEFVAL { false }
38      ::= { lldpV2Xdot1ConfigPortVlanEntry 1 }
39
40
41  --
42  -- lldpV2Xdot1ConfigVlanNameTable : configure the transmission of the
43  --                                VLAN name instances on set of ports.
44  --
45
46  lldpV2Xdot1ConfigVlanNameTable OBJECT-TYPE
47      SYNTAX      SEQUENCE OF LldpV2Xdot1ConfigVlanNameEntry
48      MAX-ACCESS  not-accessible
49      STATUS      current
50      DESCRIPTION
51          "The table that controls selection of LLDP VLAN name TLV
52          instances to be transmitted on individual ports."
53      ::= { lldpV2Xdot1Config 2 }
54
55  lldpV2Xdot1ConfigVlanNameEntry OBJECT-TYPE
56      SYNTAX      LldpV2Xdot1ConfigVlanNameEntry
57      MAX-ACCESS  not-accessible
58      STATUS      current
59      DESCRIPTION
60          "LLDP configuration information that specifies the set of
61          ports (represented as a PortList) on which the Local System
62          VLAN name instance is transmitted.
63
64          This configuration object augments the lldpV2LocVlanEntry,
65          therefore it is only present along with the VLAN Name
66          instance contained in the associated lldpV2LocVlanNameEntry
67          entry.
68
69          Each active lldpV2Xdot1ConfigVlanNameEntry is restored
70          from non-volatile storage (along with the corresponding
71          lldpV2Xdot1LocVlanNameEntry) after a re-initialization of
72          the management system."

```

```

1  AUGMENTS { lldpV2Xdot1LocVlanNameEntry }
2  ::= { lldpV2Xdot1ConfigVlanNameTable 1 }
3
4
5
6  lldpV2Xdot1ConfigVlanNameEntry ::= SEQUENCE {
7      lldpV2Xdot1ConfigVlanNameTxEnable  TruthValue
8  }
9
10 lldpV2Xdot1ConfigVlanNameTxEnable  OBJECT-TYPE
11     SYNTAX          TruthValue
12     MAX-ACCESS      read-write
13     STATUS          current
14     DESCRIPTION
15         "The boolean value that indicates whether the corresponding
16         Local System VLAN name instance is transmitted on the
17         port defined by the given lldpV2Xdot1LocVlanNameEntry.
18
19         The value of this object is restored from non-volatile
20         storage after a re-initialization of the management
21         system."
22     REFERENCE
23         "9.1.2.1 of IEEE Std 802.1AB"
24     DEFVAL { false }
25     ::= { lldpV2Xdot1ConfigVlanNameEntry 1 }
26
27
28 --
29 -- lldpV2Xdot1ConfigProtoVlanTable : configure the transmission of the
30 --                                protocol VLAN instances on set
31 --                                of ports.
32 --
33
34 lldpV2Xdot1ConfigProtoVlanTable OBJECT-TYPE
35     SYNTAX          SEQUENCE OF LldpV2Xdot1ConfigProtoVlanEntry
36     MAX-ACCESS      not-accessible
37     STATUS          current
38     DESCRIPTION
39         "The table that controls selection of LLDP Port And
40         Protocol VLAN ID TLV instances to be transmitted on
41         individual ports."
42     ::= { lldpV2Xdot1Config 3 }
43
44 lldpV2Xdot1ConfigProtoVlanEntry OBJECT-TYPE
45     SYNTAX          LldpV2Xdot1ConfigProtoVlanEntry
46     MAX-ACCESS      not-accessible
47     STATUS          current
48     DESCRIPTION
49         "LLDP configuration information that specifies the set of
50         ports (represented as a PortList) on which the Local System
51         Protocol VLAN instance is transmitted.
52
53         This configuration object augments the
54         lldpV2Xdot1LocVlanEntry, therefore it is only present along
55         with the Port and Protocol VLAN ID instance contained in
56         the associated lldpV2Xdot1LocVlanEntry entry.
57
58         Each active lldpV2Xdot1ConfigProtoVlanEntry is restored
59         from non-volatile storage (along with the corresponding
60         lldpV2Xdot1LocProtoVlanEntry) after a re-initialization of
61         the management system."
62
63     AUGMENTS { lldpV2Xdot1LocProtoVlanEntry }
64     ::= { lldpV2Xdot1ConfigProtoVlanTable 1 }
65
66
67 lldpV2Xdot1ConfigProtoVlanEntry ::= SEQUENCE {
68     lldpV2Xdot1ConfigProtoVlanTxEnable  TruthValue
69 }
70
71 lldpV2Xdot1ConfigProtoVlanTxEnable OBJECT-TYPE
72     SYNTAX          TruthValue

```



```

1  MAX-ACCESS      read-write
2  STATUS          current
3  DESCRIPTION
4      "The boolean value that indicates whether the corresponding
5      Local System Port and Protocol VLAN instance is
6      transmitted on the port defined by the given
7      lldpV2Xdot1LocProtoVlanEntry.
8
9      The value of this object is restored from non-volatile
10     storage after a re-initialization of the management system."
11  REFERENCE
12     "9.1.2.1 of IEEE Std 802.1AB"
13  DEFVAL { false }
14  ::= { lldpV2Xdot1ConfigProtoVlanEntry 1 }
15
16
17 --
18 -- lldpV2Xdot1ConfigProtocolTable : configure the transmission of the
19 --                                protocol instances on set
20 --                                of ports.
21 --
22
23 lldpV2Xdot1ConfigProtocolTable OBJECT-TYPE
24     SYNTAX      SEQUENCE OF LldpV2Xdot1ConfigProtocolEntry
25     MAX-ACCESS  not-accessible
26     STATUS      current
27     DESCRIPTION
28         "The table that controls selection of LLDP Protocol
29         TLV instances to be transmitted on individual ports."
30     ::= { lldpV2Xdot1Config 4 }
31
32 lldpV2Xdot1ConfigProtocolEntry OBJECT-TYPE
33     SYNTAX      LldpV2Xdot1ConfigProtocolEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "LLDP configuration information that specifies the set of
38         ports (represented as a PortList) on which the Local System
39         Protocol instance is transmitted.
40
41         This configuration object augments the
42         lldpV2Xdot1LocProtoEntry, therefore it is only present
43         along with the Protocol instance contained in the
44         associated lldpV2Xdot1LocProtoEntry entry.
45
46         Each active lldpV2Xdot1ConfigProtocolEntry is restored
47         from non-volatile storage (along with the corresponding
48         lldpV2Xdot1LocProtocolEntry) after a re-initialization of
49         the management system."
50     AUGMENTS { lldpV2Xdot1LocProtocolEntry }
51     ::= { lldpV2Xdot1ConfigProtocolTable 1 }
52
53
54 LldpV2Xdot1ConfigProtocolEntry ::= SEQUENCE {
55     lldpV2Xdot1ConfigProtocolTxEnable  TruthValue
56 }
57
58 lldpV2Xdot1ConfigProtocolTxEnable OBJECT-TYPE
59     SYNTAX      TruthValue
60     MAX-ACCESS  read-write
61     STATUS      current
62     DESCRIPTION
63         "The boolean value that indicates whether the corresponding
64         Local System Protocol Identity instance is transmitted
65         on the port defined by the given
66         lldpV2Xdot1LocProtocolEntry.
67
68         The value of this object is restored from non-volatile
69         storage after a re-initialization of the management
70         system."
71     REFERENCE
72         "9.1.2.1 of IEEE Std 802.1AB"

```

```

1  DEFVAL { false }
2  ::= { lldpV2Xdot1ConfigProtocolEntry 1 }
3
4 --
5 -- lldpV2Xdot1ConfigVidUsageDigestTable: configure the transmission
6 -- of the VID Usage Digest TLVs on set of ports.
7 --
8 lldpV2Xdot1ConfigVidUsageDigestTable OBJECT-TYPE
9     SYNTAX SEQUENCE OF LldpV2Xdot1ConfigVidUsageDigestEntry
10    MAX-ACCESS not-accessible
11    STATUS current
12    DESCRIPTION
13        "A table that controls selection of LLDP VID Usage Digest
14        TLVs to be transmitted on individual ports."
15 ::= { lldpV2Xdot1Config 5 }
16
17 lldpV2Xdot1ConfigVidUsageDigestEntry OBJECT-TYPE
18     SYNTAX LldpV2Xdot1ConfigVidUsageDigestEntry
19     MAX-ACCESS not-accessible
20     STATUS current
21     DESCRIPTION
22         "LLDP configuration information that specifies the set of
23         ports (represented as a PortList) on which the local
24         system VID Usage Digest instance will be transmitted.
25         This configuration object augments the
26         lldpLocVidUsageDigestEntry, therefore it is only present
27         along with the VID Usage Digest instance
28         contained in the associated lldpV2Xdot1LocVidUsageDigestEntry
29         entry. Each active lldpConfigVidUsageDigestEntry must be
30         restored from non-volatile storage and re-created (along with
31         the corresponding lldpV2Xdot1LocVidUsageDigestEntry) after
32         a re-initialization of the management system."
33     AUGMENTS { lldpV2Xdot1LocVidUsageDigestEntry }
34 ::= { lldpV2Xdot1ConfigVidUsageDigestTable 1 }
35
36 LldpV2Xdot1ConfigVidUsageDigestEntry ::= SEQUENCE {
37     lldpV2Xdot1ConfigVidUsageDigestTxEnable TruthValue
38 }
39
40 lldpV2Xdot1ConfigVidUsageDigestTxEnable OBJECT-TYPE
41     SYNTAX TruthValue
42     MAX-ACCESS read-write
43     STATUS current
44     DESCRIPTION
45         "The boolean value that indicates whether the corresponding
46         Local System VID Usage Digest instance will be transmitted
47         on the port defined by the given
48         lldpV2Xdot1LocVidUsageDigestEntry. The value of this object
49         must be restored from non-volatile storage after a
50         reinitialization of the management system."
51     REFERENCE
52         "9.1.2.1 of IEEE Std 802.1AB"
53     DEFVAL { false }
54 ::= { lldpV2Xdot1ConfigVidUsageDigestEntry 1 }
55
56
57 --
58 -- lldpV2Xdot1ConfigManVidTable : configure the transmission of the
59 -- Management VID TLVs on set of ports.
60 --
61 lldpV2Xdot1ConfigManVidTable OBJECT-TYPE
62     SYNTAX SEQUENCE OF LldpV2Xdot1ConfigManVidEntry
63     MAX-ACCESS not-accessible
64     STATUS current
65     DESCRIPTION
66         "A table that controls selection of LLDP Management VID
67         TLVs to be transmitted on individual ports."
68 ::= { lldpV2Xdot1Config 6 }
69
70 lldpV2Xdot1ConfigManVidEntry OBJECT-TYPE
71     SYNTAX LldpV2Xdot1ConfigManVidEntry
72     MAX-ACCESS not-accessible

```

```
1 STATUS current
2 DESCRIPTION
3     "LLDP configuration information that specifies the set of
4     port/destination address pairs on which the Local
5     System Management VID will be transmitted.
6     This configuration object augments the
7     lldpV2Xdot1LocManVidEntry, therefore it is
8     only present along with the Management VID contained
9     in the associated lldpV2Xdot1LocManVidEntry entry.
10    Each active lldpV2Xdot1ConfigManVidEntry must be
11    restored from non-volatile storage (along with the
12    corresponding lldpV2Xdot1LocManVidEntry) after a
13    re-initialization of the management system."
14    AUGMENTS { lldpV2Xdot1LocManVidEntry }
15 ::= { lldpV2Xdot1ConfigManVidTable 1 }
16
17 lldpV2Xdot1ConfigManVidEntry ::= SEQUENCE {
18     lldpV2Xdot1ConfigManVidTxEnable TruthValue
19 }
20
21 lldpV2Xdot1ConfigManVidTxEnable OBJECT-TYPE
22     SYNTAX TruthValue
23     MAX-ACCESS read-write
24     STATUS current
25     DESCRIPTION
26         "The lldpV2Xdot1ConfigManVidTxEnable, which is defined as a
27         truth value and configured by the network management,
28         determines whether the IEEE 802.1 organizationally
29         defined Management VID TLV transmission is allowed on a given
30         LLDP transmission-capable port.
31         The value of this object must be restored from
32         non-volatile storage after a re-initialization of the
33         management system."
34     REFERENCE
35         "9.1.2.1 of IEEE Std 802.1AB"
36     DEFVAL { false }
37 ::= { lldpV2Xdot1ConfigManVidEntry 1 }
38
39
40 -----
41 -- IEEE 802.1 - Local System Information
42 -----
43
44 --
45 -- lldpV2Xdot1LocTable - indexed by ifIndex.
46 --
47
48 lldpV2Xdot1LocTable OBJECT-TYPE
49     SYNTAX SEQUENCE OF LldpV2Xdot1LocEntry
50     MAX-ACCESS not-accessible
51     STATUS current
52     DESCRIPTION
53         "This table contains one row per port for IEEE 802.1
54         organizationally defined LLDP extension on the local system
55         known to this agent."
56     ::= { lldpV2Xdot1LocalData 1 }
57
58 lldpV2Xdot1LocEntry OBJECT-TYPE
59     SYNTAX LldpV2Xdot1LocEntry
60     MAX-ACCESS not-accessible
61     STATUS current
62     DESCRIPTION
63         "Information about IEEE 802.1 organizationally defined
64         LLDP extension."
65     INDEX { lldpV2LocPortIfIndex }
66     ::= { lldpV2Xdot1LocTable 1 }
67
68 lldpV2Xdot1LocEntry ::= SEQUENCE {
69     lldpV2Xdot1LocPortVlanId Unsigned32
70 }
71
72 lldpV2Xdot1LocPortVlanId OBJECT-TYPE
```

```

1  SYNTAX      Unsigned32(0|1..4094)
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "The integer value used to identify the port's VLAN
6      Identifier associated with the local system. A value
7      of zero shall be used if the system either does not know
8      the PVID or does
9      not support Port-based VLAN operation."
10 REFERENCE
11     "D.2.1.1"
12 ::= { lldpV2Xdot1LocEntry 1 }
13
14
15
16 --
17 -- lldpV2Xdot1LocProtoVlanTable: Port and Protocol VLAN information
18 -- re-indexed by ifIndex.
19 --
20
21 lldpV2Xdot1LocProtoVlanTable OBJECT-TYPE
22     SYNTAX      SEQUENCE OF LldpV2Xdot1LocProtoVlanEntry
23     MAX-ACCESS  not-accessible
24     STATUS      current
25     DESCRIPTION
26         "This table contains one or more rows per Port and Protocol
27         VLAN information about the local system."
28     ::= { lldpV2Xdot1LocalData 2 }
29
30 lldpV2Xdot1LocProtoVlanEntry OBJECT-TYPE
31     SYNTAX      LldpV2Xdot1LocProtoVlanEntry
32     MAX-ACCESS  not-accessible
33     STATUS      current
34     DESCRIPTION
35         "Port and protocol VLAN ID Information about a particular
36         port component. There may be multiple port and protocol
37         VLANs, identified by a particular
38         lldpV2Xdot1LocProtoVlanId, configured on the given port."
39     INDEX      { lldpV2LocPortIfIndex,
40                 lldpV2Xdot1LocProtoVlanId }
41     ::= { lldpV2Xdot1LocProtoVlanTable 1 }
42
43 lldpV2Xdot1LocProtoVlanEntry ::= SEQUENCE {
44     lldpV2Xdot1LocProtoVlanId      Unsigned32,
45     lldpV2Xdot1LocProtoVlanSupported TruthValue,
46     lldpV2Xdot1LocProtoVlanEnabled TruthValue
47 }
48
49 lldpV2Xdot1LocProtoVlanId OBJECT-TYPE
50     SYNTAX      Unsigned32(0|1..4094)
51     MAX-ACCESS  not-accessible
52     STATUS      current
53     DESCRIPTION
54         "The integer value used to identify the port and protocol
55         VLANs associated with the given port associated with the
56         local system. A value of zero shall be used if the system
57         either does not know the protocol VLAN ID (PPVID) or does
58         not support port and protocol VLAN operation."
59     REFERENCE
60         "D.2.2.2"
61     ::= { lldpV2Xdot1LocProtoVlanEntry 1 }
62
63 lldpV2Xdot1LocProtoVlanSupported OBJECT-TYPE
64     SYNTAX      TruthValue
65     MAX-ACCESS  read-only
66     STATUS      current
67     DESCRIPTION
68         "The truth value used to indicate whether the given port
69         (associated with the local system) supports port and
70         protocol VLANs."
71     REFERENCE
72         "D.2.2.1"

```

```

1      ::= { lldpV2Xdot1LocProtoVlanEntry 2 }
2
3 lldpV2Xdot1LocProtoVlanEnabled OBJECT-TYPE
4     SYNTAX      TruthValue
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "The truth value used to indicate whether the port and
9         protocol VLANs are enabled on the given port associated
10        with the local system."
11    REFERENCE
12        "D.2.2.1"
13    ::= { lldpV2Xdot1LocProtoVlanEntry 3 }
14
15
16
17 --
18 -- lldpV2Xdot1LocVlanNameTable : VLAN name information about the local
19 -- system indexed by ifIndex.
20 --
21
22 lldpV2Xdot1LocVlanNameTable OBJECT-TYPE
23     SYNTAX      SEQUENCE OF LldpV2Xdot1LocVlanNameEntry
24     MAX-ACCESS  not-accessible
25     STATUS      current
26     DESCRIPTION
27         "This table contains one or more rows per IEEE 802.1Q VLAN
28         name information on the local system known to this agent."
29     ::= { lldpV2Xdot1LocalData 3 }
30
31 lldpV2Xdot1LocVlanNameEntry OBJECT-TYPE
32     SYNTAX      LldpV2Xdot1LocVlanNameEntry
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "VLAN name Information about a particular port component.
37         There may be multiple VLANs, identified by a particular
38         lldpV2Xdot1LocVlanId, configured on the given port."
39     INDEX      { lldpV2LocPortIfIndex,
40                 lldpV2Xdot1LocVlanId }
41     ::= { lldpV2Xdot1LocVlanNameTable 1 }
42
43 LldpV2Xdot1LocVlanNameEntry ::= SEQUENCE {
44     lldpV2Xdot1LocVlanId      VlanId,
45     lldpV2Xdot1LocVlanName    SnmpAdminString
46 }
47
48 lldpV2Xdot1LocVlanId OBJECT-TYPE
49     SYNTAX      VlanId
50     MAX-ACCESS  not-accessible
51     STATUS      current
52     DESCRIPTION
53         "The integer value used to identify the IEEE 802.1Q
54         VLAN IDs with which the given port is compatible."
55     REFERENCE
56         "D.2.3.2"
57     ::= { lldpV2Xdot1LocVlanNameEntry 1 }
58
59 lldpV2Xdot1LocVlanName OBJECT-TYPE
60     SYNTAX      SnmpAdminString (SIZE(1..32))
61     MAX-ACCESS  read-only
62     STATUS      current
63     DESCRIPTION
64         "The string value used to identify VLAN name identified
65         by the Vlan Id associated with the given port on the
66         local system.
67
68         This object should contain the value of the
69         dot1QVlanStaticName object (defined in IETF RFC 4363)
70         identified with the given lldpV2Xdot1LocVlanId."
71     REFERENCE
72         "D.2.3.4"

```

```

1 ::= { lldpV2Xdot1LocVlanNameEntry 2 }
2
3
4
5 --
6 -- lldpV2Xdot1LocProtocolTable : Protocol Identity information
7 -- re-indexed by ifIndex and destination address
8 --
9
10 lldpV2Xdot1LocProtocolTable OBJECT-TYPE
11     SYNTAX      SEQUENCE OF LldpV2Xdot1LocProtocolEntry
12     MAX-ACCESS   not-accessible
13     STATUS       current
14     DESCRIPTION
15         "This table contains one or more rows per protocol identity
16         information on the local system known to this agent."
17     REFERENCE
18         "D.2.4"
19     ::= { lldpV2Xdot1LocalData 4 }
20
21 lldpV2Xdot1LocProtocolEntry OBJECT-TYPE
22     SYNTAX      LldpV2Xdot1LocProtocolEntry
23     MAX-ACCESS   not-accessible
24     STATUS       current
25     DESCRIPTION
26         "Information about particular protocols that are accessible
27         through the given port component.
28
29         There may be multiple protocols, identified by particular
30         lldpV2Xdot1ProtocolIndex, lldpV2LocPortIfIndex"
31     REFERENCE
32         "D.2.4"
33     INDEX      { lldpV2LocPortIfIndex,
34                  lldpV2Xdot1LocProtocolIndex }
35     ::= { lldpV2Xdot1LocProtocolTable 1 }
36
37 LldpV2Xdot1LocProtocolEntry ::= SEQUENCE {
38     lldpV2Xdot1LocProtocolIndex Unsigned32,
39     lldpV2Xdot1LocProtocolId    OCTET STRING
40 }
41
42
43 lldpV2Xdot1LocProtocolIndex OBJECT-TYPE
44     SYNTAX      Unsigned32(1..2147483647)
45     MAX-ACCESS   not-accessible
46     STATUS       current
47     DESCRIPTION
48         "This object represents an arbitrary local integer value
49         used by this agent to identify a particular protocol
50         identity."
51     ::= { lldpV2Xdot1LocProtocolEntry 1 }
52
53 lldpV2Xdot1LocProtocolId OBJECT-TYPE
54     SYNTAX      OCTET STRING (SIZE (1..255))
55     MAX-ACCESS   read-only
56     STATUS       current
57     DESCRIPTION
58         "The octet string value used to identify the protocols
59         associated with the given port of the local system."
60     REFERENCE
61         "D.2.4.3"
62     ::= { lldpV2Xdot1LocProtocolEntry 2 }
63
64 --
65 -- lldpV2Xdot1LocVidUsageDigestTable: Table of hash values of
66 -- system VID Usage Table transmitted
67 -- via VID Usage Digest TLV.
68 --
69
70
71 lldpV2Xdot1LocVidUsageDigestTable OBJECT-TYPE
72     SYNTAX      SEQUENCE OF LldpV2Xdot1LocVidUsageDigestEntry

```

```

1  MAX-ACCESS    not-accessible
2  STATUS       current
3  DESCRIPTION
4      "This table contains one row per ifIndex/
5      destination MAC address pair for usage digest
6      information on the local system known to this agent."
7  REFERENCE
8      "D.2.5"
9  ::= { lldpV2Xdot1LocalData 5 }
10
11 lldpV2Xdot1LocVidUsageDigestEntry OBJECT-TYPE
12     SYNTAX      LldpV2Xdot1LocVidUsageDigestEntry
13     MAX-ACCESS  not-accessible
14     STATUS      current
15     DESCRIPTION
16         "Usage digest information to be transmitted
17         through the given port."
18     REFERENCE
19         "D.2.5"
20     INDEX       { lldpV2LocPortIfIndex }
21     ::= { lldpV2Xdot1LocVidUsageDigestTable 1 }
22
23 lldpV2Xdot1LocVidUsageDigestEntry ::= SEQUENCE {
24     lldpV2Xdot1LocVidUsageDigest Unsigned32
25 }
26
27
28 lldpV2Xdot1LocVidUsageDigest OBJECT-TYPE
29     SYNTAX      Unsigned32
30     MAX-ACCESS  read-only
31     STATUS      current
32     DESCRIPTION
33         "The integer value obtained by applying the CRC32 function
34         to the 128-octet VID Usage Table. A bit of the VID Usage
35         Table contains the value PBB-TE-USAGE (binary 1) if the
36         corresponding element of the MST Configuration Table
37         (IEEE Std 802.1Q 8.9.1) contains the value PBB-TE MSTID
38         (hex FFE) and otherwise contains the value NON-PBB-TE-USAGE
39         (binary 0)."
```

40 REFERENCE

41 "D.2.5.1"

42 ::= { lldpV2Xdot1LocVidUsageDigestEntry 1 }

43

44 --

45 -- lldpV2Xdot1LocManVidTable: Table of values configured on the Local

46 -- system for the Management VID, or the value 0 if a Management VID

47 -- has not been provisioned.

48 --

49

50 lldpV2Xdot1LocManVidTable OBJECT-TYPE

51 SYNTAX SEQUENCE OF LldpV2Xdot1LocManVidEntry

52 MAX-ACCESS not-accessible

53 STATUS current

54 DESCRIPTION

55 "This table contains one row per ifIndex/

56 destination MAC address pair for usage digest

57 information on the local system known to this agent."

58 REFERENCE

59 "D.2.6"

60 ::= { lldpV2Xdot1LocalData 6 }

61

62 lldpV2Xdot1LocManVidEntry OBJECT-TYPE

63 SYNTAX LldpV2Xdot1LocManVidEntry

64 MAX-ACCESS not-accessible

65 STATUS current

66 DESCRIPTION

67 "Usage digest information to be transmitted

68 through the given port."

69 REFERENCE

70 "D.2.6"

71 INDEX { lldpV2LocPortIfIndex }

72 ::= { lldpV2Xdot1LocManVidTable 1 }

```

1
2 LldpV2Xdot1LocManVidEntry ::= SEQUENCE {
3     lldpV2Xdot1LocManVid Unsigned32
4 }
5
6 lldpV2Xdot1LocManVid OBJECT-TYPE
7     SYNTAX      Unsigned32 (0|1..4094)
8     MAX-ACCESS  read-only
9     STATUS      current
10    DESCRIPTION
11        "The integer value configured on the Local system for
12         the Management VID, or
13         the value 0 if a Management VID has not been provisioned."
14    REFERENCE
15        "D.2.6.1"
16 ::= { lldpV2Xdot1LocManVidEntry 1 }
17
18
19 -----
20 -- IEEE 802.1 - Local System Information - Link Aggregation
21 -----
22
23 ---
24 ---
25 --- lldpV2Xdot1LocLinkAggTable: Link Aggregation Information Table
26 ---
27 ---
28 lldpV2Xdot1LocLinkAggTable OBJECT-TYPE
29     SYNTAX      SEQUENCE OF LldpV2Xdot1LocLinkAggEntry
30     MAX-ACCESS  not-accessible
31     STATUS      current
32     DESCRIPTION
33         "This table contains one row per port of link aggregation
34         information (as a part of the LLDP 802.1 organizational
35         extension) on the local system known to this agent."
36     ::= { lldpV2Xdot1LocalData 7 }
37
38 lldpV2Xdot1LocLinkAggEntry OBJECT-TYPE
39     SYNTAX      LldpV2Xdot1LocLinkAggEntry
40     MAX-ACCESS  not-accessible
41     STATUS      current
42     DESCRIPTION
43         "Link Aggregation information about a particular port
44         component."
45     INDEX      { lldpV2LocPortIfIndex }
46     ::= { lldpV2Xdot1LocLinkAggTable 1 }
47
48 LldpV2Xdot1LocLinkAggEntry ::= SEQUENCE {
49     lldpV2Xdot1LocLinkAggStatus      LldpV2XLinkAggStatusMap,
50     lldpV2Xdot1LocLinkAggPortId      Unsigned32
51 }
52
53 lldpV2Xdot1LocLinkAggStatus OBJECT-TYPE
54     SYNTAX      LldpV2XLinkAggStatusMap
55     MAX-ACCESS  read-only
56     STATUS      current
57     DESCRIPTION
58         "The bitmap value contains the link aggregation
59         capabilities and the current aggregation status of the
60         link."
61     REFERENCE
62         "IEEE Std 802.1AX"
63     ::= { lldpV2Xdot1LocLinkAggEntry 1 }
64
65 lldpV2Xdot1LocLinkAggPortId OBJECT-TYPE
66     SYNTAX      Unsigned32 (0|1..2147483647)
67     MAX-ACCESS  read-only
68     STATUS      current
69     DESCRIPTION
70         "This object contains the IEEE 802.1 aggregated port
71         identifier, aAggPortID (IEEE Std 802.1AX, 6.3.2.1.1),
72         derived from the ifNumber of the ifIndex for the port

```



```
1         component in link aggregation.
2
3         If the port is not in link aggregation state and/or it
4         does not support link aggregation, this value should be set
5         to zero."
6     REFERENCE
7         "IEEE Std 802.1AX"
8     ::= { lldpV2Xdot1LocLinkAggEntry 2 }
9
10
11
12 -----
13 -- IEEE 802.1 - Remote System Information
14 -----
15
16 --
17 -- lldpV2Xdot1RemTable - re-indexed for ifIndex and destination MAC
18 -- address
19
20 lldpV2Xdot1RemTable OBJECT-TYPE
21     SYNTAX      SEQUENCE OF LldpV2Xdot1RemEntry
22     MAX-ACCESS  not-accessible
23     STATUS      current
24     DESCRIPTION
25         "This table contains one or more rows per physical network
26         connection known to this agent. The agent may wish to
27         ensure that only one lldpV2Xdot1RemEntry is present for
28         each local port, or it may choose to maintain multiple
29         lldpV2Xdot1RemEntries for the same local port."
30     ::= { lldpV2Xdot1RemoteData 1 }
31
32 lldpV2Xdot1RemEntry OBJECT-TYPE
33     SYNTAX      LldpV2Xdot1RemEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "Information about a particular port component."
38     INDEX      { lldpV2RemTimeMark,
39                 lldpV2RemLocalIfIndex,
40                 lldpV2RemLocalDestMACAddress,
41                 lldpV2RemIndex }
42     ::= { lldpV2Xdot1RemTable 1 }
43
44 LldpV2Xdot1RemEntry ::= SEQUENCE {
45     lldpV2Xdot1RemPortVlanId      Unsigned32
46 }
47
48 lldpV2Xdot1RemPortVlanId OBJECT-TYPE
49     SYNTAX      Unsigned32(0|1..4094)
50     MAX-ACCESS  read-only
51     STATUS      current
52     DESCRIPTION
53         "The integer value used to identify the port's VLAN
54         Identifier associated with the remote system. If the
55         remote system either does not know the PVID or does not
56         support Port-based VLAN operation, the value of
57         lldpV2Xdot1RemPortVlanId should be zero."
58     REFERENCE
59         "D.2.1.1"
60     ::= { lldpV2Xdot1RemEntry 1 }
61
62
63
64 --
65 -- lldpV2Xdot1RemProtoVlanTable - re-indexed by ifIndex and
66 -- destination MAC address
67 --
68
69 lldpV2Xdot1RemProtoVlanTable OBJECT-TYPE
70     SYNTAX      SEQUENCE OF LldpV2Xdot1RemProtoVlanEntry
71     MAX-ACCESS  not-accessible
72     STATUS      current
```

```

1  DESCRIPTION
2      "This table contains one or more rows per Port and Protocol
3      VLAN information about the remote system, received on the
4      given port."
5      ::= { lldpV2Xdot1RemoteData 2 }
6
7  lldpV2Xdot1RemProtoVlanEntry OBJECT-TYPE
8      SYNTAX      LldpV2Xdot1RemProtoVlanEntry
9      MAX-ACCESS  not-accessible
10     STATUS      current
11     DESCRIPTION
12         "Port and protocol VLAN name Information about a particular
13         port component. There may be multiple protocol VLANs,
14         identified by a particular lldpV2Xdot1RemProtoVlanId,
15         configured on the remote system."
16     INDEX      { lldpV2RemTimeMark,
17                 lldpV2RemLocalIfIndex,
18                 lldpV2RemLocalDestMACAddress,
19                 lldpV2RemIndex,
20                 lldpV2Xdot1RemProtoVlanId }
21     ::= { lldpV2Xdot1RemProtoVlanTable 1 }
22
23 lldpV2Xdot1RemProtoVlanEntry ::= SEQUENCE {
24     lldpV2Xdot1RemProtoVlanId      Unsigned32,
25     lldpV2Xdot1RemProtoVlanSupported TruthValue,
26     lldpV2Xdot1RemProtoVlanEnabled TruthValue
27 }
28
29 lldpV2Xdot1RemProtoVlanId OBJECT-TYPE
30     SYNTAX      Unsigned32(0|1..4094)
31     MAX-ACCESS  not-accessible
32     STATUS      current
33     DESCRIPTION
34         "The integer value used to identify the port and protocol
35         VLANs associated with the given port associated with the
36         remote system.
37
38         If port and protocol VLANs are not supported on the given
39         port associated with the remote system, or if the port is
40         not enabled with any port and protocol VLAN, the value of
41         lldpV2Xdot1RemProtoVlanId should be zero."
42     REFERENCE
43         "D.2.2.2"
44     ::= { lldpV2Xdot1RemProtoVlanEntry 1 }
45
46 lldpV2Xdot1RemProtoVlanSupported OBJECT-TYPE
47     SYNTAX      TruthValue
48     MAX-ACCESS  read-only
49     STATUS      current
50     DESCRIPTION
51         "The truth value used to indicate whether the given port
52         (associated with the remote system) is capable of
53         supporting port and protocol VLANs."
54     REFERENCE
55         "D.2.2.1"
56     ::= { lldpV2Xdot1RemProtoVlanEntry 2 }
57
58 lldpV2Xdot1RemProtoVlanEnabled OBJECT-TYPE
59     SYNTAX      TruthValue
60     MAX-ACCESS  read-only
61     STATUS      current
62     DESCRIPTION
63         "The truth value used to indicate whether the port and
64         protocol VLANs are enabled on the given port associated
65         with
66         the remote system."
67     REFERENCE
68         "D.2.2.1"
69     ::= { lldpV2Xdot1RemProtoVlanEntry 3 }
70
71
72

```

```

1 --
2 -- lldpV2Xdot1RemVlanNameTable : VLAN name information of the remote
3 --                               systems
4 -- Re-indexed by ifIndex and destination MAC address
5 --
6
7 lldpV2Xdot1RemVlanNameTable OBJECT-TYPE
8     SYNTAX      SEQUENCE OF LldpV2Xdot1RemVlanNameEntry
9     MAX-ACCESS  not-accessible
10    STATUS      current
11    DESCRIPTION
12        "This table contains one or more rows per IEEE 802.1Q VLAN
13        name information about the remote system, received on the
14        given port."
15    REFERENCE
16        "D.2.3"
17    ::= { lldpV2Xdot1RemoteData 3 }
18
19 lldpV2Xdot1RemVlanNameEntry OBJECT-TYPE
20     SYNTAX      LldpV2Xdot1RemVlanNameEntry
21     MAX-ACCESS  not-accessible
22     STATUS      current
23     DESCRIPTION
24         "VLAN name Information about a particular port component.
25         There may be multiple VLANs, identified by a particular
26         lldpV2Xdot1RemVlanId, received on the given port."
27     INDEX       { lldpV2RemTimeMark,
28                   lldpV2RemLocalIfIndex,
29                   lldpV2RemLocalDestMACAddress,
30                   lldpV2RemIndex,
31                   lldpV2Xdot1RemVlanId }
32     ::= { lldpV2Xdot1RemVlanNameTable 1 }
33
34 lldpV2Xdot1RemVlanNameEntry ::= SEQUENCE {
35     lldpV2Xdot1RemVlanId      VlanId,
36     lldpV2Xdot1RemVlanName    SnmpAdminString
37 }
38
39
40 lldpV2Xdot1RemVlanId OBJECT-TYPE
41     SYNTAX      VlanId
42     MAX-ACCESS  not-accessible
43     STATUS      current
44     DESCRIPTION
45         "The integer value used to identify the IEEE 802.1Q
46         VLAN IDs with which the given port of the remote system
47         is compatible."
48     REFERENCE
49         "D.2.3.2"
50     ::= { lldpV2Xdot1RemVlanNameEntry 1 }
51
52 lldpV2Xdot1RemVlanName OBJECT-TYPE
53     SYNTAX      SnmpAdminString (SIZE(1..32))
54     MAX-ACCESS  read-only
55     STATUS      current
56     DESCRIPTION
57         "The string value used to identify VLAN name identified
58         by the VLAN Id associated with the remote system."
59     REFERENCE
60         "D.2.3.4"
61     ::= { lldpV2Xdot1RemVlanNameEntry 2 }
62
63
64
65 --
66 -- lldpV2Xdot1RemProtocolTable : Protocol information of the remote
67 -- systems Re-indexed by ifIndex and destination MAC address
68 --
69
70 lldpV2Xdot1RemProtocolTable OBJECT-TYPE
71     SYNTAX      SEQUENCE OF LldpV2Xdot1RemProtocolEntry
72     MAX-ACCESS  not-accessible

```

```

1  STATUS      current
2  DESCRIPTION
3      "This table contains one or more rows per protocol
4      information about the remote system, received on
5      the given port."
6  ::= { lldpV2Xdot1RemoteData 4 }
7
8  lldpV2Xdot1RemProtocolEntry OBJECT-TYPE
9  SYNTAX      LldpV2Xdot1RemProtocolEntry
10 MAX-ACCESS  not-accessible
11 STATUS      current
12 DESCRIPTION
13     "Protocol information about a particular port component.
14     There may be multiple protocols, identified by a particular
15     lldpV2Xdot1ProtocolIndex, received on the given port."
16 INDEX      { lldpV2RemTimeMark,
17              lldpV2RemLocalIfIndex,
18              lldpV2RemLocalDestMACAddress,
19              lldpV2RemIndex,
20              lldpV2Xdot1RemProtocolIndex }
21 ::= { lldpV2Xdot1RemProtocolTable 1 }
22
23 lldpV2Xdot1RemProtocolEntry ::= SEQUENCE {
24     lldpV2Xdot1RemProtocolIndex  Unsigned32,
25     lldpV2Xdot1RemProtocolId     OCTET STRING
26 }
27
28 lldpV2Xdot1RemProtocolIndex OBJECT-TYPE
29 SYNTAX      Unsigned32(1..2147483647)
30 MAX-ACCESS  not-accessible
31 STATUS      current
32 DESCRIPTION
33     "This object represents an arbitrary local integer value
34     used by this agent to identify a particular protocol
35     identity."
36 ::= { lldpV2Xdot1RemProtocolEntry 1 }
37
38 lldpV2Xdot1RemProtocolId OBJECT-TYPE
39 SYNTAX      OCTET STRING (SIZE (1..255))
40 MAX-ACCESS  read-only
41 STATUS      current
42 DESCRIPTION
43     "The octet string value used to identify the protocols
44     associated with the given port of remote system."
45 REFERENCE
46     "D.2.4.3"
47 ::= { lldpV2Xdot1RemProtocolEntry 2 }
48
49
50
51 --
52 -- lldpV2Xdot1RemVidUsageDigestTable: Table of hash values of
53 -- system VID Usage Table received
54 -- via VID Usage Digest TLV.
55 -- This version replaced by a reindexed version (V2).
56 --
57
58
59 lldpV2Xdot1RemVidUsageDigestTable OBJECT-TYPE
60 SYNTAX      SEQUENCE OF LldpV2Xdot1RemVidUsageDigestEntry
61 MAX-ACCESS  not-accessible
62 STATUS      deprecated
63 DESCRIPTION
64     "This table contains one row per ifIndex/
65     destination MAC address pair for usage digest
66     information received by the local system."
67 REFERENCE
68     "D.2.5"
69 ::= { lldpV2Xdot1RemoteData 5 }
70
71 lldpV2Xdot1RemVidUsageDigestEntry OBJECT-TYPE
72 SYNTAX      LldpV2Xdot1RemVidUsageDigestEntry

```

```

1  MAX-ACCESS    not-accessible
2  STATUS        deprecated
3  DESCRIPTION
4      "Usage digest information received on
5      the given port/destination address pair."
6  REFERENCE
7      "D.2.5"
8  INDEX { lldpV2RemTimeMark,
9          lldpV2RemLocalIfIndex,
10         lldpV2RemLocalDestMACAddress }
11 ::= { lldpV2Xdot1RemVidUsageDigestTable 1 }
12
13 LldpV2Xdot1RemVidUsageDigestEntry ::= SEQUENCE {
14     lldpV2Xdot1RemVidUsageDigest Unsigned32
15 }
16
17
18 lldpV2Xdot1RemVidUsageDigest OBJECT-TYPE
19     SYNTAX      Unsigned32
20     MAX-ACCESS  read-only
21     STATUS      deprecated
22     DESCRIPTION
23         "The integer value obtained by applying the CRC32 function
24         to the 128-octet VID Usage Table. A bit of the VID Usage
25         Table contains the value PBB-TE-USAGE (binary 1) if the
26         corresponding element of the MST Configuration Table
27         (IEEE Std 802.1Q 8.9.1) contains the value PBB-TE MSTID
28         (hex FFE) and otherwise contains the value NON-PBB-TE-USAGE
29         (binary 0)."
```

REFERENCE

```

30     "D.2.5.1"
31 ::= { lldpV2Xdot1RemVidUsageDigestEntry 1 }
32
33
34 --
35 -- lldpV2Xdot1RemManVidTable: Table of values configured on remote
36 -- systems for the Management VID, or the value 0 if a Management
37 -- VID has not been provisioned.
38 -- This version replaced by a reindexed version (V2).
39 --
40
41
42 lldpV2Xdot1RemManVidTable OBJECT-TYPE
43     SYNTAX      SEQUENCE OF LldpV2Xdot1RemManVidEntry
44     MAX-ACCESS  not-accessible
45     STATUS      deprecated
46     DESCRIPTION
47         "This table contains one row per ifIndex/
48         destination MAC address pair for management VID
49         information received from remote systems."
50     REFERENCE
51         "D.2.6"
52 ::= { lldpV2Xdot1RemoteData 6 }
53
54 lldpV2Xdot1RemManVidEntry OBJECT-TYPE
55     SYNTAX      LldpV2Xdot1RemManVidEntry
56     MAX-ACCESS  not-accessible
57     STATUS      deprecated
58     DESCRIPTION
59         "Management VID information received
60         through the given port/destination address pair."
61     REFERENCE
62         "D.2.6"
63     INDEX { lldpV2RemTimeMark,
64           lldpV2RemLocalIfIndex,
65           lldpV2RemLocalDestMACAddress }
66 ::= { lldpV2Xdot1RemManVidTable 1 }
67
68 LldpV2Xdot1RemManVidEntry ::= SEQUENCE {
69     lldpV2Xdot1RemManVid      Unsigned32
70 }
71
72 lldpV2Xdot1RemManVid OBJECT-TYPE
```

```

1 SYNTAX Unsigned32 (0|1..4094)
2 MAX-ACCESS read-only
3 STATUS deprecated
4 DESCRIPTION
5     "The integer value configured on a system for
6     the Management VID, or
7     the value 0 if a Management VID has not been provisioned."
8 REFERENCE
9     "D.2.6.1"
10 ::= { lldpV2Xdot1RemManVidEntry 1 }
11
12
13
14 --
15 -- lldpV2Xdot1RemVidUsageDigestV2Table: Table of hash values of
16 -- system VID Usage Table received
17 -- via VID Usage Digest TLV.
18 --
19
20
21 lldpV2Xdot1RemVidUsageDigestV2Table OBJECT-TYPE
22     SYNTAX SEQUENCE OF LldpV2Xdot1RemVidUsageDigestV2Entry
23     MAX-ACCESS not-accessible
24     STATUS current
25     DESCRIPTION
26         "This table contains one row per ifIndex/
27         destination MAC address pair for usage digest
28         information received by the local system."
29     REFERENCE
30         "D.2.5"
31     ::= { lldpV2Xdot1RemoteData 8 }
32
33 lldpV2Xdot1RemVidUsageDigestV2Entry OBJECT-TYPE
34     SYNTAX LldpV2Xdot1RemVidUsageDigestV2Entry
35     MAX-ACCESS not-accessible
36     STATUS current
37     DESCRIPTION
38         "Usage digest information received on
39         the given port/destination address pair."
40     REFERENCE
41         "D.2.5"
42     INDEX { lldpV2RemTimeMark,
43             lldpV2RemLocalIfIndex,
44             lldpV2RemLocalDestMACAddress,
45             lldpV2RemIndex }
46     ::= { lldpV2Xdot1RemVidUsageDigestV2Table 1 }
47
48 LldpV2Xdot1RemVidUsageDigestV2Entry ::= SEQUENCE {
49     lldpV2Xdot1RemVidUsageDigestV2 Unsigned32
50 }
51
52
53 lldpV2Xdot1RemVidUsageDigestV2 OBJECT-TYPE
54     SYNTAX Unsigned32
55     MAX-ACCESS read-only
56     STATUS current
57     DESCRIPTION
58         "The integer value obtained by applying the CRC32 function
59         to the 128-octet VID Usage Table. A bit of the VID Usage
60         Table contains the value PBB-TE-USAGE (binary 1) if the
61         corresponding element of the MST Configuration Table
62         (IEEE Std 802.1Q 8.9.1) contains the value PBB-TE MSTID
63         (hex FFE) and otherwise contains the value NON-PBB-TE-USAGE
64         (binary 0)."
65     REFERENCE
66         "D.2.5.1"
67     ::= { lldpV2Xdot1RemVidUsageDigestV2Entry 1 }
68
69
70 --
71 -- lldpV2Xdot1RemManVidV2Table: Table of values configured on remote
72 -- systems for the Management VID, or the value 0 if a Management

```

```

1 -- VID has not been provisioned.
2 --
3
4 lldpV2Xdot1RemManVidV2Table OBJECT-TYPE
5     SYNTAX      SEQUENCE OF LldpV2Xdot1RemManVidV2Entry
6     MAX-ACCESS  not-accessible
7     STATUS      current
8     DESCRIPTION
9         "This table contains one row per ifIndex/
10         destination MAC address pair for management VID
11         information received from remote systems."
12     REFERENCE
13         "D.2.6"
14     ::= { lldpV2Xdot1RemoteData 9 }
15
16 lldpV2Xdot1RemManVidV2Entry OBJECT-TYPE
17     SYNTAX      LldpV2Xdot1RemManVidV2Entry
18     MAX-ACCESS  not-accessible
19     STATUS      current
20     DESCRIPTION
21         "Management VID information received
22         through the given port/destination address pair."
23     REFERENCE
24         "D.2.6"
25     INDEX      { lldpV2RemTimeMark,
26                 lldpV2RemLocalIfIndex,
27                 lldpV2RemLocalDestMACAddress,
28                 lldpV2RemIndex }
29     ::= { lldpV2Xdot1RemManVidV2Table 1 }
30
31 lldpV2Xdot1RemManVidV2Entry ::= SEQUENCE {
32     lldpV2Xdot1RemManVidV2      Unsigned32
33 }
34
35 lldpV2Xdot1RemManVidV2 OBJECT-TYPE
36     SYNTAX      Unsigned32 (0|1..4094)
37     MAX-ACCESS  read-only
38     STATUS      current
39     DESCRIPTION
40         "The integer value configured on a system for
41         the Management VID, or
42         the value 0 if a Management VID has not been provisioned."
43     REFERENCE
44         "D.2.6.1"
45     ::= { lldpV2Xdot1RemManVidV2Entry 1 }
46
47
48
49 -----
50 -- Remote System Information - Link Aggregation
51 -----
52
53 ---
54 ---
55 --- lldpV2Xdot1RemLinkAggTable: Link Aggregation Information Table
56 ---
57 ---
58 lldpV2Xdot1RemLinkAggTable OBJECT-TYPE
59     SYNTAX      SEQUENCE OF LldpV2Xdot1RemLinkAggEntry
60     MAX-ACCESS  not-accessible
61     STATUS      current
62     DESCRIPTION
63         "This table contains port link aggregation information
64         (as a part of the LLDP IEEE 802.1 organizational extension)
65         of the remote system."
66     ::= { lldpV2Xdot1RemoteData 7 }
67
68 lldpV2Xdot1RemLinkAggEntry OBJECT-TYPE
69     SYNTAX      LldpV2Xdot1RemLinkAggEntry
70     MAX-ACCESS  not-accessible
71     STATUS      current
72     DESCRIPTION

```

```

1      "Link Aggregation information about remote system's port
2      component."
3      INDEX      { lldpV2RemTimeMark,
4                  lldpV2RemLocalIfIndex,
5                  lldpV2RemLocalDestMACAddress,
6                  lldpV2RemIndex }
7      ::= { lldpV2Xdot1RemLinkAggTable 1 }
8
9      lldpV2Xdot1RemLinkAggEntry ::= SEQUENCE {
10         lldpV2Xdot1RemLinkAggStatus      LldpV2XLinkAggStatusMap,
11         lldpV2Xdot1RemLinkAggPortId      Unsigned32
12     }
13
14     lldpV2Xdot1RemLinkAggStatus OBJECT-TYPE
15     SYNTAX      LldpV2XLinkAggStatusMap
16     MAX-ACCESS  read-only
17     STATUS      current
18     DESCRIPTION
19         "The bitmap value contains the link aggregation capabilities
20         and the current aggregation status of the link."
21     REFERENCE
22         "IEEE Std 802.1AX"
23     ::= { lldpV2Xdot1RemLinkAggEntry 1 }
24
25     lldpV2Xdot1RemLinkAggPortId OBJECT-TYPE
26     SYNTAX      Unsigned32(0|1..2147483647)
27     MAX-ACCESS  read-only
28     STATUS      current
29     DESCRIPTION
30         "This object contains the IEEE 802.1 aggregated port
31         identifier, aAggPortID (IEEE Std 802.1AX, 7.3.2.1.1),
32         derived from the ifNumber of the ifIndex for the port
33         component associated with the remote system.
34
35         If the remote port is not in link aggregation state and/or
36         it does not support link aggregation, this value should be
37         zero."
38     REFERENCE
39         "IEEE Std 802.1AX"
40     ::= { lldpV2Xdot1RemLinkAggEntry 2 }
41
42
43 -----
44 -- Conformance Information for the basicSet TLV set
45 -----
46
47     lldpV2Xdot1Conformance
48     OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 2 }
49     lldpV2Xdot1Compliances
50     OBJECT IDENTIFIER ::= { lldpV2Xdot1Conformance 1 }
51     lldpV2Xdot1Groups
52     OBJECT IDENTIFIER ::= { lldpV2Xdot1Conformance 2 }
53
54 -- compliance statements
55
56
57     lldpV2Xdot1TxRxCompliance MODULE-COMPLIANCE
58     STATUS      current
59     DESCRIPTION
60         "A compliance statement for SNMP entities that implement
61         the IEEE 802.1 organizationally defined LLDP extension MIB.
62
63         This group is mandatory for all agents that implement the
64         LLDP 802.1 organizational extension in TX and/or RX mode
65         for the basicSet TLV set.
66
67         This version defines compliance requirements for
68         V2 of the LLDP MIB."
69     MODULE      -- this module
70         MANDATORY-GROUPS { lldpV2Xdot1ConfigGroup,
71                             ifGeneralInformationGroup
72         }

```



```
1 ::= { lldpV2Xdot1Compliances 1 }
2
3
4 lldpV2Xdot1TxCompliance MODULE-COMPLIANCE
5     STATUS current
6     DESCRIPTION
7         "A compliance statement for SNMP entities that implement
8         the IEEE 802.1 organizationally defined LLDP extension MIB.
9
10        This group is mandatory for agents that implement the
11        LLDP 802.1 organizational extension in the RX mode
12        for the basicSet TLV set.
13
14        This version defines compliance requirements for
15        V2 of the LLDP MIB."
16     MODULE -- this module
17         MANDATORY-GROUPS { lldpV2Xdot1LocSysGroup }
18
19 ::= { lldpV2Xdot1Compliances 2 }
20
21 lldpV2Xdot1RxCompliance MODULE-COMPLIANCE
22     STATUS deprecated
23     DESCRIPTION
24         "A compliance statement for SNMP entities that implement
25         the IEEE 802.1 organizationally defined LLDP extension MIB.
26
27        This group is mandatory for agents that implement the
28        LLDP 802.1 organizational extension in the RX mode
29        for the basicSet TLV set.
30
31        This version defines compliance requirements for
32        V2 of the LLDP MIB."
33     MODULE -- this module
34         MANDATORY-GROUPS { lldpV2Xdot1RemSysGroup }
35
36 ::= { lldpV2Xdot1Compliances 3 }
37
38 lldpV2Xdot1RxComplianceV2 MODULE-COMPLIANCE
39     STATUS current
40     DESCRIPTION
41         "A compliance statement for SNMP entities that implement
42         the IEEE 802.1 organizationally defined LLDP extension MIB.
43
44        This group is mandatory for agents that implement the
45        LLDP 802.1 organizational extension in the RX mode
46        for the basicSet TLV set.
47
48        This version defines compliance requirements for
49        V2 of the LLDP MIB."
50     MODULE -- this module
51         MANDATORY-GROUPS { lldpV2Xdot1RemSysV2Group }
52
53 ::= { lldpV2Xdot1Compliances 4 }
54
55
56 -- MIB groupings for the basicSet TLV set
57
58 lldpV2Xdot1ConfigGroup OBJECT-GROUP
59     OBJECTS {
60         lldpV2Xdot1ConfigPortVlanTxEnable,
61         lldpV2Xdot1ConfigVlanNameTxEnable,
62         lldpV2Xdot1ConfigProtoVlanTxEnable,
63         lldpV2Xdot1ConfigProtocolTxEnable,
64         lldpV2Xdot1ConfigVidUsageDigestTxEnable,
65         lldpV2Xdot1ConfigManVidTxEnable
66     }
67     STATUS current
68     DESCRIPTION
69         "The collection of objects that are used to configure the
70         IEEE 802.1 organizationally defined LLDP extension
71         implementation behavior for the basicSet TLV set."
72 ::= { lldpV2Xdot1Groups 1 }
```

```
1
2 lldpV2Xdot1LocSysGroup OBJECT-GROUP
3   OBJECTS {
4     lldpV2Xdot1LocPortVlanId,
5     lldpV2Xdot1LocProtoVlanSupported,
6     lldpV2Xdot1LocProtoVlanEnabled,
7     lldpV2Xdot1LocVlanName,
8     lldpV2Xdot1LocProtocolId,
9     lldpV2Xdot1LocVidUsageDigest,
10    lldpV2Xdot1LocManVid,
11    lldpV2Xdot1LocLinkAggStatus,
12    lldpV2Xdot1LocLinkAggPortId
13  }
14  STATUS current
15  DESCRIPTION
16    "The collection of objects that are used to represent
17    IEEE 802.1 organizationally defined LLDP extension
18    associated with the Local Device Information for the
19    basicSet TLV set."
20  ::= { lldpV2Xdot1Groups 2 }
21
22 lldpV2Xdot1RemSysGroup OBJECT-GROUP
23   OBJECTS {
24     lldpV2Xdot1RemPortVlanId,
25     lldpV2Xdot1RemProtoVlanSupported,
26     lldpV2Xdot1RemProtoVlanEnabled,
27     lldpV2Xdot1RemVlanName,
28     lldpV2Xdot1RemProtocolId,
29     lldpV2Xdot1RemVidUsageDigest,
30     lldpV2Xdot1RemManVid,
31     lldpV2Xdot1RemLinkAggStatus,
32     lldpV2Xdot1RemLinkAggPortId
33  }
34  STATUS deprecated
35  DESCRIPTION
36    "The collection of objects that are used to represent LLDP
37    802.1 organizational extension Remote Device Information
38    for the basicSet TLV set."
39  ::= { lldpV2Xdot1Groups 3 }
40
41 lldpV2Xdot1RemSysV2Group OBJECT-GROUP
42   OBJECTS {
43     lldpV2Xdot1RemPortVlanId,
44     lldpV2Xdot1RemProtoVlanSupported,
45     lldpV2Xdot1RemProtoVlanEnabled,
46     lldpV2Xdot1RemVlanName,
47     lldpV2Xdot1RemProtocolId,
48     lldpV2Xdot1RemVidUsageDigestV2,
49     lldpV2Xdot1RemManVidV2,
50     lldpV2Xdot1RemLinkAggStatus,
51     lldpV2Xdot1RemLinkAggPortId
52  }
53  STATUS current
54  DESCRIPTION
55    "The collection of objects that are used to represent LLDP
56    802.1 organizational extension Remote Device Information
57    for the basicSet TLV set."
58  ::= { lldpV2Xdot1Groups 4 }
59
60 -----
61 -----
62 --
63 -- Organizationally Defined Information Extension - IEEE 802.1
64 -- Definitions to support the cnSet TLV set (Table D-1)
65 -- for Congestion Notification
66 --
67 -----
68 -----
69
70 lldpXdot1CnMIB OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 3 }
71 lldpXdot1CnObjects OBJECT IDENTIFIER ::= { lldpXdot1CnMIB 1 }
72
```

```

1 -- CN 802.1 MIB Extension groups
2
3 lldpXdot1CnConfig OBJECT IDENTIFIER ::= { lldpXdot1CnObjects 1 }
4 lldpXdot1CnLocalData OBJECT IDENTIFIER ::= { lldpXdot1CnObjects 2 }
5 lldpXdot1CnRemoteData OBJECT IDENTIFIER ::= { lldpXdot1CnObjects 3 }
6
7 -----
8 -- Textual conventions for Congestion Notification
9 -----
10
11 lldpV2CnBitVector ::= TEXTUAL-CONVENTION
12     STATUS          current
13     DESCRIPTION
14         "This TC describes a bit vector used in the Congestion
15         Notification objects. Each bit represents a Boolean status
16         associated with a priority code point. A bit value of 0
17         represents FALSE, 1 represents TRUE.
18
19         The bit 'pri0status(0)' indicates the status for priority 0
20         The bit 'pri1status(1)' indicates the status for priority 1
21         The bit 'pri2status(2)' indicates the status for priority 2
22         The bit 'pri3status(3)' indicates the status for priority 3
23         The bit 'pri4status(4)' indicates the status for priority 4
24         The bit 'pri5status(5)' indicates the status for priority 5
25         The bit 'pri6status(6)' indicates the status for priority 6
26         The bit 'pri7status(7)' indicates the status for priority 7"
27
28     SYNTAX  BITS {
29         pri0status(0),
30         pri1status(1),
31         pri2status(2),
32         pri3status(3),
33         pri4status(4),
34         pri5status(5),
35         pri6status(6),
36         pri7status(7)
37     }
38
39 -----
40 -- IEEE 802.1 - Congestion Notification Configuration
41 -----
42
43 --
44 -- lldpXdot1CnConfigCnTable : configure the
45 -- transmission of the Congestion Notification TLV on a set of ports
46 --
47
48 lldpXdot1CnConfigCnTable OBJECT-TYPE
49     SYNTAX      SEQUENCE OF LldpXdot1CnConfigCnEntry
50     MAX-ACCESS  not-accessible
51     STATUS      current
52     DESCRIPTION
53         "A table that controls selection of Congestion Notification
54         TLVs to be transmitted on individual ports."
55     ::= { lldpXdot1CnConfig 1 }
56
57 lldpXdot1CnConfigCnEntry OBJECT-TYPE
58     SYNTAX      LldpXdot1CnConfigCnEntry
59     MAX-ACCESS  not-accessible
60     STATUS      current
61     DESCRIPTION
62         "LLDP configuration information that controls the
63         transmission of IEEE 802.1 organizationally defined
64         Congestion Notification TLV on LLDP transmission-capable ports.
65
66         This configuration object augments the lldpV2PortConfigEntry of
67         the LLDP-MIB, therefore it is only present along with the port
68         configuration defined by the associated lldpV2PortConfigEntry
69         entry.
70
71         Each active lldpConfigEntry is restored from non-volatile
72         storage (along with the corresponding lldpV2PortConfigEntry)

```

```

1      after a re-initialization of the management system."
2      AUGMENTS      { lldpV2PortConfigEntry }
3      ::= { lldpXdot1CnConfigCnTable 1 }
4
5      lldpXdot1CnConfigCnEntry ::= SEQUENCE {
6          lldpXdot1CnConfigCnTxEnable TruthValue
7      }
8
9      lldpXdot1CnConfigCnTxEnable OBJECT-TYPE
10     SYNTAX          TruthValue
11     MAX-ACCESS      read-write
12     STATUS          current
13     DESCRIPTION
14         "The lldpXdot1CnConfigCnTxEnable, which is
15         defined as a truth value and configured by the network
16         management, determines whether the IEEE 802.1 organizationally
17         defined Congestion Notification TLV transmission is allowed
18         on a given LLDP transmission-capable port.
19
20         The value of this object is restored from non-volatile
21         storage after a re-initialization of the management system."
22     REFERENCE
23         "D.2.7"
24     DEFVAL          { false }
25     ::= { lldpXdot1CnConfigCnEntry 1 }
26
27 -----
28 -- IEEE 802.1 - Congestion Notification Local System Information
29 -----
30
31 ---
32 ---
33 --- lldpV2Xdot1LocCnTable: Port Extension Information Table
34 ---
35 ---
36 lldpV2Xdot1LocCnTable OBJECT-TYPE
37     SYNTAX          SEQUENCE OF LldpV2Xdot1LocCnEntry
38     MAX-ACCESS      not-accessible
39     STATUS          current
40     DESCRIPTION
41         "This table contains one row per port of Congestion
42         Notification information (as a part of the LLDP
43         802.1 organizational extension) on the local system
44         known to this agent."
45     ::= { lldpXdot1CnLocalData 1 }
46
47 lldpV2Xdot1LocCnEntry OBJECT-TYPE
48     SYNTAX          LldpV2Xdot1LocCnEntry
49     MAX-ACCESS      not-accessible
50     STATUS          current
51     DESCRIPTION
52         "Congestion Notification information about a
53         particular port component."
54     INDEX          { lldpV2LocPortIfIndex }
55     ::= { lldpV2Xdot1LocCnTable 1 }
56
57 lldpV2Xdot1LocCnEntry ::= SEQUENCE {
58     lldpV2Xdot1LocCNPVIndicators LldpV2CnBitVector,
59     lldpV2Xdot1LocReadyIndicators LldpV2CnBitVector
60 }
61
62 lldpV2Xdot1LocCNPVIndicators OBJECT-TYPE
63     SYNTAX          LldpV2CnBitVector
64     MAX-ACCESS      read-only
65     STATUS          current
66     DESCRIPTION
67         "This object contains the CNPV indicators
68         for the Port."
69     REFERENCE
70         "D.2.7.3"
71     ::= { lldpV2Xdot1LocCnEntry 1 }
72

```

```

1 lldpV2Xdot1LocReadyIndicators OBJECT-TYPE
2     SYNTAX      LldpV2CnBitVector
3     MAX-ACCESS  read-only
4     STATUS      current
5     DESCRIPTION
6         "This object contains the Ready indicators
7         for the Port."
8     REFERENCE
9         "D.2.7.4"
10    ::= { lldpV2Xdot1LocCnEntry 2 }
11
12 -----
13 -- IEEE 802.1 - Congestion Notification Remote System Information
14 -----
15
16 ---
17 ---
18 --- lldpV2Xdot1RemCnTable: Port Extension Information Table
19 ---
20 ---
21 lldpV2Xdot1RemCnTable OBJECT-TYPE
22     SYNTAX      SEQUENCE OF LldpV2Xdot1RemCnEntry
23     MAX-ACCESS  not-accessible
24     STATUS      current
25     DESCRIPTION
26         "This table contains Congestion Notification information
27         (as a part of the LLDP IEEE 802.1 organizational extension)
28         of the remote system."
29     ::= { lldpXdot1CnRemoteData 1 }
30
31 lldpV2Xdot1RemCnEntry OBJECT-TYPE
32     SYNTAX      LldpV2Xdot1RemCnEntry
33     MAX-ACCESS  not-accessible
34     STATUS      current
35     DESCRIPTION
36         "Port Extension information about remote systems port
37         component."
38     INDEX      { lldpV2RemTimeMark,
39                 lldpV2RemLocalIfIndex,
40                 lldpV2RemLocalDestMACAddress,
41                 lldpV2RemIndex }
42     ::= { lldpV2Xdot1RemCnTable 1 }
43
44 lldpV2Xdot1RemCnEntry ::= SEQUENCE {
45     lldpV2Xdot1RemCNPVIndicators  LldpV2CnBitVector,
46     lldpV2Xdot1RemReadyIndicators  LldpV2CnBitVector
47 }
48
49 lldpV2Xdot1RemCNPVIndicators OBJECT-TYPE
50     SYNTAX      LldpV2CnBitVector
51     MAX-ACCESS  read-only
52     STATUS      current
53     DESCRIPTION
54         "This object contains the CNPV indicators
55         for the Port."
56     REFERENCE
57         "D.2.7.3"
58     ::= { lldpV2Xdot1RemCnEntry 1 }
59
60 lldpV2Xdot1RemReadyIndicators OBJECT-TYPE
61     SYNTAX      LldpV2CnBitVector
62     MAX-ACCESS  read-only
63     STATUS      current
64     DESCRIPTION
65         "This object contains the Ready indicators
66         for the Port."
67     REFERENCE
68         "D.2.7.4"
69     ::= { lldpV2Xdot1RemCnEntry 2 }
70
71 -----
72 -- IEEE 802.1 - Congestion Notification Conformance Information

```

```
1 -----
2
3 lldpXdot1CnConformance OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 4 }
4
5 lldpXdot1CnCompliances
6   OBJECT IDENTIFIER ::= { lldpXdot1CnConformance 1 }
7 lldpXdot1CnGroups OBJECT IDENTIFIER ::= { lldpXdot1CnConformance 2 }
8
9 --
10 -- Congestion Notification - Compliance Statements
11 --
12
13 lldpXdot1CnCompliance MODULE-COMPLIANCE
14   STATUS          current
15   DESCRIPTION
16     "A compliance statement for SNMP entities that implement
17     the IEEE 802.1 organizationally defined Congestion
18     Notification group in the LLDP extension MIB.
19
20     This group is mandatory for agents that implement the
21     Congestion Notification cnSet TLV set."
22   MODULE          -- this module
23     MANDATORY-GROUPS { lldpXdot1CnGroup,
24                       ifGeneralInformationGroup }
25   ::= { lldpXdot1CnCompliances 1 }
26
27
28 --
29 -- Congestion Notification - MIB groupings
30 --
31
32 lldpXdot1CnGroup OBJECT-GROUP
33   OBJECTS {
34     lldpXdot1CnConfigCnTxEnable,
35     lldpV2Xdot1LocCNPVIndicators,
36     lldpV2Xdot1LocReadyIndicators,
37     lldpV2Xdot1RemCNPVIndicators,
38     lldpV2Xdot1RemReadyIndicators
39   }
40   STATUS current
41   DESCRIPTION
42     "The collection of objects that support the
43     Congestion Notification cnSet TLV set."
44   ::= { lldpXdot1CnGroups 1 }
45
46 -----
47 -----
48 --
49 -- Organizationally Defined Information Extension - IEEE 802.1
50 -- Definitions to support the Data Center eXchange Protocol
51 -- (DCBX) TLV set (Table D-1)
52 --
53 -----
54 -----
55 lldpXdot1dcbxMIB OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 5 }
56 lldpXdot1dcbxObjects OBJECT IDENTIFIER ::= { lldpXdot1dcbxMIB 1 }
57
58 -- DCBX 802.1 MIB Extension groups
59
60 lldpXdot1dcbxConfig OBJECT IDENTIFIER ::= { lldpXdot1dcbxObjects 1 }
61 lldpXdot1dcbxLocalData OBJECT IDENTIFIER ::= { lldpXdot1dcbxObjects 2 }
62 lldpXdot1dcbxRemoteData OBJECT IDENTIFIER ::= { lldpXdot1dcbxObjects 3 }
63 lldpXdot1dcbxAdminData OBJECT IDENTIFIER ::= { lldpXdot1dcbxObjects 4 }
64
65 -----
66 -- IEEE 802.1 - DCBX Textual Conventions
67 -----
68
69 LldpXdot1dcbxTrafficClassValue ::= TEXTUAL-CONVENTION
70   DISPLAY-HINT "d"
71   STATUS current
72   DESCRIPTION
```

```
1      "Indicates a traffic class. Values 0-7 correspond to
2      traffic classes."
3      SYNTAX      Unsigned32 (0..7)
4
5      LldpXdot1dcbxTrafficClassBandwidthValue ::= TEXTUAL-CONVENTION
6      DISPLAY-HINT "d"
7      STATUS      current
8      DESCRIPTION
9      "Indicates the bandwidth in percent assigned to a
10     traffic class."
11     SYNTAX      Unsigned32 (0..100)
12
13     LldpXdot1dcbxAppSelector ::= TEXTUAL-CONVENTION
14     STATUS      current
15     DESCRIPTION
16     "Indicates the contents of a protocol object
17     1: EtherType
18     2: Well Known Port number over TCP, or SCTP
19     3: Well Known Port number over UDP, or DCCP
20     4: Well Known Port number over TCP, SCTP, UDP, and DCCP
21     5: Differentiated Services Code Point (DSCP) value. The
22     6 bit DSCP value is stored in the low order 6 bits of the
23     protocol object. The higher order bits are set to zero.
24     (See IETF RFC 2474 for the definition of the DSCP value.)"
25     SYNTAX INTEGER {
26         asEtherType(1),
27         asTCPPortNumber(2),
28         asUDPPortNumber(3),
29         asTCPUDPPortNumber(4),
30         asDSCPValue(5)
31     }
32
33     LldpXdot1dcbxAppProtocol ::= TEXTUAL-CONVENTION
34     DISPLAY-HINT "d"
35     STATUS      current
36     DESCRIPTION
37     "Contains the application protocol indicator the
38     type of which is specified by an object with
39     the syntax of
40     LldpXdot1dcbxAppSelector"
41     SYNTAX Unsigned32 (0..65535)
42
43     LldpXdot1dcbxSupportedCapacity ::= TEXTUAL-CONVENTION
44     DISPLAY-HINT "d"
45     STATUS      current
46     DESCRIPTION
47     "Indicates the supported capacity of a given feature,
48     for example, the number of traffic classes supported.
49     This TC is used for features that have a maximum
50     capacity of eight and a minimum of one."
51     SYNTAX Unsigned32 (1..8)
52
53     LldpXdot1dcbxTrafficSelectionAlgorithm ::= TEXTUAL-CONVENTION
54     STATUS      current
55     DESCRIPTION
56     "Indicates the Traffic Selection Algorithm
57     0: Strict Priority
58     1: Credit-based shaper
59     2: Enhanced transmission selection
60     3-254: Reserved for future standardization
61     255: Vendor specific"
62     SYNTAX INTEGER {
63         tsaStrictPriority(0),
64         tsaCreditBasedShaper(1),
65         tsaEnhancedTransmission(2),
66         tsaVendorSpecific(255)
67     }
68
69 -----
70 -- IEEE 802.1 - DCBX Configuration
71 -----
72
```

```

1 --
2 -- lldpXdotldcbxConfigETSConfigurationTable : configure the
3 -- transmission of the ETS Configuration TLV on a set of ports
4 --
5
6 lldpXdotldcbxConfigETSConfigurationTable OBJECT-TYPE
7     SYNTAX      SEQUENCE OF LldpXdotldcbxConfigETSConfigurationEntry
8     MAX-ACCESS   not-accessible
9     STATUS       current
10    DESCRIPTION
11        "A table that controls selection of ETS Configuration
12         TLVs to be transmitted on individual ports."
13    ::= { lldpXdotldcbxConfig 1 }
14
15 lldpXdotldcbxConfigETSConfigurationEntry OBJECT-TYPE
16     SYNTAX      LldpXdotldcbxConfigETSConfigurationEntry
17     MAX-ACCESS   not-accessible
18     STATUS       current
19     DESCRIPTION
20        "LLDP configuration information that controls the
21         transmission of IEEE 802.1 organizationally defined
22         ETS Configuration TLV on LLDP transmission-capable ports.
23
24         This configuration object augments the lldpV2PortConfigEntry of
25         the LLDP-MIB, therefore it is only present along with the port
26         configuration defined by the associated lldpV2PortConfigEntry
27         entry.
28
29         Each active lldpConfigEntry is restored from non-volatile
30         storage (along with the corresponding lldpV2PortConfigEntry)
31         after a re-initialization of the management system."
32     AUGMENTS     { lldpV2PortConfigEntry }
33    ::= { lldpXdotldcbxConfigETSConfigurationTable 1 }
34
35 lldpXdotldcbxConfigETSConfigurationEntry ::= SEQUENCE {
36     lldpXdotldcbxConfigETSConfigurationTxEnable TruthValue
37 }
38
39 lldpXdotldcbxConfigETSConfigurationTxEnable OBJECT-TYPE
40     SYNTAX      TruthValue
41     MAX-ACCESS   read-write
42     STATUS       current
43     DESCRIPTION
44        "The lldpXdotldcbxConfigETSConfigurationTxEnable, which is
45         defined as a truth value and configured by the network
46         management, determines whether the IEEE 802.1 organizationally
47         defined ETS Configuration TLV transmission is allowed on a
48         given LLDP transmission-capable port.
49
50         The value of this object is restored from non-volatile
51         storage after a re-initialization of the management system."
52     REFERENCE
53        "D.2.8"
54     DEFVAL      { false }
55    ::= { lldpXdotldcbxConfigETSConfigurationEntry 1 }
56
57 --
58 -- lldpXdotldcbxConfigETSRecommendationTable : configure the
59 -- transmission of the ETS Recommendation TLV on a set of ports
60 --
61
62 lldpXdotldcbxConfigETSRecommendationTable OBJECT-TYPE
63     SYNTAX      SEQUENCE OF LldpXdotldcbxConfigETSRecommendationEntry
64     MAX-ACCESS   not-accessible
65     STATUS       current
66     DESCRIPTION
67        "A table that controls selection of ETS Recommendation
68         TLVs to be transmitted on individual ports."
69    ::= { lldpXdotldcbxConfig 2 }
70
71 lldpXdotldcbxConfigETSRecommendationEntry OBJECT-TYPE
72     SYNTAX      LldpXdotldcbxConfigETSRecommendationEntry

```



```

1  MAX-ACCESS    not-accessible
2  STATUS       current
3  DESCRIPTION
4      "LLDP configuration information that controls the
5      transmission of IEEE 802.1 organizationally defined
6      ETS Recommendation TLV on LLDP transmission-capable ports.
7
8      This configuration object augments the lldpV2PortConfigEntry of
9      the LLDP-MIB, therefore it is only present along with the port
10     configuration defined by the associated lldpV2PortConfigEntry
11     entry.
12
13     Each active lldpConfigEntry is restored from non-volatile
14     storage (along with the corresponding lldpV2PortConfigEntry)
15     after a re-initialization of the management system."
16  AUGMENTS     { lldpV2PortConfigEntry }
17  ::= { lldpXdot1dcbxConfigETSRecommendationTable 1 }
18
19 lldpXdot1dcbxConfigETSRecommendationEntry ::= SEQUENCE {
20     lldpXdot1dcbxConfigETSRecommendationTxEnable TruthValue
21 }
22
23 lldpXdot1dcbxConfigETSRecommendationTxEnable OBJECT-TYPE
24     SYNTAX      TruthValue
25     MAX-ACCESS  read-write
26     STATUS      current
27     DESCRIPTION
28         "The lldpXdot1dcbxConfigETSRecommendationTxEnable, which is
29         defined as a truth value and configured by the network
30         management, determines whether the IEEE 802.1 organizationally
31         defined ETS Recommendation TLV transmission is allowed on a
32         given LLDP transmission-capable port.
33
34         The value of this object is restored from non-volatile
35         storage after a re-initialization of the management system."
36     REFERENCE
37         "D.2.9"
38     DEFVAL      { false }
39     ::= { lldpXdot1dcbxConfigETSRecommendationEntry 1 }
40 --
41 -- lldpXdot1dcbxConfigPFCTable : configure the transmission of the
42 -- Priority-based Flow Control Configuration TLV on a set of ports
43 --
44
45 lldpXdot1dcbxConfigPFCTable OBJECT-TYPE
46     SYNTAX      SEQUENCE OF LldpXdot1dcbxConfigPFCEntEntry
47     MAX-ACCESS  not-accessible
48     STATUS      current
49     DESCRIPTION
50         "A table that controls selection of Priority-based
51         Flow Control Configuration TLVs to be transmitted on individual ports."
52     ::= { lldpXdot1dcbxConfig 3 }
53
54 lldpXdot1dcbxConfigPFCEntEntry OBJECT-TYPE
55     SYNTAX      LldpXdot1dcbxConfigPFCEntEntry
56     MAX-ACCESS  not-accessible
57     STATUS      current
58     DESCRIPTION
59         "LLDP configuration information that controls the
60         transmission of IEEE 802.1 organizationally defined
61         Priority-based Flow Control Configuration TLV on LLDP
62         transmission-capable ports.
63
64         This configuration object augments the lldpV2PortConfigEntry of
65         the LLDP-MIB, therefore it is only present along with the port
66         configuration defined by the associated lldpV2PortConfigEntry
67         entry.
68
69         Each active lldpConfigEntry is restored from non-volatile
70         storage (along with the corresponding lldpV2PortConfigEntry)
71         after a re-initialization of the management system."
72     AUGMENTS    { lldpV2PortConfigEntry }

```

```

1 ::= { lldpXdot1dcbxConfigPFCTable 1 }
2
3 lldpXdot1dcbxConfigPFCEntree ::= SEQUENCE {
4     lldpXdot1dcbxConfigPFCTxEnable TruthValue
5 }
6
7 lldpXdot1dcbxConfigPFCTxEnable OBJECT-TYPE
8     SYNTAX      TruthValue
9     MAX-ACCESS   read-write
10    STATUS      current
11    DESCRIPTION
12        "The lldpXdot1dcbxConfigPFCTxEnable, which is defined
13        as a truth value and configured by the network management,
14        determines whether the IEEE 802.1 organizationally defined
15        Priority-based Flow Control Configuration TLV transmission is allowed on
16        a given LLDP transmission-capable port.
17
18        The value of this object is restored from non-volatile
19        storage after a re-initialization of the management system."
20    REFERENCE
21        "D.2.10"
22    DEFVAL      { false }
23    ::= { lldpXdot1dcbxConfigPFCEntree 1 }
24
25 --
26 -- lldpXdot1dcbxConfigApplicationPriorityTable : configure the
27 -- transmission of the Application Priority TLV on a set of ports
28 --
29
30 lldpXdot1dcbxConfigApplicationPriorityTable OBJECT-TYPE
31     SYNTAX      SEQUENCE OF
32         lldpXdot1dcbxConfigApplicationPriorityEntry
33     MAX-ACCESS   not-accessible
34     STATUS      current
35     DESCRIPTION
36         "A table that controls selection of Priority-based
37         Flow Control Configuration TLVs to be transmitted on individual ports."
38     ::= { lldpXdot1dcbxConfig 4 }
39
40 lldpXdot1dcbxConfigApplicationPriorityEntry OBJECT-TYPE
41     SYNTAX      lldpXdot1dcbxConfigApplicationPriorityEntry
42     MAX-ACCESS   not-accessible
43     STATUS      current
44     DESCRIPTION
45         "LLDP configuration information that controls the
46         transmission of IEEE 802.1 organizationally defined
47         Application Priority TLV on LLDP transmission-capable ports.
48
49         This configuration object augments the lldpV2PortConfigEntry of
50         the LLDP-MIB, therefore it is only present along with the port
51         configuration defined by the associated lldpV2PortConfigEntry
52         entry.
53
54         Each active lldpConfigEntry is restored from non-volatile
55         storage (along with the corresponding lldpV2PortConfigEntry)
56         after a re-initialization of the management system."
57     AUGMENTS    { lldpV2PortConfigEntry }
58     ::= { lldpXdot1dcbxConfigApplicationPriorityTable 1 }
59
60 lldpXdot1dcbxConfigApplicationPriorityEntry ::= SEQUENCE {
61     lldpXdot1dcbxConfigApplicationPriorityTxEnable TruthValue
62 }
63
64 lldpXdot1dcbxConfigApplicationPriorityTxEnable OBJECT-TYPE
65     SYNTAX      TruthValue
66     MAX-ACCESS   read-write
67     STATUS      current
68     DESCRIPTION
69         "The lldpXdot1dcbxConfigApplicationPriorityTxEnable, which
70         is defined as a truth value and configured by the network
71         management, determines whether the IEEE 802.1 organizationally
72         defined Application Priority TLV transmission is allowed on

```

```

1      a given LLDP transmission-capable port.
2
3      The value of this object is restored from non-volatile
4      storage after a re-initialization of the management system."
5  REFERENCE
6      "D.2.11"
7  DEFVAL      { false }
8  ::= { lldpXdot1dcbxConfigApplicationPriorityEntry 1 }
9
10 --
11 -- lldpXdot1dcbxConfigApplicationVlanTable : configure the
12 -- transmission of the Application VLAN TLV on a set of ports
13 --
14
15 lldpXdot1dcbxConfigApplicationVlanTable OBJECT-TYPE
16     SYNTAX      SEQUENCE OF
17         LldpXdot1dcbxConfigApplicationVlanEntry
18     MAX-ACCESS   not-accessible
19     STATUS       current
20     DESCRIPTION
21         "A table that controls selection of Application VLAN
22         TLVs to be transmitted on individual ports."
23     ::= { lldpXdot1dcbxConfig 5 }
24
25 lldpXdot1dcbxConfigApplicationVlanEntry OBJECT-TYPE
26     SYNTAX      LldpXdot1dcbxConfigApplicationVlanEntry
27     MAX-ACCESS   not-accessible
28     STATUS       current
29     DESCRIPTION
30         "LLDP configuration information that controls the
31         transmission of IEEE 802.1 organizationally defined
32         Application VLAN TLV on LLDP transmission-capable ports.
33
34         This configuration object augments the lldpV2PortConfigEntry of
35         the LLDP-MIB, therefore it is only present along with the port
36         configuration defined by the associated lldpV2PortConfigEntry
37         entry.
38
39         Each active lldpConfigEntry is restored from non-volatile
40         storage (along with the corresponding lldpV2PortConfigEntry)
41         after a re-initialization of the management system."
42     AUGMENTS     { lldpV2PortConfigEntry }
43     ::= { lldpXdot1dcbxConfigApplicationVlanTable 1 }
44
45 LldpXdot1dcbxConfigApplicationVlanEntry ::= SEQUENCE {
46     lldpXdot1dcbxConfigApplicationVlanTxEnable TruthValue
47 }
48
49 lldpXdot1dcbxConfigApplicationVlanTxEnable OBJECT-TYPE
50     SYNTAX      TruthValue
51     MAX-ACCESS   read-write
52     STATUS       current
53     DESCRIPTION
54         "The lldpXdot1dcbxConfigApplicationVlanTxEnable, which
55         is defined as a truth value and configured by the network
56         management, determines whether the IEEE 802.1 organizationally
57         defined Application VLAN TLV transmission is allowed on
58         a given LLDP transmission-capable port.
59
60         The value of this object is restored from non-volatile
61         storage after a re-initialization of the management system."
62     REFERENCE
63         "D.2.14"
64     DEFVAL      { false }
65     ::= { lldpXdot1dcbxConfigApplicationVlanEntry 1 }
66
67 -----
68 -- IEEE 802.1 - DCBX Local System Information
69 -----
70
71 --
72 -- lldpXdot1dcbxLocETSTConfigurationTable - Contains the information

```

```

1 -- for the ETS Configuration TLV.
2 --
3 lldpXdotldcbxLocETSConfiguration OBJECT IDENTIFIER
4 ::= { lldpXdotldcbxLocalData 1 }
5
6 lldpXdotldcbxLocETSBasicConfigurationTable OBJECT-TYPE
7 SYNTAX      SEQUENCE OF LldpXdotldcbxLocETSBasicConfigurationEntry
8 MAX-ACCESS  not-accessible
9 STATUS      current
10 DESCRIPTION
11     "This table contains one row per port for the IEEE 802.1
12     organizationally defined LLDP ETS Configuration TLV on
13     the local system known to this agent."
14 ::= { lldpXdotldcbxLocETSConfiguration 1 }
15
16 lldpXdotldcbxLocETSBasicConfigurationEntry OBJECT-TYPE
17 SYNTAX      LldpXdotldcbxLocETSBasicConfigurationEntry
18 MAX-ACCESS  not-accessible
19 STATUS      current
20 DESCRIPTION
21     "Information about the IEEE 802.1 organizational defined
22     ETS Configuration TLV LLDP extension."
23 INDEX      { lldpV2LocPortIfIndex }
24 ::= { lldpXdotldcbxLocETSBasicConfigurationTable 1 }
25
26 LldpXdotldcbxLocETSBasicConfigurationEntry ::= SEQUENCE {
27     lldpXdotldcbxLocETSConCreditBasedShaperSupport TruthValue,
28     lldpXdotldcbxLocETSConTrafficClassesSupported
29         LldpXdotldcbxSupportedCapacity,
30     lldpXdotldcbxLocETSConWilling      TruthValue
31 }
32
33 lldpXdotldcbxLocETSConCreditBasedShaperSupport OBJECT-TYPE
34 SYNTAX      TruthValue
35 MAX-ACCESS  read-only
36 STATUS      current
37 DESCRIPTION
38     "Indicates if the credit-based shaper Traffic Selection
39     Algorithm is supported on the local system."
40 REFERENCE
41     "D.2.8.4"
42 ::= { lldpXdotldcbxLocETSBasicConfigurationEntry 1 }
43
44 lldpXdotldcbxLocETSConTrafficClassesSupported OBJECT-TYPE
45 SYNTAX      LldpXdotldcbxSupportedCapacity
46 MAX-ACCESS  read-only
47 STATUS      current
48 DESCRIPTION
49     "Indicates the number of traffic classes supported."
50 REFERENCE
51     "D.2.8.5"
52 ::= { lldpXdotldcbxLocETSBasicConfigurationEntry 2 }
53
54 lldpXdotldcbxLocETSConWilling OBJECT-TYPE
55 SYNTAX      TruthValue
56 MAX-ACCESS  read-only
57 STATUS      current
58 DESCRIPTION
59     "Indicates if the local system is willing to accept the
60     ETS configuration recommended by the remote system."
61 REFERENCE
62     "D.2.8.3"
63 ::= { lldpXdotldcbxLocETSBasicConfigurationEntry 3 }
64
65 lldpXdotldcbxLocETSConPriorityAssignmentTable OBJECT-TYPE
66 SYNTAX      SEQUENCE OF
67             LldpXdotldcbxLocETSConPriorityAssignmentEntry
68 MAX-ACCESS  not-accessible
69 STATUS      current
70 DESCRIPTION
71     "This table contains one row per priority. The entry in each
72     row indicates the traffic class to which the priority is

```

```

1      assigned."
2      ::= { lldpXdot1dcbxLocETSTConfiguration 2 }
3
4 lldpXdot1dcbxLocETSTConPriorityAssignmentEntry OBJECT-TYPE
5     SYNTAX      LldpXdot1dcbxLocETSTConPriorityAssignmentEntry
6     MAX-ACCESS   not-accessible
7     STATUS       current
8     DESCRIPTION
9         "Indicates a priority to traffic class assignment."
10    INDEX        {
11                lldpV2LocPortIfIndex,
12                lldpXdot1dcbxLocETSTConPriority
13            }
14    ::= { lldpXdot1dcbxLocETSTConPriorityAssignmentTable 1 }
15
16 LldpXdot1dcbxLocETSTConPriorityAssignmentEntry ::= SEQUENCE {
17     lldpXdot1dcbxLocETSTConPriority      IEEE8021PriorityValue,
18     lldpXdot1dcbxLocETSTConPriTrafficClass
19         LldpXdot1dcbxTrafficClassValue
20 }
21
22 lldpXdot1dcbxLocETSTConPriority OBJECT-TYPE
23     SYNTAX      IEEE8021PriorityValue
24     MAX-ACCESS   not-accessible
25     STATUS       current
26     DESCRIPTION
27         "Indicates the priority that is assigned to a traffic
28         class."
29     REFERENCE
30         "D.2.8.6"
31     ::= { lldpXdot1dcbxLocETSTConPriorityAssignmentEntry 1 }
32
33 lldpXdot1dcbxLocETSTConPriTrafficClass OBJECT-TYPE
34     SYNTAX      LldpXdot1dcbxTrafficClassValue
35     MAX-ACCESS   read-only
36     STATUS       current
37     DESCRIPTION
38         "Indicates the traffic class to which this priority is
39         to be assigned."
40     REFERENCE
41         "D.2.8.6"
42     ::= { lldpXdot1dcbxLocETSTConPriorityAssignmentEntry 2 }
43
44 lldpXdot1dcbxLocETSTConTrafficClassBandwidthTable OBJECT-TYPE
45     SYNTAX      SEQUENCE OF
46         LldpXdot1dcbxLocETSTConTrafficClassBandwidthEntry
47     MAX-ACCESS   not-accessible
48     STATUS       current
49     DESCRIPTION
50         "This table contains one row per traffic class. The
51         entry in each row indicates the traffic class to
52         which the bandwidth is assigned."
53     ::= { lldpXdot1dcbxLocETSTConfiguration 3 }
54
55 lldpXdot1dcbxLocETSTConTrafficClassBandwidthEntry OBJECT-TYPE
56     SYNTAX      LldpXdot1dcbxLocETSTConTrafficClassBandwidthEntry
57     MAX-ACCESS   not-accessible
58     STATUS       current
59     DESCRIPTION
60         "Indicates a traffic class to Bandwidth assignment."
61     INDEX        {
62                lldpV2LocPortIfIndex,
63                lldpXdot1dcbxLocETSTConTrafficClass
64            }
65     ::= { lldpXdot1dcbxLocETSTConTrafficClassBandwidthTable 1 }
66
67 LldpXdot1dcbxLocETSTConTrafficClassBandwidthEntry ::= SEQUENCE {
68     lldpXdot1dcbxLocETSTConTrafficClass
69         LldpXdot1dcbxTrafficClassValue,
70     lldpXdot1dcbxLocETSTConTrafficClassBandwidth
71         LldpXdot1dcbxTrafficClassBandwidthValue
72 }

```

```

1
2 lldpXdot1dcbxLocETSTrafficClass OBJECT-TYPE
3   SYNTAX      LldpXdot1dcbxTrafficClassValue
4   MAX-ACCESS   not-accessible
5   STATUS      current
6   DESCRIPTION
7       "Indicates the traffic class to
8       which this bandwidth applies."
9   REFERENCE
10      "D.2.8.7"
11 ::= { lldpXdot1dcbxLocETSTrafficClassBandwidthEntry 1 }
12
13 lldpXdot1dcbxLocETSTrafficClassBandwidth OBJECT-TYPE
14   SYNTAX      LldpXdot1dcbxTrafficClassBandwidthValue
15   MAX-ACCESS   read-only
16   STATUS      current
17   DESCRIPTION
18       "Indicates the bandwidth assigned to this traffic class."
19   REFERENCE
20      "D.2.8.7"
21 ::= { lldpXdot1dcbxLocETSTrafficClassBandwidthEntry 2 }
22
23
24 lldpXdot1dcbxLocETSTrafficSelectionAlgorithmTable OBJECT-TYPE
25   SYNTAX      SEQUENCE OF
26               LldpXdot1dcbxLocETSTrafficSelectionAlgorithmEntry
27   MAX-ACCESS   not-accessible
28   STATUS      current
29   DESCRIPTION
30       "This table contains one row per traffic class. The entry
31       in each row indicates the traffic selection algorithm to be
32       used by the traffic class."
33 ::= { lldpXdot1dcbxLocETSTrafficSelectionAlgorithmTable 4 }
34
35 lldpXdot1dcbxLocETSTrafficSelectionAlgorithmEntry OBJECT-TYPE
36   SYNTAX      LldpXdot1dcbxLocETSTrafficSelectionAlgorithmEntry
37   MAX-ACCESS   not-accessible
38   STATUS      current
39   DESCRIPTION
40       "Indicates a traffic class to traffic selection algorithm
41       assignment."
42   INDEX      {
43               lldpV2LocPortIfIndex,
44               lldpXdot1dcbxLocETSTrafficClass
45             }
46 ::= { lldpXdot1dcbxLocETSTrafficSelectionAlgorithmTable 1 }
47
48 LldpXdot1dcbxLocETSTrafficSelectionAlgorithmEntry ::= SEQUENCE {
49   lldpXdot1dcbxLocETSTrafficClass
50   LldpXdot1dcbxTrafficClassValue,
51   lldpXdot1dcbxLocETSTrafficSelectionAlgorithm
52   LldpXdot1dcbxTrafficSelectionAlgorithm
53 }
54
55 lldpXdot1dcbxLocETSTrafficSelectionAlgorithm OBJECT-TYPE
56   SYNTAX      LldpXdot1dcbxTrafficClassValue
57   MAX-ACCESS   not-accessible
58   STATUS      current
59   DESCRIPTION
60       "Indicates the traffic class that is assigned to a traffic
61       selection algorithm."
62   REFERENCE
63      "D.2.8.8"
64 ::= { lldpXdot1dcbxLocETSTrafficSelectionAlgorithmEntry 1 }
65
66 lldpXdot1dcbxLocETSTrafficSelectionAlgorithm OBJECT-TYPE
67   SYNTAX      LldpXdot1dcbxTrafficSelectionAlgorithm
68   MAX-ACCESS   read-only
69   STATUS      current
70   DESCRIPTION
71       "Indicates the Traffic Selection Algorithm to which this
72       traffic class is to be assigned."

```

```

1  REFERENCE
2  "D.2.8.8"
3  ::= { lldpXdotldcbxLocETSConTrafficSelectionAlgorithmEntry 2 }
4
5  --
6  -- lldpXdotldcbxLocETSRecommendationTable - Contains the information for
7  -- the ETS Recommendation TLV.
8  --
9  lldpXdotldcbxLocETSReco OBJECT IDENTIFIER ::=
10 { lldpXdotldcbxLocalData 2 }
11
12 lldpXdotldcbxLocETSRecoTrafficClassBandwidthTable OBJECT-TYPE
13 SYNTAX      SEQUENCE OF
14      LldpXdotldcbxLocETSRecoTrafficClassBandwidthEntry
15 MAX-ACCESS  not-accessible
16 STATUS      current
17 DESCRIPTION
18      "This table contains one row per traffic class. The
19      entry in each row indicates the traffic class to
20      which the bandwidth is assigned."
21 ::= { lldpXdotldcbxLocETSReco 1 }
22
23 lldpXdotldcbxLocETSRecoTrafficClassBandwidthEntry OBJECT-TYPE
24 SYNTAX      LldpXdotldcbxLocETSRecoTrafficClassBandwidthEntry
25 MAX-ACCESS  not-accessible
26 STATUS      current
27 DESCRIPTION
28      "Indicates a traffic class to Bandwidth assignment."
29 INDEX      {
30      lldpV2LocPortIfIndex,
31      lldpXdotldcbxLocETSRecoTrafficClass
32 }
33 ::= { lldpXdotldcbxLocETSRecoTrafficClassBandwidthTable 1 }
34
35 lldpXdotldcbxLocETSRecoTrafficClassBandwidthEntry ::= SEQUENCE {
36      lldpXdotldcbxLocETSRecoTrafficClass
37      LldpXdotldcbxTrafficClassValue,
38      lldpXdotldcbxLocETSRecoTrafficClassBandwidth
39      LldpXdotldcbxTrafficClassBandwidthValue
40 }
41
42 lldpXdotldcbxLocETSRecoTrafficClass OBJECT-TYPE
43 SYNTAX      LldpXdotldcbxTrafficClassValue
44 MAX-ACCESS  not-accessible
45 STATUS      current
46 DESCRIPTION
47      "Indicates the traffic class to
48      which this bandwidth applies."
49 REFERENCE
50      "D.2.9.3"
51 ::= { lldpXdotldcbxLocETSRecoTrafficClassBandwidthEntry 1 }
52
53 lldpXdotldcbxLocETSRecoTrafficClassBandwidth OBJECT-TYPE
54 SYNTAX      LldpXdotldcbxTrafficClassBandwidthValue
55 MAX-ACCESS  read-only
56 STATUS      current
57 DESCRIPTION
58      "Indicates the bandwidth assigned to this traffic class."
59 REFERENCE
60      "D.2.9.4"
61 ::= { lldpXdotldcbxLocETSRecoTrafficClassBandwidthEntry 2 }
62
63 lldpXdotldcbxLocETSRecoTrafficSelectionAlgorithmTable OBJECT-TYPE
64 SYNTAX      SEQUENCE OF
65      LldpXdotldcbxLocETSRecoTrafficSelectionAlgorithmEntry
66 MAX-ACCESS  not-accessible
67 STATUS      current
68 DESCRIPTION
69      "This table contains one row per priority. The entry in each
70      row indicates the traffic selection algorithm to be used
71      by the traffic class."
72 ::= { lldpXdotldcbxLocETSReco 2 }

```

```

1
2 lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithmEntry OBJECT-TYPE
3   SYNTAX      LldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithmEntry
4   MAX-ACCESS   not-accessible
5   STATUS      current
6   DESCRIPTION
7       "Indicates a priority to traffic selection algorithm
8       assignment."
9   INDEX      {
10              lldpV2LocPortIfIndex,
11              lldpXdot1dcbxLocETSRecoTSATrafficClass
12          }
13   ::= { lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithmTable 1 }
14
15 lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithmEntry ::= SEQUENCE {
16     lldpXdot1dcbxLocETSRecoTSATrafficClass
17     LldpXdot1dcbxTrafficClassValue,
18     lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithm
19     LldpXdot1dcbxTrafficSelectionAlgorithm
20 }
21
22 lldpXdot1dcbxLocETSRecoTSATrafficClass OBJECT-TYPE
23   SYNTAX      LldpXdot1dcbxTrafficClassValue
24   MAX-ACCESS   not-accessible
25   STATUS      current
26   DESCRIPTION
27       "Indicates the traffic class that is assigned to a traffic
28       selection algorithm."
29   REFERENCE
30       "D.2.9.5"
31   ::= { lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithmEntry 1 }
32
33 lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithm OBJECT-TYPE
34   SYNTAX      LldpXdot1dcbxTrafficSelectionAlgorithm
35   MAX-ACCESS   read-only
36   STATUS      current
37   DESCRIPTION
38       "Indicates the Traffic Selection Algorithm to which this
39       traffic class is to be assigned."
40   REFERENCE
41       "D.2.9.5"
42   ::= { lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithmEntry 2 }
43
44 --
45 -- lldpXdot1dcbxLocPFCBTable - Contains the information for the PFC
46 -- Configuration TLV.
47 --
48 lldpXdot1dcbxLocPFC OBJECT IDENTIFIER ::= { lldpXdot1dcbxLocalData 3 }
49
50 lldpXdot1dcbxLocPFCBasicTable OBJECT-TYPE
51   SYNTAX      SEQUENCE OF LldpXdot1dcbxLocPFCBasicEntry
52   MAX-ACCESS   not-accessible
53   STATUS      current
54   DESCRIPTION
55       "This table contains one row per port for the IEEE 802.1
56       organizationally defined LLDP PFC TLV on the local
57       system known to this agent."
58   ::= { lldpXdot1dcbxLocPFC 1 }
59
60 lldpXdot1dcbxLocPFCBasicEntry OBJECT-TYPE
61   SYNTAX      LldpXdot1dcbxLocPFCBasicEntry
62   MAX-ACCESS   not-accessible
63   STATUS      current
64   DESCRIPTION
65       "Information about the IEEE 802.1 organizational defined
66       PFC TLV LLDP extension."
67   INDEX      { lldpV2LocPortIfIndex }
68   ::= { lldpXdot1dcbxLocPFCBasicTable 1 }
69
70 lldpXdot1dcbxLocPFCBasicEntry ::= SEQUENCE {
71     lldpXdot1dcbxLocPFCWilling      TruthValue,
72     lldpXdot1dcbxLocPFCMBC          TruthValue,

```



```

1      lldpXdot1dcbxLocPFCCap          LldpXdot1dcbxSupportedCapacity
2 }
3
4 lldpXdot1dcbxLocPFCWilling OBJECT-TYPE
5     SYNTAX          TruthValue
6     MAX-ACCESS      read-only
7     STATUS          current
8     DESCRIPTION
9         "Indicates if the local system is willing to accept the
10        PFC configuration of the remote system."
11     REFERENCE
12         "D.2.10.3"
13     ::= { lldpXdot1dcbxLocPFCBasicEntry 1}
14
15 lldpXdot1dcbxLocPFCMBC OBJECT-TYPE
16     SYNTAX          TruthValue
17     MAX-ACCESS      read-only
18     STATUS          current
19     DESCRIPTION
20         "Indicates if the local system is capable of bypassing
21        MACsec processing when MACsec is disabled."
22     REFERENCE
23         "D.2.10.4"
24     ::= { lldpXdot1dcbxLocPFCBasicEntry 2}
25
26 lldpXdot1dcbxLocPFCCap OBJECT-TYPE
27     SYNTAX          LldpXdot1dcbxSupportedCapacity
28     MAX-ACCESS      read-only
29     STATUS          current
30     DESCRIPTION
31         "Indicates the number of traffic classes on the local device
32        that may simultaneously have PFC enabled."
33     REFERENCE
34         "D.2.10.5"
35     ::= { lldpXdot1dcbxLocPFCBasicEntry 3}
36
37 lldpXdot1dcbxLocPFCEnableTable OBJECT-TYPE
38     SYNTAX          SEQUENCE OF LldpXdot1dcbxLocPFCEnableEntry
39     MAX-ACCESS      not-accessible
40     STATUS          current
41     DESCRIPTION
42         "This table contains eight entries, one entry per priority,
43        indicating if PFC is enabled on the corresponding priority."
44     ::= { lldpXdot1dcbxLocPFC 2 }
45
46 lldpXdot1dcbxLocPFCEnableEntry OBJECT-TYPE
47     SYNTAX          LldpXdot1dcbxLocPFCEnableEntry
48     MAX-ACCESS      not-accessible
49     STATUS          current
50     DESCRIPTION
51         "Each entry indicates if PFC is enabled on the
52        corresponding priority"
53     INDEX {
54         lldpV2LocPortIfIndex,
55         lldpXdot1dcbxLocPFCEnablePriority
56     }
57     ::= { lldpXdot1dcbxLocPFCEnableTable 1 }
58
59 lldpXdot1dcbxLocPFCEnableEntry ::= SEQUENCE {
60     lldpXdot1dcbxLocPFCEnablePriority IEEE8021PriorityValue,
61     lldpXdot1dcbxLocPFCEnableEnabled TruthValue
62 }
63
64 lldpXdot1dcbxLocPFCEnablePriority OBJECT-TYPE
65     SYNTAX          IEEE8021PriorityValue
66     MAX-ACCESS      not-accessible
67     STATUS          current
68     DESCRIPTION
69         "Priority for which PFC is enabled / disabled"
70     ::= { lldpXdot1dcbxLocPFCEnableEntry 1 }
71
72 lldpXdot1dcbxLocPFCEnableEnabled OBJECT-TYPE

```

```

1 SYNTAX TruthValue
2 MAX-ACCESS read-only
3 STATUS current
4 DESCRIPTION
5 "Indicates if PFC is enabled on the corresponding priority."
6 REFERENCE
7 "D.2.10.6"
8 ::= { lldpXdot1dcbxLocPFCEnableEntry 2 }
9 --
10 -- lldpXdot1dcbxLocApplicationPriorityTable - Contains the information
11 -- for the Application Priority TLV.
12 --
13
14 lldpXdot1dcbxLocApplicationPriorityAppTable OBJECT-TYPE
15 SYNTAX SEQUENCE OF
16 LldpXdot1dcbxLocApplicationPriorityAppEntry
17 MAX-ACCESS not-accessible
18 STATUS current
19 DESCRIPTION
20 "Table containing entries indicating the priority to be used
21 for a given application."
22 ::= { lldpXdot1dcbxLocalData 4 }
23
24 lldpXdot1dcbxLocApplicationPriorityAppEntry OBJECT-TYPE
25 SYNTAX LldpXdot1dcbxLocApplicationPriorityAppEntry
26 MAX-ACCESS not-accessible
27 STATUS current
28 DESCRIPTION
29 "Entry that indicates the priority to be used for a
30 given application."
31 INDEX
32 {
33 lldpV2LocPortIfIndex,
34 lldpXdot1dcbxLocApplicationPriorityAESelector,
35 lldpXdot1dcbxLocApplicationPriorityAEProtocol
36 }
37 ::= { lldpXdot1dcbxLocApplicationPriorityAppTable 1 }
38
39 LldpXdot1dcbxLocApplicationPriorityAppEntry ::= SEQUENCE {
40 lldpXdot1dcbxLocApplicationPriorityAESelector
41 LldpXdot1dcbxAppSelector,
42 lldpXdot1dcbxLocApplicationPriorityAEProtocol
43 LldpXdot1dcbxAppProtocol,
44 lldpXdot1dcbxLocApplicationPriorityAEPriority
45 IEEE8021PriorityValue
46 }
47
48 lldpXdot1dcbxLocApplicationPriorityAESelector OBJECT-TYPE
49 SYNTAX LldpXdot1dcbxAppSelector
50 MAX-ACCESS not-accessible
51 STATUS current
52 DESCRIPTION
53 "Indicates the contents of the protocol object
54 (lldpXdot1dcbxLocApplicationPriorityAEProtocol)
55 1: EtherType
56 2: Well Known Port number over TCP, or SCTP
57 3: Well Known Port number over UDP, or DCCP
58 4: Well Known Port number over TCP, SCTP, UDP, and DCCP
59 5: Differentiated Services Code Point (DSCP) value. The
60 6 bit DSCP value is stored in the low order 6 bits of the
61 protocol object. The higher order bits are set to zero.
62 (See IETF RFC 2474 for the definition of the DSCP value.)"
63 REFERENCE
64 "D.2.11.3"
65 ::= { lldpXdot1dcbxLocApplicationPriorityAppEntry 1 }
66
67 lldpXdot1dcbxLocApplicationPriorityAEProtocol OBJECT-TYPE
68 SYNTAX LldpXdot1dcbxAppProtocol
69 MAX-ACCESS not-accessible
70 STATUS current
71 DESCRIPTION
72 "The protocol indicator of the type indicated by
73 lldpXdot1dcbxLocApplicationPriorityAESelector."

```

```

1  REFERENCE
2  "D.2.11.3"
3  ::= { lldpXdotldcbxLocApplicationPriorityAppEntry 2 }
4
5  lldpXdotldcbxLocApplicationPriorityAEPriority OBJECT-TYPE
6  SYNTAX      IEEE8021PriorityValue
7  MAX-ACCESS  read-only
8  STATUS      current
9  DESCRIPTION
10     "The priority code point that should be used in
11     frames transporting the protocol indicated by
12     lldpXdotldcbxLocApplicationPriorityAESelector and
13     lldpXdotldcbxLocApplicationPriorityAEProtocol."
14  REFERENCE
15     "D.2.11.3"
16  ::= { lldpXdotldcbxLocApplicationPriorityAppEntry 3 }
17
18 --
19 -- lldpXdotldcbxLocApplicationVlanAppTable - Contains the information
20 -- for the Application VLAN TLV.
21 --
22
23 lldpXdotldcbxLocApplicationVlanAppTable OBJECT-TYPE
24 SYNTAX      SEQUENCE OF
25             lldpXdotldcbxLocApplicationVlanAppEntry
26 MAX-ACCESS  not-accessible
27 STATUS      current
28 DESCRIPTION
29     "Table containing entries indicating the VLAN to be used
30     for a given application."
31 ::= { lldpXdotldcbxLocalData 5 }
32
33 lldpXdotldcbxLocApplicationVlanAppEntry OBJECT-TYPE
34 SYNTAX      lldpXdotldcbxLocApplicationVlanAppEntry
35 MAX-ACCESS  not-accessible
36 STATUS      current
37 DESCRIPTION
38     "Entry that indicates the VLAN to be used for a
39     given application."
40 INDEX      {
41             lldpV2LocPortIfIndex,
42             lldpXdotldcbxLocApplicationVlanAESelector,
43             lldpXdotldcbxLocApplicationVlanAEProtocol
44         }
45 ::= { lldpXdotldcbxLocApplicationVlanAppTable 1 }
46
47 lldpXdotldcbxLocApplicationVlanAppEntry ::= SEQUENCE {
48     lldpXdotldcbxLocApplicationVlanAESelector
49     lldpXdotldcbxAppSelector,
50     lldpXdotldcbxLocApplicationVlanAEProtocol
51     lldpXdotldcbxAppProtocol,
52     lldpXdotldcbxLocApplicationVlanAEVlanId
53     VlanId
54 }
55
56 lldpXdotldcbxLocApplicationVlanAESelector OBJECT-TYPE
57 SYNTAX      lldpXdotldcbxAppSelector
58 MAX-ACCESS  not-accessible
59 STATUS      current
60 DESCRIPTION
61     "Indicates the contents of the protocol object
62     (lldpXdotldcbxLocApplicationVlanAEProtocol)
63     1: EtherType
64     2: Well Known Port number over TCP, or SCTP
65     3: Well Known Port number over UDP, or DCCP
66     4: Well Known Port number over TCP, SCTP, UDP, and DCCP
67     5: Differentiated Services Code Point (DSCP) value. The
68     6 bit DSCP value is stored in the low order 6 bits of the
69     protocol object. The higher order bits are set to zero.
70     (See IETF RFC 2474 for the definition of the DSCP value.)"
71  REFERENCE
72  "D.2.11.3"

```

```

1      ::= { lldpXdot1dcbxLocApplicationVlanAppEntry 1 }
2
3 lldpXdot1dcbxLocApplicationVlanAEProtocol OBJECT-TYPE
4     SYNTAX      LldpXdot1dcbxAppProtocol
5     MAX-ACCESS  not-accessible
6     STATUS      current
7     DESCRIPTION
8         "The protocol indicator of the type indicated by
9         lldpXdot1dcbxLocApplicationVlanAESelector."
10    REFERENCE
11        "D.2.11.3"
12    ::= { lldpXdot1dcbxLocApplicationVlanAppEntry 2 }
13
14 lldpXdot1dcbxLocApplicationVlanAEVlanId OBJECT-TYPE
15     SYNTAX      VlanId
16     MAX-ACCESS  read-only
17     STATUS      current
18     DESCRIPTION
19         "The VLAN Identifier that should be used in
20         frames transporting the protocol indicated by
21         lldpXdot1dcbxLocApplicationVlanAESelector and
22         lldpXdot1dcbxLocApplicationVlanAEProtocol."
23     REFERENCE
24        "D.2.14.3"
25    ::= { lldpXdot1dcbxLocApplicationVlanAppEntry 3 }
26
27
28 -----
29 -- IEEE 802.1 - DCBX Remote System Information
30 -----
31
32 --
33 -- lldpXdot1dcbxRemETSConfigurationTable - Contains the information
34 -- for the remote system ETS Configuration TLV.
35 --
36 lldpXdot1dcbxRemETSConfiguration OBJECT IDENTIFIER
37     ::= { lldpXdot1dcbxRemoteData 1 }
38
39 lldpXdot1dcbxRemETSBasicConfigurationTable OBJECT-TYPE
40     SYNTAX      SEQUENCE OF LldpXdot1dcbxRemETSBasicConfigurationEntry
41     MAX-ACCESS  not-accessible
42     STATUS      current
43     DESCRIPTION
44         "This table contains one row per port for the IEEE 802.1
45         organizationally defined LLDP ETS Configuration TLV on
46         the local system known to this agent."
47     ::= { lldpXdot1dcbxRemETSConfiguration 1 }
48
49 lldpXdot1dcbxRemETSBasicConfigurationEntry OBJECT-TYPE
50     SYNTAX      LldpXdot1dcbxRemETSBasicConfigurationEntry
51     MAX-ACCESS  not-accessible
52     STATUS      current
53     DESCRIPTION
54         "Information about the IEEE 802.1 organizational defined
55         ETS Configuration TLV LLDP extension."
56     INDEX
57         {
58             lldpV2RemTimeMark,
59             lldpV2RemLocalIfIndex,
60             lldpV2RemLocalDestMACAddress,
61             lldpV2RemIndex
62         }
63     ::= { lldpXdot1dcbxRemETSBasicConfigurationTable 1 }
64
65 lldpXdot1dcbxRemETSBasicConfigurationEntry ::= SEQUENCE {
66     lldpXdot1dcbxRemETSConCreditBasedShaperSupport      TruthValue,
67     lldpXdot1dcbxRemETSConTrafficClassesSupported
68         LldpXdot1dcbxSupportedCapacity,
69     lldpXdot1dcbxRemETSConWilling      TruthValue
70 }
71
72 lldpXdot1dcbxRemETSConCreditBasedShaperSupport OBJECT-TYPE
73     SYNTAX      TruthValue

```

```

1  MAX-ACCESS      read-only
2  STATUS          current
3  DESCRIPTION
4      "Indicates if the credit-based shaper Traffic Selection
5      algorithm is supported on the remote system."
6  REFERENCE
7      "D.2.8.4"
8  ::= { lldpXdot1dcbxRemETSTrafficClassesSupported 1 }
9
10 lldpXdot1dcbxRemETSTrafficClassesSupported OBJECT-TYPE
11     SYNTAX      LldpXdot1dcbxSupportedCapacity
12     MAX-ACCESS  read-only
13     STATUS      current
14     DESCRIPTION
15         "Indicates the number of traffic classes supported."
16     REFERENCE
17         "D.2.8.5"
18     ::= { lldpXdot1dcbxRemETSTrafficClassesSupported 2 }
19
20 lldpXdot1dcbxRemETSTrafficClassesSupported OBJECT-TYPE
21     SYNTAX      TruthValue
22     MAX-ACCESS  read-only
23     STATUS      current
24     DESCRIPTION
25         "Indicates if the remote system is willing to accept the
26         ETS configuration recommended by the remote system."
27     REFERENCE
28         "D.2.8.3"
29     ::= { lldpXdot1dcbxRemETSTrafficClassesSupported 3 }
30
31 lldpXdot1dcbxRemETSTrafficClassesSupported OBJECT-TYPE
32     SYNTAX      SEQUENCE OF
33         LldpXdot1dcbxRemETSTrafficClassesSupportedEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "This table contains one row per priority. The entry in
38         each row indicates the traffic class to which the
39         priority is assigned."
40     ::= { lldpXdot1dcbxRemETSTrafficClassesSupported 2 }
41
42 lldpXdot1dcbxRemETSTrafficClassesSupportedEntry OBJECT-TYPE
43     SYNTAX      LldpXdot1dcbxRemETSTrafficClassesSupportedEntry
44     MAX-ACCESS  not-accessible
45     STATUS      current
46     DESCRIPTION
47         "Indicates a priority to traffic class assignment."
48     INDEX      {
49         lldpV2RemTimeMark,
50         lldpV2RemLocalIfIndex,
51         lldpV2RemLocalDestMACAddress,
52         lldpV2RemIndex,
53         lldpXdot1dcbxRemETSTrafficClassesSupportedEntry
54     }
55     ::= { lldpXdot1dcbxRemETSTrafficClassesSupportedEntry 1 }
56
57 LldpXdot1dcbxRemETSTrafficClassesSupportedEntry ::= SEQUENCE {
58     lldpXdot1dcbxRemETSTrafficClassesSupportedEntry IEEE8021PriorityValue,
59     lldpXdot1dcbxRemETSTrafficClassesSupportedEntry TrafficClass
60     LldpXdot1dcbxTrafficClassValue
61 }
62
63 lldpXdot1dcbxRemETSTrafficClassesSupported OBJECT-TYPE
64     SYNTAX      IEEE8021PriorityValue
65     MAX-ACCESS  not-accessible
66     STATUS      current
67     DESCRIPTION
68         "Indicates the priority that is assigned to a traffic
69         class."
70     REFERENCE
71         "D.2.8.6"
72     ::= { lldpXdot1dcbxRemETSTrafficClassesSupportedEntry 1 }

```

```

1
2 lldpXdot1dcbxRemETSTrafficClass OBJECT-TYPE
3   SYNTAX      LldpXdot1dcbxTrafficClassValue
4   MAX-ACCESS   read-only
5   STATUS       current
6   DESCRIPTION
7       "Indicates the traffic class to which this priority is
8       to be assigned."
9   REFERENCE
10      "D.2.8.6"
11 ::= { lldpXdot1dcbxRemETSPriorityAssignmentEntry 2 }
12
13 lldpXdot1dcbxRemETSTrafficClassBandwidthTable OBJECT-TYPE
14   SYNTAX      SEQUENCE OF
15               LldpXdot1dcbxRemETSTrafficClassBandwidthEntry
16   MAX-ACCESS   not-accessible
17   STATUS       current
18   DESCRIPTION
19       "This table contains one row per traffic class. The
20       entry in each row indicates the traffic class to
21       which the bandwidth is assigned."
22 ::= { lldpXdot1dcbxRemETSTrafficClassBandwidthTable 3 }
23
24 lldpXdot1dcbxRemETSTrafficClassBandwidthEntry OBJECT-TYPE
25   SYNTAX      LldpXdot1dcbxRemETSTrafficClassBandwidthEntry
26   MAX-ACCESS   not-accessible
27   STATUS       current
28   DESCRIPTION
29       "Indicates a traffic class to Bandwidth assignment."
30   INDEX       {
31               lldpV2RemTimeMark,
32               lldpV2RemLocalIfIndex,
33               lldpV2RemLocalDestMACAddress,
34               lldpV2RemIndex,
35               lldpXdot1dcbxRemETSTrafficClass
36   }
37 ::= { lldpXdot1dcbxRemETSTrafficClassBandwidthTable 1 }
38
39 lldpXdot1dcbxRemETSTrafficClassBandwidthEntry ::= SEQUENCE {
40   lldpXdot1dcbxRemETSTrafficClass
41   lldpXdot1dcbxTrafficClassValue,
42   lldpXdot1dcbxRemETSTrafficClassBandwidth
43   lldpXdot1dcbxTrafficClassBandwidthValue
44 }
45
46 lldpXdot1dcbxRemETSTrafficClass OBJECT-TYPE
47   SYNTAX      LldpXdot1dcbxTrafficClassValue
48   MAX-ACCESS   not-accessible
49   STATUS       current
50   DESCRIPTION
51       "Indicates the traffic class to
52       which this bandwidth applies."
53   REFERENCE
54       "D.2.8.7"
55 ::= { lldpXdot1dcbxRemETSTrafficClassBandwidthEntry 1 }
56
57 lldpXdot1dcbxRemETSTrafficClassBandwidth OBJECT-TYPE
58   SYNTAX      LldpXdot1dcbxTrafficClassBandwidthValue
59   MAX-ACCESS   read-only
60   STATUS       current
61   DESCRIPTION
62       "Indicates the bandwidth assigned to this traffic class."
63   REFERENCE
64       "D.2.8.7"
65 ::= { lldpXdot1dcbxRemETSTrafficClassBandwidthEntry 2 }
66
67
68 lldpXdot1dcbxRemETSTrafficSelectionAlgorithmTable OBJECT-TYPE
69   SYNTAX      SEQUENCE OF
70               LldpXdot1dcbxRemETSTrafficSelectionAlgorithmEntry
71   MAX-ACCESS   not-accessible
72   STATUS       current

```

```

1  DESCRIPTION
2      "This table contains one row per traffic class. The
3      entry in each row indicates the traffic selection
4      algorithm to be used by the traffic class."
5      ::= { lldpXdot1dcbxRemETSConfiguration 4 }
6
7  lldpXdot1dcbxRemETSConTrafficSelectionAlgorithmEntry OBJECT-TYPE
8      SYNTAX      LldpXdot1dcbxRemETSConTrafficSelectionAlgorithmEntry
9      MAX-ACCESS   not-accessible
10     STATUS      current
11     DESCRIPTION
12         "Indicates a traffic class to traffic selection
13         algorithm assignment."
14     INDEX        {
15         lldpV2RemTimeMark,
16         lldpV2RemLocalIfIndex,
17         lldpV2RemLocalDestMACAddress,
18         lldpV2RemIndex,
19         lldpXdot1dcbxRemETSConTSATrafficClass
20     }
21     ::= { lldpXdot1dcbxRemETSConTrafficSelectionAlgorithmTable 1 }
22
23 lldpXdot1dcbxRemETSConTrafficSelectionAlgorithmEntry ::= SEQUENCE {
24     lldpXdot1dcbxRemETSConTSATrafficClass
25     LldpXdot1dcbxTrafficClassValue,
26     lldpXdot1dcbxRemETSConTrafficSelectionAlgorithm
27     LldpXdot1dcbxTrafficSelectionAlgorithm
28 }
29
30 lldpXdot1dcbxRemETSConTSATrafficClass OBJECT-TYPE
31     SYNTAX      LldpXdot1dcbxTrafficClassValue
32     MAX-ACCESS   not-accessible
33     STATUS      current
34     DESCRIPTION
35         "Indicates the traffic class that is assigned to a traffic
36         selection algorithm."
37     REFERENCE
38         "D.2.8.8"
39     ::= { lldpXdot1dcbxRemETSConTrafficSelectionAlgorithmEntry 1 }
40
41 lldpXdot1dcbxRemETSConTrafficSelectionAlgorithm OBJECT-TYPE
42     SYNTAX      LldpXdot1dcbxTrafficSelectionAlgorithm
43     MAX-ACCESS   read-only
44     STATUS      current
45     DESCRIPTION
46         "Indicates the Traffic Selection Algorithm to which this
47         traffic class is to be assigned."
48     REFERENCE
49         "D.2.8.8"
50     ::= { lldpXdot1dcbxRemETSConTrafficSelectionAlgorithmEntry 2 }
51
52 --
53 -- lldpXdot1dcbxRemETSRecommendationTable - Contains the information for
54 -- the remote system ETS Recommendation TLV.
55 --
56 lldpXdot1dcbxRemETSReco OBJECT IDENTIFIER ::=
57 { lldpXdot1dcbxRemoteData 2 }
58
59 lldpXdot1dcbxRemETSRecoTrafficClassBandwidthTable OBJECT-TYPE
60     SYNTAX      SEQUENCE OF
61         LldpXdot1dcbxRemETSRecoTrafficClassBandwidthEntry
62     MAX-ACCESS   not-accessible
63     STATUS      current
64     DESCRIPTION
65         "This table contains one row per traffic class. The
66         entry in each row indicates the traffic class to
67         which the bandwidth is assigned."
68     ::= { lldpXdot1dcbxRemETSReco 1 }
69
70 lldpXdot1dcbxRemETSRecoTrafficClassBandwidthEntry OBJECT-TYPE
71     SYNTAX      LldpXdot1dcbxRemETSRecoTrafficClassBandwidthEntry
72     MAX-ACCESS   not-accessible

```

```

1  STATUS      current
2  DESCRIPTION
3      "Indicates a traffic class to Bandwidth assignment."
4  INDEX      {
5              lldpV2RemTimeMark,
6              lldpV2RemLocalIfIndex,
7              lldpV2RemLocalDestMACAddress,
8              lldpV2RemIndex,
9              lldpXdot1dcbxRemETSRecoTrafficClass
10 }
11 ::= { lldpXdot1dcbxRemETSRecoTrafficClassBandwidthTable 1 }
12
13 LldpXdot1dcbxRemETSRecoTrafficClassBandwidthEntry ::= SEQUENCE {
14     lldpXdot1dcbxRemETSRecoTrafficClass
15     LldpXdot1dcbxTrafficClassValue,
16     lldpXdot1dcbxRemETSRecoTrafficClassBandwidth
17     LldpXdot1dcbxTrafficClassBandwidthValue
18 }
19
20 lldpXdot1dcbxRemETSRecoTrafficClass OBJECT-TYPE
21     SYNTAX      LldpXdot1dcbxTrafficClassValue
22     MAX-ACCESS   not-accessible
23     STATUS      current
24     DESCRIPTION
25         "Indicates the traffic class to
26         which this bandwidth applies."
27     REFERENCE
28         "D.2.9.4"
29     ::= { lldpXdot1dcbxRemETSRecoTrafficClassBandwidthEntry 1 }
30
31 lldpXdot1dcbxRemETSRecoTrafficClassBandwidth OBJECT-TYPE
32     SYNTAX      LldpXdot1dcbxTrafficClassBandwidthValue
33     MAX-ACCESS   read-only
34     STATUS      current
35     DESCRIPTION
36         "Indicates the bandwidth assigned to this traffic class."
37     REFERENCE
38         "D.2.9.4"
39     ::= { lldpXdot1dcbxRemETSRecoTrafficClassBandwidthEntry 2 }
40
41 lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmTable OBJECT-TYPE
42     SYNTAX      SEQUENCE OF
43                 LldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmEntry
44     MAX-ACCESS   not-accessible
45     STATUS      current
46     DESCRIPTION
47         "This table contains one row per traffic class. The
48         entry in each row indicates the traffic selection
49         algorithm to be used by the priority."
50     ::= { lldpXdot1dcbxRemETSReco 2 }
51
52 lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmEntry OBJECT-TYPE
53     SYNTAX      LldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmEntry
54     MAX-ACCESS   not-accessible
55     STATUS      current
56     DESCRIPTION
57         "Indicates a priority to traffic selection algorithm
58         assignment."
59     INDEX      {
60                 lldpV2RemTimeMark,
61                 lldpV2RemLocalIfIndex,
62                 lldpV2RemLocalDestMACAddress,
63                 lldpV2RemIndex,
64                 lldpXdot1dcbxRemETSRecoTSATrafficClass
65             }
66     ::= { lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmTable 1 }
67
68 LldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmEntry ::= SEQUENCE {
69     lldpXdot1dcbxRemETSRecoTSATrafficClass
70     LldpXdot1dcbxTrafficClassValue,
71     lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithm
72     LldpXdot1dcbxTrafficSelectionAlgorithm

```



```

1 }
2
3 lldpXdot1dcbxRemETSRecoTSATrafficClass OBJECT-TYPE
4     SYNTAX          LldpXdot1dcbxTrafficClassValue
5     MAX-ACCESS      not-accessible
6     STATUS          current
7     DESCRIPTION
8         "Indicates the traffic class that is assigned to a traffic
9         selection algorithm."
10    REFERENCE
11        "D.2.9.5"
12    ::= { lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmEntry 1 }
13
14 lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithm OBJECT-TYPE
15     SYNTAX          LldpXdot1dcbxTrafficSelectionAlgorithm
16     MAX-ACCESS      read-only
17     STATUS          current
18     DESCRIPTION
19         "Indicates the Traffic Selection Algorithm to which this
20         traffic class is to be assigned."
21    REFERENCE
22        "D.2.9.5"
23    ::= { lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithmEntry 2 }
24
25 --
26 -- lldpXdot1dcbxRemPFCTable - Contains the information for the remote
27 -- system PFC TLV.
28 --
29 lldpXdot1dcbxRemPFC OBJECT IDENTIFIER ::= { lldpXdot1dcbxRemoteData 3 }
30
31 lldpXdot1dcbxRemPFCBasicTable OBJECT-TYPE
32     SYNTAX          SEQUENCE OF LldpXdot1dcbxRemPFCBasicEntry
33     MAX-ACCESS      not-accessible
34     STATUS          current
35     DESCRIPTION
36         "This table contains one row per port for the IEEE 802.1
37         organizationally defined LLDP PFC TLV on the local
38         system known to this agent."
39    ::= { lldpXdot1dcbxRemPFC 1 }
40
41 lldpXdot1dcbxRemPFCBasicEntry OBJECT-TYPE
42     SYNTAX          LldpXdot1dcbxRemPFCBasicEntry
43     MAX-ACCESS      not-accessible
44     STATUS          current
45     DESCRIPTION
46         "Information about the IEEE 802.1 organizational defined
47         PFC TLV LLDP extension."
48     INDEX
49         {
50             lldpV2RemTimeMark,
51             lldpV2RemLocalIfIndex,
52             lldpV2RemLocalDestMACAddress,
53             lldpV2RemIndex
54         }
55     ::= { lldpXdot1dcbxRemPFCBasicTable 1 }
56
57 lldpXdot1dcbxRemPFCBasicEntry ::= SEQUENCE {
58     lldpXdot1dcbxRemPFCWilling      TruthValue,
59     lldpXdot1dcbxRemPFCMBC          TruthValue,
60     lldpXdot1dcbxRemPFCCap          LldpXdot1dcbxSupportedCapacity
61 }
62
63 lldpXdot1dcbxRemPFCWilling OBJECT-TYPE
64     SYNTAX          TruthValue
65     MAX-ACCESS      read-only
66     STATUS          current
67     DESCRIPTION
68         "Indicates if the remote system is willing to accept the
69         PFC configuration of the local system."
70    REFERENCE
71        "D.2.10.3"
72    ::= { lldpXdot1dcbxRemPFCBasicEntry 1 }

```

```

1 lldpXdot1dcbxRemPFCMBC OBJECT-TYPE
2     SYNTAX      TruthValue
3     MAX-ACCESS  read-only
4     STATUS      current
5     DESCRIPTION
6         "Indicates if the remote system is capable of bypassing
7         MACsec processing when MACsec is disabled."
8     REFERENCE
9         "D.2.10.4"
10    ::= { lldpXdot1dcbxRemPFCBasicEntry 2}
11
12 lldpXdot1dcbxRemPFCCap OBJECT-TYPE
13     SYNTAX      LldpXdot1dcbxSupportedCapacity
14     MAX-ACCESS  read-only
15     STATUS      current
16     DESCRIPTION
17         "Indicates the number of traffic classes on the remote device
18         that may simultaneously have PFC enabled."
19     REFERENCE
20         "D.2.10.5"
21    ::= { lldpXdot1dcbxRemPFCBasicEntry 3}
22
23 lldpXdot1dcbxRemPFCEnableTable OBJECT-TYPE
24     SYNTAX      SEQUENCE OF LldpXdot1dcbxRemPFCEnableEntry
25     MAX-ACCESS  not-accessible
26     STATUS      current
27     DESCRIPTION
28         "This table contains eight entries, one entry per priority,
29         indicating if PFC is enabled on the corresponding priority."
30    ::= { lldpXdot1dcbxRemPFC 2 }
31
32 lldpXdot1dcbxRemPFCEnableEntry OBJECT-TYPE
33     SYNTAX      LldpXdot1dcbxRemPFCEnableEntry
34     MAX-ACCESS  not-accessible
35     STATUS      current
36     DESCRIPTION
37         "Each entry indicates if PFC is enabled on the
38         corresponding priority."
39     INDEX      {
40         lldpV2RemTimeMark,
41         lldpV2RemLocalIfIndex,
42         lldpV2RemLocalDestMACAddress,
43         lldpV2RemIndex,
44         lldpXdot1dcbxRemPFCEnablePriority
45     }
46    ::= { lldpXdot1dcbxRemPFCEnableTable 1 }
47
48 LldpXdot1dcbxRemPFCEnableEntry ::= SEQUENCE {
49     lldpXdot1dcbxRemPFCEnablePriority  IEEE8021PriorityValue,
50     lldpXdot1dcbxRemPFCEnableEnabled  TruthValue
51 }
52
53 lldpXdot1dcbxRemPFCEnablePriority OBJECT-TYPE
54     SYNTAX      IEEE8021PriorityValue
55     MAX-ACCESS  not-accessible
56     STATUS      current
57     DESCRIPTION
58         "Priority for which PFC is enabled / disabled"
59    ::= { lldpXdot1dcbxRemPFCEnableEntry 1 }
60
61 lldpXdot1dcbxRemPFCEnableEnabled OBJECT-TYPE
62     SYNTAX      TruthValue
63     MAX-ACCESS  read-only
64     STATUS      current
65     DESCRIPTION
66         "Indicates if PFC is enabled on the corresponding priority."
67     REFERENCE
68         "D.2.10.6"
69    ::= { lldpXdot1dcbxRemPFCEnableEntry 2 }
70
71 --
72 -- lldpXdot1dcbxRemApplicationPriorityTable - Contains the information

```

```

1 -- for the remote system Application Priority TLV.
2 --
3
4 lldpXdot1dcbxRemApplicationPriorityAppTable OBJECT-TYPE
5     SYNTAX          SEQUENCE OF
6         LldpXdot1dcbxRemApplicationPriorityAppEntry
7     MAX-ACCESS      not-accessible
8     STATUS          current
9     DESCRIPTION
10        "Table containing entries indicating the priority to be used
11        for a given application."
12    ::= { lldpXdot1dcbxRemoteData 4 }
13
14 lldpXdot1dcbxRemApplicationPriorityAppEntry OBJECT-TYPE
15     SYNTAX          LldpXdot1dcbxRemApplicationPriorityAppEntry
16     MAX-ACCESS      not-accessible
17     STATUS          current
18     DESCRIPTION
19        "Entry that indicates the priority to be used for a
20        given application."
21     INDEX
22        {
23            lldpV2RemTimeMark,
24            lldpV2RemLocalIfIndex,
25            lldpV2RemLocalDestMACAddress,
26            lldpV2RemIndex,
27            lldpXdot1dcbxRemApplicationPriorityAESelector,
28            lldpXdot1dcbxRemApplicationPriorityAEProtocol
29        }
30    ::= { lldpXdot1dcbxRemApplicationPriorityAppTable 1 }
31
32 lldpXdot1dcbxRemApplicationPriorityAppEntry ::= SEQUENCE {
33     lldpXdot1dcbxRemApplicationPriorityAESelector
34     LldpXdot1dcbxAppSelector,
35     lldpXdot1dcbxRemApplicationPriorityAEProtocol
36     LldpXdot1dcbxAppProtocol,
37     lldpXdot1dcbxRemApplicationPriorityAEPriority
38     IEEE8021PriorityValue
39 }
40
41 lldpXdot1dcbxRemApplicationPriorityAESelector OBJECT-TYPE
42     SYNTAX          LldpXdot1dcbxAppSelector
43     MAX-ACCESS      not-accessible
44     STATUS          current
45     DESCRIPTION
46        "Indicates the contents of the protocol object
47        (lldpXdot1dcbxRemApplicationPriorityAEProtocol)
48        1: EtherType
49        2: Well Known Port number over TCP, or SCTP
50        3: Well Known Port number over UDP, or DCCP
51        4: Well Known Port number over TCP, SCTP, UDP, and DCCP
52        5: Differentiated Services Code Point (DSCP) value. The
53        6 bit DSCP value is stored in the low order 6 bits of the
54        protocol object. The higher order bits are set to zero.
55        (See IETF RFC 2474 for the definition of the DSCP value.)"
56     REFERENCE
57        "D.2.11.3"
58    ::= { lldpXdot1dcbxRemApplicationPriorityAppEntry 1 }
59
60 lldpXdot1dcbxRemApplicationPriorityAEProtocol OBJECT-TYPE
61     SYNTAX          LldpXdot1dcbxAppProtocol
62     MAX-ACCESS      not-accessible
63     STATUS          current
64     DESCRIPTION
65        "The protocol indicator of the type indicated by
66        lldpXdot1dcbxRemApplicationPriorityAESelector."
67     REFERENCE
68        "D.2.11.3"
69    ::= { lldpXdot1dcbxRemApplicationPriorityAppEntry 2 }
70
71 lldpXdot1dcbxRemApplicationPriorityAEPriority OBJECT-TYPE
72     SYNTAX          IEEE8021PriorityValue
73     MAX-ACCESS      read-only

```

```

1  STATUS      current
2  DESCRIPTION
3      "The priority code point that should be used in
4      frames transporting the protocol indicated by
5      lldpXdotldcbxRemApplicationPriorityAESelector and
6      lldpXdotldcbxRemApplicationPriorityAEProtocol."
7  REFERENCE
8      "D.2.11.3"
9      ::= { lldpXdotldcbxRemApplicationPriorityAppEntry 3 }
10
11 --
12 -- lldpXdotldcbxRemApplicationVlanAppTable - Contains the information
13 -- for the remote system Application VLAN TLV.
14 --
15
16 lldpXdotldcbxRemApplicationVlanAppTable OBJECT-TYPE
17     SYNTAX      SEQUENCE OF
18         LldpXdotldcbxRemApplicationVlanAppEntry
19     MAX-ACCESS   not-accessible
20     STATUS       current
21     DESCRIPTION
22         "Table containing entries indicating the VLAN to be used
23         for a given application."
24     ::= { lldpXdotldcbxRemoteData 5 }
25
26 lldpXdotldcbxRemApplicationVlanAppEntry OBJECT-TYPE
27     SYNTAX      LldpXdotldcbxRemApplicationVlanAppEntry
28     MAX-ACCESS   not-accessible
29     STATUS       current
30     DESCRIPTION
31         "Entry that indicates the VLAN to be used for a
32         given application."
33     INDEX
34         {
35             lldpV2RemTimeMark,
36             lldpV2RemLocalIfIndex,
37             lldpV2RemLocalDestMACAddress,
38             lldpV2RemIndex,
39             lldpXdotldcbxRemApplicationVlanAESelector,
40             lldpXdotldcbxRemApplicationVlanAEProtocol
41         }
42     ::= { lldpXdotldcbxRemApplicationVlanAppTable 1 }
43
44 lldpXdotldcbxRemApplicationVlanAppEntry ::= SEQUENCE {
45     lldpXdotldcbxRemApplicationVlanAESelector
46     LldpXdotldcbxAppSelector,
47     lldpXdotldcbxRemApplicationVlanAEProtocol
48     LldpXdotldcbxAppProtocol,
49     lldpXdotldcbxRemApplicationVlanAEVlanId
50     VlanId
51 }
52
53 lldpXdotldcbxRemApplicationVlanAESelector OBJECT-TYPE
54     SYNTAX      LldpXdotldcbxAppSelector
55     MAX-ACCESS   not-accessible
56     STATUS       current
57     DESCRIPTION
58         "Indicates the contents of the protocol object
59         (lldpXdotldcbxRemApplicationVlanAEProtocol)
60         1: EtherType
61         2: Well Known Port number over TCP, or SCTP
62         3: Well Known Port number over UDP, or DCCP
63         4: Well Known Port number over TCP, SCTP, UDP, and DCCP
64         5: Differentiated Services Code Point (DSCP) value. The
65         6 bit DSCP value is stored in the low order 6 bits of the
66         protocol object. The higher order bits are set to zero.
67         (See IETF RFC 2474 for the definition of the DSCP value.)"
68     REFERENCE
69         "D.2.11.3"
70     ::= { lldpXdotldcbxRemApplicationVlanAppEntry 1 }
71
72 lldpXdotldcbxRemApplicationVlanAEProtocol OBJECT-TYPE
73     SYNTAX      LldpXdotldcbxAppProtocol

```

```

1  MAX-ACCESS    not-accessible
2  STATUS        current
3  DESCRIPTION
4      "The protocol indicator of the type indicated by
5      lldpXdot1dcbxRemApplicationVlanAESelector."
6  REFERENCE
7      "D.2.11.3"
8  ::= { lldpXdot1dcbxRemApplicationVlanAppEntry 2 }
9
10 lldpXdot1dcbxRemApplicationVlanAEVlanId OBJECT-TYPE
11     SYNTAX      VlanId
12     MAX-ACCESS   read-only
13     STATUS       current
14     DESCRIPTION
15         "The VLAN Identifier that should be used in
16         frames transporting the protocol indicated by
17         lldpXdot1dcbxRemApplicationVlanAESelector and
18         lldpXdot1dcbxRemApplicationVlanAEProtocol."
19     REFERENCE
20         "D.2.14.3"
21     ::= { lldpXdot1dcbxRemApplicationVlanAppEntry 3 }
22
23
24 -----
25 -- IEEE 802.1 - DCBX Administrative Information
26 -----
27
28 --
29 -- lldpXdot1dcbxAdminETSTConfigurationTable - Contains the information
30 -- for the ETS Configuration TLV.
31 --
32 lldpXdot1dcbxAdminETSTConfiguration OBJECT IDENTIFIER
33     ::= { lldpXdot1dcbxAdminData 1 }
34
35 lldpXdot1dcbxAdminETSTBasicConfigurationTable OBJECT-TYPE
36     SYNTAX      SEQUENCE OF
37         LldpXdot1dcbxAdminETSTBasicConfigurationEntry
38     MAX-ACCESS   not-accessible
39     STATUS       current
40     DESCRIPTION
41         "This table contains one row per port for the IEEE 802.1
42         organizationally defined LLDP ETS Configuration TLV
43         on the local system known to this agent."
44     ::= { lldpXdot1dcbxAdminETSTConfiguration 1 }
45
46 lldpXdot1dcbxAdminETSTBasicConfigurationEntry OBJECT-TYPE
47     SYNTAX      LldpXdot1dcbxAdminETSTBasicConfigurationEntry
48     MAX-ACCESS   not-accessible
49     STATUS       current
50     DESCRIPTION
51         "Information about the IEEE 802.1 organizational defined
52         ETS Configuration TLV LLDP extension."
53     INDEX       { lldpV2LocPortIfIndex }
54     ::= { lldpXdot1dcbxAdminETSTBasicConfigurationTable 1 }
55
56 LldpXdot1dcbxAdminETSTBasicConfigurationEntry ::= SEQUENCE {
57     lldpXdot1dcbxAdminETSTConCreditBasedShaperSupport      TruthValue,
58     lldpXdot1dcbxAdminETSTConTrafficClassesSupported
59         LldpXdot1dcbxSupportedCapacity,
60     lldpXdot1dcbxAdminETSTConWilling      TruthValue
61 }
62
63 lldpXdot1dcbxAdminETSTConCreditBasedShaperSupport OBJECT-TYPE
64     SYNTAX      TruthValue
65     MAX-ACCESS   read-only
66     STATUS       current
67     DESCRIPTION
68         "Indicates support for the credit-based shaper Traffic
69         Selection Algorithm."
70     REFERENCE
71         "D.2.8.4"
72     ::= { lldpXdot1dcbxAdminETSTBasicConfigurationEntry 1 }

```

```

1
2 lldpXdot1dcbxAdminETSTrafficClassesSupported OBJECT-TYPE
3   SYNTAX      LldpXdot1dcbxSupportedCapacity
4   MAX-ACCESS   read-only
5   STATUS       current
6   DESCRIPTION
7       "Indicates the number of traffic classes supported."
8   REFERENCE
9       "D.2.8.5"
10  ::= { lldpXdot1dcbxAdminETSTrafficClassesSupported 2 }
11
12 lldpXdot1dcbxAdminETSTrafficClassesSupported OBJECT-TYPE
13   SYNTAX      TruthValue
14   MAX-ACCESS   read-write
15   STATUS       current
16   DESCRIPTION
17       "Indicates if the local system is willing to accept the
18       ETS configuration recommended by the remote system."
19   REFERENCE
20       "D.2.8.3"
21   DEFVAL      { false }
22  ::= { lldpXdot1dcbxAdminETSTrafficClassesSupported 3 }
23
24 lldpXdot1dcbxAdminETSPriorityAssignmentTable OBJECT-TYPE
25   SYNTAX      SEQUENCE OF
26       LldpXdot1dcbxAdminETSPriorityAssignmentEntry
27   MAX-ACCESS   not-accessible
28   STATUS       current
29   DESCRIPTION
30       "This table contains one row per priority. The entry in each
31       row indicates the traffic class to which the priority is
32       assigned."
33  ::= { lldpXdot1dcbxAdminETSPriorityAssignmentTable 2 }
34
35 lldpXdot1dcbxAdminETSPriorityAssignmentEntry OBJECT-TYPE
36   SYNTAX      LldpXdot1dcbxAdminETSPriorityAssignmentEntry
37   MAX-ACCESS   not-accessible
38   STATUS       current
39   DESCRIPTION
40       "Indicates a priority to traffic class assignment."
41   INDEX
42       {
43           lldpV2LocPortIfIndex,
44           lldpXdot1dcbxAdminETSPriorityAssignmentEntry
45       }
46  ::= { lldpXdot1dcbxAdminETSPriorityAssignmentTable 1 }
47
48 lldpXdot1dcbxAdminETSPriorityAssignmentEntry ::= SEQUENCE {
49     lldpXdot1dcbxAdminETSPriorityAssignmentEntry IEEE8021PriorityValue,
50     lldpXdot1dcbxAdminETSPriorityAssignmentEntryTrafficClass
51     LldpXdot1dcbxTrafficClassValue
52 }
53
54 lldpXdot1dcbxAdminETSPriorityAssignmentEntry OBJECT-TYPE
55   SYNTAX      IEEE8021PriorityValue
56   MAX-ACCESS   not-accessible
57   STATUS       current
58   DESCRIPTION
59       "Indicates the priority that is assigned to a traffic
60       class."
61   REFERENCE
62       "D.2.8.6"
63  ::= { lldpXdot1dcbxAdminETSPriorityAssignmentEntry 1 }
64
65 lldpXdot1dcbxAdminETSPriorityAssignmentEntryTrafficClass OBJECT-TYPE
66   SYNTAX      LldpXdot1dcbxTrafficClassValue
67   MAX-ACCESS   read-write
68   STATUS       current
69   DESCRIPTION
70       "Indicates the traffic class to which this priority is
71       to be assigned."
72   REFERENCE
73       "D.2.8.6"

```

```

1  DEFVAL      { 0 }
2  ::= { lldpXdot1dcbxAdminETSTrafficClassBandwidthTable 2 }
3
4  lldpXdot1dcbxAdminETSTrafficClassBandwidthTable OBJECT-TYPE
5  SYNTAX      SEQUENCE OF
6  lldpXdot1dcbxAdminETSTrafficClassBandwidthEntry
7  MAX-ACCESS  not-accessible
8  STATUS      current
9  DESCRIPTION
10     "This table contains one row per traffic class. The
11     entry in each row indicates the traffic class to
12     which the bandwidth is assigned."
13     ::= { lldpXdot1dcbxAdminETSTrafficClassBandwidthTable 3 }
14
15  lldpXdot1dcbxAdminETSTrafficClassBandwidthEntry OBJECT-TYPE
16  SYNTAX      lldpXdot1dcbxAdminETSTrafficClassBandwidthEntry
17  MAX-ACCESS  not-accessible
18  STATUS      current
19  DESCRIPTION
20     "Indicates a traffic class to Bandwidth assignment."
21  INDEX       {
22              lldpV2LocPortIfIndex,
23              lldpXdot1dcbxAdminETSTrafficClass
24          }
25  ::= { lldpXdot1dcbxAdminETSTrafficClassBandwidthTable 1 }
26
27  lldpXdot1dcbxAdminETSTrafficClassBandwidthEntry ::= SEQUENCE {
28      lldpXdot1dcbxAdminETSTrafficClass
29      lldpXdot1dcbxTrafficClassValue,
30      lldpXdot1dcbxAdminETSTrafficClassBandwidth
31      lldpXdot1dcbxTrafficClassBandwidthValue
32  }
33
34  lldpXdot1dcbxAdminETSTrafficClass OBJECT-TYPE
35  SYNTAX      lldpXdot1dcbxTrafficClassValue
36  MAX-ACCESS  not-accessible
37  STATUS      current
38  DESCRIPTION
39     "Indicates the traffic class to
40     which this bandwidth applies."
41  REFERENCE
42     "D.2.8.7"
43  ::= { lldpXdot1dcbxAdminETSTrafficClassBandwidthEntry 1 }
44
45  lldpXdot1dcbxAdminETSTrafficClassBandwidth OBJECT-TYPE
46  SYNTAX      lldpXdot1dcbxTrafficClassBandwidthValue
47  MAX-ACCESS  read-write
48  STATUS      current
49  DESCRIPTION
50     "Indicates the bandwidth assigned to this traffic class.
51     The sum of the bandwidths assigned to a given port is
52     required at all times to equal 100. An operation that
53     attempts to change this table such that the bandwidth
54     entries do not total 100 shall be rejected. An implication
55     of this is that modification of this table requires that
56     multiple set operations be included in a single SNMP PDU,
57     commonly referred to as an MSET operation, to perform
58     simultaneous set operations to keep the sum at 100. Any
59     attempt to change a single entry in this table will result
60     in the operation being rejected since entries in the
61     table referring to the given port will no longer
62     sum to 100."
63  REFERENCE
64     "D.2.8.7"
65  ::= { lldpXdot1dcbxAdminETSTrafficClassBandwidthEntry 2 }
66
67  lldpXdot1dcbxAdminETSTrafficSelectionAlgorithmTable OBJECT-TYPE
68  SYNTAX      SEQUENCE OF
69  lldpXdot1dcbxAdminETSTrafficSelectionAlgorithmEntry
70  MAX-ACCESS  not-accessible
71  STATUS      current
72  DESCRIPTION

```

```

1      "This table contains one row per traffic class. The entry
2      in each row indicates the traffic selection algorithm to
3      be used by the priority."
4      ::= { lldpXdot1dcbxAdminETSConfiguration 4 }
5
6 lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithmEntry OBJECT-TYPE
7     SYNTAX      LldpXdot1dcbxAdminETSConTrafficSelectionAlgorithmEntry
8     MAX-ACCESS   not-accessible
9     STATUS       current
10    DESCRIPTION
11        "Indicates a traffic class to traffic selection
12        algorithm assignment."
13    INDEX        {
14        lldpV2LocPortIfIndex,
15        lldpXdot1dcbxAdminETSConTSATrafficClass
16    }
17    ::= { lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithmTable 1 }
18
19 lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithmEntry ::= SEQUENCE {
20     lldpXdot1dcbxAdminETSConTSATrafficClass
21     LldpXdot1dcbxTrafficClassValue,
22     lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithm
23     LldpXdot1dcbxTrafficSelectionAlgorithm
24 }
25
26 lldpXdot1dcbxAdminETSConTSATrafficClass OBJECT-TYPE
27     SYNTAX      LldpXdot1dcbxTrafficClassValue
28     MAX-ACCESS   not-accessible
29     STATUS       current
30     DESCRIPTION
31         "Indicates the traffic class that is assigned
32         to a traffic selection algorithm."
33     REFERENCE
34         "D.2.8.8"
35     ::= { lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithmEntry 1 }
36
37 lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithm OBJECT-TYPE
38     SYNTAX      LldpXdot1dcbxTrafficSelectionAlgorithm
39     MAX-ACCESS   read-write
40     STATUS       current
41     DESCRIPTION
42         "Indicates the Traffic Selection Algorithm to which this
43         traffic class is to be assigned."
44     REFERENCE
45         "D.2.8.8"
46     ::= { lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithmEntry 2 }
47
48
49 --
50 -- lldpXdot1dcbxAdminETSRecommendationTable - Contains the information
51 -- for the ETS Recommendation TLV.
52 --
53 lldpXdot1dcbxAdminETSReco OBJECT IDENTIFIER ::=
54 { lldpXdot1dcbxAdminData 2 }
55
56 lldpXdot1dcbxAdminETSRecoTrafficClassBandwidthTable OBJECT-TYPE
57     SYNTAX      SEQUENCE OF
58         LldpXdot1dcbxAdminETSRecoTrafficClassBandwidthEntry
59     MAX-ACCESS   not-accessible
60     STATUS       current
61     DESCRIPTION
62         "This table contains one row per traffic class. The
63         entry in each row indicates the traffic class to
64         which the bandwidth is assigned."
65     ::= { lldpXdot1dcbxAdminETSReco 1 }
66
67 lldpXdot1dcbxAdminETSRecoTrafficClassBandwidthEntry OBJECT-TYPE
68     SYNTAX      LldpXdot1dcbxAdminETSRecoTrafficClassBandwidthEntry
69     MAX-ACCESS   not-accessible
70     STATUS       current
71     DESCRIPTION
72         "Indicates a traffic class to Bandwidth assignment."

```



```

1     INDEX          {
2         lldpV2LocPortIfIndex,
3         lldpXdot1dcbxAdminETSRecoTrafficClass
4     }
5     ::= { lldpXdot1dcbxAdminETSRecoTrafficClassBandwidthTable 1 }
6
7 lldpXdot1dcbxAdminETSRecoTrafficClassBandwidthEntry ::= SEQUENCE {
8     lldpXdot1dcbxAdminETSRecoTrafficClass
9     lldpXdot1dcbxTrafficClassValue,
10    lldpXdot1dcbxAdminETSRecoTrafficClassBandwidth
11    lldpXdot1dcbxTrafficClassBandwidthValue
12 }
13
14 lldpXdot1dcbxAdminETSRecoTrafficClass OBJECT-TYPE
15     SYNTAX          LldpXdot1dcbxTrafficClassValue
16     MAX-ACCESS      not-accessible
17     STATUS          current
18     DESCRIPTION
19         "Indicates the traffic class to
20         which this bandwidth applies."
21     REFERENCE
22         "D.2.9.4"
23     ::= { lldpXdot1dcbxAdminETSRecoTrafficClassBandwidthEntry 1 }
24
25 lldpXdot1dcbxAdminETSRecoTrafficClassBandwidth OBJECT-TYPE
26     SYNTAX          LldpXdot1dcbxTrafficClassBandwidthValue
27     MAX-ACCESS      read-write
28     STATUS          current
29     DESCRIPTION
30         "Indicates the bandwidth assigned to this traffic class.
31         The sum of the bandwidths assigned to a given port is
32         required at all times to equal 100. An operation that
33         attempts to change this table such that the bandwidth
34         entries do not total 100 shall be rejected. An implication
35         of this is that modification of this table requires that
36         multiple set operations be included in a single SNMP PDU,
37         commonly referred to as an MSET operation, to perform
38         simultaneous set operations to keep the sum at 100. Any
39         attempt to change a single entry in this table will result
40         in the operation being rejected since entries in the
41         table referring to the given port will no longer
42         sum to 100."
43
44     REFERENCE
45         "D.2.9.4"
46     ::= { lldpXdot1dcbxAdminETSRecoTrafficClassBandwidthEntry 2 }
47
48 lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmTable OBJECT-TYPE
49     SYNTAX          SEQUENCE OF
50         LldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmEntry
51     MAX-ACCESS      not-accessible
52     STATUS          current
53     DESCRIPTION
54         "This table contains one row per traffic class. The entry
55         in each row indicates the traffic selection algorithm to
56         be used by the traffic class."
57     ::= { lldpXdot1dcbxAdminETSReco 2 }
58
59 lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmEntry OBJECT-TYPE
60     SYNTAX          LldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmEntry
61     MAX-ACCESS      not-accessible
62     STATUS          current
63     DESCRIPTION
64         "Indicates a traffic class to traffic selection
65         algorithm assignment."
66     INDEX          {
67         lldpV2LocPortIfIndex,
68         lldpXdot1dcbxAdminETSRecoTSATrafficClass
69     }
70     ::= { lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmTable 1 }
71
72 lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmEntry ::= SEQUENCE {

```

```

1  lldpXdot1dcbxAdminETSRecoTSATrafficClass
2      LldpXdot1dcbxTrafficClassValue,
3  lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithm
4      LldpXdot1dcbxTrafficSelectionAlgorithm
5 }
6
7 lldpXdot1dcbxAdminETSRecoTSATrafficClass OBJECT-TYPE
8     SYNTAX      LldpXdot1dcbxTrafficClassValue
9     MAX-ACCESS   not-accessible
10    STATUS      current
11    DESCRIPTION
12        "Indicates the traffic class that is assigned to a traffic
13        selection algorithm."
14    REFERENCE
15        "D.2.9.5"
16    ::= { lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmEntry 1 }
17
18 lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithm OBJECT-TYPE
19     SYNTAX      LldpXdot1dcbxTrafficSelectionAlgorithm
20     MAX-ACCESS   read-write
21     STATUS      current
22     DESCRIPTION
23         "Indicates the Traffic Selection Algorithm to which this
24         traffic class is to be assigned."
25     REFERENCE
26         "D.2.9.5"
27     ::= { lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithmEntry 2 }
28
29 --
30 -- lldpXdot1dcbxAdminPFCTable - Contains the information for the PFC
31 -- Configuration TLV.
32 --
33 lldpXdot1dcbxAdminPFC OBJECT IDENTIFIER ::= { lldpXdot1dcbxAdminData 3 }
34
35 lldpXdot1dcbxAdminPFCBasicTable OBJECT-TYPE
36     SYNTAX      SEQUENCE OF LldpXdot1dcbxAdminPFCBasicEntry
37     MAX-ACCESS   not-accessible
38     STATUS      current
39     DESCRIPTION
40         "This table contains one row per port for the IEEE 802.1
41         organizationally defined LLDP PFC TLV on the local
42         system known to this agent."
43     ::= { lldpXdot1dcbxAdminPFC 1 }
44
45 lldpXdot1dcbxAdminPFCBasicEntry OBJECT-TYPE
46     SYNTAX      LldpXdot1dcbxAdminPFCBasicEntry
47     MAX-ACCESS   not-accessible
48     STATUS      current
49     DESCRIPTION
50         "Information about the IEEE 802.1 organizational defined
51         PFC TLV LLDP extension."
52     INDEX      { lldpV2LocPortIfIndex }
53     ::= { lldpXdot1dcbxAdminPFCBasicTable 1 }
54
55 lldpXdot1dcbxAdminPFCBasicEntry ::= SEQUENCE {
56     lldpXdot1dcbxAdminPFCWilling      TruthValue,
57     lldpXdot1dcbxAdminPFCMBC          TruthValue,
58     lldpXdot1dcbxAdminPFCCap          LldpXdot1dcbxSupportedCapacity
59 }
60
61 lldpXdot1dcbxAdminPFCWilling OBJECT-TYPE
62     SYNTAX      TruthValue
63     MAX-ACCESS   read-write
64     STATUS      current
65     DESCRIPTION
66         "Indicates if the local system is willing to accept the
67         PFC configuration of the remote system."
68     REFERENCE
69         "D.2.10.3"
70     DEFVAL      { false }
71     ::= { lldpXdot1dcbxAdminPFCBasicEntry 1 }
72

```

```

1 lldpXdot1dcbxAdminPFCMBC OBJECT-TYPE
2     SYNTAX      TruthValue
3     MAX-ACCESS  read-only
4     STATUS      current
5     DESCRIPTION
6         "Indicates if the local system is capable of bypassing
7         MACsec processing when MACsec is disabled."
8     REFERENCE
9         "D.2.10.4"
10    ::= { lldpXdot1dcbxAdminPFCBasicEntry 2}
11
12 lldpXdot1dcbxAdminPFCCap OBJECT-TYPE
13     SYNTAX      LldpXdot1dcbxSupportedCapacity
14     MAX-ACCESS  read-only
15     STATUS      current
16     DESCRIPTION
17         "Indicates the number of traffic classes on the local device
18         that may simultaneously have PFC enabled.
19
20         Note that this typically indicates a physical limitation of the
21         device. However, some devices may allow this parameter to be
22         administratively configured, in which case the MAX-ACCESS
23         should be changed to read-write with and an appropriate
24         DEFVAL added."
25     REFERENCE
26         "D.2.10.5"
27    ::= { lldpXdot1dcbxAdminPFCBasicEntry 3}
28
29 lldpXdot1dcbxAdminPFCEnableTable OBJECT-TYPE
30     SYNTAX      SEQUENCE OF LldpXdot1dcbxAdminPFCEnableEntry
31     MAX-ACCESS  not-accessible
32     STATUS      current
33     DESCRIPTION
34         "This table contains eight entries, one entry per priority,
35         indicating if PFC is enabled on the corresponding priority."
36    ::= { lldpXdot1dcbxAdminPFC 2 }
37
38 lldpXdot1dcbxAdminPFCEnableEntry OBJECT-TYPE
39     SYNTAX      LldpXdot1dcbxAdminPFCEnableEntry
40     MAX-ACCESS  not-accessible
41     STATUS      current
42     DESCRIPTION
43         "Each entry indicates if PFC is enabled on the
44         corresponding priority."
45     INDEX {
46         lldpV2LocPortIfIndex,
47         lldpXdot1dcbxAdminPFCEnablePriority
48     }
49    ::= { lldpXdot1dcbxAdminPFCEnableTable 1 }
50
51 lldpXdot1dcbxAdminPFCEnableEntry ::= SEQUENCE {
52     lldpXdot1dcbxAdminPFCEnablePriority    IEEE8021PriorityValue,
53     lldpXdot1dcbxAdminPFCEnableEnabled    TruthValue
54 }
55
56 lldpXdot1dcbxAdminPFCEnablePriority OBJECT-TYPE
57     SYNTAX      IEEE8021PriorityValue
58     MAX-ACCESS  not-accessible
59     STATUS      current
60     DESCRIPTION
61         "Priority for which PFC is enabled / disabled"
62    ::= { lldpXdot1dcbxAdminPFCEnableEntry 1 }
63
64 lldpXdot1dcbxAdminPFCEnableEnabled OBJECT-TYPE
65     SYNTAX      TruthValue
66     MAX-ACCESS  read-write
67     STATUS      current
68     DESCRIPTION
69         "Indicates if PFC is enabled on the corresponding priority."
70     REFERENCE
71         "D.2.10.6"
72     DEFVAL      { false }

```

```

1 ::= { lldpXdot1dcbxAdminPFCEnableEntry 2 }
2
3 --
4 -- lldpXdot1dcbxAdminApplicationPriorityTable - Contains the
5 -- information for the Application Priority TLV.
6 --
7
8 lldpXdot1dcbxAdminApplicationPriorityAppTable OBJECT-TYPE
9     SYNTAX      SEQUENCE OF
10        LldpXdot1dcbxAdminApplicationPriorityAppEntry
11     MAX-ACCESS  not-accessible
12     STATUS      current
13     DESCRIPTION
14         "Table containing entries indicating the priority to be used
15         for a given application."
16     ::= { lldpXdot1dcbxAdminData 4 }
17
18 lldpXdot1dcbxAdminApplicationPriorityAppEntry OBJECT-TYPE
19     SYNTAX      LldpXdot1dcbxAdminApplicationPriorityAppEntry
20     MAX-ACCESS  not-accessible
21     STATUS      current
22     DESCRIPTION
23         "Entry that indicates the priority to be used for a
24         given application."
25     INDEX      {
26         lldpV2LocPortIfIndex,
27         lldpXdot1dcbxAdminApplicationPriorityAESelector,
28         lldpXdot1dcbxAdminApplicationPriorityAEProtocol
29     }
30     ::= { lldpXdot1dcbxAdminApplicationPriorityAppTable 1 }
31
32 LldpXdot1dcbxAdminApplicationPriorityAppEntry ::= SEQUENCE {
33     lldpXdot1dcbxAdminApplicationPriorityAESelector
34         LldpXdot1dcbxAppSelector,
35     lldpXdot1dcbxAdminApplicationPriorityAEProtocol
36         LldpXdot1dcbxAppProtocol,
37     lldpXdot1dcbxAdminApplicationPriorityAEPriority
38         IEEE8021PriorityValue
39 }
40
41 lldpXdot1dcbxAdminApplicationPriorityAESelector OBJECT-TYPE
42     SYNTAX      LldpXdot1dcbxAppSelector
43     MAX-ACCESS  not-accessible
44     STATUS      current
45     DESCRIPTION
46         "Indicates the contents of the protocol object
47         (lldpXdot1dcbxAdminApplicationPriorityAEProtocol)
48         1: EtherType
49         2: Well Known Port number over TCP, or SCTP
50         3: Well Known Port number over UDP, or DCCP
51         4: Well Known Port number over TCP, SCTP, UDP, and DCCP
52         5: Differentiated Services Code Point (DSCP) value. The
53         6 bit DSCP value is stored in the low order 6 bits of the
54         protocol object. The higher order bits are set to zero.
55         (See IETF RFC 2474 for the definition of the DSCP value.)"
56     REFERENCE
57         "D.2.10.6"
58     ::= { lldpXdot1dcbxAdminApplicationPriorityAppEntry 1 }
59
60 lldpXdot1dcbxAdminApplicationPriorityAEProtocol OBJECT-TYPE
61     SYNTAX      LldpXdot1dcbxAppProtocol
62     MAX-ACCESS  not-accessible
63     STATUS      current
64     DESCRIPTION
65         "The protocol indicator of the type indicated by
66         lldpXdot1dcbxAdminApplicationPriorityAESelector."
67     REFERENCE
68         "D.2.10.6"
69     ::= { lldpXdot1dcbxAdminApplicationPriorityAppEntry 2 }
70
71 lldpXdot1dcbxAdminApplicationPriorityAEPriority OBJECT-TYPE
72     SYNTAX      IEEE8021PriorityValue

```

```

1  MAX-ACCESS      read-create
2  STATUS          current
3  DESCRIPTION
4      "The priority code point that should be used in
5      frames transporting the protocol indicated by
6      lldpXdotlcbxAdminApplicationPriorityAESelector and
7      lldpXdotlcbxAdminApplicationPriorityAEProtocol."
8  REFERENCE
9      "D.2.10.6"
10 ::= { lldpXdotlcbxAdminApplicationPriorityAppEntry 3 }
11
12 --
13 -- lldpXdotlcbxAdminApplicationVlanAppTable - Contains the
14 -- information for the Application VLAN TLV.
15 --
16
17 lldpXdotlcbxAdminApplicationVlanAppTable OBJECT-TYPE
18     SYNTAX      SEQUENCE OF
19                 LldpXdotlcbxAdminApplicationVlanAppEntry
20     MAX-ACCESS   not-accessible
21     STATUS       current
22     DESCRIPTION
23         "Table containing entries indicating the VLAN to be used
24         for a given application."
25     ::= { lldpXdotlcbxAdminData 5 }
26
27 lldpXdotlcbxAdminApplicationVlanAppEntry OBJECT-TYPE
28     SYNTAX      LldpXdotlcbxAdminApplicationVlanAppEntry
29     MAX-ACCESS   not-accessible
30     STATUS       current
31     DESCRIPTION
32         "Entry that indicates the VLAN to be used for a
33         given application."
34     INDEX
35         {
36             lldpV2LocPortIfIndex,
37             lldpXdotlcbxAdminApplicationVlanAESelector,
38             lldpXdotlcbxAdminApplicationVlanAEProtocol
39         }
40     ::= { lldpXdotlcbxAdminApplicationVlanAppTable 1 }
41
42 lldpXdotlcbxAdminApplicationVlanAppEntry ::= SEQUENCE {
43     lldpXdotlcbxAdminApplicationVlanAESelector
44     LldpXdotlcbxAppSelector,
45     lldpXdotlcbxAdminApplicationVlanAEProtocol
46     LldpXdotlcbxAppProtocol,
47     lldpXdotlcbxAdminApplicationVlanAEVlanId
48     VlanId
49 }
50
51 lldpXdotlcbxAdminApplicationVlanAESelector OBJECT-TYPE
52     SYNTAX      LldpXdotlcbxAppSelector
53     MAX-ACCESS   not-accessible
54     STATUS       current
55     DESCRIPTION
56         "Indicates the contents of the protocol object
57         (lldpXdotlcbxAdminApplicationVlanAEProtocol)
58         1: EtherType
59         2: Well Known Port number over TCP, or SCTP
60         3: Well Known Port number over UDP, or DCCP
61         4: Well Known Port number over TCP, SCTP, UDP, and DCCP
62         5: Differentiated Services Code Point (DSCP) value. The
63         6 bit DSCP value is stored in the low order 6 bits of the
64         protocol object. The higher order bits are set to zero.
65         (See IETF RFC 2474 for the definition of the DSCP value.)"
66     REFERENCE
67         "D.2.12.3"
68     ::= { lldpXdotlcbxAdminApplicationVlanAppEntry 1 }
69
70 lldpXdotlcbxAdminApplicationVlanAEProtocol OBJECT-TYPE
71     SYNTAX      LldpXdotlcbxAppProtocol
72     MAX-ACCESS   not-accessible
73     STATUS       current

```

```

1  DESCRIPTION
2      "The protocol indicator of the type indicated by
3      lldpXdot1dcbxAdminApplicationVlanAESelector."
4  REFERENCE
5      "D.2.14.3"
6      ::= { lldpXdot1dcbxAdminApplicationVlanAppEntry 2 }
7
8  lldpXdot1dcbxAdminApplicationVlanAEVlanId OBJECT-TYPE
9      SYNTAX      VlanId
10     MAX-ACCESS   read-create
11     STATUS       current
12     DESCRIPTION
13         "The VLAN Identifier that should be used in
14         frames transporting the protocol indicated by
15         lldpXdot1dcbxAdminApplicationVlanAESelector and
16         lldpXdot1dcbxAdminApplicationVlanAEProtocol."
17     REFERENCE
18         "D.2.14.3"
19     ::= { lldpXdot1dcbxAdminApplicationVlanAppEntry 3 }
20
21 -----
22 -- IEEE 802.1 - DCBX Conformance Information
23 -----
24 lldpXdot1dcbxConformance OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 6 }
25 lldpXdot1dcbxCompliances
26     OBJECT IDENTIFIER ::= { lldpXdot1dcbxConformance 1 }
27 lldpXdot1dcbxGroups
28     OBJECT IDENTIFIER ::= { lldpXdot1dcbxConformance 2 }
29
30 --
31 -- Compliance Statements
32 --
33
34 lldpXdot1dcbxCompliance MODULE-COMPLIANCE
35     STATUS       current
36     DESCRIPTION
37         "A compliance statement for SNMP entities that implement
38         the IEEE 802.1 organizationally defined DCBX group in the
39         LLDP extension MIB.
40
41         This group is mandatory for agents that implement Enhanced
42         Transmission Selection."
43     MODULE      -- this module
44         MANDATORY-GROUPS { lldpXdot1dcbxETSGroup,
45                             lldpXdot1dcbxPFCGroup,
46                             lldpXdot1dcbxApplicationPriorityGroup,
47                             lldpXdot1dcbxApplicationVlanGroup,
48                             ifGeneralInformationGroup
49                         }
50     ::= { lldpXdot1dcbxCompliances 1 }
51
52 --
53 -- MIB Groupings
54 --
55
56 lldpXdot1dcbxETSGroup OBJECT-GROUP
57     OBJECTS {
58         lldpXdot1dcbxConfigETSConfigurationTxEnable,
59         lldpXdot1dcbxConfigETSRecommendationTxEnable,
60         lldpXdot1dcbxLocETSConCreditBasedShaperSupport,
61         lldpXdot1dcbxLocETSConTrafficClassesSupported,
62         lldpXdot1dcbxLocETSConWilling,
63         lldpXdot1dcbxLocETSConPriTrafficClass,
64         lldpXdot1dcbxLocETSConTrafficClassBandwidth,
65         lldpXdot1dcbxLocETSConTrafficSelectionAlgorithm,
66         lldpXdot1dcbxLocETSRecoTrafficClassBandwidth,
67         lldpXdot1dcbxLocETSRecoTrafficSelectionAlgorithm,
68         lldpXdot1dcbxRemETSConCreditBasedShaperSupport,
69         lldpXdot1dcbxRemETSConTrafficClassesSupported,
70         lldpXdot1dcbxRemETSConWilling,
71         lldpXdot1dcbxRemETSConPriTrafficClass,
72         lldpXdot1dcbxRemETSConTrafficClassBandwidth,

```

```

1      lldpXdot1dcbxRemETSConTrafficSelectionAlgorithm,
2      lldpXdot1dcbxRemETSRecoTrafficClassBandwidth,
3      lldpXdot1dcbxRemETSRecoTrafficSelectionAlgorithm,
4      lldpXdot1dcbxAdminETSConCreditBasedShaperSupport,
5      lldpXdot1dcbxAdminETSConTrafficClassesSupported,
6      lldpXdot1dcbxAdminETSConWilling,
7      lldpXdot1dcbxAdminETSConPriTrafficClass,
8      lldpXdot1dcbxAdminETSConTrafficClassBandwidth,
9      lldpXdot1dcbxAdminETSConTrafficSelectionAlgorithm,
10     lldpXdot1dcbxAdminETSRecoTrafficClassBandwidth,
11     lldpXdot1dcbxAdminETSRecoTrafficSelectionAlgorithm
12 }
13 STATUS current
14 DESCRIPTION
15     "The collection of objects used for Enhanced
16     Transmission Selection."
17 ::= { lldpXdot1dcbxGroups 1 }
18
19 lldpXdot1dcbxPFCGroup OBJECT-GROUP
20 OBJECTS {
21     lldpXdot1dcbxConfigPFCTxEnable,
22     lldpXdot1dcbxLocPFCWilling,
23     lldpXdot1dcbxLocPFCMBC,
24     lldpXdot1dcbxLocPFCCap,
25     lldpXdot1dcbxLocPFCEnableEnabled,
26     lldpXdot1dcbxRemPFCWilling,
27     lldpXdot1dcbxRemPFCMBC,
28     lldpXdot1dcbxRemPFCCap,
29     lldpXdot1dcbxRemPFCEnableEnabled,
30     lldpXdot1dcbxAdminPFCWilling,
31     lldpXdot1dcbxAdminPFCMBC,
32     lldpXdot1dcbxAdminPFCCap,
33     lldpXdot1dcbxAdminPFCEnableEnabled
34 }
35 STATUS current
36 DESCRIPTION
37     "The collection of objects used for Priority-
38     base Flow Control."
39 ::= { lldpXdot1dcbxGroups 2 }
40
41 lldpXdot1dcbxApplicationPriorityGroup OBJECT-GROUP
42 OBJECTS {
43     lldpXdot1dcbxConfigApplicationPriorityTxEnable,
44     lldpXdot1dcbxLocApplicationPriorityAEPriority,
45     lldpXdot1dcbxRemApplicationPriorityAEPriority,
46     lldpXdot1dcbxAdminApplicationPriorityAEPriority
47 }
48 STATUS current
49 DESCRIPTION
50     "The collection of objects used for Application
51     priority."
52 ::= { lldpXdot1dcbxGroups 3 }
53
54 lldpXdot1dcbxApplicationVlanGroup OBJECT-GROUP
55 OBJECTS {
56     lldpXdot1dcbxConfigApplicationVlanTxEnable,
57     lldpXdot1dcbxLocApplicationVlanAEVlanId,
58     lldpXdot1dcbxRemApplicationVlanAEVlanId,
59     lldpXdot1dcbxAdminApplicationVlanAEVlanId
60 }
61 STATUS current
62 DESCRIPTION
63     "The collection of objects used for Application
64     VLAN."
65 ::= { lldpXdot1dcbxGroups 4 }
66
67 -----
68 -----
69 -----
70 --
71 -- Organizationally Defined Information Extension - IEEE 802.1
72 -- Definitions to support the ciSet TLV set (Table D-1)

```

```

1 -- for Congestion Isolation
2 --
3 -----
4 -----
5
6 lldpXdot1CiMIB OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 8 }
7 lldpXdot1CiObjects OBJECT IDENTIFIER ::= { lldpXdot1CiMIB 1 }
8
9 -- CI 802.1 MIB Extension groups
10
11 lldpXdot1CiConfig OBJECT IDENTIFIER ::= { lldpXdot1CiObjects 1 }
12 lldpXdot1CiLocalData OBJECT IDENTIFIER ::= { lldpXdot1CiObjects 2 }
13 lldpXdot1CiRemoteData OBJECT IDENTIFIER ::= { lldpXdot1CiObjects 3 }
14
15 -----
16 -- Textual conventions for Congestion Isolation
17 -----
18
19 CiQueueType ::= TEXTUAL-CONVENTION
20     STATUS      current
21     DESCRIPTION
22         "Indicates whether a queue is a monitored queue, a congesting
23         queue or not participating in Congestion Isolation. The
24         queue type is used when building the queue map field of the
25         Congestion Isolation TLV.
26         0: The queue is not participating in Congestion Isolation
27         1: The queue is a monitored queue
28         2: The queue is a congesting queue"
29     SYNTAX      INTEGER {
30         ciQNone(0),
31         ciQMonitor(1),
32         ciQCongesting(2)
33     }
34
35 -----
36 -- IEEE 802.1 - Congestion Isolation Configuration
37 -----
38
39 --
40 -- lldpXdot1CiConfigCiTable : configure the
41 -- transmission of the Congestion Isolation TLV on a set of ports
42 --
43
44 lldpXdot1CiConfigCiTable OBJECT-TYPE
45     SYNTAX      SEQUENCE OF LldpXdot1CiConfigCiEntry
46     MAX-ACCESS   not-accessible
47     STATUS      current
48     DESCRIPTION
49         "A table that controls selection of Congestion Isolation
50         TLVs to be transmitted on individual ports."
51     ::= { lldpXdot1CiConfig 1 }
52
53 lldpXdot1CiConfigCiEntry OBJECT-TYPE
54     SYNTAX      LldpXdot1CiConfigCiEntry
55     MAX-ACCESS   not-accessible
56     STATUS      current
57     DESCRIPTION
58         "LLDP configuration information that controls the
59         transmission of IEEE 802.1 organizationally defined
60         Congestion Isolation TLV on LLDP transmission-capable ports.
61
62         This configuration object augments the lldpV2PortConfigEntry of
63         the LLDP-MIB, therefore it is only present along with the port
64         configuration defined by the associated lldpV2PortConfigEntry
65         entry.
66
67         Each active lldpConfigEntry is restored from non-volatile
68         storage (along with the corresponding lldpV2PortConfigEntry)
69         after a re-initialization of the management system."
70     AUGMENTS     { lldpV2PortConfigEntry }
71     ::= { lldpXdot1CiConfigCiTable 1 }
72

```



```

1 LldpXdot1CiConfigCiEntry ::= SEQUENCE {
2   lldpXdot1CiConfigCiTxEnable TruthValue
3 }
4
5 lldpXdot1CiConfigCiTxEnable OBJECT-TYPE
6     SYNTAX      TruthValue
7     MAX-ACCESS   read-write
8     STATUS       current
9     DESCRIPTION
10        "The lldpXdot1CiConfigCiTxEnable, which is
11        defined as a truth value and configured by the network
12        management, determines whether the IEEE 802.1 organizationally
13        defined Congestion Isolation TLV transmission is allowed
14        on a given LLDP transmission-capable port.
15
16        The value of this object is restored from non-volatile
17        storage after a re-initialization of the management system."
18    REFERENCE
19        "D.2.15"
20    DEFVAL       { false }
21    ::= { lldpXdot1CiConfigCiEntry 1 }
22
23 -----
24 -- IEEE 802.1 - Congestion Isolation Local System Information
25 -----
26
27 ---
28 ---
29 --- lldpXdot1CiLocalData: Contains the information for the CI TLV
30 ---
31 ---
32 lldpXdot1LocCiBasicTable OBJECT-TYPE
33     SYNTAX      SEQUENCE OF LldpXdot1LocCiBasicEntry
34     MAX-ACCESS   not-accessible
35     STATUS       current
36     DESCRIPTION
37        "This table contains one row per port of basic Congestion Isolation
38        information (as a part of the LLDP 802.1 organizational extension)
39        on the local system known to this agent."
40    ::= { lldpXdot1CiLocalData 1 }
41
42 lldpXdot1LocCiBasicEntry OBJECT-TYPE
43     SYNTAX      LldpXdot1LocCiBasicEntry
44     MAX-ACCESS   not-accessible
45     STATUS       current
46     DESCRIPTION
47        "Basic Congestion Isolation information about a particular port
48        component."
49     INDEX       { lldpV2LocPortIfIndex }
50    ::= { lldpXdot1LocCiBasicTable 1 }
51
52 LldpXdot1LocCiBasicEntry ::= SEQUENCE {
53     lldpXdot1LocCiCIMEncapLen      Unsigned32,
54     lldpXdot1LocCiMacAddress       MacAddress,
55     lldpXdot1LocCiNetAddressType   AddressFamilyNumbers,
56     lldpXdot1LocCiNetAddress       OCTET STRING,
57     lldpXdot1LocCiUDPPort          InetPortNumber
58 }
59
60 lldpXdot1LocCiCIMEncapLen OBJECT-TYPE
61     SYNTAX      Unsigned32 (60..512)
62     MAX-ACCESS   read-only
63     STATUS       current
64     DESCRIPTION
65        "The requested length in octets of the data encapsulated into
66        a CIM by a peer. The default and minimum required value is 60.
67        The maximum value is 512."
68     REFERENCE
69        "D.2.15.4"
70    ::= { lldpXdot1LocCiBasicEntry 1 }
71
72 lldpXdot1LocCiMacAddress OBJECT-TYPE

```

```

1  SYNTAX      MacAddress
2  MAX-ACCESS  read-only
3  STATUS      current
4  DESCRIPTION
5      "The MAC address that is to be used as the destination MAC address
6      for a CIM to the Bridge or end station."
7  REFERENCE
8      "D.2.15.5"
9      ::= { lldpXdot1LocCiBasicEntry 2 }
10
11 lldpXdot1LocCiNetAddressType OBJECT-TYPE
12     SYNTAX      AddressFamilyNumbers
13     MAX-ACCESS  read-only
14     STATUS      current
15     DESCRIPTION
16         "The type of network address identifier encoding used in the
17         associated 'lldpXdot1LocCiNetAddress' object. Only the IPv4
18         and IPv6 address type encodings enumerated in AddressFamilyNumbers
19         are used in the Congestion Isolation TLV."
20     REFERENCE
21         "D.2.15.7"
22     ::= { lldpXdot1LocCiBasicEntry 3 }
23
24 lldpXdot1LocCiNetAddress OBJECT-TYPE
25     SYNTAX      OCTET STRING (SIZE(1..31))
26     MAX-ACCESS  read-only
27     STATUS      current
28     DESCRIPTION
29         "The string value used to identify the network address component
30         associated with the local system. This address is to be used as
31         the destination network address of a layer-3 CIM."
32     REFERENCE
33         "D.2.15.8"
34     ::= { lldpXdot1LocCiBasicEntry 4 }
35
36 lldpXdot1LocCiUDPPort OBJECT-TYPE
37     SYNTAX      InetPortNumber
38     MAX-ACCESS  read-only
39     STATUS      current
40     DESCRIPTION
41         "The UDP port number to be used in layer 3 CIMS sent by the peer."
42     REFERENCE
43         "D.2.15.6"
44     ::= { lldpXdot1LocCiBasicEntry 5 }
45
46 --
47 -- The table specifying, the queue map to advertise in the
48 -- Congestion Isolation TLV.
49 --
50
51 lldpXdot1LocCiQueueMapTable OBJECT-TYPE
52     SYNTAX      SEQUENCE OF LldpXdot1LocCiQueueMapEntry
53     MAX-ACCESS  not-accessible
54     STATUS      current
55     DESCRIPTION
56         "A table specifying whether the queue of a priority is to
57         be a monitored queue, congesting queue, or not participating.
58         For each participating queue there is a mapped queue. This
59         information can be transmitted in the Congestion Isolation TLV.
60         "
61     REFERENCE
62         "D.2.15.3"
63     ::= { lldpXdot1CiLocalData 2 }
64
65 lldpXdot1LocCiQueueMapEntry OBJECT-TYPE
66     SYNTAX      LldpXdot1LocCiQueueMapEntry
67     MAX-ACCESS  not-accessible
68     STATUS      current
69     DESCRIPTION
70         "A table specifying whether the queue of a priority is to
71         be a monitored queue, congesting queue, or not participating.
72         For each participating queue there is a mapped queue. This

```

```

1      information can be transmitted in the Congestion Isolation TLV.
2      "
3      INDEX { lldpV2LocPortIfIndex,
4              lldpXdot1LocCiQueueId }
5      ::= { lldpXdot1LocCiQueueMapTable 1 }
6
7      lldpXdot1LocCiQueueMapEntry ::= SEQUENCE {
8          lldpXdot1LocCiQueueId      IEEE8021PriorityValue,
9          lldpXdot1LocCiQueueType    CiQueueType,
10         lldpXdot1LocCiMappedQueue  IEEE8021PriorityValue
11     }
12
13     lldpXdot1LocCiQueueId OBJECT-TYPE
14         SYNTAX      IEEE8021PriorityValue
15         MAX-ACCESS  not-accessible
16         STATUS      current
17         DESCRIPTION
18             "The priority for a frame received on this port."
19         REFERENCE
20             "D.2.15.3"
21         ::= { lldpXdot1LocCiQueueMapEntry 1 }
22
23     lldpXdot1LocCiQueueType OBJECT-TYPE
24         SYNTAX      CiQueueType
25         MAX-ACCESS  read-write
26         STATUS      current
27         DESCRIPTION
28             "monitored, congesting, or not used
29             The value of this object is restored from non-volatile
30             storage after a re-initialization of the management system.
31             "
32         REFERENCE
33             "D.2.15.3"
34         ::= { lldpXdot1LocCiQueueMapEntry 2 }
35
36     lldpXdot1LocCiMappedQueue OBJECT-TYPE
37         SYNTAX      IEEE8021PriorityValue
38         MAX-ACCESS  read-write
39         STATUS      current
40         DESCRIPTION
41             "The remapped priority for frames received on this port
42             at the received priority.
43
44             The value of this object is restored from non-volatile
45             storage after a re-initialization of the management system.
46             "
47         REFERENCE
48             "D.2.15.3"
49         ::= { lldpXdot1LocCiQueueMapEntry 3 }
50
51 -----
52 -- IEEE 802.1 - Congestion Isolation Remote System Information
53 -----
54
55 ---
56 ---
57 --- lldpXdot1CiRemoteData: Contains the received information for the CI TLV
58 ---
59 ---
60     lldpXdot1RemCiBasicTable OBJECT-TYPE
61         SYNTAX      SEQUENCE OF LldpXdot1RemCiBasicEntry
62         MAX-ACCESS  not-accessible
63         STATUS      current
64         DESCRIPTION
65             "This table contains one row per port of basic Congestion Isolation
66             information (as a part of the LLDP 802.1 organizational extension)
67             of the remote system."
68         ::= { lldpXdot1CiRemoteData 1 }
69
70     lldpXdot1RemCiBasicEntry OBJECT-TYPE
71         SYNTAX      LldpXdot1RemCiBasicEntry
72         MAX-ACCESS  not-accessible

```

```

1  STATUS      current
2  DESCRIPTION
3      "Basic Congestion Isolation information about a particular port
4      component."
5  INDEX      { llDPV2RemTimeMark,
6              llDPV2RemLocalIfIndex,
7              llDPV2RemLocalDestMACAddress,
8              llDPV2RemIndex }
9  ::= { llDPXdot1RemCiBasicTable 1 }
10
11 llDPXdot1RemCiBasicEntry ::= SEQUENCE {
12     llDPXdot1RemCiCIMEncapLen      Unsigned32,
13     llDPXdot1RemCiMacAddress       MacAddress,
14     llDPXdot1RemCiNetAddressType   AddressFamilyNumbers,
15     llDPXdot1RemCiNetAddress       OCTET STRING,
16     llDPXdot1RemCiUDPPort          InetPortNumber
17 }
18
19 llDPXdot1RemCiCIMEncapLen OBJECT-TYPE
20     SYNTAX      Unsigned32 (60..512)
21     MAX-ACCESS  read-only
22     STATUS      current
23     DESCRIPTION
24         "The requested length in octets of the data encapsulated into
25         a CIM by a peer. The default and minimum required value is 60.
26         The maximum value is 512."
27     REFERENCE
28         "D.2.15.4"
29     ::= { llDPXdot1RemCiBasicEntry 1 }
30
31 llDPXdot1RemCiMacAddress OBJECT-TYPE
32     SYNTAX      MacAddress
33     MAX-ACCESS  read-only
34     STATUS      current
35     DESCRIPTION
36         "The MAC address that is to be used as the destination MAC address
37         for a CIM to the Bridge or end station."
38     REFERENCE
39         "D.2.15.5"
40     ::= { llDPXdot1RemCiBasicEntry 2 }
41
42 llDPXdot1RemCiNetAddressType OBJECT-TYPE
43     SYNTAX      AddressFamilyNumbers
44     MAX-ACCESS  read-only
45     STATUS      current
46     DESCRIPTION
47         "The type of network address identifier encoding used in the
48         associated 'llDPXdot1RemCiNetAddress' object. Only the IPv4
49         and IPv6 address type encodings enumerated in AddressFamilyNumbers
50         are used in the Congestion Isolation TLV."
51     REFERENCE
52         "D.2.15.7"
53     ::= { llDPXdot1RemCiBasicEntry 3 }
54
55 llDPXdot1RemCiNetAddress OBJECT-TYPE
56     SYNTAX      OCTET STRING (SIZE(1..31))
57     MAX-ACCESS  read-only
58     STATUS      current
59     DESCRIPTION
60         "The string value used to identify the network address component
61         associated with the local system. This address is to be used as
62         the destination network address of a layer-3 CIM."
63     REFERENCE
64         "D.2.15.8"
65     ::= { llDPXdot1RemCiBasicEntry 4 }
66
67 llDPXdot1RemCiUDPPort OBJECT-TYPE
68     SYNTAX      InetPortNumber
69     MAX-ACCESS  read-only
70     STATUS      current
71     DESCRIPTION
72         "The UDP port number to be used in layer 3 CIMS sent by the peer."

```

```
1  REFERENCE
2  "D.2.15.6"
3  ::= { lldpXdot1RemCiBasicEntry 5 }
4
5 --
6 -- The table specifying, the queue map to advertise in the
7 -- Congestion Isolation TLV.
8 --
9
10 lldpXdot1RemCiQueueMapTable OBJECT-TYPE
11     SYNTAX      SEQUENCE OF LldpXdot1RemCiQueueMapEntry
12     MAX-ACCESS  not-accessible
13     STATUS      current
14     DESCRIPTION
15         "A table specifying whether the queue of a priority is to
16         be a monitored queue, congesting queue, or not participating.
17         For each participating queue there is a mapped queue. This
18         information can be transmitted in the Congestion Isolation TLV.
19         "
20     REFERENCE
21         "D.2.15.3"
22     ::= { lldpXdot1CiRemoteData 2 }
23
24 lldpXdot1RemCiQueueMapEntry OBJECT-TYPE
25     SYNTAX      LldpXdot1RemCiQueueMapEntry
26     MAX-ACCESS  not-accessible
27     STATUS      current
28     DESCRIPTION
29         "A table specifying whether the queue of a priority is to
30         be a monitored queue, congesting queue, or not participating.
31         For each participating queue there is a mapped queue. This
32         information can be transmitted in the Congestion Isolation TLV.
33         "
34     INDEX      { lldpV2RemTimeMark,
35                 lldpV2RemLocalIfIndex,
36                 lldpV2RemLocalDestMACAddress,
37                 lldpV2RemIndex,
38                 lldpXdot1RemCiQueueId }
39     ::= { lldpXdot1RemCiQueueMapTable 1 }
40
41 lldpXdot1RemCiQueueMapEntry ::= SEQUENCE {
42     lldpXdot1RemCiQueueId      IEEE8021PriorityValue,
43     lldpXdot1RemCiQueueType    CiQueueType,
44     lldpXdot1RemCiMappedQueue  IEEE8021PriorityValue
45 }
46
47 lldpXdot1RemCiQueueId OBJECT-TYPE
48     SYNTAX      IEEE8021PriorityValue
49     MAX-ACCESS  not-accessible
50     STATUS      current
51     DESCRIPTION
52         "The priority for a frame received on this port."
53     REFERENCE
54         "D.2.15.3"
55     ::= { lldpXdot1RemCiQueueMapEntry 1 }
56
57 lldpXdot1RemCiQueueType OBJECT-TYPE
58     SYNTAX      CiQueueType
59     MAX-ACCESS  read-write
60     STATUS      current
61     DESCRIPTION
62         "monitored, congesting, or not used
63         The value of this object is restored from non-volatile
64         storage after a re-initialization of the management system.
65         "
66     REFERENCE
67         "D.2.15.3"
68     ::= { lldpXdot1RemCiQueueMapEntry 2 }
69
70 lldpXdot1RemCiMappedQueue OBJECT-TYPE
71     SYNTAX      IEEE8021PriorityValue
72     MAX-ACCESS  read-write
```

```

1  STATUS      current
2  DESCRIPTION
3      "The remapped priority for frames received on this port
4      at the received priority.
5
6      The value of this object is restored from non-volatile
7      storage after a re-initialization of the management system.
8      "
9  REFERENCE
10     "D.2.15.3"
11     ::= { lldpXdot1RemCiQueueMapEntry 3 }
12
13
14 -----
15 -- IEEE 802.1 - Congestion Isolation Conformance Information
16 -----
17
18 lldpXdot1CiConformance OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 9 }
19
20 lldpXdot1CiCompliances
21     OBJECT IDENTIFIER ::= { lldpXdot1CiConformance 1 }
22 lldpXdot1CiGroups OBJECT IDENTIFIER ::= { lldpXdot1CiConformance 2 }
23
24 --
25 -- Congestion Isolation - Compliance Statements
26 --
27
28 lldpXdot1CiCompliance MODULE-COMPLIANCE
29     STATUS      current
30     DESCRIPTION
31         "A compliance statement for SNMP entities that implement
32         the IEEE 802.1 organizationally defined Congestion
33         Isolation group in the LLDP extension MIB.
34
35         This group is mandatory for agents that implement the
36         Congestion Isolation cnSet TLV set."
37     MODULE      -- this module
38         MANDATORY-GROUPS { lldpXdot1CiGroup,
39                             ifGeneralInformationGroup }
40     ::= { lldpXdot1CiCompliances 1 }
41
42
43 --
44 -- Congestion Isolation - MIB groupings
45 --
46
47 lldpXdot1CiGroup OBJECT-GROUP
48     OBJECTS {
49         lldpXdot1CiConfigCiTxEnable,
50         lldpXdot1LocCiCIMEncapLen,
51         lldpXdot1LocCiMacAddress,
52         lldpXdot1LocCiNetAddressType,
53         lldpXdot1LocCiNetAddress,
54         lldpXdot1LocCiUDPPort,
55         lldpXdot1LocCiQueueType,
56         lldpXdot1LocCiMappedQueue,
57         lldpXdot1RemCiCIMEncapLen,
58         lldpXdot1RemCiMacAddress,
59         lldpXdot1RemCiNetAddressType,
60         lldpXdot1RemCiNetAddress,
61         lldpXdot1RemCiUDPPort,
62         lldpXdot1RemCiQueueType,
63         lldpXdot1RemCiMappedQueue
64     }
65     STATUS      current
66     DESCRIPTION
67         "The collection of objects that support the
68         Congestion Isolation ciSet TLV set."
69     ::= { lldpXdot1CiGroups 1 }
70
71 -----
72 -----

```

```

1 --
2 -- Organizationally Defined Information Extension - IEEE 802.1
3 -- Definitions to support the trSet TLV set (Table D-1)
4 -- for Topology Recognition
5 --
6 -----
7 -----
8
9 lldpXdot1TrMIB OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 10 }
10 lldpXdot1TrObjects OBJECT IDENTIFIER ::= { lldpXdot1TrMIB 1 }
11
12 -- TR 802.1 MIB Extension groups
13
14 lldpXdot1TrConfig OBJECT IDENTIFIER ::= { lldpXdot1TrObjects 1 }
15 lldpXdot1TrLocalData OBJECT IDENTIFIER ::= { lldpXdot1TrObjects 2 }
16 lldpXdot1TrRemoteData OBJECT IDENTIFIER ::= { lldpXdot1TrObjects 3 }
17
18 -----
19 -- Textual conventions for Topology Recognition
20 -----
21
22 LldpXdot1TrDeviceType ::= TEXTUAL-CONVENTION
23     STATUS      current
24     DESCRIPTION
25         "Indicates the type of device for the Topology Recognition
26         algorithm. Used in the Device Type field of the Topology
27         Recognition TLV
28         0: End-station / Server
29         1: Bridge
30         2: Router
31         255: Unknown"
32     SYNTAX      INTEGER {
33         trEndStation(0),
34         trBridge(1),
35         trRouter(2),
36         trUnknown(255)
37     }
38
39 LldpXdot1TrPortOrientation ::= TEXTUAL-CONVENTION
40     STATUS      current
41     DESCRIPTION
42         "Indicates the orientation of the port in the Topology Recognition
43         algorithm. Used in the Port Orientation field of the Topology
44         Recognition TLV
45         0: uplink
46         1: downlink
47         2: crosslink
48         255: Unknown"
49     SYNTAX      INTEGER {
50         trPortUplink(0),
51         trPortDownlink(1),
52         trPortCrosslink(2),
53         trPortUnknown(255)
54     }
55
56 -----
57 -- IEEE 802.1 - Topology Recognition Configuration
58 -----
59
60 --
61 -- lldpXdot1TrConfigTrTable : configure the
62 -- transmission of the Topology Recognition TLV on a set of ports
63 --
64
65 lldpXdot1TrConfigTrTable OBJECT-TYPE
66     SYNTAX      SEQUENCE OF LldpXdot1TrConfigTrEntry
67     MAX-ACCESS  not-accessible
68     STATUS      current
69     DESCRIPTION
70         "A table that controls selection of Topology Recognition
71         TLVs to be transmitted on individual ports."
72     ::= { lldpXdot1TrConfig 1 }

```

```

1
2 lldpXdot1TrConfigTrEntry OBJECT-TYPE
3     SYNTAX          LldpXdot1TrConfigTrEntry
4     MAX-ACCESS      not-accessible
5     STATUS          current
6     DESCRIPTION
7         "LLDP configuration information that controls the
8         transmission of IEEE 802.1 organizationally defined
9         Topology Recognition TLV on LLDP transmission-capable ports.
10
11         This configuration object augments the lldpV2PortConfigEntry of
12         the LLDP-MIB, therefore it is only present along with the port
13         configuration defined by the associated lldpV2PortConfigEntry
14         entry.
15
16         Each active lldpConfigEntry is restored from non-volatile
17         storage (along with the corresponding lldpV2PortConfigEntry)
18         after a re-initialization of the management system."
19     AUGMENTS         { lldpV2PortConfigEntry }
20     ::= { lldpXdot1TrConfigTrTable 1 }
21
22 lldpXdot1TrConfigTrTxEnable ::= SEQUENCE {
23     lldpXdot1TrConfigTrTxEnable TruthValue
24 }
25
26 lldpXdot1TrConfigTrTxEnable OBJECT-TYPE
27     SYNTAX          TruthValue
28     MAX-ACCESS      read-write
29     STATUS          current
30     DESCRIPTION
31         "The lldpXdot1TrConfigTrTxEnable, which is
32         defined as a truth value and configured by the network
33         management, determines whether the IEEE 802.1 organizationally
34         defined Topology Recognition TLV transmission is allowed
35         on a given LLDP transmission-capable port.
36
37         The value of this object is restored from non-volatile
38         storage after a re-initialization of the management system."
39     REFERENCE
40         "D.2.16"
41     DEFVAL          { false }
42     ::= { lldpXdot1TrConfigTrEntry 1 }
43
44 -----
45 -- IEEE 802.1 - Topology Recognition Local System Information
46 -----
47
48 ---
49 ---
50 --- lldpXdot1TrLocalData: Contains the information for the TR TLV
51 ---
52 ---
53 lldpXdot1LocTrTable OBJECT-TYPE
54     SYNTAX          SEQUENCE OF LldpXdot1LocTrEntry
55     MAX-ACCESS      not-accessible
56     STATUS          current
57     DESCRIPTION
58         "This table contains one row per port of Topology Recognition
59         information (as a part of the LLDP 802.1 organizational extension)
60         on the local system known to this agent."
61     ::= { lldpXdot1TrLocalData 1 }
62
63 lldpXdot1LocTrEntry OBJECT-TYPE
64     SYNTAX          LldpXdot1LocTrEntry
65     MAX-ACCESS      not-accessible
66     STATUS          current
67     DESCRIPTION
68         "Topology Recognition information about a particular port
69         component."
70     INDEX          { lldpV2LocPortIfIndex }
71     ::= { lldpXdot1LocTrTable 1 }
72

```



```

1 LldpXdot1LocTrEntry ::= SEQUENCE {
2     lldpXdot1LocTrDeviceType      LldpXdot1TrDeviceType,
3     lldpXdot1LocTrTopoLevel        Unsigned32,
4     lldpXdot1LocTrPortOrientation  LldpXdot1TrPortOrientation
5 }
6
7 lldpXdot1LocTrDeviceType OBJECT-TYPE
8     SYNTAX      LldpXdot1TrDeviceType
9     MAX-ACCESS  read-only
10    STATUS      current
11    DESCRIPTION
12        "Indicates the type of device sending the Topology Recognition TLV.
13        End-stations or servers that are non-relay systems at the edge
14        of the topology use the value trEndStation(0). A layer-2 relay
15        system uses the value trBridge(1) and layer-3 relay systems use
16        the value trRouter(2)."

```

```

1 lldpXdot1RemTrEntry OBJECT-TYPE
2   SYNTAX      LldpXdot1RemTrEntry
3   MAX-ACCESS  not-accessible
4   STATUS      current
5   DESCRIPTION
6       "Basic Topology Recognition information about a particular port
7       component."
8   INDEX      { lldpV2RemTimeMark,
9               lldpV2RemLocalIfIndex,
10              lldpV2RemLocalDestMACAddress,
11              lldpV2RemIndex }
12   ::= { lldpXdot1RemTrTable 1 }
13
14 lldpXdot1RemTrEntry ::= SEQUENCE {
15     lldpXdot1RemTrDeviceType  LldpXdot1TrDeviceType,
16     lldpXdot1RemTrTopoLevel    Unsigned32,
17     lldpXdot1RemTrPortOrientation LldpXdot1TrPortOrientation
18 }
19
20 lldpXdot1RemTrDeviceType OBJECT-TYPE
21   SYNTAX      LldpXdot1TrDeviceType
22   MAX-ACCESS  read-only
23   STATUS      current
24   DESCRIPTION
25       "Indicates the type of device sending the Topology Recognition TLV.
26       End-stations or servers that are non-relay systems at the edge
27       of the topology use the value trEndStation(0). A layer-2 relay
28       system uses the value trBridge(1) and layer-3 relay systems use
29       the value trRouter(2)."
```

REFERENCE

```

31       "D.2.16.3"
32   ::= { lldpXdot1RemTrEntry 1 }
33
34 lldpXdot1RemTrTopoLevel OBJECT-TYPE
35   SYNTAX      Unsigned32 (0..255)
36   MAX-ACCESS  read-only
37   STATUS      current
38   DESCRIPTION
39       "Indicates the system understanding of its current level in
40       the topology. The value of 0 indicates the edge of the topology
41       and the value 255 indicates the level is currently unknown. Other
42       non-zero values between 1 and 254 indicate the minimum number
43       of links between the system and the edge of the topology."
44   REFERENCE
45       "D.2.16.4"
46   ::= { lldpXdot1RemTrEntry 2 }
47
48 lldpXdot1RemTrPortOrientation OBJECT-TYPE
49   SYNTAX      LldpXdot1TrPortOrientation
50   MAX-ACCESS  read-only
51   STATUS      current
52   DESCRIPTION
53       "Indicates whether the port is facing a trUplink(0), trDownlink(1),
54       trCrosslink(2), or the orientation is TrUnknown(255). An uplink is a
55       port that is facing a system that is deeper in the topology (i.e.,
56       has a Topology Level greater than the sending system). A downlink
57       is a port that is facing a system closer to the edge of the topology
58       (i.e., has a Topology Level less than the sending system). A crosslink
59       is a port that is facing a system at the same level of the topology.
60       Systems that are end-stations initialize this value to trUplink(0).
61       Systems that are not end-stations initialize this value to
62       TrUnknown(255)."
```

REFERENCE

```

63       "D.2.16.5"
64   ::= { lldpXdot1RemTrEntry 3 }
65
66 -----
67 -- IEEE 802.1 - Topology Recognition Conformance Information
68 -----
69
70
71 lldpXdot1TrConformance OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 11 }
72

```

```
1 lldpXdot1TrCompliances
2   OBJECT IDENTIFIER ::= { lldpXdot1TrConformance 1 }
3 lldpXdot1TrGroups OBJECT IDENTIFIER ::= { lldpXdot1TrConformance 2 }
4
5 --
6 -- Topology Recognition - Compliance Statements
7 --
8
9 lldpXdot1TrCompliance MODULE-COMPLIANCE
10   STATUS          current
11   DESCRIPTION
12     "A compliance statement for SNMP entities that implement
13     the IEEE 802.1 organizationally defined Topology
14     Recognition group in the LLDP extension MIB.
15
16     This group is mandatory for agents that implement the
17     Topology Recognition cnSet TLV set."
18   MODULE          -- this module
19     MANDATORY-GROUPS { lldpXdot1TrGroup,
20                       ifGeneralInformationGroup }
21   ::= { lldpXdot1TrCompliances 1 }
22
23
24 --
25 -- Topology Recognition - MIB groupings
26 --
27
28 lldpXdot1TrGroup OBJECT-GROUP
29   OBJECTS {
30     lldpXdot1TrConfigTrTxEnable,
31     lldpXdot1LocTrDeviceType,
32     lldpXdot1LocTrTopoLevel,
33     lldpXdot1LocTrPortOrientation,
34     lldpXdot1RemTrDeviceType,
35     lldpXdot1RemTrTopoLevel,
36     lldpXdot1RemTrPortOrientation
37   }
38   STATUS          current
39   DESCRIPTION
40     "The collection of objects that support the
41     Topology Recognition trSet TLV set."
42   ::= { lldpXdot1TrGroups 1 }
43 END
44
45
```

1 D.5.6 EVB extensions to the IEEE 802.1 LLDP extension MIB module

2 In the following MIB definition, should any discrepancy between the DESCRIPTION text and the
3 corresponding definition in D.2.1 through D.5 occur, the definition in D.2.1 through D.5 shall take
4 precedence.

```
5 LLDP-EXT-DOT1-EVB-EXTENSIONS-MIB DEFINITIONS ::= BEGIN
6
7 IMPORTS
8     MODULE-IDENTITY,
9     OBJECT-TYPE
10    FROM SNMPv2-SMI
11    TruthValue
12    FROM SNMPv2-TC
13    MODULE-COMPLIANCE,
14    OBJECT-GROUP
15    FROM SNMPv2-CONF
16    ifGeneralInformationGroup
17    FROM IF-MIB
18    lldpV2LocPortIfIndex,
19    lldpV2RemTimeMark,
20    lldpV2RemLocalIfIndex,
21    lldpV2RemLocalDestMACAddress,
22    lldpV2RemIndex,
23    lldpV2PortConfigEntry
24    FROM LLDP-V2-MIB
25    lldpV2Xdot1MIB
26    FROM LLDP-EXT-DOT1-V2-MIB;
27
28 -- Define the MIB module
29     lldpXdot1EvbExtensions MODULE-IDENTITY
30     LAST-UPDATED "202309210000Z" -- September 21, 2023
31     ORGANIZATION "IEEE 802.1 Working Group"
32     CONTACT-INFO
33         " WG-URL: http://www.ieee802.org/1/
34         WG-E-Mail: stds-802-1-l@ieee.org
35         Contact: IEEE 802.1 Working Group Chair
36         Postal: C/O IEEE 802.1 Working Group
37             IEEE Standards Association
38             445 Hoes Lane
39             Piscataway, NJ 08854
40             USA
41         E-mail: stds-802-1-chairs@ieee.org"
42     DESCRIPTION
43         "The LLDP Management Information Base extension module for
44         IEEE 802.1 organizationally defined discovery information
45         for the EVB extension and PBBN Auto Attach objects.
46
47         This MIB module is rooted under the lldpXdot1StandAloneExtensions
48         OID arc, in order to allow it to be defined independently
49         of other 802.1 LLDP extension MIBs.
50
51         Unless otherwise indicated, the references in this MIB module are to
52         IEEE Std 802.1Q Bridges and Bridged Networks:
53         IEEE Std 802.1Q-2022 as amended by IEEE Std 802.1Qcz, IEEE Std 802.1Qcw,
54         and IEEE Std 802.1Qcj.
55
56         Copyright (C) IEEE (2023).
57         This version of this MIB module is part of IEEE Std 802.1Q;
58         see that standard for full legal notices."
59
60     REVISION "202309210000Z" -- September 21, 2023
61     DESCRIPTION
62         "Published as part of IEEE Std 802.1Qcj.
63         Auto Attach objects added, references updated."
64
65     REVISION "202211080000Z" -- November 8, 2022
66     DESCRIPTION
67         "Published as part of IEEE Std 802.1Q-2022.
68         Cross references and contact information updated."
```

```
1
2 REVISION "201807010000Z" -- July 1, 2018
3 DESCRIPTION
4     "Published as part of IEEE Std 802.1Q 2018 revision.
5     Cross references updated and corrected."
6
7 REVISION "201412150000Z" -- December 15, 2014
8 DESCRIPTION
9     "Published as part of IEEE Std 802.1Q 2014 revision.
10    Cross references updated and corrected."
11
12 REVISION "201202150000Z" -- February 15, 2012
13 DESCRIPTION
14     "Initial version published as part of IEEE Std 802.1Qbg"
15
16 -- Hang this MIB module under the stand-alone extension MIBs arc:
17 ::= { lldpXdot1StandAloneExtensions 1 }
18
19 -- Define the root arc for stand-alone extension MIBs in 802.1
20 lldpXdot1StandAloneExtensions OBJECT IDENTIFIER ::= { lldpV2Xdot1MIB 7 }
21
22 -----
23 -----
24 --
25 -- Organizationally Defined Information Extension - IEEE 802.1
26 -- Definitions to support the evbSet TLV set (Table D-1)
27 -- for Edge Virtual Bridging
28 --
29 -----
30 -----
31
32 lldpXdot1EvbMIB OBJECT IDENTIFIER ::= { lldpXdot1EvbExtensions 1 }
33 lldpXdot1EvbObjects OBJECT IDENTIFIER ::= { lldpXdot1EvbMIB 1 }
34
35 -- EVB 802.1 MIB Extension groups
36
37 lldpXdot1EvbConfig OBJECT IDENTIFIER ::= { lldpXdot1EvbObjects 1 }
38 lldpXdot1EvbLocalData OBJECT IDENTIFIER ::= { lldpXdot1EvbObjects 2 }
39 lldpXdot1EvbRemoteData OBJECT IDENTIFIER ::= { lldpXdot1EvbObjects 3 }
40
41 -----
42 -- IEEE 802.1 - EVB Configuration
43 -----
44 --
45 -- lldpXdot1EvbConfigEvbTable : configure the
46 -- transmission of the EVB TLV on a set of ports
47 --
48 lldpXdot1EvbConfigEvbTable OBJECT-TYPE
49     SYNTAX      SEQUENCE OF LldpXdot1EvbConfigEvbEntry
50     MAX-ACCESS   not-accessible
51     STATUS      current
52     DESCRIPTION
53         "A table that controls selection of EVB
54         TLVs to be transmitted on individual ports."
55     ::= { lldpXdot1EvbConfig 1 }
56
57 lldpXdot1EvbConfigEvbEntry OBJECT-TYPE
58     SYNTAX      LldpXdot1EvbConfigEvbEntry
59     MAX-ACCESS   not-accessible
60     STATUS      current
61     DESCRIPTION
62         "LLDP configuration information that controls the
63         transmission of IEEE 802.1 organizationally defined
64         EVB TLV on LLDP transmission-capable ports.
65
66         This configuration object augments the lldpV2PortConfigEntry of
67         the LLDP-MIB, therefore it is only present along with the port
68         configuration defined by the associated lldpV2PortConfigEntry
69         entry.
70
71         Each active lldpConfigEntry is restored from non-volatile
72         storage (along with the corresponding lldpV2PortConfigEntry)
```

```

1      after a re-initialization of the management system."
2  AUGMENTS      { lldpV2PortConfigEntry }
3  ::= { lldpXdot1EvbConfigEvbTable 1 }
4
5  lldpXdot1EvbConfigEvbEntry ::= SEQUENCE {
6      lldpXdot1EvbConfigEvbTxEnable TruthValue
7  }
8
9  lldpXdot1EvbConfigEvbTxEnable OBJECT-TYPE
10     SYNTAX      TruthValue
11     MAX-ACCESS   read-write
12     STATUS       current
13     DESCRIPTION
14         "The lldpXdot1EvbConfigEvbTxEnable, which is
15         defined as a truth value and configured by the network
16         management, determines whether the IEEE 802.1 organizationally
17         defined EVB TLV transmission is allowed
18         on a given LLDP transmission-capable port.
19
20         The value of this object is restored from non-volatile
21         storage after a re-initialization of the management system."
22     REFERENCE
23         "D.2.12"
24     DEFVAL       { false }
25     ::= { lldpXdot1EvbConfigEvbEntry 1 }
26 --
27 -- lldpXdot1EvbConfigCdcTable : configure the
28 -- transmission of the CDCP TLV on a set of ports
29 --
30 lldpXdot1EvbConfigCdcTable OBJECT-TYPE
31     SYNTAX      SEQUENCE OF LldpXdot1EvbConfigCdcEntry
32     MAX-ACCESS   not-accessible
33     STATUS       current
34     DESCRIPTION
35         "A table that controls selection of EVB
36         TLVs to be transmitted on individual ports."
37     ::= { lldpXdot1EvbConfig 2 }
38
39 lldpXdot1EvbConfigCdcEntry OBJECT-TYPE
40     SYNTAX      LldpXdot1EvbConfigCdcEntry
41     MAX-ACCESS   not-accessible
42     STATUS       current
43     DESCRIPTION
44         "LLDP configuration information that controls the
45         transmission of IEEE 802.1 organizationally defined
46         CDCP TLV on LLDP transmission-capable ports.
47
48         This configuration object augments the lldpV2PortConfigEntry of
49         the LLDP-MIB, therefore it is only present along with the port
50         configuration defined by the associated lldpV2PortConfigEntry
51         entry.
52
53         Each active lldpConfigEntry is restored from non-volatile
54         storage (along with the corresponding lldpV2PortConfigEntry)
55         after a re-initialization of the management system."
56     AUGMENTS      { lldpV2PortConfigEntry }
57     ::= { lldpXdot1EvbConfigCdcTable 1 }
58
59 lldpXdot1EvbConfigCdcEntry ::= SEQUENCE {
60     lldpXdot1EvbConfigCdcTxEnable TruthValue
61 }
62
63 lldpXdot1EvbConfigCdcTxEnable OBJECT-TYPE
64     SYNTAX      TruthValue
65     MAX-ACCESS   read-write
66     STATUS       current
67     DESCRIPTION
68         "The lldpXdot1EvbConfigCdcTxEnable, which is
69         defined as a truth value and configured by the network
70         management, determines whether the IEEE 802.1 organizationally
71         defined CDCP TLV transmission is allowed
72         on a given LLDP transmission-capable port.

```

```

1
2     The value of this object is restored from non-volatile
3     storage after a re-initialization of the management system."
4 REFERENCE
5     "D.2.13"
6 DEFVAL      { false }
7 ::= { lldpXdot1EvbConfigCdcEntry 1 }
8
9 -----
10 -- IEEE 802.1 - EVB Local System Information
11 -----
12 ---
13 --- lldpV2Xdot1LocEvbTlvTable: EVB TLV Information Table
14 ---
15 lldpV2Xdot1LocEvbTlvTable OBJECT-TYPE
16     SYNTAX      SEQUENCE OF LldpV2Xdot1LocEvbTlvEntry
17     MAX-ACCESS  not-accessible
18     STATUS      current
19     DESCRIPTION
20         "This table contains one row per port of EVB
21         TLV information (as a part of the LLDP
22         802.1 organizational extension) on the local system
23         known to this agent."
24     ::= { lldpXdot1EvbLocalData 1 }
25
26 lldpV2Xdot1LocEvbTlvEntry OBJECT-TYPE
27     SYNTAX      LldpV2Xdot1LocEvbTlvEntry
28     MAX-ACCESS  not-accessible
29     STATUS      current
30     DESCRIPTION
31         "EVB TLV information about a
32         particular port component."
33     INDEX      { lldpV2LocPortIfIndex }
34     ::= { lldpV2Xdot1LocEvbTlvTable 1 }
35
36 LldpV2Xdot1LocEvbTlvEntry ::= SEQUENCE {
37     lldpV2Xdot1LocEvbTlvString      OCTET STRING
38 }
39
40 lldpV2Xdot1LocEvbTlvString OBJECT-TYPE
41     SYNTAX      OCTET STRING (SIZE (0..514))
42     MAX-ACCESS  read-only
43     STATUS      current
44     DESCRIPTION
45         "This object contains the EVB TLV information string
46         for the Port, as defined in D.2.12.
47         As the elements within the string are not individually
48         manipulated via SNMP (they are of concern only to the
49         state machines), the sub-structure of the string
50         is not visible as separate objects within the
51         local database."
52     REFERENCE
53         "D.2.12"
54     ::= { lldpV2Xdot1LocEvbTlvEntry 1 }
55 ---
56 --- lldpV2Xdot1LocCdcEntryTable: CDCP TLV Information Table
57 ---
58 lldpV2Xdot1LocCdcEntryTable OBJECT-TYPE
59     SYNTAX      SEQUENCE OF LldpV2Xdot1LocCdcEntry
60     MAX-ACCESS  not-accessible
61     STATUS      current
62     DESCRIPTION
63         "This table contains one row per port of CDCP
64         TLV information (as a part of the LLDP
65         802.1 organizational extension) on the local system
66         known to this agent."
67     ::= { lldpXdot1EvbLocalData 2 }
68
69 lldpV2Xdot1LocCdcEntry OBJECT-TYPE
70     SYNTAX      LldpV2Xdot1LocCdcEntry
71     MAX-ACCESS  not-accessible
72     STATUS      current

```

```

1      DESCRIPTION
2          "CDCP TLV information about a
3          particular port component."
4      INDEX    { lldpV2LocPortIfIndex }
5      ::= { lldpV2Xdot1LocCdcPtlvTable 1 }
6
7      lldpV2Xdot1LocCdcPtlvEntry ::= SEQUENCE {
8          lldpV2Xdot1LocCdcPtlvString    OCTET STRING
9      }
10
11 lldpV2Xdot1LocCdcPtlvString OBJECT-TYPE
12     SYNTAX      OCTET STRING (SIZE(0..514))
13     MAX-ACCESS  read-only
14     STATUS      current
15     DESCRIPTION
16         "This object contains the CDCP TLV information string
17         for the Port, as defined in D.2.13.
18         As the elements within the string are not individually
19         manipulated via SNMP (they are of concern only to the
20         state machines), the sub-structure of the string
21         is not visible as separate objects within the
22         local database."
23     REFERENCE
24         "D.2.13"
25     ::= { lldpV2Xdot1LocCdcPtlvEntry 1 }
26
27 -----
28 -- IEEE 802.1 - EVB Remote System Information
29 -----
30 ---
31 --- lldpV2Xdot1RemEvbTlvTable: EVB TLV Information Table
32 ---
33 lldpV2Xdot1RemEvbTlvTable OBJECT-TYPE
34     SYNTAX      SEQUENCE OF LldpV2Xdot1RemEvbTlvEntry
35     MAX-ACCESS  not-accessible
36     STATUS      current
37     DESCRIPTION
38         "This table contains one row per port of EVB
39         TLV information (as a part of the LLDP
40         802.1 organizational extension) on the remote system
41         known to this agent."
42     ::= { lldpXdot1EvbRemoteData 1 }
43
44 lldpV2Xdot1RemEvbTlvEntry OBJECT-TYPE
45     SYNTAX      LldpV2Xdot1RemEvbTlvEntry
46     MAX-ACCESS  not-accessible
47     STATUS      current
48     DESCRIPTION
49         "EVB TLV information about a
50         particular port component."
51     INDEX    { lldpV2RemTimeMark,
52               lldpV2RemLocalIfIndex,
53               lldpV2RemLocalDestMACAddress,
54               lldpV2RemIndex }
55     ::= { lldpV2Xdot1RemEvbTlvTable 1 }
56
57 lldpV2Xdot1RemEvbTlvEntry ::= SEQUENCE {
58     lldpV2Xdot1RemEvbTlvString    OCTET STRING
59 }
60
61 lldpV2Xdot1RemEvbTlvString OBJECT-TYPE
62     SYNTAX      OCTET STRING (SIZE (0..514))
63     MAX-ACCESS  read-only
64     STATUS      current
65     DESCRIPTION
66         "This object contains the EVB TLV information string
67         for the Port, as defined in D.2.12.
68         As the elements within the string are not individually
69         manipulated via SNMP (they are of concern only to the
70         state machines), the sub-structure of the string
71         is not visible as separate objects within the
72         local database."

```



```

1  REFERENCE
2      "D.2.12"
3  ::= { lldpV2Xdot1RemEvbTlvEntry 1 }
4  ---
5  --- lldpV2Xdot1RemCdcplTlvTable: CDCP TLV Information Table
6  ---
7  lldpV2Xdot1RemCdcplTlvTable OBJECT-TYPE
8      SYNTAX      SEQUENCE OF LldpV2Xdot1RemCdcplTlvEntry
9      MAX-ACCESS   not-accessible
10     STATUS      current
11     DESCRIPTION
12         "This table contains one row per port of CDCP
13         TLV information (as a part of the LLDP
14         802.1 organizational extension) on the remote system
15         known to this agent."
16     ::= { lldpXdot1EvbRemoteData 2 }
17
18 lldpV2Xdot1RemCdcplTlvEntry OBJECT-TYPE
19     SYNTAX      LldpV2Xdot1RemCdcplTlvEntry
20     MAX-ACCESS   not-accessible
21     STATUS      current
22     DESCRIPTION
23         "CDCP TLV information about a
24         particular port component."
25     INDEX      { lldpV2RemTimeMark,
26                 lldpV2RemLocalIfIndex,
27                 lldpV2RemLocalDestMACAddress,
28                 lldpV2RemIndex }
29     ::= { lldpV2Xdot1RemCdcplTlvTable 1 }
30
31 lldpV2Xdot1RemCdcplTlvEntry ::= SEQUENCE {
32     lldpV2Xdot1RemCdcplTlvString      OCTET STRING
33 }
34
35 lldpV2Xdot1RemCdcplTlvString OBJECT-TYPE
36     SYNTAX      OCTET STRING (SIZE (0..514))
37     MAX-ACCESS   read-only
38     STATUS      current
39     DESCRIPTION
40         "This object contains the CDCP TLV information string
41         for the Port, as defined in D.2.13.
42         As the elements within the string are not individually
43         manipulated via SNMP (they are of concern only to the
44         state machines), the sub-structure of the string
45         is not visible as separate objects within the
46         local database."
47     REFERENCE
48         "D.2.13"
49     ::= { lldpV2Xdot1RemCdcplTlvEntry 1 }
50
51 -----
52 -- IEEE 802.1 - EVB Conformance Information
53 -----
54
55 lldpXdot1EvbConformance OBJECT IDENTIFIER ::= { lldpXdot1EvbExtensions 2 }
56
57 lldpXdot1EvbCompliances
58     OBJECT IDENTIFIER ::= { lldpXdot1EvbConformance 1 }
59 lldpXdot1EvbGroups
60     OBJECT IDENTIFIER ::= { lldpXdot1EvbConformance 2 }
61 --
62 -- EVB - Compliance Statements
63 --
64 lldpXdot1EvbCompliance MODULE-COMPLIANCE
65     STATUS      current
66     DESCRIPTION
67         "A compliance statement for SNMP entities that implement
68         the IEEE 802.1 organizationally defined Congestion
69         Notification LLDP extension MIB.
70
71         This group is mandatory for agents that implement the
72         EVB evbSet TLV set."

```

```

1  MODULE          -- this module
2      MANDATORY-GROUPS { lldpXdot1EvbGroup,
3                          ifGeneralInformationGroup }
4  ::= { lldpXdot1EvbCompliances 1 }
5  --
6  -- EVB - MIB groupings
7  --
8  lldpXdot1EvbGroup OBJECT-GROUP
9      OBJECTS {
10         lldpXdot1EvbConfigEvbTxEnable,
11         lldpXdot1EvbConfigCdcpxTxEnable,
12         lldpV2Xdot1LocEvbTlvString,
13         lldpV2Xdot1LocCdcpxTlvString,
14         lldpV2Xdot1RemEvbTlvString,
15         lldpV2Xdot1RemCdcpxTlvString
16     }
17     STATUS current
18     DESCRIPTION
19         "The collection of objects that support the
20         EVB evbSet TLV set."
21     ::= { lldpXdot1EvbGroups 1 }
22
23 -----
24 -----
25 --
26 -- Organizationally Defined Information Extension - IEEE 802.1
27 -- Definitions to support the aaSet TLV set (Table D-1)
28 -- for PBBN Auto Attach
29 --
30 -----
31 -----
32
33 lldpXdot1AaMIB OBJECT IDENTIFIER ::= { lldpXdot1EvbExtensions 3 }
34 lldpXdot1AaObjects OBJECT IDENTIFIER ::= { lldpXdot1AaMIB 1 }
35
36 -- Auto Attach MIB extension groups
37
38 lldpXdot1AaConfig OBJECT IDENTIFIER ::= { lldpXdot1AaObjects 1 }
39 lldpXdot1AaLocalData OBJECT IDENTIFIER ::= { lldpXdot1AaObjects 2 }
40 lldpXdot1AaRemoteData OBJECT IDENTIFIER ::= { lldpXdot1AaObjects 3 }
41
42 -----
43 -- IEEE 802.1 - PBBN Auto Attach configuration
44 -----
45 --
46 -- lldpXdot1AaConfigAaTable : configure the
47 -- transmission of the PBBN Auto Attach TLVs on a set of ports
48 --
49 lldpXdot1AaConfigAaTable OBJECT-TYPE
50     SYNTAX      SEQUENCE OF LldpXdot1AaConfigAaEntry
51     MAX-ACCESS   not-accessible
52     STATUS       current
53     DESCRIPTION
54         "A table that controls selection of PBBN Auto Attach
55         TLVs to be transmitted on individual ports."
56     ::= { lldpXdot1AaConfig 1 }
57
58 lldpXdot1AaConfigAaEntry OBJECT-TYPE
59     SYNTAX      LldpXdot1AaConfigAaEntry
60     MAX-ACCESS   not-accessible
61     STATUS       current
62     DESCRIPTION
63         "LLDP configuration information that controls the
64         transmission of IEEE 802.1 organizationally defined
65         PBBN Auto Attach System and Assignment TLVs on LLDP
66         transmission-capable ports.
67
68         This configuration object augments the lldpV2PortConfigEntry of
69         the LLDP-MIB, therefore it is only present along with the port
70         configuration defined by the associated lldpV2PortConfigEntry
71         entry.
72

```

```

1      Each active lldpConfigEntry is restored from non-volatile
2      storage (along with the corresponding lldpV2PortConfigEntry)
3      after a re-initialization of the management system."
4  AUGMENTS      { lldpV2PortConfigEntry }
5  ::= { lldpXdot1AaConfigAaTable 1 }
6
7  lldpXdot1AaConfigAaEntry ::= SEQUENCE {
8      lldpXdot1AaConfigSysTxEnable TruthValue,
9      lldpXdot1AaConfigAsgnsTxEnable TruthValue
10 }
11
12 lldpXdot1AaConfigSysTxEnable OBJECT-TYPE
13     SYNTAX      TruthValue
14     MAX-ACCESS   read-only
15     STATUS      current
16     DESCRIPTION
17         "The lldpXdot1AaConfigSysTxEnable, which is
18         defined as a truth value and configured by the PBBN Auto Attach
19         state machines, determines whether the IEEE 802.1 organizationally
20         defined PBBN Auto Attach System TLV transmission is allowed
21         on a given LLDP transmission-capable port.
22
23         The value of this object is returned to FALSE after a re-initialization
24         of the management system."
25     REFERENCE
26         "D.2.17"
27     DEFVAL      { false }
28     ::= { lldpXdot1AaConfigAaEntry 1 }
29
30 lldpXdot1AaConfigAsgnsTxEnable OBJECT-TYPE
31     SYNTAX      TruthValue
32     MAX-ACCESS   read-only
33     STATUS      current
34     DESCRIPTION
35         "The lldpXdot1AaConfigAsgnsTxEnable, which is
36         defined as a truth value and configured by the PBBN Auto Attach
37         state machines, determines whether the IEEE 802.1 organizationally
38         defined PBBN Auto Attach Assignment TLV transmission is allowed
39         on a given LLDP transmission-capable port.
40
41         The value of this object is returned to FALSE after a re-initialization
42         of the management system."
43     REFERENCE
44         "D.2.18"
45     DEFVAL      { false }
46     ::= { lldpXdot1AaConfigAaEntry 2 }
47
48 -----
49 -- IEEE 802.1 - PBBN Auto Attach local system information
50 -----
51 ---
52 --- lldpV2Xdot1LocAaTlvTable: PBBN Auto Attach TLVs information table
53 ---
54 lldpV2Xdot1LocAaTlvTable OBJECT-TYPE
55     SYNTAX      SEQUENCE OF LldpV2Xdot1LocAaTlvEntry
56     MAX-ACCESS   not-accessible
57     STATUS      current
58     DESCRIPTION
59         "This table contains one row per port of PBBN Auto Attach
60         TLV information (as a part of the LLDP
61         802.1 organizational extension) on the local system
62         known to this agent."
63     ::= { lldpXdot1AaLocalData 1 }
64
65 lldpV2Xdot1LocAaTlvEntry OBJECT-TYPE
66     SYNTAX      LldpV2Xdot1LocAaTlvEntry
67     MAX-ACCESS   not-accessible
68     STATUS      current
69     DESCRIPTION
70         "PBBN Auto Attach TLV information about a
71         particular port component."
72     INDEX      { lldpV2LocPortIfIndex }

```

```

1 ::= { lldpV2Xdot1LocAaTlvTable 1 }
2
3 lldpV2Xdot1LocAaTlvEntry ::= SEQUENCE {
4     lldpV2Xdot1LocAaSysTlvString      OCTET STRING,
5     lldpV2Xdot1LocAaAsgnsTlvString    OCTET STRING
6 }
7
8 lldpV2Xdot1LocAaSysTlvString OBJECT-TYPE
9     SYNTAX      OCTET STRING (SIZE (0..514))
10    MAX-ACCESS   read-only
11    STATUS       current
12    DESCRIPTION
13        "This object contains the PBBN Auto Attach System TLV information
14        string for the Port, as defined in D.2.17.
15        As the elements within the string are not individually
16        manipulated via SNMP (they are of concern only to the
17        state machines), the sub-structure of the string
18        is not visible as separate objects within the
19        local database."
20    REFERENCE
21        "D.2.17"
22 ::= { lldpV2Xdot1LocAaTlvEntry 1 }
23
24 lldpV2Xdot1LocAaAsgnsTlvString OBJECT-TYPE
25     SYNTAX      OCTET STRING (SIZE (0..514))
26     MAX-ACCESS   read-only
27     STATUS       current
28     DESCRIPTION
29         "This object contains the PBBN Auto Attach Assignment TLV
30         information string for the Port, as defined in D.2.18.
31         As the elements within the string are not individually
32         manipulated via SNMP (they are of concern only to the
33         state machines), the sub-structure of the string
34         is not visible as separate objects within the
35         local database."
36     REFERENCE
37         "D.2.18"
38 ::= { lldpV2Xdot1LocAaTlvEntry 2 }
39
40 -----
41 -- IEEE 802.1 - PBBN Auto Attach remote system information
42 -----
43 ---
44 --- lldpV2Xdot1RemAaTlvTable: PBBN Auto Attach TLV information table
45 ---
46 lldpV2Xdot1RemAaTlvTable OBJECT-TYPE
47     SYNTAX      SEQUENCE OF LldpV2Xdot1RemAaTlvEntry
48     MAX-ACCESS   not-accessible
49     STATUS       current
50     DESCRIPTION
51         "This table contains one row per port of PBBN Auto Attach
52         TLV information (as a part of the LLDP
53         802.1 organizational extension) on the remote system
54         known to this agent."
55 ::= { lldpXdot1AaRemoteData 1 }
56
57 lldpV2Xdot1RemAaTlvEntry OBJECT-TYPE
58     SYNTAX      LldpV2Xdot1RemAaTlvEntry
59     MAX-ACCESS   not-accessible
60     STATUS       current
61     DESCRIPTION
62         "PBBN Auto Attach TLV information about a
63         particular port component."
64     INDEX      { lldpV2RemTimeMark,
65                 lldpV2RemLocalIfIndex,
66                 lldpV2RemLocalDestMACAddress,
67                 lldpV2RemIndex }
68 ::= { lldpV2Xdot1RemAaTlvTable 1 }
69
70 lldpV2Xdot1RemAaTlvEntry ::= SEQUENCE {
71     lldpV2Xdot1RemAaSysTlvString      OCTET STRING,
72     lldpV2Xdot1RemAaAsgnsTlvString    OCTET STRING

```

```

1      }
2
3 lldpV2Xdot1RemAaSysTlvString OBJECT-TYPE
4     SYNTAX      OCTET STRING (SIZE (0..514))
5     MAX-ACCESS  read-only
6     STATUS      current
7     DESCRIPTION
8         "This object contains the PBBN Auto Attach System TLV
9         information string for the Port, as defined in D.2.17.
10        As the elements within the string are not individually
11        manipulated via SNMP (they are of concern only to the
12        state machines), the sub-structure of the string
13        is not visible as separate objects within the
14        local database."
15     REFERENCE
16         "D.2.17"
17     ::= { lldpV2Xdot1RemAaTlvEntry 1 }
18
19 lldpV2Xdot1RemAaAsgnsTlvString OBJECT-TYPE
20     SYNTAX      OCTET STRING (SIZE (0..514))
21     MAX-ACCESS  read-only
22     STATUS      current
23     DESCRIPTION
24         "This object contains the PBBN Auto Attach Assignment TLV
25         information string for the Port, as defined in D.2.18.
26        As the elements within the string are not individually
27        manipulated via SNMP (they are of concern only to the
28        state machines), the sub-structure of the string
29        is not visible as separate objects within the
30        local database."
31     REFERENCE
32         "D.2.18"
33     ::= { lldpV2Xdot1RemAaTlvEntry 2 }
34
35 -----
36 -- IEEE 802.1 - PBBN Auto Attach conformance information
37 -----
38
39 lldpXdot1AaConformance OBJECT IDENTIFIER ::= { lldpXdot1EvbExtensions 4 }
40
41 lldpXdot1AaCompliances
42     OBJECT IDENTIFIER ::= { lldpXdot1AaConformance 1 }
43 lldpXdot1AaGroups
44     OBJECT IDENTIFIER ::= { lldpXdot1AaConformance 2 }
45 --
46 -- PBBN Auto Attach - compliance statements
47 --
48 lldpXdot1AaCompliance MODULE-COMPLIANCE
49     STATUS      current
50     DESCRIPTION
51         "A compliance statement for SNMP entities that implement
52         the IEEE 802.1 organizationally defined PBBN Auto Attach LLDP
53         MIB extensions.
54
55         This group is mandatory for agents that implement the
56         PBBN Auto Attach aaSet TLV set."
57     MODULE      -- this module
58         MANDATORY-GROUPS { lldpXdot1AaGroup,
59                             ifGeneralInformationGroup }
60     ::= { lldpXdot1AaCompliances 1 }
61 --
62 -- PBBN Auto Attach - MIB groupings
63 --
64 lldpXdot1AaGroup OBJECT-GROUP
65     OBJECTS {
66         lldpXdot1AaConfigSysTxEnable,
67         lldpXdot1AaConfigAsgnsTxEnable,
68         lldpV2Xdot1LocAaSysTlvString,
69         lldpV2Xdot1LocAaAsgnsTlvString,
70         lldpV2Xdot1RemAaSysTlvString,
71         lldpV2Xdot1RemAaAsgnsTlvString
72     }

```

```
1  STATUS   current
2  DESCRIPTION
3      "The collection of objects that support the
4      PBBN Auto Attach aaSet TLV set."
5      ::= { lldpXdot1AaGroups 1 }
6
7  END
```

1 D.6 IEEE 802.1/LLDP extension YANG

2 This clause specifies YANG data models that provide objects for managing the operation of IEEE 802.1
3 Organizationally Specific TLVs. The YANG modules defined here extend the basic YANG module for
4 controlling and monitoring the LLDP agent defined in IEEE Std 802.1AB.

5 D.6.1 YANG framework

6 This clause has been developed in a manner consistent with the principles of the Internet Standard
7 Management Framework and according to the YANG guidelines published in IETF RFC 7950 as applicable
8 to IEEE standards.

9 The YANG framework for the Organizationally Specific TLV YANG follows the framework described in
10 48.1 and as extended by IEEE Std 802.1AB. Each of the TLV sets shown in Table D-1 has an associated
11 YANG module that augments the base and per-port LLDP YANG defined in IEEE Std 802.1AB.
12 Figure D-19 shows the high-level arrangement of IEEE 802.1 Organizationally Specific TLV YANG in the
13 general YANG framework.

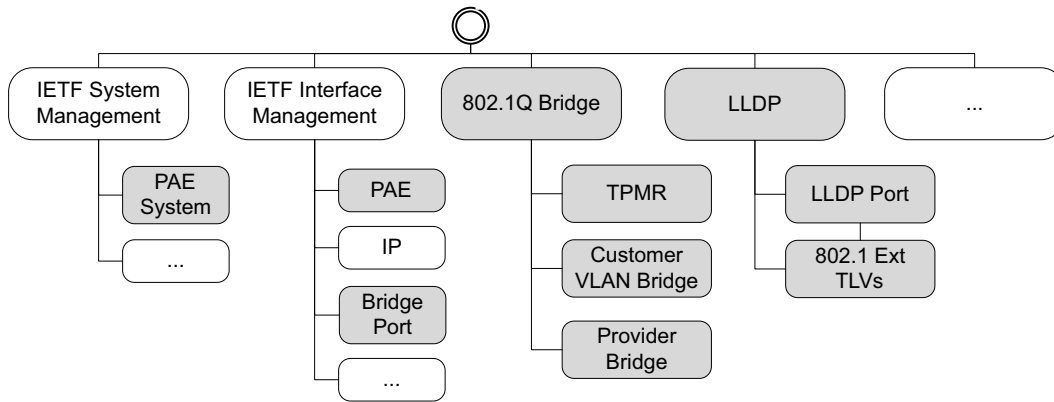


Figure D-19—YANG hierarchy with IEEE 802.1Q Extension TLV YANG

14 D.6.2 IEEE 802.1 Organizationally Specific TLV YANG data models

15 The YANG data models are based on the management model outline in D.3. A UML representation of the
16 management model for each TLV set is provided in D.6.2.1 to D.6.2.6.

1 D.6.2.1 IEEE 802.1/LLDP extension basicSet TLV model

2 The attributes for the TLVs in the basicSet are obtained from both system wide and per-port managed
3 objects. The model augments both the LLDP base model and the LLDP port model. The UML for the
4 IEEE 802.1/LLDP extension basicSet is derived from the UML specified in IEEE Std 802.1AB and is
5 shown in Figure D-20. The highlighted portions of the UML show how the LLDP models have been
6 augmented.

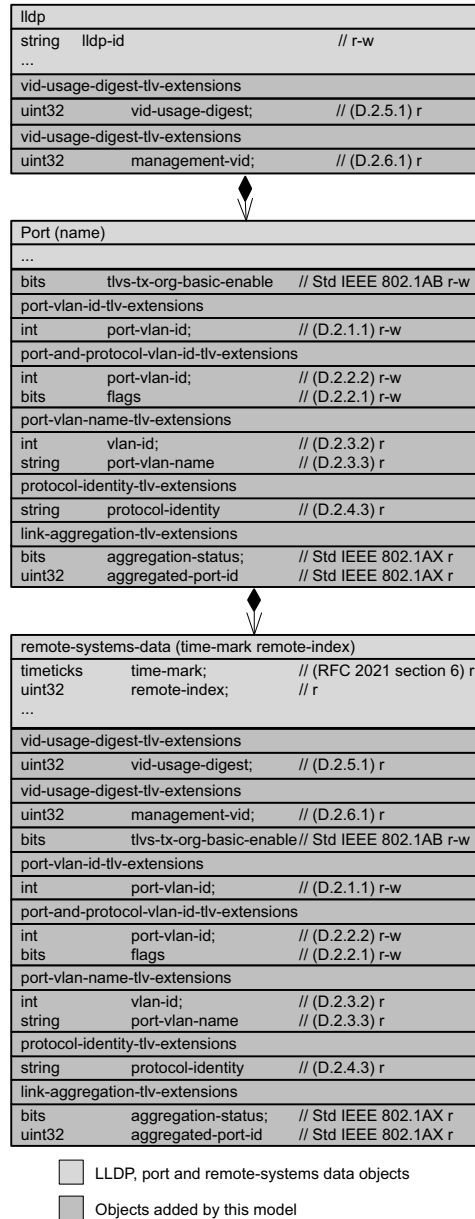


Figure D-20—basicSet TLV model

1 **D.6.2.2 IEEE 802.1/LLDP extension cnSet YANG model**

2 The attributes for the TLV in the cnSet are obtained from per-port managed objects. The model augments the
3 LLDP port model. The UML for the IEEE 802.1/LLDP extension cnSet is derived from the UML specified
4 in IEEE Std 802.1AB and is shown in Figure D-21. The highlighted portions of the UML show how the
5 LLDP model has been augmented.

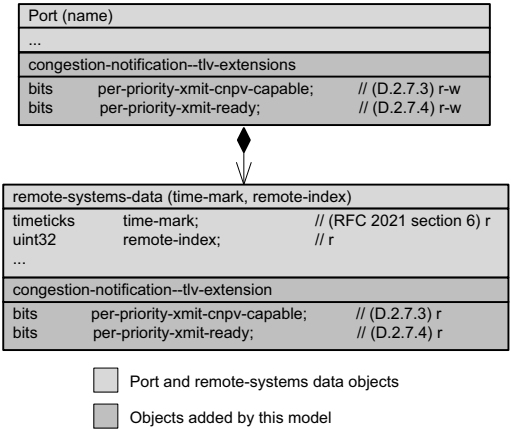


Figure D-21—cnSet TLV model

1 D.6.2.3 IEEE 802.1/LLDP extension dcbxSet YANG model

2 The attributes for the TLV in the dcbxSet are obtained from per-port managed objects. The model augments
3 the LLDP port model. The UML for the IEEE 802.1/LLDP extension dcbxSet is derived from the UML
4 specified in IEEE Std 802.1AB and is shown in Figure D-22. The highlighted portions of the UML show
5 how the LLDP model has been augmented.

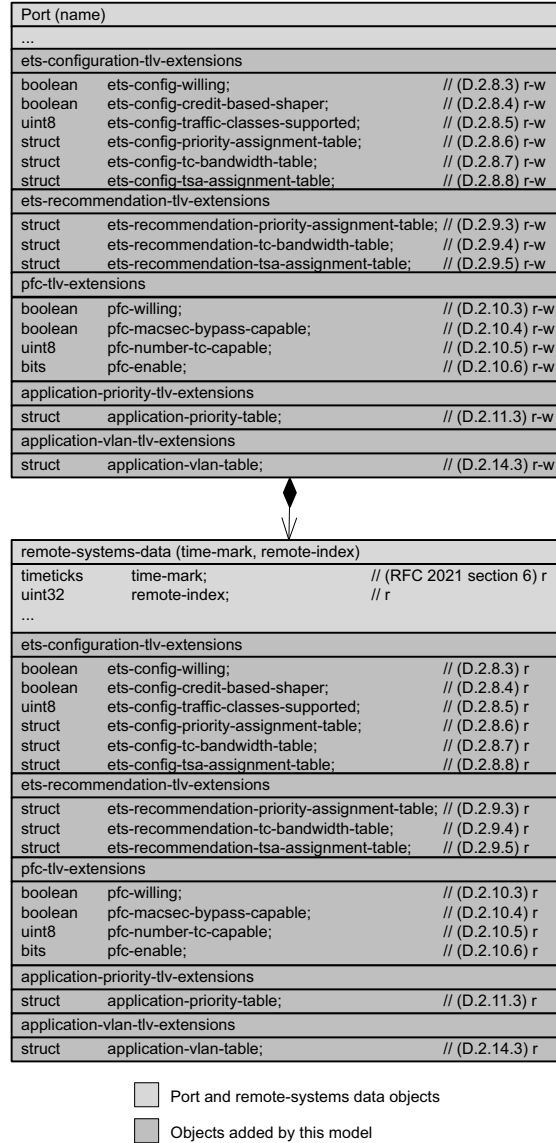


Figure D-22—dcbxSet TLV model

1 D.6.2.4 IEEE 802.1/LLDP extension evbSet YANG model

2 The attributes for the TLVs in the evbSet are obtained from per-port managed objects. The model augments
3 the LLDP port model. The UML for the IEEE 802.1/LLDP extension evbSet is derived from the UML
4 specified in IEEE Std 802.1AB and is shown in Figure D-23. The highlighted portions of the UML show
5 how the LLDP model has been augmented.

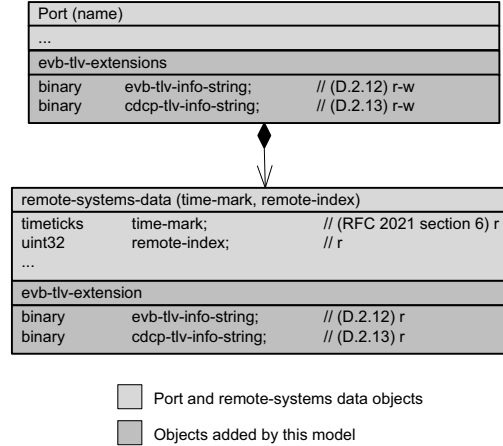


Figure D-23—evbSet TLV model

6 D.6.2.5 IEEE 802.1/LLDP extension ciSet TLV model

7 The attributes for the Congestion Isolation TLV in the ciSet are obtained from per-port managed objects and
8 augment the LLDP port. The UML for the IEEE 802.1/LLDP extension ciSet is derived from the UML
9 specified in IEEE Std 802.1AB and is shown in Figure D-24. The highlighted portions of the UML show
10 how the LLDP models have been augmented..

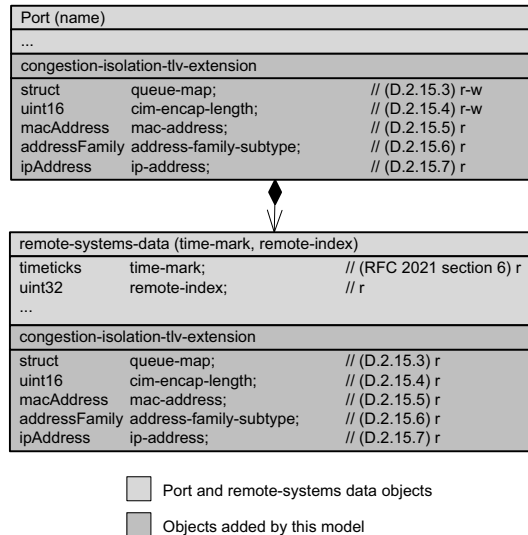


Figure D-24—ciSet TLV model

1 **D.6.2.6 IEEE 802.1/LLDP extension trSet TLV model**

2 The attributes for the Topology Recognition TLV in the trSet are obtained from both system wide and
3 per-port managed objects. The model augments both the LLDP system and the LLDP port. The UML for the
4 IEEE 802.1/LLDP extension trSet is derived from the UML specified in IEEE Std 802.1AB and is shown in
5 Figure D-25. The highlighted portions of the UML show how the LLDP models have been augmented.
6

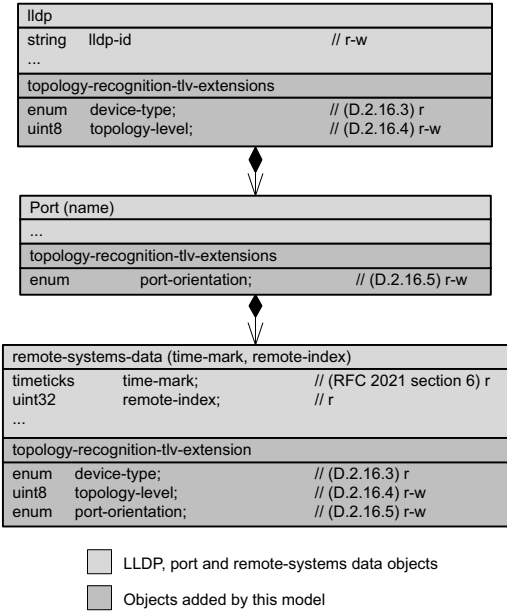


Figure D-25—trSet TLV model

1 D.6.2.7 IEEE 802.1/LLDP extension aaSet TLV model

2 The attributes for the TLVs in the aaSet are obtained from both system wide and per-port managed objects of
3 the IEEE8021-PBBN-AA-MIB. The model augments the LLDP port model. The UML for the IEEE
4 802.1/LLDP extension aaSet is derived from the UML specified in IEEE Std 802.1AB and is shown in
5 Figure D-26. The highlighted portions of the UML show how the LLDP model has been augmented.

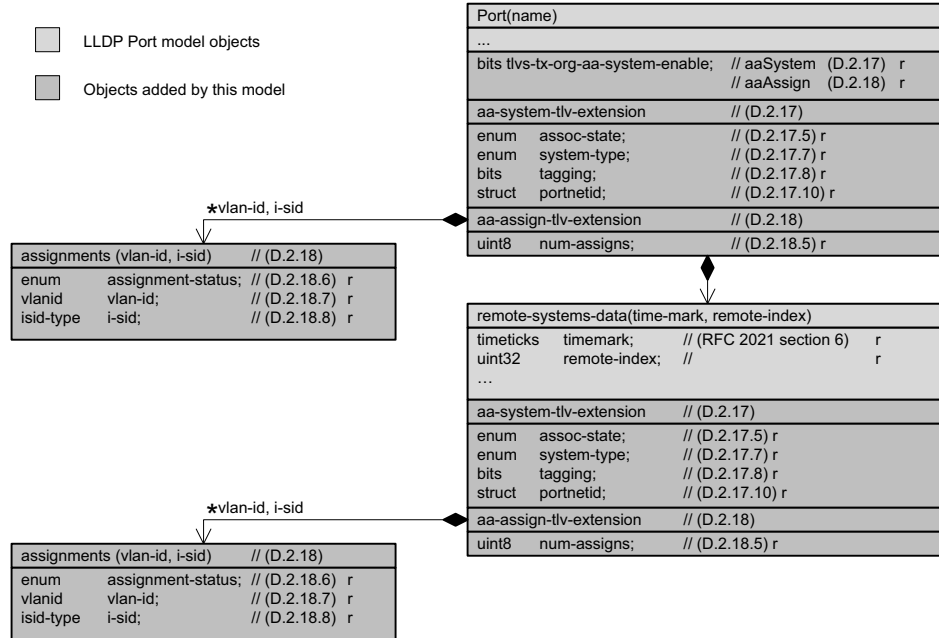


Figure D-26—PBBN Auto Attach aaSet TLV model

6 D.6.3 Structure of the IEEE 802.1/LLDP extension YANG models

7 IEEE 802.1/LLDP extension YANG models are divided into a number of YANG modules, one for each TLV
8 set defined in Table D-1. Each YANG module augments the LLDP base module and/or the LLDP port
9 module with objects needed to generate the IEEE 802.1 Organizationally Specific TLVs. Each YANG
10 module augments the LLDP port module's remote-systems-data to create space to hold the objects received
11 from a peer LLDP agent. Table D-23 summarizes the modules in this Annex.

12 D.6.4 Security considerations

13 The general security considerations specified in 48.4 apply to the YANG modules defined in this clause.
14 There are a number of management objects defined in the Organizationally Specific TLV extension YANG
15 modules that are configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects may be
16 considered sensitive or vulnerable in some network environments. A network configuration protocol, such
17 as NETCONF (IETF RFC 6241 [B51]), can support protocol operations that can edit or delete YANG
18 module configuration data (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure
19 environment without proper protection, then negative effects on the network operation is possible.

20 The following subclauses define module specific security considerations for each of the Organizationally
21 Specific TLV sets.

Table D-23—Summary of the YANG modules

Module	References	Notes
ieee802-dot1q-lldp-basic-tlv	D.6.6.1	Basic set of IEEE 802.1 Organizationally Specific TLVs.
ieee802-dot1q-lldp-cn-tlv	D.6.6.2	Congestion Notification set of IEEE 802.1 Organizationally Specific TLVs.
ieee802-dot1q-lldp-dcbx-tlv	D.6.6.3	DCBX set of IEEE 802.1 Organizationally Specific TLVs.
ieee802-dot1q-lldp-evb-tlv	D.6.6.4	Edge Virtual Bridging set of IEEE 802.1 Organizationally Specific TLVs.
ieee802-dot1q-lldp-ci-tlv	D.6.6.5	Congestion Isolation set of IEEE 802.1 Organizationally Specific TLVs.
ieee802-dot1q-lldp-tr-tlv	D.6.6.6	Topology Recognition set of IEEE 802.1 Organizationally Specific TLVs.
ieee802-dot1q-lldp-pbbn-aa-tlv	D.6.6.7	PBBN Auto Attach set of IEEE 802.1 Organizationally Specific TLVs.

1 D.6.4.1 Security considerations of the ieee802-dot1q-lldp-basic-tlv YANG module

2 The following objects in the ieee802-dot1q-lldp-basic-tlv YANG module could be manipulated to interfere
3 with the operation of IEEE 802.1 organizationally specific functionality using LLDP on a port and, for
4 example, be used to cause network instability:

5 /lldp/vid-usage-digest-tlv-extension/vid-usage-digest
6 /lldp/management-vid-tlv-extension/management-vid
7 /lldp/port/tlvs-tx-org-basic-enable
8 /lldp/port/port-vlan-id-tlv-extension/port-vlan-id
9 /lldp/port/port-and-protocol-vlan-id-extension/port-vlan-id
10 /lldp/port/port-and-protocol-vlan-id-extension/flags
11 /lldp/port/port-vlan-name-tlv-extension/vlan-id
12 /lldp/port/port-vlan-name-tlv-extension/port-vlan-name
13 /lldp/port/protocol-identity-tlv-extension/protocol-identity
14 /lldp/port/link-aggregation-tlv-extension/aggregation-status
15 /lldp/port/link-aggregation-tlv-extension/aggregated-port-id

16 D.6.4.2 Security considerations of the ieee802-dot1q-lldp-cn-tlv YANG module

17 The following objects in the ieee802-dot1q-lldp-cn-tlv YANG module could be manipulated to interfere
18 with the operation of IEEE 802.1 organizationally specific functionality using LLDP on a port and, for
19 example, be used to cause network instability:

20 /lldp/port/congestion-notification-tlv-extension/per-priority-xmit-cnpv-enable
21 /lldp/port/congestion-notification-tlv-extension/per-priority-xmit-ready

1 **D.6.4.3 Security considerations of the ieee802-dot1q-lldp-dcbx-tlv YANG module**

2 The following objects in the ieee802-dot1q-lldp-dcbx-tlv YANG module could be manipulated to interfere
3 with the operation of IEEE 802.1 organizationally specific functionality using LLDP on a port and, for
4 example, be used to cause network instability:

5 /lldp/port/ets-configuration-tlv-extension/ets-config-willing
6 /lldp/port/ets-configuration-tlv-extension/ets-config-credit-based-shaper
7 /lldp/port/ets-configuration-tlv-extension/ets-config-traffic-classes-supported
8 /lldp/port/ets-configuration-tlv-extension/ets-config-priority-assignment-table
9 /lldp/port/ets-configuration-tlv-extension/ets-config-tc-bandwidth-table
10 /lldp/port/ets-configuration-tlv-extension/ets-config-tsa-assignment-table
11 /lldp/port/ets-recommendation-tlv-extension/ets-recommend-priority-assignment-table
12 /lldp/port/ets-recommendation-tlv-extension/ets-recommend-tc-bandwidth-table
13 /lldp/port/ets-recommendation-tlv-extension/ets-recommend-tsa-assignment-table
14 /lldp/port/pfc-tlv-extension/pfc-willing
15 /lldp/port/pfc-tlv-extension/pfc-macsec-bypass-capable
16 /lldp/port/pfc-tlv-extension/pfc-number-tc-capable
17 /lldp/port/pfc-tlv-extension/pfc-enable
18 /lldp/port/application-priority-tlv-extension/application-priority-table
19 /lldp/port/application-vlan-tlv-extension/application-vlan-table

20 **D.6.4.4 Security considerations of the ieee802-dot1q-lldp-evb-tlv YANG module**

21 The following objects in the ieee802-dot1q-lldp-evb-tlv YANG module could be manipulated to interfere
22 with the operation of IEEE 802.1 organizationally specific functionality using LLDP on a port and, for
23 example, be used to cause network instability:

24 /lldp/port/evb-tlv-extension/evb-tlv-info-string
25 /lldp/port/evb-tlv-extension/cdcp-tlv-info-string

26 **D.6.4.5 Security considerations of the ieee802-dot1q-lldp-ci-tlv YANG module**

27 The following objects in the ieee802-dot1q-lldp-ci-tlv YANG module could be manipulated to interfere with
28 the operation of IEEE 802.1 organizationally specific functionality using LLDP on a port and, for example,
29 be used to cause network instability:

30 /lldp/port/tlvs-tx-org-ci-enable
31 /lldp/port/congestion-isolation-tlv-extension/queue-map
32 /lldp/port/congestion-isolation-tlv-extension/cim-encap-length
33 /lldp/port/congestion-isolation-tlv-extension/address-family-subtype
34 /lldp/port/congestion-isolation-tlv-extension/ip-address

35 **D.6.4.6 Security considerations of the ieee802-dot1q-lldp-tr-tlv YANG module**

36 The following objects in the ieee802-dot1q-lldp-tr-tlv YANG module could be manipulated to interfere with
37 the operation of IEEE 802.1 organizationally specific functionality using LLDP on a port and, for example,
38 be used to cause network instability:

39 /lldp/topology-recognition-tlv-extension/topology-level
40 /lldp/port/tlvs-tx-org-tr-enable
41 /lldp/port/topology-recognition-tlv-extension/port-orientation

1 *Insert D.6.4.7 after D.6.4.6 as follows:*

2 **D.6.4.7 Security considerations of the ieee802-dot1q-lldp-pbbn-aa-tlv YANG module**

3 All objects in ieee802-dot1q-lldp-pbbn-aa-tlv are read only and so cannot be manipulated. The objects:

- 4 /lldp/port/aa-system-tlv-extension/assoc-state
- 5 /lldp/port/aa-system-tlv-extension/system-type
- 6 /lldp/port/aa-system-tlv-extension/tagging
- 7 /lldp/port/aa-system-tlv-extension/portnetid
- 8 /lldp/port/aa-assign-tlv-extension/assignments

9 provide information about attachments between AADs and AABs which could be useful to an attacker.

10 **D.6.5 Definition of the IEEE 802.1/LLDP extension YANG modules**

11 The structure of the IEEE 802.1/LLDP extension YANG modules is described in D.6.4. In the following
12 YANG module definitions, if any discrepancy between the DESCRIPTION text and the corresponding
13 definition in any other part of this standard occur, the definitions outside this subclause take precedence.

14 The simplified graphical representation of the data model described in 48.5 is used in the following
15 subclauses to describe the data schema for the IEEE 802.1/LLDP extension YANG modules.

16 **D.6.5.1 Schema for the ieee802-dot1q-lldp-basic-tlv YANG module**

```

17 module: ieee802-dot1q-lldp-basic-tlv
18
19   augment /lldp:lldp:
20     +--rw vid-usage-digest-tlv-extension
21     | +--ro vid-usage-digest?  uint32
22     +--rw management-vid-tlv-extension
23     | +--rw management-vid?   dot1qtypes:vlanid
24   augment /lldp:lldp/lldp:port:
25     +--rw tlvs-tx-org-basic-enable?          bits
26     +--rw port-vlan-id-tlv-extension
27     | +--rw port-vlan-id?   dot1qtypes:vlanid
28     +--rw port-and-protocol-vlan-id-extension
29     | +--rw port-vlan-id?   dot1qtypes:vlanid
30     | +--rw flags?          bits
31     +--rw port-vlan-name-tlv-extension
32     | +--rw vlan-id?        dot1qtypes:vlanid
33     | +--rw port-vlan-name?  string
34     +--rw protocol-identity-tlv-extension
35     | +--rw protocol-identity? string
36     +--rw link-aggregation-tlv-extension
37     | +--rw aggregation-status? bits
38     | +--rw aggregated-port-id?  uint32
39   augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
40     +--ro port-vlan-id-tlv-extension
41     | +--ro port-vlan-id?   dot1qtypes:vlanid
42     +--ro port-and-protocol-vlan-id-extension
43     | +--ro port-vlan-id?   dot1qtypes:vlanid
44     | +--ro flags?          bits
45     +--ro port-vlan-name-tlv-extension
46     | +--ro vlan-id?        dot1qtypes:vlanid
47     | +--ro port-vlan-name?  string
48     +--ro protocol-identity-tlv-extension
49     | +--ro protocol-identity? string
50     +--ro link-aggregation-tlv-extension
51     | +--ro aggregation-status? bits
52     | +--ro aggregated-port-id?  uint32
53     +--ro vid-usage-digest-tlv-extension
54     | +--ro vid-usage-digest?  uint32
55     +--ro management-vid-tlv-extension
56     | +--ro management-vid?   dot1qtypes:vlanid

```


1

2 D.6.5.2 Schema for the ieee802-dot1q-ldp-cn-tlv YANG module

```
3 module: ieee802-dot1q-ldp-cn-tlv
4
5   augment /lldp:lldp/lldp:port:
6     +--rw tlvs-tx-org-congestion-notification-enable?   boolean
7     +--rw congestion-notification-tlv-extension
8       |
8       +--rw per-priority-xmit-cnpv-capable?
9         |
9         +--rw congestion-notification-bit-vector
10      +--rw per-priority-xmit-ready?
11        |
11        +--rw congestion-notification-bit-vector
12  augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
13    +--ro congestion-notification-tlv-extension
14      +--ro per-priority-xmit-cnpv-capable?
15        |
15        +--rw congestion-notification-bit-vector
16      +--ro per-priority-xmit-ready?
17        |
17        +--rw congestion-notification-bit-vector
18
```

19 D.6.5.3 Schema for the ieee802-dot1q-ldp-dcbx-tlv YANG module

```
20 module: ieee802-dot1q-ldp-dcbx-tlv
21
22   augment /lldp:lldp/lldp:port:
23     +--rw tlvs-tx-org-dcbx-enable?                     bits
24     +--rw ets-configuration-tlv-extension
25       |
25       +--rw willing?                                   boolean
26       |
26       +--rw credit-based-shaper?                       boolean
27       |
27       +--rw traffic-classes-supported?
28         |
28         +--rw dot1q-types:num-traffic-class-type
29       +--rw priority-assignment-table* [priority]
30         |
30         +--rw priority
31         |
31         +--rw dot1q-types:priority-type
32         |
32         +--rw priority-traffic-class?
33           |
33           +--rw dot1q-types:traffic-class-type
34         +--rw transmission-selection-algorithm?        identityref
35       +--rw tc-bandwidth-table* [traffic-class]
36         |
36         +--rw traffic-class                             dot1q-types:traffic-class-type
37         |
37         +--rw percentage-bandwidth?                     uint8
38       +--rw tsa-assignment-table* [tsa-traffic-class]
39         |
39         +--rw tsa-traffic-class
40         |
40         +--rw dot1q-types:traffic-class-type
41       +--rw transmission-selection-algorithm?          identityref
42     +--rw ets-recommendation-tlv-extension
43       |
43       +--rw priority-assignment-table* [priority]
44         |
44         +--rw priority                                 dot1q-types:priority-type
45         |
45         +--rw priority-traffic-class?                  dot1q-types:traffic-class-type
46       +--rw tc-bandwidth-table* [traffic-class]
47         |
47         +--rw traffic-class                             dot1q-types:traffic-class-type
48         |
48         +--rw percentage-bandwidth?                     uint8
49       +--rw tsa-assignment-table* [tsa-traffic-class]
50         |
50         +--rw tsa-traffic-class
51         |
51         +--rw dot1q-types:traffic-class-type
52       +--rw transmission-selection-algorithm?          identityref
53     +--rw pfc-tlv-extension
54       |
54       +--rw willing?                                   boolean
55       |
55       +--rw macsec-bypass-capable?                     boolean
56       |
56       +--rw number-tc-capable?                         dot1q-types:num-traffic-class-type
57       |
57       +--rw enable?                                    bits
58     +--rw application-priority-tlv-extension
59       |
59       +--rw application-priority-table* [application-priority]
60         |
60         +--rw application-priority
61         |
61         +--rw dot1q-types:priority-type
62       +--rw application-priority-selector?              identityref
63       +--rw application-priority-protocol?              uint16
64     +--rw application-vlan-tlv-extension
65       |
65       +--rw application-vlan-table* [application-vlan]
66         |
66         +--rw application-vlan                         dot1q-types:vlanid
67         |
67         +--rw application-vlan-selector?                identityref
68         |
68         +--rw application-vlan-protocol?                uint16
```

```

1  augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
2  +--ro ets-configuration-tlv-extension
3  | +--ro willing? boolean
4  | +--ro credit-based-shaper? boolean
5  | +--ro traffic-classes-supported?
6  | | dot1q-types:num-traffic-class-type
7  | +--ro priority-assignment-table* [priority]
8  | | +--ro priority
9  | | | dot1q-types:priority-type
10 | | +--ro priority-traffic-class?
11 | | | dot1q-types:traffic-class-type
12 | | +--ro transmission-selection-algorithm? identityref
13 | +--ro tc-bandwidth-table* [traffic-class]
14 | | +--ro traffic-class dot1q-types:traffic-class-type
15 | | +--ro percentage-bandwidth? uint8
16 | +--ro tsa-assignment-table* [tsa-traffic-class]
17 | +--ro tsa-traffic-class
18 | | dot1q-types:traffic-class-type
19 | +--ro transmission-selection-algorithm? identityref
20 +--ro ets-recommendation-tlv-extension
21 | +--ro priority-assignment-table* [priority]
22 | | +--ro priority dot1q-types:priority-type
23 | | +--ro priority-traffic-class? dot1q-types:traffic-class-type
24 | +--ro tc-bandwidth-table* [traffic-class]
25 | | +--ro traffic-class dot1q-types:traffic-class-type
26 | | +--ro percentage-bandwidth? uint8
27 | +--ro tsa-assignment-table* [tsa-traffic-class]
28 | +--ro tsa-traffic-class
29 | | dot1q-types:traffic-class-type
30 | +--ro transmission-selection-algorithm? identityref
31 +--ro pfc-tlv-extension
32 | +--ro willing? boolean
33 | +--ro macsec-bypass-capable? boolean
34 | +--ro number-tc-capable? dot1q-types:num-traffic-class-type
35 | +--ro enable? bits
36 +--ro application-priority-tlv-extension
37 | +--ro application-priority-table* [application-priority]
38 | +--ro application-priority
39 | | dot1q-types:priority-type
40 | +--ro application-priority-selector? identityref
41 | +--ro application-priority-protocol? uint16
42 +--ro application-vlan-tlv-extension
43 | +--ro application-vlan-table* [application-vlan]
44 | +--ro application-vlan dot1q-types:vlanid
45 | +--ro application-vlan-selector? identityref
46 | +--ro application-vlan-protocol? uint16
47

```

48 D.6.5.4 Schema for the ieee802-dot1q-lldp-evb-tlv YANG module

```

49 module: ieee802-dot1q-lldp-evb-tlv
50
51 augment /lldp:lldp/lldp:port:
52 +--rw tlvs-tx-org-evb-enable? bits
53 +--rw evb-tlv-extension
54 | +--ro evb-tlv-info-string? binary
55 +--rw cdcp-tlv-extension
56 | +--ro cdcp-tlv-info-string? binary
57 augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
58 +--ro evb-tlv-extension
59 | +--ro evb-tlv-info-string? binary
60 +--ro cdcp-tlv-extension
61 | +--ro cdcp-tlv-info-string? binary
62

```

63 D.6.5.5 Schema for the ieee802-dot1q-lldp-ci-tlv YANG module

```

64 module: ieee802-dot1q-lldp-ci-tlv
65
66 augment /lldp:lldp/lldp:port:
67 +--rw tlvs-tx-org-ci-enable? bits
68 +--rw congestion-isolation-tlv-extension

```

```

1      +--rw queue-map* [priority]
2      | +--rw priority          dot1q-types:priority-type
3      | +--rw queue-config?     dot1q-ci:abs-traffic-class-plus-one-type
4      +--rw cim-encap-length?   uint16
5      +--ro mac-address?        ieee:mac-address
6      +--ro udp-port-number?    inet:port-number
7      +--ro ip-address?         inet:ip-address
8  augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
9      +--ro congestion-isolation-tlv-extension
10     +--ro queue-map* [priority]
11     | +--ro priority          dot1q-types:priority-type
12     | +--ro queue-config?     dot1q-ci:abs-traffic-class-plus-one-type
13     +--ro cim-encap-length?   uint16
14     +--ro mac-address?        ieee:mac-address
15     +--ro udp-port-number?    inet:port-number
16     +--ro ip-address?         inet:ip-address
17

```

18 D.6.5.6 Schema for the ieee802-dot1q-lldp-tr-tlv YANG module

```

19 module: ieee802-dot1q-lldp-tr-tlv
20
21 augment /lldp:lldp:
22     +--rw topology-recognition-tlv-extension
23     +--ro device-type?        enumeration
24     +--rw topology-level?     uint8
25 augment /lldp:lldp/lldp:port:
26     +--rw tlvs-tx-org-tr-enable? bits
27     +--rw topology-recognition-tlv-extension
28     +--rw port-orientation?   enumeration
29 augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
30     +--ro congestion-isolation-tlv-extension
31     +--ro device-type?        enumeration
32     +--ro topology-level?     uint8
33     +--ro port-orientation?   enumeration
34
35

```

36 D.6.5.7 Schema for the ieee802-dot1q-lldp-pbbn-aa-tlv YANG module

```

37 module: ieee802-dot1q-lldp-pbbn-aa-tlv
38
39 augment /lldp:lldp/lldp:port:
40     +--ro tlvs-tx-org-aa-system-enable? bits
41     +--rw aa-system-tlv-extension
42     | +--ro assoc-state?      identityref
43     | +--ro system-type?      identityref
44     | +--ro tagging?          identityref
45     | +--ro portnetid?        port-netid-type
46     +--rw aa-assign-tlv-extension
47     +--ro num-assigns?        uint8
48     +--ro assignments* [vlan-id i-sid]
49     | +--ro assignment-status? identityref
50     | +--ro vlan-id           dot1qt-types:vlanid
51     | +--ro i-sid             dot1qt-types:isid-type
52 augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
53     +--ro aa-system-tlv-extension
54     | +--ro assoc-state?      identityref
55     | +--ro system-type?      identityref
56     | +--ro tagging?          identityref
57     | +--ro portnetid?        port-netid-type
58     +--rw aa-assign-tlv-extension
59     +--ro num-assigns?        uint8
60     +--ro assignments* [vlan-id i-sid]
61     | +--ro assignment-status? identityref
62     | +--ro vlan-id           dot1qt-types:vlanid
63     | +--ro i-sid             dot1qt-types:isid-type

```

1 D.6.6 IEEE 802.1/LLDP extension YANG modules

2 D.6.6.1 The ieee802-dot1q-lldp-basic-tlv YANG module

```
3 module ieee802-dot1q-lldp-basic-tlv {
4   yang-version "1.1";
5   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-lldp-basic-tlv;
6   prefix lldp-basic-tlv;
7   import ieee802-dot1q-lldp {
8     prefix lldp;
9   }
10  import ieee802-dot1q-types {
11    prefix dot1qtypes;
12  }
13  organization
14    "Institute of Electrical and Electronics Engineers";
15  contact
16    "WG-URL: http://ieee802.org/1/
17    WG-EMail: stds-802-1-1@ieee.org
18    Contact: IEEE 802.1 Working Group Chair
19    Postal: C/O IEEE 802.1 Working Group
20    IEEE Standards Association
21    445 Hoes Lane
22    Piscataway, NJ 08854
23    USA
24
25    E-mail: stds-802-1-chairs@ieee.org";
26  description
27    "IEEE Std 802.1Q extension TLVs for LLDP
28
29    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
30    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
31
32    Copyright (C) IEEE (2023).
33
34    This version of this YANG module is part of IEEE Std 802.1Q; see
35    the standard itself for full legal notices.";
36  revision 2023-07-03 {
37    description
38      "LLDP extension TLVs for the basicSet. Published as part of IEEE
39      Std 802.1Qcz-2023.";
40    reference
41      "Annex D of IEEE Std 802.1Q";
42  }
43  grouping port-vlan-id-tlv {
44    description
45      "Port VLAN ID TLV";
46    reference
47      "D.2.1 of IEEE Std 802.1Q";
48    leaf port-vlan-id {
49      type dot1qtypes:vlanid;
50      description
51        "Advertise port VLAN Identifier, Send 0 if unknown or unsupported";
52      reference
53        "D.2.1 of IEEE Std 802.1Q";
54    }
55  }
56  grouping port-and-protocol-vlan-id-tlv {
57    description
58      "Port and Protocol VLAN ID TLV";
59    reference
60      "D.2.2 of IEEE Std 802.1Q";
61    uses port-vlan-id-tlv {
62      reference
63        "D.2.2 of IEEE Std 802.1Q";
64    }
65    leaf flags {
66      type bits {
67        bit port-and-protocol-vlan-supported {
68          position 1;
69          description
```

```

1         "1 = supported, 0 = not supported";
2     }
3     bit port-and-protocol-vlan-enabled {
4         position 2;
5         description
6             "1 = enabled, 0 = not enabled";
7     }
8 }
9 description
10    "Port and Protocol VLAN capability and status. Bit positions 0
11    and 3-7 are reserved for future standardization";
12 reference
13    "D.2.2.1 of IEEE Std 802.1Q";
14 }
15 }
16 grouping port-vlan-name-tlv {
17     description
18         "Port VLAN NAME TLV";
19     reference
20         "D.2.3 of IEEE Std 802.1Q";
21     leaf vlan-id {
22         type dot1qtotypes:vlanid;
23         description
24             "The VLAN ID associated with the VLAN Name";
25         reference
26             "D.2.3.2 of IEEE Std 802.1Q";
27     }
28     leaf port-vlan-name {
29         type string {
30             length "0..32";
31         }
32         description
33             "Advertise port VLAN Name, Send 0 length if unknown or
34             unsupported";
35         reference
36             "D.2.3.3 of IEEE Std 802.1Q";
37     }
38 }
39 grouping protocol-identity-tlv {
40     description
41         "Protocol Identity TLV";
42     reference
43         "D.2.4 of IEEE Std 802.1Q";
44     leaf protocol-identity {
45         type string {
46             length "1..255";
47         }
48         description
49             "Advertise ProtocolIdentity";
50         reference
51             "D.2.4.3 of IEEE Std 802.1Q";
52     }
53 }
54 grouping vid-usage-digest-tlv {
55     description
56         "VID Usage Digest TLV";
57     reference
58         "D.2.5 of IEEE Std 802.1Q";
59     leaf vid-usage-digest {
60         type uint32;
61         config false;
62         description
63             "Advertise VID Usage Digest";
64         reference
65             "D.2.5.1 of IEEE Std 802.1Q";
66     }
67 }
68 grouping management-vid-tlv {
69     description
70         "Management VID TLV";
71     reference
72         "D.2.6 of IEEE Std 802.1Q";

```

```

1  leaf management-vid {
2      type dot1qttype:vlanid;
3      description
4          "Advertise Management VID";
5      reference
6          "D.2.6.1 of IEEE Std 802.1Q";
7  }
8  }
9  grouping link-aggregation-tlv {
10     description
11         "Link Aggregation TLV";
12     reference
13         "F.2 of IEEE Std 802.1AX-2020";
14     leaf aggregation-status {
15         type bits {
16             bit aggregation-capability {
17                 position 0;
18                 description
19                     "1 = supported, 0 = not supported";
20             }
21             bit aggregation-status {
22                 position 1;
23                 description
24                     "1 = currently aggregated, 0 = not currently aggregated";
25             }
26             bit port-typeLS {
27                 position 2;
28                 description
29                     "Least Significant Bit of 2 bit port type value";
30             }
31             bit port-typeMS {
32                 position 3;
33                 description
34                     "Most Significant Bit of 2 bit port type value";
35             }
36         }
37     description
38         "Link Aggregation capability and status. Bit positions 4-7 are
39         reserved for future standardization";
40     reference
41         "F.2.1 of IEEE Std 802.1AX-2020";
42 }
43 leaf aggregated-port-id {
44     type uint32;
45     description
46         "Advertise aggregated port identifier";
47     reference
48         "F.2.2 of IEEE Std 802.1AX-2020";
49 }
50 }
51 augment "/lldp:lldp" {
52     description
53         "Augments lldp with information needed for system level extension
54         TLVs";
55     container vid-usage-digest-tlv-extension {
56         description
57             "The VID Usage Digest TLV";
58         uses vid-usage-digest-tlv;
59     }
60     container management-vid-tlv-extension {
61         description
62             "The Management VID TLV";
63         uses management-vid-tlv;
64     }
65 }
66 augment "/lldp:lldp/lldp:port" {
67     description
68         "Augments port with basic extension TLVs";
69     leaf tlvs-tx-org-basic-enable {
70         type bits {
71             bit port-vlan-id {
72                 position 0;

```

```

1      description
2          "D.2.1 of IEEE Std 802.1Q";
3      }
4      bit port-protocol-vlan-id {
5          position 1;
6          description
7              "D.2.2 of IEEE Std 802.1Q";
8      }
9      bit vlan-name {
10         position 2;
11         description
12             "D.2.3 of IEEE Std 802.1Q";
13     }
14     bit protocol-identity {
15         position 3;
16         description
17             "D.2.4 of IEEE Std 802.1Q";
18     }
19     bit vid-user-digest {
20         position 4;
21         description
22             "D.2.5 of IEEE Std 802.1Q";
23     }
24     bit management-vid {
25         position 5;
26         description
27             "D.2.6 of IEEE Std 802.1Q";
28     }
29     bit link-aggregation {
30         position 6;
31         description
32             "Annex F of IEEE Std 802.1AX-2020";
33     }
34 }
35 description
36     "Bitmap that includes the basicSet of TLVs from Table D-1 of
37     IEEE Std 802.1Q";
38 reference
39     "D.2 of IEEE Std 802.1Q";
40 }
41 container port-vlan-id-tlv-extension {
42     description
43         "The Port VLAN ID TLV";
44     uses port-vlan-id-tlv;
45 }
46 container port-and-protocol-vlan-id-extension {
47     description
48         "The Port and Protocol VLAN ID TLV";
49     uses port-and-protocol-vlan-id-tlv;
50 }
51 container port-vlan-name-tlv-extension {
52     description
53         "The Port VLAN Name TLV";
54     uses port-vlan-name-tlv;
55 }
56 container protocol-identity-tlv-extension {
57     description
58         "The Protocol Identity TLV";
59     uses protocol-identity-tlv;
60 }
61 container link-aggregation-tlv-extension {
62     description
63         "The Link Aggregation TLV";
64     uses link-aggregation-tlv;
65 }
66 }
67 augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
68     description
69         "Augments port remote-systems-data with received basic extension
70         TLVs";
71     container port-vlan-id-tlv-extension {
72         description

```

```

1      "Holds a received Port VLAN ID TLV";
2      uses port-vlan-id-tlv;
3  }
4  container port-and-protocol-vlan-id-extension {
5      description
6          "Holds a received Port and Protocol VLAN ID TLV";
7      uses port-and-protocol-vlan-id-tlv;
8  }
9  container port-vlan-name-tlv-extension {
10     description
11         "Holds a received VLAN Name TLV";
12     uses port-vlan-name-tlv;
13 }
14 container protocol-identity-tlv-extension {
15     description
16         "Holds a received Protocol Identity TLV";
17     uses protocol-identity-tlv;
18 }
19 container link-aggregation-tlv-extension {
20     description
21         "Holds a received Link Aggregation TLV";
22     uses link-aggregation-tlv;
23 }
24 container vid-usage-digest-tlv-extension {
25     description
26         "Holds a received VID Usage TLV";
27     uses vid-usage-digest-tlv;
28 }
29 container management-vid-tlv-extension {
30     description
31         "Holds a received Management VID TLV";
32     uses management-vid-tlv;
33 }
34 }
35 }
36

```

37 D.6.6.2 The ieee802-dot1q-lldp-cn-tlv YANG module

```

38 module ieee802-dot1q-lldp-cn-tlv {
39     yang-version "1.1";
40     namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-lldp-cn-tlv;
41     prefix lldp-cn-tlv;
42     import ieee802-dot1q-llab-lldp {
43         prefix lldp;
44     }
45     organization
46         "Institute of Electrical and Electronics Engineers";
47     contact
48         "WG-URL: http://ieee802.org/1/
49         WG-EMail: stds-802-1-1@ieee.org
50         Contact: IEEE 802.1 Working Group Chair
51         Postal: C/O IEEE 802.1 Working Group
52         IEEE Standards Association
53             445 Hoes Lane
54             Piscataway, NJ 08854
55             USA
56
57         E-mail: stds-802-1-chairs@ieee.org";
58     description
59         "IEEE Std 802.1Q extension TLVs for LLDP
60
61         References in this YANG module to IEEE Std 802.1Q are to IEEE Std
62         802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
63
64         Copyright (C) IEEE (2023).
65
66         This version of this YANG module is part of IEEE Std 802.1Q; see
67         the standard itself for full legal notices.";
68     revision 2023-07-03 {
69         description
70             "LLDP extension tlv for congestion notification.";
71     }
72 }

```



```

1   reference
2   "Annex D of IEEE Std 802.1Q";
3 }
4 grouping congestion-notification-tlv {
5   description
6   "Congestion Notification TLV";
7   reference
8   "D.2.7 of IEEE Std 802.1Q";
9   typedef congestion-notification-bit-vector {
10    type bits {
11      bit priority0 {
12        position 0;
13        description
14        "status for priority0, 0 is FALSE, 1 is TRUE";
15      }
16      bit priority1 {
17        position 1;
18        description
19        "status for priority1, 0 is FALSE, 1 is TRUE";
20      }
21      bit priority2 {
22        position 2;
23        description
24        "status for priority2, 0 is FALSE, 1 is TRUE";
25      }
26      bit priority3 {
27        position 3;
28        description
29        "status for priority3, 0 is FALSE, 1 is TRUE";
30      }
31      bit priority4 {
32        position 4;
33        description
34        "status for priority4, 0 is FALSE, 1 is TRUE";
35      }
36      bit priority5 {
37        position 5;
38        description
39        "status for priority5, 0 is FALSE, 1 is TRUE";
40      }
41      bit priority6 {
42        position 6;
43        description
44        "status for priority6, 0 is FALSE, 1 is TRUE";
45      }
46      bit priority7 {
47        position 7;
48        description
49        "status for priority7, 0 is FALSE, 1 is TRUE";
50      }
51    }
52    description
53    "Describes a bit vector used in Congestion Notification Objects";
54    reference
55    "32.4.7, 32.4.8 of IEEE Std 802.1Q";
56  }
57  leaf per-priority-xmit-cnpv-capable {
58    type congestion-notification-bit-vector;
59    description
60    "Indicates if a priority on this Port is operating as a CNPV";
61    reference
62    "D.2.7.3, 32.4.7 of IEEE Std 802.1Q";
63  }
64  leaf per-priority-xmit-ready {
65    type congestion-notification-bit-vector;
66    description
67    "Indicates if the priority remap defenses for this Port and CNPV
68    have been disabled";
69    reference
70    "D.2.7.4, 32.4.8 of IEEE Std 802.1Q";
71  }
72 }

```

```
1 augment "/lldp:lldp/lldp:port" {
2   description
3     "Augments port with the congestion notification extension tlv";
4   leaf tlvs-tx-org-congestion-notification-enable {
5     type boolean;
6     description
7       "Leaf that indicates if congestion notification tlv is enabled
8        from Table D-1 of IEEE Std 802.1Q";
9     reference
10      "D.2 of IEEE Std 802.1Q";
11   }
12   container congestion-notification-tlv-extension {
13     description
14       "The Congestion Notification TLV";
15     uses congestion-notification-tlv;
16   }
17 }
18 augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
19   description
20     "Augments port remote-systems-data with received cn extension TLVs";
21   container congestion-notification-tlv-extension {
22     description
23       "Holds a received Congestion Notification TLV";
24     uses congestion-notification-tlv;
25   }
26 }
27 }
28
```

1 D.6.6.3 The ieee802-dot1q-lldp-dcbx-tlv YANG module

```
2 module ieee802-dot1q-lldp-dcbx-tlv {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-lldp-dcbx-tlv;
5   prefix lldp-dcbx-tlv;
6   import ieee802-dot1q-types {
7     prefix dot1q-types;
8   }
9   import ieee802-dot1q-llab-lldp {
10    prefix lldp;
11  }
12  organization
13    "Institute of Electrical and Electronics Engineers";
14  contact
15    "WG-URL: http://ieee802.org/1/
16    WG-EMail: stds-802-1-1@ieee.org
17    Contact: IEEE 802.1 Working Group Chair
18    Postal: C/O IEEE 802.1 Working Group
19    IEEE Standards Association
20    445 Hoes Lane
21    Piscataway, NJ 08854
22    USA
23
24    E-mail: stds-802-1-chairs@ieee.org";
25  description
26    "IEEE Std 802.1Q extension TLVs for LLDP
27
28    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
29    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
30
31    Copyright (C) IEEE (2023).
32
33    This version of this YANG module is part of IEEE Std 802.1Q; see
34    the standard itself for full legal notices.";
35  revision 2023-07-03 {
36    description
37      "LLDP extension TLVs for DCBX. Published as part of IEEE Std
38      802.1Qcz-2023.";
39    reference
40      "Annex D of IEEE Std 802.1Q";
41  }
42  identity application-priority-selector {
43    description
44      "Specify the application priority selection of IEEE Std 802.1Q
45      D.2.11 and Table D-8";
46  }
47  identity default {
48    base application-priority-selector;
49    description
50      "Indicates the default application selection of the Application
51      Priority Table field of the Application Priority TLV specified in
52      D.2.11 of IEEE Std 802.1Q. Signaled as value 1.";
53  }
54  identity pri-tcp-sctp {
55    base application-priority-selector;
56    description
57      "Indicates TCP or SCTP application selection of the Application
58      Priority Table field of the Application Priority TLV specified in
59      D.2.11 of IEEE Std 802.1Q. Signaled as value 2.";
60  }
61  identity pri-udp-dccp {
62    base application-priority-selector;
63    description
64      "Indicates UDP or DCCP application selection of the Application
65      Priority Table field of the Application Priority TLV specified in
66      D.2.11 of IEEE Std 802.1Q. Signaled as value 3.";
67  }
68  identity pri-tcp-sctp-udp-dccp {
69    base application-priority-selector;
70    description
71      "Indicates TCP, SCTP, UDP or DCCP application selection of the
```

```

1      Application Priority Table field of the Application Priority TLV
2      specified in D.2.11 of IEEE Std 802.1Q. Signaled as value 4.";
3  }
4  identity pri-dscp {
5      base application-priority-selector;
6      description
7          "Indicates DSCP application selection of the Application Priority
8          Table field of the Application Priority TLV specified in D.2.11 of
9          IEEE Std 802.1Q. Signaled as value 5.";
10 }
11 identity application-vlan-selector {
12     description
13         "Specify the application VLAN selection of D.2.14, Table D-12, and
14         Table D-13 of IEEE Std 802.1Q";
15 }
16 identity vlan-pvid-ethertype {
17     base application-vlan-selector;
18     description
19         "Indicates PVID or EtherType selection of the Application VLAN
20         Table field of the Application VLAN TLV specified in D.2.14 of
21         IEEE Std 802.1Q. Signaled as value 1.";
22 }
23 identity vlan-tcp-sctp {
24     base application-vlan-selector;
25     description
26         "Indicates TCP or SCTP application selection of the Application
27         VLAN Table field of the Application VLAN TLV specified in D.2.14 of
28         IEEE Std 802.1Q. Signaled as value 2.";
29 }
30 identity vlan-udp-dccp {
31     base application-vlan-selector;
32     description
33         "Indicates UDP or DCCP application selection of the Application
34         VLAN Table field of the Application VLAN TLV specified in D.2.14 of
35         IEEE Std 802.1Q. Signaled as value 3.";
36 }
37 identity vlan-tcp-sctp-udp-dccp {
38     base application-vlan-selector;
39     description
40         "Indicates TCP, SCTP, UDP or DCCP application selection of the
41         Application VLAN Table field of the Application VLAN TLV specified
42         in D.2.14 of IEEE Std 802.1Q. Signaled as value 4.";
43 }
44 identity vlan-dscp {
45     base application-vlan-selector;
46     description
47         "Indicates DSCP application selection of the Application VLAN Table
48         field of the Application VLAN TLV specified in D.2.14 of IEEE Std
49         802.1Q. Signaled as value 5.";
50 }
51 grouping ets-configuration-tlv {
52     description
53         "The Enhanced Transmission Selection configuration TLV";
54     reference
55         "D.2.8 of IEEE Std 802.1Q";
56     leaf willing {
57         type boolean;
58         description
59             "True indicates willing to accept configurations from remote
60             station";
61         reference
62             "D.2.8.3 of IEEE Std 802.1Q";
63     }
64     leaf credit-based-shaper {
65         type boolean;
66         description
67             "True indicates station supports the Credit-based Shaper
68             transmission selection algorithm";
69         reference
70             "D.2.8.4 of IEEE Std 802.1Q";
71     }
72     leaf traffic-classes-supported {

```

```

1     type dot1q-types:num-traffic-class-type;
2     description
3         "Indicates number of traffic classes supported. The value of 8 is
4         encoded as 0 in the TLV since 3-bits are used to specify the
5         number";
6     reference
7         "D.2.8.5 of IEEE Std 802.1Q";
8 }
9 list priority-assignment-table {
10     key "priority";
11     description
12         "Maps a priority to a traffic class";
13     leaf priority {
14         type dot1q-types:priority-type;
15         description
16             "Indicates priority";
17         reference
18             "D.2.8.6 of IEEE Std 802.1Q";
19     }
20     leaf priority-traffic-class {
21         type dot1q-types:traffic-class-type;
22         description
23             "Indicates mapped traffic class for priority";
24         reference
25             "D.2.8.6 of IEEE Std 802.1Q";
26     }
27     leaf transmission-selection-algorithm {
28         type identityref {
29             base dot1q-types:transmission-selection-algorithm;
30         }
31         description
32             "Transmission selection algorithm";
33         reference
34             "8.6.8 and Table 8-6 of IEEE Std 802.1Q";
35     }
36 }
37 list tc-bandwidth-table {
38     key "traffic-class";
39     description
40         "Indicates the current bandwidth percentage for each traffic
41         class";
42     leaf traffic-class {
43         type dot1q-types:traffic-class-type;
44         description
45             "Indicates traffic class";
46         reference
47             "D.2.8.7 of IEEE Std 802.1Q";
48     }
49     leaf percentage-bandwidth {
50         type uint8 {
51             range "0..100";
52         }
53         description
54             "Percentage configured for the traffic class";
55         reference
56             "D.2.8.7 of IEEE Std 802.1Q";
57     }
58 }
59 list tsa-assignment-table {
60     key "tsa-traffic-class";
61     description
62         "Indicates the transmission selection algorithm used for a
63         traffic class";
64     leaf tsa-traffic-class {
65         type dot1q-types:traffic-class-type;
66         description
67             "Indicates traffic class";
68         reference
69             "D.2.8.8 of IEEE Std 802.1Q";
70     }
71     leaf transmission-selection-algorithm {
72         type identityref {

```

```

1      base dot1q-types:transmission-selection-algorithm;
2    }
3    description
4      "Transmission selection algorithm";
5    reference
6      "8.6.8 and Table 8-6 of IEEE Std 802.1Q";
7  }
8 }
9 }
10 grouping ets-recommendation-tlv {
11   description
12     "Recommendation of Enhanced Transmission Selection configuration
13     TLV";
14   reference
15     "D.2.9 of IEEE Std 802.1Q";
16   list priority-assignment-table {
17     key "priority";
18     description
19       "Maps a priority to a traffic class";
20     leaf priority {
21       type dot1q-types:priority-type;
22       description
23         "Indicates priority";
24       reference
25         "D.2.9.3 of IEEE Std 802.1Q";
26     }
27     leaf priority-traffic-class {
28       type dot1q-types:traffic-class-type;
29       description
30         "Indicates mapped traffic class for priority";
31       reference
32         "D.2.9.3 of IEEE Std 802.1Q";
33     }
34   }
35   list tc-bandwidth-table {
36     key "traffic-class";
37     description
38       "Indicates the current bandwidth percentage for each traffic
39       class";
40     leaf traffic-class {
41       type dot1q-types:traffic-class-type;
42       description
43         "Indicates traffic class";
44       reference
45         "D.2.9.4 of IEEE Std 802.1Q";
46     }
47     leaf percentage-bandwidth {
48       type uint8 {
49         range "0..100";
50       }
51       description
52         "Percentage configured for the traffic class";
53       reference
54         "D.2.9.4 of IEEE Std 802.1Q";
55     }
56   }
57   list tsa-assignment-table {
58     key "tsa-traffic-class";
59     description
60       "Indicates the transmission selection algorithm used for a
61       traffic class";
62     leaf tsa-traffic-class {
63       type dot1q-types:traffic-class-type;
64       description
65         "Indicates traffic class";
66       reference
67         "D.2.9.5 of IEEE Std 802.1Q";
68     }
69     leaf transmission-selection-algorithm {
70       type identityref {
71         base dot1q-types:transmission-selection-algorithm;
72       }

```

```
1      description
2      "Transmission selection algorithm";
3      reference
4      "8.6.8 and Table 8-6 of IEEE Std 802.Q";
5  }
6  }
7  }
8  grouping pfc-tlv {
9      description
10     "The Priority-based Flow Control configuration TLV";
11     reference
12     "D.2.10 of IEEE Std 802.1Q";
13     leaf willing {
14         type boolean;
15         description
16         "True indicates willing to accept configurations from remote
17         station";
18         reference
19         "D.2.10.3 of IEEE Std 802.1Q";
20     }
21     leaf macsec-bypass-capable {
22         type boolean;
23         description
24         "True indicates sending station is not capable of bypassing
25         MACsec";
26         reference
27         "D.2.10.4 of IEEE Std 802.1Q";
28     }
29     leaf number-tc-capable {
30         type dot1q-types:num-traffic-class-type;
31         description
32         "Indicates how many traffic classes may simultaneously support
33         PFC.";
34         reference
35         "D.2.10.5 of IEEE Std 802.1Q";
36     }
37     leaf enable {
38         type bits {
39             bit p0 {
40                 position 0;
41                 description
42                 "1 indicates PFC is enabled on the priority";
43             }
44             bit p1 {
45                 position 1;
46                 description
47                 "1 indicates PFC is enabled on the priority";
48             }
49             bit p2 {
50                 position 2;
51                 description
52                 "1 indicates PFC is enabled on the priority";
53             }
54             bit p3 {
55                 position 3;
56                 description
57                 "1 indicates PFC is enabled on the priority";
58             }
59             bit p4 {
60                 position 4;
61                 description
62                 "1 indicates PFC is enabled on the priority";
63             }
64             bit p5 {
65                 position 5;
66                 description
67                 "1 indicates PFC is enabled on the priority";
68             }
69             bit p6 {
70                 position 6;
71                 description
72                 "1 indicates PFC is enabled on the priority";
```

```

1      }
2      bit p7 {
3          position 7;
4          description
5              "1 indicates PFC is enabled on the priority";
6      }
7  }
8  description
9      "PFC enabled per priority";
10 reference
11     "D.2.10.6 of IEEE Std 802.1Q";
12 }
13 }
14 grouping application-priority-tlv {
15     description
16         "The application priority table TLV";
17     reference
18         "D.2.11 of IEEE Std 802.1Q";
19     list application-priority-table {
20         key "application-priority";
21         description
22             "Specifies a 3-bit priority for a Protocol ID";
23         leaf application-priority {
24             type dot1q-types:priority-type;
25             description
26                 "Priority for which the Protocol ID is being used";
27             reference
28                 "D.2.11.3 of IEEE Std 802.1Q";
29         }
30         leaf application-priority-selector {
31             type identityref {
32                 base application-priority-selector;
33             }
34             description
35                 "Selector to determine what the Protocol ID means.";
36             reference
37                 "D.2.11.3 of IEEE Std 802.1Q";
38         }
39         leaf application-priority-protocol {
40             type uint16 {
41                 range "0..65535";
42             }
43             description
44                 "Protocol ID of the type indicated by the selector";
45             reference
46                 "D.2.11.3 of IEEE Std 802.1Q";
47         }
48     }
49 }
50 grouping application-vlan-tlv {
51     description
52         "The Application VLAN table TLV";
53     reference
54         "D.2.14 of IEEE Std 802.1Q";
55     list application-vlan-table {
56         key "application-vlan";
57         description
58             "Specifies a VLAN ID priority for a Protocol ID";
59         leaf application-vlan {
60             type dot1q-types:vlanid;
61             description
62                 "Vlan for which the Protocol ID is being used";
63             reference
64                 "D.2.14.3 of IEEE Std 802.1Q";
65         }
66         leaf application-vlan-selector {
67             type identityref {
68                 base application-vlan-selector;
69             }
70             description
71                 "Selector to determine what the Protocol ID means.";
72             reference

```



```

1      "D.2.14.3 of IEEE Std 802.1Q";
2    }
3    leaf application-vlan-protocol {
4      type uint16 {
5        range "0..65535";
6      }
7      description
8        "Protocol ID of the type indicated by the selector";
9      reference
10     "D.2.14.3 of IEEE Std 802.1Q";
11   }
12 }
13 }
14 augment "/lldp:lldp:lldp:port" {
15   description
16     "Augments port with the dcbx extension tlv";
17   leaf tlvs-tx-org-dcbx-enable {
18     type bits {
19       bit ets-configuration {
20         position 0;
21         description
22           "D.2.8 of IEEE Std 802.1Q";
23       }
24       bit ets-recommendation {
25         position 1;
26         description
27           "D.2.9 of IEEE Std 802.1Q";
28       }
29       bit pfc {
30         position 2;
31         description
32           "D.2.10 of IEEE Std 802.1Q";
33       }
34       bit application-priority {
35         position 3;
36         description
37           "D.2.11 of IEEE Std 802.1Q";
38       }
39       bit application-vlan {
40         position 4;
41         description
42           "D.2.14 of IEEE Std 802.1Q";
43       }
44     }
45     description
46       "Bitmap including the dcbxSet of TLVs from Table D-1 of IEEE Std
47       802.1Q";
48     reference
49       "D.1 of IEEE Std 802.1Q";
50   }
51   container ets-configuration-tlv-extension {
52     description
53       "The ETS Configuration TLV";
54     uses ets-configuration-tlv;
55   }
56   container ets-recommendation-tlv-extension {
57     description
58       "The ETS Recommendation TLV";
59     uses ets-recommendation-tlv;
60   }
61   container pfc-tlv-extension {
62     description
63       "The Priority-based Flow Control Configuration TLV";
64     uses pfc-tlv;
65   }
66   container application-priority-tlv-extension {
67     description
68       "The Application Priority TLV";
69     uses application-priority-tlv;
70   }
71   container application-vlan-tlv-extension {
72     description

```

```
1      "The Application VLAN TLV";
2      uses application-vlan-tlv;
3  }
4  }
5  augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
6      description
7          "Augments port remote-systems-data with received dcbx extension
8          TLVs";
9      container ets-configuration-tlv-extension {
10         description
11             "Holds a received ETS Configuration TLV";
12         uses ets-configuration-tlv;
13     }
14     container ets-recommendation-tlv-extension {
15         description
16             "Holds a received ETS Recommendation TLV";
17         uses ets-recommendation-tlv;
18     }
19     container pfc-tlv-extension {
20         description
21             "Holds a received Priority-based Flow Control Configuration TLV";
22         uses pfc-tlv;
23     }
24     container application-priority-tlv-extension {
25         description
26             "Holds a received Application Priority TLV";
27         uses application-priority-tlv;
28     }
29     container application-vlan-tlv-extension {
30         description
31             "Holds a received Application VLAN TLV";
32         uses application-vlan-tlv;
33     }
34 }
35 }
36
```

1 D.6.6.4 The ieee802-dot1q-lldp-evb-tlv YANG module

```
2 module ieee802-dot1q-lldp-evb-tlv {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-lldp-evb-tlv;
5   prefix lldp-evb-tlv;
6   import ieee802-dot1q-lldp {
7     prefix lldp;
8   }
9   organization
10    "Institute of Electrical and Electronics Engineers";
11   contact
12    "WG-URL: http://ieee802.org/1/
13     WG-EMail: stds-802-1-1@ieee.org
14     Contact: IEEE 802.1 Working Group Chair
15     Postal: C/O IEEE 802.1 Working Group
16     IEEE Standards Association
17     445 Hoes Lane
18     Piscataway, NJ 08854
19     USA
20
21    E-mail: stds-802-1-chairs@ieee.org";
22   description
23    "IEEE Std 802.1Q extension TLVs for LLDP
24
25    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
26    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
27
28    Copyright (C) IEEE (2023).
29
30    This version of this YANG module is part of IEEE Std 802.1Q; see the
31    standard itself for full legal notices.";
32   revision 2023-07-03 {
33     description
34      "LLDP extension tlv for EVB. Published as part of IEEE Std
35      802.1Qcz-2023.";
36     reference
37      "Annex D of IEEE Std 802.1Q";
38   }
39   grouping evb-tlv {
40     description
41      "Edge Virtual Bridging (EVB) TLV";
42     reference
43      "D.2.12 of IEEE Std 802.1Q";
44     leaf evb-tlv-info-string {
45       type binary {
46         length "0..9";
47       }
48     }
49     config false;
50     description
51      "Opaque string containing EVB TLV information string. As the
52      elements within the string are not individually manipulated by
53      management (they are of concern only to the state machines), the
54      sub-structure of the string is not visible as separate objects
55      within the local database.";
56     reference
57      "D.2.12 of IEEE Std 802.1Q";
58   }
59   grouping cdcv-tlv {
60     description
61      "Channel Discovery and Configuration TLV";
62     reference
63      "D.2.13 of IEEE Std 802.1Q";
64     leaf cdcv-tlv-info-string {
65       type binary {
66         length "0..511";
67       }
68     }
69     config false;
70     description
71      "Opaque string containing CDCV TLV information. Up to 167
72      S-channel numbers are supported. As the elements within the
```

```

1      string are not individually manipulated by management (they are
2      of concern only to the state machines), the sub-structure of the
3      string is not visible as separate objects within the local
4      database.";
5      reference
6      "D.2.13 and D.2.13.8 of IEEE Std 802.1Q";
7  }
8  }
9  augment "/lldp:lldp/lldp:port" {
10     description
11     "Augments port with the EVB TLV";
12     leaf tlvs-tx-org-evb-enable {
13         type bits {
14             bit evb {
15                 position 0;
16                 description
17                 "D.2.12 of IEEE Std 802.1Q";
18             }
19             bit cdcv {
20                 position 1;
21                 description
22                 "D.2.13 of IEEE Std 802.1Q";
23             }
24         }
25         description
26         "Bitmap that includes the evbSet of TLVs from Table D-1 of
27         IEEE Std 802.1Q";
28         reference
29         "D.1 of IEEE Std 802.1Q";
30     }
31     container evb-tlv-extension {
32         description
33         "The EVB TLV";
34         uses evb-tlv;
35     }
36     container cdcv-tlv-extension {
37         description
38         "The CDCV TLV";
39         uses cdcv-tlv;
40     }
41 }
42 augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
43     description
44     "Augments port remote-systems-data with received EVB TLVs";
45     container evb-tlv-extension {
46         description
47         "Holds a received EVB TLV";
48         uses evb-tlv;
49     }
50     container cdcv-tlv-extension {
51         description
52         "Holds a received CDCV TLV";
53         uses cdcv-tlv;
54     }
55 }
56 }
57

```

1 D.6.6.5 The ieee802-dot1q-lldp-ci-tlv YANG module

```

2 module ieee802-dot1q-lldp-ci-tlv {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-lldp-ci-tlv;
5   prefix lldp-ci-tlv;
6   import ieee802-dot1q-lldp {
7     prefix lldp;
8   }
9   import ieee802-dot1q-congestion-isolation {
10    prefix dot1q-ci;
11  }
12  import ietf-inet-types {
13    prefix inet;
14  }
15  import ieee802-types {
16    prefix ieee;
17  }
18  import ieee802-dot1q-types {
19    prefix dot1q-types;
20  }
21  organization
22    "Institute of Electrical and Electronics Engineers";
23  contact
24    "WG-URL: http://ieee802.org/1/
25    WG-EMail: stds-802-1-1@ieee.org
26    Contact: IEEE 802.1 Working Group Chair
27    Postal: C/O IEEE 802.1 Working Group
28    IEEE Standards Association
29    445 Hoes Lane
30    Piscataway, NJ 08854
31    USA
32
33    E-mail: stds-802-1-chairs@ieee.org";
34  description
35    "IEEE Std 802.1Q extension TLVs for LLDP from the Congestion
36    Isolation set (ciSet).
37
38    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
39    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
40
41    Copyright (C) IEEE (2023).
42
43    This version of this YANG module is part of IEEE Std 802.1Q; see the
44    standard itself for full legal notices.";
45  revision 2023-07-03 {
46    description
47      "LLDP extension TLVs for the ciSet. Published as part of IEEE Std
48      802.1Qcz-2023.";
49    reference
50      "Annex D of IEEE Std 802.1Q";
51  }
52  grouping congestion-isolation-tlv {
53    description
54      "Congestion Isolation TLV";
55    reference
56      "D.2.15 of IEEE Std 802.1Q";
57    list queue-map {
58      key "priority";
59      description
60        "Eight integer octets, one entry for each traffic class indexed
61        by priority. Each octet specifies a value that can be translated
62        to the numeric value of the traffic class to be used as either
63        the congesting traffic class or the monitored traffic class for
64        the traffic class specified by the index. The octets range in
65        value from -8 to 8. A value of 0 in the list specifies that the
66        traffic class is not participating in congestion isolation. A
67        positive number specifies a traffic class for a monitored queue
68        that is one less than the value (e.g., a value of 5 represents
69        traffic class 4). A negative number specifies a traffic class for
70        a congesting queue that is one less than the absolute value
71        (e.g., a value of -4 represents traffic class 3).";

```

```

1  reference
2  "D.2.15.3 of IEEE Std 802.1Q";
3  leaf priority {
4      type dot1q-types:priority-type;
5      description
6          "There are eight values of Priority that map to an absolute
7           value that is a traffic class plus one, or the value 0 which
8           indicates the traffic class is not used by congestion
9           isolation.";
10     reference
11     "D.2.15.3 of IEEE Std 802.1Q";
12 }
13 leaf queue-config {
14     type dot1q-ci:abs-traffic-class-plus-one-type;
15     description
16         "A value that can be translated to represent a traffic class or
17         an indication of non-use. A value of 0 specifies that the
18         traffic class is not participating in congestion isolation. A
19         positive number specifies a traffic class for a monitored
20         queue, and a negative number specifies a traffic class for a
21         congesting queue.";
22     reference
23     "D.2.15.3 of IEEE Std 802.1Q";
24 }
25 }
26 leaf cim-encap-length {
27     type uint16;
28     description
29         "The minimum number of octets to include in the Encapsulated MSDU
30         field of each CIM generated. The default value is 48.";
31     reference
32     "D.2.15.4 of IEEE Std 802.1Q";
33 }
34 leaf mac-address {
35     type ieee:mac-address;
36     config false;
37     description
38         "The MAC address to be used as the destination MAC address of a
39         CIM sent by the peer to reach this station.";
40     reference
41     "D.2.15.5 of IEEE Std 802.1Q";
42 }
43 leaf udp-port-number {
44     type inet:port-number;
45     config false;
46     description
47         "The UDP port number to be used as the destination port number of
48         a layer-3 CIM sent by the peer to reach this station.";
49     reference
50     "D.2.15.6 of IEEE Std 802.1Q";
51 }
52 leaf ip-address {
53     type inet:ip-address;
54     config false;
55     description
56         "This leaf holds the IP address that will be used to populate
57         both the address family and IP address fields of the TLV. The IP
58         address field in the TLV is an octet string to be encoded in
59         network octet order with length of 0, 4, or 16 octets dependent
60         upon the address family. If this leaf is an IPv4 address, the
61         address family field is 1 and the IP address field is 4 octets
62         representing the IPv4 address (e.g., IPv4 address 192.0.2.10
63         would be encoded as C0-00-02-0A). If this leaf is an IPv6
64         address, the address family field is 2 and the IP address field
65         is 16 octets representing the IPv6 address. No address shall be
66         provided for any other address families.";
67     reference
68     "D.2.15.7 and D.2.15.8 of IEEE Std 802.1Q";
69 }
70 }
71 augment "/lldp:lldp/lldp:port" {
72     description

```

```
1      "Augments port with Congestion Isolation extension TLVs";
2  leaf tlvs-tx-org-ci-enable {
3      type bits {
4          bit congestion-isolation {
5              position 0;
6              description
7                  "Enables transmission of the Congestion Isolation TLV in the
8                  ciSet";
9          }
10     }
11     description
12         "Bitmap that includes the ciSet of TLVs from Table D-1 of
13         IEEE Std 802.1Q";
14     reference
15         "D.2 of IEEE Std 802.1Q";
16 }
17 container congestion-isolation-tlv-extension {
18     description
19         "The Congestion Isolation TLV";
20     uses congestion-isolation-tlv;
21 }
22 }
23 augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
24     description
25         "Augments port remote-systems-data with received Congestion
26         Isolation extension TLVs";
27     container congestion-isolation-tlv-extension {
28         description
29             "Holds a received Congestion Isolation TLV";
30         uses congestion-isolation-tlv;
31     }
32 }
33 }
34
```

1 D.6.6.6 The ieee802-dot1q-lldp-tr-tlv YANG module

```
2 module ieee802-dot1q-lldp-tr-tlv {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-lldp-tr-tlv;
5   prefix lldp-tr-tlv;
6   import ieee802-dot1ab-lldp {
7     prefix lldp;
8   }
9   organization
10    "Institute of Electrical and Electronics Engineers";
11   contact
12    "WG-URL: http://ieee802.org/1/
13    WG-EMail: stds-802-1-1@ieee.org
14    Contact: IEEE 802.1 Working Group Chair
15    Postal: C/O IEEE 802.1 Working Group
16    IEEE Standards Association
17    445 Hoes Lane
18    Piscataway, NJ 08854
19    USA
20
21    E-mail: stds-802-1-chairs@ieee.org";
22   description
23    "IEEE Std 802.1Q extension TLVs for LLDP from the Topology
24    Recognition set (trSet)
25
26    References in this YANG module to IEEE Std 802.1Q are to IEEE Std
27    802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023.
28
29    Copyright (C) IEEE (2023).
30
31    This version of this YANG module is part of IEEE Std 802.1Q; see the
32    standard itself for full legal notices.";
33   revision 2023-07-03 {
34     description
35      "LLDP extension TLVs for the trSet. Published as part of IEEE Std
36      802.1Qcz-2023.";
37     reference
38      "Annex D of IEEE Std 802.1Q";
39   }
40   grouping topology-recognition-tlv-lldp {
41     description
42      "System level components of Topology Recognition TLV";
43     reference
44      "D.2.16 of IEEE Std 802.1Q";
45     leaf device-type {
46       type enumeration {
47         enum end-station {
48           value 0;
49           description
50            "Indicates that the device is a non-relay end-station or
51            server.";
52         }
53         enum bridge {
54           value 1;
55           description
56            "Indicates that the device is a layer-2 bridge.";
57         }
58         enum router {
59           value 2;
60           description
61            "Indicates that the device is a layer-3 router.";
62         }
63         enum unknown {
64           value 255;
65           description
66            "Indicates that the device type is unknown.";
67         }
68       }
69     }
70     config false;
71     description
72      "Identifies the type of system performing Topology Recognition";
```



```

1      reference
2      "D.2.16.3 of IEEE Std 802.1Q";
3  }
4  leaf topology-level {
5      type uint8;
6      description
7          "A single octet unsigned integer that indicates the system's
8           understanding of its current level in the topology. The value of
9           0 indicates the edge of the topology and the value 255 indicates
10          the level is currently unknown. Other non-zero values indicate
11          the minimum number of links between the system and the edge of
12          the topology. Initially systems may not know their position in
13          the topology and will use the value of unknown by default. As
14          systems discover the device type and topology level of their
15          peers the topology level of the sending system may change.";
16      reference
17          "D.2.16.4 of IEEE Std 802.1Q";
18  }
19 }
20 grouping topology-recognition-tlv-port {
21     description
22         "Port level components of Topology Recognition TLV";
23     reference
24         "D.2.16 of IEEE Std 802.1Q";
25     leaf port-orientation {
26         type enumeration {
27             enum uplink {
28                 value 0;
29                 description
30                     "Indicates that the port is facing a system deeper in the
31                     topology.";
32             }
33             enum downlink {
34                 value 1;
35                 description
36                     "Indicates that the port is facing a system closer to the
37                     edge of the topology.";
38             }
39             enum crosslink {
40                 value 2;
41                 description
42                     "Indicates that the port is facing a system at the same level
43                     in the topology.";
44             }
45             enum unknown {
46                 value 255;
47                 description
48                     "Indicates that the port orientation is unknown.";
49             }
50         }
51     description
52         "Indicates whether the port is facing an uplink, downlink,
53         crosslink or the orientation is currently unknown. Initially
54         systems may not know the port orientation and will use the value
55         of unknown by default. As systems discover the device type and
56         topology level of their peers the port orientation of the sending
57         system may change.";
58     reference
59         "D.2.16.5 of IEEE Std 802.1Q";
60 }
61 }
62 augment "/lldp:lldp" {
63     description
64         "Augments lldp with information at the system level needed to
65         construct Topology Recognition TLVs";
66     reference
67         "D.2.16 of IEEE Std 802.1Q";
68     container topology-recognition-tlv-extension {
69         description
70             "The system level components of the Topology Recognition TLV";
71         uses topology-recognition-tlv-lldp;
72     }

```

```

1  }
2  augment "/lldp:lldp/lldp:port" {
3    when
4      "lldp:name";
5    description
6      "Augments port with Topology Recognition extension TLVs";
7    leaf tlvs-tx-org-tr-enable {
8      type bits {
9        bit topology-recognition {
10         position 0;
11         description
12           "Enables transmission of the Topology Recognition TLV in the
13            trSet";
14       }
15     }
16    description
17      "Bitmap that includes the trSet of TLVs from Table D-1 of
18       IEEE Std 802.1Q";
19    reference
20      "D.2 of IEEE Std 802.1Q";
21  }
22  container topology-recognition-tlv-extension {
23    description
24      "The port level components of the Topology Recognition TLV";
25    uses topology-recognition-tlv-port;
26  }
27 }
28 augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
29   description
30     "Augments port remote-systems-data with received Topology
31      Recognition extension TLVs";
32   container congestion-isolation-tlv-extension {
33     description
34       "Holds a received Topology Recognition TLV";
35     uses topology-recognition-tlv-lldp;
36     uses topology-recognition-tlv-port;
37   }
38 }
39 }
40

```

1 D.6.6.7 The ieee802-dot1q-lldp-pbbn-aa-tlv YANG module

```
2 module ieee802-dot1q-lldp-pbbn-aa-tlv {
3   yang-version "1.1";
4   namespace urn:ieee:std:802.1Q:yang:ieee802-dot1q-lldp-pbbn-aa-tlv;
5   prefix lldp-aa-tlv;
6   import ieee802-dot1q-lldp {
7     prefix lldp;
8   }
9   import ieee802-dot1q-types {
10    prefix dot1qtypes;
11  }
12  organization
13    "Institute of Electrical and Electronics Engineers";
14  contact
15    "WG-URL: http://ieee802.org/1/
16    WG-EMail: stds-802-1-1@ieee.org
17    Contact: IEEE 802.1 Working Group Chair
18    Postal: C/O IEEE 802.1 Working Group
19            IEEE Standards Association
20            445 Hoes Lane
21            Piscataway, NJ 08854
22            USA
23
24    E-mail: stds-802-1-chairs@ieee.org";
25  description
26    "LLDP PBBN Auto Attach TLVs
27
28    Copyright (C) IEEE (2023).
29
30    This version of this YANG module is part of IEEE Std 802.1Q; see the
31    standard itself for full legal notices.";
32  revision 2023-10-26 {
33    description
34      "Published as part of IEEE Std 802.1Qcj-2023.
35
36      The following reference statement identifies each referenced IEEE
37      Standard as updated by applicable amendments.";
38    reference
39      "IEEE Std 802.1Q Bridges and Bridged Networks:
40      IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023,
41      IEEE Std 802.1Qcj-2023.
42      IEEE Std 802 Overview and Architecture: IEEE Std 802-2014.";
43  }
44  identity assoc-state-selector {
45    description
46      "Specify the state of the association between the AAS entities as
47      described by D.2.17.5 of IEEE Std 802.1Q";
48  }
49  identity not-ready {
50    base assoc-state-selector;
51    description
52      "Indicates the association state of NOT_READY as specified in D.2.17.5
53      of IEEE Std 802.1Q. Signaled as the value 0x00.";
54  }
55  identity ready-to-assoc {
56    base assoc-state-selector;
57    description
58      "Indicates the association state of READY_TO_ASSOC as specified in
59      D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x01.";
60  }
61  identity ready-to-attach {
62    base assoc-state-selector;
63    description
64      "Indicates the association state of READY_TO_ATTACH as specified in
65      D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x02.";
66  }
67  identity assoc-failed-types {
68    base assoc-state-selector;
69    description
70      "Indicates the association state of ASSOC_FAILED_TYPES as specified
71      in D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x12.";
```

```
1 }
2 identity assoc-failed-tags {
3     base assoc-state-selector;
4     description
5         "Indicates the association state of ASSOC_FAILED_TAGS as specified
6         in D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x22.";
7 }
8 identity assoc-failed-topo {
9     base assoc-state-selector;
10    description
11        "Indicates the association state of ASSOC_FAILED_TOPO as specified
12        in D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x32.";
13 }
14 identity assoc-failed-other {
15     base assoc-state-selector;
16     description
17         "Indicates the association state of ASSOC_FAILED_OTHER as specified
18         in D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x42.";
19 }
20 identity assoc-attached {
21     base assoc-state-selector;
22     description
23         "Indicates the association state of ASSOC_ATTACHED as specified in
24         D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x03.";
25 }
26 identity assoc-standby {
27     base assoc-state-selector;
28     description
29         "Indicates the association state of ASSOC_STANDBY as specified in
30         D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x13.";
31 }
32 identity assoc-invalid {
33     base assoc-state-selector;
34     description
35         "Indicates the association state of ASSOC_INVALID as specified in
36         D.2.17.5 of IEEE Std 802.1Q. Signaled as the value 0x23.";
37 }
38 identity system-type-selector {
39     description
40         "Specify the selection of the PPBN Auto-attach system type as
41         specified in D.2.17.7 of IEEE Std 802.1Q";
42 }
43 identity aab-system {
44     base system-type-selector;
45     description
46         "Indicates the system type of an Auto Attach BEB (AAB) in the PPBN
47         Auto Attach System TLV specified in D.2.17 of IEEE Std 802.1Q.
48         Signaled as the value 1.";
49 }
50 identity aad-cvlan-system {
51     base system-type-selector;
52     description
53         "Indicates the system type of a C-VLAN aware Auto Attach Device
54         (AAD) in the PPBN Auto Attach System TLV specified in D.2.17 of
55         IEEE Std 802.1Q. Signaled as the value 2.";
56 }
57 identity aad-vlan-unaware-system {
58     base system-type-selector;
59     description
60         "Indicates the system type of a VLAN unaware Auto Attach Device
61         (AAD) in the PPBN Auto Attach System TLV specified in D.2.17 of
62         IEEE Std 802.1Q. Signaled as the value 3.";
63 }
64 identity aad-svlan-system {
65     base system-type-selector;
66     description
67         "Indicates the system type of a S-VLAN aware Auto Attach Device
68         (AAD) in the PPBN Auto Attach System TLV specified in D.2.17 of
69         IEEE Std 802.1Q. Signaled as the value 4.";
70 }
71 identity tagging-selector {
72     description
```

```

1      "Specify the tagging field selection of the PPBN Auto Attach System
2      TLV as specified in D.2.17.8 of IEEE Std 802.1Q";
3  }
4  identity tagged-only {
5      base tagging-selector;
6      description
7          "Indicates the AAD link tagging requirement of all VLAN tagged in
8          AAD sourced frames. Signaled as the value 0 in the PPBN Auto Attach
9          System TLV specified in D.2.17 of IEEE Std 802.1Q.";
10 }
11 identity untagged-or-tagged {
12     base tagging-selector;
13     description
14         "Indicates the AAD link tagging requirement of untagged and VLAN
15         tagged in AAD sourced frames. Signaled as the value 1 in the PPBN
16         Auto Attach System TLV specified in D.2.17 of IEEE Std 802.1Q.";
17 }
18 identity untagged-only {
19     base tagging-selector;
20     description
21         "Indicates the AAD link tagging requirement of all untagged in AAD
22         sourced frames. Signaled as the value 2 in the PPBN Auto Attach
23         System TLV specified in D.2.17 of IEEE Std 802.1Q.";
24 }
25 identity assignment-status-selector {
26     description
27         "Specify the status of an assignment in the PPBN Auto Attach
28         Assignment TLV sent by the AAB for each VID/I-SID assignment request.";
29 }
30 identity pending {
31     base assignment-status-selector;
32     description
33         "Indicates the AAB is processing the assignment. This status is
34         used by the AAD while it is waiting for a response from the AAB as
35         described in D.2.18.6 of IEEE Std 802.1Q. Signaled as the value 1.";
36 }
37 identity accepted {
38     base assignment-status-selector;
39     description
40         "Indicates the AAB assignment processing request is complete for
41         the VID/I-SID and the VLAN to BSI connection has been established
42         as described in D.2.18.6 of IEEE Std 802.1Q. Signaled as the value
43         2.";
44 }
45 identity rejected-generic {
46     base assignment-status-selector;
47     description
48         "Indicates an undefined rejection has occurred as described in
49         D.2.18.6 of IEEE Std 802.1Q. Signaled as the value 3.";
50 }
51 identity rejected-resource {
52     base assignment-status-selector;
53     description
54         "Indicates a rejection has occurred due to system resources being
55         unavailable as described in D.2.18.6 of IEEE Std 802.1Q. Signaled
56         as the value 4.";
57 }
58 identity rejected-invalid-vlan {
59     base assignment-status-selector;
60     description
61         "Indicates a rejection has occurred because the VID value is
62         outside the range of 1 to 4094 as described in D.2.18.6 of IEEE Std
63         802.1Q. Signaled as the value 5.";
64 }
65 identity rejected-vlan-resource {
66     base assignment-status-selector;
67     description
68         "Indicates a rejection has occurred due to maximum VLAN resource
69         limits have been reached as described in D.2.18.6 of IEEE Std
70         802.1Q. Signaled as the value 6.";
71 }
72 identity rejected-invalid-isid {

```

```

1  base assignment-status-selector;
2  description
3      "Indicates a rejection has occurred because the I-SID value is
4      outside of the range 1 or 256 through 16777214 as described in
5      D.2.18.6 of IEEE Std 802.1Q. Signaled as the value 7.";
6  }
7  identity rejected-isid-resource {
8      base assignment-status-selector;
9      description
10         "Indicates a rejection has occurred due to maximum I-SID resource
11         limits have been reached as described in D.2.18.6 of IEEE Std
12         802.1Q. Signaled as the value 8.";
13  }
14  identity rejected-application {
15      base assignment-status-selector;
16      description
17         "Indicates a rejection has occurred because of an issue with Auto
18         Attach agent functions on the AAB as described in D.2.18.6 of IEEE
19         Std 802.1Q. Signaled as the value 9.";
20  }
21  identity rejected-policy {
22      base assignment-status-selector;
23      description
24         "Indicates a rejection has occurred because the Auto Attach
25         assignment processing is subject to a policy or rule on the AAB
26         where the assignment requested is not permitted or denied as
27         described in D.2.18.6 of IEEE Std 802.1Q. Signaled as the value 10.";
28  }
29  typedef aa-isid-type {
30      type dot1qttype:isid-type {
31          range "1 | 256..16777214";
32      }
33      description
34         "The aa-i-sid type represents a Backbone Service Instance
35         Identifier (I-SID). This is the 24-bit I-SID field used in the
36         I-TAG TCI of a Provider Backbone Bridging frame. The values 0,
37         2-254, and 16777215 are reserved for future standardization. The
38         value 255 is dedicated to use by the SPB protocol. For Auto Attach
39         values 2-255 are not allowed.";
40      reference
41         "D.2.18.8 of IEEE Std 802.1Q.";
42  }
43  typedef port-netid-type {
44      type binary {
45          length "12";
46      }
47      description
48         "A 12-octet binary string representing the PortNetID fields of the
49         PPBN Auto Attach System TLV as specified in D.2.17 of IEEE Std
50         802.1Q. The 12 octet field uniquely identifies an Auto Attach port
51         within the administrative domain used for Auto Attach. The first 6
52         octets of the string are the binary representation of the system's
53         MAC address. The next 2 octets are reserved and have the value 0.
54         The last 4 octets are the integer IfIndex for the port.";
55      reference
56         "D.2.17.10 of IEEE Std 802.1Q";
57  }
58  grouping aa-system-tlv {
59      description
60         "PPBN Auto Attach System TLV";
61      reference
62         "D.2.17 of IEEE Std 802.1Q";
63      leaf assoc-state {
64          type identityref {
65              base assoc-state-selector;
66          }
67          config false;
68          description
69             "Association state between AAS entities on the link";
70          reference
71             "D.2.17.5 of IEEE Std 802.1Q.";
72  }

```

```

1  leaf system-type {
2      type identityref {
3          base system-type-selector;
4      }
5      config false;
6      description
7          "Identifies the capability of the advertising system type.";
8      reference
9          "D.2.17.7 of IEEE Std 802.1Q.";
10 }
11 leaf tagging {
12     type identityref {
13         base tagging-selector;
14     }
15     config false;
16     description
17         "Indicates AAD link tagging requirements in AAD-sourced frames
18         and current provisioning mode information.";
19     reference
20         "D.2.17.8 of IEEE Std 802.1Q.";
21 }
22 leaf portnetid {
23     type port-netid-type;
24     config false;
25     description
26         "Uniquely identifies an Auto Attach port within the
27         administrative domain used for Auto Attach connection management.";
28     reference
29         "D.2.17.10 of IEEE Std 802.1Q.";
30 }
31 }
32 grouping aa-assign-tlv {
33     description
34         "PBBN Auto Attach Assignment TLV";
35     reference
36         "D.2.18 of IEEE Std 802.1Q";
37     leaf num-assigns {
38         type uint8 {
39             range "0..101";
40         }
41         config false;
42         description
43             "Contains the number of Status/VID/I-SID triples in the PBBN Auto
44             Attach assignment TLV.";
45         reference
46             "D.2.18.5 of IEEE Std 802.1Q.";
47     }
48     list assignments {
49         key "vlan-id i-sid";
50         config false;
51         description
52             "A triplet of fields in the PBBN Auto Attach Assignment TLV that
53             indicates the mapping of VLAN ID to I-SID and the status of that
54             mapping.";
55         leaf assignment-status {
56             type identityref {
57                 base assignment-status-selector;
58             }
59             config false;
60             reference
61                 "D.2.18.6 of IEEE Std 802.1Q.";
62         }
63         leaf vlan-id {
64             type dot1qttype:vlanid;
65             config false;
66             description
67                 "Advertises the VLAN ID of the VLAN being mapped by the
68                 assignment. If this is an association with a VLAN unaware AAD
69                 this field is transmitted as zero and ignored on receive. If
70                 this is a VLAN aware AAD, then this field is a valid VID.";
71             reference
72                 "D.2.18.7 of IEEE Std 802.1Q.";

```

```

1      }
2      leaf i-sid {
3          type aa-isid-type;
4          config false;
5          description
6              "Advertises the value of the PBBN Backbone Service
7              Instance Identifier (I-SID) mapped by this assignment.";
8          reference
9              "D.2.18.8 of IEEE Std 802.1Q.";
10     }
11 }
12 }
13 augment "/lldp:lldp/lldp:port" {
14     description
15         "Augments port with the PBBN Auto Attach TLVs";
16     leaf tlvs-tx-org-aa-system-enable {
17         type bits {
18             bit aaSystem {
19                 position 0;
20                 description
21                     "D.2.17 of IEEE Std 802.1Q";
22             }
23             bit aaAssign {
24                 position 1;
25                 description
26                     "D.2.18 of IEEE Std 802.1Q";
27             }
28         }
29         config false;
30         description
31             "Bitmap that includes the aaSet of TLVs from Table D-1 of IEEE
32             Std 802.1Q";
33         reference
34             "D.1 of IEEE Std 802.1Q";
35     }
36     container aa-system-tlv-extension {
37         description
38             "The PBBN Auto Attach System TLV";
39         uses aa-system-tlv;
40     }
41     container aa-assign-tlv-extension {
42         description
43             "The PBBN Auto Attach Assignment TLV";
44         uses aa-assign-tlv;
45     }
46 }
47 augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
48     description
49         "Augments port remote-systems-data with received PBBN Auto Attach
50         TLVs";
51     container aa-system-tlv-extension {
52         description
53             "Holds a received PBBN Auto Attach System TLV";
54         uses aa-system-tlv;
55     }
56     container aa-assign-tlv-extension {
57         description
58             "Holds a received PBBN Auto Attach Assignment TLV";
59         uses aa-assign-tlv;
60     }
61 }
62 }
63
64

```


Annex E

(normative)

Notational conventions used in state diagrams

State diagrams are used to represent the operation of the protocol by a number of cooperating state machines each comprising a group of connected, mutually exclusive states. Only one state of each machine can be active at any given time.

Each state is represented in the state diagram as a rectangular box, divided into two parts by a horizontal line. The upper part contains the state identifier, written in upper case letters. The lower part contains any procedures that are executed on entry to the state.

All permissible transitions between states are represented by arrows, the arrowhead denoting the direction of the possible transition. Labels attached to arrows denote the condition(s) that must be met in order for the transition to take place. All conditions are expressions that evaluate to TRUE or FALSE; if a condition evaluates to TRUE, then the condition is met. The label UCT denotes an unconditional transition (i.e., UCT always evaluates to TRUE). A transition that is global in nature (i.e., a transition that occurs from any of the possible states if the condition attached to the arrow is met) is denoted by an open arrow, i.e., no specific state is identified as the origin of the transition. When the condition associated with a global transition is met, it supersedes all other exit conditions including UCT. The special global condition BEGIN supersedes all other global conditions, and once asserted remains asserted until all state blocks have executed to the point that variable assignments and other consequences of their execution remain unchanged.

On entry to a state, the procedures defined for the state (if any) are executed exactly once, in the order that they appear on the page. Each action is deemed to be atomic, i.e., execution of a procedure completes before the next sequential procedure starts to execute. No procedures execute outside of a state block. The procedures in only one state block execute at a time, even if the conditions for execution of state blocks in different state machines are satisfied. All procedures in an executing state block complete execution before the transition to and execution of any other state block occurs, i.e., the execution of any state block appears to be atomic with respect to the execution of any other state block and the transition condition to that state from the previous state is TRUE when execution commences. The order of execution of state blocks in different state machines is undefined except as constrained by their transition conditions. A variable that is set to a particular value in a state block retains this value until a subsequent state block executes a procedure that modifies the value.

On completion of all of the procedures within a state, all exit conditions for the state (including all conditions associated with global transitions) are evaluated continuously until one of the conditions is met. The label ELSE denotes a transition that occurs if none of the other conditions for transitions from the state are met (i.e., ELSE evaluates to TRUE if all other possible exit conditions from the state evaluate to FALSE). Where two or more exit conditions with the same level of precedence become TRUE simultaneously, the choice about which exit condition causes the state transition to take place is arbitrary.

Where it is necessary to split a state machine description across more than one diagram, a transition between two states that appear on different diagrams is represented by an exit arrow drawn with dashed lines, plus a reference to the diagram that contains the destination state. Similarly, dashed arrows and a dashed state box are used on the destination diagram to show the transition to the destination state. In a state machine that has been split in this way, any global transitions that can cause entry to states defined in one of the diagrams are deemed to be potential exit conditions for all of the states of the state machine, regardless of which diagram the state boxes appear in.

Should a conflict exist between a state diagram and either the corresponding global transition tables or the textual description associated with the state machine, the state diagram takes precedence. The interpretation of the special symbols and operators used in the state diagrams is as defined in Table E-1; these symbols and operators are derived from the notation of the “C++” programming language, ISO/IEC 14882. If a Boolean variable is described in this clause as being set it has or is assigned the value TRUE, if reset or clear the value FALSE.

Table E-1—State machine symbols

Symbol	Interpretation
()	Used to force the precedence of operators in Boolean expressions and to delimit the argument(s) of actions and conditional actions within state boxes.
{ }	Used to group multiple actions.
;	Used as a terminating delimiter for actions within state boxes. Where a state box contains multiple actions, the order of execution follows the normal English language conventions for reading text.
=	Assignment action. The value of the expression to the right of the operator is assigned to the variable to the left of the operator. Where this operator is used to define multiple assignments, e.g., a = b = X the action causes the value of the expression following the right-most assignment operator to be assigned to all of the variables that appear to the left of the right-most assignment operator.
!	Logical NOT operator.
&&	Logical AND operator.
	Logical OR operator.
if <expr> <action>	Conditional action. If the Boolean expression, <expr>, evaluates to TRUE, then the <action> following the expression is executed. If multiple actions are to be executed, they are grouped together by means of { }.
!=	Inequality. Evaluates to TRUE if the expression to the left of the operator is not equal in value to the expression to the right.
==	Equality. Evaluates to TRUE if the expression to the left of the operator is equal in value to the expression to the right.
*	Arithmetic multiplication operator.
-	Arithmetic subtraction operator.

1 **Annex F**

2 (informative)

3 **Shared and Independent VLAN Learning (SVL and IVL)**

4 This standard provides for a variety of approaches to the implementation of Bridges from the point of view
5 of the way that individual MAC addresses are learned, and how that learned information is used in
6 subsequent forwarding/filtering decisions. Two mechanisms are used as a basis for these variants:

- 7 a) Making use of address information learned across a number of VIDs in order to make learning
8 decisions relative to any one of those VIDs. This is referred to as Shared VLAN Learning (SVL,
9 3.247).
- 10 b) Making use of address information learned in association with one VID only in order to make
11 learning decisions relative to that VID, and ensuring that it is not used in learning decisions relative
12 to any other VID. This is referred to as *Independent VLAN Learning* (IVL, 3.120).

13 These mechanisms lead to the SVL/IVL model for how a Bridge implements learning and filtering for MAC
14 addresses. Using the terminology of 8.8.8, an SVL/IVL Bridge supports multiple FIDs (which effectively
15 equates to supporting multiple FDBs), and multiple VIDs can use each FID. By varying the number of FIDs
16 supported, and the number of VIDs that can share each FID, the following simplifications of the SVL/IVL
17 model can be created:

- 18 c) *SVL only*. The implementation supports only one FID, so all VIDs share the same learned MAC
19 address information, regardless of which VID the information was learned in.
- 20 d) *IVL only*. Multiple FIDs are supported, but each FID can support only one VID, so each VID makes
21 use only of MAC address information learned within the context of that VID.

22 All three approaches are permitted by this standard, and each has advantages in particular circumstances.
23 The remainder of this annex discusses the following:

- 24 e) The requirements for IVL, SVL, or both.
- 25 f) How Bridges are made aware of the requirement for particular VLANs to be “shared” or
26 “independent.”
- 27 g) How Bridges based on one of these models can interoperate with Bridges based on a different
28 model, in the same network.

29 **F.1 Requirements for Shared and Independent Learning**

30 Under most circumstances, the SVL and IVL approaches work equally well, and Bridges adopting either
31 approach can be freely intermixed within a network. There are, however, a small number of configuration
32 cases where, in order to prevent undue flooding of unicast frames, and in some cases, to make
33 communication between the affected end systems possible, it is necessary to make specific choices about
34 how Bridges that adopt these different learning models are deployed in a network. The following subclauses
35 give examples of some of these configurations and provide a generic statement of the requirements that must
36 be met in order for each learning model to be successfully deployed.

1 F.1.1 Connecting independent VLANs

2 Figure F-1 illustrates how a device that connects two VLANs together, and which therefore itself shares
3 learning between those VLANs, creates a need for those VLANs to be independent in other Bridges.

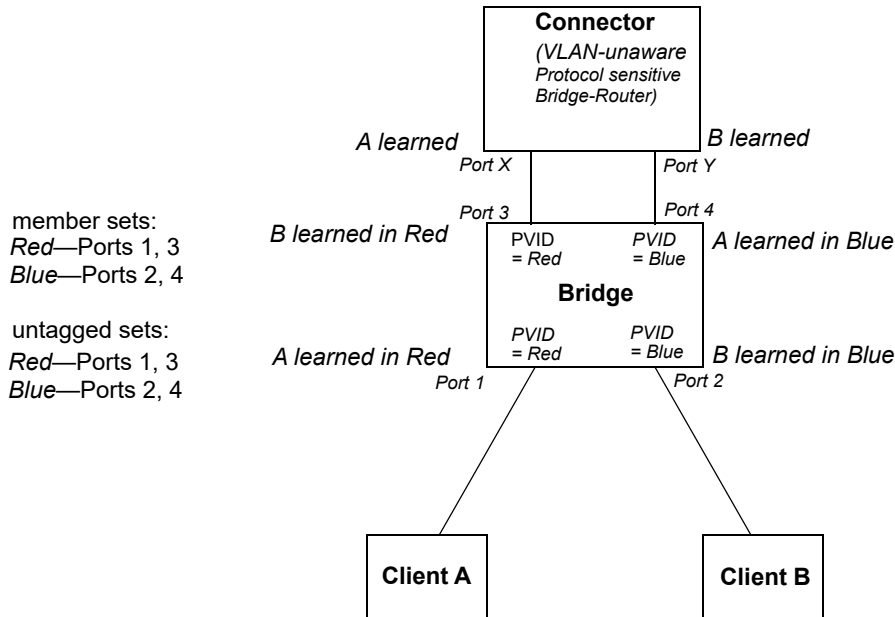


Figure F-1—Connecting independent VLANs—1

4 Clients A and B are connected via the protocol sensitive Bridge-Router (Connector), with an intervening
5 VLAN Bridge. The fact that all Ports of the Bridge carry untagged traffic neatly conceals the fact that the
6 Connector has the effect of connecting VLANs Red and Blue together with regard to bridged traffic. The
7 Connector learns A and B in the same database, as it has no knowledge of VLANs Red and Blue. This
8 prevents any traffic transmitted on the Red VLAN (Port X of the Connector) that is destined for A, from
9 being bridged to Port Y and transmitted on the Blue VLAN.

10 The VLAN Bridge must keep its learning separate for Red and Blue; otherwise, the addresses of the two
11 clients would be alternately learned on diagonally opposite Ports as, for example, traffic sourced by A
12 reenters the Bridge on Port 4 having previously been seen on Port 1.

13 NOTE—This example assumes that the spanning tree protocol is disabled in the Connector, so that the VLAN Bridge
14 does not attempt to suppress the loop that apparently exists if VLANs are not taken into account.

15 A simpler example can be constructed, with a single Port connecting the Connector and the VLAN Bridge, if
16 the Connector is itself VLAN aware and transmits and receives only VLAN-tagged traffic. In this case, the
17 Connector would allocate a single FID for use by Red and Blue. This is shown in Figure F-2.

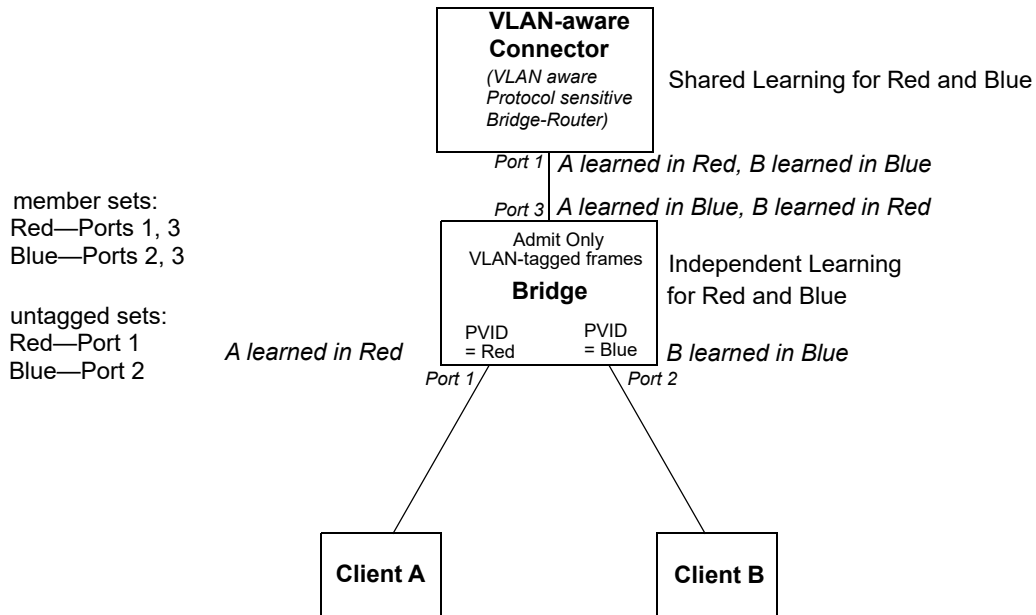


Figure F-2—Connecting independent VLANs—2

1 F.1.2 Duplicate MAC addresses

2 The simplest example of a need for IVL occurs where two (or more) distinct devices in different parts of the
3 network reuse the same individual MAC address, or where a single device is connected to multiple LANs,
4 and all of its LAN interfaces use the same individual MAC address. This is shown in Figure F-3.

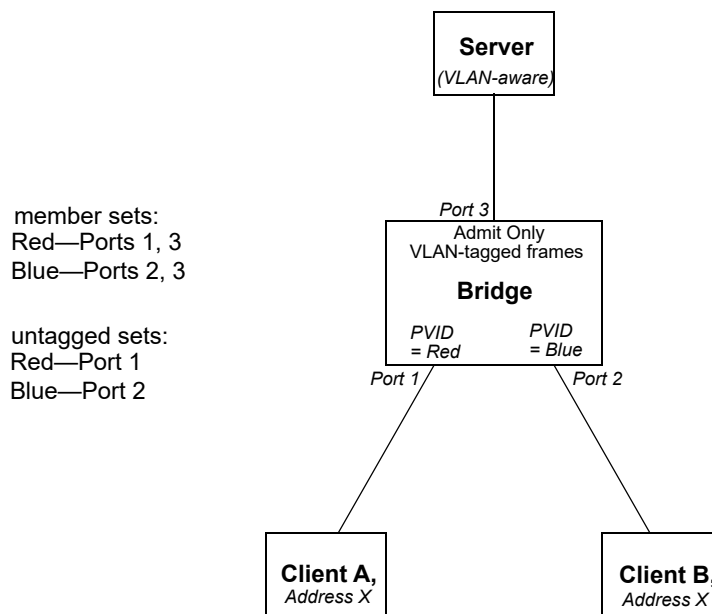


Figure F-3—Duplicate MAC addresses

The example shows two clients with access to the same server; both clients are using the same individual MAC address, X. If the Bridge shares learning between VLAN Red (which serves Client A) and VLAN Blue (which serves Client B), i.e., the Bridge uses the same FID for both VLANs, then Address X will appear to move between Ports 1 and 2 of the Bridge, depending on which client has most recently transmitted a frame. Communication between these Clients and the server will therefore be seriously disrupted. Assignment of distinct FIDs for Red and Blue ensures that communication can take place correctly.

Hence, in order to construct this particular VLAN configuration, either an IVL Bridge or an SVL/IVL Bridge would be required.

F.1.3 Asymmetric VLANs and Rooted-Multipoint connectivity

A primary example of the requirement for SVL is found in “asymmetric” uses of VLANs. Under normal circumstances, a pair of devices communicating in a VLAN environment will both send and receive using the same VID; however, there are some circumstances in which it is convenient to make use of two distinct VLANs, one used for A to transmit to B and the other used for B to transmit to A. Examples of such applications include configuring multi-netted servers, and configuring rooted-multipoint services.

F.1.3.1 Multi-netted Server

An example of Multi-netted Server application of asymmetric VLANs is shown in Figure F-4. Note that:

- In the example, the server and both clients are assumed to be VLAN-unaware devices, i.e., they transmit and receive untagged frames only.
- The ingress classification rules assumed by the example are as defined in this standard, i.e., Port-based classification only.
- The configuration shown can only be achieved by management configuration of appropriate values in Static VLAN Registration Entries (8.8.10) in order to configure the indicated member sets and untagged sets.

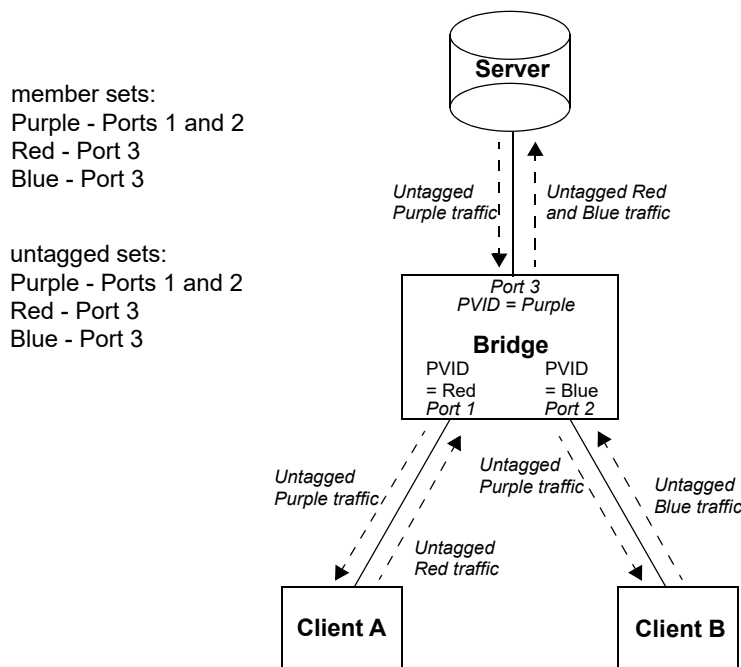


Figure F-4—Asymmetric VID use: “multi-netted server”

1 In the example, Port-based tagging and an asymmetric VLAN configuration is used in order to permit
2 Clients A and B access to a common server, but to prohibit Clients A and B from talking to each other.
3 Examples of where this type of configuration might be required are if the clients are on distinct IP subnets,
4 or if there is some confidentiality-related need to segregate traffic between the clients.

5 Client A transmits to the server via Port 1, which will classify this traffic as belonging to VLAN Red; the
6 Bridge therefore learns Client A's MAC address on Port 1 in association with VLAN Red's VID. The Server
7 transmits its responses to Client A via Port 3, which classifies the return traffic as belonging to VLAN
8 Purple. If individual MAC address learning is configured in the Bridge such that learning is independent
9 between Red and Purple (the VIDs for Red and Purple are allocated to distinct FIDs), then the Bridge will
10 have no knowledge of A in VLAN Purple and will therefore flood the server's responses to Client A on both
11 Port 1 and Port 2. Conversely, if the VIDs for Red and Purple are defined to share the same FID, then the
12 address information learned in Red will be available for use in forwarding the Purple traffic, and the
13 responses to Client A are forwarded only through Port 1.

14 Similarly, there is a need in this configuration for Blue and Purple to share learning information; hence, in
15 order for this configuration to achieve its objectives, the Red, Blue, and Purple VIDs must be allocated to the
16 same FID in the Bridge.

17 Hence, in order to construct this particular VLAN configuration, either an SVL Bridge or an SVL/IVL
18 Bridge would be required.

19 NOTE—The example has been deliberately simplified; in practical applications, the central Bridge would likely be
20 replaced by a number of VLAN Bridges, interconnected with links that would carry the traffic between clients and server
21 as VLAN-tagged frames, with VLAN-tagging and untagging occurring only at the "edge" Ports of the network. An
22 alternative approach to the one described here could also be achieved either by using a VLAN-aware server or by use of
23 more sophisticated ingress classification rules.

24 F.1.3.2 Rooted-Multipoint

25 An example of a rooted-multipoint application of asymmetric VLANs is shown in Figure F-5. The defining
26 characteristic of a rooted-multipoint service is that all Bridge Ports that provide access points to the service
27 are designated as either a Root Port or a Leaf Port, and that the Root Ports can communicate with all other
28 Root Ports and all Leaf Ports, whereas the Leaf Ports can communicate with all Root Ports but not with any
29 other Leaf Ports. Services based on rooted-multipoint connectivity are also known as "Rooted-Multipoint
30 EVC" services as defined in MEF 10.3. Rooted-multipoint connectivity would typically be used by an
31 Internet access provider or a video content provider, where the provider of the service connects to Root
32 Ports, and the subscribers of the service connect to Leaf Ports. This connectivity can be achieved with
33 the "multi-netted server" configuration shown in Figure F-4, but it consumes one VID per subscriber.
34 Figure F-5 shows an alternative that uses just two VIDs: a "Trunk" VID for traffic from Root Ports to Leaf
35 Ports (and to other Root Ports) and a "Branch" VID for traffic from Leaf Ports to Root Ports.

36 To allow the servers to communicate with all other servers and all subscribers, and to allow the subscribers
37 to communicate with all servers but not with other subscribers, the rooted-multipoint connectivity requires
38 the following configuration:

- 39 a) The server(s) and all subscribers transmit and receive untagged frames only. Note that for a rooted-
40 multipoint service on a PBN, this means the frames do not contain S-TAGs at the customer service
41 interface (in other words, the User Network Interface (UNI)). Therefore, rooted-multipoint can be
42 supported at Port-based service interfaces (15.3) and C-tagged service interfaces (15.4).
- 43 b) The ingress C-VLAN classification may be Port-based or Port-and-Protocol-based for rooted-
44 multipoint on CBNs. The ingress S-VLAN classification may be Port-based or C-TAG-based for
45 rooted-multipoint on PBNs.
- 46 c) In Customer Bridges and S-VLAN Bridges, the indicated values of the member sets and untagged
47 sets (8.8.10) are achieved by management configuration of Static VLAN Registration Entries

- (8.8.2). In Provider Edge Bridges the Trunk and Branch VIDs are also mapped to the appropriate CNP/PEP pair by management configuration of the PEP Table (12.13.2.4, ieee8021PbEdgePortTable 17.6.1.2). Note that connectivity between Leaf Ports is blocked by not including Leaf Ports in the member set associated with the Branch VID.
- d) In Customer Bridges and S-VLAN Bridges the indicated values of the PVID parameters are configured by management. In Provider Edge Bridges the equivalent configuration is achieved by management configuration of the C-VID Registration Table (12.13.2.2, ieee8021PbCVidRegistrationTable 17.6.1.2).
- e) The Enable Ingress Filtering parameter (8.6.2) is reset, i.e., ingress VLAN filtering is disabled, on Leaf Ports.
- f) The Trunk and Branch VIDs share the same FID (to enable SVL).

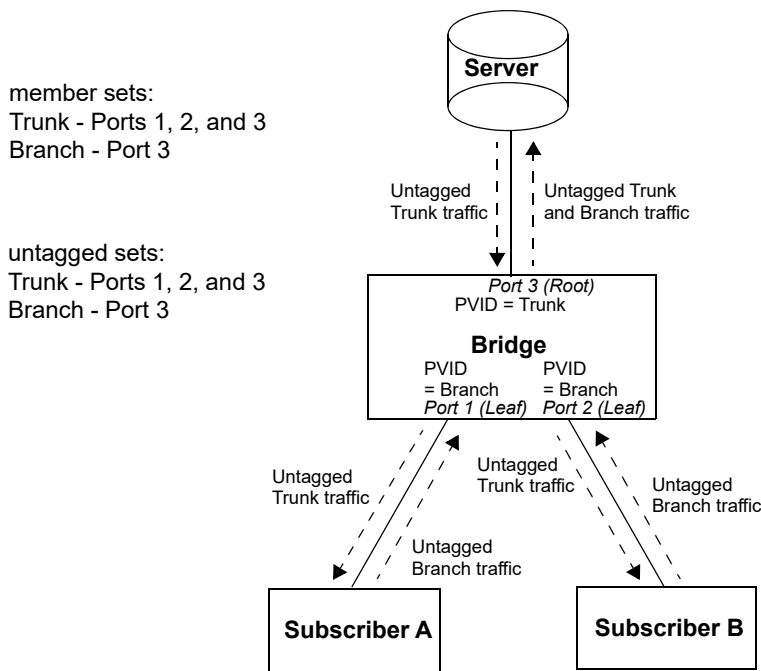


Figure F-5—Asymmetric VLAN use: “Rooted-Multipoint”

NOTE 1—The example has been deliberately simplified; in practical applications, the central Bridge would likely be replaced by a number of VLAN Bridges, interconnected with links that would carry the traffic between subscribers and servers as VLAN-tagged frames, with only the Root and Leaf Ports of the network carrying untagged frames.

Rooted-multipoint services can be carried on PBBNs by configuring Trunk and Branch S-VIDs as described above, with the Root and Leaf Ports at CNPs either at an I-component of a BEB, or at a Provider Edge Bridge of a PBN that then connects to the PBBN. The I-component of a BEB can then either map the Trunk and Branch S-VIDs to separate backbone service instances (i.e., mapped to separate I-SIDs) or bundle them into a single backbone service instance.

Rooted-multipoint connectivity can also be created on a PBBN by configuring Trunk and Branch B-VIDs as described above, with the Root and Leaf Ports at CBPs of a B-component in a BEB. One or more backbone service instances can then be mapped to the Trunk and Branch B-VIDs to take advantage of the rooted-multipoint connectivity.

As noted in item a) above, the configurations described above apply to scenarios where the ingress and egress frames at a Root or Leaf Port on a Customer Bridge or S-VLAN Bridge are untagged. Customer Bridges and S-VLAN Bridges that support both the VID Translation Table and Egress VID Translation Table (6.9) can support tagged ingress and egress frames at a Root or Leaf Port. Figure F-6 shows the configuration where the server(s) and subscribers transmit and receive tagged frames. The server uses VID R for frames associated with the rooted-multipoint service, and the subscribers use VIDs L1 and L2. The values of R, L1, and L2 could be, but are not necessarily, the same. The configuration of the Bridge(s) with Root and Leaf Ports are shown in the diagram. All Root and Leaf Ports are included in the member set of the Trunk and Branch VIDs, but not in the untagged set (including the Leaf Ports in the member set of the Branch VID allows ingress VLAN filtering to be enabled at these ports if desired). The VID translation at a Root Port ensures that ingress frames go to the Trunk VID, and egress frames come from both the Trunk and Branch VIDs. The VID translation at a Leaf Port ensures that ingress frames go to the Branch VID, and egress frames come from the Trunk VID (translating the Branch VID to 0xFFFF on egress frames at Leaf Ports causes these frames to be discarded).

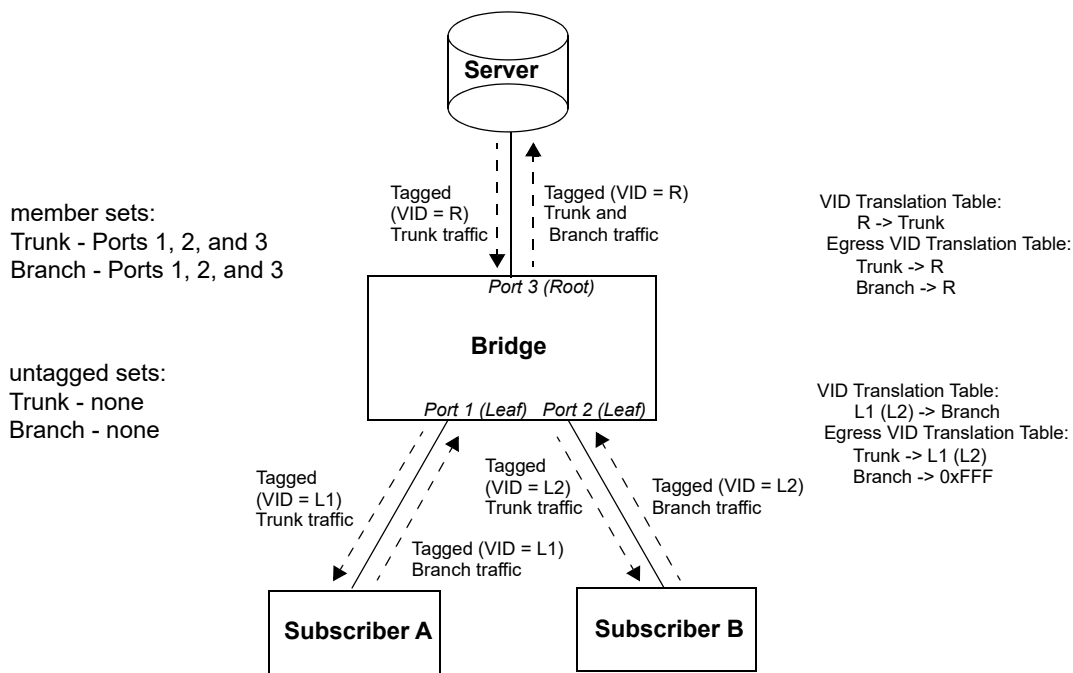


Figure F-6—Rooted-Multipoint with tagged interfaces

NOTE 2—The example has been deliberately simplified; in practical applications, the central Bridge would likely be replaced by a number of VLAN Bridges, interconnected with links that would carry the traffic between subscribers and servers as frames tagged with either the Trunk or Branch VID. VLAN translation occurs only at the Root and Leaf Ports of the network.

F.1.4 Shared learning and Shortest Path Bridging VID (SPBV) mode

SPBV mode uses shared learning among the set of SPVIDs that support a given SPBV VLAN. Each SPT Bridge supporting an SPBV VLAN is assigned a unique VID (SPVID) for transmitting VLAN frames into the SPT Region. This SPVID is registered by the ISIS-SPB control plane along an SPT rooted at the SPT Bridge to which it is assigned. The SPTs determined by ISIS-SPB provide symmetric bidirectional paths (i.e., congruent paths) between any pair of SPT Bridges within an SPT Region. The congruency of the paths is essential to allow shared learning. Shared learning is required because frames forwarded from a given SPT Bridge will contain a different SPVID from frames being forwarded to that SPT Bridge.

Figure F-7 shows an example of a simple SPBV VLAN operating in an SPT Region comprising three Bridges A, B, and C. Each of the Bridges is assigned an SPVID, in this example using the same letter designation as the Bridge to which it is assigned. ISIS-SPB maintains the VID member sets such that the SPVID of an SPT Bridge follows an SPT rooted at that particular SPT Bridge. The SPT includes all boundary ports participating in the SPBV VLAN (in this example the ports connected to stations A, B, and C).

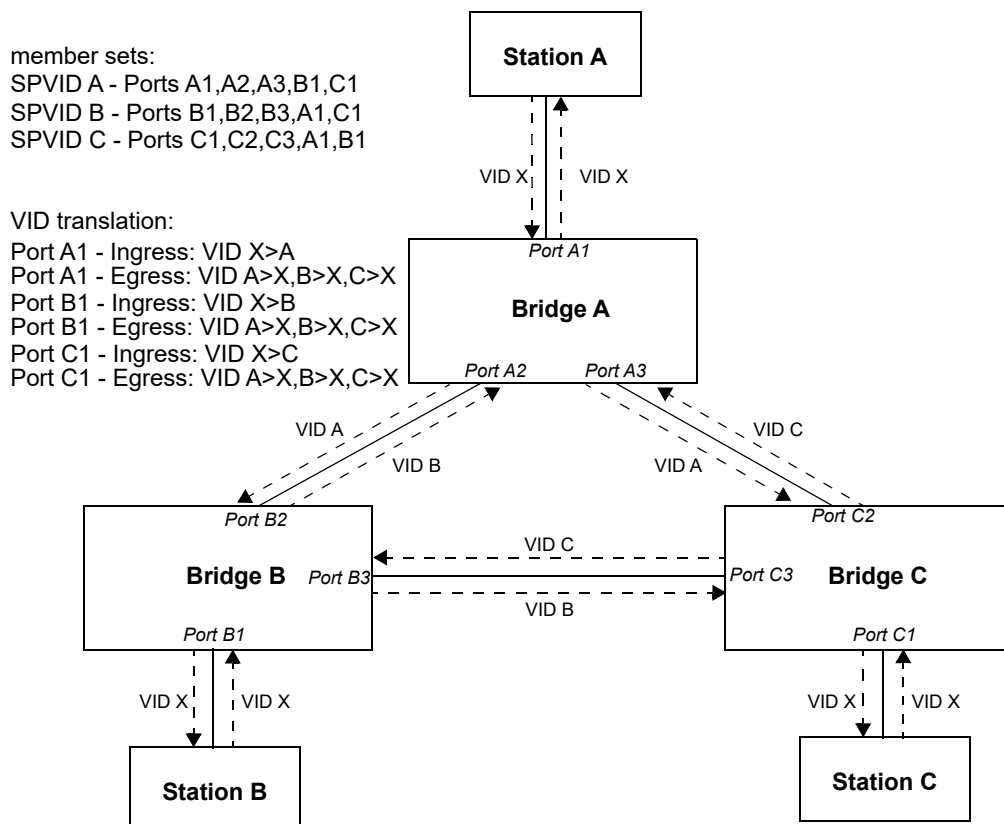


Figure F-7—SPBV VLAN Shared Learning and VID Translation

Shared learning is used among all the SPVIDs belonging to the SPBV VLAN. This enables the filtering information learned from frames transmitted from a Bridge (with that Bridge's SPVID) to be applied to frames being transmitted to that Bridge from other Bridges using the other Bridges' SPVIDs. For example, a frame sent from station A to station B will be transmitted by Bridge A using SPVID A. At Bridge B the frame's source address will be learned at port B2. When station B later sends a frame to station A, Bridge B will assign the frame SPVID B. Shared learning enables Bridge B to use the MAC address previously learned from the frame with SPVID A to filter the frame with SPVID B to transmit only on port B2.

Enhanced VID translation capabilities allow a single VID to be used for VLANs on boundary ports connected to an SPT Region even though many SPVIDs with shared learning are used within the SPT Region. In this example stations A, B, and C all use VID X. At each boundary port ingress translation is set to translate the external VID (usually the Base VID) to the SPVID for the Bridge on which the boundary port resides. Egress VID translation is set to translate all the SPVIDs belonging to the SPBV VLAN to the external VID.

1 F.1.5 Generic constraints on SVL and IVL use

2 This subclause describes the general constraints on the mapping of VIDs to FIDs, from the point of view of
3 a given Bridge that learns from or forwards frames on a set of VIDs (the Bridge’s “active set” of VIDs). If:

- 4 a) The individual MAC addresses associated with each point of attachment to the active set of VIDs
5 are unique (i.e., the “Duplicate MAC Address problem” is not present), and
- 6 b) There is no Bridge or Bridge-like device that takes frames from one VID in the active set and
7 subsequently transmits them on another VID in the active set, then

8 every VID in the active set may share the same FID; in other words, individual MAC address information
9 learned in the context of any one VID may be used in forwarding decisions taken relative to any of the
10 others, so the SVL approach can be used in that Bridge.

11 Furthermore, if:

- 12 c) Each bidirectional, individual MAC-Addressed, conversation between pairs of end stations makes
13 use of the same VID in both directions, then

14 every VID in the active set may be allocated a distinct FID (with the possibility of a little extra flooding as
15 learning of addresses in the context of one VID does not contribute to forwarding decisions for that address
16 relative to any other VID). Under these circumstances, rule b) may also be relaxed and restated as follows:

- 17 d) Frames associated with one VID in the active set may be received by (up to) one Bridge and
18 transmitted using another VID in the active set, provided that there is no loop in such VLAN to
19 VLAN forwarding, e.g., for a set of VIDs Red, Blue, and Green, there is no logical loop in copying
20 frames between VIDs, such as copying from Red to Green by one Bridge, Green to Blue by another,
21 and Blue back to Red by a third.

22 So:

- 23 e) If rules a), b), and c) are true, and d) is false, for all VIDs in the active set, then either an SVL or an
24 IVL Bridge can be deployed;
- 25 f) If rules a) and b) are true, and c) and d) are false for all VIDs in the active set, then only an SVL
26 Bridge can be deployed;
- 27 g) If rules a) or b) or d) are false, and c) is true for all VIDs in the active set, then only an IVL Bridge
28 can be deployed.

29 These conditions are all on the basis that they apply “for all VIDs in the active set.” Clearly, in more
30 complex scenarios, some VIDs in the active set will have requirements that dictate SVL behavior on the part
31 of a given Bridge, while others will have requirements that dictate IVL behavior. Under such circumstances,
32 an SVL/IVL Bridge is required, allowing those VIDs that need to be shared to be mapped to a single FID,
33 while those that need to be independent are mapped to distinct FIDs. Needless to say, wherever an SVL or
34 IVL Bridge can be deployed, it can successfully be replaced by an appropriately configured SVL/IVL
35 Bridge.

Annex G

(informative)

MAC method-dependent aspects of VLAN support

G.1 Example tagged IEEE 802.3 EtherType-encoded frame format

Figure G-1 provides an illustrative example of a single tagged EtherType-encoded frame format as used in an Ethernet to Ethernet Bridge. This illustration is only of the simplest case, that is, a single-level, fixed-size tagging between identical MACs.⁶²

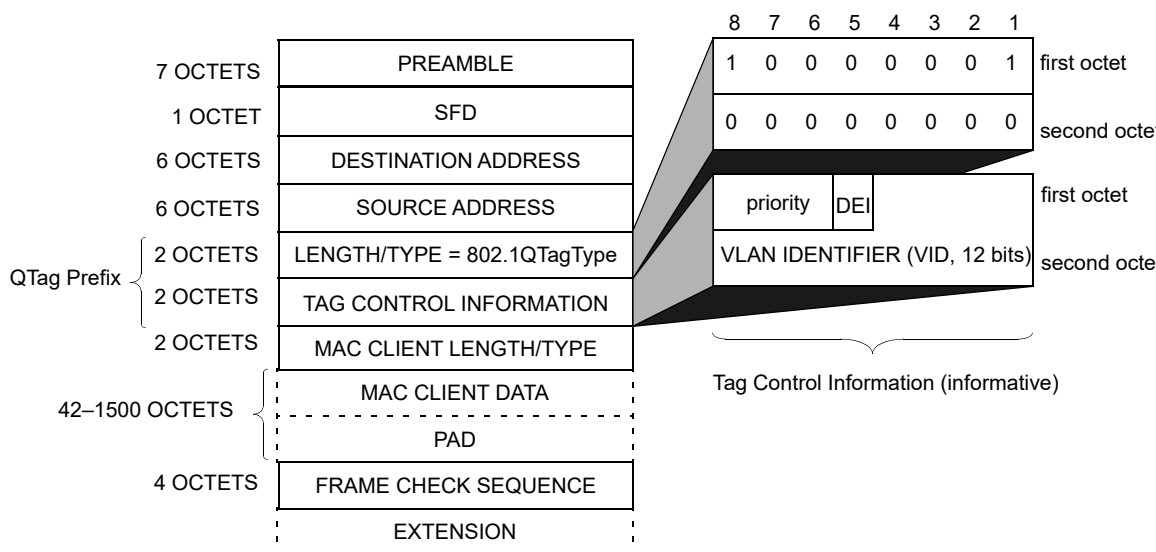


Figure G-1—Example of IEEE 802.3 MAC frame format

G.2 Padding and frame size considerations

G.2.1 Treatment of PAD fields in IEEE 802.3 frames

The minimum frame size constraint placed on IEEE 802.3/Ethernet frames requires frames to carry zero or more pad octets following the MAC client data, in order to ensure that no frame of total length less than 64 octets is transmitted on the medium. This requirement means that frames whose overall length would otherwise be less than 64 octets in length have (64-len) octets of padding added after the MAC client data, where len is the size of the frame before padding.

⁶² Figure G-1 was part of Clause 3 of IEEE Std 802.3-2005.

1 When tagged frames are transmitted by a Bridge on an IEEE 802.3 MAC, there are two permissible
2 approaches, as follows:

- 3 a) Keep the minimum frame size generated by the Bridge equal to 64 octets. This implies that the
4 number of pad octets in a received untagged IEEE 802.3 frame would be reduced by up to 4 octets
5 when that frame was tagged.
- 6 b) Adopt a minimum tagged frame length of 68 octets. This implies that the number of pad octets in a
7 received untagged IEEE 802.3 frame would not be adjusted when tagging such frames; equally, if
8 subsequently untagged, no pad adjustment would be necessary before transmission on IEEE 802.3/
9 Ethernet.

10 There is a similar choice to be made in end stations that generate tagged frames:

- 11 c) In some existing implementations, the decision about whether pad octets are needed will be made at
12 a point where it is impractical to distinguish between tagged and untagged frames. In these cases, the
13 end station will use a minimum frame size of 64 octets for all frames.
- 14 d) In other cases, the padding decision will be taken at a point before it is known whether the frame will
15 be transmitted tagged or untagged. In these cases, the end station will use a minimum tagged frame
16 size of 68 octets and a minimum of 64 octets for untagged frames.

17 These approaches are all consistent with the IEEE 802.3 frame specification.

18 The implication is that, for correct operation on IEEE 802.3/Ethernet, all devices have to be capable of
19 correctly handling tagged frames of less than 68 octets in length (G.2.3).

20 **G.2.2 Maximum PDU size**

21 VLAN tagging of an untagged frame, or relaying frames in tagged frame format, can result in an increase in
22 the length of the original frame. If transmission of a given frame in tagged frame format through a given
23 destination Port would result in violation of the maximum PDU size for the destination MAC method, the
24 Bridge discards the frame for that destination Port.

25 NOTE—Violation of the maximum PDU sizes for destination MAC methods can produce undefined results in networks
26 that contain devices that adhere strictly to these maxima, or in MAC methods where these maxima are inherently
27 constrained by the operation of the MAC method itself (e.g., constrained by timing considerations in the MAC state
28 machines).

29 IEEE Std 802.3 defines an extension to the normal IEEE 802.3 maximum frame size for the specific purpose
30 of accommodating the additional octets of the VLAN tag header. The example frame translations in this
31 annex make use of this extension to the IEEE 802.3 frame size.

32 **G.2.3 Minimum PDU size**

33 VLAN untagging of a tagged frame results in the original frame decreasing in length.

34 Where the destination MAC is Ethernet:

- 35 a) If untagging a given frame would result in violation of the minimum frame length requirements of
36 Ethernet, the Bridge is required to adjust the PAD field to ensure that the frame length equals the
37 minimum length of 64 octets (G.2.1).
- 38 b) If a frame is transmitted in tagged frame format, the Bridge may adopt a minimum tagged frame
39 length of either 64 or 68 octets, as an implementation option. If the latter is chosen, the Bridge
40 adjusts the size of the PAD field on transmission for any tagged frame that is less than 68 octets in
41 length (G.2.1).

1 **G.3 Tag insertion and removal for LLC media**

2 In IEEE Std 802.1Q-2014 and all previous revisions of IEEE Std 802.1Q, the method of inserting and
3 deleting tags was independent of whether a Length/Type field is used for protocol type encoding. The tag
4 was inserted into or deleted from the first octets of the `mac_service_data_unit` and the remainder of the
5 `mac_service_data_unit` was unchanged.

6 That deletion/insertion method did not provide for interoperability between VLAN-aware end stations
7 where only one used a Length/Type field to encode protocol identifiers. It specified that the station on the
8 LLC media used a SNAP-encoded VLAN tag followed by a SNAP-encoded data unit, whereas on the
9 station using Length/Type, the VLAN tag was followed by an EtherType protocol identifier. These multiple
10 encodings required a bridge to translate both the VLAN tag and the following protocol identifier, along with
11 any other tags defined previously or subsequently. Translation of the second and any subsequent protocol
12 identifiers is impossible if the contents of the frame are protected against modification or rendered
13 confidential, as in the case of a frame protected by MAC Security (IEEE Std 802.1AE).

14 To avoid this problem, revisions of IEEE Std 802.1Q subsequent to 2014, and IEEE Std 802.1AC
15 subsequent to 2012, specify that following the first protocol identifier, all subsequent protocol identifiers (if
16 any) use Length/Type encoding.

17 The net result was that the expected format for a tagged frame changed. For example, on LLC media, the
18 first 18 octets of a VLAN-tagged SNAP-encoded MSDU carrying an EtherType of 08-00 (hexadecimal) for
19 the user data, priority 0, VLAN 123, was, in revisions prior to IEEE Std 802.1Q-2018:

20 AA-AA-03-00-00-00-81-00-01-23-AA-AA-03-00-00-00-08-00

21 and subsequently, is:

22 AA-AA-03-00-00-00-81-00-01-23-08-00

23 The underlined octets are no longer present. That is, on LLC media, all tags and protocol identifiers after the
24 first are Length/Type encoded.

25 **G.4 IEEE 802.11 and PMPN media**

26 Figure G-2 illustrates an example of a five-port Bridge that uses three of the methods defined by
27 IEEE Std 802.1AC whereby a Bridge can connect to an IEEE 802.11 medium, or to a Point-to-Multipoint
28 Network (PMPN). The following subclauses describe these methods.

29 **G.4.1 IEEE 802.11 Portal convergence**

30 A Bridge, along with the portal convergence function defined in 13.2.1 of IEEE Std 802.1AC-2016 [B10],
31 together constitute an example of an IEEE 802.11 Portal.

32 **G.4.2 Point-to-Multipoint Network convergence: multiple connections**

33 The PMPN convergence function, defined in 13.6.2 of IEEE Std 802.1AC-2016 [B10], provides a set of
34 virtual point-to-point instances of the ISS, one to each PMPN node to which an underlying PMPN node is
35 associated. The convergence function maps this set of ISS instances to a single instance of the service
36 defined for the PMPN node. This interface supports PMPN media interior to a Bridged Network or Virtual
37 Bridged Network. In Figure G-2, the center block in the diagram has three Bridge Ports connecting to three
38 of these instances.

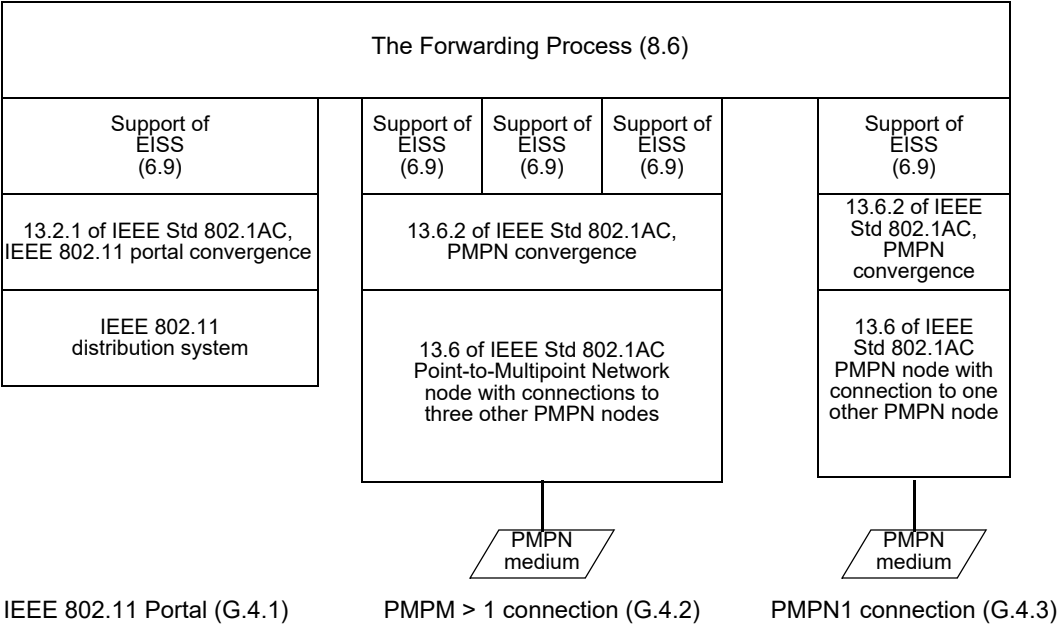


Figure G-2—Methods for Bridge access to IEEE 802.11 and PMPN media: example

1 G.4.3 Point-to-Multipoint Network convergence: single connection

2 The PMPN convergence function, defined in 13.6.2 of IEEE Std 802.1AC-2016 [B10], is also used between
3 the ISS and the MAC service interface defined for a PMPN node that has, or is capable of, only a single
4 connection.

1 Annex H

2 (informative)

3 Interoperability considerations

4 VLAN Bridges are able to interoperate in networks with other VLAN Bridges. However, the VLAN-based
5 filtering service defined in this standard, as provided in the context of a single spanning tree for the network,
6 involves some constraints on the network topology and individual device configurations that differ from the
7 set of constraints that apply to the building and configuration of networks based only on MAC Bridges.

8 In addition, VLAN Bridges are able to interoperate with MAC Bridges, as well as with both priority-aware
9 and VLAN-aware end systems. Both the VLAN-based filtering service and the tag insertion and removal
10 service specified in this standard cause constraints on intermixed network topologies and device
11 configurations that again differ from the building and configuration of networks containing MAC Bridges.

12 The implications of certain device configurations may not be immediately apparent from the technical detail
13 of this standard. In order to clarify the nature of the additional constraints, H.1 through H.4

- 14 a) Describe the basic requirements for interoperability.
- 15 b) Discuss those requirements in the context of homogeneous and heterogeneous configurations, with
16 examples of some of the problems that can occur if these requirements are not adhered to.

17 H.1 Requirements for interoperability

18 Two primary aspects of the configuration of a network are of concern from the point of view of
19 interoperability:

- 20 a) Establishing a consistent view of the static filtering configuration of Bridges in the network.
- 21 b) Ensuring that untagged frames are VLAN-tagged (and that the tag is subsequently removed)
22 consistently regardless of spanning tree reconfigurations.

23 H.1.1 Static filtering requirements

24 Static filtering controls allow the network administrator to impose a level of control over the permitted
25 connectivity in the network, by setting static MAC address filters in the FDBs of Bridges, and by controlling
26 the extent of particular VLANs by manipulation of Static VLAN Registration Entries (8.8.2).

27 In order to ensure that end station-to-end station connectivity (or the lack of it) is consistent in all possible
28 spanning tree configurations, any static filters need to be established taking account of the full mesh
29 topology of the physical interconnections between Bridges in the network, not just the “normal” spanning
30 tree topology to which the network is configured when all Bridges and LANs are operating correctly. An
31 example of the consequences of failure to establish consistent controls for static VLAN filtering is given in
32 H.2.1.

33 H.1.2 Configuration requirements for VLAN-tagging

34 VLAN Bridges classify incoming untagged frames by applying either a Port-based tagging rule on ingress
35 that uses the PVID for the reception Port as the VLAN classification for such frames or a
36 Port-and-Protocol-based rule that uses the frame’s ULP to select one of a port’s VIDs. Maintaining

1 consistent connectivity between any pair of end stations that are on the same VLAN, and where one or both
2 of those end stations is VLAN-unaware, requires that:

- 3 a) All VLAN Bridge Ports that are connected to the same LAN apply a consistent set of ingress rules
4 (8.6).
- 5 b) All VLAN Bridge Ports that are connected to the same *legacy region* of a network apply a consistent
6 set of ingress rules.
- 7 c) All VLAN Bridge Ports that serve LANs to which members of the same VLAN are (or can be)
8 attached apply a consistent set of ingress rules.

9 A legacy region of a network consists of any set of LANs that are physically interconnected via MAC
10 Bridges. A legacy region has the property that, by appropriate configuration of the spanning tree, a spanning
11 tree path could be created between any pair of LANs in the region such that the path would pass only
12 through MAC Bridges.

13 NOTE—In case b), spanning tree reconfiguration within the legacy region can change the logical connectivity between
14 the VLAN Bridge Ports and the LANs that they (directly or indirectly) serve. Hence, a spanning tree reconfiguration
15 could result in any end stations connected to the legacy region being serviced via any VLAN Bridge Port. In effect, such
16 a reconfiguration reduces case b) to case a). Figure H-2 and Figure H-3 give examples of this type of configuration. In
17 Figure H-2, the legacy region consists of all three LANs and both MAC Bridges. In Figure H-3, the legacy region
18 consists of the MAC Bridge and both LANs to which it is attached. An example of case c) is where an end station
19 attached to a leaf LAN is in the same VLAN as a server that is attached to a distinct LAN, i.e., all possible spanning tree
20 paths between the two stations pass through a VLAN-aware region of the network.

21 The essence of what these rules express is that if a given untagged frame belongs on a given VLAN, then the
22 classification and tagging behavior of any VLAN Bridges that are required to tag that frame needs to be the
23 same, regardless of the logical connectivity that is created by the spanning tree configuration of the network.
24 Examples of the consequences of failure to apply these rules appear in H.3 and H.4.

25 H.2 Homogeneous VLAN-aware networks

26 This standard requires new considerations in building a network in which all Bridges are VLAN Bridges.
27 The arbitrary plug-and-play capability of MAC Bridges in creating a network topology is restricted when
28 making use of the VLAN extensions defined in this standard.

29 H.2.1 Consistency of static VLAN filtering

30 In order for stations that are members of a given VLAN to be able to reach other members of the same
31 VLAN elsewhere in the network, all Bridge Ports that are part of the spanning tree active topology (i.e., all
32 Bridge Ports that are in a forwarding state) connecting the stations must be included in the member set
33 (8.8.10) for the VID that identifies that VLAN. In order for this connectivity to be independent of any
34 reconfiguration of the spanning tree topology, all paths among those stations, both forwarding and blocked,
35 must have this characteristic. Use of management controls to manipulate the member set (e.g., filters for
36 security) must be applied in a manner consistent with requirements of the full mesh topology of the network.

37 An inconsistency occurs, for example, if a VID is restricted from an active path, but not from a redundant
38 path currently blocked by the operation of spanning tree. Should a spanning tree reconfiguration enable the
39 previously blocked path, the restriction will no longer be in place. In the reverse, a spanning tree
40 reconfiguration may suddenly impose a restriction that had not existed. A common use of such management
41 restriction will likely arise from managers who make use of an “access” port construct. An access port may
42 be a port that is absent from the member set (8.8.10) for all VIDs but the untagged, default VID. Should such
43 an access port become the active connection between two portions of the network as a result of a spanning
44 tree reconfiguration, all VLANs but the one identified by the default VID will be partitioned at that point in
45 the topology.

1 In Figure H-1, the trunk style link and access style link cause a loop through the left and right portions of the
2 network. STP will block one or the other. Should the spanning tree block at point #1, all 2000 users may
3 communicate on any of the 50 VLANs. However, should the spanning tree block at point #2, the left and
4 right portions of the network will be partitioned on all VLANs excepting VLAN 3 (the VLAN carried via
5 the access style link.)

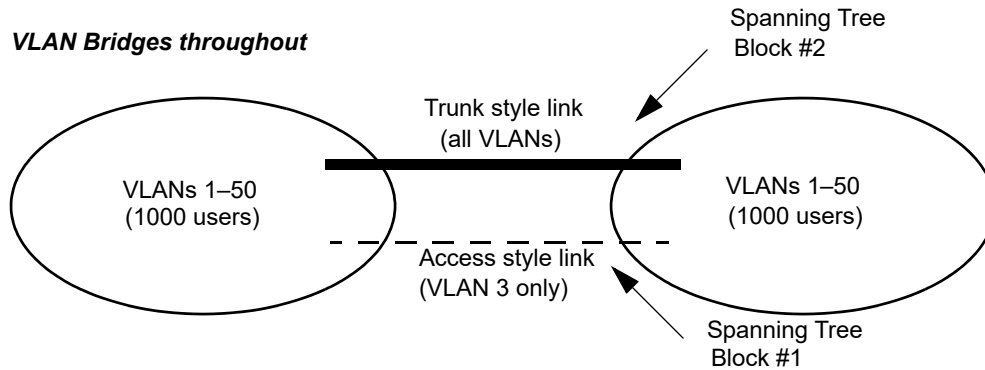


Figure H-1—Static filtering inconsistency

6 H.2.2 Consistent view of the “untagged VLAN(s)” on a given LAN

7 In the Port-based VLAN model defined in this standard, the PVID for a Bridge Port provides the VLAN
8 classification for all frames on the attached LAN that are received in untagged form. Any LAN with more
9 than one VLAN Bridge attached has such an “untagged VLAN” for each VLAN Bridge. No explicit
10 requirement that PVID be consistent for all VLAN Bridges on the same LAN, nor mechanism to ensure
11 such, has been included in this standard.

12 Similarly, in the Port-and-Protocol-based VLAN model defined in this standard, one member of the VID Set
13 for a Bridge Port provides the VLAN classification for all frames on the attached LAN that are received in
14 untagged form with a particular ULP. Any LAN with more than one such VLAN Bridge attached has such a
15 set of “untagged VLANs” for each VLAN Bridge. No explicit requirement that these be consistent for all
16 VLAN Bridges on the same LAN, nor mechanism to ensure such, has been included in this standard.

17 Consider the case of a LAN to which are attached three VLAN Bridges, each of which is using Port-based
18 classification and is capable of transmission of untagged frames onto the LAN. An untagged frame placed
19 on that LAN by any one of the VLAN Bridges will be associated by each of the other two VLAN Bridges
20 with its own configured PVID for its reception Port on that LAN. The VLAN model requires that each frame
21 have a unique VLAN association, and that association is represented by a single, global VID value.
22 Therefore, it follows that all VLAN Bridges on that LAN must make use of the same classification rules (in
23 this case, the same PVID) for their ports connected to that LAN.

24 It has been suggested that in the special case of a direct point-to-point connection between two VLAN
25 Bridges or other VLAN-aware devices, other rules might apply. No mechanism for identifying such links
26 has yet been suggested.

27 This creates a configuration challenge for installers of VLAN Bridges that conform to this standard. Initial
28 management configuration of the VLAN Bridges (the setting of PVIDs, VID Sets, and Protocol Group
29 Databases) must be made consistent among the VLAN Bridges, in a manner that takes into account the
30 actual physical topology. Changes to the physical topology may require specific changes to the configuration
31 of all affected switches. These requirements effectively disallow a plug-and-play installation as supported by
32 MAC Bridges, unless all VLAN Bridges are left with their default Port-based classification rules and with
33 each PVID = 1.

1 H.3 Heterogeneous networks: Intermixing MAC Bridges (M) and VLAN 2 Bridges (V)

3 This subclause discusses networks in which VLAN Bridges are intermixed with MAC Bridges.

4 A principal limitation in intermixing VLAN Bridges with MAC Bridges is that the VLAN filtering services
5 are not universally available throughout the network. Also, services for the insertion and removal of tags are
6 not universally available. Furthermore, spanning tree reconfigurations may cause filtering services, as well
7 as tag insertion and removal services, to become available or become unavailable independent of actions of
8 affected users.

9 H.3.1 Example: Adding a VLAN Bridge to provide filtering to a MAC Bridged 10 Network

11 Example problems can be shown with the following topology diagrams. Figure H-2 includes one V Bridge
12 and two M Bridges.

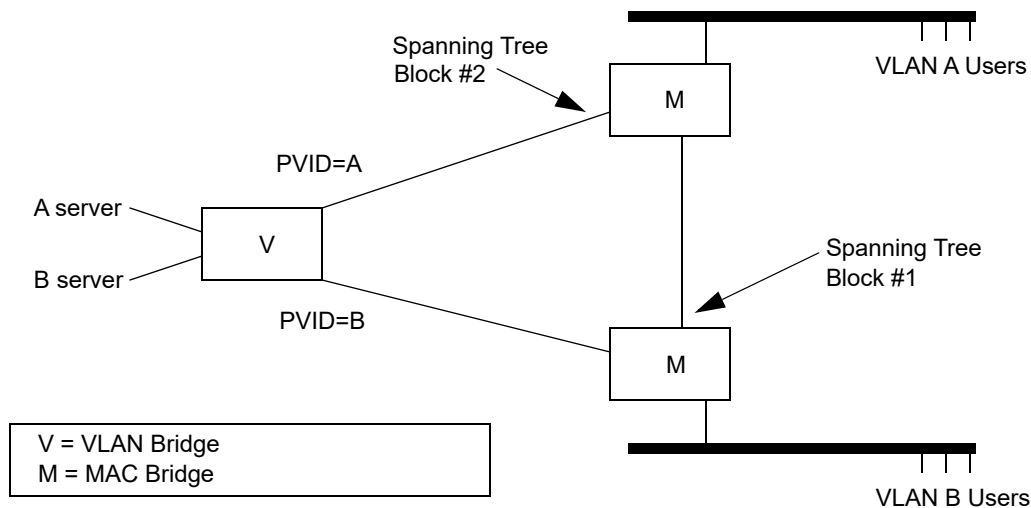


Figure H-2—Interoperability with MAC Bridges: example 1

13 If the spanning tree protocol determines to break the loop among the three Bridges by blocking at point #1,
14 connectivity within each VLAN is as desired. However, should the block occur at point #2, traffic from
15 VLAN A users will pass through both M Bridges and be treated as VLAN B traffic upon arrival in the V
16 Bridge. Connectivity to the A server will be lost for the A users.

1 H.3.2 Example: Adding a MAC Bridge to a (previously) Homogeneous VLAN 2 Bridged Network

3 A similar problem, demonstrating the impact of placing an M Bridge within an otherwise homogeneous V
4 topology, can be shown by the configuration in Figure H-3. This figure includes two V Bridges and adds a
5 single redundant M Bridge.

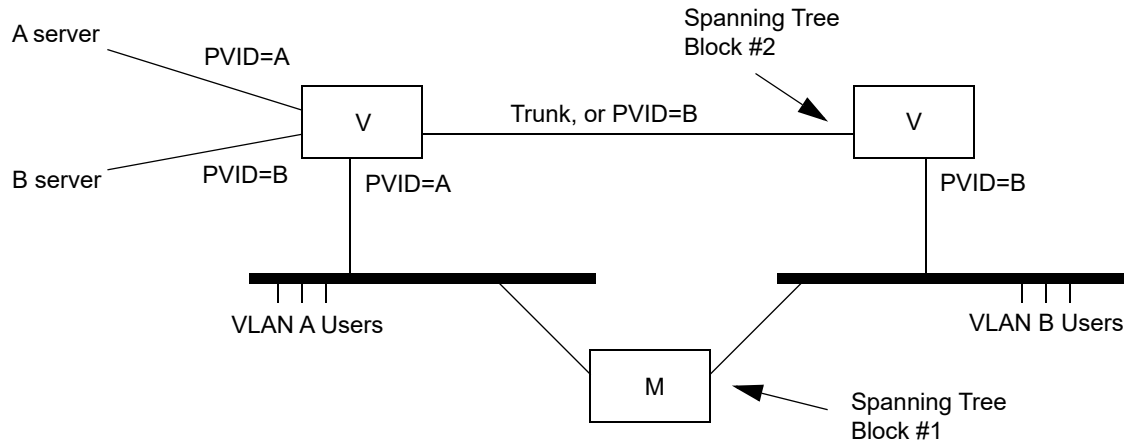


Figure H-3—Interoperability with MAC Bridges: example 2

6 If STP determines to break the loop among the three Bridges by blocking at point #1, connectivity within
7 each VLAN is as desired. The two V switches operate as expected. A and B VLAN frames are
8 VLAN-tagged on arrival in either V Bridge and forwarded only to the appropriate servers. Now suppose an
9 STP reconfiguration results in a block at point #2, but not at #1. This redirects VLAN B user traffic through
10 the M Bridge. VLAN B users no longer have their traffic identifiably distinct from VLAN A. An immediate
11 consequence is that the VLAN B users will no longer have access to the “B server.”

12 H.4 Intermixing Port-based classification and Port-and-Protocol-based 13 classification or future enhancements in VLAN Bridges

14 Intermixing V Bridges with M Bridges has a direct analogue in the mixing of Bridges implementing only
15 Port-based and Port-and-Protocol-based classification of frames in VLAN Bridged networks. This revision
16 and potential subsequent editions of this standard extend the VLAN classification capabilities to support
17 more sophisticated ingress rules for frame classification.

18 In VLAN configurations that use both Port-based and Port-and-Protocol-based VLAN classification, a
19 Bridge that supports only Port-based VLAN classification will merge VLANs that would otherwise be
20 classified separately by a Bridge that supports Port-and-Protocol-based VLAN classification. To get around
21 this problem, it may be possible to dedicate specific Ports to specific protocols in Bridges that support only
22 Port-based VLAN classification, as in the example shown in Figure H-4. However, such solutions may not
23 be possible where there are multiprotocol end stations in the network.

1 H.4.1 Example: Intermixing Protocol-based ingress rules

2 Consider the case where Bridges implement a configuration mechanism to select between Port-based
3 classification rules (a “V-Port Bridge”) and Port-and-Protocol-based rules (a “V-Port/Protocol Bridge”).
4 This case would allow, for example, for support for IP and IPX as distinct VLANs. The topology shown in
5 Figure H-4 might apply when a V-Port/Protocol Bridge is added to a topology; otherwise, using only V-Port
6 Bridges allows users of two protocols to participate in two separate VLANs.

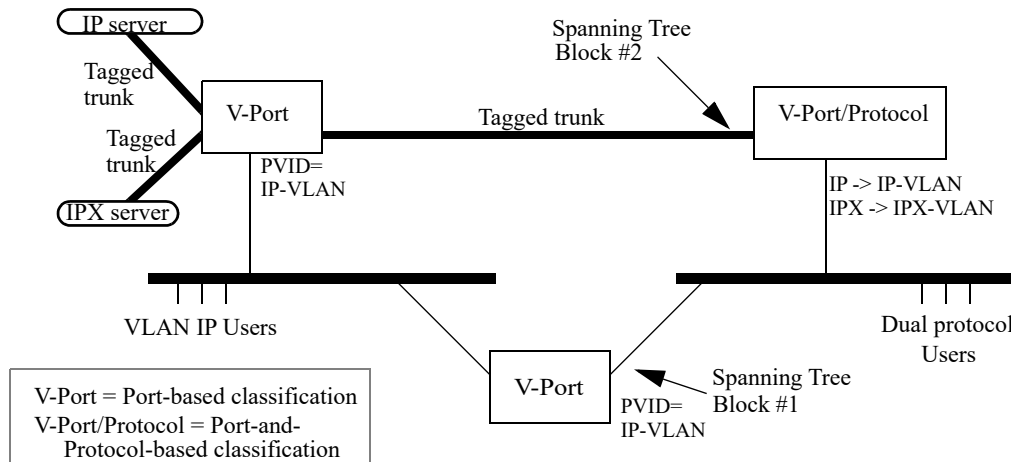


Figure H-4—Interoperability between Port-based and Port-and-Protocol-based classification

7 Consider this network, when STP has blocked at point #1, and not at #2. The upper V-Port Bridge operates
8 as expected, and the V-Port/Protocol Bridge provides Port-and-Protocol-based classification for the frames
9 received from the dual protocol users on the right-hand LAN. IP-VLAN and IPX-VLAN frames are
10 VLAN-tagged on the trunks to and from the uppermost Bridges and servers. But if a STP reconfiguration
11 should result in a block at point #2, but not at #1, activating traffic through the lower V-Port Bridge, dual
12 protocol users will have all their traffic treated as part of the IP-VLAN. An immediate consequence is that
13 the uppermost Bridge will no longer provide them access to the “IPX server.” It is the fact that the lower
14 V-Port Bridge must have just one of the two VLANs configured for all untagged traffic, regardless of its
15 protocol, that leads to this lack of connectivity.

16 H.4.2 Differing views of untagged traffic on a given LAN

17 Further challenges arise when one considers the case where several V Bridges, some implementing
18 Port-based and some implementing Port-and-Protocol-based classification, all attach to the same LAN.
19 Again, the rule that any given frame exists in exactly one VLAN requires that all of these Bridges be
20 configured with consistent ingress rules. In this case, the V-Port Bridges will provide a least common
21 capability, and this further requires common configuration of the PVID in the V-Port and V-Port/Protocol
22 Bridges.

Annex I

(informative)

Priority and drop precedence

This standard allows management of priorities, flow metering, queue assignment, and queue servicing. This annex documents the rationale for the recommended and default priority to traffic class mappings in Table 8-5 and the encoding of priority and drop eligibility in Table 6-2 and Table 6-3.

Classification of user data frames into a small number of behavior aggregates, together with aggregate dependent forwarding behavior in each Bridge, allows signaling of application requirements to the network. Frame classification, aggregate bandwidth metering, policing, and drop precedence marking also facilitate network scaling and provision of services to independent customers through allocation of those functions to appropriate Bridges in the network.

While there are many possible ways to classify frames and to specify forwarding behaviors, it is widely appreciated that a set of well-known and easily understood defaults can facilitate interoperability and the deployment of services. The defaults described in this annex and supported by this standard were chosen to support integrated and differentiated services, to minimize the burden of management, to reduce the possibility of misconfiguration and out-of-order frame delivery, and to provide useful service without management in many networks.

This standard mandates support for strict priority frame transmission (8.6.6) but permits the use of additional traffic class-based transmission selection algorithms. The default assignments of frames to traffic classes on the basis of frame priority, as described in this annex, also support the use of frame priority to select general traffic class-based forwarding behavior.

NOTE—Annex X provides references to the IETF work on integrated and differentiated services ([B21] through [B25] and [B26] through [B33]).

I.1 Traffic types

A full description of the QoS needs of applications and network services is too complex to be represented by a simple number 0 through 7. The pragmatic aim of traffic classification is to simplify requirements to preserve the high-speed, low-cost characteristics of Bridges. At the margin, potential bandwidth efficiency is traded for simplicity and higher speed operation—historically a good decision in the LAN.

The following list of traffic types, each of which can benefit from simple segregation from the others, are of general interest:

- a) Network Control—characterized by a guaranteed delivery requirement to support configuration and maintenance of the network infrastructure.
- b) Internetwork Control—in large networks comprising separate administrative domains there is typically a requirement to distinguish traffic supporting the network as a concatenation of those domains from the Network Control of the immediate domain.
- c) Voice—characterized by less than 10 ms delay and, hence, maximum jitter (one way transmission through the LAN infrastructure of a single campus).
- d) Video—characterized by less than 100 ms delay, or other applications with low latency as the primary QoS requirement.
- e) Critical Applications—characterized by having a guaranteed minimum bandwidth as their primary QoS requirement and subject to some form of admission control to ensure that one system or application does not consume bandwidth at the expense of others. The admission control mechanism

- 1 can range from preplanning of the network requirement at one extreme to bandwidth reservation per
2 flow at the time the flow is started at the other.
- 3 f) Excellent Effort—or “CEO’s best effort,” the best-effort type services that an information services
4 organization would deliver to its most important customers.
- 5 g) Best Effort—for default use by unprioritized applications with fairness only regulated by the effects
6 of TCP’s dynamic windowing and retransmission strategy.
- 7 h) Background—bulk transfers and other activities that are permitted on the network but that should
8 not impact the use of the network by other users and applications.

9 I.2 Managing latency and throughput

10 Use of priorities and queuing by traffic classes, each class encompassing one or more priorities, facilitates
11 improvement and management of latency and throughput, allowing QoS goals to be supported at higher
12 levels of network loading than would otherwise be possible.

13 Congestion, resulting in QoS degradation, is not equally likely at all Bridges in a network. Transient traffic
14 patterns are likely to result in congestion in only a few Bridges at a time, while over an extended period,
15 momentary congestion is more likely to occur in the network core than at Bridge Ports attached to one or a
16 relatively small number of end stations. Use of fewer traffic classes for those Ports can lower the cost of
17 implementation and management, and this standard facilitates the use of Bridges supporting differing
18 numbers of classes within a single network that delivers a consistent set of QoS parameters for each frame
19 priority level. Although the number of traffic classes supported by each Bridge Port along the path taken by
20 a given flow of data can vary, the default mappings of priorities to classes ensures that frame ordering is
21 preserved as required by 8.6.6.

22 With few classes, the focus is on meeting latency requirements—the bandwidth surplus required in a bursty
23 data environment to guarantee sub-10 ms delays without a distinct traffic classification is uneconomically
24 large. As the number of traffic classes that can be used increases, the focus shifts to managing throughput.

25 The simple default queue servicing policy defined in this standard, strict priority, supports latency
26 management. Active management of bandwidth sharing necessarily requires some management.

27 I.3 Traffic type to traffic class mapping

28 Table I-1 is an example of how traffic types can be grouped to match the number of traffic class queues
29 supported by a Bridge Port. Each grouping of types is shown as *{Distinguishing type, Type, Type, . . .}*. The
30 “distinguishing type” is not treated in any way differently in a Bridge but is italicized here to illustrate, for
31 any given number of queues, which traffic types have driven the allocation of types to classes.

32 The step-by-step breaking out of traffic types as more classes are available proceeds as follows:

- 33 a) With a single queue, there are no choices. All traffic is Best Effort.
- 34 b) To support integrated services in the presence of bursty best effort data, it is necessary to segregate
35 all time-critical traffic. The amount of high-priority traffic will be restricted by the need to support
36 low latency for Voice, which becomes the defining type for the additional queue.
- 37 c) Two queues may be adequate for Bridge Ports attaching to end stations. The stability of the network
38 as a whole may be unaffected by the performance of configuration protocols on those Ports, and in-
39 band management of the Bridge is likely to occur through another Port. For Bridges within the
40 network infrastructure, a further queue is used to isolate Network Control from the user data traffic.
- 41 d) Traffic for business Critical Applications is separated from Best Effort to allow a bandwidth
42 guarantee to be provided.

- 1 e) The queue separation so far provided can support a large network. The next queue is allocated to
2 distinguishing Internetwork Control traffic from local Network Control.
- 3 f) Background is separated from Best Effort to minimize the effect of bulk transfers on ordinary
4 network use.
- 5 g) Excellent Effort is separated from Critical Applications, either to provide a simple superior service
6 based on policy controlled access or to provide an additional segregated bandwidth guarantee.
- 7 h) The final provides increased network utilization as the higher bandwidth traffic associated with
8 Video is no longer given the same latency guarantee as Voice. This description is illustrative rather
9 than definitive of the logic of allocating traffic types to classes. The mappings in Table 8-5 support
10 the assignment of other semantics to each traffic type identified by priority values, e.g., the
11 identification of all three illustrative types “Video,” “Critical Applications,” and “Excellent Effort”
12 with ensured forwarding classes that provide segregated bandwidth guarantees. However, alternate
13 semantics should take into account the service provided by Bridges with limited traffic class
14 queuing; e.g., of the foregoing, only “Video” would receive priority treatment by default at a Bridge
15 Port supporting queuing for only two classes.

Table I-1—Traffic type to traffic class mapping

Number of queues	Traffic types
1	{ <i>Best Effort</i> , Background, Excellent effort, Critical Applications, Voice, Video, Internetwork Control, Network Control}
2	{ <i>Best Effort</i> , Background, Excellent effort, Critical Applications} { <i>Voice</i> , Video, Internetwork Control, Network Control}
3	{ <i>Best Effort</i> , Background, Excellent effort, Critical Applications} { <i>Voice</i> , Video} { <i>Network Control</i> , Internetwork Control}
4	{ <i>Best Effort</i> , Background} { <i>Critical Applications</i> , Excellent effort} { <i>Voice</i> , Video} { <i>Network Control</i> , Internetwork Control}
5	{ <i>Best Effort</i> , Background} { <i>Critical Applications</i> , Excellent effort} { <i>Voice</i> , Video} { <i>Internetwork Control</i> } { <i>Network Control</i> }
6	{ <i>Background</i> } { <i>Best Effort</i> } { <i>Critical Applications</i> , Excellent effort} { <i>Voice</i> , Video} { <i>Internetwork Control</i> } { <i>Network Control</i> }
7	{ <i>Background</i> } { <i>Best Effort</i> } { <i>Excellent effort</i> } { <i>Critical Applications</i> } { <i>Voice</i> , Video} { <i>Internetwork Control</i> } { <i>Network Control</i> }
8	{ <i>Background</i> } { <i>Best Effort</i> } { <i>Excellent effort</i> } { <i>Critical Applications</i> } { <i>Video</i> } { <i>Voice</i> } { <i>Internetwork Control</i> } { <i>Network Control</i> }

1.4 Traffic types and priority values

Table I-2 shows the correspondence between traffic types and priority values used to select the defaults in Table 8-5. The default priority used for transmission by end stations is 0. Changing this default would result in confusion and likely in interoperability problems. At the same time, the default traffic type is definitely Best Effort. 0 is thus used both for default priority and for Best Effort, and Background is associated with a priority value of 1. This means that the value 1 effectively communicates a lower priority than 0.

Table I-2—Traffic type acronyms

Priority	Acronym	Traffic type
1	BK	Background
0 (Default)	BE	Best Effort
2	EE	Excellent Effort
3	CA	Critical Applications
4	VI	“Video,” < 100 ms latency and jitter
5	VO	“Voice,” < 10 ms latency and jitter
6	IC	Internetwork Control
7	NC	Network Control

Table I-3 summarizes Table I-1, showing just the defining traffic types. By maintaining the groupings of types established for a given number of queues for all less numbers, the table preserves the order of frames of any given type, independent of the number of queues.

Table I-3—Defining traffic types

Number of queues	Defining traffic type							
1	BE							
2	VO				BE			
3	NC		VO		BE			
4	NC		VO		CA	BE		
5	NC	IC	VO		CA	BE		
6	NC	IC	VO		CA	BE	BK	
7	NC	IC	VO		CA	EE	BE	BK
8	NC	IC	VO	VI	CA	EE	BE	BK

This discussion of traffic types, and the suggested association of each with a priority value, differs from the similar discussion in Annex G of IEEE Std 802.1D-2004 [B14] and prior revisions of that standard. The latter was developed contemporaneously with IETF Intserv and predates Diffserv. The discussion in this

1 annex better aligns with current practice; in particular, Voice is associated with priority 5, matching the
2 setting of the relevant bits for Expedited Forwarding (EF) in the DSCP (Differentiated Services Code Point)
3 for IP and in the common use of the EXP bits for MPLS. Standards for DSCPs are believed to be the prime
4 reference for use of priority by end stations, and there is no direct change to the behavior of Bridge
5 implementations conforming to this standard as a result of this change.

6 The priority to traffic class mappings in Table 8-5 differ in one minor respect from those specified in prior
7 revisions of this standard and in IEEE Std 802.1D-2004 [B14] and its prior revisions. Priority value 2 was
8 previously described as ‘Spare’ and positioned lower than 0 (Best Effort) in priority order. This change may
9 result in networks, including Bridges, conformant to prior revisions of this standard, and implementing four
10 or more traffic classes, providing less-than-expected priority to traffic described in this annex as Excellent
11 Effort, and misordering drop eligible traffic for Critical Applications. The change allows better use of the
12 available traffic classes, given the low demand for two distinct priorities of lesser importance than Best
13 Effort, and the best use of Priority Code Point when encoding drop eligibility.

14 I.5 Supporting the credit-based shaper algorithm

15 Table 34-1 and Table 34-2 define a set of recommended priority to traffic class mappings where the credit-
16 based shaper algorithm (8.6.8.1) is supported by one or two of the available traffic classes; the recommended
17 mappings shown are intended for use where priority 3 is used to support SR class A and priority 2 is used to
18 support SR class B.

19 In order for the credit-based shaper algorithm to operate in the way it is intended, it is necessary to ensure
20 that the shaper algorithm is supported on the numerically highest traffic class(es). Hence, if two traffic
21 classes are used to support the shaper algorithm, it follows that the minimum useful number of traffic classes
22 that a Port could support is three.

23 Table I-4 redraws Table I-3 for the case where only SR class B (SR-B) is supported. The defining traffic type
24 for the numerically highest traffic class is SR-B in all cases. VI does not appear in this table since SR-B is
25 used for the video traffic.

Table I-4—Defining traffic types—Credit-based shaper support of SR class B only

Number of queues	Defining traffic type							
2	SR-B	BE						
3	SR-B	NC			BE			
4	SR-B	NC			CA		BE	
5	SR-B	NC	IC		CA		BE	
6	SR-B	NC	IC		CA		BE	BK
7	SR-B	NC	IC		CA	EE	BE	BK
8	SR-B	NC	IC	VO	CA	EE	BE	BK

26 NOTE—When one or more SR classes are supported, the primary distinction being made is between traffic that requires
27 bandwidth reservation and latency guarantees on the one hand, and different types of “best effort” traffic on the other.
28 The audio and video traffic types identified in Table I-1 are therefore not applicable.

1 Table I-5 redraws Table I-3 for the case where SR classes A (SR-A) and B (SR-B) are supported in place of
2 VO and VI. The defining traffic types for the two numerically highest traffic classes are SR-A and SR-B in
3 all cases; for 4 through 8 traffic classes, the remaining traffic types are broken out in the same order as in
4 Table I-3.

Table I-5—Defining traffic types—Credit-based shaper support of SR classes A and B

Number of queues	Defining traffic type							
3	SR-A	SR-B	BE					
4	SR-A	SR-B	NC		BE			
5	SR-A	SR-B	NC		CA		BE	
6	SR-A	SR-B	NC	IC	CA		BE	
7	SR-A	SR-B	NC	IC	CA		BE	BK
8	SR-A	SR-B	NC	IC	CA	EE	BE	BK

5 I.6 Supporting drop precedence

6 It is often desirable to meter traffic from different users to ensure fairness or to meet bandwidth guarantees;
7 however, dropping all traffic in excess of a committed rate is likely to result in severe under-utilization of the
8 networks, because most traffic sources are bursty in nature. At the same time, it is burdensome to meter
9 traffic at all points in the network where bandwidth contention occurs. One solution is to mark those frames
10 in excess of the committed rate as drop eligible on admission to the network.

11 This standard allows drop eligibility to be conveyed separately from priority in VLAN tags so that all
12 previously introduced traffic types can be marked as drop eligible. To provide compatibility with previous
13 revisions of this standard while allowing drop eligibility to be conveyed in C-TAGs, this standard also
14 allows a subset of the priorities to be conveyed along with drop eligibility marking for some of those
15 priorities within the PCP field of both S-TAGs and C-TAGs.

16 I.7 Priority Code Point allocation

17 The Priority Code Point of VLAN tags allows encoding of five, six, seven, or eight distinct priorities, with a
18 single level of drop eligibility on three, two, one, or zero of those priorities, respectively. Table 6-2 and
19 Table 6-3 specify encoding and decoding for five through eight priorities. The tables are consistent with the
20 following step-by-step reduction in the number of distinct priorities to provide drop eligibility for certain
21 traffic types:

- 22 a) If eight distinct priorities are required, drop eligibility cannot be encoded in the Priority Code Point
23 (but may be encoded in VLAN tags using the DEI).
- 24 b) Drop eligibility in support of QoS maintenance for traffic conforming to a committed rate is most
25 effective when used to support time-critical traffic. If seven priorities, one of which can be marked
26 as drop eligible, are required, then the traffic class queuing distinction between Voice and Video is
27 sacrificed to providing drop eligibility for the combined traffic types. This does not preclude
28 marking all Video traffic as drop eligible upon ingress to a network to provide the same guarantee to
29 Voice as a distinct priority.

- 1 c) The distinctions between Critical Applications and Excellent Effort, and between Best Effort and
2 Background traffic types, is removed to provide drop eligibility for Critical Applications and for
3 Best Effort.
- 4 d) Although the use of four priorities, each with drop eligibility, is possible, it is not recommended.
5 Combining Network Control with Internetwork Control could only serve to increase the guarantees
6 provided to the latter at the expense of the former, which if not delivered threatens the stability of the
7 overall network in any case. Moreover, both traffic types should be supportable with guaranteed
8 bandwidth if the network is to be operated successfully.

9 Choosing first to combine Video and Voice, and then Critical Applications with Excellent Effort (for six
10 queues), provides consistency with the allocation of priorities to traffic classes in the absence of drop
11 eligibility. Bridges that do not implement drop eligibility, but are configured to use the same number or
12 fewer traffic classes, will not misorder frames. If such a Bridge is configured to use only five traffic classes,
13 and in accordance with Table 8-5, it will not misorder frames with a Priority Code Point encoded using any
14 of the alternatives provided by Table 6-2.

15 **1.8 Interoperability**

16 Encoding of drop eligibility within the Priority Code Point, as opposed to explicitly with the DEI, provides
17 interoperability with Bridges that are not capable of supporting the DEI. It also provides compatibility with
18 the use of the MPLS EXP bits to convey priority and drop eligibility.

19 However, the requirement to provide different combinations of priorities with drop eligibility within the
20 confines of the Priority Code Point means that priority and drop eligibility information can be lost for frames
21 traversing a network if the combinations used on individual LANs differ. Use of the DEI does not suffer
22 from this problem. In some cases, loss of drop eligibility information at the boundary between
23 administrative domains risks impacting profile conformant traffic from some users with out-of-profile drop
24 eligible traffic from others; in other cases, the drop eligible marking has already done its job (the next LAN,
25 for example, might deliver traffic to a single customer). Without explicit management, a Bridge Port cannot
26 decide, so the Require Drop Encoding parameter (8.6.7) has been provided.

27 If Bridges attached to the same LAN encode and decode the Priority Code Point differently, then incorrect
28 priority values can be attributed and subsequent misordering of frames can occur. Misordering will not
29 occur, with the recommended priority to traffic class mappings of Table 8-5 and the recommended Priority
30 Code Point encoding and decoding in Table 6-2 and Table 6-3, if the Bridge performing the incorrect
31 decoding assumes fewer priorities than are actually encoded or if all Bridges subsequently transited by the
32 frame use the same number or fewer traffic classes than those used for the encoding. However, incorrect
33 decoding will in all probability affect other service guarantees that the network is intended to support. If a
34 Bridge can be used in a network that encodes drop eligibility in the Priority Code Point, and there is any
35 likelihood of the Bridge being brought into service prior to the network-dependent service-level
36 configuration, then five priorities, three with drop eligibility (5P3D encoding and decoding), should be used.
37 Bridges that do not support drop precedence should be configured to support five or fewer traffic classes in
38 the same circumstances.

39 The use of separate Priority Code Point Encoding and Priority Code Point Decoding Tables for each Bridge
40 Port allows adaptation between the encoding scheme in one domain of the network and the encoding scheme
41 used in another to be accomplished in only one of any pair of Bridges, each serving as a boundary of its
42 domain, connected by a point-to-point LAN. However, if more than two Bridges are attached to a LAN, all
43 need to use the same encoding so that each of its recipients can assign the correct priority to the frame.

1 The default Priority Code Point encoding and decoding, as documented in Table 6-2 and Table 6-3, are
2 reproduced in Table I-6 and Table I-7, with the addition of the default allocation of priorities to traffic
3 classes to the latter.

4 NOTE—The sequence of the columns for the priority values of 0 and 1 in Table I-6 and Table I-7 are reversed from the
5 sequence in Table 6-2 and Table 6-3 in order to show the alignment with the traffic classes in Table I-3.

Table I-6—Priority Code Point encoding

priority drop_eligible		7	7DE	6	6DE	5	5DE	4	4DE	3	3DE	2	2DE	0	0DE	1	1DE
PCP	8P0D	7	7	6	6	5	5	4	4	3	3	2	2	0	0	1	1
	7P1D	7	7	6	6	5	4	5	4	3	3	2	2	0	0	1	1
	6P2D	7	7	6	6	5	4	5	4	3	2	3	2	0	0	1	1
	5P3D	7	7	6	6	5	4	5	4	3	2	3	2	1	0	1	0

Table I-7—Priority Code Point decoding

PCP		7	6	5	4	3	2	0	1
priority drop_eligible	8P0D	7	6	5	4	3	2	0	1
	7P1D	7	6	4	4DE	3	2	0	1
	6P2D	7	6	4	4DE	2	2DE	0	1
	5P3D	7	6	4	4DE	2	2DE	0DE	0
number of traffic classes	1	BE							
	2	VO				BE			
	3	NC		VO		BE			
	4	NC		VO		CA		BE	
	5	NC	IC	VO		CA		BE	
	6	NC	IC	VO		CA		BE	BK
	7	NC	IC	VO		CA	EE	BE	BK
	8	NC	IC	VO	VI	CA	EE	BE	BK

1 **Annex J**

2 (informative)

3 **CFM protocol design and use**

4 **J.1 Origin of CFM**

5 Operations, Administration, and Maintenance (OAM) functions are a product of the world of telephony. As
6 Ethernet, which has traditionally been an Enterprise network technology, expands into the realm of service
7 providers, such useful notions as OAM become important. The OAM work conducted in various standards
8 bodies and in various vendors' products can be classified into five groups:

- 9 a) Provider Edge OAM, as defined by IEEE Std 802.3.
- 10 b) Ethernet Local Management Interface (E-LMI), as defined by MEF 16 [B54].
- 11 c) Underlying layers of OAM, such as MPLS OAM [B59] or ATM OAM [B2], used on various sorts of
12 emulated Ethernet links, carrying Ethernet frames as a higher layer.
- 13 d) Ethernet frames carrying End-to-End CFM PDUs, defined by this standard and by
14 ITU-T G.8013/Y.1731.
- 15 e) Network performance measurement, defined by the MEF Forum in MEF 35.1 [B56] and ITU-T,
16 including ITU-T G.8013/Y.1731.

17 CFM is one part of the whole OAM picture. It defines group in item d), Ethernet frames carrying
18 End-to-End CFM PDUs, defined by this standard and by ITU-T G.8013/Y.1731. This standard does not
19 explicitly address the subject of the configuration or provisioning of MEF services, but it does generate
20 significant requirements on configuration and provisioning. Certain CFM PDU formats are also useful for
21 network performance measurement [item e)] and are defined in this standard. Additional PDUs for these
22 operations are defined in ITU-T G.8013/Y.1731.

23 Finally, unlike OAM in the telephony world, CFM does not address recovery from a loss of connectivity.
24 That is the function of the various Layer 2 control protocols such as spanning tree (see Clause 13).

25 **J.2 Deployment of CFM**

26 It is critical to the early deployment of CFM that no hardware changes be required to achieve a significant
27 level of CFM functionality. However, if the MEPs in a large number of large MAs issue Continuity Check
28 Messages (CCMs) at the highest allowed rates, a Provider Bridge's ability to generate and/or absorb these
29 CCMs could be overwhelmed. To achieve its full potential, CFM could require hardware modifications to
30 existing Provider Bridges.

- 31 a) The CFM shims shown in Figure 22-4 are an abstract concept. They have been defined so that it
32 should be possible to emulate their functions on most existing platforms, though perhaps not at high
33 rates of CFM PDU transmission and reception. Hardware assistance can enable higher CFM PDU
34 rates and can enable the implementation of the Shared MP address model of operation instead of the
35 Individual MP address model (J.6).
- 36 b) Directing CFM PDUs to the appropriate ports of a Provider Bridge could necessitate a separate
37 MAC Address Registration Entry for each VID in the FDB. Some means whereby this large number
38 of entries could be abstracted would reduce the load on the FDB and enable the use of a larger
39 number of MD Levels.
- 40 c) The reception of an LBM and generation of the corresponding LBR have been defined in a manner
41 that is amenable to implementation in hardware. If so implemented, then the providers' ability to
42 support a particular requirement for bandwidth in a service instance can easily be tested.

1 J.3 MD Level allocation alternative

2 There are eight Maintenance Domain Levels (MD Levels) as shown in Table J-1. In order to allow
3 administrations to configure different Maintenance Domains in the same Bridge without having to resort to
4 detailed inter-organizational agreements about which MD Levels are available for use, an administrator of a
5 Maintenance Domain can decide which of three roles that Maintenance Domain will play: the “customer”
6 role, the “service provider” role, or the “operator” role. The administrator would then be free to assign an
7 MD Level to a Maintenance Domain within the range shown in Table J-1 according to the Maintenance
8 Domain’s role. If two administrations can operate Maintenance Domains using the same role in one Bridge
9 at different MD Levels.

Table J-1—Provider MD Level allocation

MD Level	7	6	5	4	3	2	1	0
Use	customer			service provider		operator		
	Highest	<- Higher lower->						Lowest

10 This method of MD Level allocation attempts to minimize the likelihood of incompatibilities caused by MD
11 Level selection among separate administrative organizations that interconnect in a manner such that their
12 MD Levels are visible to each other. Due to the addition of encapsulations such as VLAN tags that can hide
13 one organization’s MD Levels from another, it is considered unlikely that the allocation method shown in
14 Table J-1 will be necessary and that the method described in 22.3 will be sufficient.

15 J.4 Relationship of IEEE Std 802.1Q CFM to other standards

16 ITU-T G.8013/Y.1731 is a compatible extension of CFM. Certain terminology is different between ITU-T
17 G.8013/Y.1731 and this standard. These differences are summarized in Table J-2 .

Table J-2—IEEE / ITU-T terminology differences

IEEE Std 802.1Q	ITU-T G.8013/Y.1731
Maintenance Association (MA)	Maintenance Entity Group (MEG)
Maintenance Association Identifier (MAID)	Maintenance Entity Group Identifier (MEGID)
Maintenance Domain	(No such construct present)
Maintenance Domain Level (MD Level)	Maintenance Entity Group Level (MEG Level)

18 ITU-T G.8013/Y.1731 includes a number of OpCodes in addition to the CCM, LBM, LBR, LTM, and LTR
19 defined in this standard. ITU-T G.8013/Y.1731 also specifies that the LBM can have a Group
20 destination_address, in addition to the ability of using an Individual destination_address, as defined in this
21 standard. It is for this reason that the MP Loopback Responder’s ProcessLBM() procedure (20.28.1) is
22 required to respond to an LBM whose destination_address is the appropriate CCM Group address, even
23 though this standard specifies no means for generating such a frame.

1 The MIP is defined as two MHFs in this standard, but not in ITU-T G.8013/Y.1731. This is because
2 ITU-T G.8013/Y.1731 is not concerned with the details of the implementation.

3 ITU-T G.8013/Y.1731 does not include an LTM Egress Identifier TLV (21.8.8) or LTR Egress Identifier
4 TLV (21.9.7). For compatibility with early implementations of ITU-T G.8013/Y.1731, this standard does not
5 require that these TLVs be present on received LTMs or LTRs. However, this standard does require these
6 TLVs to be transmitted in LTMs and LTRs.

7 ITU-T G.8013/Y.1731 specifies that, if an LTM passes through MPs on both the ingress and egress ports, the
8 TTL field of the LTM will be decremented twice. This standard requires, in that case, that the Bridge
9 decrement the TTL field only once, and return a single LTR. The ITU-T G.8013/Y.1731 behavior is
10 compatible with the LTR analysis presented in J.5, as long as each of the MPs that decrement the LTM's
11 TTL field also return an LTR. If a Bridge decremented the TTL twice, but returned only a single LTR, it
12 would be impossible for the originating MEP to tell whether an LTR was lost in transit.

13 The Flags field (21.9.1) returned in the LTR contains two extra bits of information in this standard that are
14 not in ITU-T G.8013/Y.1731.

15 J.5 Interpreting Linktrace results

16 The LTR entries are retrieved from the Linktrace Database in the order the LTRs were received by the MEP.
17 There is no way to determine when the last LTR has been received, except to wait for some number of
18 seconds (e.g., long enough for a worst-case delay through the Bridged Network plus a few seconds), after
19 the successful completion of the Transmit Linktrace Message command, before assuming that no further
20 LTRs are likely to be received.

21 Because of the timer used by the LTR Transmitter state machine (20.5.1), the order in which the LTRs are
22 received (time order) is likely to be different from the order of the MPs reached by the LTM as it was
23 forwarded, MP by MP (path order). It is possible for an LTM to take more than one path, so that the LTRs
24 report a tree, rooted at the originating MEP, rather than a simple linear series of zero or more MHFs
25 terminating at a MEP or an MHF. If there are no network topology changes during the Linktrace operation,
26 and if all systems' behavior conforms to this standard, the following can be said about constructing this path
27 tree from the list of LTR entries returned by the Read Linktrace Reply command:

- 28 a) Sort the returned LTR entries by decreasing value of `ltrReplyTTL` (20.41.2.2). The highest reported
29 value is one less than the initial LTM TTL field value [item d) in 12.14.7.4.2] used in the Transmit
30 Linktrace Message command, and corresponds to the first Linktrace Responder reached by the
31 initial LTM. This Linktrace Responder resides in the same Bridge as the MEP, in the case of an LTM
32 originated by an Up MEP. The next-lower value corresponds to the MHF(s) or MEP(s) reached by
33 the first forwarded copy of that LTM, and so on. Any gaps in the sequence of Reply TTL values
34 indicate lost LTRs.
- 35 b) The `ltrNextEgressId` variable (20.41.2.4) in every LTR entry returned from a given Transmit
36 Linktrace Message command is unique among the LTR entries in a given LTM entry in the Linktrace
37 database, since no Linktrace Responder transmits more than one copy of an LTM.
- 38 c) The `ltrLastEgressId` variable (20.41.2.3) in any LTR entry is unique only over those LTR entries
39 with matching values of the `ltrReplyTTL` variable.
- 40 d) The `ltrLastEgressId` variable (20.41.2.3) in any LTR entry whose `ltrReplyTTL` variable is one less
41 than the initial LTM TTL field also matches the original LTM's LTM Egress Identifier TLV value, as
42 returned from the Transmit Linktrace Message command [item c) in 12.14.7.4.3].
- 43 e) In the absence of lost LTRs, the `ltrLastEgressId` variable of any LTR whose `ltrReplyTTL` variable
44 contains x will match the `ltrNextEgressId` variable of one of the LTRs whose `ltrReplyTTL` variable
45 contains $(x + 1)$. For LTR entry x , this match indicates which of the Linktrace Responders $(x + 1)$
46 forwarded the LTM that was received by the Linktrace Responder that returned LTR entry x , and
47 thus turns the list of LTR entries at each `ltrReplyTTL` value into a tree.

- 1 f) Along any given path from the root of the reply tree, the last reply can report any of the following
2 conditions that terminate the forwarding of the LTM:
- 3 1) FwdYes bit of the ltrFlags variable is reset (false); or
4 2) TerminalMEP of the ltrFlags variable is set (true); or
5 3) ltrRelayAction variable contains the value, RlyHit; or
6 4) ltrReplyTTL variable contains 0.
- 7 g) If any path does not terminate with one of the conditions 1) through 3) in the previous item f), then
8 either the initial LTM TTL field was insufficiently large to trace the complete path (and the last
9 ltrReplyTTL variable of the end MP in the path contains 0), or a data frame addressed to the target
10 MAC address of the LTM would be flooded after being forwarded from the Bridge that returned the
11 last LTR in the path.
- 12 h) If the first Bridge encountered by the original LTM (which is the Bridge on which the LTM was
13 originated, if originated from an Up MEP) would have flooded a data frame addressed to the target
14 MAC address of the LTM, no LTRs will be returned in response to the LTM.
- 15 i) A gap in the ltrReplyTTL value sequence indicates the loss of (or failure to transmit) one or more
16 LTRs. Similarly, a gap in the chain of ltrNextEgressId and ltrLastEgressId variables can be caused
17 by the loss of LTRs. These conditions can make the construction of a reply tree impossible, if
18 multiple LTRs have been returned with the same ltrReplyTTL value. If the last LTR along the path is
19 lost, there is no gap in the ltrReplyTTL sequence to indicate the loss.
- 20 j) The Transmit Linktrace Message command can be reissued, and the resultant LTRs correlated with
21 the results of a previous LTM, to improve the reliability of the results.

22 Diagnosing whether an anomalous result from a Read Linktrace Reply command, i.e., one that violates any
23 of the prior assertions, is the result of ordinary frame loss (caused, e.g., by congestion), transitory network
24 behavior (e.g., the loss of a LAN during the Linktrace procedure), a permanent malfunction (e.g., a Bridge is
25 behaving in a noncompliant manner due to a hardware or software malfunction), or some other reason, is
26 beyond the scope of this standard.

27 J.6 MP addressing: Individual and Shared MP addresses

28 It is stated in 21.3.2 that the source MAC address for a frame carrying a CFM PDU is the MAC address of
29 the MP transmitting the PDU. This does not necessarily mean that the MP's physical implementation is tied
30 to a particular Bridge Port, and that the MAC addresses used in the frames carrying CFM PDUs are the
31 MAC addresses of those Bridge Ports. While this method of MAC address assignment produces the best
32 protection that this standard can provide for a network, it is not required that every MP use its Bridge Port's
33 MAC address.

34 CFM does not require the MPs configured on a single Bridge Port, e.g., all of the MPs illustrated in
35 Figure 22-4, to have MAC addresses that are different from each other, nor does CFM require that they be
36 the same. If the MPs in Figure 22-4 all have the same MAC address, CFM frames can be directed to the
37 different MPs on that Bridge Port based on VID and MD Level. Choices can be made, however, with regard
38 to whether MPs on different Bridge Ports can share the same MAC address. Two models are presented in
39 this subclause for the assignment of MAC addresses to MPs:

- 40 a) **Individual MP address model:** The individual MAC address assigned to an MP associated with a
41 particular MA, and hence that MA's MD Level and set of VIDs, is unique over all service instances
42 associated with that same set of VIDs, such that those service instances' CFM PDUs can be
43 distinguished only by their MD Levels that can pass through that MP's Bridge Port.
- 44 b) **Shared MP address model:** The same as the Individual MP address model, except that Up MPs
45 (but not Down) in the same MA and in the same Bridge can share a (i.e., can all have the same)
46 MAC address.

1 Down MPs in different Bridge Ports have different MAC addresses. If two Bridge Ports are connected to the
2 same LAN, but have the same MAC address, then both would respond to an LBM sent to that MAC
3 address.⁶³ This is true, even though one of those two Bridge Ports would be Blocked (it would be either a
4 Backup or an Alternate Port, see 13.12), because Down MEPs are between the LAN and the Port filtering
5 entities (8.6.1, 8.6.2, 8.6.4) that enforce the blocking. This placement allows Down MEPs to test their
6 service instances even when their Port is blocked, so that the system administrator knows that the failover
7 links will be ready to carry data when a network event causes the Port to become Forwarding.

8 A Bridge can use the Shared MP address model, or the Individual MP address model, on any given Bridge
9 Port. This is an implementation decision; no managed objects are provided to select one or the other model
10 of operation.

11 A Bridge using either of these two models can also make use of a third scheme for configuring MPs:

- 12 c) **Management Port MEPs:** Up MEPs (not Up MHFs, not Down MPs) are configured on a
13 Management Port, a Bridge Port that does not connect to any LAN exterior to the Bridge. (See 8.3
14 and Figure 8-8.) Typically, this would be the same MAC address, and the same Bridge Port, that
15 used for the Bridge's Management Port (see Figure 8-8).

16 The Individual MP address model is described in J.6.1, Shared MP address model in J.6.2, and Management
17 Port MEPs in 22.7.

18 J.6.1 Individual MP address model

19 The purpose of CFM is to detect and diagnose connectivity errors. If a Bridge implements its Up MPs as
20 close to the physical layer hardware as possible, it maximizes the capability of CFM to detect connectivity
21 errors caused by malfunctions in that Bridge. In particular, giving each Up MP a MAC address that is tied to
22 its particular Bridge Port means that an LBM/LBR exchange tests the Frame filtering entity (8.6.3) at both
23 ends of the path. Figure 22-5 illustrates this fact. Any LBMs or LBRs passing between the Up MEPs and
24 MHFs at the ends of the grayed service instance traverse exactly the same path through the Frame filtering
25 entity as ordinary data frames carried from the left-most LAN to the right-most LAN.

26 J.6.2 Shared MP address model and the CFM Port

27 Although the Individual MP address model (J.6.1) gives the best detection and diagnostic capabilities, not
28 all Bridge implementations are capable of inserting and removing CFM PDUs at points close to the physical
29 layer. For example, the reflection of a high-speed LTM/LTR test stream by an Up MEP configured on a Port
30 that is also receiving normal traffic from its LAN implies the presence of prioritization, data path, queuing,
31 and bandwidth resources not otherwise required by this standard. This does not mean that Bridges lacking
32 those resources cannot support CFM. The Shared MP address model provides one way for these Bridges to
33 maximize their CFM capabilities.

34 Consider Figure 20-16, case c), but assume that the PDU entering Port 5, instead of being an LTM, is an
35 LBM addressed to the Up MEP on Port 3. One cannot discern, from observing the external behavior of a
36 Bridge, whether that LBM was actually delivered to the Up MEP on Port 3 or to some other entity within the
37 Bridge. The Shared MP address model takes advantage of this fact by allowing Up MPs in different Bridge
38 Ports to share the same MAC address. The Up MPs are still configured on different Bridge Ports according
39 to the managed objects in Clause 12. For the most part, they have distinct identities. They share their MAC
40 addresses and, as discussed more fully in 20.53, certain state machines and variables.

⁶³ Bridge implementations are known that use the same source MAC address for different Bridge Ports. As long as the Bridge Port or Management Port to which higher layer entities (e.g., an IP/TCP stack) are attached has its own unique MAC address, and as long as those Bridge Ports that share an address transmit only frames, such as BPDUs that do not traverse other Bridges, and as long as those Ports are not running any protocols that expect unicast frames to pass through a Bridge to reach them, Bridges of this description can interoperate each other and can interoperate with Bridges that follow the requirement to have a different MAC address for every Port. However, CFM PDUs do traverse other Bridges, and the LBM protocol does utilize individual MAC addresses. Therefore, the CFM protocols do not meet the goals of Clause 18 in all network topologies if Bridge Ports' Down MPs share MAC addresses.

1 In a Bridge configured according to the Shared MP address model, whether an LBM (or other PDU) that
2 enters Port 5 in Figure 20-16 is addressed to the Up MEP on Port 3, or the Up MHF on Port 2, it is delivered
3 to the same physical Bridge Port, called a “CFM Port.” This could be a normal Bridge Port, e.g., one with
4 more capability than the Bridge Port on which the addressed MP is configured, or it could be a Port that is
5 attached to no LAN, in the manner of a Management Port (8.3, 22.7). A Bridge can have more than one
6 CFM Port, but within that Bridge, each CFM Port has a MAC address that is different from the other CFM
7 Ports. In the limiting case, each CFM Port is also a Bridge Port, and is associated with only its own Up MPs;
8 this is indistinguishable from the Individual MP address model. A CFM Port, for which the MPs of three
9 different Bridge Ports are residing, is illustrated in Figure J-1.

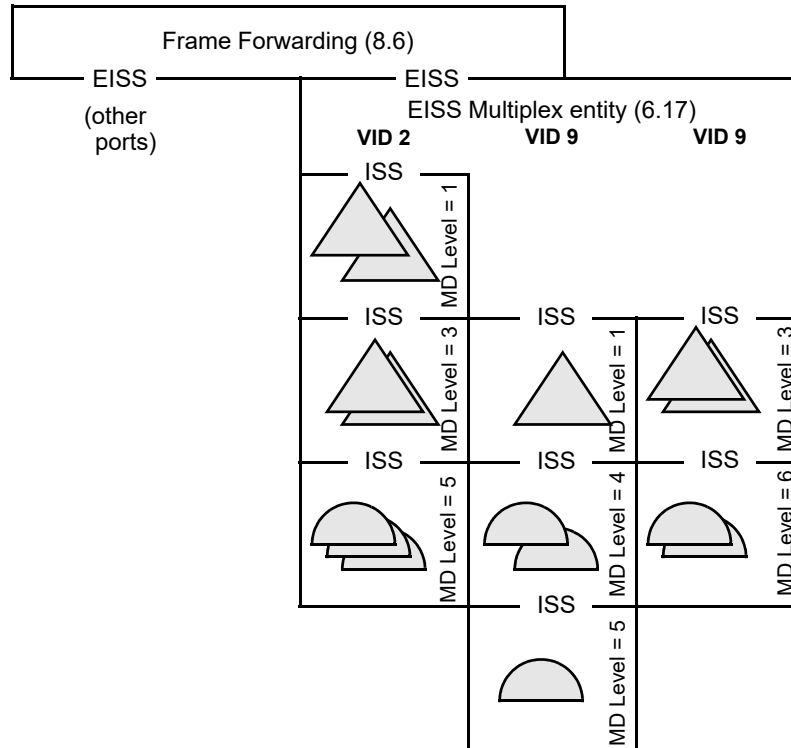


Figure J-1—Up MPs in a CFM Port

10 NOTE 1—Unlike normal Bridge Ports or a Management Port, multiple levels of MHFs can be present on a CFM Port
11 for the same service instance, because they belong to different MAs on different Bridge Ports.

12 MPs on this CFM Port are distinguished by VID and by MD Level. However, those criteria are insufficient
13 when Up MPs configured on different Bridge Ports are in the same MA, and thus in the same VID and at the
14 same MD Level. There is no way of distinguishing, in that case, to which of those Up MPs a given CFM
15 PDU is addressed. For the PDUs defined in this standard, this is not a problem.

16 Each MEP, even when multiple MEPs reside on a single CFM Port, has its own MEP Continuity Check
17 Initiator state machine (20.12) and associated variables. Each MEP is responsible for transmitting its own
18 CCMs, carrying information (e.g., the optional Sender ID TLV, 21.5.3) peculiar to its configured Bridge
19 Port.

20 On the other hand, otherwise indistinguishable MEPs sharing a MAC address share a single LTR
21 Transmitter state machine (20.50) and a single MEP Loopback Initiator transmit state machine (20.32), and
22 their associated variables. The managed objects in 12.14.7.1.3 support this sharing.

1 When any CFM PDU arrives at the CFM Port, it is copied and delivered to each of the MPs that share the
2 same MA. Each of these MPs responds to the CFM PDU separately and responds as if it resided in its
3 configured Bridge Port. Since a multicast would be delivered to all of the MPs anyway, there is no harm in
4 placing all of them in a single CFM Port, with a single MAC address.

5 Unicast CCMs, LTMs, LBRs, and LTRs present no issues. CCMs require no response. The LTMs all go to
6 the Bridge's single Linktrace Responder. Even when an LTM is multicast, only one LTR and/or forward
7 LTM is produced, so delivering a unicast LTM to multiple MPs causes no additional problems. Since the
8 MEPs sharing a single CFM Port also share a single MEP Loopback Initiator transmit state machine (20.32)
9 and a single MEP Loopback Initiator receive state machine (20.34), and since they all share a single set of
10 variables, particularly those related to the LTM Transaction Identifier fields in these PDUs, there is no
11 confusion about which MEP the LBR or LTR is directed.

12 The remaining PDU type is the LBM. Consider what would happen if the MEPs and their associated MP
13 Loopback Responders (19.2.10) resided on separate Bridge Ports, but all shared the same MAC address.
14 Assuming that the Frame filtering entity had learned that MAC address, the LBM would be forwarded to
15 exactly one of those MEPs, namely the one that, by chance, last happened to transmit a frame, thus causing
16 the Learning Process (8.7) to associate the shared MAC address with that transmitter's Bridge Port. Thus,
17 exactly one LBR would be returned, not one per Bridge Port. The MEP receiving that LBR would not know
18 from which MP the LBR was returned, because there is nothing in the LBR to indicate the identity of the
19 responding MP. Therefore, only one response to a unicast LBM is returned from a CFM Port, no matter how
20 many MPs on that LBM's MA reside on the CFM Port.

21 On the other hand, a multicast LBM could cause any number of Up MPs configured in the same MA, but on
22 different Bridge Ports in a single Bridge, to respond. Thus, a high-speed stream of multicast LBMs could
23 impose an arbitrarily large burden upon a single CFM Port that is supporting a large number of Bridge Ports'
24 MPs, if each MP responded to that LBM. Therefore, among a set of MPs for a single MA that all share the
25 same MAC address (i.e., reside on the same CFM Port), one or more, but not all, of their MP Loopback
26 Responders (19.2.10) can be disabled by not configuring any group MAC address for them to recognize.
27 This configuration is accomplished automatically by a Bridge that utilizes CFM Ports and is not controlled
28 by any managed object.

29 The externally visible behaviors that can be discerned from outside a Bridge that uses the Shared MP
30 address model are as follows:

- 31 a) An external MEP can regularly receive CFM PDUs other than LTRs, from two different MPs in the
32 same MA that have the same source_address.
- 33 b) A multicast LBM can regularly receive only one reply from among a set of MPs that normally all
34 respond to unicast LBMs, but all with the same source_address, without implying the presence of an
35 intermittent network failure.
- 36 c) A system administrator can see unexpected advances in a MEP's managed objects that count LBMs
37 and LBRs, apparently due to activity on another MEP.
- 38 d) If conceptual rows in the Interface MIB module of IETF RFC 2863 are instantiated for individual
39 MPs, the packet and frame counts of those interfaces count both CFM PDUs and ordinary data
40 frames passing through the Active and Passive SAPs, and thus can indicate whether an MP is
41 instantiated on the Bridge Port on which it is configured, or on a separate CFM Port through which
42 no ordinary data frames pass.

43 NOTE 2—None of these behaviors is impossible for a Bridge using the Individual MP address model; MAC addresses
44 can change faster than CCMs time out, momentary congestion can cause frame loss, multiple administrators can access
45 different MEPs' managed objects, and ordinary data frames can be absent. It is only the frequency of these behaviors
46 that distinguishes the Individual MP address model from the Shared MP address model.

47

Annex K

(informative)

TPMR use cases

This material is intended to assist the reader of this standard in putting the operation of a TPMR in context with the operation of Bridged Networks as a whole, by illustrating some of the uses of a TPMR in conjunction with other types of Bridge. The set of cases is not, and is not intended to be, exhaustive in terms of all of the possible usage scenarios.

K.1 Use case 1—TPMR as User to Network Interface (UNI) demarcation device

Figure K-1 illustrates a use case where the TPMR performs the function of a UNI demarcation device between an customer network and a provider network. In this illustration, the user link between the customer Bridge and the TPMR, and the extension link between the TPMR and the provider Bridge, are both assumed to be IEEE 802.3 links, although, with the exception of the use of IEEE 802.3 EFM OAM on the extension link, these links could in principle be provided by use of any IEEE 802 LAN technology.

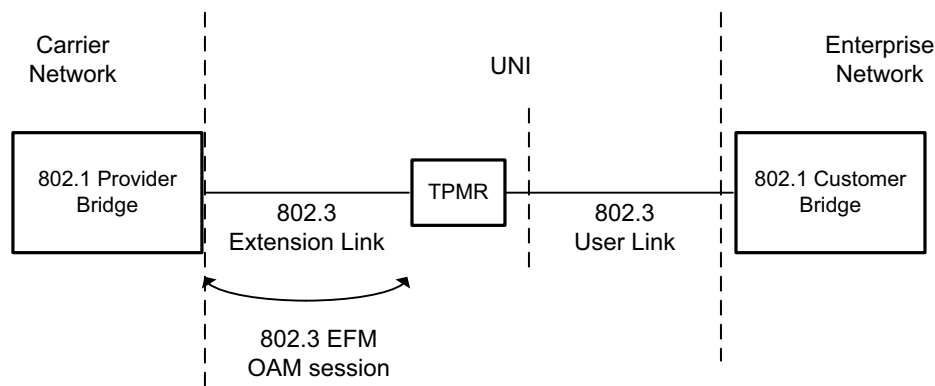


Figure K-1—TPMR as UNI demarcation device

The use of EFM OAM on the extension link allows the service provider to:

- a) Put the TPMR into an intrusive loopback for out-of-service testing.
- b) Query the IEEE 802.3 MIB in the TPMR.
- c) Be notified of critical events (Link Fault, Dying Gasp, Critical Event).
- d) Be notified of link performance events (Errored Symbol Period event, Errored Frame event, Errored Frame Period event, Errored Frame Seconds Summary event).

The potential exists with EFM OAM (on the extension link) to make use of extensions, via organization-specific OAMPDUs, to manage the Port of the TPMR that faces the user link.

Alternatively, the TPMR might be managed using SNMP over UDP/IP either in-band on the IEEE 802.3 extension link or out-of-band through separate management connector on the TPMR.

1 K.2 Use case 2—TPMRs with aggregated links

2 Figure K-2 illustrates a use case where two TPMRs are used within paired extension/user links that are
3 aggregated at their endpoints (the Bridges). The TPMRs are transparent to the operation of the IEEE
4 802.1AX Link Aggregation Control Protocol (LACP), because the Provider Bridge and Customer Bridge
5 are configured to use the “Nearest non-TPMR Bridge group address” as the destination address for LACP;
6 from the point of view of Link Aggregation, each paired extension link and user link, with the TPMR joining
7 them together, appears to be a single IEEE 802.3 LAN.

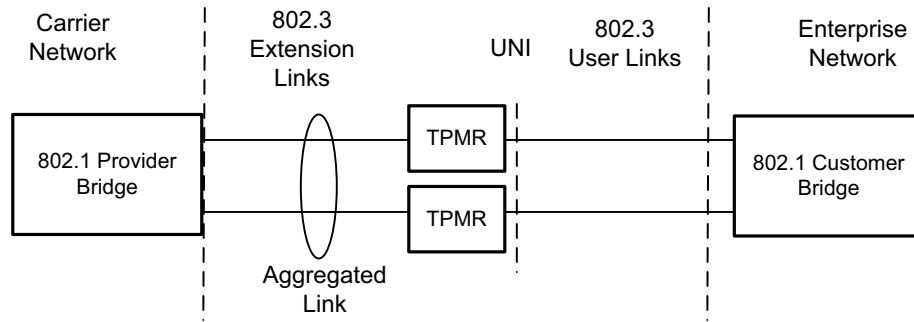


Figure K-2—TPMRs with aggregated links

8 The TPMRs would be managed separately, via EFM OAM, as in use case 1 (K.1). It should be noted that the
9 use of SNMP to manage the TPMRs in this scenario is problematic, as the aggregated link appears to the
10 higher layers to be a single link, but individual conversations are allocated by the frame distributor in the
11 aggregator to the links in the aggregation in a nondeterministic manner.

12 K.3 Use case 3—Multiple TPMRs

13 Figure K-3 shows a variant of use case 1 (K.1) where two extension links and a user link are used,
14 interconnected using two TPMRs.

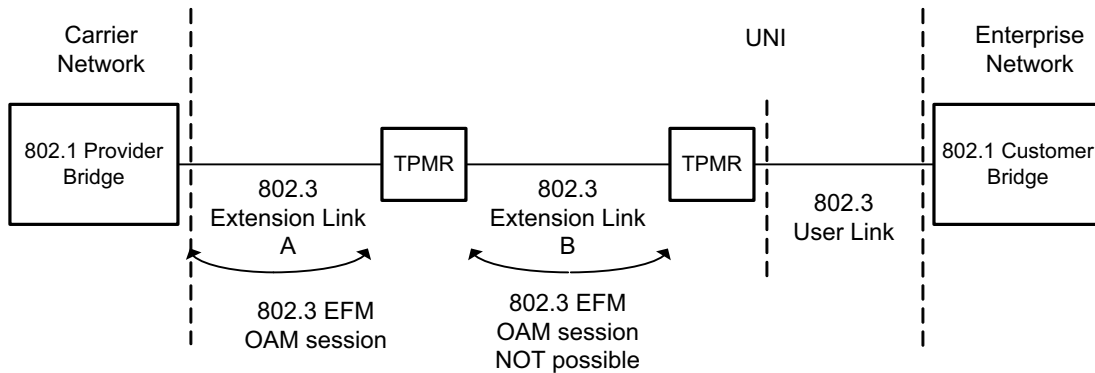


Figure K-3—Multiple TPMRs

15 Some commercial TPMRs used as demarcation devices power up in EFM OAM Passive mode, waiting to
16 hear from the Active mode master device at the edge of the provider network. A passive-to-passive EFM
17 OAM session, as would occur on extension link B, would never activate, making it impossible to manage
18 the customer-facing TPMR by this means, unless significant changes were made to the capabilities of EFM
19 OAM to allow some kind of relay capability. On the other hand, by using SNMP as the management
20 protocol, this problem would be avoided.

1 K.4 Special cases

2 Figure K-4 (new connectivity) and Figure K-6 (connectivity failure) represent the target cases for the MAC
3 status propagation design.

4 Figure K-4 shows recovery of the individual LAN at one end of at TPMR chain. The result is to “blip” the
5 MAC_Operational status at the other end of the chain, delaying availability of the recovered link by slightly
6 more than $T_w + T_d$. The figure also shows a possible effect of timing relationships between T_r and T_d , as the
7 initiator of the change retries its Add message transmission just as the TPMR at the other end of the chain
8 returns a confirm. The crossing of messages shown is unlikely, as the time sequence diagrams
9 overemphasize the delay experienced by the Add and Confirm messages, but the effect is benign in any case.

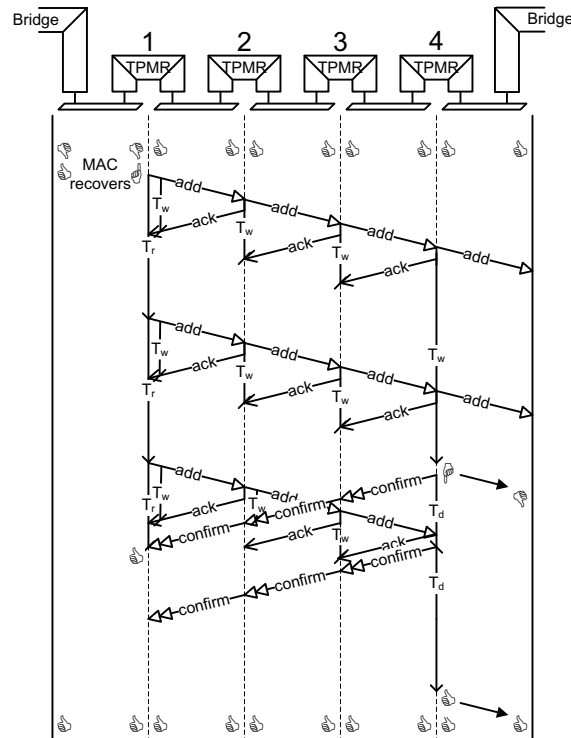


Figure K-4—Recovery at the end of a chain

10 If the LAN between the end system and the first TPMR in the chain (TPMR 1) had failed instead of
11 recovering, the rest of Figure K-4 would have looked the same, with the exception of the final state of that
12 first LAN. The net effect of the status notification protocol is to transfer the change in MAC_Operational to
13 the other end of the chain so that both ends see the transition. For best effects the transfer should complete in
14 a short enough time so that the end system protocols at the initiating end of the link are still executing their
15 initial state.

16 Figure K-5 shows what happens if both ends of the link come up at the same time, the net effect is simply to
17 delay the simultaneous start.

18 Figure K-6 shows simultaneous transitions of one LAN in the chain to OperUp and the other to OperDown,
19 and illustrates a number of points about how the protocol or protocols do or should operate.

20 First, signaling of addition or loss can be somewhat arbitrary and depend on the relative timing of transitions
21 on separate individual LANs. The same final connectivity can be represented by a loss followed by an

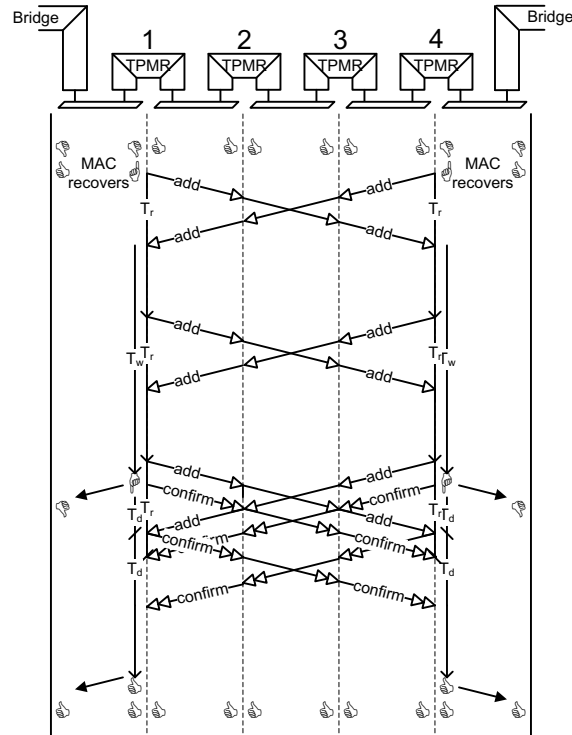


Figure K-5—Near simultaneous recoveries

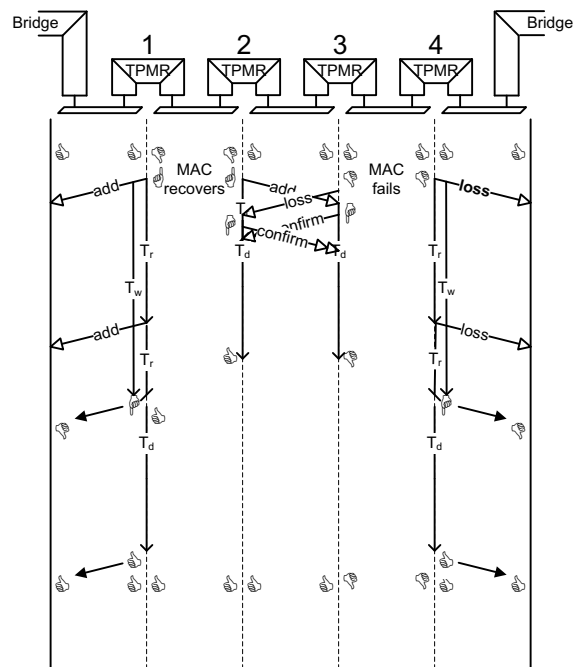


Figure K-6—Near simultaneous failure and recovery

1 addition, or an addition followed by a loss. If the implied resulting state is to mean anything then the
2 information “connected as far as,” or “connected to and beyond” also needs to be communicated. However,
3 while this is a candidate for inclusion in any new protocol, support by existing protocols is unlikely.
4 However, this clause persists with distinguishing Add and Loss messages, and their confirmations, because
5 that ensures that closely spaced transitions, which might otherwise disguise unusual loss of higher layer

1 protocol messages, are not missed by the systems connected to the TPMR chain. Figure K-7 shows a loss
2 followed by recovery of the same LAN.

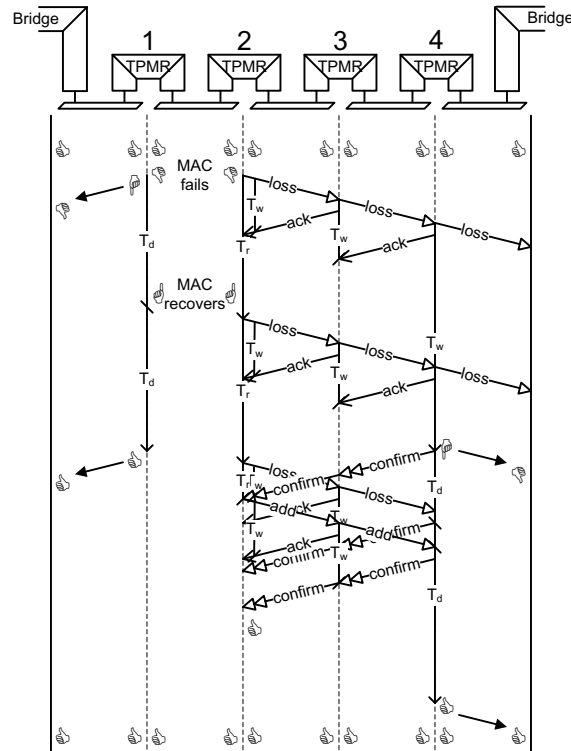


Figure K-7—Loss with quick recovery

3 Second, no message is sent across the LAN between TPMRs 1 and 2 when the MAC is OperDown from the
4 protocol client's point of view. That allows support of the protocol to be architected such that a message that
5 signals "add(ition)" or "loss" is forwarded through the normal bridged path so its speed of propagation does
6 not depend on scheduling or notifying a status propagation process in each TPMR. Also, users have a free
7 choice about what additional functionality can reside between the entities responsible for status propagation
8 and the "real" MAC.

9 One last common case of simultaneous transitions is what happens when the two ports of a TPMR power up
10 or are enabled at the same time. Each tries to send a link notification through the other, and each link
11 notification is discarded as the Port is not yet available (mssOperUp FALSE). The result is to "blip" each
12 LAN. The ends of the TPMR chain see an Add Loss Add sequence, if they are protocol capable, and a blip
13 in MAC_Operational otherwise. If T_w is less than T_d , this will be a single blip.

1 Annex L

2 (informative)

3 Operation of the credit-based shaper algorithm

4 This annex contains a more detailed analysis of the way the credit-based shaper algorithm (8.6.8.2) operates,
5 and how its operation affects the performance of the network, from the point of view of traffic that uses a
6 credit-based shaped queue, and also from the point of view of other queues associated with the Port. The
7 analysis and description in this annex does not take into account any interruption to bandwidth availability
8 resulting from the operation of stream gates.

9 L.1 Overview of credit-based shaper operation

10 The credit-based shaper algorithm has a single externally determined parameter, *idleSlope* (see 8.6.8.2), that
11 determines the maximum fraction of the *portTransmitRate* that is available to the queue associated with a
12 traffic class (*bandwidthFraction*), as follows in Equation (L-1):

$$13 \quad \textit{bandwidthFraction} = \textit{idleSlope} / \textit{portTransmitRate} \quad (\text{L-1})$$

14 where *portTransmitRate* is the maximum transmit data rate that the port supporting the outbound queue is
15 able to deliver. The detailed derivation of Equation (L-1) can be seen in Equation (L-5) through
16 Equation (L-8).

17 If a traffic class supported by the credit-based shaper algorithm uses less than the bandwidth allocated to it,
18 then the unused bandwidth can be used by other traffic classes, in accordance with the relative priorities of
19 the traffic classes and the transmission selection algorithms that are associated with them.

20 NOTE 1—It is a natural consequence of the way transmission selection operates that if a given traffic class does not
21 have a frame available for transmission, then a lower priority traffic class that does have a frame available for
22 transmission is able to transmit. Hence, if a given traffic class that uses the credit-based shaper algorithm has been
23 allocated X% of the available bandwidth on a Port, but only uses (X-Y)%, then the unused portion of its allocation (Y%)
24 is available for other traffic classes to use if they are in a position to do so. Whether this unused bandwidth can be used
25 use by a traffic class depends upon the transmission selection algorithm that is associated with it. For example, a traffic
26 class that has been allocated a specific bandwidth, such as one that uses the credit-based shaper algorithm, would be
27 unable to use any bandwidth allocation not used by a higher priority traffic class, as that would result in it exceeding its
28 bandwidth allocation; however, a traffic class that uses the strict priority algorithm, which is not associated with a
29 bandwidth allocation, would be able to use bandwidth allocated to, but not used by, a higher priority traffic class.

30 The *idleSlope* parameter, in conjunction with the size of the frames that are being transmitted using the
31 queue, and the maximum time delay that a queue can experience before it is able to transmit a queued frame,
32 places an upper bound on the burst size that can be transmitted from queues that use the algorithm.

33 In order to describe the operation of the algorithm, it is useful to define the following values, in addition to
34 the formal parameters of the algorithm described in 8.6.8.2:

- 35 a) ***maxFrameSize***. The maximum sized frame that can be transmitted through the Port for the traffic
36 class concerned. This value can be determined by admission control algorithms, or can be simply a
37 function of the transmission behavior of the traffic source. Either way, the value can be smaller than
38 would be allowed by the normal operation of the underlying MAC Service.
- 39 b) ***hiCredit***. The maximum value that can be accumulated in the *credit* parameter. This parameter is a
40 function of the operation of the algorithm, and cannot be controlled by management.

c) **loCredit**. The minimum value that can be accumulated in the *credit* parameter. The value of *loCredit* is a function of the values of *maxFrameSize*, *portTransmitRate*, and *sendSlope*, as follows in Equation (L-2).

$$loCredit = maxFrameSize \times (sendSlope/portTransmitRate) \quad (L-2)$$

d) **maxInterferenceSize**. The maximum size, in bits, of any burst of traffic that can delay the transmission of a frame that is available for transmission for this traffic class. For the numerically highest traffic class, *maxInterferenceSize* is exactly equal to the maximum sized frame that can be transmitted through the Port (the maximum frame size supported by the underlying MAC). For all other traffic classes, the derivation of *maxInterferenceSize* becomes more complex, owing to the possibility that any higher traffic classes that support the credit-based shaper algorithm can delay the transmission of a frame. *maxInterferenceSize* must also account for any media access delay, such as the recovery time in Energy Efficient Ethernet. A detailed analysis can be found in L.3.1.1.

NOTE 2—The definition of *maxInterferenceSize* assumes that any traffic classes that support the shaper algorithm are numerically higher than any traffic classes that support the default strict priority algorithm (8.6.8.1). If this is not true, then the value of *maxInterferenceSize* is infinite, and the operation of the credit-based shaper algorithm cannot provide the bandwidth that has been reserved for it. A detailed analysis of how *maxInterferenceSize* can be determined for a given traffic class is contained in L.3.

The value of *hiCredit* is determined by the worst-case interfering traffic, as follows in Equation (L-3).

$$hiCredit = maxInterferenceSize \times (idleSlope/portTransmitRate) \quad (L-3)$$

The value of *hiCredit* can therefore be considered as a “high water mark”; the operation of the algorithm is such that the value of the *credit* parameter will never exceed *hiCredit*.

The choice of value for *idleSlope*, and the calculated value of *hiCredit*, can be used to determine the maximum burst size that can be output from the queue, as follows in Equation (L-4).

$$maxBurstSize = (portTransmitRate \times ((hiCredit - loCredit)/(-sendSlope))) \quad (L-4)$$

NOTE 3—There is an interrelationship between the maximum burst size and the buffer size available to the traffic class. There is no point in accumulating more credit than the traffic class can handle in queued frames, as the limiting factor is the discarding of frames when the queue overflows. This means that the amount of bandwidth that can be reserved on a given Port for a given traffic class depends upon the amount of buffering available to that traffic class. This in turn, combined with the port’s transmission rate, will determine the maximum value of idle slope that can be supported for that traffic class on that Port.

The operation of the credit-based shaper algorithm is illustrated in the examples shown in Figure L-1 through Figure L-3. In Figure L-1, a frame is queued at a time when the *credit* value is zero, and there is no conflicting traffic (there is no higher priority traffic awaiting transmission, and there is no frame being transmitted on the Port). The frame is immediately selected for transmission, and *credit* decreases at the rate of *sendSlope* as the transmission proceeds. Once the frame transmission is complete, *credit* increases back to zero at the rate of *idleSlope*, at which point, a further frame can be selected for transmission.

If a continuous stream of frames is made available to the shaper algorithm, i.e., there is always one frame queued awaiting transmission when the *credit* value reaches zero, then the fraction of the *portTransmitRate* that is available to the queue (*bandwidthFraction*) is equal to the fraction of time that frames are being transmitted from the queue. Assuming that *credit* decreases to value *loCredit* at the end of each frame transmission, then the following shows the detailed derivation of Equation (L-1):

$$bandwidthFraction = (-loCredit/sendSlope)/[(-loCredit/sendSlope) + (loCredit/idleSlope)] \quad (L-5)$$

$$= (-1/sendSlope)/[(1/idleSlope) - (1/sendSlope)] \quad (L-6)$$

$$= (-idleSlope)/(sendSlope - idleSlope) \quad (L-7)$$

$$= idleSlope/portTransmitRate \quad (L-8)$$

This fraction of the bandwidth is available to the stream even in the presence of conflicting traffic, as illustrated in the following examples.

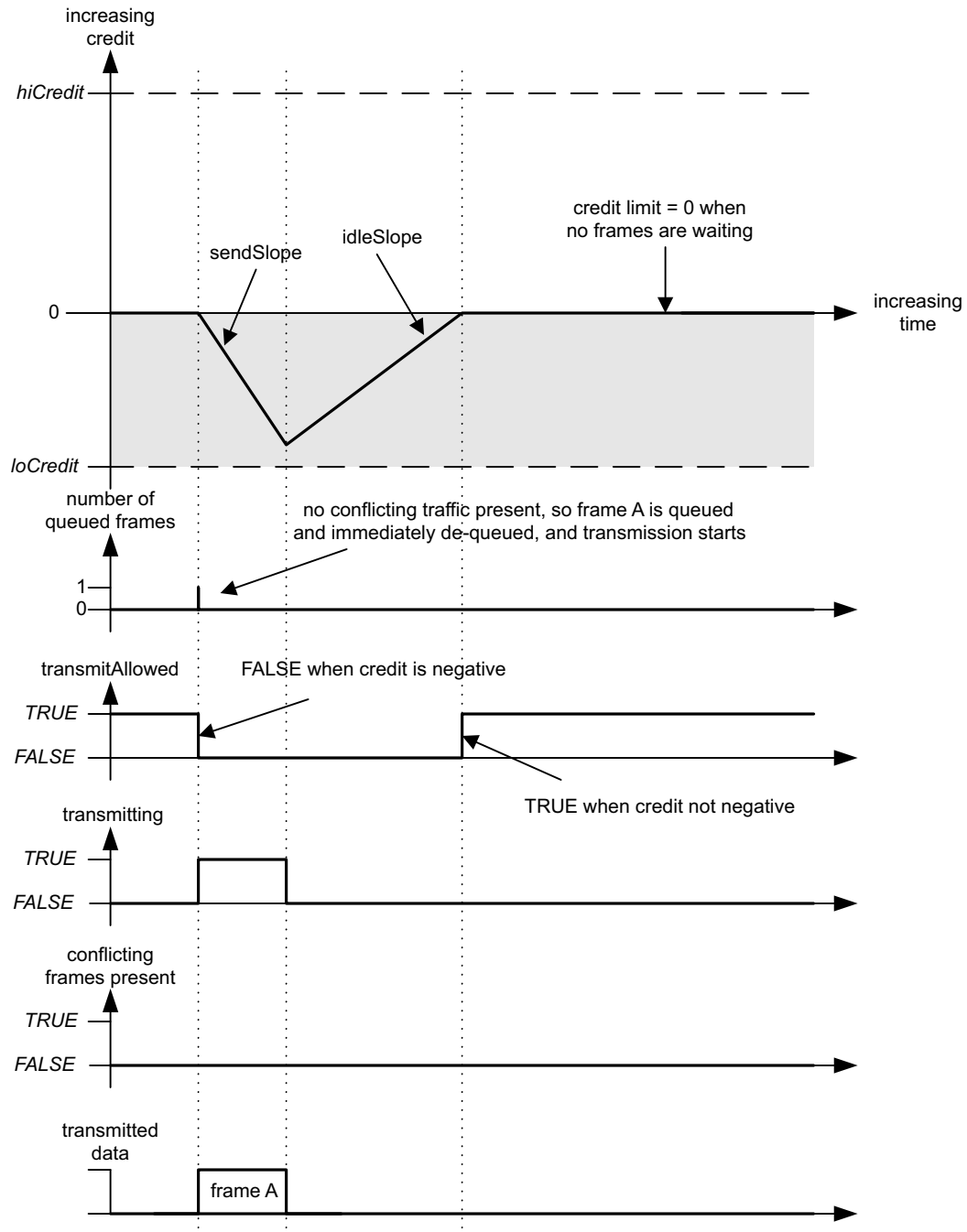


Figure L-1—Credit-based shaper operation—no conflicting traffic

1 In Figure L-2, a frame is queued at a time when the Port is transmitting conflicting traffic. The value of
2 *credit* increases at the rate of *idleSlope* while the queued frame waits for the Port to become available.
3 Transmission of the conflicting traffic completes before the value of *credit* is limited by *hiCredit*, and
4 transmission of the queued frame starts. The value of *credit* starts to decrease at the rate of *sendSlope*;
5 however, as the frame is not large enough to consume all of the available *credit*, and there are no further
6 frames queued, *credit* is reduced to zero on completion of the transmission.

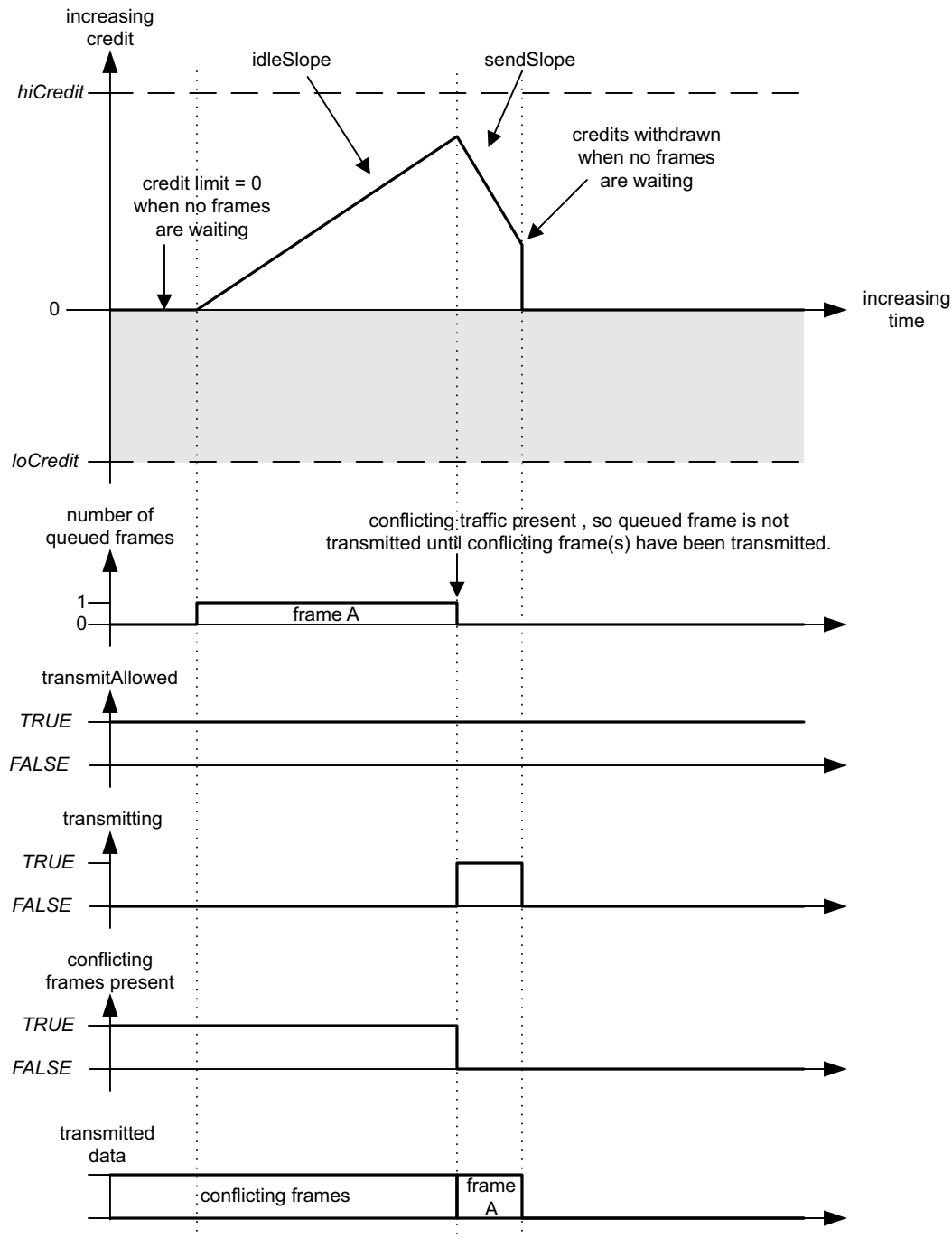


Figure L-2—Credit-based shaper operation—conflicting traffic

1 In Figure L-3, three frames are queued while the Port is transmitting conflicting traffic, and *credit*
2 accumulates at the rate of *idleSlope*. Once the conflicting traffic has been transmitted, the first and second
3 frames are transmitted back-to-back, because transmitting the first frame leaves $credit \geq 0$. However, as
4 transmitting the second frame causes credit to become negative, transmission of the third frame is delayed
5 until *credit* returns to zero.

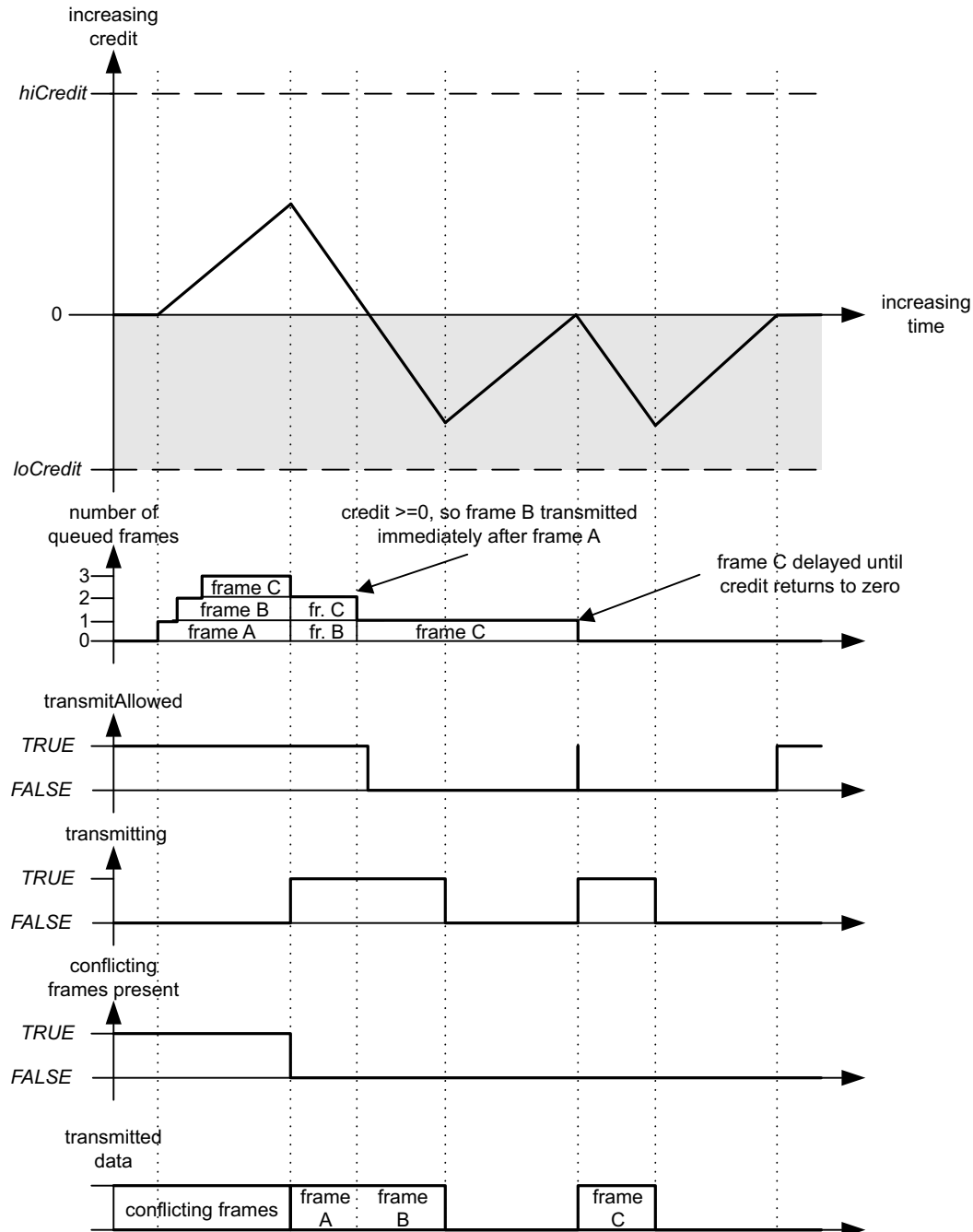


Figure L-3—Credit-based shaper operation—burst traffic

1 L.2 “Class measurement intervals” in Bridges

2 The “class measurement interval” defines acceptable Talker behavior with respect to per-stream
3 transmission (see 34.6.1.1); however, it does not directly impact the operation of the credit-based shaper
4 algorithm in Bridges.

5 The maximum fraction of the available bandwidth that a queue operating the credit-based shaper algorithm
6 consumes is determined by *idleSlope*, as discussed in L.1; if *idleSlope* is 75% of the link transmission rate,
7 then the algorithm gets a maximum of 75% of the link, averaged over some time period (the class
8 measurement interval) at the Talker. For a Bridge, where transmission becomes bursty due to interfering
9 transmission from other traffic classes, the minimum measurement interval that is meaningful (i.e., the
10 minimum period over which it is still possible to observe that the 75% maximum bandwidth is not exceeded)
11 is the time taken for the traffic class concerned to transmit a maximum sized burst, multiplied by 100/75. In
12 the case of the 125 μs class measurement interval for SR class A traffic, on 100 Mb/s Ethernet, and with
13 75% reservation, the approximate calculation is (ignoring interframe gaps)

- 14 a) The maximum interference size (*maxInterferenceSize*) is one maximum sized Ethernet frame, which
15 is 2000 octets, or 16 000 bits, so *hiCredit* would be 75% of that, or 12 000 bits.

16 NOTE 1—Media access delays, such as those imposed by Energy Efficient Ethernet, could be more significant than the
17 interference created by a maximum sized frame.

18 NOTE 2—The figure of 2000 octets is a maximum for Ethernet and therefore a worst case for an interfering frame; in a
19 specific implementation, the maximum frame size that is used could be smaller. In implementations where the maximum
20 frame size is known to be less than 2000, the smaller value could be used.

- 21 b) The maximum frame size for this traffic class is 75% of the number of bits that can be transmitted in
22 125 μs, so 75% of 12 500 bits in the case of a 100 Mb/s LAN, or 9375 bits (including up to 500 or so
23 possible “overhead” octets—used for, e.g., Tag headers, physical layer overhead). As this is not an
24 integral number of octets, this figure is rounded down to 9368. *loCredit* is 25% of that, or –2342.
- 25 c) The maximum burst size, for frames no greater than 9368 bits long, would occur if *credit* reached
26 12 000, then a series of frames were transmitted that exactly took credit to 0, followed by a final
27 9368 bit frame that would take credit down to –2342. This maximum burst is therefore calculated as
28 $(12\,000 + 2342) \cdot (\text{portTransmitRate}/(-\text{sendSlope}))$ or 57 368 bits.
- 29 d) The minimum measurement interval over which one can expect to be able to measure the bandwidth
30 utilization for SR class A, and observe that it does not exceed its 75% allocation, is therefore
31 57 368 bit times multiplied by 100/75, which is 76 491 bit times, rounded up to the nearest integer,
32 or 764.91 μs.

33 NOTE 3—The maximum sized burst will be preceded by a *maxInterferenceSize* event, and at the end of the burst, *credit*
34 will have been reduced to –2342; it will therefore take $2342/\text{idleSlope}$ seconds, or nearly 3123 bit times, for *credit* to
35 return to zero, and for SR Class A to be allowed to transmit again. In addition, the actual utilization for SR Class A is
36 $57368/(57368+16000+(2342/0.75)) = 75\%$.

37 For the SR class B traffic, all that really changes is the potential size of any interference, which increases
38 from the SR class A *maxInterferenceSize* to the maximum burst size for SR class A plus the class A
39 *maxInterferenceSize*.

40 So, the smallest possible measurement interval over which the ratio $\text{idleSlope}/\text{portTransmitRate}$ will be seen
41 to hold good is dependent on the maximum interference size for the SR class, the maximum frame size for
42 the SR class, and the value of *idleSlope*.

1 L.3 Determining worst-case latency contribution and buffering requirements

2 From the point of view of the operation of AV Bridges, it is important to be able to assess the contribution
3 that each Bridge makes to the worst-case latency that stream data frames can be subject to during their
4 transmission along the path from Talker to Listener. Closely related to the latency contribution that a Bridge
5 makes is the buffering requirement that is required in the Bridge in order for the delay not to result in frame
6 discard; if a Bridge can potentially delay a frame of a given SR class for a given period of time, then it must
7 be capable of buffering the worst-case number of frames of that same SR class that could be received for
8 onward transmission on each Port. The following discussion looks at the factors that contribute to the delay
9 that could be experienced by a frame as it passes through a Bridge, and how the delay can be accurately
10 determined.

11 NOTE 1—This discussion is also useful to guide designers of Talker stations in determining the delay added by the
12 Talker's network interface.

13 The worst-case latency for a single hop from Bridge to Bridge, measured from arrival of the last bit at Port n
14 of Bridge A to the arrival of the last bit at Port m of Bridge B, can be broken out into the following
15 components:

- 16 a) Input queuing delay. (There are no input queues in the IEEE 802.1 architecture, but if present, the
17 implementation must account for them.)
- 18 b) Interference delay (L.3.1).
- 19 c) Frame transmission delay. (The time taken to transmit one maximum frame at *portTransmitRate*.)
- 20 d) LAN propagation delay. (A variable delay that depends on the length of the LAN connection to the
21 next Bridge; this is measurable using the mechanisms defined in IEEE Std 802.1AS.)
- 22 e) Store-and-forward delay. This includes all other elements of forwarding delay that are a
23 consequence of the internal processing of the Bridge, assuming that the input and output queues are
24 empty, such as:
 - 25 1) The time needed to pass a frame from the input port to the output port, assuming empty queues.
 - 26 2) The time delay between a frame being available for transmission on a Port and the Port being
27 ready to transmit the frame.

28 NOTE 2—For example, in the case where the MAC/PHY has entered a power saving mode, there may be a delay
29 incurred in switching the Port back to normal operation.

- 30 3) The difference, if any, in the delay incurred by a frame that bypasses an empty queue, vs. that
31 incurred by a frame that must be enqueued.
- 32 4) The time added (subtracted) by the lengthening (shortening) of the frame due to addition
33 (removal) of frame headers such as Q-tags or MACSec-tags.
- 34 5) The time needed to encrypt a MACSec frame.

35 In the remainder of this clause, the following abbreviations are used:

- | | | |
|----|-----------------|--|
| 36 | M_0 | Maximum sized frame for non-SR classes |
| 37 | M_x | Maximum sized frame for SR class x |
| 38 | R_0 | <i>portTransmitRate</i> |
| 39 | R_x | <i>idleSlope</i> for SR class x |
| 40 | T_{xy} | Time interval between events x and y |
| 41 | W_x | <i>-sendSlope_x</i> |
| 42 | $variable_{<x}$ | The sum of the values of the named <i>variable</i> for all SR classes with a higher priority
43 (and therefore earlier letter in the collating sequence) than x |

1 L.3.1 Interference delay

2 The interference delay for frame X can be broken out into the following components:

- 3 a) **Queuing delay:** The delay caused by the frame that was selected for transmission an arbitrarily
4 small time before frame X became eligible for transmission selection, plus the delay caused by
5 queued-up frames from all stream frames with higher priority than frame X's class (i.e., the
6 "*maxBurstSize*" for SR Classes with higher priority than X—see L.1). This is what is referred to as
7 *maxInterferenceSize* in L.1. Queuing delay is analyzed in detail in L.3.1.1.
- 8 b) **Fan-in delay:** The delay caused by other frames in the same class as frame X that arrive at
9 more-or-less the same time from different input Ports. Fan-in delay is analyzed in detail in L.3.1.2.
- 10 c) **Permanent delay:** The delay caused by frames that reside in a buffer for a long time, relative to the
11 output queuing delay, because of the history of activity in the network. Permanent delay is analyzed
12 in detail in L.3.1.3.

13 L.3.1.1 Queuing delay

14 The queuing delay for frame X can be broken out into the following components:

- 15 a) The delay, *maxInterferenceTime*, before a frame X queued for an SR class, and that is eligible for
16 transmission, can actually be transmitted. This delay is experienced when a frame, of any priority, is
17 selected for transmission an arbitrarily small time before frame X became eligible for transmission;
18 it is also experienced in LAN media, such as Ethernet, where there can be a low energy consumption
19 mode that is entered under idle conditions, and where there is a significant delay,
20 *mediumAccessDelay*, before the LAN returns to normal operation and is able to transmit. The value
21 of *maxInterferenceTime* is therefore equal to whichever is the greater of the following:
 - 22 1) The time taken to transmit the maximum frame size supported by the medium, i.e.,
23 $\text{maxFrameSize}_{(m)} / \text{portTransmitRate}$.
 - 24 2) The *mediumAccessDelay* for the medium.
- 25 b) The delay caused by any queued-up frames associated with the next higher priority SR class than
26 frame X's SR class (i.e., *maxBurstSize* for the next higher priority SR class), if frame X does not
27 belong to the highest SR class.

28 For SR class A, only the first of these components applies, as there is no higher SR class.

29 Suppose that the queue for SR class A is full, and it has accumulated the maximum amount of *credit*
30 (*hiCredit*). Because SR class A frames have priority over all other traffic (even BPDUs), *hiCredit* for SR
31 class A is merely the credit accumulated during the *maxInterferenceTime* required to transmit a
32 lower-priority frame. Until that accumulated credit is exhausted, SR class B (C, D,...) frames cannot be
33 transmitted.

34 If SR class A were permitted to use 100% of the LAN bandwidth, then the SR class A queue would never
35 catch up, because it would use credit as fast as it was gained, and there would never be a chance for any
36 other traffic class to transmit a frame. If SR class A were permitted to use 99% of the LAN bandwidth, then
37 SR class A can transmit back-to-back frames and the accumulated *credit* would be drained at 1% of the LAN
38 bandwidth until it goes negative; at that point, SR class A cannot transmit further frames until its *credit*
39 returns to a non-negative value, which ensures that at least 1% of the bandwidth is available to other traffic
40 classes.

41 NOTE—In general, if SR Class A is permitted to use X% of the bandwidth, then only (100-X)% remains available for
42 use by lower priority traffic classes. In practice, as 75% is the default value for the highest percentage of
43 *portTransmitRate* that can be used by all SR classes, there is always at least 25% of *portTransmitRate* available for use
44 by the non-SR traffic classes.

Figure L-4 illustrates this burst transmission behavior, and the effect of *maxInterferenceTime* on the latency of SR class frames. At point *a* on the time line, the queues for SR classes A and B are empty, and a maximum sized frame (M_0 bits long) belonging to a non-SR traffic class starts transmission. An instant later, SR class A and B frames arrive that are all of the maximum size that the classes are currently experiencing (M_A and M_B bits long, respectively); however, as there is a frame already being transmitted, they are queued. At point *b*, transmission of the non-SR class frame completes, and the first of a burst of four SR class A frames are transmitted. By point *d*, SR class A's credit has been taken negative by the transmission of the last of its burst of frames, and therefore no further SR Class A frames can be transmitted until its credit is no longer negative; this allows SR Class B to transmit a frame. What happens at the end of the class B frame will depend on what frames are queued waiting for transmission, and whether the credit available to either of the SR classes is non-negative.

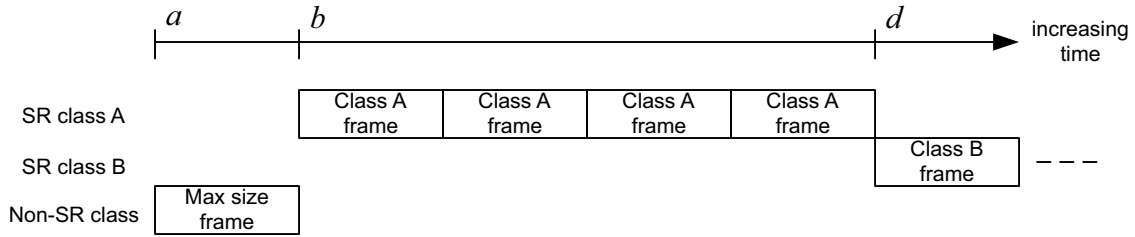


Figure L-4—Interference and latency

The queuing delay experienced by the first SR Class A frame is therefore the time interval between *a* and *b*, T_{ab} . If R_0 is the transmission rate (*portTransmitRate*), then:

$$T_{ab} = M_0 / R_0 = \text{maxInterferenceTime} \quad (\text{L-9})$$

Figure L-5 shows how credit changes for SR Class A during this sequence.

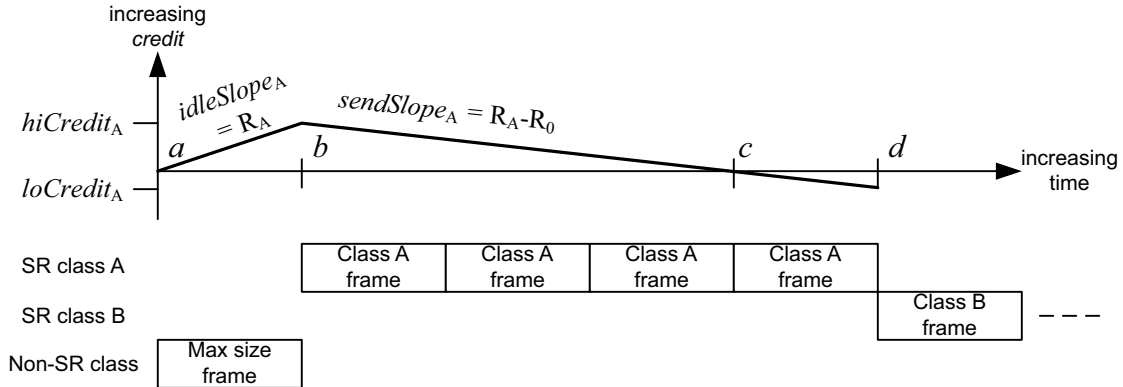


Figure L-5—Burst behavior and credit

R_A is the reserved data rate for SR Class A (which is the *idleSlope* for class A), R_B is the reserved data rate for SR Class B, and so on. The value of *credit*_A accumulates up to *hiCredit*_A during the transmission of the maximum sized non-SR frame so that:

$$\text{hiCredit}_A = R_A \times M_0 / R_0 = \text{idleSlope}_A \times \text{maxInterferenceTime} \quad (\text{L-10})$$

credit is drained during the transmission of the SR Class A frames, at the rate *sendSlope*_A, as follows:

$$\text{sendSlope}_A = (R_A - R_0) \quad (\text{L-11})$$

1 *credit* reaches zero at point *c* in Figure L-5, which, for the purposes of this discussion, is assumed to happen
2 exactly at the point where transmission of the third class A frame finishes. Since a frame can be transmitted
3 when *credit* = 0, Class A's credits continue to drain to the following value:

$$4 \quad loCredit_A = (R_A - R_0) \times M_A / R_0 \quad (L-12)$$

5 during the transmission of the fourth class A frame (M_A bits long, transmitted in time M_A / R_0). At point *d*,
6 transmission of the fourth frame completes; as *credit*_A is now negative, no more class A frames can be
7 transmitted, and transmission of a class B frame starts.

8 So, for class A, the maximum burst size is equal to the transmission rate, R_0 , multiplied by the time interval
9 between points *b* and *d* in Figure L-5:

$$10 \quad \text{max Class A burst size} = R_0 \times (- (hiCredit_A - loCredit_A) / sendSlope_A) \quad (L-13)$$

11 Substituting $hiCredit_A$, $sendSlope_A$, and $loCredit_A$ from Equation (L-10), Equation (L-11), and
12 Equation (L-12), respectively, results in the following:

$$13 \quad \text{max Class A burst size} = R_0 \times (- (R_A \times M_0 / R_0 - (R_A - R_0) \times M_A / R_0) / (R_A - R_0)) \quad (L-14)$$

$$14 \quad = R_0 \times (R_A \times M_0 / R_0) / (R_0 - R_A) + M_A / R_0 \quad (L-15)$$

$$15 \quad = (R_A \times M_0) / (R_0 - R_A) + M_A \quad (L-16)$$

16 The queuing delay experienced by SR class B is the time interval between *a* and *d* in Figure L-5, T_{ad} :

$$17 \quad T_{ad} = T_{ab} + T_{bc} + T_{cd} \quad (L-17)$$

$$18 \quad = M_0 / R_0 + hiCredit_A / (-sendSlope_A) + M_A / R_0 \quad (L-18)$$

$$19 \quad = M_0 / R_0 + (R_A \times M_0 / R_0) / (R_0 - R_A) + M_A / R_0 \quad (L-19)$$

$$20 \quad = M_0 / (R_0 - R_A) + M_A / R_0 \quad (L-20)$$

21 This can be generalized to calculate the queuing delay experienced by any SR class, *X*, by using the sum of
22 the credits available to all higher priority SR classes. In the following, the subscript "<*X*" is used to indicate
23 the sum of the values for all SR classes with higher priority (and therefore, earlier letters in the collating
24 sequence) than *X*.

25 The credit acquisition rate $idleSlope_{<X}$ for SR classes A through *X* – 1 is the sum of the data rates for all
26 higher priority classes so that:

$$27 \quad idleSlope_{<X} = \sum_{K < X} R_K \quad (L-21)$$

28 This is just the sum of the $idleSlope_K$ values for the higher classes. However, the combined classes
29 accumulate credits only until the single interfering non-SR interfering frame stops transmitting, and then the
30 credits start decreasing linearly. So, the upper limit for the combined credits is not the sum of the individual
31 Class's credits; it is the number of bits divided by the slope, or:

$$32 \quad hiCredit_{<X} = (\sum_{K < X} R_K) \times M_0 / R_0 \quad (L-22)$$

33 Similarly, the combined $sendSlope_{<X}$ is:

$$34 \quad sendSlope_{<X} = - (R_0 - \sum_{K < X} R_K) \quad (L-23)$$

35 These combined rates hold true until some class's buffer empties and its credits are forced to 0. In the
36 worst-case scenarios examined here, this does not happen.

1 Defining:

$$2 \quad W_{<X} = -\text{sendSlope}_{<X} = R_0 - \sum_{K<X} R_K \quad (\text{L-24})$$

3 leads to:

$$4 \quad \text{sendSlope}_{<X} = -W_{<X} \quad (\text{L-25})$$

5 and:

$$6 \quad \text{hiCredit}_{<X} = (R_0 - W_{<X}) \times M_0 / R_0 \quad (\text{L-26})$$

7 For all combined classes, T_{ab} is the same as for SR class A, the time for the non-SR frame to transmit:

$$8 \quad T_{ab} = M_0 / R_0 \quad (\text{L-27})$$

9 The combined classes A through X – 1 drain from $\text{hiCredit}_{<X}$ to 0 in time $\text{hiCredit}_{<X} / W_{<X}$ so that:

$$10 \quad T_{bc} = ((R_0 - W_{<X}) \times M_0 / R_0) / W_{<X} \quad (\text{L-28})$$

11 The worst-case value of $\text{loCredit}_{<X}$ for the combined $<X$ classes is not the sum $\sum_{K<X} \text{loCredit}_K$ as this
12 would overestimate the worst case. Only one of the $<X$ classes can be transmitting a frame at a given time,
13 so the end of the burst for the $<X$ classes occurs when one of the $<X$ classes, Q say, starts transmitting a
14 frame at a point where all the other $<X$ classes have negative credit, and their credit is therefore rising. If, by
15 the time Q's frame transmission finishes, the other $<X$ classes still have negative credit, then Q's frame is the
16 last frame of the $<X$ burst; but at that point, all the other $<X$ classes must, by definition, have more credit
17 than their respective loCredit values (i.e., for these other classes, $\text{credit} > \text{loCredit}$) because even if they were
18 at their respective loCredit values when Q started transmitting, they have been gaining credit during the
19 transmission of Q's frame.

20 It is also not possible to decide that $\text{loCredit}_{<X}$ is simply the time needed to transmit one copy of each
21 Class's maximum-length frame; for some classes it could be possible transmit more than a single last frame
22 after the total credits = 0, even if all Classes' credits reach 0 simultaneously. For example, classes A and B
23 reach $\text{credit} = 0$ at the same time; A transmits a frame, then B transmits a frame. By the end of the B frame,
24 A's credit has returned to zero, and so A can transmit a further "last" frame.

25 Some class must transmit the last frame, and for it to be the worst case, that last frame is a maximum length
26 frame. If it were not, then either:

- 27 c) Extending it to the maximum length still leaves the other classes at negative credit ; or
- 28 d) Extending it to the maximum length leaves one or more other class with 0 or positive credit , in
29 which case they will transmit more frames.

30 So, for class Q to be the class that transmits the last frame of the $<X$ burst, the following condition must be
31 true for all other $<X$ classes, P:

$$32 \quad M_q \times R_p > (R_0 - R_p) \times M_p \quad (\text{L-29})$$

33 Equation (L-29) follows from the fact that the increase in credits of class P during the transmission of
34 M_q , which is equal to $(M_q/R_0) \times R_p$ is greater than the absolute value of the loCredit of P, which is equal to
35 $(R_0 - R_p) \times M_p$. So,

$$36 \quad (M_q/R_0) \times R_p > (R_0 - R_p) \times M_p / R_0 \quad (\text{L-30})$$

37 This statement is true because the other classes P must have low enough credit that their credit cannot climb
38 above 0 during the transmission of M_q .

1 The very lowest that $credit_{<X}$ might reach for three SR classes, Q, P, and S is shown in Equation (L-31).

$$2 \quad loCredit_{<X} \geq loCredit_Q + loCredit_P + loCredit_S + R_S \times M_q/R_0 + R_P \times M_q/R_0 \quad (L-31)$$

3 However, even this is pessimistic, as it assumes that more than one class can be at their $loCredit$ at the same
4 time, which is not the case. What this shows, however, is that if R_K is small for all $K < X$ (i.e., $R_K \ll R_0$), then
5 the $R_K \times M_q/R_0$ terms all approach zero, and therefore $loCredit_{<X}$ approaches the sum $\sum_{K < X} loCredit_K$. So,

$$6 \quad loCredit_{<X} = \sum_{K < X} loCredit_K \quad (L-32)$$

$$7 \quad = \sum_{K < X} (R_K - R_0) \times M_K / R_0 \quad (L-33)$$

8 But in this worst case, $R_K \ll R_0$; therefore,

$$9 \quad loCredit_{<X} = \sum_{K < X} (-M_K) \quad (L-34)$$

10 Putting together these various components, the queuing delay for SR Class X, $qDelay_X$, is as follows:

$$11 \quad qDelay_X = M_0 / R_0 + (hiCredit_{<X} - loCredit_{<X}) / |sendSlope_{<X}| \quad (L-35)$$

$$12 \quad = M_0 / R_0 + ((R_0 - W_{<X}) \times M_0 / R_0 + \sum_{K < X} M_K) / W_{<X} \quad (L-36)$$

$$13 \quad qDelay_X = (M_0 + \sum_{K < X} M_K) / W_{<X} \quad (L-37)$$

14 For SR class A, this equation reduces to Equation (L-38):

$$15 \quad qDelay_A = M_0 / W_{<A} \quad (L-38)$$

16 As there is not a “<A” class, A being the highest priority class, $W_{<A}$ is simply R_0 , so this reduces to
17 Equation (L-39):

$$18 \quad qDelay_A = M_0 / R_0 \quad (L-39)$$

19 This agrees with the earlier discussion in L.1.

20 For SR class B, the equation reduces to Equation (L-40):

$$21 \quad qDelay_B = (M_0 + M_A) / W_A \quad (L-40)$$

22 As W_A is $-sendSlope_A$, which is just $R_0 - R_A$, this can be rewritten as shown in Equation (L-41):

$$23 \quad qDelay_B = (M_0 + M_A) / (R_0 - R_A) \quad (L-41)$$

24 The maximum burst size for a given class, X, is the number of bits that can be transmitted in “back-to-back”
25 frames by class X. The maximum burst occurs when that class has experienced the maximum queuing delay
26 for that class, $qDelay_X$, and class X is then able to transmit frames, uninterrupted by any higher priority
27 classes, until it reaches $loCredit_X$. The size of this burst, $maxBurst_X$, is equal to the number of bits that can
28 be transmitted at line rate, R_0 , in the time it takes for $credit_X$ to reduce from $hiCredit_X$ to $-loCredit_X$, as
29 follows in Equation (L-42):

$$30 \quad maxBurst_X = (hiCredit_X - loCredit_X) \times R_0 / W_X \quad (L-42)$$

31 From the earlier results, the worst case for $loCredit_X$ is just $(-M_X)$ so that:

$$32 \quad = hiCredit_X \times R_0 / W_X + M_X \times R_0 / W_X \quad (L-43)$$

1 and $hiCredit_X$ is the amount of credit that can be accumulated during $qDelay_X$ seconds, which is $qDelay_X$
2 multiplied by the idle slope, R_X , so that:

$$3 \quad \maxBurst_X = (M_0 + \sum_{K < X} M_K) \times R_X \times R_0 / (W_{<X} \times W_X) + M_X \times R_0 / W_X \quad (L-44)$$

4 L.3.1.2 Fan-in delay

5 The delay caused by other frames in the same class as frame X that arrive at more-or-less the same time from
6 different input Ports. Consider the scenario in Figure L-6, where a Bridge has a number of Ports on which it
7 receives frames for a given SR Class and the frames will be queued on a single outbound Port. Imagine that
8 the “upstream” Bridge Ports feeding every reception Port of the Bridge all encounter a maximum
9 interference event for Class X at the same time. The output Port will be starved of data. Then, once the
10 interference events complete, the “upstream” Bridges start to transmit frames and the input Ports all receive
11 a frame at the same moment; all received frames are delivered to the output queue.

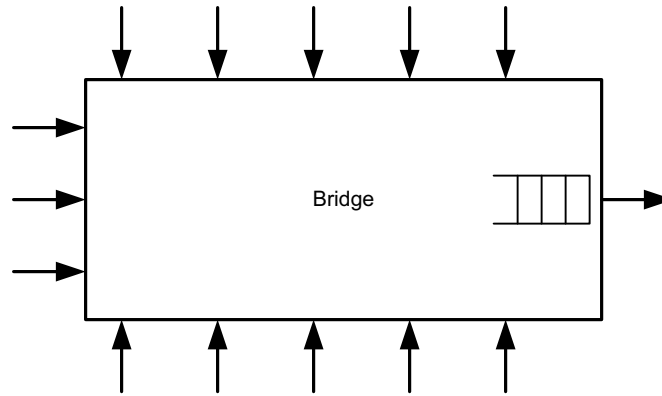


Figure L-6—Fan-in scenario

12 Assuming that all of the Ports of the Bridge support the same data rate, and the outbound queues of the
13 upstream Bridges are all configured the same as the outbound queue of this Bridge, then the outbound Port
14 would not be able to transmit frames from its queue at the desired, combined data rate of the incoming
15 traffic, as this would require the outbound Port (in this example) to transmit at 20 times its reserved data rate,
16 which would not be permitted by the Stream Reservation Protocol (SRP). It follows that whatever the
17 reserved data rate is on the outbound Port, this must be the same as the sum of the reserved data rates being
18 fed into the inbound Ports.

19 By allocating the majority of the reserved bandwidth to one of the inbound Ports (shown in gray in
20 Figure L-6) and a very small amount of bandwidth to all of the other inbound Ports, the amount of fan-in
21 data that will be received can be maximized; the gray Port will receive $\maxBurst_X$ bits of burst data, and all
22 other Ports will receive a single frame of length M_X . Hence, the data that the outbound queue has to handle
23 in this scenario is $\maxBurst_X + 12 \times M_X$ bits of fan-in data.

24 NOTE 1—The reason that this choice maximizes the fan-in data burst size is that regardless of how little bandwidth is
25 allocated to the black Ports, they still get to burst a complete frame.

1 The following is a procedure (not a formula) for determining the maximum amount of fan-in data for a given
2 SR class and output port:

- 3 a) Determine B_O , the maximum possible bandwidth for class X on the output port.
- 4 b) For each possible input port i , determine the maximum possible bandwidth B_i for Class X—it is
5 assumed that this information is obtained, by some means, from the transmitting Bridge that is the
6 source of the frames received by i .
- 7 c) For input port i , calculate $\max\text{Burst}_{X,i}$ using $\max(B_O, B_i)$ for the bandwidth. (In the $\max\text{Burst}$
8 formula, use $W_X = R_0 - \max(B_O, B_i)$.)
- 9 d) Add $\max(\max\text{Burst}_{X,i})$ to the total fan-in data.
- 10 e) Set $B_O = B_O - B_i$ and repeat from step c) until $B_O = 0$.
- 11 f) For each remaining port, add $M_{X,i}$ to the total fan-in data.

12 The fan-in delay is then equal to the time taken for the total fan-in burst data, computed as shown above, to
13 be output at the line rate of the output port.

14 NOTE 2—The definition of a measurement interval for stream TSPECs implies a minimum frame transmission rate
15 (8000 frames per second for class A's 125 μs measurement interval). At this frame transmission rate, the frames are
16 smaller and, due to the overhead in small frames, fewer streams are possible (no more than 13 through a 100 Mb port).
17 These constraints should be given due consideration when determining the value to be used for M_X and for determining
18 the maximum number of ports contributing to fan-in delay.

19 Example calculations of fan-in delay can be found in the paper “Calculating the Delay Added by Qav
20 Stream Queue” [B3].

21 L.3.1.3 Permanent delay

22 Permanent delay reflects the fact that, if a path through the network is being used continuously at its full
23 capacity for a particular SR class, interfering traffic can cause “permanent” buffer occupancy at the point of
24 congestion. This is illustrated in Figure L-7 through Figure L-10 and the accompanying text.

25 In Figure L-7 a Talker station, T, has reserved the highest possible bandwidth, R_X , for an SR class X stream
26 on the path through 3 Bridges to Listener L. For the purposes of the discussion, R_X is the maximum
27 reservable data rate for class X on the outbound Ports of all three Bridges, and Talker T starts transmitting
28 the stream, at the reserved data rate R_X . Assuming that the LANs are short and there is no interfering traffic
29 (i.e., the only traffic being handled is this particular stream), the buffer occupancy in each Bridge will
30 approach zero, as frames can be transmitted onwards as soon as they are received.

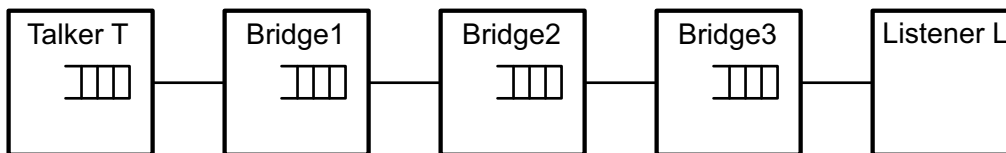


Figure L-7—Permanent delay scenario

31 In Figure L-8, a second transmission, Q, generates the maximum possible interference to class X (i.e., class
32 X's next frame for transmission is delayed by $q\text{Delay}_X$). Talker T's class X queue fills, and its *credit*
33 increases to $hi\text{Credit}_X$ (shown on the figure as $C=C_X$). Because T is no longer transmitting class X frames,
34 the downstream Bridges are starved of frames to transmit, their buffers are empty, and their credit is zero
35 ($C = 0$).

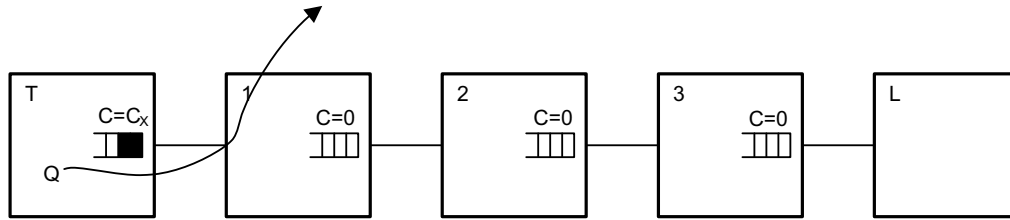


Figure L-8—Building up buffer occupancy—1

1 In Figure L-9, the interfering traffic goes away, and T is able to transmit a maximum sized class X burst of
2 frames at line rate; once the burst has finished, T reverts to transmitting the stream at the reserved rate R_X .
3 As Bridge 1 has no credit, it will have to buffer most of the burst of frames, and as T is sending frames as
4 fast as Bridge 1 can get rid of them, its buffer occupancy will remain at that level unless something else
5 affects that steady state. Bridge 1 ends up with a credit value that hovers around the *hiCredit* level; the credit
6 in the other Bridges will have *credit* of 0 or some value between 0 and *loCredit* as the Bridges receive and
7 transmit frames.

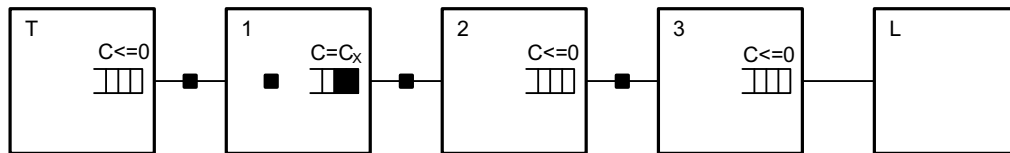


Figure L-9—Building up buffer occupancy—2

8 If the interfering traffic, Q, starts up again, Talker T's buffer occupancy will build up again; however, as
9 Bridge 1 still has frames buffered, it can continue to transmit, and the downstream Bridges are not starved of
10 frames, and Bridge 1's buffer occupancy and *credit* returns to around 0. When the interference stops, T
11 transmits a burst at line rate, as before, which simply restores Bridge 1's queue to the state shown in
12 Figure L-9.

13 If a new interfering frame source, S, starts up that affects class X in Bridge 1, then Bridges 2 and 3 could
14 again be starved of traffic, and the buffer occupancy and credit in Bridge 1 could increase temporarily to
15 $2C_X$ (see Figure L-10). When S stops transmitting, then Bridge 2 can create a line rate burst of traffic that
16 will restore its buffer to the previous steady state (Figure L-9), but now Bridge 2 has a permanent buffer
17 loading as shown in Figure L-10.

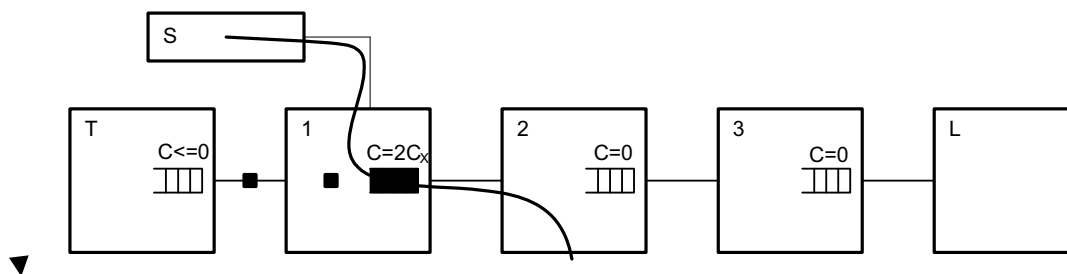


Figure L-10—Building up buffer occupancy—3

1 This can of course happen for other Bridges in the path, leading eventually to some level of permanent
2 buffer occupancy in each Bridge (Table L-11). The problem does not get any worse from that point on,
3 because no Bridge ever starves (i.e., no Bridge's credit is set to zero because it runs out of frames to send);
4 starvation is a requirement in order for permanent partial occupancy to be created.

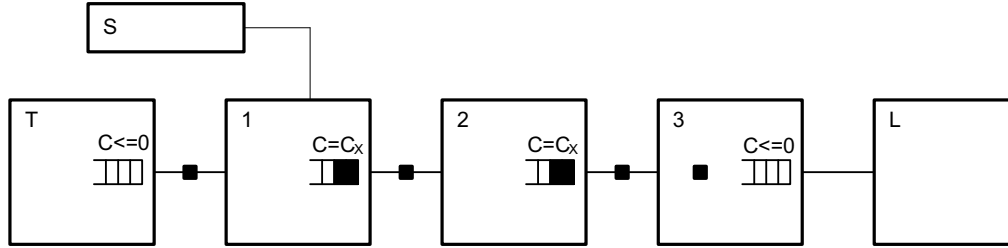


Figure L-11—Building up buffer occupancy—4

5 It may, therefore, be considered desirable for all Bridges to reserve a little bit more bandwidth than the actual
6 total reserved by the Talkers, so that this “permanent” buffer build-up can drain away. However, this would
7 not make the problem significantly better on the timescales of the output queuing delay, so it does not factor
8 into those calculations, and, of course, this additional capacity would increase the queuing delay at each hop.
9 NOTE—Many applications have media clocks that are asynchronous to the free-running clock used to determine class
10 measurement intervals. As the two clocks could be operating at the extremes of their permissible variation, it is
11 necessary to reserve more bandwidth than is nominally needed in order to account for the potential difference in clock
12 rate.

13 Since this kind of event could happen on multiple input ports at the same time, the “permanently buffered”
14 data equals the worst-case fan-in data.

15 L.3.2 Maximum interference delay and maximum buffer requirement

16 As discussed in L.3.1, the worst-case interference delay for SR class X on a given Port is the sum of the
17 following three parts:

- 18 a) $qDelay_X$ [see Equation (L-35) in L.3.1.1]
- 19 b) The fan-in delay (see L.3.1.2)
- 20 c) The “permanent” buffer contents contribution (see L.3.1.3), which is equal to the worst-case fan-in
21 delay

22 The buffering requirements for SR class X consist of the following three components:

- 23 d) A contribution from the queuing delay, equal to $maxBurst_X$
- 24 e) The fan-in burst data size, computed as described in L.3.1.2
- 25 f) The “permanent” buffer contents, which is equal to the fan-in burst data size

26 The fan-in burst size and the permanent buffer contents do not count twice; if a fan-in burst happens, then it
27 becomes the permanent buffer contents. If it happens a second time, it can only happen after a pause that has
28 allowed the permanent buffer loading to drain. So the worst-case buffering requirement for SR class X is
29 simply the sum of the first two elements, $maxBurst_X$ [Equation (L-42)] and fan-in burst data size; therefore,

$$30 \quad maxBuffering_X = maxBurst_X + (fan-in burst data size)_X \quad (L-45)$$

1 The worst-case total buffering requirement for all of the SR classes on a given Port is the sum of the
2 following two parts:

- 3 g) The worst-case buffering requirement for the lowest priority SR class on that Port
- 4 h) One max-size frame $M_{i,Z}$ for each class Z of higher priority than Class X for each input port i

5 Since the worst-case buffer requirement for Class X is when it is taking almost all of the bandwidth from the
6 higher-priority classes, the SR priority levels can share buffering space. The only additional requirement is
7 for the fan-in from higher priorities on other Ports. Thus, the SR classes would do well to share their buffer
8 space on each Port.

9 **L.4 Operation of credit-based shaper in Coordinated Shared Network (CSN)**

10 A CSN is a contention-free, QoS-capable, time division multiplexed access network. One of the nodes of the
11 network acts as the Network Coordinator (NC) node granting transmission opportunities to the other nodes
12 of the network. The NC node also acts as the QoS manager of the network.

13 From the perspective of the specification of the credit-based shaper, a CSN is equivalent to a distributed
14 Bridge. Rather than shaping the AV traffic in the egress nodes, which could lack the buffering resources to
15 do so, the shaping could instead be performed by the CSN's NC per Class of Service when it schedules
16 transmissions opportunities from the CSN's ingress to the CSN egress nodes. The behavior of CSN's egress
17 to another MAC technology should be consistent with the behavior specified in this standard, regardless of
18 how the shaper is implemented.

1 Annex M

2 (normative)

3 Support for PFC in link layers without MAC Control

4 M.1 Overview

5 Priority-based Flow Control (PFC) is a function defined for only point-to-point full-duplex links in terms of
6 the M_CONTROL primitives (11.4 of IEEE Std 802.1AC-2016 [B10]). For IEEE 802.3 link layers the
7 M_CONTROL primitives are mapped into the MAC Control MA_CONTROL primitives
8 (IEEE Std 802.1AC), that use the PDU format defined in IEEE Std 802.3. Other link layers supporting
9 point-to-point full-duplex operations need to define their mapping of the M_CONTROL primitives. This
10 annex describes a PDU format suitable to support PFC.

11 M.2 PFC PDU format

12 Figure M-1 shows a PDU format suitable to support PFC.

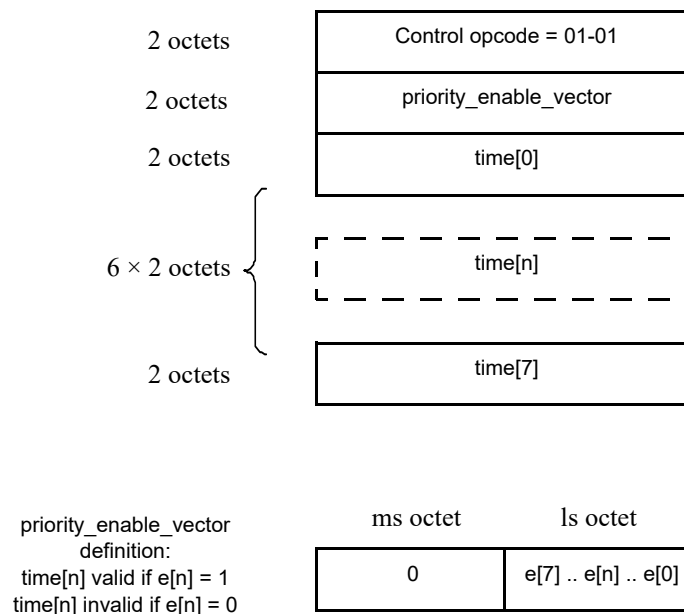


Figure M-1—PFC PDU format

13 The Control opcode field contains a 2-octet operation code indicating the Control function.

14 The remaining fields contain the parameters defined in 36.1.2.

Annex N

(informative)

Buffer requirements for PFC

N.1 Overview

To ensure that data frames are not lost due to lack of receive buffer space, receivers must ensure that a PFC M_CONTROL.request primitive is invoked while there is sufficient receive buffer to absorb the data that can continue to be received during the time needed by the remote system to react to the PFC operation. The precise calculation of this buffer requirement is highly implementation dependent. This annex provides an example of how it can be calculated based on a hypothetical delay model. Setting the PFCLinkDelayAllowance (see 12.22.6) to less than the round-trip delay value can result in frames loss.

Figure N-1 provides an high-level view of the various delays to consider:

- a) Processing and queuing delay of the PFC request
- b) Propagation delay of the PFC frame across the media
- c) Response time to the PFC indication at the far end
- d) Propagation delay across the media on the return path

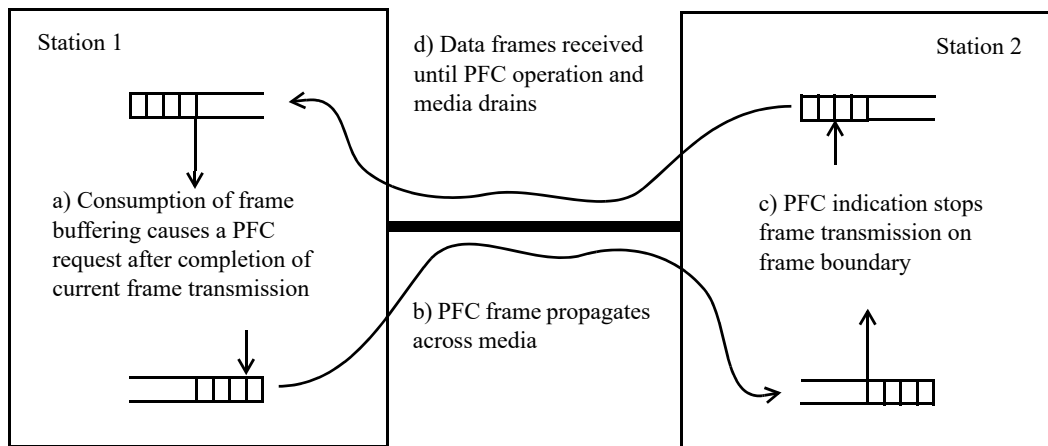


Figure N-1—PFC delays

N.2 Delay model

Figure N-2 shows how to model the various delays between two stations connected by a point-to-point full-duplex IEEE 802.3 link.

The main delay components shown in Figure N-2 are as follows:

- a) **PFC transmission delay:** the time needed by a station to request transmission of a PFC frame after a PFC M_CONTROL.request has been invoked (e.g., because a maximum length data frame can be transmitted).
- b) **Interface Delay (ID):** the sum of MAC Control, MAC/RS, PCS, PMA, and PMD delays. Interface Delay is dependent on the MAC and physical layer in use.

- 1 c) **Cable Delay:** the number of bits in flight stored in the transmission medium. This delay value is
2 dependent on the selected technology and on the medium length.
- 3 d) **Higher Layer Delay (HD):** the time needed for a queue to go into paused state after the reception of
4 a PFC M_CONTROL.indication that paused its priority. A substantial portion of this delay
5 component is implementation specific.

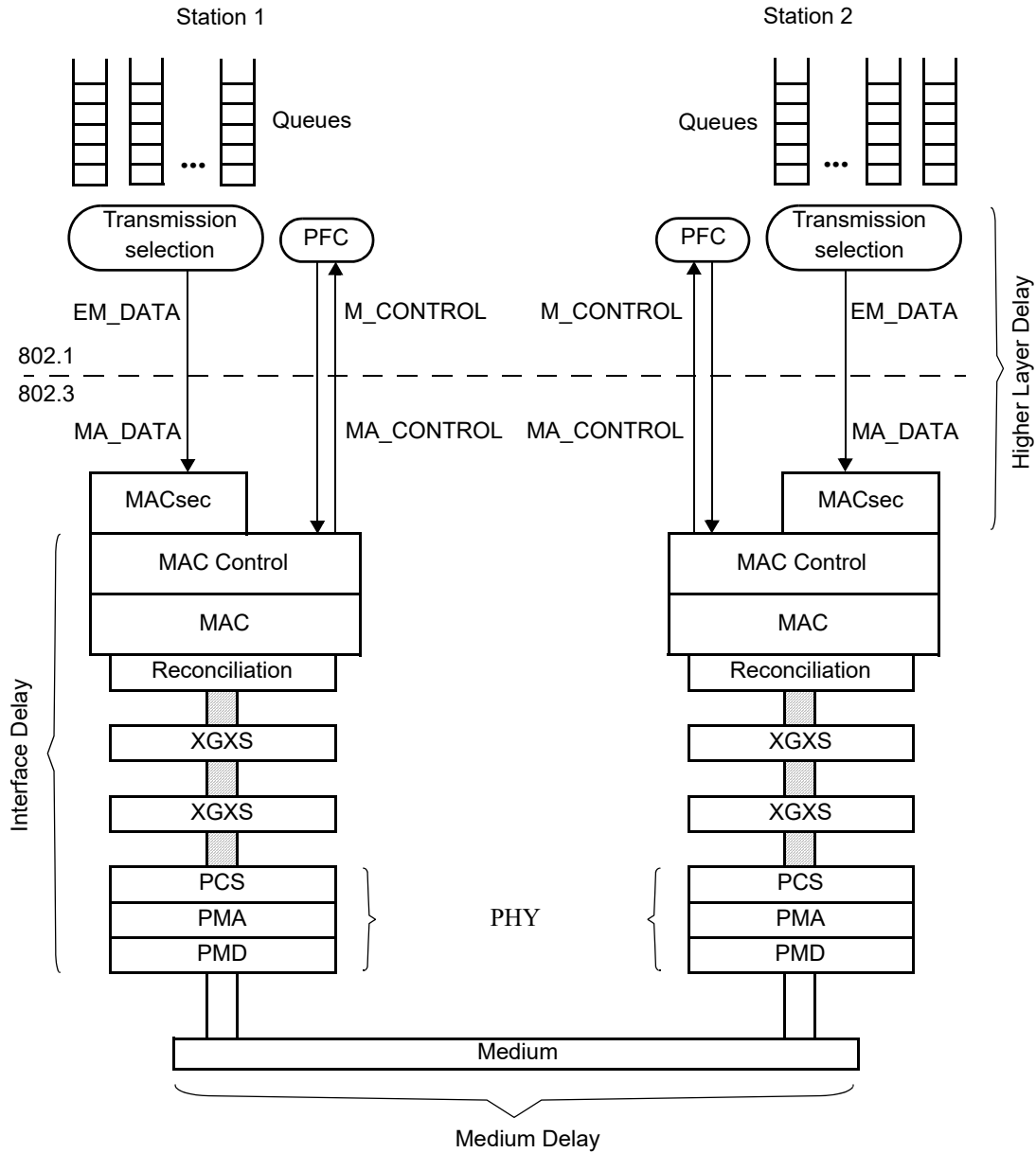


Figure N-2—Delay model

6 Figure N-3 shows a possible worst-case delay example.

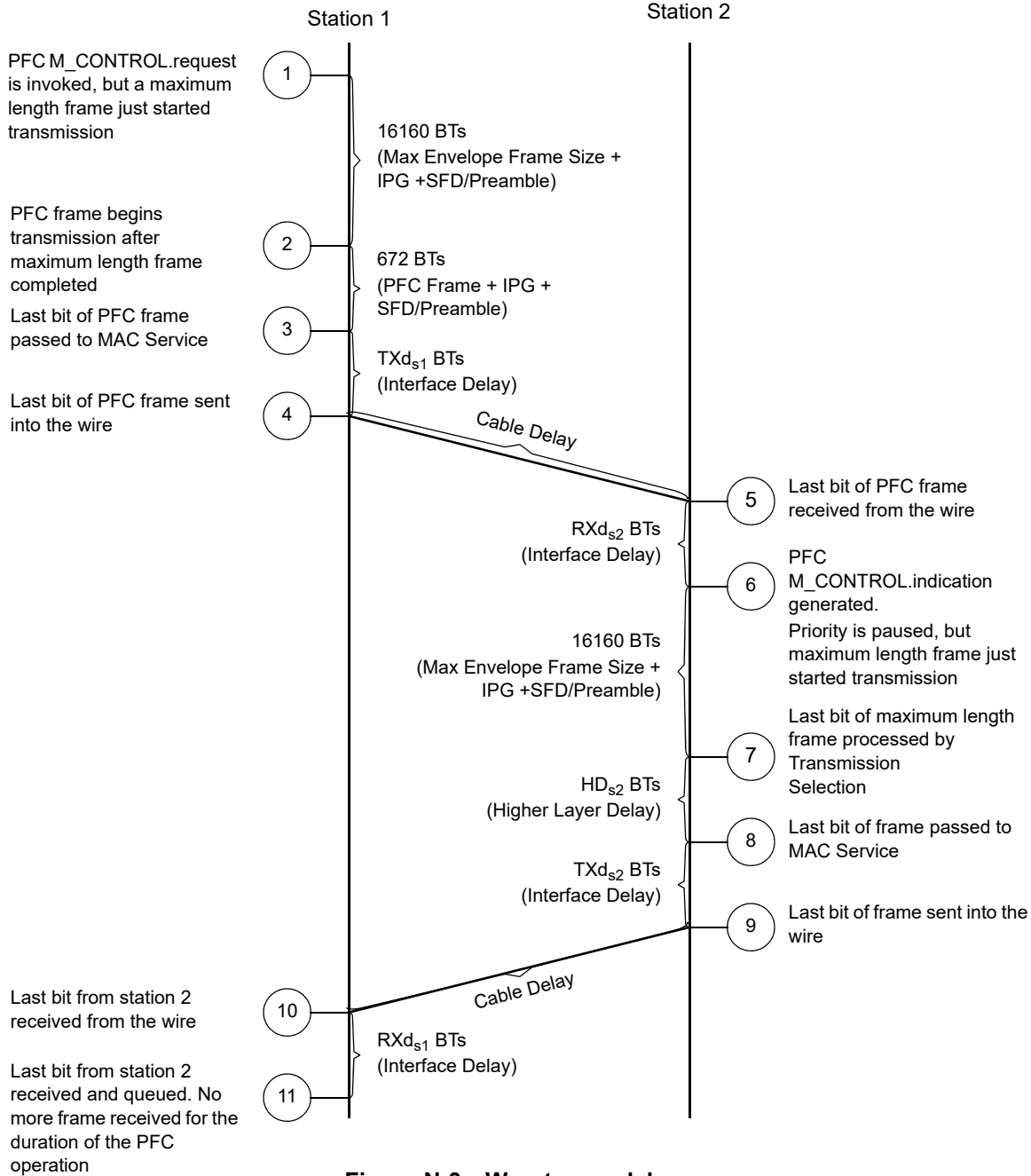


Figure N-3—Worst-case delay

1 The total Delay Value (DV) is the sum of all delays shown in Figure N-3:

$$2 \quad DV = 2 \times (\text{Max Frame}) + (\text{PFC Frame}) + 2 \times (\text{Cable Delay}) + \text{TXd}_{s1} + \text{RXd}_{s2} + \text{HD}_{s2} + \text{TXd}_{s2} + \text{RXd}_{s1}$$

4 For any given station the Interface Delay includes both transmit and receive paths (i.e., ID = TXd + RXd).
5 Therefore:

$$6 \quad DV = 2 \times (\text{Max Frame}) + (\text{PFC Frame}) + 2 \times (\text{Cable Delay}) + \text{ID}_{s1} + \text{ID}_{s2} + \text{HD}_{s2}$$

7 Usually the peer stations connected by a point-to-point link use the same technology, therefore $\text{ID}_{s1} = \text{ID}_{s2}$:

$$8 \quad DV = 2 \times (\text{Max Frame}) + (\text{PFC Frame}) + 2 \times (\text{Cable Delay}) + 2 \times \text{ID} + \text{HD}_{s2}$$

1 N.3 Interface Delay

2 The Interface Delay comprises all delay components below the MAC Control Client, excluding the cable
3 delay. Table N-1 shows the Interface Delay constraints for some IEEE 802.3 interfaces.

Table N-1—IEEE 802.3 Interface Delays

Sublayer	Maximum RTT (bit times)	Maximum RTT (pause quanta)	Reference (subclause of IEEE Std 802.3-2018 [B15])
10G MAC Control, MAC, and RS	8192	16	46.1.4
XGXS and XAUI	2048	4	48.5
10GBASE-X PCS	2048	4	49.2.15
10GBASE-R PCS	3584	7	50.3.7
LX4 PMD	512	1	53.2
CX4 PMD	512	1	54.3
Serial PMA and PMD	512	1	52.2
10GBASE-T	25 600	50	55.11

4 N.4 Cable Delay

5 The Cable Delay is the propagation delay over the transmission medium and can be approximated by the
6 following equation:

$$7 \quad \text{Cable Delay} = \text{Medium Length} \times \frac{1}{BT \times v}$$

8 where v is the signal propagation speed in the medium and BT is the bit time of the medium.

9 N.5 Higher Layer Delay

10 The Higher Layer Delay comprises the delay components between the MAC Control Client and the port
11 Transmission Selection. Example of these delays are MACsec and implementation specific delays.

12 For link speeds of up to 10Gb/s, MACsec constrains each of the transmit delay and the receive delay to a
13 maximum of 19 360 bit times (see 36.1.3.3).

14 This standard constrains the implementation specific delays to be less than 614.4 ns (see 36.1.3.3). This
15 delay is equivalent to 6144 bit times at the speed of 10Gb/s.

1 N.6 Computation example

2 A station needs to be capable of buffering DV bit times of data to ensure no frame loss due to congestion.
3 The worst case is with a 10GBASE-T PHY. Assuming MACsec is not supported, this results in the
4 following:

- 5 — Maximum envelope frame size: 2000 octets, 16 160 bit times;
- 6 — PFC frame size: 64 octets, 672 bit times;
- 7 — XGMII MAC/RS and XAUI interface: $8192 + 2 \times 2048 = 12\,288$ bit times;
- 8 — 10GBASE-T Delay: 25 600 bit times;
- 9 — 100 meters Cat6 cable: 5556 bit times (computed assuming $v = 0.6 \times c$, where c is the speed of the
10 light in meters per second);
- 11 — HD = 6144.

12 The total Delay Value in this scenario results as follows:

$$\begin{aligned} 13 \quad DV &= 2 \times (\text{Max Frame}) + (\text{PFC Frame}) + 2 \times (\text{Cable Delay}) + 2 \times \text{ID} + \text{HD}_{s2} \\ 14 \quad DV &= 2 \times (16\,160) + (672) + 2 \times (5556) + 2 \times (25\,600) + 2 \times (12\,288) + 6144 = 126\,024 \text{ bit times} \end{aligned}$$

15 For this case, the amount of buffering needed to ensure no frame loss due to congestion results to be
16 126 024 bit times, roughly equivalent to 15.5 kB.

17 If MACsec is used, the High Layer Delay is incremented by 19 360 bit times; therefore, the total Delay
18 Value results as follows:

$$19 \quad DV = 2 \times (16\,160) + (672) + 2 \times (5556) + 2 \times (25\,600) + 2 \times (12\,288) + 25\,504 = 145\,384 \text{ bit times}$$

20 For this case, the amount of buffering needed to ensure no frame loss due to congestion results to be
21 145 384 bit times, roughly equivalent to 18 kB.

1 **Annex O**

2 (informative)

3 **Preserving the integrity of FCS fields in MAC Bridges**

4 **O.1 Background**

5 When relaying frames between Ports of a MAC Bridge, one of the functions of the Bridge is to regenerate
6 the FCS field in accordance with the MAC procedures that apply to the medium access method through
7 which the frame will be relayed. One of the requirements of the MAC Bridge standard is that any such
8 regeneration of the FCS shall not increase the level of undetected FCS errors that are experienced on the
9 transmitting MAC over that which would be experienced by preserving the FCS. The point at issue here is
10 only the *undetected* error rate; a conformant Bridge will discard frames that are received with an FCS error;
11 however, there is a finite possibility of undetectable corruption taking place, i.e., the frame is corrupted, but
12 the FCS algorithm does not reveal the error.

13 This annex looks at the various cases that apply to the operation of the MAC Bridge with respect to FCS
14 regeneration and the alternative approaches that can be taken in order to address the cases. The following
15 cases need to be considered:

16 a) The source and destination media access methods are identical from the point of view of the
17 operation of the FCS algorithm.

18 NOTE—This is the case, for example, where the source and destination media access methods are identical. It could also
19 happen with dissimilar media access methods if the FCS coverage, the FCS algorithm, the definitions of the fields
20 covered by the FCS, and the bit/octet ordering are the same in both media access methods.

21 b) The source and destination media access methods differ from the point of view of the operation of
22 the FCS algorithm.

23 c) The data covered by the FCS is not modified by the operation of the relay function of the Bridge.

24 d) The data covered by the FCS is modified by the operation of the relay function of the Bridge.

25 In each case, it is important to ensure that there is a low probability of additional undetected errors being
26 generated by corruption of the data portion of the frame between removal of the old FCS and computation of
27 the new one, which would result in the transmitted frame carrying invalid data and a valid FCS. Such
28 corruption might occur, for example, as a result of the effect of environmental noise on the operation of the
29 Bridge hardware.

30 If both item a) and item c) are true, then the FCS carried in the received frame is still valid, and the ideal
31 approach would be to reuse this value as the FCS for the transmitted frame.

32 If either item b) or item d) are true, then it will be necessary for the Bridge to recalculate the FCS on
33 transmission, and may therefore need to take additional precautions against in-memory corruption causing
34 increased undetected FCS error levels.

1 O.2 Basic mathematical ideas behind CRC and FCS

2 The standard CRC algorithm is based on the following ideas:

- 3 a) An n -bit message is regarded as a polynomial, $M(x)$, of degree $n - 1$.
- 4 b) In order to generate a CRC value of length r bits, a generator polynomial, $G(x)$, is used, of degree r
- 5 c) The value of the last r bits of $M(x)$ are chosen such that $M(x) \div G(x)$ has a remainder of 0 [i.e., $M(x) = 0 \bmod G(x)$].

7 Messages can be added, in which each coefficient is 0 or 1:

- 8 d) $M3(x) = M1(x) + M2(x)$. Messages are added together by bit-wise addition (XOR) of coefficients
- 9 with the same bit position (i.e., coefficients with the same exponent).

10 Subtraction of messages:

- 11 e) Addition and subtraction are equivalent operations (as $A = A \text{ XOR } B \text{ XOR } B$); hence, using the
- 12 messages in item d) above, $M1(x)$ can be regenerated from $M3(x)$ by adding $M2(x)$.

13 Linearity:

- 14 f) If $M(x) = 0 \bmod G(x)$, then $(M(x) + E(x)) = 0 \bmod G(x)$ IFF $E(x) = 0 \bmod G(x)$. In other words, if
- 15 an error pattern, $E(x)$, is added to a message, $M(x)$, then the resultant message will give no
- 16 remainder when divided by the generator polynomial if both $M(x)$ and $E(x)$ gave no remainder when
- 17 divided by the generator polynomial.
- 18 g) $x^m M(x) = 0 \bmod G(x)$ IFF $M(x) = 0 \bmod G(x)$. A message can be shifted by adding zero padding
- 19 without affecting the integrity of the CRC.
- 20 h) Thus, the CRC algorithm has the property that it will detect any burst error, of length r bits or less,
- 21 applied to the message, as such an error must result in a nonzero remainder when the message is
- 22 divided by the generator polynomial. Alternatively, adding two messages together, both of which
- 23 have valid CRCs, results in a message with a valid CRC.

24 In Ethernet, the algorithm used differs from the standard CRC, and is known as an FCS. The FCS is based

25 on the CRC but with the following differences:

- 26 — The first 32 bits of $M(x)$ are complemented before the FCS value is calculated.
- 27 — A CRC value is calculated, as described above, by dividing the first $n - 32$ bits of $M(x)$ by $G(x)$ and
- 28 taking the remainder as the CRC value.
- 29 — The CRC value is complemented and inserted as the last 32 bits of $M(x)$.

30 By application of the addition and linearity rules, item d) and item f) above, the Ethernet frame with its FCS

31 can be viewed as consisting of the following component messages, added together:

- 32 — An n -bit message, $M(x)$, carrying a standard CRC as the last 32 bits.
- 33 — An n -bit “FCS adjustment factor,” $A_n(x)$, which adjusts the CRC to form an FCS.

34 Figure O-1 illustrates the conversion of a message with a CRC to the equivalent message with an FCS, by

35 the addition of the adjustment factor. The adjustment factor consists of the following two components:

- 36 — The first component, when added to $M(x)$, contributes an adjustment to the CRC, which is
- 37 equivalent to the effect of complementing the first 32 bits of $M(x)$ before calculating the CRC bits,
- 38 and complements the first 32 bits of $M(x)$.
- 39 — The second component, when added to the partially adjusted $M(x)$, restores the first 32 bits of $M(x)$
- 40 to their original value, and complements the (adjusted) CRC to form the final FCS.

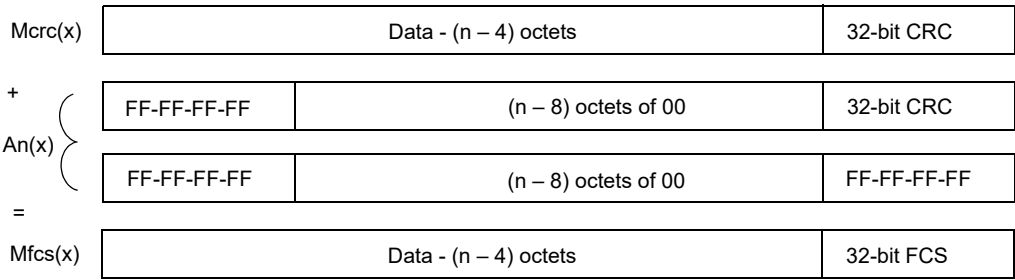


Figure O-1—Converting a CRC to an FCS

1 O.3 Detection Lossless Circuit approach

2 This approach is illustrated in Figure O-2. The basis of this approach is that the FCS used in the modified
3 message is always recomputed from scratch using the normal FCS generation algorithm (Check Generator
4 B); however, the original message data and FCS are also used as an input to an FCS checker (Checker A) in
5 order to check that the inputs to the FCS generator were correct. Hence, if errors are introduced into the
6 message while it is in memory in the Bridge in the time period since the message was received and FCS-
7 checked, then Checker A will detect the error at the same time as the new FCS for the modified message is
8 being generated.

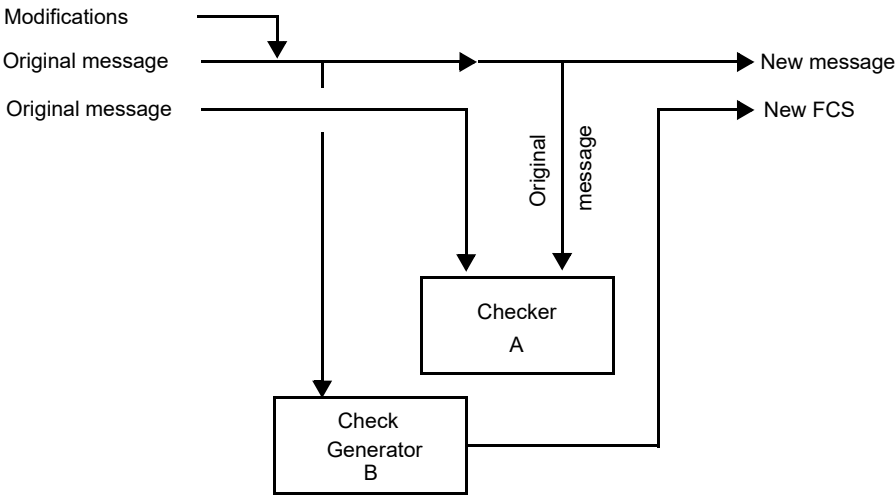


Figure O-2—Detection Lossless Circuit

9 Checker A is fed both with the original message in its unmodified form, and with the original message as
10 used in the construction of the new message; this occurs after (or simultaneously with) the generation of the
11 new FCS by Check Generator B. Any discrepancies detected by Checker A indicate that the information
12 used to generate the new message and its FCS are suspect, and that the message should therefore be
13 discarded.

1 O.4 Algorithmic modification of an FCS

2 In cases where the need to recalculate the FCS comes about as a result of changes to the data rather than by
3 changes in the operation of the FCS algorithm, one option, rather than recalculating an FCS from scratch for
4 a given message, is to examine the changes that have been made to the message and to calculate an FCS
5 adjustment factor that reflects those changes. There are two cases, as follows:

- 6 a) The overall length of the message is unchanged, but the contents of one or more octets of data
7 covered by the FCS have changed.
- 8 b) The overall length of the message has increased or decreased, as a result of the addition or removal
9 of octets covered by the FCS, but the contents of the original octets is unchanged (although some
10 may have been shifted in position as a result of the insertion/deletion).

11 O.4.1 Data changed, length unchanged

12 Adjustment of a message, $MF_1(x)$, in order to change the value of a field F from current value F_1 to new
13 value F_2 consists of the following steps:

- 14 — Remove the FCS adjustment factor for a message of length n , $An(x)$.
- 15 — Remove the contribution to the message and its FCS provided by the current value of the field,
16 $F_1(x)$.
- 17 — Add the contribution to the message and its FCS provided by the new value of the field, $F_2(x)$.
- 18 — Add the FCS adjustment factor for a message of length n , $An(x)$.

19 In other words,

$$20 \quad MF_2(x) = MF_1(x) - An(x) - F_1(x) + F_2(x) + An(x)$$

21 The addition and subtraction of $An(x)$ cancel each other out, so,

$$22 \quad MF_2(x) = MF_1(x) - F_1(x) + F_2(x)$$

23 Also, the correction factors $F_1(x)$ and $F_2(x)$ can be replaced by a single factor, $F_3(x)$, which is the
24 contribution to $M(x)$ and its FCS that would be provided by the value of Field F that would be produced by
25 F_1 XOR F_2 . So,

$$26 \quad MF_2(x) = MF_1(x) + F_3(x)$$

27 This is shown in Figure O-3.

28 Alternatively,

$$29 \quad \text{FCS of } MF_2(x) = (\text{FCS of } MF_1(x)) + \text{adjustment}$$

30 where

$$31 \quad \text{Adjustment} = (\text{FCS of } F_1(x)) + (\text{FCS of } F_2(x))$$

32 or equivalently,

$$33 \quad \text{FCS of } MF_2(x) = (\text{FCS of } MF_1(x)) + (\text{FCS of } F_3(x))$$

34 The adjustment factor (FCS of $F_3(x)$) is incidentally the same factor that would be used if the value of F
35 were to be changed back from F_2 to F_1 .

$MF_1(x)$	x data octets	F_1	y data octets	32-bit FCS
$- F_1(x)$	x octets of 00	F_1	y octets of 00	32-bit CRC
$+ F_2(x)$	x octets of 00	F_2	y octets of 00	32-bit CRC
$= MF_2(x)$	x data octets	F_2	y data octets	32-bit FCS

Note also that

$F_1(x)$	x octets of 00	F_1	y octets of 00	32-bit CRC
$+ F_2(x)$	x octets of 00	F_2	y octets of 00	32-bit CRC
$= F_3(x)$	x octets of 00	$F_1 \text{ XOR } F_2$	y octets of 00	32-bit FCS

Figure O-3—Field change adjustment

1 The relative simplicity of this case (relative to the case where the length changes) is a consequence of
2 linearity and the fact that the FCS adjustment factor, $An(x)$, is constant for any given message length n , not a
3 function of the data carried in the message.

4 **O.4.2 Length changed, original data unchanged**

5 Adjustment of a message $M(x)$ of length n to cater for an inserted field, I , of length i involves the following
6 steps:

- 7 — Remove the FCS adjustment factor for a message of length n , $An(x)$;
- 8 — Remove the contribution to the message and FCS that is provided by the header portion of the
9 message (the portion before the inserted field), $H(x)$;
- 10 — Add the contribution to the message and FCS that is provided by the header plus the inserted field I ,
11 $HI(x)$;
- 12 — Add the FCS adjustment factor for a message of length $n+i$, $An+i(x)$.

13 So,

14
$$MI(x) = M(x) - An(x) - H(x) + HI(x) + An+i(x)$$

15 This is shown in Figure O-4.

16 Alternatively,

17
$$\text{FCS of } MI(x) = (\text{FCS of } M(x)) + \text{adjustment}$$

18 where

19
$$\text{adjustment} = (\text{CRC of } An(x)) + (\text{CRC of } H(x)) + (\text{CRC of } HI(x)) + (\text{CRC of } (An+i(x)))$$

20 As with the field change example, the value of *adjustment* is the same adjustment factor that would be used
21 to adjust the FCS of $MI(x)$ if field I were to be removed from the message.

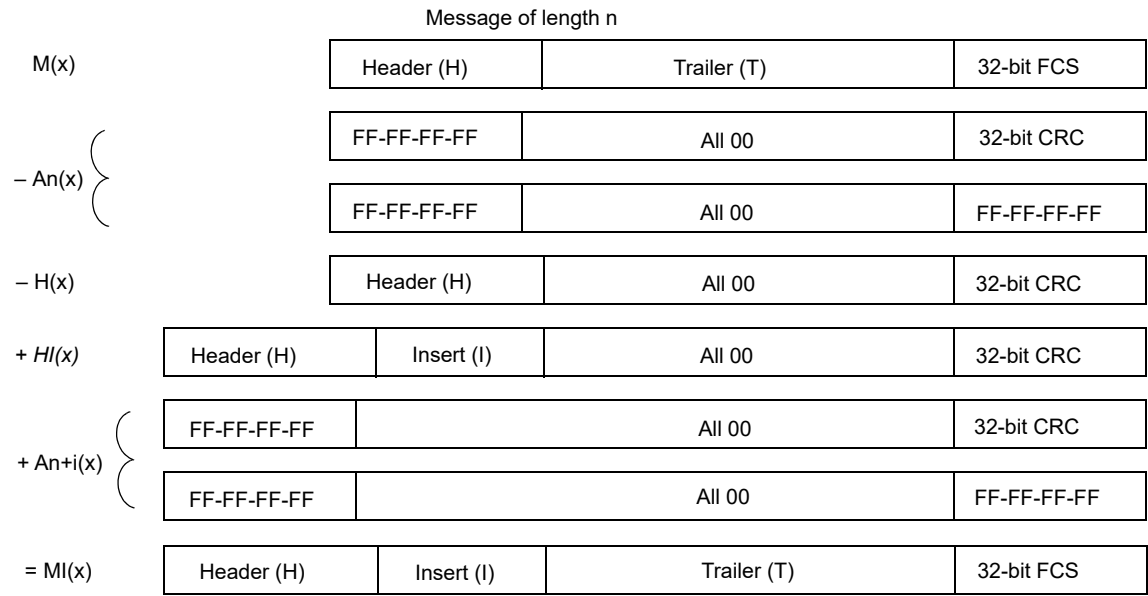


Figure O-4—Field insertion adjustment

1 O.4.3 Preservation of detectability

2 With either of the algorithmic adjustment methods described here, the important question is whether the
3 ability of the FCS to detect errors in the message is preserved across these transformations; in other words, if
4 a detectable error pattern, E(x), is added to the message M(x) before the modification, addition, or removal
5 of a field, is that error still detectable after the message has been adjusted.

6 NOTE—The following analysis assumes that E(x) is a detectable error pattern, of length 32 bits or less. Analysis of the
7 performance of this method with longer error patterns has not been attempted here.

8 In the cases where the error component is in those parts of the message that are not affected (changed or
9 shifted) by the transformation, it is reasonably apparent that, as the transformation takes no account of the
10 portion of the message that is in error, the detectability of that error is unchanged.

11 In the case of a field change, where field F₁ is to be replaced with F₂, and E(x) had been applied in some part
12 of the message before changing the field, the new FCS value would be calculated as follows:

13
$$\text{FCS of } MF_2(x) = (\text{FCS of } MF_1(x)) + (\text{FCS of } F_1(x)) + (\text{FCS of } F_2(x))$$

14 However, the correction factor that is applied to the FCS contains components based only on F₁ and F₂, and
15 no contribution related to the error pattern E(x). Therefore, after the value of the field has been changed to
16 F₂, the new FCS value as calculated above will still be capable of detecting error pattern E(x), regardless of
17 where the error pattern appears in the message.

18 In the case of a field insertion (or removal, as insertion and removal have already been shown to be
19 equivalent), where E(x) has been applied to H(x), the correction factor that is applied to the FCS will contain
20 two components related to E(x); the first based on E(x) in its original position, and the second based on E(x)
21 shifted by i, the length of the inserted field, as follows:

22
$$\text{FCS of } MI(x) = (\text{FCS of } M(x)) + \text{adjustment}$$

1 where

$$\begin{aligned} \text{adjustment} = & (\text{CRC of } A_n(x)) + (\text{CRC of } H(x)) + (\text{CRC of } H_I(x)) + (\text{CRC of } (A_{n+i}(x)) \\ & + (\text{CRC of } E(x)) + (\text{CRC of } (E(x) \text{ shifted by } i)) \end{aligned}$$

4 The presence of the (CRC of $E(x)$) component ensures that the final message will still indicate an FCS error.
5 (Note that, even if (CRC of $E(x)$) and (CRC of $(E(x)$ shifted by i)) turn out to be the same value, the final
6 message will still indicate an FCS error, as its FCS is now correct for the value of $M_I(x)$ without the error
7 component.)

8 The above does not exhaustively analyze all the cases of field insertion and removal, and in particular, does
9 not analyze the cases where the error pattern crosses the insertion/removal boundaries; however, it can be
10 shown that the error detection characteristics of the FCS are preserved in these cases as well.

11 The conclusion from the above examples is that, as long as the field values used in the calculation of these
12 FCS correction factors are the same as the field values present in the original and final messages (e.g., that
13 the same value of $H+E$ is used both in its original and shifted forms), then the properties of the FCS are
14 preserved. However, if different values are used for calculation of the correction factor from those present in
15 the messages (as could happen, for example, as a result of in-memory corruption), then the properties of the
16 FCS are no longer preserved. Hence, if these algorithmic approaches are used, it is necessary to design the
17 implementation in a manner that ensures that these conditions are met.

18 **O.5 Conclusions**

19 Where it is necessary to recalculate the FCS before transmission, then either the algorithmic FCS
20 modification approaches or the Detection Lossless Circuit approach offer a basis for this to be achieved
21 while still preserving the error detection coverage that was provided by the original FCS.

22 The Detection Lossless Circuit approach requires access to the entire contents of the original message, to
23 compute the necessary FCS adjustments or to check the integrity of the data that is being used to create the
24 new message and its FCS. In contrast, the algorithmic approach requires access only to the header, insert,
25 and FCS of the original message.

26 With the algorithmic approach, care needs to be taken in the implementation in order to ensure that the
27 inputs to the correction algorithms are consistent with the contents of the original and modified messages,
28 whereas in the Detection Lossless Circuit, such consistency checking is inherent in the way that the original
29 FCS is used to check the generation of the new FCS.

Annex P

(informative)

Frame duplication and misordering

Although RSTP provides a single loop-free path between communicating stations at any instant, RSTP can reconfigure the active topology of the Bridged Network in less time than it can take a frame to transit from its source to its destination(s). This annex explains how this can result in user data frame duplication and misordering and examines the possible QoS impact.

NOTE—The ForceVersion parameter provided by RSTP allows network administrators to choose slower reconfiguration, thereby trading off service availability against the risk of duplication and misordering.

P.1 Background

In a network of Bridges implementing STP, specified in the 1998 and earlier editions of IEEE Std 802.1D [B13], delays in transitioning Ports to Forwarding ensure that frames in transit prior to a network reconfiguration are delivered or discarded before being forwarded on the new active topology. In such a network, the only source of misordered or duplicated frames is a “magically healed” LAN between two Bridges, for example, as a result of accidental interconnections between shared media hubs.

RSTP, by design, can transition a Port to Forwarding very rapidly. A Root Port transition can occur as rapidly as the hardware can manage it, and a Designated Port transition in the time it takes to transmit two frames from separate stations on a single LAN. It is conceivable that a reconfiguration can take place, and a Port be made Forwarding as a result, while frames forwarded on the prior active topology are still in transit.

P.2 Frame duplication

A unicast frame whose destination address has been learned by the Bridges that can forward it, can only be buffered for transmission on one Port of one Bridge at any one time, and cannot be duplicated.

A unicast frame whose destination address has not been learned, can be flooded by some Bridges, and therefore buffered on multiple outbound Ports at the same time, and can be duplicated during network reconfiguration. Figure P-1 provides an example. In this network fragment, the Root Bridge is assumed to be somewhere to the left of Bridge A’s Port A1, and the Port Path Costs of the three Bridges result in the active topology shown, with Port B3 Discarding and the remaining Ports all Forwarding.

If the configuration is stable, a unicast frame arriving at A1 and destined for an unlearned unicast destination reachable through B2 is flooded by Bridge A to its Ports A2 and A3, and by Bridge C to its Ports C2 and C3. As B3 is Discarding, only the frame reaching B1 is transmitted through B2, and the destination receives a single copy of the frame.

During reconfiguration the following sequence of events can occur:

- a) A receives the frame through A1 and transmits copies of it through A2 and A3.
- b) B and C each receive a copy and queue them for transmission through B2, C2, and C3.
- c) B2 transmits its copy of the frame to its destination.
- d) B detects that LAN A2-B1 has failed, and immediately makes B3 Forwarding as its new Root Port.
- e) C2 transmits its copy of the frame, B receives and transmits this second copy through B2.

1 The destination station, reachable through B2, receives the same frame twice. However, any subsequent
2 response from the destination station would cause the address to be learned. While this reconfiguration will
3 itself cause addresses to be flushed from the Bridge's FDBs of Bridges, the flushing commences after the
4 Port becomes Forwarding, and will not increase frame duplication unless there are further changes in the
5 active topology.

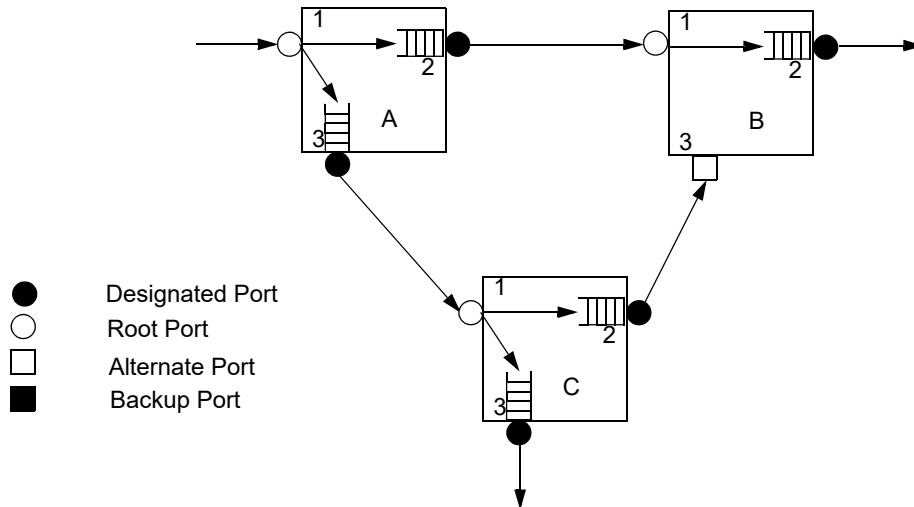


Figure P-1—Frame duplication scenario

6 Multicast frames are all potentially subject to duplication on reconfiguration events, as they can be buffered
7 at multiple outbound Ports. Frame duplication does not cause problems in known multicast LAN protocols.

8 The risk of frame duplication depends upon network configuration, frequency of reconfiguration, equipment
9 implementation details, and the characteristics of the network traffic that determine how long each frame is
10 likely to be buffered in each Bridge, and therefore cannot be quantified without reference to the a particular
11 network.

12 P.3 Frame misordering

13 A change in the active topology between two communicating end stations can result in frame misordering,
14 as a frame sent after reconfiguration can experience a lower transit delay, being queued at fewer Bridge Ports
15 or for less time. Figure P-2 provides an example. In this network fragment a station connected to A1 is
16 communicating with a station connected to B2. Prior to reconfiguration, A2 is an Alternate Port and is
17 Discarding; all traffic between the two stations is relayed by Bridge C. During reconfiguration the following
18 sequence of events can occur:

- 19 a) Frame F1 is received through A1, buffered and transmitted through A3, received through C1 and
20 buffered awaiting transmission through C2.
- 21 b) A detects that LAN A3-C1 has failed, and immediately makes A2 Forwarding as its new Root Port.
- 22 c) Frame F2 is received through A1, transmitted through A2 to B1, and then transmitted through B2.
- 23 d) F1 is transmitted through C2 to B3, and then transmitted through B2.

24 Clearly, the receiving station connected to B2 sees the frames in the reverse of the order in which the
25 originating station transmitted them.

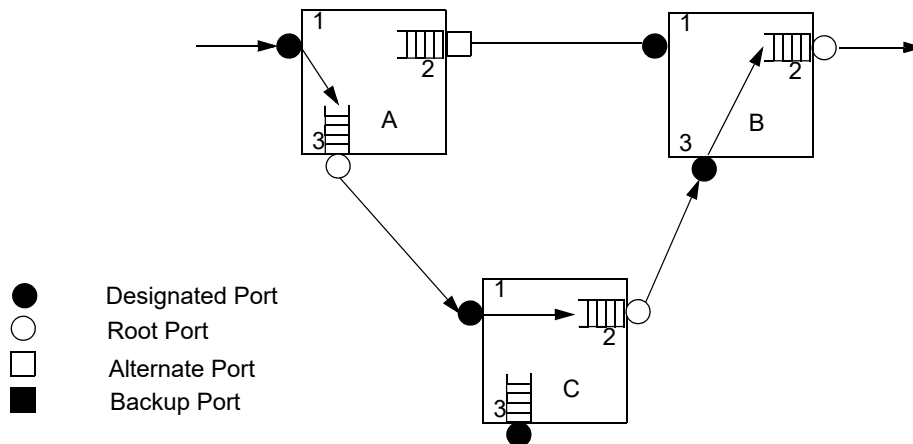


Figure P-2—Frame misordering scenario

1 As with frame duplication, the overall risk of frame misordering depends upon network configuration,
2 frequency of reconfiguration, equipment implementation details, and the characteristics of the network
3 traffic that determine how long each frame is likely to be buffered in each Bridge, and therefore cannot be
4 quantified without reference to the a particular network.

5 Some LAN protocols, for example LAT, LLC2, and NETBEUI, are sensitive to misordering, so even a low
6 incidence of misordering could result in perceived problems in networks that support these protocols.

7 **P.4 Other considerations**

8 The possibility that frames may be stored “in transit” on old routes after a reconfiguration has completed
9 means that there is also a possibility that addresses will be mis-learned, leading to a risk of denial of service
10 for the addresses concerned. Hence, it is necessary to run short aging timers for a period after a
11 reconfiguration to allow such frames to be delivered or discarded.

12 The Bridge model described in this standard does not include the concept of queuing on input to a Port;
13 however, practical Bridge designs generally include some input queuing. While this is not a solution to the
14 effects described above, a Bridge should flush any input queue associated with a Port that becomes
15 Disabled. This behavior is consistent with the Bridge model; if a frame is in an input queue, it has not been
16 received as far as the Bridge Port is concerned, and therefore should not be received after the Port becomes
17 Disabled.

1 **Annex Q**

2 (informative)

3 **Traffic scheduling**

4 This annex provides some background to the mechanisms provided in this standard for traffic scheduling,
5 with the intent of providing some insight into the motivation for the provision of mechanisms for traffic
6 scheduling and some of the ways that the mechanisms might be used.

7 **Q.1 Motivation**

8 Some applications have a need for frame delivery that is highly predictable in terms of the time at which
9 frame transmission will occur, and the overall latency and jitter that will be experienced as the frame is
10 propagated to its destination. Examples include industrial and automotive control applications, where data
11 transmitted over the network is used to feed the parameters of control loops that are critical to the operation
12 of the plant or machinery involved, and where frames carrying control data are transmitted on a repeating
13 time schedule; late delivery of such frames can result in instability, inaccuracy, or failure of the operation of
14 the control loops concerned. In some implementations, this need has been met by the provision of dedicated,
15 highly engineered networks that are used solely for the transmission of time-critical control traffic; however,
16 as the bandwidth occupied by such traffic is often low, and the cost of providing a dedicated control network
17 can be high, it can be desirable to mix time-critical traffic with other classes of traffic in the same network,
18 as long as such mixing can be achieved while still meeting the timing requirements of the time-critical
19 traffic.

20 Prioritization alone is insufficient to address the needs of this kind of traffic; if a low priority frame is
21 already being transmitted, then that transmission will complete before a higher priority frame can access the
22 transmission medium, so there could be a delay of up to a maximum-sized frame before a high priority
23 transmission can start. If such delays occur at every hop, then the accumulated latency could be
24 unacceptably large.

25 Interfering traffic can itself be time-critical; for example, there could be multiple time-critical traffic streams
26 related to different control systems. The ability to coordinate different time-critical traffic streams in such a
27 way that they do not interfere with each other is also important.

28 One approach to meeting these objectives is to ensure that, at specific times, only one traffic class (or set of
29 traffic classes) has access to the network; in effect to create a protected “channel” that is used by that traffic
30 class alone. However, in order to ensure that the remaining unprotected traffic cannot affect the transmission
31 of protected traffic, it is necessary to stop the transmission of unprotected traffic sufficiently far in advance
32 of the protected time slot to be certain that the last unprotected transmission has completed before protected
33 transmission starts. In the worst case, this would mean that the last unprotected transmission would start a
34 maximum-sized frame transmission time before the start of the protected traffic “window.” In effect, a guard
35 band is created before the time that the protected traffic transmission is due to start; transmission of
36 unprotected traffic is not permitted between the start of the guard band and the start of the protected traffic
37 window. The simplest approach is for the guard band to be as long as a maximum-sized frame transmission;
38 however, the start of the guard band need not be fixed if the implementation is able to determine, from the
39 size of the frames that it has queued, that there is sufficient time for a frame to be transmitted in its entirety
40 before the start of the protected traffic window. This approach is illustrated in Figure Q-1. If the simple
41 approach were taken, then the time difference between T0 and T1 would be equal to the transmission time of
42 a maximum-sized frame on that port (Figure Q-1, part A); if the implementation is able to determine frame
43 transmission times on the fly, then T0 could vary between a maximum frame transmission time before T1
44 and a minimum frame transmission time before T1, depending upon what unprotected frames were available
45 for transmission on the Port (Figure Q-1, part B).



Figure Q-1—Establishing a guard band

1 Q.2 Using gate operations to create protected windows

2 The enhancements for scheduled traffic described in 8.6.8.4 allow transmission to be switched on and off on
3 a timed basis for each traffic class that is implemented on a port. This switching is achieved by means of
4 individual on/off transmission gates associated with each traffic class and a list of gate operations that
5 control the gates; an individual SetGateStates operation has a time delay parameter that indicates the delay
6 after the gate operation is executed until the next operation is to occur, and a GateState parameter that
7 defines a vector of up to eight state values (open or closed) that is to be applied to each gate when the
8 operation is executed. The gate operations allow any combination of open/closed states to be defined, and
9 the mechanism makes no assumptions about which traffic classes are being “protected” and which are
10 “unprotected”; any such assumptions are left to the designer of the sequence of gate operations. The
11 sequence of gate operations terminates at the end of the list, and restarts when OperCycleTime (8.6.9.4.19)
12 nanoseconds have elapsed since the start of the sequence, thus providing a repeating cycle of gate state
13 changes. If OperCycleTime is shorter than the total execution time of the sequence of operations, the
14 sequence is truncated and restarted at OperCycleTime.

15 There is no necessity to explicitly define the start of a guard band, as the specification of how the gates
16 operate states that a frame for a given traffic class can only be transmitted if the gate associated with that
17 traffic class is open and there is sufficient time for the whole of the frame to be transmitted before the gate is
18 due to close. Figure Q-2 illustrates how gate operations could be used to create a protected window for
19 traffic class 3 in order to achieve the result illustrated in Figure Q-1. A SetGateStates operation at time T1
20 sets the state of the gates for traffic classes 7, 6, 5, 4, 2, 1, and 0 to closed (C) and sets the state of the gate for
21 traffic class 3 to open (o), so only traffic class 3 can transmit after T1. At time T2, the SetGateStates
22 operation reverses the state for all eight traffic classes, so traffic class 3 can no longer transmit after T1, but
23 all other traffic classes can.

24 There is no need to explicitly define the position of T0, because the “guard band” behavior is inherent in the
25 defined behavior of the gates, as previously stated.

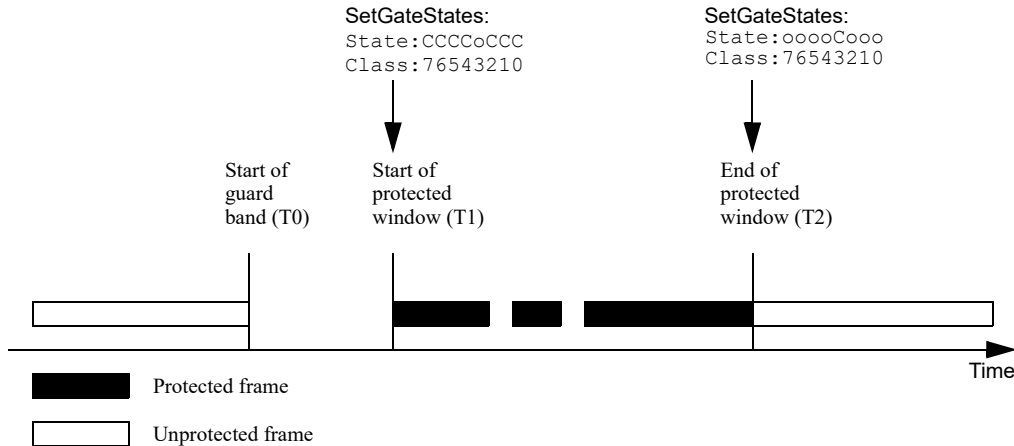


Figure Q-2—Using gate operations

1 Q.3 Availability of PTP

2 The timing of gate operations as specified in 8.6.8.4 assumes that PTP is operating and therefore PTP time is
3 available to the Bridge or end station. However, that is clearly not the case on initial startup or when PTP
4 communication is interrupted for some reason; particular implementations could also choose to use
5 alternative (non-PTP) mechanisms for ensuring that a common view of time is available across a number of
6 systems. The approach adopted in these circumstances will be heavily dependent on the application, and so
7 is an implementation choice; the mechanisms described in 8.6.8.4 are not themselves affected by that choice.

8 Q.4 Scheduled traffic and end stations

9 It is possible for Bridges to support the enhancements for scheduled traffic while end stations connected to
10 those Bridges do not. In general, the scheduling mechanism is not needed in end stations that are only
11 recipients of scheduled traffic.

12 Q.5 CycleTimeExtension variables

13 The AdminCycleTimeExtension (8.6.9.4.4) and OperCycleTimeExtension (8.6.9.4.20) state machine
14 variables provide a means for an operator to control how a new gating cycle is configured into a system that
15 is already running an existing gating cycle.

16 If the choice of cycle time for the new gating cycle (AdminCycleTime) is unchanged from the cycle time for
17 the old gating cycle (OperCycleTime), and if the base time chosen for the new gating cycle
18 (AdminBaseTime) is an integer multiple of the OperCycleTime different to OperBaseTime, then the new
19 gating cycle will exactly “line up” with the old gating cycle, i.e., the cycle start times for the new gating
20 cycle will be the same as they would have been for the old configuration. This could be considered to be the
21 ideal case and allows the new gating cycle to be installed and executed with no timing issues.

22 However, if AdminCycleTime differs from OperCycleTime, and/or AdminBaseTime is in the future and is
23 not an integer multiple of OperCycleTime different to OperBaseTime, then the old and new cycles do not
24 line up, and when AdminBaseTime is reached (i.e., when the new configuration is installed and starts to
25 execute), the last old cycle would normally be truncated in order to start the first new cycle. This could be
26 undesirable if it results in a very short last old cycle; arguably it would be better to simply extend the
27 penultimate old cycle by that small amount, rather than starting a very short cycle. The CycleTimeExtension
28 variables allow this extension of the last old cycle to be done in a defined way; if the last complete old cycle
29 would normally end less than OperCycleTimeExtension nanoseconds before the new base time, then the last
30 complete cycle before AdminBaseTime is reached is extended so that it ends at AdminBaseTime.

1 **Annex R**

2 (informative)

3 **Preemption and IEEE 802.1AE MAC Security**

4 This annex answers questions about the use of MAC Security (MACsec) that arose in the development of
5 the IEEE Std 802.1Qbu (Frame Preemption) amendment to this standard. It provides background for readers
6 not familiar with MACsec. For the normative specification of MACsec and its supporting authentication and
7 key management framework refer to IEEE Std 802.1AE and IEEE Std 802.1X.

8 NOTE—While these issues arose during development of preemption, they can also apply to other situations, such as a
9 customer link running MACsec over a provider bridged network.

10 Preemption allows one or more higher priority (express) frames to interrupt the transmission of a lower
11 priority (preemptible) frame, the preemptible frame transmission being resumed and completed once the
12 express frame(s) have been transmitted. A preemptible frame can thus be broken into two or more
13 fragments. A MACsec Cipher Suite implementation can protect each frame in its entirety before requesting
14 its transmission, or can transmit the octets of the protected frame as soon as they are made available by the
15 operation of the block by block symmetric cryptography. The latter is often preferred as it reduces the delay
16 due to protection, and some implementations impose a constant delay in support of time-sensitive
17 networking objectives. Similarly validation of a received frame can be delayed until its reception is
18 complete, or can proceed as each block of octets becomes available. These implementation details lie
19 outside the normative scope of IEEE Std 802.1AE, but naturally give rise to questions as to how frames
20 should be protected when they are fragmented, particularly when the symmetric keys used to protect frames
21 are being changed.

22 Each time a Cipher Suite's symmetric cryptography is used to protect a frame with a given key, a unique
23 value, known as a 'nonce' or IV (initial value), has to be included in the computation. A PN (packet number)
24 that is incremented for each frame transmitted forms part of the MACsec IV. A replayProtect control with a
25 configurable replay window allows the receiver to use the PN to protect against misordered, replayed, and
26 much delayed frames. A frame whose PN value is less than the highest received with a successfully
27 validated frame (from the same source, protected by the same key) minus the replay window size is
28 discarded. Setting the replay window size to zero and enabling replayProtect enforces strict replay
29 protection, no out of order or replayed frames are accepted. The transmitter of each MACsec frame is
30 identified by a Secure Channel Identifier (SCI), and the Secure Channel (SC) that allows it to communicate
31 securely with its peers in a secure Connectivity Association (CA) is supported by a sequence of Secure
32 Associations (SAs). Each SA is replaced by its successor when its associated secret Secure Association Key
33 (SAK) can no longer be used, either because all or most of the PN space has been used with one of the SAs
34 in the CA, or because the network administrator has a security policy that mandates regular key changes
35 (often once a week). New SAs and SAKs are also required after reauthentication and reauthorization, when
36 the (possibly changed) members of a CA acquire a fresh secure Connectivity Association Key (CAK), the
37 shared secret key whose possession permits participation in the CA and that is used by the MAC Security
38 Key Agreement protocol (MKA) to identify the CA members to each other and to distribute SAKs.

39 MACsec connectivity is not interrupted or delayed by SA and SAK changes. The SCI and the Association
40 Number (AN), a two-bit circular sequence number that identifies the SA within the SC, are communicated
41 by the MAC Security TAG (SecTAG) and form part of the IV. Their values thus have to be decided and
42 encoded in the frame to be transmitted prior to encipherment. The MACsec operational and management
43 models reflect the fact that, at any instant, the frame whose SecTAG is currently being encoded can be using
44 a different SA and SAK than the frame currently being cryptographically processed. This accommodates
45 implementations that buffer user transmit requests for performance or implementation reasons (e.g., to allow
46 parallel processing). A transmit SA/SAK is not deleted until sometime after it has ceased to be used.
47 Similarly an implementation that is enciphering a preemptible frame block by block can suspend its

1 cryptographic operation, encipher an express frame that has been assigned to a subsequent SA, and resume
2 processing the preempted frame using the IV already encoded, the SAK identified by the SCI and AN for
3 that IV, and the intermediate results produced by the Cipher Suite up to the point that the frame was
4 preempted. The Integrity Check Value (ICV) generated, and appended to the MACsec protected frame,
5 protects the entire frame. An attacker cannot omit a fragment or substitute one for another. The ICV also has
6 exactly the same value that would be obtained by protecting the entire frame prior to transmitting any of its
7 octets, which is important for reasons that go beyond facilitating interoperability between different styles of
8 implementation. While MACsec protects communication between neighboring peer users of the MAC
9 Service, those service instances can be supported by bridging at a lower layer. Communication between two
10 C-VLAN Bridges can be supported by individual LANs connected by TPMRs, or by S-VLAN Bridges, for
11 example. The receiving MAC Security Entity (SecY) that validates a preemptible frame can receive it in
12 fragments that differ from those transmitted by the port and SecY that originally protected the frame, and is
13 not necessarily aware that the received frame was fragmented en-route. The first of two TPMRs supporting
14 a link between two C-VLAN Bridges (for example) might reassemble all received frames prior to
15 forwarding, thus allowing express frames to overtake preemptible frames on the link connecting the TPMRs
16 without fragmenting the latter.

17 Just as the use of an SA can be overlapped with the use of its successor on transmission, each receive SA and
18 its successor are enabled for reception for 3 seconds after an MKA Key Server tells the other member(s) of
19 the CA to start transmitting using a new SAK and SA(s). Thus the receipt of an express frame protected with
20 a new SAK does not preclude the receipt and validation of a preempted frame, protected with the prior SAK,
21 after its final fragment (including the protected frame's ICV) has been received.

22 IEEE Std 802.1AE describes the use of MACsec in a number of different systems, interface stacks, and
23 interworking scenarios. When MACsec is used to secure connectivity across a provider network, the
24 protected frame can be priority-tagged (6.13) so the provider can forward frames of different priorities
25 appropriately, even if the contents of those frames is otherwise confidential, and they can be received and
26 validated out of order just as they might be if some frames had been preempted en-route. The receiving
27 SecY's replay window size can be managed to accommodate these frames, though it is not always easy to
28 determine or predict an optimal value for the size of replay window that is needed. MACsec's strict replay
29 protection can also be retained by assigning transmitted frames for different service access priorities (or for
30 express and preemptible frames) to different SCs, each supported by its own sequence of SAs. From the
31 point of view of a receiving SecY, the separate SCs behave just as if they had been transmitted by separate
32 SecYs. IEEE Std 802.1AE allows the choice of SCI and access priority⁶⁴ to be determined by the frame's
33 user priority.⁶⁵ (See Seaman [B62].)

⁶⁴ The priority value accompanying the service request made at the SecY's lower Common Port to transmit the protected frame.

⁶⁵ The priority value provided by the service request, to protect and transmit the frame, made at the SecY's upper Controlled Port.

1 **Annex S**

2 **(informative)**

3 **Preemption and scheduled traffic**

4 The enhancements for scheduled traffic specified in 8.6.8.4 and the support for frame preemption specified
5 in 6.7.2 and 8.6.8 can be used individually or in combination in order to provide different degrees of
6 protection to “time-sensitive traffic” from interference caused by traffic with less stringent timing
7 requirements. The choice of which features to use, and in what combination, will depend upon the latency
8 and jitter requirements of the application concerned. There are four permutations of interest. They are as
9 follows:

- 10 a) Scheduling used in isolation (S.1)
- 11 b) Preemption used in isolation (S.2)
- 12 c) Scheduling and preemption used in combination, but without use of HOLD and RELEASE (S.3)
- 13 d) Scheduling and preemption used in combination, with use of HOLD and RELEASE (S.4)

14 NOTE—As the only MAC standard for preemption at the time this annex was developed was IEEE Std 802.3, it reflects
15 constraints that exist in that particular approach to preemption.

16 **S.1 Scheduling used in isolation**

17 This subject is dealt with in Annex Q.

18 **S.2 Preemption used in isolation**

19 Preemption allows one or more higher priority (express) frames to interrupt the transmission of a lower
20 priority (preemptable) frame, the preemptable frame transmission being resumed and completed once the
21 express frame(s) have been transmitted.

22 The ability to designate one or more priority as express (and the remaining priorities as preemptable) means
23 that, when an express frame is ready for transmission and a preemptable frame is already in the process of
24 being transmitted, the delay before the express frame transmission starts is at worst 123 octet times and will
25 often be 64 octet times or less. Preemption, used without scheduling, is therefore a useful tool for reducing
26 the jitter experienced by time-sensitive frames. These frames could be transmitted using any of the available
27 transmission selection algorithms (Table 8-6), but the likely choice would be an algorithm that is intended
28 for use with time-sensitive data, such as the credit-based shaper (8.6.8.2). In common use cases, it would be
29 expected that express traffic would have a higher priority than preemptable traffic.

30 NOTE 1—IEEE Std 802.3 defines a minimum final fragment size of 64 octets and allows the receiver to request that any
31 non-final fragments are larger than this by 0, 64, 128, or 192 octets. Therefore, the minimum non-final fragment size is
32 64, 128, 192, or 256 octets, depending upon the non-final fragment size stipulated by the receiving MAC. For the
33 purposes of the discussion in this annex, it is assumed that the minimum non-final fragment size selected is 64 octets.

34 NOTE 2—The 123 octet times worst-case delay is because non-final fragments have to be 64 octets in length, including
35 the mCRC that is added by the MAC Merge sublayer, so a fragment that is 124 (60 + 64) octets in length can be
36 preempted, but a fragment that is 123 or fewer octets in length cannot be preempted. This occurs when transmission has
37 just started for a fragment of a frame with less than 124 octets left to transmit, or for a preemptable frame of less than
38 124 octets in total. The more likely case is that the delay would be between 0 and 64 octet times. The actual delay
39 experienced also has to take account of the extra overheads imposed by the need for preamble, start of frame delimiter
40 (SFD), inter-frame gap (IFG), and the latency of the MAC itself.

41 It should be noted that, other things being equal, designating a priority as “express” effectively increases its
42 priority above that of any priority designated as “preemptable”. Also, in cases where the number of traffic

1 classes supported on a port is less than 8, all priority values that map to the same traffic class should have the
2 same designation (all express or all preemptable).

3 If preemptable and express priorities are assigned to the same traffic class, an express frame might not be
4 able to preempt a preemptable frame that is ahead of it in the queue.

5 **S.3 Scheduling and preemption used in combination, no HOLD/RELEASE**

6 The discussion of scheduling in Annex Q points out that there is no need for an explicitly scheduled “guard
7 band” when using the scheduling mechanism in isolation, because the specification of the operation of the
8 transmission gates is such that, in a conformant implementation, a frame cannot start transmission if the
9 transmission gate for its traffic class is due to close before transmission of the frame would complete.
10 However, the requirement not to transmit a frame if transmission cannot complete before the gate closes
11 means that, depending upon the size of frames in the queue, there would be a significant length of time
12 during which frames cannot be transmitted.

13 If the schedule that applies to the preemptable traffic classes was designed to allow preemptable traffic to be
14 transmitted at any time (i.e., there is no “gate close” event for the preemptable traffic), then preemptable
15 frame transmission can occur at any time up to the start of an express transmission, and depending upon the
16 timings, can continue up to 123 octet times after the express traffic gate opens. This dramatically reduces the
17 delay experienced by preemptable traffic, at the expense of increasing the delay experienced by express
18 traffic. While that may be acceptable for some applications, it is clearly not acceptable for applications
19 where it is desired to reduce or remove any delay experienced by express frames.

20 **S.4 Scheduling and preemption used in combination with HOLD/RELEASE**

21 The HOLD and RELEASE mechanism provided by preemption (6.7.2, 12.30.1) allows an explicit “guard
22 band” to be implemented around a protected transmission window, but one that is rather smaller than would
23 otherwise be needed. As with S.3, this scenario assumes that the preemptable traffic’s gate is scheduled to be
24 open at all times. If an MM_CTL.request(hold_req) primitive is scheduled 124 octet times before the
25 express traffic window opens, then preemptable frame transmission can start up to 124 octet times before the
26 start of the protected window, and can potentially continue up to (but not beyond) the start of the window.

27 The result of using scheduling, HOLD/RELEASE and preemption in combination is that the express traffic’s
28 protected window can be completely protected from interference from preemptable traffic, while at the same
29 time reducing the impact of the protected window on the amount of bandwidth that is available to
30 preemptable traffic.

31 A further reason to use HOLD and RELEASE is that a schedule will often be intended for a string of express
32 frames to be sent within an open window. If HOLD/RELEASE is not used and there is a greater than IPG
33 gap between transmitting express frames (e.g., because of minor jitter in their arrival time), a preemptable
34 frame or fragment could be started between express frames and cause up to 124 octets of latency in the
35 middle of the string of express frames. Using HOLD ensures that this will not happen.

36 **S.5 Bandwidth allocation and express traffic**

37 The implementation of preemption in an IEEE 802.3 MAC (see IEEE Std 802.3) does not enforce a limit on
38 how long a preemptable frame can be preempted by express traffic. This is at odds with the following:

- 39 a) Transit delay requirements of a bridge, if there is any preempted traffic present.
- 40 b) The impossibility, as shown by queuing theory, of arranging express traffic to occupy exactly 100%
41 of the available bandwidth in the absence of unbounded queues.

1 For effective use of preemption, system-wide mechanisms, such as bandwidth allocation, must operate so as
2 to keep the bandwidth allocated to express traffic below 100% as judged on timescales comparable to the
3 delays tolerated by time-sensitive traffic.

4 The fact that preemption can add delay that is invisible to transit delay protection could be addressed by
5 schedules being set such that express traffic windows are short enough and gaps between them are short
6 enough that the worst case time to send a preemptable frame is much less than transit delay requirements.
7 For example, if transit delay is one second, the time between express windows is greater than the maximum
8 transmission time for a single frame, and express window size is small compared to one second (e.g., <50
9 ms), then preemption will not add significantly to worst-case transit delay.

10 For preemption without scheduling, there is no guarantee on the added latency, but that alternative should be
11 used when the volume of express traffic is expected to be small. If the express traffic is using a shaper, such
12 as the credit-based shaper, that could put a similar reasonable bound on transit delay.

Annex T

(informative)

Cyclic queuing and forwarding⁶⁶

T.1 Overview of CQF

Cyclic queuing and forwarding (CQF) is a method of traffic shaping that can deliver deterministic, and easily calculated, latency for time-sensitive traffic streams. As the name implies, the principle underlying CQF is that stream traffic is transmitted and queued for transmission along a network path in a cyclic manner. Time is divided into numbered time intervals i , $i+1$, $i+2$, ... $i+N$, each of duration d . Frames transmitted by a Bridge, *Alice*, during time interval i are received by a downstream Bridge, *Bob*, during time interval i and are transmitted onwards by *Bob* towards Bridge *Charlie* during time interval $i+1$, and so on. A starting assumption is that, for a given traffic class, all Bridges and all end stations connected to a given bridge have a common understanding (to a known accuracy) of the start time of cycle i , and the cycle duration, d .

Frames transmitted by *Alice* during interval i are transmitted by *Bob* in interval $i+1$; the maximum possible delay experienced by a given frame is from the beginning of i to the end of $i+1$, or twice d . Similarly, the minimum possible delay experienced is from the end of i to the beginning of $i+1$, which is zero. More generally, the maximum delay experienced by a given frame is:

$$(h+1) \times d$$

and the minimum delay experienced by a given frame is:

$$(h-1) \times d$$

where h is the number of hops.

This illustrates the attraction of CQF as a technique for handling time-sensitive traffic; the latency introduced as a frame transits the network is completely described by the cycle time and the number of hops, and is unaffected by any other topology considerations, including interference from other non time-sensitive traffic. This only holds, however, if frames are kept to their allotted cycles; if, for example, some of the frames that were expected to be received by *Bob* during cycle i do not appear until cycle $i+1$ has started, then the stated assumptions about maximum latency calculation no longer hold. Careful choice of cycle times, alignment of cycle times among the Bridges in the network, and the timing of first and last transmissions within a cycle are required in order to ensure that the desired latency bounds are achieved.

Any delays through a particular intermediate relay (for example, *Bob*) do not affect the end-to-end delay so long as *Bob*'s performance does not affect the correct assignment of frames to time intervals.

Since one of the goals for the handling of time-sensitive streams is zero frame loss (assuming that no unrecognizable non-conformant traffic is present), it is prudent to assume that reception is continuous—a frame received by a downstream system will always be assigned to one interval or another. This places most of the burden of correct interval assignment on the transmitting system; frames should not be transmitted if incorrect interval assignment is possible upon reception. It is therefore necessary to define the anticipated (and accommodated) errors in reception assignment with respect to the point in interval time, t , where interval $i-1$ becomes interval i . A relay (such as *Bob*) can of course choose when to start reception assignment to i in relation to t ; it is assumed that *Bob*'s intent is that the earliest frame to be assigned to i is the first whose very last octet (or other frame transmission encoding symbol) is still on the transmission medium (or other definable external event to what is considered to be *Bob* reference point) at t , thus placing any accommodation of known implementation dependent delays within *Bob* under *Bob*'s control.

⁶⁶ In early discussions, CQF was known as the “Peristaltic Shaper” [B67].

1 While *Bob* attempts to start *i* reception with a frame coming off the medium at *i*, and may factor known and
2 repeatable internal delays into the way he goes about that intent, his actual start time depends on:

- 3 a) The error in *Bob*'s time sync (i.e., the error in his determination as to when *t* actually occurs).
- 4 b) The maximum deviation (jitter) in *Bob*'s use of that time.
- 5 c) Additional delays that *Bob* does not account for, such as delays in selecting the output queue to be
6 used for *i*.

7 *Alice* has to stop transmitting frames for *i*–1 before *t*, by a time that is the sum of *Bob*'s possible early start of
8 *i* as a consequence of a) through c), and the following:

- 9 d) The error in *Alice*'s time sync (i.e., the error in her determination as to when *t* occurs).
- 10 e) The maximum deviation (jitter) in *Alice*'s use of that time.
- 11 f) The time between *Alice* deciding to commit a frame for transmission and the appearance of the last
12 octet/symbol “on the medium” at *Alice*'s end.
- 13 g) The length of “the medium” in transmission time, i.e., the time for the last octet/symbol to leave
14 *Alice* and reach *Bob*, including any consideration of the effect of interfering frames or fragments.

15 The description of CQF in terms of a number of consecutive intervals (as opposed to their support by “odd/
16 even” queues, as discussed in T.2 onwards) gives easy answers to what to do with traffic still queued when
17 its selected transmission interval has expired—discard it, or mark it down (discard eligible or priority
18 change) and generate an alarm. In an environment where the stream bandwidth is allocated appropriately
19 (i.e., the bandwidth allocated per time interval is less than can be received/transmitted in the chosen interval
20 duration), this will be a rare occurrence, the traffic that follows will be conformant, and the overall system
21 performance will be recoverable.

22 The discussion so far has assumed that all link speeds are the same; however, the situation becomes more
23 complicated when links of different speeds are considered. One typical arrangement might comprise low
24 speed links at the start and end of the path (network periphery to periphery), another with the high speed
25 towards one end (periphery to core or vice versa). Taking the first of these, and placing *Alice* at the first
26 transition from slow to fast, *Bob* as her fast neighbor, *Charlie* as his fast neighbor, and *Donald* at the
27 transition from fast to slow, the important thing (treating the fast core of the network as a CQF black box) is
28 that all conformant traffic received by *Alice* in interval *i* (say) is transmitted by *Donald* in a later interval
29 *i*+*n*. A number of internal arrangements might be made between *Alice*, *Bob*, *Charlie*, and *Donald* to make
30 this happen and would be valid from an external CQF perspective. It is also possible to consider fractional *n*,
31 where *n* is still > 1, as *Alice* may need to collect the entirety of any slow cycle before transmitting that in a
32 more compressed burst into the rest of the fast network. More complex possibilities are equivalent to
33 redefining the slow cycle time. Some of the less elaborate possibilities for the use of links of different speeds
34 are discussed in T.5.

35 T.2 An approach to CQF implementation

36 In essence, the approach involves the use of two transmission queues and a cycle timer. During even
37 numbered cycles (intervals), queue 1 accumulates received frames from the Bridge's reception Ports (and
38 does not transmit them), while queue 2 transmits any queued frames from the previous odd-numbered cycle
39 (and does not receive any frames). During odd-numbered cycles, queue 2 accumulates received frames from
40 the Bridge's reception Ports (and does not transmit them), while queue 1 transmits any queued frames
41 from the previous even-numbered cycle (and does not receive any frames). With appropriate choice of
42 receive and transmit cycle times (see T.5), such that, for any given stream, the cycle is at least long enough
43 to accommodate all of the time-sensitive traffic that will need to be transmitted on the Bridge Port during the
44 class measurement interval for that stream (see 34.6.1.1, also known as the observation interval in
45 IEEE Std 802.1BA™ [B11]), plus a maximum-sized interfering frame (or frame fragment, if preemption is
46 supported), then all of the stream's traffic will be accumulated during the cycle time in queues that are in
47 receive mode, and it will all be transmitted during the cycle time when the queues switch to transmit mode.

1 CQF is implemented by configuring a combination of the stream gate control mechanisms defined in 8.6.5.4
2 and the traffic scheduling mechanisms defined in 8.6.8.4 and 8.6.9. Per-stream filtering is used to direct
3 received frames to one of a pair of outbound queues on a timed basis, determined by the cycle time of the
4 per-stream filter, and traffic scheduling is used to ensure that frames are transmitted from the appropriate
5 queue using the same cycle time, as described in the rest of this annex.

6 **T.3 Use of Per-Stream Filtering and Policing for CQF**

7 The first step in establishing the filtering and queuing structures needed for CQF is to set up one or more
8 stream filters (8.6.5.3) and a stream gate instance (8.6.5.4) that will be receiving incoming time-sensitive
9 frames. The stream filter(s) are configured so that all time-sensitive frames received on a given Port are
10 directed to the same stream gate instance; in turn, the stream gate instance is configured so that the internal
11 priority value (IPV) associated with the time-sensitive frames will direct them to one of two outbound
12 queues on a timed basis. The use of the IPV allows this direction of frames to outbound queues to be
13 independent of the received priority, and also does not affect the priority associated with the frame on
14 transmission.

15 **T.3.1 Stream filter configuration**

16 The simplest stream filter configuration would be achieved where the same priority is used for all time-
17 sensitive frames (and this priority is not used for any other frames); for example, the default priority
18 assigned to SR class A (see Clause 34) could be used, in which case, the priority associated with the time-
19 sensitive frames would be 3. The parameters that would define the stream filter for the time-sensitive frames
20 would then be as follows:

- 21 a) The *stream handle specification* would take the wild-card value.
- 22 b) The *priority specification* would take the priority value 3.
- 23 c) The *stream gate instance identifier* would take the value of the instance identifier for the stream gate
24 (T.3.2).
- 25 d) In the simplest case, there would be no further per-stream classification and metering operations
26 (8.6.5.2); however, these could be added as appropriate, for example if the maximum SDU size
27 (8.6.5.3.1) for the time-sensitive traffic is bounded at a value less than the maximum SDU size for
28 the medium.

29 This stream filter configuration results in all frames that carry a priority value of 3 being submitted to the
30 stream gate. As the operation of stream filters is such that received frames that do not match a stream filter
31 are handled as if subsequent per-stream classification and metering operations were not implemented, there
32 is no need for further stream filter configuration to handle frames that carry priorities other than 3 unless
33 there are other filtering or gating decisions that need to be taken for such frames.

34 **T.3.2 Stream gate configuration**

35 The *stream gate instance* (8.6.5.4) needed to support the stream filter described in T.3.1 has a *stream gate*
36 *control list* that contains two entries, each containing a SetGateAndIPV operation, with parameters as
37 follows:

- 38 1) StreamGateState = *open*, IPV = 7, TimeInterval = T
- 39 2) StreamGateState = *open*, IPV = 6, TimeInterval = T

40 This control list has the effect of directing any traffic that passes the stream filter specified in T.3.1 to one of
41 two different outbound queues (assuming that the outbound Ports support 8 queues, and that the default
42 assignments for priorities to traffic classes follows the recommendation shown in Table 34-1); in the first
43 time interval T, traffic is directed to queue 7, in the second time interval T, to queue 6, in the third time

interval to queue 7, in the fourth time interval, to queue 6, and so on. The choice of time interval T is discussed in T.5; the cycle time (OperCycleTime, see 8.6.9.4.19) for the stream gate state machines would need to be set to $2T$ in order to accommodate the sum of the time intervals for the two gate operations. See Figure T-1.

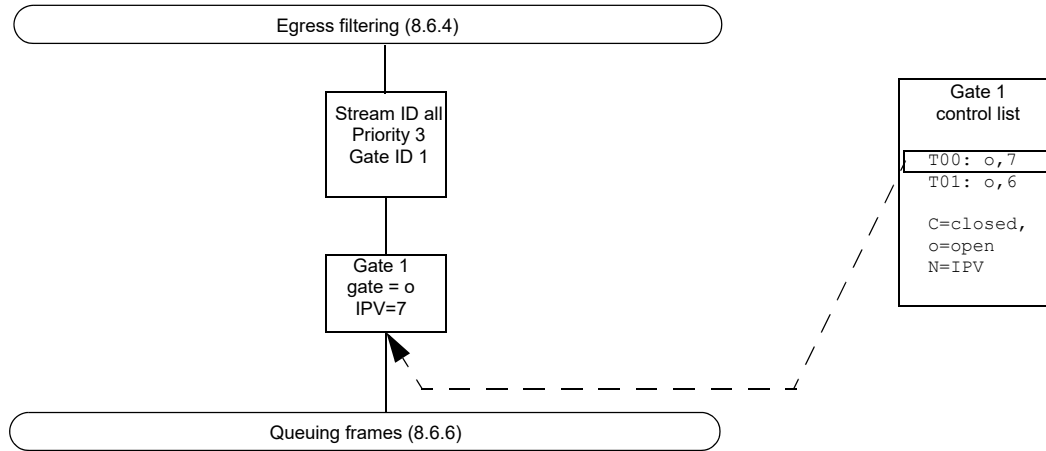


Figure T-1—Example Stream Filer and Stream Gate configuration for CQF

T.4 Use of traffic scheduling for CQF

The traffic scheduling support needed on each outbound Port in order to support the PSFP configuration described in T.3 is to execute a gate control list that will set the GateState to *open* for queue 6 and *closed* for queue 7 for a TimeInterval of T , and then set the GateState to *open* for queue 7 and *closed* for queue 6 for a TimeInterval of T , repeating ad infinitum. If there are no other traffic scheduling considerations, this can be achieved with a gate control list that contains just two SetGateStates gate operations, with parameters as follows:

- 1) GateState: 0, 1, 2, 3, 4, 5, 6 *open*, 7 *closed*, TimeInterval = T
- 2) GateState: 0, 1, 2, 3, 4, 5, 7 *open*, 6 *closed*, TimeInterval = T

This sequence of gate operations has the effect that during the initial time period T , the GateState for queue 7 is closed while queue 7 is being filled, and queue 6 is open to allow any queued frames to be transmitted; during the second time period T , the GateState for queue 6 is closed while queue 6 is being filled, and queue 7 is open to allow any queued frames to be transmitted. The gates for all other queues are open. The choice of time interval T is discussed in T.5; the cycle time (OperCycleTime; see 8.6.9.4.19) for the scheduled traffic state machines would be set to $2T$ in order to accommodate the sum of the time intervals for the two gate operations.

If there are traffic scheduling requirements for any of the other queues, then the gate control list could be extended to accommodate those requirements; however, the time interval between the changes of state of the gates for queues 6 and 7 has to be T , and consequently, OperCycleTime has to be a multiple of $2T$, in order for the CQF requirements to be met. Figure T-2 illustrates the simplest possible traffic scheduling configuration for the case that traffic scheduling is only needed to support CQF.

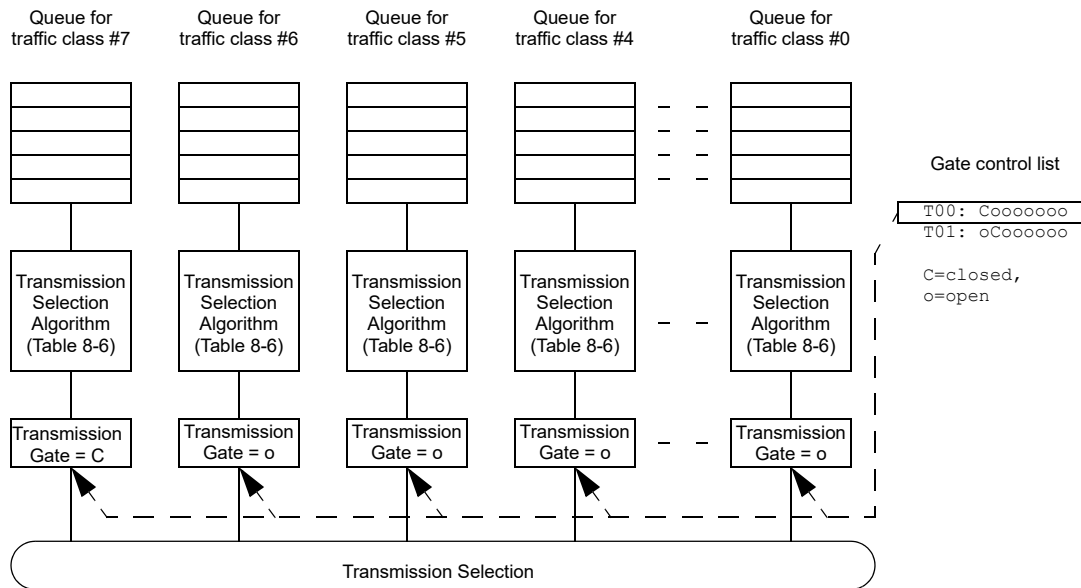


Figure T-2—Traffic scheduling example for CQF

1 T.5 Timing considerations

2 T.5.1 Choice of T

3 T should be chosen such that it is large enough to accommodate the stream data that can be received during
4 the class measurement interval for the stream(s) concerned, plus at least one maximal interfering frame or
5 frame fragment. This is important in order to ensure the key performance aspect associated with the class
6 measurement interval; namely, that if a stream or set of streams is observed over time, the reserved data rate
7 for that stream or set of streams will not be exceeded during any observed time period equal to the class
8 measurement interval associated with those streams. This effectively places a lower bound on the choice of
9 T, that it should not be smaller than the class measurement interval, and also places a restriction on larger
10 values of T, that they should be integer multiples of the class measurement interval.

11 If streams associated with two different observation intervals are being handled, for example if streams that
12 use SR classes A and B pass through the Bridge, then the OperCycleTime used for the transmit traffic
13 scheduling has to be a common multiple of the two class measurement intervals that are in use in order to
14 make it possible for the transmission cycles to properly match the two values of T that are chosen. Figure T-
15 3 and Figure T-4 illustrate how the Stream Filters, Stream Gates, and traffic scheduling could be configured
16 in the case where SR classes A and B are active; in Figure T-3, incoming frames that carry SR Class A
17 (priority 3) are handled using Gate 1, and the cycle time for the stream gate control list is twice the class
18 measurement interval for SR Class A, which is $2 \times 125 \mu\text{s}$. Gate 1 alternately tags these frames with an IPV
19 of 7 or 6. Incoming frames that carry SR Class B (priority 2) are handled using Gate 2, and the cycle time for
20 the stream gate control list is twice the class measurement interval for SR Class B, which is $2 \times 250 \mu\text{s}$. Gate
21 2 alternately tags these frames with an IPV of 5 or 4.

22 The traffic schedule is based on the smaller of the two class measurement intervals, $125 \mu\text{s}$, but now has four
23 entries in the gate control list (as opposed to 2 entries in Figure T-2), giving an overall cycle time of $500 \mu\text{s}$.
24 The gate control list switches the gate states for traffic classes 7 and 6 every $125 \mu\text{s}$, and switches the gate
25 states for traffic classes 5 and 4 every $250 \mu\text{s}$.

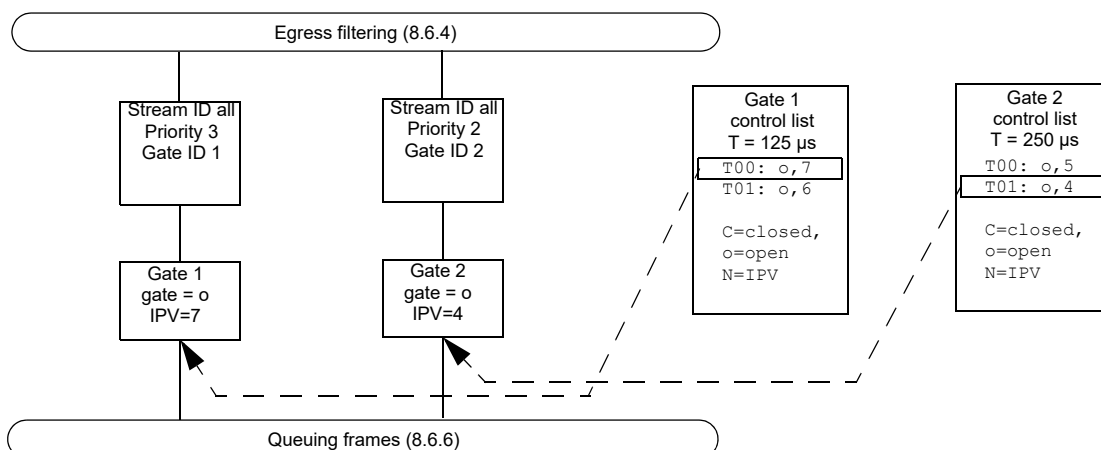


Figure T-3—Example Stream Filter and Stream Gate configuration with two values of T

1

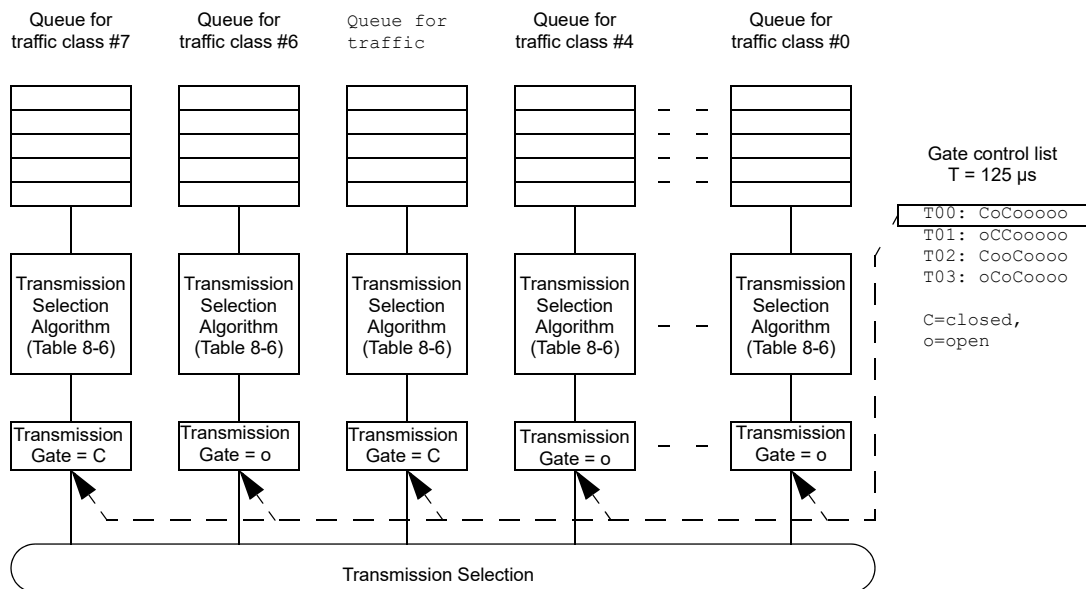


Figure T-4—Traffic scheduling example with two values of T

2 T.5.2 Cycle interleaving

3 In some circumstances, particularly where the data rates differ between reception and transmission Ports, it
4 can be desirable to interleave cycles on the faster Port so that the best use is made of the higher bandwidth
5 available, and also to reduce the latency that is added as a stream passes through faster parts of the
6 transmission path. Because there is a delay imposed on the transmission of received frames, caused by the
7 cyclic switching of reception and transmission between a pair of queues, when a queue is allowed to
8 transmit, all of its received frames will have been enqueued, and all of them will therefore be transmitted in
9 a burst, assuming that priorities permit, and that the transmission queue uses the strict priority transmission
10 selection algorithm (8.6.8.1). Hence, if the received traffic from a given Port was spread out over the time
11 interval T, and it is all sent to the same queue, the transmitted traffic will be compressed into a burst. If

the transmission data rate is, say, ten times the reception data rate, then the maximum length of that burst is $T/10$, so there is the potential to fit 9 more such bursts into the bandwidth available on that transmission Port. With appropriate timing on reception Ports and transmission Ports, the reception and transmission cycles can be interleaved such that those additional transmission bursts can occur. In the example illustrated in Figure T-5, it is assumed that:

- a) There are two Ports on which stream data is being received, Rx1 and Rx2.
- b) There is a single Port on which stream data is being transmitted, Tx1.
- c) Rx1 and Rx2 operate at half of the data rate of Tx1 (or less).
- d) All stream traffic is SR Class A, and is received with priority 3.

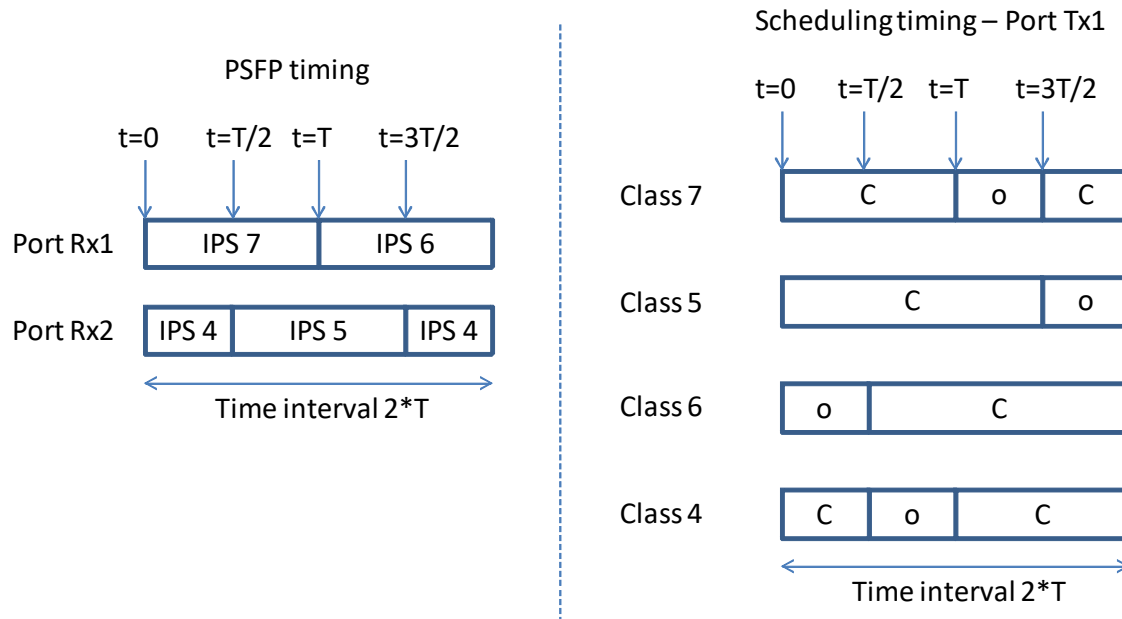


Figure T-5—Interleaving example—factor of 2

On the reception side, traffic received on Rx1 with priority 3 is sent to traffic class queue 7 during the odd cycles and to traffic class queue 6 during the even cycles. Similarly, traffic received on Rx2 with priority 3 is sent to traffic class queue 5 during the odd cycles and to traffic class queue 4 during the even cycles; however, these cycles are offset with respect to the cycles for Rx1 by $T/2$.

On the transmission side, each queue is in the open (transmitting) state for $T/2$, and in the closed (receiving) state for $3T/2$. The timings of the gate open/gate closed events are arranged so that only one queue is transmitting at any one time.

It should be noted that with a scheme like this, although the transmit side appears to be operating on the basis of a value of T that is half that used on the Rx side, the effect from the point of view of the streams originating from a given reception Port is that the Rx and Tx timing is the same, as frames received on that Port are transmitted once in every time period T , and as assumed in the preceding point c), the transmit rate is twice the receive rate (or more), so the available transmit bandwidth is the same as or more than the receive bandwidth. The interleaving therefore meets the requirements stated in T.5.1. The reception Port of the Bridge downstream of Tx1 can operate using $T/2$, and if it is possible to carry this through to the transmission Ports as well, the contribution to the latency for streams passing through this Bridge will be $T/2$, rather than the T contributed by the first Bridge.

1 More complex schemes can be envisaged; for example, using more than two Rx Ports, and these can be
2 made to work as long as the received bandwidth can be shared equally between the pairs of traffic classes
3 that are used. Interleaving of this kind can also be defined for larger interleave factors; the only limitation is
4 the number of available outbound queues. It is also possible to define interleaving schemes where the
5 received bandwidth is not shared equally among the pairs of traffic classes, as long as the bandwidth
6 allocated to a given pair of traffic classes does not exceed the bandwidth available during the time intervals
7 when those traffic classes are able to transmit.

8 NOTE—Although the number of traffic classes described in this standard is limited to 8, the value of the IPV does not
9 have such a limitation placed upon it. If a system supported more than 8 traffic classes, it would therefore be possible to
10 define interleaving factors greater than 4.

11 T.5.3 Cycle alignment between adjacent Ports

12 The examples so far assume a perfect world where the transmission from transmission Port to reception Port
13 is instantaneous and the internal timings of the transmitting and receiving systems are perfectly
14 synchronized. In reality, transmission takes time, and synchronization is not perfect; therefore, it would be
15 possible that a transmission Port launches the last frame(s) of one transmission burst just after the reception
16 Port downstream has switched reception and transmission queues, which would mean that those last frames
17 are placed in the wrong queue. Similarly, if the timing misalignment worked the other way, it would be
18 possible for the transmission Port to finish transmitting its burst early, and switch to transmitting the next
19 burst before the downstream Port had changed state.

20 In order to avoid this problem, the timings must be adjusted such that there is a very high degree of
21 probability that when a reception Port changes the state of the stream filters to direct incoming frames to a
22 different outbound queue, there are no frames still to be transmitted, or in flight, from the upstream
23 transmission Port. This can be achieved by slightly delaying the start of the transmission window, and
24 slightly advancing the end of the transmission window. The value of “slightly” depends on a number of
25 factors, including the following:

- 26 a) Any error in the time synchronization between the adjacent systems.
- 27 b) Jitter in the propagation time of a frame from starting to leave the transmit queue in the upstream
28 system to being presented to the downstream policing function.
- 29 c) Jitter in the propagation time of a frame between the downstream policing function and the
30 appropriate transmission queue in the downstream system.
- 31 d) The size of any potential interfering frame or frame fragment.
- 32 e) Difference in the resolution of the clocks that are maintained by adjacent systems.

33 The effect of this adjustment factor, S , on the timings shown on the transmission Port in the earlier examples
34 would be that the time slots where the gate is in the “open” state would be shorter by a factor of $2S$, and
35 would start S later. Hence, the transmission phase for traffic class 7 in Figure T-5 would start at $T+S$, and
36 would end at $(3T/2)-S$. S should be set to the sum of the errors or jitter values from all sources given in the
37 preceding list.

Annex U

(informative)

TSN configuration examples

This annex provides implementation examples for Time-Sensitive Networking (TSN) configuration (Clause 46).

The examples in this annex are not intended to be comprehensive, but to serve as informative background to assist in understanding of Clause 46.

U.1 Examples for time-aware talker

One of the goals of TSN configuration is avoid forcing the user's application to be subservient to the network's configuration. The user describes what it does (e.g., `TrafficSpecification`) and what it needs from the network (e.g., `UserToNetworkRequirements`). The user also describes the configuration that it is willing to accept from the network (e.g., `InterfaceCapabilities`), but the goals are to keep that configuration to a minimum and to avoid topics that change aspects of the user's application, such as the timing of the application's execution and interaction with the physical world (e.g., sensors, actuators).

Aligned with these goals, it is possible for the Talker to transmit frames in a manner that is not aware of time synchronization protocols like IEEE Std 802.1AS. Using the time-aware features of `TrafficSpecification` and `InterfaceConfiguration`, it is also possible for a Talker to use awareness of time synchronization for transmission of frames. Nevertheless, TSN configuration avoids assumptions of a specific implementation of time-aware transmit, and TSN configuration avoids changes to application timing.

In order to demonstrate these concepts, consider an example in which Talkers and Listeners use IEEE Std 802.1AS to execute synchronized application loops at an interval of 500 μ s. The code for the application consists of two stages: a 250 μ s computation stage that generates data and a 250 μ s stage to transmit that data in TSN frames. The `MaxLatency` requirements can span one or more iterations of the application loop.

This example uses IEEE Std 802.1AS for time synchronization. The application begins execution on September 13, 2020, at 12:26:40 pm TAI, which corresponds to 1 600 000 000 000 000 μ s in IEEE 802.1AS time.

The application loops begin in alignment with IEEE 802.1AS time; in other words, the 250 μ s transmit stage begins at a base offset of 250 μ s in IEEE 802.1AS time. For example, after the application has been running for 6 s, a sequence of transmit stages can execute at 1 600 000 006 000 750 μ s, 1 600 000 006 001 250 μ s, 1 600 000 006 001 750 μ s, and so on.

This example focuses on a single Talker that transmits two Streams, J and K. The application generates a single frame of data for J and/or K at arbitrary times in the computation. In other words, during a single iteration data for K can follow J, and in the next iteration data for J can follow K, and in the next iteration data can be J only.

The frame for Stream J uses 120 μ s in time. The frame for Stream K uses 80 μ s in time. This implies a 100 Mb/s LAN technology.

1 U.1.1 Using enhancements for scheduled traffic

2 One option for implementation of this time-aware Talker is the enhancements for scheduled traffic (8.6.8.4).
3 The Talker has two queues for transmit: one for TSN frames and another for non-TSN frames (e.g., best
4 effort). The scheduled traffic enhancements use IEEE 802.1AS time to open and close gates for each queue
5 (traffic class).

6 The enhancements of 8.6.8.4 schedule traffic per queue, and not per stream. Given the preceding application
7 assumptions, therefore, the frame for Stream J can precede the frame for Stream K, and in the next cycle K
8 can precede J.

9 For a time-aware Talker, the Jitter of TrafficSpecification (46.2.3.5) represents the variance in time for the
10 single Stream. Since the enhancements of 8.6.8.4 are per queue, the Jitter must include the variance that is
11 introduced by other Streams from this Talker.

12 Using the scheduled traffic enhancements of 8.6.8.4, the time-aware Talker needs an open TSN window of
13 $120+80=200\ \mu\text{s}$ for both J and K. The network (CNC) can locate that TSN window from $250\ \mu\text{s}$ to $450\ \mu\text{s}$
14 through $300\ \mu\text{s}$ to $500\ \mu\text{s}$. The Talker uses the TimeAwareOffset returned in InterfaceConfiguration
15 (46.2.5.3) to determine where to locate the TSN window.

16 For this implementation, the TrafficSpecifications are as follows:

- 17 — Stream J
 - 18 — MaxFrameSize = (120 μs in time)
 - 19 — MaxFramesPerInterval = 1
 - 20 — Interval = 500 μs
 - 21 — Jitter = 40 μs (Stream K can occur before J, such that J is 80 μs in window.
22 Stream K can occur after J, such that J is 0 μs in window.
23 Jitter represents the midpoint of this variance.
24 Use the returned TimeAwareOffset-Jitter as the start of
25 the TSN window.)
 - 26 — EarliestTransmitOffset = $250\ \mu\text{s} + \text{Jitter} = 290\ \mu\text{s}$
 - 27 — LatestTransmitOffset = $500\ \mu\text{s} - \langle \text{MaxFrameSize in time} \rangle - \text{Jitter} = 340\ \mu\text{s}$
- 28 — Stream K
 - 29 — MaxFrameSize = (80 μs in time)
 - 30 — MaxFramesPerInterval = 1
 - 31 — Interval = 500 μs
 - 32 — Jitter = 60 μs (Stream J can occur before or after K)
 - 33 EarliestTransmitOffset = $250\ \mu\text{s} + \text{Jitter} = 310\ \mu\text{s}$
 - 34 — LatestTransmitOffset = $500\ \mu\text{s} - \langle \text{MaxFrameSize in time} \rangle - \text{Jitter} = 360\ \mu\text{s}$

35 Assume that the CNC decides to use EarliestTransmitOffset, presumably because that can be aligned with
36 the schedules in Bridges. This results in InterfaceConfigurations of:

- 37 — Stream J
 - 38 — TimeAwareOffset = 290 μs
- 39 — Stream K
 - 40 — TimeAwareOffset = 310 μs

41 The Talker subtracts the Jitter from these TimeAwareOffset values and uses the lower of the results to
42 configure AdminBaseTime and OperBaseTime of 8.6.8.4. The resulting configuration of the scheduled
43 traffic enhancements of 8.6.8.4 is shown in Figure U-1 along with example traffic.

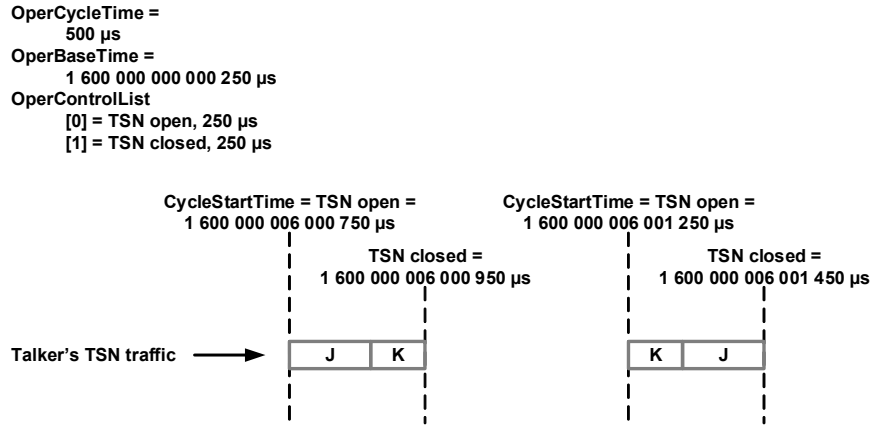


Figure U-1—Example of enhancements for scheduled traffic

NOTE—For this example, the Talker could also decide to use OperBaseTime of 0 μ s and swap OperControlList elements 0 and 1 (i.e., 250 μ s TSN closed followed by 250 μ s TSN open). That configuration is equivalent from the network perspective.

1 U.1.2 Using strict priority

2 If the Talker has no hardware to assist in scheduling of traffic, it can take advantage of its application timing
 3 for TSN configuration of its Streams. This implementation assumes that the Talker has two queues for
 4 transmit, but each queue uses strict priority alone for transmission selection (8.6.8.1). The TSN queue uses a
 5 higher priority traffic class than the non-TSN queue.

6 For this implementation, assume that the Talker disables the TSN queue during its computation stage, but
 7 continues to place frames for J and K into the TSN queue. When the Talker transitions to its transmit stage
 8 (i.e., 250 μ s base offset in IEEE 802.1AS time), it enables the TSN queue for transmit.

9 When the TSN queue is enabled, a non-TSN frame (e.g., best effort) can interfere such that it has just begun
 10 its transmit. For 100 Mb/s full-duplex IEEE Std 802.3 and a maximum frame size of 1522, this interference
 11 is 123.36 μ s. This interference must be incorporated into the Jitter of TrafficSpecification, much like the
 12 Jitter of Streams J and K in U.1.1.

13 This implementation has no flexibility in the location of its TSN window, which always starts at a 250 μ s
 14 offset.

15 For this implementation, the TrafficSpecifications are as follows:

- 16 — Stream J
- 17 — MaxFrameSize = (120 μ s in time)
- 18 — MaxFramesPerInterval = 1
- 19 — Interval = 500 μ s
- 20 — Jitter = (80 μ s + 123.36 μ s) / 2 = 101.68 μ s
- 21 (Stream K and max non-TSN can occur before J.)
- 22 — EarliestTransmitOffset = 250 μ s + Jitter = 351.68 μ s
- 23 — LatestTransmitOffset = 250 μ s + Jitter = 351.68 μ s
- 24 (No flexibility. TSN window always starts at 250 μ s.)

- 1 — Stream K
- 2 — MaxFrameSize = (80 μ s in time)
- 3 — MaxFramesPerInterval = 1
- 4 — Interval = 500 μ s
- 5 — Jitter = (120 μ s + 123.36 μ s) / 2 = 121.68 μ s
- 6 — EarliestTransmitOffset = 250 μ s + Jitter = 371.68 μ s
- 7 — LatestTransmitOffset = 250 μ s + Jitter = 371.68 μ s

8 This results in the following InterfaceConfigurations (or a failure):

- 9 — Stream J
- 10 — TimeAwareOffset = 351.68 μ s
- 11 — Stream K
- 12 — TimeAwareOffset = 371.68 μ s

13 Due to the high Jitter and inflexibility of the scheduling window, this implementation has the potential for
14 more Stream configuration failures as compared to the implementation described in U.1.1.

15 U.1.3 Using per-stream scheduling

16 Some Talker implementations provide additional hardware assistance for time-aware transmission, such that
17 each individual Stream has its own scheduled gating (i.e., per stream).

18 One such implementation is similar to the scheduled traffic enhancements of 8.6.8.4, but each Stream has a
19 dedicated transmit queue and gate control list. An alternative implementation has a single transmit queue for
20 TSN traffic, but each frame has an associated timestamp to specify its time for transmit. A variety of
21 implementations are possible. For this example, assume that scheduled transmit is per stream, but do not
22 assume a specific hardware implementation.

23 Assume that the per-stream scheduling hardware uses IEEE 802.1AS time directly. Therefore, the time-
24 aware transmit has no additional variance beyond what IEEE Std 802.1AS provides, and the Jitter of
25 TrafficSpecification is zero.

26 Given the application requirements, the CNC can locate Stream J anywhere in the window from 250 μ s to
27 500 μ s. The CNC can also locate Stream K anywhere in the window from 250 μ s to 500 μ s, but it cannot
28 overlap with Stream J.

29 For this implementation, the TrafficSpecifications are as follows:

- 30 — Stream J
 - 31 — MaxFrameSize = (120 μ s in time)
 - 32 — MaxFramesPerInterval = 1
 - 33 — Interval = 500 μ s
 - 34 — Jitter = 0 μ s
 - 35 — EarliestTransmitOffset = 250 μ s + Jitter = 250 μ s
 - 36 — LatestTransmitOffset = 500 μ s – <MaxFrameSize in time> – Jitter = 380 μ s
- 37 — Stream K
 - 38 — MaxFrameSize = (80 μ s in time)
 - 39 — MaxFramesPerInterval = 1
 - 40 — Interval = 500 μ s
 - 41 — Jitter = 0 μ s
 - 42 — EarliestTransmitOffset = 250 μ s + Jitter = 250 μ s
 - 43 — LatestTransmitOffset = 500 μ s – <MaxFrameSize in time> – Jitter = 420 μ s

1 Assuming that the CNC can locate Stream J at 250 μ s and Stream K immediately after, this results in the
2 following InterfaceConfigurations:

- 3 — Stream J
- 4 — TimeAwareOffset = 250 μ s
- 5 — Stream K
- 6 — TimeAwareOffset = 370 μ s

7 Due to the zero Jitter and improved flexibility of the scheduling window, this implementation has the
8 potential for more Stream configuration success as compared to the implementation described in U.1.1. In
9 addition, given that each Stream has a distinct offset in time, this implementation enables the CNC to
10 configure the scheduled traffic enhancements of 8.6.8.4 in Bridges such that multiple Streams do not overlap
11 upon ingress from multiple Ports. In turn, the overall variance in latency is reduced.

12 U.2 Example of workflow for fully centralized models

13 This subclause provides an example workflow for the centralized TSN configuration models of Clause 46.
14 The example is not intended to be comprehensive, but to help the reader understand how aspects of the
15 configuration can be addressed (e.g., TSN domain detection and enforcement).

16 The example assumes the fully centralized model (46.1.3.3) with a single Centralized User Configuration
17 (CUC) entity and a single Centralized Network Configuration (CNC) entity.

18 Many user-level protocol standards exist for the purpose of discovering and configuring end stations in order
19 to design a distributed application. The application is usually specific to a particular market segment (e.g.,
20 industrial automation, professional audio/video). A centralized software entity is used for the application's
21 design, possibly including programming of software in end stations (e.g., industrial Programmable Logic
22 Controller). The centralized entity uses the user-level protocol to communicate with the relevant end stations
23 for its application. When TSN configuration is applied to these existing standards, the goal is to integrate
24 TSN functionality without significant changes. The existing application design software is represented by
25 the CUC concept, and the end stations are represented by the Talker and Listener concepts. This example
26 does not designate a specific user-level protocol, but it does assume the existence of a single CUC.

27 The CNC discovers capabilities of network infrastructure (e.g., Bridges) and configures those features.

28 The example provides a workflow as a framework for describing each step in the TSN configuration
29 procedure. The steps do not necessarily need to occur sequentially. Although the example assumes a single
30 CUC, its CNC workflow (i.e., step 4 through step 8 below) can be applied to other examples that assume
31 multiple CUCs, such as the following:

- 32 a) Multiple application design tools are used (i.e., multiple user-level protocols), and therefore multiple
33 CUCs are used with a single CNC.
- 34 b) No application design tool is used, and instead each CUC represents a single Talker or Listener that
35 communicates directly with the CNC.
- 36 c) Each CUC represents a single Talker or Listener that uses the Nearest Bridge as a proxy to the CNC
37 (i.e., centralized network/distributed user model).

38 The workflow for this example consists of the following steps:

39 1) CUC discovers Talker and Listener end stations.

40 The protocol used by the CUC is outside the scope of this standard. When the communication
41 between CUC and end stations uses IP, protocols such as mDNS (IETF RFC 6762 [B42]) are
42 sometimes used.

2) CUC reads the capabilities of each end station.

This step includes TSN capabilities, but also many other capabilities that are not related to network technology. For example, if the end station is a sensor, the CUC reads the capabilities of the sensor as it applies to the physical world. The CUC also reads the frame size that is needed to transfer the sensor's measured value and associated diagnostics.

Although the communication between CUC and end station is not directly related to Clause 46, the CUC needs to obtain the following information:

- Number of Ports and MAC address of each Port
- InterfaceCapabilities: IEEE 802.1 capabilities of each Port
- TrafficSpecification.MaxFrameSize and MaxFramesPerInterval: Size of end station's data

The user-level protocol used by the CUC is outside the scope of this standard.

3) End-user designs the distributed application using the CUC.

The end user of the CUC decides which end stations communicate with one another as part of the distributed application. In other words, in terms of this standard, the CUC is used to design the Streams, including selection of the Talker and Listeners for each Stream. The CUC is also used to design the application's timing requirements.

The CUC creates the following information for each Stream, and its end stations are not directly aware of this information:

- StreamID: This identifier is primarily used between CUC and CNC.
- StreamRank: The importance of each Stream is related to its use in the application.
- UserToNetworkRequirements: MaxLatency is tied to application timing requirements.

The periodic execution of application code is designed using the CUC. This translates to the interval (i.e., period, rate) at which a Talker transmits its data. The CUC sends this interval to its end stations using the user-level protocol.

4) CNC discovers the physical network topology.

To discover the physical links between end stations and Bridges, the CNC uses IEEE Std 802.1AB (LLDP) along with a remote management protocol.

5) CNC reads the TSN capabilities of each Bridge.

The CNC uses a remote management protocol to read the TSN capabilities of each Bridge.

Whereas the previous step used the MIB/YANG of IEEE Std 802.1AB, this step uses the MIB/YANG of standards for TSN, such as IEEE Std 802.1Q, IEEE Std 802.1AS, and IEEE Std 802.1CB. The CNC uses the Chassis (Bridge) and Port identifications from the IEEE 802.1AB MIB/YANG to find the corresponding Port capabilities in the MIB/YANG of other IEEE 802.1 standards.

For example, the CNC will read the Bridge Delay (12.32.1) and Propagation Delay (12.32.2) from each Bridge in order to compute AccumulatedLatency (for step 9).

The CNC of this fully centralized model coexists with the fully distributed model (46.1.3.1). If the CNC finds a Bridge with msrpEnabledStatus TRUE (12.22.1) and MRP externalControl FALSE (12.32.4.1), the CNC avoids use of MSRP's VLAN ID (12.22.2, 12.22.4) for Streams configured by the CNC. The CNC also avoids use of MSRP's Priority (12.20.4) and associated traffic class. Although there might be methodologies for sharing the same resources (VLANs and traffic classes) between both models, this CNC takes the simpler approach of keeping the resources separate.

6) CUC sends Talker/Listener groups to the CNC.

The CUC requests configuration of Streams by sending Join requests (46.2.2) that specify the Talker group and Listener groups for each Stream.

The EndStationInterfaces group of each Talker and Listener identifies the associated end station in the physical topology, using the end station's MAC address. The MAC address is persistent (i.e., does not change due to software or protocols) and is assumed to be unique within the network, which serves as excellent identification in the physical topology. For this example, assume that end

stations use MAC address as their LLDP Chassis ID, such that the CNC can associate each Talker/Listener group to its neighboring Bridge Port.

7) CNC configures TSN domain(s).

Using the information learned in step 4 through step 6, the CNC knows the Bridges in the path(s) between each Talker and its Listeners. The CNC also knows the Bridge Ports that do not transmit frames for a Stream. This effectively defines the TSN domain for the collection of Streams.

The CNC uses destination MAC address and VLAN ID as tools to configure TSN features within the TSN domain. The InterfaceConfiguration group provides a mechanism for the CNC to allocate destination MAC addresses and VLAN IDs from its own pool and to provide the allocated values to the CUC (and the end stations). If end stations do not support InterfaceConfiguration, the CNC learns the destination MAC addresses and VLAN IDs from the DataFrameSpecification of each Stream.

i) Inside the TSN domain

When Bridges in the TSN domain are using spanning tree protocols (e.g., MSTP, ISIS-SPB) to forward Streams, the CNC uses management of the Static Filtering Entries (12.7) in order to avoid spanning tree protocols updating the active topology of the VLAN used by Streams.

When the CNC uses IEEE Std 802.1CB for redundant paths for Streams, the CNC uses management of Static Trees (12.32.3) and the associated TE-MSTID in order to avoid spanning tree operation on the VLAN ID used by Streams. The CNC uses management of IEEE Std 802.1CB to configure the redundant paths in Bridges between each Talker and its Listeners.

ii) Outside the TSN domain

The CNC uses management of Static Filtering Entries (12.7) to prevent transmit of TSN frames to Bridge Ports outside the TSN domain.

The CNC uses management of the Priority Regeneration Table (12.6) to prevent interference by non-TSN frames received on Bridge Ports outside the TSN domain, in that the priority of the non-TSN frame can be downgraded to zero. This is similar to the use of the Priority Regeneration Override Table (12.20.3) by SRP (6.9.4).

The accuracy/precision of IEEE Std 802.1AS degrades slightly for each Bridge that transfers time, so it makes sense for the CNC to limit the size of the IEEE 802.1AS domain as much as possible. As its default mode of operation, the CNC of this example uses management of IEEE Std 802.1AS to disable operation on Ports outside of the TSN domain. For example, assume that the CNC detects two distinct TSN domains G and H with respect to Talkers and Listeners (i.e., no Stream from G to H or vice versa). There is a Bridge between TSN domains G and H, and that Bridge supports IEEE Std 802.1AS. Rather than use a single IEEE 802.1AS domain that spans G and H, the CNC disables IEEE Std 802.1AS on the Ports between G and H; this action effectively creates two distinct IEEE 802.1AS domains, one for each TSN domain.

8) CNC configures TSN features for Streams.

The CNC uses the management of IEEE Std 802.1Q to configure TSN features in the path(s) from Talker to Listeners. Examples include enhancements for scheduled traffic (8.6.8.4), frame preemption (6.7.2), and the credit-based shaper (8.6.8.2).

9) CNC returns the Status of each Stream to the CUC.

This step includes the success/failure of each Stream's configuration, the AccumulatedLatency, and the InterfaceConfiguration for each end station.

1 **10) CUC configures each end station.**

2 If one or more Streams fail configuration, the CUC might decide to adjust its requirements and try
3 again (i.e., return to step 3).

4 Assuming that the Stream status is sufficient to proceed, the CUC configures each end station to
5 prepare it for application execution. Although the communication between CUC and end station is
6 not directly related to Clause 46, the CUC provides the following information for each Stream:

7 — InterfaceConfiguration: Provides the destination MAC address and/or VLAN ID (if needed).

8 **11) CUC executes the distributed application.**

9 If there are other CUCs for other applications, the CNC can continue to configure Bridge resources
10 for those while this application runs.

11 **12) CUC stops the distributed application.**

12 The CUC can send Leave requests (46.2.2) for Streams that are no longer needed.

13 If the application is changed, this workflow can return to step 1, step 2, or step 3.

1 **Annex V**

2 (informative)

3 **Asynchronous Traffic Shaping delay analysis framework**

4 The framework in this annex provides methods for worst-case delay analysis in static networks with static
5 configurations. General assumptions of this framework are listed in V.1. The end-to-end delay modeling
6 approach is described in V.2. An upper bound on buffering delays is described in V.3. Subsequent clauses
7 (V.4 through V.8) cover additional sources of delays. The combined delay bounds are shown in V.9.

8 **V.1 General assumptions**

9 The following assumptions are made throughout the remainder of this framework:

- 10 a) The Transmission Selection Table (8.6.8) of all transmission Ports under consideration assigns the
11 ATS Transmission Selection Algorithm (Table 8-6, 8.6.8.5) to one or more numerically highest
12 traffic classes (i.e., no other algorithm than the ATS Transmission Selection Algorithm is assigned to
13 a numerically higher traffic class than the traffic classes to which the ATS Transmission Selection
14 Algorithm is assigned).
- 15 b) The transmission gates (8.6.8.4) associated with all traffic classes using the ATS Transmission
16 Selection Algorithm of all transmission Ports under consideration reside permanently in state Open.
- 17 c) All streams are associated with ATS traffic classes in all transmission Ports.
- 18 d) If frame preemption (6.7.2) is supported by a transmission Port under consideration, it is assumed
19 that the frame preemption status in the frame preemption status table associated with the
20 transmission Port in the network is either express, or preemptable for all traffic classes using the
21 ATS Transmission Selection Algorithm.
- 22 e) The underlying MAC Service of all transmission Ports under consideration transmits at a constant
23 data rate.
- 24 f) The committed information rate parameters in Bridges under consideration are set consistently with
25 the constraints in V.8.
- 26 g) The data rate of a Port under consideration is greater than the sum of associated committed
27 information rates.
- 28 h) Frames at all transmission Ports have the same media-dependent overhead (12.4.2.2). Likewise,
29 there is no variation in frame lengths (i.e., tag addition or removal along a path is not considered).

30 **V.2 End-to-end delay modeling approach**

31 The path from the Talker station to a single Listener station is subdivided into the contained n hops, where
32 the 1st hop is the hop from the Talker to the first Bridge on the path, and the n th hop is the hop from the last
33 Bridge on the path to the Listener.

34 The end-to-end delay bound $d_{\max}(f)$ of a stream of interest is given by per hop delay bounds $d_{\max}(k, f)$ of
35 the n subsequent hops along the path of stream f as shown in Equation (V-1).

$$36 \quad d_{\max}(f) = \sum_{k=1}^n d_{\max}(k, f) \quad (\text{V-1})$$

1 where

- 2 f is the stream of interest
 3 k is the hop index at the path from the Talker to the Listener ($1 \leq k \leq n$)
 4 $d_{\max}(k, f)$ is the per hop delay bound of stream f at the k th hop

5 Figure V-1 illustrates the path of stream f across the relevant mechanisms and interfaces of two subsequent
 6 bridges along the path, and the associated delay bounds [e.g., $d_{AT, \max}(k)$, introduced in V.5] described in
 7 the following subclauses.

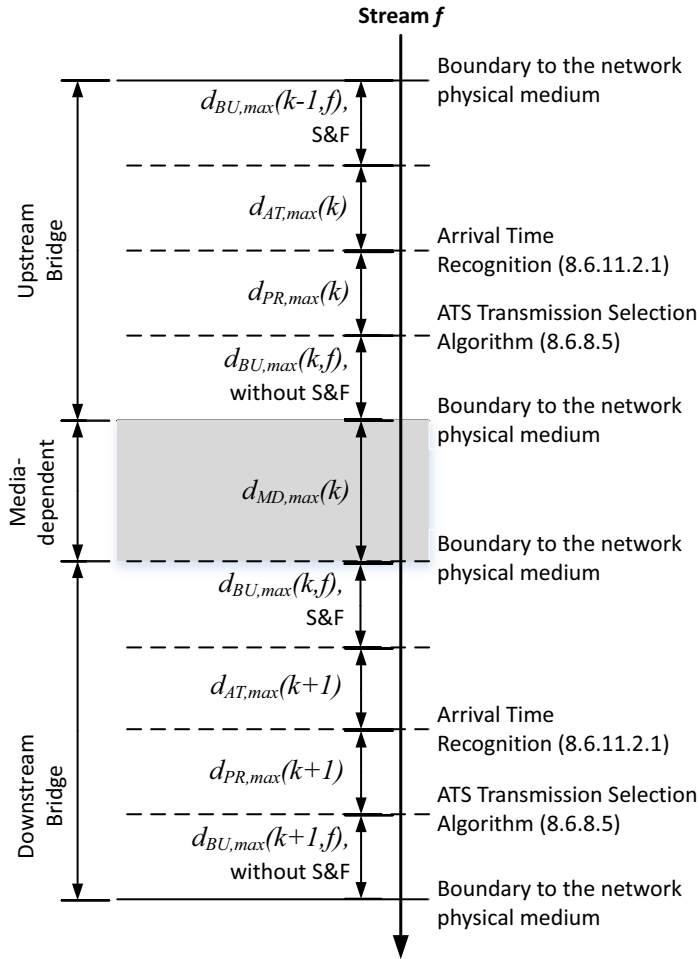


Figure V-1—Path of frames along a single hop with index k with two Bridges

8 V.3 Buffering delays

9 The buffering delays comprise all delays on frames of a stream f resulting from their residence in queues of
 10 associated traffic classes in the transmission Ports along the path. For a transmission Port on the path, these
 11 delays are caused by:

- 12 a) Competing frames in queues of traffic classes different than the traffic class of stream f .
- 13 b) Competing frames in the queue of the traffic class of stream f competing for transmission.
- 14 c) The enforcement of traffic envelopes by the operation of the ATS scheduler state machines (8.6.11).
- 15 d) Store and forward operation in reception Ports.

A delay bound $d_{BU, \max}(k, f)$ on the buffering delay for a single hop in absence of the effects covered in subsequent clauses is known (see Specht and Samii [B66]). For $k < n$ (i.e., all but the last hop to the Listener), the delay bound $d_{BU, \max}(k, f)$ is given by Equation (V-2).

$$d_{BU, \max}(k, f) = \max_{h \in F_S(k, f)} \left\{ \frac{\sum_{g \in F_H(k, h) \cup F_S(k, h)} b_{\max}(k, g) - l_{\min}(h) + l_{LP, \max}(k, h)}{R(k) - \sum_{g \in F_H(k, h)} r_{\max}(k, g)} + \frac{l_{\min}(h)}{R(k)} \right\} \quad (V-2)$$

For $k = n$ (i.e., the last hop to the Listener), this bound is given by Equation (V-3).

$$d_{BU, \max}(n, f) = \frac{\sum_{g \in F_H(n, f) \cup F_S(n, f)} b_{\max}(n, g) - l_{\min}(f) + l_{LP, \max}(n, f)}{R(n) - \sum_{g \in F_H(n, f)} r_{\max}(n, g)} + \frac{l_{\min}(f)}{R(n)} \quad (V-3)$$

where

$F_H(k, f)$ and $F_H(k, h)$	denote the set of streams transmitted in a numerically higher traffic class (8.6.8) than stream f and a stream h , respectively, at the upstream transmission Port of the k th hop
$F_S(k, f)$ and $F_S(k, h)$	denote the set of streams transmitted in the same traffic class as stream f , including stream f and stream h , respectively, at the upstream transmission Port of the k th hop
$l_{LP, \max}(k, f)$ and $l_{LP, \max}(k, h)$	denote the maximum interference length, in bits, by any numerically lower traffic class than the class of stream f and a stream h , respectively, at the upstream transmission Port of the k th hop
$l_{\min}(f)$ and $l_{\min}(h)$	denote the minimum frame length of stream f and a stream h , respectively, in bits, including all media-dependent overhead (8.6.11.3.11, 12.4.2.2)
$b_{\max}(k, g)$	is the maximum burst size associated with a stream g at the k th hop, in bits
$r_{\max}(k, g)$	is the committed information rate of stream g at the upstream device of the k th hop, in bits per second
$R(k)$	is the transmission rate, in bits per second, that the underlying MAC Service that supports transmission through the upstream transmission Port of the k th hop provides

NOTE 1—The relationship between $b_{\max}(k, g)$ and the committed burst size at the Talker station is further discussed in V.7.

NOTE 2—The relationship between $r_{\max}(k, g)$ and the committed information rate at the Talker station (47.2) is further discussed in V.8.

NOTE 3—If frame preemption is not supported, $l_{LP, \max}(k, f)$ is at most one maximum frame length, including all media-dependent overhead, supported by the upstream transmission Port of the k th hop. If frame preemption is supported, and all classes with the ATS transmission selection algorithm can preempt all numerically lower traffic classes, $l_{LP, \max}(k, f)$ is at most a maximum fragment length (S.2), including all media-dependent overhead.

1 V.4 Media-dependent delays

2 Frames of stream f experience a media-dependent delay between the upstream transmission Port of this hop
3 and the downstream reception Port of a hop. This delay is measured between:

- 4 a) The time at which a particular octet of a frame passed the boundary from the upstream transmission
5 Port to the network physical medium and
- 6 b) The time at which this particular octet passed the boundary from the network physical medium to
7 the downstream Port.

8 The maximum media-dependent delay of the k th hop is denoted as $d_{MD, \max}(k)$ and assumed to be known
9 for analysis.

10 Variations in media-dependent delays do not affect the combined delay bounds in V.9. Such variations
11 happen before the arrival time recognition by the associated ATS scheduler clock and the processing by the
12 associated ATS scheduler instance. A frame that experiences a lower media-dependent delay than an earlier
13 frame processed by the same scheduler instance would be delayed by scheduler, if required by envelope
14 enforcement.

15 V.5 Bridge—Internal arrival time recognition delays

16 The maximum arrival time recognition delay in the upstream Bridge of the k th hop ($k > 1$) for any frame of
17 a stream is given by the associated ArrivalRecognitionDelayMax parameter (8.6.11.3.1, 12.31.8.5). For
18 compact representation and notational consistency, ArrivalRecognitionDelayMax is subsequently denoted
19 as $d_{AT, \max}(k)$.

20 Variations in arrival time recognition delays do not affect the combined delay bounds in V.9, similar to
21 variations in media-dependent delays (V.4).

22 V.6 Bridge—Internal processing delays

23 The bounds on the processing delays in the upstream Bridge of the k th hop ($k > 1$) for any frame of a stream
24 are given by the associated ProcessingDelayMin and ProcessingDelayMax parameters (8.6.11.3.2,
25 12.31.8.6, 12.31.8.7). The associated ProcessingDelayMin and ProcessingDelayMax parameters are
26 subsequently denoted as $d_{PR, \min}(k)$ and $d_{PR, \max}(k)$, respectively.

27 Variations in Bridge-internal processing delays can increase the burst sizes in buffering delays (V.3). Based
28 on the specified computation of the assigned eligibility time (8.6.11.3.2), these variations reside within the
29 associated clock offset variations (V.7).

30 NOTE—The total of the maximum arrival time recognition delay and the maximum processing delay is different from
31 the associated maximum independent delay found in the Bridge Delay attributes (12.32.1.1). This total can be larger.

1 V.7 Bridge—Internal clock offset variations

2 The maximum clock offset variation in the upstream Bridge of the k th hop ($k > 1$) for any frame of a stream
3 is given by the associated ClockOffsetVariationMax parameter (8.6.11.2, 12.31.8.3). The associated
4 ClockOffsetVariationMax is subsequently denoted as $\Delta_{CO,max}(k, g)$.

5 Clock offset variations of up to $d_{AT,max}(k)$ between ATS scheduler clock and transmission selection clock
6 instances (8.6.11.2) associated with a stream g can increase the burst size of streams subsequently competing
7 in the transmission Port.

8 This impact can be taken into account by definition of $b_{max}(k, g)$ for $k > 1$, as shown in Equation (V-4).

$$9 \quad b_{max}(k, g) = r_{max}(k, g)\Delta_{CO,max}(k, g) + b_{max}(1, g) \quad (V-4)$$

10 where $b_{max}(1, g)$ is the committed burst size of stream g at the Talker station (47.2).

11 V.8 Inter-device clock rate deviations

12 The clock constraints in 8.6.11 limit ATS scheduler clock instances and transmission selection clock
13 instances in a Bridge to effectively operate at the same rate, although differences within
14 [ClockOffsetMin, ClockOffsetMax] are permitted (8.6.11.2). As a basic property of asynchronous
15 mechanisms such as ATS, no such limitation exists between different devices (i.e., clocks in different
16 devices are not synchronized).

17 Deviations of clocks from their nominal rates (e.g., within oscillator tolerances) affect the spacing between
18 successive frames according to the assigned eligibility times (8.6.11.3.2, 8.6.8.5). If the upstream device at a
19 hop runs faster than nominal (e.g., +100 ppm), and a connected downstream Bridge at this hop runs slower
20 than nominal (e.g., -100 ppm), the backlog as well as the per hop delay in the downstream Bridge could
21 grow under peak load conditions.

22 The situation can be prevented by management constraints (12.31.5) on $r_{max}(k, g)$ for $1 < k < n$, such that
23 Equation (V-5) holds for any stream g .

$$24 \quad r_{max}(k, g) \geq \frac{1 + 10^{-6}\Delta_{CR,max}(k-1)}{1 - 10^{-6}\Delta_{CR,max}(k)} r_{max}(k-1, g) \quad (V-5)$$

25 where

26 $\Delta_{CR,max}(k)$ is the upper bound over all absolute rate deviations of all ATS scheduler clocks and
27 transmission selection clocks from their nominal rate in the upstream device of the k th hop
28 available via the associated ClockRateDeviationMax parameter (12.31.8.4), in ppm (e.g.,
29 $\Delta_{CR,max}(k) = 100$)
30 $r_{max}(1, g)$ is the committed information rate of stream g at the Talker station (47.2)

V.9 Combined delay bounds

For a path from a Talker station to a Listener station with at least one Bridge ($n > 1$), a combined end-to-end delay bound $d_{\max}(f)$ of a stream of interest f can be summarized by Equation (V-6)

$$d_{\max}(f) = \sum_{k=1}^n d_{BU, \max}(k, f) + \sum_{k=1}^n d_{MD, \max}(k) + \sum_{k=2}^n d_{AT, \max}(k) + \sum_{k=2}^n d_{PR, \max}(k) \quad (V-6)$$

with Equation (V-7), Equation (V-8), and Equation (V-9) for $1 < k < n$.

$$d_{BU, \max}(1, f) = \max_{h \in F_S(1, f)} \left\{ \frac{\sum_{g \in F_H(1, h) \cup F_S(1, h)} b_{\max}(1, g) - l_{\min}(h) + l_{LP, \max}(1, h)}{R(1) - \sum_{g \in F_H(1, h)} r_{\max}(1, g)} + \frac{l_{\min}(h)}{R(1)} \right\} \quad (V-7)$$

$$d_{BU, \max}(n, f) = \frac{\sum_{g \in F_H(n, f) \cup F_S(n, f)} (r_{\max}(n, g) \Delta_{CO, \max}(n, g) + b_{\max}(1, g)) - l_{\min}(f) + l_{LP, \max}(n, f)}{R(n) - \sum_{g \in F_H(n, f)} r_{\max}(n, g)} + \frac{l_{\min}(f)}{R(n)} \quad (V-8)$$

$$d_{BU, \max}(k, f) = \max_{h \in F_S(k, f)} \left\{ \frac{\sum_{g \in F_H(k, h) \cup F_S(k, h)} (r_{\max}(k, g) \Delta_{CO, \max}(k, g) + b_{\max}(1, g)) - l_{\min}(h) + l_{LP, \max}(k, h)}{R(k) - \sum_{g \in F_H(k, h)} r_{\max}(k, g)} + \frac{l_{\min}(h)}{R(k)} \right\} \quad (V-9)$$

In absence of Bridges on the path from the Talker station to the Listener station (i.e., $n = 1$), the end-to-end delay bound can be summarized by Equation (V-10).

$$d_{\max}(1, f) = \frac{\sum_{g \in F_H(1, f) \cup F_S(1, f)} b_{\max}(1, g) - l_{\min}(f) + l_{LP, \max}(1, f)}{R(1) - \sum_{g \in F_H(1, f)} r_{\max}(1, g)} + \frac{l_{\min}(f)}{R(1)} + d_{MD, \max}(1) \quad (V-10)$$

1 Annex W

2 (informative)

3 Maintaining frame order with Congestion Isolation

4 The process of congestion isolation involves identifying the frames of a congesting flow and subsequently
5 modifying the egress traffic class of those frames based on the level of congestion in the monitored and
6 congesting queues. During this process, it is possible that frames for the same flow can exist in multiple
7 queues at the same time, resulting in the possibility of an out-of-order frame delivery. As an illustration,
8 consider the following example depicted in Figure W-1.

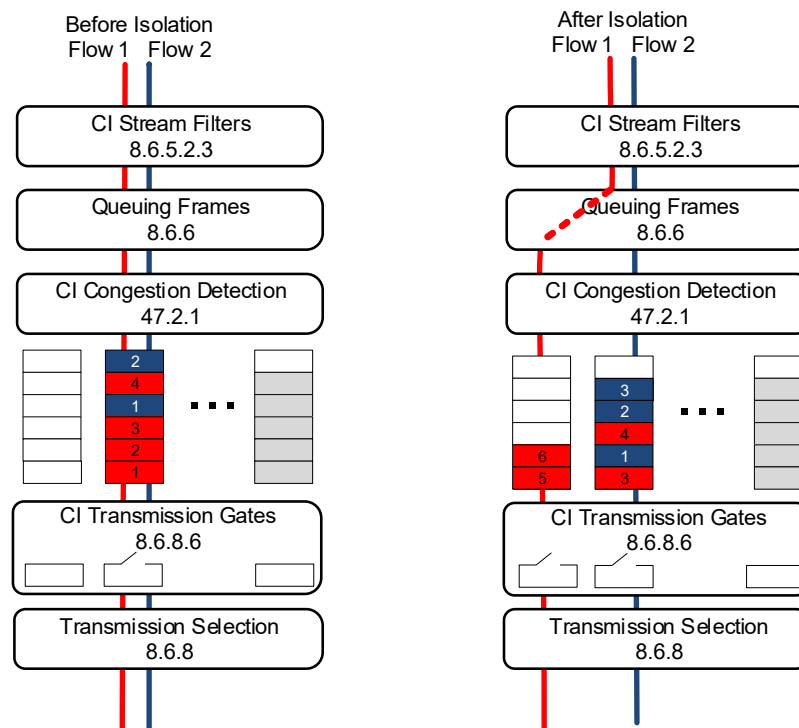


Figure W-1—Isolation out-of-order frame example

9 In the example in Figure W-1, the frames of two flows, red and blue, are intermixed while traversing a
10 common monitored queue. As the monitored queue fills, the red flow is determined to be a congesting flow
11 and subsequent frames of that flow will be reclassified and queued in the congesting queue. Previously
12 received frames for that flow, numbered 1 through 4, may reside in the monitored queue. Since the
13 congesting queue is empty, the subsequent frames, numbered 5 and 6, are placed at the head of the
14 congesting queue. Depending upon the traffic selection algorithm, it may be possible for frames 5 and 6 to
15 be selected for transmission before some of frames 1 through 4.

16 The priority of the congesting queue is lower than the priority of the monitored queue. The strict priority
17 transmission selection algorithm (8.6.8.1) will ensure that no out-of-order frame delivery occurs, however
18 there is a risk of starvation for congesting flows and alternative traffic selection algorithms may be desired.
19 The enhanced transmission selection algorithm (8.6.8.3) or other vendor specific algorithms may not ensure
20 no out-of-order delivery without special consideration.

While the strict priority transmission selection algorithm can ensure in-order delivery when congesting flows are isolated to the congesting queue, it does not ensure in-order delivery when those flows are returned to non-congesting status except in the case where both the congesting and monitored queues themselves become empty. Implementations may want to free resources and reduce flow completion time by returning flows to non-congesting status as soon as possible. This, however, may create another situation where frames from the same flow exist in both the congesting and monitored queue at the same time. Figure W-2 shows an example of possible out-of-order frame delivery when de-isolating a flow (i.e., returning the flow to non-congesting status).

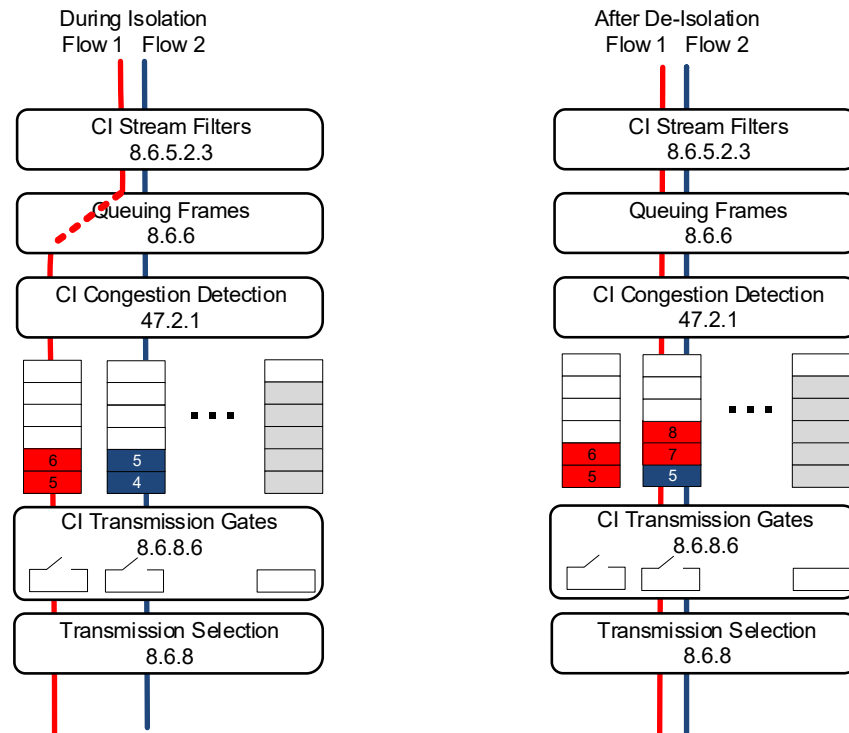


Figure W-2—De-isolation out-of-order frame example

In the example in Figure W-2, a red flow has been isolated to the congesting queue and the blue flow is traversing the monitored queue. Since the queue occupancy is low in both queues, the implementation may decide to de-isolate the red flow. Subsequent frames 7 and 8 will now be placed in the monitored queue resulting in frames from the red flow existing in both queues at the same time. The monitored queue is scheduled at a higher priority or with a greater frequency than the congesting queue and it may become possible for frames 7 and 8 to exit the system ahead of frames 5 and 6. Waiting for the congesting queue to completely empty before de-isolating any congesting flows can ensure in-order delivery during de-isolation, but it may leave flows in a congesting state longer than necessary and makes the transition of one congesting flow dependent upon all other congesting flows.

Congestion isolation defines a transmission gate (8.6.8.6) for the monitored and congesting queues that make those queues available to the transmission selection algorithm. When the transmission selection algorithm is strict priority, the transmission gate is permanently *open*. The transmission gate is controlled by the `cipGateControl` variable for other transmission selection algorithms that can not ensure in-order delivery at all times. The management of `cipGateControl` is implementation dependent, but must be asserted in a way to ensure the externally visible behavior of the Bridge supporting congestion isolation is to maintain frame order.

The following informative description provides example mechanisms to preserve frame order when isolating and de-isolating flows.

1 W.1 Queue markers for order preservation

2 The mechanism below provides control of the cipGateControl variable for a congesting queue in order to
3 preserve the order of frames for a contested flow. It involves a queue position marker and marker counter for
4 both the monitored and congesting queues. The mechanism is described using the example shown in
5 Figure W-3.

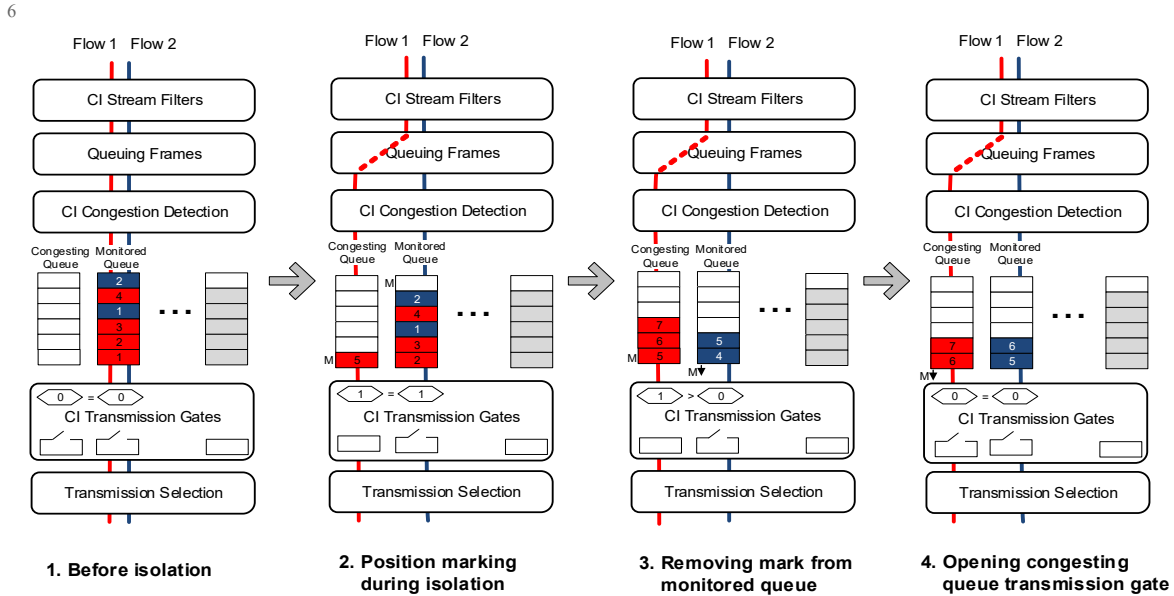


Figure W-3—Using queue markers and counters to preserve order when isolating

7 The example in Figure W-3 shows the state of the position markers and marker counters during four
8 different phases of congestion isolation operation; before a flow is isolated, position marking during the
9 isolation of a flow, the closing of the congesting queue transmission gate, and the opening of the congesting
10 queue transmission gate. When a queue transmission gate is open, that queue is available to the traffic
11 selection algorithm. When it is closed, the queue is not available to the traffic selection algorithm.

12 In the example, the frames of two flows are intermixed in the monitored queue as they traverse the Bridge.
13 Since no flows have been isolated yet, the position marker counters of both the congesting queue and the
14 monitored queue are set to 0. Once a flow has been isolated and subsequent frames of that flow are placed in
15 the congesting queue, a marker is placed in both the congesting queue and the monitored queue. The marker
16 counter is incremented for both queues once the position is marked. The congesting queue is empty at the
17 time the first flow is isolated, so the marker will be at the head of the queue. When a marker is at the head of
18 the congesting queue and the marker counters are equal, the cipGateControl variable for the congesting
19 queue is set to closed. The monitored queue continues to drain and eventually the position marker will reach
20 the head of the monitored queue. The marker counter for the monitored queue will be decremented when the
21 frame associated with the position marker is scheduled for transmission and exits the queue. When the value
22 of the congesting queue marker counter is greater than the monitored queue position counter, it is possible to
23 set the cipGateControl variable to open and begin to schedule the congesting queue. Whenever a frame that
24 aligns with a position marker is scheduled for transmission, the associated marker counter is decremented.

25 A variation of the mechanism above can be used to ensure in-order delivery when de-isolating a flow
26 (i.e., returning a flow to non-congesting status). A separate set of counters and markers can be used to
27 identify the positions in the queues when a congesting flow is allowed to begin to use the monitored queue
28 again.

1

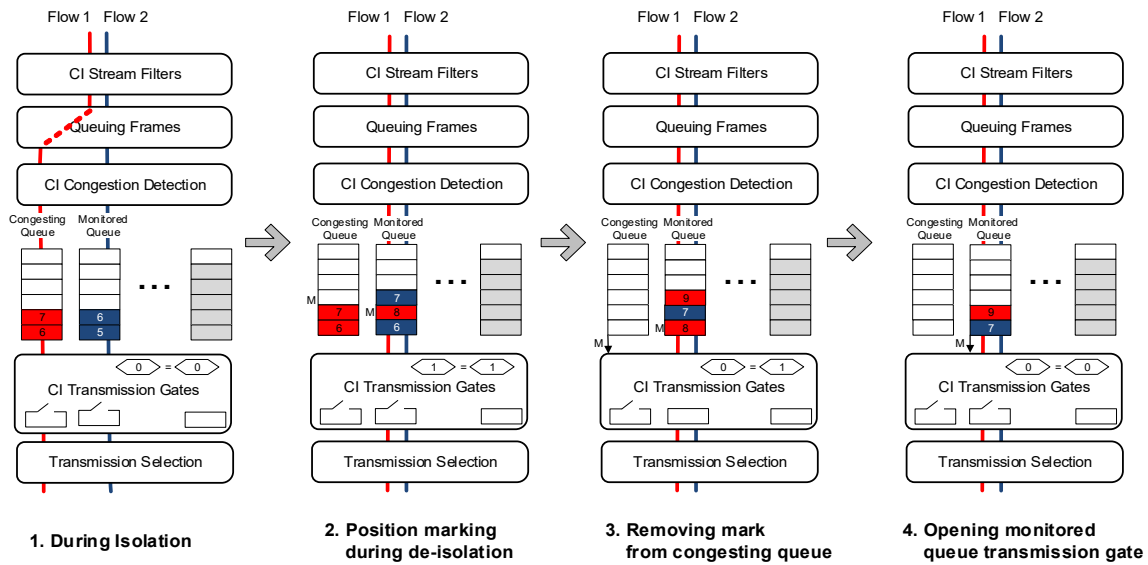


Figure W-4—Using queue markers and counters to preserve order when de-isolating

2 The example in Figure W-4 shows the state of a separate set of position markers and marker counters during
3 four different phases of the congestion de-isolation process; during isolation, during initial de-isolation,
4 closing the monitored queue, and opening the monitored queue once order has been ensured. In the example,
5 the red flow is currently isolated to the congesting queue and the blue flow is traversing the monitored
6 queue. Since there are no flow ordering issues the counters are initialized to 0. When the implementation
7 decides to de-isolate the red flow, it places a marker after the last frame of the red flow in the congesting
8 queue and with the next frame of the congesting flow in the monitored queue. The counters are incremented
9 each time a marker is placed in the queue. The queue scheduling continues when the counters are equal and
10 no markers are at the head of the monitored queue. Once the marker of the monitored queue reaches the head
11 of the queue, the transmission gate for the monitored queue is closed by setting the `cipGateControl` variable
12 to *closed*. The lower priority congesting queue is scheduled, allowing the previously isolated frames to be
13 transmitted. Once the marker in the congesting queue is removed and the counter is decremented, the
14 monitored queue gate is opened by setting `cipGateControl` to *open* and the normal scheduling continues.

15 The marker and counter mechanism described in these examples is fairly simple, but care must be taken
16 when combining the mechanisms for isolation and de-isolation to ensure no deadlocks exist. Implementation
17 details depend on queue and memory architectures and are beyond the scope of this standard.

18 W.2 Congestion Isolation queuing and Priority-based Flow Control

19 Priority-based Flow Control (PFC) (Clause 36) is used to create a lossless network on the traffic classes for
20 which it is enabled. PFC, as defined by this standard, is intentionally ambiguous about when the PFC
21 Initiator entity (36.2.1) generates `M_CONTROL` PFC requests using the `M_CONTROL.request` primitive.
22 This allows implementation flexibility, as there are many possible buffering and queuing approaches used in
23 practice. Some Bridges may be input buffered, others may be primarily output buffered with small ingress
24 buffers, some may have shared memory and others may use combinations of different memory architectures.
25 In each implementation, the events that cause the PFC Initiator entity to generate an `M_CONTROL` PFC
26 request may be different. In all cases, however, the purpose of PFC is to avoid dropping frames, not only
27 within a single Bridge but across the entire network. The PFC Initiator is expected to generate an
28 `M_CONTROL` PFC request when internally Bridge buffering is not able to absorb the reception of
29 additional frames. The `M_CONTROL` PFC request is expected to eliminate the chance of frame loss.

PFC is known to cause congestion spreading and has recommended use within the data center because of its limited extent (36.1.1). One of the key objectives for congestion isolation is to reduce the frequency of PFC requests and avoid head-of-line blocking in lossless data center networks. By reducing the frequency of PFC requests the impact of congestion spreading can be reduced. Congestion isolation does this by isolating congesting flows to a separate congesting queue, reducing the need to invoke PFC requests caused by congesting flows. During the isolation process, however, it is possible for frames of the same flow to be in multiple queues at the same time, both within the local system and the upstream peer. This scenario can require special consideration for the design of the PFC Initiator.

Consider the Bridge architecture shown in Figure W-5. In the figure, the 4-port Bridge has small ingress buffers, shown on port 1, designed to absorb received frames while forwarding decisions are made. There are two priorities represented in the figure. In this example, priority 3 is the congesting queue and priority 4 is the monitored queue. In the example, frames are being forwarded from port 1 to the egress buffers of port 3. The shaded buffers are assumed to be occupied.

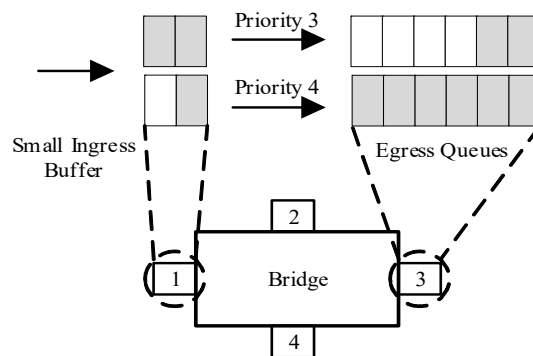


Figure W-5—Example Bridge buffering supporting PFC and CI

When congestion isolation occurs, as seen in Figure W-6, frames received by the downstream Bridge will begin to be placed in the congesting queue (priority 3). A CIM is sent upstream to inform the peer to isolate the same flow. The upstream peer has several frames queued in the monitored queue and will continue transmission while the CIM is in flight.

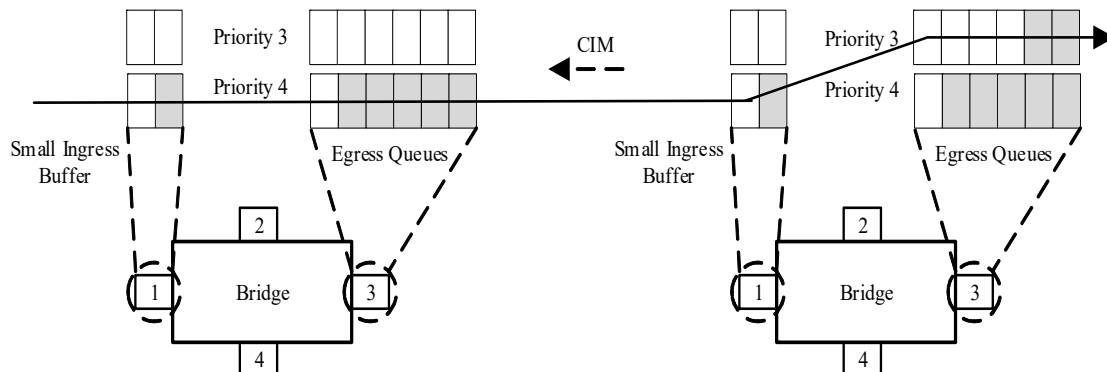


Figure W-6—Example CI initiation by downstream peer

During the isolation process, as seen in Figure W-7, frames will continue to drain from the monitored queue of the upstream peer and will begin to accumulate in the congesting queue. Since the monitored queue has higher priority than the congesting queue, and to avoid delivering out-of-order frames, the scheduling of the congesting queue in the upstream peer is blocked. Frames from the congesting flow accumulate in the congesting queue of the downstream Bridge while previous frames begin to drain from the monitored queue.

1 When multiple flows are sharing the monitored queue the number of frames accumulating in the congesting
2 queue can increase. Since frames for the congesting flow are still in the monitored queue upstream, they will
3 egress the upstream peer at priority 4.

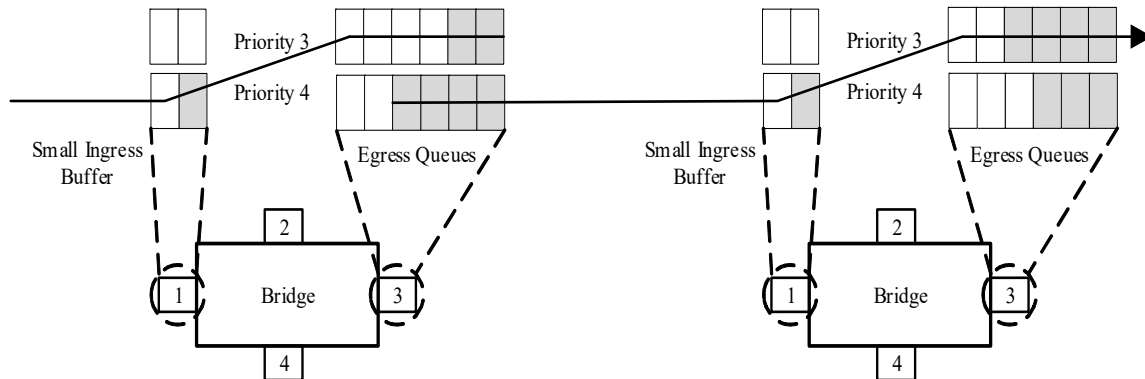


Figure W-7—Example CI in process

4 If the congesting queue is not allowed to drain at a rate faster than it is being filled, the congesting queue can
5 become exhausted, as seen in Figure W-8. Since the congesting queue is at priority 3, the natural response to
6 an exhausted queue in a lossless environment would be to issue a PFC request on priority 3. However, in the
7 example, there are still frames for the congesting flow egressing the upstream peer at priority 4. A pause on
8 priority 3 will have no effect. Additional frames received at priority 4 on the downstream Bridge may be
9 dropped unless a PFC request is issued at priority 4.

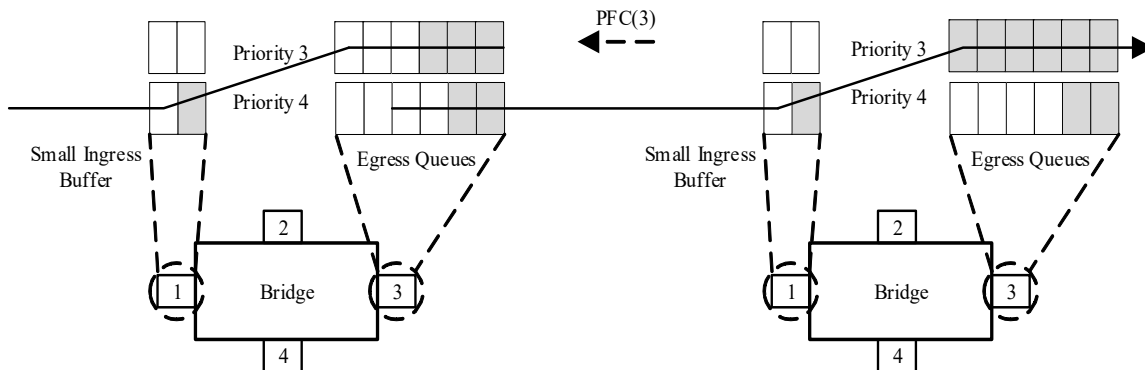


Figure W-8—Example PFC request for congesting queue

10 The problem with the example in Figure W-8 is that the PFC Initiator is considering the occupancy of the
11 egress queue as the stimulus for issuing the PFC without considering the frame's priority at ingress. The
12 PFC design in this example is not taking into account that a frame's priority may be changed by congestion
13 isolation after reception.

14 One solution to the problem is for the Bridge to hold the frame at ingress until there is space at egress to
15 queue it. The PFC Initiator can be solely based on the occupancy of the ingress buffers in this case. A PFC
16 request would only be issued for the priority of received frames that have no space in their ultimate egress
17 queue. Obviously it is critical that the downstream Bridge can determine the traffic class used by the
18 upstream peer to transmit the frame.

1

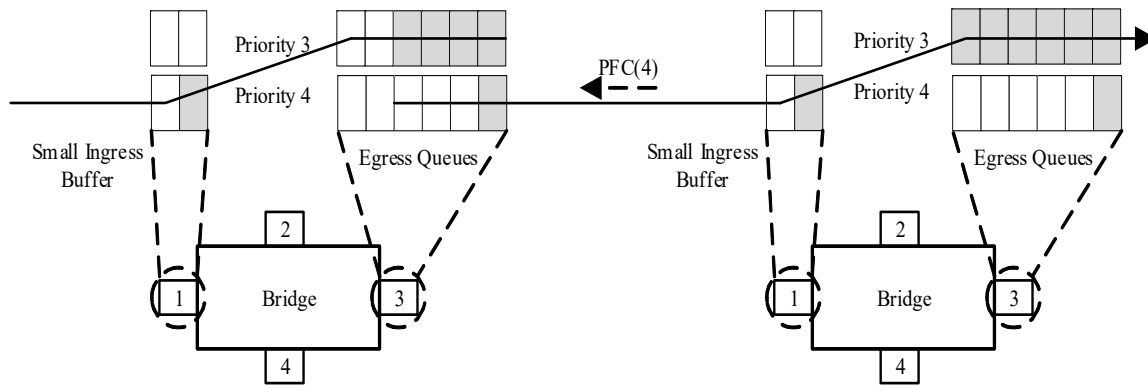


Figure W-9—Example PFC request to avoid packet loss with CI enabled

2 Figure W-9 shows the scenario where the PFC request is issued on priority 4 to prevent frame loss within the
3 downstream Bridge even though those frames will egress the downstream Bridge at priority 3. This scenario
4 avoids internal packet loss, but has the unfortunate consequence of blocking the monitored queue and
5 potentially creating head-of-line blocking for other non-isolated flows. While this unfortunate scenario only
6 exists in a lossless environment, it defeats the purpose of congestion isolation. The scenario does not create
7 an erroneous situation and is expected to be rare.

1 **Annex X**

2 (informative)

3 **Bibliography**

4 Bibliographical references are resources that provide additional or helpful material but do not need to be
5 understood or used to implement this standard. Reference to these resources is made for informational use
6 only.

7 [B1] Alizadeh, M., B. Atikoglu, A. Kabbani, A. Lakshmikantha, R. Pan, B. Prabhakar, and M. Seaman,
8 “Data Center Transport Mechanisms: Congestion Control Theory and IEEE Standardization,” *Proceedings*
9 *of the 46th Annual Allerton Conference on Communication, Control and Computing*, Urbana-Champaign,
10 Sept. 2008.

11 [B2] Asynchronous Transfer Mode (ATM): A collection of equipment and standards used for
12 telecommunications and data transfer, <https://www.itu.int/ITU-T/> and <https://www.broadband-forum.org>.

13 [B3] Calculating the Delay Added by Qav Stream Queue,
14 <https://www.ieee802.org/1/files/public/docs2009/av-fuller-queue-delay-calculation-0809-v02.pdf>.

15 [B4] Duato, J., “A Necessary and Sufficient Condition for Deadlock-Free Routing in Cut-Through and
16 Store-and-Forward Networks,” *IEEE Transactions on Parallel and Distributed Systems*,” vol. 7, no. 8, pp.
17 841–854, Aug. 1996. doi:10.1109/71.532115.

18 [B5] Hu, S., Y. Zhu, P. Cheng, C. Guo, K. Tan, J. Padhye, and K. Chen “Tagger: Practical PFC Deadlock
19 Prevention in Data Center Networks,” *Proceedings of the 13th International Conference on emerging*
20 *Networking EXperiments and Technologies (CoNEXT '17)*, ACM, New York, NY, USA, pp. 451–463.
21 doi:10.1145/3143361.3143382.

22 [B6] IEC 62439-3:2016, Industrial communications networks—High availability automation
23 networks—Part 3: Parallel Redundancy Protocol (PRP) and High-availability Seamless Redundancy
24 (HSR).⁶⁷

25 [B7] IEEE Std 802[®]-2014, IEEE Standard for Local and Metropolitan Area Networks—Overview and
26 Architecture.^{68, 69}

27 [B8] IEEE Std 802.1AB™-2005, IEEE Standard for Local and metropolitan area networks—Station and
28 Media Access Control Connectivity Discovery.

29 [B9] IEEE Std 802.1AB™-2009, IEEE Standard for Local and metropolitan area networks—Station and
30 Media Access Control Connectivity Discovery.

31 [B10] IEEE Std 802.1AC™-2016, IEEE Standard for Local and metropolitan area networks—Media Access
32 Control (MAC) Service Definition.

33 [B11] IEEE Std 802.1BA™, IEEE Standard for Local and Metropolitan Area Networks—Audio Video
34 Bridging (AVB) Systems.

⁶⁷ IEC publications are available from the International Electrotechnical Commission (<https://www.iec.ch>) and the American National Standards Institute (<https://www.ansi.org/>).

⁶⁸ The IEEE standards or products referred to in Annex X are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

⁶⁹ IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>).

- 1 [B12] IEEE Std 802.1D™, 1993 Edition [ISO/IEC 10038:1993], IEEE Standard for Information
2 technology—Telecommunications and information exchange between systems—Local area
3 networks—Media Access Control (MAC) bridges.
- 4 [B13] IEEE Std 802.1D™, 1998 Edition [ISO/IEC 15802-3:1998], IEEE Standard for Information
5 technology—Telecommunications and information exchange between systems—Local and metropolitan
6 area networks—Common specifications—Part 3: Media Access Control (MAC) Bridges.
- 7 [B14] IEEE Std 802.1D™-2004, IEEE Standard for Local and metropolitan area networks—Media Access
8 Control (MAC) Bridges.
- 9 [B15] IEEE Std 802.3™-2018, IEEE Standard for Ethernet.
- 10 [B16] IEEE Std 802.11™-2016, IEEE Standard for Information technology—Telecommunications and
11 information exchange between systems—Local and metropolitan area networks—Specific
12 requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY)
13 Specifications.
- 14 [B17] IEEE Std 1588-2019, IEEE Standard for a Precision Clock Synchronization Protocol for Networked
15 Measurements and Control Systems.
- 16 [B18] IEEE Std 1722™, IEEE Standard for a Transport Protocol for Time-Sensitive Applications in Bridged
17 Local Area Networks.
- 18 NOTE—See also IEC 61883-6:2005, Consumer audio/video equipment—Digital Interface—Part 6: Audio and music
19 data transmission protocol and IEC 61883-4:2004, Consumer audio/video equipment—Digital Interface—Part 4:
20 MPEG2-TS data transmission.
- 21 [B19] IETF RFC 793 (STD0007), Transmission Control Protocol, September 1981.⁷⁰
- 22 [B20] IETF RFC 1321, The MD5 Message-Digest Algorithm, April 1992.
- 23 [B21] IETF RFC 1633, Integrated Services in the Internet Architecture: an Overview, June 1994.
- 24 [B22] IETF RFC 2210, The Use of RSVP with IETF Integrated Services, September 1997.
- 25 [B23] IETF RFC 2211, Specification of the Controlled-Load Network Element Service, September 1997.
- 26 [B24] IETF RFC 2212, Specification of Guaranteed Quality of Service, September 1997.
- 27 [B25] IETF RFC 2215, General Characterization Parameters for Integrated Service Network Elements,
28 September 1997.
- 29 [B26] IETF RFC 2475, An Architecture for Differentiated Services, December 1998.
- 30 [B27] IETF RFC 2597, Assured Forwarding PHB Group, June 1999.
- 31 [B28] IETF RFC 2814, SBM (Subnet Bandwidth Manager): A Protocol for RSVP-based Admission Control
32 over IEEE 802-style Networks, May 2000.
- 33 [B29] IETF RFC 2815, Integrated Service Mappings on IEEE 802 Networks, May 2000.
- 34 [B30] IETF RFC 2816, A Framework for Integrated Services Over Shared and Switched IEEE 802 LAN
35 Technologies, May 2000.
- 36 [B31] IETF RFC 3031, Multiprotocol Label Switching Architecture, January 2001.
- 37 [B32] IETF RFC 3246, An Expedited Forwarding PHB (Per-Hop Behavior), March 2002.

⁷⁰ IETF RFCs are available from the Internet Engineering Task Force (<https://www.ietf.org/>).

- ¹ [B33] IETF RFC 3270, Multi-Protocol Label Switching (MPLS) Support of Differentiated Services,
² May 2002.
- ³ [B34] IETF RFC 4655, A Path Computation Element (PCE)-Based Architecture, August 2006.
- ⁴ [B35] IETF RFC 4663, Transferring MIB Work from IETF Bridge MIB WG to IEEE 802.1 WG,
⁵ September 2006.
- ⁶ [B36] IETF RFC 4960, Stream Control Transmission Protocol, September 2007.
- ⁷ [B37] IETF RFC 5306, Restart Signaling for IS-IS, October 2008.
- ⁸ [B38] IETF RFC 6087, Guidelines for Authors and Reviewers of YANG Data Model Documents,
⁹ January 2011.
- ¹⁰ [B39] IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011.
- ¹¹ [B40] IETF RFC 6242, Using the NETCONF Protocol over Secure Shell (SSH), June 2011.
- ¹² [B41] IETF RFC 6536, Network Configuration Protocol (NETCONF) Access Control Model, March 2012.
- ¹³ [B42] IETF RFC 6762, Multicast DNS, February 2013.
- ¹⁴ [B43] IETF RFC 7223, A YANG Data Model for Interface Management, May 2014.
- ¹⁵ [B44] IETF RFC 7319, IANA Considerations for Connectivity Fault Management (CFM) Code Points,
¹⁶ July 2014.
- ¹⁷ [B45] IETF RFC 7567, IETF Recommendations Regarding Active Queue Management, July 2015.
- ¹⁸ [B46] IETF RFC 8014, An Architecture for Data-Center Network Virtualization over Layer 3 (NVO3),
¹⁹ December 2016.
- ²⁰ [B47] IETF RFC 8040, RESTCONF Protocol, January 2017.
- ²¹ [B48] IETF RFC 8341 (STD 91), Network Configuration Access Control Model, March 2018.
- ²² [B49] ITU-T Recommendation G.806, Characteristics of transport equipment—Description methodology
²³ and generic functionality.⁷¹
- ²⁴ [B50] ITU-T Recommendation G.8031/Y.1342, Ethernet linear protection switching.
- ²⁵ [B51] ITU-T Recommendation I.610 (02/1999), B-ISDN operation and maintenance principles and
²⁶ functions.
- ²⁷ [B52] ITU-T Recommendation X.25 (10/1996), Interface between Data Terminal Equipment (DTE) and
²⁸ Data Circuit-terminating Equipment (DCE) for terminals operating in the packet mode and connected to
²⁹ public data networks by dedicated circuit.
- ³⁰ [B53] MEF Technical Specification 4 (MEF 4), Metro Ethernet Network Architecture Framework—Part 1:
³¹ Generic Framework, May 2004.⁷²
- ³² [B54] MEF Technical Specification 16 (MEF 16), Ethernet Local Management Interface (E-LMI),
³³ January 2006.

⁷¹ ITU-T publications are available from the International Telecommunications Union (<https://www.itu.int/>).

⁷² MEF publications are available from the MEF Forum (<https://www.mef.net/>).

- ¹ [B55] MEF Technical Specification 26 (MEF 26), External Network Network Interface (ENNI)—Phase 1,
² January 2010.
- ³ [B56] MEF 35.1, Service OAM Performance Monitoring Implementation Agreement.
- ⁴ [B57] MoCA MAC/PHY Specification Extensions v1.1, MoCA-M/P-SPEC-V1.1-06162009, June 2009.⁷³
- ⁵ [B58] MoCA MAC/PHY Specification v2.0, MoCA_Specification_v2-131121, November 2013.
- ⁶ [B59] Multiprotocol Label Switching (MPLS): A standard for label-based forwarding in an IP network. The
⁷ standard is specified in several RFCs, (see <https://datatracker.ietf.org/doc/charter-ietf-mpls/>) and ITU-T
⁸ recommendations (see <https://www.itu.int/ITU-T/>).
- ⁹ [B60] OMG Unified Modeling Language (OMG UML), Version 2.5.1, December 2017.
- ¹⁰ [B61] Rocher-Gonzalez, J., J. Escudero-Sahuquillo, P. J. Garcia, F. J. Quiles, “On the Impact of Routing
¹¹ Algorithms in the Effectiveness of Queuing Schemes in High-Performance Interconnection Networks,”
¹² 2017 *IEEE 25th Annual Symposium on High-Performance Interconnects (HOTI)*, Santa Clara, CA, 2017,
¹³ pp. 65–72. doi:10.1109/HOTI.2017.16.
- ¹⁴ [B62] Seaman, M., “Preemption and MACsec replay protection,” available at
¹⁵ <https://www.ieee802.org/1/files/public/docs2014/ae-seaman-preemption-1114-v04.pdf>, November 2014.
- ¹⁶ [B63] SMPTE 259M-2008, SMPTE Standard for Television—SDTV Digital Signal/Data—Serial Digital
¹⁷ Interface, 2008. See section 8.⁷⁴
- ¹⁸ [B64] SMPTE 292M-2008, SMPTE Standard 1.5 Gb/s Signal/Data Serial Interface, 2008.
- ¹⁹ [B65] SMPTE 424M-2012, SMPTE Standard for Television—3 Gb/s Signal/Data Serial Interface.
- ²⁰ [B66] Specht, J., and S. Samii, “Urgency-Based Scheduler for Time-Sensitive Switched Ethernet
²¹ Networks,” 28th *Euromicro Conference on Real-Time Systems (ECRTS)*, pp. 75–85, 2016.
- ²² [B67] Teener, M. J., “Peristaltic Shaper: updates, multiple speeds,” available at
²³ <https://www.ieee802.org/1/files/public/docs2014/new-tsn-mjt-peristaltic-shaper-0114.pdf>, January 2014.
- ²⁴ [B68] Xu, L., K. Harfoush, and I. Rhee, “Binary increase congestion control (BIC) for fast long-distance
²⁵ networks,” *IEEE INFOCOM 2004*, vol. 4, pp. 2514–2524, Mar. 2004.
- ²⁶ [B69] Zhu, Y., H.Eran, D. Firestone, C. Guo, M. Lipshteyn, Y. Liron, J. Padhye, S. Raindel, M. H. Yahia,
²⁷ and M. Zhang, 2015. “Congestion Control for Large-Scale RDMA Deployments,” *ACM SIGCOMM*
²⁸ *Computer Communication Review*, 45(4), pp. 523–536. doi:10.1145/2785956.2787484.

29

⁷³ MoCa publications are available from the Multimedia over Coax Alliance (<https://mocalliance.org>).

⁷⁴ SMPTE publications are available from the Society of Motion Picture and Television Engineers (<https://www.smpte.org>).